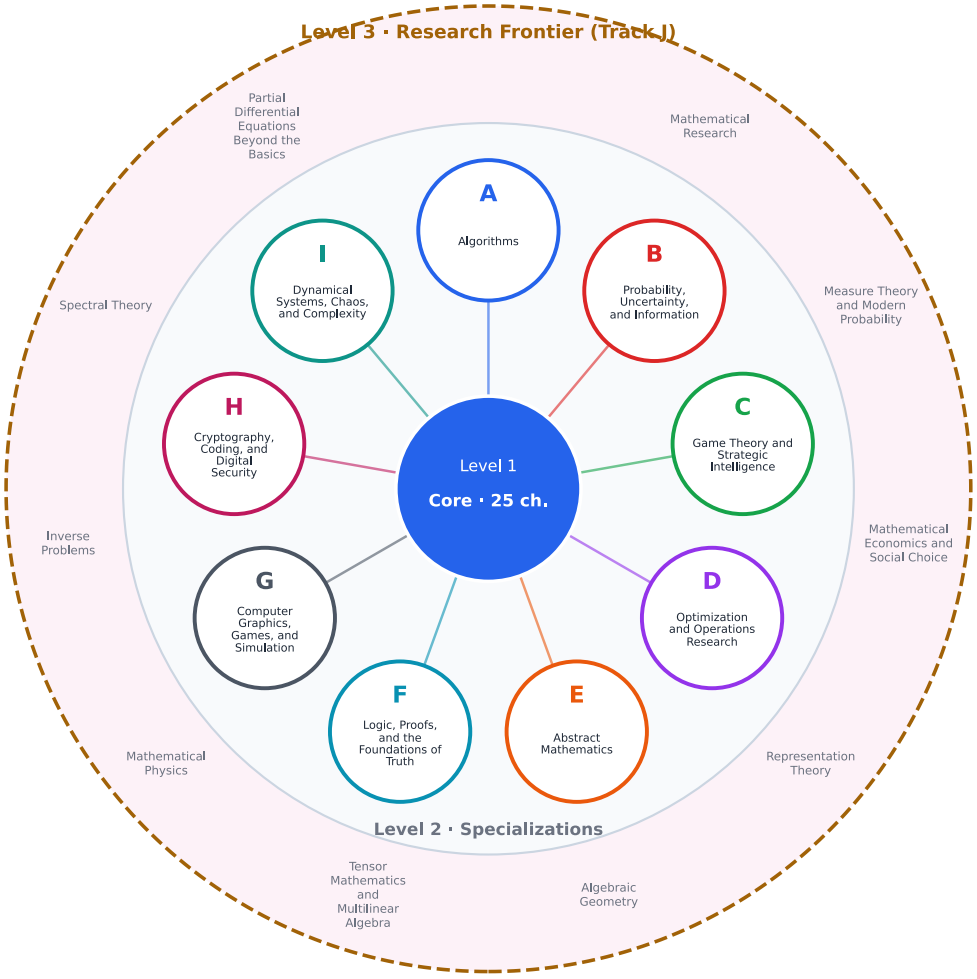
The Mathematics Universe

Engineering Mathematics Core and Beyond



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Notation and Conventions

These conventions hold throughout the book unless a chapter states otherwise.

Numbers and sets

Symbol	Meaning
$\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$	naturals, integers, rationals, reals, complex numbers
j	imaginary unit, $j^2 = -1$ (engineering convention; mathematics texts use i)
z^* or $\bar{z}$	complex conjugate
$\lvert z\rvert,\ \angle z$	magnitude and argument (phase) of z
$\mathrm{Re}\{z\},\ \mathrm{Im}\{z\}$	real and imaginary parts

Scalars, vectors, matrices

Symbol	Meaning
$a,\ x,\ \lambda$	scalars (italic)
$\mathbf{x},\ \mathbf{E}$	vectors (bold lowercase or field quantities)
$\hat{\mathbf{x}},\ \hat{\mathbf{n}}$	unit vectors
$\mathbf{A},\ \mathbf{L}$	matrices (bold uppercase)
$\mathbf{A}^{\mathsf{T}},\ \mathbf{A}^{\mathsf{H}}$	transpose, conjugate (Hermitian) transpose
$\mathbf{I}$	identity matrix
$\langle \mathbf{u}, \mathbf{v} \rangle$	inner product
$\lVert \mathbf{x} \rVert$	norm (Euclidean unless subscripted)

Coordinate systems

System	Coordinates	Notes
Cartesian	(x, y, z)	
Cylindrical	(r, ϕ, z)	r = distance from z-axis; keeps ρ free for charge density
Spherical	(R, θ, ϕ)	θ = polar angle from $+z$, ϕ = azimuth (physics/Cheng convention)

Calculus and operators

Symbol	Meaning
$\frac{dy}{dt},\ \dot{y},\ y'$	derivative (dot = time derivative)
$\frac{\partial f}{\partial x},\ f_x$	partial derivative
$\nabla f,\ \nabla \cdot \mathbf{F},\ \nabla \times \mathbf{F},\ \nabla^2 f$	gradient, divergence, curl, Laplacian
$\oint_C$	integral around a closed contour/curve

Signals and transforms

Symbol	Meaning
t, n	continuous time, discrete-time index
$u(t), u[n]$	unit step
$\delta(t), \delta[n]$	Dirac impulse, Kronecker (unit) impulse
$\omega = 2\pi f$	angular frequency (rad/s), f in Hz
$X(s) = \mathcal{L}\{x(t)\}$	Laplace transform, $s = \sigma + j\omega$
$X(\omega) = \mathcal{F}\{x(t)\}$	Fourier transform (angular-frequency convention, see Ch. 16)
$X(z) = \mathcal{Z}\{x[n]\}$	Z-transform
$x * h$	convolution

Probability

Symbol	Meaning
$P(A)$, $P(A \mid B)$	probability, conditional probability
X, Y	random variables (uppercase); x, y their values
$E[X]$, μ	expectation
$\mathrm{Var}(X)$, σ^2	variance
$X \sim \mathcal{N}(\mu, \sigma^2)$	X is Gaussian
$R_{XX}(\tau)$, $S_{XX}(\omega)$	autocorrelation, power spectral density

Units and dimensions

SI units throughout. Dimensions are written in brackets: $[V] = \mathrm{M\,L^2\,T^{-3}\,I^{-1}}$. Every equation in this book should be dimensionally consistent — checking this is the cheapest error detector you have (Ch. 1, Ch. 25).

Callouts

The book uses Obsidian callouts (they degrade to blockquotes on GitHub):

> **Definition**
>
> A precise statement of a concept.

> **Theorem**
>
> A result you can rely on.

> **Worked Example**
>
> A solved problem.

> **Engineering Note**
>
> Where the method shows up in EE practice.

> **Common Pitfall**
>
> A mistake worth avoiding.

> **? Missing clue**
>
> A question the chapter raises on purpose and leaves open, with a pointer to where it is answered (track chapters only).

Level 1 — Core

The original 25 chapters: the engineering-mathematics foundation every track builds on.

Part I — Mathematical Foundations

Chapter 1 — Mathematical Language and Algebra

> **Prerequisites:** Secondary-school arithmetic and algebra; read Notation & Conventions first. **Learning outcomes:** After this chapter you can
>
> - classify numbers, reason about precision and rounding, and use interval, set and absolute-value notation correctly;
> - rearrange, factor and simplify algebraic and rational expressions, including partial-fraction decomposition;
> - solve linear, quadratic, radical, exponential, logarithmic and absolute-value equations and inequalities, and recognise extraneous or lost roots;
> - work fluently with exponential, logarithmic, hyperbolic and trigonometric functions and their identities, including decibels and harmonic addition;
> - convert points and equations between Cartesian, polar, cylindrical (r, ϕ, z) and spherical (R, θ, ϕ) coordinates;
> - state domains and ranges, build inverse and composite functions, read mathematical notation precisely, and check any formula for dimensional consistency.

Every later chapter in this book is written in the language set out here. Very little of it is new mathematics, but experience in design reviews, laboratories and examination halls is unanimous: engineering errors are far more often algebra slips, sign errors, misplaced factors of ten and unit mistakes than misunderstandings of calculus. A transfer function with one wrong sign predicts a stable amplifier that oscillates; a decibel figure computed with 10 log instead of 20 log is off by a factor of ten in power. This chapter is therefore deliberately thorough. It revisits the familiar material with the precision that later chapters assume — exact definitions, the reasons the rules hold, and the traps that catch even experienced engineers — and it points forward to where each tool is used. Treat it as the toolkit you will reach for on every page that follows.

1.1 Real and complex number systems

Why the number system matters

An engineer measures a voltage as 3.30 V, a computer stores it as a 64-bit floating-point pattern, a theorem about limits assumes it is an exact real number, and an AC analysis represents its amplitude and phase as a single complex number. These are four different number systems, and moving between them without noticing is the source of many subtle mistakes. Knowing what each system can and cannot represent — and what operations are legal in each — is the first piece of mathematical language.

The standard number sets

The standard number sets

$$\mathbb{N} = \{1, 2, 3, \dots\} \subset \mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\} \subset \mathbb{Q} = \left\{ \tfrac{p}{q} : p, q \in \mathbb{Z},\ q \neq 0 \right\} \subset \mathbb{R} \subset \mathbb{C}$$

- $\mathbb{N}$, the **natural numbers**, are the counting numbers.
- $\mathbb{Z}$, the **integers**, add zero and the negatives, so subtraction is always possible.
- $\mathbb{Q}$, the **rationals**, are ratios of integers, so division by any non-zero number is possible.
- $\mathbb{R}$, the **reals**, fill the gaps in $\mathbb{Q}$; real numbers that are not rational are **irrational** (e.g. $\sqrt{2}$, π, e).
- $\mathbb{C}$, the **complex numbers**, have the form $z = a + jb$ with $a, b \in \mathbb{R}$ and $j^2 = -1$.

Some authors include 0 in $\mathbb{N}$; this book does not, and writes $\mathbb{N}_0 = \{0, 1, 2, \dots\}$ when zero is needed (for example the index n of a discrete-time signal that starts at $n = 0$). Each enlargement of the number system is motivated by an equation that could not be solved in the smaller system: $x + 3 = 1$ has no solution in $\mathbb{N}$, $3x = 1$ none in $\mathbb{Z}$, $x^2 = 2$ none in $\mathbb{Q}$, and $x^2 = -1$ none in $\mathbb{R}$.

Decimal expansions of rationals

Decimal characterisation of $\mathbb{Q}$

A real number is rational if and only if its decimal expansion either terminates or eventually repeats.

Proof sketch. Long division of p by q produces at each step a remainder in $\{0, 1, \dots, q-1\}$. After at most q steps a remainder must recur (there are only q possibilities), and from then on the digits repeat. Conversely, a repeating decimal can be turned into a fraction by the shift-and-subtract trick in the next example. □

Worked Example 1.1 — Repeating decimals to fractions

(a) $x = 0.454545\dots$ The repeating block has length 2, so multiply by 10^2:

$$100x = 45.4545\dots, \qquad 100x - x = 45 \quad \Longrightarrow \quad x = \frac{45}{99} = \frac{5}{11}.$$

(b) $y = 1.2363636\dots$ Here one digit (2) precedes the repeating block 36. Shift so that the repeat starts at the decimal point, then shift by one block:

$$10y = 12.363636\dots, \qquad 1000y = 1236.363636\dots$$

Subtracting, $990y = 1224$, so $y = \dfrac{1224}{990} = \dfrac{68}{55}$.

Check: $68 \div 55 = 1.236363\dots$ ✓ and $5 \div 11 = 0.454545\dots$ ✓

Irrational numbers

Why $\sqrt{2}$ is irrational. Suppose $\sqrt{2} = p/q$ with p, q integers having no common factor. Then $p^2 = 2q^2$, so p^2 is even, hence p is even (the square of an odd number is odd), say $p = 2k$. Then $4k^2 = 2q^2$, so $q^2 = 2k^2$ and q is even too — contradicting "no common factor". $\square$

The same argument shows $\sqrt{n}$ is irrational whenever the integer n is not a perfect square. The numbers π and e are also irrational (indeed **transcendental**: they are not roots of any polynomial with integer coefficients), though the proofs are harder. In practice this means that π, e, $\sqrt{2}$ and $\sqrt{3}$ — which appear constantly in RMS values, three-phase power and time constants — can only ever be *approximated* in a calculation. Keep them symbolic as long as possible and round only at the end.

The real numbers as a complete ordered field

What exactly *is* $\mathbb{R}$? For engineering purposes it is enough to know the three groups of properties that every manipulation in this book relies on.

Axioms of the real numbers (summary)

Field axioms. Addition and multiplication are commutative and associative; multiplication distributes over addition, $a(b+c) = ab + ac$; there are identities 0 and 1; every a has a negative $-a$, and every $a \neq 0$ has a reciprocal a^{-1}.

Order axioms. For any a, b exactly one of $a < b$, $a = b$, $a > b$ holds; if $a < b$ then $a + c < b + c$; if $a < b$ and $c > 0$ then $ac < bc$.

Completeness. Every non-empty set of reals that is bounded above has a **least upper bound** (supremum) in $\mathbb{R}$.

$\mathbb{Q}$ satisfies the field and order axioms but not completeness: the set $\{x \in \mathbb{Q} : x^2 < 2\}$ is bounded above but has no least upper bound *in* $\mathbb{Q}$ — the "missing" bound is $\sqrt{2}$. Completeness is equivalent to the statement that every bounded, increasing sequence converges to a real limit, and it is what makes limits, continuity, the intermediate value theorem and hence all of calculus work (Ch. 4). It also guarantees that numerical algorithms such as bisection converge to an actual number.

Every rule of algebra in Section 1.2 is a consequence of the field axioms. For example, $a \cdot 0 = 0$ follows from $a \cdot 0 = a(0+0) = a \cdot 0 + a \cdot 0$ and subtracting $a \cdot 0$ from both sides; and $(-1)(-1) = 1$ follows from $0 = (-1) \cdot 0 = (-1)(1 + (-1)) = -1 + (-1)(-1)$. Nothing is arbitrary.

Intervals

Subsets of $\mathbb{R}$ that appear as domains, tolerance bands and solution sets are usually intervals.

Notation	Set	Notation	Set
$[a, b]$	$a \le x \le b$ (closed)	(a, b)	$a < x < b$ (open)
$[a, b)$	$a \le x < b$ (half-open)	$(a, b]$	$a < x \le b$
(a, ∞)	$x > a$	$[a, \infty)$	$x \ge a$
$(-\infty, b)$	$x < b$	$\mathbb{R}$	$(-\infty, \infty)$

The symbol ∞ is never a real number and never sits next to a square bracket. Unions such as $(-\infty, -2) \cup [1, \infty)$ describe solution sets of inequalities (Section 1.3).

Absolute value

Absolute value

$$|x| = \begin{cases} x, & x \ge 0 \\ -x, & x < 0 \end{cases}$$

Equivalently $|x| = \sqrt{x^2}$. Geometrically, $|x - a|$ is the distance between x and a on the number line.

Properties of absolute value

For all real x, y:

$$|x| \ge 0, \qquad |xy| = |x|\,|y|, \qquad \left|\frac{x}{y}\right| = \frac{|x|}{|y|} \ (y \neq 0),$$

$$|x + y| \le |x| + |y| \ \text{(triangle inequality)}, \qquad \big||x| - |y|\big| \le |x - y| \ \text{(reverse triangle inequality)}.$$

Proof of the triangle inequality. Since $-|x| \le x \le |x|$ and $-|y| \le y \le |y|$, adding gives $-(|x| + |y|) \le x + y \le |x| + |y|$, which is exactly $|x + y| \le |x| + |y|$. For the reverse form, write $x = (x - y) + y$ so $|x| \le |x - y| + |y|$, i.e. $|x| - |y| \le |x - y|$; swapping x and y gives $|y| - |x| \le |x - y|$, and together these are the claim. $\square$

The triangle inequality is the source of every **worst-case tolerance** bound in circuit design: if two errors are at most ε_1 and ε_2 in magnitude, their sum is at most $\varepsilon_1 + \varepsilon_2$ in magnitude — and no better guarantee is possible without further information.

Worked Example 1.2 — Worst-case tolerance

A 1.00 kΩ resistor with ±5% tolerance is placed in series with a 2.20 kΩ resistor with ±1% tolerance. Bound the error in the total resistance.

Step 1 — individual error bounds. $|e_1| \le 0.05 \times 1000 = 50\ \Omega$ and $|e_2| \le 0.01 \times 2200 = 22\ \Omega$.

Step 2 — triangle inequality. The total is $R = 3200 + e_1 + e_2$, and

$$|e_1 + e_2| \le |e_1| + |e_2| \le 72\ \Omega.$$

Step 3 — relative bound. $72/3200 = 2.25\%$. So $R \in [3128,\ 3272]\ \Omega$.

Check: the extreme values $1050 + 2222 = 3272$ and $950 + 2178 = 3128$ are attained, so the bound is tight.

Precision, significant figures and engineering notation

Measured quantities carry finite precision. The number of **significant figures** is the number of digits known reliably: 3.30 V has three, $0.0047\ \mu$F has two. **Scientific notation** writes a number as $m \times 10^k$ with $1 \le |m| < 10$; **engineering notation** restricts k to multiples of 3 so that the SI prefixes apply directly: 4.7×10^{-9} F $= 4.7$ nF, $2.2 \times 10^4\ \Omega = 22$ kΩ. A result should not be quoted to more significant figures than the least precise input justifies, but intermediate calculations should carry two or three extra digits to avoid accumulating rounding error.

Real numbers in a computer

Computers do not store real numbers; they store a finite set of **floating-point** numbers of the form $\pm m \times 2^e$ with a fixed number of binary digits in m. IEEE 754 double precision uses a 53-bit significand, giving about 15–16 significant decimal digits; the gap between 1 and the next representable number is the **machine epsilon** $\varepsilon_{\text{mach}} = 2^{-52} \approx 2.2 \times 10^{-16}$.

Most decimal fractions are not representable exactly in binary, just as $\frac{1}{3}$ is not representable exactly in decimal. In binary, $0.1 = 0.000110011001100\ldots_2$ repeats forever, so the stored value is slightly off.

Worked Example 1.3 — Floating-point surprises

(a) In double precision, `0.1 + 0.2` evaluates to 0.30000000000000004, so the test `0.1 + 0.2 == 0.3` is **false**. Every term was rounded to the nearest binary fraction and the rounding errors did not cancel. Comparisons of floating-point results must use a tolerance: $|a - b| \le \varepsilon\ \max(|a|, |b|)$.

(b) Subtracting nearly equal numbers destroys relative accuracy. If $a = 1.0000001$ and $b = 1.0000000$ are each known to 8 significant figures, then $a - b = 1 \times 10^{-7}$ is known to only **one** significant figure. This is **catastrophic cancellation**; Section 1.3 shows how it ruins the textbook quadratic formula, and Ch. 22 treats it systematically.

Engineering Note — fixed point and quantisation

Microcontrollers and DSP chips often use **fixed-point** arithmetic: an N-bit integer interpreted as a fraction with a fixed binary point. An analogue-to-digital converter maps a continuous voltage range onto 2^N integer codes, so it replaces a real number by the nearest of a finite set — a quantisation error of at most half a step. An ideal 12-bit ADC spanning 3.3 V has step $3.3/4096 = 0.806$ mV and therefore a worst-case quantisation error of 0.403 mV. Rational and real numbers are idealisations; the hardware works with integers.

Complex numbers — a preview

The equation $x^2 + 1 = 0$ has no real solution, because $x^2 \ge 0$ for every real x. Adjoining a new number j with $j^2 = -1$ and insisting that the field axioms still hold produces $\mathbb{C} = \{a + jb : a, b \in \mathbb{R}\}$, with

$$(a + jb) + (c + jd) = (a + c) + j(b + d), \qquad (a + jb)(c + jd) = (ac - bd) + j(ad + bc).$$

The payoff is remarkable: in $\mathbb{C}$, **every** polynomial equation has a solution (the fundamental theorem of algebra, Section 1.4). Complex numbers are the natural language of AC circuits, filters and signals and are developed fully in Ch. 2. For now note only that $z = a + jb$ has real part $\mathrm{Re}\{z\} = a$ and imaginary part $\mathrm{Im}\{z\} = b$ (both real), that $z^* = a - jb$ is its **conjugate**, and that $|z| = \sqrt{a^2 + b^2}$ is its **magnitude**, so that $zz^* = |z|^2$ is real and non-negative.

Common Pitfall — $\mathbb{C}$ is not ordered

There is no way to define "<" on $\mathbb{C}$ that respects the order axioms: if $j > 0$ then $j \cdot j = -1 > 0$, absurd; if $j < 0$ then $-j > 0$ and $(-j)(-j) = -1 > 0$, equally absurd. An inequality between complex numbers, such as $Z_1 < Z_2$ for two impedances, is meaningless. Compare **magnitudes** $|Z_1| < |Z_2|$ or **real parts** instead, and say which.

Common Pitfall — the letter j

Mathematics texts and many programming languages write i for $\sqrt{-1}$; electrical engineering uses j because i denotes current. Python writes `1j`, MATLAB accepts both `1i` and `1j`. Never use j simultaneously as a summation index and the imaginary unit.

1.2 Algebraic manipulation

Why fluency matters

Engineering formulas are rarely used in the form in which they are derived. Ohm's law is written $V = IR$ but used as $R = V/I$; a resonance formula gives frequency in terms of L and C but the designer needs C in terms of frequency; a transfer function must be put over a common denominator before its poles can be read off. Algebraic manipulation is the skill of transforming an expression or equation into an *equivalent* one that is more useful. Every transformation must be justified by the field axioms of Section 1.1 — and most mistakes come from "rules" that are not among them.

Order of operations and the distributive law

Expressions are evaluated in the conventional order: brackets, then powers and roots, then multiplication and division (left to right), then addition and subtraction (left to right). Two conventions trip people up:

- $-x^2$ means $-(x^2)$, not $(-x)^2$. So $-3^2 = -9$.
- a/bc is ambiguous in print. In this book a fraction bar or explicit brackets are always used: $\dfrac{a}{bc}$ or $(a/b)c$.

The **distributive law** $a(b+c) = ab + ac$ is the only axiom that links addition and multiplication, so it is behind every expansion and every factorisation. Its two directions are **expanding** (left to right) and **factoring** (right to left). Repeated use gives $(a+b)(c+d) = ac + ad + bc + bd$ and, more generally, the product of two sums is the sum of all pairwise products of their terms.

Laws of exponents

For a positive integer n, a^n means $a \cdot a \cdots a$ (n factors). The laws below are obvious for positive integer exponents by counting factors; the definitions $a^0 = 1$, $a^{-n} = 1/a^n$ and $a^{p/q} = \sqrt[q]{a^p}$ are then *chosen* precisely so that the laws continue to hold.

Laws of exponents

For $a, b > 0$ and real m, n:

$$a^m a^n = a^{m+n}, \qquad \frac{a^m}{a^n} = a^{m-n}, \qquad (a^m)^n = a^{mn}, \qquad (ab)^n = a^n b^n, \qquad \left(\frac{a}{b}\right)^n = \frac{a^n}{b^n},$$

$$a^0 = 1, \qquad a^{-n} = \frac{1}{a^n}, \qquad a^{1/n} = \sqrt[n]{a}, \qquad a^{m/n} = \left(\sqrt[n]{a}\right)^m = \sqrt[n]{a^m}.$$

For example, requiring $a^{1/2}a^{1/2} = a^1$ forces $a^{1/2} = \sqrt{a}$, and requiring $a^0 a^n = a^n$ forces $a^0 = 1$. Irrational exponents such as $2^{\sqrt{2}}$ are defined by continuity — as the limit of 2^q over rationals $q \to \sqrt{2}$ — or, more cleanly, as $e^{\sqrt{2}\ln 2}$ (Section 1.5).

Common Pitfall — negative bases

The laws above require $a, b > 0$. With negative bases they fail: $\sqrt{(-2)^2} = 2 \neq -2$, and $\left((-8)^2\right)^{1/6} = 2$ while $\left((-8)^{1/6}\right)^2$ is not even real. In general $\sqrt{x^2} = |x|$, not x. Calculators and programming languages also differ on whether $(-8)^{1/3}$ returns the real cube root -2 or a complex number.

Expansion and factoring identities

The following identities should be recognisable instantly in both directions:

$$\begin{aligned}
(a \pm b)^2 &= a^2 \pm 2ab + b^2 \\
(a+b)^3 &= a^3 + 3a^2 b + 3ab^2 + b^3 \\
a^2 - b^2 &= (a-b)(a+b) \\
a^3 \pm b^3 &= (a \pm b)(a^2 \mp ab + b^2) \\
a^n - b^n &= (a-b)\left(a^{n-1} + a^{n-2}b + \cdots + ab^{n-2} + b^{n-1}\right)
\end{aligned}$$

The last identity, verified by multiplying out (all middle terms cancel in pairs), contains the geometric-series formula of Section 1.10 as the special case $a = 1$, $b = r$.

The binomial theorem

Binomial theorem

For any positive integer n,

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k, \qquad \binom{n}{k} = \frac{n!}{k!\,(n-k)!},$$

where $n! = 1 \cdot 2 \cdots n$ and $0! = 1$.

Proof sketch. Expanding $(a+b)(a+b)\cdots(a+b)$ (n factors) means choosing a or b from each factor and multiplying. A term $a^{n-k}b^k$ arises once for each way of choosing which k of the n factors supply b, and there are $\binom{n}{k}$ such choices. □

The coefficients form **Pascal's triangle**, in which each entry is the sum of the two above it, $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$:

n	coefficients $\binom{n}{0}, \dots, \binom{n}{n}$
0	1
1	1, 1
2	1, 2, 1
3	1, 3, 3, 1
4	1, 4, 6, 4, 1
5	1, 5, 10, 10, 5, 1

The binomial coefficients reappear as counts of outcomes in the binomial probability distribution (Ch. 19).

Worked Example 1.4 — A specific binomial term

Find the term independent of x in $\left(2x - \dfrac{1}{x}\right)^6$.

Step 1 — general term. With $a = 2x$, $b = -1/x$:

$$T_k = \binom{6}{k}(2x)^{6-k}\left(-\frac{1}{x}\right)^k = \binom{6}{k}2^{6-k}(-1)^k\,x^{6-2k}.$$

Step 2 — choose k. The power of x is zero when $6 - 2k = 0$, i.e. $k = 3$.

Step 3 — evaluate. $T_3 = \binom{6}{3}\,2^3(-1)^3 = 20 \cdot 8 \cdot (-1) = -160$.

Check: substituting $x = 1$ in the full expansion gives $(2-1)^6 = 1$; the seven coefficients $64, -192, 240, -160, 60, -12, 1$ do sum to 1. ✓

The binomial approximation

For small x the binomial expansion is dominated by its first two terms:

$$(1+x)^n \approx 1 + nx, \qquad |x| \ll 1.$$

Remarkably, this holds for **any real** n — negative and fractional included — as the Taylor series of Ch. 4 shows. The next term is $\frac{n(n-1)}{2}x^2$, which estimates the error. Special cases used constantly:

$$\frac{1}{1+x} \approx 1 - x, \qquad \sqrt{1+x} \approx 1 + \frac{x}{2}, \qquad \frac{1}{\sqrt{1+x}} \approx 1 - \frac{x}{2}.$$

Worked Example 1.5 — Loading of a voltage divider

A divider $R_1 = R_2 = 10\ \text{k}\Omega$ feeds a load $R_L = 100\ \text{k}\Omega$. Its output with the load attached can be written

$$V_{\text{out}} = V_{\text{in}}\frac{R_2}{R_1 + R_2}\cdot\frac{1}{1 + R_{\text{th}}/R_L}, \qquad R_{\text{th}} = \frac{R_1 R_2}{R_1 + R_2} = 5\ \text{k}\Omega.$$

Step 1. $x = R_{\text{th}}/R_L = 0.05$ is small, so $\dfrac{1}{1+x} \approx 1 - x = 0.95$.

Step 2. The loaded output is about 5% below the unloaded value $V_{\text{in}}/2$.

Check: the exact factor is $1/1.05 = 0.95238$; the approximation error 0.0024 matches the size of the neglected term $x^2 = 0.0025$. ✓ The approximation turns a messy expression into a rule of thumb: *loading error* $\approx R_{\text{th}}/R_L$.

Completing the square

Completing the square

For $a \neq 0$,

$$ax^2 + bx + c = a\left(x + \frac{b}{2a}\right)^2 + c - \frac{b^2}{4a}.$$

Derivation. Factor out a: $a\left(x^2 + \frac{b}{a}x\right) + c$. Since $\left(x + \frac{b}{2a}\right)^2 = x^2 + \frac{b}{a}x + \frac{b^2}{4a^2}$, adding and subtracting $\frac{b^2}{4a^2}$ inside the bracket gives the result. □

This single manipulation yields the quadratic formula (Section 1.3), locates the vertex $\left(-\frac{b}{2a},\ c - \frac{b^2}{4a}\right)$ of a parabola, converts the equation of a circle to centre–radius form (Section 1.7), and in Ch. 14 turns a second-order denominator $s^2 + 2\zeta\omega_n s + \omega_n^2$ into $(s + \zeta\omega_n)^2 + \omega_d^2$, the form that identifies a damped sinusoid.

Worked Example 1.6 — Completing the square

(a) $2x^2 - 8x + 3 = 2(x^2 - 4x) + 3 = 2\big[(x-2)^2 - 4\big] + 3 = 2(x-2)^2 - 5$. Minimum value -5 at $x = 2$.

(b) $s^2 + 6s + 13 = (s+3)^2 - 9 + 13 = (s+3)^2 + 2^2$. As a sum of squares it never vanishes for real s; its complex zeros are $s = -3 \pm j2$.

Check (a): expanding, $2(x^2 - 4x + 4) - 5 = 2x^2 - 8x + 3$. ✓ **Check (b):** $(-3 + j2)^2 + 6(-3 + j2) + 13 = (5 - j12) + (-18 + j12) + 13 = 0$. ✓

Algebraic fractions

The rules for fractions follow from the field axioms:

$$\frac{a}{b}+\frac{c}{d}=\frac{ad+bc}{bd}, \qquad \frac{a}{b}\cdot\frac{c}{d}=\frac{ac}{bd}, \qquad \frac{a/b}{c/d}=\frac{ad}{bc}, \qquad \frac{ac}{bc}=\frac{a}{b}\ (c\neq 0).$$

Three working principles:

1. **Combine over the lowest common denominator**, factoring denominators first so you do not create needlessly large products.
2. **Cancel only common factors**, never common terms. In $\dfrac{x(x+1)}{x(x+2)}$ the factor x cancels (for $x\neq 0$); in $\dfrac{x+1}{x+2}$ nothing cancels.
3. **Rationalise** surd denominators by multiplying by the conjugate, using $(\sqrt{a}-\sqrt{b})(\sqrt{a}+\sqrt{b})=a-b$:

$$\frac{1}{\sqrt{a}+\sqrt{b}}=\frac{\sqrt{a}-\sqrt{b}}{a-b}.$$

The same conjugate trick, applied to $a+jb$, is how complex division is done in Ch. 2.

Worked Example 1.7 — Simplifying fractions

(a) Compound fraction. Simplify $\dfrac{\frac{1}{x}-\frac{1}{y}}{\frac{1}{x^2}-\frac{1}{y^2}}$. Numerator $=\dfrac{y-x}{xy}$; denominator $=\dfrac{y^2-x^2}{x^2y^2}=\dfrac{(y-x)(y+x)}{x^2y^2}$. Dividing,

$$\frac{y-x}{xy}\cdot\frac{x^2y^2}{(y-x)(y+x)}=\frac{xy}{x+y} \qquad (x\neq y).$$

Check with $x=2, y=3$: original $=\dfrac{1/2-1/3}{1/4-1/9}=\dfrac{1/6}{5/36}=1.2$; result $=6/5=1.2$. ✓ (The result is the "parallel combination" of x and y.)

(b) Rationalising. $\dfrac{1}{\sqrt{5}-\sqrt{3}}=\dfrac{\sqrt{5}+\sqrt{3}}{5-3}=\dfrac{\sqrt{5}+\sqrt{3}}{2}\approx 1.98406$. **Check:** $1/(2.23607-1.73205)=1/0.50402=1.98406$. ✓

Worked Example 1.8 — Parallel resistors

Two resistors in parallel satisfy $\dfrac{1}{R}=\dfrac{1}{R_1}+\dfrac{1}{R_2}$. (a) Express R explicitly. (b) What R_2 must be paralleled with $R_1=10\ \text{k}\Omega$ to give $R=4\ \text{k}\Omega$?

(a) Common denominator: $\dfrac{1}{R}=\dfrac{R_2+R_1}{R_1R_2}$, so the familiar "product over sum"

$$R=\frac{R_1R_2}{R_1+R_2}.$$

(b) Solve $\dfrac{1}{R_2}=\dfrac{1}{R}-\dfrac{1}{R_1}=\dfrac{R_1-R}{RR_1}$, giving

$$R_2=\frac{RR_1}{R_1-R}=\frac{4\times 10}{10-4}\ \text{k}\Omega=6.667\ \text{k}\Omega.$$

Check: $\dfrac{10\times 6.667}{16.667}=4.000\ \text{k}\Omega$. ✓ Note that the formula requires $R<R_1$ — a parallel combination is always smaller than either branch.

Changing the subject of a formula

To make a variable the **subject**, apply the same operation to both sides repeatedly, undoing the operations that act on the variable in reverse order. Keep track of conditions (division by zero, square roots of negatives, sign of quantities) as you go.

Worked Example 1.9 — Two rearrangements

(a) Resonance. An LC tank resonates at $f_0=\dfrac{1}{2\pi\sqrt{LC}}$. Find C for $f_0=1$ MHz with $L=10\ \mu$H. Square both sides: $f_0^2=\dfrac{1}{4\pi^2LC}$, then solve: $C=\dfrac{1}{4\pi^2 f_0^2 L}$. Numerically

$$C=\frac{1}{4\pi^2\,(10^6)^2\,(10\times 10^{-6})}=2.533\times 10^{-9}\ \text{F}=2.533\ \text{nF}.$$

(b) Feedback. A negative-feedback amplifier has closed-loop gain $A_f=\dfrac{A}{1+A\beta}$. Find β for $A=10^5$ and $A_f=10$. Multiply out: $A_f+A_fA\beta=A$, so $\beta=\dfrac{A-A_f}{AA_f}=\dfrac{1}{A_f}-\dfrac{1}{A}=0.1-0.00001=0.09999$.

Checks: (a) $1/(2\pi\sqrt{10^{-5}\times 2.533\times 10^{-9}})=1.000\times 10^6$ Hz ✓; (b) $10^5/(1+10^5\times 0.09999)=10^5/10000=10$ ✓. The form $\beta=1/A_f-1/A$ also shows why $\beta\approx 1/A_f$ when A is huge.

Common Pitfall — illegal "simplifications"

$\dfrac{a+b}{a+c}\neq\dfrac{b}{c}$, $\sqrt{a^2+b^2}\neq a+b$, $(a+b)^{-1}\neq a^{-1}+b^{-1}$, $(a+b)^2\neq a^2+b^2$. Each of these cancels or distributes an operation across a **sum**, which is never allowed. A two-second numerical spot check (put $a=1, b=2, c=3$) exposes all of them.

Engineering Note — keep it symbolic

Rearrange symbolically first and substitute numbers last. The symbolic answer can be dimension-checked (Section 1.10), tested at limiting cases ($R_2 \to \infty$ in a parallel combination should give $R \to R_1$), and reused for other component values. Plugging numbers in early hides all of these checks.

1.3 Equations and inequalities

Equivalent equations

Solving an equation means finding every value of the unknown that makes it true — the **solution set**. The method is to replace the equation by a chain of simpler **equivalent** equations (equations with the same solution set). The safe operations are:

- adding the same quantity to both sides;
- multiplying or dividing both sides by the same **non-zero** quantity;
- applying a **one-to-one** function to both sides (e.g. ln to positive quantities, exp, cubing).

Two operations are *not* equivalence-preserving and must be handled with care:

- **Squaring** both sides (or applying any function that is not one-to-one) can introduce **extraneous roots**: $x = 2$ has one solution, but $x^2 = 4$ has two.
- **Dividing by an expression that may be zero** can **lose roots**: dividing $x^2 = 3x$ by x gives $x = 3$ and loses $x = 0$. Factor instead: $x(x - 3) = 0$.

Whenever a non-equivalent step is used, every candidate must be checked in the *original* equation.

Linear equations

A linear equation $ax + b = 0$ with $a \neq 0$ has the single root $x = -b/a$. If $a = 0$ there is no solution ($b \neq 0$) or every x is a solution ($b = 0$). This trivial trichotomy — one, none, or infinitely many — reappears unchanged for systems of linear equations in Ch. 8.

Quadratic equations

Quadratic formula

The roots of $ax^2 + bx + c = 0$, $a \neq 0$, are

$$x = \frac{-b \pm \sqrt{\Delta}}{2a}, \qquad \Delta = b^2 - 4ac.$$

- $\Delta > 0$: two distinct real roots;
- $\Delta = 0$: one repeated real root $x = -b/(2a)$;
- $\Delta < 0$: a complex-conjugate pair $x = \dfrac{-b}{2a} \pm j\dfrac{\sqrt{-\Delta}}{2a}$.

Derivation. By completing the square (Section 1.2), $ax^2 + bx + c = 0$ becomes $a\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a}$, i.e. $\left(x + \frac{b}{2a}\right)^2 = \frac{\Delta}{4a^2}$. Taking square roots, $x + \frac{b}{2a} = \pm\frac{\sqrt{\Delta}}{2a}$. □

Vieta's relations for a quadratic

If x_1, x_2 are the roots of $ax^2 + bx + c = 0$, then

$$x_1 + x_2 = -\frac{b}{a}, \qquad x_1 x_2 = \frac{c}{a}.$$

These follow from comparing coefficients in $a(x - x_1)(x - x_2) = ax^2 - a(x_1 + x_2)x + ax_1x_2$. They are the cheapest check on any computed pair of roots.

The sign of the discriminant decides whether a second-order circuit is over-, critically or under-damped (Ch. 12) — the same three cases, because the characteristic equation $LCs^2 + RCs + 1 = 0$ of a series RLC circuit is a quadratic.

Worked Example 1.10 — RLC characteristic roots

A series RLC circuit with $R = 200\ \Omega$, $L = 0.1$ H, $C = 2\ \mu$F has characteristic equation $s^2 + \frac{R}{L}s + \frac{1}{LC} = 0$, i.e. $s^2 + 2000s + 5 \times 10^6 = 0$. Solve it.

Step 1. $\Delta = 2000^2 - 4 \cdot 5 \times 10^6 = 4 \times 10^6 - 2 \times 10^7 = -1.6 \times 10^7 < 0$: complex roots, an under-damped response.

Step 2. $\sqrt{-\Delta} = 4000$, so

$$s = \frac{-2000 \pm j4000}{2} = -1000 \pm j2000\ \text{s}^{-1}.$$

Check (Vieta): sum $= -2000 = -b/a$ ✓; product $= 1000^2 + 2000^2 = 5 \times 10^6 = c/a$ ✓. The real part -1000 is the decay rate and the imaginary part 2000 rad/s the ringing frequency.

Numerically stable roots

The textbook formula is exact in exact arithmetic but can be disastrous in floating point. When $b^2 \gg |4ac|$, $\sqrt{\Delta} \approx |b|$, and one of $-b \pm \sqrt{\Delta}$ subtracts two nearly equal numbers — catastrophic cancellation (Section 1.1).

Stable quadratic formula

For real coefficients with $\Delta \geq 0$, compute

$$q = -\tfrac{1}{2}\left(b + \operatorname{sgn}(b)\sqrt{\Delta}\right), \qquad x_1 = \frac{q}{a}, \qquad x_2 = \frac{c}{q}.$$

The sum inside q never cancels, and x_2 follows from Vieta's product $x_1 x_2 = c/a$.

Worked Example 1.11 — Cancellation in the quadratic formula

Solve $x^2 - 10^8 x + 1 = 0$ in double precision.

Naive formula. $\sqrt{\Delta} = \sqrt{10^{16} - 4}$ rounds to exactly 10^8 in double precision. The small root $\left(10^8 - \sqrt{\Delta}\right)/2$ is computed as 7.45×10^{-9} — wrong by 25% (the true value is 1.0000×10^{-8}).

Stable formula. $q = -\tfrac{1}{2}\left(-10^8 - 10^8\right) = 10^8$, so $x_1 = 10^8$ and $x_2 = 1/10^8 = 10^{-8}$, correct to full precision.

Check: product $x_1 x_2 = 1 = c/a$ ✓; sum $\approx 10^8 = -b/a$ ✓.

Equations reducible to quadratics; radical equations

Many equations become quadratic after a substitution: $x^4 - 5x^2 + 4 = 0$ with $u = x^2$ gives $u^2 - 5u + 4 = 0$, so $u = 1$ or $u = 4$ and $x \in \{-2, -1, 1, 2\}$. Equations in e^x (Section 1.5) and $\sin x$ (Section 1.6) yield to the same trick.

Equations containing square roots are solved by isolating the root and squaring — a non-equivalent step, so every candidate must be checked.

Worked Example 1.12 — An extraneous root

Solve $\sqrt{x + 5} = x - 1$.

Step 1 — square. $x + 5 = x^2 - 2x + 1$, i.e. $x^2 - 3x - 4 = 0$, so $(x - 4)(x + 1) = 0$ and the candidates are $x = 4$ and $x = -1$.

Step 2 — check in the original. $x = 4$: $\sqrt{9} = 3 = 4 - 1$ ✓. $x = -1$: $\sqrt{4} = 2$ but $-1 - 1 = -2$ ✗.

Solution: $x = 4$ only. The root $x = -1$ solves $\sqrt{x + 5} = -(x - 1)$, which squares to the same quadratic. Note the original equation also requires $x - 1 \geq 0$, which rules out $x = -1$ in advance.

Simultaneous equations

Two linear equations in two unknowns are solved by **substitution** (solve one for a variable, substitute in the other) or **elimination** (add multiples of the equations to remove a variable). Eliminating in general gives **Cramer's rule** for the 2×2 case:

$$\begin{aligned} a_1 x + b_1 y &= c_1 \\ a_2 x + b_2 y &= c_2 \end{aligned} \quad \Longrightarrow \quad x = \frac{c_1 b_2 - c_2 b_1}{a_1 b_2 - a_2 b_1}, \quad y = \frac{a_1 c_2 - a_2 c_1}{a_1 b_2 - a_2 b_1},$$

provided the **determinant** $D = a_1 b_2 - a_2 b_1 \neq 0$. Geometrically each equation is a line; $D \neq 0$ means the lines cross at one point, while $D = 0$ means they are parallel (no solution) or coincident (infinitely many). Larger systems, and the matrix language that organises them, are the subject of Ch. 8.

Worked Example 1.13 — Two-mesh circuit

Mesh analysis of a two-loop resistive circuit ($R_1 = 2\ \Omega$, $R_2 = 4\ \Omega$ shared, $R_3 = 6\ \Omega$, source 10 V in loop 1) gives

$$\begin{aligned} 6I_1 - 4I_2 &= 10 \\ -4I_1 + 10I_2 &= 0 \end{aligned}$$

Step 1 — determinant. $D = 6 \cdot 10 - (-4)(-4) = 60 - 16 = 44$.

Step 2 — Cramer. $I_1 = \dfrac{10 \cdot 10 - 0 \cdot (-4)}{44} = \dfrac{100}{44} = 2.273\ \text{A}$, $\quad I_2 = \dfrac{6 \cdot 0 - (-4) \cdot 10}{44} = \dfrac{40}{44} = 0.909\ \text{A}$.

Check: $6(2.273) - 4(0.909) = 13.636 - 3.636 = 10.000$ ✓ and $-4(2.273) + 10(0.909) = 0.000$ ✓.

Nonlinear simultaneous equations are usually reduced by substitution to a single equation in one unknown.

Worked Example 1.14 — Line meets circle

Find the intersections of $y = x + 1$ and $x^2 + y^2 = 25$.

Substitute: $x^2 + (x + 1)^2 = 25 \Rightarrow 2x^2 + 2x - 24 = 0 \Rightarrow x^2 + x - 12 = 0 \Rightarrow (x + 4)(x - 3) = 0$. So $x = 3,\ y = 4$ or $x = -4,\ y = -3$: the points $(3, 4)$ and $(-4, -3)$.

Check: $9 + 16 = 25$ ✓, $16 + 9 = 25$ ✓. A negative discriminant here would have meant the line misses the circle.

Inequalities

Inequalities obey the same rules as equations **except**:

1. multiplying or dividing by a **negative** number reverses the inequality ($-2 < 3$ but $4 > -6$);
2. taking reciprocals of two quantities of the same sign reverses it ($0 < a < b \Rightarrow 1/a > 1/b$);
3. squaring preserves an inequality only between **non-negative** quantities ($0 \le a < b \Rightarrow a^2 < b^2$, but $-3 < 2$ while $9 > 4$).

Each rule is a consequence of the order axioms in Section 1.1. For instance, if $a < b$ and $c < 0$, then $-c > 0$ so $-ac < -bc$, i.e. $ac > bc$.

Inequality	Solution set
$\lvert x - a\rvert < \delta$	$a - \delta < x < a + \delta$
$\lvert x - a\rvert > \delta$	$x < a - \delta$ or $x > a + \delta$
$(x - r_1)(x - r_2) < 0,\ r_1 < r_2$	$r_1 < x < r_2$ (between the roots)
$(x - r_1)(x - r_2) > 0,\ r_1 < r_2$	$x < r_1$ or $x > r_2$ (outside the roots)

For polynomial or rational inequalities use a **sign chart**: bring everything to one side as a single fraction compared with 0, find all zeros of numerator and denominator, split the real line at them, and determine the sign on each piece (each linear factor changes sign at its own zero; factors with even multiplicity do not).

Worked Example 1.15 — Three inequalities

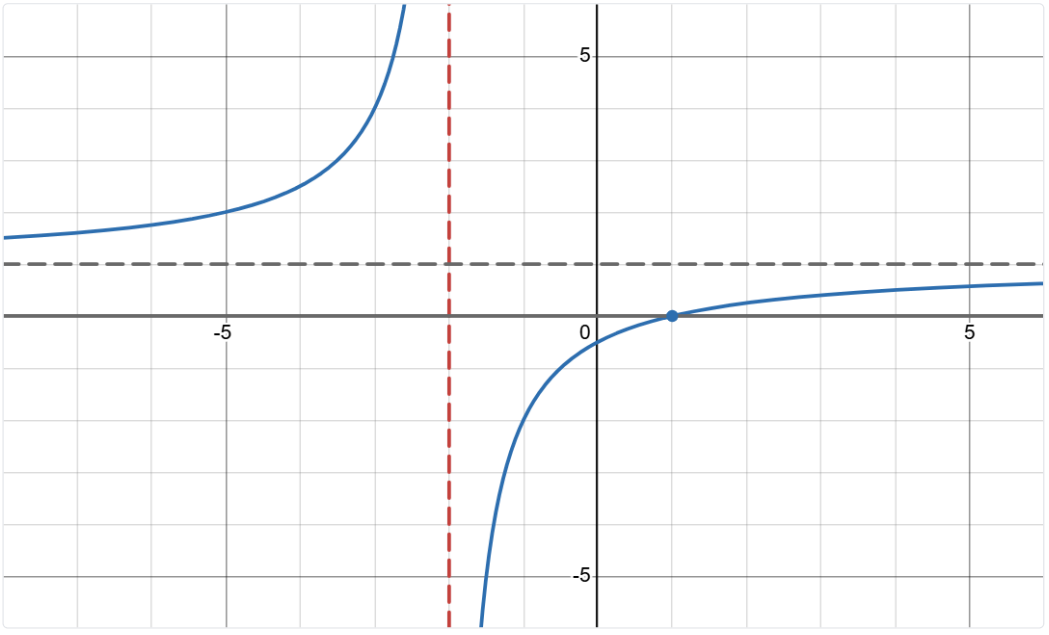
(a) $\lvert x - 2\rvert < 3 \iff -3 < x - 2 < 3 \iff -1 < x < 5$, i.e. $x \in (-1, 5)$.

(b) $x^2 - 5x + 6 \le 0 \iff (x - 2)(x - 3) \le 0 \iff x \in [2, 3]$.

(c) $\dfrac{x - 1}{x + 2} \ge 0$. Critical points $x = 1$ (zero of numerator, included) and $x = -2$ (zero of denominator, always excluded).

Interval	$x - 1$	$x + 2$	quotient
$x < -2$	$-$	$-$	$+$
$-2 < x < 1$	$-$	$+$	$-$
$x > 1$	$+$	$+$	$+$

Solution: $x \in (-\infty, -2) \cup [1, \infty)$. We may **not** multiply through by $x + 2$ without knowing its sign — doing so blindly gives $x \ge 1$ and loses the whole interval $x < -2$.



$y = (x - 1)/(x + 2)$: the curve is on or above the axis for $x < -2$ and $x \ge 1$, matching the sign chart. The red dashed line is the vertical asymptote where the denominator vanishes.

Worked Example 1.16 — Two absolute values

Solve $\lvert x - 1\rvert + \lvert x + 2\rvert \le 5$.

The expressions inside change sign at $x = 1$ and $x = -2$, so treat three cases.

- $x \ge 1$: $(x - 1) + (x + 2) = 2x + 1 \le 5 \Rightarrow x \le 2$. Combined: $1 \le x \le 2$.
- $-2 \le x < 1$: $-(x - 1) + (x + 2) = 3 \le 5$, always true. Combined: $-2 \le x < 1$.
- $x < -2$: $-(x - 1) - (x + 2) = -2x - 1 \le 5 \Rightarrow x \ge -3$. Combined: $-3 \le x < -2$.

Solution: $x \in [-3, 2]$. **Check:** at $x = 2$, $1 + 4 = 5$ ✓; at $x = -3$, $4 + 1 = 5$ ✓; at $x = 0$, $1 + 2 = 3 \le 5$ ✓. Geometrically, the sum of distances from x to 1 and to -2 is at most 5.

The AM–GM inequality and a design bound

Arithmetic–geometric mean inequality

For $a, b \geq 0$,

$$\frac{a+b}{2} \geq \sqrt{ab},$$

with equality if and only if $a = b$.

Proof. $(\sqrt{a} - \sqrt{b})^2 \geq 0$ expands to $a - 2\sqrt{ab} + b \geq 0$. Equality holds only when $\sqrt{a} = \sqrt{b}$. □

Inequalities like this one prove that a design is optimal *without calculus*.

Worked Example 1.17 — Maximum power transfer

A source of EMF V and internal resistance R_s drives a load R_L, delivering $P = \dfrac{V^2 R_L}{(R_s + R_L)^2}$. Show $P \leq \dfrac{V^2}{4R_s}$ and find when equality holds.

Step 1. By AM–GM with $a = R_s$, $b = R_L$: $R_s + R_L \geq 2\sqrt{R_s R_L}$, so $(R_s + R_L)^2 \geq 4R_s R_L$.

Step 2. Therefore $P = \dfrac{V^2 R_L}{(R_s + R_L)^2} \leq \dfrac{V^2 R_L}{4R_s R_L} = \dfrac{V^2}{4R_s}$, with equality iff $R_L = R_s$.

Numerical check: $V = 10$ V, $R_s = 50\ \Omega$: the bound is $100/200 = 0.5$ W, attained at $R_L = 50\ \Omega$ ($100 \cdot 50/100^2 = 0.5$ ✓). At $R_L = 100\ \Omega$: $100 \cdot 100/150^2 = 0.444$ W < 0.5 ✓.

Inequalities as specifications

Worked Example 1.18 — Choosing a component range

An RC timer needs $\tau = RC \in [0.85,\ 1.15]$ ms for **every** capacitor in a batch of nominal 100 nF $\pm$ 10%. Which resistances guarantee this?

Step 1. $C \in [90, 110]$ nF and $R > 0$, so τ ranges over $[90R,\ 110R]$ (nF times R).

Step 2. Both ends must satisfy the spec: $90 \times 10^{-9} R \geq 0.85 \times 10^{-3}$ and $110 \times 10^{-9} R \leq 1.15 \times 10^{-3}$.

Step 3. $R \geq 9.444$ kΩ and $R \leq 10.455$ kΩ. So $R \in [9.44,\ 10.45]$ kΩ, and a 10.0 kΩ $\pm$ 1% resistor (range 9.9–10.1 kΩ) qualifies.

Check: worst cases $9.9\ \text{k}\Omega \times 90\ \text{nF} = 0.891$ ms ≥ 0.85 ✓ and $10.1\ \text{k}\Omega \times 110\ \text{nF} = 1.111$ ms ≤ 1.15 ✓.

Engineering Note

Design specifications are inequalities: "gain within ±0.5 dB", "ripple below 50 mV", "phase margin at least 45°". Solving them gives the admissible range of component values, and the worst-case analysis above (check every combination of extremes) is the systematic way to guarantee them. When there are many parameters, the same problem becomes constrained optimisation (Ch. 23).

Common Pitfall

Never multiply both sides of an inequality by an expression whose sign you do not know, and never square both sides unless both are known to be non-negative. When in doubt, move everything to one side and use a sign chart.

1.4 Polynomials and rational expressions

Why polynomials and rational functions

Polynomials are the simplest functions that can be computed with nothing but addition and multiplication, which is why they are used to approximate everything else (Taylor series, Ch. 4; curve fitting, Ch. 22). Ratios of polynomials — **rational functions** — are even more central to electrical engineering: the impedance of any network of resistors, capacitors and inductors, and the transfer function of any linear time-invariant circuit, is a rational function of the complex frequency s. Its **zeros** and **poles** (roots of numerator and denominator) determine the circuit's frequency response and stability. Factoring polynomials and decomposing rational functions are therefore everyday skills.

Definitions and arithmetic

Polynomial

A **polynomial** of **degree** n in x is

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0, \qquad a_n \neq 0.$$

The a_k are the **coefficients**, a_n the **leading coefficient** (the polynomial is **monic** if $a_n = 1$), and a_0 the **constant term**. A **root** or **zero** is a number r with $p(r) = 0$.

Sums and products of polynomials are polynomials, with $\deg(pq) = \deg p + \deg q$. Division is like integer division: it leaves a quotient and a remainder.

Division algorithm

For polynomials p and d with $d \neq 0$ there are unique polynomials q (quotient) and r (remainder) with

$$p(x) = d(x)\,q(x) + r(x), \qquad \deg r < \deg d \ \ (\text{or } r = 0).$$

The quotient and remainder are found by **long division**: repeatedly divide the leading term of what remains by the leading term of d, multiply back, and subtract. When $d(x) = x - r$ the bookkeeping shrinks to **synthetic division**, which keeps only the coefficients.

Worked Example 1.19 — Long and synthetic division

(a) Divide $p(x) = x^3 + 2$ by $d(x) = x^2 - 1$. Leading terms: $x^3/x^2 = x$. Subtract $x(x^2 - 1) = x^3 - x$ from $x^3 + 0x^2 + 0x + 2$ to leave $x + 2$, whose degree is below 2. So

$$x^3 + 2 = (x^2 - 1)\,x + (x + 2), \qquad q(x) = x, \quad r(x) = x + 2.$$

(b) Divide $2x^3 - 3x^2 - 11x + 6$ by $x - 3$ synthetically. Write the coefficients, bring down the leading one, and repeatedly multiply by 3 and add:

	2	−3	−11	6
×3		6	9	−6
sum	2	3	−2	**0**

Quotient $2x^2 + 3x - 2$, remainder 0.

Check: $(x - 3)(2x^2 + 3x - 2) = 2x^3 + 3x^2 - 2x - 6x^2 - 9x + 6 = 2x^3 - 3x^2 - 11x + 6$ ✓.

Remainder and factor theorems

Remainder and factor theorems

- **Remainder theorem:** dividing $p(x)$ by $(x - r)$ leaves the constant remainder $p(r)$.
- **Factor theorem:** $(x - r)$ is a factor of $p(x)$ if and only if $p(r) = 0$.

Proof. By the division algorithm $p(x) = (x - r)q(x) + c$ with c constant (degree below 1). Setting $x = r$ gives $p(r) = c$. The factor theorem is the case $c = 0$. □

So the remainder of $x^5 - 3x + 2$ on division by $x + 1$ is $p(-1) = -1 + 3 + 2 = 4$, found without dividing.

The fundamental theorem of algebra

Fundamental theorem of algebra

Every polynomial of degree $n \geq 1$ with complex (in particular, real) coefficients has exactly n complex roots counted with multiplicity, and factors completely as

$$p(x) = a_n(x - r_1)(x - r_2)\cdots(x - r_n).$$

If all coefficients are **real**, non-real roots occur in **conjugate pairs** $\alpha \pm j\beta$.

The existence of at least one root is the deep part (it needs completeness of $\mathbb{R}$ and is proved with complex analysis in Ch. 18); the full factorisation then follows by applying the factor theorem repeatedly. The conjugate-pair property has a one-line proof: conjugating $\sum a_k r^k = 0$ with real a_k gives $\sum a_k (r^*)^k = 0$, so r^* is a root too.

Consequently every **real** polynomial factors into real linear factors and real **irreducible quadratics** $x^2 + bx + c$ with $b^2 < 4c$, since $(x - \alpha - j\beta)(x - \alpha + j\beta) = (x - \alpha)^2 + \beta^2$. A real polynomial of odd degree therefore always has at least one real root.

A root r has **multiplicity** m if $(x - r)^m$ divides p but $(x - r)^{m+1}$ does not. On a graph, the curve crosses the axis at a root of odd multiplicity and touches it (turning back) at a root of even multiplicity; higher multiplicity makes the curve flatter at the root.

Vieta's relations

For $p(x) = a_n x^n + \cdots + a_0$ with roots $r_1, \dots, r_n$:

$$\sum_{i} r_i = -\frac{a_{n-1}}{a_n}, \qquad \sum_{i<k} r_i r_k = \frac{a_{n-2}}{a_n}, \qquad \prod_{i} r_i = (-1)^n \frac{a_0}{a_n}.$$

These follow from expanding $a_n \prod (x - r_i)$ and comparing coefficients. In a control system, the sum of the closed-loop poles is fixed by one coefficient, which is why moving one pole to the left forces another to the right.

Finding roots: the rational-root test

Rational-root test

If p has integer coefficients and a rational root u/v in lowest terms, then u divides a_0 and v divides a_n.

Proof. Multiply $p(u/v)=0$ by v^n: $a_n u^n + a_{n-1}u^{n-1}v + \cdots + a_0 v^n = 0$. Every term except $a_0 v^n$ contains u, so u divides $a_0 v^n$; as u, v share no factor, u divides a_0. Similarly v divides $a_n u^n$, hence a_n. □

Worked Example 1.20 — Factoring a cubic

Factor $p(x) = 2x^3 - 3x^2 - 11x + 6$ completely.

Step 1 — candidates. $u \mid 6$ and $v \mid 2$: $\pm1, \pm2, \pm3, \pm6, \pm\tfrac{1}{2}, \pm\tfrac{3}{2}$.

Step 2 — test. $p(1) = -6$, $p(-1) = 12$, $p(3) = 54 - 27 - 33 + 6 = 0$. ✓

Step 3 — deflate. By the synthetic division of Example 1.19(b), $p(x) = (x-3)(2x^2 + 3x - 2)$, and $2x^2 + 3x - 2 = (2x-1)(x+2)$.

Result: $p(x) = (x-3)(2x-1)(x+2)$, roots $3,\ \tfrac{1}{2},\ -2$.

Check (Vieta): sum $3 + \tfrac{1}{2} - 2 = \tfrac{3}{2} = -(-3)/2$ ✓; product $3 \cdot \tfrac{1}{2} \cdot (-2) = -3 = (-1)^3 \cdot 6/2$ ✓.

Worked Example 1.21 — Characteristic polynomial with complex roots

A third-order system has characteristic polynomial $D(s) = s^3 + 4s^2 + 6s + 4$. Find its roots.

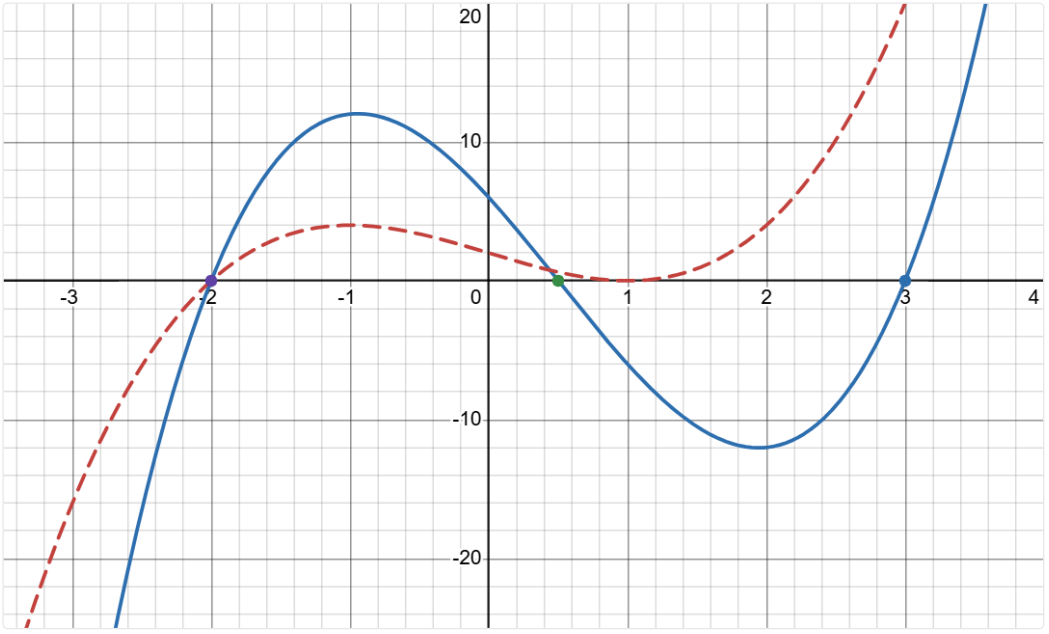
Step 1. Rational candidates $\pm1, \pm2, \pm4$. All coefficients are positive, so no positive real root is possible; try negatives: $D(-1) = -1 + 4 - 6 + 4 = 1$, $D(-2) = -8 + 16 - 12 + 4 = 0$. ✓

Step 2. Deflate: $D(s) = (s+2)(s^2 + 2s + 2)$.

Step 3. Complete the square: $s^2 + 2s + 2 = (s+1)^2 + 1$, so $s = -1 \pm j$.

Roots: $-2,\ -1 + j,\ -1 - j$ — all with negative real part, so the system is stable (Ch. 14). **Check:** sum $= -4 = -a_2/a_3$ ✓; product $(-2)\big((-1)^2 + 1^2\big) = -4 = (-1)^3 \cdot 4$ ✓.

For degree 5 and above there is no general formula in radicals (Abel–Ruffini theorem), and even the cubic and quartic formulas are rarely worth using. In practice, high-degree roots are found numerically — Newton's method or the eigenvalues of a companion matrix (Ch. 22).



The cubic of Example 1.20 (blue) crosses the axis at its three simple roots. The red dashed cubic $(x-1)^2(x+2)$ touches the axis at its double root $x = 1$ without crossing.

Rational functions

Rational function

A **rational function** is $R(x) = \dfrac{N(x)}{D(x)}$ with polynomials N, D and $D \not\equiv 0$. It is **proper** if $\deg N < \deg D$ and **improper** otherwise. After cancelling common factors, zeros of N are the **zeros** of R and zeros of D are its **poles**.

An improper rational function is reduced by long division to a polynomial plus a proper part: $R(x) = q(x) + \dfrac{r(x)}{D(x)}$. This also identifies the **asymptotes** of its graph:

- **vertical asymptotes** at the real poles, where $|R(x)| \to \infty$;
- a **horizontal asymptote** $y = a_n/b_n$ if $\deg N = \deg D$, or $y = 0$ if $\deg N < \deg D$;
- an **oblique** (slant) asymptote $y = q(x)$ if $\deg N = \deg D + 1$.

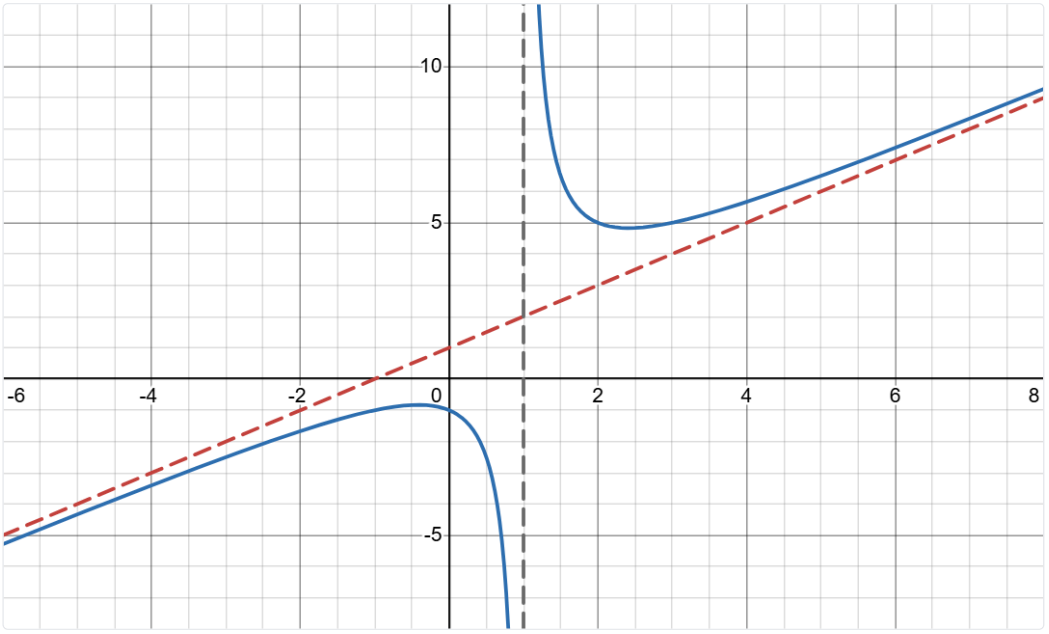
Worked Example 1.22 — Asymptotes

$R(x) = \dfrac{x^2+1}{x-1}$. Division gives $x^2 + 1 = (x-1)(x+1) + 2$, so

$$R(x) = x + 1 + \frac{2}{x-1}.$$

Vertical asymptote $x = 1$ (pole); oblique asymptote $y = x + 1$, approached from above as $x \to +\infty$ (since $\frac{2}{x-1} > 0$) and from below as $x \to -\infty$. No real zeros, as $x^2 + 1 > 0$.

Check: at $x = 3$, $R = 10/2 = 5$ and $3 + 1 + 2/2 = 5$ ✓.



$R(x) = (x^2+1)/(x-1)$ (blue) hugs its oblique asymptote $y = x + 1$ (red) far from the pole at $x = 1$.

Common Pitfall — cancelled factors

$\dfrac{x^2-1}{x-1} = x + 1$ only for $x \neq 1$. The left side is undefined at $x = 1$ (a "hole", not a pole). In a transfer function, cancelling a pole against a zero is mathematically legal but can hide an internal mode that still exists in the hardware — an unstable cancelled pole is still unstable.

Partial-fraction decomposition

Adding fractions over a common denominator is easy; partial fractions run the process **backwards**, splitting a complicated proper rational function into a sum of simple terms, each of which can be integrated, inverse-transformed or interpreted on its own.

Partial-fraction decomposition

Every proper real rational function $N(x)/D(x)$ can be written uniquely as a sum of terms, one group for each factor of D:

Factor in $D(x)$	Terms in the decomposition
$(x-a)$	$\dfrac{A}{x-a}$
$(x-a)^m$	$\dfrac{A_1}{x-a} + \dfrac{A_2}{(x-a)^2} + \cdots + \dfrac{A_m}{(x-a)^m}$
x^2+bx+c (irreducible)	$\dfrac{Bx+C}{x^2+bx+c}$
$(x^2+bx+c)^m$	$\displaystyle\sum_{k=1}^{m} \frac{B_k x + C_k}{(x^2+bx+c)^k}$

The number of unknown constants always equals $\deg D$, which is why the decomposition is unique: multiplying out gives $\deg D$ linear equations in $\deg D$ unknowns. There are three ways to find them, often combined:

1. **Cover-up rule** (distinct linear factor). Multiply both sides by $(x-a)$ and set $x=a$; every other term vanishes:

$$A = \Big[(x-a)R(x)\Big]_{x=a} = \left.\frac{N(x)}{D(x)/(x-a)}\right|_{x=a}.$$

2. **Substituting convenient values** of x into the cleared-of-fractions identity.
3. **Equating coefficients** of like powers of x.

For a repeated factor $(x-a)^m$, the cover-up rule gives the top coefficient $A_m = \big[(x-a)^m R(x)\big]_{x=a}$, and the others follow by differentiation:

$$A_{m-k} = \frac{1}{k!}\frac{d^k}{dx^k}\Big[(x-a)^m R(x)\Big]_{x=a}, \qquad k = 1, \dots, m-1.$$

(Differentiation is introduced in Ch. 4; until then, method 3 always works.)

Worked Example 1.23 — Distinct and repeated linear factors

(a) $\dfrac{s+3}{(s+1)(s+2)} = \dfrac{A}{s+1} + \dfrac{B}{s+2}$. Cover-up: $A = \dfrac{-1+3}{-1+2} = 2$, $B = \dfrac{-2+3}{-2+1} = -1$. **Check:** $2(s+2) - (s+1) = s+3$ ✓.

(b) $\dfrac{1}{s(s+1)^2} = \dfrac{A}{s} + \dfrac{B}{s+1} + \dfrac{C}{(s+1)^2}$. Clear fractions: $1 = A(s+1)^2 + Bs(s+1) + Cs$. $s=0$: $A=1$. $s=-1$: $C=-1$. Coefficient of s^2: $0 = A + B \Rightarrow B = -1$. (By the derivative formula: $B = \frac{d}{ds}\big[1/s\big]_{s=-1} = -1/s^2\big|_{-1} = -1$ ✓.) **Check** at $s=1$: LHS $= \frac{1}{4}$; RHS $= 1 - \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$ ✓.

Worked Example 1.24 — An irreducible quadratic factor

Decompose $\dfrac{s+2}{(s+1)(s^2+2s+5)}$.

Step 1. $s^2+2s+5 = (s+1)^2 + 4$ has $\Delta = 4 - 20 < 0$, so it is irreducible. Write

$$\frac{s+2}{(s+1)(s^2+2s+5)} = \frac{A}{s+1} + \frac{Bs+C}{s^2+2s+5}.$$

Step 2 — cover-up for A. $A = \dfrac{-1+2}{1-2+5} = \dfrac{1}{4}$.

Step 3 — equate coefficients in $s+2 = \frac{1}{4}(s^2+2s+5) + (Bs+C)(s+1)$: s^2: $0 = \frac{1}{4} + B \Rightarrow B = -\frac{1}{4}$. Constant: $2 = \frac{5}{4} + C \Rightarrow C = \frac{3}{4}$.

Step 4 — check the unused coefficient (s^1): $\frac{2}{4} + B + C = \frac{1}{2} - \frac{1}{4} + \frac{3}{4} = 1$ ✓.

Result, with the quadratic term arranged for later use in Laplace inversion:

$$\frac{1/4}{s+1} + \frac{-\frac{1}{4}s + \frac{3}{4}}{(s+1)^2+4} = \frac{1/4}{s+1} - \frac{1}{4}\cdot\frac{s+1}{(s+1)^2+2^2} + \frac{1}{2}\cdot\frac{2}{(s+1)^2+2^2}.$$

Worked Example 1.25 — An improper fraction

Decompose $\dfrac{x^3+2}{x^2-1}$.

Step 1 — divide first (Example 1.19a): $\dfrac{x^3+2}{x^2-1} = x + \dfrac{x+2}{(x-1)(x+1)}$.

Step 2 — cover-up on the proper part: at $x=1$, $\dfrac{1+2}{1+1} = \dfrac{3}{2}$; at $x=-1$, $\dfrac{-1+2}{-1-1} = -\dfrac{1}{2}$.

Result: $\dfrac{x^3+2}{x^2-1} = x + \dfrac{3/2}{x-1} - \dfrac{1/2}{x+1}$.

Check at $x=2$: LHS $= 10/3 = 3.333$; RHS $= 2 + 1.5 - \frac{1}{6} = 3.333$ ✓.

Partial fractions are the workhorse of Laplace and Z-transform inversion (Ch. 14, Ch. 17) and of integrating rational functions (Ch. 5); in transfer functions the variable is conventionally s, and the coefficients A are called **residues**.

Engineering Note — partial fractions as parallel decomposition

Each term $\frac{A}{s-p}$ corresponds to a first-order mode Ae^{pt} in the time response. A partial-fraction expansion therefore splits a high-order system into a **parallel bank of simple subsystems**, one per pole. Software such as MATLAB's `residue` or SciPy's `scipy.signal.residue` performs the decomposition numerically; the hand method remains essential for checking it and for symbolic work.

Common Pitfall

Partial fractions apply only to **proper** fractions: $\dfrac{x^2}{x^2-1}$ must first become $1 + \dfrac{1}{x^2-1}$. Also, an irreducible quadratic needs a **linear** numerator $Bx + C$, not a constant — using only C leaves too few unknowns and the equations become inconsistent.

1.5 Exponential and logarithmic functions

Why exponentials and logarithms

Whenever the rate of change of a quantity is proportional to the quantity itself, the result is exponential: a capacitor discharging through a resistor, current building up in an inductor, a radioactive source decaying, a population of charge carriers being swept out of a junction. The diode current grows exponentially with voltage, and the open-loop growth of an oscillator is exponential until it saturates. Logarithms, the inverse operation, turn multiplication into addition, which is why engineers measure gains in decibels, plot frequency responses on logarithmic axes, and quote signal-to-noise ratios as logarithms. These two functions are as fundamental to engineering as the four arithmetic operations.

Exponential functions

Exponential function

For a fixed base $a > 0$, $a \neq 1$, the **exponential function** $f(x) = a^x$ is defined for all real x by the exponent laws of Section 1.2 (extended to irrational x by continuity). It satisfies

$$a^{x+y} = a^x a^y, \qquad a^0 = 1, \qquad a^x > 0 \text{ for all } x.$$

For $a > 1$ it is strictly increasing, with $a^x \to \infty$ as $x \to \infty$ and $a^x \to 0$ as $x \to -\infty$; for $0 < a < 1$ it is strictly decreasing.

The functional equation $f(x+y) = f(x)f(y)$ is the essential property: equal steps in x multiply f by equal factors. A quantity that halves every 5 ms is $f(t) = f(0)\,2^{-t/5\,\text{ms}}$.

The natural base e

Among all bases one is singled out by calculus. Consider interest at a nominal rate of 100% per year compounded n times a year: one unit grows to $\left(1 + \tfrac{1}{n}\right)^n$.

n	1	10	100	1000	10^6
$\left(1+\frac{1}{n}\right)^n$	2	2.59374	2.70481	2.71692	2.71828

Euler's number

$$e = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n = \sum_{k=0}^{\infty}\frac{1}{k!} = 1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} + \cdots = 2.718281828\ldots$$

The **natural exponential** is $\exp(x) = e^x$, and more generally $e^x = \lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n = \sum_{k=0}^{\infty}\frac{x^k}{k!}$.

The series converges very fast: the first eight terms already give 2.71825. What makes e natural is that e^x is the unique function equal to its own derivative with value 1 at $x = 0$ (Ch. 4). Consequently the solution of $\dot{y} = -y/\tau$ is $y = y(0)e^{-t/\tau}$, which is why e — not 2 or 10 — appears in every first-order circuit (Ch. 11). Any other exponential can be rewritten in base e: $a^x = e^{x \ln a}$.

Logarithms

Logarithm

For $a > 0$, $a \neq 1$, the **logarithm to base** a is the inverse of a^x:

$$y = a^x \iff x = \log_a y, \qquad y > 0.$$

The **natural logarithm** is $\ln x = \log_e x$; the **common logarithm** is $\log_{10} x$; $\log_2 x$ is used in information theory and digital design.

So $\log_a y$ answers the question "to what power must a be raised to give y?" Immediately, $\log_a 1 = 0$, $\log_a a = 1$, $a^{\log_a y} = y$ and $\log_a(a^x) = x$. The domain of $\log_a$ is $(0,\infty)$ and its range is all of $\mathbb{R}$.

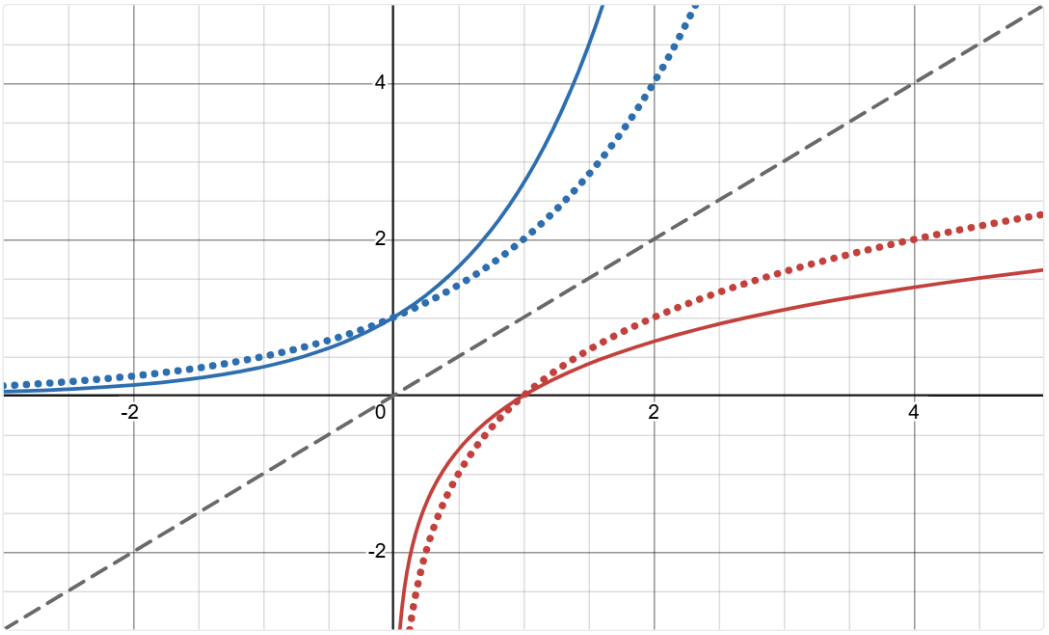
Laws of logarithms ($x, y > 0$, any real p)

$$\log_a(xy) = \log_a x + \log_a y, \qquad \log_a\frac{x}{y} = \log_a x - \log_a y, \qquad \log_a x^p = p\log_a x,$$

$$\log_a x = \frac{\log_b x}{\log_b a} \quad \text{(change of base)}, \qquad \text{in particular} \quad \log_a x = \frac{\ln x}{\ln a}.$$

Proof. Let $u = \log_a x$ and $v = \log_a y$, so $x = a^u$, $y = a^v$. Then $xy = a^{u+v}$, so $\log_a(xy) = u + v$; $x^p = a^{pu}$, so $\log_a x^p = pu$. For change of base, take $\log_b$ of $x = a^u$: $\log_b x = u \log_b a$. $\square$

Each law is just an exponent law read backwards. Two useful consequences: $\log_{10} 2 = 0.30103$, so $2^{10} = 1024 \approx 10^3$ ("ten bits per factor of a thousand"), and the number of bits needed to index N items is $\lceil \log_2 N \rceil$; a million items need $\lceil 19.93 \rceil = 20$ bits.



e^x (blue) and $\ln x$ (red) are mirror images in $y = x$, as every function and its inverse must be (Section 1.9). Base-2 versions (dotted) have the same shapes. Exponentials all pass through $(0, 1)$ and logarithms through $(1, 0)$.

Solving exponential and logarithmic equations

The strategy is to isolate the exponential (or logarithm), then apply the inverse function. Because exp and ln are one-to-one, these steps are equivalences — provided arguments of logarithms are positive.

Worked Example 1.26 — Exponential equations

(a) Solve $2^{x+1} = 3^x$. Take natural logs: $(x+1)\ln 2 = x \ln 3$, so $x(\ln 3 - \ln 2) = \ln 2$ and

$$x = \frac{\ln 2}{\ln 3 - \ln 2} = \frac{0.69315}{0.40547} = 1.7095.$$

Check: $2^{2.7095} = 6.541$ and $3^{1.7095} = 6.541$ ✓.

(b) Solve $e^{2x} - 3e^x + 2 = 0$. With $u = e^x > 0$: $u^2 - 3u + 2 = (u-1)(u-2) = 0$, so $e^x = 1$ or $e^x = 2$, i.e. $x = 0$ or $x = \ln 2 = 0.6931$. (Had a root $u \le 0$ appeared, it would be rejected, since $e^x > 0$.)

(c) Solve $\log_{10}(x) + \log_{10}(x-3) = 1$. Combine: $\log_{10}\big(x(x-3)\big) = 1$, so $x^2 - 3x - 10 = 0$, $(x-5)(x+2) = 0$. The candidate $x = -2$ is rejected (logarithm of a negative number), leaving $x = 5$. **Check:** $\log_{10} 5 + \log_{10} 2 = \log_{10} 10 = 1$ ✓.

First-order transients

The step and natural responses of every first-order circuit are built from $e^{-t/\tau}$, where the **time constant** is $\tau = RC$ or $\tau = L/R$. Useful landmarks follow directly from the logarithm:

Elapsed time	$e^{-t/\tau}$	$1 - e^{-t/\tau}$
τ	0.368	0.632
2τ	0.135	0.865
3τ	0.050	0.950
5τ	0.0067	0.9933
$\tau \ln 2 = 0.693\tau$	0.5	0.5

Worked Example 1.27 — Capacitor discharge and rise time

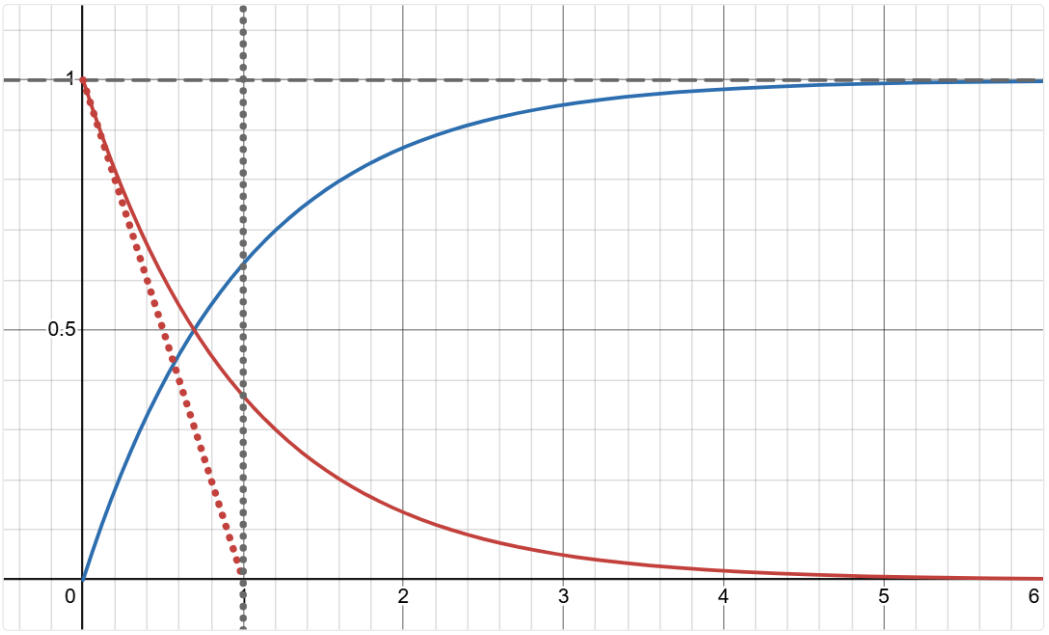
(a) A capacitor discharges as $v(t) = V_0 e^{-t/\tau}$ with $R = 10\ \text{k}\Omega$, $C = 100\ \text{nF}$. How long until v falls to 10% of V_0? $\tau = 10^4 \times 10^{-7} = 10^{-3}$ s. Set $e^{-t/\tau} = 0.1$ and take ln: $-t/\tau = \ln 0.1 = -\ln 10$, so

$$t = \tau \ln 10 = 2.303\ \text{ms}.$$

(b) The **10–90 % rise time** of a charging response $v = V_\infty(1 - e^{-t/\tau})$: solve $1 - e^{-t_1/\tau} = 0.1$ and $1 - e^{-t_2/\tau} = 0.9$, giving $t_1 = \tau \ln \frac{1}{0.9}$ and $t_2 = \tau \ln 10$. Hence

$$t_r = t_2 - t_1 = \tau\left(\ln 10 - \ln \tfrac{10}{9}\right) = \tau \ln 9 = 2.197\,\tau.$$

Check: $e^{-2.303} = 0.100$ ✓ and $\ln 9 = 2.1972$ ✓. Combined with the bandwidth of the same circuit, $f_{3\text{dB}} = 1/(2\pi\tau)$, this gives the well-known rule $t_r f_{3\text{dB}} \approx 0.35$.



Charging $1 - e^{-t/\tau}$ (blue) and discharging $e^{-t/\tau}$ (red) against t/τ. The dotted tangent at $t = 0$ reaches zero at exactly one time constant; after 5τ the transient is within 1% of its final value.

Worked Example 1.28 — Inverting the diode equation

The Shockley diode equation is $I = I_S\left(e^{V/(nV_T)} - 1\right)$. Find V for $I = 1$ mA when $I_S = 10^{-14}$ A, $n = 1$, $V_T = 25.85$ mV, and find how much V changes per decade of current.

Step 1 — isolate the exponential. $e^{V/(nV_T)} = 1 + I/I_S$, so

$$V = nV_T \ln\left(1 + \frac{I}{I_S}\right).$$

Step 2 — evaluate. $I/I_S = 10^{11}$, $\ln(10^{11} + 1) = 25.328$, so $V = 0.02585 \times 25.328 = 0.6547$ V.

Step 3 — per decade. For $I \gg I_S$, $V \approx nV_T \ln(I/I_S)$, and multiplying I by 10 adds $nV_T \ln 10 = 59.5$ mV.

Check: $10^{-14}\left(e^{0.6547/0.02585} - 1\right) \approx 1.00 \times 10^{-3}$ A ✓. The "60 mV per decade" rule of analogue design is just $\ln 10$ in disguise.

Decibels and logarithmic scales

Power ratios in electronics span many orders of magnitude — a radio receiver may handle signals from 10^{-15} W to 10^{-3} W — so engineers use the **decibel**:

Decibel

$$G_{\text{dB}} = 10\log_{10}\frac{P_2}{P_1} = 20\log_{10}\frac{V_2}{V_1} \quad \text{(voltages across equal resistances)}.$$

Absolute levels use a stated reference: dBm is relative to 1 mW, dBW to 1 W, dBV to 1 V, dBμV to 1 μV.

The factor 20 for voltage comes from $P \propto V^2$ and the law $\log x^2 = 2\log x$. Useful anchors: $\times 2$ in power $= +3.01$ dB; $\times 2$ in voltage $= +6.02$ dB; $\times 10$ in voltage $= +20$ dB; $1/\sqrt{2}$ in voltage $= -3.01$ dB (the "half-power" or "-3 dB" point that defines bandwidth). Since 0 dBm $= 1$ mW, $+30$ dBm $= 1$ W.

Gains in cascade **multiply**, so their dB values **add** — the whole point of the logarithm.

Worked Example 1.29 — A receiver chain

A signal of -90 dBm passes through a low-noise amplifier ($+15$ dB), a mixer (-7 dB conversion loss) and an IF amplifier ($+25$ dB). Find the output power in dBm and in watts, and the overall voltage gain factor.

Step 1 — add. $-90 + 15 - 7 + 25 = -57$ dBm.

Step 2 — convert. $P = 10^{-57/10}$ mW $= 1.995 \times 10^{-6}$ mW $= 1.995$ nW.

Step 3 — overall gain. $+33$ dB; as a power ratio $10^{3.3} = 1995$; as a voltage ratio (equal impedances) $10^{33/20} = 44.7$.

Check: multiplying the linear power gains $31.62 \times 0.1995 \times 316.2 = 1995$ ✓.

Logarithmic axes. Plotting $\log y$ against $\log x$ turns a **power law** $y = ax^b$ into a straight line, $\log y = \log a + b\log x$, of slope b. Plotting $\log y$ against x (semi-log) turns an exponential $y = ae^{kx}$ into a straight line of slope $k\log e$. Bode plots (Ch. 16) use dB against $\log\omega$, so that a pole contributes a straight-line slope of -20 dB per decade.

Worked Example 1.30 — Fitting a power law from two points

A measured quantity gives $y = 5.0$ at $x = 2$ and $y = 158.1$ at $x = 20$. Assuming $y = ax^b$, find a and b.

Step 1 — slope on log–log axes. $b = \dfrac{\log_{10} 158.1 - \log_{10} 5.0}{\log_{10} 20 - \log_{10} 2} = \dfrac{\log_{10} 31.62}{1} = 1.500.$

Step 2 — intercept. $a = 5.0/2^{1.5} = 1.768.$

Check: $1.768 \times 20^{1.5} = 1.768 \times 89.44 = 158.1$ ✓.

Hyperbolic functions

Certain combinations of exponentials occur so often — in transmission-line theory, catenary cables, and the solution of $y'' = k^2 y$ — that they have names.

Hyperbolic functions

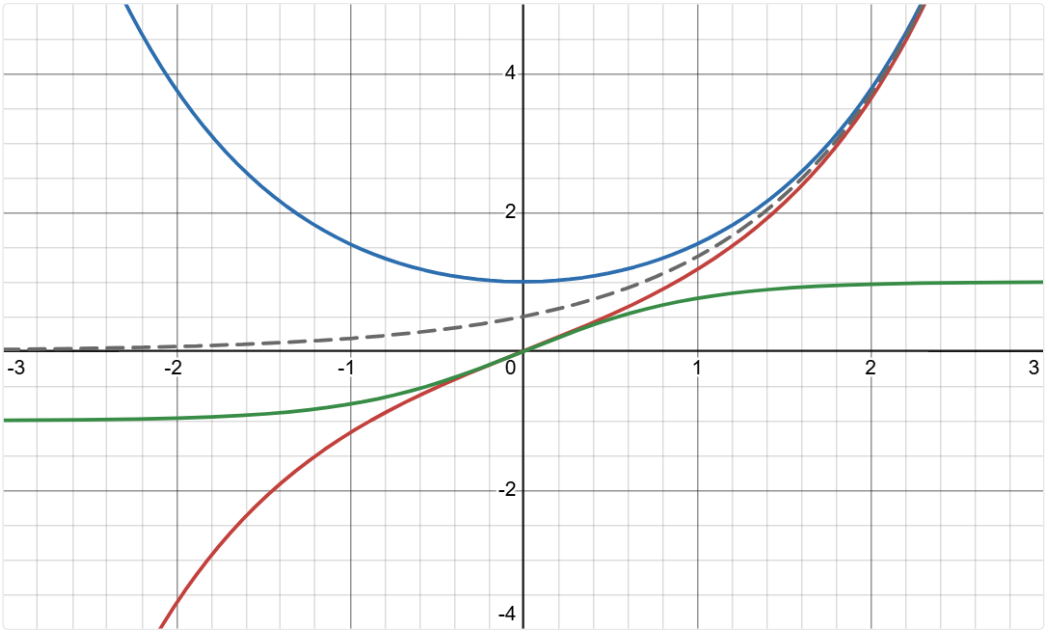
$$\cosh x = \frac{e^x + e^{-x}}{2}, \qquad \sinh x = \frac{e^x - e^{-x}}{2}, \qquad \tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}.$$

Hyperbolic identities

$$\cosh^2 x - \sinh^2 x = 1, \qquad \cosh x + \sinh x = e^x, \qquad \sinh 2x = 2\sinh x \cosh x, \qquad \cosh 2x = \cosh^2 x + \sinh^2 x.$$

Proof of the first. $\cosh^2 x - \sinh^2 x = \frac{1}{4}\left[(e^{2x} + 2 + e^{-2x}) - (e^{2x} - 2 + e^{-2x})\right] = 1.$ □

They mirror the trigonometric identities with some sign changes — no coincidence: with complex arguments $\cosh(jx) = \cos x$ and $\sinh(jx) = j\sin x$ (Ch. 2). The parametrisation $(\cosh t, \sinh t)$ traces the hyperbola $x^2 - y^2 = 1$, just as $(\cos t, \sin t)$ traces the circle. Note cosh is even, sinh and tanh are odd, $\cosh x \geq 1$, and $\tanh x \in (-1, 1)$ — the soft-saturation shape used to model a differential pair (Section 1.8). Numerically, $\cosh 1 = 1.5431$, $\sinh 1 = 1.1752$, $\tanh 1 = 0.7616$.



cosh x (blue), sinh x (red) and tanh x (green). For large x both cosh x and sinh x approach $e^x/2$ (grey dashed), while tanh x saturates at ± 1.

Common Pitfall

$\ln(x + y) \neq \ln x + \ln y$, $\ln(xy) \neq \ln x \cdot \ln y$, and ln is undefined for $x \leq 0$. Before taking logs of both sides of an equation, make sure both sides are positive. In decibels, never use 20 log for a power ratio or 10 log for a voltage ratio, and remember that "dB" alone is a ratio while "dBm" is an absolute power.

Engineering Note — logarithms of dimensional quantities

Strictly, only dimensionless numbers have logarithms (Section 1.10). An expression like $\ln I$ is shorthand for $\ln(I/I_0)$ with some reference I_0, and $\ln I_1 - \ln I_2 = \ln(I_1/I_2)$ is then reference-independent. This is exactly why decibel levels always state a reference (dBm, dBV): without it the number is meaningless.

1.6 Trigonometric functions and identities

Why trigonometry

Alternating current, radio carriers, audio tones, rotating machines and the components of a Fourier series are all sinusoids. Angles also enter through geometry: the phase difference between two phasors, the direction of a field vector, the bearing of an antenna beam. Trigonometric identities are the algebra of these quantities — they let a product of two signals be rewritten as a sum of tones (the principle of every mixer), a sum of a sine and a cosine be rewritten as a single shifted sinusoid, and a squared signal be rewritten in terms of its average power. This section gathers the functions and identities used throughout the book; Ch. 3 develops sinusoids and periodic signals in depth, and Ch. 2 shows how Euler's formula derives every identity below from one exponential law.

Angles and radians

Radian

The **radian** measure of an angle is the length of the arc it subtends on a circle divided by the radius: $\theta = s/r$. A full turn is 2π rad, so

$$\pi \text{ rad} = 180^\circ, \qquad 1 \text{ rad} = 57.2958^\circ, \qquad 1^\circ = 0.0174533 \text{ rad}.$$

Being a ratio of two lengths, the radian is dimensionless, which is what allows $\sin\theta$ to have a power-series expansion (Section 1.10). In calculus, angles are **always** in radians: the derivative $\frac{d}{d\theta}\sin\theta = \cos\theta$ holds only in radians (in degrees it acquires a factor $\pi/180$). The arc length $s = r\theta$ and the sector area $A = \frac{1}{2}r^2\theta$ also assume radians.

Definitions on the unit circle

Sine, cosine and tangent

Let the ray at angle θ, measured anticlockwise from the positive x-axis, meet the unit circle $x^2 + y^2 = 1$ at the point P. Then

$$P = (\cos\theta,\ \sin\theta), \qquad \tan\theta = \frac{\sin\theta}{\cos\theta} \quad (\cos\theta \neq 0).$$

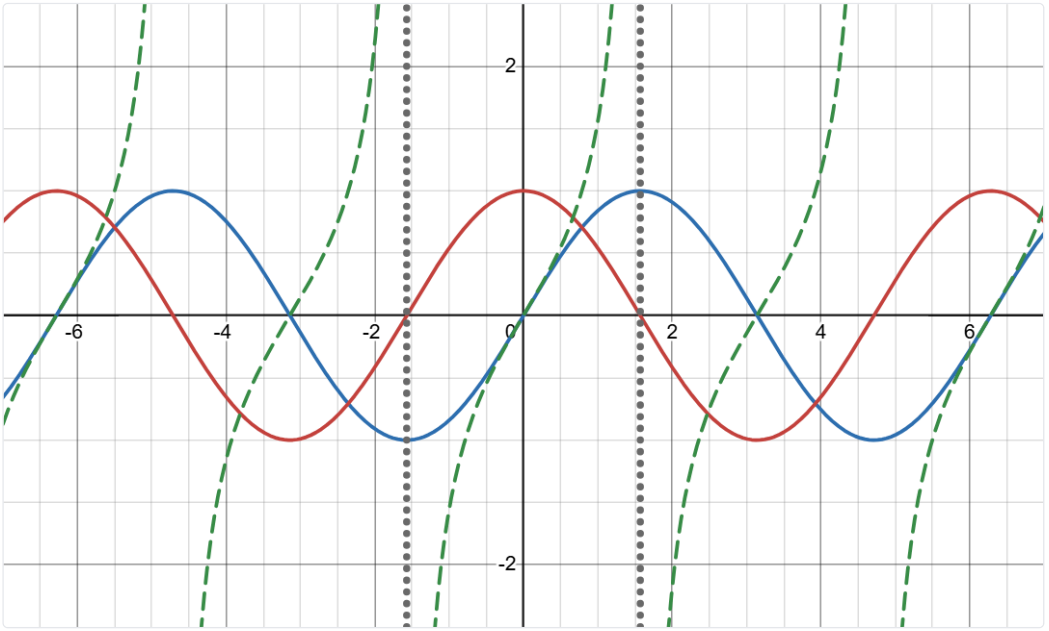
The reciprocal functions are $\sec\theta = 1/\cos\theta$, $\csc\theta = 1/\sin\theta$, $\cot\theta = 1/\tan\theta$.

For an acute angle in a right triangle this agrees with "opposite over hypotenuse" etc., but the unit-circle definition works for every real θ, including negative angles and angles beyond 2π. Immediately:

- sin and cos are **periodic** with period 2π; tan has period π.
- $-1 \le \sin\theta \le 1$ and $-1 \le \cos\theta \le 1$; tan takes every real value.
- cos is **even** and sin, tan are **odd**: $\cos(-\theta) = \cos\theta$, $\sin(-\theta) = -\sin\theta$.
- Shifts: $\sin\left(\theta + \frac{\pi}{2}\right) = \cos\theta$, $\cos\left(\theta + \frac{\pi}{2}\right) = -\sin\theta$, $\sin(\theta + \pi) = -\sin\theta$, $\cos(\pi - \theta) = -\cos\theta$, $\sin(\pi - \theta) = \sin\theta$.
- Signs by quadrant: all positive in the first; only sin in the second; only tan in the third; only cos in the fourth.

The exact values below follow from the 45° isosceles right triangle and the half-equilateral 30°–60° triangle.

θ	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$	$2\pi/3$	$3\pi/4$	π
$\sin\theta$	0	$1/2$	$\sqrt{2}/2$	$\sqrt{3}/2$	1	$\sqrt{3}/2$	$\sqrt{2}/2$	0
$\cos\theta$	1	$\sqrt{3}/2$	$\sqrt{2}/2$	$1/2$	0	$-1/2$	$-\sqrt{2}/2$	-1
$\tan\theta$	0	$1/\sqrt{3}$	1	$\sqrt{3}$	undef.	$-\sqrt{3}$	-1	0



$\sin x$ (blue), $\cos x$ (red, the sine shifted left by $\pi/2$) and $\tan x$ (green, period π, vertical asymptotes where $\cos x = 0$, marked at $\pm\pi/2$).

The fundamental identities

Pythagorean identities

$$\sin^2\theta + \cos^2\theta = 1, \qquad 1 + \tan^2\theta = \sec^2\theta, \qquad 1 + \cot^2\theta = \csc^2\theta.$$

Proof. The point $(\cos\theta, \sin\theta)$ lies on $x^2 + y^2 = 1$. Divide by $\cos^2\theta$ or $\sin^2\theta$ for the other two. □

Compound-angle formulas

$$\begin{aligned}\cos(A \pm B) &= \cos A\cos B \mp \sin A\sin B\\ \sin(A \pm B) &= \sin A\cos B \pm \cos A\sin B\\ \tan(A \pm B) &= \frac{\tan A \pm \tan B}{1 \mp \tan A\tan B}\end{aligned}$$

Proof of $\cos(A-B)$. The points $P = (\cos A, \sin A)$ and $Q = (\cos B, \sin B)$ on the unit circle are separated by the angle $A - B$. By the distance formula,

$$|PQ|^2 = (\cos A - \cos B)^2 + (\sin A - \sin B)^2 = 2 - 2(\cos A\cos B + \sin A\sin B).$$

Rotating the picture so that Q lies at $(1, 0)$ moves P to $(\cos(A-B), \sin(A-B))$ without changing distances, so also $|PQ|^2 = (\cos(A-B) - 1)^2 + \sin^2(A-B) = 2 - 2\cos(A-B)$. Comparing gives $\cos(A-B) = \cos A\cos B + \sin A\sin B$. Replacing B by $-B$ gives $\cos(A+B)$; replacing A by $\frac{\pi}{2} - A$ converts cosines to sines and gives the sine formulas; dividing gives the tangent formula. □

Every other identity in this section follows from these by algebra.

Double-angle and power-reduction formulas

$$\begin{aligned}\sin 2\theta &= 2\sin\theta\cos\theta\\ \cos 2\theta &= \cos^2\theta - \sin^2\theta = 1 - 2\sin^2\theta = 2\cos^2\theta - 1\\ \sin^2\theta &= \tfrac{1}{2}(1 - \cos 2\theta), \qquad \cos^2\theta = \tfrac{1}{2}(1 + \cos 2\theta)\end{aligned}$$

(Set $A = B = \theta$ in the compound formulas, then use $\sin^2 + \cos^2 = 1$.)

Product-to-sum and sum-to-product

$$\begin{aligned}\cos A\cos B &= \tfrac{1}{2}[\cos(A-B) + \cos(A+B)]\\ \sin A\sin B &= \tfrac{1}{2}[\cos(A-B) - \cos(A+B)]\\ \sin A\cos B &= \tfrac{1}{2}[\sin(A-B) + \sin(A+B)]\\ \sin P + \sin Q &= 2\sin\tfrac{P+Q}{2}\cos\tfrac{P-Q}{2}, \qquad \cos P + \cos Q = 2\cos\tfrac{P+Q}{2}\cos\tfrac{P-Q}{2}\end{aligned}$$

Each product formula follows by adding or subtracting a pair of compound-angle formulas (e.g. $\cos(A-B) + \cos(A+B) = 2\cos A\cos B$). The sum-to-product forms are the same identities read backwards with $P = A + B$, $Q = A - B$.

Worked Example 1.31 — Exact values from identities

(a) $\cos 75^\circ = \cos(45^\circ + 30^\circ) = \dfrac{\sqrt{2}}{2}\cdot\dfrac{\sqrt{3}}{2} - \dfrac{\sqrt{2}}{2}\cdot\dfrac{1}{2} = \dfrac{\sqrt{6}-\sqrt{2}}{4}$. **Check:** $(2.44949 - 1.41421)/4 = 0.25882 = \cos 75^\circ$ ✓.

(b) If $\sin\theta = \frac{3}{5}$ with θ in the second quadrant, find $\cos\theta$ and $\sin 2\theta$. From $\cos^2\theta = 1 - \frac{9}{25} = \frac{16}{25}$ and $\cos\theta < 0$ in quadrant II, $\cos\theta = -\frac{4}{5}$. Then $\sin 2\theta = 2\cdot\frac{3}{5}\cdot\left(-\frac{4}{5}\right) = -\frac{24}{25}$. **Check:** $\theta = \pi - \arcsin 0.6 = 2.4981$ rad, $\sin(4.9962) = -0.96 = -24/25$ ✓.

Harmonic addition: $a\cos\theta + b\sin\theta$

A sum of a cosine and a sine at the **same** frequency is a single sinusoid with a phase shift. This identity is the bridge between the rectangular and polar descriptions of a sinusoid.

Harmonic addition

For real a, b not both zero,

$$a\cos\theta + b\sin\theta = R\cos(\theta - \alpha), \qquad R = \sqrt{a^2 + b^2}, \quad \alpha = \operatorname{atan2}(b, a).$$

Derivation. Expand the right side: $R\cos\theta\cos\alpha + R\sin\theta\sin\alpha$. Matching coefficients requires $R\cos\alpha = a$ and $R\sin\alpha = b$, i.e. (a, b) has polar coordinates (R, α) (Section 1.7). Squaring and adding gives $R^2 = a^2 + b^2$; the quadrant of α is fixed by the signs of a and b, which is exactly what atan2 provides. □

Worked Example 1.32 — Combining two components

Write $v(t) = 3\cos\omega t + 4\sin\omega t$ as a single cosine.

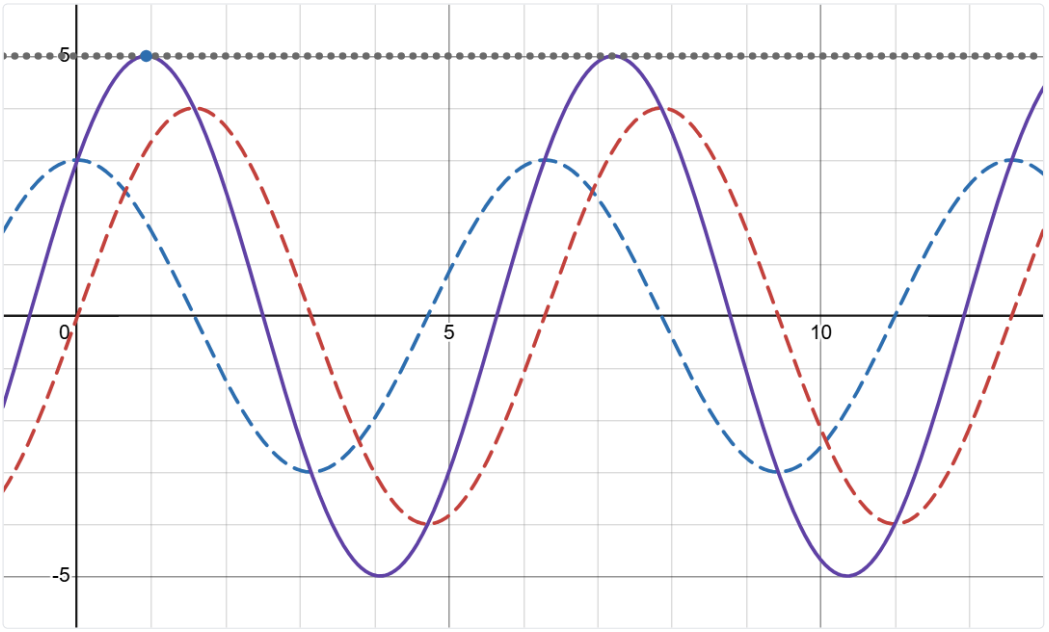
Step 1. $R = \sqrt{9+16} = 5$.

Step 2. $\alpha = \operatorname{atan2}(4, 3) = 0.9273$ rad $= 53.13°$ (first quadrant, since $a, b > 0$).

Result: $v(t) = 5\cos(\omega t - 0.9273)$: amplitude 5, lagging the reference cosine by $53.13°$.

Check at $\omega t = 0.5$: $3\cos 0.5 + 4\sin 0.5 = 4.5504$; $5\cos(0.5 - 0.9273) = 4.5504$ ✓.

Had the signal been $-3\cos\omega t + 4\sin\omega t$, $\alpha = \operatorname{atan2}(4, -3) = 126.87°$ — **not** $\arctan(4/(-3)) = -53.13°$, which would put the phase in the wrong quadrant.



$3\cos x$ (blue) plus $4\sin x$ (red) is a single sinusoid (purple) of amplitude 5, peaking at $x = \alpha = 0.927$ rad: harmonic addition made visible.

Products of sinusoids: mixing, beats and power

Worked Example 1.33 — A mixer and beats

(a) Mixing. Multiplying two tones,

$$\cos(\omega_1 t)\cos(\omega_2 t) = \tfrac{1}{2}\cos\big((\omega_1 - \omega_2)t\big) + \tfrac{1}{2}\cos\big((\omega_1 + \omega_2)t\big),$$

produces components at the **difference** and **sum** frequencies. With $f_1 = 10.7$ MHz and $f_2 = 10.0$ MHz the outputs are at 0.7 MHz and 20.7 MHz. This identity is the entire mathematical content of frequency conversion in a superheterodyne receiver.

(b) Beats. Adding two nearby tones, sum-to-product gives

$$\cos(2\pi \cdot 440\, t) + \cos(2\pi \cdot 444\, t) = 2\cos(2\pi \cdot 2\, t)\cos(2\pi \cdot 442\, t)\ :$$

a 442 Hz tone whose envelope $|2\cos(2\pi \cdot 2t)|$ swells and fades **4 times per second** (the envelope magnitude has period 0.25 s).

Check (b) at $t = 0.1$ s: LHS $= \cos(88\pi) + \cos(88.8\pi) = 1 + \cos(0.8\pi) = 1 - 0.80902 = 0.19098$; RHS $= 2\cos(0.4\pi)\cos(88.4\pi) = 2(0.30902)(0.30902) = 0.19098$ ✓.

Worked Example 1.34 — Average power of a sinusoid

A voltage $v(t) = V_m\cos\omega t$ across a resistor R delivers instantaneous power

$$p(t) = \frac{v^2}{R} = \frac{V_m^2}{R}\cos^2\omega t = \frac{V_m^2}{2R}\big(1 + \cos 2\omega t\big)$$

by power reduction. The $\cos 2\omega t$ term averages to zero over a period, so the mean power is $\bar{P} = \dfrac{V_m^2}{2R}$. Writing $\bar{P} = V_{\text{rms}}^2/R$ defines the **RMS value** $V_{\text{rms}} = V_m/\sqrt{2}$. For the 230 V (RMS) mains, $V_m = 230\sqrt{2} = 325.3$ V. Note also that the power fluctuates at **twice** the supply frequency — the origin of the 100/120 Hz hum in poorly filtered supplies. (Averages of functions are treated formally in Ch. 5.)

Solving trigonometric equations

Because trigonometric functions are periodic, a solvable trigonometric equation has **infinitely many** solutions. Find all solutions in one period, then add multiples of the period.

General solutions

For an integer k:

$$\sin x = \sin\alpha \iff x = \alpha + 2k\pi \ \text{ or } \ x = \pi - \alpha + 2k\pi; \qquad \cos x = \cos\alpha \iff x = \pm\alpha + 2k\pi; \qquad \tan x = \tan\alpha \iff x = \alpha + k\pi.$$

Worked Example 1.35 — A quadratic in $\sin x$

Solve $2\sin^2 x - \sin x - 1 = 0$.

Step 1. With $u = \sin x$: $2u^2 - u - 1 = (2u+1)(u-1) = 0$, so $\sin x = 1$ or $\sin x = -\tfrac{1}{2}$.

Step 2. $\sin x = 1 \Rightarrow x = \tfrac{\pi}{2} + 2k\pi$. $\sin x = -\tfrac{1}{2} = \sin\left(-\tfrac{\pi}{6}\right) \Rightarrow x = -\tfrac{\pi}{6} + 2k\pi$ or $x = \pi + \tfrac{\pi}{6} + 2k\pi = \tfrac{7\pi}{6} + 2k\pi$.

In $[0, 2\pi)$: $x \in \left\{\tfrac{\pi}{2},\ \tfrac{7\pi}{6},\ \tfrac{11\pi}{6}\right\}$.

Check $x = 7\pi/6$: $\sin x = -0.5$, $2(0.25) + 0.5 - 1 = 0$ ✓.

Small-angle approximations

For $|\theta| \ll 1$ (in radians),

$$\sin\theta \approx \theta, \qquad \tan\theta \approx \theta, \qquad \cos\theta \approx 1 - \frac{\theta^2}{2}.$$

These are the first terms of the Taylor series (Ch. 4). At $\theta = 0.1$ rad (5.7°): $\sin 0.1 = 0.099833$ (error 0.17%), $\tan 0.1 = 0.100335$ (error 0.33%), $\cos 0.1 = 0.995004$ versus 0.995. They linearise pendulums, phase-locked loops near lock, and small phase errors in modulators.

Triangles: the laws of sines and cosines

For a triangle with sides a, b, c opposite angles A, B, C:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}, \qquad c^2 = a^2 + b^2 - 2ab\cos C.$$

The law of cosines generalises Pythagoras ($C = 90^\circ$) and is the quickest way to add two phasors graphically.

Worked Example 1.36 — Adding two phasors geometrically

Two voltages of magnitude 10 V and 6 V differ in phase by 60°. Find the magnitude and phase of their sum relative to the 10 V phasor.

Step 1 — triangle. Placing the phasors head to tail, the angle *inside* the triangle between them is $180^\circ - 60^\circ = 120^\circ$. The law of cosines gives

$$|V|^2 = 10^2 + 6^2 - 2(10)(6)\cos 120^\circ = 100 + 36 + 60 = 196, \qquad |V| = 14 \text{ V}.$$

Step 2 — angle. By the law of sines, $\sin\beta = \dfrac{6\sin 120^\circ}{14} = 0.3712$, so $\beta = 21.79^\circ$.

Check by components: $10 + 6\cos 60^\circ = 13$, $6\sin 60^\circ = 5.196$; $\sqrt{13^2 + 5.196^2} = 14.00$ ✓ and $\operatorname{atan2}(5.196, 13) = 21.79^\circ$ ✓. Ch. 2 does such sums with complex numbers, which avoids the triangle entirely.

Common Pitfall

Calculators default to degrees or radians unpredictably — $\sin(30)$ is 0.5 in degree mode and -0.988 in radian mode. Programming languages always use radians. Also, $\sin(A+B) \neq \sin A + \sin B$ and $\sin 2\theta \neq 2\sin\theta$; and an equation like $\sin x = 0.5$ has infinitely many solutions, not just the one the calculator returns.

Engineering Note — identities as circuit operations

Product-to-sum is what a nonlinear device (a diode or transistor mixer) does to two inputs; power reduction explains why squaring a sinusoid produces DC plus a second harmonic, the basis of RMS detectors and frequency doublers; harmonic addition is how two quadrature signals ($I\cos\omega t + Q\sin\omega t$) carry amplitude and phase in a digital radio.

1.7 Coordinate systems

Why several coordinate systems

A coordinate system is a rule that labels each point of the plane or of space by numbers. The Cartesian system is the default, but it is not always the most convenient: the field of a long straight wire depends only on the distance from the wire, the radiation pattern of an antenna depends on direction, and a complex number is often best described by a length and an angle. Choosing coordinates that match the **symmetry** of a problem can turn an intractable calculation into a short one. This section fixes the conventions used throughout the book — cylindrical (r, ϕ, z) and spherical (R, θ, ϕ), as in Notation & Conventions — and gives the conversion formulas; the calculus in these coordinates (Jacobians, gradient, divergence) comes in Ch. 6 and Ch. 7.

The Cartesian plane

A point P is labelled (x, y) by its signed distances from two perpendicular axes. The basic formulas follow from Pythagoras:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}, \qquad M = \left(\frac{x_1 + x_2}{2}, \, \frac{y_1 + y_2}{2} \right),$$

for the distance between two points and their midpoint.

Straight lines

- **Slope** through (x_1, y_1) and (x_2, y_2): $m = \dfrac{y_2 - y_1}{x_2 - x_1}$ (undefined for a vertical line).
- **Slope–intercept form:** $y = mx + c$. **Point–slope form:** $y - y_1 = m(x - x_1)$. **General form:** $ax + by + c = 0$.
- Lines are **parallel** iff $m_1 = m_2$ and **perpendicular** iff $m_1 m_2 = -1$.
- **Distance from a point** (x_0, y_0) to the line $ax + by + c = 0$:

$$d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}.$$

Circle

The circle with centre (h, k) and radius $\rho_0 > 0$ is the set of points at distance ρ_0 from the centre:

$$(x - h)^2 + (y - k)^2 = \rho_0^2.$$

Worked Example 1.37 — Lines and circles

(a) Load line. A transistor's collector circuit satisfies $V_{CE} = V_{CC} - I_C R_C$. With $V_{CC} = 12$ V, $R_C = 2\ \text{k}\Omega$, plotted with I_C (mA) on the vertical axis and V_{CE} (V) on the horizontal, this is the line $I_C = 6 - 0.5 V_{CE}$: slope $-1/R_C = -0.5$ mA/V, intercepts $(0, 6)$ and $(12, 0)$. The operating point must lie on it.

(b) Circle from a general equation. Find the centre and radius of $x^2 + y^2 - 4x + 6y - 12 = 0$. Complete the square in each variable:

$$(x^2 - 4x + 4) + (y^2 + 6y + 9) = 12 + 4 + 9 \quad \Longrightarrow \quad (x - 2)^2 + (y + 3)^2 = 25.$$

Centre $(2, -3)$, radius 5.

(c) Point-to-line distance. The distance from $(1, 2)$ to $3x + 4y - 1 = 0$ is $\dfrac{|3 + 8 - 1|}{\sqrt{9 + 16}} = \dfrac{10}{5} = 2$.

Check (b): the point $(7, -3)$ is 5 units right of the centre; $49 + 9 - 28 - 18 - 12 = 0$ ✓.

Plane polar coordinates

Polar coordinates

A point is labelled (r, θ), where $r \geq 0$ is its distance from the origin (the **pole**) and θ the angle from the positive x-axis, anticlockwise positive.

$$x = r\cos\theta, \quad y = r\sin\theta; \qquad r = \sqrt{x^2 + y^2}, \quad \theta = \operatorname{atan2}(y, x).$$

The angle is not unique — (r, θ) and $(r, \theta + 2k\pi)$ are the same point — so a **principal value** is chosen, usually $\theta \in (-\pi, \pi]$. The origin has $r = 0$ and any θ.

Two-argument arctangent

$$\operatorname{atan2}(y, x) = \begin{cases} \arctan(y/x), & x > 0 \\ \arctan(y/x) + \pi, & x < 0,\ y \geq 0 \\ \arctan(y/x) - \pi, & x < 0,\ y < 0 \\ +\pi/2, & x = 0,\ y > 0 \\ -\pi/2, & x = 0,\ y < 0 \end{cases}$$

It returns the angle in $(-\pi, \pi]$ in the **correct quadrant** (undefined at the origin).

The single-argument $\arctan(y/x)$ only sees the ratio y/x, so it cannot distinguish $(1, 1)$ from $(-1, -1)$; it always returns an angle in $(-\pi/2, \pi/2)$. Every programming language provides `atan2(y, x)` — note the order of arguments.

Worked Example 1.38 — Polar conversion

(a) $(3, 4)$: $r = 5$, $\theta = \operatorname{atan2}(4, 3) = \arctan(4/3) = 0.9273$ rad $= 53.13°$.

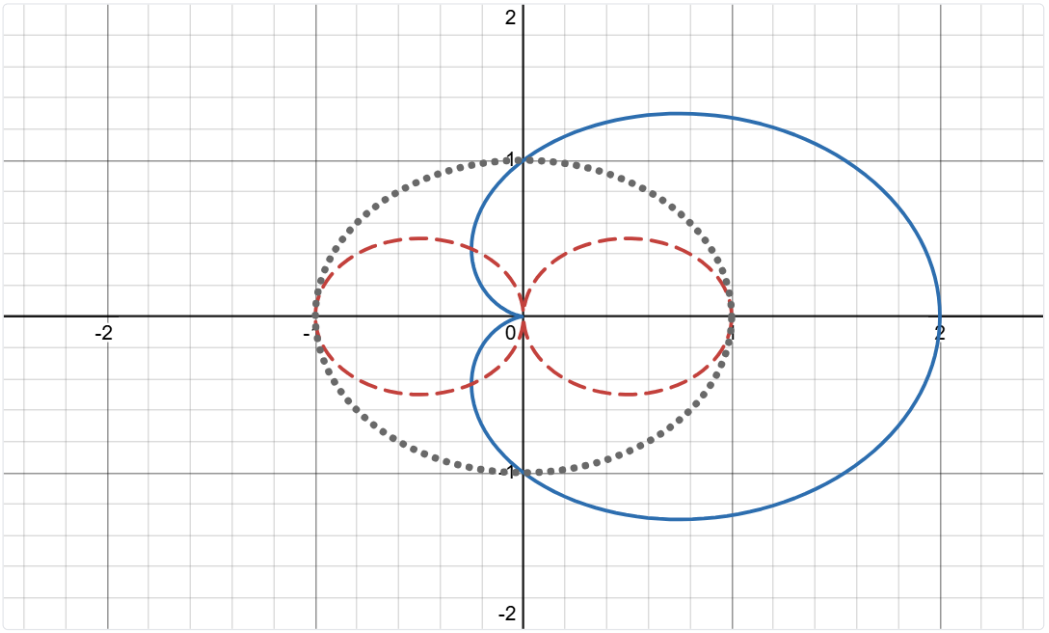
(b) $(-3, -4)$: $r = 5$, but $\theta = \operatorname{atan2}(-4, -3) = \arctan(4/3) - \pi = -2.2143$ rad $= -126.87°$, not $53.13°$.

(c) $(r, \theta) = (2,\ 5\pi/6)$: $x = 2\cos 150° = -\sqrt{3} = -1.732$, $y = 2\sin 150° = 1$.

Check (b): $5\cos(-2.2143) = -3.000$, $5\sin(-2.2143) = -4.000$ ✓.

This is exactly the rectangular-to-polar conversion of complex numbers in Ch. 2, where $(x, y) \leftrightarrow x + jy$ and $(r, \theta) \leftrightarrow re^{j\theta}$.

Curves in polar form. Equations convert using $x = r\cos\theta$, $y = r\sin\theta$, $x^2 + y^2 = r^2$. For instance the circle $x^2 + y^2 = 4x$ (centre $(2, 0)$, radius 2) becomes $r^2 = 4r\cos\theta$, i.e. $r = 4\cos\theta$. Polar plots are the natural way to show **radiation patterns**: the cardioid $r = 1 + \cos\theta$ is the pattern of a simple directional (cardioid) microphone or of two suitably phased antenna elements, with full response forward ($\theta = 0$) and a null behind ($\theta = \pi$).



Polar patterns: the cardioid $r = 1 + \cos\theta$ (blue) has a single null at $\theta = \pi$; the figure-eight $r = |\cos\theta|$ (red) of a dipole-like element has nulls at $\pm\pi/2$; the grey circle is an omnidirectional pattern.

Three-dimensional Cartesian coordinates

A point in space is (x, y, z) with respect to three mutually perpendicular axes forming a **right-handed** system: curling the fingers of the right hand from $+x$ towards $+y$, the thumb points along $+z$. The distance formula extends directly: $d = \sqrt{\Delta x^2 + \Delta y^2 + \Delta z^2}$.

Cylindrical coordinates (r, ϕ, z)

Cylindrical coordinates

$r \geq 0$ is the perpendicular distance from the z-axis, $\phi \in [0, 2\pi)$ (or $(-\pi, \pi]$) the azimuthal angle measured from the $+x$ axis in the xy-plane, and z the usual height.

$$x = r\cos\phi, \quad y = r\sin\phi, \quad z = z; \qquad r = \sqrt{x^2 + y^2}, \quad \phi = \operatorname{atan2}(y, x).$$

Cylindrical coordinates are plane polar coordinates with a z axis added. The surfaces $r = \text{const}$ are coaxial cylinders, $\phi = \text{const}$ half-planes hinged on the z-axis, and $z = \text{const}$ horizontal planes. This book uses r (not ρ) for the cylindrical radius so that ρ remains free for charge density.

Spherical coordinates (R, θ, ϕ)

Spherical coordinates

$R \geq 0$ is the distance from the origin, $\theta \in [0, \pi]$ the **polar angle** measured from the $+z$ axis, and ϕ the **azimuth**, the same angle as in cylindrical coordinates.

$$x = R\sin\theta\cos\phi, \quad y = R\sin\theta\sin\phi, \quad z = R\cos\theta;$$

$$R = \sqrt{x^2 + y^2 + z^2}, \qquad \theta = \arccos\frac{z}{R}, \qquad \phi = \operatorname{atan2}(y, x).$$

Derivation. Drop a perpendicular from P to the xy-plane. The right triangle formed by the origin, P and the z-axis has hypotenuse R and angle θ at the origin, so $z = R\cos\theta$ and the distance from the z-axis is $r = R\sin\theta$. Within the xy-plane, that distance r and the azimuth ϕ give $x = r\cos\phi$, $y = r\sin\phi$. □

The bridge between the two systems is therefore

$$r = R\sin\theta, \qquad z = R\cos\theta, \qquad \phi \text{ unchanged.}$$

The surfaces $R = \text{const}$ are spheres, $\theta = \text{const}$ cones about the z-axis (the plane $z = 0$ when $\theta = \pi/2$), and $\phi = \text{const}$ half-planes. On the Earth, θ is the **colatitude** (90° minus latitude) and ϕ the longitude.

System	Coordinates	To Cartesian	Line element ds^2	Volume element
Cartesian	(x, y, z)	—	$dx^2 + dy^2 + dz^2$	$dx\,dy\,dz$

System	Coordinates	To Cartesian	Line element ds^2	Volume element
Cylindrical	(r, ϕ, z)	$x = r\cos\phi,\ y = r\sin\phi,\ z = z$	$dr^2 + r^2 d\phi^2 + dz^2$	$r\, dr\, d\phi\, dz$
Spherical	(R, θ, ϕ)	$x = R\sin\theta\cos\phi,\ y = R\sin\theta\sin\phi,\ z = R\cos\theta$	$dR^2 + R^2 d\theta^2 + R^2\sin^2\theta\, d\phi^2$	$R^2\sin\theta\, dR\, d\theta\, d\phi$

The line elements follow geometrically: in cylindrical coordinates a small change $d\phi$ moves a point along an arc of radius r, a distance $r\, d\phi$; in spherical coordinates $d\theta$ moves it along a circle of radius R (distance $R\, d\theta$), and $d\phi$ along a circle of radius $R\sin\theta$ (distance $R\sin\theta\, d\phi$). Multiplying the three perpendicular displacements gives the volume elements. For example, integrating $R^2\sin\theta$ over $\theta \in [0, \pi]$ and $\phi \in [0, 2\pi)$ gives the sphere's surface area $4\pi R^2$ (Ch. 6).

Worked Example 1.39 — Converting points

(a) Cartesian to cylindrical. $(x, y, z) = (-2,\ 2\sqrt{3},\ 5)$: $r = \sqrt{4 + 12} = 4$, $\phi = \operatorname{atan2}(2\sqrt{3}, -2) = 2\pi/3$ (second quadrant), $z = 5$. So $(r, \phi, z) = (4,\ 2\pi/3,\ 5)$.

(b) Cartesian to spherical. $(1, 1, \sqrt{2})$: $R = \sqrt{1 + 1 + 2} = 2$; $\theta = \arccos(\sqrt{2}/2) = \pi/4$; $\phi = \operatorname{atan2}(1, 1) = \pi/4$.

(c) Spherical to cylindrical. $(R, \theta, \phi) = (6,\ \pi/3,\ \pi/2)$: $r = 6\sin\frac{\pi}{3} = 5.196$, $z = 6\cos\frac{\pi}{3} = 3$, $\phi = \pi/2$.

Checks: (a) $4\cos(2\pi/3) = -2$, $4\sin(2\pi/3) = 3.464 = 2\sqrt{3}$ ✓. (b) $x = 2\sin\frac{\pi}{4}\cos\frac{\pi}{4} = 2\cdot\frac{1}{2} = 1$ ✓, $z = 2\cos\frac{\pi}{4} = \sqrt{2}$ ✓. (c) $r^2 + z^2 = 27 + 9 = 36 = R^2$ ✓.

Worked Example 1.40 — Distance between two points given in spherical form

Find the distance between $P_1 = (R, \theta, \phi) = (2,\ \pi/2,\ 0)$ and $P_2 = (3,\ \pi/3,\ \pi/2)$.

Step 1 — convert. $P_1 = (2\cdot 1\cdot 1,\ 0,\ 0) = (2, 0, 0)$. $P_2 = (3\sin\frac{\pi}{3}\cos\frac{\pi}{2},\ 3\sin\frac{\pi}{3}\sin\frac{\pi}{2},\ 3\cos\frac{\pi}{3}) = (0,\ 2.598,\ 1.5)$.

Step 2 — distance. $d = \sqrt{2^2 + 2.598^2 + 1.5^2} = \sqrt{4 + 6.75 + 2.25} = \sqrt{13} = 3.606$.

Check by the spherical law of cosines. The angle γ between the two position vectors satisfies $\cos\gamma = \sin\theta_1\sin\theta_2\cos(\phi_1 - \phi_2) + \cos\theta_1\cos\theta_2 = 1\cdot 0.866\cdot 0 + 0 = 0$, so $d^2 = R_1^2 + R_2^2 - 2R_1R_2\cos\gamma = 4 + 9 = 13$ ✓.

Note that one may **not** subtract spherical coordinates component by component: $(3 - 2,\ \pi/3 - \pi/2,\ \pi/2 - 0)$ is meaningless as a displacement.

Worked Example 1.41 — Converting surfaces

Surface	Cartesian	Cylindrical	Spherical
Sphere of radius a	$x^2 + y^2 + z^2 = a^2$	$r^2 + z^2 = a^2$	$R = a$
Cylinder of radius 3 about z	$x^2 + y^2 = 9$	$r = 3$	$R\sin\theta = 3$
Cone, half-angle 45°	$z = \sqrt{x^2 + y^2}$	$z = r$	$\theta = \pi/4$
Half-plane $y = x,\ x > 0$	$y = x,\ x > 0$	$\phi = \pi/4$	$\phi = \pi/4$

For the cone: $z = r$ means $R\cos\theta = R\sin\theta$, so $\tan\theta = 1$ and $\theta = \pi/4$. Each surface has the simplest equation in the system matched to its symmetry.

Common Pitfall — conventions differ

Many mathematics texts swap the roles of θ and ϕ in spherical coordinates (using θ for azimuth), and some use ρ for the spherical radius. This book follows the physics/electromagnetics convention: θ is the polar angle from $+z$, ϕ the azimuth. Also, the cylindrical r (distance from the z-axis) is **not** the spherical R (distance from the origin): $r = R\sin\theta$. Always check a source's convention before using its formulas.

Engineering Note — let symmetry choose

Choose coordinates to match the geometry: cylindrical for coaxial cables, long wires, solenoids and optical fibres; spherical for point charges, antennas and far-field radiation; Cartesian for parallel-plate capacitors and rectangular waveguides. The field of a line charge, a messy expression in (x, y, z), is simply proportional to $1/r$ in cylindrical coordinates; the far field of an antenna is a function of (θ, ϕ) times e^{-jkR}/R.

1.8 Functions, domains, and ranges

Why the function concept

A circuit, a sensor, an amplifier or a filter is, to a mathematician, a **function**: a rule that takes an input (a voltage, a temperature, a signal) and produces exactly one output. Signals themselves are functions of time. Speaking precisely about functions — what inputs are allowed, what outputs can occur, how graphs are transformed by delays and gains — is the vocabulary of signals and systems. Questions like "for what input range is this amplifier linear?" or "can this sensor reading be inverted to give a unique temperature?" are questions about domains, ranges and one-to-one functions.

Definition

Function

A **function** $f : A \to B$ assigns to each element x of the set A (the **domain**) exactly one element $f(x)$ of the set B (the **codomain**). The **range** (or image) is the set of values actually taken,

$$f(A) = \{f(x) : x \in A\} \subseteq B.$$

x is the **independent variable** (argument, input) and $y = f(x)$ the **dependent variable** (value, output).

The essential word is *exactly one*. The relation $y^2 = x$ does not define y as a function of x, because $x = 4$ gives $y = \pm 2$; it defines two functions, $y = \sqrt{x}$ and $y = -\sqrt{x}$. Graphically, a curve in the xy-plane is the graph of a function of x if and only if every vertical line meets it at most once — the **vertical-line test**.

A function is not the same thing as a formula. The same formula on different domains gives different functions (with different ranges and different inverses), and one function may need several formulas (a piecewise definition). Notation such as $f : \mathbb{R} \to \mathbb{R},\ x \mapsto x^2$ ("x maps to x^2") states domain, codomain and rule together.

Natural domain

When a formula is given without a stated domain, the **natural domain** is the largest set of reals for which it makes sense. Exclude:

- zeros of denominators;
- negative arguments of even roots ($\sqrt{\ }$, $\sqrt[4]{\ }$, …);
- non-positive arguments of logarithms;
- points where tan or sec is undefined ($\cos = 0$), or cot, csc ($\sin = 0$);
- arguments outside $[-1, 1]$ for arcsin and arccos.

In a physical model the domain is further restricted by physics: a resistance is positive, a probability lies in $[0, 1]$, a model of a transistor is valid only in its active region.

Worked Example 1.42 — Natural domains

(a) $f(x) = \dfrac{\ln(4 - x^2)}{\sqrt{x + 1}}$ needs $4 - x^2 > 0$, i.e. $-2 < x < 2$, and $x + 1 > 0$ (strictly, since the root is in a denominator), i.e. $x > -1$. Intersection: domain $(-1, 2)$.

(b) $g(x) = \sqrt{\dfrac{x - 1}{x + 2}}$ needs $\dfrac{x - 1}{x + 2} \geq 0$, which by the sign chart of Example 1.15(c) gives $(-\infty, -2) \cup [1, \infty)$.

(c) $h(x) = \arcsin(2x - 1)$ needs $-1 \leq 2x - 1 \leq 1$, i.e. $0 \leq x \leq 1$.

Finding the range

The range is usually harder than the domain. Three systematic methods:

1. **Solve for x.** Set $y = f(x)$ and ask for which y the equation has a solution x in the domain.
2. **Complete the square** or otherwise rewrite f in a form whose extreme values are obvious.
3. **Use monotonicity**: a continuous increasing function on $[a, b]$ has range $[f(a), f(b)]$. (Calculus gives a general method in Ch. 4.)

Worked Example 1.43 — Three ranges

(a) By completing the square. $f(x) = x^2 - 4x + 7$ on $[0, 5]$. Since $f(x) = (x - 2)^2 + 3$, the minimum is 3 at $x = 2$; the maximum over $[0, 5]$ is at the end farther from 2, $f(5) = 12$ (while $f(0) = 7$). Range $[3, 12]$.

(b) By solving for x. $f(x) = \dfrac{x}{x^2 + 1}$ on $\mathbb{R}$. Setting $y = \dfrac{x}{x^2 + 1}$ gives $yx^2 - x + y = 0$. If $y = 0$, then $x = 0$ works. If $y \neq 0$, this quadratic in x has a real root iff its discriminant $1 - 4y^2 \geq 0$, i.e. $|y| \leq \frac{1}{2}$. Range $\left[-\frac{1}{2}, \frac{1}{2}\right]$, with $y = \frac{1}{2}$ at $x = 1$ and $y = -\frac{1}{2}$ at $x = -1$.

(c) From bounded functions. $g(x) = 2 + 3\sin x$: since $-1 \leq \sin x \leq 1$, the range is $[-1, 5]$.

Check (b): $f(1) = 1/2$ ✓; $f(3) = 0.3 < 0.5$ ✓.

Properties worth naming

Even, odd, periodic, monotonic, bounded

- **Even:** $f(-x) = f(x)$ for all x (graph symmetric about the y-axis; x^2, $\cos x$, $|x|$, $\cosh x$).
- **Odd:** $f(-x) = -f(x)$ (graph symmetric under a half-turn about the origin; x^3, $\sin x$, $\tan x$, $\sinh x$).
- **Periodic** with period $T > 0$: $f(x + T) = f(x)$ for all x; the smallest such T is the **fundamental period**.
- **Increasing** on an interval: $x_1 < x_2 \Rightarrow f(x_1) \leq f(x_2)$ (**strictly** if $<$); **decreasing** similarly. **Monotonic** means one or the other.
- **Bounded:** $|f(x)| \leq M$ for some M and all x in the domain.

Even–odd decomposition

Every function f defined on a symmetric domain splits uniquely as $f = f_e + f_o$ with f_e even and f_o odd:

$$f_e(x) = \tfrac{1}{2}\big[f(x) + f(-x)\big], \qquad f_o(x) = \tfrac{1}{2}\big[f(x) - f(-x)\big].$$

Proof. The formulas clearly give an even and an odd function summing to f. For uniqueness, if $f = g_e + g_o$, then $f(-x) = g_e(x) - g_o(x)$; adding and subtracting recovers the formulas. □

Products follow sign rules: even×even and odd×odd are even, even×odd is odd. This is used constantly in Fourier analysis (Ch. 15), where an even signal has only cosine terms and an odd one only sine terms.

Worked Example 1.44 — Even and odd parts

(a) For $f(x) = e^x$: $f_e = \tfrac{1}{2}(e^x + e^{-x}) = \cosh x$ and $f_o = \tfrac{1}{2}(e^x - e^{-x}) = \sinh x$, recovering $e^x = \cosh x + \sinh x$ (Section 1.5).

(b) For the decaying exponential pulse $x(t) = e^{-t}u(t)$ (zero for $t < 0$):

$$x_e(t) = \tfrac{1}{2}e^{-|t|}, \qquad x_o(t) = \tfrac{1}{2}\operatorname{sgn}(t)\, e^{-|t|},$$

where $\operatorname{sgn}(t)$ is $+1$ for $t > 0$ and -1 for $t < 0$.

Check at $t = 1$: $x_e + x_o = \tfrac{1}{2}e^{-1} + \tfrac{1}{2}e^{-1} = e^{-1} = x(1)$ ✓; at $t = -1$: $\tfrac{1}{2}e^{-1} - \tfrac{1}{2}e^{-1} = 0 = x(-1)$ ✓.

Transformations of graphs

Given the graph of $y = f(x)$, the graph of

$$y = A\, f\big(b(x - c)\big) + d$$

is obtained by: horizontal scaling by a factor $1/b$ (compression if $|b| > 1$, with reflection in the y-axis if $b < 0$); horizontal **shift** right by c; vertical scaling by A (reflection in the x-axis if $A < 0$); and vertical shift up by d. Changes **inside** the function act on x and work "backwards" ($f(x - 3)$ moves the graph *right*); changes **outside** act on y and work as expected.

In signal language with t for time: $x(t - t_0)$ is a **delay** by t_0 (for $t_0 > 0$), $x(at)$ is time compression ($a > 1$) or expansion ($0 < a < 1$), $x(-t)$ is time reversal, and $Ax(t)$ is amplification.

Worked Example 1.45 — Order of shift and scale

Describe $y(t) = x(2t - 4)$ in terms of $x(t)$.

Step 1 — factor the inner coefficient. $x(2t - 4) = x\big(2(t - 2)\big)$.

Step 2 — interpret. Compress x by a factor 2 in time, then delay by 2 — **not** by 4. A feature of x at time $t = \tau$ appears in y at the time where $2t - 4 = \tau$, i.e. $t = (\tau + 4)/2$.

Check: if x has a peak at $t = 0$, then y peaks where $2t - 4 = 0$, i.e. $t = 2$ ✓.

Piecewise-defined functions

Many engineering functions are defined by different formulas on different sub-domains. Two that matter everywhere in EE are the unit step and the saturation (clipping) function:

$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases} \qquad \operatorname{sat}(x) = \begin{cases} -1, & x < -1 \\ x, & |x| \leq 1 \\ 1, & x > 1 \end{cases}$$

(The value of $u(0)$ is a convention; some texts use $\tfrac{1}{2}$. It never affects integrals.) Steps are the building blocks of piecewise signals: the rectangular pulse of width T starting at 0 is $u(t) - u(t - T)$, and the ramp is $r(t) = t\,u(t)$. Absolute value, sgn, and $\max(0, x)$ (the ideal rectifier, and the "ReLU" of machine learning) are other common piecewise functions: $\max(0, x) = \tfrac{1}{2}(x + |x|)$.

Worked Example 1.46 — Building a signal from steps

Express the signal that is 0 for $t < 0$, rises linearly to 2 at $t = 1$, stays at 2 until $t = 3$, then drops to 0.

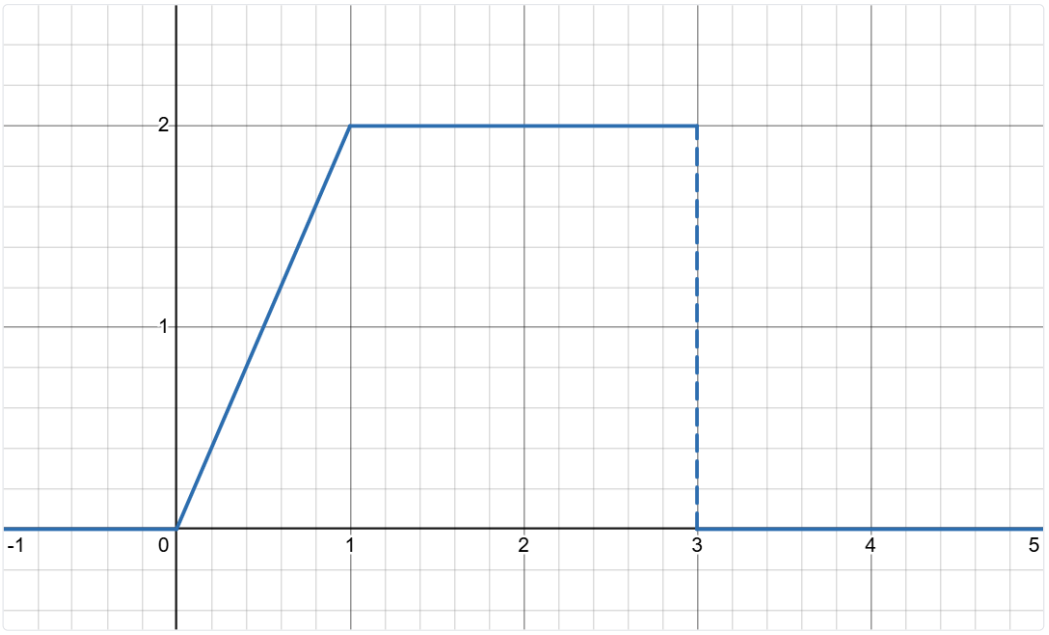
Step 1. The ramp $2t\,u(t)$ produces the rise but keeps going.

Step 2. At $t = 1$ cancel the slope with $-2(t - 1)u(t - 1)$; for $t > 1$ the sum is $2t - 2(t - 1) = 2$ ✓.

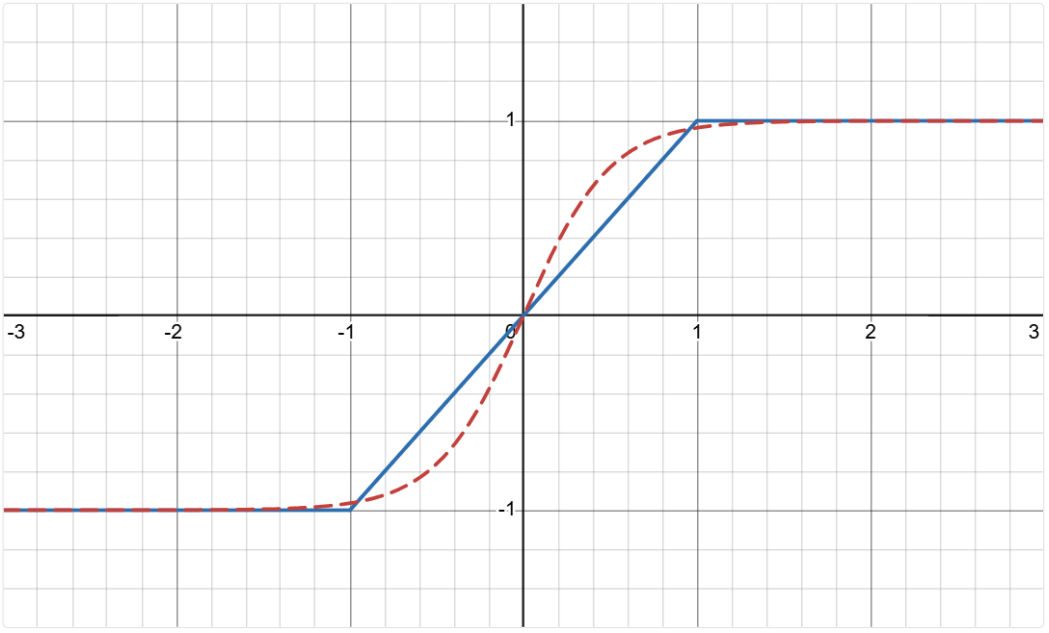
Step 3. At $t = 3$ subtract the level: $-2u(t - 3)$.

Result: $x(t) = 2t\,u(t) - 2(t - 1)\,u(t - 1) - 2\,u(t - 3)$.

Check at $t = 0.5$: $1 - 0 - 0 = 1$ ✓; at $t = 2$: $4 - 2 - 0 = 2$ ✓; at $t = 4$: $8 - 6 - 2 = 0$ ✓.



The trapezoidal-ramp signal of Example 1.46, built from a ramp, a delayed ramp and a delayed step. The dashed segment marks the jump at $t = 3$.



The ideal saturation function (blue) and the smooth $\tanh(2x)$ (red), often used to model an amplifier driven into its supply rails. Domain $\mathbb{R}$, range $[-1, 1]$; the linear region is the domain sub-interval $|x| \le 1$.

Functions of several variables

A function may take several inputs: the power $P(V, R) = V^2/R$ has domain $\{(V, R) : R > 0\}$ and the potential of a point charge $\Phi(x, y, z)$ is defined everywhere except at the charge. The ideas of domain and range carry over unchanged; differentiation and integration of such functions are the subject of Ch. 6.

Common Pitfall — the formula is not the domain

Simplifying a formula can silently enlarge its domain. $\dfrac{x^2 - 1}{x - 1}$ and $x + 1$ agree except at $x = 1$, where the first is undefined; $(\sqrt{x})^2$ and x agree only for $x \ge 0$; $e^{\ln x}$ and x agree only for $x > 0$. State the domain before simplifying, and keep it afterwards.

Engineering Note — domains as operating regions

A device model is a function with a restricted domain. The small-signal model of a transistor is valid only near its bias point; the Shockley equation fails at reverse breakdown; an op-amp's linear model $v_o = A(v_+ - v_-)$ holds only while $|v_o|$ is below the rails. Many design failures are evaluations of a function outside its domain.

1.9 Inverse and composite functions

Why composition and inversion

Engineering systems are built by connecting blocks in series: a sensor converts temperature to resistance, a bridge converts resistance to voltage, an amplifier scales the voltage, an ADC converts it to a number. The overall input–output relation is a **composite function**. Measurement is the reverse problem: from the number read out, recover the temperature. That requires the **inverse** of each block — and it is possible only if each block is one-to-one on the range of interest. Composition also underlies the chain rule of differentiation (Ch. 4) and substitution in integration (Ch. 5), both of which require recognising a function as "an outer function of an inner function".

Composite functions

Composite function

The **composition** of f with g is

$$(f \circ g)(x) = f\big(g(x)\big),$$

read "f after g". Its domain is the set of x in the domain of g for which $g(x)$ lies in the domain of f.

Composition is **associative**, $(f \circ g) \circ h = f \circ (g \circ h)$, but in general **not commutative**: $f \circ g \neq g \circ f$. A signal chain $x \to g \to f \to y$ is the composite $y = (f \circ g)(x)$ — note that the block applied first is written on the right.

Worked Example 1.47 — Composition and its domain

Let $f(x) = \sqrt{x}$ and $g(x) = 1 - x^2$.

(a) $(f \circ g)(x) = \sqrt{1 - x^2}$. Need $g(x) \geq 0$: $1 - x^2 \geq 0$, so the domain is $[-1, 1]$; the range is $[0, 1]$ (the upper unit semicircle).

(b) $(g \circ f)(x) = 1 - (\sqrt{x})^2 = 1 - x$, but only for x in the domain of f: the domain is $[0, \infty)$, even though the simplified formula $1 - x$ looks defined everywhere.

(c) $f \circ g \neq g \circ f$: at $x = 0.5$, $(f \circ g)(0.5) = \sqrt{0.75} = 0.866$ while $(g \circ f)(0.5) = 0.5$.

Worked Example 1.48 — Decomposing a function

Write $h(t) = e^{-(t-2)^2}$ and $k(x) = \ln\left(1 + \cos^2 x\right)$ as compositions of simpler functions.

$h = f_1 \circ f_2 \circ f_3$ with $f_3(t) = t - 2$ (delay), $f_2(u) = -u^2$, $f_1(w) = e^w$.

$k = F_1 \circ F_2 \circ F_3$ with $F_3(x) = \cos x$, $F_2(u) = 1 + u^2$, $F_1(w) = \ln w$. The domain is all of $\mathbb{R}$, since $1 + \cos^2 x \geq 1 > 0$.

Recognising this nesting — innermost to outermost — is exactly the skill needed to apply the chain rule.

One-to-one functions

One-to-one (injective), onto (surjective)

$f : A \to B$ is **one-to-one** if $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$ — distinct inputs give distinct outputs. Graphically, every horizontal line meets the graph at most once (the **horizontal-line test**). f is **onto** B if its range is all of B. A function that is both is a **bijection**.

Strict monotonicity implies one-to-one

A function that is strictly increasing (or strictly decreasing) on an interval is one-to-one there.

Proof. If $x_1 \neq x_2$, say $x_1 < x_2$, then strict increase gives $f(x_1) < f(x_2)$, so $f(x_1) \neq f(x_2)$. $\square$

So x^3, e^x, $\ln x$ and $\tanh x$ are one-to-one on their whole domains, while x^2 and $\sin x$ are not — but become so when restricted to $[0, \infty)$ and $[-\pi/2, \pi/2]$ respectively.

Inverse functions

Inverse function

If f is one-to-one with domain A and range B, its **inverse** $f^{-1} : B \to A$ is defined by

$$f^{-1}(y) = x \iff f(x) = y.$$

Equivalently, $f^{-1}\big(f(x)\big) = x$ for all $x \in A$ and $f\big(f^{-1}(y)\big) = y$ for all $y \in B$. The domain of f^{-1} is the range of f, and vice versa.

The graph of f^{-1} is the reflection of the graph of f in the line $y = x$, because the point (a, b) lies on the graph of f exactly when (b, a) lies on the graph of f^{-1}. If f is increasing, so is f^{-1}.

Procedure. To find f^{-1}: write $y = f(x)$; solve for x in terms of y (if the solution is not unique, f is not one-to-one — restrict its domain); state the domain of f^{-1} as the range of f; and, if desired, rename the variable.

Worked Example 1.49 — A fractional-linear inverse

$f(x) = \dfrac{2x+1}{x-3}$, $x \neq 3$.

Step 1. $y(x-3) = 2x+1 \Rightarrow xy - 3y = 2x+1 \Rightarrow x(y-2) = 3y+1$.

Step 2. For $y \neq 2$,

$$f^{-1}(y) = \frac{3y+1}{y-2}.$$

Step 3 — domain/range. The step fails only at $y = 2$, so the range of f is $\mathbb{R} \setminus \{2\}$ — the horizontal asymptote of f, which it never reaches.

Check: $f(0) = -\frac{1}{3}$ and $f^{-1}\left(-\frac{1}{3}\right) = \dfrac{-1+1}{-\frac{7}{3}} = 0$ ✓.

Worked Example 1.50 — Inverting the logistic and decibel functions

(a) The logistic (sigmoid) function $\sigma(x) = \dfrac{1}{1+e^{-x}}$ maps $\mathbb{R}$ onto $(0,1)$ and is strictly increasing. Solve $y = \dfrac{1}{1+e^{-x}}$: $1 + e^{-x} = \dfrac{1}{y}$, $e^{-x} = \dfrac{1-y}{y}$, so

$$\sigma^{-1}(y) = \ln \frac{y}{1-y}, \qquad 0 < y < 1.$$

Check: $\sigma^{-1}(0.8) = \ln 4 = 1.3863$ and $\sigma(1.3863) = 1/(1+0.25) = 0.8$ ✓.

(b) Voltage gain in dB is $G = 20 \log_{10} A$ for $A > 0$. Its inverse is $A = 10^{G/20}$; a gain of 26 dB is a voltage ratio $10^{1.3} = 19.95$.

Worked Example 1.51 — Inverting a measurement chain

A platinum RTD has $R(T) = R_0(1 + \alpha T)$ with $R_0 = 100\ \Omega$, $\alpha = 0.00385\ ^\circ\text{C}^{-1}$. It forms a divider with a fixed 100 Ω resistor across 5 V, giving $V = \dfrac{5R}{R+100}$, read by a 10-bit ADC as the integer $N = \lfloor 1024\, V/5 \rfloor$.

(a) Forward (composite). At $T = 50.39\ ^\circ$C: $R = 119.4\ \Omega$, $V = 5 \times 119.4/219.4 = 2.7211$ V, $N = \lfloor 557.27 \rfloor = 557$.

(b) Inverse. Each block is strictly increasing on its operating range, so invert them in reverse order:

$$V \approx \frac{5N}{1024}, \qquad R = \frac{100\,V}{5-V}, \qquad T = \frac{R/R_0 - 1}{\alpha}.$$

From $N = 557$: $V = 2.7197$ V, $R = 119.27\ \Omega$, $T = 50.06\ ^\circ$C.

Interpretation. The recovered $50.06\ ^\circ$C differs from the true $50.39\ ^\circ$C because the floor function in the ADC is **not** one-to-one: every voltage in a 4.9 mV band maps to the same code. Its "inverse" can only return a representative value — here a $0.33\ ^\circ$C quantisation error. The general rule: $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$ — undo the last operation first.

Inverse of a composite

If f and g are one-to-one and $f \circ g$ is defined, then

$$(f \circ g)^{-1} = g^{-1} \circ f^{-1}.$$

(To undo "put on socks, then shoes", take off the shoes first.)

Inverse trigonometric functions

sin, cos and tan are periodic, hence far from one-to-one. Their inverses are defined by restricting each to a **principal** interval on which it is strictly monotonic and takes every value once.

Function	Domain	Range (principal values)	Restriction inverted
$\arcsin x$	$[-1, 1]$	$[-\pi/2, \pi/2]$	sin on $[-\pi/2, \pi/2]$
$\arccos x$	$[-1, 1]$	$[0, \pi]$	cos on $[0, \pi]$
$\arctan x$	$\mathbb{R}$	$(-\pi/2, \pi/2)$	tan on $(-\pi/2, \pi/2)$

Useful identities, valid on the stated domains:

$$\arcsin x + \arccos x = \frac{\pi}{2}, \qquad \arctan x + \arctan \frac{1}{x} = \frac{\pi}{2}\ (x > 0), \qquad \arcsin(-x) = -\arcsin x, \qquad \arccos(-x) = \pi - \arccos x.$$

Worked Example 1.52 — Compositions with inverse trig functions

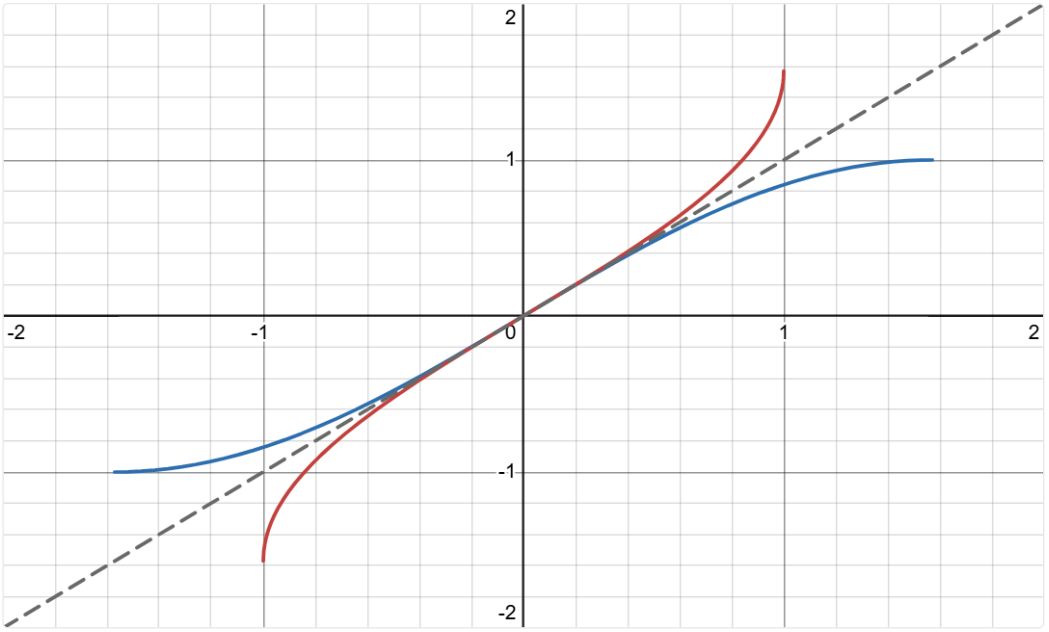
(a) Simplify $\cos(\arctan x)$. Let $\theta = \arctan x \in (-\pi/2, \pi/2)$, so $\tan\theta = x$ and $\cos\theta > 0$. From $1 + \tan^2\theta = \sec^2\theta$,

$$\cos(\arctan x) = \frac{1}{\sqrt{1+x^2}}.$$

(A right triangle with opposite side x and adjacent side 1 gives the same result.) **Check** at $x = 2$: $\cos(1.1071) = 0.4472 = 1/\sqrt{5}$ ✓.

(b) Similarly $\sin(\arccos x) = \sqrt{1-x^2}$, positive because $\arccos x \in [0, \pi]$.

(c) $\arcsin(\sin 2.5)$ is **not** 2.5, since $2.5 \notin [-\pi/2, \pi/2]$. Using $\sin(\pi - \theta) = \sin\theta$, $\sin 2.5 = \sin(\pi - 2.5)$ with $\pi - 2.5 = 0.6416$ in the principal range, so $\arcsin(\sin 2.5) = 0.6416$.



$\sin x$ restricted to $[-\pi/2, \pi/2]$ (blue) and its inverse $\arcsin x$ (red) are reflections of each other in $y = x$.

Inverse hyperbolic functions

sinh is strictly increasing on $\mathbb{R}$ and tanh on $\mathbb{R}$ (onto $(-1, 1)$); cosh is one-to-one on $[0, \infty)$. Since they are built from exponentials, their inverses are logarithms.

Inverse hyperbolic functions

$$\operatorname{arsinh} x = \ln\left(x + \sqrt{x^2+1}\right), \qquad \operatorname{arcosh} x = \ln\left(x + \sqrt{x^2-1}\right)\ (x \geq 1), \qquad \operatorname{artanh} x = \frac{1}{2}\ln\frac{1+x}{1-x}\ (|x| < 1).$$

Derivation of arsinh. Set $x = \sinh y = \frac{1}{2}(e^y - e^{-y})$ and multiply by $2e^y$: $e^{2y} - 2xe^y - 1 = 0$, a quadratic in $u = e^y$. Its roots are $u = x \pm \sqrt{x^2+1}$; only the $+$ root is positive, so $e^y = x + \sqrt{x^2+1}$ and $y = \ln\left(x + \sqrt{x^2+1}\right)$. □

Check: $\operatorname{arsinh} 2 = \ln(2 + \sqrt{5}) = 1.4436$ and $\sinh(1.4436) = 2.000$ ✓. Inverse hyperbolic functions appear in the capacitance of two parallel wires, $C' = \pi\varepsilon/\operatorname{arcosh}(d/2a)$, and in the design of Chebyshev filters.

Common Pitfall — the -1 superscript

$\sin^{-1} x$ means $\arcsin x$, **not** $1/\sin x = \csc x$; yet $\sin^2 x$ means $(\sin x)^2$. Likewise $f^{-1}(x)$ is the inverse function, which is in general **not** $1/f(x)$. Prefer the unambiguous names arcsin, arccos, arctan. Also, $f^{-1}(f(x)) = x$ holds only on the domain on which f was inverted.

Engineering Note — calibration and linearisation

Inverting a sensor characteristic is called **calibration** or **linearisation**. When the forward model has no closed-form inverse (a thermocouple polynomial, the full diode equation with series resistance), the inverse is computed numerically — by a lookup table with interpolation, or by Newton's method (Ch. 22). Either way the forward function must be one-to-one on the measuring range, or the reading is ambiguous.

1.10 Mathematical notation and dimensional consistency

Why notation and dimensions

Mathematics is a language with a compact, precise grammar. Datasheets, standards, textbooks and papers state results as "for all $\varepsilon > 0$ there exists $\delta > 0$ such that…", or "$P \Rightarrow Q$", or with nested sums; misreading a quantifier or the direction of an implication changes the meaning entirely. At the same time, every equation an engineer writes describes physical quantities, and physical quantities carry **dimensions**. Requiring both sides of every equation to have the same

dimensions is the cheapest, fastest error check there is — it catches most algebra slips in seconds, before any numbers are substituted.

Logic and set notation

Symbol	Meaning	Symbol	Meaning
$\forall$	for all	$\exists$	there exists
$\exists!$	there exists exactly one	$\neg P$	not P
$P \Rightarrow Q$	P implies Q	$P \iff Q$	P if and only if Q
$\wedge$, $\vee$	and, or (inclusive)	$:=$ or $\equiv$	is defined as
$\in$, $\notin$	is / is not an element of	$\subseteq$, $\subset$	subset of, proper subset of
$\cup$, $\cap$	union, intersection	$A \setminus B$	elements of A not in B
$\varnothing$	empty set	$A \times B$	Cartesian product (ordered pairs)
$\approx$	approximately equal	$\propto$	proportional to
$\sim$	asymptotically equal / of the order of	$\ll$, $\gg$	much less, much greater than

Set-builder notation describes a set by a property: $\{x \in \mathbb{R} : x^2 < 4\} = (-2, 2)$, read "the set of real x such that $x^2 < 4$". **Cartesian products** build multi-dimensional spaces: $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$ is the plane, $\mathbb{R}^3$ is space.

Implication. $P \Rightarrow Q$ says that whenever P is true, Q is true. Then P is **sufficient** for Q and Q is **necessary** for P. The implication is equivalent to its **contrapositive** $\neg Q \Rightarrow \neg P$, but **not** to its **converse** $Q \Rightarrow P$. For example, "if a system is linear and time-invariant, its response to a sinusoid is a sinusoid at the same frequency" is true; the converse is false. "$P \iff Q$" asserts both directions.

Quantifiers. The order matters. "$\forall x \, \exists y : y > x$" (every number has a larger one) is true; "$\exists y \, \forall x : y > x$" (some number is larger than every number) is false. To negate a quantified statement, swap $\forall$ and $\exists$ and negate the inner statement: $\neg(\forall x : P(x)) \iff \exists x : \neg P(x)$. So disproving "all transistors in this batch meet spec" needs only one counter-example.

Proof techniques

Four patterns cover nearly every argument in this book:

- **Direct proof:** assume P, deduce Q (the proof of the triangle inequality in Section 1.1).
- **Contrapositive:** assume $\neg Q$, deduce $\neg P$.
- **Contradiction:** assume the statement false and derive an absurdity (the irrationality of $\sqrt{2}$).
- **Induction:** to prove $P(n)$ for all $n \in \mathbb{N}$, prove $P(1)$ (base case) and $P(n) \Rightarrow P(n+1)$ (inductive step).

Worked Example 1.53 — Proof by induction

Prove $\displaystyle\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$ for all $n \in \mathbb{N}$.

Base case. $n = 1$: LHS $= 1$, RHS $= \frac{1 \cdot 2}{2} = 1$ ✓.

Inductive step. Assume the formula for n. Then

$$\sum_{k=1}^{n+1} k = \frac{n(n+1)}{2} + (n+1) = \frac{(n+1)(n+2)}{2},$$

which is the formula for $n+1$. By induction it holds for all n. □

Check at $n = 100$: 5050 — the sum Gauss is said to have found as a schoolboy by pairing $1 + 100$, $2 + 99$, … .

Sums and products

$$\sum_{k=1}^{n} a_k = a_1 + a_2 + \cdots + a_n, \qquad \prod_{k=1}^{n} a_k = a_1 a_2 \cdots a_n.$$

The index k is a **dummy variable**: $\sum_{k=1}^{n} a_k = \sum_{i=1}^{n} a_i$. Rules that follow from the field axioms:

$$\sum_{k} (c a_k + b_k) = c \sum_{k} a_k + \sum_{k} b_k, \qquad \sum_{k=1}^{n} a_k = \sum_{m=0}^{n-1} a_{m+1} \ \ (\text{shift } m = k-1), \qquad \sum_{k=1}^{n} c = nc.$$

But $\sum a_k b_k \neq \left(\sum a_k\right)\left(\sum b_k\right)$ in general. Useful closed forms:

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}, \qquad \sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}, \qquad \sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2.$$

(Check: $\sum_{k=1}^{10} k^2 = 385 = \frac{10 \cdot 11 \cdot 21}{6}$ ✓.)

Geometric series

For $r \neq 1$,

$$\sum_{k=0}^{n-1} r^k = 1 + r + r^2 + \cdots + r^{n-1} = \frac{1 - r^n}{1 - r},$$

and for $|r| < 1$ the infinite series converges:

$$\sum_{k=0}^{\infty} r^k = \frac{1}{1 - r}.$$

Proof. Let $S = 1 + r + \cdots + r^{n-1}$. Then $rS = r + r^2 + \cdots + r^n$, and subtracting, $S - rS = 1 - r^n$ — all other terms cancel ("telescope"). Divide by $1 - r$. If $|r| < 1$ then $r^n \to 0$ as $n \to \infty$, giving the infinite sum. □

Worked Example 1.54 — Geometric sums in digital systems

(a) Binary full scale. An 8-bit unsigned register holds at most $\sum_{k=0}^{7} 2^k = \dfrac{1 - 2^8}{1 - 2} = 255$.

(b) Echoes. A signal is reflected repeatedly so that each echo has half the amplitude of the previous one. The total of the original plus all echoes is $\sum_{k=0}^{\infty} 0.5^k = \dfrac{1}{1 - 0.5} = 2$ times the original amplitude; the first ten terms already give 1.998.

(c) Feedback. In a loop with forward gain A and feedback β, a signal circulating repeatedly sums to $A \sum_{k=0}^{\infty} (-A\beta)^k = \dfrac{A}{1 + A\beta}$ when $|A\beta| < 1$ — the closed-loop gain of Example 1.9(b) seen as a geometric series.

Check (b): $(1 - 0.5^{10})/0.5 = 1.998047$ ✓.

The geometric series underlies discrete-time systems, the Z-transform (Ch. 17) and the analysis of convergence of iterative algorithms.

Order-of-magnitude notation

Big-O notation states how fast a quantity grows or shrinks: $f(x) = O(g(x))$ as $x \to 0$ means $|f(x)| \leq M|g(x)|$ for some constant M and all small enough x. Thus $\sin x = x + O(x^3)$ and $(1 + x)^n = 1 + nx + O(x^2)$, which makes precise the size of the error in the approximations of Sections 1.2 and 1.6. As $n \to \infty$, Big-O describes algorithm cost: Gaussian elimination needs $O(n^3)$ operations, the FFT $O(n \log n)$ (Ch. 24).

Dimensions and units

Every physical quantity has a **dimension** built from the seven SI base dimensions: M (mass), L (length), T (time), I (electric current), Θ (temperature), N (amount of substance), J (luminous intensity). The dimension of a quantity Q is written $[Q]$. Derived dimensions follow from defining equations:

Quantity	Defining relation	Dimension	SI unit
Charge Q	$Q = It$	$\mathrm{T\,I}$	C
Force F	$F = ma$	$\mathrm{M\,L\,T^{-2}}$	N
Energy W	force × distance	$\mathrm{M\,L^2\,T^{-2}}$	J
Power P	W/t	$\mathrm{M\,L^2\,T^{-3}}$	W
Voltage V	P/I	$\mathrm{M\,L^2\,T^{-3}\,I^{-1}}$	V
Resistance R	V/I	$\mathrm{M\,L^2\,T^{-3}\,I^{-2}}$	Ω
Conductance G	$1/R$	$\mathrm{M^{-1}\,L^{-2}\,T^{3}\,I^{2}}$	S
Capacitance C	Q/V	$\mathrm{M^{-1}\,L^{-2}\,T^{4}\,I^{2}}$	F
Inductance L	$V\,t/I$	$\mathrm{M\,L^2\,T^{-2}\,I^{-2}}$	H
Magnetic flux Φ	$V\,t$	$\mathrm{M\,L^2\,T^{-2}\,I^{-1}}$	Wb
Electric field E	V/length	$\mathrm{M\,L\,T^{-3}\,I^{-1}}$	V/m
Permittivity ε	C·length/area	$\mathrm{M^{-1}\,L^{-3}\,T^{4}\,I^{2}}$	F/m
Permeability μ	L·length/area	$\mathrm{M\,L\,T^{-2}\,I^{-2}}$	H/m
Conductivity σ	G·length/area	$\mathrm{M^{-1}\,L^{-3}\,T^{3}\,I^{2}}$	S/m
Frequency f, ω	cycles or radians per second	$\mathrm{T^{-1}}$	Hz, rad/s

Rules of dimensional consistency

1. Only quantities of the **same dimension** may be added, subtracted, equated or compared.
2. Dimensions **multiply and divide** like algebraic symbols, and powers apply to them: $[x^n] = [x]^n$.
3. The arguments of exp, ln, sin, cos, tanh and of any other function defined by a power series must be **dimensionless**, and so are the values of these functions.
4. Pure numbers (2, π, $\frac{1}{2}$) and angles in radians are dimensionless.

Why rule 3 holds. $e^x = 1 + x + \frac{x^2}{2} + \dots$ adds 1, x and x^2; by rule 1 this is possible only if x is dimensionless. The same argument applies to every function with a power series. □

Worked Example 1.55 — Checking time constants and energy

(a) $[RC] = \mathrm{M\,L^2\,T^{-3}\,I^{-2}} \cdot \mathrm{M^{-1}L^{-2}T^4I^2} = \mathrm{T}$, so $e^{-t/RC}$ has a dimensionless exponent ✓.

(b) $[L/R] = \dfrac{\mathrm{M\,L^2\,T^{-2}\,I^{-2}}}{\mathrm{M\,L^2\,T^{-3}\,I^{-2}}} = \mathrm{T}$ ✓.

(c) $[LC] = \mathrm{T^2}$, so $\omega_0 = 1/\sqrt{LC}$ has dimension $\mathrm{T^{-1}}$ — a frequency ✓.

(d) Capacitor energy $W = \frac{1}{2}CV^2$: $[C][V]^2 = \mathrm{M^{-1}L^{-2}T^4I^2} \cdot \mathrm{M^2L^4T^{-6}I^{-2}} = \mathrm{M\,L^2\,T^{-2}}$, an energy ✓. Numerically $\frac{1}{2}(100\ \mu\mathrm{F})(20\ \mathrm{V})^2 = 0.02\ \mathrm{J}$.

(e) A formula claiming $\tau = R/C$ fails immediately: $[R/C] = \mathrm{M^2L^4T^{-7}I^{-4}} \neq \mathrm{T}$.

Worked Example 1.56 — Checking an electromagnetic formula

(a) Speed of light. $[\mu\varepsilon] = \mathrm{M\,L\,T^{-2}I^{-2}} \cdot \mathrm{M^{-1}L^{-3}T^4I^2} = \mathrm{L^{-2}T^2}$, so $1/\sqrt{\mu_0\varepsilon_0}$ is a speed. Numerically $1/\sqrt{(8.854\times10^{-12})(4\pi\times10^{-7})} = 2.998\times10^8\ \mathrm{m/s}$ ✓.

(b) Skin depth. The claimed formula $\delta = \sqrt{2/(\omega\mu\sigma)}$ should be a length. $[\omega\mu\sigma] = \mathrm{T^{-1}} \cdot \mathrm{M\,L\,T^{-2}I^{-2}} \cdot \mathrm{M^{-1}L^{-3}T^3I^2} = \mathrm{L^{-2}}$, so $[\delta] = \mathrm{L}$ ✓. For copper ($\sigma = 5.8\times10^7\ \mathrm{S/m}$, $\mu = \mu_0$) at 60 Hz, $\delta = 8.53$ mm; at 1 MHz, $\delta = 66\ \mu$m.

Dimensional analysis as a design tool

Dimensional consistency can do more than check formulas: it can predict their form. If a quantity depends only on a known list of others, its formula must be a product of powers with matching dimensions.

Worked Example 1.57 — Predicting the resonance formula

Suppose the resonant angular frequency of an ideal LC circuit depends only on L and C: $\omega_0 = k\,L^aC^b$ with k dimensionless. Find a and b.

Step 1 — dimensions. $\mathrm{T^{-1}} = \left(\mathrm{M\,L^2T^{-2}I^{-2}}\right)^a \left(\mathrm{M^{-1}L^{-2}T^4I^2}\right)^b$.

Step 2 — match exponents. M: $a - b = 0$. L: $2a - 2b = 0$. I: $-2a + 2b = 0$. T: $-2a + 4b = -1$.

Step 3 — solve. $a = b$ and $2a = -1$, so $a = b = -\frac{1}{2}$, giving $\omega_0 = k/\sqrt{LC}$.

The analysis cannot find k (circuit analysis gives $k = 1$), but it pins down the structure. Its systematic form, the Buckingham π theorem, is developed in Ch. 25.

Units, prefixes and conversions

Dimensions say *what kind* of quantity; **units** say *how much*. SI prefixes scale units by powers of 10^3:

Prefix	p	n	μ	m	k	M	G
Factor	10^{-12}	10^{-9}	10^{-6}	10^{-3}	10^3	10^6	10^9

Convert by multiplying by factors equal to one, such as $\frac{10^{-9}\ \mathrm{F}}{1\ \mathrm{nF}}$, so that units cancel algebraically. Prefixes bind to the unit before powers apply: $1\ \mathrm{cm^2} = (10^{-2}\ \mathrm{m})^2 = 10^{-4}\ \mathrm{m^2}$, not $10^{-2}\ \mathrm{m^2}$. Treat units as algebraic symbols throughout a calculation and the final unit appears automatically — if it is not the expected one, there is an error.

Common Pitfall — dimensionless is not unitless in practice

Angles are dimensionless, but radians and degrees are different units: ωt must be in radians for $\sin\omega t$ to be correct with ω in rad/s. Hertz and rad/s have the same dimension $\mathrm{T^{-1}}$ but differ by 2π — a dimensional check cannot catch f written for ω. Likewise decibels are dimensionless but need a stated reference (Section 1.5).

Engineering Note

Dimensional checking cannot detect missing dimensionless factors (2π, $\frac{1}{2}$, $\sqrt{2}$), but it catches the majority of algebra slips in seconds. Make it a habit: check the dimensions of every derived formula before substituting numbers, check limiting cases (what happens as $R \to 0$ or $\omega \to \infty$?), and estimate the order of magnitude of the answer before trusting a calculator.

Summary

Topic	Key result
Number sets	$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$; $\mathbb{C}$ is not ordered
Triangle inequality	$\lvert x+y\rvert \le \lvert x\rvert + \lvert y\rvert$, $\big\lvert\lvert x\rvert - \lvert y\rvert\big\rvert \le \lvert x-y\rvert$
Machine epsilon (double)	$\varepsilon_{\text{mach}} = 2^{-52} \approx 2.2\times10^{-16}$

Topic	Key result
Binomial theorem	$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k,\ \binom{n}{k} = \dfrac{n!}{k!(n-k)!}$
Binomial approximation	$(1+x)^n \approx 1 + nx,\ \lvert x\rvert \ll 1$, any real n
Completing the square	$ax^2 + bx + c = a\left(x + \frac{b}{2a}\right)^2 + c - \frac{b^2}{4a}$
Quadratic formula	$x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$; stable form $q = -\frac{1}{2}\big(b + \operatorname{sgn}(b)\sqrt{\Delta}\big)$, $x_1 = q/a$, $x_2 = c/q$
Vieta (quadratic)	$x_1 + x_2 = -b/a,\ x_1 x_2 = c/a$
Cramer (2×2)	$x = \dfrac{c_1 b_2 - c_2 b_1}{a_1 b_2 - a_2 b_1},\ y = \dfrac{a_1 c_2 - a_2 c_1}{a_1 b_2 - a_2 b_1}$
AM–GM	$\frac{1}{2}(a+b) \geq \sqrt{ab}$, equality iff $a = b$
Remainder / factor theorem	$p(x) \bmod (x-r) = p(r);\ p(r) = 0 \iff (x-r) \mid p(x)$
Fundamental theorem of algebra	$p(x) = a_n \prod_{i=1}^{n} (x - r_i)$; real coefficients $\Rightarrow$ conjugate pairs
Cover-up rule	$A = \Big[(x-a)R(x)\Big]_{x=a}$
Repeated pole	$A_{m-k} = \dfrac{1}{k!}\dfrac{d^k}{dx^k}\Big[(x-a)^m R(x)\Big]_{x=a}$
Euler's number	$e = \lim_{n\to\infty}(1+1/n)^n = \sum 1/k! = 2.71828\ldots$
Logarithms	$\log(xy) = \log x + \log y,\ \log x^p = p \log x,\ \log_a x = \ln x / \ln a$
First-order transient	$e^{-t/\tau}$; $t_{10\%} = \tau \ln 10$; 10–90 % rise time $= \tau \ln 9 = 2.197\tau$
Decibels	$10 \log_{10}(P_2/P_1) = 20 \log_{10}(V_2/V_1)$; 0 dBm $= 1$ mW
Hyperbolic	$\cosh x = \frac{1}{2}(e^x + e^{-x})$, $\sinh x = \frac{1}{2}(e^x - e^{-x})$, $\cosh^2 x - \sinh^2 x = 1$
Pythagorean identity	$\sin^2\theta + \cos^2\theta = 1,\ 1 + \tan^2\theta = \sec^2\theta$
Compound angle	$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B,\ \sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$
Double angle / power reduction	$\sin 2\theta = 2\sin\theta\cos\theta,\ \cos^2\theta = \frac{1}{2}(1 + \cos 2\theta)$
Product-to-sum	$\cos A \cos B = \frac{1}{2}[\cos(A-B) + \cos(A+B)]$
Harmonic addition	$a\cos\theta + b\sin\theta = \sqrt{a^2+b^2}\cos\big(\theta - \operatorname{atan2}(b,a)\big)$
Law of cosines	$c^2 = a^2 + b^2 - 2ab\cos C$
Polar	$r = \sqrt{x^2+y^2},\ \theta = \operatorname{atan2}(y,x)$
Cylindrical (r,ϕ,z)	$x = r\cos\phi,\ y = r\sin\phi;\ dV = r\,dr\,d\phi\,dz$
Spherical (R,θ,ϕ)	$x = R\sin\theta\cos\phi,\ y = R\sin\theta\sin\phi,\ z = R\cos\theta;\ dV = R^2\sin\theta\,dR\,d\theta\,d\phi$
Even–odd split	$f_e = \frac{1}{2}[f(x) + f(-x)],\ f_o = \frac{1}{2}[f(x) - f(-x)]$
Graph transformation	$A\,f\big(b(x-c)\big) + d$: shift right c, scale by $1/b$ horizontally, A vertically, up d
Inverse of composite	$(f \circ g)^{-1} = g^{-1} \circ f^{-1}$
Inverse hyperbolic	$\operatorname{arsinh} x = \ln\left(x + \sqrt{x^2+1}\right),\ \operatorname{artanh} x = \frac{1}{2}\ln\frac{1+x}{1-x}$
Geometric series	$\sum_{k=0}^{n-1} r^k = \dfrac{1-r^n}{1-r};\ \sum_{k=0}^{\infty} r^k = \dfrac{1}{1-r},\ \lvert r\rvert < 1$
Dimensions	$[V] = \mathrm{M\,L^2T^{-3}I^{-1}}$; $[RC] = [L/R] = \mathrm{T}$; $[LC] = \mathrm{T^2}$; arguments of exp, ln, sin dimensionless

Further reading

- E. Kreyszig, *Advanced Engineering Mathematics*, 10th ed., Wiley, 2011 — Appendix 3 (formulas and identities) for reference.
- K. A. Stroud and D. J. Booth, *Engineering Mathematics*, 8th ed., Palgrave Macmillan, 2020 — Foundation programmes F.1–F.11 and Programmes 1–3 give extensive practice in algebra, logarithms, trigonometry and functions.
- G. James et al., *Modern Engineering Mathematics*, 6th ed., Pearson, 2020 — Chapters 1–2 (number, algebra, functions, coordinate geometry).
- G. Strang, *Calculus*, 3rd ed., Wellesley-Cambridge Press, 2017 — Chapter 1 on functions and their graphs (freely available from MIT OpenCourseWare).

Chapter 2 — Complex Numbers and Engineering Representation

> **Prerequisites:** Ch. 1 (algebra, polynomials, exponentials and logarithms, trigonometric identities, polar coordinates) **Learning outcomes:** After this chapter you can
>
> - move freely between rectangular, polar and exponential forms of a complex number, getting the quadrant of the angle right every time;
> - add, multiply, divide, conjugate and take magnitudes of complex numbers, choosing the form that makes each operation easiest;
> - use Euler's formula to turn trigonometric identities, sums and integrals into algebra with exponentials;
> - compute integer powers and all n-th roots of a complex number, and use roots of unity;
> - interpret complex arithmetic geometrically and describe curves and regions of the complex plane;
> - represent sinusoids as phasors, solve sinusoidal steady-state problems algebraically, and express impedances and transfer functions as complex functions of $j\omega$ or s.

A sinusoidal signal is described by two numbers — an amplitude and a phase. A complex number is also two numbers, and complex multiplication combines magnitudes and angles in exactly the way that linear circuits combine amplitudes and phases. That coincidence, which turns out to be no coincidence at all, is why every alternating-current calculation in electrical engineering, every frequency response, every Fourier and Laplace transform and every digital filter is written in complex numbers.

Historically, complex numbers were not invented to solve $x^2 + 1 = 0$; mathematicians were content to say that equation had no solution. They were forced on sixteenth-century algebraists by the *cubic* formula, which for equations with three perfectly real roots (such as $x^3 - 15x - 4 = 0$, whose roots include $x = 4$) produces intermediate expressions like $\sqrt{-121}$. The square roots of negative numbers cancel at the end, but there is no way to avoid them on the way. The lesson carries straight into engineering: complex numbers are a *computational device* that often lets you reach a real answer by the shortest route. A circuit's voltages are real, its transient responses are real, its measured gains are real — yet the shortest path to all of them runs through $\mathbb{C}$.

This chapter is the mathematical foundation for that path. Sections 2.1–2.3 build the algebra, Sections 2.4–2.6 the exponential machinery that makes it powerful, Section 2.7 the geometry, and Sections 2.8–2.9 show how the whole apparatus becomes the phasor method and the language of impedance and transfer functions. Complex *functions* — differentiation, contour integration and residues — are deferred to Ch. 18.

2.1 Rectangular and polar forms

Why two forms?

An engineer meets complex numbers in two guises. A network analyser reports an impedance as "$40 - j30\ \Omega$" — a resistance and a reactance, the **rectangular** description. An oscilloscope comparing two sinusoids reports "amplitude ratio 0.8, phase $-36.9°$" — a magnitude and an angle, the **polar** description. They are the same complex number seen in two coordinate systems, exactly as a point in the plane can be given by (x, y) or by (r, θ) (Ch. 1, Section 1.7). Fluency means converting between them without thinking and knowing which one a given operation prefers.

Definition and the complex number system

Complex number

A **complex number** is an expression $z = a + jb$ with $a, b \in \mathbb{R}$, where the **imaginary unit** j satisfies $j^2 = -1$. The real number $a = \mathrm{Re}\{z\}$ is the **real part** and the real number $b = \mathrm{Im}\{z\}$ is the **imaginary part** (note: b, not jb). The writing $a + jb$ is the **rectangular** (Cartesian) form. The set of all complex numbers is $\mathbb{C}$.

A number with $b = 0$ is **real**; a number with $a = 0$, $b \neq 0$ is **purely imaginary**.

Equality

Two complex numbers are equal if and only if their real parts are equal **and** their imaginary parts are equal:

$$a + jb = c + jd \iff a = c \ \text{ and } \ b = d.$$

The equality rule is more powerful than it looks: **one complex equation is two real equations.** Many derivations in circuit theory exploit exactly this — a single complex KVL equation delivers both an amplitude condition and a phase condition.

A rigorous construction, for readers who distrust a symbol "whose square is -1": define $\mathbb{C}$ as the set of ordered pairs (a, b) of reals with

$$(a, b) + (c, d) = (a + c,\ b + d), \qquad (a, b) \cdot (c, d) = (ac - bd,\ ad + bc).$$

One checks that these operations obey all the usual laws of arithmetic (commutative, associative, distributive, with $(0, 0)$ and $(1, 0)$ as additive and multiplicative identities). Pairs $(a, 0)$ behave exactly like the real numbers a, and the pair $j = (0, 1)$ satisfies $j \cdot j = (0 \cdot 0 - 1 \cdot 1,\ 0) = (-1, 0) = -1$. Writing $(a, b) = a + jb$ recovers the familiar notation. So complex numbers are nothing more mysterious than points of the plane with a cleverly chosen multiplication rule — the rule is what makes them useful.

Common Pitfall — $\mathbb{C}$ is not ordered

There is no meaningful "$z_1 < z_2$" for complex numbers. Statements such as "$3 + j4 > 2$" are meaningless. Only real quantities derived from complex numbers — magnitudes, real parts, imaginary parts, angles — can be compared. When an engineer says "this impedance is larger", they mean $|Z|$ is larger.

The complex plane and polar coordinates

Plot $z = a + jb$ as the point (a, b), with the real axis horizontal and the imaginary axis vertical. Its distance from the origin and its angle measured anticlockwise from the positive real axis give the **polar form**.

Magnitude, argument, polar form

For $z = a + jb \neq 0$,

$$r = |z| = \sqrt{a^2 + b^2}, \qquad \theta = \arg z = \angle z, \quad \text{where } \cos\theta = \frac{a}{r},\ \sin\theta = \frac{b}{r}.$$

Then

$$z = r(\cos\theta + j\sin\theta), \qquad a = r\cos\theta, \quad b = r\sin\theta.$$

Engineers abbreviate this as $z = r\angle\theta$. The quantity $r \geq 0$ is the **magnitude** (modulus, absolute value) and θ the **argument** (angle, phase). The argument is defined only up to multiples of 2π; the **principal argument** $\operatorname{Arg} z$ is the value in $(-\pi, \pi]$. For $z = 0$ the magnitude is 0 and the argument is undefined.

Both $\cos\theta = a/r$ **and** $\sin\theta = b/r$ are needed to fix θ; either one alone leaves two candidate angles. The ratio $\tan\theta = b/a$ also leaves two candidates, π apart, which is the source of the most common error in this whole subject.

The quadrant problem and atan2

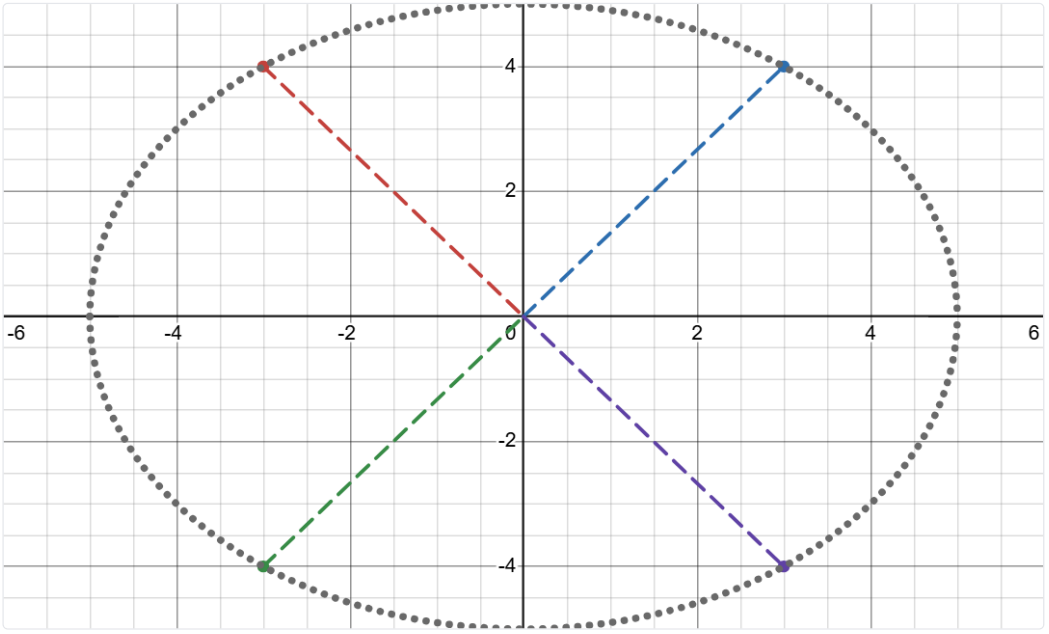
The calculator function arctan returns values in $(-\pi/2, \pi/2)$ — the right half-plane only. It therefore gives the correct argument only when $a > 0$. The correct rule is the two-argument arctangent:

Two-argument arctangent

$$\operatorname{atan2}(b, a) = \begin{cases} \arctan(b/a), & a > 0, \\ \arctan(b/a) + \pi, & a < 0,\ b \geq 0, \\ \arctan(b/a) - \pi, & a < 0,\ b < 0, \\ +\pi/2, & a = 0,\ b > 0, \\ -\pi/2, & a = 0,\ b < 0. \end{cases}$$

It returns $\operatorname{Arg}(a + jb) \in (-\pi, \pi]$. It is available as `atan2(b, a)` in C, Python, MATLAB and spreadsheets (spreadsheets often reverse the argument order — check).

The geometric content is simple: points in the second and third quadrants have arguments of magnitude greater than 90°, and arctan can never return such a value, so a correction of $\pm 180^\circ$ is required there.



The four points $\pm 3 \pm j4$ all have magnitude 5 and $|\tan\theta| = 4/3$, but their arguments are 53.13°, 126.87°, -126.87° and -53.13°. Only the sign pattern of (a, b) distinguishes them.

Degrees and radians

Engineers quote phase angles in degrees ($10\angle -30^\circ$) because instruments do; mathematics and every exponential formula require radians. Convert with $\theta_{\text{rad}} = \theta_{\text{deg}} \cdot \pi/180$. The convention in this book: when an angle carries a degree sign it is in degrees; otherwise it is in radians. Inside an exponent ($e^{j\theta}$, Section 2.4) the angle is **always** in radians.

Worked Example 2.1 — Rectangular to polar

Convert $3+j4$, $-3+j4$, $-1-j$, -2, $j5$ and $-5-j12$ to polar form.

Step 1 — magnitude. $|3+j4| = \sqrt{9+16} = 5$; $|-3+j4| = 5$; $|-1-j| = \sqrt{2} \approx 1.4142$; $|-2| = 2$; $|j5| = 5$; $|-5-j12| = \sqrt{25+144} = 13$.

Step 2 — locate the quadrant, then the angle.

- $3+j4$: first quadrant, $\theta = \arctan\frac{4}{3} = 53.13^\circ$ (0.9273 rad).
- $-3+j4$: second quadrant, $\arctan\frac{4}{-3} = -53.13^\circ$ is wrong; add 180°: $\theta = 126.87^\circ$.
- $-1-j$: third quadrant, $\arctan\frac{-1}{-1} = 45^\circ$ is wrong; subtract 180°: $\theta = -135^\circ = -3\pi/4$.
- -2: negative real axis, $\theta = 180^\circ = \pi$ (principal value $+\pi$, not $-\pi$).
- $j5$: positive imaginary axis, $\theta = 90^\circ$; $\arctan(5/0)$ is undefined, so use the special case.
- $-5-j12$: third quadrant, $\arctan\frac{12}{5} = 67.38^\circ$, so $\theta = 67.38^\circ - 180^\circ = -112.62^\circ$.

z	$\lvert z\rvert$	Arg z
$3+j4$	5	53.13°
$-3+j4$	5	126.87°
$-1-j$	1.4142	-135°
-2	2	180°
$j5$	5	90°
$-5-j12$	13	-112.62°

Check one of them backwards: $13\cos(-112.62^\circ) = 13(-0.3846) = -5.00$ and $13\sin(-112.62^\circ) = 13(-0.9231) = -12.00$. ✓

Worked Example 2.2 — Polar to rectangular

Convert $10\angle -30^\circ$, $25\angle 143.13^\circ$, $4\angle\frac{5\pi}{6}$ and $2\angle 200^\circ$.

Use $a = r\cos\theta$, $b = r\sin\theta$ — no quadrant care is needed in this direction, because cosine and sine carry the signs automatically.

- $10\angle -30^\circ = 10(0.8660) + j10(-0.5) = 8.660 - j5.000$.
- $25\angle 143.13^\circ = 25(-0.8000) + j25(0.6000) = -20.00 + j15.00$.
- $4\angle\frac{5\pi}{6} = 4\left(-\frac{\sqrt{3}}{2}\right) + j4\left(\frac{1}{2}\right) = -2\sqrt{3} + j2 \approx -3.464 + j2.000$.
- $2\angle 200^\circ = 2(-0.9397) + j2(-0.3420) = -1.879 - j0.684$. Note that 200° is not a principal argument; the same number is $2\angle -160^\circ$.

Worked Example 2.3 — Using equality of complex numbers

Find real x and y such that $(x+jy)(2-j) = 7+j4$.

Expand the left side: $(x+jy)(2-j) = 2x - jx + j2y - j^2y = (2x+y) + j(2y-x)$.

Equate real and imaginary parts — one complex equation gives two real ones:

$$2x + y = 7, \qquad -x + 2y = 4.$$

From the second, $x = 2y - 4$; substituting, $4y - 8 + y = 7$, so $y = 3$ and $x = 2$.

Check: $(2+j3)(2-j) = 4 - j2 + j6 - j^2 3 = 4 + 3 + j4 = 7 + j4$. ✓

(Section 2.3 gives a quicker route: $x + jy = (7+j4)/(2-j)$.)

Worked Example 2.4 — An impedance measurement

An impedance analyser reports $|Z| = 50\ \Omega$ and $\angle Z = -36.87^\circ$ for an unknown component at 1 kHz. Find the series resistance and reactance, and say whether the component behaves capacitively or inductively.

$$Z = 50\cos(-36.87^\circ) + j50\sin(-36.87^\circ) = 40.0 - j30.0\ \Omega.$$

The resistance is $R = 40\ \Omega$ and the reactance $X = -30\ \Omega$. A negative reactance is capacitive (Section 2.9: $Z_C = -j/(\omega C)$). If the reactance comes from a pure capacitor, $1/(\omega C) = 30$, so $C = 1/(2\pi \cdot 1000 \cdot 30) = 5.31\ \mu\text{F}$.

Check: $\sqrt{40^2 + 30^2} = 50$ ✓ and $\operatorname{atan2}(-30, 40) = -36.87^\circ$ ✓.

Common Pitfall — the calculator arctangent

A calculator's $\tan^{-1}$ gives $\arctan\frac{-1}{-1} = \arctan 1 = 45^\circ$ for $-1-j$, which is the angle of $1+j$ — the diametrically opposite point. **Always sketch the point first** (or at least note the signs of a and b), or use `atan2`. A phase error of exactly 180° in a circuit result is almost always this mistake: it flips the sign of the entire signal.

Engineering Note — which form do instruments use?

Network and impedance analysers can display either form: "$R + jX$" (rectangular), "$|Z|, \theta$" (polar), or on a Smith chart (a geometric map, Section 2.7). Oscilloscopes and lock-in amplifiers naturally measure magnitude and phase, or the **in-phase** (I) and **quadrature** (Q) components a and b. In digital communications, a received symbol is literally a complex number $I + jQ$, and a constellation diagram is a plot of the complex plane. Every one of these displays is just one of the two forms of this section.

2.2 Complex conjugates

Motivation

Division by a complex number, extracting real and imaginary parts, computing a magnitude without square roots, finding average power in an AC circuit, and proving that a real filter's poles come in mirror-image pairs — all of these rest on one operation: changing the sign of j. Because j and $-j$ are algebraically indistinguishable (both square to -1), replacing every j by $-j$ in a true equation produces another true equation. That symmetry is the complex conjugate.

Complex conjugate

The **complex conjugate** of $z = a + jb = r\angle\theta$ is

$$z^* = a - jb = r\angle(-\theta).$$

Geometrically it is the **reflection of z in the real axis**. Mathematics texts often write $\bar{z}$; this book uses z^*, which matches the conjugate-transpose notation $\mathbf{A}^{\mathsf{H}}$ of linear algebra.

Algebraic properties

Properties of the conjugate

For all $z, z_1, z_2 \in \mathbb{C}$:

(i) $(z^*)^* = z$, (ii) $(z_1 \pm z_2)^* = z_1^* \pm z_2^*$, (iii) $(z_1 z_2)^* = z_1^* z_2^*$, (iv) $\left(\dfrac{z_1}{z_2}\right)^* = \dfrac{z_1^*}{z_2^*}$ $(z_2 \neq 0)$, (v) $(z^n)^* = (z^*)^n$, (vi) $zz^* = a^2 + b^2 = |z|^2$,

Proof sketch. Each is a one-line computation in rectangular form. For (iii), with $z_1 = a + jb$, $z_2 = c + jd$:

$$z_1 z_2 = (ac - bd) + j(ad + bc) \;\Rightarrow\; (z_1 z_2)^* = (ac - bd) - j(ad + bc),$$

$$z_1^* z_2^* = (a - jb)(c - jd) = (ac - bd) - j(ad + bc). \checkmark$$

Property (v) follows from (iii) by induction, (iv) from (iii) applied to $z_1 = (z_1/z_2)\, z_2$, and (vi) from $(a + jb)(a - jb) = a^2 - j^2 b^2 = a^2 + b^2$. For (viii) and (ix), add and subtract $z = a + jb$ and $z^* = a - jb$. □

Property (vi) deserves emphasis: **the product of a number and its conjugate is real and non-negative**. It is the identity behind division (Section 2.3), behind the energy of a signal $\int |x(t)|^2 dt = \int x\, x^*\, dt$, and behind power in circuits.

Conjugate of a rational expression

Let f be built from z and **real** constants by addition, subtraction, multiplication, division and integer powers (for example a polynomial or rational function with real coefficients). Then

$$f(z^*) = \big(f(z)\big)^*.$$

Proof. Properties (ii)–(v) let the conjugate pass through every operation, and a real constant c satisfies $c^* = c$. □

The most important consequence:

Conjugate root theorem

If $p(x) = c_n x^n + \cdots + c_1 x + c_0$ has **real** coefficients and $p(z_0) = 0$, then $p(z_0^*) = 0$. Non-real roots of real polynomials occur in **conjugate pairs**, and a real polynomial of odd degree has at least one real root.

Proof. $p(z_0^*) = \big(p(z_0)\big)^* = 0^* = 0$. Pairing off the non-real roots leaves an odd number of roots to be real when n is odd. □

Each conjugate pair $\alpha \pm j\beta$ combines into a real quadratic factor:

$$\big(x - (\alpha + j\beta)\big)\big(x - (\alpha - j\beta)\big) = (x - \alpha)^2 + \beta^2 = x^2 - 2\alpha x + (\alpha^2 + \beta^2).$$

So every real polynomial factors into real linear and real irreducible quadratic factors — exactly the factors that appear in the partial-fraction decompositions of Ch. 1 and Ch. 14.

Worked Example 2.5 — Basic conjugate manipulations

Let $z = 2 - j3$. Find z^*, zz^*, $\text{Re}\{z\}$ via (viii), and $1/z$.

- $z^* = 2 + j3$.
- $zz^* = (2 - j3)(2 + j3) = 4 + 9 = 13 = |z|^2$, so $|z| = \sqrt{13} \approx 3.606$.
- $\dfrac{z + z^*}{2} = \dfrac{4}{2} = 2$. ✓
- $\dfrac{1}{z} = \dfrac{z^*}{zz^*} = \dfrac{2 + j3}{13} = 0.1538 + j0.2308$.

Check: $(2 - j3)(0.1538 + j0.2308) = 0.3077 + 0.6923 + j(0.4615 - 0.4615) = 1.000$. ✓

Worked Example 2.6 — An equation involving z and z^*

Solve $z + 2z^* = 3 - j4$.

An equation containing both z and z^* is **not** a polynomial equation in z, so substitute $z = x + jy$ and use equality:

$$(x + jy) + 2(x - jy) = 3x - jy = 3 - j4.$$

Real parts: $3x = 3$, so $x = 1$. Imaginary parts: $-y = -4$, so $y = 4$. Hence $z = 1 + j4$.

Check: $(1 + j4) + 2(1 - j4) = 3 - j4$. ✓

Worked Example 2.7 — Completing a factorisation

The real cubic $p(z) = z^3 - 5z^2 + 9z - 5$ has a root $z = 2 + j$. Find the others.

Step 1. Coefficients are real, so $2 - j$ is also a root.

Step 2. The pair gives the real quadratic factor $(z - 2)^2 + 1 = z^2 - 4z + 5$.

Step 3. Divide: $z^3 - 5z^2 + 9z - 5 = (z^2 - 4z + 5)(z - 1)$ — check by expanding: $z^3 - z^2 - 4z^2 + 4z + 5z - 5 = z^3 - 5z^2 + 9z - 5$. ✓

The roots are $1, 2 + j, 2 - j$. Their sum is 5 (the negative of the z^2 coefficient) and their product is $1 \cdot \big((2 + j)(2 - j)\big) = 5$ (the negative of the constant term) — both consistent with Vieta's relations. ✓

Worked Example 2.8 — Unit magnitude and the all-pass property

(a) Show that if $|z| = 1$ then $1/z = z^*$.

From (vi), $zz^* = |z|^2 = 1$, so $z^* = 1/z$. Points on the unit circle are inverted by conjugation — a fact used constantly in discrete-time signal processing, where $z = e^{j\Omega}$ lies on the unit circle (Ch. 17).

(b) Let a be any complex number with $|a| \neq 1$. Show that

$$A(z) = \frac{z - a}{1 - a^* z}$$

satisfies $|A(z)| = 1$ whenever $|z| = 1$.

Step 1. Use $|z| = 1$ to write $1 = zz^*$ in the denominator: $1 - a^* z = zz^* - a^* z = z\,(z^* - a^*)$.

Step 2. Take magnitudes: $|1 - a^* z| = |z|\,|z^* - a^*| = 1 \cdot |(z - a)^*| = |z - a|$, by property (ii) and (vii).

Step 3. Hence $|A(z)| = |z - a| / |z - a| = 1$. □

Numerical check: $a = 0.5 + j0.3$, $z = e^{j1} = 0.5403 + j0.8415$: $z - a = 0.0403 + j0.5415$, $|z - a| = 0.5430$; $a^* z = (0.5 - j0.3)(0.5403 + j0.8415) = 0.5226 + j0.2587$, $1 - a^* z = 0.4774 - j0.2587$, $|1 - a^* z| = 0.5430$. ✓

This $A(z)$ is the **first-order all-pass** section of digital filtering: it changes phase but passes every frequency with unit gain.

Engineering Note — where conjugates appear

- **Power:** with amplitude phasors, complex power is $S = \frac{1}{2}\mathbf{V}\mathbf{I}^*$ (Section 2.9). The conjugate makes the angle of S equal to the voltage–current phase difference.
- **Real signals:** the Fourier transform of a real signal satisfies $X(-\omega) = X(\omega)^*$ (Ch. 16) — a direct consequence of the theorem on conjugates of expressions with real constants.
- **Real systems:** poles and zeros of any circuit built from real R, L, C and controlled sources are real or occur in conjugate pairs, so pole–zero plots are always symmetric about the real axis.
- **Matched filtering and correlation** use $x(t)\,y^*(t)$; **maximum power transfer** requires a load equal to the conjugate of the source impedance (Section 2.9).

Common Pitfall — conjugating only part of an expression

To conjugate $\dfrac{3+j2}{(1-j)\,e^{j\pi/5}}$ you must change the sign of **every** j: $\dfrac{3-j2}{(1+j)\,e^{-j\pi/5}}$. A frequent error is to conjugate the numerator only. Conversely, do **not** conjugate real symbols that merely look complex: if ω, R, L, C are real, then $(R+j\omega L)^* = R - j\omega L$, with ω untouched.

2.3 Complex arithmetic

Motivation

Circuit analysis in the sinusoidal steady state is ordinary DC circuit analysis — Ohm's law, Kirchhoff's laws, series and parallel combination, nodal and mesh equations — carried out with complex numbers instead of real ones (Section 2.9). Every one of those operations reduces to the four arithmetic operations of this section. The arithmetic itself is easy; what takes practice is doing it **reliably** and choosing the form that minimises effort.

The four operations

Rectangular arithmetic

For $z_1 = a + jb$ and $z_2 = c + jd$:

$$\begin{aligned} z_1 \pm z_2 &= (a \pm c) + j(b \pm d), \\ z_1 z_2 &= (ac - bd) + j(ad + bc), \\ \frac{z_1}{z_2} &= \frac{z_1 z_2^*}{z_2 z_2^*} = \frac{(ac+bd) + j(bc-ad)}{c^2 + d^2}, \qquad z_2 \neq 0. \end{aligned}$$

Addition is component-wise — vector addition in the plane. **Multiplication** is ordinary expansion of brackets followed by $j^2 = -1$:

$$(a+jb)(c+jd) = ac + jad + jbc + j^2 bd = (ac - bd) + j(ad + bc).$$

Division uses the conjugate trick: multiplying numerator and denominator by z_2^* makes the denominator the real number $|z_2|^2$, after which division is just scaling. In particular the **reciprocal** is

$$\frac{1}{c+jd} = \frac{c - jd}{c^2 + d^2}, \qquad \text{e.g.} \quad \frac{1}{j} = \frac{-j}{1} = -j.$$

Polar arithmetic

For $z_1 = r_1 \angle \theta_1$ and $z_2 = r_2 \angle \theta_2$:

$$z_1 z_2 = r_1 r_2 \angle(\theta_1 + \theta_2), \qquad \frac{z_1}{z_2} = \frac{r_1}{r_2}\angle(\theta_1 - \theta_2), \qquad \frac{1}{z_2} = \frac{1}{r_2}\angle(-\theta_2).$$

Magnitudes multiply, angles add.

Proof. Multiply the polar forms and use the compound-angle identities of Ch. 1:

$$\begin{aligned} z_1 z_2 &= r_1 r_2 \big[(\cos\theta_1 \cos\theta_2 - \sin\theta_1 \sin\theta_2) + j(\sin\theta_1 \cos\theta_2 + \cos\theta_1 \sin\theta_2)\big] \\ &= r_1 r_2 \big[\cos(\theta_1 + \theta_2) + j \sin(\theta_1 + \theta_2)\big]. \end{aligned}$$

The quotient rule follows by checking that $\frac{r_1}{r_2}\angle(\theta_1 - \theta_2)$ times $r_2 \angle \theta_2$ gives $r_1 \angle \theta_1$. $\square$

This proof is a little laborious; Section 2.5 gives a one-line version using exponentials. Notice that the trigonometric identities and the multiplication rule are *the same fact* in two notations — a theme Section 2.4 makes systematic.

Powers of j

Because $j^2 = -1$, the powers of j cycle with period 4:

$$j^0 = 1, \quad j^1 = j, \quad j^2 = -1, \quad j^3 = -j, \quad j^4 = 1, \quad j^5 = j, \ \ldots$$

so $j^n = j^{\,n \bmod 4}$. Negative powers follow from $1/j = -j$: $j^{-1} = -j$, $j^{-2} = -1$, $j^{-3} = j$. In polar form $j = 1\angle 90^\circ$, and $j^n = 1\angle(90n)^\circ$ — each multiplication by j is a quarter-turn.

Field properties

$\mathbb{C}$ is a field

Addition and multiplication of complex numbers are commutative and associative, multiplication distributes over addition, 0 and 1 are identities, every z has an additive inverse $-z$, and every $z \neq 0$ has a multiplicative inverse $1/z$.

The practical meaning: **every algebraic manipulation you know for real numbers is valid for complex numbers** — expanding brackets, collecting terms, cross-multiplying, solving linear systems by elimination or Cramer's rule, factorising polynomials, the quadratic formula. What does *not* carry over are statements involving order ($<$, $>$) and some rules for multivalued functions such as square roots and logarithms (pitfall below, and Section 2.5).

Magnitude rules and the triangle inequality

Magnitude rules

$$|z_1 z_2| = |z_1|\,|z_2|, \qquad \left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}, \qquad |z^n| = |z|^n, \qquad |\operatorname{Re}\{z\}| \le |z|, \quad |\operatorname{Im}\{z\}| \le |z|.$$

Triangle inequality:

$$|z_1 + z_2| \le |z_1| + |z_2|, \qquad \text{and} \qquad \big||z_1| - |z_2|\big| \le |z_1 - z_2|.$$

Proof of the product rule without polar form. $|z_1 z_2|^2 = (z_1 z_2)(z_1 z_2)^* = z_1 z_1^*\, z_2 z_2^* = |z_1|^2 |z_2|^2$.

Proof of the triangle inequality.

$$\begin{aligned} |z_1 + z_2|^2 &= (z_1 + z_2)(z_1^* + z_2^*) = |z_1|^2 + |z_2|^2 + z_1 z_2^* + z_1^* z_2 \\ &= |z_1|^2 + |z_2|^2 + 2\operatorname{Re}\{z_1 z_2^*\} \\ &\le |z_1|^2 + |z_2|^2 + 2|z_1 z_2^*| = (|z_1| + |z_2|)^2. \end{aligned}$$

Take square roots. Equality holds exactly when $z_1 z_2^*$ is real and non-negative, i.e. when z_1 and z_2 point in the same direction. The reverse inequality follows by applying the triangle inequality to $z_1 = (z_1 - z_2) + z_2$. □

The middle line of that proof is worth remembering on its own:

$$|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2 + 2|z_1||z_2|\cos(\theta_1 - \theta_2),$$

which is the **law of cosines**. When two sinusoids of the same frequency are added (Section 2.8), the amplitude of the sum is given by exactly this formula: maximal ($|z_1| + |z_2|$) when in phase, minimal ($\big||z_1| - |z_2|\big|$) when in antiphase.

Engineering Note — choose the form to suit the operation

Rectangular for sums and differences (series impedances, KCL/KVL, adding phasors), polar for products, quotients, powers and roots (gains, voltage dividers, Ohm's law, cascaded stages). A mixed calculation such as $\dfrac{z_1 z_2}{z_3 + z_4}$ is done by forming $z_3 + z_4$ in rectangular form, converting it to polar, then multiplying and dividing in polar. Numerical packages store complex numbers in rectangular form and convert on demand, so on a computer you simply write the expression; by hand, the choice of form is half the skill.

Worked Example 2.9 — The four operations

Let $z_1 = 3 + j4$ and $z_2 = 1 - j2$.

- $z_1 + z_2 = 4 + j2$; $\quad z_1 - z_2 = 2 + j6$.
- $z_1 z_2 = 3 - j6 + j4 - j^2 8 = 3 + 8 + j(-6 + 4) = 11 - j2$.
- $\dfrac{z_1}{z_2} = \dfrac{(3 + j4)(1 + j2)}{1^2 + 2^2} = \dfrac{3 + j6 + j4 + j^2 8}{5} = \dfrac{-5 + j10}{5} = -1 + j2$.

Check the quotient by multiplying back: $(-1 + j2)(1 - j2) = -1 + j2 + j2 - j^2 4 = 3 + j4$. ✓

Check with polar form: $z_1 = 5\angle 53.13^\circ$, $z_2 = \sqrt{5}\angle -63.43^\circ$. Then $z_1 z_2 = 5\sqrt{5}\angle(-10.30^\circ) = 11.18\angle -10.30^\circ$, and $|11 - j2| = \sqrt{125} = 11.18$, $\operatorname{atan2}(-2, 11) = -10.30^\circ$. ✓ Likewise $z_1/z_2 = \sqrt{5}\angle 116.57^\circ = -1 + j2$. ✓

Worked Example 2.10 — A compound expression

Evaluate $w = \dfrac{(2 + j)(3 - j2)}{1 + j} - j^7$ in rectangular form.

Step 1 — numerator. $(2 + j)(3 - j2) = 6 - j4 + j3 - j^2 2 = 8 - j$.

Step 2 — divide by $1 + j$. $\dfrac{8 - j}{1 + j} = \dfrac{(8 - j)(1 - j)}{2} = \dfrac{8 - j8 - j + j^2}{2} = \dfrac{7 - j9}{2} = 3.5 - j4.5$.

Step 3 — the power of j. $7 \bmod 4 = 3$, so $j^7 = j^3 = -j$, and $-j^7 = +j$.

Step 4. $w = 3.5 - j4.5 + j = 3.5 - j3.5 = 3.5\sqrt{2}\angle -45^\circ \approx 4.950\angle -45^\circ$.

Polar check of Step 2: $|8 - j|/|1 + j| = \sqrt{65}/\sqrt{2} = 5.701$ and $|3.5 - j4.5| = \sqrt{32.5} = 5.701$. ✓

Worked Example 2.11 — A complex linear system

Two coupled loop currents satisfy

$$(3+j4)I_1 - j4\,I_2 = 10, \qquad -j4\,I_1 + (2+j2)I_2 = 0.$$

(This is the form mesh analysis produces for two loops sharing an inductive element.) Solve for I_1 and I_2.

Step 1 — determinant. Cramer's rule works exactly as for real systems:

$$\Delta = (3+j4)(2+j2) - (-j4)(-j4) = (6+j6+j8-8) - (j^2\,16) = (-2+j14) + 16 = 14 + j14.$$

Step 2 — I_1.

$$I_1 = \frac{10(2+j2) - (-j4)(0)}{\Delta} = \frac{20+j20}{14+j14} = \frac{20}{14} = 1.4286\angle 0^\circ.$$

Step 3 — I_2.

$$I_2 = \frac{(3+j4)(0) - (-j4)(10)}{\Delta} = \frac{j40}{14+j14} = \frac{40\angle 90^\circ}{19.799\angle 45^\circ} = 2.0203\angle 45^\circ = 1.4286 + j1.4286.$$

Check the second equation: $-j4(1.4286) + (2+j2)(1.4286+j1.4286) = -j5.714 + (2.857 - 2.857) + j(2.857+2.857) = 0$. ✓

Common Pitfall — $\sqrt{a}\sqrt{b} \neq \sqrt{ab}$ for negative numbers

The rule $\sqrt{a}\sqrt{b} = \sqrt{ab}$ holds for $a, b \geq 0$ but fails otherwise: $\sqrt{-4}\sqrt{-9} = (j2)(j3) = -6$, whereas $\sqrt{(-4)(-9)} = \sqrt{36} = 6$. Square roots of complex numbers are two-valued (Section 2.6), and "laws" that silently pick one value can break. Safe practice: write $\sqrt{-a} = j\sqrt{a}$ for $a > 0$ **first**, then multiply.

Engineering Note — numerical care

Computing $|z| = \sqrt{a^2+b^2}$ naively can overflow or underflow when a or b is very large or very small, even if $|z|$ itself is representable. Library functions (`abs` for complex types, `hypot`) rescale internally. Similarly, robust complex division (Smith's algorithm) divides through by the larger of $|c|, |d|$ before forming $c^2 + d^2$. You rarely need to implement these yourself, but you should know why `abs(z)` is preferred to `sqrt(z.real**2 + z.imag**2)`.

2.4 Euler's formula

Motivation

Polar multiplication says angles **add** when numbers multiply. Exponentials have the same property: $e^x e^y = e^{x+y}$, exponents add when quantities multiply. Euler's formula makes this analogy exact — the angle of a complex number behaves precisely like an imaginary exponent. The payoff is enormous: sinusoids, which are awkward to differentiate, integrate, multiply and add, become exponentials, which are the easiest functions in all of calculus. This single identity underlies phasors, Fourier series and transforms, the Laplace transform, and the solution of every linear differential equation with constant coefficients.

Euler's formula

For every real θ,

$$e^{j\theta} = \cos\theta + j\sin\theta.$$

Before proving it, be clear about what is being claimed. For real x, e^x is defined (Ch. 1, Ch. 4) and has the series $e^x = \sum_{n\geq 0} x^n/n!$. For imaginary arguments $e^{j\theta}$ has no meaning until we *give* it one. There are three equivalent ways to do so, and each gives a different insight.

Derivation 1 — power series

Define e^z for any complex z by the same series that works for real arguments,

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!} = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \cdots,$$

which converges for every z (Ch. 4, Ch. 18). Put $z = j\theta$ and use the cycle $j^2 = -1$, $j^3 = -j$, $j^4 = 1$:

$$\begin{aligned} e^{j\theta} &= 1 + j\theta + \frac{(j\theta)^2}{2!} + \frac{(j\theta)^3}{3!} + \frac{(j\theta)^4}{4!} + \frac{(j\theta)^5}{5!} + \cdots \\ &= 1 + j\theta - \frac{\theta^2}{2!} - j\frac{\theta^3}{3!} + \frac{\theta^4}{4!} + j\frac{\theta^5}{5!} - \cdots \\ &= \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots\right) + j\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots\right) \\ &= \cos\theta + j\sin\theta. \end{aligned}$$

The even powers are real and reproduce the Maclaurin series of $\cos\theta$; the odd powers are imaginary and reproduce that of $\sin\theta$. (Rearranging the terms is legitimate because the series converges absolutely.)

Derivation 2 — a differential equation

Any sensible definition of $e^{j\theta}$ should satisfy $\frac{d}{d\theta}e^{j\theta} = je^{j\theta}$ and $e^{j0} = 1$. Let $g(\theta) = \cos\theta + j\sin\theta$. Then

$$g'(\theta) = -\sin\theta + j\cos\theta = j(\cos\theta + j\sin\theta) = j\,g(\theta), \qquad g(0) = 1.$$

So g satisfies the same initial-value problem. To see that the two must coincide, consider $f(\theta) = e^{-j\theta}g(\theta)$. By the product rule $f'(\theta) = -je^{-j\theta}g + e^{-j\theta}jg = 0$, so f is constant, and $f(0) = 1$. Hence $g(\theta) = e^{j\theta}$. (This uniqueness argument is the prototype of the uniqueness theorems for ODEs in Ch. 11.)

Derivation 3 — geometry of motion

Think of $z(\theta) = e^{j\theta}$ as the position of a particle at "time" θ. The equation $z' = jz$ says the velocity is the position rotated by 90° — always perpendicular to the radius — and has the same magnitude as the position. Motion perpendicular to the radius keeps the distance from the origin constant, so the particle moves on the circle $|z| = 1$; with unit speed, after time θ it has travelled arc length θ, i.e. it sits at angle θ radians. That point is $(\cos\theta, \sin\theta)$. This picture explains *why* the angle in $e^{j\theta}$ must be in radians: radians are arc length on the unit circle.

Immediate consequences

Basic properties of $e^{j\theta}$

For real θ, ϕ and integer k:

$$\begin{aligned}
&|e^{j\theta}| = 1, \qquad \angle e^{j\theta} = \theta, \qquad (e^{j\theta})^* = e^{-j\theta} = \frac{1}{e^{j\theta}},\\
&e^{j\theta}e^{j\phi} = e^{j(\theta+\phi)}, \qquad e^{j(\theta+2\pi k)} = e^{j\theta},\\
&e^{j0} = 1, \quad e^{j\pi/2} = j, \quad e^{j\pi} = -1, \quad e^{j3\pi/2} = e^{-j\pi/2} = -j, \quad e^{j2\pi} = 1.
\end{aligned}$$

The modulus follows from $\cos^2\theta + \sin^2\theta = 1$. The addition rule is the polar multiplication theorem of Section 2.3 restated; periodicity follows from that of cosine and sine. The case $e^{j\pi} + 1 = 0$ — linking e, π, j, 1 and 0 — is famous, but the engineer's version is more useful: **multiplying by $e^{j\pi} = -1$ is a 180° phase shift, and multiplying by $e^{j\pi/2} = j$ is a 90° phase advance.**

The inverse Euler formulas

Replace θ by $-\theta$: $e^{-j\theta} = \cos\theta - j\sin\theta$. Adding and subtracting this from Euler's formula isolates the cosine and the sine.

Cosine and sine as exponentials

$$\cos\theta = \frac{e^{j\theta} + e^{-j\theta}}{2} = \mathrm{Re}\{e^{j\theta}\}, \qquad \sin\theta = \frac{e^{j\theta} - e^{-j\theta}}{2j} = \mathrm{Im}\{e^{j\theta}\}.$$

These are the workhorses of signal analysis. They express the two real trigonometric functions as combinations of a **single** kind of function, the complex exponential, which differentiates, integrates and multiplies trivially:

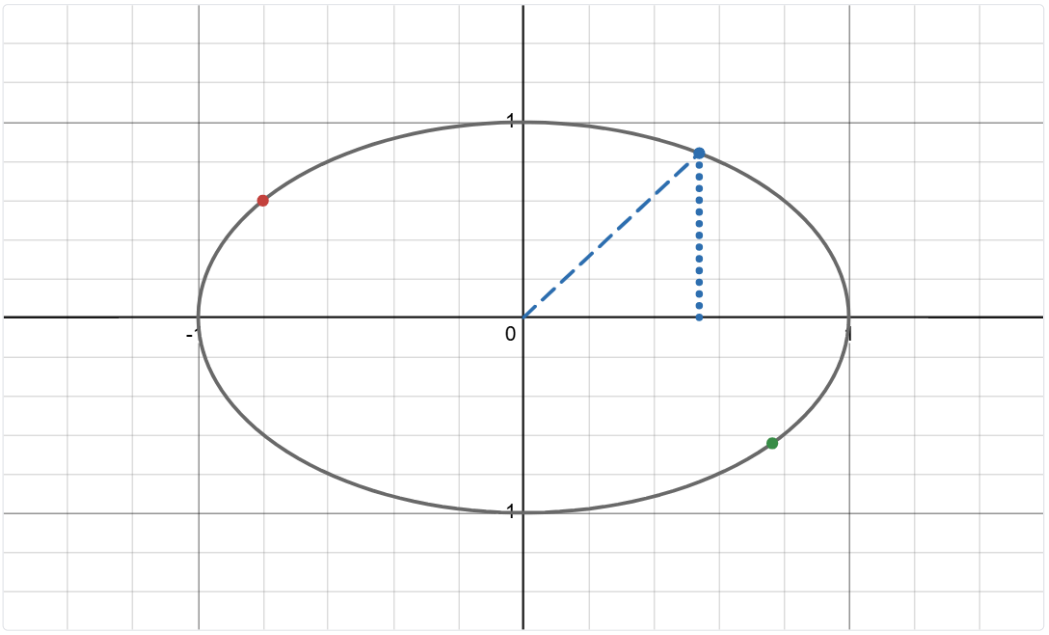
$$\frac{d}{d\theta}e^{ja\theta} = ja\,e^{ja\theta}, \qquad \int e^{ja\theta}\,d\theta = \frac{e^{ja\theta}}{ja} + C\ (a \neq 0), \qquad e^{ja\theta}e^{jb\theta} = e^{j(a+b)\theta}.$$

The general strategy, used again and again in this book, is therefore:

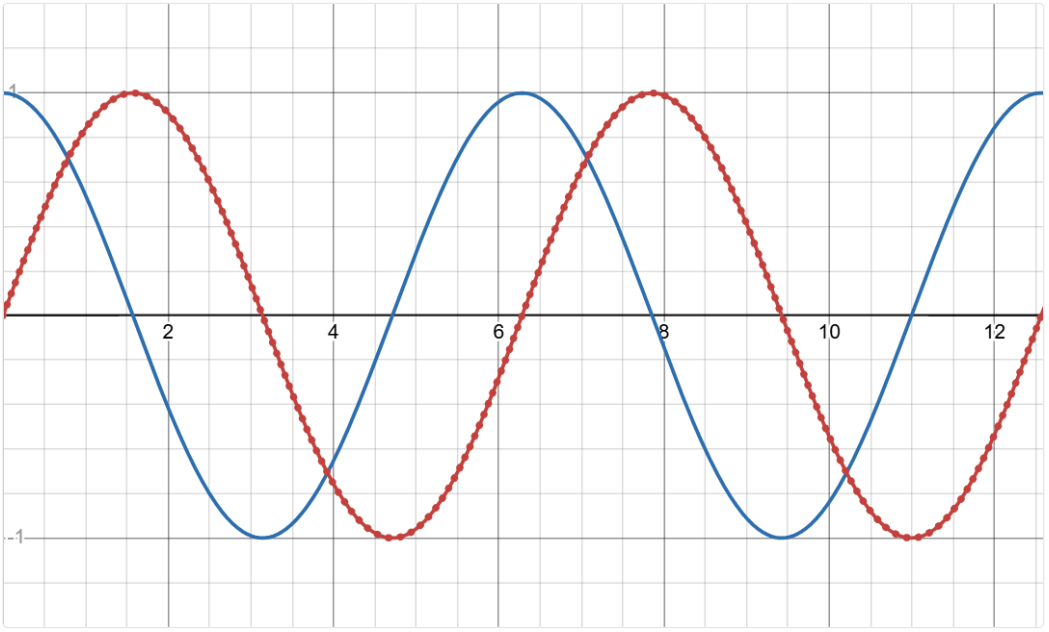
1. Rewrite sines and cosines as exponentials (or as real/imaginary parts of exponentials).
2. Do the algebra or calculus with exponentials.
3. Convert back with Euler's formula, or take a real/imaginary part.

Engineering Note — rotating phasors and negative frequency

As t increases, $e^{j\omega t}$ traces the unit circle anticlockwise at ω rad/s, completing a revolution every $T = 2\pi/\omega$ seconds. Its horizontal projection is $\cos\omega t$ and its vertical projection is $\sin\omega t$. The inverse Euler formula says a real cosine is the average of two **counter-rotating** exponentials, $e^{+j\omega t}$ and $e^{-j\omega t}$, whose imaginary parts cancel. This is the origin of the "negative frequencies" in two-sided spectra (Ch. 15–16): a real signal at 100 Hz has spectral lines at both $+100$ Hz and -100 Hz, each carrying half the amplitude.



Euler's formula places $e^{j\theta}$ on the unit circle at angle θ radians. For $\theta = 1$ the dashed radius and dotted drop line show the real part $\cos 1 = 0.540$ and imaginary part $\sin 1 = 0.841$.



The real part (blue, $\cos\theta$) and imaginary part (red, $\sin\theta$) of $e^{j\theta}$. The sine is the cosine delayed by $\pi/2$ — the dotted curve, $\cos(\theta - \pi/2)$, lies exactly on it — reflecting $\mathrm{Im}\{e^{j\theta}\} = \mathrm{Re}\{e^{j(\theta-\pi/2)}\} = \mathrm{Re}\{-je^{j\theta}\}$.

Trigonometry from exponentials

All the identities of Ch. 1, Section 1.6 can be derived mechanically, which means you never need to memorise more than Euler's formula.

Worked Example 2.12 — Compound-angle formulas

Derive $\cos(A+B)$ and $\sin(A+B)$.

Step 1. $e^{j(A+B)} = e^{jA}e^{jB}$.

Step 2. Expand the right side with Euler's formula:

$$(\cos A + j\sin A)(\cos B + j\sin B) = (\cos A\cos B - \sin A\sin B) + j(\sin A\cos B + \cos A\sin B).$$

Step 3. The left side is $\cos(A+B) + j\sin(A+B)$. Equate real and imaginary parts (one complex equation = two real ones):

$$\cos(A+B) = \cos A\cos B - \sin A\sin B, \qquad \sin(A+B) = \sin A\cos B + \cos A\sin B.$$

Check with $A = 0.4$, $B = 0.9$: $\cos 1.3 = 0.2675$, and $\cos 0.4\cos 0.9 - \sin 0.4\sin 0.9 = 0.9211(0.6216) - 0.3894(0.7833) = 0.5726 - 0.3050 = 0.2675$. ✓

Worked Example 2.13 — Power reduction and the mixer identity

(a) Express $\cos^3\theta$ in terms of cosines of multiple angles (no powers).

$$\cos^3\theta = \left(\frac{e^{j\theta}+e^{-j\theta}}{2}\right)^3 = \frac{1}{8}\left(e^{j3\theta}+3e^{j\theta}+3e^{-j\theta}+e^{-j3\theta}\right)$$
$$= \frac{1}{8}\left(2\cos 3\theta + 6\cos\theta\right) = \frac{3}{4}\cos\theta + \frac{1}{4}\cos 3\theta.$$

Check at $\theta = 0.5$: $\cos^3 0.5 = 0.8776^3 = 0.6759$; $0.75(0.8776) + 0.25\cos 1.5 = 0.6582 + 0.25(0.0707) = 0.6759$. ✓

Engineering reading: passing a pure cosine through a cubic nonlinearity $y = x^3$ produces a fundamental component plus a **third harmonic** of one-third its size — harmonic distortion, computed in one line.

(b) A mixer multiplies a received signal $\cos\omega_1 t$ by a local oscillator $\cos\omega_2 t$. Show what frequencies emerge.

$$\cos\omega_1 t\cos\omega_2 t = \frac{1}{4}\left(e^{j\omega_1 t}+e^{-j\omega_1 t}\right)\left(e^{j\omega_2 t}+e^{-j\omega_2 t}\right) = \frac{1}{2}\cos(\omega_1-\omega_2)t + \frac{1}{2}\cos(\omega_1+\omega_2)t.$$

With $f_1 = 10.7$ MHz and $f_2 = 10.0$ MHz the product contains 0.7 MHz and 20.7 MHz components — the sum and difference frequencies on which every superheterodyne receiver relies.

Worked Example 2.14 — Summing a trigonometric series

Find a closed form for $C_N(\theta) = \sum_{k=0}^{N-1}\cos k\theta$ for θ not a multiple of 2π.

Step 1 — embed in an exponential sum. $C_N = \mathrm{Re}\{S_N\}$ where $S_N = \sum_{k=0}^{N-1} e^{jk\theta}$, a geometric series with ratio $e^{j\theta} \neq 1$:

$$S_N = \frac{1-e^{jN\theta}}{1-e^{j\theta}}.$$

Step 2 — factor out half-angles (a standard trick that turns $1-e^{jx}$ into a sine):

$$1-e^{jx} = e^{jx/2}\left(e^{-jx/2}-e^{jx/2}\right) = -2j\,e^{jx/2}\sin\frac{x}{2}.$$

Applying this to numerator and denominator:

$$S_N = \frac{e^{jN\theta/2}\sin(N\theta/2)}{e^{j\theta/2}\sin(\theta/2)} = e^{j(N-1)\theta/2}\,\frac{\sin(N\theta/2)}{\sin(\theta/2)}.$$

Step 3 — take the real part.

$$\sum_{k=0}^{N-1}\cos k\theta = \cos\left(\frac{(N-1)\theta}{2}\right)\frac{\sin(N\theta/2)}{\sin(\theta/2)}.$$

Check with $N=5$, $\theta = 0.3$: the direct sum is $1+0.9553+0.8253+0.6216+0.3624 = 3.7646$; the formula gives $\cos(0.6)\,\sin(0.75)/\sin(0.15) = 0.8253\times 0.6816/0.1494 = 3.7646$. ✓

The ratio $\sin(N\theta/2)/\sin(\theta/2)$ is the **Dirichlet kernel**; it describes the beam pattern of an N-element antenna array and the frequency response of an N-point moving-average filter.

Worked Example 2.15 — An integral via exponentials

Evaluate $I = \int_0^\infty e^{-2t}\cos 3t\,dt$.

Step 1. $\cos 3t = \mathrm{Re}\{e^{j3t}\}$, and since e^{-2t} is real and integration is linear, $I = \mathrm{Re}\left\{\int_0^\infty e^{(-2+j3)t}dt\right\}$.

Step 2. For $a = -2+j3$, $\int_0^\infty e^{at}dt = \left[\frac{e^{at}}{a}\right]_0^\infty = 0 - \frac{1}{a} = \frac{1}{2-j3}$, because $|e^{at}| = e^{-2t}\to 0$.

Step 3. $\dfrac{1}{2-j3} = \dfrac{2+j3}{13}$, so $I = \dfrac{2}{13} \approx 0.1538$.

As a bonus the imaginary part gives $\int_0^\infty e^{-2t}\sin 3t\,dt = \frac{3}{13}$ for free. Compare with the alternative — integrating by parts twice (Ch. 5). This calculation is, in fact, the Laplace transform of $\cos 3t$ evaluated at $s = 2$ (Ch. 14): $\mathcal{L}\{\cos\omega t\} = s/(s^2+\omega^2) = 2/13$. ✓

Common Pitfall — degrees inside an exponent

e^{j30} means $e^{j\cdot 30\ \mathrm{rad}}$, which is $\cos 30 + j\sin 30 = 0.154 - j0.988$, not $0.866 + j0.5$. The notation $1\angle 30^\circ$ may use degrees; the exponential $e^{j\pi/6}$ must use radians. Calculators set to degree mode will happily give wrong answers to complex-exponential calculations.

Common Pitfall — $\cos\theta \neq e^{j\theta}$

A cosine is the **real part** of $e^{j\theta}$, not equal to it. Replacing $\cos\omega t$ by $e^{j\omega t}$ in a calculation is legitimate only if every operation that follows is **linear with real coefficients** (addition, multiplication by real constants, differentiation, integration), so that "take the real part" can be postponed to the end. Products of two signals do not satisfy this: $\mathrm{Re}\{z_1\}\,\mathrm{Re}\{z_2\} \neq \mathrm{Re}\{z_1 z_2\}$ in general. This is why power calculations need the conjugate formula $\frac{1}{2}\mathrm{Re}\{\mathbf{V}\mathbf{I}^*\}$ rather than $\mathrm{Re}\{\mathbf{V}\mathbf{I}\}$ (Section 2.9).

2.5 Exponential representation

Motivation

Combining the polar form of Section 2.1 with Euler's formula gives the most compact and most powerful way to write a complex number. In exponential form the rules for multiplication, division, powers and roots are simply the laws of exponents you already know, and a complex number can be fed directly into calculus. It is also the form in which signals are written: a damped oscillation, a growing instability and a pure tone are all e^{st} for different complex s.

Exponential form

Every non-zero complex number can be written

$$z = re^{j\theta} = r(\cos\theta + j\sin\theta) = r\angle\theta, \qquad r = |z| > 0,\ \theta = \arg z.$$

The notation $r\angle\theta$ is engineering shorthand for $re^{j\theta}$. Angles written with $\angle$ are often in degrees; inside $e^{j\theta}$ they must be in radians.

Polar arithmetic from the laws of exponents

Exponential-form arithmetic

$$z_1 z_2 = r_1 r_2\, e^{j(\theta_1+\theta_2)}, \quad \frac{z_1}{z_2} = \frac{r_1}{r_2}\, e^{j(\theta_1-\theta_2)},$$
$$z^* = r\, e^{-j\theta}, \qquad \frac{1}{z} = \frac{1}{r}\, e^{-j\theta}, \qquad z^n = r^n e^{jn\theta}\ (n \in \mathbb{Z}).$$

Proof. $z_1 z_2 = r_1 e^{j\theta_1} r_2 e^{j\theta_2} = r_1 r_2 e^{j\theta_1} e^{j\theta_2} = r_1 r_2 e^{j(\theta_1+\theta_2)}$, using $e^{j\theta_1}e^{j\theta_2} = e^{j(\theta_1+\theta_2)}$ from Section 2.4. The others follow similarly. □

Compare this one-line proof with the trigonometric proof in Section 2.3: the exponential notation has absorbed the compound-angle identities.

An important special case: since $|e^{j\alpha}| = 1$, multiplying by $e^{j\alpha}$ changes only the angle. $e^{j\alpha}$ **is a pure rotation (phase shift) by** α, and a general multiplier $Ae^{j\alpha}$ is a scaling by A combined with a rotation by α. Every linear time-invariant system acting on a sinusoid does exactly this (Section 2.9).

Worked Example 2.16 — Products and quotients in exponential form

Evaluate $w = \dfrac{(2e^{j\pi/3})(6e^{-j\pi/6})}{3e^{j\pi/2}}$ and give it in rectangular form.

Magnitude: $|w| = \dfrac{2\cdot 6}{3} = 4.$

Angle: $\angle w = \dfrac{\pi}{3} - \dfrac{\pi}{6} - \dfrac{\pi}{2} = \dfrac{2\pi - \pi - 3\pi}{6} = -\dfrac{\pi}{3}.$

Rectangular: $w = 4e^{-j\pi/3} = 4\left(\frac{1}{2} - j\frac{\sqrt{3}}{2}\right) = 2 - j2\sqrt{3} \approx 2 - j3.464.$

Check by direct rectangular computation: $2e^{j\pi/3} = 1 + j1.732$, $6e^{-j\pi/6} = 5.196 - j3$, their product is $5.196 + 5.196 + j(9-3) = 10.392 + j6$; dividing by $3e^{j\pi/2} = j3$ gives $(10.392 + j6)/(j3) = 2 - j3.464$. ✓

Worked Example 2.17 — A sum in exponential form

Evaluate $w = 4e^{j\pi/6} + 3e^{-j\pi/3}$.

Exponential form has no addition rule, so convert to rectangular:

$$4e^{j\pi/6} = 3.464 + j2.000, \qquad 3e^{-j\pi/3} = 1.500 - j2.598, \qquad w = 4.964 - j0.598.$$

Then $|w| = \sqrt{4.964^2 + 0.598^2} = 5.000$ and $\angle w = \mathrm{atan2}(-0.598, 4.964) = -6.87^\circ$.

Geometric check: the two terms point at $+30^\circ$ and -60°, which are 90° apart, so they form the legs of a right triangle: $|w| = \sqrt{4^2 + 3^2} = 5$ ✓, and w lies $\arctan\frac{3}{4} = 36.87^\circ$ clockwise of the $4e^{j\pi/6}$ term: $30^\circ - 36.87^\circ = -6.87^\circ$. ✓

The complex exponential function

For a general complex exponent $z = \sigma + j\omega$, the law $e^{a+b} = e^a e^b$ (which holds for the series definition) gives

Exponential of a complex number

$$e^{\sigma+j\omega} = e^{\sigma}e^{j\omega} = e^{\sigma}(\cos\omega + j\sin\omega), \qquad |e^{z}| = e^{\mathrm{Re}\{z\}}, \quad \angle e^{z} = \mathrm{Im}\{z\}.$$

Moreover $e^{z} \neq 0$ for every z, and $e^{z+j2\pi k} = e^{z}$ for every integer k: the complex exponential is **periodic with period** $j2\pi$.

So the real part of the exponent controls the **size** and the imaginary part the **angle**. This splitting is the key to the Laplace variable $s = \sigma + j\omega$ (Ch. 14). Replace σ, ω by $\sigma t, \omega t$:

$$e^{st} = e^{\sigma t}e^{j\omega t}, \qquad \mathrm{Re}\{e^{st}\} = e^{\sigma t}\cos\omega t.$$

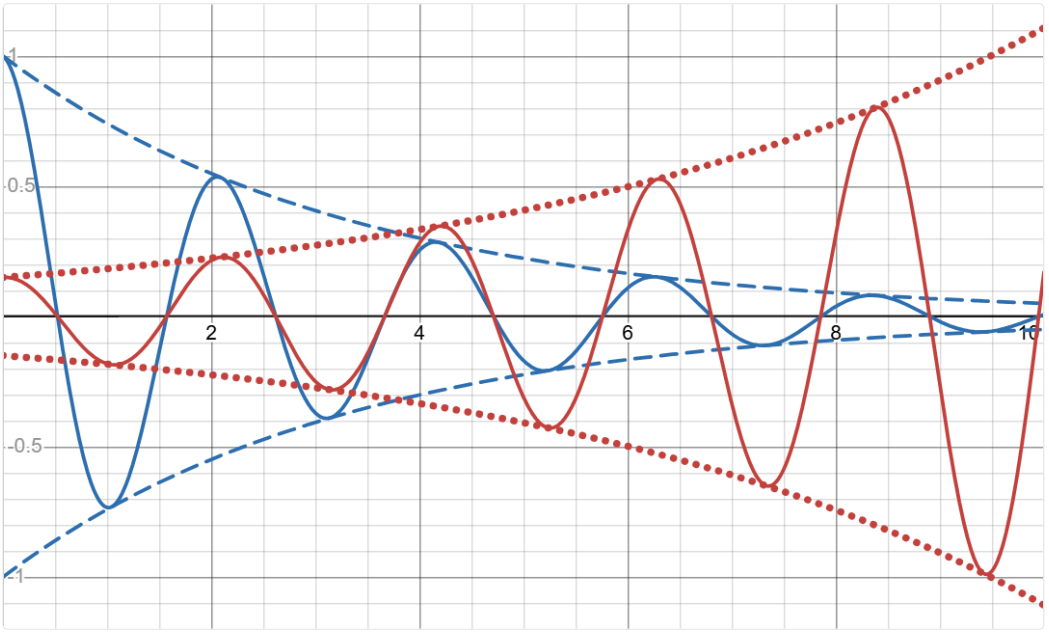
The single expression e^{st} covers every elementary signal shape:

s	$\mathrm{Re}\{e^{st}\}$	**Behaviour**
0	1	constant (DC)
$\sigma < 0$ real	$e^{\sigma t}$	exponential decay
$\sigma > 0$ real	$e^{\sigma t}$	exponential growth
$j\omega$	$\cos\omega t$	steady oscillation
$\sigma + j\omega, \sigma < 0$	$e^{\sigma t}\cos\omega t$	damped oscillation
$\sigma + j\omega, \sigma > 0$	$e^{\sigma t}\cos\omega t$	growing oscillation (instability)

More generally, a damped sinusoid with arbitrary amplitude and phase is

$$A\,e^{\sigma t}\cos(\omega t + \phi) = \mathrm{Re}\left\{Ae^{j\phi}\,e^{(\sigma+j\omega)t}\right\},$$

a complex constant $Ae^{j\phi}$ (which will be called a phasor in Section 2.8) times a complex exponential.



$\mathrm{Re}\{e^{st}\}$ for $s = -0.3 + j3$ (blue, decaying inside its envelope $\pm e^{-0.3t}$) and $0.15\,\mathrm{Re}\{e^{st}\}$ for $s = 0.2 + j3$ (red, growing inside $\pm 0.15e^{0.2t}$). The imaginary part of s sets the oscillation rate; the real part sets the envelope.

Worked Example 2.18 — Evaluating e^{z} and writing a signal as $\mathrm{Re}\{Ae^{st}\}$

(a) Evaluate e^{1+j2}.

$$e^{1+j2} = e^{1}(\cos 2 + j\sin 2) = 2.7183(-0.4161 + j0.9093) = -1.1312 + j2.4717.$$

Its magnitude is $e^{1} = 2.718$ and its angle 2 rad $= 114.6^{\circ}$ — read straight off the exponent.

(b) Write $x(t) = 3e^{-2t}\cos(5t - 40^{\circ})$ in the form $\mathrm{Re}\{Ae^{st}\}$.

Match to $\mathrm{Re}\{Ae^{j\phi}e^{(\sigma+j\omega)t}\}$: $A = 3\angle -40^{\circ} = 2.298 - j1.928$ and $s = -2 + j5\ \mathrm{s}^{-1}$. So

$$x(t) = \mathrm{Re}\left\{(2.298 - j1.928)\,e^{(-2+j5)t}\right\}.$$

Check at $t = 0$: $\mathrm{Re}\{A\} = 2.298$ and $3\cos(-40^{\circ}) = 2.298$. ✓

(c) Solve $e^{z} = -2$. Write $z = x + jy$; then $e^{x}e^{jy} = 2e^{j\pi}$. Equating magnitudes, $e^{x} = 2$, so $x = \ln 2$; equating angles (up to 2π), $y = \pi + 2\pi k$. Hence $z = \ln 2 + j(2k+1)\pi$, $k \in \mathbb{Z}$ — infinitely many solutions, because e^{z} is periodic.

The complex logarithm

Part (c) shows that inverting the exponential is ambiguous. Solving $e^w = z$ for $z = re^{j\theta} \neq 0$ in general gives $w = \ln r + j(\theta + 2\pi k)$.

Complex logarithm

For $z \neq 0$, the **logarithm** is the multivalued expression

$$\log z = \ln|z| + j(\operatorname{Arg} z + 2\pi k), \qquad k \in \mathbb{Z},$$

and the **principal logarithm** is the single value

$$\operatorname{Log} z = \ln|z| + j \operatorname{Arg} z, \qquad -\pi < \operatorname{Arg} z \leq \pi.$$

The real part of a logarithm is the log of the magnitude; the imaginary part is the angle. In engineering this is exactly the content of a Bode plot: $\ln H = \ln|H| + j\angle H$, so plotting $\ln|H|$ (or $20\log_{10}|H|$, in dB) and $\angle H$ is plotting the real and imaginary parts of $\ln H$, and the gains and phases of cascaded stages add because logarithms turn products into sums.

Complex powers

For $z \neq 0$ and complex c, define $z^c = e^{c \log z}$. This is multivalued in general (single-valued when c is an integer, n-valued when $c = 1/n$ — the n-th roots of Section 2.6). The **principal value** uses $\operatorname{Log} z$.

Worked Example 2.19 — Logarithms and complex powers

(a) $\operatorname{Log}(-3 + j4)$. Here $|z| = 5$ and $\operatorname{Arg} z = 126.87^\circ = 2.2143$ rad, so

$$\operatorname{Log}(-3 + j4) = \ln 5 + j2.2143 = 1.6094 + j2.2143.$$

Check: $e^{1.6094} e^{j2.2143} = 5(\cos 2.2143 + j \sin 2.2143) = 5(-0.6 + j0.8) = -3 + j4$. ✓

(b) j^j. Since $\log j = j(\pi/2 + 2\pi k)$,

$$j^j = e^{j \log j} = e^{j \cdot j(\pi/2 + 2\pi k)} = e^{-\pi/2 - 2\pi k}.$$

Every value is **real**; the principal value ($k = 0$) is $e^{-\pi/2} = 0.2079$.

(c) Principal value of $(1 + j)^j$. $\operatorname{Log}(1 + j) = \ln\sqrt{2} + j\pi/4 = 0.3466 + j0.7854$. Then

$$(1 + j)^j = e^{j(0.3466 + j0.7854)} = e^{-0.7854} e^{j0.3466} = 0.4559(0.9405 + j0.3397) = 0.4288 + j0.1549.$$

Common Pitfall — logarithm laws with complex numbers

The rule $\operatorname{Log}(z_1 z_2) = \operatorname{Log} z_1 + \operatorname{Log} z_2$ can fail for principal values, because the sum of two principal arguments may fall outside $(-\pi, \pi]$. Example: $z_1 = z_2 = -1$ gives $\operatorname{Log}(1) = 0$ but $\operatorname{Log}(-1) + \operatorname{Log}(-1) = j2\pi$. The identity holds **modulo** $j2\pi$. In phase calculations this appears as the familiar "phase wrapping": a phase computed by adding contributions may differ from the instrument's principal value by a multiple of 360°, and both are correct.

Hyperbolic functions and complex arguments

Replacing θ by jx in the inverse Euler formulas links trigonometric and hyperbolic functions:

$$\cos(jx) = \frac{e^{-x} + e^{x}}{2} = \cosh x, \qquad \sin(jx) = \frac{e^{-x} - e^{x}}{2j} = j \sinh x.$$

Combined with the compound-angle formula this gives the real and imaginary parts of the sine and cosine of any complex number:

$$\sin(x + jy) = \sin x \cosh y + j \cos x \sinh y, \qquad \cos(x + jy) = \cos x \cosh y - j \sin x \sinh y.$$

For example $\sin(\pi/2 + j1) = \cosh 1 = 1.5431$ — a sine can exceed 1 when its argument is complex. These formulas appear in transmission-line theory, where the voltage along a lossy line involves $\cosh(\gamma\ell)$ and $\sinh(\gamma\ell)$ with a complex propagation constant $\gamma = \alpha + j\beta$, and in waveguides below cut-off, where a "sinusoidal" field becomes an exponentially decaying one.

Engineering Note — the exponential form in software

Numerical libraries provide `exp`, `log`, `abs` and `angle` / `arg` for complex arguments, and they return **principal** values. When a quantity is built up by multiplying many complex factors (a cascade of filter stages, a long chain of transmission-line sections), it is often more robust to accumulate $\ln|H|$ and $\angle H$ separately — adding logarithms of magnitudes and adding (unwrapped) phases — than to multiply complex numbers directly, which can overflow or lose track of full turns.

2.6 Roots and powers of complex numbers

Motivation

Raising a complex number to a high power by repeated rectangular multiplication is tedious and error-prone; in exponential form it is a single step. The inverse problem — finding all solutions of $w^n = z$ — is even more important in engineering, because the poles of standard filter families, the frequencies of a discrete Fourier transform, the phases of a polyphase power system and the symbols of a phase-shift-keyed modulator are all n-th roots of some complex number. The central fact is that a non-zero complex number has exactly n distinct n-th roots, arranged symmetrically on a circle.

Integer powers: De Moivre's theorem

De Moivre's theorem

For every integer n (positive, zero or negative) and real θ,

$$(\cos\theta + j\sin\theta)^n = \cos n\theta + j\sin n\theta, \qquad \text{equivalently} \qquad (re^{j\theta})^n = r^n e^{jn\theta}.$$

Proof. For $n \geq 0$, by induction: the case $n = 0$ is $1 = 1$. If true for n, then

$$(\cos\theta + j\sin\theta)^{n+1} = (\cos n\theta + j\sin n\theta)(\cos\theta + j\sin\theta) = \cos(n+1)\theta + j\sin(n+1)\theta$$

by the polar multiplication rule (angles add). For negative $n = -m$, $(\cos\theta + j\sin\theta)^{-m} = 1/(\cos m\theta + j\sin m\theta) = \cos m\theta - j\sin m\theta = \cos(-m\theta) + j\sin(-m\theta)$, using $1/e^{j\phi} = e^{-j\phi}$. With exponentials the whole theorem is just $(e^{j\theta})^n = e^{jn\theta}$. $\square$

Recipe for z^n: convert to polar, raise the magnitude to the n-th power, multiply the angle by n, reduce the angle to the principal range if desired, convert back.

Worked Example 2.20 — High powers

(a) $(1+j)^8$. Here $1 + j = \sqrt{2}\,e^{j\pi/4}$, so $(1+j)^8 = (\sqrt{2})^8 e^{j2\pi} = 16$.

Check: $(1+j)^2 = 2j$, so $(1+j)^8 = (2j)^4 = 16j^4 = 16$. ✓

(b) $(1 + j\sqrt{3})^{10}$. Here $1 + j\sqrt{3} = 2e^{j\pi/3}$, so

$$(1+j\sqrt{3})^{10} = 2^{10}e^{j10\pi/3} = 1024\,e^{j(10\pi/3 - 4\pi)} = 1024\,e^{-j2\pi/3} = 1024\left(-\tfrac{1}{2} - j\tfrac{\sqrt{3}}{2}\right) = -512 - j886.8.$$

Expanding by the binomial theorem would have required eleven terms.

(c) A negative power: $(1 - j\sqrt{3})^{-4}$. With $1 - j\sqrt{3} = 2e^{-j\pi/3}$,

$$(1-j\sqrt{3})^{-4} = 2^{-4}e^{j4\pi/3} = \tfrac{1}{16}\left(-\tfrac{1}{2} - j\tfrac{\sqrt{3}}{2}\right) = -0.03125 - j0.05413.$$

Multiple-angle formulas

Expanding the left side of De Moivre's theorem with the binomial theorem and equating real and imaginary parts produces formulas for $\cos n\theta$ and $\sin n\theta$ as polynomials in $\cos\theta$ and $\sin\theta$ — the reverse of the power-reduction of Example 2.13.

Worked Example 2.21 — Triple-angle formulas

Write $c = \cos\theta$, $s = \sin\theta$. Then

$$(c + js)^3 = c^3 + 3c^2(js) + 3c(js)^2 + (js)^3 = (c^3 - 3cs^2) + j(3c^2 s - s^3).$$

By De Moivre this equals $\cos 3\theta + j\sin 3\theta$. Equate parts and use $s^2 = 1 - c^2$, $c^2 = 1 - s^2$:

$$\cos 3\theta = c^3 - 3c(1-c^2) = 4\cos^3\theta - 3\cos\theta, \qquad \sin 3\theta = 3(1-s^2)s - s^3 = 3\sin\theta - 4\sin^3\theta.$$

Check at $\theta = 0.7$: $4\cos^3 0.7 - 3\cos 0.7 = -0.5048$ and $\cos 2.1 = -0.5048$. ✓

The polynomial $T_3(x) = 4x^3 - 3x$ with $\cos 3\theta = T_3(\cos\theta)$ is a **Chebyshev polynomial**; the family T_n defined by $\cos n\theta = T_n(\cos\theta)$ is the basis of Chebyshev filter design.

n-th roots

n-th roots of a complex number

Let $z = re^{j\theta} \neq 0$ and $n \geq 1$ an integer. The equation $w^n = z$ has exactly n distinct solutions

$$w_k = r^{1/n}\exp\left(j\,\frac{\theta + 2\pi k}{n}\right), \qquad k = 0, 1, \ldots, n-1,$$

where $r^{1/n}$ is the ordinary positive real root. They lie on the circle $|w| = r^{1/n}$, equally spaced at angular intervals of $2\pi/n$, forming the vertices of a regular n-gon.

Proof. Write the unknown as $w = \rho e^{j\phi}$. Then $w^n = \rho^n e^{jn\phi}$, and $w^n = z$ requires equal magnitudes and angles equal up to multiples of 2π:

$$\rho^n = r \;\Rightarrow\; \rho = r^{1/n}; \qquad n\phi = \theta + 2\pi k \;\Rightarrow\; \phi = \frac{\theta + 2\pi k}{n}, \quad k \in \mathbb{Z}.$$

Values of k that differ by a multiple of n give angles differing by a multiple of 2π — the same point. So exactly the n values $k = 0, \dots, n-1$ give distinct roots. □

The crucial step is keeping the "$+2\pi k$" **before** dividing by n. Adding 2π to θ does not change z, but adding $2\pi/n$ to the root's angle does change the root — that is where the extra roots come from. The root with $k = 0$ and principal θ is the **principal root**.

Recipe for the n-th roots of z: (1) write $z = re^{j\theta}$; (2) the first root is $r^{1/n}e^{j\theta/n}$; (3) the others follow by repeatedly rotating by $2\pi/n$ (i.e. multiplying by $e^{j2\pi/n}$).

Worked Example 2.22 — Cube roots of $8j$

Solve $w^3 = 8j$.

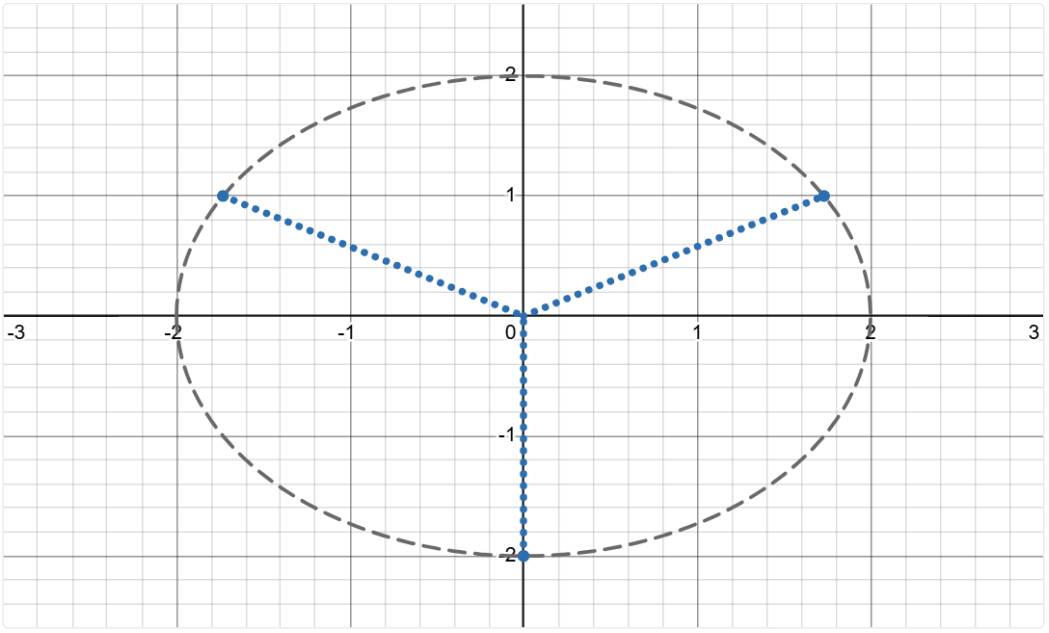
Step 1. $8j = 8e^{j\pi/2}$, so $r^{1/3} = 2$ and the angles are $\frac{\pi/2 + 2\pi k}{3} = \frac{\pi}{6} + \frac{2\pi k}{3}$.

Step 2.

$$w_0 = 2e^{j\pi/6} = \sqrt{3} + j, \qquad w_1 = 2e^{j5\pi/6} = -\sqrt{3} + j, \qquad w_2 = 2e^{j3\pi/2} = -2j.$$

Check w_2: $(-2j)^3 = -8j^3 = -8(-j) = 8j$. ✓ And w_0: $(\sqrt{3}+j)^2 = 3 + 2\sqrt{3}j - 1 = 2 + j2\sqrt{3}$; times $(\sqrt{3}+j)$ gives $2\sqrt{3} + j2 + j6 - 2\sqrt{3} = 8j$. ✓

Structural check: the roots of $w^3 - 8j = 0$ sum to zero (the coefficient of w^2 is zero): $\sqrt{3} + j - \sqrt{3} + j - 2j = 0$. ✓



The three cube roots of $8j$ lie on the circle of radius 2, 120° apart, starting at 30° (one-third of the angle 90° of $8j$).

Square roots in rectangular form

For square roots it is sometimes quicker to avoid polar form. Seek $w = x + jy$ with $w^2 = a + jb$: equating parts of $x^2 - y^2 + j2xy = a + jb$ gives $x^2 - y^2 = a$, $2xy = b$, and taking magnitudes gives $x^2 + y^2 = r = \sqrt{a^2 + b^2}$. Adding and subtracting:

Square root of $a + jb$

$$\sqrt{a + jb} = \pm\left(\sqrt{\frac{r + a}{2}} + j\,\operatorname{sgn}(b)\sqrt{\frac{r - a}{2}}\right), \qquad r = \sqrt{a^2 + b^2},$$

with $\operatorname{sgn}(b) = 1$ if $b \geq 0$ and -1 if $b < 0$ (the sign makes $2xy$ have the sign of b).

Worked Example 2.23 — A quadratic with complex coefficients

Solve $z^2 - (3+j)z + (4+j3) = 0$.

Step 1 — discriminant. The quadratic formula is valid in $\mathbb{C}$ (it is pure field algebra):

$$D = (3+j)^2 - 4(4+j3) = (9+j6-1) - 16 - j12 = -8 - j6.$$

Step 2 — square root of D. $a=-8, b=-6, r=10$: $\sqrt{(10-8)/2}=1$, $\sqrt{(10+8)/2}=3$, and $b<0$, so $\sqrt{D} = \pm(1-j3)$. *Check:* $(1-j3)^2 = 1 - j6 - 9 = -8-j6$. ✓

Step 3 — roots.

$$z = \frac{(3+j) \pm (1-j3)}{2}: \qquad z_1 = \frac{4-j2}{2} = 2-j, \qquad z_2 = \frac{2+j4}{2} = 1+j2.$$

Check with Vieta: $z_1+z_2 = 3+j$ ✓ and $z_1 z_2 = (2-j)(1+j2) = 2+j4-j+2 = 4+j3$. ✓

Note that the roots are **not** conjugates of each other: the conjugate root theorem needs real coefficients.

Roots of unity

Roots of unity

The n solutions of $w^n = 1$ are the **n-th roots of unity**

$$W^k = e^{j2\pi k/n}, \qquad k = 0, 1, \ldots, n-1, \qquad W = e^{j2\pi/n}.$$

W is the **primitive** n-th root of unity; the others are its powers.

The n-th roots of any z are the principal root multiplied by each root of unity: $w_k = w_0 W^k$. Roots of unity therefore carry all the "rotational" structure of root-finding.

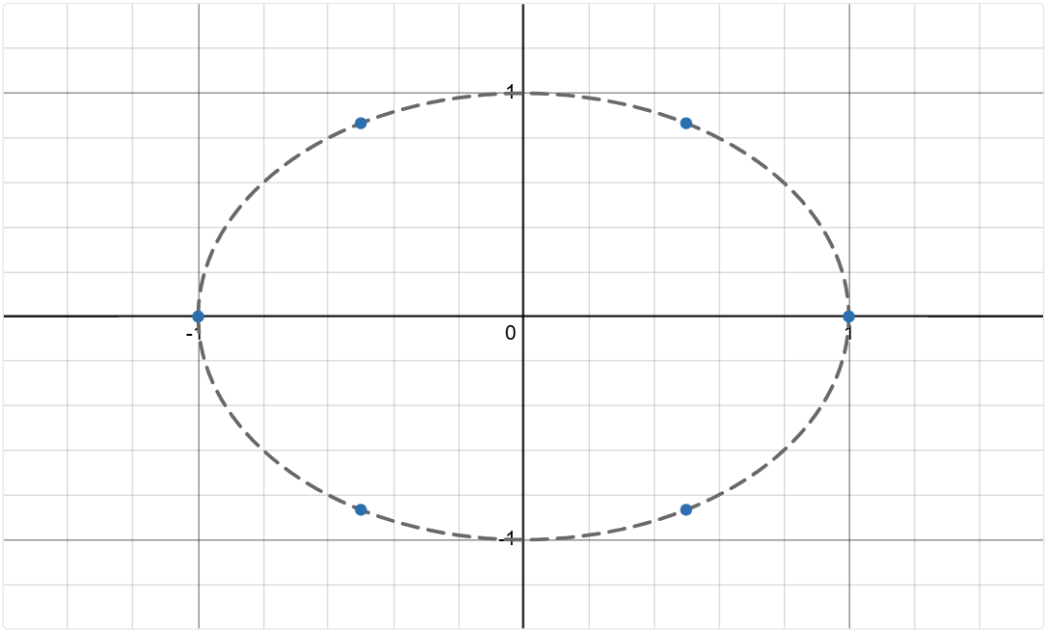
Sums of roots of unity

For $n \geq 2$,

$$\sum_{k=0}^{n-1} W^k = 0, \qquad \text{and more generally} \qquad \sum_{k=0}^{n-1} W^{mk} = \begin{cases} n, & m \equiv 0 \pmod{n}, \\ 0, & \text{otherwise.} \end{cases}$$

Proof. If m is a multiple of n, every term is $W^{mk} = e^{j2\pi mk/n} = 1$ and the sum is n. Otherwise $W^m \neq 1$ and the geometric series gives $\dfrac{1-W^{mn}}{1-W^m} = \dfrac{1-1}{1-W^m} = 0$, since $W^{mn} = (W^n)^m = 1$. □

Geometrically: the roots are the vertices of a regular polygon centred at the origin, and arrows from the centre to the vertices of a regular polygon balance out. The second form is the **orthogonality relation** of the discrete Fourier transform (Ch. 16, Ch. 24), where $W_N = e^{-j2\pi/N}$ is called the twiddle factor; it is what allows the DFT to be inverted.



The six sixth roots of unity, $e^{j2\pi k/6}$, form a regular hexagon on the unit circle; their sum is zero.

Engineering Note — three-phase power and PSK

The cube roots of unity 1, $a = e^{j2\pi/3}$, $a^2 = e^{-j2\pi/3}$ describe the 120° spacing of a balanced three-phase supply. Because $1 + a + a^2 = 0$, the three phase voltages (or balanced load currents) sum to zero at every instant, so the neutral conductor of a balanced system carries no current. Power engineers use the operator "a" throughout symmetrical-component analysis of faults. In digital communications, M-ary phase-shift keying transmits one of the M-th roots of unity (scaled) per symbol: QPSK uses $e^{j(\pi/4 + k\pi/2)}$, the four fourth roots of -1.

Factoring polynomials with roots

Worked Example 2.24 — Factoring $z^4 + 1$ over the reals

Step 1 — roots. $z^4 = -1 = e^{j\pi}$ gives $z_k = e^{j(\pi + 2\pi k)/4}$, i.e. $e^{\pm j\pi/4}$ and $e^{\pm j3\pi/4}$, or $\frac{\pm 1 \pm j}{\sqrt{2}}$.

Step 2 — pair conjugates. For a pair $e^{\pm j\alpha}$ the real quadratic factor is $z^2 - 2\cos\alpha\, z + 1$:

$$\alpha = \tfrac{\pi}{4}: \; z^2 - \sqrt{2}z + 1, \qquad \alpha = \tfrac{3\pi}{4}: \; z^2 + \sqrt{2}z + 1.$$

Step 3. $z^4 + 1 = (z^2 - \sqrt{2}z + 1)(z^2 + \sqrt{2}z + 1)$. *Check:* the product is $(z^2+1)^2 - 2z^2 = z^4 + 2z^2 + 1 - 2z^2 = z^4 + 1$. ✓

Such factorisations are needed when integrating rational functions by partial fractions (Ch. 5) and when building filters from second-order sections.

Worked Example 2.25 — Butterworth filter poles

An n-th order Butterworth low-pass filter with cut-off 1 rad/s has $|H(j\omega)|^2 = \dfrac{1}{1 + \omega^{2n}}$. Its poles are the left-half-plane roots of $1 + (-s^2)^n = 0$, obtained by substituting $\omega^2 = -s^2$ (since $s = j\omega$). Find the third-order Butterworth polynomial.

Step 1. For $n = 3$: $1 + (-s^2)^3 = 1 - s^6 = 0$, so $s^6 = 1$: the sixth roots of unity, $s_k = e^{jk\pi/3}$.

Step 2 — keep the stable ones. The roots with $\mathrm{Re}\{s\} < 0$ are $k = 2, 3, 4$: $e^{j2\pi/3} = -\frac{1}{2} + j\frac{\sqrt{3}}{2}$, -1, and $e^{j4\pi/3} = -\frac{1}{2} - j\frac{\sqrt{3}}{2}$. (The right-half-plane roots belong to $H(-s)$, the mirror factor of $|H|^2 = H(s)H(-s)$.)

Step 3 — multiply out. The conjugate pair gives $s^2 + s + 1$ (since $-2\cos(2\pi/3) = 1$), and

$$B_3(s) = (s+1)(s^2 + s + 1) = s^3 + 2s^2 + 2s + 1, \qquad H(s) = \frac{1}{s^3 + 2s^2 + 2s + 1}.$$

Check at $\omega = 1$: $B_3(j) = -j - 2 + 2j + 1 = -1 + j$, so $|H(j1)|^2 = 1/|-1+j|^2 = 1/2$, matching $1/(1+1^6)$. ✓ The poles lie on the unit circle at 120°, 180° and 240° — evenly spaced by $\pi/n = 60^\circ$, the defining geometry of the Butterworth family.

Common Pitfall — keeping only the principal root

A calculator's $\sqrt[n]{z}$, or `z**(1/n)` in software, returns only the principal root ($k = 0$). Equations such as $w^4 = -16$ have four roots ($\sqrt{2}(\pm 1 \pm j)$); forgetting the other three loses poles of filters, branches of a root locus, or valid solutions of a characteristic equation. Always write "$+2\pi k$" before dividing the angle by n.

Common Pitfall — $(z^{1/n})^n$ versus $(z^n)^{1/n}$

$(z^{1/n})^n = z$ for every root, but $(z^n)^{1/n}$ need not return z: $\left((-1)^2\right)^{1/2} = 1^{1/2}$ has principal value 1, not -1. Fractional powers do not commute with integer powers once principal values are used, for the same reason the logarithm laws fail (Section 2.5).

2.7 Complex plane and geometric interpretation

Motivation

Many engineering judgements are made by looking at the complex plane rather than by computing: whether a system is stable (where are its poles?), how much phase margin a feedback loop has (how close does the Nyquist curve pass to -1?), whether an antenna is matched (where does its reflection coefficient sit on the Smith chart?), how noisy a received constellation is (how spread are the points?). Each of these is a geometric statement about complex numbers. This section collects the dictionary between algebra and geometry so that such pictures can be read fluently.

Complex numbers as vectors

Complex (Argand) plane

The **complex plane** identifies $z = x + jy$ with the point (x, y), or with the vector from the origin to that point. The horizontal axis is the **real axis**, the vertical axis the **imaginary axis**. The magnitude $|z|$ is the length of the vector and $\arg z$ its direction.

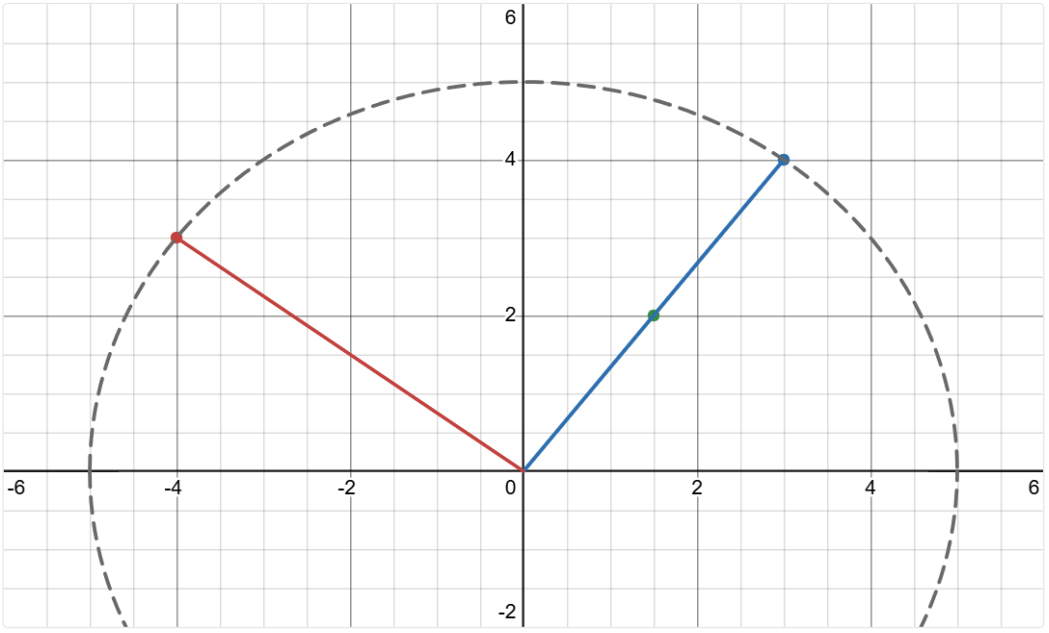
Under this identification addition, subtraction and real scaling of complex numbers are exactly vector addition, subtraction and scaling in $\mathbb{R}^2$. What the plane adds to $\mathbb{R}^2$ is **multiplication**, which has no counterpart for ordinary 2-D vectors and which is a rotation-plus-scaling.

Geometric meaning of the operations

Operation	Geometric effect
$z + w$	translation of z by the vector w (parallelogram rule)
$z - w$	vector from w to z
$\lvert z - w \rvert$	distance between the points z and w
$\arg(z - w)$	direction of the segment from w to z
kz, $k > 0$ real	scaling by k about the origin
$-z$	rotation by 180° (point reflection in the origin)
jz	rotation by $+90^\circ$ (anticlockwise)
$e^{j\alpha}z$	rotation by α about the origin
$(re^{j\alpha})z$	scaling by r combined with rotation by α (a "spiral similarity")
z^*	reflection in the real axis
$1/z$	$\frac{1}{r}e^{-j\theta}$: inversion in the unit circle ($r \to 1/r$) followed by reflection in the real axis
$w_0 + e^{j\alpha}(z - w_0)$	rotation by α about the point w_0

The last entry is built from simpler ones: translate w_0 to the origin, rotate, translate back. Composite geometric transformations are obtained by composing complex functions.

Why multiplication by j is a quarter-turn, without polar form: $j(x + jy) = -y + jx$, and the map $(x, y) \mapsto (-y, x)$ is the standard 90° rotation matrix $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ of linear algebra (Ch. 9). More generally, multiplication by $a + jb$ is the linear map with matrix $\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$, which is r times a rotation matrix.



Multiplying by j rotates $z = 3 + j4$ by 90° to $-4 + j3$; both lie on the circle $|z| = 5$. Multiplying by the real number 0.5 only shrinks it.

Worked Example 2.26 — Completing an equilateral triangle

Two vertices of an equilateral triangle are 0 and $4 + j2$. Find the third vertex.

Idea. The third vertex is obtained by rotating the side vector $4 + j2$ by $\pm 60^\circ$ about 0.

Step 1. $e^{j\pi/3} = \frac{1}{2} + j\frac{\sqrt{3}}{2} = 0.5 + j0.8660$.

Step 2. $(4 + j2)(0.5 + j0.866) = 2 + j3.464 + j1 - 1.732 = 0.268 + j4.464$.

Step 3. The other orientation: $(4 + j2)(0.5 - j0.866) = 2 - j3.464 + j1 + 1.732 = 3.732 - j2.464$.

Check the first solution: its distance from 0 is $\sqrt{0.268^2 + 4.464^2} = 4.472 = \sqrt{20} = |4 + j2|$, and its distance from $4 + j2$ is $|-3.732 + j2.464| = \sqrt{13.93 + 6.07} = \sqrt{20}$. ✓ All three sides equal.

The same idea computes the line-to-line voltages of a three-phase system (Section 2.8), whose phasors form an equilateral triangle.

Distances, midpoints and centroids

Distances between points are magnitudes of differences, $|z_1 - z_2|$. The midpoint of the segment from z_1 to z_2 is $\frac{1}{2}(z_1 + z_2)$, and the point dividing it in ratio $t : (1-t)$ is $(1-t)z_1 + tz_2$, $0 \le t \le 1$ — the **parametric equation of a segment**. The centroid of points $z_1, \dots, z_n$ is their average $\frac{1}{n}\sum z_k$; for instance the centroid of the triangle $1 + j2$, $5 - j$, $3 + j4$ is $\frac{1}{3}(9 + j5) = 3 + j1.667$. In a received constellation, the centroid of many noisy samples estimates the transmitted symbol.

Curves and regions

Simple equations and inequalities in z describe the curves and regions that recur in engineering. Here $z = x + jy$.

Standard loci

Equation	Locus
$\lvert z - z_0\rvert = R$	circle, centre z_0, radius R
$\lvert z - z_0\rvert < R$	open disc (interior of the circle)
$\lvert z - z_1\rvert = \lvert z - z_2\rvert$	perpendicular bisector of the segment $z_1 z_2$
$\lvert z - z_1\rvert = k\lvert z - z_2\rvert$, $k > 0$, $k \neq 1$	a circle (circle of Apollonius)
$\arg(z - z_0) = \alpha$	ray from z_0 at angle α (excluding z_0)
$\mathrm{Re}\{z\} = c$, $\mathrm{Im}\{z\} = c$	vertical / horizontal line
$\mathrm{Re}\{z\} < 0$	open left half-plane
$\alpha < \arg z < \beta$	open sector (wedge)
$z = z_0 + Re^{jt}$, $0 \le t < 2\pi$	parametric form of the circle $\lvert z - z_0\rvert = R$

To identify an unfamiliar locus, the reliable method is to substitute $z = x + jy$, square any magnitudes (using $|w|^2 = ww^*$ or simply $|a + jb|^2 = a^2 + b^2$), and recognise the resulting real equation in x and y.

Worked Example 2.27 — A half-plane

Describe $\{z : |z - 1| \le |z + 1|\}$.

Geometric reading. Points no farther from 1 than from -1. The boundary $|z - 1| = |z + 1|$ is the set of points equidistant from ± 1: the perpendicular bisector, which is the imaginary axis. Testing $z = 2$: $|1| \le |3|$ holds, so the region is the side containing 2.

Algebraic confirmation. Both sides are non-negative, so square: $(x-1)^2 + y^2 \le (x+1)^2 + y^2 \iff -2x \le 2x \iff x \ge 0$.

The region is the closed right half-plane $\mathrm{Re}\{z\} \ge 0$. ✓

Worked Example 2.28 — A circle of Apollonius

Identify the locus $|z - j| = 2|z + 1|$.

Step 1 — substitute and square.

$$x^2 + (y-1)^2 = 4\left[(x+1)^2 + y^2\right].$$

Step 2 — expand and collect. $x^2 + y^2 - 2y + 1 = 4x^2 + 8x + 4 + 4y^2$, so $3x^2 + 3y^2 + 8x + 2y + 3 = 0$, i.e.

$$x^2 + y^2 + \tfrac{8}{3}x + \tfrac{2}{3}y + 1 = 0.$$

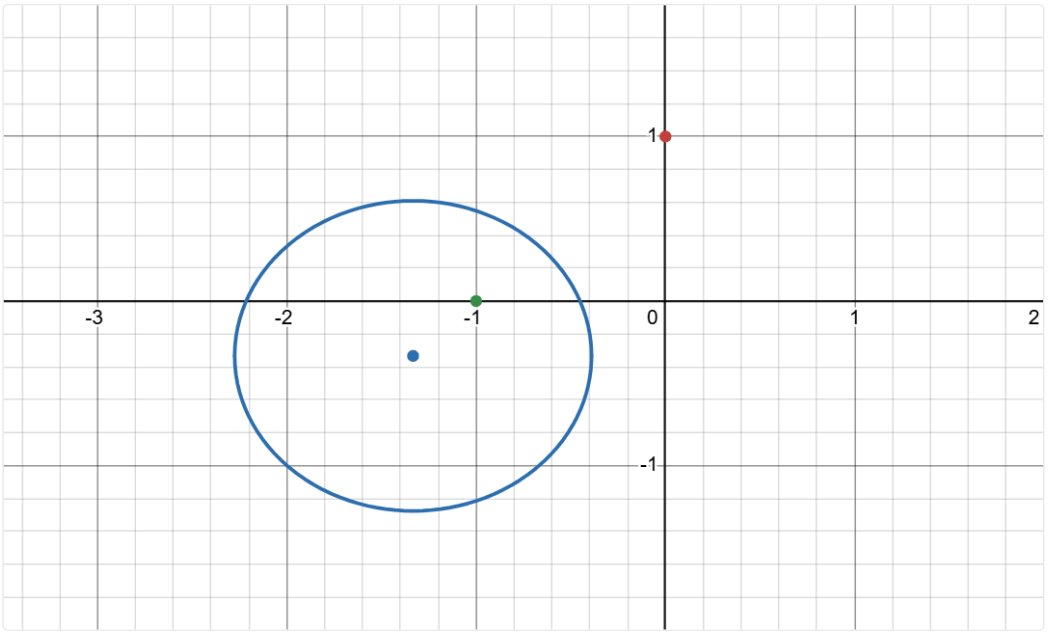
Step 3 — complete the square.

$$\left(x + \tfrac{4}{3}\right)^2 + \left(y + \tfrac{1}{3}\right)^2 = \tfrac{16}{9} + \tfrac{1}{9} - 1 = \tfrac{8}{9}.$$

A circle with centre $-\frac{4}{3} - j\frac{1}{3}$ and radius $\sqrt{8}/3 \approx 0.943$.

Check a point: the rightmost point of the circle is $z = -\frac{4}{3} + 0.943 - j\frac{1}{3} = -0.391 - j0.333$. Then $|z - j| = |-0.391 - j1.333| = 1.389$ and $|z + 1| = |0.609 - j0.333| = 0.694$; ratio 2.00. ✓

The circle encloses -1 (the point the ratio favours) but not j. Circles of this kind are the constant-gain contours of a single-pole, single-zero system: $|z - j|/|z + 1|$ is the magnitude of $(z - j)/(z + 1)$.



The locus $|z - j| = 2|z + 1|$ is a circle around -1; every point on it is twice as far from j as from -1.

Mappings: what $1/z$ does to lines

The map $w = 1/z$ is a good first example of a complex function viewed as a transformation of the plane. Its key property — lines and circles map to lines and circles — is the basis of the Smith chart.

Worked Example 2.29 — Image of a line under inversion

Show that $w = 1/z$ maps the vertical line $\mathrm{Re}\{z\} = 1$ onto the circle $|w - \frac{1}{2}| = \frac{1}{2}$ (minus the origin).

Step 1 — parametrise the line. $z = 1 + jy$, $y \in \mathbb{R}$.

Step 2 — compute w.

$$w = \frac{1}{1+jy} = \frac{1-jy}{1+y^2}, \qquad u = \mathrm{Re}\{w\} = \frac{1}{1+y^2}, \quad v = \mathrm{Im}\{w\} = \frac{-y}{1+y^2}.$$

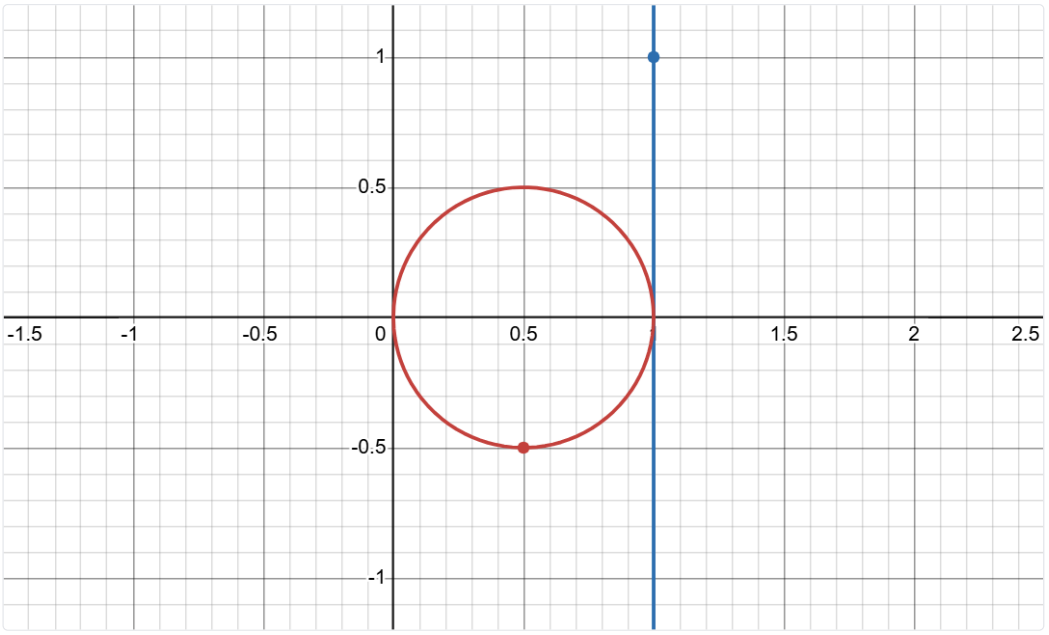
Step 3 — eliminate y. $u^2 + v^2 = \dfrac{1+y^2}{(1+y^2)^2} = \dfrac{1}{1+y^2} = u$, so $u^2 - u + v^2 = 0$, i.e.

$$\left(u - \tfrac{1}{2}\right)^2 + v^2 = \tfrac{1}{4}.$$

As $y \to \pm\infty$, $w \to 0$, which is approached but never reached.

Check: $y = 1$ gives $w = \frac{1-j}{2}$, and $|w - \frac{1}{2}| = |-j/2| = \frac{1}{2}$. ✓

Engineering reading: the line $\mathrm{Re}\{Z\} = 1\ \Omega$ is "all impedances with 1 Ω resistance". Its image under $Y = 1/Z$ shows that the corresponding admittances fill a circle in the admittance plane — exactly the constant-resistance circles of impedance/admittance charts.



Inversion $w = 1/z$ maps the vertical line $\mathrm{Re}\{z\} = 1$ (blue) onto the circle of diameter 1 through the origin (red). The point $1 + j$ maps to $\frac{1}{2} - j\frac{1}{2}$: inversion also reflects in the real axis.

Engineering Note — the Smith chart and stability regions

Smith chart. The reflection coefficient of a load Z_L on a line of characteristic impedance Z_0 is $\Gamma = \dfrac{z-1}{z+1}$ with $z = Z_L/Z_0$. This map (a **bilinear** or Möbius transformation, studied in Ch. 18) sends the entire right half-plane $\mathrm{Re}\{z\} \geq 0$ — every passive load — into the unit disc $|\Gamma| \leq 1$. A matched load $z = 1$ lands at the centre $\Gamma = 0$; any pure reactance $z = jx$ lands on the rim $|\Gamma| = 1$ (total reflection). For example $z = 1 + j$ gives $\Gamma = \dfrac{j}{2+j} = 0.2 + j0.4$, $|\Gamma| = 0.447$. The Smith chart is the unit disc with the images of the lines $\mathrm{Re}\{z\} = \text{const}$ and $\mathrm{Im}\{z\} = \text{const}$ drawn on it — all circles, by the line-to-circle property.

Stability. A continuous-time linear time-invariant system is stable when every pole of its transfer function lies in the open left half-plane $\mathrm{Re}\{s\} < 0$; a discrete-time system is stable when every pole lies inside the unit circle $|z| < 1$ (Ch. 14, 17). Lines of constant damping ratio are rays $\arg s = \pm(\pi - \cos^{-1}\zeta)$; contours of constant settling time are vertical lines $\mathrm{Re}\{s\} = -\sigma$. Root locus, Nyquist and pole–zero plots are all geometry in this plane.

Common Pitfall — reading "arg" loci

The equation $\arg(z - z_0) = \alpha$ describes a **ray**, not a full line: the opposite ray has argument $\alpha \pm \pi$. Likewise $\arg\left(\frac{z-z_1}{z-z_2}\right) = \alpha$ is an **arc** of a circle through z_1 and z_2, not the whole circle (the other arc has argument $\alpha - \pi$). Squaring or taking tangents of an argument equation silently adds the missing pieces, so check candidate points against the original equation.

2.8 Applications to AC circuits and phasors

Motivation

A linear circuit driven by a sinusoidal source is governed by a linear differential equation. Solving it directly — guessing $A\cos\omega t + B\sin\omega t$, substituting, and grinding through trigonometric identities — is laborious even for a single loop and impractical for a network. The **phasor method** replaces the differential equation by an algebraic equation in complex numbers. It rests on three facts already established: a sinusoid is the real part of a complex exponential (Section 2.4), complex arithmetic follows all the ordinary rules (Section 2.3), and differentiating $e^{j\omega t}$ merely multiplies it by $j\omega$.

The method applies to the **sinusoidal steady state**: the response that remains after transients have died out, when every voltage and current is a sinusoid at the source frequency ω. For a stable linear time-invariant circuit, such a steady state always exists and differs from the input only in amplitude and phase. Transients are the subject of Ch. 11–12 and the Laplace transform (Ch. 14).

Sinusoids as real parts

A sinusoid is specified by an amplitude $V_m > 0$, an angular frequency ω and a phase ϕ (Ch. 3 treats these in detail). By Euler's formula

$$v(t) = V_m\cos(\omega t + \phi) = \mathrm{Re}\left\{V_m e^{j(\omega t+\phi)}\right\} = \mathrm{Re}\left\{\left(V_m e^{j\phi}\right)e^{j\omega t}\right\}.$$

The factor $e^{j\omega t}$ is common to every signal in the circuit. All the information that distinguishes one signal from another is in the complex constant $V_m e^{j\phi}$.

Phasor

The **phasor** of the sinusoid $v(t) = V_m \cos(\omega t + \phi)$ is the complex constant

$$\mathbf{V} = V_m e^{j\phi} = V_m \angle \phi.$$

The sinusoid is recovered by

$$v(t) = \operatorname{Re}\left\{\mathbf{V} e^{j\omega t}\right\}.$$

The magnitude of the phasor is the amplitude; its angle is the phase. The phasor does not contain ω — the frequency is a shared, separately remembered parameter.

Conventions. This book uses **amplitude (peak) phasors** and the **cosine reference**. A sine is first converted to a cosine: $V_m \sin(\omega t + \phi) = V_m \cos(\omega t + \phi - 90^\circ)$, so its phasor is $V_m \angle(\phi - 90^\circ) = -jV_m e^{j\phi}$. A negative amplitude is absorbed into the phase: $-V_m \cos(\omega t + \phi) = V_m \cos(\omega t + \phi \pm 180^\circ)$. Power-systems texts commonly use **RMS phasors** with $|\mathbf{V}| = V_m/\sqrt{2}$; for example the UK mains is "230 V" RMS, a peak of $230\sqrt{2} = 325.3$ V. Every formula in this section holds for either convention, except power formulas, which lose the factor $\frac{1}{2}$ with RMS phasors (Section 2.9). State which convention you are using.

The phasor transform is one-to-one and linear

Uniqueness of phasors

Let $\mathbf{A}, \mathbf{B}$ be complex constants and $\omega \neq 0$. If

$$\operatorname{Re}\{\mathbf{A} e^{j\omega t}\} = \operatorname{Re}\{\mathbf{B} e^{j\omega t}\} \quad \text{for all } t,$$

then $\mathbf{A} = \mathbf{B}$.

Proof. Put $\mathbf{C} = \mathbf{A} - \mathbf{B} = c_1 + jc_2$; we have $\operatorname{Re}\{\mathbf{C} e^{j\omega t}\} = c_1 \cos \omega t - c_2 \sin \omega t = 0$ for all t. At $t = 0$ this gives $c_1 = 0$; at $\omega t = \pi/2$ it gives $-c_2 = 0$. So $\mathbf{C} = 0$. □

This theorem is what licenses the method: if two sinusoidal expressions at the same frequency are equal for all time, their phasors are equal, and conversely. Combined with the following rules, it converts any linear constant-coefficient relation between same-frequency sinusoids into the same relation between phasors.

Phasor properties

Let $x(t) \leftrightarrow \mathbf{X}$ and $y(t) \leftrightarrow \mathbf{Y}$ be sinusoids at a common frequency ω. Then:

- **Linearity:** $a\,x(t) + b\,y(t) \;\leftrightarrow\; a\mathbf{X} + b\mathbf{Y}$ for real constants a, b.
- **Differentiation:** $\dfrac{dx}{dt} \;\leftrightarrow\; j\omega\mathbf{X}$, and $\dfrac{d^n x}{dt^n} \;\leftrightarrow\; (j\omega)^n \mathbf{X}$.
- **Integration** (the zero-mean, steady-state antiderivative): $\displaystyle\int x\,dt \;\leftrightarrow\; \frac{\mathbf{X}}{j\omega}$.
- **Time delay:** $x(t - \tau) \;\leftrightarrow\; e^{-j\omega\tau}\mathbf{X}$.

Proof of the differentiation rule. The real part commutes with differentiation with respect to the real variable t, so

$$\frac{d}{dt}\operatorname{Re}\{\mathbf{X} e^{j\omega t}\} = \operatorname{Re}\left\{\mathbf{X}\frac{d}{dt}e^{j\omega t}\right\} = \operatorname{Re}\{(j\omega\mathbf{X}) e^{j\omega t}\}.$$

Linearity follows because Re is additive and commutes with multiplication by **real** constants. The delay rule: $\operatorname{Re}\{\mathbf{X} e^{j\omega(t-\tau)}\} = \operatorname{Re}\{(\mathbf{X} e^{-j\omega\tau}) e^{j\omega t}\}$. □

In words: **differentiation multiplies the amplitude by ω and advances the phase by** 90°, since $j\omega = \omega \angle 90^\circ$. Check directly: $\frac{d}{dt}\cos \omega t = -\omega \sin \omega t = \omega \cos(\omega t + 90^\circ)$. ✓ A delay of τ seconds is a phase lag of $\omega\tau$ radians.

The method

1. Convert every source to a cosine and write its phasor.
2. Replace every d/dt by $j\omega$ (equivalently, replace each circuit element by its impedance — Section 2.9).
3. Solve the resulting **algebraic** equations with complex arithmetic.
4. Convert each phasor answer back: $\mathbf{X} = X_m \angle \phi \;\Rightarrow\; x(t) = X_m \cos(\omega t + \phi)$.

Worked Example 2.30 — Adding sinusoids

Express $v(t) = 10\cos(\omega t + 30^\circ) + 5\cos(\omega t - 45^\circ)$ as a single sinusoid.

Step 1 — phasors in rectangular form (addition needs rectangular):

$$\begin{aligned} 10\angle 30^\circ &= 8.660 + j5.000 \\ 5\angle -45^\circ &= 3.536 - j3.536 \\ \text{sum} &= 12.196 + j1.464 \end{aligned}$$

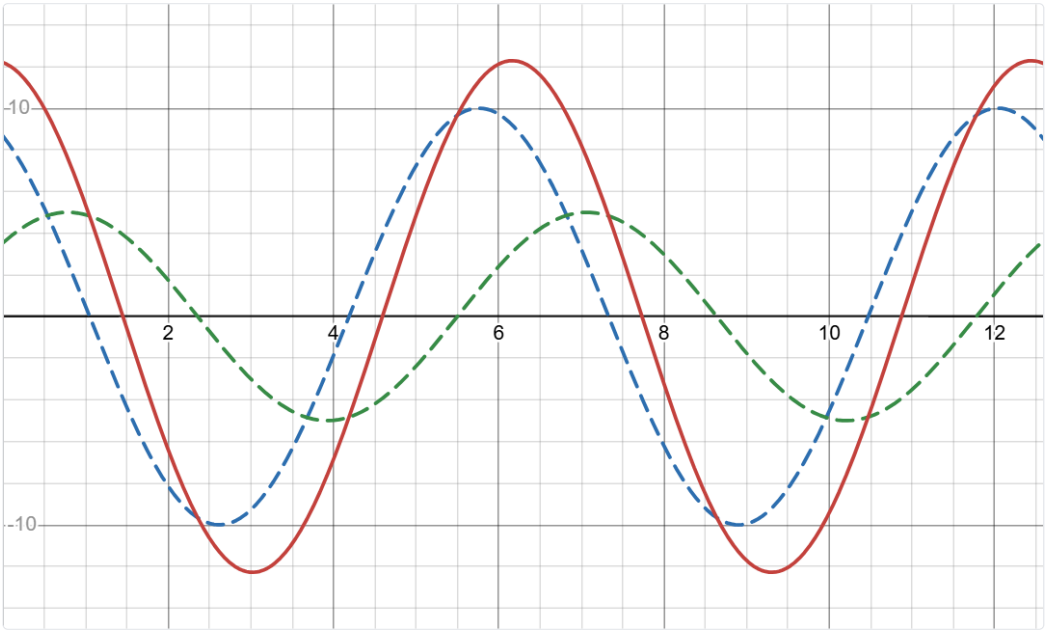
Step 2 — back to polar. $\sqrt{12.196^2 + 1.464^2} = 12.28$; $\operatorname{atan2}(1.464, 12.196) = 6.85^\circ$.

Step 3.

$$v(t) = 12.28\cos(\omega t + 6.85^\circ).$$

Checks. At $t = 0$: $10\cos 30^\circ + 5\cos 45^\circ = 8.660 + 3.536 = 12.196$ and $12.28\cos 6.85^\circ = 12.196$. ✓ Law of cosines (Section 2.3): $\sqrt{100 + 25 + 2(10)(5)\cos 75^\circ} = \sqrt{150.88} = 12.28$. ✓

Note $12.28 < 10 + 5$: same-frequency sinusoids add as **vectors**, not as amplitudes.



The sum (red) of $10\cos(x + 30^\circ)$ (blue) and $5\cos(x - 45^\circ)$ (green) is again a sinusoid of the same frequency, with amplitude 12.28 and phase 6.85° — exactly as the phasor sum predicts.

Worked Example 2.31 — Mixing sines and cosines

Express $v(t) = 4\sin\omega t - 3\cos(\omega t + 60^\circ)$ as a single cosine.

Step 1 — cosine reference. $4\sin\omega t = 4\cos(\omega t - 90^\circ) \leftrightarrow 4\angle -90^\circ = -j4$. The second term has phasor $3\angle 60^\circ = 1.500 + j2.598$.

Step 2 — subtract. $\mathbf{V} = -j4 - (1.500 + j2.598) = -1.500 - j6.598$.

Step 3 — polar. $|\mathbf{V}| = \sqrt{2.25 + 43.53} = 6.766$. The point is in the **third quadrant**, so $\angle\mathbf{V} = \arctan\frac{6.598}{1.5} - 180^\circ = 77.19^\circ - 180^\circ = -102.81^\circ$.

$$v(t) = 6.766\cos(\omega t - 102.81^\circ).$$

Check at $t = 0$: $0 - 3\cos 60^\circ = -1.5$ and $6.766\cos(-102.81^\circ) = -1.500$. ✓ At $\omega t = 90^\circ$: $4 - 3\cos 150^\circ = 4 + 2.598 = 6.598$, and $6.766\cos(-12.81^\circ) = 6.598$. ✓

Worked Example 2.32 — A first-order differential equation

A series RL circuit with $R = 30\ \Omega$, $L = 0.1$ H is driven by $v(t) = 100\cos 400t$ V. Its current obeys

$$L\frac{di}{dt} + Ri = v(t).$$

Find the steady-state current, then the complete response for $i(0) = 0$.

Step 1 — phasor equation. With $\mathbf{V} = 100\angle 0^\circ$ and $d/dt \to j\omega = j400$:

$$(j\omega L + R)\mathbf{I} = \mathbf{V} \;\Rightarrow\; \mathbf{I} = \frac{100}{30 + j40} = \frac{100\angle 0^\circ}{50\angle 53.13^\circ} = 2\angle -53.13^\circ \text{ A}.$$

Step 2 — back to time. $i_{ss}(t) = 2\cos(400t - 53.13^\circ)$ A.

Step 3 — verify by substitution. In rectangular form $\mathbf{I} = 1.2 - j1.6$, i.e. $i_{ss} = 1.2\cos 400t + 1.6\sin 400t$. Then

$$L\frac{di_{ss}}{dt} = 0.1(-480\sin 400t + 640\cos 400t) = 64\cos 400t - 48\sin 400t,$$

$$Ri_{ss} = 36\cos 400t + 48\sin 400t,$$

and the sum is $100\cos 400t$. ✓ The phasor method did in one division what substitution did in several lines.

Step 4 — complete response. The homogeneous solution of $Li' + Ri = 0$ is $Ke^{-(R/L)t} = Ke^{-300t}$ (Ch. 11). Then $i(0) = 1.2 + K = 0$ gives $K = -1.2$:

$$i(t) = 2\cos(400t - 53.13^\circ) - 1.2\,e^{-300t} \text{ A}.$$

The transient decays with time constant $L/R = 3.33$ ms; after about five time constants only the phasor solution remains.

Worked Example 2.33 — A second-order differential equation

Find the steady-state solution of

$$\frac{d^2v}{dt^2} + 3\frac{dv}{dt} + 2v = 10\cos 2t.$$

Step 1. With $\omega = 2$, $d/dt \to j2$ and $d^2/dt^2 \to (j2)^2 = -4$:

$$(-4 + j6 + 2)\mathbf{V} = 10 \;\Rightarrow\; \mathbf{V} = \frac{10}{-2 + j6}.$$

Step 2. $\mathbf{V} = \dfrac{10(-2 - j6)}{4 + 36} = -0.5 - j1.5 = 1.581\angle -108.43^\circ$ (third quadrant).

Step 3. $v_{ss}(t) = 1.581\cos(2t - 108.43^\circ)$.

Check (rectangular): $v_{ss} = -0.5\cos 2t + 1.5\sin 2t$, $v' = \sin 2t + 3\cos 2t$, $v'' = 2\cos 2t - 6\sin 2t$. Then $v'' + 3v' + 2v = (2 + 9 - 1)\cos 2t + (-6 + 3 + 3)\sin 2t = 10\cos 2t$. ✓

The complex number $(j\omega)^2 + 3(j\omega) + 2$ is the characteristic polynomial $p(s) = s^2 + 3s + 2$ evaluated at $s = j\omega$. In general the steady-state response of $p(D)y = x$ to $X_m\cos\omega t$ has phasor $X_m/p(j\omega)$ — the frequency response $H(j\omega) = 1/p(j\omega)$ of Section 2.9.

Phasor diagrams

Drawing phasors as arrows in the complex plane shows phase relationships at a glance. The whole diagram rotates anticlockwise at ω rad/s ($\mathbf{V}e^{j\omega t}$), and the instantaneous value of each signal is the projection of its rotating arrow onto the real axis. Relative angles never change, so the frozen diagram at $t = 0$ carries all the information.

A phasor at a larger angle **leads**: its peaks occur earlier in time. If $\mathbf{V} = V_m\angle\phi_v$ and $\mathbf{I} = I_m\angle\phi_i$, the voltage leads the current by $\phi_v - \phi_i$ (a negative value means it lags). For a capacitor, $i = C\,dv/dt$ gives $\mathbf{I} = j\omega C\mathbf{V}$, so the current **leads** the voltage by 90°; for an inductor, $\mathbf{V} = j\omega L\mathbf{I}$, so the current **lags** by 90°. (The mnemonic "ELI the ICE man": voltage E before current I in an inductor L; current before voltage in a capacitor C.)

Worked Example 2.34 — Three-phase line voltages

A balanced three-phase supply has phase voltages (RMS phasors) $\mathbf{V}_a = V\angle 0^\circ$, $\mathbf{V}_b = V\angle -120^\circ$, $\mathbf{V}_c = V\angle 120^\circ$. Find the line voltage $\mathbf{V}_{ab} = \mathbf{V}_a - \mathbf{V}_b$.

Step 1. $\mathbf{V}_{ab} = V\left(1 - e^{-j2\pi/3}\right) = V\left(1 + \frac{1}{2} + j\frac{\sqrt{3}}{2}\right) = V\left(\frac{3}{2} + j\frac{\sqrt{3}}{2}\right)$.

Step 2. $\left|\frac{3}{2} + j\frac{\sqrt{3}}{2}\right| = \sqrt{\frac{9}{4} + \frac{3}{4}} = \sqrt{3}$ and the angle is $\arctan\frac{1}{\sqrt{3}} = 30^\circ$:

$$\mathbf{V}_{ab} = \sqrt{3}\,V\angle 30^\circ.$$

With $V = 230$ V the line voltage is $\sqrt{3} \times 230 = 398.4$ V, quoted as the familiar "400 V" three-phase supply. Geometrically, the tips of $\mathbf{V}_a$, $\mathbf{V}_b$, $\mathbf{V}_c$ form an equilateral triangle and the line voltages are its sides (compare Example 2.26). Also $\mathbf{V}_a + \mathbf{V}_b + \mathbf{V}_c = V(1 + a^2 + a) = 0$ (Section 2.6). ✓

Common Pitfall — different frequencies

Phasors only combine signals of the **same** frequency. $\cos\omega_1 t + \cos\omega_2 t$ with $\omega_1 \neq \omega_2$ cannot be written as one phasor, because the factor $e^{j\omega t}$ is no longer common. For a circuit driven at several frequencies (or by a periodic non-sinusoidal waveform), solve a separate phasor problem at each frequency and **add the time-domain results** (superposition) — the basis of Fourier-series circuit analysis (Ch. 15).

Common Pitfall — mixing phasor and time domains

Expressions such as "$i(t) = 2\angle -53.13°$" or "$\mathbf{I} = 2\cos(400t - 53.13°)$" are category errors: a phasor is a time-independent complex number, a signal is a real function of time. Keep the arrow $\leftrightarrow$ between them and convert explicitly. Likewise, never multiply two phasors and expect the phasor of the product of the signals — the product of two sinusoids at ω contains frequencies 0 and 2ω (Example 2.13b).

Engineering Note — what phasors buy you

Every DC circuit-analysis technique — nodal and mesh analysis, superposition, source transformation, Thévenin and Norton equivalents — carries over unchanged to the sinusoidal steady state with resistances replaced by complex impedances and real sources by phasors. A network of fifty components at one frequency becomes a complex linear system $\mathbf{Y}\mathbf{v} = \mathbf{i}$, solved by the same Gaussian elimination as a real one (Ch. 8). This is precisely what circuit simulators do in "AC analysis" mode, sweeping ω and solving one complex linear system per frequency.

2.9 Complex impedance and transfer functions

Motivation

Section 2.8 converted differential equations into algebra one equation at a time. Engineers go one step further and attach the conversion to the **components**: each resistor, inductor and capacitor is assigned a complex number, its impedance, and the circuit is then analysed exactly like a resistive network. The ratio of an output phasor to an input phasor, viewed as a function of frequency, is the **frequency response**; extended to the complex variable s it becomes the **transfer function**, the central object of circuit theory, filtering and control. Both are nothing but complex-valued functions evaluated with the arithmetic of this chapter.

Impedance and admittance

Impedance and admittance

For a linear two-terminal element or network in sinusoidal steady state at angular frequency ω, with terminal voltage phasor $\mathbf{V}$ and current phasor $\mathbf{I}$ (passive sign convention), the **impedance** is

$$Z(j\omega) = \frac{\mathbf{V}}{\mathbf{I}} = R + jX \quad [\Omega],$$

and the **admittance** is

$$Y(j\omega) = \frac{1}{Z} = \frac{\mathbf{I}}{\mathbf{V}} = G + jB \quad [\mathrm{S}].$$

$R = \mathrm{Re}\{Z\}$ is the **resistance**, $X = \mathrm{Im}\{Z\}$ the **reactance**, G the **conductance** and B the **susceptance**. $X > 0$ is called inductive, $X < 0$ capacitive.

In polar form $Z = |Z|\angle\theta_Z$, Ohm's law $\mathbf{V} = Z\mathbf{I}$ says two things at once: the voltage amplitude is $|Z|$ times the current amplitude, and the voltage leads the current by θ_Z. Impedance is a complex number, **not** a phasor: it does not represent a sinusoid, it relates two of them.

Note that $G \neq 1/R$ in general. From $Y = 1/(R + jX) = (R - jX)/(R^2 + X^2)$,

$$G = \frac{R}{R^2 + X^2}, \qquad B = \frac{-X}{R^2 + X^2}.$$

For example $Z = 30 + j40\ \Omega$ gives $Y = (30 - j40)/2500 = 0.012 - j0.016$ S: the conductance is 0.012 S, not $1/30 = 0.0333$ S.

Element impedances

Apply the phasor differentiation rule ($d/dt \leftrightarrow j\omega$) to each element law:

Element	Time-domain law	Phasor law	Impedance	$\angle Z$
Resistor	$v = Ri$	$\mathbf{V} = R\mathbf{I}$	$Z_R = R$	$0°$
Inductor	$v = L\dfrac{di}{dt}$	$\mathbf{V} = j\omega L\,\mathbf{I}$	$Z_L = j\omega L$	$+90°$
Capacitor	$i = C\dfrac{dv}{dt}$	$\mathbf{I} = j\omega C\,\mathbf{V}$	$Z_C = \dfrac{1}{j\omega C} = -\dfrac{j}{\omega C}$	$-90°$

The inductive reactance $X_L = \omega L$ rises with frequency (an inductor is a short circuit at DC and an open circuit at very high frequency); the capacitive reactance $X_C = -1/(\omega C)$ has magnitude falling with frequency (open at DC, short at high frequency). Dimensional check: ωL has units $\mathrm{s}^{-1}\cdot\mathrm{H} = \mathrm{s}^{-1}\cdot\Omega\,\mathrm{s} = \Omega$, and $1/(\omega C)$ has units $1/(\mathrm{s}^{-1}\cdot\mathrm{F}) = \mathrm{s}/(\mathrm{A\,s/V}) = \Omega$. ✓

Combining impedances

Because KVL and KCL are linear relations with real (± 1) coefficients, they hold for phasors (Section 2.8). Hence impedances combine exactly as resistances do:

Series, parallel and divider rules

$$Z_{\text{series}} = Z_1 + Z_2 + \cdots, \qquad \frac{1}{Z_{\text{par}}} = \frac{1}{Z_1} + \frac{1}{Z_2} + \cdots \quad \left(Z_{\text{par}} = \frac{Z_1 Z_2}{Z_1 + Z_2} \text{ for two}\right),$$

$$\mathbf{V}_2 = \frac{Z_2}{Z_1 + Z_2}\mathbf{V} \quad \text{(voltage divider)}, \qquad \mathbf{I}_2 = \frac{Z_1}{Z_1 + Z_2}\mathbf{I} \quad \text{(current divider)}.$$

Admittances add in parallel: $Y_{\text{par}} = Y_1 + Y_2 + \cdots$.

Worked Example 2.35 — Series RL: current and power

$R = 30\ \Omega$, $L = 0.1$ H, source $v(t) = 100\cos 400t$ V (the circuit of Example 2.32). Find $i(t)$ and the power drawn.

Impedance. $X_L = \omega L = 400(0.1) = 40\ \Omega$, so $Z = 30 + j40 = 50\angle 53.13^\circ\ \Omega$.

Current. $\mathbf{I} = \mathbf{V}/Z = \dfrac{100\angle 0^\circ}{50\angle 53.13^\circ} = 2\angle -53.13^\circ$ A, so $i(t) = 2\cos(400t - 53.13^\circ)$ A — the same as Example 2.32, now with no differential equation in sight. The current lags the voltage by 53.13° (inductive).

Power (formulas derived below). $S = \tfrac{1}{2}\mathbf{V}\mathbf{I}^* = \tfrac{1}{2}(100)(2)\angle 53.13^\circ = 100\angle 53.13^\circ = 60 + j80$. Average power $P = 60$ W, reactive power $Q = 80$ var.

Check: $P = \tfrac{1}{2}|\mathbf{I}|^2 R = \tfrac{1}{2}(4)(30) = 60$ W ✓ and $Q = \tfrac{1}{2}|\mathbf{I}|^2 X_L = \tfrac{1}{2}(4)(40) = 80$ var. ✓ Only the resistor dissipates power.

Worked Example 2.36 — A series–parallel network

At $\omega = 1000$ rad/s, a $10\ \Omega$ resistor is in series with the parallel combination of a $20\ \Omega$ resistor and a $50\ \mu$F capacitor. A source $\mathbf{V} = 100\angle 0^\circ$ V drives the network. Find the input impedance, the source current and the voltage across the parallel part.

Step 1 — capacitor. $Z_C = \dfrac{1}{j(1000)(50\times 10^{-6})} = \dfrac{1}{j0.05} = -j20\ \Omega$.

Step 2 — parallel part.

$$Z_p = \frac{20(-j20)}{20 - j20} = \frac{-j400}{20 - j20} = \frac{-j400(20 + j20)}{800} = \frac{8000 - j8000}{800} = 10 - j10\ \Omega.$$

Step 3 — total. $Z = 10 + (10 - j10) = 20 - j10 = 22.36\angle -26.57^\circ\ \Omega$ (capacitive).

Step 4 — current. $\mathbf{I} = \dfrac{100}{22.36\angle -26.57^\circ} = 4.472\angle 26.57^\circ$ A: the current **leads** the voltage, as expected of a capacitive load.

Step 5 — divider. $\mathbf{V}_p = Z_p\mathbf{I} = (14.14\angle -45^\circ)(4.472\angle 26.57^\circ) = 63.25\angle -18.43^\circ = 60 - j20$ V.

Check by KVL: the series resistor drops $10\mathbf{I} = 44.72\angle 26.57^\circ = 40 + j20$ V, and $(40 + j20) + (60 - j20) = 100$. ✓ Note that $|\mathbf{V}_p| + |10\mathbf{I}| = 63.25 + 44.72 = 107.97 > 100$: voltage **magnitudes** around a loop need not sum to the source magnitude — only the phasors do.

Resonance

Worked Example 2.37 — Series RLC resonance, bandwidth and Q

A series RLC circuit has $R = 10\ \Omega$, $L = 10$ mH, $C = 1\ \mu$F. Its impedance is

$$Z(j\omega) = R + j\left(\omega L - \frac{1}{\omega C}\right).$$

Resonance. The reactance vanishes when $\omega L = 1/(\omega C)$, i.e. at

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{10^{-2}\cdot 10^{-6}}} = 10^4 \text{ rad/s} \quad (f_0 = 1.592 \text{ kHz}).$$

There $Z = R = 10\ \Omega$: purely resistive and minimal in magnitude, so the current for a given voltage is maximal and in phase with it.

Off resonance. At $\omega = 2\times 10^4$: $\omega L = 200$, $1/(\omega C) = 50$, $Z = 10 + j150 = 150.3\angle 86.19^\circ\ \Omega$ — strongly inductive, and the current is 15 times smaller than at resonance.

Half-power frequencies. The power $\tfrac{1}{2}|\mathbf{I}|^2 R$ falls to half its peak when $|Z| = \sqrt{2}R$, i.e. when the reactance equals $\pm R$: $\omega L - \dfrac{1}{\omega C} = \pm R$. Multiplying by ω/L gives $\omega^2 \mp \frac{R}{L}\omega - \omega_0^2 = 0$, whose positive roots are

$$\omega_{1,2} = \mp\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \omega_0^2} = \mp 500 + \sqrt{500^2 + 10^8} = 9512.5,\ 10512.5 \text{ rad/s}.$$

Bandwidth $\Delta\omega = \omega_2 - \omega_1 = R/L = 1000$ rad/s, and **quality factor** $Q = \omega_0/\Delta\omega = \omega_0 L/R = 10$.

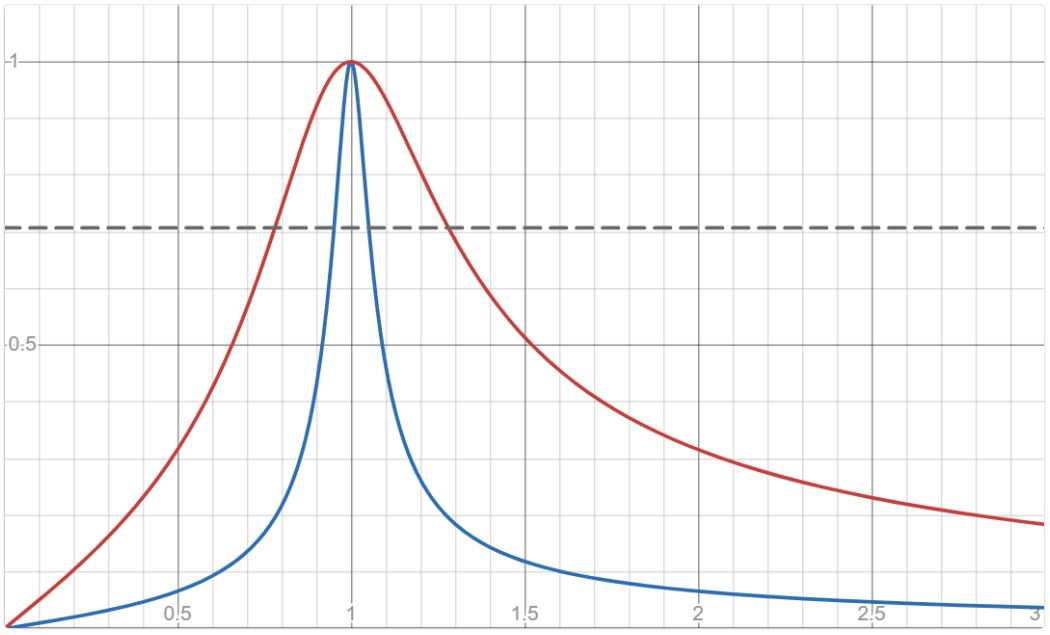
Check at $\omega_1 = 9512.5$: $\omega_1 L = 95.125$, $1/(\omega_1 C) = 105.125$, reactance $-10.00\ \Omega = -R$. ✓

Voltage magnification. At resonance, with source amplitude V, $|\mathbf{I}| = V/R$ and the capacitor voltage is $|\mathbf{V}_C| = \dfrac{V}{R\,\omega_0 C} = QV = 10V$: the capacitor sees ten times the source voltage — a design hazard and the basis of tuned circuits.

Normalising $x = \omega/\omega_0$, the current relative to its peak is

$$\frac{|\mathbf{I}(j\omega)|}{|\mathbf{I}|_{\max}} = \frac{R}{|Z|} = \frac{1}{\sqrt{1 + Q^2\left(x - \dfrac{1}{x}\right)^2}},$$

which shows that the sharpness of the peak depends only on Q.



Normalised series-RLC current $R/|Z|$ against $x = \omega/\omega_0$ for $Q = 10$ (blue, narrow) and $Q = 2$ (red, broad). The dashed line marks the half-power level $1/\sqrt{2}$; its intersections give the bandwidth ω_0/Q.

Complex power

Average and complex power

With amplitude phasors $\mathbf{V} = V_m\angle\phi_v$ and $\mathbf{I} = I_m\angle\phi_i$, the instantaneous power $p(t) = v(t)i(t)$ has average

$$P = \frac{1}{T}\int_0^T p(t)\,dt = \tfrac{1}{2}V_m I_m\cos(\phi_v - \phi_i) = \operatorname{Re}\{S\}, \qquad S = \tfrac{1}{2}\mathbf{V}\mathbf{I}^* = P + jQ.$$

S is the **complex power** [VA], P the **average (real) power** [W], $Q = \tfrac{1}{2}V_m I_m\sin(\phi_v - \phi_i)$ the **reactive power** [var], $|S|$ the **apparent power** and $\cos(\phi_v - \phi_i)$ the **power factor**. For a load Z: $S = \tfrac{1}{2}|\mathbf{I}|^2 Z$, so $P = \tfrac{1}{2}|\mathbf{I}|^2 R$ and $Q = \tfrac{1}{2}|\mathbf{I}|^2 X$. With RMS phasors, drop the factor $\tfrac{1}{2}$: $S = \mathbf{V}_{\text{rms}}\mathbf{I}_{\text{rms}}^*$.

Derivation. Using the product-to-sum identity (Example 2.13b),

$$p(t) = V_m I_m\cos(\omega t + \phi_v)\cos(\omega t + \phi_i) = \tfrac{1}{2}V_m I_m\left[\cos(\phi_v - \phi_i) + \cos(2\omega t + \phi_v + \phi_i)\right].$$

The second term oscillates at 2ω and averages to zero over a period, leaving $P = \tfrac{1}{2}V_m I_m\cos(\phi_v - \phi_i)$. Since $\mathbf{V}\mathbf{I}^* = V_m I_m\angle(\phi_v - \phi_i)$, this is $\operatorname{Re}\{\tfrac{1}{2}\mathbf{V}\mathbf{I}^*\}$. The conjugate is essential: it makes the angle of S the *difference* of phases, the only thing that physically matters. For $S = \tfrac{1}{2}\mathbf{V}\mathbf{I}^*$ with $\mathbf{V} = Z\mathbf{I}$: $S = \tfrac{1}{2}Z\mathbf{I}\mathbf{I}^* = \tfrac{1}{2}|\mathbf{I}|^2 Z$. □

Worked Example 2.38 — Power factor and its correction

A 50 Hz, 230 V (RMS) supply feeds a load $Z = 30 + j40\ \Omega$. Find P, Q and the power factor, then the parallel capacitor that raises the power factor to 1.

Step 1 — current (RMS phasors). $\mathbf{I} = \dfrac{230\angle 0^\circ}{50\angle 53.13^\circ} = 4.6\angle -53.13^\circ$ A.

Step 2 — power. $S = \mathbf{V}\mathbf{I}^* = 230(4.6)\angle 53.13^\circ = 1058\angle 53.13^\circ = 634.8 + j846.4$ VA. So $P = 634.8$ W, $Q = 846.4$ var (inductive), power factor $\cos 53.13^\circ = 0.6$ lagging. *Check:* $|\mathbf{I}|^2 R = 21.16(30) = 634.8$ W ✓, $|\mathbf{I}|^2 X = 21.16(40) = 846.4$ var. ✓

Step 3 — correction. A parallel capacitor draws $S_C = \mathbf{V}\mathbf{I}_C^*$ with $\mathbf{I}_C = j\omega C\mathbf{V}$, so $S_C = \mathbf{V}(-j\omega C\mathbf{V}^*) = -j\omega C|\mathbf{V}|^2$: purely negative reactive power. To cancel Q:

$$\omega C|\mathbf{V}|^2 = 846.4 \;\Rightarrow\; C = \frac{846.4}{(2\pi\cdot 50)(230^2)} = 50.9\ \mu\text{F}.$$

The supply then delivers the same 634.8 W with current $634.8/230 = 2.76$ A instead of 4.6 A, reducing the resistive losses in the supply cables by a factor $0.6^2 = 0.36$.

Maximum power transfer

A source with phasor $\mathbf{V}_s$ and internal impedance $Z_s = R_s + jX_s$ delivers the greatest average power to a load Z_L when

$$Z_L = Z_s^* = R_s - jX_s, \qquad P_{\max} = \frac{|\mathbf{V}_s|^2}{8R_s} \text{ (amplitude phasor)}.$$

Proof sketch. $P = \tfrac{1}{2}|\mathbf{I}|^2 R_L = \dfrac{\tfrac{1}{2}|\mathbf{V}_s|^2 R_L}{(R_s+R_L)^2 + (X_s+X_L)^2}$. For any R_L the denominator is smallest when $X_L = -X_s$; then $P = \tfrac{1}{2}|\mathbf{V}_s|^2 R_L/(R_s+R_L)^2$, maximised over R_L at $R_L = R_s$ (set the derivative to zero, Ch. 4). □ For example $Z_s = 50 + j30\ \Omega$ and $|\mathbf{V}_s| = 10$ V give $Z_L = 50 - j30\ \Omega$, current $10/100 = 0.1$ A, and $P_{\max} = \tfrac{1}{2}(0.1)^2(50) = 0.25$ W $= 100/(8\cdot 50)$. ✓ This **conjugate match** is the target of every RF matching network.

Frequency response and transfer functions

Frequency response

For a stable linear time-invariant circuit with sinusoidal input $x(t) = X_m\cos(\omega t + \phi_x) \leftrightarrow \mathbf{X}$ and steady-state output $\mathbf{Y}$, the **frequency response** is

$$H(j\omega) = \frac{\mathbf{Y}}{\mathbf{X}}.$$

The steady-state output is then

$$y(t) = |H(j\omega)|\, X_m \cos\big(\omega t + \phi_x + \angle H(j\omega)\big).$$

$|H(j\omega)|$ is the **gain** (often quoted as $20\log_{10}|H|$ dB) and $\angle H(j\omega)$ the **phase shift**.

So an LTI system acts on a sinusoid by **one complex multiplication**: scale by $|H|$, rotate by $\angle H$. H is computed from the circuit by impedance methods, or from the governing differential equation by $d/dt \to j\omega$ (Example 2.33).

Transfer function, poles and zeros

When $H(j\omega)$ is a ratio of polynomials in $j\omega$, replacing $j\omega$ by the complex variable $s = \sigma + j\omega$ gives the **transfer function**

$$H(s) = \frac{N(s)}{D(s)} = K\,\frac{(s-z_1)(s-z_2)\cdots(s-z_m)}{(s-p_1)(s-p_2)\cdots(s-p_n)}.$$

The roots z_i of N are the **zeros** and the roots p_i of D the **poles**. For real circuits N and D have real coefficients, so poles and zeros are real or occur in conjugate pairs (Section 2.2). In terms of impedances, each element is assigned $Z_R = R$, $Z_L = sL$, $Z_C = 1/(sC)$.

The interpretation of $H(s)$ for complex s — the response to e^{st} — is developed with the Laplace transform in Ch. 14. Here what matters is that the factored form turns the evaluation of gain and phase into geometry.

Gain and phase from poles and zeros

At $s = j\omega$,

$$|H(j\omega)| = |K|\,\frac{\prod_i |j\omega - z_i|}{\prod_k |j\omega - p_k|}, \qquad \angle H(j\omega) = \angle K + \sum_i \angle(j\omega - z_i) - \sum_k \angle(j\omega - p_k).$$

Each factor $j\omega - z_i$ is the vector from the zero z_i to the point $j\omega$ on the imaginary axis.

Proof. Magnitudes of products and quotients multiply and divide, angles add and subtract (Section 2.3). □ As ω moves up the imaginary axis, the gain rises near zeros and peaks near poles that lie close to the axis — the geometric explanation of resonance.

Worked Example 2.39 — RC low-pass filter

A series resistor R feeds an output taken across a capacitor C. Find $H(j\omega) = \mathbf{V}_{\text{out}}/\mathbf{V}_{\text{in}}$, its gain and phase, and its pole.

Step 1 — divider.

$$H(j\omega) = \frac{Z_C}{R + Z_C} = \frac{1/(j\omega C)}{R + 1/(j\omega C)} = \frac{1}{1 + j\omega RC}, \qquad H(s) = \frac{1}{1+sRC}.$$

Step 2 — polar form.

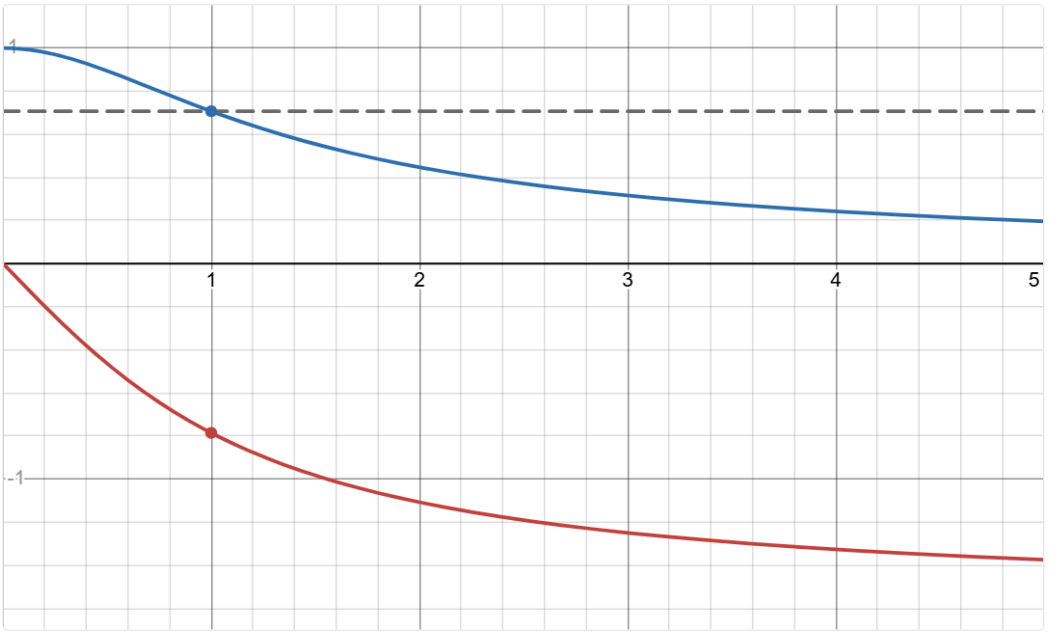
$$|H(j\omega)| = \frac{1}{\sqrt{1 + (\omega RC)^2}}, \qquad \angle H(j\omega) = -\arctan(\omega RC).$$

(The denominator $1 + j\omega RC$ is in the first quadrant, so arctan is safe here.)

Step 3 — cut-off. At $\omega_c = 1/(RC)$: $|H| = 1/\sqrt{2} = 0.7071$, i.e. $20\log_{10} 0.7071 = -3.01$ dB, and $\angle H = -45°$. With $R = 1$ kΩ, $C = 1$ μF: $\omega_c = 1000$ rad/s, $f_c = 159.2$ Hz.

Step 4 — a specific input. For $v_{\text{in}} = 10\cos 2000t$ V, $\omega RC = 2$: $H = 1/(1+j2) = 0.4472\angle -63.43°$ (-6.99 dB), so $v_{\text{out}} = 4.472\cos(2000t - 63.43°)$ V.

Step 5 — pole. $D(s) = 1 + sRC = 0$ at $s = -1/(RC) = -1000$ s^{-1}, on the negative real axis (left half-plane: stable). Geometrically, $|H(j\omega)| = \omega_c/|j\omega + \omega_c|$ is ω_c divided by the distance from the pole to $j\omega$; that distance is $\sqrt{2}\,\omega_c$ when $\omega = \omega_c$, which is where the -3 dB point comes from.



RC low-pass response against normalised frequency $x = \omega RC$: magnitude $|H|$ (blue) and phase $\angle H$ in radians (red), tending to $-\pi/2$. Both shapes follow directly from the polar form of $1/(1 + jx)$.

Worked Example 2.40 — Gain and phase from a pole–zero plot

Evaluate the gain and phase of

$$H(s) = \frac{10(s+1)}{(s+2)(s+10)}$$

at $\omega = 2$ rad/s.

Step 1 — poles and zeros. Zero at -1; poles at -2 and -10; $K = 10$.

Step 2 — vectors to $j2$.

$$\begin{aligned} j2 - (-1) &= 1 + j2 = 2.236\angle 63.43^\circ, \\ j2 - (-2) &= 2 + j2 = 2.828\angle 45.00^\circ, \\ j2 - (-10) &= 10 + j2 = 10.198\angle 11.31^\circ. \end{aligned}$$

Step 3 — combine.

$$|H(j2)| = \frac{10 \times 2.236}{2.828 \times 10.198} = 0.7752 \ (-2.21 \text{ dB}), \qquad \angle H(j2) = 63.43^\circ - 45.00^\circ - 11.31^\circ = 7.12^\circ.$$

Check by direct rectangular evaluation: $(2 + j2)(10 + j2) = 20 + j4 + j20 - 4 = 16 + j24$, and $\dfrac{10(1 + j2)}{16 + j24} = \dfrac{10(1 + j2)(16 - j24)}{832} = \dfrac{10(64 + j8)}{832} = 0.7692 + j0.0962$, whose magnitude is 0.7752 and angle 7.13°. ✓ (The last-digit difference is rounding in the angles.)

Common Pitfall — phase of a factor in the left half of its plane

When a factor such as $j\omega - p$ or $1 - \omega^2 LC + j\omega RC$ has a **negative real part**, its angle exceeds 90°, and $\arctan(\text{Im}/\text{Re})$ is off by 180° (Section 2.1). Second-order denominators $1 - (\omega/\omega_0)^2 + j \cdot (\dots)$ change quadrant at $\omega = \omega_0$, which is why a naive formula makes the phase of a resonant filter jump from -90° to $+90^\circ$ instead of continuing smoothly to -180°. Use `atan2` and then unwrap.

Engineering Note — from here to Bode, filters and control

Bode plots are $20 \log_{10} |H(j\omega)|$ and $\angle H(j\omega)$ against $\log \omega$. Because the log of a product is the sum of logs (Section 2.5), each pole and zero contributes an independent, easily sketched term, and the total is their sum. Filter design chooses pole and zero locations (Butterworth on a circle, Example 2.25; Chebyshev on an ellipse); feedback-loop stability margins are read from $|H|$ and $\angle H$ near the crossover frequency. The mathematics needed is exactly this chapter plus logarithms; the design methods belong to circuits and control texts.

Summary

Topic	Key formula
Rectangular form	$z = a + jb, a = \text{Re}\{z\}, b = \text{Im}\{z\}, j^2 = -1$
Rectangular ↔ polar	$\lvert z\rvert = \sqrt{a^2 + b^2}, \angle z = \text{atan2}(b, a); a = r\cos\theta, b = r\sin\theta$
Equality	$a + jb = c + jd \iff a = c,\ b = d$ (one complex equation = two real equations)

Topic	Key formula
Conjugate	$z^* = a - jb = re^{-j\theta}$; $zz^* = \lvert z\rvert^2$; $\operatorname{Re}\{z\} = \frac{z+z^*}{2}$, $\operatorname{Im}\{z\} = \frac{z-z^*}{2j}$
Conjugate roots	real coefficients $\Rightarrow$ non-real roots in pairs $\alpha \pm j\beta$, factor $x^2 - 2\alpha x + \alpha^2 + \beta^2$
Product	$(a+jb)(c+jd) = (ac-bd) + j(ad+bc)$; $r_1\angle\theta_1 \cdot r_2\angle\theta_2 = r_1 r_2 \angle(\theta_1+\theta_2)$
Division	$\frac{z_1}{z_2} = \frac{z_1 z_2^*}{\lvert z_2\rvert^2} = \frac{r_1}{r_2}\angle(\theta_1 - \theta_2)$
Powers of j	$j^n = j^{\,n \bmod 4}$; $1/j = -j$
Triangle inequality	$\lvert z_1 + z_2\rvert \le \lvert z_1\rvert + \lvert z_2\rvert$; $\lvert z_1+z_2\rvert^2 = \lvert z_1\rvert^2 + \lvert z_2\rvert^2 + 2\operatorname{Re}\{z_1 z_2^*\}$
Euler	$e^{j\theta} = \cos\theta + j\sin\theta$; $e^{j\pi} = -1$, $e^{j\pi/2} = j$
Inverse Euler	$\cos\theta = \frac{e^{j\theta}+e^{-j\theta}}{2}$, $\sin\theta = \frac{e^{j\theta}-e^{-j\theta}}{2j}$
Exponential form	$z = re^{j\theta}$; $z_1 z_2 = r_1 r_2 e^{j(\theta_1+\theta_2)}$
Complex exponential	$e^{\sigma+j\omega} = e^\sigma(\cos\omega + j\sin\omega)$, $\lvert e^z\rvert = e^{\operatorname{Re}\{z\}}$, period $j2\pi$
Logarithm	$\operatorname{Log} z = \ln\lvert z\rvert + j\operatorname{Arg} z$; $\log z = \operatorname{Log} z + j2\pi k$
Complex power of z	$z^c = e^{c\log z}$ (multivalued; principal value uses Log)
De Moivre	$(re^{j\theta})^n = r^n e^{jn\theta}$, $n \in \mathbb{Z}$
n-th roots	$w_k = r^{1/n} e^{j(\theta+2\pi k)/n}$, $k = 0, \dots, n-1$
Square root	$\sqrt{a+jb} = \pm\left(\sqrt{\frac{r+a}{2}} + j\,\operatorname{sgn}(b)\sqrt{\frac{r-a}{2}}\right)$
Roots of unity	$W = e^{j2\pi/n}$; $\sum_{k=0}^{n-1} W^{mk} = n$ if $n \mid m$, else 0
Rotation	multiply by $e^{j\alpha}$; by j for $+90^\circ$; about w_0: $w_0 + e^{j\alpha}(z - w_0)$
Circle	$\lvert z - z_0\rvert = R$; Apollonius $\lvert z - z_1\rvert = k\lvert z - z_2\rvert$
Phasor	$V_m\cos(\omega t + \phi) \leftrightarrow V_m\angle\phi$; $v(t) = \operatorname{Re}\{\mathbf{V}e^{j\omega t}\}$
Phasor calculus	$\frac{d}{dt} \leftrightarrow j\omega$, $\int dt \leftrightarrow \frac{1}{j\omega}$, delay $\tau \leftrightarrow e^{-j\omega\tau}$
Impedances	$Z_R = R$, $Z_L = j\omega L$, $Z_C = \dfrac{1}{j\omega C}$; series add, admittances add in parallel
Series RLC	$\omega_0 = 1/\sqrt{LC}$, $Q = \omega_0 L/R$, bandwidth R/L
Complex power	$S = \frac{1}{2}\mathbf{V}\mathbf{I}^* = P + jQ = \frac{1}{2}\lvert\mathbf{I}\rvert^2 Z$ (amplitude); $S = \mathbf{V}\mathbf{I}^*$ (RMS)
Max power transfer	$Z_L = Z_s^*$, $P_{\max} = \lvert\mathbf{V}_s\rvert^2/(8R_s)$ (amplitude)
Frequency response	$y(t) = \lvert H\rvert X_m\cos(\omega t + \phi_x + \angle H)$
Pole–zero evaluation	$\lvert H\rvert = \lvert K\rvert\frac{\prod\lvert j\omega - z_i\rvert}{\prod\lvert j\omega - p_k\rvert}$, $\angle H = \angle K + \sum\angle(j\omega - z_i) - \sum\angle(j\omega - p_k)$
RC low-pass	$H = \dfrac{1}{1 + j\omega RC}$, $\lvert H\rvert = \dfrac{1}{\sqrt{1+(\omega RC)^2}}$, $\angle H = -\arctan\omega RC$

Further reading

- E. Kreyszig, *Advanced Engineering Mathematics*, 10th ed., Wiley, 2011 — Chapter 13 (complex numbers and functions, the complex exponential and logarithm).
- K. A. Stroud and D. J. Booth, *Engineering Mathematics*, 8th ed., Palgrave Macmillan, 2020 — Programmes on complex numbers, with extensive step-by-step drill.
- G. James et al., *Modern Engineering Mathematics*, 6th ed., Pearson, 2020 — Chapter 3 (complex numbers), including phasors and loci.
- A. V. Oppenheim and A. S. Willsky, *Signals and Systems*, 2nd ed., Prentice Hall, 1997 — Chapter 1 on complex exponential signals and Chapter 3 on eigenfunctions of LTI systems.
- C. K. Alexander and M. N. O. Sadiku, *Fundamentals of Electric Circuits*, 7th ed., McGraw-Hill, 2021 — Chapters 9–11 for the phasor method, impedance and AC power in full circuit-analysis detail.

Chapter 3 — Trigonometry and Periodic Functions

> **Prerequisites:** Ch. 1 (functions, graphs, basic trigonometric ratios), Ch. 2 (Euler's formula, phasors) **Learning outcomes:** After this chapter you can
>
> - measure angles in radians, degrees and revolutions, convert between them, and explain why calculus insists on radians;
> - read amplitude, offset, frequency and phase directly from a sinusoidal expression or graph, and convert any $a\cos\omega t + b\sin\omega t$ into amplitude–phase form;
> - compute phase differences, time delays and leads/lags between sinusoids, and add same-frequency sinusoids exactly;
> - decide whether a sum of sinusoids (continuous or discrete-time) is periodic and find its fundamental period;
> - use the compound-angle, power-reduction and product-to-sum identities to move between products, powers and harmonics;
> - describe periodic waveforms (square, triangle, sawtooth, rectified sine) by formula, symmetry, mean and RMS, and explain how they are built from harmonics.

Almost every quantity an electrical engineer measures either *is* a sinusoid or is analysed as a sum of sinusoids. The mains supply is a 50 Hz or 60 Hz sinusoid; a radio carrier is a sinusoid whose amplitude, frequency or phase is nudged to carry information; the steady-state response of any linear circuit to a sinusoid is another sinusoid of the same frequency; and a distorted current drawn by a rectifier is described by the list of sinusoidal *harmonics* it contains. Trigonometry is the language of all of this.

This chapter builds the working fluency that later chapters assume. It is not a chapter on triangle-solving: the emphasis is on the circular functions as *functions of time*, on the three numbers (amplitude, frequency, phase) that describe a sinusoid, on the identities that convert products and powers of sinusoids into sums of sinusoids, and on the idea that a periodic waveform is a superposition of harmonically related sinusoids. The full theory of that last idea — Fourier series, convergence and spectra — is developed in Ch. 15; here we build the intuition and the algebra that make Ch. 15 easy to read.

Throughout, t is time in seconds, ω is angular frequency in rad/s, f is frequency in Hz, and j is the imaginary unit.

3.1 Angular measurement

Why angle measure needs care

An angle measures *rotation*: how far a ray has turned about a point. In electrical engineering, rotation appears literally (the shaft of a motor or generator, the rotor position read by an encoder, the bearing of an antenna) and figuratively (the phase of a sinusoid, which is the angle of a rotating phasor). Getting the *unit* of angle right is the single most common source of numerical error in trigonometric calculations, because the same number — say 1.5 — is a modest angle in radians, a tiny angle in degrees and an enormous one in revolutions, and a calculator or software library will silently accept any of them.

Units of angle

Radian

Let a circle of radius r be centred at the vertex of an angle, and let s be the length of arc the angle cuts off. The angle in **radians** is

$$\theta = \frac{s}{r}.$$

Because it is a ratio of two lengths, the radian is dimensionless: $[\theta] = 1$. One radian is the angle that cuts off an arc equal to the radius.

A full turn cuts off the whole circumference $2\pi r$, so one revolution is 2π rad. All other units are defined by dividing the full turn into a conventional number of parts:

Unit	Full turn	Right angle	Typical use
radian (rad)	$2\pi \approx 6.2832$	$\pi/2$	all of calculus, ωt, software libraries
degree (°)	360	90	phase on datasheets and phasor diagrams
revolution (rev, turn)	1	$1/4$	shaft speed (rpm), encoder counts
gradian (gon)	400	100	surveying; a trap on some calculators
milliradian (mrad)	≈ 6283	≈ 1571	small pointing errors, optics

The conversion rules follow from equating full turns:

$$\theta_{\text{deg}} = \frac{180}{\pi}\,\theta_{\text{rad}}, \qquad \theta_{\text{rad}} = \frac{\pi}{180}\,\theta_{\text{deg}}, \qquad \theta_{\text{rev}} = \frac{\theta_{\text{rad}}}{2\pi}.$$

A useful anchor is $1\text{ rad} \approx 57.296^\circ$ and $1^\circ \approx 0.017453\text{ rad}$.

Worked Example 3.1 — Unit conversions

(a) Convert 2.5 rad to degrees. (b) Convert 37° to radians. (c) A shaft turns through 1350°. How many revolutions is that, and what angle in $[0^\circ, 360^\circ)$ does it end at?

(a) $\theta = 2.5 \times 180/\pi = 143.24^\circ$.

(b) $\theta = 37 \times \pi/180 = 0.6458$ rad.

(c) $1350/360 = 3.75$ rev. Three whole turns (1080°) bring the shaft back to where it started, leaving $1350 - 1080 = 270^\circ$. The final position is 270° (equivalently $3\pi/2$ rad, or -90°).

Check: $143.24^\circ / 57.296 = 2.500$; $0.6458 \times 57.296 = 37.00^\circ$.

Arc length, sector area and angular speed

The radian definition gives three formulas with no conversion factors — which is precisely why radians are the natural unit:

$$s = r\theta, \qquad A_{\text{sector}} = \tfrac{1}{2} r^2 \theta, \qquad v = r\,\Omega,$$

where $\Omega = d\theta/dt$ is the angular speed in rad/s and v the linear speed of a point at radius r. The sector area follows by proportion: the sector is the fraction $\theta/(2\pi)$ of the disc of area πr^2, so $A = \frac{\theta}{2\pi}\pi r^2 = \frac{1}{2} r^2 \theta$. If θ were in degrees each formula would acquire a factor $\pi/180$.

Shaft speeds are usually quoted in revolutions per minute (rpm). Since one revolution is 2π rad and one minute is 60 s,

$$\Omega\ [\text{rad/s}] = \frac{2\pi}{60}\, N\ [\text{rpm}] \approx 0.10472\, N.$$

Worked Example 3.2 — Encoder resolution and tip speed

An incremental encoder on a motor shaft gives 2048 counts per revolution. The shaft drives a pulley of radius $r = 40$ mm and runs at $N = 3000$ rpm.

(a) What angle does one count represent, in degrees and in milliradians? (b) What arc length of belt passes per count? (c) What is the belt speed?

(a) One count is $1/2048$ rev, i.e. $360/2048 = 0.17578^\circ$, or $2\pi/2048 = 3.0680 \times 10^{-3}$ rad $= 3.068$ mrad.

(b) $s = r\theta = 0.040 \times 3.0680 \times 10^{-3} = 1.227 \times 10^{-4}$ m $= 0.1227$ mm.

(c) $\Omega = 2\pi \times 3000/60 = 314.16$ rad/s, so $v = r\Omega = 0.040 \times 314.16 = 12.57$ m/s.

Check: At 3000 rpm the shaft makes 50 rev/s; the circumference is $2\pi(0.04) = 0.2513$ m; $50 \times 0.2513 = 12.57$ m/s. ✓

Electrical and mechanical degrees

In a rotating machine with p pole *pairs*, the magnetic field pattern repeats p times around the air gap. One mechanical revolution therefore carries the generated voltage through p complete electrical cycles. Engineers distinguish

$$\theta_e = p\,\theta_m, \qquad \omega_e = p\,\Omega_m,$$

where the subscript e denotes *electrical* angle/frequency and m *mechanical*. A four-pole machine ($p = 2$) running at 1500 rpm has $\Omega_m = 157.08$ rad/s and produces $\omega_e = 314.16$ rad/s, i.e. $f = 50$ Hz. Phase angles of voltages and currents are always electrical angles.

Engineering Note — Which "degree" is on the datasheet?

Motor-control datasheets quote commutation angles in electrical degrees; encoder datasheets quote resolution in mechanical counts. Mixing them up in a field-oriented controller for a p-pole-pair motor produces a commutation error multiplied by p, which shows up as a motor that vibrates, runs hot, or will not start.

The circular functions on the unit circle

The six trigonometric functions are defined for *any* real angle, not only acute angles of right triangles, by placing the angle in standard position (vertex at the origin, initial ray along the positive x-axis, positive angles anticlockwise) and reading off the point where the terminal ray meets the unit circle.

Circular functions

Let the terminal ray of the angle θ (in radians) meet the unit circle $x^2 + y^2 = 1$ at $P = (x, y)$. Then

$$\cos\theta = x, \qquad \sin\theta = y, \qquad \tan\theta = \frac{y}{x}\ (x \neq 0),$$

$$\sec\theta = \frac{1}{x}, \qquad \csc\theta = \frac{1}{y}, \qquad \cot\theta = \frac{x}{y}.$$

For a point at distance r from the origin, $(x, y) = (r\cos\theta,\ r\sin\theta)$.

Several facts follow at once from the picture:

- **Range.** $-1 \le \cos\theta \le 1$ and $-1 \le \sin\theta \le 1$, because P lies on the unit circle.
- **Pythagorean identity.** $\cos^2\theta + \sin^2\theta = 1$, because $x^2 + y^2 = 1$.

- **Periodicity.** Adding 2π to θ returns P to the same point, so $\sin(\theta+2\pi)=\sin\theta$ and $\cos(\theta+2\pi)=\cos\theta$. The tangent repeats after only π, because $\theta+\pi$ sends P to $-P$ and $(-y)/(-x)=y/x$.
- **Symmetry.** Reflecting in the x-axis replaces θ by $-\theta$ and y by $-y$: $\cos(-\theta)=\cos\theta$ (cosine is *even*), $\sin(-\theta)=-\sin\theta$ (sine is *odd*).
- **Complementarity.** Reflecting in the line $y=x$ swaps x and y and sends θ to $\pi/2-\theta$: $\sin(\pi/2-\theta)=\cos\theta$, $\cos(\pi/2-\theta)=\sin\theta$.

The signs of sin and cos in each quadrant follow from the signs of y and x:

Quadrant	θ range	$\sin\theta$	$\cos\theta$	$\tan\theta$
I	$0<\theta<\pi/2$	$+$	$+$	$+$
II	$\pi/2<\theta<\pi$	$+$	$-$	$-$
III	$\pi<\theta<3\pi/2$	$-$	$-$	$+$
IV	$3\pi/2<\theta<2\pi$	$-$	$+$	$-$

Special angles and reference angles

The values at multiples of 30° and 45° come from the equilateral triangle (halved) and the isosceles right triangle, and are worth knowing without a calculator:

θ	0	$\pi/6$ (30°)	$\pi/4$ (45°)	$\pi/3$ (60°)	$\pi/2$ (90°)
$\sin\theta$	0	$1/2$	$\sqrt{2}/2$	$\sqrt{3}/2$	1
$\cos\theta$	1	$\sqrt{3}/2$	$\sqrt{2}/2$	$1/2$	0
$\tan\theta$	0	$1/\sqrt{3}$	1	$\sqrt{3}$	undefined

For any other quadrant, the **reference angle** $\theta_r\in[0,\pi/2]$ is the acute angle between the terminal ray and the x-axis. Then $|\sin\theta|=\sin\theta_r$, $|\cos\theta|=\cos\theta_r$, and the sign comes from the quadrant table. For example, $\theta=240^\circ$ lies in quadrant III with $\theta_r=60^\circ$, so $\sin 240^\circ=-\sqrt{3}/2$ and $\cos 240^\circ=-1/2$. These two numbers reappear constantly in three-phase work, where phases are 0°, -120° and -240° (equivalently $+120^\circ$).

Inverse functions and the quadrant problem

Because sin, cos and tan are periodic they are not one-to-one, so their inverses are defined by restricting the domain to a **principal branch**:

Principal inverse trigonometric functions

$$\begin{aligned}\arcsin x &\in [-\pi/2,\ \pi/2], && -1\le x\le 1,\\ \arccos x &\in [0,\ \pi], && -1\le x\le 1,\\ \arctan x &\in (-\pi/2,\ \pi/2), && x\in\mathbb{R}.\end{aligned}$$

Each returns the *one* angle in its range with the given sine, cosine or tangent. Every other solution differs by the symmetries above: $\sin\theta=s$ has solutions $\theta=\arcsin s+2\pi k$ and $\theta=\pi-\arcsin s+2\pi k$; $\cos\theta=c$ has $\theta=\pm\arccos c+2\pi k$; $\tan\theta=t$ has $\theta=\arctan t+\pi k$.

The engineering problem is usually: *given the coordinates (x,y) of a point (or the real and imaginary parts of a phasor), find its angle.* The ratio y/x loses the signs of x and y separately, so $\arctan(y/x)$ cannot distinguish $(1,1)$ from $(-1,-1)$. The two-argument arctangent fixes this.

Two-argument arctangent

For $(x,y)\neq(0,0)$, $\operatorname{atan2}(y,x)$ is the unique $\theta\in(-\pi,\pi]$ with $\cos\theta=x/\sqrt{x^2+y^2}$ and $\sin\theta=y/\sqrt{x^2+y^2}$. In terms of arctan:

$$\operatorname{atan2}(y,x)=\begin{cases}\arctan(y/x), & x>0,\\ \arctan(y/x)+\pi, & x<0,\ y\ge 0,\\ \arctan(y/x)-\pi, & x<0,\ y<0,\\ +\pi/2, & x=0,\ y>0,\\ -\pi/2, & x=0,\ y<0.\end{cases}$$

Worked Example 3.3 — Solving trigonometric equations on a full period

(a) Find all $\theta\in[0,2\pi)$ with $\sin\theta=-0.5$. (b) Find the angle of the point $(-4,-3)$. (c) Solve $\cos(2\theta)=0.3$ for $\theta\in[0,2\pi)$.

(a) $\arcsin(-0.5)=-\pi/6$. The two families are $\theta=-\pi/6+2\pi k$ and $\theta=\pi-(-\pi/6)+2\pi k=7\pi/6+2\pi k$. In $[0,2\pi)$: $\theta=7\pi/6$ (210°) and $\theta=11\pi/6$ (330°). Both are in quadrants III and IV, where sine is negative. ✓

(b) $\arctan(-3/-4)=\arctan(0.75)=36.87^\circ$ — a first-quadrant answer for a third-quadrant point. Since $x<0,\ y<0$: $\operatorname{atan2}(-3,-4)=36.87^\circ-180^\circ=-143.13^\circ$ (or 216.87°). Check: $5\cos(-143.13^\circ)=-4.00$, $5\sin(-143.13^\circ)=-3.00$. ✓

(c) Let $u=2\theta$; as θ ranges over $[0,2\pi)$, u ranges over $[0,4\pi)$ — *two* full turns, so expect four solutions. $\arccos 0.3=1.2661$ rad. Solutions $u=\pm1.2661+2\pi k$ in $[0,4\pi)$: $u=1.2661,\ 5.0171,\ 7.5493,\ 11.3003$. Dividing by 2: $\theta=0.6331,\ 2.5086,\ 3.7746,\ 5.6502$ rad.

Check: $\cos(2\times 2.5086)=\cos(5.0171)=0.300$. ✓

Common Pitfall — Calculator and library angle modes

`sin(30)` is -0.988 in radian mode and 0.5 in degree mode. Every mainstream programming library (C `math.h`, Python `math`, NumPy, MATLAB `sin`) works in radians; MATLAB's `sind`, `cosd` and spreadsheet `RADIANS()` exist precisely because people forget. A second trap is `atan` vs `atan2`: using `atan(y/x)` to compute the phase of a phasor gives an answer off by exactly 180° whenever the real part is negative.

Why calculus insists on radians

The decisive reason for radians is the limit

$$\lim_{\theta \to 0} \frac{\sin\theta}{\theta} = 1 \qquad (\theta \text{ in radians}).$$

Proof sketch. For $0 < \theta < \pi/2$ draw the unit circle, the point $P = (\cos\theta, \sin\theta)$ and the point $Q = (1, \tan\theta)$ on the tangent line $x = 1$. The triangle OAP (with $A = (1,0)$), the sector OAP and the triangle OAQ are nested, so their areas satisfy

$$\tfrac{1}{2}\sin\theta \;\le\; \tfrac{1}{2}\theta \;\le\; \tfrac{1}{2}\tan\theta.$$

The middle term uses the radian sector-area formula $\tfrac{1}{2}r^2\theta$. Dividing by $\tfrac{1}{2}\sin\theta > 0$ and taking reciprocals gives $\cos\theta \le \dfrac{\sin\theta}{\theta} \le 1$. As $\theta \to 0^+$, $\cos\theta \to 1$, and the squeeze gives the limit; evenness of $\sin\theta/\theta$ covers $\theta \to 0^-$. $\square$

This limit is exactly what makes $\dfrac{d}{d\theta}\sin\theta = \cos\theta$ (derived in Ch. 4). If θ were measured in degrees, every derivative would carry a factor $\pi/180$: $\frac{d}{d\theta^\circ}\sin\theta^\circ = \frac{\pi}{180}\cos\theta^\circ$. The exponential form $e^{j\theta} = \cos\theta + j\sin\theta$ of Ch. 2 likewise holds only with θ in radians.

Small-angle approximations

The limit above says $\sin\theta \approx \theta$ for small θ. Combined with the Taylor series (developed in Ch. 4)

$$\sin\theta = \theta - \frac{\theta^3}{6} + \cdots, \qquad \cos\theta = 1 - \frac{\theta^2}{2} + \cdots, \qquad \tan\theta = \theta + \frac{\theta^3}{3} + \cdots,$$

we have the working approximations $\sin\theta \approx \tan\theta \approx \theta$ and $\cos\theta \approx 1 - \theta^2/2$, with relative error of $\sin\theta \approx \theta$ about $\theta^2/6$.

Worked Example 3.4 — How small is "small"?

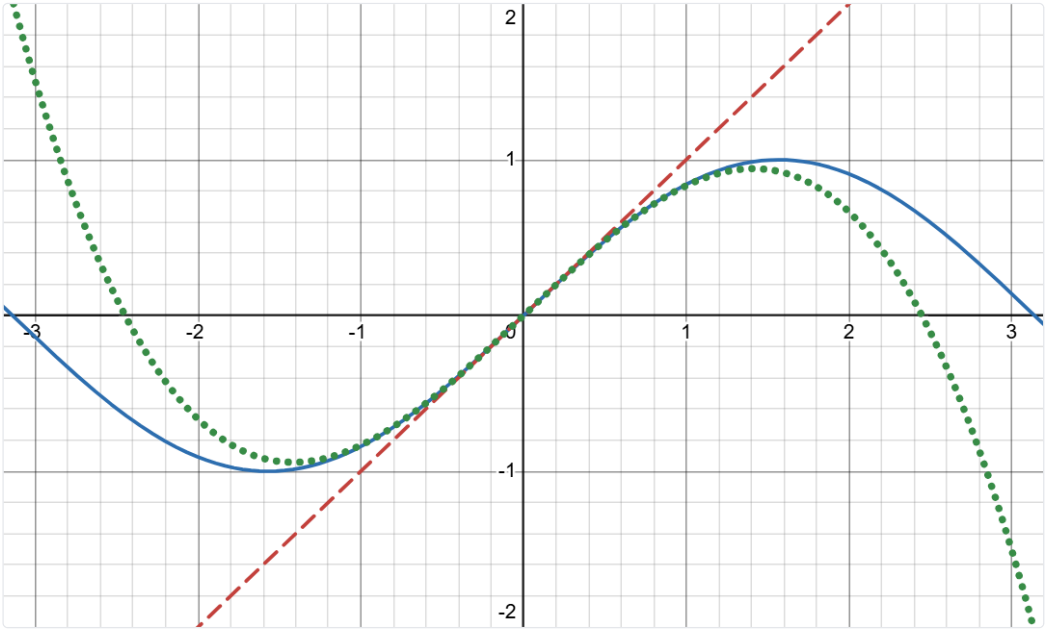
For what angle is the relative error of $\sin\theta \approx \theta$ equal to 1%? Evaluate the approximation at $\theta = 0.1$ rad and at 10°.

Solution. Relative error $\approx \theta^2/6 = 0.01$ gives $\theta \approx \sqrt{0.06} = 0.245$ rad $\approx 14^\circ$.

At $\theta = 0.1$: $\sin 0.1 = 0.099833$, error $0.1 - 0.099833 = 1.67 \times 10^{-4}$, relative error 0.167%, matching $\theta^2/6 = 0.00167$. ✓

At $10^\circ = 0.17453$ rad: $\sin 10^\circ = 0.17365$; relative error $(0.17453 - 0.17365)/0.17365 = 0.51\%$; prediction $0.17453^2/6 = 0.51\%$. ✓

So the approximation is good to about 1 % up to roughly 14°, which is why it is used freely for small phase errors, small pointing errors and small-signal analysis of phase-locked loops.



sin x (blue) with its linear approximation x (red) and cubic approximation $x - x^3/6$ (green): both hug the sine near the origin and peel away for larger angles.

3.2 Sinusoidal functions

Motivation

A sinusoid is the only periodic waveform whose shape survives every linear, time-invariant operation: scaling it, delaying it, differentiating it, integrating it, or adding it to another sinusoid of the same frequency all produce a sinusoid of the same frequency. Resistors scale, capacitors and inductors differentiate or integrate, transmission lines delay — so in a linear circuit driven by a sinusoid, *every* voltage and current is a sinusoid of the source frequency, differing only in amplitude and phase. That is the mathematical reason why AC analysis, phasors (Ch. 2) and frequency response (Ch. 14, Ch. 16) all reduce to the bookkeeping of two numbers per signal.

The general sinusoid

Sinusoid

A **sinusoid** (sinusoidal function of time) is any function of the form

$$x(t) = A\cos(\omega t + \phi) + C,$$

with

- **amplitude** (peak value) $A > 0$, in the units of x;
- **angular frequency** $\omega > 0$, in rad/s;
- **phase** (phase angle, initial phase) ϕ, in rad or degrees, conventionally reduced to $(-\pi, \pi]$;
- **offset** (DC component, mean value) C, in the units of x.

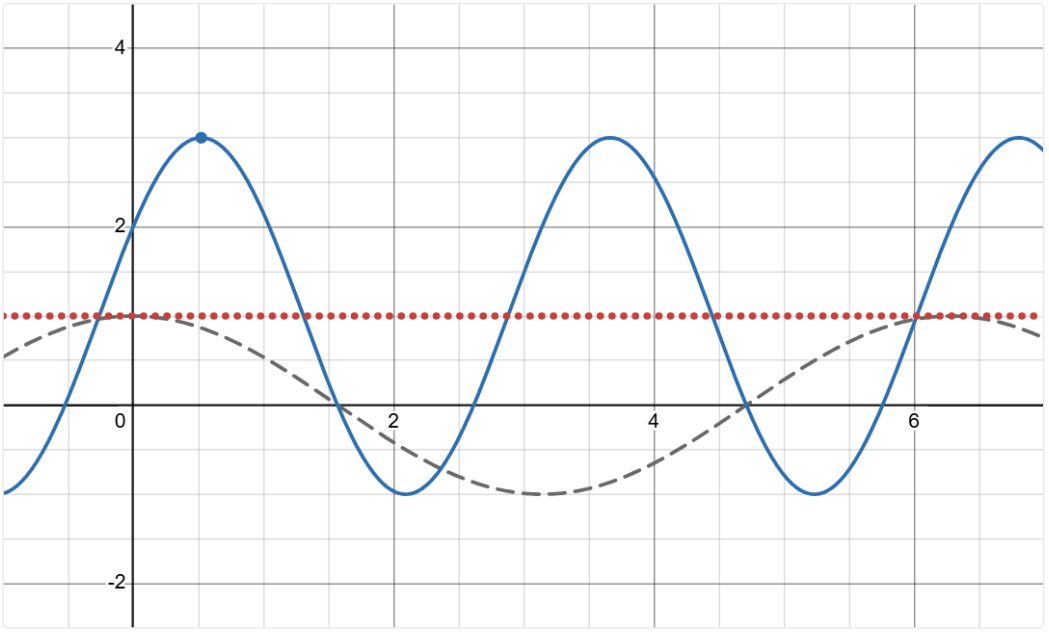
The quantity $\Theta(t) = \omega t + \phi$ is the **instantaneous phase** (total phase angle) at time t; $\phi = \Theta(0)$.

The electrical-engineering convention is to use the **cosine** as the reference sinusoid, because $\cos(\omega t + \phi) = \mathrm{Re}\{e^{j(\omega t + \phi)}\}$ connects directly to the phasor $Ae^{j\phi}$. Many physics and mathematics texts use sin as the reference; both describe the same family of functions, and you must convert before comparing phases (§3.3).

The four parameters act on the graph of $\cos t$ as elementary transformations (Ch. 1):

Parameter	Effect on the graph of $\cos t$
A	vertical stretch: peaks at $C \pm A$
C	vertical shift: the waveform oscillates about the line $x = C$
ω	horizontal compression by the factor ω: one cycle takes $2\pi/\omega$ seconds
ϕ	horizontal shift by $-\phi/\omega$: the peak that $\cos t$ has at $t = 0$ moves to $t = -\phi/\omega$

Note that the horizontal shift is ϕ/ω, **not** ϕ: writing $\cos(\omega t + \phi) = \cos\big(\omega(t + \phi/\omega)\big)$ shows that the graph is shifted left by ϕ/ω seconds. A positive phase moves the waveform *earlier* in time.



The reference $\cos t$ (grey) and $1 + 2\cos(2t - \pi/3)$ (blue): amplitude 2, offset 1 (red line), angular frequency 2 so the period halves to π, and a phase of $-\pi/3$ that delays the first peak to $t = \pi/6$.

Elementary properties

For $x(t) = A\cos(\omega t + \phi) + C$:

- **Extremes.** Maximum $C + A$ where $\omega t + \phi = 2\pi k$; minimum $C - A$ where $\omega t + \phi = \pi + 2\pi k$.

- **Peak-to-peak value.** $x_{pp} = 2A$.
- **Crossings of the mean.** $x = C$ where $\omega t + \phi = \pi/2 + \pi k$, i.e. twice per cycle, a quarter-cycle after each extreme.
- **Period.** $T = 2\pi/\omega$ (§3.4).
- **Boundedness and smoothness.** x is bounded and infinitely differentiable; all of its derivatives are sinusoids of the same frequency.

Converting between sine and cosine and normalising the phase

Because the sine curve is the cosine curve delayed by a quarter cycle, every sinusoid can be written in the cosine reference with a positive amplitude. The conversion rules, read off the unit circle, are:

$$\begin{aligned} \sin\theta &= \cos(\theta - \tfrac{\pi}{2}), & -\sin\theta &= \cos(\theta + \tfrac{\pi}{2}), \\ -\cos\theta &= \cos(\theta \pm \pi), & \cos\theta &= \sin(\theta + \tfrac{\pi}{2}). \end{aligned}$$

In degrees: $\sin\theta = \cos(\theta - 90^\circ)$, $-\sin\theta = \cos(\theta + 90^\circ)$, $-\cos\theta = \cos(\theta \pm 180^\circ)$. A negative amplitude is absorbed into the phase as $\pm 180^\circ$, chosen so that the resulting phase lands in $(-180^\circ, 180^\circ]$.

Standard (canonical) form

A sinusoid is in **standard form** when written $A\cos(\omega t + \phi) + C$ with $A > 0$, $\omega > 0$ and $-\pi < \phi \le \pi$. In standard form the parameters are unique, so two sinusoids are equal if and only if their standard-form parameters agree.

A negative frequency can likewise be removed, using evenness of cosine: $\cos(-\omega t + \phi) = \cos(\omega t - \phi)$.

Worked Example 3.5 — Reading parameters and putting a sinusoid into standard form

Write each signal in standard form and state amplitude, angular frequency, frequency, phase and offset.

(a) $v(t) = -170\sin(377t + 20^\circ)$ V. (b) $i(t) = 5 - 3\cos(2000\pi t - 60^\circ)$ mA. (c) $x(t) = 4\sin(-50t + 0.3)$.

(a) $-\sin\theta = \cos(\theta + 90^\circ)$, so $v = 170\cos(377t + 110^\circ)$ V. Amplitude 170 V, $\omega = 377$ rad/s, $f = 377/2\pi = 60.0$ Hz, phase $+110^\circ$, offset 0. (This is a 120 V RMS, 60 Hz supply: $170/\sqrt{2} = 120.2$ V.)

(b) $-\cos\theta = \cos(\theta + 180^\circ)$: $-3\cos(2000\pi t - 60^\circ) = 3\cos(2000\pi t + 120^\circ)$. So $i = 5 + 3\cos(2000\pi t + 120^\circ)$ mA: amplitude 3 mA, $\omega = 2000\pi = 6283$ rad/s, $f = 1$ kHz, phase 120°, offset 5 mA. The current never reverses: it swings between 2 mA and 8 mA.

(c) Sine is odd: $4\sin(-50t + 0.3) = -4\sin(50t - 0.3) = 4\cos(50t - 0.3 + \pi/2) = 4\cos(50t + 1.2708)$. Amplitude 4, $\omega = 50$ rad/s, $f = 7.958$ Hz, phase 1.2708 rad $= 72.8^\circ$.

Check (c) at $t = 0.01$: original $4\sin(-0.5 + 0.3) = 4\sin(-0.2) = -0.7947$; standard form $4\cos(0.5 + 1.2708) = 4\cos(1.7708) = -0.7947$. ✓

Amplitude–phase form: combining $a\cos\omega t + b\sin\omega t$

Solutions of differential equations (Ch. 11, Ch. 12) and Fourier series (Ch. 15) naturally come out as $a\cos\omega t + b\sin\omega t$. For interpretation we want a single sinusoid.

Amplitude–phase form

For real a, b not both zero,

$$a\cos\omega t + b\sin\omega t = R\cos(\omega t - \alpha), \qquad R = \sqrt{a^2 + b^2}, \quad \alpha = \operatorname{atan2}(b, a).$$

Equivalently $a\cos\omega t + b\sin\omega t = R\cos(\omega t + \phi)$ with $\phi = -\alpha = \operatorname{atan2}(-b, a)$. Conversely, $R\cos(\omega t + \phi) = a\cos\omega t + b\sin\omega t$ with $a = R\cos\phi$, $b = -R\sin\phi$.

Derivation 1 (compound angle). Expand the right-hand side with $\cos(X - Y) = \cos X\cos Y + \sin X\sin Y$ (proved in §3.6):

$$R\cos(\omega t - \alpha) = R\cos\alpha\,\cos\omega t + R\sin\alpha\,\sin\omega t.$$

This equals $a\cos\omega t + b\sin\omega t$ for all t if and only if $R\cos\alpha = a$ and $R\sin\alpha = b$ (compare values at $t = 0$ and at $\omega t = \pi/2$). These say that (a, b) is the point with polar coordinates (R, α), so $R = \sqrt{a^2 + b^2}$ and $\alpha = \operatorname{atan2}(b, a)$.

Derivation 2 (phasor). Since $\cos\omega t = \operatorname{Re}\{e^{j\omega t}\}$ and $\sin\omega t = \operatorname{Re}\{-je^{j\omega t}\}$,

$$a\cos\omega t + b\sin\omega t = \operatorname{Re}\{(a - jb)e^{j\omega t}\}.$$

The phasor is $a - jb = Re^{-j\alpha}$, so the sum is $\operatorname{Re}\{Re^{j(\omega t - \alpha)}\} = R\cos(\omega t - \alpha)$. The phasor route makes the sign of the phase automatic: **the sine coefficient enters the phasor with a minus sign.**

Worked Example 3.6 — Amplitude–phase conversions

(a) $3\cos\omega t + 4\sin\omega t$. (b) $5\cos\omega t - 12\sin\omega t$. (c) $-2\sqrt{3}\cos\omega t - 2\sin\omega t$.

(a) $R = \sqrt{9+16} = 5$, $\alpha = \operatorname{atan2}(4,3) = 53.13°$: $3\cos\omega t + 4\sin\omega t = 5\cos(\omega t - 53.13°)$.

(b) $R = \sqrt{25+144} = 13$, $\alpha = \operatorname{atan2}(-12,5) = -67.38°$: $5\cos\omega t - 12\sin\omega t = 13\cos(\omega t + 67.38°)$.

(c) $R = \sqrt{12+4} = 4$. Here $a<0, b<0$, so the point is in quadrant III: $\alpha = \operatorname{atan2}(-2,-2\sqrt{3}) = -150°$. Hence $-2\sqrt{3}\cos\omega t - 2\sin\omega t = 4\cos(\omega t + 150°)$.

Check (c): $4\cos(\omega t + 150°) = 4\cos 150°\cos\omega t - 4\sin 150°\sin\omega t = 4(-0.8660)\cos\omega t - 4(0.5)\sin\omega t = -3.464\cos\omega t - 2\sin\omega t$. ✓ A naive $\arctan(b/a) = \arctan(1/\sqrt{3}) = 30°$ would have given $4\cos(\omega t - 30°)$ — the negative of the correct answer.

Derivatives and integrals of sinusoids

Using $\frac{d}{dt}\cos(\omega t+\phi) = -\omega\sin(\omega t+\phi)$ and the conversion rule $-\sin\theta = \cos(\theta+\pi/2)$:

$$\frac{d}{dt}\big[A\cos(\omega t+\phi)\big] = \omega A\cos\big(\omega t+\phi+\tfrac{\pi}{2}\big), \qquad \int A\cos(\omega t+\phi)\,dt = \frac{A}{\omega}\cos\big(\omega t+\phi-\tfrac{\pi}{2}\big) + K.$$

Differentiation multiplies the amplitude by ω and advances the phase by $90°$; integration divides by ω and retards the phase by $90°$. In phasor language these are multiplication and division by $j\omega$, since $j = e^{j\pi/2}$. Differentiating twice gives back the original sinusoid times $-\omega^2$, so every sinusoid of angular frequency ω (without offset) satisfies

$$\ddot{x} + \omega^2 x = 0,$$

the equation of simple harmonic motion (§3.5).

Worked Example 3.7 — Capacitor current from capacitor voltage

A $C = 1\ \mu$F capacitor has $v(t) = 10\cos(1000t + 30°)$ V. Find $i(t) = C\,dv/dt$ and state which leads.

Solution. $dv/dt = 1000 \times 10\cos(1000t + 30° + 90°) = 10^4\cos(1000t + 120°)$ V/s. So $i = 10^{-6} \times 10^4\cos(1000t + 120°) = 10\cos(1000t + 120°)$ mA.

The current phase ($120°$) exceeds the voltage phase ($30°$) by $90°$: **current leads voltage by** $90°$ in a capacitor. The amplitude ratio $V/I = 10/0.01 = 1000\ \Omega = 1/(\omega C)$ is the capacitive reactance. ✓

Mean, RMS and related measures

The **mean** (average) of a periodic function over one period T is $\bar{x} = \frac{1}{T}\int_0^T x\,dt$, and the **root-mean-square** value is $X_{\text{rms}} = \sqrt{\frac{1}{T}\int_0^T x^2\,dt}$. RMS matters because the average power delivered to a resistor is $\overline{v^2}/R = V_{\text{rms}}^2/R$: the RMS value is the DC value that would deliver the same power. The general theory of mean and RMS values belongs to Ch. 5; here we need the sinusoidal case.

Mean and RMS of a sinusoid

Over any whole number of periods,

$$\overline{\cos(\omega t+\phi)} = 0, \qquad \overline{\cos^2(\omega t+\phi)} = \tfrac{1}{2}.$$

Hence for $x = A\cos(\omega t+\phi) + C$:

$$\bar{x} = C, \qquad X_{\text{rms}} = \sqrt{C^2 + \tfrac{1}{2}A^2}.$$

For a pure sinusoid ($C = 0$), $X_{\text{rms}} = A/\sqrt{2} \approx 0.7071A$.

Proof. The integral of $\cos(\omega t+\phi)$ over one period is $\frac{1}{\omega}[\sin(\omega t+\phi)]_0^T = \frac{1}{\omega}[\sin(2\pi+\phi) - \sin\phi] = 0$. For the square, use the power-reduction identity $\cos^2\theta = \frac{1}{2}(1+\cos 2\theta)$ (§3.6): the constant $\frac{1}{2}$ averages to $\frac{1}{2}$ and $\cos 2\theta$ averages to 0. Then

$$\overline{x^2} = \overline{C^2 + 2AC\cos(\cdot) + A^2\cos^2(\cdot)} = C^2 + 0 + \tfrac{1}{2}A^2. \qquad \square$$

There is a neat way to see $\overline{\cos^2} = \frac{1}{2}$ without any identity: $\cos^2$ and $\sin^2$ have the same graph shifted by a quarter period, so they have the same average over a full period; and they add to 1. Each must average $\frac{1}{2}$.

Two further ratios describe waveform shape and are printed on meter datasheets:

$$\text{crest factor} = \frac{X_{\text{peak}}}{X_{\text{rms}}}, \qquad \text{form factor} = \frac{X_{\text{rms}}}{\overline{|x|}}.$$

For a pure sinusoid, the rectified average is $\overline{|A\cos\omega t|} = 2A/\pi$ (computed in §3.8), so crest factor $= \sqrt{2} \approx 1.414$ and form factor $= \dfrac{A/\sqrt{2}}{2A/\pi} = \dfrac{\pi}{2\sqrt{2}} \approx 1.1107$.

Worked Example 3.8 — Mains voltage and a biased sinusoid

(a) A 230 V (RMS), 50 Hz supply: what is the peak voltage, the peak-to-peak voltage, and the instantaneous voltage 3 ms after a positive peak? (b) A sensor output is $v(t) = 2 + 3\sin(100t)$ V. Find its mean, its RMS value and the power it would deliver to a 50 Ω resistor.

(a) $V_p = \sqrt{2} \times 230 = 325.3$ V; $V_{pp} = 650.5$ V. With the peak at $t = 0$, $v = 325.3\cos(2\pi \cdot 50\,t)$. At $t = 3$ ms: $\omega t = 2\pi(50)(0.003) = 0.9425$ rad $= 54°$, so $v = 325.3\cos 54° = 191.2$ V.

(b) Mean $= 2$ V. RMS $= \sqrt{2^2 + 3^2/2} = \sqrt{8.5} = 2.915$ V. Power $= V_{\text{rms}}^2/R = 8.5/50 = 0.170$ W. Note that the DC and AC powers simply add: $2^2/50 + (3/\sqrt{2})^2/50 = 0.08 + 0.09 = 0.17$ W — a first glimpse of the orthogonality result of §3.7.

Worked Example 3.9 — Recovering a sinusoid from two observations

An oscilloscope shows a sinusoid with a maximum of 7 V at $t = 2$ ms and the next minimum of -1 V at $t = 7$ ms. Write it in standard form.

Offset and amplitude. $C = \frac{7+(-1)}{2} = 3$ V, $A = \frac{7-(-1)}{2} = 4$ V.

Frequency. A maximum and the next minimum are half a period apart: $T/2 = 5$ ms, so $T = 10$ ms, $f = 100$ Hz, $\omega = 200\pi = 628.3$ rad/s.

Phase. A maximum occurs where $\omega t + \phi = 0 \pmod{2\pi}$: $\phi = -\omega(0.002) = -0.4\pi$ rad $= -72°$.

Result. $v(t) = 3 + 4\cos(200\pi t - 0.4\pi)$ V.

Check: at $t = 7$ ms, $200\pi(0.007) - 0.4\pi = 1.0\pi$, $\cos\pi = -1$, $v = 3 - 4 = -1$ V. ✓

Common Pitfall — Peak, peak-to-peak or RMS?

"10 V AC" is ambiguous. Function generators often display peak-to-peak (into a matched load!), multimeters display RMS, and the formula $A\cos(\omega t + \phi)$ uses peak. Convert explicitly: $V_{pp} = 2V_p = 2\sqrt{2}\,V_{\text{rms}} \approx 2.83\,V_{\text{rms}}$. Also note that the $1/\sqrt{2}$ rule is for *sinusoids only*: an "average-responding, RMS-calibrated" meter multiplies the rectified average by 1.1107 and is wrong for every non-sinusoidal waveform.

3.3 Phase and phase difference

Motivation

Amplitude tells you *how big* a sinusoid is; phase tells you *when* it peaks. In a single isolated signal the phase is arbitrary — it depends only on where you chose $t = 0$. What carries physical meaning is the **phase difference** between two sinusoids of the same frequency: the angle by which the current in a motor lags its voltage (which fixes the power factor), the phase shift through an amplifier (which decides whether a feedback loop oscillates), the phase between the three lines of a power system, or the phase between signals at two antennas (which steers a beam). This section makes "leads", "lags" and "phase difference" precise and shows how to compute them without sign errors.

Phase of a single sinusoid

For $x(t) = A\cos(\omega t + \phi)$ the phase ϕ is the instantaneous phase at $t = 0$. Moving the time origin by t_0 (that is, writing $t = t' + t_0$) changes the phase to $\phi + \omega t_0$ but leaves the waveform itself untouched. Hence:

- the phase of one sinusoid is meaningful only relative to a stated time reference (or a reference signal);
- phases are defined modulo 2π (360°), and are normally reported in the principal interval $(-180°, 180°]$ — this reduction is called **phase wrapping**.

Phase difference, lead and lag

Phase difference, lead, lag

Let $x_1(t) = A_1\cos(\omega t + \phi_1)$ and $x_2(t) = A_2\cos(\omega t + \phi_2)$ have the **same** angular frequency ω and be written in standard form ($A_1, A_2 > 0$, both cosines). The **phase difference** of x_1 relative to x_2 is

$$\Delta\phi = \phi_1 - \phi_2, \qquad \text{reduced to } (-\pi, \pi].$$

If $0 < \Delta\phi < \pi$, x_1 **leads** x_2 by $\Delta\phi$; if $-\pi < \Delta\phi < 0$, x_1 **lags** x_2 by $|\Delta\phi|$. The signals are **in phase** if $\Delta\phi = 0$, in **quadrature** if $\Delta\phi = \pm\pi/2$, and in **antiphase** (phase opposition) if $\Delta\phi = \pi$.

"Leads" means *happens earlier*: if x_1 leads x_2 by $\Delta\phi$, each feature of x_1 (peak, upward zero crossing) occurs $\Delta\phi/\omega$ seconds before the corresponding feature of x_2. Indeed

$$x_1(t) = A_1\cos\big(\omega t + \phi_2 + \Delta\phi\big) = \frac{A_1}{A_2}\,x_2\!\left(t + \frac{\Delta\phi}{\omega}\right),$$

so x_1 is a scaled copy of x_2 advanced by $\Delta\phi/\omega$.

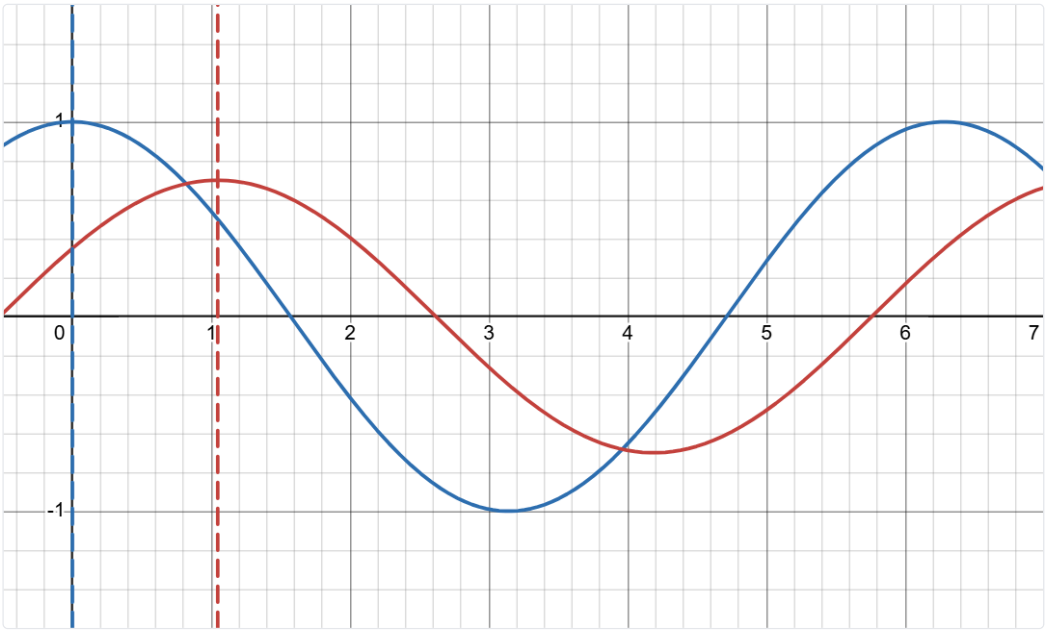
The restriction to $(-\pi, \pi]$ matters: a lead of 270° is the same physical relationship as a lag of 90°, and engineers always quote the smaller one. A phase difference of exactly 180° is neither a lead nor a lag.

Phase difference as time delay

For two sinusoids of the same frequency f (period T), a phase difference $\Delta\phi$ corresponds to a time shift

$$\Delta t = \frac{\Delta\phi}{\omega} = \frac{\Delta\phi}{2\pi f} = \frac{\Delta\phi_{\text{deg}}}{360^\circ}\,T.$$

Conversely a pure time delay τ produces a phase lag $\omega\tau$, which grows linearly with frequency.



$\cos t$ (blue) leads $0.7\cos(t-\pi/3)$ (red) by 60°: every red peak arrives $\pi/3$ rad of ωt after the corresponding blue peak (dashed lines).

Worked Example 3.10 — Comparing phases written in different forms

$v(t) = 10\sin(\omega t + 30^\circ)$ V and $i(t) = -2\cos(\omega t - 20^\circ)$ A. Find the phase of i relative to v.

Step 1 — convert both to standard cosine form. $v = 10\cos(\omega t + 30^\circ - 90^\circ) = 10\cos(\omega t - 60^\circ)$. $i = 2\cos(\omega t - 20^\circ + 180^\circ) = 2\cos(\omega t + 160^\circ)$.

Step 2 — subtract. $\Delta\phi = \phi_i - \phi_v = 160^\circ - (-60^\circ) = 220^\circ$.

Step 3 — wrap. $220^\circ - 360^\circ = -140^\circ$. So i **lags** v **by** 140° (equivalently, v leads i by 140°).

Check. Comparing the original expressions directly (30° vs -20°, "v leads by 50°") is wrong on two counts: one is a sine and the other a cosine, and i has a negative amplitude. Numerically, v has a positive peak at $\omega t = 60^\circ$; i has a positive peak at $\omega t = -160^\circ \equiv 200^\circ$, which is 140° later. ✓

Worked Example 3.11 — From phase to time and back

(a) Two 50 Hz voltages differ in phase by 36°. What is the time between their peaks? (b) A coaxial cable delays a signal by 5 ns per metre. What phase lag does 2 m of cable introduce at 10 MHz, at 100 MHz and at 150 MHz? (c) An RC low-pass filter with $R = 1\ \text{k}\Omega$, $C = 100$ nF has output phase $-\arctan(\omega RC)$ relative to its input. At 1 kHz, what are the phase lag and the corresponding time delay?

(a) $T = 20$ ms; $\Delta t = (36/360)\times 20 = 2$ ms.

(b) $\tau = 10$ ns. Lag $= 360^\circ f\tau$: at 10 MHz, 36°; at 100 MHz, 360° (indistinguishable from 0°!); at 150 MHz, $540^\circ \equiv 180^\circ$.

(c) $\omega RC = 2\pi(1000)(1000)(10^{-7}) = 0.6283$. Lag $= \arctan 0.6283 = 32.14^\circ$. Delay $= \frac{32.14}{360}\times 1\text{ ms} = 89.3\ \mu\text{s}$.

Interpretation. Part (b) shows that a measured phase cannot distinguish delays that differ by whole periods; part (c) shows that the phase lag of a filter, unlike that of a cable, is *not* proportional to frequency, so different frequencies are delayed by different amounts.

Common Pitfall — Phase ambiguity

A phase measurement determines a time delay only modulo one period. Phase meters, network analysers and FFT routines report a wrapped phase in $(-180^\circ, 180^\circ]$; to recover total delay you must *unwrap* — track the phase continuously as frequency is swept and add 360° at each jump. NumPy's `unwrap` and MATLAB's `unwrap` do exactly this, assuming that consecutive samples differ by less than 180°.

Adding sinusoids of the same frequency

The sum of two sinusoids of the same frequency is a sinusoid of that frequency. The cleanest proof uses phasors (Ch. 2):

$$A_1\cos(\omega t + \phi_1) + A_2\cos(\omega t + \phi_2) = \text{Re}\left\{(A_1 e^{j\phi_1} + A_2 e^{j\phi_2})e^{j\omega t}\right\} = R\cos(\omega t + \psi),$$

where $Re^{j\psi} = A_1 e^{j\phi_1} + A_2 e^{j\phi_2}$. Writing out real and imaginary parts gives explicit formulas:

Sum of two same-frequency sinusoids

$$R^2 = A_1^2 + A_2^2 + 2A_1A_2\cos(\phi_1 - \phi_2), \qquad \psi = \operatorname{atan2}\left(A_1\sin\phi_1 + A_2\sin\phi_2,\ A_1\cos\phi_1 + A_2\cos\phi_2\right).$$

The magnitude formula is the law of cosines applied to the phasor triangle. It shows that the resultant amplitude depends only on the *phase difference*: it ranges from $A_1 + A_2$ (in phase, **constructive** interference) down to $|A_1 - A_2|$ (antiphase, **destructive** interference), and equals $\sqrt{A_1^2 + A_2^2}$ in quadrature.

Worked Example 3.12 — Adding two currents at a node

Two branch currents enter a node: $i_1 = 10\cos(\omega t + 30^\circ)$ A and $i_2 = 6\cos(\omega t - 45^\circ)$ A. Find $i_1 + i_2$.

Rectangular components. $10\angle 30^\circ = 8.660 + j5.000$; $6\angle -45^\circ = 4.243 - j4.243$. Sum: $12.903 + j0.757$.

Polar. $R = \sqrt{12.903^2 + 0.757^2} = 12.925$; $\psi = \operatorname{atan2}(0.757, 12.903) = 3.36^\circ$.

Result. $i_1 + i_2 = 12.93\cos(\omega t + 3.36^\circ)$ A.

Check with the law of cosines: $R^2 = 100 + 36 + 120\cos 75^\circ = 136 + 31.06 = 167.06$, $R = 12.925$. ✓ Also at $\omega t = 0.3$ rad the original sum evaluates to 12.103 and the result to 12.103. ✓ Note that the resultant (12.9 A) is less than the arithmetic sum (16 A) — an ammeter reading RMS values on each branch does *not* add.

Three-phase sets

A **balanced three-phase set** consists of three sinusoids of equal amplitude and frequency displaced by 120°:

$$v_a = V\cos\omega t, \qquad v_b = V\cos(\omega t - 120^\circ), \qquad v_c = V\cos(\omega t + 120^\circ).$$

Balanced sets sum to zero

$v_a + v_b + v_c = 0$ for all t. More generally, for any integer $N \geq 2$, $\displaystyle\sum_{k=0}^{N-1}\cos\left(\omega t + \phi - \frac{2\pi k}{N}\right) = 0$.

Proof. The phasor sum is $V\sum_{k=0}^{N-1} e^{-j2\pi k/N}$, a geometric series with ratio $r = e^{-j2\pi/N} \neq 1$: it equals $V\frac{1-r^N}{1-r} = 0$ because $r^N = e^{-j2\pi} = 1$. (These are the N-th roots of unity of Ch. 2, which sum to zero.) □

This is why a balanced three-phase load needs no neutral return current.

Worked Example 3.13 — Line-to-line voltage

With v_a and v_b as above, find the line-to-line voltage $v_{ab} = v_a - v_b$.

Phasors. $V\angle 0^\circ - V\angle -120^\circ = V\left[1 - \left(-\tfrac{1}{2} - j\tfrac{\sqrt{3}}{2}\right)\right] = V\left(\tfrac{3}{2} + j\tfrac{\sqrt{3}}{2}\right)$.

Magnitude $V\sqrt{9/4 + 3/4} = \sqrt{3}\,V$; angle $\arctan\left(\frac{\sqrt{3}/2}{3/2}\right) = 30^\circ$.

Result. $v_{ab} = \sqrt{3}\,V\cos(\omega t + 30^\circ)$: the line voltage is $\sqrt{3}$ times the phase voltage and leads v_a by 30°. For a 230 V phase supply, the line voltage is $\sqrt{3} \times 230 = 398 \approx 400$ V.

Measuring phase: zero crossings and Lissajous figures

Two practical measurement methods follow directly from the definitions.

Zero-crossing timing. Measure the time Δt between corresponding upward zero crossings of the two signals and the period T; then $|\Delta\phi| = 360^\circ\,\Delta t/T$. The signal whose crossing comes first leads.

Lissajous (X–Y) method. Apply $x = X\sin\omega t$ to the horizontal input of an oscilloscope and $y = Y\sin(\omega t + \phi)$ to the vertical. Eliminating t gives an ellipse:

$$\frac{x^2}{X^2} - \frac{2xy\cos\phi}{XY} + \frac{y^2}{Y^2} = \sin^2\phi.$$

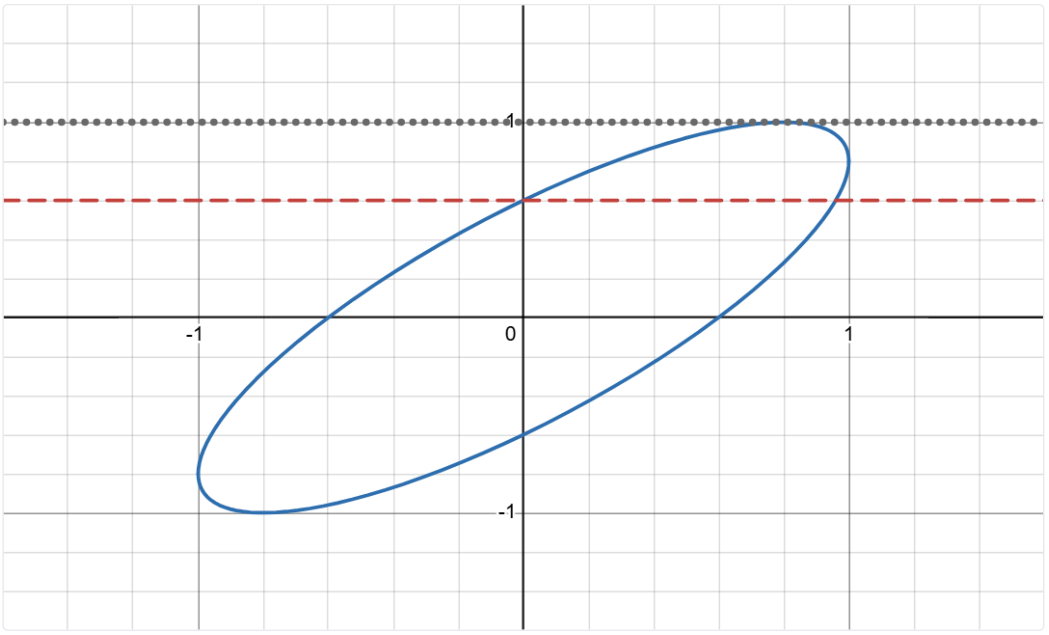
Derivation. Write $s = \sin\omega t = x/X$ and expand $y/Y = s\cos\phi + \cos\omega t\,\sin\phi$, so $\cos\omega t\,\sin\phi = y/Y - (x/X)\cos\phi$. Squaring and using $\cos^2\omega t = 1 - x^2/X^2$:

$$\left(1 - \frac{x^2}{X^2}\right)\sin^2\phi = \frac{y^2}{Y^2} - \frac{2xy}{XY}\cos\phi + \frac{x^2}{X^2}\cos^2\phi,$$

which rearranges to the ellipse equation. Setting $x = 0$ gives $y = \pm Y|\sin\phi|$. Hence

$$|\sin\phi| = \frac{y_0}{Y},$$

where y_0 is the height at which the ellipse crosses the vertical axis and Y is its maximum height. The ellipse collapses to a line of positive slope at $\phi = 0$, to a line of negative slope at $\phi = 180^\circ$, and becomes an axis-aligned ellipse at $\phi = \pm 90^\circ$. The sign of ϕ (lead or lag) cannot be read from a static picture; the tilt distinguishes $|\phi| < 90^\circ$ (major axis in quadrants I/III) from $|\phi| > 90^\circ$ (quadrants II/IV).



Lissajous ellipse for equal amplitudes and $\phi = 36.87°$: it crosses the vertical axis at $y_0 = 0.6$ (red), its maximum height is $Y = 1$ (grey), and $\sin\phi = 0.6$.

Worked Example 3.14 — Reading a Lissajous figure

An X–Y display shows an ellipse tilted into quadrants I and III, with maximum height 5.0 divisions and vertical-axis intercepts at ± 3.0 divisions. Find the magnitude of the phase difference.

Solution. $|\sin\phi| = 3.0/5.0 = 0.6$, so $|\phi| = \arcsin 0.6 = 36.87°$ or $180° - 36.87° = 143.13°$. The tilt into quadrants I/III means $|\phi| < 90°$: $|\phi| = 36.9°$.

Check: at $x = 0$ we have $\sin\omega t = 0$, $\cos\omega t = \pm 1$, so $y = \pm Y \sin\phi = \pm 5(0.6) = \pm 3$. ✓

Engineering Note — Phase in feedback and power systems

Two phase numbers dominate EE practice. In power systems the **power factor** $\cos\Delta\phi$ (with $\Delta\phi$ the voltage–current phase difference) sets how much of the volt-amperes delivered is real power: $P = V_{\text{rms}} I_{\text{rms}} \cos\Delta\phi$ — a consequence of the product-to-sum identity in §3.6. In feedback amplifiers the **phase margin** is $180°$ plus the loop phase at the frequency where loop gain is 1; when the loop phase reaches $-180°$ negative feedback has become positive feedback and the circuit oscillates. Both are phase *differences* of same-frequency sinusoids.

3.4 Period, frequency, and angular frequency

Motivation

"How often does it repeat?" is the first question asked of any signal. The answer comes in three interchangeable currencies — period in seconds, frequency in hertz, angular frequency in radians per second — and each is natural in a different place: the oscilloscope shows periods, the spectrum analyser and the datasheet show hertz, and every formula derived from calculus or phasors uses rad/s. A factor of 2π lost between them is the most common numerical error in circuit and signal calculations. Beyond single sinusoids, we need to know when a *combination* of sinusoids repeats, because that decides whether Fourier series (Ch. 15) or Fourier transforms (Ch. 16) is the right tool, and how sampled (discrete-time) sinusoids differ from continuous ones.

Periodic functions

Periodic function, fundamental period

A function $x(t)$ defined for all real t is **periodic** with period $T > 0$ if

$$x(t + T) = x(t) \qquad \text{for all } t.$$

If T is a period then so is kT for every positive integer k. The smallest positive period, when one exists, is the **fundamental period** T_0; its reciprocal $f_0 = 1/T_0$ is the **fundamental frequency** and $\omega_0 = 2\pi f_0$ the **fundamental angular frequency**.

A constant function satisfies $x(t + T) = x(t)$ for every $T > 0$ and so has no smallest period; by convention it is periodic with any period but has no fundamental period. (Its "frequency" is zero, which is why a constant is called the DC component.)

Period, frequency and angular frequency of a sinusoid

For $x(t) = A\cos(\omega t + \phi)$, the cosine repeats when its argument advances by 2π. Solving $\omega(t + T) + \phi = \omega t + \phi + 2\pi$ gives

$$T = \frac{2\pi}{\omega}, \qquad f = \frac{1}{T} = \frac{\omega}{2\pi}, \qquad \omega = 2\pi f = \frac{2\pi}{T}.$$

Quantity	Symbol	Unit	Meaning
period	T	s	time for one cycle
frequency	f	$\text{Hz} = \text{s}^{-1}$ (cycles per second)	cycles per unit time
angular frequency	ω	rad/s	rate of change of phase, $\omega = d\Theta/dt$

Dimensionally $[f] = [\omega] = \text{T}^{-1}$; the radian is dimensionless, so the unit "rad/s" is formally s^{-1} as well. This is exactly why a factor of 2π is never caught by a dimensional check — you must track it by keeping the units "Hz" and "rad/s" distinct in your working. The product ωt is dimensionless, as the argument of any function like cos must be.

Worked Example 3.15 — Conversions in practice

(a) North American mains has $\omega = 377$ rad/s: find f and T. (b) A Wi-Fi carrier is at 2.4 GHz: find T and ω. (c) A scope shows 4 complete cycles across 10 divisions at 50 μs/div: find f and ω.

(a) $f = 377/2\pi = 60.0$ Hz, $T = 1/60 = 16.67$ ms.

(b) $T = 1/(2.4 \times 10^9) = 416.7$ ps, $\omega = 2\pi \times 2.4 \times 10^9 = 1.508 \times 10^{10}$ rad/s.

(c) 10 divisions $= 500$ μs contains 4 periods, so $T = 125$ μs, $f = 8$ kHz, $\omega = 5.027 \times 10^4$ rad/s.

Time scaling and related functions

If $x(t)$ has fundamental period T_0, then $x(at)$ with $a > 0$ has fundamental period T_0/a: compressing time by a multiplies all frequencies by a. Several functions built from sinusoids have periods *shorter* than the sinusoid they come from:

Function	Fundamental period	Reason
$\cos(\omega t + \phi)$	$2\pi/\omega$	definition
$\tan \omega t$	π/ω	$\tan(\theta + \pi) = \tan\theta$
$\lvert\cos \omega t\rvert$	π/ω	full-wave rectification removes the sign
$\cos^2 \omega t$	π/ω	$\cos^2\theta = \frac{1}{2} + \frac{1}{2}\cos 2\theta$
$\cos \omega t \, \sin \omega t$	π/ω	$= \frac{1}{2}\sin 2\omega t$

The full-wave-rectifier entry is the reason the ripple on a 50 Hz full-wave rectifier's output is at 100 Hz.

When is a sum of sinusoids periodic?

Periodicity of a sum

Let x_1 and x_2 be periodic with fundamental periods T_1 and T_2.

1. If $T_1/T_2 = p/q$ is **rational**, with p, q coprime positive integers, then $x_1 + x_2$ is periodic with period $T = qT_1 = pT_2$ (the least common multiple of T_1 and T_2). Equivalently, the common frequency is the greatest common divisor of f_1 and f_2.
2. For sums of sinusoids, if T_1/T_2 is **irrational**, $x_1 + x_2$ is **not** periodic.

In case 1, T is the fundamental period of the sum unless cancellations occur (for example $x_2 = -x_1$); for a sum of sinusoids with nonzero amplitudes and distinct frequencies the fundamental frequency is exactly $\gcd(f_1, f_2)$.

Proof of 1. qT_1 is a whole number of periods of x_1, and $pT_2 = qT_1$ is a whole number of periods of x_2, so both terms, hence their sum, repeat after T.

Proof of 2 (for $\cos \omega_1 t + \cos \omega_2 t$). The sum equals its maximum value 2 at $t = 0$. It equals 2 at any other time t only if both cosines equal 1, i.e. $\omega_1 t = 2\pi m$ and $\omega_2 t = 2\pi n$ for integers m, n with $t \neq 0$. Dividing, $\omega_1/\omega_2 = m/n$ would be rational — contradiction. So the value 2 is attained only at $t = 0$. A periodic function attains each of its values infinitely often (at $t, t + T, t + 2T, \ldots$), so the sum is not periodic. □

Such sums are called **almost periodic** (or quasi-periodic): they come arbitrarily close to repeating but never do exactly. In practice, any measured frequency ratio is rational to within instrument precision, but the theoretical distinction matters: Fourier *series* need exact periodicity, whereas the Fourier *transform* (Ch. 16) handles any combination of frequencies.

Worked Example 3.16 — Fundamental periods of combinations

Find the fundamental period, or show none exists.

(a) $x(t) = \sin 3t + \cos 4.5t$. (b) $v(t) = \cos(100\pi t) + 0.2\cos(300\pi t) + 0.1\cos(500\pi t)$. (c) $y(t) = \cos t + \cos\sqrt{2}\,t$. (d) $p(t) = \cos^2(100\pi t)$.

(a) $\omega_1 = 3$, $\omega_2 = 4.5$; ratio $3/4.5 = 2/3$, rational. The common angular frequency is $\gcd(3, 4.5) = 1.5$ rad/s ($3 = 2 \times 1.5$, $4.5 = 3 \times 1.5$). So $T_0 = 2\pi/1.5 = 4\pi/3 \approx 4.189$ s. Check: in $4\pi/3$ s the first term completes $3 \times \frac{4\pi}{3}/2\pi = 2$ cycles and the second $4.5 \times \frac{4\pi}{3}/2\pi = 3$ cycles. ✓

(b) Frequencies 50, 150, 250 Hz; gcd $= 50$ Hz, so $T_0 = 20$ ms. This is a 50 Hz waveform with 3rd and 5th harmonic distortion (§3.5, §3.7).

(c) $1/\sqrt{2}$ is irrational: **not periodic**.

(d) $\cos^2(100\pi t) = \frac{1}{2} + \frac{1}{2}\cos(200\pi t)$: fundamental period $2\pi/(200\pi) = 10$ ms — half the 20 ms period of $\cos(100\pi t)$. This is the instantaneous power in a resistor fed from 50 Hz: it pulses at 100 Hz.

Spatial periodicity: wavelength and wavenumber

Signals on transmission lines, in waveguides and in free space vary in both time and position. A sinusoidal wave travelling in the $+z$ direction is

$$u(z,t) = A\cos(\omega t - kz + \phi).$$

At a fixed position it is a sinusoid in time with angular frequency ω; at a fixed instant it is a sinusoid in space with **wavenumber** (spatial angular frequency) k in rad/m and spatial period, the **wavelength**,

$$\lambda = \frac{2\pi}{k}.$$

A point of constant phase satisfies $\omega t - kz = \text{const}$, so it moves at the **phase velocity** $v_p = dz/dt = \omega/k$. Combining,

$$v_p = \frac{\omega}{k} = f\lambda, \qquad \lambda = \frac{v_p}{f}.$$

The sign convention is worth memorising: $\omega t - kz$ travels towards $+z$; $\omega t + kz$ travels towards $-z$.

Worked Example 3.17 — Wavelengths and electrical length

(a) Find the free-space wavelength ($v_p = c = 3.00\times10^8$ m/s) at 50 Hz and at 2.4 GHz. (b) A PCB trace has $v_p = 0.5c$. How long is a trace that is a quarter-wavelength at 2.4 GHz, and what phase shift does a 10 mm trace introduce at that frequency?

(a) At 50 Hz: $\lambda = 3\times10^8/50 = 6000$ km. At 2.4 GHz: $\lambda = 0.125$ m $= 12.5$ cm.

(b) On the trace $\lambda = 0.5\times12.5 = 6.25$ cm, so $\lambda/4 = 15.6$ mm. A 10 mm trace has $kz = 2\pi(10/62.5) = 1.005$ rad $= 57.6^\circ$ of phase lag.

Interpretation. A circuit is "electrically small" — phase differences across it negligible, lumped-element analysis valid — when its dimensions are a small fraction of λ. Mains wiring is tiny compared with 6000 km; a 10 mm PCB trace at 2.4 GHz is not small at all.

Discrete-time sinusoids

When a sinusoid $\cos(\omega t + \phi)$ is sampled every T_s seconds, the samples are

$$x[n] = \cos(\omega n T_s + \phi) = \cos(\Omega n + \phi), \qquad \Omega = \omega T_s = 2\pi\frac{f}{f_s},$$

where $f_s = 1/T_s$ is the sampling rate and Ω is the **digital (normalised) angular frequency** in rad/sample. Discrete-time sinusoids behave differently from continuous ones in two important ways.

Periodicity of a discrete-time sinusoid

$x[n] = \cos(\Omega n + \phi)$ is periodic in n if and only if $\Omega/(2\pi)$ is rational. If $\Omega/(2\pi) = m/N$ in lowest terms ($N > 0$), the fundamental period is N samples.

Proof. $x[n+N] = x[n]$ for all n requires $\Omega N = 2\pi m$ for some integer m, i.e. $\Omega/2\pi = m/N$. The smallest such positive N is the denominator in lowest terms. □

Frequency ambiguity (aliasing)

$\cos\big((\Omega + 2\pi k)n + \phi\big) = \cos(\Omega n + \phi)$ for every integer k, and $\cos(-\Omega n - \phi) = \cos(\Omega n + \phi)$. Hence discrete-time frequencies are only distinguishable within $0 \le \Omega \le \pi$; in terms of the original analogue frequency, tones at f and $|f - k f_s|$ produce identical samples.

The highest distinguishable frequency, $\Omega = \pi$ (i.e. $f = f_s/2$), is the **Nyquist frequency**; sampling theory proper is in Ch. 16–17.

Worked Example 3.18 — Periods and aliases of sampled sinusoids

(a) Find the fundamental periods of $\cos(\pi n/6)$, $\cos(3\pi n/8)$ and $\cos(0.5n)$. (b) A 1 kHz tone and a 7 kHz tone are each sampled at $f_s = 8$ kHz. Compare the samples.

(a) $\frac{\pi/6}{2\pi} = \frac{1}{12}$: period 12 samples. $\frac{3\pi/8}{2\pi} = \frac{3}{16}$: period 16 samples (it completes 3 cycles of the underlying cosine in 16 samples). $\frac{0.5}{2\pi} = \frac{1}{4\pi}$ is irrational: **not periodic**, even though the continuous-time $\cos(0.5t)$ is periodic with period 4π.

(b) $\Omega_1 = 2\pi(1/8) = \pi/4$; $\Omega_2 = 2\pi(7/8) = 7\pi/4 = 2\pi - \pi/4$. Since $\cos(7\pi n/4) = \cos(2\pi n - \pi n/4) = \cos(\pi n/4)$, the two sample sequences are **identical**: the 7 kHz tone aliases to 1 kHz ($|7-8| = 1$ kHz). Both have period 8 samples.

Check: computing $\cos(2\pi\cdot 7000\,n/8000)$ and $\cos(2\pi\cdot 1000\,n/8000)$ for $n = 0,\dots,11$ gives identical values to machine precision. ✓

Common Pitfall — Hz versus rad/s in formulas

Reactance is $X_C = 1/(\omega C) = 1/(2\pi f C)$, not $1/(fC)$; a resonant frequency $\omega_0 = 1/\sqrt{LC}$ is in rad/s, so $f_0 = 1/(2\pi\sqrt{LC})$; filter "corner frequencies" on datasheets are usually in Hz while transfer functions $H(j\omega)$ use rad/s. Because the radian is dimensionless, no unit check will catch a missing 2π — always label which one a number is.

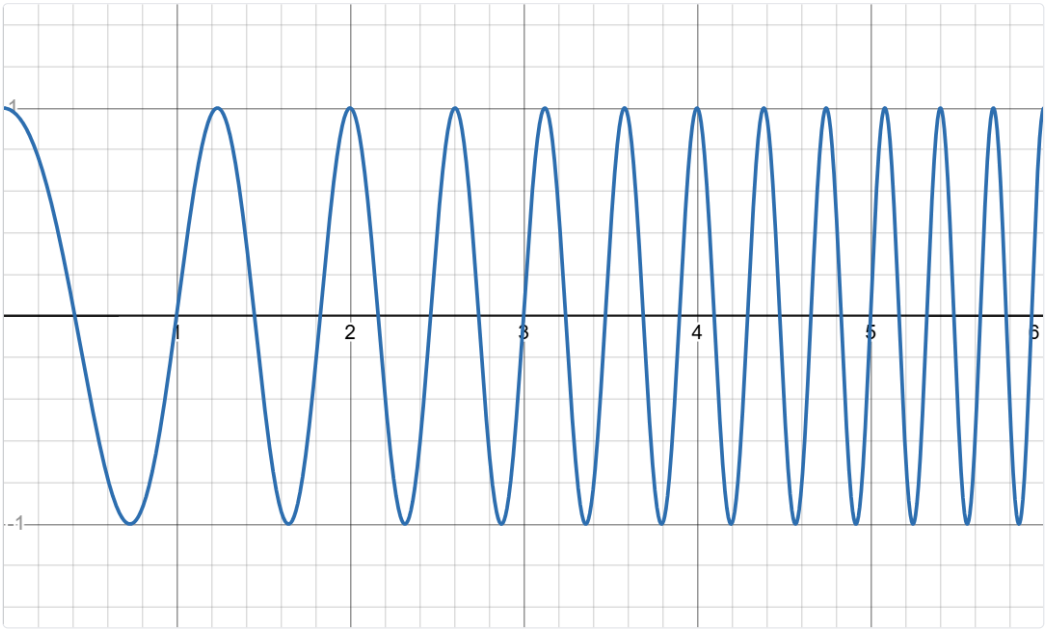
Instantaneous frequency

When the phase of a signal is not a linear function of time — as in frequency modulation or a radar "chirp" — we generalise frequency to the rate of change of phase.

Instantaneous angular frequency

For $x(t) = A\cos\Theta(t)$, the **instantaneous angular frequency** is $\omega_i(t) = \dfrac{d\Theta}{dt}$ and the instantaneous frequency is $f_i(t) = \dfrac{1}{2\pi}\dfrac{d\Theta}{dt}$.

For a pure sinusoid $\Theta = \omega t + \phi$ and $\omega_i = \omega$, as it should. For the linear chirp $\Theta(t) = 2\pi\left(f_0 t + \tfrac{1}{2}\beta t^2\right)$, $f_i(t) = f_0 + \beta t$: the frequency sweeps linearly at β Hz/s. With $f_0 = 1$ kHz and $\beta = 10$ kHz/s, at $t = 0.5$ s the instantaneous frequency is $1 + 10(0.5) = 6$ kHz. Note that the "frequency" of a chirp is *not* the coefficient of t inside the cosine at a given instant: writing $\cos(2\pi f(t)\,t)$ with $f(t) = f_0 + \beta t$ would give $f_i = f_0 + 2\beta t$, twice the intended sweep rate.



A linear chirp $\cos\left(2\pi(0.5t + 0.25t^2)\right)$: the instantaneous frequency $0.5 + 0.5t$ Hz rises steadily, so the oscillations crowd together as t increases.

3.5 Harmonic functions

Two meanings of "harmonic"

The word *harmonic* carries two related meanings in engineering mathematics, and this section is about both:

1. A **harmonic function of time** (a *harmonic oscillation*) is a sinusoid $A\cos(\omega t + \phi)$ — the motion of an ideal mass on a spring, the voltage across an ideal LC tank, the projection of a uniformly rotating vector. It is characterised as the solution of the simple-harmonic equation $\ddot{x} + \omega^2 x = 0$.
2. The **harmonics** of a periodic signal with fundamental frequency f_0 are the sinusoids at the integer multiples $n f_0$. A distorted 50 Hz current "contains a 5th harmonic" if it has a 250 Hz sinusoidal component.

(A third, unrelated, meaning — a function of several variables satisfying Laplace's equation $\nabla^2 u = 0$ — appears in Ch. 13 and Ch. 18. The shared word comes from the fact that Laplace's equation in a disc is solved by sums of $r^n\cos n\phi$ — harmonics again.)

Simple harmonic motion

Simple harmonic oscillator

A quantity $x(t)$ undergoes **simple harmonic motion** (SHM) with natural angular frequency $\omega_0 > 0$ if

$$\ddot{x} + \omega_0^2 x = 0.$$

The acceleration is proportional to the displacement and directed back towards zero.

Solutions of the simple-harmonic equation

Every solution of $\ddot{x} + \omega_0^2 x = 0$ can be written in each of the equivalent forms

$$x(t) = c_1\cos\omega_0 t + c_2\sin\omega_0 t = A\cos(\omega_0 t + \phi) = \operatorname{Re}\{\mathbf{X}e^{j\omega_0 t}\},$$

with $\mathbf{X} = Ae^{j\phi} = c_1 - jc_2$. Given initial values $x(0) = x_0$ and $\dot{x}(0) = v_0$,

$$c_1 = x_0, \quad c_2 = \frac{v_0}{\omega_0}, \qquad A = \sqrt{x_0^2 + \left(\frac{v_0}{\omega_0}\right)^2}, \quad \phi = \operatorname{atan2}\!\left(-\frac{v_0}{\omega_0},\, x_0\right).$$

Derivation. *That these are solutions* follows from §3.2: differentiating $A\cos(\omega_0 t+\phi)$ twice multiplies it by $-\omega_0^2$. *That they are all the solutions* follows from an energy argument. Suppose x is any solution and let $y = x - (c_1\cos\omega_0 t + c_2\sin\omega_0 t)$ with $c_1 = x(0)$, $c_2 = \dot x(0)/\omega_0$. Then y solves the same equation with $y(0) = \dot y(0) = 0$. Multiply $\ddot y + \omega_0^2 y = 0$ by $\dot y$:

$$\frac{d}{dt}\left[\tfrac12 \dot y^2 + \tfrac12\omega_0^2 y^2\right] = \dot y\,\ddot y + \omega_0^2 y\,\dot y = 0.$$

So $E = \frac12\dot y^2 + \frac12\omega_0^2 y^2$ is constant; it is zero at $t=0$, hence zero for all t, so $y \equiv 0$. The initial-value formulas come from $x(0)=c_1$, $\dot x(0) = \omega_0 c_2$ and the amplitude–phase conversion of §3.2. The general theory of such equations is in Ch. 11–12. □

The quantity E in the proof is (up to a constant factor) the physical energy, which in SHM sloshes between two forms — kinetic and potential, or electric and magnetic — while its total stays fixed.

> **Worked Example 3.19 — Fitting initial conditions**
>
> Solve $\ddot x + 9x = 0$ with $x(0) = 2$, $\dot x(0) = -6$ and write the answer in amplitude–phase form.
>
> **Solution.** $\omega_0 = 3$. $c_1 = 2$, $c_2 = -6/3 = -2$, so $x = 2\cos 3t - 2\sin 3t$. Amplitude $A = \sqrt{4+4} = 2\sqrt2 = 2.828$; phase $\phi = \operatorname{atan2}(2,2) = 45^\circ$. $x(t) = 2\sqrt2\cos(3t + 45^\circ)$.
>
> **Check:** $x(0) = 2\sqrt2\cos 45^\circ = 2$ ✓; $\dot x(0) = -3(2\sqrt2)\sin 45^\circ = -6$ ✓.

The LC tank: an electrical harmonic oscillator

For an ideal inductor L connected across an ideal capacitor C, the capacitor voltage v and the current i (taken as the current leaving the capacitor's positive plate and flowing through L) satisfy $i = -C\,\dot v$ and $v = L\,\dot i$. Eliminating i:

$$\ddot v + \frac{1}{LC}v = 0, \qquad \omega_0 = \frac{1}{\sqrt{LC}}, \qquad f_0 = \frac{1}{2\pi\sqrt{LC}}.$$

The stored energy $W = \frac12 C v^2 + \frac12 L i^2$ is constant. With $v = V\cos(\omega_0 t+\phi)$, the capacitor energy is $\frac12 CV^2\cos^2(\cdot) = \frac14 CV^2[1+\cos(2\omega_0 t + 2\phi)]$: **each stored energy oscillates at twice the natural frequency**, the two oscillating parts cancelling exactly.

> **Worked Example 3.20 — Free oscillation of an LC tank**
>
> $L = 10$ mH, $C = 100$ nF. At $t=0$ the capacitor voltage is 5 V and the current through the inductor is $i(0) = 20$ mA (so $C\dot v(0) = -20$ mA). Find $v(t)$, the peak voltage, and verify energy conservation.
>
> **Natural frequency.** $\omega_0 = 1/\sqrt{10^{-2}\times 10^{-7}} = 1/\sqrt{10^{-9}} = 31\,623$ rad/s; $f_0 = 5.033$ kHz.
>
> **Initial slope.** $\dot v(0) = -0.020/10^{-7} = -2.0\times10^5$ V/s, so $v_0/\omega_0 = -6.325$ V.
>
> **Amplitude and phase.** $V = \sqrt{5^2 + 6.325^2} = \sqrt{65} = 8.062$ V; $\phi = \operatorname{atan2}(6.325, 5) = 51.67^\circ$.
>
> **Result.** $v(t) = 8.062\cos(31\,623\,t + 51.67^\circ)$ V.
>
> **Energy check.** Initially $\frac12(10^{-7})(25) + \frac12(10^{-2})(0.02)^2 = 1.25 + 2.00 = 3.25$ μJ. At a voltage peak the current is zero and $W = \frac12(10^{-7})(65) = 3.25$ μJ. ✓

The rotating phasor picture

Euler's formula gives a geometric picture of every harmonic oscillation. The complex number $\mathbf{X}e^{j\omega t} = Ae^{j(\omega t+\phi)}$ moves anticlockwise around a circle of radius A, completing one revolution every $T = 2\pi/\omega$ seconds. Its projection on the real axis is $A\cos(\omega t+\phi)$ and on the imaginary axis $A\sin(\omega t+\phi)$:

- **amplitude** = radius of the circle,
- **angular frequency** = angular speed of rotation,
- **phase** = angle of the vector at $t=0$,
- **phase difference** between two same-frequency signals = the fixed angle between their rotating vectors.

Since both vectors rotate at the same rate, the angle between them never changes; this is why a phasor diagram can be drawn as a static picture "frozen" at $t=0$.

A real cosine can also be written as the sum of *two* counter-rotating vectors:

$$A\cos(\omega t+\phi) = \frac{A}{2}e^{j(\omega t+\phi)} + \frac{A}{2}e^{-j(\omega t+\phi)}.$$

The second term rotates clockwise — it has *negative frequency* $-\omega$. The imaginary parts cancel and the real parts add. This two-sided view is the basis of the complex Fourier series (Ch. 15) and the Fourier transform (Ch. 16), in which every real sinusoid appears as a pair of spectral lines at $\pm\omega$ with conjugate complex amplitudes $\frac{A}{2}e^{\pm j\phi}$.

Damped harmonic oscillations

Real oscillators lose energy. Adding a resistance to the LC tank, or friction to the spring, gives the damped equation

$$\ddot x + 2\alpha\dot x + \omega_0^2 x = 0, \qquad \alpha > 0,$$

whose solution for light damping ($\alpha < \omega_0$) is derived in Ch. 12:

Underdamped oscillation

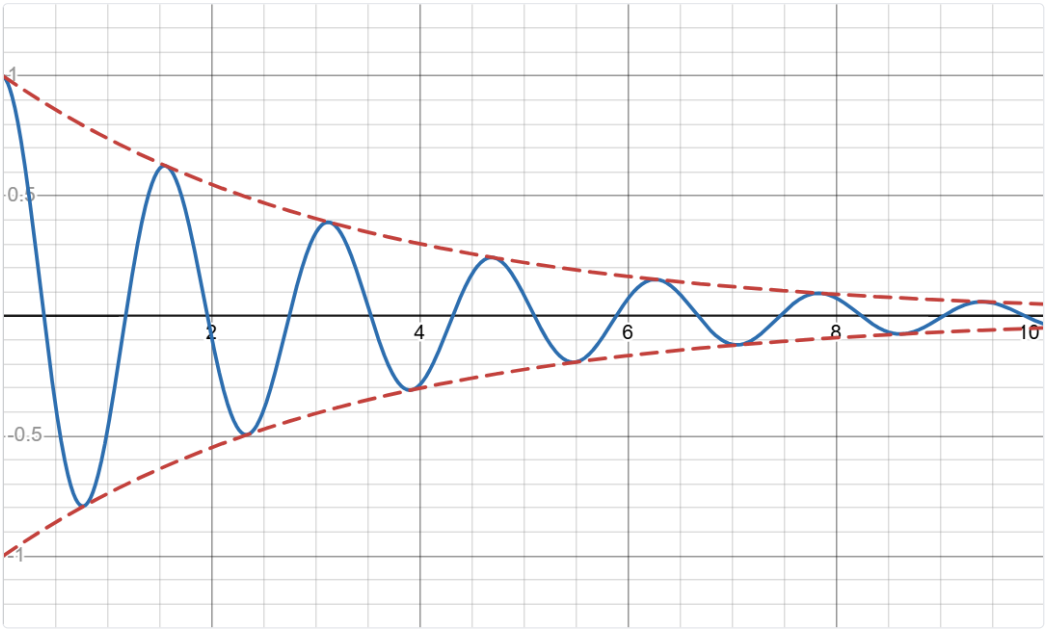
For $0 < \alpha < \omega_0$ every solution has the form

$$x(t) = A e^{-\alpha t}\cos(\omega_d t + \phi), \qquad \omega_d = \sqrt{\omega_0^2 - \alpha^2}.$$

It is a sinusoid of angular frequency ω_d (the **damped natural frequency**) whose amplitude decays inside the **envelope** $\pm A e^{-\alpha t}$. The **quality factor** is $Q = \omega_0/(2\alpha)$.

The verification is a direct (if tedious) differentiation; more elegantly, $x = \operatorname{Re}\{\mathbf{X} e^{st}\}$ with $s = -\alpha + j\omega_d$, and s is a root of $s^2 + 2\alpha s + \omega_0^2 = 0$.

Such a signal is **not periodic** — it never repeats exactly — but its zero crossings are evenly spaced, half a *pseudo-period* $T_d = 2\pi/\omega_d$ apart, and the ratio of successive peaks is the constant $e^{-\alpha T_d}$. The natural logarithm of that ratio, $\delta = \alpha T_d$, is the **logarithmic decrement**, a standard way to measure damping from an oscilloscope trace.



An underdamped oscillation $e^{-0.3t}\cos 4t$ (blue) confined within its exponential envelope $\pm e^{-0.3t}$ (red).

Worked Example 3.21 — Ringing of a series RLC circuit

A series RLC circuit with $R = 20\ \Omega$, $L = 10$ mH, $C = 100$ nF rings after a disturbance. For a series circuit $\alpha = R/(2L)$ and $\omega_0 = 1/\sqrt{LC}$. Find ω_d, Q, the logarithmic decrement, and the fraction of the initial amplitude left after 10 cycles.

Parameters. $\alpha = 20/(0.02) = 1000\ \text{s}^{-1}$; $\omega_0 = 31\,623$ rad/s; $\omega_d = \sqrt{31\,623^2 - 1000^2} = 31\,607$ rad/s (only 0.05 % below ω_0).

Quality factor. $Q = 31\,623/2000 = 15.8$.

Decrement. $T_d = 2\pi/\omega_d = 198.8\ \mu$s; $\delta = \alpha T_d = 0.1988$, so each peak is $e^{-0.1988} = 0.820$ of the previous one.

After 10 cycles. $e^{-10\delta} = e^{-1.988} = 0.137$: about 14 % of the initial amplitude remains.

Check: a useful rule of thumb is that the amplitude falls by $e^{-\pi} \approx 4.3\%$ in Q cycles; here $Q \approx 16$ cycles gives $e^{-16\times 0.1988} = e^{-3.14}$. ✓

Harmonics of a fundamental

Harmonics

Given a fundamental frequency f_0 (angular frequency ω_0), the **n-th harmonic** is a sinusoid at $n f_0$, $n = 1, 2, 3, \ldots$:

$$h_n(t) = A_n \cos(n\omega_0 t + \phi_n).$$

The 1st harmonic is the fundamental itself; n is the **harmonic number** (harmonic order). Harmonics with even n are **even harmonics**, those with odd n **odd harmonics**. A component at a frequency that is not an integer multiple of f_0 is an **interharmonic**; one at f_0/m is a **subharmonic**.

Musicians call the n-th harmonic the $(n-1)$-th *overtone*; engineers almost always count harmonics, so the "3rd harmonic" of 50 Hz is 150 Hz.

A sum of harmonics is periodic with the fundamental period

Any finite sum

$$x(t) = A_0 + \sum_{n=1}^{N} A_n \cos(n\omega_0 t + \phi_n)$$

is periodic with period $T_0 = 2\pi/\omega_0$. If $A_1 \neq 0$ it is the fundamental period; in general the fundamental period is T_0/g, where g is the greatest common divisor of the harmonic numbers n with $A_n \neq 0$.

Proof. In time T_0 the n-th harmonic advances its phase by $n\omega_0 T_0 = 2\pi n$, a whole number of cycles, so every term repeats. The statement about g follows from §3.4: the frequencies present are multiples of $g f_0$ and their greatest common divisor is exactly $g f_0$. □

So, for example, $\cos 2\omega_0 t + \cos 4\omega_0 t$ repeats every $T_0/2$: it "contains only even harmonics of f_0", but is better described as the 1st and 2nd harmonics of $2f_0$.

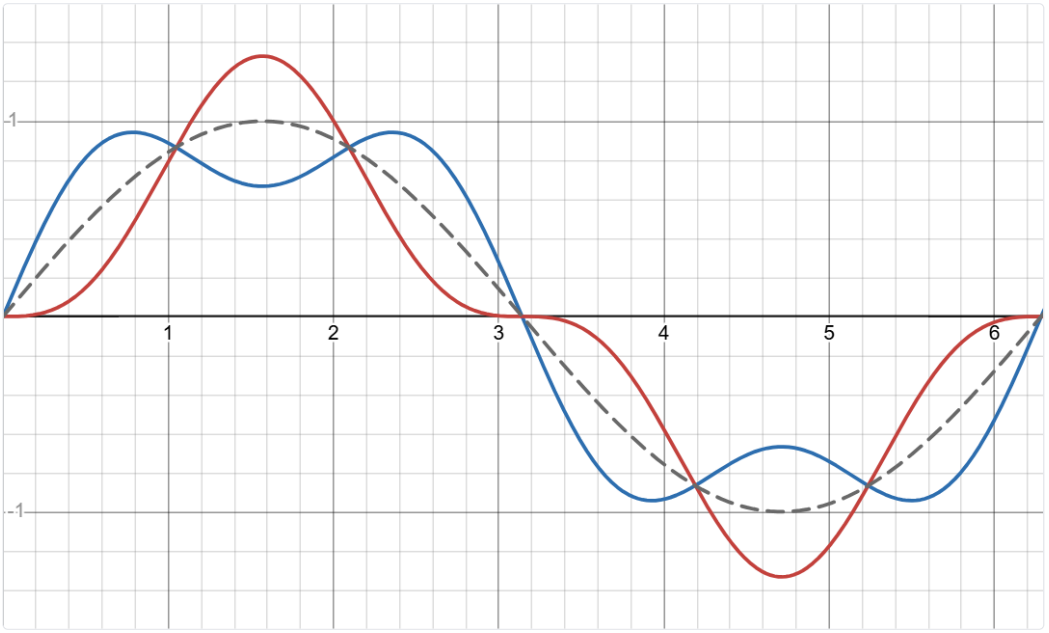
Worked Example 3.22 — Identifying harmonics

A power-quality analyser on a 60 Hz feeder reports components at 60, 180, 300, 420, 540 and 210 Hz. Classify each.

Solution. $180/60 = 3$, $300/60 = 5$, $420/60 = 7$, $540/60 = 9$: the 3rd, 5th, 7th and 9th harmonics (all odd — typical of loads with half-wave symmetric current, §3.8). $210/60 = 3.5$ is not an integer: an **interharmonic**, often produced by variable-frequency drives or arc furnaces, and not part of the 60 Hz harmonic series.

Harmonic content versus waveform shape

The amplitudes A_n determine the *power* in each harmonic (§3.7) but the phases ϕ_n strongly affect the *shape*. Adding a third harmonic of amplitude $\frac{1}{3}$ to a fundamental gives a flattened, nearly square top if the third harmonic is in the right phase, and a peaked waveform if it is inverted:



Fundamental plus a one-third-amplitude third harmonic: in phase (blue) the peaks flatten towards a square wave; inverted (red) the peaks sharpen. Both have exactly the same RMS value.

Engineering Note — Harmonics in power systems and RF

In power systems, nonlinear loads (rectifiers, switch-mode supplies, LED drivers) draw currents rich in odd harmonics; standards such as IEC 61000-3-2 limit each harmonic current by order. In RF systems, a power amplifier driven at f_0 radiates harmonics at $2f_0, 3f_0, \ldots$, which regulations require to be filtered. In both cases the engineer works with the *list* of harmonic amplitudes, which is why the rest of this chapter develops the algebra of harmonics.

3.6 Trigonometric expansions

Motivation

Linear systems are analysed one frequency at a time, so the natural currency of signal analysis is the *sum* of sinusoids. But physics keeps handing us *products* and *powers*: instantaneous power is the product $v(t)\,i(t)$; a mixer multiplies two signals; a diode or transistor responds to v^2, v^3, …; a modulator multiplies a carrier by a message. Trigonometric identities are the dictionary that translates products and powers back into sums, and the frequencies that appear in those sums are exactly the frequencies a spectrum analyser will show. This section collects the identities, derives them all from one source — Euler's formula — and shows how to apply them systematically.

The master method: Euler's formula

From $e^{j\theta} = \cos\theta + j\sin\theta$ (Ch. 2) and its conjugate,

$$\cos\theta = \frac{e^{j\theta} + e^{-j\theta}}{2}, \qquad \sin\theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}.$$

Every identity in this section can be proved by (i) replacing each cos and sin by these exponentials, (ii) multiplying out using $e^{ja}e^{jb} = e^{j(a+b)}$, and (iii) regrouping conjugate pairs back into cosines and sines. Products of exponentials *add* their exponents, which is the whole reason products of sinusoids become sinusoids at sum and difference frequencies.

Compound-angle formulas

Compound-angle (addition) formulas

$$\begin{aligned}\cos(A \pm B) &= \cos A\cos B \mp \sin A\sin B,\\ \sin(A \pm B) &= \sin A\cos B \pm \cos A\sin B,\\ \tan(A \pm B) &= \frac{\tan A \pm \tan B}{1 \mp \tan A\tan B}.\end{aligned}$$

Proof. $e^{j(A+B)} = e^{jA}e^{jB}$. Expand both sides:

$$\cos(A+B) + j\sin(A+B) = (\cos A + j\sin A)(\cos B + j\sin B) = (\cos A\cos B - \sin A\sin B) + j(\sin A\cos B + \cos A\sin B).$$

Equating real and imaginary parts gives the $+$ formulas; replacing B by $-B$ and using evenness of cos and oddness of sin gives the $-$ formulas. Dividing the sine formula by the cosine formula, then numerator and denominator by $\cos A\cos B$, gives the tangent formula. $\square$

A purely geometric proof (rotating a unit vector by A and then by B) is also standard; the rotation view is worth keeping in mind because it says that the compound-angle formula *is* the rule for multiplying by the rotation matrix $\begin{pmatrix}\cos B & -\sin B\\ \sin B & \cos B\end{pmatrix}$ of Ch. 9.

Worked Example 3.23 — Exact values from compound angles

Find exact values of $\sin 75^\circ$, $\cos 15^\circ$ and $\tan 75^\circ$.

Solution. $75^\circ = 45^\circ + 30^\circ$:

$$\sin 75^\circ = \sin 45^\circ\cos 30^\circ + \cos 45^\circ\sin 30^\circ = \frac{\sqrt{2}}{2}\cdot\frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2}\cdot\frac{1}{2} = \frac{\sqrt{6}+\sqrt{2}}{4}.$$

$\cos 15^\circ = \cos(90^\circ - 75^\circ) = \sin 75^\circ = \dfrac{\sqrt{6}+\sqrt{2}}{4}$.

$$\tan 75^\circ = \frac{\tan 45^\circ + \tan 30^\circ}{1 - \tan 45^\circ\tan 30^\circ} = \frac{1 + 1/\sqrt{3}}{1 - 1/\sqrt{3}} = \frac{\sqrt{3}+1}{\sqrt{3}-1} = 2 + \sqrt{3}.$$

Check: $(\sqrt{6}+\sqrt{2})/4 = 0.965926 = \sin 75^\circ$; $2+\sqrt{3} = 3.73205 = \tan 75^\circ$. ✓

Double-angle, half-angle and power-reduction formulas

Setting $B = A$ in the compound-angle formulas:

$$\sin 2A = 2\sin A\cos A, \qquad \cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A, \qquad \tan 2A = \frac{2\tan A}{1 - \tan^2 A}.$$

Solving the two alternative forms of $\cos 2A$ for the squares gives the **power-reduction** formulas, which convert a square (a "power") into a constant plus a second harmonic:

$$\cos^2 A = \frac{1 + \cos 2A}{2}, \qquad \sin^2 A = \frac{1 - \cos 2A}{2}, \qquad \sin A\cos A = \frac{\sin 2A}{2}.$$

Replacing A by $A/2$ gives the **half-angle** formulas $\cos\frac{A}{2} = \pm\sqrt{\frac{1+\cos A}{2}}$, $\sin\frac{A}{2} = \pm\sqrt{\frac{1-\cos A}{2}}$, with the sign fixed by the quadrant of $A/2$, and

$$\tan\frac{A}{2} = \frac{\sin A}{1 + \cos A} = \frac{1 - \cos A}{\sin A}.$$

The **tangent half-angle substitution** $u = \tan(A/2)$ expresses everything rationally,

$$\sin A = \frac{2u}{1+u^2}, \qquad \cos A = \frac{1-u^2}{1+u^2},$$

which turns any rational function of $\sin A$ and $\cos A$ into a rational function of u — a standard integration technique (Ch. 5) and, in signal processing, the origin of the bilinear transform (Ch. 17).

Applying the double-angle formula twice gives triple angles:

$$\cos 3A = 4\cos^3 A - 3\cos A, \qquad \sin 3A = 3\sin A - 4\sin^3 A,$$

and inverting these gives the cubic power-reduction formulas

$$\cos^3 A = \frac{3\cos A + \cos 3A}{4}, \qquad \sin^3 A = \frac{3\sin A - \sin 3A}{4}.$$

These are the identities that explain harmonic distortion. If a device has a cubic term in its transfer characteristic, a pure input $\cos \omega t$ produces $\cos^3 \omega t$, which contains a component at 3ω (§3.7).

Worked Example 3.24 — Reducing a fourth power

Express $\sin^4 \theta$ as a sum of cosines of multiple angles, and deduce its average value.

Solution. Square the power-reduction formula:

$$\sin^4 \theta = \left(\frac{1 - \cos 2\theta}{2}\right)^2 = \frac{1 - 2\cos 2\theta + \cos^2 2\theta}{4}.$$

Reduce $\cos^2 2\theta = \frac{1+\cos 4\theta}{2}$:

$$\sin^4 \theta = \frac{1}{4} - \frac{1}{2}\cos 2\theta + \frac{1 + \cos 4\theta}{8} = \frac{3}{8} - \frac{1}{2}\cos 2\theta + \frac{1}{8}\cos 4\theta.$$

The cosine terms average to zero over a period, so $\overline{\sin^4 \theta} = 3/8$.

Check: at $\theta = \pi/2$: LHS $= 1$; RHS $= \frac{3}{8} + \frac{1}{2} + \frac{1}{8} = 1$ ✓. At $\theta = \pi/4$: LHS $= \frac{1}{4}$; RHS $= \frac{3}{8} - 0 - \frac{1}{8} = \frac{1}{4}$ ✓. A numerical comparison at 50 points agrees to 10^{-16}.

Powers of cosine in general

Harmonic expansion of $\cos^n \theta$

For a positive integer n,

$$\cos^n \theta = \frac{1}{2^n} \sum_{k=0}^{n} \binom{n}{k} \cos\big((n - 2k)\theta\big).$$

Grouping the terms k and $n - k$ (which give equal cosines), $\cos^n \theta$ contains only the harmonics $n,\ n-2,\ n-4, \ldots$ — all odd if n is odd, and all even (including a constant) if n is even. The constant term for even n is $\binom{n}{n/2}/2^n$.

Proof. By the binomial theorem,

$$\cos^n \theta = \frac{(e^{j\theta} + e^{-j\theta})^n}{2^n} = \frac{1}{2^n}\sum_{k=0}^{n} \binom{n}{k} e^{j(n-k)\theta} e^{-jk\theta} = \frac{1}{2^n}\sum_{k=0}^{n} \binom{n}{k} e^{j(n-2k)\theta}.$$

The left side is real, so it equals the real part of the right, which replaces each exponential by a cosine. □

The same argument with $(e^{j\theta} - e^{-j\theta})^n/(2j)^n$ handles $\sin^n \theta$: odd powers of sine give sines of odd multiples, even powers give cosines of even multiples.

Multiple angles and Chebyshev polynomials

Going the other way, De Moivre's theorem $(\cos\theta + j\sin\theta)^n = \cos n\theta + j \sin n\theta$ (Ch. 2) expresses $\cos n\theta$ as a polynomial in $\cos\theta$.

Chebyshev polynomials

For each $n \geq 0$ there is a polynomial T_n of degree n with

$$\cos n\theta = T_n(\cos\theta).$$

They satisfy $T_0(x) = 1$, $T_1(x) = x$ and the recurrence

$$T_{n+1}(x) = 2x\,T_n(x) - T_{n-1}(x),$$

giving $T_2 = 2x^2 - 1$, $T_3 = 4x^3 - 3x$, $T_4 = 8x^4 - 8x^2 + 1$, $T_5 = 16x^5 - 20x^3 + 5x$.

Proof of the recurrence. Add the compound-angle formulas $\cos(n+1)\theta = \cos n\theta\cos\theta - \sin n\theta \sin\theta$ and $\cos(n-1)\theta = \cos n\theta\cos\theta + \sin n\theta\sin\theta$:

$$\cos(n+1)\theta + \cos(n-1)\theta = 2\cos\theta\cos n\theta.$$

With $x = \cos\theta$ this is $T_{n+1}(x) + T_{n-1}(x) = 2xT_n(x)$. Since T_0 and T_1 are polynomials, induction shows every T_n is a polynomial of degree n with leading coefficient 2^{n-1} $(n \geq 1)$. □

Because $|T_n(x)| = |\cos n\theta| \leq 1$ on $-1 \leq x \leq 1$ while growing rapidly outside, Chebyshev polynomials are the basis of Chebyshev filters (equal-ripple passband) and of near-optimal polynomial approximation (Ch. 22).

Worked Example 3.25 — Two directions: $\cos 5\theta$ and $\cos^5\theta$

(a) Express $\cos 5\theta$ as a polynomial in $c = \cos\theta$. (b) Express $\cos^5\theta$ as a sum of harmonics.

(a) From the recurrence: $T_4 = 2x(4x^3 - 3x) - (2x^2 - 1) = 8x^4 - 8x^2 + 1$, then $T_5 = 2x(8x^4 - 8x^2 + 1) - (4x^3 - 3x) = 16x^5 - 20x^3 + 5x$. So $\cos 5\theta = 16c^5 - 20c^3 + 5c$.

(b) With $n = 5$: $\binom{5}{0}, \dots, \binom{5}{5} = 1, 5, 10, 10, 5, 1$ multiply $\cos 5\theta, \cos 3\theta, \cos\theta, \cos(-\theta), \cos(-3\theta), \cos(-5\theta)$. Pairing:

$$\cos^5\theta = \frac{1}{32}\big(2\cos 5\theta + 10\cos 3\theta + 20\cos\theta\big) = \frac{10\cos\theta + 5\cos 3\theta + \cos 5\theta}{16}.$$

Consistency check. Substitute (b)'s $\cos 5\theta = 16\cos^5\theta - 5\cos 3\theta - 10\cos\theta$ and $\cos 3\theta = 4c^3 - 3c$: $16c^5 - 20c^3 + 15c - 10c = 16c^5 - 20c^3 + 5c$, agreeing with (a). ✓ At $\theta = 0$ both sides of (b) equal $16/16 = 1$. ✓

Product-to-sum and sum-to-product formulas

Adding and subtracting the compound-angle formulas in pairs gives:

Product-to-sum formulas

$$\begin{aligned}
\cos A\cos B &= \tfrac{1}{2}\big[\cos(A-B) + \cos(A+B)\big],\\
\sin A\sin B &= \tfrac{1}{2}\big[\cos(A-B) - \cos(A+B)\big],\\
\sin A\cos B &= \tfrac{1}{2}\big[\sin(A+B) + \sin(A-B)\big].
\end{aligned}$$

For example, the first comes from adding $\cos(A-B) = \cos A\cos B + \sin A\sin B$ and $\cos(A+B) = \cos A\cos B - \sin A\sin B$. Applied to signals, $A = \omega_1 t$ and $B = \omega_2 t$: **the product of two sinusoids is the sum of two sinusoids at the sum and difference frequencies, each with half the product amplitude.** This single fact underlies mixing, modulation, demodulation and the computation of average power.

Setting $A = (P+Q)/2$, $B = (P-Q)/2$ reverses the formulas:

Sum-to-product formulas

$$\begin{aligned}
\cos P + \cos Q &= 2\cos\tfrac{P+Q}{2}\cos\tfrac{P-Q}{2}, & \cos P - \cos Q &= -2\sin\tfrac{P+Q}{2}\sin\tfrac{P-Q}{2},\\
\sin P + \sin Q &= 2\sin\tfrac{P+Q}{2}\cos\tfrac{P-Q}{2}, & \sin P - \sin Q &= 2\cos\tfrac{P+Q}{2}\sin\tfrac{P-Q}{2}.
\end{aligned}$$

Sum-to-product is the tool for equations (a product is zero iff a factor is) and for the beat phenomenon (§3.7).

Worked Example 3.26 — Instantaneous and average power

A load has $v(t) = 325\cos\omega t$ V and $i(t) = 10\cos(\omega t - 30^\circ)$ A. Find the instantaneous power $p(t) = v\,i$, its average, and the frequency at which it pulsates.

Solution. Product-to-sum with $A = \omega t$, $B = \omega t - 30^\circ$:

$$p(t) = 3250\cdot\tfrac{1}{2}\big[\cos 30^\circ + \cos(2\omega t - 30^\circ)\big] = 1407 + 1625\cos(2\omega t - 30^\circ)\ \text{W}.$$

The average (real) power is $P = 1407$ W and $p(t)$ pulsates at **twice** the supply frequency with amplitude 1625 W. Because $1625 > 1407$, p is briefly negative each half-cycle: energy flows back from the load's reactive part to the source.

General result. For $v = V_m\cos(\omega t + \phi_v)$, $i = I_m\cos(\omega t + \phi_i)$:

$$P = \overline{p} = \tfrac{1}{2}V_m I_m\cos(\phi_v - \phi_i) = V_{\text{rms}}I_{\text{rms}}\cos(\phi_v - \phi_i),$$

the power-factor formula quoted in §3.3. Check: $\frac{325}{\sqrt{2}}\cdot\frac{10}{\sqrt{2}}\cos 30^\circ = 1625\times 0.8660 = 1407$ W. ✓

Worked Example 3.27 — Solving an equation with sum-to-product

Solve $\sin 3x + \sin x = \cos x$ on $[0, \pi]$.

Solution. $\sin 3x + \sin x = 2\sin 2x\cos x$. The equation becomes $2\sin 2x\cos x - \cos x = 0$, i.e. $\cos x\,(2\sin 2x - 1) = 0$.

- $\cos x = 0$: $x = \pi/2$.
- $\sin 2x = \tfrac{1}{2}$: $2x = \pi/6$ or $5\pi/6$ (within $[0, 2\pi]$), so $x = \pi/12$ or $5\pi/12$.

Solutions: $x = \pi/12,\ 5\pi/12,\ \pi/2$.

Check $x = \pi/12 = 15^\circ$: $\sin 45^\circ + \sin 15^\circ = 0.7071 + 0.2588 = 0.9659 = \cos 15^\circ$. ✓

Sums of sinusoids with phases in arithmetic progression

A sum that appears in antenna arrays, multi-tap filters and the discrete Fourier transform is

$$S = \sum_{k=0}^{N-1} e^{jk\delta}.$$

Geometric phasor sum

For δ not a multiple of 2π,

$$\sum_{k=0}^{N-1} e^{jk\delta} = e^{j(N-1)\delta/2}\,\frac{\sin(N\delta/2)}{\sin(\delta/2)},$$

and consequently

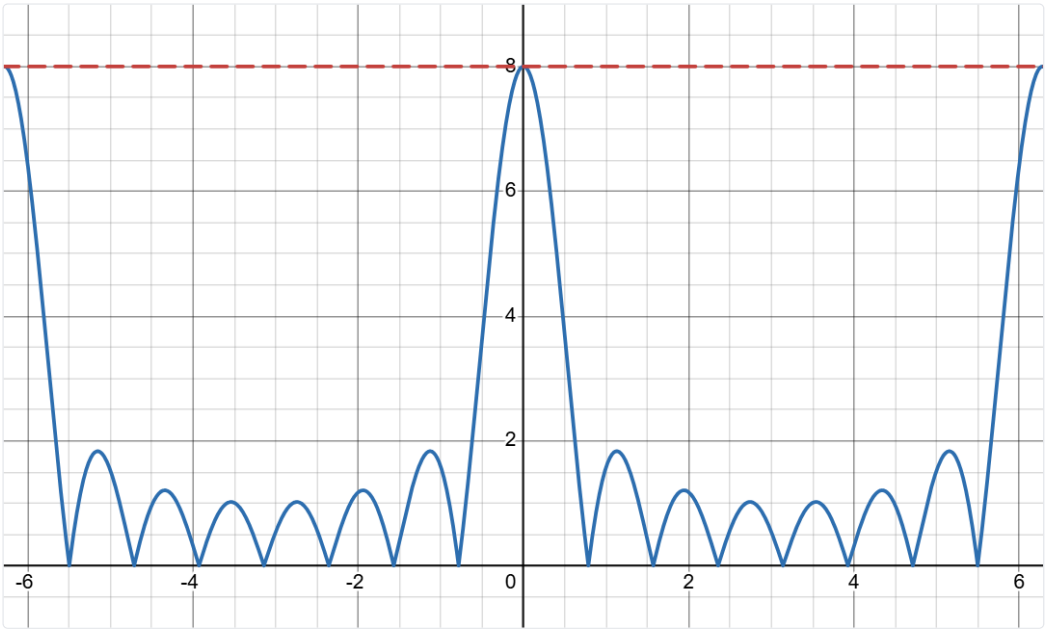
$$\sum_{k=0}^{N-1} \cos(\theta + k\delta) = \frac{\sin(N\delta/2)}{\sin(\delta/2)}\cos\!\Big(\theta + \frac{(N-1)\delta}{2}\Big).$$

Proof. The geometric-series formula gives $S = \dfrac{e^{jN\delta}-1}{e^{j\delta}-1}$. Factor $e^{jN\delta/2}$ from the numerator and $e^{j\delta/2}$ from the denominator:

$$S = \frac{e^{jN\delta/2}\big(e^{jN\delta/2} - e^{-jN\delta/2}\big)}{e^{j\delta/2}\big(e^{j\delta/2} - e^{-j\delta/2}\big)} = e^{j(N-1)\delta/2}\,\frac{2j\sin(N\delta/2)}{2j\sin(\delta/2)}.$$

For the cosine sum, multiply by $e^{j\theta}$ and take real parts. $\square$

The magnitude $|S| = |\sin(N\delta/2)/\sin(\delta/2)|$ equals N at $\delta = 0$ (all phasors aligned) and vanishes when $N\delta/2$ is a nonzero multiple of π (phasors spread evenly round a full circle, as in the three-phase theorem of §3.3).



The array factor $|\sin(N\delta/2)/\sin(\delta/2)|$ for $N = 8$ against the inter-element phase δ: a main lobe of height $N = 8$ at $\delta = 0$, nulls at multiples of $\delta = 2\pi/8$, and smaller side lobes between them.

Worked Example 3.28 — Eight-element array

Eight equal antenna elements are fed with a progressive phase δ between neighbours. Find the magnitude of the combined phasor (relative to one element) for $\delta = 0^\circ$, 10° and 45°.

Solution. $|S| = |\sin(4\delta)/\sin(\delta/2)|$.

- $\delta = 0$: limit $= N = 8$ (full coherent addition).
- $\delta = 10^\circ$: $\sin 40^\circ / \sin 5^\circ = 0.6428/0.08716 = 7.375$.
- $\delta = 45^\circ$: $\sin 180^\circ = 0$: complete cancellation — the eight phasors are spaced 45° apart round the full circle.

Check for $\delta = 10^\circ$: summing $\cos(10^\circ k)$ and $\sin(10^\circ k)$ for $k = 0..7$ directly gives 6.041 and 4.230, magnitude $\sqrt{6.041^2 + 4.230^2} = 7.375$. ✓

Series expansions

The circular functions also have power-series expansions, valid for every real (indeed complex) θ:

$$\sin\theta = \sum_{m=0}^{\infty} \frac{(-1)^m \theta^{2m+1}}{(2m+1)!}, \qquad \cos\theta = \sum_{m=0}^{\infty} \frac{(-1)^m \theta^{2m}}{(2m)!}.$$

They are derived as Maclaurin series in Ch. 4; substituting them into $e^{j\theta} = \sum (j\theta)^k/k!$ and separating even and odd powers is one standard proof of Euler's formula. For computation they converge quickly only for small $|\theta|$, so libraries first reduce the argument into $[-\pi/4, \pi/4]$ using periodicity and the identities above, then apply a short polynomial — a direct engineering use of this section.

Common Pitfall — Squaring away the sign

Half-angle formulas involve a square root, and solving $\sin^2 x = k$ or $\cos 2x = c$ yields *all* sign choices; check which roots are genuine. Conversely, identities like $\sqrt{1-\cos^2\theta} = \sin\theta$ are only true where $\sin\theta \geq 0$. In an algorithm that computes $\sin\phi$ from $\cos\phi$, the lost sign is a lost lead/lag distinction — exactly the error that a power meter makes if it computes reactive power as $\sqrt{S^2 - P^2}$ instead of $V_{\text{rms}} I_{\text{rms}} \sin\Delta\phi$.

3.7 Harmonic relationships

Motivation

Section 3.5 introduced harmonics as sinusoids at integer multiples of a fundamental; §3.6 supplied the identities that turn products and powers into sums. This section puts the two together and answers the questions engineers actually ask about collections of sinusoids:

- How do the powers of several sinusoids combine? (Orthogonality.)
- Where do harmonics come from, and how big are they? (Nonlinearity.)
- What new frequencies appear when two signals pass through the same nonlinear device or are multiplied? (Intermodulation, mixing, modulation.)
- What does the sum of two *nearly* equal frequencies look like? (Beats.)
- How is "how distorted" quantified? (Total harmonic distortion.)
- How do harmonics behave in three-phase systems? (Phase sequence of harmonics.)

Orthogonality of harmonics

Orthogonality relations

Let $\omega_0 = 2\pi/T_0$ and let m, n be non-negative integers. Over one period,

$$\int_0^{T_0} \cos m\omega_0 t \, \cos n\omega_0 t \, dt = \begin{cases} 0, & m \neq n, \\ T_0/2, & m = n \geq 1, \\ T_0, & m = n = 0, \end{cases}$$

$$\int_0^{T_0} \sin m\omega_0 t \, \sin n\omega_0 t \, dt = \begin{cases} 0, & m \neq n, \\ T_0/2, & m = n \geq 1, \end{cases}$$

$$\int_0^{T_0} \sin m\omega_0 t \, \cos n\omega_0 t \, dt = 0 \quad \text{for all } m, n.$$

The same results hold over any interval of length T_0.

Proof. By product-to-sum, $\cos m\omega_0 t \cos n\omega_0 t = \tfrac{1}{2}[\cos(m-n)\omega_0 t + \cos(m+n)\omega_0 t]$. A cosine $\cos k\omega_0 t$ with integer $k \neq 0$ completes exactly $|k|$ cycles in T_0, so its integral is zero; for $k = 0$ it is the constant 1 with integral T_0. If $m \neq n$ both $m-n$ and $m+n$ are nonzero and the integral vanishes. If $m = n \geq 1$, only $\cos 0 = 1$ survives, giving $\tfrac{1}{2}T_0$; if $m = n = 0$ the integrand is 1. The sine–sine case is identical with $\tfrac{1}{2}[\cos(m-n)\omega_0 t - \cos(m+n)\omega_0 t]$. For sine–cosine, $\tfrac{1}{2}[\sin(m+n)\omega_0 t + \sin(m-n)\omega_0 t]$ integrates to zero in every case (including $\sin 0 = 0$). □

In the language of Ch. 9, the functions $1, \cos\omega_0 t, \sin\omega_0 t, \cos 2\omega_0 t, \ldots$ are mutually **orthogonal** with respect to the inner product $\langle x, y\rangle = \int_0^{T_0} x(t)y(t)\,dt$. That is the structural fact behind Fourier series: it is what lets us *extract* each harmonic's coefficient from a waveform by a single integral (§3.8, Ch. 15).

In complex form the relations are even more compact:

$$\frac{1}{T_0}\int_0^{T_0} e^{jm\omega_0 t}\, e^{-jn\omega_0 t}\, dt = \begin{cases} 1, & m = n, \\ 0, & m \neq n. \end{cases}$$

Power of a sum of harmonics

Mean-square value of a harmonic sum

If $x(t) = A_0 + \sum_{n=1}^{N} A_n \cos(n\omega_0 t + \phi_n)$, then

$$\overline{x^2} = \frac{1}{T_0}\int_0^{T_0} x^2\, dt = A_0^2 + \sum_{n=1}^{N} \frac{A_n^2}{2}, \qquad X_{\text{rms}} = \sqrt{A_0^2 + \sum_{n=1}^{N} \left(\frac{A_n}{\sqrt{2}}\right)^2}.$$

The mean-square value — the average power in a 1 Ω resistor — is the **sum of the powers of the individual harmonics**, independent of their phases.

Proof. Squaring x produces squared terms and cross terms. Each cross term $2A_m A_n \cos(m\omega_0 t + \phi_m)\cos(n\omega_0 t + \phi_n)$ with $m \neq n$ is, after expanding the phases with the compound-angle formula, a combination of the orthogonal products above, so it averages to zero; the cross terms with the constant average to zero because each harmonic does. Each squared harmonic averages to $A_n^2/2$ (§3.2). □

This is the finite version of **Parseval's theorem** (Ch. 15). Two warnings: the result needs *distinct* frequencies (two components at the same frequency must first be combined with phasors, §3.3, because their cross term does *not* vanish); and it concerns power, not peak value — the peak of x depends strongly on the phases (recall the two waveforms in §3.5 with identical RMS but different peaks).

Worked Example 3.29 — RMS of a multi-harmonic waveform

Find the RMS value of $v(t) = 10 + 8\cos\omega t + 6\cos(3\omega t + 40^\circ)$ V and the average power it delivers to $R = 5\ \Omega$.

Solution. $V_{\text{rms}}^2 = 10^2 + \frac{8^2}{2} + \frac{6^2}{2} = 100 + 32 + 18 = 150\ \text{V}^2$, so $V_{\text{rms}} = 12.25$ V. Power $= 150/5 = 30$ W, of which 20 W is DC, 6.4 W fundamental, 3.6 W third harmonic. The 40° phase plays no role.

Check: numerical averaging of v^2 over a period on 200 000 points gives $V_{\text{rms}} = 12.2474$ V. ✓

Nonlinearity generates harmonics

A linear system cannot create new frequencies. A nonlinear one — a diode, a transistor driven hard, a saturating transformer core, an amplifier clipping at its supply rails — can and does. Many weakly nonlinear devices are well modelled by a polynomial (the first few terms of a Taylor expansion about the operating point, Ch. 4):

$$y = a_1 x + a_2 x^2 + a_3 x^3.$$

Drive it with $x = A\cos\omega t$ and reduce the powers with §3.6:

$$x^2 = \frac{A^2}{2} + \frac{A^2}{2}\cos 2\omega t, \qquad x^3 = \frac{3A^3}{4}\cos\omega t + \frac{A^3}{4}\cos 3\omega t.$$

Harmonics from a cubic nonlinearity

For $y = a_1 x + a_2 x^2 + a_3 x^3$ and $x = A\cos\omega t$,

$$y = \underbrace{\frac{a_2 A^2}{2}}_{\text{DC}} + \underbrace{\left(a_1 A + \frac{3a_3 A^3}{4}\right)}_{\text{fundamental}}\cos\omega t + \underbrace{\frac{a_2 A^2}{2}}_{\text{2nd}}\cos 2\omega t + \underbrace{\frac{a_3 A^3}{4}}_{\text{3rd}}\cos 3\omega t.$$

Four features generalise to higher-order polynomials:

1. A term x^k produces harmonics up to the k-th, of the same parity as k (the $\cos^n$ theorem of §3.6).
2. **Even-order terms** produce even harmonics and a **DC shift** — the principle of rectification and of envelope detection.
3. **Odd-order terms** produce odd harmonics and also change the fundamental: if $a_3 < 0$ (a compressive, saturating device) the fundamental gain falls as A grows — **gain compression**.
4. The k-th harmonic grows as A^k: halving the drive level (a 6 dB reduction) lowers the fundamental by 6 dB, the 2nd harmonic by 12 dB and the 3rd by 18 dB, so *relative to the fundamental* the 2nd harmonic improves by 6 dB and the 3rd by 12 dB. Plotting harmonic levels against drive level on log axes gives straight lines of slope k — the basis of intercept-point specifications.

A device with odd symmetry, $y(-x) = -y(x)$ (a symmetric clipper, a push–pull amplifier), has $a_2 = a_4 = \cdots = 0$ and generates **only odd harmonics**; this is the waveform-symmetry result of §3.8 seen from the device side.

Worked Example 3.30 — Harmonic distortion of a weakly nonlinear amplifier

An amplifier has $y = x + 0.1x^2 + 0.05x^3$ (volts) and is driven by $x = 2\cos\omega t$ V. Find each output component and the second- and third-harmonic distortion relative to the output fundamental.

Solution. With $A = 2$, $a_1 = 1$, $a_2 = 0.1$, $a_3 = 0.05$:

- DC: $a_2 A^2/2 = 0.1(4)/2 = 0.20$ V,
- fundamental: $a_1 A + \frac{3}{4}a_3 A^3 = 2 + \frac{3}{4}(0.05)(8) = 2.30$ V,
- 2nd harmonic: $a_2 A^2/2 = 0.20$ V,
- 3rd harmonic: $a_3 A^3/4 = 0.05(8)/4 = 0.10$ V.

$\text{HD}_2 = 0.20/2.30 = 8.7\%$, $\text{HD}_3 = 0.10/2.30 = 4.3\%$.

Check: at $\omega t = 0$, $x = 2$ and $y = 2 + 0.4 + 0.4 = 2.8$; the harmonic sum gives $0.2 + 2.3 + 0.2 + 0.1 = 2.8$. ✓ Numerical projection onto $\cos n\omega t$ also returns $0.2,\ 2.3,\ 0.2,\ 0.1$.

Two tones in a nonlinearity: intermodulation

Now drive the same nonlinearity with two tones, $x = A\cos\omega_1 t + A\cos\omega_2 t$. The square term gives, by product-to-sum,

$$x^2 = A^2 + \frac{A^2}{2}\cos 2\omega_1 t + \frac{A^2}{2}\cos 2\omega_2 t + A^2\cos(\omega_1 + \omega_2)t + A^2\cos(\omega_1 - \omega_2)t.$$

Besides the harmonics $2\omega_1$, $2\omega_2$ there are new **intermodulation (IM) products** at $\omega_1 \pm \omega_2$. The cube term, expanded the same way, produces components at

$$\omega_1,\ \omega_2,\ 3\omega_1,\ 3\omega_2,\ 2\omega_1 \pm \omega_2,\ 2\omega_2 \pm \omega_1.$$

In general an order-k nonlinearity produces every frequency $|p\omega_1 + q\omega_2|$ with integers p, q satisfying $|p| + |q| \le k$ and of the same parity as k. The **order** of a product is $|p| + |q|$.

Worked Example 3.31 — Third-order intermodulation

Two equal carriers at $f_1 = 100$ MHz and $f_2 = 101$ MHz pass through an amplifier with second- and third-order nonlinearity. List the output frequencies and identify those that fall near the original band.

Second order ($|p| + |q| = 2$): $2f_1 = 200$, $2f_2 = 202$, $f_1 + f_2 = 201$, $f_2 - f_1 = 1$ MHz, plus DC.

Third order ($|p| + |q| = 3$): $3f_1 = 300$, $3f_2 = 303$, $2f_1 + f_2 = 301$, $2f_2 + f_1 = 302$, $2f_1 - f_2 = 99$, $2f_2 - f_1 = 102$ MHz, plus contributions at f_1 and f_2 themselves.

Interpretation. Every second-order product and every harmonic lies far outside a 99–102 MHz band and can be filtered out. The **third-order IM products at 99 MHz and 102 MHz** sit right next to the wanted signals, cannot be filtered, and land on neighbouring channels. This is why RF amplifiers and mixers are specified by their third-order intercept point rather than by their harmonic distortion.

Mixing and modulation

A **mixer** multiplies two signals. With product-to-sum,

$$\cos\omega_1 t \cdot \cos\omega_2 t = \tfrac{1}{2}\cos(\omega_1 - \omega_2)t + \tfrac{1}{2}\cos(\omega_1 + \omega_2)t,$$

so multiplication *shifts* frequencies: a signal at ω_1 mixed with a local oscillator at ω_2 reappears at the **difference** (intermediate) frequency $|\omega_1 - \omega_2|$ and the **sum** frequency $\omega_1 + \omega_2$. A filter then keeps the one that is wanted. This is the superheterodyne principle behind essentially every radio receiver.

Amplitude modulation is mixing of a carrier with $(1 + \text{message})$:

$$s(t) = A_c\big[1 + m\cos\omega_m t\big]\cos\omega_c t = A_c\cos\omega_c t + \frac{mA_c}{2}\cos(\omega_c - \omega_m)t + \frac{mA_c}{2}\cos(\omega_c + \omega_m)t.$$

The spectrum is a carrier with two **sidebands** at $\omega_c \pm \omega_m$; $0 < m \le 1$ is the **modulation index**.

Worked Example 3.32 — Spectrum and power of an AM signal

A 1 MHz carrier of amplitude 10 V is amplitude-modulated by a 5 kHz tone with $m = 0.5$. Find the spectral components and the fraction of the total power in the sidebands.

Components. Carrier: 1000 kHz, 10 V. Sidebands: 995 kHz and 1005 kHz, each $mA_c/2 = 2.5$ V.

Power (into 1 Ω, using orthogonality): carrier $10^2/2 = 50$ W; each sideband $2.5^2/2 = 3.125$ W; total 56.25 W. Sideband fraction $6.25/56.25 = 11.1\%$.

General formula. Sideband fraction $= \dfrac{m^2/2}{1 + m^2/2}$; at $m = 0.5$ this is $0.125/1.125 = 11.1\%$ ✓, and even at $m = 1$ only one third of the power carries information — the reason for suppressed-carrier and single-sideband systems.

Beats

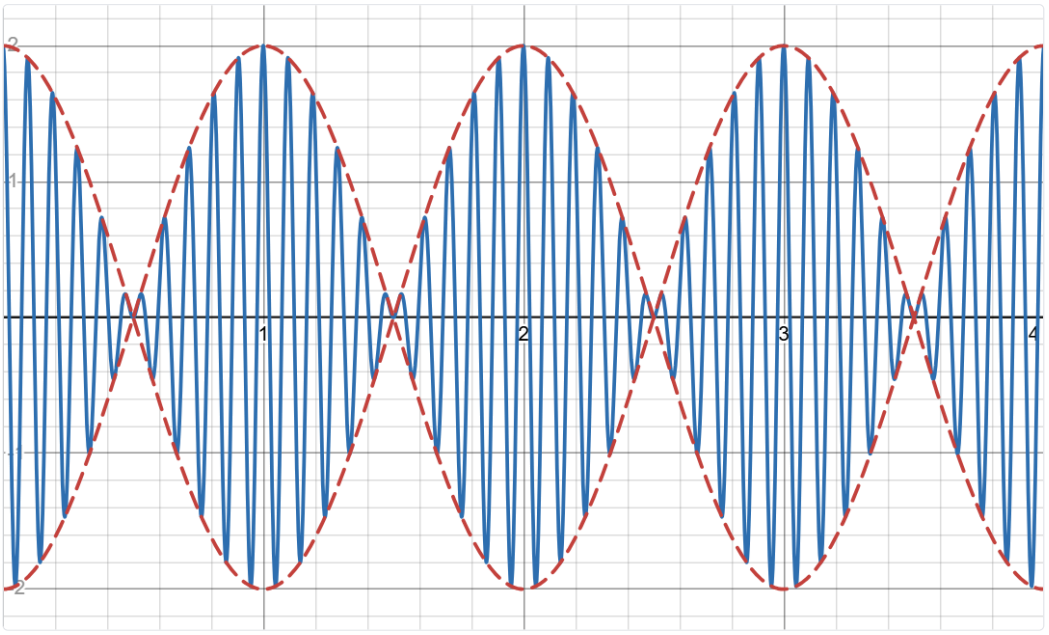
The sum of two equal-amplitude sinusoids at *close* frequencies is best understood through sum-to-product:

$$\cos\omega_1 t + \cos\omega_2 t = 2\cos\left(\frac{\omega_1 - \omega_2}{2}t\right)\cos\left(\frac{\omega_1 + \omega_2}{2}t\right).$$

When $|\omega_1 - \omega_2| \ll \omega_1$, this is a fast oscillation at the **average** frequency $\bar\omega = (\omega_1 + \omega_2)/2$, whose amplitude is modulated by the slow envelope $|2\cos(\Delta\omega\, t/2)|$, with $\Delta\omega = \omega_1 - \omega_2$. The envelope magnitude peaks twice per cycle of $\cos(\Delta\omega t/2)$, so the loudness (or the reading of an envelope detector) pulses at the **beat frequency**

$$f_{\text{beat}} = |f_1 - f_2|.$$

Physically: the two phasors rotate at slightly different speeds; once per beat period they line up (constructive, amplitude $2A$) and once they oppose (destructive, amplitude 0).



Beats: $\cos(2\pi\,10t) + \cos(2\pi\,11t)$ (blue) inside the envelope $\pm 2\cos(\pi t)$ (red). The envelope magnitude peaks once per second, the beat frequency $|11 - 10| = 1$ Hz.

Worked Example 3.33 — Tuning by beats, and the difference with AM

(a) A reference tone at 440 Hz and a slightly sharp tone at 444 Hz sound together. What is heard? (b) Compare $\cos\omega_1 t + \cos\omega_2 t$ with the AM signal of Example 3.32. Why do they look similar?

(a) A 442 Hz tone whose loudness pulses $|444 - 440| = 4$ times per second. Adjusting until the beats slow to zero brings the tones into unison — the classical way to tune an instrument, and the principle of a *zero-beat* frequency comparison.

(b) By sum-to-product, the beat signal is a product $2\cos(\Delta\omega t/2)\cos\bar{\omega}t$: a carrier at $\bar{\omega}$ multiplied by a sinusoid at $\Delta\omega/2$. That is AM with *suppressed* carrier (double-sideband suppressed carrier, DSB-SC): two sidebands at $\bar{\omega} \pm \Delta\omega/2$ and no component at $\bar{\omega}$ itself. The "envelope" crosses zero and the fast oscillation flips sign there, unlike true AM with $m < 1$, whose envelope $1 + m\cos\omega_m t$ stays positive.

Total harmonic distortion

Total harmonic distortion

For a periodic signal with fundamental RMS amplitude X_1 and harmonic RMS amplitudes $X_2, X_3, \ldots$,

$$\mathrm{THD} = \frac{\sqrt{X_2^2 + X_3^2 + X_4^2 + \cdots}}{X_1}.$$

It is often quoted in percent or in dB ($20\log_{10}$ THD). By orthogonality the total RMS value is $X_{\mathrm{rms}} = X_1\sqrt{1 + \mathrm{THD}^2}$ when there is no DC component.

(Some instruments normalise by the total RMS instead of the fundamental — "THD-R" or THD+N — so state which definition is used.)

Worked Example 3.34 — THD of a rectifier load current

A single-phase rectifier draws a 50 Hz current with RMS harmonic content $I_1 = 10$ A, $I_3 = 3$ A, $I_5 = 2$ A, $I_7 = 1$ A. Find the THD, the total RMS current, and the RMS current a meter responding only to the fundamental would read.

Solution. $\mathrm{THD} = \sqrt{9 + 4 + 1}/10 = \sqrt{14}/10 = 37.4\%$. $I_{\mathrm{rms}} = \sqrt{100 + 9 + 4 + 1} = \sqrt{114} = 10.68$ A $= 10\sqrt{1 + 0.374^2}$ ✓. A fundamental-only meter reads 10 A, under-reading the heating current by 6.4 %.

Engineering consequence. The harmonic currents carry no average power with a sinusoidal supply voltage (by orthogonality, $V_1 I_n \cos(\cdot)$ averages to zero for $n \neq 1$), yet they heat cables and transformers. The *true* power factor therefore drops below $\cos\Delta\phi_1$ by the factor $1/\sqrt{1 + \mathrm{THD}^2}$.

Harmonics in three-phase systems

Take a balanced set of identical, distorted phase currents, each a copy of the previous one delayed by one-third of a period: $i_b(t) = i_a(t - T_0/3)$, $i_c(t) = i_a(t - 2T_0/3)$. A delay of $T_0/3$ shifts the n-th harmonic by $n\omega_0 T_0/3 = n \cdot 120^\circ$. So the n-th harmonics of the three phases have phases 0°, $-120^\circ n$ and $-240^\circ n$, which reduce as follows:

Harmonic n	Phases (mod 360°)	Sequence	Sum of the three phases
$1, 4, 7, \ldots$ $(n = 3k + 1)$	$0^\circ, -120^\circ, +120^\circ$	positive (abc)	0
$2, 5, 8, \ldots$ $(n = 3k + 2)$	$0^\circ, +120^\circ, -120^\circ$	negative (acb)	0
$3, 6, 9, \ldots$ $(n = 3k)$	$0^\circ, 0^\circ, 0^\circ$	zero (in phase)	$3\times$ one phase

Triplen harmonics

In a balanced three-phase set, the harmonics of order $3, 6, 9, \ldots$ (**triplens**) are in phase in all three lines. They do not cancel in the neutral: the neutral current at each triplen frequency is three times the per-phase value. All other harmonics form balanced sets (positive or negative sequence) and sum to zero.

Worked Example 3.35 — Neutral current from single-phase loads

A four-wire 50 Hz system feeds three identical banks of switch-mode supplies, one per phase. Each phase current has $I_1 = 20$ A, $I_3 = 12$ A, $I_5 = 6$ A (RMS). Find the RMS neutral current.

Solution. Fundamentals and 5th harmonics form balanced sets and cancel in the neutral. The 3rd harmonics add in phase: $I_N = 3 \times 12 = 36$ A at 150 Hz.

Comparison. Each phase carries $\sqrt{400 + 144 + 36} = 24.1$ A RMS; the neutral carries 36 A — **more than any phase conductor**, although the load is "balanced". This is why neutral conductors in buildings full of computers are oversized, and why transformers feeding such loads use a delta winding, in which triplen currents circulate instead of reaching the supply.

Engineering Note — Thinking in harmonics

A good habit when meeting any nonlinear or time-varying operation is to ask "what does it do to a single sinusoid?" and then "what does it do to two?". Squaring doubles frequencies and creates DC; cubing triples them and creates $2f_1 - f_2$; multiplying shifts; rectifying (§3.8) creates all even harmonics. Once you know the frequencies a device produces, filter design, channel planning and EMC compliance become bookkeeping.

3.8 Periodic function representation

Motivation

Real periodic signals are rarely pure sinusoids: a clock is a square wave, a switching converter's inductor current is a triangle, a scanning circuit produces a sawtooth, a rectifier delivers humps of sine. We need to (i) write such waveforms down unambiguously, (ii) compute their mean and RMS values, (iii) recognise their symmetries, and (iv) express them as sums of harmonics so that the sinusoidal machinery of the previous sections — phasors, filtering, power by orthogonality — can be applied to them. Items (i)–(iii) are completed here; item (iv) is introduced here and developed fully as Fourier series in Ch. 15.

Defining a periodic function from one period

A periodic function is completely determined by its values on any interval of length T (one period). The standard way to specify it is to give a formula on, say, $0 \le t < T$ (or $-T/2 \le t < T/2$) and then declare the **periodic extension**

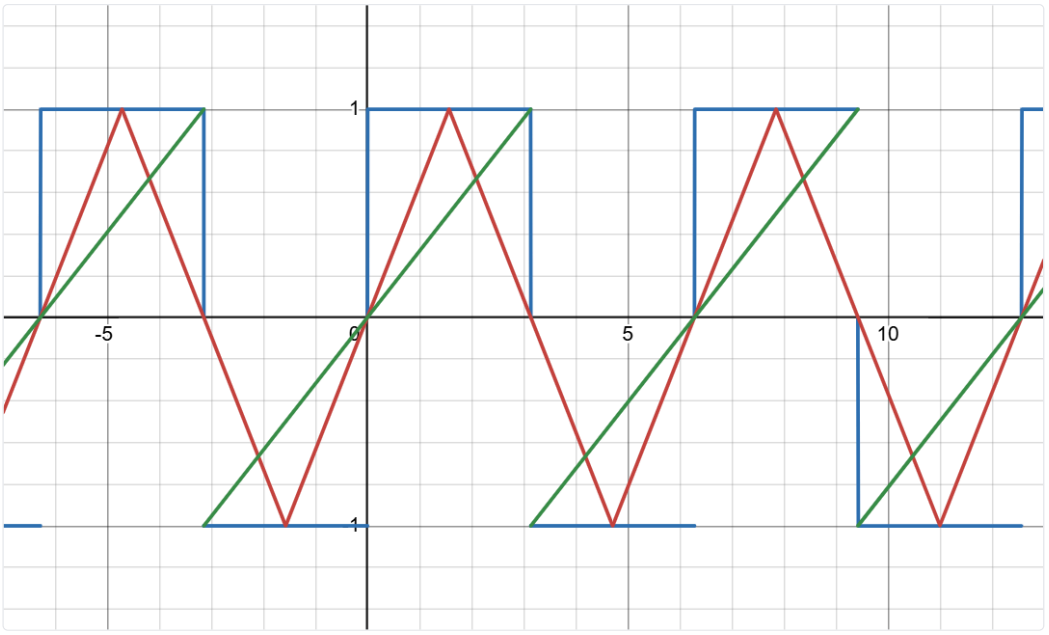
$$x(t + kT) = x(t), \qquad k \in \mathbb{Z}.$$

To evaluate the extension at an arbitrary t, reduce t into the base interval with the floor function: $t \mapsto t - T\lfloor t/T \rfloor \in [0, T)$, which is what `mod(t, T)` computes for $T > 0$.

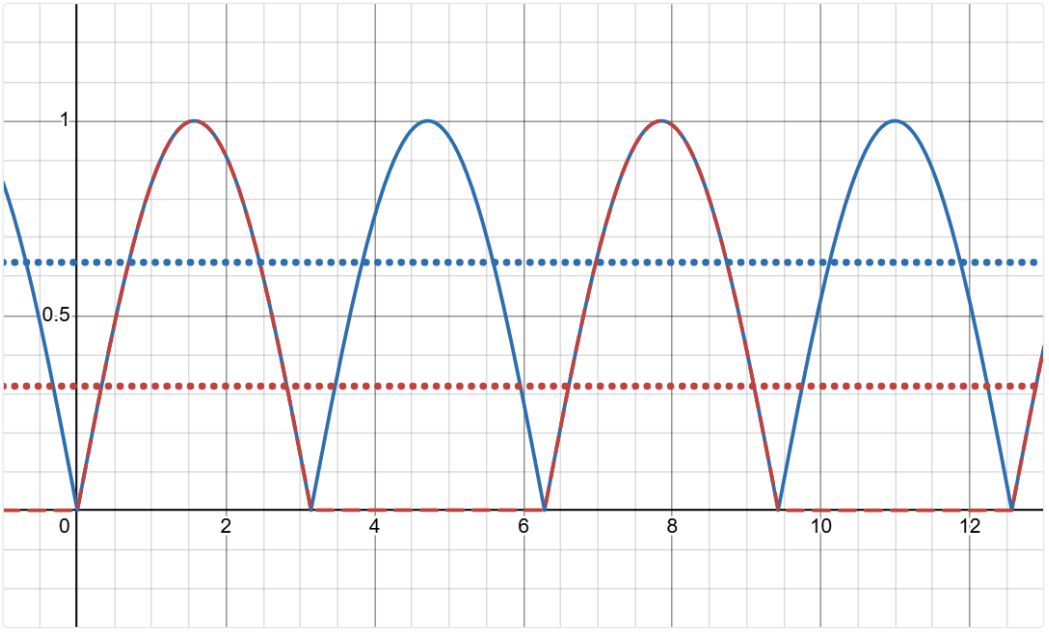
The common engineering waveforms, with amplitude A, period T and $\omega_0 = 2\pi/T$:

Waveform	One period	Closed form for all t
square wave ($\pm A$)	A on $(0, T/2)$, $-A$ on $(T/2, T)$	$A\,\mathrm{sgn}(\sin \omega_0 t)$
sawtooth ($-A$ to A)	$2At/T$ on $(-T/2, T/2)$	$2A\left(\frac{t}{T} - \left\lfloor \frac{t}{T} + \frac{1}{2} \right\rfloor\right)$
triangle ($\pm A$)	linear between $-A$ and A	$\frac{2A}{\pi} \arcsin(\sin \omega_0 t)$
full-wave rectified sine	$A\lvert\sin \omega_0 t\rvert$	$A\lvert\sin \omega_0 t\rvert$ (period $T/2$)
half-wave rectified sine	$A \sin \omega_0 t$ on $(0, T/2)$, 0 on $(T/2, T)$	$\frac{A}{2}\left(\sin \omega_0 t + \lvert\sin \omega_0 t\rvert\right)$
pulse train, duty cycle D	A on $(0, DT)$, 0 on (DT, T)	$A \cdot \mathbf{1}\left[\mathrm{mod}(t, T) < DT\right]$

The triangle-wave formula works because $\arcsin(\sin\theta)$ equals θ on $[-\pi/2, \pi/2]$ and folds back linearly outside it — an application of the principal branch of §3.1.



Three standard waveforms of period 2π and amplitude 1: square (blue), triangle (red) and sawtooth (green), all generated from closed-form expressions.



Full-wave (blue) and half-wave (red, dashed) rectified sines with their mean values $2/\pi$ and $1/\pi$ (dotted). The full-wave output repeats every π — twice the input frequency.

Worked Example 3.36 — Evaluating a periodic extension

A sawtooth generator produces a waveform of amplitude 5 V and period 2 ms that rises linearly from -5 V to $+5$ V. Find its value at $t = 3.7$ ms and at $t = -0.3$ ms.

Solution. $x(t) = 10\big(t/T - \lfloor t/T + \tfrac{1}{2}\rfloor\big)$ V with $T = 2$ ms.

- $t = 3.7$ ms: $t/T = 1.85$, $\lfloor 2.35\rfloor = 2$, so $x = 10(1.85 - 2) = -1.5$ V.
- $t = -0.3$ ms: $t/T = -0.15$, $\lfloor 0.35\rfloor = 0$, so $x = 10(-0.15) = -1.5$ V.

Check: the two times differ by 4 ms $= 2T$, so the values must agree. ✓ Note that $\lfloor\cdot\rfloor$ rounds *down* for negative arguments ($\lfloor -0.65\rfloor = -1$), which is why the floor form handles negative t correctly where truncation would not.

Mean and RMS of standard waveforms

Using $\bar{x} = \frac{1}{T}\int_0^T x\,dt$ and $X_{\text{rms}}^2 = \frac{1}{T}\int_0^T x^2\,dt$ (Ch. 5):

Waveform (amplitude A)	Mean	RMS	Crest factor
sine	0	$A/\sqrt{2}$	$\sqrt{2} = 1.414$
square ($\pm A$)	0	A	1

Waveform (amplitude A)	Mean	RMS	Crest factor
triangle or sawtooth ($\pm A$)	0	$A/\sqrt{3}$	$\sqrt{3} = 1.732$
full-wave rectified sine	$2A/\pi$	$A/\sqrt{2}$	$\sqrt{2}$
half-wave rectified sine	A/π	$A/2$	2
pulse train, 0 to A, duty D	DA	$A\sqrt{D}$	$1/\sqrt{D}$

Derivations. *Full-wave mean:* over one hump, $\frac{1}{\pi}\int_0^{\pi} A\sin\theta\,d\theta = \frac{A}{\pi}[-\cos\theta]_0^{\pi} = \frac{2A}{\pi}$. *Full-wave RMS:* $|\sin|^2 = \sin^2$, so the RMS is that of the sine. *Half-wave:* the waveform equals the sine on half the period and zero on the other half, so its mean is half the full-wave mean, A/π, and its mean square is half of $A^2/2$, giving RMS $A/2$. *Sawtooth:* on $(-T/2, T/2)$, $x = 2At/T$, so

$$X_{\text{rms}}^2 = \frac{1}{T}\int_{-T/2}^{T/2} \frac{4A^2t^2}{T^2}\,dt = \frac{4A^2}{T^3}\cdot\frac{2}{3}\left(\frac{T}{2}\right)^3 = \frac{A^2}{3}.$$

The triangle wave consists of the same linear ramps rearranged in time, so it has the same RMS. *Pulse train:* $x^2 = A^2$ for a fraction D of the time, so $X_{\text{rms}}^2 = DA^2$.

Worked Example 3.37 — Rectifier and PWM levels

(a) A half-wave rectifier fed from 230 V RMS mains (peak 325 V) drives a resistive load. What DC voltmeter and true-RMS voltmeter readings are expected across the load? (b) A PWM output switches between 0 and 5 V at 25 % duty cycle. Find its mean and RMS, and the power delivered to a 10 Ω heater.

(a) Mean $= 325/\pi = 103.5$ V. RMS $= 325/2 = 162.5$ V.

(b) Mean $= 0.25 \times 5 = 1.25$ V. RMS $= 5\sqrt{0.25} = 2.5$ V. Power $= 2.5^2/10 = 0.625$ W — which is $D \times 5^2/10$, as it must be, and *not* $1.25^2/10 = 0.156$ W. Heating depends on the RMS value, not the mean.

Symmetry

Symmetries of a waveform constrain which harmonics it can contain, often halving the work of analysis and giving an instant sanity check.

Even, odd and half-wave symmetry

A periodic function x with period T is

- **even** if $x(-t) = x(t)$ (mirror symmetry about $t = 0$);
- **odd** if $x(-t) = -x(t)$ (point symmetry about the origin);
- **half-wave symmetric** if $x(t + T/2) = -x(t)$ (the second half-cycle is the first one inverted).

Every function splits uniquely into even and odd parts,

$$x(t) = x_e(t) + x_o(t), \qquad x_e(t) = \frac{x(t) + x(-t)}{2}, \qquad x_o(t) = \frac{x(t) - x(-t)}{2}.$$

Symmetry and harmonic content

For a periodic function written as a sum of harmonics $a_0 + \sum_{n\ge1}(a_n\cos n\omega_0 t + b_n \sin n\omega_0 t)$:

1. If x is **even**, all $b_n = 0$: only cosines (and possibly DC) appear.
2. If x is **odd**, $a_0 = 0$ and all $a_n = 0$: only sines appear.
3. If x has **half-wave symmetry**, $a_0 = 0$ and all **even** harmonics vanish: only odd harmonics appear.

Shifting the time origin changes even/odd symmetry but not half-wave symmetry, so property 3 is the more robust one.

Proof of 3. If $x(t + T/2) = -x(t)$, then the mean over a period is minus itself, so $a_0 = 0$. For the n-th harmonic coefficient (formula below), split $\int_0^T$ into $\int_0^{T/2} + \int_{T/2}^{T}$ and substitute $t = \tau + T/2$ in the second:

$$\int_{T/2}^{T} x(t)\cos n\omega_0 t\,dt = \int_0^{T/2} x(\tau + \tfrac{T}{2})\cos(n\omega_0\tau + n\pi)\,d\tau = (-1)^{n+1}\int_0^{T/2} x(\tau)\cos n\omega_0\tau\,d\tau.$$

So $a_n \propto \left(1 + (-1)^{n+1}\right)\int_0^{T/2} x\cos n\omega_0 t\,dt$, which is zero for even n; the same holds for b_n. Properties 1 and 2 follow because cos is even, sin is odd, and the integral over a symmetric interval of an odd function is zero. □

Worked Example 3.38 — Predicting harmonic content from symmetry

Without computing any integrals, state which harmonics can appear in: (a) the square wave $A\,\mathrm{sgn}(\sin\omega_0 t)$; (b) the triangle wave; (c) the sawtooth; (d) the full-wave rectified sine $A|\sin\omega_0 t|$ (period T of the *original* sine); (e) the half-wave rectified sine.

- **(a)** Odd and half-wave symmetric: only sines of odd harmonics.
- **(b)** $\frac{2A}{\pi}\arcsin(\sin\omega_0 t)$ is odd and half-wave symmetric: only sines of odd harmonics.
- **(c)** Odd but *not* half-wave symmetric: sines of all harmonics.
- **(d)** Even, and it repeats every $T/2$, so in terms of ω_0 it contains a DC term and only **even** harmonics $2\omega_0, 4\omega_0, \ldots$, all cosines. (Its own fundamental is $2\omega_0$.)
- **(e)** Neither even nor odd and not half-wave symmetric. But $\frac{A}{2}(\sin\omega_0 t + |\sin\omega_0 t|)$ shows it is half a sine plus half of (d): DC, the fundamental $\sin\omega_0 t$, and even cosine harmonics — and **no odd harmonics other than the fundamental**.

These predictions are confirmed by the expansions in the summary table below and by numerical projection.

Representing a periodic function by harmonics

The orthogonality relations of §3.7 suggest how a periodic function might be built from harmonics. *Suppose* that

$$x(t) = a_0 + \sum_{n=1}^{\infty}\left(a_n\cos n\omega_0 t + b_n\sin n\omega_0 t\right).$$

Multiply both sides by $\cos m\omega_0 t$ and integrate over a period. By orthogonality every term on the right vanishes except the $n=m$ cosine term, which contributes $a_m T/2$. Doing the same with $\sin m\omega_0 t$ and with 1 gives:

Harmonic (Fourier) coefficients

If x has a harmonic representation as above, its coefficients are

$$a_0 = \frac{1}{T}\int_0^T x(t)\,dt, \qquad a_n = \frac{2}{T}\int_0^T x(t)\cos n\omega_0 t\,dt, \qquad b_n = \frac{2}{T}\int_0^T x(t)\sin n\omega_0 t\,dt.$$

In amplitude–phase form (§3.2) the n-th harmonic is $A_n\cos(n\omega_0 t+\phi_n)$ with $A_n=\sqrt{a_n^2+b_n^2}$ and $\phi_n = \operatorname{atan2}(-b_n, a_n)$. In complex form, $x(t)=\sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t}$ with $c_n = \frac{1}{T}\int_0^T x(t)e^{-jn\omega_0 t}\,dt$, $c_0 = a_0$ and $c_{\pm n} = \frac{1}{2}(a_n \mp jb_n)$ for $n\ge 1$.

Each coefficient is a *correlation*: a_n measures how much x "looks like" $\cos n\omega_0 t$. The questions of *which* functions have such representations, in what sense the infinite series converges, and what happens at discontinuities (the Gibbs phenomenon) are the subject of Ch. 15. For all the waveforms in this section the series converges to $x(t)$ at every point of continuity, and to the midpoint of the jump at a discontinuity.

Worked Example 3.39 — Harmonics of a square wave

Find the harmonic representation of the square wave $x = A$ on $(0, T/2)$, $x = -A$ on $(T/2, T)$, its THD, and check Parseval's relation.

Symmetry first. Odd and half-wave symmetric, so $a_0 = a_n = 0$ and only odd-n sines appear.

Coefficients. Directly from the definition:

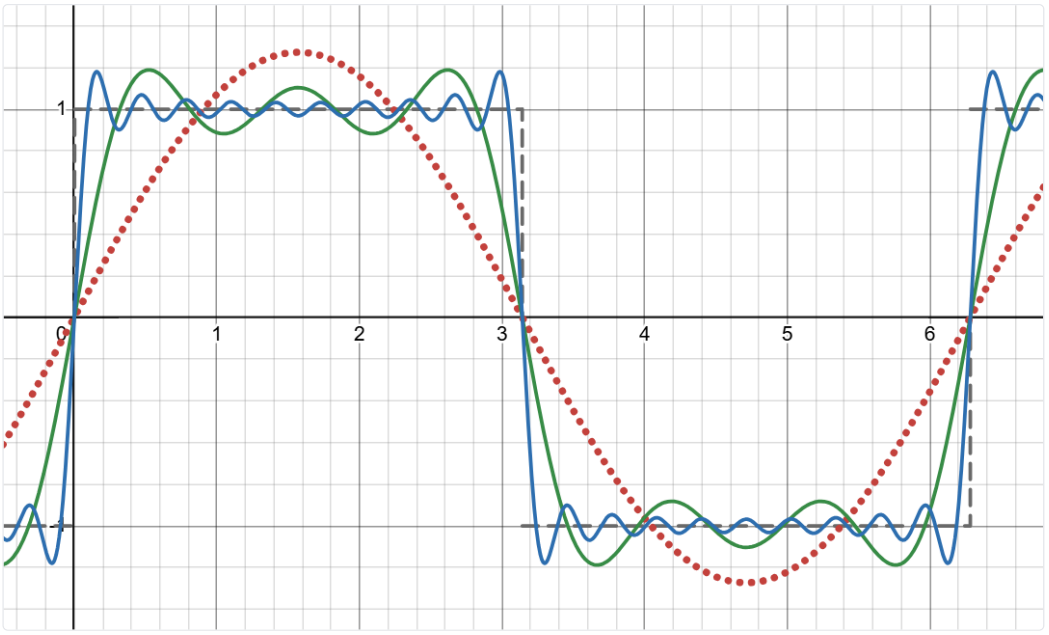
$$b_n = \frac{2}{T}\left[\int_0^{T/2} A\sin n\omega_0 t\,dt - \int_{T/2}^{T} A\sin n\omega_0 t\,dt\right] = \frac{2A}{T}\cdot\frac{1}{n\omega_0}\left[(1-\cos n\pi) - (\cos n\pi - \cos 2n\pi)\right] = \frac{2A}{n\pi}\left(1-(-1)^n\right).$$

So $b_n = \dfrac{4A}{n\pi}$ for odd n and 0 for even n:

$$x(t) = \frac{4A}{\pi}\left(\sin\omega_0 t + \frac{1}{3}\sin 3\omega_0 t + \frac{1}{5}\sin 5\omega_0 t + \cdots\right).$$

THD. The fundamental has RMS $\frac{4A}{\pi\sqrt{2}} = 0.9003A$; the total RMS is A. So $\mathrm{THD} = \sqrt{A^2 - 0.8106A^2}/(0.9003A) = \sqrt{\pi^2/8 - 1} = 48.3\%$.

Parseval check. $\sum_{n\ \text{odd}} \frac{1}{2}\left(\frac{4A}{n\pi}\right)^2 = \frac{8A^2}{\pi^2}\sum_{n\ \text{odd}}\frac{1}{n^2} = \frac{8A^2}{\pi^2}\cdot\frac{\pi^2}{8} = A^2 = X_{\mathrm{rms}}^2$ ✓, using the classical sum $1+\frac{1}{9}+\frac{1}{25}+\cdots = \pi^2/8$ (itself provable from this very example in Ch. 15). Numerical projection gives $b_1 = 1.2732A = 4A/\pi$, $b_3 = 0.4244A$, $b_5 = 0.2546A$. ✓



Building a square wave (grey) from harmonics: the fundamental alone (red), the sum through the 5th harmonic (green), and through the 19th (blue). The ripple narrows toward the jumps but its overshoot stays near 9 % of the jump — the Gibbs phenomenon of Ch. 15.

Worked Example 3.40 — Harmonics of a full-wave rectified sine

Find the DC and the first two nonzero harmonics of $x(t) = A|\sin \omega_0 t|$, and interpret for a full-wave rectifier fed at 50 Hz.

Symmetry. Even with period $T/2$: only DC and cosines at $2\omega_0, 4\omega_0, \dots$ (Example 3.38d).

DC. $a_0 = 2A/\pi$ (computed above).

Harmonics. By evenness and the $T/2$ periodicity, $a_n = \frac{2}{T}\int_0^T A|\sin \omega_0 t| \cos n\omega_0 t\, dt = \frac{2A}{\pi}\int_0^\pi \sin\theta \cos n\theta\, d\theta$ for even n (with $\theta = \omega_0 t$, using that the two half-periods contribute equally). Product-to-sum: $\sin\theta\cos n\theta = \frac{1}{2}[\sin(n+1)\theta - \sin(n-1)\theta]$, and $\int_0^\pi \sin k\theta\, d\theta = \frac{1-\cos k\pi}{k} = \frac{2}{k}$ for odd k. With n even, $n \pm 1$ are odd:

$$a_n = \frac{2A}{\pi}\cdot\frac{1}{2}\left[\frac{2}{n+1} - \frac{2}{n-1}\right] = \frac{2A}{\pi}\cdot\frac{-2}{n^2-1} = -\frac{4A}{\pi(n^2-1)}.$$

So $a_2 = -\frac{4A}{3\pi} = -0.4244A$ and $a_4 = -\frac{4A}{15\pi} = -0.0849A$:

$$A|\sin \omega_0 t| = \frac{2A}{\pi} - \frac{4A}{3\pi}\cos 2\omega_0 t - \frac{4A}{15\pi}\cos 4\omega_0 t - \cdots$$

Interpretation. From a 50 Hz supply the rectifier output is DC ($0.637A$) plus a dominant 100 Hz ripple of amplitude $0.424A$ and a 200 Hz component one-fifth as large. A smoothing filter is designed against the 100 Hz term.

Check: numerical projection gives $a_2 = -0.42441A$, $a_4 = -0.08488A$, $a_6 = -0.03638A$ $(= -4A/35\pi)$. ✓

Effect of time shift, scaling and smoothness on the harmonics

Three general rules let you reuse known expansions:

- **Time shift.** If $x(t)$ has harmonics $A_n\cos(n\omega_0 t + \phi_n)$, then $x(t - t_0)$ has harmonics $A_n\cos(n\omega_0 t + \phi_n - n\omega_0 t_0)$: amplitudes are unchanged; the n-th phase shifts by $-n\omega_0 t_0$ — proportional to n, the "linear phase" of a pure delay seen in §3.3. Example: the square wave shifted by $T/4$ becomes even, and its expansion becomes $\frac{4A}{\pi}\left(\cos\omega_0 t - \frac{1}{3}\cos 3\omega_0 t + \frac{1}{5}\cos 5\omega_0 t - \cdots\right)$.
- **Amplitude scaling and offset.** $\alpha x(t) + \beta$ multiplies every harmonic by α and adds β to the DC term. A 0-to-A square wave is $\frac{A}{2} + \frac{1}{2}(\pm A$ square wave$)$.
- **Smoothness and decay.** The smoother the waveform, the faster the harmonic amplitudes decay with n. A waveform with jumps (square, sawtooth) has $A_n \sim 1/n$; a continuous waveform with corners (triangle, rectified sine) has $A_n \sim 1/n^2$; each further continuous derivative buys another factor of $1/n$. Precise statements are in Ch. 15.

Waveform	Harmonic expansion	Decay
square $\pm A$	$\frac{4A}{\pi}\sum_{n\ \text{odd}}\frac{\sin n\omega_0 t}{n}$	$1/n$
sawtooth $-A$ to A	$\frac{2A}{\pi}\sum_{n\ge 1}\frac{(-1)^{n+1}\sin n\omega_0 t}{n}$	$1/n$
triangle $\pm A$ (odd, peak at $T/4$)	$\frac{8A}{\pi^2}\sum_{n\ \text{odd}}\frac{(-1)^{(n-1)/2}\sin n\omega_0 t}{n^2}$	$1/n^2$
full-wave $A\lvert\sin \omega_0 t\rvert$	$\frac{2A}{\pi} - \frac{4A}{\pi}\sum_{k\ge 1}\frac{\cos 2k\omega_0 t}{4k^2-1}$	$1/n^2$

Waveform	Harmonic expansion	Decay
half-wave rectified	$\frac{A}{\pi}+\frac{A}{2}\sin\omega_0 t-\frac{2A}{\pi}\sum_{k\geq 1}\frac{\cos 2k\omega_0 t}{4k^2-1}$	$1/n^2$

Engineering Note — Rise time, bandwidth and EMI

The $1/n$ decay of a square wave means that a digital clock edge contains significant energy at many harmonics of the clock frequency; a 100 MHz clock with fast edges produces radiated emissions at 300, 500, 700 MHz and beyond. Slowing the edges (rounding the corners) converts the spectrum to $1/n^2$ decay above a frequency of roughly $1/(\pi t_r)$, where t_r is the rise time — a standard EMC design rule. Conversely, an oscilloscope or channel that passes only the first few harmonics turns a square wave into the rippled shape in the partial-sum graph above.

Common Pitfall — Which fundamental?

When a waveform has a hidden shorter period (such as $|\sin\omega_0 t|$, whose period is $T/2$), state whether harmonic numbers refer to the original ω_0 or to the waveform's own fundamental $2\omega_0$. "The 2nd harmonic of the rectifier output" is ambiguous: it is 200 Hz if counted from the 100 Hz ripple, and 100 Hz if counted from the 50 Hz supply. Power-system practice counts from the supply frequency.

Summary

Topic	Key formulas
Angle units	$\theta_{\text{rad}}=s/r$; π rad $=180^\circ$; $\Omega=2\pi N/60$ (rpm); $\theta_e=p\,\theta_m$
Arc, sector	$s=r\theta$; $A=\frac{1}{2}r^2\theta$; $v=r\Omega$
Fundamental limit	$\lim_{\theta\to 0}\sin\theta/\theta=1$ (radians); $\sin\theta\approx\theta$, $\cos\theta\approx 1-\theta^2/2$
Quadrant-safe angle	$\operatorname{atan2}(y,x)\in(-\pi,\pi]$
General sinusoid	$x=A\cos(\omega t+\phi)+C$; shift $=-\phi/\omega$
Sine/cosine conversion	$\sin\theta=\cos(\theta-90^\circ)$; $-\cos\theta=\cos(\theta\pm 180^\circ)$
Amplitude–phase form	$a\cos\omega t+b\sin\omega t=R\cos(\omega t-\alpha)$, $R=\sqrt{a^2+b^2}$, $\alpha=\operatorname{atan2}(b,a)$
Derivative / integral	$\frac{d}{dt}$: $\times\omega$, $+90^\circ$; $\int dt$: $\div\omega$, -90°
Mean, RMS of sinusoid	$\bar{x}=C$; $X_{\text{rms}}=\sqrt{C^2+A^2/2}$; pure sine $A/\sqrt{2}$
Phase difference	$\Delta\phi=\phi_1-\phi_2\in(-180^\circ,180^\circ]$; $\Delta t=\Delta\phi/\omega$
Same-frequency sum	$R^2=A_1^2+A_2^2+2A_1A_2\cos(\phi_1-\phi_2)$
Three-phase	$\sum_{k=0}^{2}\cos(\omega t-120^\circ k)=0$; $V_{LL}=\sqrt{3}V_{ph}\angle 30^\circ$
Lissajous	$\lvert\sin\phi\rvert=y_0/Y$
Period, frequency	$T=2\pi/\omega=1/f$; $\omega=2\pi f$
Sum periodicity	periodic iff f_1/f_2 rational; $f_0=\gcd(f_1,f_2)$
Waves	$\cos(\omega t-kz)$; $\lambda=2\pi/k=v_p/f$; $v_p=\omega/k$
Discrete sinusoid	$\Omega=\omega T_s$; periodic iff $\Omega/2\pi=m/N$; $\Omega\equiv\Omega+2\pi$
Instantaneous frequency	$\omega_i=d\Theta/dt$
SHM	$\ddot{x}+\omega_0^2x=0\Rightarrow x=A\cos(\omega_0 t+\phi)$; $A=\sqrt{x_0^2+(v_0/\omega_0)^2}$
LC tank	$\omega_0=1/\sqrt{LC}$
Damped oscillation	$Ae^{-\alpha t}\cos(\omega_d t+\phi)$, $\omega_d=\sqrt{\omega_0^2-\alpha^2}$, $Q=\omega_0/2\alpha$
Compound angle	$\cos(A\pm B)=\cos A\cos B\mp\sin A\sin B$; $\sin(A\pm B)=\sin A\cos B\pm\cos A\sin B$
Power reduction	$\cos^2 A=\frac{1+\cos 2A}{2}$; $\sin^2 A=\frac{1-\cos 2A}{2}$; $\cos^3 A=\frac{3\cos A+\cos 3A}{4}$
Powers of cosine	$\cos^n\theta=2^{-n}\sum_k\binom{n}{k}\cos(n-2k)\theta$
Chebyshev	$\cos n\theta=T_n(\cos\theta)$, $T_{n+1}=2xT_n-T_{n-1}$
Product-to-sum	$\cos A\cos B=\frac{1}{2}[\cos(A-B)+\cos(A+B)]$
Sum-to-product	$\cos P+\cos Q=2\cos\frac{P+Q}{2}\cos\frac{P-Q}{2}$
Phasor-array sum	$\left\lvert\sum_{k=0}^{N-1}e^{jk\delta}\right\rvert=\lvert\sin(N\delta/2)/\sin(\delta/2)\rvert$
Average power	$P=V_{\text{rms}}I_{\text{rms}}\cos(\phi_v-\phi_i)$
Orthogonality	$\frac{2}{T}\int_0^T\cos m\omega_0 t\cos n\omega_0 t\,dt=\delta_{mn}$ $(m,n\geq 1)$
Power of harmonic sum	$X_{\text{rms}}^2=A_0^2+\sum A_n^2/2$
Cubic nonlinearity	$x^3\to\frac{3A^3}{4}\cos\omega t+\frac{A^3}{4}\cos 3\omega t$
IM products	$\lvert pf_1+qf_2\rvert$, order $\lvert p\rvert+\lvert q\rvert$; IM3 at $2f_1-f_2$, $2f_2-f_1$
Beats	$f_{\text{beat}}=\lvert f_1-f_2\rvert$
AM	$A_c(1+m\cos\omega_m t)\cos\omega_c t$: sidebands $mA_c/2$ at $\omega_c\pm\omega_m$

Topic	Key formulas
THD	$\sqrt{\sum_{n\geq 2} X_n^2}/X_1$
Triplens	harmonics $3k$ are zero-sequence: neutral current $= 3\times$ per phase
Standard waveform RMS	square A; triangle/sawtooth $A/\sqrt{3}$; half-wave $A/2$; pulse $A\sqrt{D}$
Symmetry	even → cosines; odd → sines; half-wave → odd harmonics only
Harmonic coefficients	$a_0 = \bar{x}$; $a_n, b_n = \frac{2}{T}\int_0^T x\,\{\cos,\sin\}\,n\omega_0 t\,dt$
Square wave	$\frac{4A}{\pi}\sum_{n\ \text{odd}} \frac{\sin n\omega_0 t}{n}$; THD 48.3%

Further reading

- E. Kreyszig, *Advanced Engineering Mathematics*, 10th ed., Wiley, 2011 — Appendix on trigonometric identities and Ch. 11 on Fourier series, the natural continuation of §3.7–3.8.
- K. A. Stroud and D. J. Booth, *Engineering Mathematics*, 8th ed., Palgrave/Bloomsbury, 2020 — patient, example-driven treatment of trigonometry, sinusoids and phasors.
- A. V. Oppenheim and A. S. Willsky, *Signals and Systems*, 2nd ed., Prentice Hall, 1997 — Ch. 1 on continuous- and discrete-time sinusoids, periodicity and aliasing; Ch. 3 on harmonic representation.
- C. K. Alexander and M. N. O. Sadiku, *Fundamentals of Electric Circuits*, 7th ed., McGraw-Hill, 2021 — sinusoids, phase, average power, three-phase systems and harmonics from the circuit side.

Part II — Calculus

Chapter 4 — Differential Calculus

> **Prerequisites:** Ch. 1 (functions, algebra, inequalities), Ch. 2 (complex exponentials, j), Ch. 3 (trigonometric identities, radians) **Learning outcomes:** After this chapter you can
>
> - evaluate limits (algebraically, with the squeeze theorem, with L'Hôpital's rule and with series) and decide whether a function is continuous;
> - differentiate any elementary function using the definition, the product, quotient and chain rules, implicit and logarithmic differentiation;
> - compute partial and higher-order derivatives and use them to describe how a quantity responds to each of its inputs;
> - build Taylor and Maclaurin polynomials, bound their error, and use them to linearize nonlinear devices around an operating point;
> - solve rate-of-change and related-rates problems and locate optimal design values with first- and second-derivative tests;
> - quantify how component tolerances propagate to circuit performance with absolute and relative sensitivities.

Electrical engineering is built on rates. Current is the rate at which charge passes a point, $i = dq/dt$. The voltage across an inductor is proportional to the rate of change of its current, $v = L\,di/dt$; the current into a capacitor is proportional to the rate of change of its voltage, $i = C\,dv/dt$. A transistor amplifier works because a small change in input voltage produces a larger change in output current, and that ratio is a derivative (the transconductance). Designing a filter that tolerates 5 % resistors is a question about derivatives too. Differential calculus is the mathematics of *how fast* and *how sensitive*. This chapter develops it from limits, the idea underneath every derivative, through to engineering optimization and sensitivity analysis.

Throughout, angles are in **radians** unless stated otherwise. Every derivative formula for trigonometric functions in this chapter depends on that.

4.1 Limits and continuity

Why limits?

Suppose a charge meter reports $q(t)$, the charge that has passed a cross-section of a wire. Over an interval $[t, t+\Delta t]$ the *average* current is

$$\bar{i} = \frac{q(t+\Delta t) - q(t)}{\Delta t}.$$

The *instantaneous* current at time t is what this ratio approaches as Δt shrinks to zero. We cannot simply set $\Delta t = 0$, because that gives $0/0$. We need a precise way to say what a quantity *approaches* without ever reaching the forbidden point. That is the limit. Limits also make precise what "a capacitor voltage cannot jump" means (continuity), what a filter does "at very high frequency" (limits at infinity), and when an equation must have a root (the intermediate value theorem).

Definition of a limit

Informally, $\lim_{x \to a} f(x) = L$ means that $f(x)$ can be made as close to L as we like by taking x close enough to a (but not equal to a). The value $f(a)$ itself, or whether it exists, is irrelevant.

Limit (ε–δ)

Let f be defined on an open interval containing a, except possibly at a. We write

$$\lim_{x \to a} f(x) = L$$

if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$0 < |x - a| < \delta \quad \Longrightarrow \quad |f(x) - L| < \varepsilon.$$

Read it as a game. An adversary names a tolerance ε on the output. You must reply with a tolerance δ on the input that guarantees the output stays within ε of L. If you can always win, the limit is L. This is the same logic as a specification: "the output must be within ± 1 mV (that is ε), so how tightly must the input be held (that is δ)?"

Worked Example 4.1 — An ε–δ proof

Prove that $\lim_{x \to 2} (3x + 1) = 7$.

Step 1 (scratch work). We need $|(3x+1) - 7| < \varepsilon$, i.e. $|3x - 6| = 3|x - 2| < \varepsilon$, i.e. $|x - 2| < \varepsilon/3$.

Step 2 (choose δ). Take $\delta = \varepsilon/3$.

Step 3 (verify). If $0 < |x - 2| < \delta$ then $|(3x+1) - 7| = 3|x-2| < 3\delta = \varepsilon$. ■

Check. With $\varepsilon = 0.03$, $\delta = 0.01$: $x = 2.009$ gives $3x + 1 = 7.027$, which is within 0.03 of 7. ✓

Nobody evaluates practical limits this way. The definition matters because every limit law below is proved from it, and because it pins down exactly what "approaches" means when intuition fails.

One-sided limits

The **right-hand limit** $\lim_{x \to a^+} f(x)$ only lets x approach from above ($x > a$); the **left-hand limit** $\lim_{x \to a^-} f(x)$ only from below. The two-sided limit exists if and only if both one-sided limits exist and are equal.

The unit step $u(t)$ (equal to 0 for $t<0$ and 1 for $t>0$) has $\lim_{t\to 0^-}u(t)=0$ and $\lim_{t\to 0^+}u(t)=1$. These differ, so $\lim_{t\to 0}u(t)$ does not exist. Circuit analysts use one-sided limits constantly: $v_C(0^-)$ is the capacitor voltage just before a switch operates and $v_C(0^+)$ the value just after. The principle that a capacitor voltage is continuous, $v_C(0^+)=v_C(0^-)$, is a statement about one-sided limits.

Limit laws

Limit laws

If $\lim_{x\to a}f(x)=L$ and $\lim_{x\to a}g(x)=M$ (both finite), then

$$\lim_{x\to a}[f(x)\pm g(x)]=L\pm M,\quad \lim_{x\to a}[c\,f(x)]=cL,$$
$$\lim_{x\to a}[f(x)\,g(x)]=LM,\qquad \lim_{x\to a}\frac{f(x)}{g(x)}=\frac{L}{M}\quad (M\neq 0),$$

and $\lim_{x\to a}[f(x)]^n=L^n$ for any positive integer n.

Proof sketch of the sum law. Given ε, choose δ_1 so that $|f-L|<\varepsilon/2$ and δ_2 so that $|g-M|<\varepsilon/2$. With $\delta=\min(\delta_1,\delta_2)$, the triangle inequality gives $|(f+g)-(L+M)|\le|f-L|+|g-M|<\varepsilon$. The product law uses $fg-LM=f(g-M)+M(f-L)$ in the same way.

Because polynomials are built from sums and products of x and constants, the laws give $\lim_{x\to a}p(x)=p(a)$ for every polynomial, and $\lim_{x\to a}p(x)/q(x)=p(a)/q(a)$ for rational functions whenever $q(a)\neq 0$. The interesting cases are the ones where direct substitution gives an **indeterminate form** such as $0/0$ or ∞/∞.

Worked Example 4.2 — Removing a $0/0$ by factoring

Evaluate $\displaystyle\lim_{x\to 2}\frac{x^2-4}{x-2}$ and $\displaystyle\lim_{h\to 0}\frac{\sqrt{9+h}-3}{h}$.

First limit. Substitution gives $0/0$. Factor: $\dfrac{(x-2)(x+2)}{x-2}=x+2$ for $x\neq 2$. Since the limit ignores the point $x=2$ itself, the limit is $\lim_{x\to 2}(x+2)=4$.

Second limit. Multiply by the conjugate:

$$\frac{\sqrt{9+h}-3}{h}\cdot\frac{\sqrt{9+h}+3}{\sqrt{9+h}+3}=\frac{(9+h)-9}{h(\sqrt{9+h}+3)}=\frac{1}{\sqrt{9+h}+3}\longrightarrow\frac{1}{6}.$$

Numerical check. $h=10^{-4}$: $(\sqrt{9.0001}-3)/10^{-4}=0.166666\ldots\checkmark$. (This second limit is the derivative of $\sqrt{x}$ at $x=9$; see §4.2.)

The squeeze theorem and the fundamental trigonometric limit

Squeeze (sandwich) theorem

If $g(x)\le f(x)\le h(x)$ for all x near a (except possibly at a) and $\lim_{x\to a}g(x)=\lim_{x\to a}h(x)=L$, then $\lim_{x\to a}f(x)=L$.

The most important application in this book is

$$\lim_{x\to 0}\frac{\sin x}{x}=1.$$

Proof. For $0<x<\pi/2$, draw the unit circle with an angle x. The triangle with vertices $(0,0)$, $(1,0)$, $(\cos x,\sin x)$ has area $\tfrac12\sin x$; the circular sector has area $\tfrac12 x$; the larger right triangle with legs 1 and $\tan x$ has area $\tfrac12\tan x$. These regions are nested, so

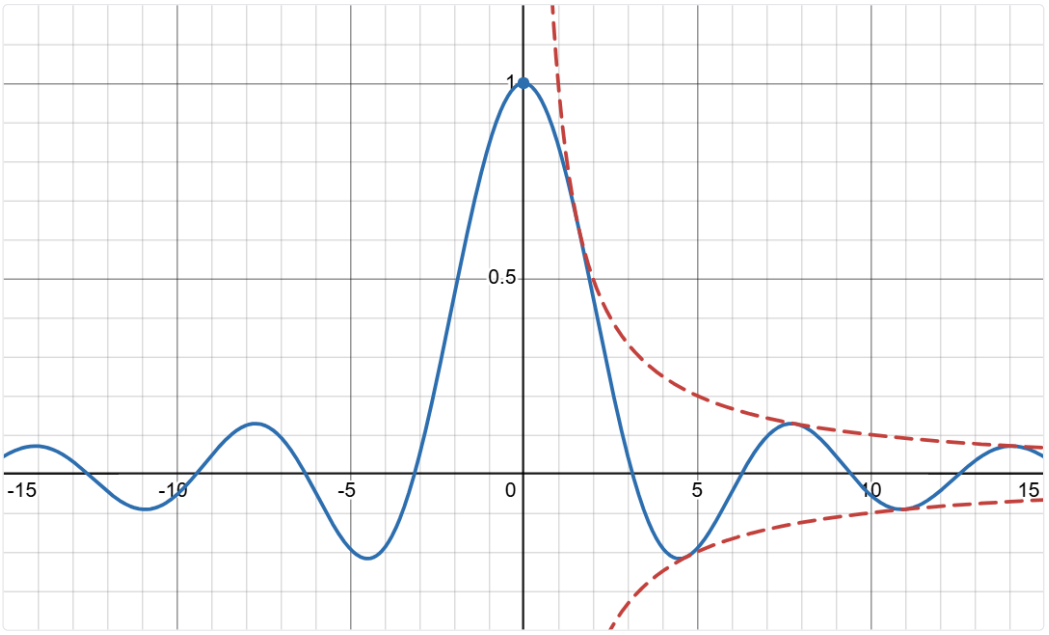
$$\tfrac12\sin x\le\tfrac12 x\le\tfrac12\tan x\quad\Longrightarrow\quad\cos x\le\frac{\sin x}{x}\le 1.$$

As $x\to 0^+$, $\cos x\to 1$, so the squeeze theorem gives the right-hand limit 1. Because $\sin x/x$ is an even function, the left-hand limit is also 1. $\blacksquare$

A consequence used repeatedly below:

$$\lim_{x\to 0}\frac{1-\cos x}{x}=\lim_{x\to 0}\frac{\sin^2 x}{x(1+\cos x)}=\lim_{x\to 0}\frac{\sin x}{x}\cdot\frac{\sin x}{1+\cos x}=1\cdot 0=0,$$

and in the same way $\lim_{x\to 0}(1-\cos x)/x^2=\tfrac12$.



The sinc-type function $\sin x/x$ is undefined at $x=0$ but tends to 1 there; for large x it is squeezed between $\pm 1/x$ and so tends to 0.

Engineering Note — the sinc function

The function $\operatorname{sinc}(x) = \sin(\pi x)/(\pi x)$ (with $\operatorname{sinc}(0)=1$ by the limit above) is the Fourier transform of a rectangular pulse (Ch. 16) and the ideal interpolation kernel in sampling theory. Defining $\operatorname{sinc}(0)=1$ is not a convention pulled from the air; it is the unique value that makes the function continuous.

Worked Example 4.3 — Trigonometric limits

Evaluate (a) $\displaystyle\lim_{x\to 0}\frac{\sin 3x}{x}$, (b) $\displaystyle\lim_{x\to 0}\frac{\tan 5x}{\sin 2x}$, (c) $\displaystyle\lim_{x\to 0}\frac{1-\cos 4x}{x^2}$.

(a) Write $\dfrac{\sin 3x}{x} = 3\cdot\dfrac{\sin 3x}{3x}$. With $u = 3x \to 0$, the limit is $3\cdot 1 = 3$.

(b) $\dfrac{\tan 5x}{\sin 2x} = \dfrac{\sin 5x}{5x}\cdot\dfrac{2x}{\sin 2x}\cdot\dfrac{1}{\cos 5x}\cdot\dfrac{5}{2} \to 1\cdot 1\cdot 1\cdot\tfrac{5}{2} = \tfrac{5}{2}$.

(c) With $u = 4x$, $\dfrac{1-\cos u}{x^2} = 16\,\dfrac{1-\cos u}{u^2} \to 16\cdot\tfrac{1}{2} = 8$.

Numerical check at $x = 10^{-3}$: (a) 2.9999955, (b) 2.50002, (c) 7.99999. ✓

Limits at infinity and infinite limits

$\lim_{x\to\infty} f(x) = L$ means $f(x)$ can be made arbitrarily close to L by taking x large enough; the line $y = L$ is then a **horizontal asymptote**. $\lim_{x\to a} f(x) = \infty$ means $f(x)$ exceeds any bound as $x \to a$; the line $x = a$ is then a **vertical asymptote**. "$=\infty$" here is shorthand for a particular way of *not* having a finite limit.

For rational functions, divide numerator and denominator by the highest power of x in the denominator. The useful facts are

$$\lim_{x\to\infty}\frac{1}{x^p} = 0\ (p>0), \qquad \lim_{x\to\infty} x^n e^{-x} = 0\ (\text{any } n), \qquad \lim_{x\to\infty}\frac{\ln x}{x^p} = 0\ (p>0).$$

In words: exponentials beat powers, and powers beat logarithms. We prove the last two with L'Hôpital's rule in §4.2.

The number e is itself a limit:

$$e = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n = 2.718281828\ldots, \qquad \text{and more generally} \qquad \lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n = e^x.$$

Worked Example 4.4 — Asymptotic behaviour of a transfer function

A first-order low-pass filter has magnitude response

$$|H(\omega)| = \frac{1}{\sqrt{1 + (\omega/\omega_c)^2}}.$$

Find $\lim_{\omega \to 0} |H|$, $\lim_{\omega \to \infty} |H|$, and $\lim_{\omega \to \infty} \omega\,|H(\omega)|$.

Low frequency. Direct substitution: $|H(0)| = 1$. The function is continuous, so the limit is 1 (the filter passes DC).

High frequency. The denominator grows without bound, so $|H| \to 0$.

The product. Divide inside the root by ω^2:

$$\omega |H| = \frac{\omega}{\sqrt{1 + \omega^2/\omega_c^2}} = \frac{1}{\sqrt{1/\omega^2 + 1/\omega_c^2}} \longrightarrow \omega_c.$$

So for $\omega \gg \omega_c$, $|H| \approx \omega_c/\omega$: the response falls in proportion to $1/\omega$, which is -20 dB per decade on a Bode plot.

Check. $\omega = 1000\,\omega_c$: $\omega|H|/\omega_c = 1000/\sqrt{1 + 10^6} = 0.9999995$. ✓

Continuity

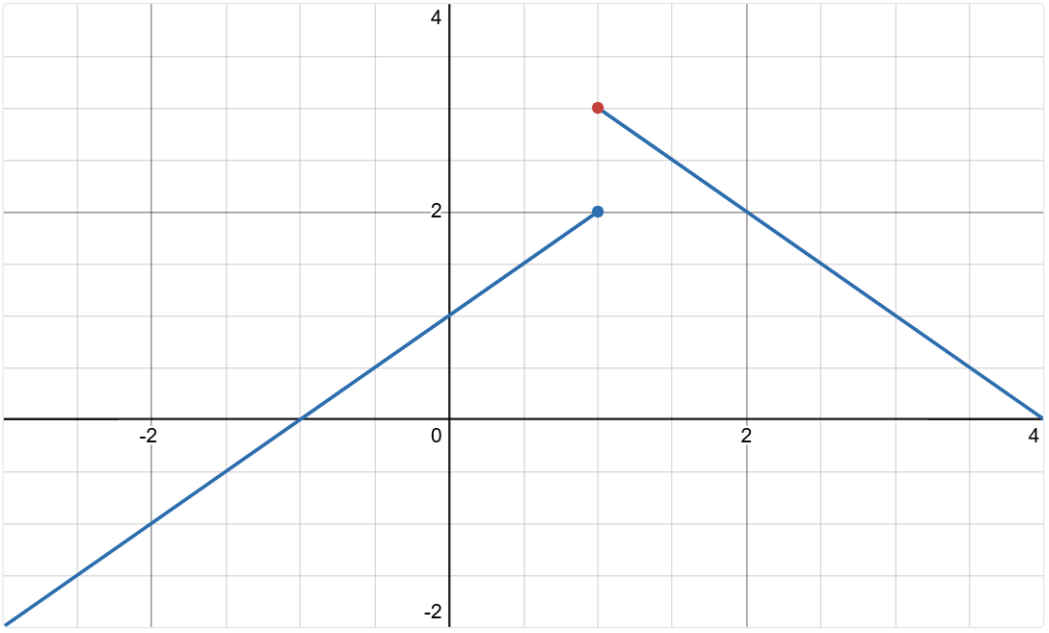
Continuity

f is **continuous at** a if (i) $f(a)$ is defined, (ii) $\lim_{x \to a} f(x)$ exists, and (iii) $\lim_{x \to a} f(x) = f(a)$. f is continuous on an interval if it is continuous at every point of it (using one-sided limits at closed endpoints).

Sums, products, quotients (where the denominator is nonzero) and compositions of continuous functions are continuous. Polynomials, rational functions (on their domains), e^x, $\ln x$ ($x > 0$), $\sin x$, $\cos x$, and $\sqrt{x}$ ($x \geq 0$) are all continuous. So nearly every formula you meet is continuous wherever it is defined, and the action is at the points where it is not.

Types of discontinuity:

Type	What fails	Example
Removable	limit exists but $\neq f(a)$ or $f(a)$ undefined	$\sin x / x$ at 0
Jump	one-sided limits exist but differ	$u(t)$ at 0; a switched source
Infinite	function unbounded near a	$1/(x-1)$ at $x = 1$; an undamped resonance $1/(\omega_0^2 - \omega^2)$
Oscillatory	no limit, no blow-up	$\sin(1/x)$ at 0



A jump discontinuity at $x = 1$: the left-hand limit is 2 and the right-hand limit is 3, so no choice of $f(1)$ can make the function continuous there.

Worked Example 4.5 — Choosing a constant for continuity

For what value of k is

$$f(x) = \begin{cases} \dfrac{e^{2x} - 1}{x}, & x \neq 0 \\ k, & x = 0 \end{cases}$$

continuous at 0?

We need $k = \lim_{x \to 0} (e^{2x} - 1)/x$. With $u = 2x$, this is $2 \lim_{u \to 0} (e^u - 1)/u$. The limit $\lim_{u \to 0} (e^u - 1)/u = 1$ is the defining property of e (it says that the slope of e^u at $u = 0$ is 1; see §4.2). Hence $k = 2$.

Check. $x = 10^{-5}$: $(e^{2 \times 10^{-5}} - 1)/10^{-5} = 2.00002$. ✓

Two theorems about continuous functions

Intermediate value theorem (IVT)

If f is continuous on $[a, b]$ and N lies strictly between $f(a)$ and $f(b)$, then there is at least one $c \in (a, b)$ with $f(c) = N$. In particular, if $f(a)$ and $f(b)$ have opposite signs, f has a root in (a, b).

Extreme value theorem (EVT)

If f is continuous on a closed, bounded interval $[a, b]$, then f attains an absolute maximum and an absolute minimum on $[a, b]$.

The IVT is the basis of the **bisection method** for solving equations (Ch. 22): if $f(a) f(b) < 0$, evaluate f at the midpoint and keep whichever half still has a sign change. Each step halves the interval, so n steps shrink an interval of length L to $L/2^n$. The EVT guarantees that the optimization problems of §4.10 have answers when the design variable lives on a closed interval.

Worked Example 4.6 — Bracketing a root

Show that $x^3 - x - 1 = 0$ has a root in $(1, 2)$ and locate it to within 0.1.

$f(x) = x^3 - x - 1$ is a polynomial, hence continuous. $f(1) = -1 < 0$ and $f(2) = 5 > 0$, so by the IVT there is a root in $(1, 2)$.

Bisection: $f(1.5) = 0.875 > 0 \Rightarrow (1, 1.5)$; $f(1.25) = -0.297 < 0 \Rightarrow (1.25, 1.5)$; $f(1.375) = 0.224 > 0 \Rightarrow (1.25, 1.375)$; $f(1.3125) = -0.0515 < 0 \Rightarrow (1.3125, 1.375)$. The interval now has length $0.0625 < 0.1$; the root is 1.34 ± 0.03.

Check. The root to 9 digits is 1.324717957, inside the final bracket. ✓

Common Pitfall — the IVT needs continuity on the whole interval

$f(x) = 1/x$ has $f(-1) = -1$ and $f(1) = 1$, yet there is no root in $(-1, 1)$. The IVT does not apply because f is discontinuous at 0. A numerical root finder that brackets a sign change across a pole will happily "converge" to the pole. Always check for singularities (for example, poles of a transfer function) inside a bracket.

Engineering Note — continuity of state variables

In a circuit with finite voltages and currents, capacitor voltages and inductor currents are continuous functions of time. The reason is that $i = C\,dv/dt$ is finite, so v cannot jump (a jump would need infinite current); dually for $v = L\,di/dt$. This continuity, $v_C(0^+) = v_C(0^-)$ and $i_L(0^+) = i_L(0^-)$, supplies the initial conditions for every switched-circuit problem in Ch. 11 and Ch. 12. Currents through capacitors and voltages across inductors are *not* state variables and may jump.

4.2 Derivatives and differentiation rules

The derivative as a limit

Derivative

The **derivative** of f at x is

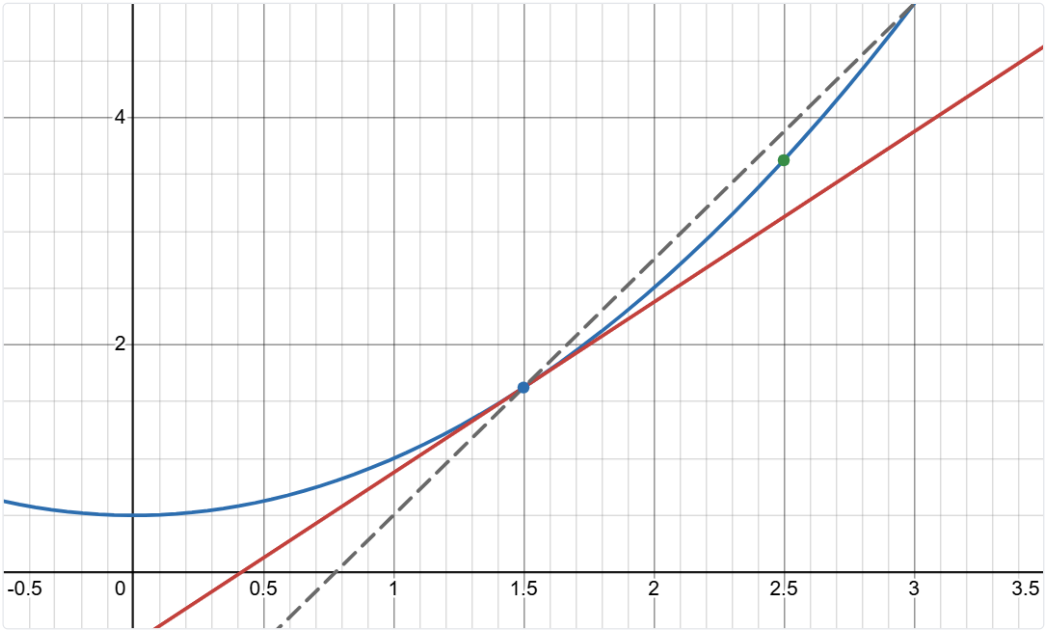
$$f'(x) = \lim_{h \to 0} \frac{f(x + h) - f(x)}{h},$$

provided the limit exists; f is then **differentiable** at x. Equivalent notations: $f'(x)$, $\dfrac{df}{dx}$, $\dfrac{d}{dx} f(x)$, $Df(x)$, and $\dot{f}$ when the variable is time.

The **difference quotient** $\frac{f(x+h) - f(x)}{h}$ is the slope of the secant line through $(x, f(x))$ and $(x + h, f(x + h))$. As $h \to 0$ the secant rotates into the **tangent line**, whose equation at $x = a$ is

$$y = f(a) + f'(a)(x - a).$$

Physically the derivative is an **instantaneous rate of change**, with units of (units of f)/(units of x). If q is in coulombs and t in seconds, dq/dt is in C/s = A.



The curve $y = \tfrac{1}{2}x^2 + \tfrac{1}{2}$, its tangent at $x = 1.5$ (slope 1.5), and a secant through $x = 1.5$ and $x = 2.5$ (slope 2). As the second point slides toward the first, the secant slope tends to the tangent slope.

Worked Example 4.7 — Derivatives from first principles

Find $f'(x)$ from the definition for (a) $f(x) = x^2$, (b) $f(x) = 1/x$, (c) $f(x) = \sqrt{x}$.

(a) $\dfrac{(x+h)^2 - x^2}{h} = \dfrac{2xh + h^2}{h} = 2x + h \to 2x$.

(b) $\dfrac{1}{h}\left(\dfrac{1}{x+h} - \dfrac{1}{x}\right) = \dfrac{x - (x+h)}{h\,x(x+h)} = \dfrac{-1}{x(x+h)} \to -\dfrac{1}{x^2}$.

(c) Rationalize as in Example 4.2: $\dfrac{\sqrt{x+h} - \sqrt{x}}{h} = \dfrac{1}{\sqrt{x+h} + \sqrt{x}} \to \dfrac{1}{2\sqrt{x}}$ (for $x > 0$). At $x = 9$ this is $1/6$, the limit computed in Example 4.2.

All three agree with the power rule $\frac{d}{dx}x^n = nx^{n-1}$ for $n = 2, -1, \tfrac{1}{2}$.

Differentiability and continuity

Differentiable implies continuous

If f is differentiable at a, then f is continuous at a.

Proof. $f(a+h) - f(a) = \dfrac{f(a+h) - f(a)}{h} \cdot h \to f'(a) \cdot 0 = 0$. ∎

The converse is false. $f(x) = |x|$ is continuous at 0, but its difference quotient $|h|/h$ is $+1$ for $h > 0$ and -1 for $h < 0$, so the one-sided limits disagree and $f'(0)$ does not exist. Graphically, a **corner** (or a vertical tangent, as for $x^{1/3}$ at 0, or a cusp) blocks differentiability. An ideal diode's i–v characteristic and the output of a full-wave rectifier $|\sin \omega t|$ both have corners. At a corner the left and right derivatives, defined with one-sided limits, exist but differ.

Derivatives of the basic functions

Powers. For a positive integer n, the binomial theorem gives $(x+h)^n = x^n + nx^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \cdots$, so

$$\frac{(x+h)^n - x^n}{h} = nx^{n-1} + \binom{n}{2}x^{n-2}h + \cdots + h^{n-1} \to nx^{n-1}.$$

The rule $\frac{d}{dx}x^r = rx^{r-1}$ holds for every real r (for $x > 0$); we prove the general case with logarithmic differentiation in §4.3.

Sine and cosine. Using $\sin(x+h) = \sin x \cos h + \cos x \sin h$,

$$\frac{\sin(x+h) - \sin x}{h} = \sin x\,\frac{\cos h - 1}{h} + \cos x\,\frac{\sin h}{h} \;\longrightarrow\; \sin x \cdot 0 + \cos x \cdot 1 = \cos x.$$

A similar computation gives $\frac{d}{dx}\cos x = -\sin x$. Both rely on the two trigonometric limits from §4.1, which in turn rely on measuring x in radians.

Exponential. $\dfrac{e^{x+h} - e^x}{h} = e^x\,\dfrac{e^h - 1}{h} \to e^x$, because e is precisely the base for which $\lim_{h\to 0}(e^h - 1)/h = 1$. The exponential is the function equal to its own derivative, which is why it solves every first-order linear circuit.

Logarithm. From $\ln(x+h) - \ln x = \ln(1 + h/x)$ and the substitution $n = x/h$,

$$\frac{\ln(1+h/x)}{h} = \frac{1}{x}\ln\left(1+\frac{1}{n}\right)^{n} \to \frac{1}{x}\ln e = \frac{1}{x}.$$

The differentiation rules

Rules of differentiation

If f and g are differentiable and c is a constant:

$$\begin{aligned}
&\text{Linearity:} && (cf \pm g)' = cf' \pm g' \\
&\text{Product rule:} && (fg)' = f'g + fg' \\
&\text{Quotient rule:} && \left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2} \quad (g \neq 0) \\
&\text{Power rule:} && (x^r)' = r x^{r-1}
\end{aligned}$$

Proof of the product rule. Add and subtract $f(x+h)g(x)$:

$$\frac{f(x+h)g(x+h) - f(x)g(x)}{h} = f(x+h)\frac{g(x+h)-g(x)}{h} + g(x)\frac{f(x+h)-f(x)}{h}.$$

As $h \to 0$, $f(x+h) \to f(x)$ (differentiable implies continuous), and the two quotients tend to g' and f'. $\blacksquare$

Quotient rule. Write $f = (f/g)\,g$ and apply the product rule: $f' = (f/g)'g + (f/g)g'$. Solve for $(f/g)'$.

A useful mnemonic for the product rule is in differentials: $d(fg) = g\,df + f\,dg$. If f and g each change by a small amount, the product changes by "each one's change times the other". For a capacitor with charge $q = Cv$ whose capacitance *and* voltage vary, this gives $i = \dot q = \dot C v + C\dot v$ (see §4.9).

Table of standard derivatives

$f(x)$	$f'(x)$	$f(x)$	$f'(x)$
x^r	$r x^{r-1}$	e^{ax}	$a e^{ax}$
a^x	$a^x \ln a$	$\ln\lvert x\rvert$	$1/x$
$\log_a x$	$1/(x \ln a)$	$\sin x$	$\cos x$
$\cos x$	$-\sin x$	$\tan x$	$\sec^2 x = 1 + \tan^2 x$
$\cot x$	$-\csc^2 x$	$\sec x$	$\sec x \tan x$
$\arcsin x$	$1/\sqrt{1-x^2}$	$\arccos x$	$-1/\sqrt{1-x^2}$
$\arctan x$	$1/(1+x^2)$	$\sinh x$	$\cosh x$
$\cosh x$	$\sinh x$	$\tanh x$	$\operatorname{sech}^2 x = 1 - \tanh^2 x$

The inverse-trigonometric entries follow from the inverse-function rule in §4.3. The hyperbolic ones follow from $\sinh x = (e^x - e^{-x})/2$, $\cosh x = (e^x + e^{-x})/2$.

Complex exponentials. For a real variable t and a complex constant s, $\frac{d}{dt}e^{st} = s e^{st}$. The proof is identical to the real case (or follows by differentiating $e^{\sigma t}\cos\omega t$ and $e^{\sigma t}\sin\omega t$ separately). With $s = j\omega$:

$$\frac{d}{dt}e^{j\omega t} = j\omega\, e^{j\omega t}.$$

This single line is why **differentiation becomes multiplication by $j\omega$** in phasor analysis (Ch. 2) and multiplication by s in the Laplace domain (Ch. 14). An inductor's $v = L\,di/dt$ becomes $V = j\omega L\, I$, giving the impedance $j\omega L$.

Worked Example 4.8 — Product and quotient rules: power into a load

A source of EMF V and internal resistance R_s drives a load R_L. The load power is

$$P(R_L) = \frac{V^2 R_L}{(R_s + R_L)^2}.$$

Find dP/dR_L.

Quotient rule with numerator $N = V^2 R_L$ ($N' = V^2$) and denominator $D = (R_s + R_L)^2$ ($D' = 2(R_s + R_L)$), by the chain rule of §4.3 or by expanding):

$$\frac{dP}{dR_L} = \frac{V^2(R_s+R_L)^2 - V^2 R_L \cdot 2(R_s+R_L)}{(R_s+R_L)^4} = \frac{V^2\left[(R_s+R_L) - 2R_L\right]}{(R_s+R_L)^3} = \frac{V^2(R_s - R_L)}{(R_s+R_L)^3}.$$

Interpretation. The derivative is positive for $R_L < R_s$ and negative for $R_L > R_s$: power rises, peaks at $R_L = R_s$, then falls. This is the maximum power transfer theorem, revisited in §4.10.

Units check. $V^2\,\Omega/\Omega^3 = V^2/\Omega^2 = \mathrm{W}/\Omega$. ✓

Numerical check. $V = 10\,\mathrm{V}$, $R_s = 50\,\Omega$, $R_L = 25\,\Omega$: formula gives $100 \cdot 25/75^3 = 5.926 \times 10^{-3}\ \mathrm{W}/\Omega$; central difference with $h = 10^{-3}\,\Omega$ gives 5.926×10^{-3}. ✓

Worked Example 4.9 — Inductor voltage

The current in a $L = 10\,\text{mH}$ inductor is $i(t) = I_0\,(t/\tau)\,e^{-t/\tau}$ with $I_0 = 2\,\text{A}$ and $\tau = 1\,\text{ms}$. Find $v(t) = L\,di/dt$ and the time at which $v = 0$.

Product rule. With $f = t/\tau$ and $g = e^{-t/\tau}$, $f' = 1/\tau$ and $g' = -e^{-t/\tau}/\tau$:

$$\frac{di}{dt} = \frac{I_0}{\tau}e^{-t/\tau} - \frac{I_0 t}{\tau^2}e^{-t/\tau} = \frac{I_0}{\tau}\left(1 - \frac{t}{\tau}\right)e^{-t/\tau}.$$

So $v(t) = \dfrac{LI_0}{\tau}\left(1 - \dfrac{t}{\tau}\right)e^{-t/\tau} = 20\,(1 - t/\tau)\,e^{-t/\tau}\ \text{V}$.

Zero. $v = 0$ at $t = \tau = 1\,\text{ms}$, which is exactly when the current reaches its peak $I_0/e = 0.736\,\text{A}$. At $t = 0$, $v = 20\,\text{V}$.

Units. $\text{H}\cdot\text{A/s} = \text{V}$. ✓

The mean value theorem

Rolle's theorem and the mean value theorem (MVT)

Let f be continuous on $[a, b]$ and differentiable on (a, b).

- (Rolle) If $f(a) = f(b)$, there is $c \in (a, b)$ with $f'(c) = 0$.
- (MVT) There is $c \in (a, b)$ with

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Proof sketch. Rolle: by the EVT f has a max and a min on $[a, b]$; if both are at the endpoints, f is constant and $f' \equiv 0$; otherwise an interior extremum has $f' = 0$ (Fermat's theorem, §4.10). MVT: apply Rolle to $g(x) = f(x) - \frac{f(b)-f(a)}{b-a}(x - a)$, which has $g(a) = g(b) = f(a)$.

In words: somewhere, the instantaneous rate equals the average rate. If a capacitor voltage rises from 0 to 5 V in 2 μs, then at some instant $dv/dt = 2.5\,\text{V}/\mu\text{s}$ exactly, and with $C = 100\,\text{pF}$ the current at that instant is $i = C\,dv/dt = 0.25\,\text{mA}$, whatever the shape of the waveform.

Three consequences used throughout the book:

1. If $f'(x) = 0$ on an interval, f is constant there.
2. If $f' = g'$ on an interval, $f - g$ is constant (the basis of "$+C$" in Ch. 5).
3. If $f' > 0$ on an interval, f is strictly increasing there; if $f' < 0$, strictly decreasing.

The MVT also gives error bounds: $|f(b) - f(a)| \le M|b - a|$ where $M = \max|f'|$. A sensor with transfer function slope at most M amplifies an input uncertainty Δx to at most $M\Delta x$.

L'Hôpital's rule

L'Hôpital's rule

Suppose f and g are differentiable near a (except possibly at a), $g'(x) \neq 0$ near a, and $\lim f = \lim g = 0$ or $\lim f = \pm\lim g = \pm\infty$. If $\lim_{x\to a} f'(x)/g'(x) = L$ (finite or infinite), then

$$\lim_{x\to a}\frac{f(x)}{g(x)} = L.$$

The same holds for one-sided limits and for $a = \pm\infty$.

Why it works (for $0/0$ with f', g' continuous and $g'(a) \neq 0$). Since $f(a) = g(a) = 0$,

$$\frac{f(x)}{g(x)} = \frac{f(x) - f(a)}{g(x) - g(a)} = \frac{[f(x) - f(a)]/(x - a)}{[g(x) - g(a)]/(x - a)} \to \frac{f'(a)}{g'(a)}.$$

The general case uses Cauchy's version of the MVT.

Other indeterminate forms ($0\cdot\infty$, $\infty - \infty$, 1^∞, 0^0, ∞^0) are first rewritten as $0/0$ or ∞/∞, usually by moving a factor to the denominator or by taking logarithms.

Worked Example 4.10 — L'Hôpital's rule

(a) $\lim_{x\to 0}\dfrac{e^x-1-x}{x^2}$. Form $0/0$. Differentiate top and bottom: $\dfrac{e^x-1}{2x}$, still $0/0$. Again: $\dfrac{e^x}{2}\to\dfrac{1}{2}$.

(b) $\lim_{x\to\infty} x^3e^{-x}=\lim\dfrac{x^3}{e^x}$. Form ∞/∞. Three applications: $\dfrac{3x^2}{e^x}\to\dfrac{6x}{e^x}\to\dfrac{6}{e^x}\to 0$. The same argument works for any power x^n: exponentials dominate.

(c) $\lim_{x\to 0^+} x\ln x$. Form $0\cdot(-\infty)$. Rewrite as $\dfrac{\ln x}{1/x}$ $(-\infty/\infty)$: $\dfrac{1/x}{-1/x^2}=-x\to 0$.

(d) $\lim_{x\to\infty}\left(1+\dfrac{a}{x}\right)^x$. Form 1^∞. Take logs: $x\ln(1+a/x)=\dfrac{\ln(1+a/x)}{1/x}$, form $0/0$. Differentiate: $\dfrac{-a/x^2\cdot 1/(1+a/x)}{-1/x^2}=\dfrac{a}{1+a/x}\to a$. So the limit is e^a.

Checks. (a) at $x=10^{-3}$: 0.500167; (b) at $x=50$: 2.4×10^{-17}; (c) at $x=10^{-6}$: -1.4×10^{-5}; (d) $a=1$, $x=10^6$: 2.7182805. ✓

Worked Example 4.11 — The critically damped limit

Two first-order stages in cascade with time constants $1/a$ and $1/b$ give an impulse response proportional to

$$h(t)=\frac{e^{-at}-e^{-bt}}{b-a}\qquad(a\neq b).$$

What happens as $a\to b$ (equal poles)?

For fixed t, treat this as a function of a. Numerator and denominator both tend to 0. Differentiate with respect to a: numerator derivative $-te^{-at}$, denominator derivative -1. So

$$\lim_{a\to b}h(t)=\lim_{a\to b}\frac{-te^{-at}}{-1}=t\,e^{-bt}.$$

This explains the appearance of the factor t in the solution of an ODE with a repeated characteristic root (Ch. 11): it is the limit of two nearly equal exponentials.

Check. $b=1$, $a=1.001$, $t=2$: $(e^{-2.002}-e^{-2})/(-0.001)=0.27040$; $te^{-t}=0.27067$ (the difference is $O(a-b)$). ✓

Common Pitfall — applying L'Hôpital where it does not apply

L'Hôpital's rule is only for $0/0$ or ∞/∞. $\lim_{x\to 0}\frac{\cos x}{x+1}=1$ by substitution, but "differentiating" gives $\frac{-\sin x}{1}\to 0$, which is wrong. Also, the rule differentiates numerator and denominator **separately**; it is not the quotient rule. Finally, if $\lim f'/g'$ fails to exist the rule says nothing: $\lim_{x\to\infty}(x+\sin x)/x=1$, although $(1+\cos x)/1$ has no limit.

Engineering Note — derivatives from measured data

Instruments give samples, not formulas. A measured current $i[n]=i(nT_s)$ is differentiated numerically by $\big(i[n+1]-i[n-1]\big)/(2T_s)$, the central difference, which is the definition of the derivative with $h=T_s$ (made symmetric for accuracy; see §4.7). Differentiation amplifies high-frequency noise, because $j\omega$ grows with ω: a 1 mV noise component at 1 MHz contributes $2\pi\times10^6\times10^{-3}\approx 6.3\times10^3$ V/s to dv/dt. Practical differentiators therefore band-limit the signal first.

4.3 Chain rule

Motivation

Most functions in engineering are compositions. A decaying sinusoid $e^{-t/\tau}\cos(\omega t+\phi)$ has the time variable buried inside an exponential and a cosine. The gain of a filter in decibels is a logarithm of a square root of a rational function of ω. A diode current is an exponential of a voltage that is itself a function of time. The chain rule differentiates compositions, and it is the most frequently used rule in calculus.

Statement

Chain rule

If g is differentiable at x and f is differentiable at $u=g(x)$, then $f\circ g$ is differentiable at x and

$$\frac{d}{dx}f\big(g(x)\big)=f'\big(g(x)\big)\,g'(x).$$

In Leibniz notation, with $y=f(u)$ and $u=g(x)$:

$$\frac{dy}{dx}=\frac{dy}{du}\,\frac{du}{dx}.$$

The Leibniz form looks like cancelling du, and that is a reliable mnemonic. Rates multiply along a chain: if y changes 3 times as fast as u, and u changes 2 times as fast as x, then y changes 6 times as fast as x. A cascade of amplifiers with small-signal gains A_1 and A_2 has overall gain A_1A_2 for exactly this reason.

Proof sketch. The naive argument writes

$$\frac{f(g(x+h))-f(g(x))}{h}=\frac{f(g(x+h))-f(g(x))}{g(x+h)-g(x)}\cdot\frac{g(x+h)-g(x)}{h},$$

and lets $h\to 0$. This fails if $g(x+h)=g(x)$ for small $h\neq 0$ (division by zero). The fix is to define

$$\phi(k)=\begin{cases}\dfrac{f(u+k)-f(u)}{k}, & k\neq 0\\[2mm] f'(u), & k=0\end{cases}$$

which is continuous at $k=0$ because f is differentiable at u. Then $f(u+k)-f(u)=\phi(k)\,k$ holds for *all* k, including $k=0$. Put $k=g(x+h)-g(x)$:

$$\frac{f(g(x+h))-f(g(x))}{h}=\phi(k)\,\frac{g(x+h)-g(x)}{h}\to f'(u)\,g'(x),$$

since $k\to 0$ as $h\to 0$ (because g is continuous). $\blacksquare$

Standard consequences

Applying the chain rule to each entry in the table of §4.2 gives the "generalized" rules, with $u=u(x)$:

$$\frac{d}{dx}u^n=nu^{n-1}u',\qquad \frac{d}{dx}e^u=e^u u',\qquad \frac{d}{dx}\ln|u|=\frac{u'}{u},$$
$$\frac{d}{dx}\sin u=\cos u\;u',\quad \frac{d}{dx}\cos u=-\sin u\;u',\quad \frac{d}{dx}\arctan u=\frac{u'}{1+u^2}.$$

For a sinusoid with angular frequency ω and phase ϕ:

$$\frac{d}{dt}A\sin(\omega t+\phi)=A\omega\cos(\omega t+\phi)=A\omega\sin\!\left(\omega t+\phi+\frac{\pi}{2}\right).$$

Differentiating a sinusoid multiplies its amplitude by ω and advances its phase by 90°. That is the time-domain meaning of the factor $j\omega=\omega\angle 90^\circ$. For an inductor, the voltage leads the current by 90°; for a capacitor, $i=C\,dv/dt$ means the current leads the voltage by 90°.

Worked Example 4.12 — A damped sinusoid

The voltage across an underdamped RLC circuit is $v(t)=V_0e^{-t/\tau}\cos(\omega_d t)$ with $V_0=10\,\text{V}$, $\tau=1\,\text{ms}$, $\omega_d=2\pi\times 1000\,\text{rad/s}$. Find dv/dt and evaluate it at $t=0$.

Product rule, then chain rule on each factor. $\frac{d}{dt}e^{-t/\tau}=-\frac{1}{\tau}e^{-t/\tau}$ and $\frac{d}{dt}\cos\omega_d t=-\omega_d\sin\omega_d t$. So

$$\frac{dv}{dt}=-V_0e^{-t/\tau}\left[\frac{1}{\tau}\cos\omega_d t+\omega_d\sin\omega_d t\right].$$

At $t=0$. $\dfrac{dv}{dt}(0)=-V_0/\tau=-10/10^{-3}=-10^4\,\text{V/s}=-10\,\text{mV/µs}$.

Compact form. With $\tan\psi=\omega_d\tau$, the bracket is $\sqrt{1/\tau^2+\omega_d^2}\,\cos(\omega_d t-\psi)$. The derivative is another damped sinusoid with the same decay, scaled by $\sqrt{\tau^{-2}+\omega_d^2}=6362\,\text{s}^{-1}$ and phase-shifted. In complex form this is immediate: $v=\text{Re}\{V_0e^{st}\}$ with $s=-1/\tau+j\omega_d$, so $\dot v=\text{Re}\{sV_0e^{st}\}$ and $|s|=6362\,\text{s}^{-1}$.

Logarithmic differentiation

For products, quotients and powers of many factors, take logarithms first. If $y=f_1^{a_1}f_2^{a_2}\cdots f_n^{a_n}$ then $\ln|y|=\sum a_k\ln|f_k|$, and differentiating with the chain rule:

$$\frac{y'}{y}=a_1\frac{f_1'}{f_1}+a_2\frac{f_2'}{f_2}+\cdots+a_n\frac{f_n'}{f_n}.$$

The quantity $y'/y=(\ln y)'$ is the **logarithmic derivative**: the fractional (relative) rate of change. It turns products into sums and is the engine of the relative sensitivity in §4.11.

The general power rule. For $y=x^r$ with real r and $x>0$: $\ln y=r\ln x$, so $y'/y=r/x$ and $y'=rx^{r-1}$. This completes the proof of the power rule for all real exponents.

Variable exponents. $y=x^x$ $(x>0)$: $\ln y=x\ln x$, so $y'/y=\ln x+1$ and $y'=x^x(1+\ln x)$. Similarly $\frac{d}{dx}a^x=a^x\ln a$, from $a^x=e^{x\ln a}$.

Worked Example 4.13 — Slope of a Bode magnitude plot

For the first-order low-pass filter $|H| = \big(1 + (\omega/\omega_c)^2\big)^{-1/2}$, the gain in decibels is $G = 20\log_{10}|H| = -10\log_{10}\big(1 + (\omega/\omega_c)^2\big)$. A Bode plot uses $x = \log_{10}\omega$ on the horizontal axis. Find the slope dG/dx in dB per decade.

Chain rule. $dG/dx = (dG/d\omega)(d\omega/dx)$. Since $\omega = 10^x$, $d\omega/dx = 10^x \ln 10 = \omega \ln 10$. And

$$\frac{dG}{d\omega} = -\frac{10}{\ln 10}\cdot\frac{2\omega/\omega_c^2}{1 + (\omega/\omega_c)^2}.$$

Multiplying,

$$\frac{dG}{dx} = -20\,\frac{(\omega/\omega_c)^2}{1 + (\omega/\omega_c)^2} \quad \text{dB/decade}.$$

Interpretation. For $\omega \ll \omega_c$ the slope is ≈ 0; at $\omega = \omega_c$ it is exactly -10 dB/decade; for $\omega \gg \omega_c$ it tends to -20 dB/decade. The asymptotic straight-line Bode plot replaces this smooth curve by its two limits, which is accurate except near the corner.

Check. $\omega = 10\omega_c$: $-20 \times 100/101 = -19.80$ dB/decade. Numerically, $G(10.01\omega_c) - G(9.99\omega_c)$ divided by $\log_{10}(10.01/9.99)$ gives -19.80. ✓

Engineering Note — why decibels and log axes

The logarithmic derivative explains the habit of plotting gain in dB against $\log\omega$. If $|H| \propto \omega^n$ over some band, then $\log|H| = n\log\omega + \text{const}$, a straight line of slope n (i.e. $20n$ dB/decade). Each pole contributes -20 dB/decade and each zero $+20$ dB/decade beyond its corner frequency because logarithms turn the product of factors in H into a sum of their contributions. The slope $d\ln|H|/d\ln\omega$ is exactly the relative sensitivity $S_\omega^{|H|}$ of §4.11.

Derivatives of inverse functions

If $y = f(x)$ is differentiable and strictly monotonic with $f'(x) \neq 0$, its inverse $x = f^{-1}(y)$ is differentiable and

$$\frac{d}{dy}f^{-1}(y) = \frac{1}{f'(x)}\bigg|_{x = f^{-1}(y)}, \qquad \text{i.e.} \qquad \frac{dx}{dy} = \frac{1}{dy/dx}.$$

Derivation. Differentiate the identity $f\big(f^{-1}(y)\big) = y$ with respect to y using the chain rule: $f'\big(f^{-1}(y)\big)\cdot(f^{-1})'(y) = 1$. Graphically, the graph of f^{-1} is the mirror image of the graph of f in the line $y = x$, which turns a slope m into $1/m$.

Arctangent. $y = \arctan x$ means $x = \tan y$ with $y \in (-\pi/2, \pi/2)$. Then $dx/dy = \sec^2 y = 1 + \tan^2 y = 1 + x^2$, so $dy/dx = 1/(1 + x^2)$.

Arcsine. $x = \sin y$, $y \in [-\pi/2, \pi/2]$: $dx/dy = \cos y = +\sqrt{1 - x^2}$ (positive on that interval), so $dy/dx = 1/\sqrt{1 - x^2}$.

Logarithm. $x = e^y$: $dx/dy = e^y = x$, so $\frac{d}{dx}\ln x = 1/x$ (a second proof).

Worked Example 4.14 — Inverse of a device characteristic

A nonlinear sensor has output $y = f(x) = x^3 + x$ (in normalized units). It is used "in reverse": the measured y is converted to $x = f^{-1}(y)$. How sensitive is the recovered x to errors in y when $y = 2$?

$f(1) = 2$, so $f^{-1}(2) = 1$. $f'(x) = 3x^2 + 1$, so $f'(1) = 4$. Hence $(f^{-1})'(2) = 1/4$: an error of 0.01 in y produces an error of only about 0.0025 in x.

Check. Solve $x^3 + x = 2.01$ numerically: $x = 1.0024953$; the change is $0.0024953 \approx 0.01/4$. ✓

We never needed a formula for f^{-1}, which would require solving a cubic.

Parametric curves

If a curve is given as $x = x(t)$, $y = y(t)$ (a phasor tip, a Lissajous figure on an oscilloscope, a trajectory in a phase plane), then wherever $dx/dt \neq 0$,

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}.$$

This is the chain rule $\frac{dy}{dt} = \frac{dy}{dx}\frac{dx}{dt}$ solved for dy/dx. The second derivative needs one more application:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{dx/dt}.$$

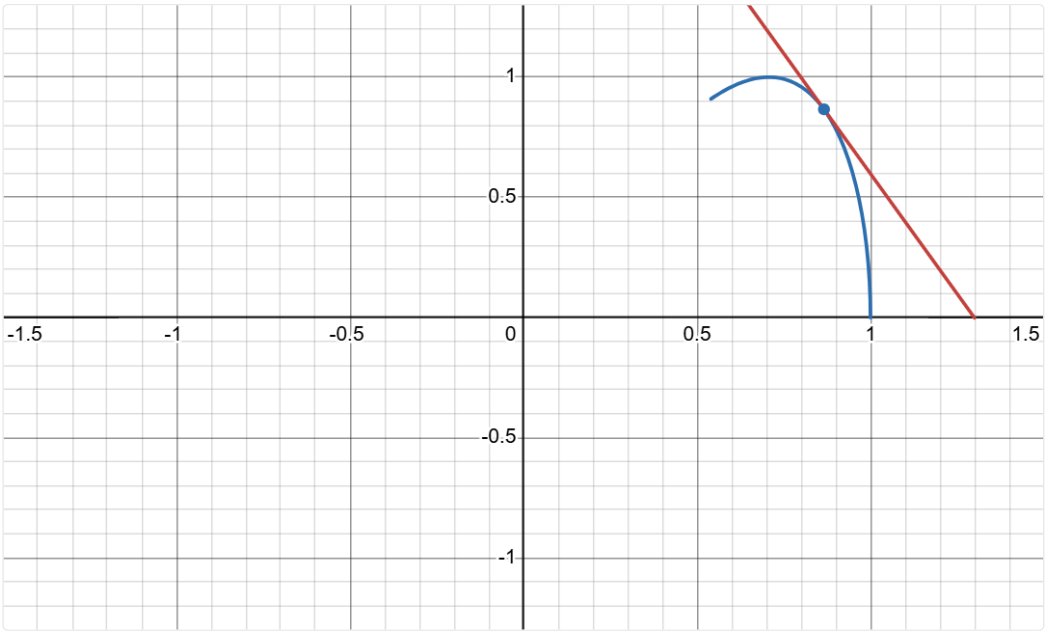
Worked Example 4.15 — Slope of a Lissajous figure

An oscilloscope in X–Y mode displays $x = \cos t$, $y = \sin 2t$. Find the slope of the trace at $t = \pi/6$ and the points where the tangent is horizontal or vertical.

$dx/dt = -\sin t$, $dy/dt = 2\cos 2t$. At $t = \pi/6$: $dx/dt = -\frac{1}{2}$, $dy/dt = 2\cdot\frac{1}{2} = 1$. Slope $= 1/(-\frac{1}{2}) = -2$, at the point $(\cos\frac{\pi}{6}, \sin\frac{\pi}{3}) = (0.866, 0.866)$.

Horizontal tangents need $dy/dt = 0$ with $dx/dt \neq 0$: $\cos 2t = 0 \Rightarrow t = \pi/4, 3\pi/4, \ldots$, giving $(\pm 0.707, \pm 1)$, the four peaks of the figure-eight.

Vertical tangents need $dx/dt = 0$ with $dy/dt \neq 0$: $t = 0, \pi$, where $dy/dt = 2$. The tangent is vertical at $(\pm 1, 0)$.



The Lissajous figure $x = \cos t$, $y = \sin 2t$ (a 1:2 frequency ratio) and its tangent at $t = \pi/6$.

Worked Example 4.16 — Small-signal conductance of a diode

The Shockley diode equation is $I = I_s\left(e^{V/(nV_T)} - 1\right)$. Find the incremental conductance $g_d = dI/dV$ and evaluate it at $I = 1\,\text{mA}$ with $nV_T = 26\,\text{mV}$.

Chain rule on $e^{V/(nV_T)}$ (inner derivative $1/(nV_T)$):

$$g_d = \frac{dI}{dV} = \frac{I_s}{nV_T} e^{V/(nV_T)} = \frac{I + I_s}{nV_T} \approx \frac{I}{nV_T}.$$

The last step uses $I_s \ll I$ (typically $I_s \sim 10^{-14}$ A). Note that g_d depends only on the bias current, not on I_s: a universal result.

Numbers. $g_d = 10^{-3}/0.026 = 38.5\,\text{mS}$, so the small-signal resistance is $r_d = 1/g_d = 26\,\Omega$.

If the voltage varies in time, the chain rule again gives $\dfrac{dI}{dt} = g_d \dfrac{dV}{dt}$: the diode responds to small, slow changes like a resistor r_d. We turn this into a full linear model in §4.8.

Worked Example 4.17 — Rate of change of an impedance magnitude

A series RLC circuit has $|Z(\omega)| = \sqrt{R^2 + X^2}$ with reactance $X(\omega) = \omega L - \dfrac{1}{\omega C}$. Find $d|Z|/d\omega$ and show that $|Z|$ has a stationary point at resonance.

Outer function $\sqrt{\cdot}$, inner function $R^2 + X^2$, and X itself depends on ω:

$$\frac{d|Z|}{d\omega} = \frac{1}{2\sqrt{R^2 + X^2}} \cdot 2X \frac{dX}{d\omega} = \frac{X}{|Z|}\left(L + \frac{1}{\omega^2 C}\right).$$

The bracket is always positive, so the sign of $d|Z|/d\omega$ is the sign of X. At resonance $\omega_0 = 1/\sqrt{LC}$, $X = 0$, so $d|Z|/d\omega = 0$: $|Z|$ has its minimum value R there.

Numbers. $R = 10\,\Omega$, $L = 1\,\text{mH}$, $C = 1\,\mu\text{F}$: $\omega_0 = 31\,623\,\text{rad/s}$. At $\omega = 1.1\omega_0$: $X = 34.785 - 28.748 = 6.037\,\Omega$, $|Z| = 11.68\,\Omega$, and $d|Z|/d\omega = (6.037/11.68)(10^{-3} + 0.826 \times 10^{-3}) = 9.44 \times 10^{-4}\,\Omega\,\text{s/rad}$.

Common Pitfall — forgetting the inner derivative, and degrees

$\frac{d}{dx}\sin(x^2) = 2x\cos(x^2)$, not $\cos(x^2)$ and not $\cos(2x)$. Every layer of a composition contributes a factor. A subtler instance: $\frac{d}{dx}\sin(x^\circ)$, with x in degrees, is $\frac{\pi}{180}\cos(x^\circ)$, because $x^\circ = \pi x/180$ radians. Calculators set to degrees silently break every derivative formula in this chapter.

4.4 Implicit differentiation

Motivation

Many relations in engineering cannot be solved explicitly for the quantity of interest. The current in a resistor–diode circuit satisfies

$$V_s = IR + nV_T \ln\!\left(1 + \frac{I}{I_s}\right),$$

which has no closed-form solution for I in elementary functions. Yet we still want dI/dV_s: how much the current moves when the supply moves. Implicit differentiation finds derivatives directly from an equation $F(x, y) = 0$ without solving it.

Method

Treat y as an (unknown) differentiable function of x, differentiate **both sides** of the equation with respect to x, applying the chain rule to every term containing y (so $\frac{d}{dx}y^2 = 2y\,y'$, $\frac{d}{dx}\sin y = \cos y\,y'$, $\frac{d}{dx}(xy) = y + xy'$), then solve the resulting linear equation for y'.

The result is always linear in y', because the chain rule produces exactly one factor y' per occurrence of y.

Worked Example 4.18 — A circle

Find dy/dx and d^2y/dx^2 on $x^2 + y^2 = r^2$.

First derivative. $2x + 2y\,y' = 0 \Rightarrow y' = -x/y$ (valid where $y \neq 0$). At $(r/\sqrt{2}, r/\sqrt{2})$ the slope is -1. The tangent is perpendicular to the radius, whose slope is y/x; the product of slopes is -1. ✓

Second derivative. Differentiate $y' = -x/y$ by the quotient rule, then substitute for y':

$$y'' = -\frac{y - xy'}{y^2} = -\frac{y + x^2/y}{y^2} = -\frac{x^2 + y^2}{y^3} = -\frac{r^2}{y^3}.$$

On the upper half ($y > 0$) the circle is concave down ($y'' < 0$), as it should be.

Check against the explicit form $y = \sqrt{r^2 - x^2}$: $y' = -x/\sqrt{r^2 - x^2} = -x/y$. ✓

Worked Example 4.19 — Tangent to the folium of Descartes

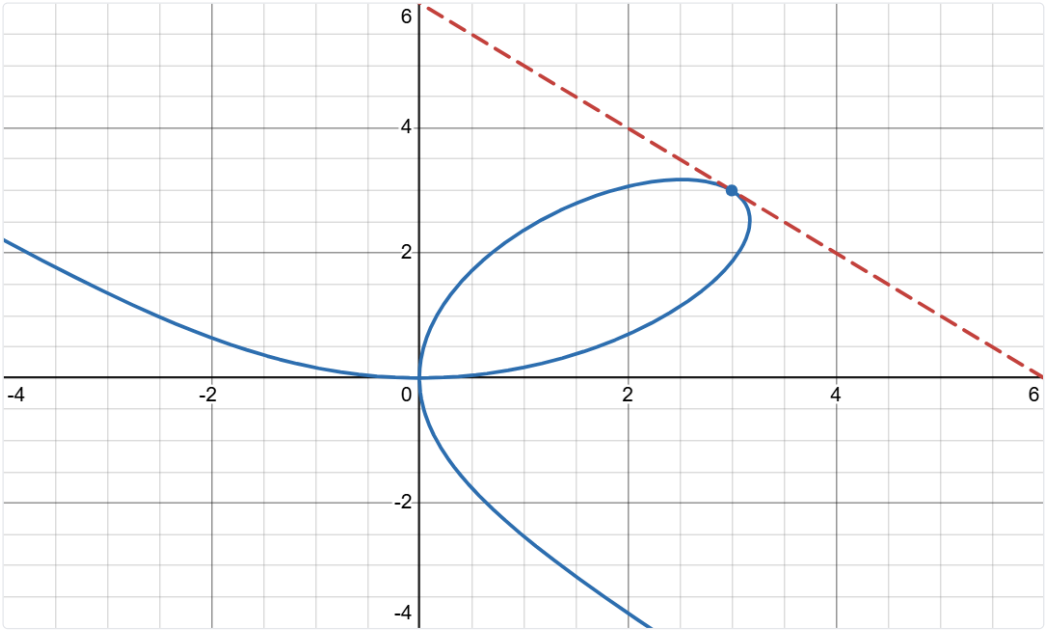
Find the tangent line to $x^3 + y^3 = 6xy$ at $(3, 3)$.

First check the point: $27 + 27 = 54 = 6 \cdot 9$. ✓ Differentiate:

$$3x^2 + 3y^2 y' = 6y + 6xy' \quad\Longrightarrow\quad y'(3y^2 - 6x) = 6y - 3x^2 \quad\Longrightarrow\quad y' = \frac{2y - x^2}{y^2 - 2x}.$$

At $(3, 3)$: $y' = (6 - 9)/(9 - 6) = -1$. Tangent: $y - 3 = -(x - 3)$, i.e. $y = 6 - x$.

Where is y' undefined? Where $y^2 = 2x$. Combined with the curve equation this gives the origin and the point $(2^{5/3}, 2^{4/3}) \approx (3.17, 2.52)$, where the tangent is vertical.



The folium of Descartes $x^3 + y^3 = 6xy$ is not the graph of a single function $y(x)$, yet implicit differentiation gives its slope at every regular point; the tangent at $(3, 3)$ is $y = 6 - x$.

The general formula and when it is valid

For a relation $F(x, y) = 0$, differentiating with respect to x (using the multivariable chain rule of §4.5) gives $F_x + F_y\,y' = 0$, so

$$\frac{dy}{dx} = -\frac{F_x}{F_y} = -\frac{\partial F/\partial x}{\partial F/\partial y} \qquad (F_y \neq 0).$$

Implicit function theorem (one-variable form, stated)

If F has continuous partial derivatives near (a, b), $F(a, b) = 0$ and $F_y(a, b) \neq 0$, then near $x = a$ the equation $F(x, y) = 0$ defines a unique differentiable function $y = y(x)$ with $y(a) = b$, and $y' = -F_x/F_y$.

The condition $F_y \neq 0$ matters. At points where $F_y = 0$ (the leftmost and rightmost points of a circle, the self-intersection of the folium at the origin), the curve may have a vertical tangent, a branch point, or may not be a function of x at all. Ch. 6 proves the theorem and extends it to systems of equations via the Jacobian.

Worked Example 4.20 — Resistor–diode circuit

A source V_s drives a resistor $R = 1\,\text{k}\Omega$ in series with a diode ($nV_T = 26\,\text{mV}$, $I_s = 10^{-14}\,\text{A}$). KVL gives

$$V_s = IR + nV_T \ln\left(1 + \frac{I}{I_s}\right).$$

Find dI/dV_s at the operating point $I = 1\,\text{mA}$.

Differentiate both sides with respect to V_s, treating $I = I(V_s)$:

$$1 = R\frac{dI}{dV_s} + nV_T \cdot \frac{1/I_s}{1 + I/I_s}\frac{dI}{dV_s} = \left(R + \frac{nV_T}{I_s + I}\right)\frac{dI}{dV_s}.$$

Hence

$$\frac{dI}{dV_s} = \frac{1}{R + r_d}, \qquad r_d = \frac{nV_T}{I + I_s} \approx \frac{nV_T}{I}.$$

Numbers. $r_d = 0.026/10^{-3} = 26\,\Omega$, so $dI/dV_s = 1/1026\,\Omega = 0.975\,\text{mS}$. A 10 mV ripple on the supply produces about $9.75\,\mu\text{A}$ of current ripple.

Interpretation. The diode behaves, for small changes, like a $26\,\Omega$ resistor in series with R: exactly the small-signal model of Example 4.16, obtained here without ever solving the nonlinear equation.

Numerical check. Solving the KVL equation numerically at $V_s \pm 1\,\text{mV}$ around the operating point and forming the central difference gives $0.9747\,\text{mS}$. ✓

Worked Example 4.21 — Horizontal tangents of an ellipse

The rotated ellipse $x^2 + xy + y^2 = 7$ passes through $(1, 2)$. Find the slope there and the points with horizontal tangents.

Differentiate: $2x + (y + xy') + 2yy' = 0 \Rightarrow y' = -\dfrac{2x + y}{x + 2y}$. At $(1, 2)$: $y' = -4/5$.

Horizontal tangents: $2x + y = 0$, i.e. $y = -2x$. Substitute: $x^2 - 2x^2 + 4x^2 = 3x^2 = 7$, so $x = \pm\sqrt{7/3} = \pm 1.528$, $y = \mp 3.055$. The ellipse's highest point is $(-1.528, 3.055)$ and its lowest $(1.528, -3.055)$.

Check. $1.528^2 + 1.528 \cdot (-3.055) + 3.055^2 = 2.333 - 4.667 + 9.333 = 7.000$. ✓

Engineering Note — implicit relations are everywhere in circuits

Any circuit containing a nonlinear element, solved by nodal analysis, produces an implicit equation (or system of equations) for the node voltages. Circuit simulators such as SPICE do not solve these explicitly. They linearize them (§4.8) and iterate with Newton's method, and the derivatives they need are exactly the implicit derivatives of this section, assembled into a Jacobian matrix (Ch. 6, Ch. 22).

Common Pitfall — differentiating y as if it were constant

When differentiating implicitly with respect to x, $\frac{d}{dx}(y^3)$ is $3y^2y'$, not $3y^2$ and not 0. Likewise, $\frac{d}{dx}(xy)$ needs the product rule: $y + xy'$. Forgetting the y' factor is the most common error in this topic and always produces an equation that cannot be solved for y' correctly.

4.5 Partial derivatives

Motivation

Engineering quantities rarely depend on one variable. The power dissipated in a resistor, $P = V^2/R$, depends on both voltage and resistance. The cutoff frequency of an RC filter, $f_c = 1/(2\pi RC)$, depends on two components. A potential $\varphi(x, y, z)$ depends on three coordinates, and a travelling wave $u(x, t)$ on position and time. To ask "how does P change if only R drifts?" we hold the other variables fixed and differentiate with respect to one. That is a partial derivative. This section is an introduction, enough for the rest of this chapter (implicit differentiation, linearization in several variables, sensitivity analysis). Ch. 6 develops multivariable calculus fully: total differentials, gradients, directional derivatives, Jacobians and differentiability.

Definition

Partial derivative

Let $f(x, y)$ be defined near (a, b). The **partial derivatives** of f at (a, b) are

$$\frac{\partial f}{\partial x}(a, b) = \lim_{h \to 0} \frac{f(a + h, b) - f(a, b)}{h}, \qquad \frac{\partial f}{\partial y}(a, b) = \lim_{h \to 0} \frac{f(a, b + h) - f(a, b)}{h},$$

when the limits exist. Other notations: f_x, $\partial_x f$, $D_1 f$; and $\left(\dfrac{\partial f}{\partial x}\right)_y$ when it is important to say which variable is held fixed. The definition extends to any number of variables: vary one, freeze all the others.

The symbol ∂ ("partial", "del", or "dee") distinguishes a partial derivative from an ordinary one. Computing $\partial f/\partial x$ needs no new rules: treat every other variable as a constant and use the one-variable rules of §4.2–4.3.

Geometric meaning. The graph of $f(x,y)$ is a surface $z = f(x,y)$. Freezing $y = b$ slices the surface with the vertical plane $y = b$, leaving a curve $z = f(x,b)$. Then $f_x(a,b)$ is the slope of that curve at $x = a$: the slope of the surface in the x-direction. Similarly, f_y is the slope in the y-direction. A topographic analogy: f_x is how steeply the ground rises as you walk due east, f_y as you walk due north.

Worked Example 4.22 — Computing partial derivatives

Find f_x and f_y for (a) $f = x^2y^3 + \sin(xy)$, (b) $f = e^{-x}\cos(3y)$, (c) $f = \sqrt{x^2+y^2}$.

(a) Holding y fixed, x^2y^3 is (y^3 times) x^2, and $\sin(xy)$ needs the chain rule with inner derivative $\partial(xy)/\partial x = y$:

$$f_x = 2xy^3 + y\cos(xy), \qquad f_y = 3x^2y^2 + x\cos(xy).$$

(b) $f_x = -e^{-x}\cos 3y$, $f_y = -3e^{-x}\sin 3y$.

(c) With $r = \sqrt{x^2+y^2}$: $f_x = \dfrac{x}{\sqrt{x^2+y^2}} = \dfrac{x}{r}$, $f_y = \dfrac{y}{r}$ (for $r \neq 0$). At $(3,4)$: $f_x = 0.6$, $f_y = 0.8$. These are the direction cosines of the point: the distance from the origin grows fastest in the radial direction.

Worked Example 4.23 — Partial derivatives of circuit formulas

(a) For $P = V^2/R$: $\dfrac{\partial P}{\partial V} = \dfrac{2V}{R}$ (units W/V = A) and $\dfrac{\partial P}{\partial R} = -\dfrac{V^2}{R^2}$ (W/Ω). At $V = 12\,\text{V}$, $R = 6\,\Omega$: $\partial P/\partial V = 4\,\text{A}$ and $\partial P/\partial R = -4\,\text{W}/\Omega$.

(b) Two resistors in parallel, $R_p = \dfrac{R_1R_2}{R_1+R_2}$. By the quotient rule with R_2 constant:

$$\frac{\partial R_p}{\partial R_1} = \frac{R_2(R_1+R_2) - R_1R_2}{(R_1+R_2)^2} = \frac{R_2^2}{(R_1+R_2)^2}, \qquad \frac{\partial R_p}{\partial R_2} = \frac{R_1^2}{(R_1+R_2)^2}.$$

With $R_1 = 1\,\text{k}\Omega$, $R_2 = 4\,\text{k}\Omega$ ($R_p = 800\,\Omega$): $\partial R_p/\partial R_1 = 16/25 = 0.64$ and $\partial R_p/\partial R_2 = 1/25 = 0.04$. The parallel combination is dominated by the smaller resistor; a $10\,\Omega$ change in R_1 moves R_p by about $6.4\,\Omega$, the same change in R_2 by only $0.4\,\Omega$.

Check. $R_1 = 1010$: $R_p = 1010 \cdot 4000/5010 = 806.39\,\Omega$, an increase of $6.39\,\Omega$. ✓

The first-order change: a preview of the total differential

If both V and R change a little, the change in P is approximately the sum of the separate effects:

$$\Delta P \approx \frac{\partial P}{\partial V}\Delta V + \frac{\partial P}{\partial R}\Delta R.$$

Why. Change V first and then R:

$$\Delta P = \big[P(V+\Delta V, R+\Delta R) - P(V, R+\Delta R)\big] + \big[P(V, R+\Delta R) - P(V,R)\big].$$

By the mean value theorem, each bracket is a one-variable derivative times the step, evaluated at an intermediate point. If the partials are continuous, the intermediate points can be replaced by (V,R) at the cost of an error that is small compared with $|\Delta V| + |\Delta R|$.

With $V = 12\,\text{V}$, $R = 6\,\Omega$, $\Delta V = +0.2\,\text{V}$ and $\Delta R = -0.1\,\Omega$: $\Delta P \approx 4(0.2) + (-4)(-0.1) = 1.2\,\text{W}$. The exact change is $12.2^2/5.9 - 24 = 1.227\,\text{W}$. The small discrepancy comes from second-order terms. The expression $df = f_x\,dx + f_y\,dy$ is the **total differential**; it underlies linearization (§4.8) and sensitivity analysis (§4.11), and is studied properly in Ch. 6.

The chain rule for several variables

If $z = f(x,y)$ and both $x = x(t)$ and $y = y(t)$ depend on t, dividing the first-order change by Δt and letting $\Delta t \to 0$ gives

$$\frac{dz}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}.$$

Each path from t to z contributes a product of derivatives, and the contributions add. Applied to $F(x, y(x)) = 0$ with $t = x$, this gives $F_x + F_y\,y' = 0$, the implicit-differentiation formula of §4.4.

Worked Example 4.24 — A heating resistor on a drifting supply

A resistor is fed from a supply whose voltage is rising at $dV/dt = 0.1\,\text{V/s}$ while the resistor heats and its resistance rises at $dR/dt = 0.02\,\Omega/\text{s}$. At the instant when $V = 12\,\text{V}$ and $R = 6\,\Omega$, how fast is the dissipated power changing?

$$\frac{dP}{dt} = \frac{\partial P}{\partial V}\frac{dV}{dt} + \frac{\partial P}{\partial R}\frac{dR}{dt} = \frac{2V}{R}(0.1) - \frac{V^2}{R^2}(0.02) = 4(0.1) - 4(0.02) = 0.32\ \text{W/s}.$$

The voltage rise increases the power by $0.40\,\text{W/s}$; the resistance rise reduces it by $0.08\,\text{W/s}$.

Units. A·(V/s) = W/s and (W/Ω)(Ω/s) = W/s. ✓

Higher partial derivatives and Clairaut's theorem

Partial derivatives can be differentiated again. With two variables there are four second partials:

$$f_{xx} = \frac{\partial^2 f}{\partial x^2}, \qquad f_{yy} = \frac{\partial^2 f}{\partial y^2}, \qquad f_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial y\,\partial x}, \qquad f_{yx} = \frac{\partial^2 f}{\partial x\,\partial y}.$$

Clairaut's (Schwarz's) theorem

If f_{xy} and f_{yx} are both continuous near (a,b), then $f_{xy}(a,b) = f_{yx}(a,b)$. The order of differentiation does not matter.

For the function of Example 4.22(a): $f_{xy} = \partial_y\big(2xy^3 + y\cos(xy)\big) = 6xy^2 + \cos(xy) - xy\sin(xy)$, and $f_{yx} = \partial_x\big(3x^2y^2 + x\cos(xy)\big) = 6xy^2 + \cos(xy) - xy\sin(xy)$. They agree.

Clairaut's theorem is behind several facts in field theory. For example, the curl of a gradient vanishes, $\nabla \times \nabla\varphi = \mathbf{0}$, because each component is a difference of equal mixed partials (Ch. 7).

Worked Example 4.25 — Checking that a function solves a PDE

(a) Show that $\varphi = \ln(x^2 + y^2)$ satisfies Laplace's equation $\varphi_{xx} + \varphi_{yy} = 0$ for $(x,y) \neq (0,0)$.

$\varphi_x = \dfrac{2x}{x^2+y^2}$. Quotient rule: $\varphi_{xx} = \dfrac{2(x^2+y^2) - 2x\cdot 2x}{(x^2+y^2)^2} = \dfrac{2(y^2-x^2)}{(x^2+y^2)^2}$. By the symmetry $x \leftrightarrow y$, $\varphi_{yy} = \dfrac{2(x^2-y^2)}{(x^2+y^2)^2}$. The sum is 0. ✓ (Up to a constant factor this is the potential of an infinite line charge.)

(b) Show that $u(x,t) = \sin(x - ct)$ satisfies the wave equation $u_{tt} = c^2 u_{xx}$.

$u_x = \cos(x-ct)$, $u_{xx} = -\sin(x-ct)$; $u_t = -c\cos(x-ct)$, $u_{tt} = -c^2\sin(x-ct)$. So $u_{tt} = c^2 u_{xx}$. ✓ The same holds for *any* twice-differentiable $g(x-ct)$, by the chain rule: a waveform travelling at speed c.

(c) Show that $u = e^{-\alpha k^2 t}\sin(kx)$ satisfies the heat (diffusion) equation $u_t = \alpha u_{xx}$.

$u_t = -\alpha k^2 e^{-\alpha k^2 t}\sin kx$ and $u_{xx} = -k^2 e^{-\alpha k^2 t}\sin kx$, so $u_t = \alpha u_{xx}$. ✓ High spatial frequencies (k large) decay fastest, as $e^{-\alpha k^2 t}$.

These checks are the starting point for Ch. 13.

Engineering Note — fields from potentials

The electric field is the negative gradient of the potential: $E_x = -\partial\varphi/\partial x$, $E_y = -\partial\varphi/\partial y$, $E_z = -\partial\varphi/\partial z$. For a point charge, $\varphi = \dfrac{q}{4\pi\varepsilon_0}(x^2+y^2+z^2)^{-1/2}$, and the chain rule gives $E_x = \dfrac{q}{4\pi\varepsilon_0}\dfrac{x}{R^3}$ with $R = \sqrt{x^2+y^2+z^2}$, and similarly for E_y, E_z. Together these give the familiar inverse-square law $|\mathbf{E}| = q/(4\pi\varepsilon_0 R^2)$, pointing radially. Partial derivatives turn one scalar function into the three components of a vector field.

Common Pitfall — partial derivatives existing is not enough

In one variable, differentiable implies continuous. In two variables, the mere existence of f_x and f_y does **not** imply continuity. The function $f(x,y) = \dfrac{xy}{x^2+y^2}$ (with $f(0,0) = 0$) has $f_x(0,0) = f_y(0,0) = 0$, because f is identically zero on both axes. Yet along the line $y = x$, $f = \tfrac{1}{2}$ everywhere except the origin, so f is not continuous at $(0,0)$. The proper notion of differentiability in several variables, and the condition (continuous partials) that guarantees it, is in Ch. 6. Also be explicit about which variables are held fixed: in thermodynamics and device physics, $(\partial f/\partial x)_y$ and $(\partial f/\partial x)_z$ are different quantities.

4.6 Higher-order derivatives

Definitions and meaning

The derivative f' is itself a function and can be differentiated again. The **second derivative** is $f'' = (f')'$, written $\dfrac{d^2 f}{dx^2}$ or $\ddot{f}$ for time derivatives; the **n-th derivative** is $f^{(n)} = \dfrac{d^n f}{dx^n}$.

Physically, if $x(t)$ is position, $\dot{x}$ is velocity, $\ddot{x}$ acceleration and $\dddot{x}$ jerk. In circuits, if $q(t)$ is the charge on a capacitor, $\dot{q} = i$ is current and $\ddot{q} = di/dt$ determines the inductor voltage $L\ddot{q}$. The series RLC equation

$$L\ddot{q} + R\dot{q} + \frac{1}{C}q = v_s(t)$$

is a statement about the zeroth, first and second derivatives of one function. Every ODE of Ch. 11 and Ch. 12 is such a statement.

Geometrically, f'' measures how the slope changes: **concavity**.

Concavity and inflection points

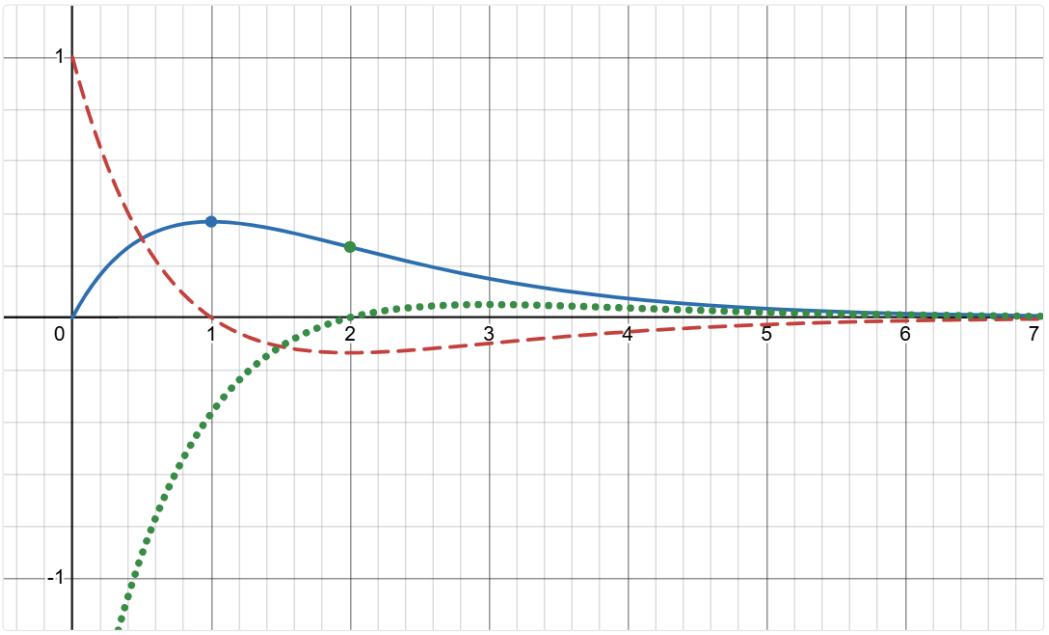
On an interval where $f'' > 0$, the slope f' is increasing and the graph is **concave up** (it lies above its tangent lines). Where $f'' < 0$, the graph is **concave down**. A point where f is continuous and the concavity changes is an **inflection point**; there $f'' = 0$ or f'' fails to exist.

Worked Example 4.26 — Shape of te^{-t}

The pulse $f(t) = te^{-t}$ (the critically damped response of Example 4.11 with $b = 1$) appears constantly in circuit transients. Find f', f'', the peak and the inflection point.

$$f' = (1 - t)e^{-t}, \qquad f'' = -e^{-t} - (1 - t)e^{-t} = (t - 2)e^{-t}.$$

- $f' = 0$ at $t = 1$; $f''(1) = -e^{-1} < 0$, so this is a maximum, $f(1) = 1/e = 0.368$.
- $f'' = 0$ at $t = 2$ and changes sign there (from $-$ to $+$), so $t = 2$ is an inflection point, with $f(2) = 2e^{-2} = 0.271$. After $t = 2$ the pulse is concave up and decays towards zero.
- At the inflection point $f'(2) = -e^{-2} = -0.135$: the pulse is falling fastest there.



The pulse te^{-t} (blue), its first derivative (red, dashed; zero at the peak) and its second derivative (green, dotted; zero at the inflection point, where the decay is steepest).

Patterns for n-th derivatives

$f(x)$	$f^{(n)}(x)$
x^m (m a positive integer)	$\dfrac{m!}{(m-n)!}x^{m-n}$ for $n \le m$; 0 for $n > m$
e^{ax}	$a^n e^{ax}$
$\sin(ax)$	$a^n \sin\left(ax + \dfrac{n\pi}{2}\right)$
$\cos(ax)$	$a^n \cos\left(ax + \dfrac{n\pi}{2}\right)$
$\dfrac{1}{1-x}$	$\dfrac{n!}{(1-x)^{n+1}}$
$\ln x$	$(-1)^{n-1}\dfrac{(n-1)!}{x^n}$ $(n \ge 1)$

Each row is proved by induction: compute the first few derivatives, guess the pattern, and check that differentiating the n-th gives the $(n+1)$-th. For example, $\frac{d}{dx}\, n!(1-x)^{-(n+1)} = n!(n+1)(1-x)^{-(n+2)} = (n+1)!(1-x)^{-(n+2)}$. The sine row says that each differentiation multiplies by a and shifts the phase by $90°$, the same fact as $(j\omega)^n$ in phasor form.

Leibniz's rule for products

General Leibniz rule

If f and g are n times differentiable,

$$(fg)^{(n)} = \sum_{k=0}^{n} \binom{n}{k} f^{(k)} g^{(n-k)}.$$

Proof sketch. For $n = 1$ this is the product rule. Assume it for n and differentiate each term with the product rule; the coefficients combine by Pascal's identity $\binom{n}{k} + \binom{n}{k-1} = \binom{n+1}{k}$, exactly as in the proof of the binomial theorem. The rule is the binomial theorem with powers replaced by derivatives.

Worked Example 4.27 — n-th derivative of $x^2 e^{2x}$

With $f = x^2$ (only f, f', f'' nonzero: $x^2, 2x, 2$) and $g = e^{2x}$ ($g^{(m)} = 2^m e^{2x}$), only three terms of Leibniz's rule survive:

$$\frac{d^n}{dx^n}\left(x^2 e^{2x}\right) = x^2\, 2^n e^{2x} + n\,(2x)\, 2^{n-1} e^{2x} + \binom{n}{2} 2 \cdot 2^{n-2} e^{2x} = 2^{n-2} e^{2x}\left(4x^2 + 4nx + n(n-1)\right).$$

Checks. $n = 1$: $2^{-1} e^{2x}(4x^2 + 4x) = e^{2x}(2x^2 + 2x)$, which matches the product rule. $n = 3$: $2e^{2x}(4x^2 + 12x + 6) = e^{2x}(8x^2 + 24x + 12)$, which matches direct differentiation (verified symbolically for $n = 1, \dots, 5$). ✓

Worked Example 4.28 — Verifying a circuit response

Show that $q(t) = e^{-t} \sin 2t$ satisfies $\ddot{q} + 2\dot{q} + 5q = 0$, the source-free equation of a series RLC circuit with (normalized) $L = 1$, $R = 2$, $1/C = 5$.

$$\dot{q} = e^{-t}(-\sin 2t + 2\cos 2t),$$
$$\ddot{q} = e^{-t}\big(\sin 2t - 2\cos 2t\big) + e^{-t}\big(-2\cos 2t - 4\sin 2t\big) = e^{-t}(-3\sin 2t - 4\cos 2t).$$

Then

$$\ddot{q} + 2\dot{q} + 5q = e^{-t}\big[(-3 - 2 + 5)\sin 2t + (-4 + 4 + 0)\cos 2t\big] = 0. \checkmark$$

The faster route uses complex exponentials: $q = \operatorname{Im}\{e^{st}\}$ with $s = -1 + 2j$, and each derivative multiplies by s, so the equation becomes $s^2 + 2s + 5 = 0$. Indeed $(-1 + 2j)^2 + 2(-1 + 2j) + 5 = (1 - 4j - 4) + (-2 + 4j) + 5 = 0$. Turning derivatives into powers of s is the central idea of Ch. 12 and Ch. 14.

Smoothness classes

A function is of class C^k on an interval if its first k derivatives exist and are continuous there; C^∞ means all derivatives exist (polynomials, e^x, $\sin x$). The hierarchy is strict:

- $|x|$ is C^0 but not differentiable at 0;
- $x|x|$ has derivative $2|x|$, so it is C^1 but not C^2;
- $x^2|x|$ is C^2 but not C^3.

Smoothness matters in practice: a waveform whose k-th derivative jumps has a Fourier spectrum that decays like $1/\omega^{k+1}$ (Ch. 15). A square wave (a jump in the function itself) has harmonics falling as $1/n$; a triangle wave (a jump in the first derivative) as $1/n^2$. Smoother edges mean less high-frequency interference, which is why digital edge rates are deliberately controlled.

Curvature

The second derivative alone does not measure how sharply a curve bends, because it depends on how steep the curve is. The **curvature** of $y = f(x)$ is

$$\kappa = \frac{|f''(x)|}{\left(1 + f'(x)^2\right)^{3/2}},$$

and the radius of curvature is $1/\kappa$. For $y = x^2$ at the origin, $\kappa = 2$, so the best-fitting circle there has radius $\tfrac{1}{2}$. For a circle of radius r, Example 4.18 gave $y'' = -r^2/y^3$ and $y' = -x/y$, so $\kappa = \dfrac{r^2/|y|^3}{(1 + x^2/y^2)^{3/2}} = \dfrac{r^2/|y|^3}{r^3/|y|^3} = \dfrac{1}{r}$, as it must be.

Engineering Note — second derivatives in measurement

The second derivative is a sensitive detector of features that are hard to see in the raw data: the "knee" of a diode or MOSFET characteristic, the onset of saturation in a magnetic core, or a peak position in a spectrum. Numerically, it is estimated by the central second difference

$$f''(x) \approx \frac{f(x + h) - 2f(x) + f(x - h)}{h^2},$$

whose error is $O(h^2)$ (derived from Taylor series in §4.7). Because of the h^2 in the denominator, noise of size σ in f produces an error of order σ/h^2 in f'': second derivatives of measured data must be heavily smoothed.

Common Pitfall — $f''(c) = 0$ does not guarantee an inflection point

For $f = x^4$, $f''(0) = 0$, but $f'' = 12x^2 \geq 0$ on both sides: no change of concavity, so no inflection point ($x = 0$ is in fact a minimum). Always check that f'' changes sign. Also note that $\dfrac{d^2y}{dx^2} \neq \left(\dfrac{dy}{dx}\right)^2$ and, for parametric curves, $\dfrac{d^2y}{dx^2} \neq \dfrac{d^2y/dt^2}{d^2x/dt^2}$ (use the formula in §4.3).

4.7 Taylor and Maclaurin series

Motivation

Polynomials are the only functions a processor can evaluate directly: they need nothing but additions and multiplications. Taylor's idea is to replace a complicated function, near a chosen point, by the polynomial that matches its value and as many derivatives as we like. This single idea gives:

- the small-signal models of diodes and transistors (keep the linear term),
- harmonic-distortion estimates (keep the quadratic and cubic terms),
- the approximations $\sin\theta \approx \theta$, $\cos\theta \approx 1 - \theta^2/2$, $(1 + x)^n \approx 1 + nx$ used throughout engineering,

- how calculators and DSP libraries compute e^x, $\sin x$ and $\ln x$,
- the error analysis of every numerical method in Ch. 22,
- a proof of Euler's formula $e^{j\theta} = \cos\theta + j\sin\theta$.

Building the Taylor polynomial

We look for a polynomial $p_n(x) = c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots + c_n(x-a)^n$ with $p_n^{(k)}(a) = f^{(k)}(a)$ for $k = 0, 1, \dots, n$. Differentiating p_n k times kills all terms of degree below k and turns $c_k(x-a)^k$ into $k!\,c_k$; the higher terms still contain a factor $(x-a)$ and vanish at $x = a$. So $p_n^{(k)}(a) = k!\,c_k$, and matching gives $c_k = f^{(k)}(a)/k!$.

Taylor polynomial and series

The **Taylor polynomial of degree** n of f about $x = a$ is

$$p_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

Letting $n \to \infty$ gives the **Taylor series**. With $a = 0$ it is called a **Maclaurin series**.

The degree-1 polynomial $p_1(x) = f(a) + f'(a)(x-a)$ is the tangent line: the linearization of §4.8. Each further term corrects the curvature, then the rate of change of curvature, and so on.

Taylor's theorem: how big is the error?

Taylor's theorem with Lagrange remainder

If f has $n+1$ continuous derivatives on an interval containing a and x, then

$$f(x) = p_n(x) + R_n(x), \qquad R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$$

for some c between a and x. Consequently, if $|f^{(n+1)}| \le M$ between a and x,

$$|R_n(x)| \le \frac{M\,|x-a|^{n+1}}{(n+1)!}.$$

Proof sketch. For $n = 0$ this is the mean value theorem: $f(x) = f(a) + f'(c)(x-a)$. For general n, fix x and choose the constant K so that $g(t) = f(t) - p_n(t) - K(t-a)^{n+1}$ vanishes at $t = x$. Then g and its first n derivatives vanish at $t = a$, and $g(x) = 0$ too. Rolle's theorem gives a zero of g' between a and x; applying Rolle repeatedly to $g', g'', \dots$ gives a point c with $g^{(n+1)}(c) = 0$, i.e. $f^{(n+1)}(c) = (n+1)!\,K$. Hence $R_n(x) = K(x-a)^{n+1}$ has the stated form. ∎

An equivalent integral form, $R_n(x) = \frac{1}{n!}\int_a^x f^{(n+1)}(t)(x-t)^n\,dt$, follows by repeated integration by parts (Ch. 5).

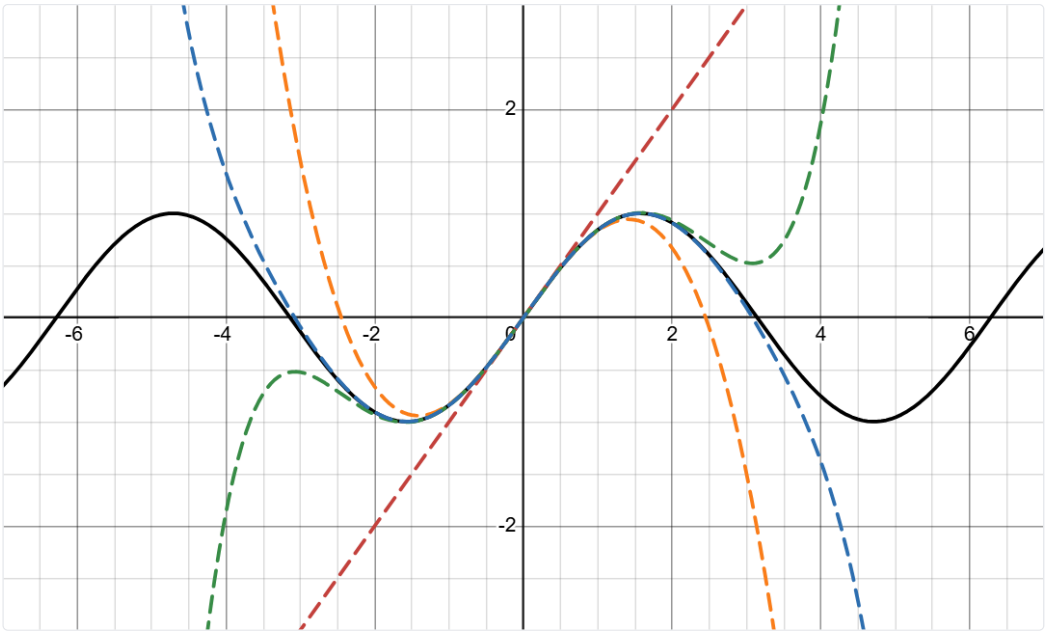
The remainder says two things. Near a, the error behaves like $|x-a|^{n+1}$: halving the distance cuts the error of a degree-n approximation by 2^{n+1}. Far from a, the factorial in the denominator usually wins in the end, but only if the derivatives do not grow faster.

The standard series

Computing derivatives at 0 gives the following Maclaurin series, valid in the stated ranges:

Function	Maclaurin series	Converges for
e^x	$1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = \sum \frac{x^n}{n!}$	all x
$\sin x$	$x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$	all x
$\cos x$	$1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$	all x
$\sinh x$, $\cosh x$	as sin, cos but all signs $+$	all x
$\frac{1}{1-x}$	$1 + x + x^2 + x^3 + \cdots$	$\lvert x\rvert < 1$
$\ln(1+x)$	$x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$	$-1 < x \le 1$
$(1+x)^\alpha$	$1 + \alpha x + \frac{\alpha(\alpha-1)}{2!}x^2 + \frac{\alpha(\alpha-1)(\alpha-2)}{3!}x^3 + \cdots$	$\lvert x\rvert < 1$
$\arctan x$	$x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$	$\lvert x\rvert \le 1$
$\tan x$	$x + \frac{x^3}{3} + \frac{2x^5}{15} + \cdots$	$\lvert x\rvert < \pi/2$

Derivations. For e^x, every derivative at 0 is 1. For $\sin x$, the derivatives at 0 cycle $0, 1, 0, -1$, so only odd powers appear, with alternating signs. For $(1+x)^\alpha$, the k-th derivative at 0 is $\alpha(\alpha-1)\cdots(\alpha-k+1)$; when α is a positive integer the series terminates and is the binomial theorem. The $\ln(1+x)$ and $\arctan x$ series are obtained by integrating $1/(1+x) = 1 - x + x^2 - \cdots$ and $1/(1+x^2) = 1 - x^2 + x^4 - \cdots$ term by term.



Maclaurin polynomials of $\sin x$ of degree 1, 3, 5 and 7. Each added term extends the interval over which the polynomial hugs the curve; all of them fail eventually, since a polynomial is unbounded and $\sin x$ is not.

Radius of convergence

A power series $\sum c_n (x-a)^n$ converges for $|x-a| < \rho$ and diverges for $|x-a| > \rho$, where the **radius of convergence** ρ can often be found from the ratio test:

$$\rho = \lim_{n\to\infty} \left| \frac{c_n}{c_{n+1}} \right| \quad \text{(when this limit exists)}.$$

For e^x, $c_n = 1/n!$ and $c_n/c_{n+1} = n+1 \to \infty$: it converges everywhere. For $1/(1-x)$, $c_n = 1$ and $\rho = 1$. Inside the radius, a power series can be differentiated and integrated term by term, and the result has the same radius.

> **Engineering Note — why $1/(1+x^2)$ has radius 1**
>
> The function $1/(1+x^2)$ is perfectly smooth for every real x, yet its Maclaurin series $1 - x^2 + x^4 - \cdots$ diverges for $|x| > 1$. The reason is invisible on the real line: as a function of a complex variable, $1/(1+z^2)$ has poles at $z = \pm j$, a distance 1 from the origin. **The radius of convergence is the distance to the nearest singularity in the complex plane.** This is why the pole locations of a transfer function control how far a series expansion (a low-frequency approximation, say) can be trusted. The full story is in Ch. 18.

Euler's formula from series

Substitute $x = j\theta$ into the exponential series and separate even and odd powers, using $j^2 = -1$, $j^3 = -j$, $j^4 = 1$:

$$e^{j\theta} = 1 + j\theta - \frac{\theta^2}{2!} - j\frac{\theta^3}{3!} + \frac{\theta^4}{4!} + \cdots = \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots\right) + j\left(\theta - \frac{\theta^3}{3!} + \cdots\right) = \cos\theta + j\sin\theta.$$

The rearrangement is legitimate because the exponential series converges absolutely for every complex argument.

> **Worked Example 4.29 — Computing with a remainder bound**
>
> (a) Estimate $e^{0.1}$ with the degree-3 Maclaurin polynomial and bound the error.
>
> $p_3(0.1) = 1 + 0.1 + 0.005 + 0.000166\overline{6} = 1.1051667$. The remainder is $R_3 = e^c (0.1)^4/4!$ with $0 < c < 0.1$, and $e^c < e^{0.1} < 1.2$, so $|R_3| < 1.2 \times 10^{-4}/24 = 5 \times 10^{-6}$.
>
> **Check.** $e^{0.1} = 1.1051709$; actual error $4.25 \times 10^{-6} < 5 \times 10^{-6}$. ✓
>
> (b) Estimate $\sin 0.5$ with p_5 and bound the error.
>
> $p_5(0.5) = 0.5 - 0.0208333 + 0.0002604 = 0.4794271$. Because the x^6 coefficient is zero, $p_5 = p_6$, so we may use the degree-6 remainder: $|R_6| \le |0.5|^7/7! = 1.550 \times 10^{-6}$ (all derivatives of sin are bounded by 1).
>
> **Check.** $\sin 0.5 = 0.4794255$; actual error 1.545×10^{-6}, just inside the bound. ✓

Manipulating series

Finding high derivatives is tedious. In practice series are built from the table by substitution, multiplication, division, differentiation and integration.

- **Substitution.** $e^{-x^2} = 1 - x^2 + \dfrac{x^4}{2!} - \dfrac{x^6}{3!} + \cdots$ (replace x by $-x^2$ in e^x). This is how the Gaussian error function is evaluated (Ch. 19).
- **Multiplication.** $e^x \sin x = \left(1 + x + \frac{x^2}{2} + \cdots\right)\left(x - \frac{x^3}{6} + \cdots\right) = x + x^2 + \frac{x^3}{3} + \cdots$.

- **Division.** For $\tan x = \sin x/\cos x$, write $\tan x = b_1 x + b_3 x^3 + b_5 x^5 + \cdots$ (odd function) and require $(b_1 x + b_3 x^3 + b_5 x^5)(1 - \frac{x^2}{2} + \frac{x^4}{24}) = x - \frac{x^3}{6} + \frac{x^5}{120}$. Matching powers: $b_1 = 1$; $b_3 - \frac{1}{2} = -\frac{1}{6} \Rightarrow b_3 = \frac{1}{3}$; $b_5 - \frac{b_3}{2} + \frac{1}{24} = \frac{1}{120} \Rightarrow b_5 = \frac{2}{15}$.
- **Limits.** Series settle indeterminate forms instantly: $\dfrac{x - \sin x}{x^3} = \dfrac{x^3/6 - x^5/120 + \cdots}{x^3} \to \dfrac{1}{6}$. (Three applications of L'Hôpital's rule give the same answer more slowly.)

Worked Example 4.30 — A Taylor series about a point other than 0

Expand $f(x) = \sqrt{x}$ about $a = 4$ to second order and use it to estimate $\sqrt{4.1}$.

$f = x^{1/2}$, $f' = \frac{1}{2}x^{-1/2}$, $f'' = -\frac{1}{4}x^{-3/2}$. At $x = 4$: $f = 2$, $f' = \frac{1}{4}$, $f'' = -\frac{1}{32}$. So

$$\sqrt{x} \approx 2 + \frac{x-4}{4} - \frac{(x-4)^2}{64}.$$

At $x = 4.1$: $2 + 0.025 - 0.00015625 = 2.02484375$.

Check. $\sqrt{4.1} = 2.02484567$; the error is 1.9×10^{-6}, consistent with the next term $f'''(4)(0.1)^3/6 = \frac{3}{8} \cdot 4^{-5/2} \cdot \frac{10^{-3}}{6} = 1.95 \times 10^{-6}$. ✓

Worked Example 4.31 — Harmonic distortion of an exponential device

A forward-biased diode (or a bipolar transistor) carries $i = I_Q e^{v/V_T}$, where v is the small signal added to the bias, $V_T = 26\,\text{mV}$. Let $v = V_m \cos \omega t$. Estimate the second-harmonic distortion HD_2 (amplitude of the 2ω component divided by that of the ω component) when $V_m = 2.6\,\text{mV}$.

Expand with $x = v/V_T = a\cos\omega t$, $a = V_m/V_T$:

$$\frac{i}{I_Q} = 1 + a\cos\omega t + \frac{a^2}{2}\cos^2\omega t + \frac{a^3}{6}\cos^3\omega t + \cdots$$

Use $\cos^2\theta = \frac{1}{2}(1 + \cos 2\theta)$ (Ch. 3):

$$\frac{i}{I_Q} \approx \underbrace{1 + \frac{a^2}{4}}_{\text{DC shift}} + \underbrace{a\cos\omega t}_{\text{fundamental}} + \underbrace{\frac{a^2}{4}\cos 2\omega t}_{\text{2nd harmonic}} + \cdots$$

So $\text{HD}_2 \approx \dfrac{a^2/4}{a} = \dfrac{a}{4} = \dfrac{V_m}{4V_T}$.

Numbers. $a = 2.6/26 = 0.1$, so $\text{HD}_2 \approx 2.5\,\%$. To keep distortion below $1\,\%$ the input must stay below $4 \times 0.01 \times 26 = 1.04\,\text{mV}$ peak.

Check. The exact Fourier coefficients of $e^{a\cos\omega t}$ are modified Bessel functions, giving $\text{HD}_2 = I_2(a)/I_1(a) = 0.02499$ at $a = 0.1$. ✓ (At $a = 0.5$ the estimate 0.125 is still within $1\,\%$ of the exact 0.1237.)

The DC shift $a^2/4$ is the mechanism of diode detection and mixing: a nonlinear device converts part of an AC signal into DC.

Finite-difference formulas from Taylor series

Write the Taylor expansions at $x \pm h$:

$$f(x \pm h) = f(x) \pm h f'(x) + \frac{h^2}{2}f''(x) \pm \frac{h^3}{6}f'''(x) + \frac{h^4}{24}f^{(4)}(x) \pm \cdots$$

Subtracting, adding, or using one of them gives the standard numerical derivatives and their errors:

$$\begin{aligned}
\text{forward:} \quad & \frac{f(x+h) - f(x)}{h} = f'(x) + \frac{h}{2}f''(x) + O(h^2), \\
\text{central:} \quad & \frac{f(x+h) - f(x-h)}{2h} = f'(x) + \frac{h^2}{6}f'''(x) + O(h^4), \\
\text{second:} \quad & \frac{f(x+h) - 2f(x) + f(x-h)}{h^2} = f''(x) + \frac{h^2}{12}f^{(4)}(x) + O(h^4).
\end{aligned}$$

For $f = e^x$ at $x = 0$ with $h = 0.1$, the forward difference is off by 0.0517 (predicted $h/2 \approx 0.05$), the central difference by 0.00167 (predicted $h^2/6 = 0.00167$) and the second difference by 0.00083 (predicted $h^2/12 = 0.00083$). The central difference is far better for the same number of function evaluations, because the even-order error terms cancel by symmetry.

Common Pitfall — using a series outside its radius

$\ln(1+x) = x - x^2/2 + x^3/3 - \cdots$ is useless at $x = 2$: the terms grow and the partial sums oscillate wildly. To compute $\ln 3$, use $\ln 3 = \ln\frac{1+y}{1-y}$ with $y = \frac{1}{2}$, and the rapidly convergent series $\ln\frac{1+y}{1-y} = 2\left(y + \frac{y^3}{3} + \frac{y^5}{5} + \cdots\right)$. Similarly, the binomial approximation $(1+x)^\alpha \approx 1 + \alpha x$ is only good for $|x| \ll 1$, and it is a statement about $|x|$, not $|\alpha x|$.

4.8 Linearization

The tangent-line approximation

Linearization

The **linearization** of f at a is the degree-1 Taylor polynomial

$$L(x) = f(a) + f'(a)(x - a).$$

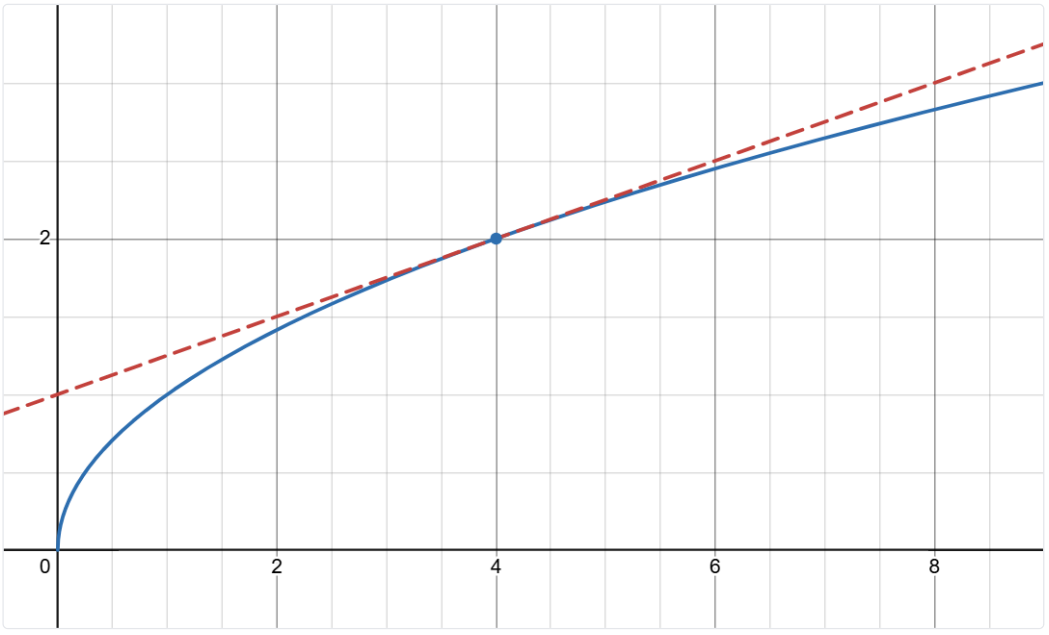
For x near a, $f(x) \approx L(x)$, with error $f(x) - L(x) = \tfrac{1}{2} f''(c)(x-a)^2$ for some c between a and x.

In terms of increments, with $\Delta x = x - a$ and $\Delta f = f(x) - f(a)$:

$$\Delta f \approx f'(a)\,\Delta x.$$

The **differential** notation writes this as $df = f'(x)\,dx$: dx is an independent increment, and df is the corresponding change *along the tangent line*. The true change Δf and the differential df differ by a term of order $(\Delta x)^2$.

Linearization is the most important approximation in engineering. Nonlinear devices (diodes, transistors, magnetic cores, sensors) are analysed by choosing an **operating point** (a bias, or quiescent point, Q) and replacing the device by its tangent line there. The slope becomes a small-signal parameter: a conductance, a transconductance, an incremental inductance.



$\sqrt{x}$ and its linearization at $x = 4$. Near the operating point the line is an excellent substitute; the error grows quadratically with distance, and the line always lies above this concave-down curve.

Worked Example 4.32 — Quick numerical estimates

(a) $\sqrt{4.1}$: with $f = \sqrt{x}$, $a = 4$, $f'(4) = \tfrac{1}{4}$: $L(4.1) = 2 + 0.1/4 = 2.025$. True: 2.024846 (error 1.5×10^{-4}, which matches $\tfrac{1}{2}|f''(4)|(0.1)^2 = 1.6 \times 10^{-4}$).

(b) $\sin 31^\circ$: with $f = \sin x$, $a = 30^\circ = \pi/6$, $\Delta x = 1^\circ = \pi/180 = 0.017453$ rad: $L = \sin\tfrac{\pi}{6} + \cos\tfrac{\pi}{6} \cdot \tfrac{\pi}{180} = 0.5 + 0.866 \times 0.017453 = 0.51511$. True: 0.51504. (Converting Δx to radians is essential.)

(c) 1.02^{10}: with $(1+x)^n \approx 1 + nx$: $1 + 10(0.02) = 1.20$. True: 1.2190. The error, $\approx \binom{10}{2}(0.02)^2 = 0.018$, is the next binomial term.

(d) Small angles: $\sin\theta \approx \theta$ at $\theta = 10^\circ = 0.17453$ rad gives 0.17453 against the true 0.17365, an error of $0.5\,\%$. The relative error of $\sin\theta \approx \theta$ is about $\theta^2/6$.

Small-signal models

Worked Example 4.33 — MOSFET transconductance

A MOSFET in saturation obeys the square law $i_D = \tfrac{k}{2}(v_{GS} - V_t)^2$ with $k = 2\ \text{mA/V}^2$ and $V_t = 0.5\ \text{V}$. It is biased at $V_{GS} = 1.0\ \text{V}$. Find I_D, the transconductance $g_m = di_D/dv_{GS}$ at the Q-point, and the error of the linear model for a $10\ \text{mV}$ signal.

Bias. Overdrive $V_{ov} = V_{GS} - V_t = 0.5\ \text{V}$, so $I_D = \tfrac{2}{2}(0.5)^2 = 0.25\ \text{mA}$.

Linearize. $g_m = k(V_{GS} - V_t) = 2 \times 0.5 = 1\ \text{mA/V}$. Equivalently $g_m = 2I_D/V_{ov} = 2(0.25)/0.5 = 1\ \text{mA/V}$.

Small-signal model. $i_D \approx I_D + g_m v_{gs}$, where $v_{gs} = v_{GS} - V_{GS}$ is the signal. For $v_{gs} = 10\ \text{mV}$, the model predicts $\Delta i_D = 10\ \mu\text{A}$.

Exact. $\Delta i_D = \tfrac{k}{2}\left[(0.51)^2 - (0.5)^2\right] = 10^{-3}(0.2601 - 0.25) = 10.1\ \mu\text{A}$. The relative error is $v_{gs}/(2V_{ov}) = 1\,\%$. Because the square law has f'' constant, the error is exactly the quadratic term $\tfrac{k}{2}v_{gs}^2 = 0.1\ \mu\text{A}$.

Worked Example 4.34 — Linearizing in two variables

The magnitude of an impedance $Z = R + jX$ is $|Z| = \sqrt{R^2 + X^2}$. Linearize it about $R = 3\,\Omega$, $X = 4\,\Omega$ and estimate $|Z|$ at $R = 3.1$, $X = 3.9$.

The two-variable linearization uses both partial derivatives (§4.5):

$$L(R, X) = |Z|_0 + \frac{\partial |Z|}{\partial R}\Delta R + \frac{\partial |Z|}{\partial X}\Delta X, \qquad \frac{\partial |Z|}{\partial R} = \frac{R}{|Z|},\ \frac{\partial |Z|}{\partial X} = \frac{X}{|Z|}.$$

At $(3, 4)$: $|Z|_0 = 5$, partials 0.6 and 0.8. So $L = 5 + 0.6(0.1) + 0.8(-0.1) = 4.98\,\Omega$.

Check. $\sqrt{3.1^2 + 3.9^2} = \sqrt{24.82} = 4.98197\,\Omega$; error $0.002\,\Omega$. ✓ Geometrically, $L(R, X)$ is the tangent plane to the cone $z = \sqrt{R^2 + X^2}$ (Ch. 6).

Linearizing a differential equation about an equilibrium

Consider a first-order system $\dot{x} = f(x)$. An **equilibrium** x^* satisfies $f(x^*) = 0$. Write $x = x^* + \delta$ with δ small. Since $\dot{x}^* = 0$,

$$\dot{\delta} = f(x^* + \delta) \approx f(x^*) + f'(x^*)\,\delta = f'(x^*)\,\delta.$$

This linear equation has solution $\delta(t) = \delta(0) e^{f'(x^*)t}$. Hence:

- if $f'(x^*) < 0$, small disturbances decay: the equilibrium is **stable**, with time constant $\tau = -1/f'(x^*)$;
- if $f'(x^*) > 0$, they grow: the equilibrium is **unstable**;
- if $f'(x^*) = 0$, linearization is inconclusive.

For systems of equations the derivative becomes a Jacobian matrix and its eigenvalues decide stability (Ch. 10).

Worked Example 4.35 — A capacitor charged by a current source into a diode

A constant current $I_0 = 1\,\text{mA}$ charges a capacitor $C = 1\,\text{nF}$ that is shunted by a diode ($I_s = 10^{-14}\,\text{A}$, $V_T = 25.85\,\text{mV}$). The capacitor voltage obeys

$$C\frac{dv}{dt} = I_0 - I_s\left(e^{v/V_T} - 1\right) \equiv C\,f(v).$$

Find the equilibrium, decide its stability and find the time constant of small deviations.

Equilibrium. $I_s(e^{v^*/V_T} - 1) = I_0 \Rightarrow v^* = V_T \ln(1 + I_0/I_s) = 0.02585 \ln(10^{11} + 1) = 0.6547\,\text{V}$. (All of I_0 flows through the diode.)

Linearize. $f'(v) = -\dfrac{I_s}{CV_T}e^{v/V_T}$, so $f'(v^*) = -\dfrac{I_0 + I_s}{CV_T} < 0$: stable.

Time constant. $\tau = -1/f'(v^*) = \dfrac{CV_T}{I_0 + I_s} \approx C\,r_d$ with $r_d = V_T/I_0 = 25.85\,\Omega$. Numerically $\tau = 10^{-9} \times 25.85 = 25.9\,\text{ns}$.

The linearized circuit is the capacitor in parallel with the diode's small-signal resistance: an RC circuit. Linearization has turned a nonlinear ODE into the most familiar circuit of all.

Newton's method

Linearization also solves equations. To solve $f(x) = 0$ starting from a guess x_n, replace f by its tangent line at x_n and solve the *linear* equation $f(x_n) + f'(x_n)(x - x_n) = 0$:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

Quadratic convergence of Newton's method

If f is C^2 near a simple root r ($f(r) = 0$, $f'(r) \neq 0$) and x_0 is close enough to r, the errors $e_n = x_n - r$ satisfy

$$e_{n+1} \approx \frac{f''(r)}{2f'(r)}\,e_n^2.$$

The number of correct digits roughly doubles with each step.

Proof sketch. Taylor's theorem about x_n: $0 = f(r) = f(x_n) + f'(x_n)(r - x_n) + \tfrac{1}{2}f''(c)(r - x_n)^2$. Divide by $f'(x_n)$ and rearrange: $x_{n+1} - r = \dfrac{f''(c)}{2f'(x_n)}(x_n - r)^2$.

Worked Example 4.36 — Newton's method for a diode bias point

A 5 V source drives a 1 kΩ resistor in series with a diode ($I_s = 10^{-14}$ A, $V_T = 25.852$ mV). Find the diode voltage V.

KCL at the diode: $f(V) = \dfrac{5 - V}{1000} - I_s\left(e^{V/V_T} - 1\right) = 0$, with $f'(V) = -\dfrac{1}{1000} - \dfrac{I_s}{V_T}e^{V/V_T}$.

Starting from $V_0 = 0.7$ V:

n	V_n (V)	error $V_n - V^*$
0	0.700000	7.5×10^{-3}
1	0.693518	9.7×10^{-4}
2	0.692562	1.8×10^{-5}
3	0.6925436394	6.3×10^{-9}
4	0.6925436332	$< 10^{-15}$

The diode voltage is $V^* = 0.69254$ V and the current $(5 - 0.69254)/1000 = 4.3075$ mA. The error is roughly squared at each step, as the theorem predicts.

Common Pitfall — Newton's method far from the root

Started at $V_0 = 0$ V, the iteration above jumps to $V_1 \approx 5.0$ V (at 0 V the diode is nearly an open circuit, so the tangent line only sees the resistor) and then creeps back down by about one V_T per step: it needs roughly 170 iterations, and along the way $e^{V/V_T} \approx e^{193} \approx 10^{84}$. With a slightly larger source voltage the exponential overflows in floating point. Circuit simulators therefore **limit** the voltage step per iteration on junctions. Newton's method is excellent locally and unreliable globally; combine it with bracketing (§4.1) or step limiting.

Engineering Note — the limits of small-signal analysis

A linear model is valid only while the neglected quadratic term is small compared with the linear one: for the exponential device, $|v| \ll 2V_T \approx 50$ mV; for the square-law MOSFET, $|v_{gs}| \ll 2V_{ov}$. Example 4.31 quantified exactly what "small" costs in distortion. Large-signal behaviour (clipping, slewing, switching) requires the full nonlinear model, usually solved numerically.

4.9 Rates of change

Average and instantaneous rates

For a quantity $y = f(t)$, the **average rate of change** over $[t_1, t_2]$ is $\dfrac{f(t_2) - f(t_1)}{t_2 - t_1}$, the slope of a secant. The **instantaneous rate of change** at t is $f'(t)$, the slope of the tangent. The mean value theorem (§4.2) links them: somewhere in the interval the instantaneous rate equals the average.

Almost every constitutive law in circuit theory is a rate law:

Law	Rate	Units
Current	$i = \dfrac{dq}{dt}$	C/s = A
Capacitor	$i = C\dfrac{dv}{dt}$ (more generally $i = \dfrac{d(Cv)}{dt}$)	F·V/s = A
Inductor	$v = L\dfrac{di}{dt}$ (more generally $v = \dfrac{d\lambda}{dt}$, λ = flux linkage)	H·A/s = V
Faraday's law	$\mathcal{E} = -\dfrac{d\Phi}{dt}$	Wb/s = V
Power	$p = \dfrac{dW}{dt}$	J/s = W
Slew rate	$\mathrm{SR} = \max\left\lvert\dfrac{dv_o}{dt}\right\rvert$	V/s (usually V/µs)

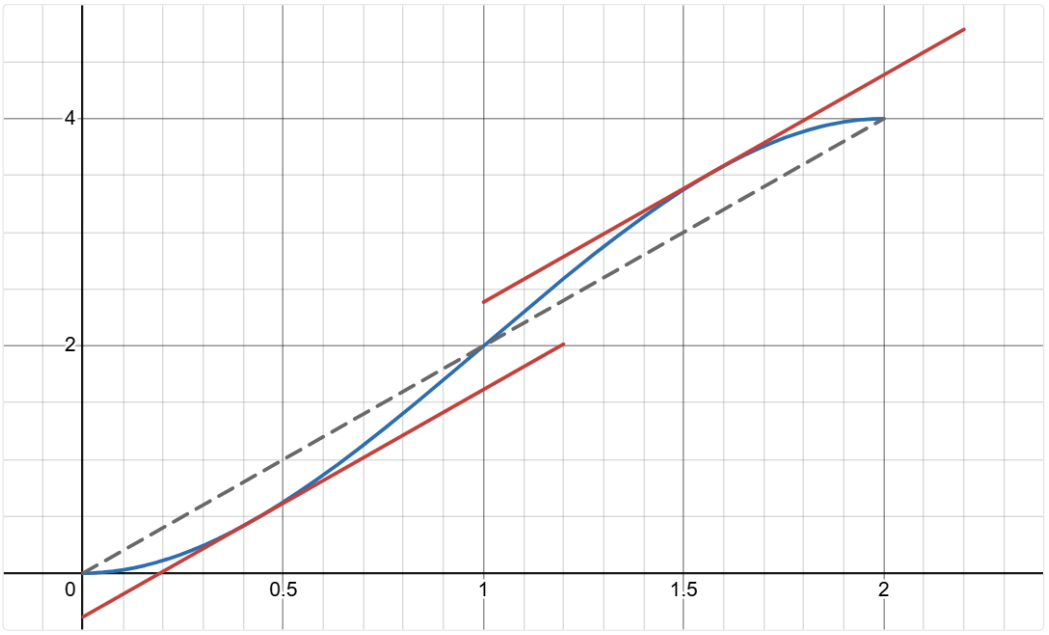
Worked Example 4.37 — Average versus instantaneous current

The charge through a conductor is $q(t) = 3t^2 - t^3$ mC for $0 \le t \le 2$ s. Find the average current over $[0, 2]$, the instantaneous current at $t = 1$ s, and the times at which the instantaneous current equals the average.

Average. $\bar{i} = \dfrac{q(2) - q(0)}{2} = \dfrac{12 - 8}{2} = 2$ mA.

Instantaneous. $i(t) = q'(t) = 6t - 3t^2$ mA, so $i(1) = 3$ mA (the maximum, since $i'(t) = 6 - 6t = 0$ there).

MVT points. $6c - 3c^2 = 2 \Rightarrow 3c^2 - 6c + 2 = 0 \Rightarrow c = 1 \pm \dfrac{1}{\sqrt{3}} = 0.423$ s or 1.577 s. Both lie in $(0, 2)$, as the MVT guarantees at least one must.



The charge $q(t) = 3t^2 - t^3$ (blue), the secant over $[0, 2]$ with slope 2 (the average current, dashed) and the two tangent lines with the same slope (red) at $t = 1 \mp 1/\sqrt{3}$.

Relative (logarithmic) rates

The **relative rate of change** is $\dfrac{\dot y}{y} = \dfrac{d}{dt}\ln y$, the fraction by which y changes per unit time. For an exponential $y = Y_0 e^{-t/\tau}$ it is constant: $\dot y / y = -1/\tau$. That is the defining property of exponential decay: the *relative* rate never changes, so the capacitor in an RC circuit loses the same *fraction* of its remaining voltage in every equal time interval. With $\tau = 2\,\text{ms}$, the instantaneous relative rate is $-500\,\text{s}^{-1}$, i.e. $-0.05\,\%$ per µs. Note that the voltage falls by $1 - e^{-1/2} = 39\,\%$, not $50\,\%$, in the first millisecond: a relative rate is instantaneous, not a percentage per finite interval.

Related rates

In a **related-rates** problem, several quantities vary with time and are linked by an equation. Differentiating the equation with respect to t (with the chain rule, as in implicit differentiation) links their rates. The procedure:

1. Name the variables and write the equation that connects them, valid at **all** times.
2. Differentiate both sides with respect to t.
3. Only **then** substitute the values at the instant of interest, and solve for the unknown rate.

Substituting before differentiating is the classic mistake: it freezes quantities that are in fact changing.

Worked Example 4.38 — A MEMS capacitive sensor

A parallel-plate capacitor has $C = \varepsilon_0 A/d$ with $A = 1\,\text{cm}^2$, held at a constant $V = 5\,\text{V}$. At the instant when $d = 10\,\mu\text{m}$, the moving plate recedes at $\dot d = 1$ mm/s. Find the current.

Step 1. $q = Cv$ and $C = \varepsilon_0 A/d$.

Step 2. Product rule: $i = \dot q = \dot C v + C\dot v$. Here $\dot v = 0$. Chain rule: $\dot C = -\dfrac{\varepsilon_0 A}{d^2}\dot d = -\dfrac{C}{d}\dot d$.

Step 3. $C = 8.854\times10^{-12}\times10^{-4}/10^{-5} = 88.54\,\text{pF}$. $\dot C = -88.54\times10^{-12}\times(10^{-3}/10^{-5}) = -8.854\,\text{nF/s}$. So

$$i = V\dot C = 5\times(-8.854\times10^{-9}) = -44.3\,\text{nA}.$$

The negative sign means charge flows *out* of the capacitor as the plates separate. This current is the output signal of a capacitive microphone or accelerometer.

Units. $\text{V}\cdot\text{F/s} = \text{C/s} = \text{A}$. ✓

Worked Example 4.39 — EMF in an expanding loop

A circular conducting loop lies in a uniform magnetic field $B = 0.5\,\text{T}$ perpendicular to its plane. Its radius increases at $\dot{r} = 2\,\text{cm/s}$. Find the induced EMF when $r = 10\,\text{cm}$.

Equation. $\Phi = B\pi r^2$, valid at all times.

Differentiate. $\dot{\Phi} = 2\pi B r\dot{r}$ (B constant), so $|\mathcal{E}| = 2\pi B r\dot{r}$.

Substitute. $|\mathcal{E}| = 2\pi(0.5)(0.1)(0.02) = 6.28\,\text{mV}$.

Interpretation. $2\pi r\dot{r}$ is the rate at which area is added (the circumference times the outward speed), and B times that is the rate at which flux is swept: the motional EMF.

Worked Example 4.40 — Slew-rate limit of an amplifier

An op-amp has slew rate $\text{SR} = 0.5\,\text{V}/\mu\text{s}$. What is the highest frequency at which it can output an undistorted 10 V-peak sine wave?

For $v_o = V_m \sin(2\pi f t)$, $\dot{v}_o = 2\pi f V_m \cos(2\pi f t)$, whose maximum magnitude is $2\pi f V_m$ (at the zero crossings, where the sine is steepest). Require $2\pi f V_m \le \text{SR}$:

$$f_{\max} = \frac{\text{SR}}{2\pi V_m} = \frac{0.5 \times 10^6}{2\pi \times 10} = 7.96\,\text{kHz}.$$

This is the **full-power bandwidth**. Above it the output becomes a triangle wave, even if the amplifier's small-signal bandwidth is far higher. Note that the limit depends on amplitude: at 1 V peak, $f_{\max} = 79.6\,\text{kHz}$.

Worked Example 4.41 — Power in reactive elements

Show that the rate of change of stored energy in an inductor equals the power delivered to it.

$W = \tfrac{1}{2}Li^2$. Chain rule: $\dfrac{dW}{dt} = \tfrac{1}{2}L \cdot 2i\dfrac{di}{dt} = i\left(L\dfrac{di}{dt}\right) = iv = p$. ✓ Likewise for a capacitor, $W = \tfrac{1}{2}Cv^2$ gives $\dot{W} = Cv\dot{v} = vi$.

For $i = I_m \sin\omega t$: $p = \tfrac{1}{2}\omega L I_m^2 \sin 2\omega t$, which averages to zero. An ideal inductor stores and returns energy; it does not dissipate it.

Common Pitfall — signs, units and constants that are not constant

Decide a sign convention first (passive sign convention: current entering the $+$ terminal) and keep it; a negative rate is information, not an error. Convert units before substituting (1 mm/s $= 10^{-3}$ m/s, μs in the denominator of a slew rate means $\times 10^6$). And remember that "$i = C\,dv/dt$" assumes C is constant: for a varying capacitance, as in Example 4.38, the full rule is $i = d(Cv)/dt$.

Engineering Note — rates as specifications

Many specifications are bounds on derivatives: slew rate (V/µs), dv/dt immunity of thyristors and gate drivers (V/ns), di/dt limits on power semiconductors (A/µs), ramp rates of power-supply sequencing, and the rate of change of frequency (RoCoF, Hz/s) in power grids. The inductor law also explains why a fast current edge through a small parasitic inductance causes trouble: 1 A switched in 1 ns through 10 nH of trace inductance produces $L\,di/dt = 10\,\text{V}$ of ground bounce.

4.10 Engineering optimization

Motivation

Design is optimization: the load that extracts the most power, the switching frequency that minimizes total loss, the cable geometry with least attenuation, the damping that gives the flattest response. When a performance measure depends smoothly on one design variable, calculus finds the best value exactly. Multivariable and constrained problems (Lagrange multipliers) are in Ch. 6; numerical optimization algorithms are in Ch. 23.

Critical points

Extrema and critical points

f has a **local maximum** at c if $f(c) \ge f(x)$ for all x near c, and an **absolute (global) maximum** on a set D if $f(c) \ge f(x)$ for all $x \in D$; similarly for minima. A **critical point** of f is an interior point c of its domain where $f'(c) = 0$ or $f'(c)$ does not exist.

Fermat's theorem

If f has a local extremum at an interior point c and $f'(c)$ exists, then $f'(c) = 0$.

Proof. Suppose c is a local maximum. For small $h > 0$, $f(c+h) - f(c) \le 0$, so $\frac{f(c+h)-f(c)}{h} \le 0$ and the right-hand limit gives $f'(c) \le 0$. For small $h < 0$ the quotient is ≥ 0, giving $f'(c) \ge 0$. Hence $f'(c) = 0$. ■

The converse fails: $f = x^3$ has $f'(0) = 0$ but no extremum at 0. Critical points are *candidates*, which must then be classified.

Classifying critical points

First-derivative test. If f' changes sign from $+$ to $-$ at c, f has a local maximum at c; from $-$ to $+$, a local minimum; no sign change, no extremum.

Second-derivative test. If $f'(c) = 0$ and $f''(c) < 0$, c is a local maximum; if $f''(c) > 0$, a local minimum; if $f''(c) = 0$, the test is inconclusive.

Why the second-derivative test works. By Taylor's theorem, $f(c+h) = f(c) + 0 \cdot h + \tfrac{1}{2}f''(c)h^2 + o(h^2)$. For small h the quadratic term dominates, and its sign is that of $f''(c)$, whichever sign h has.

The closed-interval method

When the design variable lies in a closed interval $[a, b]$ (a component range, a physical limit), the EVT guarantees an absolute maximum and minimum, and they occur either at critical points or at the endpoints. So:

1. Find the critical points in (a, b).
2. Evaluate f at them and at a and b.
3. The largest value is the absolute maximum, the smallest the absolute minimum.

For example, $f(x) = x^3 - 3x$ on $[-2, 3]$ has critical points $x = \pm 1$ (from $3x^2 - 3 = 0$). The values are $f(-2) = -2$, $f(-1) = 2$, $f(1) = -2$, $f(3) = 18$. The absolute maximum 18 is at the endpoint $x = 3$, not at the local maximum $x = -1$; the absolute minimum -2 is attained twice.

A strategy for applied problems

1. Identify the design variable and the objective, with units; write the objective as a function of one variable (use constraints to eliminate others).
2. State the feasible range (resistances are positive, frequencies lie in a band).
3. Differentiate, solve $f' = 0$, and classify.
4. Compare with endpoints and check that the answer is physically sensible.
5. Ask how sensitive the optimum is (§4.11): a flat optimum is forgiving.

Worked Example 4.42 — Maximum power transfer

Choose $R_L > 0$ to maximize $P = \dfrac{V^2 R_L}{(R_s + R_L)^2}$.

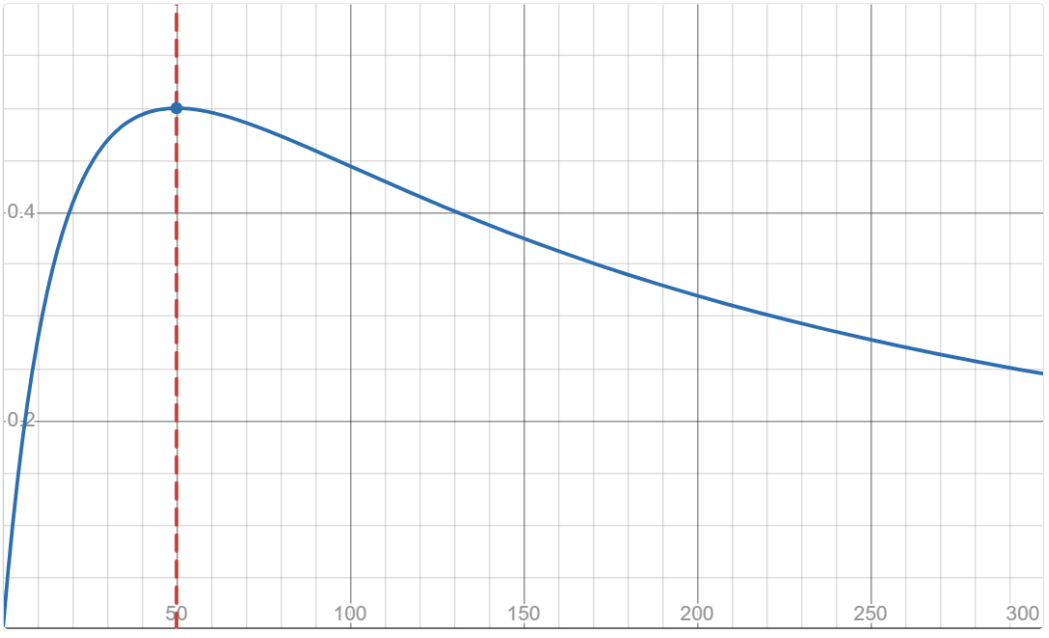
From Example 4.8, $P'(R_L) = \dfrac{V^2(R_s - R_L)}{(R_s + R_L)^3}$. The only critical point in $(0, \infty)$ is $R_L = R_s$. The derivative changes from $+$ to $-$ there, so it is a maximum; since $P \to 0$ as $R_L \to 0$ and as $R_L \to \infty$, it is the global maximum.

$$P_{\max} = \frac{V^2 R_s}{(2R_s)^2} = \frac{V^2}{4R_s}.$$

Numbers. $V = 10\,\text{V}$, $R_s = 50\,\Omega$: $P_{\max} = 0.5\,\text{W}$ at $R_L = 50\,\Omega$.

Efficiency. At the optimum, the source resistance dissipates as much as the load: efficiency is 50 %. Maximum power transfer is the right objective for weak signals (antennas, sensors, RF stages), never for power distribution, where $R_L \gg R_s$ is used to maximize efficiency.

AC version. For a source impedance $Z_s = R_s + jX_s$ and load $Z_L = R_L + jX_L$, $P = \tfrac{1}{2}|V|^2 R_L / \left[(R_s + R_L)^2 + (X_s + X_L)^2\right]$. Setting $\partial P/\partial X_L = 0$ gives $X_L = -X_s$; then $\partial P/\partial R_L = 0$ gives $R_L = R_s$. So $Z_L = Z_s^*$: **conjugate matching**.



Load power versus load resistance for $V = 10\,\text{V}$, $R_s = 50\,\Omega$. The peak is broad: the curve is flat near the optimum because $P' = 0$ there, which is the subject of Example 4.49.

Worked Example 4.43 — Optimal switching frequency

In a switching converter, losses that scale with ripple (magnetics, capacitor ESR) fall as $1/f$ while switching losses rise as f. A model is $P(f) = \dfrac{a}{f} + bf$ with $a = 2 \times 10^5$ W·Hz and $b = 2 \times 10^{-6}$ W/Hz. Find the frequency of minimum loss.

$P'(f) = -\dfrac{a}{f^2} + b = 0 \Rightarrow f^* = \sqrt{a/b}$. $P''(f) = 2a/f^3 > 0$, so it is a minimum (and the only critical point for $f > 0$, with $P \to \infty$ at both ends, so it is global).

$$f^* = \sqrt{\frac{2 \times 10^5}{2 \times 10^{-6}}} = \sqrt{10^{11}} = 316 \text{ kHz}, \qquad P(f^*) = 2\sqrt{ab} = 2\sqrt{0.4} = 1.26 \text{ W}.$$

At the optimum the two loss components are equal ($a/f^* = bf^* = \sqrt{ab}$). This **balance principle** holds whenever a cost is the sum of a term growing like x and one falling like $1/x$: copper versus core loss in a transformer, conduction versus switching loss, wire size versus cost.

Worked Example 4.44 — Resonant peak of a second-order low-pass filter

A second-order low-pass filter has

$$|H(u)| = \frac{1}{\sqrt{(1-u^2)^2 + (2\zeta u)^2}}, \qquad u = \omega/\omega_0,$$

with damping ratio $\zeta > 0$. Find the frequency and height of the resonant peak.

Simplify first. $|H|$ is largest where $D(w) = (1-w)^2 + 4\zeta^2 w$ is smallest, with $w = u^2 \geq 0$. (Maximizing a positive function is equivalent to minimizing its reciprocal; squaring is monotonic for positive quantities, so working with D avoids square roots.)

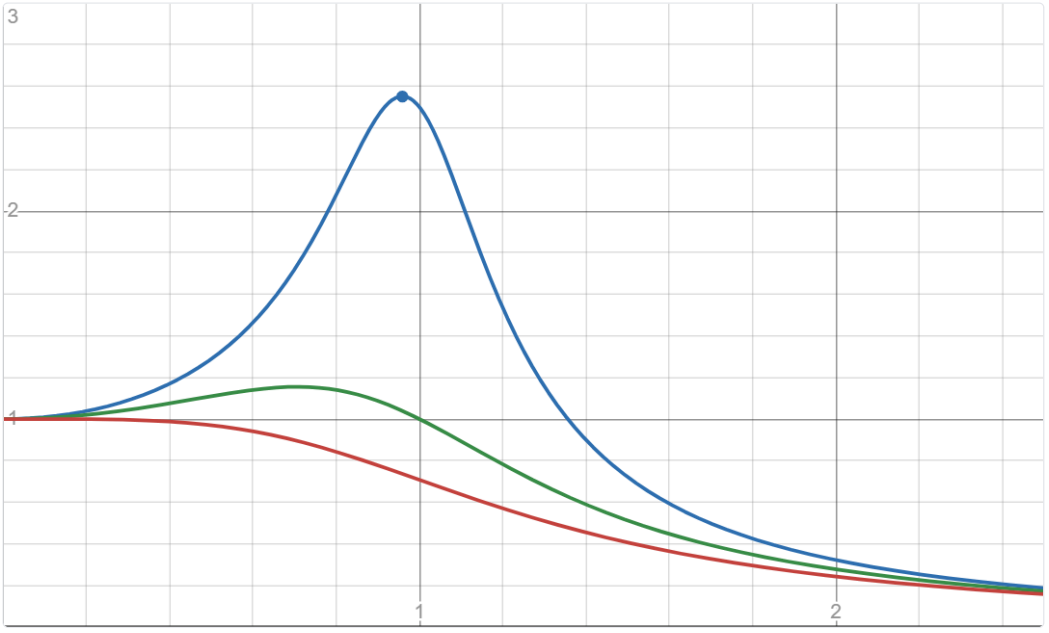
Differentiate. $D'(w) = -2(1-w) + 4\zeta^2 = 0 \Rightarrow w^* = 1 - 2\zeta^2$. $D'' = 2 > 0$: minimum.

Feasibility. $w^* \geq 0$ requires $\zeta \leq 1/\sqrt{2}$. For $\zeta > 1/\sqrt{2}$, D is increasing on $w \geq 0$, the minimum is at the endpoint $w = 0$, and there is no peak: the response falls monotonically. $\zeta = 1/\sqrt{2}$ is the **Butterworth** (maximally flat) boundary.

Peak. For $\zeta < 1/\sqrt{2}$:

$$u_{\text{peak}} = \sqrt{1 - 2\zeta^2}, \qquad D(w^*) = 4\zeta^2 - 4\zeta^4 = 4\zeta^2(1-\zeta^2), \qquad |H|_{\max} = \frac{1}{2\zeta\sqrt{1-\zeta^2}}.$$

Numbers. $\zeta = 0.2$: $u_{\text{peak}} = 0.959$ and $|H|_{\max} = 2.55$ (+8.1 dB). $\zeta = 0.5$: $u_{\text{peak}} = 0.707$, $|H|_{\max} = 1.155$ (+1.25 dB).



Second-order low-pass magnitude for $\zeta = 0.2$ (blue), 0.5 (green) and $1/\sqrt{2}$ (red, Butterworth). The peak moves below ω_0 and disappears at $\zeta = 1/\sqrt{2}$, as found in Example 4.44.

Worked Example 4.45 — Why coaxial cable is $75\,\Omega$

For a coaxial line with inner radius a and fixed outer radius b (set by the cable size), the conductor loss per unit length is proportional to

$$\alpha(a) \propto \frac{1/a + 1/b}{\ln(b/a)}.$$

Find the ratio $x = b/a$ that minimizes the loss.

One variable. With $1/a = x/b$, $\alpha \propto \dfrac{1}{b} \cdot \dfrac{x+1}{\ln x}$; minimize $g(x) = \dfrac{x+1}{\ln x}$ for $x > 1$.

Differentiate. Quotient rule:

$$g'(x) = \frac{\ln x - (x+1)/x}{(\ln x)^2} = 0 \quad \Longrightarrow \quad \ln x = 1 + \frac{1}{x}.$$

This transcendental equation has no closed-form solution. Newton's method (§4.8) on $h(x) = \ln x - 1 - 1/x$, $h'(x) = 1/x + 1/x^2$, from $x_0 = 3$: $x_1 = 3.5281$, $x_2 = 3.5904$, $x_3 = 3.59112$. So $x^* = 3.5911$.

Check it is a minimum. $g \to \infty$ as $x \to 1^+$ (denominator $\to 0$) and as $x \to \infty$ (since $x/\ln x \to \infty$), and x^* is the only critical point, so it is the global minimum.

Consequence. The characteristic impedance of an air-filled coax is $Z_0 = 60 \ln(b/a)\,\Omega = 60 \ln 3.5911 = 76.7\,\Omega$. That is why low-loss video and RF distribution cable is $75\,\Omega$. (The $50\,\Omega$ standard is a compromise between this and the $30\,\Omega$ ratio that maximizes power handling.)

Common Pitfall — forgetting endpoints and the domain

A critical point is only a candidate. The global optimum may be at an endpoint (as for $x^3 - 3x$ on $[-2, 3]$ above), may not exist on an open interval (minimize $1/x$ on $(0, \infty)$), or the critical point may be infeasible (the resonant peak of Example 4.44 for $\zeta > 1/\sqrt{2}$). Points where f' fails to exist are critical points too: the minimum of $|x - 2|$ is at the corner $x = 2$. Finally, $f''(c) = 0$ does not mean "no extremum": x^4 has a minimum at 0. Use the first-derivative test when the second-derivative test is inconclusive.

Engineering Note — optimizing a simpler function

Because $\sqrt{\cdot}$, $\ln$, squaring of positive quantities and $x \mapsto 1/x$ (on positive values, reversing the order) are monotonic, the optimizer of f is also the optimizer of f^2, $\ln f$ or $1/f$. Maximize $|H|^2$ instead of $|H|$, $\ln L$ instead of a likelihood L (Ch. 20), and gain in dB instead of linear gain. The location of the optimum is unchanged, and the algebra is usually much lighter.

4.11 Sensitivity analysis

Motivation

Real components are not their nominal values. Resistors come in 1 % or 5 % tolerances, capacitors in 10 % or 20 %; values drift with temperature and age; an op-amp's open-loop gain varies by a factor of several between units. A designer must answer: *how much does the performance change when a component changes?* The answer is a derivative, usually normalized so that percentage changes map to percentage changes. Sensitivity analysis tells you which components need tight tolerances, which can be cheap, and whether a design is robust.

Absolute and relative sensitivity

Sensitivity

Let a performance measure y depend on a parameter x (other parameters fixed). The **absolute sensitivity** is $\dfrac{\partial y}{\partial x}$ (units of y per unit of x). The **relative (normalized, Bode) sensitivity** is

$$S_x^y = \frac{x}{y}\frac{\partial y}{\partial x} = \frac{\partial y/y}{\partial x/x} = \frac{\partial \ln y}{\partial \ln x}.$$

It is dimensionless: for small changes, a 1 % change in x produces approximately an S_x^y % change in y.

The first-order change (from the linearization of §4.8, in relative form) is

$$\frac{\Delta y}{y} \approx S_x^y \frac{\Delta x}{x}.$$

Because S_x^y is a logarithmic derivative (§4.3), it obeys simple algebraic rules that make most sensitivities computable at sight.

Properties of relative sensitivity

For functions $u(x)$, $v(x)$ and constants c, n:

$$S_x^{cu} = S_x^u, \qquad S_x^{uv} = S_x^u + S_x^v, \qquad S_x^{u/v} = S_x^u - S_x^v,$$
$$S_x^{u^n} = n S_x^u, \qquad S_x^{x^n} = n, \qquad S_x^{1/u} = -S_x^u,$$
$$S_x^{y} = S_u^y \, S_x^u \ \text{(chain rule)}, \qquad S_x^{u+v} = \frac{u\,S_x^u + v\,S_x^v}{u+v}.$$

Proofs. Each follows from $S_x^y = d\ln y / d\ln x$: $\ln(uv) = \ln u + \ln v$ gives the product rule; $\ln u^n = n \ln u$ gives the power rule; and the chain rule is $\dfrac{d\ln y}{d\ln x} = \dfrac{d\ln y}{d\ln u}\dfrac{d\ln u}{d\ln x}$. For the sum, $S_x^{u+v} = \dfrac{x(u'+v')}{u+v} = \dfrac{u\cdot x u'/u + v \cdot x v'/v}{u+v}$.

So for any **monomial formula** $y = c\,x_1^{n_1} x_2^{n_2}\cdots$, the relative sensitivity to x_k is just the exponent n_k. For example, for $f_c = \dfrac{1}{2\pi RC} = \dfrac{1}{2\pi} R^{-1} C^{-1}$: $S_R^{f_c} = S_C^{f_c} = -1$.

Several parameters: worst case and statistical tolerance

If y depends on $x_1, \dots, x_n$, the total differential of §4.5 gives

$$\frac{\Delta y}{y} \approx \sum_{k=1}^{n} S_{x_k}^{y} \, \frac{\Delta x_k}{x_k}.$$

Two ways of combining tolerances follow.

- **Worst case.** If each relative error is bounded, $|\Delta x_k / x_k| \le t_k$, then

$$\left|\frac{\Delta y}{y}\right| \lesssim \sum_k \left|S_{x_k}^{y}\right| t_k.$$

This is guaranteed (to first order) but pessimistic: it assumes every component sits at the worst extreme of its tolerance simultaneously.

- **Statistical (root-sum-square, RSS).** If the parameter errors are independent random variables with relative standard deviations σ_k, then the relative standard deviation of y is approximately

$$\sigma_y \approx \sqrt{\sum_k \left(S_{x_k}^{y}\right)^2 \sigma_k^2}.$$

This is the first-order **propagation of uncertainty** (Ch. 20). It describes the spread of a production batch, and is smaller than the worst case.

Both rest on linearization, so they are reliable only when the relative changes are small enough that second-order terms are negligible.

Worked Example 4.46 — Voltage divider

$V_o = V_i \dfrac{R_2}{R_1 + R_2}$ with $R_1 = R_2 = 10\,\text{k}\Omega$, both $\pm 1\,\%$. Find the sensitivities and the tolerance of V_o.

Sensitivities. Using the quotient and sum rules, or differentiating directly:

$$S_{R_2}^{V_o} = 1 - \frac{R_2}{R_1+R_2} = \frac{R_1}{R_1+R_2}, \qquad S_{R_1}^{V_o} = -\frac{R_1}{R_1+R_2}, \qquad S_{V_i}^{V_o} = 1.$$

For equal resistors, $S_{R_2} = +0.5$ and $S_{R_1} = -0.5$.

Tolerance (taking V_i as exact). Worst case: $0.5(1\,\%) + 0.5(1\,\%) = 1\,\%$. RSS: $\sqrt{0.5^2 + 0.5^2}\,(1\,\%) = 0.71\,\%$.

Exact worst case. $R_1 = 9.9\,\text{k}\Omega$, $R_2 = 10.1\,\text{k}\Omega$ gives $V_o/V_i = 0.505$, i.e. $+1.00\,\%$. ✓

Insight. $S_{R_1} + S_{R_2} = 0$: the ratio is unchanged if both resistors drift by the same fraction (for example, with a common temperature coefficient). This is why precision dividers use matched resistor networks: tracking matters more than absolute accuracy.

Worked Example 4.47 — Cutoff frequency of an RC filter

$f_c = 1/(2\pi RC)$ with $R = 10\,\text{k}\Omega \pm 5\,\%$ and $C = 10\,\text{nF} \pm 10\,\%$ (nominal $f_c = 1.592\,\text{kHz}$). Estimate the spread of f_c.

$S_R^{f_c} = S_C^{f_c} = -1$. Worst case: $5\,\% + 10\,\% = 15\,\%$. RSS: $\sqrt{5^2 + 10^2}\,\% = 11.2\,\%$.

Exact extremes. R and C both high: $f_c/f_{c0} = 1/(1.05 \times 1.10) = 0.866$ ($-13.4\,\%$). Both low: $1/(0.95 \times 0.90) = 1.170$ ($+17.0\,\%$). The linear estimate $\pm 15\,\%$ is the average of these; the asymmetry is the second-order effect of the reciprocal. For tolerances this large, check the extremes exactly.

Design consequence. The capacitor dominates the error. Buying a $2\,\%$ capacitor (RSS $5.4\,\%$) helps far more than buying a $1\,\%$ resistor (RSS $10.05\,\%$).

Worked Example 4.48 — Series RLC: resonant frequency and Q

A series RLC circuit has $\omega_0 = \dfrac{1}{\sqrt{LC}}$ and $Q = \dfrac{1}{R}\sqrt{\dfrac{L}{C}}$. With $L = 10\,\text{mH} \pm 5\,\%$, $C = 100\,\text{nF} \pm 10\,\%$, $R = 10\,\Omega \pm 1\,\%$, find the sensitivities and tolerances.

Nominal values. $\omega_0 = 1/\sqrt{10^{-9}} = 31\,623\,\text{rad/s}$ ($f_0 = 5.03\,\text{kHz}$); $Q = 0.1\sqrt{10^5} = 31.6$.

Sensitivities (read off the exponents). $\omega_0 = L^{-1/2}C^{-1/2}$: $S_L^{\omega_0} = S_C^{\omega_0} = -\tfrac{1}{2}$, $S_R^{\omega_0} = 0$. $Q = R^{-1}L^{1/2}C^{-1/2}$: $S_R^{Q} = -1$, $S_L^{Q} = +\tfrac{1}{2}$, $S_C^{Q} = -\tfrac{1}{2}$.

Tolerances.

Quantity	Worst case	RSS
ω_0	$\tfrac{1}{2}(5) + \tfrac{1}{2}(10) = 7.5\,\%$	$\sqrt{2.5^2 + 5^2} = 5.6\,\%$
Q	$1 + 2.5 + 5 = 8.5\,\%$	$\sqrt{1^2 + 2.5^2 + 5^2} = 5.7\,\%$

The square root in ω_0 halves the effect of L and C errors: a nice property, but a tuned circuit with $Q = 31.6$ has a -3 dB bandwidth of only $\omega_0/Q \approx 3.2\,\%$ of ω_0. A $5.6\,\%$ spread in ω_0 is larger than the passband itself, so such a filter must be trimmed or built with tighter parts.

Worked Example 4.49 — The optimum is insensitive: second-order sensitivity

At an optimum, $\partial y/\partial x = 0$, so every first-order sensitivity vanishes. The change is then governed by the second derivative. For the maximum-power problem of Example 4.42, let $r = R_L/R_s$:

$$\frac{P}{P_{\max}} = \frac{4r}{(1+r)^2}.$$

At $r = 1$ the first derivative is 0. The second derivative of $4r(1+r)^{-2}$ is $\dfrac{8(r-2)}{(1+r)^4}$, which equals $-\tfrac{1}{2}$ at $r = 1$, so by Taylor's theorem

$$\frac{P}{P_{\max}} \approx 1 - \frac{1}{4}(r-1)^2.$$

Numbers. A $20\,\%$ mismatch ($r = 1.2$) is predicted to cost $0.25 \times 0.04 = 1\,\%$ of the power; exactly, $4(1.2)/2.2^2 = 0.9917$, a loss of $0.83\,\%$. ✓

This is a general and useful fact: **near an optimum, errors in the design variable cost only quadratically**. Precise matching is rarely worth paying for when the goal is maximum power; it matters much more when the goal is minimum *reflection*, which depends on mismatch linearly.

Worked Example 4.50 — Negative feedback desensitizes gain

A non-inverting amplifier has closed-loop gain $G = \dfrac{A}{1 + A\beta}$, where A is the op-amp's open-loop gain and β the feedback fraction set by resistors. With $A = 10^5$ and $\beta = 0.01$, find S_A^G and S_β^G.

Sensitivity to A. $G = A \cdot (1 + A\beta)^{-1}$, so by the product, power and sum rules

$$S_A^G = 1 - S_A^{1+A\beta} = 1 - \frac{A\beta}{1 + A\beta} = \frac{1}{1 + A\beta}.$$

Sensitivity to β. $S_\beta^G = -S_\beta^{1+A\beta} = -\dfrac{A\beta}{1 + A\beta}$.

Numbers. Loop gain $A\beta = 1000$: $G = 99.90$, $S_A^G = 1/1001 \approx 0.001$, $S_\beta^G \approx -0.999$.

Interpretation. A $10\,\%$ change in the op-amp gain changes the closed-loop gain by only about $0.01\,\%$: feedback divides the sensitivity by $1 + A\beta$. Even a $50\,\%$ drop ($A = 5 \times 10^4$, a large change for which the linear estimate predicts $0.05\,\%$) gives $G = 99.80$, a change of only $0.10\,\%$. The price is that the gain now depends almost entirely on β, i.e. on two resistors, which are cheap to make accurate. This is the central reason for negative feedback in amplifier design.

Worked Example 4.51 — Sensitivity of a strain-gauge bridge

A Wheatstone bridge excited by $V = 5\,\text{V}$ has one active strain gauge $R = R_0(1 + x)$ and three fixed resistors R_0. Its output is

$$v_o(x) = V\left(\frac{R}{R + R_0} - \frac{1}{2}\right) = V\left(\frac{1+x}{2+x} - \frac{1}{2}\right) = \frac{V}{2}\cdot\frac{x}{2+x}.$$

Find the absolute sensitivity dv_o/dx at balance and the output for a strain of 10^{-3} with gauge factor 2 (so $x = 2 \times 10^{-3}$).

$\dfrac{dv_o}{dx} = \dfrac{V}{2}\cdot\dfrac{(2+x) - x}{(2+x)^2} = \dfrac{V}{(2+x)^2}$, which is $V/4 = 1.25\,\text{V}$ per unit x at $x = 0$. Hence $v_o \approx 1.25 \times 2 \times 10^{-3} = 2.50\,\text{mV}$.

Exact. $v_o = 2.5 \times \dfrac{0.002}{2.002} = 2.4975\,\text{mV}$. The nonlinearity error is $-0.1\,\%$, about $x/2$ in relative terms, which is why bridges are operated near balance (and why half- and full-bridge arrangements, which cancel this term, are preferred).

Here the *relative* sensitivity $S_x^{v_o}$ is not useful, because the nominal output is zero; absolute sensitivity (V per unit strain) is the meaningful figure of merit. Choose the measure that matches the specification.

Engineering Note — temperature coefficients are sensitivities

A resistor's temperature coefficient of resistance (TCR) is $\alpha = \dfrac{1}{R}\dfrac{dR}{dT}$, a relative rate per kelvin, often quoted in ppm/K. Copper has $\alpha \approx 3.9 \times 10^{-3}\,\mathrm{K}^{-1}$; precision metal-film resistors reach ± 5 ppm/K. By the chain rule of sensitivities, the temperature coefficient of any performance measure is $\dfrac{1}{y}\dfrac{dy}{dT} = \sum_k S^y_{x_k}\,\alpha_k$. For the RC filter of Example 4.47, f_c drifts at $-(\alpha_R + \alpha_C)$; choosing a capacitor whose TC is equal and opposite to the resistor's ($\alpha_C = -\alpha_R$) makes the cutoff frequency first-order temperature-independent. Larger tolerance studies with many parameters and nonlinear effects use Monte Carlo simulation (Ch. 24), but the sensitivities computed here tell you which parameters matter before you start.

Common Pitfall — linear sensitivity has a range of validity

First-order sensitivity assumes small relative changes. With $\pm 10\,\%$ or $\pm 20\,\%$ tolerances, or near an optimum where the first derivative vanishes, include second-order terms or evaluate the extremes exactly (Examples 4.47 and 4.49). The RSS formula also assumes **independent** errors; components from the same reel, or resistors on one thin-film network, are correlated, and their errors may add (worst case) or cancel (Example 4.46) far more than RSS predicts. Finally, do not confuse absolute and relative sensitivity: $\partial y/\partial x$ has units and depends on the units chosen, S^y_x does not.

Summary

Topic	Key result
Limit	$\lim_{x\to a} f(x) = L$: $\forall \varepsilon > 0\ \exists \delta > 0, 0 < \lvert x-a\rvert < \delta \Rightarrow \lvert f(x) - L\rvert < \varepsilon$
Fundamental limits	$\lim_{x\to 0}\frac{\sin x}{x} = 1$, $\lim_{x\to 0}\frac{1-\cos x}{x^2} = \frac{1}{2}$, $\lim_{h\to 0}\frac{e^h-1}{h} = 1$, $\lim_{n\to\infty}(1+\frac{x}{n})^n = e^x$
Continuity	$\lim_{x\to a} f(x) = f(a)$; IVT (roots by sign change), EVT (extrema on $[a,b]$)
Derivative	$f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$; differentiable $\Rightarrow$ continuous
Product, quotient	$(fg)' = f'g + fg'$, $(f/g)' = (f'g - fg')/g^2$
Complex exponential	$\frac{d}{dt}e^{st} = se^{st}$; $\frac{d}{dt} \leftrightarrow j\omega$ for phasors
Mean value theorem	$f'(c) = \frac{f(b)-f(a)}{b-a}$ for some $c \in (a,b)$
L'Hôpital	$0/0$ or ∞/∞: $\lim f/g = \lim f'/g'$
Chain rule	$\frac{d}{dx}f(g(x)) = f'(g(x))g'(x)$, $\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$
Logarithmic derivative	$\frac{y'}{y} = (\ln y)'$; for $y = \prod f_k^{a_k}$, $\frac{y'}{y} = \sum a_k \frac{f_k'}{f_k}$
Inverse function	$\frac{dx}{dy} = 1/\frac{dy}{dx}$
Parametric	$\frac{dy}{dx} = \frac{\dot y}{\dot x}$
Implicit	$F(x,y) = 0 \Rightarrow y' = -F_x/F_y$ $(F_y \neq 0)$
Partial derivative	$f_x = \lim_{h\to 0}\frac{f(x+h,y)-f(x,y)}{h}$; $f_{xy} = f_{yx}$ if continuous
Multivariable chain rule	$\frac{dz}{dt} = f_x\dot x + f_y\dot y$; $\Delta f \approx f_x\Delta x + f_y\Delta y$
Leibniz rule	$(fg)^{(n)} = \sum_{k=0}^{n}\binom{n}{k}f^{(k)}g^{(n-k)}$
Curvature	$\kappa = \lvert f''\rvert/(1+f'^2)^{3/2}$
Taylor series	$f(x) = \sum_{k=0}^{n}\frac{f^{(k)}(a)}{k!}(x-a)^k + \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$
Key series	$e^x = \sum\frac{x^n}{n!}$, $\sin x = x - \frac{x^3}{6} + \cdots$, $\cos x = 1 - \frac{x^2}{2} + \cdots$, $(1+x)^\alpha = 1 + \alpha x + \cdots$
Finite differences	central: $\frac{f(x+h)-f(x-h)}{2h} = f' + O(h^2)$; second: $\frac{f(x+h)-2f(x)+f(x-h)}{h^2} = f'' + O(h^2)$
Linearization	$f(x) \approx f(a) + f'(a)(x-a)$; diode $g_d = I_D/(nV_T)$; MOSFET $g_m = 2I_D/V_{ov}$
Equilibrium stability	$\dot x = f(x)$, $f(x^*) = 0$: stable if $f'(x^*) < 0$, $\tau = -1/f'(x^*)$
Newton's method	$x_{n+1} = x_n - f(x_n)/f'(x_n)$; $e_{n+1} \approx \frac{f''}{2f'}e_n^2$
Rates	$i = \dot q$, $i = \frac{d(Cv)}{dt}$, $v = L\frac{di}{dt}$, $\mathcal{E} = -\dot\Phi$, max $\lvert\dot v\rvert$ of a sine $= 2\pi f V_m$
Optimization	$f'(c) = 0$; $f''(c) < 0$ max, > 0 min; check endpoints; max power at $R_L = R_s$, $P_{\max} = V^2/(4R_s)$
Relative sensitivity	$S^y_x = \frac{x}{y}\frac{\partial y}{\partial x} = \frac{\partial \ln y}{\partial \ln x}$; for $y = cx^n$, $S = n$; $S^{uv} = S^u + S^v$
Tolerance	worst case $\sum\lvert S_k\rvert t_k$; RSS $\sqrt{\sum S_k^2\sigma_k^2}$; feedback $S^G_A = 1/(1+A\beta)$

Further reading

- J. Stewart, *Calculus: Early Transcendentals*, Cengage — clear, thorough treatment of limits, derivatives, Taylor series and applications, with many geometric explanations.
- E. Kreyszig, *Advanced Engineering Mathematics*, Wiley — the standard engineering reference; useful for the bridge from this chapter to multivariable calculus and ODEs.
- M. Spivak, *Calculus*, Cambridge University Press — rigorous proofs of the theorems sketched here (limits, MVT, Taylor's theorem), for the reader who wants every detail.
- A. S. Sedra and K. C. Smith, *Microelectronic Circuits*, Oxford University Press — small-signal modelling and distortion of diodes and transistors, the main engineering consumer of linearization and Taylor expansion.

Chapter 5 — Integral Calculus

> **Prerequisites:** Ch. 1, Ch. 3, Ch. 4 **Learning outcomes:** After this chapter you can
>
> - explain the definite integral as a limit of Riemann sums and use the Fundamental Theorem of Calculus to evaluate it through antiderivatives;
> - choose and apply the standard techniques — algebraic manipulation, trigonometric identities, partial fractions, trigonometric substitution, substitution and integration by parts — to reduce an integral to a table form;
> - decide whether an improper integral converges, using direct evaluation, comparison and limit-comparison tests;
> - set up and evaluate double and triple integrals over simple regions as iterated integrals;
> - approximate integrals with the trapezoidal and Simpson's rules and estimate their error;
> - compute charge, energy, work and other accumulated quantities, and the mean and RMS values of engineering waveforms.

Differentiation answers the question *how fast is it changing right now?* Integration answers the reverse question: *given the rate, how much has accumulated?* An electrical engineer meets this question constantly. The charge on a capacitor is the accumulated current, $q(t) = q(t_0) + \int_{t_0}^{t} i(\tau)\,d\tau$. The energy delivered to a load is the accumulated power, $W = \int p(t)\,dt$. The voltage a heater "feels" is the root-mean-square value of a waveform, which is the square root of an average, and an average is an integral divided by a length. The flux through a coil, the total charge in a cloud of electrons, the expected value of a noise voltage — all are integrals.

This chapter develops single-variable integration thoroughly and introduces integrals over regions of the plane and of space. Chapter 6 takes multiple integrals further with changes of variables (polar, cylindrical, spherical, general Jacobians), and Chapter 22 treats numerical integration in depth (Romberg, Gaussian quadrature, adaptive schemes). Here we lay the foundations those chapters build on.

5.1 Indefinite and definite integrals

5.1.1 Motivation: area and accumulation

Suppose a current $i(t)$ flows into a capacitor. Over a short interval Δt during which the current is nearly constant, the charge delivered is approximately $i(t)\,\Delta t$. Over a long interval $[a, b]$ we chop time into many short pieces and add:

$$Q \approx \sum_{k=1}^{n} i(t_k^*)\,\Delta t_k.$$

As the pieces shrink, the approximation becomes exact. Geometrically, each term $i(t_k^*)\,\Delta t_k$ is the area of a thin rectangle, so the limit is the *area under the curve* $i(t)$ between $t = a$ and $t = b$ (counted negative where $i < 0$). This limit is the definite integral. It is a *number* attached to a function and an interval.

Separately, differentiation (Chapter 4) suggests the question: which functions F have a given derivative f? The answer is a *family of functions*, the indefinite integral. The **Fundamental Theorem of Calculus** is the remarkable statement that these two ideas — area and anti-differentiation — are the same thing.

5.1.2 Riemann sums and the definite integral

Partition and Riemann sum

A **partition** P of $[a, b]$ is a finite set of points $a = x_0 < x_1 < \cdots < x_n = b$. Write $\Delta x_k = x_k - x_{k-1}$ and $\|P\| = \max_k \Delta x_k$ (the **mesh**). Choosing a **sample point** $x_k^* \in [x_{k-1}, x_k]$ in each subinterval, the **Riemann sum** of f is

$$S(f, P, \{x_k^*\}) = \sum_{k=1}^{n} f(x_k^*)\,\Delta x_k.$$

Definite integral

f is **(Riemann) integrable** on $[a, b]$ if there is a number I such that for every $\varepsilon > 0$ there is a $\delta > 0$ with $|S(f, P, \{x_k^*\}) - I| < \varepsilon$ for *every* partition with $\|P\| < \delta$ and *every* choice of sample points. Then

$$\int_a^b f(x)\,dx = I = \lim_{\|P\| \to 0} \sum_{k=1}^{n} f(x_k^*)\,\Delta x_k.$$

f is the **integrand**, a and b the **limits**, and x a **dummy variable**: $\int_a^b f(x)\,dx = \int_a^b f(t)\,dt$.

The notation is a fossil of the sum: $\int$ is an elongated S (*summa*), and dx recalls Δx. Dimensionally, $[\int f\,dx] = [f]\,[x]$; integrating a current in amperes over time in seconds gives coulombs.

Which functions are integrable? Not every function — the function equal to 1 on rationals and 0 on irrationals is the standard counterexample — but every function an engineer is likely to write down is.

Sufficient conditions for integrability

If f is continuous on $[a, b]$, or bounded on $[a, b]$ with only finitely many discontinuities, or monotonic on $[a, b]$, then f is Riemann integrable on $[a, b]$.

Thus square waves, sawtooth waves, rectified sinusoids and pulse trains — all piecewise continuous — can be integrated over any finite interval.

Two common special choices of Riemann sum use uniform width $h = (b-a)/n$ and sample at the left or right ends:

$$L_n = h\sum_{k=0}^{n-1} f(a+kh), \qquad R_n = h\sum_{k=1}^{n} f(a+kh).$$

For an increasing function $L_n \le \int_a^b f \le R_n$, and $R_n - L_n = h\,[f(b)-f(a)] \to 0$, which proves directly that monotone functions are integrable.

Worked Example 5.1 — An integral from the definition

Compute $\displaystyle\int_0^2 x^2\,dx$ as a limit of right-endpoint sums.

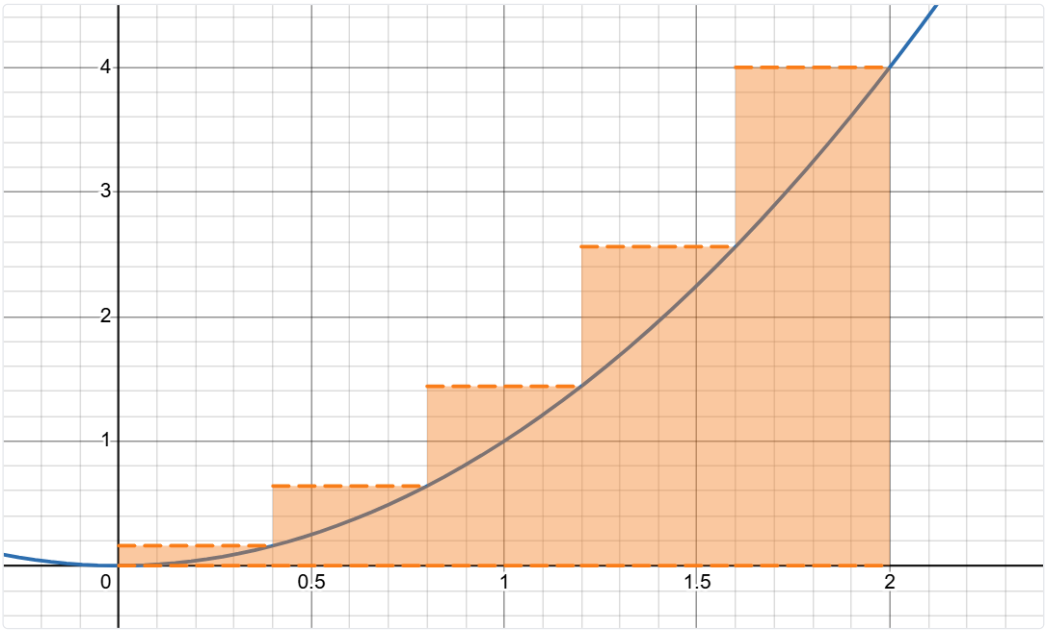
Step 1 — partition. $h = 2/n$, $x_k = 2k/n$.

Step 2 — sum. Using $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$,

$$R_n = \sum_{k=1}^{n} \left(\frac{2k}{n}\right)^2 \frac{2}{n} = \frac{8}{n^3}\cdot\frac{n(n+1)(2n+1)}{6} = \frac{4}{3}\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right).$$

Step 3 — limit. As $n\to\infty$, $R_n \to \frac{4}{3}\cdot 2 = \frac{8}{3} \approx 2.6667$.

Check. $n=10$: $R_{10} = \frac{4}{3}(1.1)(2.1) = 3.08$; $n=100$: $R_{100} = \frac{4}{3}(1.01)(2.01) = 2.7068$; $n=1000$: 2.6707. The error shrinks like $1/n$, as expected for an endpoint rule (§5.7).



Right-endpoint Riemann sum with $n=5$ for $\int_0^2 x^2\,dx$: the rectangles overshoot the exact area $8/3$ because x^2 is increasing.

5.1.3 Properties of the definite integral

All of the following follow from the corresponding properties of finite sums, by passing to the limit.

Algebraic properties

For integrable f, g and constants α, β:

1. **Linearity:** $\displaystyle\int_a^b (\alpha f + \beta g)\,dx = \alpha\int_a^b f\,dx + \beta\int_a^b g\,dx$.
2. **Additivity over intervals:** $\displaystyle\int_a^c f\,dx = \int_a^b f\,dx + \int_b^c f\,dx$ for any a, b, c.
3. **Orientation:** $\displaystyle\int_b^a f\,dx = -\int_a^b f\,dx$, and $\displaystyle\int_a^a f\,dx = 0$.
4. **Monotonicity:** if $f \le g$ on $[a,b]$ $(a<b)$ then $\displaystyle\int_a^b f\,dx \le \int_a^b g\,dx$.
5. **Triangle inequality:** $\displaystyle\left|\int_a^b f\,dx\right| \le \int_a^b |f|\,dx$.
6. **Bounds:** if $m \le f(x) \le M$ on $[a,b]$ then $m(b-a) \le \displaystyle\int_a^b f\,dx \le M(b-a)$.

Property 5 is the integral form of the fact that a phasor sum can never exceed the sum of magnitudes; it is used constantly to bound errors. Property 6 leads to the following.

Mean Value Theorem for integrals

If f is continuous on $[a,b]$, there is a $c \in (a,b)$ with

$$\int_a^b f(x)\,dx = f(c)\,(b-a).$$

The number $\bar{f} = \dfrac{1}{b-a}\displaystyle\int_a^b f(x)\,dx$ is the **mean value** of f on $[a,b]$.

Proof sketch. By property 6 with $m = \min f$, $M = \max f$, the mean $\bar{f}$ lies between m and M. A continuous function takes every value between its minimum and maximum (Intermediate Value Theorem, Ch. 4), so $\bar{f} = f(c)$ for some c. $\square$

The mean value is the "DC value" of a waveform and is developed fully in §5.9.

Symmetry. If f is **even** ($f(-x) = f(x)$) then $\int_{-a}^{a} f\,dx = 2\int_0^a f\,dx$; if f is **odd** ($f(-x) = -f(x)$) then $\int_{-a}^{a} f\,dx = 0$. If f is **periodic** with period T, then $\int_c^{c+T} f\,dx$ is the same for every c — you may integrate over any convenient full period. These three facts eliminate a large share of the work in Fourier analysis (Ch. 15).

5.1.4 Antiderivatives and the indefinite integral

Antiderivative and indefinite integral

F is an **antiderivative** of f on an interval I if $F'(x) = f(x)$ for all $x \in I$. The set of all antiderivatives is the **indefinite integral**

$$\int f(x)\,dx = F(x) + C,$$

where C is an arbitrary **constant of integration**.

Why only a constant? If $F' = G' = f$ on an interval, then $(F-G)' = 0$, and by the Mean Value Theorem of differential calculus a function with zero derivative on an interval is constant. On a domain made of several disjoint intervals, the "constant" may differ on each piece: $\int \frac{dx}{x} = \ln|x| + C$ really means $\ln x + C_1$ for $x > 0$ and $\ln(-x) + C_2$ for $x < 0$.

Every differentiation rule read backwards gives an integration rule. The table below is the working vocabulary of this chapter; every entry can be checked by differentiating the right-hand side.

$f(x)$	$\int f(x)\,dx$	Condition
x^n	$\dfrac{x^{n+1}}{n+1} + C$	$n \neq -1$
$\dfrac{1}{x}$	$\ln\lvert x\rvert + C$	$x \neq 0$
e^{ax}	$\dfrac{1}{a}e^{ax} + C$	$a \neq 0$ (real or complex)
a^x	$\dfrac{a^x}{\ln a} + C$	$a > 0,\ a \neq 1$
$\sin ax$	$-\dfrac{1}{a}\cos ax + C$	
$\cos ax$	$\dfrac{1}{a}\sin ax + C$	
$\sec^2 x$	$\tan x + C$	
$\tan x$	$-\ln\lvert\cos x\rvert + C$	
$\sec x$	$\ln\lvert\sec x + \tan x\rvert + C$	
$\dfrac{1}{a^2 + x^2}$	$\dfrac{1}{a}\arctan\dfrac{x}{a} + C$	$a > 0$
$\dfrac{1}{\sqrt{a^2 - x^2}}$	$\arcsin\dfrac{x}{a} + C$	$\lvert x\rvert < a$
$\dfrac{1}{\sqrt{x^2 \pm a^2}}$	$\ln\left\lvert x + \sqrt{x^2 \pm a^2}\right\rvert + C$	
$\dfrac{1}{x^2 - a^2}$	$\dfrac{1}{2a}\ln\left\lvert\dfrac{x-a}{x+a}\right\rvert + C$	$\lvert x\rvert \neq a$
$\sinh ax,\ \cosh ax$	$\dfrac{1}{a}\cosh ax,\ \dfrac{1}{a}\sinh ax$	

The entry $\int e^{ax}dx = e^{ax}/a$ holds for complex a as well, which is why phasor methods integrate so easily: $\int e^{j\omega t}dt = \dfrac{e^{j\omega t}}{j\omega}$. Integrating a sinusoid divides its phasor by $j\omega$ — a -90° phase shift and a scaling by $1/\omega$. This is exactly the capacitor law $V = I/(j\omega C)$ (Ch. 2).

5.1.5 The Fundamental Theorem of Calculus

Fundamental Theorem of Calculus (FTC)

Let f be continuous on $[a, b]$.

Part 1. The accumulation function $A(x) = \displaystyle\int_a^x f(t)\,dt$ is differentiable on (a, b) and

$$A'(x) = f(x).$$

Part 2. If F is any antiderivative of f on $[a, b]$, then

$$\int_a^b f(x)\,dx = F(b) - F(a) \equiv \Big[F(x)\Big]_a^b.$$

Proof of Part 1. For $h \neq 0$,

$$\frac{A(x+h) - A(x)}{h} = \frac{1}{h}\int_x^{x+h} f(t)\,dt = f(c_h)$$

for some c_h between x and $x + h$, by the Mean Value Theorem for integrals. As $h \to 0$, $c_h \to x$, and continuity of f gives $f(c_h) \to f(x)$. $\square$

Proof of Part 2. A and F are both antiderivatives of f, so $F = A + C$. Then $F(b) - F(a) = A(b) - A(a) = \int_a^b f - 0$. $\square$

Part 1 says integration followed by differentiation returns the original function: integration *creates* antiderivatives, so every continuous function has one (even if, like e^{-x^2}, it cannot be written with elementary functions). Part 2 turns the computation of a limit of sums into a subtraction.

Combining Part 1 with the chain rule gives **Leibniz's rule** for variable limits:

$$\frac{d}{dx}\int_{u(x)}^{v(x)} f(t)\,dt = f\big(v(x)\big)\,v'(x) - f\big(u(x)\big)\,u'(x).$$

Worked Example 5.2 — Evaluating definite integrals with the FTC

(a) $\displaystyle\int_0^2 x^2\,dx = \Big[\frac{x^3}{3}\Big]_0^2 = \frac{8}{3}$, agreeing with Example 5.1 with no limits needed.

(b) $\displaystyle\int_0^{\pi} \sin x\,dx = \big[-\cos x\big]_0^{\pi} = -(-1) - (-1) = 2.$

(c) $\displaystyle\int_1^e \frac{3}{x}\,dx = 3\big[\ln x\big]_1^e = 3.$

(d) $\displaystyle\int_0^1 \frac{dx}{1+x^2} = \big[\arctan x\big]_0^1 = \frac{\pi}{4} \approx 0.785398.$

(e) $\displaystyle\frac{d}{dx}\int_0^{x^2} \cos t\,dt = \cos(x^2)\cdot 2x$ by Leibniz's rule. Check: the integral equals $\sin(x^2)$, whose derivative is $2x\cos(x^2)$. ✓

Worked Example 5.3 — Capacitor voltage from a current waveform

A $C = 10\ \mu\text{F}$ capacitor, initially at $v(0) = 2$ V, is driven by $i(t) = 5e^{-200t}$ mA for $t \geq 0$. Find $v(t)$ and the final voltage.

Step 1 — accumulation law. $v(t) = v(0) + \dfrac{1}{C}\displaystyle\int_0^t i(\tau)\,d\tau$.

Step 2 — integrate.

$$\int_0^t 5\times 10^{-3} e^{-200\tau}\,d\tau = 5\times 10^{-3}\Big[\frac{e^{-200\tau}}{-200}\Big]_0^t = 2.5\times 10^{-5}\big(1 - e^{-200t}\big)\ \text{C}.$$

Step 3 — divide by C. $v(t) = 2 + \dfrac{2.5\times 10^{-5}}{10^{-5}}(1 - e^{-200t}) = 2 + 2.5(1 - e^{-200t})$ V.

Step 4 — final value. As $t \to \infty$, $v \to 4.5$ V. Units: $[\text{A}\cdot\text{s/F}] = [\text{C/F}] = [\text{V}]$ ✓. At $t = 5$ ms, $v = 2 + 2.5(1 - e^{-1}) = 3.580$ V.

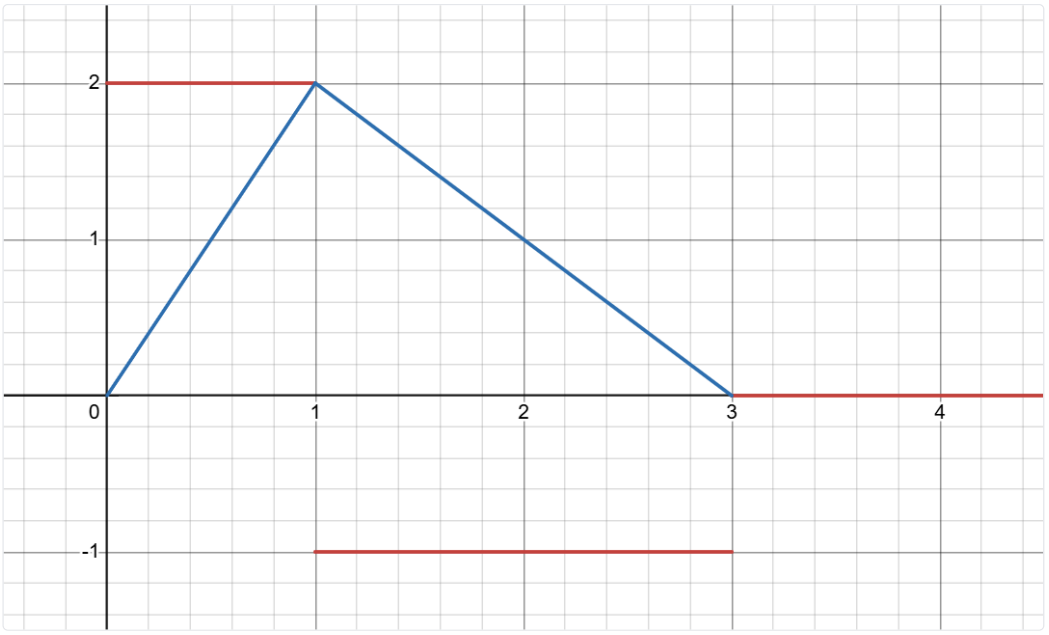
Worked Example 5.4 — Integrating a piecewise waveform

A current is $i(t) = 2$ A for $0 \leq t < 1$ ms, -1 A for $1 \leq t < 3$ ms, and 0 afterwards. Find the net charge $q(t)$ delivered from $t = 0$.

Split the integral at the breakpoints (additivity):

- $0 \leq t < 1$ ms: $q = 2t$, reaching 2 mC at $t = 1$ ms.
- $1 \leq t < 3$ ms: $q = 2\ \text{mC} - 1\cdot(t - 1\ \text{ms})$, reaching $2 - 2 = 0$ mC at $t = 3$ ms.
- $t \geq 3$ ms: $q = 0$.

The charge is a continuous piecewise-linear function even though the current jumps. **Integration smooths**: it turns jumps into kinks and kinks into smooth bends. This is why a capacitor voltage cannot jump while its current can.



Current $i(t)$ in A (red) and accumulated charge $q(t)$ in mC (blue), time in ms: the integral of a discontinuous function is continuous.

Common Pitfall — Applying the FTC across a discontinuity or singularity

$\int_{-1}^{1} \frac{dx}{x^2} = \left[-\frac{1}{x}\right]_{-1}^{1} = -2$ is **wrong**: a positive integrand cannot have a negative integral. The FTC requires $F' = f$ on the *whole* interval, and $1/x^2$ blows up at $x = 0$. The integral is improper and in fact diverges (§5.3). Always scan the interval for points where the integrand is undefined before subtracting antiderivative values.

Engineering Note — Integrators in hardware

An op-amp integrator produces $v_o(t) = -\frac{1}{RC}\int_0^t v_i(\tau)\,d\tau + v_o(0)$. The "constant of integration" is the initial capacitor voltage, which in a real circuit drifts because of input offset voltage: integrating a constant offset V_{os} gives a ramp $-V_{os}t/(RC)$ that eventually saturates the amplifier. Practical integrators add a large resistor across the capacitor to bleed this off — mathematically, replacing $\int$ by a leaky integrator with a finite time constant.

5.2 Integration techniques

5.2.1 Why technique matters

Differentiation is mechanical: the product, quotient and chain rules dispose of any elementary expression. Integration is not. There is no product rule for integrals, and some innocent-looking integrands — e^{-x^2}, $\frac{\sin x}{x}$, $\sqrt{1 - k^2 \sin^2 x}$ — have no elementary antiderivative at all. What we have instead is a toolkit of *transformations* that convert an integral into one already in the table. This section surveys the toolkit and the algebraic and trigonometric moves; the two most powerful tools, **integration by parts** and **integration by substitution**, each get a full section (§5.5 and §5.6), and the numerical fallback when everything fails is §5.7.

Even in an age of computer algebra, an engineer needs these techniques: to recognise the *form* of an answer (logarithm? arctangent? decaying exponential?), to sanity-check software output, to carry symbolic parameters such as R, L, C through a calculation, and to see the structure that a numerical answer hides. Partial fractions, in particular, are the engine of inverse Laplace and Z-transforms (Ch. 14, Ch. 17).

5.2.2 A strategy

Faced with $\int f(x)\,dx$, work down this list:

1. **Simplify algebraically.** Expand products, divide out improper fractions, split sums, cancel common factors, use $\sqrt{x^2} = |x|$ carefully.
2. **Match the table** (§5.1.4) after linearity, possibly with a linear change $x \to ax + b$.
3. **Spot a derivative.** If the integrand has the form $g(h(x))\,h'(x)$, substitute $u = h(x)$ (§5.6). Special case: $\int \frac{h'(x)}{h(x)}dx = \ln|h(x)| + C$.
4. **Trigonometric identities** for powers and products of sines and cosines (§5.2.4).
5. **Partial fractions** for rational functions (§5.2.5).
6. **Trigonometric or hyperbolic substitution** for $\sqrt{a^2 - x^2}$, $\sqrt{a^2 + x^2}$, $\sqrt{x^2 - a^2}$ (§5.6.4).
7. **Integration by parts** for products of unlike functions — polynomial × exponential, polynomial × trig, logarithms, inverse trig (§5.5).
8. **Complex exponentials** for products of exponentials and sinusoids (§5.2.6).
9. **Numerical integration** when no closed form exists or only a number is needed (§5.7).

5.2.3 Algebraic manipulation

Many integrals yield to algebra alone.

Worked Example 5.5 — Algebra first

(a) $\displaystyle\int \frac{x^3 + 2x - \sqrt{x}}{x^2}\,dx = \int \left(x + \frac{2}{x} - x^{-3/2}\right)dx = \frac{x^2}{2} + 2\ln|x| + 2x^{-1/2} + C.$

(b) $\displaystyle\int \frac{x^2}{x+1}\,dx$. The fraction is *improper* (degree of numerator $\geq$ degree of denominator), so divide first: $x^2 = (x+1)(x-1) + 1$, hence

$$\int \frac{x^2}{x+1}dx = \int \left(x - 1 + \frac{1}{x+1}\right)dx = \frac{x^2}{2} - x + \ln|x+1| + C.$$

(c) $\displaystyle\int \frac{dx}{x^2 + 4x + 13}$. **Complete the square:** $x^2 + 4x + 13 = (x+2)^2 + 9$, so

$$\int \frac{dx}{(x+2)^2 + 3^2} = \frac{1}{3}\arctan\frac{x+2}{3} + C.$$

(d) $\displaystyle\int_0^1 \frac{2x+1}{x^2+x+1}dx$. The numerator is exactly the derivative of the denominator, so the answer is $\big[\ln(x^2+x+1)\big]_0^1 = \ln 3 \approx 1.0986$.

Completing the square deserves emphasis because it is how quadratic denominators — which in circuit theory come from second-order (RLC) characteristic polynomials $s^2 + 2\zeta\omega_n s + \omega_n^2$ — are put into a recognisable form:

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = (s + \zeta\omega_n)^2 + \omega_n^2(1 - \zeta^2).$$

When $\zeta < 1$ the second term is positive and an arctangent (in the time domain: a damped sinusoid) appears; when $\zeta > 1$ it is negative, the quadratic factors into real roots, and logarithms (decaying exponentials) appear instead.

5.2.4 Trigonometric integrals

The integrals of products and powers of sinusoids are the backbone of power calculations and Fourier analysis. The identities that drive them are (Ch. 3):

$$\cos^2\theta = \frac{1+\cos 2\theta}{2}, \qquad \sin^2\theta = \frac{1-\cos 2\theta}{2}, \qquad \sin\theta\cos\theta = \frac{\sin 2\theta}{2},$$

$$\begin{aligned}
\sin A\cos B &= \tfrac{1}{2}\big[\sin(A+B) + \sin(A-B)\big],\\
\cos A\cos B &= \tfrac{1}{2}\big[\cos(A-B) + \cos(A+B)\big],\\
\sin A\sin B &= \tfrac{1}{2}\big[\cos(A-B) - \cos(A+B)\big].
\end{aligned}$$

Rules for $\int \sin^m x \cos^n x\,dx$:

- If m is **odd**, save one $\sin x$, convert the remaining even power of sin to cos using $\sin^2 = 1 - \cos^2$, and substitute $u = \cos x$.
- If n is **odd**, save one $\cos x$, convert the rest to sin, and substitute $u = \sin x$.
- If both are **even**, reduce powers with the half-angle identities, repeatedly if necessary.

Worked Example 5.6 — Powers of sinusoids

(a) **Odd power.** $\displaystyle\int \sin^3 x\,dx = \int (1 - \cos^2 x)\sin x\,dx$. With $u = \cos x$, $du = -\sin x\,dx$:

$$-\int (1 - u^2)\,du = -u + \frac{u^3}{3} + C = -\cos x + \frac{\cos^3 x}{3} + C.$$

(b) **Even powers.** $\displaystyle\int \sin^2 x\cos^2 x\,dx = \int \frac{\sin^2 2x}{4}dx = \int \frac{1-\cos 4x}{8}dx = \frac{x}{8} - \frac{\sin 4x}{32} + C.$

(c) **Over a period.** For $\omega = 2\pi/T$,

$$\int_0^T \sin^2(\omega t)\,dt = \int_0^T \frac{1-\cos 2\omega t}{2}dt = \frac{T}{2} - \Big[\frac{\sin 2\omega t}{4\omega}\Big]_0^T = \frac{T}{2}.$$

The cosine term integrates to zero over a whole number of periods. This single result gives the RMS value $V_m/\sqrt{2}$ of a sinusoid (§5.9).

Check (b) numerically: $\int_0^1 \sin^2 x\cos^2 x\,dx = \frac{1}{8} - \frac{\sin 4}{32} = 0.125 + 0.02365 = 0.14865$; adaptive quadrature gives 0.148650 ✓.

Orthogonality of sinusoids

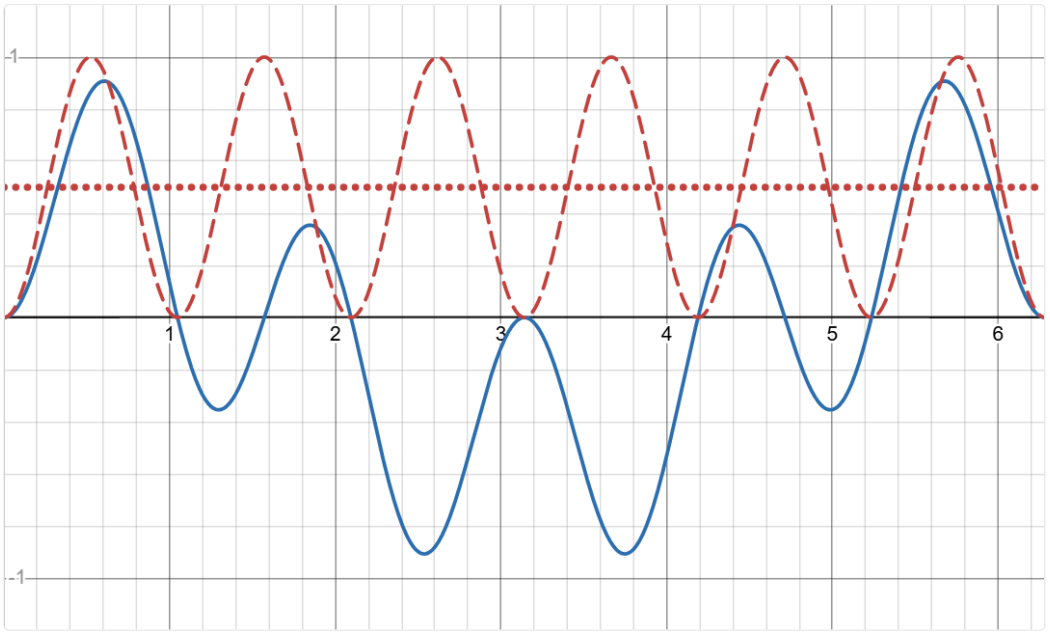
For integers $m, n \geq 1$ and $T = 2\pi/\omega_0$:

$$\int_0^T \sin(m\omega_0 t)\sin(n\omega_0 t)\,dt = \int_0^T \cos(m\omega_0 t)\cos(n\omega_0 t)\,dt = \begin{cases} T/2, & m = n \\ 0, & m \neq n \end{cases}$$

and $\displaystyle\int_0^T \sin(m\omega_0 t)\cos(n\omega_0 t)\,dt = 0$ for all m, n.

Proof. Use the product-to-sum identity. For $m \neq n$, $\sin m\theta \sin n\theta = \frac{1}{2}[\cos(m-n)\theta - \cos(m+n)\theta]$, and each cosine of a nonzero integer multiple of $\theta = \omega_0 t$ completes a whole number of cycles in $[0, T]$, integrating to zero. For $m = n$ the first cosine becomes $\cos 0 = 1$, giving $T/2$. The other cases are identical. □

Orthogonality is the reason power in different harmonics simply *adds* (Parseval's theorem, §5.9) and the reason Fourier coefficients can be computed one at a time (Ch. 15).



$\sin 2x \sin 3x$ (blue) has equal positive and negative area over $[0, 2\pi]$; $\sin^2 3x$ (red) oscillates about its mean $1/2$.

5.2.5 Partial fractions

A **rational function** $P(x)/Q(x)$ (ratio of polynomials) can always be integrated in elementary terms. The method:

1. If $\deg P \geq \deg Q$, divide to get a polynomial plus a **proper** fraction.
2. Factor Q over the reals into linear factors $(x-a)$ and irreducible quadratics (x^2+bx+c) with $b^2 < 4c$. (Over $\mathbb{C}$ every factor is linear; the real quadratics pair complex-conjugate roots.)
3. Write the proper fraction as a sum of simple terms according to this table.

Factor of Q	Terms in the expansion
$(x-a)$, simple	$\dfrac{A}{x-a}$
$(x-a)^k$, repeated	$\dfrac{A_1}{x-a} + \dfrac{A_2}{(x-a)^2} + \cdots + \dfrac{A_k}{(x-a)^k}$
x^2+bx+c, irreducible	$\dfrac{Bx+C}{x^2+bx+c}$
$(x^2+bx+c)^k$, repeated	$\displaystyle\sum_{i=1}^{k} \frac{B_i x + C_i}{(x^2+bx+c)^i}$

4. Find the constants. Fastest for simple linear factors is the **cover-up (Heaviside) rule**:

$$A = \Big[(x-a)\frac{P(x)}{Q(x)}\Big]_{x=a}.$$

For repeated factors, cover-up gives the highest-power coefficient A_k; the others follow by comparing coefficients, substituting convenient x values, or differentiating: $A_{k-i} = \dfrac{1}{i!}\dfrac{d^i}{dx^i}\Big[(x-a)^k P/Q\Big]_{x=a}$. 5. Integrate each term: $\int \frac{dx}{x-a} = \ln|x-a|$, $\int \frac{dx}{(x-a)^k} = \frac{(x-a)^{1-k}}{1-k}$ for $k \geq 2$, and quadratic terms by completing the square (split into a ln part and an arctan part).

Why the cover-up rule works. If $\frac{P}{Q} = \frac{A}{x-a} + R(x)$ where R has no pole at a, multiplying by $(x-a)$ gives $(x-a)\frac{P}{Q} = A + (x-a)R(x)$; setting $x = a$ kills the second term.

Worked Example 5.7 — Distinct linear factors

$\displaystyle\int \frac{3x+5}{(x+1)(x+2)}dx.$

Cover-up: $A = \dfrac{3(-1)+5}{-1+2} = 2$, $B = \dfrac{3(-2)+5}{-2+1} = 1$. So

$$\int \Big(\frac{2}{x+1} + \frac{1}{x+2}\Big)dx = 2\ln|x+1| + \ln|x+2| + C.$$

Check at $x=0$: $\frac{2}{1} + \frac{1}{2} = 2.5 = \frac{5}{2}$ ✓.

Worked Example 5.8 — A repeated factor

$\int \frac{x^2+1}{x(x-1)^2}dx$. Write $\frac{x^2+1}{x(x-1)^2} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{(x-1)^2}$.

- Cover x, set $x=0$: $A = \frac{1}{(-1)^2} = 1$.
- Cover $(x-1)^2$, set $x=1$: $C = \frac{1+1}{1} = 2$.
- Compare x^2 coefficients in $x^2+1 = A(x-1)^2 + Bx(x-1) + Cx$: $1 = A + B$, so $B = 0$.

$$\int \Big(\frac{1}{x} + \frac{2}{(x-1)^2}\Big)dx = \ln|x| - \frac{2}{x-1} + C.$$

Check at $x=2$: integrand $= 5/2$; $\frac{1}{2} + \frac{2}{1} = 2.5$ ✓.

Worked Example 5.9 — An irreducible quadratic

$\int_0^1 \frac{5x+3}{(x+1)(x^2+1)}dx$.

Step 1 — form. $\frac{5x+3}{(x+1)(x^2+1)} = \frac{A}{x+1} + \frac{Bx+C}{x^2+1}$.

Step 2 — constants. Cover-up at $x=-1$: $A = \frac{-5+3}{2} = -1$. Then $5x+3 = -(x^2+1) + (Bx+C)(x+1)$. Comparing x^2: $0 = -1 + B \Rightarrow B = 1$. Constants: $3 = -1 + C \Rightarrow C = 4$. (Check x terms: $5 = B + C$ ✓.)

Step 3 — integrate.

$$\int_0^1 \Big(-\frac{1}{x+1} + \frac{x}{x^2+1} + \frac{4}{x^2+1}\Big)dx = \Big[-\ln(x+1) + \tfrac{1}{2}\ln(x^2+1) + 4\arctan x\Big]_0^1.$$

Step 4 — evaluate. $-\ln 2 + \frac{1}{2}\ln 2 + 4\cdot\frac{\pi}{4} = \pi - \frac{1}{2}\ln 2 \approx 3.14159 - 0.34657 = 2.79502$.

Check: numerical quadrature gives 2.795019 ✓.

Worked Example 5.10 — A circuit-flavoured rational integral

The integral $I = \int_0^\infty \frac{d\omega}{(\omega^2+a^2)(\omega^2+b^2)}$ $(a, b > 0,\ a \neq b)$ appears in noise-bandwidth calculations for cascaded RC filters. Treat $u = \omega^2$ as the variable: cover-up on $\frac{1}{(u+a^2)(u+b^2)}$ gives

$$\frac{1}{(\omega^2+a^2)(\omega^2+b^2)} = \frac{1}{b^2-a^2}\Big(\frac{1}{\omega^2+a^2} - \frac{1}{\omega^2+b^2}\Big).$$

Each term integrates to an arctangent; on $[0,\infty)$, $\int_0^\infty \frac{d\omega}{\omega^2+a^2} = \frac{\pi}{2a}$ (an improper integral, §5.3). Hence

$$I = \frac{1}{b^2-a^2}\Big(\frac{\pi}{2a} - \frac{\pi}{2b}\Big) = \frac{\pi}{2ab(a+b)}.$$

Check with $a=1, b=2$: $\pi/12 = 0.261799$; numerical quadrature agrees.

Engineering Note — Partial fractions and system response

In the Laplace domain the response of a linear circuit is a rational function $Y(s) = P(s)/Q(s)$. The roots of Q are the **poles** (natural frequencies). Expanding Y in partial fractions decomposes the response into **modes**: each simple real pole p contributes Ae^{pt}, each repeated pole contributes $t^{k-1}e^{pt}$ terms, and each conjugate pair contributes a damped sinusoid. The cover-up coefficient A is called the **residue**, a name that returns in Ch. 18.

5.2.6 The complex-exponential method

Integrals of the form $\int e^{at}\cos bt\,dt$ and $\int e^{at}\sin bt\,dt$ can be done by parts twice (§5.5), but the fastest route uses Euler's formula. Since $e^{at}\cos bt = \mathrm{Re}\{e^{(a+jb)t}\}$ and integration commutes with taking real parts,

$$\int e^{at}\cos bt\,dt = \mathrm{Re}\left\{\frac{e^{(a+jb)t}}{a+jb}\right\} = \mathrm{Re}\left\{\frac{(a-jb)\,e^{at}(\cos bt + j\sin bt)}{a^2+b^2}\right\},$$

so

$$\int e^{at}\cos bt\,dt = \frac{e^{at}(a\cos bt + b\sin bt)}{a^2+b^2} + C, \qquad \int e^{at}\sin bt\,dt = \frac{e^{at}(a\sin bt - b\cos bt)}{a^2+b^2} + C,$$

the second from the imaginary part.

Worked Example 5.11 — Charge in a damped oscillation

The current in an underdamped RLC circuit is $i(t) = e^{-2t}\sin 4t$ A (time in seconds). Find $\int_0^\infty i(t)\,dt$, the net charge transferred.

With $a = -2$, $b = 4$, $a^2 + b^2 = 20$:

$$\int_0^\infty e^{-2t}\sin 4t\,dt = \Big[\frac{e^{-2t}(-2\sin 4t - 4\cos 4t)}{20}\Big]_0^\infty = 0 - \frac{-4}{20} = 0.2\text{ C}.$$

Shortcut: $\int_0^\infty e^{(a+jb)t}dt = \frac{-1}{a+jb}$ for $a<0$, so the answer is $\mathrm{Im}\{\frac{1}{2-4j}\} = \mathrm{Im}\{\frac{2+4j}{20}\} = 0.2$ ✓. This is the Laplace transform $\frac{4}{s^2+4s+20}$ evaluated at $s=0$ — an early glimpse of Ch. 14.

Common Pitfall — "Re" and products

$\mathrm{Re}\{\cdot\}$ commutes with integration and with multiplication by *real* quantities, but not with multiplication of two complex quantities: $\mathrm{Re}\{z_1 z_2\} \neq \mathrm{Re}\{z_1\}\,\mathrm{Re}\{z_2\}$ in general. When computing average power $\frac{1}{T}\int v(t)i(t)\,dt$ from phasors you cannot simply multiply phasors; the correct result is $\frac{1}{2}\,\mathrm{Re}\{\mathbf{V}\mathbf{I}^*\}$ (§5.9).

5.2.7 Integrals with no elementary antiderivative

Liouville's theory (19th century) shows that certain integrals cannot be expressed using finitely many elementary functions. Important examples define *new* special functions:

Integral	Name	Where it appears
$\frac{2}{\sqrt{\pi}}\int_0^x e^{-t^2}dt$	error function $\mathrm{erf}(x)$	Gaussian noise, bit-error rate, diffusion
$\int_0^x \frac{\sin t}{t}dt$	sine integral $\mathrm{Si}(x)$	Gibbs phenomenon, ideal-filter step response
$\int_0^{\pi/2}\frac{d\theta}{\sqrt{1-k^2\sin^2\theta}}$	complete elliptic integral $K(k)$	pendulum period, coil inductance, line geometries
$\int_x^\infty \frac{e^{-t}}{t}dt$	exponential integral $E_1(x)$	radiation, transient heating

These are tabulated and built into every numerical library. Recognising that an integral *is* one of them is a useful skill; trying to "find" an elementary antiderivative for them is not.

5.3 Improper integrals

5.3.1 Motivation

The Riemann integral is defined for a *bounded* function on a *bounded* interval. Engineering keeps breaking both conditions. The total energy dissipated as a capacitor discharges is an integral from $t=0$ to $t=\infty$. The Laplace and Fourier transforms are integrals over infinite ranges. The field of a line charge involves $\int \frac{dz}{(r^2+z^2)^{3/2}}$ over the whole z-axis. The probability that Gaussian noise exceeds a threshold is a tail integral to infinity. And the potential near a point charge involves integrands that blow up like $1/R$. **Improper integrals** extend the definition to these cases by taking limits.

5.3.2 Definitions

Improper integral of Type I (infinite interval)

If f is integrable on $[a,b]$ for every $b > a$,

$$\int_a^\infty f(x)\,dx = \lim_{b\to\infty}\int_a^b f(x)\,dx,$$

provided the limit exists and is finite; the integral then **converges**, otherwise it **diverges**. Similarly $\int_{-\infty}^b f = \lim_{a\to-\infty}\int_a^b f$, and

$$\int_{-\infty}^\infty f(x)\,dx = \int_{-\infty}^c f(x)\,dx + \int_c^\infty f(x)\,dx,$$

which converges only if **both** halves converge independently (the choice of c does not matter).

Improper integral of Type II (unbounded integrand)

If f is integrable on $[a+\varepsilon, b]$ for every $\varepsilon > 0$ but unbounded near a,

$$\int_a^b f(x)\,dx = \lim_{\varepsilon\to 0^+}\int_{a+\varepsilon}^b f(x)\,dx.$$

Likewise at the right end. If the singularity is at an interior point c, split: $\int_a^b = \int_a^c + \int_c^b$, and require both to converge.

An integral can be improper for several reasons at once ($\int_0^\infty \frac{dx}{\sqrt{x}}$ is improper at both ends); split it so that each piece has only one difficulty.

5.3.3 The benchmark p-integrals

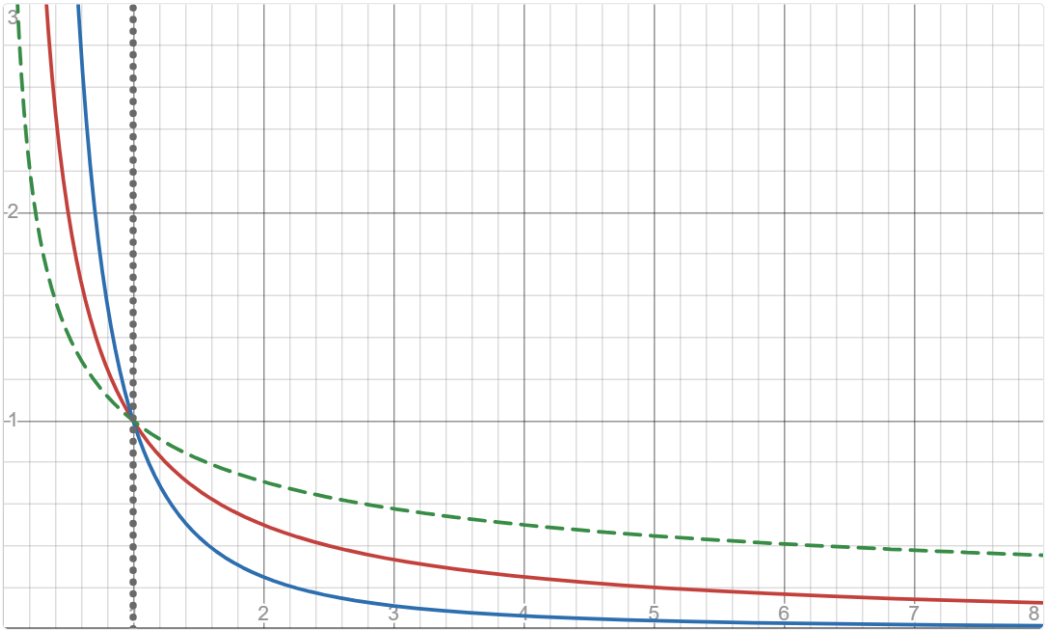
Almost every convergence question is settled by comparison with powers of x.

p-integrals

$$\int_1^\infty \frac{dx}{x^p} \begin{cases} = \dfrac{1}{p-1}, & p > 1 \\ \text{diverges}, & p \le 1 \end{cases} \qquad \int_0^1 \frac{dx}{x^p} \begin{cases} = \dfrac{1}{1-p}, & p < 1 \\ \text{diverges}, & p \ge 1 \end{cases}$$

Proof. For $p \neq 1$, $\int_1^b x^{-p}dx = \frac{b^{1-p}-1}{1-p}$. As $b \to \infty$, $b^{1-p} \to 0$ if $p > 1$ and $\to \infty$ if $p < 1$. For $p = 1$, $\int_1^b \frac{dx}{x} = \ln b \to \infty$. The $[0,1]$ case is the same computation with ε^{1-p} as $\varepsilon \to 0^+$. $\square$

The memory aid: **at infinity a function must decay faster than $1/x$; at a singularity it must blow up slower than $1/x$.** The function $1/x$ itself fails both tests — its integral diverges logarithmically at both ends. Exponentials decay faster than any power, so $\int_0^\infty x^n e^{-ax}dx$ converges for every $n \ge 0$, $a > 0$.



$1/x$ (red) sits on the borderline: its area diverges both near 0 and towards ∞. $1/x^2$ (blue) has finite area on $[1,\infty)$; $1/\sqrt{x}$ (green) has finite area on $(0,1]$.

Worked Example 5.12 — Direct evaluation

(a) $\displaystyle\int_0^\infty e^{-t/\tau}dt = \lim_{b\to\infty}\left[-\tau e^{-t/\tau}\right]_0^b = \tau$ for $\tau > 0$. The area under a decaying exponential equals its time constant — the basis of the "area = height × time constant" rule of thumb.

(b) $\displaystyle\int_0^1 \frac{dx}{\sqrt{x}} = \lim_{\varepsilon\to 0^+}\left[2\sqrt{x}\right]_\varepsilon^1 = 2$. A function can be infinite at a point and still enclose finite area.

(c) $\displaystyle\int_0^1 \ln x\,dx = \lim_{\varepsilon\to 0^+}\left[x\ln x - x\right]_\varepsilon^1 = -1 - \lim_{\varepsilon\to 0^+}(\varepsilon\ln\varepsilon - \varepsilon) = -1$, using $\varepsilon\ln\varepsilon \to 0$ (L'Hôpital, Ch. 4).

(d) $\displaystyle\int_{-\infty}^\infty \frac{dx}{1+x^2} = \lim_{a\to-\infty}\left[\arctan x\right]_a^0 + \lim_{b\to\infty}\left[\arctan x\right]_0^b = \frac{\pi}{2} + \frac{\pi}{2} = \pi$.

Worked Example 5.13 — Energy dissipated by a discharging capacitor

A capacitor C charged to V_0 discharges through R: $v(t) = V_0 e^{-t/RC}$. Find the total energy dissipated in R.

Step 1 — power. $p(t) = v^2/R = \dfrac{V_0^2}{R}e^{-2t/RC}$.

Step 2 — improper integral. Using Example 5.12(a) with $\tau = RC/2$,

$$W = \int_0^\infty \frac{V_0^2}{R}e^{-2t/RC}dt = \frac{V_0^2}{R}\cdot\frac{RC}{2} = \frac{1}{2}CV_0^2.$$

Interpretation. All of the stored energy $\frac{1}{2}CV_0^2$ ends up as heat, independent of R — a larger resistor dissipates less power but for longer. With $C = 100\ \mu$F, $V_0 = 12$ V: $W = 7.2$ mJ.

Worked Example 5.14 — A divergent integral in disguise

Is $\displaystyle\int_0^2 \frac{dx}{(x-1)^2}$ finite?

The integrand is singular at the interior point $x = 1$. Split at 1:

$$\int_0^1 \frac{dx}{(x-1)^2} = \lim_{\varepsilon\to 0^+}\Big[-\frac{1}{x-1}\Big]_0^{1-\varepsilon} = \lim_{\varepsilon\to 0^+}\Big(\frac{1}{\varepsilon} - 1\Big) = \infty.$$

One half diverges, so the whole integral diverges. Blindly applying the FTC gives $\big[-\frac{1}{x-1}\big]_0^2 = -1 - 1 = -2$, a negative "area" for a positive function — the same trap as in §5.1.

5.3.4 Comparison tests

Often we only need to know *whether* an improper integral converges — for example, to confirm that a Fourier or Laplace transform exists — without computing it.

Direct comparison test

Suppose $0 \le f(x) \le g(x)$ for $x \ge a$.

- If $\int_a^\infty g\,dx$ converges, so does $\int_a^\infty f\,dx$ (and $\int_a^\infty f \le \int_a^\infty g$).
- If $\int_a^\infty f\,dx$ diverges, so does $\int_a^\infty g\,dx$.

The same holds for Type II integrals near a singularity.

Proof sketch. $F(b) = \int_a^b f$ is non-decreasing in b (since $f \ge 0$) and bounded above by $\int_a^\infty g$. A bounded monotone function has a limit. □

Limit comparison test

If $f, g > 0$ for $x \ge a$ and $\displaystyle\lim_{x\to\infty}\frac{f(x)}{g(x)} = L$ with $0 < L < \infty$, then $\int_a^\infty f$ and $\int_a^\infty g$ either both converge or both diverge.

Limit comparison formalises "only the dominant behaviour matters". To test $\int_1^\infty \frac{3x+2}{x^3+5}dx$, note the integrand behaves like $3x/x^3 = 3/x^2$ for large x; the ratio to $1/x^2$ tends to 3, and $\int_1^\infty x^{-2}$ converges, so the original converges.

Absolute convergence

If $\int_a^\infty |f|\,dx$ converges then $\int_a^\infty f\,dx$ converges, and $\big|\int_a^\infty f\big| \le \int_a^\infty |f|$.

An integral that converges but not absolutely is **conditionally convergent**. The classic example is the Dirichlet integral

$$\int_0^\infty \frac{\sin x}{x}\,dx = \frac{\pi}{2},$$

which converges because the positive and negative lobes of $\sin x/x$ shrink and alternate, yet $\int_0^\infty \big|\frac{\sin x}{x}\big|dx = \infty$ (each lobe has area roughly $2/(k\pi)$, and $\sum 1/k$ diverges). This integral is the area under the sinc function and underlies the Gibbs phenomenon and the ideal low-pass filter (Ch. 16).

Worked Example 5.15 — Convergence decisions

(a) $\displaystyle\int_1^\infty e^{-x^2}dx$: for $x \ge 1$, $x^2 \ge x$, so $0 < e^{-x^2} \le e^{-x}$, and $\int_1^\infty e^{-x}dx = e^{-1}$. **Converges**, with value at most 0.3679 (actual: 0.1394).

(b) $\displaystyle\int_1^\infty \frac{\sqrt{x}+1}{x^2}dx$: dominant term $x^{1/2}/x^2 = x^{-3/2}$, $p = 3/2 > 1$. **Converges** (in fact it equals $2 + 1 = 3$).

(c) $\displaystyle\int_2^\infty \frac{dx}{\ln x}$: for $x \ge 2$, $\ln x < x$, so $\frac{1}{\ln x} > \frac{1}{x}$, and $\int_2^\infty \frac{dx}{x}$ diverges. **Diverges.**

(d) $\displaystyle\int_0^1 \frac{\cos x}{x^{2/3}}dx$: near 0, $\cos x \to 1$, so the integrand behaves like $x^{-2/3}$ with $p = 2/3 < 1$. **Converges.**

(e) $\displaystyle\int_0^\infty \frac{\sin^2 x}{x^2}dx$: near 0 the integrand $\to 1$ (no singularity at all); for large x it is bounded by $1/x^2$. **Converges** absolutely (its value is $\pi/2$).

5.3.5 Cauchy principal value

Some integrals over symmetric ranges diverge in the strict sense but have a natural "balanced" value.

Cauchy principal value

$$\mathrm{PV}\int_{-\infty}^{\infty} f(x)\,dx = \lim_{R\to\infty}\int_{-R}^{R} f(x)\,dx, \qquad \mathrm{PV}\int_a^b f(x)\,dx = \lim_{\varepsilon\to 0^+}\Big(\int_a^{c-\varepsilon} f + \int_{c+\varepsilon}^b f\Big)$$

for a singularity at $c \in (a, b)$.

For example $\int_{-\infty}^{\infty} x\,dx$ diverges, but its principal value is 0; and $\mathrm{PV}\int_{-1}^{2}\frac{dx}{x} = \lim_{\varepsilon\to 0^+}\left(\ln\varepsilon - 0 + \ln 2 - \ln\varepsilon\right) = \ln 2$. Principal values appear in the Hilbert transform (single-sideband modulation, Kramers–Kronig relations between the real and imaginary parts of a causal response) and are evaluated systematically with contour integration in Ch. 18.

> **Common Pitfall — Symmetric limits are not the definition**
>
> $\int_{-\infty}^{\infty} f\,dx$ is defined by two *independent* limits. Computing $\lim_{R\to\infty}\int_{-R}^{R} f$ gives the principal value, which can be finite even when the integral diverges. Only when the integral is known to converge do the two agree.

5.3.6 Two important improper integrals

The Gaussian integral.

$$\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}.$$

Convergence follows from Example 5.15(a). The value cannot be found with single-variable antiderivatives; the standard trick squares it into a double integral over the plane and switches to polar coordinates — a computation done in Ch. 6. Scaling $x \to x/(\sigma\sqrt{2})$ gives the normalisation of the Gaussian density, $\int_{-\infty}^{\infty} e^{-x^2/2\sigma^2}\,dx = \sigma\sqrt{2\pi}$ (Ch. 19).

The Gamma function.

> **Gamma function**
>
> For $x > 0$,
>
> $$\Gamma(x) = \int_0^{\infty} t^{x-1}e^{-t}\,dt.$$

It converges at ∞ because the exponential wins, and at 0 because t^{x-1} has $p = 1 - x < 1$. Integration by parts (§5.5) gives $\Gamma(x+1) = x\,\Gamma(x)$, and $\Gamma(1) = 1$, so

$$\Gamma(n+1) = n! \qquad (n = 0, 1, 2, \dots),$$

and $\Gamma(\tfrac{1}{2}) = \sqrt{\pi}$ (substitute $t = u^2$ and use the Gaussian integral). The Gamma function makes "$\int_0^{\infty} t^n e^{-st}dt = n!/s^{n+1}$" — the Laplace transform of t^n — a one-line result.

> **Engineering Note — Existence of transforms**
>
> The Laplace transform $\int_0^{\infty} f(t)e^{-st}dt$ converges absolutely when $|f(t)| \le Me^{\alpha t}$ and $\mathrm{Re}\,s > \alpha$: compare with $\int_0^{\infty} Me^{-(\mathrm{Re}\,s-\alpha)t}dt$. That is the entire content of the "exponential order" existence theorem in Ch. 14 — a direct comparison test.

5.4 Multiple integrals

5.4.1 Motivation

Charge is rarely concentrated on a line. A semiconductor wafer carries a surface charge density $\sigma(x, y)$ in C/m²; a depletion region holds a volume charge density $\rho(x, y, z)$ in C/m³; a printed coil's inductance depends on how current is distributed across a cross-section. The total charge is obtained exactly as in §5.1 — chop, multiply by the size of each piece, add, take a limit — except that the pieces are now small rectangles of area ΔA or small boxes of volume ΔV. The result is a **double** or **triple integral**.

This section introduces multiple integrals over simple regions in Cartesian coordinates and shows how to compute them as *iterated* single integrals, so that everything from §§5.1–5.3 carries over. Chapter 6 develops the subject further: general changes of variables, the Jacobian, and polar, cylindrical and spherical coordinates.

5.4.2 Double integrals over a rectangle

> **Double integral**
>
> Let f be bounded on the rectangle $R = [a, b] \times [c, d]$. Partition $[a, b]$ and $[c, d]$, giving sub-rectangles R_{ij} of area $\Delta A_{ij} = \Delta x_i\,\Delta y_j$, and pick sample points $(x_{ij}^*, y_{ij}^*) \in R_{ij}$. If the double Riemann sum
>
> $$\sum_i \sum_j f(x_{ij}^*, y_{ij}^*)\,\Delta A_{ij}$$
>
> tends to a limit I as the largest sub-rectangle diagonal tends to zero, independent of the sample points, then
>
> $$\iint_R f(x, y)\,dA = I.$$

If $f \ge 0$, the double integral is the **volume** under the surface $z = f(x, y)$ above R. If f is a surface density (charge, mass, power per area), it is the **total amount**. With $f \equiv 1$ it is the **area** of R. Linearity, additivity over regions, monotonicity and the triangle inequality all carry over from §5.1.3. Continuous functions — and bounded functions whose discontinuities lie on finitely many smooth curves — are integrable.

Computing a two-dimensional limit directly is hopeless, but the following theorem reduces it to two one-dimensional integrals.

Fubini's theorem (rectangle)

If f is continuous on $R = [a,b] \times [c,d]$, then

$$\iint_R f(x,y)\,dA = \int_a^b \Big[\int_c^d f(x,y)\,dy\Big]dx = \int_c^d \Big[\int_a^b f(x,y)\,dx\Big]dy.$$

Proof idea. Arrange the double Riemann sum in a grid. Summing first down each column (fixed x_i) gives approximately $\big[\int_c^d f(x_i,y)\,dy\big]\Delta x_i$; summing those columns across gives a Riemann sum for the outer x-integral. Summing rows first gives the other order. Both are rearrangements of the same finite sum, and continuity ensures both limits exist and agree. □

In the inner integral the outer variable is held **constant**, exactly as in partial differentiation. The bracket is a function of the outer variable alone.

Separable integrands. If $f(x,y) = g(x)h(y)$ on a rectangle, the double integral factorises:

$$\iint_{[a,b]\times[c,d]} g(x)h(y)\,dA = \Big(\int_a^b g(x)\,dx\Big)\Big(\int_c^d h(y)\,dy\Big).$$

Worked Example 5.16 — A rectangle

Evaluate $\iint_R x^2 y\,dA$ over $R = [0,2]\times[1,3]$.

Order dy then dx:

$$\int_0^2 \Big[\int_1^3 x^2 y\,dy\Big]dx = \int_0^2 x^2\Big[\frac{y^2}{2}\Big]_1^3 dx = \int_0^2 4x^2\,dx = \frac{32}{3}.$$

Check by factorising: $\big(\int_0^2 x^2dx\big)\big(\int_1^3 y\,dy\big) = \frac{8}{3}\cdot 4 = \frac{32}{3} \approx 10.667$ ✓.

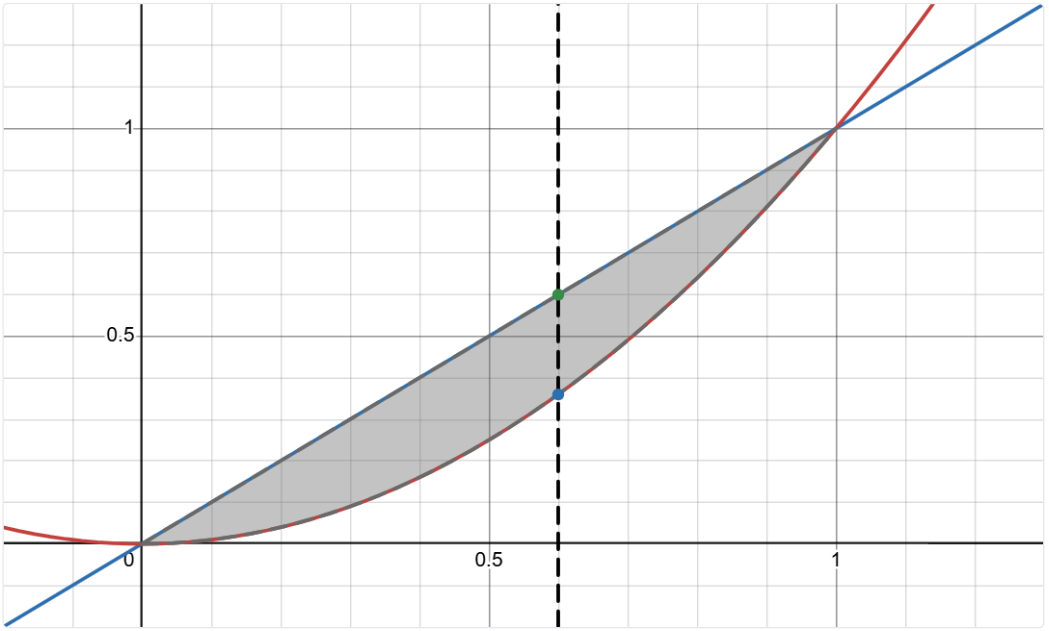
5.4.3 Double integrals over general regions

For a region D that is not a rectangle, extend f by zero outside D and integrate over an enclosing rectangle. In practice we describe D by its bounding curves.

Type I and Type II regions

- **Type I (vertically simple):** $D = \{(x,y) : a \le x \le b,\ g_1(x) \le y \le g_2(x)\}$. Then
$$\iint_D f\,dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y)\,dy\,dx.$$
- **Type II (horizontally simple):** $D = \{(x,y) : c \le y \le d,\ h_1(y) \le x \le h_2(y)\}$. Then
$$\iint_D f\,dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x,y)\,dx\,dy.$$

Recipe for setting limits. (1) Sketch D. (2) Choose the inner variable, say y. (3) Draw a vertical line through D at a typical x: it enters D on the lower curve $y = g_1(x)$ and leaves on the upper curve $y = g_2(x)$ — those are the inner limits, and they may depend on x. (4) The outer limits are the extreme x-values of D and must be **constants**. A region that is neither type is split into pieces that are.



The region between $y = x^2$ and $y = x$. A vertical line at $x = 0.6$ enters on the parabola and leaves on the line, fixing the inner y-limits.

Worked Example 5.17 — A region between two curves, both orders

Evaluate $\iint_D xy\,dA$ where D is bounded by $y = x$ and $y = x^2$.

Step 1 — sketch. The curves meet where $x = x^2$, i.e. $x = 0$ and $x = 1$. For $0 < x < 1$, $x^2 < x$.

Step 2 — Type I.

$$\int_0^1 \int_{x^2}^{x} xy\,dy\,dx = \int_0^1 x \cdot \frac{x^2 - x^4}{2}\,dx = \frac{1}{2}\left(\frac{1}{4} - \frac{1}{6}\right) = \frac{1}{24}.$$

Step 3 — Type II, as a check. A horizontal line at height y enters at $x = y$ and leaves at $x = \sqrt{y}$:

$$\int_0^1 \int_{y}^{\sqrt{y}} xy\,dx\,dy = \int_0^1 y \cdot \frac{y - y^2}{2}\,dy = \frac{1}{2}\left(\frac{1}{3} - \frac{1}{4}\right) = \frac{1}{24}\ \checkmark$$

Area of D (integrand 1): $\int_0^1 (x - x^2)dx = \frac{1}{6}$. So the **mean value** of xy over D is $\frac{1/24}{1/6} = \frac{1}{4}$.

5.4.4 Reversing the order of integration

Sometimes the inner integral in one order has no elementary antiderivative, but the other order is easy. Reversing the order means re-describing the *same region* the other way — the limits do not simply swap.

Worked Example 5.18 — Reversal rescues an impossible integral

Evaluate $I = \int_0^1 \int_x^1 e^{y^2}\,dy\,dx$.

Step 1 — the obstacle. $\int e^{y^2}\,dy$ has no elementary antiderivative.

Step 2 — describe the region. $0 \le x \le 1$, $x \le y \le 1$: the triangle with vertices $(0,0)$, $(0,1)$, $(1,1)$, lying above the line $y = x$.

Step 3 — redescribe as Type II. For fixed $y \in [0,1]$, x runs from 0 to y.

Step 4 — integrate.

$$I = \int_0^1 \int_0^y e^{y^2}\,dx\,dy = \int_0^1 y\,e^{y^2}\,dy = \left[\tfrac{1}{2}e^{y^2}\right]_0^1 = \frac{e - 1}{2} \approx 0.85914.$$

The inner integration produced the factor y, which is precisely the derivative needed for the substitution $u = y^2$ (§5.6).

Common Pitfall — Variable outer limits

The outermost limits must be constants. Writing $\int_x^1 \int_0^1 f\,dx\,dy$ is meaningless — after both integrations an x would survive. When reversing order, *always* sketch the region and read the new limits off the sketch.

5.4.5 Triple integrals

The triple integral $\iiint_E f(x,y,z)\,dV$ is defined by Riemann sums over small boxes of volume $\Delta V = \Delta x\,\Delta y\,\Delta z$. With $f \equiv 1$ it gives the volume of E; with $f = \rho$ (charge density) it gives the enclosed charge — the right-hand side of Gauss's law in integral form (Ch. 7). Fubini's theorem extends: for a solid described by

$$E = \{(x,y,z) : a \le x \le b,\ g_1(x) \le y \le g_2(x),\ u_1(x,y) \le z \le u_2(x,y)\},$$

$$\iiint_E f\,dV = \int_a^b \int_{g_1(x)}^{g_2(x)} \int_{u_1(x,y)}^{u_2(x,y)} f(x,y,z)\,dz\,dy\,dx.$$

Work from the inside out: integrate in z treating x, y as constants, then in y treating x as constant, then in x. The limits at each stage may depend only on variables that are still to be integrated.

Worked Example 5.19 — Charge in a tetrahedron

A charge density $\rho = \rho_0\, z/h$ (with ρ_0 in C/m³) fills the tetrahedron bounded by the coordinate planes and the plane $x + y + z = h$. Find the total charge.

Step 1 — limits. The base in the xy-plane is the triangle $x \ge 0$, $y \ge 0$, $x + y \le h$. Above a point (x,y), z runs from 0 to $h - x - y$.

Step 2 — innermost. $\int_0^{h-x-y} \frac{\rho_0 z}{h}dz = \frac{\rho_0}{2h}(h - x - y)^2$.

Step 3 — middle (y from 0 to $h - x$). With $w = h - x$: $\int_0^{w} \frac{\rho_0}{2h}(w - y)^2 dy = \frac{\rho_0}{2h} \cdot \frac{w^3}{3}$.

Step 4 — outer. $\int_0^h \frac{\rho_0}{6h}(h - x)^3 dx = \frac{\rho_0}{6h} \cdot \frac{h^4}{4} = \frac{\rho_0 h^3}{24}$.

Check. The volume is $h^3/6$, so the mean density is $\rho_0/4$, i.e. the mean of z/h over the tetrahedron is $1/4$ — consistent with the centroid of a tetrahedron lying a quarter of the way up from the base. Units: $[\mathrm{C/m^3}][\mathrm{m^3}] = [\mathrm{C}]$ ✓. With $h = 1$, a Monte Carlo estimate of $\iiint z\,dV$ gives $0.0417 \approx 1/24$.

5.4.6 A first look at polar coordinates

Circular regions are awkward in Cartesian form: the disc $x^2 + y^2 \le a^2$ needs limits $-\sqrt{a^2 - x^2} \le y \le \sqrt{a^2 - x^2}$. In polar coordinates (r, ϕ) it is simply $0 \le r \le a$, $0 \le \phi \le 2\pi$. The area element is **not** $dr\,d\phi$: a small polar "rectangle" has radial side dr and arc side $r\,d\phi$, so

$$dA = r\,dr\,d\phi.$$

The extra factor r is the simplest example of a **Jacobian**; Chapter 6 derives it in general and extends it to cylindrical ($dV = r\,dr\,d\phi\,dz$) and spherical ($dV = R^2 \sin\theta\,dR\,d\theta\,d\phi$) coordinates.

Worked Example 5.20 — Charge on a disc

A disc of radius a carries surface charge density $\sigma(r) = \sigma_0\, r/a$. Find the total charge.

$$Q = \int_0^{2\pi}\int_0^{a} \sigma_0 \frac{r}{a}\, r\,dr\,d\phi = 2\pi \cdot \frac{\sigma_0}{a} \cdot \frac{a^3}{3} = \frac{2\pi\sigma_0 a^2}{3}.$$

Check. A uniform density σ_0 would give $\pi a^2\sigma_0$; our density averages to $\tfrac{2}{3}\sigma_0$ over the disc (more area sits near the rim, where σ is largest), giving $\tfrac{2}{3}\pi a^2\sigma_0$ ✓. Forgetting the factor r in dA would give $\pi\sigma_0 a$ — dimensionally wrong (C/m instead of C), which is how the error is caught.

Engineering Note — Choosing coordinates

The effort of a multiple integral is set mostly by how well the coordinates fit the boundary. Parallel-plate geometries suit Cartesian coordinates; coaxial cables, wires and solenoids suit cylindrical; point charges and antennas suit spherical. Picking coordinates in which every boundary is "coordinate = constant" turns every limit into a constant and often makes the integrand separable.

5.5 Integration by parts

5.5.1 The rule and where it comes from

Substitution (§5.6) undoes the chain rule. **Integration by parts** undoes the product rule, and it is the tool for integrands that are products of *unlike* functions: $t\,e^{-t/\tau}$, $t\sin\omega t$, $\ln x$, $x^2\cos x$. It is also the mechanism behind reduction formulas, the Gamma-function recursion, the derivative rule of the Laplace transform, and — in higher dimensions — the divergence theorem and Green's identities (Ch. 7).

Start from the product rule $(uv)' = u'v + uv'$ and integrate both sides:

$$uv = \int u'v\,dx + \int uv'\,dx.$$

Rearranging gives:

Integration by parts

If u and v have continuous derivatives,

$$\int u\,dv = uv - \int v\,du, \qquad \text{i.e.} \qquad \int u(x)v'(x)\,dx = u(x)v(x) - \int u'(x)v(x)\,dx,$$

and for definite integrals

$$\int_a^b u\,dv = \Big[uv\Big]_a^b - \int_a^b v\,du.$$

The rule *trades* one integral for another. It helps when $\int v\,du$ is simpler than $\int u\,dv$, which happens when differentiating u simplifies it (a polynomial loses a degree; $\ln x$ becomes $1/x$) while integrating dv does not make it worse (exponentials and sinusoids stay the same kind).

Choosing u. A long-standing heuristic is **LIATE**: choose u as the first type present in the list

> **L**ogarithmic, **I**nverse trigonometric, **A**lgebraic (polynomials), **T**rigonometric, **E**xponential,

and let dv be the rest. Functions early in the list get simpler when differentiated; those late in the list are easy to integrate.

Worked Example 5.21 — The basic pattern

(a) $\int xe^x\,dx$. Take $u = x$, $dv = e^x dx$, so $du = dx$, $v = e^x$:

$$\int xe^x dx = xe^x - \int e^x dx = (x-1)e^x + C.$$

Check: $\frac{d}{dx}(x-1)e^x = e^x + (x-1)e^x = xe^x$ ✓.

(b) $\int x\cos 3x\,dx$. $u = x$, $dv = \cos 3x\,dx$, $v = \frac{1}{3}\sin 3x$:

$$\int x\cos 3x\,dx = \frac{x\sin 3x}{3} - \frac{1}{3}\int \sin 3x\,dx = \frac{x\sin 3x}{3} + \frac{\cos 3x}{9} + C.$$

(c) $\int \ln x\,dx$. There is no obvious product — take $u = \ln x$, $dv = dx$, so $du = dx/x$, $v = x$:

$$\int \ln x\,dx = x\ln x - \int x\cdot\frac{1}{x}dx = x\ln x - x + C.$$

(d) $\int \arctan x\,dx$. $u = \arctan x$, $dv = dx$:

$$\int \arctan x\,dx = x\arctan x - \int \frac{x}{1+x^2}dx = x\arctan x - \tfrac{1}{2}\ln(1+x^2) + C.$$

Parts (c) and (d) show the trick of taking $dv = dx$: it converts the integral of a function whose *derivative* is algebraic into an algebraic integral.

Worked Example 5.22 — Mean delay of an RC response

The impulse response of an RC low-pass filter is $h(t) = \frac{1}{\tau}e^{-t/\tau}$, $t \ge 0$, with $\int_0^\infty h\,dt = 1$. Its **mean delay** (first moment, the basis of the Elmore delay used to estimate interconnect delay in integrated circuits) is $\bar{t} = \int_0^\infty t\,h(t)\,dt$.

Step 1 — parts. $u = t$, $dv = \frac{1}{\tau}e^{-t/\tau}dt$, $v = -e^{-t/\tau}$:

$$\int_0^\infty \frac{t}{\tau}e^{-t/\tau}dt = \Big[-te^{-t/\tau}\Big]_0^\infty + \int_0^\infty e^{-t/\tau}dt.$$

Step 2 — boundary term. $te^{-t/\tau} \to 0$ as $t \to \infty$ (exponential beats polynomial) and is 0 at $t = 0$.

Step 3 — remaining integral. $\int_0^\infty e^{-t/\tau}dt = \tau$ (Example 5.12a). Hence $\bar{t} = \tau$.

The mean delay equals the time constant. With $R = 1\ \text{k}\Omega$ and $C = 1\ \text{nF}$, $\bar{t} = 1\ \mu\text{s}$.

5.5.2 Repeated parts and the tabular method

When u is a polynomial of degree n, parts must be applied n times. The bookkeeping is compressed by the **tabular method**: list u and its successive derivatives in one column (until zero), dv and its successive integrals in another, and combine diagonally with alternating signs $+, -, +, \ldots$

Worked Example 5.23 — Tabular integration

$\int x^3 e^{2x}dx$.

sign	D: derivatives of x^3	I: integrals of e^{2x}
$+$	x^3	e^{2x}
$-$	$3x^2$	$\frac{1}{2}e^{2x}$
$+$	$6x$	$\frac{1}{4}e^{2x}$
$-$	6	$\frac{1}{8}e^{2x}$
	0	$\frac{1}{16}e^{2x}$

Multiply each D entry by the I entry one row lower, with the sign of its row:

$$\int x^3e^{2x}dx = e^{2x}\Big(\frac{x^3}{2} - \frac{3x^2}{4} + \frac{6x}{8} - \frac{6}{16}\Big) + C = e^{2x}\Big(\frac{x^3}{2} - \frac{3x^2}{4} + \frac{3x}{4} - \frac{3}{8}\Big) + C.$$

Check with $\int_0^1$: formula gives $e^2(0.5 - 0.75 + 0.75 - 0.375) + 0.375 = 0.125e^2 + 0.375 = 1.29863$; numerical quadrature gives 1.298632 ✓.

5.5.3 Cyclic integrals

For $\int e^{ax}\sin bx\,dx$ two applications of parts return the original integral, which can then be solved for algebraically.

Worked Example 5.24 — Solving for the integral

$I = \int e^x \sin x \, dx$.

First parts ($u = \sin x$, $dv = e^x dx$): $I = e^x \sin x - \int e^x \cos x \, dx$.

Second parts ($u = \cos x$, $dv = e^x dx$): $\int e^x \cos x \, dx = e^x \cos x + \int e^x \sin x \, dx = e^x \cos x + I$.

So $I = e^x \sin x - e^x \cos x - I$, giving

$$I = \frac{e^x(\sin x - \cos x)}{2} + C,$$

which matches the complex-exponential formula of §5.2.6 with $a = b = 1$. Keep the *same type* of choice for u (trig both times, or exponential both times); switching sides undoes the first step and yields the useless identity $I = I$.

5.5.4 Reduction formulas

Parts can lower an exponent in a whole family of integrals at once.

Reduction formula for powers of sine

For $n \geq 2$,

$$\int \sin^n x \, dx = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} \int \sin^{n-2} x \, dx,$$

and on $[0, \pi/2]$, where the boundary term vanishes, $I_n = \displaystyle\int_0^{\pi/2} \sin^n x \, dx$ satisfies $I_n = \dfrac{n-1}{n} I_{n-2}$ with $I_0 = \pi/2$, $I_1 = 1$.

Derivation. Write $\sin^n x = \sin^{n-1} x \cdot \sin x$, take $u = \sin^{n-1} x$, $dv = \sin x \, dx$. Then $du = (n-1)\sin^{n-2} x \cos x \, dx$, $v = -\cos x$, and

$$\int \sin^n x \, dx = -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x \cos^2 x \, dx.$$

Replace $\cos^2 x = 1 - \sin^2 x$: the right side contains $(n-1)\int \sin^{n-2} x \, dx - (n-1)\int \sin^n x \, dx$. Move the last term to the left, collect $n \int \sin^n x \, dx$, and divide by n. $\square$

The resulting **Wallis formulas** are

$$I_{2m} = \frac{(2m-1)(2m-3)\cdots 1}{2m(2m-2)\cdots 2} \cdot \frac{\pi}{2}, \qquad I_{2m+1} = \frac{2m(2m-2)\cdots 2}{(2m+1)(2m-1)\cdots 3}.$$

For instance $\int_0^{\pi/2} \sin^4 x \, dx = \frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2} = \frac{3\pi}{16} \approx 0.58905$. The identical formulas hold for $\cos^n x$ on $[0, \pi/2]$.

The same idea applied to $\Gamma(x+1) = \int_0^\infty t^x e^{-t} dt$ with $u = t^x$, $dv = e^{-t}dt$ gives

$$\Gamma(x+1) = \Big[-t^x e^{-t}\Big]_0^\infty + x\int_0^\infty t^{x-1} e^{-t} dt = x\,\Gamma(x),$$

the recursion quoted in §5.3.6.

5.5.5 Fourier coefficients by parts

Fourier coefficients of polynomial-shaped waveforms (ramps, sawtooth and triangle waves) are always integrals of polynomial × sinusoid, i.e. integration by parts.

Worked Example 5.25 — Sawtooth coefficients

The sawtooth $f(t) = t$ on $(-\pi, \pi)$, extended with period 2π, is odd, so only sine terms appear, with

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} t \sin nt \, dt = \frac{2}{\pi}\int_0^{\pi} t \sin nt \, dt.$$

Parts with $u = t$, $dv = \sin nt \, dt$, $v = -\frac{\cos nt}{n}$:

$$\int_0^{\pi} t \sin nt \, dt = \Big[-\frac{t\cos nt}{n}\Big]_0^{\pi} + \frac{1}{n}\int_0^{\pi} \cos nt \, dt = -\frac{\pi\cos n\pi}{n} + 0 = \frac{\pi(-1)^{n+1}}{n}.$$

Hence $b_n = \dfrac{2(-1)^{n+1}}{n}$: $b_1 = 2$, $b_2 = -1$, $b_3 = \frac{2}{3}$, … The $1/n$ decay reflects the jump discontinuity of the sawtooth; the triangle wave, which is continuous with a kinked derivative, has coefficients decaying like $1/n^2$ — one more integration by parts gains one more power of n. This link between smoothness and spectral decay is developed in Ch. 15.

Common Pitfall — Dropping or mishandling the boundary term

In definite integrals the term $[uv]_a^b$ is easy to forget, and for improper integrals it must be evaluated as a *limit*. In Example 5.22, $[-te^{-t/\tau}]_0^\infty = 0$ needed the fact that $te^{-t/\tau} \to 0$. Had the integrand been $t\,e^{+t}$ the boundary term would diverge and the whole computation would be meaningless.

Engineering Note — Parts in transform theory

The Laplace-transform rule $\mathcal{L}\{f'(t)\} = sF(s) - f(0)$ is a single integration by parts:

$$\int_0^\infty f'(t)e^{-st}dt = \Big[f(t)e^{-st}\Big]_0^\infty + s\int_0^\infty f(t)e^{-st}dt = -f(0) + sF(s).$$

The boundary term $-f(0)$ is how initial conditions — initial capacitor voltages and inductor currents — enter circuit analysis in the s-domain (Ch. 14).

5.6 Integration by substitution

5.6.1 The rule

The chain rule says $\frac{d}{dx}F(g(x)) = F'(g(x))\,g'(x)$. Reading it backwards: if $F' = f$, then $\int f(g(x))\,g'(x)\,dx = F(g(x)) + C$. Writing $u = g(x)$ and $du = g'(x)\,dx$ makes this look like an algebraic change of symbols.

Substitution rule

If $u = g(x)$ has a continuous derivative and f is continuous on the range of g,

$$\int f(g(x))\,g'(x)\,dx = \int f(u)\,du\Big|_{u=g(x)},$$

and for definite integrals

$$\int_a^b f(g(x))\,g'(x)\,dx = \int_{g(a)}^{g(b)} f(u)\,du.$$

Proof of the definite form. Let $F' = f$. Both sides equal $F(g(b)) - F(g(a))$: the left by the FTC applied to $(F\circ g)' = f(g)g'$, the right by the FTC applied to F. $\square$

Note that in the definite form **the limits change** and there is no need to convert back to x. Dimensionally, du carries the units of u; a correct substitution keeps $f(u)\,du$ dimensionally equal to the original integrand times dx.

How to spot it. Look for a function *inside* another function whose derivative (up to a constant factor) also appears as a factor. Typical patterns:

Pattern	Substitution	Result
$\int f(ax+b)\,dx$	$u = ax+b$	$\frac{1}{a}F(ax+b)$
$\int \frac{g'(x)}{g(x)}dx$	$u = g(x)$	$\ln\lvert g(x)\rvert$
$\int g(x)^n g'(x)\,dx$	$u = g(x)$	$\frac{g^{n+1}}{n+1}, n\neq -1$
$\int e^{g(x)}g'(x)\,dx$	$u = g(x)$	$e^{g(x)}$
$\int x\,f(x^2)\,dx$	$u = x^2$	$\frac{1}{2}F(x^2)$
$\int \frac{f(\ln x)}{x}dx$	$u = \ln x$	$F(\ln x)$

Worked Example 5.26 — Recognising the inner function

(a) $\int 2x\cos(x^2)\,dx$: $u = x^2$, $du = 2x\,dx$, giving $\int \cos u\,du = \sin(x^2) + C$.

(b) $\int \frac{x}{(x^2+a^2)^{3/2}}dx$: $u = x^2 + a^2$, $du = 2x\,dx$:

$$\frac{1}{2}\int u^{-3/2}du = -u^{-1/2} + C = -\frac{1}{\sqrt{x^2+a^2}} + C.$$

This integral gives the field on the axis of a charged ring or disc.

(c) $\int \tan x\,dx = \int \frac{\sin x}{\cos x}dx$: $u = \cos x$, $du = -\sin x\,dx$, giving $-\ln\lvert\cos x\rvert + C$.

(d) $\int \frac{e^x}{1+e^{2x}}dx$: $u = e^x$, $du = e^x dx$, giving $\int \frac{du}{1+u^2} = \arctan(e^x) + C$.

Worked Example 5.27 — Definite integrals with changed limits

(a) $\displaystyle\int_0^1 x\sqrt{1-x^2}\,dx$. Let $u = 1-x^2$, $du = -2x\,dx$; $x=0 \mapsto u=1$, $x=1 \mapsto u=0$:

$$\int_1^0 \sqrt{u}\left(-\frac{du}{2}\right) = \frac{1}{2}\int_0^1 u^{1/2}du = \frac{1}{2}\cdot\frac{2}{3} = \frac{1}{3}.$$

(b) $\displaystyle\int_1^e \frac{(\ln x)^2}{x}dx$. Let $u = \ln x$, $du = dx/x$; limits 0 to 1: $\int_0^1 u^2 du = \tfrac{1}{3}$.

(c) $\displaystyle\int_0^\infty t\,e^{-t^2/2\sigma^2}\,dt$ (a Rayleigh-type integral from fading-channel statistics). Let $u = t^2/2\sigma^2$, $du = t\,dt/\sigma^2$:

$$\sigma^2\int_0^\infty e^{-u}du = \sigma^2.$$

Common Pitfall — Forgetting to change the limits (or the dx)

In a definite integral either change the limits to u-values *or* convert the antiderivative back to x before substituting the original limits — never mix the two. Equally, every x must disappear: if after writing $du = g'(x)dx$ some stray x remains that cannot be expressed in terms of u, the substitution was the wrong one.

5.6.2 Linear substitutions: shifting and scaling

The simplest substitutions, $u = t - t_0$ and $u = at$, are used so often in signal analysis that their effects should be automatic:

$$\int_{-\infty}^{\infty} f(t-t_0)\,dt = \int_{-\infty}^{\infty} f(u)\,du, \qquad \int_{-\infty}^{\infty} f(at)\,dt = \frac{1}{|a|}\int_{-\infty}^{\infty} f(u)\,du \quad (a \neq 0).$$

A time shift does not change the area of a pulse; compressing time by a factor a divides the area by $|a|$. (For $a<0$ the limits swap, which produces the absolute value.) The same scaling argument shows why the Dirac impulse obeys $\delta(at) = \delta(t)/|a|$, and why compressing a signal in time spreads its spectrum (Ch. 16).

For a T-periodic function, the substitution $u = t - c$ proves that the integral over any full period is the same (§5.1.3), and $u = \omega t$ converts integrals over one period to the standard interval $[0, 2\pi]$:

$$\frac{1}{T}\int_0^T g(\omega t)\,dt = \frac{1}{2\pi}\int_0^{2\pi} g(\theta)\,d\theta, \qquad \omega = \frac{2\pi}{T}.$$

This is why averages of sinusoidal waveforms do not depend on frequency.

5.6.3 Symmetry by substitution

The substitution $x \to -x$ proves the even/odd rules of §5.1.3: for odd f, $\int_{-a}^{0} f(x)dx = \int_a^0 f(-u)(-du) = -\int_0^a f(u)du$, so the halves cancel. A less obvious use is the "king's rule" $x \to a+b-x$:

$$\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx,$$

which evaluates, for example, $\int_0^{\pi/2} \frac{\sin x}{\sin x+\cos x}dx$: adding the integral to its transformed self gives $\int_0^{\pi/2} 1\,dx = \frac{\pi}{2}$, so the integral is $\frac{\pi}{4}$.

5.6.4 Trigonometric and hyperbolic substitution

When the integrand contains $\sqrt{a^2-x^2}$, $\sqrt{a^2+x^2}$ or $\sqrt{x^2-a^2}$, an **inverse substitution** $x = g(\theta)$ chosen to turn the radicand into a perfect square removes the root. Here the new variable is introduced through x, and $dx = g'(\theta)\,d\theta$.

Radical	Substitution	Identity used	dx
$\sqrt{a^2-x^2}$	$x = a\sin\theta$, $\lvert\theta\rvert \le \frac{\pi}{2}$	$1-\sin^2 = \cos^2$	$a\cos\theta\,d\theta$
$\sqrt{a^2+x^2}$	$x = a\tan\theta$, $\lvert\theta\rvert < \frac{\pi}{2}$	$1+\tan^2 = \sec^2$	$a\sec^2\theta\,d\theta$
$\sqrt{a^2+x^2}$	$x = a\sinh u$	$1+\sinh^2 = \cosh^2$	$a\cosh u\,du$
$\sqrt{x^2-a^2}$	$x = a\sec\theta$ or $x = a\cosh u$	$\sec^2-1=\tan^2$, $\cosh^2-1=\sinh^2$	

The restricted ranges of θ make the substitution invertible and fix the sign of the square root (e.g. $\cos\theta \ge 0$ for $|\theta| \le \pi/2$).

Worked Example 5.28 — Area of a circle

Show that $\displaystyle\int_{-a}^{a}\sqrt{a^2-x^2}\,dx = \frac{\pi a^2}{2}$.

Let $x = a\sin\theta$; $x = \pm a \mapsto \theta = \pm\frac{\pi}{2}$; $\sqrt{a^2-x^2} = a\cos\theta$; $dx = a\cos\theta\,d\theta$:

$$\int_{-\pi/2}^{\pi/2} a^2\cos^2\theta\,d\theta = a^2\int_{-\pi/2}^{\pi/2}\frac{1+\cos 2\theta}{2}d\theta = a^2\cdot\frac{\pi}{2}.$$

The area of a half-disc, as it must be.

Worked Example 5.29 — Field of a finite line charge

A uniform line charge λ (C/m) lies on the z-axis from $z=-L$ to $z=L$. The radial field at distance r in the mid-plane is

$$E_r = \frac{\lambda r}{4\pi\varepsilon_0}\int_{-L}^{L}\frac{dz}{(r^2+z^2)^{3/2}}.$$

Step 1 — substitute $z = r\tan\theta$, $dz = r\sec^2\theta\,d\theta$, $r^2+z^2 = r^2\sec^2\theta$:

$$\int \frac{r\sec^2\theta\,d\theta}{r^3\sec^3\theta} = \frac{1}{r^2}\int\cos\theta\,d\theta = \frac{\sin\theta}{r^2}.$$

Step 2 — back-substitute. From the triangle with opposite side z and adjacent side r: $\sin\theta = \dfrac{z}{\sqrt{r^2+z^2}}$. So

$$\int_{-L}^{L}\frac{dz}{(r^2+z^2)^{3/2}} = \frac{2L}{r^2\sqrt{r^2+L^2}}, \qquad E_r = \frac{\lambda}{4\pi\varepsilon_0}\cdot\frac{2L}{r\sqrt{r^2+L^2}}.$$

Step 3 — limits. As $L\to\infty$ (an improper integral), $E_r \to \dfrac{\lambda}{2\pi\varepsilon_0 r}$, the infinite-line result. For $L \ll r$, $E_r \approx \dfrac{2L\lambda}{4\pi\varepsilon_0 r^2}$ — the field of a point charge $Q = 2L\lambda$, as it should be.

Check: with $r=1$, $L=2$, the integral is $4/\sqrt{5} = 1.78885$; quadrature gives 1.788854 ✓.

Worked Example 5.30 — A hyperbolic substitution

$\displaystyle\int \frac{dx}{\sqrt{x^2+a^2}}$. Let $x = a\sinh u$, $dx = a\cosh u\,du$, $\sqrt{x^2+a^2} = a\cosh u$:

$$\int du = u + C = \operatorname{arsinh}\frac{x}{a} + C = \ln\left(x+\sqrt{x^2+a^2}\right) + C',$$

the table entry of §5.1.4. The hyperbolic route avoids the awkward $\int\sec\theta\,d\theta$ that the tangent substitution produces. The same integral gives the capacitance per length of a two-wire line, $C' = \pi\varepsilon/\operatorname{arcosh}(D/d)$, where an inverse hyperbolic function appears for exactly this reason.

5.6.5 The tangent half-angle substitution

Any rational function of $\sin\theta$ and $\cos\theta$ becomes a rational function of t under

$$t = \tan\frac{\theta}{2}, \qquad \sin\theta = \frac{2t}{1+t^2}, \qquad \cos\theta = \frac{1-t^2}{1+t^2}, \qquad d\theta = \frac{2\,dt}{1+t^2},$$

after which partial fractions (§5.2.5) finish the job. It is a last resort — often clumsy — but it always works.

Worked Example 5.31 — An integral over a full period

Evaluate $\displaystyle I = \int_0^{2\pi}\frac{d\theta}{2+\cos\theta}$.

Step 1 — avoid the singular point. $t = \tan(\theta/2)$ blows up at $\theta=\pi$, so use symmetry (cos is even about π): $I = 2\int_0^{\pi}\frac{d\theta}{2+\cos\theta}$, and $\theta: 0\to\pi$ maps to $t: 0\to\infty$.

Step 2 — transform. $2+\cos\theta = \dfrac{2(1+t^2)+1-t^2}{1+t^2} = \dfrac{3+t^2}{1+t^2}$, so

$$\frac{d\theta}{2+\cos\theta} = \frac{1+t^2}{3+t^2}\cdot\frac{2\,dt}{1+t^2} = \frac{2\,dt}{3+t^2}.$$

Step 3 — integrate.

$$I = 2\int_0^{\infty}\frac{2\,dt}{t^2+3} = 4\cdot\frac{1}{\sqrt{3}}\cdot\frac{\pi}{2} = \frac{2\pi}{\sqrt{3}} \approx 3.6276.$$

Check: a 64-point trapezoidal sum over the period gives 3.627599 — and for smooth periodic integrands the trapezoidal rule is spectacularly accurate (§5.7.5). Integrals of this form are evaluated in one line by residues in Ch. 18.

Engineering Note — Substitution as change of physical variable

Engineers substitute all the time without calling it that: converting an integral over time to one over phase ($\theta=\omega t$), over frequency to one over normalised frequency ($u=\omega/\omega_c$), or over distance to one over angle as in Example 5.29. Normalising first — making the variable dimensionless — exposes the shape of the integral and leaves the physical constants outside as a prefactor whose units are easy to check.

5.7 Numerical integration

5.7.1 Why numerical integration

Three situations force us to integrate numerically:

1. **No elementary antiderivative exists** — $\int_0^1 e^{-x^2}dx$, the bit-error probability of a digital link, the inductance of a real coil.

2. **The integrand is known only at samples** — an oscilloscope or ADC record of current, voltage or temperature; the energy of a measured pulse is $\sum$ of samples, not a formula.
3. **A closed form exists but is not worth deriving** — when only a number is required, a few lines of code beat a page of algebra.

All practical methods are **quadrature rules** of the form

$$\int_a^b f(x)\,dx \approx \sum_{k=0}^{n} w_k\, f(x_k),$$

a weighted sum of samples at **nodes** x_k with **weights** w_k. This section develops the three basic Newton–Cotes rules on equally spaced nodes and their error behaviour. Chapter 22 continues with Romberg integration, Gaussian quadrature, adaptive schemes and multidimensional methods.

Throughout, $h = (b-a)/n$ and $x_k = a + kh$, $f_k = f(x_k)$.

5.7.2 The midpoint and trapezoidal rules

The **midpoint rule** replaces f on each subinterval by its value at the centre; the **trapezoidal rule** replaces it by the straight line through the end values:

$$M_n = h\sum_{k=1}^{n} f\Big(x_{k-1} + \frac{h}{2}\Big), \qquad T_n = h\Big[\tfrac12 f_0 + f_1 + f_2 + \cdots + f_{n-1} + \tfrac12 f_n\Big].$$

Each trapezoid has area $\frac{h}{2}(f_{k-1}+f_k)$; summing them gives T_n, in which interior samples appear twice and therefore get weight h.

Error of the midpoint and trapezoidal rules

If f'' is continuous on $[a,b]$, there exist $\xi, \eta \in (a,b)$ with

$$\int_a^b f\,dx - T_n = -\frac{(b-a)h^2}{12} f''(\xi), \qquad \int_a^b f\,dx - M_n = +\frac{(b-a)h^2}{24} f''(\eta).$$

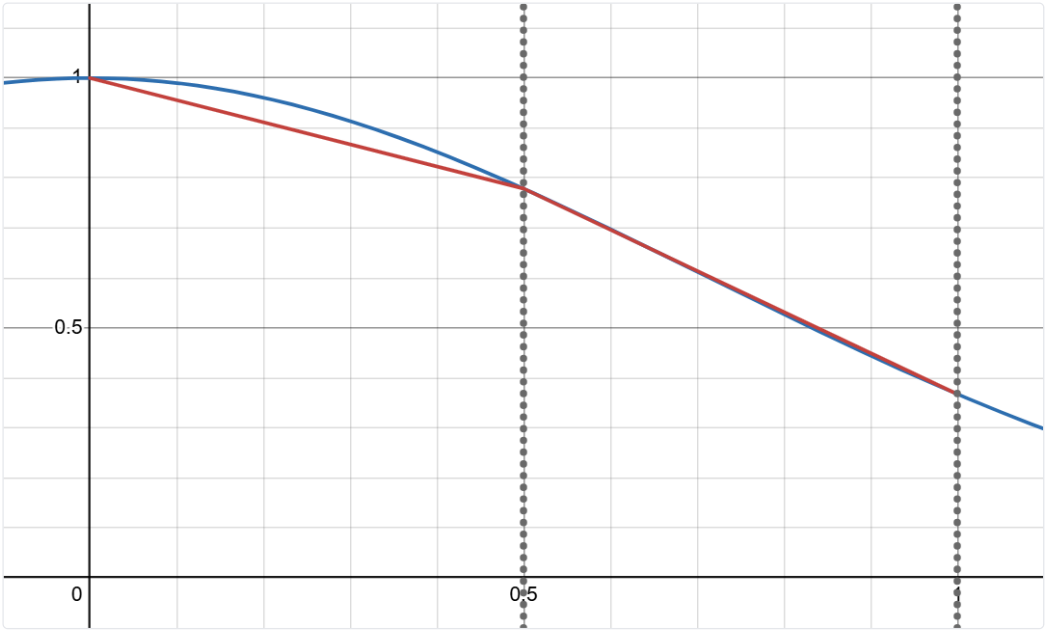
Both rules are **second order**: halving h divides the error by about 4. Both are exact for straight lines.

Derivation sketch (single panel). Expand f about the panel midpoint m with $s = x - m$: $f = f(m) + sf'(m) + \tfrac12 s^2 f''(m) + \dots$. Integrating over $s \in [-h/2, h/2]$, the odd term vanishes:

$$\int f\,dx = hf(m) + \frac{h^3}{24} f''(m) + O(h^5).$$

So the midpoint rule has local error $+\frac{h^3}{24}f''$. For the trapezoid, average the endpoint expansions $f(m \pm h/2) = f(m) \pm \frac{h}{2}f' + \frac{h^2}{8}f'' + \dots$: $\frac{h}{2}(f_{\text{left}} + f_{\text{right}}) = hf(m) + \frac{h^3}{8}f'' + \dots$, so its local error is $\frac{h^3}{24}f'' - \frac{h^3}{8}f'' = -\frac{h^3}{12}f''$. Summing $n = (b-a)/h$ panels multiplies the local error by n, giving global error $\propto (b-a)h^2$. $\square$

Two practical consequences follow. First, the midpoint error is about half the trapezoidal error and of **opposite sign**, so for a convex function ($f'' > 0$) the trapezoid overestimates and the midpoint underestimates, bracketing the true value. Second, the weighted combination $\frac{1}{3}(2M_n + T_n)$ cancels the leading error — that combination is Simpson's rule.



Trapezoidal rule with $n = 2$ for e^{-x^2} on $[0,1]$: the chords (red) lie below the concave part of the curve and above the convex tail, so the errors partly cancel.

5.7.3 Simpson's rule

Fitting a **parabola** through three consecutive samples and integrating it exactly gives, over a pair of panels,

$$\int_{x_0}^{x_2} f\,dx \approx \frac{h}{3}\bigl(f_0 + 4f_1 + f_2\bigr).$$

Derivation. Put $x_1 = 0$, so the nodes are $-h, 0, h$. For $p(x) = \alpha + \beta x + \gamma x^2$, $\int_{-h}^{h} p\,dx = 2\alpha h + \tfrac{2}{3}\gamma h^3$. From the samples, $\alpha = f_1$ and $f_0 + f_2 = 2\alpha + 2\gamma h^2$, so $\gamma h^2 = \frac{f_0+f_2}{2} - f_1$. Substituting: $2hf_1 + \frac{2h}{3}\bigl(\frac{f_0+f_2}{2} - f_1\bigr) = \frac{h}{3}(f_0 + 4f_1 + f_2)$. $\square$

Applying this to consecutive pairs requires n **even** and gives the composite rule

$$S_n = \frac{h}{3}\Bigl[f_0 + 4f_1 + 2f_2 + 4f_3 + 2f_4 + \cdots + 2f_{n-2} + 4f_{n-1} + f_n\Bigr].$$

Error of Simpson's rule

If $f^{(4)}$ is continuous on $[a,b]$ and n is even,

$$\int_a^b f\,dx - S_n = -\frac{(b-a)h^4}{180} f^{(4)}(\xi)$$

for some $\xi \in (a,b)$. Simpson's rule is **fourth order** — halving h divides the error by about 16 — and, surprisingly, exact for **cubics**, not just quadratics (the cubic term is odd about the panel centre and integrates to zero).

Worked Example 5.32 — Comparing the rules on $\int_0^1 e^{-x^2}dx$

The exact value is $\frac{\sqrt{\pi}}{2}\operatorname{erf}(1) = 0.746824133$. With $n = 4$, $h = 0.25$, the samples are

x	0	0.25	0.5	0.75	1
f	1.000000	0.939413	0.778801	0.569783	0.367879

Trapezoidal: $T_4 = 0.25\bigl[\tfrac{1}{2}(1 + 0.367879) + 0.939413 + 0.778801 + 0.569783\bigr] = 0.742984$.

Simpson: $S_4 = \frac{0.25}{3}\bigl[1 + 4(0.939413) + 2(0.778801) + 4(0.569783) + 0.367879\bigr] = 0.746855$.

Midpoint (samples at $0.125, 0.375, 0.625, 0.875$): $M_4 = 0.748747$.

Rule	$n=4$	error	$n=8$	error	ratio
Trapezoidal	0.742984	-3.84×10^{-3}	0.745866	-9.59×10^{-4}	4.0
Midpoint	0.748747	$+1.92\times10^{-3}$	0.747304	$+4.79\times10^{-4}$	4.0
Simpson	0.746855	$+3.12\times10^{-5}$	0.746826	$+1.99\times10^{-6}$	15.7

The error ratios on halving h — 4 and 16 — confirm the orders h^2 and h^4. Simpson with five samples beats the trapezoidal rule with nine by a factor of about 30. Also note M and T bracket the true value with $|E_M| \approx \tfrac{1}{2}|E_T|$.

5.7.4 Choosing n from an error bound

The error formulas give a guaranteed step size when a bound on the relevant derivative is available.

Worked Example 5.33 — How many samples for six decimals?

Find n so that the error in $\int_0^1 e^{-x^2}dx$ is below 10^{-6}.

Trapezoidal. $f'' = (4x^2 - 2)e^{-x^2}$; on $[0,1]$, $|f''| \le 2$ (at $x = 0$). Need $\frac{1\cdot h^2}{12}\cdot 2 \le 10^{-6}$, i.e. $h \le 2.449\times10^{-3}$, so $n \ge 409$. (Actual error at $n = 410$: 3.6×10^{-7}.)

Simpson. $f^{(4)} = (16x^4 - 48x^2 + 12)e^{-x^2}$; $|f^{(4)}| \le 12$ on $[0,1]$. Need $\frac{h^4}{180}\cdot 12 \le 10^{-6}$, i.e. $h \le 0.0622$, so $n \ge 16.07$; take $n = 18$ (even). (Actual error: 7.8×10^{-8}.)

The bounds are conservative — the worst-case derivative occurs only at one end — but they are *guarantees*. Twenty-odd samples for Simpson versus four hundred for the trapezoidal rule is the practical meaning of "fourth order versus second order".

5.7.5 Error estimation without derivatives: Richardson extrapolation

Derivative bounds are rarely available for measured data or complicated formulas. Instead, compute with h and $h/2$ and use the known order. If $I = T_n + Ch^2 + O(h^4)$, then $I = T_{2n} + C\frac{h^2}{4} + O(h^4)$, and eliminating C:

$$I \approx T_{2n} + \frac{T_{2n} - T_n}{3}, \qquad \text{error of } T_{2n} \approx \frac{T_{2n} - T_n}{3}.$$

From Example 5.32, $\frac{T_8 - T_4}{3} = \frac{0.745866 - 0.742984}{3} = 9.61\times10^{-4}$, an excellent estimate of the actual error 9.59×10^{-4}; the extrapolated value 0.746826 is identical to S_8 — Richardson extrapolation of the trapezoidal rule *is* Simpson's rule. Repeating the idea to eliminate $h^4, h^6, \ldots$ gives Romberg integration (Ch. 22).

Engineering Note — The trapezoidal rule and periodic signals

For a smooth **periodic** function integrated over a whole period, the trapezoidal rule is not second order but converges *exponentially* fast: the error terms of the Euler–Maclaurin expansion are differences of derivatives at the two ends, which cancel by periodicity. Example 5.31's 64-point sum was accurate to 15 digits. This is why the DFT — essentially the trapezoidal rule applied to $\frac{1}{T}\int_0^T x(t)e^{-jk\omega_0 t}dt$ — computes Fourier coefficients of band-limited periodic signals exactly, and why sampled RMS and power meters work so well on steady-state waveforms.

5.7.6 Sampled data

Worked Example 5.34 — Energy of a measured pulse

An ADC samples the current through a 10 Ω resistor every 1 ms:

t (ms)	0	1	2	3	4	5	6
i (A)	0	2.1	3.4	3.9	3.6	2.5	1.0
$p = Ri^2$ (W)	0	44.1	115.6	152.1	129.6	62.5	10.0

Estimate the energy delivered, $W = \int_0^{6\,\text{ms}} p\,dt$.

Trapezoidal: $W \approx 10^{-3}\left[\frac{1}{2}(0+10) + 44.1 + 115.6 + 152.1 + 129.6 + 62.5\right] = 0.5089\ \text{J}$.

Simpson ($n = 6$, even): $W \approx \frac{10^{-3}}{3}\left[0 + 4(44.1 + 152.1 + 62.5) + 2(115.6 + 129.6) + 10\right] = 0.5117\ \text{J}$.

The 0.5% difference indicates the sampling is adequate for an energy estimate to about two significant figures. Note that we integrated $p = Ri^2$, **not** $R\,(\int i\,dt)^2$ or $R\,(\bar{i})^2$ — the square must be taken sample by sample before integrating (§5.9).

5.7.7 Difficult integrands

The error formulas assume smooth integrands. When that fails:

- **Discontinuities or kinks** (rectified waves, PWM): place nodes at the breakpoints and integrate each smooth piece separately; otherwise the error drops only like h.
- **Integrable singularities** such as $\int_0^1 \frac{\cos x}{\sqrt{x}}dx$: remove the singularity by substitution ($x = u^2$ gives $\int_0^1 2\cos(u^2)\,du$, which is smooth), or subtract it off analytically.
- **Infinite ranges:** truncate where the tail is provably small (bound it by a comparison integral, §5.3.4), or map to a finite interval with a substitution such as $x = \frac{u}{1-u}$.
- **Rapid oscillation** ($\int f(x)\cos(\omega x)\,dx$ with large ω): ensure several samples per oscillation period, or use specialised (Filon-type, FFT-based) methods.

Common Pitfall — Simpson's rule needs an even number of panels

With an odd number of panels the composite Simpson pattern $1, 4, 2, 4, \ldots, 4, 1$ does not close. Either use Simpson's $\frac{3}{8}$ rule, $\frac{3h}{8}(f_0 + 3f_1 + 3f_2 + f_3)$, on the last three panels, or resample. Silently applying the even-n formula to odd n gives an answer that can be badly wrong.

5.8 Applications to charge, energy, and accumulated quantities

5.8.1 The accumulation principle

Every application in this section follows one pattern, worth stating explicitly because it is how integrals are *set up* rather than evaluated:

The differential-element method

To find a total Q that is built from a varying density or rate:

1. **Slice** the domain (time, length, area, volume) into small elements dt, dx, dA or dV.
2. On one element the density is essentially constant, so write the **contribution** $dQ = (\text{density}) \times (\text{size of element})$.
3. **Integrate** dQ over the domain, choosing limits that cover each element exactly once.
4. **Check** units, limiting cases and signs.

The principle covers charge from current ($dq = i\,dt$), charge from charge density ($dQ = \lambda\,d\ell$, $\sigma\,dA$, $\rho\,dV$), energy from power ($dW = p\,dt$), work from force ($dW = F\,dx$), resistance of a non-uniform conductor ($dR = \rho_e\,dx/A(x)$), flux from flux density ($d\Phi = B\,dA$), and probability from a density ($dP = f_X(x)\,dx$, Ch. 19). The table summarises the most common cases.

Accumulated quantity	Element	Integral	Units check
Charge from current	$dq = i\,dt$	$q = \int i\,dt$	A·s = C
Charge from line density	$dQ = \lambda\,d\ell$	$Q = \int \lambda\,d\ell$	(C/m)·m
Charge from surface density	$dQ = \sigma\,dA$	$Q = \iint \sigma\,dA$	(C/m²)·m²
Charge from volume density	$dQ = \rho\,dV$	$Q = \iiint \rho\,dV$	(C/m³)·m³
Energy from power	$dW = p\,dt$	$W = \int p\,dt$	W·s = J

Accumulated quantity	Element	Integral	Units check
Work against a force	$dW = F\,dx$	$W = \int F\,dx$	N·m = J
Flux linkage from voltage	$d\lambda = v\,dt$	$\lambda = \int v\,dt$	V·s = Wb
Resistance, non-uniform bar	$dR = \rho_e\,dx/A(x)$	$R = \int \rho_e\,dx/A$	(Ω·m)·m/m² = Ω
Battery charge	$dq = i\,dt$	$q = \int i\,dt$	A·h = 3600 C

5.8.2 Charge from current

Since $i = dq/dt$, the FTC gives

$$q(t) = q(t_0) + \int_{t_0}^{t} i(\tau)\,d\tau,$$

and the **net** charge passing a point between t_1 and t_2 is $\int_{t_1}^{t_2} i\,dt$, which may be zero even when large currents flow (pure AC). The **total** charge moved irrespective of direction is $\int |i|\,dt$ — the quantity that matters for electroplating, electrode wear and battery throughput.

Worked Example 5.35 — Charge in a half-cycle

A diode passes the positive half-cycles of $i(t) = I_m \sin \omega t$ with $I_m = 10$ A at $f = 50$ Hz. Find the charge passed per half-cycle and the average current.

Step 1 — integrate over one half-cycle $0 \le t \le \pi/\omega$:

$$Q = \int_0^{\pi/\omega} I_m \sin \omega t\,dt = \frac{I_m}{\omega}\big[-\cos \omega t\big]_0^{\pi/\omega} = \frac{2I_m}{\omega}.$$

Step 2 — numbers. $\omega = 100\pi$ rad/s: $Q = \dfrac{20}{100\pi} = 0.06366$ C $= 63.66$ mC.

Step 3 — average over a full period $T = 20$ ms (the diode blocks during the negative half):

$$I_{\text{avg}} = \frac{Q}{T} = \frac{0.06366}{0.02} = 3.183 \text{ A} = \frac{I_m}{\pi}.$$

This is the DC component of a half-wave rectified current (§5.9).

Worked Example 5.36 — Battery capacity under a load profile

A sensor node draws $i(t) = 20$ mA for 5 s while transmitting, then 50 μA for the remaining 595 s of each 10-minute cycle. How long does a 2000 mAh battery last (ignoring self-discharge)?

Charge per cycle: $q = 20\,\text{mA} \times 5\,\text{s} + 0.05\,\text{mA} \times 595\,\text{s} = 100 + 29.75 = 129.75$ mA·s.

Average current: $\bar{i} = 129.75/600 = 0.21625$ mA.

Lifetime: $2000\ \text{mAh}/0.21625\ \text{mA} = 9249$ h ≈ 385 days.

The integral of a piecewise-constant current is a sum of rectangles; the insight is that the "small" sleep current contributes almost a quarter of the total charge because it flows 119 times longer.

5.8.3 Energy stored in capacitors and inductors

The power into any two-terminal element is $p = vi$, so the energy delivered between t_1 and t_2 is $W = \int_{t_1}^{t_2} v\,i\,dt$. For a capacitor, $i = C\,dv/dt$; for an inductor, $v = L\,di/dt$. Substituting and changing the variable of integration (§5.6) gives two of the most-used formulas in electrical engineering.

Stored energy

A linear capacitor charged from 0 to V and a linear inductor carrying current rising from 0 to I store

$$W_C = \int_0^t v\,C\frac{dv}{dt}\,dt = C\int_0^V v\,dv = \tfrac{1}{2}CV^2, \qquad W_L = \int_0^t L\frac{di}{dt}\,i\,dt = L\int_0^I i\,di = \tfrac{1}{2}LI^2.$$

The substitution $dv = \frac{dv}{dt}dt$ removes time entirely: the stored energy depends only on the final state, not on how fast or by what path it was reached. That is the mathematical signature of a **conservative** (lossless) element.

For a *nonlinear* capacitor described by $q = q(v)$, the same argument gives $W = \int_0^V v\,\frac{dq}{dv}\,dv = \int_0^Q v\,dq$, the area between the q–v curve and the q-axis. Similarly, for a nonlinear inductor (iron core) $W = \int i\,d\lambda$, and over one cycle of an AC magnetisation the loop integral $\oint i\,d\lambda$ is the **hysteresis loss** per cycle — the area of the B–H loop scaled by core volume.

Worked Example 5.37 — Charging a capacitor through a resistor: where does the energy go?

A capacitor C is charged from 0 through R by a step V_s. The current is $i = \frac{V_s}{R} e^{-t/RC}$.

Energy supplied by the source:

$$W_s = \int_0^\infty V_s\, i\, dt = \frac{V_s^2}{R} \int_0^\infty e^{-t/RC} dt = \frac{V_s^2}{R} \cdot RC = CV_s^2.$$

Energy lost in the resistor:

$$W_R = \int_0^\infty i^2 R\, dt = \frac{V_s^2}{R} \int_0^\infty e^{-2t/RC} dt = \tfrac{1}{2} CV_s^2.$$

Energy stored: $W_C = \frac{1}{2}CV_s^2$, and $W_s = W_R + W_C$ ✓.

Exactly half the source energy is always lost, whatever R is. This is a fundamental limit of charging a capacitor from a fixed voltage, and the reason switched-capacitor and adiabatic logic circuits charge capacitances in small voltage steps: charging in N equal steps reduces the loss to $\frac{1}{2N}CV_s^2$.

Worked Example 5.38 — Energy into an inductor with a non-trivial waveform

A 2 mH inductor carries $i(t) = 3t^2$ A for $0 \le t \le 2$ s (an unusually slow ramp, for clarity). Compute the energy delivered by integrating power, then check against $\frac{1}{2}LI^2$.

$v = L\, di/dt = 2 \times 10^{-3} \cdot 6t = 0.012t$ V, so $p = vi = 0.012t \cdot 3t^2 = 0.036t^3$ W and

$$W = \int_0^2 0.036 t^3 dt = 0.036 \cdot \frac{16}{4} = 0.144 \text{ J}.$$

Final current $I = 3 \cdot 4 = 12$ A: $\frac{1}{2}LI^2 = \frac{1}{2} \cdot 0.002 \cdot 144 = 0.144$ J ✓.

5.8.4 Work and potential

Moving a charge q against an electric force requires work $W = -\int \mathbf{F} \cdot d\boldsymbol{\ell}$; along a radial line this is a single-variable integral. The **potential** of a point is the work per unit charge to bring a test charge from a reference point (usually infinity), so potential is an *improper integral* of field.

Worked Example 5.39 — Potential of a point charge

The field of a point charge Q is $E(R) = \dfrac{Q}{4\pi\varepsilon_0 R^2}$, radial. The potential at distance a relative to infinity is

$$V(a) = -\int_\infty^a E(R)\, dR = \frac{Q}{4\pi\varepsilon_0} \int_a^\infty \frac{dR}{R^2} = \frac{Q}{4\pi\varepsilon_0} \Big[-\frac{1}{R} \Big]_a^\infty = \frac{Q}{4\pi\varepsilon_0 a}.$$

The integral converges because $1/R^2$ has $p = 2 > 1$ (§5.3.3). In two dimensions the field of a line charge falls only as $1/r$, and the analogous integral **diverges** — which is why the potential of an infinite line charge can only be defined relative to a finite reference radius r_0: $V(r) = \frac{\lambda}{2\pi\varepsilon_0} \ln \frac{r_0}{r}$.

5.8.5 Charge in non-uniform distributions

When a density depends on one coordinate only, a well-chosen element reduces a multiple integral to a single one: thin strips for a planar density varying in one direction, thin **rings** ($dA = 2\pi r\, dr$) for a disc, thin **shells** ($dV = 4\pi R^2 dR$) for a sphere, thin **cylindrical shells** ($dV = 2\pi r \ell\, dr$) for a cylinder of length ℓ.

Worked Example 5.40 — Charge in a sphere with radially varying density

A sphere of radius a holds $\rho(R) = \rho_0 \big(1 - \frac{R}{a}\big)$. Find the total charge and the radius $R_{1/2}$ enclosing half of it.

Step 1 — shell element. $dQ = \rho(R)\, 4\pi R^2 dR$.

Step 2 — integrate.

$$Q = 4\pi\rho_0 \int_0^a \Big(R^2 - \frac{R^3}{a} \Big) dR = 4\pi\rho_0 \Big(\frac{a^3}{3} - \frac{a^3}{4} \Big) = \frac{\pi \rho_0 a^3}{3}.$$

Step 3 — enclosed charge as a function of radius. $Q(R) = 4\pi\rho_0 \big(\frac{R^3}{3} - \frac{R^4}{4a} \big)$. Setting $Q(R) = \frac{Q}{2}$ with $s = R/a$: $4\big(\frac{s^3}{3} - \frac{s^4}{4}\big) = \frac{1}{6}$, i.e. $3s^4 - 4s^3 + \frac{1}{2} = 0$. Newton's method (Ch. 4) gives $s \approx 0.614$.

Check. A uniform sphere with ρ_0 would hold $\frac{4}{3}\pi\rho_0 a^3$; ours holds a quarter of that, since the density averages to $\rho_0/4$ when weighted by R^2. Gauss's law then gives the field at any radius as $E = Q(R)/(4\pi\varepsilon_0 R^2)$ (Ch. 7).

Worked Example 5.41 — Energy stored in a coaxial cable

A coaxial line with inner radius a, outer radius b and dielectric permittivity ε carries charge λ per unit length. The field between the conductors is $E(r) = \dfrac{\lambda}{2\pi\varepsilon r}$, and the energy density is $w = \tfrac{1}{2}\varepsilon E^2$ (J/m³). Find the stored energy per unit length and deduce the capacitance per unit length.

Step 1 — element. A cylindrical shell of radius r, thickness dr and unit length has volume $2\pi r\,dr$.

Step 2 — integrate.

$$W' = \int_a^b \frac{1}{2}\varepsilon\Big(\frac{\lambda}{2\pi\varepsilon r}\Big)^2 2\pi r\,dr = \frac{\lambda^2}{4\pi\varepsilon}\int_a^b \frac{dr}{r} = \frac{\lambda^2}{4\pi\varepsilon}\ln\frac{b}{a}.$$

Step 3 — capacitance. Equating $W' = \dfrac{\lambda^2}{2C'}$ (the energy of a capacitor with charge λ per length) gives

$$C' = \frac{2\pi\varepsilon}{\ln(b/a)}.$$

For RG-58-like dimensions, $b/a = 3.5$, polyethylene $\varepsilon = 2.25\varepsilon_0$: $C' = \dfrac{2\pi(2.25)(8.854\times10^{-12})}{\ln 3.5} = 99.9$ pF/m, close to the datasheet value of about 100 pF/m. The logarithm comes directly from $\int dr/r$ — the borderline $p = 1$ integrand of §5.3.

5.8.6 Resistance of a non-uniform conductor

A conductor whose cross-section $A(x)$ varies slowly along its length behaves as a series chain of thin slices, each with resistance $dR = \rho_e\,dx/A(x)$. Series resistances add, so

$$R = \int_0^L \frac{\rho_e\,dx}{A(x)}.$$

Worked Example 5.42 — A tapered wire

A wire of resistivity ρ_e and length L tapers linearly in radius from a to b. Find R.

Step 1 — area. $r(x) = a + \frac{(b-a)x}{L}$, $A = \pi r^2$.

Step 2 — substitute $u = r(x)$, $dx = \frac{L}{b-a}du$:

$$R = \frac{\rho_e}{\pi}\cdot\frac{L}{b-a}\int_a^b \frac{du}{u^2} = \frac{\rho_e L}{\pi(b-a)}\Big(\frac{1}{a}-\frac{1}{b}\Big) = \frac{\rho_e L}{\pi ab}.$$

Check. For $b \to a$ this reduces to $\rho_e L/(\pi a^2)$ ✓. Note that the answer uses the *geometric* mean of the end areas, πab, not the arithmetic mean — averaging the area first and then computing $\rho_e L/\bar{A}$ underestimates R. (Copper, $\rho_e = 1.72\times10^{-8}$ Ω·m, $L = 1$ m, $a = 0.5$ mm, $b = 1$ mm: $R = 10.95$ mΩ.)

Common Pitfall — Averaging the wrong quantity

Examples 5.42 and 5.34 make the same point: the integral of a nonlinear function of a variable is not the function of the integral. $\int \frac{dx}{A} \neq \frac{L^2}{\int A\,dx}$ and $\int i^2 dt \neq \frac{1}{T}\big(\int i\,dt\big)^2$. Always build dQ from the local physics and integrate *that*.

Engineering Note — Accumulated quantities in instrumentation

Energy meters, coulomb counters (battery "fuel gauges"), and flux meters are hardware integrators. Digital versions accumulate samples, which is a rectangular or trapezoidal rule (§5.7). Their main error sources are exactly those of numerical integration — sampling too slowly for the waveform's bandwidth — plus offset drift (§5.1 Engineering Note), which an integrator turns into an ever-growing error. Coulomb counters therefore periodically re-calibrate against a known state such as "fully charged".

5.9 Mean and RMS values of functions

5.9.1 Motivation

A mains socket is "230 V", a USB charger "5 V", a class-D amplifier output "20 V RMS". These single numbers summarise waveforms that vary continuously in time. Two summaries dominate engineering practice: the **mean** (average, DC) value, which tells how much net charge or flux a waveform delivers, and the **root-mean-square (RMS)** value, which tells how much *power* it delivers to a resistor. Both are integrals, and both appear on every multimeter, datasheet and power-supply specification.

5.9.2 The mean value

Mean (average) value

The mean value of f over $[a,b]$ is

$$\bar f = \langle f \rangle = \frac{1}{b-a}\int_a^b f(t)\,dt.$$

For a T-periodic signal the **DC value** is the mean over one period, $V_{\text{dc}} = \dfrac{1}{T}\displaystyle\int_{t_0}^{t_0+T} v(t)\,dt$ (any t_0). For a non-periodic signal on $(-\infty,\infty)$ the **time average** is $\displaystyle\lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T} v(t)\,dt$ when the limit exists.

Geometrically $\bar f$ is the height of the rectangle on $[a,b]$ with the same area as f — the Mean Value Theorem for integrals (§5.1.3) guarantees that a continuous f actually attains it. Physically, the mean current $\bar i$ is the steady current that would carry the same charge in the same time (Example 5.35).

Basic properties follow from linearity: $\langle \alpha f + \beta g\rangle = \alpha\langle f\rangle + \beta\langle g\rangle$; the mean of a constant is itself; and the mean of any pure sinusoid over a whole number of periods is **zero**. Hence a signal $v(t) = V_0 + \sum_{n} c_n\cos(n\omega_0 t + \theta_n)$ has DC value exactly V_0. A capacitor in series blocks precisely this term; an ideal inductor in steady state has zero *mean* voltage (otherwise its current would grow without bound, since $\Delta i = \frac{1}{L}\int v\,dt$), and a capacitor in steady state has zero *mean* current. These "volt-second balance" and "charge balance" conditions are the main analytic tools for switched-mode power converters.

Worked Example 5.43 — Volt-second balance in a buck converter

In a buck converter the inductor voltage is $V_{\text{in}} - V_o$ for a fraction D of each switching period T and $-V_o$ for the rest. In steady state the mean inductor voltage is zero:

$$\frac{1}{T}\Big[(V_{\text{in}} - V_o)DT + (-V_o)(1-D)T\Big] = 0 \quad\Longrightarrow\quad V_o = D V_{\text{in}}.$$

With $V_{\text{in}} = 12$ V and $D = 0.25$, $V_o = 3$ V. One integral — a sum of two rectangles — gives the converter's fundamental conversion ratio.

5.9.3 The RMS value

Feed a waveform $v(t)$ into a resistor R. The instantaneous power is v^2/R and the mean power over a period is $\frac{1}{T}\int_0^T \frac{v^2}{R}dt$. The **effective** or RMS voltage is the DC voltage that would produce the same mean power:

$$\frac{V_{\text{rms}}^2}{R} = \frac{1}{T}\int_0^T \frac{v(t)^2}{R}\,dt.$$

RMS value

$$F_{\text{rms}} = \sqrt{\frac{1}{b-a}\int_a^b f(t)^2\,dt} = \sqrt{\langle f^2\rangle},$$

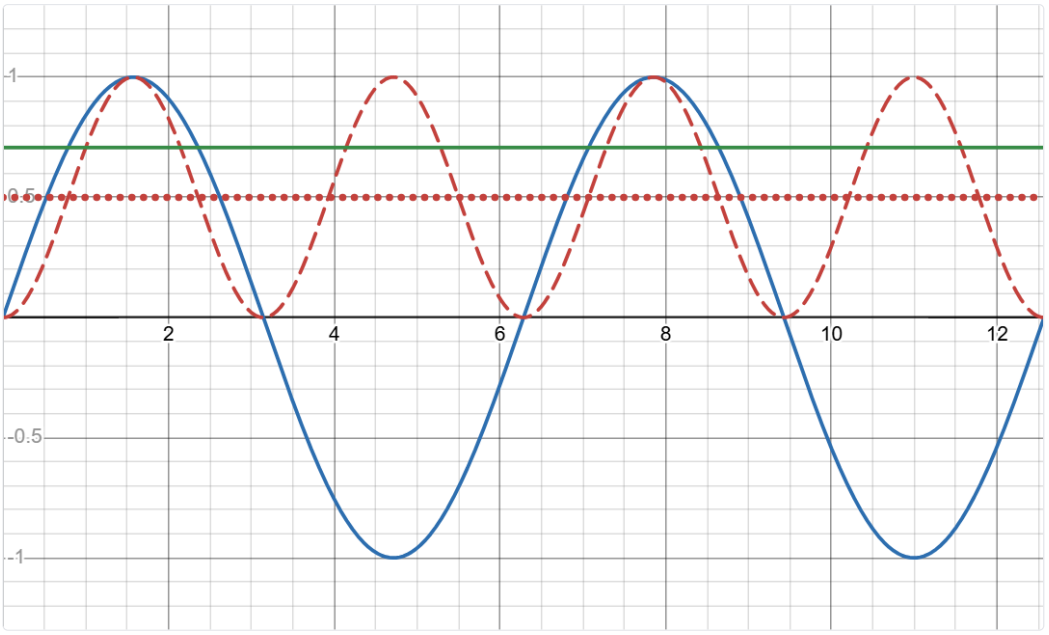
literally the **root** of the **mean** of the **square**. For periodic signals take $[a,b]$ as one period. For complex-valued signals use $|f|^2$.

Since $f^2 \ge 0$, the RMS value is always non-negative, is zero only for a signal that is zero almost everywhere, and satisfies $F_{\text{rms}} \ge |\bar f|$ (because $\langle (f-\bar f)^2\rangle \ge 0$ expands to $\langle f^2\rangle - \bar f^2 \ge 0$). Dimensionally, F_{rms} has the same units as f.

RMS of a sinusoid. For $v = V_m\sin(\omega t + \theta)$, Example 5.6(c) gives $\frac{1}{T}\int_0^T \sin^2(\omega t+\theta)\,dt = \frac{1}{2}$, so

$$V_{\text{rms}} = \frac{V_m}{\sqrt{2}} \approx 0.7071\,V_m,$$

independent of frequency and phase. A "230 V" mains supply therefore has peak $230\sqrt{2} = 325$ V.



sin t (blue), its square (red, dashed) with mean $1/2$ (red, dotted), and the RMS level $1/\sqrt{2} \approx 0.707$ (green).

5.9.4 Mean and RMS of standard waveforms

Every entry below is a short integral over one period; amplitude V_m means the peak value.

Waveform	Mean V_{dc}	Mean of $\lvert v\rvert$	RMS	Crest factor V_m/V_{rms}
Sine $V_m \sin \omega t$	0	$\dfrac{2V_m}{\pi}$	$\dfrac{V_m}{\sqrt{2}}$	$\sqrt{2} = 1.414$
Full-wave rectified sine $V_m \lvert\sin \omega t\rvert$	$\dfrac{2V_m}{\pi}$	$\dfrac{2V_m}{\pi}$	$\dfrac{V_m}{\sqrt{2}}$	1.414
Half-wave rectified sine	$\dfrac{V_m}{\pi}$	$\dfrac{V_m}{\pi}$	$\dfrac{V_m}{2}$	2
Square wave $\pm V_m$	0	V_m	V_m	1
Triangle or sawtooth, $\pm V_m$	0	$\dfrac{V_m}{2}$	$\dfrac{V_m}{\sqrt{3}}$	$\sqrt{3} = 1.732$
Pulse train, 0 to V_m, duty D	DV_m	DV_m	$\sqrt{D}\,V_m$	$\dfrac{1}{\sqrt{D}}$

Worked Example 5.44 — Deriving three table entries

(a) **Half-wave rectified sine.** Over $T = 2\pi/\omega$, $v = V_m \sin \omega t$ for $0 < t < T/2$ and 0 otherwise.

$$V_{\text{dc}} = \frac{1}{T}\int_0^{T/2} V_m \sin \omega t \, dt = \frac{1}{T}\cdot\frac{2V_m}{\omega} = \frac{V_m}{\pi}, \qquad V_{\text{rms}}^2 = \frac{1}{T}\int_0^{T/2} V_m^2 \sin^2 \omega t \, dt = \frac{1}{T}\cdot\frac{V_m^2}{2}\cdot\frac{T}{2} = \frac{V_m^2}{4}.$$

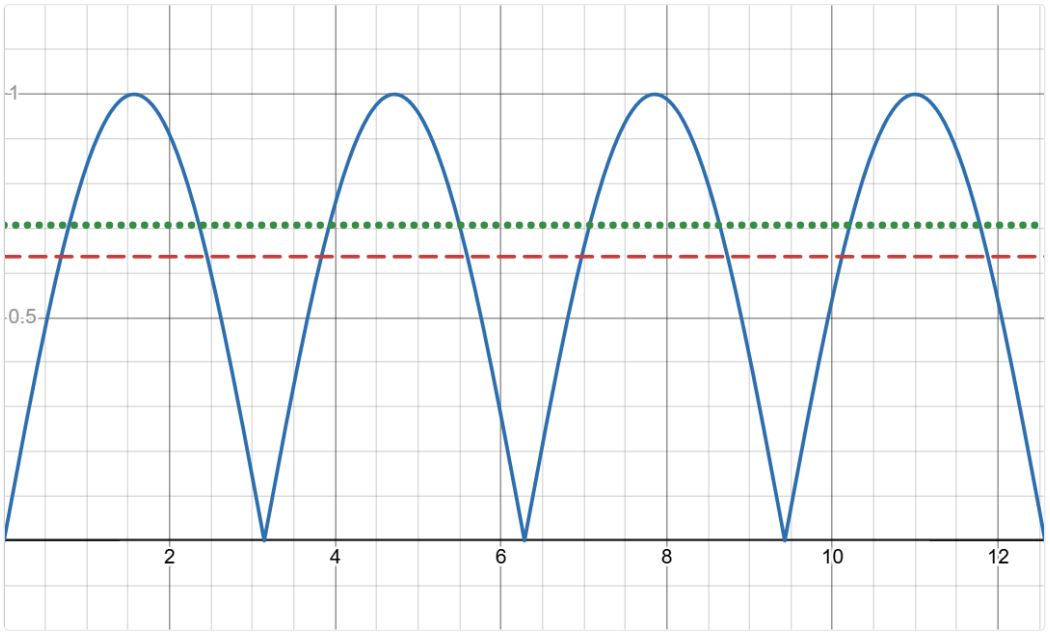
So $V_{\text{rms}} = V_m/2$: half the period carries the same power density as a full sine, so the mean square halves.

(b) **Sawtooth** $v = V_m\,\dfrac{2t - T}{T}$ on $[0, T]$ (from $-V_m$ to V_m). Substitute $u = \frac{2t-T}{T}$, $dt = \frac{T}{2}du$, $u \in [-1, 1]$:

$$V_{\text{rms}}^2 = \frac{1}{T}\int_{-1}^{1} V_m^2 u^2 \,\frac{T}{2}\,du = \frac{V_m^2}{2}\cdot\frac{2}{3} = \frac{V_m^2}{3}.$$

A triangle wave is made of the same linear pieces rearranged, so it has the same RMS value $V_m/\sqrt{3}$. RMS depends only on the *distribution of values*, not on their order in time.

(c) **Pulse train** of height V_m and duty cycle D: $v^2 = V_m^2$ for a fraction D of the time, so $\langle v^2\rangle = DV_m^2$ and $V_{\text{rms}} = \sqrt{D}\,V_m$, while $V_{\text{dc}} = DV_m$. A 12 V PWM signal at $D = 0.25$ has mean 3 V but RMS 6 V: a resistive heater driven by it dissipates $(6)^2/R$, four times what a 3 V DC supply would produce.



Full-wave rectified sine (blue) with its mean $2/\pi \approx 0.637$ (red) and RMS $1/\sqrt{2} \approx 0.707$ (green). The RMS exceeds the mean, as it always does unless the waveform is constant.

5.9.5 Mean, RMS and their combination

Write a periodic signal as its DC value plus an AC part with zero mean: $v = V_{\text{dc}} + v_{\text{ac}}$. Then

$$\langle v^2 \rangle = V_{\text{dc}}^2 + 2V_{\text{dc}}\langle v_{\text{ac}} \rangle + \langle v_{\text{ac}}^2 \rangle = V_{\text{dc}}^2 + V_{\text{ac,rms}}^2,$$

because $\langle v_{\text{ac}} \rangle = 0$. Thus

$$V_{\text{rms}} = \sqrt{V_{\text{dc}}^2 + V_{\text{ac,rms}}^2}.$$

In statistical language (Ch. 19), V_{dc} is the mean, $V_{\text{ac,rms}}$ is the standard deviation, and this is the identity $E[X^2] = \mu^2 + \sigma^2$. The RMS of a zero-mean noise voltage is its standard deviation.

Extending the argument to several harmonics uses orthogonality (§5.2.4): every cross term $\langle \cos m\omega_0 t \cos n\omega_0 t \rangle$ with $m \neq n$ vanishes.

RMS of a sum of harmonics (Parseval)

If $v(t) = V_0 + \sum_{n=1}^{N} V_n \cos(n\omega_0 t + \theta_n)$, then

$$V_{\text{rms}}^2 = V_0^2 + \sum_{n=1}^{N} \frac{V_n^2}{2} = V_0^2 + \sum_{n=1}^{N} V_{n,\text{rms}}^2.$$

Powers of different frequencies add; amplitudes do not.

Common Pitfall — Adding RMS values

RMS values of different-frequency components add **in quadrature**, not linearly: 3 V RMS at 1 kHz plus 4 V RMS at 3 kHz is 5 V RMS, not 7 V. Only components at the *same* frequency combine as phasors — and then their phase matters: two 3 V RMS sinusoids of the same frequency give anywhere from 0 to 6 V RMS.

Worked Example 5.45 — A signal with DC and harmonics

$v(t) = 5 + 10\sin\omega t + 4\sin 3\omega t$ V. Find V_{dc}, V_{rms} and the power into 8 Ω.

DC: $V_{\text{dc}} = 5$ V.

RMS: $V_{\text{rms}}^2 = 5^2 + \frac{10^2}{2} + \frac{4^2}{2} = 25 + 50 + 8 = 83$, so $V_{\text{rms}} = 9.110$ V.

Power: $P = V_{\text{rms}}^2/R = 83/8 = 10.375$ W, of which $25/8 = 3.125$ W is due to the DC component.

The **total harmonic distortion** of the AC part, $\text{THD} = \frac{\sqrt{\sum_{n\geq 2} V_{n,\text{rms}}^2}}{V_{1,\text{rms}}} = \frac{4/\sqrt{2}}{10/\sqrt{2}} = 40\%$, is itself an RMS ratio.

Worked Example 5.46 — Ripple on a DC supply

A 12 V supply carries a triangular ripple of 1 V peak-to-peak. Its AC part is a triangle of amplitude 0.5 V, with RMS $0.5/\sqrt{3} = 0.2887$ V. Then

$$V_{\text{rms}} = \sqrt{12^2 + 0.2887^2} = 12.0035 \text{ V}.$$

The ripple raises the RMS by only 0.03%; RMS is dominated by the largest component. That is why ripple is specified separately (in mV RMS or peak-to-peak) rather than inferred from a total RMS reading.

5.9.6 Average power and power factor

For a two-terminal element with periodic $v(t)$ and $i(t)$, the **average (real) power** is the mean of the instantaneous power:

$$P = \frac{1}{T}\int_0^T v(t)\, i(t)\, dt.$$

Average power for sinusoids

If $v = V_m \cos \omega t$ and $i = I_m \cos(\omega t - \varphi)$, then

$$P = \tfrac{1}{2} V_m I_m \cos\varphi = V_{\text{rms}} I_{\text{rms}} \cos\varphi = \tfrac{1}{2}\,\text{Re}\{\mathbf{V}\mathbf{I}^*\},$$

where $\mathbf{V} = V_m$ and $\mathbf{I} = I_m e^{-j\varphi}$ are peak phasors.

Derivation. By the product-to-sum identity, $v\,i = \tfrac{1}{2}V_m I_m\big[\cos\varphi + \cos(2\omega t - \varphi)\big]$. The second term oscillates at 2ω and averages to zero over a period, leaving $\tfrac{1}{2}V_m I_m \cos\varphi$. In phasor form $\mathbf{V}\mathbf{I}^* = V_m I_m e^{j\varphi}$, whose real part is $V_m I_m \cos\varphi$. □

The factor $\cos\varphi$ is the **power factor**. More generally the power factor is $P/(V_{\text{rms}} I_{\text{rms}})$, which also accounts for harmonic distortion: by orthogonality, only voltage and current harmonics of the *same* order contribute to P, while every harmonic increases I_{rms}.

Worked Example 5.47 — Real power of an inductive load

A load on 230 V RMS mains ($V_m = 325$ V) draws $I_m = 10$ A lagging by 30°.

$$P = \tfrac{1}{2}(325)(10)\cos 30^\circ = 1407 \text{ W}, \qquad S = V_{\text{rms}} I_{\text{rms}} = \tfrac{1}{2}(325)(10) = 1625 \text{ VA}.$$

Check by direct integration: numerically averaging $325\cos\omega t \cdot 10\cos(\omega t - \pi/6)$ over one period gives 1407.3 W ✓.

5.9.7 Form factor, crest factor and meters

Two dimensionless ratios describe waveform shape:

$$\text{form factor} = \frac{V_{\text{rms}}}{\langle |v| \rangle}, \qquad \text{crest factor} = \frac{V_{\text{peak}}}{V_{\text{rms}}}.$$

For a sine, the form factor is $\frac{V_m/\sqrt{2}}{2V_m/\pi} = \frac{\pi}{2\sqrt{2}} = 1.1107$ and the crest factor is $\sqrt{2}$.

Inexpensive "average-responding" AC meters rectify the signal, measure $\langle |v| \rangle$ (an easy analogue integral), and multiply by 1.1107. They read correctly only for sinusoids.

Worked Example 5.48 — Meter error on non-sinusoidal waveforms

(a) **Square wave** $\pm V_m$**.** $\langle |v| \rangle = V_m$, so the meter displays $1.1107 V_m$; the true RMS is V_m. The reading is 11.1% high.

(b) **Triangle wave** $\pm V_m$**.** $\langle |v| \rangle = V_m/2$, display $0.5554 V_m$; true RMS $V_m/\sqrt{3} = 0.5774 V_m$. The reading is 3.8% low.

(c) **PWM,** $D = 0.1$**.** $\langle v \rangle = 0.1 V_m$, display $0.111 V_m$; true RMS $\sqrt{0.1}\, V_m = 0.316 V_m$ — the meter reads barely a third of the true value. Signals with high crest factor (here $\sqrt{10} = 3.16$) are where average-responding meters fail worst, and where a "true-RMS" meter, which computes $\sqrt{\langle v^2 \rangle}$ by squaring and integrating, is essential.

5.9.8 Mean and RMS over a finite window

For non-periodic signals the value depends on the averaging window, which must therefore be stated.

Worked Example 5.49 — A decaying pulse

$v(t) = 10 e^{-t/\tau}$ V with $\tau = 1$ ms, averaged over the window $0 \le t \le T = 5$ ms.

$$\bar{v} = \frac{1}{T}\int_0^T 10 e^{-t/\tau} dt = \frac{10\tau}{T}\left(1 - e^{-T/\tau}\right) = 1.987 \text{ V},$$

$$V_{\text{rms}}^2 = \frac{1}{T}\int_0^T 100 e^{-2t/\tau} dt = \frac{100\tau}{2T}\left(1 - e^{-2T/\tau}\right) = 9.9995, \qquad V_{\text{rms}} = 3.162 \text{ V}.$$

Doubling the window to 10 ms roughly halves $\bar{v}$ and divides V_{rms} by about $\sqrt{2}$ — a transient's "average" is meaningful only together with its window. For such signals the window-independent quantities are the **total** integral $\int_0^\infty v\, dt = 10\tau = 10$ mV·s and the **energy** $\int_0^\infty v^2 dt = 50\tau = 0.05$ V²·s (the energy delivered to 1 Ω), which lead to the energy-signal concepts of Ch. 16.

Engineering Note — RMS in noise and measurement

For random signals the time-average RMS equals the ensemble standard deviation (for ergodic processes, Ch. 21). Thermal noise in a resistor has RMS voltage $\sqrt{4kTRB}$ — the square root of an *integral* of a flat power spectral density over the bandwidth B. Uncorrelated noise sources add in quadrature, for exactly the orthogonality reason that harmonics do: the mean of the product of two independent zero-mean signals is zero.

Summary

Topic	Key formula or result
Definite integral	$\displaystyle\int_a^b f\,dx = \lim_{\|P\|\to 0}\sum f(x_k^*)\,\Delta x_k$
FTC	$\displaystyle\frac{d}{dx}\int_a^x f(t)\,dt = f(x);\ \int_a^b f\,dx = F(b)-F(a)$
Leibniz rule	$\displaystyle\frac{d}{dx}\int_{u(x)}^{v(x)} f\,dt = f(v)v' - f(u)u'$
Mean value	$\displaystyle\bar f = \frac{1}{b-a}\int_a^b f\,dx = f(c)$ for some c
Complex exponential	$\displaystyle\int e^{(a+jb)t}dt = \frac{e^{(a+jb)t}}{a+jb}$
Orthogonality	$\displaystyle\int_0^T \sin m\omega_0 t\,\sin n\omega_0 t\,dt = \tfrac{T}{2}\,\delta_{mn}$
Cover-up rule	$A = \big[(x-a)P/Q\big]_{x=a}$
p-integrals	$\int_1^\infty x^{-p}dx<\infty \iff p>1;\ \int_0^1 x^{-p}dx<\infty \iff p<1$
Gaussian, Gamma	$\int_{-\infty}^{\infty} e^{-x^2}dx=\sqrt{\pi};\ \Gamma(n+1)=n!$
Fubini	$\iint_R f\,dA = \int_a^b\int_c^d f\,dy\,dx = \int_c^d\int_a^b f\,dx\,dy$
Polar area element	$dA = r\,dr\,d\phi$
Integration by parts	$\int u\,dv = uv - \int v\,du$
Substitution	$\int_a^b f(g(x))g'(x)\,dx = \int_{g(a)}^{g(b)} f(u)\,du$
Scaling	$\int_{-\infty}^{\infty} f(at)\,dt = \frac{1}{\|a\|}\int_{-\infty}^{\infty} f(u)\,du$
Trapezoidal rule	$T_n = h[\frac12 f_0 + f_1 + \cdots + f_{n-1} + \frac12 f_n]$, error $-\frac{(b-a)h^2}{12}f''(\xi)$
Simpson's rule	$S_n = \frac{h}{3}[f_0 + 4f_1 + 2f_2 + \cdots + 4f_{n-1} + f_n]$, error $-\frac{(b-a)h^4}{180}f^{(4)}(\xi)$
Richardson	$I \approx T_{2n} + (T_{2n} - T_n)/3$
Charge, energy	$q=\int i\,dt;\ W=\int p\,dt;\ W_C=\frac12 CV^2;\ W_L=\frac12 LI^2$
Non-uniform resistance	$R=\int_0^L \rho_e\,dx/A(x)$
RMS	$F_{\text{rms}}=\sqrt{\frac1T\int_0^T f^2dt}$; sine: $V_m/\sqrt2$
DC + AC	$V_{\text{rms}}^2 = V_{\text{dc}}^2 + V_{\text{ac,rms}}^2 = V_0^2 + \sum V_n^2/2$
Average power	$P=\frac1T\int_0^T v\,i\,dt = \frac12 V_m I_m\cos\varphi = \frac12\,\text{Re}\{\mathbf{V}\mathbf{I}^*\}$

Further reading

- E. Kreyszig, *Advanced Engineering Mathematics*, 10th ed., Wiley — appendix review of integration and the chapters on multiple integrals.
- G. B. Thomas, M. D. Weir and J. Hass, *Thomas' Calculus*, Pearson — thorough treatment of techniques of integration, improper integrals and multiple integrals with many worked examples.
- K. A. Stroud and D. J. Booth, *Engineering Mathematics*, Palgrave Macmillan — programmed-learning approach to integration techniques, mean and RMS values, and numerical integration.
- R. L. Burden and J. D. Faires, *Numerical Analysis*, Cengage — derivation and error analysis of Newton–Cotes rules and Richardson extrapolation (continued in Ch. 22).

Chapter 6 — Multivariable Calculus

> **Prerequisites:** Ch. 4, Ch. 5; basic vectors (dot product) and 2×2 determinants. **Learning outcomes:** After this chapter you can
>
> - describe functions of several variables through their domains, level curves and level surfaces, and decide whether limits exist;
> - compute partial derivatives, total differentials and gradients, and use them for linear approximation, sensitivity and tolerance analysis;
> - evaluate directional derivatives and build Jacobian matrices for linearisation, small-signal models and Newton's method;
> - set up and evaluate double and triple integrals, changing the order of integration when that helps;
> - change variables in integrals, in particular to polar, cylindrical and spherical coordinates;
> - locate and classify extrema of functions of several variables, with and without constraints, using the Hessian test and Lagrange multipliers.

Almost every quantity an engineer measures depends on more than one thing. The power in a resistor depends on voltage *and* resistance; the potential in a region of space depends on x, y and z; the collector current of a transistor depends on v_{BE} and v_{CE}; the cutoff frequency of a filter depends on every component value in it. Chapters 4 and 5 built derivatives and integrals for functions of one variable. This chapter extends both ideas to functions of several variables, and the extension is not merely notational: new phenomena appear (limits that depend on the direction of approach, derivatives in every direction at once, regions of integration with shape), and new tools (the gradient, the Jacobian, the Hessian, the Lagrange multiplier) become the everyday language of circuit sensitivity, field theory, nonlinear circuit solvers and optimisation.

The chapter has three movements. Sections 6.1–6.6 are **differential**: they describe how a function of several variables changes locally, culminating in the Jacobian matrix as *the* derivative of a vector function. Sections 6.7–6.9 are **integral**: they add up quantities over areas and volumes and show how to choose coordinates that fit the geometry. Sections 6.10–6.11 combine the two to find **optimal** points, with and without constraints. Chapter 7 then adds direction to the picture with vector fields, and almost every result there rests on something proved here.

6.1 Functions of several variables

Motivation

A single-variable function $y = f(x)$ is a curve; you can sketch it and read off slopes and areas. An engineer's quantities are rarely that simple. The resonant frequency of a series RLC circuit, $\omega_0 = 1/\sqrt{LC}$, is a function of two variables. The electrostatic potential $V(x, y, z)$ in a device is a function of three. The bit-error rate of a link depends on signal power, noise power, bandwidth and modulation order. Before we can differentiate or integrate such functions we need a vocabulary for describing them and a way to *see* them.

Definitions

Function of several variables

A real function of n variables assigns to each point $\mathbf{x} = (x_1, \dots, x_n)$ of a set $D \subseteq \mathbb{R}^n$ exactly one real number $f(\mathbf{x})$. The set D is the **domain** and the set of values taken is the **range**. For $n = 2$ we usually write $z = f(x, y)$; the **graph** $\{(x, y, f(x, y))\}$ is a surface in $\mathbb{R}^3$. A function of position $f(x, y, z)$ is also called a **scalar field**.

Unless stated otherwise, the domain is the largest set on which the formula makes sense: no division by zero, no square roots of negative numbers, no logarithms of non-positive numbers. Physical constraints may shrink it further (a resistance is positive, a probability lies in $[0, 1]$).

A few terms about sets in $\mathbb{R}^n$ will be needed later. A **neighbourhood** (open ball) of $\mathbf{a}$ is the set of points within distance $\delta > 0$ of $\mathbf{a}$: $\|\mathbf{x} - \mathbf{a}\| < \delta$. A point is an **interior point** of D if some neighbourhood of it lies entirely in D, and a **boundary point** if every neighbourhood contains points both in and out of D. D is **open** if all its points are interior, **closed** if it contains all its boundary points, and **bounded** if it fits inside some ball. "Closed and bounded" regions are the ones on which continuous functions are guaranteed to have a maximum and a minimum (§6.10).

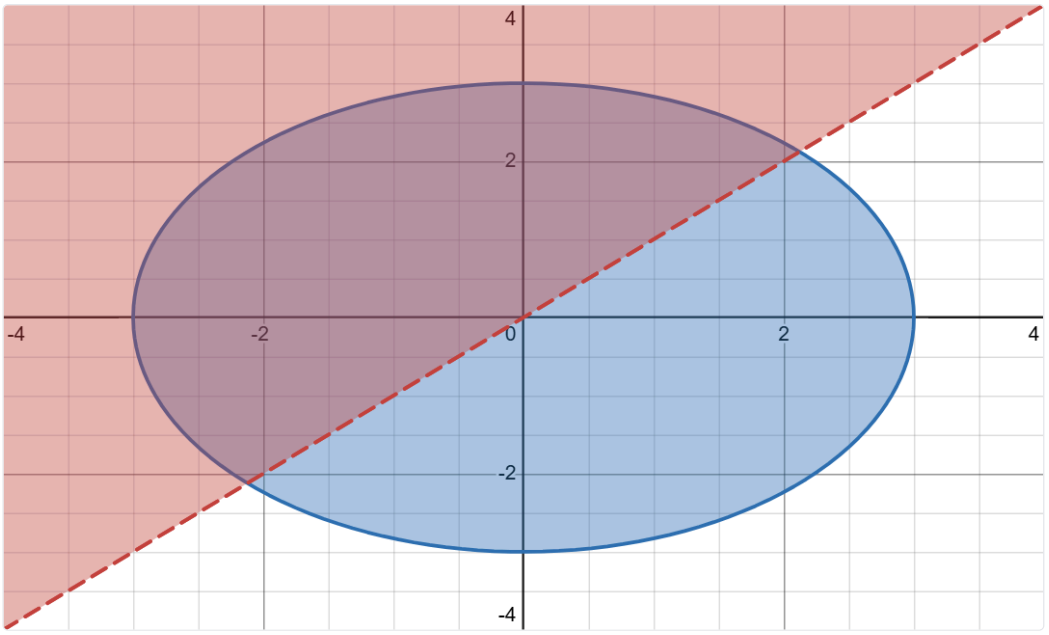
Worked Example — finding and sketching a domain

Find the domain of $f(x, y) = \sqrt{9 - x^2 - y^2} + \ln(y - x)$.

Step 1 (square root): need $9 - x^2 - y^2 \geq 0$, i.e. $x^2 + y^2 \leq 9$ — the closed disc of radius 3.

Step 2 (logarithm): need $y - x > 0$, i.e. the open half-plane above the line $y = x$.

Step 3 (intersect): $D = \{(x, y) : x^2 + y^2 \leq 9,\ y > x\}$, a half-disc. It is bounded but neither open (it contains part of the circle) nor closed (it excludes the diameter on $y = x$). Check a point: $(0, 1)$ gives $\sqrt{8} + \ln 1 = 2.828$, defined; $(1, 0)$ gives $\ln(-1)$, undefined. ✓



The domain of $\sqrt{9-x^2-y^2}+\ln(y-x)$ is where the blue disc and the red half-plane overlap; the dashed diameter itself is excluded.

Level curves and level surfaces

Because a surface is hard to draw, we usually draw **level curves** (contours) $f(x,y)=c$ for several constants c, the way a topographic map shows height. Where the contours crowd together the surface is steep; where they spread out it is flat. For $f(x,y,z)$ the analogous objects are **level surfaces** $f(x,y,z)=c$; in electrostatics these are the equipotential surfaces.

Level set

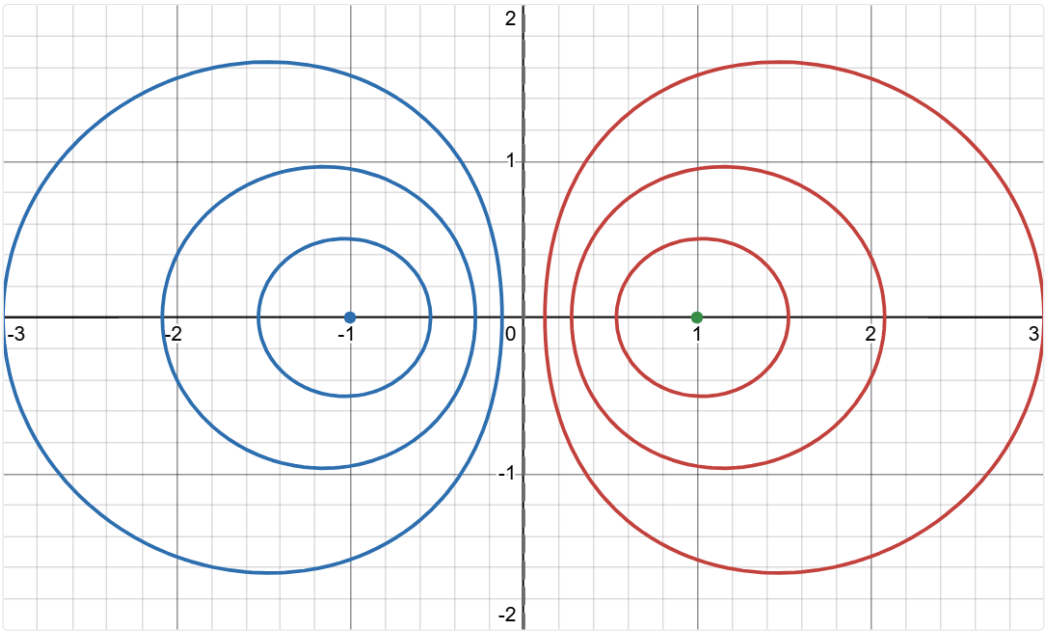
The level set of f at height c is $\{\mathbf{x}\in D : f(\mathbf{x})=c\}$. For $n=2$ it is (generically) a curve, for $n=3$ a surface. Level sets for different values of c never intersect, because f has only one value at each point.

Function	Level sets	Engineering reading
$f=x^2+y^2$	circles centred on the origin	equal-distance contours
$f=ax+by$	parallel straight lines	equipotentials of a uniform field
$f=xy$	rectangular hyperbolas	constant-power curves $P=VI$
$V=\dfrac{q}{4\pi\varepsilon_0\sqrt{x^2+y^2+z^2}}$	spheres	equipotentials of a point charge
$P=V^2/R$	curves $V^2=cR$ (parabolas in the RV-plane)	constant-dissipation curves
$\omega_0=1/\sqrt{LC}$	hyperbolas $LC=\text{const}$	all (L,C) pairs tuned to one frequency

The last row is worth a moment's thought: a whole curve of (L,C) pairs gives the same resonant frequency, and the designer moves along that curve to trade other properties (characteristic impedance $\sqrt{L/C}$, component cost). Level sets turn such design freedom into geometry.

The equipotentials of a pair of equal and opposite point charges at $(\mp1,0)$ (in the $z=0$ plane, constants dropped) are the level curves of

$$V(x,y)=\frac{1}{\sqrt{(x+1)^2+y^2}}-\frac{1}{\sqrt{(x-1)^2+y^2}}.$$



Level curves of the dipole potential: positive contours (blue) circle the positive charge, negative contours (red) the negative one, and the $V = 0$ contour is the bisecting line $x = 0$.

Three-variable functions cannot be graphed at all (the graph would live in $\mathbb{R}^4$), so level surfaces are the main visual tool. The **quadric surfaces** are the level sets of quadratic functions and appear constantly as constraint surfaces and equipotentials:

Equation	Surface
$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$	ellipsoid (sphere if $a = b = c$)
$x^2 + y^2 = a^2$	circular cylinder along z
$z = x^2 + y^2$	paraboloid of revolution
$z^2 = x^2 + y^2$	double cone
$z = x^2 - y^2$	hyperbolic paraboloid ("saddle")
$x^2 + y^2 - z^2 = 1$	hyperboloid of one sheet

Limits and continuity

Limit and continuity

$\lim_{\mathbf{x}\to\mathbf{a}} f(\mathbf{x}) = L$ means: for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|f(\mathbf{x}) - L| < \varepsilon$ whenever $0 < \|\mathbf{x} - \mathbf{a}\| < \delta$ and $\mathbf{x} \in D$. The function is **continuous** at $\mathbf{a}$ if $\lim_{\mathbf{x}\to\mathbf{a}} f(\mathbf{x}) = f(\mathbf{a})$.

The definition looks exactly like the one-variable definition in Chapter 4, but $\|\mathbf{x} - \mathbf{a}\| < \delta$ is now a disc (or ball), not an interval. That means f must approach L **along every path** into $\mathbf{a}$ — straight lines from every direction, parabolas, spirals, anything. In one variable there are only two directions of approach (left and right); in two there are infinitely many. Two consequences follow:

- If two paths give different limiting values, the limit does **not** exist.
- To prove that a limit **does** exist, you must bound $|f - L|$ by something that tends to zero independently of the path. Polar coordinates centred on $\mathbf{a}$ are the standard tool: write $x - a = r\cos\theta$, $y - b = r\sin\theta$ and show $|f - L| \le g(r)$ with $g(r) \to 0$.

Worked Example 1 — a limit that does not exist

Let $f(x, y) = \dfrac{xy}{x^2 + y^2}$ for $(x, y) \neq (0, 0)$. Along the line $y = mx$,

$$f(x, mx) = \frac{mx^2}{x^2 + m^2x^2} = \frac{m}{1 + m^2},$$

which depends on m: it is 0 along the axes ($m = 0$) and $\frac{1}{2}$ along $y = x$ ($m = 1$). Different paths give different values, so the limit at the origin does not exist, and no choice of $f(0, 0)$ makes f continuous there. In polar form $f = \frac{1}{2}\sin 2\theta$: the function depends only on the *direction* from the origin, not on the distance, so it cannot settle to a single value as $r \to 0$.

Worked Example 2 — all straight lines agree, yet no limit

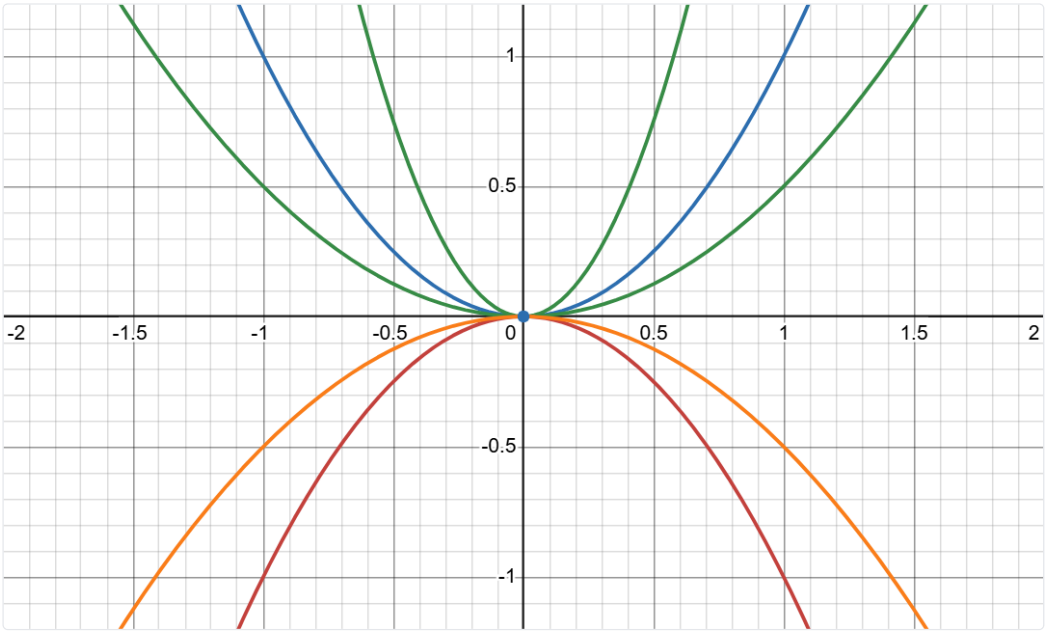
Let $f(x,y) = \dfrac{x^2 y}{x^4 + y^2}$. Along $y = mx$ $(m \neq 0)$,

$$f(x, mx) = \frac{mx^3}{x^4 + m^2 x^2} = \frac{mx}{x^2 + m^2} \to 0 \quad (x \to 0),$$

and along the axes $f = 0$ identically. Every straight line gives 0. But along the parabola $y = x^2$,

$$f(x, x^2) = \frac{x^4}{x^4 + x^4} = \frac{1}{2}.$$

So the limit does not exist. More generally, on $y = kx^2$ the value is $k/(1+k^2)$: the level curves of f are parabolas through the origin, and every one of them reaches the origin carrying its own value.



Level curves of $x^2y/(x^4+y^2)$ are parabolas $y = kx^2$ with value $k/(1+k^2)$: blue ($k = 1$) carries 0.5, green ($k = 0.5$ and $k = 3$) carry 0.4 and 0.3, red ($k = -1$) and orange ($k = -0.5$) carry -0.5 and -0.4. Every one of them enters the origin, so no single limit exists — although every straight line through the origin gives 0.

Worked Example 3 — proving a limit exists

Show that $\displaystyle\lim_{(x,y)\to(0,0)} \frac{x^3 + 2x^2y^2}{x^2 + y^2} = 0$.

In polar coordinates $x = r\cos\theta$, $y = r\sin\theta$, the denominator is r^2 and

$$f = \frac{r^3\cos^3\theta + 2r^4\cos^2\theta\sin^2\theta}{r^2} = r\cos^3\theta + 2r^2\cos^2\theta\sin^2\theta.$$

Since $|\cos\theta| \le 1$ and $|\sin\theta| \le 1$, $|f| \le r + 2r^2$, which tends to 0 as $r \to 0$ regardless of θ. Hence the limit is 0, and defining $f(0,0) = 0$ makes f continuous. Numerically, at $(0.01, 0.01)$: $f = (10^{-6} + 2\times10^{-8})/(2\times10^{-4}) = 0.0051$, small as expected.

Common Pitfall

Checking straight-line paths can prove that a limit does **not** exist, but never that it does — Worked Example 2 passes every straight-line test and still fails. Likewise, a bound in polar form must be independent of θ: $|f| \le r/|\cos\theta|$ is *not* good enough, because $\cos\theta$ can be arbitrarily small.

Building continuous functions

Polynomials in $x_1, \ldots, x_n$ are continuous everywhere. Sums, products and compositions of continuous functions are continuous; quotients are continuous wherever the denominator is non-zero. Elementary functions (e^u, $\sin u$, $\ln u$, $\sqrt{u}$, …) of continuous expressions are continuous wherever they are defined.

So $e^{-x^2-y^2}\cos(xy)$ is continuous on all of $\mathbb{R}^2$, and $\ln(1 - x^2 - y^2)$ on the open unit disc. Trouble only arises at points excluded from the formula's natural domain, as at the origin above.

Engineering Note

Real devices are continuous functions of their inputs, but their *models* are not always: piecewise device models (a diode switching between "off" and "on" regions, a MOSFET between triode and saturation) can be discontinuous or have discontinuous derivatives at region boundaries. Circuit simulators fail to converge exactly at such seams, which is why modern compact models are built from smooth functions that blend the regions together.

6.2 Partial differentiation

Motivation

The first question an engineer asks of a multi-input function is: *if I nudge one input and hold the rest, how much does the output move?* That is a partial derivative. It answers "which resistor matters most in this filter?", "how fast does the potential change as I move east?", and "what is the transistor's transconductance at this bias?". Partial derivatives are also the building blocks of everything else in the chapter — the gradient, the Jacobian and the Hessian are just partial derivatives arranged in vectors and matrices.

Definition and geometric meaning

Partial derivative

The partial derivative of $f(x, y)$ with respect to x at (a, b) is the ordinary derivative obtained by holding y fixed at b:

$$\frac{\partial f}{\partial x}(a, b) = f_x(a, b) = \lim_{h \to 0} \frac{f(a + h, b) - f(a, b)}{h}.$$

f_y is defined likewise with x held fixed. For $f(x_1, \dots, x_n)$ there are n first partial derivatives $\partial f / \partial x_i$, each obtained by varying one variable and freezing all the others.

Other notations in common use: $\partial_x f$, $D_1 f$, $\left(\dfrac{\partial f}{\partial x}\right)_y$. The last one, with the held-constant variable written as a subscript, is used in thermodynamics and wherever there is real ambiguity about what is held fixed (see the pitfall below).

Geometrically, $f_x(a, b)$ is the slope of the surface $z = f(x, y)$ in the x-direction at (a, b): slice the surface with the vertical plane $y = b$, and $f_x(a, b)$ is the slope of the resulting curve $z = f(x, b)$. Similarly $f_y(a, b)$ is the slope of the curve cut out by the plane $x = a$. The two partials therefore describe two tangent lines to the surface; together they span the tangent plane (§6.3).

Computing. Because $\partial f / \partial x$ is an ordinary derivative in disguise, every single-variable rule (power, product, quotient, chain) applies unchanged, treating the other variables as constants. For example, with $f(x, y) = x^2 y^3 + \sin(xy)$:

$$f_x = 2xy^3 + y\cos(xy), \qquad f_y = 3x^2 y^2 + x\cos(xy).$$

The y in front of $\cos(xy)$ in f_x is the inner derivative $\partial(xy)/\partial x = y$ from the chain rule.

Checking numerically. A partial derivative can always be estimated by a **finite difference**, varying one input in a simulation or on the bench:

$$f_x(a, b) \approx \frac{f(a + h, b) - f(a - h, b)}{2h} \qquad \text{(central difference, error } O(h^2)\text{)}.$$

At $(1, 2)$ the formula above gives $f_x = 16 + 2\cos 2 = 15.1677$; the central difference with $h = 10^{-3}$ gives 15.1677 to all digits shown, while the one-sided difference $[f(a + h, b) - f(a, b)]/h$ gives 15.1739, correct only to about three figures (error $O(h)$). This is the method circuit simulators fall back on for sensitivity analysis when analytic derivatives are not available, and the reason central differences are preferred (Ch. 22).

Worked Example 1 — partial derivatives of an RC low-pass response

The magnitude response of a first-order RC low-pass filter is

$$M(\omega, \tau) = \frac{1}{\sqrt{1 + \omega^2 \tau^2}}, \qquad \tau = RC.$$

Partial with respect to frequency (hold τ): write $M = (1 + \omega^2\tau^2)^{-1/2}$ and use the chain rule:

$$\frac{\partial M}{\partial \omega} = -\tfrac{1}{2}(1 + \omega^2\tau^2)^{-3/2} \cdot 2\omega\tau^2 = -\frac{\omega\tau^2}{(1 + \omega^2\tau^2)^{3/2}}.$$

Partial with respect to time constant (hold ω): by the same steps, $\partial M/\partial \tau = -\omega^2 \tau / (1 + \omega^2\tau^2)^{3/2}$.

At the cutoff $\omega\tau = 1$: $\partial M/\partial \omega = -\tau/2^{3/2} = -0.354\,\tau$ and $\partial M/\partial \tau = -\omega/2^{3/2} = -0.354\,\omega$. Units: M is dimensionless, so $\partial M/\partial\omega$ must have units of seconds (it does: τ) and $\partial M/\partial\tau$ units of s^{-1} (it does: ω). The symmetry $\omega\,\partial M/\partial\omega = \tau\,\partial M/\partial\tau$ is no accident: M depends only on the product $\omega\tau$.

Worked Example 2 — sensitivity of a parallel combination

For two resistors in parallel, $R = \dfrac{R_1 R_2}{R_1 + R_2}$. Holding R_2 fixed and using the quotient rule,

$$\frac{\partial R}{\partial R_1} = \frac{R_2(R_1 + R_2) - R_1 R_2 \cdot 1}{(R_1 + R_2)^2} = \frac{R_2^2}{(R_1 + R_2)^2},$$

and by symmetry $\partial R/\partial R_2 = R_1^2/(R_1 + R_2)^2$. With $R_1 = 1\,\mathrm{k\Omega}$, $R_2 = 4\,\mathrm{k\Omega}$: $R = 0.8\,\mathrm{k\Omega}$, $\partial R/\partial R_1 = 16/25 = 0.64$ and $\partial R/\partial R_2 = 1/25 = 0.04$.

The combination is 16 times more sensitive to the *smaller* resistor — a fact worth knowing when choosing which part needs the tighter tolerance. Both derivatives are dimensionless, as they must be (Ω/Ω). Check by finite difference: raising R_1 to $1.01\,\mathrm{k\Omega}$ gives $R = 4.04/5.01 = 0.806387\,\mathrm{k\Omega}$, an increase of $6.39\ \Omega$ for a $10\ \Omega$ change, ratio 0.639. ✓

Normalised sensitivity

Absolute partial derivatives mix units, which makes it hard to compare the influence of, say, a capacitor and a resistor. Circuit designers therefore use the **normalised (relative) sensitivity**

$$S_x^f = \frac{x}{f}\frac{\partial f}{\partial x} = \frac{\partial \ln f}{\partial \ln x} \approx \frac{\Delta f / f}{\Delta x / x},$$

the percentage change in f per percentage change in x. It is dimensionless, and for a power law $f = k x^a y^b$ it is simply the exponent: $S_x^f = a$, $S_y^f = b$.

Worked Example 3 — sensitivities of a series RLC circuit

A series RLC circuit has resonant frequency and quality factor

$$\omega_0 = (LC)^{-1/2}, \qquad Q = \frac{1}{R}\sqrt{\frac{L}{C}} = R^{-1} L^{1/2} C^{-1/2}.$$

Both are power laws, so the normalised sensitivities are read off the exponents:

$$S_L^{\omega_0} = S_C^{\omega_0} = -\tfrac{1}{2}, \quad S_R^{\omega_0} = 0; \qquad S_R^{Q} = -1, \quad S_L^{Q} = +\tfrac{1}{2}, \quad S_C^{Q} = -\tfrac{1}{2}.$$

Verify one from the definition: $\partial \omega_0 / \partial L = -\tfrac{1}{2} L^{-3/2} C^{-1/2}$, so $S_L^{\omega_0} = \dfrac{L}{(LC)^{-1/2}} \cdot \left(-\tfrac{1}{2} L^{-3/2} C^{-1/2}\right) = -\tfrac{1}{2}$. ✓

Reading: a 2 % high capacitor lowers ω_0 by about 1 % and Q by about 1 %; a 2 % high resistor leaves ω_0 alone but lowers Q by 2 %. A designer who needs an accurate centre frequency should spend money on L and C, not on R.

Common Pitfall — what is being held constant?

A partial derivative is only defined once you know which *other* variables are held fixed. For $P = VI$ with I held fixed, $\partial P / \partial V = I$. For the same resistor written as $P = V^2/R$ with R held fixed, $\partial P / \partial V = 2V/R = 2I$. Both are correct — they answer different questions (fixed current versus fixed resistance). When the context does not make this obvious, write $(\partial P / \partial V)_I$ or $(\partial P / \partial V)_R$.

Higher-order partial derivatives

Each first partial is itself a function of (x, y) and can be differentiated again, giving four second-order partials of $f(x, y)$:

$$f_{xx} = \frac{\partial^2 f}{\partial x^2}, \quad f_{yy} = \frac{\partial^2 f}{\partial y^2}, \quad f_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial y\, \partial x}, \quad f_{yx} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial x\, \partial y}.$$

(The subscript notation is read left to right — f_{xy} means "first x, then y" — while the fraction notation is read right to left. Fortunately the next theorem usually makes the order irrelevant.) f_{xx} measures the curvature of the x-slice; f_{xy} measures how the x-slope changes as you move in y, i.e. the *twist* of the surface.

Clairaut (Schwarz) theorem

If f_{xy} and f_{yx} both exist and are continuous on a neighbourhood of a point, they are equal there: the order of mixed differentiation does not matter. The same holds for higher derivatives: any two mixed partials involving the same number of differentiations in each variable agree if they are continuous.

Proof sketch. Consider the "second difference"

$$\Delta(h, k) = f(a+h, b+k) - f(a+h, b) - f(a, b+k) + f(a, b).$$

Grouping it as $g(a+h) - g(a)$ with $g(x) = f(x, b+k) - f(x, b)$ and applying the mean value theorem twice (first in x, then in y) gives $\Delta = hk\, f_{xy}(\xi, \eta)$ for some point (ξ, η) in the small rectangle. Grouping it the other way round, as a difference in y first, gives $\Delta = hk\, f_{yx}(\xi', \eta')$. Dividing by hk and letting $h, k \to 0$, continuity forces both mixed partials to the same value $\lim \Delta/(hk)$ at (a, b). □

Worked Example 4 — second partials and a numerical check

For $f = x^2 y^3 + \sin(xy)$, from the first partials above:

$$\begin{aligned} f_{xx} &= 2y^3 - y^2 \sin(xy), \qquad f_{yy} = 6x^2 y - x^2 \sin(xy), \\ f_{xy} &= \frac{\partial}{\partial y}\left[2xy^3 + y\cos(xy)\right] = 6xy^2 + \cos(xy) - xy\sin(xy), \\ f_{yx} &= \frac{\partial}{\partial x}\left[3x^2 y^2 + x\cos(xy)\right] = 6xy^2 + \cos(xy) - xy\sin(xy). \end{aligned}$$

The mixed partials agree, as Clairaut promises. At $(1, 2)$: $f_{xy} = 24 + \cos 2 - 2\sin 2 = 24 - 0.4161 - 1.8186 = 21.7653$. The four-point finite difference $\dfrac{f(a+h,b+h) - f(a+h,b-h) - f(a-h,b+h) + f(a-h,b-h)}{4h^2}$ with $h = 10^{-3}$ gives 21.7653. ✓

The hypotheses of Clairaut's theorem matter. The standard counterexample $f = xy(x^2 - y^2)/(x^2 + y^2)$, $f(0,0) = 0$, has $f_{xy}(0,0) = -1$ but $f_{yx}(0,0) = +1$; its second partials exist near the origin but are not continuous there. Such functions do not arise from smooth physical models, which is why engineers can (and do) swap the order of differentiation freely.

Common Pitfall — partial derivatives do not guarantee continuity

In one variable, a differentiable function is continuous. In several variables, the mere *existence* of partial derivatives guarantees nothing of the kind. The function $xy/(x^2+y^2)$ of §6.1 (with value 0 at the origin) has $f_x(0,0) = f_y(0,0) = 0$, because it is identically zero along both axes — yet it is not even continuous at the origin. Partial derivatives only probe two directions. The right multivariable notion of "differentiable" is the subject of §6.3.

Partial differential equations

Equations relating partial derivatives are **partial differential equations** (PDEs), the subject of Ch. 13. Here we only need to *check* that a given function satisfies one, which is pure partial differentiation. The three classical equations are:

Equation	Form (2-D, or 1-D plus time)	Typical physics
Laplace	$u_{xx} + u_{yy} = 0$	electrostatic potential in a charge-free region
Heat (diffusion)	$u_t = k\,u_{xx}$	heat flow, carrier diffusion, RC transmission lines
Wave	$u_{tt} = c^2 u_{xx}$	voltage on a lossless line, sound, light

Worked Example 5 — checking Laplace's equation

The potential of an infinite line charge, up to constants, is $u = \ln(x^2+y^2)$. Then

$$u_x = \frac{2x}{x^2+y^2}, \qquad u_{xx} = \frac{2(x^2+y^2) - 2x\cdot 2x}{(x^2+y^2)^2} = \frac{2y^2 - 2x^2}{(x^2+y^2)^2},$$

and by the symmetry $x \leftrightarrow y$, $u_{yy} = \dfrac{2x^2 - 2y^2}{(x^2+y^2)^2}$. Hence $u_{xx} + u_{yy} = 0$ away from the origin: u satisfies **Laplace's equation**, as every electrostatic potential in a charge-free region must. (At the origin, where the line charge sits, u is not even defined.) A function satisfying Laplace's equation is called **harmonic**.

Worked Example 6 — travelling waves and decaying modes

(a) Let $u(x,t) = g(x - ct)$ for any twice-differentiable g. With $s = x - ct$, the chain rule gives $u_x = g'(s)$, $u_{xx} = g''(s)$, $u_t = -c\,g'(s)$, $u_{tt} = c^2 g''(s)$. Hence $u_{tt} = c^2 u_{xx}$: **any** shape moving at speed c without distortion solves the wave equation. This is why a lossless transmission line carries pulses of any shape intact.

(b) Let $u(x,t) = e^{-k\beta^2 t}\sin(\beta x)$. Then $u_t = -k\beta^2 u$ and $u_{xx} = -\beta^2 u$, so $u_t = k\,u_{xx}$: a sinusoidal temperature (or charge) profile of spatial frequency β decays exponentially with time constant $1/(k\beta^2)$. Finer ripples (large β) die out faster, which is why diffusion smooths things. Units: k is in m^2/s and β in m^{-1}, so $k\beta^2$ is in s^{-1}. ✓

Engineering Note

Partial derivatives of device equations *are* the small-signal parameters on a datasheet. A MOSFET in saturation with $i_D = \tfrac{1}{2}k(v_{GS} - V_t)^2(1 + \lambda v_{DS})$ has transconductance $g_m = \partial i_D/\partial v_{GS} = k(v_{GS} - V_t)(1 + \lambda v_{DS})$ and output conductance $g_{ds} = \partial i_D/\partial v_{DS} = \tfrac{1}{2}k(v_{GS} - V_t)^2\lambda$, each evaluated at the bias point with the other voltage held fixed. Section 6.6 assembles such derivatives into a matrix.

6.3 Total differentials

Motivation

Partial derivatives answer "what if I change one input?". In practice everything changes at once: every component is off-nominal by its own tolerance, temperature shifts several parameters together, a sensor reading depends on several imperfect measurements. The **total differential** answers "what happens when all the inputs change a little, simultaneously?". It is the multivariable version of the linear approximation $\Delta y \approx f'(x)\,\Delta x$ from Chapter 4, and it leads directly to tolerance analysis, the multivariable chain rule and implicit differentiation.

Differentiability and the tangent plane

Differentiability and total differential

$f(x,y)$ is **differentiable** at (a,b) if there are constants A, B such that

$$f(a+\Delta x,\, b+\Delta y) = f(a,b) + A\,\Delta x + B\,\Delta y + \varepsilon\sqrt{\Delta x^2 + \Delta y^2},$$

with $\varepsilon \to 0$ as $(\Delta x, \Delta y) \to (0,0)$. Then necessarily $A = f_x(a,b)$ and $B = f_y(a,b)$, and the linear part

$$df = \frac{\partial f}{\partial x}\,dx + \frac{\partial f}{\partial y}\,dy$$

is called the **total differential** of f. For n variables, $df = \sum_{i=1}^{n} \dfrac{\partial f}{\partial x_i}\,dx_i$.

In words: near a point of differentiability, f is a plane plus an error that shrinks *faster than the distance* from the point. Setting $\Delta y = 0$ and dividing by Δx shows $A = f_x$; similarly $B = f_y$. So the coefficients are forced to be the partial derivatives — differentiability is the extra requirement that this plane approximates f well in **all** directions, not just along the axes.

A practical test for differentiability

If f_x and f_y exist near (a,b) and are continuous at (a,b), then f is differentiable at (a,b). A differentiable function is continuous.

Proof sketch. Write $f(a+\Delta x, b+\Delta y) - f(a,b) = [f(a+\Delta x, b+\Delta y) - f(a, b+\Delta y)] + [f(a, b+\Delta y) - f(a,b)]$. Apply the mean value theorem to each bracket: the first is $f_x(\xi, b+\Delta y)\Delta x$, the second $f_y(a,\eta)\Delta y$, with ξ between a and $a+\Delta x$ and η between b and $b+\Delta y$. Continuity of the partials lets us replace $f_x(\xi, b+\Delta y)$ by $f_x(a,b)$ and $f_y(a,\eta)$ by $f_y(a,b)$ with errors that tend to zero, which is exactly the definition. $\square$

All the functions in this book's applications have continuous partial derivatives wherever they are defined, so in practice "differentiable" means "the partials exist and are continuous", and functions such as $xy/(x^2+y^2)$ at the origin are exotic exceptions.

Geometrically, the linear part defines the **tangent plane** to the surface $z = f(x,y)$ at $(a, b, f(a,b))$:

$$z = f(a,b) + f_x(a,b)\,(x-a) + f_y(a,b)\,(y-b).$$

The **linearisation** $L(x,y)$ is the right-hand side; near (a,b), $f \approx L$.

Worked Example 1 — linear approximation of a phasor magnitude

An impedance $Z = R + jX$ has magnitude $|Z| = f(R,X) = \sqrt{R^2+X^2}$. At $R = 3\ \Omega$, $X = 4\ \Omega$, $|Z| = 5\ \Omega$. Estimate $|Z|$ when $R = 3.02\ \Omega$ and $X = 3.97\ \Omega$.

Partials: $f_R = R/\sqrt{R^2+X^2} = 3/5 = 0.6$, $f_X = X/\sqrt{R^2+X^2} = 4/5 = 0.8$ (both dimensionless).

Differential: $df = 0.6(0.02) + 0.8(-0.03) = 0.012 - 0.024 = -0.012\ \Omega$.

Estimate: $|Z| \approx 5 - 0.012 = 4.988\ \Omega$. The exact value is $\sqrt{3.02^2 + 3.97^2} = 4.98812\ \Omega$; the linearisation error, $1.2 \times 10^{-4}\ \Omega$, is of second order in the changes, as the definition promises.

Note the pattern: $f_R = \cos\angle Z$ and $f_X = \sin\angle Z$. The magnitude responds to R and X in proportion to how much each contributes to the phasor's direction.

Error and tolerance propagation

Interpreting dx_i as small errors or tolerances Δx_i turns the total differential into an error-propagation formula. Two estimates are standard:

$$\Delta f_{\text{worst}} = \sum_i \left|\frac{\partial f}{\partial x_i}\right| \Delta x_i, \qquad \Delta f_{\text{RSS}} = \sqrt{\sum_i \left(\frac{\partial f}{\partial x_i}\Delta x_i\right)^2}.$$

The **worst-case** sum assumes every error conspires in the same direction; it is guaranteed (to first order) and is what you use for safety-critical limits. The **root-sum-square** (RSS) value treats the errors as independent random variables with standard deviations Δx_i; the variance of a linear combination is the sum of the squared-weighted variances (Ch. 19, Ch. 20), so RSS gives the *typical* spread. RSS is never larger than worst case, and the gap widens as the number of contributing parameters grows.

For products of powers, $f = k x^a y^b z^c$, dividing df by f gives the handy **relative-error rule**

$$\frac{df}{f} = a\frac{dx}{x} + b\frac{dy}{y} + c\frac{dz}{z},$$

whose coefficients are the normalised sensitivities of §6.2. In words: for products and quotients, *relative* errors add, weighted by the exponents.

Worked Example 2 — power measurement uncertainty

A resistor $R = 100 \pm 1\ \Omega$ carries $V = 12 \pm 0.1$ V. Find $P = V^2/R$ and its uncertainty.

Nominal: $P = 144/100 = 1.44$ W.

Differential:

$$dP = \frac{2V}{R}\,dV - \frac{V^2}{R^2}\,dR = (0.24\ \text{A})\,dV - (0.0144\ \text{A}^2)\,dR.$$

Worst case: $\Delta P = 0.24(0.1) + 0.0144(1) = 0.024 + 0.0144 = 0.0384$ W, i.e. $2.67\,\%$. Check with the relative rule: $2(0.1/12) + 1(1/100) = 0.0167 + 0.0100 = 0.0267$. ✓

RSS: $\Delta P = \sqrt{0.024^2 + 0.0144^2} = 0.0280$ W ($1.94\,\%$).

The voltage term dominates because V enters squared: improving the voltmeter is worth more than buying a better resistor. Unit check: $(\text{V}/\Omega)\cdot\text{V} = \text{W}$ and $(\text{V}^2/\Omega^2)\cdot\Omega = \text{W}$. ✓

Worked Example 3 — tolerance of an RC cutoff frequency

A low-pass filter uses $R = 10\ \text{k}\Omega \pm 1\,\%$ and $C = 10\ \text{nF} \pm 5\,\%$. Its cutoff is $f_c = 1/(2\pi RC) = 1591.5$ Hz.

Since $f_c \propto R^{-1}C^{-1}$, the relative rule gives $\dfrac{df_c}{f_c} = -\dfrac{dR}{R} - \dfrac{dC}{C}$.

Worst case: $1\,\% + 5\,\% = 6\,\%$, so $f_c = 1591.5 \pm 95.5$ Hz.

RSS: $\sqrt{1^2 + 5^2}\,\% = 5.10\,\%$, so ± 81.2 Hz.

The capacitor tolerance is almost the whole story: replacing the 1 % resistor by a 0.1 % one would change the RSS figure only from 5.10 % to 5.00 %. The exact worst case, $1/(2\pi \cdot 0.99R \cdot 0.95C)$, is 6.33 % high; the first-order estimate (6 %) is slightly optimistic because the true dependence is curved — differentials are accurate only for *small* changes.

Common Pitfall

Two mistakes recur in tolerance work. First, adding signed terms: in Example 2 the two contributions have opposite signs, and $0.024 - 0.0144$ would grossly understate the uncertainty; take absolute values (worst case) or squares (RSS). Second, using RSS for errors that are *not* independent — two resistors from the same reel, or two parameters that both drift with temperature, are correlated, and RSS can then badly underestimate the spread.

The multivariable chain rule

Suppose $z = f(x, y)$ and the inputs themselves depend on time: $x = x(t)$, $y = y(t)$. Dividing the total differential by dt suggests the following, and the differentiability definition makes it rigorous.

Chain rule

If f is differentiable and $x(t), y(t)$ are differentiable, then

$$\frac{dz}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}.$$

If instead $x = x(u, v)$ and $y = y(u, v)$, then

$$\frac{\partial z}{\partial u} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial u}, \qquad \frac{\partial z}{\partial v} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial v}.$$

Proof sketch. From the definition of differentiability, $\Delta z = f_x \Delta x + f_y \Delta y + \varepsilon\sqrt{\Delta x^2 + \Delta y^2}$. Divide by Δt and let $\Delta t \to 0$: $\Delta x/\Delta t \to \dot{x}$, $\Delta y/\Delta t \to \dot{y}$, the square root over $|\Delta t|$ stays bounded (it tends to $\sqrt{\dot{x}^2 + \dot{y}^2}$), and $\varepsilon \to 0$. $\square$

A **tree diagram** keeps the terms straight: draw z at the top, branches to each intermediate variable (x, y), and from each of those branches down to the independent variables. Each path from z to the variable of interest contributes the product of the derivatives along it; add over all paths.

Worked Example 4 — rate of change of dissipated power

A resistor's voltage rises at $0.5\ \text{V/s}$ while self-heating raises its resistance at $2\ \Omega/\text{s}$. At the instant $V = 12$ V, $R = 100\ \Omega$, how fast is $P = V^2/R$ changing?

$$\frac{dP}{dt} = \frac{\partial P}{\partial V}\dot{V} + \frac{\partial P}{\partial R}\dot{R} = \frac{2V}{R}\dot{V} - \frac{V^2}{R^2}\dot{R} = 0.24(0.5) - 0.0144(2) = 0.120 - 0.0288 = 0.0912\ \text{W/s}.$$

The rising voltage increases the power; the rising resistance reduces it; the net effect is the sum of both paths of the tree.

Worked Example 5 — Cartesian to polar derivatives

Let $f(x, y) = x^2 + xy$ and change to polar coordinates, $x = r\cos\theta$, $y = r\sin\theta$. By the chain rule,

$$\frac{\partial f}{\partial r} = f_x\cos\theta + f_y\sin\theta, \qquad \frac{\partial f}{\partial \theta} = -f_x\, r\sin\theta + f_y\, r\cos\theta.$$

At $r = 2$, $\theta = \pi/6$: $x = \sqrt{3}$, $y = 1$, $f_x = 2x + y = 2\sqrt{3} + 1 = 4.4641$, $f_y = x = 1.7321$. Then

$$\frac{\partial f}{\partial r} = 4.4641(0.8660) + 1.7321(0.5) = 4.7321, \qquad \frac{\partial f}{\partial \theta} = -4.4641(1) + 1.7321(1.7321) = -1.4641.$$

Check directly: in polar form $f = r^2(\cos^2\theta + \sin\theta\cos\theta)$, so $f_r = 2r(\cos^2\theta + \sin\theta\cos\theta) = 4(0.75 + 0.4330) = 4.7321$ and $f_\theta = r^2(\cos 2\theta - \sin 2\theta) = 4(0.5 - 0.8660) = -1.4641$. ✓

Note that f_θ has the units of f (it is a rate per radian), whereas f_r has the units of f per metre; the factor r in the θ-equation carries the difference. This observation is the seed of the scale factors in §6.9.

Common Pitfall — cancelling partials

In one variable, $\frac{dz}{dx}\frac{dx}{dt}$ "cancels" to $\frac{dz}{dt}$. With partial derivatives the cancellation is wrong: $\frac{\partial z}{\partial x}\frac{\partial x}{\partial u}$ is only *one* of the terms in $\partial z/\partial u$. Likewise $\partial x/\partial r = \cos\theta$, but $\partial r/\partial x = x/r = \cos\theta$ too, not $1/\cos\theta$ — the reciprocal rule fails because different variables are held constant in the two derivatives (θ in one, y in the other).

Implicit differentiation

Often a relationship between variables is given implicitly, $F(x, y) = 0$, and cannot (or need not) be solved for y. Along the curve F is constant, so its total differential vanishes: $F_x\,dx + F_y\,dy = 0$, and therefore

$$\frac{dy}{dx} = -\frac{F_x}{F_y} \qquad (F_y \neq 0).$$

With three variables and $F(x, y, z) = 0$ defining $z = z(x, y)$, holding y fixed gives $F_x + F_z\,\partial z/\partial x = 0$, so

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z}, \qquad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z} \qquad (F_z \neq 0).$$

Implicit function theorem (two-variable form)

If F has continuous partial derivatives near (a, b), $F(a, b) = 0$ and $F_y(a, b) \neq 0$, then near a the equation $F(x, y) = 0$ defines a unique differentiable function $y = g(x)$ with $g(a) = b$, and $g'(x) = -F_x/F_y$.

The condition $F_y \neq 0$ is the important part: where $F_y = 0$ the curve has a vertical tangent (or a self-intersection) and y cannot be written as a single-valued function of x there. For example, on $x^2 + xy + y^2 = 7$ at $(1, 2)$: $F_x = 2x + y = 4$, $F_y = x + 2y = 5$, so $dy/dx = -4/5$.

Worked Example 6 — small-signal conductance of a real diode

A diode with series resistance obeys

$$F(I, V) = I - I_s\left(e^{(V - IR_s)/V_T} - 1\right) = 0,$$

which cannot be solved for I in elementary functions. Implicit differentiation still gives the small-signal conductance exactly. With $E = e^{(V - IR_s)/V_T}$:

$$F_V = -\frac{I_s E}{V_T}, \qquad F_I = 1 + \frac{I_s E R_s}{V_T}, \qquad \frac{dI}{dV} = -\frac{F_V}{F_I} = \frac{I_s E / V_T}{1 + R_s I_s E / V_T}.$$

From the equation itself, $I_s E = I + I_s$. Writing $r_d = V_T/(I + I_s)$ (the junction's own small-signal resistance), this simplifies to

$$\frac{dI}{dV} = \frac{1}{r_d + R_s}.$$

At $I = 1$ mA with $V_T = 25$ mV and $R_s = 10\ \Omega$ (I_s negligible): $r_d = 25\ \Omega$ and $dI/dV = 1/35$ S $= 28.6$ mS. The answer is physically obvious in hindsight — the junction resistance and the series resistance add — but the calculus delivered it without ever solving for I.

Exact differentials

The total differential of any $\phi(x, y)$ has the form $M\,dx + N\,dy$ with $M = \phi_x$, $N = \phi_y$. Conversely, given an expression $M\,dx + N\,dy$, is it the differential of something? If so, $M_y = \phi_{xy} = \phi_{yx} = N_x$ by Clairaut's theorem. The converse also holds on regions without holes:

Test for an exact differential

On a simply connected domain, with M, N continuously differentiable, $M\,dx + N\,dy$ is the total differential of some function ϕ if and only if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$

Worked Example 7 — recovering a potential function

Is $(2xy + 3)\,dx + (x^2 + 4y)\,dy$ exact? $M_y = 2x$, $N_x = 2x$: yes. To find ϕ, integrate M in x treating y as constant: $\phi = x^2y + 3x + h(y)$. Differentiate in y and match N: $x^2 + h'(y) = x^2 + 4y$, so $h' = 4y$, $h = 2y^2$. Thus $\phi = x^2y + 3x + 2y^2$ (plus any constant). Check: $\phi_x = 2xy + 3$, $\phi_y = x^2 + 4y$. ✓

This test reappears as "curl-free implies conservative" in Ch. 7 — the reason a static electric field can be written as the gradient of a potential — and in exact first-order ODEs in Ch. 11.

Taylor's theorem in two variables

The tangent plane is the first-order Taylor approximation. Adding second-order terms captures curvature, which we will need to classify maxima and minima in §6.10. Applying the one-variable Taylor series to $g(t) = f(a + th,\ b + tk)$ and using the chain rule twice gives

$$f(a + h, b + k) \approx f + f_x h + f_y k + \tfrac{1}{2}\left(f_{xx}h^2 + 2f_{xy}hk + f_{yy}k^2\right),$$

with all derivatives evaluated at (a, b). The bracket is a **quadratic form**, $\begin{bmatrix} h & k \end{bmatrix}\mathbf{H}\begin{bmatrix} h \\ k \end{bmatrix}$, built from the **Hessian matrix** $\mathbf{H}$ of second partials.

Worked Example 8 — second-order approximation

Approximate $f = e^x \cos y$ near $(0,0)$. All needed derivatives at the origin: $f = 1$, $f_x = e^x \cos y = 1$, $f_y = -e^x \sin y = 0$, $f_{xx} = 1$, $f_{xy} = -e^x \sin y = 0$, $f_{yy} = -e^x \cos y = -1$. Hence

$$e^x \cos y \approx 1 + x + \tfrac{1}{2}\left(x^2 - y^2\right).$$

At $(0.1, 0.2)$: first order gives 1.1, second order 1.085, exact 1.08314. The remaining error, 0.0019, matches the third-order terms $x^3/6 - xy^2/2 = -0.0018$. Each order improves the estimate by roughly a factor of the step size, just as in one variable.

6.4 Gradient

Motivation

The total differential $df = f_x\,dx + f_y\,dy + f_z\,dz$ has the shape of a dot product: one vector made of partial derivatives, the other made of the displacement (dx, dy, dz). Packaging the partial derivatives into a single vector — the gradient — turns a list of numbers into a geometric object with a direction and a length. That direction is the direction of steepest ascent, it is perpendicular to the level sets, and in electrostatics it *is* (minus) the electric field. The gradient is the first of the three differential operators of vector calculus, and the only one that acts on a scalar field.

Definition

Gradient

For a differentiable scalar function $f(x, y, z)$, the gradient is the vector of its first partial derivatives

$$\nabla f = \frac{\partial f}{\partial x}\hat{\mathbf{x}} + \frac{\partial f}{\partial y}\hat{\mathbf{y}} + \frac{\partial f}{\partial z}\hat{\mathbf{z}} = (f_x,\ f_y,\ f_z).$$

For $f(x_1, \dots, x_n)$, $\nabla f = (\partial f/\partial x_1, \dots, \partial f/\partial x_n)$. The symbol ∇ ("del" or "nabla") is the operator $\hat{\mathbf{x}}\,\partial_x + \hat{\mathbf{y}}\,\partial_y + \hat{\mathbf{z}}\,\partial_z$.

With $d\mathbf{r} = (dx, dy, dz)$, the total differential becomes

$$df = \nabla f \cdot d\mathbf{r},$$

and the linearisation of §6.3 becomes $f(\mathbf{r}) \approx f(\mathbf{a}) + \nabla f(\mathbf{a}) \cdot (\mathbf{r} - \mathbf{a})$. **Units:** ∇ carries m^{-1}, so the gradient of a temperature is in $\mathrm{K/m}$ and of a potential in $\mathrm{V/m}$.

Geometric properties

Properties of the gradient

Let f be differentiable at $\mathbf{a}$ with $\nabla f(\mathbf{a}) \neq \mathbf{0}$.

1. **Steepest ascent.** $\nabla f(\mathbf{a})$ points in the direction in which f increases most rapidly, and $\|\nabla f(\mathbf{a})\|$ is that maximum rate of increase (per unit distance).
2. **Normal to level sets.** $\nabla f(\mathbf{a})$ is perpendicular to the level curve (or level surface) $f = f(\mathbf{a})$ through $\mathbf{a}$.
3. **Tangent plane.** The tangent plane to the level surface $f(x, y, z) = c$ at $\mathbf{a}$ is $\nabla f(\mathbf{a}) \cdot (\mathbf{r} - \mathbf{a}) = 0$.

Proof. (1) Move a small distance ds in the unit direction $\hat{\mathbf{u}}$: $df = \nabla f \cdot \hat{\mathbf{u}}\,ds = \|\nabla f\| \cos\theta\,ds$, where θ is the angle between $\hat{\mathbf{u}}$ and ∇f. This is largest when $\theta = 0$, giving rate $\|\nabla f\|$. (2) Let $\mathbf{r}(t)$ be any curve lying in the level set, through $\mathbf{a}$ at $t = 0$. Since $f(\mathbf{r}(t)) = c$ for all t, the chain rule gives $0 = \frac{d}{dt} f(\mathbf{r}(t)) = \nabla f \cdot \mathbf{r}'(0)$. So ∇f is perpendicular to the tangent vector of *every* curve in the level set, i.e. to the level set itself. (3) The tangent plane consists of points $\mathbf{r}$ with $\mathbf{r} - \mathbf{a}$ perpendicular to the normal $\nabla f(\mathbf{a})$. $\square$

Property 2 explains a picture every electrical engineer knows: field lines cross equipotentials at right angles. Property 1 explains why field strength is greatest where equipotentials crowd together — closely spaced contours mean a large change of f per unit distance.

Worked Example 1 — normal and tangent to a level curve

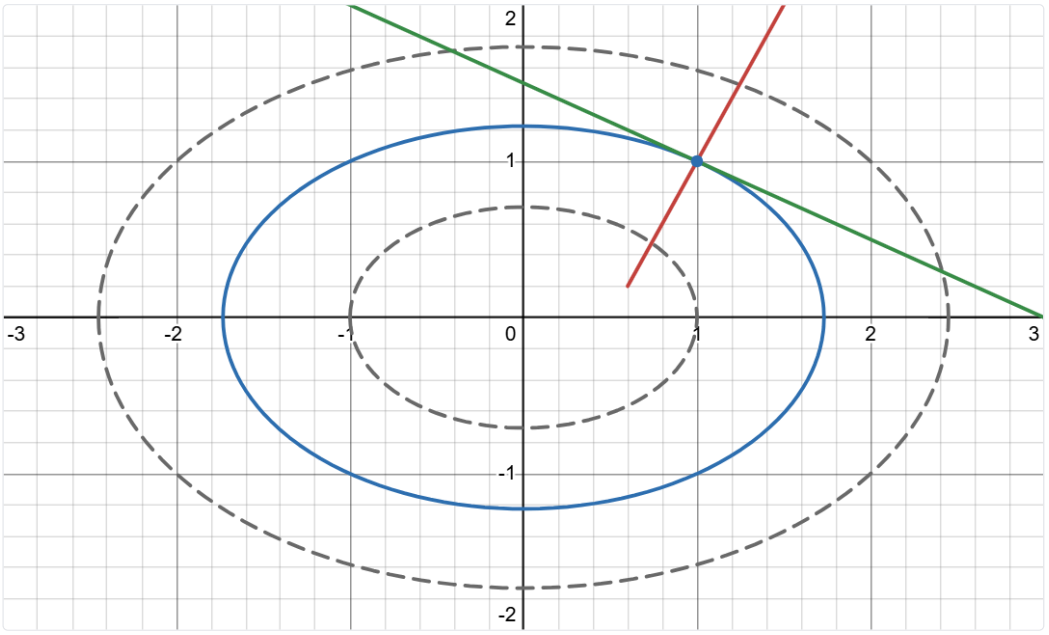
Let $f(x, y) = x^2 + 2y^2$ and take the point $(1, 1)$, which lies on the level curve $f = 3$ (an ellipse). Then $\nabla f = (2x, 4y) = (2, 4)$ at $(1, 1)$.

Tangent line: $\nabla f \cdot (x - 1,\ y - 1) = 0$, i.e. $2(x - 1) + 4(y - 1) = 0$, or $x + 2y = 3$.

Normal line (along ∇f): $(x, y) = (1, 1) + t(2, 4)$, i.e. $y - 1 = 2(x - 1)$.

Maximum rate of increase at $(1, 1)$: $\|(2, 4)\| = \sqrt{20} \approx 4.47$ units of f per unit distance, in the direction $(2, 4)/\sqrt{20} = (0.447,\ 0.894)$.

Check by implicit differentiation (§6.3): $dy/dx = -f_x/f_y = -2/4 = -\tfrac{1}{2}$, which is the slope of $x + 2y = 3$. ✓



The gradient (red, along the normal) crosses each level curve at right angles; the tangent line (green) is perpendicular to it.

Worked Example 2 — tangent plane to a level surface

Find the tangent plane and normal line to the ellipsoid $\dfrac{x^2}{4} + y^2 + \dfrac{z^2}{9} = 3$ at $(2, 1, 3)$.

Check the point: $1 + 1 + 1 = 3$. ✓

Gradient: with $F = \frac{1}{4}x^2 + y^2 + \frac{1}{9}z^2$, $\nabla F = \left(\frac{x}{2},\ 2y,\ \frac{2z}{9}\right) = \left(1,\ 2,\ \frac{2}{3}\right)$.

Tangent plane: $1(x-2) + 2(y-1) + \frac{2}{3}(z-3) = 0$; multiplying by 3, $3x + 6y + 2z = 18$. The point satisfies it: $6 + 6 + 6 = 18$. ✓

Normal line: $(x, y, z) = (2, 1, 3) + t\left(1, 2, \frac{2}{3}\right)$.

The same recipe gives the tangent plane to a graph $z = f(x, y)$: write it as the level surface $F = f(x, y) - z = 0$, whose gradient $(f_x, f_y, -1)$ reproduces the tangent-plane formula of §6.3.

Algebra of the gradient

All of the following follow at once from the corresponding single-variable rules applied component by component:

$$\nabla(af + bg) = a\nabla f + b\nabla g, \qquad \nabla(fg) = f\nabla g + g\nabla f, \qquad \nabla\left(\frac{f}{g}\right) = \frac{g\nabla f - f\nabla g}{g^2}, \qquad \nabla h(f) = h'(f)\,\nabla f.$$

The last one (chain rule) is especially useful for functions of distance. With $\mathbf{R} = (x, y, z)$ and $R = \|\mathbf{R}\| = \sqrt{x^2 + y^2 + z^2}$, $\partial R/\partial x = x/R$ and so on, giving

$$\nabla R = \frac{\mathbf{R}}{R} = \hat{\mathbf{R}}, \qquad \nabla R^n = nR^{n-1}\hat{\mathbf{R}}, \qquad \nabla\frac{1}{R} = -\frac{\hat{\mathbf{R}}}{R^2}.$$

Any function of distance alone has a gradient pointing radially, and its level sets are spheres — the geometry of every point source.

Worked Example 3 — Coulomb's law from a potential

The potential of a point charge q at the origin is $V = \dfrac{q}{4\pi\varepsilon_0 R}$. Using $\nabla(1/R) = -\hat{\mathbf{R}}/R^2$,

$$\mathbf{E} = -\nabla V = -\frac{q}{4\pi\varepsilon_0}\nabla\frac{1}{R} = \frac{q}{4\pi\varepsilon_0 R^2}\hat{\mathbf{R}},$$

which is Coulomb's law. Units: V in volts, ∇ in m^{-1}, so $\mathbf{E}$ in $\mathrm{V/m}$. ✓ Component check at $(x, y, z) = (1, 2, 2)$ m, $R = 3$ m: $\partial(1/R)/\partial x = -x/R^3 = -1/27$, matching $-\hat{\mathbf{R}}_x/R^2 = -(1/3)/9 = -1/27$. ✓

Worked Example 4 — field of the dipole on its bisector

For the 2-D dipole potential of §6.1, $V = \dfrac{1}{r_1} - \dfrac{1}{r_2}$ with $r_1 = \sqrt{(x+1)^2 + y^2}$ and $r_2 = \sqrt{(x-1)^2 + y^2}$, find $\mathbf{E} = -\nabla V$ at $(0, 1)$.

Using $\partial(1/r_1)/\partial x = -(x+1)/r_1^3$, $\partial(1/r_1)/\partial y = -y/r_1^3$, and similarly for r_2:

$$V_x = -\frac{x+1}{r_1^3} + \frac{x-1}{r_2^3}, \qquad V_y = -\frac{y}{r_1^3} + \frac{y}{r_2^3}.$$

At $(0, 1)$, $r_1 = r_2 = \sqrt{2}$, $r^3 = 2^{3/2} = 2.828$: $V_x = -\dfrac{1}{2.828} - \dfrac{1}{2.828} = -0.707$, $V_y = 0$. Hence $\mathbf{E} = (0.707,\ 0)$.

The field on the bisector points in $+x$, from the positive charge towards the negative one, and parallel to the axis — perpendicular to the $V = 0$ equipotential $x = 0$, exactly as property 2 requires.

The gradient in other coordinates (preview)

The definition $df = \nabla f \cdot d\mathbf{r}$ is coordinate-free, so it can be used to *derive* the gradient in any coordinate system. In polar coordinates a small displacement is $d\mathbf{r} = dr\,\hat{\mathbf{r}} + r\,d\theta\,\hat{\boldsymbol{\theta}}$ (moving $d\theta$ in angle covers a distance $r\,d\theta$), while $df = f_r\,dr + f_\theta\,d\theta$. Matching the two forms,

$$\nabla f = \frac{\partial f}{\partial r}\hat{\mathbf{r}} + \frac{1}{r}\frac{\partial f}{\partial \theta}\hat{\boldsymbol{\theta}}.$$

The factor $1/r$ is forced by the units: f_θ is per radian, the gradient must be per metre. The cylindrical and spherical versions follow the same way from the scale factors of §6.9 and are tabulated in Ch. 7.

Common Pitfall

Writing $\nabla f = (f_r,\ f_\theta)$ in polar coordinates — simply listing partial derivatives — is wrong, for the units reason just given. The "list of partials" recipe works only in Cartesian coordinates, where every coordinate is a length.

Gradient descent

Since $-\nabla f$ points in the direction of steepest *decrease*, a natural minimisation algorithm is to step repeatedly downhill:

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha\,\nabla f(\mathbf{x}_k),$$

with a step size $\alpha > 0$. This is the core of adaptive filtering (the LMS algorithm), neural-network training and many circuit-optimisation tools.

Worked Example 5 — gradient descent and its step-size limit

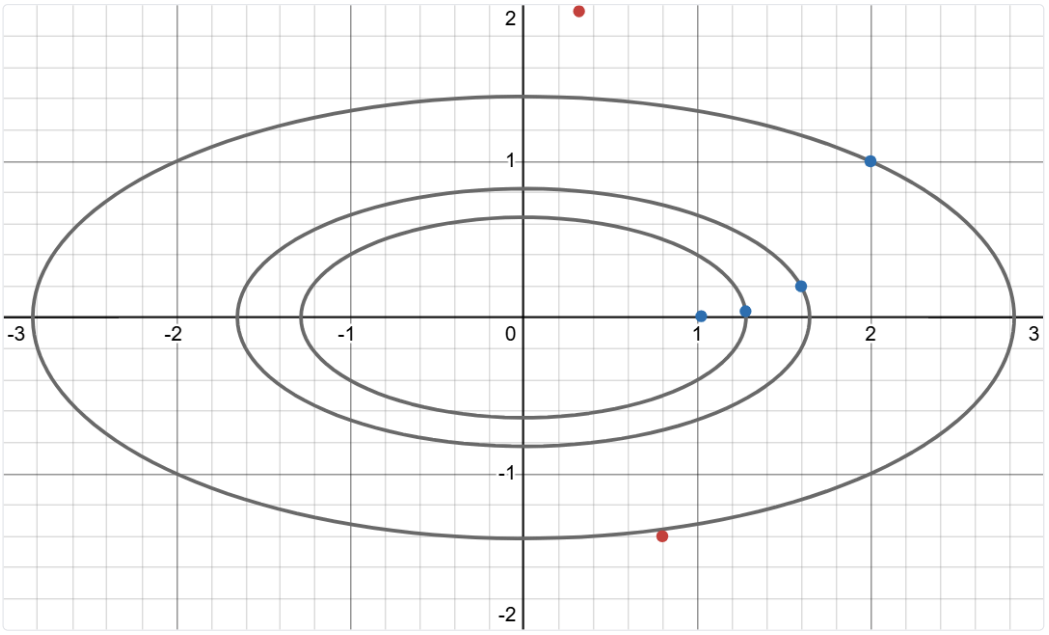
Minimise $f(x, y) = x^2 + 4y^2$ (minimum 0 at the origin) from $(2, 1)$, where $f = 8$. Here $\nabla f = (2x,\ 8y)$, so one step is

$$x_{k+1} = (1 - 2\alpha)x_k, \qquad y_{k+1} = (1 - 8\alpha)y_k.$$

With $\alpha = 0.1$: x shrinks by 0.8 and y by 0.2 per step: $(2, 1) \to (1.6, 0.2) \to (1.28, 0.04) \to (1.024, 0.008)$, with $f = 8 \to 2.72 \to 1.645 \to 1.049$. The steep y-direction converges fast; the shallow x-direction slowly.

With $\alpha = 0.3$: the y factor is $1 - 2.4 = -1.4$: y overshoots and grows, $1 \to -1.4 \to 1.96 \to -2.744$, and f *increases*, $8 \to 8.48 \to 15.47 \to 30.13$. The iteration diverges.

Stability requires $|1 - 8\alpha| < 1$, i.e. $\alpha < 0.25$, set by the *steepest* direction, while convergence speed is limited by the *shallowest* one. The ratio of curvatures ($8/2 = 4$ here) — the condition number of the Hessian — governs how badly gradient descent zig-zags (Ch. 23).



Gradient descent on $x^2 + 4y^2$: with $\alpha = 0.1$ (blue) the iterates slide down towards the minimum, each step perpendicular to the contour it leaves; with $\alpha = 0.3$ (red) they bounce across the valley onto ever higher contours.

6.5 Directional derivatives

Motivation

Partial derivatives measure change along the coordinate axes, but the axes are an arbitrary human choice. A probe moving diagonally across a circuit board, a satellite crossing a radiation pattern at an angle, an optimiser deciding which way to adjust two component values at once — each needs the rate of change in a *chosen* direction. The directional derivative supplies it, and the remarkable fact is that the gradient alone determines it for every direction at once.

Definition and the gradient formula

Directional derivative

The rate of change of f at $\mathbf{a}$ in the direction of a **unit** vector $\hat{\mathbf{u}}$ is

$$D_{\hat{\mathbf{u}}} f(\mathbf{a}) = \lim_{h \to 0} \frac{f(\mathbf{a} + h\hat{\mathbf{u}}) - f(\mathbf{a})}{h}.$$

It is the slope of the curve obtained by slicing the graph of f with the vertical plane through $\mathbf{a}$ in the direction $\hat{\mathbf{u}}$. The partial derivatives are the special cases $\hat{\mathbf{u}} = \hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$.

Directional derivative from the gradient

If f is differentiable at $\mathbf{a}$, then for every unit vector $\hat{\mathbf{u}}$

$$D_{\hat{\mathbf{u}}} f(\mathbf{a}) = \nabla f(\mathbf{a}) \cdot \hat{\mathbf{u}} = \|\nabla f(\mathbf{a})\| \cos\theta,$$

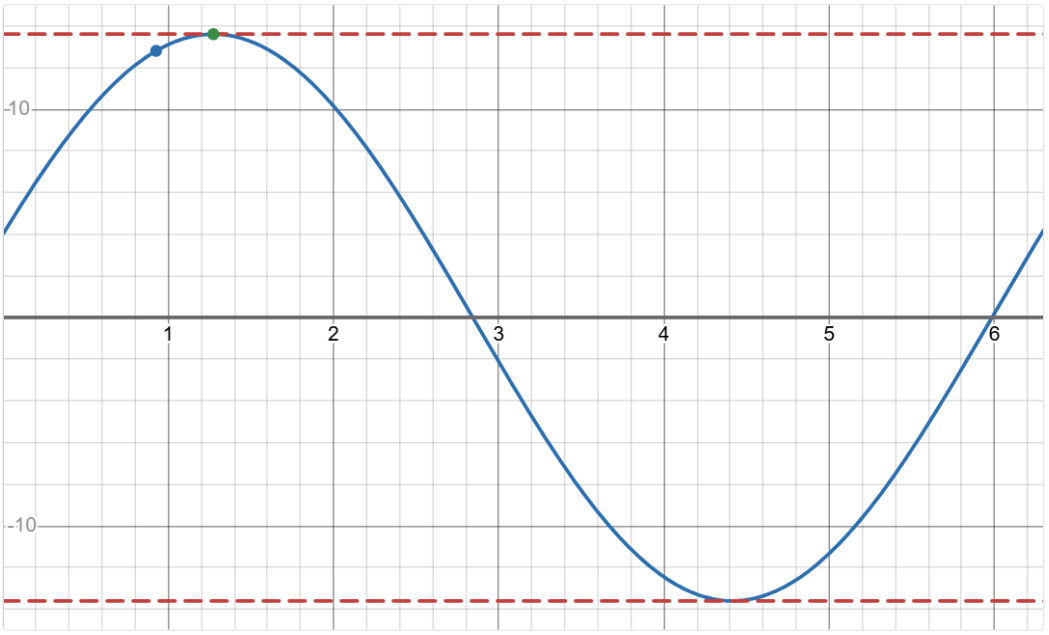
where θ is the angle between $\hat{\mathbf{u}}$ and $\nabla f(\mathbf{a})$. In two dimensions, with $\hat{\mathbf{u}} = (\cos\alpha, \sin\alpha)$, $D_{\hat{\mathbf{u}}} f = f_x \cos\alpha + f_y \sin\alpha$.

Proof. Let $g(h) = f(\mathbf{a} + h\hat{\mathbf{u}})$, a function of one variable. By definition $D_{\hat{\mathbf{u}}} f(\mathbf{a}) = g'(0)$. The point $\mathbf{a} + h\hat{\mathbf{u}}$ has coordinates $x_i = a_i + h u_i$, so $dx_i/dh = u_i$, and the chain rule of §6.3 gives $g'(0) = \sum_i f_{x_i}(\mathbf{a})\, u_i = \nabla f(\mathbf{a}) \cdot \hat{\mathbf{u}}$. $\square$

The cosine form packs the geometry of §6.4 into one line:

- the maximum value $+\|\nabla f\|$ occurs for $\hat{\mathbf{u}}$ along ∇f ($\theta = 0$);
- the minimum $-\|\nabla f\|$ for $\hat{\mathbf{u}}$ opposite to ∇f ($\theta = \pi$);
- $D_{\hat{\mathbf{u}}} f = 0$ for $\hat{\mathbf{u}}$ perpendicular to ∇f, i.e. tangent to the level set ($\theta = \pi/2$);
- reversing the direction reverses the sign: $D_{-\hat{\mathbf{u}}} f = -D_{\hat{\mathbf{u}}} f$.

So the n numbers in the gradient encode the slopes in infinitely many directions; as $\hat{\mathbf{u}}$ rotates once around the circle, $D_{\hat{\mathbf{u}}} f$ traces one period of a sinusoid of amplitude $\|\nabla f\|$.



Directional derivative of $f = x^2y + y^3$ at $(1, 2)$ (gradient $(4, 13)$) versus the direction angle α: a sinusoid of amplitude $\|\nabla f\| = \sqrt{185} \approx 13.6$, peaking along the gradient and vanishing in the two directions tangent to the level curve.

Worked Example 1 — the basic calculation

Find the rate of change of $f(x, y) = x^2y + y^3$ at $(1, 2)$ in the direction of $\mathbf{v} = (3, 4)$.

Gradient: $\nabla f = (2xy,\ x^2 + 3y^2) = (4,\ 13)$ at $(1, 2)$.

Normalise: $\|\mathbf{v}\| = 5$, so $\hat{\mathbf{u}} = (0.6,\ 0.8)$.

Dot product:

$$D_{\hat{\mathbf{u}}} f = 4(0.6) + 13(0.8) = 2.4 + 10.4 = 12.8.$$

This is less than the maximum possible rate $\|\nabla f\| = \sqrt{185} \approx 13.6$, as it must be — $(3, 4)$ is only 19.8° away from the gradient direction ($\cos\theta = 12.8/13.6 = 0.941$).

Numerical check: $[f(1.0006, 2.0008) - f(0.9994, 1.9992)]/0.002 = 12.8000$. ✓

Common Pitfall

Forgetting to normalise the direction vector. Using $\mathbf{v} = (3, 4)$ directly would give $\nabla f \cdot \mathbf{v} = 64$, five times too large. The quantity $\nabla f \cdot \mathbf{v}$ for a non-unit $\mathbf{v}$ is still meaningful — it is the rate of change *per unit time* for a point moving with velocity $\mathbf{v}$ — but it is not the slope per unit distance.

Worked Example 2 — hot spot on a circuit board

Near a power device, the board temperature is modelled by $T(x, y) = 80 - x^2 - 2y^2$ (°C, with x, y in cm, device at the origin). A probe sits at $(3, 2)$, where $T = 80 - 9 - 8 = 63\ ^\circ\text{C}$.

Gradient: $\nabla T = (-2x,\ -4y) = (-6,\ -8)\ ^\circ\text{C/cm}$, with $\|\nabla T\| = 10\ ^\circ\text{C/cm}$.

(a) *Fastest heating:* towards $\nabla T/\|\nabla T\| = (-0.6, -0.8)$, at $10\ ^\circ\text{C/cm}$. Note that this is **not** straight towards the device at the origin, which would be the direction $(-3, -2)/\sqrt{13} = (-0.832, -0.555)$; the steeper y-dependence pulls the gradient towards the y-direction.

(b) *Fastest cooling:* the opposite direction, $(0.6, 0.8)$, at $-10\ ^\circ\text{C/cm}$.

(c) *No change:* perpendicular to the gradient, $\pm(0.8, -0.6)$ — along the isotherm $x^2 + 2y^2 = 17$.

(d) *Heating at exactly* $5\ ^\circ\text{C/cm}$*:* need $10\cos\theta = 5$, $\theta = \pm 60^\circ$ from the gradient. The gradient direction has angle $\alpha_0 = \operatorname{atan2}(-8, -6) = -126.87^\circ$, so $\alpha = -66.87^\circ$ or -186.87°, i.e. $\hat{\mathbf{u}} = (0.393, -0.920)$ or $(-0.993, 0.120)$. Check the first: $-6(0.393) - 8(-0.920) = -2.36 + 7.36 = 5.0$. ✓

(e) *Towards the device:* $D = (-6, -8) \cdot (-0.832, -0.555) = 4.99 + 4.44 = 9.43\ ^\circ\text{C/cm}$, close to but below the maximum.

Worked Example 3 — a three-dimensional potential

A potential is $V(x,y,z) = xy^2 + yz^3$ (volts, coordinates in metres). Find the rate of change of V at $P=(2,-1,1)$ moving towards $Q=(4,1,2)$, and the electric field there.

Gradient: $\nabla V = (y^2,\ 2xy+z^3,\ 3yz^2) = (1,\ -3,\ -3)$ V/m at P.

Direction: $\overrightarrow{PQ} = (2,2,1)$, length 3, so $\hat{\mathbf{u}} = \tfrac{1}{3}(2,2,1)$.

Rate: $D_{\hat{\mathbf{u}}}V = \tfrac{1}{3}(2-6-3) = -\tfrac{7}{3} \approx -2.33$ V/m: the potential falls as the probe moves towards Q.

Field: $\mathbf{E} = -\nabla V = (-1,\ 3,\ 3)$ V/m, magnitude $\sqrt{19} = 4.36$ V/m. Since $D_{\hat{\mathbf{u}}}V = -\mathbf{E}\cdot\hat{\mathbf{u}}$, the result says the field has a component $+2.33$ V/m along $\overrightarrow{PQ}$, so a positive test charge would be pushed towards Q: positive charges move "downhill" in potential.

Rate of change along a curve

If a point moves along a curve $\mathbf{r}(t)$, the chain rule gives

$$\frac{d}{dt} f(\mathbf{r}(t)) = \nabla f \cdot \mathbf{r}'(t) = \|\mathbf{r}'(t)\|\, D_{\hat{\mathbf{T}}} f,$$

where $\hat{\mathbf{T}} = \mathbf{r}'/\|\mathbf{r}'\|$ is the unit tangent. The rate per unit *time* is the speed times the directional derivative along the path; the rate per unit *arc length* is the directional derivative itself. Integrating $\nabla f \cdot d\mathbf{r}$ along a path gives the total change $f(\text{end}) - f(\text{start})$, whatever the path — the fundamental theorem for line integrals of Ch. 7, and the reason the work done moving a charge between two points depends only on the potential difference.

Worked Example 4 — a sensor on a circular track

A sensor moves on the circle $\mathbf{r}(t) = (2\cos t,\ 2\sin t)$ through the temperature field $T = 80 - x^2 - 2y^2$ of Worked Example 2. How fast does its reading change, and where is it hottest?

Directly: $T(\mathbf{r}(t)) = 80 - 4\cos^2 t - 8\sin^2 t = 76 - 4\sin^2 t$, so $dT/dt = -8\sin t\cos t = -4\sin 2t$.

By the chain rule: $\nabla T = (-4\cos t,\ -8\sin t)$, $\mathbf{r}' = (-2\sin t,\ 2\cos t)$, and $\nabla T\cdot\mathbf{r}' = 8\sin t\cos t - 16\sin t\cos t = -8\sin t\cos t$. ✓

The reading is stationary where $\sin 2t = 0$: hottest ($76\ ^\circ$C) at $t = 0, \pi$ (the points $(\pm 2, 0)$) and coolest ($72\ ^\circ$C) at $t = \pm\pi/2$. At those points $\mathbf{r}'$ is tangent to an isotherm — a first glimpse of the Lagrange condition of §6.11.

Second directional derivative

Differentiating $g(h) = f(\mathbf{a} + h\hat{\mathbf{u}})$ twice gives the curvature of f along $\hat{\mathbf{u}}$:

$$D^2_{\hat{\mathbf{u}}} f = g''(0) = \sum_{i,j} f_{x_i x_j} u_i u_j = \hat{\mathbf{u}}^{\mathsf{T}} \mathbf{H}\, \hat{\mathbf{u}},$$

with $\mathbf{H}$ the Hessian. For $T = 80 - x^2 - 2y^2$, $\mathbf{H} = \operatorname{diag}(-2,-4)$, and along $\hat{\mathbf{u}} = (\cos\alpha, \sin\alpha)$ the curvature is $-2\cos^2\alpha - 4\sin^2\alpha$, between -4 and -2: the surface curves downward in every direction, which is what makes the origin a maximum (§6.10).

Common Pitfall — directional derivatives without differentiability

The formula $D_{\hat{\mathbf{u}}} f = \nabla f \cdot \hat{\mathbf{u}}$ requires f to be differentiable. For $f = x^2 y/(x^4+y^2)$ of §6.1 (with $f(0,0)=0$), every directional derivative at the origin exists: with $\hat{\mathbf{u}} = (a,b)$, $b \neq 0$,

$$D_{\hat{\mathbf{u}}} f(\mathbf{0}) = \lim_{h\to 0} \frac{h^3 a^2 b}{h\,(h^4 a^4 + h^2 b^2)} = \frac{a^2}{b},$$

and 0 when $b = 0$. But $\nabla f(\mathbf{0}) = (0,0)$, which would predict 0 in every direction. The formula fails because f is not even continuous at the origin. For smooth engineering functions this never bites, but it explains why "all directional derivatives exist" is not the definition of differentiability.

Engineering Note

In a phased-array antenna or a MIMO receiver, the received power is a function $P(\boldsymbol{\theta})$ of several steering parameters, and tracking loops adjust them along ∇P (gradient ascent). When only measurements are available, the gradient is estimated by probing two or more directions ("dithering") and solving $D_{\hat{\mathbf{u}}_k} P = \nabla P \cdot \hat{\mathbf{u}}_k$ for ∇P — two independent directions suffice in two dimensions, precisely because the gradient formula is linear in $\hat{\mathbf{u}}$.

6.6 Jacobian matrices

Motivation

So far f has produced a single number. Many engineering maps produce several: a transistor maps two terminal voltages to two currents; a coordinate change maps (r,θ) to (x,y); a nonlinear circuit's node equations map a vector of node voltages to a vector of current imbalances. For such vector functions the derivative is no longer a number or a vector but a **matrix** — the Jacobian. It is the object that makes "linearise about an operating point" precise, it is the engine of Newton's method in every circuit simulator, and its determinant is the area-scaling factor that §6.8 needs for changing variables in integrals.

Definition

A vector function $\mathbf{F}:\mathbb{R}^n\to\mathbb{R}^m$ has m component functions $F_1,\dots,F_m$, each depending on $x_1,\dots,x_n$.

Jacobian matrix

The Jacobian of $\mathbf{F}$ is the $m\times n$ matrix of first partials,

$$\mathbf{J}_{\mathbf{F}}=\frac{\partial(F_1,\dots,F_m)}{\partial(x_1,\dots,x_n)}=\begin{bmatrix}\dfrac{\partial F_1}{\partial x_1} & \cdots & \dfrac{\partial F_1}{\partial x_n}\\ \vdots & \ddots & \vdots\\ \dfrac{\partial F_m}{\partial x_1} & \cdots & \dfrac{\partial F_m}{\partial x_n}\end{bmatrix},\qquad (\mathbf{J}_{\mathbf{F}})_{ij}=\frac{\partial F_i}{\partial x_j}.$$

Row i is $(\nabla F_i)^{\mathsf T}$; column j is the rate of change of the whole output vector with respect to input x_j. When $m=n$ the matrix is square and its determinant $\det\mathbf{J}=\dfrac{\partial(F_1,\dots,F_n)}{\partial(x_1,\dots,x_n)}$ is called the **Jacobian determinant** (often just "the Jacobian").

Special cases tie together everything so far. For $m=n=1$ the Jacobian is the ordinary derivative $f'(x)$. For $m=1$ it is the gradient written as a row, $\mathbf{J}_f=(\nabla f)^{\mathsf T}$. For $n=1$ (a curve $\mathbf{r}(t)$) it is the velocity column $\mathbf{r}'(t)$.

The Jacobian as the best linear approximation

Applying the total differential to each component and stacking the results:

$$\mathbf{F}(\mathbf{x}_0+d\mathbf{x})\approx\mathbf{F}(\mathbf{x}_0)+\mathbf{J}_{\mathbf{F}}(\mathbf{x}_0)\,d\mathbf{x},\qquad\text{i.e.}\qquad d\mathbf{F}=\mathbf{J}_{\mathbf{F}}\,d\mathbf{x}.$$

Near $\mathbf{x}_0$, a smooth nonlinear map behaves like the linear map $d\mathbf{x}\mapsto\mathbf{J}\,d\mathbf{x}$ (Ch. 9). Small circles of inputs map to (approximately) small ellipses of outputs; the matrix says how they are stretched, rotated and sheared.

Chain rule in matrix form

If $\mathbf{G}:\mathbb{R}^p\to\mathbb{R}^n$ is differentiable at $\mathbf{u}$ and $\mathbf{F}:\mathbb{R}^n\to\mathbb{R}^m$ is differentiable at $\mathbf{G}(\mathbf{u})$, then

$$\mathbf{J}_{\mathbf{F}\circ\mathbf{G}}(\mathbf{u})=\mathbf{J}_{\mathbf{F}}\big(\mathbf{G}(\mathbf{u})\big)\,\mathbf{J}_{\mathbf{G}}(\mathbf{u}),$$

and for square Jacobians, $\det\mathbf{J}_{\mathbf{F}\circ\mathbf{G}}=\det\mathbf{J}_{\mathbf{F}}\cdot\det\mathbf{J}_{\mathbf{G}}$.

Proof sketch. Linearise each map: $d\mathbf{G}=\mathbf{J}_{\mathbf{G}}\,d\mathbf{u}$ and $d\mathbf{F}=\mathbf{J}_{\mathbf{F}}\,d\mathbf{G}$; substitute. Entry by entry, $(\mathbf{J}_{\mathbf{F}}\mathbf{J}_{\mathbf{G}})_{ik}=\sum_j\dfrac{\partial F_i}{\partial x_j}\dfrac{\partial x_j}{\partial u_k}$ — exactly the "sum over all paths" of the tree diagram in §6.3. The determinant statement is the product rule for determinants (Ch. 8). □

So the multivariable chain rule, with all its terms, is just matrix multiplication.

Worked Example 1 — the polar map

For $x=r\cos\theta$, $y=r\sin\theta$,

$$\frac{\partial(x,y)}{\partial(r,\theta)}=\begin{bmatrix}x_r & x_\theta\\ y_r & y_\theta\end{bmatrix}=\begin{bmatrix}\cos\theta & -r\sin\theta\\ \sin\theta & r\cos\theta\end{bmatrix},\qquad \det\mathbf{J}=r\cos^2\theta+r\sin^2\theta=r.$$

Interpretation: a small change dr moves the point dr metres radially (first column, a unit vector); a small change $d\theta$ moves it $r\,d\theta$ metres tangentially (second column, length r). The two columns are perpendicular, so a small $dr\times d\theta$ rectangle maps to a small rectangle of area $r\,dr\,d\theta$. The determinant r is exactly the area factor in $dA=r\,dr\,d\theta$ (§6.8). At the origin $\det\mathbf{J}=0$: the whole line $r=0$ (every θ) collapses to a single point.

Worked Example 2 — composing maps: squaring a complex number

The complex map $w=z^2$, with $z=u+jv$ and $w=p+jq$, is $p=u^2-v^2$, $q=2uv$. Its Jacobian is

$$\mathbf{J}_{\mathbf{F}}=\begin{bmatrix}2u & -2v\\ 2v & 2u\end{bmatrix},\qquad \det\mathbf{J}_{\mathbf{F}}=4(u^2+v^2)=4|z|^2.$$

(The pattern — equal diagonal entries, opposite off-diagonal ones — is the Cauchy–Riemann structure of an analytic function, and $\det\mathbf{J}=|dw/dz|^2=|2z|^2$; see Ch. 18.)

Now feed in polar coordinates, $u=r\cos\theta$, $v=r\sin\theta$. By the chain rule,

$$\frac{\partial(p,q)}{\partial(r,\theta)}=\begin{bmatrix}2r\cos\theta & -2r\sin\theta\\ 2r\sin\theta & 2r\cos\theta\end{bmatrix}\begin{bmatrix}\cos\theta & -r\sin\theta\\ \sin\theta & r\cos\theta\end{bmatrix}=\begin{bmatrix}2r\cos2\theta & -2r^2\sin2\theta\\ 2r\sin2\theta & 2r^2\cos2\theta\end{bmatrix}.$$

Check directly: $w=r^2e^{j2\theta}$, so $p=r^2\cos2\theta$, $q=r^2\sin2\theta$, and differentiating gives the same matrix. ✓ Determinants: $4r^2\cdot r=4r^3$ from the product, and $2r\cdot2r^2(\cos^2 2\theta+\sin^2 2\theta)=4r^3$ directly. ✓

Small-signal models

Worked Example 3 — the BJT as a two-port Jacobian

A simple model of an NPN transistor in the active region is

$$i_C = I_s e^{v_{BE}/V_T}\left(1 + \frac{v_{CE}}{V_A}\right), \qquad i_B = \frac{I_s}{\beta} e^{v_{BE}/V_T},$$

a map $(v_{BE}, v_{CE}) \mapsto (i_B, i_C)$. Its Jacobian is

$$\frac{\partial(i_B, i_C)}{\partial(v_{BE}, v_{CE})} = \begin{bmatrix} \dfrac{i_B}{V_T} & 0 \\ \dfrac{i_C}{V_T} & \dfrac{I_s e^{v_{BE}/V_T}}{V_A} \end{bmatrix} = \begin{bmatrix} 1/r_\pi & 0 \\ g_m & 1/r_o \end{bmatrix}.$$

Numbers. Bias at $I_C = 1$ mA, $V_{CE} = 5$ V, with $V_T = 25$ mV, $V_A = 100$ V, $\beta = 100$. Then $g_m = I_C/V_T = 40$ mS; $r_\pi \approx \beta/g_m = 2.5$ kΩ; and since $I_s e^{v_{BE}/V_T} = I_C/(1 + V_{CE}/V_A)$, $1/r_o = I_C/(V_A + V_{CE})$, so $r_o = 105$ kΩ.

Using it. A 1 mV change in v_{BE} changes i_C by $g_m \cdot 1\text{ mV} = 40\ \mu$A (linearised). The exact change is $1\text{ mA}\,(e^{0.04} - 1) = 40.8\ \mu$A; the 2 % discrepancy is the curvature of the exponential, and it is the origin of distortion in amplifiers driven too hard. The Jacobian entries are precisely the hybrid-π small-signal parameters, and this matrix is the device's y**-parameter** matrix at the bias point.

Engineering Note

Every two-port parameter set (y, z, h, g, $ABCD$) of a nonlinear device is a Jacobian at its bias point, expressed with different choices of which port variables are inputs. Converting between parameter sets is inverting or rearranging these matrices, which is why small-signal analysis is linear algebra (Ch. 8).

The inverse function theorem

Inverse function theorem

Let $\mathbf{F} : \mathbb{R}^n \to \mathbb{R}^n$ have continuous partial derivatives near $\mathbf{x}_0$, with $\det \mathbf{J}_\mathbf{F}(\mathbf{x}_0) \neq 0$. Then $\mathbf{F}$ has a differentiable inverse near $\mathbf{y}_0 = \mathbf{F}(\mathbf{x}_0)$, and

$$\mathbf{J}_{\mathbf{F}^{-1}}(\mathbf{y}_0) = \left[\mathbf{J}_\mathbf{F}(\mathbf{x}_0)\right]^{-1}, \qquad \det \mathbf{J}_{\mathbf{F}^{-1}} = \frac{1}{\det \mathbf{J}_\mathbf{F}}.$$

The formula follows from the chain rule applied to $\mathbf{F}^{-1} \circ \mathbf{F} = \text{identity}$, whose Jacobian is $\mathbf{I}$. The theorem says that a non-zero Jacobian determinant is what makes a change of variables locally reversible; where it vanishes (the origin in polar coordinates) the coordinates break down.

Worked Example 4 — inverse polar derivatives without differentiating

Inverting the matrix of Worked Example 1 (determinant r),

$$\frac{\partial(r,\theta)}{\partial(x,y)} = \frac{1}{r}\begin{bmatrix} r\cos\theta & r\sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\dfrac{\sin\theta}{r} & \dfrac{\cos\theta}{r} \end{bmatrix}.$$

Check directly: $r = \sqrt{x^2 + y^2}$ gives $r_x = x/r = \cos\theta$ and $r_y = y/r = \sin\theta$; $\theta = \operatorname{atan2}(y, x)$ gives $\theta_x = -y/(x^2 + y^2) = -\sin\theta/r$ and $\theta_y = x/(x^2 + y^2) = \cos\theta/r$. ✓ This resolves the puzzle of §6.3: $x_r = \cos\theta$ and $r_x = \cos\theta$ are not reciprocals; it is the *matrices* that are inverses, not their individual entries.

Sensitivity of solutions: the implicit function theorem

Suppose a system of n equations $\mathbf{F}(\mathbf{x}, \mathbf{p}) = \mathbf{0}$ determines n unknowns $\mathbf{x}$ (node voltages, say) in terms of parameters $\mathbf{p}$ (component values). Differentiating the identity $\mathbf{F}(\mathbf{x}(\mathbf{p}), \mathbf{p}) = \mathbf{0}$:

$$\mathbf{J}_\mathbf{x}\, d\mathbf{x} + \mathbf{J}_\mathbf{p}\, d\mathbf{p} = \mathbf{0} \quad \Longrightarrow \quad \frac{\partial \mathbf{x}}{\partial \mathbf{p}} = -\mathbf{J}_\mathbf{x}^{-1}\mathbf{J}_\mathbf{p},$$

valid wherever $\det \mathbf{J}_\mathbf{x} \neq 0$ (the vector **implicit function theorem**). This is the matrix version of $dy/dx = -F_x/F_y$, and it is how circuit simulators compute the sensitivity of an operating point to every component in one linear solve, reusing the Jacobian they already built for Newton's method.

Newton's method for nonlinear systems

To solve $\mathbf{F}(\mathbf{x}) = \mathbf{0}$, replace $\mathbf{F}$ by its linearisation at the current guess and solve the resulting *linear* system for the step:

$$\mathbf{J}_\mathbf{F}(\mathbf{x}_k)\,\Delta\mathbf{x} = -\mathbf{F}(\mathbf{x}_k), \qquad \mathbf{x}_{k+1} = \mathbf{x}_k + \Delta\mathbf{x}.$$

Geometrically, each component F_i is replaced by its tangent plane, and the step goes to where all the tangent planes cross zero simultaneously. Near a root with non-singular Jacobian the error is roughly *squared* at each step (**quadratic convergence**), so the number of correct digits about doubles per iteration. This is precisely how SPICE finds the DC operating point of a nonlinear circuit: each diode and transistor is replaced by its small-signal "companion model" (a Jacobian entry plus a source), the resulting linear circuit is solved, and the process repeats.

Worked Example 5 — Newton iteration for two equations

Solve $x^2 + y^2 = 4$, $xy = 1$, starting from $(x_0, y_0) = (2,\ 0.5)$.

Set up: $\mathbf{F} = (x^2 + y^2 - 4,\ xy - 1)$, $\mathbf{J} = \begin{bmatrix} 2x & 2y \\ y & x \end{bmatrix}$.

Step 1: $\mathbf{F}(2, 0.5) = (0.25,\ 0)$ and

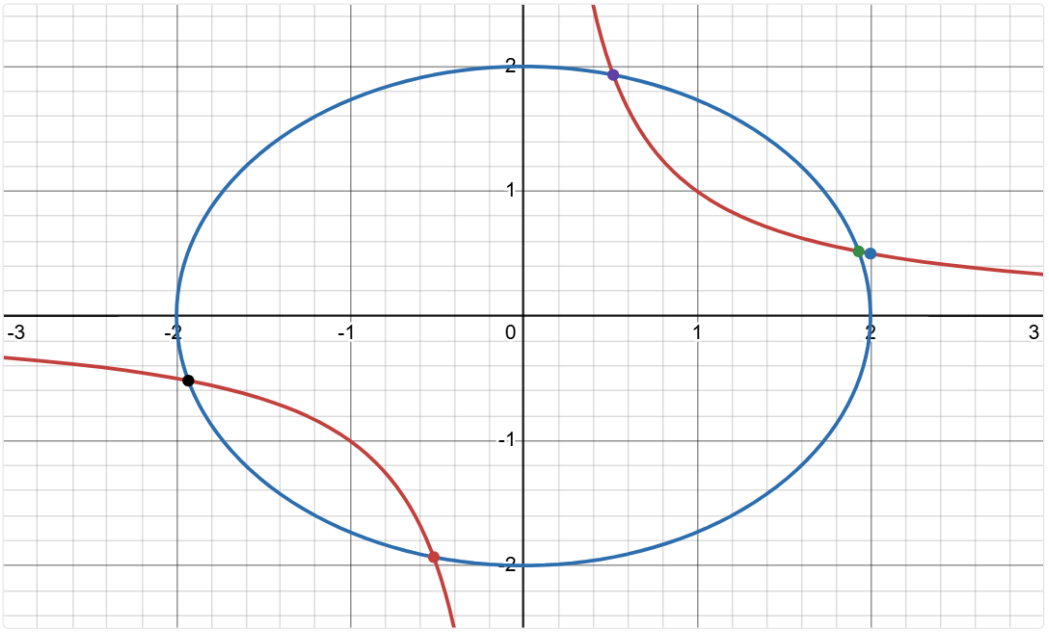
$$\mathbf{J} = \begin{bmatrix} 4 & 1 \\ 0.5 & 2 \end{bmatrix}, \qquad \det \mathbf{J} = 7.5, \qquad \Delta\mathbf{x} = \frac{1}{7.5}\begin{bmatrix} 2 & -1 \\ -0.5 & 4 \end{bmatrix}\begin{bmatrix} -0.25 \\ 0 \end{bmatrix} = \begin{bmatrix} -0.0667 \\ 0.0167 \end{bmatrix}.$$

So $(x_1, y_1) = (1.93333,\ 0.51667)$.

Further steps (same recipe):

k	x_k	y_k	error $\lVert \mathbf{x}_k - \mathbf{x}^* \rVert$
0	2.00000	0.50000	7.0×10^{-2}
1	1.93333	0.51667	1.8×10^{-3}
2	1.93185274	0.51763705	1.5×10^{-6}
3	1.93185165	0.51763809	1.1×10^{-12}

The exact root is $x^* = (\sqrt{6} + \sqrt{2})/2 = 1.93185165$, $y^* = (\sqrt{6} - \sqrt{2})/2 = 0.51763809$. Each error is roughly a constant times the square of the previous one ($1.8 \times 10^{-3} \approx 0.36\,(7.0 \times 10^{-2})^2$, $1.5 \times 10^{-6} \approx 0.47\,(1.8 \times 10^{-3})^2$): quadratic convergence.



The circle and hyperbola intersect in four points; Newton's method converges to the one nearest the starting guess. Starting near the line $y = x$, where $\det \mathbf{J} = 2(x^2 - y^2)$ vanishes, would send the first step far away.

Worked Example 6 — how the solution moves with a parameter

Replace the circle by $x^2 + y^2 = c$. How fast does the root found above move as c increases through 4?

With $\mathbf{F} = (x^2 + y^2 - c,\ xy - 1)$, $\mathbf{J}_\mathbf{x} = \begin{bmatrix} 2x & 2y \\ y & x \end{bmatrix}$ and $\mathbf{J}_c = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$. The sensitivity formula gives

$$\frac{d\mathbf{x}}{dc} = -\mathbf{J}_\mathbf{x}^{-1}\mathbf{J}_c = \frac{1}{2(x^2 - y^2)}\begin{bmatrix} x & -2y \\ -y & 2x \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{2(x^2 - y^2)}\begin{bmatrix} x \\ -y \end{bmatrix}.$$

At the root, $x^2 - y^2 = (x + y)(x - y) = \sqrt{6} \cdot \sqrt{2} = 3.4641$, so $dx/dc = 1.93185/6.9282 = 0.2788$ and $dy/dc = -0.0747$. Re-solving with $c = 4.01$ and differencing gives $(0.2786,\ -0.0745)$. ✓ Enlarging the circle slides the intersection outward along the hyperbola, as the picture suggests.

Common Pitfall

Newton's method needs a good starting point and a non-singular Jacobian. Near a point where $\det \mathbf{J} \approx 0$ the step $\Delta\mathbf{x} = -\mathbf{J}^{-1}\mathbf{F}$ becomes huge and the iteration can jump to a different root or diverge. Circuit simulators protect themselves with damping (limiting the step, especially across exponential junctions), source stepping (ramping supplies up from zero) and "gmin" stepping (adding small conductances that keep $\mathbf{J}$ well conditioned). Never compute $\mathbf{J}^{-1}$ explicitly in practice: solve $\mathbf{J}\Delta\mathbf{x} = -\mathbf{F}$ by elimination (Ch. 8).

The Hessian as a Jacobian

The gradient ∇f is itself a vector function $\mathbb{R}^n \to \mathbb{R}^n$, and its Jacobian is the symmetric matrix of second partials:

$$\mathbf{H}_f = \mathbf{J}_{\nabla f} = \begin{bmatrix} f_{x_1x_1} & \cdots & f_{x_1x_n} \\ \vdots & \ddots & \vdots \\ f_{x_nx_1} & \cdots & f_{x_nx_n} \end{bmatrix}, \qquad \mathbf{H}^{\mathsf{T}} = \mathbf{H} \ (\text{Clairaut}).$$

Newton's method applied to $\nabla f = \mathbf{0}$ — the condition for an extremum — reads $\mathbf{H}\,\Delta\mathbf{x} = -\nabla f$, the basis of Newton-type optimisers. The Hessian also classifies critical points (§6.10).

6.7 Multiple integration

Motivation

Integration in one variable adds up a quantity distributed along a line: charge on a wire, energy over time. In two and three dimensions the same idea adds up quantities spread over areas and volumes: the total charge on a plate with non-uniform surface density, the current through a conductor cross-section with non-uniform current density, the energy stored in a field, the probability that two random noise voltages both exceed a threshold. The new ingredient is the *shape* of the region, which determines the limits of integration — and most of the skill in multiple integration lies in describing that shape correctly.

Double integrals: definition and properties

Let f be bounded on a bounded region R of the plane. Cover R with a grid of small rectangles; let the k-th rectangle inside R have area ΔA_k and contain a sample point (x_k, y_k).

Double integral

$$\iint_R f(x,y)\,dA = \lim_{\max \Delta A_k \to 0} \sum_k f(x_k, y_k)\,\Delta A_k,$$

when the limit exists and is independent of the grid and the sample points. It exists whenever f is continuous on a closed, bounded region R with a reasonably smooth boundary.

Interpretations depend on what f is:

Integrand f	$\iint_R f\,dA$ means	Units
1	area of R	m^2
height $f \geq 0$	volume under the surface $z = f(x,y)$	m^3
surface charge density ρ_s	total charge on R	C
current density J_z across a cross-section	current through it	A
joint probability density p_{XY}	probability that $(X,Y) \in R$	dimensionless

Properties of double integrals

1. **Linearity:** $\iint_R (af + bg)\,dA = a\iint_R f\,dA + b\iint_R g\,dA$.
2. **Additivity:** if $R = R_1 \cup R_2$ with R_1, R_2 overlapping only on boundaries, $\iint_R = \iint_{R_1} + \iint_{R_2}$.
3. **Monotonicity:** if $f \leq g$ on R, then $\iint_R f\,dA \leq \iint_R g\,dA$; in particular $m\,A(R) \leq \iint_R f\,dA \leq M\,A(R)$ when $m \leq f \leq M$.
4. **Mean value theorem:** if f is continuous and R connected, $\iint_R f\,dA = f(\mathbf{c})\,A(R)$ for some point $\mathbf{c} \in R$.

Iterated integrals and Fubini's theorem

The Riemann-sum definition is not a practical way to compute. Instead, we slice: on a rectangle $[a,b] \times [c,d]$, add up the rectangles in each vertical column first (an integral in y at fixed x), then add the columns (an integral in x).

Fubini's theorem

If f is continuous on the rectangle $[a,b] \times [c,d]$,

$$\iint_R f\,dA = \int_a^b \left[\int_c^d f(x,y)\,dy\right] dx = \int_c^d \left[\int_a^b f(x,y)\,dx\right] dy.$$

More generally, for a **type I** region $R = \{a \leq x \leq b,\ g_1(x) \leq y \leq g_2(x)\}$,

$$\iint_R f\,dA = \int_a^b \left[\int_{g_1(x)}^{g_2(x)} f(x,y)\,dy\right] dx,$$

and for a **type II** region $\{c \leq y \leq d,\ h_1(y) \leq x \leq h_2(y)\}$ integrate in x first, between $h_1(y)$ and $h_2(y)$.

Why it works. The inner integral $A(x) = \int_{g_1(x)}^{g_2(x)} f(x,y)\,dy$ is the area of the cross-section of the solid under f at position x. Integrating cross-sectional area along x gives volume — the slicing method of Chapter 5 — and that volume is the double integral. Slicing the other way must give the same volume.

Recipe for setting up limits.

1. Sketch the region and label its boundary curves.
2. Choose the inner variable (say y). Draw a line across the region in the y-direction at a typical x.
3. The inner limits are the y-values where the line *enters* and *leaves* the region — in general functions of x.
4. The outer limits are the extreme values of x over the whole region — always constants.
5. If the entry or exit curve changes partway across, split the region into pieces.

Worked Example 1 — a rectangle, both orders

Evaluate $\iint_R (x^2 + xy)\,dA$ over $R = [0,2] \times [0,1]$.

y first:

$$\int_0^2 \left[\int_0^1 (x^2 + xy)\,dy\right] dx = \int_0^2 \left[x^2 y + \tfrac{1}{2}xy^2\right]_{y=0}^{1} dx = \int_0^2 \left(x^2 + \tfrac{1}{2}x\right) dx = \tfrac{8}{3} + 1 = \tfrac{11}{3}.$$

x first:

$$\int_0^1 \left[\int_0^2 (x^2 + xy)\,dx\right] dy = \int_0^1 \left(\tfrac{8}{3} + 2y\right) dy = \tfrac{8}{3} + 1 = \tfrac{11}{3}.\checkmark$$

Numerically 3.6667. The average value of the integrand over the rectangle is $(11/3)/2 = 1.833$.

Worked Example 2 — area and centroid of a region between curves

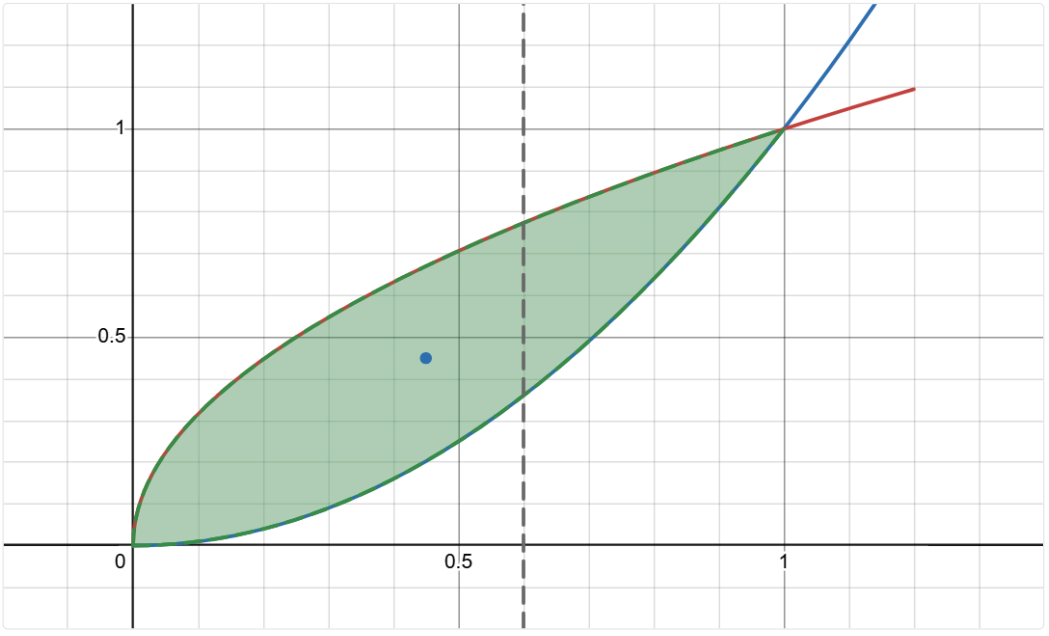
The region between $y = x^2$ and $y = \sqrt{x}$ for $0 \le x \le 1$ is type I (vertical strips from $y = x^2$ up to $y = \sqrt{x}$).

Area:

$$A = \int_0^1 \int_{x^2}^{\sqrt{x}} dy\,dx = \int_0^1 \left(\sqrt{x} - x^2\right) dx = \frac{2}{3} - \frac{1}{3} = \frac{1}{3}.$$

Centroid: $\bar{x} = \dfrac{1}{A}\iint x\,dA = 3\int_0^1 x\left(\sqrt{x} - x^2\right) dx = 3\left(\tfrac{2}{5} - \tfrac{1}{4}\right) = \tfrac{9}{20} = 0.45$. The region is symmetric about $y = x$ (swapping x and y swaps the two curves), so $\bar{y} = \bar{x} = 0.45$ without further work; direct integration, $\bar{y} = 3\int_0^1 \tfrac{1}{2}(x - x^4)\,dx = 3 \cdot \tfrac{1}{2}\left(\tfrac{1}{2} - \tfrac{1}{5}\right) = 0.45$, confirms it.

As a type II region the same set is $0 \le y \le 1$, $y^2 \le x \le \sqrt{y}$ — horizontal strips from the parabola $x = y^2$ to $x = \sqrt{y}$.



The shaded region is type I (vertical strips such as the dashed one, entering at $y = x^2$ and leaving at $y = \sqrt{x}$) and also type II (horizontal strips from $x = y^2$ to $x = \sqrt{y}$). Its centroid lies on the symmetry line $y = x$.

Worked Example 3 — charge on a triangular plate

A triangular plate occupies $0 \le y \le x \le a$ and carries surface charge density $\rho_s = \rho_0\, xy/a^2$ (ρ_0 in $\mathrm{C/m^2}$). Find the total charge.

Region: vertical strips at fixed x run from $y = 0$ to $y = x$, for $0 \le x \le a$.

$$Q = \frac{\rho_0}{a^2}\int_0^a \int_0^x xy\,dy\,dx = \frac{\rho_0}{a^2}\int_0^a x \cdot \frac{x^2}{2}\,dx = \frac{\rho_0}{a^2} \cdot \frac{a^4}{8} = \frac{\rho_0 a^2}{8}.$$

Units: $\mathrm{C/m^2} \times \mathrm{m^2} = \mathrm{C}$. ✓ *Sanity check:* the plate area is $a^2/2$ and the density ranges from 0 to ρ_0, so Q must lie between 0 and $\rho_0 a^2/2$; the answer is a quarter of the upper bound, reasonable since the density is small over most of the plate.

Reversing the order of integration

Sometimes the inner integral cannot be done in the given order, but the other order is easy. Reversing is a matter of re-describing the *same region* the other way round — never of swapping the limits mechanically.

Worked Example 4 — reversing to remove a non-elementary integral

(a) $I = \displaystyle\int_0^1 \int_y^1 e^{x^2}\,dx\,dy$ cannot be done as written, since e^{x^2} has no elementary antiderivative.

Region: $0 \le y \le 1$ and $y \le x \le 1$, i.e. the triangle $0 \le y \le x \le 1$ with vertices $(0,0)$, $(1,0)$, $(1,1)$.

Re-describe as type I: for each $x \in [0,1]$, y runs from 0 to x. Then

$$I = \int_0^1 \int_0^x e^{x^2}\,dy\,dx = \int_0^1 x\,e^{x^2}\,dx = \Big[\tfrac{1}{2}e^{x^2}\Big]_0^1 = \frac{e-1}{2} \approx 0.859.$$

(b) The same trick handles $J = \displaystyle\int_0^\pi \int_y^\pi \frac{\sin x}{x}\,dx\,dy$, whose inner integral is the non-elementary sine integral. The region is $0 \le y \le x \le \pi$, so

$$J = \int_0^\pi \int_0^x \frac{\sin x}{x}\,dy\,dx = \int_0^\pi \frac{\sin x}{x}\cdot x\,dx = \int_0^\pi \sin x\,dx = 2.$$

Numerical double integration in the original order gives 2.0000. ✓ The pattern: the inner integration in the new order produces a factor (x) that cancels the troublesome part.

Common Pitfall

The outer limits must be **constants** and the inner limits may depend only on the **outer** variable. A setup like $\int_0^1 \int_0^x f\,dx\,dy$ (inner limit depending on the inner variable) is meaningless. When reversing the order, always sketch the region: swapping $\int_0^1 \int_y^1$ into $\int_y^1 \int_0^1$ is a common and fatal error.

Triple integrals

For $f(x,y,z)$ on a solid V, the triple integral $\iiint_V f\,dV$ is defined by Riemann sums over small boxes and evaluated, by Fubini's theorem again, as three nested single integrals. A typical setup is

$$\iiint_V f\,dV = \int_a^b \int_{g_1(x)}^{g_2(x)} \int_{h_1(x,y)}^{h_2(x,y)} f(x,y,z)\,dz\,dy\,dx,$$

where for each (x,y) in the "shadow" of V on the xy-plane, z runs from the bottom surface h_1 to the top surface h_2, and the shadow itself is described as a 2-D region. With $f = 1$ the integral gives volume; with $f = \rho_v$ (volume charge density, $\mathrm{C/m^3}$) it gives charge; with $f = w_e = \tfrac{1}{2}\varepsilon\|\mathbf{E}\|^2$ it gives stored electric energy.

Worked Example 5 — separable integrand on a cube

A cube $0 \le x, y, z \le a$ carries $\rho_v = \rho_0\,xyz/a^3$. Both the region and the integrand separate, so the triple integral is a product of single integrals:

$$Q = \frac{\rho_0}{a^3}\int_0^a x\,dx \int_0^a y\,dy \int_0^a z\,dz = \frac{\rho_0}{a^3}\left(\frac{a^2}{2}\right)^3 = \frac{\rho_0 a^3}{8}.$$

Units: $\mathrm{C/m^3} \times \mathrm{m^3} = \mathrm{C}$. ✓ Because the density is a product of three linear ramps, its average over the cube is $\rho_0/8$, and $Q = (\rho_0/8)\cdot a^3$.

Worked Example 6 — the tetrahedron

Let V be the tetrahedron bounded by the coordinate planes and $x + y + z = 1$. Find its volume and the height $\bar{z}$ of its centroid.

Limits: top surface $z = 1 - x - y$, bottom $z = 0$; the shadow is the triangle $x \ge 0$, $y \ge 0$, $x + y \le 1$, i.e. $0 \le x \le 1$, $0 \le y \le 1 - x$.

Volume:

$$\int_0^1 \int_0^{1-x} (1 - x - y)\,dy\,dx = \int_0^1 \frac{(1-x)^2}{2}\,dx = \frac{1}{6}.$$

Moment:

$$\iiint_V z\,dV = \int_0^1 \int_0^{1-x} \frac{(1-x-y)^2}{2}\,dy\,dx = \int_0^1 \frac{(1-x)^3}{6}\,dx = \frac{1}{24}.$$

Hence $\bar{z} = (1/24)/(1/6) = 1/4$, and by symmetry the centroid is $\left(\tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4}\right)$ — one quarter of the way from the base face to the opposite vertex, the standard result for any tetrahedron. Numerical triple integration gives $0.041667 = 1/24$. ✓

Averages, centroids and moments

$$\bar{f} = \frac{1}{A}\iint_R f\,dA, \qquad \bar{x} = \frac{1}{A}\iint_R x\,dA, \qquad \bar{y} = \frac{1}{A}\iint_R y\,dA,$$

with the obvious extensions to volumes, and to *weighted* centres when a density ρ multiplies the integrand and the total "mass" $\iint \rho\,dA$ replaces A. **Second moments** $\iint x^2\rho\,dA$ appear as moments of inertia in mechanics and as variances in probability (Ch. 19); the root-mean-square value of a field over an area, $\sqrt{\frac{1}{A}\iint f^2\,dA}$, is the quantity that determines average power.

Numerical double integration

When no antiderivative exists, the iterated form still allows numerical evaluation: apply any one-dimensional rule (Ch. 22) to the inner integral and again to the outer one. The two-dimensional **midpoint rule** on an $n \times n$ grid of cells of size $h \times k$ is

$$\iint_R f\,dA \approx hk \sum_{i=1}^{n}\sum_{j=1}^{n} f(x_i^*, y_j^*),$$

with (x_i^*, y_j^*) the cell centres.

Worked Example 7 — midpoint rule for a Gaussian on the unit square

$I = \iint_{[0,1]^2} e^{-(x^2+y^2)}\,dA = \left(\int_0^1 e^{-x^2}dx\right)^2 = 0.746824^2 = 0.557746$ (separable, using the tabulated 1-D value).

grid	midpoint estimate	error
2×2	0.56942	1.17×10^{-2}
4×4	0.56062	2.88×10^{-3}
8×8	0.55846	7.16×10^{-4}

Halving the cell size divides the error by about 4: the midpoint rule is second-order accurate in 2-D, as in 1-D. But the cost — the number of function evaluations — grows as n^2 in 2-D and n^d in d dimensions. For high-dimensional integrals (bit-error rates over many noise variables, yield over many component tolerances) this "curse of dimensionality" is why Monte Carlo methods take over (Ch. 24).

Engineering Note

Improper double integrals — over unbounded regions or with singular integrands — are routine in EE: the probability that a Gaussian noise vector lands outside a decision region, the energy of a field extending to infinity, the potential of a charged sheet. They are defined as limits of integrals over growing (or shrinking-hole) regions, and for non-negative integrands Fubini's theorem still allows any order. The most important one, $\iint_{\mathbb{R}^2} e^{-(x^2+y^2)}dA = \pi$, is evaluated in §6.8.

6.8 Coordinate transformations

Motivation

Cartesian coordinates make rectangles easy and everything else awkward. A disc, a sector, a region bounded by hyperbolas, an integrand containing $x^2 + y^2$ — each becomes simple in the right coordinates. Changing variables in a multiple integral is the multivariable analogue of substitution, $\int f(x)\,dx = \int f(x(u))\,x'(u)\,du$, and the role of $x'(u)$ is played by the Jacobian determinant of §6.6. The idea also has a probabilistic face: transforming random variables (from in-phase and quadrature noise to envelope and phase, say) uses exactly the same Jacobian factor.

How a map scales area

Let $x = x(u,v)$, $y = y(u,v)$ be a one-to-one, continuously differentiable map from a region S of the uv-plane onto a region R of the xy-plane, and write $\mathbf{r}(u,v) = (x(u,v), y(u,v))$. A small coordinate rectangle with corners (u,v), $(u+du,v)$, $(u,v+dv)$ is carried to an approximate **parallelogram** with sides

$$\mathbf{r}(u+du,v) - \mathbf{r}(u,v) \approx \mathbf{r}_u\,du = (x_u,\ y_u)\,du, \qquad \mathbf{r}(u,v+dv) - \mathbf{r}(u,v) \approx \mathbf{r}_v\,dv = (x_v,\ y_v)\,dv.$$

The area of a parallelogram spanned by (a_1,a_2) and (b_1,b_2) is $|a_1b_2 - a_2b_1|$, the absolute value of a 2×2 determinant. Hence

$$dA = |x_u y_v - x_v y_u|\,du\,dv = \left|\det \frac{\partial(x,y)}{\partial(u,v)}\right| du\,dv.$$

The Jacobian determinant is the local **area magnification factor** of the map. Summing over all the little parallelograms and taking the limit gives the theorem.

Change of variables in multiple integrals

Under the conditions above (one-to-one, continuously differentiable, Jacobian determinant non-zero except possibly on a set of zero area),

$$\iint_R f(x,y)\,dx\,dy = \iint_S f\big(x(u,v), y(u,v)\big)\left|\frac{\partial(x,y)}{\partial(u,v)}\right| du\,dv.$$

In three dimensions, with $(x,y,z) = \mathbf{r}(u,v,w)$, the small box becomes a parallelepiped of volume $|\det \mathbf{J}|\,du\,dv\,dw$, and

$$\iiint_V f\,dx\,dy\,dz = \iiint_W f\big(\mathbf{r}(u,v,w)\big)\left|\frac{\partial(x,y,z)}{\partial(u,v,w)}\right| du\,dv\,dw.$$

Three practical facts:

- **The absolute value is essential.** Area and volume are positive; a negative Jacobian determinant only means the map reverses orientation (like a mirror).
- **Inverse Jacobians are reciprocals** (inverse function theorem, §6.6): $\dfrac{\partial(x,y)}{\partial(u,v)} = 1 \Big/ \dfrac{\partial(u,v)}{\partial(x,y)}$. When the substitution is given as $u = u(x,y)$, $v = v(x,y)$, compute the easy Jacobian and invert it — there is no need to solve for x and y.

- **Choose the substitution to simplify the region first.** The integrand usually follows. Good choices make the boundary curves of R into coordinate lines $u = \text{const}$, $v = \text{const}$, so that S is a rectangle.

Worked Example 1 — a linear change of variables

Evaluate $\iint_D (x+y)\,dA$ over the parallelogram D bounded by $x+y=0$, $x+y=2$, $x-y=0$, $x-y=1$.

Substitution suggested by the boundaries: $u = x+y \in [0,2]$, $v = x-y \in [0,1]$; S is a rectangle.

Jacobian: $\dfrac{\partial(u,v)}{\partial(x,y)} = \det\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix} = -2$, so $\left|\dfrac{\partial(x,y)}{\partial(u,v)}\right| = \dfrac{1}{2}$.

Integral:

$$\iint_D (x+y)\,dA = \int_0^1 \int_0^2 u \cdot \tfrac{1}{2}\,du\,dv = \tfrac{1}{2}\cdot 2 \cdot 1 = 1.$$

Check: the area of D is $\iint_S \frac{1}{2}\,du\,dv = \frac{1}{2}\cdot 2\cdot 1 = 1$, and the average of $u = x+y$ over D is clearly 1 (it runs uniformly from 0 to 2), so the integral is $1 \times 1 = 1$. ✓

Worked Example 2 — the ellipse

Find the area of the ellipse $x^2/a^2 + y^2/b^2 \le 1$, and $\iint x^2\,dA$ over it.

Substitution: $x = au$, $y = bv$ maps the unit disc $u^2+v^2 \le 1$ onto the ellipse, with $\partial(x,y)/\partial(u,v) = ab$.

Area: $\iint_{\text{disc}} ab\,du\,dv = ab\cdot\pi = \pi ab$.

Second moment: $\iint x^2\,dA = \iint_{\text{disc}} a^2u^2 \cdot ab\,du\,dv = a^3b\iint_{\text{disc}} u^2\,du\,dv = a^3b\cdot\dfrac{\pi}{4}$, using the polar result $\iint_{\text{disc}} u^2 = \int_0^{2\pi}\int_0^1 r^2\cos^2\theta\, r\,dr\,d\theta = \frac{1}{4}\cdot\pi$. For $a=2$, $b=1$: $2\pi = 6.2832$, matching direct numerical integration. ✓ The linear stretch turned an ellipse problem into a circle problem.

Worked Example 3 — a region bounded by hyperbolas and lines

Find the area of the region R in the first quadrant bounded by $xy=1$, $xy=4$, $y=x$ and $y=2x$.

Substitution: $u = xy \in [1,4]$, $v = y/x \in [1,2]$ — again the boundaries become coordinate lines.

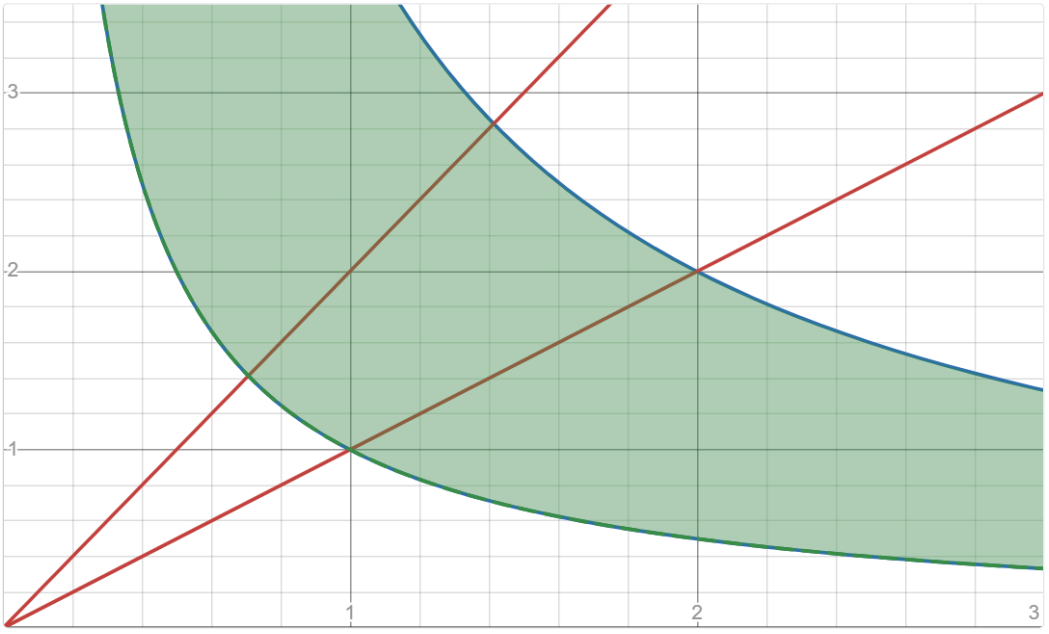
Jacobian of the easy direction:

$$\frac{\partial(u,v)}{\partial(x,y)} = \det\begin{bmatrix} y & x \\ -y/x^2 & 1/x\end{bmatrix} = \frac{y}{x} + \frac{y}{x} = 2v, \qquad \text{so} \qquad \left|\frac{\partial(x,y)}{\partial(u,v)}\right| = \frac{1}{2v}.$$

Area:

$$A = \int_1^4 \int_1^2 \frac{1}{2v}\,dv\,du = 3\cdot\tfrac{1}{2}\ln 2 = \tfrac{3}{2}\ln 2 \approx 1.0397.$$

A direct Cartesian computation would need the region split into three pieces (the entry and exit curves change at $x = 1/\sqrt{2}$, 1, $\sqrt{2}$ and 2); numerically it gives 1.0397. ✓



The region of Worked Example 3 is bounded by two hyperbolas (blue) and two lines (red); in the coordinates $u = xy$, $v = y/x$ it becomes the rectangle $[1,4]\times[1,2]$ — only the part between the red lines counts.

Polar coordinates

With $x = r\cos\theta$, $y = r\sin\theta$, the Jacobian determinant is r (§6.6), so

$$dA = r\,dr\,d\theta.$$

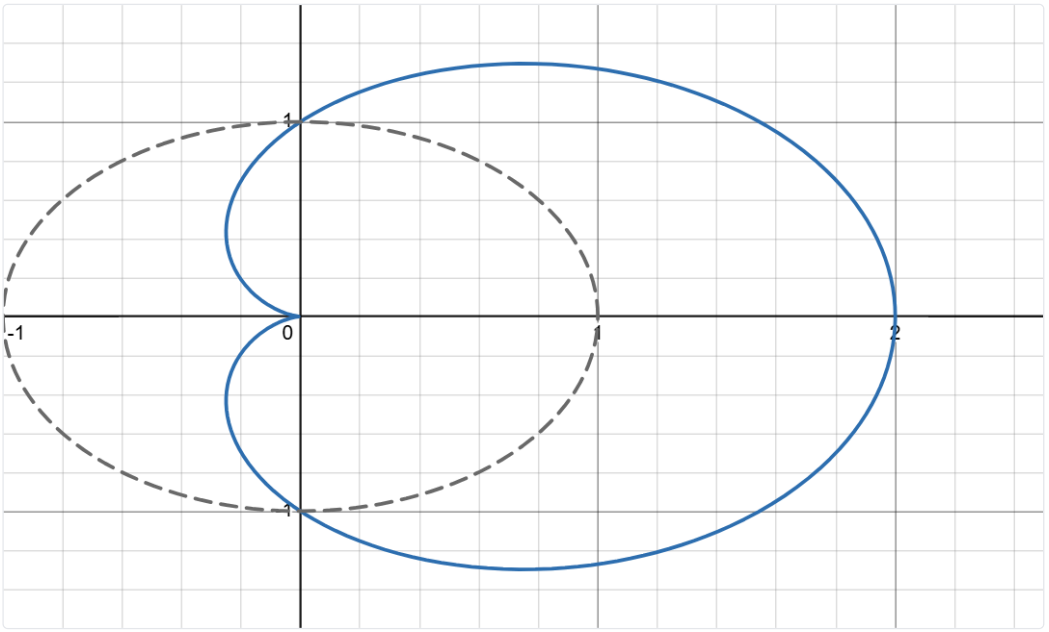
The factor r has a direct geometric meaning: the small polar "rectangle" between radii r and $r + dr$ and angles θ and $\theta + d\theta$ has radial side dr and arc side $r\,d\theta$. Use polar coordinates whenever the region is a disc, annulus, sector or is bounded by a polar curve $r = g(\theta)$, or when the integrand contains $x^2 + y^2$.

Worked Example 4 — area inside a cardioid

The cardioid $r = 1 + \cos\theta$ ($0 \le \theta < 2\pi$) — the shape of a cardioid microphone's polar pattern — encloses the region $0 \le r \le 1 + \cos\theta$. Its area is

$$A = \int_0^{2\pi}\int_0^{1+\cos\theta} r\,dr\,d\theta = \int_0^{2\pi}\frac{(1+\cos\theta)^2}{2}\,d\theta = \frac{1}{2}\int_0^{2\pi}\left(1 + 2\cos\theta + \cos^2\theta\right)d\theta = \frac{1}{2}\left(2\pi + 0 + \pi\right) = \frac{3\pi}{2}.$$

Numerically 4.712, which sits sensibly between the area of the circle of radius 1 (π) and the circle of radius 2 that just encloses the cardioid (4π).



The cardioid $r = 1 + \cos\theta$ (blue) compared with the unit circle; integrating $r\,dr$ from 0 out to the curve at each angle sweeps out its area $3\pi/2$.

Worked Example 5 — the Gaussian integral

Let $I = \int_{-\infty}^{\infty} e^{-x^2}dx$. No antiderivative exists, but the *square* of I is a double integral that polar coordinates make trivial:

$$I^2 = \int_{-\infty}^{\infty}e^{-x^2}dx\int_{-\infty}^{\infty}e^{-y^2}dy = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)}\,dx\,dy = \int_0^{2\pi}\int_0^{\infty} e^{-r^2}\,r\,dr\,d\theta.$$

The factor r from the Jacobian is exactly what is needed: with $w = r^2$, $\int_0^{\infty} e^{-r^2} r\,dr = \tfrac{1}{2}\int_0^{\infty} e^{-w}dw = \tfrac{1}{2}$. Hence $I^2 = 2\pi\cdot\tfrac{1}{2} = \pi$ and

$$\int_{-\infty}^{\infty} e^{-x^2}dx = \sqrt{\pi}, \qquad \int_{-\infty}^{\infty} e^{-x^2/(2\sigma^2)}dx = \sigma\sqrt{2\pi}.$$

The second form (substitute $x \to x/(\sigma\sqrt{2})$) normalises the Gaussian (normal) distribution of thermal noise (Ch. 19) and underlies the Fourier transform of a Gaussian pulse (Ch. 16).

Worked Example 6 — the Rayleigh distribution of a noise envelope

The in-phase and quadrature components X, Y of narrowband noise are independent Gaussians with zero mean and variance σ^2, with joint density

$$p(x,y) = \frac{1}{2\pi\sigma^2}e^{-(x^2+y^2)/(2\sigma^2)}.$$

What is the probability that the envelope $R = \sqrt{X^2 + Y^2}$ exceeds a threshold ρ?

The event $R \le \rho$ is the disc of radius ρ, so in polar coordinates

$$P(R \le \rho) = \int_0^{2\pi}\int_0^{\rho}\frac{1}{2\pi\sigma^2}e^{-r^2/(2\sigma^2)}\,r\,dr\,d\theta = \left[-e^{-r^2/(2\sigma^2)}\right]_0^{\rho} = 1 - e^{-\rho^2/(2\sigma^2)}.$$

Differentiating in ρ gives the **Rayleigh density** $p_R(\rho) = \dfrac{\rho}{\sigma^2}e^{-\rho^2/(2\sigma^2)}$ — and the leading factor ρ is nothing but the polar Jacobian. For a threshold $\rho = 3\sigma$, $P(R > 3\sigma) = e^{-4.5} = 0.0111$: about 1.1 % of the time, noise alone crosses a 3σ envelope threshold. This is the basic false-alarm calculation of a radar or envelope detector (Ch. 21).

Common Pitfall

Forgetting the Jacobian. "Area of the unit disc $= \int_0^{2\pi}\int_0^1 dr\,d\theta = 2\pi$" is wrong by a factor of 2; the correct $\int_0^{2\pi}\int_0^1 r\,dr\,d\theta = \pi$. A quick check that catches this every time: integrate $f = 1$ and see whether you get the right area or volume. A second trap is a map that is not one-to-one: letting θ run over $[0, 4\pi]$ counts every point twice.

Engineering Note

The same Jacobian factor transforms probability densities. If random variables (X, Y) with joint density p_{XY} are mapped one-to-one to (U, V), then $p_{UV}(u,v) = p_{XY}(x,y)\,|\partial(x,y)/\partial(u,v)|$, because probability (like area times density) must be conserved. Worked Example 6 is the special case $(X, Y) \to (R, \Theta)$; it shows at once that the phase Θ is uniform on $[0, 2\pi)$ and independent of the envelope.

6.9 Cylindrical and spherical coordinates

Motivation

Wires, coaxial cables, solenoids and circular waveguides have an axis of symmetry; point charges, antennas seen from far away and spherical shells have a centre. Cartesian coordinates fight these geometries: the surface of a wire becomes $x^2 + y^2 = a^2$, and its cross-section a region with curved limits. In **cylindrical** and **spherical** coordinates the same surfaces become $r = a$ and $R = a$, the limits of integration become constants, and the integrals often separate into products of single integrals. This section sets up both systems, derives their volume elements from the Jacobian, and introduces the scale factors that Chapter 7 uses to write gradient, divergence and curl in curvilinear form.

Definitions and conventions

To avoid clashing with charge density ρ, this book uses (r, ϕ, z) for cylindrical and (R, θ, ϕ) for spherical coordinates, as in Cheng's field-theory text.

Cylindrical coordinates (r, ϕ, z)

$r \geq 0$ is the perpendicular distance from the z-axis, $\phi \in [0, 2\pi)$ the azimuth measured from the $+x$ axis towards $+y$, and z the usual height:

$$x = r\cos\phi, \qquad y = r\sin\phi, \qquad z = z; \qquad r = \sqrt{x^2 + y^2}, \quad \phi = \operatorname{atan2}(y, x).$$

Coordinate surfaces: $r =$ const is a circular cylinder, $\phi =$ const a half-plane hinged on the z-axis, $z =$ const a horizontal plane.

Spherical coordinates (R, θ, ϕ)

$R \geq 0$ is the distance from the origin, $\theta \in [0, \pi]$ the **polar angle** measured from the $+z$ axis, and $\phi \in [0, 2\pi)$ the same azimuth as in cylindrical coordinates:

$$x = R\sin\theta\cos\phi, \qquad y = R\sin\theta\sin\phi, \qquad z = R\cos\theta;$$

$$R = \sqrt{x^2 + y^2 + z^2}, \qquad \theta = \arccos\frac{z}{R}, \qquad \phi = \operatorname{atan2}(y, x).$$

Coordinate surfaces: $R =$ const is a sphere, $\theta =$ const a cone about the z-axis (the plane $z = 0$ when $\theta = \pi/2$), $\phi =$ const a half-plane.

The two systems are linked by $r = R\sin\theta$ and $z = R\cos\theta$: the cylindrical radius is the projection of the spherical radius onto the xy-plane. On the Earth, θ is the colatitude (90° minus latitude) and ϕ the longitude.

Worked Example 1 — converting a point

Express $P = (1,\ 1,\ \sqrt{2})$ in cylindrical and spherical coordinates.

Cylindrical: $r = \sqrt{1+1} = \sqrt{2}$, $\phi = \operatorname{atan2}(1,1) = \pi/4$, $z = \sqrt{2}$; so $P = (\sqrt{2},\ \pi/4,\ \sqrt{2})$.

Spherical: $R = \sqrt{1+1+2} = 2$, $\theta = \arccos(\sqrt{2}/2) = \pi/4$, $\phi = \pi/4$; so $P = (2,\ \pi/4,\ \pi/4)$.

Check back: $x = 2\sin\frac{\pi}{4}\cos\frac{\pi}{4} = 2\cdot\frac{1}{2} = 1$, $y = 1$, $z = 2\cos\frac{\pi}{4} = \sqrt{2}$. ✓ And $r = R\sin\theta = 2\cdot\frac{\sqrt{2}}{2} = \sqrt{2}$. ✓

For the point $(-1, -1, 0)$, $\tan\phi = y/x = 1$ would suggest $\phi = \pi/4$, but the point is in the third quadrant: $\operatorname{atan2}(-1,-1) = -3\pi/4$, i.e. $\phi = 5\pi/4$.

Common Pitfall

Mathematics texts (and some software) swap the roles of θ and ϕ in spherical coordinates, using θ for the azimuth and ϕ for the polar angle. Always check which angle is measured from the z-axis before using a formula from another source — the volume element $R^2\sin\theta$ contains the sine of the *polar* angle. Also, $\tan\phi = y/x$ alone does not fix the quadrant; use the signs of x and y (the `atan2` function).

Jacobians and volume elements

The cylindrical Jacobian is the polar one with a trivial z row and column added, so its determinant is r. For spherical coordinates,

$$\frac{\partial(x,y,z)}{\partial(R,\theta,\phi)} = \begin{bmatrix} \sin\theta\cos\phi & R\cos\theta\cos\phi & -R\sin\theta\sin\phi \\ \sin\theta\sin\phi & R\cos\theta\sin\phi & R\sin\theta\cos\phi \\ \cos\theta & -R\sin\theta & 0 \end{bmatrix}.$$

Expanding along the bottom row,

$$\begin{aligned}\det \mathbf{J} &= \cos\theta\left(R^2\sin\theta\cos\theta\cos^2\phi + R^2\sin\theta\cos\theta\sin^2\phi\right) + R\sin\theta\left(R\sin^2\theta\cos^2\phi + R\sin^2\theta\sin^2\phi\right)\\ &= R^2\sin\theta\cos^2\theta + R^2\sin^3\theta = R^2\sin\theta.\end{aligned}$$

Since $0 \le \theta \le \pi$, $\sin\theta \ge 0$ and no absolute value is needed. The determinant vanishes on the z-axis ($\theta = 0, \pi$) and at the origin, where the coordinates are degenerate (every ϕ describes the same point).

Scale factors and the geometric picture

The Jacobian can be read off without any determinant, from geometry. Each column of the Jacobian is the displacement per unit change of one coordinate, and in both systems the three columns are **mutually perpendicular** (the systems are *orthogonal*). Their lengths are the **scale factors** h_i: the distance moved per unit change of coordinate i.

- Changing r by dr moves you dr; changing z by dz moves you dz; changing ϕ by $d\phi$ moves you along a circle of radius r, a distance $r\,d\phi$.
- In spherical coordinates, changing R by dR moves you dR; changing θ by $d\theta$ moves you along a great circle (a meridian) of radius R, a distance $R\,d\theta$; changing ϕ by $d\phi$ moves you along a circle of latitude, whose radius is $r = R\sin\theta$, a distance $R\sin\theta\,d\phi$.

A small coordinate box is therefore (to first order) a rectangular box with edges h_1du_1, h_2du_2, h_3du_3, and

$$dV = h_1h_2h_3\,du_1\,du_2\,du_3, \qquad d\ell^2 = h_1^2du_1^2 + h_2^2du_2^2 + h_3^2du_3^2.$$

	Cylindrical (r,ϕ,z)	**Spherical** (R,θ,ϕ)
to Cartesian	$x = r\cos\phi,\ y = r\sin\phi,\ z = z$	$x = R\sin\theta\cos\phi,\ y = R\sin\theta\sin\phi,\ z = R\cos\theta$
from Cartesian	$r = \sqrt{x^2+y^2},\ \phi = \operatorname{atan2}(y,x)$	$R = \sqrt{x^2+y^2+z^2},\ \cos\theta = z/R$
ranges	$r \ge 0,\ 0 \le \phi < 2\pi$	$R \ge 0,\ 0 \le \theta \le \pi,\ 0 \le \phi < 2\pi$
scale factors (h_1,h_2,h_3)	$(1,\ r,\ 1)$	$(1,\ R,\ R\sin\theta)$
Jacobian determinant	r	$R^2\sin\theta$
volume element dV	$r\,dr\,d\phi\,dz$	$R^2\sin\theta\,dR\,d\theta\,d\phi$
line element $d\ell^2$	$dr^2 + r^2d\phi^2 + dz^2$	$dR^2 + R^2d\theta^2 + R^2\sin^2\theta\,d\phi^2$
surface element on $r = a$ / $R = a$	$a\,d\phi\,dz$	$a^2\sin\theta\,d\theta\,d\phi$
surface element on $z = \text{const}$ / $\theta = \text{const}$	$r\,dr\,d\phi$	$R\sin\theta\,dR\,d\phi$

The scale factors also generate the curvilinear forms of grad, div and curl in Ch. 7; for instance the gradient in spherical coordinates is $\nabla f = f_R\hat{\mathbf{R}} + \frac{1}{R}f_\theta\hat{\boldsymbol{\theta}} + \frac{1}{R\sin\theta}f_\phi\hat{\boldsymbol{\phi}}$, each partial divided by its scale factor, exactly as in the polar case of §6.4.

Unit vectors depend on position

The unit vectors of curvilinear coordinates point along the directions of increasing coordinates, and those directions change from point to point:

$$\hat{\mathbf{r}} = \cos\phi\,\hat{\mathbf{x}} + \sin\phi\,\hat{\mathbf{y}}, \qquad \hat{\boldsymbol{\phi}} = -\sin\phi\,\hat{\mathbf{x}} + \cos\phi\,\hat{\mathbf{y}}, \qquad \hat{\mathbf{z}} = \hat{\mathbf{z}};$$

$$\hat{\mathbf{R}} = \sin\theta\cos\phi\,\hat{\mathbf{x}} + \sin\theta\sin\phi\,\hat{\mathbf{y}} + \cos\theta\,\hat{\mathbf{z}}, \quad \hat{\boldsymbol{\theta}} = \cos\theta\cos\phi\,\hat{\mathbf{x}} + \cos\theta\sin\phi\,\hat{\mathbf{y}} - \sin\theta\,\hat{\mathbf{z}}, \quad \hat{\boldsymbol{\phi}} = -\sin\phi\,\hat{\mathbf{x}} + \cos\phi\,\hat{\mathbf{y}}.$$

(They are the Jacobian columns divided by the scale factors.) Each triple is orthonormal and right-handed at every point, but $\hat{\mathbf{r}}$ at $\phi = 0$ is $\hat{\mathbf{x}}$ while at $\phi = \pi$ it is $-\hat{\mathbf{x}}$.

> **Common Pitfall — integrating vectors in curvilinear coordinates**
>
> Because $\hat{\mathbf{r}}$, $\hat{\boldsymbol{\phi}}$, $\hat{\mathbf{R}}$, $\hat{\boldsymbol{\theta}}$ vary with position, they cannot be taken outside an integral. The total field of a uniformly charged ring at its centre is $\oint \hat{\mathbf{r}}\,d\phi\cdots = \mathbf{0}$, not "$2\pi\hat{\mathbf{r}}\cdots$". To integrate a vector quantity, convert the unit vectors to the constant Cartesian $\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$ first (using the formulas above), then integrate component by component. Scalar integrands — charge, energy, power — have no such problem.

Worked examples in cylindrical coordinates

> **Worked Example 2 — current in a wire with non-uniform density**
>
> At high frequency, current crowds towards a conductor's surface (the skin effect). Model it in a wire of radius a by $J_z = J_0\,(r/a)^2$. The cross-section is $0 \le r \le a$, $0 \le \phi < 2\pi$, with $dA = r\,dr\,d\phi$, so
>
> $$I = \int_0^{2\pi}\!\!\int_0^{a} J_0\frac{r^2}{a^2}\,r\,dr\,d\phi = 2\pi\frac{J_0}{a^2}\cdot\frac{a^4}{4} = \frac{\pi J_0 a^2}{2},$$
>
> half what a uniform density J_0 would carry. Units: $\mathrm{A/m^2}\times\mathrm{m^2} = \mathrm{A}$. ✓ The factor r in dA weights the outer annuli more heavily — there is more area at large r — which is why the result is not simply "the average of $(r/a)^2$ over $[0,1]$", which would be $1/3$.

Worked Example 3 — energy stored in a coaxial cable

Between the conductors of a coaxial cable (inner radius a, outer radius b, voltage V_0, dielectric permittivity ε) the field magnitude is $E = \dfrac{V_0}{r\ln(b/a)}$. Find the stored energy per unit length and hence the capacitance per unit length.

Energy density: $w_e = \tfrac{1}{2}\varepsilon E^2 = \dfrac{\varepsilon V_0^2}{2r^2\ln^2(b/a)}$, a function of r only.

Integrate over a length ℓ: $dV = r\,dr\,d\phi\,dz$, so

$$W = \int_0^{\ell}\int_0^{2\pi}\int_a^b \frac{\varepsilon V_0^2}{2r^2\ln^2(b/a)}\, r\,dr\,d\phi\,dz = \ell\cdot 2\pi\cdot\frac{\varepsilon V_0^2}{2\ln^2(b/a)}\int_a^b\frac{dr}{r} = \frac{\pi\varepsilon\ell V_0^2}{\ln(b/a)}.$$

Capacitance: equating $W = \tfrac{1}{2}CV_0^2$ gives $C/\ell = 2\pi\varepsilon/\ln(b/a)$.

Numbers: $a = 0.5$ mm, $b = 1.75$ mm, polyethylene $\varepsilon = 2.25\varepsilon_0$: $C/\ell = 2\pi(2.25)(8.854\times10^{-12})/\ln 3.5 = 99.9$ pF/m, the familiar figure for such cables. Units: $(\mathrm{F/m})\cdot\mathrm{V}^2\cdot\mathrm{m}/1 = \mathrm{J}$ for W. ✓ The Jacobian r cancelled one power of $1/r^2$, leaving the logarithm.

Worked examples in spherical coordinates

Worked Example 4 — charge in a sphere

A sphere of radius a has $\rho_v = \rho_0(1 - R/a)$. With nothing depending on θ or ϕ, the angular integrals give

$$\int_0^{2\pi}\int_0^{\pi}\sin\theta\,d\theta\,d\phi = 2\pi\cdot 2 = 4\pi,$$

and

$$Q = 4\pi\rho_0\int_0^a\left(1-\frac{R}{a}\right)R^2\,dR = 4\pi\rho_0\left(\frac{a^3}{3}-\frac{a^3}{4}\right) = \frac{\pi\rho_0 a^3}{3}.$$

This is a quarter of what a uniform density ρ_0 would give ($\tfrac{4}{3}\pi a^3\rho_0$). For a spherically symmetric density, $dV = 4\pi R^2\,dR$: the volume of a thin shell of radius R and thickness dR.

Worked Example 5 — volume of an "ice-cream cone"

Find the volume of the solid inside the sphere $R = a$ and inside the cone $\theta = \alpha$ (i.e. $0 \le \theta \le \alpha$).

$$V = \int_0^{2\pi}\int_0^{\alpha}\int_0^{a} R^2\sin\theta\,dR\,d\theta\,d\phi = 2\pi\cdot(1-\cos\alpha)\cdot\frac{a^3}{3} = \frac{2\pi a^3}{3}(1-\cos\alpha).$$

Checks: $\alpha = \pi$ gives $\tfrac{4}{3}\pi a^3$, the whole sphere; $\alpha = \pi/2$ gives the hemisphere $\tfrac{2}{3}\pi a^3$. ✓ For $\alpha = \pi/3$, $V = \pi a^3/3$. In Cartesian coordinates the same solid would need limits involving both $\sqrt{a^2-x^2-y^2}$ and $\sqrt{x^2+y^2}\cot\alpha$; in spherical coordinates all three limits are constants.

Solid angle and antenna patterns

The angular part of the spherical surface element, $d\Omega = \sin\theta\,d\theta\,d\phi$, is the element of **solid angle**: the area of a patch on the unit sphere, measured in steradians (sr). The full sphere subtends $\int d\Omega = 4\pi$ sr, and a cone of half-angle α subtends

$$\Omega(\alpha) = \int_0^{2\pi}\int_0^{\alpha}\sin\theta\,d\theta\,d\phi = 2\pi(1-\cos\alpha)\ \text{sr},$$

which gives 2π sr for a hemisphere. Any function on the sphere — an antenna pattern, the angular distribution of scattered light — is integrated against $d\Omega$.

Worked Example 6 — radiated power and directivity

An antenna's radiation intensity $U(\theta,\phi)$ is the power radiated per steradian (W/sr); the total radiated power is $P = \oint U\,d\Omega = \int_0^{2\pi}\int_0^{\pi} U\sin\theta\,d\theta\,d\phi$, and the **directivity** is $D = 4\pi U_{\max}/P$, the ratio of peak intensity to the average over all directions.

(a) *Short (Hertzian) dipole,* $U = U_0\sin^2\theta$:

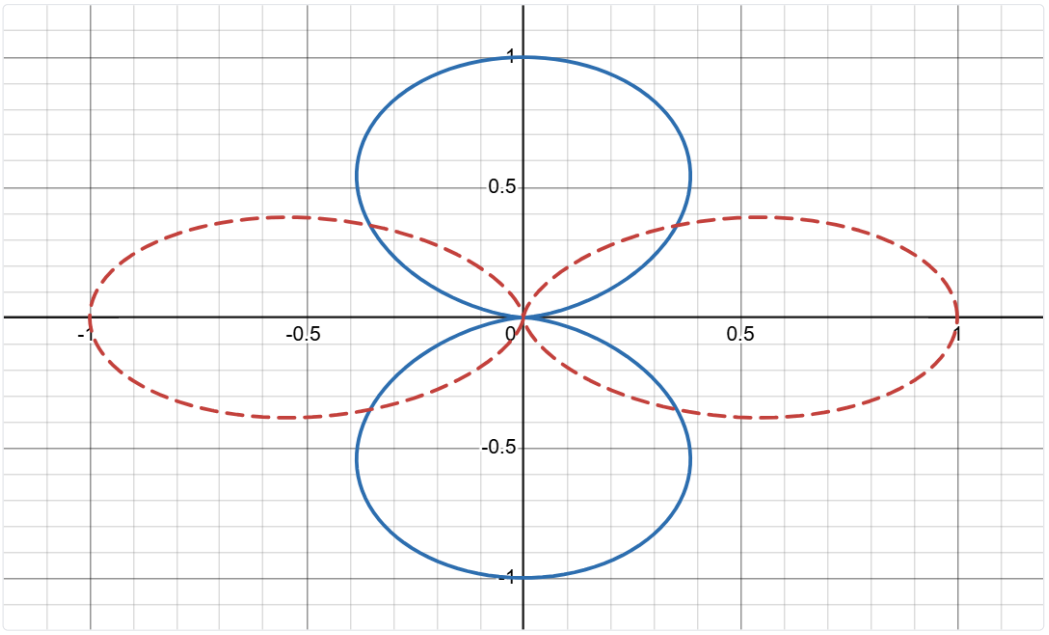
$$P = 2\pi U_0\int_0^{\pi}\sin^3\theta\,d\theta = 2\pi U_0\cdot\frac{4}{3} = \frac{8\pi U_0}{3},$$

using $\sin^3\theta = (1-\cos^2\theta)\sin\theta$ and $w = \cos\theta$: $\int_{-1}^{1}(1-w^2)\,dw = \tfrac{4}{3}$. Then $D = 4\pi U_0/(8\pi U_0/3) = 1.5$ (1.76 dBi), the textbook value for a short dipole.

(b) *A pattern radiating only into the upper hemisphere,* $U = U_0\cos^2\theta$ for $0 \le \theta \le \pi/2$ and zero below:

$$P = 2\pi U_0\int_0^{\pi/2}\cos^2\theta\sin\theta\,d\theta = 2\pi U_0\cdot\frac{1}{3} = \frac{2\pi U_0}{3}, \qquad D = \frac{4\pi U_0}{2\pi U_0/3} = 6\ (7.8\ \text{dBi}).$$

Concentrating the same peak intensity into a narrower beam reduces the total power needed — or, for fixed power, raises the peak — and directivity measures exactly that. Numerical quadrature of both angular integrals gives 1.3333 and 0.3333. ✓



Polar cuts through the z-axis, drawn with the antenna's z-axis along the horizontal: $\sin^2\theta$ (blue) is the figure-eight cross-section of the short dipole's doughnut, with nulls along the axis; $\cos^2\theta$ (red) points along the axis — in Example 6(b) only its right-hand lobe ($0 \le \theta \le \pi/2$) is present.

Engineering Note

The choice of coordinates is part of the modelling. A cylindrical problem (a long wire) in spherical coordinates is as awkward as a spherical one (a point charge) in Cartesian coordinates. Before setting up any multiple integral, ask: *on which coordinate surfaces are the boundaries, and on which coordinates does the integrand depend?* When both answers point to the same system, the integral usually separates into a product of three single integrals.

6.10 Constrained optimization

Motivation

Design is optimisation: maximise the power delivered to a load, minimise the loss in a network, fit a model to measured data with the least error, choose dimensions that use the least material. In one variable (Chapter 4) we set $f'(x) = 0$ and checked the sign of f''. With several variables the same ideas work, but "the sign of the second derivative" becomes a property of a matrix, and a new kind of critical point — the saddle — appears. Real problems almost always carry **constraints** as well: a fixed power budget, a fixed total current, a component that must fit in a box. This section treats unconstrained extrema first, then constraints handled by substitution and by searching boundaries; §6.11 gives the general method.

Local and global extrema

Extrema and critical points

f has a **local maximum** at $\mathbf{a}$ if $f(\mathbf{x}) \le f(\mathbf{a})$ for all $\mathbf{x}$ in some neighbourhood of $\mathbf{a}$, and a **global maximum** if this holds for all $\mathbf{x}$ in the domain (similarly for minima). A **critical point** (stationary point) is an interior point where $\nabla f = \mathbf{0}$, or where ∇f does not exist.

First-order necessary condition

If f is differentiable at an interior point $\mathbf{a}$ and has a local extremum there, then $\nabla f(\mathbf{a}) = \mathbf{0}$: every partial derivative vanishes, and the tangent plane is horizontal.

Proof. If f has a local maximum at $\mathbf{a}$, then the one-variable function $g(t) = f(\mathbf{a} + t\hat{\mathbf{x}}_i)$ has a local maximum at $t = 0$, so $g'(0) = f_{x_i}(\mathbf{a}) = 0$ by the one-variable result. This holds for each i. □

The condition is necessary, not sufficient. At a **saddle point** $\nabla f = \mathbf{0}$, yet f increases in some directions and decreases in others — like the middle of a mountain pass, or the origin for $f = x^2 - y^2$.

The second-derivative test

Near a critical point $\mathbf{a}$ the first-order terms of Taylor's formula (§6.3) vanish, leaving

$$f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) \approx \tfrac{1}{2}\,\mathbf{h}^{\mathsf{T}}\mathbf{H}\,\mathbf{h} = \tfrac{1}{2}\left(f_{xx}h^2 + 2f_{xy}hk + f_{yy}k^2\right), \qquad \mathbf{h} = (h,k).$$

The sign of this **quadratic form** for small $\mathbf{h}$ in every direction decides the type of the point. Write $A = f_{xx}$, $B = f_{xy}$, $C = f_{yy}$ and, if $A \neq 0$, complete the square:

$$Ah^2 + 2Bhk + Ck^2 = A\left(h + \frac{B}{A}k\right)^2 + \frac{AC - B^2}{A}k^2.$$

If $D = AC - B^2 > 0$, both terms have the sign of A for every $(h, k) \neq \mathbf{0}$: the form is definite, positive (minimum) if $A > 0$ and negative (maximum) if $A < 0$. If $D < 0$, the two terms have opposite signs, so the form takes both signs: a saddle. If $D = 0$, the form vanishes along some direction and higher-order terms decide.

Second-derivative test (two variables)

Let $\nabla f(\mathbf{a}) = \mathbf{0}$, let f have continuous second partials near $\mathbf{a}$, and let $\mathbf{H} = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{bmatrix}$ with determinant $D = f_{xx} f_{yy} - f_{xy}^2$ at $\mathbf{a}$.

Condition at the critical point	Conclusion
$D > 0,\ f_{xx} > 0$	local minimum
$D > 0,\ f_{xx} < 0$	local maximum
$D < 0$	saddle point
$D = 0$	test inconclusive

In n variables the same reasoning applies to $\mathbf{h}^{\mathsf{T}}\mathbf{H}\mathbf{h}$: a critical point is a local minimum if $\mathbf{H}$ is **positive definite** (all eigenvalues positive), a local maximum if it is negative definite (all eigenvalues negative), and a saddle if $\mathbf{H}$ has eigenvalues of both signs. Equivalent tests via leading principal minors (Sylvester's criterion) and the eigenvalue theory are in Ch. 10; in two variables, $D > 0$ says the two eigenvalues have the same sign and f_{xx} says which.

Worked Example 1 — a minimum and a saddle

$f(x, y) = x^3 - 3x + y^2$.

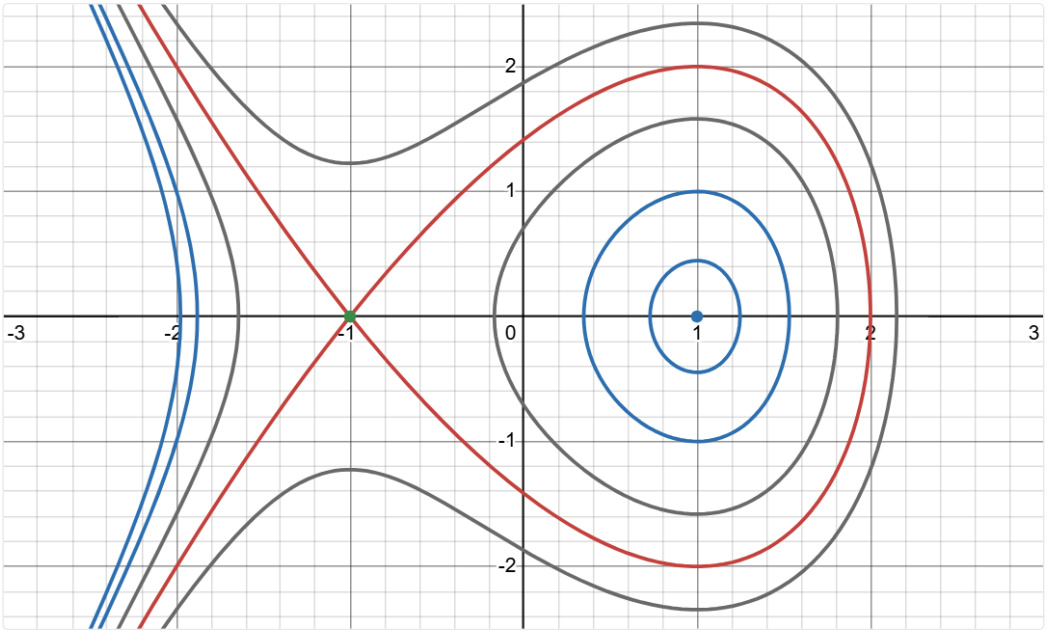
Critical points: $f_x = 3x^2 - 3 = 0$ and $f_y = 2y = 0$ give $(1, 0)$ and $(-1, 0)$.

Second partials: $f_{xx} = 6x$, $f_{yy} = 2$, $f_{xy} = 0$.

At $(1, 0)$: $D = 6 \cdot 2 - 0 = 12 > 0$ and $f_{xx} = 6 > 0$: **local minimum**, $f = -2$.

At $(-1, 0)$: $D = -6 \cdot 2 = -12 < 0$: **saddle point**, $f = 2$. Along the x-axis f has a local *maximum* at $x = -1$; along the line $x = -1$ it has a *minimum* in y.

Neither point is a global extremum: $f \to -\infty$ as $x \to -\infty$.



Level curves of $x^3 - 3x + y^2$: small closed loops (blue) surround the minimum at $(1, 0)$, while the level curve through the saddle (red, $f = 2$) crosses itself at $(-1, 0)$ — the signature of a saddle on a contour map.

Worked Example 2 — three critical points

$f(x,y) = x^4 + y^4 - 4xy$.

Critical points: $f_x = 4x^3 - 4y = 0$ gives $y = x^3$; $f_y = 4y^3 - 4x = 0$ gives $x = y^3 = x^9$. So $x(x^8 - 1) = 0$: $x = 0, \pm 1$, giving $(0,0)$, $(1,1)$, $(-1,-1)$.

Hessian: $\mathbf{H} = \begin{bmatrix} 12x^2 & -4 \\ -4 & 12y^2 \end{bmatrix}$.

At $(0,0)$: $D = 0 - 16 = -16 < 0$: saddle, $f = 0$.

At $(\pm 1, \pm 1)$: $D = 144 - 16 = 128 > 0$, $f_{xx} = 12 > 0$: local minima, $f = 1 + 1 - 4 = -2$.

Since $f \to +\infty$ in every direction (the quartic terms dominate), these two minima are also the **global** minima. Eigenvalue check at $(1,1)$: $\mathbf{H} = \begin{bmatrix} 12 & -4 \\ -4 & 12 \end{bmatrix}$ has eigenvalues 8 and 16, both positive. ✓

Worked Example 3 — least-squares line fit

A sensor is calibrated at four input levels, giving output voltages $(x_i, y_i) = (0,1), (1,3), (2,4), (3,7)$. Find the line $y = mx + b$ that minimises the sum of squared errors

$$S(m,b) = \sum_{i=1}^{4} (y_i - mx_i - b)^2 .$$

Critical point:

$$\frac{\partial S}{\partial m} = -2\sum x_i(y_i - mx_i - b) = 0, \qquad \frac{\partial S}{\partial b} = -2\sum (y_i - mx_i - b) = 0,$$

which are the **normal equations**

$$m\sum x_i^2 + b\sum x_i = \sum x_i y_i, \qquad m\sum x_i + nb = \sum y_i.$$

With $\sum x_i = 6$, $\sum x_i^2 = 14$, $\sum y_i = 15$, $\sum x_i y_i = 32$, $n = 4$: $14m + 6b = 32$ and $6m + 4b = 15$, so $m = 1.9$, $b = 0.9$.

Is it a minimum? $\mathbf{H} = 2\begin{bmatrix} \sum x_i^2 & \sum x_i \\ \sum x_i & n \end{bmatrix} = \begin{bmatrix} 28 & 12 \\ 12 & 8 \end{bmatrix}$, $D = 224 - 144 = 80 > 0$, $S_{mm} = 28 > 0$: yes — and since S is a quadratic with positive-definite Hessian, it is the global minimum.

Residuals: $0.1,\ 0.2,\ -0.7,\ 0.4$, which sum to zero (that is the second normal equation), with $S_{\min} = 0.70\ \text{V}^2$. The same calculation with many parameters is linear least squares, treated with matrices in Ch. 9 and statistically in Ch. 20.

Worked Example 4 — maximum power transfer to a complex load

A source with open-circuit RMS voltage V_s and internal impedance $Z_s = R_s + jX_s$ drives a load $Z_L = R_L + jX_L$. The load power is

$$P(R_L, X_L) = \frac{V_s^2 R_L}{(R_s + R_L)^2 + (X_s + X_L)^2}, \qquad R_L > 0.$$

Reactance first: $P_{X_L} = -\dfrac{2V_s^2 R_L (X_s + X_L)}{[(R_s + R_L)^2 + (X_s + X_L)^2]^2} = 0$ gives $X_L = -X_s$.

Then resistance: with $X_L = -X_s$, $P = V_s^2 R_L / (R_s + R_L)^2$ and $P_{R_L} = V_s^2 \dfrac{(R_s + R_L) - 2R_L}{(R_s + R_L)^3} = 0$ gives $R_L = R_s$.

So the only critical point is $Z_L = Z_s^*$ — the **conjugate match** — with $P_{\max} = V_s^2/(4R_s)$. For $V_s = 10$ V and $Z_s = 50 + j30\ \Omega$: $Z_L = 50 - j30\ \Omega$ and $P_{\max} = 100/200 = 0.5$ W.

Classification: at the critical point one finds $P_{R_L R_L} = P_{X_L X_L} = -V_s^2/(8R_s^3) < 0$ and $P_{R_L X_L} = 0$, so $D = V_s^4/(64R_s^6) > 0$ with $P_{R_L R_L} < 0$: a maximum. (Units: W/Ω^2, as a second derivative of power with respect to resistance must be.) Physically obvious too — $P \to 0$ as $R_L \to 0$, as $R_L \to \infty$ or as $|X_L| \to \infty$, so the single critical point must be the global maximum.

Common Pitfall

Checking only one variable at a time. A point can be a minimum along the x-direction and along the y-direction and still be a saddle: $f = x^2 + y^2 - 3xy$ has $f_{xx} = f_{yy} = 2 > 0$ at the origin, yet $D = 4 - 9 < 0$, and along $y = x$, $f = -x^2$ decreases. The mixed derivative f_{xy} — the twist — must be included, which is exactly what D does.

Constraints by substitution

A **constrained** problem reads

$$\text{optimise } f(\mathbf{x}) \quad \text{subject to} \quad g(\mathbf{x}) = c.$$

If the constraint can be solved for one variable, substitute and reduce the problem to an unconstrained one in fewer variables.

Worked Example 5 — an enclosure of minimum material

An open-top rectangular enclosure must hold 32 cm^3. Minimise the sheet-metal area $A = xy + 2xz + 2yz$ subject to $xyz = 32$.

Substitute $z = 32/(xy)$:

$$A(x, y) = xy + \frac{64}{y} + \frac{64}{x}, \qquad x, y > 0.$$

Critical point: $A_x = y - 64/x^2 = 0$, $A_y = x - 64/y^2 = 0$. So $yx^2 = 64$ and $xy^2 = 64$; dividing, $x = y$, then $x^3 = 64$, $x = y = 4$ cm, $z = 2$ cm.

Classify: $A_{xx} = 128/x^3 = 2$, $A_{yy} = 2$, $A_{xy} = 1$; $D = 4 - 1 = 3 > 0$, $A_{xx} > 0$: minimum, $A = 16 + 32 + 32 = 48$ cm^2. The open top makes the optimal box half as tall as it is wide.

Global extrema on closed, bounded regions

Extreme value theorem

A continuous function on a **closed and bounded** region attains a global maximum and a global minimum there. They occur either at interior critical points or on the boundary.

Hence the **closed-region method**:

1. Find the critical points of f inside the region and evaluate f there.
2. Find the extreme values of f on the boundary — a constrained problem, solved by parametrising or substituting (or by Lagrange multipliers, §6.11). Include corners and endpoints of boundary pieces.
3. The largest value from steps 1–2 is the global maximum, the smallest the global minimum.

Worked Example 6 — extremes on a disc

Find the extreme values of $T(x, y) = x^2 + 2y^2 - x$ on the disc $x^2 + y^2 \le 1$.

Interior: $T_x = 2x - 1 = 0$, $T_y = 4y = 0$ give $(\tfrac{1}{2}, 0)$, with $T = -\tfrac{1}{4}$.

Boundary: substitute $y^2 = 1 - x^2$: $T = x^2 + 2(1 - x^2) - x = 2 - x - x^2$ for $-1 \le x \le 1$. Then $dT/dx = -1 - 2x = 0$ gives $x = -\tfrac{1}{2}$, $y = \pm\tfrac{\sqrt{3}}{2}$, $T = 2 + \tfrac{1}{2} - \tfrac{1}{4} = \tfrac{9}{4}$. The endpoints of the x-interval give $T(1, 0) = 0$ and $T(-1, 0) = 2$.

Compare: $\{-\tfrac{1}{4},\ \tfrac{9}{4},\ 0,\ 2\}$: global maximum $\tfrac{9}{4}$ at $(-\tfrac{1}{2}, \pm\tfrac{\sqrt{3}}{2})$; global minimum $-\tfrac{1}{4}$ at $(\tfrac{1}{2}, 0)$.

Alternatively, parametrise the circle as $(\cos t, \sin t)$: $T = \cos^2 t + 2\sin^2 t - \cos t = 1 + \sin^2 t - \cos t$, and $dT/dt = \sin t(2\cos t + 1) = 0$ gives the same four boundary candidates.

Substitution becomes awkward when the constraint cannot be solved explicitly (try solving $x^3 + y^3 + xy = 5$ for y), when it breaks the symmetry between variables, or when there are several constraints. Lagrange's method, next, avoids all three problems.

Engineering Note

Most practical optimisation problems are solved numerically (Ch. 23), but the analysis of this section remains essential: it tells you what the optimiser should find, whether the problem has one solution or several, and whether the point it returns is a minimum or merely a saddle where the gradient happens to vanish. Convex problems — those whose Hessian is positive semidefinite everywhere, like least squares — have no saddles and no spurious local minima, which is why engineers work hard to formulate problems in convex form.

6.11 Lagrange multipliers

Motivation

Substitution solves a constrained problem by eliminating variables, which requires solving the constraint and usually destroys any symmetry the problem had. Lagrange's method keeps all the variables and treats them even-handedly, handles several constraints at once, and — as a bonus — produces a number, the multiplier, that measures how much the optimum would improve if the constraint were relaxed. In engineering that number is often the most useful output of the whole calculation: it is the marginal value of an extra watt of power budget, an extra square centimetre of board area, an extra volt of supply.

The geometric idea

Consider maximising $f(x, y)$ along the curve $g(x, y) = c$. Walk along the constraint curve and watch the level curves of f that you cross. If at some point you are *crossing* a level curve of f transversally, then moving one way along the constraint increases f and the other way decreases it — so that point is not an extremum. At a constrained extremum, therefore, the level curve of f through the point must be **tangent** to the constraint curve. Tangent curves have parallel normals, and the normals are the gradients (§6.4). Hence at the extremum

$$\nabla f = \lambda \nabla g$$

for some scalar λ. The same argument in three dimensions says that ∇f must be perpendicular to the constraint surface, i.e. parallel to its normal ∇g.

An equivalent analytic view: let $\mathbf{r}(t)$ be any curve lying in the constraint set and passing through the extremum $\mathbf{x}^*$ at $t = 0$. Then $f(\mathbf{r}(t))$ has an extremum at $t = 0$, so $\nabla f(\mathbf{x}^*) \cdot \mathbf{r}'(0) = 0$. Thus ∇f is perpendicular to every tangent direction of the constraint set — and the only vectors with that property are multiples of ∇g (or, with several constraints, combinations of the ∇g_k).

The method

Method of Lagrange multipliers

Let f and g have continuous first partial derivatives. If f has a local extremum at $\mathbf{x}^*$ subject to $g(\mathbf{x}) = c$, and $\nabla g(\mathbf{x}^*) \neq \mathbf{0}$, then there is a scalar λ such that

$$\nabla f(\mathbf{x}^*) = \lambda \, \nabla g(\mathbf{x}^*), \qquad g(\mathbf{x}^*) = c.$$

Equivalently, $(\mathbf{x}^*, \lambda)$ is a stationary point of the **Lagrangian**

$$\mathcal{L}(\mathbf{x}, \lambda) = f(\mathbf{x}) - \lambda \, \big(g(\mathbf{x}) - c\big) \, :$$

setting $\partial \mathcal{L} / \partial x_i = 0$ reproduces $\nabla f = \lambda \nabla g$, and $\partial \mathcal{L} / \partial \lambda = 0$ reproduces the constraint.

With several constraints $g_1 = c_1, \dots, g_m = c_m$ (and the ∇g_k linearly independent at $\mathbf{x}^*$), the condition becomes

$$\nabla f = \lambda_1 \nabla g_1 + \cdots + \lambda_m \nabla g_m, \qquad g_k = c_k.$$

For n variables and m constraints this is $n + m$ equations in $n + m$ unknowns. The procedure:

1. Write down $\nabla f = \lambda \nabla g$ component by component, together with the constraint.
2. Solve the system. There is no general algorithm — the equations are nonlinear — but useful tricks are: divide equations to eliminate λ; consider separately the cases where a factor is zero; exploit symmetry.
3. Evaluate f at **every** candidate and compare. Also check points where $\nabla g = \mathbf{0}$ and, for constraint curves with ends, the endpoints.

Worked Example 1 — a linear function on a circle

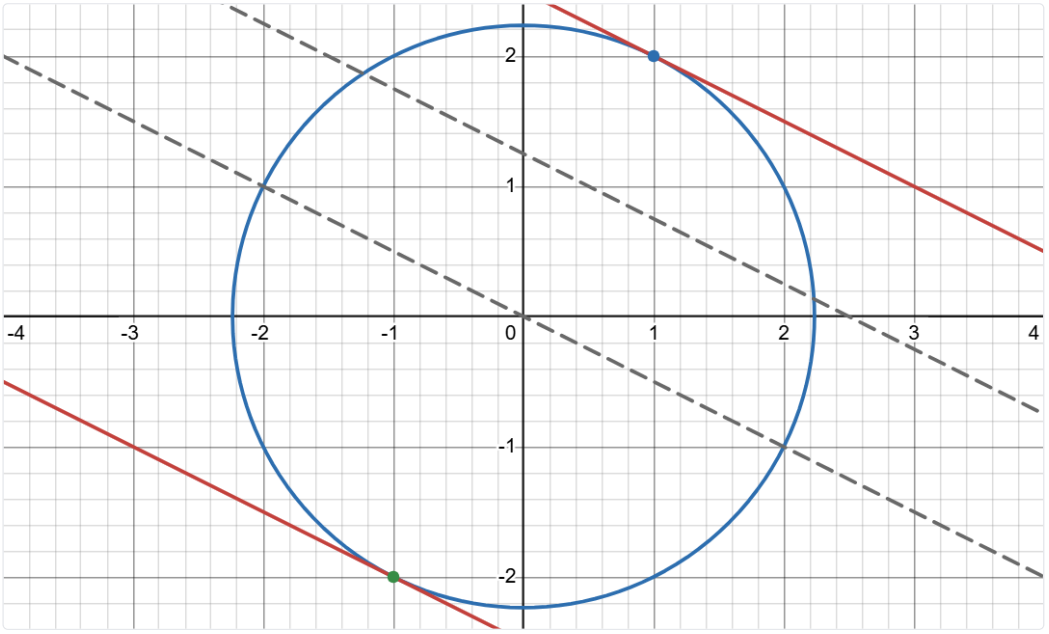
Extremise $f = x + 2y$ subject to $x^2 + y^2 = 5$.

Conditions: $\nabla f = (1, 2) = \lambda(2x, 2y)$ gives $1 = 2\lambda x$ and $2 = 2\lambda y$. Neither x nor λ can be zero, so dividing, $y/x = 2$, i.e. $y = 2x$.

Constraint: $x^2 + 4x^2 = 5$, $x = \pm 1$.

Candidates: $(1, 2)$ with $f = 5$ (maximum) and $(-1, -2)$ with $f = -5$ (minimum); $\lambda = 1/(2x) = \pm\frac{1}{2}$.

The answer is the Cauchy–Schwarz inequality in action: $|(1, 2) \cdot (x, y)| \leq \|(1, 2)\| \, \|(x, y)\| = \sqrt{5} \cdot \sqrt{5} = 5$, with equality when (x, y) is parallel to $(1, 2)$.



Level lines of $f = x + 2y$ (red and grey) sweep across the circle; the extremes occur where a level line just touches it — there the gradients of f and g are parallel.

Worked Example 2 — the disc boundary revisited

On $x^2 + y^2 = 1$, extremise $T = x^2 + 2y^2 - x$ (the boundary step of §6.10, Worked Example 6).

Conditions: $(2x - 1,\ 4y) = \lambda(2x,\ 2y)$. The second component gives $4y = 2\lambda y$, so **either** $y = 0$ **or** $\lambda = 2$.

Case $y = 0$: the constraint gives $x = \pm 1$, with $T(1, 0) = 0$ and $T(-1, 0) = 2$.

Case $\lambda = 2$: the first component gives $2x - 1 = 4x$, $x = -\tfrac{1}{2}$, and the constraint $y = \pm\tfrac{\sqrt{3}}{2}$, with $T = \tfrac{9}{4}$.

These are the same four candidates found by substitution, but obtained without breaking the symmetry of the constraint — and the case split "$y = 0$ or $\lambda = 2$" is the standard way such systems open up. Note that the substitution method found $(\pm 1, 0)$ only as *endpoints* of the x-interval; Lagrange's method finds them as genuine critical points of the constrained problem.

Worked Example 3 — closest point on a plane

Find the point of the plane $x + 2y + 2z = 9$ closest to the origin.

Minimise the squared distance $f = x^2 + y^2 + z^2$ (same minimiser as the distance, simpler derivatives).

Conditions: $(2x, 2y, 2z) = \lambda(1, 2, 2)$, so $(x, y, z) = \tfrac{\lambda}{2}(1, 2, 2)$: the closest point lies along the plane's normal, as geometry demands.

Constraint: $\tfrac{\lambda}{2}(1 + 4 + 4) = 9$, $\lambda = 2$, giving $(1, 2, 2)$ at distance 3.

Check: the point-to-plane distance formula gives $9/\sqrt{1^2 + 2^2 + 2^2} = 3$. ✓ No comparison of candidates is needed: there is only one, and f grows without bound on the plane, so it must be the minimum.

Worked Example 4 — the current divider minimises dissipation

A current I splits between resistors R_1 and R_2 in parallel. Ask which split $I_1 + I_2 = I$ **minimises** the total dissipation $P = R_1 I_1^2 + R_2 I_2^2$.

Conditions:

$$2R_1 I_1 = \lambda, \qquad 2R_2 I_2 = \lambda, \qquad I_1 + I_2 = I.$$

Hence $R_1 I_1 = R_2 I_2$ — equal voltages across both branches — and

$$I_1 = \frac{R_2}{R_1 + R_2} I, \qquad I_2 = \frac{R_1}{R_1 + R_2} I, \qquad P_{\min} = \frac{R_1 R_2}{R_1 + R_2} I^2.$$

That is exactly the current-divider rule: in a resistive network, currents distribute themselves so as to minimise heat (Maxwell's minimum-heat theorem). Here $\lambda = 2R_1 I_1 = 2IR_1R_2/(R_1 + R_2)$, twice the common branch voltage, in volts.

Numbers: $R_1 = 1\ \text{k}\Omega$, $R_2 = 4\ \text{k}\Omega$, $I = 10\ \text{mA}$: $I_1 = 8\ \text{mA}$, $I_2 = 2\ \text{mA}$, $P_{\min} = 800\ \Omega \times 10^{-4}\ \text{A}^2 = 80\ \text{mW}$. An equal split (5 mA each) would dissipate $25 + 100 = 125\ \text{mW}$.

The meaning of λ

Sensitivity interpretation of the multiplier

Let $\mathbf{x}^*(c)$ be the constrained optimum as a function of the constraint level c, with multiplier $\lambda(c)$, and let $f^*(c) = f(\mathbf{x}^*(c))$ be the optimal value. If $\mathbf{x}^*(c)$ depends differentiably on c, then

$$\lambda = \frac{df^*}{dc}.$$

Proof. By the chain rule, $\dfrac{df^*}{dc} = \nabla f(\mathbf{x}^*) \cdot \dfrac{d\mathbf{x}^*}{dc} = \lambda\, \nabla g(\mathbf{x}^*) \cdot \dfrac{d\mathbf{x}^*}{dc}$. But $g(\mathbf{x}^*(c)) = c$ for all c, so differentiating, $\nabla g \cdot \dfrac{d\mathbf{x}^*}{dc} = 1$. Hence $df^*/dc = \lambda$. □

So λ is the **shadow price** of the constraint: how much the optimal value improves per unit relaxation of the constraint. Its units are (units of f)/(units of g).

Check on Worked Example 1: with $x^2 + y^2 = c$, the maximum is $f^* = \sqrt{5c}$, so $df^*/dc = \sqrt{5}/(2\sqrt{c}) = \tfrac{1}{2}$ at $c = 5$ — and indeed $\lambda = \tfrac{1}{2}$ at $(1, 2)$. ✓

Check on Worked Example 4: $dP_{\min}/dI = 2IR_1R_2/(R_1 + R_2) = \lambda$. ✓ With the numbers above, $\lambda = 16\ \text{V}$: each extra milliampere of total current costs about 16 mW of extra dissipation.

Check on the enclosure of §6.10: with $\mathcal{L} = xy + 2xz + 2yz - \lambda(xyz - V)$, the conditions $y + 2z = \lambda yz$, $x + 2z = \lambda xz$, $2x + 2y = \lambda xy$ at $(4, 4, 2)$ give $\lambda = 1\ \text{cm}^{-1}$. Scaling says the optimal area grows as $A^* = 48\,(V/32)^{2/3}$, so $dA^*/dV = \tfrac{2}{3} \cdot \tfrac{48}{32} = 1\ \text{cm}^2/\text{cm}^3$. ✓

Two constraints

Worked Example 5 — extremes on the intersection of a plane and a cylinder

Find the maximum and minimum of $f = x + 2y + 3z$ on the curve where the plane $x - y + z = 1$ meets the cylinder $x^2 + y^2 = 1$ (an ellipse).

Conditions: $\nabla f = \lambda \nabla g_1 + \mu \nabla g_2$ with $g_1 = x - y + z$, $g_2 = x^2 + y^2$:

$$1 = \lambda + 2\mu x, \qquad 2 = -\lambda + 2\mu y, \qquad 3 = \lambda.$$

Solve: $\lambda = 3$, so $2\mu x = -2$ and $2\mu y = 5$: $x = -1/\mu$, $y = 5/(2\mu)$. The cylinder gives $\dfrac{1}{\mu^2} + \dfrac{25}{4\mu^2} = 1$, $\mu^2 = \dfrac{29}{4}$, $\mu = \pm\dfrac{\sqrt{29}}{2}$.

Then $x = \mp\dfrac{2}{\sqrt{29}}$, $y = \pm\dfrac{5}{\sqrt{29}}$, and the plane gives $z = 1 - x + y = 1 \pm \dfrac{7}{\sqrt{29}}$.

Values: $f = x + 2y + 3z = \pm\dfrac{-2 + 10 + 21}{\sqrt{29}} + 3 = 3 \pm \sqrt{29}$.

So the maximum is $3 + \sqrt{29} \approx 8.385$ at $(-0.3714,\ 0.9285,\ 2.2999)$ and the minimum $3 - \sqrt{29} \approx -2.385$. A numerical constrained optimiser started from $(1, 0, 0)$ returns the same maximiser to six figures. ✓

Second-order check: the bordered Hessian

Lagrange's condition is necessary, not sufficient. When comparing candidate values is impractical, a second-order test is available. For two variables and one constraint, form the **bordered Hessian**

$$\bar{\mathbf{H}} = \begin{bmatrix} 0 & g_x & g_y \\ g_x & \mathcal{L}_{xx} & \mathcal{L}_{xy} \\ g_y & \mathcal{L}_{xy} & \mathcal{L}_{yy} \end{bmatrix}$$

at the candidate. If $\det \bar{\mathbf{H}} > 0$ the point is a constrained local maximum; if $\det \bar{\mathbf{H}} < 0$, a constrained local minimum. (The test asks whether the Hessian of $\mathcal{L}$ is negative or positive definite *along the constraint's tangent direction*, which is all that matters.) For Worked Example 1 at $(1, 2)$ with $\lambda = \frac{1}{2}$: $\mathcal{L}_{xx} = \mathcal{L}_{yy} = -2\lambda = -1$, $\mathcal{L}_{xy} = 0$, $g_x = 2$, $g_y = 4$, and $\det \bar{\mathbf{H}} = 0 - 2(2 \cdot (-1) - 0) + 4(0 - 4 \cdot (-1)) = 4 + 16 = 20 > 0$: a maximum. ✓

Common Pitfall

Three ways to lose a solution. (1) Dividing by a quantity that might be zero — in Worked Example 2, dividing $4y = 2\lambda y$ by y loses the candidates with $y = 0$; factor instead and treat each case. (2) Ignoring points where $\nabla g = \mathbf{0}$: the theorem says nothing there (minimise $f = x$ on the cusp curve $y^2 = x^3$; the minimum at the origin, where $\nabla g = \mathbf{0}$, does not satisfy $\nabla f = \lambda \nabla g$). (3) Forgetting endpoints of a constraint curve that is not closed. And always remember that a Lagrange point may be a maximum, a minimum or neither — compare values.

Inequality constraints and water-filling

Engineering limits are usually inequalities — a power budget $\sum p_i \le P$, a current limit, non-negative powers $p_i \ge 0$. The extension of Lagrange's method is the set of **Karush–Kuhn–Tucker (KKT) conditions**: each inequality gets a multiplier that must be non-negative, and that multiplier is zero unless its constraint is **active** (holds with equality) — "complementary slackness". In effect, inactive constraints are ignored and active ones are treated as equalities; the work lies in finding which are active. The full theory is in Ch. 23; the classic EE example shows the idea.

Worked Example 6 — water-filling power allocation

A transmitter shares total power P between two parallel channels with noise powers $N_1 = 1$ and $N_2 = 3$ (in the same units as power). The total capacity in bit/s/Hz is

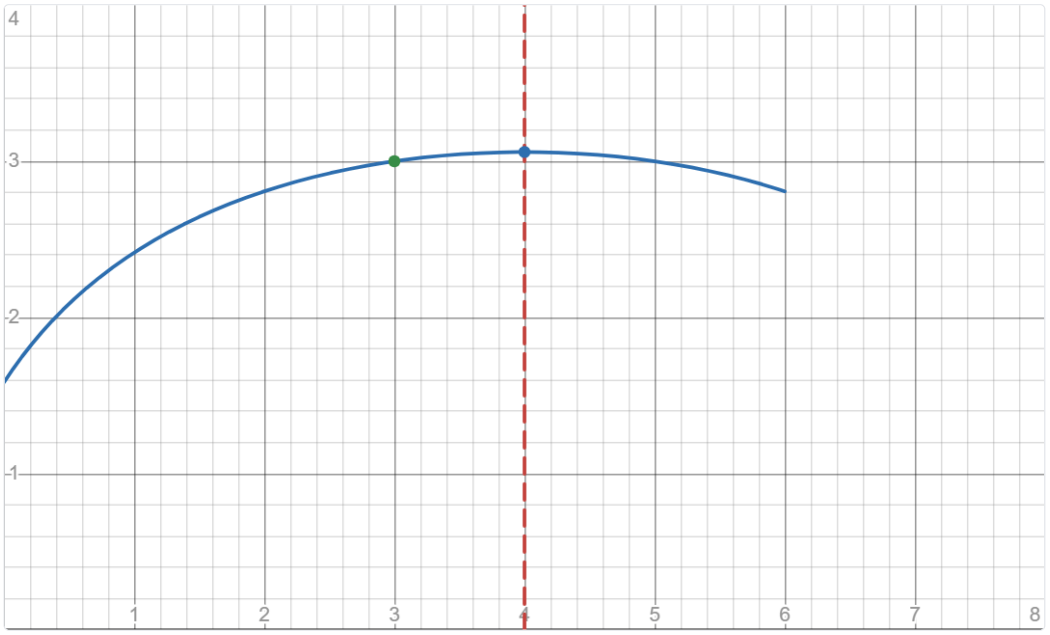
$$C = \log_2\!\left(1 + \frac{p_1}{N_1}\right) + \log_2\!\left(1 + \frac{p_2}{N_2}\right), \qquad p_1 + p_2 = P,\ \ p_1, p_2 \ge 0.$$

Lagrange conditions (both channels active): $\dfrac{1}{\ln 2} \cdot \dfrac{1}{N_i + p_i} = \lambda$, so $N_i + p_i = \nu$ with the same "water level" $\nu = 1/(\lambda \ln 2)$ for both channels.

With $P = 6$: $p_1 = \nu - 1$, $p_2 = \nu - 3$, sum $2\nu - 4 = 6$, $\nu = 5$: $p_1 = 4$, $p_2 = 2$, both non-negative, so both constraints $p_i \ge 0$ are inactive and the solution stands. Then $C = \log_2 5 + \log_2 \frac{5}{3} = 2.322 + 0.737 = 3.059$ bit/s/Hz, better than an equal split ($p_1 = p_2 = 3$: $\log_2 4 + \log_2 2 = 3$ bit/s/Hz).

With $P = 1$: the same formulas would give $\nu = 2.5$ and $p_2 = -0.5 < 0$ — impossible. So the constraint $p_2 \ge 0$ is active: $p_2 = 0$, and all power goes to the quieter channel, $p_1 = 1$, with water level $\nu = N_1 + p_1 = 2$, below the noise "floor" $N_2 = 3$ of channel 2. Then $C = \log_2 2 = 1$ bit/s/Hz.

The picture: pour total power P like water into a vessel whose floor heights are the noise levels N_i; it settles at a common level ν, and channels whose floor lies above the water get nothing. The rule "pour into the quietest channels first" is how DSL, Wi-Fi (OFDM) and MIMO systems allocate power across sub-channels.



Capacity versus the power p_1 given to the quieter channel (total $P = 6$): the curve is concave with a single maximum at $p_1 = 4$, where the marginal capacity per watt — the multiplier λ — is the same in both channels.

Engineering Note

The multiplier is often the quantity of real interest. In economic dispatch of power stations, the multiplier on the "generation equals demand" constraint is the system marginal cost (the price of one more MWh). In the water-filling example it is the marginal capacity per watt. In circuit design, it tells you which constraint — area, power, a noise specification — is binding and how much it is costing you, so you know where relaxing a specification would pay off most.

Summary

Concept	Key formula
Limit along all paths	polar test: $\lvert f - L\rvert \le g(r) \to 0$ independent of θ
Partial derivative	$f_x = \lim_{h\to 0} \dfrac{f(x+h,y) - f(x,y)}{h}$ (other variables fixed)
Normalised sensitivity	$S_x^f = \dfrac{x}{f}\dfrac{\partial f}{\partial x} = \dfrac{\partial \ln f}{\partial \ln x}$
Clairaut	$f_{xy} = f_{yx}$ (continuous second partials)
Total differential	$df = f_x\,dx + f_y\,dy = \nabla f \cdot d\mathbf{r}$
Tangent plane	$z = f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b)$
Worst-case / RSS error	$\sum \lvert f_{x_i}\rvert \Delta x_i \,/\, \sqrt{\sum (f_{x_i}\Delta x_i)^2}$
Relative error, $f = kx^a y^b$	$df/f = a\,dx/x + b\,dy/y$
Chain rule	$\dfrac{dz}{dt} = f_x\dot{x} + f_y\dot{y}$; $\mathbf{J}_{\mathbf{F}\circ\mathbf{G}} = \mathbf{J}_\mathbf{F}\mathbf{J}_\mathbf{G}$
Implicit derivative	$dy/dx = -F_x/F_y$; $\partial z/\partial x = -F_x/F_z$
Exact differential test	$M\,dx + N\,dy$ exact $\iff M_y = N_x$
Second-order Taylor	$f(\mathbf{a}+\mathbf{h}) \approx f + \nabla f\cdot\mathbf{h} + \tfrac{1}{2}\mathbf{h}^\mathsf{T}\mathbf{H}\mathbf{h}$
Gradient	$\nabla f = (f_x, f_y, f_z)$; normal to level sets; $\lVert\nabla f\rVert$ = max rate
Useful gradients	$\nabla R = \hat{\mathbf{R}}$, $\nabla(1/R) = -\hat{\mathbf{R}}/R^2$, $\mathbf{E} = -\nabla V$
Gradient descent	$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha\nabla f(\mathbf{x}_k)$
Directional derivative	$D_{\hat{\mathbf{u}}} f = \nabla f\cdot\hat{\mathbf{u}} = \lVert\nabla f\rVert\cos\theta$
Jacobian / linearisation	$(\mathbf{J})_{ij} = \partial F_i/\partial x_j$; $d\mathbf{F} = \mathbf{J}\,d\mathbf{x}$
Inverse function	$\mathbf{J}_{\mathbf{F}^{-1}} = \mathbf{J}_\mathbf{F}^{-1}$
Solution sensitivity	$\partial\mathbf{x}/\partial\mathbf{p} = -\mathbf{J}_\mathbf{x}^{-1}\mathbf{J}_\mathbf{p}$
Newton's method	$\mathbf{J}(\mathbf{x}_k)\Delta\mathbf{x} = -\mathbf{F}(\mathbf{x}_k)$
Fubini (type I)	$\iint_R f\,dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f\,dy\,dx$
Change of variables	$dx\,dy = \left\lvert\dfrac{\partial(x,y)}{\partial(u,v)}\right\rvert du\,dv$
Polar	$dA = r\,dr\,d\theta$; $\int_{-\infty}^{\infty} e^{-x^2}dx = \sqrt{\pi}$

Concept	Key formula
Cylindrical	$dV = r\,dr\,d\phi\,dz$; scale factors $(1, r, 1)$
Spherical	$dV = R^2 \sin\theta\,dR\,d\theta\,d\phi$; scale factors $(1, R, R\sin\theta)$
Solid angle	$d\Omega = \sin\theta\,d\theta\,d\phi$; cone: $2\pi(1-\cos\alpha)$; sphere: 4π
Second-derivative test	$D = f_{xx}f_{yy} - f_{xy}^2$: $D>0$ extremum (sign of f_{xx}), $D<0$ saddle
Lagrange multipliers	$\nabla f = \lambda \nabla g,\ g = c$; several: $\nabla f = \sum \lambda_k \nabla g_k$
Multiplier sensitivity	$\lambda = df^*/dc$
Water-filling	$p_i = \max(0,\ \nu - N_i),\ \sum p_i = P$

Further reading

- E. Kreyszig, *Advanced Engineering Mathematics*, 10th ed., Wiley — Ch. 9–10 for multivariable and vector calculus at engineering level.
- K. A. Stroud and D. J. Booth, *Advanced Engineering Mathematics*, 6th ed., Palgrave Macmillan — programme-style treatment of partial differentiation, multiple integrals and Lagrange multipliers with many drills.
- J. Stewart, *Multivariable Calculus*, 8th ed., Cengage — clear geometric treatment of gradients, change of variables and constrained optimisation.
- D. K. Cheng, *Field and Wave Electromagnetics*, 2nd ed., Addison-Wesley — Ch. 2 for orthogonal curvilinear coordinates, scale factors and unit vectors as used in field theory.
- S. Boyd and L. Vandenberghe, *Convex Optimization*, Cambridge University Press — Lagrange duality, KKT conditions and water-filling, from the engineering viewpoint.

Chapter 7 — Vector Calculus

> **Prerequisites:** Ch. 4, Ch. 5, Ch. 6 **Learning outcomes:** After this chapter you can
>
> - manipulate vectors in Cartesian, cylindrical and spherical bases, including dot, cross and triple products;
> - differentiate vector-valued functions and describe curves, velocities and field lines;
> - compute gradient, divergence, curl and Laplacian in all three coordinate systems and interpret them physically;
> - set up and evaluate line, surface and volume integrals, choosing parametrisations and orientations correctly;
> - state, apply and sketch the proofs of the theorems of Green, Stokes and Gauss (divergence theorem);
> - convert Maxwell's equations between integral and differential form, and use them to solve symmetric field problems.

Chapter 6 gave us functions of several variables and a first look at the gradient. This chapter adds the missing ingredient for field theory: **direction**. An electric field, a current density, a heat flux and the velocity of a fluid each attach a *vector* to every point of space. Engineers need to ask three kinds of question about such fields:

1. **Local questions** — at a point, how fast does a potential change, how much does a field spread out, how much does it swirl? The answers are the gradient, divergence and curl.
2. **Global questions** — how much work does a field do along a wire, how much of it flows through a surface, how much charge sits in a volume? The answers are line, surface and volume integrals.
3. **The bridge** — Green's, Stokes' and the divergence theorem say that a global integral over a boundary equals the integral of a local derivative over the inside. They are the multivariable versions of the fundamental theorem of calculus.

Put the three together and Maxwell's equations fall out in two equivalent forms — the integral form you measure, and the differential form you solve. That is the payoff in Section 7.12. Throughout, the methods are general; electromagnetics is the running motivation.

Conventions. Cartesian unit vectors are $\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$. Cylindrical coordinates are (r, ϕ, z) with unit vectors $\hat{\mathbf{r}}, \hat{\boldsymbol{\phi}}, \hat{\mathbf{z}}$; spherical coordinates are (R, θ, ϕ) with θ the polar angle from $+z$ and unit vectors $\hat{\mathbf{R}}, \hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\phi}}$ (as in Ch. 6, §6.9). The position vector of a point is written $\mathbf{R} = x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}$, with length $R = \|\mathbf{R}\|$, so that $\mathbf{R} = R\hat{\mathbf{R}}$ in spherical coordinates. A curve is written $\mathbf{r}(t)$ when it is parametrised by a variable t.

7.1 Scalars and vectors

Why the distinction matters

A **scalar** is a quantity fully described by one number (with a unit): temperature, voltage, charge, energy, resistance. A **vector** needs a magnitude *and* a direction: force, velocity, electric field, current density, magnetic flux density. Mixing them up is not a matter of taste — adding a scalar to a vector is meaningless, and the direction of a field is often the whole point (it decides, for example, whether a force accelerates or decelerates an electron, or whether flux enters or leaves a surface).

Vector

A **vector** in three-dimensional space is a quantity with magnitude and direction, which adds by the parallelogram rule. Relative to a right-handed orthonormal basis $\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$ it is written

$$\mathbf{A} = A_x\hat{\mathbf{x}} + A_y\hat{\mathbf{y}} + A_z\hat{\mathbf{z}} = (A_x, A_y, A_z),$$

where the **components** A_x, A_y, A_z are scalars. Its **magnitude** is $\|\mathbf{A}\| = A = \sqrt{A_x^2 + A_y^2 + A_z^2}$ and, if $A \neq 0$, its **unit vector** is $\hat{\mathbf{a}}_A = \mathbf{A}/A$.

A vector is a geometric object; its components depend on the basis you choose. Rotating the axes changes the three numbers but not the arrow. This is why physical laws are written in vector form ($\mathbf{F} = q\mathbf{E}$) — they then hold in every coordinate system.

Algebra. Addition and scalar multiplication act component by component:

$$\mathbf{A} + \mathbf{B} = (A_x + B_x,\ A_y + B_y,\ A_z + B_z), \qquad c\mathbf{A} = (cA_x,\ cA_y,\ cA_z).$$

These obey the familiar laws: $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$, $(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C})$, $c(\mathbf{A} + \mathbf{B}) = c\mathbf{A} + c\mathbf{B}$. The zero vector $\mathbf{0}$ has no direction. Geometrically, $\mathbf{A} + \mathbf{B}$ is the diagonal of the parallelogram spanned by $\mathbf{A}$ and $\mathbf{B}$, or equivalently the result of placing the tail of $\mathbf{B}$ at the head of $\mathbf{A}$.

Position and separation vectors

The **position vector** $\mathbf{R} = (x, y, z)$ points from the origin to the point $P(x, y, z)$. The vector from a *source point* P' at $\mathbf{R}'$ to a *field point* P at $\mathbf{R}$ is the **separation vector**

$$\mathbf{R} - \mathbf{R}' = (x - x',\ y - y',\ z - z'), \qquad \|\mathbf{R} - \mathbf{R}'\| = \sqrt{(x - x')^2 + (y - y')^2 + (z - z')^2}.$$

Every inverse-square law in field theory is written in terms of it. Coulomb's law for the force on charge q at $\mathbf{R}$ due to charge q' at $\mathbf{R}'$ is

$$\mathbf{F} = \frac{qq'}{4\pi\varepsilon_0}\,\frac{\mathbf{R} - \mathbf{R}'}{\|\mathbf{R} - \mathbf{R}'\|^3}.$$

The cube in the denominator is not a typo: one power of the distance normalises the numerator to a unit vector, and the other two give the inverse-square law.

Worked Example — separation vector and unit vector

A point charge sits at $P'(1, 2, 0)$ m and we observe at $P(3, -1, 6)$ m. Find the separation vector, its length and its unit vector.

Step 1. $\mathbf{R} - \mathbf{R}' = (3-1,\ -1-2,\ 6-0) = (2, -3, 6)$ m.

Step 2. $\|\mathbf{R} - \mathbf{R}'\| = \sqrt{4 + 9 + 36} = \sqrt{49} = 7$ m.

Step 3. $\hat{\mathbf{a}} = (2, -3, 6)/7 = (0.2857,\ -0.4286,\ 0.8571)$. Check: $0.2857^2 + 0.4286^2 + 0.8571^2 = 0.0816 + 0.1837 + 0.7347 = 1.000$.

Worked Example — superposition of Coulomb forces

Charges $q_1 = +2$ nC at $(0, 0, 0)$ and $q_2 = -1$ nC at $(2, 0, 0)$ m act on a test charge $q = +1$ nC at $(0, 1, 0)$ m. Find the net force. Use $1/(4\pi\varepsilon_0) = 8.988 \times 10^9\ \mathrm{N\,m^2/C^2}$.

Step 1 — force from q_1. Separation $(0, 1, 0)$, length 1. $\mathbf{F}_1 = 8.988 \times 10^9 \cdot (2 \times 10^{-9})(1 \times 10^{-9})\,(0, 1, 0)/1^3 = (0,\ 17.98,\ 0)$ nN.

Step 2 — force from q_2. Separation $(0-2,\ 1-0,\ 0) = (-2, 1, 0)$, length $\sqrt{5}$, cube $5\sqrt{5} = 11.180$. $\mathbf{F}_2 = 8.988 \times 10^9 \cdot (-1 \times 10^{-9})(1 \times 10^{-9})\,(-2, 1, 0)/11.180 = (1.608,\ -0.804,\ 0)$ nN.

Step 3 — superpose. $\mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 = (1.608,\ 17.17,\ 0)$ nN, magnitude 17.25 nN, at $\arctan(17.17/1.608) = 84.6^\circ$ from $+x$.

The negative q_2 *attracts* the test charge towards $(2, 0, 0)$, which is why $F_{2x} > 0$ and $F_{2y} < 0$. Forces add as vectors, never as magnitudes.

Curvilinear bases

In cylindrical and spherical coordinates we also use orthonormal bases, but the basis vectors **change direction from point to point**. At a point with azimuth ϕ,

$$\hat{\mathbf{r}} = \cos\phi\,\hat{\mathbf{x}} + \sin\phi\,\hat{\mathbf{y}}, \qquad \hat{\boldsymbol{\phi}} = -\sin\phi\,\hat{\mathbf{x}} + \cos\phi\,\hat{\mathbf{y}}, \qquad \hat{\mathbf{z}} = \hat{\mathbf{z}},$$

and in spherical coordinates

$$\begin{aligned}
\hat{\mathbf{R}} &= \sin\theta\cos\phi\,\hat{\mathbf{x}} + \sin\theta\sin\phi\,\hat{\mathbf{y}} + \cos\theta\,\hat{\mathbf{z}},\\
\hat{\boldsymbol{\theta}} &= \cos\theta\cos\phi\,\hat{\mathbf{x}} + \cos\theta\sin\phi\,\hat{\mathbf{y}} - \sin\theta\,\hat{\mathbf{z}},\\
\hat{\boldsymbol{\phi}} &= -\sin\phi\,\hat{\mathbf{x}} + \cos\phi\,\hat{\mathbf{y}}.
\end{aligned}$$

Each unit vector points in the direction in which its own coordinate increases. Both triads are right-handed: $\hat{\mathbf{r}} \times \hat{\boldsymbol{\phi}} = \hat{\mathbf{z}}$ and $\hat{\mathbf{R}} \times \hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\phi}}$.

Because each triad is orthonormal, the component of $\mathbf{A}$ along a basis vector is a dot product (Section 7.2), and converting components is a rotation:

$$\begin{bmatrix} A_r \\ A_\phi \\ A_z \end{bmatrix} = \begin{bmatrix} \cos\phi & \sin\phi & 0 \\ -\sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix},$$

$$\begin{bmatrix} A_R \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \sin\theta\cos\phi & \sin\theta\sin\phi & \cos\theta \\ \cos\theta\cos\phi & \cos\theta\sin\phi & -\sin\theta \\ -\sin\phi & \cos\phi & 0 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}.$$

The matrices are orthogonal (rows are orthonormal), so the inverse conversion uses the transpose (Ch. 9).

Worked Example — Cartesian to cylindrical components

Express $\mathbf{A} = y\,\hat{\mathbf{x}} - x\,\hat{\mathbf{y}} + z\,\hat{\mathbf{z}}$ in cylindrical coordinates, and evaluate it at $(x, y, z) = (1, \sqrt{3}, 2)$.

Step 1. Substitute $x = r\cos\phi$, $y = r\sin\phi$: $A_x = r\sin\phi$, $A_y = -r\cos\phi$, $A_z = z$.

Step 2. Rotate: $A_r = \cos\phi\,(r\sin\phi) + \sin\phi\,(-r\cos\phi) = 0$, $A_\phi = -\sin\phi\,(r\sin\phi) + \cos\phi\,(-r\cos\phi) = -r$, $A_z = z$. So $\mathbf{A} = -r\,\hat{\boldsymbol{\phi}} + z\,\hat{\mathbf{z}}$ — a clockwise swirl (seen from $+z$) plus a vertical part. The cylindrical form makes the geometry obvious, which the Cartesian form hid.

Step 3. At $(1, \sqrt{3}, 2)$: $r = 2$, $\phi = 60^\circ$, so $\mathbf{A} = -2\hat{\boldsymbol{\phi}} + 2\hat{\mathbf{z}}$. Check in Cartesian: $\mathbf{A} = (\sqrt{3}, -1, 2)$, $\|\mathbf{A}\| = \sqrt{3 + 1 + 4} = \sqrt{8}$; cylindrical: $\sqrt{4 + 4} = \sqrt{8}$. ✓

Common Pitfall

You may **not** add the cylindrical or spherical components of vectors located at *different points*. $\hat{\mathbf{r}}$ at $\phi = 0$ is $\hat{\mathbf{x}}$, but at $\phi = \pi$ it is $-\hat{\mathbf{x}}$. Two vectors "$1\,\hat{\mathbf{r}}$" at those two points sum to zero, not to $2\hat{\mathbf{r}}$. Convert to Cartesian components before superposing contributions from many source points — this is exactly why field integrals such as the field of a ring of charge are done in Cartesian components.

Engineering Note

Field solvers and antenna-pattern software report fields in whichever basis matches the symmetry: a coaxial cable's $\mathbf{E}$ is purely $\hat{\mathbf{r}}$ and its $\mathbf{H}$ purely $\hat{\boldsymbol{\phi}}$; an antenna's far field is purely $\hat{\boldsymbol{\theta}}$ and $\hat{\boldsymbol{\phi}}$. Choosing the basis that matches the symmetry reduces a three-component problem to a one-component one.

7.2 Vector operations

There are two ways to multiply two vectors, and each answers a different geometric question. The dot product measures *how much two vectors agree*; the cross product measures *how much they span*.

The dot (scalar) product

Dot product

$$\mathbf{A}\cdot\mathbf{B} = AB\cos\theta_{AB} = A_xB_x + A_yB_y + A_zB_z,$$

where $\theta_{AB}\in[0,\pi]$ is the angle between the vectors.

Why the two forms agree. Apply the law of cosines to the triangle with sides $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{A}-\mathbf{B}$: $\|\mathbf{A}-\mathbf{B}\|^2 = A^2 + B^2 - 2AB\cos\theta_{AB}$. Expanding the left side in components gives $A^2+B^2-2(A_xB_x+A_yB_y+A_zB_z)$. Comparing the two proves the identity.

Properties: commutative, distributive, $\mathbf{A}\cdot\mathbf{A} = A^2$, and $\mathbf{A}\cdot\mathbf{B}=0$ iff the vectors are perpendicular (or one is zero). For orthonormal bases $\hat{\mathbf{x}}\cdot\hat{\mathbf{x}}=1$, $\hat{\mathbf{x}}\cdot\hat{\mathbf{y}}=0$, and so on.

Projection. The scalar component of $\mathbf{A}$ along a unit vector $\hat{\mathbf{u}}$ is $\mathbf{A}\cdot\hat{\mathbf{u}}$, and the vector projection is $(\mathbf{A}\cdot\hat{\mathbf{u}})\hat{\mathbf{u}}$. The rest, $\mathbf{A}-(\mathbf{A}\cdot\hat{\mathbf{u}})\hat{\mathbf{u}}$, is perpendicular to $\hat{\mathbf{u}}$. This decomposition into normal and tangential parts is used constantly at material boundaries.

Physically, the dot product appears whenever only the *aligned* part of a vector matters: work $dW = \mathbf{F}\cdot d\mathbf{l}$, flux $d\Phi = \mathbf{B}\cdot d\mathbf{S}$, power $p = \mathbf{F}\cdot\mathbf{v}$.

The cross (vector) product

Cross product

$$\mathbf{A}\times\mathbf{B} = AB\sin\theta_{AB}\,\hat{\mathbf{n}} = \begin{vmatrix}\hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}}\\ A_x & A_y & A_z\\ B_x & B_y & B_z\end{vmatrix} = (A_yB_z - A_zB_y,\; A_zB_x - A_xB_z,\; A_xB_y - A_yB_x),$$

where $\hat{\mathbf{n}}$ is the unit normal to the plane of $\mathbf{A}$ and $\mathbf{B}$ chosen by the **right-hand rule** (fingers from $\mathbf{A}$ to $\mathbf{B}$, thumb along $\hat{\mathbf{n}}$).

Key facts:

- $\|\mathbf{A}\times\mathbf{B}\| = AB\sin\theta_{AB}$ is the **area of the parallelogram** spanned by $\mathbf{A}$ and $\mathbf{B}$.
- $\mathbf{A}\times\mathbf{B}$ is perpendicular to both $\mathbf{A}$ and $\mathbf{B}$.
- **Anticommutative:** $\mathbf{B}\times\mathbf{A} = -\mathbf{A}\times\mathbf{B}$, hence $\mathbf{A}\times\mathbf{A}=\mathbf{0}$; parallel vectors have zero cross product.
- Distributive over addition, but **not associative**: $(\mathbf{A}\times\mathbf{B})\times\mathbf{C} \neq \mathbf{A}\times(\mathbf{B}\times\mathbf{C})$ in general.
- Cyclic basis rule: $\hat{\mathbf{x}}\times\hat{\mathbf{y}}=\hat{\mathbf{z}}$, $\hat{\mathbf{y}}\times\hat{\mathbf{z}}=\hat{\mathbf{x}}$, $\hat{\mathbf{z}}\times\hat{\mathbf{x}}=\hat{\mathbf{y}}$; the same cyclic rule holds for $(\hat{\mathbf{r}},\hat{\boldsymbol{\phi}},\hat{\mathbf{z}})$ and $(\hat{\mathbf{R}},\hat{\boldsymbol{\theta}},\hat{\boldsymbol{\phi}})$ at a given point.

Physically, the cross product appears whenever a "turning" or "sideways" effect occurs: torque $\boldsymbol{\tau}=\mathbf{r}\times\mathbf{F}$, the magnetic force $\mathbf{F}=q\mathbf{v}\times\mathbf{B}$, the Biot–Savart element $d\mathbf{B}\propto I\,d\mathbf{l}\times\hat{\mathbf{R}}/R^2$, the Poynting vector $\mathbf{S}=\mathbf{E}\times\mathbf{H}$, and the normal to a surface $\mathbf{r}_u\times\mathbf{r}_v$ (Section 7.7).

Triple products

Scalar and vector triple products

Scalar triple product.

$$\mathbf{A}\cdot(\mathbf{B}\times\mathbf{C}) = \begin{vmatrix}A_x & A_y & A_z\\ B_x & B_y & B_z\\ C_x & C_y & C_z\end{vmatrix} = \mathbf{B}\cdot(\mathbf{C}\times\mathbf{A}) = \mathbf{C}\cdot(\mathbf{A}\times\mathbf{B}).$$

Its absolute value is the volume of the parallelepiped with edges $\mathbf{A},\mathbf{B},\mathbf{C}$; it vanishes iff the three vectors are coplanar.

Vector triple product ("BAC–CAB").

$$\mathbf{A}\times(\mathbf{B}\times\mathbf{C}) = \mathbf{B}(\mathbf{A}\cdot\mathbf{C}) - \mathbf{C}(\mathbf{A}\cdot\mathbf{B}).$$

Proof sketch of BAC–CAB. Both sides are linear in each of $\mathbf{A},\mathbf{B},\mathbf{C}$, so it suffices to check basis vectors. The left side is perpendicular to $\mathbf{B}\times\mathbf{C}$, hence lies in the plane of $\mathbf{B}$ and $\mathbf{C}$; e.g. $\hat{\mathbf{x}}\times(\hat{\mathbf{x}}\times\hat{\mathbf{y}}) = \hat{\mathbf{x}}\times\hat{\mathbf{z}} = -\hat{\mathbf{y}}$, while the right side gives $\hat{\mathbf{x}}(0)-\hat{\mathbf{y}}(1) = -\hat{\mathbf{y}}$. The remaining cases are checked the same way. The identity reappears later with ∇ in place of $\mathbf{A}$ and $\mathbf{B}$: $\nabla\times(\nabla\times\mathbf{F}) = \nabla(\nabla\cdot\mathbf{F}) - \nabla^2\mathbf{F}$ (Section 7.5), the step that turns Maxwell's equations into the wave equation.

Worked Example — angle, projection and area

Let $\mathbf{A}=(1,2,2)$ and $\mathbf{B}=(3,0,4)$.

Angle. $\mathbf{A}\cdot\mathbf{B} = 3+0+8 = 11$, $A=3$, $B=5$, so $\cos\theta = 11/15$ and $\theta = 42.83^\circ$.

Projection of A on B. $\mathbf{A}\cdot\hat{\mathbf{B}} = 11/5 = 2.2$; vector projection $2.2\,(0.6,0,0.8) = (1.32,0,1.76)$. The perpendicular remainder is $(1,2,2) - (1.32,0,1.76) = (-0.32,2,0.24)$; check $(-0.32)(3)+0+(0.24)(4) = 0$. ✓

Cross product. $\mathbf{A}\times\mathbf{B} = (2\cdot4-2\cdot0,\; 2\cdot3-1\cdot4,\; 1\cdot0-2\cdot3) = (8,2,-6)$. Its length $\sqrt{64+4+36} = \sqrt{104} = 10.198$ should equal $AB\sin\theta = 15\sin42.83^\circ = 15\times0.6799 = 10.198$. ✓ So the triangle with sides $\mathbf{A}$ and $\mathbf{B}$ has area $\tfrac{1}{2}\sqrt{104} = 5.10$.

Worked Example — volume and coplanarity

Are $\mathbf{A}=(1,0,1)$, $\mathbf{B}=(2,1,0)$, $\mathbf{C}=(0,1,-2)$ coplanar?

$$\begin{vmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 0 & 1 & -2 \end{vmatrix} = 1(1\cdot(-2)-0\cdot 1)-0+1(2\cdot 1-1\cdot 0) = -2+2 = 0.$$

Yes: $\mathbf{C}=\mathbf{B}-2\mathbf{A}$. Replacing $\mathbf{C}$ by $(0,1,2)$ gives determinant $1(2-0)+1(2)=4$, a parallelepiped of volume 4.

Worked Example — Lorentz force on an electron

An electron ($q=-1.602\times10^{-19}$ C) moves with $\mathbf{v}=(2\times10^{5},\ 0,\ 1\times10^{5})$ m/s through $\mathbf{E}=(0,\ 500,\ 0)$ V/m and $\mathbf{B}=(0,\ 0,\ 0.01)$ T. Find $\mathbf{F}=q(\mathbf{E}+\mathbf{v}\times\mathbf{B})$.

Step 1. $\mathbf{v}\times\mathbf{B}=(v_yB_z-v_zB_y,\ v_zB_x-v_xB_z,\ v_xB_y-v_yB_x)=(0,\ -2\times10^{3},\ 0)$ V/m. (Units: $\mathrm{m/s}\cdot\mathrm{T}=\mathrm{V/m}$.)

Step 2. $\mathbf{E}+\mathbf{v}\times\mathbf{B}=(0,\ 500-2000,\ 0)=(0,-1500,0)$ V/m.

Step 3. $\mathbf{F}=-1.602\times10^{-19}\cdot(0,-1500,0)=(0,\ 2.403\times10^{-16},\ 0)$ N.

The v_z component, parallel to $\mathbf{B}$, contributes nothing to the magnetic force — a cross product only sees the perpendicular part. Choosing $\mathbf{E}=(0,-v_xB_z,0)$ would cancel the force entirely: this is the velocity-selector (Wien filter) condition $E=vB$.

Common Pitfall

The cross product is defined only in three dimensions and depends on handedness. If you set up a left-handed axis system (for instance by drawing $\hat{\mathbf{y}}$ the wrong way on a page), every cross product — and every curl — changes sign. Always check $\hat{\mathbf{x}}\times\hat{\mathbf{y}}=\hat{\mathbf{z}}$ for your drawing.

Engineering Note

The dot product is the continuous cousin of the inner product used in signal processing (Ch. 9): correlating two signals is "dotting" them. For complex phasor fields the time-average power uses a conjugate, $\langle\mathbf{S}\rangle=\tfrac{1}{2}\,\mathrm{Re}\{\mathbf{E}\times\mathbf{H}^{*}\}$; forgetting the conjugate is a classic error.

7.3 Vector-valued functions

Definition and derivative

A trajectory, a wire, the tip of a rotating phasor, the path of an electron in a magnetron — each is a point moving through space as a parameter changes. We describe it with a vector-valued function.

Vector-valued function

A **vector-valued function** of one real variable assigns a vector to each t in an interval:

$$\mathbf{r}(t)=x(t)\hat{\mathbf{x}}+y(t)\hat{\mathbf{y}}+z(t)\hat{\mathbf{z}}.$$

Its **derivative** is

$$\mathbf{r}'(t)=\lim_{h\to0}\frac{\mathbf{r}(t+h)-\mathbf{r}(t)}{h}=x'(t)\hat{\mathbf{x}}+y'(t)\hat{\mathbf{y}}+z'(t)\hat{\mathbf{z}},$$

and its integral is likewise taken component by component. The curve is **smooth** if $\mathbf{r}'$ is continuous and never zero.

The componentwise rule works *only because* $\hat{\mathbf{x}},\hat{\mathbf{y}},\hat{\mathbf{z}}$ are constant. Geometrically, $\mathbf{r}(t+h)-\mathbf{r}(t)$ is a chord of the curve; dividing by h and letting $h\to0$ turns it into a **tangent vector**. If t is time, $\mathbf{r}'=\mathbf{v}$ is the velocity, $\|\mathbf{v}\|$ the speed, and $\mathbf{r}''=\mathbf{a}$ the acceleration.

Differentiation rules

For differentiable vector functions $\mathbf{A}(t),\mathbf{B}(t)$ and scalar $f(t)$:

$$\begin{aligned}(\mathbf{A}+\mathbf{B})'&=\mathbf{A}'+\mathbf{B}', & (f\mathbf{A})'&=f'\mathbf{A}+f\mathbf{A}',\\ (\mathbf{A}\cdot\mathbf{B})'&=\mathbf{A}'\cdot\mathbf{B}+\mathbf{A}\cdot\mathbf{B}', & (\mathbf{A}\times\mathbf{B})'&=\mathbf{A}'\times\mathbf{B}+\mathbf{A}\times\mathbf{B}',\\ \frac{d}{dt}\mathbf{A}(u(t))&=\mathbf{A}'(u)\,u'(t).\end{aligned}$$

In the cross-product rule the **order matters**.

Each rule follows by writing the product in components and applying the scalar product rule. A useful corollary: if $\|\mathbf{A}(t)\|$ is constant then $\mathbf{A}\cdot\mathbf{A}=\text{const}$, so $2\mathbf{A}\cdot\mathbf{A}'=0$ — **a vector of constant length is perpendicular to its own derivative.** A unit vector can only rotate, never stretch.

Arc length, unit tangent and curvature

The distance travelled along the curve from $t=a$ to $t=b$ is the **arc length**

$$s = \int_a^b \|\mathbf{r}'(t)\|\,dt = \int_a^b \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2}\,dt,$$

because a small step $d\mathbf{r} = \mathbf{r}'\,dt$ has length $ds = \|\mathbf{r}'\|\,dt$. The **unit tangent** is $\hat{\mathbf{T}} = \mathbf{r}'/\|\mathbf{r}'\|$, and the **curvature** $\kappa = \|d\hat{\mathbf{T}}/ds\|$ measures how fast the direction turns per metre. A convenient formula is

$$\kappa = \frac{\|\mathbf{r}' \times \mathbf{r}''\|}{\|\mathbf{r}'\|^3}.$$

A straight line has $\kappa = 0$; a circle of radius a has $\kappa = 1/a$. The arc length is independent of how the curve is parametrised: substituting $t = g(u)$ with $g' > 0$ multiplies $\|\mathbf{r}'\|$ by $g'(u)$ and dt by $1/g'(u)$ in reverse — the same cancellation that will make line integrals parametrisation-independent in Section 7.6.

Worked Example — length of a solenoid winding

A solenoid has radius $a = 5$ mm and pitch $p = 1$ mm per turn, with 100 turns. Find the wire length and the curvature of the wire.

Step 1 — parametrise. One turn is 2π of angle and advances p along z: $\mathbf{r}(t) = (a\cos t,\ a\sin t,\ bt)$ with $b = p/(2\pi) = 0.15915$ mm.

Step 2 — speed. $\mathbf{r}' = (-a\sin t,\ a\cos t,\ b)$, $\|\mathbf{r}'\| = \sqrt{a^2 + b^2} = \sqrt{25 + 0.02533} = 5.00253$ mm (constant).

Step 3 — length. $s = \int_0^{200\pi} \sqrt{a^2+b^2}\,dt = 200\pi \times 5.00253 = 3143.2$ mm $= 3.1432$ m. A flat-circle estimate $100 \times 2\pi a = 3.1416$ m is only 0.05 % low, because the pitch is small compared with the circumference.

Step 4 — curvature. $\mathbf{r}'' = (-a\cos t, -a\sin t, 0)$ and $\mathbf{r}' \times \mathbf{r}'' = (ab\sin t,\ -ab\cos t,\ a^2)$ with length $a\sqrt{a^2+b^2}$. So $\kappa = a/(a^2+b^2) = 0.19980$ mm^{-1}, a radius of curvature 5.005 mm — very slightly larger than the coil radius.

Worked Example — arc length that needs care

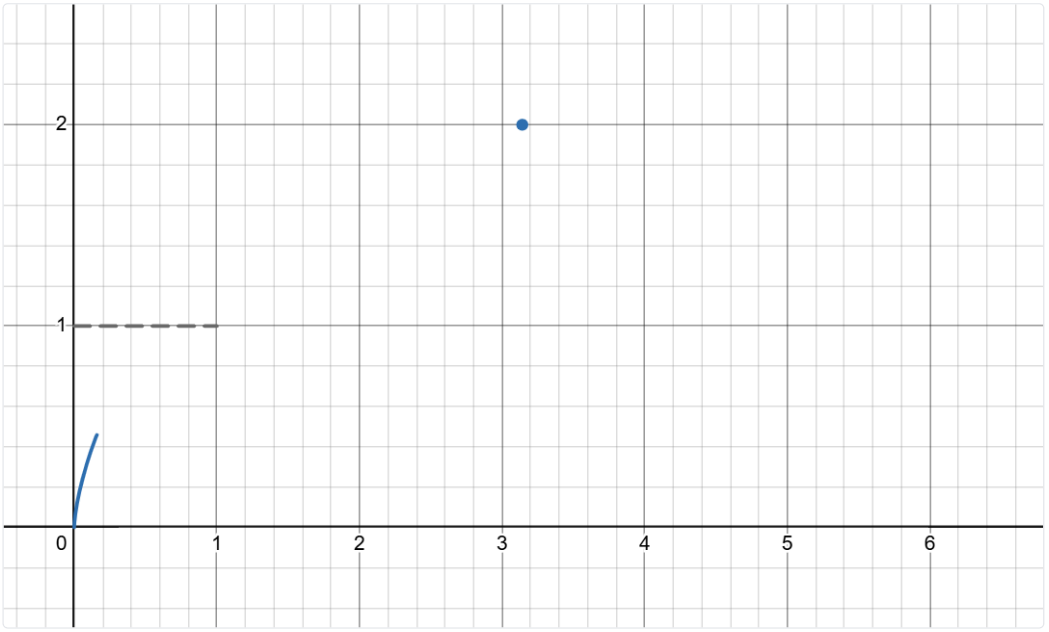
Find the length of one arch of the cycloid $\mathbf{r}(t) = (t - \sin t,\ 1 - \cos t)$, $0 \le t \le 2\pi$ — the path of a point on the rim of a rolling unit wheel.

Step 1. $\mathbf{r}' = (1 - \cos t,\ \sin t)$, so $\|\mathbf{r}'\|^2 = 1 - 2\cos t + \cos^2 t + \sin^2 t = 2 - 2\cos t = 4\sin^2(t/2)$.

Step 2. On $[0, 2\pi]$, $\sin(t/2) \ge 0$, so $\|\mathbf{r}'\| = 2\sin(t/2)$ — no absolute-value trouble.

Step 3. $s = \int_0^{2\pi} 2\sin(t/2)\,dt = \big[-4\cos(t/2)\big]_0^{2\pi} = 4 + 4 = 8$.

Note that $\mathbf{r}'(0) = \mathbf{0}$: the cycloid has a cusp there and is not smooth at the ends, but the arc-length integral is still perfectly finite.



One arch of the cycloid traced by a point on a rolling unit wheel (axle path dashed). Its length is exactly 8, four times the wheel's diameter.

Moving bases: velocity in polar and cylindrical coordinates

If a point moves with cylindrical coordinates $(r(t), \phi(t), z(t))$, its position is $\mathbf{r} = r\hat{\mathbf{r}} + z\hat{\mathbf{z}}$ — but $\hat{\mathbf{r}}$ depends on ϕ, which depends on t. Differentiating the expressions of Section 7.1 with the chain rule,

$$\frac{d\hat{\mathbf{r}}}{dt} = \dot\phi\,(-\sin\phi\,\hat{\mathbf{x}} + \cos\phi\,\hat{\mathbf{y}}) = \dot\phi\,\hat{\boldsymbol{\phi}}, \qquad \frac{d\hat{\boldsymbol{\phi}}}{dt} = -\dot\phi\,\hat{\mathbf{r}}.$$

(Both derivatives are perpendicular to the vector being differentiated, as the constant-length corollary demands.) Hence

$$\mathbf{v} = \dot r\,\hat{\mathbf{r}} + r\dot\phi\,\hat{\boldsymbol{\phi}} + \dot z\,\hat{\mathbf{z}},$$
$$\mathbf{a} = (\ddot r - r\dot\phi^2)\,\hat{\mathbf{r}} + (r\ddot\phi + 2\dot r\dot\phi)\,\hat{\boldsymbol{\phi}} + \ddot z\,\hat{\mathbf{z}}.$$

The term $-r\dot\phi^2\hat{\mathbf{r}}$ is the centripetal acceleration and $2\dot r\dot\phi\,\hat{\boldsymbol\phi}$ the Coriolis term. Neither appears if you forget that the basis vectors move. The speed formula $\|\mathbf{v}\|^2 = \dot r^2 + r^2\dot\phi^2 + \dot z^2$ is the time-derivative version of the cylindrical line element $ds^2 = dr^2 + r^2 d\phi^2 + dz^2$.

Worked Example — an electron in a uniform magnetic field

A particle of charge q and mass m moves in $\mathbf{B} = B_0\hat{\mathbf{z}}$ with no electric field. Newton's law is $m\,\dot{\mathbf{v}} = q\,\mathbf{v}\times\mathbf{B}$. Show that the motion is a helix and find its radius and frequency for an electron with $v_\perp = 10^6$ m/s in $B_0 = 0.01$ T.

Step 1 — energy is conserved. $\frac{d}{dt}\left(\frac12 m\,\mathbf{v}\cdot\mathbf{v}\right) = m\,\dot{\mathbf{v}}\cdot\mathbf{v} = q(\mathbf{v}\times\mathbf{B})\cdot\mathbf{v} = 0$ by the scalar triple product (two equal vectors). A magnetic field never changes the speed.

Step 2 — components. $\mathbf{v}\times\mathbf{B} = (v_y B_0,\ -v_x B_0,\ 0)$. With $\omega_c = qB_0/m$: $\dot v_x = \omega_c v_y$, $\dot v_y = -\omega_c v_x$, $\dot v_z = 0$.

Step 3 — solve. Try $v_x = v_\perp\cos\omega_c t$, $v_y = -v_\perp\sin\omega_c t$: then $\dot v_x = -v_\perp\omega_c\sin\omega_c t = \omega_c v_y$ ✓ and $\dot v_y = -v_\perp\omega_c\cos\omega_c t = -\omega_c v_x$ ✓. Integrating, the orbit is a circle of radius $r_c = v_\perp/|\omega_c|$ while z grows uniformly — a helix like the solenoid winding above.

Step 4 — numbers. $|\omega_c| = 1.602\times10^{-19}\times0.01/9.109\times10^{-31} = 1.759\times10^{9}$ rad/s, i.e. $f_c = |\omega_c|/2\pi = 279.9$ MHz. Radius $r_c = 10^6/1.759\times10^9 = 0.569$ mm.

Step 5 — check with the polar formula. For uniform circular motion $\dot r = \ddot r = 0$, $\ddot\phi = 0$, so $\mathbf{a} = -r\dot\phi^2\hat{\mathbf{r}}$: magnitude $r_c\omega_c^2 = v_\perp|\omega_c| = 1.759\times10^{15}$ m/s², which equals $|q|v_\perp B_0/m$ ✓.

The cyclotron frequency $\omega_c = qB/m$ is independent of speed — the fact that makes cyclotrons and magnetrons work.

Common Pitfall

"Differentiate component by component" is valid only for a **fixed** basis. Writing $\frac{d}{dt}(r\hat{\mathbf{r}}) = \dot r\hat{\mathbf{r}}$ loses the $r\dot\phi\,\hat{\boldsymbol\phi}$ term — the whole of the velocity for circular motion.

Engineering Note

A phasor $V_m e^{j\omega t}$ is a vector-valued function in the complex plane: $\mathbf{r}(t) = (V_m\cos\omega t,\ V_m\sin\omega t)$. Its derivative $j\omega V_m e^{j\omega t}$ is the tangent vector — rotated 90° ahead and scaled by ω — which is why differentiation becomes multiplication by $j\omega$ in AC analysis (Ch. 2). The polarisation of a plane wave is likewise the curve traced by the tip of $\mathbf{E}(t)$ at a fixed point: a line, circle or ellipse.

7.4 Vector fields

Definition and examples

Scalar and vector fields

A **scalar field** assigns a number $f(\mathbf{R})$ to each point of a region of space. A **vector field** assigns a vector

$$\mathbf{F}(x,y,z) = F_x(x,y,z)\,\hat{\mathbf{x}} + F_y(x,y,z)\,\hat{\mathbf{y}} + F_z(x,y,z)\,\hat{\mathbf{z}}$$

to each point. A field is **static** if it does not depend on time and **time-varying** otherwise; it is **uniform** if it is the same at every point.

Potential V, temperature T and charge density ρ are scalar fields; $\mathbf{E}$, $\mathbf{B}$, current density $\mathbf{J}$ and heat flux $\mathbf{q}$ are vector fields. Picture a vector field as an arrow attached to every point, with length showing magnitude. The components are ordinary scalar functions of several variables, so everything in Ch. 6 — continuity, partial derivatives, the chain rule — applies to each component separately.

A short catalogue of fields that recur throughout the chapter:

Field	Formula	Picture
Uniform	$\mathbf{F} = E_0\hat{\mathbf{x}}$	parallel arrows, all equal (parallel-plate capacitor)
Radial (3-D)	$\mathbf{F} = \mathbf{R} = x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}$	arrows point outward, growing with distance
Point charge	$\mathbf{E} = \dfrac{q}{4\pi\varepsilon_0}\dfrac{\hat{\mathbf{R}}}{R^2}$	outward, decaying as $1/R^2$
Line charge	$\mathbf{E} = \dfrac{\rho_\ell}{2\pi\varepsilon_0 r}\hat{\mathbf{r}}$	outward from the z-axis, decaying as $1/r$
Rigid rotation	$\mathbf{F} = -y\hat{\mathbf{x}} + x\hat{\mathbf{y}} = r\hat{\boldsymbol\phi}$	circles, speed growing with r
Line current	$\mathbf{B} = \dfrac{\mu_0 I}{2\pi r}\hat{\boldsymbol\phi}$	circles, strength decaying as $1/r$

The last two both circulate around the z-axis, but (as we shall see) one has curl and the other does not. Pictures alone can mislead; the differential operators of Section 7.5 make the distinction precise.

Field lines

Field line (streamline, flux line)

A **field line** of $\mathbf{F}$ is a curve that is tangent to $\mathbf{F}$ at every point. If $\mathbf{r}(s)$ is a field line then $d\mathbf{r}/ds$ is parallel to $\mathbf{F}(\mathbf{r})$, i.e.

$$\frac{dx}{F_x} = \frac{dy}{F_y} = \frac{dz}{F_z}.$$

In two dimensions this is the first-order ODE $dy/dx = F_y/F_x$, solved by the methods of Ch. 11. Field lines never cross where $\mathbf{F} \neq \mathbf{0}$ (the tangent direction would be ambiguous); they can start or stop only where $\mathbf{F}$ vanishes or is singular — at charges, for instance. Their density is drawn proportional to the field magnitude, which is why "lines of force" became Faraday's mental model of the field.

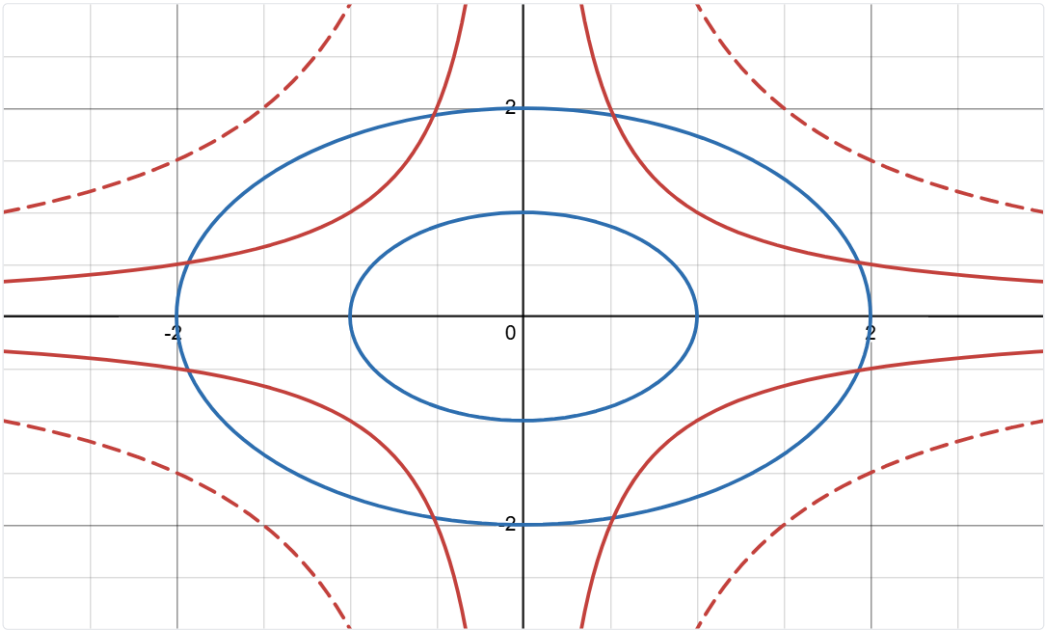
Worked Example — field lines of three planar fields

Find the field lines of (a) $\mathbf{F} = -y\hat{\mathbf{x}} + x\hat{\mathbf{y}}$, (b) $\mathbf{G} = x\hat{\mathbf{x}} - y\hat{\mathbf{y}}$, (c) $\mathbf{H} = \hat{\mathbf{x}} + 2x\hat{\mathbf{y}}$.

(a) $dy/dx = x/(-y)$, so $y\,dy = -x\,dx$ and $x^2 + y^2 = C$: **circles** about the origin, traversed counter-clockwise (at $(1,0)$ the field is $+\hat{\mathbf{y}}$).

(b) $dy/dx = -y/x$, so $dy/y = -dx/x$, $\ln|y| = -\ln|x| + c$ and $xy = C$: **rectangular hyperbolas**. Flow comes in along the y-axis and leaves along the x-axis — the field near a saddle point of a potential, e.g. the centre of a quadrupole lens.

(c) $dy/dx = 2x$, so $y = x^2 + C$: a family of **parabolas** shifted vertically.



Field lines of the rotation field $-y\hat{\mathbf{x}} + x\hat{\mathbf{y}}$ (blue circles) and of the saddle field $x\hat{\mathbf{x}} - y\hat{\mathbf{y}}$ (red hyperbolas $xy = C$).

Worked Example — field lines and equipotentials of a dipole

A small electric dipole on the z-axis has, in spherical coordinates,

$$\mathbf{E} = \frac{p}{4\pi\varepsilon_0 R^3}\left(2\cos\theta\,\hat{\mathbf{R}} + \sin\theta\,\hat{\boldsymbol{\theta}}\right), \qquad V = \frac{p\cos\theta}{4\pi\varepsilon_0 R^2}.$$

Find the field lines.

Step 1 — the field-line condition in spherical coordinates. A displacement along a curve in the plane $\phi = \text{const}$ is $dR\,\hat{\mathbf{R}} + R\,d\theta\,\hat{\boldsymbol{\theta}}$ (the scale factors of Ch. 6). Parallel to $\mathbf{E}$ means

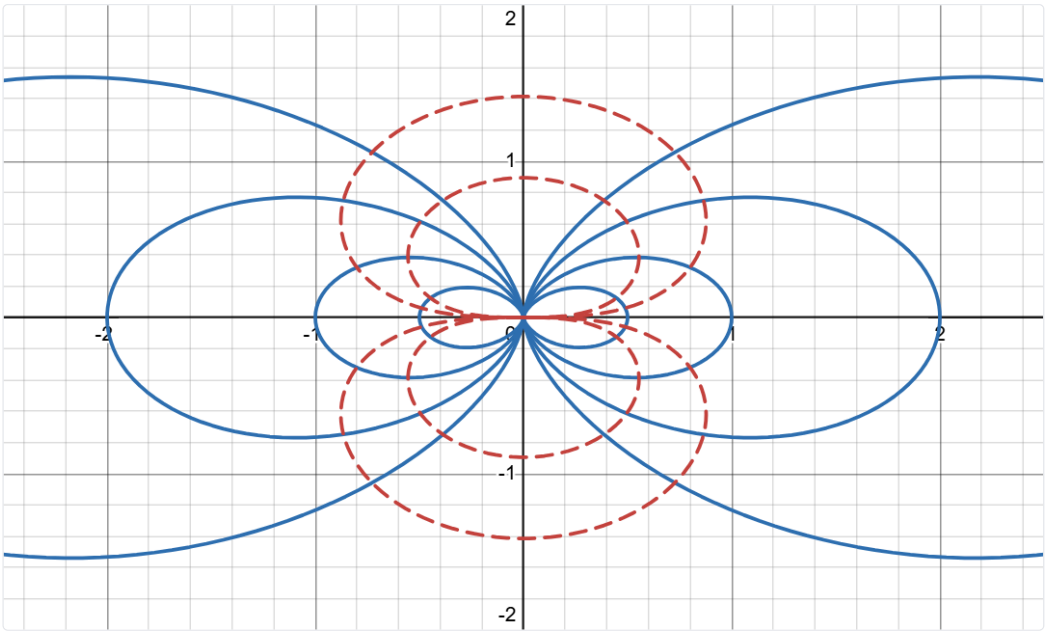
$$\frac{dR}{E_R} = \frac{R\,d\theta}{E_\theta} \quad\Longrightarrow\quad \frac{dR}{2\cos\theta} = \frac{R\,d\theta}{\sin\theta}.$$

Step 2 — separate. $\dfrac{dR}{R} = \dfrac{2\cos\theta}{\sin\theta}d\theta$, so $\ln R = 2\ln\sin\theta + c$ and

$$R = C\sin^2\theta.$$

Step 3 — check orthogonality. Equipotentials are $R^2 = K\cos\theta$. On them $2R\,dR = -K\sin\theta\,d\theta$, so $dR/(R\,d\theta) = -\tfrac{1}{2}\tan\theta$; on field lines $dR/(R\,d\theta) = 2\cot\theta$. The product of these two slopes is -1, so the families cross at right angles, as $\mathbf{E} = -\nabla V$ requires.

In the Desmos plot below the dipole axis is drawn vertically; with Desmos's polar angle ϑ measured from the horizontal, $\sin\theta = \cos\vartheta$ and $\cos\theta = \sin\vartheta$.



Dipole with its axis vertical: field lines $R = C\sin^2\theta$ (blue) leave the upper (positive) end and return to the lower end, crossing the equipotentials $R^2 = K\cos\theta$ (red, dashed) at right angles.

Worked Example — the magnetic field of a wire in Cartesian form

Convert $\mathbf{B} = \dfrac{\mu_0 I}{2\pi r}\hat{\boldsymbol{\phi}}$ to Cartesian components and evaluate it at $(3, 4, 0)$ cm for $I = 10$ A.

Step 1. $\hat{\boldsymbol{\phi}} = (-\sin\phi, \cos\phi, 0) = (-y/r,\ x/r,\ 0)$, so

$$\mathbf{B} = \frac{\mu_0 I}{2\pi}\,\frac{-y\,\hat{\mathbf{x}} + x\,\hat{\mathbf{y}}}{x^2 + y^2}.$$

Step 2. At $(0.03, 0.04)$ m, $r = 0.05$ m and $\mu_0 I/(2\pi r) = (4\pi \times 10^{-7} \times 10)/(2\pi \times 0.05) = 40\ \mu$T. Direction $\hat{\boldsymbol{\phi}} = (-0.8, 0.6, 0)$, so $\mathbf{B} = (-32,\ 24,\ 0)\ \mu$T.

The field lines are again circles, exactly like the rotation field $-y\hat{\mathbf{x}} + x\hat{\mathbf{y}}$, but the magnitude falls off as $1/r$ instead of growing as r. That difference will decide whether the curl is zero (Section 7.5). For comparison, the Earth's field is about 50 μT — which is why a 10 A cable a few centimetres from a compass upsets it.

Common Pitfall

A field-line diagram shows direction and (by density) magnitude, but **not** the path a charged particle follows. A particle's acceleration, not its velocity, is along $\mathbf{E}$; except when it starts from rest in a straight-line field, its trajectory leaves the field line.

Engineering Note

Finite-element field solvers (for PCB crosstalk, motor design, antenna feeds) compute a field at mesh nodes and display it as arrow plots, streamlines and colour maps of magnitude. Reading those plots is reading a vector field: where lines crowd, the field (and the risk of dielectric breakdown) is high; where they circulate, there is curl, hence current or changing flux.

7.5 Gradient, divergence, and curl

The del operator

All three local derivatives of a field are built from one symbolic vector, the **del** (or nabla) operator

$$\nabla = \hat{\mathbf{x}}\frac{\partial}{\partial x} + \hat{\mathbf{y}}\frac{\partial}{\partial y} + \hat{\mathbf{z}}\frac{\partial}{\partial z}.$$

∇ is not a vector in its own right — it has no magnitude until it acts on something — but in Cartesian coordinates it can be combined with a field using the three products we already know:

Operation	Acts on	Produces	Cartesian form	Measures
gradient ∇f	scalar	vector	$(f_x,\ f_y,\ f_z)$	rate and direction of steepest increase
divergence $\nabla \cdot \mathbf{F}$	vector	scalar	$\partial_x F_x + \partial_y F_y + \partial_z F_z$	net outflow per unit volume
curl $\nabla \times \mathbf{F}$	vector	vector	determinant below	circulation per unit area (rotation)
Laplacian $\nabla^2 f = \nabla \cdot \nabla f$	scalar	scalar	$f_{xx} + f_{yy} + f_{zz}$	how f differs from its local average

Each operator has a dimension of m^{-1} (or m^{-2} for the Laplacian). If V is in volts, ∇V is in V/m; if $\mathbf{D}$ is in $\mathrm{C/m^2}$, $\nabla\cdot\mathbf{D}$ is in $\mathrm{C/m^3}$, a charge density — the first hint of Gauss's law.

The best way to understand the three operators is through their **coordinate-free definitions** as limits of integrals over shrinking regions. These definitions explain what the operators mean, tell you how to compute them in any coordinate system, and are exactly the seeds from which the integral theorems of Sections 7.9–7.11 grow.

Gradient

The gradient was introduced in Ch. 6, §6.4: $df = \nabla f\cdot d\mathbf{r}$, ∇f points uphill, is normal to level surfaces, and its length is the maximum rate of increase. In integral language, for any path from $\mathbf{a}$ to $\mathbf{b}$,

$$f(\mathbf{b}) - f(\mathbf{a}) = \int_{\mathbf{a}}^{\mathbf{b}} \nabla f\cdot d\mathbf{r},$$

which is the fundamental theorem for line integrals (Section 7.6).

In curvilinear coordinates. A step du_i in coordinate u_i moves a distance $h_i\,du_i$, where the h_i are the **scale factors** of Ch. 6: $(1, r, 1)$ for cylindrical (r,ϕ,z) and $(1, R, R\sin\theta)$ for spherical (R,θ,ϕ). The component of ∇f along $\hat{\mathbf{u}}_i$ is the rate of change of f *per metre* in that direction, $\frac{1}{h_i}\frac{\partial f}{\partial u_i}$. Hence

$$\nabla f = \frac{\partial f}{\partial r}\hat{\mathbf{r}} + \frac{1}{r}\frac{\partial f}{\partial \phi}\hat{\boldsymbol{\phi}} + \frac{\partial f}{\partial z}\hat{\mathbf{z}} \qquad \text{(cylindrical)},$$
$$\nabla f = \frac{\partial f}{\partial R}\hat{\mathbf{R}} + \frac{1}{R}\frac{\partial f}{\partial \theta}\hat{\boldsymbol{\theta}} + \frac{1}{R\sin\theta}\frac{\partial f}{\partial \phi}\hat{\boldsymbol{\phi}} \qquad \text{(spherical)}.$$

Divergence

Divergence

The **divergence** of $\mathbf{F}$ at a point P is the net outward flux per unit volume of a small closed surface S enclosing P:

$$\nabla\cdot\mathbf{F} = \lim_{\Delta V\to 0}\frac{1}{\Delta V}\oint_S \mathbf{F}\cdot d\mathbf{S}.$$

In Cartesian coordinates

$$\nabla\cdot\mathbf{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}.$$

Derivation of the Cartesian formula. Take a small box with one corner at (x,y,z) and sides $\Delta x, \Delta y, \Delta z$. The flux out through the face at $x+\Delta x$ is approximately $F_x(x+\Delta x, y, z)\,\Delta y\,\Delta z$; through the opposite face at x the outward normal is $-\hat{\mathbf{x}}$, giving $-F_x(x,y,z)\,\Delta y\,\Delta z$. Together

$$\big[F_x(x+\Delta x,y,z) - F_x(x,y,z)\big]\Delta y\,\Delta z \approx \frac{\partial F_x}{\partial x}\,\Delta x\,\Delta y\,\Delta z.$$

The other two pairs of faces contribute $\partial F_y/\partial y\,\Delta V$ and $\partial F_z/\partial z\,\Delta V$. Dividing by ΔV and letting the box shrink gives the formula. The idea: only the *change* of F_x across the box in the x direction produces net outflow; a uniform field carries in as much as it carries out.

Interpretation. $\nabla\cdot\mathbf{F} > 0$ at a **source** (field lines start there, net outflow), < 0 at a **sink**, and $= 0$ where the field is **solenoidal** (whatever flows in flows out). For an electric field, sources are positive charges; for the velocity of an incompressible fluid, the divergence is zero everywhere; for current density, $\nabla\cdot\mathbf{J} = -\partial\rho/\partial t$ (charge conservation, Section 7.11).

In curvilinear coordinates. Repeat the box argument with a curvilinear box whose faces have areas $h_2h_3\,du_2\,du_3$, and so on. The faces now have different areas on the two sides, so the area factors go *inside* the derivatives:

$$\nabla\cdot\mathbf{F} = \frac{1}{h_1h_2h_3}\left[\frac{\partial}{\partial u_1}(h_2h_3F_1) + \frac{\partial}{\partial u_2}(h_3h_1F_2) + \frac{\partial}{\partial u_3}(h_1h_2F_3)\right].$$

In particular

$$\nabla\cdot\mathbf{F} = \frac{1}{r}\frac{\partial}{\partial r}(rF_r) + \frac{1}{r}\frac{\partial F_\phi}{\partial\phi} + \frac{\partial F_z}{\partial z} \qquad \text{(cylindrical)},$$
$$\nabla\cdot\mathbf{F} = \frac{1}{R^2}\frac{\partial}{\partial R}(R^2F_R) + \frac{1}{R\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\,F_\theta) + \frac{1}{R\sin\theta}\frac{\partial F_\phi}{\partial\phi} \qquad \text{(spherical)}.$$

The factor R^2 in the spherical radial term is the area of a sphere growing as R^2: a field $\propto 1/R^2$ spreads over exactly that growing area and so has zero divergence away from the origin.

Curl

Curl

The component of the **curl** of $\mathbf{F}$ along a unit vector $\hat{\mathbf{n}}$ at P is the circulation per unit area around a small loop C through P in the plane perpendicular to $\hat{\mathbf{n}}$, traversed counter-clockwise when viewed from the tip of $\hat{\mathbf{n}}$ (right-hand rule):

$$(\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}} = \lim_{\Delta S \to 0} \frac{1}{\Delta S} \oint_C \mathbf{F} \cdot d\mathbf{r}.$$

In Cartesian coordinates

$$\nabla \times \mathbf{F} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial_x & \partial_y & \partial_z \\ F_x & F_y & F_z \end{vmatrix} = \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z}\right)\hat{\mathbf{x}} + \left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x}\right)\hat{\mathbf{y}} + \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}\right)\hat{\mathbf{z}}.$$

Derivation of the z-component. Take a small rectangle in the plane $z = \text{const}$ with corner (x, y) and sides $\Delta x, \Delta y$, traversed counter-clockwise seen from $+z$. The four sides contribute approximately

$$F_x(x, y)\Delta x + F_y(x + \Delta x, y)\Delta y - F_x(x, y + \Delta y)\Delta x - F_y(x, y)\Delta y.$$

Pairing the opposite sides,

$$\big[F_y(x + \Delta x, y) - F_y(x, y)\big]\Delta y - \big[F_x(x, y + \Delta y) - F_x(x, y)\big]\Delta x \approx \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}\right)\Delta x\,\Delta y.$$

Divide by the area $\Delta x\,\Delta y$ and take the limit. The x- and y-components follow by cyclic permutation $x \to y \to z \to x$.

Interpretation. Put a tiny paddle wheel in a flowing fluid with velocity $\mathbf{v}$. It spins if the flow on one side is faster than on the other; its axis aligns with $\nabla \times \mathbf{v}$ and its angular velocity is $\tfrac{1}{2}\|\nabla \times \mathbf{v}\|$. Curl is about **shear across the flow**, not about the field lines being curved: a field can circulate without curl (the wire field, below) or have curl with straight field lines (a shear flow $\mathbf{F} = y\hat{\mathbf{x}}$ has curl $-\hat{\mathbf{z}}$).

In curvilinear coordinates. The loop argument with sides $h_i\,du_i$ gives

$$\nabla \times \mathbf{F} = \frac{1}{h_1 h_2 h_3}\begin{vmatrix} h_1\hat{\mathbf{u}}_1 & h_2\hat{\mathbf{u}}_2 & h_3\hat{\mathbf{u}}_3 \\ \partial_{u_1} & \partial_{u_2} & \partial_{u_3} \\ h_1 F_1 & h_2 F_2 & h_3 F_3 \end{vmatrix},$$

which written out is

$$\nabla \times \mathbf{F} = \left(\frac{1}{r}\frac{\partial F_z}{\partial \phi} - \frac{\partial F_\phi}{\partial z}\right)\hat{\mathbf{r}} + \left(\frac{\partial F_r}{\partial z} - \frac{\partial F_z}{\partial r}\right)\hat{\boldsymbol{\phi}} + \frac{1}{r}\left(\frac{\partial (r F_\phi)}{\partial r} - \frac{\partial F_r}{\partial \phi}\right)\hat{\mathbf{z}} \quad \text{(cylindrical)},$$

$$\nabla \times \mathbf{F} = \frac{1}{R\sin\theta}\left(\frac{\partial (\sin\theta\, F_\phi)}{\partial \theta} - \frac{\partial F_\theta}{\partial \phi}\right)\hat{\mathbf{R}} + \frac{1}{R}\left(\frac{1}{\sin\theta}\frac{\partial F_R}{\partial \phi} - \frac{\partial (R F_\phi)}{\partial R}\right)\hat{\boldsymbol{\theta}} + \frac{1}{R}\left(\frac{\partial (R F_\theta)}{\partial R} - \frac{\partial F_R}{\partial \theta}\right)\hat{\boldsymbol{\phi}} \quad \text{(spherical)}.$$

The Laplacian

The divergence of a gradient is the **Laplacian**:

$$\nabla^2 f = \nabla \cdot \nabla f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}.$$

Combining the curvilinear gradient and divergence,

$$\nabla^2 f = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial f}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 f}{\partial \phi^2} + \frac{\partial^2 f}{\partial z^2} \qquad \text{(cylindrical)},$$

$$\nabla^2 f = \frac{1}{R^2}\frac{\partial}{\partial R}\left(R^2\frac{\partial f}{\partial R}\right) + \frac{1}{R^2\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta\frac{\partial f}{\partial \theta}\right) + \frac{1}{R^2\sin^2\theta}\frac{\partial^2 f}{\partial \phi^2} \quad \text{(spherical)}.$$

Interpretation. In one dimension, $f''(x) \approx [f(x + h) + f(x - h) - 2f(x)]/h^2$: the second derivative compares f with the average of its neighbours. In three dimensions, $\nabla^2 f$ at a point is proportional to (average of f on a small sphere) $-$ (f at the centre). A function with $\nabla^2 f = 0$ — a **harmonic** function — equals the average of its surroundings, so it can have no local maximum or minimum inside a region. That is why the electrostatic potential in a charge-free region has its extremes on the electrodes.

For a vector field, the **vector Laplacian** in Cartesian coordinates is the Laplacian of each component: $\nabla^2\mathbf{F} = (\nabla^2 F_x, \nabla^2 F_y, \nabla^2 F_z)$. In curvilinear coordinates it is *not* the Laplacian of each component (the basis vectors vary); use the identity $\nabla^2\mathbf{F} = \nabla(\nabla \cdot \mathbf{F}) - \nabla \times (\nabla \times \mathbf{F})$ instead.

Identities

Vector-calculus identities

For twice continuously differentiable f, g and $\mathbf{A}, \mathbf{B}$:

1. $\nabla \times (\nabla f) = \mathbf{0}$ — *the curl of a gradient is zero.*
2. $\nabla \cdot (\nabla \times \mathbf{A}) = 0$ — *the divergence of a curl is zero.*
3. $\nabla(fg) = f\nabla g + g\nabla f$.
4. $\nabla \cdot (f\mathbf{A}) = f\,\nabla\cdot\mathbf{A} + \mathbf{A}\cdot\nabla f$.
5. $\nabla \times (f\mathbf{A}) = f\,\nabla\times\mathbf{A} + \nabla f \times \mathbf{A}$.
6. $\nabla \cdot (\mathbf{A}\times\mathbf{B}) = \mathbf{B}\cdot(\nabla\times\mathbf{A}) - \mathbf{A}\cdot(\nabla\times\mathbf{B})$.
7. $\nabla \times (\nabla\times\mathbf{A}) = \nabla(\nabla\cdot\mathbf{A}) - \nabla^2\mathbf{A}$.

Proof of 1 and 2. For 1, the z-component is $\partial_x\partial_y f - \partial_y\partial_x f = 0$ by equality of mixed partials (Clairaut, Ch. 6); the other components are the same. For 2,

$$\nabla\cdot(\nabla\times\mathbf{A}) = \partial_x(\partial_y A_z - \partial_z A_y) + \partial_y(\partial_z A_x - \partial_x A_z) + \partial_z(\partial_x A_y - \partial_y A_x) = 0,$$

since the six terms cancel in pairs. Formally both are "$\mathbf{a}\times\mathbf{a} = \mathbf{0}$" and "$\mathbf{a}\cdot(\mathbf{a}\times\mathbf{b}) = 0$" with $\mathbf{a} = \nabla$. Identities 4–6 are product rules: each term has ∇ acting on one factor while the other is held fixed. Identity 6 is the one behind Poynting's theorem; identity 7 is BAC–CAB with ∇ and is the step to the wave equation.

These two "zero" identities have converses that are the foundation of potential theory:

Potentials

On a **simply connected** region (every closed loop can be shrunk to a point without leaving it):

- if $\nabla\times\mathbf{F} = \mathbf{0}$, there is a scalar potential f with $\mathbf{F} = \nabla f$ (the field is **irrotational** or **conservative**);
- on a region where every closed surface encloses only points of the region (e.g. all of space), if $\nabla\cdot\mathbf{F} = 0$, there is a vector potential $\mathbf{A}$ with $\mathbf{F} = \nabla\times\mathbf{A}$ (the field is **solenoidal**).

Helmholtz theorem. A field that is smooth and vanishes suitably fast at infinity is uniquely determined by its divergence and its curl, and can be split as $\mathbf{F} = -\nabla V + \nabla\times\mathbf{A}$ — an irrotational part plus a solenoidal part.

The Helmholtz theorem explains the shape of Maxwell's equations: to fix $\mathbf{E}$ and $\mathbf{B}$ you need their divergences (the charges and the absence of magnetic charge) and their curls (changing fields and currents) — exactly four equations.

Worked Example — divergence and curl in Cartesian coordinates

For $\mathbf{F} = x^2y\,\hat{\mathbf{x}} + 2yz\,\hat{\mathbf{y}} + xz^2\,\hat{\mathbf{z}}$ compute $\nabla\cdot\mathbf{F}$ and $\nabla\times\mathbf{F}$ at $(1,2,3)$, and confirm identity 2.

Divergence. $\partial_x(x^2y) + \partial_y(2yz) + \partial_z(xz^2) = 2xy + 2z + 2xz$. At $(1,2,3)$: $4 + 6 + 6 = 16$.

Curl. $(\partial_yF_z - \partial_zF_y,\ \partial_zF_x - \partial_xF_z,\ \partial_xF_y - \partial_yF_x) = (0 - 2y,\ 0 - z^2,\ 0 - x^2) = (-2y, -z^2, -x^2)$. At $(1,2,3)$: $(-4,-9,-1)$.

Identity check. $\nabla\cdot(-2y,-z^2,-x^2) = \partial_x(-2y) + \partial_y(-z^2) + \partial_z(-x^2) = 0$. ✓

So near $(1,2,3)$ the field is a strong source (net outflow 16 units per unit volume) and its local rotation axis points along $(-4,-9,-1)/\sqrt{98}$.

Worked Example — rotation field versus wire field

Compare $\mathbf{F}_1 = r\,\hat{\boldsymbol{\phi}}$ (rigid rotation) and $\mathbf{F}_2 = \dfrac{1}{r}\hat{\boldsymbol{\phi}}$ (the shape of the field of a line current).

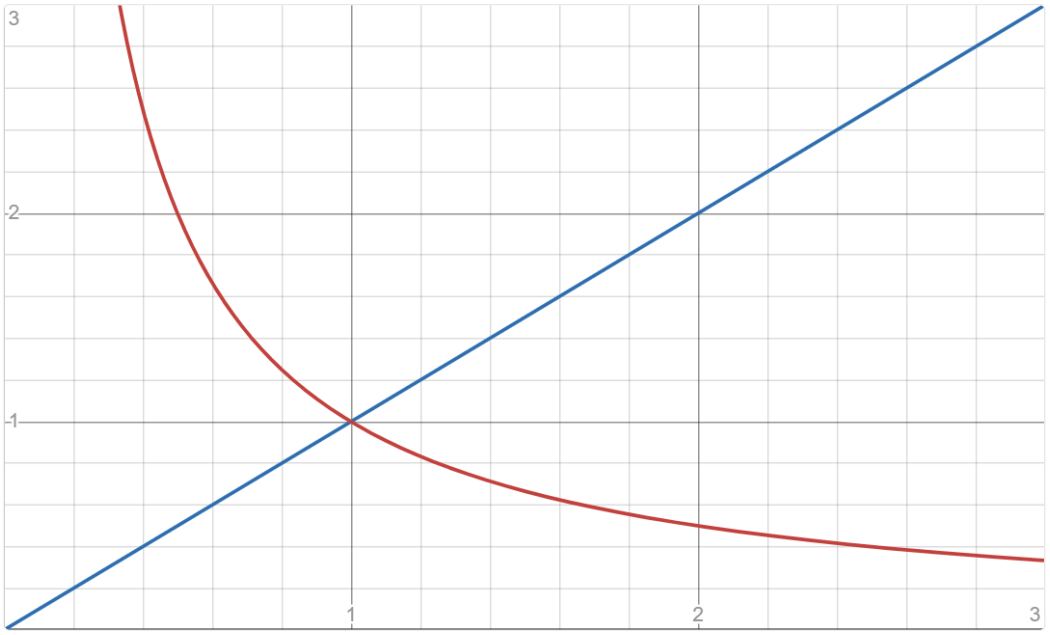
Both have only a ϕ-component that depends only on r, so $\nabla\cdot\mathbf{F} = \frac{1}{r}\partial_\phi F_\phi = 0$ for both: neither has sources.

For the curl, only the $\hat{\mathbf{z}}$ term survives: $(\nabla\times\mathbf{F})_z = \frac{1}{r}\frac{\partial}{\partial r}(rF_\phi)$.

- $\mathbf{F}_1$: $\frac{1}{r}\frac{d}{dr}(r^2) = 2$, so $\nabla\times\mathbf{F}_1 = 2\hat{\mathbf{z}}$ — uniform rotation everywhere (a paddle wheel spins at angular rate 1 = the rotation rate of the fluid, consistent with $\frac{1}{2}\|\nabla\times\mathbf{v}\|$).
- $\mathbf{F}_2$: $\frac{1}{r}\frac{d}{dr}(1) = 0$ for $r > 0$ — **irrotational**, even though its field lines are circles. The faster flow near the axis exactly compensates for the shorter path: a paddle wheel is carried around the axis without spinning.

Cartesian cross-check for $\mathbf{F}_2$. $\mathbf{F}_2 = (-y,x)/(x^2+y^2)$. $\partial_xF_y = \dfrac{(x^2+y^2) - 2x^2}{(x^2+y^2)^2} = \dfrac{y^2-x^2}{(x^2+y^2)^2}$ and $\partial_yF_x = \dfrac{-(x^2+y^2) + 2y^2}{(x^2+y^2)^2} = \dfrac{y^2-x^2}{(x^2+y^2)^2}$. Their difference is 0. ✓

Yet in Section 7.6 we will find that the circulation of $\mathbf{F}_2$ around the unit circle is 2π, not zero. The resolution is that $\mathbf{F}_2$ is undefined on the z-axis, so the region is not simply connected — all its curl is concentrated on the axis, where the current flows.



Field magnitude against distance r from the axis: rigid rotation $\|\mathbf{F}_1\| = r$ (blue) has uniform curl; the wire-type field $\|\mathbf{F}_2\| = 1/r$ (red) is curl-free for $r > 0$, although both have circular field lines.

Worked Example — the inverse-square field and $\nabla^2(1/R)$

Show that $\nabla \cdot (\hat{\mathbf{R}}/R^2) = 0$ and $\nabla^2(1/R) = 0$ for $R \neq 0$, first in spherical and then in Cartesian coordinates.

Spherical. $\nabla \cdot (\hat{\mathbf{R}}/R^2) = \dfrac{1}{R^2}\dfrac{\partial}{\partial R}\left(R^2 \cdot \dfrac{1}{R^2}\right) = \dfrac{1}{R^2}\dfrac{\partial}{\partial R}(1) = 0$. Since $\nabla(1/R) = -\hat{\mathbf{R}}/R^2$, also $\nabla^2(1/R) = 0$.

Cartesian. With $R^2 = x^2 + y^2 + z^2$, $\partial_x(1/R) = -x/R^3$ and

$$\frac{\partial^2}{\partial x^2}\frac{1}{R} = -\frac{1}{R^3} + \frac{3x^2}{R^5} = \frac{3x^2 - R^2}{R^5}.$$

Adding the y and z terms: $\nabla^2(1/R) = \dfrac{3(x^2+y^2+z^2) - 3R^2}{R^5} = 0$. ✓

The spherical calculation took one line, the Cartesian one five. The flux of $\hat{\mathbf{R}}/R^2$ through a sphere is nevertheless 4π (Section 7.7): all the divergence sits at $R = 0$. Section 7.11 makes this precise with the Dirac delta, $\nabla^2(1/R) = -4\pi\delta(\mathbf{R})$.

Worked Example — gradient and divergence in spherical coordinates

(a) Show that the dipole potential $V = \dfrac{p\cos\theta}{4\pi\varepsilon_0 R^2}$ gives the field quoted in Section 7.4. (b) Inside a sphere of uniform charge density ρ_0, $\mathbf{E} = \dfrac{\rho_0 R}{3\varepsilon_0}\hat{\mathbf{R}}$; find $\nabla \cdot \mathbf{E}$ and $\nabla \times \mathbf{E}$.

(a) $E_R = -\partial V/\partial R = \dfrac{2p\cos\theta}{4\pi\varepsilon_0 R^3}$, $E_\theta = -\dfrac{1}{R}\dfrac{\partial V}{\partial \theta} = \dfrac{p\sin\theta}{4\pi\varepsilon_0 R^3}$, $E_\phi = 0$. ✓

(b) $\nabla \cdot \mathbf{E} = \dfrac{1}{R^2}\dfrac{\partial}{\partial R}\left(R^2 \cdot \dfrac{\rho_0 R}{3\varepsilon_0}\right) = \dfrac{1}{R^2}\cdot\dfrac{3\rho_0 R^2}{3\varepsilon_0} = \dfrac{\rho_0}{\varepsilon_0}$ — exactly the charge density divided by ε_0 (Gauss's law in differential form, Section 7.12). The curl has components built from $\partial_\theta E_R$ and $\partial_\phi E_R$ and from E_θ, E_ϕ, all zero, so $\nabla \times \mathbf{E} = \mathbf{0}$: the field is conservative, as every electrostatic field must be.

Units. ρ_0/ε_0 is $(\mathrm{C/m^3})/(\mathrm{F/m}) = \mathrm{V/m^2}$, which is $[\mathbf{E}]/\mathrm{m}$ ✓.

Common Pitfall

Applying the *Cartesian* formulas to curvilinear components. For $\mathbf{F} = \hat{\mathbf{R}}$ (all $F_R = 1$) the naive "$\partial F_R/\partial R$" gives zero divergence, but the correct result is $\frac{1}{R^2}\partial_R(R^2) = 2/R$: unit arrows pointing outward everywhere *do* spread apart. Likewise $\nabla^2\mathbf{F}$ is not $(\nabla^2 F_r, \nabla^2 F_\phi, \nabla^2 F_z)$.

Common Pitfall

$\nabla \cdot \mathbf{F}$ is a scalar and $\nabla \times \mathbf{F}$ a vector; $\nabla \cdot f$ (divergence of a scalar) and $\nabla \times f$ are meaningless, and $\mathbf{F} \cdot \nabla$ is an *operator*, not the divergence: $(\mathbf{F}\cdot\nabla)g = F_x\partial_x g + F_y\partial_y g + F_z\partial_z g$ is the directional derivative of g along $\mathbf{F}$, scaled by $\|\mathbf{F}\|$.

Engineering Note

Finite-difference field solvers replace $\nabla^2 V$ by the "five-point stencil" $[V_{i+1,j} + V_{i-1,j} + V_{i,j+1} + V_{i,j-1} - 4V_{i,j}]/h^2$. Setting it to zero says each node equals the average of its four neighbours — the discrete version of the mean-value property — and iterating that averaging (Jacobi or Gauss–Seidel relaxation, Ch. 22) solves Laplace's equation for the potential between electrodes.

7.6 Line integrals

Motivation

The voltage between two points is the work per unit charge done *against* the electric field along a path: $V_{ab} = -\int_b^a \mathbf{E}\cdot d\mathbf{l}$. The EMF around a loop is $\oint \mathbf{E}\cdot d\mathbf{l}$. The total charge on a curved wire with non-uniform line density is $\int \rho_\ell \, ds$. All are integrals along a curve — **line integrals** — and they come in two kinds: integrals of a scalar with respect to arc length, and integrals of the tangential component of a vector field.

Scalar line integrals

Line integral of a scalar field

Let C be a smooth curve parametrised by $\mathbf{r}(t)$, $a \le t \le b$. The line integral of f along C with respect to arc length is

$$\int_C f\, ds = \lim_{\max \Delta s_k \to 0} \sum_k f(\mathbf{r}_k)\,\Delta s_k = \int_a^b f(\mathbf{r}(t))\,\|\mathbf{r}'(t)\|\,dt.$$

The value does not depend on the parametrisation *or on the direction* of travel (ds is a length, always positive). With $f = 1$ it gives the arc length; with $f = \rho_\ell$ the total charge; with f a linear resistance per metre, the total resistance.

Worked Example — charge on a curved wire

A quarter-circle wire of radius a in the first quadrant carries line charge density $\rho_\ell = \rho_0\, xy/a^2$. Find its total charge.

Step 1 — parametrise. $\mathbf{r}(\phi) = (a\cos\phi,\ a\sin\phi)$, $0 \le \phi \le \pi/2$; $\|\mathbf{r}'\| = a$, so $ds = a\,d\phi$.

Step 2 — integrand. $\rho_\ell = \rho_0 \cos\phi \sin\phi$.

Step 3 — integrate. $Q = \int_0^{\pi/2} \rho_0 \cos\phi\sin\phi\, a\, d\phi = \rho_0 a\left[\tfrac{1}{2}\sin^2\phi\right]_0^{\pi/2} = \tfrac{1}{2}\rho_0 a$.

Units. ρ_0 in C/m times a in m gives coulombs ✓.

Line integrals of vector fields

Line integral of a vector field (work, circulation)

For a vector field $\mathbf{F}$ and an **oriented** curve C parametrised by $\mathbf{r}(t)$, $a \le t \le b$ (direction of increasing t),

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \int_C \mathbf{F}\cdot\hat{\mathbf{T}}\, ds = \int_a^b \mathbf{F}(\mathbf{r}(t))\cdot\mathbf{r}'(t)\,dt = \int_C (F_x\,dx + F_y\,dy + F_z\,dz).$$

If C is closed the integral is called the **circulation** and written $\oint_C \mathbf{F}\cdot d\mathbf{r}$.

Only the tangential component $\mathbf{F}\cdot\hat{\mathbf{T}}$ contributes. Reversing the orientation of C reverses $\hat{\mathbf{T}}$ and so **changes the sign**: $\int_{-C}\mathbf{F}\cdot d\mathbf{r} = -\int_C \mathbf{F}\cdot d\mathbf{r}$. Any other parametrisation with the same orientation gives the same value: if $t = g(u)$ with $g' > 0$, then $\mathbf{r}'(g(u))g'(u)\,du = \mathbf{r}'(t)\,dt$. For a curve made of smooth pieces (a polygon, say) add the integrals over the pieces.

The procedure — almost every line-integral computation follows these four steps:

1. Parametrise each piece of C as $\mathbf{r}(t)$, with t increasing in the direction of travel.
2. Compute $d\mathbf{r} = \mathbf{r}'(t)\,dt$.
3. Substitute $\mathbf{r}(t)$ into $\mathbf{F}$ and form the scalar $\mathbf{F}(\mathbf{r}(t))\cdot\mathbf{r}'(t)$.
4. Integrate in t from the start value to the end value.

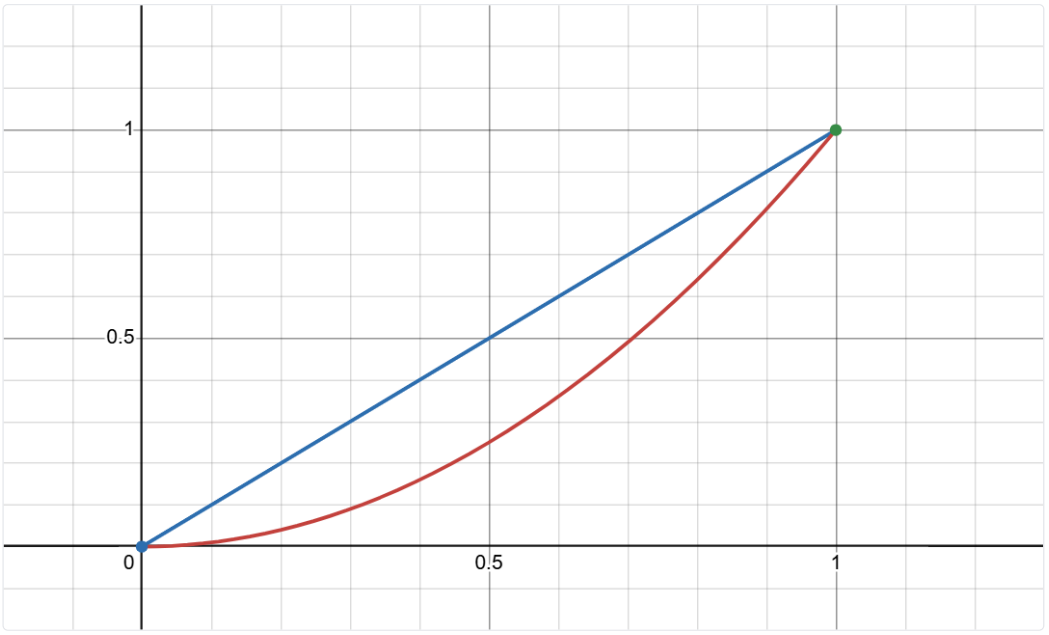
Worked Example — path dependence

Compute $\int_C \mathbf{F}\cdot d\mathbf{r}$ from $(0,0)$ to $(1,1)$ for (a) $\mathbf{F} = y\hat{\mathbf{x}} + x\hat{\mathbf{y}}$ and (b) $\mathbf{G} = -y\hat{\mathbf{x}} + x\hat{\mathbf{y}}$, along C_1: the line $y = x$, and C_2: the parabola $y = x^2$.

C_1**:** $\mathbf{r} = (t,t)$, $\mathbf{r}' = (1,1)$, $0 \le t \le 1$. (a) $\mathbf{F}\cdot\mathbf{r}' = t + t = 2t$, integral 1. (b) $\mathbf{G}\cdot\mathbf{r}' = -t + t = 0$, integral 0.

C_2**:** $\mathbf{r} = (t, t^2)$, $\mathbf{r}' = (1, 2t)$. (a) $\mathbf{F} = (t^2, t)$, $\mathbf{F}\cdot\mathbf{r}' = t^2 + 2t^2 = 3t^2$, integral 1. (b) $\mathbf{G} = (-t^2, t)$, $\mathbf{G}\cdot\mathbf{r}' = -t^2 + 2t^2 = t^2$, integral $\tfrac{1}{3}$.

$\mathbf{F}$ gives the same answer on both paths; $\mathbf{G}$ does not. The reason: $\mathbf{F} = \nabla(xy)$ is a gradient, while $\nabla\times\mathbf{G} = 2\hat{\mathbf{z}} \neq \mathbf{0}$. Going out along C_1 and back along C_2 gives a circulation $0 - \tfrac{1}{3} = -\tfrac{1}{3}$ for $\mathbf{G}$: by Green's theorem (Section 7.9) this is $-2\times$ (area between the curves $= \tfrac{1}{6}$), the minus sign because that loop runs clockwise.



Two paths from $(0,0)$ to $(1,1)$: the line C_1 (blue) and the parabola C_2 (red). A gradient field gives the same work along both; the rotation field $\mathbf{G}$ does not.

The fundamental theorem for line integrals and conservative fields

Fundamental theorem for line integrals

If f is continuously differentiable on a region containing a smooth curve C from $\mathbf{a}$ to $\mathbf{b}$, then

$$\int_C \nabla f \cdot d\mathbf{r} = f(\mathbf{b}) - f(\mathbf{a}).$$

Proof. By the chain rule, $\frac{d}{dt} f(\mathbf{r}(t)) = \nabla f \cdot \mathbf{r}'(t)$, so the integral is $\int_a^b \frac{d}{dt} f(\mathbf{r}(t))\, dt = f(\mathbf{r}(b)) - f(\mathbf{r}(a))$ by the ordinary fundamental theorem of calculus. ∎

Conservative fields — equivalent statements

For a continuous field $\mathbf{F}$ on an open connected region D, the following are equivalent:

1. $\mathbf{F} = \nabla f$ for some scalar f on D;
2. $\int_C \mathbf{F} \cdot d\mathbf{r}$ depends only on the endpoints of C (path independence);
3. $\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$ for every closed curve C in D.

Each implies $\nabla \times \mathbf{F} = \mathbf{0}$; conversely $\nabla \times \mathbf{F} = \mathbf{0}$ implies 1–3 **if D is simply connected**.

Sketch. 1⇒2 is the fundamental theorem. 2⇔3: two paths with the same ends form a closed loop, and the difference of their integrals is the circulation. 2⇒1: fix $\mathbf{a}$ and define $f(\mathbf{x}) = \int_{\mathbf{a}}^{\mathbf{x}} \mathbf{F} \cdot d\mathbf{r}$ (well defined by path independence); moving the endpoint a little in the x direction along a straight segment shows $\partial f/\partial x = F_x$, and likewise for y, z. The curl statement is Stokes' theorem (Section 7.10).

Finding a potential. Given an irrotational $\mathbf{F}$ on a simply connected region, integrate one component at a time:

1. $f = \int F_x\, dx + g(y,z)$, where g is an unknown "constant" of integration;
2. differentiate with respect to y, set equal to F_y, and solve for $\partial g/\partial y$;
3. repeat for z; the leftover constant is arbitrary (only potential *differences* are physical).

Worked Example — finding and using a potential

Show that $\mathbf{F} = (2xy + z)\hat{\mathbf{x}} + x^2\hat{\mathbf{y}} + (x + 3z^2)\hat{\mathbf{z}}$ is conservative, find f with $\mathbf{F} = \nabla f$, and evaluate the work from $(0,0,0)$ to $(1,2,1)$.

Step 1 — curl test. $\partial_y F_z - \partial_z F_y = 0 - 0$, $\partial_z F_x - \partial_x F_z = 1 - 1$, $\partial_x F_y - \partial_y F_x = 2x - 2x$. All zero, and the domain ($\mathbb{R}^3$) is simply connected, so $\mathbf{F}$ is conservative.

Step 2 — integrate F_x. $f = \int (2xy + z)\, dx = x^2 y + xz + g(y,z)$.

Step 3 — match F_y. $\partial_y f = x^2 + \partial_y g = x^2$, so $g = h(z)$.

Step 4 — match F_z. $\partial_z f = x + h'(z) = x + 3z^2$, so $h = z^3 + c$. Thus $f = x^2 y + xz + z^3$.

Step 5 — work. $f(1,2,1) - f(0,0,0) = 2 + 1 + 1 = 4$.

Check along the straight line $\mathbf{r} = (t, 2t, t)$: $\mathbf{F} = (4t^2 + t,\ t^2,\ t + 3t^2)$, $\mathbf{r}' = (1,2,1)$, $\mathbf{F} \cdot \mathbf{r}' = 9t^2 + 2t$, $\int_0^1 (9t^2 + 2t)\, dt = 3 + 1 = 4$. ✓

Worked Example — circulation and the non-simply-connected trap

Compute the circulation counter-clockwise around the circle $x^2+y^2=a^2$ of (a) the rotation field $\mathbf{G}=-y\hat{\mathbf{x}}+x\hat{\mathbf{y}}$ and (b) the magnetic field of a long wire on the z-axis, $\mathbf{B}=\dfrac{\mu_0 I}{2\pi r}\hat{\boldsymbol{\phi}}$.

Parametrise. $\mathbf{r}=(a\cos\phi, a\sin\phi, 0)$, $\mathbf{r}'=a(-\sin\phi,\cos\phi,0)=a\hat{\boldsymbol{\phi}}$, so $d\mathbf{r}=a\,d\phi\,\hat{\boldsymbol{\phi}}$.

(a) $\mathbf{G}=a\hat{\boldsymbol{\phi}}$ on the circle, so $\oint \mathbf{G}\cdot d\mathbf{r}=\int_0^{2\pi} a\cdot a\,d\phi=2\pi a^2$. This equals $(\nabla\times\mathbf{G})_z\times\text{area}=2\times\pi a^2$, as the curl definition predicts.

(b) $\oint \mathbf{B}\cdot d\mathbf{r}=\int_0^{2\pi}\dfrac{\mu_0 I}{2\pi a}\,a\,d\phi=\mu_0 I$ **— independent of the radius**. This is Ampère's circuital law.

So $\mathbf{B}$ has zero curl wherever it is defined ($r>0$, Section 7.5), yet its circulation around the wire is not zero. There is no contradiction: the region $r>0$ has a hole (the wire), loops around it cannot be shrunk to a point, and "curl-free ⇒ conservative" fails. A loop that does **not** encircle the wire has zero circulation. The "potential" $\mu_0 I\phi/(2\pi)$ exists only locally; it increases by $\mu_0 I$ on every trip around the wire, like a multi-storey car park ramp.

Worked Example — voltage as a line integral

A point charge q at the origin produces $\mathbf{E}=\dfrac{q}{4\pi\varepsilon_0 R^2}\hat{\mathbf{R}}$. Find the voltage $V_{ab}=V_a-V_b=-\int_b^a \mathbf{E}\cdot d\mathbf{l}$ between a at radius R_a and b at radius R_b, along any path.

Step 1. In spherical coordinates $d\mathbf{l}=dR\,\hat{\mathbf{R}}+R\,d\theta\,\hat{\boldsymbol{\theta}}+R\sin\theta\,d\phi\,\hat{\boldsymbol{\phi}}$, so $\mathbf{E}\cdot d\mathbf{l}=\dfrac{q}{4\pi\varepsilon_0}\dfrac{dR}{R^2}$ — only the radial motion counts.

Step 2. $V_{ab}=-\dfrac{q}{4\pi\varepsilon_0}\int_{R_b}^{R_a}\dfrac{dR}{R^2}=\dfrac{q}{4\pi\varepsilon_0}\left(\dfrac{1}{R_a}-\dfrac{1}{R_b}\right)$.

Step 3 — numbers. $q=1$ nC, $R_a=1$ cm, $R_b=10$ cm: $V_{ab}=8.988\times10^{9}\times10^{-9}\times(100-10)=809$ V.

Letting $R_b\to\infty$ gives the familiar $V=q/(4\pi\varepsilon_0 R)$. The path never mattered: $\mathbf{E}=-\nabla V$ is a gradient.

Common Pitfall

Writing $d\mathbf{r}=-dR\,\hat{\mathbf{R}}$ "because the path goes inward" *and* also integrating from the larger to the smaller radius. The parametrisation and the limits together encode the direction — put it in once. If you integrate from R_b to R_a, use $d\mathbf{l}=dR\,\hat{\mathbf{R}}$ and let the limits do the work.

Engineering Note

Kirchhoff's voltage law, $\sum v_k=0$ around a loop, is the statement $\oint\mathbf{E}\cdot d\mathbf{l}=0$. It holds when the loop links no changing magnetic flux. When it does (a loop around a transformer core, or a large ground loop near a switching converter), $\oint\mathbf{E}\cdot d\mathbf{l}=-d\Phi/dt\neq0$ and "voltage between two points" becomes path-dependent: two voltmeters with differently routed leads read different values. This is the Faraday-law effect behind much measurement interference (Section 7.12).

7.7 Surface integrals

Parametrised surfaces

A surface in space is described by two parameters, just as a curve is described by one:

$$\mathbf{r}(u,v)=x(u,v)\hat{\mathbf{x}}+y(u,v)\hat{\mathbf{y}}+z(u,v)\hat{\mathbf{z}},\qquad (u,v)\in D.$$

Holding v fixed and varying u traces a curve on the surface with tangent $\mathbf{r}_u=\partial\mathbf{r}/\partial u$; similarly $\mathbf{r}_v$. A small parameter rectangle $du\times dv$ maps to a small parallelogram with edges $\mathbf{r}_u\,du$ and $\mathbf{r}_v\,dv$, whose **vector area** is

$$d\mathbf{S}=(\mathbf{r}_u\times\mathbf{r}_v)\,du\,dv,\qquad dS=\|\mathbf{r}_u\times\mathbf{r}_v\|\,du\,dv.$$

The vector $\mathbf{N}=\mathbf{r}_u\times\mathbf{r}_v$ is normal to the surface; the surface is smooth where $\mathbf{N}\neq\mathbf{0}$. The standard surfaces of field theory:

Surface	Parameters	$\mathbf{r}_u\times\mathbf{r}_v$	dS
graph $z=g(x,y)$	x,y	$(-g_x,\,-g_y,\,1)$ (upward)	$\sqrt{1+g_x^2+g_y^2}\,dx\,dy$
plane $z=\text{const}$	x,y	$\hat{\mathbf{z}}$	$dx\,dy$
cylinder $r=a$	ϕ,z	$a\,\hat{\mathbf{r}}$ (outward)	$a\,d\phi\,dz$
sphere $R=a$	θ,ϕ	$a^2\sin\theta\,\hat{\mathbf{R}}$ (outward)	$a^2\sin\theta\,d\theta\,d\phi$
disk/annulus $z=\text{const}$	r,ϕ	$r\,\hat{\mathbf{z}}$	$r\,dr\,d\phi$

Derivation for the sphere. $\mathbf{r}=a(\sin\theta\cos\phi,\ \sin\theta\sin\phi,\ \cos\theta)$. Then $\mathbf{r}_\theta=a\hat{\boldsymbol{\theta}}$ and $\mathbf{r}_\phi=a\sin\theta\,\hat{\boldsymbol{\phi}}$ (lengths $h_\theta=a$, $h_\phi=a\sin\theta$), and $\hat{\boldsymbol{\theta}}\times\hat{\boldsymbol{\phi}}=\hat{\mathbf{R}}$, so $\mathbf{r}_\theta\times\mathbf{r}_\phi=a^2\sin\theta\,\hat{\mathbf{R}}$.

Scalar surface integrals and flux

Surface integrals

For a smooth surface S parametrised by $\mathbf{r}(u,v)$ over D:

- **scalar surface integral:** $\displaystyle\iint_S f\,dS = \iint_D f(\mathbf{r}(u,v))\,\|\mathbf{r}_u \times \mathbf{r}_v\|\,du\,dv$;
- **flux** of $\mathbf{F}$ through S with unit normal $\hat{\mathbf{n}}$: $\displaystyle\iint_S \mathbf{F}\cdot d\mathbf{S} = \iint_S \mathbf{F}\cdot\hat{\mathbf{n}}\,dS = \iint_D \mathbf{F}(\mathbf{r}(u,v))\cdot(\mathbf{r}_u\times\mathbf{r}_v)\,du\,dv$,

the last form being valid when $\mathbf{r}_u \times \mathbf{r}_v$ points along the chosen $\hat{\mathbf{n}}$ (otherwise change the sign). For a closed surface the flux is written $\oint_S \mathbf{F}\cdot d\mathbf{S}$ and $\hat{\mathbf{n}}$ is taken **outward**.

Flux counts how much of the field passes *through* the surface: only the normal component $\mathbf{F}\cdot\hat{\mathbf{n}}$ matters; a field tangent to the surface contributes nothing. If $\mathbf{F}$ is the velocity of a fluid, the flux is the volume flow rate (m^3/s); if it is the current density $\mathbf{J}$, the flux is the current $I = \iint \mathbf{J}\cdot d\mathbf{S}$; if it is $\mathbf{B}$, the flux Φ is in webers.

Orientation. A flux needs a choice of side. An **orientable** surface admits a continuous choice of unit normal; a closed surface is oriented outward by convention; an open surface bounded by a curve is oriented consistently with the curve's direction by the right-hand rule (Section 7.10). The Möbius strip is the standard example of a non-orientable surface — you cannot define the flux through it.

Worked Example — areas: spherical cap and paraboloid

(a) Find the area of the cap $0 \le \theta \le \alpha$ of a sphere of radius a.

$$A = \int_0^{2\pi}\int_0^{\alpha} a^2 \sin\theta\,d\theta\,d\phi = 2\pi a^2(1-\cos\alpha).$$

For $\alpha = \pi$ this is $4\pi a^2$ ✓; for $\alpha = \pi/2$, the hemisphere $2\pi a^2$ ✓. The cap's **solid angle** is $A/a^2 = 2\pi(1-\cos\alpha)$ steradians, the quantity used for antenna beam widths.

(b) Find the area of the paraboloid $z = x^2 + y^2$ below $z = 1$. Here $g_x = 2x$, $g_y = 2y$, so $dS = \sqrt{1+4x^2+4y^2}\,dx\,dy = \sqrt{1+4r^2}\,r\,dr\,d\phi$ over the unit disk:

$$A = 2\pi\int_0^1 \sqrt{1+4r^2}\,r\,dr = 2\pi\left[\frac{(1+4r^2)^{3/2}}{12}\right]_0^1 = \frac{\pi}{6}\left(5\sqrt{5}-1\right) = 5.330.$$

Sanity check: larger than the unit disk ($\pi = 3.14$) it sits above, as a curved surface must be.

Worked Example — outward flux through a sphere

Find the outward flux of $\mathbf{F} = x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}} = \mathbf{R}$ through the sphere $R = a$.

On the sphere $\mathbf{F} = a\hat{\mathbf{R}}$ and $d\mathbf{S} = a^2\sin\theta\,d\theta\,d\phi\,\hat{\mathbf{R}}$, so $\mathbf{F}\cdot d\mathbf{S} = a^3\sin\theta\,d\theta\,d\phi$ and

$$\oint_S \mathbf{F}\cdot d\mathbf{S} = a^3\int_0^{2\pi}\int_0^{\pi}\sin\theta\,d\theta\,d\phi = 4\pi a^3.$$

Since $\nabla\cdot\mathbf{F} = 3$ and the ball has volume $\frac{4}{3}\pi a^3$, the flux is $3\times$ volume — a preview of the divergence theorem.

By contrast, for the inverse-square field $\hat{\mathbf{R}}/R^2$ the flux is $\frac{1}{a^2}\cdot 4\pi a^2 = 4\pi$, **independent of** a. Every sphere around the origin catches all the field lines.

Worked Example — flux through a curved open surface

Find the upward flux of $\mathbf{F} = z\,\hat{\mathbf{z}}$ through the paraboloid cap S: $z = 1 - x^2 - y^2$, $z \ge 0$.

Step 1 — normal. With $g = 1 - x^2 - y^2$, $\mathbf{N} = (-g_x, -g_y, 1) = (2x, 2y, 1)$, which points upward ✓.

Step 2 — integrand. $\mathbf{F}\cdot\mathbf{N} = z\cdot 1 = 1 - x^2 - y^2 = 1 - r^2$.

Step 3 — integrate over the projection (the unit disk):

$$\iint_S \mathbf{F}\cdot d\mathbf{S} = \int_0^{2\pi}\int_0^1 (1-r^2)\,r\,dr\,d\phi = 2\pi\left(\tfrac{1}{2} - \tfrac{1}{4}\right) = \frac{\pi}{2}.$$

We will recover this from the divergence theorem in Section 7.11.

Worked Example — flux of a point charge through one face of a cube

A charge q sits at the centre of a cube of side $2d$. Find the flux of $\mathbf{E}$ through the top face, both by symmetry and directly.

By symmetry. The total outward flux through any closed surface around q is q/ε_0 (Gauss's law, Section 7.12); the six faces are equivalent, so each receives $q/(6\varepsilon_0)$.

Directly. On the top face $z = d$, $\hat{\mathbf{n}} = \hat{\mathbf{z}}$, and $\mathbf{E}\cdot\hat{\mathbf{z}} = \dfrac{q}{4\pi\varepsilon_0}\dfrac{d}{(x^2+y^2+d^2)^{3/2}}$. So

$$\Phi_E = \frac{q}{4\pi\varepsilon_0}\int_{-d}^{d}\int_{-d}^{d}\frac{d\,dx\,dy}{(x^2+y^2+d^2)^{3/2}} = \frac{q}{4\pi\varepsilon_0}\cdot\Omega,$$

where Ω is the solid angle the face subtends. Substituting $x = du$, $y = dv$ removes d, and the standard result for a rectangle $\int_{-a}^{a}\int_{-b}^{b}\frac{du\,dv}{(u^2+v^2+1)^{3/2}} = 4\arctan\dfrac{ab}{\sqrt{a^2+b^2+1}}$ with $a = b = 1$ gives

$$\Omega = 4\arctan\frac{1}{\sqrt{3}} = 4\cdot\frac{\pi}{6} = \frac{2\pi}{3}.$$

Hence $\Phi_E = \dfrac{q}{4\pi\varepsilon_0}\cdot\dfrac{2\pi}{3} = \dfrac{q}{6\varepsilon_0}$ ✓ (and $2\pi/3 = 2.0944$ was confirmed by numerical quadrature).

Worked Example — current through a conductor cross-section

A round wire of radius a carries current density $\mathbf{J} = J_0(1 - r^2/a^2)\hat{\mathbf{z}}$ (peaked on the axis). Find the total current, and the radius inside which half of it flows.

Total. $I = \iint \mathbf{J}\cdot\hat{\mathbf{z}}\,dS = \int_0^{2\pi}\int_0^a J_0\left(1-\frac{r^2}{a^2}\right) r\,dr\,d\phi = 2\pi J_0\left(\frac{a^2}{2}-\frac{a^2}{4}\right) = \frac{\pi a^2 J_0}{2}$ — half of what a uniform J_0 would give.

Half-current radius. $I(\rho) = 2\pi J_0\left(\frac{\rho^2}{2}-\frac{\rho^4}{4a^2}\right) = \frac{\pi a^2 J_0}{4}$ gives, with $s = \rho^2/a^2$, $s - \frac{s^2}{2} = \frac{1}{4}$, i.e. $s^2 - 2s + \frac{1}{2} = 0$, $s = 1 - 1/\sqrt{2} = 0.2929$, so $\rho = 0.541a$.

Common Pitfall

Forgetting the area factor. On a sphere $dS = a^2\sin\theta\,d\theta\,d\phi$, not $d\theta\,d\phi$; on a cylinder $dS = a\,d\phi\,dz$. The quickest check is to integrate 1 and see whether you get the known area. A second pitfall is projecting onto the xy-plane without the factor $\sqrt{1+g_x^2+g_y^2}$ (for scalar integrals) — for *flux* the vector $(-g_x, -g_y, 1)$ already contains the right factor.

Engineering Note

Radiated power is a flux: an antenna's total radiated power is $P = \oint\langle\mathbf{S}\rangle\cdot d\mathbf{S}$ over a large sphere, and its **directivity** compares the power per unit solid angle in the main beam with the average $P/(4\pi)$. The $\sin\theta$ in dS is why a beam near the pole of the coordinate system carries less power than one of the same angular width near the equator.

7.8 Volume integrals

Definition and the three volume elements

Volume integrals were set up in Ch. 6 as triple integrals. In field theory they answer questions such as: how much charge is in this region, how much energy is stored in this field, where is the centre of this charge distribution?

Volume integral

For a scalar field f over a region V,

$$\iiint_V f\,dV = \lim_{\max\Delta V_k\to 0}\sum_k f(\mathbf{R}_k)\,\Delta V_k,$$

evaluated as an iterated integral with

$$dV = dx\,dy\,dz = r\,dr\,d\phi\,dz = R^2\sin\theta\,dR\,d\theta\,d\phi.$$

A **vector** volume integral $\iiint_V \mathbf{F}\,dV$ is computed component by component **in Cartesian components**.

The volume elements are $h_1h_2h_3\,du_1\,du_2\,du_3$: the product of the three edge lengths of an infinitesimal coordinate box. The choice of coordinates is dictated by the shape of the region more than by the integrand: spheres and cones in spherical coordinates, cylinders and pipes in cylindrical, boxes in Cartesian.

Quantity	Volume integral
total charge	$Q = \iiint \rho\,dV$
mass, from density ρ_m	$M = \iiint \rho_m\,dV$
electrostatic energy	$W_e = \frac{1}{2}\iiint \varepsilon\|\mathbf{E}\|^2 dV$
magnetic energy	$W_m = \frac{1}{2}\iiint \mu\|\mathbf{H}\|^2 dV$
ohmic power	$P = \iiint \mathbf{J}\cdot\mathbf{E}\,dV = \iiint \sigma\|\mathbf{E}\|^2 dV$

Quantity	Volume integral
dipole moment	$\mathbf{p} = \iiint \mathbf{R}\,\rho\,dV$
potential of a distribution	$V(\mathbf{R}) = \dfrac{1}{4\pi\varepsilon_0}\iiint \dfrac{\rho(\mathbf{R}')}{\lVert\mathbf{R}-\mathbf{R}'\rVert}dV'$

Worked Example — charge in a non-uniform sphere

A sphere of radius a has charge density $\rho = \rho_0(1 - R/a)$. Find the total charge.

Step 1. The integrand depends only on R, so the angular integrals give $\int_0^{2\pi}\int_0^{\pi}\sin\theta\,d\theta\,d\phi = 4\pi$.

Step 2. $Q = 4\pi\rho_0\int_0^a\left(1-\dfrac{R}{a}\right)R^2\,dR = 4\pi\rho_0\left(\dfrac{a^3}{3}-\dfrac{a^3}{4}\right) = \dfrac{\pi\rho_0 a^3}{3}$.

Check. A uniform density ρ_0 would give $\frac{4}{3}\pi\rho_0 a^3$; the tapered sphere holds a quarter of that. Plausible: most of a ball's volume is near its surface, where this density is small.

Worked Example — charge in a cylinder (separable integrand)

A cylinder $0 \le r \le a$, $0 \le z \le L$ holds $\rho = \rho_0\,\dfrac{r}{a}\,e^{-z/L}$. Find Q.

The integrand is a product of functions of r, ϕ and z separately and the limits are constants, so the triple integral is a product of three single integrals:

$$Q = \rho_0\underbrace{\int_0^{2\pi}d\phi}_{2\pi}\underbrace{\int_0^a\frac{r}{a}\,r\,dr}_{a^2/3}\underbrace{\int_0^L e^{-z/L}dz}_{L(1-e^{-1})} = \frac{2\pi}{3}\left(1-e^{-1}\right)\rho_0 a^2 L = 1.324\,\rho_0 a^2 L.$$

Do not forget the r in dV: it turns $\int r\,dr$ into $\int r^2\,dr$ here.

Worked Example — energy stored in the field of a charged sphere

A sphere of radius a carries total charge Q uniformly through its volume. Its field (derived in Section 7.12) is $E = \dfrac{QR}{4\pi\varepsilon_0 a^3}$ inside and $E = \dfrac{Q}{4\pi\varepsilon_0 R^2}$ outside. Find the stored energy $W = \frac{\varepsilon_0}{2}\iiint E^2\,dV$ over all space.

Inside.

$$W_{\text{in}} = \frac{\varepsilon_0}{2}\left(\frac{Q}{4\pi\varepsilon_0 a^3}\right)^2\int_0^a R^2\cdot 4\pi R^2\,dR = \frac{\varepsilon_0}{2}\cdot\frac{Q^2}{16\pi^2\varepsilon_0^2 a^6}\cdot\frac{4\pi a^5}{5} = \frac{Q^2}{40\pi\varepsilon_0 a}.$$

Outside.

$$W_{\text{out}} = \frac{\varepsilon_0}{2}\cdot\frac{Q^2}{16\pi^2\varepsilon_0^2}\int_a^{\infty}\frac{4\pi R^2}{R^4}\,dR = \frac{Q^2}{8\pi\varepsilon_0 a}.$$

Total. $W = \dfrac{Q^2}{8\pi\varepsilon_0 a}\left(\dfrac{1}{5}+1\right) = \dfrac{3Q^2}{20\pi\varepsilon_0 a}$.

Numbers. $Q = 1$ nC, $a = 1$ cm: $W = 3\times10^{-18}/(20\pi\times 8.854\times10^{-12}\times 0.01) = 5.39\times10^{-7}$ J. Five-sixths of the energy is stored *outside* the charge — the field, not the charge, holds the energy.

Worked Example — a vector volume integral: dipole moment

A sphere of radius a has $\rho = \rho_0\cos\theta$ (positive in the upper half, negative in the lower). Find the total charge and the dipole moment $\mathbf{p} = \iiint \mathbf{R}\rho\,dV$.

Charge. $Q = \rho_0\int_0^a R^2 dR\int_0^{\pi}\cos\theta\sin\theta\,d\theta\int_0^{2\pi}d\phi = 0$, since $\int_0^{\pi}\cos\theta\sin\theta\,d\theta = 0$.

Dipole moment. $\mathbf{R} = R\hat{\mathbf{R}}$, but $\hat{\mathbf{R}}$ varies, so use Cartesian components: $\mathbf{R} = (R\sin\theta\cos\phi,\ R\sin\theta\sin\phi,\ R\cos\theta)$. The x- and y-components contain $\int_0^{2\pi}\cos\phi\,d\phi = 0$ and $\int_0^{2\pi}\sin\phi\,d\phi = 0$. The z-component is

$$p_z = \rho_0\int_0^a R^3\,dR\int_0^{\pi}\cos^2\theta\sin\theta\,d\theta\int_0^{2\pi}d\phi = \rho_0\cdot\frac{a^4}{4}\cdot\frac{2}{3}\cdot 2\pi = \frac{\pi\rho_0 a^4}{3}.$$

So $\mathbf{p} = \frac{1}{3}\pi\rho_0 a^4\,\hat{\mathbf{z}}$, pointing from the negative half to the positive half, as expected. Far away, this neutral sphere looks like the dipole of Section 7.4.

Common Pitfall

Integrating the spherical *components* of a vector field over a volume, e.g. writing $\iiint\hat{\mathbf{R}}\,dV = \hat{\mathbf{R}}\iiint dV$. $\hat{\mathbf{R}}$ is not a constant and cannot leave the integral; over a full sphere $\iiint\hat{\mathbf{R}}\,dV = \mathbf{0}$ by symmetry. Convert to Cartesian components first.

Engineering Note

Capacitance and inductance can be computed from energy: $C = 2W_e/V^2$ and $L = 2W_m/I^2$. Field solvers often extract L and C of interconnects this way, by integrating $\frac{1}{2}\varepsilon E^2$ and $\frac{1}{2}\mu H^2$ over the mesh — a large, numerical volume integral (Ch. 24).

7.9 Green's theorem

Statement

Green's theorem is the two-dimensional prototype of all the integral theorems. It relates a line integral around a closed plane curve to a double integral over the region inside.

Green's theorem

Let D be a bounded region of the plane whose boundary C consists of finitely many piecewise-smooth simple closed curves, oriented **positively** (as you walk along C, D is on your left — counter-clockwise for an outer boundary, clockwise for the boundary of a hole). If $P(x,y)$ and $Q(x,y)$ have continuous first partial derivatives on an open set containing D, then

$$\oint_C (P\,dx + Q\,dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA.$$

With $\mathbf{F} = P\hat{\mathbf{x}} + Q\hat{\mathbf{y}}$, the left side is the circulation $\oint \mathbf{F}\cdot d\mathbf{r}$ and the integrand on the right is $(\nabla \times \mathbf{F})\cdot\hat{\mathbf{z}}$. In words: **the circulation around the boundary equals the total curl inside.**

Proof sketch

Step 1 — the P term on a "vertically simple" region. Suppose $D = \{a \le x \le b,\ g_1(x) \le y \le g_2(x)\}$. Then

$$-\iint_D \frac{\partial P}{\partial y}\,dA = -\int_a^b \Big[P(x,g_2(x)) - P(x,g_1(x))\Big]\,dx.$$

Now compute $\oint_C P\,dx$ directly. On the bottom curve $y = g_1(x)$, traversed left to right, it gives $\int_a^b P(x,g_1(x))\,dx$. On the top curve, traversed right to left, it gives $-\int_a^b P(x,g_2(x))\,dx$. On any vertical sides $dx = 0$. The two expressions agree.

Step 2 — the Q term on a "horizontally simple" region is identical with the roles of x and y exchanged (and a sign flip that produces $+\partial Q/\partial x$).

Step 3 — general regions. Cut D into pieces that are both vertically and horizontally simple. Apply the theorem to each piece and add. The integrals along each internal cut appear twice, in opposite directions, and cancel, leaving only the outer boundary. ∎

The cancellation in Step 3 is the idea behind every integral theorem: **interior contributions cancel in pairs and only the boundary survives.** It is also how the coordinate-free definition of curl (circulation per unit area of a tiny loop) adds up to the theorem: tile D with tiny squares, each contributing $(\nabla \times \mathbf{F})_z\,\Delta A$, and the internal edges cancel.

The normal (flux) form

Apply Green's theorem to $P = -F_y$, $Q = F_x$. The outward unit normal to a counter-clockwise curve is $\hat{\mathbf{n}} = (dy, -dx)/ds$, so $\mathbf{F}\cdot\hat{\mathbf{n}}\,ds = F_x\,dy - F_y\,dx$. Hence

$$\oint_C \mathbf{F}\cdot\hat{\mathbf{n}}\,ds = \iint_D \left(\frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y}\right) dA = \iint_D \nabla\cdot\mathbf{F}\,dA\,:$$

the outward flux through the boundary equals the total divergence inside — the two-dimensional divergence theorem.

Area by a line integral

Choosing P, Q with $\partial_x Q - \partial_y P = 1$ turns the right side into the area of D. Three convenient choices:

$$A = \oint_C x\,dy = -\oint_C y\,dx = \frac{1}{2}\oint_C (x\,dy - y\,dx).$$

For a polygon with vertices $(x_1,y_1),\dots,(x_n,y_n)$ listed counter-clockwise, applying the last formula edge by edge gives the **shoelace formula**

$$A = \frac{1}{2}\sum_{k=1}^{n} (x_k y_{k+1} - x_{k+1} y_k)\,, \qquad (x_{n+1},y_{n+1}) = (x_1,y_1).$$

This is how CAD and PCB-layout tools compute the area of an arbitrary copper pour, and how a mechanical planimeter measures area by tracing the boundary.

Worked Example — verifying Green's theorem on a square

Let $P = x - y$, $Q = x^2$ on the unit square $[0,1]\times[0,1]$.

Double integral. $\partial_x Q - \partial_y P = 2x - (-1) = 2x + 1$, and $\int_0^1\int_0^1 (2x+1)\,dx\,dy = 1 + 1 = 2$.

Line integral, counter-clockwise from the origin:

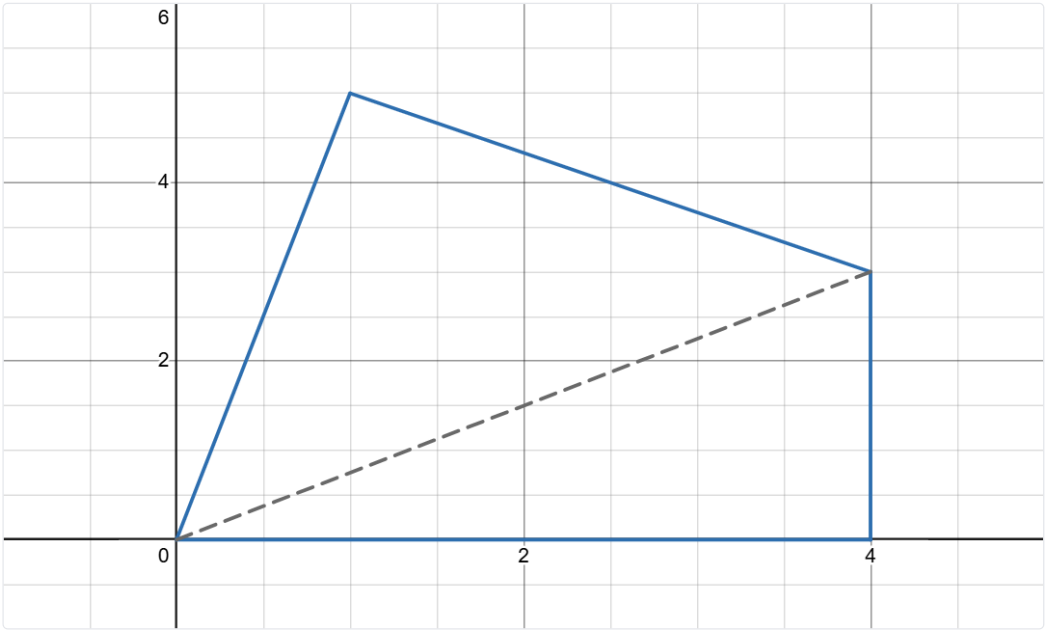
- bottom ($y = 0$, $x: 0 \to 1$, $dy = 0$): $\int_0^1 x\,dx = \frac{1}{2}$;
- right ($x = 1$, $y: 0 \to 1$, $dx = 0$): $\int_0^1 Q\,dy = \int_0^1 1\,dy = 1$;
- top ($y = 1$, $x: 1 \to 0$, $dy = 0$): $\int_1^0 (x-1)\,dx = \frac{1}{2}$;
- left ($x = 0$, $y: 1 \to 0$, $dx = 0$): $Q = 0$, contribution 0.

Total $\frac{1}{2} + 1 + \frac{1}{2} + 0 = 2$ ✓.

Worked Example — area of an ellipse and of a polygon

Ellipse. $x = a\cos t$, $y = b\sin t$, $0 \le t \le 2\pi$ (counter-clockwise). Then $x\,dy - y\,dx = (ab\cos^2 t + ab\sin^2 t)\,dt = ab\,dt$, so $A = \frac{1}{2}\int_0^{2\pi} ab\,dt = \pi ab$.

Polygon. A copper pad has vertices $(0,0)$, $(4,0)$, $(4,3)$, $(1,5)$ mm, in counter-clockwise order. Shoelace terms $x_k y_{k+1} - x_{k+1} y_k$: $(0)(0) - (4)(0) = 0$; $(4)(3) - (4)(0) = 12$; $(4)(5) - (1)(3) = 17$; $(1)(0) - (0)(5) = 0$. $A = \frac{1}{2}(0 + 12 + 17 + 0) = 14.5$ mm^2. **Check by splitting** along the diagonal from $(0,0)$ to $(4,3)$: triangle $(0,0)$, $(4,0)$, $(4,3)$ has area $\frac{1}{2}\cdot 4 \cdot 3 = 6$, and triangle $(0,0)$, $(4,3)$, $(1,5)$ has area $\frac{1}{2}|4\cdot 5 - 3\cdot 1| = 8.5$ (half the cross product of its edge vectors, Section 7.2); total 14.5 ✓. Listing the vertices clockwise would give -14.5: the sign of the shoelace sum reports the orientation.



The pad of the shoelace example. The dashed diagonal splits it into triangles of area 6 and 8.5, confirming $A = 14.5$.

Worked Example — a region with a hole

Let $\mathbf{F} = \dfrac{-y\,\hat{\mathbf{x}} + x\,\hat{\mathbf{y}}}{x^2 + y^2}$ (the wire field of Section 7.5, without constants). Evaluate $\oint \mathbf{F}\cdot d\mathbf{r}$ counter-clockwise around the ellipse C: $x^2/4 + y^2 = 1$.

Why the direct approach is unpleasant. On the ellipse $x^2 + y^2$ is not constant, and the resulting integral $\int_0^{2\pi} \frac{2\,dt}{4\cos^2 t + \sin^2 t}$ needs a substitution.

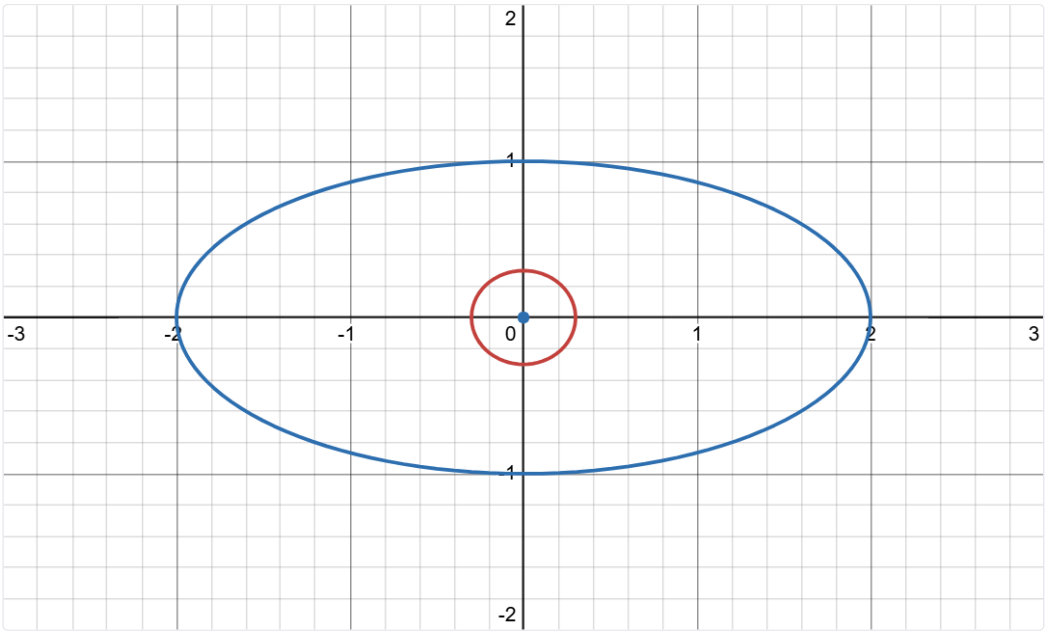
Green's theorem with a hole. $\nabla \times \mathbf{F} = \mathbf{0}$ everywhere except the origin, where $\mathbf{F}$ is undefined, so we cannot apply Green's theorem to the inside of the ellipse. Instead cut out a small circle C_ϵ of radius ϵ and apply it to the region D between C_ϵ and C. Its positively oriented boundary is C counter-clockwise plus C_ϵ clockwise:

$$\oint_C \mathbf{F}\cdot d\mathbf{r} - \oint_{C_\epsilon,\,\circlearrowleft} \mathbf{F}\cdot d\mathbf{r} = \iint_D 0\,dA = 0.$$

So the integral around the ellipse equals the integral around the small circle, which is $\int_0^{2\pi} \frac{\epsilon^2}{\epsilon^2}\,d\phi = 2\pi$.

Numerical check. $\int_0^{2\pi} \frac{2\,dt}{4\cos^2 t + \sin^2 t} = 2\cdot\frac{2\pi}{\sqrt{4\cdot 1}} = 2\pi$ using $\int_0^{2\pi} \frac{dt}{A\cos^2 t + B\sin^2 t} = \frac{2\pi}{\sqrt{AB}}$ ✓.

Any closed curve winding once around the origin gives 2π; any curve not enclosing it gives 0. The integral counts how many times the curve **winds** around the singularity — the same idea as the residue theorem of complex analysis (Ch. 18).



The ellipse (blue) and a small circle (red) around the singular point: the region between them is curl-free, so the two circulations are equal (2π).

Common Pitfall

Applying Green's theorem when P or Q is singular inside C. The hypotheses demand continuous partial derivatives on the **whole** region. Ignoring the singularity of the wire field would "prove" that $\oint \mathbf{F}\cdot d\mathbf{r} = 0$, contradicting the direct calculation.

Engineering Note

In two-dimensional field problems (long transmission lines, the cross-section of a motor), Green's theorem is the tool that turns a circulation around a conductor's cross-section into the current it carries, and a flux through a closed contour into the charge per unit length inside. Boundary-element methods for 2-D electrostatics are built on its identities.

7.10 Stokes' theorem

Statement

Green's theorem lives in a plane. Stokes' theorem lifts it to any curved surface in space: the circulation around the rim of a surface equals the flux of the curl through the surface.

Stokes' theorem

Let S be a piecewise-smooth oriented surface bounded by a piecewise-smooth simple closed curve C, with the orientation of C **induced** by that of S: if your right thumb points along the normal $\hat{\mathbf{n}}$, your fingers curl in the direction of C (equivalently, walking along C with your head along $\hat{\mathbf{n}}$, the surface is on your left). If $\mathbf{F}$ has continuous first partial derivatives on an open region containing S, then

$$\oint_C \mathbf{F}\cdot d\mathbf{r} = \iint_S (\nabla\times\mathbf{F})\cdot d\mathbf{S}.$$

For a flat surface in the xy-plane with $\hat{\mathbf{n}} = \hat{\mathbf{z}}$ this is exactly Green's theorem.

Proof sketches

Sketch 1 — tiling. Cover S with a fine mesh of small, nearly flat patches ΔS_k with normals $\hat{\mathbf{n}}_k$. By the definition of curl (Section 7.5), the circulation around the edge of patch k is approximately $(\nabla\times\mathbf{F})\cdot\hat{\mathbf{n}}_k\,\Delta S_k$. Add over all patches. Each internal edge is shared by two neighbouring patches and is traversed in opposite directions, so its contributions cancel; only the edges on the rim C survive. In the limit the sum of the right-hand sides becomes $\iint_S(\nabla\times\mathbf{F})\cdot d\mathbf{S}$. ■

Sketch 2 — reduction to Green. For a graph $z = g(x,y)$ over a region D, the parametrisation $(x,y,g(x,y))$ turns $\oint_C \mathbf{F}\cdot d\mathbf{r}$ into a plane line integral around the boundary of D, with $dz = g_x\,dx + g_y\,dy$:

$$\oint_{\partial D}\big[(F_x + F_z g_x)\,dx + (F_y + F_z g_y)\,dy\big].$$

Green's theorem converts this into a double integral over D. Expanding with the chain rule (where F_x means $F_x(x,y,g(x,y))$, so ∂_y acting on it produces $\partial_y F_x + \partial_z F_x\, g_y$), the terms with g_{xy} cancel by Clairaut and what remains is precisely $(\nabla\times\mathbf{F})\cdot(-g_x,-g_y,1)$ integrated over D — the flux of the curl through the graph. ■

Consequences

Corollaries of Stokes' theorem

1. **Surface independence.** $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$ depends only on the boundary curve C, not on which surface spanning C you choose. Pick the easiest one (usually a flat disk).
2. **Closed surfaces.** For a closed surface (no boundary), $\oint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = 0$ — consistent with $\nabla \cdot (\nabla \times \mathbf{F}) = 0$ and the divergence theorem.
3. **Curl-free implies conservative** on a simply connected region: every loop bounds a surface inside the region, through which the flux of $\nabla \times \mathbf{F} = \mathbf{0}$ is zero, so every circulation vanishes (Section 7.6).
4. **The curl definition is recovered.** Applying the theorem to a tiny flat disk of area ΔS and normal $\hat{\mathbf{n}}$ and dividing by ΔS gives back $(\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}} = \lim \frac{1}{\Delta S} \oint \mathbf{F} \cdot d\mathbf{r}$.

Corollary 1 has a vivid electromagnetic meaning: the current "through a loop" is well defined only because $\nabla \cdot \mathbf{J} = 0$ for steady currents — whichever surface you stretch across the loop, the same current crosses it. When charge accumulates (a charging capacitor), two surfaces spanning the same loop can carry different conduction currents, which is the puzzle Maxwell solved with the displacement current (Section 7.12).

Worked Example — verifying Stokes' theorem on a hemisphere

Let $\mathbf{F} = z\,\hat{\mathbf{x}} + x\,\hat{\mathbf{y}} + y\,\hat{\mathbf{z}}$ and S the upper unit hemisphere $x^2 + y^2 + z^2 = 1$, $z \geq 0$, with outward (upward) normal. The boundary is the unit circle in the xy-plane, counter-clockwise seen from above.

Line integral. $\mathbf{r} = (\cos t, \sin t, 0)$, $\mathbf{r}' = (-\sin t, \cos t, 0)$, and on C, $\mathbf{F} = (0, \cos t, \sin t)$. So $\mathbf{F} \cdot \mathbf{r}' = \cos^2 t$ and $\oint = \int_0^{2\pi} \cos^2 t\, dt = \pi$.

Curl. $\nabla \times \mathbf{F} = (\partial_y y - \partial_z x,\ \partial_z z - \partial_x y,\ \partial_x x - \partial_y z) = (1, 1, 1)$.

Flux of the curl through the hemisphere. $d\mathbf{S} = \sin\theta\, d\theta\, d\phi\, \hat{\mathbf{R}}$ and $(1,1,1) \cdot \hat{\mathbf{R}} = \sin\theta\cos\phi + \sin\theta\sin\phi + \cos\theta$. The $\cos\phi$ and $\sin\phi$ terms integrate to zero over $0 \leq \phi \leq 2\pi$, leaving

$$\int_0^{2\pi} \int_0^{\pi/2} \cos\theta \sin\theta\, d\theta\, d\phi = 2\pi \cdot \tfrac{1}{2} = \pi \ \checkmark$$

Using surface independence instead. The flat unit disk also spans C, with normal $\hat{\mathbf{z}}$: $\iint (1,1,1) \cdot \hat{\mathbf{z}}\, dA = \pi$. Same answer in one line.

Worked Example — a line integral made easy

Evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = z^2\,\hat{\mathbf{x}} + y^2\,\hat{\mathbf{y}} + x\,\hat{\mathbf{z}}$, where C is the triangle $(1,0,0) \to (0,1,0) \to (0,0,1) \to (1,0,0)$.

Step 1 — curl. $\nabla \times \mathbf{F} = (\partial_y x - \partial_z y^2,\ \partial_z z^2 - \partial_x x,\ \partial_x y^2 - \partial_y z^2) = (0,\ 2z - 1,\ 0)$.

Step 2 — surface and orientation. Take the flat triangle in the plane $x + y + z = 1$, i.e. $z = 1 - x - y$ over the projected triangle T: $x, y \geq 0$, $x + y \leq 1$. The edge vectors $(0,1,0) - (1,0,0) = (-1,1,0)$ and $(0,0,1) - (1,0,0) = (-1,0,1)$ have cross product $(1,1,1)$, so the given order of vertices corresponds to the upward normal. For the graph, $\mathbf{N} = (-g_x, -g_y, 1) = (1,1,1)$ ✓.

Step 3 — integrate. $(\nabla \times \mathbf{F}) \cdot \mathbf{N} = 2z - 1 = 2(1 - x - y) - 1 = 1 - 2x - 2y$. Over T (area $\frac{1}{2}$, with $\iint_T x\, dA = \iint_T y\, dA = \frac{1}{6}$):

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \tfrac{1}{2} - 2 \cdot \tfrac{1}{6} - 2 \cdot \tfrac{1}{6} = -\tfrac{1}{6}.$$

Check by computing the three edge integrals directly (each a polynomial in t): they sum to $-\frac{1}{6}$ ✓. One surface integral replaced three line integrals.

Worked Example — the vector potential of a solenoid

Inside a long solenoid of radius a the magnetic flux density is uniform, $\mathbf{B} = B_0\hat{\mathbf{z}}$, and outside it is (nearly) zero. A vector potential with $\nabla \times \mathbf{A} = \mathbf{B}$ is sought in the form $\mathbf{A} = A_\phi(r)\,\hat{\boldsymbol{\phi}}$. Find it inside and outside.

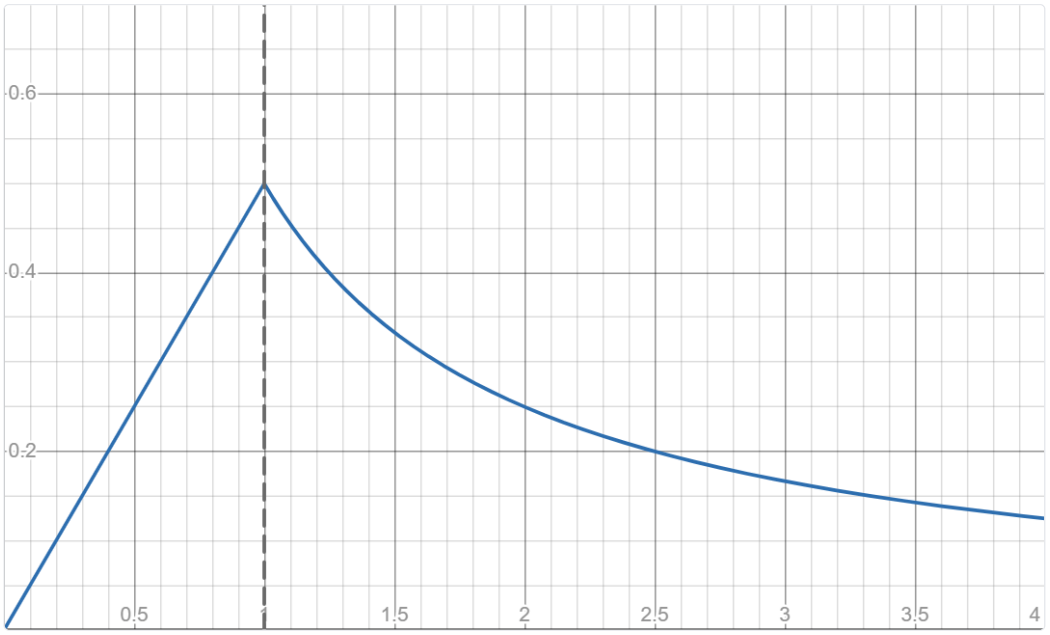
Step 1 — Stokes on a circle of radius r. $\oint \mathbf{A} \cdot d\mathbf{r} = 2\pi r A_\phi(r)$, and this equals $\iint (\nabla \times \mathbf{A}) \cdot d\mathbf{S} = \iint \mathbf{B} \cdot d\mathbf{S} = \Phi(r)$, the flux enclosed.

Step 2 — inside ($r < a$). $\Phi = B_0 \pi r^2$, so $A_\phi = B_0 r/2$.

Step 3 — outside ($r > a$). $\Phi = B_0 \pi a^2$ (all the flux), so $A_\phi = \dfrac{B_0 a^2}{2r}$.

Check with the cylindrical curl. Inside: $\frac{1}{r}\partial_r(r \cdot B_0 r/2) = B_0$ ✓. Outside: $\frac{1}{r}\partial_r(B_0 a^2/2) = 0$ ✓.

The vector potential is non-zero outside even though $\mathbf{B}$ is zero there — it "remembers" the enclosed flux through its circulation. Faraday's law (Section 7.12) uses this: a changing enclosed flux drives an EMF in a loop that lies entirely in the field-free region.



Vector potential $A_\phi(r)$ of an ideal solenoid ($B_0 = 1$, $a = 1$): it grows linearly inside and decays as $1/r$ outside, where $\mathbf{B} = \mathbf{0}$ but the circulation of $\mathbf{A}$ still equals the enclosed flux.

Common Pitfall

Mismatched orientation. If the surface normal points down but you traverse C counter-clockwise as seen from above, Stokes' theorem gives the wrong sign. Fix the curve direction first, then use the right-hand rule to choose $\hat{\mathbf{n}}$ — or vice versa — never independently.

Engineering Note

Stokes' theorem is why a current probe (Rogowski coil) works: a coil wound around a closed path measures $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}}$ regardless of where the conductor sits inside the loop. Likewise the voltage induced in a PCB loop depends only on the flux through the loop area, so minimising loop area (return path directly under the signal trace) minimises inductive pickup.

7.11 Divergence theorem

Statement

Divergence theorem (Gauss's theorem)

Let V be a bounded solid region whose boundary S is a closed piecewise-smooth surface with **outward** unit normal. If $\mathbf{F}$ has continuous first partial derivatives on an open region containing V, then

$$\oiint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_V \nabla \cdot \mathbf{F} \, dV.$$

In words: **the net outward flux through a closed surface equals the total source strength inside.** It is the three-dimensional analogue of the flux form of Green's theorem, and the mathematical content of Gauss's law.

Proof sketches

Sketch 1 — tiling. Divide V into tiny boxes. By the definition of divergence, the outward flux of each box is $\nabla \cdot \mathbf{F}\, \Delta V_k$. Summing over all boxes, the flux through every internal face is counted twice with opposite normals and cancels; only the faces on the outer boundary S survive. ∎

Sketch 2 — one component at a time. Write $\mathbf{F} = F_x\hat{\mathbf{x}} + F_y\hat{\mathbf{y}} + F_z\hat{\mathbf{z}}$; it suffices to prove $\oint F_z\hat{\mathbf{z}} \cdot d\mathbf{S} = \iiint \partial_z F_z \, dV$ and its two companions. Let $V = \{(x,y) \in D,\ g_1(x,y) \le z \le g_2(x,y)\}$. Then

$$\iiint_V \frac{\partial F_z}{\partial z}\, dV = \iint_D \Big[F_z(x,y,g_2) - F_z(x,y,g_1)\Big]\, dx\, dy.$$

On the top surface the upward normal vector is $(-g_{2x}, -g_{2y}, 1)\, dx\, dy$, so $F_z\hat{\mathbf{z}} \cdot d\mathbf{S} = F_z(x,y,g_2)\, dx\, dy$; on the bottom the outward normal points down, giving $-F_z(x,y,g_1)\, dx\, dy$; on any vertical side walls $\hat{\mathbf{z}} \cdot \hat{\mathbf{n}} = 0$. The flux equals the volume integral. ∎

Consequences

Solenoidal fields. If $\nabla \cdot \mathbf{F} = 0$ in V, the flux into V equals the flux out: what enters through one part of the surface leaves through another. For $\mathbf{B}$ this is "no magnetic monopoles"; it is why magnetic flux through a transformer core is the same at every cross-section.

Conservation laws. Let ρ be the density of something (charge, say) and $\mathbf{J}$ its flux density. Conservation says the rate of decrease of the amount inside V equals the outflow:

$$-\frac{d}{dt}\iiint_V \rho\, dV = \oint_S \mathbf{J}\cdot d\mathbf{S} = \iiint_V \nabla\cdot\mathbf{J}\, dV.$$

Since V is arbitrary, the integrands must agree, giving the **continuity equation**

$$\nabla\cdot\mathbf{J} + \frac{\partial\rho}{\partial t} = 0.$$

The step "integral is zero for every region ⇒ integrand is zero" (valid for continuous integrands: if the integrand were positive at a point it would be positive on a small ball around it) is the device that converts every integral law into a differential one. We use it four more times in Section 7.12.

Green's identities. Apply the theorem to $\mathbf{F} = f\nabla g$, using identity 4 of Section 7.5, $\nabla\cdot(f\nabla g) = f\nabla^2 g + \nabla f\cdot\nabla g$:

$$\iiint_V \left(f\nabla^2 g + \nabla f\cdot\nabla g\right) dV = \oint_S f\,\nabla g\cdot d\mathbf{S} \qquad \text{(first identity)}.$$

Swapping f and g and subtracting:

$$\iiint_V \left(f\nabla^2 g - g\nabla^2 f\right) dV = \oint_S \left(f\nabla g - g\nabla f\right)\cdot d\mathbf{S} \qquad \text{(second identity)}.$$

Uniqueness of solutions of Poisson's equation

If $\nabla^2 V = -\rho/\varepsilon$ in V and V is prescribed on the boundary S, the solution is unique.

Proof. Let V_1, V_2 be two solutions and $w = V_1 - V_2$. Then $\nabla^2 w = 0$ inside and $w = 0$ on S. The first identity with $f = g = w$ gives $\iiint \|\nabla w\|^2 dV = \oint w\nabla w\cdot d\mathbf{S} = 0$, so $\nabla w = \mathbf{0}$, w is constant, and it is zero because it vanishes on S. ■ This theorem is what licenses the method of images and every "guess the solution and check it" approach in electrostatics (Ch. 13).

Point sources and the delta function

The inverse-square field $\hat{\mathbf{R}}/R^2$ has zero divergence for $R \neq 0$, yet its flux through any sphere about the origin is 4π. The divergence theorem cannot be applied naively to a region containing the origin because the field is singular there. Two ways to handle it:

- **Excise the singularity** (as with the hole in Green's theorem): for any closed surface S enclosing the origin, apply the theorem to the region between S and a small sphere S_ϵ. The divergence is zero there, so the flux through S equals the flux through S_ϵ, which is 4π. If S does not enclose the origin, the flux is 0.
- **Use the Dirac delta** (Ch. 14, Ch. 16): define $\delta(\mathbf{R}) = \delta(x)\delta(y)\delta(z)$, with $\iiint_V \delta(\mathbf{R})\, dV = 1$ if V contains the origin and 0 otherwise. Then the flux results above are summarised as

$$\nabla\cdot\frac{\hat{\mathbf{R}}}{R^2} = 4\pi\,\delta(\mathbf{R}), \qquad \nabla^2\frac{1}{R} = -4\pi\,\delta(\mathbf{R}).$$

The second formula says that $G(\mathbf{R},\mathbf{R}') = \dfrac{1}{4\pi\|\mathbf{R}-\mathbf{R}'\|}$ is the **Green's function** of $-\nabla^2$: the potential of a unit point source. Superposing point sources gives the solution of Poisson's equation in free space, $V(\mathbf{R}) = \dfrac{1}{4\pi\varepsilon_0}\iiint \dfrac{\rho(\mathbf{R}')}{\|\mathbf{R}-\mathbf{R}'\|}dV'$ — the same superposition-of-impulse-responses idea as convolution in circuit theory.

Worked Example — verifying the theorem on a cube

Let $\mathbf{F} = x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}$ and $V = [0,1]^3$.

Volume side. $\nabla\cdot\mathbf{F} = 3$, so $\iiint 3\, dV = 3$.

Surface side. On the face $x = 1$, $\hat{\mathbf{n}} = \hat{\mathbf{x}}$ and $\mathbf{F}\cdot\hat{\mathbf{n}} = x = 1$: flux 1. On $x = 0$, $\mathbf{F}\cdot(-\hat{\mathbf{x}}) = -x = 0$: flux 0. The y and z faces behave the same way. Total $1 + 1 + 1 = 3$ ✓.

Worked Example — a flux that is easier as a volume integral

Find the outward flux of $\mathbf{F} = x^3\hat{\mathbf{x}} + y^3\hat{\mathbf{y}} + z^3\hat{\mathbf{z}}$ through the sphere $R = a$.

Direct route (hard). $\mathbf{F}\cdot\hat{\mathbf{R}} = (x^4+y^4+z^4)/a$ on the sphere — an awkward angular integral.

Divergence theorem. $\nabla\cdot\mathbf{F} = 3(x^2+y^2+z^2) = 3R^2$, so

$$\oint_S \mathbf{F}\cdot d\mathbf{S} = \int_0^{2\pi}\int_0^{\pi}\int_0^{a} 3R^2\cdot R^2\sin\theta\, dR\, d\theta\, d\phi = 3\cdot\frac{a^5}{5}\cdot 2\cdot 2\pi = \frac{12\pi a^5}{5}.$$

Cross-check of the direct route. By symmetry $\oint x^4\, dS = \oint y^4\, dS = \oint z^4\, dS$, and $\oint z^4\, dS = a^4\cdot a^2\int_0^{2\pi}\int_0^{\pi}\cos^4\theta\sin\theta\, d\theta\, d\phi = a^6\cdot\frac{2}{5}\cdot 2\pi = \frac{4\pi a^6}{5}$. So the flux is $\frac{3}{a}\cdot\frac{4\pi a^6}{5} = \frac{12\pi a^5}{5}$ ✓.

Worked Example — closing an open surface

Recover the flux $\pi/2$ of $\mathbf{F} = z\hat{\mathbf{z}}$ through the paraboloid cap $z = 1 - x^2 - y^2$, $z \geq 0$ (Section 7.7), using the divergence theorem.

Step 1 — close the surface with the disk $z = 0$, $r \leq 1$. The closed surface encloses the solid V under the cap.

Step 2 — volume integral. $\nabla \cdot \mathbf{F} = 1$, so $\iiint_V 1\, dV$ is the volume: $\int_0^{2\pi} \int_0^1 (1 - r^2)\, r\, dr\, d\phi = \pi/2$.

Step 3 — subtract the disk. On the disk $z = 0$, so $\mathbf{F} = \mathbf{0}$ and its flux is 0.

Result. Flux through the cap $= \pi/2 - 0 = \pi/2$ ✓. "Close the surface, use the volume integral, subtract the easy lid" is a standard trick.

Worked Example — Gauss's law for an arbitrary surface

A point charge q at the origin has $\mathbf{E} = \dfrac{q}{4\pi\varepsilon_0} \dfrac{\hat{\mathbf{R}}}{R^2}$. Find the flux of $\mathbf{E}$ out of (a) the cube $|x|, |y|, |z| \leq 1$; (b) the cube $1 \leq x \leq 3$, $|y|, |z| \leq 1$.

(a) The cube encloses the origin. By the excision argument the flux equals that through a small sphere: $\dfrac{q}{4\pi\varepsilon_0} \cdot 4\pi = \dfrac{q}{\varepsilon_0}$. (Consistent with the per-face result $q/(6\varepsilon_0)$ of Section 7.7.)

(b) The origin lies outside; $\nabla \cdot \mathbf{E} = 0$ throughout, so the flux is 0 — the field lines that enter through the face $x = 1$ all leave through the other five faces.

Common Pitfall

Using an inward normal, or forgetting a piece of the boundary. The divergence theorem needs the **whole** closed boundary with **outward** normals. For a region with an internal cavity (a hollow sphere), the boundary includes the inner surface with its normal pointing *into the cavity* — away from the material.

Engineering Note

The finite-volume method used in thermal simulators and in many TCAD semiconductor-device solvers is the divergence theorem applied cell by cell: each mesh cell's stored quantity changes by the sum of fluxes through its faces. Because each face flux is shared by two cells with opposite signs, the scheme conserves charge (or heat) exactly — the discrete version of the interior-cancellation argument.

7.12 Applications to electromagnetic field theory

Everything in this chapter was built for this section. Maxwell's equations are four statements about the fields $\mathbf{E}, \mathbf{D}, \mathbf{H}, \mathbf{B}$; in **integral form** they are statements about line integrals around loops and fluxes through surfaces — the form in which they were discovered and are measured. Stokes' theorem and the divergence theorem turn them into **differential form** — local partial differential equations that hold at every point and can be solved. The mathematics is the subject here; the physics is the motivation.

The field quantities

Symbol	Name	SI unit
$\mathbf{E}$	electric field intensity	V/m
$\mathbf{D}$	electric flux density	$\mathrm{C/m^2}$
$\mathbf{H}$	magnetic field intensity	A/m
$\mathbf{B}$	magnetic flux density	$\mathrm{T} = \mathrm{Wb/m^2}$
$\mathbf{J}$	(free) current density	$\mathrm{A/m^2}$
ρ	(free) volume charge density	$\mathrm{C/m^3}$

In a linear, isotropic medium the **constitutive relations** are $\mathbf{D} = \varepsilon\mathbf{E}$, $\mathbf{B} = \mu\mathbf{H}$, $\mathbf{J} = \sigma\mathbf{E}$ (Ohm's law), with $\varepsilon = \varepsilon_r\varepsilon_0$, $\mu = \mu_r\mu_0$, $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m and $\mu_0 \approx 4\pi \times 10^{-7}$ H/m.

Maxwell's equations in integral form

Maxwell's equations — integral form

For any closed surface S enclosing a volume V, and any open surface S bounded by a closed curve C (orientations related by the right-hand rule):

$$\begin{aligned}
&\text{Gauss's law:} && \oint_S \mathbf{D} \cdot d\mathbf{S} = \iiint_V \rho\, dV = Q_{\text{enc}}, \\
&\text{no magnetic charge:} && \oint_S \mathbf{B} \cdot d\mathbf{S} = 0, \\
&\text{Faraday's law:} && \oint_C \mathbf{E} \cdot d\mathbf{l} = -\frac{d}{dt} \iint_S \mathbf{B} \cdot d\mathbf{S}, \\
&\text{Ampère–Maxwell law:} && \oint_C \mathbf{H} \cdot d\mathbf{l} = \iint_S \left(\mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \right) \cdot d\mathbf{S}.
\end{aligned}$$

Read with the vocabulary of this chapter:

- **Gauss:** the outward flux of $\mathbf{D}$ through a closed surface is the enclosed charge.
- **No monopoles:** the net flux of $\mathbf{B}$ out of any closed surface is zero; flux lines of $\mathbf{B}$ close on themselves.
- **Faraday:** the circulation of $\mathbf{E}$ (the EMF) around a loop is minus the rate of change of magnetic flux through it.
- **Ampère–Maxwell:** the circulation of $\mathbf{H}$ around a loop is the total current — conduction plus displacement — through it.

From integral form to differential form

The recipe is the same each time: (1) turn the boundary integral into an integral over the inside using Stokes or Gauss; (2) move everything to one side under a single integral; (3) since the region is arbitrary, the integrand vanishes.

Gauss's law. By the divergence theorem, $\oint_S \mathbf{D}\cdot d\mathbf{S} = \iiint_V \nabla\cdot\mathbf{D}\,dV$. Hence $\iiint_V (\nabla\cdot\mathbf{D}-\rho)\,dV = 0$ for every V, so

$$\nabla\cdot\mathbf{D} = \rho.$$

No magnetic charge. The same argument gives $\nabla\cdot\mathbf{B} = 0$.

Faraday's law. For a stationary surface the time derivative passes inside the integral. By Stokes' theorem $\oint_C \mathbf{E}\cdot d\mathbf{l} = \iint_S (\nabla\times\mathbf{E})\cdot d\mathbf{S}$, so $\iint_S \left(\nabla\times\mathbf{E} + \dfrac{\partial\mathbf{B}}{\partial t}\right)\cdot d\mathbf{S} = 0$ for every surface S. Choosing tiny flat surfaces with each of the three normal directions shows that each component of the bracket vanishes:

$$\nabla\times\mathbf{E} = -\frac{\partial\mathbf{B}}{\partial t}.$$

Ampère–Maxwell law. Identically,

$$\nabla\times\mathbf{H} = \mathbf{J} + \frac{\partial\mathbf{D}}{\partial t}.$$

Maxwell's equations — differential form

$$\nabla\cdot\mathbf{D} = \rho, \quad \nabla\times\mathbf{E} = -\frac{\partial\mathbf{B}}{\partial t},$$
$$\nabla\cdot\mathbf{B} = 0, \quad \nabla\times\mathbf{H} = \mathbf{J} + \frac{\partial\mathbf{D}}{\partial t}.$$

For time-harmonic fields with phasor convention $\mathbf{E}(\mathbf{R},t) = \mathrm{Re}\{\mathbf{E}(\mathbf{R})e^{j\omega t}\}$, $\partial/\partial t \to j\omega$:

$$\nabla\times\mathbf{E} = -j\omega\mathbf{B}, \qquad \nabla\times\mathbf{H} = \mathbf{J} + j\omega\mathbf{D}.$$

Law	Integral form	Theorem used	Differential form
Gauss	$\oint \mathbf{D}\cdot d\mathbf{S} = Q_{\text{enc}}$	divergence	$\nabla\cdot\mathbf{D} = \rho$
No monopoles	$\oint \mathbf{B}\cdot d\mathbf{S} = 0$	divergence	$\nabla\cdot\mathbf{B} = 0$
Faraday	$\oint \mathbf{E}\cdot d\mathbf{l} = -\dfrac{d\Phi}{dt}$	Stokes	$\nabla\times\mathbf{E} = -\partial_t\mathbf{B}$
Ampère–Maxwell	$\oint \mathbf{H}\cdot d\mathbf{l} = I_{\text{enc}} + \dfrac{d}{dt}\iint \mathbf{D}\cdot d\mathbf{S}$	Stokes	$\nabla\times\mathbf{H} = \mathbf{J} + \partial_t\mathbf{D}$

The integral forms are best for problems with high symmetry (where the field can be pulled out of the integral) and for circuits and measurements; the differential forms are best for deriving wave equations, boundary-value problems and numerical methods. The two divergence equations are, by the Helmholtz theorem, exactly what is needed alongside the two curl equations to fix the fields.

Why the displacement current is required

Take the divergence of Ampère's law without Maxwell's term, $\nabla\times\mathbf{H} = \mathbf{J}$. Since the divergence of a curl is zero (identity 2 of Section 7.5), this would force $\nabla\cdot\mathbf{J} = 0$ — but the continuity equation (Section 7.11) says $\nabla\cdot\mathbf{J} = -\partial\rho/\partial t$, which is non-zero whenever charge accumulates (as on a charging capacitor plate). With Maxwell's term,

$$0 = \nabla\cdot(\nabla\times\mathbf{H}) = \nabla\cdot\mathbf{J} + \frac{\partial}{\partial t}\nabla\cdot\mathbf{D} = \nabla\cdot\mathbf{J} + \frac{\partial\rho}{\partial t},$$

which is precisely charge conservation. A vector identity, not an experiment, demanded the extra term — and that term is what makes electromagnetic waves possible.

Potentials, Poisson's and Laplace's equations

Since $\nabla\cdot\mathbf{B} = 0$, there is a vector potential with $\mathbf{B} = \nabla\times\mathbf{A}$ (Section 7.5). Substituting into Faraday's law gives $\nabla\times(\mathbf{E} + \partial_t\mathbf{A}) = \mathbf{0}$, so the bracket is a gradient:

$$\mathbf{E} = -\nabla V - \frac{\partial\mathbf{A}}{\partial t}.$$

In electrostatics $\mathbf{E} = -\nabla V$; inserting into $\nabla\cdot(\varepsilon\mathbf{E}) = \rho$ with constant ε gives **Poisson's equation**, and in charge-free regions **Laplace's equation**:

$$\nabla^2 V = -\frac{\rho}{\varepsilon}, \qquad \nabla^2 V = 0.$$

These are solved by separation of variables in Ch. 13 and numerically in Ch. 22; uniqueness was proved in Section 7.11.

The wave equation

In a source-free ($\rho = 0$, $\mathbf{J} = \mathbf{0}$), linear, homogeneous medium, take the curl of Faraday's law and use Ampère–Maxwell:

$$\nabla \times (\nabla \times \mathbf{E}) = -\frac{\partial}{\partial t}\nabla \times \mathbf{B} = -\mu\frac{\partial}{\partial t}\left(\varepsilon\frac{\partial \mathbf{E}}{\partial t}\right) = -\mu\varepsilon\frac{\partial^2 \mathbf{E}}{\partial t^2}.$$

By identity 7, $\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2\mathbf{E} = -\nabla^2\mathbf{E}$, because $\nabla \cdot \mathbf{E} = 0$ here. Therefore

$$\nabla^2 \mathbf{E} = \mu\varepsilon\frac{\partial^2 \mathbf{E}}{\partial t^2}, \qquad u = \frac{1}{\sqrt{\mu\varepsilon}}.$$

In vacuum $u = 1/\sqrt{\mu_0\varepsilon_0} = 2.998 \times 10^8$ m/s — the speed of light, predicted from two constants measured with capacitors and coils. In phasor form ($\partial_t \to j\omega$) it becomes the **Helmholtz equation** $\nabla^2\mathbf{E} + k^2\mathbf{E} = \mathbf{0}$ with wavenumber $k = \omega\sqrt{\mu\varepsilon}$. The same equation holds for $\mathbf{H}$.

Boundary conditions from the integral forms

At an interface between two media the fields can jump, and the differential equations (which need derivatives) do not apply on the interface itself. The integral forms do.

- **Normal components — pillbox.** Apply Gauss's law to a thin pillbox straddling the interface, faces of area ΔS on either side, height $\to 0$. The side-wall flux vanishes, leaving $(\mathbf{D}_1 - \mathbf{D}_2)\cdot\hat{\mathbf{n}}\,\Delta S = \rho_s \Delta S$, where $\hat{\mathbf{n}}$ points from medium 2 into medium 1 and ρ_s is surface charge. Likewise for $\mathbf{B}$ with zero right side.
- **Tangential components — thin loop.** Apply Faraday's law to a thin rectangular loop straddling the interface, sides Δl parallel to it, height $\to 0$. The enclosed flux vanishes with the area, leaving $(\mathbf{E}_1 - \mathbf{E}_2)\cdot\hat{\mathbf{t}}\,\Delta l = 0$. For $\mathbf{H}$ the right side is a surface current $J_s \Delta l$.

$$\hat{\mathbf{n}}\cdot(\mathbf{D}_1 - \mathbf{D}_2) = \rho_s, \quad \hat{\mathbf{n}}\cdot(\mathbf{B}_1 - \mathbf{B}_2) = 0, \quad \hat{\mathbf{n}}\times(\mathbf{E}_1 - \mathbf{E}_2) = \mathbf{0}, \quad \hat{\mathbf{n}}\times(\mathbf{H}_1 - \mathbf{H}_2) = \mathbf{J}_s.$$

These are exactly the "shrink the region onto the boundary" arguments by which we defined divergence and curl, now used on a surface where the fields are discontinuous.

Energy flow: Poynting's theorem

Identity 6 of Section 7.5 and the two curl equations give

$$\nabla\cdot(\mathbf{E}\times\mathbf{H}) = \mathbf{H}\cdot(\nabla\times\mathbf{E}) - \mathbf{E}\cdot(\nabla\times\mathbf{H}) = -\mathbf{H}\cdot\frac{\partial\mathbf{B}}{\partial t} - \mathbf{E}\cdot\mathbf{J} - \mathbf{E}\cdot\frac{\partial\mathbf{D}}{\partial t}.$$

For a linear medium $\mathbf{H}\cdot\partial_t\mathbf{B} = \partial_t(\tfrac{1}{2}\mu H^2)$ and $\mathbf{E}\cdot\partial_t\mathbf{D} = \partial_t(\tfrac{1}{2}\varepsilon E^2)$. Integrating over a volume and applying the divergence theorem:

$$-\oint_S (\mathbf{E}\times\mathbf{H})\cdot d\mathbf{S} = \frac{d}{dt}\iiint_V \left(\tfrac{1}{2}\varepsilon E^2 + \tfrac{1}{2}\mu H^2\right) dV + \iiint_V \mathbf{E}\cdot\mathbf{J}\, dV.$$

Power flowing **in** through the surface (flux of the Poynting vector $\mathbf{S} = \mathbf{E}\times\mathbf{H}$, in $\mathrm{W/m^2}$) equals the rate of increase of stored field energy plus the power dissipated. It is the field version of "power in = $\frac{d}{dt}$(energy stored) + I^2R".

Worked examples

Worked Example — Gauss's law: the uniformly charged sphere

A sphere of radius a in free space holds charge Q uniformly ($\rho_0 = 3Q/(4\pi a^3)$). Find $\mathbf{E}$ everywhere and check the differential form.

Step 1 — symmetry. By spherical symmetry $\mathbf{E} = E_R(R)\hat{\mathbf{R}}$. On a Gaussian sphere of radius R, $\mathbf{D}\cdot d\mathbf{S} = \varepsilon_0 E_R\, dS$ with E_R constant, so $\oint \mathbf{D}\cdot d\mathbf{S} = 4\pi R^2\varepsilon_0 E_R$.

Step 2 — enclosed charge. For $R \ge a$, $Q_{\text{enc}} = Q$. For $R < a$, $Q_{\text{enc}} = Q\,R^3/a^3$.

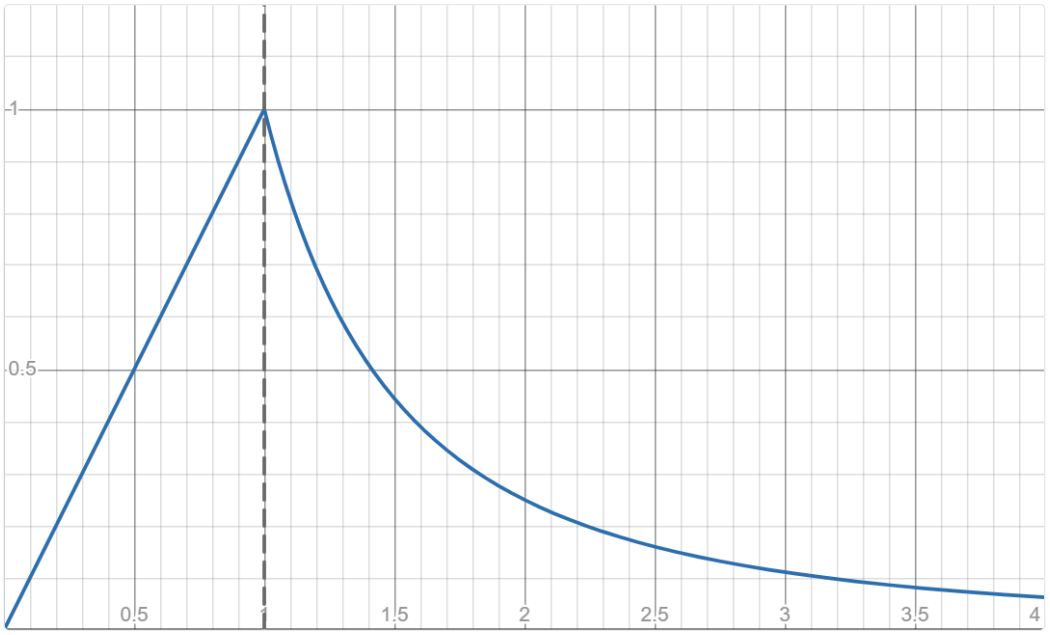
Step 3 — solve.

$$E_R = \begin{cases} \dfrac{QR}{4\pi\varepsilon_0 a^3}, & R < a, \\[2mm] \dfrac{Q}{4\pi\varepsilon_0 R^2}, & R \ge a. \end{cases}$$

Both branches give $Q/(4\pi\varepsilon_0 a^2)$ at $R = a$: the field is continuous (no surface charge).

Step 4 — differential check. Inside, $\nabla\cdot\mathbf{E} = \frac{1}{R^2}\partial_R\big(R^2\cdot\frac{QR}{4\pi\varepsilon_0 a^3}\big) = \frac{3Q}{4\pi\varepsilon_0 a^3} = \rho_0/\varepsilon_0$ ✓. Outside, $\frac{1}{R^2}\partial_R(\text{const}) = 0$ ✓.

Numbers. $Q = 1$ nC, $a = 1$ cm: peak field $E(a) = 8.988\times10^9 \times 10^{-9}/10^{-4} = 8.99\times10^4$ V/m ≈ 90 kV/m, about 3 % of the breakdown strength of air.



Field of a uniformly charged sphere, normalised to its surface value, against R/a: linear inside (where enclosed charge grows as R^3), inverse-square outside.

Worked Example — Ampère's law: the field in and around a thick wire

A long straight wire of radius a carries current I uniformly ($J = I/(\pi a^2)$, along $\hat{\mathbf{z}}$). Find $\mathbf{H}$ and $\mathbf{B}$ ($\mu = \mu_0$), and verify $\nabla \times \mathbf{H} = \mathbf{J}$.

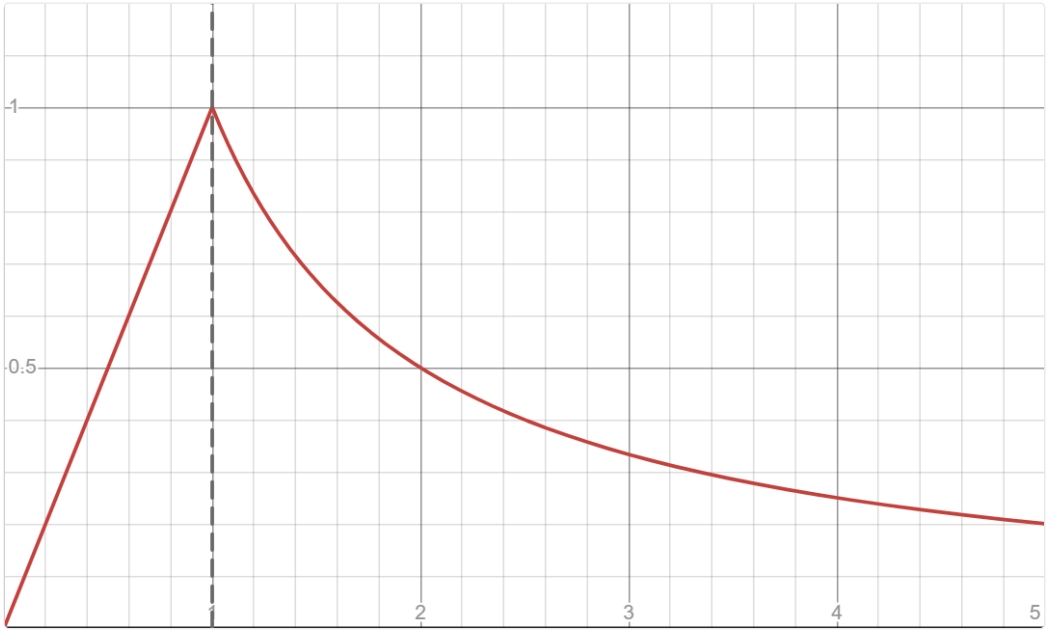
Step 1 — symmetry. $\mathbf{H} = H_\phi(r)\hat{\boldsymbol{\phi}}$. Around a circle of radius r, $\oint \mathbf{H} \cdot d\mathbf{l} = 2\pi r H_\phi$.

Step 2 — enclosed current. $I_{\text{enc}} = I\,r^2/a^2$ for $r < a$, and I for $r \geq a$ (static, so no displacement current).

Step 3 — solve. $H_\phi = \dfrac{Ir}{2\pi a^2}$ inside, $\dfrac{I}{2\pi r}$ outside.

Step 4 — differential check. Inside: $(\nabla \times \mathbf{H})_z = \frac{1}{r}\partial_r\left(r \cdot \frac{Ir}{2\pi a^2}\right) = \frac{I}{\pi a^2} = J$ ✓. Outside: $\frac{1}{r}\partial_r(I/2\pi) = 0$ ✓ — the curl-free wire field of Sections 7.5–7.6, now seen as the field of a current that occupies only $r < a$.

Numbers. $I = 10$ A, $a = 1$ mm: $B(a) = \mu_0 I/(2\pi a) = 2.0$ mT; $B(0.5\text{ mm}) = 1.0$ mT; $B(1\text{ cm}) = 0.20$ mT.



$\|\mathbf{B}\|$ of a thick wire normalised to its surface value, against r/a: linear inside, $1/r$ outside. Compare with the sphere: the outside decay is slower because the source is a line, not a point.

Worked Example — capacitance per metre of a coaxial cable

A coaxial cable has inner radius a, outer radius b, and dielectric ε. Find the capacitance per metre and evaluate it for $a = 0.45$ mm, $b = 1.47$ mm, $\varepsilon_r = 2.25$ (polyethylene).

Step 1 — Gauss. Put charge ρ_ℓ per metre on the inner conductor. A coaxial Gaussian cylinder of radius r and length L ($a < r < b$) has flux only through its curved side: $\varepsilon E_r \cdot 2\pi r L = \rho_\ell L$, so $E_r = \dfrac{\rho_\ell}{2\pi\varepsilon r}$.

Step 2 — line integral for the voltage.

$$V = -\int_b^a E_r\, dr = \frac{\rho_\ell}{2\pi\varepsilon}\int_a^b \frac{dr}{r} = \frac{\rho_\ell}{2\pi\varepsilon}\ln\frac{b}{a}.$$

Step 3 — capacitance. $C' = \rho_\ell / V = \dfrac{2\pi\varepsilon}{\ln(b/a)}$.

Numbers. $\ln(1.47/0.45) = 1.1838$, $2\pi \times 2.25 \times 8.854 \times 10^{-12} = 1.2517 \times 10^{-10}$, so $C' = 105.7$ pF/m — the right order for common 50 Ω cable (about 100 pF/m).

Gauss's law (a surface integral) gave the field; a line integral gave the voltage. That pair of steps solves every symmetric capacitor problem.

Worked Example — Faraday's law: induced EMF and induced field

A uniform field $\mathbf{B} = B_0 \sin(\omega t)\,\hat{\mathbf{z}}$ fills a region. (a) Find the EMF around a circular loop of radius a in the xy-plane centred on the axis. (b) Find the induced electric field and check $\nabla \times \mathbf{E} = -\partial_t \mathbf{B}$. (c) Evaluate the peak EMF for $a = 5$ cm, $B_0 = 10$ mT, $f = 50$ Hz.

(a) Flux $\Phi = \pi a^2 B_0 \sin\omega t$, so EMF $= \oint \mathbf{E}\cdot d\mathbf{l} = -d\Phi/dt = -\pi a^2 B_0 \omega \cos\omega t$.

(b) By symmetry $\mathbf{E} = E_\phi(r)\hat{\boldsymbol{\phi}}$, so the EMF around radius r is $2\pi r E_\phi = -\pi r^2 \dot{B}$, giving $E_\phi = -\frac{r}{2}\dot{B} = -\frac{r}{2}B_0\omega\cos\omega t$. Check: $(\nabla\times\mathbf{E})_z = \frac{1}{r}\partial_r(rE_\phi) = \frac{1}{r}\partial_r\left(-\frac{r^2}{2}\dot{B}\right) = -\dot{B}$ ✓.

(c) Peak EMF $= \pi(0.05)^2(0.01)(2\pi\cdot 50) = 24.7$ mV.

This induced $\mathbf{E}$ has circular field lines and non-zero curl: it is **not** a gradient, and no scalar potential can describe it — Kirchhoff's voltage law fails for a loop around changing flux.

Worked Example — displacement current in a capacitor

A parallel-plate air capacitor has plate area $A = 1$ cm^2 and gap $d = 0.1$ mm, driven by $v(t) = V_0 \sin\omega t$ with $V_0 = 10$ V, $f = 1$ MHz. Show that the displacement current between the plates equals the conduction current in the leads.

Step 1. Between the plates $E = v/d$, so the displacement current density is $\partial D/\partial t = \varepsilon_0 \dot{v}/d$.

Step 2. Its flux through a surface between the plates is $I_d = A\varepsilon_0\dot{v}/d = C\dot{v}$ with $C = \varepsilon_0 A/d$ — exactly the lead current $i = C\,dv/dt$ ✓.

Step 3 — numbers. $C = 8.854\times10^{-12}\times10^{-4}/10^{-4} = 8.854$ pF; peak $I = C\omega V_0 = 8.854\times10^{-12}\times 2\pi\times10^6\times10 = 0.556$ mA.

So a loop around the lead "sees" the same $\oint \mathbf{H}\cdot d\mathbf{l}$ whether the surface it bounds is pierced by the wire (conduction current) or bulges between the plates (displacement current) — Stokes' surface independence restored.

Worked Example — Poynting's theorem for a resistive wire

A straight wire of radius a, length L and conductivity σ carries a DC current I. Show that the Poynting flux into its surface equals I^2R.

Step 1 — fields at the surface. $\mathbf{E} = \mathbf{J}/\sigma = \dfrac{I}{\sigma\pi a^2}\hat{\mathbf{z}}$; tangential $\mathbf{E}$ is continuous, so this is also the field just outside. From the Ampère example, $\mathbf{H} = \dfrac{I}{2\pi a}\hat{\boldsymbol{\phi}}$ at $r = a$.

Step 2 — Poynting vector. $\mathbf{S} = \mathbf{E}\times\mathbf{H} = EH\,(\hat{\mathbf{z}}\times\hat{\boldsymbol{\phi}}) = -EH\,\hat{\mathbf{r}}$: it points **into** the wire.

Step 3 — inward flux. $P = EH\cdot 2\pi aL = \dfrac{I}{\sigma\pi a^2}\cdot\dfrac{I}{2\pi a}\cdot 2\pi aL = I^2\dfrac{L}{\sigma\pi a^2} = I^2R$ ✓.

The energy dissipated in the wire flows in from the surrounding field, not along the inside of the conductor — a striking conclusion reached purely by a cross product and a surface integral.

Common Pitfall

Using Gauss's or Ampère's law to "find" a field without enough symmetry. The integral forms are always true, but they determine the field only when symmetry lets you pull E or H out of the integral as a constant. For a finite wire, a square loop or a charged cube, $\oint \mathbf{H}\cdot d\mathbf{l} = I$ is still correct but does not tell you $\mathbf{H}$ at any point; use Biot–Savart, superposition or a numerical solver instead.

Common Pitfall

Mixing time-domain and phasor forms. In the phasor equations $\mathbf{E}$ is a complex vector and ∂_t has become $j\omega$; quantities like energy and power, which are quadratic, need $\tfrac{1}{2}\,\mathrm{Re}\{\cdot\}$ and conjugates ($\langle \mathbf{S} \rangle = \tfrac{1}{2}\,\mathrm{Re}\{\mathbf{E} \times \mathbf{H}^*\}$), not the time-domain formulas applied to phasors.

Engineering Note

Circuit theory is the low-frequency limit of these equations. KCL is $\nabla \cdot \mathbf{J} = 0$ (continuity with no charge accumulation at nodes, or with capacitor displacement currents counted as branch currents); KVL is $\oint \mathbf{E} \cdot d\mathbf{l} = 0$ (Faraday with flux changes lumped into inductors); $v = L\,di/dt$ and $i = C\,dv/dt$ are Faraday's and the displacement term of Ampère–Maxwell confined to components. When a circuit's dimensions approach a tenth of a wavelength these lumped approximations fail and the full field equations — via transmission-line theory and the wave equation — take over.

Summary

Concept	Key formula
Dot product	$\mathbf{A} \cdot \mathbf{B} = AB\cos\theta = A_xB_x + A_yB_y + A_zB_z$
Cross product	$\lVert \mathbf{A} \times \mathbf{B} \rVert = AB\sin\theta$; determinant with $\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$
Triple products	$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$ = volume; $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$
Arc length, curvature	$s = \int \lVert \mathbf{r}' \rVert dt$, $\kappa = \lVert \mathbf{r}' \times \mathbf{r}'' \rVert / \lVert \mathbf{r}' \rVert^3$
Polar velocity	$\mathbf{v} = \dot{r}\hat{\mathbf{r}} + r\dot{\phi}\hat{\boldsymbol{\phi}} + \dot{z}\hat{\mathbf{z}}$
Field lines	$dx/F_x = dy/F_y = dz/F_z$
Gradient	$\nabla f = \sum \frac{1}{h_i}\partial_{u_i} f\,\hat{\mathbf{u}}_i$; $h = (1, r, 1)$ cyl., $(1, R, R\sin\theta)$ sph.
Divergence (def.)	$\nabla \cdot \mathbf{F} = \lim \frac{1}{\Delta V} \oint \mathbf{F} \cdot d\mathbf{S}$
Divergence (sph.)	$\frac{1}{R^2}\partial_R(R^2F_R) + \frac{1}{R\sin\theta}\partial_\theta(\sin\theta F_\theta) + \frac{1}{R\sin\theta}\partial_\phi F_\phi$
Curl (def.)	$(\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}} = \lim \frac{1}{\Delta S} \oint \mathbf{F} \cdot d\mathbf{r}$
Curl z-comp. (cyl.)	$\frac{1}{r}[\partial_r(rF_\phi) - \partial_\phi F_r]$
Laplacian (sph.)	$\frac{1}{R^2}\partial_R(R^2\partial_R f) + \frac{1}{R^2\sin\theta}\partial_\theta(\sin\theta\,\partial_\theta f) + \frac{1}{R^2\sin^2\theta}\partial_\phi^2 f$
Zero identities	$\nabla \times \nabla f = \mathbf{0}$, $\nabla \cdot (\nabla \times \mathbf{A}) = 0$
Curl curl	$\nabla \times \nabla \times \mathbf{A} = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2\mathbf{A}$
Line integral	$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)\,dt$
Fundamental theorem	$\int_C \nabla f \cdot d\mathbf{r} = f(\mathbf{b}) - f(\mathbf{a})$
Surface element	$d\mathbf{S} = \mathbf{r}_u \times \mathbf{r}_v\,du\,dv$; sphere $a^2\sin\theta\,d\theta\,d\phi\,\hat{\mathbf{R}}$
Volume element	$dV = r\,dr\,d\phi\,dz = R^2\sin\theta\,dR\,d\theta\,d\phi$
Green	$\oint (P\,dx + Q\,dy) = \iint (Q_x - P_y)\,dA$; $A = \frac{1}{2}\oint (x\,dy - y\,dx)$
Stokes	$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$
Divergence theorem	$\oint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_V \nabla \cdot \mathbf{F}\,dV$
Continuity	$\nabla \cdot \mathbf{J} + \partial_t \rho = 0$
Point source	$\nabla \cdot (\hat{\mathbf{R}}/R^2) = 4\pi\delta(\mathbf{R})$, $\nabla^2(1/R) = -4\pi\delta(\mathbf{R})$
Maxwell (differential)	$\nabla \cdot \mathbf{D} = \rho$, $\nabla \cdot \mathbf{B} = 0$, $\nabla \times \mathbf{E} = -\partial_t\mathbf{B}$, $\nabla \times \mathbf{H} = \mathbf{J} + \partial_t\mathbf{D}$
Potentials	$\mathbf{B} = \nabla \times \mathbf{A}$, $\mathbf{E} = -\nabla V - \partial_t\mathbf{A}$; $\nabla^2 V = -\rho/\varepsilon$
Wave / Helmholtz	$\nabla^2\mathbf{E} = \mu\varepsilon\,\partial_t^2\mathbf{E}$; $\nabla^2\mathbf{E} + k^2\mathbf{E} = \mathbf{0}$, $k = \omega\sqrt{\mu\varepsilon}$
Poynting	$\mathbf{S} = \mathbf{E} \times \mathbf{H}$; $\langle \mathbf{S} \rangle = \frac{1}{2}\,\mathrm{Re}\{\mathbf{E} \times \mathbf{H}^*\}$

Further reading

- E. Kreyszig, *Advanced Engineering Mathematics*, 10th ed., Wiley — Ch. 9–10: vector differential and integral calculus, with the integral theorems.
- H. M. Schey, *Div, Grad, Curl, and All That: An Informal Text on Vector Calculus*, 4th ed., Norton — builds the operators from flux and circulation, exactly as in Section 7.5.
- D. K. Cheng, *Field and Wave Electromagnetics*, 2nd ed., Addison-Wesley — Ch. 2 (vector analysis in the coordinate conventions used here) and Ch. 7 (Maxwell's equations).
- D. J. Griffiths, *Introduction to Electrodynamics*, 4th ed., Cambridge University Press — Ch. 1 is a compact, careful treatment of vector calculus, including the Dirac delta and the Helmholtz theorem.

Part III — Linear Algebra

Chapter 8 — Matrices and Linear Systems

> **Prerequisites:** Ch. 1, Ch. 2; the Jacobian matrix of Ch. 6 is helpful but not essential. **Learning outcomes:** After this chapter you can
>
> - manipulate matrices fluently (sums, products, transposes, conjugate transposes, block forms) and read circuit equations as matrix equations;
> - evaluate determinants by cofactor expansion and by row reduction, and use them to test invertibility and to apply Cramer's rule;
> - compute inverses, determine rank, and classify a linear system as having a unique solution, no solution or infinitely many solutions;
> - solve systems by Gaussian elimination with partial pivoting, by Gauss–Jordan elimination, and by $\mathbf{LU}$ and Cholesky factorisation, and count the work each needs;
> - measure the sensitivity of a solution with norms and the condition number, and recognise ill-conditioned problems;
> - exploit sparsity (banded and tridiagonal solvers, compressed storage, iterative methods) in the large systems produced by circuit simulators and field solvers.

A linear circuit with n unknown node voltages produces n linear equations in n unknowns. A finite-difference model of the potential in a capacitor produces ten thousand. A power-flow study of a national grid produces a hundred thousand, solved again at every iteration of Newton's method. Whatever the size, the mathematical object is the same: a **linear system** $\mathbf{A}\mathbf{x} = \mathbf{b}$. This chapter develops the algebra of matrices needed to write such systems compactly, the theory that tells us when they can be solved, and the algorithms — elimination and factorisation — that every circuit simulator, field solver and estimation routine actually runs. Chapter 9 then looks at the same objects geometrically, as vector spaces and linear maps, and Chapter 10 studies their eigenvalues.

Throughout the chapter one small circuit serves as a running example. It has three non-reference nodes and six resistive branches:

Branch	Connects	Conductance
a	node 1 – ground	1 mS
b	node 1 – node 2	2 mS
c	node 2 – node 3	1 mS
d	node 2 – ground	1 mS
e	node 3 – ground	2 mS
f	node 1 – node 3	1 mS

Independent current sources inject 8 mA, 2 mA and 1 mA into nodes 1, 2 and 3. Kirchhoff's current law (KCL) at each node, with currents leaving the node counted positive, gives

$$\begin{aligned} (1+2+1)v_1 - 2v_2 - 1v_3 &= 8,\\ -2v_1 + (2+1+1)v_2 - 1v_3 &= 2,\\ -1v_1 - 1v_2 + (1+2+1)v_3 &= 1, \end{aligned}$$

with conductances in mS, currents in mA and therefore voltages in V (since $\mathrm{mA}/\mathrm{mS} = \mathrm{V}$). In matrix form,

$$\underbrace{\begin{bmatrix} 4 & -2 & -1 \\ -2 & 4 & -1 \\ -1 & -1 & 4 \end{bmatrix}}_{\mathbf{G}} \underbrace{\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}}_{\mathbf{v}} = \underbrace{\begin{bmatrix} 8 \\ 2 \\ 1 \end{bmatrix}}_{\mathbf{i}_s}.$$

We will find that $\mathbf{v} = (4, 3, 2)^{\mathsf{T}}$ V in at least six different ways before the chapter is over, and each way teaches something different about $\mathbf{G}$.

8.1 Matrix operations

Why matrices

Writing the nodal equations above as three separate lines is fine for three nodes. For three thousand it is hopeless — and, more importantly, it hides structure. The matrix $\mathbf{G}$ is symmetric; its diagonal entries dominate its rows; its off-diagonal entries are non-positive; most of its entries in a large circuit are zero. Each of these properties has a name, a theorem attached to it, and an algorithmic payoff. Matrix notation is what lets us see them.

Definitions

Matrix

An $m \times n$ **matrix** $\mathbf{A}$ is a rectangular array of mn numbers (real or complex) arranged in m rows and n columns. The entry in row i, column j is written a_{ij} or $[\mathbf{A}]_{ij}$:

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} = [a_{ij}].$$

The pair (m, n) is the **size** (or shape). A matrix with $m = n$ is **square** of **order** n. An $n \times 1$ matrix is a **column vector**, a $1 \times n$ matrix a **row vector**. The set of real $m \times n$ matrices is written $\mathbb{R}^{m \times n}$, the complex ones $\mathbb{C}^{m \times n}$.

Vectors in this book are columns unless stated otherwise: $\mathbf{x} = (x_1, \dots, x_n)^{\mathsf{T}}$ is a convenient way to print a column on one line.

Some square matrices occur so often they have names.

Name	Definition	Example (3×3 pattern)
Diagonal	$a_{ij} = 0$ for $i \neq j$	$\operatorname{diag}(d_1, d_2, d_3)$
Identity $\mathbf{I}$ (or $\mathbf{I}_n$)	diagonal with all $d_i = 1$	δ_{ij} (Kronecker delta)
Zero $\mathbf{0}$	all entries zero	
Upper triangular	$a_{ij} = 0$ for $i > j$	nonzeros on and above diagonal
Lower triangular	$a_{ij} = 0$ for $i < j$	nonzeros on and below diagonal
Tridiagonal	$a_{ij} = 0$ for $\lvert i - j \rvert > 1$	ladder networks, 1-D finite differences
Symmetric	$a_{ij} = a_{ji}$	nodal matrix of a reciprocal circuit
Skew-symmetric	$a_{ij} = -a_{ji}$ (so $a_{ii} = 0$)	gyrator, cross-product matrix
Hermitian	$a_{ij} = a_{ji}^{*}$	power/covariance matrices with complex entries

The **main diagonal** of a square matrix is $a_{11}, a_{22}, \dots, a_{nn}$. The **trace** is the sum of the diagonal entries, $\operatorname{tr}\mathbf{A} = \sum_i a_{ii}$; for $\mathbf{G}$ above, $\operatorname{tr}\mathbf{G} = 12$ mS, which is twice the total conductance of branches b, c, f plus once that of a, d, e (each inter-node branch is counted at both its ends).

Equality, addition and scalar multiplication

Linear operations on matrices

Two matrices are **equal** if they have the same size and all corresponding entries are equal. For $\mathbf{A}, \mathbf{B} \in \mathbb{C}^{m \times n}$ and a scalar c:

$$[\mathbf{A} + \mathbf{B}]_{ij} = a_{ij} + b_{ij}, \qquad [c\mathbf{A}]_{ij} = c\, a_{ij}.$$

Addition of matrices of different sizes is undefined.

These operations inherit every rule of ordinary arithmetic, entry by entry:

$$\begin{aligned} &\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}, && (\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C}), \\ &\mathbf{A} + \mathbf{0} = \mathbf{A}, && \mathbf{A} + (-\mathbf{A}) = \mathbf{0}, \\ &c(\mathbf{A} + \mathbf{B}) = c\mathbf{A} + c\mathbf{B}, && (c + d)\mathbf{A} = c\mathbf{A} + d\mathbf{A}, \quad (cd)\mathbf{A} = c(d\mathbf{A}). \end{aligned}$$

In the language of Chapter 9, $\mathbb{R}^{m \times n}$ is a vector space of dimension mn.

Addition has a direct circuit meaning. Each branch of the running example contributes a small "stamp" to $\mathbf{G}$, and $\mathbf{G}$ is the sum of the stamps. Branch b (2 mS between nodes 1 and 2) contributes

$$\mathbf{G}_b = 2\begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

branch a (1 mS from node 1 to ground) contributes 1 at position $(1,1)$ only, and so on. Summing the six stamps reproduces $\mathbf{G}$. This **stamping** procedure is exactly how SPICE-type simulators assemble their matrices: loop over the elements, add each element's stamp into the global matrix.

Transpose and conjugate transpose

Transpose, conjugate transpose

The **transpose** of $\mathbf{A} \in \mathbb{C}^{m \times n}$ is the $n \times m$ matrix $\mathbf{A}^{\mathsf{T}}$ with $[\mathbf{A}^{\mathsf{T}}]_{ij} = a_{ji}$ (rows become columns). The **conjugate transpose** (Hermitian transpose) is $\mathbf{A}^{\mathsf{H}} = (\mathbf{A}^{*})^{\mathsf{T}}$, i.e. $[\mathbf{A}^{\mathsf{H}}]_{ij} = a_{ji}^{*}$. For real matrices $\mathbf{A}^{\mathsf{H}} = \mathbf{A}^{\mathsf{T}}$.

Properties, each checked entry by entry:

$$(\mathbf{A}^{\mathsf{T}})^{\mathsf{T}} = \mathbf{A}, \qquad (\mathbf{A} + \mathbf{B})^{\mathsf{T}} = \mathbf{A}^{\mathsf{T}} + \mathbf{B}^{\mathsf{T}}, \qquad (c\mathbf{A})^{\mathsf{T}} = c\mathbf{A}^{\mathsf{T}}, \qquad (c\mathbf{A})^{\mathsf{H}} = c^{*}\mathbf{A}^{\mathsf{H}}.$$

The rule for products, $(\mathbf{AB})^{\mathsf{T}} = \mathbf{B}^{\mathsf{T}}\mathbf{A}^{\mathsf{T}}$, is proved in §8.2.

A square matrix is symmetric exactly when $\mathbf{A}^{\mathsf{T}} = \mathbf{A}$, skew-symmetric when $\mathbf{A}^{\mathsf{T}} = -\mathbf{A}$, and Hermitian when $\mathbf{A}^{\mathsf{H}} = \mathbf{A}$. A Hermitian matrix has a real diagonal, since $a_{ii} = a_{ii}^{*}$.

Symmetric/skew decomposition

Every square matrix splits uniquely into a symmetric and a skew-symmetric part:

$$\mathbf{A} = \underbrace{\tfrac{1}{2}(\mathbf{A} + \mathbf{A}^{\mathsf{T}})}_{\text{symmetric}} + \underbrace{\tfrac{1}{2}(\mathbf{A} - \mathbf{A}^{\mathsf{T}})}_{\text{skew-symmetric}}.$$

Likewise every complex square matrix is a Hermitian part plus j times another Hermitian part: $\mathbf{A} = \mathbf{H}_1 + j\mathbf{H}_2$ with $\mathbf{H}_1 = \tfrac{1}{2}(\mathbf{A} + \mathbf{A}^{\mathsf{H}})$, $\mathbf{H}_2 = \tfrac{1}{2j}(\mathbf{A} - \mathbf{A}^{\mathsf{H}})$.

Proof sketch. Transposing the first bracket returns it unchanged; transposing the second negates it; their sum is $\mathbf{A}$. For uniqueness, if $\mathbf{A} = \mathbf{S} + \mathbf{K}$ with $\mathbf{S}$ symmetric and $\mathbf{K}$ skew, then $\mathbf{A}^{\mathsf{T}} = \mathbf{S} - \mathbf{K}$; adding and subtracting recovers the two brackets. $\square$

Worked Example 8.1 — Basic operations

Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 0 & 1 & -2 \\ 5 & 2 & 1 \end{bmatrix}.$$

Compute $2\mathbf{A} - 3\mathbf{B}$ and $(\mathbf{A} + \mathbf{B})^{\mathsf{T}}$.

Step 1. Both are 2×3, so linear combinations are defined. Entry by entry,

$$2\mathbf{A} - 3\mathbf{B} = \begin{bmatrix} 2-0 & 4-3 & 0+6 \\ 6-15 & -2-6 & 8-3 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 6 \\ -9 & -8 & 5 \end{bmatrix}.$$

Step 2. $\mathbf{A} + \mathbf{B} = \begin{bmatrix} 1 & 3 & -2 \\ 8 & 1 & 5 \end{bmatrix}$, so

$$(\mathbf{A} + \mathbf{B})^{\mathsf{T}} = \begin{bmatrix} 1 & 8 \\ 3 & 1 \\ -2 & 5 \end{bmatrix},$$

a 3×2 matrix. Check: $\mathbf{A}^{\mathsf{T}} + \mathbf{B}^{\mathsf{T}}$ gives the same thing.

Worked Example 8.2 — A nonreciprocal circuit and its symmetric part

A transistor small-signal model adds a voltage-controlled current source $g_m v_1$ flowing *out* of node 2 to the running example with $g_m = 3$ mS (the source is controlled by v_1). KCL at node 2 gains a term $+3v_1$, so the nodal matrix becomes

$$\mathbf{Y} = \begin{bmatrix} 4 & -2 & -1 \\ 1 & 4 & -1 \\ -1 & -1 & 4 \end{bmatrix}.$$

Split it into symmetric and skew parts.

Step 1. $\mathbf{Y}^{\mathsf{T}} = \begin{bmatrix} 4 & 1 & -1 \\ -2 & 4 & -1 \\ -1 & -1 & 4 \end{bmatrix}$.

Step 2.

$$\mathbf{S} = \tfrac{1}{2}(\mathbf{Y} + \mathbf{Y}^{\mathsf{T}}) = \begin{bmatrix} 4 & -\frac{1}{2} & -1 \\ -\frac{1}{2} & 4 & -1 \\ -1 & -1 & 4 \end{bmatrix}, \qquad \mathbf{K} = \tfrac{1}{2}(\mathbf{Y} - \mathbf{Y}^{\mathsf{T}}) = \begin{bmatrix} 0 & -\frac{3}{2} & 0 \\ \frac{3}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Step 3 (check). $\mathbf{S} + \mathbf{K} = \mathbf{Y}$, $\mathbf{S}^{\mathsf{T}} = \mathbf{S}$, $\mathbf{K}^{\mathsf{T}} = -\mathbf{K}$.

Reading. The skew part is entirely due to the controlled source: a passive reciprocal network has a symmetric nodal matrix, and the skew part measures how far a network is from reciprocity. In §8.2 we will see that the power absorbed by the network, $\mathbf{v}^{\mathsf{T}}\mathbf{Y}\mathbf{v}$, depends only on $\mathbf{S}$.

Worked Example 8.3 — Hermitian part of an AC admittance matrix

At angular frequency ω a two-node network has phasor nodal equations $\mathbf{Y}\mathbf{V} = \mathbf{I}$ with

$$\mathbf{Y} = \begin{bmatrix} 2 + j3 & -j3 \\ -j3 & 1 + j4 \end{bmatrix} \text{ mS}.$$

Find $\mathbf{Y}^{\mathsf{H}}$ and the Hermitian part $\mathbf{H}_1 = \tfrac{1}{2}(\mathbf{Y} + \mathbf{Y}^{\mathsf{H}})$.

Step 1. Conjugate every entry, then transpose: $\mathbf{Y}^{*} = \begin{bmatrix} 2 - j3 & j3 \\ j3 & 1 - j4 \end{bmatrix}$, which is already symmetric, so $\mathbf{Y}^{\mathsf{H}} = \mathbf{Y}^{*}$.

Step 2. $\mathbf{H}_1 = \tfrac{1}{2}(\mathbf{Y} + \mathbf{Y}^{*}) = \operatorname{Re}\{\mathbf{Y}\} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$ mS.

Reading. For a reciprocal network $\mathbf{Y}$ is complex *symmetric* ($\mathbf{Y}^{\mathsf{T}} = \mathbf{Y}$) but not Hermitian. Its Hermitian part is the conductance matrix: average power is $P = \tfrac{1}{2}\operatorname{Re}\{\mathbf{V}^{\mathsf{H}}\mathbf{I}\} = \tfrac{1}{2}\mathbf{V}^{\mathsf{H}}\mathbf{H}_1\mathbf{V}$, which only involves the resistive branches. The $j3$ susceptance (a capacitor between the nodes) and the extra $j1$ at node 2 (a capacitor to ground) store energy but dissipate none.

Common Pitfall

For complex matrices, $\mathbf{A}^{\mathsf{T}}$ and $\mathbf{A}^{\mathsf{H}}$ are different, and MATLAB's `'` operator is the *conjugate* transpose; `.'` is the plain transpose. Using `'` on a complex symmetric admittance matrix silently conjugates it — a classic source of sign errors in reactive power. NumPy's `.T` is the plain transpose; use `.conj().T` for $\mathbf{A}^{\mathsf{H}}$.

Engineering Note

Matrix symmetry is not a mathematical nicety; it is reciprocity. Tellegen's theorem implies that any network made of resistors, capacitors, inductors and coupled inductors has a symmetric nodal admittance matrix. Simulators check symmetry to choose faster solvers (Cholesky rather than general $\mathbf{LU}$, §8.9), and a non-symmetric matrix in a model that should be passive is a quick bug detector.

8.2 Matrix multiplication

Motivation: composition

Multiplication of matrices is defined the way it is because it represents *composition* of linear maps. If $\mathbf{y} = \mathbf{B}\mathbf{x}$ and $\mathbf{z} = \mathbf{A}\mathbf{y}$, we want a single matrix $\mathbf{C}$ with $\mathbf{z} = \mathbf{C}\mathbf{x}$. Writing out the sums,

$$z_i = \sum_{k} a_{ik} y_k = \sum_{k} a_{ik} \sum_{j} b_{kj} x_j = \sum_{j} \Big(\sum_{k} a_{ik} b_{kj} \Big) x_j,$$

so the right definition is $c_{ij} = \sum_k a_{ik} b_{kj}$. Two-port networks in cascade, coordinate rotations followed by scalings, and a filter followed by a sampler are all compositions, and all become products.

Matrix product

If $\mathbf{A}$ is $m \times p$ and $\mathbf{B}$ is $p \times n$, the product $\mathbf{C} = \mathbf{A}\mathbf{B}$ is the $m \times n$ matrix

$$c_{ij} = \sum_{k=1}^{p} a_{ik} b_{kj}, \qquad i = 1, \dots, m,\ j = 1, \dots, n.$$

The product is defined only when the inner dimensions agree: $(m \times p)(p \times n) \to (m \times n)$.

The entry c_{ij} is the **dot product of row i of $\mathbf{A}$ with column j of $\mathbf{B}$**. Three other readings are equally important:

1. **Column view.** Column j of $\mathbf{A}\mathbf{B}$ is $\mathbf{A}$ times column j of $\mathbf{B}$. In particular $\mathbf{A}\mathbf{x} = x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 + \cdots + x_n \mathbf{a}_n$ is a **linear combination of the columns** $\mathbf{a}_k$ of $\mathbf{A}$. Solving $\mathbf{A}\mathbf{x} = \mathbf{b}$ means finding the weights that combine the columns into $\mathbf{b}$.
2. **Row view.** Row i of $\mathbf{A}\mathbf{B}$ is row i of $\mathbf{A}$ times $\mathbf{B}$: a linear combination of the rows of $\mathbf{B}$. Row operations in elimination (§8.7) are left-multiplications.
3. **Outer-product view.** $\mathbf{A}\mathbf{B} = \sum_k \mathbf{a}_k \mathbf{b}_k^{\mathsf{T}}$, a sum of rank-one matrices, column k of $\mathbf{A}$ times row k of $\mathbf{B}$. This is the view behind $\mathbf{LU}$ factorisation and the Sherman–Morrison formula.

For the running example the column view reads

$$\mathbf{G}\mathbf{v} = 4 \begin{bmatrix} 4 \\ -2 \\ -1 \end{bmatrix} + 3 \begin{bmatrix} -2 \\ 4 \\ -1 \end{bmatrix} + 2 \begin{bmatrix} -1 \\ -1 \\ 4 \end{bmatrix} = \begin{bmatrix} 16 - 6 - 2 \\ -8 + 12 - 2 \\ -4 - 3 + 8 \end{bmatrix} = \begin{bmatrix} 8 \\ 2 \\ 1 \end{bmatrix},$$

which confirms the claimed solution $\mathbf{v} = (4, 3, 2)^{\mathsf{T}}$. Column k of $\mathbf{G}$ is the vector of currents that must be injected to raise node k to $1\ \mathrm{V}$ with every other node held at $0\ \mathrm{V}$ — the circuit meaning of the column view.

Algebraic rules

Properties of the matrix product

Whenever the sizes allow the products:

1. Associative: $(\mathbf{A}\mathbf{B})\mathbf{C} = \mathbf{A}(\mathbf{B}\mathbf{C})$.
2. Distributive: $\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{A}\mathbf{B} + \mathbf{A}\mathbf{C}$, $(\mathbf{A} + \mathbf{B})\mathbf{C} = \mathbf{A}\mathbf{C} + \mathbf{B}\mathbf{C}$.
3. Scalars commute: $c(\mathbf{A}\mathbf{B}) = (c\mathbf{A})\mathbf{B} = \mathbf{A}(c\mathbf{B})$.
4. Identity: $\mathbf{I}_m \mathbf{A} = \mathbf{A} = \mathbf{A}\mathbf{I}_n$ for $\mathbf{A} \in \mathbb{C}^{m \times n}$.
5. Transpose reverses order: $(\mathbf{A}\mathbf{B})^{\mathsf{T}} = \mathbf{B}^{\mathsf{T}}\mathbf{A}^{\mathsf{T}}$, $(\mathbf{A}\mathbf{B})^{\mathsf{H}} = \mathbf{B}^{\mathsf{H}}\mathbf{A}^{\mathsf{H}}$.
6. Trace is cyclic: $\operatorname{tr}(\mathbf{A}\mathbf{B}) = \operatorname{tr}(\mathbf{B}\mathbf{A})$ (when both products are square).

But in general $\mathbf{A}\mathbf{B} \neq \mathbf{B}\mathbf{A}$ — matrix multiplication is **not commutative**.

Proofs. (1) The (i,j) entry of both sides is $\sum_k \sum_l a_{ik} b_{kl} c_{lj}$; only the order of summation differs. (5) $[(\mathbf{A}\mathbf{B})^{\mathsf{T}}]_{ij} = [\mathbf{A}\mathbf{B}]_{ji} = \sum_k a_{jk} b_{ki} = \sum_k [\mathbf{B}^{\mathsf{T}}]_{ik} [\mathbf{A}^{\mathsf{T}}]_{kj} = [\mathbf{B}^{\mathsf{T}}\mathbf{A}^{\mathsf{T}}]_{ij}$. (6) $\operatorname{tr}(\mathbf{A}\mathbf{B}) = \sum_i \sum_k a_{ik} b_{ki} = \sum_k \sum_i b_{ki} a_{ik} = \operatorname{tr}(\mathbf{B}\mathbf{A})$. $\square$

Other familiar rules of arithmetic also fail:

- $\mathbf{A}\mathbf{B} = \mathbf{0}$ does **not** imply $\mathbf{A} = \mathbf{0}$ or $\mathbf{B} = \mathbf{0}$. Example: $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \mathbf{0}$.
- $\mathbf{A}\mathbf{B} = \mathbf{A}\mathbf{C}$ does **not** imply $\mathbf{B} = \mathbf{C}$ unless $\mathbf{A}$ is invertible (§8.4).
- $(\mathbf{A} + \mathbf{B})^2 = \mathbf{A}^2 + \mathbf{A}\mathbf{B} + \mathbf{B}\mathbf{A} + \mathbf{B}^2$, which equals $\mathbf{A}^2 + 2\mathbf{A}\mathbf{B} + \mathbf{B}^2$ only if $\mathbf{A}$ and $\mathbf{B}$ commute.

Quadratic forms and power

For a square $\mathbf{A}$ and vector $\mathbf{x}$ the scalar $\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} = \sum_i \sum_j a_{ij} x_i x_j$ is a **quadratic form**. Because $\mathbf{x}^{\mathsf{T}}\mathbf{K}\mathbf{x}$ is a scalar equal to its own transpose $\mathbf{x}^{\mathsf{T}}\mathbf{K}^{\mathsf{T}}\mathbf{x} = -\mathbf{x}^{\mathsf{T}}\mathbf{K}\mathbf{x}$ for skew $\mathbf{K}$, it vanishes, and only the symmetric part of $\mathbf{A}$ contributes to a quadratic form. For a resistive network, the total power absorbed is

$$P = \mathbf{v}^{\mathsf{T}} \mathbf{i}_s = \mathbf{v}^{\mathsf{T}} \mathbf{G} \mathbf{v}.$$

For the running example $P = (4, 3, 2) \cdot (8, 2, 1) = 32 + 6 + 2 = 40\ \mathrm{mW}$. Since every resistor absorbs non-negative power, $\mathbf{v}^{\mathsf{T}}\mathbf{G}\mathbf{v} > 0$ for every nonzero $\mathbf{v}$ when the circuit is connected to ground: $\mathbf{G}$ is **positive definite**. That property is studied in Chapter 10 and exploited by the Cholesky factorisation in §8.9.

Incidence matrices: building $\mathbf{G}$ by multiplication

The cleanest derivation of nodal analysis is a matrix product. Number the nodes and branches and give each branch a reference direction (from its first to its second node in the table above).

Reduced incidence matrix

For a network with n non-reference nodes and b branches, the **reduced incidence matrix** $\mathbf{A}$ is $n \times b$ with

$$a_{ik} = \begin{cases} +1 & \text{branch } k \text{ leaves node } i, \\ -1 & \text{branch } k \text{ enters node } i, \\ 0 & \text{otherwise.} \end{cases}$$

The row for the ground node is omitted.

With branch currents $\mathbf{i}_b$, branch voltages $\mathbf{v}_b$ and branch conductances collected in $\mathbf{G}_b = \operatorname{diag}(g_1, \dots, g_b)$, the three laws of circuit theory are three matrix equations:

$$\begin{aligned} &\text{KCL:} && \mathbf{A}\mathbf{i}_b = \mathbf{i}_s, \\ &\text{KVL:} && \mathbf{v}_b = \mathbf{A}^{\mathsf{T}}\mathbf{v}, \\ &\text{Ohm:} && \mathbf{i}_b = \mathbf{G}_b\mathbf{v}_b. \end{aligned}$$

Substituting, $\mathbf{A}\mathbf{G}_b\mathbf{A}^{\mathsf{T}}\mathbf{v} = \mathbf{i}_s$, so

$$\mathbf{G} = \mathbf{A}\mathbf{G}_b\mathbf{A}^{\mathsf{T}}.$$

Symmetry is now automatic: $\mathbf{G}^{\mathsf{T}} = (\mathbf{A}^{\mathsf{T}})^{\mathsf{T}}\mathbf{G}_b^{\mathsf{T}}\mathbf{A}^{\mathsf{T}} = \mathbf{A}\mathbf{G}_b\mathbf{A}^{\mathsf{T}} = \mathbf{G}$. And positivity is automatic too: $\mathbf{v}^{\mathsf{T}}\mathbf{G}\mathbf{v} = \mathbf{v}_b^{\mathsf{T}}\mathbf{G}_b\mathbf{v}_b = \sum_k g_k v_{b,k}^2 \geq 0$.

Worked Example 8.4 — Assembling the running example from its incidence matrix

Branches a, b, c, d, e, f in that order, directed from the first-named node to the second:

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & -1 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 1 & -1 \end{bmatrix}, \qquad \mathbf{G}_b = \operatorname{diag}(1, 2, 1, 1, 2, 1).$$

Step 1. $\mathbf{A}\mathbf{G}_b$ scales column k of $\mathbf{A}$ by g_k (right-multiplying by a diagonal matrix scales columns):

$$\mathbf{A}\mathbf{G}_b = \begin{bmatrix} 1 & 2 & 0 & 0 & 0 & 1 \\ 0 & -2 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 2 & -1 \end{bmatrix}.$$

Step 2. Multiply by $\mathbf{A}^{\mathsf{T}}$ (6×3). Entry $(1,1)$ is row 1 of $\mathbf{A}\mathbf{G}_b$ dotted with row 1 of $\mathbf{A}$: $1 + 2 + 1 = 4$. Entry $(1,2)$: row 1 of $\mathbf{A}\mathbf{G}_b$ with row 2 of $\mathbf{A}$: $2 \cdot (-1) = -2$. Entry $(1,3)$: $1 \cdot (-1) = -1$. Entry $(2,2)$: $(-2)(-1) + 1 + 1 = 4$. Entry $(2,3)$: $1 \cdot (-1) = -1$. Entry $(3,3)$: $1 + 2 + 1 = 4$. By symmetry the rest follow:

$$\mathbf{A}\mathbf{G}_b\mathbf{A}^{\mathsf{T}} = \begin{bmatrix} 4 & -2 & -1 \\ -2 & 4 & -1 \\ -1 & -1 & 4 \end{bmatrix} = \mathbf{G}. \checkmark$$

Step 3. Branch voltages: $\mathbf{v}_b = \mathbf{A}^{\mathsf{T}}\mathbf{v} = (4,\ 1,\ 1,\ 3,\ 2,\ 2)^{\mathsf{T}}$ V. Branch currents $\mathbf{i}_b = \mathbf{G}_b\mathbf{v}_b = (4, 2, 1, 3, 4, 2)^{\mathsf{T}}$ mA. KCL check: $\mathbf{A}\mathbf{i}_b = (4 + 2 + 2,\ -2 + 1 + 3,\ -1 + 4 - 2)^{\mathsf{T}} = (8, 2, 1)^{\mathsf{T}}$. ✓ Power: $\sum g_k v_{b,k}^2 = 16 + 2 + 1 + 9 + 8 + 4 = 40$ mW, matching $\mathbf{v}^{\mathsf{T}}\mathbf{i}_s$.

Non-commutativity in practice: cascaded two-ports

A two-port is described by its **ABCD (transmission) matrix**, relating input voltage and current to output voltage and current:

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ I_2 \end{bmatrix}.$$

Because the output of one stage is the input of the next, a cascade is a product, taken in the physical order of the stages. A series impedance Z has $\begin{bmatrix} 1 & Z \\ 0 & 1 \end{bmatrix}$, a shunt admittance Y has $\begin{bmatrix} 1 & 0 \\ Y & 1 \end{bmatrix}$.

Worked Example 8.5 — Order matters (an L-section both ways round)

A series resistor $R = 2\ \Omega$ is followed by a shunt conductance $G = 0.5$ S. Find the cascade ABCD matrix, then repeat with the order reversed.

Series then shunt:

$$\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0.5 & 1 \end{bmatrix} = \begin{bmatrix} 1+1 & 0+2 \\ 0+0.5 & 0+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 0.5 & 1 \end{bmatrix}.$$

Shunt then series:

$$\begin{bmatrix} 1 & 0 \\ 0.5 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0.5 & 2 \end{bmatrix}.$$

Reading. The open-circuit voltage gain is $V_2/V_1 = 1/A$ (set $I_2 = 0$): 0.5 in the first case (a divider) and 1 in the second (the shunt element sits across the source and does nothing to V_2). Different physical circuits, different products. Both matrices have determinant $AD - BC = 1$, the hallmark of a reciprocal two-port (§8.3).

Block matrices

Partitioning matrices into blocks and multiplying the blocks as if they were scalars is legitimate provided the partitions are conformable and the order of block factors is kept:

$$\begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix} \begin{bmatrix} \mathbf{B}_{11} & \mathbf{B}_{12} \\ \mathbf{B}_{21} & \mathbf{B}_{22} \end{bmatrix} = \begin{bmatrix} \mathbf{A}_{11}\mathbf{B}_{11} + \mathbf{A}_{12}\mathbf{B}_{21} & \mathbf{A}_{11}\mathbf{B}_{12} + \mathbf{A}_{12}\mathbf{B}_{22} \\ \mathbf{A}_{21}\mathbf{B}_{11} + \mathbf{A}_{22}\mathbf{B}_{21} & \mathbf{A}_{21}\mathbf{B}_{12} + \mathbf{A}_{22}\mathbf{B}_{22} \end{bmatrix}.$$

Block structure appears naturally in **modified nodal analysis** (MNA), where unknown node voltages $\mathbf{v}$ and unknown voltage-source currents $\mathbf{j}$ are stacked:

$$\begin{bmatrix} \mathbf{G} & \mathbf{B} \\ \mathbf{B}^{\mathsf{T}} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{v} \\ \mathbf{j} \end{bmatrix} = \begin{bmatrix} \mathbf{i}_s \\ \mathbf{e} \end{bmatrix},$$

where $\mathbf{B}$ says which nodes each voltage source connects and $\mathbf{e}$ holds the source voltages. Eliminating one block in terms of the other gives the **Schur complement**, a block version of Gaussian elimination (§8.7).

Worked Example 8.6 — Matrix powers and a recursive network

A ladder of identical L-sections from Worked Example 8.5 (series 2 Ω then shunt 0.5 S) has $\mathbf{T} = \begin{bmatrix} 2 & 2 \\ 0.5 & 1 \end{bmatrix}$. Find the ABCD matrix of three sections and the open-circuit voltage gain.

Step 1. $\mathbf{T}^2 = \begin{bmatrix} 2 & 2 \\ 0.5 & 1 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 0.5 & 1 \end{bmatrix} = \begin{bmatrix} 4+1 & 4+2 \\ 1+0.5 & 1+1 \end{bmatrix} = \begin{bmatrix} 5 & 6 \\ 1.5 & 2 \end{bmatrix}$.

Step 2. $\mathbf{T}^3 = \mathbf{T}^2\mathbf{T} = \begin{bmatrix} 10+3 & 10+6 \\ 3+1 & 3+2 \end{bmatrix} = \begin{bmatrix} 13 & 16 \\ 4 & 5 \end{bmatrix}$.

Step 3. Gain $V_2/V_1 = 1/A = 1/13 \approx 0.0769$. Check reciprocity: $\det \mathbf{T}^3 = 65 - 64 = 1$. ✓

Powers of a matrix are computed far more cleverly via eigenvalues (Chapter 10); here the attenuation per section tends to the reciprocal of the larger eigenvalue of $\mathbf{T}$, which is $(3 + \sqrt{5})/2 \approx 2.618$.

Cost of multiplication

Each entry of an $m \times p$ by $p \times n$ product costs p multiplications and $p - 1$ additions, so the whole product costs about $2mpn$ floating-point operations (**flops**). For two $n \times n$ matrices this is $2n^3$; a matrix–vector product is $2n^2$. These counts set the scale for everything in §§8.7–8.11: a dense $n = 10^4$ matrix–matrix product is 2×10^{12} flops, a couple of seconds on a modern CPU core with vector units, while a dense $n = 10^6$ problem is out of reach — which is why sparsity (§8.11) matters.

Associativity lets us choose the cheapest bracketing. For $\mathbf{A} \in \mathbb{R}^{n\times n}$, $\mathbf{B} \in \mathbb{R}^{n\times n}$ and $\mathbf{x} \in \mathbb{R}^n$, $(\mathbf{AB})\mathbf{x}$ costs $2n^3 + 2n^2$ but $\mathbf{A}(\mathbf{Bx})$ costs only $4n^2$.

Common Pitfall

"Multiplying matrices entry by entry" (the **Hadamard product** $\mathbf{A} \odot \mathbf{B}$, MATLAB `.*`, NumPy `*`) is a different operation from the matrix product (MATLAB `*`, NumPy `@`). In NumPy, `A*x` with a 2-D array `A` and 1-D `x` silently broadcasts and returns a matrix, not $\mathbf{Ax}$. Always check the shape of the result.

Engineering Note

Chain matrices (ABCD), scattering-transfer matrices (T-parameters) in RF design, Jones matrices in optics and the ray-transfer matrices of lens systems are all multiplied in the physical order in which the signal meets the elements. Write the product from left to right in the order the wave travels from input to output, and check the first and last factors against a sketch.

8.3 Determinants

Motivation

The determinant is a single number computed from a square matrix that answers the most important yes/no question about it: *is the matrix invertible?* — equivalently, does $\mathbf{A}\mathbf{x} = \mathbf{b}$ have exactly one solution for every $\mathbf{b}$? It also measures how the linear map $\mathbf{x} \mapsto \mathbf{A}\mathbf{x}$ scales areas and volumes (the Jacobian determinant of Chapter 6 is exactly this, applied to the local linearisation of a nonlinear map), it gives closed-form solutions of small systems (Cramer's rule), and its vanishing defines the eigenvalues of Chapter 10 through $\det(\mathbf{A} - \lambda\mathbf{I}) = 0$. In circuit theory the determinant of the nodal or mesh matrix is the denominator of every network function, so its zeros as a function of s are the natural frequencies (poles) of the network.

Definition

For small matrices:

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc,$$

$$\det \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31}).$$

The 3×3 formula is a pattern that generalises.

Minor, cofactor, determinant

Let $\mathbf{A}$ be $n \times n$. The **minor** M_{ij} is the determinant of the $(n-1) \times (n-1)$ matrix obtained by deleting row i and column j. The **cofactor** is $C_{ij} = (-1)^{i+j} M_{ij}$. The **determinant** is defined recursively by $\det[a] = a$ for 1×1 and, for $n \geq 2$, by **cofactor (Laplace) expansion along the first row**:

$$\det \mathbf{A} = \sum_{j=1}^{n} a_{1j} C_{1j}.$$

We also write $|\mathbf{A}|$ or Δ.

The sign factor $(-1)^{i+j}$ follows a chessboard pattern beginning with $+$ in the top-left corner:

$$\begin{bmatrix} + & - & + & \cdots \\ - & + & - & \cdots \\ + & - & + & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}.$$

Laplace expansion along any row or column

For any fixed row i or any fixed column j,

$$\det \mathbf{A} = \sum_{k=1}^{n} a_{ik} C_{ik} = \sum_{k=1}^{n} a_{kj} C_{kj}.$$

Moreover, expanding a row against the cofactors of a *different* row gives zero: $\sum_k a_{ik} C_{lk} = 0$ for $i \neq l$.

The equivalent closed form, which makes the theorem transparent, is the **Leibniz formula**

$$\det \mathbf{A} = \sum_{\sigma} \operatorname{sgn}(\sigma)\, a_{1\sigma(1)} a_{2\sigma(2)} \cdots a_{n\sigma(n)},$$

summed over all $n!$ permutations σ of $\{1, \dots, n\}$, with $\operatorname{sgn}\sigma = \pm 1$ according to whether σ is an even or odd number of swaps. Each term takes exactly one entry from every row and every column. Expanding along any row or column just groups these $n!$ terms differently. The "different row gives zero" statement holds because $\sum_k a_{ik} C_{lk}$ is the determinant of a matrix whose row l has been replaced by a copy of row i — a matrix with two equal rows, whose determinant is zero (property 3 below).

The $n!$ terms make the definition useless for computation beyond $n \approx 5$: for $n = 25$, $25! \approx 1.6 \times 10^{25}$ products would take longer than the age of the universe at 10^9 products per second. Row reduction (below) needs only about $\tfrac{2}{3}n^3$ flops.

Properties

Properties of determinants

For $n \times n$ matrices $\mathbf{A}, \mathbf{B}$ and scalar c:

1. $\det \mathbf{A}^{\mathsf{T}} = \det \mathbf{A}$; $\det \mathbf{A}^{\mathsf{H}} = (\det \mathbf{A})^{*}$.
2. **Row swap** changes the sign of the determinant.
3. If two rows (or columns) are equal, or a row is all zeros, $\det \mathbf{A} = 0$.
4. **Row scaling**: multiplying one row by c multiplies the determinant by c. Hence $\det(c\mathbf{A}) = c^{n} \det \mathbf{A}$.
5. **Row replacement**: adding a multiple of one row to another leaves the determinant unchanged.
6. The determinant of a triangular (in particular diagonal) matrix is the product of its diagonal entries.
7. **Product rule**: $\det(\mathbf{AB}) = \det \mathbf{A} \, \det \mathbf{B}$.
8. $\mathbf{A}$ is invertible if and only if $\det \mathbf{A} \neq 0$, and then $\det(\mathbf{A}^{-1}) = 1/\det \mathbf{A}$.
9. The determinant is linear in each row separately (multilinearity).

Proof sketches. (1) Transposing maps each Leibniz term to a term of the same sign. (2) Swapping two rows composes every permutation with a transposition, flipping every sign. (3) Swapping two equal rows leaves $\mathbf{A}$ unchanged but negates $\det \mathbf{A}$, so $\det \mathbf{A} = -\det \mathbf{A} = 0$; a zero row puts a zero factor in every Leibniz term. (4) and (9) Each Leibniz term contains exactly one factor from each row. (5) By multilinearity, $\det(\dots, \mathbf{r}_i + c\mathbf{r}_l, \dots) = \det(\dots, \mathbf{r}_i, \dots) + c \det(\dots, \mathbf{r}_l, \dots)$, and the second determinant has two equal rows. (6) Expand repeatedly down the first column: only the diagonal entry survives each time. (7) Row-reduce $\mathbf{A}$ with elementary matrices whose determinants are known (§8.4), then use (6). (8) follows from (7), $\det \mathbf{I} = 1$, and the fact that row reduction of a singular matrix produces a zero row. □

Properties 2, 4 and 5 are exactly the three row operations of Gaussian elimination. That gives the practical algorithm:

> **Determinant by elimination.** Reduce $\mathbf{A}$ to upper triangular $\mathbf{U}$ using row replacements and s row swaps (no scaling). Then $\det \mathbf{A} = (-1)^{s} \, u_{11} u_{22} \cdots u_{nn}$, the signed product of the pivots.

Geometric meaning

For a 2×2 matrix with columns $\mathbf{a}_1 = (a, c)^{\mathsf{T}}$ and $\mathbf{a}_2 = (b, d)^{\mathsf{T}}$, $|ad - bc|$ is the area of the parallelogram spanned by the columns; the sign records orientation (positive if $\mathbf{a}_1$ turns anticlockwise to reach $\mathbf{a}_2$). For 3×3, $|\det \mathbf{A}|$ is the volume of the parallelepiped spanned by the columns — the scalar triple product $\mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)$ of Chapter 7. In general the linear map $\mathbf{x} \mapsto \mathbf{Ax}$ multiplies every n-dimensional volume by $|\det \mathbf{A}|$. A zero determinant means the columns are squashed into a lower-dimensional set — they are linearly dependent — and information is lost; no inverse map can recover it. This is the geometric content of property 8, and it also explains the product rule: applying $\mathbf{B}$ then $\mathbf{A}$ scales volumes by $|\det \mathbf{B}|$ and then by $|\det \mathbf{A}|$.

Cramer's rule

Cramer's rule

If $\det \mathbf{A} \neq 0$, the unique solution of $\mathbf{Ax} = \mathbf{b}$ is

$$x_k = \frac{\det \mathbf{A}_k}{\det \mathbf{A}}, \qquad k = 1, \dots, n,$$

where $\mathbf{A}_k$ is $\mathbf{A}$ with column k replaced by $\mathbf{b}$.

Proof. Write $\mathbf{b} = \sum_j x_j \mathbf{a}_j$ (column view). Replace column k of $\mathbf{A}$ by this sum and use multilinearity in columns: $\det \mathbf{A}_k = \sum_j x_j \det(\mathbf{a}_1, \dots, \mathbf{a}_j, \dots, \mathbf{a}_n)$ with $\mathbf{a}_j$ in slot k. Every term with $j \neq k$ has a repeated column and vanishes, leaving $\det \mathbf{A}_k = x_k \det \mathbf{A}$. □

Cramer's rule is the right tool for 2×2 and 3×3 systems with symbolic entries — for example, to obtain a transfer function in terms of s and component values — and the wrong tool for numerical systems larger than that, where it costs $n + 1$ determinants and is numerically inferior to elimination.

Worked Example 8.7 — Determinant of the running example, two ways

Find $\det \mathbf{G}$ for $\mathbf{G} = \begin{bmatrix} 4 & -2 & -1 \\ -2 & 4 & -1 \\ -1 & -1 & 4 \end{bmatrix}$.

Method 1: cofactor expansion along row 1.

$$\begin{aligned} \det \mathbf{G} &= 4 \begin{vmatrix} 4 & -1 \\ -1 & 4 \end{vmatrix} - (-2) \begin{vmatrix} -2 & -1 \\ -1 & 4 \end{vmatrix} + (-1) \begin{vmatrix} -2 & 4 \\ -1 & -1 \end{vmatrix} \\ &= 4(16 - 1) + 2(-8 - 1) - (2 + 4) = 60 - 18 - 6 = 36. \end{aligned}$$

Method 2: elimination. $R_2 \leftarrow R_2 + \tfrac{1}{2} R_1$ and $R_3 \leftarrow R_3 + \tfrac{1}{4} R_1$ give rows $(0, 3, -\tfrac{3}{2})$ and $(0, -\tfrac{3}{2}, \tfrac{15}{4})$. Then $R_3 \leftarrow R_3 + \tfrac{1}{2} R_2$ gives $(0, 0, 3)$. No swaps, so

$$\det \mathbf{G} = 4 \cdot 3 \cdot 3 = 36.$$

The units are mS^3. Method 2 is the one that scales.

Worked Example 8.8 — A determinant needing a row swap

The modified nodal formulation of §8.7 (Worked Example 8.22) leads to

$$\mathbf{M} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 4 & -2 & -1 \\ 0 & -2 & 4 & -1 \\ 0 & -1 & -1 & 4 \end{bmatrix}.$$

Find $\det \mathbf{M}$.

Step 1. $m_{11} = 0$, so swap $R_1 \leftrightarrow R_2$ ($s = 1$):

$$\begin{bmatrix} 1 & 4 & -2 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & -2 & 4 & -1 \\ 0 & -1 & -1 & 4 \end{bmatrix}.$$

Step 2. $R_3 \leftarrow R_3 + 2R_2$, $R_4 \leftarrow R_4 + R_2$: rows 3 and 4 become $(0, 0, 4, -1)$ and $(0, 0, -1, 4)$.

Step 3. $R_4 \leftarrow R_4 + \tfrac{1}{4} R_3$: row 4 becomes $(0, 0, 0, \tfrac{15}{4})$.

Step 4. $\det \mathbf{M} = (-1)^1 \cdot 1 \cdot 1 \cdot 4 \cdot \tfrac{15}{4} = -15$.

Check by expansion along row 1: only $m_{12} = 1$ is nonzero, so $\det \mathbf{M} = (-1)^{1+2} M_{12}$, and $M_{12} = \det \begin{bmatrix} 1 & -2 & -1 \\ 0 & 4 & -1 \\ 0 & -1 & 4 \end{bmatrix} = 1 \cdot (16 - 1) = 15$. Hence -15. ✓

(Choosing the sparsest row for the expansion is always worth a glance.)

Two-mesh example and its geometry

Consider a two-mesh resistive circuit: mesh 1 contains a 2 kΩ resistor and a 7 V source, mesh 2 contains a 4 kΩ resistor and a 1 V source, and the two meshes share a 3 kΩ resistor. With clockwise mesh currents i_1, i_2 in mA, KVL gives the **mesh equations** $\mathbf{R}\mathbf{i} = \mathbf{v}_s$:

$$\begin{bmatrix} 5 & -3 \\ -3 & 7 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 7 \\ 1 \end{bmatrix}.$$

The diagonal entries are the total resistance around each mesh, the off-diagonal entries minus the shared resistance — the mesh-analysis mirror image of the nodal rule. Units: $\text{k}\Omega \times \text{mA} = \text{V}$.

Worked Example 8.9 — Cramer's rule for the two-mesh circuit

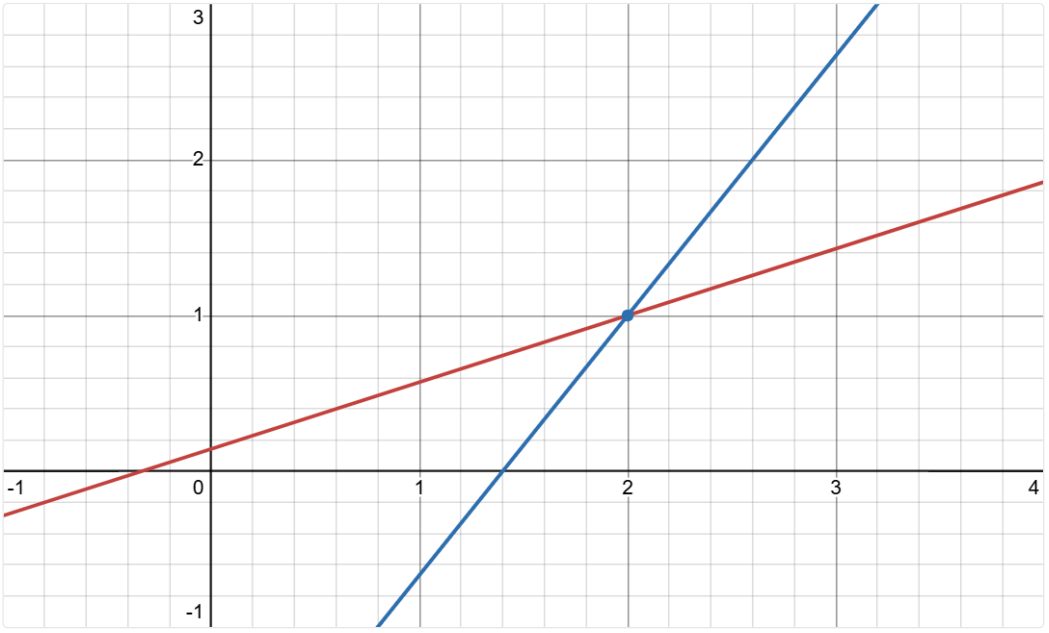
Step 1. $\det \mathbf{R} = 5 \cdot 7 - (-3)(-3) = 35 - 9 = 26\ \text{k}\Omega^2$.

Step 2.

$$i_1 = \frac{\begin{vmatrix} 7 & -3 \\ 1 & 7 \end{vmatrix}}{26} = \frac{49 + 3}{26} = 2\ \text{mA}, \qquad i_2 = \frac{\begin{vmatrix} 5 & 7 \\ -3 & 1 \end{vmatrix}}{26} = \frac{5 + 21}{26} = 1\ \text{mA}.$$

Step 3 (check). $5(2) - 3(1) = 7$ ✓, $-3(2) + 7(1) = 1$ ✓. The shared resistor carries $i_1 - i_2 = 1$ mA.

Geometry. Each equation is a straight line in the (i_1, i_2) plane; the solution is their intersection. Because $\det \mathbf{R} \neq 0$ the lines are not parallel and meet in exactly one point.



The two mesh equations as lines in the (i_1, i_2) plane (horizontal axis i_1, vertical axis i_2, both in mA). They cross at the unique solution $(2, 1)$; a nonzero determinant guarantees the slopes $5/3$ and $3/7$ differ.

Worked Example 8.10 — Cramer's rule for one node voltage, symbolically

In the running example, suppose only node 1 is driven, by a current I, and we want the transfer resistance v_3/I.

Step 1. $\mathbf{i}_s = (I, 0, 0)^{\mathsf{T}}$. Replace column 3 of $\mathbf{G}$:

$$\det \mathbf{G}_3 = \begin{vmatrix} 4 & -2 & I \\ -2 & 4 & 0 \\ -1 & -1 & 0 \end{vmatrix} = I \begin{vmatrix} -2 & 4 \\ -1 & -1 \end{vmatrix} = I(2 + 4) = 6I,$$

expanding along column 3, where only the top entry is nonzero and its sign factor is $(-1)^{1+3} = +1$.

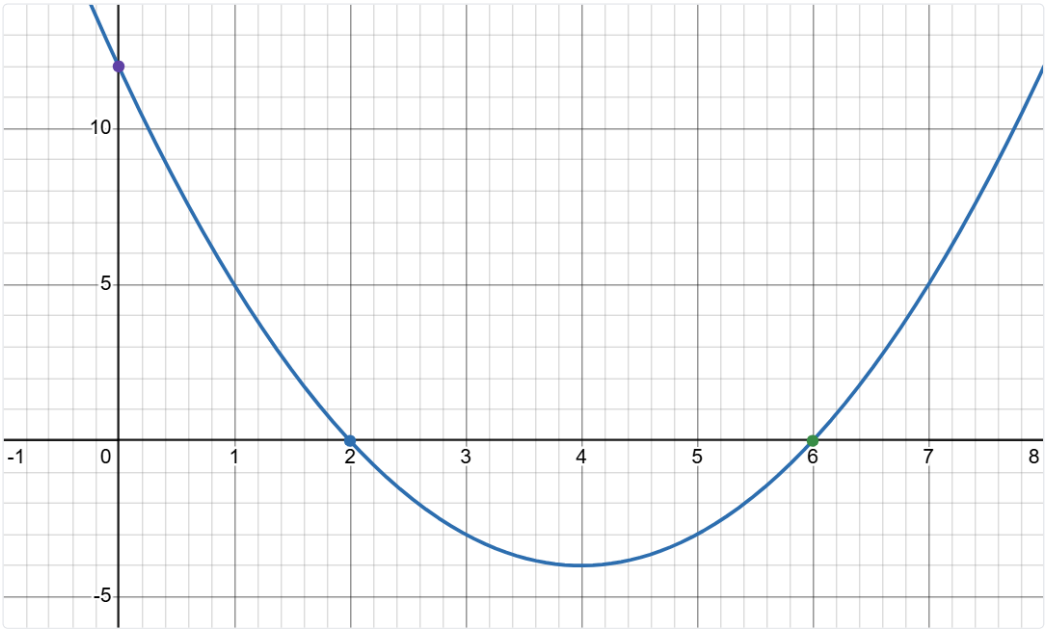
Step 2. $v_3 = 6I/36 = I/6$, so the transfer resistance is $v_3/I = 1/6\ \text{k}\Omega \approx 167\ \Omega$ (since $1/\text{mS} = \text{k}\Omega$).

Step 3 (full solution check). With the original sources $(8, 2, 1)$, Cramer gives $\det \mathbf{G}_1 = 144$, $\det \mathbf{G}_2 = 108$, $\det \mathbf{G}_3 = 72$, so $\mathbf{v} = (144, 108, 72)/36 = (4, 3, 2)\ \text{V}$. ✓

Determinant as a function of a parameter

When a component value varies, the determinant becomes a polynomial in that value, and its zeros mark the values at which the circuit equations become singular. Consider $\mathbf{A} - \lambda\mathbf{I}$ with $\mathbf{A} = \begin{bmatrix} 4 & -2 \\ -2 & 4 \end{bmatrix}$ (the nodal matrix of nodes 1 and 2 of a circuit like the running example with node 3 tied to ground):

$$p(\lambda) = \det \begin{bmatrix} 4 - \lambda & -2 \\ -2 & 4 - \lambda \end{bmatrix} = (4 - \lambda)^2 - 4 = \lambda^2 - 8\lambda + 12 = (\lambda - 2)(\lambda - 6).$$



$\det(\mathbf{A} - \lambda\mathbf{I})$ as a function of λ (horizontal axis). At $\lambda = 0$ it equals $\det \mathbf{A} = 12$; it vanishes at $\lambda = 2$ and $\lambda = 6$, where $\mathbf{A} - \lambda\mathbf{I}$ is singular. These are the eigenvalues of $\mathbf{A}$ (Chapter 10).

Common Pitfall

A small determinant does **not** mean a matrix is nearly singular. $\det(10^{-3}\mathbf{I}_{10}) = 10^{-30}$, yet $10^{-3}\mathbf{I}$ is perfectly conditioned; and the determinant changes with units ($\mathbf{G}$ in S instead of mS has a determinant 10^9 times smaller, with no change in the difficulty of solving the system). Use the condition number (§8.10), not the determinant, to judge numerical near-singularity.

Engineering Note

For a network with impedances depending on s, $\Delta(s) = \det \mathbf{Y}(s)$ is, up to a factor, the characteristic polynomial of the network; its roots are the natural frequencies. Every node voltage is $\Delta_k(s)/\Delta(s)$ by Cramer's rule, which is why all transfer functions of a network share the same denominator. Mason's gain formula for signal-flow graphs is a combinatorial way of evaluating these same determinants.

8.4 Matrix inverse

Motivation

For a scalar equation $ax = b$ we write $x = a^{-1}b$ provided $a \neq 0$. The matrix analogue $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$ is the natural way to *express* the solution of a linear system and to reason about it, even though (as we shall stress) it is rarely the way to *compute* it. The inverse also has direct physical meaning: the inverse of a nodal conductance matrix is the **nodal resistance (impedance) matrix**, whose entries are driving-point and transfer resistances; the inverse of a mesh resistance matrix is a conductance matrix; the inverse of a two-port's $\mathbf{Z}$-parameter matrix is its $\mathbf{Y}$-parameter matrix.

Definition and uniqueness

Inverse

A square matrix $\mathbf{A}$ is **invertible** (nonsingular) if there is a matrix $\mathbf{A}^{-1}$ with

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}.$$

A square matrix with no inverse is **singular**.

The inverse, when it exists, is unique: if $\mathbf{B}$ and $\mathbf{C}$ are both inverses then $\mathbf{B} = \mathbf{B}\mathbf{I} = \mathbf{B}(\mathbf{A}\mathbf{C}) = (\mathbf{B}\mathbf{A})\mathbf{C} = \mathbf{I}\mathbf{C} = \mathbf{C}$. For square matrices a one-sided inverse is automatically two-sided: if $\mathbf{B}\mathbf{A} = \mathbf{I}$ then $\mathbf{A}\mathbf{B} = \mathbf{I}$ too (this follows from the rank arguments of §8.5). Non-square matrices never have two-sided inverses; their one-sided and generalised inverses (the pseudo-inverse) belong to least-squares theory in Chapter 9.

If $\mathbf{A}$ is invertible, $\mathbf{A}\mathbf{x} = \mathbf{b}$ has the unique solution $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$: it is a solution because $\mathbf{A}(\mathbf{A}^{-1}\mathbf{b}) = \mathbf{b}$, and it is the only one because any solution satisfies $\mathbf{x} = \mathbf{A}^{-1}\mathbf{A}\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$.

Formulas

For a 2×2 matrix the inverse is: *swap the diagonal, negate the off-diagonal, divide by the determinant*,

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}, \qquad ad - bc \neq 0.$$

In general:

Adjugate formula

Let $\mathbf{C} = [C_{ij}]$ be the matrix of cofactors of $\mathbf{A}$ and define the **adjugate** $\operatorname{adj}\mathbf{A} = \mathbf{C}^{\mathsf{T}}$. Then

$$\mathbf{A}\,\operatorname{adj}\mathbf{A} = \operatorname{adj}\mathbf{A}\,\mathbf{A} = (\det\mathbf{A})\,\mathbf{I},$$

so if $\det\mathbf{A} \neq 0$,

$$\mathbf{A}^{-1} = \frac{1}{\det\mathbf{A}}\operatorname{adj}\mathbf{A}.$$

Proof. The (i,l) entry of $\mathbf{A}\operatorname{adj}\mathbf{A}$ is $\sum_k a_{ik}C_{lk}$. By the Laplace expansion theorem of §8.3 this equals $\det\mathbf{A}$ when $i = l$ and 0 when $i \neq l$. $\square$

The adjugate formula is the matrix version of Cramer's rule — indeed $(\operatorname{adj}\mathbf{A}\,\mathbf{b})_k = \det\mathbf{A}_k$ — and has the same uses (symbolic 2×2 and 3×3 work, theory) and the same drawback (cost) for anything larger.

Algebra of inverses

If $\mathbf{A}$ and $\mathbf{B}$ are invertible $n \times n$ matrices and $c \neq 0$:

1. $(\mathbf{A}^{-1})^{-1} = \mathbf{A}$.
2. $(\mathbf{A}\mathbf{B})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$ (order reverses — "socks and shoes").
3. $(\mathbf{A}^{\mathsf{T}})^{-1} = (\mathbf{A}^{-1})^{\mathsf{T}}$ and $(\mathbf{A}^{\mathsf{H}})^{-1} = (\mathbf{A}^{-1})^{\mathsf{H}}$; hence the inverse of a symmetric matrix is symmetric.
4. $(c\mathbf{A})^{-1} = c^{-1}\mathbf{A}^{-1}$.
5. $\mathbf{A}^{-1}$ of a diagonal matrix is the diagonal matrix of reciprocals; the inverse of an upper (lower) triangular matrix is upper (lower) triangular.

Proof of 2 and 3. $(\mathbf{A}\mathbf{B})(\mathbf{B}^{-1}\mathbf{A}^{-1}) = \mathbf{A}\mathbf{I}\mathbf{A}^{-1} = \mathbf{I}$. Transposing $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$ gives $(\mathbf{A}^{-1})^{\mathsf{T}}\mathbf{A}^{\mathsf{T}} = \mathbf{I}$. $\square$

Elementary matrices

Each row operation of elimination is a left-multiplication by an **elementary matrix**, obtained by doing that operation to $\mathbf{I}$:

Row operation	Elementary matrix (3×3 example)	Inverse	Determinant
$R_2 \leftarrow R_2 + mR_1$	$\mathbf{E} = \begin{bmatrix} 1 & 0 & 0 \\ m & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	same with $-m$	1
$R_2 \leftarrow cR_2$	$\operatorname{diag}(1, c, 1)$	$\operatorname{diag}(1, 1/c, 1)$	c
$R_1 \leftrightarrow R_2$	$\mathbf{P} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\mathbf{P}$ itself	-1

Every elementary matrix is invertible, and its inverse undoes the operation. If a sequence of row operations reduces $\mathbf{A}$ to $\mathbf{I}$, say $\mathbf{E}_k \cdots \mathbf{E}_2 \mathbf{E}_1 \mathbf{A} = \mathbf{I}$, then $\mathbf{A}^{-1} = \mathbf{E}_k \cdots \mathbf{E}_1$ — the same operations applied to $\mathbf{I}$. This is the Gauss–Jordan method for the inverse (§8.8), and it also proves the product rule for determinants claimed in §8.3.

The invertible matrix theorem

Invertible matrix theorem

For an $n \times n$ matrix $\mathbf{A}$ the following are equivalent:

1. $\mathbf{A}$ is invertible.
2. $\det \mathbf{A} \neq 0$.
3. $\mathbf{A}\mathbf{x} = \mathbf{b}$ has a unique solution for every $\mathbf{b}$.
4. $\mathbf{A}\mathbf{x} = \mathbf{0}$ has only the trivial solution $\mathbf{x} = \mathbf{0}$.
5. $\operatorname{rank} \mathbf{A} = n$ (§8.5); the columns (and rows) are linearly independent.
6. Gaussian elimination produces n nonzero pivots (with row swaps if needed).
7. $\mathbf{A}$ is a product of elementary matrices.
8. 0 is not an eigenvalue of $\mathbf{A}$ (Chapter 10).

For circuits, statement 4 has a direct reading: the nodal equations with all independent sources switched off have only the zero solution. A network in which some node "floats" (no DC path to ground) violates it — the floating part can sit at any voltage — and its nodal matrix is singular (§8.5, Worked Example 8.17).

Worked Example 8.11 — Inverse of the mesh matrix

For $\mathbf{R} = \begin{bmatrix} 5 & -3 \\ -3 & 7 \end{bmatrix}$ kΩ (§8.3), find $\mathbf{R}^{-1}$ and use it to solve $\mathbf{R}\mathbf{i} = (7, 1)^{\mathsf{T}}$.

Step 1. $\det \mathbf{R} = 26$. Swap diagonal, negate off-diagonal:

$$\mathbf{R}^{-1} = \frac{1}{26}\begin{bmatrix} 7 & 3 \\ 3 & 5 \end{bmatrix} \text{ mS}.$$

Step 2. $\mathbf{i} = \mathbf{R}^{-1}\mathbf{v}_s = \frac{1}{26}(49 + 3,\ 21 + 5)^{\mathsf{T}} = (2, 1)^{\mathsf{T}}$ mA. ✓

Step 3 (check). $\mathbf{R}\mathbf{R}^{-1} = \frac{1}{26}\begin{bmatrix} 35 - 9 & 15 - 15 \\ -21 + 21 & -9 + 35 \end{bmatrix} = \mathbf{I}$. ✓

Reading. $[\mathbf{R}^{-1}]_{12} = 3/26$ mS is the current in mesh 1 per volt of source in mesh 2. Its equality with $[\mathbf{R}^{-1}]_{21}$ is reciprocity, and it follows from property 3 of the inverse.

Worked Example 8.12 — The nodal resistance matrix of the running example

Find $\mathbf{Z} = \mathbf{G}^{-1}$ by the adjugate formula and interpret its entries.

Step 1 (cofactors). Using $\mathbf{G} = \begin{bmatrix} 4 & -2 & -1 \\ -2 & 4 & -1 \\ -1 & -1 & 4 \end{bmatrix}$:

$$C_{11} = +\begin{vmatrix} 4 & -1 \\ -1 & 4 \end{vmatrix} = 15, \quad C_{12} = -\begin{vmatrix} -2 & -1 \\ -1 & 4 \end{vmatrix} = 9, \quad C_{13} = +\begin{vmatrix} -2 & 4 \\ -1 & -1 \end{vmatrix} = 6,$$
$$C_{22} = +\begin{vmatrix} 4 & -1 \\ -1 & 4 \end{vmatrix} = 15, \quad C_{23} = -\begin{vmatrix} 4 & -2 \\ -1 & -1 \end{vmatrix} = 6, \quad C_{33} = +\begin{vmatrix} 4 & -2 \\ -2 & 4 \end{vmatrix} = 12,$$

and by symmetry $C_{21} = 9$, $C_{31} = 6$, $C_{32} = 6$.

Step 2. $\det \mathbf{G} = 36$ (Worked Example 8.7), so

$$\mathbf{Z} = \mathbf{G}^{-1} = \frac{1}{36}\begin{bmatrix} 15 & 9 & 6 \\ 9 & 15 & 6 \\ 6 & 6 & 12 \end{bmatrix} = \begin{bmatrix} \frac{5}{12} & \frac{1}{4} & \frac{1}{6} \\ \frac{1}{4} & \frac{5}{12} & \frac{1}{6} \\ \frac{1}{6} & \frac{1}{6} & \frac{1}{3} \end{bmatrix} \text{ kΩ}.$$

Step 3 (check). Row 1 of $\mathbf{G}$ times column 1 of $\operatorname{adj}\mathbf{G}$: $4(15) - 2(9) - 1(6) = 36$ ✓; row 1 times column 2: $4(9) - 2(15) - 6 = 0$ ✓.

Step 4 (solve). $\mathbf{v} = \mathbf{Z}\mathbf{i}_s = \frac{1}{36}(120 + 18 + 6,\ 72 + 30 + 6,\ 48 + 12 + 12)^{\mathsf{T}} = (4, 3, 2)^{\mathsf{T}}$ V. ✓

Reading. $Z_{11} = 5/12$ kΩ ≈ 417 Ω is the resistance seen looking into node 1 from ground (the Thévenin resistance at node 1); $Z_{31} = 1/6$ kΩ is the transfer resistance found by Cramer's rule in Worked Example 8.10. All entries are positive and each diagonal entry is the largest in its column — typical of resistive networks: injecting current at a node raises that node most.

Updating an inverse: Sherman–Morrison

Design work often changes one component and re-solves. Changing a conductance g between nodes p and q by δ changes $\mathbf{G}$ by a **rank-one** matrix $\delta\,\mathbf{u}\mathbf{u}^{\mathsf{T}}$ with $\mathbf{u} = \mathbf{e}_p - \mathbf{e}_q$ (or $\mathbf{u} = \mathbf{e}_p$ for a branch to ground), where $\mathbf{e}_k$ is column k of $\mathbf{I}$.

Sherman–Morrison formula

If $\mathbf{A}$ is invertible and $1 + \mathbf{v}^{\mathsf T}\mathbf{A}^{-1}\mathbf{u} \neq 0$, then

$$(\mathbf{A} + \mathbf{u}\mathbf{v}^{\mathsf T})^{-1} = \mathbf{A}^{-1} - \frac{\mathbf{A}^{-1}\mathbf{u}\,\mathbf{v}^{\mathsf T}\mathbf{A}^{-1}}{1 + \mathbf{v}^{\mathsf T}\mathbf{A}^{-1}\mathbf{u}}.$$

Consequently the new solution of $(\mathbf{A} + \mathbf{u}\mathbf{v}^{\mathsf T})\mathbf{x}' = \mathbf{b}$ is

$$\mathbf{x}' = \mathbf{x} - \frac{\mathbf{v}^{\mathsf T}\mathbf{x}}{1 + \mathbf{v}^{\mathsf T}\mathbf{w}}\,\mathbf{w}, \qquad \mathbf{x} = \mathbf{A}^{-1}\mathbf{b},\ \ \mathbf{w} = \mathbf{A}^{-1}\mathbf{u}.$$

Proof. Multiply $\mathbf{A} + \mathbf{u}\mathbf{v}^{\mathsf T}$ by the claimed inverse. With $\beta = \mathbf{v}^{\mathsf T}\mathbf{A}^{-1}\mathbf{u}$ (a scalar),

$$(\mathbf{A} + \mathbf{u}\mathbf{v}^{\mathsf T})\Big(\mathbf{A}^{-1} - \frac{\mathbf{A}^{-1}\mathbf{u}\mathbf{v}^{\mathsf T}\mathbf{A}^{-1}}{1+\beta}\Big) = \mathbf{I} + \mathbf{u}\mathbf{v}^{\mathsf T}\mathbf{A}^{-1} - \frac{\mathbf{u}\mathbf{v}^{\mathsf T}\mathbf{A}^{-1} + \beta\,\mathbf{u}\mathbf{v}^{\mathsf T}\mathbf{A}^{-1}}{1+\beta} = \mathbf{I}. \qquad \square$$

The update costs $O(n^2)$ given $\mathbf{w}$ (one extra solve with the existing factorisation), instead of $O(n^3)$ for a fresh solve. Its generalisation to rank-k changes is the **Woodbury identity** $(\mathbf{A} + \mathbf{U}\mathbf{C}\mathbf{V}^{\mathsf T})^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{U}(\mathbf{C}^{-1} + \mathbf{V}^{\mathsf T}\mathbf{A}^{-1}\mathbf{U})^{-1}\mathbf{V}^{\mathsf T}\mathbf{A}^{-1}$, used in Kalman filtering and in contingency analysis of power networks (removing a line is a rank-one change with $\delta = -g$).

Worked Example 8.13 — Doubling one resistor's conductance

In the running example, branch f (between nodes 1 and 3) is changed from 1 mS to 2 mS. Find the new node voltages without re-solving.

Step 1. $\delta = 1$, $\mathbf{u} = \mathbf{v} = \mathbf{e}_1 - \mathbf{e}_3 = (1, 0, -1)^{\mathsf T}$, so $\mathbf{G}' = \mathbf{G} + \mathbf{u}\mathbf{u}^{\mathsf T}$.

Step 2. $\mathbf{w} = \mathbf{Z}\mathbf{u} =$ column 1 minus column 3 of $\mathbf{Z} = (\tfrac{5}{12} - \tfrac{1}{6},\ \tfrac{1}{4} - \tfrac{1}{6},\ \tfrac{1}{6} - \tfrac{1}{3})^{\mathsf T} = (\tfrac{1}{4},\ \tfrac{1}{12},\ -\tfrac{1}{6})^{\mathsf T}$.

Step 3. $\mathbf{u}^{\mathsf T}\mathbf{w} = \tfrac{1}{4} + \tfrac{1}{6} = \tfrac{5}{12}$ — this is the resistance (in kΩ) seen by the branch between nodes 1 and 3 — and $\mathbf{u}^{\mathsf T}\mathbf{v}_{\text{old}} = 4 - 2 = 2$ V.

Step 4.

$$\mathbf{v}' = \begin{bmatrix}4\\3\\2\end{bmatrix} - \frac{2}{1 + \frac{5}{12}}\begin{bmatrix}\frac{1}{4}\\ \frac{1}{12}\\ -\frac{1}{6}\end{bmatrix} = \begin{bmatrix}4\\3\\2\end{bmatrix} - \frac{24}{17}\begin{bmatrix}\frac{1}{4}\\ \frac{1}{12}\\ -\frac{1}{6}\end{bmatrix} = \frac{1}{17}\begin{bmatrix}62\\49\\38\end{bmatrix} \approx \begin{bmatrix}3.647\\2.882\\2.235\end{bmatrix}\ \text{V}.$$

Step 5 (check). $\mathbf{G}' = \begin{bmatrix}5 & -2 & -2\\ -2 & 4 & -1\\ -2 & -1 & 5\end{bmatrix}$; row 1 times $(62, 49, 38)/17$ is $(310 - 98 - 76)/17 = 136/17 = 8$ ✓; row 3: $(-124 - 49 + 190)/17 = 17/17 = 1$ ✓.

Reading. Stiffening the link between nodes 1 and 3 pulls their voltages together: the difference falls from 2 V to $24/17 \approx 1.41$ V.

Block inverses, Schur complements and Kron reduction

Partition a system into the unknowns we care about ($\mathbf{x}_1$) and those we would like to eliminate ($\mathbf{x}_2$):

$$\begin{bmatrix}\mathbf{A}_{11} & \mathbf{A}_{12}\\ \mathbf{A}_{21} & \mathbf{A}_{22}\end{bmatrix}\begin{bmatrix}\mathbf{x}_1\\ \mathbf{x}_2\end{bmatrix} = \begin{bmatrix}\mathbf{b}_1\\ \mathbf{b}_2\end{bmatrix}.$$

If $\mathbf{A}_{22}$ is invertible, the second block row gives $\mathbf{x}_2 = \mathbf{A}_{22}^{-1}(\mathbf{b}_2 - \mathbf{A}_{21}\mathbf{x}_1)$; substituting into the first,

$$\underbrace{(\mathbf{A}_{11} - \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{A}_{21})}_{\mathbf{S},\ \text{Schur complement}}\mathbf{x}_1 = \mathbf{b}_1 - \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{b}_2.$$

Moreover $\det\mathbf{A} = \det\mathbf{A}_{22}\,\det\mathbf{S}$. In network theory this is **Kron reduction**: eliminating internal nodes leaves an equivalent network seen from the retained nodes. The star–delta transformation is the special case of eliminating a single node with three neighbours.

Worked Example 8.14 — Kron reduction of node 3

Eliminate v_3 from the running example and solve the reduced 2×2 system.

Step 1. $\mathbf{A}_{11} = \begin{bmatrix} 4 & -2 \\ -2 & 4 \end{bmatrix}$, $\mathbf{A}_{12} = \begin{bmatrix} -1 \\ -1 \end{bmatrix} = \mathbf{A}_{21}^{\mathsf{T}}$, $\mathbf{A}_{22} = [4]$, $\mathbf{b}_1 = (8, 2)^{\mathsf{T}}$, $b_2 = 1$.

Step 2.

$$\mathbf{S} = \begin{bmatrix} 4 & -2 \\ -2 & 4 \end{bmatrix} - \frac{1}{4}\begin{bmatrix} -1 \\ -1 \end{bmatrix}\begin{bmatrix} -1 & -1 \end{bmatrix} = \begin{bmatrix} 3.75 & -2.25 \\ -2.25 & 3.75 \end{bmatrix}, \qquad \mathbf{b}_1' = \begin{bmatrix} 8 \\ 2 \end{bmatrix} - \frac{1}{4}\begin{bmatrix} -1 \\ -1 \end{bmatrix}(1) = \begin{bmatrix} 8.25 \\ 2.25 \end{bmatrix}.$$

Step 3. $\det \mathbf{S} = 3.75^2 - 2.25^2 = 9$, and $\det \mathbf{A}_{22} \det \mathbf{S} = 4 \cdot 9 = 36 = \det \mathbf{G}$ ✓. Then

$$v_1 = \frac{8.25(3.75) + 2.25(2.25)}{9} = \frac{36}{9} = 4, \qquad v_2 = \frac{3.75(2.25) + 2.25(8.25)}{9} = \frac{27}{9} = 3 \text{ V}.$$

Step 4. Back-substitute: $v_3 = \frac{1}{4}(1 + v_1 + v_2) = \frac{1}{4}(8) = 2$ V. ✓

Reading. $\mathbf{S}$ is the nodal matrix of an equivalent two-node network: the off-diagonal -2.25 says the direct 2 mS link between nodes 1 and 2 is supplemented by an effective 0.25 mS through the eliminated node (the series combination of the two 1 mS branches meeting there, weighted by node 3's 4 mS total — exactly the star–delta rule $g_1 g_2 / \sum g$). The modified source vector shows that node 3's 1 mA source is redistributed as 0.25 mA into each of nodes 1 and 2.

Common Pitfall

Do not compute $\mathbf{A}^{-1}$ in order to solve $\mathbf{A}\mathbf{x} = \mathbf{b}$. Forming the inverse costs about $2n^3$ flops versus $\frac{2}{3}n^3$ for an $\mathbf{LU}$ solve, is less accurate in floating point, and for sparse matrices is catastrophic: the inverse of a sparse nodal matrix is typically completely full (every entry of $\mathbf{Z}$ above is nonzero although $\mathbf{G}$ could be sparse). Write $\mathbf{A}^{-1}\mathbf{b}$ in formulas; compute it as `A\b` (MATLAB) or `numpy.linalg.solve(A, b)`.

Engineering Note

The nodal resistance matrix $\mathbf{Z} = \mathbf{G}^{-1}$ is the natural object in **sensitivity analysis**. By differentiating $\mathbf{G}\mathbf{v} = \mathbf{i}_s$, $\partial \mathbf{v}/\partial g = -\mathbf{Z}(\partial \mathbf{G}/\partial g)\mathbf{v}$; with the stamp $\partial \mathbf{G}/\partial g = \mathbf{u}\mathbf{u}^{\mathsf{T}}$ this is the derivative form of Sherman–Morrison. Circuit simulators compute such sensitivities with one extra *adjoint* solve using $\mathbf{G}^{\mathsf{T}}$, reusing the factorisation — the same trick that underlies back-propagation in neural networks.

8.5 Rank

Motivation

A square matrix is either invertible or not; the determinant is a yes/no test. But engineers constantly meet matrices that are rectangular (more measurements than unknowns, more branches than nodes) or singular for structural reasons (a floating subcircuit, a redundant sensor, a dependent equation written twice). The **rank** is the single number that says how much independent information a matrix carries. It decides whether a linear system is solvable, how many free parameters its solution has, how many independent KCL or KVL equations a network supplies, and — in Chapter 9 — the dimensions of the column space and null space.

Echelon form

Row echelon form, pivots

A matrix is in **row echelon form (REF)** if

1. all zero rows are at the bottom, and
2. the first nonzero entry (the **pivot** or leading entry) of each nonzero row lies strictly to the right of the pivot of the row above.

The columns containing pivots are **pivot columns**; the others are **free columns**.

A typical REF pattern, with ■ a nonzero pivot and $*$ anything:

$$\begin{bmatrix} \blacksquare & * & * & * & * \\ 0 & 0 & \blacksquare & * & * \\ 0 & 0 & 0 & \blacksquare & * \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Gaussian elimination (§8.7) reduces every matrix to an REF using row operations. The REF itself is not unique (it depends on the operations chosen), but the *number* of pivots and the *positions* of the pivot columns are.

Rank

The **rank** of $\mathbf{A}$, written $\operatorname{rank} \mathbf{A}$, is the number of pivots in any row echelon form of $\mathbf{A}$. Equivalently, it is

- the maximum number of linearly independent rows of $\mathbf{A}$ (the **row rank**);
- the maximum number of linearly independent columns of $\mathbf{A}$ (the **column rank**);
- the order of the largest square submatrix with nonzero determinant (the largest nonzero **minor**).

A set of vectors $\mathbf{a}_1, \dots, \mathbf{a}_k$ is **linearly independent** if $c_1\mathbf{a}_1 + \cdots + c_k\mathbf{a}_k = \mathbf{0}$ forces $c_1 = \cdots = c_k = 0$; otherwise one of them is a linear combination of the others. Chapter 9 develops this notion fully; here we need only the computational test supplied by elimination.

Why the characterisations agree (sketch). Row operations replace rows by combinations of rows and are reversible, so they preserve the set of all combinations of rows and hence the row rank; in an REF the nonzero rows are visibly independent, so row rank equals the number of pivots. Row operations also preserve every linear relation among the columns (if $\mathbf{Ac} = \mathbf{0}$ then $\mathbf{EAc} = \mathbf{0}$, and conversely since $\mathbf{E}$ is invertible), so the pivot columns of $\mathbf{A}$ are independent and every free column is a combination of them: column rank equals the number of pivots too. Finally, a nonzero $r \times r$ minor exists in the REF (the pivot rows and columns form a triangular block with nonzero diagonal), and no larger one can, since any larger set of rows is dependent. $\square$

The equality of row rank and column rank is one of the genuinely surprising facts of linear algebra: a 3×100 matrix cannot have more than three independent columns, because it has only three rows.

Properties

Properties of rank

For $\mathbf{A} \in \mathbb{C}^{m\times n}$:

1. $0 \le \operatorname{rank}\mathbf{A} \le \min(m,n)$; $\operatorname{rank}\mathbf{A} = 0$ only for $\mathbf{A} = \mathbf{0}$. If $\operatorname{rank}\mathbf{A} = \min(m,n)$, $\mathbf{A}$ has **full rank**.
2. $\operatorname{rank}\mathbf{A}^{\mathsf{T}} = \operatorname{rank}\mathbf{A}^{\mathsf{H}} = \operatorname{rank}\mathbf{A}$.
3. $\operatorname{rank}(\mathbf{AB}) \le \min(\operatorname{rank}\mathbf{A}, \operatorname{rank}\mathbf{B})$.
4. If $\mathbf{P}$ and $\mathbf{Q}$ are invertible, $\operatorname{rank}(\mathbf{PAQ}) = \operatorname{rank}\mathbf{A}$.
5. A nonzero outer product $\mathbf{uv}^{\mathsf{T}}$ has rank 1; conversely every rank-1 matrix is an outer product.
6. $\operatorname{rank}(\mathbf{A}+\mathbf{B}) \le \operatorname{rank}\mathbf{A} + \operatorname{rank}\mathbf{B}$.
7. **Rank–nullity:** the solutions of $\mathbf{Ax} = \mathbf{0}$ form a set described by exactly $n - \operatorname{rank}\mathbf{A}$ free parameters (the *nullity*; see Chapter 9).
8. A square $n \times n$ matrix is invertible if and only if its rank is n.

Property 3 follows from the column view: every column of $\mathbf{AB}$ is a combination of the columns of $\mathbf{A}$, and (transposing) every row of $\mathbf{AB}$ is a combination of the rows of $\mathbf{B}$. It explains why $\mathbf{G} = \mathbf{A}\mathbf{G}_b\mathbf{A}^{\mathsf{T}}$ can be no better than the incidence matrix $\mathbf{A}$ it is built from.

Worked Example 8.15 — Rank of a rectangular matrix

Find the rank of

$$\mathbf{M} = \begin{bmatrix} 1 & 2 & 1 & 3 \\ 2 & 4 & 0 & 2 \\ 3 & 6 & 1 & 5 \end{bmatrix}$$

and express the free columns in terms of the pivot columns.

Step 1. $R_2 \leftarrow R_2 - 2R_1$, $R_3 \leftarrow R_3 - 3R_1$:

$$\begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & 0 & -2 & -4 \\ 0 & 0 & -2 & -4 \end{bmatrix}.$$

Step 2. $R_3 \leftarrow R_3 - R_2$ gives a zero third row. The REF has pivots in columns 1 and 3, so $\operatorname{rank}\mathbf{M} = 2$.

Step 3. Row 3 of $\mathbf{M}$ is row 1 plus row 2 — the dependency the elimination discovered. Columns: $\mathbf{m}_2 = 2\mathbf{m}_1$ by inspection; for $\mathbf{m}_4 = \alpha\mathbf{m}_1 + \beta\mathbf{m}_3$, the second component gives $2\alpha = 2$, so $\alpha = 1$, and the first gives $1 + \beta = 3$, so $\beta = 2$. Check the third: $3 + 2(1) = 5$ ✓.

Reading. Although $\mathbf{M}$ has four columns in $\mathbb{R}^3$, they span only a plane. A system $\mathbf{Mx} = \mathbf{b}$ can be solved only for $\mathbf{b}$ in that plane (those with $b_3 = b_1 + b_2$), and then with $4 - 2 = 2$ free parameters.

Worked Example 8.16 — Rank depending on a parameter

For which k does $\mathbf{B} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & k \end{bmatrix}$ have rank 3? What is the rank otherwise?

Step 1. $R_2 \leftarrow R_2 - R_1$, $R_3 \leftarrow R_3 - R_1$: rows $(0,1,2)$ and $(0,2,k-1)$.

Step 2. $R_3 \leftarrow R_3 - 2R_2$: row $(0,0,k-5)$.

Step 3. The pivots are $1, 1, k-5$, so $\det\mathbf{B} = k - 5$. For $k \neq 5$, $\operatorname{rank}\mathbf{B} = 3$; for $k = 5$ the third row vanishes and $\operatorname{rank}\mathbf{B} = 2$ (the 2×2 minor $\begin{vmatrix} 1 & 1 \\ 1 & 2 \end{vmatrix} = 1 \neq 0$ confirms it is not lower). At $k = 5$, row 3 $=$ 2 row 2 $-$ row 1. We use this matrix again in §8.6.

Rank in network topology

For a connected network with N nodes (including ground) and b branches, the full $N \times b$ incidence matrix has every column containing one $+1$ and one -1, so its rows sum to zero: the rows are dependent and the rank is at most $N - 1$. In fact it is exactly $N - 1$ for a connected graph — which is why we delete the ground row and keep the $(N-1)\times b$ **reduced** incidence matrix, of full row rank. The number of independent KCL equations is therefore $N - 1$, and the number of independent KVL (loop/mesh) equations is $b - (N - 1)$, the nullity of the reduced incidence matrix. For the running example $N = 4$, $b = 6$: three nodal equations, three independent loops.

Worked Example 8.17 — A floating network has a singular nodal matrix

Remove the three branches to ground (a, d, e) from the running example, leaving branches b (2 mS, 1–2), c (1 mS, 2–3), f (1 mS, 1–3), with no connection to ground. Write the nodal matrix and find its rank.

Step 1. Stamping the three branches:

$$\mathbf{L} = \begin{bmatrix} 3 & -2 & -1 \\ -2 & 3 & -1 \\ -1 & -1 & 2 \end{bmatrix}.$$

Step 2. Every row sums to zero: $\mathbf{L}(1,1,1)^{\mathsf{T}} = \mathbf{0}$. A nonzero solution of $\mathbf{L}\mathbf{x} = \mathbf{0}$ exists, so $\mathbf{L}$ is singular and $\operatorname{rank}\mathbf{L} \le 2$.

Step 3. Elimination: $R_2 \leftarrow R_2 + \tfrac{2}{3}R_1$ gives $(0, \tfrac{5}{3}, -\tfrac{5}{3})$; $R_3 \leftarrow R_3 + \tfrac{1}{3}R_1$ gives $(0, -\tfrac{5}{3}, \tfrac{5}{3})$; then $R_3 \leftarrow R_3 + R_2$ gives $\mathbf{0}$. Two pivots: $\operatorname{rank}\mathbf{L} = 2$.

Reading. Physically, adding the same voltage to every node of a floating network changes no branch voltage and therefore no current; the solution can only ever be determined up to that common offset. Choosing any node as the reference (deleting its row and column) restores full rank. Matrices with zero row sums and non-positive off-diagonals like $\mathbf{L}$ are **graph Laplacians**; their rank is $N - c$ where c is the number of connected pieces.

Numerical rank

In exact arithmetic rank is an integer and elimination finds it. In floating point, or with measured data, "zero" pivots come out as 10^{-14} or as noise, and the question becomes *how close is the matrix to one of lower rank?* The matrix $\begin{bmatrix} 1 & 1 \\ 1 & 1.0001 \end{bmatrix}$ has rank 2, but a perturbation of size 10^{-4} in one entry makes it rank 1; for most engineering purposes it behaves like a rank-1 matrix. The robust tool is the singular value decomposition (Chapter 10): the singular values of this matrix are about 2.00005 and 5.0×10^{-5}, and the **numerical rank** is the number of singular values above a tolerance tied to the data accuracy. The ratio of the largest to the smallest singular value is the 2-norm condition number of §8.10.

Common Pitfall

Rank is not the number of nonzero rows of the *original* matrix, nor the number of nonzero entries on its diagonal. $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ has no zero rows and a nonzero diagonal but rank 1. Always reduce to echelon form (or use singular values) before counting.

Engineering Note

Rank deficiency in a circuit simulator usually means a modelling error, and SPICE reports it as a "singular matrix" error. The usual culprits: a node with no DC path to ground (e.g. the junction of two series capacitors), a loop consisting only of ideal voltage sources and inductors (KVL makes one of the source equations redundant or contradictory), or a cut-set of only current sources and capacitors. Each is a rank defect of a structural submatrix and can be detected from the network topology before any number is computed.

8.6 Systems of linear equations

Formulation

Linear system

A system of m linear equations in n unknowns,

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1, \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m, \end{aligned}$$

is written $\mathbf{A}\mathbf{x} = \mathbf{b}$ with **coefficient matrix** $\mathbf{A} \in \mathbb{C}^{m\times n}$ and **right-hand side** $\mathbf{b} \in \mathbb{C}^m$. The **augmented matrix** is $[\mathbf{A} \mid \mathbf{b}]$. The system is **homogeneous** if $\mathbf{b} = \mathbf{0}$. It is **consistent** if it has at least one solution, **inconsistent** otherwise.

Systems are classified by shape: **square** ($m = n$, the usual case for well-posed circuit equations), **overdetermined** ($m > n$, typical of measurements; usually inconsistent and solved in the least-squares sense of Chapter 9) and **underdetermined** ($m < n$; consistent systems have infinitely many solutions, and a particular one is chosen by an extra criterion such as minimum norm).

Existence and uniqueness

Rouché–Capelli (Kronecker–Capelli) theorem

Let $\mathbf{A}$ be $m \times n$ and $r = \operatorname{rank}\mathbf{A}$.

1. $\mathbf{A}\mathbf{x} = \mathbf{b}$ is consistent if and only if $\operatorname{rank}[\mathbf{A} \mid \mathbf{b}] = \operatorname{rank}\mathbf{A}$.
2. If it is consistent and $r = n$, the solution is unique.
3. If it is consistent and $r < n$, there are infinitely many solutions, described by $n - r$ free parameters.

Proof sketch. Reduce $[\mathbf{A} \mid \mathbf{b}]$ to echelon form. If the augmented column contains a pivot, some row reads $0 = c$ with $c \neq 0$: inconsistent, and the augmented rank exceeds r. Otherwise the r nonzero rows can be solved by back substitution for the r pivot variables in terms of the $n - r$ free variables, which may be chosen arbitrarily. □

In terms of the column view: $\mathbf{A}\mathbf{x} = \mathbf{b}$ is solvable exactly when $\mathbf{b}$ is a combination of the columns of $\mathbf{A}$, and appending such a $\mathbf{b}$ cannot raise the rank.

Structure of the solution set

If $\mathbf{x}_p$ is any one solution of $\mathbf{A}\mathbf{x} = \mathbf{b}$, every solution has the form

$$\mathbf{x} = \mathbf{x}_p + \mathbf{x}_h,$$

where $\mathbf{x}_h$ ranges over all solutions of the homogeneous system $\mathbf{A}\mathbf{x}_h = \mathbf{0}$.

Proof. If $\mathbf{A}\mathbf{x} = \mathbf{b}$ then $\mathbf{A}(\mathbf{x} - \mathbf{x}_p) = \mathbf{b} - \mathbf{b} = \mathbf{0}$; conversely $\mathbf{A}(\mathbf{x}_p + \mathbf{x}_h) = \mathbf{b} + \mathbf{0}$. □

This "particular plus homogeneous" structure reappears verbatim for linear differential equations (Chapters 11–12), where it is the split into forced and natural response. Its companion is **superposition**: if $\mathbf{A}\mathbf{x}_1 = \mathbf{b}_1$ and $\mathbf{A}\mathbf{x}_2 = \mathbf{b}_2$ then $\mathbf{A}(\alpha\mathbf{x}_1 + \beta\mathbf{x}_2) = \alpha\mathbf{b}_1 + \beta\mathbf{b}_2$. A circuit's response to several sources is the sum of its responses to each — which is just the linearity of $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$ in $\mathbf{b}$.

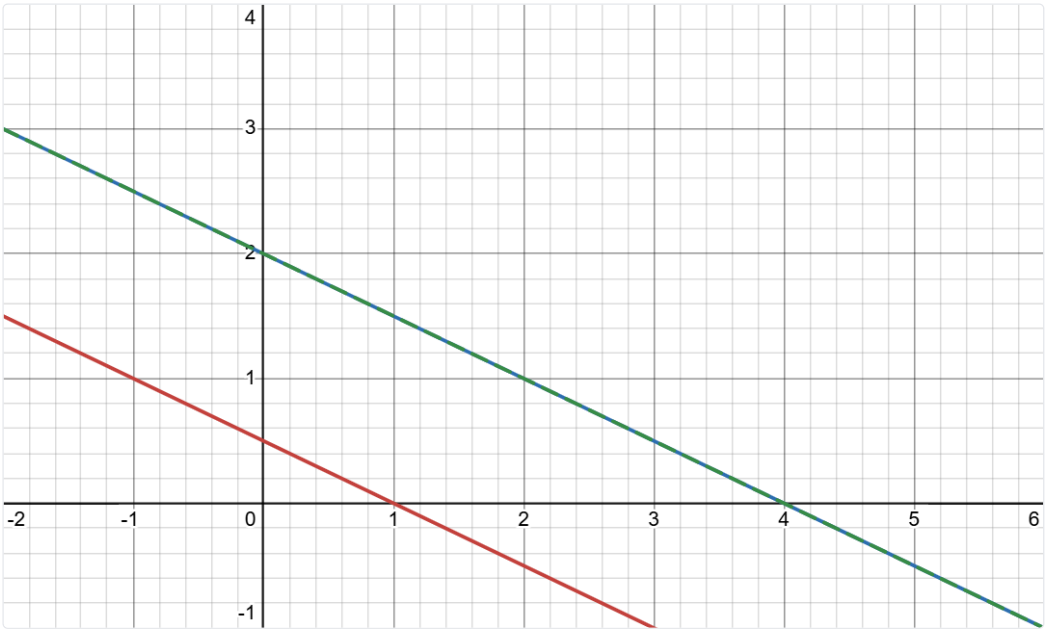
Homogeneous systems

A homogeneous system always has the **trivial solution** $\mathbf{x} = \mathbf{0}$. By Rouché–Capelli it has nontrivial solutions if and only if $\operatorname{rank}\mathbf{A} < n$. Two consequences:

- A homogeneous system with **more unknowns than equations** ($n > m$) always has nontrivial solutions, since $r \leq m < n$.
- A **square** homogeneous system has nontrivial solutions if and only if $\det\mathbf{A} = 0$. This is how eigenvalue problems $(\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = \mathbf{0}$ lead to the characteristic equation $\det(\mathbf{A} - \lambda\mathbf{I}) = 0$ (Chapter 10), and how a circuit's natural frequencies arise from $\det\mathbf{Y}(s) = 0$: a nonzero response with no sources needs a singular system matrix.

Geometry

With two unknowns, each equation $a_{i1}x + a_{i2}y = b_i$ is a line in the plane. Two lines either cross at one point (unique solution, $\det\mathbf{A} \neq 0$, the case drawn in §8.3), are parallel and distinct (no solution), or coincide (infinitely many). With three unknowns each equation is a plane in space; three planes in general position meet in a single point, but they may also share a common line (rank 2, consistent), be arranged like the faces of a triangular prism with no common point (rank 2, inconsistent), contain two parallel planes, or all coincide (rank 1).



Singular 2×2 systems. The blue line $x + 2y = 4$ and red line $x + 2y = 1$ are parallel: the system has no solution ($\operatorname{rank}\mathbf{A} = 1$, $\operatorname{rank}[\mathbf{A} \mid \mathbf{b}] = 2$). The dashed green line $2x + 4y = 8$ lies exactly on the blue one: together they give infinitely many solutions (both ranks 1, one free parameter).

Worked Example 8.18 — The three cases for two unknowns

Classify (a) $x + 2y = 4,\ 3x - y = 5$; (b) $x + 2y = 4,\ x + 2y = 1$; (c) $x + 2y = 4,\ 2x + 4y = 8$.

(a) $\det\begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix} = -7 \neq 0$: unique solution. Cramer: $x = \dfrac{-4 - 10}{-7} = 2$, $y = \dfrac{5 - 12}{-7} = 1$.

(b) $R_2 \leftarrow R_2 - R_1$ gives $[\,0 \;\; 0 \mid -3\,]$, i.e. $0 = -3$. $\operatorname{rank}\mathbf{A} = 1 < \operatorname{rank}[\mathbf{A} \mid \mathbf{b}] = 2$: inconsistent.

(c) $R_2 \leftarrow R_2 - 2R_1$ gives $[\,0 \;\; 0 \mid 0\,]$. Both ranks are 1, $n - r = 1$ free parameter. Let $y = t$; then $x = 4 - 2t$, so $(x, y) = (4, 0) + t(-2, 1)$ — a particular solution plus any multiple of the homogeneous solution $(-2, 1)$, which satisfies $x + 2y = 0$.

Worked Example 8.19 — A system with parameters

Discuss, in terms of k and m, the solutions of

$$\begin{aligned} x + y + z &= 3, \\ x + 2y + 3z &= 6, \\ x + 3y + kz &= m. \end{aligned}$$

Step 1. The coefficient matrix is $\mathbf{B}$ of Worked Example 8.16. Apply the same row operations to the augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 1 & 2 & 3 & 6 \\ 1 & 3 & k & m \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & 2 & 3 \\ 0 & 2 & k-1 & m-3 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & k-5 & m-9 \end{array}\right].$$

Step 2 (classify).

- $k \neq 5$: three pivots, rank $3 = n$: unique solution $z = \dfrac{m-9}{k-5}$, $y = 3 - 2z$, $x = 3 - y - z = z$.
- $k = 5$, $m \neq 9$: last row reads $0 = m - 9 \neq 0$; rank $\mathbf{A} = 2 <$ rank$[\mathbf{A} \mid \mathbf{b}] = 3$: no solution.
- $k = 5$, $m = 9$: both ranks 2, one free parameter. With $z = t$: $y = 3 - 2t$, $x = 3 - (3 - 2t) - t = t$, so $(x, y, z) = (0, 3, 0) + t(1, -2, 1)$.

Step 3 (check). For $k = 6$, $m = 10$: $z = 1$, $y = 1$, $x = 1$; indeed $1 + 1 + 1 = 3$, $1 + 2 + 3 = 6$, $1 + 3 + 6 = 10$ ✓. For the infinite family, $(t, 3 - 2t, t)$ in the third equation with $k = 5$: $t + 9 - 6t + 5t = 9$ ✓.

Geometry. For $k = 5$ the three planes share a common line when $m = 9$ and form a "prism" (pairwise intersecting lines, no common point) when $m \neq 9$.

Worked Example 8.20 — Solving the floating network: offsets and superposition

For the floating network $\mathbf{L}$ of Worked Example 8.17, inject 1 mA into node 1 and extract it from node 3: $\mathbf{L}\mathbf{v} = (1, 0, -1)^{\mathsf{T}}$. Find all solutions.

Step 1 (consistency). Since every column of $\mathbf{L}$ sums to zero, so does every vector $\mathbf{L}\mathbf{v}$: the system is solvable only if the injected currents sum to zero. Here $1 + 0 - 1 = 0$ — physically, current into a floating network must also leave it. Reduction of $[\mathbf{L} \mid \mathbf{b}]$ confirms both ranks are 2.

Step 2 (particular solution). Fix the free variable by grounding node 3, $v_3 = 0$; the first two equations become $3v_1 - 2v_2 = 1$, $-2v_1 + 3v_2 = 0$, with solution $v_1 = \frac{3}{5}$, $v_2 = \frac{2}{5}$ V. Check row 3: $-\frac{3}{5} - \frac{2}{5} + 0 = -1$ ✓.

Step 3 (general solution). The homogeneous solutions are $t(1, 1, 1)^{\mathsf{T}}$, so

$$\mathbf{v} = \begin{bmatrix} 0.6 \\ 0.4 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \qquad t \in \mathbb{R}.$$

Every physically meaningful quantity — the branch voltages $v_1 - v_2 = 0.2$, $v_2 - v_3 = 0.4$, $v_1 - v_3 = 0.6$ V — is independent of t. The resistance between nodes 1 and 3 is $0.6\ \text{V} / 1\ \text{mA} = 0.6\ \text{k}\Omega$, which matches the series–parallel calculation: $(\frac{1}{2} + 1)\ \text{k}\Omega$ in parallel with $1\ \text{k}\Omega$ is $\frac{1.5}{2.5} = 0.6\ \text{k}\Omega$ ✓.

Step 4 (superposition). Adding a second injection pattern $(0, 1, -1)^{\mathsf{T}}$ (with solution $(0.4, 0.6, 0)^{\mathsf{T}} + t(1, 1, 1)^{\mathsf{T}}$ by the same method — check: $3(0.4) - 2(0.6) = 0$, $-2(0.4) + 3(0.6) = 1$ ✓) gives the response to $(1, 1, -2)^{\mathsf{T}}$ as the sum $(1, 1, 0)^{\mathsf{T}} + t(1, 1, 1)^{\mathsf{T}}$.

Common Pitfall

Counting equations and unknowns does not settle solvability. Three equations in three unknowns may have no solution or infinitely many (Worked Example 8.19 with $k = 5$); five equations in three unknowns may have exactly one (if two are redundant). Only the ranks of $\mathbf{A}$ and $[\mathbf{A} \mid \mathbf{b}]$ decide.

Engineering Note

Well-posed circuit formulations (nodal, mesh, modified nodal) are constructed to give square nonsingular systems when the circuit itself is well-posed. Overdetermined systems arise on purpose in **state estimation**: a power-system control centre measures more line flows and voltages than there are unknown bus voltages, precisely so that bad measurements can be detected from the inconsistency, and solves the system in the weighted least-squares sense. Underdetermined systems arise in **compressed sensing** and in redundant actuator allocation, where the extra freedom is spent on minimising energy or sparsity.

8.7 Gaussian elimination

Motivation

Gaussian elimination is the workhorse algorithm of computational linear algebra. Every dense linear solver — MATLAB's backslash, LAPACK's `dgesv`, the DC operating-point and transient solvers inside SPICE — is a carefully engineered version of it. The idea is the one taught in school for two equations: use one equation to eliminate a variable from the others, repeat until one equation in one unknown remains, then substitute back. Doing this systematically turns $\mathbf{A}\mathbf{x} = \mathbf{b}$ into an equivalent **upper triangular** system $\mathbf{U}\mathbf{x} = \mathbf{c}$, which is trivial to solve.

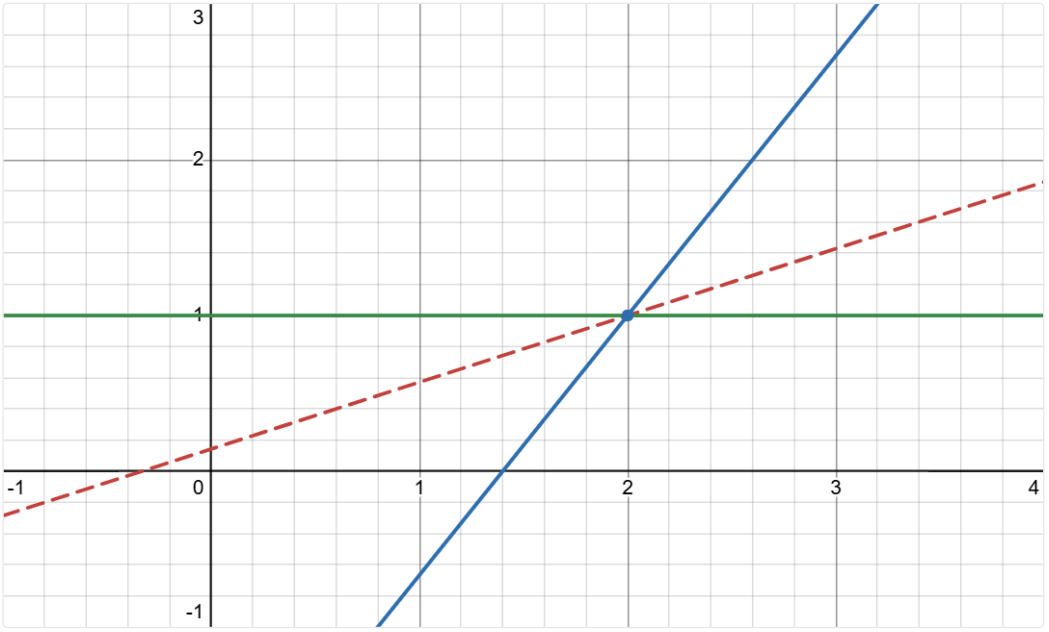
Elementary row operations and equivalence

Elementary row operations

1. **Replacement:** add a multiple of one row to another, $R_i \leftarrow R_i + cR_k$.
2. **Interchange:** swap two rows, $R_i \leftrightarrow R_k$.
3. **Scaling:** multiply a row by a nonzero constant, $R_i \leftarrow cR_i$.

Applied to the augmented matrix $[\mathbf{A} \mid \mathbf{b}]$, they produce an **equivalent** system: one with exactly the same solution set.

Equivalence holds because each operation is a left-multiplication by an invertible elementary matrix $\mathbf{E}$ (§8.4): $\mathbf{Ax} = \mathbf{b} \iff \mathbf{EAx} = \mathbf{Eb}$. In the plane, a replacement swaps one line for a different line through the *same* intersection point — the solution never moves.



Elimination in pictures for the mesh system of §8.3. The replacement $R_2 \leftarrow R_2 + \frac{3}{5}R_1$ turns the red dashed line $-3x + 7y = 1$ into the green line $\frac{26}{5}y = \frac{26}{5}$, i.e. $y = 1$, which passes through the same point $(2, 1)$ but no longer involves x. Back substitution then reads off x from the blue line.

The algorithm

Forward elimination. For $k = 1, 2, \dots, n-1$ (the current pivot column):

1. The **pivot** is $a_{kk}^{(k)}$, the current (k, k) entry. (If it is zero, or small, interchange rows first — see pivoting below.)
2. For each row $i = k+1, \dots, n$ compute the **multiplier** $m_{ik} = a_{ik}^{(k)} / a_{kk}^{(k)}$.
3. Replace $R_i \leftarrow R_i - m_{ik} R_k$, which makes $a_{ik}^{(k+1)} = 0$:

$$a_{ij}^{(k+1)} = a_{ij}^{(k)} - m_{ik} a_{kj}^{(k)}, \quad j = k+1, \dots, n, \qquad b_i^{(k+1)} = b_i^{(k)} - m_{ik} b_k^{(k)}.$$

After $n-1$ stages the matrix is upper triangular, $\mathbf{U} = [u_{ij}]$ with $u_{ij} = a_{ij}^{(i)}$, and the right-hand side has become $\mathbf{c}$.

Back substitution. Solve $\mathbf{Ux} = \mathbf{c}$ from the bottom up:

$$x_n = \frac{c_n}{u_{nn}}, \qquad x_i = \frac{1}{u_{ii}}\Big(c_i - \sum_{j=i+1}^{n} u_{ij} x_j\Big), \quad i = n-1, \dots, 1.$$

The process succeeds — in exact arithmetic, with row interchanges allowed — exactly when $\mathbf{A}$ is nonsingular, and the pivots $u_{11}, \dots, u_{nn}$ multiply to $\pm \det \mathbf{A}$.

Operation count

At stage k there are $n-k$ rows to update, each with $n-k$ matrix entries (plus one right-hand-side entry), each costing one multiply and one subtract. Ignoring lower-order terms,

$$\text{forward elimination} \approx \sum_{k=1}^{n-1} 2(n-k)^2 = 2\sum_{p=1}^{n-1} p^2 \approx \frac{2n^3}{3} \text{ flops},$$

while back substitution costs $\sum_i 2(n-i) \approx n^2$ flops, as does applying the elimination to one extra right-hand side. Doubling n multiplies the elimination time by eight. For $n = 1000$, $\frac{2}{3}n^3 \approx 6.7 \times 10^8$ flops: well under a second. For $n = 10^5$ dense it is 6.7×10^{14} — minutes to hours, and 80 GB just to store the matrix in double precision. Large problems are only tractable because they are sparse (§8.11).

Worked Example 8.21 — The running example by Gaussian elimination

Solve $\mathbf{G}\mathbf{v} = \mathbf{i}_s$ with the augmented matrix

$$\left[\begin{array}{ccc|c} 4 & -2 & -1 & 8 \\ -2 & 4 & -1 & 2 \\ -1 & -1 & 4 & 1 \end{array}\right].$$

Stage 1 (pivot 4). Multipliers $m_{21} = -2/4 = -\tfrac{1}{2}$, $m_{31} = -\tfrac{1}{4}$.

$$\begin{aligned} R_2 \leftarrow R_2 + \tfrac{1}{2}R_1: &\quad (0,\ 4-1,\ -1-\tfrac{1}{2} \mid 2+4) = (0,\ 3,\ -\tfrac{3}{2} \mid 6), \\ R_3 \leftarrow R_3 + \tfrac{1}{4}R_1: &\quad (0,\ -1-\tfrac{1}{2},\ 4-\tfrac{1}{4} \mid 1+2) = (0,\ -\tfrac{3}{2},\ \tfrac{15}{4} \mid 3). \end{aligned}$$

Stage 2 (pivot 3). Multiplier $m_{32} = (-\tfrac{3}{2})/3 = -\tfrac{1}{2}$.

$$R_3 \leftarrow R_3 + \tfrac{1}{2}R_2: \quad (0,\ 0,\ \tfrac{15}{4} - \tfrac{3}{4} \mid 3+3) = (0,\ 0,\ 3 \mid 6).$$

Triangular system.

$$\left[\begin{array}{ccc|c} 4 & -2 & -1 & 8 \\ 0 & 3 & -\frac{3}{2} & 6 \\ 0 & 0 & 3 & 6 \end{array}\right].$$

Back substitution. $v_3 = 6/3 = 2$; $v_2 = (6 + \tfrac{3}{2}\cdot 2)/3 = 9/3 = 3$; $v_1 = (8 + 2\cdot 3 + 1\cdot 2)/4 = 16/4 = 4$. So $\mathbf{v} = (4, 3, 2)^{\mathsf{T}}$ V. ✓

By-products. $\det \mathbf{G} = 4 \cdot 3 \cdot 3 = 36$. The multipliers $-\tfrac{1}{2}, -\tfrac{1}{4}, -\tfrac{1}{2}$ will reappear as the entries of $\mathbf{L}$ in §8.9.

Circuit reading. Stage 1 is Kron reduction of node 1: the 2×2 block $\begin{bmatrix} 3 & -\frac{3}{2} \\ -\frac{3}{2} & \frac{15}{4} \end{bmatrix}$ is the nodal matrix, seen from nodes 2 and 3, of the network with node 1 eliminated, and $(6, 3)$ are the correspondingly redistributed source currents. Gaussian elimination *is* repeated node elimination.

Pivoting

If a pivot is zero the algorithm cannot proceed as written, even though the matrix may be perfectly nonsingular; if a pivot is merely small, the multipliers are large and rounding errors are amplified.

Partial pivoting

At stage k, choose the row $p \geq k$ for which $|a_{pk}^{(k)}|$ is largest, and interchange rows p and k before eliminating. Then every multiplier satisfies $|m_{ik}| \leq 1$.

Complete pivoting searches the whole remaining submatrix and swaps columns as well as rows; it is more stable in theory but rarely needed. The quantity that controls stability is the **growth factor** $\rho = \max_{i,j,k} |a_{ij}^{(k)}| / \max_{i,j} |a_{ij}|$: backward error analysis (Wilkinson) shows that the computed solution is the exact solution of a nearby system $(\mathbf{A} + \delta\mathbf{A})\hat{\mathbf{x}} = \mathbf{b}$ with $\|\delta\mathbf{A}\| \lesssim c\, n\, \rho\, u \|\mathbf{A}\|$, where $u \approx 1.1 \times 10^{-16}$ is the double-precision unit roundoff. With partial pivoting $\rho \leq 2^{n-1}$ in the worst case, but in practice ρ is almost always small, which is why partial pivoting is the universal default.

When pivoting is unnecessary

If $\mathbf{A}$ is **strictly diagonally dominant by columns** ($|a_{jj}| > \sum_{i \neq j} |a_{ij}|$ for every j), or **symmetric positive definite**, then Gaussian elimination without interchanges never meets a zero pivot, the growth factor is bounded ($\rho \leq 2$ and $\rho \leq 1$ respectively), and the matrix is nonsingular. Diagonal dominance by rows also guarantees nonsingularity and nonzero pivots.

Sketch for diagonal dominance. After one elimination stage, the reduced $(n-1) \times (n-1)$ block is again diagonally dominant (a short calculation with the triangle inequality), so induction shows every pivot is nonzero. □

Nodal matrices of resistive networks with at least one branch to ground at every node are strictly diagonally dominant (diagonal = sum of all conductances at the node, off-diagonal magnitudes = the conductances to other nodes) and symmetric positive definite, so they can be eliminated in any order without pivoting for stability. Simulators exploit this freedom to choose an elimination order that preserves sparsity (§8.11).

Worked Example 8.22 — A zero pivot from a voltage source

Replace the 8 mA source at node 1 of the running example by an ideal 4 V voltage source from ground to node 1, and let j be the current (mA) flowing from node 1 into the source. With unknowns ordered (j, v_1, v_2, v_3) and the source equation written first, modified nodal analysis gives

$$\left[\begin{array}{cccc|c} 0 & 1 & 0 & 0 & 4 \\ 1 & 4 & -2 & -1 & 0 \\ 0 & -2 & 4 & -1 & 2 \\ 0 & -1 & -1 & 4 & 1 \end{array}\right].$$

Step 1. The $(1,1)$ pivot is 0. Interchange $R_1 \leftrightarrow R_2$:

$$\left[\begin{array}{cccc|c} 1 & 4 & -2 & -1 & 0 \\ 0 & 1 & 0 & 0 & 4 \\ 0 & -2 & 4 & -1 & 2 \\ 0 & -1 & -1 & 4 & 1 \end{array}\right].$$

Column 1 is already clear below the pivot.

Step 2 (pivot 1 in column 2; strict partial pivoting would pick the -2 in row 3, which gives the same exact answer). $R_3 \leftarrow R_3 + 2R_2$, $R_4 \leftarrow R_4 + R_2$: rows 3, 4 become $(0, 0, 4, -1 \mid 10)$ and $(0, 0, -1, 4 \mid 5)$.

Step 3. $R_4 \leftarrow R_4 + \tfrac{1}{4}R_3$: $(0, 0, 0, \tfrac{15}{4} \mid 7.5)$.

Step 4 (back substitution). $v_3 = 7.5/3.75 = 2$; row 3 reads $4v_2 - v_3 = 10$, so $v_2 = 3$; row 2 gives $v_1 = 4$; and $j + 4(4) - 2(3) - 1(2) = 0 \Rightarrow j = -8$ mA.

Reading. The node voltages are unchanged, as they must be: the voltage source holds node 1 at exactly the voltage the current source produced. The source current $j = -8$ mA means 8 mA flows *out* of the source into node 1 — the same 8 mA the current source injected. The determinant is $(-1)^1 \cdot 1 \cdot 1 \cdot 4 \cdot \tfrac{15}{4} = -15$, agreeing with Worked Example 8.8. MNA matrices are symmetric but **indefinite** (zero diagonal blocks), so they need pivoting where pure nodal matrices do not.

Worked Example 8.23 — Why small pivots are dangerous

Solve

$$\begin{aligned} 0.0001\,x_1 + x_2 &= 1, \\ x_1 + x_2 &= 2 \end{aligned}$$

in **three-significant-figure** arithmetic (each result rounded to 3 significant figures), first without and then with pivoting. The exact solution is $x_1 = 1/0.9999 = 1.00010\ldots$, $x_2 = 0.9998/0.9999 = 0.99990\ldots$.

Without pivoting. $m_{21} = 1/0.0001 = 10000$. Then $a_{22} = 1 - 10000 = -9999 \to -1.00 \times 10^4$ and $b_2 = 2 - 10000 = -9998 \to -1.00 \times 10^4$. So $x_2 = 1.00$, and $x_1 = (1 - x_2)/0.0001 = 0/0.0001 = 0$. The answer $x_1 = 0$ is completely wrong: the information in the second equation ("$x_1 + x_2 = 2$") was swamped by the huge multiple of the first.

With partial pivoting. $|1| > |0.0001|$, so swap the equations. Now $m_{21} = 0.0001$, $a_{22} = 1 - 0.0001 = 0.9999 \to 1.00$, $b_2 = 1 - 0.0002 = 0.9998 \to 1.00$. So $x_2 = 1.00$ and $x_1 = 2 - 1.00 = 1.00$. Both components are correct to the working precision.

Moral. The matrix is well conditioned (condition number about 4, §8.10); the failure without pivoting is entirely the algorithm's fault. In double precision the same thing happens with a pivot of 10^{-20}.

Worked Example 8.24 — Elimination with complex (phasor) entries

A two-node AC network has a 1 S conductance from each node to ground and a capacitive susceptance $j1$ S between the nodes. A phasor current $1\angle 0^\circ$ A is injected at node 1. Find V_1 and V_2.

Step 1. $\mathbf{Y} = \begin{bmatrix} 1+j & -j \\ -j & 1+j \end{bmatrix}$, $\mathbf{I} = (1, 0)^{\mathsf{T}}$.

Step 2. $m_{21} = \dfrac{-j}{1+j} = \dfrac{-j(1-j)}{2} = \dfrac{-1-j}{2}$. Then

$$u_{22} = (1+j) - m_{21}(-j) = (1+j) + \frac{(-1-j)j}{2} = (1+j) + \frac{1-j}{2} = \frac{3+j}{2}, \qquad c_2 = 0 - m_{21} \cdot 1 = \frac{1+j}{2}.$$

Step 3.

$$V_2 = \frac{(1+j)/2}{(3+j)/2} = \frac{(1+j)(3-j)}{10} = \frac{4+2j}{10} = 0.4 + j0.2 \text{ V},$$

$$V_1 = \frac{1 + jV_2}{1+j} = \frac{1 + j0.4 - 0.2}{1+j} = \frac{(0.8 + j0.4)(1-j)}{2} = 0.6 - j0.2 \text{ V}.$$

Step 4 (check). $(1+j)(0.6 - j0.2) - j(0.4 + j0.2) = (0.8 + j0.4) + (0.2 - j0.4) = 1$ ✓; $-j(0.6 - j0.2) + (1+j)(0.4 + j0.2) = (-0.2 - j0.6) + (0.2 + j0.6) = 0$ ✓.

Reading. Complex arithmetic changes nothing in the algorithm; each complex flop costs about four real ones. $|V_2| \approx 0.447$ V with phase $+26.6^\circ$: the capacitor couples a leading current into node 2.

Common Pitfall

When doing elimination by hand, apply every row operation to the right-hand side as well (or carry the augmented matrix), and do **not** scale rows by convenience factors in the middle of a determinant calculation without tracking the factor. Scaling is harmless for solving but changes the determinant.

Engineering Note

Elimination order has a physical meaning in circuits: eliminating a node is Kron reduction, and the "fill-in" it creates is the set of new branches the star–mesh transformation adds between the node's neighbours. A node with many neighbours (a ground plane, a power rail) creates a dense clique when eliminated, which is why simulators eliminate such nodes last.

8.8 Gauss–Jordan elimination

Motivation

Gaussian elimination stops at a triangular matrix and finishes with back substitution. **Gauss–Jordan elimination** keeps going: it scales every pivot to 1 and clears the entries *above* each pivot as well as below, until the coefficient part is as simple as possible. For a nonsingular square system that simplest form is the identity, and the solution can be read off directly. The method costs more than Gaussian elimination for a single solve, but it has three uses that make it indispensable: it computes inverses (by reducing $[\mathbf{A} \mid \mathbf{I}]$), it produces the unique **reduced row echelon form** that settles questions of rank and dependence unambiguously, and it writes the complete solution of an underdetermined system in terms of free parameters.

Reduced row echelon form

Reduced row echelon form (RREF)

A matrix is in **reduced row echelon form** if it is in row echelon form and, in addition,

1. every pivot equals 1, and
2. every pivot is the only nonzero entry in its column.

For example,

$$\begin{bmatrix} 1 & 2 & 0 & 1 & 0 \\ 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

is in RREF, with pivot columns 1, 3, 5.

Uniqueness of the RREF

Every matrix is row-equivalent to exactly one matrix in reduced row echelon form, written $\operatorname{rref}(\mathbf{A})$.

Proof sketch. Row operations preserve all linear relations among columns (§8.5). In an RREF those relations are visible: each pivot column is a standard basis vector $\mathbf{e}_1, \mathbf{e}_2, \dots$ (in order) and each free column lists the coefficients expressing it in terms of the earlier pivot columns. Since the relations are determined by $\mathbf{A}$, so is the RREF. $\square$

Uniqueness is what makes the RREF a reliable bookkeeping device: two people reducing the same matrix in different orders reach the same answer (whereas their REFs may differ).

The algorithm

For each column from left to right:

1. Find a row at or below the current row with a nonzero entry in this column (with partial pivoting, the one of largest magnitude); swap it into the current row. If there is none, the column is free: move to the next column, staying on the same row.
2. **Normalise:** divide the current row by its pivot so the pivot becomes 1.
3. **Eliminate:** subtract multiples of the current row from *every other row*, above and below, to make the rest of the column zero.
4. Move to the next row and column.

Applied to $[\mathbf{A} \mid \mathbf{b}]$ with $\mathbf{A}$ nonsingular, the result is $[\mathbf{I} \mid \mathbf{x}]$.

Cost. Clearing above as well as below the pivots roughly doubles the work in the elimination phase: about n^3 flops for one right-hand side, versus $\tfrac{2}{3}n^3 + n^2$ for Gaussian elimination with back substitution. For solving a single system Gauss–Jordan is therefore 50% more expensive and is not used in production solvers; its value is conceptual and for the special tasks below.

Worked Example 8.25 — The running example by Gauss–Jordan

Reduce $\left[\begin{array}{ccc|c} 4 & -2 & -1 & 8 \\ -2 & 4 & -1 & 2 \\ -1 & -1 & 4 & 1 \end{array}\right]$ to RREF.

Column 1. $R_1 \leftarrow \tfrac14 R_1$: $(1, -\tfrac12, -\tfrac14 \mid 2)$. Then $R_2 \leftarrow R_2 + 2R_1$: $(0, 3, -\tfrac32 \mid 6)$; $R_3 \leftarrow R_3 + R_1$: $(0, -\tfrac32, \tfrac{15}{4} \mid 3)$.

Column 2. $R_2 \leftarrow \tfrac13 R_2$: $(0, 1, -\tfrac12 \mid 2)$. Then $R_1 \leftarrow R_1 + \tfrac12 R_2$: $(1, 0, -\tfrac12 \mid 3)$; $R_3 \leftarrow R_3 + \tfrac32 R_2$: $(0, 0, 3 \mid 6)$.

Column 3. $R_3 \leftarrow \tfrac13 R_3$: $(0, 0, 1 \mid 2)$. Then $R_1 \leftarrow R_1 + \tfrac12 R_3$: $(1, 0, 0 \mid 4)$; $R_2 \leftarrow R_2 + \tfrac12 R_3$: $(0, 1, 0 \mid 3)$.

Result. $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 2 \end{array}\right]$, so $\mathbf{v} = (4, 3, 2)^{\mathsf T}$ V with no back substitution. ✓

Computing the inverse

Since $\mathbf{A}\mathbf{X} = \mathbf{I}$ is n systems $\mathbf{A}\mathbf{x}_j = \mathbf{e}_j$ sharing one matrix, reduce them all at once:

$$[\,\mathbf{A} \mid \mathbf{I}\,] \xrightarrow{\text{Gauss–Jordan}} [\,\mathbf{I} \mid \mathbf{A}^{-1}\,].$$

If the left block cannot be reduced to $\mathbf{I}$ (a zero row appears), $\mathbf{A}$ is singular. In terms of elementary matrices: the operations that turn $\mathbf{A}$ into $\mathbf{I}$, accumulated, are $\mathbf{E}_k \cdots \mathbf{E}_1 = \mathbf{A}^{-1}$, and the right block records exactly that product applied to $\mathbf{I}$. The cost is about $2n^3$ flops (fewer if one exploits the zeros of $\mathbf{I}$).

Worked Example 8.26 — Inverting $\mathbf{G}$ by Gauss–Jordan

Reduce $[\mathbf{G} \mid \mathbf{I}]$.

Start.

$$\left[\begin{array}{ccc|ccc} 4 & -2 & -1 & 1 & 0 & 0 \\ -2 & 4 & -1 & 0 & 1 & 0 \\ -1 & -1 & 4 & 0 & 0 & 1 \end{array}\right]$$

Column 1. $R_1 \leftarrow \tfrac14 R_1$, $R_2 \leftarrow R_2 + 2R_1$, $R_3 \leftarrow R_3 + R_1$:

$$\left[\begin{array}{ccc|ccc} 1 & -\tfrac12 & -\tfrac14 & \tfrac14 & 0 & 0 \\ 0 & 3 & -\tfrac32 & \tfrac12 & 1 & 0 \\ 0 & -\tfrac32 & \tfrac{15}{4} & \tfrac14 & 0 & 1 \end{array}\right]$$

Column 2. $R_2 \leftarrow \tfrac13 R_2$, $R_1 \leftarrow R_1 + \tfrac12 R_2$, $R_3 \leftarrow R_3 + \tfrac32 R_2$:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -\tfrac12 & \tfrac13 & \tfrac16 & 0 \\ 0 & 1 & -\tfrac12 & \tfrac16 & \tfrac13 & 0 \\ 0 & 0 & 3 & \tfrac12 & \tfrac12 & 1 \end{array}\right]$$

Column 3. $R_3 \leftarrow \tfrac13 R_3$, $R_1 \leftarrow R_1 + \tfrac12 R_3$, $R_2 \leftarrow R_2 + \tfrac12 R_3$:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \tfrac{5}{12} & \tfrac14 & \tfrac16 \\ 0 & 1 & 0 & \tfrac14 & \tfrac{5}{12} & \tfrac16 \\ 0 & 0 & 1 & \tfrac16 & \tfrac16 & \tfrac13 \end{array}\right]$$

The right block is $\mathbf{Z} = \mathbf{G}^{-1}$, agreeing with the adjugate calculation of Worked Example 8.12. ✓ Each column of $\mathbf{Z}$ is the vector of node voltages produced by a 1 mA source at one node.

Several right-hand sides

Gauss–Jordan (and Gaussian elimination) can process any number of right-hand sides in one pass: reduce $[\mathbf{A} \mid \mathbf{b}_1\ \mathbf{b}_2\ \cdots\ \mathbf{b}_k]$. Each extra column costs only $O(n^2)$. In circuit terms this computes the response to several source patterns at once. In production codes the same economy is obtained more flexibly by storing an $\mathbf{LU}$ factorisation (§8.9), since the right-hand sides are often not all known in advance.

General solutions of consistent singular systems

When $\operatorname{rank}\mathbf{A} = r < n$, the RREF of $[\mathbf{A} \mid \mathbf{b}]$ splits the unknowns into r **basic** (pivot) variables and $n - r$ **free** variables. Each nonzero row of the RREF contains exactly one basic variable, so it expresses that variable directly in terms of the free ones. Setting the free variables to parameters $s, t, \ldots$ gives the solution in the form

$$\mathbf{x} = \mathbf{x}_p + s\,\mathbf{n}_1 + t\,\mathbf{n}_2 + \cdots,$$

where $\mathbf{x}_p$ (all free variables zero) is a particular solution and $\mathbf{n}_1, \mathbf{n}_2, \ldots$ are homogeneous solutions forming a basis of the **null space** of $\mathbf{A}$ (Chapter 9). This is the constructive content of the Rouché–Capelli theorem.

Worked Example 8.27 — An underdetermined system

Solve $\mathbf{M}\mathbf{x} = \mathbf{b}$ with $\mathbf{M}$ from Worked Example 8.15 and $\mathbf{b} = (2, 2, 4)^{\mathsf{T}}$.

Step 1 (consistency). $\mathbf{b}$ satisfies $b_3 = b_1 + b_2$, the compatibility condition found in Worked Example 8.15, so a solution exists.

Step 2 (reduce).

$$\left[\begin{array}{cccc|c} 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 0 & 2 & 2 \\ 3 & 6 & 1 & 5 & 4 \end{array}\right] \to \left[\begin{array}{cccc|c} 1 & 2 & 1 & 3 & 2 \\ 0 & 0 & -2 & -4 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \to \left[\begin{array}{cccc|c} 1 & 2 & 0 & 1 & 1 \\ 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

(The last step is $R_2 \leftarrow -\tfrac{1}{2}R_2$, then $R_1 \leftarrow R_1 - R_2$.)

Step 3 (read off). Basic variables x_1, x_3; free variables $x_2 = s$, $x_4 = t$:

$$x_1 = 1 - 2s - t, \qquad x_3 = 1 - 2t,$$

$$\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + s \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ -2 \\ 1 \end{bmatrix}.$$

Step 4 (check). $\mathbf{M}(1,0,1,0)^{\mathsf{T}} = (2,2,4)^{\mathsf{T}}$ ✓; $\mathbf{M}(-2,1,0,0)^{\mathsf{T}} = (0,0,0)^{\mathsf{T}}$ ✓; $\mathbf{M}(-1,0,-2,1)^{\mathsf{T}} = (-1-2+3,\ -2+0+2,\ -3-2+5)^{\mathsf{T}} = \mathbf{0}$ ✓. The two free parameters match $n - r = 4 - 2$.

Reading. The RREF also displays the column relations of Worked Example 8.15: the free column 2 reads $(2, 0)$, i.e. $\mathbf{m}_2 = 2\mathbf{m}_1$; the free column 4 reads $(1, 2)$, i.e. $\mathbf{m}_4 = \mathbf{m}_1 + 2\mathbf{m}_3$.

Common Pitfall

Inverting by Gauss–Jordan in floating point still needs pivoting. And having $\mathbf{A}^{-1}$ in hand is rarely useful by itself: if you need $\mathbf{A}^{-1}\mathbf{B}$, solve $\mathbf{A}\mathbf{X} = \mathbf{B}$; if you need one entry of the inverse (say a transfer impedance), solve for one column. Compute the full inverse only when its entries are themselves the deliverable — a nodal impedance matrix for a short-circuit study, or a covariance matrix.

Engineering Note

In power-system fault analysis the **bus impedance matrix** $\mathbf{Z}_{\text{bus}} = \mathbf{Y}_{\text{bus}}^{-1}$ is genuinely wanted: its diagonal entries are the Thévenin impedances at each bus, which set the short-circuit currents. For large networks it is built not by Gauss–Jordan but by adding one branch at a time with rank-one updates — Sherman–Morrison again — or only the needed columns are computed from an $\mathbf{LU}$ factorisation.

8.9 LU decomposition

Motivation

In practice the same coefficient matrix meets many right-hand sides. A transient simulation with fixed time step solves $(\mathbf{G} + \mathbf{C}/h)\mathbf{v}_{k+1} = \mathbf{i}_{k+1} + (\mathbf{C}/h)\mathbf{v}_k$ at every step with the *same* matrix; a sweep over source positions, a Newton iteration with a frozen Jacobian, or an adjoint sensitivity computation all have the same pattern. Gaussian elimination spends $\tfrac{2}{3}n^3$ flops on the matrix and only $O(n^2)$ on each right-hand side. **LU decomposition** packages the expensive part — the record of the elimination — so it can be stored and reused.

Elimination as factorisation

Each stage of Gaussian elimination without interchanges is a left-multiplication by a unit lower triangular matrix. For $n = 3$:

$$\mathbf{L}_1 = \begin{bmatrix} 1 & 0 & 0 \\ -m_{21} & 1 & 0 \\ -m_{31} & 0 & 1 \end{bmatrix}, \qquad \mathbf{L}_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -m_{32} & 1 \end{bmatrix}, \qquad \mathbf{L}_2\mathbf{L}_1\mathbf{A} = \mathbf{U}.$$

Inverting, $\mathbf{A} = \mathbf{L}_1^{-1}\mathbf{L}_2^{-1}\mathbf{U}$. The inverses simply flip the signs of the multipliers, and — the pleasant surprise — their product just collects the multipliers in place:

$$\mathbf{L} = \mathbf{L}_1^{-1}\mathbf{L}_2^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ m_{21} & 1 & 0 \\ m_{31} & m_{32} & 1 \end{bmatrix}.$$

LU factorisation

If Gaussian elimination on $\mathbf{A}$ ($n \times n$) succeeds without row interchanges, then

$$\mathbf{A} = \mathbf{L}\mathbf{U},$$

where $\mathbf{L}$ is **unit lower triangular** (ones on the diagonal) with $\ell_{ik} = m_{ik}$, the elimination multipliers, and $\mathbf{U}$ is the upper triangular result of elimination. This happens if and only if the leading principal minors $\det \mathbf{A}_{1:k,1:k}$, $k = 1, \dots, n-1$, are nonzero, and then (for nonsingular $\mathbf{A}$) the factorisation is unique.

For **any** nonsingular $\mathbf{A}$ there is a permutation matrix $\mathbf{P}$ (recording the row interchanges) such that

$$\mathbf{P}\mathbf{A} = \mathbf{L}\mathbf{U}.$$

With partial pivoting, all $|\ell_{ik}| \le 1$.

Why the leading minors. The pivots satisfy $u_{11}u_{22}\cdots u_{kk} = \det \mathbf{A}_{1:k,1:k}$, because row replacements using rows $1, \dots, k$ do not change the leading $k \times k$ determinant. So the k-th pivot is nonzero exactly when the k-th leading minor is. Uniqueness: if $\mathbf{L}_1\mathbf{U}_1 = \mathbf{L}_2\mathbf{U}_2$ then $\mathbf{L}_2^{-1}\mathbf{L}_1 = \mathbf{U}_2\mathbf{U}_1^{-1}$ is both unit lower and upper triangular, hence $\mathbf{I}$. $\square$

Solving with the factors

With $\mathbf{P}\mathbf{A} = \mathbf{L}\mathbf{U}$, the system $\mathbf{A}\mathbf{x} = \mathbf{b}$ becomes $\mathbf{L}\mathbf{U}\mathbf{x} = \mathbf{P}\mathbf{b}$, solved in two triangular sweeps:

$$\begin{aligned} &\text{forward substitution:} && \mathbf{L}\mathbf{y} = \mathbf{P}\mathbf{b}, && y_i = (\mathbf{P}\mathbf{b})_i - \sum_{j<i} \ell_{ij} y_j, \\ &\text{back substitution:} && \mathbf{U}\mathbf{x} = \mathbf{y}, && x_i = \frac{1}{u_{ii}}\Big(y_i - \sum_{j>i} u_{ij} x_j\Big). \end{aligned}$$

Each sweep costs about n^2 flops. So: **factor once for $\tfrac{2}{3}n^3$, then solve each right-hand side for $2n^2$.** For $n = 1000$ that is 6.7×10^8 flops for the factorisation and 2×10^6 per solve — a ratio of over 300. The determinant also comes free: $\det \mathbf{A} = \det \mathbf{P}^{-1} \det \mathbf{L} \det \mathbf{U} = \pm \prod_i u_{ii}$.

Direct (Doolittle) formulas

Equating entries of $\mathbf{A} = \mathbf{L}\mathbf{U}$ row by row gives the factors without writing out elimination steps. For $k = 1, \dots, n$:

$$u_{kj} = a_{kj} - \sum_{p=1}^{k-1} \ell_{kp} u_{pj} \quad (j \ge k), \qquad \ell_{ik} = \frac{1}{u_{kk}}\Big(a_{ik} - \sum_{p=1}^{k-1} \ell_{ip} u_{pk}\Big) \quad (i > k).$$

This is the **Doolittle** algorithm ($\mathbf{L}$ unit diagonal). **Crout**'s variant puts the ones on the diagonal of $\mathbf{U}$ instead. Both perform the same arithmetic as elimination in a different order, which matters for memory access patterns on real hardware but not for the result.

Worked Example 8.28 — LU of the running example and reuse

Step 1 (factor). From Worked Example 8.21 the multipliers are $m_{21} = -\tfrac{1}{2}$, $m_{31} = -\tfrac{1}{4}$, $m_{32} = -\tfrac{1}{2}$ and the final triangle is $\mathbf{U}$:

$$\mathbf{G} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ -\tfrac{1}{2} & 1 & 0 \\ -\tfrac{1}{4} & -\tfrac{1}{2} & 1 \end{bmatrix}}_{\mathbf{L}} \underbrace{\begin{bmatrix} 4 & -2 & -1 \\ 0 & 3 & -\tfrac{3}{2} \\ 0 & 0 & 3 \end{bmatrix}}_{\mathbf{U}}.$$

Check row 3: $-\tfrac{1}{4}(4,-2,-1) - \tfrac{1}{2}(0,3,-\tfrac{3}{2}) + (0,0,3) = (-1,\ \tfrac{1}{2} - \tfrac{3}{2},\ \tfrac{1}{4} + \tfrac{3}{4} + 3) = (-1,-1,4)$ ✓.

Step 2 (solve for the original sources). $\mathbf{L}\mathbf{y} = (8,2,1)^{\mathsf{T}}$: $y_1 = 8$, $y_2 = 2 + \tfrac{1}{2}(8) = 6$, $y_3 = 1 + \tfrac{1}{4}(8) + \tfrac{1}{2}(6) = 6$. Then $\mathbf{U}\mathbf{v} = \mathbf{y}$ gives $v_3 = 2$, $v_2 = (6 + 3)/3 = 3$, $v_1 = (8 + 6 + 2)/4 = 4$ ✓ — the same arithmetic as before, now split into two clean sweeps.

Step 3 (reuse for a new source pattern). A 1 mA source at node 1 only: $\mathbf{L}\mathbf{y} = (1,0,0)^{\mathsf{T}}$ gives $y = (1, \tfrac{1}{2}, \tfrac{1}{4} + \tfrac{1}{4}) = (1, \tfrac{1}{2}, \tfrac{1}{2})$. Then $v_3 = \tfrac{1}{2}/3 = \tfrac{1}{6}$, $v_2 = (\tfrac{1}{2} + \tfrac{3}{2}\cdot\tfrac{1}{6})/3 = (\tfrac{1}{2} + \tfrac{1}{4})/3 = \tfrac{1}{4}$, $v_1 = (1 + 2\cdot\tfrac{1}{4} + \tfrac{1}{6})/4 = \tfrac{5/3}{4} = \tfrac{5}{12}$. This is the first column of $\mathbf{Z}$ from Worked Example 8.12 ✓, obtained for about $2n^2$ operations.

Step 4. $\det \mathbf{G} = 4 \cdot 3 \cdot 3 = 36$ ✓.

Worked Example 8.29 — A factorisation that needs a permutation

Factor $\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 5 \\ 4 & 6 & 8 \end{bmatrix}$.

Step 1. Eliminate column 1 with $m_{21} = 2$, $m_{31} = 4$: rows 2 and 3 become $(0, 0, 3)$ and $(0, 2, 4)$.

Step 2. The $(2,2)$ pivot is 0, so no $\mathbf{LU}$ factorisation of $\mathbf{A}$ exists (indeed the leading minor $\begin{vmatrix} 1 & 1 \\ 2 & 2 \end{vmatrix} = 0$). Interchange rows 2 and 3 — together with their stored multipliers — to get pivot 2; column 2 is then already clear below it.

Step 3. With $\mathbf{P}$ swapping rows 2 and 3,

$$\mathbf{PA} = \begin{bmatrix} 1 & 1 & 1 \\ 4 & 6 & 8 \\ 2 & 2 & 5 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}}_{\mathbf{L}} \underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 4 \\ 0 & 0 & 3 \end{bmatrix}}_{\mathbf{U}}.$$

Check row 2: $4(1,1,1) + (0,2,4) = (4,6,8)$ ✓; row 3: $2(1,1,1) + (0,0,3) = (2,2,5)$ ✓.

Step 4. $\det \mathbf{A} = (-1)^1 \cdot 1 \cdot 2 \cdot 3 = -6$. Direct check by cofactors: $1(16-30) - 1(16-20) + 1(12-8) = -14 + 4 + 4 = -6$ ✓.

(A library routine with partial pivoting would instead start by swapping row 3, whose entry 4 is the largest in column 1, to the top; it produces different but equally valid $\mathbf{P}, \mathbf{L}, \mathbf{U}$ with all $|\ell_{ik}| \le 1$.)

Symmetric matrices: $\mathbf{LDL}^{\mathsf{T}}$ and Cholesky

If $\mathbf{A}$ is symmetric and has an $\mathbf{LU}$ factorisation, write $\mathbf{U} = \mathbf{D}\tilde{\mathbf{U}}$ with $\mathbf{D} = \operatorname{diag}(u_{11}, \dots, u_{nn})$ and $\tilde{\mathbf{U}}$ unit upper triangular. Then $\mathbf{A} = \mathbf{L}\mathbf{D}\tilde{\mathbf{U}}$, and symmetry plus uniqueness force $\tilde{\mathbf{U}} = \mathbf{L}^{\mathsf{T}}$:

$$\mathbf{A} = \mathbf{L}\mathbf{D}\mathbf{L}^{\mathsf{T}}.$$

If moreover $\mathbf{A}$ is **symmetric positive definite** (SPD: $\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$), all pivots are positive and we can split $\mathbf{D} = \mathbf{D}^{1/2}\mathbf{D}^{1/2}$.

Cholesky factorisation

A real symmetric matrix $\mathbf{A}$ is positive definite if and only if it can be written

$$\mathbf{A} = \mathbf{R}\mathbf{R}^{\mathsf{T}}$$

with $\mathbf{R}$ lower triangular with positive diagonal entries; $\mathbf{R}$ is then unique. (For complex Hermitian positive definite $\mathbf{A}$, $\mathbf{A} = \mathbf{R}\mathbf{R}^{\mathsf{H}}$.) The entries are computed column by column:

$$r_{kk} = \sqrt{a_{kk} - \sum_{p<k} r_{kp}^2}, \qquad r_{ik} = \frac{1}{r_{kk}}\Big(a_{ik} - \sum_{p<k} r_{ip} r_{kp}\Big) \quad (i > k).$$

Cholesky needs about $\tfrac{1}{3}n^3$ flops — half of $\mathbf{LU}$ — and half the storage, requires no pivoting for stability, and doubles as a **test for positive definiteness**: if a square root of a non-positive number is ever requested, $\mathbf{A}$ is not SPD. Nodal matrices of passive resistive networks with a ground connection are SPD (§8.2), so Cholesky is the natural direct solver for them.

Worked Example 8.30 — $\mathbf{LDL}^{\mathsf{T}}$ and Cholesky for $\mathbf{G}$

Step 1 ($\mathbf{LDL}^{\mathsf{T}}$). From Worked Example 8.28, the pivots are $4, 3, 3$, so

$$\mathbf{G} = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 \\ -\frac{1}{4} & -\frac{1}{2} & 1 \end{bmatrix} \begin{bmatrix} 4 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{4} \\ 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 1 \end{bmatrix}.$$

Indeed $\mathbf{D}\mathbf{L}^{\mathsf{T}} = \mathbf{U}$: row 2 of $\mathbf{D}\mathbf{L}^{\mathsf{T}}$ is $3(0, 1, -\tfrac{1}{2}) = (0, 3, -\tfrac{3}{2})$ ✓.

Step 2 (Cholesky by the formulas).

$$\begin{aligned} r_{11} &= \sqrt{4} = 2, & r_{21} &= -2/2 = -1, & r_{31} &= -1/2, \\ r_{22} &= \sqrt{4 - (-1)^2} = \sqrt{3}, & r_{32} &= \frac{-1 - (-\tfrac{1}{2})(-1)}{\sqrt{3}} = -\frac{3/2}{\sqrt{3}} = -\frac{\sqrt{3}}{2}, \\ r_{33} &= \sqrt{4 - \tfrac{1}{4} - \tfrac{3}{4}} = \sqrt{3}. \end{aligned}$$

$$\mathbf{R} = \begin{bmatrix} 2 & 0 & 0 \\ -1 & \sqrt{3} & 0 \\ -\frac{1}{2} & -\frac{\sqrt{3}}{2} & \sqrt{3} \end{bmatrix} = \mathbf{L}\mathbf{D}^{1/2}.$$

Step 3 (check). Row 3 of $\mathbf{R}$ dotted with itself: $\frac{1}{4} + \frac{3}{4} + 3 = 4 = g_{33}$ ✓; row 3 with row 2: $\frac{1}{2} - \frac{3}{2} + 0 = -1 = g_{32}$ ✓.

Reading. All square-root arguments were positive, confirming $\mathbf{G}$ is positive definite — the algebraic face of "every resistor absorbs power". Units: $\mathbf{R}$ has units $\mathrm{mS}^{1/2}$.

Worked Example 8.31 — Factor once, step many times

A node with conductance $G = 1$ mS to ground and capacitance $C = 1\ \mu$F to ground, coupled through 2 mS to a second identical node, is simulated with backward Euler at step $h = 1$ ms. Find the system matrix and explain the cost per step.

Step 1. Backward Euler replaces $\mathbf{C}\dot{\mathbf{v}}$ by $\mathbf{C}(\mathbf{v}_{k+1} - \mathbf{v}_k)/h$, giving $(\mathbf{G} + \mathbf{C}/h)\mathbf{v}_{k+1} = \mathbf{i}_{k+1} + (\mathbf{C}/h)\mathbf{v}_k$. Here $C/h = 10^{-6}/10^{-3} = 1$ mS, so

$$\mathbf{G} + \mathbf{C}/h = \begin{bmatrix} 1+2+1 & -2 \\ -2 & 1+2+1 \end{bmatrix} = \begin{bmatrix} 4 & -2 \\ -2 & 4 \end{bmatrix} \text{ mS}.$$

Step 2. Factor once: $\ell_{21} = -\tfrac{1}{2}$, $u_{22} = 4 - 1 = 3$, so $\mathbf{L} = \begin{bmatrix} 1 & 0 \\ -\tfrac{1}{2} & 1 \end{bmatrix}$, $\mathbf{U} = \begin{bmatrix} 4 & -2 \\ 0 & 3 \end{bmatrix}$.

Step 3. Each time step is now one forward and one back substitution. With a 3 mA step input at node 1 and $\mathbf{v}_0 = \mathbf{0}$: the right-hand side is $(3, 0)$; $\mathbf{y} = (3, 1.5)$; $v_2 = 0.5$, $v_1 = (3 + 1)/4 = 1$ V. Next step, right-hand side $(3, 0) + 1 \cdot (1, 0.5) = (4, 0.5)$, $\mathbf{y} = (4, 2.5)$, $v_2 = \tfrac{5}{6} \approx 0.833$, $v_1 = (4 + \tfrac{5}{3})/4 = \tfrac{17}{12} \approx 1.417$ V. The voltages approach the DC solution of $\begin{bmatrix} 3 & -2 \\ -2 & 3 \end{bmatrix}\mathbf{v} = (3, 0)^{\mathsf{T}}$, namely $(1.8, 1.2)$ V.

Cost. For an n-node circuit simulated over N steps the work is $\tfrac{2}{3}n^3 + 2Nn^2$ instead of $N \cdot \tfrac{2}{3}n^3$. Changing h changes the matrix and forces a refactorisation — one reason adaptive-step simulators change h reluctantly.

Common Pitfall

The $\mathbf{L}$ and $\mathbf{U}$ returned by a library are tied to its permutation: `scipy.linalg.lu` returns $\mathbf{P}, \mathbf{L}, \mathbf{U}$ with $\mathbf{A} = \mathbf{PLU}$, while MATLAB's `[L,U,P]=lu(A)` satisfies $\mathbf{PA} = \mathbf{LU}$. Forgetting to apply the permutation to $\mathbf{b}$ gives a wrong answer with no error message. Use the library's paired factor/solve routines (`lu_factor` / `lu_solve`, `decomposition`) rather than handling $\mathbf{P}$ yourself.

Engineering Note

SPICE's inner loop is: stamp the linearised companion models into the MNA matrix, $\mathbf{LU}$-factor it (sparse, §8.11), forward/back substitute, update the nonlinear devices, repeat until Newton converges, then advance time. Because the sparsity *pattern* never changes, the symbolic part of the factorisation (which entries will be nonzero, which pivot order to use) is computed once at the start of the run, and only the numerical values are refactored at each iteration.

8.10 Numerical conditioning

Motivation

Every number that enters a real linear system carries an error. Component values have tolerances of 1% or 5%; measured data carry noise; and even exact inputs are rounded to about 16 significant digits when stored in double precision. The question is how much those small input errors can change the solution. For some systems the answer is "about as much as the inputs changed"; for others a change in the fifth digit of the data changes the first digit of the answer. This sensitivity is a property of the *problem*, not of the algorithm used to solve it, and it is measured by the **condition number**. Distinguishing an ill-conditioned problem from an unstable algorithm (Worked Example 8.23 was the latter) is the central skill of numerical linear algebra.

Vector and matrix norms

To talk about "small" errors in vectors and matrices we need sizes.

Vector norms

For $\mathbf{x} \in \mathbb{C}^n$:

$$\|\mathbf{x}\|_1 = \sum_i |x_i|, \qquad \|\mathbf{x}\|_2 = \Big(\sum_i |x_i|^2\Big)^{1/2}, \qquad \|\mathbf{x}\|_\infty = \max_i |x_i|.$$

Each satisfies $\|\mathbf{x}\| \geq 0$ with equality only for $\mathbf{x} = \mathbf{0}$, $\|c\mathbf{x}\| = |c|\|\mathbf{x}\|$, and the triangle inequality $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$.

Induced matrix norm

Given a vector norm, the **induced** (operator) norm of $\mathbf{A}$ is the largest factor by which $\mathbf{A}$ can stretch a vector:

$$\|\mathbf{A}\| = \max_{\mathbf{x} \neq \mathbf{0}} \frac{\|\mathbf{A}\mathbf{x}\|}{\|\mathbf{x}\|}.$$

For the three common vector norms this gives

$$\|\mathbf{A}\|_1 = \max_j \sum_i |a_{ij}| \ \text{(max column sum)}, \qquad \|\mathbf{A}\|_\infty = \max_i \sum_j |a_{ij}| \ \text{(max row sum)},$$

$$\|\mathbf{A}\|_2 = \sigma_{\max}(\mathbf{A}) = \sqrt{\lambda_{\max}(\mathbf{A}^{\mathsf{H}}\mathbf{A})} \ \text{(spectral norm)}.$$

The **Frobenius norm** $\|\mathbf{A}\|_F = \big(\sum_{i,j} |a_{ij}|^2\big)^{1/2}$ is not induced but is easy to compute and satisfies $\|\mathbf{A}\|_2 \leq \|\mathbf{A}\|_F$.

Why the max-row-sum formula. For $\|\mathbf{x}\|_\infty = 1$, $|(\mathbf{A}\mathbf{x})_i| \le \sum_j |a_{ij}||x_j| \le \sum_j |a_{ij}|$, so $\|\mathbf{A}\|_\infty \le \max_i \sum_j |a_{ij}|$; and choosing $x_j = \operatorname{sgn} a_{pj}$ for the row p attaining the maximum achieves it. The 1-norm case is the same argument on columns. □

Induced norms satisfy $\|\mathbf{A}\mathbf{x}\| \le \|\mathbf{A}\|\|\mathbf{x}\|$ and are **submultiplicative**, $\|\mathbf{A}\mathbf{B}\| \le \|\mathbf{A}\|\|\mathbf{B}\|$, with $\|\mathbf{I}\| = 1$. For the running example $\|\mathbf{G}\|_1 = \|\mathbf{G}\|_\infty = 4 + 2 + 1 = 7$ mS and, $\mathbf{G}$ being symmetric, $\|\mathbf{G}\|_2$ equals its largest eigenvalue, 6 mS.

The condition number

Condition number

For a nonsingular square matrix,

$$\kappa(\mathbf{A}) = \|\mathbf{A}\|\,\|\mathbf{A}^{-1}\|,$$

with a subscript naming the norm used ($\kappa_1, \kappa_2, \kappa_\infty$). For singular $\mathbf{A}$ one sets $\kappa = \infty$.

Basic facts: $\kappa(\mathbf{A}) \ge \|\mathbf{A}\mathbf{A}^{-1}\| = 1$; $\kappa(c\mathbf{A}) = \kappa(\mathbf{A})$, so unlike the determinant it is **independent of overall scale and units**; $\kappa_2(\mathbf{A}) = \sigma_{\max}/\sigma_{\min}$, the ratio of the largest to smallest singular values; for symmetric positive definite $\mathbf{A}$, $\kappa_2 = \lambda_{\max}/\lambda_{\min}$. Geometrically, $\mathbf{A}$ maps the unit sphere to an ellipsoid, and κ_2 is the ratio of its longest to shortest axis: a very flat ellipsoid means some directions are almost annihilated, and recovering them (inverting) amplifies errors enormously. Moreover $1/\kappa(\mathbf{A})$ is the relative distance from $\mathbf{A}$ to the nearest singular matrix: $\min\{\|\delta\mathbf{A}\|/\|\mathbf{A}\| : \ \mathbf{A} + \delta\mathbf{A} \text{ singular}\} = 1/\kappa(\mathbf{A})$ in the 2-norm.

Perturbation theorems

Sensitivity to the right-hand side

If $\mathbf{A}\mathbf{x} = \mathbf{b}$ with $\mathbf{b} \neq \mathbf{0}$ and $\mathbf{A}(\mathbf{x} + \delta\mathbf{x}) = \mathbf{b} + \delta\mathbf{b}$, then

$$\frac{\|\delta\mathbf{x}\|}{\|\mathbf{x}\|} \le \kappa(\mathbf{A})\,\frac{\|\delta\mathbf{b}\|}{\|\mathbf{b}\|},$$

and for every $\mathbf{A}$ there are $\mathbf{b}, \delta\mathbf{b}$ for which equality holds.

Proof. Subtracting, $\mathbf{A}\,\delta\mathbf{x} = \delta\mathbf{b}$, so $\|\delta\mathbf{x}\| = \|\mathbf{A}^{-1}\delta\mathbf{b}\| \le \|\mathbf{A}^{-1}\|\|\delta\mathbf{b}\|$. Also $\|\mathbf{b}\| = \|\mathbf{A}\mathbf{x}\| \le \|\mathbf{A}\|\|\mathbf{x}\|$, i.e. $1/\|\mathbf{x}\| \le \|\mathbf{A}\|/\|\mathbf{b}\|$. Multiply the two inequalities. Equality needs $\mathbf{x}$ to be a direction that $\mathbf{A}$ stretches most and $\delta\mathbf{b}$ a direction that $\mathbf{A}^{-1}$ stretches most. □

Sensitivity to the matrix

If $(\mathbf{A} + \delta\mathbf{A})(\mathbf{x} + \delta\mathbf{x}) = \mathbf{b}$ and $\kappa(\mathbf{A})\|\delta\mathbf{A}\|/\|\mathbf{A}\| < 1$, then

$$\frac{\|\delta\mathbf{x}\|}{\|\mathbf{x}\|} \le \frac{\kappa(\mathbf{A})\,\|\delta\mathbf{A}\|/\|\mathbf{A}\|}{1 - \kappa(\mathbf{A})\,\|\delta\mathbf{A}\|/\|\mathbf{A}\|} \approx \kappa(\mathbf{A})\frac{\|\delta\mathbf{A}\|}{\|\mathbf{A}\|}.$$

Sketch. Expand: $\mathbf{A}\,\delta\mathbf{x} = -\delta\mathbf{A}(\mathbf{x} + \delta\mathbf{x})$, so $\|\delta\mathbf{x}\| \le \|\mathbf{A}^{-1}\|\|\delta\mathbf{A}\|(\|\mathbf{x}\| + \|\delta\mathbf{x}\|)$; collect the $\|\delta\mathbf{x}\|$ terms and divide by $\|\mathbf{x}\|$. □

Rule of thumb. If the data are accurate to d significant digits and $\kappa \approx 10^k$, expect the solution to be accurate to about $d - k$ digits. With double precision ($d \approx 16$) a system with $\kappa = 10^{10}$ still yields about 6 good digits; with component values known to 1% ($d = 2$) even $\kappa = 100$ leaves nothing trustworthy in the worst case.

Residual versus error

Having computed $\hat{\mathbf{x}}$, we can always evaluate the **residual** $\mathbf{r} = \mathbf{b} - \mathbf{A}\hat{\mathbf{x}}$, but not the **error** $\mathbf{e} = \mathbf{x} - \hat{\mathbf{x}}$ (we do not know $\mathbf{x}$). Since $\mathbf{A}\mathbf{e} = \mathbf{r}$, the first perturbation theorem applies with $\delta\mathbf{b} = \mathbf{r}$:

$$\frac{\|\mathbf{e}\|}{\|\mathbf{x}\|} \le \kappa(\mathbf{A})\frac{\|\mathbf{r}\|}{\|\mathbf{b}\|}.$$

A small residual guarantees a small error only when κ is modest. Gaussian elimination with partial pivoting is **backward stable**: it always produces a small residual, $\|\mathbf{r}\| \lesssim n\rho u\|\mathbf{A}\|\|\hat{\mathbf{x}}\|$. So for a well-conditioned system its answer is accurate, and for an ill-conditioned one its answer is as accurate as the problem allows — no algorithm can do substantially better with the same data.

An ill-conditioned 2×2 system: nearly parallel lines $x + y = 2$ (blue) and $x + 1.05y = 2.05$ (red) meet at $(1, 1)$. Raising the second right-hand side by only 0.05 (dashed red) slides the intersection along the blue line to $(0, 2)$. The narrow angle between the lines is what a large condition number looks like.

Worked Example 8.32 — Condition number of the running example

Compute $\kappa_1(\mathbf{G})$ and $\kappa_2(\mathbf{G})$.

Step 1. $\|\mathbf{G}\|_1 = \max(4 + 2 + 1,\ 2 + 4 + 1,\ 1 + 1 + 4) = 7$.

Step 2. From Worked Example 8.12, the column sums of $|\mathbf{Z}|$ are $\frac{30}{36}, \frac{30}{36}, \frac{24}{36}$, so $\|\mathbf{Z}\|_1 = \frac{5}{6}$.

Step 3. $\kappa_1(\mathbf{G}) = 7 \cdot \frac{5}{6} = \frac{35}{6} \approx 5.83$.

Step 4 (2-norm). $\mathbf{G}$ is symmetric positive definite with eigenvalues $3 - \sqrt{3} \approx 1.268$, $3 + \sqrt{3} \approx 4.732$ and 6 (Chapter 10). So $\kappa_2 = 6/(3 - \sqrt{3}) = 3 + \sqrt{3} \approx 4.73$.

Reading. $\kappa \approx 5$: a 1% uncertainty in the source currents produces at most about 5% relative uncertainty in the node voltages. Diagonally dominant nodal matrices of modest size are typically this benign.

Worked Example 8.33 — Nearly parallel lines

For $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1.05 \end{bmatrix}$, $\mathbf{b} = (2,\ 2.05)^{\mathsf{T}}$, verify the perturbation bound in the ∞-norm for $\delta\mathbf{b} = (0,\ 0.05)^{\mathsf{T}}$, and show that a small residual can hide a large error.

Step 1. $\det \mathbf{A} = 0.05$, $\mathbf{A}^{-1} = \dfrac{1}{0.05}\begin{bmatrix} 1.05 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 21 & -20 \\ -20 & 20 \end{bmatrix}$.

Step 2. $\|\mathbf{A}\|_\infty = 2.05$, $\|\mathbf{A}^{-1}\|_\infty = 41$, so $\kappa_\infty = 84.05$.

Step 3. $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b} = (42 - 41,\ -40 + 41)^{\mathsf{T}} = (1, 1)^{\mathsf{T}}$. $\delta\mathbf{x} = \mathbf{A}^{-1}\delta\mathbf{b} = (-1,\ 1)^{\mathsf{T}}$, so $\mathbf{x} + \delta\mathbf{x} = (0, 2)^{\mathsf{T}}$.

Step 4 (bound). Relative input change $\|\delta\mathbf{b}\|_\infty / \|\mathbf{b}\|_\infty = 0.05/2.05 \approx 0.0244$; relative output change $\|\delta\mathbf{x}\|_\infty / \|\mathbf{x}\|_\infty = 1$, i.e. 100%. The amplification is $1/0.0244 = 41$, within the bound 84.05 ✓.

Step 5 (residual). Suppose some method returned $\hat{\mathbf{x}} = (0, 2)^{\mathsf{T}}$ for the original system. Its residual $\mathbf{b} - \mathbf{A}\hat{\mathbf{x}} = (2 - 2,\ 2.05 - 2.1)^{\mathsf{T}} = (0, -0.05)^{\mathsf{T}}$ is about 2% of $\|\mathbf{b}\|$, yet the error is 100%. Checking that "the answer satisfies the equations" is not enough for an ill-conditioned system.

Worked Example 8.34 — Wide spread of conductances

Two nodes are joined by a low-resistance strap of 1000 mS (1 Ω), and each has 1 mS to ground. Analyse

$$\mathbf{G}_s = \begin{bmatrix} 1001 & -1000 \\ -1000 & 1001 \end{bmatrix} \text{ mS}.$$

Step 1. The eigenvalues are 1001 ∓ 1000, i.e. 1 and 2001, with eigenvectors $(1,1)$ (both nodes moving together, opposed only by the ground conductances) and $(1,-1)$ (the nodes moving apart, opposed by the strap). Hence $\kappa_2 = 2001$.

Step 2. With 1 mA into node 1, $\mathbf{v} = \frac{1}{2001}(1001, 1000)^{\mathsf{T}} \approx (0.50025,\ 0.49975)$ V.

Step 3. The *sum* $v_1 + v_2$ is well determined, but the *difference* $v_1 - v_2 = 1/2001 \approx 0.5$ mV — the voltage across the strap — is the small result of subtracting two nearly equal numbers. A 1% change in the strap conductance changes $v_1 - v_2$ by about 1%, while a 10^{-6} relative rounding error in v_1 and v_2 changes it by about 0.1%: the relative accuracy lost is roughly the factor κ.

Reading. Circuits with conductances spanning many decades (milliohm interconnect next to megaohm bias resistors, or the GΩ "gmin" conductances SPICE adds to ground) have large condition numbers. Engineers often replace an extremely stiff element by an ideal short (merging the nodes), which removes the ill-conditioning analytically.

Worked Example 8.35 — Scaling can remove apparent ill-conditioning

The system of Worked Example 8.23, with its first equation multiplied by 10^4, reads

$$\begin{bmatrix} 1 & 10^4 \\ 1 & 1 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 10^4 \\ 2 \end{bmatrix}.$$

Step 1. $\|\mathbf{A}\|_\infty = 10001$, $\mathbf{A}^{-1} = \dfrac{1}{1-10^4}\begin{bmatrix} 1 & -10^4 \\ -1 & 1 \end{bmatrix}$, $\|\mathbf{A}^{-1}\|_\infty = 10001/9999 \approx 1.0002$, so $\kappa_\infty \approx 1.0 \times 10^4$.

Step 2. Dividing row 1 by 10^4 (**row equilibration**, making the largest entry in each row equal to 1) gives back $\begin{bmatrix} 10^{-4} & 1 \\ 1 & 1 \end{bmatrix}$ with $\kappa_\infty \approx 4.0$.

Reading. The solution $(1.0001,\ 0.9999)$ is the same either way; only the bookkeeping changed. Part of a large condition number may therefore come from poor scaling — for example, MNA equations mixing currents in amperes with voltage constraints in volts, or conductances in siemens next to capacitances in farads — and is cured by rescaling rows or columns (equilibration), which good solvers do automatically. The part that survives every scaling, as in Worked Example 8.33, is intrinsic to the problem.

A notoriously ill-conditioned family

The **Hilbert matrix** $\mathbf{H}_n$, with $h_{ij} = 1/(i+j-1)$, arises in least-squares fitting of polynomials $1, t, t^2, \dots$ on $[0,1]$ — a reason never to fit high-degree polynomials in the monomial basis. Its condition numbers grow explosively:

n	3	5	8	10
$\kappa_2(\mathbf{H}_n)$	5.2×10^2	4.8×10^5	1.5×10^{10}	1.6×10^{13}

At $n = 10$ a double-precision solve keeps only about three correct digits. The cure is a better-conditioned formulation (orthogonal polynomials, QR factorisation — Chapter 9), not a cleverer elimination.

Common Pitfall

Do not judge conditioning by the determinant (§8.3) or by the size of the entries. And do not confuse an **ill-conditioned problem** (large κ: every method struggles) with an **unstable algorithm** (elimination without pivoting on Worked Example 8.23: a good method would have succeeded). The remedies are different: reformulate or measure better for the first; pivot or use a stable factorisation for the second.

Engineering Note

Libraries estimate κ_1 cheaply from an existing $\mathbf{LU}$ factorisation (LAPACK `dgecon`, MATLAB `rcond`, which returns $1/\kappa_1$); MATLAB's warning "Matrix is close to singular or badly scaled. RCOND = 1e-17" is this estimate. In tolerance analysis the condition number gives a quick worst-case bound before running a Monte Carlo study: if $\kappa \approx 10^3$ and components are 1% parts, some outputs may be completely unreliable, and the design should be changed (for instance, a bridge measurement should be read as a difference directly rather than computed from two large node voltages).

8.11 Sparse matrices

Motivation

A circuit with n nodes has an $n \times n$ nodal matrix, but each node connects to only a few others — typically three to five. The matrix therefore has about $5n$ nonzero entries out of n^2. For a post-layout simulation with $n = 10^6$ nodes, the dense matrix would have 10^{12} entries (8 terabytes) and dense elimination would need $\frac{2}{3} \times 10^{18}$ flops; the sparse matrix has perhaps 5×10^6 nonzeros (tens of megabytes) and can be factored in seconds. Finite-difference and finite-element field solvers, power-grid models and the normal equations of large estimation problems are all sparse in the same way. **Sparse matrix methods** store and operate on only the nonzeros, and choose algorithms that keep it that way.

Sparse matrix

A matrix is **sparse** when the number of nonzero entries, $\operatorname{nnz}(\mathbf{A})$, is small enough compared with mn that exploiting the zeros pays off — in practice $\operatorname{nnz} = O(n)$ for an $n \times n$ matrix. The **sparsity pattern** is the set of positions of the nonzeros. A matrix is **banded** with bandwidth w if $a_{ij} = 0$ for $|i-j| > w$; **tridiagonal** is $w = 1$.

The sparsity pattern of a nodal matrix *is* the circuit graph: $g_{ij} \neq 0$ ($i \neq j$) exactly when nodes i and j are joined by a branch.

Storage formats

- **Coordinate (COO, "triplet")**: three arrays of length nnz holding (row, column, value). Easy to build by stamping — duplicates are simply summed afterwards — and the usual assembly format.
- **Compressed sparse row (CSR)**: the values and column indices row by row, plus a row-pointer array of length $n+1$ marking where each row starts. Storage $2\,\text{nnz} + n + 1$; matrix–vector products touch each nonzero once ($2\,\text{nnz}$ flops).
- **Compressed sparse column (CSC)**: the same by columns; the native format of MATLAB and most direct solvers.
- **Banded storage**: the $2w+1$ diagonals as dense vectors — ideal for ladder networks and 1-D discretisations.

Worked Example 8.36 — CSR storage of a ladder network

A four-node resistive ladder has 1 S series links between consecutive nodes and 1 S to ground at every node, with the left end driven through a further 1 S from a 10 V source (Norton equivalent: 10 A into node 1). Its nodal matrix is

$$\mathbf{T} = \begin{bmatrix} 3 & -1 & 0 & 0 \\ -1 & 3 & -1 & 0 \\ 0 & -1 & 3 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 10 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

Give its CSR representation (0-based indices).

Solution. Reading row by row,

Array	Contents
values	$3, -1,\ -1, 3, -1,\ -1, 3, -1,\ -1, 2$
column index	$0, 1,\ 0, 1, 2,\ 1, 2, 3,\ 2, 3$
row pointer	$0, 2, 5, 8, 10$

Row i occupies positions $\text{ptr}[i]$ to $\text{ptr}[i+1]-1$; e.g. row 2 is positions 5–7: values $-1, 3, -1$ in columns $1, 2, 3$. Ten nonzeros out of sixteen entries here; for a 1000-node ladder it would be 2998 out of 10^6.

Fill-in and ordering

Eliminating a variable creates nonzeros wherever two of its neighbours were not already connected: the update $a_{ij} \leftarrow a_{ij} - a_{ik}a_{kj}/a_{kk}$ makes a_{ij} nonzero whenever a_{ik} and a_{kj} are. In graph terms, eliminating node k joins all its neighbours into a clique — exactly the star–mesh transformation of circuit theory. These new nonzeros are **fill-in**. The factors $\mathbf{L}$ and $\mathbf{U}$ can be far denser than $\mathbf{A}$, and the amount of fill depends dramatically on the **elimination order** (the symmetric permutation $\mathbf{P}\mathbf{A}\mathbf{P}^{\mathsf{T}}$).

Worked Example 8.37 — A star network: order is everything

A hub node connects through 1 S to each of four leaf nodes, and each leaf has 1 S to ground. Compare elimination with the hub numbered first and last.

Hub first.

$$\mathbf{A} = \begin{bmatrix} 4 & -1 & -1 & -1 & -1 \\ -1 & 2 & 0 & 0 & 0 \\ -1 & 0 & 2 & 0 & 0 \\ -1 & 0 & 0 & 2 & 0 \\ -1 & 0 & 0 & 0 & 2 \end{bmatrix}.$$

Eliminating the hub replaces the leaf block by $2\mathbf{I} - \tfrac{1}{4}\mathbf{1}\mathbf{1}^{\mathsf{T}}$ (where $\mathbf{1}$ is the all-ones vector): every one of its 12 off-diagonal zeros becomes $-\tfrac{1}{4}$. The factor $\mathbf{L}$ is completely full — the eliminated hub has "meshed" all four leaves together.

Hub last. The matrix is the "arrowhead" $\begin{bmatrix} 2\mathbf{I}_4 & -\mathbf{1} \\ -\mathbf{1}^{\mathsf{T}} & 4 \end{bmatrix}$. Eliminating leaf k affects only the hub's diagonal: $4 \leftarrow 4 - (-1)(-1)/2$. After all four leaves the hub pivot is $4 - 4 \cdot \tfrac{1}{2} = 2$, and **no fill-in** occurs: $\mathbf{L}$ has exactly the nonzeros of the lower triangle of $\mathbf{A}$.

Check. Both orders must give the same determinant: hub last, $2^4 \cdot 2 = 32$; hub first, $4 \cdot \det(2\mathbf{I} - \tfrac{1}{4}\mathbf{1}\mathbf{1}^{\mathsf{T}}) = 4 \cdot 2^3(2 - \tfrac{4}{4}) = 4 \cdot 8 \cdot 1 = 32$ ✓ (using $\det(c\mathbf{I} - \alpha\mathbf{1}\mathbf{1}^{\mathsf{T}}) = c^{n-1}(c - n\alpha)$).

Reading. Eliminate low-degree nodes first and high-degree nodes (hubs, supply rails) last. This greedy rule is the **minimum-degree** ordering; its refinements (approximate minimum degree, AMD) and **nested dissection** (recursively cutting the graph with small separators, which are numbered last) are what production sparse solvers use.

For a nodal matrix that is SPD and diagonally dominant, any symmetric ordering is numerically safe, so the ordering can be chosen purely to minimise fill. For MNA and other indefinite matrices the solver must balance sparsity against pivoting for stability (threshold pivoting: accept a pivot if it is not much smaller than the largest candidate).

Banded and tridiagonal systems

If $\mathbf{A}$ has bandwidth w, elimination without pivoting creates no fill outside the band, and the factorisation costs about $2nw^2$ flops instead of $\tfrac{2}{3}n^3$. The tridiagonal case ($w = 1$) is the **Thomas algorithm**: for the system with subdiagonal a_i, diagonal d_i, superdiagonal c_i, right-hand side b_i,

$$\begin{aligned} &\text{forward:} \quad c'_1 = \frac{c_1}{d_1}, \quad b'_1 = \frac{b_1}{d_1}, \quad c'_i = \frac{c_i}{d_i - a_i c'_{i-1}}, \quad b'_i = \frac{b_i - a_i b'_{i-1}}{d_i - a_i c'_{i-1}}, \\ &\text{backward:} \quad x_n = b'_n, \quad x_i = b'_i - c'_i x_{i+1}. \end{aligned}$$

About $8n$ flops in total — linear in n. It is stable without pivoting when the matrix is diagonally dominant.

Worked Example 8.38 — Solving the ladder by the Thomas algorithm

Solve $\mathbf{T}\mathbf{v} = \mathbf{b}$ from Worked Example 8.36. Here $d = (3, 3, 3, 2)$, $a_i = c_i = -1$.

Forward sweep.

$$\begin{aligned} c'_1 &= -\tfrac{1}{3}, & b'_1 &= \tfrac{10}{3}, \\ d_2 - a_2 c'_1 &= 3 - \tfrac{1}{3} = \tfrac{8}{3}, & c'_2 &= -\tfrac{3}{8}, & b'_2 &= \frac{0 + \frac{10}{3}}{8/3} = \tfrac{5}{4}, \\ d_3 - a_3 c'_2 &= 3 - \tfrac{3}{8} = \tfrac{21}{8}, & c'_3 &= -\tfrac{8}{21}, & b'_3 &= \frac{0 + \frac{5}{4}}{21/8} = \tfrac{10}{21}, \\ d_4 - a_4 c'_3 &= 2 - \tfrac{8}{21} = \tfrac{34}{21}, & & & b'_4 &= \frac{0 + \frac{10}{21}}{34/21} = \tfrac{5}{17}. \end{aligned}$$

Backward sweep. $v_4 = \tfrac{5}{17}$; $v_3 = \tfrac{10}{21} + \tfrac{8}{21} \cdot \tfrac{5}{17} = \tfrac{210}{357} = \tfrac{10}{17}$; $v_2 = \tfrac{5}{4} + \tfrac{3}{8} \cdot \tfrac{10}{17} = \tfrac{25}{17}$; $v_1 = \tfrac{10}{3} + \tfrac{1}{3} \cdot \tfrac{25}{17} = \tfrac{65}{17}$.

So $\mathbf{v} = \tfrac{1}{17}(65, 25, 10, 5)^{\mathsf{T}} \approx (3.824,\ 1.471,\ 0.588,\ 0.294)$ V.

Check. Row 1: $3 \cdot 65 - 25 = 170 = 17 \cdot 10$ ✓. Row 4: $-10 + 2 \cdot 5 = 0$ ✓. The pivots $3, \tfrac{8}{3}, \tfrac{21}{8}, \tfrac{34}{21}$ multiply to $\det \mathbf{T} = 34$. Successive node voltages fall by factors 2.6, 2.5 and 2 (the last is an end effect of the open termination); far from the ends of a long ladder the ratio approaches r with $r + 1/r = 3$, i.e. $(3 + \sqrt{5})/2 \approx 2.618$ of an infinite ladder, the same number met in Worked Example 8.6.

Iterative methods

For very large sparse systems — especially 3-D field problems, where even the best ordering produces heavy fill — **iterative methods** replace factorisation by repeated cheap matrix–vector products. Split $\mathbf{A} = \mathbf{D} + \mathbf{L}_A + \mathbf{U}_A$ into diagonal, strictly lower and strictly upper parts.

Jacobi and Gauss–Seidel iterations

Starting from a guess $\mathbf{x}^{(0)}$,

$$\text{Jacobi:} \quad x_i^{(k+1)} = \frac{1}{a_{ii}}\Big(b_i - \sum_{j \neq i} a_{ij} x_j^{(k)}\Big),$$

$$\text{Gauss–Seidel:} \quad x_i^{(k+1)} = \frac{1}{a_{ii}}\Big(b_i - \sum_{j<i} a_{ij} x_j^{(k+1)} - \sum_{j>i} a_{ij} x_j^{(k)}\Big).$$

In matrix form $\mathbf{x}^{(k+1)} = \mathbf{M}\mathbf{x}^{(k)} + \mathbf{c}$ with $\mathbf{M}_J = -\mathbf{D}^{-1}(\mathbf{L}_A + \mathbf{U}_A)$ or $\mathbf{M}_{GS} = -(\mathbf{D} + \mathbf{L}_A)^{-1}\mathbf{U}_A$.

Gauss–Seidel uses each new value as soon as it is available. In circuit terms, Jacobi sets every node voltage to the conductance-weighted average of its neighbours' old voltages plus its source contribution — a relaxation process.

Convergence of stationary iterations

The iteration $\mathbf{x}^{(k+1)} = \mathbf{M}\mathbf{x}^{(k)} + \mathbf{c}$ converges for every starting vector if and only if the **spectral radius** $\rho(\mathbf{M}) = \max|\lambda_i(\mathbf{M})| < 1$; the error then shrinks asymptotically by the factor $\rho(\mathbf{M})$ per iteration. Sufficient conditions: if $\mathbf{A}$ is strictly diagonally dominant, both Jacobi and Gauss–Seidel converge; if $\mathbf{A}$ is symmetric positive definite, Gauss–Seidel converges.

Sketch. The error $\mathbf{e}^{(k)} = \mathbf{x}^{(k)} - \mathbf{x}$ obeys $\mathbf{e}^{(k+1)} = \mathbf{M}\mathbf{e}^{(k)}$, so $\mathbf{e}^{(k)} = \mathbf{M}^k \mathbf{e}^{(0)}$, and $\mathbf{M}^k \to \mathbf{0}$ exactly when $\rho(\mathbf{M}) < 1$ (Chapter 10). For diagonal dominance, $\|\mathbf{M}_J\|_\infty = \max_i \sum_{j \neq i} |a_{ij}|/|a_{ii}| < 1$ directly. $\square$

Worked Example 8.39 — Jacobi and Gauss–Seidel on the running example

Solve $\mathbf{G}\mathbf{v} = (8, 2, 1)^{\mathsf{T}}$ from $\mathbf{v}^{(0)} = \mathbf{0}$.

Jacobi, iteration 1. $v_1 = \tfrac{1}{4}(8 + 2\cdot 0 + 0) = 2$, $v_2 = \tfrac{1}{4}(2 + 0 + 0) = 0.5$, $v_3 = \tfrac{1}{4}(1 + 0 + 0) = 0.25$.

Jacobi, iteration 2. $v_1 = \tfrac{1}{4}(8 + 2(0.5) + 0.25) = 2.3125$, $v_2 = \tfrac{1}{4}(2 + 2(2) + 0.25) = 1.5625$, $v_3 = \tfrac{1}{4}(1 + 2 + 0.5) = 0.875$.

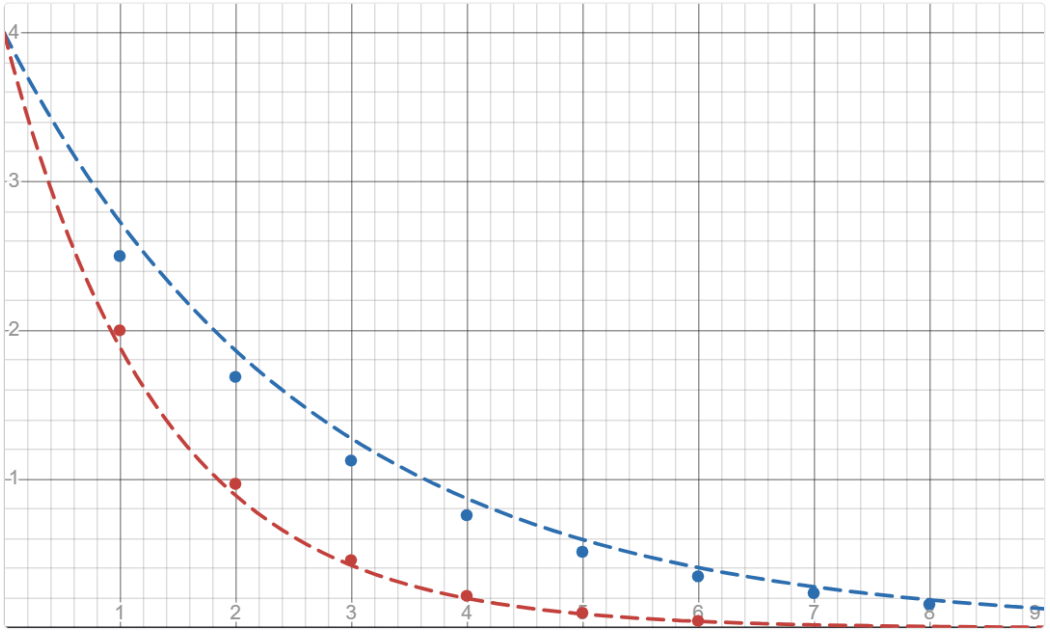
Gauss–Seidel, iteration 1. $v_1 = \tfrac{1}{4}(8) = 2$; then, using the new v_1, $v_2 = \tfrac{1}{4}(2 + 2\cdot 2 + 0) = 1.5$; then $v_3 = \tfrac{1}{4}(1 + 2 + 1.5) = 1.125$.

Gauss–Seidel, iteration 2. $v_1 = \tfrac{1}{4}(8 + 3 + 1.125) = 3.03125$, $v_2 = \tfrac{1}{4}(2 + 6.0625 + 1.125) = 2.296875$, $v_3 = \tfrac{1}{4}(1 + 3.03125 + 2.296875) = 1.58203$.

Progress. Maximum error $\|\mathbf{v}^{(k)} - (4, 3, 2)^{\mathsf{T}}\|_\infty$:

k	1	2	3	4	5	6
Jacobi	2.500	1.688	1.125	0.758	0.512	0.347
Gauss–Seidel	2.000	0.969	0.456	0.216	0.102	0.048

Rates. $\rho(\mathbf{M}_J) = (1 + \sqrt{3})/4 \approx 0.683$ and $\rho(\mathbf{M}_{GS}) \approx 0.472$: Gauss–Seidel removes roughly twice as many digits per iteration. Reaching 10^{-6} accuracy takes about $\ln(10^{-6}/4)/\ln 0.683 \approx 40$ Jacobi or 21 Gauss–Seidel iterations — far more work than a 3×3 elimination, which is why iterative methods are only worthwhile for large sparse systems.



Maximum error against iteration number k for Jacobi (blue) and Gauss–Seidel (red) on the running example. The dashed curves $4\rho^k$ with $\rho = 0.683$ and 0.472 show the geometric decay set by the spectral radius of each iteration matrix.

Modern practice goes well beyond these classical methods. **Successive over-relaxation** (SOR) extrapolates the Gauss–Seidel step by a factor $1 < \omega < 2$; the **conjugate gradient** method (for SPD matrices) and **GMRES** (general matrices) are Krylov-subspace methods that converge in a number of iterations governed by $\sqrt{\kappa}$ or by the eigenvalue distribution; and **preconditioning** — solving $\mathbf{P}^{-1}\mathbf{A}\mathbf{x} = \mathbf{P}^{-1}\mathbf{b}$ with $\mathbf{P} \approx \mathbf{A}$ cheap to invert, such as an incomplete $\mathbf{LU}$ factorisation that discards fill — is what makes them fast. These are treated in Chapter 22.

Common Pitfall

Never form the inverse, or any dense intermediate, of a sparse matrix: $\mathbf{A}^{-1}$ is generally full (Worked Example 8.12 shows a dense $\mathbf{Z}$). Likewise, building a sparse matrix by inserting entries one at a time into a CSR/CSC structure is quadratic in cost; assemble in COO (triplet) form and convert once. And beware of accidental densification: in NumPy/SciPy, `A + 0` or `A * dense_array` may silently return a dense result.

Engineering Note

SPICE-class simulators use sparse $\mathbf{LU}$ with a Markowitz-type ordering chosen once per circuit (KLU, designed specifically for circuit matrices, exploits their near-block-triangular structure). Field solvers using finite elements produce SPD sparse matrices solved by preconditioned conjugate gradients or multigrid. Power-system load flow uses sparse $\mathbf{LU}$ of the Jacobian with ordering by minimum degree — introduced for exactly this purpose by Tinney and Walker in 1967. In every case the decisive step is the same: recognise the graph behind the matrix and order the unknowns to keep the factors sparse.

Summary

Concept	Key formula or fact
Product	$c_{ij} = \sum_k a_{ik}b_{kj}$; $\mathbf{Ax} = \sum_k x_k \mathbf{a}_k$; $\mathbf{AB} \neq \mathbf{BA}$ in general

Concept	Key formula or fact
Transpose rules	$(\mathbf{AB})^{\mathsf{T}} = \mathbf{B}^{\mathsf{T}}\mathbf{A}^{\mathsf{T}}$, $(\mathbf{AB})^{\mathsf{H}} = \mathbf{B}^{\mathsf{H}}\mathbf{A}^{\mathsf{H}}$
Nodal matrix	$\mathbf{G} = \mathbf{A}\mathbf{G}_b\mathbf{A}^{\mathsf{T}}$ (symmetric, positive definite if grounded)
2×2 determinant / inverse	$ad - bc$; $\dfrac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$
Cofactor expansion	$\det \mathbf{A} = \sum_k a_{ik}C_{ik}$, $C_{ik} = (-1)^{i+k}M_{ik}$
Determinant rules	$\det(\mathbf{AB}) = \det \mathbf{A} \det \mathbf{B}$; swap $\to -$; replacement $\to$ unchanged; $\det = \pm \prod u_{ii}$
Cramer's rule	$x_k = \det \mathbf{A}_k / \det \mathbf{A}$
Adjugate inverse	$\mathbf{A}^{-1} = \operatorname{adj} \mathbf{A} / \det \mathbf{A}$
Product inverse	$(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$
Sherman–Morrison	$(\mathbf{A} + \mathbf{uv}^{\mathsf{T}})^{-1} = \mathbf{A}^{-1} - \dfrac{\mathbf{A}^{-1}\mathbf{uv}^{\mathsf{T}}\mathbf{A}^{-1}}{1 + \mathbf{v}^{\mathsf{T}}\mathbf{A}^{-1}\mathbf{u}}$
Schur complement (Kron)	$\mathbf{S} = \mathbf{A}_{11} - \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{A}_{21}$, $\det \mathbf{A} = \det \mathbf{A}_{22} \det \mathbf{S}$
Rank	number of pivots = independent rows = independent columns
Rouché–Capelli	consistent $\iff \operatorname{rank} \mathbf{A} = \operatorname{rank}[\mathbf{A} \mid \mathbf{b}]$; $n - r$ free parameters
Solution structure	$\mathbf{x} = \mathbf{x}_p + \mathbf{x}_h$, $\mathbf{A}\mathbf{x}_h = \mathbf{0}$
Gaussian elimination	$m_{ik} = a_{ik}/a_{kk}$; cost $\frac{2}{3}n^3$; back substitution n^2
Partial pivoting	largest $\lvert a_{pk}\rvert$ as pivot; $\lvert m_{ik}\rvert \le 1$
Gauss–Jordan	$[\mathbf{A} \mid \mathbf{I}] \to [\mathbf{I} \mid \mathbf{A}^{-1}]$; cost $\approx n^3$ per solve
LU	$\mathbf{PA} = \mathbf{LU}$; $\mathbf{Ly} = \mathbf{Pb}$, $\mathbf{Ux} = \mathbf{y}$; $2n^2$ per extra right-hand side
Cholesky	$\mathbf{A} = \mathbf{R}\mathbf{R}^{\mathsf{T}}$ for SPD; $r_{kk} = \sqrt{a_{kk} - \sum_{p<k} r_{kp}^2}$; cost $\frac{1}{3}n^3$
Induced norms	$\lVert\mathbf{A}\rVert_1$ = max column sum, $\lVert\mathbf{A}\rVert_\infty$ = max row sum, $\lVert\mathbf{A}\rVert_2 = \sigma_{\max}$
Condition number	$\kappa = \lVert\mathbf{A}\rVert\lVert\mathbf{A}^{-1}\rVert$; $\kappa_2 = \sigma_{\max}/\sigma_{\min}$
Perturbation bound	$\lVert\delta\mathbf{x}\rVert/\lVert\mathbf{x}\rVert \le \kappa\, \lVert\delta\mathbf{b}\rVert/\lVert\mathbf{b}\rVert$; lose $\approx \log_{10}\kappa$ digits
Thomas algorithm	tridiagonal solve in $\approx 8n$ flops
Stationary iteration	converges $\iff \rho(\mathbf{M}) < 1$; diagonal dominance suffices

Further reading

- E. Kreyszig, *Advanced Engineering Mathematics*, 10th ed., Wiley — Ch. 7 (matrices, determinants, linear systems) and Ch. 20 (numerical linear algebra).
- G. Strang, *Introduction to Linear Algebra*, 5th ed., Wellesley-Cambridge Press — the column picture, elimination as $\mathbf{LU}$, and an engineering-minded treatment of rank and solvability.
- L. N. Trefethen and D. Bau, *Numerical Linear Algebra*, SIAM — conditioning, stability, pivoting and the backward-error viewpoint, in short self-contained lectures.
- T. A. Davis, *Direct Methods for Sparse Linear Systems*, SIAM — sparse storage, fill-reducing orderings and sparse $\mathbf{LU}$/Cholesky; with C. W. Ho and A. E. Ruehli's 1975 paper on modified nodal analysis as the classic circuit-side reference.

Chapter 9 — Vector Spaces and Linear Transformations

> **Prerequisites:** Ch. 2 (complex numbers), Ch. 5 (definite integrals), Ch. 8 (matrices, Gaussian elimination, rank) **Learning outcomes:** After this chapter you can
>
> - recognise a vector space — whether its elements are arrows, columns of numbers, polynomials, matrices or signals — and test whether a subset is a subspace;
> - decide linear independence, find a basis and the dimension of a space, and compute coordinates and change-of-basis matrices;
> - represent a linear transformation by a matrix, and compute bases for its null space, column space, row space and left null space using the rank–nullity theorem;
> - work with inner products on $\mathbb{R}^n$, $\mathbb{C}^n$ and on function spaces, and use orthogonality, norms and the Cauchy–Schwarz inequality;
> - project onto lines and subspaces, build orthonormal bases with Gram–Schmidt (including QR and Legendre polynomials);
> - solve overdetermined systems by least squares and fit lines, polynomials and exponentials to measured data.

Chapter 8 treated a matrix as a table of coefficients and a linear system as something to be solved by elimination. This chapter changes the point of view. A matrix is a *machine* that maps one vector space into another; the questions "does $\mathbf{A}\mathbf{x} = \mathbf{b}$ have a solution?" and "is it unique?" become questions about two subspaces attached to that machine. The same language then applies, almost word for word, to objects that do not look like columns of numbers at all: the set of solutions of a linear differential equation, the set of polynomials of degree at most n, the set of finite-energy signals. That is the real pay-off. When you later expand a periodic waveform in a Fourier series (Ch. 15), you will be doing exactly what this chapter does with arrows in $\mathbb{R}^3$: projecting a vector onto an orthogonal basis. When you fit a calibration curve to noisy measurements, you will be projecting a data vector onto the column space of a matrix.

The chapter is organised in two halves. Sections 9.1–9.6 build the *algebraic* structure: spaces, subspaces, independence, bases, linear maps and the four fundamental subspaces. Sections 9.7–9.11 add *geometry* — lengths, angles and orthogonality — and use it for approximation: projection, Gram–Schmidt and least squares.

9.1 Vector spaces

Motivation

An engineer meets many different objects that can be *added* and *scaled*: forces and field vectors, node-voltage vectors of a circuit, sampled signals $x[0], \dots, x[N-1]$, continuous waveforms $x(t)$, polynomials used in filter design, matrices of network parameters. In each case the same handful of algebraic rules hold — addition is commutative, there is a zero, scaling distributes over addition, and so on. Rather than prove every theorem separately for arrows, for sampled signals and for waveforms, we prove it once for any set obeying those rules. That set is a **vector space**.

The most important consequence is **superposition**. If $x_1(t)$ and $x_2(t)$ are responses of a linear circuit, so is $a x_1(t) + b x_2(t)$. The set of all possible responses is therefore closed under linear combinations — it is a vector space — and everything we learn about bases and dimension tells us how many "independent" responses we need to describe them all.

Definition

Vector space

A **vector space** over a field $\mathbb{F}$ (in this book $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$) is a set V of objects called **vectors**, together with two operations

- vector addition $\mathbf{u} + \mathbf{v} \in V$ for all $\mathbf{u}, \mathbf{v} \in V$,
- scalar multiplication $c\,\mathbf{u} \in V$ for all $c \in \mathbb{F}$, $\mathbf{u} \in V$,

such that for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and $a, b \in \mathbb{F}$:

1. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ (commutativity)
2. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$ (associativity)
3. there is a **zero vector** $\mathbf{0} \in V$ with $\mathbf{u} + \mathbf{0} = \mathbf{u}$
4. each $\mathbf{u}$ has an **additive inverse** $-\mathbf{u}$ with $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$
5. $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$
6. $(a + b)\mathbf{u} = a\mathbf{u} + b\mathbf{u}$
7. $a(b\mathbf{u}) = (ab)\mathbf{u}$
8. $1\,\mathbf{u} = \mathbf{u}$

When $\mathbb{F} = \mathbb{R}$ we speak of a **real** vector space, when $\mathbb{F} = \mathbb{C}$ of a **complex** vector space.

The two **closure** requirements (sums and scalar multiples stay inside V) are the ones that fail in practice; the eight axioms are almost always inherited from ordinary arithmetic of numbers or functions. A few consequences follow directly from the axioms and are used constantly:

$$0\,\mathbf{u} = \mathbf{0}, \qquad c\,\mathbf{0} = \mathbf{0}, \qquad (-1)\mathbf{u} = -\mathbf{u}, \qquad c\,\mathbf{u} = \mathbf{0} \;\Rightarrow\; c = 0 \text{ or } \mathbf{u} = \mathbf{0}.$$

Proof of the first. By axiom 6, $0\mathbf{u} = (0 + 0)\mathbf{u} = 0\mathbf{u} + 0\mathbf{u}$. Add $-(0\mathbf{u})$ to both sides: $\mathbf{0} = 0\mathbf{u}$. The others are similar.

The standard examples

Space	Vectors	Addition and scaling	Zero vector
$\mathbb{R}^n$	columns $(x_1, \dots, x_n)^{\mathsf{T}}$ of reals	componentwise	$(0, \dots, 0)^{\mathsf{T}}$
$\mathbb{C}^n$	columns of complex numbers (phasor vectors)	componentwise, complex scalars	$(0, \dots, 0)^{\mathsf{T}}$
$\mathbb{R}^{m \times n}$	$m \times n$ real matrices	entrywise	zero matrix
$\mathcal{P}_n$	polynomials $a_0 + a_1 t + \cdots + a_n t^n$ of degree $\le n$	add/scale coefficients	the zero polynomial
$C[a, b]$	continuous real functions on $[a, b]$	$(f+g)(t) = f(t) + g(t)$, $(cf)(t) = cf(t)$	$f(t) \equiv 0$
ℓ^2 / $\mathbb{R}^N$	sampled signals $x[n]$	samplewise	the all-zero sequence
$L^2(a, b)$	functions with $\int_a^b \lvert f(t)\rvert^2\, dt < \infty$ (finite-energy signals)	pointwise	$f \equiv 0$

The table shows how broad the idea is. A polynomial is "a vector" because two polynomials can be added and scaled and the axioms hold; it does not matter that we never draw it as an arrow. The function spaces $C[a, b]$ and $L^2(a, b)$ are the setting for Fourier analysis: a periodic signal will be treated as a single point in an infinite-dimensional space.

Worked Example 9.1 — Checking the axioms for $\mathcal{P}_2$

Show that $\mathcal{P}_2 = \{a_0 + a_1 t + a_2 t^2 : a_k \in \mathbb{R}\}$ is a real vector space.

Closure. If $p(t) = a_0 + a_1 t + a_2 t^2$ and $q(t) = b_0 + b_1 t + b_2 t^2$, then

$$(p + q)(t) = (a_0 + b_0) + (a_1 + b_1)t + (a_2 + b_2)t^2 \in \mathcal{P}_2, \qquad (cp)(t) = ca_0 + ca_1 t + ca_2 t^2 \in \mathcal{P}_2.$$

Axioms. Each axiom reduces to the same axiom for real numbers applied coefficient by coefficient. For instance, commutativity: $(p + q)(t)$ and $(q + p)(t)$ have coefficients $a_k + b_k = b_k + a_k$. The zero vector is $0 + 0t + 0t^2$; the inverse of p is $-a_0 - a_1 t - a_2 t^2$.

Remark. The set of polynomials of degree *exactly* 2 is **not** a vector space: $t^2 + t$ and $-t^2$ both have degree 2 but their sum t does not, and the zero polynomial is missing.

Non-examples

Most failures are failures of closure or of the zero vector:

- **The first quadrant** $\{(x, y) : x \ge 0,\ y \ge 0\}$: closed under addition but not under scaling by -1.
- **A line not through the origin**, e.g. $\{(x, y) : y = 2x + 1\}$: $(0, 1)$ and $(1, 3)$ are in it but their sum $(1, 4)$ is not; and $(0, 0)$ is not in it. Such a set is an **affine** set — a vector space shifted away from the origin.
- **Unit vectors** $\{\mathbf{x} : \|\mathbf{x}\| = 1\}$: the sum of two unit vectors is generally not a unit vector.
- **Positive voltages** $\{v(t) : v(t) > 0\}$: not closed under multiplication by -1, no zero vector.
- **Solutions of a non-homogeneous equation**, e.g. all $y(t)$ with $y'' + 3y' + 2y = 1$ (worked example below).

Worked Example 9.2 — Solution sets of differential equations

(a) Show that the set S_0 of twice-differentiable functions satisfying $y'' + 3y' + 2y = 0$ is a vector space. (b) Show that the set S_1 satisfying $y'' + 3y' + 2y = 1$ is not.

(a) S_0 is a subset of the vector space of all twice-differentiable functions, so the axioms are inherited; we need only closure (this is the subspace test of Section 9.2). Let $y_1, y_2 \in S_0$ and $a, b \in \mathbb{R}$, and put $y = ay_1 + by_2$. Because differentiation is linear,

$$y'' + 3y' + 2y = a\,(y_1'' + 3y_1' + 2y_1) + b\,(y_2'' + 3y_2' + 2y_2) = a \cdot 0 + b \cdot 0 = 0.$$

So $y \in S_0$. The zero function is in S_0. Hence S_0 is a vector space. (In Ch. 12 you will find $S_0 = \{c_1 e^{-t} + c_2 e^{-2t}\}$.)

(b) Take $y_1 = y_2 = \tfrac{1}{2}$ (a constant: $0 + 0 + 2 \cdot \tfrac{1}{2} = 1$, so it is in S_1). Then $y_1 + y_2 = 1$ gives $0 + 0 + 2 = 2 \neq 1$. Closure fails; also $y \equiv 0$ gives $0 \neq 1$, so the zero vector is absent. S_1 is an affine set: $S_1 = \tfrac{1}{2} + S_0$, a particular solution plus the homogeneous space.

This "particular solution plus vector space" structure is the general shape of the solution set of every consistent linear problem $L(\mathbf{x}) = \mathbf{b}$, whether L is a matrix or a differential operator. It reappears in Section 9.6.

Sinusoids of a fixed frequency

The set

$$\mathcal{S}_\omega = \{A\cos(\omega t + \phi) : A \ge 0,\ \phi \in \mathbb{R}\}$$

of all sinusoids at a fixed angular frequency ω is closed under addition — the sum of two sinusoids at the same frequency is another sinusoid at that frequency — and under real scaling (a negative scale factor is a phase shift by π). With the zero signal ($A = 0$) included it is a real vector space. Writing $A\cos(\omega t + \phi) = a\cos\omega t + b\sin\omega t$ with $a = A\cos\phi$, $b = -A\sin\phi$ shows that each sinusoid is fixed by the pair $(a, b) \in \mathbb{R}^2$; equivalently by the **phasor** $\mathbf{X} = Ae^{j\phi} = a - jb \in \mathbb{C}$. Adding sinusoids corresponds to adding phasors. This is the vector-space reason why phasor analysis (Ch. 2) works: the map "sinusoid $\mapsto$ phasor" respects addition and real scaling, so it is a linear bijection between $\mathcal{S}_\omega$ and $\mathbb{C}$ viewed as the real space $\mathbb{R}^2$ (Section 9.5 calls this an *isomorphism*).

Worked Example 9.3 — Adding sinusoids as vectors

Write $x(t) = 3\cos(\omega t) + 4\cos(\omega t - \pi/2)$ as a single sinusoid.

As vectors in the basis $\{\cos\omega t,\ \sin\omega t\}$**.** $\cos(\omega t - \pi/2) = \sin\omega t$, so $x = 3\cos\omega t + 4\sin\omega t$, i.e. coordinates $(a, b) = (3, 4)$.

Convert. $A = \sqrt{a^2 + b^2} = 5$ and, from $a = A\cos\phi$, $b = -A\sin\phi$: $\cos\phi = 0.6$, $\sin\phi = -0.8$, so $\phi = -0.9273$ rad $= -53.13°$.

$$x(t) = 5\cos(\omega t - 53.13°).$$

Phasor check. $3e^{j0} + 4e^{-j\pi/2} = 3 - 4j = 5\angle -53.13°$. Same answer, because the phasor map is linear.

Common Pitfall — "vector" does not mean "arrow"

In this chapter a vector is *any element of a vector space*. The word says nothing about geometry. A polynomial, a 3×3 matrix and a voltage waveform are all vectors in suitable spaces. Conversely, a list of numbers is only a vector if the operations you intend to perform on it are componentwise addition and scaling: a list of (resistance, temperature, date) readings is a data record, not a meaningful element of $\mathbb{R}^3$.

Engineering Note — why EE lives in complex vector spaces

Steady-state AC analysis replaces each node voltage by a phasor, so the unknown vector of nodal analysis lies in $\mathbb{C}^n$ and the admittance matrix has complex entries. Every definition in this chapter works over $\mathbb{C}$; the one change needed is in the inner product (Section 9.8), where complex conjugation is required so that lengths remain real and positive.

9.2 Subspaces

Motivation

Inside a big vector space we constantly meet smaller sets that are vector spaces in their own right: the plane of all current vectors satisfying Kirchhoff's current law, the set of band-limited signals, the polynomials that vanish at a given point, the set of all outputs a system can produce. Recognising these as **subspaces** lets us use all of the vector-space machinery inside them.

Definition and the subspace test

Subspace

A subset W of a vector space V is a **subspace** if W is itself a vector space under the operations of V.

Checking eight axioms again is unnecessary: they are inherited from V. Only closure and the presence of the zero vector can fail.

Subspace test

A subset $W \subseteq V$ is a subspace if and only if

1. $\mathbf{0} \in W$;
2. $\mathbf{u}, \mathbf{v} \in W \Rightarrow \mathbf{u} + \mathbf{v} \in W$ (closed under addition);
3. $\mathbf{u} \in W,\ c \in \mathbb{F} \Rightarrow c\mathbf{u} \in W$ (closed under scaling).

Conditions 2 and 3 can be merged: W is a subspace iff $\mathbf{0} \in W$ and $a\mathbf{u} + b\mathbf{v} \in W$ for all $\mathbf{u}, \mathbf{v} \in W$ and scalars a, b.

Proof sketch. Necessity is clear. For sufficiency, axioms 1, 2 and 5–8 hold in W because they hold in V and the operations are the same. Axiom 3 is condition 1. For axiom 4, if $\mathbf{u} \in W$ then $-\mathbf{u} = (-1)\mathbf{u} \in W$ by condition 3.

Every vector space V has two **trivial subspaces**: $\{\mathbf{0}\}$ and V itself. The subspaces of $\mathbb{R}^3$ are exactly: $\{\mathbf{0}\}$, lines through the origin, planes through the origin, and $\mathbb{R}^3$. The phrase "through the origin" is essential — it is condition 1.

Worked Example 9.4 — Four subspace tests

Decide which are subspaces.

(a) $W_a = \{(x, y, z) \in \mathbb{R}^3 : x + 2y - z = 0\}$. Zero: $0 + 0 - 0 = 0$ ✓. If $\mathbf{u} = (x_1, y_1, z_1)$ and $\mathbf{v} = (x_2, y_2, z_2)$ satisfy the equation, then $a\mathbf{u} + b\mathbf{v}$ gives $a(x_1 + 2y_1 - z_1) + b(x_2 + 2y_2 - z_2) = 0$ ✓. **Subspace** (a plane through the origin).

(b) $W_b = \{(x, y, z) : x + 2y - z = 4\}$. $(0, 0, 0)$ gives $0 \neq 4$. **Not a subspace** (a parallel plane shifted off the origin).

(c) $W_c = \{p \in \mathcal{P}_3 : p(1) = 0\}$. The zero polynomial vanishes at 1 ✓. $(ap + bq)(1) = a\,p(1) + b\,q(1) = 0$ ✓. **Subspace.**

(d) $W_d = \{(x, y) \in \mathbb{R}^2 : xy = 0\}$ (the union of the two axes). $(1, 0)$ and $(0, 1)$ are in W_d but $(1, 1)$ is not. **Not a subspace.** A union of subspaces is usually not a subspace.

A useful pattern emerges: **the solution set of a homogeneous linear condition is always a subspace; an inhomogeneous condition (right-hand side $\neq 0$) or a nonlinear condition usually is not.**

Span

Linear combination and span

A **linear combination** of $\mathbf{v}_1, \dots, \mathbf{v}_k$ is any vector $c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k$. The **span** of the set is the collection of all linear combinations:

$$\operatorname{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} = \{c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k : c_i \in \mathbb{F}\}.$$

If $\operatorname{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} = W$, the vectors **span** (or **generate**) W.

A span is a subspace

$\operatorname{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is a subspace of V, and it is the *smallest* subspace containing all the $\mathbf{v}_i$.

Proof. The zero vector is the combination with all $c_i = 0$. A linear combination of two linear combinations is again a linear combination (collect coefficients). Any subspace containing every $\mathbf{v}_i$ must, by closure, contain every combination of them.

Deciding whether $\mathbf{b} \in \operatorname{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ in $\mathbb{R}^n$ is the same as deciding whether the linear system

$$\begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_k \end{bmatrix} \begin{bmatrix} c_1 \\ \vdots \\ c_k \end{bmatrix} = \mathbf{b}$$

is consistent — a Chapter 8 problem solved by Gaussian elimination.

Worked Example 9.5 — Membership in a span

Is $\mathbf{b} = (1, 4, 7)^{\mathsf{T}}$ in $\operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2\}$ with $\mathbf{v}_1 = (1, 2, 3)^{\mathsf{T}}$, $\mathbf{v}_2 = (0, 1, 2)^{\mathsf{T}}$? What about $\mathbf{d} = (1, 4, 8)^{\mathsf{T}}$?

Row-reduce the augmented matrix with both right-hand sides:

$$\left[\begin{array}{cc|cc} 1 & 0 & 1 & 1 \\ 2 & 1 & 4 & 4 \\ 3 & 2 & 7 & 8 \end{array}\right] \xrightarrow{R_2 - 2R_1,\ R_3 - 3R_1} \left[\begin{array}{cc|cc} 1 & 0 & 1 & 1 \\ 0 & 1 & 2 & 2 \\ 0 & 2 & 4 & 5 \end{array}\right] \xrightarrow{R_3 - 2R_2} \left[\begin{array}{cc|cc} 1 & 0 & 1 & 1 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 1 \end{array}\right].$$

For $\mathbf{b}$ the last row reads $0 = 0$: consistent, $c_1 = 1$, $c_2 = 2$. Check: $1(1, 2, 3) + 2(0, 1, 2) = (1, 4, 7)$ ✓.

For $\mathbf{d}$ the last row reads $0 = 1$: inconsistent, so $\mathbf{d} \notin$ the span. Geometrically, the span is the plane through the origin with normal $\mathbf{v}_1 \times \mathbf{v}_2 = (1, -2, 1)$; indeed $1 - 8 + 7 = 0$ for $\mathbf{b}$ but $1 - 8 + 8 = 1 \neq 0$ for $\mathbf{d}$.

Sums and intersections of subspaces

If U and W are subspaces of V, then

- $U \cap W$ is a subspace (check the three conditions — each holds in both);
- $U + W = \{\mathbf{u} + \mathbf{w} : \mathbf{u} \in U,\ \mathbf{w} \in W\}$ is a subspace, the smallest containing both;
- $U \cup W$ is a subspace only if one contains the other (Worked Example 9.4(d)).

When $U \cap W = \{\mathbf{0}\}$ the sum is called **direct**, written $U \oplus W$, and every vector in it splits *uniquely* as $\mathbf{u} + \mathbf{w}$. Example: every function on $[-a, a]$ splits uniquely into an even and an odd part,

$$f(t) = \underbrace{\tfrac{1}{2}\big[f(t) + f(-t)\big]}_{\text{even}} + \underbrace{\tfrac{1}{2}\big[f(t) - f(-t)\big]}_{\text{odd}},$$

and the only function that is both even and odd is 0. So the function space is the direct sum of the even subspace and the odd subspace — the fact behind the cosine-only and sine-only Fourier series of Ch. 15.

Worked Example 9.6 — Intersection of two planes

Let $U = \{x + 2y - z = 0\}$ and $W = \{x - y + z = 0\}$ in $\mathbb{R}^3$. Find $U \cap W$.

Solve both equations together. Adding them: $2x + y = 0$, so $y = -2x$; then $z = x + 2y = -3x$. Thus

$$U \cap W = \operatorname{span}\{(1, -2, -3)^{\mathsf{T}}\},$$

a line through the origin. Check: $1 - 4 + 3 = 0$ ✓ and $1 + 2 - 3 = 0$ ✓. Equivalently, the line is along $\mathbf{n}_U \times \mathbf{n}_W = (1, 2, -1) \times (1, -1, 1) = (1, -2, -3)$.

Signal subspaces

Function spaces have many natural subspaces that matter in signal processing:

- **Zero-mean signals** on $[0, T]$: $\{x : \int_0^T x\,dt = 0\}$ — a homogeneous linear condition, so a subspace (the "AC part" of signals).
- **T-periodic signals**: $x(t + T) = x(t)$ is preserved by sums and scalings.
- **Trigonometric polynomials of order N**: $\operatorname{span}\{1, \cos\omega_0 t, \sin\omega_0 t, \dots, \cos N\omega_0 t, \sin N\omega_0 t\}$ — the space in which a truncated Fourier series lives.
- **Band-limited signals**: those whose Fourier transform vanishes for $|\omega| > W$ (Ch. 16); closed under addition because the transform is linear.
- **Causal signals** $x(t) = 0$ for $t < 0$.

Signals with **peak amplitude at most 1 V**, by contrast, do not form a subspace (scaling by 2 breaks the bound).

Common Pitfall — forgetting the origin

Before doing any work, check whether $\mathbf{0}$ belongs to the set. If it does not, the set is not a subspace and you are done. Most "almost-subspaces" met in practice (operating points, biased signals, regions bounded by tolerances) are affine sets or cones, not subspaces.

Engineering Note — KCL defines a subspace

In a circuit with b branches, collect the branch currents in $\mathbf{i} \in \mathbb{R}^b$. Kirchhoff's current law at each node is a homogeneous linear equation, so the set of all KCL-compatible current vectors is a subspace of $\mathbb{R}^b$. Its dimension (Section 9.6) is the number of independent loop currents — the number of unknowns in mesh analysis.

9.3 Linear independence

Motivation

A spanning set may contain redundant vectors. If a third probe signal is just a combination of the first two, measuring it tells us nothing new; if one KVL equation is the sum of two others, writing it down adds no information and makes the system singular. **Linear independence** is the precise notion of "no redundancy". It decides whether a set of node equations has a unique solution, whether two waveforms can be separated by a receiver, and whether a set of functions can serve as a basis for a general solution of a differential equation.

Definition

Linear independence

Vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ are **linearly independent** if the only solution of

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k = \mathbf{0}$$

is the **trivial** one $c_1 = c_2 = \cdots = c_k = 0$. Otherwise they are **linearly dependent**, and a nontrivial solution is called a **dependence relation**.

Equivalent formulations, each useful in a different setting:

- The set is dependent iff **at least one vector is a linear combination of the others**. (If $c_j \neq 0$ in a dependence relation, solve for $\mathbf{v}_j$.)
- The set is independent iff **every vector in its span has a unique representation** $c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k$. (If two representations existed, their difference would be a nontrivial dependence relation.)
- In $\mathbb{F}^n$, with $\mathbf{V} = [\mathbf{v}_1 \ \cdots \ \mathbf{v}_k]$, the set is independent iff $\mathbf{V}\mathbf{c} = \mathbf{0}$ has only $\mathbf{c} = \mathbf{0}$, iff $\operatorname{rank}\mathbf{V} = k$, iff every column of the echelon form of $\mathbf{V}$ has a pivot.
- For $k = n$ (square $\mathbf{V}$): independent iff $\det\mathbf{V} \neq 0$.

Some quick facts:

- Any set containing $\mathbf{0}$ is dependent ($1 \cdot \mathbf{0} = \mathbf{0}$).
- Two vectors are dependent iff one is a scalar multiple of the other.
- **More than n vectors in $\mathbb{F}^n$ are always dependent**: $\mathbf{V}\mathbf{c} = \mathbf{0}$ then has more unknowns than equations, so a free variable exists.
- A subset of an independent set is independent; a superset of a dependent set is dependent.

Worked Example 9.7 — Testing vectors in $\mathbb{R}^3$

(a) Are $\mathbf{v}_1 = (1,2,0)^{\mathsf{T}}$, $\mathbf{v}_2 = (0,1,1)^{\mathsf{T}}$, $\mathbf{v}_3 = (1,0,-2)^{\mathsf{T}}$ independent?

Row-reduce $\mathbf{V} = [\mathbf{v}_1\ \mathbf{v}_2\ \mathbf{v}_3]$:

$$\begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 0 & 1 & -2 \end{bmatrix} \xrightarrow{R_2 - 2R_1} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 1 & -2 \end{bmatrix} \xrightarrow{R_3 - R_2} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix}.$$

Column 3 has no pivot, so the set is **dependent**. Back-substitute with $c_3 = 1$: $c_2 = 2$, $c_1 = -1$. Dependence relation:

$$-\mathbf{v}_1 + 2\mathbf{v}_2 + \mathbf{v}_3 = \mathbf{0} \quad\Longleftrightarrow\quad \mathbf{v}_3 = \mathbf{v}_1 - 2\mathbf{v}_2.$$

Check: $(1,2,0) - 2(0,1,1) = (1,0,-2)$ ✓. (Also $\det\mathbf{V} = 0$.)

(b) Are $\mathbf{w}_1 = (1,1,0)^{\mathsf{T}}$, $\mathbf{w}_2 = (1,0,1)^{\mathsf{T}}$, $\mathbf{w}_3 = (0,1,1)^{\mathsf{T}}$ independent?

$$\det\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} = 1(0-1) - 1(1-0) + 0 = -2 \neq 0.$$

Independent. Three independent vectors in $\mathbb{R}^3$ — they will turn out to be a basis.

Independence of functions

For vectors in a function space, the coefficients c_i are constants and the relation $c_1 f_1(t) + \cdots + c_k f_k(t) = 0$ must hold **for every** t in the interval. Two practical tests:

1. Evaluate at sample points. If the functions are dependent, then for any points $t_1, \dots, t_k$ the matrix $[f_j(t_i)]$ is singular. So finding *one* set of points where $\det[f_j(t_i)] \neq 0$ proves independence.

2. The Wronskian. For k functions with $k-1$ derivatives, define

$$W(f_1, \dots, f_k)(t) = \det \begin{bmatrix} f_1 & f_2 & \cdots & f_k \\ f_1' & f_2' & \cdots & f_k' \\ \vdots & & & \vdots \\ f_1^{(k-1)} & f_2^{(k-1)} & \cdots & f_k^{(k-1)} \end{bmatrix}.$$

Wronskian test

If $W(f_1, \dots, f_k)(t_0) \neq 0$ at some point t_0 of the interval, then $f_1, \dots, f_k$ are linearly independent there.

If $f_1, \dots, f_k$ are solutions of the same kth-order linear homogeneous ODE with continuous coefficients, then the converse also holds: they are independent iff $W \neq 0$, and in that case $W(t) \neq 0$ for **every** t.

Proof of the first part. Differentiate a dependence relation $\sum c_i f_i(t) = 0$ repeatedly ($k-1$ times); at t_0 this gives the square system $\mathbf{W}(t_0)\mathbf{c} = \mathbf{0}$. If $\det \mathbf{W}(t_0) \neq 0$ then $\mathbf{c} = \mathbf{0}$.

Common Pitfall — $W \equiv 0$ does not prove dependence in general

The functions t^2 and $t|t|$ on $[-1, 1]$ have Wronskian identically zero, yet they are independent (on $t > 0$ they agree, on $t < 0$ they are negatives of each other, so no single constant ratio works). The converse of the Wronskian test needs the extra hypothesis that the functions solve a common linear ODE.

Worked Example 9.8 — Exponentials and trigonometric functions

(a) Show e^t, e^{2t}, e^{3t} are independent on $\mathbb{R}$.

$$W = \det \begin{bmatrix} e^t & e^{2t} & e^{3t} \\ e^t & 2e^{2t} & 3e^{3t} \\ e^t & 4e^{2t} & 9e^{3t} \end{bmatrix} = e^{6t} \det \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{bmatrix} = 2e^{6t} \neq 0.$$

The 3×3 determinant is a Vandermonde determinant $(2-1)(3-1)(3-2) = 2$. In general $e^{s_1 t}, \dots, e^{s_k t}$ are independent whenever the s_i are distinct — the reason distinct characteristic roots give independent natural-response modes (Ch. 12).

(b) Are $1, \cos^2 t, \sin^2 t$ independent? No: $1 \cdot 1 - 1 \cdot \cos^2 t - 1 \cdot \sin^2 t = 0$ for all t. Likewise $\{1, \cos^2 t, \cos 2t\}$ is dependent because $\cos 2t = 2\cos^2 t - 1$.

(c) Are $\cos \omega t$ and $\sin \omega t$ independent? $W = \cos \omega t \cdot \omega \cos \omega t - \sin \omega t \cdot (-\omega \sin \omega t) = \omega \neq 0$ for $\omega \neq 0$. Yes — which is why every sinusoid at frequency ω needs exactly two numbers (amplitude and phase, or a and b).

Worked Example 9.9 — Independence in $\mathcal{P}_2$

Are $p_1 = 1 + t$, $p_2 = 1 - t$, $p_3 = t^2$ independent in $\mathcal{P}_2$?

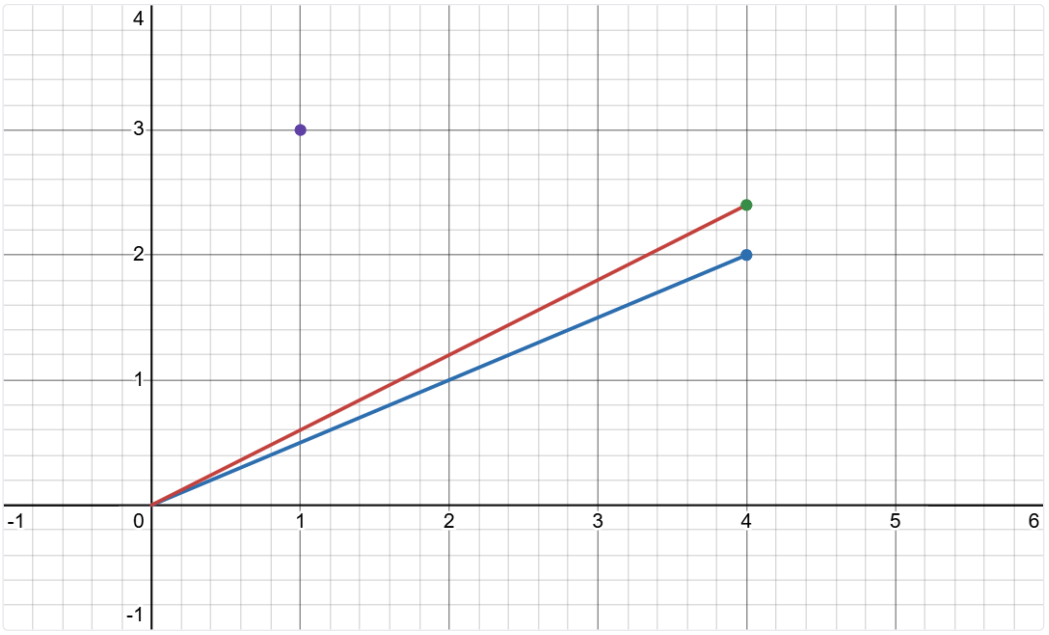
Suppose $c_1(1+t) + c_2(1-t) + c_3 t^2 = 0$ for all t. Collect powers of t:

$$(c_1 + c_2) + (c_1 - c_2)t + c_3 t^2 = 0 \;\Rightarrow\; c_1 + c_2 = 0, \quad c_1 - c_2 = 0, \quad c_3 = 0.$$

Hence $c_1 = c_2 = c_3 = 0$: **independent**. The method — matching coefficients — converts a question about functions into a question about the coefficient vectors $(1, 1, 0)$, $(1, -1, 0)$, $(0, 0, 1)$ in $\mathbb{R}^3$. This translation is made systematic by *coordinates* in Section 9.4.

Geometric picture

In $\mathbb{R}^2$ two vectors are independent iff they are not collinear; in $\mathbb{R}^3$ three vectors are independent iff they are not coplanar. The determinant measures the area or volume of the parallelogram/parallelepiped they span, so "nearly dependent" vectors give a small determinant relative to the product of their lengths. The graph below shows two nearly parallel vectors; any vector in the plane can still be written as a combination of them, but the coefficients become large and very sensitive to noise.



Nearly parallel vectors $\mathbf{v}_1 = (4, 2)$ and $\mathbf{v}_2 = (4, 2.4)$ are independent, but $\mathbf{b} = (1, 3) = -6\,\mathbf{v}_1 + 6.25\,\mathbf{v}_2$ needs large, cancelling coefficients — the hallmark of ill-conditioning.

Engineering Note — near-dependence in measurements

In practice the question is rarely "independent or not?" but "how independent?". Two antenna-array steering vectors at nearly the same angle, two sensor channels that are almost proportional, or regressor columns t and t^2 sampled on $[100, 101]$ are all *technically* independent but numerically almost dependent. The relevant measure is the condition number of the matrix of columns, introduced in Ch. 8 and made precise with singular values in Ch. 10. It reappears in least squares (Section 9.11).

9.4 Basis and dimension

Motivation

A basis is a minimal set of "building blocks" for a space: enough vectors to span it, with no redundancy. Once a basis is chosen, every vector is described by a unique list of numbers — its **coordinates** — and abstract vectors (polynomials, sinusoids, signals) can be handled by matrix computations. The number of vectors in a basis, the **dimension**, is the number of independent degrees of freedom: the number of initial conditions a circuit needs, the number of loop currents, the number of samples needed to pin down a band-limited signal over an interval.

Definition

Basis

A set $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ in V is a **basis** of V if

1. $\mathcal{B}$ is linearly independent, and
2. $\mathcal{B}$ spans V.

Unique representation

$\mathcal{B}$ is a basis of V iff every $\mathbf{v} \in V$ can be written in exactly one way as

$$\mathbf{v} = c_1\mathbf{b}_1 + \cdots + c_n\mathbf{b}_n.$$

The numbers $c_1, \dots, c_n$ are the **coordinates** of $\mathbf{v}$ relative to $\mathcal{B}$, written $[\mathbf{v}]_{\mathcal{B}} = (c_1, \dots, c_n)^{\mathsf{T}}$.

Spanning gives existence; independence gives uniqueness. Note that a basis is an *ordered* list: swapping $\mathbf{b}_1$ and $\mathbf{b}_2$ swaps the first two coordinates.

Standard bases.

Space	Standard basis	Dimension
$\mathbb{R}^n$, $\mathbb{C}^n$	$\mathbf{e}_1 = (1, 0, \dots, 0)^{\mathsf{T}}, \dots, \mathbf{e}_n$	n
$\mathcal{P}_n$	$1, t, t^2, \dots, t^n$	$n + 1$
$\mathbb{R}^{m \times n}$	$\mathbf{E}_{ij}$ (1 in position (i, j), zeros elsewhere)	mn
$\mathcal{S}_\omega$ (sinusoids at ω)	$\cos \omega t,\ \sin \omega t$	2
trig. polynomials of order N	$1, \cos k\omega_0 t, \sin k\omega_0 t\ (k = 1, \dots, N)$	$2N + 1$

Space	Standard basis	Dimension
solutions of an nth-order linear homogeneous ODE	any n independent solutions	n

Dimension

The definition of dimension rests on a counting theorem.

Exchange (replacement) lemma

If V is spanned by m vectors, then any linearly independent set in V has at most m vectors.

Proof sketch. Let $\mathbf{w}_1, \dots, \mathbf{w}_m$ span V and let $\mathbf{u}_1, \dots, \mathbf{u}_k$ be independent with $k > m$. Each $\mathbf{u}_j = \sum_i a_{ij}\mathbf{w}_i$, i.e. $[\mathbf{u}_1 \cdots \mathbf{u}_k] = [\mathbf{w}_1 \cdots \mathbf{w}_m]\mathbf{A}$ with $\mathbf{A}$ of size $m \times k$. Since $k > m$, the homogeneous system $\mathbf{A}\mathbf{c} = \mathbf{0}$ has a nonzero solution. Then $\sum_j c_j \mathbf{u}_j = [\mathbf{w}_1 \cdots \mathbf{w}_m]\mathbf{A}\mathbf{c} = \mathbf{0}$ — a nontrivial dependence among the $\mathbf{u}_j$, contradiction.

Invariance of dimension

All bases of a vector space V have the same number of elements. This number is the **dimension** $\dim V$. By convention $\dim\{\mathbf{0}\} = 0$. If no finite set spans V, V is **infinite-dimensional**.

Proof. If $\mathcal{B}_1$ has n_1 elements and $\mathcal{B}_2$ has n_2, apply the lemma both ways: $\mathcal{B}_2$ independent and $\mathcal{B}_1$ spanning gives $n_2 \le n_1$; the reverse gives $n_1 \le n_2$.

The function spaces $C[a,b]$ and $L^2(a,b)$ are infinite-dimensional: for every n the functions $1, t, \dots, t^n$ are independent in them, so no finite set can span. This is why a Fourier series needs infinitely many terms in general, and why "convergence" becomes an issue there (Ch. 15) that never arises in $\mathbb{R}^n$.

Consequences in an n-dimensional space

Let $\dim V = n$.

1. Any $n+1$ or more vectors are dependent; any $n-1$ or fewer cannot span.
2. Any n independent vectors form a basis; any n spanning vectors form a basis. (So for exactly n vectors, check *one* of the two conditions.)
3. Every independent set can be **extended** to a basis; every spanning set can be **pruned** to a basis.
4. If W is a subspace of V then $\dim W \le n$, with equality iff $W = V$.

Property 2 is a great labour-saver: three independent vectors in $\mathbb{R}^3$, such as $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$ of Worked Example 9.7(b), automatically form a basis.

For two subspaces U, W of a finite-dimensional space there is also a counting formula,

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W),$$

the linear-algebra analogue of $|A \cup B| = |A| + |B| - |A \cap B|$. For the two planes of Worked Example 9.6: $\dim(U+W) = 2 + 2 - 1 = 3$, so together they span $\mathbb{R}^3$.

Worked Example 9.10 — Basis and dimension of a plane

Find a basis and the dimension of $W = \{(x, y, z) : x + 2y - z = 0\}$.

The single equation has one pivot (x) and two free variables (y, z). Write $x = -2y + z$:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = y \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

So $W = \operatorname{span}\{(-2,1,0)^{\mathsf T}, (1,0,1)^{\mathsf T}\}$. The two vectors are independent (look at the 2nd and 3rd components: each has a 1 where the other has a 0). Hence they form a basis and $\dim W = 2$.

General rule. The solution space of $\mathbf{A}\mathbf{x} = \mathbf{0}$ has dimension equal to the number of free variables, and the "special solutions" obtained by setting one free variable to 1 and the others to 0 form a basis.

Worked Example 9.11 — Coordinates in a non-standard polynomial basis

Show that $\mathcal{B} = \{1,\ 1+t,\ 1+t+t^2\}$ is a basis of $\mathcal{P}_2$ and find the coordinates of $p(t) = 3 + 2t - t^2$.

Basis. Three vectors in a 3-dimensional space: we only need independence. Their coefficient vectors relative to $\{1, t, t^2\}$ are the columns of an upper-triangular matrix with unit diagonal,

$$\mathbf{P} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \qquad \det \mathbf{P} = 1 \neq 0.$$

Coordinates. Solve $a \cdot 1 + b(1+t) + c(1+t+t^2) = 3 + 2t - t^2$. Matching coefficients:

$$t^2:\ c = -1; \qquad t:\ b + c = 2 \Rightarrow b = 3; \qquad 1:\ a + b + c = 3 \Rightarrow a = 1.$$

So $[p]_{\mathcal{B}} = (1, 3, -1)^{\mathsf T}$. Check: $1 + 3(1+t) - (1+t+t^2) = 3 + 2t - t^2$ ✓.

Change of basis

Let $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ be a basis of $\mathbb{F}^n$ and $\mathbf{P}_{\mathcal{B}} = [\mathbf{b}_1 \ \cdots \ \mathbf{b}_n]$. The equation $\mathbf{x} = c_1\mathbf{b}_1 + \cdots + c_n\mathbf{b}_n$ reads

$$\mathbf{x} = \mathbf{P}_{\mathcal{B}}\,[\mathbf{x}]_{\mathcal{B}}, \qquad [\mathbf{x}]_{\mathcal{B}} = \mathbf{P}_{\mathcal{B}}^{-1}\mathbf{x}.$$

$\mathbf{P}_{\mathcal{B}}$ converts $\mathcal{B}$-coordinates to standard coordinates; its inverse goes the other way. Between two non-standard bases $\mathcal{B}$ and $\mathcal{C}$,

$$[\mathbf{x}]_{\mathcal{C}} = \mathbf{P}_{\mathcal{C}\leftarrow\mathcal{B}}[\mathbf{x}]_{\mathcal{B}}, \qquad \mathbf{P}_{\mathcal{C}\leftarrow\mathcal{B}} = \mathbf{P}_{\mathcal{C}}^{-1}\mathbf{P}_{\mathcal{B}},$$

whose jth column is $[\mathbf{b}_j]_{\mathcal{C}}$.

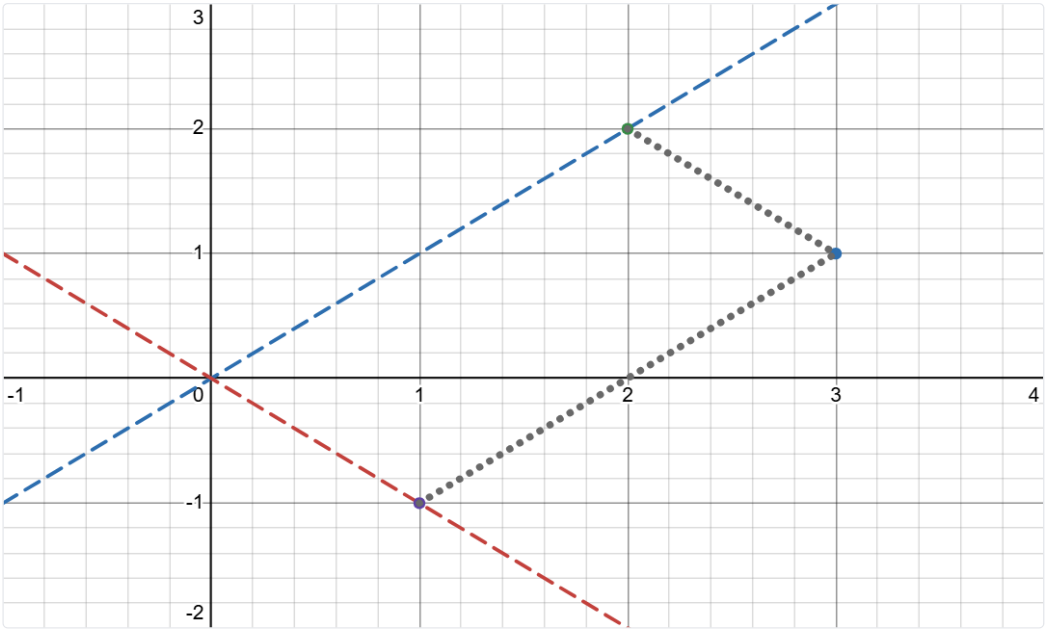
Worked Example 9.12 — Common-mode / differential-mode coordinates

Two-wire signals are often described not by the individual wire voltages (v_1, v_2) but by **common-mode** and **differential-mode** components. Use the basis $\mathbf{b}_1 = (1,1)^{\mathsf{T}}$ (common mode) and $\mathbf{b}_2 = (1,-1)^{\mathsf{T}}$ (differential mode). Find the coordinates of $\mathbf{v} = (3,1)^{\mathsf{T}}$ V.

$$\mathbf{P}_{\mathcal{B}} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \qquad \mathbf{P}_{\mathcal{B}}^{-1} = \frac{1}{-2}\begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

$$[\mathbf{v}]_{\mathcal{B}} = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

So $\mathbf{v} = 2(1,1) + 1(1,-1)$: a common-mode level $v_{cm} = (v_1+v_2)/2 = 2$ V and a differential half-amplitude $(v_1 - v_2)/2 = 1$ V (the full differential voltage is $v_1 - v_2 = 2$ V). A differential amplifier with ideal common-mode rejection responds only to the second coordinate.



The vector $\mathbf{v} = (3,1)$ resolved along the common-mode axis (blue, $y = x$) and the differential-mode axis (red, $y = -x$): $\mathbf{v} = 2\mathbf{b}_1 + 1\mathbf{b}_2$; the dotted lines complete the parallelogram.

Worked Example 9.13 — Dimension of familiar spaces

(a) The 2×2 real symmetric matrices $\begin{bmatrix} a & b \\ b & d \end{bmatrix} = a\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + d\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ form a space of dimension $\mathbf{3}$.

(b) $W_c = \{p \in \mathcal{P}_3 : p(1) = 0\}$ from Worked Example 9.4. By the factor theorem $p(t) = (t-1)q(t)$ with $q \in \mathcal{P}_2$, so a basis is $\{t-1,\ t(t-1),\ t^2(t-1)\}$ and $\dim W_c = 3 = \dim \mathcal{P}_3 - 1$: one linear constraint removes one dimension.

(c) Solutions of $y'' + 3y' + 2y = 0$: basis $\{e^{-t}, e^{-2t}\}$ (independent by the Wronskian $W = e^{-t}(-2e^{-2t}) - e^{-2t}(-e^{-t}) = -e^{-3t} \neq 0$), dimension $\mathbf{2}$ — matching the two initial conditions $y(0)$, $y'(0)$ a second-order circuit needs.

Common Pitfall — dimension is not the number of components

The plane $x + 2y - z = 0$ consists of vectors with three components, yet its dimension is 2. Dimension counts *degrees of freedom*, not entries. Likewise $\mathcal{P}_2$ has dimension 3 (three coefficients), not 2.

Engineering Note — counting unknowns in circuit analysis

For a connected circuit graph with n nodes and b branches, the space of KCL-compatible branch-current vectors has dimension $b - n + 1$ and the space of branch-voltage vectors that obey KVL has dimension $n - 1$. These are exactly the number of mesh (loop) currents and of node voltages (with one reference node) that mesh and nodal analysis use. Choosing "mesh currents" or "node voltages" is choosing a **basis** for one of these subspaces. Section 9.6 proves the counts with the rank–nullity theorem.

9.5 Linear transformations

Motivation

An amplifier, a filter, a differentiator, a sampler, a coordinate rotation, a Fourier transform — each takes an input from one vector space and produces an output in another, and each obeys **superposition**: the response to a weighted sum of inputs is the same weighted sum of responses. Mappings with this property are **linear transformations** (also called linear maps or linear operators). Superposition is the single most exploited property in electrical engineering: it lets us analyse a circuit one source at a time, decompose a signal into sinusoids and treat each separately, and represent the whole system by a matrix or a transfer function.

Definition

Linear transformation

Let V and W be vector spaces over the same field $\mathbb{F}$. A map $T : V \to W$ is **linear** if for all $\mathbf{u}, \mathbf{v} \in V$ and $c \in \mathbb{F}$

1. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ (additivity),
2. $T(c\mathbf{u}) = cT(\mathbf{u})$ (homogeneity).

Equivalently, $T(a\mathbf{u} + b\mathbf{v}) = aT(\mathbf{u}) + bT(\mathbf{v})$ — the **superposition principle**. V is the **domain**, W the **codomain**. A linear map $V \to V$ is often called a linear **operator** on V.

Immediate consequences:

$$T(\mathbf{0}) = \mathbf{0}, \qquad T\Big(\sum_{i} c_i \mathbf{v}_i\Big) = \sum_{i} c_i T(\mathbf{v}_i).$$

The first gives a one-line non-linearity test: **if $T(\mathbf{0}) \neq \mathbf{0}$, T is not linear.** The second says a linear map is completely determined by what it does to a basis: if $\{\mathbf{b}_j\}$ is a basis of V and we know $T(\mathbf{b}_j)$ for each j, we know T everywhere.

Examples and non-examples

Map	Linear?	Reason
$T(\mathbf{x}) = \mathbf{A}\mathbf{x}$, $\mathbf{A} \in \mathbb{F}^{m\times n}$	yes	matrix algebra
rotation of $\mathbb{R}^2$ by angle θ	yes	the rotated parallelogram of $\mathbf{u}, \mathbf{v}$ is the parallelogram of the rotated vectors
$D : \mathcal{P}_n \to \mathcal{P}_{n-1}$, $D(p) = p'$	yes	$(ap + bq)' = ap' + bq'$
$J : C[0,T] \to C[0,T]$, $J(f)(t) = \int_0^t f(\tau)\,d\tau$	yes	linearity of the integral
time shift $(S_\tau x)(t) = x(t - \tau)$	yes	shifts commute with sums
convolution with a fixed h: $x \mapsto h * x$	yes	integral of a linear expression
evaluation $x \mapsto x(t_0)$, sampling $x \mapsto (x(0), x(T_s), \dots)$	yes	pointwise operations
$T(x) = ax + b$ with $b \neq 0$ ("linear" in school algebra)	**no**	$T(0) = b \neq 0$; it is **affine**
squaring $v \mapsto v^2$, rectifier $v \mapsto \lvert v\rvert$, clipper	**no**	$(2v)^2 \neq 2v^2$
$\mathbf{x} \mapsto \lVert\mathbf{x}\rVert$	**no**	$\lVert -\mathbf{x}\rVert \neq -\lVert\mathbf{x}\rVert$
complex conjugation $z \mapsto z^*$ on $\mathbb{C}$ as a *complex* space	**no**	$(jz)^* = -jz^* \neq jz^*$ (but it is linear on $\mathbb{C}$ viewed as the real space $\mathbb{R}^2$)

Worked Example 9.14 — Testing linearity

(a) $T : \mathbb{R}^2 \to \mathbb{R}^2$, $T(x, y) = (x + 2y,\ 3x)$. Write $T(\mathbf{x}) = \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}\mathbf{x}$: linear.

(b) $T(x, y) = (x + 1,\ y)$. $T(0, 0) = (1, 0) \neq \mathbf{0}$: **not linear**.

(c) $T(x, y) = (xy,\ x)$. $T(2(1, 1)) = T(2, 2) = (4, 2)$ but $2T(1, 1) = (2, 2)$: **not linear**.

(d) The RC low-pass filter initially at rest, $RC\,\dot{y} + y = x$, $y(0) = 0$, defines $y = T(x)$ with

$$y(t) = \frac{1}{RC}\int_0^t e^{-(t-\tau)/RC}\,x(\tau)\,d\tau.$$

For inputs x_1, x_2 and scalars a, b, the integral of $ax_1 + bx_2$ splits into a times the first integral plus b times the second: $T(ax_1 + bx_2) = aT(x_1) + bT(x_2)$. **Linear.**

(e) The same filter with a nonzero initial voltage $y(0) = V_0$: now $T(0) = V_0 e^{-t/RC} \neq 0$. **Not linear** in the strict sense — it is affine (zero-input response plus a linear zero-state response). This is why "linear system" in circuit theory always means *zero initial state* when superposition is invoked.

The matrix of a linear map

Every linear map $\mathbb{F}^n \to \mathbb{F}^m$ is a matrix

If $T : \mathbb{F}^n \to \mathbb{F}^m$ is linear, then $T(\mathbf{x}) = \mathbf{A}\mathbf{x}$ for the unique $m \times n$ matrix

$$\mathbf{A} = \begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \cdots & T(\mathbf{e}_n) \end{bmatrix},$$

called the **standard matrix** of T.

Proof. $\mathbf{x} = \sum_j x_j \mathbf{e}_j$, so $T(\mathbf{x}) = \sum_j x_j T(\mathbf{e}_j)$, which is the matrix–vector product of $[T(\mathbf{e}_1) \cdots T(\mathbf{e}_n)]$ with $\mathbf{x}$.

Recipe: *the jth column of the matrix is the image of the jth basis vector.*

More generally, for $T : V \to W$ with bases $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ of V and $\mathcal{C}$ of W, the **matrix of T relative to $\mathcal{B}$ and $\mathcal{C}$** is

$$[T]_{\mathcal{C} \leftarrow \mathcal{B}} = \begin{bmatrix} [T(\mathbf{b}_1)]_{\mathcal{C}} & \cdots & [T(\mathbf{b}_n)]_{\mathcal{C}} \end{bmatrix}, \qquad [T(\mathbf{v})]_{\mathcal{C}} = [T]_{\mathcal{C} \leftarrow \mathcal{B}} \, [\mathbf{v}]_{\mathcal{B}}.$$

So *any* linear map between finite-dimensional spaces — differentiation of polynomials, a filter acting on a finite set of sinusoids — becomes a matrix once bases are chosen.

Worked Example 9.15 — Rotation and projection in the plane

(a) **Rotation by θ.** $\mathbf{e}_1 = (1, 0)$ rotates to $(\cos\theta, \sin\theta)$ and $\mathbf{e}_2 = (0, 1)$ to $(-\sin\theta, \cos\theta)$, so

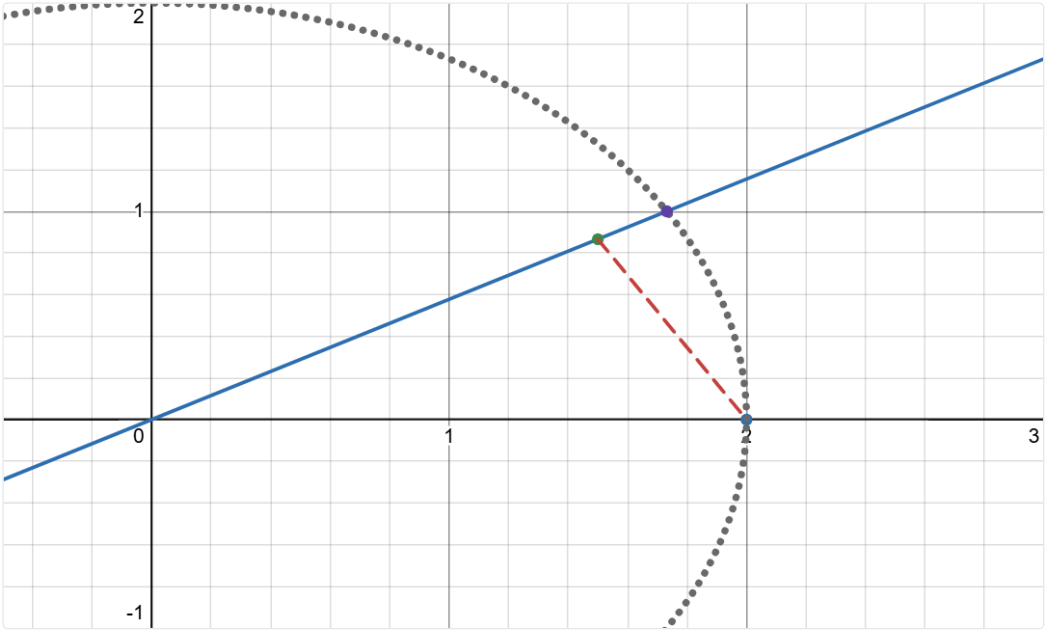
$$\mathbf{R}_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}.$$

This is the matrix form of multiplying a phasor by $e^{j\theta}$: $(x + jy)e^{j\theta} = (x\cos\theta - y\sin\theta) + j(x\sin\theta + y\cos\theta)$.

(b) **Orthogonal projection onto the line at angle θ** through the origin, with unit direction $\hat{\mathbf{u}} = (\cos\theta, \sin\theta)$. Section 9.9 shows $P(\mathbf{x}) = (\hat{\mathbf{u}} \cdot \mathbf{x})\hat{\mathbf{u}}$. Images of the basis vectors: $P(\mathbf{e}_1) = \cos\theta\,\hat{\mathbf{u}}$, $P(\mathbf{e}_2) = \sin\theta\,\hat{\mathbf{u}}$, so

$$\mathbf{P}_\theta = \begin{bmatrix} \cos^2\theta & \cos\theta\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta \end{bmatrix}.$$

For $\theta = 30^\circ$: $\mathbf{P} = \begin{bmatrix} 3/4 & \sqrt{3}/4 \\ \sqrt{3}/4 & 1/4 \end{bmatrix}$ and $\mathbf{P}(2, 0)^{\mathsf{T}} = (1.5,\ 0.866)^{\mathsf{T}}$. Check: this point lies on $y = x\tan 30^\circ = 0.5774x$ ($1.5 \times 0.5774 = 0.866$ ✓), and the residual $(0.5, -0.866)$ is perpendicular to $\hat{\mathbf{u}}$: $0.5 \cdot 0.866 - 0.866 \cdot 0.5 = 0$ ✓.



The vector $\mathbf{x} = (2, 0)$, its rotation by 30° (on the dotted circle of radius 2) and its orthogonal projection onto the 30° line; the dashed segment is perpendicular to the line.

Worked Example 9.16 — The differentiation matrix

Find the matrix of $D : \mathcal{P}_3 \to \mathcal{P}_2$, $D(p) = p'$, relative to the monomial bases $\{1, t, t^2, t^3\}$ and $\{1, t, t^2\}$, and use it to differentiate $p(t) = 2 - t + 4t^2 + t^3$.

Images of the basis vectors: $D(1) = 0$, $D(t) = 1$, $D(t^2) = 2t$, $D(t^3) = 3t^2$. In coordinates these are $(0,0,0)$, $(1,0,0)$, $(0,2,0)$, $(0,0,3)$, so

$$[D] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}.$$

With $[p] = (2, -1, 4, 1)^{\mathsf{T}}$:

$$[D][p] = \begin{bmatrix} 0\cdot 2 + 1\cdot(-1) \\ 2 \cdot 4 \\ 3 \cdot 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 8 \\ 3 \end{bmatrix} \Rightarrow\ p'(t) = -1 + 8t + 3t^2.$$

Direct check: $\frac{d}{dt}(2 - t + 4t^2 + t^3) = -1 + 8t + 3t^2$ ✓. The matrix has a zero first column: constants are sent to zero, which is the null space of D (Section 9.6).

Composition, inverses and isomorphism

If $S : U \to V$ and $T : V \to W$ are linear, so is the composition $T \circ S$, and in coordinates

$$[T \circ S] = [T]\,[S].$$

Matrix multiplication is *defined* so that this holds — it is the reason the row-times-column rule looks the way it does. A cascade of linear stages (pre-amplifier, filter, ADC) is represented by the product of their matrices, applied **right to left**.

A linear map $T : V \to W$ is

- **one-to-one** (injective) if $T(\mathbf{u}) = T(\mathbf{v}) \Rightarrow \mathbf{u} = \mathbf{v}$;
- **onto** (surjective) if every $\mathbf{w} \in W$ equals $T(\mathbf{v})$ for some $\mathbf{v}$;
- an **isomorphism** if it is both; then T^{-1} exists and is linear.

Coordinates are an isomorphism

If $\dim V = n$ and $\mathcal{B}$ is a basis, the coordinate map $\mathbf{v} \mapsto [\mathbf{v}]_{\mathcal{B}}$ is an isomorphism $V \to \mathbb{F}^n$. Consequently any two vector spaces of the same finite dimension over the same field are isomorphic.

This theorem is what licenses replacing $\mathcal{P}_2$ by $\mathbb{R}^3$, sinusoids at ω by $\mathbb{R}^2$ (or by phasors in $\mathbb{C}$), and so on: every finite-dimensional calculation can be done with columns of numbers.

Change of basis for a linear operator

Let $T : \mathbb{F}^n \to \mathbb{F}^n$ have standard matrix $\mathbf{A}$, and let $\mathcal{B}$ be another basis with $\mathbf{P} = \mathbf{P}_{\mathcal{B}}$. Since $\mathbf{x} = \mathbf{P}[\mathbf{x}]_{\mathcal{B}}$,

$$[T(\mathbf{x})]_{\mathcal{B}} = \mathbf{P}^{-1}\mathbf{A}\mathbf{x} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}\,[\mathbf{x}]_{\mathcal{B}} \quad\Longrightarrow\quad [T]_{\mathcal{B}} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}.$$

Matrices related by $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ are **similar**: they represent the *same* operator in different coordinates, so they share every basis-independent property — determinant, trace, rank and (Chapter 10) eigenvalues.

Worked Example 9.17 — A basis that makes an operator simple

Let $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$ and $\mathcal{B} = \{(1,0)^{\mathsf{T}}, (1,1)^{\mathsf{T}}\}$. Find $[T]_{\mathcal{B}}$.

$$\mathbf{P} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{P}^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{A}\mathbf{P} = \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix},$$

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}.$$

In the new basis the operator simply scales the first coordinate by 1 and the second by 2: $\mathbf{A}(1,0) = (1,0)$ and $\mathbf{A}(1,1) = (2,2)$. These special basis vectors are **eigenvectors**; finding them systematically is the subject of Ch. 10. Check invariants: $\det \mathbf{A} = 2 = \det \operatorname{diag}(1,2)$ and $\operatorname{tr}\mathbf{A} = 3 = 1 + 2$ ✓.

Common Pitfall — "linear" versus "affine"

The straight-line function $y = mx + c$ is called linear in school algebra and in device data sheets, but with $c \neq 0$ it is **affine**, not linear: doubling the input does not double the output. A biased transistor stage is affine about its operating point; small-signal analysis subtracts the operating point so that the remaining map *is* linear. Always remove offsets (DC bias, initial conditions) before invoking superposition.

Engineering Note — LTI systems are linear operators on signal spaces

A linear time-invariant (LTI) system is a linear operator T on a signal space that commutes with every time shift: $T \circ S_\tau = S_\tau \circ T$. One can show that every such operator is a convolution $y = h * x$ with its impulse response h. Complex exponentials $e^{j\omega t}$ satisfy $T(e^{j\omega t}) = H(j\omega)e^{j\omega t}$ — they are the "eigenvectors" of every LTI operator, the infinite-dimensional analogue of Worked Example 9.17. Expanding inputs in that basis is exactly Fourier analysis (Ch. 15, Ch. 16).

9.6 Null space and column space

Motivation

For a linear map $T(\mathbf{x}) = \mathbf{A}\mathbf{x}$ two questions dominate:

1. **Existence:** for which outputs $\mathbf{b}$ does $\mathbf{A}\mathbf{x} = \mathbf{b}$ have a solution?
2. **Uniqueness:** if a solution exists, how much freedom is left in it?

The answer to the first is a subspace of the output space, the **column space**; the answer to the second is a subspace of the input space, the **null space**. Their dimensions are tied together by the **rank–nullity theorem**, which in circuit theory counts independent node voltages and loop currents, in control theory counts uncontrollable modes, and in signal processing counts the components of a signal a measurement system cannot see.

Definitions

The four fundamental subspaces

For $\mathbf{A} \in \mathbb{R}^{m\times n}$:

- **Null space (kernel):** $\operatorname{null}(\mathbf{A}) = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{A}\mathbf{x} = \mathbf{0}\} \subseteq \mathbb{R}^n$.
- **Column space (range):** $\operatorname{col}(\mathbf{A}) = \{\mathbf{A}\mathbf{x} : \mathbf{x} \in \mathbb{R}^n\} = \operatorname{span}\{\text{columns of } \mathbf{A}\} \subseteq \mathbb{R}^m$.
- **Row space:** $\operatorname{row}(\mathbf{A}) = \operatorname{col}(\mathbf{A}^{\mathsf{T}}) \subseteq \mathbb{R}^n$.
- **Left null space:** $\operatorname{null}(\mathbf{A}^{\mathsf{T}}) = \{\mathbf{y} \in \mathbb{R}^m : \mathbf{A}^{\mathsf{T}}\mathbf{y} = \mathbf{0}\} \subseteq \mathbb{R}^m$.

For a general linear map $T : V \to W$, $\ker T = \{\mathbf{v} : T(\mathbf{v}) = \mathbf{0}\}$ and $\operatorname{range} T = \{T(\mathbf{v})\}$. The dimension of the column space is the **rank** r; the dimension of the null space is the **nullity**. (Over $\mathbb{C}$ replace T by H.)

Both are subspaces: $\mathbf{A}\mathbf{0} = \mathbf{0}$; if $\mathbf{A}\mathbf{x} = \mathbf{A}\mathbf{y} = \mathbf{0}$ then $\mathbf{A}(a\mathbf{x} + b\mathbf{y}) = \mathbf{0}$; and a span is always a subspace.

The column space viewpoint comes from reading $\mathbf{A}\mathbf{x}$ **column by column**:

$$\mathbf{A}\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n.$$

So $\mathbf{A}\mathbf{x} = \mathbf{b}$ asks: *which combination of the columns produces* $\mathbf{b}$*?*

Existence and structure of solutions

1. $\mathbf{A}\mathbf{x} = \mathbf{b}$ is solvable iff $\mathbf{b} \in \operatorname{col}(\mathbf{A})$.
2. If $\mathbf{x}_p$ is one solution, the complete solution set is
$$\{\mathbf{x}_p + \mathbf{x}_n : \ \mathbf{x}_n \in \operatorname{null}(\mathbf{A})\}.$$
3. The solution is unique iff $\operatorname{null}(\mathbf{A}) = \{\mathbf{0}\}$, i.e. iff the columns of $\mathbf{A}$ are independent. More generally, T is one-to-one iff $\ker T = \{\mathbf{0}\}$.

Proof of 2. If $\mathbf{A}\mathbf{x} = \mathbf{b}$ then $\mathbf{A}(\mathbf{x} - \mathbf{x}_p) = \mathbf{b} - \mathbf{b} = \mathbf{0}$, so $\mathbf{x} - \mathbf{x}_p \in \operatorname{null}(\mathbf{A})$; conversely $\mathbf{A}(\mathbf{x}_p + \mathbf{x}_n) = \mathbf{b} + \mathbf{0}$.

This is the same "particular plus homogeneous" structure as the solution of a linear ODE (Worked Example 9.2): the null space of the operator $L = D^2 + 3D + 2$ is the space of natural responses, and any forced response is one particular solution plus an element of that null space.

Computing bases

All four subspaces are read off from the reduced row echelon form $\mathbf{R} = \operatorname{rref}(\mathbf{A})$ (Ch. 8).

Subspace	Basis	Dimension
$\operatorname{col}(\mathbf{A})$	the **pivot columns of the original** $\mathbf{A}$	r
$\operatorname{row}(\mathbf{A})$	the nonzero rows of $\mathbf{R}$	r
$\operatorname{null}(\mathbf{A})$	special solutions: one per free variable	$n - r$
$\operatorname{null}(\mathbf{A}^{\mathsf{T}})$	null space of $\mathbf{A}^{\mathsf{T}}$ (or last $m - r$ rows of $\mathbf{E}$ where $\mathbf{E}\mathbf{A} = \mathbf{R}$)	$m - r$

Why the pivot columns of $\mathbf{A}$ (not of $\mathbf{R}$)? Row operations change the column space, but they do **not** change linear relations among the columns, because $\mathbf{A}\mathbf{x} = \mathbf{0} \iff \mathbf{R}\mathbf{x} = \mathbf{0}$. In $\mathbf{R}$ the pivot columns are obviously independent and every other column is a combination of them; the same relations hold among the columns of $\mathbf{A}$.

Row operations replace rows by combinations of rows, so the row space *is* unchanged, and the nonzero rows of $\mathbf{R}$ are independent (each has a leading 1 where the others have 0).

Rank–nullity

Rank–nullity theorem

For $\mathbf{A} \in \mathbb{F}^{m \times n}$ (or a linear map $T : V \to W$ with $\dim V = n$),

$$\operatorname{rank}(\mathbf{A}) + \dim \operatorname{null}(\mathbf{A}) = n \quad \text{(number of columns)}.$$

Moreover $\dim \operatorname{row}(\mathbf{A}) = \dim \operatorname{col}(\mathbf{A}) = r$ ("row rank equals column rank"), and $\dim \operatorname{null}(\mathbf{A}^{\mathsf{T}}) = m - r$.

Proof (matrix form). Every column of $\mathbf{R}$ is either a pivot column (r of them) or a free column. The null space has one special solution per free column, and these are independent (each has a 1 in its own free position and 0 in the other free positions), so the nullity is $n - r$. The column space has the r pivot columns as basis. The row space has the r nonzero rows as basis. Applying the result to $\mathbf{A}^{\mathsf{T}}$ (n rows, m columns, rank r) gives $\dim \operatorname{null}(\mathbf{A}^{\mathsf{T}}) = m - r$.

Proof (abstract form). Choose a basis $\mathbf{k}_1, \dots, \mathbf{k}_q$ of $\ker T$ and extend it to a basis $\mathbf{k}_1, \dots, \mathbf{k}_q, \mathbf{v}_1, \dots, \mathbf{v}_s$ of V, so $q + s = n$. Then $T(\mathbf{v}_1), \dots, T(\mathbf{v}_s)$ span the range (the $T(\mathbf{k}_i)$ are zero) and are independent (a relation $\sum c_i T(\mathbf{v}_i) = \mathbf{0}$ puts $\sum c_i \mathbf{v}_i$ in the kernel, which forces all $c_i = 0$ by independence of the full basis). So $\dim \operatorname{range} T = s = n - q$.

In words: **each input dimension either survives to the output or is lost in the null space.** The map cannot create dimensions.

Worked Example 9.18 — All four subspaces of a 3×4 matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 0 & 2 \\ 3 & 6 & 1 & 2 \end{bmatrix}.$$

Row-reduce. $R_2 - 2R_1$, $R_3 - 3R_1$ give rows $(0, 0, -2, 2)$ and $(0, 0, -2, 2)$; then $R_3 - R_2$ kills row 3; scale row 2 by $-\frac{1}{2}$ and clear above:

$$\mathbf{R} = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Pivots in columns 1 and 3, so $r = 2$; free variables x_2, x_4.

Column space (pivot columns of $\mathbf{A}$): $\{(1, 2, 3)^{\mathsf{T}}, (1, 0, 1)^{\mathsf{T}}\}$, dimension 2 — a plane in $\mathbb{R}^3$.

Row space: $\{(1, 2, 0, 1), (0, 0, 1, -1)\}$, dimension 2.

Null space. From $\mathbf{R}$: $x_1 = -2x_2 - x_4$, $x_3 = x_4$. Special solutions:

$$x_2 = 1, x_4 = 0 : \ \mathbf{n}_1 = (-2, 1, 0, 0)^{\mathsf{T}}; \qquad x_2 = 0, x_4 = 1 : \ \mathbf{n}_2 = (-1, 0, 1, 1)^{\mathsf{T}}.$$

Dimension $2 = n - r = 4 - 2$ ✓. Check: $\mathbf{A}\mathbf{n}_2 = (-1 + 1, \ -2 + 2, \ -3 + 1 + 2) = \mathbf{0}$ ✓.

Left null space. Solve $\mathbf{A}^{\mathsf{T}}\mathbf{y} = \mathbf{0}$; one finds $\mathbf{y} = (-1, -1, 1)^{\mathsf{T}}$ (row 3 = row 1 + row 2). Dimension $1 = m - r = 3 - 2$ ✓.

Solvability. $\mathbf{A}\mathbf{x} = \mathbf{b}$ is solvable iff $\mathbf{b} \in \operatorname{col}(\mathbf{A})$, which here is equivalent to $\mathbf{y}^{\mathsf{T}}\mathbf{b} = 0$, i.e. $b_3 = b_1 + b_2$. For $\mathbf{b} = (2, 2, 4)^{\mathsf{T}}$: $4 = 2 + 2$ ✓. Reducing $[\mathbf{A} \mid \mathbf{b}]$ gives $x_1 + 2x_2 + x_4 = 1$, $x_3 - x_4 = 1$; with the free variables zero, $\mathbf{x}_p = (1, 0, 1, 0)^{\mathsf{T}}$ (check: $\mathbf{A}\mathbf{x}_p = (2, 2, 4)^{\mathsf{T}}$ ✓). Complete solution:

$$\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + s \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \qquad s, t \in \mathbb{R}.$$

For $\mathbf{b} = (1, 1, 1)^{\mathsf{T}}$: $1 \neq 1 + 1$, no solution — $\mathbf{b}$ lies off the column-space plane. Section 9.11 finds the best approximate solution.

The left null space acts as a set of **consistency checks** on the right-hand side: $\mathbf{A}\mathbf{x} = \mathbf{b}$ is solvable iff $\mathbf{y}^{\mathsf{T}}\mathbf{b} = 0$ for every $\mathbf{y} \in \operatorname{null}(\mathbf{A}^{\mathsf{T}})$. (Proof: $\mathbf{y}^{\mathsf{T}}\mathbf{A}\mathbf{x} = (\mathbf{A}^{\mathsf{T}}\mathbf{y})^{\mathsf{T}}\mathbf{x} = 0$ for every $\mathbf{x}$, so every $\mathbf{b}$ in the column space is orthogonal to the left null space; Section 9.7 shows the converse.)

Worked Example 9.19 — The incidence matrix of a circuit graph

A circuit has 4 nodes and 5 branches: $1\colon\ 1 \to 2$, $2\colon\ 2 \to 3$, $3\colon\ 3 \to 1$, $4\colon\ 3 \to 4$, $5\colon\ 4 \to 1$. The **node–branch incidence matrix** $\mathbf{A}$ (4×5) has $a_{kb} = +1$ if branch b leaves node k, -1 if it enters, 0 otherwise:

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & -1 & 0 & -1 \\ -1 & 1 & 0 & 0 & 0 \\ 0 & -1 & 1 & 1 & 0 \\ 0 & 0 & 0 & -1 & 1 \end{bmatrix}.$$

KCL says $\mathbf{A}\mathbf{i} = \mathbf{0}$ for the branch-current vector $\mathbf{i}$: allowed currents form $\operatorname{null}(\mathbf{A})$.

Rank. Every column has one $+1$ and one -1, so the four rows sum to zero: $\mathbf{y} = (1,1,1,1)^{\mathsf{T}}$ is in the left null space and $r \le 3$. Deleting the reference node (row 4) leaves three independent rows, so $r = 3 = n - 1$.

Null space. Dimension $b - r = 5 - 3 = 2$: two independent loop currents. A basis:

$$\boldsymbol{\ell}_1 = (1,1,1,0,0)^{\mathsf{T}} \text{ (loop } 1 \to 2 \to 3 \to 1), \qquad \boldsymbol{\ell}_2 = (1,1,0,1,1)^{\mathsf{T}} \text{ (loop } 1 \to 2 \to 3 \to 4 \to 1).$$

Check $\mathbf{A}\boldsymbol{\ell}_2$: row 1: $1 - 0 - 1 = 0$; row 2: $-1 + 1 = 0$; row 3: $-1 + 1 = 0$; row 4: $-1 + 1 = 0$ ✓.

Interpretation. Every KCL-compatible current distribution is $\mathbf{i} = I_1\boldsymbol{\ell}_1 + I_2\boldsymbol{\ell}_2$ — mesh analysis needs exactly $b - n + 1 = 2$ unknowns. Dually, branch voltages derived from node potentials, $\mathbf{v} = \mathbf{A}^{\mathsf{T}}\mathbf{e}$, fill $\operatorname{row}(\mathbf{A})$ of dimension $n - 1 = 3$, and the node potentials are determined only up to the left-null-space vector $(1,1,1,1)$ — a common offset, removed by grounding one node.

Worked Example 9.20 — Kernel and range of differentiation

For $D : \mathcal{P}_3 \to \mathcal{P}_2$ (Worked Example 9.16): $\ker D = \{\text{constants}\} = \operatorname{span}\{1\}$, nullity 1. Every $q \in \mathcal{P}_2$ is the derivative of some cubic (integrate it), so range $D = \mathcal{P}_2$, rank 3. Rank–nullity: $3 + 1 = 4 = \dim \mathcal{P}_3$ ✓. The matrix $[D]$ of Worked Example 9.16 shows the same thing: three pivots, one free (first) column. The "constant of integration" of calculus is precisely the freedom given by the null space.

The invertible matrix theorem (vector-space form)

For a square $\mathbf{A} \in \mathbb{F}^{n \times n}$ the following are equivalent: $\mathbf{A}$ is invertible; $\operatorname{rank}\mathbf{A} = n$; $\operatorname{null}(\mathbf{A}) = \{\mathbf{0}\}$; $\operatorname{col}(\mathbf{A}) = \mathbb{F}^n$; the columns form a basis of $\mathbb{F}^n$; the rows form a basis; $\det\mathbf{A} \neq 0$; $\mathbf{A}\mathbf{x} = \mathbf{b}$ has exactly one solution for every $\mathbf{b}$; $T(\mathbf{x}) = \mathbf{A}\mathbf{x}$ is one-to-one; T is onto.

For a square matrix, one-to-one and onto are *equivalent* — a direct consequence of rank–nullity ($r = n \iff n - r = 0$). For rectangular matrices they are not: a tall matrix ($m > n$) can be one-to-one but never onto; a wide matrix ($m < n$) can be onto but never one-to-one.

Common Pitfall — reading the column space from the wrong matrix

The pivot columns of $\mathbf{R} = \operatorname{rref}(\mathbf{A})$ are standard unit vectors such as $(1,0,0)^{\mathsf{T}}$ and $(0,1,0)^{\mathsf{T}}$; they span a *different* plane from $\operatorname{col}(\mathbf{A})$ in general. Use the pivot **positions** from $\mathbf{R}$ but take the **columns** from $\mathbf{A}$. (For the row space it is the other way round: the rows of $\mathbf{R}$ are fine.)

Engineering Note — what a measurement cannot see

If a sensor array produces readings $\mathbf{y} = \mathbf{H}\mathbf{x}$ of an unknown source vector $\mathbf{x}$, any component of $\mathbf{x}$ in $\operatorname{null}(\mathbf{H})$ is invisible: adding it changes nothing in $\mathbf{y}$. In tomography, phased arrays and state estimation (observability in control), the first design question is therefore "what is the null space of the measurement matrix?" A wide $\mathbf{H}$ (fewer measurements than unknowns) always has a nontrivial null space, so extra assumptions — minimum norm, sparsity, smoothness — are needed to choose among the infinitely many consistent $\mathbf{x}$.

9.7 Orthogonality

Motivation

So far nothing has been said about lengths or angles; a vector space is pure algebra. Yet the most useful bases in engineering are **orthogonal** ones: the $\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$ axes of field theory, the in-phase and quadrature carriers $\cos\omega_c t$ and $\sin\omega_c t$ of a modulator, the harmonics of a Fourier series, the subcarriers of OFDM. With an orthogonal basis, coordinates are found by a single dot product each instead of by solving a linear system, energy splits into a sum of independent parts (Pythagoras), and signals sent along orthogonal directions do not interfere. This section develops orthogonality in $\mathbb{R}^n$ with the ordinary dot product; Section 9.8 generalises it to complex vectors and to functions.

Dot product, length and angle

Dot product, norm, distance

For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$,

$$\mathbf{x} \cdot \mathbf{y} = \mathbf{x}^{\mathsf{T}}\mathbf{y} = \sum_{k=1}^{n} x_k y_k, \qquad \|\mathbf{x}\| = \sqrt{\mathbf{x} \cdot \mathbf{x}}, \qquad d(\mathbf{x},\mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|.$$

A **unit vector** has $\|\mathbf{x}\| = 1$; **normalising** $\mathbf{x} \neq \mathbf{0}$ means forming $\hat{\mathbf{x}} = \mathbf{x}/\|\mathbf{x}\|$.

Cauchy–Schwarz inequality

$$|\mathbf{x}\cdot\mathbf{y}| \le \|\mathbf{x}\|\,\|\mathbf{y}\|,$$

with equality iff $\mathbf{x}$ and $\mathbf{y}$ are parallel (one is a scalar multiple of the other).

Proof. If $\mathbf{y}=\mathbf{0}$ both sides vanish. Otherwise, for every real s,

$$0 \le \|\mathbf{x}-s\mathbf{y}\|^2 = \|\mathbf{x}\|^2 - 2s\,\mathbf{x}\cdot\mathbf{y} + s^2\|\mathbf{y}\|^2.$$

Choose the minimising value $s=\mathbf{x}\cdot\mathbf{y}/\|\mathbf{y}\|^2$: $0 \le \|\mathbf{x}\|^2 - (\mathbf{x}\cdot\mathbf{y})^2/\|\mathbf{y}\|^2$, which rearranges to the inequality. Equality means $\mathbf{x}-s\mathbf{y}=\mathbf{0}$.

The proof is worth remembering, because the same "minimise $\|\mathbf{x}-s\mathbf{y}\|^2$" step is a projection (Section 9.9) and it works verbatim in any inner-product space.

Cauchy–Schwarz guarantees $-1 \le \dfrac{\mathbf{x}\cdot\mathbf{y}}{\|\mathbf{x}\|\|\mathbf{y}\|} \le 1$, so the **angle** between nonzero vectors can be *defined* in any dimension by

$$\cos\theta = \frac{\mathbf{x}\cdot\mathbf{y}}{\|\mathbf{x}\|\,\|\mathbf{y}\|}, \qquad 0 \le \theta \le \pi.$$

It also gives the **triangle inequality** $\|\mathbf{x}+\mathbf{y}\| \le \|\mathbf{x}\| + \|\mathbf{y}\|$: expand $\|\mathbf{x}+\mathbf{y}\|^2 = \|\mathbf{x}\|^2 + 2\mathbf{x}\cdot\mathbf{y} + \|\mathbf{y}\|^2 \le (\|\mathbf{x}\|+\|\mathbf{y}\|)^2$.

Orthogonal vectors and sets

Orthogonality

$\mathbf{x}$ and $\mathbf{y}$ are **orthogonal**, $\mathbf{x}\perp\mathbf{y}$, if $\mathbf{x}\cdot\mathbf{y}=0$. A set $\{\mathbf{u}_1,\dots,\mathbf{u}_k\}$ is **orthogonal** if $\mathbf{u}_i\cdot\mathbf{u}_j=0$ for all $i\neq j$, and **orthonormal** if in addition every $\|\mathbf{u}_i\|=1$, i.e. $\mathbf{u}_i\cdot\mathbf{u}_j=\delta_{ij}$.

Pythagoras

$\mathbf{x}\perp\mathbf{y} \iff \|\mathbf{x}+\mathbf{y}\|^2 = \|\mathbf{x}\|^2+\|\mathbf{y}\|^2$. For an orthogonal set, $\left\|\sum_i \mathbf{u}_i\right\|^2 = \sum_i \|\mathbf{u}_i\|^2$.

Orthogonal sets are independent

An orthogonal set of nonzero vectors is linearly independent.

Proof. Take the dot product of $\sum_i c_i\mathbf{u}_i=\mathbf{0}$ with $\mathbf{u}_j$: all cross terms vanish, leaving $c_j\|\mathbf{u}_j\|^2=0$, so $c_j=0$.

Hence n mutually orthogonal nonzero vectors in $\mathbb{R}^n$ are automatically a basis. Their real advantage is how easy coordinates become:

Coordinates in an orthogonal basis

If $\{\mathbf{u}_1,\dots,\mathbf{u}_n\}$ is an orthogonal basis then

$$\mathbf{x} = \sum_{j=1}^{n} \frac{\mathbf{x}\cdot\mathbf{u}_j}{\mathbf{u}_j\cdot\mathbf{u}_j}\,\mathbf{u}_j,$$

and if it is orthonormal, $\mathbf{x}=\sum_j (\mathbf{x}\cdot\mathbf{u}_j)\,\mathbf{u}_j$ and $\|\mathbf{x}\|^2=\sum_j(\mathbf{x}\cdot\mathbf{u}_j)^2$ (Parseval).

Proof. Write $\mathbf{x}=\sum c_i\mathbf{u}_i$ and dot with $\mathbf{u}_j$: $\mathbf{x}\cdot\mathbf{u}_j = c_j\,\mathbf{u}_j\cdot\mathbf{u}_j$.

Compare: in a general basis we must solve an $n\times n$ system; in an orthogonal basis each coordinate is computed independently. This is *exactly* the structure of the Fourier coefficient formulas of Ch. 15, where the dot product becomes an integral.

Worked Example 9.21 — Lengths and angles

Find the angle between $\mathbf{x}=(1,2,2)^{\mathsf T}$ and $\mathbf{y}=(2,0,1)^{\mathsf T}$ and verify Cauchy–Schwarz.

$\mathbf{x}\cdot\mathbf{y} = 2+0+2 = 4$, $\|\mathbf{x}\| = \sqrt{1+4+4} = 3$, $\|\mathbf{y}\| = \sqrt{5} = 2.236$.

$$\cos\theta = \frac{4}{3\sqrt{5}} = 0.5963, \qquad \theta = 53.4^\circ.$$

Cauchy–Schwarz: $4 \le 3\times 2.236 = 6.708$ ✓. Triangle inequality: $\|\mathbf{x}+\mathbf{y}\| = \|(3,2,3)\| = \sqrt{22} = 4.690 \le 3 + 2.236$ ✓.

Worked Example 9.22 — Expanding in an orthonormal basis

Verify that

$$\mathbf{u}_1 = \tfrac{1}{\sqrt{3}}(1,1,1), \qquad \mathbf{u}_2 = \tfrac{1}{\sqrt{2}}(1,-1,0), \qquad \mathbf{u}_3 = \tfrac{1}{\sqrt{6}}(1,1,-2)$$

is an orthonormal basis of $\mathbb{R}^3$, and expand $\mathbf{x} = (3,1,2)$.

Orthogonality. $(1,1,1)\cdot(1,-1,0) = 0$, $(1,1,1)\cdot(1,1,-2) = 0$, $(1,-1,0)\cdot(1,1,-2) = 0$ ✓. Each vector has been divided by its length ($\sqrt{3}$, $\sqrt{2}$, $\sqrt{6}$), so all are unit vectors.

Coordinates by dot products:

$$\mathbf{x}\cdot\mathbf{u}_1 = \tfrac{6}{\sqrt{3}} = 2\sqrt{3}, \qquad \mathbf{x}\cdot\mathbf{u}_2 = \tfrac{2}{\sqrt{2}} = \sqrt{2}, \qquad \mathbf{x}\cdot\mathbf{u}_3 = \tfrac{3+1-4}{\sqrt{6}} = 0.$$

So $\mathbf{x} = 2\sqrt{3}\,\mathbf{u}_1 + \sqrt{2}\,\mathbf{u}_2 = 2(1,1,1) + (1,-1,0) = (3,1,2)$ ✓.

Parseval. $(2\sqrt{3})^2 + (\sqrt{2})^2 + 0^2 = 12 + 2 = 14 = \|\mathbf{x}\|^2 = 9 + 1 + 4$ ✓.

Interpretation. $\mathbf{u}_1$ is the "average" (DC) direction: $\mathbf{x}\cdot\mathbf{u}_1\,\mathbf{u}_1 = (2,2,2)$ is the mean value of the entries repeated. The rest, $(1,-1,0)$, is the zero-mean (AC) part. Energy splits as 12 (DC) $+$ 2 (AC).

Orthogonal matrices

Orthogonal matrix

A square real matrix $\mathbf{Q}$ is **orthogonal** if its columns are orthonormal, equivalently

$$\mathbf{Q}^{\mathsf{T}}\mathbf{Q} = \mathbf{I}, \qquad \text{i.e.} \qquad \mathbf{Q}^{-1} = \mathbf{Q}^{\mathsf{T}}.$$

(The complex analogue, $\mathbf{U}^{\mathsf{H}}\mathbf{U} = \mathbf{I}$, is a **unitary** matrix.)

Orthogonal matrices preserve dot products, lengths and angles:

$$(\mathbf{Q}\mathbf{x})\cdot(\mathbf{Q}\mathbf{y}) = \mathbf{x}^{\mathsf{T}}\mathbf{Q}^{\mathsf{T}}\mathbf{Q}\mathbf{y} = \mathbf{x}\cdot\mathbf{y}, \qquad \|\mathbf{Q}\mathbf{x}\| = \|\mathbf{x}\|.$$

They are the rotations and reflections, and $\det\mathbf{Q} = \pm 1$. The rotation $\mathbf{R}_\theta$ of Worked Example 9.15 is orthogonal; so is the matrix $[\mathbf{u}_1\ \mathbf{u}_2\ \mathbf{u}_3]$ of Worked Example 9.22. Numerically they are ideal: $\kappa(\mathbf{Q}) = 1$, so solving $\mathbf{Q}\mathbf{x} = \mathbf{b}$ by $\mathbf{x} = \mathbf{Q}^{\mathsf{T}}\mathbf{b}$ never amplifies errors. This is why orthogonal transforms (the DFT scaled by $1/\sqrt{N}$, the discrete cosine transform, Householder reflections in QR) dominate numerical signal processing.

Orthogonal complements

Orthogonal complement

For a subspace W of $\mathbb{R}^n$, its **orthogonal complement** is

$$W^{\perp} = \{\mathbf{x} \in \mathbb{R}^n : \ \mathbf{x}\cdot\mathbf{w} = 0 \text{ for all } \mathbf{w} \in W\}.$$

$W^{\perp}$ is a subspace, and it suffices to check orthogonality against a *spanning set* of W. Key properties:

$$W \cap W^{\perp} = \{\mathbf{0}\}, \qquad \dim W + \dim W^{\perp} = n, \qquad (W^{\perp})^{\perp} = W, \qquad \mathbb{R}^n = W \oplus W^{\perp}.$$

Fundamental theorem of linear algebra (orthogonality part)

For $\mathbf{A} \in \mathbb{R}^{m\times n}$:

$$\operatorname{null}(\mathbf{A}) = \operatorname{row}(\mathbf{A})^{\perp} \text{ in } \mathbb{R}^n, \qquad \operatorname{null}(\mathbf{A}^{\mathsf{T}}) = \operatorname{col}(\mathbf{A})^{\perp} \text{ in } \mathbb{R}^m.$$

Proof. $\mathbf{A}\mathbf{x} = \mathbf{0}$ says that the dot product of $\mathbf{x}$ with every row of $\mathbf{A}$ is zero, i.e. $\mathbf{x}$ is orthogonal to the row space. Apply this to $\mathbf{A}^{\mathsf{T}}$ for the second statement.

Together with rank–nullity this gives the complete picture: $\mathbb{R}^n$ splits orthogonally into the row space ($\dim r$) and the null space ($\dim n - r$); $\mathbb{R}^m$ splits into the column space ($\dim r$) and the left null space ($\dim m - r$). $\mathbf{A}$ maps the row space one-to-one onto the column space and sends the null space to zero. The second statement also completes the solvability test of Section 9.6: $\mathbf{b} \in \operatorname{col}(\mathbf{A})$ **iff** $\mathbf{b} \perp \operatorname{null}(\mathbf{A}^{\mathsf{T}})$.

Worked Example 9.23 — Complements and the four subspaces

(a) Find $W^\perp$ for $W = \text{span}\{(1,1,1)\}$.

$\mathbf{x} \perp (1,1,1) \iff x+y+z=0$. So $W^\perp$ is the plane $x+y+z=0$, with basis $\{(-1,1,0),(-1,0,1)\}$; $\dim W + \dim W^\perp = 1+2=3$ ✓. In the language of Worked Example 9.22: the orthogonal complement of the DC direction is the subspace of zero-mean vectors.

(b) For $\mathbf{A}$ of Worked Example 9.18, check $\text{null}(\mathbf{A}) \perp \text{row}(\mathbf{A})$:

$$(-2,1,0,0)\cdot(1,2,0,1)=0,\quad (-2,1,0,0)\cdot(0,0,1,-1)=0,\quad (-1,0,1,1)\cdot(1,2,0,1)=0,\quad (-1,0,1,1)\cdot(0,0,1,-1)=0\ \checkmark.$$

And $\text{null}(\mathbf{A}^\mathsf{T}) = \text{span}\{(-1,-1,1)\} \perp \text{col}(\mathbf{A})$: $(-1,-1,1)\cdot(1,2,3)=0$ and $(-1,-1,1)\cdot(1,0,1)=0$ ✓. Dimensions $2+2=4=n$ and $2+1=3=m$ ✓.

Common Pitfall — orthogonal is not the same as independent

Orthogonal nonzero vectors are independent, but independent vectors need not be orthogonal: $(1,0)$ and $(1,1)$ are independent and meet at 45°. Also, "orthogonal matrix" means ortho*normal* columns; a matrix with merely orthogonal columns satisfies $\mathbf{Q}^\mathsf{T}\mathbf{Q}$ = diagonal, not $\mathbf{I}$, and its inverse is *not* its transpose.

Engineering Note — I/Q modulation and orthogonal channels

Over one symbol period T containing an integer number of carrier cycles, $\cos\omega_c t$ and $\sin\omega_c t$ are orthogonal (Section 9.8). A QAM transmitter sends $s(t) = a\cos\omega_c t - b\sin\omega_c t$; the receiver recovers a and b separately by taking inner products with each carrier — exactly the orthonormal-coordinate formula of this section. OFDM extends this to hundreds of mutually orthogonal subcarriers, and CDMA to orthogonal spreading codes (rows of a Hadamard matrix, which satisfies $\mathbf{H}^\mathsf{T}\mathbf{H} = N\mathbf{I}$).

9.8 Inner products

Motivation

The dot product gave $\mathbb{R}^n$ lengths, angles and orthogonality. To do the same for complex phasor vectors, for vectors whose components have different importance (weights), and above all for **signals**, we abstract its essential properties. The result, an **inner product**, turns the function space of finite-energy signals into a geometric space in which a signal has a length (its RMS value or the square root of its energy), two signals have an angle (their correlation), and the harmonics $\cos k\omega_0 t$, $\sin k\omega_0 t$ are mutually perpendicular axes. This is the bridge from linear algebra to Fourier analysis.

Definition

Inner product

An **inner product** on a vector space V over $\mathbb{F}$ assigns to each pair $\mathbf{u},\mathbf{v}$ a scalar $\langle \mathbf{u},\mathbf{v}\rangle \in \mathbb{F}$ such that for all $\mathbf{u},\mathbf{v},\mathbf{w}$ and scalars a,b:

1. **Linearity in the first argument:** $\langle a\mathbf{u}+b\mathbf{w},\mathbf{v}\rangle = a\langle\mathbf{u},\mathbf{v}\rangle + b\langle\mathbf{w},\mathbf{v}\rangle$;
2. **Conjugate symmetry:** $\langle\mathbf{u},\mathbf{v}\rangle = \overline{\langle\mathbf{v},\mathbf{u}\rangle}$ (plain symmetry when $\mathbb{F}=\mathbb{R}$);
3. **Positivity:** $\langle\mathbf{u},\mathbf{u}\rangle > 0$ for $\mathbf{u}\neq\mathbf{0}$ (and $\langle\mathbf{0},\mathbf{0}\rangle = 0$).

A vector space with an inner product is an **inner-product space**. The induced **norm** is $\|\mathbf{u}\| = \sqrt{\langle\mathbf{u},\mathbf{u}\rangle}$, and $\mathbf{u}\perp\mathbf{v}$ means $\langle\mathbf{u},\mathbf{v}\rangle = 0$.

Properties 1 and 2 together give **conjugate linearity in the second argument**: $\langle\mathbf{u},a\mathbf{v}\rangle = \bar{a}\langle\mathbf{u},\mathbf{v}\rangle$. This book follows the engineering and signal-processing convention (linear in the first slot, conjugated in the second); physics texts often use the opposite convention. Both give the same norms and the same notion of orthogonality.

Results that hold in every inner-product space

- **Cauchy–Schwarz:** $|\langle\mathbf{u},\mathbf{v}\rangle| \le \|\mathbf{u}\|\,\|\mathbf{v}\|$, equality iff $\mathbf{u},\mathbf{v}$ are parallel.
- **Triangle inequality:** $\|\mathbf{u}+\mathbf{v}\| \le \|\mathbf{u}\| + \|\mathbf{v}\|$.
- **Pythagoras:** if $\mathbf{u}\perp\mathbf{v}$ then $\|\mathbf{u}+\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$.
- **Parallelogram law:** $\|\mathbf{u}+\mathbf{v}\|^2 + \|\mathbf{u}-\mathbf{v}\|^2 = 2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2$.
- Orthogonal nonzero vectors are independent, and coordinates in an orthogonal basis are $c_j = \langle\mathbf{x},\mathbf{u}_j\rangle / \langle\mathbf{u}_j,\mathbf{u}_j\rangle$.

The proofs of Section 9.7 go through unchanged, with $\mathbf{x}\cdot\mathbf{y}$ replaced by $\langle\mathbf{x},\mathbf{y}\rangle$. (In the complex case of Cauchy–Schwarz, minimise $\|\mathbf{u}-s\mathbf{v}\|^2$ over complex s; the minimiser is $s = \langle\mathbf{u},\mathbf{v}\rangle/\|\mathbf{v}\|^2$.)

Inner products on $\mathbb{R}^n$ and $\mathbb{C}^n$

Standard complex inner product.

$$\langle\mathbf{u},\mathbf{v}\rangle = \sum_{k=1}^{n} u_k\,\overline{v_k} = \mathbf{v}^\mathsf{H}\mathbf{u}, \qquad \|\mathbf{u}\|^2 = \sum_k |u_k|^2.$$

The conjugate is essential. Without it, $(1,j)$ would have "squared length" $1^2 + j^2 = 0$, violating positivity.

Weighted inner product. For a symmetric positive-definite matrix $\mathbf{W}$ (Ch. 10), $\langle \mathbf{u}, \mathbf{v} \rangle_{\mathbf{W}} = \mathbf{v}^{\mathsf{T}} \mathbf{W} \mathbf{u}$. The diagonal case $\mathbf{W} = \operatorname{diag}(w_1, \dots, w_n)$ with $w_k > 0$ weights each component by its reliability — the basis of weighted least squares (Section 9.11).

Worked Example 9.24 — A complex inner product

Let $\mathbf{u} = (1 + j,\ 2)^{\mathsf{T}}$ and $\mathbf{v} = (j,\ 1 - j)^{\mathsf{T}}$ in $\mathbb{C}^2$. Compute $\langle \mathbf{u}, \mathbf{v} \rangle$, both norms, and check Cauchy–Schwarz.

$$\langle \mathbf{u}, \mathbf{v} \rangle = (1 + j)(j)^* + 2(1 - j)^* = (1 + j)(-j) + 2(1 + j) = (-j + 1) + (2 + 2j) = 3 + j.$$

$$\|\mathbf{u}\|^2 = |1 + j|^2 + |2|^2 = 2 + 4 = 6, \qquad \|\mathbf{v}\|^2 = |j|^2 + |1 - j|^2 = 1 + 2 = 3.$$

Cauchy–Schwarz: $|3 + j| = \sqrt{10} = 3.162 \le \sqrt{6}\sqrt{3} = \sqrt{18} = 4.243$ ✓. Conjugate symmetry: $\langle \mathbf{v}, \mathbf{u} \rangle = j(1 - j) + (1 - j) \cdot 2 = (j + 1) + (2 - 2j) = 3 - j = \langle \mathbf{u}, \mathbf{v} \rangle^*$ ✓.

Inner products of functions and signals

For real continuous functions on $[a, b]$, the natural inner product replaces the sum over components by an integral over t:

$$\langle f, g \rangle = \int_a^b f(t)\, g(t)\, dt, \qquad \|f\|^2 = \int_a^b f(t)^2\, dt.$$

For complex-valued signals, $\langle f, g \rangle = \int_a^b f(t)\, g^*(t)\, dt$. Positivity holds because a continuous function with $\int f^2 = 0$ is identically zero. (On L^2, functions that differ only on a set of zero length — say at isolated points — are regarded as the same vector, so that positivity still holds.)

The analogy with $\mathbb{R}^n$ is exact if you think of a function as a vector with a continuum of components indexed by t: sampling f and g at spacing Δt gives vectors whose dot product times Δt is a Riemann sum for $\int f g\, dt$.

Two inner products are standard in signal analysis:

Signal class	Inner product	Squared norm	Physical meaning
energy signals on $\mathbb{R}$	$\langle x, y \rangle = \int_{-\infty}^{\infty} x(t) y^*(t)\, dt$	$E_x = \int \lvert x \rvert^2 dt$	energy delivered to a 1 Ω load
T-periodic (power) signals	$\langle x, y \rangle_T = \frac{1}{T} \int_T x(t) y^*(t)\, dt$	$P_x = \frac{1}{T} \int_T \lvert x \rvert^2 dt$	average power in 1 Ω; $\|x\|_T$ = RMS value

So **the RMS value of a periodic waveform is its norm** in the power inner product, and the energy of a pulse is its squared norm.

Orthogonality of harmonics

Orthogonality of the trigonometric system

Let $\omega_0 = 2\pi / T$. For integers $m, n \ge 0$, over any interval of length T:

$$\int_T \cos m\omega_0 t\, \cos n\omega_0 t\, dt = \begin{cases} 0, & m \ne n \\ T/2, & m = n \ge 1 \\ T, & m = n = 0 \end{cases} \qquad \int_T \sin m\omega_0 t\, \sin n\omega_0 t\, dt = \begin{cases} 0, & m \ne n \\ T/2, & m = n \ge 1 \end{cases}$$

$$\int_T \cos m\omega_0 t\, \sin n\omega_0 t\, dt = 0 \quad \text{for all } m, n.$$

In complex form, with the power inner product,

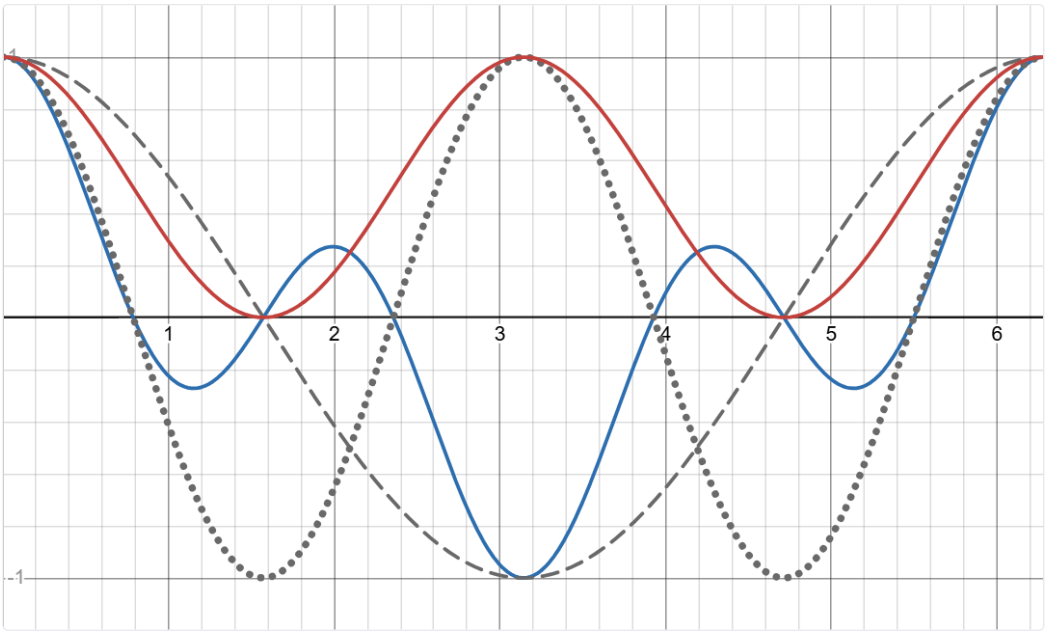
$$\left\langle e^{jm\omega_0 t}, e^{jn\omega_0 t} \right\rangle_T = \frac{1}{T} \int_T e^{j(m-n)\omega_0 t}\, dt = \delta_{mn} \qquad (m, n \in \mathbb{Z}),$$

so $\{ e^{jn\omega_0 t} \}_{n \in \mathbb{Z}}$ is an **orthonormal** set.

Derivation. The complex form is the cleanest: if $m = n$ the integrand is 1 and the integral is T. If $m \ne n$, put $k = m - n \ne 0$:

$$\int_{t_0}^{t_0 + T} e^{jk\omega_0 t}\, dt = \frac{e^{jk\omega_0 t_0} \left(e^{jk\omega_0 T} - 1 \right)}{jk\omega_0} = 0,$$

because $k\omega_0 T = 2\pi k$. The real formulas follow from product-to-sum identities, e.g. $\cos m\omega_0 t \cos n\omega_0 t = \tfrac{1}{2}[\cos(m - n)\omega_0 t + \cos(m + n)\omega_0 t]$: each cosine integrates to zero over a period unless its frequency is zero.



Over one period $[0, 2\pi]$, the product $\cos t \cos 2t$ (blue) has equal positive and negative areas, so the inner product is 0; the product $\cos^2 t$ (red) is never negative and integrates to $\pi = T/2$.

Consequently, if a signal is a finite trigonometric sum $x(t) = \sum_{n=-N}^{N} c_n e^{jn\omega_0 t}$, its coefficients are its coordinates in an orthonormal basis:

$$c_n = \langle x, e^{jn\omega_0 t} \rangle_T = \frac{1}{T}\int_T x(t)\, e^{-jn\omega_0 t}\, dt, \qquad P_x = \sum_n |c_n|^2.$$

These are the **Fourier coefficient formula** and **Parseval's relation** of Ch. 15 — here obtained purely from the orthonormal-coordinate theorem. What Chapter 15 adds is the analysis of what happens when $N \to \infty$.

Worked Example 9.25 — Angles between polynomials

(a) On $[-1, 1]$ with $\langle f, g \rangle = \int_{-1}^{1} fg\, dt$: $\langle 1, t \rangle = \int_{-1}^{1} t\, dt = 0$ and $\langle t, t^2 \rangle = \int_{-1}^{1} t^3\, dt = 0$. Any even function is orthogonal to any odd function on a symmetric interval — the subspaces of even and odd functions are orthogonal complements. Norms: $\|1\|^2 = 2$, $\|t\|^2 = 2/3$, $\|t^2\|^2 = 2/5$. But $\langle 1, t^2 \rangle = 2/3 \neq 0$: the monomials are *not* an orthogonal set (Section 9.10 fixes this).

(b) On $[0, 1]$ the same functions behave very differently:

$$\langle t, t^2 \rangle = \int_0^1 t^3\, dt = \tfrac{1}{4}, \quad \|t\| = \sqrt{\int_0^1 t^2 dt} = \tfrac{1}{\sqrt{3}}, \quad \|t^2\| = \tfrac{1}{\sqrt{5}},$$

$$\cos\theta = \frac{1/4}{(1/\sqrt{3})(1/\sqrt{5})} = \frac{\sqrt{15}}{4} = 0.9682, \qquad \theta = 14.5^\circ.$$

On $[0, 1]$, t and t^2 are nearly parallel. This geometric fact is the root of the ill-conditioning of polynomial fitting with monomials (Section 9.11). The inner product — and hence orthogonality — depends on the interval.

Worked Example 9.26 — Energy, correlation and distance of two pulses

Let $x_1(t) = e^{-t}u(t)$ and $x_2(t) = e^{-2t}u(t)$ (volts, t in seconds, 1 Ω reference).

$$E_1 = \|x_1\|^2 = \int_0^\infty e^{-2t} dt = \tfrac{1}{2}\ \text{J}, \qquad E_2 = \int_0^\infty e^{-4t} dt = \tfrac{1}{4}\ \text{J}, \qquad \langle x_1, x_2 \rangle = \int_0^\infty e^{-3t} dt = \tfrac{1}{3}.$$

Correlation coefficient (cosine of the angle):

$$\rho = \frac{\langle x_1, x_2 \rangle}{\|x_1\|\|x_2\|} = \frac{1/3}{\sqrt{1/8}} = \frac{2\sqrt{2}}{3} = 0.943.$$

Distance (energy of the difference): $\|x_1 - x_2\|^2 = E_1 + E_2 - 2\langle x_1, x_2 \rangle = \tfrac{1}{2} + \tfrac{1}{4} - \tfrac{2}{3} = \tfrac{1}{12}$ J. A receiver deciding between these two pulses in noise has a hard job: the pulses are nearly parallel and close together. In digital communications, error probability is governed by exactly this distance between signal vectors.

Worked Example 9.27 — RMS value by Pythagoras

Find the RMS value of $v(t) = 3 + 4\cos\omega_0 t - 2\sin 3\omega_0 t$ V.

The three terms are mutually orthogonal in the power inner product, so their powers add:

$$P = \|3\|_T^2 + \|4\cos\omega_0 t\|_T^2 + \|2\sin 3\omega_0 t\|_T^2 = 9 + \frac{16}{2} + \frac{4}{2} = 19\ \text{V}^2,$$

using $\frac{1}{T}\int_T \cos^2 = \frac{1}{T}\int_T \sin^2 = \frac{1}{2}$. Hence $V_{\text{rms}} = \sqrt{19} = 4.359$ V. Without orthogonality we would have to expand v^2 and integrate nine cross terms; orthogonality says the six cross terms all vanish.

Cauchy–Schwarz and the matched filter

Cauchy–Schwarz has a celebrated engineering reading. A known pulse $s(t)$ arrives in white noise and is passed through a filter whose output at the sampling instant is an inner product $\langle s, h\rangle$ with some template h. The signal-to-noise ratio turns out to be proportional to

$$\frac{|\langle s, h\rangle|^2}{\|h\|^2} \le \frac{\|s\|^2\|h\|^2}{\|h\|^2} = \|s\|^2 = E_s,$$

with equality iff h is parallel to s. The optimal template is the signal itself (the **matched filter**), and the best achievable SNR depends only on the pulse energy, not its shape. The derivation of the noise side belongs to Ch. 21; the optimisation step is pure Cauchy–Schwarz.

Common Pitfall — forgetting the conjugate, or the interval

With complex vectors or signals, $\sum u_k v_k$ (no conjugate) is *not* an inner product, and "orthogonality" computed with it is meaningless. Similarly, $\cos t$ and $\cos 2t$ are orthogonal over $[0, 2\pi]$ but not over $[0, 1]$: orthogonality of sinusoids requires an interval containing a whole number of periods of each (more precisely, of their difference frequency). Truncating a measurement to a non-integer number of periods destroys orthogonality and causes spectral leakage in the DFT.

Engineering Note — signal space

Digital modulation is analysed in **signal space**: choose an orthonormal basis $\phi_1(t), \dots, \phi_N(t)$ for the set of transmitted waveforms (for QPSK, $\sqrt{2/T}\cos\omega_c t$ and $\sqrt{2/T}\sin\omega_c t$ over one symbol). Each waveform becomes a point in $\mathbb{R}^N$ — the constellation diagram — energies become squared lengths, and the receiver's correlators compute the coordinates $\langle r, \phi_k\rangle$. Every geometric statement of this chapter has a direct meaning on that diagram.

9.9 Projection

Motivation

Most approximation problems in engineering have the same shape: we want to represent a complicated object (a measured data vector, a distorted waveform, a field) by the closest member of a simpler family (a straight line, a few harmonics, a low-order model). When the simpler family is a subspace and "closest" is measured by an inner-product norm — energy, RMS error, sum of squared residuals — the answer is always the **orthogonal projection**, and it is characterised by one geometric fact: *the error is perpendicular to the subspace*. Least squares (Section 9.11), truncated Fourier series (Ch. 15) and Wiener filtering are all instances.

Projection onto a line

Let $\mathbf{a} \neq \mathbf{0}$ span a line through the origin. The point $\mathbf{p} = \hat{c}\,\mathbf{a}$ closest to $\mathbf{b}$ is determined by requiring the error $\mathbf{e} = \mathbf{b} - \hat{c}\mathbf{a}$ to be orthogonal to $\mathbf{a}$:

$$\langle \mathbf{b} - \hat{c}\,\mathbf{a}, \mathbf{a}\rangle = 0 \quad\Longrightarrow\quad \hat{c} = \frac{\langle \mathbf{b}, \mathbf{a}\rangle}{\langle \mathbf{a}, \mathbf{a}\rangle}, \qquad \mathbf{p} = \operatorname{proj}_{\mathbf{a}}\mathbf{b} = \frac{\langle \mathbf{b}, \mathbf{a}\rangle}{\langle \mathbf{a}, \mathbf{a}\rangle}\,\mathbf{a}.$$

In $\mathbb{R}^n$ the projection is a matrix acting on $\mathbf{b}$:

$$\mathbf{p} = \frac{\mathbf{a}\mathbf{a}^{\mathsf{T}}}{\mathbf{a}^{\mathsf{T}}\mathbf{a}}\,\mathbf{b} = \mathbf{P}\mathbf{b}, \qquad \mathbf{P} = \frac{\mathbf{a}\mathbf{a}^{\mathsf{T}}}{\mathbf{a}^{\mathsf{T}}\mathbf{a}} \ \ (\text{rank } 1).$$

The scalar $\langle \mathbf{b}, \mathbf{a}\rangle / \|\mathbf{a}\|$ is the **scalar component** of $\mathbf{b}$ along $\mathbf{a}$, equal to $\|\mathbf{b}\|\cos\theta$.

Worked Example 9.28 — Projecting onto the DC direction

Project $\mathbf{b} = (1, 2, 3)^{\mathsf{T}}$ onto $\mathbf{a} = (1, 1, 1)^{\mathsf{T}}$.

$\hat{c} = \dfrac{\mathbf{b}\cdot\mathbf{a}}{\mathbf{a}\cdot\mathbf{a}} = \dfrac{6}{3} = 2$, so $\mathbf{p} = (2, 2, 2)^{\mathsf{T}}$ and $\mathbf{e} = \mathbf{b} - \mathbf{p} = (-1, 0, 1)^{\mathsf{T}}$.

Check: $\mathbf{e}\cdot\mathbf{a} = -1 + 0 + 1 = 0$ ✓. **Pythagoras:** $\|\mathbf{b}\|^2 = 14 = \|\mathbf{p}\|^2 + \|\mathbf{e}\|^2 = 12 + 2$ ✓.

$\hat{c} = 2$ is the **mean** of the entries of $\mathbf{b}$: projecting onto the all-ones vector extracts the average, and the error is the zero-mean residual. The distance from $\mathbf{b}$ to the line is $\|\mathbf{e}\| = \sqrt{2} - \sqrt{3}$ times the (population) standard deviation of the entries $1, 2, 3$, which is $\sqrt{2/3}$.

Projection onto a subspace

Best approximation (orthogonal projection) theorem

Let W be a finite-dimensional subspace of an inner-product space V and $\mathbf{b} \in V$. There is a unique $\mathbf{p} \in W$ closest to $\mathbf{b}$:

$$\|\mathbf{b} - \mathbf{p}\| < \|\mathbf{b} - \mathbf{w}\| \quad \text{for every } \mathbf{w} \in W,\ \mathbf{w} \neq \mathbf{p}.$$

It is characterised by the **orthogonality principle**: $\mathbf{b} - \mathbf{p} \perp W$. We write $\mathbf{p} = \operatorname{proj}_W \mathbf{b}$.

Proof. Suppose $\mathbf{p} \in W$ with $\mathbf{e} = \mathbf{b} - \mathbf{p} \perp W$. For any $\mathbf{w} \in W$, $\mathbf{p} - \mathbf{w} \in W$, so by Pythagoras

$$\|\mathbf{b} - \mathbf{w}\|^2 = \|(\mathbf{b} - \mathbf{p}) + (\mathbf{p} - \mathbf{w})\|^2 = \|\mathbf{e}\|^2 + \|\mathbf{p} - \mathbf{w}\|^2 \geq \|\mathbf{e}\|^2,$$

with equality only if $\mathbf{w} = \mathbf{p}$. Existence: construct $\mathbf{p}$ with an orthogonal basis (next formula) or by the normal equations.

With an orthogonal basis $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ of W the projection is the sum of the projections onto each basis vector:

$$\operatorname{proj}_W \mathbf{b} = \sum_{j=1}^{k} \frac{\langle \mathbf{b}, \mathbf{u}_j \rangle}{\langle \mathbf{u}_j, \mathbf{u}_j \rangle}\, \mathbf{u}_j \qquad \Big(= \sum_j \langle \mathbf{b}, \mathbf{u}_j \rangle \mathbf{u}_j \text{ if orthonormal} \Big).$$

Check of orthogonality: $\langle \mathbf{b} - \mathbf{p}, \mathbf{u}_i \rangle = \langle \mathbf{b}, \mathbf{u}_i \rangle - \frac{\langle \mathbf{b}, \mathbf{u}_i \rangle}{\langle \mathbf{u}_i, \mathbf{u}_i \rangle} \langle \mathbf{u}_i, \mathbf{u}_i \rangle = 0$, the cross terms having vanished by orthogonality.

Common Pitfall — the sum formula needs an orthogonal basis

Adding up the projections onto each basis vector gives the projection onto their span **only** if the basis is orthogonal. For a non-orthogonal basis the individual projections "double count" the overlap. Either orthogonalise first (Gram–Schmidt, Section 9.10) or use the normal-equation formula below.

With an arbitrary basis — the columns of $\mathbf{A} \in \mathbb{R}^{m \times k}$, assumed independent — write $\mathbf{p} = \mathbf{A}\hat{\mathbf{x}}$. Orthogonality of the error to every column means $\mathbf{A}^{\mathsf{T}}(\mathbf{b} - \mathbf{A}\hat{\mathbf{x}}) = \mathbf{0}$:

$$\mathbf{A}^{\mathsf{T}}\mathbf{A}\,\hat{\mathbf{x}} = \mathbf{A}^{\mathsf{T}}\mathbf{b} \qquad \text{(normal equations)},$$

$$\hat{\mathbf{x}} = (\mathbf{A}^{\mathsf{T}}\mathbf{A})^{-1}\mathbf{A}^{\mathsf{T}}\mathbf{b}, \qquad \mathbf{p} = \mathbf{P}\mathbf{b}, \qquad \mathbf{P} = \mathbf{A}(\mathbf{A}^{\mathsf{T}}\mathbf{A})^{-1}\mathbf{A}^{\mathsf{T}}.$$

$\mathbf{A}^{\mathsf{T}}\mathbf{A}$ is invertible precisely when the columns of $\mathbf{A}$ are independent: if $\mathbf{A}^{\mathsf{T}}\mathbf{A}\mathbf{x} = \mathbf{0}$ then $\mathbf{x}^{\mathsf{T}}\mathbf{A}^{\mathsf{T}}\mathbf{A}\mathbf{x} = \|\mathbf{A}\mathbf{x}\|^2 = 0$, so $\mathbf{A}\mathbf{x} = \mathbf{0}$, so $\mathbf{x} = \mathbf{0}$. For complex data replace T by H.

Properties of projection matrices

The orthogonal projection matrix $\mathbf{P}$ onto a subspace W satisfies

1. $\mathbf{P}^2 = \mathbf{P}$ (**idempotent**: projecting twice changes nothing);
2. $\mathbf{P}^{\mathsf{T}} = \mathbf{P}$ (**symmetric**);
3. $\operatorname{col}(\mathbf{P}) = W$, $\operatorname{null}(\mathbf{P}) = W^{\perp}$, $\operatorname{rank}\mathbf{P} = \dim W$;
4. $\mathbf{I} - \mathbf{P}$ is the projection onto $W^{\perp}$, and $\mathbf{b} = \mathbf{P}\mathbf{b} + (\mathbf{I} - \mathbf{P})\mathbf{b}$ is the orthogonal splitting $\mathbb{R}^m = W \oplus W^{\perp}$.

Conversely, any matrix with $\mathbf{P}^2 = \mathbf{P} = \mathbf{P}^{\mathsf{T}}$ is the orthogonal projection onto its column space. If W has an orthonormal basis forming the columns of $\mathbf{Q}$, then simply $\mathbf{P} = \mathbf{Q}\mathbf{Q}^{\mathsf{T}}$.

Proof of 1. $\mathbf{P}^2 = \mathbf{A}(\mathbf{A}^{\mathsf{T}}\mathbf{A})^{-1}(\mathbf{A}^{\mathsf{T}}\mathbf{A})(\mathbf{A}^{\mathsf{T}}\mathbf{A})^{-1}\mathbf{A}^{\mathsf{T}} = \mathbf{P}$. Property 2 follows because $(\mathbf{A}^{\mathsf{T}}\mathbf{A})^{-1}$ is symmetric.

A matrix with $\mathbf{P}^2 = \mathbf{P}$ but $\mathbf{P}^{\mathsf{T}} \neq \mathbf{P}$ is an **oblique** projection: it maps onto a subspace along a direction that is not perpendicular to it.

Worked Example 9.29 — Projection onto a plane in $\mathbb{R}^3$

Project $\mathbf{b} = (6, 0, 0)^{\mathsf{T}}$ onto the column space of

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

Normal equations.

$$\mathbf{A}^{\mathsf{T}}\mathbf{A} = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}, \qquad \mathbf{A}^{\mathsf{T}}\mathbf{b} = \begin{bmatrix} 6 \\ 0 \end{bmatrix}, \qquad \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}\begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \end{bmatrix}.$$

Subtracting the first equation from the second: $2\hat{x}_2 = -6$, $\hat{x}_2 = -3$; then $3\hat{x}_1 - 9 = 6$, $\hat{x}_1 = 5$.

Projection and error. $\mathbf{p} = \mathbf{A}\hat{\mathbf{x}} = 5(1,1,1) - 3(0,1,2) = (5, 2, -1)^{\mathsf{T}}$, $\mathbf{e} = \mathbf{b} - \mathbf{p} = (1, -2, 1)^{\mathsf{T}}$.

Checks. $\mathbf{A}^{\mathsf{T}}\mathbf{e} = (1 - 2 + 1,\ 0 - 2 + 2) = (0, 0)$ ✓ — the error is orthogonal to both columns. $\|\mathbf{b}\|^2 = 36 = \|\mathbf{p}\|^2 + \|\mathbf{e}\|^2 = 30 + 6$ ✓.

Projection matrix. $(\mathbf{A}^{\mathsf{T}}\mathbf{A})^{-1} = \frac{1}{6}\begin{bmatrix} 5 & -3 \\ -3 & 3 \end{bmatrix}$ and

$$\mathbf{P} = \mathbf{A}(\mathbf{A}^{\mathsf{T}}\mathbf{A})^{-1}\mathbf{A}^{\mathsf{T}} = \frac{1}{6}\begin{bmatrix} 5 & 2 & -1 \\ 2 & 2 & 2 \\ -1 & 2 & 5 \end{bmatrix}.$$

Its first column is $\mathbf{P}\mathbf{e}_1 = \frac{1}{6}(5, 2, -1)$, consistent with $\mathbf{P}\mathbf{b} = 6\,\mathbf{P}\mathbf{e}_1 = (5, 2, -1)$ ✓. Its trace is $12/6 = 2 = \operatorname{rank}\mathbf{P}$ (the trace of a projection equals the dimension of the subspace). This same matrix reappears in Section 9.11 as fitting a straight line to three data points.

Worked Example 9.30 — Projection with an orthonormal basis

Project $\mathbf{x} = (3, 1, 2)^{\mathsf{T}}$ onto the plane $W:\ x + y + z = 0$.

From Worked Example 9.22, $\mathbf{u}_2 = \frac{1}{\sqrt{2}}(1, -1, 0)$ and $\mathbf{u}_3 = \frac{1}{\sqrt{6}}(1, 1, -2)$ form an orthonormal basis of W (both are orthogonal to $(1, 1, 1)$). Then

$$\operatorname{proj}_W \mathbf{x} = (\mathbf{x}\cdot\mathbf{u}_2)\mathbf{u}_2 + (\mathbf{x}\cdot\mathbf{u}_3)\mathbf{u}_3 = \sqrt{2}\cdot\tfrac{1}{\sqrt{2}}(1, -1, 0) + 0 = (1, -1, 0).$$

Alternative via the complement: $W^{\perp} = \operatorname{span}\{(1, 1, 1)\}$, so $\operatorname{proj}_W \mathbf{x} = \mathbf{x} - \operatorname{proj}_{(1,1,1)}\mathbf{x} = (3, 1, 2) - (2, 2, 2) = (1, -1, 0)$ ✓. When the complement is smaller than the subspace, projecting onto the complement and subtracting is quicker.

Projection in function spaces and Bessel's inequality

Everything above works in a function space with the integral inner product. If $\{\phi_1, \dots, \phi_N\}$ is an orthogonal set of functions, the best approximation of f in their span (minimum $\int |f - \hat{f}|^2$) is

$$\hat{f} = \sum_{n=1}^{N} c_n \phi_n, \qquad c_n = \frac{\langle f, \phi_n \rangle}{\langle \phi_n, \phi_n \rangle},$$

and Pythagoras gives the error energy and **Bessel's inequality**:

$$\|f - \hat{f}\|^2 = \|f\|^2 - \sum_{n=1}^{N} |c_n|^2 \|\phi_n\|^2 \ \geq 0 \quad \Longrightarrow \quad \sum_{n=1}^{N} |c_n|^2 \|\phi_n\|^2 \leq \|f\|^2.$$

Two important consequences:

1. **The coefficients do not change when N grows.** Adding ϕ_{N+1} to the basis adds a term but leaves $c_1, \dots, c_N$ alone — a property unique to orthogonal bases, and the reason Fourier coefficients can be computed one at a time.
2. **The error energy decreases monotonically with N.** If it tends to zero for every f, the orthogonal set is **complete** and Bessel's inequality becomes **Parseval's identity** $\sum |c_n|^2 \|\phi_n\|^2 = \|f\|^2$ — the central convergence statement of Ch. 15.

A simple but important special case: the projection of f onto the constant function 1 on $[a, b]$ is $\langle f, 1\rangle / \langle 1, 1\rangle = \frac{1}{b-a}\int_a^b f\,dt$, the **average value**. The DC component of a signal is its projection onto the constants.

Worked Example 9.31 — Best sinusoidal approximation of a square wave

Let $q(t) = \operatorname{sgn}(\sin t)$ on $[-\pi, \pi]$ (a ± 1 square wave, period 2π). Find its best approximation (least error energy) in $\operatorname{span}\{\sin t\}$ and then in $\operatorname{span}\{\sin t, \sin 3t\}$, and the fraction of energy left in the error.

Inner products. $\|q\|^2 = \int_{-\pi}^{\pi} 1\, dt = 2\pi$, $\|\sin nt\|^2 = \pi$, and since $q(t)\sin nt$ is even,

$$\langle q, \sin nt\rangle = 2\int_0^{\pi} \sin nt\, dt = \frac{2(1-\cos n\pi)}{n} = \begin{cases} 4/n, & n \text{ odd} \\ 0, & n \text{ even.} \end{cases}$$

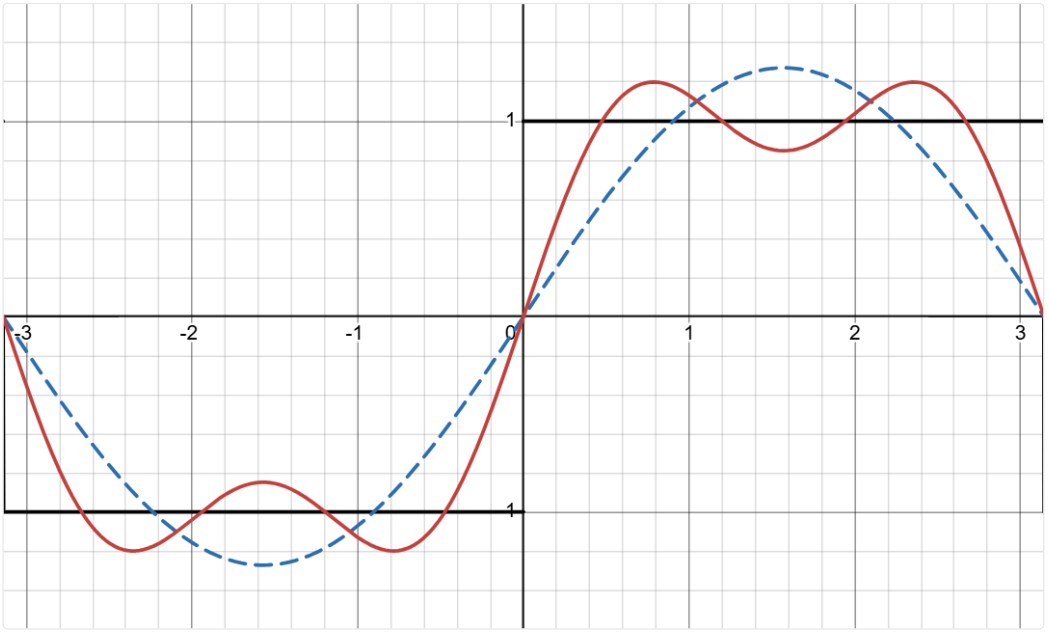
One term. $c_1 = \dfrac{4}{\pi} = 1.2732$, so $\hat{q}_1(t) = \dfrac{4}{\pi}\sin t$. Note that the best amplitude is *larger* than the square wave's amplitude 1.

Error energy: $\|q\|^2 - c_1^2\|\sin t\|^2 = 2\pi - \dfrac{16}{\pi^2}\pi = 2\pi - \dfrac{16}{\pi} = 6.2832 - 5.0930 = 1.1902$. Fraction of energy in the error: $1 - \dfrac{8}{\pi^2} = 0.1894$ (18.9 %).

Two terms. $\sin t \perp \sin 3t$, so c_1 is unchanged and $c_3 = \dfrac{4/3}{\pi} = \dfrac{4}{3\pi} = 0.4244$. Error fraction:

$$1 - \frac{8}{\pi^2}\left(1 + \frac{1}{9}\right) = 0.0994 \quad (9.9\ \%).$$

Adding the third harmonic halves the error energy, and c_1 did not have to be recomputed.



A ± 1 square wave (black), its orthogonal projection onto $\operatorname{span}\{\sin t\}$ (blue dashed, amplitude $4/\pi$) and onto $\operatorname{span}\{\sin t, \sin 3t\}$ (red). Each added orthogonal direction reduces the error energy without altering the earlier coefficients.

Engineering Note — the orthogonality principle in estimation

The condition "error $\perp$ every allowed direction" is the **orthogonality principle** of estimation theory. In a Wiener filter or a linear minimum-mean-square-error estimator, the vector space consists of random variables with inner product $\langle X, Y\rangle = E[XY]$, and the optimal linear estimate makes the estimation error uncorrelated with every observation used. Only the inner product has changed; the geometry is that of this section (Ch. 21).

9.10 Gram–Schmidt orthogonalization

Motivation

Orthogonal bases make coordinates, projections and approximations trivial — but the bases that arise naturally (the columns of a data matrix, the monomials $1, t, t^2, \ldots$, a set of measured waveforms) are rarely orthogonal. The **Gram–Schmidt process** converts any independent set into an orthonormal set with the same span, one vector at a time. In matrix form it is the **QR factorisation** $\mathbf{A} = \mathbf{QR}$, the standard numerical route to least squares; applied to polynomials it produces the **Legendre polynomials**, and applied to other weights it produces the other classical orthogonal polynomials used in filter design and spectral methods.

The algorithm

Given independent $\mathbf{a}_1, \ldots, \mathbf{a}_k$ in an inner-product space, subtract from each new vector its projections onto the orthogonal vectors already built:

$$\begin{aligned}
\mathbf{v}_1 &= \mathbf{a}_1,\\
\mathbf{v}_2 &= \mathbf{a}_2 - \frac{\langle \mathbf{a}_2,\mathbf{v}_1\rangle}{\langle \mathbf{v}_1,\mathbf{v}_1\rangle}\mathbf{v}_1,\\
\mathbf{v}_3 &= \mathbf{a}_3 - \frac{\langle \mathbf{a}_3,\mathbf{v}_1\rangle}{\langle \mathbf{v}_1,\mathbf{v}_1\rangle}\mathbf{v}_1 - \frac{\langle \mathbf{a}_3,\mathbf{v}_2\rangle}{\langle \mathbf{v}_2,\mathbf{v}_2\rangle}\mathbf{v}_2,\\
&\;\;\vdots\\
\mathbf{v}_j &= \mathbf{a}_j - \sum_{i=1}^{j-1}\frac{\langle \mathbf{a}_j,\mathbf{v}_i\rangle}{\langle \mathbf{v}_i,\mathbf{v}_i\rangle}\mathbf{v}_i = \mathbf{a}_j - \operatorname{proj}_{W_{j-1}}\mathbf{a}_j,
\end{aligned}$$

where $W_{j-1}=\operatorname{span}\{\mathbf{a}_1,\dots,\mathbf{a}_{j-1}\}$. Finally normalise: $\mathbf{q}_j=\mathbf{v}_j/\|\mathbf{v}_j\|$.

Gram–Schmidt

If $\mathbf{a}_1,\dots,\mathbf{a}_k$ are linearly independent, the vectors $\mathbf{q}_1,\dots,\mathbf{q}_k$ produced above are orthonormal and, for every j,

$$\operatorname{span}\{\mathbf{q}_1,\dots,\mathbf{q}_j\}=\operatorname{span}\{\mathbf{a}_1,\dots,\mathbf{a}_j\}.$$

Consequently every finite-dimensional inner-product space has an orthonormal basis.

Proof by induction. Suppose $\mathbf{v}_1,\dots,\mathbf{v}_{j-1}$ are orthogonal, nonzero and span W_{j-1}. Then $\mathbf{v}_j=\mathbf{a}_j-\operatorname{proj}_{W_{j-1}}\mathbf{a}_j$ is orthogonal to W_{j-1} by the best-approximation theorem (the error of a projection is perpendicular to the subspace). It is nonzero, because otherwise $\mathbf{a}_j\in W_{j-1}$, contradicting independence. And $\mathbf{v}_j$ is $\mathbf{a}_j$ minus an element of W_{j-1}, so $\operatorname{span}\{\mathbf{v}_1,\dots,\mathbf{v}_j\}=\operatorname{span}\{\mathbf{a}_1,\dots,\mathbf{a}_j\}$.

If the input vectors are dependent, some $\mathbf{v}_j$ comes out as $\mathbf{0}$; dropping it and continuing yields an orthonormal basis of the span. In numerical work a "nearly zero" $\mathbf{v}_j$ is a warning of near-dependence.

Worked Example 9.32 — Gram–Schmidt in $\mathbb{R}^3$

Orthonormalise $\mathbf{a}_1=(1,1,0)$, $\mathbf{a}_2=(1,0,1)$, $\mathbf{a}_3=(0,1,1)$ (independent by Worked Example 9.7(b)).

Step 1. $\mathbf{v}_1=(1,1,0)$, $\|\mathbf{v}_1\|^2=2$.

Step 2. $\langle \mathbf{a}_2,\mathbf{v}_1\rangle=1$:

$$\mathbf{v}_2=(1,0,1)-\tfrac12(1,1,0)=\left(\tfrac12,-\tfrac12,1\right),\qquad \|\mathbf{v}_2\|^2=\tfrac14+\tfrac14+1=\tfrac32.$$

Step 3. $\langle \mathbf{a}_3,\mathbf{v}_1\rangle=1$, $\langle \mathbf{a}_3,\mathbf{v}_2\rangle=-\tfrac12+1=\tfrac12$:

$$\mathbf{v}_3=(0,1,1)-\tfrac12(1,1,0)-\frac{1/2}{3/2}\left(\tfrac12,-\tfrac12,1\right)=\left(-\tfrac12-\tfrac16,\ \tfrac12+\tfrac16,\ 1-\tfrac13\right)=\left(-\tfrac23,\tfrac23,\tfrac23\right),$$

$\|\mathbf{v}_3\|^2=\tfrac43$.

Checks. $\mathbf{v}_1\cdot\mathbf{v}_2=\tfrac12-\tfrac12+0=0$; $\mathbf{v}_1\cdot\mathbf{v}_3=-\tfrac23+\tfrac23=0$; $\mathbf{v}_2\cdot\mathbf{v}_3=-\tfrac13-\tfrac13+\tfrac23=0$ ✓.

Normalise.

$$\mathbf{q}_1=\tfrac{1}{\sqrt2}(1,1,0),\qquad \mathbf{q}_2=\tfrac{1}{\sqrt6}(1,-1,2),\qquad \mathbf{q}_3=\tfrac{1}{\sqrt3}(-1,1,1).$$

(Scaling $\mathbf{v}_2$ by 2 and $\mathbf{v}_3$ by $\tfrac32$ before normalising avoids fractions — any positive rescaling of a $\mathbf{v}_j$ is harmless.)

QR factorisation

Gram–Schmidt can be recorded as a matrix factorisation. Since each $\mathbf{a}_j$ is a combination of $\mathbf{q}_1,\dots,\mathbf{q}_j$ only,

$$\mathbf{a}_j=r_{1j}\mathbf{q}_1+r_{2j}\mathbf{q}_2+\cdots+r_{jj}\mathbf{q}_j,\qquad r_{ij}=\langle \mathbf{a}_j,\mathbf{q}_i\rangle\ (i<j),\quad r_{jj}=\|\mathbf{v}_j\|>0.$$

QR factorisation

Every $\mathbf{A}\in\mathbb{R}^{m\times n}$ with independent columns can be written

$$\mathbf{A}=\mathbf{Q}\mathbf{R},$$

where $\mathbf{Q}\in\mathbb{R}^{m\times n}$ has orthonormal columns ($\mathbf{Q}^{\mathsf T}\mathbf{Q}=\mathbf{I}_n$) spanning $\operatorname{col}(\mathbf{A})$, and $\mathbf{R}\in\mathbb{R}^{n\times n}$ is upper triangular with positive diagonal. With these sign conventions the factorisation is unique.

Since $\mathbf{Q}^{\mathsf T}\mathbf{Q}=\mathbf{I}$, the triangular factor is obtained as $\mathbf{R}=\mathbf{Q}^{\mathsf T}\mathbf{A}$, and the projection onto the column space is $\mathbf{P}=\mathbf{Q}\mathbf{Q}^{\mathsf T}$.

Worked Example 9.33 — QR of a 3×2 matrix

Factor $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$ (the matrix of Worked Example 9.29).

$\mathbf{a}_1 = (1,1,1)$: $r_{11} = \sqrt{3}$, $\mathbf{q}_1 = \frac{1}{\sqrt{3}}(1,1,1)$.

$\mathbf{a}_2 = (0,1,2)$: $r_{12} = \langle \mathbf{a}_2, \mathbf{q}_1 \rangle = \frac{3}{\sqrt{3}} = \sqrt{3}$. Then $\mathbf{v}_2 = \mathbf{a}_2 - r_{12}\mathbf{q}_1 = (0,1,2) - (1,1,1) = (-1,0,1)$, $r_{22} = \sqrt{2}$, $\mathbf{q}_2 = \frac{1}{\sqrt{2}}(-1,0,1)$.

$$\mathbf{A} = \underbrace{\begin{bmatrix} 1/\sqrt{3} & -1/\sqrt{2} \\ 1/\sqrt{3} & 0 \\ 1/\sqrt{3} & 1/\sqrt{2} \end{bmatrix}}_{\mathbf{Q}} \underbrace{\begin{bmatrix} \sqrt{3} & \sqrt{3} \\ 0 & \sqrt{2} \end{bmatrix}}_{\mathbf{R}}.$$

Check the second column: $\sqrt{3}\,\mathbf{q}_1 + \sqrt{2}\,\mathbf{q}_2 = (1,1,1) + (-1,0,1) = (0,1,2)$ ✓. Notice that $\mathbf{v}_2 = (-1,0,1)$ is the second column "centred" by removing its mean — Gram–Schmidt against the all-ones column is mean removal.

Orthogonal polynomials: Legendre

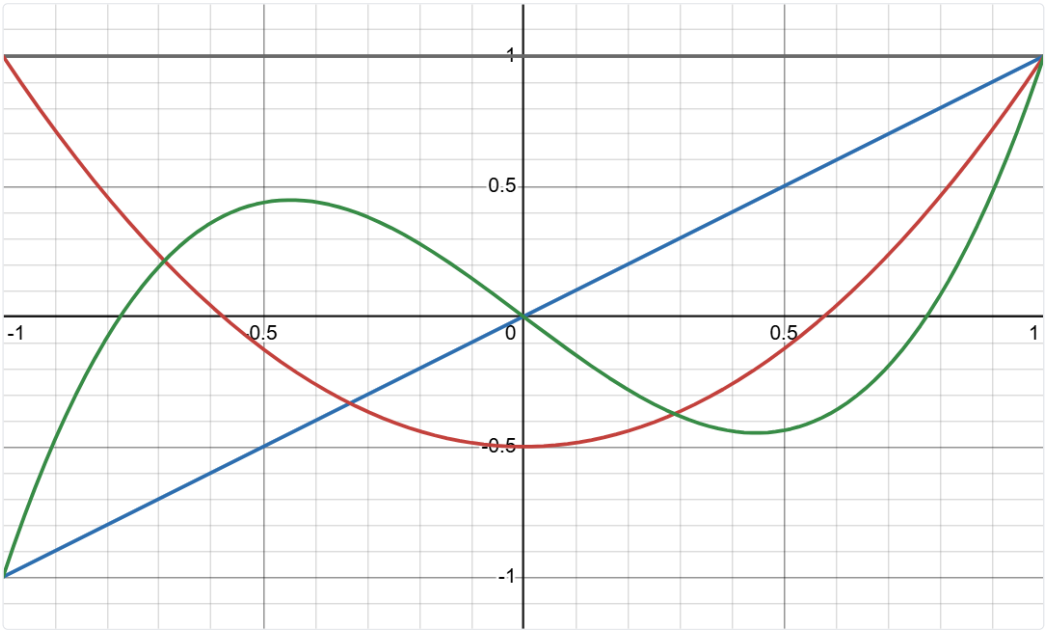
Apply Gram–Schmidt to $1, t, t^2, t^3$ in $C[-1,1]$ with $\langle f, g \rangle = \int_{-1}^{1} f g \, dt$. Useful integrals: $\int_{-1}^{1} t^k \, dt = \frac{2}{k+1}$ for even k, 0 for odd k.

- $v_0 = 1$, $\|v_0\|^2 = 2$.
- $v_1 = t - \frac{\langle t, 1 \rangle}{2} \cdot 1 = t$ (since $\langle t, 1 \rangle = 0$), $\|v_1\|^2 = \frac{2}{3}$.
- $v_2 = t^2 - \frac{\langle t^2, 1 \rangle}{2} \cdot 1 - \frac{\langle t^2, t \rangle}{2/3} t = t^2 - \frac{2/3}{2} = t^2 - \frac{1}{3}$.
- $v_3 = t^3 - \frac{\langle t^3, 1 \rangle}{2} - \frac{\langle t^3, t \rangle}{2/3} t - \frac{\langle t^3, v_2 \rangle}{\|v_2\|^2} v_2 = t^3 - \frac{2/5}{2/3} t = t^3 - \frac{3}{5} t$ (the first and last inner products vanish by odd symmetry).

Rescaled so that $P_n(1) = 1$, these are the **Legendre polynomials**:

$$P_0 = 1, \qquad P_1 = t, \qquad P_2 = \tfrac{1}{2}(3t^2 - 1), \qquad P_3 = \tfrac{1}{2}(5t^3 - 3t), \qquad \int_{-1}^{1} P_m P_n \, dt = \frac{2}{2n+1} \delta_{mn}.$$

They satisfy the three-term recurrence $(n+1)P_{n+1} = (2n+1)tP_n - nP_{n-1}$, which generates them far more cheaply than Gram–Schmidt. Changing the interval or inserting a weight function $w(t)$ in the inner product produces the other classical families: Chebyshev ($w = 1/\sqrt{1-t^2}$ — used in Chebyshev filters and minimax approximation), Laguerre ($w = e^{-t}$ on $[0, \infty)$), Hermite ($w = e^{-t^2}$ on $\mathbb{R}$).



Legendre polynomials P_0 (grey), P_1 (blue), P_2 (red) and P_3 (green) on $[-1,1]$: each has exactly n zeros in the interval and they are mutually orthogonal under $\int_{-1}^{1} f g \, dt$.

Worked Example 9.34 — Checking Legendre orthogonality and norms

Verify $\langle P_0, P_2\rangle = 0$, $\langle P_1, P_3\rangle = 0$ and $\|P_2\|^2 = \tfrac{2}{5}$.

$$\langle P_0, P_2\rangle = \tfrac{1}{2}\int_{-1}^{1} (3t^2 - 1)\,dt = \tfrac{1}{2}(2 - 2) = 0 \checkmark$$

$$\langle P_1, P_3\rangle = \tfrac{1}{2}\int_{-1}^{1} (5t^4 - 3t^2)\,dt = \tfrac{1}{2}\left(5\cdot\tfrac{2}{5} - 3\cdot\tfrac{2}{3}\right) = \tfrac{1}{2}(2 - 2) = 0 \checkmark$$

$$\|P_2\|^2 = \tfrac{1}{4}\int_{-1}^{1} (9t^4 - 6t^2 + 1)\,dt = \tfrac{1}{4}\left(\tfrac{18}{5} - 4 + 2\right) = \tfrac{1}{4}\cdot\tfrac{8}{5} = \tfrac{2}{5} = \tfrac{2}{2\cdot 2+1} \checkmark$$

(Pairs of opposite parity, such as P_0, P_1 or P_2, P_3, are orthogonal automatically by symmetry.)

Numerical behaviour: classical versus modified Gram–Schmidt

In floating-point arithmetic the formula above (**classical** Gram–Schmidt, CGS) computes all the coefficients $\langle \mathbf{a}_j, \mathbf{v}_i\rangle$ from the *original* vector $\mathbf{a}_j$. When the columns are nearly dependent, rounding errors make the computed $\mathbf{q}_j$ lose orthogonality badly. **Modified** Gram–Schmidt (MGS) subtracts the projections one at a time, using the partially updated vector:

$$\mathbf{w} \leftarrow \mathbf{a}_j; \quad \text{for } i = 1, \dots, j-1: \;\; r_{ij} = \langle \mathbf{w}, \mathbf{q}_i\rangle, \;\; \mathbf{w} \leftarrow \mathbf{w} - r_{ij}\mathbf{q}_i; \quad r_{jj} = \|\mathbf{w}\|, \;\; \mathbf{q}_j = \mathbf{w}/r_{jj}.$$

In exact arithmetic CGS and MGS are identical; in floating point MGS loses orthogonality roughly in proportion to $\kappa(\mathbf{A})\,\varepsilon_{\text{mach}}$ rather than $\kappa(\mathbf{A})^2\varepsilon_{\text{mach}}$. Library QR routines use Householder reflections, which are more stable still (Ch. 22).

Common Pitfall — projecting onto the original vectors

A frequent hand-calculation error is to subtract projections onto the *original* $\mathbf{a}_i$ instead of onto the already-orthogonalised $\mathbf{v}_i$. The $\mathbf{a}_i$ are not orthogonal, so the sum-of-projections formula does not apply to them and the result is not orthogonal. Always project onto the $\mathbf{v}_i$ (or $\mathbf{q}_i$), and check each new vector against all previous ones.

Engineering Note — orthogonalising measured signals

Gram–Schmidt is used directly in adaptive signal processing: a **Gram–Schmidt (lattice) canceller** in a radar or antenna array orthogonalises the auxiliary channels one stage at a time, so that interference can be removed from the main channel stage by stage. The same idea appears as "prewhitening" and "decorrelation" in multichannel receivers, and in the QR-based recursive least-squares (QR-RLS) adaptive filter.

9.11 Least-squares approximation

Motivation

Measurements are noisy and models are idealised, so an engineer usually has **more equations than unknowns** and no exact solution. Calibrating a sensor from twenty readings with a two-parameter straight-line model, estimating a resistor's value and an amplifier's offset from many voltage–current pairs, identifying the time constant of an RC network from a sampled decay, fitting a polynomial to a thermistor curve — each is an overdetermined linear system $\mathbf{A}\mathbf{x} = \mathbf{b}$ with $\mathbf{b} \notin \operatorname{col}(\mathbf{A})$. The **least-squares** solution chooses $\mathbf{x}$ to make the residual as small as possible in the Euclidean norm. Geometrically it is a projection (Section 9.9); computationally it is solved by the normal equations or, better, by QR (Section 9.10).

The least-squares problem

Least-squares solution

For $\mathbf{A} \in \mathbb{R}^{m\times n}$ and $\mathbf{b} \in \mathbb{R}^m$, a **least-squares solution** of $\mathbf{A}\mathbf{x} = \mathbf{b}$ is any $\hat{\mathbf{x}}$ minimising

$$J(\mathbf{x}) = \|\mathbf{b} - \mathbf{A}\mathbf{x}\|^2 = \sum_{i=1}^{m} \big(b_i - (\mathbf{A}\mathbf{x})_i\big)^2.$$

The vector $\mathbf{r} = \mathbf{b} - \mathbf{A}\hat{\mathbf{x}}$ is the **residual**, and $\|\mathbf{r}\|^2$ is the **sum of squared errors** (SSE).

Normal equations

$\hat{\mathbf{x}}$ is a least-squares solution of $\mathbf{A}\mathbf{x} = \mathbf{b}$ if and only if

$$\mathbf{A}^{\mathsf{T}}\mathbf{A}\,\hat{\mathbf{x}} = \mathbf{A}^{\mathsf{T}}\mathbf{b}.$$

The normal equations are always consistent. The least-squares solution is unique iff the columns of $\mathbf{A}$ are independent ($\operatorname{rank}\mathbf{A} = n$), and then

$$\hat{\mathbf{x}} = (\mathbf{A}^{\mathsf{T}}\mathbf{A})^{-1}\mathbf{A}^{\mathsf{T}}\mathbf{b}, \qquad \mathbf{A}\hat{\mathbf{x}} = \operatorname{proj}_{\operatorname{col}(\mathbf{A})}\mathbf{b}.$$

Geometric derivation. As $\mathbf{x}$ ranges over $\mathbb{R}^n$, $\mathbf{A}\mathbf{x}$ ranges over $\operatorname{col}(\mathbf{A})$. The closest point of that subspace to $\mathbf{b}$ is the orthogonal projection $\mathbf{p}$, characterised by $\mathbf{b} - \mathbf{p} \perp \operatorname{col}(\mathbf{A})$, i.e. $\mathbf{b} - \mathbf{A}\hat{\mathbf{x}} \in \operatorname{col}(\mathbf{A})^{\perp} = \operatorname{null}(\mathbf{A}^{\mathsf{T}})$ (Section 9.7). That is exactly $\mathbf{A}^{\mathsf{T}}(\mathbf{b} - \mathbf{A}\hat{\mathbf{x}}) = \mathbf{0}$. Since $\mathbf{p}$ always exists, the normal equations are always consistent; if $\operatorname{null}(\mathbf{A}) \neq \{\mathbf{0}\}$, any null-space vector can be added to $\hat{\mathbf{x}}$ without changing $\mathbf{A}\hat{\mathbf{x}}$.

Calculus derivation. Expand

$$J(\mathbf{x}) = \mathbf{b}^{\mathsf{T}}\mathbf{b} - 2\mathbf{x}^{\mathsf{T}}\mathbf{A}^{\mathsf{T}}\mathbf{b} + \mathbf{x}^{\mathsf{T}}\mathbf{A}^{\mathsf{T}}\mathbf{A}\mathbf{x}.$$

Setting the gradient (Ch. 6) to zero gives $\nabla J = -2\mathbf{A}^{\mathsf{T}}\mathbf{b} + 2\mathbf{A}^{\mathsf{T}}\mathbf{A}\mathbf{x} = \mathbf{0}$, the normal equations. The Hessian $2\mathbf{A}^{\mathsf{T}}\mathbf{A}$ is positive semidefinite ($\mathbf{x}^{\mathsf{T}}\mathbf{A}^{\mathsf{T}}\mathbf{A}\mathbf{x} = \|\mathbf{A}\mathbf{x}\|^2 \geq 0$), so the stationary point is a minimum — strictly so when the columns are independent. The two derivations agree, as they must: minimising a squared distance *is* projecting.

The minimum value follows from Pythagoras, $\|\mathbf{b}\|^2 = \|\mathbf{A}\hat{\mathbf{x}}\|^2 + \|\mathbf{r}\|^2$, and can be computed as $\|\mathbf{r}\|^2 = \mathbf{b}^{\mathsf{T}}\mathbf{b} - \hat{\mathbf{x}}^{\mathsf{T}}\mathbf{A}^{\mathsf{T}}\mathbf{b}$.

Fitting a straight line

Given data (t_i, y_i), $i = 1, \dots, m$, the model $y \approx c_0 + c_1 t$ produces one equation per data point:

$$\underbrace{\begin{bmatrix} 1 & t_1 \\ 1 & t_2 \\ \vdots & \vdots \\ 1 & t_m \end{bmatrix}}_{\mathbf{A}} \begin{bmatrix} c_0 \\ c_1 \end{bmatrix} \approx \underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}}_{\mathbf{b}}, \qquad \mathbf{A}^{\mathsf{T}}\mathbf{A} = \begin{bmatrix} m & \sum t_i \\ \sum t_i & \sum t_i^2 \end{bmatrix}, \quad \mathbf{A}^{\mathsf{T}}\mathbf{b} = \begin{bmatrix} \sum y_i \\ \sum t_i y_i \end{bmatrix}.$$

Solving the 2×2 system gives the familiar regression formulas

$$c_1 = \frac{\sum (t_i - \bar{t})(y_i - \bar{y})}{\sum (t_i - \bar{t})^2}, \qquad c_0 = \bar{y} - c_1 \bar{t},$$

where $\bar{t}, \bar{y}$ are the sample means. (They follow most quickly from the QR remark of Worked Example 9.33: orthogonalising the t column against the ones column replaces t_i by $t_i - \bar{t}$, which decouples the two unknowns.) In particular **the least-squares line passes through the centroid** $(\bar{t}, \bar{y})$, and the residuals sum to zero (the residual is orthogonal to the column of ones).

Worked Example 9.35 — The smallest line-fitting problem

Fit $y = c_0 + c_1 t$ to the points $(0, 6)$, $(1, 0)$, $(2, 0)$.

This is exactly Worked Example 9.29 with $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$ and $\mathbf{b} = (6, 0, 0)^{\mathsf{T}}$. The normal equations gave $c_0 = 5$, $c_1 = -3$:

$$\hat{y} = 5 - 3t, \qquad \text{fitted values } (5, 2, -1), \qquad \text{residuals } (1, -2, 1), \qquad \text{SSE} = 6.$$

Checks. Residuals sum to zero ✓. The line passes through the centroid $(\bar{t}, \bar{y}) = (1, 2)$: $5 - 3 = 2$ ✓. The regression formula: $\bar{t} = 1$, $\sum (t_i - \bar{t})(y_i - \bar{y}) = (-1)(4) + 0 + (1)(-2) = -6$, $\sum (t_i - \bar{t})^2 = 2$, so $c_1 = -3$ ✓.

Via QR (Worked Example 9.33): $\mathbf{Q}^{\mathsf{T}}\mathbf{b} = \left(\frac{6}{\sqrt{3}}, -\frac{6}{\sqrt{2}}\right) = (2\sqrt{3}, -3\sqrt{2})$. Back-substitute in $\mathbf{R}\hat{\mathbf{x}} = \mathbf{Q}^{\mathsf{T}}\mathbf{b}$: $\sqrt{2}\, c_1 = -3\sqrt{2} \Rightarrow c_1 = -3$; $\sqrt{3}\, c_0 + \sqrt{3}(-3) = 2\sqrt{3} \Rightarrow c_0 = 5$ ✓.

Worked Example 9.36 — Resistance and offset from V–I measurements

A resistor is measured with a voltmeter that may have an offset. The data are

I (mA)	0	1	2	3	4
V (V)	0.1	1.9	4.2	5.8	8.1

Fit $V = V_{\text{off}} + RI$.

Sums. $m = 5$, $\sum I = 10$, $\sum I^2 = 30$, $\sum V = 20.1$, $\sum IV = 0 + 1.9 + 8.4 + 17.4 + 32.4 = 60.1$.

Normal equations.

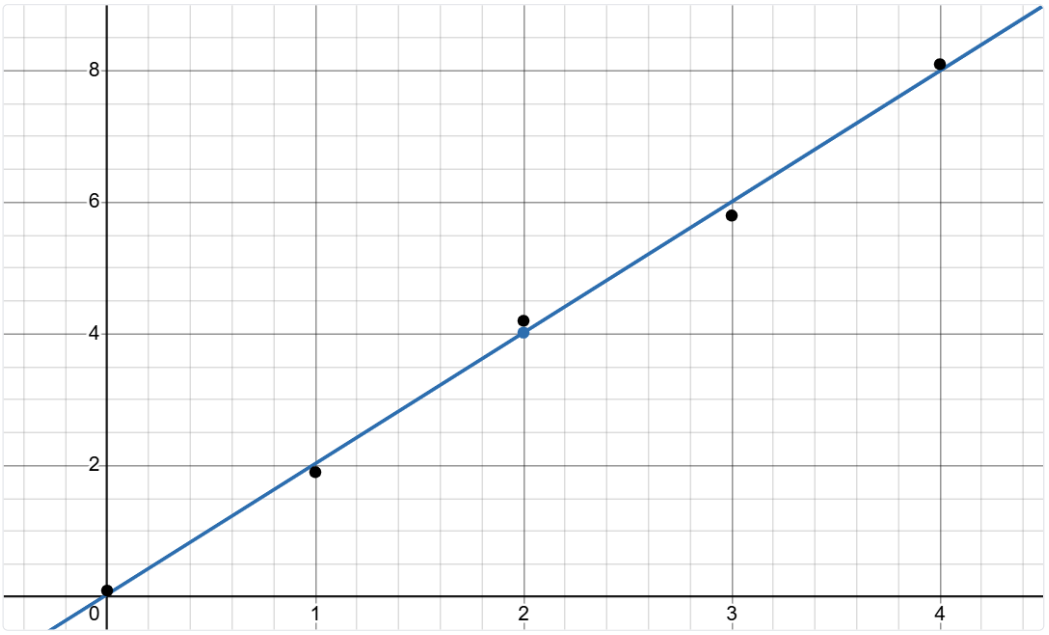
$$\begin{bmatrix} 5 & 10 \\ 10 & 30 \end{bmatrix} \begin{bmatrix} V_{\text{off}} \\ R \end{bmatrix} = \begin{bmatrix} 20.1 \\ 60.1 \end{bmatrix}.$$

Determinant $5 \cdot 30 - 10 \cdot 10 = 50$. By Cramer's rule,

$$V_{\text{off}} = \frac{30(20.1) - 10(60.1)}{50} = \frac{603 - 601}{50} = 0.04 \text{ V}, \qquad R = \frac{5(60.1) - 10(20.1)}{50} = \frac{300.5 - 201}{50} = 1.99 \text{ V/mA} = 1.99 \text{ k}\Omega.$$

Residuals. Fitted values $0.04, 2.03, 4.02, 6.01, 8.00$; residuals $0.06, -0.13, 0.18, -0.21, 0.10$ (sum 0 ✓); SSE $= 0.107$ V^2.

Quality of fit. The total variation about the mean is $\sum (V_i - \bar{V})^2 = \sum V_i^2 - m\bar{V}^2 = 120.51 - 5(4.02)^2 = 39.71$, so the coefficient of determination is $R^2 = 1 - 0.107/39.71 = 0.9973$. **Units check:** V_{off} in volts, R in V/mA — each column of $\mathbf{A}$ carries its own units, and each unknown takes the units that make $\mathbf{A}\mathbf{x}$ volts.



Least-squares line $V = 0.04 + 1.99I$ through the five measured points; it passes through the centroid $(\bar{I}, \bar{V}) = (2, 4.02)$, and the vertical residuals are what is minimised.

Polynomial and general linear-in-the-parameters models

The same machinery fits *any* model that is **linear in its unknown parameters**, however nonlinear it is in t:

$$y(t) \approx c_1\phi_1(t) + c_2\phi_2(t) + \cdots + c_n\phi_n(t), \qquad \mathbf{A} = \big[\phi_j(t_i)\big]_{m\times n}.$$

Examples: polynomials ($\phi_j = t^{j-1}$, giving a **Vandermonde** matrix), sinusoids of known frequency ($\phi = \cos\omega t, \sin\omega t$, which estimates amplitude and phase), exponentials with known rates, splines. The column j of $\mathbf{A}$ is simply the jth basis function sampled at the measurement times.

Worked Example 9.37 — A quadratic fit

Fit $y = c_0 + c_1 t + c_2 t^2$ to

t	-2	-1	0	1	2
y	8.2	2.4	0.9	3.6	10.1

Matrix. Rows $(1, t_i, t_i^2)$. Because the t_i are symmetric about 0, all odd power sums vanish: $\sum t_i = \sum t_i^3 = 0$, $\sum t_i^2 = 10$, $\sum t_i^4 = 34$.

$$\mathbf{A}^{\mathsf T}\mathbf{A} = \begin{bmatrix} 5 & 0 & 10 \\ 0 & 10 & 0 \\ 10 & 0 & 34 \end{bmatrix}, \qquad \mathbf{A}^{\mathsf T}\mathbf{b} = \begin{bmatrix} \sum y_i \\ \sum t_i y_i \\ \sum t_i^2 y_i \end{bmatrix} = \begin{bmatrix} 25.2 \\ 5.0 \\ 79.2 \end{bmatrix}.$$

($\sum t_i y_i = -16.4 - 2.4 + 0 + 3.6 + 20.2 = 5.0$; $\sum t_i^2 y_i = 32.8 + 2.4 + 0 + 3.6 + 40.4 = 79.2$.)

Solve. The middle equation decouples: $10c_1 = 5 \Rightarrow c_1 = 0.5$. From the first, $c_0 = 5.04 - 2c_2$; substitute in the third: $10(5.04 - 2c_2) + 34c_2 = 79.2 \Rightarrow 14c_2 = 28.8 \Rightarrow c_2 = 2.0571$, and $c_0 = 0.9257$.

$$\hat{y} = 0.9257 + 0.5t + 2.0571t^2.$$

Residuals $(0.046, -0.083, -0.026, 0.117, -0.054)$, SSE $= 0.0263$. Symmetric sample points made t orthogonal to both 1 and t^2 — a small taste of the simplification orthogonal bases bring.

Worked Example 9.38 — Time constant of an RC discharge (linearised fit)

A capacitor discharging through a resistor is sampled:

t (ms)	0	1	2	3	4
v (V)	5.00	3.02	1.85	1.10	0.68

Estimate V_0 and τ in $v = V_0 e^{-t/\tau}$.

Linearise. The model is not linear in τ, but $\ln v = \ln V_0 - t/\tau$ is linear in the parameters $a = \ln V_0$ and $b = -1/\tau$. Take logs: $\ln v = (1.6094,\ 1.1053,\ 0.6152,\ 0.0953,\ -0.3857)$.

Normal equations (same $\mathbf{A}^{\mathsf T}\mathbf{A}$ as Worked Example 9.36 since the t_i are again $0,\dots,4$):

$$\begin{bmatrix} 5 & 10 \\ 10 & 30 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} \sum \ln v_i \\ \sum t_i \ln v_i \end{bmatrix} = \begin{bmatrix} 3.0395 \\ 1.0789 \end{bmatrix}.$$

$$a = \frac{30(3.0395) - 10(1.0789)}{50} = 1.6079, \qquad b = \frac{5(1.0789) - 10(3.0395)}{50} = -0.5000 \ \text{ms}^{-1}.$$

Result. $V_0 = e^{1.6079} = 4.99$ V and $\tau = -1/b = 2.00$ ms. Fitted values $4.99, 3.03, 1.84, 1.11, 0.68$ V agree with the data to within 0.02 V.

Caveat. Minimising squared errors in $\ln v$ is *not* the same as minimising squared errors in v: the log transform gives small voltages (late samples, where relative noise is largest) as much weight as large ones. When that matters, use weighted least squares (below) with weights $w_i \propto v_i^2$, or a genuinely nonlinear least-squares fit (Ch. 23).

Weighted least squares

If some measurements are more reliable than others — say measurement i has noise standard deviation σ_i — minimise the weighted sum

$$J_{\mathbf{W}}(\mathbf{x}) = \sum_{i} w_i \big(b_i - (\mathbf{A}\mathbf{x})_i\big)^2 = (\mathbf{b} - \mathbf{A}\mathbf{x})^{\mathsf T}\mathbf{W}(\mathbf{b} - \mathbf{A}\mathbf{x}), \qquad \mathbf{W} = \operatorname{diag}(w_i),\ w_i = 1/\sigma_i^2.$$

This is ordinary least squares in the weighted inner product $\langle \mathbf{u}, \mathbf{v}\rangle_{\mathbf{W}} = \mathbf{v}^{\mathsf T}\mathbf{W}\mathbf{u}$ of Section 9.8, so the orthogonality principle gives the **weighted normal equations**

$$\mathbf{A}^{\mathsf T}\mathbf{W}\mathbf{A}\,\hat{\mathbf{x}} = \mathbf{A}^{\mathsf T}\mathbf{W}\mathbf{b}.$$

Equivalently, scale row i of $\mathbf{A}$ and $\mathbf{b}$ by $\sqrt{w_i}$ and solve an ordinary problem. The statistical justification (best linear unbiased estimation) and confidence intervals for the fitted parameters belong to Ch. 20.

Worked Example 9.39 — Averaging measurements of unequal quality

A voltage is measured three times: 5.02 V and 4.98 V with a meter of $\sigma = 0.02$ V, and 5.20 V with a meter of $\sigma = 0.10$ V. Find the weighted least-squares estimate.

The model is $v \approx x$ for each reading: $\mathbf{A} = (1,1,1)^{\mathsf T}$, $\mathbf{b} = (5.02, 4.98, 5.20)^{\mathsf T}$, weights $w = (2500, 2500, 100)$ V^{-2}.

$$\hat{x} = \frac{\mathbf{A}^{\mathsf T}\mathbf{W}\mathbf{b}}{\mathbf{A}^{\mathsf T}\mathbf{W}\mathbf{A}} = \frac{2500(5.02) + 2500(4.98) + 100(5.20)}{2500 + 2500 + 100} = \frac{25\,520}{5100} = 5.0039 \ \text{V}.$$

The unweighted mean would be 5.0667 V, pulled up by the poor reading. Weighted least squares with a single constant column is simply the **inverse-variance-weighted mean**.

Numerical aspects: normal equations versus QR

Forming $\mathbf{A}^{\mathsf T}\mathbf{A}$ squares the condition number:

$$\kappa(\mathbf{A}^{\mathsf T}\mathbf{A}) = \kappa(\mathbf{A})^2.$$

If $\kappa(\mathbf{A}) = 10^6$, the normal equations have $\kappa = 10^{12}$ and, in double precision, only about four correct digits can be expected. The QR route avoids the squaring. With $\mathbf{A} = \mathbf{Q}\mathbf{R}$,

$$\mathbf{A}^{\mathsf T}\mathbf{A}\hat{\mathbf{x}} = \mathbf{A}^{\mathsf T}\mathbf{b} \implies \mathbf{R}^{\mathsf T}\mathbf{Q}^{\mathsf T}\mathbf{Q}\mathbf{R}\hat{\mathbf{x}} = \mathbf{R}^{\mathsf T}\mathbf{Q}^{\mathsf T}\mathbf{b} \implies \mathbf{R}\hat{\mathbf{x}} = \mathbf{Q}^{\mathsf T}\mathbf{b},$$

a triangular system solved by back substitution (Worked Example 9.35). This is what `numpy.linalg.lstsq`, MATLAB's backslash and similar routines do (some use the SVD of Ch. 10, which also handles rank deficiency gracefully).

The classic ill-conditioned case is polynomial fitting with the monomial basis. In the continuous version — approximating f on $[0,1]$ by $\sum_{j=0}^{n-1} c_j t^j$ — the normal-equation matrix has entries $\langle t^i, t^j\rangle = \int_0^1 t^{i+j}\,dt = \frac{1}{i+j+1}$: the **Hilbert matrix**. Its condition number explodes:

n	3	6	10
$\kappa(\mathbf{H}_n)$	5.2×10^2	1.5×10^7	1.6×10^{13}

The cause is the near-parallelism of t^i and t^j seen in Worked Example 9.25(b). The cure is to use an orthogonal basis, which makes the normal-equation matrix diagonal.

Continuous least squares and orthogonal polynomials

To approximate a function f on $[a,b]$ by an element of a finite-dimensional subspace W in the sense of least integrated squared error, project in the function-space inner product (Section 9.9). With an orthogonal basis $\{\phi_k\}$ of W there is no system to solve at all:

$$\hat f=\sum_k c_k\phi_k,\qquad c_k=\frac{\langle f,\phi_k\rangle}{\langle \phi_k,\phi_k\rangle},\qquad \|f-\hat f\|^2=\|f\|^2-\sum_k |c_k|^2\|\phi_k\|^2.$$

Worked Example 9.40 — Best quadratic approximation of e^t on $[-1,1]$

Find the quadratic $q(t)$ minimising $\int_{-1}^{1}(e^t-q(t))^2\,dt$ and compare it with the Taylor polynomial $1+t+t^2/2$.

Use Legendre polynomials (Section 9.10), $\|P_k\|^2=\frac{2}{2k+1}$, so $c_k=\frac{2k+1}{2}\int_{-1}^{1}e^tP_k\,dt$. The integrals ($\int te^t dt=(t-1)e^t$, $\int t^2e^t dt=(t^2-2t+2)e^t$):

$$\int_{-1}^{1}e^t\,dt=e-e^{-1},\qquad \int_{-1}^{1}te^t\,dt=\frac{2}{e},\qquad \int_{-1}^{1}t^2e^t\,dt=e-\frac{5}{e}.$$

Hence

$$c_0=\tfrac12(e-e^{-1})=\sinh 1=1.1752,\qquad c_1=\tfrac32\cdot\tfrac{2}{e}=\tfrac{3}{e}=1.1036,$$

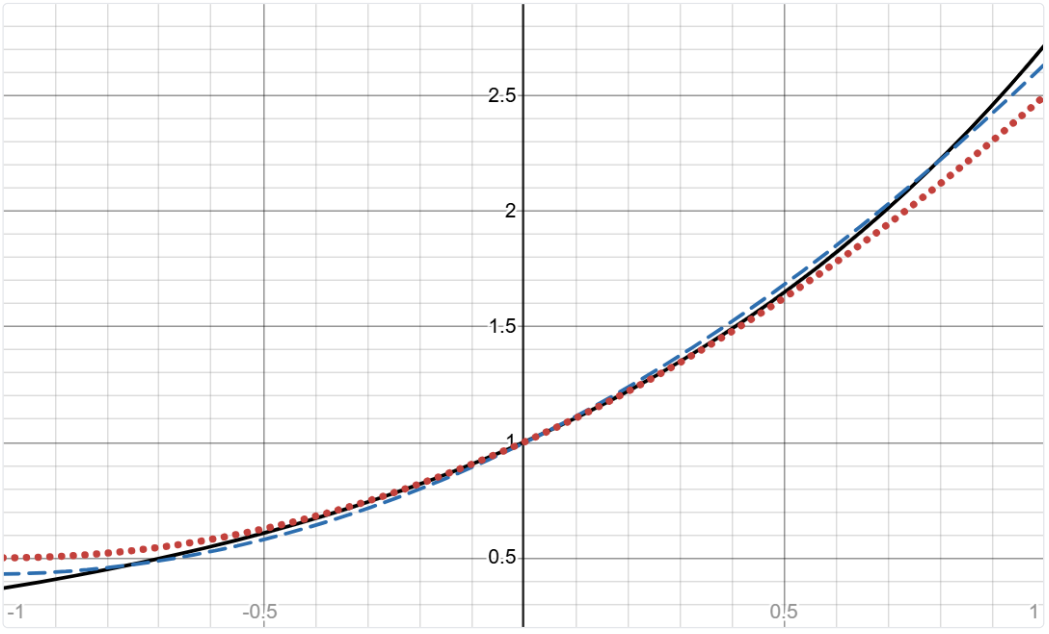
$$c_2=\tfrac52\int_{-1}^{1}\tfrac{3t^2-1}{2}e^t\,dt=\tfrac52\Big[\tfrac32\big(e-\tfrac5e\big)-\tfrac12\big(e-\tfrac1e\big)\Big]=\tfrac52\Big(e-\tfrac{7}{e}\Big)=0.3578.$$

Assemble. $q=c_0+c_1t+c_2\cdot\frac{3t^2-1}{2}$:

$$q(t)=\big(1.1752-0.1789\big)+1.1036\,t+0.5367\,t^2=0.9963+1.1036\,t+0.5367\,t^2.$$

Error. Numerically, $\int_{-1}^{1}(e^t-q)^2dt=1.44\times10^{-3}$ (RMS error 0.027 over the interval), whereas the Taylor polynomial gives 8.99×10^{-3} — more than six times larger. Taylor is perfect at $t=0$ and poor at the ends; least squares spreads the error over the whole interval (errors -0.061, 0.004, 0.082 at $t=-1,0,1$).

Monomial check. Solving the 3×3 normal equations in the basis $1,t,t^2$ (Gram matrix entries $\int_{-1}^{1}t^{i+j}dt$) gives the same q, as it must — the projection does not depend on the basis — but requires solving a coupled system.



e^t (black), its least-squares quadratic on $[-1,1]$ (blue dashed, nearly indistinguishable) and the Taylor quadratic about 0 (red dotted), which drifts away towards $t=\pm1$.

Fourier series as least squares

Take $W=\operatorname{span}\{1,\cos\omega_0t,\sin\omega_0t,\dots,\cos N\omega_0t,\sin N\omega_0t\}$ on one period. This basis is orthogonal (Section 9.8), so the best approximation of a periodic f in W — the trigonometric polynomial of order N with least mean-square error — has coefficients

$$a_0=\frac{1}{T}\int_T f\,dt,\qquad a_k=\frac{2}{T}\int_T f\cos k\omega_0t\,dt,\qquad b_k=\frac{2}{T}\int_T f\sin k\omega_0t\,dt.$$

These are precisely the Fourier coefficients. In other words, **the truncated Fourier series is the least-squares approximation of f by harmonics up to order N**, the error is orthogonal to every retained harmonic, and adding more harmonics never changes the earlier coefficients (Worked Example 9.31). In discrete form, fitting N samples with the DFT basis is a least-squares problem whose matrix has orthogonal columns, which is why the DFT can be inverted by a conjugate transpose. Ch. 15 takes up convergence as $N\to\infty$, Gibbs' phenomenon and power relations; Ch. 16 extends the idea to aperiodic signals.

Common Pitfall — extrapolating and over-fitting

A least-squares polynomial is only as good as the range of the data. A high-degree polynomial fitted to m points can pass close to every one of them while oscillating wildly between and beyond them; with degree $m-1$ it interpolates exactly, the residual is zero, and noise is fitted along with signal. Choose the model order from physics or from residual behaviour on held-out data, not by driving the SSE to zero, and never trust a polynomial fit outside the measured interval.

Engineering Note — where least squares runs in hardware

Least squares is computed continuously inside modern electronics: channel estimation in a Wi-Fi or LTE receiver solves a small least-squares problem against known pilot symbols on every frame; GPS receivers solve linearised least squares for position and clock bias from four or more pseudoranges; battery-management ICs fit equivalent-circuit parameters; adaptive equalisers run recursive least squares (RLS), which updates $\hat{\mathbf{x}}$ as each new row of $\mathbf{A}$ arrives without re-solving from scratch. The mathematics is always this section's: project the measurement vector onto the column space of the model.

Summary

Concept	Key formula / statement
Vector space	closed under $\mathbf{u}+\mathbf{v}$ and $c\mathbf{u}$; eight axioms; contains $\mathbf{0}$
Subspace test	$\mathbf{0}\in W$ and $a\mathbf{u}+b\mathbf{v}\in W$ for all $\mathbf{u},\mathbf{v}\in W$
Span	$\operatorname{span}\{\mathbf{v}_i\}=\{\sum c_i\mathbf{v}_i\}$, the smallest subspace containing the $\mathbf{v}_i$
Linear independence	$\sum c_i\mathbf{v}_i=\mathbf{0}\Rightarrow$ all $c_i=0$; in $\mathbb{F}^m$: $\operatorname{rank}[\mathbf{v}_1\cdots\mathbf{v}_k]=k$
Wronskian	$W(f_1,\dots,f_k)(t_0)\neq 0\Rightarrow$ independent
Basis, coordinates	independent + spanning; $\mathbf{x}=\mathbf{P}_{\mathcal{B}}[\mathbf{x}]_{\mathcal{B}}$
Dimension	number of vectors in any basis; $\dim\mathcal{P}_n=n+1$, $\dim\mathbb{R}^{m\times n}=mn$
Linear map	$T(a\mathbf{u}+b\mathbf{v})=aT(\mathbf{u})+bT(\mathbf{v})$; matrix columns $=T(\mathbf{e}_j)$
Change of basis	$[T]_{\mathcal{B}}=\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ (similar matrices)
Rank–nullity	$\operatorname{rank}\mathbf{A}+\dim\operatorname{null}\mathbf{A}=n$
Four subspaces	$\dim\operatorname{col}=\dim\operatorname{row}=r$, $\dim\operatorname{null}=n-r$, $\dim\operatorname{null}(\mathbf{A}^{\mathsf{T}})=m-r$
Orthogonal complements	$\operatorname{null}(\mathbf{A})=\operatorname{row}(\mathbf{A})^{\perp}$, $\operatorname{null}(\mathbf{A}^{\mathsf{T}})=\operatorname{col}(\mathbf{A})^{\perp}$
Solution structure	$\mathbf{x}=\mathbf{x}_p+\mathbf{x}_n$, $\mathbf{x}_n\in\operatorname{null}(\mathbf{A})$; solvable iff $\mathbf{b}\in\operatorname{col}(\mathbf{A})$
Cauchy–Schwarz	$\lvert\langle\mathbf{u},\mathbf{v}\rangle\rvert\le\lVert\mathbf{u}\rVert\lVert\mathbf{v}\rVert$; $\cos\theta=\langle\mathbf{u},\mathbf{v}\rangle/(\lVert\mathbf{u}\rVert\lVert\mathbf{v}\rVert)$
Complex inner product	$\langle\mathbf{u},\mathbf{v}\rangle=\mathbf{v}^{\mathsf{H}}\mathbf{u}=\sum u_kv_k^*$
Signal inner products	$\langle x,y\rangle=\int x\,y^*\,dt$ (energy); $\langle x,y\rangle_T=\frac{1}{T}\int_T x\,y^*\,dt$ (power, norm $=$ RMS)
Orthogonal harmonics	$\frac{1}{T}\int_T e^{jm\omega_0 t}e^{-jn\omega_0 t}dt=\delta_{mn}$
Orthogonal coordinates	$c_j=\langle\mathbf{x},\mathbf{u}_j\rangle/\langle\mathbf{u}_j,\mathbf{u}_j\rangle$; Parseval $\lVert\mathbf{x}\rVert^2=\sum\lvert c_j\rvert^2\lVert\mathbf{u}_j\rVert^2$
Orthogonal matrix	$\mathbf{Q}^{\mathsf{T}}\mathbf{Q}=\mathbf{I}$, $\mathbf{Q}^{-1}=\mathbf{Q}^{\mathsf{T}}$, $\lVert\mathbf{Q}\mathbf{x}\rVert=\lVert\mathbf{x}\rVert$
Projection onto a line	$\operatorname{proj}_{\mathbf{a}}\mathbf{b}=\dfrac{\langle\mathbf{b},\mathbf{a}\rangle}{\langle\mathbf{a},\mathbf{a}\rangle}\mathbf{a}$
Projection onto $\operatorname{col}(\mathbf{A})$	$\mathbf{P}=\mathbf{A}(\mathbf{A}^{\mathsf{T}}\mathbf{A})^{-1}\mathbf{A}^{\mathsf{T}}=\mathbf{Q}\mathbf{Q}^{\mathsf{T}}$; $\mathbf{P}^2=\mathbf{P}=\mathbf{P}^{\mathsf{T}}$
Best approximation	$\mathbf{b}-\operatorname{proj}_W\mathbf{b}\perp W$; Bessel $\sum\lvert c_k\rvert^2\lVert\phi_k\rVert^2\le\lVert f\rVert^2$
Gram–Schmidt	$\mathbf{v}_j=\mathbf{a}_j-\sum_{i<j}\frac{\langle\mathbf{a}_j,\mathbf{v}_i\rangle}{\langle\mathbf{v}_i,\mathbf{v}_i\rangle}\mathbf{v}_i$; $\mathbf{A}=\mathbf{Q}\mathbf{R}$
Legendre polynomials	$1,\ t,\ \frac{1}{2}(3t^2-1),\ \frac{1}{2}(5t^3-3t)$; $\int_{-1}^{1}P_mP_n\,dt=\frac{2}{2n+1}\delta_{mn}$
Normal equations	$\mathbf{A}^{\mathsf{T}}\mathbf{A}\hat{\mathbf{x}}=\mathbf{A}^{\mathsf{T}}\mathbf{b}$; via QR: $\mathbf{R}\hat{\mathbf{x}}=\mathbf{Q}^{\mathsf{T}}\mathbf{b}$
Line fit	$c_1=\dfrac{\sum(t_i-\bar{t})(y_i-\bar{y})}{\sum(t_i-\bar{t})^2}$, $c_0=\bar{y}-c_1\bar{t}$
Weighted LS	$\mathbf{A}^{\mathsf{T}}\mathbf{W}\mathbf{A}\hat{\mathbf{x}}=\mathbf{A}^{\mathsf{T}}\mathbf{W}\mathbf{b}$, $w_i=1/\sigma_i^2$
Conditioning	$\kappa(\mathbf{A}^{\mathsf{T}}\mathbf{A})=\kappa(\mathbf{A})^2$ — prefer QR or SVD

Further reading

- G. Strang, *Introduction to Linear Algebra*, 6th ed., Wellesley-Cambridge Press, 2023 — the four fundamental subspaces, projections and least squares, with exceptionally clear geometry.
- D. C. Lay, S. R. Lay and J. J. McDonald, *Linear Algebra and Its Applications*, 6th ed., Pearson, 2021 — careful treatment of vector spaces, bases, coordinates and inner-product spaces.
- E. Kreyszig, *Advanced Engineering Mathematics*, 10th ed., Wiley, 2011 — linear algebra and orthogonal functions in the broader engineering-mathematics setting.
- T. K. Moon and W. C. Stirling, *Mathematical Methods and Algorithms for Signal Processing*, Prentice Hall, 2000 — vector spaces, Hilbert spaces, projection and least squares applied to signals and estimation.
- L. N. Trefethen and D. Bau, *Numerical Linear Algebra*, SIAM, 1997 — QR, Gram–Schmidt stability and least-squares conditioning.

Chapter 10 — Eigenvalues and Eigenvectors

> **Prerequisites:** Ch. 2 (complex numbers, polar form), Ch. 8 (determinants, inverses, Gaussian elimination, rank), Ch. 9 (bases, change of basis, null space, inner products, orthogonality, Gram–Schmidt) **Learning outcomes:** After this chapter you can
>
> - form and solve the characteristic equation of a square matrix, and use trace, determinant and the Cayley–Hamilton theorem as checks and shortcuts;
> - compute eigenvalues, eigenvectors and eigenspaces (real and complex), and decide whether a matrix is diagonalizable;
> - diagonalize a matrix and use the factorization to compute powers $\mathbf{A}^k$, the matrix exponential $e^{\mathbf{A}t}$ and the long-run behaviour of difference equations;
> - apply the spectral theorem to symmetric and Hermitian matrices, find principal axes of quadratic forms, and test a matrix for positive definiteness;
> - compute and interpret the singular value decomposition, including matrix norm, condition number, rank, low-rank approximation and the pseudoinverse;
> - decouple vibrating and oscillating systems into normal modes, and judge the stability of linear (and linearized) continuous- and discrete-time systems from their eigenvalues.

Chapters 8 and 9 asked what a matrix *does* to a vector: it maps $\mathbf{x}$ to $\mathbf{A}\mathbf{x}$, generally rotating it, stretching it and mixing its components. This chapter asks a sharper question: **are there special directions that the matrix does not rotate at all?** Along such a direction the matrix acts like a plain number, $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$. The number λ is an *eigenvalue* (German *eigen*, "own" or "characteristic") and $\mathbf{v}$ an *eigenvector*.

The idea sounds modest, but it is one of the most productive in applied mathematics. If a matrix has enough eigenvectors to form a basis, then in that basis the matrix is *diagonal*: a coupled problem in n unknowns falls apart into n independent scalar problems. A system of coupled differential equations $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ becomes n separate first-order equations $\dot{y}_i = \lambda_i y_i$, each solved by an exponential $e^{\lambda_i t}$. Coupled masses or coupled resonant circuits separate into *normal modes*, each vibrating at its own frequency. Repeated multiplication $\mathbf{A}^k$ — the heart of every discrete-time recursion, Markov chain and iterative algorithm — becomes repeated multiplication of numbers, λ_i^k. And the question "is this system stable?" reduces to "do all the eigenvalues lie in the left half-plane (or inside the unit circle)?"

For an electrical engineer, eigenvalues are hiding in plain sight. The **natural frequencies** of a circuit, the **poles** of its transfer function, the **resonant modes** of a cavity or a coupled-line filter, the **principal axes** of a polarisation ellipse or of an anisotropic dielectric tensor, the **principal components** of a set of sensor measurements, the **singular values** that measure how close a MIMO channel is to losing a spatial stream — all are eigenvalue problems. Chapters 11–14 use the results of this chapter constantly.

The plan is as follows. Sections 10.1–10.4 develop the general theory: the characteristic equation, eigenvectors, diagonalization and matrix powers (including the matrix exponential). Sections 10.5–10.7 specialise to the symmetric and Hermitian matrices that arise from energy, power and covariance, culminating in the singular value decomposition, which works for *every* matrix. Sections 10.8–10.9 turn the theory onto dynamics: normal-mode analysis and stability.

10.1 Characteristic equations

Motivation

Suppose we want a nonzero vector $\mathbf{v}$ and a scalar λ with $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$. Unlike $\mathbf{A}\mathbf{x} = \mathbf{b}$, this is not a linear system for $\mathbf{v}$ in the usual sense, because λ is unknown too and multiplies $\mathbf{v}$. The trick is to separate the two unknowns. We first ask *for which* λ the equation has a nonzero solution at all; that condition turns out to be a single polynomial equation in λ — the **characteristic equation**. Only after the eigenvalues are known do we go back and find the eigenvectors, which is then an ordinary homogeneous linear system.

The characteristic equation is not new to you. When you solved $a\ddot{y} + b\dot{y} + cy = 0$ by trying $y = e^{st}$ and obtained $as^2 + bs + c = 0$, that quadratic was the characteristic equation of a 2×2 matrix in disguise (Example 10.3 below makes the connection exact). The roots of the denominator of a transfer function — the poles — are likewise eigenvalues of the system's state matrix (Section 10.9).

Derivation

Rewrite $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ as

$$\mathbf{A}\mathbf{v} - \lambda\mathbf{I}\mathbf{v} = \mathbf{0} \quad \Longleftrightarrow \quad (\mathbf{A} - \lambda\mathbf{I})\,\mathbf{v} = \mathbf{0}.$$

The identity $\mathbf{I}$ is needed: $\mathbf{A} - \lambda$ (matrix minus number) is meaningless. For a fixed number λ this *is* a homogeneous linear system. It always has the trivial solution $\mathbf{v} = \mathbf{0}$, and by the invertible matrix theorem of Ch. 8 it has a nonzero solution if and only if the coefficient matrix $\mathbf{A} - \lambda\mathbf{I}$ is singular, i.e. if and only if its determinant vanishes.

> **Characteristic polynomial and characteristic equation**
>
> Let $\mathbf{A}$ be an $n \times n$ matrix (real or complex). Its **characteristic polynomial** is
>
> $$p(\lambda) = \det(\mathbf{A} - \lambda\mathbf{I}),$$
>
> a polynomial of degree exactly n in λ. The equation $p(\lambda) = 0$ is the **characteristic equation**, and its roots are the **eigenvalues** of $\mathbf{A}$. The multiset of eigenvalues is the **spectrum** of $\mathbf{A}$, and $\rho(\mathbf{A}) = \max_i |\lambda_i|$ is the **spectral radius**.

Some texts use $\det(\lambda\mathbf{I} - \mathbf{A})$ instead; the two differ by the factor $(-1)^n$ and have the same roots. The form $\det(\lambda\mathbf{I} - \mathbf{A})$ has leading coefficient $+1$ (it is *monic*), which is convenient for control-system work where it matches the denominator of a transfer function. We shall use whichever is tidier and say which.

Why is the degree exactly n? Expanding the determinant as a sum of signed products, one per permutation (Ch. 8), the only product that contains n factors of λ is the product of the diagonal entries $(a_{11}-\lambda)(a_{22}-\lambda)\cdots(a_{nn}-\lambda)$, which contributes $(-\lambda)^n$. Every other permutation leaves out at least two diagonal entries and therefore has degree at most $n-2$.

Coefficients: trace, determinant and principal minors

The same expansion gives two coefficients immediately. The coefficient of λ^{n-1} also comes only from the diagonal product, and equals $(-1)^{n-1}(a_{11}+\cdots+a_{nn})$. The constant term is $p(0)=\det\mathbf{A}$. On the other hand, by the fundamental theorem of algebra p factors over $\mathbb{C}$ as

$$p(\lambda)=(\lambda_1-\lambda)(\lambda_2-\lambda)\cdots(\lambda_n-\lambda),$$

whose λ^{n-1} coefficient is $(-1)^{n-1}(\lambda_1+\cdots+\lambda_n)$ and whose constant term is $\lambda_1\lambda_2\cdots\lambda_n$. Comparing:

Trace and determinant

If $\lambda_1,\dots,\lambda_n$ are the eigenvalues of $\mathbf{A}$, repeated according to algebraic multiplicity, then

$$\lambda_1+\lambda_2+\cdots+\lambda_n=\operatorname{tr}\mathbf{A}=\sum_{i=1}^{n}a_{ii},\qquad \lambda_1\lambda_2\cdots\lambda_n=\det\mathbf{A}.$$

In particular $\mathbf{A}$ is singular if and only if 0 is an eigenvalue.

More generally, the coefficient of λ^{n-k} in $\det(\lambda\mathbf{I}-\mathbf{A})$ is $(-1)^k E_k$, where E_k is the sum of all $k\times k$ **principal minors** of $\mathbf{A}$ (determinants of submatrices obtained by keeping the same k rows and columns). This gives the very useful closed forms

$$\begin{aligned} n=2:&\quad \det(\lambda\mathbf{I}-\mathbf{A})=\lambda^2-(\operatorname{tr}\mathbf{A})\,\lambda+\det\mathbf{A},\\ n=3:&\quad \det(\lambda\mathbf{I}-\mathbf{A})=\lambda^3-(\operatorname{tr}\mathbf{A})\,\lambda^2+E_2\,\lambda-\det\mathbf{A},\end{aligned}$$

with $E_2=(a_{11}a_{22}-a_{12}a_{21})+(a_{11}a_{33}-a_{13}a_{31})+(a_{22}a_{33}-a_{23}a_{32})$.

For a 2×2 matrix the quadratic formula then gives the eigenvalues in one line:

$$\lambda_{1,2}=\frac{\operatorname{tr}\mathbf{A}}{2}\pm\sqrt{\left(\frac{\operatorname{tr}\mathbf{A}}{2}\right)^2-\det\mathbf{A}}.$$

The sign of the discriminant $(\operatorname{tr}\mathbf{A})^2-4\det\mathbf{A}$ decides whether the eigenvalues are real and distinct, real and repeated, or a complex-conjugate pair — the same three cases (overdamped, critically damped, underdamped) you meet for second-order circuits in Ch. 12. Section 10.9 turns this observation into a complete classification of two-dimensional dynamics.

Algebraic multiplicity and complex roots

Algebraic multiplicity

If $p(\lambda)=(\lambda_0-\lambda)^m q(\lambda)$ with $q(\lambda_0)\neq0$, the eigenvalue λ_0 has **algebraic multiplicity** m. The multiplicities of the distinct eigenvalues add up to n.

Two facts follow from elementary polynomial algebra (Ch. 1, Ch. 2):

- **Every $n\times n$ matrix has exactly n eigenvalues in $\mathbb{C}$**, counted with multiplicity. A real matrix may have *no* real eigenvalues (a rotation of the plane is the standard example), so eigenvalue theory naturally lives over the complex numbers.
- **A real matrix has real characteristic-polynomial coefficients, so its non-real eigenvalues occur in conjugate pairs** λ,λ^*. A real 3×3 matrix therefore always has at least one real eigenvalue.

Special structures

Several structures let you read eigenvalues off without solving anything:

- **Triangular and diagonal matrices.** If $\mathbf{A}$ is upper or lower triangular, so is $\mathbf{A}-\lambda\mathbf{I}$, and its determinant is the product of its diagonal entries. Hence *the eigenvalues of a triangular matrix are its diagonal entries.*
- **Block-triangular matrices.** If $\mathbf{A}=\begin{bmatrix}\mathbf{A}_{11}&\mathbf{A}_{12}\\ \mathbf{0}&\mathbf{A}_{22}\end{bmatrix}$ with square diagonal blocks, then $\det(\mathbf{A}-\lambda\mathbf{I})=\det(\mathbf{A}_{11}-\lambda\mathbf{I})\det(\mathbf{A}_{22}-\lambda\mathbf{I})$: the spectrum is the union of the spectra of the diagonal blocks. This is how a system made of a cascade of subsystems inherits the natural frequencies of each stage.
- **Transpose.** $\det(\mathbf{A}^{\mathsf T}-\lambda\mathbf{I})=\det\left((\mathbf{A}-\lambda\mathbf{I})^{\mathsf T}\right)=\det(\mathbf{A}-\lambda\mathbf{I})$, so $\mathbf{A}$ and $\mathbf{A}^{\mathsf T}$ have the same eigenvalues (but usually different eigenvectors).
- **Similar matrices.** If $\mathbf{B}=\mathbf{S}^{-1}\mathbf{A}\mathbf{S}$ for some invertible $\mathbf{S}$, then

$$\det(\mathbf{B}-\lambda\mathbf{I})=\det\left(\mathbf{S}^{-1}(\mathbf{A}-\lambda\mathbf{I})\mathbf{S}\right)=\det(\mathbf{S}^{-1})\det(\mathbf{A}-\lambda\mathbf{I})\det(\mathbf{S})=\det(\mathbf{A}-\lambda\mathbf{I}).$$

Similar matrices have the same characteristic polynomial, hence the same eigenvalues, trace and determinant. Since $\mathbf{S}$ is a change of basis (Ch. 9), this says that **eigenvalues belong to the linear transformation, not to the particular matrix that represents it.** Choosing different state variables for the same circuit changes the state matrix but never its eigenvalues — the natural frequencies are physical.

Companion matrices

Every monic polynomial is the characteristic polynomial of some matrix. For $q(\lambda)=\lambda^n+c_{n-1}\lambda^{n-1}+\cdots+c_1\lambda+c_0$, the **companion matrix**

$$\mathbf{C} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -c_0 & -c_1 & -c_2 & \cdots & -c_{n-1} \end{bmatrix}$$

satisfies $\det(\lambda \mathbf{I} - \mathbf{C}) = q(\lambda)$ (expand along the last row, or verify for $n = 2$ and induct). The companion matrix is exactly what you obtain when you convert the scalar ODE $y^{(n)} + c_{n-1} y^{(n-1)} + \cdots + c_0 y = 0$ to first-order form with the state $\mathbf{x} = (y, \dot{y}, \ldots, y^{(n-1)})$. It is also how numerical software finds polynomial roots: `numpy.roots` builds the companion matrix and calls an eigenvalue routine, because eigenvalue algorithms are more robust than direct root-finding.

The Cayley–Hamilton theorem

Cayley–Hamilton

Every square matrix satisfies its own characteristic equation: if $p(\lambda) = \det(\lambda \mathbf{I} - \mathbf{A}) = \lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_0$, then

$$p(\mathbf{A}) = \mathbf{A}^n + c_{n-1}\mathbf{A}^{n-1} + \cdots + c_1 \mathbf{A} + c_0 \mathbf{I} = \mathbf{0}.$$

Proof sketch. The adjugate identity of Ch. 8 gives $(\lambda \mathbf{I} - \mathbf{A})\,\operatorname{adj}(\lambda \mathbf{I} - \mathbf{A}) = p(\lambda)\mathbf{I}$. The entries of the adjugate are polynomials of degree at most $n-1$ in λ, so we can write $\operatorname{adj}(\lambda \mathbf{I} - \mathbf{A}) = \mathbf{B}_{n-1}\lambda^{n-1} + \cdots + \mathbf{B}_1 \lambda + \mathbf{B}_0$ with constant matrices $\mathbf{B}_k$. Multiplying out and equating coefficients of each power of λ gives $n+1$ matrix equations: $\mathbf{B}_{n-1} = \mathbf{I}$, $\mathbf{B}_{k-1} - \mathbf{A}\mathbf{B}_k = c_k \mathbf{I}$, and $-\mathbf{A}\mathbf{B}_0 = c_0 \mathbf{I}$. Multiply the equation for λ^k on the left by $\mathbf{A}^k$ and add them all: the left side telescopes to $\mathbf{0}$ and the right side is $p(\mathbf{A})$. (Simply "substituting $\lambda = \mathbf{A}$ into $\det(\lambda \mathbf{I} - \mathbf{A})$" is *not* a valid proof, because the determinant of a matrix with matrix entries is not defined that way.) $\square$

Cayley–Hamilton has three practical uses. First, $\mathbf{A}^n$ — and therefore every higher power — is a linear combination of $\mathbf{I}, \mathbf{A}, \ldots, \mathbf{A}^{n-1}$, so any polynomial or power series in $\mathbf{A}$ (including $e^{\mathbf{A}t}$, Section 10.4) can be reduced to a polynomial of degree at most $n-1$. Second, if $c_0 = \pm \det \mathbf{A} \neq 0$ we can solve for the inverse:

$$\mathbf{A}^{-1} = -\frac{1}{c_0}\left(\mathbf{A}^{n-1} + c_{n-1}\mathbf{A}^{n-2} + \cdots + c_1 \mathbf{I}\right).$$

Third, it underlies the controllability and observability tests of state-space control theory, where powers of $\mathbf{A}$ beyond $n-1$ never add new directions.

Worked examples

Worked Example 10.1 — A 2×2 characteristic equation

Find the eigenvalues of $\mathbf{A} = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$.

Step 1 — form $\mathbf{A} - \lambda \mathbf{I}$. Subtract λ from the diagonal only:

$$\mathbf{A} - \lambda \mathbf{I} = \begin{bmatrix} 4 - \lambda & 1 \\ 2 & 3 - \lambda \end{bmatrix}.$$

Step 2 — take the determinant.

$$p(\lambda) = (4 - \lambda)(3 - \lambda) - (1)(2) = \lambda^2 - 7\lambda + 12 - 2 = \lambda^2 - 7\lambda + 10.$$

Step 3 — check with the shortcut. $\operatorname{tr} \mathbf{A} = 7$ and $\det \mathbf{A} = 12 - 2 = 10$, so $\lambda^2 - 7\lambda + 10$ is confirmed.

Step 4 — solve. $\lambda^2 - 7\lambda + 10 = (\lambda - 5)(\lambda - 2)$, so $\lambda_1 = 5$, $\lambda_2 = 2$.

Check: $\lambda_1 + \lambda_2 = 7 = \operatorname{tr} \mathbf{A}$ and $\lambda_1 \lambda_2 = 10 = \det \mathbf{A}$. ✓

Worked Example 10.2 — A 3×3 tridiagonal matrix

Find the eigenvalues of

$$\mathbf{B} = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}.$$

Step 1 — expand along the first row.

$$\det(\mathbf{B} - \lambda \mathbf{I}) = (2-\lambda)\begin{vmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{vmatrix} - 1 \cdot \begin{vmatrix} 1 & 1 \\ 0 & 2-\lambda \end{vmatrix} = (2-\lambda)\big[(2-\lambda)^2 - 1\big] - (2-\lambda).$$

Step 2 — factor before expanding. The common factor $(2-\lambda)$ is the key labour-saver:

$$\det(\mathbf{B} - \lambda \mathbf{I}) = (2-\lambda)\big[(2-\lambda)^2 - 2\big].$$

Step 3 — solve. Either $\lambda = 2$, or $(2-\lambda)^2 = 2$, i.e. $\lambda = 2 \pm \sqrt{2}$. The eigenvalues are

$$\lambda_1 = 2 + \sqrt{2} \approx 3.4142, \qquad \lambda_2 = 2, \qquad \lambda_3 = 2 - \sqrt{2} \approx 0.5858.$$

Step 4 — check with principal minors. $\operatorname{tr}\mathbf{B} = 6$; $E_2 = (4-1) + (4-0) + (4-1) = 10$; $\det \mathbf{B} = 2(4-1) - 1(2-0) = 4$. So $\det(\lambda\mathbf{I} - \mathbf{B}) = \lambda^3 - 6\lambda^2 + 10\lambda - 4$. Sum of roots $= 6$ ✓; product $= 2\,(4-2) = 4$ ✓; and substituting $\lambda = 2$: $8 - 24 + 20 - 4 = 0$ ✓.

Matrices of this "2 on the diagonal, 1 off it" type (with sign changes) arise whenever identical elements are chained — an LC ladder, a discretised transmission line, a finite-difference Laplacian (Ch. 13). Their eigenvalues are known in closed form: for the $n \times n$ version, $\lambda_k = 2 + 2\cos\frac{k\pi}{n+1}$, $k = 1, \dots, n$, which for $n = 3$ gives $2 + \sqrt{2},\ 2,\ 2 - \sqrt{2}$.

Worked Example 10.3 — The companion matrix of a third-order ODE

The differential equation $\dddot{y} + 6\ddot{y} + 11\dot{y} + 6y = 0$ is put into state form with $x_1 = y$, $x_2 = \dot{y}$, $x_3 = \ddot{y}$:

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \mathbf{x} = \mathbf{C}\mathbf{x}.$$

Step 1 — characteristic polynomial. Expand $\det(\lambda\mathbf{I} - \mathbf{C})$ along the first column:

$$\det \begin{bmatrix} \lambda & -1 & 0 \\ 0 & \lambda & -1 \\ 6 & 11 & \lambda + 6 \end{bmatrix} = \lambda\big[\lambda(\lambda+6) + 11\big] + 6\big[(-1)(-1) - 0 \cdot \lambda\big] = \lambda^3 + 6\lambda^2 + 11\lambda + 6.$$

This is precisely the characteristic polynomial you would obtain by substituting $y = e^{\lambda t}$ into the ODE.

Step 2 — roots. Trying $\lambda = -1$: $-1 + 6 - 11 + 6 = 0$. Dividing, $\lambda^3 + 6\lambda^2 + 11\lambda + 6 = (\lambda+1)(\lambda^2 + 5\lambda + 6) = (\lambda+1)(\lambda+2)(\lambda+3)$.

The eigenvalues $-1, -2, -3$ are the natural frequencies (in s^{-1}, if t is in seconds) of the system; all are negative, so every solution decays (Section 10.9). **Check:** $\operatorname{tr}\mathbf{C} = -6 = -1 - 2 - 3$ ✓, $\det\mathbf{C} = -6 = (-1)(-2)(-3)$ ✓.

Worked Example 10.4 — Inverse and powers by Cayley–Hamilton

For $\mathbf{A} = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$ of Example 10.1, Cayley–Hamilton gives $\mathbf{A}^2 - 7\mathbf{A} + 10\mathbf{I} = \mathbf{0}$.

Inverse. Rearranging, $\mathbf{A}(7\mathbf{I} - \mathbf{A}) = 10\mathbf{I}$, so

$$\mathbf{A}^{-1} = \frac{1}{10}(7\mathbf{I} - \mathbf{A}) = \frac{1}{10}\begin{bmatrix} 3 & -1 \\ -2 & 4 \end{bmatrix}.$$

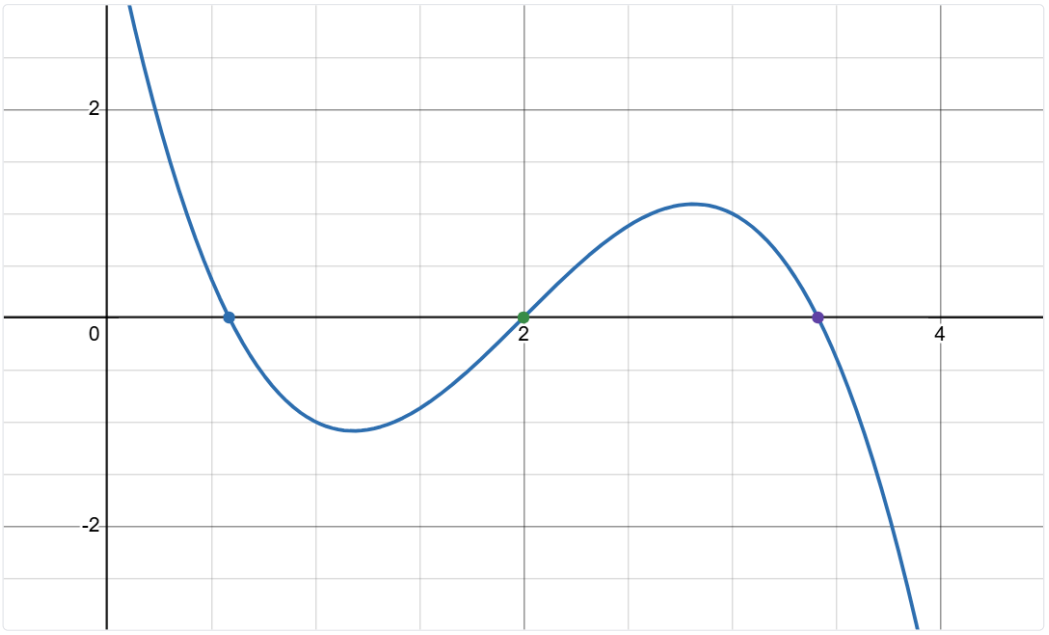
Check: $\mathbf{A}\mathbf{A}^{-1} = \frac{1}{10}\begin{bmatrix} 12-2 & -4+4 \\ 6-6 & -2+12 \end{bmatrix} = \mathbf{I}$ ✓.

Cube. $\mathbf{A}^2 = 7\mathbf{A} - 10\mathbf{I}$, so $\mathbf{A}^3 = 7\mathbf{A}^2 - 10\mathbf{A} = 7(7\mathbf{A} - 10\mathbf{I}) - 10\mathbf{A} = 39\mathbf{A} - 70\mathbf{I}$:

$$\mathbf{A}^3 = \begin{bmatrix} 156 - 70 & 39 \\ 78 & 117 - 70 \end{bmatrix} = \begin{bmatrix} 86 & 39 \\ 78 & 47 \end{bmatrix},$$

which agrees with direct multiplication. Every power $\mathbf{A}^k$ is of the form $\alpha_k\mathbf{A} + \beta_k\mathbf{I}$; Section 10.4 finds α_k, β_k in closed form.

The characteristic polynomial of Example 10.2 is worth seeing as a graph. Its three real roots are where the cubic crosses the axis; for a symmetric matrix (Section 10.5) such crossings are guaranteed to be real.



The characteristic polynomial $p(\lambda) = \det(\mathbf{B} - \lambda\mathbf{I}) = -\lambda^3 + 6\lambda^2 - 10\lambda + 4$ of Example 10.2 (horizontal axis λ); its zeros are the eigenvalues and its intercept is $\det \mathbf{B}$.

Common Pitfall — subtracting λ everywhere, or expanding too early

Two errors account for most wrong characteristic polynomials. (i) λ is subtracted from the **diagonal entries only**; writing $a_{12} - \lambda$ is wrong. (ii) Multiplying everything out and then hunting for roots of a cubic is laborious and error-prone; look for a common factor first (Example 10.2) or row-reduce cleverly. Always finish with the trace and determinant checks — they cost seconds and catch almost every arithmetic slip.

Common Pitfall — row reduction changes eigenvalues

Gaussian elimination preserves the *solution set* of $\mathbf{A}\mathbf{x} = \mathbf{b}$ but not the eigenvalues of $\mathbf{A}$. The echelon form of $\begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$ is $\begin{bmatrix} 4 & 1 \\ 0 & 2.5 \end{bmatrix}$, with diagonal $4, 2.5$ — not the eigenvalues $5, 2$. You may row-reduce $\mathbf{A} - \lambda\mathbf{I}$ to find *eigenvectors* once λ is fixed, but never row-reduce $\mathbf{A}$ to read off eigenvalues. Only *similarity* transformations $\mathbf{S}^{-1}\mathbf{A}\mathbf{S}$ preserve them.

Engineering Note — never compute large characteristic polynomials numerically

For hand work with $n \le 3$ the characteristic polynomial is the right tool. For n in the tens or thousands — a power-system network, a finite-element model, a circuit simulator's Jacobian — the roots of a polynomial are extraordinarily sensitive to rounding of its coefficients (Wilkinson's famous example: perturbing one coefficient of $\prod_{k=1}^{20}(\lambda - k)$ by 2^{-23} makes several roots complex). Production software (LAPACK, MATLAB `eig`, NumPy) never forms $p(\lambda)$; it applies a sequence of orthogonal similarity transformations — the QR algorithm — that drive $\mathbf{A}$ to triangular form, whose diagonal holds the eigenvalues. The method is treated in Ch. 22; this chapter is about what the answers mean.

10.2 Eigenvalues and eigenvectors

Motivation

Once the eigenvalues are known, the eigenvectors tell us *where* the matrix acts as pure scaling. Geometrically: a 2×2 matrix turns the unit circle into an ellipse and generally rotates every direction; an eigenvector is a direction that comes out parallel to itself — stretched by λ, shrunk, or reversed if $\lambda < 0$. Physically: in a vibrating structure an eigenvector is a *mode shape*, the pattern of relative amplitudes with which all parts move together at one frequency; in a Markov chain the eigenvector for $\lambda = 1$ is the long-run probability distribution; in an antenna array or a MIMO channel the dominant eigenvector of the correlation matrix is the best beam-forming direction.

Definitions

Eigenvalue, eigenvector, eigenspace

Let $\mathbf{A}$ be $n \times n$. A scalar $\lambda \in \mathbb{C}$ is an **eigenvalue** of $\mathbf{A}$ if there is a **nonzero** vector $\mathbf{v} \in \mathbb{C}^n$ with

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}.$$

Such a $\mathbf{v}$ is an **eigenvector** belonging to λ, and $(\lambda, \mathbf{v})$ is an **eigenpair**. The set

$$E_\lambda = \operatorname{null}(\mathbf{A} - \lambda\mathbf{I}) = \{\mathbf{v} : \mathbf{A}\mathbf{v} = \lambda\mathbf{v}\}$$

is the **eigenspace** of λ; it consists of all eigenvectors for λ together with $\mathbf{0}$. Its dimension $g = \dim E_\lambda = n - \operatorname{rank}(\mathbf{A} - \lambda\mathbf{I})$ is the **geometric multiplicity** of λ.

Three remarks clarify the definition.

1. **The zero vector is excluded as an eigenvector** (otherwise every λ would be an eigenvalue), but $\lambda = 0$ is a perfectly good eigenvalue: it means $\mathbf{A}\mathbf{v} = \mathbf{0}$ for some $\mathbf{v} \neq \mathbf{0}$, i.e. $\mathbf{A}$ is singular and $\mathbf{v}$ lies in its null space.
2. **Eigenvectors are never unique.** If $\mathbf{v}$ is an eigenvector, so is $c\mathbf{v}$ for every $c \neq 0$ (including complex c). An eigenvector really specifies a *direction* (a line through the origin), and when the eigenspace has dimension $g > 1$, any nonzero vector in that g-dimensional subspace qualifies. Different software normalises differently — unit length, largest entry 1, first entry 1 — so do not be alarmed when your hand answer differs from MATLAB's by a scale factor or a sign.
3. **The eigenspace is a subspace** (Ch. 9): it is the null space of a matrix, so it is closed under addition and scaling.

Finding eigenvectors

The procedure is a direct application of null-space computation:

1. Solve $\det(\mathbf{A} - \lambda\mathbf{I}) = 0$ for the eigenvalues.
2. For each distinct eigenvalue λ_i, row-reduce $\mathbf{A} - \lambda_i\mathbf{I}$. Because λ_i makes this matrix singular, at least one row must reduce to zero — **if it doesn't, your eigenvalue is wrong.**
3. Express the basic variables in terms of the free variables; one free variable per dimension of E_{λ_i}. Setting each free variable to 1 in turn (others 0) gives a basis of the eigenspace.
4. Check $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ by direct multiplication.

For a 2×2 matrix step 2 is trivial. If $\mathbf{A} - \lambda\mathbf{I} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is singular, its rows are proportional, and $\mathbf{v} = \begin{bmatrix} b \\ -a \end{bmatrix}$ (or $\begin{bmatrix} d \\ -c \end{bmatrix}$) is an eigenvector whenever it is nonzero — it is perpendicular to the first row.

Algebraic versus geometric multiplicity

Multiplicity inequality

For every eigenvalue λ_0 of $\mathbf{A}$,

$$1 \le g(\lambda_0) \le m(\lambda_0),$$

where g is the geometric and m the algebraic multiplicity.

Proof sketch. $g \ge 1$ because $\mathbf{A} - \lambda_0\mathbf{I}$ is singular. For the upper bound, take a basis $\mathbf{v}_1, \dots, \mathbf{v}_g$ of E_{λ_0} and extend it to a basis of $\mathbb{C}^n$; put these vectors as the columns of $\mathbf{S}$. In the new basis the first g columns of $\mathbf{S}^{-1}\mathbf{A}\mathbf{S}$ are $\lambda_0\mathbf{e}_1, \dots, \lambda_0\mathbf{e}_g$, so $\mathbf{S}^{-1}\mathbf{A}\mathbf{S} = \begin{bmatrix} \lambda_0\mathbf{I}_g & * \\ \mathbf{0} & \mathbf{C} \end{bmatrix}$. This block-triangular matrix has characteristic polynomial $(\lambda_0 - \lambda)^g \det(\mathbf{C} - \lambda\mathbf{I})$, and similar matrices share characteristic polynomials, so $(\lambda_0 - \lambda)^g$ divides $p(\lambda)$: $g \le m$. $\square$

An eigenvalue with $g < m$ is called **defective**, and so is a matrix having one. The standard example is the **shear**

$$\mathbf{N} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}: \qquad p(\lambda) = (1-\lambda)^2, \quad m = 2, \qquad \mathbf{N} - \mathbf{I} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad g = 1.$$

The only eigenvector direction is $\mathbf{e}_1 = (1, 0)$: the shear slides every horizontal line along itself, so only horizontal vectors keep their direction. Defective matrices are exactly the ones that cannot be diagonalized (Section 10.3); in dynamics they produce the $te^{\lambda t}$ terms familiar from critically damped circuits.

Independence of eigenvectors

Eigenvectors for distinct eigenvalues are independent

If $\mathbf{v}_1, \dots, \mathbf{v}_k$ are eigenvectors of $\mathbf{A}$ belonging to **distinct** eigenvalues $\lambda_1, \dots, \lambda_k$, then $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is linearly independent.

Proof. Suppose not, and let r be the smallest number such that $\mathbf{v}_{r+1}$ is a combination of the independent set $\mathbf{v}_1, \dots, \mathbf{v}_r$: $\mathbf{v}_{r+1} = c_1\mathbf{v}_1 + \cdots + c_r\mathbf{v}_r$. Apply $\mathbf{A}$: $\lambda_{r+1}\mathbf{v}_{r+1} = c_1\lambda_1\mathbf{v}_1 + \cdots + c_r\lambda_r\mathbf{v}_r$. Subtract λ_{r+1} times the first relation:

$$\mathbf{0} = c_1(\lambda_1 - \lambda_{r+1})\mathbf{v}_1 + \cdots + c_r(\lambda_r - \lambda_{r+1})\mathbf{v}_r.$$

Since $\mathbf{v}_1, \dots, \mathbf{v}_r$ are independent and every $\lambda_i - \lambda_{r+1} \neq 0$, all $c_i = 0$, so $\mathbf{v}_{r+1} = \mathbf{0}$ — impossible for an eigenvector. $\square$

The same argument shows that if we take bases of several eigenspaces for distinct eigenvalues and pool them, the pooled set is still independent. This is the engine of the diagonalization theorem in Section 10.3.

Eigenvalues of related matrices

If $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$, then repeated application gives $\mathbf{A}^k\mathbf{v} = \lambda^k\mathbf{v}$, and linear combinations give the following table. The eigenvector $\mathbf{v}$ is the **same** in every row except the last.

Matrix	Eigenvalue	Condition
$\mathbf{A}^k$, $k = 1, 2, \dots$	λ^k	—
$\mathbf{A}^{-1}$	$1/\lambda$	$\mathbf{A}$ invertible ($\lambda \neq 0$)
$\mathbf{A} + c\mathbf{I}$	$\lambda + c$	— (spectral shift)
$c\mathbf{A}$	$c\lambda$	—

Matrix	Eigenvalue	Condition
$q(\mathbf{A}) = \sum_k q_k \mathbf{A}^k$	$q(\lambda)$	any polynomial q
$(\mathbf{A} - \mu\mathbf{I})^{-1}$	$1/(\lambda - \mu)$	μ not an eigenvalue
$e^{\mathbf{A}t}$	$e^{\lambda t}$	(Section 10.4)
$\mathbf{A}^{\mathsf{T}}$	λ	eigenvector generally different

The shifted-inverse row is the basis of the *inverse iteration* algorithm: choosing μ close to a wanted eigenvalue makes $1/(\lambda - \mu)$ huge, so that eigenvalue dominates. The last row has an important companion: vectors with $\mathbf{w}^{\mathsf{T}}\mathbf{A} = \lambda\mathbf{w}^{\mathsf{T}}$, i.e. eigenvectors of $\mathbf{A}^{\mathsf{T}}$, are called **left eigenvectors**. A left eigenvector for λ_i is orthogonal to every right eigenvector for a different eigenvalue λ_j, since $\lambda_i \mathbf{w}_i^{\mathsf{T}}\mathbf{v}_j = \mathbf{w}_i^{\mathsf{T}}\mathbf{A}\mathbf{v}_j = \lambda_j \mathbf{w}_i^{\mathsf{T}}\mathbf{v}_j$ forces $\mathbf{w}_i^{\mathsf{T}}\mathbf{v}_j = 0$. Left eigenvectors reappear in modal analysis (Section 10.8) as the tools that extract modal amplitudes.

Complex eigenvalues of real matrices

When a real matrix has a complex eigenvalue $\lambda = \sigma + j\omega$ ($\omega \neq 0$), its eigenvector is necessarily complex, and conjugating $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ (with $\mathbf{A}$ real) gives $\mathbf{A}\mathbf{v}^* = \lambda^*\mathbf{v}^*$: **the conjugate eigenvalue has the conjugate eigenvector**, so only one of each pair needs to be computed. Geometrically, a complex pair signals *rotation*: no real direction is preserved. The prototype is

$$\mathbf{A} = \begin{bmatrix} \sigma & -\omega \\ \omega & \sigma \end{bmatrix} = r \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}, \qquad r = \sqrt{\sigma^2 + \omega^2},\ \theta = \angle(\sigma + j\omega),$$

with eigenvalues $\sigma \pm j\omega = re^{\pm j\theta}$ and eigenvectors $(1, \mp j)$. This matrix is multiplication by the complex number $re^{j\theta}$ written in real coordinates (Ch. 2): rotate by θ, scale by r. Section 10.3 shows that *every* real 2×2 matrix with complex eigenvalues is similar to this rotation-scaling form.

Worked examples

Worked Example 10.5 — Eigenvectors of a 2×2 matrix

Find the eigenvectors of $\mathbf{A} = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$ (eigenvalues 5 and 2, Example 10.1).

$\lambda_1 = 5$. $\mathbf{A} - 5\mathbf{I} = \begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix}$. The second row is -2 times the first, as it must be. The first row says $-v_1 + v_2 = 0$, so $v_2 = v_1$:

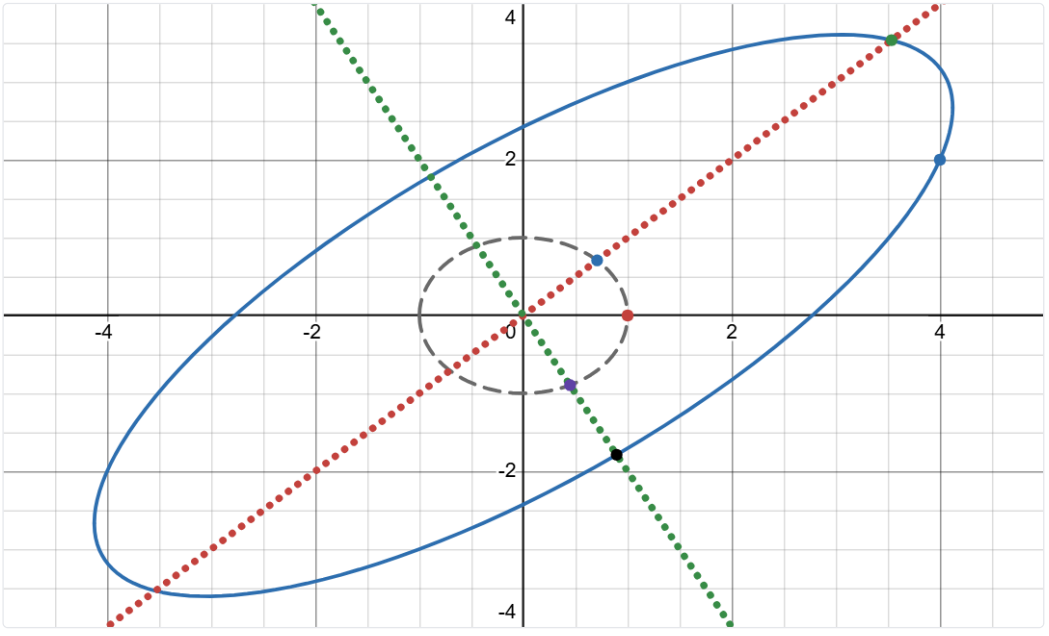
$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad \mathbf{A}\mathbf{v}_1 = \begin{bmatrix} 5 \\ 5 \end{bmatrix} = 5\mathbf{v}_1 \ \checkmark$$

$\lambda_2 = 2$. $\mathbf{A} - 2\mathbf{I} = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix}$, giving $2v_1 + v_2 = 0$:

$$\mathbf{v}_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \qquad \mathbf{A}\mathbf{v}_2 = \begin{bmatrix} 2 \\ -4 \end{bmatrix} = 2\mathbf{v}_2 \ \checkmark$$

The eigenvectors are independent (distinct eigenvalues) but **not** orthogonal: $\mathbf{v}_1 \cdot \mathbf{v}_2 = 1 - 2 = -1$. Orthogonality is a privilege of symmetric matrices (Section 10.5).

The graph below shows the geometry. The matrix maps the unit circle onto an ellipse: the image of $\|\mathbf{x}\| = 1$ is the set of $\mathbf{y}$ with $\|\mathbf{A}^{-1}\mathbf{y}\| = 1$, i.e. $\mathbf{y}^{\mathsf{T}}(\mathbf{A}\mathbf{A}^{\mathsf{T}})^{-1}\mathbf{y} = 1$, which here is $13x^2 - 22xy + 17y^2 = 100$. Points of the circle on the eigenvector lines land on the same lines, scaled by 5 and 2; the point $(1, 0)$ is rotated off its line to $(4, 2)$.



The unit circle (dashed) and its image under $\mathbf{A} = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$ (blue ellipse). Vectors on the eigenvector lines $y = x$ (red, $\lambda = 5$) and $y = -2x$ (green, $\lambda = 2$) are only stretched; every other direction, such as $\mathbf{e}_1 \mapsto (4, 2)$, is also turned. The eigenvector lines are not the ellipse's axes — that happens only for symmetric matrices.

Worked Example 10.6 — A repeated eigenvalue with a two-dimensional eigenspace

Find all eigenvalues and eigenspaces of

$$\mathbf{L} = \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix}.$$

Step 1 — eigenvalues. $\operatorname{tr}\mathbf{L} = 13$, $E_2 = (4+2) + (32-12) + (8+6) = 40$, $\det\mathbf{L} = 4(8+6) + 1(16-12) + 6(-2-2) = 56 + 4 - 24 = 36$. Hence

$$\det(\lambda\mathbf{I} - \mathbf{L}) = \lambda^3 - 13\lambda^2 + 40\lambda - 36.$$

Trying $\lambda = 2$: $8 - 52 + 80 - 36 = 0$. Dividing out, $\lambda^3 - 13\lambda^2 + 40\lambda - 36 = (\lambda - 2)(\lambda^2 - 11\lambda + 18) = (\lambda - 2)^2(\lambda - 9)$. So $\lambda = 2$ with $m = 2$ and $\lambda = 9$ with $m = 1$.

Step 2 — eigenspace for $\lambda = 2$.

$$\mathbf{L} - 2\mathbf{I} = \begin{bmatrix} 2 & -1 & 6 \\ 2 & -1 & 6 \\ 2 & -1 & 6 \end{bmatrix} \longrightarrow \begin{bmatrix} 2 & -1 & 6 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Rank 1, so $g = 3 - 1 = 2$. The single equation $2v_1 - v_2 + 6v_3 = 0$ gives $v_2 = 2v_1 + 6v_3$ with v_1, v_3 free. Choosing $(v_1, v_3) = (1, 0)$ and $(0, 1)$:

$$E_2 = \operatorname{span}\left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 6 \\ 1 \end{bmatrix} \right\}.$$

(Equally valid: $(1, 2, 0)$ and $(-3, 0, 1)$ — any two independent solutions of the plane equation.)

Step 3 — eigenspace for $\lambda = 9$.

$$\mathbf{L} - 9\mathbf{I} = \begin{bmatrix} -5 & -1 & 6 \\ 2 & -8 & 6 \\ 2 & -1 & -1 \end{bmatrix}.$$

The third row gives $v_2 = 2v_1 - v_3$; substituting into the first, $-5v_1 - 2v_1 + v_3 + 6v_3 = 0$, so $v_3 = v_1$ and then $v_2 = v_1$. Thus $E_9 = \operatorname{span}\{(1, 1, 1)\}$. Check with the second row: $2 - 8 + 6 = 0$ ✓, and $\mathbf{L}(1,1,1)^{\mathsf{T}} = (9, 9, 9)^{\mathsf{T}}$ ✓.

Conclusion. Here $g(2) = m(2) = 2$, so the repeated eigenvalue is *not* defective, and the three eigenvectors $(1, 2, 0)$, $(0, 6, 1)$, $(1, 1, 1)$ form a basis of $\mathbb{R}^3$. Section 10.3 uses this to diagonalize $\mathbf{L}$. Contrast the shear $\mathbf{N}$ above, where the double eigenvalue had only a one-dimensional eigenspace.

Worked Example 10.7 — Complex eigenvalues and eigenvectors

Find the eigenpairs of $\mathbf{R} = \begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}$.

Step 1 — eigenvalues. $\operatorname{tr}\mathbf{R} = 4$, $\det\mathbf{R} = 3 + 2 = 5$, so $\lambda^2 - 4\lambda + 5 = 0$ and

$$\lambda = 2 \pm \sqrt{4 - 5} = 2 \pm j.$$

Step 2 — eigenvector for $\lambda = 2 + j$.

$$\mathbf{R} - (2+j)\mathbf{I} = \begin{bmatrix} -1 - j & -2 \\ 1 & 1 - j \end{bmatrix}.$$

Use the simpler second row: $v_1 + (1 - j)v_2 = 0$. With $v_2 = 1$, $v_1 = -1 + j$, so

$$\mathbf{v} = \begin{bmatrix} -1 + j \\ 1 \end{bmatrix}.$$

Check the first row, which must also vanish: $(-1 - j)(-1 + j) - 2 = (1 + 1) - 2 = 0$ ✓.

Step 3 — the conjugate pair. For $\lambda^* = 2 - j$ the eigenvector is $\mathbf{v}^* = (-1 - j,\ 1)$, with no further work.

Interpretation. $|\lambda| = \sqrt{5}$ and $\angle\lambda = \arctan(1/2) = 26.57°$: in a suitable (non-orthogonal) coordinate system, $\mathbf{R}$ rotates by $26.57°$ and scales by $\sqrt{5} \approx 2.236$. In continuous-time dynamics $\dot{\mathbf{x}} = \mathbf{R}\mathbf{x}$, the pair $2 \pm j$ means oscillation at 1 rad/s inside a growing envelope e^{2t} — an unstable spiral (Section 10.9).

Worked Example 10.8 — Eigenvalues of functions of a matrix

With $\mathbf{A} = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$ (eigenvalues $5, 2$), find the eigenvalues of $\mathbf{B} = \mathbf{A}^2 - 3\mathbf{A} + \mathbf{I}$, of $\mathbf{A}^{-1}$ and of $\mathbf{A} - 6\mathbf{I}$, and decide whether $\mathbf{B}$ is invertible — without computing $\mathbf{B}$.

- $q(\lambda) = \lambda^2 - 3\lambda + 1$: $q(5) = 11$, $q(2) = -1$. Eigenvalues of $\mathbf{B}$: $11, -1$. Since neither is zero, $\mathbf{B}$ is invertible, with $\det \mathbf{B} = -11$.
- $\mathbf{A}^{-1}$: $1/5,\ 1/2$ (compare Example 10.4: $\operatorname{tr} \mathbf{A}^{-1} = (3+4)/10 = 0.7 = 0.2 + 0.5$ ✓).
- $\mathbf{A} - 6\mathbf{I}$: $-1,\ -4$. Shifting moves the whole spectrum without changing any eigenvector.

Direct check for B. $\mathbf{A}^2 = \begin{bmatrix} 18 & 7 \\ 14 & 11 \end{bmatrix}$, so $\mathbf{B} = \begin{bmatrix} 18 - 12 + 1 & 7 - 3 \\ 14 - 6 & 11 - 9 + 1 \end{bmatrix} = \begin{bmatrix} 7 & 4 \\ 8 & 3 \end{bmatrix}$; $\operatorname{tr} \mathbf{B} = 10 = 11 + (-1)$ ✓, $\det \mathbf{B} = 21 - 32 = -11$ ✓. And $\mathbf{B}(1,1)^{\mathsf{T}} = (11, 11)^{\mathsf{T}}$: same eigenvector as for $\mathbf{A}$ ✓.

Common Pitfall — eigenvalues of sums and products

The table above concerns functions of a **single** matrix. In general the eigenvalues of $\mathbf{A} + \mathbf{B}$ are **not** sums of eigenvalues of $\mathbf{A}$ and $\mathbf{B}$, and those of $\mathbf{AB}$ are not products — unless $\mathbf{A}$ and $\mathbf{B}$ share a full set of eigenvectors (which happens, for diagonalizable matrices, exactly when $\mathbf{AB} = \mathbf{BA}$). Example: $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ and $\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$ both have only the eigenvalue 0, but their sum $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ has eigenvalues ± 1. What *is* true: $\operatorname{tr}(\mathbf{A} + \mathbf{B}) = \operatorname{tr} \mathbf{A} + \operatorname{tr} \mathbf{B}$, and $\mathbf{AB}$ and $\mathbf{BA}$ have the same eigenvalues.

Engineering Note — what eigenvalues mean physically

When $\mathbf{A}$ maps a physical state to its rate of change ($\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$), the eigenvalues have units of s^{-1}: a real eigenvalue $-1/\tau$ is a decay rate with time constant τ, and a complex pair $-\alpha \pm j\omega_d$ is a damped oscillation at ω_d rad/s. When $\mathbf{A}$ maps a state at one sample to the next ($\mathbf{x}[k+1] = \mathbf{A}\mathbf{x}[k]$), the eigenvalues are dimensionless multipliers per step. When $\mathbf{A}$ is a covariance matrix of voltages, its eigenvalues are in V^2 — the variance (power) carried along each eigenvector direction. Always ask what $\mathbf{A}$ maps from and to; the units of the eigenvalues follow.

10.3 Diagonalization

Motivation

A diagonal matrix is the simplest matrix there is: it scales the i-th coordinate by d_i and leaves the coordinates uncoupled. Its powers, inverse, exponential and every other function are computed entry by entry. A general matrix couples the coordinates — the rate of change of the capacitor voltage depends on the inductor current and vice versa — and that coupling is what makes systems hard to analyse. **Diagonalization** removes the coupling by changing to a basis of eigenvectors. In that basis the matrix is diagonal with the eigenvalues on the diagonal, and every computation becomes n scalar computations. Afterwards we change back.

This is the matrix version of a strategy used all through this book: transform a hard problem into a domain where it is easy (phasors, Laplace, Fourier), solve it there, and transform back. Here the "transform" is multiplication by $\mathbf{P}^{-1}$, the "easy domain" is modal coordinates, and the "inverse transform" is multiplication by $\mathbf{P}$.

The factorization

Suppose $\mathbf{A}$ has n linearly independent eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ with eigenvalues $\lambda_1, \dots, \lambda_n$ (not necessarily distinct). Place the eigenvectors as columns of $\mathbf{P}$ and the eigenvalues, **in the same order**, on the diagonal of $\mathbf{D}$. Then, column by column,

$$\mathbf{A}\mathbf{P} = \mathbf{A}\begin{bmatrix} \mathbf{v}_1 & \cdots & \mathbf{v}_n \end{bmatrix} = \begin{bmatrix} \lambda_1\mathbf{v}_1 & \cdots & \lambda_n\mathbf{v}_n \end{bmatrix} = \begin{bmatrix} \mathbf{v}_1 & \cdots & \mathbf{v}_n \end{bmatrix} \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} = \mathbf{P}\mathbf{D}.$$

Since the columns of $\mathbf{P}$ are independent, $\mathbf{P}$ is invertible, and

$$\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}, \qquad \mathbf{D} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}.$$

Conversely, if $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$ with $\mathbf{D}$ diagonal, then $\mathbf{A}\mathbf{P} = \mathbf{P}\mathbf{D}$ says exactly that each column of $\mathbf{P}$ is an eigenvector with eigenvalue the corresponding diagonal entry of $\mathbf{D}$.

Diagonalizable matrix

A square matrix $\mathbf{A}$ is **diagonalizable** if it is similar to a diagonal matrix: there exist an invertible $\mathbf{P}$ and a diagonal $\mathbf{D}$ with $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$. The columns of $\mathbf{P}$ form an **eigenbasis**.

Diagonalization theorem

For an $n \times n$ matrix $\mathbf{A}$ the following are equivalent:

1. $\mathbf{A}$ is diagonalizable;
2. $\mathbf{A}$ has n linearly independent eigenvectors;
3. for every eigenvalue, the geometric multiplicity equals the algebraic multiplicity, $g(\lambda_i) = m(\lambda_i)$;
4. the dimensions of the eigenspaces add up to n.

In particular, **an $n \times n$ matrix with n distinct eigenvalues is diagonalizable.**

Proof sketch. (1)⇔(2) is the computation above. For (2)⇔(3)⇔(4): the sum of algebraic multiplicities is always n and $g_i \le m_i$, so $\sum g_i = n$ if and only if $g_i = m_i$ for all i. Pooling bases of the eigenspaces gives an independent set (Section 10.2) with $\sum g_i$ vectors, and no independent set of eigenvectors can be larger. The final statement follows because each distinct eigenvalue has $g = m = 1$. □

The "distinct eigenvalues" criterion is sufficient but not necessary: the identity matrix has a single eigenvalue repeated n times yet is already diagonal, and $\mathbf{L}$ of Example 10.6 has a double eigenvalue but is diagonalizable.

Interpretation as a change of basis

Read $\mathbf{A}\mathbf{x} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}\mathbf{x}$ from right to left:

1. $\mathbf{y} = \mathbf{P}^{-1}\mathbf{x}$ — find the coordinates of $\mathbf{x}$ in the eigenbasis, i.e. write $\mathbf{x} = y_1\mathbf{v}_1 + \cdots + y_n\mathbf{v}_n$ (Ch. 9);
2. $\mathbf{D}\mathbf{y}$ — scale each coordinate by its eigenvalue: $y_i \mapsto \lambda_i y_i$;
3. $\mathbf{P}(\mathbf{D}\mathbf{y})$ — reassemble: $\mathbf{A}\mathbf{x} = \lambda_1 y_1 \mathbf{v}_1 + \cdots + \lambda_n y_n \mathbf{v}_n$.

The same reading applied to the linear system $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ is the whole of modal analysis. Substituting $\mathbf{x} = \mathbf{P}\mathbf{y}$ gives $\mathbf{P}\dot{\mathbf{y}} = \mathbf{A}\mathbf{P}\mathbf{y}$, i.e.

$$\dot{\mathbf{y}} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}\,\mathbf{y} = \mathbf{D}\mathbf{y} \quad \Longleftrightarrow \quad \dot{y}_i = \lambda_i y_i, \quad i = 1, \dots, n,$$

n uncoupled first-order equations with solutions $y_i(t) = y_i(0)e^{\lambda_i t}$. The coordinates y_i are the **modal coordinates** (Section 10.8).

Diagonalizing with complex eigenvalues; the real block form

A real matrix with complex eigenvalues is diagonalizable over $\mathbb{C}$ (if its eigenvalues are distinct), with complex $\mathbf{P}$ and $\mathbf{D}$. Often we prefer to stay real. For a 2×2 real matrix with eigenvalues $\sigma \pm j\omega$ ($\omega \neq 0$) and eigenvector $\mathbf{v} = \mathbf{a} + j\mathbf{b}$ for $\sigma + j\omega$ (with $\mathbf{a}, \mathbf{b}$ real), separate $\mathbf{A}(\mathbf{a} + j\mathbf{b}) = (\sigma + j\omega)(\mathbf{a} + j\mathbf{b})$ into real and imaginary parts:

$$\mathbf{A}\mathbf{a} = \sigma\mathbf{a} - \omega\mathbf{b}, \qquad \mathbf{A}\mathbf{b} = \omega\mathbf{a} + \sigma\mathbf{b}.$$

With $\mathbf{P} = \begin{bmatrix}\mathbf{a} & \mathbf{b}\end{bmatrix}$ (real and invertible, since $\mathbf{a}, \mathbf{b}$ are independent when $\omega \neq 0$), these two lines say

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{bmatrix} \sigma & \omega \\ -\omega & \sigma \end{bmatrix}.$$

This **real block form** is a rotation-scaling matrix. For an $n \times n$ real matrix, each complex pair contributes such a 2×2 block and each real eigenvalue a 1×1 block; the result is block-diagonal and entirely real. In state-space terms the two real coordinates of a block are the in-phase and quadrature components of an oscillating mode.

When diagonalization fails: the Jordan form

A defective matrix has too few eigenvectors to form a basis. The best that similarity can achieve is the **Jordan canonical form**: $\mathbf{A} = \mathbf{P}\mathbf{J}\mathbf{P}^{-1}$ with $\mathbf{J}$ block-diagonal, each block of the form

$$\mathbf{J}_k(\lambda) = \begin{bmatrix} \lambda & 1 & & \\ & \lambda & \ddots & \\ & & \ddots & 1 \\ & & & \lambda \end{bmatrix} \quad (k \times k).$$

The columns of $\mathbf{P}$ for a 2×2 Jordan block are an eigenvector $\mathbf{v}$ and a **generalized eigenvector** $\mathbf{w}$ satisfying

$$(\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}, \qquad (\mathbf{A} - \lambda\mathbf{I})\mathbf{w} = \mathbf{v}.$$

Then $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ and $\mathbf{A}\mathbf{w} = \mathbf{v} + \lambda\mathbf{w}$, which is the column statement of $\mathbf{A}\mathbf{P} = \mathbf{P}\mathbf{J}_2(\lambda)$. Jordan forms are numerically fragile — an arbitrarily small perturbation makes a defective matrix diagonalizable with nearly parallel eigenvectors — so they are a theoretical tool rather than a computational one. Their practical message is the appearance of polynomial-times-exponential terms, $te^{\lambda t}$ and $k\lambda^{k-1}$ (Section 10.4).

Simultaneous diagonalization

Two diagonalizable matrices $\mathbf{A}$ and $\mathbf{B}$ can be diagonalized by the **same** $\mathbf{P}$ if and only if they commute, $\mathbf{A}\mathbf{B} = \mathbf{B}\mathbf{A}$. One direction is immediate: diagonal matrices commute, so $\mathbf{P}\mathbf{D}_1\mathbf{P}^{-1}\mathbf{P}\mathbf{D}_2\mathbf{P}^{-1} = \mathbf{P}\mathbf{D}_2\mathbf{D}_1\mathbf{P}^{-1}$. This is why, for instance, all circulant matrices (which commute with the cyclic shift) are diagonalized by the same matrix — the DFT matrix — a fact that underlies fast convolution (Ch. 16, Ch. 24).

Worked examples

Worked Example 10.9 — Diagonalizing a 2×2 matrix

Diagonalize $\mathbf{A} = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$.

Step 1 — eigenpairs (Examples 10.1, 10.5): $(5, (1,1))$ and $(2, (1,-2))$.

Step 2 — assemble.

$$\mathbf{P} = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}, \qquad \mathbf{D} = \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix}.$$

Step 3 — invert $\mathbf{P}$. $\det \mathbf{P} = -2 - 1 = -3$, so

$$\mathbf{P}^{-1} = \frac{1}{-3}\begin{bmatrix} -2 & -1 \\ -1 & 1 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}.$$

Step 4 — verify.

$$\mathbf{P}\mathbf{D}\mathbf{P}^{-1} = \begin{bmatrix} 5 & 2 \\ 5 & -4 \end{bmatrix} \cdot \frac{1}{3}\begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 12 & 3 \\ 6 & 9 \end{bmatrix} = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} \checkmark$$

Remarks. Reordering the columns of $\mathbf{P}$ (and correspondingly the diagonal of $\mathbf{D}$) or rescaling any column of $\mathbf{P}$ gives an equally valid factorization; software might return $\mathbf{P} = \begin{bmatrix} -1 & 1 \\ 2 & 1 \end{bmatrix}$, $\mathbf{D} = \operatorname{diag}(2, 5)$. The rows of $\mathbf{P}^{-1}$, $(2,1)/3$ and $(1,-1)/3$, are left eigenvectors of $\mathbf{A}$: indeed $(2,1)\mathbf{A} = (10, 5) = 5(2,1)$.

Worked Example 10.10 — A 3×3 matrix with a repeated eigenvalue

Diagonalize $\mathbf{L} = \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix}$ from Example 10.6.

Step 1 — check diagonalizability. $m(2) = 2 = g(2)$ and $m(9) = 1 = g(9)$, so the eigenspace dimensions sum to 3: diagonalizable.

Step 2 — assemble (eigenvectors in the order $2, 2, 9$):

$$\mathbf{P} = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 6 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \qquad \mathbf{D} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 9 \end{bmatrix}.$$

Step 3 — invert. $\det \mathbf{P} = 1(6-1) - 0 + 1(2-0) = 7$, and the adjugate gives

$$\mathbf{P}^{-1} = \frac{1}{7}\begin{bmatrix} 5 & 1 & -6 \\ -2 & 1 & 1 \\ 2 & -1 & 6 \end{bmatrix}.$$

Step 4 — verify without inverting (cheaper and just as conclusive): check $\mathbf{L}\mathbf{P} = \mathbf{P}\mathbf{D}$.

$$\mathbf{L}\mathbf{P} = \begin{bmatrix} 4-2 & 0-6+6 & 4-1+6 \\ 2+2 & 0+6+6 & 2+1+6 \\ 2-2 & 0-6+8 & 2-1+8 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 9 \\ 4 & 12 & 9 \\ 0 & 2 & 9 \end{bmatrix} = \mathbf{P}\mathbf{D} \checkmark$$

(The columns of $\mathbf{P}\mathbf{D}$ are $2\times$, $2\times$ and $9\times$ the columns of $\mathbf{P}$.)

Worked Example 10.11 — A defective matrix and its Jordan form

Show that $\mathbf{A} = \begin{bmatrix} 3 & 1 \\ -1 & 1 \end{bmatrix}$ is not diagonalizable and find its Jordan form.

Step 1 — eigenvalues. $\operatorname{tr} \mathbf{A} = 4$, $\det \mathbf{A} = 3 + 1 = 4$: $\lambda^2 - 4\lambda + 4 = (\lambda - 2)^2$. Double eigenvalue $\lambda = 2$, $m = 2$.

Step 2 — eigenspace. $\mathbf{A} - 2\mathbf{I} = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$ has rank 1, so $g = 1 < m$. Defective: not diagonalizable. Eigenvector $\mathbf{v} = (1, -1)$.

Step 3 — generalized eigenvector. Solve $(\mathbf{A} - 2\mathbf{I})\mathbf{w} = \mathbf{v}$: $w_1 + w_2 = 1$. Take $\mathbf{w} = (1, 0)$.

Step 4 — Jordan form.

$$\mathbf{P} = \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}, \qquad \mathbf{J} = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}, \qquad \mathbf{A} = \mathbf{P}\mathbf{J}\mathbf{P}^{-1}.$$

Check $\mathbf{A}\mathbf{P} = \begin{bmatrix} 2 & 3 \\ -2 & -1 \end{bmatrix}$ and $\mathbf{P}\mathbf{J} = \begin{bmatrix} 2 & 1+2 \\ -2 & -1+0 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ -2 & -1 \end{bmatrix}$ ✓.

Consequence. The system $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ has solutions containing te^{2t}; indeed (Section 10.4) $e^{\mathbf{A}t} = e^{2t}\begin{bmatrix} 1+t & t \\ -t & 1-t \end{bmatrix}$. In circuit language: a critically damped response.

Worked Example 10.12 — Real block form of a rotation-scaling matrix

Bring $\mathbf{R} = \begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}$ (Example 10.7, eigenvalues $2 \pm j$) to real block form.

Step 1 — split the eigenvector. For $\lambda = 2 + j$, $\mathbf{v} = (-1 + j,\ 1) = \underbrace{(-1, 1)}_{\mathbf{a}} + j\,\underbrace{(1, 0)}_{\mathbf{b}}$.

Step 2 — form $\mathbf{P} = [\mathbf{a}\ \mathbf{b}]$. $\mathbf{P} = \begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix}$, $\det \mathbf{P} = -1$, $\mathbf{P}^{-1} = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$.

Step 3 — compute.

$$\mathbf{RP} = \begin{bmatrix} -3 & 1 \\ 2 & 1 \end{bmatrix}, \qquad \mathbf{P}^{-1}\mathbf{RP} = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} -3 & 1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} \sigma & \omega \\ -\omega & \sigma \end{bmatrix}$$

with $\sigma = 2$, $\omega = 1$, as predicted. In the coordinates defined by $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{R}$ is exactly "scale by $\sqrt{5}$, rotate by $26.57°$" (clockwise, given this sign arrangement).

Common Pitfall — mismatched order, and "distinct" misread

The i-th column of $\mathbf{P}$ must be the eigenvector for the i-th diagonal entry of $\mathbf{D}$; swapping one without the other produces a matrix that is not $\mathbf{A}$. Also, "distinct eigenvalues $\Rightarrow$ diagonalizable" is one-way: a repeated eigenvalue does **not** imply non-diagonalizable. Always compute $\operatorname{rank}(\mathbf{A} - \lambda\mathbf{I})$ for each repeated eigenvalue before deciding.

Engineering Note — nearly defective means sensitive

Matrices that are close to defective have eigenvector matrices $\mathbf{P}$ that are close to singular (nearly parallel eigenvectors). The condition number $\kappa(\mathbf{P})$ then bounds how much eigenvalues can move under a perturbation of $\mathbf{A}$ (the Bauer–Fike theorem: each eigenvalue of $\mathbf{A} + \mathbf{E}$ lies within $\kappa(\mathbf{P})\|\mathbf{E}\|$ of an eigenvalue of $\mathbf{A}$). In control design, closed-loop poles placed with a nearly defective $\mathbf{A}$ are very sensitive to component tolerances, and large $\kappa(\mathbf{P})$ also predicts large transient overshoot before an eventually stable response decays (Section 10.9). Symmetric matrices, whose $\mathbf{P}$ can be orthogonal with $\kappa = 1$, are the opposite extreme.

10.4 Matrix powers

Motivation

Many engineering processes advance in steps: a digital filter updates its internal state once per sample, an iterative solver refines its estimate once per sweep, a Markov model of a communication channel moves between "good" and "bad" states once per packet, a power-flow solver repeats a Newton step. If each step is linear, $\mathbf{x}[k+1] = \mathbf{A}\mathbf{x}[k]$, then after k steps

$$\mathbf{x}[k] = \mathbf{A}^k \mathbf{x}[0].$$

Everything about the long-run behaviour — growth, decay, oscillation, convergence to a steady state — is encoded in $\mathbf{A}^k$. Multiplying out $\mathbf{A}^k$ directly costs $k-1$ matrix products and reveals nothing about the trend. Diagonalization turns it into n scalar powers and makes the trend obvious. The same machinery then gives the **matrix exponential** $e^{\mathbf{A}t}$, which plays for continuous-time systems $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ exactly the role that $\mathbf{A}^k$ plays for discrete-time ones.

Powers by diagonalization

If $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$, the inner factors cancel in pairs:

$$\mathbf{A}^k = (\mathbf{P}\mathbf{D}\mathbf{P}^{-1})(\mathbf{P}\mathbf{D}\mathbf{P}^{-1})\cdots(\mathbf{P}\mathbf{D}\mathbf{P}^{-1}) = \mathbf{P}\mathbf{D}^k\mathbf{P}^{-1}, \qquad \mathbf{D}^k = \begin{bmatrix} \lambda_1^k & & \\ & \ddots & \\ & & \lambda_n^k \end{bmatrix}.$$

The formula also holds for negative k when $\mathbf{A}$ is invertible, and suggests a definition of $\mathbf{A}^{1/2}$ when all $\lambda_i \geq 0$ (Section 10.6).

An equivalent and often more illuminating form expands the initial state in the eigenbasis. If $\mathbf{x}[0] = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n$, i.e. $\mathbf{c} = \mathbf{P}^{-1}\mathbf{x}[0]$, then

$$\mathbf{x}[k] = \mathbf{A}^k\mathbf{x}[0] = c_1\lambda_1^k\mathbf{v}_1 + c_2\lambda_2^k\mathbf{v}_2 + \cdots + c_n\lambda_n^k\mathbf{v}_n.$$

Each eigen-direction evolves independently, multiplied by λ_i at every step. This **modal expansion** answers the qualitative questions at a glance:

- terms with $|\lambda_i| < 1$ die out geometrically, those with $|\lambda_i| > 1$ grow, those with $|\lambda_i| = 1$ persist;
- a negative real λ_i makes its term alternate in sign every step; a complex pair $re^{\pm j\theta}$ makes the pair rotate by θ per step while scaling by r;
- for large k, the term with the largest $|\lambda_i|$ (if unique, and if $c_i \neq 0$) **dominates**: $\mathbf{x}[k] \approx c_1\lambda_1^k\mathbf{v}_1$, so the state lines up with the dominant eigenvector and grows or decays like $\rho(\mathbf{A})^k$.

The last observation is the **power method** for computing a dominant eigenpair: start from any vector, multiply repeatedly by $\mathbf{A}$ and normalise. It converges at the rate $|\lambda_2/\lambda_1|^k$ and is the basis of Google's PageRank and of many large-scale eigen-solvers (Ch. 22).

Powers of a Jordan block and the spectral-radius theorem

For a defective matrix we use the Jordan form instead. Write $\mathbf{J}_2(\lambda) = \lambda\mathbf{I} + \mathbf{N}$ with $\mathbf{N} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $\mathbf{N}^2 = \mathbf{0}$. Because $\lambda\mathbf{I}$ commutes with $\mathbf{N}$, the binomial theorem applies and stops after two terms:

$$\mathbf{J}_2(\lambda)^k = \lambda^k \mathbf{I} + k\lambda^{k-1}\mathbf{N} = \begin{bmatrix} \lambda^k & k\lambda^{k-1} \\ 0 & \lambda^k \end{bmatrix}.$$

For an $m \times m$ block, entries $\binom{k}{i}\lambda^{k-i}$ appear for $i < m$. Polynomial factors like k are eventually overwhelmed by $|\lambda|^k$ when $|\lambda| < 1$, but they can make $\mathbf{A}^k$ grow substantially before it decays. Combining the diagonalizable and Jordan cases:

Convergence of matrix powers

For any square matrix $\mathbf{A}$:

1. $\mathbf{A}^k \to \mathbf{0}$ as $k \to \infty$ **if and only if** $\rho(\mathbf{A}) < 1$ (all eigenvalues strictly inside the unit circle).
2. If $\rho(\mathbf{A}) > 1$, $\|\mathbf{A}^k \mathbf{x}\| \to \infty$ for almost every starting vector $\mathbf{x}$.
3. If $\rho(\mathbf{A}) = 1$, the powers stay bounded if and only if every eigenvalue with $|\lambda| = 1$ is non-defective.
4. (Gelfand) For any matrix norm, $\|\mathbf{A}^k\|^{1/k} \to \rho(\mathbf{A})$.

Part 1 is the stability condition for discrete-time linear systems (Section 10.9) and the convergence condition for stationary iterative solvers such as Jacobi and Gauss–Seidel, whose error obeys $\mathbf{e}[k+1] = \mathbf{G}\mathbf{e}[k]$ for an iteration matrix $\mathbf{G}$.

The matrix exponential

For a scalar, $\dot{x} = ax$ has solution $x(t) = e^{at}x(0)$. For the system $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ we want a matrix that does the same job. Define, by analogy with the Taylor series of Ch. 4,

$$e^{\mathbf{A}t} = \mathbf{I} + \mathbf{A}t + \frac{(\mathbf{A}t)^2}{2!} + \frac{(\mathbf{A}t)^3}{3!} + \cdots = \sum_{k=0}^{\infty} \frac{\mathbf{A}^k t^k}{k!}.$$

The series converges for every square $\mathbf{A}$ and every t (each entry is bounded by the scalar series for $e^{\|\mathbf{A}\|t}$). Differentiating term by term,

$$\frac{d}{dt}e^{\mathbf{A}t} = \mathbf{A} + \mathbf{A}^2 t + \frac{\mathbf{A}^3 t^2}{2!} + \cdots = \mathbf{A}e^{\mathbf{A}t} = e^{\mathbf{A}t}\mathbf{A},$$

so $\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}(0)$ satisfies $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ with the right initial value; by uniqueness (Ch. 11) it is *the* solution. In control texts $e^{\mathbf{A}t}$ is called the **state-transition matrix** $\mathbf{\Phi}(t)$.

Properties of the matrix exponential

For square $\mathbf{A}, \mathbf{B}$ and scalars t, s:

1. $e^{\mathbf{0}} = \mathbf{I}$ and $e^{\mathbf{A}(t+s)} = e^{\mathbf{A}t}e^{\mathbf{A}s}$;
2. $e^{\mathbf{A}t}$ is always invertible, with $(e^{\mathbf{A}t})^{-1} = e^{-\mathbf{A}t}$;
3. if $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$ then $e^{\mathbf{A}t} = \mathbf{P}e^{\mathbf{D}t}\mathbf{P}^{-1}$ with $e^{\mathbf{D}t} = \operatorname{diag}(e^{\lambda_1 t}, \dots, e^{\lambda_n t})$;
4. the eigenvalues of $e^{\mathbf{A}t}$ are $e^{\lambda_i t}$, so $\det e^{\mathbf{A}t} = e^{(\operatorname{tr}\mathbf{A})t}$;
5. $e^{(\mathbf{A}+\mathbf{B})t} = e^{\mathbf{A}t}e^{\mathbf{B}t}$ **if** $\mathbf{A}\mathbf{B} = \mathbf{B}\mathbf{A}$ — but not in general.

Property 3 follows from $\mathbf{A}^k = \mathbf{P}\mathbf{D}^k\mathbf{P}^{-1}$ applied term by term. With the modal expansion $\mathbf{x}(0) = \sum c_i \mathbf{v}_i$ it gives the continuous-time analogue of the discrete result:

$$\mathbf{x}(t) = c_1 e^{\lambda_1 t}\mathbf{v}_1 + c_2 e^{\lambda_2 t}\mathbf{v}_2 + \cdots + c_n e^{\lambda_n t}\mathbf{v}_n.$$

Now the dividing line is $\operatorname{Re}\lambda_i = 0$ instead of $|\lambda_i| = 1$: $e^{\lambda t}$ decays if $\operatorname{Re}\lambda < 0$. The two pictures are linked by sampling: if $\mathbf{x}(t)$ is sampled every T seconds, $\mathbf{x}[k+1] = e^{\mathbf{A}T}\mathbf{x}[k]$, and the eigenvalue λ maps to $z = e^{\lambda T}$ — the left half of the s-plane maps to the inside of the unit circle of the z-plane, exactly the $s \to z$ correspondence of Ch. 17.

For a Jordan block, $e^{\mathbf{J}_2(\lambda)t} = e^{\lambda t}e^{\mathbf{N}t} = e^{\lambda t}(\mathbf{I} + \mathbf{N}t) = e^{\lambda t}\begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}$ — the origin of the $te^{\lambda t}$ terms. A rotation generator gives the familiar oscillation:

$$\exp\left(\begin{bmatrix} 0 & -\omega \\ \omega & 0 \end{bmatrix} t\right) = \begin{bmatrix} \cos\omega t & -\sin\omega t \\ \sin\omega t & \cos\omega t \end{bmatrix},$$

the real-matrix form of $e^{j\omega t}$ — a rotating phasor.

Functions of a matrix via Cayley–Hamilton (Sylvester interpolation)

By Cayley–Hamilton, every power $\mathbf{A}^k$ with $k \geq n$ reduces to a combination of $\mathbf{I}, \mathbf{A}, \dots, \mathbf{A}^{n-1}$, and so does any convergent power series. Hence for an analytic function f (such as λ^k or $e^{\lambda t}$),

$$f(\mathbf{A}) = \alpha_0\mathbf{I} + \alpha_1\mathbf{A} + \cdots + \alpha_{n-1}\mathbf{A}^{n-1}$$

for some scalars α_i. Applying both sides to an eigenvector shows that the polynomial $r(\lambda) = \alpha_0 + \alpha_1\lambda + \cdots + \alpha_{n-1}\lambda^{n-1}$ must agree with f at each eigenvalue:

$$f(\lambda_i) = r(\lambda_i), \qquad i = 1, \dots, n.$$

With distinct eigenvalues this is n linear equations for the n unknowns α_i. For an eigenvalue of multiplicity m, also match the first $m-1$ derivatives, $f^{(q)}(\lambda_i) = r^{(q)}(\lambda_i)$. This method avoids eigenvectors altogether and is convenient for 2×2 and 3×3 hand calculations.

Worked examples

Worked Example 10.13 — A tenth power

Compute $\mathbf{A}^{10}$ for $\mathbf{A} = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$.

Step 1 — use Example 10.9. $\mathbf{P} = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}$, $\mathbf{D} = \operatorname{diag}(5, 2)$, $\mathbf{P}^{-1} = \frac{1}{3}\begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}$.

Step 2 — multiply.

$$\mathbf{A}^{10} = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 5^{10} & 0 \\ 0 & 2^{10} \end{bmatrix} \frac{1}{3} \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 \cdot 5^{10} + 2^{10} & 5^{10} - 2^{10} \\ 2 \cdot 5^{10} - 2 \cdot 2^{10} & 5^{10} + 2 \cdot 2^{10} \end{bmatrix}.$$

Step 3 — numbers. $5^{10} = 9\,765\,625$, $2^{10} = 1024$:

$$\mathbf{A}^{10} = \begin{bmatrix} 6\,510\,758 & 3\,254\,867 \\ 6\,509\,734 & 3\,255\,891 \end{bmatrix}.$$

Checks. $\operatorname{tr} \mathbf{A}^{10} = 9\,766\,649 = 5^{10} + 2^{10}$ ✓. The general formula at $k = 1$ returns $\frac{1}{3}\begin{bmatrix} 12 & 3 \\ 6 & 9 \end{bmatrix} = \mathbf{A}$ ✓. Notice how the 5^{10} term dominates: the columns of $\mathbf{A}^{10}$ are both nearly parallel to $\mathbf{v}_1 = (1, 1)$ — the power method at work.

Sylvester alternative. Write $\mathbf{A}^{10} = \alpha_0 \mathbf{I} + \alpha_1 \mathbf{A}$ with $5^{10} = \alpha_0 + 5\alpha_1$, $2^{10} = \alpha_0 + 2\alpha_1$. Then $\alpha_1 = (5^{10} - 2^{10})/3 = 3\,254\,867$ and $\alpha_0 = 2^{10} - 2\alpha_1 = -6\,508\,710$. The $(1, 1)$ entry is $\alpha_0 + 4\alpha_1 = 6\,510\,758$ ✓.

Worked Example 10.14 — Fibonacci numbers in closed form

The recursion $F_{k+1} = F_k + F_{k-1}$, $F_0 = 0$, $F_1 = 1$, is the prototype of a second-order linear difference equation (the same structure as an IIR filter with no input). With $\mathbf{x}[k] = (F_{k+1}, F_k)$,

$$\mathbf{x}[k+1] = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \mathbf{x}[k] = \mathbf{F}\mathbf{x}[k], \qquad \mathbf{x}[0] = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Step 1 — eigenvalues. $\lambda^2 - \lambda - 1 = 0$: $\varphi = \frac{1+\sqrt{5}}{2} \approx 1.618$ and $\psi = \frac{1-\sqrt{5}}{2} \approx -0.618$.

Step 2 — eigenvectors. $\mathbf{F} - \lambda \mathbf{I} = \begin{bmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{bmatrix}$; the second row gives $\mathbf{v} = (\lambda, 1)$. So $\mathbf{v}_\varphi = (\varphi, 1)$, $\mathbf{v}_\psi = (\psi, 1)$.

Step 3 — expand the initial state. $(1, 0) = c_1(\varphi, 1) + c_2(\psi, 1)$ gives $c_2 = -c_1$ and $c_1(\varphi - \psi) = 1$, so $c_1 = 1/\sqrt{5}$.

Step 4 — read off the second component.

$$F_k = c_1 \varphi^k + c_2 \psi^k = \frac{\varphi^k - \psi^k}{\sqrt{5}} \qquad \text{(Binet's formula)}.$$

Check. $F_{10} = (1.618034^{10} - (-0.618034)^{10})/\sqrt{5} = (122.9919 - 0.0081)/2.23607 = 55.000$ ✓, and directly $\mathbf{F}^{10} = \begin{bmatrix} 89 & 55 \\ 55 & 34 \end{bmatrix}$. Because $|\psi| < 1$, F_k is the integer nearest to $\varphi^k/\sqrt{5}$, and $F_{k+1}/F_k \to \varphi$: the dominant eigenvalue sets the growth rate.

Worked Example 10.15 — A two-state Markov channel and its steady state

A burst-error channel (Gilbert–Elliott model) is either Good (G) or Bad (B) during each packet. From G it stays G with probability 0.9; from B it returns to G with probability 0.4. With $\mathbf{x}[k] = (P(\mathrm{G}), P(\mathrm{B}))$ at packet k,

$$\mathbf{x}[k+1] = \mathbf{M}\mathbf{x}[k], \qquad \mathbf{M} = \begin{bmatrix} 0.9 & 0.4 \\ 0.1 & 0.6 \end{bmatrix}$$

(each column sums to 1). Find $\mathbf{x}[k]$ if the channel starts Bad, and the long-run fraction of Good packets.

Step 1 — eigenvalues. $\operatorname{tr} \mathbf{M} = 1.5$, $\det \mathbf{M} = 0.54 - 0.04 = 0.5$; $\lambda^2 - 1.5\lambda + 0.5 = (\lambda - 1)(\lambda - 0.5)$. A column-stochastic matrix always has $\lambda = 1$, because $(1, 1)\mathbf{M} = (1, 1)$: the all-ones row is a left eigenvector.

Step 2 — eigenvectors. $\lambda = 1$: $\mathbf{M} - \mathbf{I} = \begin{bmatrix} -0.1 & 0.4 \\ 0.1 & -0.4 \end{bmatrix}$ gives $\mathbf{v}_1 \propto (4, 1)$, normalised as a probability vector $\boldsymbol{\pi} = (0.8, 0.2)$. $\lambda = 0.5$: $\mathbf{M} - 0.5\mathbf{I} = \begin{bmatrix} 0.4 & 0.4 \\ 0.1 & 0.1 \end{bmatrix}$ gives $\mathbf{v}_2 = (1, -1)$ (entries sum to zero, as they must for a mode that does not change total probability).

Step 3 — expand $\mathbf{x}[0] = (0, 1)$. $(0, 1) = \boldsymbol{\pi} + c_2(1, -1)$ forces $c_2 = -0.8$.

Step 4 — solution.

$$\mathbf{x}[k] = \begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix} - 0.8\,(0.5)^k \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \qquad P_k(\mathrm{G}) = 0.8\big(1 - 0.5^k\big).$$

Check. $k = 1$: $P(\mathrm{G}) = 0.4$, matching one step from B ✓. $k = 10$: $0.8(1 - 1/1024) = 0.79922$ (numerical $\mathbf{M}^{10}(0, 1)^{\mathsf{T}}$ agrees). Long run: **80 % of packets are Good**, whatever the initial state. The second eigenvalue 0.5 sets the **mixing rate**: the memory of the initial state halves every packet. A channel with λ_2 close to 1 has long error bursts — which is why interleaver depth is chosen from the second eigenvalue.

Worked Example 10.16 — The state-transition matrix $e^{\mathbf{A}t}$

Compute $e^{\mathbf{A}t}$ for $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$ (the companion matrix of $\ddot{y} + 3\dot{y} + 2y = 0$) and find $\mathbf{x}(t)$ for $\mathbf{x}(0) = (1, 0)$.

Method 1 — diagonalization. $\lambda^2 + 3\lambda + 2 = (\lambda + 1)(\lambda + 2)$. Eigenvectors of a companion matrix are $(1, \lambda)$: $\mathbf{v}_1 = (1, -1)$, $\mathbf{v}_2 = (1, -2)$. With $\mathbf{P} = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix}$, $\det \mathbf{P} = -1$, $\mathbf{P}^{-1} = \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}$:

$$e^{\mathbf{A}t} = \mathbf{P} \begin{bmatrix} e^{-t} & 0 \\ 0 & e^{-2t} \end{bmatrix} \mathbf{P}^{-1} = \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix}.$$

Method 2 — Sylvester. $e^{\mathbf{A}t} = \alpha_0 \mathbf{I} + \alpha_1 \mathbf{A}$ with $e^{-t} = \alpha_0 - \alpha_1$ and $e^{-2t} = \alpha_0 - 2\alpha_1$, so $\alpha_1 = e^{-t} - e^{-2t}$, $\alpha_0 = 2e^{-t} - e^{-2t}$. Then $e^{\mathbf{A}t} = \begin{bmatrix} \alpha_0 & \alpha_1 \\ -2\alpha_1 & \alpha_0 - 3\alpha_1 \end{bmatrix}$, identical to Method 1 (e.g. $\alpha_0 - 3\alpha_1 = -e^{-t} + 2e^{-2t}$ ✓).

Response. $\mathbf{x}(t) = e^{\mathbf{A}t}(1, 0)^{\mathsf{T}}$ is the first column: $y(t) = 2e^{-t} - e^{-2t}$, $\dot{y}(t) = -2e^{-t} + 2e^{-2t}$.

Checks. $t = 0$ gives $\mathbf{I}$ ✓. $\det e^{\mathbf{A}t} = e^{-3t} = e^{(\operatorname{tr}\mathbf{A})t}$ (expand: $(2e^{-t} - e^{-2t})(-e^{-t} + 2e^{-2t}) + (e^{-t} - e^{-2t})(2e^{-t} - 2e^{-2t}) = e^{-3t}$) ✓. At $t = 1$: $2e^{-1} - e^{-2} = 0.6004$ and $e^{-1} - e^{-2} = 0.2325$, matching a numerical `expm` ✓.

Worked Example 10.17 — Exponential of a defective matrix

For the defective $\mathbf{A} = \begin{bmatrix} 3 & 1 \\ -1 & 1 \end{bmatrix}$ of Example 10.11 (double eigenvalue 2), Sylvester's method needs the derivative condition. Write $e^{\mathbf{A}t} = \alpha_0 \mathbf{I} + \alpha_1 \mathbf{A}$ and match $f(\lambda) = e^{\lambda t}$ and $f'(\lambda) = te^{\lambda t}$ (derivative with respect to λ) at $\lambda = 2$:

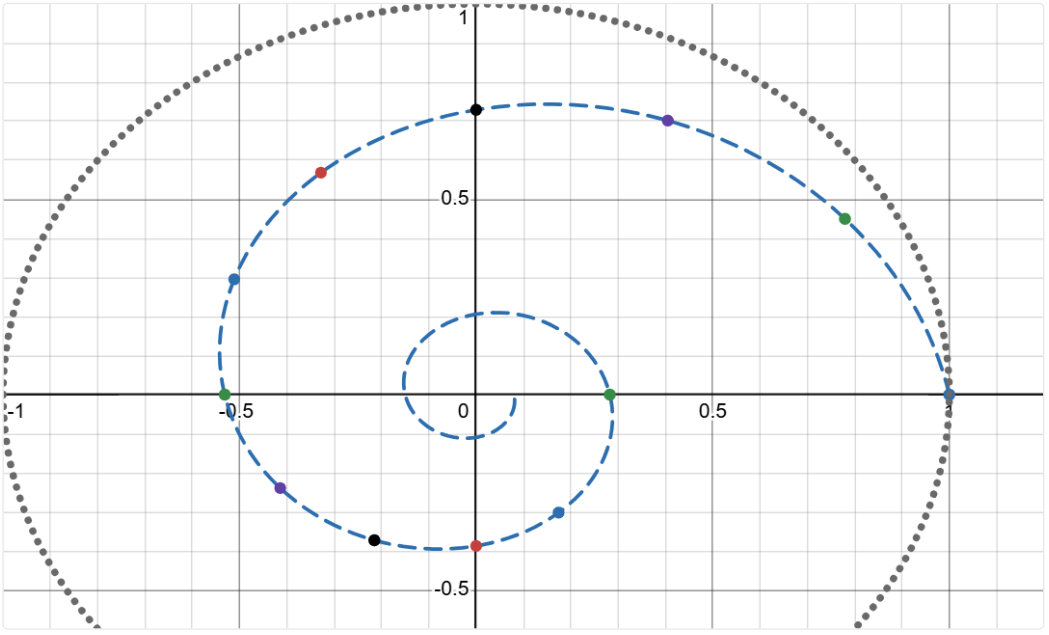
$$e^{2t} = \alpha_0 + 2\alpha_1, \qquad te^{2t} = \alpha_1.$$

So $\alpha_1 = te^{2t}$, $\alpha_0 = (1 - 2t)e^{2t}$, and

$$e^{\mathbf{A}t} = (1 - 2t)e^{2t}\mathbf{I} + te^{2t} \begin{bmatrix} 3 & 1 \\ -1 & 1 \end{bmatrix} = e^{2t} \begin{bmatrix} 1 + t & t \\ -t & 1 - t \end{bmatrix}.$$

Check via Jordan form: $\mathbf{P}e^{\mathbf{J}t}\mathbf{P}^{-1}$ with $e^{\mathbf{J}t} = e^{2t}\begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}$ gives the same matrix, and $\frac{d}{dt}$ at $t = 0$ returns $\begin{bmatrix} 2 + 1 & 1 \\ -1 & 2 - 1 \end{bmatrix} = \mathbf{A}$ ✓.

The graph shows the discrete-time counterpart of a complex eigenvalue pair. For $\mathbf{A} = 0.9 \begin{bmatrix} \cos 30^\circ & -\sin 30^\circ \\ \sin 30^\circ & \cos 30^\circ \end{bmatrix}$ (eigenvalues $0.9e^{\pm j\pi/6}$), the iterates $\mathbf{x}[k] = \mathbf{A}^k(1, 0)$ rotate 30° per step and shrink by 10%; they lie on the logarithmic spiral $r = 0.9^{6\theta/\pi} = e^{-0.2012\theta}$.



Iterates of a discrete-time system with eigenvalues $0.9e^{\pm j\pi/6}$: each step rotates by $\angle\lambda = 30^\circ$ and scales by $|\lambda| = 0.9$, so the state spirals into the origin along $r = e^{-0.2012\theta}$; the dotted unit circle is the stability boundary $|\lambda| = 1$.

Common Pitfall — exponentiating entry by entry

$e^{\mathbf{A}t}$ is **not** the matrix of $e^{a_{ij}t}$ (except for diagonal $\mathbf{A}$), and $\mathbf{A}^k$ is not the matrix of a_{ij}^k. Equally, $e^{(\mathbf{A}+\mathbf{B})t} \neq e^{\mathbf{A}t}e^{\mathbf{B}t}$ unless $\mathbf{A}$ and $\mathbf{B}$ commute — which matters when a state matrix is split into "nominal" and "perturbation" parts, or when switching between circuit topologies in a switched-mode converter.

Engineering Note — computing $e^{\mathbf{A}T}$ in practice

Discretising a continuous-time model for a digital controller or a circuit simulator needs $\mathbf{\Phi} = e^{\mathbf{A}T}$ for the sample period T. Diagonalization is fine for small, well-conditioned problems, but robust software (MATLAB `expm`, SciPy `scipy.linalg.expm`) uses **scaling and squaring** with a Padé approximant: compute $e^{\mathbf{A}T/2^s}$ for an s that makes the argument small, then square s times. This avoids both the ill-conditioning of nearly defective $\mathbf{P}$ and the catastrophic cancellation of the raw Taylor series for large $\|\mathbf{A}T\|$ (Moler and Van Loan's classic survey lists nineteen methods and why most of them are dubious).

10.5 Symmetric matrices

Motivation

A remarkable fraction of the matrices an engineer meets are symmetric, $\mathbf{A}^{\mathsf{T}} = \mathbf{A}$, or their complex cousins, Hermitian, $\mathbf{A}^{\mathsf{H}} = \mathbf{A}$. The nodal conductance matrix of a resistive network is symmetric (reciprocity: the current at node j due to a voltage at node i equals the current at i due to the same voltage at j). Inductance and capacitance matrices of coupled conductors are symmetric because they come from stored energy. Covariance and correlation matrices of noise are symmetric (real signals) or Hermitian (complex baseband). Hessian matrices of second derivatives are symmetric (Ch. 6). The impedance matrix of a lossless reciprocal network at a given frequency is j times a real symmetric matrix.

For all of these, eigen-theory is as good as it can possibly be: the eigenvalues are real, the eigenvectors are orthogonal, and diagonalization never fails. This section proves these facts and draws out their geometry.

Real eigenvalues and orthogonal eigenvectors

Symmetric and Hermitian matrices: real eigenvalues

If $\mathbf{A}$ is Hermitian ($\mathbf{A}^{\mathsf{H}} = \mathbf{A}$; in particular if $\mathbf{A}$ is real symmetric), every eigenvalue of $\mathbf{A}$ is real.

Proof. Let $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$, possibly complex. Multiply on the left by $\mathbf{v}^{\mathsf{H}}$: $\mathbf{v}^{\mathsf{H}}\mathbf{A}\mathbf{v} = \lambda\,\mathbf{v}^{\mathsf{H}}\mathbf{v} = \lambda\|\mathbf{v}\|^2$. The scalar $\mathbf{v}^{\mathsf{H}}\mathbf{A}\mathbf{v}$ is real, because it equals its own conjugate: $(\mathbf{v}^{\mathsf{H}}\mathbf{A}\mathbf{v})^* = (\mathbf{v}^{\mathsf{H}}\mathbf{A}\mathbf{v})^{\mathsf{H}} = \mathbf{v}^{\mathsf{H}}\mathbf{A}^{\mathsf{H}}\mathbf{v} = \mathbf{v}^{\mathsf{H}}\mathbf{A}\mathbf{v}$. Since $\|\mathbf{v}\|^2 > 0$ is real too, $\lambda = \mathbf{v}^{\mathsf{H}}\mathbf{A}\mathbf{v}/\|\mathbf{v}\|^2$ is real. $\square$

For a real symmetric matrix, real λ means $\mathbf{A} - \lambda\mathbf{I}$ is real, so the eigenvectors can be chosen real.

Orthogonality of eigenvectors

Eigenvectors of a Hermitian matrix belonging to distinct eigenvalues are orthogonal.

Proof. Let $\mathbf{A}\mathbf{v}_1 = \lambda_1\mathbf{v}_1$, $\mathbf{A}\mathbf{v}_2 = \lambda_2\mathbf{v}_2$, $\lambda_1 \neq \lambda_2$ (both real). Then

$$\lambda_1\,\mathbf{v}_2^{\mathsf{H}}\mathbf{v}_1 = \mathbf{v}_2^{\mathsf{H}}\mathbf{A}\mathbf{v}_1 = (\mathbf{A}^{\mathsf{H}}\mathbf{v}_2)^{\mathsf{H}}\mathbf{v}_1 = (\mathbf{A}\mathbf{v}_2)^{\mathsf{H}}\mathbf{v}_1 = \lambda_2\,\mathbf{v}_2^{\mathsf{H}}\mathbf{v}_1,$$

so $(\lambda_1 - \lambda_2)\langle\mathbf{v}_1, \mathbf{v}_2\rangle = 0$ and $\langle\mathbf{v}_1, \mathbf{v}_2\rangle = 0$. $\square$

Within a repeated eigenvalue's eigenspace the eigenvectors are not automatically orthogonal, but the eigenspace is a subspace, so Gram–Schmidt (Ch. 9) produces an orthonormal basis of it — and the orthogonalised vectors are still eigenvectors.

The spectral theorem

Spectral theorem

A real $n \times n$ matrix $\mathbf{A}$ is symmetric **if and only if** it can be factored as

$$\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^{\mathsf{T}},$$

where $\mathbf{\Lambda} = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$ is real and $\mathbf{Q}$ is **orthogonal** ($\mathbf{Q}^{\mathsf{T}}\mathbf{Q} = \mathbf{I}$, so $\mathbf{Q}^{-1} = \mathbf{Q}^{\mathsf{T}}$) with orthonormal eigenvectors $\mathbf{q}_1, \dots, \mathbf{q}_n$ as columns. Likewise a complex matrix is Hermitian if and only if $\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^{\mathsf{H}}$ with $\mathbf{U}$ unitary ($\mathbf{U}^{\mathsf{H}}\mathbf{U} = \mathbf{I}$) and $\mathbf{\Lambda}$ real.

Proof sketch. "If": $(\mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^{\mathsf{T}})^{\mathsf{T}} = \mathbf{Q}\mathbf{\Lambda}^{\mathsf{T}}\mathbf{Q}^{\mathsf{T}} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^{\mathsf{T}}$. "Only if", by induction on n: $\mathbf{A}$ has a real eigenvalue λ_1 with a unit eigenvector $\mathbf{q}_1$. Extend $\mathbf{q}_1$ to an orthonormal basis and let $\mathbf{Q}_1$ have these as columns. The first column of $\mathbf{Q}_1^{\mathsf{T}}\mathbf{A}\mathbf{Q}_1$ is $\lambda_1\mathbf{e}_1$; because $\mathbf{Q}_1^{\mathsf{T}}\mathbf{A}\mathbf{Q}_1$ is again symmetric, its first row is $\lambda_1\mathbf{e}_1^{\mathsf{T}}$ too:

$$\mathbf{Q}_1^{\mathsf{T}}\mathbf{A}\mathbf{Q}_1 = \begin{bmatrix} \lambda_1 & \mathbf{0}^{\mathsf{T}} \\ \mathbf{0} & \mathbf{A}_1 \end{bmatrix},$$

with $\mathbf{A}_1$ symmetric of size $n-1$. Apply the induction hypothesis to $\mathbf{A}_1$ and assemble. Note what symmetry bought: the zero block in the *top right*, which is what prevents Jordan blocks. $\square$

So a symmetric matrix is never defective, and its eigenvector matrix is as well-conditioned as a matrix can be: $\kappa(\mathbf{Q}) = 1$.

Spectral decomposition

Multiplying out $\mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^{\mathsf{T}}$ column-by-row gives the **spectral decomposition**

$$\mathbf{A} = \lambda_1\mathbf{q}_1\mathbf{q}_1^{\mathsf{T}} + \lambda_2\mathbf{q}_2\mathbf{q}_2^{\mathsf{T}} + \cdots + \lambda_n\mathbf{q}_n\mathbf{q}_n^{\mathsf{T}}.$$

Each $\mathbf{P}_i = \mathbf{q}_i\mathbf{q}_i^{\mathsf{T}}$ is the orthogonal projector onto the line through $\mathbf{q}_i$ (Ch. 9); these projectors satisfy $\mathbf{P}_i^2 = \mathbf{P}_i$, $\mathbf{P}_i\mathbf{P}_k = \mathbf{0}$ for $i \neq k$ and $\sum_i \mathbf{P}_i = \mathbf{I}$. A symmetric matrix is therefore a weighted sum of mutually perpendicular projections: to apply $\mathbf{A}$, resolve $\mathbf{x}$ into its components along the eigenvectors, scale each by its eigenvalue, and add. Functions of $\mathbf{A}$ follow at once: $f(\mathbf{A}) = \sum_i f(\lambda_i)\mathbf{q}_i\mathbf{q}_i^{\mathsf{T}}$, e.g. $\mathbf{A}^{-1} = \sum \lambda_i^{-1}\mathbf{q}_i\mathbf{q}_i^{\mathsf{T}}$.

Quadratic forms and principal axes

A **quadratic form** is a function $Q(\mathbf{x}) = \mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} = \sum_{i,k} a_{ik}x_i x_k$. Only the symmetric part of $\mathbf{A}$ contributes (since $\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} = \mathbf{x}^{\mathsf{T}}\mathbf{A}^{\mathsf{T}}\mathbf{x}$), so we always take $\mathbf{A}$ symmetric: for $Q = ax^2 + 2bxy + cy^2$, $\mathbf{A} = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$ — note the **half** of the cross coefficient off the diagonal. Energy expressions are quadratic forms: $\tfrac{1}{2}\mathbf{i}^{\mathsf{T}}\mathbf{L}\mathbf{i}$ for coupled inductors, $\tfrac{1}{2}\mathbf{v}^{\mathsf{T}}\mathbf{C}\mathbf{v}$ for a capacitance matrix, $\mathbf{v}^{\mathsf{T}}\mathbf{G}\mathbf{v}$ for power dissipated in a conductance network.

Substituting $\mathbf{x} = \mathbf{Q}\mathbf{y}$ (a rotation, possibly with a reflection) and using $\mathbf{Q}^{\mathsf{T}}\mathbf{A}\mathbf{Q} = \mathbf{\Lambda}$:

$$\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} = \mathbf{y}^{\mathsf{T}}\mathbf{\Lambda}\mathbf{y} = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \cdots + \lambda_n y_n^2.$$

Principal axes theorem

Every real quadratic form can be reduced by an orthogonal change of variables $\mathbf{x} = \mathbf{Q}\mathbf{y}$ to a sum of squares $\sum \lambda_i y_i^2$ with no cross terms. The new axes — the eigenvectors $\mathbf{q}_i$ — are the **principal axes** of the form. For $n = 2$, the level curve $\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} = 1$ is an ellipse with semi-axes $1/\sqrt{\lambda_1}$, $1/\sqrt{\lambda_2}$ along $\mathbf{q}_1, \mathbf{q}_2$ when both eigenvalues are positive, and a hyperbola when they have opposite signs.

The Rayleigh quotient and extremal eigenvalues

Rayleigh quotient

For symmetric $\mathbf{A}$ and $\mathbf{x} \neq \mathbf{0}$, the **Rayleigh quotient** is

$$R(\mathbf{x}) = \frac{\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x}}{\mathbf{x}^{\mathsf{T}}\mathbf{x}}.$$

Expand $\mathbf{x} = \sum y_i \mathbf{q}_i$ with eigenvalues ordered $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$:

$$R(\mathbf{x}) = \frac{\lambda_1 y_1^2 + \cdots + \lambda_n y_n^2}{y_1^2 + \cdots + y_n^2},$$

a weighted average of the eigenvalues with weights $y_i^2 \geq 0$. Hence:

Rayleigh's principle

$$\lambda_{\min} = \lambda_n \leq R(\mathbf{x}) \leq \lambda_1 = \lambda_{\max} \quad \text{for all } \mathbf{x} \neq \mathbf{0},$$

with $\max R = \lambda_1$ attained at $\mathbf{x} = \mathbf{q}_1$ and $\min R = \lambda_n$ at $\mathbf{x} = \mathbf{q}_n$. More generally (Courant–Fischer), λ_k is the maximum of R over vectors orthogonal to $\mathbf{q}_1, \ldots, \mathbf{q}_{k-1}$.

Two consequences are used constantly. First, R **is stationary at every eigenvector**, so an approximate eigenvector with error ε gives an eigenvalue estimate with error of order ε^2 — the reason Rayleigh's method gives excellent natural-frequency estimates from rough guesses of mode shape. Second, maximising a quadratic form on the unit sphere is an eigenvalue problem: the direction of maximum output power of a beamformer, the direction of largest variance in data (principal component analysis), the polarisation that maximises received power — each is the top eigenvector of a suitable symmetric or Hermitian matrix.

Worked examples

Worked Example 10.18 — Orthogonal diagonalization of a 2×2 symmetric matrix

Diagonalize $\mathbf{S} = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$ orthogonally and write its spectral decomposition.

Step 1 — eigenvalues. $\lambda^2 - 6\lambda + 8 = 0$: $\lambda_1 = 4$, $\lambda_2 = 2$.

Step 2 — unit eigenvectors. $\lambda = 4$: $\mathbf{S} - 4\mathbf{I} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$, $\mathbf{q}_1 = \frac{1}{\sqrt{2}}(1, 1)$. $\lambda = 2$: $\mathbf{S} - 2\mathbf{I} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, $\mathbf{q}_2 = \frac{1}{\sqrt{2}}(1, -1)$. They are orthogonal, as guaranteed.

Step 3 — factor.

$$\mathbf{Q} = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad \mathbf{\Lambda} = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}, \quad \mathbf{S} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^{\mathsf{T}}.$$

No matrix inversion was needed: $\mathbf{Q}^{-1} = \mathbf{Q}^{\mathsf{T}}$.

Step 4 — spectral decomposition.

$$\mathbf{S} = 4 \cdot \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + 2 \cdot \frac{1}{2}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} + \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \checkmark$$

Interpretation. This is the matrix of a symmetric two-port or a pair of identical coupled lines. The eigenvectors are the **common mode** $(1, 1)$ and **differential mode** $(1, -1)$ of Ch. 9; any symmetric structure with two identical halves has them as eigenvectors, which is why even/odd-mode analysis works.

Worked Example 10.19 — A repeated eigenvalue and Gram–Schmidt

Orthogonally diagonalize $\mathbf{A} = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$.

Step 1 — eigenvalues. Write $\mathbf{A} = \mathbf{I} + \mathbf{J}$ with $\mathbf{J}$ the all-ones matrix. $\mathbf{J}$ has rank 1, so eigenvalue 0 twice, and $\operatorname{tr}\mathbf{J} = 3$ is its third eigenvalue (eigenvector $(1,1,1)$). Adding 1: the eigenvalues of $\mathbf{A}$ are $4, 1, 1$. (Check: $\operatorname{tr}\mathbf{A} = 6$ ✓; $\det\mathbf{A} = 2(3) - 1(1) + 1(-1) = 4$ ✓.)

Step 2 — $\lambda = 4$. $\mathbf{q}_1 = \frac{1}{\sqrt{3}}(1,1,1)$.

Step 3 — $\lambda = 1$. $\mathbf{A} - \mathbf{I} = \mathbf{J}$, so the eigenspace is the plane $x_1 + x_2 + x_3 = 0$. A natural basis is $\mathbf{w}_1 = (1,-1,0)$, $\mathbf{w}_2 = (1,0,-1)$ — valid eigenvectors but **not orthogonal** ($\mathbf{w}_1 \cdot \mathbf{w}_2 = 1$). Apply Gram–Schmidt inside the eigenspace:

$$\mathbf{q}_2 = \frac{\mathbf{w}_1}{\|\mathbf{w}_1\|} = \frac{1}{\sqrt{2}}(1,-1,0), \qquad \mathbf{w}_2 - (\mathbf{w}_2 \cdot \mathbf{q}_2)\mathbf{q}_2 = (1,0,-1) - \tfrac{1}{2}(1,-1,0) = \left(\tfrac{1}{2}, \tfrac{1}{2}, -1\right),$$

normalised: $\mathbf{q}_3 = \frac{1}{\sqrt{6}}(1,1,-2)$. It still satisfies $x_1 + x_2 + x_3 = 0$, so it is still an eigenvector.

Step 4 — result.

$$\mathbf{Q} = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{3} & -1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{3} & 0 & -2/\sqrt{6} \end{bmatrix}, \qquad \mathbf{\Lambda} = \operatorname{diag}(4,1,1).$$

Check via projectors. $\mathbf{q}_1\mathbf{q}_1^{\mathsf{T}} = \frac{1}{3}\mathbf{J}$ and, since the three projectors sum to $\mathbf{I}$, $\mathbf{q}_2\mathbf{q}_2^{\mathsf{T}} + \mathbf{q}_3\mathbf{q}_3^{\mathsf{T}} = \mathbf{I} - \frac{1}{3}\mathbf{J}$. So $\mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^{\mathsf{T}} = 4 \cdot \frac{1}{3}\mathbf{J} + 1 \cdot (\mathbf{I} - \frac{1}{3}\mathbf{J}) = \mathbf{I} + \mathbf{J} = \mathbf{A}$ ✓. In a balanced three-phase system this same matrix structure gives the **zero-sequence** vector $(1,1,1)$ and a two-dimensional space of positive/negative-sequence vectors — the symmetrical-components decomposition is a spectral decomposition.

Rayleigh check. $\mathbf{x} = (1,1,0)$: $\mathbf{A}\mathbf{x} = (3,3,2)$, $R = 6/2 = 3$, which lies in $[\lambda_{\min}, \lambda_{\max}] = [1,4]$ ✓.

Worked Example 10.20 — Principal axes of a conic

Identify and sketch the curve $5x^2 + 4xy + 2y^2 = 6$.

Step 1 — matrix. $Q = \mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x}$ with $\mathbf{A} = \begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix}$ (half of 4 off the diagonal).

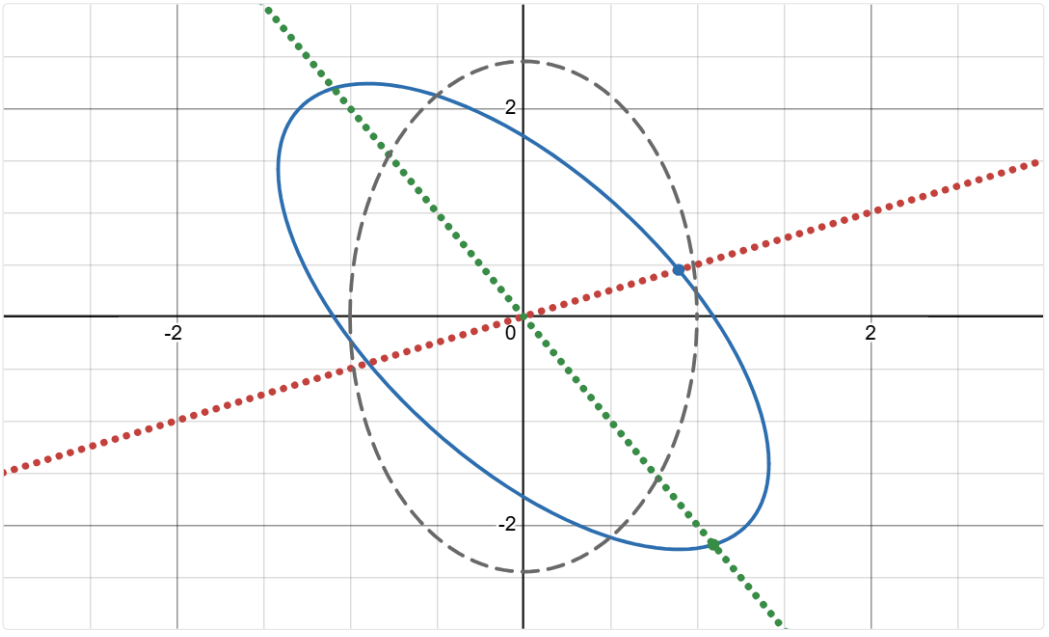
Step 2 — eigen-decomposition. $\lambda^2 - 7\lambda + 6 = 0$: $\lambda_1 = 6$, $\lambda_2 = 1$. $\lambda = 6$: $\mathbf{A} - 6\mathbf{I} = \begin{bmatrix} -1 & 2 \\ 2 & -4 \end{bmatrix}$, $\mathbf{q}_1 = \frac{1}{\sqrt{5}}(2,1)$. $\lambda = 1$: $\mathbf{q}_2 = \frac{1}{\sqrt{5}}(1,-2)$.

Step 3 — rotate. With $\mathbf{x} = u\mathbf{q}_1 + v\mathbf{q}_2$, the equation becomes $6u^2 + v^2 = 6$, i.e.

$$\frac{u^2}{1} + \frac{v^2}{6} = 1 :$$

an ellipse with semi-major axis $\sqrt{6} \approx 2.449$ along $\mathbf{q}_2$ (the direction of the **smaller** eigenvalue) and semi-minor axis 1 along $\mathbf{q}_1$.

Step 4 — check a vertex. $\sqrt{6}\,\mathbf{q}_2 = (1.0954, -2.1909)$: $5(1.2) + 4(1.0954)(-2.1909) + 2(4.8) = 6 - 9.6 + 9.6 = 6$ ✓. The tilt of the major axis is $\arctan(-2) = -63.43°$.



The conic $5x^2 + 4xy + 2y^2 = 6$ (blue) and its principal axes along the eigenvectors $\mathbf{q}_1 \propto (2,1)$ ($\lambda = 6$, red) and $\mathbf{q}_2 \propto (1,-2)$ ($\lambda = 1$, green). The dashed ellipse $6u^2 + v^2 = 6$ is the same curve drawn in principal-axis coordinates before rotation: the eigenvector change of variables simply rotates it into place.

Worked Example 10.21 — A Hermitian matrix

Find the eigenvalues and orthonormal eigenvectors of $\mathbf{H} = \begin{bmatrix} 2 & 1+j \\ 1-j & 3 \end{bmatrix}$.

Step 1 — eigenvalues. $\operatorname{tr}\mathbf{H} = 5$, $\det\mathbf{H} = 6 - (1+j)(1-j) = 6 - 2 = 4$. So $\lambda^2 - 5\lambda + 4 = 0$: $\lambda = 4, 1$ — real, as promised, even though $\mathbf{H}$ is complex.

Step 2 — eigenvectors. $\lambda = 4$: $\mathbf{H} - 4\mathbf{I} = \begin{bmatrix} -2 & 1+j \\ 1-j & -1 \end{bmatrix}$. From the first row, $\mathbf{v}_1 = (1+j,\ 2)$; check the second row: $(1-j)(1+j) - 2 = 0$ ✓. $\lambda = 1$: first row $v_1 + (1+j)v_2 = 0$, so $\mathbf{v}_2 = (1+j,\ -1)$.

Step 3 — orthogonality with the Hermitian inner product. $\mathbf{v}_1^{\mathsf{H}}\mathbf{v}_2 = (1-j)(1+j) + 2(-1) = 2 - 2 = 0$ ✓. Norms: $\|\mathbf{v}_1\|^2 = 2 + 4 = 6$, $\|\mathbf{v}_2\|^2 = 2 + 1 = 3$.

$$\mathbf{U} = \begin{bmatrix} (1+j)/\sqrt{6} & (1+j)/\sqrt{3} \\ 2/\sqrt{6} & -1/\sqrt{3} \end{bmatrix}, \qquad \mathbf{H} = \mathbf{U}\begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}\mathbf{U}^{\mathsf{H}}.$$

Check. $\mathbf{H}\mathbf{v}_1 = \big(2(1+j) + 2(1+j),\ (1-j)(1+j) + 6\big) = (4+4j,\ 8) = 4\mathbf{v}_1$ ✓. Hermitian matrices of this kind are the spatial covariance matrices of two-antenna receivers; the eigenvalue 4 is the power captured by the best beamformer $\mathbf{u}_1$, and 1 the power in the orthogonal direction.

Worked Example 10.22 — Rayleigh's quotient as an eigenvalue estimator

Estimate the largest eigenvalue of the tridiagonal $\mathbf{B}$ of Example 10.2 ($\lambda_{\max} = 2 + \sqrt{2} = 3.414214$) from trial vectors.

- $\mathbf{x} = (1, 1, 1)$: $\mathbf{B}\mathbf{x} = (3, 4, 3)$, $R = 10/3 = 3.3333$ — error 0.081.
- $\mathbf{x} = (1, 1.4, 1)$ (a better mode-shape guess, centre bulging): $\mathbf{B}\mathbf{x} = (3.4, 4.8, 3.4)$, $\mathbf{x}^{\mathsf{T}}\mathbf{B}\mathbf{x} = 3.4 + 6.72 + 3.4 = 13.52$, $\mathbf{x}^{\mathsf{T}}\mathbf{x} = 3.96$, $R = 3.414141$ — error 7×10^{-5}.

The exact eigenvector is $(1, \sqrt{2}, 1)$; the second trial vector has its middle entry about 1% too small, yet the eigenvalue is accurate to five significant figures. Both estimates are **below** $\lambda_{\max}$, as Rayleigh's principle guarantees. This quadratic accuracy is why a rough guess of a mode shape gives a good resonant frequency.

Common Pitfall — "symmetric" complex matrices

A complex matrix with $\mathbf{A}^{\mathsf{T}} = \mathbf{A}$ (without conjugation) is **complex symmetric**, not Hermitian, and none of the theorems of this section apply: $\begin{bmatrix} 1 & j \\ j & -1 \end{bmatrix}$ is complex symmetric with the double eigenvalue 0 and only one eigenvector — it is defective. Impedance and admittance matrices of *lossy* reciprocal networks are complex symmetric. Their eigenvalues are generally complex and the eigenvectors are orthogonal only in the unconjugated sense $\mathbf{v}_i^{\mathsf{T}}\mathbf{v}_k = 0$. Check which transpose the theorem needs before using it.

Engineering Note — principal component analysis of sensor data

Given N zero-mean measurement vectors $\mathbf{x}_k \in \mathbb{R}^n$ (for example, the outputs of n microphones), the sample covariance $\mathbf{C} = \frac{1}{N}\sum_k \mathbf{x}_k\mathbf{x}_k^{\mathsf{T}}$ is symmetric with non-negative eigenvalues. Its spectral decomposition sorts the signal space into orthogonal directions by power: λ_i is the average power along $\mathbf{q}_i$. Projecting onto the top few eigenvectors keeps most of the energy and discards mostly noise — principal component analysis, also known as the Karhunen–Loève transform. The statistics side is in Ch. 20 and Ch. 21; the mathematics is the spectral theorem.

10.6 Positive-definite matrices

Motivation

Energy is never negative. The energy stored in a set of coupled inductors, $W = \frac{1}{2}\mathbf{i}^{\mathsf{T}}\mathbf{L}\mathbf{i}$, is positive for every nonzero current vector; so is the electrostatic energy $\frac{1}{2}\mathbf{v}^{\mathsf{T}}\mathbf{C}\mathbf{v}$ of a system of conductors, and the power $\mathbf{v}^{\mathsf{T}}\mathbf{G}\mathbf{v}$ dissipated in a passive resistive network. The variance of any linear combination $\mathbf{w}^{\mathsf{T}}\mathbf{x}$ of random voltages is $\mathbf{w}^{\mathsf{T}}\mathbf{C}_{xx}\mathbf{w} \geq 0$. A smooth function has a strict local minimum where its gradient vanishes and its Hessian quadratic form is positive for every displacement (Ch. 6, Ch. 23). In all these cases a symmetric matrix $\mathbf{A}$ has the property $\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} > 0$ — it is **positive definite**. Such matrices are the "positive numbers" of matrix algebra: they have square roots, they can be factored as $\mathbf{R}^{\mathsf{T}}\mathbf{R}$, linear systems with them can be solved stably without pivoting, and they define inner products and distances.

Definitions

Definiteness

A real symmetric (or complex Hermitian) $n \times n$ matrix $\mathbf{A}$ is

- **positive definite** (PD, $\mathbf{A} \succ 0$) if $\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} > 0$ for every $\mathbf{x} \neq \mathbf{0}$;
- **positive semidefinite** (PSD, $\mathbf{A} \succeq 0$) if $\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} \geq 0$ for every $\mathbf{x}$;
- **negative definite / semidefinite** if $-\mathbf{A}$ is positive definite / semidefinite;
- **indefinite** if $\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x}$ takes both signs.

For complex Hermitian matrices replace $\mathbf{x}^{\mathsf{T}}$ by $\mathbf{x}^{\mathsf{H}}$; the form $\mathbf{x}^{\mathsf{H}}\mathbf{A}\mathbf{x}$ is then real.

Definiteness is normally defined only for symmetric matrices, because the quadratic form only sees the symmetric part $\frac{1}{2}(\mathbf{A} + \mathbf{A}^{\mathsf{T}})$.

Equivalent tests

Tests for positive definiteness

For a real symmetric $\mathbf{A}$, the following are equivalent:

1. $\mathbf{x}^\mathsf{T}\mathbf{A}\mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$;
2. **all eigenvalues of A are positive**;
3. **all leading principal minors are positive** (Sylvester's criterion): $\Delta_1 = a_{11} > 0$, $\Delta_2 = \det \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} > 0$, ..., $\Delta_n = \det \mathbf{A} > 0$;
4. **all pivots** of Gaussian elimination without row exchanges are positive;
5. $\mathbf{A} = \mathbf{R}^\mathsf{T}\mathbf{R}$ for some matrix $\mathbf{R}$ with independent columns — in particular, the **Cholesky factorization** $\mathbf{A} = \mathbf{L}\mathbf{L}^\mathsf{T}$ exists with $\mathbf{L}$ lower triangular with positive diagonal.

Proof sketch. (1)⇔(2): with $\mathbf{x} = \mathbf{Q}\mathbf{y}$, $\mathbf{x}^\mathsf{T}\mathbf{A}\mathbf{x} = \sum \lambda_i y_i^2$; this is positive for all $\mathbf{y} \neq \mathbf{0}$ exactly when every $\lambda_i > 0$ (take $\mathbf{y} = \mathbf{e}_i$ for necessity). (2)⇒(3): the upper-left $k \times k$ block $\mathbf{A}_k$ is itself PD (restrict (1) to vectors whose last $n-k$ entries are zero), so its eigenvalues are positive and $\Delta_k = \det \mathbf{A}_k$ — their product — is positive. (3)⇔(4): elimination without row exchanges gives pivots $d_k = \Delta_k / \Delta_{k-1}$ (with $\Delta_0 = 1$), because the determinant of each leading block is the product of its pivots. (4)⇒(5): elimination gives $\mathbf{A} = \mathbf{L}_1\mathbf{D}\mathbf{L}_1^\mathsf{T}$ with unit lower-triangular $\mathbf{L}_1$ and $\mathbf{D} = \operatorname{diag}(d_k)$; set $\mathbf{L} = \mathbf{L}_1\mathbf{D}^{1/2}$. (5)⇒(1): $\mathbf{x}^\mathsf{T}\mathbf{R}^\mathsf{T}\mathbf{R}\mathbf{x} = \|\mathbf{R}\mathbf{x}\|^2 \geq 0$, with equality only if $\mathbf{R}\mathbf{x} = \mathbf{0}$, i.e. $\mathbf{x} = \mathbf{0}$ because $\mathbf{R}$ has independent columns. □

For semidefiniteness the tests weaken to: all eigenvalues ≥ 0; $\mathbf{A} = \mathbf{R}^\mathsf{T}\mathbf{R}$ for *some* $\mathbf{R}$; or **all** principal minors (not just the leading ones) ≥ 0.

In practice: for a 2×2 matrix, check $a_{11} > 0$ and $\det \mathbf{A} > 0$. For 3×3 by hand, use leading minors or pivots. For large matrices, attempt a Cholesky factorization — it succeeds if and only if the matrix is PD, and it costs about half as much as an LU factorization.

Gram matrices and the energy form

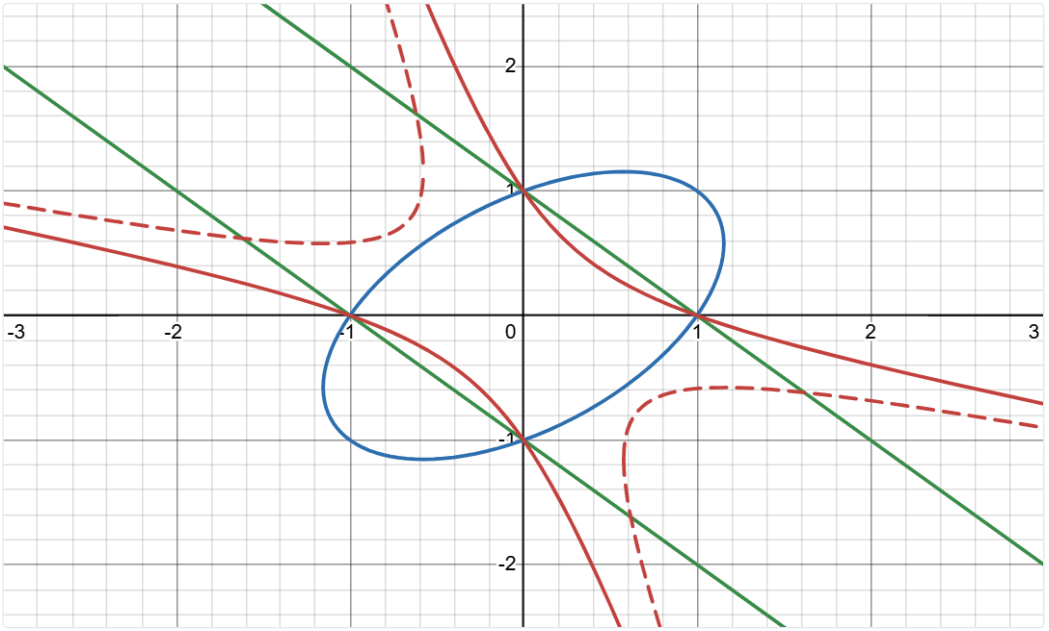
The test (5) has a very useful reading. For **any** $m \times n$ matrix $\mathbf{R}$, the **Gram matrix** $\mathbf{R}^\mathsf{T}\mathbf{R}$ is symmetric positive semidefinite, and it is positive definite iff the columns of $\mathbf{R}$ are linearly independent. The normal-equation matrix $\mathbf{A}^\mathsf{T}\mathbf{A}$ of least squares (Ch. 9), sample covariance matrices and the matrices $\mathbf{A}^\mathsf{T}\mathbf{A}$, $\mathbf{A}\mathbf{A}^\mathsf{T}$ of the SVD (Section 10.7) are all Gram matrices.

Physical matrices often arrive in exactly this form. The conductance matrix of a resistor network is $\mathbf{G} = \mathbf{B}^\mathsf{T} \operatorname{diag}(g_e)\mathbf{B}$, where $\mathbf{B}$ is the branch–node incidence matrix (each branch voltage is a difference of node voltages) and $g_e > 0$ are the branch conductances; the dissipated power $\mathbf{v}^\mathsf{T}\mathbf{G}\mathbf{v} = \sum_e g_e (\text{branch voltage})^2$ is a sum of squares. Grounding at least one node makes $\mathbf{B}$'s columns independent and $\mathbf{G}$ positive definite.

Geometry: level curves

The level sets $\mathbf{x}^\mathsf{T}\mathbf{A}\mathbf{x} = c$ show definiteness at a glance. In principal-axis coordinates they read $\sum \lambda_i y_i^2 = c$:

- **PD**: ellipses (ellipsoids) centred at the origin; Q is a bowl with a unique minimum.
- **PSD but singular**: a zero eigenvalue makes the ellipse degenerate into a pair of parallel lines (a cylinder in 3-D); Q is a trough, flat along the null-space direction.
- **Indefinite**: hyperbolas; Q is a saddle.



Level curves of three quadratic forms: positive definite $x^2 - xy + y^2 = 1$ (blue ellipse, eigenvalues 1.5 and 0.5); positive semidefinite $(x+y)^2 = 1$ (green parallel lines, eigenvalues 2 and 0); indefinite $x^2 + 4xy + y^2 = \pm 1$ (red hyperbolas, eigenvalues 3 and -1).

Square roots and the energy norm

A PSD matrix has a unique PSD **square root** $\mathbf{A}^{1/2} = \mathbf{Q}\mathbf{\Lambda}^{1/2}\mathbf{Q}^{\mathsf{T}}$, with $\mathbf{\Lambda}^{1/2} = \operatorname{diag}(\sqrt{\lambda_i})$; indeed $\mathbf{A}^{1/2}\mathbf{A}^{1/2} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^{\mathsf{T}} = \mathbf{A}$. Both $\mathbf{A}^{1/2}$ and the Cholesky factor $\mathbf{L}^{\mathsf{T}}$ are "square roots" in the sense $\mathbf{A} = \mathbf{R}^{\mathsf{T}}\mathbf{R}$; they differ by an orthogonal factor. A PD matrix also defines an inner product $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{A}} = \mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{y}$ and an **energy norm** $\|\mathbf{x}\|_{\mathbf{A}} = \sqrt{\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x}}$; the Mahalanobis distance of statistics is the energy norm of the inverse covariance, and "whitening" noise means multiplying by $\mathbf{C}^{-1/2}$ or $\mathbf{L}^{-1}$ so that the noise covariance becomes $\mathbf{I}$.

Worked examples

Worked Example 10.23 — Three tests on one matrix

Show that $\mathbf{K} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ is positive definite, in three ways.

Leading minors. $\Delta_1 = 2$, $\Delta_2 = 4 - 1 = 3$, $\Delta_3 = 2(4-1) - (-1)(-2-0) = 6 - 2 = 4$. All positive $\Rightarrow$ PD.

Pivots. $d_k = \Delta_k / \Delta_{k-1}$: $2,\ 3/2,\ 4/3$ — all positive. (Elimination directly: $R_2 + \frac{1}{2}R_1$ gives pivot $2 - \frac{1}{2} = \frac{3}{2}$; $R_3 + \frac{2}{3}R_2$ gives $2 - \frac{2}{3} = \frac{4}{3}$ ✓.)

Eigenvalues. $\mathbf{K} = \mathbf{S}\mathbf{B}\mathbf{S}$ with $\mathbf{S} = \operatorname{diag}(1,-1,1) = \mathbf{S}^{-1}$ and $\mathbf{B}$ the matrix of Example 10.2, so $\mathbf{K}$ is similar to $\mathbf{B}$ and has eigenvalues $2 - \sqrt{2},\ 2,\ 2 + \sqrt{2}$, all positive. (Check: product $= 4 = \Delta_3$ ✓.)

Energy. Better than any test is to *see* the sum of squares:

$$\mathbf{x}^{\mathsf{T}}\mathbf{K}\mathbf{x} = 2x_1^2 + 2x_2^2 + 2x_3^2 - 2x_1x_2 - 2x_2x_3 = x_1^2 + (x_1 - x_2)^2 + (x_2 - x_3)^2 + x_3^2,$$

which vanishes only if $x_1 = 0$, $x_1 = x_2$, $x_2 = x_3$, $x_3 = 0$. This is the stiffness (or conductance) matrix of a chain of four identical springs (resistors) between two fixed walls (grounded ends): the four squares are the four element energies.

Worked Example 10.24 — Cholesky factorization

Factor $\mathbf{A} = \begin{bmatrix} 4 & 2 & -2 \\ 2 & 10 & 2 \\ -2 & 2 & 6 \end{bmatrix}$ as $\mathbf{L}\mathbf{L}^{\mathsf{T}}$.

Write $\mathbf{L} = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix}$ and match entries of $\mathbf{L}\mathbf{L}^{\mathsf{T}}$ column by column:

- $l_{11}^2 = 4 \Rightarrow l_{11} = 2$; $l_{21}l_{11} = 2 \Rightarrow l_{21} = 1$; $l_{31}l_{11} = -2 \Rightarrow l_{31} = -1$.
- $l_{21}^2 + l_{22}^2 = 10 \Rightarrow l_{22} = \sqrt{9} = 3$; $l_{31}l_{21} + l_{32}l_{22} = 2 \Rightarrow l_{32} = (2+1)/3 = 1$.
- $l_{31}^2 + l_{32}^2 + l_{33}^2 = 6 \Rightarrow l_{33} = \sqrt{6 - 1 - 1} = 2$.

$$\mathbf{L} = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 3 & 0 \\ -1 & 1 & 2 \end{bmatrix}.$$

Every square root had a positive argument, which certifies that $\mathbf{A}$ is PD. **By-products:** $\det \mathbf{A} = (2 \cdot 3 \cdot 2)^2 = 144$, and the pivots are $l_{kk}^2 = 4, 9, 4$. To solve $\mathbf{A}\mathbf{x} = \mathbf{b}$, solve $\mathbf{L}\mathbf{y} = \mathbf{b}$ forward and $\mathbf{L}^{\mathsf{T}}\mathbf{x} = \mathbf{y}$ backward. **Check:** row 3 of $\mathbf{L}$ dotted with itself, $1 + 1 + 4 = 6 = a_{33}$ ✓; with row 2, $-1 + 3 + 0 = 2 = a_{32}$ ✓.

Worked Example 10.25 — For which parameter values is a matrix PD?

For which real k is $\mathbf{A}(k) = \begin{bmatrix} 1 & k & 0 \\ k & 1 & k \\ 0 & k & 1 \end{bmatrix}$ positive definite?

Leading minors. $\Delta_1 = 1 > 0$; $\Delta_2 = 1 - k^2 > 0 \iff |k| < 1$; $\Delta_3 = 1(1-k^2) - k(k-0) = 1 - 2k^2 > 0 \iff |k| < 1/\sqrt{2}$. The binding condition is the last:

$$\mathbf{A}(k) \succ 0 \iff |k| < \tfrac{1}{\sqrt{2}} \approx 0.7071.$$

Eigenvalue check. Expanding $\det(\mathbf{A} - \lambda\mathbf{I}) = (1-\lambda)[(1-\lambda)^2 - 2k^2]$ gives $\lambda = 1,\ 1 \pm \sqrt{2}\,k$. The smallest, $1 - \sqrt{2}|k|$, is positive exactly when $|k| < 1/\sqrt{2}$ ✓. At $k = 1/\sqrt{2}$ the matrix is PSD and singular, with null vector $(1, -\sqrt{2}, 1)$.

Moral. Positivity of all entries, or diagonal entries larger than off-diagonal ones (here $1 > k$ for $k = 0.8$), does **not** imply definiteness; at $k = 0.8$, $\Delta_3 = -0.28 < 0$.

Worked Example 10.26 — Coupled inductors: the energy condition $k < 1$

Two coupled coils have inductance matrix $\mathbf{L} = \begin{bmatrix} L_1 & M \\ M & L_2 \end{bmatrix}$ and stored energy $W = \tfrac{1}{2}\mathbf{i}^{\mathsf{T}}\mathbf{L}\mathbf{i} = \tfrac{1}{2}(L_1 i_1^2 + 2M i_1 i_2 + L_2 i_2^2)$.

Condition. Passivity requires $W > 0$ for all nonzero currents: $\mathbf{L} \succ 0 \iff L_1 > 0$ and $L_1 L_2 - M^2 > 0$, i.e. the **coupling coefficient** $k = |M|/\sqrt{L_1 L_2} < 1$.

Numbers. $L_1 = 4$ mH, $L_2 = 9$ mH, $M = 5$ mH: $k = 5/6 = 0.833$, $\det \mathbf{L} = 36 - 25 = 11$ mH$^2 > 0$. Eigenvalues: $\lambda = \tfrac{1}{2}\left(13 \pm \sqrt{169 - 44}\right) = 12.09$ mH and 0.910 mH. For opposing currents $\mathbf{i} = (1, -1)$ A, $W = \tfrac{1}{2}(4 - 10 + 9) = 1.5$ mJ > 0, and indeed no current vector gives less than $\tfrac{1}{2}(0.910\text{ mH})\|\mathbf{i}\|^2$.

The limit $k = 1$. With $M = 6$ mH, $\det \mathbf{L} = 0$: PSD, not PD. The null vector $\mathbf{i} = (3, -2)$ A gives $W = \tfrac{1}{2}(4 \cdot 9 - 2 \cdot 6 \cdot 6 + 9 \cdot 4) = 0$: the two coils' fluxes cancel exactly. An ideal transformer is this singular limit; a value $k > 1$ in a simulator netlist describes a device that could deliver energy from nothing, and the simulation will typically blow up.

Common Pitfall — leading minors and semidefiniteness

Sylvester's criterion with $\geq$ instead of $>$ does **not** test semidefiniteness. $\begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}$ has leading minors 0 and 0, both ≥ 0, yet it is negative semidefinite. For PSD you must check *all* principal minors (here the 1×1 minor -1 fails) or the eigenvalues. Also, for **negative** definiteness the leading minors must *alternate* in sign, $\Delta_1 < 0,\ \Delta_2 > 0,\ \Delta_3 < 0, \ldots$ — not all be negative.

Engineering Note — positive definiteness as a modelling check

Numerically extracted parasitic matrices (field-solver $\mathbf{L}$, $\mathbf{C}$ and $\mathbf{G}$ for packages and interconnect), measured covariance matrices with missing samples, and estimated Hessians in optimisation all lose positive definiteness through rounding or estimation error. A non-PD inductance or capacitance matrix makes a passive model non-passive, and transient simulators then produce growing oscillations. Standard practice is to test with a Cholesky attempt and, if it fails, repair the matrix by clipping negative eigenvalues to a small positive floor: $\mathbf{A} \leftarrow \mathbf{Q}\max(\mathbf{\Lambda}, \epsilon\mathbf{I})\mathbf{Q}^{\mathsf{T}}$, the nearest PD matrix in the Frobenius norm.

10.7 Singular value decomposition

Motivation

Eigen-decomposition has three limitations: it applies only to square matrices; it may fail (defective matrices); and, for non-symmetric matrices, the eigenvectors are not orthogonal, so $\mathbf{P}$ can be badly conditioned. Many engineering matrices are not square — a 4×2 MIMO channel matrix from 2 transmit to 4 receive antennas, a 1000×3 regression matrix of measurements against model parameters, an $m \times n$ image. The **singular value decomposition** (SVD) repairs all three defects at once. Every matrix, of any shape, factors as

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\mathsf{T}}$$

with **orthogonal** $\mathbf{U}$ and $\mathbf{V}$ and a non-negative diagonal $\mathbf{\Sigma}$. The price is that we use two different orthonormal bases — one for the input space and one for the output space — instead of one eigenbasis. The SVD is arguably the most useful single factorization in applied linear algebra: it gives the rank, the norm and the condition number of a matrix, the best low-rank approximation, the pseudoinverse and the least-squares solution, and the capacity-achieving decomposition of a MIMO channel into parallel independent sub-channels.

Statement

Singular value decomposition

Every real $m \times n$ matrix $\mathbf{A}$ of rank r can be factored as

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\mathsf{T}},$$

where $\mathbf{U}$ ($m \times m$) and $\mathbf{V}$ ($n \times n$) are orthogonal and $\mathbf{\Sigma}$ ($m \times n$) is zero except for its leading diagonal entries

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0 = \sigma_{r+1} = \ldots .$$

The σ_i are the **singular values**, the columns $\mathbf{u}_i$ of $\mathbf{U}$ the **left singular vectors** and the columns $\mathbf{v}_i$ of $\mathbf{V}$ the **right singular vectors**. They satisfy

$$\mathbf{A}\mathbf{v}_i = \sigma_i \mathbf{u}_i, \qquad \mathbf{A}^{\mathsf{T}}\mathbf{u}_i = \sigma_i \mathbf{v}_i,$$

and σ_i^2 are the nonzero eigenvalues of both $\mathbf{A}^{\mathsf{T}}\mathbf{A}$ and $\mathbf{A}\mathbf{A}^{\mathsf{T}}$. For complex $\mathbf{A}$, replace transposes by conjugate transposes and "orthogonal" by "unitary"; the σ_i are still real and non-negative.

The singular values are unique; the singular vectors are unique up to sign (or a phase factor, in the complex case) when the σ_i are distinct.

Derivation

The construction uses only the spectral theorem. The $n \times n$ matrix $\mathbf{A}^{\mathsf{T}}\mathbf{A}$ is symmetric and positive semidefinite (a Gram matrix, Section 10.6), so it has orthonormal eigenvectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$ with eigenvalues $\mu_1 \geq \cdots \geq \mu_n \geq 0$. Exactly $r = \operatorname{rank}\mathbf{A}$ of them are positive, because $\operatorname{null}(\mathbf{A}^{\mathsf{T}}\mathbf{A}) = \operatorname{null}(\mathbf{A})$. Define

$$\sigma_i = \sqrt{\mu_i}, \qquad \mathbf{u}_i = \frac{\mathbf{A}\mathbf{v}_i}{\sigma_i} \quad (i = 1, \ldots, r).$$

These $\mathbf{u}_i$ are orthonormal:

$$\mathbf{u}_i^{\mathsf{T}}\mathbf{u}_k = \frac{\mathbf{v}_i^{\mathsf{T}}\mathbf{A}^{\mathsf{T}}\mathbf{A}\mathbf{v}_k}{\sigma_i\sigma_k} = \frac{\mu_k\,\mathbf{v}_i^{\mathsf{T}}\mathbf{v}_k}{\sigma_i\sigma_k} = \begin{cases} 1 & i = k \\ 0 & i \neq k. \end{cases}$$

Extend $\mathbf{u}_1, \dots, \mathbf{u}_r$ to an orthonormal basis of $\mathbb{R}^m$ (Gram–Schmidt). Then $\mathbf{A}\mathbf{v}_i = \sigma_i\mathbf{u}_i$ for $i \le r$ and $\mathbf{A}\mathbf{v}_i = \mathbf{0}$ for $i > r$; in matrix form $\mathbf{A}\mathbf{V} = \mathbf{U}\boldsymbol{\Sigma}$, and multiplying on the right by $\mathbf{V}^{\mathsf{T}} = \mathbf{V}^{-1}$ gives the SVD. □

The derivation is also a hand-computation recipe: eigen-decompose the smaller of $\mathbf{A}^{\mathsf{T}}\mathbf{A}$ and $\mathbf{A}\mathbf{A}^{\mathsf{T}}$, then get the other set of singular vectors from $\mathbf{u}_i = \mathbf{A}\mathbf{v}_i/\sigma_i$ (or $\mathbf{v}_i = \mathbf{A}^{\mathsf{T}}\mathbf{u}_i/\sigma_i$). Do **not** compute $\mathbf{U}$ and $\mathbf{V}$ independently from the two Gram matrices: each eigenvector is only determined up to sign, and independently chosen signs may not match.

Geometry

Reading $\mathbf{A}\mathbf{x} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^{\mathsf{T}}\mathbf{x}$ right to left, **every linear map is a rotation (or reflection), followed by axis-aligned stretching, followed by another rotation**. In particular, $\mathbf{A}$ maps the unit sphere in $\mathbb{R}^n$ onto an ellipsoid (possibly flattened) in $\mathbb{R}^m$ whose semi-axes are $\sigma_i\mathbf{u}_i$; the right singular vectors $\mathbf{v}_i$ are the orthogonal input directions that map onto those axes. Compare the graph in Section 10.2: eigenvectors are directions that are not rotated; singular vectors are orthogonal directions that *stay* orthogonal.

The SVD and the four fundamental subspaces

The SVD supplies orthonormal bases for all four fundamental subspaces of Ch. 9 at once:

Subspace	Orthonormal basis	Dimension
column space $\mathcal{C}(\mathbf{A})$	$\mathbf{u}_1, \dots, \mathbf{u}_r$	r
left null space $\mathcal{N}(\mathbf{A}^{\mathsf{T}})$	$\mathbf{u}_{r+1}, \dots, \mathbf{u}_m$	$m - r$
row space $\mathcal{C}(\mathbf{A}^{\mathsf{T}})$	$\mathbf{v}_1, \dots, \mathbf{v}_r$	r
null space $\mathcal{N}(\mathbf{A})$	$\mathbf{v}_{r+1}, \dots, \mathbf{v}_n$	$n - r$

Keeping only the first r columns gives the **compact** (or "thin") SVD and the **outer-product expansion**

$$\mathbf{A} = \mathbf{U}_r\boldsymbol{\Sigma}_r\mathbf{V}_r^{\mathsf{T}} = \sigma_1\mathbf{u}_1\mathbf{v}_1^{\mathsf{T}} + \sigma_2\mathbf{u}_2\mathbf{v}_2^{\mathsf{T}} + \cdots + \sigma_r\mathbf{u}_r\mathbf{v}_r^{\mathsf{T}},$$

a sum of r rank-one matrices in decreasing order of importance.

Norms, condition number and rank

The SVD measures matrices. Because orthogonal matrices preserve lengths,

$$\|\mathbf{A}\|_2 = \max_{\mathbf{x}\neq\mathbf{0}} \frac{\|\mathbf{A}\mathbf{x}\|}{\|\mathbf{x}\|} = \sigma_1, \qquad \|\mathbf{A}\|_F = \sqrt{\sum_{i,k} a_{ik}^2} = \sqrt{\sigma_1^2 + \cdots + \sigma_r^2},$$

the spectral norm (maximum gain) and the Frobenius norm. For a square invertible matrix, $|\det\mathbf{A}| = \sigma_1\sigma_2\cdots\sigma_n$, $\|\mathbf{A}^{-1}\|_2 = 1/\sigma_n$, and the **condition number** introduced in Ch. 8 is

$$\kappa_2(\mathbf{A}) = \|\mathbf{A}\|_2\|\mathbf{A}^{-1}\|_2 = \frac{\sigma_{\max}}{\sigma_{\min}}.$$

It bounds error amplification in solving $\mathbf{A}\mathbf{x} = \mathbf{b}$: a relative perturbation ϵ in $\mathbf{b}$ can cause a relative change of up to $\kappa\epsilon$ in $\mathbf{x}$. Rule of thumb: you lose about $\log_{10}\kappa$ significant digits.

The **numerical rank** is the number of singular values above a tolerance (typically $\max(m,n)\,\sigma_1\,\epsilon_{\text{mach}}$). Unlike the exact rank, which jumps discontinuously under perturbation, the singular values move continuously: $|\sigma_i(\mathbf{A}+\mathbf{E}) - \sigma_i(\mathbf{A})| \le \|\mathbf{E}\|_2$. The smallest singular value is the **distance to the nearest singular matrix**: $\min\{\|\mathbf{E}\|_2 : \mathbf{A}+\mathbf{E} \text{ singular}\} = \sigma_n$.

Eckart–Young (best low-rank approximation)

For $k < r$, the truncated sum $\mathbf{A}_k = \sum_{i=1}^{k}\sigma_i\mathbf{u}_i\mathbf{v}_i^{\mathsf{T}}$ is the closest matrix of rank at most k to $\mathbf{A}$, in both the spectral and Frobenius norms:

$$\min_{\operatorname{rank}\mathbf{B}\le k}\|\mathbf{A}-\mathbf{B}\|_2 = \sigma_{k+1}, \qquad \min_{\operatorname{rank}\mathbf{B}\le k}\|\mathbf{A}-\mathbf{B}\|_F = \sqrt{\sigma_{k+1}^2 + \cdots + \sigma_r^2}.$$

This underlies image and data compression, noise reduction by truncating small singular values, and model-order reduction of large circuits (balanced truncation discards states associated with small *Hankel* singular values).

The pseudoinverse and least squares

Moore–Penrose pseudoinverse

If $\mathbf{A} = \mathbf{U}_r\boldsymbol{\Sigma}_r\mathbf{V}_r^{\mathsf{T}}$ (compact SVD), its **pseudoinverse** is

$$\mathbf{A}^{+} = \mathbf{V}_r\boldsymbol{\Sigma}_r^{-1}\mathbf{U}_r^{\mathsf{T}} = \sum_{i=1}^{r}\frac{1}{\sigma_i}\mathbf{v}_i\mathbf{u}_i^{\mathsf{T}}.$$

For any $\mathbf{b}$, $\mathbf{x}^+ = \mathbf{A}^+\mathbf{b}$ is the **minimum-norm least-squares solution**: among all $\mathbf{x}$ that minimise $\|\mathbf{A}\mathbf{x} - \mathbf{b}\|$, it is the one with smallest $\|\mathbf{x}\|$. *Why:* in the rotated coordinates $\mathbf{y} = \mathbf{V}^\mathsf{T}\mathbf{x}$, $\mathbf{c} = \mathbf{U}^\mathsf{T}\mathbf{b}$, the residual is $\|\boldsymbol{\Sigma}\mathbf{y} - \mathbf{c}\|^2 = \sum_{i \le r}(\sigma_i y_i - c_i)^2 + \sum_{i>r} c_i^2$. The first sum is zeroed by $y_i = c_i/\sigma_i$; the second cannot be changed; the remaining y_i ($i > r$) do not affect the residual and are set to zero to minimise the norm. Special cases: if $\mathbf{A}$ has independent columns, $\mathbf{A}^+ = (\mathbf{A}^\mathsf{T}\mathbf{A})^{-1}\mathbf{A}^\mathsf{T}$ (the normal-equation solution of Ch. 9); if it has independent rows, $\mathbf{A}^+ = \mathbf{A}^\mathsf{T}(\mathbf{A}\mathbf{A}^\mathsf{T})^{-1}$; if it is square and invertible, $\mathbf{A}^+ = \mathbf{A}^{-1}$.

The formula also exposes the danger of tiny singular values: noise in c_i is amplified by $1/\sigma_i$. **Truncated SVD** (drop terms with σ_i below a threshold) and **Tikhonov regularisation** (replace $1/\sigma_i$ by $\sigma_i/(\sigma_i^2 + \alpha^2)$) tame this amplification, at the cost of a small bias.

Singular values versus eigenvalues

For a symmetric matrix the SVD and the spectral decomposition coincide up to signs: $\sigma_i = |\lambda_i|$, and a negative λ_i is absorbed by flipping $\mathbf{u}_i = -\mathbf{q}_i$. For a symmetric PSD matrix they are identical. For non-symmetric matrices the two sets of numbers can be very different. For $\mathbf{A} = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$ the eigenvalues are $5, 2$ but the singular values are $5.117, 1.954$ (square roots of the eigenvalues $15 \pm \sqrt{125}$ of $\mathbf{A}\mathbf{A}^\mathsf{T}$); their products agree ($|\det \mathbf{A}| = 10$) but nothing else need. Always $\sigma_{\min} \le |\lambda_i| \le \sigma_{\max}$. Eigenvalues govern what happens under *repeated* application of $\mathbf{A}$ (powers, dynamics); singular values govern what happens under a *single* application (gain, sensitivity, rank).

Worked examples

Worked Example 10.27 — SVD of a 2×2 matrix

Find the SVD of $\mathbf{A} = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}$.

Step 1 — $\mathbf{A}^\mathsf{T}\mathbf{A}$.

$$\mathbf{A}^\mathsf{T}\mathbf{A} = \begin{bmatrix} 3 & 4 \\ 0 & 5 \end{bmatrix}\begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} 25 & 20 \\ 20 & 25 \end{bmatrix}.$$

Step 2 — its eigen-decomposition. Eigenvalues 25 ± 20: $\mu_1 = 45$, $\mu_2 = 5$, with $\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1,1)$, $\mathbf{v}_2 = \frac{1}{\sqrt{2}}(-1,1)$. Singular values: $\sigma_1 = \sqrt{45} = 3\sqrt{5} \approx 6.708$, $\sigma_2 = \sqrt{5} \approx 2.236$.

Step 3 — left singular vectors from $\mathbf{u}_i = \mathbf{A}\mathbf{v}_i/\sigma_i$.

$$\mathbf{u}_1 = \frac{1}{3\sqrt{5}} \cdot \frac{1}{\sqrt{2}}\begin{bmatrix} 3 \\ 9 \end{bmatrix} = \frac{1}{\sqrt{10}}\begin{bmatrix} 1 \\ 3 \end{bmatrix}, \qquad \mathbf{u}_2 = \frac{1}{\sqrt{5}} \cdot \frac{1}{\sqrt{2}}\begin{bmatrix} -3 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{10}}\begin{bmatrix} -3 \\ 1 \end{bmatrix}.$$

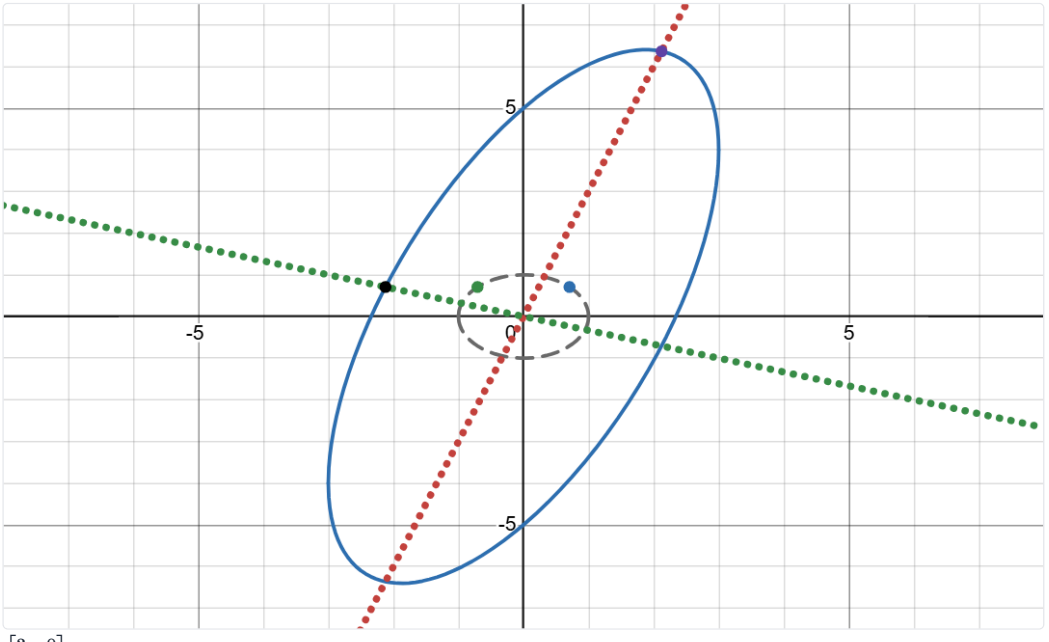
They are orthonormal: $\mathbf{u}_1 \cdot \mathbf{u}_2 = (-3+3)/10 = 0$ ✓.

Step 4 — assemble and verify by outer products.

$$\sigma_1\mathbf{u}_1\mathbf{v}_1^\mathsf{T} + \sigma_2\mathbf{u}_2\mathbf{v}_2^\mathsf{T} = \frac{3\sqrt{5}}{\sqrt{20}}\begin{bmatrix} 1 & 1 \\ 3 & 3 \end{bmatrix} + \frac{\sqrt{5}}{\sqrt{20}}\begin{bmatrix} 3 & -3 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 1.5 & 1.5 \\ 4.5 & 4.5 \end{bmatrix} + \begin{bmatrix} 1.5 & -1.5 \\ -0.5 & 0.5 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix} \checkmark$$

Derived quantities. $\|\mathbf{A}\|_2 = 6.708$; $\|\mathbf{A}\|_F = \sqrt{45 + 5} = \sqrt{50}$ (directly: $\sqrt{9 + 16 + 25}$ ✓); $|\det \mathbf{A}| = 15 = \sigma_1\sigma_2$ ✓; $\kappa_2 = 3$. Note that the eigenvalues of $\mathbf{A}$ are 3 and 5 (triangular), different from the singular values.

Best rank-1 approximation. $\mathbf{A}_1 = \begin{bmatrix} 1.5 & 1.5 \\ 4.5 & 4.5 \end{bmatrix}$, with $\|\mathbf{A} - \mathbf{A}_1\|_2 = \sigma_2 = \sqrt{5}$: no rank-1 matrix is closer.



The matrix $\mathbf{A} = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}$ maps the unit circle (dashed) onto the blue ellipse $41x^2 - 24xy + 9y^2 = 225$. The orthogonal input directions $\mathbf{v}_1, \mathbf{v}_2$ land on the ellipse's semi-axes $\sigma_1\mathbf{u}_1$ (length 6.708, red line) and $\sigma_2\mathbf{u}_2$ (length 2.236, green line), which are again orthogonal.

Worked Example 10.28 — SVD of a rectangular matrix

Find the SVD of $\mathbf{A} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$ (2×3).

Step 1 — use the smaller Gram matrix. $\mathbf{A}\mathbf{A}^{\mathsf{T}} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, eigenvalues 3 and 1, eigenvectors $\mathbf{u}_1 = \frac{1}{\sqrt{2}}(1,1)$, $\mathbf{u}_2 = \frac{1}{\sqrt{2}}(1,-1)$. So $\sigma_1 = \sqrt{3}$, $\sigma_2 = 1$, rank 2.

Step 2 — right singular vectors from $\mathbf{v}_i = \mathbf{A}^{\mathsf{T}}\mathbf{u}_i/\sigma_i$.

$$\mathbf{v}_1 = \frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \qquad \mathbf{v}_2 = \frac{1}{1} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}.$$

Step 3 — complete $\mathbf{V}$. The third right singular vector spans $\mathcal{N}(\mathbf{A})$: $x_1 + x_2 = 0$, $x_2 + x_3 = 0$ gives $\mathbf{v}_3 = \frac{1}{\sqrt{3}}(1,-1,1)$. Check orthogonality: $\mathbf{v}_1 \cdot \mathbf{v}_3 \propto 1 - 2 + 1 = 0$, $\mathbf{v}_2 \cdot \mathbf{v}_3 \propto 1 - 1 = 0$ ✓.

Step 4 — assemble.

$$\mathbf{A} = \underbrace{\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}}_{\mathbf{U}} \underbrace{\begin{bmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}}_{\mathbf{\Sigma}} \underbrace{\begin{bmatrix} 1/\sqrt{6} & 2/\sqrt{6} & 1/\sqrt{6} \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 1/\sqrt{3} & -1/\sqrt{3} & 1/\sqrt{3} \end{bmatrix}}_{\mathbf{V}^{\mathsf{T}}}.$$

Check an entry: $(\mathbf{A})_{12} = \sigma_1 u_{11} v_{21} + \sigma_2 u_{12} v_{22} = \sqrt{3} \cdot \frac{1}{\sqrt{2}} \cdot \frac{2}{\sqrt{6}} + 1 \cdot \frac{1}{\sqrt{2}} \cdot 0 = \frac{2\sqrt{3}}{\sqrt{12}} = 1$ ✓.

Minimum-norm solution of $\mathbf{A}\mathbf{x} = (1,2)$. Independent rows, so $\mathbf{A}^{+} = \mathbf{A}^{\mathsf{T}}(\mathbf{A}\mathbf{A}^{\mathsf{T}})^{-1}$ with $(\mathbf{A}\mathbf{A}^{\mathsf{T}})^{-1} = \frac{1}{3}\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$:

$$(\mathbf{A}\mathbf{A}^{\mathsf{T}})^{-1}\begin{bmatrix} 1 \\ 2 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 0 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad \mathbf{x}^{+} = \mathbf{A}^{\mathsf{T}}\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

$\mathbf{A}\mathbf{x}^{+} = (1,2)$ ✓. Every other solution is $\mathbf{x}^{+} + t\mathbf{v}_3$ with $\|\mathbf{x}\|^2 = 2 + t^2 \cdot 1$ (since $\mathbf{x}^{+} \perp \mathbf{v}_3$), so $\mathbf{x}^{+}$ with norm $\sqrt{2}$ is the shortest. In an under-determined estimation problem (more unknowns than measurements), this is the "least-energy" estimate.

Worked Example 10.29 — Condition number and sensitivity

Solve $\mathbf{A}\mathbf{x} = \mathbf{b}$ with $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1.0001 \end{bmatrix}$ for $\mathbf{b} = (2,\ 2.0001)$ and for the slightly perturbed $\mathbf{b}' = (2,\ 2.0002)$.

Solutions. Subtracting the equations: $0.0001 x_2 = b_2 - b_1$. For $\mathbf{b}$: $x_2 = 1$, $x_1 = 1$. For $\mathbf{b}'$: $x_2 = 2$, $x_1 = 0$.

Sizes. $\|\delta\mathbf{b}\|/\|\mathbf{b}\| = 10^{-4}/2.83 = 3.5 \times 10^{-5}$, while $\|\delta\mathbf{x}\|/\|\mathbf{x}\| = \sqrt{2}/\sqrt{2} = 1$: an amplification of about 28 000.

Explanation by SVD. $\mathbf{A}$ is symmetric with eigenvalues $\lambda = \frac{1}{2}\big(2.0001 \pm \sqrt{2.0001^2 - 4 \times 0.0001}\big)$, i.e. $\sigma_1 = 2.00005$, $\sigma_2 = 0.00005$, so $\kappa_2 = 40\,002$. The observed amplification (28 000) is below the bound κ, as it must be. The perturbation $\delta\mathbf{b} = (0, 10^{-4})$ has a large component along the weak direction $\mathbf{u}_2 \approx \frac{1}{\sqrt{2}}(1,-1)$, which $\mathbf{A}^{-1}$ amplifies by $1/\sigma_2 = 20\,000$. Two sensors measuring almost the same combination of two unknowns produce exactly this situation (compare the near-dependence note in Ch. 9).

Worked Example 10.30 — MIMO channel as parallel sub-channels

A 2×2 narrowband MIMO link has $\mathbf{y} = \mathbf{H}\mathbf{s} + \mathbf{n}$, and suppose $\mathbf{H} = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}$ (in a real channel the entries are complex gains; the same steps apply with H in place of T). With $\mathbf{H} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\mathsf{T}}$ from Example 10.27, let the transmitter send $\mathbf{s} = \mathbf{V}\tilde{\mathbf{s}}$ (precoding) and the receiver compute $\tilde{\mathbf{y}} = \mathbf{U}^{\mathsf{T}}\mathbf{y}$:

$$\tilde{\mathbf{y}} = \mathbf{U}^{\mathsf{T}}\mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\mathsf{T}}\mathbf{V}\tilde{\mathbf{s}} + \mathbf{U}^{\mathsf{T}}\mathbf{n} = \mathbf{\Sigma}\tilde{\mathbf{s}} + \tilde{\mathbf{n}}, \qquad \text{i.e.} \qquad \tilde{y}_1 = 6.708\,\tilde{s}_1 + \tilde{n}_1, \quad \tilde{y}_2 = 2.236\,\tilde{s}_2 + \tilde{n}_2.$$

Because $\mathbf{U}$ is orthogonal, $\tilde{\mathbf{n}}$ has the same white statistics as $\mathbf{n}$. The coupled 2×2 channel has become **two independent scalar channels** with power gains $\sigma_1^2 = 45$ and $\sigma_2^2 = 5$ (16.5 dB and 7.0 dB). Allocating transmit power across them ("water-filling") is then a one-variable problem per sub-channel. The number of nonzero singular values is the number of spatial streams the channel can carry; a rank-deficient (keyhole) channel has only one.

Common Pitfall — confusing singular values with eigenvalues, and sign mismatches

Singular values are always real and non-negative; eigenvalues need not be either. "The matrix is nearly singular because an eigenvalue is small" is correct, but "the matrix is well conditioned because its eigenvalues are similar in size" is **not** — $\begin{bmatrix} 1 & 1000 \\ 0 & 1 \end{bmatrix}$ has both eigenvalues equal to 1 but $\kappa_2 \approx 10^6$. And when computing an SVD by hand, find one set of singular vectors from a Gram matrix and derive the other from $\mathbf{A}\mathbf{v}_i = \sigma_i\mathbf{u}_i$; computing both sets independently risks inconsistent signs and a factorization that does not reproduce $\mathbf{A}$.

Engineering Note — the SVD in practice

SVD algorithms (Golub–Kahan bidiagonalization followed by an implicit QR iteration) never form $\mathbf{A}^{\mathsf{T}}\mathbf{A}$, which would square the condition number and lose half the significant digits. Typical engineering uses: computing rank and null spaces robustly (e.g. finding the number of independent loops or the controllable subspace of a large model); least-squares fitting when the normal equations are ill-conditioned; direction-of-arrival estimation (MUSIC and ESPRIT split the SVD of the array data matrix into signal and noise subspaces); principal component analysis; and reduced-order modelling of interconnect and power-delivery networks.

10.8 Modal analysis

Motivation

Couple two identical pendulums with a weak spring, start one swinging and leave the other at rest: the motion slowly transfers to the second pendulum and back again. Couple two identical LC resonators and the single resonance peak splits into two. Excite one conductor of a multi-conductor transmission line and signals appear on its neighbours with two different propagation speeds. Each of these puzzling behaviours becomes simple once the system is viewed in its **normal modes** — the eigenvectors of the system matrices. In a normal mode every part of the system moves at the same frequency with fixed relative amplitudes, as if the system were a single oscillator. Any motion is a superposition of modes, each evolving independently. **Modal analysis** is the systematic use of this fact: find the modes (an eigenvalue problem), project the initial conditions and forcing onto them, solve n scalar problems, and superpose.

Undamped second-order systems: the generalized eigenproblem

Lossless mechanical structures and lossless LC networks obey equations of the form

$$\mathbf{M}\ddot{\mathbf{x}} + \mathbf{K}\mathbf{x} = \mathbf{0},$$

where $\mathbf{M}$ is symmetric positive definite (mass, or a capacitance matrix) and $\mathbf{K}$ is symmetric positive semidefinite (stiffness, or a reciprocal-inductance matrix $\mathbf{\Gamma} = \mathbf{L}^{-1}$). Both symmetries come from energy: kinetic energy $\tfrac{1}{2}\dot{\mathbf{x}}^{\mathsf{T}}\mathbf{M}\dot{\mathbf{x}}$ and potential energy $\tfrac{1}{2}\mathbf{x}^{\mathsf{T}}\mathbf{K}\mathbf{x}$.

Seek synchronous solutions $\mathbf{x}(t) = \boldsymbol{\phi}\cos(\omega t + \theta)$, in which all coordinates oscillate in phase with a fixed shape $\boldsymbol{\phi}$. Substituting, $(-\omega^2\mathbf{M}\boldsymbol{\phi} + \mathbf{K}\boldsymbol{\phi})\cos(\omega t + \theta) = \mathbf{0}$ for all t, so

$$\mathbf{K}\boldsymbol{\phi} = \omega^2\mathbf{M}\boldsymbol{\phi}, \qquad \text{with} \qquad \det(\mathbf{K} - \omega^2\mathbf{M}) = 0.$$

This is a **generalized eigenvalue problem** with eigenvalue ω^2; the determinant condition is the **frequency equation**. Multiplying by $\mathbf{M}^{-1}$ turns it into the standard problem $\mathbf{M}^{-1}\mathbf{K}\boldsymbol{\phi} = \omega^2\boldsymbol{\phi}$, but $\mathbf{M}^{-1}\mathbf{K}$ is generally not symmetric, so it is better to keep the symmetric pair.

Normal modes of a conservative system

Let $\mathbf{M} \succ 0$ and $\mathbf{K} = \mathbf{K}^{\mathsf{T}} \succeq 0$ be $n \times n$. Then the generalized problem $\mathbf{K}\boldsymbol{\phi} = \omega^2\mathbf{M}\boldsymbol{\phi}$ has n real non-negative eigenvalues $\omega_1^2 \le \cdots \le \omega_n^2$ and real eigenvectors (mode shapes) that can be chosen **$\mathbf{M}$-orthonormal**:

$$\boldsymbol{\phi}_i^{\mathsf{T}}\mathbf{M}\boldsymbol{\phi}_k = \delta_{ik}, \qquad \boldsymbol{\phi}_i^{\mathsf{T}}\mathbf{K}\boldsymbol{\phi}_k = \omega_i^2\,\delta_{ik}.$$

With the modal matrix $\mathbf{\Phi} = [\boldsymbol{\phi}_1 \ \cdots \ \boldsymbol{\phi}_n]$: $\mathbf{\Phi}^{\mathsf{T}}\mathbf{M}\mathbf{\Phi} = \mathbf{I}$ and $\mathbf{\Phi}^{\mathsf{T}}\mathbf{K}\mathbf{\Phi} = \mathbf{\Omega}^2 = \operatorname{diag}(\omega_i^2)$.

Proof sketch. Factor $\mathbf{M} = \mathbf{L}\mathbf{L}^{\mathsf{T}}$ (Cholesky) and put $\boldsymbol{\psi} = \mathbf{L}^{\mathsf{T}}\boldsymbol{\phi}$. Then $\mathbf{K}\boldsymbol{\phi} = \omega^2\mathbf{M}\boldsymbol{\phi}$ becomes $\tilde{\mathbf{K}}\boldsymbol{\psi} = \omega^2\boldsymbol{\psi}$ with $\tilde{\mathbf{K}} = \mathbf{L}^{-1}\mathbf{K}\mathbf{L}^{-\mathsf{T}}$, which is symmetric PSD. The spectral theorem (Section 10.5) gives real $\omega_i^2 \ge 0$ and orthonormal $\boldsymbol{\psi}_i$; transforming back, $\boldsymbol{\psi}_i^{\mathsf{T}}\boldsymbol{\psi}_k = \boldsymbol{\phi}_i^{\mathsf{T}}\mathbf{L}\mathbf{L}^{\mathsf{T}}\boldsymbol{\phi}_k = \boldsymbol{\phi}_i^{\mathsf{T}}\mathbf{M}\boldsymbol{\phi}_k$. $\square$

$\mathbf{M}$-orthogonality is orthogonality in the kinetic-energy inner product (Section 10.6): modes do not share energy. A zero eigenvalue $\omega^2 = 0$ is a **rigid-body mode** (or, in a circuit, a DC loop current or floating node voltage that the network does not restore).

Modal coordinates and the general solution

Substitute $\mathbf{x} = \mathbf{\Phi}\mathbf{q}$ into $\mathbf{M}\ddot{\mathbf{x}} + \mathbf{K}\mathbf{x} = \mathbf{f}(t)$ (now with a forcing term) and multiply on the left by $\mathbf{\Phi}^{\mathsf{T}}$:

$$\underbrace{\mathbf{\Phi}^{\mathsf{T}}\mathbf{M}\mathbf{\Phi}}_{\mathbf{I}}\ddot{\mathbf{q}} + \underbrace{\mathbf{\Phi}^{\mathsf{T}}\mathbf{K}\mathbf{\Phi}}_{\mathbf{\Omega}^2}\mathbf{q} = \mathbf{\Phi}^{\mathsf{T}}\mathbf{f}(t) \quad \Longleftrightarrow \quad \ddot{q}_i + \omega_i^2 q_i = \boldsymbol{\phi}_i^{\mathsf{T}}\mathbf{f}(t), \quad i = 1, \dots, n.$$

The **modal coordinates** q_i obey uncoupled single-degree-of-freedom equations. For free vibration,

$$\mathbf{x}(t) = \sum_{i=1}^{n} \boldsymbol{\phi}_i\big(a_i\cos\omega_i t + b_i\sin\omega_i t\big), \qquad a_i = \boldsymbol{\phi}_i^{\mathsf{T}}\mathbf{M}\mathbf{x}(0), \quad b_i = \frac{\boldsymbol{\phi}_i^{\mathsf{T}}\mathbf{M}\dot{\mathbf{x}}(0)}{\omega_i},$$

where the formulas for a_i, b_i use $\mathbf{M}$-orthonormality to invert $\mathbf{x}(0) = \mathbf{\Phi}\mathbf{a}$ without solving a linear system: $\mathbf{a} = \mathbf{\Phi}^{-1}\mathbf{x}(0) = \mathbf{\Phi}^{\mathsf{T}}\mathbf{M}\mathbf{x}(0)$.

The forced equations reveal two key facts. **Resonance**: a sinusoidal force at frequency ω drives mode i with amplitude proportional to $\boldsymbol{\phi}_i^{\mathsf{T}}\mathbf{f}/(\omega_i^2 - \omega^2)$, which blows up (in the undamped model) as $\omega \to \omega_i$. **Selective excitation**: if $\boldsymbol{\phi}_i^{\mathsf{T}}\mathbf{f} = 0$ — for instance a symmetric force on a symmetric structure and an antisymmetric mode — that mode is not excited at all, regardless of frequency. Likewise a sensor placed at a node of a mode shape (where $\phi_{ik} = 0$) cannot see that mode.

Damping

With a damping matrix, $\mathbf{M}\ddot{\mathbf{x}} + \mathbf{C}\dot{\mathbf{x}} + \mathbf{K}\mathbf{x} = \mathbf{f}$, the transformation $\mathbf{\Phi}^{\mathsf{T}}\mathbf{C}\mathbf{\Phi}$ is in general not diagonal and the modes couple. Two cases are standard:

- **Proportional (Rayleigh) damping**, $\mathbf{C} = \alpha\mathbf{M} + \beta\mathbf{K}$: then $\mathbf{\Phi}^{\mathsf{T}}\mathbf{C}\mathbf{\Phi} = \alpha\mathbf{I} + \beta\mathbf{\Omega}^2$ is diagonal, and each mode obeys $\ddot{q}_i + 2\zeta_i\omega_i\dot{q}_i + \omega_i^2 q_i = \boldsymbol{\phi}_i^{\mathsf{T}}\mathbf{f}$ with modal damping ratio $\zeta_i = \frac{\alpha}{2\omega_i} + \frac{\beta\omega_i}{2}$. The same mode shapes survive; each decays at its own rate.

- **General damping**: convert to first-order state form with $\mathbf{z} = (\mathbf{x}, \dot{\mathbf{x}})$, $\dot{\mathbf{z}} = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{M}^{-1}\mathbf{K} & -\mathbf{M}^{-1}\mathbf{C} \end{bmatrix} \mathbf{z}$, and apply the first-order modal analysis below. The eigenvalues come in pairs $-\zeta_i\omega_i \pm j\omega_i\sqrt{1-\zeta_i^2}$, and the mode shapes become complex (different parts of the structure have different phases).

First-order systems: modal decomposition with left and right eigenvectors

For $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$ with diagonalizable $\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}^{-1}$, let $\mathbf{v}_i$ be the columns of $\mathbf{P}$ (right eigenvectors) and $\mathbf{w}_i^{\mathsf{T}}$ the rows of $\mathbf{W} = \mathbf{P}^{-1}$ (left eigenvectors, normalised so that $\mathbf{w}_i^{\mathsf{T}}\mathbf{v}_k = \delta_{ik}$). Then the zero-input response is

$$\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}(0) = \sum_{i=1}^{n} \mathbf{v}_i \left(\mathbf{w}_i^{\mathsf{T}}\mathbf{x}(0)\right) e^{\lambda_i t}.$$

Each term has a clear reading: $\mathbf{w}_i^{\mathsf{T}}\mathbf{x}(0)$ is **how much the initial state excites** mode i, $e^{\lambda_i t}$ is **how the mode evolves**, and $\mathbf{v}_i$ is **how the mode appears** in the state variables. With input and output $\mathbf{y} = \mathbf{C}\mathbf{x}$, mode i contributes $\dfrac{(\mathbf{C}\mathbf{v}_i)(\mathbf{w}_i^{\mathsf{T}}\mathbf{B})}{s - \lambda_i}$ to the transfer matrix — the partial-fraction expansion of $\mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}$ (Ch. 14), with residues formed from eigenvectors. If $\mathbf{w}_i^{\mathsf{T}}\mathbf{B} = \mathbf{0}$ the mode is **uncontrollable** from the input; if $\mathbf{C}\mathbf{v}_i = \mathbf{0}$ it is **unobservable** at the output; either way it cancels from the transfer function. In power-system small-signal stability, the products $p_{ki} = v_{ki}w_{ik}$ are the **participation factors** that tell which generator state participates in which electromechanical oscillation mode.

Worked examples

Worked Example 10.31 — Two masses, three springs

Two equal masses $m = 1$ kg are connected wall–spring–mass–spring–mass–spring–wall, all springs $k = 100$ N/m. Find the natural frequencies and mode shapes.

Step 1 — equations. Mass 1 feels $-kx_1$ from the left spring and $-k(x_1 - x_2)$ from the middle one; similarly for mass 2:

$$m\ddot{\mathbf{x}} + \mathbf{K}\mathbf{x} = \mathbf{0}, \qquad \mathbf{K} = k\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}.$$

Step 2 — frequency equation. With $\mathbf{M} = m\mathbf{I}$, $\det(\mathbf{K} - \omega^2 m\mathbf{I}) = 0$ gives $(2k - m\omega^2)^2 = k^2$, so $2k - m\omega^2 = \pm k$:

$$\omega_1^2 = \frac{k}{m}, \qquad \omega_2^2 = \frac{3k}{m}.$$

Numerically $\omega_1 = 10$ rad/s (1.59 Hz) and $\omega_2 = 17.32$ rad/s (2.76 Hz).

Step 3 — mode shapes. ω_1: $(\mathbf{K} - k\mathbf{I})\boldsymbol{\phi} = k\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}\boldsymbol{\phi} = \mathbf{0}$, $\boldsymbol{\phi}_1 \propto (1, 1)$: **in phase** — the middle spring is never stretched, so each mass sees only one spring, $\omega^2 = k/m$. ω_2: $\boldsymbol{\phi}_2 \propto (1, -1)$: **antiphase** — the middle spring's midpoint is stationary, so each mass sees k plus a half-length spring of stiffness $2k$, $\omega^2 = 3k/m$.

Step 4 — M-normalise. $\boldsymbol{\phi}_i^{\mathsf{T}} m \boldsymbol{\phi}_i = 1$ gives $\boldsymbol{\phi}_1 = \frac{1}{\sqrt{2m}}(1, 1)$, $\boldsymbol{\phi}_2 = \frac{1}{\sqrt{2m}}(1, -1)$. **Check:** $\boldsymbol{\phi}_2^{\mathsf{T}}\mathbf{K}\boldsymbol{\phi}_2 = \frac{k}{2m}(2 + 2 + 2) = 3k/m = \omega_2^2$ ✓. The physical reasoning in Step 3 — recognising which springs carry force in each mode — is a valuable check on any computed mode.

Worked Example 10.32 — Two capacitively coupled LC resonators

Two identical parallel tanks ($L = 10$ μH, $C = 1$ nF) have their top nodes joined by a coupling capacitor $C_c = 0.1$ nF. Find the resonant frequencies and the mode shapes.

Step 1 — node equations. KCL at node 1: $C\dot{v}_1 + C_c(\dot{v}_1 - \dot{v}_2) + i_{L1} = 0$ with $\dot{i}_{L1} = v_1/L$. Differentiating once to eliminate i_{L1}, and similarly at node 2:

$$\mathbf{C}_n\ddot{\mathbf{v}} + \frac{1}{L}\mathbf{v} = \mathbf{0}, \qquad \mathbf{C}_n = \begin{bmatrix} C + C_c & -C_c \\ -C_c & C + C_c \end{bmatrix}.$$

This is $\mathbf{M}\ddot{\mathbf{x}} + \mathbf{K}\mathbf{x} = \mathbf{0}$ with the **capacitance matrix as mass** and the **reciprocal inductance as stiffness**.

Step 2 — modes by symmetry. Both matrices commute with the swap of nodes 1 and 2, so the even and odd vectors are eigenvectors of both (Section 10.3):

- **Even mode** $(1, 1)$: $\mathbf{C}_n(1, 1) = C(1, 1)$; no current in C_c. $\omega_e^2 = \dfrac{1}{LC}$.
- **Odd mode** $(1, -1)$: $\mathbf{C}_n(1, -1) = (C + 2C_c)(1, -1)$; the midpoint of C_c is a virtual ground, so each tank sees an extra $2C_c$. $\omega_o^2 = \dfrac{1}{L(C + 2C_c)}$.

Step 3 — numbers. $f_e = \dfrac{1}{2\pi\sqrt{10^{-5} \cdot 10^{-9}}} = 1.5915$ MHz; $f_o = \dfrac{1}{2\pi\sqrt{10^{-5} \cdot 1.2 \times 10^{-9}}} = 1.4529$ MHz. The single resonance of an isolated tank (1.5915 MHz) has **split** into two, separated by 139 kHz.

Step 4 — coupling coefficient. The standard filter-design measure is $k = \dfrac{f_e^2 - f_o^2}{f_e^2 + f_o^2} = \dfrac{1 - 1/1.2}{1 + 1/1.2} = 0.0909 = \dfrac{C_c}{C + C_c}$ ✓. Coupled-resonator bandpass filters are designed by choosing these mode splittings; the even/odd decomposition is exactly the eigen-decomposition of the symmetric pair $(\mathbf{K}, \mathbf{M})$.

Worked Example 10.33 — Beating and energy exchange between weakly coupled oscillators

Two unit masses on springs $k = 1$ N/m, coupled by a weak spring $k_c = 0.105$ N/m, have $\mathbf{K} = \begin{bmatrix} 1+k_c & -k_c \\ -k_c & 1+k_c \end{bmatrix}$. Mass 1 is displaced by 1 m and released; mass 2 starts at rest. Find $\mathbf{x}(t)$.

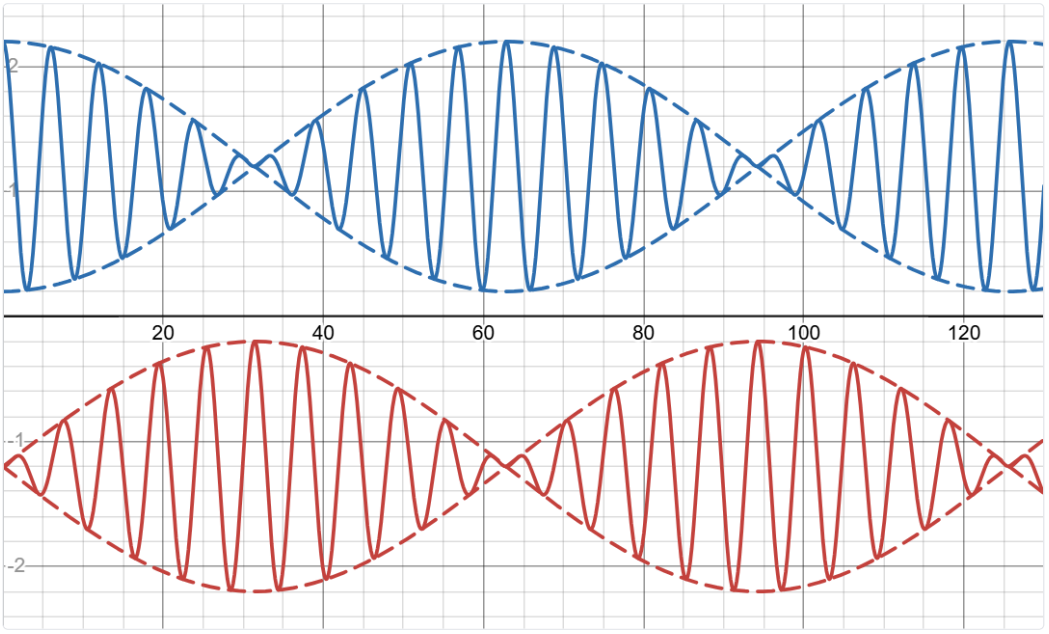
Step 1 — modes. As in Example 10.31: in-phase $(1,1)$ with $\omega_1^2 = 1$, so $\omega_1 = 1$ rad/s; antiphase $(1,-1)$ with $\omega_2^2 = 1 + 2k_c = 1.21$, so $\omega_2 = 1.1$ rad/s.

Step 2 — modal amplitudes. $\mathbf{x}(0) = (1,0) = \tfrac{1}{2}(1,1) + \tfrac{1}{2}(1,-1)$ and $\dot{\mathbf{x}}(0) = \mathbf{0}$, so only cosine terms appear, each with amplitude $\tfrac{1}{2}$.

Step 3 — superpose and simplify with sum-to-product identities (Ch. 3):

$$x_1(t) = \tfrac{1}{2}(\cos t + \cos 1.1t) = \cos(0.05t)\cos(1.05t),$$
$$x_2(t) = \tfrac{1}{2}(\cos t - \cos 1.1t) = \sin(0.05t)\sin(1.05t).$$

Interpretation. Both masses oscillate at the mean frequency 1.05 rad/s under slowly varying envelopes $|\cos 0.05t|$ and $|\sin 0.05t|$. At $t = \pi/0.1 = 31.4$ s mass 1 is momentarily still and mass 2 carries all the motion; at 62.8 s it has all come back. The exchange time $\pi/(\omega_2 - \omega_1)$ is set by the **mode splitting**: the weaker the coupling, the slower the exchange. The same mathematics describes energy transfer in a directional coupler and between coupled optical waveguides, with distance in place of time.



Free motion of two weakly coupled oscillators (Example 10.33), plotted against t in seconds: $x_1(t)$ (blue, offset $+1.2$) and $x_2(t)$ (red, offset -1.2). The dashed envelopes $\pm\cos 0.05t$ and $\pm\sin 0.05t$ show energy passing completely from one oscillator to the other every 31.4 s — the beat between the normal-mode frequencies 1 and 1.1 rad/s.

Worked Example 10.34 — Modes of an RC ladder (first-order modal analysis)

A two-section RC ladder ($R = 1$ kΩ, $C = 1$ μF per section, so $RC = 1$ ms) has its input shorted and both capacitors initially at 1 V. Find $v_1(t)$ and $v_2(t)$.

Step 1 — state equations. KCL at node 1: $C\dot{v}_1 = -v_1/R - (v_1 - v_2)/R$; at node 2: $C\dot{v}_2 = (v_1 - v_2)/R$. With t in ms,

$$\dot{\mathbf{v}} = \mathbf{A}\mathbf{v}, \qquad \mathbf{A} = \begin{bmatrix} -2 & 1 \\ 1 & -1 \end{bmatrix}.$$

Step 2 — eigenvalues. $\lambda^2 + 3\lambda + 1 = 0$: $\lambda = \dfrac{-3 \pm \sqrt{5}}{2}$, i.e. $\lambda_s = -0.382$ and $\lambda_f = -2.618$ ms^{-1}. Time constants: $\tau_s = 2.618$ ms (slow) and $\tau_f = 0.382$ ms (fast).

Step 3 — eigenvectors. $\mathbf{A}$ is symmetric, so they are orthogonal. From the second row, $v_1 = (1 + \lambda)v_2$: $\mathbf{v}_s \propto (0.618, 1) \propto (1, 1.618)$ (both capacitors discharging together, the far one higher) and $\mathbf{v}_f \propto (1, -0.618)$ (charge sloshing between the capacitors). Normalised: $\mathbf{q}_s = (0.5257, 0.8507)$, $\mathbf{q}_f = (0.8507, -0.5257)$.

Step 4 — project the initial state. Orthonormal eigenvectors make the left eigenvectors equal to the right ones: $c_s = \mathbf{q}_s \cdot (1, 1) = 1.3764$, $c_f = \mathbf{q}_f \cdot (1, 1) = 0.3249$.

Step 5 — response.

$$\begin{aligned} v_1(t) &= 0.7236\, e^{-t/2.618} + 0.2764\, e^{-t/0.382}, \\ v_2(t) &= 1.1708\, e^{-t/2.618} - 0.1708\, e^{-t/0.382} \qquad (t \text{ in ms, } v \text{ in V}). \end{aligned}$$

Checks. $t = 0$: $v_1 = 1.0000$, $v_2 = 1.0000$ ✓. At $t = 1$ ms: $v_1 = 0.7236(0.6825) + 0.2764(0.0729) = 0.5140$ V and $v_2 = 0.7866$ V, matching a numerical $e^{\mathbf{A}t}\mathbf{v}(0)$ ✓. After about 1 ms the fast mode has essentially vanished and the circuit behaves as a single-time-constant system with $\tau = 2.618$ ms — the basis of **dominant-pole approximation**. Note that neither time constant equals RC: the sections interact, which is why "n cascaded RC sections" is not n identical poles.

Common Pitfall — normalising modes and projecting initial conditions

When the mass (or capacitance) matrix is not a multiple of $\mathbf{I}$, mode shapes are orthogonal **with respect to** $\mathbf{M}$, not in the ordinary dot product. The modal amplitude is $a_i = \boldsymbol{\phi}_i^{\mathsf{T}}\mathbf{M}\mathbf{x}(0)$ for $\mathbf{M}$-normalised modes; using $\boldsymbol{\phi}_i^{\mathsf{T}}\mathbf{x}(0)$ gives wrong answers. For non-symmetric first-order systems, use the **left** eigenvectors (rows of $\mathbf{P}^{-1}$) to project; the right eigenvectors are not orthogonal.

Engineering Note — modal analysis at scale

Finite-element models of PCBs, packages, enclosures and antennas have 10^4–10^7 degrees of freedom, but only the few modes in the frequency band of interest matter. Solvers (Lanczos and Arnoldi methods, shift-invert about a target frequency) extract just those eigenpairs from the sparse pair $(\mathbf{K}, \mathbf{M})$. Cavity resonances of a metal enclosure, which can turn a shielded box into an efficient radiator at specific frequencies, are found this way; so are the even and odd propagation modes of coupled microstrip lines (eigenvectors of the product of per-unit-length $\mathbf{L}$ and $\mathbf{C}$ matrices), whose different velocities cause far-end crosstalk.

10.9 Applications to system dynamics and stability

Motivation

Every circuit, control loop and oscillator must answer one question before any other: **if it is disturbed, does it return to rest, settle into a steady oscillation, or run away?** An amplifier that oscillates, a phase-locked loop that loses lock, a switched-mode converter whose output voltage rings ever larger, a power grid whose generators swing apart — all are stability failures, and for linear (or linearized) models all are diagnosed by eigenvalues. This section collects the results: stability criteria for continuous- and discrete-time systems, the complete geometric classification of two-dimensional systems, linearization of nonlinear systems about an equilibrium, and Lyapunov's energy-like test, which brings positive-definite matrices back into the story. Chapters 11 and 12 develop the differential-equation side; here we concentrate on what the matrix tells us.

Stability of linear systems

Consider $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$, whose equilibrium is $\mathbf{x} = \mathbf{0}$. From Section 10.4 every solution is a combination of terms $t^k e^{\lambda_i t}\mathbf{v}$, with polynomial factors t^k only for defective eigenvalues. Since $|e^{\lambda t}| = e^{(\operatorname{Re}\lambda)t}$, the real parts decide everything.

Stability of an equilibrium

The equilibrium $\mathbf{x} = \mathbf{0}$ of $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ is

- **asymptotically stable** if every solution tends to $\mathbf{0}$ as $t \to \infty$;
- **marginally stable** (stable but not asymptotically) if every solution stays bounded but some do not tend to $\mathbf{0}$;
- **unstable** if some solution grows without bound.

Eigenvalue criteria for stability

Continuous time, $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$:

- asymptotically stable $\iff$ $\operatorname{Re}\lambda_i < 0$ for every eigenvalue (all eigenvalues in the open left half-plane; $\mathbf{A}$ is then called **Hurwitz**);
- marginally stable $\iff$ $\operatorname{Re}\lambda_i \le 0$ for all i, and every eigenvalue on the imaginary axis is non-defective;
- unstable otherwise.

Discrete time, $\mathbf{x}[k+1] = \mathbf{A}\mathbf{x}[k]$: the same statements with "$\operatorname{Re}\lambda < 0$" replaced by "$|\lambda| < 1$" and "the imaginary axis" by "the unit circle" ($\mathbf{A}$ is then called **Schur**).

For a linear time-invariant system with input and output, the eigenvalues of $\mathbf{A}$ are the **poles** of the transfer function $\mathbf{C}(s\mathbf{I}-\mathbf{A})^{-1}\mathbf{B}+\mathbf{D}$ — except for any modes that are uncontrollable or unobservable and therefore cancel (Section 10.8). This is why "all poles in the left half-plane" is the same condition as "all eigenvalues in the left half-plane" for a minimal realisation, and why a hidden unstable mode is dangerous: it does not appear in the transfer function, yet the internal state still grows.

Classifying two-dimensional systems: the trace–determinant plane

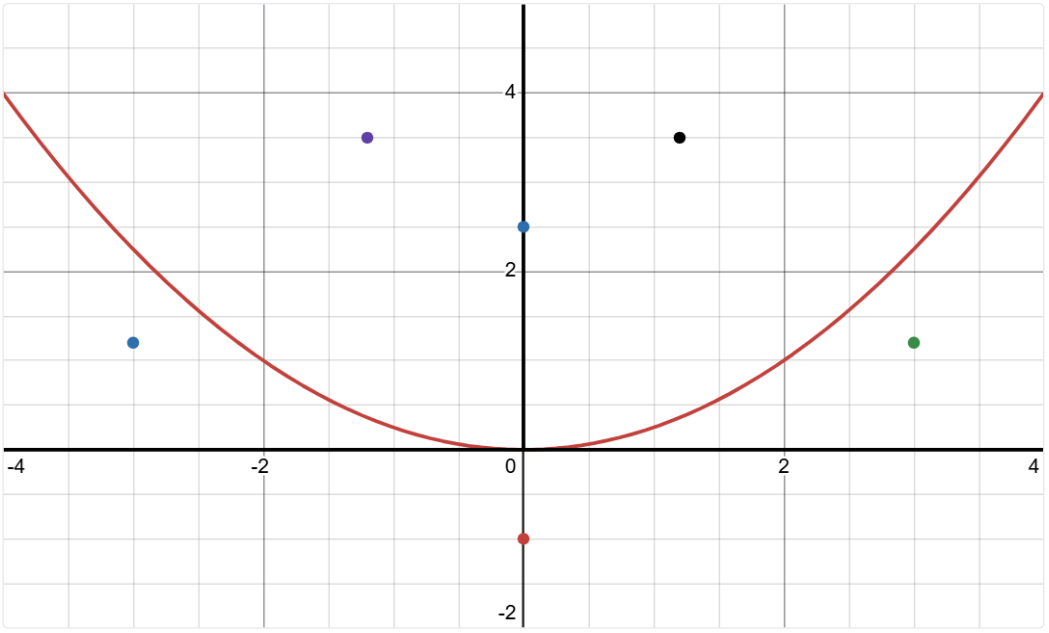
For a real 2×2 matrix, write $\tau = \operatorname{tr}\mathbf{A}$ and $\Delta = \det\mathbf{A}$. The eigenvalues are

$$\lambda_{1,2} = \frac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2}, \qquad \lambda_1+\lambda_2 = \tau, \quad \lambda_1\lambda_2 = \Delta,$$

so the two numbers (τ, Δ) determine the qualitative behaviour completely:

Region of the (τ,Δ) plane	Eigenvalues	Equilibrium type
$\Delta < 0$	real, opposite signs	**saddle** (unstable)
$\Delta > 0,\ \tau^2 > 4\Delta,\ \tau < 0$	real, both negative	**stable node**
$\Delta > 0,\ \tau^2 > 4\Delta,\ \tau > 0$	real, both positive	**unstable node**
$\Delta > 0,\ \tau^2 < 4\Delta,\ \tau < 0$	complex, $\operatorname{Re} < 0$	**stable spiral** (focus)
$\Delta > 0,\ \tau^2 < 4\Delta,\ \tau > 0$	complex, $\operatorname{Re} > 0$	**unstable spiral**
$\Delta > 0,\ \tau = 0$	$\pm j\sqrt{\Delta}$	**centre** (marginally stable)
$\tau^2 = 4\Delta$	real, repeated	degenerate (star) node — boundary case
$\Delta = 0$	one zero eigenvalue	line of equilibria — boundary case

The compact stability statement for $n=2$ is: **asymptotically stable** $\iff$ $\tau < 0$ **and** $\Delta > 0$. For a second-order circuit with characteristic polynomial $\lambda^2 + 2\zeta\omega_0\lambda + \omega_0^2$, $\tau = -2\zeta\omega_0$ and $\Delta = \omega_0^2$, and the parabola $\tau^2 = 4\Delta$ is the critical damping $\zeta = 1$: overdamped circuits are stable nodes, underdamped ones stable spirals, and a lossless LC tank is a centre.



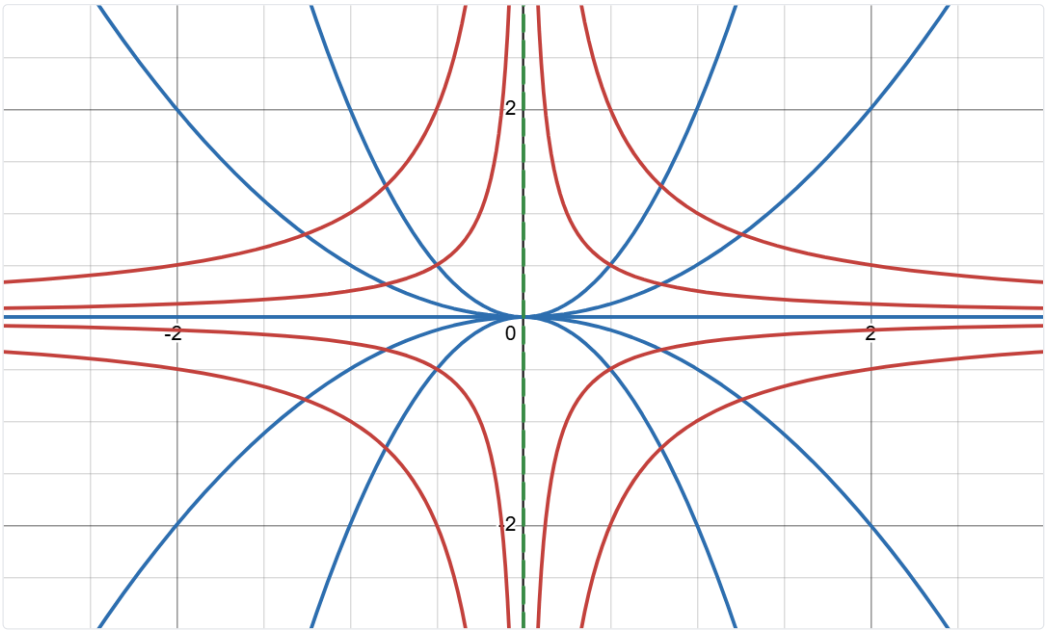
The trace–determinant plane (horizontal axis $\tau = \operatorname{tr}\mathbf{A}$, vertical axis $\Delta = \det\mathbf{A}$). Below the τ-axis lie saddles; above it, the parabola $\Delta = \tau^2/4$ (red, critical damping) separates nodes (outside) from spirals (inside); the positive Δ-axis carries centres. The stable region is the open upper-left quadrant $\tau < 0,\ \Delta > 0$.

Phase portraits from eigenvectors

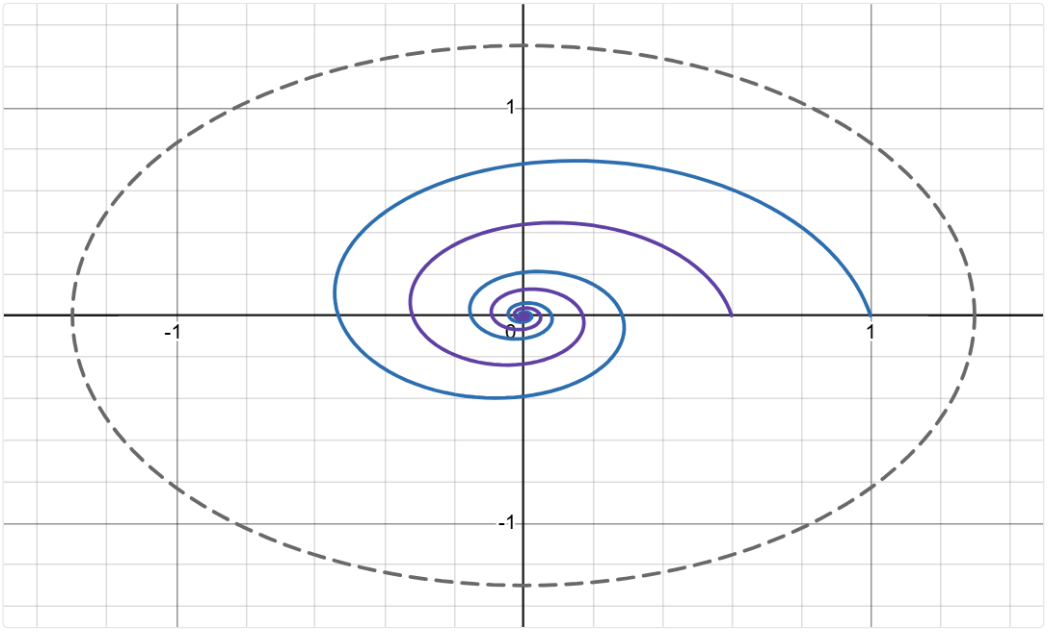
The **phase portrait** draws solution trajectories in the (x_1, x_2) plane. Eigenvectors organise it: a trajectory that starts on a real eigenvector line stays on it (it is the solution $c\,e^{\lambda t}\mathbf{v}$), moving inward if $\lambda < 0$ and outward if $\lambda > 0$. In eigenvector coordinates (y_1, y_2) a system with real eigenvalues is $\dot{y}_1 = \lambda_1 y_1$, $\dot{y}_2 = \lambda_2 y_2$, so $y_i = c_i e^{\lambda_i t}$, and eliminating t gives the **explicit trajectory curves**

$$y_2 = C\,|y_1|^{\lambda_2/\lambda_1}.$$

For a stable node with $\lambda_1 = -1$, $\lambda_2 = -2$: $y_2 = C y_1^2$, a family of parabolas all tangent at the origin to the **slow** eigenvector (y_1-axis). For a saddle with $\lambda_1 = 1$, $\lambda_2 = -1$: $y_1 y_2 = C$, hyperbolas whose asymptotes are the eigenvector lines. For a complex pair $\sigma \pm j\omega$, the real block form $\begin{bmatrix} \sigma & -\omega \\ \omega & \sigma \end{bmatrix}$ in polar coordinates gives $\dot r = \sigma r$, $\dot\theta = \omega$, hence $r = r_0 e^{(\sigma/\omega)(\theta-\theta_0)}$: **logarithmic spirals**, shrinking if $\sigma < 0$. In the original coordinates these pictures are mapped through $\mathbf{P}$, which skews them but preserves their type.



Explicit trajectory families in eigenvector coordinates. Blue: stable node with $\lambda_1 = -1$, $\lambda_2 = -2$, trajectories $y_2 = C y_1^2$, all entering the origin tangent to the slow eigen-direction (the y_1-axis); the fast direction is the y_2-axis (green dashed). Red: saddle with $\lambda_1 = 1$, $\lambda_2 = -1$, trajectories $y_1 y_2 = C$, whose asymptotes are the two eigenvector axes.



Trajectories of a stable spiral with eigenvalues $-0.2 \pm j$ (real block form): $r = r_0 e^{-0.2\theta}$, two starting radii shown. Each turn shrinks the radius by $e^{-0.2\cdot 2\pi} = 0.285$; with $\operatorname{Re}\lambda = 0$ the curves would be circles (a centre, dashed reference), and with $\operatorname{Re}\lambda > 0$ they would spiral outward.

Routh–Hurwitz: stability without eigenvalues

Often only the characteristic polynomial $\det(s\mathbf{I} - \mathbf{A}) = s^n + a_{n-1}s^{n-1} + \cdots + a_0$ is needed, and only the *signs* of the real parts of its roots. The Routh–Hurwitz criterion decides this from the coefficients. Its lowest-order cases are worth memorising:

- $n = 1, 2$: all roots in the left half-plane $\iff$ all coefficients positive.
- $n = 3$: $s^3 + a_2 s^2 + a_1 s + a_0$ is Hurwitz $\iff$ $a_2, a_1, a_0 > 0$ **and** $a_2 a_1 > a_0$.

Positivity of all coefficients is **necessary** for every n (a product of factors $(s+p)$ and $(s^2 + bs + c)$ with $p, b, c > 0$ has only positive coefficients) but not sufficient for $n \geq 3$: $s^3 + s^2 + s + 6$ has positive coefficients but $a_2 a_1 = 1 < 6$, and it has the unstable roots $0.5 \pm j1.66$. The full Routh array for higher orders is part of control theory and is treated with the Laplace transform in Ch. 14.

Linearization of nonlinear systems

Most real systems are nonlinear: $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$. An **equilibrium** $\mathbf{x}^*$ satisfies $\mathbf{f}(\mathbf{x}^*) = \mathbf{0}$ — the DC operating point of a circuit. For small deviations $\boldsymbol{\xi} = \mathbf{x} - \mathbf{x}^*$, the multivariable Taylor expansion of Ch. 6 gives

$$\dot{\boldsymbol{\xi}} = \mathbf{f}(\mathbf{x}^* + \boldsymbol{\xi}) \approx \mathbf{J}\boldsymbol{\xi}, \qquad \mathbf{J} = \left.\frac{\partial \mathbf{f}}{\partial \mathbf{x}}\right|_{\mathbf{x}^*} = \left[\frac{\partial f_i}{\partial x_k}\right]_{\mathbf{x}^*},$$

the **Jacobian matrix**. This is exactly the small-signal model of circuit analysis: transistors and diodes are replaced by their incremental conductances at the operating point.

Linearized stability (Lyapunov's indirect method; Hartman–Grobman)

Let $\mathbf{J}$ be the Jacobian at an equilibrium $\mathbf{x}^*$ of $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$, with $\mathbf{f}$ continuously differentiable.

- If every eigenvalue of $\mathbf{J}$ has $\operatorname{Re}\lambda < 0$, $\mathbf{x}^*$ is locally asymptotically stable.
- If some eigenvalue has $\operatorname{Re}\lambda > 0$, $\mathbf{x}^*$ is unstable.
- If no eigenvalue has zero real part (a *hyperbolic* equilibrium), the nonlinear phase portrait near $\mathbf{x}^*$ has the same type (node, saddle, spiral) as the linearization.

If some eigenvalue has $\operatorname{Re}\lambda = 0$ and none is positive, the linearization is **inconclusive**: the neglected nonlinear terms decide.

Lyapunov's equation: stability through energy

A physical way to prove stability is to exhibit an "energy" that always decreases. For a linear system try a quadratic form $V(\mathbf{x}) = \mathbf{x}^{\mathsf{T}}\mathbf{P}\mathbf{x}$ with $\mathbf{P} \succ 0$. Along solutions,

$$\dot{V} = \dot{\mathbf{x}}^{\mathsf{T}}\mathbf{P}\mathbf{x} + \mathbf{x}^{\mathsf{T}}\mathbf{P}\dot{\mathbf{x}} = \mathbf{x}^{\mathsf{T}}(\mathbf{A}^{\mathsf{T}}\mathbf{P} + \mathbf{P}\mathbf{A})\mathbf{x}.$$

Lyapunov stability theorem (linear case)

$\mathbf{A}$ is Hurwitz (all $\operatorname{Re}\lambda < 0$) **if and only if** for some (equivalently, every) symmetric $\mathbf{Q} \succ 0$ the **Lyapunov equation**

$$\mathbf{A}^{\mathsf{T}}\mathbf{P} + \mathbf{P}\mathbf{A} = -\mathbf{Q}$$

has a symmetric positive-definite solution $\mathbf{P}$. The solution is then unique and equals $\mathbf{P} = \int_0^\infty e^{\mathbf{A}^{\mathsf{T}}t}\mathbf{Q}e^{\mathbf{A}t}\,dt$.

Why it works: with such a $\mathbf{P}$, $V > 0$ away from the origin while $\dot{V} = -\mathbf{x}^{\mathsf{T}}\mathbf{Q}\mathbf{x} < 0$, so V — and with it $\mathbf{x}$ — decays to zero; the level curves of V are nested ellipses that trajectories cross only inwards. The discrete-time counterpart is $\mathbf{A}^{\mathsf{T}}\mathbf{P}\mathbf{A} - \mathbf{P} = -\mathbf{Q}$. The Lyapunov approach matters because it extends to nonlinear and time-varying systems, where eigenvalues alone say little, and because $\mathbf{P}$ itself has meanings: for a stable system driven by white noise, the solution of a closely related Lyapunov equation is the steady-state **covariance** of the state (Ch. 21).

Worked examples

Worked Example 10.35 — Natural frequencies of a series RLC circuit

A series RLC circuit with $R = 100\ \Omega$, $L = 10$ mH, $C = 1\ \mu$F is left to discharge. With state $\mathbf{x} = (i, v_C)$, find the eigenvalues and characterise the response.

Step 1 — state equations. KVL: $L\,di/dt = -Ri - v_C$; capacitor: $C\,dv_C/dt = i$.

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}, \qquad \mathbf{A} = \begin{bmatrix} -R/L & -1/L \\ 1/C & 0 \end{bmatrix} = \begin{bmatrix} -10^4 & -100 \\ 10^6 & 0 \end{bmatrix}.$$

(Units: R/L in s^{-1}; $1/L$ in A/(V·s); $1/C$ in V/(A·s). The off-diagonal units differ, but their product $1/(LC)$ is in s^{-2}, as an entry of $\det \mathbf{A}$ must be.)

Step 2 — trace and determinant. $\tau = -R/L = -10^4\ \text{s}^{-1}$, $\Delta = 1/(LC) = 10^8\ \text{s}^{-2}$. Since $\tau < 0$ and $\Delta > 0$: **asymptotically stable**. Since $\tau^2 = 10^8 < 4\Delta = 4 \times 10^8$: **spiral** (underdamped).

Step 3 — eigenvalues.

$$\lambda = -5000 \pm j\sqrt{10^8 - 2.5 \times 10^7} = -5000 \pm j\,8660\ \text{s}^{-1}.$$

Interpretation. $\omega_0 = \sqrt{\Delta} = 10^4$ rad/s, $\zeta = -\tau/(2\omega_0) = 0.5$; the current rings at $\omega_d = 8660$ rad/s (1.378 kHz) inside an envelope $e^{-t/0.2\,\text{ms}}$. Doubling R to $200\ \Omega$ gives $\tau^2 = 4\Delta$: critical damping, a defective double eigenvalue at $-10^4\ \text{s}^{-1}$, and $te^{-10^4 t}$ terms (compare Example 10.11).

Worked Example 10.36 — Reading the type off trace and determinant

Classify the equilibrium of $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ for each matrix.

$\mathbf{A}$	τ	Δ	$\tau^2 - 4\Delta$	Eigenvalues	Type
$\begin{bmatrix} -3 & 1 \\ 1 & -3 \end{bmatrix}$	-6	8	4	$-2,\ -4$	stable node
$\begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix}$	-3	-10	49	$2,\ -5$	saddle (unstable)
$\begin{bmatrix} -1 & -4 \\ 1 & -1 \end{bmatrix}$	-2	5	-16	$-1 \pm 2j$	stable spiral
$\begin{bmatrix} 0 & 1 \\ -4 & 0 \end{bmatrix}$	0	4	-16	$\pm 2j$	centre
$\begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}$	4	5	-4	$2 \pm j$	unstable spiral

Notes. The saddle has negative trace yet is unstable: a negative trace alone proves nothing. Its eigenvector for $\lambda = 2$ is $(2, 1)$ (from $\mathbf{A} - 2\mathbf{I} = \begin{bmatrix} -1 & 2 \\ 3 & -6 \end{bmatrix}$) — the single direction along which almost every trajectory eventually escapes. For the spiral, the sense of rotation follows from any one velocity: at $\mathbf{x} = (1, 0)$, $\dot{\mathbf{x}} = (-1, 1)$ has positive $\dot{x}_2$, so the rotation is counter-clockwise.

Worked Example 10.37 — Linearizing the damped pendulum

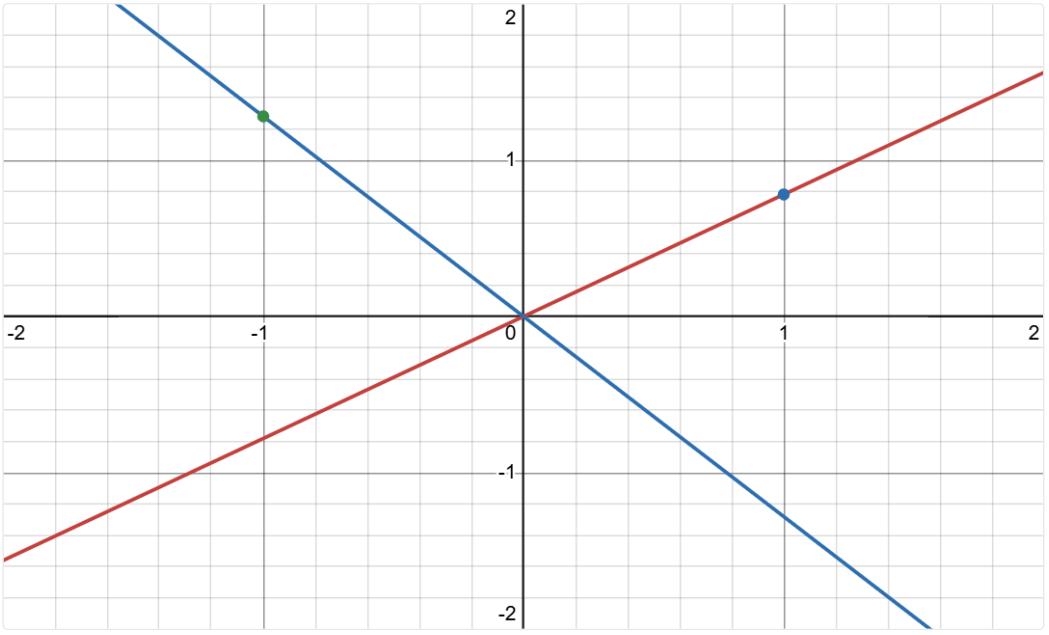
A pendulum (or, equivalently, the swing equation of a synchronous generator against an infinite bus) obeys, in normalised units, $\ddot{\theta} + 0.5\dot{\theta} + \sin\theta = 0$. With $\mathbf{x} = (\theta, \omega)$, $\omega = \dot{\theta}$:

$$\mathbf{f}(\mathbf{x}) = \begin{bmatrix} \omega \\ -\sin\theta - 0.5\,\omega \end{bmatrix}, \qquad \mathbf{J}(\theta, \omega) = \begin{bmatrix} 0 & 1 \\ -\cos\theta & -0.5 \end{bmatrix}.$$

Equilibria. $\omega = 0$, $\sin\theta = 0$: $\theta = 0$ (hanging down) and $\theta = \pi$ (inverted).

At $(0, 0)$. $\mathbf{J} = \begin{bmatrix} 0 & 1 \\ -1 & -0.5 \end{bmatrix}$, $\tau = -0.5$, $\Delta = 1$, $\tau^2 - 4\Delta = -3.75 < 0$. Eigenvalues $-0.25 \pm j\,0.968$: a **stable spiral** (decaying swings). Hyperbolic, so the nonlinear pendulum behaves the same way near the bottom.

At $(\pi, 0)$. $\mathbf{J} = \begin{bmatrix} 0 & 1 \\ 1 & -0.5 \end{bmatrix}$, $\Delta = -1 < 0$: a **saddle**. Eigenvalues from $\lambda^2 + 0.5\lambda - 1 = 0$: $\lambda = \frac{-0.5 \pm \sqrt{4.25}}{2} = 0.781,\ -1.281$. Eigenvectors of this companion-type matrix are $(1, \lambda)$: the unstable direction is $(1, 0.781)$, the stable direction $(1, -1.281)$. The stable eigen-line is the linear approximation of the **separatrix** — the knife-edge set of initial conditions that just reach the inverted position. In power systems the analogous saddle (the unstable equilibrium of the swing equation) defines the boundary of the region from which a generator can recover after a fault.



Eigenvector lines of the linearized pendulum at the inverted equilibrium, plotted in deviation coordinates $(\theta - \pi,\ \omega)$: trajectories approach along the blue stable line and leave along the red unstable line, the local skeleton of the saddle.

Worked Example 10.38 — Start-up condition of a negative-resistance oscillator

A parallel LC tank loaded by a nonlinear negative-resistance device (tunnel diode, cross-coupled transistor pair) obeys the van der Pol equation $\ddot{v} - \varepsilon(1 - v^2)\dot{v} + \omega_0^2 v = 0$, with $\varepsilon > 0$ set by how much negative conductance exceeds the tank losses. Will the oscillation start from noise?

Linearize at $v = \dot{v} = 0$. With $\mathbf{x} = (v, \dot{v})$, the term $\varepsilon v^2 \dot{v}$ is cubic and drops out:

$$\mathbf{J} = \begin{bmatrix} 0 & 1 \\ -\omega_0^2 & \varepsilon \end{bmatrix}, \qquad \tau = \varepsilon > 0,\ \Delta = \omega_0^2 > 0.$$

Classify. $\tau > 0$, $\Delta > 0$: unstable. For $\varepsilon < 2\omega_0$ the eigenvalues $\frac{\varepsilon}{2} \pm j\sqrt{\omega_0^2 - \varepsilon^2/4}$ are complex: an **unstable spiral**. Any tiny noise voltage grows as $e^{\varepsilon t/2}$ while oscillating at nearly ω_0 — the oscillator starts.

What eigenvalues cannot tell. The growth does not continue for ever: as v grows, the nonlinear term reduces the effective negative conductance and the trajectory settles onto a **limit cycle** of amplitude about 2. The linearization predicts *start-up* (Barkhausen's loop-gain condition is the same statement) but not the final amplitude, which is a nonlinear question (Ch. 11).

Worked Example 10.39 — Stability of a digital resonator

The recursion $y[n] = 1.8\,y[n-1] - 0.9\,y[n-2]$ is the homogeneous part of a second-order IIR filter. Is it stable, and at what frequency does it ring?

Step 1 — state form. With $\mathbf{x}[n] = (y[n], y[n-1])$:

$$\mathbf{x}[n+1] = \begin{bmatrix} 1.8 & -0.9 \\ 1 & 0 \end{bmatrix} \mathbf{x}[n], \qquad \det(\lambda\mathbf{I} - \mathbf{A}) = \lambda^2 - 1.8\lambda + 0.9.$$

Step 2 — eigenvalues. $\lambda = 0.9 \pm j\sqrt{0.9 - 0.81} = 0.9 \pm j\,0.3$, so $|\lambda| = \sqrt{0.9} = 0.9487$ and $\angle\lambda = \arctan(0.3/0.9) = 0.3218$ rad $= 18.43^\circ$.

Step 3 — conclusions. $|\lambda| < 1$: **stable**; the ringing decays by a factor 0.9487 per sample (to 1% in $\ln 0.01 / \ln 0.9487 \approx 87$ samples) and oscillates at 0.3218 rad/sample, i.e. $0.3218/(2\pi) = 0.0512$ cycles/sample — 2.46 kHz at a 48 kHz sample rate. Note $|\lambda|^2 = \det \mathbf{A} = 0.9$: for any second-order section $y[n] = a_1 y[n-1] - a_2 y[n-2]$ with complex poles, the pole radius is $\sqrt{a_2}$, so $a_2 < 1$ is required.

Step 4 — the boundary. Changing 0.9 to 1.0 gives $\lambda = e^{\pm j\,0.451}$ on the unit circle: a marginally stable digital oscillator. In fixed-point arithmetic, rounding of the coefficient can push the poles slightly outside the unit circle — a practical reason for using structures (such as coupled-form or lattice filters) whose eigenvalues are less sensitive to coefficient quantisation.

Worked Example 10.40 — Solving a Lyapunov equation

For $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$ (eigenvalues $-1, -2$) solve $\mathbf{A}^{\mathsf{T}}\mathbf{P} + \mathbf{P}\mathbf{A} = -\mathbf{I}$ and confirm that $\mathbf{P} \succ 0$.

Step 1 — unknowns. $\mathbf{P} = \begin{bmatrix} p_1 & p_2 \\ p_2 & p_3 \end{bmatrix}$. Compute $\mathbf{A}^{\mathsf{T}}\mathbf{P} = \begin{bmatrix} -2p_2 & -2p_3 \\ p_1 - 3p_2 & p_2 - 3p_3 \end{bmatrix}$; $\mathbf{P}\mathbf{A}$ is its transpose.

Step 2 — equate entries of the symmetric sum to $-\mathbf{I}$.

$$(1,1):\ -4p_2 = -1, \qquad (2,2):\ 2p_2 - 6p_3 = -1, \qquad (1,2):\ p_1 - 3p_2 - 2p_3 = 0.$$

So $p_2 = 0.25$, $p_3 = (1 + 0.5)/6 = 0.25$, $p_1 = 3(0.25) + 2(0.25) = 1.25$:

$$\mathbf{P} = \begin{bmatrix} 1.25 & 0.25 \\ 0.25 & 0.25 \end{bmatrix}.$$

Step 3 — definiteness. $p_1 = 1.25 > 0$, $\det \mathbf{P} = 0.3125 - 0.0625 = 0.25 > 0$: $\mathbf{P} \succ 0$, confirming that $\mathbf{A}$ is Hurwitz without computing an eigenvalue. (A numerical `solve_continuous_lyapunov` returns the same $\mathbf{P}$.)

Step 4 — interpretation. $V(\mathbf{x}) = 1.25x_1^2 + 0.5x_1x_2 + 0.25x_2^2$ is an energy-like function with $\dot{V} = -(x_1^2 + x_2^2)$ along every trajectory. Its level sets are ellipses that every trajectory crosses inward.

Common Pitfall — stable eigenvalues, large transients

Eigenvalues describe the behaviour as $t \to \infty$, not the size of the response on the way. For the stable but strongly **non-normal** matrix $\mathbf{A} = \begin{bmatrix} -1 & 50 \\ 0 & -2 \end{bmatrix}$ (eigenvalues $-1, -2$),

$$e^{\mathbf{A}t} = \begin{bmatrix} e^{-t} & 50(e^{-t} - e^{-2t}) \\ 0 & e^{-2t} \end{bmatrix},$$

and the $(1,2)$ entry reaches $50(0.5 - 0.25) = 12.5$ at $t = \ln 2$: a unit initial condition on x_2 produces a twelve-fold excursion in x_1 before decaying. If the model came from linearization, such an excursion can drive the real system out of the region where the linear model is valid — or saturate an amplifier. The eigenvector matrix of such an $\mathbf{A}$ is ill-conditioned (Section 10.3); singular values of $e^{\mathbf{A}t}$, or a Lyapunov function, quantify the transient.

Engineering Note — stability margins and eigenvalue location

Asymptotic stability is a yes/no property, but designs need margins. Requiring all eigenvalues to satisfy $\operatorname{Re}\lambda \le -\alpha$ guarantees decay at least as fast as $e^{-\alpha t}$ (settling time about $4/\alpha$); requiring them inside a sector $|\angle(-\lambda)| \le \arccos\zeta_{\min}$ guarantees a minimum damping ratio and hence bounded overshoot. In power electronics and power systems, the eigenvalue plot of the linearized model versus a parameter (load, gain, line impedance) — a **root locus** — shows how close each mode is to crossing the imaginary axis; the crossing point, where a complex pair reaches $\operatorname{Re}\lambda = 0$, is a **Hopf bifurcation**, the birth of a sustained oscillation as in Example 10.38.

Summary

Concept	Key formula or statement
Eigenpair	$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$, $\mathbf{v} \neq \mathbf{0}$; eigenspace $E_\lambda = \operatorname{null}(\mathbf{A} - \lambda\mathbf{I})$
Characteristic equation	$\det(\mathbf{A} - \lambda\mathbf{I}) = 0$, degree n, n roots in $\mathbb{C}$
2×2 shortcut	$\lambda^2 - (\operatorname{tr}\mathbf{A})\lambda + \det\mathbf{A} = 0$
3×3 shortcut	$\lambda^3 - (\operatorname{tr}\mathbf{A})\lambda^2 + E_2\lambda - \det\mathbf{A} = 0$, E_2 = sum of 2×2 principal minors
Trace and determinant	$\sum\lambda_i = \operatorname{tr}\mathbf{A}$, $\prod\lambda_i = \det\mathbf{A}$
Cayley–Hamilton	$p(\mathbf{A}) = \mathbf{0}$; e.g. $\mathbf{A}^2 - (\operatorname{tr}\mathbf{A})\mathbf{A} + (\det\mathbf{A})\mathbf{I} = \mathbf{0}$
Multiplicities	$1 \le g(\lambda) \le m(\lambda)$; defective if $g < m$
Functions of $\mathbf{A}$	eigenvalues of $q(\mathbf{A})$, $\mathbf{A}^{-1}$, $e^{\mathbf{A}t}$ are $q(\lambda)$, $1/\lambda$, $e^{\lambda t}$ (same eigenvectors)
Diagonalization	$\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$ iff n independent eigenvectors; distinct eigenvalues suffice
Real block form	$\lambda = \sigma \pm j\omega$, $\mathbf{v} = \mathbf{a} + j\mathbf{b}$: $[\mathbf{a}\ \mathbf{b}]^{-1}\mathbf{A}[\mathbf{a}\ \mathbf{b}] = \begin{bmatrix}\sigma & \omega\\ -\omega & \sigma\end{bmatrix}$
Jordan block powers	$\mathbf{J}_2(\lambda)^k = \begin{bmatrix}\lambda^k & k\lambda^{k-1}\\ 0 & \lambda^k\end{bmatrix}$, $e^{\mathbf{J}_2 t} = e^{\lambda t}\begin{bmatrix}1 & t\\ 0 & 1\end{bmatrix}$
Matrix powers	$\mathbf{A}^k = \mathbf{P}\mathbf{D}^k\mathbf{P}^{-1}$; $\mathbf{x}[k] = \sum c_i\lambda_i^k\mathbf{v}_i$; $\mathbf{A}^k \to \mathbf{0} \iff \rho(\mathbf{A}) < 1$
Matrix exponential	$e^{\mathbf{A}t} = \sum\mathbf{A}^k t^k/k! = \mathbf{P}e^{\mathbf{D}t}\mathbf{P}^{-1}$; $\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}(0) = \sum c_i e^{\lambda_i t}\mathbf{v}_i$
Sylvester interpolation	$f(\mathbf{A}) = \sum_{k<n}\alpha_k\mathbf{A}^k$ with $f(\lambda_i) = \sum\alpha_k\lambda_i^k$ (plus derivatives for repeated λ_i)
Spectral theorem	$\mathbf{A} = \mathbf{A}^{\mathsf{T}} \iff \mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^{\mathsf{T}} = \sum\lambda_i\mathbf{q}_i\mathbf{q}_i^{\mathsf{T}}$, λ_i real, $\mathbf{Q}^{\mathsf{T}}\mathbf{Q} = \mathbf{I}$
Rayleigh quotient	$\lambda_{\min} \le \mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x}/\mathbf{x}^{\mathsf{T}}\mathbf{x} \le \lambda_{\max}$
Principal axes	$\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} = \sum\lambda_i y_i^2$ with $\mathbf{y} = \mathbf{Q}^{\mathsf{T}}\mathbf{x}$; ellipse semi-axes $1/\sqrt{\lambda_i}$ for $\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} = 1$
Positive definite	$\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} > 0 \iff$ all $\lambda_i > 0 \iff$ all leading minors $> 0 \iff$ all pivots $> 0 \iff \mathbf{A} = \mathbf{L}\mathbf{L}^{\mathsf{T}}$
2×2 PD test	$a_{11} > 0$ and $a_{11}a_{22} - a_{12}^2 > 0$ (coupled inductors: $M^2 < L_1L_2$)
SVD	$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\mathsf{T}} = \sum\sigma_i\mathbf{u}_i\mathbf{v}_i^{\mathsf{T}}$, $\sigma_i^2 = \lambda_i(\mathbf{A}^{\mathsf{T}}\mathbf{A})$, $\mathbf{A}\mathbf{v}_i = \sigma_i\mathbf{u}_i$
Norms and conditioning	$\|\mathbf{A}\|_2 = \sigma_1$, $\|\mathbf{A}\|_F^2 = \sum\sigma_i^2$, $\kappa_2 = \sigma_{\max}/\sigma_{\min}$
Eckart–Young	$\min_{\operatorname{rank}\mathbf{B}\le k}\|\mathbf{A} - \mathbf{B}\|_2 = \sigma_{k+1}$, attained by truncated SVD
Pseudoinverse	$\mathbf{A}^{+} = \sum_{i\le r}\sigma_i^{-1}\mathbf{v}_i\mathbf{u}_i^{\mathsf{T}}$; $\mathbf{A}^{+}\mathbf{b}$ = minimum-norm least-squares solution
Normal modes	$\mathbf{K}\boldsymbol{\phi} = \omega^2\mathbf{M}\boldsymbol{\phi}$; $\mathbf{\Phi}^{\mathsf{T}}\mathbf{M}\mathbf{\Phi} = \mathbf{I}$, $\mathbf{\Phi}^{\mathsf{T}}\mathbf{K}\mathbf{\Phi} = \mathbf{\Omega}^2$; $\ddot{q}_i + \omega_i^2 q_i = \boldsymbol{\phi}_i^{\mathsf{T}}\mathbf{f}$
Modal response (1st order)	$\mathbf{x}(t) = \sum\mathbf{v}_i(\mathbf{w}_i^{\mathsf{T}}\mathbf{x}(0))e^{\lambda_i t}$, $\mathbf{w}_i^{\mathsf{T}}$ = rows of $\mathbf{P}^{-1}$
Continuous-time stability	asymptotically stable $\iff \operatorname{Re}\lambda_i < 0$ for all i
Discrete-time stability	asymptotically stable $\iff |\lambda_i| < 1$ for all i
2×2 classification	stable $\iff \tau < 0,\ \Delta > 0$; saddle if $\Delta < 0$; spiral if $\tau^2 < 4\Delta$; node if $\tau^2 > 4\Delta$
Third-order Routh–Hurwitz	$s^3 + a_2s^2 + a_1s + a_0$ stable $\iff a_i > 0$ and $a_2a_1 > a_0$
Linearization	$\dot{\boldsymbol{\xi}} = \mathbf{J}\boldsymbol{\xi}$, $\mathbf{J} = \partial\mathbf{f}/\partial\mathbf{x}$ at equilibrium; conclusive if no $\operatorname{Re}\lambda = 0$
Lyapunov equation	$\mathbf{A}$ Hurwitz $\iff \mathbf{A}^{\mathsf{T}}\mathbf{P} + \mathbf{P}\mathbf{A} = -\mathbf{Q}$ has $\mathbf{P} \succ 0$ for $\mathbf{Q} \succ 0$

Further reading

- G. Strang, *Introduction to Linear Algebra*, 6th ed., Wellesley–Cambridge Press, 2023 — chapters on eigenvalues, symmetric and positive-definite matrices and the SVD, with exceptional geometric intuition.
- E. Kreyszig, *Advanced Engineering Mathematics*, 10th ed., Wiley, 2011 — matrix eigenvalue problems, quadratic forms and systems of ODEs from the engineering side.
- C. D. Meyer, *Matrix Analysis and Applied Linear Algebra*, 2nd ed., SIAM, 2023 — rigorous treatment of Jordan forms, matrix functions, the SVD and Perron–Frobenius theory for Markov chains.
- G. H. Golub and C. F. Van Loan, *Matrix Computations*, 4th ed., Johns Hopkins University Press, 2013 — the reference for how eigenvalues, the SVD and matrix exponentials are actually computed.
- C.-T. Chen, *Linear System Theory and Design*, 4th ed., Oxford University Press, 2013 — state-space modal decomposition, Lyapunov equations and stability of linear systems for electrical engineers.

Part IV — Differential Equations

Chapter 11 — Ordinary Differential Equations

> **Prerequisites:** Ch. 4 (Differential Calculus), Ch. 5 (Integral Calculus), Ch. 6 (Multivariable Calculus), Ch. 8–10 (Matrices, Vector Spaces, Eigenvalues)
>
> **Learning outcomes:** After this chapter you can
>
> - classify an ODE by order, linearity, autonomy and homogeneity, and predict the kind of solution to expect before computing anything;
> - solve first-order equations that are separable, linear, exact (possibly after an integrating factor), Bernoulli or Riccati, and read their qualitative behaviour from a direction field or phase line;
> - use the structure theory of linear nth-order equations — fundamental sets, the Wronskian and Abel's formula, reduction of order, Cauchy–Euler equations, $y = y_h + y_p$ and variation of parameters;
> - state and apply the existence–uniqueness theorems for initial-value problems, and recognise blow-up and non-uniqueness when the hypotheses fail;
> - decide whether a boundary-value problem has one, none or infinitely many solutions, and solve simple Sturm–Liouville eigenvalue problems;
> - write any nth-order equation or lumped network as a first-order system $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{f}$, solve the constant-coefficient case by eigenvectors, and classify equilibria in the phase plane.

A differential equation relates an unknown function to its own derivatives. It is the natural language of anything that *stores* something — charge on a capacitor, flux in an inductor, heat in a package, carriers in a semiconductor, momentum in a motor shaft — because a storage law says how fast the stored quantity changes, not what its value is. Kirchhoff's laws plus the element laws $i_C = C\,dv/dt$ and $v_L = L\,di/dt$ turn every lumped circuit into a set of ODEs.

This chapter gives the **general theory** of ordinary differential equations (ODEs): how they are classified, which solution methods apply to which class, and when a solution exists, is unique and lasts for all time. The emphasis is on methods that work for *variable* coefficients and *nonlinear* equations, because those are what the special-purpose tools cannot handle. The special case that dominates circuit analysis — linear equations with **constant coefficients**, their characteristic equations, natural and forced responses and state-variable form — is developed fully in Chapter 12; the transform route to the same equations is Chapter 14; numerical integration is Chapter 22. Where those chapters take over, this one gives a pointer rather than a duplicate.

11.1 First-order ODEs

Why start with first order

A first-order equation is one storage element, one rate law: an RC or RL network, a thermal mass, a single-pole amplifier, a charging battery, a population. Every higher-order equation can be rewritten as a *system* of first-order equations (§11.5, §11.9), so the ideas met here — direction fields, equilibria, existence and uniqueness — are the ideas of the whole subject in their simplest setting.

Definitions and classification

Ordinary differential equation

An **ordinary differential equation** of order n is a relation

$$F\big(t, y, y', y'', \dots, y^{(n)}\big) = 0$$

between one independent variable t (time) or x (position), an unknown function y and its derivatives up to order n. The **order** is the highest derivative present. "Ordinary" means a single independent variable; with several, the equation is a *partial* differential equation (Ch. 13).

A **first-order** ODE is $F(t, y, y') = 0$. Whenever it can be solved for y' we use the **normal form**

$$y' = f(t, y).$$

The vocabulary below is used throughout the book.

- **Solution on an interval.** A function $y(t)$ is a solution on an open interval I if it is differentiable on I and $y'(t) = f(t, y(t))$ for every $t \in I$. The interval is part of the answer: a formula may satisfy the equation only on a piece of the real line.
- **General solution.** A family $y = \varphi(t, C)$ containing one arbitrary constant, such that (almost) every solution is obtained for some value of C. Its graphs form a one-parameter family of **integral curves**.
- **Particular solution.** One member of the family, usually picked out by an **initial condition** $y(t_0) = y_0$.
- **Singular solution.** A solution not obtainable from the general solution for any finite C. For $y' = y^2$ the general solution is $y = 1/(C - t)$ and $y \equiv 0$ is singular.
- **Explicit vs implicit.** An explicit solution gives y as a function of t; an implicit one is a relation $G(t, y) = 0$ that defines $y(t)$ only implicitly (by the implicit-function theorem of Ch. 6, wherever $G_y \neq 0$).
- **Linear vs nonlinear.** The equation is **linear** if it can be written $a_1(t)\,y' + a_0(t)\,y = r(t)$ — the unknown and its derivative appear to the first power, are not multiplied together, and are not inside other functions. Otherwise it is **nonlinear**.
- **Autonomous.** The equation is autonomous if f does not depend explicitly on t: $y' = f(y)$. Time-invariant circuits with constant sources are autonomous.

Equation	Order	Linear?	Autonomous?
$RC\,\dot v + v = V_s$	1	yes	yes
$\dot y + t\,y = \sin t$	1	yes	no
$C\,\dot v = -I_S\big(e^{v/V_T} - 1\big)$	1	no	yes

Equation	Order	Linear?	Autonomous?
$y' = y^2 + t$	1	no	no
$LC\,\ddot{v} + RC\,\dot{v} + v = v_s(t)$	2	yes	if v_s constant

Worked Example 11.1 — Checking a solution and its interval

Show that $y = 1/(C - t)$ solves $y' = y^2$, and find the largest interval on which the solution through $y(0) = 2$ exists.

Step 1 (verify). $y' = \dfrac{d}{dt}(C - t)^{-1} = (C - t)^{-2} = y^2$. ✓

Step 2 (fit the condition). $y(0) = 1/C = 2$ gives $C = 1/2$, so $y = 1/(\tfrac{1}{2} - t)$.

Step 3 (interval). The formula is undefined at $t = 1/2$. A solution must be differentiable on an *interval* containing $t_0 = 0$, so the solution is defined on $(-\infty, \tfrac{1}{2})$. The branch for $t > \tfrac{1}{2}$ is a *different* solution (it is negative and passes through no point with $t = 0$).

Check. At $t = 0.4$: $y = 1/0.1 = 10$, and the numerical slope $[y(0.401) - y(0.399)]/0.002 = (10.101 - 9.901)/0.002 \approx 100 = y^2$. ✓

Notice that $f(t, y) = y^2$ is perfectly smooth everywhere; the blow-up at $t = 1/2$ is invisible in the equation. §11.7 explains why this can happen only in nonlinear equations.

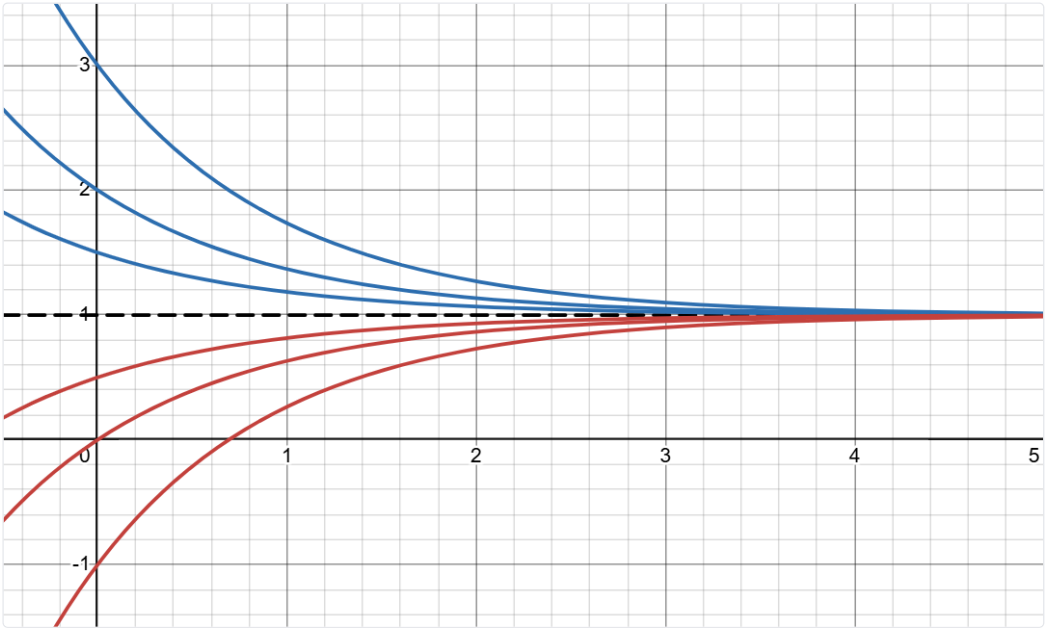
Direction fields

The equation $y' = f(t, y)$ assigns a slope to every point of the (t, y) plane. Drawing a short segment of slope $f(t, y)$ at each point of a grid gives the **direction field** (slope field). Every solution curve is tangent to the field at every point, so the field is a picture of all solutions at once — available before any formula is found, and even when no formula exists.

Two auxiliary ideas make sketching fast:

- **Isoclines** are the curves $f(t, y) = k$ along which all segments have the same slope k. The isocline $k = 0$ (the **nullcline**) is where solutions have horizontal tangents — maxima and minima of $y(t)$.
- **Equilibrium (constant) solutions** are values y^* with $f(t, y^*) = 0$ for all t. Their graphs are horizontal lines that other solutions can approach or leave but — when uniqueness holds — never cross.

For example $y' = 1 - y$ has the single equilibrium $y^* = 1$. Above it $y' < 0$ and below it $y' > 0$, so all solutions move towards $y = 1$. The general solution (derived in §11.3) is $y = 1 + Ce^{-t}$:



Integral curves of $y' = 1 - y$ for $C = \pm 0.5, \pm 1, \pm 2$: every curve approaches the stable equilibrium $y = 1$ (dashed). An RC circuit charging or discharging towards its source voltage behaves the same way.

Autonomous equations and the phase line

For an autonomous equation $y' = f(y)$ the slope depends only on height, so the direction field is the same on every vertical line. All the information fits on a single y-axis — the **phase line** — marked with the equilibria and arrows showing the sign of f between them.

Stability of an equilibrium of $y' = f(y)$

Let f be continuously differentiable and $f(y^*) = 0$.

- If $f'(y^*) < 0$, the equilibrium is **asymptotically stable**: solutions starting close enough to y^* tend to y^* as $t \to \infty$.
- If $f'(y^*) > 0$, it is **unstable**: nearby solutions move away.
- If $f'(y^*) = 0$, the test is inconclusive; look at the sign of f on each side.

Proof sketch (linearisation). Put $y = y^* + \eta$ with η small. Taylor's theorem (Ch. 4) gives $\eta' = f(y^* + \eta) = f'(y^*)\,\eta + O(\eta^2)$. Dropping the quadratic term, $\eta \approx \eta_0\, e^{f'(y^*)\,t}$, which decays if $f'(y^*) < 0$ and grows if $f'(y^*) > 0$. The quantity $-1/f'(y^*)$ is the *local time constant* of the equilibrium. Making the argument rigorous uses the fact that solutions cannot cross the equilibrium line (uniqueness, §11.7). □

Worked Example 11.2 — Phase line of a cubic

Analyse $y' = y - y^3$ without solving it.

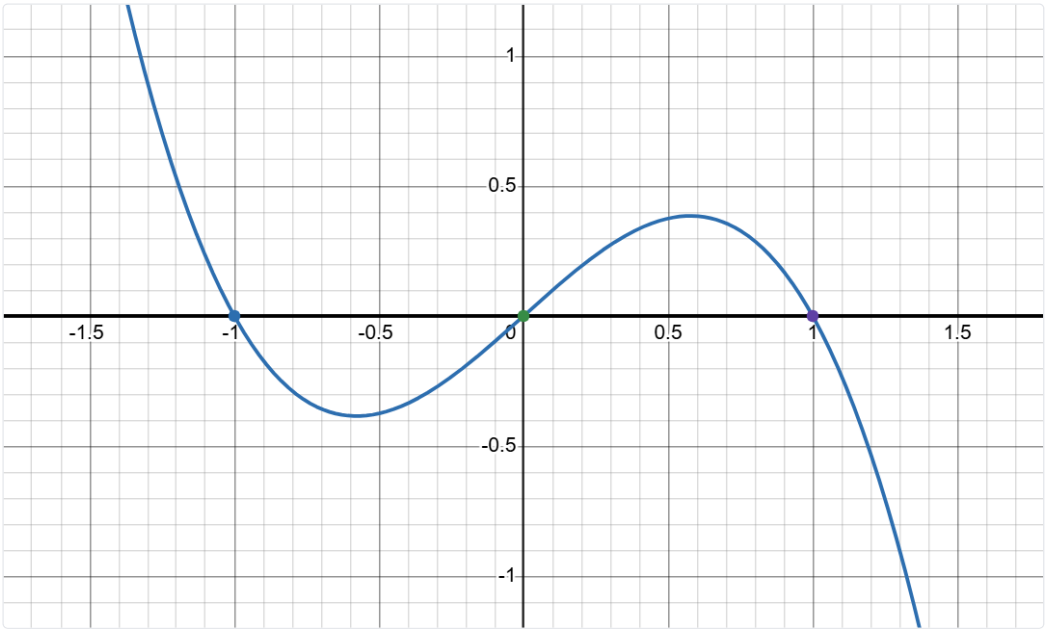
Step 1 (equilibria). $f(y) = y(1-y)(1+y) = 0$ gives $y^* = -1, 0, 1$.

Step 2 (linearise). $f'(y) = 1 - 3y^2$, so $f'(\pm 1) = -2 < 0$ (stable) and $f'(0) = 1 > 0$ (unstable).

Step 3 (signs). $f > 0$ on $(-\infty, -1)$ and $(0, 1)$; $f < 0$ on $(-1, 0)$ and $(1, \infty)$. So every solution with $y(0) > 0$ tends to $+1$, every solution with $y(0) < 0$ tends to -1, and only $y \equiv 0$ stays at zero.

Step 4 (time constant). Near $y^* = 1$, perturbations decay like e^{-2t}, so the local time constant is 0.5. Near 0 they grow like e^{t}.

This is the behaviour of a **bistable latch**: the cross-coupled inverter pair in an SRAM cell has two stable states and an unstable metastable point between them. A latch caught near the middle leaves it at a rate set by $f'(0)$ — the origin of the metastability time constant in synchroniser design.



The function $f(y) = y - y^3$ plotted against y (horizontal axis). Where the curve is above the axis solutions move right (increase); where it is below they move left. Zeros with negative slope attract, the zero with positive slope repels.

Exact equations

Write a first-order equation in **differential form**

$$M(t, y)\, dt + N(t, y)\, dy = 0,$$

which is equivalent to $y' = -M/N$. Suppose there is a function $\Phi(t, y)$ with $\Phi_t = M$ and $\Phi_y = N$. Then along any solution $y(t)$ the chain rule gives

$$\frac{d}{dt}\Phi\big(t, y(t)\big) = \Phi_t + \Phi_y\, y' = M + N y' = 0,$$

so Φ is constant along solutions and the solutions are the **level curves** $\Phi(t, y) = C$.

Test for exactness

Let M, N, M_y, N_t be continuous on a rectangle (or any simply connected region). Then $M\,dt + N\,dy$ is the differential of some Φ — the equation is **exact** — if and only if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial t}.$$

Necessity is just equality of mixed partials, $\Phi_{ty} = \Phi_{yt}$. Sufficiency is the construction below. This is exactly the test for a conservative two-dimensional vector field (M, N) in Ch. 7, and Φ is its potential.

Construction. Integrate M in t holding y fixed: $\Phi = \int M\,dt + g(y)$. Differentiate in y, set equal to N, and solve for $g'(y)$; exactness guarantees that g' comes out independent of t.

Worked Example 11.3 — An exact equation

Solve $(2ty + 1)\,dt + (t^2 + 3y^2)\,dy = 0$ with $y(1) = 1$.

Step 1 (test). $M_y = 2t$ and $N_t = 2t$. Exact. ✓

Step 2 (integrate M). $\Phi = \int (2ty + 1)\,dt = t^2y + t + g(y)$.

Step 3 (match N). $\Phi_y = t^2 + g'(y) = t^2 + 3y^2$, so $g' = 3y^2$ and $g = y^3$.

Step 4 (solution). $t^2y + t + y^3 = C$. The condition gives $1 + 1 + 1 = 3$, so the solution is the implicit curve $t^2y + t + y^3 = 3$.

Check. $d(t^2y + t + y^3) = (2ty + 1)\,dt + (t^2 + 3y^2)\,dy$. ✓ Numerically, at $t = 2$ the curve gives $y^3 + 4y - 1 = 0$, whose real root is $y \approx 0.2463$; implicit differentiation predicts slope $-M/N = -(2 \cdot 2 \cdot 0.2463 + 1)/(4 + 3 \cdot 0.0607) \approx -0.474$, which matches a finite-difference slope of the root as t varies.

Integrating factors for non-exact equations

If $M_y \neq N_t$, multiplying by a suitable nonzero function $\mu(t, y)$ may make $\mu M\,dt + \mu N\,dy = 0$ exact. The exactness condition for $\mu M, \mu N$ is a PDE for μ, usually harder than the original problem, but two special cases are routine.

Integrating factors depending on one variable

- If $\dfrac{M_y - N_t}{N}$ depends on t only, call it $R(t)$; then $\mu(t) = e^{\int R(t)\,dt}$.
- If $\dfrac{N_t - M_y}{M}$ depends on y only, call it $S(y)$; then $\mu(y) = e^{\int S(y)\,dy}$.

Derivation of the first case. With $\mu = \mu(t)$, exactness of $\mu M\,dt + \mu N\,dy$ requires $(\mu M)_y = (\mu N)_t$, i.e. $\mu M_y = \mu' N + \mu N_t$, i.e. $\mu'/\mu = (M_y - N_t)/N$. This is consistent only if the right side is a function of t alone, and then $\ln \mu = \int R\,dt$. The second case is symmetric. □

Worked Example 11.4 — Making an equation exact

Solve $(3ty + y^2)\,dt + (t^2 + ty)\,dy = 0$ for $t > 0$.

Step 1 (test). $M_y = 3t + 2y$, $N_t = 2t + y$. Not exact.

Step 2 (look for $\mu(t)$). $\dfrac{M_y - N_t}{N} = \dfrac{t + y}{t(t + y)} = \dfrac{1}{t}$, a function of t only. So $\mu = e^{\int dt/t} = t$.

Step 3 (exact form). $(3t^2y + ty^2)\,dt + (t^3 + t^2y)\,dy = 0$. Now $M_y = 3t^2 + 2ty = N_t$. ✓

Step 4 (potential). $\Phi = \int (3t^2y + ty^2)\,dt = t^3y + \tfrac{1}{2}t^2y^2 + g(y)$, and $\Phi_y = t^3 + t^2y + g'(y)$ forces $g' = 0$. The solution is

$$t^3y + \tfrac{1}{2}t^2y^2 = C.$$

Step 5 (explicit form). This is quadratic in y: $y = -t \pm \sqrt{t^2 + 2C/t^2}$.

Check. Take $C = 1/2$ and the $+$ root: at $t = 1$, $y = -1 + \sqrt{2} = 0.4142$. The ODE slope is $-M/N = -(3(0.4142) + 0.1716)/(1 + 0.4142) = -1.0000$; differentiating the explicit formula, $y' = -1 + (t - 2C/t^3)/\sqrt{t^2 + 2C/t^2} = -1 + 0/\sqrt{2} = -1$. ✓

Common pitfall — multiplying can add or remove solutions

An integrating factor that vanishes or blows up somewhere (here $\mu = t$ at $t = 0$) can introduce spurious solutions or hide genuine ones along that curve. State the region ($t > 0$ above) and check the curves where $\mu = 0$ or $\mu = \infty$ separately against the original equation.

Engineering note — conserved quantities

An exact equation says that some function $\Phi(t, y)$ is conserved along trajectories. In circuits and mechanics this is usually an **energy**: for a lossless LC tank $\tfrac{1}{2}Li^2 + \tfrac{1}{2}Cv^2$ is constant, and writing dv/di from the two element laws gives an exact equation whose level curves are the energy ellipses. Spotting a conserved quantity is often the fastest route to the qualitative behaviour of a nonlinear system.

11.2 Separable equations

Motivation

Separable equations are the first-order equations whose solution reduces to *two ordinary integrals*. They cover every autonomous first-order equation $y' = f(y)$ — a capacitor discharging into any nonlinear load, a saturating population, a cooling body — and many time-varying ones. They are also the template for the substitution methods that follow: a clever change of variable often turns an awkward equation into a separable one.

Separable equation

A first-order ODE is **separable** if it can be written

$$\frac{dy}{dt} = g(t)\,h(y),$$

the product of a function of t alone and a function of y alone.

The method and why it works

Solution of a separable equation

Let g be continuous on an interval I and h continuous with $h \neq 0$ on an interval J. Let H be an antiderivative of $1/h$ on J and G an antiderivative of g on I. Then $y(t)$ with values in J solves $y' = g(t)h(y)$ if and only if

$$H\big(y(t)\big) = G(t) + C$$

for some constant C. Informally, $\displaystyle\int \frac{dy}{h(y)} = \int g(t)\,dt + C$.

Proof. If y solves the equation, the chain rule gives $\dfrac{d}{dt}H(y(t)) = H'(y)\,y' = \dfrac{y'}{h(y)} = g(t) = G'(t)$, so $H(y) - G$ is constant. Conversely, differentiating $H(y) = G + C$ reproduces the equation. Because $H' = 1/h$ has one sign on J, H is strictly monotone and can in principle be inverted to give $y = H^{-1}(G(t) + C)$ explicitly. □

The informal manipulation "multiply by dt, divide by $h(y)$" is therefore legitimate — it is shorthand for the chain-rule argument.

Procedure.

1. Find all constant solutions: every root y^* of $h(y^*) = 0$ gives $y \equiv y^*$.
2. On each interval where $h \neq 0$, separate and integrate both sides.
3. Apply the initial condition, solve for y if possible, and state the interval of validity.

Common pitfall — lost solutions

Dividing by $h(y)$ discards every constant solution $y = y^*$ with $h(y^*) = 0$. They are often the equilibria — the most important solutions physically. Sometimes they are recovered by letting the constant take a special value (often $C = 0$); sometimes they are singular solutions that no value of C reproduces. Always list them in step 1.

Worked Example 11.5 — A Gaussian solution

Solve $y' = -2ty$ with $y(0) = 3$.

Step 1. $h(y) = y$ vanishes at $y = 0$, so $y \equiv 0$ is a solution.

Step 2. For $y \neq 0$: $\dfrac{dy}{y} = -2t\,dt$, so $\ln|y| = -t^2 + C_1$, i.e. $|y| = e^{C_1}e^{-t^2}$. Removing the absolute value, $y = Ce^{-t^2}$ with $C = \pm e^{C_1} \neq 0$; allowing $C = 0$ restores the lost solution.

Step 3. $y(0) = C = 3$, so $y = 3e^{-t^2}$, valid for all t.

Check. $y' = -6te^{-t^2} = -2t\,(3e^{-t^2})$. ✓ At $t = 1$, $y = 3e^{-1} = 1.1036$.

Worked Example 11.6 — Capacitor discharging through a square-law device

A capacitor C charged to V_0 discharges through a device whose current is $i = kv^2$ (a crude model of a space-charge-limited or square-law load). KCL at the capacitor node gives $C\,\dot v = -kv^2$, $v(0) = V_0 > 0$. Find $v(t)$ and compare with a resistor of the same initial current.

Step 1. $v \equiv 0$ is a solution; since $V_0 > 0$ and solutions cannot cross it, $v > 0$ for all t.

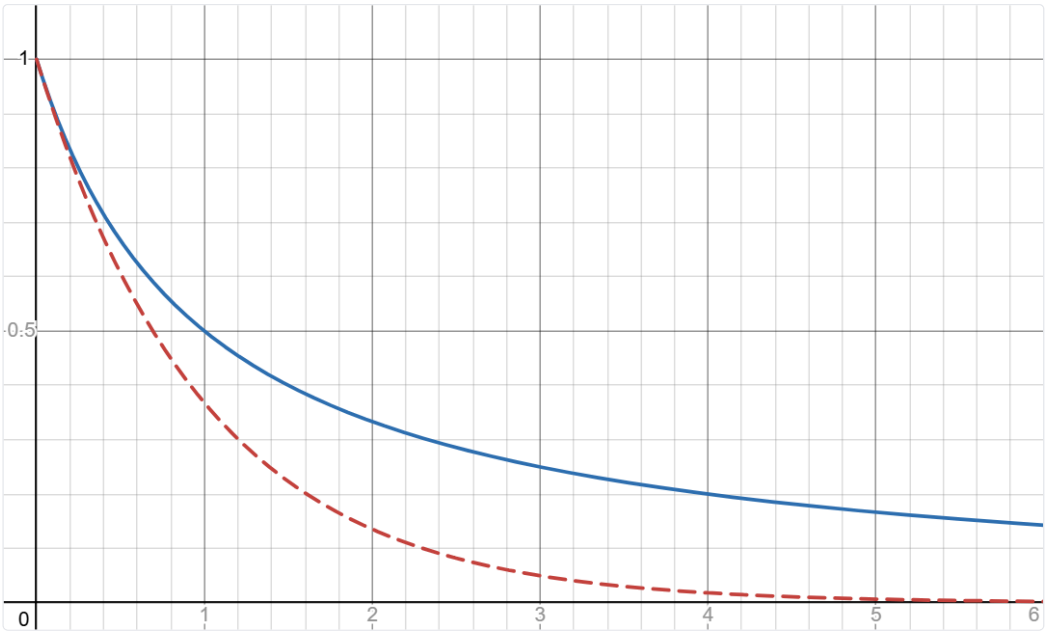
Step 2. $\dfrac{dv}{v^2} = -\dfrac{k}{C}\,dt$, so $-\dfrac{1}{v} = -\dfrac{k}{C}t + C_1$, and $v(0) = V_0$ gives $C_1 = -1/V_0$.

Step 3. Solving for v:

$$v(t) = \frac{V_0}{1 + t/T}, \qquad T = \frac{C}{kV_0}.$$

Units. k is in $\mathrm{A/V^2}$ and C in $\mathrm{A\,s/V}$, so $T = C/(kV_0)$ has units $(\mathrm{A\,s/V})/(\mathrm{A/V^2 \cdot V}) = \mathrm{s}$. ✓

Comparison. A resistor drawing the same initial current kV_0^2 has $R = 1/(kV_0)$ and gives $v = V_0e^{-t/RC} = V_0e^{-t/T}$, the same initial slope. At $t = 5T$ the resistor has left $V_0e^{-5} = 0.0067\,V_0$ while the square-law device has left $V_0/6 = 0.167\,V_0$. The nonlinear element changes the *form* of the transient — an algebraic $1/t$ tail instead of an exponential — not just its speed.



Normalised discharge v/V_0 against t/T: square-law load (blue, $1/(1+t/T)$) and a resistor with the same initial current (red dashed, $e^{-t/T}$). Same initial slope, very different tails.

Worked Example 11.7 — Capacitor discharging through a diode

A capacitor $C = 1$ nF charged to $V_0 = 0.7$ V discharges through an ideal-exponential diode, $i = I_S\left(e^{v/V_T} - 1\right)$ with $I_S = 10^{-14}$ A and $V_T = 25$ mV. Find $v(t)$ and the times at which v reaches 0.6, 0.5, 0.4 and 0.3 V.

Step 1 (set up). KCL: $C\,\dot v = -I_S\left(e^{v/V_T} - 1\right)$. Autonomous, hence separable; $v \equiv 0$ is the only equilibrium.

Step 2 (integrate). Multiply numerator and denominator by e^{-v/V_T}:

$$\int \frac{dv}{e^{v/V_T} - 1} = \int \frac{e^{-v/V_T}\,dv}{1 - e^{-v/V_T}} = V_T \ln\left(1 - e^{-v/V_T}\right).$$

So $V_T \ln\left(1 - e^{-v/V_T}\right) = -\dfrac{I_S}{C}t + C_1$.

Step 3 (solve). With $\tau = CV_T/I_S$ and the initial condition,

$$1 - e^{-v/V_T} = \left(1 - e^{-V_0/V_T}\right)e^{-t/\tau}, \qquad v(t) = -V_T \ln\left[1 - \left(1 - e^{-V_0/V_T}\right)e^{-t/\tau}\right].$$

Step 4 (numbers). $\tau = 10^{-9} \cdot 0.025/10^{-14} = 2500$ s. Inverting, $t = -\tau \ln\left[(1 - e^{-v/V_T})/(1 - e^{-V_0/V_T})\right]$:

v (V)	0.6	0.5	0.4	0.3
t	92.6 ns	5.15 µs	0.281 ms	15.4 ms

Interpretation. Each 0.1 V drop takes about $e^{0.1/V_T} = e^4 \approx 55$ times longer than the previous one. While $v \gg V_T$ the solution is approximately $v \approx -V_T \ln\left(e^{-V_0/V_T} + t/\tau\right)$ — a *logarithmic* decay. The huge τ governs only the final approach once $v \lesssim V_T$.

Check. A stiff numerical integration (Radau, relative tolerance 10^{-10}) of the original ODE from 0.7 V for 15.36 ms ends at 0.30000 V. ✓

Equations reducible to separable form

Two substitutions cover most textbook cases.

(a) $y' = F(y/t)$ — "homogeneous of degree zero". The right side is unchanged when t and y are scaled together. Put $v = y/t$, so $y = vt$ and $y' = v + tv'$. The equation becomes

$$t\,\frac{dv}{dt} = F(v) - v,$$

which is separable in v and t.

(b) $y' = G(at + by + c)$. Put $u = at + by + c$. Then $u' = a + b\,G(u)$, which is autonomous and therefore separable. For example $y' = (t + y)^2$ becomes $u' = 1 + u^2$ with $u = t + y$, so $u = \tan(t + C)$ and $y = \tan(t + C) - t$.

Worked Example 11.8 — A scale-invariant equation

Solve $y' = \dfrac{t^2 + y^2}{ty}$ with $y(1) = 2$, for $t > 0$.

Step 1 (recognise). Dividing top and bottom by t^2, $y' = \dfrac{1 + (y/t)^2}{y/t} = F(y/t)$ with $F(v) = (1 + v^2)/v$.

Step 2 (substitute). $tv' = \dfrac{1 + v^2}{v} - v = \dfrac{1}{v}$, so $v\,dv = \dfrac{dt}{t}$.

Step 3 (integrate). $\tfrac{1}{2}v^2 = \ln t + C$, i.e. $y^2 = t^2(2\ln t + 2C)$.

Step 4 (initial condition). $y(1) = 2$ gives $4 = 2C$, so $y = t\sqrt{2\ln t + 4}$ (positive root, since $y(1) > 0$).

Step 5 (interval). The square root needs $2\ln t + 4 > 0$, i.e. $t > e^{-2} \approx 0.135$. At $t = e^{-2}$ the solution reaches $y = 0$, where the equation itself is singular. The solution lives on (e^{-2}, ∞).

Check. At $t = e$: $y = e\sqrt{6} = 6.658$. The ODE gives $y' = (e^2 + 6e^2)/(e \cdot e\sqrt{6}) = 7/\sqrt{6} = 2.858$; differentiating the formula, $y' = \sqrt{2\ln t + 4} + 1/\sqrt{2\ln t + 4} = \sqrt{6} + 1/\sqrt{6} = 2.858$. ✓

The logistic equation

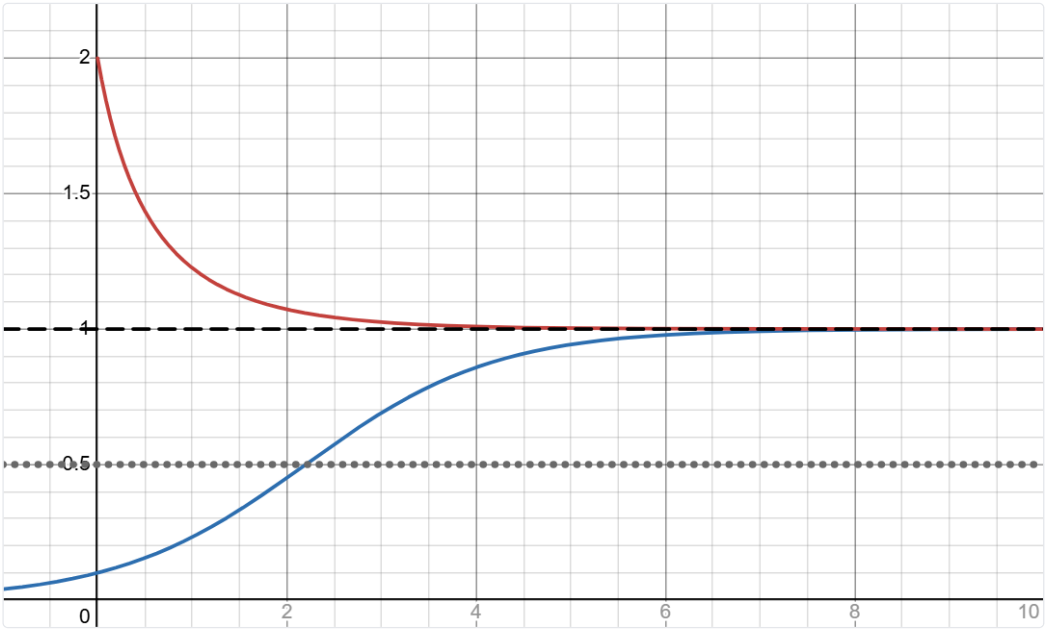
The logistic equation $y' = ry(1 - y/K)$, with growth rate $r > 0$ and capacity $K > 0$, is the standard model of growth that saturates. Its equilibria are $y = 0$ (unstable, $f'(0) = r$) and $y = K$ (stable, $f'(K) = -r$). For $0 < y < K$, partial fractions (Ch. 5) give

$$\int \frac{K\,dy}{y(K - y)} = \int \left(\frac{1}{y} + \frac{1}{K - y}\right) dy = \ln\frac{y}{K - y} = rt + C,$$

and solving with $y(0) = y_0$,

$$y(t) = \frac{K}{1 + \left(\dfrac{K}{y_0} - 1\right)e^{-rt}}.$$

The curve is an S-shape with its steepest point (inflection) at $y = K/2$, where $y'' = f'(y)\,y' = 0$ because $f'(y) = r(1 - 2y/K)$ vanishes there. Logistic behaviour appears in soft-limiting amplifiers, in the build-up of oscillation amplitude in a self-starting oscillator, and in technology-adoption curves (Ch. 25).



Logistic solutions with $r = 1$, $K = 1$: from $y_0 = 0.1$ (blue, S-shaped, steepest at $y = K/2$) and from $y_0 = 2$ (red, decaying to K). Both approach the stable equilibrium $y = K$.

Engineering note — when separation is the right tool

Any single-state circuit with constant sources is autonomous: $C\,\dot{v} = I - g(v)$. It is always separable, so the time to reach a voltage, $t(v) = \int_{V_0}^{v} C\,dv/\big(I - g(v)\big)$, can be written down even when the integral has no closed form — evaluate it numerically. This is how the delay of a nonlinear RC timing stage, or the droop time of a hold capacitor through a leakage diode, is computed in practice.

11.3 Linear first-order equations

Motivation

Linear first-order equations are the one large class of first-order equations that can *always* be solved in closed form (up to evaluating integrals). They describe every single-storage linear circuit — including those with time-varying parameters such as a switched-capacitor stage, a varactor, a thermistor-dependent resistance or a modulated source — and their solution formula already displays the structure of all linear systems: response = (decaying effect of the initial state) + (weighted memory of the input).

Linear first-order ODE

The standard form is

$$y' + p(t)\,y = r(t),$$

with p and r continuous on an interval I. It is **homogeneous** if $r \equiv 0$ and **nonhomogeneous** otherwise. An equation $a_1(t)y' + a_0(t)y = b(t)$ is put in standard form by dividing by a_1, which is allowed only where $a_1 \neq 0$; points where $a_1 = 0$ are **singular points** of the equation.

The homogeneous equation

$y' + p(t)y = 0$ is separable: $dy/y = -p\,dt$, so

$$y_h(t) = C\,e^{-P(t)}, \qquad P(t) = \int p(t)\,dt.$$

Because e^{-P} is never zero, the homogeneous solution never changes sign: a solution that starts positive stays positive. For constant $p = 1/\tau$ this is the familiar $Ce^{-t/\tau}$.

Integrating factor

Solution of the linear first-order equation

Let p, r be continuous on I and $\mu(t) = e^{P(t)}$ with $P' = p$. Then every solution of $y' + py = r$ on I is

$$y(t) = \frac{1}{\mu(t)}\left[\int \mu(t)\,r(t)\,dt + C\right].$$

With the initial condition $y(t_0) = y_0$ and $P(t) = \int_{t_0}^{t} p(s)\,ds$, the unique solution is

$$y(t) = y_0\,e^{-P(t)} + \int_{t_0}^{t} e^{-[P(t)-P(s)]}\,r(s)\,ds,$$

and it exists on the **whole** interval I.

Derivation. The integrating factor is chosen so that the left side becomes an exact derivative. By the product rule, $(\mu y)' = \mu y' + \mu' y$. If $\mu' = p\mu$ — that is, $\mu = e^{P}$ — then $(\mu y)' = \mu(y' + py) = \mu r$. Integrating, $\mu y = \int \mu r\,dt + C$. For the definite form integrate from t_0 to t with $\mu(t_0) = 1$: $\mu(t)y(t) - y_0 = \int_{t_0}^{t} \mu(s)r(s)\,ds$, then divide by $\mu(t) = e^{P(t)}$. Every step is reversible, so there are no other solutions, and the integrals exist wherever p and r are continuous. □

(It is also the special case of §11.1's integrating factor: in differential form $(py - r)\,dt + dy = 0$, the test quantity $(M_y - N_t)/N = p(t)$ depends on t alone.)

The definite form deserves a careful reading:

- $y_0 e^{-P(t)}$ is the **zero-input response**: what the initial state does with no input.
- The integral is the **zero-state response**: each input value $r(s)$ at an earlier time s contributes $r(s)\,ds$, weighted by the factor $e^{-[P(t)-P(s)]}$ by which the system has "forgotten" it by time t.
- For constant $p = 1/\tau$ the weight is $e^{-(t-s)/\tau}$, which depends only on the elapsed time $t - s$, and the integral is a **convolution** of r with $e^{-t/\tau}$. This is the first appearance of the impulse response; Ch. 12 and Ch. 14 develop it.

Whatever the form, the general solution is $y = Ce^{-P} + y_p$: a homogeneous solution plus *one* particular solution. This is the first instance of the structure theorem of §11.6.

Worked Example 11.9 — Exponential forcing

Solve $y' + 2y = e^{-t}$, $y(0) = 0$.

Step 1. $p = 2$, so $\mu = e^{2t}$.

Step 2. $(e^{2t}y)' = e^{2t}e^{-t} = e^{t}$, hence $e^{2t}y = e^{t} + C$ and $y = e^{-t} + Ce^{-2t}$.

Step 3. $y(0) = 1 + C = 0$, so $C = -1$:

$$y(t) = e^{-t} - e^{-2t}.$$

Check. $y' = -e^{-t} + 2e^{-2t}$ and $y' + 2y = -e^{-t} + 2e^{-2t} + 2e^{-t} - 2e^{-2t} = e^{-t}$. ✓

Shape. The response rises from zero, peaks where $y' = 0$, i.e. $e^{-t} = 2e^{-2t}$, so $t = \ln 2 \approx 0.693$ with $y = \tfrac{1}{2} - \tfrac{1}{4} = \tfrac{1}{4}$, then decays like the slower exponential e^{-t}.

The solution $y = e^{-t} - e^{-2t}$ of Example 11.9: a two-time-constant pulse, typical of a first-order stage driven by the decaying output of another stage.

Worked Example 11.10 — Variable coefficient and a singular point

Solve $t\,y' + 2y = 4t^2$ with $y(1) = 2$.

Step 1 (standard form). Dividing by t (allowed for $t \neq 0$): $y' + \dfrac{2}{t}y = 4t$. The coefficient $p = 2/t$ is continuous on $(0, \infty)$, which contains $t_0 = 1$, so the solution exists and is unique on $(0, \infty)$.

Step 2. $\mu = e^{\int 2/t\,dt} = e^{2\ln t} = t^2$.

Step 3. $(t^2 y)' = 4t^3$, so $t^2 y = t^4 + C$ and $y = t^2 + C/t^2$.

Step 4. $y(1) = 1 + C = 2$ gives $C = 1$: $y = t^2 + t^{-2}$.

Check. $t\,y' + 2y = t(2t - 2t^{-3}) + 2t^2 + 2t^{-2} = 4t^2$. ✓

Remark. Every solution except the one with $C = 0$ blows up at the singular point $t = 0$. The theorem promises existence only on $(0, \infty)$; the coefficient's singularity, not the solution formula, sets the interval.

Worked Example 11.11 — Tracking a ramp

An RC low-pass stage with time constant τ is driven by a ramp: $\tau\dot{v} + v = kt$, $v(0) = 0$. Find $v(t)$ and the steady tracking error.

Step 1. Standard form $\dot{v} + v/\tau = kt/\tau$, $\mu = e^{t/\tau}$.

Step 2. $\left(e^{t/\tau}v\right)' = \dfrac{k}{\tau}te^{t/\tau}$. Integrating by parts, $\int te^{t/\tau}\,dt = \tau te^{t/\tau} - \tau^2 e^{t/\tau}$, so $e^{t/\tau}v = k(t-\tau)e^{t/\tau} + C$.

Step 3. $v(0) = -k\tau + C = 0$ gives $C = k\tau$:

$$v(t) = k(t - \tau) + k\tau\, e^{-t/\tau}.$$

Check. $\tau\dot{v} + v = \tau(k - ke^{-t/\tau}) + kt - k\tau + k\tau e^{-t/\tau} = kt$. ✓

Interpretation. After a few time constants the output is a ramp of the same slope, delayed by exactly τ: the tracking error $kt - v \to k\tau$. With $k = 1$ V/ms and $\tau = 0.2$ ms the error settles at 0.2 V; at $t = 1$ ms it is $0.2(1 - e^{-5}) = 0.1987$ V. This lag is why a first-order sample-and-hold or tracking loop always trails a ramp.

Worked Example 11.12 — A time-varying capacitor

A varactor whose capacitance grows linearly, $C(t) = C_0(1 + t/T)$, is charged from 0 V through a resistor R by a constant source V_s. Take $RC_0 = T$. Find $v(t)$.

Step 1 (model). For a time-varying capacitor the charge is $q = C(t)v$, and the current is $i = \dot{q} = C\dot{v} + \dot{C}v$, not just $C\dot{v}$. KVL gives $Ri + v = V_s$:

$$RC_0\Big(1 + \frac{t}{T}\Big)\dot{v} + \Big(\frac{RC_0}{T} + 1\Big)v = V_s.$$

Step 2 (dimensionless time). With $s = t/T$ and $RC_0 = T$, this becomes $(1 + s)\,v' + 2v = V_s$, where $' = d/ds$.

Step 3 (integrating factor). Standard form $v' + \dfrac{2}{1+s}v = \dfrac{V_s}{1+s}$, so $\mu = (1+s)^2$ and $\big((1+s)^2 v\big)' = (1+s)V_s$.

Step 4. $(1+s)^2 v = \frac{1}{2}V_s(1+s)^2 + K$; $v(0) = 0$ gives $K = -V_s/2$:

$$v(t) = \frac{V_s}{2}\left[1 - \frac{1}{(1 + t/T)^2}\right].$$

Interpretation. The voltage approaches $V_s/2$, not V_s, and algebraically rather than exponentially. Physically, holding v constant on a growing capacitor still needs a current $\dot{C}v = C_0 v/T$; at $v = V_s/2$ the resistor drop $RC_0 v/T = v$ supplies exactly that. Time-varying elements break the intuition "a capacitor eventually charges to the source" — which is the principle behind parametric amplifiers and charge pumps.

Check. $(1+s)v' + 2v = (1+s)\cdot V_s(1+s)^{-3} + V_s - V_s(1+s)^{-2} = V_s$. ✓

Bernoulli equations

Bernoulli equation

$$y' + p(t)\,y = q(t)\,y^n, \qquad n \neq 0, 1.$$

(For $n = 0$ or 1 it is already linear.)

Divide by y^n and put $v = y^{1-n}$, so $v' = (1-n)y^{-n}y'$:

$$y^{-n}y' + p\,y^{1-n} = q \quad\Longrightarrow\quad v' + (1-n)\,p(t)\,v = (1-n)\,q(t),$$

which is linear in v. If $n > 0$, $y \equiv 0$ is an extra solution lost by the division.

Worked Example 11.13 — Amplitude of a self-starting oscillator

Near its operating point the slowly varying amplitude $a(t)$ of many oscillators (van der Pol type) obeys $\dot{a} = \alpha a - \beta a^3$: linear growth from the negative-resistance element, cubic limiting from gain compression. Take $\alpha = \beta = 1$ and $a(0) = 1/2$.

Step 1 (recognise). $\dot{a} - a = -a^3$ is Bernoulli with $n = 3$. Put $v = a^{-2}$: $\dot{v} + (1-3)(-1)v = (1-3)(-1)$, i.e. $\dot{v} + 2v = 2$.

Step 2 (solve linear). $v = 1 + Ce^{-2t}$; $v(0) = a(0)^{-2} = 4$ gives $C = 3$.

Step 3 (back-substitute).

$$a(t) = \frac{1}{\sqrt{1 + 3e^{-2t}}}.$$

Check. As $t \to \infty$, $a \to 1 = \sqrt{\alpha/\beta}$, the stable equilibrium of $\dot{a} = a - a^3$ (Example 11.2). At $t = 1$: $a = 1/\sqrt{1 + 3e^{-2}} = 0.8433$. Direct differentiation confirms $\dot{a} = a - a^3$ symbolically. ✓

The amplitude grows like $e^{\alpha t}$ while small and saturates at $\sqrt{\alpha/\beta}$: the oscillator's steady amplitude is set by the ratio of small-signal gain to compression.

Riccati equations

Riccati equation

$$y' = q_0(t) + q_1(t)\,y + q_2(t)\,y^2.$$

Riccati equations generally have no elementary solution, but **if one particular solution y_1 is known**, the substitution $y = y_1 + 1/v$ turns the equation into a linear one for v:

$$v' = -\big(q_1 + 2q_2 y_1\big)v - q_2.$$

(Substitute, use the fact that y_1 satisfies the equation, and multiply by $-v^2$.) Matrix Riccati equations are central in optimal control (the LQR gain) and Kalman filtering, and the scalar case is their prototype.

Worked Example 11.14 — Riccati with a known solution

Solve $y' = 1 + t^2 - 2ty + y^2$, given that $y_1 = t$ is a solution.

Step 1 (verify y_1). $y_1' = 1$ and $1 + t^2 - 2t^2 + t^2 = 1$. ✓

Step 2 (substitute). Here $q_2 = 1$, $q_1 = -2t$, so $q_1 + 2q_2 y_1 = 0$ and the linear equation is $v' = -1$: $v = C - t$.

Step 3. $y = t + \dfrac{1}{C - t}$, plus the solution $y = t$ itself (the limit $C \to \infty$).

Check. $y' = 1 + (C-t)^{-2}$, and $1 + t^2 - 2ty + y^2 = 1 + (y-t)^2 = 1 + (C-t)^{-2}$. ✓ Each solution other than y_1 blows up at $t = C$ — another nonlinear finite escape.

Engineering note — the time constant

Any first-order linear time-invariant system — an RC or RL network, a thermal mass, a single-pole amplifier, a first-order sensor — reduces to $\tau \dot{y} + y = K\,u(t)$. The homogeneous part $e^{-t/\tau}$ falls to 36.8% in one time constant, 13.5% in two, 1.8% in four and under 1% in five. The rule "settled after 5τ" comes from this table. For time-varying coefficients the role of t/τ is played by $P(t) = \int p\,dt$, so the system "forgets" its initial state when P has grown by about 5.

Common pitfall — forgetting the standard form

The integrating factor is $e^{\int p\,dt}$ where p is the coefficient of y **after** the coefficient of y' has been made 1. For $t\,y' + 2y = 4t^2$ the factor is t^2, not e^{2t}. The same mistake in variation of parameters (§11.6) is the commonest error in second-order work.

11.4 Second-order ODEs

Motivation

Two independent energy stores — an inductor and a capacitor, a mass and a spring, an electric and a magnetic field — produce a second-order equation. Second order is the lowest order at which **oscillation** is possible, which is why the series RLC circuit, the tuned amplifier and the mechanical resonator are all second-order systems. It is also the order of the field equations of physics in one dimension (Poisson's equation, the transmission-line equations, Schrödinger's equation), which is where boundary-value problems (§11.8) come from.

Forms of a second-order equation

A second-order ODE is $F(t, y, y', y'') = 0$, or in normal form $y'' = f(t, y, y')$. A solution is fixed by **two** conditions — typically $y(t_0)$ and $y'(t_0)$ — because two integrations are needed. The most important class is the **linear** equation

$$a_2(t)\,y'' + a_1(t)\,y' + a_0(t)\,y = b(t),$$

which on any interval where $a_2 \neq 0$ can be divided through to give the **standard form**

$$y'' + p(t)\,y' + q(t)\,y = r(t).$$

It is convenient to name the left side as an operator: $L[y] = y'' + py' + qy$. The operator is **linear**: $L[c_1 y_1 + c_2 y_2] = c_1 L[y_1] + c_2 L[y_2]$, because differentiation and multiplication by a function are linear.

Existence and uniqueness for linear second-order IVPs

If p, q, r are continuous on an open interval I and $t_0 \in I$, then for any numbers y_0, y_1 there is exactly one solution of $y'' + py' + qy = r$ with $y(t_0) = y_0$, $y'(t_0) = y_1$, and it exists on all of I.

This is proved in §11.7 (by rewriting the equation as a first-order system). Everything in the structure theory below rests on it.

Superposition and the dimension of the solution space

Superposition and dimension

If y_1 and y_2 solve the homogeneous equation $L[y] = 0$ on I, so does every linear combination $c_1 y_1 + c_2 y_2$. The set of all solutions is a vector space (Ch. 9) of dimension exactly 2.

Proof. Closure under combinations is linearity of L. For the dimension, consider the map that sends a solution y to its initial data $\big(y(t_0), y'(t_0)\big) \in \mathbb{R}^2$. It is linear; it is onto by the existence theorem (every pair of initial data is achieved); and it is one-to-one by uniqueness (only $y \equiv 0$ has zero initial data). A linear bijection onto $\mathbb{R}^2$ means the solution space is isomorphic to $\mathbb{R}^2$. $\square$

So the general solution of $L[y] = 0$ is $y = c_1 y_1 + c_2 y_2$ for **any** two linearly independent solutions — a **fundamental set**, i.e. a basis.

Linear independence and the Wronskian

Two functions are linearly dependent on I if one is a constant multiple of the other there. For solutions this is tested with the **Wronskian**

$$W(y_1, y_2)(t) = \det \begin{pmatrix} y_1 & y_2 \\ y_1' & y_2' \end{pmatrix} = y_1 y_2' - y_1' y_2.$$

The determinant is exactly the one that appears when we try to fit initial conditions: solving $c_1 y_1(t_0) + c_2 y_2(t_0) = y_0$, $c_1 y_1'(t_0) + c_2 y_2'(t_0) = y_1$ for c_1, c_2 is possible for *every* right side precisely when this determinant is nonzero (Ch. 8).

Abel's formula and the Wronskian test

Let y_1, y_2 solve $y'' + p(t)y' + q(t)y = 0$ on I, with p, q continuous. Then

$$W(t) = W(t_0)\, \exp\!\left(- \int_{t_0}^{t} p(s)\, ds \right).$$

Consequently W is either **never zero** on I (and y_1, y_2 form a fundamental set) or **identically zero** (and they are dependent).

Proof. Differentiate: $W' = y_1 y_2'' - y_1'' y_2$ (the $y_1' y_2'$ terms cancel). Substituting $y_k'' = -p y_k' - q y_k$,

$$W' = y_1(-p y_2' - q y_2) - (-p y_1' - q y_1) y_2 = -p\,(y_1 y_2' - y_1' y_2) = -p\, W.$$

This is a first-order linear equation for W whose solution is Abel's formula (§11.3). The exponential is never zero, so $W(t)$ vanishes everywhere or nowhere. If $W \equiv 0$, pick t_0; the system for (c_1, c_2) with zero right side has a nontrivial solution, and $c_1 y_1 + c_2 y_2$ is then a solution with zero initial data, hence identically zero by uniqueness — the functions are dependent. $\square$

Abel's formula is useful even without knowing the solutions: it gives W directly from the coefficient p, which is the key ingredient in reduction of order and variation of parameters.

Common pitfall — the Wronskian test needs solutions

For arbitrary functions, $W \equiv 0$ does **not** imply dependence. On $(-1, 1)$, $f = t^2$ and $g = t|t|$ are independent (neither is a constant multiple of the other on the whole interval), yet $W(f, g) = 0$ for every t. The "never or always zero" dichotomy is a property of *solutions of one linear equation with continuous coefficients*; it tells you that t^2 and $t|t|$ cannot both solve such an equation on an interval containing 0.

Reduction of order

If one nonzero solution y_1 of $y'' + py' + qy = 0$ is known, a second one can always be found.

Reduction of order

$$y_2(t) = y_1(t) \int \frac{e^{-\int p\, dt}}{y_1(t)^2}\, dt$$

is a second solution, independent of y_1, on any interval where $y_1 \neq 0$.

Derivation. Seek $y_2 = v(t)\, y_1(t)$. Then $y_2' = v' y_1 + v y_1'$ and $y_2'' = v'' y_1 + 2v' y_1' + v y_1''$. Substituting,

$$L[v y_1] = v'' y_1 + v'(2y_1' + p y_1) + v \underbrace{(y_1'' + p y_1' + q y_1)}_{=0} = 0.$$

The terms in v itself cancel — this is why the method works — leaving a *first-order* linear equation for $w = v'$: $w' + (2y_1'/y_1 + p)w = 0$. Its integrating factor is $y_1^2 e^{\int p\, dt}$, so $w = e^{-\int p\, dt}/y_1^2$, and $v = \int w\, dt$. Since v is not constant, y_2/y_1 is not constant and the solutions are independent. (Alternatively: Abel's formula gives $W = y_1 y_2' - y_1' y_2 = y_1^2 (y_2/y_1)' = e^{-\int p}$, which is the same equation.) $\square$

Worked Example 11.15 — Reduction of order for a Legendre equation

$y_1 = t$ solves $(1-t^2)y'' - 2ty' + 2y = 0$ on $-1 < t < 1$ (Legendre's equation with $n = 1$). Find a second solution.

Step 1 (verify y_1). $y_1'' = 0$, and $-2t \cdot 1 + 2t = 0$. ✓

Step 2 (standard form). $p = -\dfrac{2t}{1-t^2}$, so $-\int p\,dt = \int \dfrac{2t}{1-t^2}\,dt = -\ln(1-t^2)$ and $e^{-\int p\,dt} = \dfrac{1}{1-t^2}$.

Step 3 (integrate). By partial fractions,

$$\frac{1}{t^2(1-t^2)} = \frac{1}{t^2} + \frac{1}{1-t^2}, \qquad \int \left(\frac{1}{t^2} + \frac{1}{1-t^2}\right) dt = -\frac{1}{t} + \frac{1}{2}\ln\frac{1+t}{1-t}.$$

Step 4.

$$y_2 = t\left(-\frac{1}{t} + \frac{1}{2}\ln\frac{1+t}{1-t}\right) = \frac{t}{2}\ln\frac{1+t}{1-t} - 1.$$

Although the integration was done on intervals avoiding $t = 0$, the result is smooth across 0 and solves the equation on all of $(-1, 1)$; it is the Legendre function of the second kind Q_1.

Check. Symbolic substitution gives $(1-t^2)y_2'' - 2ty_2' + 2y_2 = 0$. Numerically at $t = 0.5$: $y_2 = 0.25\ln 3 - 1 = -0.72535$. Abel predicts $W = C/(1-t^2)$, and indeed $W(t, y_2) = ty_2' - y_2 = 1/(1-t^2)$, which at $t = 0.5$ is 1.3333. ✓

y_2 blows up logarithmically at $t = \pm 1$, the singular points where $a_2 = 1 - t^2$ vanishes. This is why only the polynomial solution $P_1 = t$ is admissible in physical problems on a sphere, where $t = \cos\theta$ reaches ± 1 at the poles (Ch. 13).

Cauchy–Euler (equidimensional) equations

The most important variable-coefficient equation with elementary solutions is

$$t^2 y'' + a\,t\,y' + b\,y = 0, \qquad t > 0,$$

with constants a, b. Each term scales the same way under $t \to \lambda t$ (each derivative brings a factor $1/t$ that the power of t cancels), so a power $y = t^m$ should work. Substituting, $t^2 m(m-1)t^{m-2} + a\,t\,mt^{m-1} + bt^m = t^m[m(m-1) + am + b]$, giving the **indicial equation**

$$m^2 + (a-1)m + b = 0.$$

Why it works — the substitution $t = e^x$. Put $x = \ln t$. Then $\dfrac{dy}{dt} = \dfrac{1}{t}\dfrac{dy}{dx}$ and $\dfrac{d^2y}{dt^2} = \dfrac{1}{t^2}\left(\dfrac{d^2y}{dx^2} - \dfrac{dy}{dx}\right)$, so $ty' = \dot y$ and $t^2y'' = \ddot y - \dot y$ (dots meaning d/dx). The equation becomes the constant-coefficient equation

$$\ddot y + (a-1)\dot y + by = 0,$$

whose characteristic equation is exactly the indicial equation, and $e^{mx} = t^m$. The three cases of a constant-coefficient equation (Ch. 12) translate directly:

Roots of $m^2 + (a-1)m + b = 0$	**Fundamental set on $t > 0$**
real, distinct $m_1 \neq m_2$	$t^{m_1},\ t^{m_2}$
repeated m	$t^m,\ t^m \ln t$
complex $\alpha \pm j\beta$	$t^\alpha \cos(\beta \ln t),\ t^\alpha \sin(\beta \ln t)$

For $t < 0$ replace t by $|t|$. The point $t = 0$ is a singular point, and solutions typically blow up or lose smoothness there.

Worked Example 11.16 — Real roots, cross-checked by reduction of order

Solve $t^2y'' - 2ty' + 2y = 0$ for $t > 0$, and check with Abel's formula.

Step 1. $a = -2$, $b = 2$: $m^2 - 3m + 2 = (m-1)(m-2) = 0$, so $y = c_1 t + c_2 t^2$.

Step 2 (reduction from $y_1 = t$). Standard form has $p = -2/t$, so $e^{-\int p\,dt} = t^2$ and $y_2 = t\int t^2/t^2\,dt = t^2$. ✓

Step 3 (Wronskian). $W = t \cdot 2t - 1 \cdot t^2 = t^2 \neq 0$. Abel: $W \propto e^{\int 2/t\,dt} = t^2$. ✓

Worked Example 11.17 — Complex indicial roots

Solve $t^2y'' + ty' + 4y = 0$ with $y(1) = 1, y'(1) = 0$.

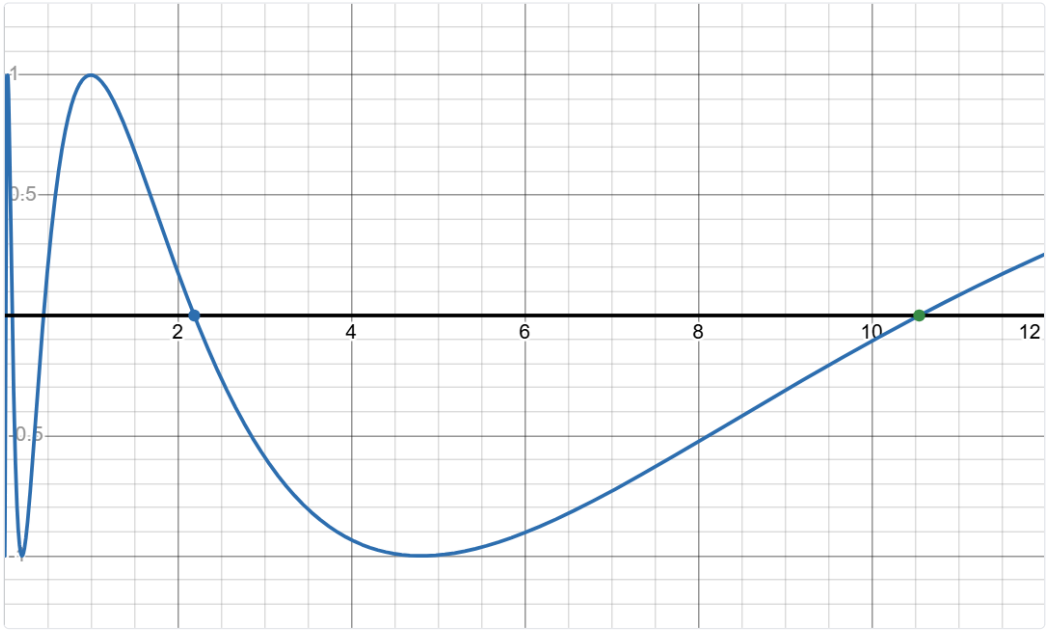
Step 1. $a = 1, b = 4$: $m^2 + 4 = 0$, so $m = \pm 2j$ ($\alpha = 0, \beta = 2$).

Step 2. $y = c_1\cos(2\ln t) + c_2\sin(2\ln t)$, and $y' = \dfrac{2}{t}\big[-c_1\sin(2\ln t) + c_2\cos(2\ln t)\big]$.

Step 3. At $t = 1$ ($\ln t = 0$): $c_1 = 1$ and $2c_2 = 0$. So $y = \cos(2\ln t)$.

Check. $y' = -\dfrac{2}{t}\sin(2\ln t)$, $y'' = \dfrac{2}{t^2}\sin(2\ln t) - \dfrac{4}{t^2}\cos(2\ln t)$. Then $t^2y'' + ty' + 4y = 2\sin - 4\cos - 2\sin + 4\cos = 0$. ✓

Shape. The solution oscillates in $\ln t$, not in t: its zeros are at $t = e^{\pi/4} = 2.193$, $e^{3\pi/4} = 10.55$, $e^{5\pi/4} = 50.75$, ... — equally spaced on a logarithmic axis. Equations of this type arise for fields in regions bounded by coaxial cylinders or wedges, where there is no natural length scale.



$y = \cos(2\ln t)$ from Example 11.17: oscillation whose period grows in proportion to t, the signature of a scale-invariant (Cauchy–Euler) equation.

Worked Example 11.18 — A repeated indicial root

Solve $t^2y'' - ty' + y = 0, t > 0$.

Step 1. $a = -1, b = 1$: $m^2 - 2m + 1 = (m-1)^2$, so $m = 1$ is a double root.

Step 2. $y = (c_1 + c_2\ln t)\,t$.

Check by reduction of order. $p = -1/t$, $e^{-\int p} = t$, so $y_2 = t\int t/t^2\,dt = t\ln t$. ✓ Direct check: $(t\ln t)' = \ln t + 1$, $(t\ln t)'' = 1/t$; $t^2/t - t(\ln t + 1) + t\ln t = 0$. ✓

Nonlinear second-order equations: two reductions

Nonlinear second-order equations rarely have closed-form solutions, but two structural features reduce the order by one.

- y **missing**, $F(t, y', y'') = 0$. Put $v = y'$; then $F(t, v, v') = 0$ is first order. Integrate v afterwards to get y.
- t **missing (autonomous)**, $F(y, y', y'') = 0$. Put $v = y'$ and regard v as a function of y. By the chain rule $y'' = \dfrac{dv}{dt} = \dfrac{dv}{dy}\dfrac{dy}{dt} = v\dfrac{dv}{dy}$, so $F\left(y, v, v\dfrac{dv}{dy}\right) = 0$ is first order in the variables (y, v) — an equation in the **phase plane** (§11.9).

For a conservative system $y'' = -U'(y)$ the second reduction gives $v\,dv = -U'(y)\,dy$, i.e. the **energy integral** $\frac{1}{2}v^2 + U(y) = E$: kinetic plus potential energy is constant.

Worked Example 11.19 — The pendulum energy integral

A pendulum obeys $\ddot\theta + (g/\ell)\sin\theta = 0$. It is released from rest at $\theta_0 = 90^\circ$ with $g/\ell = 9.81\ \text{s}^{-2}$. Find the angular speed at the bottom.

Step 1 (t missing). With $\omega = \dot\theta$ as a function of θ: $\omega\,\dfrac{d\omega}{d\theta} = -\dfrac{g}{\ell}\sin\theta$.

Step 2 (separate). $\tfrac12\omega^2 = \dfrac{g}{\ell}\cos\theta + E$. At release $\omega = 0$, $\theta = \theta_0$, so $E = -(g/\ell)\cos\theta_0$:

$$\omega^2 = \frac{2g}{\ell}\big(\cos\theta - \cos\theta_0\big).$$

Step 3. At $\theta = 0$: $\omega = \sqrt{2\cdot 9.81\cdot(1-0)} = 4.43\ \text{rad/s}$.

Check. The small-angle linearisation $\ddot\theta + (g/\ell)\theta = 0$ would give $\omega_{\max} = \theta_0\sqrt{g/\ell} = 1.571\cdot 3.132 = 4.92\ \text{rad/s}$, about 11% too high — at 90° amplitude the linear model is no longer accurate. The first-order equation in (θ,ω) can be separated once more to give the period as an integral (an elliptic integral), which has no elementary closed form.

Engineering note — constant coefficients live in Chapter 12

When p and q are constants, $y = e^{st}$ reduces $L[y] = 0$ to the algebraic **characteristic equation** $s^2 + ps + q = 0$, and the three root cases give the overdamped, critically damped and underdamped responses of RLC circuits. That theory, with damping ratio, natural frequency and forced response, is the subject of Ch. 12. The methods of this section — Wronskian, reduction of order, Cauchy–Euler — are what you use when the coefficients are **not** constant: tapered transmission lines, graded doping profiles, time-varying circuit elements, fields in curvilinear coordinates.

11.5 Higher-order ODEs

Motivation

Each independent energy store adds one to the order of a network's equation, so an nth-order filter, a multi-stage amplifier with n dominant poles, or a chain of n coupled resonators leads to an nth-order ODE. Fourth-order equations govern the bending of beams and cantilevers (MEMS accelerometers and resonators), and higher-order equations appear whenever a system of first-order equations is collapsed into a single equation for one output. The theory of §11.4 carries over almost word for word; the new ingredients are bookkeeping (an $n \times n$ Wronskian) and the systematic reduction to a first-order system, which is how such equations are actually analysed and computed.

The nth-order linear equation

Linear nth-order ODE

$$L[y] = y^{(n)} + p_{n-1}(t)\,y^{(n-1)} + \cdots + p_1(t)\,y' + p_0(t)\,y = r(t),$$

with $p_0,\dots,p_{n-1}, r$ continuous on an interval I. An **initial-value problem** prescribes $y(t_0), y'(t_0), \dots, y^{(n-1)}(t_0)$.

Structure of the nth-order homogeneous equation

1. **Existence and uniqueness.** For every $t_0 \in I$ and every vector of initial data, the IVP has exactly one solution, defined on all of I.
2. **Dimension.** The solutions of $L[y] = 0$ form a vector space of dimension n. Any n linearly independent solutions $y_1,\dots,y_n$ form a **fundamental set**, and the general solution is $y = c_1y_1 + \cdots + c_ny_n$.
3. **Wronskian.** With

$$W(t) = \det\begin{pmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & \vdots & & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{pmatrix},$$

Abel's formula reads $W(t) = W(t_0)\exp\!\Big(-\int_{t_0}^{t} p_{n-1}(s)\,ds\Big)$, so W is never zero or always zero on I, and the solutions are independent exactly when $W \neq 0$.

Proof sketch. Parts 1 and 2 follow exactly as in §11.4, with the initial-data map now onto $\mathbb{R}^n$. For Abel's formula, differentiate the determinant row by row: W' is the sum of n determinants, in each of which one row is differentiated. Differentiating row $k < n$ produces a copy of row $k+1$, so those determinants have two equal rows and vanish. Only the last row survives, becoming $\big(y_1^{(n)},\dots,y_n^{(n)}\big)$. Substituting $y_k^{(n)} = -p_{n-1}y_k^{(n-1)} - \cdots - p_0y_k$ and subtracting multiples of the earlier rows leaves $-p_{n-1}$ times the original last row. Hence $W' = -p_{n-1}W$. $\square$

The general solution of $L[y] = r$ contains n arbitrary constants, and n independent conditions are needed to fix them.

Worked Example 11.20 — A third-order constant-coefficient equation

Solve $y''' - y' = 0$ and verify independence of the solutions.

Step 1. With constant coefficients, $y = e^{st}$ gives $s^3 - s = s(s-1)(s+1) = 0$, so $y = c_1 + c_2 e^{t} + c_3 e^{-t}$.

Step 2 (Wronskian).

$$W = \det \begin{pmatrix} 1 & e^{t} & e^{-t} \\ 0 & e^{t} & -e^{-t} \\ 0 & e^{t} & e^{-t} \end{pmatrix} = 1 \cdot \left(e^{t}e^{-t} + e^{t}e^{-t} \right) = 2 \neq 0.$$

Check. The coefficient of y'' is $p_2 = 0$, so Abel predicts a constant W. ✓

Worked Example 11.21 — A third-order Cauchy–Euler IVP

Solve $t^3 y''' + t^2 y'' - 2ty' + 2y = 0$ for $t > 0$ with $y(1) = 0$, $y'(1) = 0$, $y''(1) = 6$.

Step 1 (indicial equation). For $y = t^m$, $t^3 y''' = m(m-1)(m-2)t^m$, $t^2 y'' = m(m-1)t^m$, $ty' = mt^m$:

$$m(m-1)(m-2) + m(m-1) - 2m + 2 = m^3 - 2m^2 - m + 2 = (m-2)(m-1)(m+1) = 0.$$

So $y = c_1 t + c_2 t^2 + c_3 t^{-1}$.

Step 2 (Wronskian, two ways). Directly,

$$W = \det \begin{pmatrix} t & t^2 & t^{-1} \\ 1 & 2t & -t^{-2} \\ 0 & 2 & 2t^{-3} \end{pmatrix} = \frac{6}{t}.$$

In standard form (divide by t^3) the coefficient of y'' is $p_2 = 1/t$, so Abel gives $W \propto e^{-\ln t} = 1/t$. ✓ It is nonzero on $t > 0$.

Step 3 (fit). With $y' = c_1 + 2c_2 t - c_3 t^{-2}$ and $y'' = 2c_2 + 2c_3 t^{-3}$, at $t = 1$:

$$c_1 + c_2 + c_3 = 0, \qquad c_1 + 2c_2 - c_3 = 0, \qquad 2c_2 + 2c_3 = 6.$$

The third gives $c_2 + c_3 = 3$, so $c_1 = -3$; then $2c_2 - c_3 = 3$, so $c_2 = 2$, $c_3 = 1$.

$$y = 2t^2 - 3t + \frac{1}{t}.$$

Check. $y(1) = 2 - 3 + 1 = 0$, $y'(1) = 4 - 3 - 1 = 0$, $y''(1) = 4 + 2 = 6$. ✓ The determinant of the 3×3 fitting system is $W(1) = 6 \neq 0$, which is why a unique fit was guaranteed.

Reduction to a first-order system

Companion form

Define the **state** $x_1 = y,\ x_2 = y',\ \dots,\ x_n = y^{(n-1)}$. Then $L[y] = r$ is equivalent to the first-order system

$$\dot{\mathbf{x}} = \mathbf{A}(t)\,\mathbf{x} + \mathbf{b}\,r(t), \qquad \mathbf{A} = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -p_0 & -p_1 & -p_2 & \cdots & -p_{n-1} \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}.$$

For constant coefficients, $\det(s\mathbf{I} - \mathbf{A}) = s^n + p_{n-1}s^{n-1} + \cdots + p_0$, the characteristic polynomial of the scalar equation.

The first $n-1$ rows just say "the derivative of each state is the next state"; the last row is the ODE solved for $y^{(n)}$. The initial data $y(t_0), \dots, y^{(n-1)}(t_0)$ are precisely $\mathbf{x}(t_0)$, and the Wronskian matrix is a fundamental matrix of the system (§11.9).

The conversion matters for three reasons:

- **Theory.** Existence, uniqueness and continuous dependence need only be proved once, for first-order systems (§11.7).
- **Computation.** Every numerical ODE solver (Ch. 22) — Runge–Kutta, SPICE's transient engine, MATLAB's `ode45` — accepts only first-order systems.
- **Modelling.** The physical state of a circuit (capacitor voltages, inductor currents) is a first-order system from the start; the companion form is one particular choice of state, and Ch. 12 shows how to choose others.

Worked Example 11.22 — Companion form

Write $y''' + 2y'' - y' + 3y = \sin t$ as a first-order system and confirm the characteristic polynomial.

Step 1. States $x_1 = y$, $x_2 = y'$, $x_3 = y''$. Then $\dot{x}_1 = x_2$, $\dot{x}_2 = x_3$, and from the equation $\dot{x}_3 = y''' = -3x_1 + x_2 - 2x_3 + \sin t$.

Step 2.

$$\frac{d}{dt}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -3 & 1 & -2 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}\sin t.$$

Check. Expanding along the first column, $\det(s\mathbf{I} - \mathbf{A}) = s\big(s(s+2) - 1\big) + 3 \cdot 1 = s^3 + 2s^2 - s + 3$. ✓

Operator factorisation

Write $D = d/dt$. A constant-coefficient operator is a polynomial in D, and polynomials in D **commute** and **factor** like ordinary polynomials: $D^2 - 3D + 2 = (D-1)(D-2)$. A factored equation can be solved as a chain of first-order linear equations, each by the integrating factor of §11.3. This is the cleanest way to see where the extra factors t^k for repeated roots come from.

Worked Example 11.23 — Solving by successive first-order steps

Find a particular solution of $y''' - 3y'' + 3y' - y = e^t$.

Step 1 (factor). The operator is $(D-1)^3$. Put $u = (D-1)^2 y$ and $w = (D-1)y$, so the equation is $(D-1)u = e^t$, then $(D-1)w = u$, then $(D-1)y = w$.

Step 2 (three integrating factors). Each step is $z' - z = g$, with solution $z = e^t \int e^{-t} g\,dt$ (constants dropped, since we need only one particular solution):

- $u = e^t \int e^{-t} e^t\,dt = te^t$;
- $w = e^t \int e^{-t} te^t\,dt = \tfrac{1}{2}t^2 e^t$;
- $y_p = e^t \int \tfrac{1}{2}t^2\,dt = \tfrac{1}{6}t^3 e^t$.

Check. Symbolic substitution of $y_p = \tfrac{1}{6}t^3 e^t$ returns exactly e^t. ✓

Interpretation. The input e^t coincides with a triple root of the characteristic polynomial, and each coincident factor contributes one power of t. This is *resonance* in its purest form; Ch. 12 turns the observation into the modification rule of undetermined coefficients.

Variable-coefficient operators can sometimes be factored too, but the factors no longer commute: $(D+t)D\,y = y'' + ty'$ whereas $D(D+t)\,y = y'' + ty' + y$. Factor in the right order.

Reduction of order for nth-order equations

If one solution y_1 of $L[y] = 0$ is known, substituting $y = v\,y_1$ again cancels all terms in v itself (because $L[y_1] = 0$), leaving an equation of order $n-1$ for $w = v'$. Each known solution therefore lowers the order by one. In practice this is used mostly for $n = 2$; for higher orders it is a theoretical tool showing that knowledge of solutions can be traded for reduced work.

Common pitfall — too few or too many conditions

An nth-order equation needs exactly n conditions. Fewer leaves free constants; more over-determines the problem, which usually has no solution. When a model gives you "too many" conditions, one of them is either redundant or signals that the order of the model is wrong — a classic symptom of an ignored parasitic element.

Engineering note — where higher order comes from

Each independent energy-storage element (a capacitor or inductor that cannot be merged with another by series/parallel combination, and is not in a loop of capacitors or a cutset of inductors) adds one to the order. A third-order Butterworth low-pass filter obeys a third-order ODE; a fourth-order Euler–Bernoulli beam equation $EI\,w'''' = q$ describes a MEMS cantilever. In practice engineers rarely solve high-order ODEs as single equations. They factor the characteristic polynomial (Ch. 12), work with transfer functions (Ch. 14), or integrate the first-order system numerically.

11.6 Homogeneous and nonhomogeneous equations

Motivation

A circuit left to itself after a switching event obeys a homogeneous equation; the same circuit driven by a source obeys a nonhomogeneous one. Linearity lets us treat the two effects separately and add the results — the mathematical content of the superposition theorem of circuit analysis. This section makes that precise and gives a method, **variation of parameters**, that produces a particular solution for *any* continuous input and *any* linear equation whose homogeneous solutions are known — including variable-coefficient equations where the guessing methods of Ch. 12 fail.

Homogeneous and nonhomogeneous linear equations

With $L[y] = y^{(n)} + p_{n-1}y^{(n-1)} + \cdots + p_0 y$, the equation $L[y] = 0$ is **homogeneous** and $L[y] = r(t)$ with $r \not\equiv 0$ is **nonhomogeneous**. The function $r(t)$ is called the *forcing*, *input* or *source term*. The general solution y_h of $L[y] = 0$ is the **complementary function**; any single solution y_p of $L[y] = r$ is a **particular solution**.

Common pitfall — two meanings of "homogeneous"

In §11.2, "homogeneous" meant $y' = F(y/t)$, a scaling property of a first-order equation. For linear equations it means "no forcing term". The two ideas are unrelated. In this book "homogeneous" means the linear sense unless stated otherwise.

Structure of solutions

Structure theorem

Let y_p be **any** particular solution of $L[y] = r$ on I and $y_h = c_1 y_1 + \cdots + c_n y_n$ the general solution of $L[y] = 0$. Then every solution of $L[y] = r$ on I has the form

$$y = y_h + y_p = c_1 y_1 + \cdots + c_n y_n + y_p.$$

Proof. If y is any solution, $L[y - y_p] = L[y] - L[y_p] = r - r = 0$, so $y - y_p$ is a homogeneous solution, i.e. equals $c_1 y_1 + \cdots + c_n y_n$ for some constants. Conversely every such sum is a solution. $\square$

Geometrically, the solution set of $L[y] = r$ is an **affine subspace**: the n-dimensional solution space of $L[y] = 0$ translated by y_p — the same picture as the solution set of a consistent linear system $\mathbf{A}\mathbf{x} = \mathbf{b}$ in Ch. 8. Different choices of y_p just shift the constants c_k.

Superposition of inputs

If $L[y_a] = r_a$ and $L[y_b] = r_b$, then $L[\alpha y_a + \beta y_b] = \alpha r_a + \beta r_b$ for any constants α, β.

A complicated input can therefore be split into simple pieces, a particular solution found for each, and the results added. This is the basis of the Fourier method (Ch. 15): split a periodic input into sinusoids, solve for each, and sum.

Worked Example 11.24 — Splitting the input

Find the general solution of $y'' + 4y = 8 + 5e^{t}$.

Step 1 (homogeneous). $s^2 + 4 = 0$ gives $y_h = c_1 \cos 2t + c_2 \sin 2t$.

Step 2 (input 8). A constant $y_{p1} = A$ gives $4A = 8$, so $y_{p1} = 2$.

Step 3 (input $5e^{t}$). Try $y_{p2} = Be^{t}$: $Be^{t} + 4Be^{t} = 5e^{t}$, so $B = 1$.

Step 4. $y = c_1 \cos 2t + c_2 \sin 2t + 2 + e^{t}$.

Check. $y_p'' + 4y_p = e^{t} + 8 + 4e^{t} = 8 + 5e^{t}$. ✓

The guesses in steps 2–3 are the method of **undetermined coefficients**, which works for constant-coefficient equations with inputs built from polynomials, exponentials and sinusoids. It is developed with all its special cases in Ch. 12. Here we need a method with no such restrictions.

Variation of parameters

The idea (due to Lagrange) is to look for a particular solution of the same form as y_h but with the constants replaced by functions: $y_p = u_1(t)y_1(t) + u_2(t)y_2(t)$. Two unknown functions are more freedom than one equation needs, so we may impose one extra condition of our choosing — and we choose it to keep second derivatives of u_k out of the calculation.

Variation of parameters (second order)

Let y_1, y_2 be a fundamental set for $y'' + p(t)y' + q(t)y = 0$ on I, with Wronskian $W = y_1y_2' - y_1'y_2$. Then a particular solution of $y'' + p(t)y' + q(t)y = r(t)$ is

$$y_p(t) = -y_1(t)\int \frac{y_2(t)\,r(t)}{W(t)}\,dt + y_2(t)\int \frac{y_1(t)\,r(t)}{W(t)}\,dt.$$

Derivation. With $y_p = u_1y_1 + u_2y_2$,

$$y_p' = u_1y_1' + u_2y_2' + \underbrace{(u_1'y_1 + u_2'y_2)}_{\text{set to } 0}.$$

Imposing $u_1'y_1 + u_2'y_2 = 0$, differentiate again: $y_p'' = u_1y_1'' + u_2y_2'' + u_1'y_1' + u_2'y_2'$. Substituting into the equation and grouping,

$$u_1\underbrace{(y_1'' + py_1' + qy_1)}_{0} + u_2\underbrace{(y_2'' + py_2' + qy_2)}_{0} + u_1'y_1' + u_2'y_2' = r.$$

So u_1', u_2' satisfy the linear system

$$\begin{pmatrix} y_1 & y_2 \\ y_1' & y_2' \end{pmatrix} \begin{pmatrix} u_1' \\ u_2' \end{pmatrix} = \begin{pmatrix} 0 \\ r \end{pmatrix},$$

whose determinant is the Wronskian, nonzero on I. Cramer's rule (Ch. 8) gives $u_1' = -y_2 r/W$ and $u_2' = y_1 r/W$. Integrating gives the formula. □

General order. For $L[y] = r$ of order n, impose $\sum u_k' y_k^{(j)} = 0$ for $j = 0, \dots, n-2$; the last equation becomes $\sum u_k' y_k^{(n-1)} = r$. Cramer's rule gives $u_k' = r\, W_k/W$, where W_k is W with its kth column replaced by $(0, \dots, 0, 1)^{\mathsf{T}}$.

Common pitfall — use the standard form

The $r(t)$ in the formula is the right side **after** dividing by the coefficient of y''. For $t^2 y'' - 2ty' + 2y = t^3$ it is $r = t$, not t^3. Using the raw right side is the single most common error in variation of parameters.

Worked Example 11.25 — An input undetermined coefficients cannot handle

Find a particular solution of $y'' + y = \sec t$ on $|t| < \pi/2$.

Step 1. $y_1 = \cos t$, $y_2 = \sin t$, $W = \cos^2 t + \sin^2 t = 1$.

Step 2.

$$u_1 = -\int \sin t \sec t \, dt = -\int \tan t \, dt = \ln|\cos t|,$$
$$u_2 = \int \cos t \sec t \, dt = t.$$

Step 3. $y_p = \cos t \, \ln(\cos t) + t \sin t$.

Check. $y_p' = -\sin t \ln\cos t - \sin t + \sin t + t\cos t = -\sin t \ln\cos t + t\cos t$, and $y_p'' = -\cos t \ln\cos t + \dfrac{\sin^2 t}{\cos t} + \cos t - t\sin t$. Adding y_p: $\dfrac{\sin^2 t}{\cos t} + \cos t = \dfrac{1}{\cos t} = \sec t$. ✓ The input has no finite family of derivatives, so no guess of undetermined-coefficients type can succeed.

Worked Example 11.26 — Variable coefficients with initial conditions

Solve $t^2 y'' - 2ty' + 2y = t^3$, $t > 0$, with $y(1) = 0$, $y'(1) = 0$.

Step 1 (homogeneous). From Example 11.16, $y_1 = t$, $y_2 = t^2$, $W = t^2$.

Step 2 (standard form). $y'' - \dfrac{2}{t}y' + \dfrac{2}{t^2}y = t$, so $r = t$.

Step 3 (parameters). $u_1' = -\dfrac{y_2 r}{W} = -\dfrac{t^2 \cdot t}{t^2} = -t$, so $u_1 = -\tfrac{1}{2}t^2$; $u_2' = \dfrac{y_1 r}{W} = \dfrac{t \cdot t}{t^2} = 1$, so $u_2 = t$.

Step 4. $y_p = -\tfrac{1}{2}t^2 \cdot t + t \cdot t^2 = \tfrac{1}{2}t^3$, and $y = c_1 t + c_2 t^2 + \tfrac{1}{2}t^3$.

Step 5 (conditions). $y(1) = c_1 + c_2 + \tfrac{1}{2} = 0$ and $y'(1) = c_1 + 2c_2 + \tfrac{3}{2} = 0$ give $c_2 = -1$, $c_1 = \tfrac{1}{2}$:

$$y = \tfrac{1}{2}t - t^2 + \tfrac{1}{2}t^3 = \tfrac{1}{2}\, t\,(t-1)^2.$$

Check. With $y = \tfrac{1}{2}(t^3 - 2t^2 + t)$: $y' = \tfrac{1}{2}(3t^2 - 4t + 1)$, $y'' = 3t - 2$. Then $t^2(3t-2) - t(3t^2 - 4t + 1) + (t^3 - 2t^2 + t) = t^3$. ✓

The Green's-function (impulse-response) form

Variation of parameters can be written as a single integral that builds in zero initial conditions. Taking definite integrals from t_0,

$$y_p(t) = \int_{t_0}^{t} G(t,s)\, r(s)\, ds, \qquad G(t,s) = \frac{y_1(s)\, y_2(t) - y_1(t)\, y_2(s)}{W(s)}.$$

Properties of the causal Green's function

For fixed s, $G(t,s)$ as a function of t solves the homogeneous equation with $G(s,s) = 0$ and $\partial G/\partial t\,(s,s) = 1$. Consequently y_p above is the unique solution of $L[y] = r$ with $y(t_0) = y'(t_0) = 0$, and the solution with general initial data is $y_h + y_p$ with y_h fitted to those data.

Why. $G(\cdot, s)$ is a combination of y_1, y_2, so it solves $L[y] = 0$; $G(s,s) = 0$ by inspection, and $\partial_t G(s,s) = [y_1(s)y_2'(s) - y_1'(s)y_2(s)]/W(s) = 1$. Differentiating y_p under the integral sign (Leibniz rule, Ch. 5) then gives $y_p' = \int \partial_t G\, r\, ds$ and $y_p'' = r(t) + \int \partial_t^2 G\, r\, ds$, so $L[y_p] = r$. □

Physically, $G(t,s)$ is the response at time t to a unit **impulse** of input applied at time s: the impulse leaves position unchanged and kicks the velocity by 1. The integral says the response to a general input is the superposition of responses to the impulses $r(s)\, ds$ that make it up. When the coefficients are constant, $G(t,s) = g(t-s)$ depends only on the elapsed time, and y_p is the **convolution** $g * r$ — the impulse-response description of a linear time-invariant system used throughout Ch. 12, Ch. 14 and Ch. 16.

Worked Example 11.27 — The undamped oscillator: step and resonant inputs

For $y'' + \omega_0^2 y = r(t)$ with $y(0) = y'(0) = 0$, find the response to (a) a step $r = F$, (b) a resonant input $r = \cos \omega_0 t$.

Step 1 (Green's function). $y_1 = \cos \omega_0 t$, $y_2 = \sin \omega_0 t$, $W = \omega_0$. Then

$$G(t, s) = \frac{\cos \omega_0 s \sin \omega_0 t - \cos \omega_0 t \sin \omega_0 s}{\omega_0} = \frac{\sin \omega_0 (t - s)}{\omega_0}.$$

Step 2 (step). $y = \dfrac{F}{\omega_0} \displaystyle\int_0^t \sin \omega_0 (t - s)\, ds = \frac{F}{\omega_0^2}\big(1 - \cos \omega_0 t\big)$.

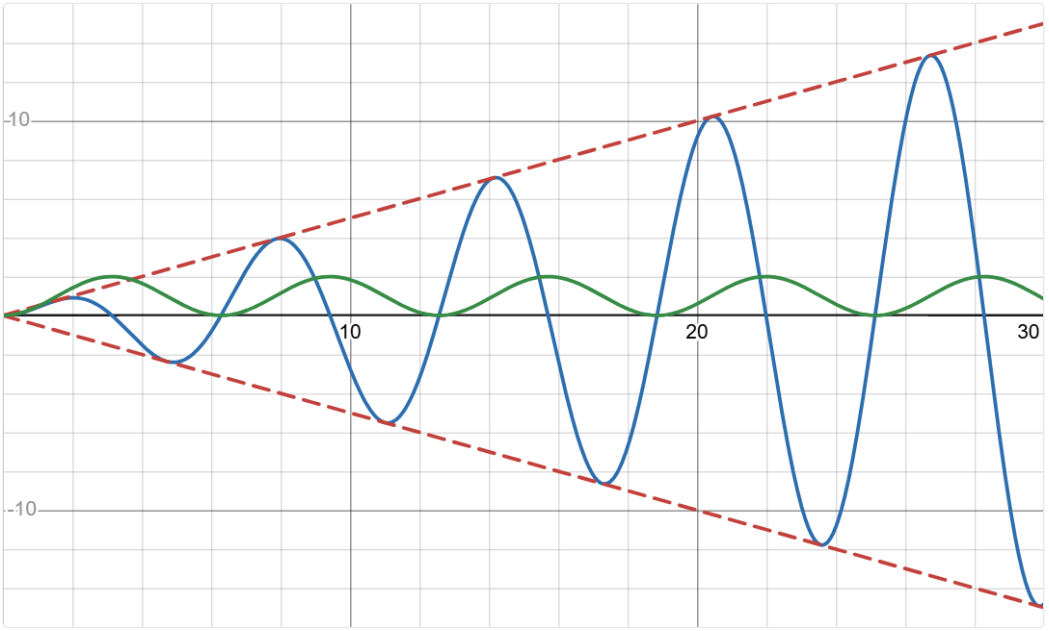
Step 3 (resonant input). Using $\sin A \cos B = \tfrac{1}{2}[\sin(A + B) + \sin(A - B)]$,

$$y = \frac{1}{\omega_0} \int_0^t \sin \omega_0 (t - s) \cos \omega_0 s \, ds = \frac{1}{2\omega_0} \int_0^t \big[\sin \omega_0 t + \sin \omega_0 (t - 2s)\big]\, ds = \frac{t \sin \omega_0 t}{2\omega_0}.$$

(The second integrand integrates to zero over $[0, t]$.)

Check. For (b), $y' = \dfrac{\sin \omega_0 t}{2\omega_0} + \dfrac{t}{2} \cos \omega_0 t$ and $y'' = \cos \omega_0 t - \dfrac{\omega_0 t}{2} \sin \omega_0 t$, so $y'' + \omega_0^2 y = \cos \omega_0 t$; $y(0) = y'(0) = 0$. ✓

Interpretation. The step response oscillates about the new equilibrium F/ω_0^2 forever (no damping). The resonant response grows linearly without bound — the envelope $t/(2\omega_0)$ — which is why lossless resonators driven at their natural frequency are dangerous, and why real circuits always contain the resistance that Ch. 12 uses to limit the peak.



Example 11.27 with $\omega_0 = 1$: resonant response $\frac{1}{2} t \sin t$ (blue) inside its linearly growing envelope (red dashed), and the bounded step response $1 - \cos t$ (green).

Engineering note — natural and forced response

In circuit language y_h is the **natural response** (shape set by the circuit, amplitude set by initial conditions) and y_p a **forced response** (shape set by the source). For stable circuits y_h decays and y_p — chosen as the steady periodic solution — is what a meter reads after the transient. Ch. 12 develops this vocabulary for constant-coefficient circuits; the structure theorem here is what guarantees that the decomposition is always possible and complete.

11.7 Initial-value problems

Motivation

An engineer who simulates a circuit is solving an initial-value problem: the state at $t = 0$ is known (capacitor voltages, inductor currents), and the simulator marches forward. Three questions must have good answers before the result means anything. Does a solution **exist**? Is it **unique** — or could the same initial state lead to different futures? Does it depend **continuously** on the data, so that small errors in component values or initial conditions produce small errors in the answer? A problem with all three properties is called **well-posed**. This section states when that is guaranteed, sketches why, and shows what goes wrong when the hypotheses fail.

Initial-value problem (IVP)

For a first-order system, find $\mathbf{y}(t)$ with

$$\mathbf{y}' = \mathbf{f}(t, \mathbf{y}), \qquad \mathbf{y}(t_0) = \mathbf{y}_0.$$

For an nth-order scalar equation, the data are $y(t_0), y'(t_0), \dots, y^{(n-1)}(t_0)$ — all at the **same** point t_0 — which is the same thing after the companion-form reduction of §11.5.

We state results for scalar first-order equations $y' = f(t, y)$; with $|\cdot|$ replaced by a vector norm $\|\cdot\|$ everything holds verbatim for systems, and hence for nth-order equations.

The Lipschitz condition

Lipschitz condition

$f(t, y)$ is **Lipschitz in** y on a region R with constant K if

$$|f(t, y) - f(t, z)| \le K\,|y - z| \quad \text{for all } (t, y), (t, z) \in R.$$

By the mean-value theorem, if $\partial f/\partial y$ is continuous on a closed rectangle then f is Lipschitz there with $K = \max|\partial f/\partial y|$. So the practical test is: **is $\partial f/\partial y$ bounded near the initial point?** The function $f = y^{2/3}$ fails at $y = 0$ (its derivative $\tfrac{2}{3}y^{-1/3}$ is unbounded); $f = |y|$ is Lipschitz with $K = 1$ even though it is not differentiable at 0.

Existence and uniqueness

Picard–Lindelöf

Let f be continuous on the rectangle $R = \{|t - t_0| \le a,\ |y - y_0| \le b\}$ and Lipschitz in y there. Let $M = \max_R |f|$ and $h = \min(a, b/M)$. Then the IVP $y' = f(t, y)$, $y(t_0) = y_0$ has **exactly one** solution on $|t - t_0| \le h$.

Proof sketch. **(1) Integral equation.** A continuous y solves the IVP if and only if

$$y(t) = y_0 + \int_{t_0}^{t} f\big(s, y(s)\big)\, ds,$$

by the fundamental theorem of calculus. This form needs no derivatives and builds in the initial condition.

(2) Picard iteration. Start from $y_0(t) \equiv y_0$ and define $y_{k+1}(t) = y_0 + \int_{t_0}^{t} f(s, y_k(s))\, ds$. The choice $h \le b/M$ keeps every iterate inside R: $|y_{k+1} - y_0| \le M|t - t_0| \le b$.

(3) Convergence. By the Lipschitz condition, $|y_{k+1}(t) - y_k(t)| \le K \int_{t_0}^{t} |y_k - y_{k-1}|\, ds$. Starting from $|y_1 - y_0| \le M|t - t_0|$ and inducting,

$$|y_{k+1}(t) - y_k(t)| \le \frac{M K^k |t - t_0|^{k+1}}{(k+1)!}.$$

The right side is the general term of a convergent exponential series, so y_k converges uniformly to a continuous limit y, and passing to the limit in the iteration shows that y satisfies the integral equation.

(4) Uniqueness. If y and z both solve it, $u = |y - z|$ satisfies $u(t) \le K \int_{t_0}^{t} u\, ds$ with $u(t_0) = 0$, and Grönwall's inequality (below) forces $u \equiv 0$. $\square$

Two weaker or complementary statements are worth knowing:

- **Peano's theorem.** Continuity of f alone guarantees that *at least one* solution exists near t_0, but not that it is unique.
- **Continuation.** A solution can be continued as long as it stays in a region where the hypotheses hold. The maximal solution either exists for all $t \ge t_0$ or leaves every bounded region — for a scalar equation on all of $\mathbb{R}^2$, it **blows up** ($|y| \to \infty$) in finite time. There is no third option such as "stops for no reason".

Worked Example 11.28 — Picard iterates for $y' = y$

Compute the Picard iterates for $y' = y$, $y(0) = 1$.

Step 1. $y_0 = 1$.

Step 2. $y_1 = 1 + \int_0^t 1\, ds = 1 + t$; $y_2 = 1 + \int_0^t (1 + s)\, ds = 1 + t + \tfrac{t^2}{2}$; and by induction

$$y_k(t) = \sum_{m=0}^{k} \frac{t^m}{m!}.$$

Step 3. The iterates are the Taylor partial sums of e^t and converge to $y = e^t$ for every t (Ch. 4).

Check against the error bound. On $|t| \le 1$ with $K = 1$, the bound on $|y_5 - y_4|$ at $t = 1$ is $MK^4/5!$ with $M \le e$ in a suitable rectangle, i.e. about 0.023; the actual difference is $1/5! = 0.0083$. ✓ The iteration is a constructive proof and, for simple f, a usable approximation method.

Worked Example 11.29 — How long is the guaranteed interval?

For $y' = 1 + y^2$, $y(0) = 0$, find the interval of existence promised by Picard–Lindelöf and compare with the true solution.

Step 1. $f = 1 + y^2$ is continuous with $f_y = 2y$ bounded on any rectangle, so the theorem applies for every a, b. Take a large; on $|y| \le b$, $M = 1 + b^2$, so $h = b/(1 + b^2)$.

Step 2 (best choice). $\dfrac{d}{db}\dfrac{b}{1+b^2} = \dfrac{1 - b^2}{(1+b^2)^2} = 0$ at $b = 1$, giving $h = \tfrac{1}{2}$.

Step 3 (true solution). Separating, $\arctan y = t$, so $y = \tan t$, which exists on $(-\pi/2, \pi/2)$ and blows up at $|t| = \pi/2 \approx 1.571$.

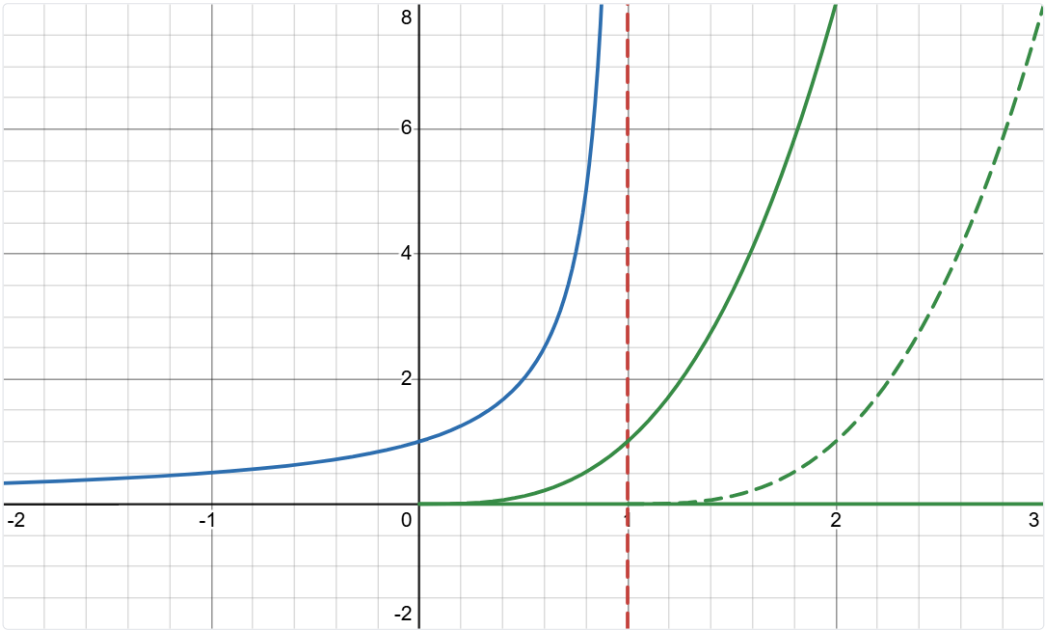
Interpretation. The theorem is conservative — it guarantees $|t| \le 0.5$ where the truth is $|t| < 1.571$ — but it is *local by nature*: no choice of rectangle could guarantee all t, because the solution genuinely escapes. Applying the theorem again from $t = 0.5$ extends the solution step by step until the blow-up.

When the hypotheses fail

Worked Example 11.30 — Blow-up and non-uniqueness

(a) Finite-time blow-up. $y' = y^2$, $y(0) = 1$. Here $f = y^2$ is smooth everywhere, and separation gives $y = 1/(1 - t)$. The solution exists only for $t < 1$, although the equation itself is defined everywhere. The cause is the superlinear growth of f: $K = 2|y|$ grows without bound as y grows.

(b) Non-uniqueness. $y' = 3y^{2/3}$, $y(0) = 0$. Both $y \equiv 0$ and $y = t^3$ are solutions: $(t^3)' = 3t^2 = 3(t^3)^{2/3}$. In fact, for any $c \ge 0$ the function equal to 0 for $t \le c$ and $(t - c)^3$ for $t > c$ is also a solution — infinitely many futures from the same initial state. Uniqueness fails because $\partial f/\partial y = 2y^{-1/3}$ is unbounded at $y = 0$; existence still holds (Peano).



Blue: $y = 1/(1 - t)$ escapes to infinity at $t = 1$ (red dashed asymptote). Green: three different solutions of $y' = 3y^{2/3}$ leave the same initial point $(0, 0)$ — $y = 0$, $y = t^3$, and y that waits until $t = 1$ before departing.

Continuous dependence and Grönwall's inequality

Grönwall's inequality

If $u(t) \ge 0$ is continuous and $u(t) \le \alpha + K\int_{t_0}^{t} u(s)\,ds$ for $t \ge t_0$, with constants $\alpha \ge 0$, $K \ge 0$, then $u(t) \le \alpha\, e^{K(t - t_0)}$.

Sketch. Let $U(t) = \alpha + K\int_{t_0}^{t} u\,ds$. Then $U' = Ku \le KU$, so $(e^{-K(t-t_0)}U)' \le 0$ and $U(t) \le U(t_0)e^{K(t-t_0)} = \alpha e^{K(t-t_0)}$; finally $u \le U$. □

Continuous dependence on initial data

If f is Lipschitz in y with constant K and y, z solve $y' = f(t, y)$ with initial values y_0, z_0, then

$$|y(t) - z(t)| \le |y_0 - z_0|\, e^{K|t - t_0|}.$$

Proof. Subtract the two integral equations and use the Lipschitz bound: $|y - z| \le |y_0 - z_0| + K\int |y - z|\,ds$. Apply Grönwall. With $y_0 = z_0$ this is the uniqueness proof. □

The bound is a worst case: errors *may* grow like e^{Kt}. For a stable system they actually shrink — for $y' = -y/\tau$ two solutions approach each other like $e^{-t/\tau}$ — but for an unstable one (an oscillator start-up, a latch near its metastable point, Example 11.2) the exponential growth of small differences is real and is exactly what limits predictability. A similar estimate shows continuous dependence on parameters in f: a small change in a component value changes the solution by a small amount on any finite time interval.

Linear equations: global existence

Linear IVPs

If $p_0, \dots, p_{n-1}, r$ (or $\mathbf{A}(t), \mathbf{f}(t)$ for a system) are continuous on an open interval I containing t_0, the linear IVP has exactly one solution, and it exists on **all** of I.

Why linear equations cannot blow up. For $\mathbf{y}' = \mathbf{A}(t)\mathbf{y} + \mathbf{f}(t)$ the Lipschitz constant on any closed subinterval $J \subset I$ is $K = \max_J \|\mathbf{A}\|$, the same for *all* $\mathbf{y}$ — no rectangle in y is needed. Grönwall then bounds $\|\mathbf{y}(t)\|$ by $\big(\|\mathbf{y}_0\| + \int_J \|\mathbf{f}\|\big) e^{K|t-t_0|}$ on J, so the solution stays bounded on every closed subinterval and, by the continuation property, extends across all of I. The only places a linear solution can misbehave are where the *coefficients* do — the singular points — and those are visible in the equation before solving it.

Worked Example 11.31 — Fitting circuit initial conditions

In a series RLC circuit with $R = 4\ \Omega$, $L = 1$ H, $C = \frac{1}{3}$ F, the capacitor holds $v_C(0^-) = 2$ V and the inductor current is $i(0^-) = 0$. At $t = 0$ a 10 V source is switched into the loop. Find $v_C(t)$ for $t > 0$.

Step 1 (equation). KVL: $Li' + Ri + v_C = 10$ with $i = C\dot{v}_C$ gives $LC\ddot{v}_C + RC\dot{v}_C + v_C = 10$, i.e. $\ddot{v}_C + 4\dot{v}_C + 3v_C = 30$.

Step 2 (initial data at 0^+). Capacitor voltage and inductor current are continuous: $v_C(0^+) = 2$ V and $i(0^+) = 0$, hence $\dot{v}_C(0^+) = i(0^+)/C = 0$.

Step 3 (general solution). The constant input gives $y_p = 10$; the homogeneous equation has $s^2 + 4s + 3 = (s+1)(s+3)$, so $v_C = 10 + Ae^{-t} + Be^{-3t}$.

Step 4 (fit). $A + B = -8$ and $-A - 3B = 0$, so $B = 4$, $A = -12$:

$$v_C(t) = 10 - 12e^{-t} + 4e^{-3t}\ \text{V}.$$

Check. $v_C(0) = 10 - 12 + 4 = 2$, $\dot{v}_C(0) = 12 - 12 = 0$. ✓ At $t = 1$ s: $v_C = 10 - 4.415 + 0.199 = 5.78$ V. The coefficients are constant, so this is the unique solution for all $t > 0$.

Engineering note — initial conditions in circuits

A capacitor voltage cannot jump unless the current is infinite, and an inductor current cannot jump unless the voltage is infinite. The natural initial conditions for a circuit are therefore $v_C(0^-)$ and $i_L(0^-)$, the stored energies just before a switching event — exactly one number per state variable, which is why state variables make the IVP automatically well-posed. Derivatives such as $\dot{v}_C(0^+) = i_C(0^+)/C$ follow from the circuit at $t = 0^+$. When an idealised circuit violates the rule (a capacitor switched directly across a voltage source, an inductor current interrupted), the model needs impulses — handled systematically by the Laplace transform in Ch. 14.

Common pitfall — trusting a simulator at a non-Lipschitz point

Models with square-root or fractional-power laws (ideal square-law MOSFET at threshold, some magnetic and friction models) can be non-Lipschitz at a point. At such points a numerical integrator may follow any of several solutions depending on step size and round-off, and the result can change with solver settings. Smoothing the model (e.g. replacing $|v|^{1/2}$ by $(v^2 + \varepsilon^2)^{1/4}$) restores uniqueness.

11.8 Boundary-value problems

Motivation

When the independent variable is **position** rather than time, conditions are naturally given at two places: the potential on both electrodes, the voltage at the source and load ends of a line, the temperature at both ends of a heat sink, the clamped and free ends of a cantilever. These are **boundary-value problems** (BVPs). They behave very differently from IVPs — existence and uniqueness are no longer automatic — and their homogeneous versions, the **eigenvalue problems**, produce the modes (standing waves, resonant frequencies, orthogonal function sets) on which separation of variables (Ch. 13) and Fourier series (Ch. 15) are built.

Two-point boundary-value problem

A second-order ODE on $a \le x \le b$ with one condition at **each end**. Common boundary conditions:

- **Dirichlet:** $y(a) = \alpha$, $y(b) = \beta$ (value prescribed — e.g. electrode potential);
- **Neumann:** $y'(a) = \alpha$, $y'(b) = \beta$ (slope prescribed — e.g. zero field or zero heat flux);
- **Robin (mixed):** $c_1 y(a) + c_2 y'(a) = \alpha$ (a linear combination — e.g. convective cooling, or a line terminated in a resistance).

The BVP is **homogeneous** if the equation and both boundary conditions are homogeneous (zero right sides).

Solvability: one, none or infinitely many

For a linear equation $y'' + p(x)y' + q(x)y = r(x)$ with fundamental set y_1, y_2 and particular solution y_p, write $y = c_1 y_1 + c_2 y_2 + y_p$ and impose the two boundary conditions. For Dirichlet conditions this gives the 2×2 system

$$\begin{pmatrix} y_1(a) & y_2(a) \\ y_1(b) & y_2(b) \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} \alpha - y_p(a) \\ \beta - y_p(b) \end{pmatrix}.$$

Everything now follows from linear algebra (Ch. 8): if the determinant $D = y_1(a)y_2(b) - y_2(a)y_1(b)$ is nonzero there is exactly one solution for every right side; if $D = 0$ there is either none or infinitely many, depending on whether the right side lies in the column space.

Fredholm alternative (two-point BVPs)

For a linear second-order BVP, exactly one of the following holds:

1. The homogeneous BVP has only the trivial solution $y \equiv 0$. Then the nonhomogeneous BVP has **exactly one** solution for every r, α, β.
2. The homogeneous BVP has a nontrivial solution. Then the nonhomogeneous BVP has **either no solution or infinitely many**, depending on the data.

Proof. The homogeneous BVP corresponds to a zero right side, and it has a nontrivial solution exactly when $D = 0$. Apply the corresponding statement for 2×2 linear systems. □

An IVP never meets case 2, because its "determinant" is the Wronskian at t_0, which is never zero. A BVP meets it whenever the interval length is "tuned" to a natural oscillation of the equation — the spatial analogue of resonance.

Worked Example 11.32 — Three possible outcomes

The equation $y'' + y = 0$ has general solution $y = c_1 \cos x + c_2 \sin x$.

1. $y(0) = 0$, $y(\pi/2) = 1$: $c_1 = 0$ and $c_2 = 1$, so the **unique** solution is $y = \sin x$. Here $D = \cos 0 \sin \frac{\pi}{2} - \sin 0 \cos \frac{\pi}{2} = 1 \neq 0$.
2. $y(0) = 0$, $y(\pi) = 1$: $c_1 = 0$, then $c_2 \sin \pi = 0 \neq 1$, so there is **no solution**.
3. $y(0) = 0$, $y(\pi) = 0$: $c_1 = 0$ and c_2 is arbitrary, so there are **infinitely many** solutions $y = c_2 \sin x$.

In cases 2 and 3, $D = \cos 0 \sin \pi - \sin 0 \cos \pi = 0$: the interval $[0, \pi]$ is exactly half a period of the natural solution $\sin x$, so the homogeneous problem has the nontrivial solution $\sin x$.

Worked Example 11.33 — A Poisson-type BVP

In one dimension the potential between two plates with a space-charge density obeys Poisson's equation $V'' = -\rho/\varepsilon$. As a dimensionless model, solve $y'' = 6x$ with $y(0) = 0$, $y(1) = 2$.

Step 1. Integrate twice: $y = x^3 + c_1 x + c_0$.

Step 2. $y(0) = c_0 = 0$ and $y(1) = 1 + c_1 = 2$, so $c_1 = 1$:

$$y = x^3 + x.$$

Check. $y'' = 6x$, $y(0) = 0$, $y(1) = 2$. ✓ Uniqueness is guaranteed because $y'' = 0$ with $y(0) = y(1) = 0$ has only $y \equiv 0$ (a straight line through two zeros).

Green's function for a BVP

As for IVPs (§11.6), the solution of a uniquely solvable BVP with homogeneous boundary conditions can be written as an integral against a kernel.

Green's function for $-y'' = f$, $y(0) = y(1) = 0$

$$y(x) = \int_0^1 G(x, s)\, f(s)\, ds, \qquad G(x, s) = \begin{cases} x(1 - s), & 0 \le x \le s, \\ s(1 - x), & s \le x \le 1. \end{cases}$$

Construction. For fixed s, G must satisfy $-G'' = 0$ for $x \neq s$ (so it is linear on each side), vanish at $x = 0$ and $x = 1$, be continuous at $x = s$, and have a slope that drops by 1 at $x = s$ (the response to a unit point source, since $-G'' = \delta(x - s)$). The linear pieces Ax and $B(1 - x)$ with $As = B(1 - s)$ and $-B - A = -1$ give $A = 1 - s$, $B = s$. □

G is **symmetric**, $G(x, s) = G(s, x)$ — the reciprocity principle of electrostatics and of passive networks: the potential at x due to a unit charge at s equals the potential at s due to a unit charge at x.

Worked Example 11.34 — Using the Green's function

Solve $-y'' = 1$, $y(0) = y(1) = 0$ (uniform space charge between grounded plates).

Step 1. Split the integral at $s = x$:

$$y(x) = \int_0^x s(1 - x)\, ds + \int_x^1 x(1 - s)\, ds = (1 - x)\frac{x^2}{2} + x\frac{(1 - x)^2}{2} = \frac{x(1 - x)}{2}.$$

Check. $y'' = -1$ ✓, $y(0) = y(1) = 0$ ✓. The maximum $1/8$ occurs midway between the plates, as symmetry demands.

Eigenvalue problems and Sturm–Liouville theory

A homogeneous BVP containing a parameter λ normally has only the trivial solution; for special values of λ — **eigenvalues** — it has nontrivial solutions, the **eigenfunctions**. This is the infinite-dimensional analogue of $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ (Ch. 10).

The model eigenvalue problem

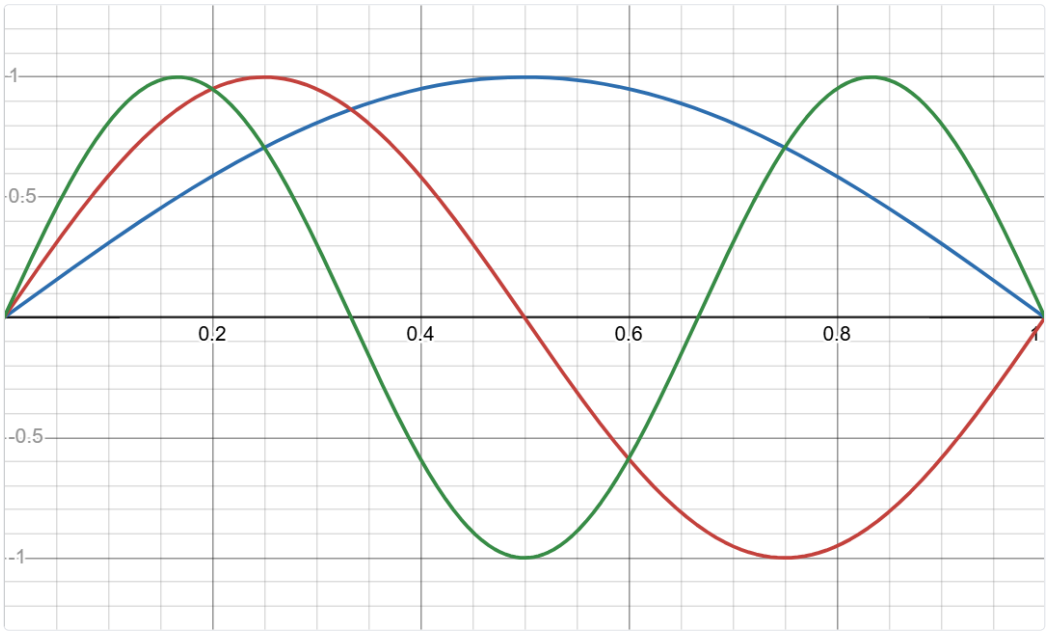
$$y'' + \lambda y = 0, \qquad y(0) = 0, \quad y(L) = 0$$

has nontrivial solutions only for

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2, \qquad y_n(x) = \sin\frac{n\pi x}{L}, \qquad n = 1, 2, 3, \dots$$

Derivation. Consider the sign of λ.

- $\lambda = -k^2 < 0$: $y = c_1 \cosh kx + c_2 \sinh kx$. Then $y(0) = c_1 = 0$ and $y(L) = c_2 \sinh kL = 0$ forces $c_2 = 0$, since $\sinh kL \neq 0$.
- $\lambda = 0$: $y = c_1 + c_2 x$, and both conditions give $c_1 = c_2 = 0$.
- $\lambda = k^2 > 0$: $y = c_1 \cos kx + c_2 \sin kx$. $y(0) = 0$ gives $c_1 = 0$; $y(L) = c_2 \sin kL = 0$ with $c_2 \neq 0$ requires $kL = n\pi$. $\square$



The first three eigenfunctions $\sin(n\pi x/L)$ for $L = 1$: the nth has $n - 1$ interior zeros. They are the standing-wave shapes of a string fixed at both ends and the voltage patterns on a lossless line section shorted at both ends.

The model problem is one member of a large family with the same properties.

Regular Sturm–Liouville problem

$$\big(P(x)\,y'\big)' + Q(x)\,y + \lambda\,w(x)\,y = 0, \qquad a < x < b,$$

$$\alpha_1 y(a) + \alpha_2 y'(a) = 0, \qquad \beta_1 y(b) + \beta_2 y'(b) = 0,$$

with P, P', Q, w continuous on $[a, b]$, $P > 0$ and $w > 0$ there, and each pair (α_1, α_2), (β_1, β_2) not both zero. The function w is the **weight**.

Any equation $y'' + p(x)y' + q(x)y + \lambda s(x)y = 0$ can be put in this form by multiplying by $P = e^{\int p\,dx}$: then $Py'' + Py'p = (Py')'$, and $Q = Pq$, $w = Ps$.

Properties of regular Sturm–Liouville problems

1. All eigenvalues are **real**, and they form an increasing sequence $\lambda_1 < \lambda_2 < \cdots \to \infty$.
2. Each eigenvalue is **simple**: its eigenfunctions are multiples of one function y_n, which can be taken real.
3. Eigenfunctions for different eigenvalues are **orthogonal with weight** w:

$$\langle y_m, y_n \rangle_w = \int_a^b y_m(x)\,y_n(x)\,w(x)\,dx = 0, \qquad m \neq n.$$

4. The nth eigenfunction has exactly $n - 1$ zeros in (a, b).
5. The eigenfunctions are **complete**: any square-integrable f on $[a, b]$ has a convergent (in the mean-square sense) expansion $f = \sum c_n y_n$ with $c_n = \langle f, y_n \rangle_w / \langle y_n, y_n \rangle_w$.

Proof of orthogonality (and reality). Write $\mathcal{S}[y] = (Py')' + Qy$. For any two functions satisfying the boundary conditions, **Lagrange's identity**

$$u\,\mathcal{S}[v] - v\,\mathcal{S}[u] = \big(P(uv' - vu')\big)'$$

(verify by expanding) integrates to

$$\int_a^b \big(u\,\mathcal{S}[v] - v\,\mathcal{S}[u]\big)\,dx = \Big[P(uv' - vu')\Big]_a^b = 0,$$

because at each end the boundary condition makes the vectors (u, u') and (v, v') both orthogonal to the nonzero vector (α_1, α_2) (resp. (β_1, β_2)), hence parallel, so $uv' - vu' = 0$ there. The operator is **self-adjoint** — the function-space counterpart of a symmetric matrix. Now take $u = y_m, v = y_n$ with $\mathcal{S}[y_k] = -\lambda_k w y_k$: the identity gives $(\lambda_m - \lambda_n)\int_a^b y_m y_n w\,dx = 0$, so the integral vanishes when $\lambda_m \neq \lambda_n$. Applying the same identity to $u = y$, $v = \bar{y}$ for a possibly complex eigenpair gives $(\lambda - \bar{\lambda})\int |y|^2 w\,dx = 0$, so λ is real. □

Parts 1, 2, 4 and 5 need more machinery (comparison theorems and functional analysis) and are quoted. Part 3 is what makes coefficients of an eigenfunction expansion computable one at a time — exactly as with the orthogonal eigenvectors of a symmetric matrix.

Worked Example 11.35 — Neumann conditions

Find the eigenvalues and eigenfunctions of $y'' + \lambda y = 0$, $y'(0) = 0$, $y'(L) = 0$.

Step 1 ($\lambda < 0$). $y = c_1 \cosh kx + c_2 \sinh kx$: $y'(0) = kc_2 = 0$, then $y'(L) = kc_1 \sinh kL = 0$ forces $c_1 = 0$. No eigenvalues.

Step 2 ($\lambda = 0$). $y = c_1 + c_2 x$: $y' = c_2 = 0$, leaving $y = c_1$. So $\lambda_0 = 0$ **is** an eigenvalue with eigenfunction $y_0 = 1$.

Step 3 ($\lambda = k^2 > 0$). $y'(0) = kc_2 = 0$; $y'(L) = -kc_1 \sin kL = 0$ needs $kL = n\pi$.

Result. $\lambda_n = (n\pi/L)^2$, $y_n = \cos(n\pi x/L)$, $n = 0, 1, 2, \ldots$ — the cosine set of Fourier series. The constant mode $n = 0$ is the "DC term"; it exists because insulating (zero-flux) ends allow any uniform level.

Worked Example 11.36 — A Robin condition with transcendental eigenvalues

Find the eigenvalues of $y'' + \lambda y = 0$, $y(0) = 0$, $y'(1) + y(1) = 0$.

Step 1. For $\lambda \leq 0$ one checks as before that only $y \equiv 0$ survives (for $\lambda = -k^2$, the condition becomes $k\cosh k + \sinh k = 0$, impossible for $k > 0$; for $\lambda = 0$, $y = c_2 x$ gives $2c_2 = 0$).

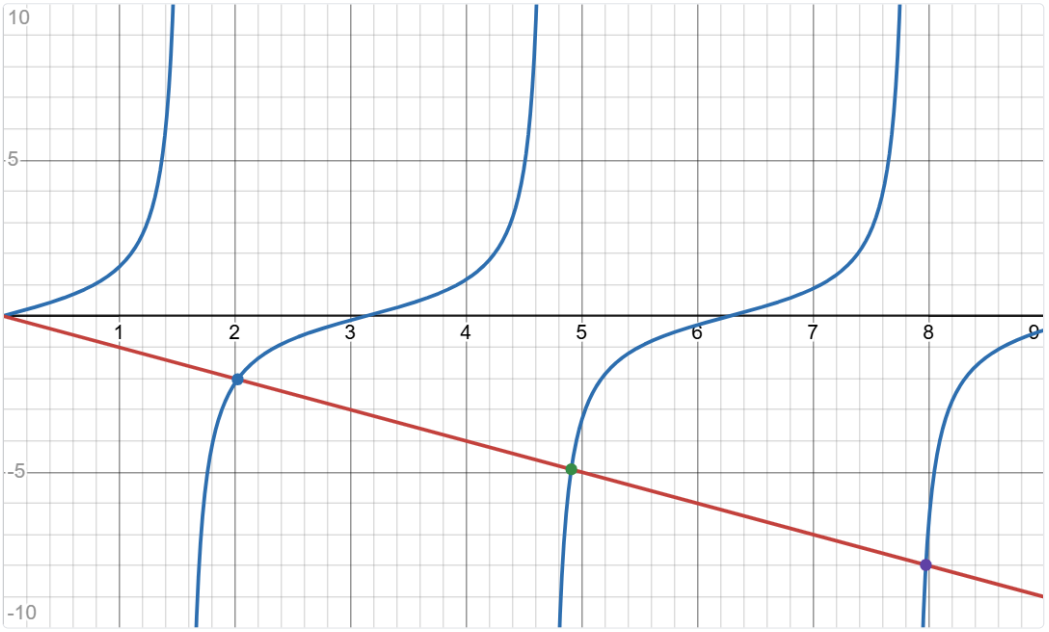
Step 2. For $\lambda = k^2 > 0$, $y(0) = 0$ gives $y = \sin kx$, and the Robin condition gives $k\cos k + \sin k = 0$, i.e.

$$\tan k = -k.$$

Step 3 (solve numerically). The roots are the intersections of $y = \tan k$ with $y = -k$, one in each interval $\left((n - \tfrac{1}{2})\pi, n\pi\right)$. Newton's method (Ch. 22) gives

$$k_1 = 2.0288, \quad k_2 = 4.9132, \quad k_3 = 7.9787, \qquad \lambda_n = k_n^2 = 4.116,\ 24.14,\ 63.66.$$

Check. $\tan(2.0288) = -2.0288$ to four figures. ✓ For large n, $k_n \to (n - \tfrac{1}{2})\pi$ from above: $(3 - 0.5)\pi = 7.854$ versus 7.979. The eigenfunctions $\sin k_n x$ are orthogonal on $[0, 1]$ even though the k_n are not equally spaced — guaranteed by the Sturm–Liouville theorem, not by any trigonometric identity.



Graphical solution of $\tan k = -k$ (Example 11.36): the eigenvalues of a Robin problem are the intersections of $\tan k$ (blue) with the line $-k$ (red), approaching $(n - \tfrac{1}{2})\pi$ as n grows.

Worked Example 11.37 — A weighted Sturm–Liouville problem

Solve $x^2y'' + xy' + \lambda y = 0$ on $1 < x < e$ with $y(1) = y(e) = 0$, and identify the weight.

Step 1 (SL form). Dividing by x: $xy'' + y' + (\lambda/x)y = 0$, i.e. $(xy')' + \lambda \dfrac{1}{x}\, y = 0$. So $P = x$, $Q = 0$, $w = 1/x$.

Step 2 (solve). This is Cauchy–Euler (§11.4) with indicial equation $m^2 + \lambda = 0$. For $\lambda = k^2 > 0$, $y = c_1 \cos(k \ln x) + c_2 \sin(k \ln x)$. $y(1) = c_1 = 0$; $y(e) = c_2 \sin k = 0$ needs $k = n\pi$. (As in the model problem, $\lambda \le 0$ gives only the trivial solution.)

Step 3. $\lambda_n = n^2\pi^2$, $y_n = \sin(n\pi \ln x)$.

Check (orthogonality with weight). With $u = \ln x$, $du = dx/x$:

$$\int_1^e \sin(m\pi \ln x)\sin(n\pi \ln x)\,\frac{dx}{x} = \int_0^1 \sin(m\pi u)\sin(n\pi u)\,du = 0, \quad m \neq n. \checkmark$$

Without the weight $1/x$ the integral is *not* zero — orthogonality is always relative to the weight of the SL form.

Common pitfall — forgetting $\lambda \le 0$

Always check $\lambda < 0$, $\lambda = 0$ and $\lambda > 0$ separately. Dirichlet problems have no zero eigenvalue, but Neumann and periodic problems do, and some Robin conditions (with the "wrong" sign) produce a negative eigenvalue. Missing the $\lambda_0 = 0$ mode loses the average value of every Fourier-cosine expansion.

Engineering note — eigenvalues are resonances

In a separation-of-variables solution of the wave equation on a line or in a cavity, $\lambda_n = (n\pi/L)^2$ becomes $(\omega_n/c)^2$: the eigenvalues fix the **resonant frequencies** $\omega_n = n\pi c/L$, and the eigenfunctions are the **mode shapes**. The Fredholm alternative is the precise statement of what happens if a structure is driven exactly at one of them: no steady solution exists unless the drive is orthogonal to that mode. Ch. 13 uses these results directly.

11.9 Systems of differential equations

Motivation

Real networks have several state variables that interact: two coupled resonators, a multi-loop filter, a motor with electrical and mechanical dynamics, a transistor amplifier with several internal capacitances. Writing each state's rate of change in terms of all states gives a **system** of first-order equations. Systems are also the universal form: every nth-order equation is one (§11.5), every simulator integrates one, and the geometry of systems — trajectories in a state space, equilibria, modes — is the clearest way to understand dynamics. This section gives the general theory and the eigenvector method for constant coefficients; the matrix exponential and the full state-variable formulation of circuits are developed in Ch. 12.

Linear first-order system

$$\dot{\mathbf{x}}(t) = \mathbf{A}(t)\,\mathbf{x}(t) + \mathbf{f}(t), \qquad \mathbf{x}(t) \in \mathbb{R}^n,$$

with $\mathbf{A}(t)$ an $n \times n$ matrix and $\mathbf{f}(t)$ an input vector, both continuous on I. The system is **homogeneous** if $\mathbf{f} \equiv \mathbf{0}$. An IVP adds $\mathbf{x}(t_0) = \mathbf{x}_0$. By §11.7, the IVP has a unique solution on all of I.

Structure theory

Everything in §11.4–§11.6 has a matrix counterpart.

- **Dimension.** Solutions of $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ form an n-dimensional vector space.
- **Fundamental matrix.** A matrix $\mathbf{\Phi}(t)$ whose n columns are independent solutions satisfies $\dot{\mathbf{\Phi}} = \mathbf{A}\mathbf{\Phi}$ and $\det \mathbf{\Phi} \neq 0$. The general homogeneous solution is $\mathbf{x} = \mathbf{\Phi}(t)\mathbf{c}$, and the IVP solution is $\mathbf{x}(t) = \mathbf{\Phi}(t)\mathbf{\Phi}(t_0)^{-1}\mathbf{x}_0$.
- **Liouville's formula** (Abel's formula for systems):

$$\det \mathbf{\Phi}(t) = \det \mathbf{\Phi}(t_0)\, \exp\!\left(\int_{t_0}^{t} \operatorname{tr} \mathbf{A}(s)\, ds \right).$$

For the companion matrix of §11.5, $\operatorname{tr}\mathbf{A} = -p_{n-1}$ and $\mathbf{\Phi}$ is the Wronskian matrix, so this reproduces Abel's formula.

- **Variation of parameters.** Seeking $\mathbf{x}_p = \mathbf{\Phi}(t)\mathbf{u}(t)$ gives $\mathbf{\Phi}\dot{\mathbf{u}} = \mathbf{f}$, so

$$\mathbf{x}(t) = \mathbf{\Phi}(t)\mathbf{\Phi}(t_0)^{-1}\mathbf{x}_0 + \mathbf{\Phi}(t)\int_{t_0}^{t} \mathbf{\Phi}(s)^{-1}\mathbf{f}(s)\, ds.$$

The first term is the zero-input response, the second the zero-state response — the matrix version of §11.3's formula.

Derivation of variation of parameters. $\dot{\mathbf{x}}_p = \dot{\mathbf{\Phi}}\mathbf{u} + \mathbf{\Phi}\dot{\mathbf{u}} = \mathbf{A}\mathbf{\Phi}\mathbf{u} + \mathbf{\Phi}\dot{\mathbf{u}}$. Setting this equal to $\mathbf{A}\mathbf{x}_p + \mathbf{f} = \mathbf{A}\mathbf{\Phi}\mathbf{u} + \mathbf{f}$ leaves $\mathbf{\Phi}\dot{\mathbf{u}} = \mathbf{f}$; $\mathbf{\Phi}$ is invertible, so $\dot{\mathbf{u}} = \mathbf{\Phi}^{-1}\mathbf{f}$. $\square$

Constant coefficients: the eigenvector method

For constant $\mathbf{A}$, try $\mathbf{x} = \mathbf{v}e^{\lambda t}$ with a constant vector $\mathbf{v} \neq \mathbf{0}$. Then $\dot{\mathbf{x}} = \lambda \mathbf{v} e^{\lambda t}$ and $\mathbf{A}\mathbf{x} = \mathbf{A}\mathbf{v}e^{\lambda t}$, so we need

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}:$$

an eigenvalue problem (Ch. 10). Each eigenpair gives a **mode** — a solution that keeps its shape $\mathbf{v}$ and only scales by $e^{\lambda t}$.

Eigen-solution of $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$

If $\mathbf{A}$ has n linearly independent eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ with eigenvalues $\lambda_1, \dots, \lambda_n$, then

$$\mathbf{x}(t) = c_1\mathbf{v}_1 e^{\lambda_1 t} + \cdots + c_n\mathbf{v}_n e^{\lambda_n t},$$

and the constants are found from $\mathbf{x}(0) = \sum c_k\mathbf{v}_k$, i.e. by solving $\mathbf{V}\mathbf{c} = \mathbf{x}_0$ with $\mathbf{V} = [\mathbf{v}_1 \cdots \mathbf{v}_n]$.

Independence of the eigensolutions follows from Liouville or directly: at $t = 0$ the fundamental matrix is $\mathbf{V}$, which is invertible. Three cases arise.

Complex eigenvalues. If $\mathbf{A}$ is real and $\lambda = \alpha + j\beta$ with eigenvector $\mathbf{v} = \mathbf{a} + j\mathbf{b}$, then $\bar{\lambda}, \bar{\mathbf{v}}$ is also an eigenpair, and the real and imaginary parts of $\mathbf{v}e^{\lambda t}$ are two real independent solutions:

$$e^{\alpha t}\big(\mathbf{a}\cos\beta t - \mathbf{b}\sin\beta t\big), \qquad e^{\alpha t}\big(\mathbf{a}\sin\beta t + \mathbf{b}\cos\beta t\big).$$

(Real and imaginary parts of a complex solution of a real linear system are solutions, because $\mathbf{A}$ is real.)

Repeated eigenvalues with too few eigenvectors (defective A). If λ is a double eigenvalue with only one eigenvector $\mathbf{v}$, a second solution is

$$\mathbf{x}_2 = (t\,\mathbf{v} + \mathbf{w})\,e^{\lambda t}, \qquad (\mathbf{A} - \lambda\mathbf{I})\mathbf{w} = \mathbf{v},$$

where $\mathbf{w}$ is a **generalised eigenvector**. (Substituting: $\dot{\mathbf{x}}_2 = (\mathbf{v} + \lambda t\mathbf{v} + \lambda\mathbf{w})e^{\lambda t}$, and $\mathbf{A}\mathbf{x}_2 = (\lambda t\mathbf{v} + \lambda\mathbf{w} + \mathbf{v})e^{\lambda t}$.) This is the matrix origin of the $te^{\lambda t}$ terms of repeated characteristic roots. The general treatment via $e^{\mathbf{A}t}$ and Jordan form is in Ch. 12.

Worked Example 11.38 — A symmetric 2×2 system

Solve $\dot{x}_1 = -3x_1 + x_2$, $\dot{x}_2 = x_1 - 3x_2$ with $x_1(0) = 2$, $x_2(0) = 0$.

Step 1 (eigenvalues). $\mathbf{A} = \begin{pmatrix} -3 & 1 \\ 1 & -3 \end{pmatrix}$, $\det(\mathbf{A} - \lambda\mathbf{I}) = (\lambda+3)^2 - 1 = 0$, so $\lambda_1 = -2$, $\lambda_2 = -4$.

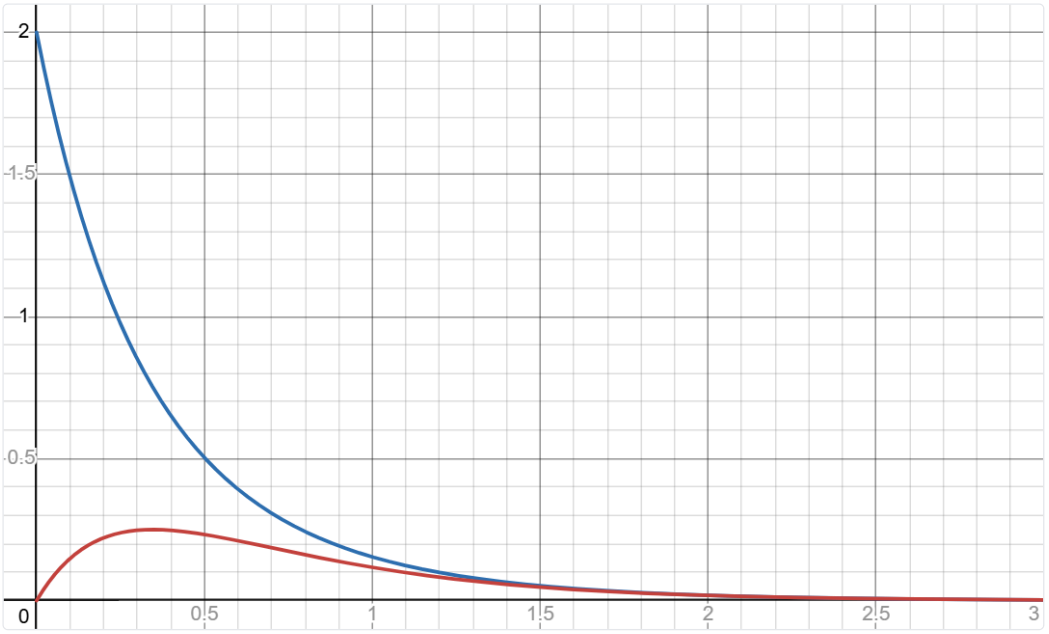
Step 2 (eigenvectors). For $\lambda_1 = -2$: $\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}\mathbf{v} = \mathbf{0}$ gives $\mathbf{v}_1 = (1, 1)^{\mathsf{T}}$. For $\lambda_2 = -4$: $\mathbf{v}_2 = (1, -1)^{\mathsf{T}}$. (Orthogonal, as for any symmetric matrix.)

Step 3 (fit). $c_1 + c_2 = 2$, $c_1 - c_2 = 0$, so $c_1 = c_2 = 1$:

$$x_1 = e^{-2t} + e^{-4t}, \qquad x_2 = e^{-2t} - e^{-4t}.$$

Check. $\dot{x}_1 = -2e^{-2t} - 4e^{-4t}$ and $-3x_1 + x_2 = -2e^{-2t} - 4e^{-4t}$. ✓ Liouville: $\operatorname{tr}\mathbf{A} = -6$, and $\det\mathbf{\Phi} \propto e^{-2t}e^{-4t} = e^{-6t}$. ✓

Modes. The symmetric mode $(1, 1)$ (both states equal) decays with time constant 0.5; the antisymmetric mode $(1, -1)$ decays twice as fast. Any initial state is a mixture of the two.



Example 11.38: x_1 (blue) and x_2 (red). The fast mode e^{-4t} dies first, after which $x_1 \approx x_2$ and both follow the slow mode e^{-2t}.

Worked Example 11.39 — Complex eigenvalues

Solve $\dot{\mathbf{x}} = \begin{pmatrix} -1 & -4 \\ 1 & -1 \end{pmatrix}\mathbf{x}$, $\mathbf{x}(0) = (1,0)^{\mathsf{T}}$.

Step 1. $\det(\mathbf{A} - \lambda\mathbf{I}) = (\lambda+1)^2 + 4 = 0$, so $\lambda = -1 \pm 2j$.

Step 2. For $\lambda = -1 + 2j$: $(\mathbf{A} - \lambda\mathbf{I}) = \begin{pmatrix} -2j & -4 \\ 1 & -2j \end{pmatrix}$; the second row gives $v_1 = 2jv_2$, so $\mathbf{v} = (2j, 1)^{\mathsf{T}}$, i.e. $\mathbf{a} = (0,1)^{\mathsf{T}}$, $\mathbf{b} = (2,0)^{\mathsf{T}}$.

Step 3 (real solutions).

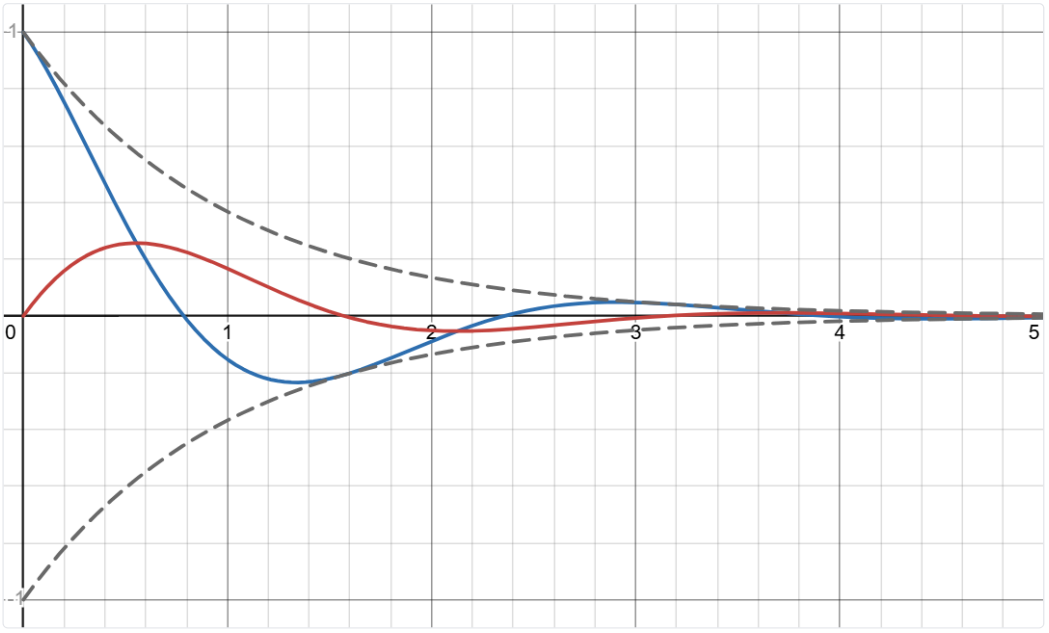
$$\mathbf{x} = e^{-t}\left[c_1\begin{pmatrix} -2\sin 2t \\ \cos 2t \end{pmatrix} + c_2\begin{pmatrix} 2\cos 2t \\ \sin 2t \end{pmatrix}\right].$$

Step 4 (fit). At $t = 0$: $(2c_2, c_1) = (1,0)$, so $c_1 = 0$, $c_2 = \tfrac{1}{2}$:

$$x_1 = e^{-t}\cos 2t, \qquad x_2 = \tfrac{1}{2}e^{-t}\sin 2t.$$

Check. $\dot{x}_1 = -e^{-t}\cos 2t - 2e^{-t}\sin 2t = -x_1 - 4x_2$ ✓; $\dot{x}_2 = -\tfrac{1}{2}e^{-t}\sin 2t + e^{-t}\cos 2t = x_1 - x_2$ ✓.

In the phase plane the state spirals into the origin (a stable focus), turning at 2 rad/s and shrinking by e^{-1} per second — the behaviour of an underdamped RLC tank.



Example 11.39: $x_1 = e^{-t}\cos 2t$ (blue) and $x_2 = \tfrac{1}{2}e^{-t}\sin 2t$ (red), a quarter-period apart inside the decaying envelope $\pm e^{-t}$ set by $\operatorname{Re}\lambda = -1$.

Worked Example 11.40 — A defective matrix

Solve $\dot{\mathbf{x}} = \begin{pmatrix} 1 & -1 \\ 1 & 3 \end{pmatrix}\mathbf{x}$, $\mathbf{x}(0) = (1,1)^{\mathsf{T}}$.

Step 1. $\operatorname{tr}\mathbf{A} = 4$, $\det\mathbf{A} = 4$, so $\lambda^2 - 4\lambda + 4 = (\lambda-2)^2$: a double eigenvalue $\lambda = 2$.

Step 2. $\mathbf{A} - 2\mathbf{I} = \begin{pmatrix} -1 & -1 \\ 1 & 1 \end{pmatrix}$ has rank 1, so there is only one eigenvector, $\mathbf{v} = (1,-1)^{\mathsf{T}}$.

Step 3 (generalised eigenvector). $(\mathbf{A} - 2\mathbf{I})\mathbf{w} = \mathbf{v}$: $-w_1 - w_2 = 1$; take $\mathbf{w} = (-1,0)^{\mathsf{T}}$.

Step 4. $\mathbf{x} = c_1\mathbf{v}e^{2t} + c_2(t\mathbf{v} + \mathbf{w})e^{2t}$. At $t = 0$: $c_1\mathbf{v} + c_2\mathbf{w} = (c_1 - c_2, -c_1) = (1,1)$, so $c_1 = -1$, $c_2 = -2$:

$$\mathbf{x}(t) = e^{2t}\begin{pmatrix} 1 - 2t \\ 1 + 2t \end{pmatrix}.$$

Check. $\dot{x}_1 = e^{2t}(2 - 4t - 2) = -4te^{2t}$ and $x_1 - x_2 = -4te^{2t}$ ✓; $\dot{x}_2 = e^{2t}(2 + 4t + 2) = (4+4t)e^{2t}$ and $x_1 + 3x_2 = (4+4t)e^{2t}$ ✓.

The phase plane for 2×2 systems

Plotting x_2 against x_1 — the **phase plane** — shows all trajectories at once. For $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ with $\det\mathbf{A} \neq 0$ the origin is the only equilibrium, and its type depends only on the eigenvalues, which in turn depend only on $T = \operatorname{tr}\mathbf{A}$ and $\Delta = \det\mathbf{A}$ through $\lambda^2 - T\lambda + \Delta = 0$:

Condition	Eigenvalues	Equilibrium type	Stability
$\Delta < 0$	real, opposite signs	saddle	unstable
$\Delta > 0, T^2 > 4\Delta, T < 0$	real, both negative	node	asymptotically stable

Condition	Eigenvalues	Equilibrium type	Stability
$\Delta > 0, T^2 > 4\Delta, T > 0$	real, both positive	node	unstable
$\Delta > 0, T^2 < 4\Delta, T < 0$	complex, $\operatorname{Re}\lambda < 0$	spiral (focus)	asymptotically stable
$\Delta > 0, T^2 < 4\Delta, T > 0$	complex, $\operatorname{Re}\lambda > 0$	spiral	unstable
$\Delta > 0, T = 0$	purely imaginary	centre	stable, not asymptotically

So a 2×2 linear system is asymptotically stable exactly when $\operatorname{tr}\mathbf{A} < 0$ and $\det\mathbf{A} > 0$ — the two-state case of the Routh–Hurwitz criterion. Examples 11.38, 11.39 and 11.40 are a stable node ($T = -6$, $\Delta = 8$), a stable spiral ($T = -2$, $\Delta = 5$) and an unstable degenerate node ($T = 4$, $\Delta = 4 = T^2/4$).

Nonlinear systems and linearisation

For an autonomous nonlinear system $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$, an **equilibrium** $\mathbf{x}^*$ satisfies $\mathbf{F}(\mathbf{x}^*) = \mathbf{0}$. Setting $\mathbf{x} = \mathbf{x}^* + \boldsymbol{\eta}$ and expanding to first order (Ch. 6),

$$\dot{\boldsymbol{\eta}} \approx \mathbf{J}\,\boldsymbol{\eta}, \qquad \mathbf{J} = \left.\frac{\partial \mathbf{F}}{\partial \mathbf{x}}\right|_{\mathbf{x}^*},$$

where $\mathbf{J}$ is the Jacobian matrix.

Linearisation (Hartman–Grobman, informal)

If no eigenvalue of $\mathbf{J}$ has zero real part (the equilibrium is **hyperbolic**), then near $\mathbf{x}^*$ the trajectories of the nonlinear system look like those of $\dot{\boldsymbol{\eta}} = \mathbf{J}\boldsymbol{\eta}$: same type (node, saddle, spiral) and same stability. If some eigenvalue has zero real part (for example a linear centre), the linearisation is inconclusive and the nonlinear terms decide.

Worked Example 11.41 — Equilibria of a damped pendulum

The damped pendulum $\ddot{\theta} + 0.5\dot{\theta} + \sin\theta = 0$ becomes, with $x_1 = \theta$, $x_2 = \dot{\theta}$,

$$\dot{x}_1 = x_2, \qquad \dot{x}_2 = -\sin x_1 - 0.5x_2.$$

Classify the equilibria $(0,0)$ and $(\pi,0)$.

Step 1 (Jacobian). $\mathbf{J} = \begin{pmatrix} 0 & 1 \\ -\cos x_1 & -0.5 \end{pmatrix}$.

Step 2 (hanging down, $(0,0)$). $\mathbf{J} = \begin{pmatrix} 0 & 1 \\ -1 & -0.5 \end{pmatrix}$: $T = -0.5$, $\Delta = 1$, $T^2 < 4\Delta$. Eigenvalues $-0.25 \pm 0.968j$: a **stable spiral** — the pendulum swings with decaying amplitude.

Step 3 (inverted, $(\pi,0)$). $\mathbf{J} = \begin{pmatrix} 0 & 1 \\ 1 & -0.5 \end{pmatrix}$: $\Delta = -1 < 0$, a **saddle**. Eigenvalues $\left(-0.5 \pm \sqrt{4.25}\right)/2 = 0.781,\ -1.281$. The unstable direction is the eigenvector for 0.781: almost every small disturbance topples the pendulum.

Check. Both equilibria are hyperbolic, so the nonlinear pendulum has the same local behaviour. The same analysis applied to a phase-locked loop (whose phase-error equation has exactly this form) shows that the loop locks at one equilibrium and is repelled from the other.

Engineering note — small-signal analysis is linearisation

The small-signal model of a transistor circuit is the Jacobian of the large-signal equations at the DC operating point: $g_m = \partial i_C/\partial v_{BE}$, r_π, r_o are entries of $\mathbf{J}$. Hartman–Grobman is the mathematical warrant for trusting small-signal stability analysis — and the warning that when a pole sits on the $j\omega$-axis (an oscillator at the edge of start-up), the linear model cannot predict the amplitude; nonlinear terms must be kept (compare Example 11.13).

11.10 Applications to electrical circuits

From a circuit to a differential equation

Every lumped-circuit ODE is built from two ingredients.

1. **Element laws.** $v_R = Ri$; $v_L = L\,\dfrac{di}{dt}$; $i_C = C\,\dfrac{dv}{dt}$; plus any nonlinear laws such as a diode $i = I_S\left(e^{v/V_T} - 1\right)$, a saturating inductor $v = \dfrac{d\lambda(i)}{dt}$, or a time-varying element $i = \dfrac{d}{dt}\big(C(t)v\big)$.
2. **Kirchhoff's laws.** KVL around loops and KCL at nodes — linear algebraic constraints (Ch. 8).

The natural unknowns are the **state variables**: capacitor voltages and inductor currents. They carry the stored energy ($\tfrac12 Cv^2$, $\tfrac12 Li^2$), they are continuous in time (§11.7), and their values at one instant, together with the sources thereafter, determine everything else. The number of independent state variables is the **order** of the circuit.

A systematic procedure.

1. Choose the state variables: one v_C per independent capacitor, one i_L per independent inductor.
2. Express each $C\,\dot{v}_C$ as a capacitor current and each $L\,\dot{i}_L$ as an inductor voltage.
3. Use KVL, KCL and the resistive element laws to write those currents and voltages in terms of the states and the sources.

4. The result is $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}, t)$ — linear, $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}(t)$, if all elements are linear and time-invariant.
5. Eliminate states if a single higher-order equation for one output is wanted (§11.5 in reverse).

Ch. 12 refines steps 1–4 into the formal state-variable method and solves the resulting constant-coefficient systems completely. The table below shows which method of *this* chapter each common circuit leads to.

Circuit	Equation	Type (this chapter)
Series RC, source v_s	$RC\,\dot{v}_C + v_C = v_s(t)$	linear first order (§11.3)
Series RL	$L\,\dot{i} + Ri = v_s(t)$	linear first order (§11.3)
C discharging into nonlinear load $i = g(v)$	$C\,\dot{v} = -g(v)$	separable (§11.2)
RC with time-varying $C(t)$ or $R(t)$	$R\,\dfrac{d}{dt}(Cv) + v = v_s$	linear, variable coefficient (§11.3)
Series RLC	$LC\,\ddot{v}_C + RC\,\dot{v}_C + v_C = v_s$	linear second order (§11.4, Ch. 12)
Multi-loop or multi-node network	$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$	linear system (§11.9)
Negative-resistance oscillator	$\ddot{v} - \mu(1 - v^2)\dot{v} + v = 0$	nonlinear system (§11.9)
Distributed line or electrode at DC	$V'' = rg\,V$ on $0 < x < \ell$	boundary-value problem (§11.8)

Worked Example 11.42 — RC charging

A 10 kΩ resistor and a 10 μF capacitor (initially uncharged) are connected to a 5 V source at $t = 0$. Find $v_C(t)$, its value at $t = 0.2$ s, and the time to reach 4.5 V.

Step 1. KVL: $RC\,\dot{v}_C + v_C = 5$, with $\tau = RC = 10^4 \cdot 10^{-5} = 0.1$ s.

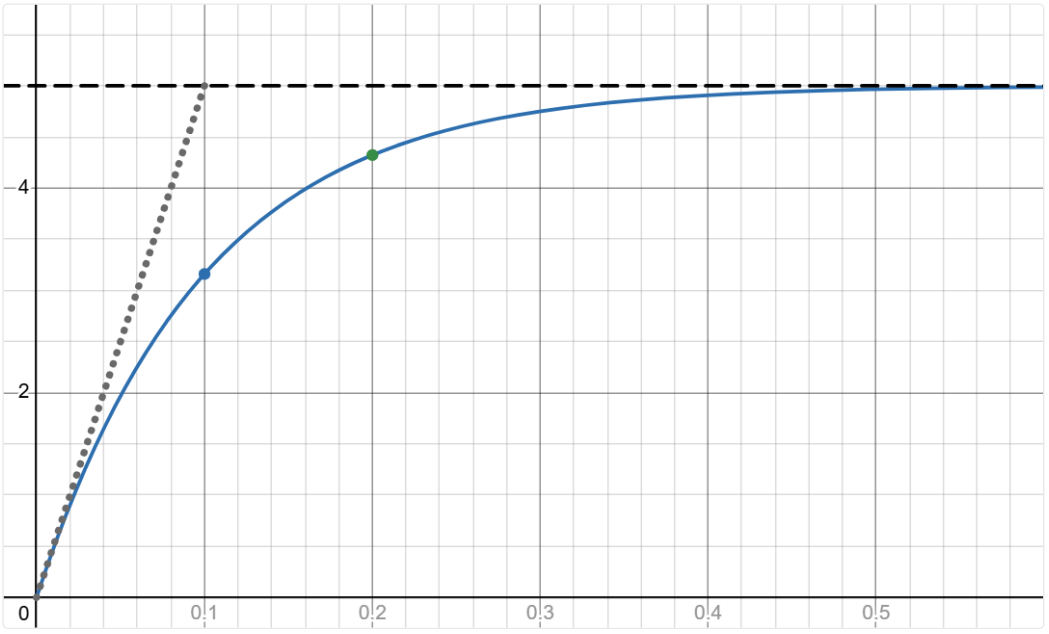
Step 2. By §11.3 with $v_C(0) = 0$:

$$v_C(t) = 5\left(1 - e^{-t/0.1}\right)\ \text{V}.$$

Step 3. At $t = 0.2$ s: $v_C = 5(1 - e^{-2}) = 5(0.8647) = 4.32$ V.

Step 4. $5(1 - e^{-t/\tau}) = 4.5$ gives $e^{-t/\tau} = 0.1$, so $t = \tau \ln 10 = 0.230$ s.

Check. Units: $\Omega \cdot \text{F} = (\text{V/A})(\text{A s/V}) = \text{s}$. ✓ Initial slope $\dot{v}_C(0) = 5/\tau = 50$ V/s equals $i(0)/C = (5/10^4)/10^{-5} = 50$ V/s. ✓



RC charging with $\tau = 0.1$ s: 63% of the final value after one time constant, 86% after two and over 98% after 4τ. The dotted initial tangent reaches the final value exactly at $t = \tau$.

Worked Example 11.43 — A two-loop network as a system

Loop 1 contains the source $v(t)$, $R_1 = 2\ \Omega$ and $L_1 = 1$ H. Loop 2 contains $R_3 = 2\ \Omega$ and $L_2 = 1$ H. The loops share $R_2 = 1\ \Omega$. Find the mesh currents when $v = 8$ V is switched on at $t = 0$ with both inductors initially unenergised.

Step 1 (KVL). With mesh currents i_1, i_2 (which here are also the inductor currents, hence states):

$$L_1\dot{i}_1 + R_1 i_1 + R_2(i_1 - i_2) = v(t),$$
$$L_2\dot{i}_2 + R_3 i_2 + R_2(i_2 - i_1) = 0.$$

Step 2 (state form).

$$\frac{d}{dt}\begin{pmatrix} i_1 \\ i_2 \end{pmatrix} = \begin{pmatrix} -3 & 1 \\ 1 & -3 \end{pmatrix}\begin{pmatrix} i_1 \\ i_2 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} v(t).$$

The matrix is that of Example 11.38: eigenvalues -2 and -4 s^{-1} (the network's **natural frequencies**), modes $(1,1)$ and $(1,-1)$.

Step 3 (particular solution). For constant input, try a constant: $\mathbf{0} = \mathbf{A}\mathbf{x}_p + \mathbf{b}\cdot 8$, so $\mathbf{x}_p = -8\mathbf{A}^{-1}\mathbf{b}$. With $\mathbf{A}^{-1} = \frac{1}{8}\begin{pmatrix} -3 & -1 \\ -1 & -3 \end{pmatrix}$, $\mathbf{x}_p = (3,1)^{\mathsf{T}}$ A.

Step 4 (fit zero initial state). $\mathbf{x} = (3,1)^{\mathsf{T}} + c_1(1,1)^{\mathsf{T}}e^{-2t} + c_2(1,-1)^{\mathsf{T}}e^{-4t}$, with $c_1 + c_2 = -3$ and $c_1 - c_2 = -1$, so $c_1 = -2$, $c_2 = -1$:

$$i_1 = 3 - 2e^{-2t} - e^{-4t}\ \text{A}, \qquad i_2 = 1 - 2e^{-2t} + e^{-4t}\ \text{A}.$$

Check. $i_1(0) = i_2(0) = 0$ ✓. DC check with inductors as shorts: loop 1, $2(3) + 1(3-1) = 8$ ✓; loop 2, $2(1) + 1(1-3) = 0$ ✓. Settling is governed by the slow mode e^{-2t}: within 1% of DC after about $t = \ln(200)/2 \approx 2.6$ s.

Worked Example 11.44 — A leaky line at DC: a boundary-value problem

A long cable (or a buried grounding conductor) has series resistance $r = 0.1\ \Omega/\text{m}$ and leakage conductance $g = 10^{-5}$ S/m to ground, length $\ell = 1$ km. A DC voltage V_0 is applied at $x = 0$ and the far end is open. Find $V(x)$, the far-end voltage and the input resistance.

Step 1 (model). On a short piece dx: the series drop is $dV = -rI\,dx$ and the leakage is $dI = -gV\,dx$. Hence $V' = -rI$, $I' = -gV$, and eliminating I,

$$V'' = \alpha^2 V, \qquad \alpha = \sqrt{rg} = \sqrt{10^{-6}} = 10^{-3}\ \text{m}^{-1}.$$

Step 2 (boundary conditions). $V(0) = V_0$ (Dirichlet). The open end carries no current: $I(\ell) = -V'(\ell)/r = 0$, i.e. $V'(\ell) = 0$ (Neumann).

Step 3 (solve). Using the basis $\cosh\alpha(\ell - x)$, $\sinh\alpha(\ell - x)$ — chosen so the Neumann condition at $x = \ell$ is easy — the condition $V'(\ell) = 0$ kills the sinh term, and $V(0) = V_0$ fixes the amplitude:

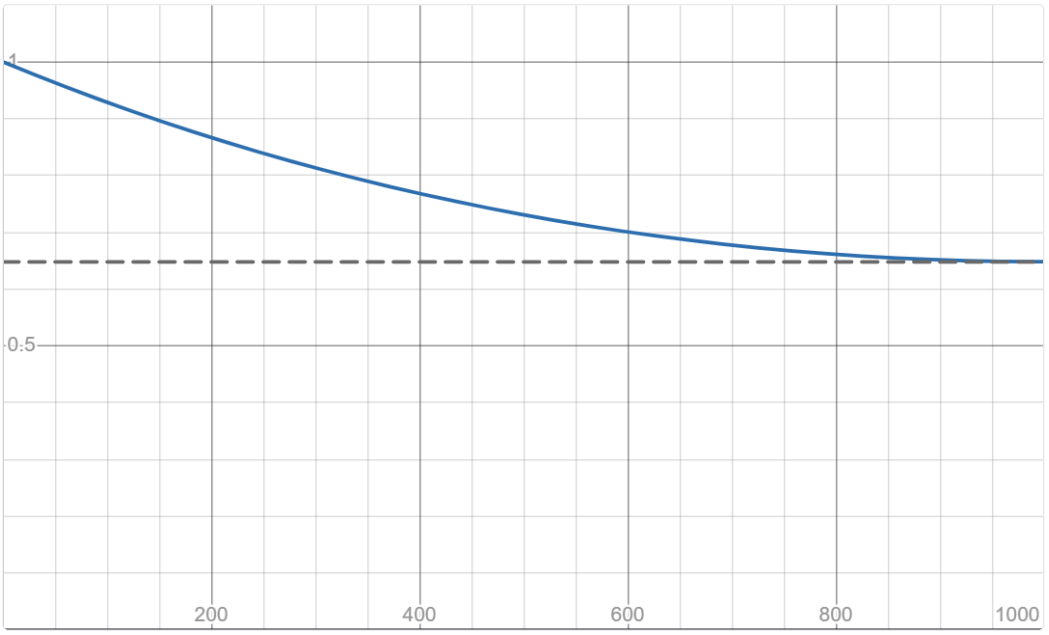
$$V(x) = V_0\,\frac{\cosh\alpha(\ell - x)}{\cosh\alpha\ell}.$$

The homogeneous problem ($V_0 = 0$) has only the trivial solution, since cosh never vanishes, so by the Fredholm alternative this is the unique solution.

Step 4 (numbers). $\alpha\ell = 1$: far-end voltage $V(\ell) = V_0/\cosh 1 = 0.648\,V_0$. Input current $I(0) = -V'(0)/r = V_0\alpha\tanh(\alpha\ell)/r$, so

$$R_{\text{in}} = \frac{r}{\alpha\tanh\alpha\ell} = \frac{0.1}{10^{-3}\cdot 0.7616} = 131\ \Omega.$$

Check. Two limits confirm the formula. For a very long line ($\alpha\ell \to \infty$), $\tanh \to 1$ and $R_{\text{in}} \to \sqrt{r/g} = 100\ \Omega$, the characteristic resistance. For a very short line, $\tanh\alpha\ell \approx \alpha\ell$ and $R_{\text{in}} \approx 1/(g\ell)$, the total leakage resistance in parallel to ground, as expected when series drop is negligible. Our line is finite and open-ended, so less leakage path is available than on an infinite line and $R_{\text{in}} = 131\ \Omega$ exceeds $100\ \Omega$. ✓ Directly: $V'' = \alpha^2 V$ holds since $\big(\cosh\alpha(\ell - x)\big)'' = \alpha^2\cosh\alpha(\ell - x)$, and $V'(\ell) = -\alpha V_0\sinh 0/\cosh\alpha\ell = 0$. ✓ The same equation, with α complex, describes AC voltage along a transmission line (Ch. 13).



Normalised DC voltage V/V_0 along the 1 km leaky line of Example 11.44. The curve meets the open end with zero slope (zero current); the far end sits at $1/\cosh 1 = 0.648$ of the input.

Worked Example 11.45 — Start-up of a negative-resistance oscillator

An LC tank in parallel with a nonlinear negative conductance (a tunnel diode, or a cross-coupled transistor pair) gives, in normalised units, the **van der Pol equation**

$$\ddot{v} - \mu\,(1 - v^2)\,\dot{v} + v = 0, \qquad \mu = 0.5.$$

Show that the oscillator starts by itself.

Step 1 (system). $x_1 = v$, $x_2 = \dot{v}$: $\dot{x}_1 = x_2$, $\dot{x}_2 = -x_1 + \mu(1 - x_1^2)x_2$. The only equilibrium is the origin.

Step 2 (linearise). $\mathbf{J} = \begin{pmatrix} 0 & 1 \\ -1 - 2\mu x_1 x_2 & \mu(1 - x_1^2) \end{pmatrix}$, which at the origin is $\begin{pmatrix} 0 & 1 \\ -1 & \mu \end{pmatrix}$: $T = 0.5 > 0$, $\Delta = 1$, $T^2 < 4\Delta$.

Step 3 (classify). Eigenvalues $\lambda = \tfrac{1}{2}\left(\mu \pm \sqrt{\mu^2 - 4}\right) = 0.25 \pm 0.968j$: an **unstable spiral**. Any noise voltage grows as $e^{0.25t}$ while oscillating at ≈ 0.97 (normalised) rad/s. The equilibrium is hyperbolic, so the nonlinear circuit does the same near the origin.

Step 4 (amplitude). Growth cannot continue forever, because for $|v| > 1$ the term $\mu(1 - v^2)$ becomes *positive damping*. Averaging the energy balance over a cycle leads to an amplitude equation of exactly the Bernoulli form of Example 11.13, whose stable equilibrium gives the steady amplitude (≈ 2 for small μ). Linear analysis predicts *whether* an oscillator starts; nonlinear analysis predicts *how large* it gets.

Common pitfall — dependent energy stores

Two capacitors in parallel (or a loop consisting only of capacitors and voltage sources), or two inductors in series (or a cutset of only inductors and current sources), do not add an order: their states are linked by an algebraic constraint. Counting them as independent gives a singular $\mathbf{A}$ or an inconsistent set of initial conditions, and in a simulator can produce "impulsive" currents at switching instants.

Engineering note — what comes next

This chapter *sets up* circuit equations and gives the general solution structure, plus the methods that survive nonlinear and time-varying elements. The systematic constant-coefficient theory — characteristic roots, damping ratio, natural and forced response, steady-state AC, state-variable form and $e^{\mathbf{A}t}$ — is Ch. 12. s-domain impedance methods that solve the same IVPs algebraically, impulses included, are Ch. 14. Nonlinear circuits beyond the reach of separation or linearisation are integrated numerically (Ch. 22), with the existence and uniqueness theory of §11.7 as the guarantee that the numerical answer means something.

Summary

Topic	Key formula / fact
Autonomous stability	$y' = f(y)$, $f(y^*) = 0$: stable if $f'(y^*) < 0$, unstable if $f'(y^*) > 0$
Exact equation	$M_y = N_t \Rightarrow \Phi(t, y) = C$ with $\Phi_t = M$, $\Phi_y = N$
Integrating factor for exactness	$(M_y - N_t)/N = R(t) \Rightarrow \mu = e^{\int R\,dt}$
Separable	$y' = g(t)h(y) \Rightarrow \int dy/h(y) = \int g\,dt + C$; check roots of h
Scale-invariant substitution	$y' = F(y/t)$: $v = y/t$, $tv' = F(v) - v$

Topic	Key formula / fact
Linear first order	$y' + py = r$: $y = y_0 e^{-P(t)} + \int_{t_0}^{t} e^{-[P(t)-P(s)]} r(s)\,ds$, $P = \int_{t_0}^{t} p$
Bernoulli	$y' + py = qy^n$: $v = y^{1-n}$, $v' + (1-n)pv = (1-n)q$
Riccati	known y_1: $y = y_1 + 1/v$, $v' = -(q_1 + 2q_2 y_1)v - q_2$
Wronskian / Abel	$W = y_1 y_2' - y_1' y_2$; $W(t) = W(t_0) e^{-\int_{t_0}^{t} p\,ds}$ (order n: use p_{n-1})
Reduction of order	$y_2 = y_1 \int e^{-\int p\,dt} / y_1^2\,dt$
Cauchy–Euler	$t^2 y'' + aty' + by = 0$: $y = t^m$, $m^2 + (a-1)m + b = 0$; $t = e^x$ gives constant coefficients
Energy integral	$y'' = -U'(y)$: $\tfrac{1}{2}(y')^2 + U(y) = E$
Companion form	$x_k = y^{(k-1)}$; $\det(s\mathbf{I} - \mathbf{A}) = s^n + p_{n-1}s^{n-1} + \cdots + p_0$
Structure theorem	$y = y_h + y_p$; y_h has n free constants
Variation of parameters	$y_p = -y_1 \int y_2 r/W\,dt + y_2 \int y_1 r/W\,dt$ (r from standard form)
Causal Green's function	$G(t,s) = [y_1(s)y_2(t) - y_1(t)y_2(s)]/W(s)$; $y_p = \int_{t_0}^{t} G\,r\,ds$
Picard–Lindelöf	f continuous, Lipschitz in y: unique solution on $\lvert t - t_0\rvert \le \min(a, b/M)$
Continuous dependence	$\lvert y - z\rvert \le \lvert y_0 - z_0\rvert e^{K\lvert t-t_0\rvert}$
Linear IVP	unique solution on the whole interval where coefficients are continuous
BVP solvability	unique $\iff$ homogeneous BVP has only $y \equiv 0$ (Fredholm alternative)
Model eigenvalue problem	$y'' + \lambda y = 0$, $y(0) = y(L) = 0$: $\lambda_n = (n\pi/L)^2$, $y_n = \sin(n\pi x/L)$
Sturm–Liouville	$(Py')' + Qy + \lambda w y = 0$: real simple eigenvalues, $\int y_m y_n w\,dx = 0$ for $m \ne n$
Systems	$\mathbf{x} = \mathbf{\Phi}\mathbf{\Phi}_0^{-1}\mathbf{x}_0 + \mathbf{\Phi}\int \mathbf{\Phi}^{-1}\mathbf{f}\,ds$; $\det\mathbf{\Phi} \propto e^{\int \operatorname{tr}\mathbf{A}}$
Eigenvector method	$\mathbf{x} = \sum c_k \mathbf{v}_k e^{\lambda_k t}$; defective: $(t\mathbf{v} + \mathbf{w})e^{\lambda t}$, $(\mathbf{A} - \lambda\mathbf{I})\mathbf{w} = \mathbf{v}$
2×2 stability	asymptotically stable $\iff$ $\operatorname{tr}\mathbf{A} < 0$ and $\det\mathbf{A} > 0$
Linearisation	$\dot{\boldsymbol{\eta}} = \mathbf{J}\boldsymbol{\eta}$, valid locally when no eigenvalue of $\mathbf{J}$ has zero real part
Circuit element laws	$v_L = L\,di/dt$, $i_C = C\,dv/dt$; states v_C, i_L are continuous; order = number of independent stores

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Further reading

- E. Kreyszig, *Advanced Engineering Mathematics*, 10th ed., Wiley — Chapters 1–5 (first-order, second-order and higher-order ODEs, systems, series solutions) and Chapter 11 (Sturm–Liouville problems).
- W. E. Boyce, R. C. DiPrima and D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 11th ed., Wiley — Chapters 2, 3, 7, 9 and 11; the standard reference for existence and uniqueness, phase-plane analysis and boundary-value problems.
- G. F. Simmons, *Differential Equations with Applications and Historical Notes*, 3rd ed., CRC Press — clear proofs of Picard's theorem, Sturm comparison and oscillation theorems.
- S. H. Strogatz, *Nonlinear Dynamics and Chaos*, 2nd ed., CRC Press — the geometric (phase-line and phase-plane) view of nonlinear systems, including oscillators and bifurcations.

Chapter 12 — Linear Differential Equations

> **Prerequisites:** Ch. 2 (Complex Numbers), Ch. 4 (Differential Calculus), Ch. 8 and Ch. 10 (Matrices, Eigenvalues), Ch. 11 (Ordinary Differential Equations)
>
> **Learning outcomes:** After this chapter you can
>
> - write a linear constant-coefficient ODE in operator form $P(D)y = r(t)$ and use linearity, superposition and time invariance to simplify it;
> - find the homogeneous solution from the roots of the characteristic polynomial in all cases (distinct, repeated, complex), and classify a second-order system as overdamped, critically damped, underdamped or undamped;
> - split a response into natural and forced parts, or into zero-input and zero-state parts, and fit initial conditions correctly;
> - find particular solutions by undetermined coefficients, the complex-exponential (phasor) method and the resonance rule;
> - separate transient from steady-state behaviour, compute frequency response, settling time and overshoot;
> - solve coupled linear equations by elimination and by modal decomposition, and write any linear lumped system as $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u$ with solution through the matrix exponential $e^{\mathbf{A}t}$.

Chapter 11 laid out the general theory of ordinary differential equations: the structure theorem $y = y_h + y_p$, the Wronskian, variation of parameters, existence and uniqueness, and the eigenvector method for first-order systems. This chapter specialises to the case that dominates electrical engineering: **linear equations with constant coefficients**. Every circuit built from ideal resistors, capacitors, inductors, coupled inductors, controlled sources and op-amps operating linearly is described by such equations; so are small-signal models of amplifiers, linearised motors, filters, control loops and transmission-line sections.

Constant coefficients change the mathematics fundamentally. Instead of searching for unknown functions, we search for **roots of a polynomial**. Calculus problems become algebra problems, and the algebra has a clear physical reading: the roots are the natural frequencies of the system, their real parts say how fast energy dissipates, and their imaginary parts say how fast it sloshes between storage elements. The same polynomial reappears as the denominator of the transfer function in Chapter 14 and as the characteristic polynomial of the state matrix in Section 12.7.

The chapter proceeds in eight sections. Sections 12.1–12.2 build the homogeneous theory. Sections 12.3–12.5 add forcing and interpret the result physically. Sections 12.6–12.7 move to several interacting unknowns and to the state-space form that modern analysis and simulation software uses. Section 12.8 pulls everything together on RLC circuits and an electromechanical system.

12.1 Constant-coefficient equations

Motivation

Write Kirchhoff's voltage law around a series loop containing a resistor R, an inductor L, a capacitor C and a source $v_s(t)$. Using the element laws $v_R = Ri$, $v_L = L\,di/dt$, $i = C\,dv_C/dt$:

$$LC\,\frac{d^2 v_C}{dt^2} + RC\,\frac{dv_C}{dt} + v_C = v_s(t).$$

The coefficients LC, RC and 1 are fixed by the component values; they do not depend on time or on v_C. That is the defining feature of the class studied here. The same form arises for a mass–spring–damper ($m\ddot{x} + b\dot{x} + kx = F$), a torsional shaft, a thermal network or the linearised dynamics of a phase-locked loop. Because the form is universal, a single toolkit solves all of them.

Definitions

Linear constant-coefficient ODE

A **linear ordinary differential equation of order n with constant coefficients** is

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_1 y' + a_0 y = r(t), \qquad a_n \neq 0,$$

where $a_0, \dots, a_n$ are real (occasionally complex) constants and $r(t)$ is a given **forcing function**, **input** or **excitation**. The equation is **homogeneous** if $r(t) \equiv 0$ and **nonhomogeneous** (or *forced*) otherwise. Dividing by a_n gives the **monic** form with $a_n = 1$.

Three words carry the content of the definition:

- *Linear*: y and its derivatives appear only to the first power and are never multiplied together or placed inside a nonlinear function.
- *Constant coefficient*: the a_k do not vary with t. Physically, the components do not age, heat up or switch during the interval considered.
- *Order n*: the highest derivative present. For lumped circuits n equals the number of independent energy-storage elements (capacitors and inductors that cannot be combined into one equivalent element).

The differential operator D

It is efficient to treat differentiation as an algebraic object. Let $D = d/dt$, so $Dy = y'$, $D^2 y = y''$ and in general $D^k y = y^{(k)}$, with $D^0 = 1$ (the identity).

Operator polynomial

For the polynomial $P(s) = a_n s^n + \cdots + a_1 s + a_0$, the **operator polynomial** $P(D)$ acts on a sufficiently differentiable function y by

$$P(D)\,y = a_n D^n y + \cdots + a_1 D y + a_0 y.$$

The equation of the definition above is then simply $P(D)\,y = r(t)$. The polynomial $P(s)$ is called the **characteristic polynomial** of the equation.

The key fact that makes D behave like a number is that constant-coefficient operators **commute** and **factor** exactly like the polynomials that define them.

Algebra of constant-coefficient operators

If $P(s)$ and $Q(s)$ are polynomials with constant coefficients, then for any sufficiently differentiable y

$$P(D)\big[Q(D)\,y\big] = Q(D)\big[P(D)\,y\big] = (PQ)(D)\,y.$$

In particular, if $P(s) = a_n(s - s_1)(s - s_2)\cdots(s - s_n)$ then $P(D) = a_n(D - s_1)(D - s_2)\cdots(D - s_n)$, with the factors in any order.

Proof sketch. By linearity it suffices to check monomials: $D^j(D^k y) = D^{j+k}y = D^k(D^j y)$. Constants pass through D because $D(cy) = cDy$ when c is constant. Expanding both sides term by term gives the same coefficients as polynomial multiplication. □

Common Pitfall

The commuting property fails as soon as a coefficient depends on t. For example $D(t\,y) = y + t\,Dy \neq t\,Dy$. So $(D + t)(D - t) \neq (D - t)(D + t)$, and the factor-and-solve techniques of this chapter do **not** apply to variable-coefficient equations such as Cauchy–Euler or Bessel equations (see Ch. 11).

Linearity and superposition

Superposition

$P(D)$ is a linear operator: for constants c_1, c_2 and functions y_1, y_2,

$$P(D)\,(c_1 y_1 + c_2 y_2) = c_1 P(D) y_1 + c_2 P(D) y_2.$$

Consequently:

1. any linear combination of solutions of the homogeneous equation $P(D)y = 0$ is again a solution;
2. if $P(D)y_1 = r_1$ and $P(D)y_2 = r_2$, then $P(D)(c_1 y_1 + c_2 y_2) = c_1 r_1 + c_2 r_2$;
3. the general solution of $P(D)y = r$ is $y = y_h + y_p$, where y_h is the general homogeneous solution (n arbitrary constants) and y_p is any one particular solution.

Item 3 is the structure theorem of Ch. 11; it is restated here because everything that follows is a recipe for finding y_h (Section 12.2) and y_p (Section 12.4) when coefficients are constant.

Time invariance

Constant coefficients bring a second structural property that variable-coefficient equations lack.

Time invariance (shift property)

If $y(t)$ solves $P(D)y = r(t)$, then for any fixed t_0 the shifted function $y(t - t_0)$ solves $P(D)y = r(t - t_0)$.

Proof. Let $w(t) = y(t - t_0)$. By the chain rule $w^{(k)}(t) = y^{(k)}(t - t_0)$, so $P(D)w(t) = \big[P(D)y\big](t - t_0) = r(t - t_0)$. The coefficients do not change under the shift because they are constants. □

Linearity plus time invariance is the defining pair of properties of an **LTI system** (linear time-invariant). Everything in signal processing that relies on convolution, transfer functions and frequency response depends on both. A circuit with a time-varying element (a switched capacitor, a mixer, a varactor driven by a pump) is linear but not time-invariant, and its equations do not have constant coefficients.

The exponential as eigenfunction

Why should exponentials be the building blocks? Apply D to e^{st}, where s is any complex constant:

$$D\,e^{st} = s\,e^{st}, \qquad D^k e^{st} = s^k e^{st}, \qquad P(D)\,e^{st} = P(s)\,e^{st}.$$

So e^{st} is an **eigenfunction** of every constant-coefficient operator, with eigenvalue $P(s)$. An LTI system therefore maps an exponential input into the same exponential, merely scaled by a complex number. This single identity drives the characteristic equation (Section 12.2), the phasor method (Section 12.4), frequency response (Section 12.5) and, after integration against e^{-st}, the Laplace transform (Ch. 14).

A useful generalisation handles exponentials multiplied by other functions.

Exponential shift theorem

For any constant a and sufficiently differentiable f,

$$P(D)\left[e^{at}f(t)\right] = e^{at}\,P(D+a)\,f(t).$$

Proof. For one factor, $D\left[e^{at}f\right] = ae^{at}f + e^{at}f' = e^{at}(D+a)f$. Applying this k times gives $D^k\left[e^{at}f\right] = e^{at}(D+a)^k f$, and summing with coefficients a_k gives the result. □

The shift theorem lets us "pull an exponential through" an operator. It explains at once why repeated roots produce factors of t (Section 12.2) and why resonant forcing produces $t\,e^{at}$ terms (Section 12.4).

Real solutions from complex ones

When the coefficients a_k are real and $y = u + jv$ is a complex-valued solution of $P(D)y = r$ with r real, then taking real and imaginary parts gives $P(D)u = r$ and $P(D)v = 0$. More generally:

Real and imaginary parts

If the coefficients of P are real and $P(D)z = w(t)$ with z, w complex, then

$$P(D)\ \mathrm{Re}\{z\} = \mathrm{Re}\{w\}, \qquad P(D)\ \mathrm{Im}\{z\} = \mathrm{Im}\{w\}.$$

This is the justification for solving with complex exponentials $e^{j\omega t}$ and then taking real parts, which is precisely what phasor analysis does.

Worked examples

Example 12.1 — First-order RC charging

A capacitor C charges through a resistor R from a DC source V_0 switched on at $t = 0$; the capacitor is initially uncharged. Find $v_C(t)$.

Model. KVL gives $Ri + v_C = V_0$ and $i = C\,dv_C/dt$, so

$$RC\,\frac{dv_C}{dt} + v_C = V_0, \qquad v_C(0) = 0.$$

With $\tau = RC$ (units: $\Omega \cdot \mathrm{F} = \mathrm{s}$) the operator form is $(\tau D + 1)v_C = V_0$, with characteristic polynomial $P(s) = \tau s + 1$ and root $s_1 = -1/\tau$.

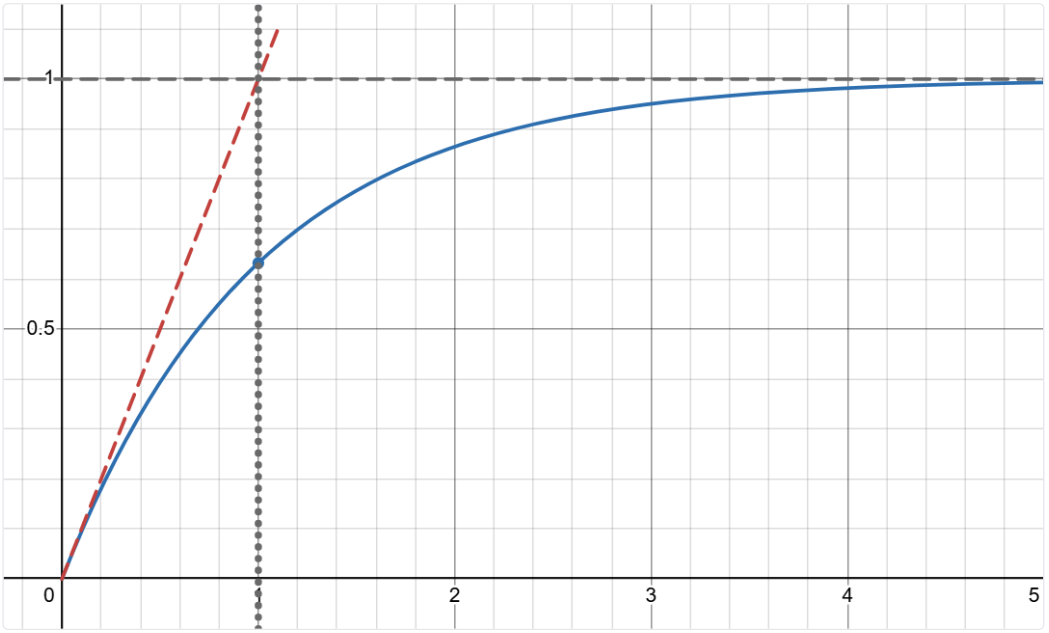
Homogeneous part. $v_h = Ae^{-t/\tau}$.

Particular part. The input is constant, so try a constant $v_p = K$: then $0 + K = V_0$, $K = V_0$.

Initial condition. $v_C = V_0 + Ae^{-t/\tau}$; $v_C(0) = V_0 + A = 0$ gives $A = -V_0$:

$$v_C(t) = V_0\left(1 - e^{-t/\tau}\right), \qquad t \ge 0.$$

Check. At $t = \tau$, $v_C = V_0(1 - e^{-1}) = 0.632\,V_0$; at $t = 5\tau$, $v_C = 0.9933\,V_0$. Substituting: $\tau \cdot (V_0/\tau)e^{-t/\tau} + V_0 - V_0e^{-t/\tau} = V_0$. ✓ With $R = 10\ \mathrm{k\Omega}$, $C = 100\ \mathrm{nF}$, $\tau = 1\ \mathrm{ms}$, the capacitor is within 1% of V_0 after 4.6 ms (since $e^{-4.6} = 0.010$).



Normalised RC charging curve v_C/V_0 against t/τ; the initial tangent (red) reaches the final value exactly at $t = \tau$.

Example 12.2 — Deriving and factoring a second-order operator

Show that $y'' + y' - 2y = 0$ factors as $(D-1)(D+2)y = 0$ and verify directly that e^{t} and e^{-2t} are solutions.

Factoring. $P(s) = s^2 + s - 2 = (s-1)(s+2)$, so by the operator algebra theorem $P(D) = (D-1)(D+2) = (D+2)(D-1)$.

Check by expansion. $(D-1)(D+2)y = (D-1)(y' + 2y) = y'' + 2y' - y' - 2y = y'' + y' - 2y$. ✓

Solutions. $(D-1)e^{t} = e^{t} - e^{t} = 0$, so $(D+2)(D-1)e^{t} = (D+2)\,0 = 0$. Likewise $(D+2)e^{-2t} = 0$, so $(D-1)(D+2)e^{-2t} = 0$. Alternatively, using the eigenfunction identity, $P(D)e^{st} = P(s)e^{st}$ vanishes exactly when $P(s) = 0$, i.e. $s = 1$ or $s = -2$.

The general solution is $y = c_1 e^{t} + c_2 e^{-2t}$; the e^{t} term grows, so this system is unstable.

Example 12.3 — Using the exponential shift theorem

Evaluate $P(D)\left[te^{-3t}\right]$ for $P(D) = D^2 + 6D + 9$, and interpret.

By the shift theorem with $a = -3$, $f = t$:

$$P(D)\left[e^{-3t}t\right] = e^{-3t}P(D-3)\,t = e^{-3t}\left[(D-3)^2 + 6(D-3) + 9\right]t = e^{-3t}D^2 t = 0.$$

The operator $(D-3)^2 + 6(D-3) + 9$ simplifies to D^2 because $P(s) = (s+3)^2$, so $P(s-3) = s^2$. Hence te^{-3t} is a solution of $y'' + 6y' + 9y = 0$, alongside e^{-3t}. This is the repeated-root case of Section 12.2, obtained here without any guesswork.

Direct check. $y = te^{-3t}$, $y' = (1-3t)e^{-3t}$, $y'' = (-6+9t)e^{-3t}$. Then $y'' + 6y' + 9y = (-6 + 9t + 6 - 18t + 9t)e^{-3t} = 0$. ✓

Example 12.4 — Series RLC in operator form

For the series RLC loop of the motivation with $R = 100\ \Omega$, $L = 10$ mH, $C = 1\ \mu$F, write the equation for v_C in monic operator form and find its characteristic polynomial.

Divide $LC\,v_C'' + RC\,v_C' + v_C = v_s$ by $LC = 10^{-8}\ \text{s}^2$:

$$\left(D^2 + \frac{R}{L}D + \frac{1}{LC}\right)v_C = \frac{1}{LC}v_s, \qquad P(s) = s^2 + 10^4 s + 10^8.$$

Dimensional check: R/L has units $\Omega/\text{H} = \text{s}^{-1}$ and $1/LC$ has units s^{-2}, matching D and D^2. The roots, $s = -5000 \pm j\,8660\ \text{s}^{-1}$, are found in Section 12.2 and the full step response in Section 12.8.

Engineering Note

Circuit simulators (SPICE) never form the single nth-order equation; they assemble first-order equations for every node and storage element (modified nodal analysis), which is the state-space viewpoint of Section 12.7. Hand analysis, by contrast, almost always reduces to one equation $P(D)y = r$, because the polynomial $P(s)$ is where the design insight lives: its roots set rise time, ringing and stability.

12.2 Characteristic equations

Motivation

Section 12.1 showed that e^{st} solves $P(D)y = 0$ whenever $P(s) = 0$. Solving a homogeneous constant-coefficient ODE therefore reduces to finding the n roots of a degree-n polynomial and handling two complications: roots may be **repeated**, and roots of a real polynomial may be **complex**. This section treats both and then studies the second-order case in detail, because nearly every practical design specification (damping, overshoot, ringing frequency, quality factor) is phrased in terms of a dominant pair of roots.

The characteristic equation

Characteristic equation and natural frequencies

For $P(D)y = 0$ with $P(s) = a_n s^n + \cdots + a_0$, the algebraic equation

$$P(s) = a_n s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0 = 0$$

is the **characteristic (auxiliary) equation**. Its roots $s_1, \ldots, s_n$ (counted with multiplicity) are the **characteristic roots**, also called the **natural frequencies**, **modes** or **poles** of the system. They have units of s^{-1} when t is in seconds.

One obtains the characteristic equation mechanically by replacing $y^{(k)}$ with s^k. By the fundamental theorem of algebra, a degree-n polynomial has exactly n complex roots counted with multiplicity, and when the coefficients are real, non-real roots occur in **conjugate pairs** $\sigma \pm j\omega$.

Case 1: distinct real roots

Distinct roots

If $P(s)$ has n distinct roots $s_1, \dots, s_n$, then $e^{s_1 t}, \dots, e^{s_n t}$ form a fundamental set of solutions of $P(D)y = 0$, and the general solution is

$$y_h(t) = c_1 e^{s_1 t} + c_2 e^{s_2 t} + \cdots + c_n e^{s_n t}.$$

Proof sketch. Each is a solution by the eigenfunction identity. Independence follows from the Wronskian at $t = 0$, which is the Vandermonde determinant $\prod_{i<k}(s_k - s_i) \neq 0$ for distinct roots (Ch. 8). By the theory of Ch. 11, n independent solutions of an nth-order linear equation span all solutions. □

The theorem holds for distinct complex roots too; Case 3 converts the complex exponentials into real functions.

Case 2: repeated roots

If s_1 is a root of multiplicity m, the list of exponentials contains $e^{s_1 t}$ only once, so we are $m - 1$ solutions short. The exponential shift theorem supplies them.

Repeated roots

If s_1 is a root of $P(s)$ of multiplicity m, then

$$e^{s_1 t},\ t e^{s_1 t},\ t^2 e^{s_1 t},\ \dots,\ t^{m-1} e^{s_1 t}$$

are m linearly independent solutions of $P(D)y = 0$. Collecting such blocks over all distinct roots gives a fundamental set of n solutions.

Derivation. Write $P(s) = (s - s_1)^m Q(s)$. Then $P(D) = Q(D)(D - s_1)^m$, and by the shift theorem

$$(D - s_1)^m \left[e^{s_1 t} t^k\right] = e^{s_1 t}\left(D + s_1 - s_1\right)^m t^k = e^{s_1 t} D^m t^k = 0 \quad \text{for } k < m,$$

because the mth derivative of a polynomial of degree $k < m$ vanishes. Applying $Q(D)$ to zero still gives zero. Independence: a relation $\sum_k c_k t^k e^{s_1 t} = 0$ for all t forces the polynomial $\sum c_k t^k$ to vanish identically, so all $c_k = 0$; mixing different roots is handled by the generalised Vandermonde argument. □

There is a second, intuitive derivation. For a double root, consider two nearby distinct roots s_1 and $s_1 + \varepsilon$. The combination $\left(e^{(s_1+\varepsilon)t} - e^{s_1 t}\right)/\varepsilon$ is a solution for every $\varepsilon \neq 0$, and as $\varepsilon \to 0$ it tends to $\partial e^{st}/\partial s\big|_{s_1} = t e^{s_1 t}$. Repeated roots are the limit of merging modes, and the factor t records the merger.

Case 3: complex-conjugate roots

For real coefficients, a complex root $s = \sigma + j\omega$ ($\omega > 0$) comes paired with $\bar{s} = \sigma - j\omega$. The complex solutions $e^{(\sigma \pm j\omega)t}$ are legitimate but inconvenient. Using Euler's formula (Ch. 2),

$$e^{(\sigma \pm j\omega)t} = e^{\sigma t}(\cos \omega t \pm j \sin \omega t),$$

and by the real/imaginary-parts theorem of Section 12.1, the real and imaginary parts are themselves real solutions.

Complex roots

If $\sigma \pm j\omega$ is a pair of simple complex-conjugate roots of a real polynomial $P(s)$, the corresponding contribution to y_h is

$$e^{\sigma t}\left(A \cos \omega t + B \sin \omega t\right) = M e^{\sigma t} \cos(\omega t - \varphi),$$

with real constants A, B, or equivalently amplitude $M = \sqrt{A^2 + B^2}$ and phase $\varphi = \operatorname{atan2}(B, A)$. If the pair has multiplicity m, multiply by $1, t, \dots, t^{m-1}$ as in Case 2.

The real part σ sets the **envelope** $e^{\sigma t}$ (decay if $\sigma < 0$, growth if $\sigma > 0$), and the imaginary part ω sets the **oscillation frequency** in rad/s. A root located in the complex s-plane can be read at a glance: distance to the left of the imaginary axis is decay rate, height above the real axis is ringing frequency.

Common Pitfall

Writing the complex-root solution as $c_1 e^{\sigma t} \cos \omega t + c_2 e^{\sigma t} \cos \omega t$ (the same function twice) or forgetting the $e^{\sigma t}$ envelope are the two most common errors. Each conjugate *pair* contributes exactly two real functions, $e^{\sigma t} \cos \omega t$ and $e^{\sigma t} \sin \omega t$. Also note: one pair accounts for two roots, so a cubic with one complex pair has one more real root to find.

Summary of the root-to-solution map

Root(s) of $P(s)$	Contribution to y_h
real s_1, simple	$c_1 e^{s_1 t}$
real s_1, multiplicity m	$(c_1 + c_2 t + \cdots + c_m t^{m-1}) e^{s_1 t}$
$\sigma \pm j\omega$, simple	$e^{\sigma t}(A \cos \omega t + B \sin \omega t)$
$\sigma \pm j\omega$, multiplicity m	$e^{\sigma t}\left[(A_1 + \cdots + A_m t^{m-1}) \cos \omega t + (B_1 + \cdots + B_m t^{m-1}) \sin \omega t\right]$
$s = 0$, multiplicity m	$c_1 + c_2 t + \cdots + c_m t^{m-1}$
$\pm j\omega$ (purely imaginary)	$A \cos \omega t + B \sin \omega t$ (sustained oscillation)

Stability from the roots

Stability of a homogeneous LTI equation

The equation $P(D)y = 0$ is

- **asymptotically stable** if every solution tends to 0 as $t \to \infty$;
- **marginally stable** if every solution stays bounded but some do not decay;
- **unstable** if some solution is unbounded.

Root criterion

$P(D)y = 0$ is asymptotically stable if and only if every characteristic root satisfies $\operatorname{Re}\{s_k\} < 0$ (all roots in the open left half-plane). It is marginally stable if all roots satisfy $\operatorname{Re}\{s_k\} \le 0$ and those on the imaginary axis are simple. Otherwise it is unstable.

Reason. $|t^k e^{st}| = t^k e^{\sigma t}$, which tends to zero for any k when $\sigma < 0$ (exponential decay beats polynomial growth), stays bounded when $\sigma = 0$ and $k = 0$, and grows otherwise. $\square$

For a real polynomial of degree two, stability is equivalent to all coefficients having the same sign. For degree three, $s^3 + a_2 s^2 + a_1 s + a_0$ is stable if and only if $a_2, a_1, a_0 > 0$ and $a_2 a_1 > a_0$. For higher degrees, the **Routh–Hurwitz criterion** tests stability from the coefficients without computing roots; positive coefficients are necessary but not sufficient beyond degree two. Detailed Routh tables belong to control theory; the point here is that stability is a property of the characteristic polynomial alone.

The standard second-order form

Every stable second-order equation with real coefficients can be written

$$y'' + 2\zeta\omega_n\, y' + \omega_n^2\, y = \omega_n^2\, K\, u(t),$$

where

- $\omega_n > 0$ is the **undamped natural frequency** (rad/s),
- $\zeta \ge 0$ is the **damping ratio** (dimensionless),
- K is the DC gain (irrelevant for the homogeneous problem).

The characteristic equation $s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$ has roots

$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}.$$

The product of the roots is ω_n^2, so both roots lie on (or, when real, on either side of) a circle of radius ω_n in the s-plane; the sum is $-2\zeta\omega_n$. The damping ratio alone decides the character of the response.

Damping cases

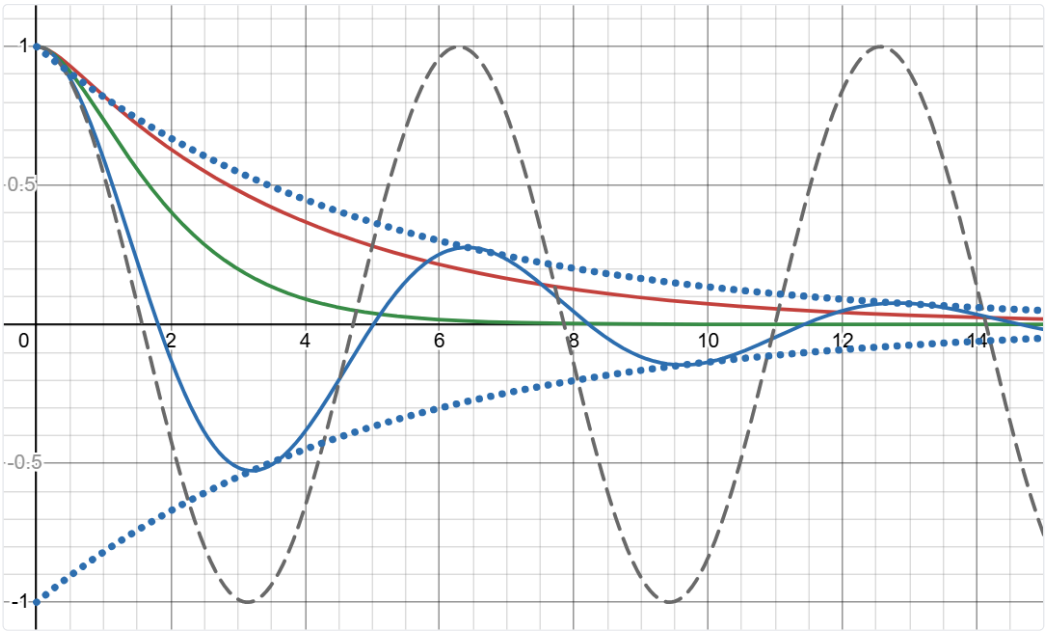
Case	Condition	Roots	Homogeneous solution
Overdamped	$\zeta > 1$	two distinct negative reals $-\alpha_1, -\alpha_2$	$c_1 e^{-\alpha_1 t} + c_2 e^{-\alpha_2 t}$
Critically damped	$\zeta = 1$	double root $-\omega_n$	$(c_1 + c_2 t)e^{-\omega_n t}$
Underdamped	$0 < \zeta < 1$	$-\alpha \pm j\omega_d$	$e^{-\alpha t}(A\cos\omega_d t + B\sin\omega_d t)$
Undamped	$\zeta = 0$	$\pm j\omega_n$	$A\cos\omega_n t + B\sin\omega_n t$

In the underdamped case, $\alpha = \zeta\omega_n$ is the **decay rate** (neper frequency) and $\omega_d = \omega_n\sqrt{1-\zeta^2}$ is the **damped natural frequency**. Note $\alpha^2 + \omega_d^2 = \omega_n^2$.

The geometric picture is useful. As ζ increases from 0, the root pair starts on the imaginary axis at $\pm j\omega_n$, travels along the circle of radius ω_n into the left half-plane, meets on the real axis at $-\omega_n$ when $\zeta = 1$, and then splits along the real axis, one root heading towards 0 and the other towards $-\infty$. The angle θ between a complex root and the negative real axis satisfies $\cos\theta = \zeta$.

Consider the response from $y(0) = 1$, $y'(0) = 0$ with $\omega_n = 1$:

- $\zeta = 2$: roots $-2 \pm \sqrt{3} = -0.2679,\ -3.7321$; $y = 1.0774\,e^{-0.2679t} - 0.0774\,e^{-3.7321t}$.
- $\zeta = 1$: $y = (1 + t)e^{-t}$.
- $\zeta = 0.2$: $\omega_d = 0.9798$; $y = e^{-0.2t}\big(\cos 0.9798t + 0.2041\sin 0.9798t\big)$.
- $\zeta = 0$: $y = \cos t$.



Free response from $y(0) = 1$, $y'(0) = 0$ with $\omega_n = 1$: overdamped $\zeta = 2$ (red), critically damped $\zeta = 1$ (green), underdamped $\zeta = 0.2$ (blue, with its $e^{-0.2t}$ envelope dotted) and undamped $\zeta = 0$ (grey). Critical damping returns to zero fastest without crossing it.

Three features of the graph are worth stating as facts:

1. **Critical damping is the fastest non-oscillatory return.** The overdamped response is dominated by its slow root -0.268; increasing damping beyond $\zeta = 1$ actually slows the system down, because the slow root moves towards the origin ($s_1 \approx -\omega_n/(2\zeta)$ for large ζ).
2. **An underdamped response crosses zero infinitely often**, at intervals of π/ω_d, while its envelope decays with time constant $1/\alpha$.
3. **An overdamped or critically damped response crosses zero at most once.** Since $c_1 e^{-\alpha_1 t} + c_2 e^{-\alpha_2 t} = 0$ means $e^{(\alpha_2 - \alpha_1)t} = -c_2/c_1$, there is at most one solution t.

Fitting initial conditions

The n constants in y_h are fixed by n initial conditions $y(0), y'(0), \dots, y^{(n-1)}(0)$. For distinct roots this gives a Vandermonde system:

$$\begin{bmatrix} 1 & 1 & \cdots & 1 \\ s_1 & s_2 & \cdots & s_n \\ \vdots & & & \vdots \\ s_1^{n-1} & s_2^{n-1} & \cdots & s_n^{n-1} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} y(0) \\ y'(0) \\ \vdots \\ y^{(n-1)}(0) \end{bmatrix}.$$

For second order with distinct roots and data $y_0 = y(0)$, $v_0 = y'(0)$ the solution is

$$c_1 = \frac{v_0 - s_2 y_0}{s_1 - s_2}, \qquad c_2 = \frac{s_1 y_0 - v_0}{s_1 - s_2}.$$

For the underdamped case, $y = e^{-\alpha t}(A \cos \omega_d t + B \sin \omega_d t)$ gives $A = y_0$ and $-\alpha A + \omega_d B = v_0$, so

$$A = y_0, \qquad B = \frac{v_0 + \alpha y_0}{\omega_d}.$$

For the critically damped case $y = (c_1 + c_2 t) e^{-\omega_n t}$: $c_1 = y_0$, $c_2 = v_0 + \omega_n y_0$.

Worked examples

Example 12.5 — Overdamped: distinct real roots

Solve $y'' + 5y' + 6y = 0$, $y(0) = 1$, $y'(0) = 0$.

Characteristic equation. $s^2 + 5s + 6 = (s+2)(s+3) = 0$, roots $s_1 = -2$, $s_2 = -3$. (Here $\omega_n = \sqrt{6} = 2.449$, $\zeta = 5/(2\sqrt{6}) = 1.021$: just overdamped.)

Constants. $c_1 = (v_0 - s_2 y_0)/(s_1 - s_2) = (0 + 3)/1 = 3$, $c_2 = (s_1 y_0 - v_0)/(s_1 - s_2) = -2$.

$$y(t) = 3e^{-2t} - 2e^{-3t}.$$

Check. $y(0) = 1$ ✓, $y'(0) = -6 + 6 = 0$ ✓. Substituting: $y'' = 12e^{-2t} - 18e^{-3t}$, $5y' = -30e^{-2t} + 30e^{-3t}$, $6y = 18e^{-2t} - 12e^{-3t}$; sum $= 0$ ✓. At $t = 1$: $y = 3(0.13534) - 2(0.04979) = 0.3064$.

Example 12.6 — Critically damped: repeated root

Solve $y'' + 4y' + 4y = 0$, $y(0) = 1$, $y'(0) = 0$.

$s^2 + 4s + 4 = (s+2)^2$: double root $s = -2$, so $y = (c_1 + c_2 t)e^{-2t}$. Then $c_1 = 1$ and $c_2 = v_0 + \omega_n y_0 = 0 + 2 = 2$:

$$y(t) = (1 + 2t)e^{-2t}.$$

Check. $y' = 2e^{-2t} - 2(1+2t)e^{-2t} = -4te^{-2t}$, so $y'(0) = 0$ ✓; $y'' = (-4 + 8t)e^{-2t}$; $y'' + 4y' + 4y = (-4 + 8t - 16t + 4 + 8t)e^{-2t} = 0$ ✓. Notice $y' < 0$ for all $t > 0$: the response decays monotonically without overshoot.

Example 12.7 — Underdamped: complex roots

Solve $y'' + 2y' + 10y = 0$, $y(0) = 1$, $y'(0) = 0$, and express the result in amplitude–phase form.

$s^2 + 2s + 10 = 0$ gives $s = -1 \pm \sqrt{1 - 10} = -1 \pm j3$. So $\alpha = 1$, $\omega_d = 3$, $\omega_n = \sqrt{10} = 3.162$, $\zeta = 1/\sqrt{10} = 0.316$.

$A = y_0 = 1$, $B = (v_0 + \alpha y_0)/\omega_d = 1/3$:

$$y(t) = e^{-t}\left(\cos 3t + \tfrac{1}{3}\sin 3t\right) = \frac{\sqrt{10}}{3}\,e^{-t}\cos(3t - 0.3218).$$

Here $M = \sqrt{1 + 1/9} = 1.0541$ and $\varphi = \arctan(1/3) = 0.3218$ rad $= 18.43°$.

Check. $y'(0) = -1 \cdot 1 + 3 \cdot \tfrac{1}{3} = 0$ ✓. The oscillation period is $2\pi/3 = 2.094$ s and the envelope falls by a factor $e^{-2.094} = 0.123$ per period; this ratio, $e^{-2\pi\zeta/\sqrt{1-\zeta^2}}$, is the basis of the **logarithmic decrement** method of measuring damping from a recorded ring-down.

Example 12.8 — Higher order with mixed roots

Find the general solution of $y^{(4)} + 2y'' + y = 0$ and of $y''' - y'' + y' - y = 0$.

First equation. $s^4 + 2s^2 + 1 = (s^2+1)^2$, so $\pm j$ are double roots. The table gives

$$y = (A_1 + A_2 t)\cos t + (B_1 + B_2 t)\sin t.$$

The $t\cos t$ and $t\sin t$ terms grow without bound, so the system is unstable even though all roots lie on the imaginary axis: repeated imaginary-axis roots violate the marginal-stability condition.

Second equation. Factor by grouping: $s^3 - s^2 + s - 1 = s^2(s-1) + (s-1) = (s-1)(s^2+1)$. Roots $1, \pm j$:

$$y = c_1 e^{t} + A\cos t + B\sin t.$$

Check of factorisation. $(s-1)(s^2+1) = s^3 + s - s^2 - 1$ ✓.

Example 12.9 — Stability of a cubic without computing roots

A third-order active filter has characteristic polynomial $s^3 + 2s^2 + 3s + K$, where $K > 0$ is a gain. For what K is it stable? What happens at the boundary?

By the cubic criterion, stability requires $2 \cdot 3 > K$, i.e. $0 < K < 6$. At $K = 6$: $s^3 + 2s^2 + 3s + 6 = s^2(s+2) + 3(s+2) = (s+2)(s^2+3)$, with roots -2 and $\pm j\sqrt{3}$. The homogeneous solution contains $A\cos\sqrt{3}\,t + B\sin\sqrt{3}\,t$, an undamped oscillation at $\sqrt{3} = 1.732$ rad/s. This is exactly how an oscillator is designed: the loop gain is set so that a root pair sits on the imaginary axis. For $K > 6$ the pair crosses into the right half-plane and the oscillation grows until nonlinearity limits it.

Engineering Note

Designers frequently specify ζ rather than component values. $\zeta = 1/\sqrt{2} \approx 0.707$ gives the second-order Butterworth response (maximally flat frequency response, about 4.3% step overshoot); $\zeta = 1$ is used where overshoot is forbidden, such as in a precision positioning stage or an ADC driver that must not exceed the converter's input range; $\zeta \approx 0.5$–0.6 is common in control loops that trade some overshoot for speed. A probe ringing on an oscilloscope is a lightly damped pair formed by ground-lead inductance and input capacitance; the fix (a shorter ground lead or a damping resistor) raises ζ.

12.3 Natural and forced responses

Motivation

When a source drives a circuit, the response always contains two ingredients with completely different origins. One is dictated by the *circuit*: the exponentials $e^{s_k t}$ whose rates are the characteristic roots, the same no matter what drives it. The other is dictated by the *source*: a DC input produces a DC component, a sinusoid produces a sinusoid at the same frequency. Engineers call these the **natural response** and the **forced response**. Recognising which is which tells you immediately which part of a waveform you can change by redesigning the circuit and which part only by changing the input.

There is a second, independent way to split the same response, according to *cause*: the part due to energy initially stored in capacitors and inductors, and the part due to the source. Both splittings are useful, they are not the same, and confusing them is a frequent source of errors.

Natural and forced decomposition

Natural and forced response

For $P(D)y = r(t)$ with general solution $y = y_h + y_p$:

- the **natural response** $y_n(t)$ is the homogeneous part, a combination of the characteristic modes $t^k e^{s_k t}$ with constants fixed by the initial conditions;
- the **forced response** $y_f(t)$ is a particular solution chosen to have the same functional form as the input (constant for constant input, sinusoid for sinusoid, and so on), containing **no** natural modes unless resonance forces them in (Section 12.4).

Then $y = y_n + y_f$.

The forced response is sometimes defined loosely as "any particular solution". The convention adopted here, that y_f is the particular solution built from the input's own functional form, makes y_f unique whenever the input's exponent does not coincide with a characteristic root. For example, for $y' + 2y = 4$ both $y_p = 2$ and $y_p = 2 + 7e^{-2t}$ are particular solutions, but only $y_f = 2$ is the forced response.

Procedure for the complete response

1. Find the characteristic roots and write the natural response with n undetermined constants.
2. Find the forced response y_f (Section 12.4).
3. Write $y = y_n + y_f$ and impose the n initial conditions **on the complete** y, not on y_n alone.

Common Pitfall

The constants in y_n must be fitted after adding y_f. Fitting y_n to the initial conditions first and then adding y_f gives a function that satisfies the ODE but violates the initial conditions, unless y_f and its derivatives happen to vanish at $t = 0$.

Zero-input and zero-state decomposition

Zero-input and zero-state response

- The **zero-input response** y_{zi} solves $P(D)y = 0$ with the actual initial conditions: the response to stored energy alone, with the source switched off.
- The **zero-state response** y_{zs} solves $P(D)y = r(t)$ with **all initial conditions zero**: the response to the source alone, starting from rest.

By linearity $y = y_{zi} + y_{zs}$.

Relation between the two decompositions

The zero-input response consists purely of natural modes. The zero-state response contains the whole forced response plus some natural modes:

$$y_{zs} = y_f + y_{n,\text{zs}}, \qquad y_n = y_{zi} + y_{n,\text{zs}}.$$

Thus the natural response is generally **not** the zero-input response; the source itself excites the natural modes when it is switched on.

The zero-state response is the one that a transfer function or a convolution integral computes: it is linear in the input, and doubling the input doubles it. The complete response is **not** linear in the input when initial conditions are nonzero; it is *affine* (y_{zi} is an offset that does not scale). That is why "linear system" in signal-processing usage tacitly assumes the system starts at rest.

The impulse response

A special natural response deserves a name. Let $h(t)$ be the zero-state response to a unit impulse $r(t) = \delta(t)$. For $t > 0$ the input is zero, so h is purely natural; the impulse merely sets its initial conditions.

Impulse response of $P(D)y = \delta(t)$

With $P(s) = a_n s^n + \cdots + a_0$, the impulse response is the solution of the homogeneous equation for $t > 0$ with

$$h(0^+) = h'(0^+) = \cdots = h^{(n-2)}(0^+) = 0, \qquad h^{(n-1)}(0^+) = \frac{1}{a_n},$$

and $h(t) = 0$ for $t < 0$. The zero-state response to any input applied at $t = 0$ is then the convolution

$$y_{zs}(t) = \int_0^t h(t - \tau)\, r(\tau)\, d\tau.$$

Derivation sketch. Integrate $a_n y^{(n)} + \cdots + a_0 y = \delta(t)$ from 0^- to 0^+. If y and its first $n-2$ derivatives are continuous and $y^{(n-1)}$ has a jump, only the highest term picks up the integral of δ, giving $a_n\left[y^{(n-1)}(0^+) - 0\right] = 1$. The lower-order terms integrate to zero over the vanishing interval because they are bounded. The convolution formula follows from writing $r(t) = \int r(\tau)\delta(t-\tau)\,d\tau$ and using linearity and time invariance: each slice $r(\tau)\,d\tau\,\delta(t-\tau)$ produces $r(\tau)\,d\tau\,h(t-\tau)$, and summing the slices gives the integral. □

The impulse response is the bridge to Ch. 14 (where $\mathcal{L}\{h\} = 1/P(s)$) and to Ch. 16 (where its Fourier transform is the frequency response). The convolution theorem is developed there; here it is enough that the zero-state response is fully determined by h.

Worked examples

Example 12.10 — First order: both decompositions

Solve $y' + 2y = 4$, $y(0) = 5$, and give the natural/forced and zero-input/zero-state splits.

Natural/forced. Root $s = -2$: $y_n = Ae^{-2t}$. Forced: constant $y_f = K$ with $2K = 4$, $K = 2$. Then $y = 2 + Ae^{-2t}$ and $y(0) = 5$ gives $A = 3$:

$$y = \underbrace{2}_{y_f} + \underbrace{3e^{-2t}}_{y_n}.$$

Zero-input/zero-state. y_{zi}: $y' + 2y = 0$, $y(0) = 5$, so $y_{zi} = 5e^{-2t}$. y_{zs}: $y' + 2y = 4$, $y(0) = 0$, so $y_{zs} = 2(1 - e^{-2t})$. Sum: $5e^{-2t} + 2 - 2e^{-2t} = 2 + 3e^{-2t}$ ✓.

The natural response $3e^{-2t}$ is the zero-input response $5e^{-2t}$ plus the natural part of the zero-state response, $-2e^{-2t}$, exactly as the relation theorem says.

Example 12.11 — Second-order step from rest

Solve $y'' + 3y' + 2y = 10$, $y(0) = 0$, $y'(0) = 0$.

Natural. $s^2 + 3s + 2 = (s + 1)(s + 2)$: $y_n = c_1 e^{-t} + c_2 e^{-2t}$.

Forced. $y_f = K$, $2K = 10$, $K = 5$.

Initial conditions on the complete response. $y = 5 + c_1 e^{-t} + c_2 e^{-2t}$:

$$\begin{aligned} y(0) &= 5 + c_1 + c_2 = 0, \\ y'(0) &= -c_1 - 2c_2 = 0, \end{aligned} \quad \Rightarrow \quad c_1 = -10,\ c_2 = 5.$$

$$y(t) = 5 - 10e^{-t} + 5e^{-2t}.$$

Check. $y(0) = 0$ ✓, $y'(0) = 10 - 10 = 0$ ✓; numerically $y(1.5) = 5 - 2.2313 + 0.2489 = 3.0176$, matching a numerical ODE solver to 10 digits. Since the system starts at rest, this complete response *is* the zero-state response.

Example 12.12 — Same system, stored energy and source together

Solve $y'' + 3y' + 2y = 10$ with $y(0) = 2$, $y'(0) = 1$, and decompose it.

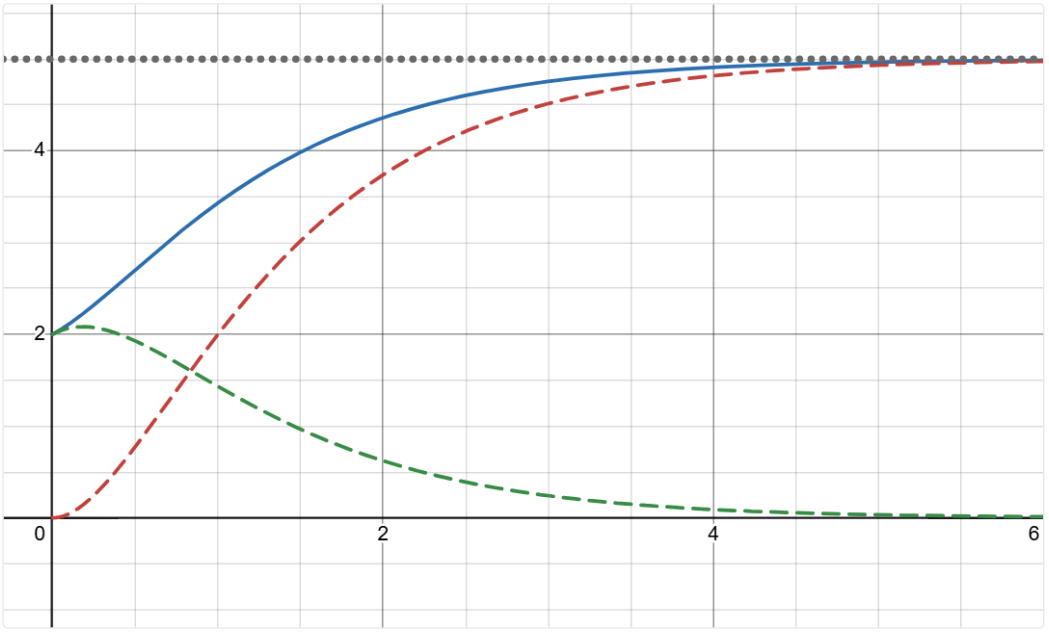
Complete response. $y = 5 + c_1 e^{-t} + c_2 e^{-2t}$; $5 + c_1 + c_2 = 2$ and $-c_1 - 2c_2 = 1$ give $c_2 = 2$, $c_1 = -5$:

$$y = 5 - 5e^{-t} + 2e^{-2t}, \qquad y_f = 5, \quad y_n = -5e^{-t} + 2e^{-2t}.$$

Zero-input. $c_1 + c_2 = 2$, $-c_1 - 2c_2 = 1$: $c_2 = -3$, $c_1 = 5$, so $y_{zi} = 5e^{-t} - 3e^{-2t}$.

Zero-state. From Example 12.11, $y_{zs} = 5 - 10e^{-t} + 5e^{-2t}$.

Consistency. $y_{zi} + y_{zs} = 5 - 5e^{-t} + 2e^{-2t}$ ✓, and $y(1.5) = 3.9839$ agrees with a numerical solver. If the source were doubled to 20, y_{zs} would double but y_{zi} would not, so the complete response would become $10 - 15e^{-t} + 7e^{-2t}$, not twice the original.



Example 12.12: the complete response (blue) is the sum of the zero-input response (green, decays to zero) and the zero-state response (red, rises from rest to the forced value 5).

Example 12.13 — Impulse response and convolution

Find the impulse response of $y'' + 3y' + 2y = r(t)$ and use convolution to recover the unit-step response.

Impulse response. For $t > 0$, $h = c_1 e^{-t} + c_2 e^{-2t}$ with $h(0^+) = 0$, $h'(0^+) = 1/a_2 = 1$: $c_1 + c_2 = 0$, $-c_1 - 2c_2 = 1$, so $c_1 = 1$, $c_2 = -1$:

$$h(t) = \left(e^{-t} - e^{-2t}\right) u(t).$$

Step response by convolution. With $r = u(t)$,

$$y_{zs}(t) = \int_0^t \left(e^{-(t-\tau)} - e^{-2(t-\tau)}\right) d\tau = \left(1 - e^{-t}\right) - \tfrac{1}{2}\left(1 - e^{-2t}\right) = \tfrac{1}{2} - e^{-t} + \tfrac{1}{2}e^{-2t}.$$

Multiplying by 10 reproduces Example 12.11 ✓. Note also that the step response is the integral of the impulse response, and h is the derivative of the step response: the derivative of $\frac{1}{2} - e^{-t} + \frac{1}{2}e^{-2t}$ is $e^{-t} - e^{-2t}$ ✓. Because $u' = \delta$, linearity and time invariance force the two responses to be related by the same derivative.

Example 12.14 — Natural response excited by switching a sinusoid

An RL branch obeys $L\,i' + R\,i = V_m \cos \omega t$ for $t \geq 0$ with $i(0) = 0$. Show that a natural component appears unless the switching instant is chosen carefully.

With $\tau = L/R$, the forced response (Section 12.4) is $i_f = I_m \cos(\omega t - \theta)$ with $I_m = V_m/\sqrt{R^2 + \omega^2 L^2}$ and $\theta = \arctan(\omega L/R)$. The natural response is $A e^{-t/\tau}$, and $i(0) = 0$ gives $A = -I_m \cos\theta$:

$$i(t) = I_m\left[\cos(\omega t - \theta) - \cos\theta\, e^{-t/\tau}\right].$$

If instead the source is switched on at a phase ψ, i.e. $V_m \cos(\omega t + \psi)$, then $A = -I_m \cos(\psi - \theta)$, which vanishes when $\psi = \theta \pm \pi/2$. Switching at that point on the wave produces no transient at all; switching at $\psi = \theta$ produces the largest. This is the origin of the **DC offset** in fault currents and transformer inrush: in a highly inductive circuit ($\theta \to 90^\circ$), switching at a voltage zero gives a natural component as large as the forced amplitude, and the first current peak approaches $2I_m$.

Engineering Note

On a bench, the zero-input response is what you see when you charge a circuit and then disconnect the source (a "ring-down" or discharge test); it measures the natural frequencies directly. The zero-state response is what a network analyser or a step-response test measures, because the instrument starts from rest. Datasheet specifications such as rise time, overshoot and settling time always refer to the zero-state step response.

12.4 Complementary and particular solutions

Motivation and vocabulary

The general solution $y = y_c + y_p$ of $P(D)y = r$ has two parts that mathematicians name differently from engineers:

Mathematics	Engineering	Found by
complementary function y_c	natural response y_n (before fitting ICs)	characteristic roots (Section 12.2)
particular integral y_p	forced response y_f	this section

Section 12.2 settled y_c completely. This section gives methods for y_p. For the inputs that occur most often in practice (constants, polynomials, exponentials, sinusoids and products of these) the **method of undetermined coefficients** is fast and purely algebraic. For sinusoids specifically, the **complex-exponential method** is faster still and leads straight to phasors. For arbitrary inputs, variation of parameters (Ch. 11) and convolution (Section 12.3) always work, and the Laplace transform (Ch. 14) automates the whole procedure.

Why guessing works: the annihilator idea

Undetermined coefficients looks like guesswork, but it rests on a precise fact. The admissible inputs are exactly the functions that are themselves solutions of some constant-coefficient homogeneous equation.

Annihilator

A constant-coefficient operator $Q(D)$ **annihilates** $r(t)$ if $Q(D)\,r = 0$. Examples:

$r(t)$	**Annihilator** $Q(D)$
t^k (or any polynomial of degree k)	D^{k+1}
e^{at}	$D - a$
$t^k e^{at}$	$(D - a)^{k+1}$
$\cos \omega t$, $\sin \omega t$	$D^2 + \omega^2$
$e^{\sigma t}\cos \omega t$, $e^{\sigma t}\sin \omega t$	$(D - \sigma)^2 + \omega^2$

If $P(D)y = r$ and $Q(D)r = 0$, apply $Q(D)$ to both sides: $Q(D)P(D)\,y = 0$. So every solution y of the forced equation solves a *higher-order homogeneous* equation whose characteristic polynomial is $Q(s)P(s)$. Its solutions are known from Section 12.2. Those coming from the roots of P are already in y_c; the remaining ones, coming from the roots of Q, give the form of y_p. If a root of Q coincides with a root of P, the multiplicities add, and the repeated-root rule automatically introduces extra factors of t. That is the whole theory behind the resonance (modification) rule below.

The method of undetermined coefficients

Trial forms for y_p

Let $r(t) = e^{at}\big[p_k(t)\cos\omega t + q_k(t)\sin\omega t\big]$, where p_k, q_k are polynomials of degree at most k (either may be zero; $\omega = 0$ covers pure exponentials and polynomials, $a = 0$ covers pure sinusoids). Let m be the multiplicity of $s_0 = a + j\omega$ as a root of $P(s)$ ($m = 0$ if it is not a root). Then there is a particular solution of the form

$$y_p = t^m e^{at}\big[A_k(t)\cos\omega t + B_k(t)\sin\omega t\big],$$

with A_k, B_k polynomials of degree k whose coefficients are determined by substitution.

Common cases in table form (with $m = 0$, i.e. no resonance):

Input $r(t)$	Trial y_p
constant K	A
t^k	$A_k t^k + \cdots + A_1 t + A_0$
e^{at}	Ae^{at}
$\cos\omega t$ or $\sin\omega t$	$A\cos\omega t + B\sin\omega t$
$e^{at}\cos\omega t$	$e^{at}(A\cos\omega t + B\sin\omega t)$
te^{at}	$(A_1 t + A_0)e^{at}$

Modification (resonance) rule

If any term of the trial form is a solution of the homogeneous equation, multiply the **entire** trial form by t^m, where m is the multiplicity of the corresponding characteristic root. After the modification no term of y_p duplicates a term of y_c.

Common Pitfall

Three errors account for most failures:

1. **Dropping a companion term.** For $r = \sin\omega t$ the trial must contain *both* $\cos\omega t$ and $\sin\omega t$; for $r = t^2$ it must contain t^2, t and 1.
2. **Multiplying only part of the trial.** For $r = te^{-t}$ with a simple root -1, the trial is $t(A_1 t + A_0)e^{-t}$, not $(A_1 t^2 + A_0)e^{-t}$.
3. **Ignoring resonance.** Substituting Ae^{at} when $P(a) = 0$ gives the impossible equation $0 = 1$; that is the signal to multiply by t.

A closed-form shortcut for exponential inputs

For $r(t) = e^{at}$ the eigenfunction identity gives the answer without solving any equations:

Exponential input formula

If $P(a) \neq 0$, then $P(D)y = e^{at}$ has the particular solution

$$y_p = \frac{e^{at}}{P(a)}.$$

If a is a root of multiplicity m, then

$$y_p = \frac{t^m e^{at}}{P^{(m)}(a)}.$$

Proof. The first case is immediate from $P(D)e^{at} = P(a)e^{at}$. For the second, write $P(s) = (s-a)^m Q(s)$ with $Q(a) \neq 0$. By the shift theorem,

$$P(D)\big[e^{at}t^m\big] = e^{at}P(D+a)\,t^m = e^{at}\,Q(D+a)\,D^m t^m = e^{at}\,Q(D+a)\,m! = m!\,Q(a)\,e^{at},$$

since $D^m t^m = m!$ is a constant and $Q(D+a)$ acting on a constant multiplies it by $Q(a)$. Finally, differentiating $P(s) = (s-a)^m Q(s)$ m times and setting $s = a$ gives $P^{(m)}(a) = m!\,Q(a)$. $\square$

For a constant input K (the case $a = 0$), the formula gives $y_p = K/P(0) = K/a_0$: the DC gain is $1/a_0$. This is the formal version of the circuit-analysis rule "at DC, capacitors are open circuits and inductors are short circuits", which amounts to setting every derivative to zero.

The complex-exponential (phasor) method

For a sinusoidal input $r(t) = R_m\cos(\omega t + \psi)$ write $r = \mathrm{Re}\{\mathbf{R}\,e^{j\omega t}\}$ with complex amplitude (phasor) $\mathbf{R} = R_m e^{j\psi}$. Solve the complex equation $P(D)z = \mathbf{R}e^{j\omega t}$ with the exponential formula, $z = \mathbf{R}e^{j\omega t}/P(j\omega)$, and take the real part.

Sinusoidal forced response

If $P(j\omega) \neq 0$ and P has real coefficients, the forced response to $R_m \cos(\omega t + \psi)$ is

$$y_f(t) = \operatorname{Re}\left\{ \frac{\mathbf{R}}{P(j\omega)} e^{j\omega t} \right\} = \frac{R_m}{|P(j\omega)|} \cos\big(\omega t + \psi - \angle P(j\omega)\big).$$

The output phasor is $\mathbf{Y} = \mathbf{R}/P(j\omega)$.

The quantity $H(j\omega) = 1/P(j\omega)$ (more generally $N(j\omega)/P(j\omega)$ when derivatives of the input appear on the right-hand side) is the **frequency response**. Section 12.5 studies it as a function of ω. The method reduces sinusoidal steady-state analysis to complex arithmetic, which is why AC circuit theory uses impedances R, $j\omega L$ and $1/(j\omega C)$: each is the eigenvalue of an element law acting on $e^{j\omega t}$.

Superposition of inputs

If $r = r_1 + r_2 + \dots$, find a particular solution for each term separately and add them. Each term may need its own resonance modification. This is valid only for the *particular* parts; the complementary function is written once, with n constants, and fitted to the initial conditions at the end.

When undetermined coefficients fails

The method requires r to have a finite-dimensional family of derivatives. It fails for inputs such as $\tan t$, $1/t$, $\ln t$, e^{t^2} or a general measured waveform. Then use:

- **variation of parameters**, which works for any continuous r once y_c is known (derived in Ch. 11);
- **convolution** with the impulse response (Section 12.3);
- **Laplace transforms** for piecewise inputs such as pulses, ramps and delayed steps (Ch. 14);
- **numerical integration** for anything else (Ch. 22).

Worked examples

Example 12.15 — Polynomial input

Find a particular solution of $y'' + 3y' + 2y = 4t^2$.

No root of $s^2 + 3s + 2$ is zero, so $m = 0$ and the trial is $y_p = At^2 + Bt + C$. Then $y_p' = 2At + B$, $y_p'' = 2A$, and

$$2A + 3(2At + B) + 2(At^2 + Bt + C) = 2A\,t^2 + (6A + 2B)\,t + (2A + 3B + 2C).$$

Matching coefficients with $4t^2 + 0t + 0$:

$$2A = 4, \qquad 6A + 2B = 0, \qquad 2A + 3B + 2C = 0 \quad \Rightarrow \quad A = 2,\ B = -6,\ C = 7.$$

$$y_p = 2t^2 - 6t + 7.$$

Check. $y_p'' = 4$, $3y_p' = 12t - 18$, $2y_p = 4t^2 - 12t + 14$; sum $= 4t^2$ ✓. Note that the constant and linear terms are essential even though the input contains only t^2.

Example 12.16 — Sinusoid by the complex-exponential method

Find the forced response of $y'' + 2y' + 5y = 10\cos t$.

$P(s) = s^2 + 2s + 5$ and $P(j1) = -1 + 2j + 5 = 4 + 2j$, which is nonzero (the roots are $-1 \pm 2j$). The output phasor is

$$\mathbf{Y} = \frac{10}{4 + 2j} = \frac{10(4 - 2j)}{20} = 2 - j.$$

$$y_f = \operatorname{Re}\{(2 - j)(\cos t + j \sin t)\} = 2\cos t + \sin t = \sqrt{5}\cos(t - 0.4636).$$

Check by substitution. $y_f' = -2\sin t + \cos t$, $y_f'' = -2\cos t - \sin t$. Then $y_f'' + 2y_f' + 5y_f = (-2 + 2 + 10)\cos t + (-1 - 4 + 5)\sin t = 10\cos t$ ✓. The amplitude $\sqrt{5} = 10/|4 + 2j| = 10/\sqrt{20}$ ✓, and the phase lag is $\arctan(2/4) = 26.57^\circ$.

The same result by undetermined coefficients requires solving $4A + 2B = 10$, $-2A + 4B = 0$ (from the cos and sin coefficients), giving $A = 2$, $B = 1$, the same answer with more algebra.

Example 12.17 — Pure resonance

Solve $y'' + 4y = 3\cos 2t$, $y(0) = 0$, $y'(0) = 0$.

Complementary. Roots $\pm 2j$: $y_c = c_1 \cos 2t + c_2 \sin 2t$.

Particular. The input frequency 2 coincides with a simple root ($m = 1$), so use the formula with $P'(s) = 2s$: the complex equation $P(D)z = 3e^{j2t}$ has

$$z_p = \frac{3\,t\,e^{j2t}}{P'(2j)} = \frac{3t\,e^{j2t}}{4j} = -\frac{3j}{4}\,t\,(\cos 2t + j\sin 2t),$$

whose real part is $y_p = \frac{3}{4}\,t\sin 2t$.

Initial conditions. $y(0) = c_1 = 0$; $y'(0) = 2c_2 + \left[\frac{3}{4}\sin 2t + \frac{3}{2}t\cos 2t\right]_{t=0} = 2c_2 = 0$. So

$$y(t) = \tfrac{3}{4}\,t\sin 2t.$$

Check. $y'' = 3\cos 2t - 3t\sin 2t$, so $y'' + 4y = 3\cos 2t - 3t\sin 2t + 3t\sin 2t = 3\cos 2t$ ✓. The amplitude grows linearly without bound: a lossless resonator driven at its natural frequency absorbs energy every cycle. Any real resonator has some damping, which caps the amplitude at a finite value (Section 12.5).

Example 12.18 — Mixed input with resonance and a full IVP

Solve $y'' - y' - 2y = e^{2t} + 3$, $y(0) = 0$, $y'(0) = 0$.

Complementary. $s^2 - s - 2 = (s-2)(s+1)$: $y_c = c_1 e^{2t} + c_2 e^{-t}$.

Particular, term by term.

- For e^{2t}: $a = 2$ is a simple root, $P'(s) = 2s - 1$, $P'(2) = 3$, so $y_{p1} = \frac{1}{3}te^{2t}$.
- For 3: $y_{p2} = 3/P(0) = 3/(-2) = -\frac{3}{2}$.

Initial conditions. $y = c_1 e^{2t} + c_2 e^{-t} + \frac{1}{3}te^{2t} - \frac{3}{2}$:

$$\begin{aligned} y(0) &= c_1 + c_2 - \tfrac{3}{2} = 0, \\ y'(0) &= 2c_1 - c_2 + \tfrac{1}{3} = 0, \end{aligned} \qquad \Rightarrow \qquad c_1 = \tfrac{7}{18}, \quad c_2 = \tfrac{10}{9}.$$

$$y(t) = \tfrac{7}{18}e^{2t} + \tfrac{10}{9}e^{-t} + \tfrac{1}{3}te^{2t} - \tfrac{3}{2}.$$

Check. At $t = 1$ the formula gives 4.24530, agreeing with a numerical solution to nine significant figures. Verifying y_{p1} directly: with $w = te^{2t}$, $w' = (1 + 2t)e^{2t}$, $w'' = (4 + 4t)e^{2t}$, and $w'' - w' - 2w = (4 + 4t - 1 - 2t - 2t)e^{2t} = 3e^{2t}$ ✓, so $w/3$ is correct.

Example 12.19 — Exponential times polynomial via the shift theorem

Find a particular solution of $y'' + y = te^t$.

Write $y_p = e^t f(t)$. By the shift theorem, $(D^2+1)\left[e^t f\right] = e^t\left[(D+1)^2 + 1\right]f = e^t\left(D^2 + 2D + 2\right)f$. We need $(D^2 + 2D + 2)f = t$. Try $f = At + B$: $2A + 2At + 2B = t$, so $A = \frac{1}{2}$, $B = -\frac{1}{2}$:

$$y_p = \tfrac{1}{2}(t-1)e^t.$$

Check. $y_p' = \frac{1}{2}te^t$, $y_p'' = \frac{1}{2}(t+1)e^t$; $y_p'' + y_p = \frac{1}{2}(t + 1 + t - 1)e^t = te^t$ ✓. The shift theorem reduces an exponential-times-polynomial problem to a pure polynomial problem, which is often less error-prone than substituting $(At + B)e^t$ directly.

Example 12.20 — Damped sinusoidal input

Find the forced response of $y'' + 4y' + 13y = e^{-2t}\cos 3t$, and explain the result.

The characteristic roots are $-2 \pm 3j$, and the input is $\mathrm{Re}\{e^{(-2+3j)t}\}$ with exponent $s_0 = -2 + 3j$, a simple root. So $m = 1$ and, with $P'(s) = 2s + 4$, $P'(s_0) = 6j$:

$$y_p = \mathrm{Re}\left\{\frac{t\,e^{(-2+3j)t}}{6j}\right\} = \mathrm{Re}\left\{-\frac{j}{6}\,t\,e^{-2t}(\cos 3t + j\sin 3t)\right\} = \frac{1}{6}\,t\,e^{-2t}\sin 3t.$$

Check. Using the shift theorem with $y_p = e^{-2t}g$: $P(D-2) = (D-2)^2 + 4(D-2) + 13 = D^2 + 9$, so we need $g'' + 9g = \cos 3t$, which by Example 12.17's pattern has $g = \frac{1}{6}t\sin 3t$ ✓. The input is itself a natural mode of the system, so it resonates even though it decays.

12.5 Transient and steady-state behavior

Motivation

Switch on a sinusoidal generator connected to a filter and watch the output on an oscilloscope. For a short time the waveform is irregular; then it settles into a clean sinusoid at the generator frequency, with a fixed amplitude and phase. The irregular part is the **transient**; the clean sinusoid is the **steady state**. Almost all AC circuit analysis, filter design and frequency-response measurement concerns the steady state, but the transient decides how long one must wait before the steady-state formulas apply, how large the switching surges are, and whether the steady state exists at all.

Definitions

Transient and steady-state response

Suppose the response of $P(D)y = r(t)$ can be written $y = y_{tr} + y_{ss}$ where

$$\lim_{t\to\infty} y_{tr}(t) = 0$$

and y_{ss} persists (for example constant, periodic, or growing like the input). Then y_{tr} is the **transient response** and y_{ss} is the **steady-state response**.

Transient = natural, steady state = forced (for stable systems)

If $P(D)y = 0$ is asymptotically stable (all characteristic roots satisfy $\mathrm{Re}\{s_k\} < 0$), then for any input of the undetermined-coefficients class:

1. the natural response y_n decays to zero for every set of initial conditions, so $y_{tr} = y_n$ and $y_{ss} = y_f$;
2. the steady state is **independent of the initial conditions**;
3. if the input is bounded, the steady state is bounded (bounded-input, bounded-output stability).

Reason. Every term of y_n has the form $t^k e^{s_k t}$ with $\mathrm{Re}\{s_k\} < 0$ and tends to zero. Initial conditions enter only through the constants of y_n. If the input contains no natural modes (stable roots never coincide with a pure sinusoid or constant, which have exponents on the imaginary axis), y_f has the same bounded form as the input. □

When the system is not asymptotically stable, the split fails in instructive ways:

- **Unstable** ($\mathrm{Re}\{s_k\} > 0$ for some root): the natural response grows; the "steady state" of the formula y_f exists on paper but is never observed, because any perturbation excites the growing mode. An unstable amplifier oscillates or latches instead of amplifying.
- **Marginally stable** (simple roots on the imaginary axis): the natural response never decays, so the initial conditions are remembered forever. A lossless LC tank rings indefinitely; an integrator ($s = 0$) holds its initial value.
- **Resonance with a marginal root**: a constant input to an integrator, or a sinusoid at an LC tank's natural frequency, produces a growing y_f (t or $t\sin\omega t$, Example 12.17) and there is no bounded steady state.

How fast does the transient die?

The slowest-decaying mode dominates the late transient. With the root closest to the imaginary axis at real part $-\sigma_{\min}$, the transient is bounded by a multiple of $e^{-\sigma_{\min} t}$ for large t (times a polynomial factor if that root is repeated).

Time constant and settling time

- For a first-order mode $e^{-t/\tau}$, the **time constant** is $\tau = 1/\sigma$.
- The **2% settling time** t_s is the time after which the response stays within 2% of its final value. For a first-order system $t_s = \tau \ln 50 \approx 3.9\tau$, usually rounded to 4τ. For an underdamped second-order system the envelope $e^{-\zeta\omega_n t}/\sqrt{1-\zeta^2}$ gives

$$t_s \approx \frac{\ln\!\left(50/\sqrt{1-\zeta^2}\right)}{\zeta\omega_n} \approx \frac{4}{\zeta\omega_n}.$$

For the 1% and 0.1% criteria replace $\ln 50 = 3.91$ by $\ln 100 = 4.61$ and $\ln 1000 = 6.91$. A rule worth memorising: **each time constant buys about 8.7 dB (a factor e)**, so reaching 16-bit accuracy ($2^{-16} \approx 1.5 \times 10^{-5}$) takes about 11τ.

Sinusoidal steady state and frequency response

For a stable system driven by $\cos\omega t$, Section 12.4 gave the steady state directly from the complex number $P(j\omega)$. In general, when the input enters through derivatives as well, the ODE reads $P(D)y = N(D)r$, and the same argument gives:

Frequency response

For the stable LTI equation $P(D)y = N(D)r$ with input $r = R_m\cos(\omega t + \psi)$, the steady-state output is

$$y_{ss}(t) = R_m\,|H(j\omega)|\cos\big(\omega t + \psi + \angle H(j\omega)\big), \qquad H(j\omega) = \frac{N(j\omega)}{P(j\omega)}.$$

$|H(j\omega)|$ is the **gain** (magnitude response) and $\angle H(j\omega)$ is the **phase shift**. In phasor form $\mathbf{Y} = H(j\omega)\,\mathbf{R}$.

The frequency response is the transfer function $H(s) = N(s)/P(s)$ of Ch. 14 evaluated on the imaginary axis. Here it arises without any transform: it is simply the eigenvalue of the system for the eigenfunction $e^{j\omega t}$.

First-order low-pass. For $\tau y' + y = r$, $H(j\omega) = 1/(1 + j\omega\tau)$:

$$|H| = \frac{1}{\sqrt{1+(\omega\tau)^2}}, \qquad \angle H = -\arctan(\omega\tau).$$

At the **corner frequency** $\omega_c = 1/\tau$ the gain is $1/\sqrt{2}$ (-3 dB) and the phase is $-45°$; well above it the gain falls as $1/(\omega\tau)$, i.e. -20 dB per decade.

Second-order low-pass. For $y'' + 2\zeta\omega_n y' + \omega_n^2 y = \omega_n^2 r$, with normalised frequency $\nu = \omega/\omega_n$,

$$H(j\omega) = \frac{1}{1 - \nu^2 + j\,2\zeta\nu}, \qquad |H| = \frac{1}{\sqrt{(1-\nu^2)^2 + (2\zeta\nu)^2}}, \qquad \angle H = -\operatorname{atan2}\left(2\zeta\nu,\ 1-\nu^2\right).$$

Resonant peak of a second-order system

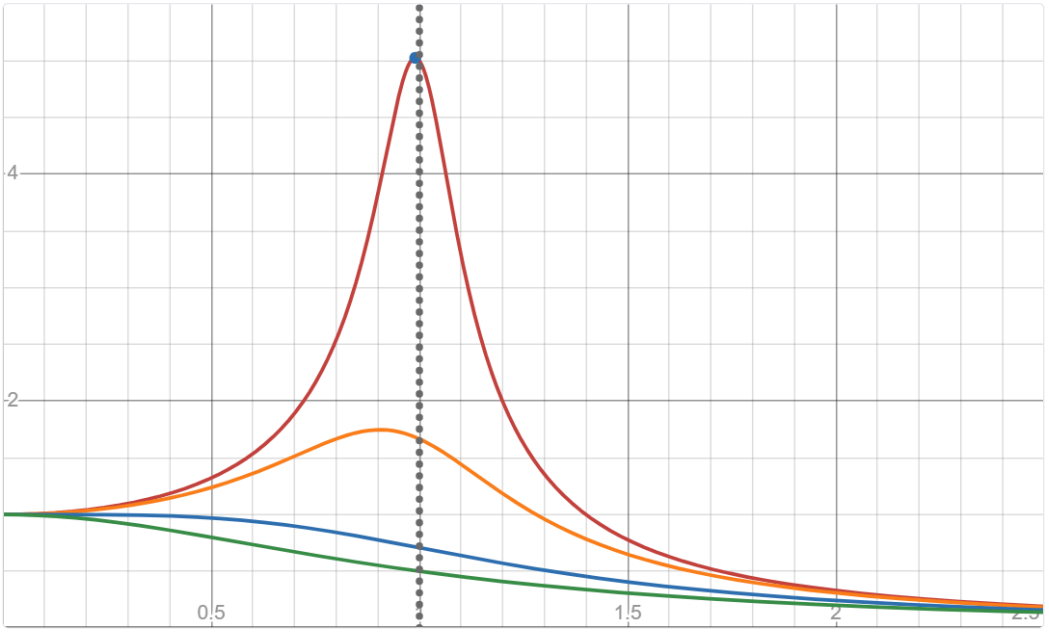
For $0 < \zeta < 1/\sqrt{2}$, $|H(j\omega)|$ has a maximum at the **resonant frequency** and with the **peak value**

$$\omega_r = \omega_n\sqrt{1 - 2\zeta^2}, \qquad M_r = \frac{1}{2\zeta\sqrt{1-\zeta^2}}.$$

For $\zeta \geq 1/\sqrt{2}$ there is no peak; the gain decreases monotonically. At $\omega = \omega_n$ the phase is exactly $-90°$ and the gain is $1/(2\zeta)$ for every ζ.

Derivation. Maximising $|H|$ means minimising $g(\nu^2) = (1-\nu^2)^2 + 4\zeta^2\nu^2$. With $x = \nu^2$, $g'(x) = -2(1-x) + 4\zeta^2 = 0$ gives $x = 1 - 2\zeta^2$, which is positive only if $\zeta^2 < 1/2$. Substituting, $g = 4\zeta^4 + 4\zeta^2(1-2\zeta^2) = 4\zeta^2(1-\zeta^2)$, so $M_r = 1/\sqrt{g} = 1/\left(2\zeta\sqrt{1-\zeta^2}\right)$. □

For light damping $M_r \approx 1/(2\zeta) = Q$, the **quality factor**, and the half-power (-3 dB) bandwidth of the peak is approximately $2\zeta\omega_n = \omega_n/Q$. For $\zeta = 0.1$ the exact peak is $M_r = 5.025$ at $\nu = 0.990$, and the exact half-power bandwidth is $0.202\,\omega_n$, close to the approximation $0.2\,\omega_n$.



Magnitude response $|H|$ against $\nu = \omega/\omega_n$ for $\zeta = 0.1$ (red), 0.3 (orange), 0.707 (blue, maximally flat) and 1 (green). Light damping produces a tall, narrow resonant peak just below ω_n.

Steady state for periodic inputs

A periodic input can be expanded in a Fourier series (Ch. 15), $r(t) = c_0 + \sum_k |c_k|\cos(k\omega_0 t + \psi_k)$. By superposition the steady state is

$$y_{ss}(t) = H(0)\,c_0 + \sum_k |c_k|\,|H(jk\omega_0)|\cos\left(k\omega_0 t + \psi_k + \angle H(jk\omega_0)\right).$$

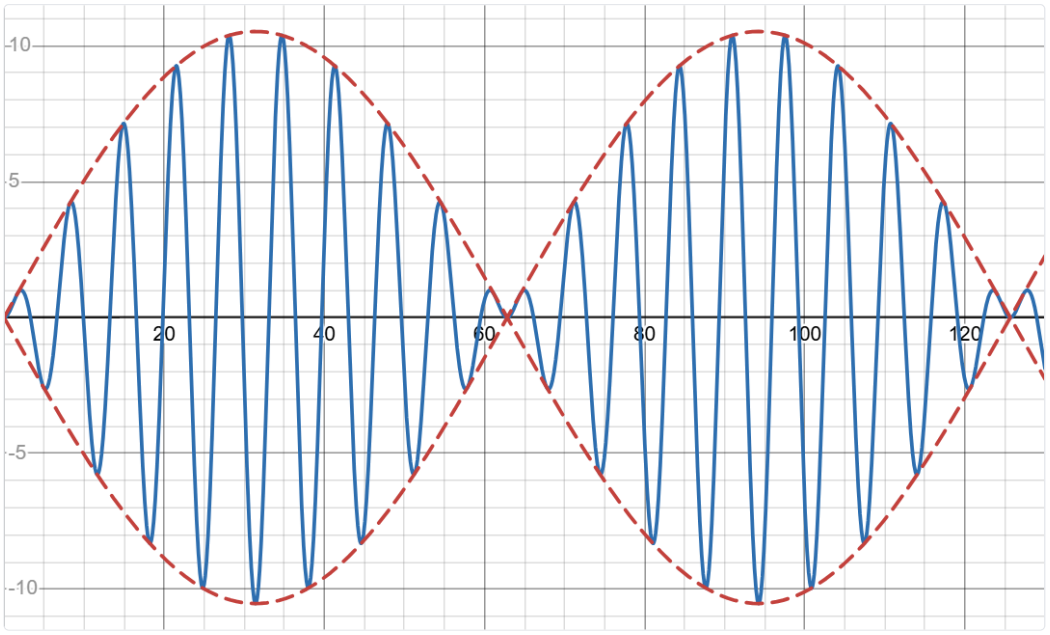
Each harmonic is scaled and shifted by the frequency response at its own frequency. A low-pass system attenuates the high harmonics, which is why a square wave through an RC filter comes out rounded. A lightly damped system can selectively amplify one harmonic that falls near its resonant peak.

Beats: the approach to resonance

For an undamped system driven near, but not at, its natural frequency, the "transient" never decays, and the sum of natural and forced responses produces **beats**. For $y'' + \omega_0^2 y = \cos\omega t$ with $y(0) = y'(0) = 0$ and $\omega \neq \omega_0$:

$$y = \frac{\cos\omega t - \cos\omega_0 t}{\omega_0^2 - \omega^2} = \frac{2}{\omega_0^2 - \omega^2}\,\sin\left(\frac{(\omega_0 - \omega)t}{2}\right)\sin\left(\frac{(\omega_0 + \omega)t}{2}\right),$$

using $\cos A - \cos B = -2\sin\frac{A+B}{2}\sin\frac{A-B}{2}$. The fast oscillation at the average frequency is modulated by a slow envelope at half the difference frequency. As $\omega \to \omega_0$, the envelope period grows without bound and, near $t = 0$, $y \to \frac{t\sin\omega_0 t}{2\omega_0}$, the resonant solution of Example 12.17. With damping, the beat pattern is a transient: it decays and leaves the steady-state sinusoid.



Beats for $\omega_0 = 1, \omega = 0.9$: the response (blue) oscillates near 0.95 rad/s inside a slow envelope $\pm 10.5\sin(0.05t)$ (red), with beat period $2\pi/0.1 = 62.8$.

Worked examples

Example 12.21 — RC low-pass switched onto a sinusoid

Solve $\tau y' + y = \cos \omega t$, $y(0) = 0$, identify transient and steady state, and evaluate for $\omega\tau = 1$.

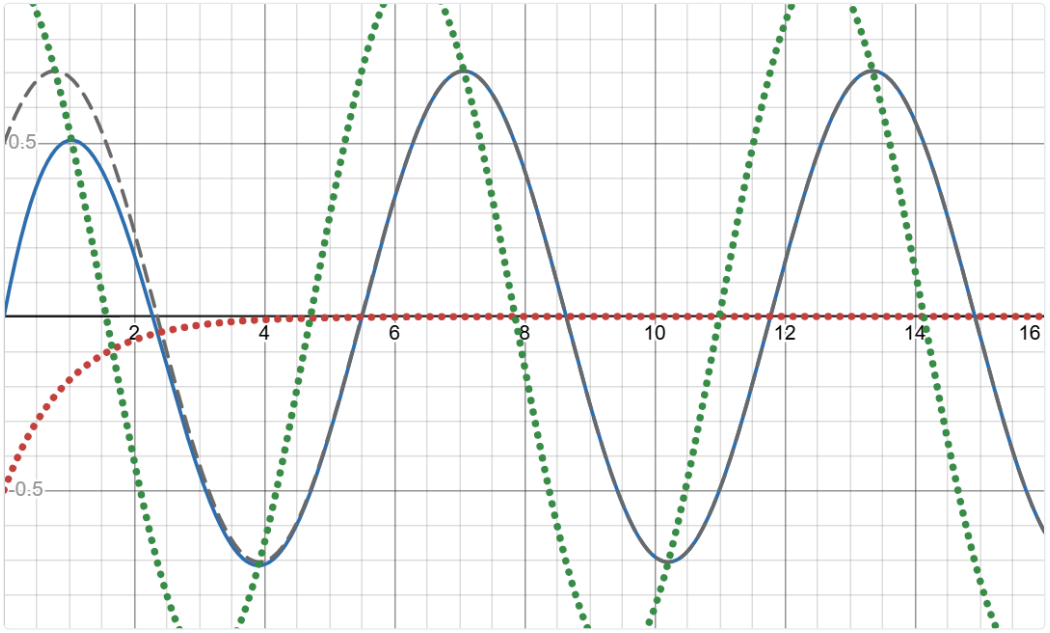
Steady state. $H(j\omega) = 1/(1 + j\omega\tau)$:

$$y_{ss} = \frac{1}{\sqrt{1+\omega^2\tau^2}}\cos\left(\omega t - \arctan\omega\tau\right) = \frac{\cos\omega t + \omega\tau\sin\omega t}{1+\omega^2\tau^2}.$$

Transient. $y_{tr} = Ae^{-t/\tau}$ with $y(0) = 0$: $A = -y_{ss}(0) = -1/(1+\omega^2\tau^2)$.

$$y(t) = \frac{\cos\omega t + \omega\tau\sin\omega t - e^{-t/\tau}}{1+\omega^2\tau^2}.$$

At $\omega\tau = 1$. $y = \tfrac{1}{2}\left(\cos\omega t + \sin\omega t - e^{-t/\tau}\right)$, with steady-state amplitude 0.7071 (-3 dB) and lag 45°. Check: $y(0) = \tfrac{1}{2}(1 + 0 - 1) = 0$ ✓. After 4τ the transient is $0.5e^{-4} = 0.009$, under 1.3% of the steady-state amplitude.



Example 12.21 with $\tau = 1, \omega = 1$: input $\cos t$ (green), complete response (blue), steady state (grey dashed) and transient (red). The output merges with the steady state within a few time constants, attenuated by 0.707 and lagging by 45°.

Example 12.22 — Second-order: transient, steady state and settling

Solve $y'' + 2y' + 5y = 10\cos t$, $y(0) = 0$, $y'(0) = 0$, and estimate when the transient is negligible.

Steady state (Example 12.16): $y_{ss} = 2\cos t + \sin t$.

Transient. Roots $-1 \pm 2j$: $y_{tr} = e^{-t}(A\cos 2t + B\sin 2t)$. Conditions on the complete response:

$$y(0) = 2 + A = 0 \Rightarrow A = -2, \qquad y'(0) = 1 + (-A + 2B) = 0 \Rightarrow B = -\tfrac{3}{2}.$$

$$y(t) = 2\cos t + \sin t - e^{-t}\left(2\cos 2t + \tfrac{3}{2}\sin 2t\right).$$

Check. At $t = 3$ the formula gives -1.91361, matching a numerical solver. The transient amplitude is $\sqrt{4 + 2.25}\,e^{-t} = 2.5e^{-t}$; it falls below 2% of the steady amplitude $\sqrt{5} = 2.236$ when $2.5e^{-t} < 0.0447$, i.e. $t > 4.0$. The transient oscillates at 2 rad/s (the natural frequency ω_d), not at the input frequency 1 rad/s: natural modes always ring at their own frequency.

Example 12.23 — Harmonics of a periodic input

The input $r(t) = \cos t + \cos 3t$ (the first two terms of a crude approximation to a pulse train) drives $y' + y = r$. Find the steady state.

With $\tau = 1$: at $\omega = 1$, $|H| = 1/\sqrt{2} = 0.7071$, $\angle H = -45°$; at $\omega = 3$, $|H| = 1/\sqrt{10} = 0.3162$, $\angle H = -71.57°$.

$$y_{ss} = 0.7071\cos(t - 45°) + 0.3162\cos(3t - 71.57°).$$

The third harmonic is attenuated more (ratio 0.447) and delayed by a different time: 45° at 1 rad/s is a delay of 0.785 s, whereas 71.57° at 3 rad/s is a delay of 0.416 s. Because different harmonics are delayed differently, the output waveform has a different *shape* from the input. This is phase distortion, which a linear-phase filter avoids.

Example 12.24 — No steady state: unstable and marginal systems

(a) $y'' - y = \cos t$; (b) $y'' + y' = 1$.

(a) Roots ± 1. The forced response is $y_f = \cos t / P(j) = \cos t/(-1 - 1) = -\frac{1}{2}\cos t$, but $y_n = c_1 e^t + c_2 e^{-t}$. Unless the initial conditions make $c_1 = 0$ exactly (a set of measure zero, destroyed by any noise), $c_1 e^t$ dominates. The formula y_f is not a physically meaningful steady state.

(b) Roots $0, -1$. The constant input matches the root $s = 0$ ($m = 1$): $y_p = t/P'(0) = t/1 = t$. The response grows as a ramp; the system integrates the constant input. In circuit terms, a capacitor charged by a constant current has no DC steady state. A DC steady state exists only if $P(0) = a_0 \neq 0$.

Common Pitfall

Phasor and impedance calculations silently assume the system is stable and that enough time has passed. Applying $H(j\omega)$ to an unstable circuit, or reading a "steady-state" measurement before 4–5 time constants of the slowest mode have elapsed, gives wrong answers. In very lightly damped circuits (Q in the hundreds, as in crystal oscillators or high-Q filters) the settling time $t_s \approx 8Q/\omega_n$ can be surprisingly long.

Engineering Note

A network analyser measures $H(j\omega)$ by sweeping a sinusoid and waiting at each frequency for the transient to die. Its IF bandwidth and dwell time are chosen from exactly the settling-time estimate above; narrow-band high-Q devices need slow sweeps. Similarly, a switching power supply's "load-step response" specification is a transient measurement: the regulator's dominant poles decide how quickly the output voltage recovers after a sudden change in current.

12.6 Coupled differential equations

Motivation

Real networks have more than one loop or node, and the unknowns interact. Two mesh currents share a resistor; two coils share magnetic flux through mutual inductance M; a motor's current produces torque and its speed produces a back-EMF that opposes the current. Each such interaction gives a system of simultaneous linear ODEs in several unknowns. This section treats them with two tools: **elimination**, which reduces the system to a single higher-order equation of the kind already solved, and **modal decomposition**, which changes variables so that the system splits into independent scalar equations. The general first-order matrix form $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{f}$ and its eigenvector solution appear in Ch. 11; here the emphasis is on equations as they arise from circuits, often second order and with derivatives on several unknowns in one equation.

Operator-matrix form

Coupled linear system with constant coefficients

A system of k linear constant-coefficient ODEs in unknowns $y_1, \dots, y_k$ can be written

$$\begin{bmatrix} P_{11}(D) & \cdots & P_{1k}(D) \\ \vdots & & \vdots \\ P_{k1}(D) & \cdots & P_{kk}(D) \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_k \end{bmatrix} = \begin{bmatrix} r_1(t) \\ \vdots \\ r_k(t) \end{bmatrix}, \qquad \text{briefly} \qquad \mathbf{P}(D)\,\mathbf{y} = \mathbf{r},$$

where each $P_{il}(D)$ is a polynomial in D. The polynomial $\Delta(s) = \det \mathbf{P}(s)$ is the **characteristic polynomial of the system**.

Because constant-coefficient operators commute (Section 12.1), the entries of $\mathbf{P}(D)$ can be manipulated like the entries of a matrix of polynomials. In particular **Cramer's rule** holds in operator form:

Elimination by operator determinants

If $\Delta(s) = \det \mathbf{P}(s)$ is not identically zero, every solution of $\mathbf{P}(D)\mathbf{y} = \mathbf{r}$ satisfies, for each component,

$$\Delta(D)\, y_i = \det \mathbf{P}_i(D),$$

where $\mathbf{P}_i$ is $\mathbf{P}$ with its ith column replaced by $\mathbf{r}$ (expanded along that column so that operators act on the r_l). Each y_i therefore has natural modes given by the roots of $\Delta(s)$. The general solution of the system contains exactly $\deg \Delta$ independent arbitrary constants.

Proof sketch. Multiply the system on the left by the adjugate operator matrix $\operatorname{adj}\mathbf{P}(D)$. Since entries commute, $\operatorname{adj}\mathbf{P}\cdot\mathbf{P} = \Delta(D)\,\mathbf{I}$, giving $\Delta(D)\mathbf{y} = \operatorname{adj}\mathbf{P}(D)\,\mathbf{r}$, which is the column-replacement formula. The count of constants is a standard result (each component individually appears to need $\deg\Delta$ constants, but the original equations tie them together so that only $\deg\Delta$ in total are free). $\square$

Common Pitfall

After eliminating, each component y_i satisfies a degree-$\deg\Delta$ equation, so writing each with its own $\deg\Delta$ constants gives $k\cdot\deg\Delta$ constants, far too many. The surplus constants are **not** free: substitute back into one of the original equations to express them in terms of the others, or, more simply, solve for one unknown and *compute* the others from the original equations (as in Example 12.25). Also: if $\Delta(s)\equiv 0$ the system is degenerate (dependent equations or a redundant variable) and must be reformulated.

The degree of Δ equals the number of independent energy-storage elements, i.e. the order of the system. For two coupled inductors, Δ has degree 2 unless the coupling is perfect ($M^2 = L_1L_2$), in which case the two inductors behave as one and the degree drops to 1.

Modal decomposition

The second method changes variables. Suppose the system can be written $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ (first order) or $\mathbf{M}\ddot{\mathbf{x}} + \mathbf{K}\mathbf{x} = \mathbf{0}$ (second order, undamped, as for LC ladders or spring–mass chains). If $\mathbf{A}$ is diagonalisable, $\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$ (Ch. 10), then the **modal coordinates** $\mathbf{q} = \mathbf{V}^{-1}\mathbf{x}$ satisfy

$$\dot{\mathbf{q}} = \mathbf{\Lambda}\mathbf{q}, \qquad \text{i.e.} \qquad \dot{q}_k = \lambda_k q_k \quad \text{independently for each } k.$$

The system has split into scalar first-order equations, each with solution $q_k = q_k(0)e^{\lambda_k t}$, and $\mathbf{x} = \mathbf{V}\mathbf{q} = \sum_k q_k(0)\,\mathbf{v}_k e^{\lambda_k t}$. Each eigenvector $\mathbf{v}_k$ is a **mode shape**: the fixed pattern of relative amplitudes in which the unknowns move together at the rate λ_k.

For the second-order form $\ddot{\mathbf{x}} + \mathbf{K}\mathbf{x} = \mathbf{0}$ (taking $\mathbf{M} = \mathbf{I}$, or after scaling) with $\mathbf{K}$ symmetric positive definite, the eigenvalues κ_k of $\mathbf{K}$ are positive and the modes are **normal modes** $\mathbf{v}_k\cos(\omega_k t + \varphi_k)$ with $\omega_k = \sqrt{\kappa_k}$. Symmetric matrices have orthogonal eigenvectors, which is what makes the decoupling exact and the mode shapes physically distinct.

With forcing, the same change of variables gives $\dot{q}_k = \lambda_k q_k + g_k(t)$ with $\mathbf{g} = \mathbf{V}^{-1}\mathbf{f}$: each mode is driven only by its own projection of the input. A mode whose projection is zero is not excited at all.

Sinusoidal steady state of coupled systems

For a stable coupled system driven sinusoidally at frequency ω, replace every D by $j\omega$: the phasor amplitudes satisfy the complex linear algebraic system

$$\mathbf{P}(j\omega)\,\mathbf{Y} = \mathbf{R},$$

solvable by Cramer's rule or Gaussian elimination (Ch. 8). This is exactly mesh or nodal analysis with impedances. Its determinant $\det\mathbf{P}(j\omega) = \Delta(j\omega)$ is small near the natural frequencies (resonances), while a numerator determinant can vanish at some frequency, making one output zero: an **antiresonance** or transmission zero.

Worked examples

Example 12.25 — Elimination for a first-order pair

Solve $x' = -2x + y$, $y' = x - 2y$, with $x(0) = 1$, $y(0) = 0$.

Eliminate y. From the first equation $y = x' + 2x$. Substituting into the second: $x'' + 2x' = x - 2(x' + 2x)$, i.e.

$$x'' + 4x' + 3x = 0, \qquad s^2 + 4s + 3 = (s+1)(s+3).$$

(Operator check: $\Delta(s) = \det \begin{bmatrix} s+2 & -1 \\ -1 & s+2 \end{bmatrix} = (s+2)^2 - 1 = s^2 + 4s + 3$ ✓.)

Initial data for x. $x(0) = 1$ and $x'(0) = -2x(0) + y(0) = -2$. With $x = c_1 e^{-t} + c_2 e^{-3t}$: $c_1 + c_2 = 1$, $-c_1 - 3c_2 = -2$, so $c_1 = c_2 = \tfrac{1}{2}$.

Recover y from the original equation (no new constants):

$$x = \tfrac{1}{2}\left(e^{-t} + e^{-3t}\right), \qquad y = x' + 2x = \tfrac{1}{2}\left(e^{-t} - e^{-3t}\right).$$

Modal view. $\mathbf{A} = \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix}$ has eigenpairs $\lambda = -1$, $\mathbf{v} = (1,1)$ and $\lambda = -3$, $\mathbf{v} = (1,-1)$. The solution is $\tfrac{1}{2}(1,1)e^{-t} + \tfrac{1}{2}(1,-1)e^{-3t}$: the sum $x + y = e^{-t}$ decays slowly and the difference $x - y = e^{-3t}$ decays fast. Check: $y(0) = 0$ ✓, and $y' = \tfrac{1}{2}(-e^{-t} + 3e^{-3t})$ equals $x - 2y = \tfrac{1}{2}(e^{-t} + e^{-3t}) - (e^{-t} - e^{-3t}) = \tfrac{1}{2}(-e^{-t} + 3e^{-3t})$ ✓.

Example 12.26 — Mutually coupled coils with a step input

Two coils with $L_1 = L_2 = 1$ H, mutual inductance $M = 0.5$ H and winding resistances $R_1 = R_2 = 1\ \Omega$ are coupled. A 1 V step is applied to coil 1 at $t = 0$; coil 2 is short-circuited. Both currents start at zero. Find $i_1(t)$ and $i_2(t)$.

Equations.

$$\begin{aligned} L_1 i_1' + M i_2' + R_1 i_1 &= 1, \\ M i_1' + L_2 i_2' + R_2 i_2 &= 0, \end{aligned} \qquad \begin{bmatrix} D+1 & 0.5D \\ 0.5D & D+1 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Characteristic polynomial. $\Delta(s) = (s+1)^2 - 0.25s^2 = 0.75s^2 + 2s + 1$, roots $s = (-2 \pm 1)/1.5$: $s_1 = -\tfrac{2}{3}$, $s_2 = -2\ \text{s}^{-1}$.

Cramer. $\Delta(D)\, i_1 = (D+1)[1] - 0.5D[0] = 1$ and $\Delta(D)\, i_2 = (D+1)[0] - 0.5D[1] = 0$. So the forced values are $i_{1f} = 1/\Delta(0) = 1$ and $i_{2f} = 0$, as expected (at DC the coils are shorts and only R_1 limits i_1).

Initial derivatives. At $t = 0^+$, $\begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix} \mathbf{i}' = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, so $i_1'(0) = \tfrac{4}{3}$, $i_2'(0) = -\tfrac{2}{3}$ A/s.

Fit. $i_1 = 1 + ae^{-2t/3} + be^{-2t}$ with $1 + a + b = 0$, $-\tfrac{2}{3}a - 2b = \tfrac{4}{3}$: $a = b = -\tfrac{1}{2}$. For $i_2 = ce^{-2t/3} + de^{-2t}$: $c + d = 0$, $-\tfrac{2}{3}c - 2d = -\tfrac{2}{3}$: $c = -\tfrac{1}{2}$, $d = \tfrac{1}{2}$.

$$i_1 = 1 - \tfrac{1}{2}e^{-2t/3} - \tfrac{1}{2}e^{-2t}, \qquad i_2 = -\tfrac{1}{2}e^{-2t/3} + \tfrac{1}{2}e^{-2t}\ \text{A}.$$

Interpretation. The sum $i_1 + i_2$ sees inductance $L + M = 1.5$ H (time constant 1.5 s, rate $2/3$), the difference $i_1 - i_2$ sees $L - M = 0.5$ H (rate 2): these are the modal coordinates. The secondary current is negative (Lenz's law: it opposes the change in flux), peaks, and returns to zero. A numerical eigen-solution of the matrix form gives the same eigenvalues -0.6667 and -2 and the same coefficients.

Example 12.27 — Coupled resonators and energy exchange

Two identical LC resonators coupled through a capacitor (or two masses joined by springs) obey, in suitable units,

$$x_1'' = -2x_1 + x_2, \qquad x_2'' = x_1 - 2x_2,$$

with $x_1(0) = 1$, $x_2(0) = 0$, $x_1'(0) = x_2'(0) = 0$. Find the normal modes and the motion.

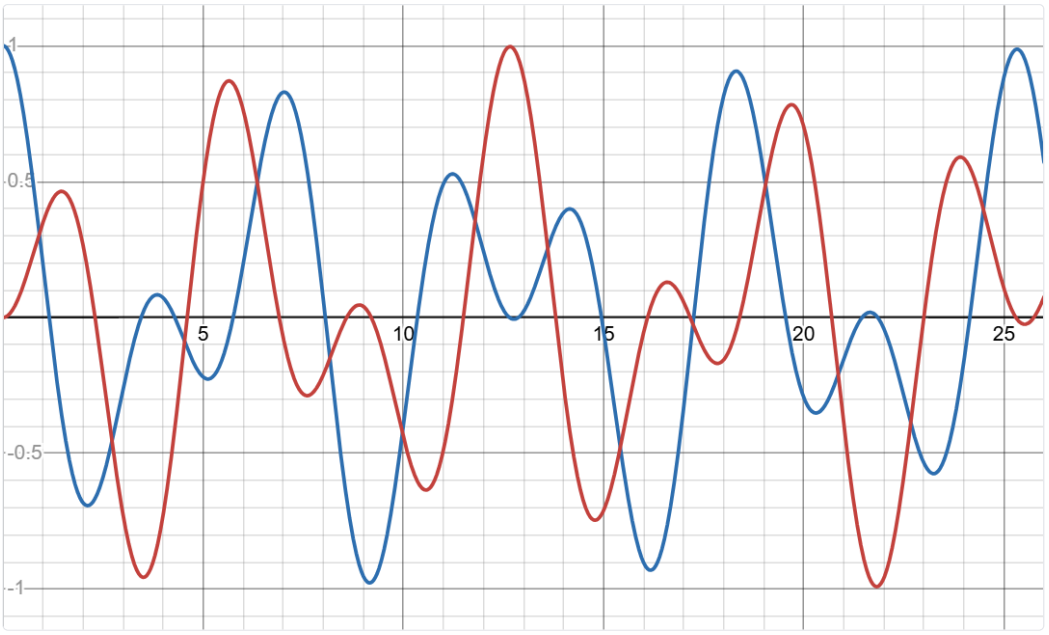
Normal modes. $\mathbf{K} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ has eigenvalues 1 and 3 with eigenvectors $(1,1)$ and $(1,-1)$. So the in-phase mode oscillates at $\omega_1 = 1$ and the anti-phase mode at $\omega_2 = \sqrt{3} = 1.732$.

Modal coordinates. $q_1 = (x_1 + x_2)/2$, $q_2 = (x_1 - x_2)/2$ satisfy $q_1'' = -q_1$, $q_2'' = -3q_2$, with $q_1(0) = q_2(0) = \tfrac{1}{2}$ and zero initial velocities. Hence $q_1 = \tfrac{1}{2}\cos t$, $q_2 = \tfrac{1}{2}\cos\sqrt{3}t$, and

$$x_1 = q_1 + q_2 = \tfrac{1}{2}\left(\cos t + \cos\sqrt{3}\,t\right), \qquad x_2 = q_1 - q_2 = \tfrac{1}{2}\left(\cos t - \cos\sqrt{3}\,t\right).$$

Check. $x_1'' = \tfrac{1}{2}(-\cos t - 3\cos\sqrt{3}t)$ and $-2x_1 + x_2 = -\cos t - \cos\sqrt{3}t + \tfrac{1}{2}\cos t - \tfrac{1}{2}\cos\sqrt{3}t = \tfrac{1}{2}(-\cos t - 3\cos\sqrt{3}t)$ ✓.

By the sum-to-product identity, $x_1 = \cos\!\left(\tfrac{\sqrt{3}-1}{2}t\right)\cos\!\left(\tfrac{\sqrt{3}+1}{2}t\right)$ and $x_2 = \sin\!\left(\tfrac{\sqrt{3}-1}{2}t\right)\sin\!\left(\tfrac{\sqrt{3}+1}{2}t\right)$. The energy moves completely from resonator 1 to resonator 2 and back, with the transfer taking $\pi/(\sqrt{3}-1) = 4.29$ time units. With weak coupling the two mode frequencies are close and the exchange is slow; this is the mechanism behind coupled-resonator bandpass filters and wireless power transfer.



Example 12.27: displacement of resonator 1 (blue) and resonator 2 (red). Energy passes back and forth between them as the two normal modes drift in and out of phase.

Example 12.28 — Driven coupled resonators and an antiresonance

Drive the system of Example 12.27 with a force on the first coordinate: $x_1'' + 2x_1 - x_2 = \cos\omega t$, $x_2'' - x_1 + 2x_2 = 0$. Find the steady-state phasor amplitudes and show that x_1 can be made to stand still.

Replace D^2 by $-\omega^2$ (strictly, the undamped system has no decaying transient; in practice a small loss makes the transient decay and leaves this steady state):

$$\begin{bmatrix} 2-\omega^2 & -1 \\ -1 & 2-\omega^2 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \Rightarrow \quad X_1 = \frac{2-\omega^2}{(2-\omega^2)^2 - 1}, \qquad X_2 = \frac{1}{(2-\omega^2)^2-1}.$$

The denominator $(2-\omega^2)^2 - 1 = (1-\omega^2)(3-\omega^2)$ vanishes at the natural frequencies $\omega = 1$ and $\sqrt{3}$ (resonances). The numerator of X_1 vanishes at $\omega = \sqrt{2}$: there $X_1 = 0$ and $X_2 = 1/(0-1) = -1$. The driven coordinate does not move; the second resonator, tuned to $\omega = \sqrt{2}$ on its own ($x_2'' + 2x_2 = x_1$), oscillates in antiphase and exactly cancels the drive. This is the principle of the mechanical vibration absorber and of the parallel-resonant **wave trap** in power-line carrier systems: a resonator tuned to an unwanted frequency blocks it.

Engineering Note

In nodal analysis of an RLC network the operator matrix $\mathbf{P}(s)$ is the admittance matrix $\mathbf{Y}(s)$, and $\Delta(s) = \det \mathbf{Y}(s)$, after clearing denominators, is the characteristic polynomial of the network. Circuit simulators find natural frequencies ("pole-zero analysis") by locating the zeros of this determinant, and they find transfer-function zeros from the cofactors, exactly the numerator determinants of Cramer's rule.

12.7 State-variable formulation

Motivation

A single nth-order equation is ideal for hand analysis of small systems, but it scales badly. Eliminating variables from a ten-node network is painful; systems with several inputs and outputs do not fit a scalar equation at all; and numerical ODE solvers only accept first-order systems. The **state-variable** (state-space) formulation writes any linear lumped system as a single first-order *vector* equation. It handles any number of inputs and outputs, it is what simulators and control-design software use internally, and its solution has a closed form, the **matrix exponential**, which is the exact matrix analogue of e^{at} for the scalar equation $\dot{x} = ax$.

State, state equation, output equation

State

The **state** of a system at time t_0 is the smallest set of numbers $x_1(t_0), \dots, x_n(t_0)$ which, together with the input $u(t)$ for $t \geq t_0$, determines the system's behaviour for all $t \geq t_0$. The vector $\mathbf{x} = (x_1, \dots, x_n)^{\mathsf{T}}$ is the **state vector**; n is the **order**.

Linear time-invariant state-space model

$$\begin{aligned}\dot{\mathbf{x}}(t) &= \mathbf{A}\,\mathbf{x}(t) + \mathbf{B}\,\mathbf{u}(t) \quad \text{(state equation)}\\ \mathbf{y}(t) &= \mathbf{C}\,\mathbf{x}(t) + \mathbf{D}\,\mathbf{u}(t) \quad \text{(output equation)}\end{aligned}$$

with $\mathbf{x} \in \mathbb{R}^n$ (state), $\mathbf{u} \in \mathbb{R}^p$ (inputs), $\mathbf{y} \in \mathbb{R}^q$ (outputs), and constant matrices $\mathbf{A}$ ($n \times n$, the **system** or **state matrix**), $\mathbf{B}$ ($n \times p$, input matrix), $\mathbf{C}$ ($q \times n$, output matrix), $\mathbf{D}$ ($q \times p$, feedthrough matrix).

The letter $\mathbf{D}$ for the feedthrough matrix is universal in control texts; context distinguishes it from the operator $D = d/dt$.

Choosing state variables

From a circuit. The natural state variables are the **capacitor voltages** and **inductor currents**: they are continuous in time (finite current cannot change a capacitor voltage instantaneously, finite voltage cannot change an inductor current), they define the stored energy $\tfrac{1}{2}Cv_C^2 + \tfrac{1}{2}Li_L^2$, and the element laws already give their derivatives:

$$\frac{dv_C}{dt} = \frac{i_C}{C}, \qquad \frac{di_L}{dt} = \frac{v_L}{L}.$$

The recipe: (1) choose every independent v_C and i_L as a state; (2) using KCL, KVL and resistor laws, express each i_C and v_L in terms of the states and the sources; (3) divide by C or L. The result is $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$ directly, with no elimination.

From an nth-order ODE. For $y^{(n)} + a_{n-1}y^{(n-1)} + \cdots + a_1 y' + a_0 y = u$ choose the **phase variables** $x_1 = y$, $x_2 = y'$, ..., $x_n = y^{(n-1)}$. Then $\dot{x}_k = x_{k+1}$ for $k < n$ and $\dot{x}_n = -a_0 x_1 - \cdots - a_{n-1}x_n + u$:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 1 & 0 & \cdots & 0 \end{bmatrix}, \quad \mathbf{D} = 0.$$

This is the **companion** (controllable canonical) form. Its characteristic polynomial $\det(s\mathbf{I} - \mathbf{A})$ equals $s^n + a_{n-1}s^{n-1} + \cdots + a_0 = P(s)$, so the eigenvalues of $\mathbf{A}$ are exactly the characteristic roots of the ODE. When the right-hand side contains input derivatives, $P(D)y = (b_{n-1}D^{n-1} + \cdots + b_0)u$, keep the same $\mathbf{A}$ and $\mathbf{B}$, solve $P(D)w = u$ for an auxiliary variable w with states $w, w', \ldots$, and take $y = b_0 w + b_1 w' + \ldots$, i.e. $\mathbf{C} = [\,b_0 \;\; b_1 \;\; \cdots \;\; b_{n-1}\,]$.

State representations are not unique

If $\mathbf{T}$ is any invertible $n \times n$ matrix and $\mathbf{z} = \mathbf{T}^{-1}\mathbf{x}$, then

$$\dot{\mathbf{z}} = \mathbf{T}^{-1}\mathbf{A}\mathbf{T}\,\mathbf{z} + \mathbf{T}^{-1}\mathbf{B}\,\mathbf{u}, \qquad \mathbf{y} = \mathbf{C}\mathbf{T}\,\mathbf{z} + \mathbf{D}\mathbf{u}$$

describes the same input–output behaviour. Similar matrices have the same eigenvalues, so the natural frequencies, stability and transfer function are invariant; only the internal coordinates change. Choosing $\mathbf{T} = \mathbf{V}$ (eigenvectors) gives the **modal** (diagonal) form of Section 12.6.

The matrix exponential

For the scalar equation $\dot{x} = ax$ the solution is $x = e^{at}x(0)$. To mimic this for $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$, define the exponential of a matrix by the same power series.

Matrix exponential

For a square matrix $\mathbf{A}$ and real t,

$$e^{\mathbf{A}t} = \mathbf{I} + \mathbf{A}t + \frac{\mathbf{A}^2t^2}{2!} + \frac{\mathbf{A}^3t^3}{3!} + \cdots = \sum_{k=0}^{\infty} \frac{\mathbf{A}^k t^k}{k!}.$$

In control theory $\boldsymbol{\Phi}(t) = e^{\mathbf{A}t}$ is called the **state-transition matrix**.

The series converges absolutely for every $\mathbf{A}$ and t, because $\|\mathbf{A}^k t^k/k!\| \le (\|\mathbf{A}\|\,|t|)^k/k!$ and the scalar exponential series converges.

Properties of $e^{\mathbf{A}t}$

1. $e^{\mathbf{A}\cdot 0} = \mathbf{I}$.
2. $\dfrac{d}{dt}e^{\mathbf{A}t} = \mathbf{A}e^{\mathbf{A}t} = e^{\mathbf{A}t}\mathbf{A}$.
3. $e^{\mathbf{A}(t+\tau)} = e^{\mathbf{A}t}e^{\mathbf{A}\tau}$ (semigroup property).
4. $\left(e^{\mathbf{A}t}\right)^{-1} = e^{-\mathbf{A}t}$; in particular $e^{\mathbf{A}t}$ is always invertible, with $\det e^{\mathbf{A}t} = e^{t\,\mathrm{tr}\,\mathbf{A}}$.
5. If $\mathbf{A} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{-1}$, then $e^{\mathbf{A}t} = \mathbf{V}e^{\boldsymbol{\Lambda}t}\mathbf{V}^{-1}$ with $e^{\boldsymbol{\Lambda}t} = \mathrm{diag}(e^{\lambda_1 t}, \ldots, e^{\lambda_n t})$.
6. $e^{(\mathbf{A}+\mathbf{B})t} = e^{\mathbf{A}t}e^{\mathbf{B}t}$ **if** $\mathbf{AB} = \mathbf{BA}$, but not in general.

Proof sketches. (2): differentiate the series term by term, $\frac{d}{dt}\frac{\mathbf{A}^k t^k}{k!} = \mathbf{A}\frac{\mathbf{A}^{k-1}t^{k-1}}{(k-1)!}$; $\mathbf{A}$ commutes with its own powers. (3): both sides satisfy $\dot{\mathbf{X}} = \mathbf{A}\mathbf{X}$ in t with value $e^{\mathbf{A}\tau}$ at $t = 0$, so they are equal by uniqueness. (4) follows from (3) with $\tau = -t$. (5): $\mathbf{A}^k = \mathbf{V}\boldsymbol{\Lambda}^k\mathbf{V}^{-1}$, so each term of the series factors. (6): the binomial expansion of $(\mathbf{A}+\mathbf{B})^k$ needs commuting factors. $\square$

Common Pitfall

$e^{\mathbf{A}t}$ is **not** the matrix of entry-wise exponentials $[e^{a_{ik}t}]$, except when $\mathbf{A}$ is diagonal. And $e^{\mathbf{A}t}e^{\mathbf{B}t} \neq e^{(\mathbf{A}+\mathbf{B})t}$ unless $\mathbf{A}$ and $\mathbf{B}$ commute; this matters, for example, in piecewise-linear switched circuits, where each switching interval has its own state matrix and the transition matrices must be multiplied in time order.

Solving the state equation

Solution of $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$

The unique solution with $\mathbf{x}(0) = \mathbf{x}_0$ is

$$\mathbf{x}(t) = \underbrace{e^{\mathbf{A}t}\mathbf{x}_0}_{\text{zero-input}} + \underbrace{\int_0^t e^{\mathbf{A}(t-\tau)}\mathbf{B}\,\mathbf{u}(\tau)\,d\tau}_{\text{zero-state}},$$

and the output is $\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}$.

Derivation (integrating factor). Multiply the state equation by $e^{-\mathbf{A}t}$ and use property 2:

$$e^{-\mathbf{A}t}\big(\dot{\mathbf{x}} - \mathbf{A}\mathbf{x}\big) = \frac{d}{dt}\Big(e^{-\mathbf{A}t}\mathbf{x}\Big) = e^{-\mathbf{A}t}\mathbf{B}\mathbf{u}.$$

Integrate from 0 to t: $e^{-\mathbf{A}t}\mathbf{x}(t) - \mathbf{x}_0 = \int_0^t e^{-\mathbf{A}\tau}\mathbf{B}\mathbf{u}(\tau)\,d\tau$. Multiply by $e^{\mathbf{A}t}$ and use property 3. Uniqueness follows from the general theory (Ch. 11). $\square$

The formula is the vector version of the first-order integrating-factor solution, and it displays the zero-input/zero-state decomposition of Section 12.3 explicitly. The matrix $e^{\mathbf{A}t}\mathbf{B}$ is the **impulse response** of the states, and $\mathbf{C}e^{\mathbf{A}t}\mathbf{B} + \mathbf{D}\delta(t)$ is the impulse response from input to output.

Constant (step) input

If $\mathbf{u}(t) = \mathbf{u}_0$ is constant for $t \geq 0$ and $\mathbf{A}$ is invertible,

$$\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}_0 + \mathbf{A}^{-1}\big(e^{\mathbf{A}t} - \mathbf{I}\big)\mathbf{B}\mathbf{u}_0.$$

If moreover $\mathbf{A}$ is stable (all eigenvalues in the open left half-plane), $e^{\mathbf{A}t} \to \mathbf{0}$ and the steady state is $\mathbf{x}_{ss} = -\mathbf{A}^{-1}\mathbf{B}\mathbf{u}_0$, the solution of $\mathbf{0} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}_0$.

Proof. $\int_0^t e^{\mathbf{A}(t-\tau)}\,d\tau = \int_0^t e^{\mathbf{A}\sigma}\,d\sigma = \mathbf{A}^{-1}\big(e^{\mathbf{A}t} - \mathbf{I}\big)$, by integrating the series term by term (or because the derivative of the right side is $e^{\mathbf{A}t}$). $\square$

Stability in state space. The eigenvalues of $\mathbf{A}$ are the characteristic roots: $\det(s\mathbf{I} - \mathbf{A})$ is the characteristic polynomial. The system is asymptotically stable if and only if every eigenvalue of $\mathbf{A}$ has negative real part, which is the root criterion of Section 12.2 in matrix language.

Frequency response in state space. For $\mathbf{u} = \mathbf{U}e^{j\omega t}$, try $\mathbf{x} = \mathbf{X}e^{j\omega t}$: $j\omega\mathbf{X} = \mathbf{A}\mathbf{X} + \mathbf{B}\mathbf{U}$, so $\mathbf{X} = (j\omega\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}\mathbf{U}$ and

$$\mathbf{H}(j\omega) = \mathbf{C}\big(j\omega\mathbf{I} - \mathbf{A}\big)^{-1}\mathbf{B} + \mathbf{D}.$$

Replacing $j\omega$ by s gives the transfer-function matrix of Ch. 14, where it is also shown that $\mathcal{L}\{e^{\mathbf{A}t}\} = (s\mathbf{I} - \mathbf{A})^{-1}$, the **resolvent**. That identity provides a fourth way to compute $e^{\mathbf{A}t}$: invert $s\mathbf{I} - \mathbf{A}$ and take inverse Laplace transforms entry by entry.

Computing $e^{\mathbf{A}t}$ by hand

Four methods cover practical needs.

1. Diagonalisation (property 5), when $\mathbf{A}$ has n independent eigenvectors.

2. Cayley–Hamilton / Sylvester interpolation. Every square matrix satisfies its own characteristic equation, so $\mathbf{A}^n$ and all higher powers are combinations of $\mathbf{I}, \mathbf{A}, \dots, \mathbf{A}^{n-1}$. Hence

$$e^{\mathbf{A}t} = \alpha_0(t)\mathbf{I} + \alpha_1(t)\mathbf{A} + \cdots + \alpha_{n-1}(t)\mathbf{A}^{n-1},$$

where the scalar functions $\alpha_k(t)$ are fixed by requiring $e^{\lambda t} = \alpha_0 + \alpha_1\lambda + \cdots + \alpha_{n-1}\lambda^{n-1}$ at every eigenvalue λ (and, for a repeated eigenvalue of multiplicity m, also the first $m-1$ derivatives of this identity with respect to λ). For 2×2 matrices this gives closed forms:

$$e^{\mathbf{A}t} = \frac{e^{\lambda_1 t}(\mathbf{A} - \lambda_2\mathbf{I}) - e^{\lambda_2 t}(\mathbf{A} - \lambda_1\mathbf{I})}{\lambda_1 - \lambda_2} \quad (\lambda_1 \neq \lambda_2), \qquad e^{\mathbf{A}t} = e^{\lambda t}\big[\mathbf{I} + (\mathbf{A} - \lambda\mathbf{I})t\big] \quad (\lambda_1 = \lambda_2 = \lambda).$$

The repeated-eigenvalue formula follows because $(\mathbf{A} - \lambda\mathbf{I})^2 = \mathbf{0}$ by Cayley–Hamilton, so $e^{\mathbf{A}t} = e^{\lambda t}e^{(\mathbf{A}-\lambda\mathbf{I})t}$ and the second series terminates after two terms.

3. Jordan / nilpotent splitting. If $\mathbf{A} = \lambda\mathbf{I} + \mathbf{N}$ with $\mathbf{N}$ nilpotent ($\mathbf{N}^m = \mathbf{0}$), then since $\lambda\mathbf{I}$ commutes with everything, $e^{\mathbf{A}t} = e^{\lambda t}\big(\mathbf{I} + \mathbf{N}t + \cdots + \mathbf{N}^{m-1}t^{m-1}/(m-1)!\big)$. This is the matrix origin of the $t^k e^{\lambda t}$ terms of Section 12.2.

4. Laplace transform: $e^{\mathbf{A}t} = \mathcal{L}^{-1}\{(s\mathbf{I} - \mathbf{A})^{-1}\}$ (Ch. 14).

For complex eigenvalues $\sigma \pm j\omega$ of a real matrix, the useful real canonical block is

$$\mathbf{A} = \begin{bmatrix} \sigma & \omega \\ -\omega & \sigma \end{bmatrix} = \sigma\mathbf{I} + \omega\mathbf{J}, \quad \mathbf{J} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \qquad e^{\mathbf{A}t} = e^{\sigma t}\begin{bmatrix} \cos\omega t & \sin\omega t \\ -\sin\omega t & \cos\omega t \end{bmatrix},$$

because $\mathbf{J}^2 = -\mathbf{I}$ makes the series for $e^{\omega\mathbf{J}t}$ split exactly like Euler's formula. Geometrically: a rotation at angular rate ω combined with a uniform contraction at rate $-\sigma$.

Worked examples

Example 12.29 — Companion form of a third-order equation

Write $y''' + 6y'' + 11y' + 6y = u$ in state-space form and confirm the eigenvalues.

With $x_1 = y$, $x_2 = y'$, $x_3 = y''$:

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u, \qquad y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \mathbf{x}.$$

Expanding $\det(s\mathbf{I} - \mathbf{A})$ along the first column: $s\big[s(s+6)+11\big] + 6 = s^3 + 6s^2 + 11s + 6 = (s+1)(s+2)(s+3)$. Eigenvalues $-1, -2, -3$: stable, overdamped, slowest time constant 1 s. If the equation were instead $y''' + 6y'' + 11y' + 6y = 2u' + 5u$, the same $\mathbf{A}$ and $\mathbf{B}$ would be kept with $\mathbf{C} = [\,5 \;\; 2 \;\; 0\,]$.

Example 12.30 — $e^{\mathbf{A}t}$ three ways

Compute $e^{\mathbf{A}t}$ for $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$ (the companion matrix of $y'' + 3y' + 2y$).

Eigenvalues. $\det(s\mathbf{I} - \mathbf{A}) = s^2 + 3s + 2$: $\lambda_1 = -1$, $\lambda_2 = -2$.

Sylvester formula. $\mathbf{A} - \lambda_2 \mathbf{I} = \mathbf{A} + 2\mathbf{I} = \begin{bmatrix} 2 & 1 \\ -2 & -1 \end{bmatrix}$, $\mathbf{A} - \lambda_1 \mathbf{I} = \mathbf{A} + \mathbf{I} = \begin{bmatrix} 1 & 1 \\ -2 & -2 \end{bmatrix}$, $\lambda_1 - \lambda_2 = 1$:

$$e^{\mathbf{A}t} = e^{-t} \begin{bmatrix} 2 & 1 \\ -2 & -1 \end{bmatrix} - e^{-2t} \begin{bmatrix} 1 & 1 \\ -2 & -2 \end{bmatrix} = \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix}.$$

Diagonalisation check. Eigenvectors $\mathbf{v}_1 = (1, -1)$, $\mathbf{v}_2 = (1, -2)$; $\mathbf{V} = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix}$, $\mathbf{V}^{-1} = \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}$. Then $\mathbf{V}\,\mathrm{diag}(e^{-t}, e^{-2t})\mathbf{V}^{-1}$ reproduces the same matrix.

Sanity checks. At $t = 0$ the matrix is $\mathbf{I}$ ✓. The derivative at $t = 0$ is $\begin{bmatrix} -2+2 & -1+2 \\ 2-4 & 1-4 \end{bmatrix} = \mathbf{A}$ ✓. Entry $(1,1)$ is the zero-input y for $y(0) = 1$, $y'(0) = 0$, and $(1,2)$ is the impulse response $e^{-t} - e^{-2t}$ of Example 12.13 ✓. At $t = 0.7$ the entries are 0.74657, 0.24999, -0.49998, -0.00339, identical to a numerical `expm` to all printed digits.

Example 12.31 — Repeated and complex eigenvalues

(a) $\mathbf{A} = \begin{bmatrix} -2 & 1 \\ 0 & -2 \end{bmatrix}$; (b) $\mathbf{A} = \begin{bmatrix} -1 & 2 \\ -2 & -1 \end{bmatrix}$.

(a) Double eigenvalue -2 with only one eigenvector, so $\mathbf{A}$ is not diagonalisable. Write $\mathbf{A} = -2\mathbf{I} + \mathbf{N}$ with $\mathbf{N} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $\mathbf{N}^2 = \mathbf{0}$:

$$e^{\mathbf{A}t} = e^{-2t}(\mathbf{I} + \mathbf{N}t) = e^{-2t} \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}.$$

The te^{-2t} entry is the matrix counterpart of a critically damped response.

(b) This is the real canonical block with $\sigma = -1$, $\omega = 2$ (eigenvalues $-1 \pm 2j$):

$$e^{\mathbf{A}t} = e^{-t} \begin{bmatrix} \cos 2t & \sin 2t \\ -\sin 2t & \cos 2t \end{bmatrix}.$$

At $t = 0.5$: $e^{-0.5}\cos 1 = 0.32771$, $e^{-0.5}\sin 1 = 0.51038$, matching numerical computation. Trajectories spiral into the origin, the phase-plane picture of an underdamped system (Ch. 11).

Example 12.32 — Forced solution by the matrix formula

For $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u$ with $\mathbf{A}$ from Example 12.30, $\mathbf{B} = (0, 1)^{\mathsf{T}}$, $u = 1$ for $t \geq 0$ and $\mathbf{x}_0 = \mathbf{0}$, find $\mathbf{x}(t)$.

$\mathbf{A}^{-1} = \tfrac{1}{2}\begin{bmatrix} -3 & -1 \\ 2 & 0 \end{bmatrix}$, so $\mathbf{A}^{-1}\mathbf{B} = (-\tfrac{1}{2}, 0)^{\mathsf{T}}$. Since $\mathbf{A}^{-1}$ commutes with $e^{\mathbf{A}t}$:

$$\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{A}^{-1}\mathbf{B} - \mathbf{A}^{-1}\mathbf{B} = -\tfrac{1}{2}\begin{bmatrix} 2e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} \end{bmatrix} + \begin{bmatrix} \tfrac{1}{2} \\ 0 \end{bmatrix} = \begin{bmatrix} \tfrac{1}{2} - e^{-t} + \tfrac{1}{2}e^{-2t} \\ e^{-t} - e^{-2t} \end{bmatrix}.$$

The first state is the unit-step response of $y'' + 3y' + 2y = u$ found by scalar methods in Example 12.13 ✓, and the second state is its derivative ✓. The steady state $-\mathbf{A}^{-1}\mathbf{B} = (\tfrac{1}{2}, 0)$ is the DC gain $1/P(0) = 1/2$ ✓.

Example 12.33 — Exact discretisation for simulation

A digital controller samples the system of Example 12.32 every $T = 0.1$ s and holds u constant between samples (zero-order hold). Derive the exact discrete-time model.

Over one interval $[kT, (k+1)T]$ with $u = u[k]$ constant, the step-input formula with $\mathbf{x}_0 = \mathbf{x}[k]$ gives

$$\mathbf{x}[k+1] = \mathbf{\Phi}\,\mathbf{x}[k] + \mathbf{\Gamma}\,u[k], \qquad \mathbf{\Phi} = e^{\mathbf{A}T}, \quad \mathbf{\Gamma} = \mathbf{A}^{-1}\big(e^{\mathbf{A}T} - \mathbf{I}\big)\mathbf{B}.$$

Numerically, from Example 12.30 with $t = 0.1$:

$$\mathbf{\Phi} = \begin{bmatrix} 0.99094 & 0.08611 \\ -0.17221 & 0.73262 \end{bmatrix}, \qquad \mathbf{\Gamma} = \begin{bmatrix} 0.004528 \\ 0.086107 \end{bmatrix}.$$

Iterating ten steps from rest with $u[k] = 1$ gives $x_1[10] = 0.199788$, equal to the exact continuous value $\tfrac{1}{2} - e^{-1} + \tfrac{1}{2}e^{-2} = 0.199788$. The discretisation is exact at the sample instants, not an approximation like Euler's method; this is how simulators and digital-control design tools convert continuous models into difference equations (Ch. 17). The eigenvalues of $\mathbf{\Phi}$ are $e^{\lambda_k T} = e^{-0.1}, e^{-0.2}$: continuous poles map to discrete poles by $z = e^{sT}$.

Engineering Note

The state-space model is the native language of modern tools. SPICE assembles nodal equations that are a (descriptor) state-space system; MATLAB/Python control libraries store models as $(\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D})$; Kalman filters, observers, optimal (LQR) controllers and model-order reduction all operate on these matrices. Numerically, $e^{\mathbf{A}T}$ is computed by scaling and squaring with a Padé approximant (the `expm` routine), not by summing the raw series, which suffers from cancellation when $\|\mathbf{A}T\|$ is large.

12.8 Applications to RLC circuits and dynamic systems

This section applies the whole chapter to the circuits and systems that motivated it. The methods are those of Sections 12.1–12.7; the new ingredients are the circuit-specific steps of **obtaining initial conditions** from the physics of switching and **interpreting** the parameters α, ω_n, ζ and Q in terms of component values.

Initial conditions at a switching instant

A switch operating at $t = 0$ changes the circuit, so the equation valid for $t > 0$ needs initial values at $t = 0^+$. They come from two continuity principles, both consequences of finite power:

Continuity of state variables

In a circuit with finite voltages and currents (no impulsive sources and no loops of capacitors or cut-sets of inductors forced to change state instantaneously):

$$v_C(0^+) = v_C(0^-), \qquad i_L(0^+) = i_L(0^-).$$

Reason. $v_C(0^+) - v_C(0^-) = \frac{1}{C}\int_{0^-}^{0^+} i_C\,dt = 0$ when i_C is bounded; similarly for i_L with v_L. Equivalently, stored energy $\tfrac{1}{2}Cv_C^2$ or $\tfrac{1}{2}Li_L^2$ cannot jump without infinite power. □

All other quantities (resistor currents, inductor voltages, capacitor currents) **may** jump. The derivative initial conditions needed for a second-order equation are found from these jump-allowed quantities at $t = 0^+$:

$$\frac{dv_C}{dt}(0^+) = \frac{i_C(0^+)}{C}, \qquad \frac{di_L}{dt}(0^+) = \frac{v_L(0^+)}{L},$$

where $i_C(0^+)$ and $v_L(0^+)$ are computed by KCL/KVL on the $t = 0^+$ circuit with the known $v_C(0^+)$ and $i_L(0^+)$. In practice, the values before switching come from the DC steady state of the old circuit, in which capacitors are open circuits and inductors are short circuits.

The complete procedure for a switched linear circuit is therefore:

1. Find $v_C(0^-), i_L(0^-)$ from the pre-switching circuit (usually its DC steady state).
2. Carry them across: $v_C(0^+) = v_C(0^-), i_L(0^+) = i_L(0^-)$.
3. Find the derivative initial values from the $t = 0^+$ circuit.
4. Find the final (forced) value from the post-switching steady state.
5. Find the characteristic roots of the post-switching circuit.
6. Assemble $y = y_f + y_n$ and fit the constants.

The series RLC circuit

For the series loop (source v_s, all elements carrying the same current i), with v_C as unknown:

$$v_C'' + \frac{R}{L}v_C' + \frac{1}{LC}v_C = \frac{1}{LC}v_s.$$

Comparing with the standard form $y'' + 2\alpha y' + \omega_n^2 y$:

$$\alpha = \frac{R}{2L}, \qquad \omega_n = \frac{1}{\sqrt{LC}}, \qquad \zeta = \frac{\alpha}{\omega_n} = \frac{R}{2}\sqrt{\frac{C}{L}}, \qquad Q = \frac{1}{2\zeta} = \frac{1}{R}\sqrt{\frac{L}{C}}.$$

The **critical resistance** $R_{\text{crit}} = 2\sqrt{L/C}$ separates underdamped ($R < R_{\text{crit}}$) from overdamped ($R > R_{\text{crit}}$) behaviour. The quantity $Z_0 = \sqrt{L/C}$ is the **characteristic impedance** of the LC pair; damping depends on how the series resistance compares with it. In the series circuit, *more* resistance means *more* damping.

Underdamped step response of the standard second-order system

For $y'' + 2\zeta\omega_n y' + \omega_n^2 y = \omega_n^2 V u(t)$ from rest, with $0 < \zeta < 1$, $\alpha = \zeta\omega_n$, $\omega_d = \omega_n\sqrt{1-\zeta^2}$:

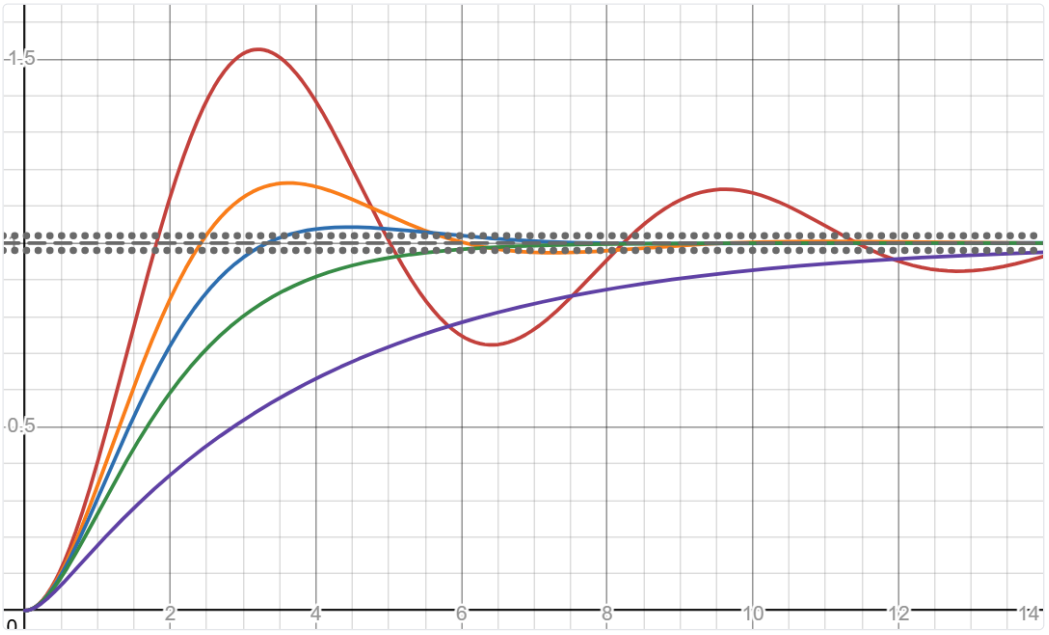
$$y(t) = V\left[1 - e^{-\alpha t}\left(\cos\omega_d t + \frac{\alpha}{\omega_d}\sin\omega_d t\right)\right] = V\left[1 - \frac{e^{-\alpha t}}{\sqrt{1-\zeta^2}}\cos(\omega_d t - \varphi)\right], \quad \sin\varphi = \zeta.$$

Its derivative is $y'(t) = V\dfrac{\omega_n^2}{\omega_d}e^{-\alpha t}\sin\omega_d t$. The response has

- **peak time** $t_p = \pi/\omega_d$,
- **maximum (percent) overshoot** $M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}}$,
- **2% settling time** $t_s \approx 4/\alpha$.

Derivation. The forced value is V. With $y = V + e^{-\alpha t}(A\cos\omega_d t + B\sin\omega_d t)$, $y(0) = 0$ gives $A = -V$, and $y'(0) = -\alpha A + \omega_d B = 0$ gives $B = \alpha A/\omega_d = -V\alpha/\omega_d$. Differentiating, the cosine terms cancel and $y' = V\left(\alpha^2/\omega_d + \omega_d\right)e^{-\alpha t}\sin\omega_d t = V(\omega_n^2/\omega_d)e^{-\alpha t}\sin\omega_d t$. This vanishes first (for $t > 0$) at $\omega_d t = \pi$, the peak. There $\cos\pi = -1$, $\sin\pi = 0$, so $y(t_p) = V(1 + e^{-\alpha\pi/\omega_d})$, and $\alpha/\omega_d = \zeta/\sqrt{1-\zeta^2}$. □

The overshoot depends **only on** ζ: 16.3% at $\zeta = 0.5$, 4.3% at $\zeta = 0.707$, 52.7% at $\zeta = 0.2$, and zero for $\zeta \geq 1$. Inverting, a measured overshoot gives the damping ratio: $\zeta = -\ln M_p/\sqrt{\pi^2 + \ln^2 M_p}$.



Normalised step responses against $\omega_n t$ for $\zeta = 0.2$ (red), 0.5 (orange), 0.707 (blue), 1 (green) and 2 (purple), with the $\pm 2\%$ band dotted. Lower damping rises faster but overshoots and rings; heavy damping is sluggish.

The parallel RLC circuit

For R, L, C in parallel driven by a current source i_s, take the common voltage v (or the inductor current) as unknown. KCL, $v/R + i_L + Cv' = i_s$ with $v = Li_L'$, gives

$$i_L'' + \frac{1}{RC}i_L' + \frac{1}{LC}i_L = \frac{1}{LC}i_s, \qquad \alpha = \frac{1}{2RC}, \qquad \zeta = \frac{1}{2R}\sqrt{\frac{L}{C}}, \qquad Q = R\sqrt{\frac{C}{L}}.$$

The parallel circuit is the **dual** of the series one: here a *larger* resistance means *less* damping, since a large parallel resistor draws little current and dissipates little energy. An ideal tank with $R \to \infty$ is undamped.

Worked examples

Example 12.34 — Series RLC step response with numbers

$R = 100\ \Omega$, $L = 10$ mH, $C = 1\ \mu$F; a 10 V step is applied at $t = 0$ with the circuit at rest. Find $v_C(t)$, $i(t)$, the peak voltage and its time, and the settling time.

Parameters. $\omega_n = 1/\sqrt{10^{-8}} = 10^4$ rad/s; $\alpha = R/2L = 5000\ \text{s}^{-1}$; $\zeta = 0.5$; $\omega_d = 10^4\sqrt{0.75} = 8660$ rad/s; $Q = 1$; $R_{\text{crit}} = 2\sqrt{L/C} = 200\ \Omega$, so the circuit is underdamped.

Initial conditions. $v_C(0^+) = 0$ and $i(0^+) = 0$ (continuity), so $v_C'(0^+) = i/C = 0$.

Response.

$$v_C(t) = 10\left[1 - e^{-5000t}\left(\cos 8660t + 0.5774 \sin 8660t\right)\right]\ \text{V},$$

$$i(t) = C\,v_C' = C \cdot 10 \cdot \frac{\omega_n^2}{\omega_d} e^{-5000t} \sin 8660t = \frac{10}{L\omega_d} e^{-5000t} \sin 8660t = 0.1155\, e^{-5000t} \sin 8660t\ \text{A}.$$

Specifications. $t_p = \pi/\omega_d = 362.8\ \mu$s; $M_p = e^{-\pi \cdot 0.5/0.866} = 0.163$, so $v_{C,\max} = 11.63$ V; $t_s \approx 4/\alpha = 0.8$ ms.

Checks. Dimension of $10/(L\omega_d)$: $\text{V}/(\text{H}\cdot\text{s}^{-1}) = \text{V}/\Omega = \text{A}$ ✓. The capacitor voltage exceeds the supply: the inductor keeps current flowing after v_C reaches 10 V. A designer who needs no overshoot would raise R to at least 200 Ω.

Example 12.35 — Source-free parallel RLC

A parallel RLC circuit with $R = 250\ \Omega$, $L = 10$ mH, $C = 1\ \mu$F has $v(0) = 10$ V on the capacitor and $i_L(0) = 0$. Find $v(t)$.

Parameters. $\omega_n = 10^4$ rad/s, $\alpha = 1/(2RC) = 2000\ \text{s}^{-1}$, $\zeta = 0.2$, $\omega_d = \sqrt{10^8 - 4 \times 10^6} = 9798$ rad/s.

Derivative initial condition. KCL at $t = 0^+$: $i_C = -v/R - i_L = -10/250 - 0 = -0.04$ A, so $v'(0^+) = i_C/C = -4 \times 10^4$ V/s.

Fit. $v = e^{-2000t}(A\cos\omega_d t + B\sin\omega_d t)$ with $A = 10$ and $-2000A + 9798B = -40000$, giving $B = -2.041$:

$$v(t) = e^{-2000t}\left(10\cos 9798t - 2.041 \sin 9798t\right)\ \text{V}.$$

Check. At $t = 100\ \mu$s the formula gives 3.1741 V; a numerical integration of the state equations $v' = -(v/R + i_L)/C$, $i_L' = v/L$ gives 3.1741 V ✓. With $Q = R\sqrt{C/L} = 2.5$, the ringing decays by a factor $e^{-2\pi\zeta/\sqrt{1-\zeta^2}} = 0.277$ per cycle.

Example 12.36 — Switching a source: critically damped transition

A series RLC circuit with $R = 4\ \Omega$, $L = 1$ H, $C = 0.25$ F has been driven by 5 V DC for a long time. At $t = 0$ the source steps to 15 V. Find $v_C(t)$.

Before switching (DC steady state): $i(0^-) = 0$, $v_C(0^-) = 5$ V.

At 0^+. $v_C(0^+) = 5$, $i(0^+) = 0$, so $v_C'(0^+) = i/C = 0$.

After switching. $\omega_n = 1/\sqrt{0.25} = 2$ rad/s, $\zeta = \frac{R}{2}\sqrt{C/L} = 2 \times 0.5 = 1$: critically damped, double root -2. Forced value 15 V.

Fit. $v_C = 15 + (A + Bt)e^{-2t}$: $15 + A = 5$ gives $A = -10$; $v_C'(0) = B - 2A = 0$ gives $B = -20$:

$$v_C(t) = 15 - 10(1 + 2t)e^{-2t}\ \text{V}, \qquad i(t) = Cv_C' = 0.25 \cdot 40\, t\, e^{-2t} = 10\, t\, e^{-2t}\ \text{A}.$$

Check. $v_C' = -20e^{-2t} + 20(1 + 2t)e^{-2t} = 40te^{-2t}$ ✓. The current peaks at $t = 0.5$ s with $i = 5e^{-1} = 1.84$ A, and v_C rises monotonically from 5 V to 15 V without overshoot, as critical damping guarantees. Only the *change* of input (10 V) appears in the transient: the zero-input part (initial 5 V) and the zero-state part combine into a single natural response.

Example 12.37 — DC motor: an electromechanical coupled system

A permanent-magnet DC motor has armature resistance $R = 2\ \Omega$, inductance $L = 0.5$ H, torque/back-EMF constant $K = 0.1\ \text{N m/A} = 0.1\ \text{V s/rad}$, rotor inertia $J = 0.02\ \text{kg m}^2$ and viscous friction $b = 0.01\ \text{N m s}$. A 12 V step is applied at rest. Find the current and speed.

Coupled equations. Electrical (KVL with back-EMF $K\omega$) and mechanical (Newton with motor torque Ki):

$$L\,i' + R\,i + K\omega = v, \qquad J\,\omega' + b\,\omega = K\,i.$$

State-space form. With $\mathbf{x} = (i, \omega)^{\mathsf{T}}$:

$$\dot{\mathbf{x}} = \begin{bmatrix} -R/L & -K/L \\ K/J & -b/J \end{bmatrix}\mathbf{x} + \begin{bmatrix} 1/L \\ 0 \end{bmatrix} v = \begin{bmatrix} -4 & -0.2 \\ 5 & -0.5 \end{bmatrix}\mathbf{x} + \begin{bmatrix} 2 \\ 0 \end{bmatrix} v.$$

Eigenvalues. $\operatorname{tr}\mathbf{A} = -4.5$, $\det\mathbf{A} = 2 + 1 = 3$: $\lambda^2 + 4.5\lambda + 3 = 0$, $\lambda = (-4.5 \pm \sqrt{8.25})/2 = -0.8139,\ -3.6861\ \text{s}^{-1}$. Both real and negative: stable, overdamped. The slow root is mostly mechanical (inertia), the fast one mostly electrical.

Steady state. $\mathbf{x}_{ss} = -\mathbf{A}^{-1}\mathbf{B}\,v$: from $\omega' = 0$, $i = b\omega/K = 0.1\omega$; from $i' = 0$, $2i + 0.1\omega = 12$, so $0.3\omega = 12$, $\omega_{ss} = 40$ rad/s, $i_{ss} = 4$ A.

Complete solution (eigenvector expansion of $\mathbf{x} - \mathbf{x}_{ss}$, fitted to $\mathbf{x}(0) = \mathbf{0}$):

$$\begin{aligned} i(t) &= 4 - 7.222\,e^{-3.686t} + 3.222\,e^{-0.814t}\ \text{A}, \\ \omega(t) &= 40 + 11.334\,e^{-3.686t} - 51.334\,e^{-0.814t}\ \text{rad/s}. \end{aligned}$$

Checks. $i(0) = 4 - 7.222 + 3.222 = 0$ ✓, $\omega(0) = 0$ ✓; $i'(0) = 7.222(3.686) - 3.222(0.814) = 24.0 = 12/L$ ✓ (initially the full voltage appears across L); $\omega'(0) = -11.334(3.686) + 51.334(0.814) = 0.0$ ✓ (no current, no torque at $t = 0$). The starting current overshoots to 5.30 A at $t = 0.81$ s, before back-EMF builds up; the speed settles within 2% after about 5.1 s, governed by the slow root ($4/0.814 = 4.9$ s).

Example 12.38 — RLC in state-variable form with a sinusoidal drive

Write the series RLC circuit of Example 12.34 in state form with states v_C and i_L, and find the steady-state capacitor voltage for $v_s = 10\cos(10^4 t)$ V.

State equations. $Cv_C' = i_L$ and $Li_L' = v_s - Ri_L - v_C$:

$$\frac{d}{dt}\begin{bmatrix} v_C \\ i_L \end{bmatrix} = \begin{bmatrix} 0 & 1/C \\ -1/L & -R/L \end{bmatrix}\begin{bmatrix} v_C \\ i_L \end{bmatrix} + \begin{bmatrix} 0 \\ 1/L \end{bmatrix} v_s, \qquad \det(s\mathbf{I} - \mathbf{A}) = s^2 + \frac{R}{L}s + \frac{1}{LC}.$$

Frequency response at $\omega = \omega_n$. $H(j\omega) = \mathbf{C}(j\omega\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}$ with $\mathbf{C} = [1\ \ 0]$ reduces to $\omega_n^2/(\omega_n^2 - \omega^2 + j2\zeta\omega_n\omega)$. At $\omega = \omega_n$ this is $1/(j2\zeta) = -jQ = -j$. So

$$v_{C,ss}(t) = 10\cos(10^4 t - 90^\circ) = 10\sin(10^4 t)\ \text{V}.$$

At resonance the capacitor voltage is Q times the source amplitude and lags it by exactly 90°. For a high-Q circuit ($Q = 50$, say) the capacitor would see $50\times$ the source voltage, which is why resonant circuits need components rated well above the drive level.

Analogies and generality

The same mathematics governs mechanical, thermal and hydraulic systems through the standard analogies:

Electrical (series)	Mechanical (translation)	Mechanical (rotation)
voltage v	force F	torque T
current i	velocity $\dot{x}$	angular velocity ω
inductance L	mass m	inertia J
resistance R	damping b	rotational damping b
elastance $1/C$	stiffness k	torsional stiffness k
$\zeta = \frac{R}{2}\sqrt{C/L}$	$\zeta = \frac{b}{2\sqrt{km}}$	$\zeta = \frac{b}{2\sqrt{kJ}}$

Any lumped linear system, of whatever physical type, reduces to $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$, and its behaviour is read from the eigenvalues of $\mathbf{A}$. Distributed systems such as transmission lines and heat conduction lead to partial differential equations (Ch. 13); after separation of variables, each spatial mode again obeys a constant-coefficient ODE of exactly the type studied here.

Common Pitfall

Three errors dominate switched-circuit problems: (1) assuming i_C or v_L is continuous (they are not; only v_C and i_L are); (2) using the *pre*-switching circuit's α and ω_n for the *post*-switching transient; (3) setting $v_C'(0^+) = 0$ by habit, when it equals $i_C(0^+)/C$, which is often nonzero (Example 12.35).

Engineering Note

The parameters of this section are design handles. A switching converter's output LC filter is designed for a chosen ω_n (well below the switching frequency) and a damping ratio set by the load resistance and capacitor ESR; a light load (large parallel R) gives low damping and visible ringing on load steps. A gate-drive loop is a series RLC (trace inductance, gate capacitance, gate resistor) and the gate resistor is chosen near $R_{\text{crit}} = 2\sqrt{L/C}$ to suppress ringing without slowing the edge too much. A PLL's loop filter is chosen to give $\zeta \approx 0.7$ for the closed loop.

Summary

Topic	Key formula / fact
Operator form	$P(D)y = r$, $P(s) = a_n s^n + \cdots + a_0$; $P(D)e^{st} = P(s)e^{st}$
Exponential shift	$P(D)[e^{at}f] = e^{at}P(D+a)f$
Distinct roots	$y_h = \sum c_k e^{s_k t}$
Repeated root s_1 (mult. m)	$(c_1 + c_2 t + \cdots + c_m t^{m-1})e^{s_1 t}$
Complex pair $\sigma \pm j\omega$	$e^{\sigma t}(A\cos\omega t + B\sin\omega t)$
Stability	all roots $\mathrm{Re}\{s_k\} < 0$; cubic $s^3 + a_2 s^2 + a_1 s + a_0$: $a_i > 0$, $a_2 a_1 > a_0$
Standard 2nd order	$y'' + 2\zeta\omega_n y' + \omega_n^2 y$; $s = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$
Damping cases	$\zeta > 1$ over, $\zeta = 1$ critical, $0 < \zeta < 1$ under ($\omega_d = \omega_n\sqrt{1-\zeta^2}$), $\zeta = 0$ undamped
Complete response	$y = y_n + y_f = y_{zi} + y_{zs}$; fit ICs to the complete y
Impulse response	homogeneous with $h^{(n-1)}(0^+) = 1/a_n$; $y_{zs} = \int_0^t h(t-\tau)r(\tau)\,d\tau$
Exponential input	$y_p = e^{at}/P(a)$; resonant: $y_p = t^m e^{at}/P^{(m)}(a)$
DC gain	$y_p = K/P(0) = K/a_0$
Sinusoidal steady state	$\mathbf{Y} = H(j\omega)\mathbf{R}$, $H = N(j\omega)/P(j\omega)$
Resonant peak	$\omega_r = \omega_n\sqrt{1-2\zeta^2}$, $M_r = 1/(2\zeta\sqrt{1-\zeta^2})$, $Q = 1/(2\zeta)$
Settling time (2%)	$t_s \approx 4\tau$ (1st order), $t_s \approx 4/(\zeta\omega_n)$ (2nd order)
Step overshoot	$M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}}$, $t_p = \pi/\omega_d$
Coupled systems	$\Delta(s) = \det\mathbf{P}(s)$; $\Delta(D)y_i = \det\mathbf{P}_i(D)$; deg Δ constants
State space	$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$, $\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}$
Matrix exponential	$e^{\mathbf{A}t} = \sum \mathbf{A}^k t^k/k! = \mathbf{V}e^{\mathbf{\Lambda}t}\mathbf{V}^{-1}$
State solution	$\mathbf{x} = e^{\mathbf{A}t}\mathbf{x}_0 + \int_0^t e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau)\,d\tau$
Step input	$\mathbf{x} = e^{\mathbf{A}t}\mathbf{x}_0 + \mathbf{A}^{-1}(e^{\mathbf{A}t} - \mathbf{I})\mathbf{B}\mathbf{u}_0$; $\mathbf{x}_{ss} = -\mathbf{A}^{-1}\mathbf{B}\mathbf{u}_0$
2×2 Sylvester	$e^{\mathbf{A}t} = [e^{\lambda_1 t}(\mathbf{A} - \lambda_2\mathbf{I}) - e^{\lambda_2 t}(\mathbf{A} - \lambda_1\mathbf{I})]/(\lambda_1 - \lambda_2)$
Discretisation (ZOH)	$\mathbf{\Phi} = e^{\mathbf{A}T}$, $\mathbf{\Gamma} = \mathbf{A}^{-1}(e^{\mathbf{A}T} - \mathbf{I})\mathbf{B}$
Series RLC	$\alpha = R/2L$, $\omega_n = 1/\sqrt{LC}$, $\zeta = \frac{R}{2}\sqrt{C/L}$, $Q = \frac{1}{R}\sqrt{L/C}$
Parallel RLC	$\alpha = 1/(2RC)$, $\zeta = \frac{1}{2R}\sqrt{L/C}$, $Q = R\sqrt{C/L}$
Switching	$v_C(0^+) = v_C(0^-)$, $i_L(0^+) = i_L(0^-)$; $v_C'(0^+) = i_C(0^+)/C$, $i_L'(0^+) = v_L(0^+)/L$

Further reading

- E. Kreyszig, *Advanced Engineering Mathematics*, 10th ed., Wiley — Chapters 2–4 (constant-coefficient equations, undetermined coefficients, systems and the matrix exponential).
- W. E. Boyce, R. C. DiPrima and D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 11th ed., Wiley — Chapters 3, 4 and 7, including fundamental matrices and the matrix exponential.
- C. K. Alexander and M. N. O. Sadiku, *Fundamentals of Electric Circuits*, 7th ed., McGraw-Hill — Chapters 7–8 on first- and second-order circuits, switching and initial conditions.
- C.-T. Chen, *Linear System Theory and Design*, 4th ed., Oxford University Press — state-space models, the matrix exponential and discretisation in depth.

Chapter 13 — Partial Differential Equations

Prerequisites: Ch. 6 (Multivariable Calculus), Ch. 7 (Vector Calculus), Ch. 11 and Ch. 12 (Ordinary and Linear Differential Equations); Ch. 15 (Fourier Series) is used heavily from Section 13.4 on and may be read in parallel.

Learning outcomes: After this chapter you can

- derive the telegrapher's equations of a transmission line and recognise the wave, diffusion and Laplace equations as its limiting cases;
- classify a second-order linear PDE as elliptic, parabolic or hyperbolic, find its characteristics and say what kind of auxiliary data makes the problem well posed;
- state Dirichlet, Neumann, Robin and periodic boundary conditions, homogenise non-homogeneous ones, and translate circuit terminations into boundary conditions;
- solve the heat, wave and Laplace equations on intervals, rectangles, disks and annuli by separation of variables and Fourier-series expansion;
- use d'Alembert's solution, characteristics and the method of images to follow pulses and reflections on lines and strings;
- apply these solutions to electrostatic fields, semiconductor diffusion, skin effect and transient heat transfer, and estimate how many series terms a given accuracy needs.

An ordinary differential equation describes a quantity that depends on one variable — usually time. Most physical quantities in electrical engineering depend on *where* as well as *when*: the voltage along a cable, the temperature across a power transistor die, the potential between the plates of a capacitor, the concentration of boron atoms below a silicon surface, the current density inside a conductor carrying high-frequency current. Each of these obeys a **partial differential equation** (PDE), a relation between an unknown function of several variables and its partial derivatives.

This chapter concentrates on the three classical second-order linear PDEs — the **heat (diffusion) equation**, the **wave equation** and **Laplace's equation** — because between them they cover almost every field problem met in an EE curriculum, and because they are the prototypes of the three types into which all second-order linear PDEs fall. The main solution technique is **separation of variables**, which turns one PDE into several ODEs (Ch. 11, Ch. 12) and then reassembles the solution as a **Fourier series** (Ch. 15). For infinite domains we also use d'Alembert's travelling-wave solution and similarity solutions built from the error function. Transform methods for PDEs (Laplace transform in time, Fourier transform in space) are touched on only briefly here; they are developed in Ch. 14 and Ch. 16, and numerical (finite-difference) solution in Ch. 22.

Motivation: the transmission line

We start with one physical system that contains all three classical equations as special cases. A two-conductor transmission line (coaxial cable, twisted pair, microstrip trace) is described by four constants **per unit length**: series resistance R (Ω/m), series inductance L ($\mathrm{H/m}$), shunt conductance G ($\mathrm{S/m}$) and shunt capacitance C ($\mathrm{F/m}$). Let $v(x,t)$ be the voltage between the conductors and $i(x,t)$ the current in the forward conductor at position x and time t.

Cut out a short piece of length Δx. Its equivalent circuit is a series $R\Delta x$, $L\Delta x$ followed by a shunt $G\Delta x$, $C\Delta x$. Kirchhoff's voltage law around the piece gives

$$v(x,t) - v(x+\Delta x,t) = R\,\Delta x\, i(x,t) + L\,\Delta x\,\frac{\partial i}{\partial t}(x,t),$$

and Kirchhoff's current law at the node $x+\Delta x$ gives

$$i(x,t) - i(x+\Delta x,t) = G\,\Delta x\, v(x+\Delta x,t) + C\,\Delta x\,\frac{\partial v}{\partial t}(x+\Delta x,t).$$

Divide by Δx and let $\Delta x \to 0$. The left-hand sides become $-\partial v/\partial x$ and $-\partial i/\partial x$:

Telegrapher's equations

$$\frac{\partial v}{\partial x} = -R\,i - L\,\frac{\partial i}{\partial t}, \qquad \frac{\partial i}{\partial x} = -G\,v - C\,\frac{\partial v}{\partial t}.$$

These are two coupled *first-order* PDEs. To get a single equation for v, differentiate the first with respect to x and substitute the second (the mixed partials $i_{xt} = i_{tx}$ commute for smooth i, Ch. 6):

$$\frac{\partial^2 v}{\partial x^2} = -R\frac{\partial i}{\partial x} - L\frac{\partial}{\partial t}\frac{\partial i}{\partial x} = R\Big(Gv + C\frac{\partial v}{\partial t}\Big) + L\Big(G\frac{\partial v}{\partial t} + C\frac{\partial^2 v}{\partial t^2}\Big).$$

Telegrapher's (transmission-line) equation

$$\frac{\partial^2 v}{\partial x^2} = LC\,\frac{\partial^2 v}{\partial t^2} + (RC + GL)\,\frac{\partial v}{\partial t} + RG\,v.$$

The current $i(x,t)$ satisfies the same equation.

Dimensional check: LC has units $\mathrm{s^2/m^2}$, so $LC\,v_{tt}$ has units $\mathrm{V/m^2}$, as does v_{xx}. RC and GL have units $\mathrm{s/m^2}$, and RG has units $\mathrm{m^{-2}}$. Every term is in $\mathrm{V/m^2}$.

Three limiting cases give the three classical equations:

1. **Lossless line** ($R = G = 0$): $v_{xx} = LC\,v_{tt}$, or

$$\frac{\partial^2 v}{\partial t^2} = c^2\frac{\partial^2 v}{\partial x^2}, \qquad c = \frac{1}{\sqrt{LC}}.$$

This is the **wave equation**. Signals travel undistorted at speed c. For a coaxial cable with polyethylene dielectric, $c \approx 2\times10^8\ \mathrm{m/s}$. 2. **RC line** ($L = G = 0$): the long submarine telegraph cables analysed by Kelvin in the 1850s, and on-chip interconnect today, are dominated by R and C. Then

$$\frac{\partial v}{\partial t} = \frac{1}{RC}\frac{\partial^2 v}{\partial x^2}.$$

This is the **diffusion (heat) equation** with diffusivity $1/(RC)$ in m^2/s. Signals do not travel; they *spread*, and the delay grows as the square of the length. 3. **DC steady state, lossless shunt** ($\partial/\partial t = 0$, $G = 0$): $v_{xx} = 0$, so v is linear in x. In two or three space dimensions the analogous statement $\nabla^2 V = 0$ is **Laplace's equation**, which governs electrostatic potential in charge-free regions.

The rest of the chapter studies each of these, first in general and then with the specific techniques that solve them.

13.1 Classification of PDEs

Basic vocabulary

Partial differential equation

A **partial differential equation** for an unknown function $u(x_1, \dots, x_n)$ of $n \geq 2$ independent variables is a relation

$$F\big(x_1, \dots, x_n,\ u,\ u_{x_1}, \dots, u_{x_n},\ u_{x_1x_1},\ u_{x_1x_2}, \dots\big) = 0$$

among the variables, the function and finitely many of its partial derivatives. Subscripts denote partial derivatives: $u_x = \partial u/\partial x$, $u_{xt} = \partial^2 u/\partial t\,\partial x$.

- The **order** is the order of the highest derivative present.
- The PDE is **linear** if F is linear in u and all its derivatives, with coefficients that depend only on the independent variables.
- A linear PDE is **homogeneous** if every term contains u or one of its derivatives; a term depending only on the independent variables (a *source* or *forcing* term) makes it **non-homogeneous**.
- The PDE is **quasilinear** if it is linear in the highest-order derivatives, with coefficients that may depend on u and lower derivatives.

So the telegrapher's equation is second order, linear and homogeneous. The heat equation with internal Joule heating, $u_t = \alpha u_{xx} + q(x)/(\rho c_p)$, is linear but non-homogeneous. The inviscid Burgers equation $u_t + u\,u_x = 0$, which describes shock formation, is first order and quasilinear but not linear. Poisson's equation $\nabla^2 V = -\rho_v/\varepsilon$ is linear, second order and non-homogeneous whenever the charge density ρ_v is non-zero.

Linearity is what makes the methods of this chapter work, through the same principle that underlies circuit analysis.

Superposition principle

Let $\mathcal{L}$ be a linear differential operator, so that $\mathcal{L}[c_1 u_1 + c_2 u_2] = c_1\mathcal{L}[u_1] + c_2\mathcal{L}[u_2]$ for constants c_1, c_2.

1. If $u_1, \dots, u_N$ solve the homogeneous equation $\mathcal{L}[u] = 0$, so does every linear combination $\sum c_k u_k$.
2. If u_p solves $\mathcal{L}[u] = f$, then every solution has the form $u = u_p + u_h$ with $\mathcal{L}[u_h] = 0$.
3. Under convergence conditions (uniform convergence of the series and of the series of the derivatives that appear in $\mathcal{L}$), an *infinite* series $\sum_{k=1}^{\infty} c_k u_k$ of solutions of $\mathcal{L}[u] = 0$ is again a solution.

Part 3 is the one that separation of variables relies on, and it is the reason convergence of Fourier series (Ch. 15) matters to us.

Second-order linear equations in two variables

The general second-order linear PDE in two independent variables x and y (one of which may be time) is

$$a_{11}\,u_{xx} + 2a_{12}\,u_{xy} + a_{22}\,u_{yy} + b_1\,u_x + b_2\,u_y + c_0\,u = f,$$

where the coefficients may depend on (x, y). The factor 2 in front of a_{12} is a convention that makes the formulas below clean. The second-order part, the **principal part**, controls the qualitative behaviour of solutions: how information travels, whether solutions are smooth, and what boundary data are appropriate. Associate with it the quadratic form $a_{11}\xi^2 + 2a_{12}\xi\eta + a_{22}\eta^2$, i.e. the symmetric matrix

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{pmatrix}.$$

Type of a second-order linear PDE

Let the **discriminant** be $\Delta = a_{12}^2 - a_{11}a_{22} = -\det \mathbf{A}$. At a point (x, y) the equation is

- **hyperbolic** if $\Delta > 0$ (eigenvalues of $\mathbf{A}$ of opposite sign),
- **parabolic** if $\Delta = 0$ (one eigenvalue zero),
- **elliptic** if $\Delta < 0$ (eigenvalues of the same sign).

If the coefficients vary, the type can change from region to region.

The names come from the conic $a_{11}\xi^2 + 2a_{12}\xi\eta + a_{22}\eta^2 = 1$, which is a hyperbola, parabola or ellipse in the three cases. The lower-order terms do not affect the type.

The three prototypes:

Equation	Standard form	a_{11}, a_{12}, a_{22}	Δ	Type
Wave	$u_{tt} - c^2 u_{xx} = 0$ (variables x, t)	$-c^2,\ 0,\ 1$	$c^2 > 0$	hyperbolic
Heat	$u_t - \alpha u_{xx} = 0$	$-\alpha,\ 0,\ 0$	0	parabolic
Laplace	$u_{xx} + u_{yy} = 0$	$1,\ 0,\ 1$	$-1 < 0$	elliptic

The telegrapher's equation $v_{xx} - LC\, v_{tt} - (RC + GL)v_t - RGv = 0$ has $a_{11} = 1$, $a_{12} = 0$, $a_{22} = -LC$, so $\Delta = LC > 0$: it is hyperbolic whenever $L \neq 0$. The loss terms are lower order and do not change the type — they cause attenuation and dispersion, but signals still have a finite front speed $1/\sqrt{LC}$. In the RC limit $L = 0$ the second-order time derivative disappears, $\Delta = 0$, and the type jumps to parabolic: information then spreads instantly (though with exponentially small amplitude) to all x.

What the type tells you

	Elliptic	Parabolic	Hyperbolic
Prototype physics	equilibrium, steady state, electrostatics	diffusion, heat conduction, RC lines	waves, lossless lines, vibrations
Typical data	boundary values on a closed curve	initial value + boundary values	initial value and initial velocity + boundary values
Speed of information	— (no time)	infinite	finite, along characteristics
Smoothness	solutions as smooth as the coefficients, even with rough boundary data	rough initial data smoothed instantly for $t > 0$	discontinuities propagate, never smoothed
Time reversal	—	ill posed backwards	reversible

Characteristics and canonical forms

The type can be understood through **characteristic curves**. Ask: along which curves $y = y(x)$ is it impossible to determine the second derivatives of u from the values of u and its first derivatives on the curve? Writing $d(u_x) = u_{xx}\,dx + u_{xy}\,dy$ and $d(u_y) = u_{xy}\,dx + u_{yy}\,dy$ together with the PDE gives a 3×3 linear system for (u_{xx}, u_{xy}, u_{yy}). Its determinant vanishes when

$$a_{11}\,(dy)^2 - 2a_{12}\,dx\,dy + a_{22}\,(dx)^2 = 0, \qquad \text{i.e.} \qquad \frac{dy}{dx} = \frac{a_{12} \pm \sqrt{\Delta}}{a_{11}}.$$

Characteristics

The solution curves of $dy/dx = (a_{12} \pm \sqrt{\Delta})/a_{11}$ are the **characteristics** of the PDE. A hyperbolic equation has two real families, a parabolic equation one, and an elliptic equation none (the roots are complex).

Data given on a characteristic cannot be continued off it uniquely; this is why initial data for the wave equation are given on a line $t = 0$ that is *not* characteristic, and why discontinuities in hyperbolic problems travel *along* characteristics.

Changing to characteristic coordinates simplifies the principal part.

Canonical forms

With constant coefficients (or locally, with variable ones) there is a change of variables $\xi = \xi(x, y)$, $\eta = \eta(x, y)$ that reduces the principal part to

- hyperbolic: $u_{\xi\eta} + (\text{lower order}) = 0$, with $\xi = \text{const}$, $\eta = \text{const}$ the two characteristic families;
- parabolic: $u_{\eta\eta} + (\text{lower order}) = 0$, with $\xi = \text{const}$ the single characteristic family and η any independent function;
- elliptic: $u_{\xi\xi} + u_{\eta\eta} + (\text{lower order}) = 0$, with ξ, η the real and imaginary parts of the complex characteristic.

Proof sketch (hyperbolic, constant coefficients). Let $m_1 \neq m_2$ be the real roots of $a_{11}m^2 - 2a_{12}m + a_{22} = 0$; these are exactly the characteristic slopes. Since $m_1 + m_2 = 2a_{12}/a_{11}$ and $m_1 m_2 = a_{22}/a_{11}$, the principal part factors as

$$a_{11}\partial_x^2 + 2a_{12}\partial_x\partial_y + a_{22}\partial_y^2 = a_{11}(\partial_x + m_1\partial_y)(\partial_x + m_2\partial_y).$$

Put $\xi = y - m_1 x$ and $\eta = y - m_2 x$. By the chain rule, $\partial_x + m_1\partial_y = (m_1 - m_2)\,\partial_\eta$ and $\partial_x + m_2\partial_y = (m_2 - m_1)\,\partial_\xi$ (each factor kills functions of "its own" characteristic variable). The principal part is therefore $-a_{11}(m_1 - m_2)^2\,\partial_\xi\partial_\eta$, a non-zero constant times $\partial_\xi\partial_\eta$. $\square$

Example 13.1 — Classifying and solving a constant-coefficient equation

Classify $u_{xx} + 4u_{xy} + 3u_{yy} = 0$ and find its general solution.

Type. $a_{11} = 1$, $2a_{12} = 4 \Rightarrow a_{12} = 2$, $a_{22} = 3$. $\Delta = 4 - 3 = 1 > 0$: hyperbolic everywhere.

Characteristics. $dy/dx = (2 \pm 1)/1$, so $dy/dx = 3$ or 1. The families are $y - 3x = \text{const}$ and $y - x = \text{const}$.

Canonical form. Put $\xi = y - x$, $\eta = y - 3x$. Then $u_x = -u_\xi - 3u_\eta$, $u_y = u_\xi + u_\eta$, and

$$\begin{aligned} u_{xx} &= u_{\xi\xi} + 6u_{\xi\eta} + 9u_{\eta\eta}, \\ u_{xy} &= -u_{\xi\xi} - 4u_{\xi\eta} - 3u_{\eta\eta}, \\ u_{yy} &= u_{\xi\xi} + 2u_{\xi\eta} + u_{\eta\eta}. \end{aligned}$$

So $u_{xx} + 4u_{xy} + 3u_{yy} = (1 - 4 + 3)u_{\xi\xi} + (6 - 16 + 6)u_{\xi\eta} + (9 - 12 + 3)u_{\eta\eta} = -4u_{\xi\eta}$. The PDE becomes $u_{\xi\eta} = 0$.

General solution. Integrating in η gives $u_\xi = h(\xi)$; integrating in ξ gives $u = F(\xi) + G(\eta)$:

$$u(x, y) = F(y - x) + G(y - 3x)$$

for arbitrary twice-differentiable F, G.

Check. For $u = G(y - 3x)$: $u_{xx} = 9G''$, $u_{xy} = -3G''$, $u_{yy} = G''$, and $9 - 12 + 3 = 0$. For $u = F(y - x)$: $1 - 4 + 3 = 0$. ✓

Notice the analogy with ODEs: a second-order ODE has two arbitrary *constants*; a second-order PDE in two variables typically has two arbitrary *functions*. This is why PDE solutions are pinned down by functions (initial profiles, boundary profiles) rather than by numbers.

Example 13.2 — A parabolic equation

Classify $u_{xx} + 2u_{xy} + u_{yy} = 0$ and find its general solution.

$a_{11} = a_{12} = a_{22} = 1$, $\Delta = 1 - 1 = 0$: parabolic. The single characteristic family is $dy/dx = 1$, i.e. $\xi = y - x$. Take $\eta = x$ as the second variable. Then $\partial_x = -\partial_\xi + \partial_\eta$, $\partial_y = \partial_\xi$, and the principal part $(\partial_x + \partial_y)^2$ becomes ∂_η^2. The PDE is $u_{\eta\eta} = 0$, so $u = F(\xi) + \eta\, G(\xi)$:

$$u(x, y) = F(y - x) + x\, G(y - x).$$

Check the second piece: with $u = xG(y - x)$, $u_{xx} = -2G' + xG''$, $u_{xy} = G' - xG''$, $u_{yy} = xG''$. Then $u_{xx} + 2u_{xy} + u_{yy} = (-2G' + 2G') + (x - 2x + x)G'' = 0$. ✓

The repeated factor x is the PDE analogue of the repeated-root solution te^{rt} of a constant-coefficient ODE (Ch. 12).

Example 13.3 — A PDE that changes type

The **Tricomi equation** $y\, u_{xx} + u_{yy} = 0$ models transonic flow. Here $a_{11} = y$, $a_{12} = 0$, $a_{22} = 1$, so $\Delta = -y$.

- For $y > 0$: $\Delta < 0$, elliptic (subsonic region).
- For $y = 0$: parabolic (sonic line).
- For $y < 0$: hyperbolic, with characteristics $dy/dx = \pm\sqrt{-y}/y$, i.e. $\tfrac{2}{3}(-y)^{3/2} = \pm x + \text{const}$.

The same physical device can therefore require different kinds of boundary data in different regions — an important warning when setting up numerical solvers.

Example 13.4 — Classifying the RC-line and the lossy line

(a) On-chip interconnect with $R = 2 \times 10^5\ \Omega/\text{m}$ (that is, $0.2\ \Omega/\mu\text{m}$) and $C = 2 \times 10^{-10}\ \text{F/m}$ ($0.2\ \text{fF}/\mu\text{m}$) and negligible L: $v_t = v_{xx}/(RC)$ with $RC = 4 \times 10^{-5}\ \text{s/m}^2$. Parabolic, diffusivity $1/(RC) = 2.5 \times 10^4\ \text{m}^2/\text{s}$. A diffusive "front" travels a distance ℓ in a time of order $RC\ell^2$; for $\ell = 1$ mm this is 4×10^{-11} s $= 40$ ps, for $\ell = 10$ mm it is 4 ns — a hundred times longer for ten times the length.

(b) If the same line has $L = 4 \times 10^{-7}\ \text{H/m}$, the equation is hyperbolic with front speed $1/\sqrt{LC} = 1/\sqrt{8 \times 10^{-17}} \approx 1.12 \times 10^8\ \text{m/s}$. The ratio of the loss term to the inertial term at frequency ω is $RC\,\omega/(LC\,\omega^2) = R/(\omega L)$; with $\omega = 2\pi \times 1$ GHz, $\omega L \approx 2.5 \times 10^3\ \Omega/\text{m}$, so $R/(\omega L) \approx 80$ and the line behaves diffusively at 1 GHz despite being formally hyperbolic. Type tells you the *ultimate* (high-frequency) behaviour; the size of the terms tells you what you will actually see.

Common pitfall — type is a property of the principal part only

Do not include first-order or zeroth-order terms when computing Δ. $u_t = \alpha u_{xx} - \beta u_x$ (diffusion with drift, as for minority carriers in an electric field) is just as parabolic as the plain heat equation. Also remember the factor 2: in $u_{xx} + 4u_{xy} + 3u_{yy}$, $a_{12} = 2$, not 4.

Engineering note — choosing a numerical method by type

Field solvers exploit the type. Elliptic problems (electrostatic capacitance extraction) are solved as one large sparse linear system for all unknowns at once. Parabolic problems (thermal transients, process simulation of diffusion) march in time but need implicit schemes to avoid tiny time steps. Hyperbolic problems (FDTD electromagnetic simulation, signal-integrity of long lines) march in time with a step limited by the Courant condition $c\,\Delta t \le \Delta x$, which is a statement that the numerical domain of dependence must contain the characteristic one (Ch. 22).

13.2 Boundary and initial conditions

A PDE on its own has far too many solutions — Example 13.1 produced two *arbitrary functions*. To single out the physically relevant one we must say what happens on the boundary of the region and, for evolution equations, at the initial instant. Getting these auxiliary conditions right is half the modelling work, and the wrong kind or the wrong number of conditions produces either no solution or a meaningless one.

Initial conditions

For an equation that is kth order in time, we prescribe the function and its first $k-1$ time derivatives at $t=0$, exactly as for an ODE.

- Heat equation (u_t first order in t): one initial condition, $u(x,0)=f(x)$ — the initial temperature profile.
- Wave and telegrapher's equations (u_{tt}): two, $u(x,0)=f(x)$ and $u_t(x,0)=g(x)$ — initial displacement and initial velocity. On a transmission line, specifying $v(x,0)$ and $i(x,0)$ is equivalent, because the second telegrapher equation gives v_t from i_x and v: $C\,v_t=-i_x-Gv$.
- Laplace's equation has no time variable and takes no initial condition.

Boundary conditions

Let the spatial region be Ω with boundary $\partial\Omega$ and outward unit normal $\hat{\mathbf{n}}$, and write $\partial u/\partial n=\hat{\mathbf{n}}\cdot\nabla u$ for the outward normal derivative. In one dimension $\Omega=(0,L)$, $\partial\Omega=\{0,L\}$, and $\partial u/\partial n$ is $-u_x$ at $x=0$ and $+u_x$ at $x=L$.

Classical boundary conditions

- **Dirichlet (first kind):** the value is prescribed, $u=h$ on $\partial\Omega$. *Examples:* an electrode held at a fixed potential; a surface clamped to a heat sink at fixed temperature; a short-circuited transmission-line end, $v(\ell,t)=0$.
- **Neumann (second kind):** the normal derivative is prescribed, $\partial u/\partial n=h$. *Examples:* a thermally insulated surface, $\partial u/\partial n=0$ (no heat flux); a prescribed heat flux $-k\,\partial u/\partial n=q_s$; the free end of a string; an open-circuited line end, $i(\ell,t)=0$, which by the first telegrapher equation (lossless case) means $v_x(\ell,t)=-L\,i_t=0$.
- **Robin (third kind, mixed):** a linear combination, $\beta\,u+\partial u/\partial n=h$. *Example:* Newton cooling at a surface in contact with a fluid at temperature u_∞, $-k\,\partial u/\partial n=h_c(u-u_\infty)$, where h_c is the convective heat-transfer coefficient ($\mathrm{W\,m^{-2}K^{-1}}$).
- **Periodic:** on an interval that represents a closed loop (angle ϕ around a ring or a cylinder), $u(-\pi)=u(\pi)$ and $u_\phi(-\pi)=u_\phi(\pi)$.

A boundary condition is **homogeneous** if $h\equiv 0$, so that the zero function satisfies it and sums of functions satisfying it also satisfy it.

Different parts of the boundary may carry different kinds of condition. For a rectangle one might hold one edge at a potential (Dirichlet) while the others are insulating symmetry planes (Neumann).

Initial–boundary-value problem

An **initial–boundary-value problem** (IBVP) consists of an evolution PDE on $\Omega\times(0,\infty)$, initial conditions at $t=0$, and boundary conditions on $\partial\Omega$ for all $t>0$. A **boundary-value problem** (BVP) consists of a steady-state PDE (typically elliptic) on Ω with conditions on $\partial\Omega$.

The model problem for the heat equation on a rod of length L with ends held at zero temperature is

$$\begin{aligned} &u_t=\alpha\,u_{xx}, && 0<x<L,\ t>0,\\ &u(0,t)=0,\quad u(L,t)=0, && t>0,\\ &u(x,0)=f(x), && 0\le x\le L. \end{aligned}$$

Well-posedness

Well-posed problem (Hadamard)

A problem is **well posed** if

1. a solution **exists**,
2. the solution is **unique**, and
3. the solution **depends continuously on the data**: small changes in the initial and boundary data produce small changes in the solution.

Condition 3 is the one engineers care most about, because measured data always carry error. A problem that fails it is **ill posed**, and any computed answer is at the mercy of noise.

The right auxiliary data depend on the type:

Type	Well-posed data	Ill-posed data
Elliptic (Laplace)	one condition (D, N or R) at each point of a *closed* boundary	Cauchy data (u and u_n) on part of the boundary only
Parabolic (heat)	initial value + one condition at each boundary point, solved *forward* in time	solving *backward* in time from a final state
Hyperbolic (wave)	u and u_t at $t=0$ + one condition at each boundary point	boundary values on a closed curve in the (x,t) plane

Example 13.5 — Hadamard's example: an ill-posed Laplace problem

Consider $u_{xx} + u_{yy} = 0$ for $y > 0$ with **Cauchy data** on the line $y = 0$: $u(x,0) = 0$, $u_y(x,0) = \dfrac{\sin(nx)}{n}$.

Try $u = X(x)Y(y)$ with $X = \sin(nx)$; then $Y'' = n^2 Y$, $Y(0) = 0$, $Y'(0) = 1/n$, so $Y = \sinh(ny)/n^2$:

$$u_n(x,y) = \frac{\sin(nx)\,\sinh(ny)}{n^2}.$$

As $n \to \infty$ the data tend to zero uniformly ($|u_y(x,0)| \le 1/n$), and the zero data have solution $u \equiv 0$. But at $y = 1$, $\max_x |u_n| = \sinh(n)/n^2 \approx e^n/(2n^2)$; for $n = 20$ this is about 6.1×10^5. Data of size 0.05 produce a solution of size 6×10^5. Continuous dependence fails: Laplace's equation with Cauchy data is ill posed.

Engineering meaning: trying to reconstruct the potential deep inside a region from measurements of V and $\partial V/\partial n$ on one face (an "inverse" problem, as in impedance tomography) amplifies measurement noise exponentially and needs regularisation.

Backward heat conduction is ill posed for the same reason: a mode $\sin(n\pi x/L)$ decays like $e^{-\alpha(n\pi/L)^2 t}$ forward in time, so running time backwards multiplies high-frequency noise by $e^{+\alpha(n\pi/L)^2 t}$ (see Section 13.4).

How many conditions?

A useful rule: a PDE that is of order 2 in a spatial variable needs **one condition at each end** of that variable's interval; a PDE of order k in time needs k **initial** conditions. Heat equation on $(0, L)$: two boundary conditions + one initial condition. Wave equation on $(0, L)$: two boundary conditions + two initial conditions. Laplace on a rectangle: one condition on each of the four sides. On an infinite interval the boundary conditions are replaced by a growth or decay condition ("u bounded as $|x| \to \infty$").

Compatibility

The initial and boundary data must agree where they meet. For the model heat problem, continuity at the corner $(x,t) = (0,0)$ requires $f(0) = 0$. If $f(0) \neq 0$ (a rod at 100 °C suddenly clamped to 0 °C ends), the solution still exists and is smooth for $t > 0$, but the series solution converges non-uniformly near the corner and the boundary condition is attained only as $t \to 0^+$. For the wave equation, incompatible data produce a discontinuity (a "kink") that travels along the characteristic from the corner and never smooths out.

Homogenising non-homogeneous boundary conditions

Separation of variables (Section 13.3) needs **homogeneous** boundary conditions. When the boundary data are constant (or independent of time), subtract a steady-state solution.

Steady-state subtraction

Suppose $u_t = \alpha u_{xx}$ on $0 < x < L$ with $u(0,t) = T_1$, $u(L,t) = T_2$ (constants) and $u(x,0) = f(x)$. Let $u_s(x)$ be the steady state: $u_s'' = 0$, $u_s(0) = T_1$, $u_s(L) = T_2$, so

$$u_s(x) = T_1 + (T_2 - T_1)\frac{x}{L}.$$

Then the **transient** $w = u - u_s$ satisfies $w_t = \alpha w_{xx}$, $w(0,t) = w(L,t) = 0$, $w(x,0) = f(x) - u_s(x)$ — a problem with homogeneous boundary conditions.

The proof is one line: u_s is independent of t and has $u_s'' = 0$, so it satisfies the PDE, and subtracting it removes the boundary values. The same idea works for Laplace's equation (split a problem with data on all four sides into four problems with data on one side each, Section 13.6), for the wave equation, and for time-dependent boundary data (subtract any function $\psi(x,t)$ that matches the boundary data; the price is a source term $-\psi_t + \alpha\psi_{xx}$ in the PDE for w, handled by eigenfunction expansion in Section 13.7).

Example 13.6 — Homogenising a Neumann–Dirichlet problem

A rod $0 < x < L$ has its end $x = 0$ held at T_0 and receives a constant heat flux q_s (W/m^2) into the end $x = L$: $k\,u_x(L,t) = q_s$. Find a steady state that removes both boundary conditions.

Steady state: $u_s'' = 0 \Rightarrow u_s = A + Bx$. $u_s(0) = T_0 \Rightarrow A = T_0$. $k u_s'(L) = q_s \Rightarrow B = q_s/k$. So

$$u_s(x) = T_0 + \frac{q_s}{k}\,x.$$

Then $w = u - u_s$ satisfies $w(0,t) = 0$, $w_x(L,t) = 0$, $w(x,0) = f(x) - T_0 - q_s x/k$.

Check units: q_s/k has units $(\mathrm{W/m^2})/(\mathrm{W\,m^{-1}K^{-1}}) = \mathrm{K/m}$, a temperature gradient. ✓ Numerically, a copper rod ($k = 400$ W m^{-1}K^{-1}, $L = 0.1$ m) receiving $q_s = 10^5$ W/m^2 (10 W/cm^2) settles with its hot end $q_s L/k = 25$ K above T_0.

Pitfall: with Neumann conditions at **both** ends, $k u_x(0,t) = q_0$, $k u_x(L,t) = q_L$, a linear steady state exists only if $q_0 = q_L$ (net heat flow in = out). Otherwise energy accumulates and there is no steady state. Subtract instead

$$\psi(x,t) = \frac{q_L - q_0}{2kL}\,x^2 + \frac{q_0}{k}\,x + \frac{(q_L - q_0)\,\alpha}{kL}\,t,$$

which matches both flux conditions and satisfies $\psi_t = \alpha\psi_{xx}$ (both sides equal $\alpha(q_L - q_0)/(kL)$), so the uniform temperature rises linearly in time.

Example 13.7 — Circuit terminations as boundary conditions

A lossless line of length ℓ ($R = G = 0$) is driven at $x = 0$ and terminated at $x = \ell$. Express the termination as a condition on $v(x,t)$ alone.

The telegrapher's equation $v_x = -L\, i_t$ links i and v.

- **Short circuit:** $v(\ell, t) = 0$ — Dirichlet.
- **Open circuit:** $i(\ell, t) = 0$ for all t, so $i_t(\ell, t) = 0$ and $v_x(\ell, t) = 0$ — Neumann.
- **Resistor R_L:** $v(\ell, t) = R_L\, i(\ell, t)$. Differentiate in time: $v_t = R_L i_t = -(R_L/L)\, v_x$. So

$$v_x(\ell, t) + \frac{L}{R_L}\, v_t(\ell, t) = 0,$$

a boundary condition that involves a *time* derivative. It is homogeneous, but it is not of the Robin type in x alone; we solve it with travelling waves in Section 13.5 and find the reflection coefficient $\Gamma = (R_L - Z_0)/(R_L + Z_0)$.

- **Source with internal resistance R_s at $x = 0$:** $v(0,t) = v_s(t) - R_s\, i(0,t)$ — non-homogeneous, time-dependent.

Check limits: $R_L \to \infty$ gives $v_x = 0$ (open). $R_L \to 0$ forces $v_t(\ell, t) = 0$, i.e. $v(\ell, t)$ stays at its initial value 0 (short). ✓ Dimensional check: L/R_L has units $(\mathrm{H/m})/\Omega = \mathrm{s/m}$, so $(L/R_L)v_t$ has units $\mathrm{V/m}$, like v_x. ✓

Common pitfall — Neumann data alone do not fix the solution

For the steady state (Laplace) with pure Neumann conditions on the whole boundary, u is determined only up to an additive constant, and a solution exists only if the net flux vanishes: $\oint_{\partial\Omega} \partial u/\partial n \, ds = 0$ (by the divergence theorem, Ch. 7). In a field solver this shows up as a singular matrix; fix one node's potential (a "ground" reference) to remove the null space, exactly as you choose a ground node in nodal circuit analysis.

Engineering note — symmetry planes are Neumann boundaries

A plane of mirror symmetry in a thermal or electrostatic problem carries zero normal flux, so it can be replaced by an insulating (Neumann, $\partial u/\partial n = 0$) boundary. A plane of *antisymmetry* (potential $+V$ on one side mirrored by $-V$ on the other) is an equipotential at 0, a Dirichlet boundary. Modelling half or a quarter of a symmetric device this way cuts the solver's work by $2\times$ or $4\times$.

Uniqueness by the energy method

Uniqueness for linear problems reduces to showing that the *difference* of two solutions, which solves the problem with zero data, is zero.

Uniqueness for the heat IBVP

The problem $u_t = \alpha u_{xx} + q(x,t)$ on $0 < x < L$, $u(x,0) = f(x)$, with Dirichlet, Neumann or Robin ($\beta \geq 0$) conditions at each end, has at most one smooth solution.

Proof. Let w be the difference of two solutions. Then $w_t = \alpha w_{xx}$ with zero initial value and homogeneous boundary conditions. Define the "energy" $E(t) = \frac{1}{2}\int_0^L w^2\, dx \geq 0$. Then, integrating by parts,

$$E'(t) = \int_0^L w\, w_t\, dx = \alpha \int_0^L w\, w_{xx}\, dx = \alpha \big[w\, w_x\big]_0^L - \alpha \int_0^L w_x^2\, dx.$$

The boundary term is zero for Dirichlet and Neumann conditions and ≤ 0 for Robin conditions with $\beta \geq 0$ (at $x = L$, $w_x = -\beta w$ gives $w\, w_x = -\beta w^2 \leq 0$; at $x = 0$ the sign flip of the outward normal gives the same conclusion). Hence $E'(t) \leq 0$. Since $E(0) = 0$ and $E \geq 0$, $E(t) = 0$ for all t, so $w \equiv 0$. □

The same argument with $E = \frac{1}{2}\int (w_t^2 + c^2 w_x^2)\, dx$ proves uniqueness for the wave equation (there E is conserved, not just non-increasing), and with $E = \int |\nabla w|^2\, dV$ for Laplace's equation.

13.3 Separation of variables

Separation of variables is the workhorse method for linear PDEs on simple domains (intervals, rectangles, disks, boxes, cylinders, spheres). Its idea is to look for solutions in which the dependence on each variable is *independent* of the others,

$$u(x,t) = X(x)\, T(t),$$

so that the PDE splits into ODEs, one per variable. Each product solution is too special to match arbitrary data, but because the problem is linear we can superpose infinitely many of them and choose the coefficients to match the data. The method is the PDE counterpart of **modal analysis**: a coupled system $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ is solved by expanding in the eigenvectors of $\mathbf{A}$ (Ch. 10, Ch. 11); here the "matrix" is the differential operator d^2/dx^2 with its boundary conditions, and the "eigenvectors" are functions.

The procedure

Separation of variables — the five steps

1. **Homogenise.** Make the PDE and the boundary conditions in all but one variable homogeneous (Section 13.2).
2. **Separate.** Substitute $u = X(x)T(t)$ (or $X(x)Y(y)$, $R(r)\Phi(\phi)$), divide by XT and put all x-dependence on one side. A function of x equal to a function of t must be a constant, the **separation constant** $-\lambda$.
3. **Solve the eigenvalue problem.** The homogeneous boundary conditions become conditions on X alone, giving a **Sturm–Liouville problem** $X'' + \lambda X = 0$ plus boundary conditions. Find all **eigenvalues** λ_n for which a non-trivial solution exists and the corresponding **eigenfunctions** X_n.
4. **Solve the remaining ODE** for each λ_n, giving $T_n(t)$ and the **product solutions** $u_n = X_n(x)T_n(t)$.
5. **Superpose and match.** Write $u = \sum_n c_n X_n(x) T_n(t)$ and determine c_n from the remaining (initial or non-homogeneous boundary) data using orthogonality of the X_n — a Fourier-series computation (Section 13.7).

Separating the heat equation

Take $u_t = \alpha u_{xx}$ on $0 < x < L$ with $u(0,t) = u(L,t) = 0$. Substituting $u = X(x)T(t)$ gives $X\,T' = \alpha X''\,T$. Dividing by αXT:

$$\frac{T'(t)}{\alpha\,T(t)} = \frac{X''(x)}{X(x)}.$$

The left side depends only on t, the right only on x. If we change x while holding t fixed, the left side does not change, so neither can the right; hence both equal a constant, which we call $-\lambda$:

$$X'' + \lambda X = 0, \qquad T' + \alpha\lambda\,T = 0.$$

The boundary conditions $u(0,t) = X(0)T(t) = 0$ for all t force $X(0) = 0$ (otherwise $T \equiv 0$ and $u \equiv 0$). Likewise $X(L) = 0$. This is the eigenvalue problem already met in Ch. 11:

Dirichlet eigenvalue problem

The problem $X'' + \lambda X = 0$, $X(0) = X(L) = 0$ has non-trivial solutions only for

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2, \qquad X_n(x) = \sin\frac{n\pi x}{L}, \qquad n = 1, 2, 3, \ldots$$

Proof. Consider the sign of λ.

- $\lambda < 0$: write $\lambda = -\mu^2$, $\mu > 0$. Then $X = A\cosh\mu x + B\sinh\mu x$. $X(0) = A = 0$; $X(L) = B\sinh\mu L = 0$ with $\sinh\mu L \neq 0$, so $B = 0$. Trivial.
- $\lambda = 0$: $X = A + Bx$; $X(0) = 0 \Rightarrow A = 0$, $X(L) = BL = 0 \Rightarrow B = 0$. Trivial.
- $\lambda > 0$: $\lambda = \mu^2$, $X = A\cos\mu x + B\sin\mu x$. $X(0) = A = 0$. $X(L) = B\sin\mu L = 0$, so a non-zero B requires $\sin\mu L = 0$, i.e. $\mu L = n\pi$. □

With λ_n known, $T_n = e^{-\alpha\lambda_n t}$, and the product solutions are

$$u_n(x,t) = e^{-\alpha(n\pi/L)^2 t}\,\sin\frac{n\pi x}{L}.$$

Each is a spatial sine pattern that keeps its shape and decays exponentially, faster for higher n. Section 13.4 assembles them into the full solution.

Common eigenvalue problems

The boundary conditions change the eigenfunctions. The following table covers nearly every case met in practice for $X'' + \lambda X = 0$ on $0 < x < L$ (on $-\pi < \phi < \pi$ for the periodic case).

Boundary conditions	Eigenvalues λ_n	Eigenfunctions X_n	n
$X(0) = 0,\ X(L) = 0$ (D–D)	$(n\pi/L)^2$	$\sin(n\pi x/L)$	$1, 2, \ldots$
$X'(0) = 0,\ X'(L) = 0$ (N–N)	$(n\pi/L)^2$	$\cos(n\pi x/L)$	$0, 1, 2, \ldots$
$X(0) = 0,\ X'(L) = 0$ (D–N)	$\big((2n-1)\pi/(2L)\big)^2$	$\sin\big((2n-1)\pi x/(2L)\big)$	$1, 2, \ldots$
$X'(0) = 0,\ X(L) = 0$ (N–D)	$\big((2n-1)\pi/(2L)\big)^2$	$\cos\big((2n-1)\pi x/(2L)\big)$	$1, 2, \ldots$
periodic on $(-\pi, \pi)$	n^2	1; $\cos n\phi$, $\sin n\phi$ (two per $n \geq 1$)	$0, 1, 2, \ldots$
$X(0) = 0,\ X'(L) + hX(L) = 0$ (D–R, $h > 0$)	μ_n^2, $\tan(\mu_n L) = -\mu_n/h$	$\sin(\mu_n x)$	$1, 2, \ldots$

Note that N–N admits $\lambda_0 = 0$ with the **constant** eigenfunction $X_0 = 1$. Physically, an insulated rod can hold any uniform temperature forever; this mode carries the average.

Sturm–Liouville theory

All the problems in the table are instances of a general theory that guarantees the properties we need.

Regular Sturm–Liouville problem

On $a < x < b$, the problem

$$\big(p(x)\,X'\big)' + q(x)\,X + \lambda\, w(x)\,X = 0,$$

$$\beta_1 X(a) + \beta_2 X'(a) = 0, \qquad \gamma_1 X(b) + \gamma_2 X'(b) = 0,$$

with p, p', q, w continuous, $p > 0$ and $w > 0$ on $[a,b]$, and $(\beta_1,\beta_2) \neq (0,0)$, $(\gamma_1,\gamma_2) \neq (0,0)$, is a **regular Sturm–Liouville problem**. w is the **weight function**.

$X'' + \lambda X = 0$ is the case $p = w = 1$, $q = 0$. Bessel's equation, which arises for the circular disk and cylinder, is the case $p = r$, $q = -n^2/r$, $w = r$ on $0 < r < a$ (singular at $r = 0$, but the conclusions below still hold with a boundedness condition there).

Sturm–Liouville theorem

For a regular Sturm–Liouville problem:

1. All eigenvalues are **real**, and they form an infinite increasing sequence $\lambda_1 < \lambda_2 < \lambda_3 < \cdots \to \infty$.
2. Each eigenvalue has a single eigenfunction (up to a constant factor), and the nth eigenfunction has exactly $n-1$ zeros in (a,b).
3. Eigenfunctions for different eigenvalues are **orthogonal with weight** w:

$$\langle X_m, X_n \rangle_w = \int_a^b X_m(x)\,X_n(x)\,w(x)\,dx = 0, \qquad m \neq n.$$

4. The eigenfunctions are **complete**: any square-integrable f on (a,b) has an expansion $f = \sum c_n X_n$ converging in the mean-square sense, with **generalised Fourier coefficients**

$$c_n = \frac{\langle f, X_n\rangle_w}{\langle X_n, X_n\rangle_w} = \frac{\int_a^b f\,X_n\,w\,dx}{\int_a^b X_n^2\,w\,dx}.$$

If f is piecewise smooth, the series converges to $\tfrac12[f(x^+) + f(x^-)]$ at each interior point, exactly as for ordinary Fourier series (Ch. 15).

Proof of orthogonality and reality. Let $\mathcal{S}[X] = (pX')' + qX$. For any two functions X, Y satisfying the boundary conditions, **Lagrange's identity** gives

$$\int_a^b \big(Y\,\mathcal{S}[X] - X\,\mathcal{S}[Y]\big)\,dx = \Big[p\,(YX' - XY')\Big]_a^b = 0,$$

the boundary term vanishing because at each end both X and Y satisfy the same homogeneous condition, so the vectors (X, X') and (Y, Y') are both orthogonal to (β_1,β_2) and are therefore parallel, making $YX' - XY' = 0$. (This "self-adjointness" is the analogue of a symmetric matrix, $\mathbf{y}^{\mathsf T}\mathbf{A}\mathbf{x} = \mathbf{x}^{\mathsf T}\mathbf{A}\mathbf{y}$.) Now if $\mathcal{S}[X_m] = -\lambda_m w X_m$ and $\mathcal{S}[X_n] = -\lambda_n w X_n$, then

$$0 = \int_a^b \big(X_n\mathcal{S}[X_m] - X_m\mathcal{S}[X_n]\big)\,dx = (\lambda_n - \lambda_m)\int_a^b X_m X_n\,w\,dx,$$

so $\lambda_m \neq \lambda_n$ forces orthogonality. If λ were complex with eigenfunction X, then $\bar\lambda$ would be an eigenvalue with eigenfunction $\bar X$, and the same identity would give $(\lambda - \bar\lambda)\int |X|^2 w\,dx = 0$, so $\lambda = \bar\lambda$ is real. $\square$

The Rayleigh quotient tells us the sign of the eigenvalues. Multiplying $(pX')' + qX = -\lambda w X$ by X and integrating by parts,

$$\lambda = \frac{-\big[pXX'\big]_a^b + \int_a^b \big(pX'^2 - qX^2\big)dx}{\int_a^b X^2 w\,dx}.$$

For $X'' + \lambda X = 0$ with Dirichlet, Neumann or Robin ($h \geq 0$) conditions, $q = 0$ and the boundary term $-[XX']_0^L$ is zero (Dirichlet, Neumann) or non-negative (Robin: $X'(L) = -hX(L)$ gives $-X(L)X'(L) = hX(L)^2 \geq 0$), so $\lambda \geq 0$: there are no negative eigenvalues, and the case analysis "$\lambda < 0$" in the proof above could have been skipped. Physically, $\lambda \geq 0$ says that no mode of a passive diffusion grows.

Example 13.8 — A Dirichlet–Neumann eigenvalue problem

A rod $0 < x < L$ is held at 0 at $x = 0$ and insulated at $x = L$. Find the eigenvalues and eigenfunctions and verify orthogonality of the first two.

$X'' + \lambda X = 0$, $X(0) = 0$, $X'(L) = 0$. From the Rayleigh quotient, $\lambda \geq 0$; $\lambda = 0$ gives $X = Bx$ with $X'(L) = B = 0$, trivial. For $\lambda = \mu^2 > 0$, $X(0) = 0$ gives $X = B\sin\mu x$, and $X'(L) = B\mu\cos\mu L = 0$ requires $\cos\mu L = 0$:

$$\mu_n = \frac{(2n-1)\pi}{2L}, \qquad X_n = \sin\frac{(2n-1)\pi x}{2L}, \qquad n = 1, 2, \ldots$$

The first eigenfunction is a quarter sine wave: zero at the clamped end, flat at the insulated end.

Orthogonality check with $L = 1$: $\int_0^1 \sin(\pi x/2)\sin(3\pi x/2)\,dx = \tfrac12\int_0^1[\cos(\pi x) - \cos(2\pi x)]dx = \tfrac12[0 - 0] = 0$. ✓ Normalisation: $\int_0^L X_n^2\,dx = L/2$.

Example 13.9 — A Robin eigenvalue problem (convective cooling)

A rod $0 < x < 1$ is held at 0 at $x = 0$ and loses heat by convection at $x = 1$: $X'(1) + hX(1) = 0$ with $h = 1$. Find the first three eigenvalues.

As before, $X = \sin \mu x$ (the $\lambda \le 0$ cases are trivial since $h > 0$). The Robin condition gives $\mu \cos \mu + h \sin \mu = 0$, i.e.

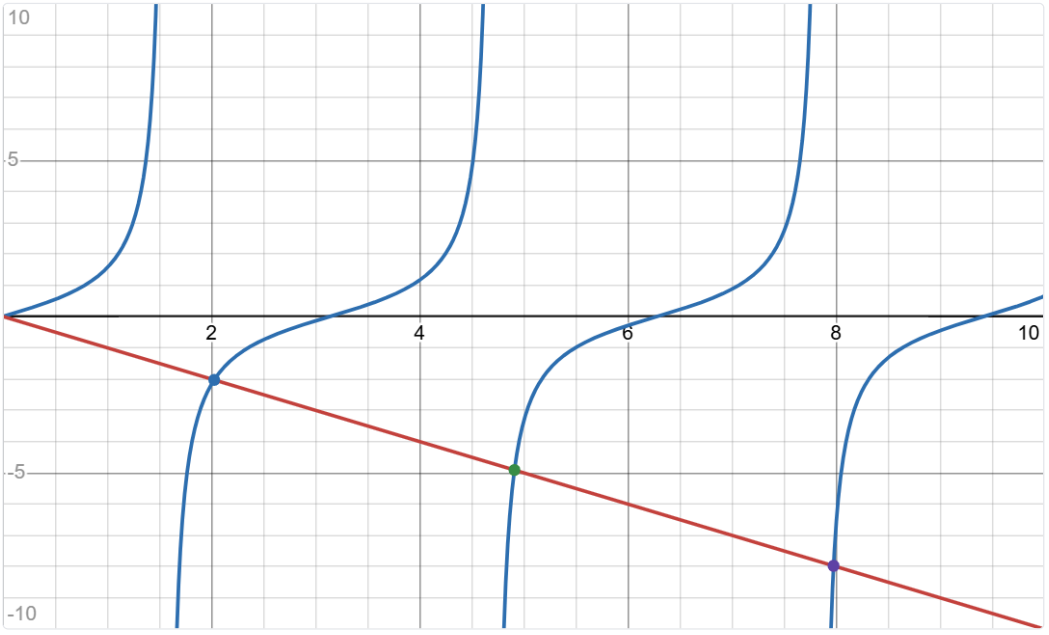
$$\tan \mu = -\frac{\mu}{h}.$$

This is **transcendental**: the eigenvalues must be found numerically, from the intersections of $y = \tan \mu$ with the line $y = -\mu/h$. One root lies in each interval $\big((n - \tfrac{1}{2})\pi,\ n\pi\big)$. Newton's method or bisection (Ch. 22) on $g(\mu) = \sin \mu + \mu \cos \mu$ gives

$$\mu_1 = 2.0288, \qquad \mu_2 = 4.9132, \qquad \mu_3 = 7.9787.$$

For large n, $\mu_n \approx (n - \tfrac{1}{2})\pi + \dfrac{h}{(n - \tfrac{1}{2})\pi}$; for $n = 3$ this gives $7.8540 + 0.1273 = 7.9813$, close to the exact 7.9787.

Limits. $h \to 0$ (insulated end) gives $\cos \mu = 0$, $\mu_1 = \pi/2 = 1.571$; $h \to \infty$ (end held at zero) gives $\sin \mu = 0$, $\mu_1 = \pi$. The Robin value 2.029 lies between, as it must. The eigenfunctions $\sin \mu_n x$ are orthogonal on $(0, 1)$ by the Sturm–Liouville theorem even though the μ_n are not equally spaced — a fact that is far from obvious by direct integration.



The Robin eigenvalues of Example 13.9 are the intersections of $y = \tan \mu$ (blue) with $y = -\mu/h$ (red, $h = 1$); they approach $(n - \tfrac{1}{2})\pi$ from above.

Separating other equations

Example 13.10 — Separating the lossy telegrapher's equation

A line of length ℓ is short-circuited at both ends, $v(0, t) = v(\ell, t) = 0$. Separate $v_{xx} = LC\, v_{tt} + (RC + GL) v_t + RGv$.

With $v = X(x)T(t)$:

$$\frac{X''}{X} = \frac{LC\, T'' + (RC + GL)\, T' + RG\, T}{T} = -\lambda.$$

The X problem is the Dirichlet problem: $\lambda_n = (n\pi/\ell)^2$, $X_n = \sin(n\pi x/\ell)$. The T equation becomes

$$T'' + \Big(\frac{R}{L} + \frac{G}{C}\Big) T' + \frac{RG + (n\pi/\ell)^2}{LC}\, T = 0,$$

a **damped harmonic oscillator** — the equation of a series RLC circuit (Ch. 12). Each spatial mode of the line behaves like its own resonant circuit with undamped natural frequency $\omega_n = \sqrt{(RG + (n\pi/\ell)^2)/(LC)}$ and damping coefficient $\zeta_n \omega_n = \tfrac{1}{2}(R/L + G/C)$.

Numbers. $\ell = 1$ m, $L = 250$ nH/m, $C = 100$ pF/m (so $c = 2 \times 10^8$ m/s, $Z_0 = 50\ \Omega$), $R = 0.5\ \Omega$/m, $G = 0$. For $n = 1$: $\omega_1 = \pi/\sqrt{LC} = \pi \times 2 \times 10^8 = 6.28 \times 10^8$ rad/s, i.e. $f_1 = 100$ MHz (half a wavelength fits in the line). Decay rate $R/(2L) = 10^6\ \text{s}^{-1}$, so the mode's quality factor is $Q = \omega_1/(2 \cdot 10^6) \approx 314$. This is the principle of a shorted-stub resonator.

Example 13.11 — Separating Laplace's equation in polar coordinates

In polar coordinates (r, ϕ), Laplace's equation is (Ch. 7)

$$u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\phi\phi} = 0.$$

Put $u = R(r)\Phi(\phi)$ and multiply by $r^2/(R\Phi)$:

$$\frac{r^2R'' + rR'}{R} = -\frac{\Phi''}{\Phi} = \lambda.$$

On a full disk or annulus ϕ is an angle, so Φ must be 2π-periodic: $\lambda = n^2$, $\Phi = a_n\cos n\phi + b_n \sin n\phi$, $n = 0, 1, 2, \ldots$. The radial equation $r^2R'' + rR' - n^2R = 0$ is a **Cauchy–Euler** equation (Ch. 11): try $R = r^m$ to get $m^2 = n^2$, so

$$R = \begin{cases} C_0 + D_0 \ln r, & n = 0, \\ C_n r^n + D_n r^{-n}, & n \geq 1. \end{cases}$$

These product solutions are used in Section 13.6.

When separation fails

Separation of variables needs three things: a separable PDE, a domain whose boundary consists of coordinate curves ($x = $ const, $r = $ const, …), and homogeneous conditions on all but one pair of opposite boundaries (or a way to split the problem so this holds).

- **Non-separable PDE.** $u_{xx} + (x + y)u_{yy} = 0$ gives $X''Y + (x + y)XY'' = 0$, and $(x + y)$ cannot be split into a function of x times or plus a function of y after division. Equations with coefficients that are products, $a(x)b(y)$, often *do* separate.
- **Non-separable domain.** An L-shaped region, a microstrip cross-section with a finite-width strip, or a region bounded by a curve that is not a coordinate line requires conformal mapping (Ch. 18) or numerical methods.
- **Non-homogeneous conditions on every side.** Split by superposition into several problems with one non-homogeneous side each (Section 13.6).

Common pitfall — choosing the sign of the separation constant

Writing the constant as $+\lambda$ rather than $-\lambda$ is not an error, but it changes which case gives the oscillatory solution, and students then reject the right case. Always examine all three cases $\lambda < 0$, $\lambda = 0$, $\lambda > 0$ (or use the Rayleigh quotient to rule some out), and remember that N–N and periodic problems *do* have a zero eigenvalue. Losing the $\lambda_0 = 0$ constant mode loses the average value of the solution.

Engineering note — modes are the natural "coordinates" of a distributed system

A distributed system (line, plate, membrane, cavity) has infinitely many degrees of freedom, but separation of variables shows that it behaves like an infinite set of *independent* lumped systems, one per eigenfunction. Engineers routinely truncate to the first few modes: a thermal model of a heat sink with one or two exponential time constants, a cavity resonator described by its lowest mode, a loudspeaker cone by its first bending modes. Sturm–Liouville orthogonality is what makes the modes decouple.

13.4 Heat equation

Derivation

Consider a rod (or a slab, or a wire) in which heat flows only along x. Let $u(x, t)$ be the temperature, ρ the density, c_p the specific heat capacity, k the thermal conductivity and $g(x, t)$ the heat generated per unit volume per unit time (Joule heating J^2/σ in a conductor, for example). Two physical laws are needed.

1. **Conservation of energy** for the slice $(x, x + \Delta x)$ of cross-section A: the rate of increase of stored heat equals heat flowing in minus heat flowing out plus heat generated:

$$\frac{\partial}{\partial t}\int_x^{x+\Delta x} \rho c_p\, u\, A\, ds = A\, q(x, t) - A\, q(x + \Delta x, t) + \int_x^{x+\Delta x} g\, A\, ds,$$

where q is the heat flux ($\mathrm{W/m^2}$) in the $+x$ direction. 2. **Fourier's law of conduction**: heat flows down the temperature gradient, $q = -k\, u_x$.

Divide by $A\,\Delta x$, let $\Delta x \to 0$ and substitute Fourier's law:

$$\rho c_p\, u_t = (k\, u_x)_x + g.$$

Heat (diffusion) equation

For constant k, ρ, c_p,

$$\frac{\partial u}{\partial t} = \alpha\,\frac{\partial^2 u}{\partial x^2} + \frac{g}{\rho c_p}, \qquad \alpha = \frac{k}{\rho c_p},$$

where α is the **thermal diffusivity** ($\mathrm{m^2/s}$). In three dimensions, $u_t = \alpha\nabla^2 u + g/(\rho c_p)$.

The same equation, with different letters, governs every process in which a conserved quantity flows down its own gradient:

Process	Quantity u	Flux law	Diffusivity
Heat conduction	temperature T	Fourier, $q = -kT_x$	$\alpha = k/(\rho c_p)$
Impurity diffusion	concentration N	Fick, $J = -DN_x$	D
Minority carriers (no field)	excess density Δn	$J_n = qD_n\,\Delta n_x$	D_n (plus recombination term $-\Delta n/\tau_n$)
RC transmission line	voltage v	Ohm, $i = -v_x/R$	$1/(RC)$
Magnetic field in a conductor	H or $\mathbf{J}$	Faraday + Ohm	$1/(\mu\sigma)$

Typical diffusivities: copper $\alpha \approx 1.1 \times 10^{-4}\ \mathrm{m^2/s}$, silicon 8.8×10^{-5}, FR-4 circuit board $\approx 1.5 \times 10^{-7}$, water $1.4 \times 10^{-7}\ \mathrm{m^2/s}$.

A diffusion process has one intrinsic scale: in time t, the quantity spreads over a distance of order $\sqrt{\alpha t}$; equivalently, crossing a distance L takes a time of order L^2/α. Doubling the thickness of a heat spreader quadruples its thermal settling time.

Solution on a finite interval with Dirichlet conditions

Section 13.3 gave the product solutions $e^{-\alpha\lambda_n t}\sin(n\pi x/L)$ of the model problem. Superposing them,

$$u(x,t) = \sum_{n=1}^{\infty} b_n\, e^{-\alpha(n\pi/L)^2 t} \sin\frac{n\pi x}{L},$$

and imposing $u(x,0) = f(x)$ gives $f(x) = \sum b_n \sin(n\pi x/L)$: the b_n are the coefficients of the **Fourier sine series** of f on $(0,L)$ (the half-range expansion of Ch. 15). Multiplying by $\sin(m\pi x/L)$ and integrating, orthogonality kills every term except $n = m$, and $\int_0^L \sin^2(m\pi x/L)\,dx = L/2$.

Heat equation on $(0,L)$, zero Dirichlet ends

If f is piecewise smooth on $[0,L]$, the IBVP $u_t = \alpha u_{xx}$, $u(0,t) = u(L,t) = 0$, $u(x,0) = f(x)$ has the unique solution

$$u(x,t) = \sum_{n=1}^{\infty} b_n\, e^{-\alpha n^2\pi^2 t/L^2}\, \sin\frac{n\pi x}{L}, \qquad b_n = \frac{2}{L}\int_0^L f(x)\sin\frac{n\pi x}{L}\,dx.$$

For $t > 0$ the series converges, together with all its derivatives, uniformly in x, and u is infinitely differentiable.

The convergence statement for $t > 0$ is easy: $|b_n| \le \frac{2}{L}\int_0^L |f|\,dx = M$, so the nth term of the series for any derivative $\partial_x^p \partial_t^q u$ is bounded by $M\,(n\pi/L)^{p+2q}\alpha^q e^{-\alpha n^2\pi^2 t/L^2}$, and the exponential beats any power: the Weierstrass M-test applies. At $t = 0$ the series is just the Fourier sine series of f, whose convergence is governed by the theory of Ch. 15 (pointwise to the mid-value at jumps, with Gibbs overshoot).

Properties of heat-equation solutions

1. **Exponential decay and the dominant mode.** For large t the $n = 1$ term dominates:

$$u(x,t) \approx b_1 e^{-t/\tau_1}\sin\frac{\pi x}{L}, \qquad \tau_1 = \frac{L^2}{\pi^2\alpha}.$$

The ratio of the second surviving term to the first decays like $e^{-(n_2^2-1)t/\tau_1}$, so after about one time constant the profile *is* a half sine wave. τ_1 is the **thermal time constant** of the rod. 2. **Smoothing.** Mode n is damped by $e^{-n^2 t/\tau_1}$. High spatial frequencies (sharp corners, jumps) die almost instantly; a discontinuous initial temperature becomes smooth for every $t > 0$. The heat operator is a spatial *low-pass filter* whose bandwidth shrinks as $1/\sqrt{t}$. 3. **Maximum principle.**

Maximum principle for the heat equation

If $u_t = \alpha u_{xx}$ in $0 < x < L$, $0 < t \le T$, then the maximum (and minimum) of u over the closed rectangle $[0,L]\times[0,T]$ is attained either at $t = 0$ or on the ends $x = 0$, $x = L$.

Physical meaning: without sources, a point in the interior cannot become hotter than the hottest initial or boundary temperature. *Proof idea:* if u had an interior maximum at (x_0,t_0), then $u_{xx} \le 0$ and $u_t \ge 0$ there; perturbing to $v = u + \varepsilon x^2$ makes the inequality $v_t - \alpha v_{xx} = -2\alpha\varepsilon < 0$ strict, contradicting $v_t \ge 0 \ge v_{xx}$ at its maximum. The principle gives uniqueness and continuous dependence on the data at once: two solutions whose data differ by at most ε differ everywhere by at most ε.

4. **Irreversibility.** Running the series backwards, $e^{+\alpha n^2\pi^2|t|/L^2}$ multiplies the nth coefficient. An error of 10^{-6} in b_{10} grows by $e^{100\,|t|/\tau_1}$; at $|t| = \tau_1$ that is $e^{100} \approx 3 \times 10^{43}$. You cannot reconstruct an earlier temperature profile from a later one.

Example 13.12 — Cooling of a uniformly heated rod

A rod of length $L = 1$ (nondimensional units, $\alpha = 1$) starts at a uniform $u = 100$ and has both ends plunged into ice at 0. Find $u(x,t)$ and the midpoint temperature at $t = 0.05,\ 0.1,\ 0.2$.

Coefficients.

$$b_n = 2\int_0^1 100\sin(n\pi x)\,dx = \frac{200}{n\pi}\bigl(1-\cos n\pi\bigr) = \begin{cases} \dfrac{400}{n\pi}, & n \text{ odd},\\[2mm] 0, & n \text{ even}. \end{cases}$$

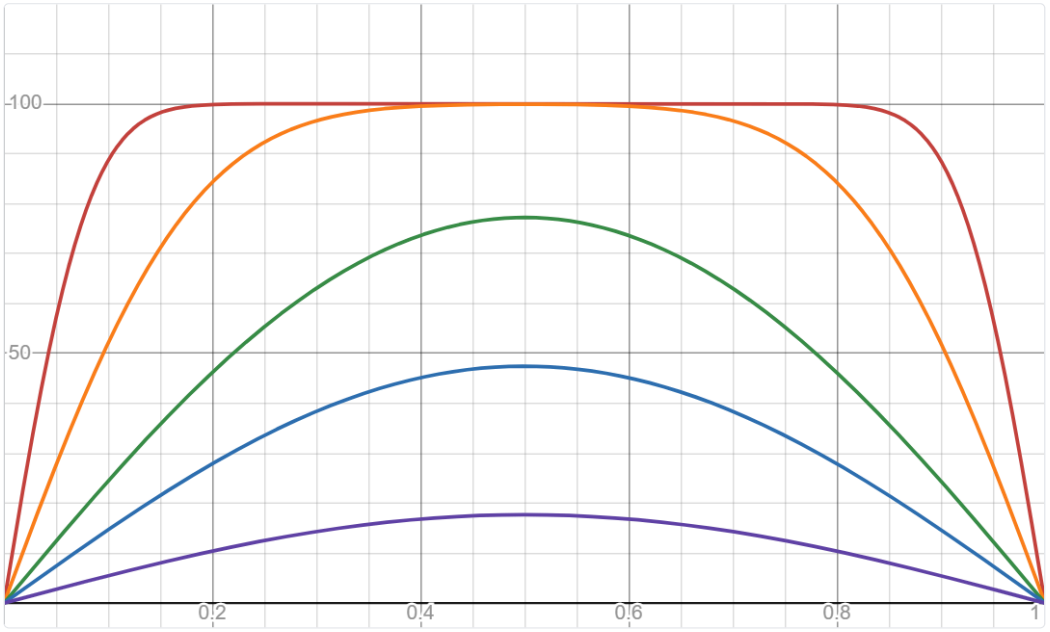
Solution.

$$u(x,t) = \frac{400}{\pi}\sum_{n\ \text{odd}} \frac{1}{n}\,e^{-n^2\pi^2 t}\sin(n\pi x).$$

Midpoint. At $x = \tfrac12$, $\sin(n\pi/2) = (-1)^{(n-1)/2}$ for odd n. Summing 1000 terms:

t	$u(\tfrac12,t)$ **(series)**	**one-term** $\tfrac{400}{\pi}e^{-\pi^2 t}$
0.01	99.92	115.36
0.05	77.23	77.73
0.1	47.45	47.45
0.2	17.69	17.69
0.5	0.916	0.916

For $t \ge 0.1$ the one-term approximation is accurate to four figures; for very small t it is useless (it even exceeds the initial temperature, violating the maximum principle) and many terms are needed. The time constant is $\tau_1 = 1/\pi^2 = 0.101$.



Snapshots of Example 13.12 at $t = 0.002$ (red), 0.01, 0.05, 0.1 and 0.2 (purple): the flat-topped profile first rounds off near the cold ends, then relaxes to a decaying half sine wave.

Example 13.13 — Thermal time constant of a copper bus bar

A copper bar of length $L = 0.10$ m ($\alpha = 1.11 \times 10^{-4}$ m^2/s), insulated along its sides, is at 100 °C above ambient when both ends are suddenly clamped to ambient. Estimate its time constant and the time for the centre excess temperature to fall to 1 K.

$\tau_1 = L^2/(\pi^2\alpha) = 0.01/(9.870 \times 1.11 \times 10^{-4}) = 9.13$ s.

From Example 13.12 (rescaled: x/L and $t/(L^2/\alpha)$ are the nondimensional variables), the centre excess temperature for $t \gtrsim \tau_1$ is $\frac{400}{\pi}e^{-t/\tau_1}$ K. Setting it equal to 1:

$$t = \tau_1 \ln\frac{400}{\pi} = 9.13 \times 4.847 = 44.2\ \text{s}.$$

For a 1 m bar the time constant is 100 times longer, 15 min — the L^2 law in action.

Insulated ends: the Neumann problem

If both ends are insulated, $u_x(0,t) = u_x(L,t) = 0$, the eigenfunctions are $\cos(n\pi x/L)$, $n = 0, 1, 2, \ldots$ (Section 13.3), and

$$u(x,t)=\frac{a_0}{2}+\sum_{n=1}^{\infty}a_n\,e^{-\alpha n^2\pi^2 t/L^2}\cos\frac{n\pi x}{L},\qquad a_n=\frac{2}{L}\int_0^L f(x)\cos\frac{n\pi x}{L}\,dx,$$

a Fourier **cosine** series. As $t\to\infty$, $u\to a_0/2=\frac{1}{L}\int_0^L f\,dx$, the **average** initial temperature: no heat escapes, so the total heat content $\int_0^L \rho c_p u\,dx$ is conserved and simply redistributes.

Example 13.14 — Equalisation in an insulated rod

An insulated rod with $L=1$, $\alpha=1$ starts with the linear profile $u(x,0)=x$. Find $u(x,t)$ and the end temperatures at $t=0.1$.

$a_0=2\int_0^1 x\,dx=1$. For $n\ge 1$, integrating by parts,

$$a_n=2\int_0^1 x\cos(n\pi x)\,dx=2\Big[\frac{x\sin n\pi x}{n\pi}\Big]_0^1-\frac{2}{n\pi}\int_0^1\sin n\pi x\,dx=\frac{2(\cos n\pi-1)}{n^2\pi^2},$$

which is $-4/(n^2\pi^2)$ for odd n and 0 for even n. So

$$u(x,t)=\frac{1}{2}-\frac{4}{\pi^2}\sum_{n\ \text{odd}}\frac{1}{n^2}e^{-n^2\pi^2 t}\cos n\pi x.$$

Checks. At $t=0$, $x=0$: $\frac12-\frac{4}{\pi^2}\sum_{\text{odd}}1/n^2=\frac12-\frac{4}{\pi^2}\cdot\frac{\pi^2}{8}=0$. ✓ At $t=0.1$: $u(0,0.1)=0.3489$, $u(1,0.1)=0.6511$; they sum to 1 by the antisymmetry of $\cos n\pi x$ (n odd) about $x=\frac12$. Both ends are converging to the average $\frac12$ with time constant $1/\pi^2$.

The coefficients decay like $1/n^2$ (compared with $1/n$ in Example 13.12) because the even periodic extension of $f(x)=x$ is continuous — a general rule from Ch. 15: smoother extension, faster coefficient decay.

Non-homogeneous boundary conditions

Example 13.15 — A rod heated at one end

A rod with $L=1$, $\alpha=1$, initially at 0, has $u(0,t)=0$ and $u(1,t)=100$ for $t>0$. Find $u(x,t)$ and $u(\frac12,t)$.

Steady state. $u_s=100x$.

Transient. $w=u-100x$ satisfies the model problem with $w(x,0)=-100x$. Its sine coefficients:

$$b_n=2\int_0^1(-100x)\sin n\pi x\,dx=-200\Big[-\frac{x\cos n\pi x}{n\pi}\Big]_0^1-200\int_0^1\frac{\cos n\pi x}{n\pi}dx=\frac{200(-1)^n}{n\pi}.$$

Solution.

$$u(x,t)=100x+\frac{200}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^n}{n}e^{-n^2\pi^2 t}\sin n\pi x.$$

Midpoint values (4000 terms): $u(\frac12,0)=0.008$ (the truncated series of a function that is 0 there; the error is the slowly converging $1/n$ tail), $u(\frac12,0.05)=11.38$, $u(\frac12,0.1)=26.28$, $u(\frac12,0.2)=41.16$, approaching the steady value 50. Only odd n contribute at $x=\frac12$ and for $t\ge 0.1$ the single term $50-\frac{200}{\pi}e^{-\pi^2 t}$ is accurate to better than 0.1: at $t=0.2$ it gives $50-63.66\times 0.1389=41.16$. ✓

Infinite and semi-infinite domains

When the rod is so long that its ends never "feel" the disturbance during the time of interest (a thick substrate under a hot spot, a semiconductor wafer seen from a surface a few micrometres deep), we replace the interval by the whole line or half-line. Fourier *series* become Fourier *integrals* (Ch. 16); the result can be written in closed form.

Heat kernel and solution on the whole line

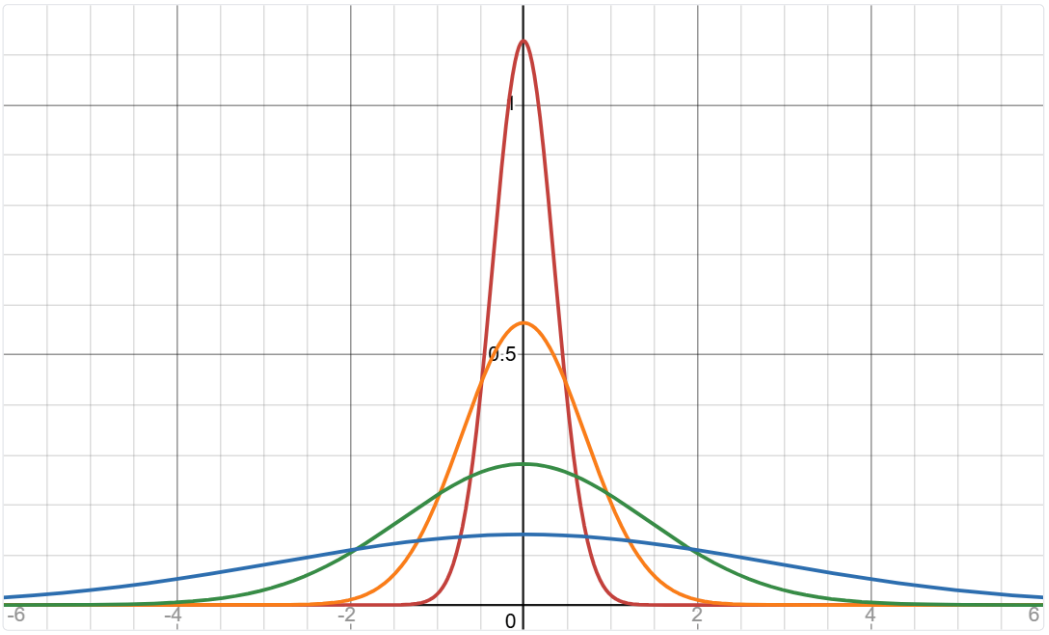
The function

$$\Phi(x,t)=\frac{1}{\sqrt{4\pi\alpha t}}\,e^{-x^2/(4\alpha t)},\qquad t>0,$$

solves $u_t=\alpha u_{xx}$, has $\int_{-\infty}^{\infty}\Phi\,dx=1$ for every $t>0$, and tends to $\delta(x)$ as $t\to 0^+$. For bounded continuous f, the solution of $u_t=\alpha u_{xx}$ on $-\infty<x<\infty$ with $u(x,0)=f(x)$ is the convolution

$$u(x,t)=\int_{-\infty}^{\infty}\Phi(x-s,t)\,f(s)\,ds.$$

Derivation sketch. The equation is unchanged by the scaling $x\to\kappa x$, $t\to\kappa^2 t$, which suggests looking for a solution of the form $t^{-1/2}F(\eta)$ with **similarity variable** $\eta=x/\sqrt{4\alpha t}$. Substituting gives the ODE $F''+2\eta F'+2F=0$, i.e. $(F'+2\eta F)'=0$, whose decaying solution is $F=Ce^{-\eta^2}$; C is fixed by unit area. Alternatively, Fourier-transform in x: $\hat u_t=-\alpha\omega^2\hat u$, so $\hat u=\hat f\,e^{-\alpha\omega^2 t}$, and the inverse transform of the Gaussian $e^{-\alpha\omega^2 t}$ is Φ (Ch. 16). In signal terms, the heat equation is a linear time-varying *Gaussian filter* whose impulse response has standard deviation $\sigma=\sqrt{2\alpha t}$.



The heat kernel ($\alpha = 1$) at $t = 1/16,\ 1/4,\ 1,\ 4$: each quadrupling of time doubles the width and halves the peak, keeping unit area.

For a semi-infinite body $x > 0$ initially at 0 whose surface is suddenly raised to T_s at $t = 0$, the similarity solution is built from the **error function** $\operatorname{erf} z = \frac{2}{\sqrt{\pi}} \int_0^z e^{-s^2}\,ds$ and its complement $\operatorname{erfc} z = 1 - \operatorname{erf} z$:

$$u(x,t) = T_s \operatorname{erfc} \frac{x}{2\sqrt{\alpha t}}.$$

Check: $\operatorname{erfc}(0) = 1$ gives $u(0,t) = T_s$; as $t \to 0^+$ for fixed $x > 0$ the argument $\to \infty$ and $u \to 0$; and u is a function of the similarity variable, so it solves the PDE by the derivation above (with F replaced by its integral). The surface heat flux $-k u_x(0,t) = k T_s / \sqrt{\pi \alpha t}$ is infinite at $t = 0^+$ and decays like $t^{-1/2}$.

Example 13.16 — Step response of a long RC interconnect

A long on-chip line with $RC = 4 \times 10^{-5}$ s/m^2 (Example 13.4) is initially discharged, and at $t = 0$ its input is stepped to V_0. Treat it as semi-infinite. When does the voltage at $x = 1$ mm reach $V_0/2$?

The RC-line equation $v_t = v_{xx}/(RC)$ is the heat equation with $\alpha = 1/(RC)$, so

$$v(x,t) = V_0 \operatorname{erfc}\left(\frac{x}{2}\sqrt{\frac{RC}{t}}\right).$$

Set $\operatorname{erfc}(z) = 0.5$: $z = 0.4769$. Then $t_{50} = RC\,x^2/(4z^2) = 1.099\,RC\,x^2$. With $x = 10^{-3}$ m: $t_{50} = 1.099 \times 4 \times 10^{-5} \times 10^{-6} = 4.4 \times 10^{-11}$ s $= 44$ ps.

The delay grows as x^2: repeaters (buffers) are inserted along long on-chip wires precisely because breaking a line of length ℓ into k segments cuts the diffusive delay from $\propto \ell^2$ to $\propto k(\ell/k)^2 = \ell^2/k$. For a *finite* open-ended line of length ℓ, the series solution (Section 13.7) gives the classic $t_{50} \approx 0.38\,RC\ell^2$; the semi-infinite estimate is pessimistic because there is no reflection from the open end to speed up charging.

Common pitfall — one-term approximations at early times

The dominant-mode approximation $b_1 e^{-t/\tau_1}\sin(\pi x/L)$ is excellent once $t \gtrsim 0.2\tau_1$ to τ_1 but grossly wrong at early times (Example 13.12 shows it exceeding the initial temperature). At early times use the semi-infinite erfc solution near each boundary instead: the two approximations are complementary, short-time and long-time.

Engineering note — thermal RC networks

Device datasheets give a transient thermal impedance $Z_{th}(t)$ as a sum of exponentials, the "Foster network" $Z_{th}(t) = \sum_i R_i(1 - e^{-t/\tau_i})$. That form is exactly a truncated eigenfunction expansion of the heat equation in the package: each τ_i is a modal time constant of the kind $L^2/(\mu_i^2\alpha)$, and the R_i are the modal weights. The $t^{1/2}$ behaviour of Z_{th} at very short times is the semi-infinite erfc regime.

13.5 Wave equation

Derivations

Lossless transmission line. With $R = G = 0$ the telegrapher's equations are $v_x = -L\,i_t$ and $i_x = -C\,v_t$, and eliminating i gives

$$v_{tt} = c^2 v_{xx}, \qquad c = \frac{1}{\sqrt{LC}}.$$

For a coaxial line with dielectric permittivity ε and permeability μ, $LC = \mu\varepsilon$, so $c = 1/\sqrt{\mu\varepsilon}$ — the speed of light in the dielectric, independent of the geometry.

Taut string. A string under tension T_s (N) with mass per unit length ρ_ℓ (kg/m), displaced by a small amount $u(x,t)$, feels a net transverse force $T_s[u_x(x+\Delta x)-u_x(x)]$ on a piece of length Δx (to first order in the slope). Newton's law $\rho_\ell \Delta x\, u_{tt} = T_s \Delta x\, u_{xx}$ gives $u_{tt} = c^2 u_{xx}$ with $c = \sqrt{T_s/\rho_\ell}$.

Electromagnetic plane waves. In a source-free, lossless medium Maxwell's equations give $\nabla^2 \mathbf{E} = \mu\varepsilon\, \mathbf{E}_{tt}$; for a field $E_y(x,t)$ that varies only with x, this is again the one-dimensional wave equation.

Wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \quad \text{(1-D)}, \qquad u_{tt} = c^2 \nabla^2 u \quad \text{(2-D, 3-D)},$$

where $c > 0$ is the **wave speed**. It is hyperbolic with characteristics $x \pm ct = \text{const}$.

d'Alembert's solution on the whole line

General solution of the 1-D wave equation

Every twice-differentiable solution of $u_{tt} = c^2 u_{xx}$ on $-\infty < x < \infty$ has the form

$$u(x,t) = F(x - ct) + G(x + ct)$$

for some functions F and G: the sum of a wave moving right at speed c and a wave moving left at speed c, each keeping its shape.

Proof. Change to characteristic variables $\xi = x - ct$, $\eta = x + ct$. Then $\partial_x = \partial_\xi + \partial_\eta$ and $\partial_t = c(\partial_\eta - \partial_\xi)$, so

$$u_{tt} - c^2 u_{xx} = c^2\big[(\partial_\eta - \partial_\xi)^2 - (\partial_\xi + \partial_\eta)^2\big]u = -4c^2\, u_{\xi\eta}.$$

The PDE becomes $u_{\xi\eta} = 0$, whose general solution (Example 13.1) is $F(\xi) + G(\eta)$. $\square$

A profile $F(x-ct)$ is constant along each line $x - ct = x_0$: the value that sat at x_0 at time 0 sits at $x_0 + ct$ at time t. Now impose the initial conditions $u(x,0) = f(x)$, $u_t(x,0) = g(x)$:

$$F(x) + G(x) = f(x), \qquad -cF'(x) + cG'(x) = g(x).$$

Integrate the second: $-F(x) + G(x) = \frac{1}{c}\int_0^x g(s)\,ds + K$. Adding and subtracting,

$$G(x) = \frac{1}{2}f(x) + \frac{1}{2c}\int_0^x g\,ds + \frac{K}{2}, \qquad F(x) = \frac{1}{2}f(x) - \frac{1}{2c}\int_0^x g\,ds - \frac{K}{2}.$$

Evaluate F at $x - ct$ and G at $x + ct$ and add; the constant K cancels:

d'Alembert's formula

The solution of $u_{tt} = c^2 u_{xx}$, $-\infty < x < \infty$, $u(x,0) = f(x)$, $u_t(x,0) = g(x)$ is

$$u(x,t) = \frac{f(x-ct) + f(x+ct)}{2} + \frac{1}{2c}\int_{x-ct}^{x+ct} g(s)\,ds.$$

If $f \in C^2$ and $g \in C^1$ this is a classical solution, and the problem is well posed: $|u_1 - u_2| \le \max|f_1 - f_2| + t\max|g_1 - g_2|$.

Three consequences follow from the formula.

- **Domain of dependence.** $u(x_0,t_0)$ depends only on the data on the interval $[x_0 - ct_0,\ x_0 + ct_0]$, the base of the backward characteristic triangle. Nothing outside can influence it.
- **Range of influence.** Data at a point x_0 influence only the forward wedge $|x - x_0| \le ct$. Signals travel at exactly speed c — in contrast to the heat equation, whose kernel is non-zero everywhere for every $t > 0$.
- **No smoothing.** A kink or jump in f is carried unchanged along the characteristics. The wave equation transmits information; it does not forget it.

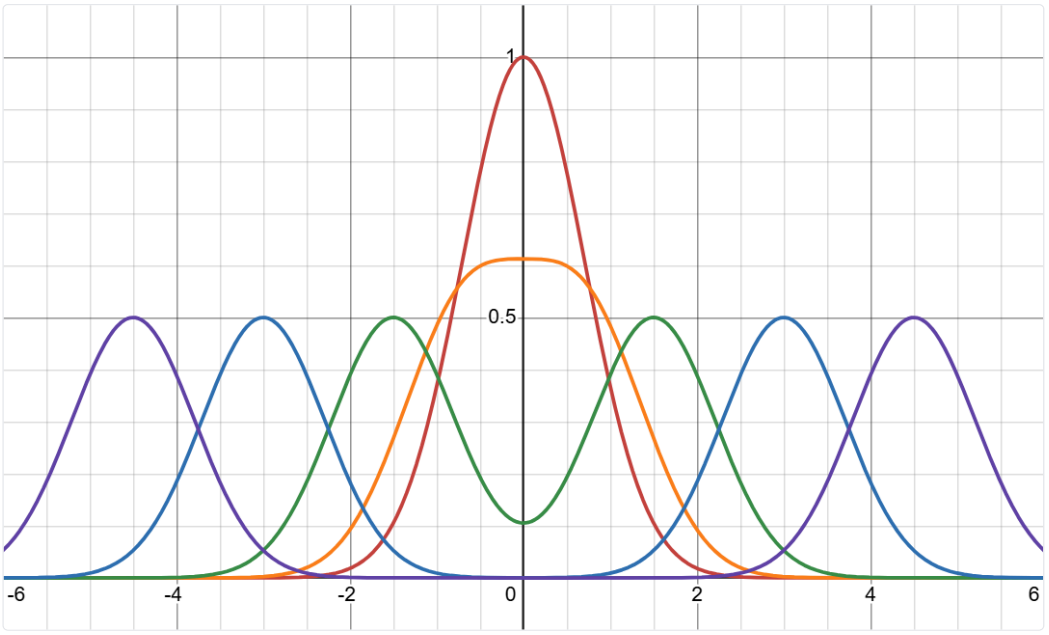
Example 13.17 — A Gaussian pulse splits in two

A string (or line) with $c = 1$ starts at rest with $f(x) = e^{-x^2}$, $g = 0$. By d'Alembert,

$$u(x,t) = \tfrac{1}{2}\big[e^{-(x-t)^2} + e^{-(x+t)^2}\big].$$

At $t = 0$ the peak is 1 at $x = 0$. By $t = 3$ the two halves are centred at $x = \pm 3$, each of height $\frac{1}{2}(1 + e^{-36}) \approx 0.5$; the overlap is negligible once $t \gtrsim 2$. At $x = 0$: $u(0,t) = e^{-t^2}$, e.g. $u(0,1) = 0.368$.

On a transmission line, the same statement reads: a voltage pulse with no accompanying current ($i(x,0) = 0$, so $v_t(x,0) = 0$) splits into equal forward and backward pulses of half the amplitude.



Snapshots of Example 13.17 at $t = 0,\ 0.7,\ 1.5,\ 3,\ 4.5$: the initial pulse separates into two half-height copies travelling in opposite directions without change of shape.

Example 13.18 — Response to an initial velocity

A string with $c = 2$ is initially straight, $f = 0$, and is struck so that $g(x) = \dfrac{1}{1+x^2}$. Find $u(x,t)$ and $u(0,t)$.

$$u(x,t) = \frac{1}{2c}\int_{x-ct}^{x+ct} \frac{ds}{1+s^2} = \frac{1}{4}\big[\arctan(x+2t) - \arctan(x-2t)\big].$$

At $x = 0$: $u(0,t) = \tfrac{1}{4}[\arctan 2t + \arctan 2t] = \tfrac{1}{2}\arctan(2t)$. So $u(0,0.5) = \tfrac{1}{2}\arctan 1 = 0.3927$, $u(0,1) = \tfrac{1}{2}\arctan 2 = 0.5536$, and $u(0,t) \to \pi/4 = 0.785$ as $t \to \infty$.

The string does **not** return to zero: an initial impulse of total momentum $\rho_\ell \int g\,dx$ produces a permanent displacement in the region the waves have passed. The total area $\int g\,dx = \pi$ is spread over a plateau of width $2ct$ and height $\frac{1}{2c}\int g = \pi/4$. On a line, the analogue is an initial current distribution leaving a residual voltage step.

Travelling waves on a transmission line

For the lossless line, d'Alembert's form applies to v, and the first telegrapher equation then determines i. Let $v = F(x-ct) + G(x+ct)$. From $i_x = -Cv_t = -C\big[-cF'(x-ct) + cG'(x+ct)\big]$, integrating in x:

$$i(x,t) = Cc\big[F(x-ct) - G(x+ct)\big] = \frac{1}{Z_0}\big[F(x-ct) - G(x+ct)\big], \qquad Z_0 = \frac{1}{Cc} = \sqrt{\frac{L}{C}}.$$

Forward and backward waves on a lossless line

Write $v^+ = F(x-ct)$ and $v^- = G(x+ct)$. Then

$$v = v^+ + v^-, \qquad i = \frac{v^+ - v^-}{Z_0},$$

where $Z_0 = \sqrt{L/C}$ is the **characteristic impedance**. A single travelling wave has $v/i = +Z_0$ (forward) or $-Z_0$ (backward); a line of finite length looks resistive, of value Z_0, to a signal until the first reflection returns.

Reflection at a resistive load. At $x = \ell$ put a resistor R_L, so $v(\ell,t) = R_L i(\ell,t)$:

$$v^+ + v^- = \frac{R_L}{Z_0}(v^+ - v^-) \quad\Longrightarrow\quad v^- = \Gamma\, v^+, \qquad \Gamma = \frac{R_L - Z_0}{R_L + Z_0}.$$

This is the solution of the dynamic boundary condition found in Example 13.7. Special cases: matched load $R_L = Z_0$, $\Gamma = 0$ (no reflection — the line looks infinite); open circuit $\Gamma = +1$; short circuit $\Gamma = -1$.

Example 13.19 — Step on a mismatched line

A $Z_0 = 50\ \Omega$ line of length $\ell = 2$ m with $c = 2 \times 10^8$ m/s is driven by a matched source (internal resistance $50\ \Omega$) whose open-circuit voltage steps from 0 to 10 V at $t = 0$, and is terminated in $R_L = 150\ \Omega$. Describe $v(x,t)$.

1. **Launch.** Initially the line looks like Z_0, so the source sees a $50\ \Omega$ divider: $v^+ = 10 \times \frac{50}{50+50} = 5$ V, and a 5 V step travels toward the load, arriving at $t_1 = \ell/c = 10$ ns.
2. **Reflection.** $\Gamma_L = (150-50)/(150+50) = 0.5$, so $v^- = 2.5$ V travels back. The load voltage jumps to $5 + 2.5 = 7.5$ V at $t = 10$ ns.
3. **Back at the source** at $t = 20$ ns: the source is matched, $\Gamma_S = 0$, so nothing more is reflected and the line voltage is 7.5 V everywhere.

Check against DC. The final state is a 10 V source driving $50\ \Omega + 150\ \Omega$: $v = 10 \times 150/200 = 7.5$ V. ✓ Current check at the load: $i = (5 - 2.5)/50 = 0.05$ A and $7.5/150 = 0.05$ A. ✓

With a mismatched source as well, the waves bounce repeatedly, multiplying by $\Gamma_L\Gamma_S$ on each round trip, and the line voltage approaches the DC value as a geometric series — the "bounce diagram" of signal-integrity analysis.

Finite string: the method of images

On a semi-infinite string $x > 0$ with a fixed end, $u(0,t) = 0$, extend the data to $x < 0$ **oddly**: $f(-x) = -f(x)$, $g(-x) = -g(x)$. The d'Alembert solution of the extended problem is odd in x for all t, so it vanishes at $x = 0$ automatically. A pulse approaching the fixed end meets its inverted mirror image coming the other way; they cancel at the end, and the image emerges as an **inverted reflected pulse** ($\Gamma = -1$, the short circuit). For a free end, $u_x(0,t) = 0$, extend **evenly**; the reflection is upright ($\Gamma = +1$, the open circuit). On $0 < x < L$ with both ends fixed, extend oddly about both ends, i.e. to an odd $2L$-periodic function, and apply d'Alembert to the extension.

Finite interval by separation of variables

For $u_{tt} = c^2 u_{xx}$ on $0 < x < L$ with $u(0,t) = u(L,t) = 0$, $u(x,0) = f(x)$, $u_t(x,0) = g(x)$, separation gives $X'' + \lambda X = 0$ (Dirichlet: $\lambda_n = (n\pi/L)^2$) and $T'' + c^2\lambda_n T = 0$, so $T_n = A_n \cos\omega_n t + B_n \sin\omega_n t$ with

$$\omega_n = \frac{n\pi c}{L}, \qquad f_n = \frac{\omega_n}{2\pi} = \frac{nc}{2L}.$$

Vibrating string / shorted line on $(0, L)$

$$u(x,t) = \sum_{n=1}^{\infty} \left(A_n \cos\frac{n\pi c t}{L} + B_n \sin\frac{n\pi c t}{L} \right) \sin\frac{n\pi x}{L},$$

$$A_n = \frac{2}{L}\int_0^L f(x)\sin\frac{n\pi x}{L}\,dx, \qquad B_n = \frac{2}{n\pi c}\int_0^L g(x)\sin\frac{n\pi x}{L}\,dx.$$

The nth term is the nth **normal mode** (standing wave), with natural frequency $f_n = nc/(2L)$, an integer multiple of the **fundamental** $f_1 = c/(2L)$. The motion is periodic with period $2L/c$.

The B_n formula comes from differentiating in t at $t = 0$: $g = \sum B_n\omega_n \sin(n\pi x/L)$, so $B_n\omega_n$ is the sine coefficient of g. Using $\sin a \cos b = \tfrac{1}{2}[\sin(a-b) + \sin(a+b)]$, each standing wave is the sum of two travelling waves, and the series reproduces d'Alembert's formula applied to the odd periodic extension.

Example 13.20 — The plucked string

A string of length $L = 1$ with $c = 1$ is pulled up at its midpoint to height h and released from rest: $f(x) = 2hx$ for $0 \le x \le \tfrac{1}{2}$, $f(x) = 2h(1-x)$ for $\tfrac{1}{2} \le x \le 1$; $g = 0$.

Coefficients. $B_n = 0$. By symmetry about $x = \tfrac{1}{2}$, $A_n = 0$ for even n. For any n, integrating by parts twice (or using the formula for the sine coefficients of a triangle),

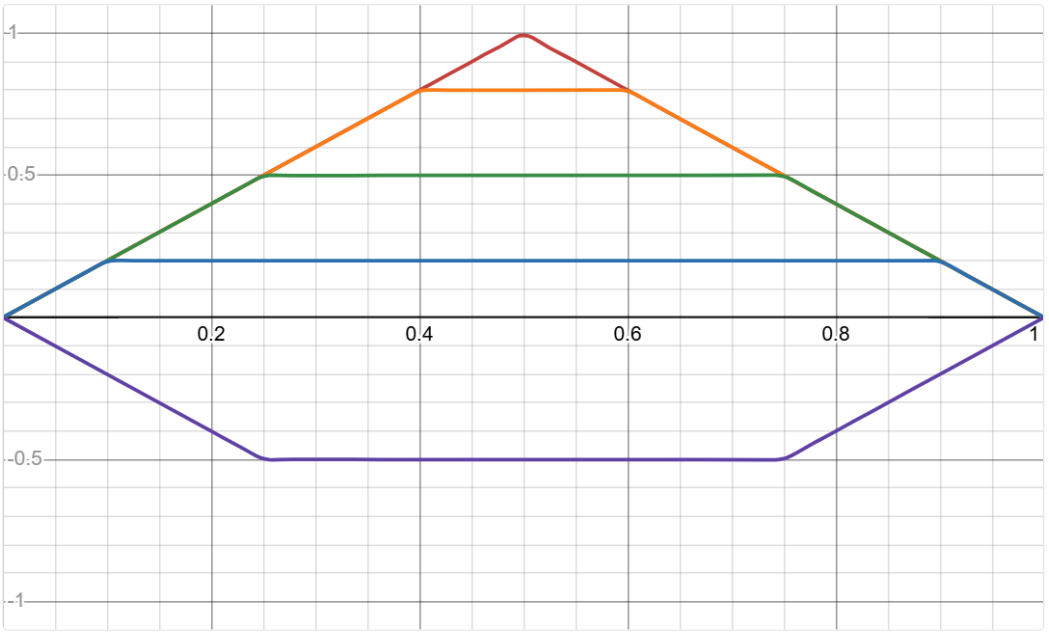
$$A_n = 2\int_0^1 f(x)\sin n\pi x\,dx = \frac{8h}{n^2\pi^2}\sin\frac{n\pi}{2}.$$

So $A_1 = 0.8106h$, $A_3 = -0.0901h$, $A_5 = 0.0324h$:

$$u(x,t) = \frac{8h}{\pi^2}\Big[\sin\pi x\cos\pi t - \frac{1}{9}\sin 3\pi x\cos 3\pi t + \frac{1}{25}\sin 5\pi x\cos 5\pi t - \cdots\Big].$$

Energy per mode. The energy of mode n is proportional to $\omega_n^2 A_n^2 \propto n^2 \cdot n^{-4} = n^{-2}$. The fraction in the fundamental is $1\big/\sum_{n\ \text{odd}} n^{-2} = 8/\pi^2 = 81.1\%$. Plucking at the centre suppresses all even harmonics; plucking nearer an end strengthens the higher harmonics, which is why a guitar sounds brighter near the bridge.

Shape. Summing 400 terms, at $t = \tfrac{1}{4}$ the profile is a trapezoid with a flat top at $h/2$ between $x = 0.25$ and 0.75; at $t = \tfrac{1}{2}$ the string is momentarily straight ($u \equiv 0$); at $t = 1$ it is the inverted triangle. The corners (kinks) travel at speed c and reflect from the ends — d'Alembert's picture.



The plucked string of Example 13.20 ($h = 1$) at $t = 0,\ 0.1,\ 0.25,\ 0.4,\ 0.75$: the initial triangle splits into two kinks that run outward, leaving a flat plateau that sinks through zero at $t = 0.5$ and reappears inverted.

Example 13.21 — Resonances of a shorted coaxial stub

A coaxial line of length $\ell = 0.75$ m with $c = 2 \times 10^8$ m/s is shorted at both ends (a closed resonator). Its voltage modes are $\sin(n\pi x/\ell)$ with

$$f_n = \frac{nc}{2\ell} = n \times \frac{2 \times 10^8}{1.5}\ \text{Hz} = 133.3\,n\ \text{MHz}.$$

If instead one end is open (D–N problem, Section 13.3), the eigenfunctions are $\sin\big((2n-1)\pi x/(2\ell)\big)$ and $f_n = (2n-1)c/(4\ell) = 66.7,\ 200,\ 333$ MHz, … — only odd multiples of the quarter-wave frequency. Only the odd multiples of the quarter-wave frequency resonate — the characteristic signature of a quarter-wave resonator, used in stub filters and in clock-distribution problems.

Energy conservation

Energy of the wave equation

For $u_{tt} = c^2 u_{xx}$ on $0 < x < L$ with fixed or free ends, the energy

$$E(t) = \frac{1}{2}\int_0^L \left(u_t^2 + c^2 u_x^2\right) dx$$

is constant in time.

Proof. $E'(t) = \int_0^L (u_t u_{tt} + c^2 u_x u_{xt})\,dx = \int_0^L u_t\,(u_{tt} - c^2 u_{xx})\,dx + c^2\big[u_x u_t\big]_0^L = 0$, because the PDE kills the integral and $u_t = 0$ (fixed end) or $u_x = 0$ (free end) kills the boundary term. □

For a transmission line the corresponding quantity is $\int_0^\ell \left(\tfrac{1}{2}Cv^2 + \tfrac{1}{2}Li^2\right)dx$, the electric plus magnetic stored energy; a resistive load makes the boundary term $-R_L i^2 \le 0$, the power it absorbs. Energy conservation gives uniqueness (as in Section 13.2) and, with Parseval's theorem (Ch. 15), the energy-per-mode decomposition used in Example 13.20.

The lossy line: distortion and the Heaviside condition

With losses, a sinusoidal wave $e^{j(\omega t - \beta x)}$ on the telegrapher's equation has a complex propagation constant $\gamma = \sqrt{(R + j\omega L)(G + j\omega C)}$, and in general both its attenuation and its phase velocity depend on frequency: a pulse spreads (disperses) as well as shrinks. There is one exception.

Distortionless line (Heaviside)

Put $v = e^{-at}w$ with $a = \tfrac{1}{2}(R/L + G/C)$. Then w satisfies

$$LC\,w_{tt} - w_{xx} = \tfrac{LC}{4}\left(\frac{R}{L} - \frac{G}{C}\right)^2 w.$$

If $R/L = G/C$ the right side vanishes, w obeys the lossless wave equation, and

$$v(x,t) = e^{-(R/L)t}\big[F(x - ct) + G(x + ct)\big], \qquad c = 1/\sqrt{LC}\,:$$

pulses travel at speed c without change of shape, merely attenuated.

Derivation. With $v = e^{-at}w$: $v_t = e^{-at}(w_t - aw)$, $v_{tt} = e^{-at}(w_{tt} - 2aw_t + a^2 w)$. Substituting into $v_{xx} = LCv_{tt} + (RC + GL)v_t + RGv$ and dividing by e^{-at}:

$$w_{xx} = LC\,w_{tt} + \big(RC + GL - 2aLC\big)w_t + \big(LCa^2 - (RC+GL)a + RG\big)w.$$

The choice $a = (RC+GL)/(2LC) = \tfrac{1}{2}(R/L + G/C)$ removes the w_t term, and the w coefficient becomes $RG - LCa^2 = -\tfrac{LC}{4}(R/L - G/C)^2$. $\square$

For a forward wave at position x the amplitude factor $e^{-(R/L)t}$, evaluated at the arrival time $t = x/c$, is $e^{-(R/L)\sqrt{LC}\,x} = e^{-(R/Z_0)x}$: the attenuation constant is $R/Z_0 = \sqrt{RG}$ nepers per metre.

Example 13.22 — Loading a line to make it distortionless

A telephone pair has $L = 0.6$ mH/km, $C = 50$ nF/km, $R = 50\ \Omega$/km, $G = 1\ \mu$S/km. Is it distortionless, and what series inductance would make it so?

$R/L = 50/(0.6\times10^{-3}) = 8.3\times10^{4}$ s^{-1}; $G/C = 10^{-6}/(50\times10^{-9}) = 20$ s^{-1}. Far from equal: $R/L \gg G/C$, the typical situation. Since adding G wastes power, Heaviside's remedy was to *increase* L: $L_{\text{new}} = RC/G = 50\times50\times10^{-9}/10^{-6} = 2.5$ H/km — impractically large, but even partial loading (Pupin coils of tens of mH every couple of km) dramatically reduced distortion on early long-distance telephone lines.

With full loading, $Z_0 = \sqrt{L/C} = \sqrt{2.5/(50\times10^{-9})} = 7071\ \Omega$ and the attenuation is $\sqrt{RG} = \sqrt{50\times10^{-6}} = 7.07\times10^{-3}$ Np/km (0.061 dB/km).

Common pitfall — applying d'Alembert directly on a finite interval

d'Alembert's formula needs f and g on the whole interval $[x-ct, x+ct]$. On a bounded interval, once ct exceeds the distance to an end, that interval sticks out of the domain: you must first extend f and g (oddly for fixed ends, evenly for free ends, periodically with period $2L$) and only then apply the formula. Forgetting this ignores the reflections.

Engineering note — time-domain reflectometry

A TDR instrument launches a fast step into a cable and watches the reflected wave. By d'Alembert, a reflection seen at time t comes from a discontinuity at distance $ct/2$, and its height and sign give the local Γ and hence the impedance there. Faults (opens, shorts, crushed sections, bad connectors) on cables and PCB traces are located this way to centimetre precision. The method relies on exactly the two properties proved above: finite propagation speed and shape preservation.

13.6 Laplace equation

Where it comes from

In electrostatics $\mathbf{E} = -\nabla V$ and Gauss's law $\nabla\cdot(\varepsilon\mathbf{E}) = \rho_v$ give, for uniform ε,

$$\nabla^2 V = -\frac{\rho_v}{\varepsilon} \quad \text{(Poisson)}, \qquad \nabla^2 V = 0 \quad \text{in charge-free regions (Laplace)}.$$

The same equation describes steady-state temperature ($u_t = 0$ in the heat equation), steady current flow in a resistive sheet or volume ($\nabla\cdot\mathbf{J} = 0$, $\mathbf{J} = -\sigma\nabla V$), magnetostatic scalar potential in current-free regions, and the steady state of any diffusion process. Laplace's equation is the **equilibrium** equation: it is what the heat and wave equations settle to once all transients have died.

Laplace's equation and harmonic functions

$$\nabla^2 u = 0, \qquad \text{in 2-D Cartesian } u_{xx} + u_{yy} = 0, \qquad \text{in polar } u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\phi\phi} = 0.$$

A twice continuously differentiable solution on a region Ω is called **harmonic** in Ω.

Examples of harmonic functions: x, xy, $x^2 - y^2$, $e^{kx}\sin ky$, $\ln r$, $r^n\cos n\phi$, $r^{-n}\sin n\phi$. In two dimensions, the real and imaginary parts of any analytic function of $z = x + jy$ are harmonic (Ch. 18); e.g. $z^2 = (x^2 - y^2) + j\,2xy$.

Properties of harmonic functions

Mean-value property

If u is harmonic in a region containing the closed disk of radius ρ centred at $\mathbf{x}_0$, then $u(\mathbf{x}_0)$ equals the average of u over the circle of radius ρ:

$$u(\mathbf{x}_0) = \frac{1}{2\pi}\int_0^{2\pi} u(\mathbf{x}_0 + \rho\,\hat{\mathbf{r}}(\phi))\,d\phi.$$

(In 3-D, the average over a sphere.)

Proof sketch. Let $M(\rho)$ be the circle average. Differentiating under the integral, $M'(\rho) = \frac{1}{2\pi}\int_0^{2\pi}\partial u/\partial r\,d\phi = \frac{1}{2\pi\rho}\oint \partial u/\partial n\,ds$, which by the divergence theorem equals $\frac{1}{2\pi\rho}\iint \nabla^2 u\,dA = 0$. So M is constant, and $M(0^+) = u(\mathbf{x}_0)$. $\square$

Maximum principle and uniqueness

A non-constant harmonic function on a bounded connected region attains its maximum and minimum only on the boundary. Consequently the Dirichlet problem $\nabla^2 u = 0$ in Ω, $u = h$ on $\partial\Omega$, has at most one solution, and if $|h_1 - h_2| \le \varepsilon$ on the boundary then $|u_1 - u_2| \le \varepsilon$ throughout.

The maximum principle follows from the mean-value property: a value cannot be an average of its neighbours and strictly larger than all of them. Physically, **there is no electrostatic potential maximum in free space**, so a charge cannot be held in stable equilibrium by electrostatic forces alone (Earnshaw's theorem), and the hottest point of an unheated plate is on its edge.

Rectangular domains

Consider the rectangle $0 < x < a$, $0 < y < b$ with Dirichlet data. By superposition, split a problem with data on all four sides into four problems, each with data on one side and zero on the other three. Take the case of data on the top side:

$$u_{xx} + u_{yy} = 0, \qquad u(0,y) = u(a,y) = 0, \quad u(x,0) = 0, \quad u(x,b) = h(x).$$

Separate, $u = X(x)Y(y)$: $X''/X = -Y''/Y = -\lambda$. The homogeneous conditions at $x = 0, a$ give the Dirichlet eigenproblem, $\lambda_n = (n\pi/a)^2$, $X_n = \sin(n\pi x/a)$. Then $Y'' - (n\pi/a)^2 Y = 0$ with $Y(0) = 0$, so $Y_n = \sinh(n\pi y/a)$. (In the direction that carries the non-homogeneous data the solutions are exponential, not oscillatory — the hallmark of Laplace's equation.) Superposing,

$$u(x,y) = \sum_{n=1}^{\infty} c_n \sinh\frac{n\pi y}{a}\sin\frac{n\pi x}{a},$$

and the top condition $u(x,b) = h(x)$ makes $c_n \sinh(n\pi b/a)$ the sine coefficients of h.

Dirichlet problem on a rectangle, data on one side

$$u(x,y) = \sum_{n=1}^{\infty} h_n \frac{\sinh(n\pi y/a)}{\sinh(n\pi b/a)}\sin\frac{n\pi x}{a}, \qquad h_n = \frac{2}{a}\int_0^a h(x)\sin\frac{n\pi x}{a}\,dx.$$

The ratio $\sinh(n\pi y/a)/\sinh(n\pi b/a) \approx e^{-n\pi(b-y)/a}$ for large n: the nth harmonic of the boundary data decays exponentially with distance from the boundary, over a length $a/(n\pi)$.

That last remark is fundamental. Fine detail in the boundary potential (period $2a/n$) is invisible a short distance away; only the smoothest components penetrate. This is why the field far from a periodic electrode array (an interdigitated capacitor, a grating) is nearly uniform, and why a grounded shield with small holes still shields well.

Example 13.23 — A single-harmonic boundary

On the square $0 < x, y < 1$, $u = 0$ on three sides and $u(x,1) = V_0 \sin \pi x$. Find u at the centre.

Only $h_1 = V_0$ is non-zero, so the series has one term:

$$u(x,y) = V_0 \frac{\sinh \pi y}{\sinh \pi}\sin \pi x.$$

At the centre: $u(\tfrac12,\tfrac12) = V_0 \sinh(\pi/2)/\sinh\pi = V_0 \times 2.3013/11.5487 = 0.1993\,V_0$.

Check harmonicity directly: $u_{xx} = -\pi^2 u$, $u_{yy} = +\pi^2 u$, sum 0. ✓ Check the maximum principle: the maximum V_0 is on the boundary at $(\tfrac12, 1)$. ✓

Example 13.24 — Trough with a raised lid

A long rectangular conducting trough of width a and height b has its three sides grounded and is closed by a lid, insulated from the sides, at potential V_0. Find $V(x,y)$ and evaluate it for a square trough.

With $h = V_0$: $h_n = \frac{2V_0}{a}\int_0^a \sin\frac{n\pi x}{a}dx = \frac{2V_0}{n\pi}(1 - \cos n\pi)$, i.e. $4V_0/(n\pi)$ for odd n, 0 for even n:

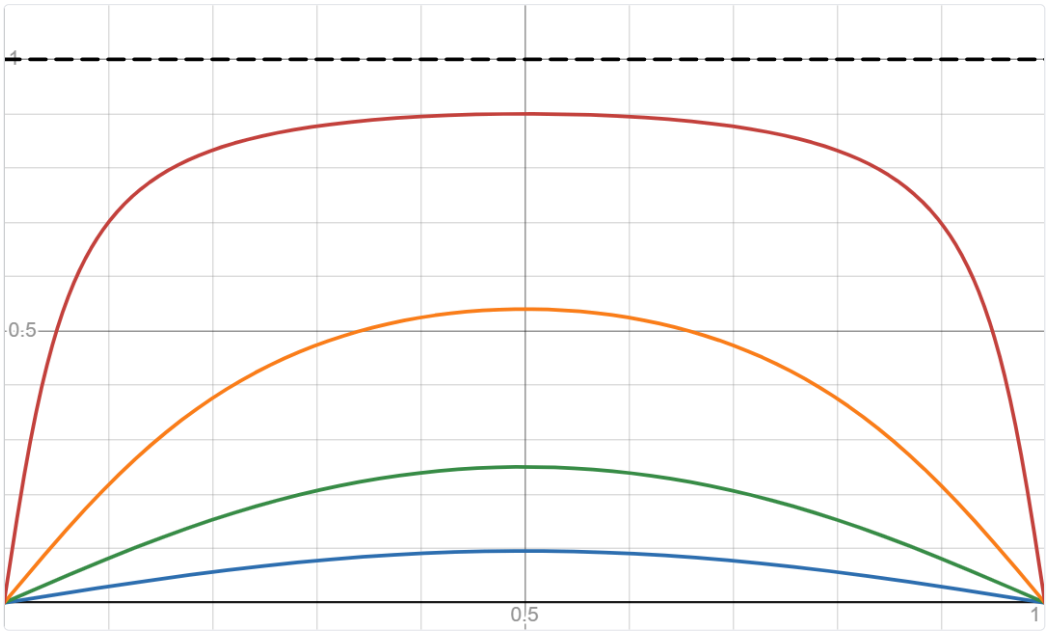
$$V(x,y) = \frac{4V_0}{\pi}\sum_{n\ \text{odd}} \frac{1}{n}\frac{\sinh(n\pi y/a)}{\sinh(n\pi b/a)}\sin\frac{n\pi x}{a}.$$

Square trough ($a = b = 1$), numerical values:

Point (x,y)	V/V_0
$(0.5,\ 0.25)$	0.0954
$(0.25,\ 0.5)$	0.1820
$(0.5,\ 0.5)$	0.2500
$(0.5,\ 0.75)$	0.5405

Exact check at the centre. Rotate the square by 90° four times: each rotation gives a problem with the lid on a different side. Their sum has V_0 on all four sides, whose solution is the constant V_0. By symmetry all four contribute equally at the centre, so $V(\tfrac12,\tfrac12) = V_0/4$ exactly. The series gives 0.25000; its first term alone gives 0.2537, already within 1.5%, because $\sinh(n\pi/2)/\sinh(n\pi) \approx e^{-n\pi/2}$ decays fast.

Near the top corners the series converges slowly, because the boundary data jump from V_0 to 0 there (the Gibbs phenomenon of Ch. 15 is present on the line $y = b$ itself, but disappears for $y < b$).



Potential profiles $V(x,y_0)/V_0$ across the square trough of Example 13.24 at heights $y_0 = 0.25,\ 0.5,\ 0.75,\ 0.95$ (blue to red). Close to the lid the profile approaches the flat boundary value with sharp edges; deeper down only the smooth fundamental survives.

Example 13.25 — Data on two sides; a semi-infinite strip

(a) A square plate $0 < x, y < 1$ has $u(x,1) = 100$ and $u(1,y) = 100$, the other two sides at 0. By superposition this is the sum of Example 13.24 (lid on top) and its reflection in the diagonal $y = x$ (lid on the right). At the centre, $u = 25 + 25 = 50$; at $(0.5, 0.25)$, $u = 100(0.0954) + 100\,V_{\text{lid}}(0.25, 0.5) = 9.54 + 18.20 = 27.74$, using the table (the reflection maps $(0.5, 0.25)$ to $(0.25, 0.5)$).

(b) **Semi-infinite strip.** $0 < x < a$, $y > 0$, sides $x = 0, a$ grounded, bottom $u(x,0) = h(x)$, u bounded as $y \to \infty$. Now $Y'' = (n\pi/a)^2 Y$ must stay bounded, so $Y_n = e^{-n\pi y/a}$ and

$$u(x,y) = \sum_{n=1}^{\infty} h_n\, e^{-n\pi y/a} \sin\frac{n\pi x}{a}.$$

With $h = V_0$ and $a = 1$, at $y = 1$ the first term is $\frac{4V_0}{\pi}e^{-\pi} = 0.055\,V_0$ and the next is $\frac{4V_0}{3\pi}e^{-3\pi} = 3.4 \times 10^{-5} V_0$: one width above the electrode only the fundamental is left. This is the basis of the rule of thumb that a grounded slot or waveguide below cut-off attenuates fields by about 27 dB ($e^{-\pi}$) per width of length.

Polar domains: disks, exteriors and annuli

From Example 13.11, the product solutions of Laplace's equation in polar coordinates are

$$1, \quad \ln r, \quad r^n\cos n\phi, \quad r^n \sin n\phi, \quad r^{-n}\cos n\phi, \quad r^{-n}\sin n\phi \qquad (n = 1, 2, \dots).$$

Which ones are allowed depends on the region:

Region	Allowed terms	Reason
Interior of disk $r < a$	$1,\ r^n\cos n\phi,\ r^n\sin n\phi$	$\ln r$ and r^{-n} blow up at $r = 0$
Exterior $r > a$, bounded at ∞	$1,\ r^{-n}\cos n\phi,\ r^{-n}\sin n\phi$	r^n and $\ln r$ grow without bound
Annulus $a < r < b$	all	$r = 0$ and ∞ excluded

Dirichlet problem for a disk

The solution of $\nabla^2 u = 0$ in $r < a$ with $u(a,\phi) = h(\phi)$ is

$$u(r,\phi) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(\frac{r}{a}\right)^n \big(a_n\cos n\phi + b_n \sin n\phi\big),$$

where a_n, b_n are the ordinary Fourier coefficients of h on $(-\pi, \pi)$: $a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} h\cos n\phi\, d\phi$, $b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} h \sin n\phi\, d\phi$. Summing the series in closed form gives **Poisson's integral formula**

$$u(r,\phi) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{a^2 - r^2}{a^2 - 2ar\cos(\phi - \psi) + r^2}\, h(\psi)\, d\psi.$$

Derivation of Poisson's formula. Substitute the coefficient integrals into the series and exchange sum and integral (legitimate for $r < a$ because of the geometric factor $(r/a)^n$):

$$u = \frac{1}{2\pi}\int_{-\pi}^{\pi} h(\psi)\Big[1 + 2\sum_{n=1}^{\infty}\Big(\frac{r}{a}\Big)^n \cos n(\phi-\psi)\Big]\,d\psi.$$

With $\rho = r/a$ and $\theta = \phi - \psi$, the bracket is $\operatorname{Re}\{1 + 2\sum_{n\ge1}(\rho e^{j\theta})^n\} = \operatorname{Re}\frac{1+\rho e^{j\theta}}{1-\rho e^{j\theta}} = \frac{1-\rho^2}{1-2\rho\cos\theta+\rho^2}$, a geometric series. □

At $r = 0$ Poisson's formula gives $u(0) = \frac{1}{2\pi}\int h\,d\psi$ — the mean-value property again. The **Poisson kernel** is positive, has unit integral and concentrates at $\psi = \phi$ as $r \to a$: it plays for the disk the role the heat kernel plays for the line.

Example 13.26 — A split cylinder

A long conducting cylinder of radius a is split lengthwise into two halves separated by thin insulating gaps; the upper half ($0 < \phi < \pi$) is held at V_0, the lower half at 0. Find the potential inside.

Coefficients. $a_0 = \frac{1}{\pi}\int_0^{\pi} V_0\,d\phi = V_0$; $a_n = \frac{V_0}{\pi}\int_0^{\pi}\cos n\phi\,d\phi = 0$; $b_n = \frac{V_0}{\pi}\int_0^{\pi}\sin n\phi\,d\phi = \frac{V_0}{n\pi}(1-\cos n\pi)$, i.e. $2V_0/(n\pi)$ for odd n:

$$V(r,\phi) = \frac{V_0}{2} + \frac{2V_0}{\pi}\sum_{n\ \text{odd}}\frac{1}{n}\Big(\frac{r}{a}\Big)^n \sin n\phi.$$

Closed form. Using $\sum_{n\ \text{odd}}\frac{z^n}{n} = \frac{1}{2}\ln\frac{1+z}{1-z}$ with $z = (r/a)e^{j\phi}$ and taking imaginary parts,

$$V(r,\phi) = \frac{V_0}{2} + \frac{V_0}{\pi}\arctan\frac{2ar\sin\phi}{a^2 - r^2}.$$

Numbers. At $r = a/2$, $\phi = \pi/2$: the arctan argument is $1/(3/4) = 4/3$, so $V = V_0[0.5 + \arctan(4/3)/\pi] = V_0(0.5 + 0.2952) = 0.7952\,V_0$; the series (200 terms) gives the same value. At the centre $V = V_0/2$, the average of the boundary, as the mean-value property demands. The equipotentials are arcs of circles through the two gaps.

Example 13.27 — Coaxial line: the annulus

Find the potential between the conductors of a coaxial line, inner radius a at V_0 and outer radius b grounded, and its capacitance per unit length.

The data do not depend on ϕ, so only the $n = 0$ terms survive: $V = A + B\ln r$. $V(b) = 0$ and $V(a) = V_0$ give

$$V(r) = V_0\frac{\ln(b/r)}{\ln(b/a)}, \qquad E_r = -\frac{dV}{dr} = \frac{V_0}{r\ln(b/a)}.$$

The surface charge on the inner conductor is $\varepsilon E_r(a)$; per unit length the charge is $Q' = 2\pi a\,\varepsilon E_r(a) = 2\pi\varepsilon V_0/\ln(b/a)$, so

$$C' = \frac{Q'}{V_0} = \frac{2\pi\varepsilon}{\ln(b/a)}.$$

Numbers (RG-58-like): $a = 0.45$ mm, $b = 1.47$ mm, $\varepsilon_r = 2.25$. $\ln(b/a) = 1.1838$, $C' = 2\pi \times 8.854\times10^{-12}\times 2.25/1.1838 = 105.7$ pF/m. With $L' = \frac{\mu_0}{2\pi}\ln(b/a) = 236.8$ nH/m, the characteristic impedance is $Z_0 = \sqrt{L'/C'} = 47.3\ \Omega$ and $c = 1/\sqrt{L'C'} = 2.0\times10^8$ m/s, closing the loop with the transmission-line equations of Section 13.5. At $V_0 = 100$ V the peak field (at $r = a$) is 1.88×10^5 V/m.

Example 13.28 — Conducting cylinder in a uniform field (exterior problem)

A grounded conducting cylinder of radius a is placed in a uniform applied field $\mathbf{E}_0 = E_0\hat{\mathbf{x}}$, whose potential is $-E_0x = -E_0r\cos\phi$. Find the potential outside.

We need $V(a,\phi) = 0$ and $V \to -E_0 r\cos\phi$ as $r\to\infty$. The applied potential is the $n = 1$ cosine term $r\cos\phi$; add the decaying partner $r^{-1}\cos\phi$:

$$V = -E_0\Big(r - \frac{a^2}{r}\Big)\cos\phi.$$

It is harmonic (sum of two product solutions), vanishes at $r = a$, and approaches the applied potential far away. The surface field is $E_r(a,\phi) = -\partial V/\partial r\big|_a = 2E_0\cos\phi$: the cylinder **doubles** the field at the points facing the applied field. (For a sphere the factor is 3.) Field enhancement at curved conductors is why high-voltage hardware uses large radii and why sharp points emit corona.

Common pitfall — forgetting the $\ln r$ and r^{-n} terms in an annulus

In an annulus or an exterior region with non-zero net charge, the term $\ln r$ (and for the annulus also r^{-n}) is essential; dropping it makes the coax problem unsolvable. Conversely, in a full disk including the origin those terms must be dropped. The rule is not "always use r^n"; it is "keep exactly those terms that are finite everywhere in the region".

Engineering note — capacitance and resistance extraction

Every capacitance or resistance calculation for an arbitrary geometry is a Laplace problem: solve $\nabla^2 V = 0$ with the electrodes at fixed potentials, compute the charge (or current) by integrating the normal field over an electrode, and divide by the voltage. Closed forms exist for coax, parallel plates, the split cylinder, and a few conformally mappable shapes; everything else (PCB traces, IC interconnect, connector pins) is done by field solvers that discretise exactly this equation.

13.7 Fourier-series solutions

Sections 13.4–13.6 all ended with the same step: expand a given function (initial temperature, initial displacement, boundary potential) in the eigenfunctions of a Sturm–Liouville problem. This section collects the general framework, explains what convergence results we can rely on, and extends the method to non-homogeneous equations (sources and forcing) and to two space dimensions. The theory of Fourier series itself — coefficient formulas, even and odd extensions, Dirichlet conditions, the Gibbs phenomenon, Parseval's theorem — is developed in Ch. 15; here we use its results.

The general recipe

For a linear evolution problem $u_t = \mathcal{A}u$ (or $u_{tt} = \mathcal{A}u$) with $\mathcal{A} = \alpha\,\partial_x^2$ (or $c^2\partial_x^2$) and homogeneous boundary conditions, let $\{X_n\}$ be the eigenfunctions of $-\partial_x^2$ with those boundary conditions, $X_n'' = -\lambda_n X_n$. Expand the unknown at every instant:

$$u(x,t) = \sum_n T_n(t)\,X_n(x), \qquad T_n(t) = \frac{\langle u(\cdot,t), X_n\rangle}{\langle X_n, X_n\rangle}.$$

Because ∂_x^2 acting on X_n just multiplies by $-\lambda_n$, the PDE becomes an *uncoupled* set of ODEs, one per mode:

PDE	Modal ODE	Modal solution
$u_t = \alpha u_{xx}$	$T_n' = -\alpha\lambda_n T_n$	$T_n(0)\,e^{-\alpha\lambda_n t}$
$u_{tt} = c^2 u_{xx}$	$T_n'' = -c^2\lambda_n T_n$	$T_n(0)\cos\omega_n t + \dfrac{T_n'(0)}{\omega_n}\sin\omega_n t$, $\omega_n = c\sqrt{\lambda_n}$
$u_{xx} + u_{yy} = 0$ (y plays "time")	$Y_n'' = \lambda_n Y_n$	$A_n\cosh\sqrt{\lambda_n}y + B_n\sinh\sqrt{\lambda_n}y$

The initial data supply $T_n(0)$, $T_n'(0)$ through the **Fourier coefficients** of f and g. In matrix language, the eigenfunctions diagonalise the spatial operator, exactly as eigenvectors diagonalise $\mathbf{A}$ in $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$.

Which Fourier series?

The boundary conditions decide which expansion is natural on $(0, L)$:

Boundary conditions	Expansion of f on $(0,L)$	Equivalent periodic extension
D–D	sine series $\sum b_n \sin(n\pi x/L)$	odd, period $2L$
N–N	cosine series $\frac{a_0}{2} + \sum a_n\cos(n\pi x/L)$	even, period $2L$
D–N	quarter-wave sine series $\sum b_n \sin\frac{(2n-1)\pi x}{2L}$	odd about 0, even about L, period $4L$
periodic on $(-\pi,\pi)$	full series $\frac{a_0}{2} + \sum(a_n\cos n\phi + b_n\sin n\phi)$	2π-periodic
Robin	$\sum c_n X_n$ with $c_n = \int f X_n / \int X_n^2$	none simple; use Sturm–Liouville orthogonality

The middle column is what you compute; the right-hand column tells you, via Ch. 15, how fast the coefficients decay and where Gibbs overshoot will appear. For example, a constant initial temperature $f = 100$ with Dirichlet ends has an odd extension that is a square wave: coefficients $\propto 1/n$, Gibbs ringing at the ends at $t = 0$. With Neumann ends the even extension of the same constant is constant: the expansion has a single term.

Convergence: what we may rely on

Convergence of series solutions

Let f be piecewise smooth on $[0, L]$.

1. **At $t = 0$** the series equals the Fourier series of the appropriate extension of f; it converges to $f(x)$ where the extension is continuous and to the average of the one-sided limits at jumps (Ch. 15). Convergence is uniform if and only if the extension is continuous (no jumps, including at the ends).
2. **Heat equation, $t > 0$:** the series and all its term-by-term derivatives converge uniformly; u is C^∞ and satisfies the PDE classically. Moreover $u(\cdot,t) \to f$ in mean square as $t \to 0^+$.
3. **Wave equation:** no smoothing. The series is a classical solution if the odd extensions of f and g are C^2 and C^1 respectively (equivalently $\sum n^2|A_n|$ and $\sum n|B_n|$ converge); otherwise it is a **weak solution** whose kinks or jumps travel along characteristics.
4. **Laplace equation:** inside the region, at positive distance from the boundary that carries the data, the factors $e^{-n\pi d/a}$ (rectangle) or $(r/a)^n$ (disk) make the series converge uniformly with all derivatives, and u is C^∞ — even real-analytic — however rough the data.

The smoothness of f controls the decay of its coefficients: jumps give $1/n$, kinks (continuous f, jump in f') give $1/n^2$, and so on, each extra derivative of continuity buying one more power of n (Ch. 15). For the heat equation this matters only at small t; for the wave equation it matters for all time.

How many terms?

Because the modal factors decay like $e^{-\alpha\lambda_n t}$, the number of terms needed for a given accuracy is set by t.

Example 13.29 — Truncation error of the cooling-rod series

In Example 13.12 ($u = \frac{400}{\pi}\sum_{n\text{ odd}} \frac{1}{n} e^{-n^2\pi^2 t}\sin n\pi x$), how many terms are needed for an error below 0.01 at $t = 0.01$ and at $t = 0.001$?

Bound. $|\sin| \le 1$, so the error after keeping all odd $n < N$ is at most the tail $\varepsilon_N(t) = \dfrac{400}{\pi}\displaystyle\sum_{n \ge N,\ \text{odd}} \frac{1}{n} e^{-n^2\pi^2 t}$. Since the terms decrease faster than geometrically, the first omitted term is a good estimate.

t	$N = 5$	$N = 11$	$N = 21$	$N = 41$
0.01	2.31	7.6×10^{-5}	$\approx 10^{-18}$	—
0.001	44.5	7.02	0.124	2.4×10^{-7}

At $t = 0.01$ the odd terms $n = 1, \dots, 9$ (five terms) suffice; at $t = 0.001$ about $n \le 29$ is needed (between $N = 21$ and $N = 41$). A rule of thumb: keep terms until $n^2\pi^2 \alpha t / L^2 \gtrsim 10$, i.e. $n \gtrsim \frac{L}{\pi}\sqrt{10/(\alpha t)}$. For very small t, switch to the erfc approximation of Section 13.4.

Non-homogeneous equations: eigenfunction expansion

Sources and forcing are handled by the same expansion. Consider

$$u_t = \alpha u_{xx} + s(x,t), \qquad u(0,t) = u(L,t) = 0, \qquad u(x,0) = f(x).$$

Expand $u = \sum T_n(t)\sin(n\pi x/L)$ and $s = \sum s_n(t)\sin(n\pi x/L)$ with $s_n(t) = \frac{2}{L}\int_0^L s(x,t)\sin(n\pi x/L)\,dx$. Substituting and matching coefficients (orthogonality),

$$T_n' + \alpha\lambda_n T_n = s_n(t), \qquad T_n(0) = b_n,$$

a first-order linear ODE for each mode (Ch. 11) with solution

$$T_n(t) = b_n e^{-\alpha\lambda_n t} + \int_0^t e^{-\alpha\lambda_n (t-\tau)} s_n(\tau)\,d\tau.$$

The second term is a **convolution** of the modal source with the modal impulse response $e^{-\alpha\lambda_n t}$ — **Duhamel's principle**. Each mode is a first-order low-pass filter with cut-off $\alpha\lambda_n$, driven by the projection of the source onto that mode. The same idea for the wave equation gives a bank of undamped resonators $T_n'' + \omega_n^2 T_n = s_n(t)$, and a source whose time dependence contains ω_n excites mode n resonantly.

Example 13.30 — Joule heating of a wire with cold ends

A wire of length $L = 1$ (nondimensional, $\alpha = 1$) starts at $u = 0$, its ends are held at 0, and a uniform current switched on at $t = 0$ produces a constant source $s = 1$. Find $u(x,t)$ and the midpoint temperature.

Modal source. $s_n = 2\int_0^1 \sin n\pi x\,dx = 4/(n\pi)$ for odd n, 0 for even n. With $b_n = 0$ and constant s_n:

$$T_n(t) = \frac{s_n}{\lambda_n}\bigl(1 - e^{-\lambda_n t}\bigr) = \frac{4}{n^3\pi^3}\bigl(1 - e^{-n^2\pi^2 t}\bigr), \qquad n \text{ odd}.$$

$$u(x,t) = \frac{4}{\pi^3}\sum_{n\text{ odd}} \frac{1 - e^{-n^2\pi^2 t}}{n^3}\sin n\pi x.$$

Steady state check. As $t \to \infty$, $u \to u_s$ with $u_s'' = -1$, $u_s(0) = u_s(1) = 0$, i.e. $u_s = x(1-x)/2$, maximum 0.125 at the centre. The series gives $\frac{4}{\pi^3}\sum_{\text{odd}} \frac{(-1)^{(n-1)/2}}{n^3} = \frac{4}{\pi^3}\cdot\frac{\pi^3}{32} = 0.125$. ✓ (The one-term value is $4/\pi^3 = 0.1290$: the $1/n^3$ decay makes even the steady-state series converge fast.)

Transient. $u(\tfrac12, 0.05) = 0.0463$, $u(\tfrac12, 0.1) = 0.0769$, $u(\tfrac12, 1) = 0.12499$. The approach to steady state is governed by the slowest mode, time constant $1/\pi^2$.

In dimensional form with $s = g/(\rho c_p)$, $g = J^2/\sigma$, the steady centre temperature rise is $gL^2/(8k)$: halving the length of a fuse element quarters its temperature rise, a key design relation for fuses and thin-film resistors.

Example 13.31 — Finite open-ended RC line (the 0.38 $RC\ell^2$ rule)

An RC line of length ℓ, open at $x=\ell$, initially discharged, has its input stepped to V_0 at $t=0$. Find $v(\ell,t)$ and the 50% delay.

Set-up. $v_t = v_{xx}/(RC)$, $v(0,t)=V_0$ (Dirichlet), $v_x(\ell,t)=0$ (open circuit, Section 13.2), $v(x,0)=0$. Steady state $v_s=V_0$; transient $w=v-V_0$ with D–N eigenfunctions $\sin\mu_n x$, $\mu_n=(2n-1)\pi/(2\ell)$, and $w(x,0)=-V_0$:

$$b_n=\frac{2}{\ell}\int_0^{\ell}(-V_0)\sin\mu_n x\,dx=-\frac{2V_0}{\ell\mu_n}=-\frac{4V_0}{(2n-1)\pi}.$$

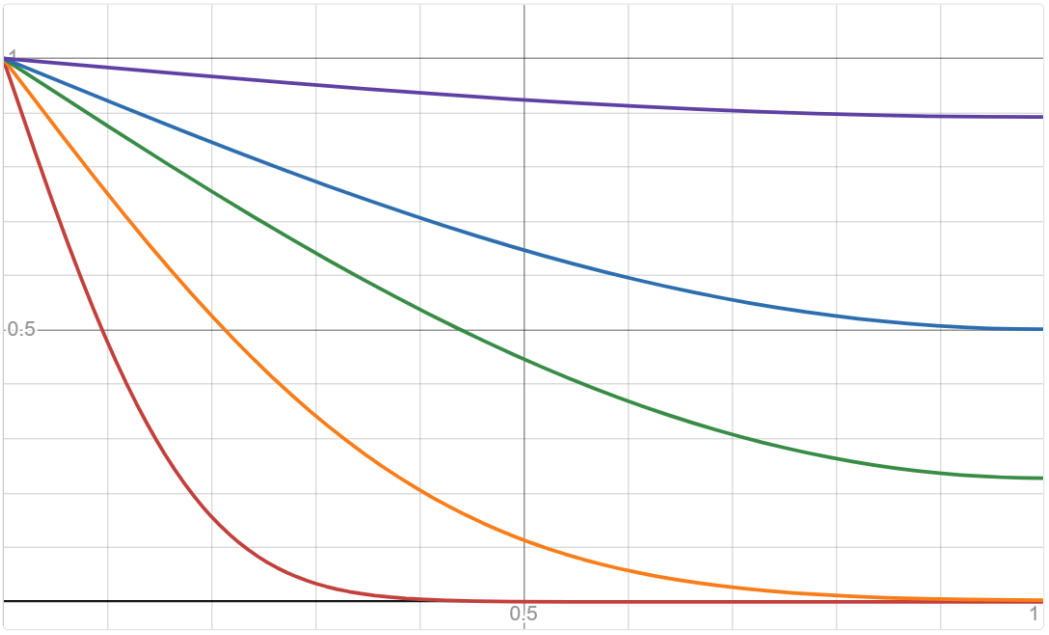
$$v(x,t)=V_0\Big[1-\frac{4}{\pi}\sum_{n=1}^{\infty}\frac{\sin\mu_n x}{2n-1}\,e^{-\mu_n^2 t/(RC)}\Big].$$

At the open end, $\sin\mu_n\ell=(-1)^{n-1}$. In terms of $\tau=t/(RC\ell^2)$:

$$\frac{v(\ell,t)}{V_0}=1-\frac{4}{\pi}\Big[e^{-\pi^2\tau/4}-\frac{1}{3}e^{-9\pi^2\tau/4}+\frac{1}{5}e^{-25\pi^2\tau/4}-\cdots\Big].$$

Solving $v(\ell,t)=V_0/2$ numerically gives $\tau_{50}=0.3787$, so $t_{50}=0.38\,RC\ell^2$ — the standard figure used in interconnect timing. The one-term formula $\tau_{50}=\frac{4}{\pi^2}\ln\frac{8}{\pi}=0.3788$ is already accurate to four figures. The 90% point is at $\tau=1.03$.

Comparison. The semi-infinite estimate of Example 13.16 gave $1.10\,RC\ell^2$; the finite open line is about three times faster because charge stops flowing out at the open end and piles up there.



Voltage $v(x,t)/V_0$ along the open-ended RC line of Example 13.31 (x in units of ℓ) at $\tau=t/(RC\ell^2)=0.01,\ 0.05,\ 0.2,\ 0.38,\ 1$. The profile has zero slope at the open end; at $\tau=0.38$ the far end reaches one half.

Two space dimensions: double Fourier series

On a rectangle $0<x<a$, $0<y<b$ with all edges at zero, the heat equation $u_t=\alpha(u_{xx}+u_{yy})$ separates as $X(x)Y(y)T(t)$ with two separation constants. The eigenfunctions are products,

$$\psi_{mn}(x,y)=\sin\frac{m\pi x}{a}\sin\frac{n\pi y}{b},\qquad \lambda_{mn}=\Big(\frac{m\pi}{a}\Big)^2+\Big(\frac{n\pi}{b}\Big)^2,$$

and the solution is a **double Fourier sine series**

$$u=\sum_{m=1}^{\infty}\sum_{n=1}^{\infty}B_{mn}\,e^{-\alpha\lambda_{mn}t}\,\psi_{mn},\qquad B_{mn}=\frac{4}{ab}\int_0^a\int_0^b f(x,y)\,\psi_{mn}\,dy\,dx.$$

For the wave equation (a rectangular membrane; the same eigenvalues give the cut-off frequencies of the TM modes of a rectangular waveguide) the frequencies are $\omega_{mn}=c\sqrt{\lambda_{mn}}$.

Example 13.32 — Cooling of a square plate; degenerate modes

A square plate of side $a = 1$ ($\alpha = 1$) with edges at zero starts at $u = 1$.

$B_{mn} = 4\left(\int_0^1 \sin m\pi x\,dx\right)\left(\int_0^1 \sin n\pi y\,dy\right) = \dfrac{16}{mn\pi^2}$ for m, n both odd, zero otherwise. The slowest mode $(1,1)$ has $\lambda_{11} = 2\pi^2$, time constant $1/(2\pi^2) = 0.0507$ — half that of a rod of the same length, because heat now escapes in two directions.

Centre temperature for large t: $u(\tfrac12, \tfrac12, t) \approx \tfrac{16}{\pi^2} e^{-2\pi^2 t}$. Note this is exactly the *square* of the 1-D one-term result $\tfrac{4}{\pi}e^{-\pi^2 t}$: for product initial data on a product domain, the 2-D solution is the product of 1-D solutions, $u(x,y,t) = u_1(x,t)\,u_1(y,t)$.

Degeneracy. $\lambda_{13} = \lambda_{31} = 10\pi^2$: distinct eigenfunctions share an eigenvalue. In a square drum or a square waveguide this means two modes with the same frequency; any combination $\alpha\psi_{13} + \beta\psi_{31}$ is also a mode, and small asymmetries split the pair. Circular regions lead to Bessel functions in r (a singular Sturm–Liouville problem; see the Further reading), with the same structure.

Common pitfall — differentiating a Fourier series term by term

Term-by-term differentiation is safe for the heat-equation series at $t > 0$ and for Laplace series away from the data boundary, but **not** at $t = 0$, and not for wave-equation series with non-smooth data. Differentiating the sine series of a square wave produces a divergent series. If you need u_x at a boundary (to compute a heat flux or a surface charge), either use $t > 0$ or interior points and take a limit, or differentiate the closed form.

Engineering note — spectral solvers

Computing the coefficients numerically with the FFT, multiplying each by its modal factor $e^{-\alpha\lambda_n t}$ or $\cos\omega_n t$, and transforming back is a **spectral method**. For simple geometries it is far more accurate per unknown than finite differences, because the error falls faster than any power of the number of modes when the solution is smooth. Spectral methods are used in thermal modelling of layered IC stacks and in optical beam propagation, which solves a heat-like (paraxial) equation.

13.8 Applications to fields, diffusion, and heat transfer

This final section applies the methods of the chapter to three problems that every electrical engineer meets. In each case the mathematics is already in hand; the work is in setting up the PDE, its domain and its auxiliary conditions, and in reading the answer. (Deeper treatment of the physics belongs to courses on electromagnetics, device physics and thermal design.)

Fields: the skin effect as a diffusion problem

Inside a good conductor ($\sigma \gg \omega\varepsilon$) the displacement current is negligible, so Ampère's and Faraday's laws read $\nabla \times \mathbf{H} = \sigma\mathbf{E}$ and $\nabla \times \mathbf{E} = -\mu\,\partial\mathbf{H}/\partial t$. Taking the curl of the first and using $\nabla \times \nabla \times \mathbf{H} = \nabla(\nabla\cdot\mathbf{H}) - \nabla^2\mathbf{H}$ with $\nabla\cdot\mathbf{H} = 0$ (Ch. 7):

$$\nabla^2 \mathbf{H} = \mu\sigma\,\frac{\partial \mathbf{H}}{\partial t}.$$

This is a **diffusion equation** with diffusivity $1/(\mu\sigma)$ — the magnetic field (and with it the current density $\mathbf{J} = \nabla \times \mathbf{H}$) *diffuses* into a conductor rather than propagating as a wave.

Sinusoidal steady state. For a conductor filling $x > 0$ with a tangential surface field $H_y(0,t) = H_0\cos\omega t$, use phasors (Ch. 2): $H_y(x,t) = \mathrm{Re}\{\tilde{H}(x)e^{j\omega t}\}$. The PDE becomes the ODE

$$\tilde{H}'' = j\omega\mu\sigma\,\tilde{H}, \qquad \tilde{H}(0) = H_0, \qquad \tilde{H} \text{ bounded as } x \to \infty.$$

Its characteristic roots are $\pm\sqrt{j\omega\mu\sigma} = \pm(1+j)/\delta$ with

$$\delta = \sqrt{\frac{2}{\omega\mu\sigma}},$$

since $\left((1+j)/\delta\right)^2 = 2j/\delta^2 = j\omega\mu\sigma$. Boundedness rejects the growing root:

Skin-effect field in a conducting half-space

$$H_y(x,t) = H_0\,e^{-x/\delta}\cos\left(\omega t - \frac{x}{\delta}\right), \qquad \delta = \sqrt{\frac{2}{\omega\mu\sigma}}\ \text{(skin depth)}.$$

The amplitude falls by e^{-1} per skin depth, and the phase lags by one radian per skin depth. The current density has the same profile, and the conductor's AC resistance per square is the **surface resistance** $R_s = 1/(\sigma\delta)$ — as if the current flowed uniformly in a layer of thickness δ.

The solution is a "critically damped" wave: a travelling wave $\cos(\omega t - x/\delta)$ whose wavelength $2\pi\delta$ equals its decay length times 2π, so it is attenuated by $e^{-2\pi} \approx 0.002$ within one wavelength. Exactly the same mathematics describes the daily or annual temperature wave penetrating into the ground, or into a heat sink under a pulsed load, with $\mu\sigma$ replaced by $1/\alpha$: the "thermal skin depth" is $\sqrt{2\alpha/\omega}$.

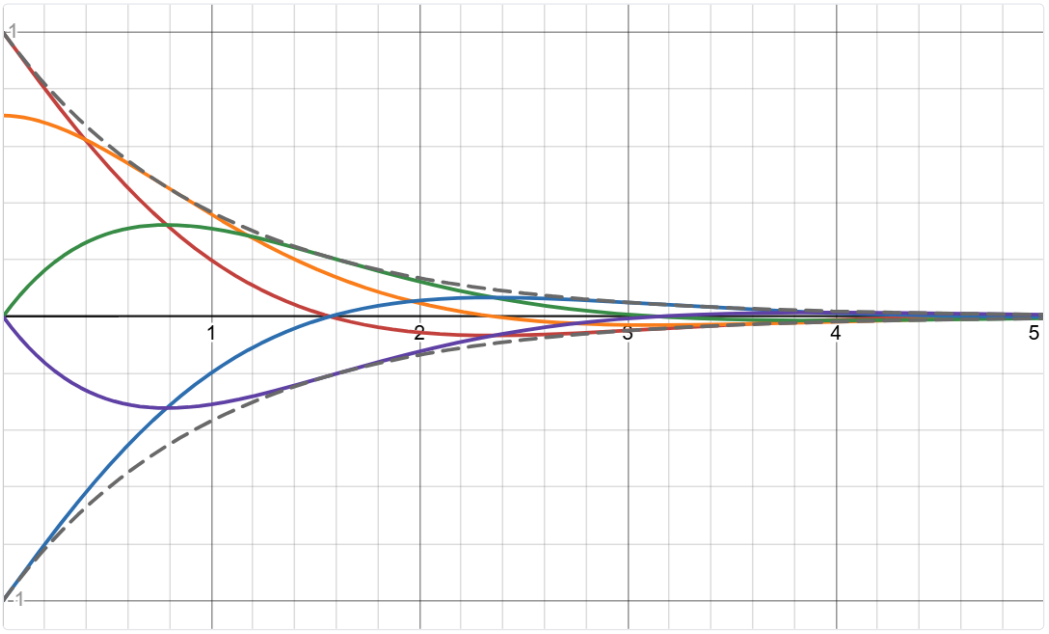
Example 13.33 — Skin depth in copper

Copper: $\sigma = 5.8 \times 10^{7}\ \mathrm{S/m}$, $\mu = \mu_0 = 4\pi \times 10^{-7}\ \mathrm{H/m}$.

f	$\delta = \sqrt{2/(2\pi f \mu_0 \sigma)}$	$R_s = 1/(\sigma\delta)$
60 Hz	8.53 mm	2.0 $\mu\Omega$
1 MHz	66.1 μm	0.26 mΩ
1 GHz	2.09 μm	8.25 mΩ

Check of units: $\omega\mu\sigma$ has units $\mathrm{s^{-1}} \cdot \mathrm{H\,m^{-1}} \cdot \mathrm{S\,m^{-1}} = \mathrm{m^{-2}}$ (since $\mathrm{H} \cdot \mathrm{S} = \mathrm{s}$), so δ is in metres. ✓

Consequences. At 60 Hz conductors thicker than about $2\delta \approx 17$ mm waste copper, which is why large bus bars are hollow or laminated. A round wire of radius $a = 1$ mm at 1 MHz ($a \gg \delta$) carries current essentially in an annulus of thickness δ, giving $R_{ac}/R_{dc} \approx a/(2\delta) = 7.6$; this motivates Litz wire in switch-mode power-supply transformers. At 1 GHz the current flows within a few micrometres of the surface, so surface roughness of PCB copper (comparable to δ) noticeably increases loss.



Skin-effect field H_y/H_0 versus depth x/δ at phases $\omega t = 0,\ \pi/4,\ \pi/2,\ \pi,\ 3\pi/2$ (red to purple), inside the envelope $\pm e^{-x/\delta}$ (dashed). Beyond about 3δ the field is negligible at every instant.

Diffusion: dopant profiles in semiconductors

Impurity atoms are introduced into silicon by diffusion at high temperature. With a constant diffusion coefficient D, the concentration $N(x,t)$ below a planar surface obeys Fick's second law,

$$\frac{\partial N}{\partial t} = D\,\frac{\partial^2 N}{\partial x^2}, \qquad x > 0,$$

on what is effectively a semi-infinite domain (diffusion lengths are micrometres, wafers are hundreds of micrometres thick). Two boundary conditions model the two standard process steps.

Standard diffusion profiles

1. **Constant-source (predeposition):** the surface is held at the solid-solubility limit, $N(0,t) = C_s$, and $N(x,0) = 0$. Then (Section 13.4)

$$N(x,t) = C_s\ \mathrm{erfc}\,\frac{x}{2\sqrt{Dt}}, \qquad Q(t) = \int_0^\infty N\,dx = \frac{2C_s\sqrt{Dt}}{\sqrt{\pi}}.$$

2. **Limited-source (drive-in):** a fixed dose Q (atoms per unit area) starts at the surface and the surface is sealed, $N_x(0,t) = 0$ (Neumann). By the method of images (even extension) the whole-line heat kernel gives

$$N(x,t) = \frac{Q}{\sqrt{\pi Dt}}\,e^{-x^2/(4Dt)},$$

a half-Gaussian whose surface concentration $Q/\sqrt{\pi Dt}$ falls as the dose spreads.

The **junction depth** x_j is where the diffused concentration equals the opposite-type background doping N_B.

Example 13.34 — Junction depths

Boron diffuses into an n-type wafer with background $N_B = 10^{16}\ \text{cm}^{-3}$ at a temperature where $D = 10^{-12}\ \text{cm}^2/\text{s}$, for $t = 1\ \text{h} = 3600\ \text{s}$. Then $\sqrt{Dt} = \sqrt{3.6\times10^{-9}} = 6.0\times10^{-5}\ \text{cm} = 0.60\ \mu\text{m}$.

(a) Predeposition with $C_s = 10^{20}\ \text{cm}^{-3}$. Solve $\operatorname{erfc}\left(x_j/(2\sqrt{Dt})\right) = N_B/C_s = 10^{-4}$: $\operatorname{erfc}^{-1}(10^{-4}) = 2.751$, so

$$x_j = 2\times 0.60\ \mu\text{m}\times 2.751 = 3.30\ \mu\text{m}.$$

The dose introduced is $Q = 2C_s\sqrt{Dt/\pi} = 2\times10^{20}\times6.0\times10^{-5}/1.7725 = 6.8\times10^{15}\ \text{cm}^{-2}$.

(b) Drive-in of an implanted dose $Q = 10^{14}\ \text{cm}^{-2}$ under the same D and t. Surface concentration $N(0) = Q/\sqrt{\pi Dt} = 10^{14}/(1.7725\times6.0\times10^{-5}) = 9.40\times10^{17}\ \text{cm}^{-3}$. Junction: $N(0)e^{-x_j^2/(4Dt)} = N_B$ gives

$$x_j = \sqrt{4Dt\,\ln\frac{N(0)}{N_B}} = \sqrt{4\times3.6\times10^{-9}\times\ln 94.0}\ \text{cm} = 2.56\ \mu\text{m}.$$

Scaling check. Both depths scale as $\sqrt{Dt}$: doubling the diffusion time deepens the junction by $\sqrt{2}$ (only approximately for (b), since the log factor also changes). This is the thermal-budget constraint of IC processing: every later high-temperature step adds to Dt and moves all earlier junctions.

Engineering note — the same erfc in carrier transport

Minority carriers injected into a long neutral region obey $\partial\Delta n/\partial t = D_n\Delta n_{xx} - \Delta n/\tau_n$, a diffusion equation with a decay term. In steady state it becomes the ODE $\Delta n'' = \Delta n/L_n^2$, $L_n = \sqrt{D_n\tau_n}$ (the diffusion length), whose decaying exponential solution underlies the ideal-diode equation. The transient (switching) behaviour of a diode is the full PDE, solved with the methods of this chapter after the substitution $\Delta n = e^{-t/\tau_n}w$, which removes the decay term exactly as in the distortionless-line trick of Section 13.5.

Heat transfer: transient cooling with convection

A plate of thickness $2L_h$ (a ceramic substrate, a heat-spreader, a slab of potting compound) is initially at uniform excess temperature θ_i above the coolant, and at $t = 0$ both faces are exposed to a fluid with heat-transfer coefficient h_c. Measure x from the mid-plane and let $\theta = u - u_\infty$. By symmetry we solve on $0 < x < L_h$:

$$\theta_t = \alpha\,\theta_{xx}, \qquad \theta_x(0,t) = 0\ \text{(symmetry)}, \qquad -k\,\theta_x(L_h,t) = h_c\,\theta(L_h,t)\ \text{(Robin)}, \qquad \theta(x,0) = \theta_i.$$

Nondimensionalise: $\xi = x/L_h$, **Fourier number** $\text{Fo} = \alpha t/L_h^2$, **Biot number** $\text{Bi} = h_cL_h/k$. The Biot number compares internal conduction resistance L_h/k with surface convection resistance $1/h_c$.

Separation (Section 13.3) with $X'(0) = 0$ gives $X = \cos\mu\xi$, and the Robin condition $X'(1) + \text{Bi}\,X(1) = 0$ gives the eigenvalue equation

$$\mu\tan\mu = \text{Bi},$$

with one root μ_n in each interval $\left((n-1)\pi,\ (n-\tfrac12)\pi\right)$. The generalised Fourier coefficients of the constant initial value (Sturm–Liouville theorem, with $\int_0^1\cos^2\mu_n\xi\,d\xi = \tfrac12 + \frac{\sin 2\mu_n}{4\mu_n}$) are

$$C_n = \frac{\int_0^1\cos\mu_n\xi\,d\xi}{\int_0^1\cos^2\mu_n\xi\,d\xi} = \frac{4\sin\mu_n}{2\mu_n + \sin 2\mu_n},$$

so

$$\frac{\theta(\xi,\text{Fo})}{\theta_i} = \sum_{n=1}^{\infty} C_n\,e^{-\mu_n^2\text{Fo}}\cos\mu_n\xi.$$

Example 13.35 — Biot number and the lumped-capacitance limit

Compute μ_1, C_1 and the centre temperature at $\mathrm{Fo} = 1$ for $\mathrm{Bi} = 0.1,\ 1,\ 10$.

Solving $\mu \tan\mu = \mathrm{Bi}$ by bisection on $(0, \pi/2)$:

Bi	μ_1	μ_1^2	C_1	μ_2	$\theta(0,1)/\theta_i$ **(one term)**
0.1	0.3111	0.0968	1.016	3.173	0.922
1	0.8603	0.7402	1.119	3.426	0.534
10	1.4289	2.042	1.262	4.306	0.164

For $\mathrm{Bi} = 1$ the full series (50 terms) gives $\theta(0,1)/\theta_i = 0.53386$ identical to five figures with the one-term value. At $\mathrm{Fo} = 0.2$ the series gives 0.9506 and one term gives 0.9651, still within 1.5%; the second mode has $\mu_2^2 = 11.7$, so it is negligible once $\mathrm{Fo} \gtrsim 0.2$.

Interpretation.

- $\mathrm{Bi} \ll 1$: $\mu_1^2 \approx \mathrm{Bi}$ (here $0.0968 \approx 0.1$) and $C_1 \approx 1$. The solution reduces to $\theta \approx \theta_i e^{-\mathrm{Bi}\,\mathrm{Fo}} = \theta_i e^{-h_c t/(\rho c_p L_h)}$: the **lumped-capacitance** (single RC) model, with thermal time constant $\rho c_p L_h / h_c$ = (heat capacity per unit area)×(surface resistance). The temperature is nearly uniform across the plate. The usual criterion for using a lumped model is $\mathrm{Bi} < 0.1$.
- $\mathrm{Bi} \gg 1$: $\mu_1 \to \pi/2$, the surface is effectively clamped at the coolant temperature (Dirichlet), and internal conduction limits the cooling.

Dimensional numbers. An alumina substrate ($k = 30\ \mathrm{W\,m^{-1}K^{-1}}$, $\rho c_p = 3.43 \times 10^6\ \mathrm{J\,m^{-3}K^{-1}}$, so $\alpha = 8.74 \times 10^{-6}\ \mathrm{m^2/s}$) of half-thickness $L_h = 5$ mm in forced water cooling with $h_c = 6000\ \mathrm{W\,m^{-2}K^{-1}}$ has $\mathrm{Bi} = 1$. Its dominant time constant is $L_h^2/(\mu_1^2\alpha) = 2.5 \times 10^{-5}/(0.7402 \times 8.74 \times 10^{-6}) = 3.86$ s; a lumped model would wrongly predict $\rho c_p L_h / h_c = 2.86$ s and a uniform temperature.

Common pitfall — lumping when Bi is not small

The single-time-constant "thermal RC" model is a one-mode truncation that is valid only when $\mathrm{Bi} \lesssim 0.1$ *and* Fo is not small. For a power device on a thick, poorly conducting substrate, Bi can exceed 1, and the lumped model underestimates both the time constant and the peak junction temperature at short times. Use the distributed solution (or a multi-mode Foster/Cauer network).

Choosing the method: a summary of the chapter's toolkit

Situation	Method	Section
Finite interval/rectangle/disk, homogeneous BCs on all but one side	separation of variables + Fourier series	13.3, 13.7
Constant non-homogeneous BCs	subtract steady state, then separate	13.2, 13.4
Sources or time-dependent BCs	eigenfunction expansion + Duhamel (modal ODEs)	13.7
Infinite line, wave equation	d'Alembert, characteristics	13.5
Semi-infinite line, wave equation with fixed/free end	method of images (odd/even extension)	13.5
Infinite/semi-infinite domain, diffusion	heat kernel, erfc similarity solution	13.4, 13.8
Sinusoidal steady state	phasors turn the PDE into an ODE in space	13.8
Short-time heat transfer	erfc near each boundary; long time: one-mode series	13.4
Lossy line with $R/L = G/C$	exponential substitution → lossless wave	13.5
Irregular geometry	numerical methods (Ch. 22), conformal maps (Ch. 18)	—

Summary

Topic	Key formula / fact
Telegrapher's equations	$v_x = -Ri - Li_t,\ i_x = -Gv - Cv_t$
Transmission-line equation	$v_{xx} = LC\,v_{tt} + (RC + GL)\,v_t + RG\,v$
Limits	lossless: $v_{tt} = c^2 v_{xx}$, $c = 1/\sqrt{LC}$; RC: $v_t = v_{xx}/(RC)$
Classification	$a_{11}u_{xx} + 2a_{12}u_{xy} + a_{22}u_{yy}$: $\Delta = a_{12}^2 - a_{11}a_{22}$; >0 hyperbolic, $=0$ parabolic, <0 elliptic
Characteristics	$dy/dx = (a_{12} \pm \sqrt{\Delta})/a_{11}$
Boundary conditions	Dirichlet $u = h$; Neumann $\partial u/\partial n = h$; Robin $\beta u + \partial u/\partial n = h$
Steady-state subtraction	$u = u_s(x) + w$, $u_s = T_1 + (T_2 - T_1)x/L$
Dirichlet eigenpairs	$\lambda_n = (n\pi/L)^2$, $X_n = \sin(n\pi x/L)$
D–N eigenpairs	$\lambda_n = ((2n-1)\pi/2L)^2$, $X_n = \sin((2n-1)\pi x/2L)$
Robin (D–R)	$\tan\mu = -\mu/h$; slab cooling $\mu\tan\mu = \mathrm{Bi}$
Sturm–Liouville coefficients	$c_n = \int f X_n w\,dx \big/ \int X_n^2 w\,dx$
Heat equation, D–D	$u = \sum b_n e^{-\alpha n^2\pi^2 t/L^2}\sin(n\pi x/L)$, $b_n = \frac{2}{L}\int_0^L f \sin\frac{n\pi x}{L}dx$

Topic	Key formula / fact
Thermal time constant	$\tau_1 = L^2/(\pi^2\alpha)$; diffusion length $\sim \sqrt{\alpha t}$
Heat kernel	$\Phi = (4\pi\alpha t)^{-1/2} e^{-x^2/(4\alpha t)}$, $u = \Phi * f$
Semi-infinite step	$u = T_s \operatorname{erfc}\left(x/(2\sqrt{\alpha t})\right)$
d'Alembert	$u = \frac{1}{2}[f(x-ct) + f(x+ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct} g\,ds$
Line waves	$v = v^+ + v^-$, $i = (v^+ - v^-)/Z_0$, $Z_0 = \sqrt{L/C}$
Reflection coefficient	$\Gamma = (R_L - Z_0)/(R_L + Z_0)$
String / shorted line modes	$f_n = nc/(2L)$; $B_n = \frac{2}{n\pi c}\int_0^L g \sin\frac{n\pi x}{L} dx$
Wave energy	$E = \frac{1}{2}\int (u_t^2 + c^2 u_x^2)\,dx$ conserved
Distortionless line	$R/L = G/C$: $v = e^{-(R/L)t}[F(x-ct) + G(x+ct)]$, attenuation $\sqrt{RG}$ Np/m
Laplace, rectangle	$u = \sum h_n \frac{\sinh(n\pi y/a)}{\sinh(n\pi b/a)} \sin\frac{n\pi x}{a}$
Laplace, polar products	$1,\ \ln r,\ r^{\pm n}\cos n\phi,\ r^{\pm n}\sin n\phi$
Disk / Poisson integral	$u = \frac{a_0}{2} + \sum (r/a)^n (a_n \cos n\phi + b_n \sin n\phi)$; kernel $\frac{a^2 - r^2}{a^2 - 2ar\cos(\phi-\psi) + r^2}$
Coax	$V = V_0 \ln(b/r)/\ln(b/a)$, $C' = 2\pi\varepsilon/\ln(b/a)$
Mean value / max principle	$u(\mathbf{x}_0)$ = circle average; extrema on the boundary
Duhamel (modal)	$T_n' + \alpha\lambda_n T_n = s_n(t)$, $T_n = b_n e^{-\alpha\lambda_n t} + \int_0^t e^{-\alpha\lambda_n(t-\tau)} s_n\,d\tau$
Open RC line delay	$t_{50} = 0.38\,RC\ell^2$
Skin depth	$\delta = \sqrt{2/(\omega\mu\sigma)}$, $H = H_0 e^{-x/\delta}\cos(\omega t - x/\delta)$, $R_s = 1/(\sigma\delta)$
Dopant profiles	$C_s \operatorname{erfc}\left(x/2\sqrt{Dt}\right)$; $\frac{Q}{\sqrt{\pi Dt}} e^{-x^2/4Dt}$

Further reading

- E. Kreyszig, *Advanced Engineering Mathematics*, 10th ed., Wiley — Chapter 12 (PDEs: separation of variables, heat, wave and Laplace equations, d'Alembert's solution).
- R. Haberman, *Applied Partial Differential Equations with Fourier Series and Boundary Value Problems*, 5th ed., Pearson — the clearest full-length treatment of Sturm–Liouville theory, eigenfunction expansions and Duhamel's principle, including Bessel functions for circular domains.
- W. A. Strauss, *Partial Differential Equations: An Introduction*, 2nd ed., Wiley — classification, characteristics, well-posedness, maximum principles and energy methods at a slightly more mathematical level.
- D. K. Cheng, *Field and Wave Electromagnetics*, 2nd ed., Addison-Wesley — boundary-value problems in electrostatics, transmission-line equations and skin effect, using the methods of this chapter.

Part V — Transforms and Mathematical Analysis

Chapter 14 — Laplace Transforms

> **Prerequisites:** Ch. 2 (complex numbers), Ch. 5 (integral calculus, improper integrals), Ch. 11 and Ch. 12 (ordinary and linear differential equations)
> **Learning outcomes:** After this chapter you can
>
> - define the unilateral Laplace transform with lower limit 0^-, decide whether it exists, and find its region of convergence;
> - use the transform properties (shifts, scaling, differentiation, integration, multiplication by t, periodicity) to build transforms without integrating;
> - invert rational transforms by partial fractions, including repeated and complex-conjugate poles and delayed terms e^{-sT};
> - apply the convolution theorem and the initial- and final-value theorems, and know exactly when the latter is valid;
> - solve linear constant-coefficient differential, integro-differential and simultaneous equations as initial-value problems in one pass;
> - model circuits directly in the s-domain, form transfer functions, read poles and zeros, and judge stability from pole locations.

A linear circuit with capacitors and inductors obeys a linear differential equation. In Ch. 12 such equations were solved by finding a complementary function, guessing a particular integral, and then fitting constants to the initial conditions — three separate steps, each with its own traps. The Laplace transform turns the whole initial-value problem into **algebra**: differentiation becomes multiplication by s, the initial conditions enter automatically, and the solution is recovered from a table after a partial-fraction expansion. Beyond solving equations, the transform gives the language in which engineers *think* about linear systems: impedances sL and $1/(sC)$, transfer functions, poles and zeros, and stability. The Fourier transform of Ch. 16 is the special case $s = j\omega$, and the Z-transform of Ch. 17 is its discrete-time sibling. The complex-variable theory behind the inversion integral is developed in Ch. 18.

14.1 Definition and existence

Motivation: from differentiation to multiplication

The exponential e^{st} is the one function that differentiation leaves essentially unchanged: $\frac{d}{dt}e^{st} = s\,e^{st}$. If a signal could be written as a superposition of exponentials e^{st} with weights $X(s)$, then differentiating the signal would simply multiply each weight by s. The Laplace transform is the recipe for computing those weights. Unlike the Fourier transform, which uses only the undamped exponentials $e^{j\omega t}$, it allows a real part σ in $s = \sigma + j\omega$. The factor $e^{-\sigma t}$ tames signals that grow — ramps, unstable responses, e^{3t} — so that many signals with no ordinary Fourier transform still have a Laplace transform.

The unilateral transform

Unilateral Laplace transform

Let $x(t)$ be defined for $t \geq 0$ (its values for $t < 0$ are ignored). Its **Laplace transform** is

$$X(s) = \mathcal{L}\{x(t)\} = \int_{0^-}^{\infty} x(t)\,e^{-st}\,dt = \lim_{\varepsilon \to 0^+,\; T \to \infty} \int_{-\varepsilon}^{T} x(t)\,e^{-st}\,dt,$$

where $s = \sigma + j\omega$ is a complex variable. The set of s for which the integral converges is the **region of convergence (ROC)**. We write $x(t) \leftrightarrow X(s)$ and call t the *time domain* and s the *s-domain* or *complex-frequency domain*.

Three points of convention deserve attention from the start.

1. **Lower limit 0^-.** The integral starts just *before* $t = 0$. An impulse $\delta(t)$ or a jump located at the origin is therefore fully inside the range of integration, so $\mathcal{L}\{\delta(t)\} = 1$, and the initial conditions that appear in the derivative rule are the values $x(0^-)$ *before* any switching event. These are exactly the values an engineer knows: the capacitor voltage and inductor current just before the switch closes. With a 0^+ lower limit, one needs instead the values just after switching, which are usually what one is trying to find. This book uses 0^- throughout.
2. **Causality.** Because only $t \geq 0$ matters, $x(t)$ and $x(t)u(t)$ have the same transform. It is good practice to write time functions with an explicit $u(t)$, for example $e^{-2t}u(t)$, whenever the distinction could matter (shifts, convolution). Tables often omit the $u(t)$; it is always understood.
3. **Units.** Since dt carries seconds, $X(s)$ has the units of x times seconds, and s has units s^{-1} (it is a complex frequency). A voltage step $V_0u(t)$ in volts has transform V_0/s in V·s. Checking that every term of an s-domain expression has the same units is as useful here as in the time domain.

The **bilateral** transform $\int_{-\infty}^{\infty} x(t)e^{-st}dt$ is used for non-causal signals in signal theory; it requires the ROC to be stated to make the inverse unique (two different signals, $e^{-at}u(t)$ and $-e^{-at}u(-t)$, both have bilateral transform $1/(s+a)$, with ROCs $\sigma > -a$ and $\sigma < -a$). For initial-value problems and circuits the unilateral transform is the right tool and all statements below refer to it.

First transforms from the definition

Worked Example 14.1 — the step and the exponential

Find $\mathcal{L}\{u(t)\}$ and $\mathcal{L}\{e^{at}\}$ for real or complex constant a, with their ROCs.

Step. For $t > 0$, $u(t) = 1$:

$$\int_{0^-}^{\infty} e^{-st}\,dt = \lim_{T\to\infty}\left[\frac{-e^{-st}}{s}\right]_0^{T} = \frac{1}{s} - \lim_{T\to\infty}\frac{e^{-\sigma T}e^{-j\omega T}}{s}.$$

The factor $e^{-j\omega T}$ has modulus 1, so the limit is zero exactly when $\sigma > 0$. Hence $\mathcal{L}\{u(t)\} = 1/s$ with ROC $\operatorname{Re}\{s\} > 0$.

Exponential. $\int_0^{\infty} e^{at}e^{-st}\,dt = \int_0^{\infty} e^{-(s-a)t}\,dt$ is the same integral with s replaced by $s - a$:

$$\mathcal{L}\{e^{at}\} = \frac{1}{s-a}, \qquad \operatorname{Re}\{s\} > \operatorname{Re}\{a\}.$$

Check. With $a = -2$: $\int_0^{\infty} e^{-2t}e^{-st}\,dt$ at $s = 1$ is $\int_0^{\infty} e^{-3t}\,dt = 1/3 = 1/(1+2)$. ✓

Worked Example 14.2 — the ramp, by parts

Find $\mathcal{L}\{t\}$.

Integrate by parts with $f = t$, $dg = e^{-st}\,dt$, valid for $\sigma > 0$:

$$\int_0^{\infty} t\,e^{-st}\,dt = \left[-\frac{t\,e^{-st}}{s}\right]_0^{\infty} + \frac{1}{s}\int_0^{\infty} e^{-st}\,dt = 0 + \frac{1}{s}\cdot\frac{1}{s} = \frac{1}{s^2}.$$

The boundary term vanishes at infinity because $te^{-\sigma t} \to 0$ for $\sigma > 0$ (exponential decay beats polynomial growth). Repeating the argument n times gives $\mathcal{L}\{t^n\} = n!/s^{n+1}$ (Section 14.3).

Check. At $s = 2$: $\int_0^{\infty} te^{-2t}\,dt = 1/4$, and $1/s^2 = 1/4$. ✓

Worked Example 14.3 — the impulse and a sifting check

Find $\mathcal{L}\{\delta(t)\}$ and $\mathcal{L}\{\delta(t-T)\}$ for $T > 0$.

By the sifting property $\int g(t)\delta(t-t_0)\,dt = g(t_0)$ for any t_0 inside the range of integration,

$$\mathcal{L}\{\delta(t)\} = \int_{0^-}^{\infty}\delta(t)e^{-st}\,dt = e^{0} = 1, \qquad \mathcal{L}\{\delta(t-T)\} = e^{-sT},$$

both for every s (ROC: the whole plane). With a 0^+ lower limit, the impulse at the origin would be *excluded* and the transform would be 0 — a clear sign that 0^- is the right choice for systems driven by impulses.

Existence: exponential order

Not every function has a Laplace transform. The integral can fail at the lower end (a non-integrable singularity) or at infinity (growth faster than any exponential). A simple sufficient condition rules out both.

Piecewise continuity and exponential order

A function $x(t)$ is **piecewise continuous** on $[0,\infty)$ if on every finite interval it has at most finitely many discontinuities, each a finite jump. It is of **exponential order** α if there are constants $M > 0$, $T_0 \ge 0$ such that

$$|x(t)| \le Me^{\alpha t} \quad \text{for all } t \ge T_0.$$

Polynomials, e^{at}, $\sin\omega t$, $\cosh t$ and all their products are of exponential order. The function e^{t^2} is not: for any α, $e^{t^2}/e^{\alpha t} = e^{t(t-\alpha)} \to \infty$.

Existence theorem

If $x(t)$ is piecewise continuous on $[0,\infty)$ and of exponential order α, then $\mathcal{L}\{x\}$ exists for all $\operatorname{Re}\{s\} > \alpha$, the integral converging absolutely. Moreover $X(s) \to 0$ as $\operatorname{Re}\{s\} \to \infty$.

Proof sketch. Split $\int_0^{\infty} = \int_0^{T_0} + \int_{T_0}^{\infty}$. The first part is the integral of a bounded piecewise continuous function over a finite interval, hence finite. For the second, with $\sigma = \operatorname{Re}\{s\} > \alpha$,

$$\int_{T_0}^{\infty}|x(t)e^{-st}|\,dt \le M\int_{T_0}^{\infty}e^{-(\sigma-\alpha)t}\,dt = \frac{Me^{-(\sigma-\alpha)T_0}}{\sigma-\alpha} < \infty.$$

Absolute convergence implies convergence. For the decay, bound $|x| \le M'e^{\alpha t}$ on all of $[0,\infty)$ (possible after enlarging M, since x is bounded on $[0,T_0]$); then $|X(s)| \le M'/(\sigma-\alpha) \to 0$. □

The infimum of all α that work is the **abscissa of convergence** σ_c; the ROC is the open half-plane $\operatorname{Re}\{s\} > \sigma_c$ (a unilateral transform always has a right half-plane as ROC). For a rational $X(s)$ the ROC begins at the right-most pole: $\sigma_c = \max \operatorname{Re}\{p_k\}$.

The conditions are **sufficient, not necessary**. The function $t^{-1/2}$ is unbounded at $t = 0$, so it is not piecewise continuous there, yet $\int_0^\infty t^{-1/2} e^{-st}\,dt = \sqrt{\pi/s}$ converges because the singularity is integrable. Impulses are not functions at all but are handled by the sifting property. In practice, every signal met in circuit analysis has a transform.

Worked Example 14.4 — deciding existence and the ROC

For each signal, decide whether the unilateral transform exists and, if so, give the ROC.

(a) $x(t) = 3e^{-t} + 2e^{4t}$. Both terms are of exponential order; the larger exponent is 4, so $|x| \le 5e^{4t}$ and the transform $\frac{3}{s+1} + \frac{2}{s-4}$ exists for $\mathrm{Re}\{s\} > 4$. The ROC of a sum is (at least) the intersection of the individual ROCs — here governed by the fastest-growing term, i.e. the right-most pole.

(b) $x(t) = e^{t^2}$. Not of exponential order. For any real s, the integrand $e^{t^2 - st}$ grows without bound once $t > s$, so the integral diverges for every s: **no transform**.

(c) $x(t) = 1/t$. Of exponential order for large t, but near the origin $\int_0^1 t^{-1} e^{-st}\,dt \ge e^{-\sigma}\int_0^1 t^{-1}\,dt = \infty$ for real $s = \sigma$. The singularity is not integrable: **no transform**. Contrast $t^{-1/2}$, whose singularity is integrable and whose transform $\sqrt{\pi/s}$ exists.

(d) $x(t) = \sin(e^{t^2})$. This oscillates ever faster, but $|x| \le 1 = e^{0\cdot t}$, so it is of exponential order 0 and the transform exists for $\mathrm{Re}\{s\} > 0$. Rapid oscillation is harmless; only growth and non-integrable singularities matter.

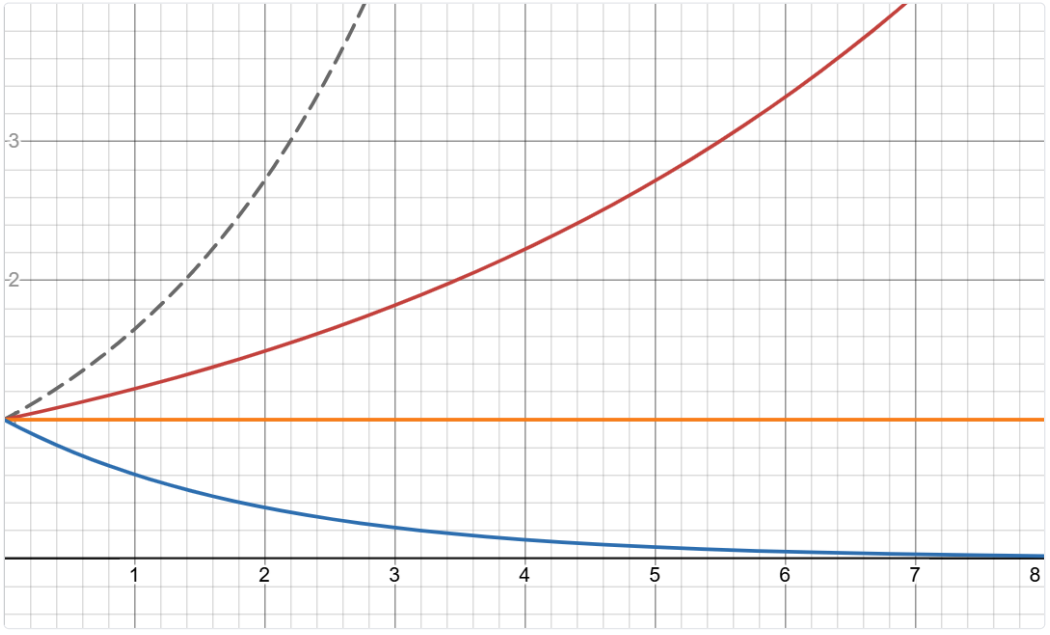
(e) $x(t) = \cosh(2t)\, e^{-3t}$. Since $\cosh 2t\, e^{-3t} = \tfrac{1}{2}(e^{-t} + e^{-5t})$, the transform $\tfrac{1}{2}\left(\frac{1}{s+1} + \frac{1}{s+5}\right)$ exists for $\mathrm{Re}\{s\} > -1$. Here the ROC includes the $j\omega$-axis, so this signal also has a Fourier transform.

Uniqueness (Lerch)

If two piecewise continuous functions of exponential order have the same Laplace transform on some half-plane $\mathrm{Re}\{s\} > \sigma_0$, they are equal at every $t > 0$ where both are continuous.

Uniqueness is what makes table look-up legitimate: if we recognise $X(s)$ as the transform of $x(t)$, then $x(t)$ is *the* answer (up to values at isolated points, which no physical measurement can detect).

The behaviour of the integrand explains the ROC graphically. For $x(t) = e^{0.5t}$, the weighted integrand $x(t)e^{-\sigma t}$ decays only when $\sigma > 0.5$:



The signal $e^{0.5t}$ (grey) multiplied by $e^{-\sigma t}$ for $\sigma = 0.3$ (red, still growing), $\sigma = 0.5$ (orange, constant — integral diverges) and $\sigma = 1$ (blue, decaying — integral converges). The ROC is $\sigma > 0.5$.

Common Pitfall — evaluating outside the ROC

The formula $1/(s-a)$ is defined for all $s \ne a$, but the *integral* converges only for $\mathrm{Re}\{s\} > \mathrm{Re}\{a\}$. For the unilateral transform this rarely causes errors, because the ROC is always a right half-plane determined by the right-most singularity. It matters when passing to the Fourier transform: $X(j\omega)$ is the Fourier transform only if the $j\omega$-axis lies in the ROC. For $e^{2t}u(t)$, $X(s) = 1/(s-2)$, but $1/(j\omega - 2)$ is **not** its Fourier transform — it has none (Ch. 16).

Engineering Note — why engineers accept a complex frequency

A pole at $s = \sigma + j\omega$ corresponds to a time behaviour $e^{\sigma t}e^{j\omega t}$: ω is the oscillation frequency in rad/s and σ is the growth rate (decay if negative) in nepers per second. Every natural response of a lumped linear circuit is a combination of such terms, so the s-plane is a map of every possible "mode" of a circuit, with stability determined by which half of the map the modes lie in (Section 14.9).

14.2 Properties

Computing every transform from the integral would be tedious. A handful of properties, each a one-line consequence of the definition, lets us build almost any transform needed in practice from a few basic pairs, and — more importantly — tells us how operations on signals (delay, differentiation, damping) appear in the s-domain. Throughout, $x(t) \leftrightarrow X(s)$ with ROC $\mathrm{Re}\{s\} > \sigma_x$, and similarly for y.

Linearity

Linearity

For constants a, b:

$$\mathcal{L}\{a\,x(t) + b\,y(t)\} = a\,X(s) + b\,Y(s),$$

valid at least on the intersection of the two ROCs.

This is inherited directly from linearity of the integral. Combined with Euler's formula it already yields the trigonometric transforms: $\cos\omega t = \tfrac{1}{2}(e^{j\omega t} + e^{-j\omega t})$ gives

$$\mathcal{L}\{\cos\omega t\} = \frac{1}{2}\left(\frac{1}{s-j\omega} + \frac{1}{s+j\omega}\right) = \frac{s}{s^2+\omega^2}, \qquad \mathcal{L}\{\sin\omega t\} = \frac{1}{2j}\left(\frac{1}{s-j\omega} - \frac{1}{s+j\omega}\right) = \frac{\omega}{s^2+\omega^2},$$

both with ROC $\mathrm{Re}\{s\} > 0$.

Frequency shift (multiplication by an exponential)

First shifting theorem (s-shift)

$$\mathcal{L}\{e^{at}x(t)\} = X(s-a), \qquad \mathrm{Re}\{s\} > \sigma_x + \mathrm{Re}\{a\}.$$

Proof. $\int_{0^-}^{\infty} e^{at}x(t)e^{-st}dt = \int_{0^-}^{\infty} x(t)e^{-(s-a)t}dt = X(s-a)$. □

Damping a signal by $e^{-\alpha t}$ slides its transform — and all its poles — a distance α to the left in the s-plane. Thus

$$e^{-\alpha t}\cos\omega t \leftrightarrow \frac{s+\alpha}{(s+\alpha)^2+\omega^2}, \qquad e^{-\alpha t}\sin\omega t \leftrightarrow \frac{\omega}{(s+\alpha)^2+\omega^2}, \qquad t^n e^{-\alpha t} \leftrightarrow \frac{n!}{(s+\alpha)^{n+1}}.$$

Time shift (delay)

Second shifting theorem (t-shift)

For $T \geq 0$,

$$\mathcal{L}\{x(t-T)\,u(t-T)\} = e^{-sT}X(s).$$

Proof. The integrand vanishes for $t < T$; substitute $\tau = t - T$:

$$\int_T^{\infty} x(t-T)e^{-st}dt = \int_0^{\infty} x(\tau)e^{-s(\tau+T)}d\tau = e^{-sT}X(s). \qquad \square$$

The factor e^{-sT} is the s-domain signature of a pure delay. The theorem applies to the *whole function* shifted, $x(t-T)u(t-T)$; it does not apply to $x(t)u(t-T)$ until that is rewritten. The safe procedure for $\mathcal{L}\{g(t)u(t-T)\}$ is

$$\mathcal{L}\{g(t)\,u(t-T)\} = e^{-sT}\,\mathcal{L}\{g(t+T)\},$$

which follows from the same substitution.

Worked Example 14.5 — piecewise signals with steps

A voltage rises linearly from 0 to 1 V during the first second and then stays at 1 V: $x(t) = t$ for $0 \leq t < 1$, $x(t) = 1$ for $t \geq 1$. Find $X(s)$.

Write with steps. A ramp that starts at $t = 0$, minus a ramp that starts at $t = 1$, levels the signal off:

$$x(t) = t\,u(t) - (t-1)\,u(t-1).$$

(Check: for $t \geq 1$, $t - (t-1) = 1$. ✓)

Transform. Each term is a ramp, one of them delayed:

$$X(s) = \frac{1}{s^2} - \frac{e^{-s}}{s^2} = \frac{1-e^{-s}}{s^2}.$$

Check by integration. $\int_0^1 te^{-st}dt + \int_1^{\infty} e^{-st}dt$ at $s = 1$: the first integral is $1 - 2e^{-1} = 0.26424$, the second $e^{-1} = 0.36788$; the sum is 0.63212. The formula gives $1 - e^{-1} = 0.63212$. ✓

Worked Example 14.6 — a shifted sinusoid

Find $\mathcal{L}\{\sin(t)\,u(t-\pi/2)\}$.

The sine is not written as a function of $t-\pi/2$. Use $g(t)u(t-T)\leftrightarrow e^{-sT}\mathcal{L}\{g(t+T)\}$ with $g=\sin$, $T=\pi/2$: $g(t+\pi/2)=\sin(t+\pi/2)=\cos t$, so

$$\mathcal{L}\{\sin t\,u(t-\pi/2)\}=e^{-\pi s/2}\,\frac{s}{s^2+1}.$$

The tempting but wrong answer $e^{-\pi s/2}/(s^2+1)$ would be the transform of $\sin(t-\pi/2)u(t-\pi/2)=-\cos t\,u(t-\pi/2)$, a signal of opposite sign.

Time scaling

Scaling

For $a>0$,

$$\mathcal{L}\{x(at)\}=\frac{1}{a}\,X\!\left(\frac{s}{a}\right).$$

Proof. Substitute $\tau=at$: $\int_0^\infty x(at)e^{-st}dt=\frac{1}{a}\int_0^\infty x(\tau)e^{-(s/a)\tau}d\tau$. $\square$

Compressing in time ($a>1$) expands in s: poles move outward by the factor a. This is the basis of *frequency scaling* of filter designs: a prototype with cut-off 1 rad/s becomes one with cut-off ω_c by replacing s with s/ω_c.

Differentiation in time

This is the property that makes the transform useful for differential equations.

Derivative rule

If x is continuous on $(0^-,\infty)$ with piecewise continuous derivative, both of exponential order, then

$$\mathcal{L}\{\dot{x}(t)\}=sX(s)-x(0^-),$$

and, applying the rule repeatedly,

$$\mathcal{L}\{x^{(n)}(t)\}=s^nX(s)-s^{n-1}x(0^-)-s^{n-2}\dot{x}(0^-)-\cdots-x^{(n-1)}(0^-).$$

Derivation. Integrate by parts:

$$\int_{0^-}^{\infty}\dot{x}(t)e^{-st}dt=\Big[x(t)e^{-st}\Big]_{0^-}^{\infty}+s\int_{0^-}^{\infty}x(t)e^{-st}dt=-x(0^-)+sX(s),$$

the upper boundary term vanishing in the ROC because x is of exponential order. For the second derivative apply the rule to $\dot{x}$: $\mathcal{L}\{\ddot{x}\}=s\mathcal{L}\{\dot{x}\}-\dot{x}(0^-)=s^2X-sx(0^-)-\dot{x}(0^-)$. Induction gives the general formula. $\square$

Jumps at the origin. With the 0^- convention, the rule remains true even when x jumps at $t=0$, provided $\dot{x}$ is understood in the generalized sense (a jump of size J contributes $J\,\delta(t)$ to $\dot{x}$). For instance $x(t)=e^{-t}u(t)$ has $x(0^-)=0$ and $\dot{x}=\delta(t)-e^{-t}u(t)$, whose transform is $1-\frac{1}{s+1}=\frac{s}{s+1}=sX(s)-0$. ✓ A version of the rule using $x(0^+)=1$ reads $sX-x(0^+)=-\frac{1}{s+1}$; it describes only the derivative for $t>0$ and silently drops the impulse. Both are self-consistent, but mixing them is a classic source of error; the 0^- form never needs to know what happens *at* the switching instant.

A jump at a later time $t_1>0$ of size J is handled the same way: $\mathcal{L}\{\dot{x}\}=sX-x(0^-)$ still holds if $\dot{x}$ includes the impulse $J\delta(t-t_1)$, while the ordinary (pointwise) derivative satisfies $\mathcal{L}\{\dot{x}_{\text{ord}}\}=sX-x(0^-)-Je^{-st_1}$.

Integration in time

Integral rule

$$\mathcal{L}\left\{\int_{0^-}^{t}x(\tau)\,d\tau\right\}=\frac{X(s)}{s}.$$

Proof. Let $g(t)=\int_{0^-}^{t}x\,d\tau$. Then $\dot{g}=x$ and $g(0^-)=0$, so the derivative rule gives $X=sG-0$. $\square$

Integration divides by s; differentiation multiplies by s. The integrator $1/s$ and the differentiator s are the two elementary dynamic operators, and $u(t)\leftrightarrow 1/s$ is the integral of $\delta(t)\leftrightarrow 1$. If the integral starts from an earlier time with a stored value, as for a capacitor voltage $v_C(t)=v_C(0^-)+\frac{1}{C}\int_{0^-}^{t}i\,d\tau$, the transform is $V_C=v_C(0^-)/s+I/(sC)$ — the form used for circuit models in Section 14.9.

Multiplication by t and division by t

Differentiation in s

$$\mathcal{L}\{t\,x(t)\}=-\frac{dX}{ds},\qquad \mathcal{L}\{t^n x(t)\}=(-1)^n\frac{d^nX}{ds^n}.$$

Proof. In the ROC the integral converges uniformly on closed sub-half-planes, so we may differentiate under the integral sign: $\frac{d}{ds}\int x(t)e^{-st}dt = \int(-t)x(t)e^{-st}dt$. $\square$

A by-product: inside its ROC, $X(s)$ is differentiable as a function of the complex variable s, i.e. **analytic** (Ch. 18). Singularities (poles) can only occur on or to the left of the boundary of the ROC.

Integration in s

If $x(t)/t$ has a finite limit as $t \to 0^+$,

$$\mathcal{L}\left\{\frac{x(t)}{t}\right\} = \int_s^\infty X(p)\,dp.$$

Proof sketch. $\int_s^\infty X(p)dp = \int_s^\infty \int_0^\infty x(t)e^{-pt}dt\,dp$; exchanging the order, $\int_s^\infty e^{-pt}dp = e^{-st}/t$, which leaves $\int_0^\infty \frac{x(t)}{t}e^{-st}dt$. $\square$

Worked Example 14.7 — combining t-multiplication and s-shift

Find $\mathcal{L}\{t\,e^{-2t}\sin 3t\}$.

Step 1 (basic pair). $\sin 3t \leftrightarrow \dfrac{3}{s^2+9}$.

Step 2 (multiply by t).

$$t\sin 3t \leftrightarrow -\frac{d}{ds}\frac{3}{s^2+9} = \frac{6s}{(s^2+9)^2}.$$

Step 3 (damp by e^{-2t}: replace s by $s+2$).

$$t\,e^{-2t}\sin 3t \leftrightarrow \frac{6(s+2)}{\big((s+2)^2+9\big)^2} = \frac{6(s+2)}{(s^2+4s+13)^2}.$$

Check. At $s=0$ the formula gives $12/169 = 0.071006$. Direct numerical integration of $\int_0^\infty te^{-2t}\sin 3t\,dt$ gives 0.071006. ✓ The order of steps 2 and 3 does not matter, but doing the s-shift last keeps the algebra simplest.

Worked Example 14.8 — the sine integral by division by t

Find $\mathcal{L}\{\sin t/t\}$ and deduce $\int_0^\infty \frac{\sin t}{t}dt$.

Since $\sin t/t \to 1$ as $t \to 0$, the rule applies:

$$\mathcal{L}\left\{\frac{\sin t}{t}\right\} = \int_s^\infty \frac{dp}{p^2+1} = \frac{\pi}{2} - \arctan s = \arctan\frac{1}{s}.$$

Letting $s \to 0^+$ (justified by Abel's theorem for Laplace integrals, since $\int_0^\infty \sin t/t\,dt$ converges) gives

$$\int_0^\infty \frac{\sin t}{t}dt = \lim_{s\to 0^+}\arctan\frac{1}{s} = \frac{\pi}{2} \approx 1.5708.$$

This integral (the area under the unnormalized sinc) is awkward by elementary methods; here it costs two lines. It reappears in Gibbs' phenomenon in Ch. 15.

Periodic functions

Transform of a periodic function

If $x(t+T) = x(t)$ for $t \geq 0$, and $x_1(t)$ denotes one period (x on $[0,T)$, zero elsewhere) with transform $X_1(s)$, then

$$X(s) = \frac{X_1(s)}{1-e^{-sT}} = \frac{1}{1-e^{-sT}}\int_0^T x(t)e^{-st}dt, \qquad \mathrm{Re}\{s\} > 0.$$

Proof. $x(t) = \sum_{k=0}^\infty x_1(t-kT)$, so by the time-shift rule $X = X_1\sum_{k=0}^\infty e^{-ksT}$, a geometric series with ratio $|e^{-sT}| = e^{-\sigma T} < 1$. $\square$

Worked Example 14.9 — square wave and full-wave rectified sine

(a) A unipolar square wave equals 1 on $[0, T/2)$ and 0 on $[T/2, T)$, repeating with period T. One period is $u(t) - u(t - T/2)$, so $X_1 = (1 - e^{-sT/2})/s$ and

$$X(s) = \frac{1 - e^{-sT/2}}{s(1 - e^{-sT})} = \frac{1}{s\,(1 + e^{-sT/2})},$$

using $1 - e^{-sT} = (1 - e^{-sT/2})(1 + e^{-sT/2})$.

(b) The full-wave rectified sine $|\sin t|$ has period π. One period:

$$\int_0^{\pi} \sin t\, e^{-st}\,dt = \frac{1 + e^{-\pi s}}{s^2 + 1} \quad\Longrightarrow\quad X(s) = \frac{1 + e^{-\pi s}}{(s^2+1)(1 - e^{-\pi s})} = \frac{1}{s^2+1}\coth\frac{\pi s}{2}.$$

Check. At $s = 1$, $\coth(\pi/2)/2 = 0.54517$; numerical integration of $\int_0^\infty |\sin t| e^{-t}dt$ gives 0.54517. ✓ As $s \to 0$, $sX(s) \to 2/\pi \approx 0.6366$, the average value of $|\sin t|$ — a preview of the final-value idea in Section 14.7.

Common Pitfall — shifting in the wrong domain

Multiplication by e^{at} in time shifts in s ($s \to s - a$); multiplication by e^{-sT} in s delays in time. The signs are easy to swap. A quick check: damping ($a < 0$) must move poles to the **left**, and a delay must leave X unchanged in magnitude on the $j\omega$-axis ($|e^{-j\omega T}| = 1$).

Engineering Note — properties as building blocks

The properties map one-to-one onto block-diagram components: $1/s$ is an integrator (an op-amp integrator stage), s a differentiator, e^{-sT} a transport delay (a cable, a sampling latency), and $s \to s/\omega_c$ a frequency scaling of a filter prototype. Circuit simulators that work in the s-domain (small-signal AC analysis) are built on exactly these rules.

14.3 Standard transforms

A working engineer inverts transforms by recognising patterns, so a reliable table is part of the toolkit. This section collects the standard pairs, derives the less obvious ones, and shows how the table is extended with the properties of Section 14.2. All time functions are understood to be multiplied by $u(t)$; a, α, ω, b are real constants unless stated otherwise, n is a non-negative integer, and $\nu > -1$ is real.

Powers of t and the Gamma function

For integer powers, repeated use of the t-multiplication rule on $1 \leftrightarrow 1/s$ gives

$$t^n = t \cdot t^{n-1} \leftrightarrow -\frac{d}{ds}\frac{(n-1)!}{s^n} = \frac{n!}{s^{n+1}}.$$

For non-integer powers, substitute $\tau = st$ (first for real $s > 0$, then extend by analytic continuation):

$$\int_0^\infty t^\nu e^{-st}dt = \frac{1}{s^{\nu+1}}\int_0^\infty \tau^\nu e^{-\tau}d\tau = \frac{\Gamma(\nu+1)}{s^{\nu+1}}, \qquad \nu > -1,$$

where $\Gamma(z) = \int_0^\infty \tau^{z-1}e^{-\tau}d\tau$ is the **Gamma function**. It satisfies $\Gamma(z+1) = z\Gamma(z)$ (integrate by parts), $\Gamma(n+1) = n!$, and $\Gamma(\tfrac12) = \sqrt{\pi}$ (from the Gaussian integral of Ch. 6).

Worked Example 14.10 — half-integer powers

Find $\mathcal{L}\{t^{-1/2}\}$, $\mathcal{L}\{t^{1/2}\}$ and $\mathcal{L}\{t^{3/2}\}$.

$$t^{-1/2} \leftrightarrow \frac{\Gamma(\frac12)}{s^{1/2}} = \sqrt{\frac{\pi}{s}}, \qquad t^{1/2} \leftrightarrow \frac{\Gamma(\frac32)}{s^{3/2}} = \frac{\frac12\sqrt{\pi}}{s^{3/2}}, \qquad t^{3/2} \leftrightarrow \frac{\Gamma(\frac52)}{s^{5/2}} = \frac{\frac34\sqrt{\pi}}{s^{5/2}}.$$

Check. At $s = 1$, $\int_0^\infty t^{3/2}e^{-t}dt = \Gamma(2.5) = 1.32934$, and $\frac34\sqrt{\pi} = 1.32934$. ✓

Transforms with $\sqrt{s}$ arise from distributed systems — diffusion in a semiconductor, heat flow, the RC transmission line — where the "impedance" of an infinitely long line is proportional to $1/\sqrt{s}$, so step responses contain $\sqrt{t}$ and $1/\sqrt{t}$ rather than exponentials.

Exponentials, sinusoids and their products

Everything in this family follows from $e^{at} \leftrightarrow 1/(s-a)$ with complex a, plus linearity and the s-shift. Two further derivations illustrate the technique.

Hyperbolic functions. $\cosh at = \frac12(e^{at} + e^{-at}) \leftrightarrow \frac12\left(\frac{1}{s-a} + \frac{1}{s+a}\right) = \frac{s}{s^2-a^2}$, and similarly $\sinh at \leftrightarrow \frac{a}{s^2-a^2}$ (ROC $\mathrm{Re}\{s\} > |a|$). Compare with cos and sin: only the sign in the denominator changes, mirroring $\cos(jat) = \cosh at$.

$t\sin$ and $t\cos$ (resonance terms). By the t-multiplication rule,

$$t\sin\omega t \leftrightarrow -\frac{d}{ds}\frac{\omega}{s^2+\omega^2} = \frac{2\omega s}{(s^2+\omega^2)^2}, \qquad t\cos\omega t \leftrightarrow -\frac{d}{ds}\frac{s}{s^2+\omega^2} = \frac{s^2-\omega^2}{(s^2+\omega^2)^2}.$$

Combining, $\sin\omega t - \omega t\cos\omega t \leftrightarrow \frac{\omega(s^2+\omega^2) - \omega(s^2-\omega^2)}{(s^2+\omega^2)^2} = \frac{2\omega^3}{(s^2+\omega^2)^2}$. A *repeated* pair of imaginary poles therefore produces an oscillation whose amplitude grows linearly in t — the signature of a lossless system driven at resonance.

Worked Example 14.11 — a general damped sinusoid

Find the transform of $x(t) = A e^{-\alpha t}\cos(\omega t + \phi)$.

Expand the cosine: $\cos(\omega t+\phi) = \cos\phi\cos\omega t - \sin\phi\sin\omega t$. Then, by the s-shift,

$$X(s) = A\,\frac{(s+\alpha)\cos\phi - \omega\sin\phi}{(s+\alpha)^2 + \omega^2}.$$

Numerical case. $A=2$, $\alpha=1$, $\omega=2$, $\phi=\pi/3$: $\cos\phi = 0.5$, $\sin\phi = 0.8660$, so

$$X(s) = \frac{(s+1) - 3.4641}{s^2+2s+5} = \frac{s - 2.4641}{s^2+2s+5}.$$

Check. $x(0) = 2\cos(\pi/3) = 1$; the initial-value theorem (Section 14.7) gives $\lim_{s\to\infty} sX = 1$. ✓ This pair, read backwards, is how complex-conjugate poles are converted to a single real cosine with amplitude and phase (Section 14.5).

Pulses, staircases and other non-smooth signals

Rectangular pulses, staircases and sawtooth waves are written with steps and handled by the time-shift rule. A rectangular pulse of height 1 from $t=a$ to $t=b$ is $u(t-a) - u(t-b) \leftrightarrow \frac{e^{-as}-e^{-bs}}{s}$. A **staircase** $\sum_{k=0}^{\infty} u(t-kT)$ (value $k+1$ on $[kT,(k+1)T)$) has transform $\frac{1}{s}\sum_{k} e^{-ksT} = \frac{1}{s(1-e^{-sT})}$. A sawtooth of slope $1/T$ resetting every T is a ramp minus a staircase: $\frac{t}{T} - \sum_{k\ge 1} u(t-kT) \leftrightarrow \frac{1}{Ts^2} - \frac{e^{-sT}}{s(1-e^{-sT})}$.

Worked Example 14.12 — a finite pulse train

A gate signal consists of three unit pulses of width 1 ms starting at $t = 0$, 3 ms and 6 ms. Find its transform (work in milliseconds, so s is in ms^{-1}).

Each pulse is $u(t-t_k) - u(t-t_k-1) \leftrightarrow e^{-st_k}(1-e^{-s})/s$. Summing over $t_k = 0, 3, 6$:

$$X(s) = \frac{1-e^{-s}}{s}\left(1 + e^{-3s} + e^{-6s}\right) = \frac{(1-e^{-s})(1-e^{-9s})}{s(1-e^{-3s})},$$

the last form using the finite geometric sum $1 + r + r^2 = (1-r^3)/(1-r)$ with $r = e^{-3s}$.

Check. As $s\to 0$, $X(s) \to$ total area: $\frac{(s)(9s)}{s\cdot 3s} = 3$ ms. Three pulses of 1 ms each. ✓

The standard transform table

Table of unilateral Laplace transforms

(All $x(t)$ are for $t \ge 0$, i.e. implicitly multiplied by $u(t)$.)

#	$x(t)$	$X(s)$	ROC
1	$\delta(t)$	1	all s
2	$\delta(t-T),\ T\ge 0$	e^{-sT}	all s
3	$\delta^{(n)}(t)$	s^n	all s
4	$u(t)$	$\dfrac{1}{s}$	$\sigma > 0$
5	$u(t-T)$	$\dfrac{e^{-sT}}{s}$	$\sigma > 0$
6	$u(t-a)-u(t-b),\ 0\le a<b$	$\dfrac{e^{-as}-e^{-bs}}{s}$	all s
7	t	$\dfrac{1}{s^2}$	$\sigma > 0$
8	t^n	$\dfrac{n!}{s^{n+1}}$	$\sigma > 0$
9	$t^{\nu},\ \nu > -1$	$\dfrac{\Gamma(\nu+1)}{s^{\nu+1}}$	$\sigma > 0$
10	$\dfrac{1}{\sqrt{\pi t}}$	$\dfrac{1}{\sqrt{s}}$	$\sigma > 0$
11	$2\sqrt{\dfrac{t}{\pi}}$	$\dfrac{1}{s^{3/2}}$	$\sigma > 0$
12	e^{-at}	$\dfrac{1}{s+a}$	$\sigma > -a$
13	$t\,e^{-at}$	$\dfrac{1}{(s+a)^2}$	$\sigma > -a$
14	$\dfrac{t^n}{n!}e^{-at}$	$\dfrac{1}{(s+a)^{n+1}}$	$\sigma > -a$
15	$\dfrac{1}{a}\left(1-e^{-at}\right)$	$\dfrac{1}{s(s+a)}$	$\sigma > \max(0,-a)$
16	$\dfrac{e^{-at}-e^{-bt}}{b-a},\ a\ne b$	$\dfrac{1}{(s+a)(s+b)}$	$\sigma > \max(-a,-b)$
17	$\dfrac{be^{-bt}-ae^{-at}}{b-a},\ a\ne b$	$\dfrac{s}{(s+a)(s+b)}$	$\sigma > \max(-a,-b)$

#	$x(t)$	$X(s)$	ROC
18	$\frac{1}{a^2}(at - 1 + e^{-at})$	$\frac{1}{s^2(s+a)}$	$\sigma > \max(0, -a)$
19	$\sin \omega t$	$\frac{\omega}{s^2+\omega^2}$	$\sigma > 0$
20	$\cos \omega t$	$\frac{s}{s^2+\omega^2}$	$\sigma > 0$
21	$\sin(\omega t + \phi)$	$\frac{s\sin\phi + \omega\cos\phi}{s^2+\omega^2}$	$\sigma > 0$
22	$\cos(\omega t + \phi)$	$\frac{s\cos\phi - \omega\sin\phi}{s^2+\omega^2}$	$\sigma > 0$
23	$e^{-\alpha t}\sin \omega t$	$\frac{\omega}{(s+\alpha)^2+\omega^2}$	$\sigma > -\alpha$
24	$e^{-\alpha t}\cos \omega t$	$\frac{s+\alpha}{(s+\alpha)^2+\omega^2}$	$\sigma > -\alpha$
25	$1 - \cos \omega t$	$\frac{\omega^2}{s(s^2+\omega^2)}$	$\sigma > 0$
26	$\omega t - \sin \omega t$	$\frac{\omega^3}{s^2(s^2+\omega^2)}$	$\sigma > 0$
27	$t\sin \omega t$	$\frac{2\omega s}{(s^2+\omega^2)^2}$	$\sigma > 0$
28	$t\cos \omega t$	$\frac{s^2-\omega^2}{(s^2+\omega^2)^2}$	$\sigma > 0$
29	$\sin \omega t - \omega t\cos \omega t$	$\frac{2\omega^3}{(s^2+\omega^2)^2}$	$\sigma > 0$
30	$\sin \omega t + \omega t\cos \omega t$	$\frac{2\omega s^2}{(s^2+\omega^2)^2}$	$\sigma > 0$
31	$\sinh at$	$\frac{a}{s^2-a^2}$	$\sigma > \lvert a\rvert$
32	$\cosh at$	$\frac{s}{s^2-a^2}$	$\sigma > \lvert a\rvert$
33	$e^{-\alpha t}\sinh at$	$\frac{a}{(s+\alpha)^2-a^2}$	$\sigma > \lvert a\rvert - \alpha$
34	$\sin^2 \omega t$	$\frac{2\omega^2}{s(s^2+4\omega^2)}$	$\sigma > 0$
35	$\frac{\sin \omega t}{t}$	$\arctan\frac{\omega}{s}$	$\sigma > 0$
36	$\frac{e^{-bt}-e^{-at}}{t}$	$\ln\frac{s+a}{s+b}$	$\sigma > \max(-a, -b)$
37	$\ln t$	$-\frac{\gamma + \ln s}{s}$, $\gamma = 0.5772\ldots$	$\sigma > 0$
38	$J_0(\omega t)$ (Bessel)	$\frac{1}{\sqrt{s^2+\omega^2}}$	$\sigma > 0$
39	$\operatorname{erfc}\frac{a}{2\sqrt{t}}$, $a > 0$	$\frac{e^{-a\sqrt{s}}}{s}$	$\sigma > 0$
40	$\frac{a}{2\sqrt{\pi t^3}}e^{-a^2/4t}$, $a > 0$	$e^{-a\sqrt{s}}$	$\sigma > 0$
41	square wave, 1 on $[0, \frac{T}{2})$, 0 on $[\frac{T}{2}, T)$	$\frac{1}{s(1+e^{-sT/2})}$	$\sigma > 0$
42	$\lvert\sin \omega t\rvert$	$\frac{\omega}{s^2+\omega^2}\coth\frac{\pi s}{2\omega}$	$\sigma > 0$
43	staircase $\sum_{k\ge 0}u(t-kT)$	$\frac{1}{s(1-e^{-sT})}$	$\sigma > 0$
44	$\sum_{k\ge 0}\delta(t-kT)$	$\frac{1}{1-e^{-sT}}$	$\sigma > 0$

Operational properties (companion table — these are the rules of Section 14.2):

Operation	$x(t)$	$X(s)$
Linearity	$ax + by$	$aX + bY$
s-shift	$e^{at}x(t)$	$X(s-a)$
Delay	$x(t-T)u(t-T)$	$e^{-sT}X(s)$
Scaling ($a>0$)	$x(at)$	$\frac{1}{a}X(s/a)$
Derivative	$\dot{x}$	$sX - x(0^-)$
Second derivative	$\ddot{x}$	$s^2X - sx(0^-) - \dot{x}(0^-)$
Integral	$\int_{0^-}^{t} x\,d\tau$	X/s
Multiply by t	$t\,x(t)$	$-X'(s)$
Divide by t	$x(t)/t$	$\int_s^\infty X(p)\,dp$
Periodic, period T	$x(t+T) = x(t)$	$\frac{1}{1-e^{-sT}}\int_0^T xe^{-st}dt$
Convolution	$(x * h)(t)$	$X(s)H(s)$

Worked Example 14.13 — extending the table: a logarithmic pair

Verify entry 36, $\dfrac{e^{-bt}-e^{-at}}{t} \leftrightarrow \ln\dfrac{s+a}{s+b}$.

The numerator $x(t)=e^{-bt}-e^{-at}$ has $X(p)=\frac{1}{p+b}-\frac{1}{p+a}$, and $x(t)/t \to a-b$ (finite) as $t\to 0$. Dividing by t:

$$\int_s^\infty \left(\frac{1}{p+b}-\frac{1}{p+a}\right)dp = \left[\ln\frac{p+b}{p+a}\right]_s^\infty = 0-\ln\frac{s+b}{s+a}=\ln\frac{s+a}{s+b}.$$

Corollary (Frullani integral). Letting $s\to 0$: $\int_0^\infty \frac{e^{-bt}-e^{-at}}{t}dt=\ln\frac{a}{b}$. For $a=2, b=1$ this is $\ln 2 = 0.693147$; numerical quadrature gives 0.693147. ✓

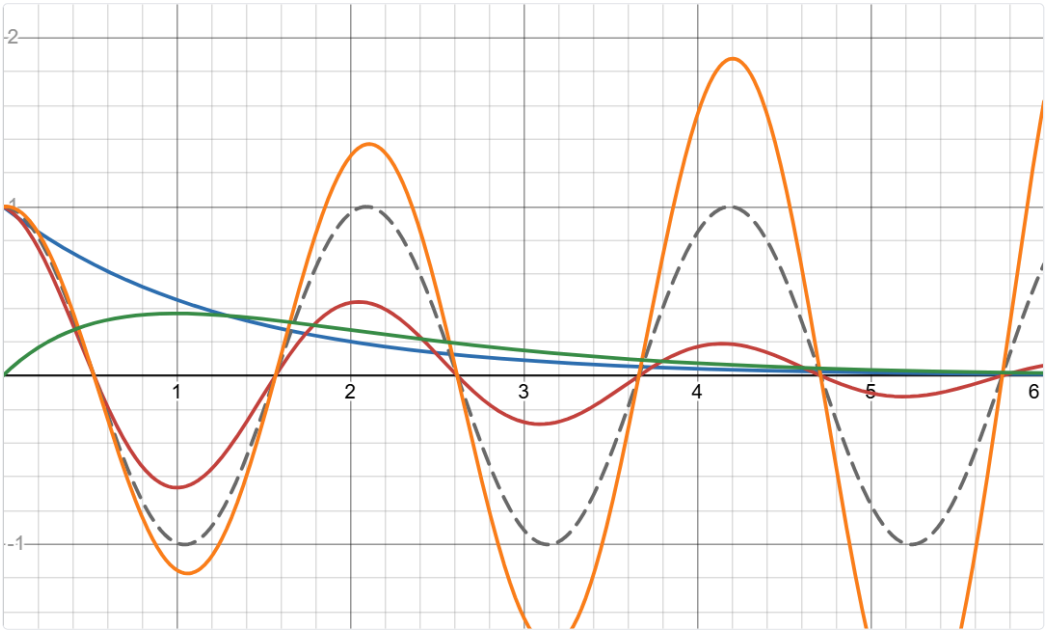
Worked Example 14.14 — $\sin^2\omega t$ two ways

Identity. $\sin^2\omega t=\frac{1}{2}(1-\cos 2\omega t)\leftrightarrow \frac{1}{2}\left(\frac{1}{s}-\frac{s}{s^2+4\omega^2}\right)=\frac{2\omega^2}{s(s^2+4\omega^2)}$.

Derivative rule. Let $x=\sin^2\omega t$: $x(0^-)=0$, $\dot{x}=2\omega\sin\omega t\cos\omega t=\omega\sin 2\omega t\leftrightarrow\frac{2\omega^2}{s^2+4\omega^2}$. Since $\mathcal{L}\{\dot{x}\}=sX$, dividing by s gives the same result. ✓

The second route is often faster when the derivative is simpler than the function.

The table shows a pattern worth internalising: **the denominator of $X(s)$ encodes the modes, the numerator encodes how much of each mode is present.** The roots of the denominator — the poles — determine the *shape* of the signal: a real pole $-a$ gives e^{-at}, a pair $-\alpha\pm j\omega$ gives $e^{-\alpha t}\cos(\omega t+\phi)$, a repeated pole multiplies by powers of t.



Pole location and time shape: real pole at -0.8 (blue); poles $-0.4\pm j3$ (red, decaying oscillation); poles $\pm j3$ (grey dashed, sustained); poles $0.15\pm j3$ (orange, growing); double pole at -1 giving te^{-t} (green).

Common Pitfall — misreading ω in the denominator

In $\dfrac{1}{s^2+9}$ the frequency is $\omega=3$, not 9, and the numerator needed for a pure sine is $\omega=3$: $\dfrac{1}{s^2+9}=\dfrac{1}{3}\cdot\dfrac{3}{s^2+9}\leftrightarrow\dfrac{1}{3}\sin 3t$. Forgetting the $\frac{1}{3}$ is one of the most frequent errors in inversion. Always check the value at $t=0^+$ or the DC gain as a sanity test.

Engineering Note — where the exotic entries appear

Entries 9–11 and 39–40 (fractional powers, $\sqrt{s}$, erfc) describe diffusion: carrier diffusion in a diode, heat spreading into a heatsink, and the step response of a long RC line (on-chip interconnect, the original transatlantic telegraph cable). Entry 38 (J_0) appears in cylindrical problems such as skin effect in round wires. Entries 41–44 describe clock and switching waveforms in power electronics.

14.4 Inverse Laplace transforms

Solving a problem in the s-domain is only half the job; the answer must be returned to the time domain, where voltages and currents are measured. The inverse transform does this. In practice we almost never evaluate an inversion integral: we rearrange $X(s)$ into a sum of table entries and read off the time function, using uniqueness (Lerch's theorem, Section 14.1) to guarantee the answer is the only one.

Definition and the inversion integral

Inverse Laplace transform

If $X(s) = \mathcal{L}\{x(t)\}$, then $x(t) = \mathcal{L}^{-1}\{X(s)\}$ is the **inverse Laplace transform** of X. By Lerch's theorem it is unique for $t > 0$ apart from values at isolated points. The inverse is linear: $\mathcal{L}^{-1}\{aX + bY\} = a\,x + b\,y$.

There is an explicit formula. Writing $s = \sigma + j\omega$ with fixed σ in the ROC, the Laplace transform of $x(t)$ is the Fourier transform of $x(t)e^{-\sigma t}u(t)$. Applying the inverse Fourier transform of Ch. 16 and multiplying by $e^{\sigma t}$ gives the following.

Bromwich inversion integral

If $X(s)$ is the transform of $x(t)$ with abscissa of convergence σ_c, then for any $\gamma > \sigma_c$,

$$x(t) = \frac{1}{2\pi j}\int_{\gamma - j\infty}^{\gamma + j\infty} X(s)\,e^{st}\,ds, \qquad t > 0,$$

where the integral is along the vertical line $\operatorname{Re}\{s\} = \gamma$ (to the right of all singularities). At a jump discontinuity the integral returns the midpoint $\tfrac{1}{2}[x(t^-) + x(t^+)]$; for $t < 0$ it returns 0.

Derivation sketch. $X(\gamma + j\omega) = \int_0^\infty \left[x(t)e^{-\gamma t}\right]e^{-j\omega t}dt$ is the Fourier transform of $g(t) = x(t)e^{-\gamma t}u(t)$, which is absolutely integrable because $\gamma > \sigma_c$. Fourier inversion gives $g(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\gamma + j\omega)e^{j\omega t}d\omega$. Multiply both sides by $e^{\gamma t}$ and change variable to $s = \gamma + j\omega$, $ds = j\,d\omega$. $\square$

This is a contour integral in the complex plane. For $t > 0$ the line can be closed by a large semicircle to the left; when $X(s)$ is rational the semicircle contributes nothing, and the **residue theorem** turns the integral into a finite sum:

$$x(t) = \sum_{\text{poles } p_k \text{ of } X} \operatorname{Res}_{s = p_k}\left[X(s)e^{st}\right], \qquad t > 0.$$

Closing the contour, Jordan's lemma, and the computation of residues are the subject of Ch. 18. The partial-fraction method of Section 14.5 is exactly this residue sum carried out with algebra, so for rational functions nothing is lost by using tables.

Inversion by table and properties

The working method is:

1. **Separate delays.** Collect terms by their factor e^{-sT}; invert each coefficient and then apply the delay $t \to t - T$ with $u(t - T)$.
2. **Reduce to proper rational form.** If the degree of the numerator is not less than that of the denominator, divide (polynomial long division); the polynomial part inverts to impulses and their derivatives.
3. **Expand in partial fractions** (Section 14.5) or, for a single quadratic factor, **complete the square**.
4. **Read the table**, using the s-shift to absorb $s + \alpha$ and adjusting constants so each term matches a table entry exactly.
5. **Check** at least one value: $x(0^+) = \lim_{s\to\infty} sX(s)$, or the final value, or X at a convenient real s.

Worked Example 14.15 — completing the square

Find $\mathcal{L}^{-1}\left\{\dfrac{2s + 12}{s^2 + 2s + 5}\right\}$.

Complete the square. $s^2 + 2s + 5 = (s + 1)^2 + 4$: the poles are $-1 \pm j2$, so expect e^{-t} times sinusoids of frequency 2 rad/s.

Rewrite the numerator in terms of $s + 1$. $2s + 12 = 2(s + 1) + 10$:

$$X(s) = 2\,\frac{s + 1}{(s + 1)^2 + 2^2} + 5 \cdot \frac{2}{(s + 1)^2 + 2^2}.$$

Table entries 23 and 24.

$$x(t) = e^{-t}\,(2\cos 2t + 5\sin 2t) = \sqrt{29}\,e^{-t}\cos(2t - 1.1903),$$

using $A\cos\theta + B\sin\theta = \sqrt{A^2 + B^2}\cos(\theta - \arctan(B/A))$ with $\arctan(5/2) = 1.1903$ rad.

Check. $x(0^+) = 2$, and $\lim_{s\to\infty} sX = \lim \frac{2s^2 + 12s}{s^2 + 2s + 5} = 2$. ✓ Also $X(0) = 12/5 = 2.4$; numerically $\int_0^\infty e^{-t}(2\cos 2t + 5\sin 2t)dt = 2 \cdot \frac{1}{5} + 5 \cdot \frac{2}{5} = 2.4$. ✓

Worked Example 14.16 — a delayed response

Find $\mathcal{L}^{-1}\left\{\dfrac{3e^{-2s}}{s(s + 3)}\right\}$.

Invert the coefficient. By entry 15 with $a = 3$: $\dfrac{3}{s(s + 3)} \leftrightarrow 1 - e^{-3t}$.

Apply the delay ($T = 2$): replace t by $t - 2$ and multiply by $u(t - 2)$:

$$x(t) = \left(1 - e^{-3(t-2)}\right)u(t - 2).$$

This is the response of a first-order system with time constant $\frac{1}{3}$ s to a step applied at $t = 2$ s. A common slip is to write $(1 - e^{-3t})u(t - 2)$, which jumps discontinuously at $t = 2$ to $1 - e^{-6}$ instead of starting smoothly from 0.

Worked Example 14.17 — repeated quadratic factor via the table

Find $\mathcal{L}^{-1}\left\{\dfrac{s}{(s^2+4)^2}\right\}$ and $\mathcal{L}^{-1}\left\{\dfrac{1}{(s^2+4)^2}\right\}$.

First. Entry 27 with $\omega = 2$: $\dfrac{4s}{(s^2+4)^2} \leftrightarrow t\sin 2t$, hence $\dfrac{s}{(s^2+4)^2} \leftrightarrow \dfrac{t\sin 2t}{4}$.

Second. Entry 29 with $\omega = 2$: $\dfrac{16}{(s^2+4)^2} \leftrightarrow \sin 2t - 2t\cos 2t$, hence

$$\frac{1}{(s^2+4)^2} \leftrightarrow \frac{\sin 2t - 2t\cos 2t}{16}.$$

Cross-check with the integral rule. $\frac{1}{(s^2+4)^2} = \frac{1}{s}\cdot\frac{s}{(s^2+4)^2}$, so the answer must equal $\int_0^t \frac{\tau\sin 2\tau}{4}\,d\tau = \frac{1}{4}\left[\frac{\sin 2\tau}{4} - \frac{\tau\cos 2\tau}{2}\right]_0^t = \frac{\sin 2t - 2t\cos 2t}{16}$. ✓

Worked Example 14.18 — several delays: a triangular pulse

Find and sketch $\mathcal{L}^{-1}\left\{\dfrac{1-2e^{-s}+e^{-2s}}{s^2}\right\}$, and identify the waveform.

Separate by delay. The coefficient of each exponential is $1/s^2 \leftrightarrow t$, so

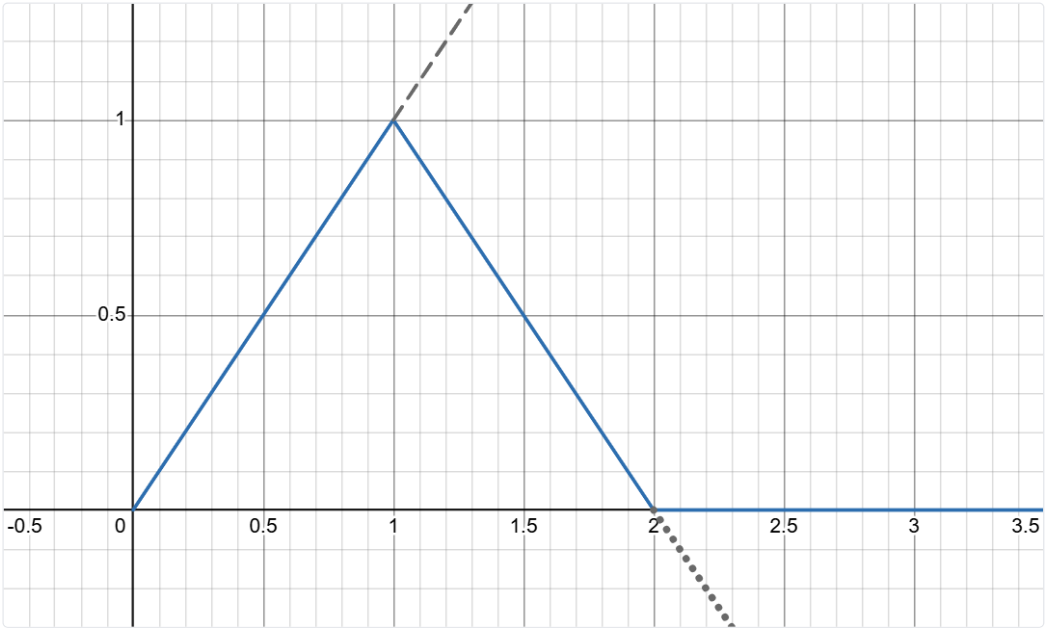
$$x(t) = t\,u(t) - 2(t-1)\,u(t-1) + (t-2)\,u(t-2).$$

Evaluate piece by piece.

- $0 \le t < 1$: $x = t$ (rising ramp).
- $1 \le t < 2$: $x = t - 2(t-1) = 2 - t$ (falling ramp).
- $t \ge 2$: $x = t - 2t + 2 + t - 2 = 0$.

The signal is a **triangular pulse** of height 1 at $t = 1$ and base 2. Note also that $\frac{(1-e^{-s})^2}{s^2} = \left(\frac{1-e^{-s}}{s}\right)^2$: the triangle is the convolution of a unit rectangular pulse with itself (Section 14.6).

Check. Area: $X(0) = \lim_{s\to 0}\frac{(1-e^{-s})^2}{s^2} = 1$, and the triangle's area is $\frac{1}{2}\cdot 2\cdot 1 = 1$ ✓.



Example 14.18: the ramp t (grey dashed beyond $t = 1$) is bent down by $-2(t-1)u(t-1)$ (grey dotted beyond $t = 2$) and levelled by $(t-2)u(t-2)$, producing the triangular pulse (blue).

Non-rational transforms

Transforms involving logarithms, inverse tangents or square roots cannot be split into partial fractions. Two tricks often work.

- **Differentiate in s.** If $X'(s)$ is rational, invert it and use $\mathcal{L}^{-1}\{X'\} = -t\,x(t)$, so $x(t) = -\frac{1}{t}\mathcal{L}^{-1}\{X'(s)\}$.
- **Recognise a scaled or shifted table entry**, e.g. $1/\sqrt{s+a} \leftrightarrow e^{-at}/\sqrt{\pi t}$ by the s-shift of entry 10.

Worked Example 14.19 — inverting an arctangent

Find $\mathcal{L}^{-1}\left\{\arctan\dfrac{2}{s}\right\}$.

Differentiate. $\dfrac{d}{ds}\arctan\dfrac{2}{s} = \dfrac{1}{1+4/s^2}\cdot\left(-\dfrac{2}{s^2}\right) = -\dfrac{2}{s^2+4} \leftrightarrow -\sin 2t$.

Undo the t-multiplication. Since $-t\,x(t) = -\sin 2t$,

$$x(t) = \frac{\sin 2t}{t},$$

agreeing with entry 35. **Check.** $x(0^+) = 2$; and $\lim_{s\to\infty} s\arctan(2/s) = \lim s\cdot\frac{2}{s} = 2$. ✓

Common Pitfall — not every function of s is a transform

By the existence theorem, a genuine transform of an ordinary function satisfies $X(s)\to 0$ as $\mathrm{Re}\{s\}\to\infty$. Functions such as s, $s^2/(s+1)$ or e^{+sT} do not; the first two contain impulses or their derivatives (handled after long division), and e^{sT} would be an *advance*, which no causal signal has. If an analysis produces e^{+sT}, a sign error has been made.

Engineering Note — numerical inversion

When $X(s)$ is known only numerically (e.g. from a measured impedance or a transmission-line model with $\sqrt{s}$), engineers use numerical Laplace inversion: the Talbot method deforms the Bromwich line into a contour where e^{st} decays rapidly, and the Stehfest algorithm uses only real values of $X(s)$. Both are standard in circuit-simulation and geophysics codes (Ch. 22).

14.5 Partial fractions

Almost every transform produced by a lumped circuit or a constant-coefficient ODE is a **rational function** — a ratio of polynomials in s:

$$X(s) = \frac{N(s)}{D(s)} = \frac{b_m s^m + \cdots + b_1 s + b_0}{a_n s^n + \cdots + a_1 s + a_0}.$$

Partial-fraction expansion rewrites it as a sum of simple terms, each of which is a table entry. Because each term corresponds to one *mode* (pole) of the system, the expansion is also a modal decomposition: it tells us how strongly each natural frequency is excited.

Proper and improper rational functions

$X(s)$ is **strictly proper** if $m < n$. Only strictly proper functions are transforms of ordinary functions (Section 14.4). If $m \ge n$, long division gives

$$X(s) = Q(s) + \frac{R(s)}{D(s)}, \qquad \deg R < \deg D,$$

and the polynomial quotient $Q(s) = c_0 + c_1 s + \cdots$ inverts to $c_0\delta(t) + c_1\dot\delta(t) + \cdots$. Physically, an improper transfer function is a differentiator: such terms appear for idealised elements (the current through an ideal capacitor driven by a voltage step), and they are a warning that the model has no high-frequency roll-off.

The expansion theorem

Factor the denominator over $\mathbb{C}$: $D(s) = a_n\prod_k (s-p_k)^{r_k}$, where the p_k are the distinct **poles** with multiplicities r_k, $\sum r_k = n$.

Partial-fraction expansion

A strictly proper rational function can be written uniquely as

$$\frac{N(s)}{D(s)} = \sum_{k}\sum_{i=1}^{r_k}\frac{K_{k,i}}{(s-p_k)^i},$$

with the coefficients of the highest power at each pole and the others given by

$$K_{k,r_k} = \Big[(s-p_k)^{r_k}X(s)\Big]_{s=p_k}, \qquad K_{k,r_k-j} = \frac{1}{j!}\frac{d^j}{ds^j}\Big[(s-p_k)^{r_k}X(s)\Big]_{s=p_k}, \quad j=1,\dots,r_k-1.$$

The inverse transform is then

$$x(t) = \sum_k e^{p_k t}\sum_{i=1}^{r_k} K_{k,i}\,\frac{t^{\,i-1}}{(i-1)!}, \qquad t>0.$$

Why the formulas work. Multiply the expansion by $(s-p_k)^{r_k}$. Every term belonging to another pole acquires a factor $(s-p_k)^{r_k}$ and vanishes at $s=p_k$, while the terms belonging to p_k become the polynomial $K_{k,r_k} + K_{k,r_k-1}(s-p_k) + \cdots + K_{k,1}(s-p_k)^{r_k-1}$. Evaluating it and its derivatives at $s=p_k$ isolates the coefficients one at a time — this is just Taylor's theorem. Existence and uniqueness follow from the fact that the n functions $(s-p_k)^{-i}$ are linearly independent and n coefficients match n degrees of freedom of N. The inverse formula uses entry 14 with the s-shift. □

For a **simple pole** ($r_k = 1$), the coefficient K_k is the **residue** of X at p_k. Since $D(p_k)=0$, L'Hôpital's rule gives a useful alternative.

Heaviside expansion formula (simple poles)

If all poles p_k of N/D are simple,

$$K_k = \left.\frac{N(s)}{D'(s)}\right|_{s=p_k}, \qquad x(t) = \sum_k \frac{N(p_k)}{D'(p_k)}\, e^{p_k t}.$$

This avoids factoring D symbolically: only its roots are needed, which a computer finds numerically.

Case 1 — distinct real poles: the cover-up rule

For a simple real pole, $K_k = [(s-p_k)X(s)]_{s=p_k}$: "cover up" the factor $(s-p_k)$ in the denominator and substitute $s=p_k$ in what remains.

Worked Example 14.20 — three real poles

Invert $X(s) = \dfrac{3s+7}{(s+1)(s+2)(s+3)}$.

Form. $X = \dfrac{K_1}{s+1} + \dfrac{K_2}{s+2} + \dfrac{K_3}{s+3}$.

Cover-up.

$$K_1 = \left.\frac{3s+7}{(s+2)(s+3)}\right|_{s=-1} = \frac{4}{(1)(2)} = 2, \quad K_2 = \left.\frac{3s+7}{(s+1)(s+3)}\right|_{s=-2} = \frac{1}{(-1)(1)} = -1, \quad K_3 = \left.\frac{3s+7}{(s+1)(s+2)}\right|_{s=-3} = \frac{-2}{(-2)(-1)} = -1.$$

Heaviside check. $D = s^3+6s^2+11s+6$, $D' = 3s^2+12s+11$: $D'(-1)=2$, $D'(-2)=-1$, $D'(-3)=2$, so $K_1 = 4/2 = 2$, $K_2 = 1/(-1) = -1$, $K_3 = -2/2 = -1$. ✓

Inverse.

$$x(t) = 2e^{-t} - e^{-2t} - e^{-3t}, \qquad t \geq 0.$$

Check. $x(0^+) = 2-1-1 = 0$, consistent with $\lim sX = \lim \frac{3s^2}{s^3} = 0$. Also $X(0) = 7/6$ and $\int_0^\infty x\,dt = 2 - \frac{1}{2} - \frac{1}{3} = \frac{7}{6}$. ✓

Case 2 — repeated poles

A pole of multiplicity r needs r terms. The top coefficient comes from the cover-up rule; the lower ones need derivatives (or, often more simply, comparison of coefficients or substitution of convenient values of s).

Worked Example 14.21 — a triple pole

Invert $X(s) = \dfrac{s^2+2s+5}{(s+1)^3(s+2)}$.

Form. $X = \dfrac{A_3}{(s+1)^3} + \dfrac{A_2}{(s+1)^2} + \dfrac{A_1}{s+1} + \dfrac{B}{s+2}$.

Simple pole. $B = \left.\dfrac{s^2+2s+5}{(s+1)^3}\right|_{s=-2} = \dfrac{5}{-1} = -5$.

Triple pole. Let $g(s) = (s+1)^3 X(s) = \dfrac{s^2+2s+5}{s+2}$. Division gives $g(s) = s + \dfrac{5}{s+2}$, which is easy to differentiate:

$$g(-1) = -1+5 = 4, \qquad g'(s) = 1 - \frac{5}{(s+2)^2} \Rightarrow g'(-1) = -4, \qquad g''(s) = \frac{10}{(s+2)^3} \Rightarrow \tfrac{1}{2}g''(-1) = 5.$$

So $A_3 = 4$, $A_2 = -4$, $A_1 = 5$.

Sanity check of coefficients. For large s, $X \approx 1/s^2$, so the $1/s$ terms must cancel: $A_1 + B = 5 - 5 = 0$. ✓

Inverse (entry 14):

$$x(t) = \left(2t^2 - 4t + 5\right) e^{-t} - 5e^{-2t}.$$

Check. $x(0^+) = 0$ ✓ (matches $\lim sX = 0$). At $s=0$: $X(0) = 5/2$; from the expansion $4 - 4 + 5 - \frac{5}{2} = \frac{5}{2}$. ✓

Case 3 — complex-conjugate poles

If D has real coefficients, complex poles occur in conjugate pairs $p, \bar{p} = -\alpha \pm j\beta$, and their coefficients are conjugates, K and $\bar{K}$. Their combined contribution is real:

$$Ke^{pt} + \bar{K}e^{\bar{p}t} = 2\,\mathrm{Re}\{Ke^{pt}\} = 2|K|e^{-\alpha t}\cos\left(\beta t + \angle K\right).$$

Complex-pole pair

If $X(s)$ has simple poles at $-\alpha \pm j\beta$ with residue $K = [(s+\alpha-j\beta)X(s)]_{s=-\alpha+j\beta}$ at the upper pole, their contribution to $x(t)$ is

$$2|K|\, e^{-\alpha t}\cos(\beta t + \angle K).$$

Two equivalent working methods exist:

- **Complex residues** (above): compute one complex number per pair and write amplitude and phase directly. Efficient, and natural for numerical work.
- **Real quadratic factor:** keep $(s+\alpha)^2+\beta^2$ intact with numerator $Bs+C$, find B, C by comparing coefficients, then complete the square as in Example 14.15. Avoids complex arithmetic and is often quicker by hand.

Worked Example 14.22 — one real pole and a complex pair, both methods

Invert $X(s) = \dfrac{s+3}{(s+1)(s^2+4s+13)}$.

The quadratic has roots $-2 \pm j3$.

Real pole (cover-up). $A = \left.\dfrac{s+3}{s^2+4s+13}\right|_{s=-1} = \dfrac{2}{10} = \dfrac{1}{5}$.

Method 1 — real quadratic. Write $X = \dfrac{1/5}{s+1} + \dfrac{Bs+C}{s^2+4s+13}$ and clear denominators:

$$s+3 = \tfrac{1}{5}(s^2+4s+13) + (Bs+C)(s+1).$$

s^2: $0 = \frac{1}{5} + B \Rightarrow B = -\frac{1}{5}$. Constant: $3 = \frac{13}{5} + C \Rightarrow C = \frac{2}{5}$. Then

$$\frac{-\frac{1}{5}s + \frac{2}{5}}{(s+2)^2+9} = -\frac{1}{5}\,\frac{(s+2)-4}{(s+2)^2+9} = -\frac{1}{5}\,\frac{s+2}{(s+2)^2+9} + \frac{4}{15}\,\frac{3}{(s+2)^2+9},$$

so

$$x(t) = \tfrac{1}{5}e^{-t} + e^{-2t}\left(-\tfrac{1}{5}\cos 3t + \tfrac{4}{15}\sin 3t\right).$$

Method 2 — complex residue. At $p = -2+j3$:

$$K = \left.\frac{s+3}{(s+1)(s+2+j3)}\right|_{s=-2+j3} = \frac{1+j3}{(-1+j3)(j6)} = \frac{1+j3}{-18-j6} = -0.1 - j0.1333.$$

$|K| = \frac{1}{6}$, $\angle K = -2.2143$ rad ($-126.87°$). The pair contributes $\frac{1}{3}e^{-2t}\cos(3t - 2.2143)$. Expanding, $\frac{1}{3}[\cos(2.2143)\cos 3t + \sin(2.2143)\sin 3t] = \frac{1}{3}[-0.6\cos 3t + 0.8\sin 3t] = -\frac{1}{5}\cos 3t + \frac{4}{15}\sin 3t$. ✓ Both methods agree.

Check. $x(0^+) = \frac{1}{5} - \frac{1}{5} = 0$ ✓; at $t = 0.7$ both forms give 0.18098.

Case 4 — improper functions

Worked Example 14.23 — long division first

Invert $X(s) = \dfrac{s^3+2s^2+3s+1}{s^2+s+1}$.

Divide. $s^3+2s^2+3s+1 = (s+1)(s^2+s+1) + s$, so

$$X(s) = s + 1 + \frac{s}{s^2+s+1}.$$

Proper part. $s^2+s+1 = (s+\frac{1}{2})^2 + \frac{3}{4}$, and $s = (s+\frac{1}{2}) - \frac{1}{2}$:

$$\frac{s}{s^2+s+1} = \frac{s+\frac{1}{2}}{(s+\frac{1}{2})^2+\frac{3}{4}} - \frac{1}{\sqrt{3}}\cdot\frac{\frac{\sqrt{3}}{2}}{(s+\frac{1}{2})^2+\frac{3}{4}}.$$

Inverse.

$$x(t) = \dot{\delta}(t) + \delta(t) + e^{-t/2}\left(\cos\frac{\sqrt{3}}{2}t - \frac{1}{\sqrt{3}}\sin\frac{\sqrt{3}}{2}t\right)u(t).$$

The impulse terms are a genuine part of the answer. If X were a transfer function, the system would differentiate its input — physically unrealisable at high frequencies, which is why practical transfer functions are proper.

Choosing a method, and checking

Situation	Fastest method
Distinct real poles	Cover-up (or N/D')
Repeated pole, low multiplicity	Cover-up for top term, then compare coefficients or substitute a value of s
Repeated pole, high multiplicity	Derivative formula, after simplifying $g(s) = (s-p)^r X$
One complex pair, hand work	Real quadratic + completing the square
Several complex pairs, numerical	Complex residues, amplitude–phase form
$\deg N \geq \deg D$	Long division first

Common Pitfall — too few terms for a repeated pole

Writing $\dfrac{1}{(s+1)^2(s+2)} = \dfrac{A}{(s+1)^2} + \dfrac{B}{s+2}$ is wrong: there are three unknowns' worth of freedom ($\deg D = 3$) and the term $\dfrac{C}{s+1}$ is missing. The correct expansion is $\dfrac{1}{(s+1)^2} - \dfrac{1}{s+1} + \dfrac{1}{s+2}$. A quick count — the number of coefficients must equal $\deg D$ — catches this every time.

Engineering Note — residues as modal amplitudes

In a circuit response, each coefficient K_k is the initial amplitude of mode $e^{p_k t}$. A pole with a tiny residue is barely excited and can often be dropped (*dominant-pole approximation*); a pole and zero that nearly coincide (a "doublet") produce a small residue and a slow, low-amplitude tail that can still matter for settling time in precision amplifiers. Computer tools (`residue` in MATLAB, `scipy.signal.residue` in Python) implement the Heaviside formula with numerical root-finding.

14.6 Convolution theorem

Partial fractions invert *sums*; the convolution theorem inverts *products*. It answers the question "if I know the time functions for $F(s)$ and $G(s)$, what time function has transform $F(s)G(s)$?" The answer — their convolution — is also the central formula of linear system theory: the output of a linear time-invariant system is the convolution of its input with its impulse response.

Definition

Convolution of causal signals

For functions f, g defined on $t \geq 0$ (zero for $t < 0$), the **convolution** is

$$(f * g)(t) = \int_{0^-}^{t} f(\tau)\, g(t-\tau)\, d\tau, \qquad t \geq 0.$$

For general (two-sided) signals the integral runs over $(-\infty, \infty)$; for causal signals the integrand vanishes outside $0 \leq \tau \leq t$, giving the finite limits above. The operation "flip g, slide it by t, multiply by f, integrate" is linear in each argument and

- commutative: $f * g = g * f$ (substitute $\tau' = t - \tau$);
- associative: $(f * g) * h = f * (g * h)$;
- distributive: $f * (g + h) = f * g + f * h$;
- has identity δ: $f * \delta = f$, and $f * \delta(t - T) = f(t - T)u(t - T)$;
- $f * u$ is the running integral: $(f * u)(t) = \int_0^t f(\tau) d\tau$.

The theorem

Convolution theorem

If $f \leftrightarrow F(s)$ and $g \leftrightarrow G(s)$, both of exponential order, then

$$\mathcal{L}\{(f * g)(t)\} = F(s)\, G(s),$$

on the intersection of the ROCs. Equivalently, $\mathcal{L}^{-1}\{F(s)G(s)\} = \int_0^t f(\tau) g(t-\tau) d\tau$.

Proof. Write the convolution with a step so that the upper limit becomes infinite:

$$\mathcal{L}\{f * g\} = \int_0^\infty e^{-st} \int_0^\infty f(\tau)\, g(t-\tau)\, u(t-\tau)\, d\tau\, dt.$$

For s in both ROCs the double integral converges absolutely, so the order of integration can be exchanged (Fubini's theorem, Ch. 6):

$$= \int_0^\infty f(\tau) \left[\int_\tau^\infty g(t-\tau) e^{-st} dt \right] d\tau = \int_0^\infty f(\tau)\, e^{-s\tau} G(s)\, d\tau = F(s)G(s),$$

where the inner integral was evaluated with $t' = t - \tau$ exactly as in the time-shift theorem. $\square$

The region of integration is the wedge $0 \leq \tau \leq t$ in the (τ, t) plane; swapping the order means integrating first over t from τ to ∞ — a picture worth drawing once.

Impulse response and the system equation

Consider an LTI system initially at rest. If the input $x(t) = \delta(t)$ produces the output $h(t)$ — the **impulse response** — then by linearity and time invariance the input $x(\tau)\delta(t-\tau)\, d\tau$ produces $x(\tau)h(t-\tau)\, d\tau$, and summing over all τ:

$$y(t) = \int_{0^-}^{t} x(\tau)\, h(t-\tau)\, d\tau = (x * h)(t) \quad \Longleftrightarrow \quad Y(s) = H(s)X(s).$$

The function $H(s) = \mathcal{L}\{h(t)\}$ is the **transfer function** (Section 14.9). Convolution in time — an integral over the whole past of the input — becomes multiplication in s. This is the deepest reason the Laplace transform simplifies linear-system analysis.

A related form, **Duhamel's integral**, uses the step response $a(t) = \int_0^t h\,d\tau$ instead: since $HX = (sA)X = A\,(sX)$, the response is $y(t) = x(0^-)a(t) + \int_0^t \dot{x}(\tau)a(t-\tau)d\tau$. It is useful when the step response has been measured experimentally.

Worked Example 14.24 — inverting a product by convolution

Find $\mathcal{L}^{-1}\left\{\dfrac{1}{s^2(s+1)}\right\}$ by convolution and confirm by partial fractions.

Factor. $F = 1/s^2 \leftrightarrow t$, $G = 1/(s+1) \leftrightarrow e^{-t}$.

Convolve (choose the order that makes the integral easiest; here put the exponential in the shifted slot):

$$\int_0^t \tau\, e^{-(t-\tau)}d\tau = e^{-t}\int_0^t \tau e^{\tau}d\tau = e^{-t}\Big[(\tau-1)e^{\tau}\Big]_0^t = e^{-t}\big[(t-1)e^t + 1\big] = t - 1 + e^{-t}.$$

Partial fractions. $\dfrac{1}{s^2(s+1)} = \dfrac{1}{s^2} - \dfrac{1}{s} + \dfrac{1}{s+1} \leftrightarrow t - 1 + e^{-t}$. ✓ (Table entry 18 with $a = 1$.)

Worked Example 14.25 — a repeated quadratic by convolution

Find $\mathcal{L}^{-1}\left\{\dfrac{1}{(s^2+1)^2}\right\}$.

$\dfrac{1}{(s^2+1)^2} = \dfrac{1}{s^2+1}\cdot\dfrac{1}{s^2+1} \leftrightarrow \sin t * \sin t$. Using $\sin A \sin B = \tfrac{1}{2}[\cos(A-B) - \cos(A+B)]$:

$$\int_0^t \sin\tau\,\sin(t-\tau)\,d\tau = \frac{1}{2}\int_0^t \big[\cos(2\tau - t) - \cos t\big]d\tau = \frac{1}{2}\left[\frac{\sin(2\tau - t)}{2}\right]_0^t - \frac{t\cos t}{2} = \frac{\sin t}{2} - \frac{t\cos t}{2}.$$

So $\mathcal{L}^{-1}\{(s^2+1)^{-2}\} = \tfrac{1}{2}(\sin t - t\cos t)$, entry 29 with $\omega = 1$. ✓ Partial fractions cannot simplify $(s^2+1)^{-2}$ further over the reals, so convolution (or the table) is the natural route.

Worked Example 14.26 — pulse response of an RC low-pass filter

An RC low-pass filter with $RC = 1$ ms has impulse response $h(t) = \frac{1}{RC}e^{-t/RC}u(t)$. Find the output for a 1 V input pulse lasting 1 ms. Work in milliseconds so $h(t) = e^{-t}$.

Time domain. The input is $x(t) = u(t) - u(t-1)$. For $0 \le t < 1$ the pulse overlaps $[0, t]$:

$$y(t) = \int_0^t 1\cdot e^{-(t-\tau)}d\tau = 1 - e^{-t}.$$

For $t \ge 1$ only $[0, 1]$ contributes:

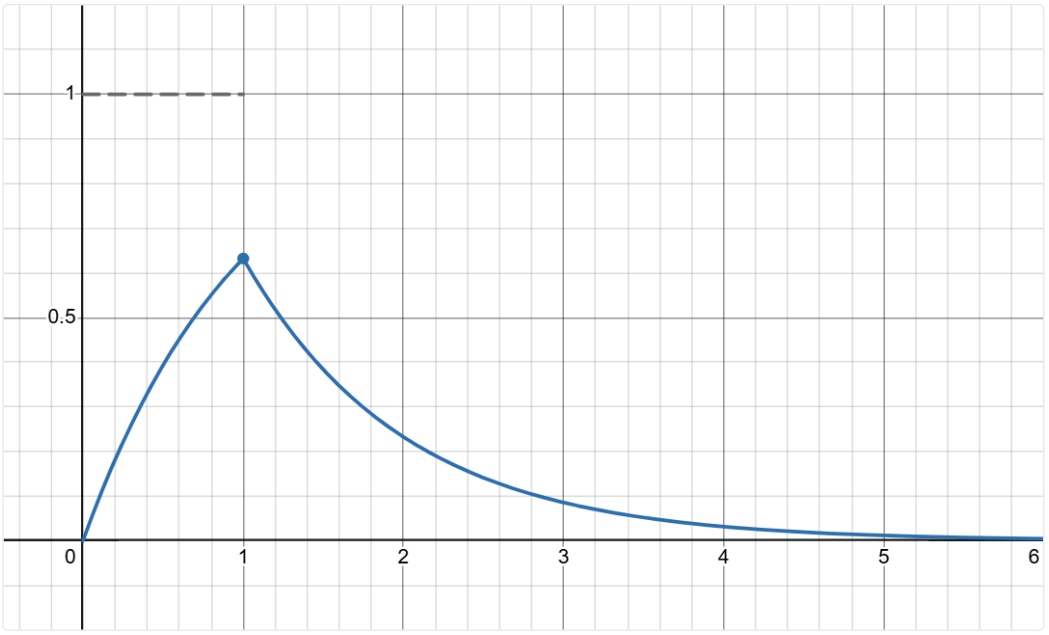
$$y(t) = \int_0^1 e^{-(t-\tau)}d\tau = e^{-t}(e-1).$$

s-domain. $Y = HX = \dfrac{1}{s+1}\cdot\dfrac{1-e^{-s}}{s} = (1-e^{-s})\left(\dfrac{1}{s} - \dfrac{1}{s+1}\right)$, so

$$y(t) = (1 - e^{-t})u(t) - \big(1 - e^{-(t-1)}\big)u(t-1).$$

For $t \ge 1$: $1 - e^{-t} - 1 + e^{-t+1} = e^{-t}(e-1)$. ✓ Both routes agree; the s-domain route needed no case analysis.

Numbers. Peak at $t = 1$ ms: $y = 1 - e^{-1} = 0.632$ V; at $t = 3$ ms: $y = (e-1)e^{-3} = 0.0855$ V.



Response of an RC low-pass filter ($RC = 1$ ms) to a 1 V, 1 ms pulse (grey): the capacitor charges toward 1 V while the pulse lasts, then discharges exponentially (Example 14.26; time in ms).

Worked Example 14.27 — a Volterra integral equation

Solve $y(t) = t + \int_0^t \sin(t-\tau)\, y(\tau)\, d\tau$.

The integral is $\sin * y$. Transforming, with $\mathcal{L}\{\sin t\} = 1/(s^2+1)$:

$$Y = \frac{1}{s^2} + \frac{Y}{s^2+1} \quad\Longrightarrow\quad Y\,\frac{s^2}{s^2+1} = \frac{1}{s^2} \quad\Longrightarrow\quad Y = \frac{s^2+1}{s^4} = \frac{1}{s^2} + \frac{1}{s^4}.$$

$$y(t) = t + \frac{t^3}{6}.$$

Check by substitution. $\int_0^t \sin(t-\tau)\left(\tau + \frac{\tau^3}{6}\right) d\tau$: using $\sin * \tau = t - \sin t$ and $\sin * \frac{\tau^3}{6} = \frac{t^3}{6} - t + \sin t$ (from $\frac{1}{s^4(s^2+1)} = \frac{1}{s^4} - \frac{1}{s^2} + \frac{1}{s^2+1}$), the sum is $t^3/6$, and $t + t^3/6 = y$. ✓

Integral equations of this convolution type arise in models with memory: hereditary (viscoelastic) materials, and circuits described by their impulse response rather than by an ODE.

Common Pitfall — the product of transforms is not the transform of the product

$\mathcal{L}\{f(t)g(t)\} \neq F(s)G(s)$. Multiplication in time corresponds to a *complex convolution* in s (a contour integral), which is rarely used in circuit work. For example $\mathcal{L}\{t \cdot t\} = 2/s^3$, while $\frac{1}{s^2}\cdot\frac{1}{s^2} = \frac{1}{s^4} \leftrightarrow t^3/6$.

Engineering Note — convolution in practice

Measuring $h(t)$ directly with an impulse is difficult, but its integral, the step response, is easy to record on an oscilloscope. Duhamel's integral then predicts the response to any input. In digital signal processing the same idea, in discrete form, is the FIR filter: the output is a weighted sum of past inputs with the impulse response as weights (Ch. 17).

14.7 Initial- and final-value theorems

Often an engineer does not need the whole response — only where it starts and where it ends. Does the output voltage jump when the switch closes? What is the steady-state error of a control loop? What is the DC gain? The initial- and final-value theorems read these numbers directly from $X(s)$, with no inversion. They are also the quickest way to check a partial-fraction result.

Statements

Initial-value theorem (IVT)

If $x(t)$ and $\dot{x}(t)$ (excluding any impulse at the origin) are transformable and $X(s)$ is strictly proper, then

$$x(0^+) = \lim_{s\to\infty} sX(s),$$

the limit taken along the positive real axis (or any path with $\operatorname{Re}\{s\} \to \infty$).

Final-value theorem (FVT)

If $x(t)$ and $\dot{x}(t)$ are transformable and **all poles of $sX(s)$ lie in the open left half-plane** $\mathrm{Re}\{s\} < 0$, then $x(\infty) = \lim_{t\to\infty} x(t)$ exists and

$$\lim_{t\to\infty} x(t) = \lim_{s\to 0} sX(s).$$

The condition on the poles of $sX(s)$ means: $X(s)$ may have at most a *simple* pole at the origin (cancelled by the factor s), and every other pole must have negative real part. Without it, the right-hand side still produces a number, but that number is meaningless.

Proofs

Both theorems come from the derivative rule. Split the derivative integral at the origin:

$$\int_{0^-}^{\infty} \dot{x}(t)e^{-st}dt = sX(s) - x(0^-).$$

The part from 0^- to 0^+ picks up the jump $x(0^+) - x(0^-)$ (the impulse in $\dot{x}$), so

$$sX(s) - x(0^+) = \int_{0^+}^{\infty} \dot{x}(t)e^{-st}dt.$$

IVT. As $s \to \infty$ along the real axis, $e^{-st} \to 0$ for every $t > 0$, and (dominated convergence, using exponential order) the right side tends to 0. Hence $sX(s) \to x(0^+)$. If X were not strictly proper, x would contain an impulse at the origin, $x(0^+)$ would not describe it, and $sX \to \infty$.

FVT. As $s \to 0$, $e^{-st} \to 1$ and the right side tends to $\int_{0^+}^{\infty} \dot{x}\,dt = x(\infty) - x(0^+)$ — *provided* this integral converges, which the pole condition guarantees (every mode of $\dot{x}$ then decays exponentially). Adding $x(0^+)$ to both sides gives $\lim_{s\to 0} sX = x(\infty)$. □

In terms of partial fractions the FVT is transparent: with all poles of X except a simple one at $s = 0$ in the left half-plane, every exponential term dies out and only the constant K_0 from the term K_0/s survives. That constant is $\lim_{s\to 0} sX(s)$ by the cover-up rule.

Higher initial derivatives

Applying the IVT to $\dot{x}$ (whose transform, for $t > 0$, is $sX - x(0^+)$) gives

$$\dot{x}(0^+) = \lim_{s\to\infty} s\big[sX(s) - x(0^+)\big], \qquad \ddot{x}(0^+) = \lim_{s\to\infty} s\big[s^2X(s) - sx(0^+) - \dot{x}(0^+)\big],$$

and so on. For a rational $X = N/D$ with $\deg D - \deg N = r$ (the **relative degree**), x and its first $r - 2$ derivatives start at zero, and $x^{(r-1)}(0^+) = b_m/a_n$, the ratio of leading coefficients. A high relative degree means a response that leaves zero gently — a smooth start.

Worked Example 14.28 — start, slope and end without inversion

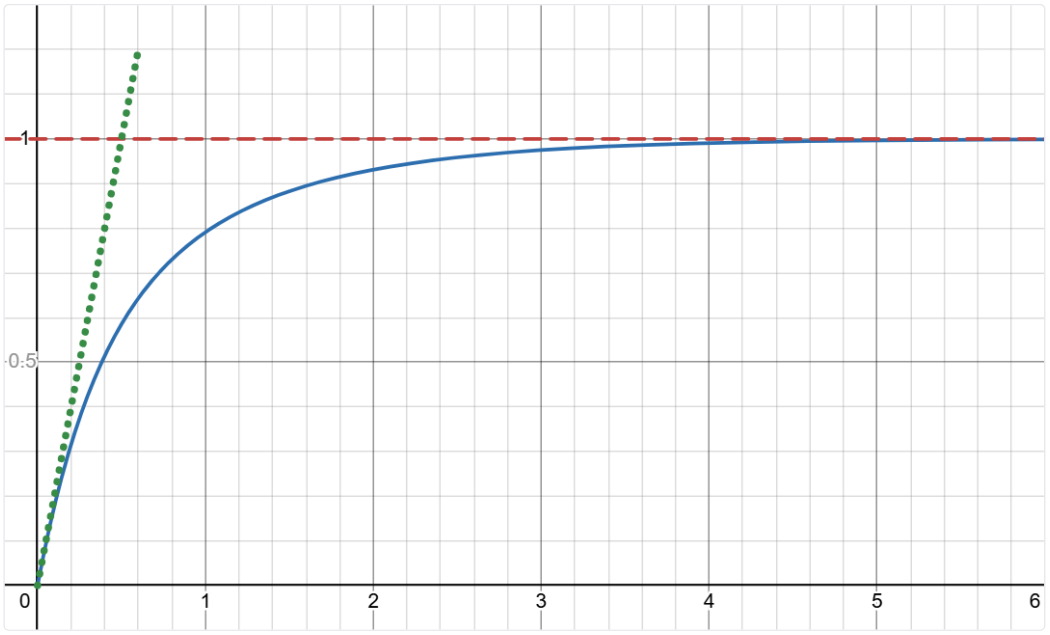
For $X(s) = \dfrac{2s+3}{s(s+1)(s+3)}$ find $x(0^+)$, $\dot{x}(0^+)$ and $x(\infty)$; then check against the inverse.

Initial value. $sX = \dfrac{2s+3}{(s+1)(s+3)} \to \dfrac{2s}{s^2} \to 0$. So $x(0^+) = 0$.

Initial slope. $\dot{x}(0^+) = \lim s \cdot sX = \lim \dfrac{s(2s+3)}{(s+1)(s+3)} = 2$. (Relative degree 2, leading-coefficient ratio $2/1$.)

Final value. Poles of sX are $-1, -3$: both in the left half-plane, so the FVT applies: $x(\infty) = \dfrac{3}{(1)(3)} = 1$.

Check. Partial fractions give $X = \dfrac{1}{s} - \dfrac{1/2}{s+1} - \dfrac{1/2}{s+3}$, so $x(t) = 1 - \tfrac{1}{2}e^{-t} - \tfrac{1}{2}e^{-3t}$: $x(0) = 0$ ✓, $\dot{x}(0) = \tfrac{1}{2} + \tfrac{3}{2} = 2$ ✓, $x(\infty) = 1$ ✓.



$x(t) = 1 - \frac{1}{2}e^{-t} - \frac{1}{2}e^{-3t}$ (blue) starts at $x(0^+) = 0$ with slope 2 (green dotted tangent) and settles to $x(\infty) = 1$ (red dashed) — all three read from $X(s)$ in Example 14.28.

Worked Example 14.29 — when the FVT lies

Apply $\lim_{s\to 0} sX(s)$ blindly to each transform, and compare with the truth.

$X(s)$	$\lim_{s\to 0} sX$	**true** $x(t)$	**verdict**
$\frac{1}{s-2}$	0	$e^{2t} \to \infty$	pole at $+2$ in RHP: FVT invalid
$\frac{\omega}{s^2+\omega^2}$	0	$\sin \omega t$, no limit	poles $\pm j\omega$ on axis: FVT invalid
$\frac{1}{s(s^2+1)}$	1	$1 - \cos t$, no limit	sX has poles $\pm j$: FVT invalid
$\frac{1}{s^2}$	∞	$t \to \infty$	double pole at 0: FVT invalid (but correctly signals unboundedness)
$\frac{4}{s(s+2)}$	2	$2 - 2e^{-2t} \to 2$	valid ✓

The third row is the dangerous one: the theorem returns a finite, plausible number (the *average* of $1 - \cos t$) for a signal that never settles. Always check the pole condition first.

Worked Example 14.30 — an impulse at the origin

$X(s) = \dfrac{s+2}{s+1}$. Find $x(0^+)$.

X is not strictly proper; $sX \to \infty$. Divide first: $X = 1 + \dfrac{1}{s+1}$, so $x(t) = \delta(t) + e^{-t}u(t)$. The IVT applies to the strictly proper part: $x(0^+) = \lim_{s\to\infty} \dfrac{s}{s+1} = 1$. The impulse lives at $t = 0$ itself, not at 0^+, and must be reported separately.

Worked Example 14.31 — DC gain and initial slope of a step response

An amplifier has transfer function $H(s) = \dfrac{10(s+5)}{s^2+3s+50}$. For a unit-step input, find the initial value, initial slope and final value of the output.

The output is $Y(s) = H(s)/s$, so $sY = H(s)$.

- $y(0^+) = \lim_{s\to\infty} H(s) = 0$ (relative degree 1 of H, so the step response is continuous).
- $\dot{y}(0^+) = \lim_{s\to\infty} sH(s) = 10$ V/s per volt of input (ratio of leading coefficients).
- Poles of H: $s^2 + 3s + 50 = 0 \Rightarrow s = -1.5 \pm j6.91$, in the LHP, so $y(\infty) = H(0) = \dfrac{50}{50} = 1$.

General rule: for a stable system, the final value of the step response is the **DC gain** $H(0)$. The step response here overshoots substantially (damping ratio $\zeta = 1.5/\sqrt{50} = 0.21$), something the FVT says nothing about — it describes only where the response ends, not how it gets there.

Common Pitfall — using the theorems on X instead of sX

Both theorems involve $sX(s)$, not $X(s)$. Forgetting the factor s gives $\lim_{s\to 0} X(s)$, which for a step response is infinite and for an impulse response equals the *area* $\int_0^\infty x\,dt$ — a different (and sometimes useful) quantity.

Engineering Note — steady-state error in control loops

In a unity-feedback loop with open-loop transfer function $G(s)$, the error for a reference $R(s)$ is $E(s) = R(s)/(1+G(s))$. The FVT gives the steady-state error $e_{ss} = \lim_{s\to 0} sR/(1+G)$. For a unit step this is $1/(1+G(0))$, which is zero exactly when G has a pole at $s=0$ (an integrator in the loop). This one-line calculation is the basis of the "system type" classification in control engineering (Section 14.9).

14.8 Solving differential equations

This is the application that justifies the whole apparatus. A linear constant-coefficient ODE with initial conditions — an **initial-value problem (IVP)** — becomes an algebraic equation for $Y(s)$. Solving it produces, in a single expression, both the natural response (fixed by the initial conditions) and the forced response (fixed by the input). Discontinuous and impulsive inputs, which are awkward for the classical method of Ch. 12, cause no extra difficulty.

The general procedure

Consider the nth-order equation

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_1 \dot{y} + a_0 y = x(t), \qquad y(0^-),\ \dot{y}(0^-),\ \ldots,\ y^{(n-1)}(0^-) \text{ given.}$$

1. **Transform** both sides, using the derivative rule for each $y^{(k)}$ and the table for $x(t)$.
2. **Collect** the terms in $Y(s)$ on the left. The coefficient of Y is the **characteristic polynomial** $P(s) = a_n s^n + \cdots + a_0$ — the same polynomial whose roots gave the exponents in the classical method.
3. **Solve** algebraically for $Y(s)$, which always takes the form shown below.
4. **Invert** by partial fractions (Section 14.5) and the table.
5. **Check** the initial conditions (IVT) and, if applicable, the final value (FVT) — or substitute back.

$$Y(s) = \underbrace{\frac{C(s)}{P(s)}}_{\text{zero-input}} + \underbrace{\frac{X(s)}{P(s)}}_{\text{zero-state}},$$

where $C(s)$ is a polynomial of degree $< n$ built from the initial conditions.

The **zero-input response** is what the system does from its initial state with no input; the **zero-state response** is what the input does to a system initially at rest. Their sum is the complete response. The ratio $H(s) = 1/P(s)$ that multiplies $X(s)$ is the transfer function of the equation.

Structure of the solution

The poles of $Y(s)$ are the roots of $P(s)$ (the **natural frequencies** or characteristic roots) together with the poles of $X(s)$ (the **forced frequencies**). The terms belonging to the roots of P form the natural (transient) response; those belonging to the poles of X form the forced (particular) response. If a pole of X coincides with a root of P, the pole of Y is repeated and terms $t^k e^{pt}$ appear — the resonance case that the classical method treats by "multiplying the trial solution by t".

Worked Example 14.32 — first-order RC charging

A capacitor voltage obeys $\dot{y} + 2y = 4$, $y(0^-) = 1$ (time in ms, voltage in V). Find $y(t)$.

Transform. $sY - 1 + 2Y = \dfrac{4}{s}$.

Solve. $Y = \dfrac{1}{s+2} + \dfrac{4}{s(s+2)} = \dfrac{s+4}{s(s+2)}$.

Partial fractions (cover-up): at $s=0$: $4/2 = 2$; at $s=-2$: $2/(-2) = -1$.

$$Y = \frac{2}{s} - \frac{1}{s+2} \quad\Longrightarrow\quad y(t) = 2 - e^{-2t}\ \text{V}.$$

Check. $y(0^+) = 1$ ✓ (no jump, as expected for a capacitor voltage); $\dot{y} + 2y = 2e^{-2t} + 4 - 2e^{-2t} = 4$ ✓; final value 2 V = forcing/coefficient ✓.

Worked Example 14.33 — second order, zero-input and zero-state parts

Solve $\ddot{y} + 3\dot{y} + 2y = e^{-3t}$ with $y(0^-) = 1$, $\dot{y}(0^-) = 0$.

Transform.

$$\big[s^2Y - s\cdot 1 - 0\big] + 3\big[sY - 1\big] + 2Y = \frac{1}{s+3}.$$

Collect. $(s^2+3s+2)Y = (s+3) + \dfrac{1}{s+3}$, with $P(s) = (s+1)(s+2)$:

$$Y = \underbrace{\frac{s+3}{(s+1)(s+2)}}_{Y_{zi}} + \underbrace{\frac{1}{(s+1)(s+2)(s+3)}}_{Y_{zs}}.$$

Zero-input. Cover-up: $\dfrac{2}{s+1} - \dfrac{1}{s+2} \leftrightarrow 2e^{-t} - e^{-2t}$.

Zero-state. Cover-up: at -1: $\frac{1}{(1)(2)} = \frac{1}{2}$; at -2: $\frac{1}{(-1)(1)} = -1$; at -3: $\frac{1}{(-2)(-1)} = \frac{1}{2}$. So $\frac{1}{2}e^{-t} - e^{-2t} + \frac{1}{2}e^{-3t}$.

Complete response.

$$y(t) = \tfrac{5}{2}e^{-t} - 2e^{-2t} + \tfrac{1}{2}e^{-3t}.$$

Check. $y(0) = \frac{5}{2} - 2 + \frac{1}{2} = 1$ ✓; $\dot{y}(0) = -\frac{5}{2} + 4 - \frac{3}{2} = 0$ ✓. Substituting into the ODE: the e^{-t} and e^{-2t} terms are annihilated by P (since $P(-1) = P(-2) = 0$), and the e^{-3t} term gives $\frac{1}{2}P(-3)e^{-3t} = \frac{1}{2}\cdot 2\,e^{-3t} = e^{-3t}$ ✓.

Note how the classical method's three steps (complementary function, particular integral, fitting constants) collapsed into one partial-fraction expansion.

Discontinuous and impulsive inputs

Worked Example 14.34 — a rectangular pulse into an undamped oscillator

An undamped LC tank, initially at rest, obeys $\ddot{y} + 4y = x(t)$ where $x(t) = 1$ for $0 \le t < \pi$ and 0 afterwards. Find $y(t)$.

Input. $x = u(t) - u(t-\pi) \leftrightarrow \dfrac{1 - e^{-\pi s}}{s}$.

Transform. $(s^2+4)Y = \dfrac{1-e^{-\pi s}}{s}$, so $Y = (1 - e^{-\pi s})\,G(s)$ with

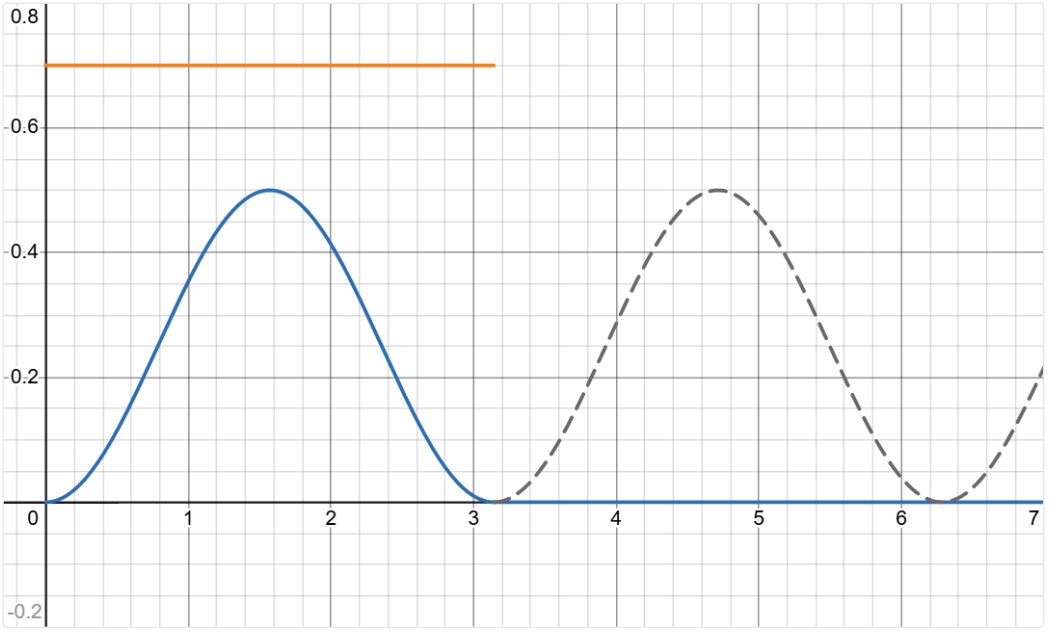
$$G(s) = \frac{1}{s(s^2+4)} = \frac{1}{4}\left(\frac{1}{s} - \frac{s}{s^2+4}\right) \leftrightarrow g(t) = \frac{1-\cos 2t}{4}.$$

Invert with the delay.

$$y(t) = \frac{1-\cos 2t}{4} - \frac{1-\cos 2(t-\pi)}{4}\,u(t-\pi).$$

For $t \ge \pi$, $\cos 2(t-\pi) = \cos 2t$, so the two terms cancel exactly: $y(t) = 0$ **for** $t \ge \pi$**.**

Interpretation. The pulse lasts exactly one natural period ($2\pi/2 = \pi$). The energy it pumps in during the first half-period is withdrawn in the second, leaving the oscillator at rest: $y(\pi) = 0$ and $\dot{y}(\pi) = \frac{1}{2}\sin 2\pi = 0$. A pulse of any other length would leave a residual oscillation. This "input shaping" idea is used to move resonant mechanical loads (disk-drive heads, cranes) without exciting vibration.



Example 14.34: during the pulse (orange, scaled) the oscillator swings once (blue) and is left exactly at rest at $t=\pi$; without the pulse ending, it would continue oscillating (grey dashed).

Worked Example 14.35 — resonance

Solve $\ddot{y}+\omega^2 y=\sin\omega t$, $y(0^-)=\dot{y}(0^-)=0$.

$$Y=\frac{\omega}{(s^2+\omega^2)^2}=\frac{1}{2\omega^2}\cdot\frac{2\omega^3}{(s^2+\omega^2)^2}\quad\Longrightarrow\quad y(t)=\frac{\sin\omega t-\omega t\cos\omega t}{2\omega^2}$$

by table entry 29. The forcing frequency coincides with the natural frequency, the poles $\pm j\omega$ are doubled, and the amplitude grows linearly as $t/(2\omega)$ without bound. In a real circuit the resistance limits the growth; the ideal model warns that a lightly damped resonator driven at resonance develops very large voltages (a factor Q above the drive).

Check ($\omega=2$, $t=1$): $y=\frac{\sin 2-2\cos 2}{8}=\frac{0.9093+0.8323}{8}=0.2177$. Numerical integration of the ODE gives 0.2177. ✓

Integro-differential equations

Equations containing both derivatives and running integrals of the unknown — typical of loop equations with capacitors — are handled just as easily, because the integral rule turns $\int_0^t y\,d\tau$ into Y/s.

Worked Example 14.36 — an integro-differential equation

Solve $\dot{y}+3y+2\displaystyle\int_0^t y(\tau)\,d\tau=u(t)$, $y(0^-)=0$.

Transform. $sY+3Y+\dfrac{2Y}{s}=\dfrac{1}{s}$. Multiply by s: $(s^2+3s+2)Y=1$.

$$Y=\frac{1}{(s+1)(s+2)}=\frac{1}{s+1}-\frac{1}{s+2}\quad\Longrightarrow\quad y(t)=e^{-t}-e^{-2t}.$$

This is the loop current of a series RLC circuit ($L=1$, $R=3$, $1/C=2$) connected to a unit step: it rises from zero, peaks at $t=\ln 2=0.693$ (value 0.25), and decays because the charged capacitor eventually blocks DC.

Simultaneous equations

Coupled circuits (mutual inductance, multi-loop networks) and state-space models give systems of ODEs. Each equation transforms to a linear algebraic equation, and the resulting linear system in $X(s), Y(s), \dots$ is solved by elimination or Cramer's rule (Ch. 8).

In matrix form, $\dot{\mathbf{x}}=\mathbf{A}\mathbf{x}+\mathbf{b}\,v(t)$ transforms to $s\mathbf{X}-\mathbf{x}(0^-)=\mathbf{A}\mathbf{X}+\mathbf{b}V$, so

$$\mathbf{X}(s)=(s\mathbf{I}-\mathbf{A})^{-1}\big[\mathbf{x}(0^-)+\mathbf{b}V(s)\big].$$

Since $(s\mathbf{I}-\mathbf{A})^{-1}=\operatorname{adj}(s\mathbf{I}-\mathbf{A})/\det(s\mathbf{I}-\mathbf{A})$, the poles of every solution component are the **eigenvalues of $\mathbf{A}$** (Ch. 10), and the matrix exponential is $e^{\mathbf{A}t}=\mathcal{L}^{-1}\{(s\mathbf{I}-\mathbf{A})^{-1}\}$.

Worked Example 14.37 — a coupled pair

Solve $\dot{x}=-2x+y$, $\dot{y}=x-2y+2$ for $t\ge 0$, with $x(0^-)=1$, $y(0^-)=0$.

Transform.

$$\begin{aligned}(s+2)X-Y&=1,\\ -X+(s+2)Y&=\frac{2}{s}.\end{aligned}$$

Cramer's rule. $\Delta=(s+2)^2-1=(s+1)(s+3)$ — the eigenvalues of $\mathbf{A}=\begin{bmatrix}-2&1\\1&-2\end{bmatrix}$ are -1 and -3.

$$X=\frac{(s+2)+\frac{2}{s}}{(s+1)(s+3)}=\frac{s^2+2s+2}{s(s+1)(s+3)},\qquad Y=\frac{(s+2)\frac{2}{s}+1}{(s+1)(s+3)}=\frac{3s+4}{s(s+1)(s+3)}.$$

Partial fractions (cover-up).

$$X=\frac{2/3}{s}-\frac{1/2}{s+1}+\frac{5/6}{s+3},\qquad Y=\frac{4/3}{s}-\frac{1/2}{s+1}-\frac{5/6}{s+3}.$$

$$x(t)=\tfrac{2}{3}-\tfrac{1}{2}e^{-t}+\tfrac{5}{6}e^{-3t},\qquad y(t)=\tfrac{4}{3}-\tfrac{1}{2}e^{-t}-\tfrac{5}{6}e^{-3t}.$$

Check. $x(0)=\frac{2}{3}-\frac{1}{2}+\frac{5}{6}=1$ ✓, $y(0)=\frac{4}{3}-\frac{1}{2}-\frac{5}{6}=0$ ✓. Steady state: setting derivatives to zero, $-2x+y=0$ and $x-2y+2=0$ give $x=\frac{2}{3}$, $y=\frac{4}{3}$ ✓.

Equations with polynomial coefficients

The transform is designed for constant coefficients, but the rule $\mathcal{L}\{t\,y\}=-dY/ds$ converts a coefficient t into a derivative in s. An ODE in t with linear coefficients becomes a *first-order* ODE in s, which is sometimes easier.

Worked Example 14.38 — Bessel's equation of order zero

Solve $t\ddot{y} + \dot{y} + ty = 0$ with $y(0) = 1$ (bounded solution).

Transform each term. $\mathcal{L}\{t\ddot{y}\} = -\frac{d}{ds}\left[s^2Y - s\,y(0) - \dot{y}(0)\right] = -2sY - s^2Y' + 1$; $\mathcal{L}\{\dot{y}\} = sY - 1$; $\mathcal{L}\{ty\} = -Y'$.

Sum. $-(s^2+1)Y' - sY = 0$, a separable first-order ODE in s:

$$\frac{Y'}{Y} = -\frac{s}{s^2+1} \quad\Longrightarrow\quad Y = \frac{C}{\sqrt{s^2+1}}.$$

Constant. IVT: $y(0^+) = \lim sY = C = 1$. So $Y = 1/\sqrt{s^2+1}$, i.e. $y(t) = J_0(t)$ (table entry 38), the Bessel function that describes, among other things, the current distribution in a round conductor at high frequency.

Common Pitfall — initial conditions at the wrong instant

If a problem states "the switch closes at $t = 0$ and $v_C(0^+) = \ldots$", the 0^- derivative rule cannot use that value directly. Either compute $v_C(0^-)$ from the circuit before switching (the usual case — capacitor voltages and inductor currents are continuous unless impulses act), or switch consistently to the 0^+ convention and exclude impulses at the origin from the input. Mixing the two produces answers that are wrong by an impulse or a step.

Engineering Note — why simulation tools think in s

SPICE's ".ac" analysis evaluates the zero-state transfer function on the $j\omega$-axis, and its ".tran" analysis integrates the ODE numerically. Laplace-domain hand analysis sits between the two: it gives the transient in closed form, exposes which poles cause ringing or slow settling, and provides the reference solution against which a numerical integrator is checked (Ch. 22).

14.9 Applications to circuits and control

The previous section transformed a differential equation that had already been written down. For circuits there is a shortcut: transform the **element laws** once and for all, draw an s-domain circuit, and apply Kirchhoff's laws and ordinary resistive-circuit techniques (series/parallel combination, voltage dividers, nodal and mesh analysis, Thévenin equivalents) directly to it. No differential equation is ever written. This section develops that method, and then the language it leads to: transfer functions, poles and zeros, stability and feedback. As elsewhere in this book, the circuits are there to motivate the mathematics; circuit theory proper belongs to its own course.

Kirchhoff's laws in the s-domain

KCL ($\sum i_k(t) = 0$ at a node) and KVL ($\sum v_k(t) = 0$ around a loop) are linear relations with constant coefficients, so by linearity of the transform they hold for the transforms too: $\sum I_k(s) = 0$ and $\sum V_k(s) = 0$. Only the element laws need translating.

s-domain element models

Resistor. $v = Ri$ transforms to $V(s) = R\,I(s)$.

Inductor. $v = L\,\dfrac{di}{dt}$ transforms, by the derivative rule, to

$$V(s) = sL\,I(s) - L\,i(0^-).$$

This is the law of an impedance sL **in series** with a voltage source $L\,i(0^-)$ (polarity opposing the stored current, as the minus sign shows). Solving instead for I:

$$I(s) = \frac{V(s)}{sL} + \frac{i(0^-)}{s},$$

an impedance sL **in parallel** with a current source $i(0^-)/s$ — the transform of a constant current $i(0^-)$.

Capacitor. $i = C\,\dfrac{dv}{dt}$ transforms to

$$I(s) = sC\,V(s) - C\,v(0^-) \qquad\Longleftrightarrow\qquad V(s) = \frac{I(s)}{sC} + \frac{v(0^-)}{s}.$$

The first form is an admittance sC in parallel with a current source $C\,v(0^-)$ (the transform of the impulse $Cv(0^-)\delta(t)$, i.e. the stored charge); the second is an impedance $1/(sC)$ in series with a voltage source $v(0^-)/s$ (a step of the stored voltage).

Impedance and admittance

For an element or two-terminal network initially at rest, the **impedance** is $Z(s) = V(s)/I(s)$ and the **admittance** $Y(s) = 1/Z(s)$:

$$Z_R = R, \qquad Z_L = sL, \qquad Z_C = \frac{1}{sC}.$$

Impedances combine like resistances: in series they add, in parallel their admittances add.

On the $j\omega$-axis these become the phasor impedances $j\omega L$ and $1/(j\omega C)$ of AC circuit analysis. The s-domain impedance is their generalisation to arbitrary (including transient and growing) signals, and it carries the initial energy through the extra sources.

Units check. sL has units $\mathrm{s}^{-1}\cdot\mathrm{H} = \Omega$ ✓; $1/(sC)$ has $1/(\mathrm{s}^{-1}\cdot\mathrm{F}) = \Omega$ ✓; $L\,i(0^-)$ has $\mathrm{H}\cdot\mathrm{A} = \mathrm{V}\cdot\mathrm{s}$, the units of a transformed voltage ✓.

Element	Time law	s-domain, series form	s-domain, parallel form
R	$v = Ri$	$V = RI$	$I = V/R$
L	$v = L\,di/dt$	$V = sLI - Li(0^-)$	$I = \dfrac{V}{sL} + \dfrac{i(0^-)}{s}$
C	$i = C\,dv/dt$	$V = \dfrac{I}{sC} + \dfrac{v(0^-)}{s}$	$I = sCV - Cv(0^-)$

Worked Example 14.39 — an inductor with stored current

A series circuit contains a source $v_s(t) = 20u(t)$ V, a resistor $R = 5\ \Omega$ and an inductor $L = 0.5$ H carrying $i(0^-) = 2$ A (from an earlier source configuration). Find $i(t)$.

s-domain circuit. Source $20/s$; impedances 5 and $0.5s$; the inductor's series source $L\,i(0^-) = 1$ V·s, aiding the existing current direction.

KVL. $\dfrac{20}{s} + 1 = (5 + 0.5s)\,I(s)$, so

$$I(s) = \frac{20/s + 1}{0.5(s + 10)} = \frac{40 + 2s}{s(s + 10)} = \frac{4}{s} - \frac{2}{s + 10}.$$

(Cover-up: at 0, $40/10 = 4$; at -10, $(40 - 20)/(-10) = -2$.)

$$i(t) = 4 - 2e^{-10t}\ \text{A}, \qquad t \geq 0.$$

Check. $i(0^+) = 2$ A — continuous, as an inductor current must be ✓. $i(\infty) = 20/5 = 4$ A, the DC value with the inductor a short circuit ✓. Time constant $L/R = 0.1$ s, matching the pole at $-10\ \text{s}^{-1}$ ✓.

Worked Example 14.40 — series RLC with a charged capacitor

A series RLC circuit has $R = 12\ \Omega$, $L = 10$ mH, $C = 100\ \mu$F. The capacitor holds $v_C(0^-) = 2$ V and $i(0^-) = 0$. At $t = 0$ a 10 V DC source is switched in. Find $i(t)$ and $v_C(t)$.

s-domain circuit. Source $10/s$; impedances R, sL, $1/(sC)$; the capacitor's series source $v_C(0^-)/s = 2/s$ opposes the drive (it is charged with the same polarity as the source would charge it).

KVL.

$$\frac{10}{s} - \frac{2}{s} = \left(R + sL + \frac{1}{sC}\right) I(s) \quad \Longrightarrow \quad I(s) = \frac{8/s}{R + sL + \frac{1}{sC}} = \frac{8}{L\left(s^2 + \frac{R}{L}s + \frac{1}{LC}\right)}.$$

With $R/L = 1200\ \text{s}^{-1}$, $1/(LC) = 10^6\ \text{s}^{-2}$, $1/L = 100\ \text{H}^{-1}$:

$$I(s) = \frac{800}{s^2 + 1200s + 10^6} = \frac{800}{(s + 600)^2 + 800^2}.$$

Inverse (entry 23):

$$i(t) = e^{-600t}\sin(800t)\ \text{A}.$$

Capacitor voltage. $V_C = \dfrac{I}{sC} + \dfrac{2}{s} = \dfrac{8 \times 10^6}{s(s^2 + 1200s + 10^6)} + \dfrac{2}{s}$. Partial fractions on the first term give $\dfrac{8}{s} - \dfrac{8(s + 600) + 6 \cdot 800}{(s + 600)^2 + 800^2}$, so

$$v_C(t) = 10 - e^{-600t}\left(8\cos 800t + 6\sin 800t\right)\ \text{V}.$$

Checks. $v_C(0^+) = 10 - 8 = 2$ V ✓ (continuous). $v_C(\infty) = 10$ V ✓ (FVT). $i(0^+) = 0$ ✓. Also $C\,dv_C/dt = 10^{-4} \cdot e^{-600t}\big[600(8\cos + 6\sin) - 800(-8\sin + 6\cos)\big] = 10^{-4}e^{-600t}(10000\sin 800t) = i$ ✓.

Transfer functions

Transfer function

For a linear time-invariant system **initially at rest**, with input $x(t)$ and output $y(t)$, the **transfer function** is

$$H(s) = \frac{Y(s)}{X(s)} = \mathcal{L}\{h(t)\},$$

where $h(t)$ is the impulse response. For a lumped system, $H(s)$ is a real rational function

$$H(s) = \frac{N(s)}{D(s)} = K\,\frac{(s - z_1)(s - z_2)\cdots(s - z_m)}{(s - p_1)(s - p_2)\cdots(s - p_n)}.$$

The roots z_i of N are the **zeros**, the roots p_k of D the **poles**, and K the gain constant. H is **proper** if $m \leq n$.

For the ODE $a_n y^{(n)} + \cdots + a_0 y = b_m x^{(m)} + \cdots + b_0 x$, transforming with zero initial conditions gives $H(s) = \dfrac{b_m s^m + \cdots + b_0}{a_n s^n + \cdots + a_0}$: the transfer function can be written down by inspection, and conversely. For a circuit it is found from the s-domain circuit with all initial-condition sources removed, using dividers and impedance combinations.

The **pole–zero plot** marks poles with × and zeros with ○ in the s-plane. Together with K, it determines H completely. Its interpretation:

- **Poles** are the natural frequencies. Each contributes a mode $e^{p_k t}$ (or $t^j e^{p_k t}$ if repeated) to the impulse response and to every transient.
- **Zeros** are frequencies at which the system blocks transmission: an input $e^{z_i t}$ produces no forced output. They do not create modes; they set the *amplitudes* (residues) of the modes and shape the frequency response.
- **Frequency response.** For a stable system, the steady-state response to $\cos \omega t$ is $|H(j\omega)| \cos(\omega t + \angle H(j\omega))$. Geometrically, $|H(j\omega)| = |K| \frac{\prod |j\omega - z_i|}{\prod |j\omega - p_k|}$: the product of distances from the point $j\omega$ to the zeros divided by that to the poles. A pole close to the $j\omega$-axis produces a sharp resonant peak; a zero on the axis produces a notch. Bode plots and the Fourier view of $H(j\omega)$ are developed in Ch. 16.

Worked Example 14.41 — zeros on the $j\omega$-axis: a notch filter

A notch filter for removing 50 Hz mains interference has

$$H(s) = \frac{s^2 + \omega_0^2}{s^2 + \frac{\omega_0}{Q} s + \omega_0^2}, \qquad \omega_0 = 2\pi \cdot 50 = 314.16 \text{ rad/s}, \quad Q = 5.$$

Find the poles and zeros, and $|H(j\omega)|$ at $\omega = 0$, ω_0, $1.1\omega_0$ and ∞.

Zeros. $s^2 + \omega_0^2 = 0 \Rightarrow s = \pm j\omega_0 = \pm j314.16$: on the $j\omega$-axis.

Poles. $s = -\frac{\omega_0}{2Q} \pm j\omega_0 \sqrt{1 - \frac{1}{4Q^2}} = -31.42 \pm j312.58$: in the LHP, so the filter is stable, and close to the zeros.

Frequency response. On the axis, $H(j\omega) = \dfrac{\omega_0^2 - \omega^2}{\omega_0^2 - \omega^2 + j\omega\omega_0/Q}$.

- $\omega = 0$: $|H| = 1$ (DC passes).
- $\omega = \omega_0$: numerator zero, $|H| = 0$ — the 50 Hz component is removed completely.
- $\omega = 1.1\omega_0$: $\omega_0^2 - \omega^2 = -0.21\omega_0^2$, $\omega\omega_0/Q = 0.22\omega_0^2$, so $|H| = \frac{0.21}{\sqrt{0.21^2 + 0.22^2}} = 0.690$.
- $\omega \to \infty$: $|H| \to 1$.

Geometric reading. At $s = j\omega$ near $j\omega_0$, the distance to the zero is tiny while the distance to the nearby pole is about $\omega_0/(2Q)$, so the ratio — and $|H|$ — dips sharply. Raising Q moves the poles closer to the axis and narrows the notch: the -3 dB width is $\omega_0/Q = 62.8$ rad/s, i.e. 10 Hz.

First- and second-order standard forms

Most systems are understood through their dominant first- and second-order factors.

First order. $H(s) = \dfrac{K}{1 + s\tau}$: one real pole at $-1/\tau$. The unit-step response is $K(1 - e^{-t/\tau})$: it reaches 63.2% of its final value at $t = \tau$ and 98.2% at 4τ. The RC low-pass filter has $\tau = RC$, $K = 1$.

Second order.

$$H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2},$$

with **natural frequency** $\omega_n > 0$ and **damping ratio** $\zeta \geq 0$. The poles are

$$p_{1,2} = -\zeta\omega_n \pm \omega_n \sqrt{\zeta^2 - 1}.$$

Damping	Poles	Step response
$\zeta > 1$ overdamped	two distinct real, negative	sum of two decaying exponentials, no overshoot
$\zeta = 1$ critically damped	double real at $-\omega_n$	$1 - (1 + \omega_n t)e^{-\omega_n t}$, fastest without overshoot
$0 < \zeta < 1$ underdamped	$-\zeta\omega_n \pm j\omega_d$, $\omega_d = \omega_n\sqrt{1 - \zeta^2}$	decaying oscillation with overshoot
$\zeta = 0$ undamped	$\pm j\omega_n$	$1 - \cos \omega_n t$, sustained oscillation

For $0 < \zeta < 1$ the poles lie on a circle of radius ω_n, at angle $\theta = \arccos \zeta$ from the negative real axis. Partial fractions on $H(s)/s$ give the underdamped step response

$$y(t) = 1 - e^{-\zeta\omega_n t}\left(\cos \omega_d t + \frac{\zeta}{\sqrt{1 - \zeta^2}} \sin \omega_d t\right) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1 - \zeta^2}} \sin(\omega_d t + \arccos \zeta)\,.$$

Setting $\dot{y} = 0$ (equivalently, the impulse response $h(t) = \frac{\omega_n}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin \omega_d t = 0$) locates the first peak at $t_p = \pi/\omega_d$, where

$$\text{overshoot} = y(t_p) - 1 = e^{-\pi\zeta/\sqrt{1-\zeta^2}}.$$

The envelope $e^{-\zeta\omega_n t}$ gives the 2% **settling time** $t_s \approx 4/(\zeta\omega_n)$: the real part of the poles sets the decay rate, the imaginary part the ringing frequency.

Worked Example 14.42 — the RLC low-pass filter

Take the series RLC circuit of Example 14.40 ($R = 12\ \Omega$, $L = 10$ mH, $C = 100\ \mu$F) with output across the capacitor. Find $H(s)$, its poles and zeros, ω_n, ζ, and the step-response overshoot and peak time.

Voltage divider (zero initial conditions):

$$H(s) = \frac{V_C}{V_s} = \frac{1/(sC)}{R + sL + 1/(sC)} = \frac{1}{LCs^2 + RCs + 1} = \frac{1/(LC)}{s^2 + \frac{R}{L}s + \frac{1}{LC}} = \frac{10^6}{s^2 + 1200s + 10^6}.$$

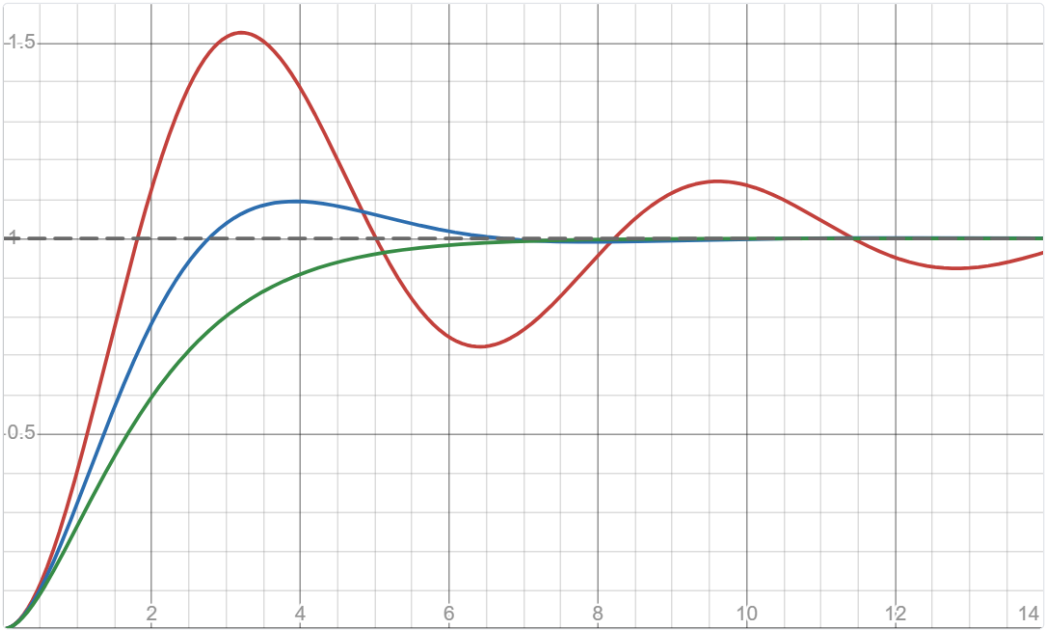
Poles and zeros. No finite zeros; poles $-600 \pm j800$ (as in Example 14.40 — the same circuit has the same natural frequencies whatever the output).

Parameters. $\omega_n = \sqrt{10^6} = 1000$ rad/s; $2\zeta\omega_n = 1200 \Rightarrow \zeta = 0.6$; $\omega_d = 800$ rad/s. In circuit terms $\zeta = \frac{R}{2}\sqrt{C/L} = 6\sqrt{0.01} = 0.6$ ✓.

Step response. $y(t) = 1 - e^{-600t}\left(\cos 800t + 0.75 \sin 800t\right)$.

- Peak time $t_p = \pi/800 = 3.93$ ms.
- Overshoot $e^{-\pi(0.6)/0.8} = e^{-2.356} = 0.0948$, i.e. 9.5%: the peak is 1.095.
- Settling time $t_s \approx 4/600 = 6.7$ ms.

Check. Evaluating $y(t)$ on a fine grid gives maximum 1.0948 at $t = 3.927$ ms ✓. DC gain $H(0) = 1$ ✓ (at DC the capacitor is open, so the full input appears across it).



Unit-step responses of the standard second-order system with $\omega_n = 1$ (time in units of $1/\omega_n$): $\zeta = 0.2$ (red, 53% overshoot), $\zeta = 0.6$ (blue, 9.5% overshoot — the RLC filter of Example 14.42 with time in ms), $\zeta = 1$ (green, critically damped).

Stability

BIBO stability

A system is **bounded-input, bounded-output (BIBO) stable** if every bounded input $|x(t)| \le M$ produces a bounded output. For an LTI system this holds if and only if the impulse response is absolutely integrable: $\int_0^\infty |h(t)|\,dt < \infty$.

Why. If $\int |h| = A < \infty$ then $|y(t)| \le \int_0^t |h(\tau)||x(t-\tau)|d\tau \le MA$. Conversely, if $\int |h| = \infty$, the bounded input $x(t-\tau) = \operatorname{sgn} h(\tau)$ makes $y(t) = \int_0^t |h|d\tau$ grow without bound.

Pole criterion for stability

A system with proper rational transfer function $H(s)$ (in lowest terms) is BIBO stable **if and only if every pole has negative real part** — all poles lie in the open left half-plane.

- Any pole with $\mathrm{Re}\{p\} > 0$, or a repeated pole on the $j\omega$-axis, gives an unbounded impulse response: **unstable**.
- Simple poles on the $j\omega$-axis (and none in the RHP) give a bounded but non-decaying impulse response: **marginally stable**, which is *not* BIBO stable (a bounded input at the pole frequency produces resonant growth, Example 14.35).

Proof sketch. By partial fractions, $h(t) = K_0\delta(t) + \sum_k \sum_i K_{k,i} \frac{t^{i-1}}{(i-1)!} e^{p_k t}$. Each term with $\mathrm{Re}\{p_k\} < 0$ is absolutely integrable (exponential decay beats any polynomial). A term with $\mathrm{Re}\{p_k\} \ge 0$ and $K \neq 0$ is not. Since the fraction is in lowest terms, no pole has zero residue at its highest power. □

Finding all roots of a high-order $D(s)$ is unnecessary to decide stability. The **Routh–Hurwitz criterion** answers it from the coefficients alone. Two consequences are easy to remember:

- **Necessary condition:** if all roots of $a_n s^n + \cdots + a_0$ (with $a_n > 0$) have negative real parts, then every coefficient is **positive**. A missing or negative coefficient means instability (or a $j\omega$-axis root).
- For **second order** positivity of all coefficients is also sufficient. For **third order**, $a_3 s^3 + a_2 s^2 + a_1 s + a_0$ with all $a_i > 0$ is stable if and only if $a_2 a_1 > a_3 a_0$.

Worked Example 14.43 — a stability range

For which gains K is the system with characteristic polynomial $D(s) = s^3 + 2s^2 + 3s + K$ stable? What happens at the boundary?

Positivity. Need $K > 0$.

Third-order condition. $a_2 a_1 > a_3 a_0$: $2 \cdot 3 > 1 \cdot K$, so $K < 6$. Stable for $0 < K < 6$.

Boundary. At $K = 6$: $D(s) = s^3 + 2s^2 + 3s + 6 = (s + 2)(s^2 + 3)$, poles -2 and $\pm j\sqrt{3} = \pm j1.732$. The system oscillates at $\sqrt{3}$ rad/s with constant amplitude — marginal stability. For $K > 6$ the complex pair crosses into the right half-plane. This is precisely how an oscillator is designed: raise the loop gain until a pole pair reaches the $j\omega$-axis.

Feedback and control

In a feedback loop with forward path $G(s)$ and feedback path $F(s)$, the error signal is $E = R - FY$ and $Y = GE$. Eliminating E:

$$T(s) = \frac{Y(s)}{R(s)} = \frac{G(s)}{1 + G(s)F(s)}.$$

The closed-loop poles are the roots of the **characteristic equation** $1 + G(s)F(s) = 0$ — they are not the poles of G. Feedback therefore *moves poles*, which is both its power (stabilising, speeding up, desensitising) and its danger (it can destabilise a stable plant, as in Example 14.43). For **unity feedback** ($F = 1$) the error is $E = R/(1 + G)$, and the final-value theorem gives the steady-state errors:

Reference	e_{ss}	Error constant
Step $1/s$	$\dfrac{1}{1 + K_p}$	$K_p = \lim_{s\to 0} G(s)$
Ramp $1/s^2$	$\dfrac{1}{K_v}$	$K_v = \lim_{s\to 0} sG(s)$
Parabola $1/s^3$	$\dfrac{1}{K_a}$	$K_a = \lim_{s\to 0} s^2 G(s)$

The number of pure integrators ($1/s$ factors) in G is the **system type**: a type-1 loop tracks a step with zero error, a type-2 loop tracks a ramp with zero error.

Worked Example 14.44 — a position servo

A motor-driven positioning system has open-loop transfer function $G(s) = \dfrac{K}{s(s + 4)}$ in a unity-feedback loop. Choose K for a damping ratio $\zeta = 0.5$; then find the overshoot, peak time, and steady-state errors for step and ramp references.

Closed loop.

$$T(s) = \frac{K/(s(s+4))}{1 + K/(s(s+4))} = \frac{K}{s^2 + 4s + K}.$$

Compare with the standard form: $\omega_n^2 = K$, $2\zeta\omega_n = 4$. With $\zeta = 0.5$: $\omega_n = 4$ rad/s, so $K = 16$.

Poles. $s^2 + 4s + 16 = 0 \Rightarrow s = -2 \pm j2\sqrt{3} = -2 \pm j3.464$.

Transient. Overshoot $e^{-\pi(0.5)/\sqrt{0.75}} = e^{-1.814} = 0.163$ (16.3%); peak time $t_p = \pi/\omega_d = \pi/3.464 = 0.907$ s; settling time $\approx 4/2 = 2$ s. Partial fractions give

$$y(t) = 1 - \frac{2}{\sqrt{3}} e^{-2t} \sin\!\left(2\sqrt{3}\,t + \frac{\pi}{3}\right).$$

A fine-grid evaluation gives a maximum of 1.1630 at $t = 0.907$ s ✓.

Steady-state errors. G has one integrator: type 1. Step: $K_p = \infty$, $e_{ss} = 0$. Ramp: $K_v = \lim sG = K/4 = 4\ \text{s}^{-1}$, so $e_{ss} = 1/4$ — the output lags a unit-slope ramp by 0.25 units.

Trade-off. Raising K reduces the ramp error ($e_{ss} = 4/K$) but lowers $\zeta = 2/\sqrt{K}$ and increases overshoot: the basic compromise of control design, visible entirely through pole locations.

Common Pitfall — pole–zero cancellation hides instability

Mathematically $\dfrac{s - 1}{(s - 1)(s + 2)} = \dfrac{1}{s + 2}$, which looks stable. Physically, if the factor $(s - 1)$ belongs to the plant and is "cancelled" by a controller zero, the unstable mode e^{t} is still present inside the system; any mismatch or initial condition excites it and an internal signal grows without bound. Stability must be judged on the internal (uncancelled) characteristic polynomial, never only on the simplified input–output transfer function.

Engineering Note — reading a system from its pole–zero plot

Experienced designers read pole–zero plots the way others read schematics. Poles far to the left are fast and usually negligible; the **dominant poles** nearest the $j\omega$-axis set the response time. The angle of a complex pair from the negative real axis fixes the damping ($\zeta = \cos\theta$, so poles at $\pm 45^\circ$ give $\zeta = 0.707$, the Butterworth value). A right-half-plane zero (common in boost converters and in amplifiers with feed-forward capacitance) produces an initial undershoot and limits the achievable bandwidth. Root-locus, Bode and Nyquist methods — the standard tools of control design — are graphical ways of tracking these pole positions as a gain varies; the Nyquist criterion itself is an application of the argument principle of Ch. 18.

Summary

Concept	Formula / statement
Unilateral transform	$X(s) = \int_{0^-}^{\infty} x(t)e^{-st}dt$, $s = \sigma + j\omega$, ROC $\mathrm{Re}\{s\} > \sigma_c$
Existence	piecewise continuous and $\lvert x\rvert \le Me^{\alpha t} \Rightarrow X$ exists for $\mathrm{Re}\{s\} > \alpha$, $X \to 0$
Uniqueness (Lerch)	equal transforms $\Rightarrow$ equal functions (at points of continuity)
Basic pairs	$\delta \leftrightarrow 1$, $u \leftrightarrow \frac{1}{s}$, $t^n \leftrightarrow \frac{n!}{s^{n+1}}$, $e^{-at} \leftrightarrow \frac{1}{s+a}$
Sinusoids	$\sin\omega t \leftrightarrow \frac{\omega}{s^2+\omega^2}$, $\cos\omega t \leftrightarrow \frac{s}{s^2+\omega^2}$
Damped sinusoids	$e^{-\alpha t}\sin\omega t \leftrightarrow \frac{\omega}{(s+\alpha)^2+\omega^2}$, $e^{-\alpha t}\cos\omega t \leftrightarrow \frac{s+\alpha}{(s+\alpha)^2+\omega^2}$
s-shift	$e^{at}x(t) \leftrightarrow X(s-a)$
Delay	$x(t-T)u(t-T) \leftrightarrow e^{-sT}X(s)$; $g(t)u(t-T) \leftrightarrow e^{-sT}\mathcal{L}\{g(t+T)\}$
Scaling	$x(at) \leftrightarrow \frac{1}{a}X(s/a)$
Derivatives	$\dot{x} \leftrightarrow sX - x(0^-)$; $\ddot{x} \leftrightarrow s^2X - sx(0^-) - \dot{x}(0^-)$
Integral	$\int_{0^-}^{t} x\,d\tau \leftrightarrow X/s$
t-multiplication / division	$tx \leftrightarrow -X'(s)$; $x/t \leftrightarrow \int_s^{\infty} X\,dp$
Periodic	$X = \frac{1}{1-e^{-sT}}\int_0^T xe^{-st}dt$
Bromwich inversion	$x(t) = \frac{1}{2\pi j}\int_{\gamma-j\infty}^{\gamma+j\infty} X(s)e^{st}ds = \sum \mathrm{Res}[Xe^{st}]$ (Ch. 18)
Simple-pole coefficient	$K_k = [(s-p_k)X]_{s=p_k} = N(p_k)/D'(p_k)$
Repeated pole	$K_{k,r-j} = \frac{1}{j!}\frac{d^j}{ds^j}[(s-p_k)^r X]_{p_k}$; $\frac{1}{(s-p)^i} \leftrightarrow \frac{t^{i-1}}{(i-1)!}e^{pt}$
Complex pair $-\alpha \pm j\beta$	$2\lvert K\rvert e^{-\alpha t}\cos(\beta t + \angle K)$
Convolution	$(f*g)(t) = \int_0^t f(\tau)g(t-\tau)d\tau \leftrightarrow F(s)G(s)$
Initial value	$x(0^+) = \lim_{s\to\infty} sX(s)$ (X strictly proper)
Final value	$x(\infty) = \lim_{s\to 0} sX(s)$ (all poles of sX in open LHP)
ODE solution	$Y = \frac{C(s)}{P(s)} + \frac{X(s)}{P(s)}$: zero-input + zero-state
State equations	$\mathbf{X} = (s\mathbf{I}-\mathbf{A})^{-1}[\mathbf{x}(0^-) + \mathbf{b}V]$, $e^{\mathbf{A}t} = \mathcal{L}^{-1}\{(s\mathbf{I}-\mathbf{A})^{-1}\}$
Element models	$Z_R = R$; $V_L = sLI - Li(0^-)$; $V_C = \frac{I}{sC} + \frac{v(0^-)}{s}$
Transfer function	$H(s) = Y/X$ (zero state) $= \mathcal{L}\{h\}$; $y = h*x$
Second-order form	$\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$; poles $-\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$
Step overshoot, peak, settling	$e^{-\pi\zeta/\sqrt{1-\zeta^2}}$; $t_p = \pi/\omega_d$; $t_s \approx 4/(\zeta\omega_n)$
BIBO stability	all poles in open LHP $\Leftrightarrow \int_0^{\infty}\lvert h\rvert dt < \infty$
Cubic Routh condition	$a_3s^3 + a_2s^2 + a_1s + a_0$, all $a_i > 0$, stable iff $a_2a_1 > a_3a_0$
Feedback	$T = \frac{G}{1+GF}$; unity-feedback step error $\frac{1}{1+G(0)}$, ramp error $\frac{1}{\lim sG}$

Further reading

- E. Kreyszig, *Advanced Engineering Mathematics*, 10th ed., Wiley — Ch. 6: a clear mathematical treatment of the transform, its properties and ODE applications, with extensive tables.
- A. V. Oppenheim, A. S. Willsky & S. H. Nawab, *Signals and Systems*, 2nd ed., Prentice Hall — Ch. 9: the bilateral transform, regions of convergence, and the system-function viewpoint.
- J. W. Nilsson & S. A. Riedel, *Electric Circuits*, Pearson — the chapters on the Laplace transform in circuit analysis: s-domain models, transfer functions and pole–zero interpretation.
- K. Ogata, *Modern Control Engineering*, 5th ed., Pearson — transient-response specifications, steady-state errors and the Routh–Hurwitz criterion.
- R. V. Churchill, *Operational Mathematics*, 3rd ed., McGraw-Hill — the classical rigorous source for existence, uniqueness and the inversion integral.

Chapter 15 — Fourier Series and Fourier Analysis

> **Prerequisites:** Ch. 2 (complex numbers), Ch. 3 (trigonometry and periodic functions), Ch. 5 (integral calculus), Ch. 9 (vector spaces and inner products), Ch. 12 (linear differential equations) **Learning outcomes:** After this chapter you can
>
> - compute trigonometric and complex (exponential) Fourier series of periodic waveforms of any period, and convert between the coefficient forms (a_n, b_n), (A_n, ϕ_n) and c_n;
> - use even, odd, half-wave and quarter-wave symmetry to predict which harmonics are present before integrating, and build half-range expansions;
> - state the Dirichlet conditions and the pointwise, uniform and mean-square convergence theorems, and explain and quantify the Gibbs phenomenon;
> - obtain coefficients quickly with the shift, differentiation, integration and jump methods, and predict how fast they decay from the smoothness of the waveform;
> - apply Parseval's theorem to compute RMS values, harmonic powers, power factor and total harmonic distortion, and to sum numerical series;
> - find the steady-state response of a linear time-invariant system to any periodic input, harmonic by harmonic.

A periodic waveform that is not a sinusoid — a clock square wave, the triangular ripple current of a buck converter, the humps of a rectified sine, the stepped output of an inverter — is still, in a precise sense, *made of* sinusoids. Chapter 3 introduced this idea informally: it established the orthogonality of harmonics (§3.7), the power of a finite sum of harmonics, the notion of total harmonic distortion, and the shapes and symmetries of the standard waveforms (§3.8). This chapter turns those observations into a complete theory. We ask, and answer, four questions:

1. **Representation.** Given a periodic function, which sinusoids does it contain, and with what amplitudes and phases? (The Fourier coefficients, §15.2–15.3, §15.7.)
2. **Structure.** How do properties of the waveform — symmetry, smoothness, discontinuities — show up in its coefficients? (§15.4, §15.5, §15.7.)
3. **Validity.** In what sense does the infinite sum of sinusoids actually equal the function, and what happens at jumps? (§15.6.)
4. **Use.** How do power, RMS values and distortion follow from the coefficients, and how does a linear system respond to a periodic input? (§15.8, §15.9.)

The next chapter lets the period tend to infinity and turns the discrete set of harmonics into the continuous Fourier transform. Everything here — orthogonality, Parseval, convolution, the trade-off between smoothness and spectral spread — carries over almost word for word.

Notation. A periodic function $x(t)$ has fundamental period T_0, fundamental frequency $f_0 = 1/T_0$ and fundamental angular frequency $\omega_0 = 2\pi/T_0$ (Ch. 3, §3.4). The harmonic index is n, so the nth harmonic has angular frequency $n\omega_0$. (Chapter 16 uses k for the same index when the discrete-time index n is also in play; the formulas are identical.) Many examples use the normalised period $T_0 = 2\pi$, $\omega_0 = 1$, in which case the time variable is written t and harmonics are $\cos nt$, $\sin nt$. The notation $\int_{T_0}$ means an integral over *any* interval of length T_0.

15.1 Periodic function decomposition

Motivation: why sinusoids, and why decompose at all?

There are infinitely many ways to write a function as a sum of simpler functions — powers of t (Taylor series, Ch. 4), polynomials, piecewise constants, wavelets. Engineers single out sums of sinusoids for one overriding reason:

> **A complex exponential $e^{j\omega t}$ is an eigenfunction of every linear time-invariant (LTI) system.**

If an LTI system with impulse response $h(t)$ is driven by $x(t) = e^{j\omega t}$, its output is

$$y(t) = \int_{-\infty}^{\infty} h(\tau)\, e^{j\omega(t-\tau)}\, d\tau = \Big[\int_{-\infty}^{\infty} h(\tau) e^{-j\omega\tau}\, d\tau\Big] e^{j\omega t} = H(j\omega)\, e^{j\omega t},$$

provided the integral converges. The input passes through *unchanged in shape*, multiplied by a single complex number $H(j\omega)$ — the frequency response, which for a circuit is the phasor transfer function of Ch. 2 and the Laplace transfer function $H(s)$ of Ch. 14 evaluated at $s = j\omega$. No other family of signals has this property for *all* LTI systems at once.

Now suppose a periodic input can be written as a sum of exponentials at the harmonic frequencies,

$$x(t) = \sum_{n=-\infty}^{\infty} c_n\, e^{jn\omega_0 t}.$$

By linearity (and, for an infinite sum, a convergence argument we postpone to §15.9) the steady-state output is

$$y(t) = \sum_{n=-\infty}^{\infty} c_n\, H(jn\omega_0)\, e^{jn\omega_0 t}.$$

A problem that in the time domain requires solving a differential equation with a discontinuous forcing function becomes, in the harmonic domain, a list of multiplications. That is the whole engineering case for Fourier series: **decompose, multiply harmonic by harmonic, recombine**. Section 15.9 carries this programme out in detail; the intervening sections build the mathematics that makes it trustworthy.

A second, independent motivation is *measurement*. Power-quality standards, audio distortion specifications, EMC emission limits and spectrum-analyser displays are all stated harmonic by harmonic. To read or write such a specification you must be able to move fluently between a waveform and its list of harmonics.

Periodic functions and their building blocks

Periodic function

A function $x:\mathbb{R}\to\mathbb{C}$ is **periodic with period** $T>0$ if $x(t+T)=x(t)$ for all t. The smallest such T, if it exists, is the **fundamental period** T_0. A function defined only on an interval of length T_0 is extended to all of $\mathbb{R}$ by **periodic extension**, $x(t+kT_0)=x(t)$, $k\in\mathbb{Z}$.

Every function in the set

$$\mathcal{B}=\{\,1,\ \cos\omega_0 t,\ \sin\omega_0 t,\ \cos 2\omega_0 t,\ \sin 2\omega_0 t,\ \dots\,\}$$

has period T_0 (the nth harmonic has period T_0/n, hence also period T_0). So does every finite linear combination of them, a **trigonometric polynomial of degree** N:

$$S_N(t)=\frac{a_0}{2}+\sum_{n=1}^{N}\bigl(a_n\cos n\omega_0 t+b_n\sin n\omega_0 t\bigr).$$

The decomposition problem is: *given a T_0-periodic $x(t)$, choose the coefficients so that S_N is as close as possible to x, and decide whether $S_N\to x$ as $N\to\infty$.*

The inner-product viewpoint

Chapter 9 treated vectors abstractly as elements of a space with an inner product. Functions of period T_0 form such a space. For real or complex functions that are square-integrable over a period, define

$$\langle x,y\rangle=\frac{1}{T_0}\int_{T_0}x(t)\,y^{*}(t)\,dt,\qquad \|x\|=\sqrt{\langle x,x\rangle}=\sqrt{\frac{1}{T_0}\int_{T_0}|x(t)|^2dt}.$$

The normalisation $1/T_0$ is the engineer's choice: it makes $\|x\|$ the **RMS value** of x and $\|x\|^2$ its **mean-square value**, the average power it would dissipate in a $1\ \Omega$ resistor. (Mathematics texts often omit $1/T_0$, which changes constants but nothing else.) The space of all such functions is denoted $L^2(T_0)$.

Orthogonality of the harmonic basis

With $\omega_0=2\pi/T_0$ and integers m,n:

$$\langle e^{jm\omega_0 t},e^{jn\omega_0 t}\rangle=\frac{1}{T_0}\int_{T_0}e^{j(m-n)\omega_0 t}\,dt=\begin{cases}1, & m=n,\\ 0, & m\neq n.\end{cases}$$

Equivalently, for the real functions with $m,n\geq 0$:

$$\frac{2}{T_0}\int_{T_0}\cos m\omega_0 t\,\cos n\omega_0 t\,dt=\delta_{mn}\quad(m,n\geq 1),\qquad \frac{2}{T_0}\int_{T_0}\sin m\omega_0 t\,\sin n\omega_0 t\,dt=\delta_{mn},$$
$$\int_{T_0}\sin m\omega_0 t\,\cos n\omega_0 t\,dt=0,\qquad \frac{1}{T_0}\int_{T_0}1\cdot 1\,dt=1,$$

where δ_{mn} is the Kronecker delta.

Proof (complex form). For $m=n$ the integrand is 1. For $m\neq n$, let $k=m-n\neq 0$; then $\int_{t_1}^{t_1+T_0}e^{jk\omega_0 t}dt=\bigl[e^{jk\omega_0 t}/(jk\omega_0)\bigr]_{t_1}^{t_1+T_0}=e^{jk\omega_0 t_1}(e^{j2\pi k}-1)/(jk\omega_0)=0$. The real relations follow by writing cos and sin as combinations of $e^{\pm j\theta}$, or directly by product-to-sum as in Ch. 3, §3.7. $\square$

So the set $\{e^{jn\omega_0 t}\}_{n\in\mathbb{Z}}$ is **orthonormal**, and the real set $\{1,\sqrt{2}\cos n\omega_0 t,\sqrt{2}\sin n\omega_0 t\}$ is orthonormal too. Each harmonic is a unit vector at "right angles" to every other.

Coefficients as projections

In $\mathbb{R}^3$ the component of a vector $\mathbf{v}$ along a unit vector $\hat{\mathbf{e}}$ is the projection $\langle\mathbf{v},\hat{\mathbf{e}}\rangle$, and the closest point to $\mathbf{v}$ in a plane spanned by orthonormal $\hat{\mathbf{e}}_1,\hat{\mathbf{e}}_2$ is $\langle\mathbf{v},\hat{\mathbf{e}}_1\rangle\hat{\mathbf{e}}_1+\langle\mathbf{v},\hat{\mathbf{e}}_2\rangle\hat{\mathbf{e}}_2$. Exactly the same holds in the function space.

Best approximation (least squares)

Let $x\in L^2(T_0)$ and let $\{\varphi_n\}_{n=-N}^{N}$ with $\varphi_n(t)=e^{jn\omega_0 t}$. Among all trigonometric polynomials $p(t)=\sum_{n=-N}^{N}d_n\varphi_n(t)$, the mean-square error $\|x-p\|^2$ is minimised uniquely by

$$d_n=c_n=\langle x,\varphi_n\rangle=\frac{1}{T_0}\int_{T_0}x(t)\,e^{-jn\omega_0 t}\,dt,$$

and the minimum error is

$$\|x-S_N\|^2=\|x\|^2-\sum_{n=-N}^{N}|c_n|^2.$$

Proof. Expand, using orthonormality:

$$\begin{aligned}\|x-p\|^2&=\|x\|^2-\sum_{n}\bigl(d_n^{*}\langle x,\varphi_n\rangle+d_n\langle x,\varphi_n\rangle^{*}\bigr)+\sum_{n}|d_n|^2\\&=\|x\|^2-\sum_{n}|c_n|^2+\sum_{n}|d_n-c_n|^2.\end{aligned}$$

(The second line follows by completing the square: $|d_n - c_n|^2 = |d_n|^2 - d_n^* c_n - d_n c_n^* + |c_n|^2$.) The last sum is non-negative and vanishes only when every $d_n = c_n$. $\square$

Three consequences are worth stating separately, because they recur throughout the chapter.

1. **Coefficients do not change when N grows.** Because the basis is orthogonal, the best coefficient for $\cos 3\omega_0 t$ is the same whether we approximate with 3 harmonics or 300. Contrast this with fitting a polynomial by least squares in the monomials $1, t, t^2, \ldots$, where adding a term changes all the earlier coefficients. This *finality of coefficients* is why a spectrum analyser can report "the 3rd harmonic" without reference to how many others it measures.
2. **Bessel's inequality.** Since the error is non-negative, $\sum_{n=-N}^{N} |c_n|^2 \le \|x\|^2$ for every N, hence $\sum_{n=-\infty}^{\infty} |c_n|^2 \le \|x\|^2 < \infty$. In particular $c_n \to 0$ as $|n| \to \infty$.
3. **Completeness and Parseval.** If the error tends to zero as $N \to \infty$, Bessel's inequality becomes an equality — Parseval's theorem (§15.8). That it does tend to zero for *every* square-integrable x is a deep theorem (the completeness of the trigonometric system) whose proof sketch appears in §15.6.

Fourier series

The **Fourier series** of a T_0-periodic function x is the formal series whose coefficients are the projections of x on the harmonics:

$$x(t) \sim \sum_{n=-\infty}^{\infty} c_n \, e^{jn\omega_0 t} = \frac{a_0}{2} + \sum_{n=1}^{\infty} \big(a_n \cos n\omega_0 t + b_n \sin n\omega_0 t\big).$$

The symbol $\sim$ means "has the Fourier series"; replacing it by $=$ requires the convergence theorems of §15.6.

Worked Example 15.1 — Least-squares approximation of a sawtooth

Let $x(t) = t$ on $(-\pi, \pi)$, extended with period 2π (a sawtooth). Find the best mean-square approximation by (a) a single term $b_1 \sin t$ and (b) $\sum_{n=1}^{3} b_n \sin nt$, and the residual mean-square error in each case. (Cosine terms and the constant are zero because x is odd; see §15.4.)

Solution. Mean-square value: $\|x\|^2 = \frac{1}{2\pi}\int_{-\pi}^{\pi} t^2 dt = \frac{\pi^2}{3} = 3.2899$.

The projection on $\sin nt$ in real form is $b_n = \frac{2}{2\pi}\int_{-\pi}^{\pi} t \sin nt \, dt$. Integrating by parts,

$$\int_{-\pi}^{\pi} t \sin nt \, dt = \Big[-\frac{t\cos nt}{n}\Big]_{-\pi}^{\pi} + \frac{1}{n}\int_{-\pi}^{\pi} \cos nt \, dt = -\frac{2\pi\cos n\pi}{n} = \frac{2\pi(-1)^{n+1}}{n},$$

so $b_n = 2(-1)^{n+1}/n$: $b_1 = 2$, $b_2 = -1$, $b_3 = 2/3$.

Each term $b_n \sin nt$ has mean-square value $b_n^2/2$, and by orthogonality the error is $\|x\|^2$ minus the sum of these.

(a) Error $= 3.2899 - \frac{4}{2} = 1.2899$: the single sine captures $2/3.2899 = 60.8\%$ of the mean-square value.

(b) Error $= 3.2899 - 2 - 0.5 - 0.2222 = 0.5676$: three terms capture 82.7%.

Check. Numerically integrating $(t - \sum_{n\le 3} b_n \sin nt)^2$ over a period and dividing by 2π gives 0.56765. ✓ Note that $b_1 = 2$ is the best coefficient in both (a) and (b) — finality of coefficients. The slow capture (still 17% error with three terms) reflects the jump of the sawtooth at $t = \pm\pi$; §15.7 explains why jumps force $|b_n| \propto 1/n$.

Worked Example 15.2 — Why a non-orthogonal basis is awkward

Approximate $x(t) = |t|$ on $(-\pi, \pi)$ in mean square by (a) a constant d_0, and (b) $d_0 + d_1 \cos t$. Compare with what happens if we instead used the non-orthogonal pair $\{1, \cos^2(t/2)\}$.

Solution. (a) The best constant is the projection on 1: $d_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} |t| dt = \frac{\pi}{2} = 1.5708$, the mean value.

(b) Because 1 and $\cos t$ are orthogonal, d_0 is unchanged and $d_1 = \frac{2}{2\pi}\int_{-\pi}^{\pi} |t| \cos t \, dt = \frac{2}{\pi}\int_0^{\pi} t\cos t \, dt = \frac{2}{\pi}[t\sin t + \cos t]_0^{\pi} = \frac{2}{\pi}(-2) = -\frac{4}{\pi} = -1.2732$.

Now note $\cos^2(t/2) = \frac{1}{2} + \frac{1}{2}\cos t$ spans the same two-dimensional space as $\{1, \cos t\}$, so the best approximation is the same function $1.5708 - 1.2732\cos t = e_0 + e_1 \cos^2(t/2)$ with $e_1 = -2.5465$ and $e_0 = 1.5708 + 1.2732 = 2.8440$. The best constant *by itself* was 1.5708, but in the combination it must change to 2.8440: with a non-orthogonal basis, adding a function alters the coefficients already found, and each coefficient is no longer a single integral but the solution of a linear system (the normal equations of Ch. 8). Orthogonality makes each coefficient an independent, final, one-line integral.

Engineering note — a spectrum analyser computes projections

A swept or FFT spectrum analyser estimates $|c_n|$ by correlating the input with $e^{-jn\omega_0 t}$ over a finite window: exactly the projection integral above. A lock-in amplifier does the same for one chosen harmonic, multiplying by a reference $\cos n\omega_0 t$ and $\sin n\omega_0 t$ and low-pass filtering (averaging) the products. The orthogonality relations are why the other harmonics average out of the measurement.

Pitfall — the decomposition needs exact periodicity

Orthogonality holds only over an *exact* period, or a whole number of periods. Integrating over a window that is not a whole number of periods (a common situation when analysing measured data) leaks energy from each harmonic into its neighbours. The Fourier series is a statement about an exactly periodic idealisation; finite-record effects belong to the Fourier transform and the DFT (Ch. 16, Ch. 24).

15.2 Trigonometric Fourier series

Motivation

The trigonometric form uses only real functions — cosines and sines — and so is the natural form when the waveform is real, when we want to see its symmetry directly, and when we want amplitudes and phases that can be read on an oscilloscope or written into a phasor calculation. It is also the form that arises from separation of variables in the heat, wave and Laplace equations (Ch. 13), where sines and cosines appear as eigenfunctions of d^2/dx^2 with boundary conditions.

The Euler (Euler–Fourier) formulas

Suppose a T_0-periodic function can be written as

$$x(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos n\omega_0 t + b_n \sin n\omega_0 t \right),$$

and that the series may be integrated term by term after multiplication by a bounded function (true, for example, when it converges uniformly; §15.6 shows the formulas below are correct in far greater generality). Multiply both sides by $\cos m\omega_0 t$, $m \geq 1$, and integrate over a period. By orthogonality every term vanishes except $a_m \int_{T_0} \cos^2 m\omega_0 t \, dt = a_m T_0/2$. Similarly with $\sin m\omega_0 t$. Integrating without a multiplier kills every harmonic and leaves $\frac{a_0}{2} T_0$.

Euler formulas for the trigonometric Fourier coefficients

For a T_0-periodic function integrable over a period, with $\omega_0 = 2\pi/T_0$,

$$a_n = \frac{2}{T_0} \int_{T_0} x(t) \cos n\omega_0 t \, dt \quad (n \geq 0), \qquad b_n = \frac{2}{T_0} \int_{T_0} x(t) \sin n\omega_0 t \, dt \quad (n \geq 1).$$

The constant term is $\dfrac{a_0}{2} = \dfrac{1}{T_0} \displaystyle\int_{T_0} x(t)\, dt$, the **mean (DC) value** of x.

The convention of writing the constant as $a_0/2$ is purely so that the formula for a_n also gives a_0; some books write a_0 for the mean value instead and adjust the formula. Always check which convention a table uses.

For the normalised period $T_0 = 2\pi$ ($\omega_0 = 1$) the formulas read

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x(t) \cos nt \, dt, \qquad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x(t) \sin nt \, dt.$$

For a function given on $(-L, L)$, so $T_0 = 2L$ and $\omega_0 = \pi/L$ — the usual form for spatial problems in Ch. 13:

$$x(s) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi s}{L} + b_n \sin \frac{n\pi s}{L} \right), \qquad a_n = \frac{1}{L} \int_{-L}^{L} x(s) \cos \frac{n\pi s}{L} \, ds, \quad b_n = \frac{1}{L} \int_{-L}^{L} x(s) \sin \frac{n\pi s}{L} \, ds.$$

Choice of interval. The integrand $x(t)\cos n\omega_0 t$ is itself T_0-periodic, and the integral of a periodic function over one period does not depend on where the period starts:

$$\int_{\alpha}^{\alpha + T_0} g(t)\, dt = \int_{0}^{T_0} g(t)\, dt \qquad \text{for every } \alpha,$$

because the piece cut from one end is exactly the piece added at the other. Choose the interval that makes the integration easiest — usually the one in which x has the fewest pieces or a symmetric form.

Dimensions. a_n and b_n carry the units of x (the factor $2/T_0$ cancels the seconds from dt), and $n\omega_0 t$ is dimensionless. A coefficient in volt-seconds signals a forgotten $1/T_0$.

Amplitude–phase (harmonic) form

Grouping each cosine–sine pair into one sinusoid (Ch. 3, §3.2):

$$a_n \cos n\omega_0 t + b_n \sin n\omega_0 t = A_n \cos(n\omega_0 t + \phi_n), \qquad A_n = \sqrt{a_n^2 + b_n^2}, \quad \phi_n = \operatorname{atan2}(-b_n,\, a_n).$$

So

$$x(t) \sim A_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t + \phi_n), \qquad A_0 = \frac{a_0}{2}.$$

The pair (A_n, ϕ_n) is the **amplitude** and **phase** of the nth harmonic; plotted against $n\omega_0$ they form the one-sided amplitude and phase spectra (§15.5). Note the sign in ϕ_n: with the cosine reference, a pure $+\sin$ term has phase $-90°$. A sine-reference form $A_n \sin(n\omega_0 t + \psi_n)$ with $\psi_n = \phi_n + 90°$ is common in power-systems literature; state the reference whenever phases are quoted.

A first full example: the square wave

Worked Example 15.3 — Square wave

Find the Fourier series of the square wave $x(t) = A$ for $0 < t < T_0/2$, $x(t) = -A$ for $T_0/2 < t < T_0$, period T_0.

Solution. Mean value: the two halves cancel, so $a_0 = 0$.

Cosine coefficients:

$$a_n = \frac{2}{T_0}\Big[\int_0^{T_0/2} A\cos n\omega_0 t\,dt - \int_{T_0/2}^{T_0} A\cos n\omega_0 t\,dt\Big] = \frac{2A}{T_0 n\omega_0}\Big[\sin n\omega_0 t\Big|_0^{T_0/2} - \sin n\omega_0 t\Big|_{T_0/2}^{T_0}\Big] = 0,$$

since $\sin(n\pi) = \sin(2n\pi) = 0$.

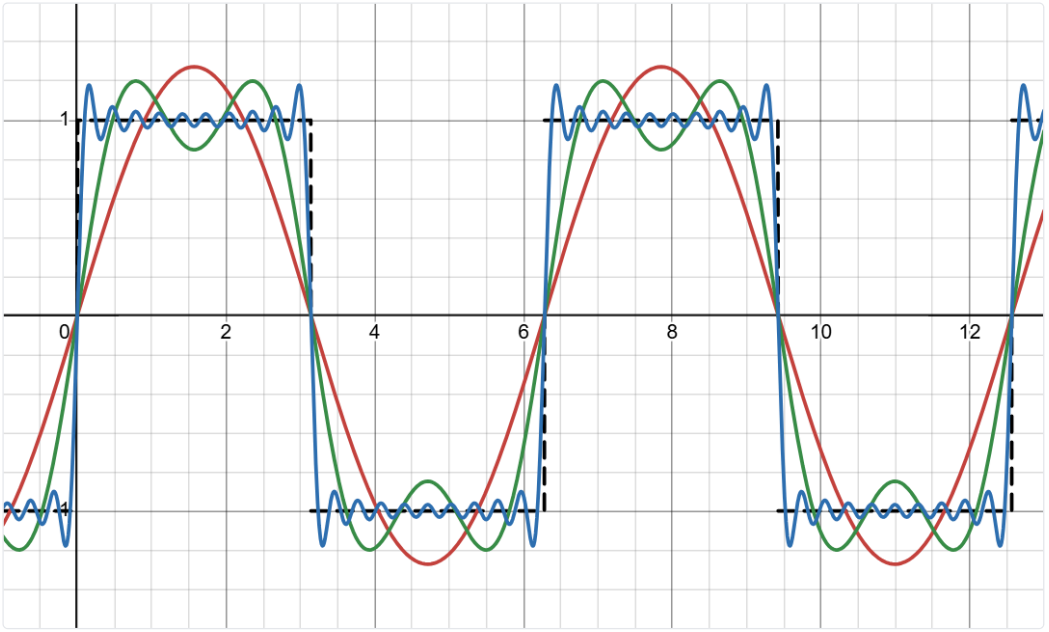
Sine coefficients, with $n\omega_0 T_0/2 = n\pi$ and $T_0\omega_0 = 2\pi$:

$$b_n = \frac{2A}{T_0 n\omega_0}\Big[-\cos n\omega_0 t\Big|_0^{T_0/2} + \cos n\omega_0 t\Big|_{T_0/2}^{T_0}\Big] = \frac{A}{n\pi}\big[(1-\cos n\pi) + (1-\cos n\pi)\big] = \frac{2A}{n\pi}\big(1-(-1)^n\big).$$

So $b_n = 4A/(n\pi)$ for odd n and 0 for even n:

$$x(t) = \frac{4A}{\pi}\Big(\sin\omega_0 t + \frac{1}{3}\sin 3\omega_0 t + \frac{1}{5}\sin 5\omega_0 t + \cdots\Big) = \frac{4A}{\pi}\sum_{k=1}^{\infty}\frac{\sin(2k-1)\omega_0 t}{2k-1}.$$

Checks. (i) Only odd harmonics — predicted by half-wave symmetry (§15.4). (ii) Only sines — predicted by odd symmetry. (iii) At $t = T_0/4$: $\frac{4A}{\pi}(1 - \frac{1}{3} + \frac{1}{5} - \cdots) = \frac{4A}{\pi}\cdot\frac{\pi}{4} = A$ by the Leibniz series. ✓ (iv) The fundamental amplitude $4A/\pi = 1.273A$ *exceeds* the square wave's peak — a reminder that harmonics can add destructively.



Partial sums of the square-wave series with $\omega_0 = 1$, $A = 1$: fundamental only (red), harmonics 1 and 3 (green), harmonics up to 19 (blue). The sum approaches the square wave (dashed) everywhere except near the jumps, where an overshoot persists (§15.6).

More standard waveforms

Worked Example 15.4 — Ramp on a non-symmetric interval

A sawtooth generator produces $x(t) = t$ volts for $0 < t < 2$ s, repeating with period $T_0 = 2$ s. Find its trigonometric Fourier series.

Solution. $\omega_0 = 2\pi/2 = \pi$ rad/s. The interval $(0, 2)$ is the natural one here.

Mean value: $\frac{a_0}{2} = \frac{1}{2}\int_0^2 t\,dt = 1$ V.

For $n \geq 1$ integrate by parts, using $\sin 2n\pi = 0$, $\cos 2n\pi = 1$:

$$a_n = \frac{2}{2}\int_0^2 t\cos n\pi t\,dt = \Big[\frac{t\sin n\pi t}{n\pi}\Big]_0^2 - \frac{1}{n\pi}\int_0^2 \sin n\pi t\,dt = 0 + \frac{1}{n^2\pi^2}\Big[\cos n\pi t\Big]_0^2 = 0,$$

$$b_n = \int_0^2 t\sin n\pi t\,dt = \Big[-\frac{t\cos n\pi t}{n\pi}\Big]_0^2 + \frac{1}{n\pi}\int_0^2 \cos n\pi t\,dt = -\frac{2}{n\pi}.$$

Hence

$$x(t) = 1 - \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{\sin n\pi t}{n}\ \text{V}.$$

Interpretation. After removing the mean, the periodic function $x(t) - 1$ is odd: $x(-t) - 1 = -(x(t) - 1)$ (check: for $0 < t < 2$, $x(-t) = x(2 - t) = 2 - t$, so $x(-t) - 1 = 1 - t$). That is why only sines survive. Every harmonic, odd and even, is present, with amplitude $2/(n\pi)$: 0.637 V, 0.318 V, 0.212 V, … — falling only as $1/n$ because of the jump at $t = 0$.

Check. Numerical quadrature gives $a_1 = 0.0000$, $b_1 = -0.6366$, $b_2 = -0.3183$, $b_3 = -0.2122$. ✓

Worked Example 15.5 — Half-wave rectified sine

A half-wave rectifier fed by $V_m \sin\omega_0 t$ produces $x(t) = V_m \sin\omega_0 t$ for $0 \leq \omega_0 t \leq \pi$ and $x(t) = 0$ for $\pi < \omega_0 t < 2\pi$. Find the Fourier series.

Solution. Work in $\theta = \omega_0 t$, period 2π, and set $V_m = 1$ (multiply by V_m at the end).

Mean: $\frac{a_0}{2} = \frac{1}{2\pi}\int_0^{\pi}\sin\theta\,d\theta = \frac{1}{\pi}$.

Cosine coefficients. Use $\sin\theta\cos n\theta = \frac{1}{2}[\sin(1+n)\theta + \sin(1-n)\theta]$. For $n \neq 1$,

$$a_n = \frac{1}{\pi}\int_0^{\pi}\sin\theta\cos n\theta\,d\theta = \frac{1}{2\pi}\Big[\frac{1-\cos(1+n)\pi}{1+n} + \frac{1-\cos(1-n)\pi}{1-n}\Big].$$

For odd $n \neq 1$, $1 \pm n$ is even, both cosines equal 1 and $a_n = 0$. For even n, $\cos(1 \pm n)\pi = -1$ and

$$a_n = \frac{1}{2\pi}\Big[\frac{2}{1+n} + \frac{2}{1-n}\Big] = \frac{1}{\pi}\cdot\frac{2}{1-n^2} = -\frac{2}{\pi(n^2-1)}.$$

For $n = 1$: $a_1 = \frac{1}{\pi}\int_0^{\pi}\sin\theta\cos\theta\,d\theta = 0$.

Sine coefficients. $b_1 = \frac{1}{\pi}\int_0^{\pi}\sin^2\theta\,d\theta = \frac{1}{2}$. For $n \geq 2$, $\sin\theta\sin n\theta = \frac{1}{2}[\cos(n-1)\theta - \cos(n+1)\theta]$ integrates over $(0, \pi)$ to $\frac{1}{2}\big[\frac{\sin(n-1)\pi}{n-1} - \frac{\sin(n+1)\pi}{n+1}\big] = 0$. So $b_n = 0$ for $n \geq 2$.

Result (writing $n = 2k$):

$$x(t) = V_m\Big[\frac{1}{\pi} + \frac{1}{2}\sin\omega_0 t - \frac{2}{\pi}\sum_{k=1}^{\infty}\frac{\cos 2k\omega_0 t}{4k^2-1}\Big] = V_m\Big[0.3183 + 0.5\sin\omega_0 t - 0.2122\cos 2\omega_0 t - 0.0424\cos 4\omega_0 t - \cdots\Big].$$

Interpretation. The waveform contains DC (the useful output of a rectifier), *half* of the original fundamental, and even harmonics only. The structure is explained in §15.4: $x(t) = \frac{V_m}{2}\big(\sin\omega_0 t + |\sin\omega_0 t|\big)$, the sum of half the input sinusoid and half of a full-wave rectified sine, which contains only DC and even harmonics.

Check. Numerical quadrature: $a_0 = 0.63662$, $b_1 = 0.5$, $a_2 = -0.212207$, $a_4 = -0.042441$, $a_3 = b_3 = 0$. ✓

Worked Example 15.6 — Rectangular pulse train (trigonometric form)

A pulse train has height A, width τ and period T_0, centred on $t = 0$: $x(t) = A$ for $|t| < \tau/2$ and 0 elsewhere in $(-T_0/2, T_0/2)$. Let $D = \tau/T_0$ be the duty cycle. Find a_n, b_n.

Solution. The function is even about $t = 0$, so every $b_n = \frac{2}{T_0}\int x(t)\sin n\omega_0 t\,dt = 0$ (odd integrand on a symmetric interval). Mean value $a_0/2 = A\tau/T_0 = AD$. For $n \geq 1$,

$$a_n = \frac{2}{T_0}\int_{-\tau/2}^{\tau/2} A\cos n\omega_0 t\,dt = \frac{4A}{T_0 n\omega_0}\sin\frac{n\omega_0\tau}{2} = \frac{2A}{n\pi}\sin(n\pi D) = 2AD\,\mathrm{sinc}(nD),$$

with the normalised $\mathrm{sinc}(u) = \sin(\pi u)/(\pi u)$ used in Ch. 16. So

$$x(t) = AD + 2AD\sum_{n=1}^{\infty}\mathrm{sinc}(nD)\cos n\omega_0 t.$$

Special cases. $D = \frac{1}{2}$ gives $a_n = \frac{2A}{n\pi}\sin\frac{n\pi}{2}$: $\frac{2A}{\pi}, 0, -\frac{2A}{3\pi}, 0, \frac{2A}{5\pi}, \ldots$ — a square wave of levels 0 and A, i.e. $\frac{A}{2}$ plus half of Worked Example 15.3 shifted by a quarter period (the sign alternation is that shift; §15.7). $D = \frac{1}{4}$ gives harmonics 4, 8, 12, … missing, because $\sin(n\pi/4) = 0$ at those n. In general, whenever $1/D$ is an integer, the harmonics at multiples of $1/D$ vanish.

The envelope of a_n is $2AD\,\mathrm{sinc}(fD/f_0)$ with first null at $f = 1/\tau$: **narrower pulses spread their power over more harmonics.** This is the discrete version of the pulse-width/bandwidth reciprocity of Ch. 16.

Engineering note — table conventions

Tables of Fourier series differ in (i) whether the constant is a_0 or $a_0/2$, (ii) the time origin of the waveform (a shift changes the sine/cosine mix but not the amplitudes A_n), (iii) whether amplitudes are peak or RMS values, and (iv) whether frequency is in rad/s or Hz. When a datasheet says "3rd harmonic 4%" it almost always means the ratio A_3/A_1, which is the same for peak and RMS values.

Pitfall — special values and the forgotten $n = 1$

Two errors account for most wrong coefficients. (1) Writing $\cos n\pi = 1$ instead of $(-1)^n$, or $\sin(n\pi/2) = 1$ for all n. Keep a table: $\sin n\pi = 0$, $\cos n\pi = (-1)^n$, $\sin\frac{n\pi}{2} = 0, 1, 0, -1, \ldots$ for $n = 0, 1, 2, 3$, $\cos\frac{n\pi}{2} = 1, 0, -1, 0, \ldots$. (2) Dividing by $n - 1$ or $n^2 - 1$ and forgetting to treat $n = 1$ separately, as in Worked Example 15.5. Whenever a formula has a factor that vanishes for a particular n, compute that coefficient directly.

15.3 Complex Fourier series

Motivation

The trigonometric form needs two families of coefficients and two families of formulas, and its algebra — shifting, differentiating, multiplying by a frequency response — keeps generating products of sines and cosines that must be untangled with identities. The complex exponential form packs each harmonic into a *single* complex number, turns differentiation into multiplication by $jn\omega_0$, turns a time shift into multiplication by a phase factor, and makes the LTI response $c_n H(jn\omega_0)$ a one-line statement. It is also the form that generalises directly to the Fourier transform (Ch. 16), the Z-transform (Ch. 17) and the DFT. In signals and systems work it is the default.

From real to complex form

Euler's formula gives $\cos\theta = \frac{1}{2}(e^{j\theta} + e^{-j\theta})$ and $\sin\theta = \frac{1}{2j}(e^{j\theta} - e^{-j\theta})$. Substituting into the trigonometric series,

$$a_n\cos n\omega_0 t + b_n\sin n\omega_0 t = \frac{a_n - jb_n}{2}e^{jn\omega_0 t} + \frac{a_n + jb_n}{2}e^{-jn\omega_0 t}.$$

Defining

$$c_0 = \frac{a_0}{2}, \qquad c_n = \frac{a_n - jb_n}{2}, \qquad c_{-n} = \frac{a_n + jb_n}{2} \qquad (n \geq 1),$$

the series becomes a single two-sided sum.

Complex (exponential) Fourier series

The complex Fourier series of a T_0-periodic function x is

$$x(t) \sim \sum_{n=-\infty}^{\infty} c_n\, e^{jn\omega_0 t}, \qquad c_n = \frac{1}{T_0}\int_{T_0} x(t)\, e^{-jn\omega_0 t}\,dt, \quad n \in \mathbb{Z}.$$

The partial sums are taken symmetrically, $S_N(t) = \sum_{n=-N}^{N} c_n e^{jn\omega_0 t}$, so that S_N is the same function as the trigonometric partial sum of degree N.

The coefficient formula can be derived directly — it is the projection $\langle x, e^{jn\omega_0 t}\rangle$ of §15.1 — or checked by combining the Euler formulas: $\frac{a_n - jb_n}{2} = \frac{1}{T_0}\int x(t)(\cos n\omega_0 t - j\sin n\omega_0 t)\,dt$. Note the single prefactor $1/T_0$, not $2/T_0$: each real harmonic of amplitude A_n is split equally between two lines of magnitude $A_n/2$ at $\pm n\omega_0$.

Conversions between coefficient forms

For $n \geq 1$:

$$a_n = c_n + c_{-n}, \qquad b_n = j(c_n - c_{-n}),$$
$$A_n = 2|c_n| \text{ (if } x \text{ real)}, \quad \phi_n = \angle c_n \text{ (if } x \text{ real)},$$

and $c_0 = a_0/2 = A_0$. For a real x, $a_n = 2\,\mathrm{Re}\{c_n\}$ and $b_n = -2\,\mathrm{Im}\{c_n\}$.

The phase relation follows because $c_n = \frac{a_n - jb_n}{2} = \frac{A_n}{2}e^{j\phi_n}$ with $\phi_n = \operatorname{atan2}(-b_n, a_n)$, exactly the phase of the amplitude–phase form in §15.2. Then $c_n e^{jn\omega_0 t} + c_{-n}e^{-jn\omega_0 t} = 2\,\mathrm{Re}\{c_n e^{jn\omega_0 t}\} = A_n\cos(n\omega_0 t + \phi_n)$: c_n **is half the phasor of the nth harmonic.** This is the bridge to Ch. 2 phasor analysis.

Real signals: conjugate symmetry

Conjugate symmetry

$x(t)$ is real if and only if $c_{-n} = c_n^*$ for all n. Then $|c_{-n}| = |c_n|$ (even magnitude spectrum) and $\angle c_{-n} = -\angle c_n$ (odd phase spectrum).

Proof. $c_{-n} = \frac{1}{T_0}\int x(t)e^{jn\omega_0 t}dt$ and $c_n^* = \frac{1}{T_0}\int x^*(t)e^{jn\omega_0 t}dt$; these agree for all n iff $x = x^*$ (by uniqueness of coefficients, §15.6). □

The negative-frequency coefficients of a real signal therefore carry no new information; they are the "clockwise-rotating" partners of Ch. 3, §3.5, needed so that the imaginary parts cancel. For a *complex* signal — the analytic signal or the baseband I/Q signal of a radio receiver, $x = I + jQ$ — the positive- and negative-frequency coefficients are independent, and a one-sided description loses information.

Line spectra, two-sided

Plotting $|c_n|$ and $\angle c_n$ against $\omega = n\omega_0$ (or $f = nf_0$) for $n \in \mathbb{Z}$ gives the **two-sided line spectrum**: a set of vertical lines at the harmonic frequencies. Because the frequencies are discrete, a periodic signal has a *line* (discrete) spectrum, while an aperiodic signal has a continuous spectrum (Ch. 16). Section 15.5 compares one- and two-sided spectra and their conventions.

Computing c_n directly

The complex formula usually needs fewer integrations than the real pair, because $\int e^{at}dt = e^{at}/a$ handles exponential, sinusoidal and constant pieces at once.

Worked Example 15.7 — Pulse train in complex form, with numbers

A clock signal is a rectangular pulse train of height $A = 5$ V, width $\tau = 1$ μs and period $T_0 = 4$ μs, with a pulse centred on $t = 0$. Find c_n, and list $c_0, \dots, c_4$ and the frequencies at which they occur.

Solution. $\omega_0 = 2\pi/T_0$, $D = \tau/T_0 = 1/4$. For $n \neq 0$,

$$c_n = \frac{1}{T_0}\int_{-\tau/2}^{\tau/2} Ae^{-jn\omega_0 t}\,dt = \frac{A}{T_0}\cdot\frac{e^{-jn\omega_0\tau/2} - e^{jn\omega_0\tau/2}}{-jn\omega_0} = \frac{A}{T_0}\cdot\frac{2\sin(n\omega_0\tau/2)}{n\omega_0} = AD\,\mathrm{sinc}(nD),$$

using $n\omega_0\tau/2 = n\pi D$. For $n = 0$ the mean is AD, consistent with $\mathrm{sinc}\,0 = 1$. The coefficients are real (because x is real and even; §15.4).

n	$f = n/T_0$	$c_n = 1.25\,\mathrm{sinc}(n/4)$ **(V)**
0	0	1.2500
±1	250 kHz	1.1254
±2	500 kHz	0.7958
±3	750 kHz	0.3751
±4	1 MHz	0

The first null is at $f = 1/\tau = 1$ MHz. Compare with Worked Example 15.6: $a_n = 2c_n$, e.g. the 250 kHz harmonic has peak amplitude $2(1.1254) = 2.251$ V.

Check. $c_1 = 1.25 \cdot \sin(\pi/4)/(\pi/4) = 1.25 \times 0.9003 = 1.1254$. ✓

Worked Example 15.8 — Periodic exponential decay

The waveform $x(t) = e^{-at}$ for $0 \le t < T_0$, repeated with period T_0, models a periodically recharged and discharged RC node. Find c_n, and evaluate for $a = 1\ \text{s}^{-1}$, $T_0 = 2\pi$ s.

Solution.

$$c_n = \frac{1}{T_0}\int_0^{T_0} e^{-at}e^{-jn\omega_0 t}\,dt = \frac{1}{T_0}\Big[\frac{e^{-(a+jn\omega_0)t}}{-(a+jn\omega_0)}\Big]_0^{T_0} = \frac{1-e^{-aT_0}e^{-jn2\pi}}{T_0(a+jn\omega_0)} = \frac{1-e^{-aT_0}}{T_0\,(a+jn\omega_0)}.$$

With $a=1$, $T_0=2\pi$, $\omega_0=1$: $\frac{1-e^{-2\pi}}{2\pi} = 0.158858$, so $c_n = 0.158858/(1+jn)$: $c_0 = 0.1589$, $c_1 = 0.0794 - 0.0794j = 0.1123\angle -45^\circ$, $c_2 = 0.0318 - 0.0635j = 0.0710\angle -63.4^\circ$.

Interpretation. $|c_n| \approx 0.159/n$ for large n: a $1/n$ decay, because the waveform jumps from e^{-aT_0} back to 1 at each period boundary. The factor $1/(a+jn\omega_0)$ is the Laplace transform $1/(s+a)$ at $s = jn\omega_0$ — a connection made general next.

Check. Numerical quadrature of the defining integral reproduces $c_1 = 0.079429 - 0.079429j$ and $c_2 = 0.031772 - 0.063543j$. ✓

Coefficients from the Laplace transform of one period

Let $x_1(t)$ be one period of x, i.e. $x_1(t) = x(t)$ on $[0, T_0)$ and 0 elsewhere, and let $X_1(s) = \mathcal{L}\{x_1\} = \int_0^{T_0} x(t)e^{-st}dt$, an entire function of s because the integral is over a finite interval. Comparing with the definition of c_n:

Coefficients via the one-period transform

$$c_n = \frac{1}{T_0}\,X_1(jn\omega_0), \qquad X_1(s) = \int_0^{T_0} x(t)\,e^{-st}\,dt.$$

Chapter 14 showed that the Laplace transform of the periodic extension (starting at $t=0$) is $X(s) = X_1(s)/(1-e^{-sT_0})$. Its poles are at the zeros of $1-e^{-sT_0}$, i.e. at $s = jn\omega_0$, and the residue there is $X_1(jn\omega_0)/T_0 = c_n$. So the Fourier coefficients are exactly the residues of the Laplace transform at its imaginary-axis poles — the steady-state part of the inverse Laplace transform. Chapter 16 obtains the same relation as $c_n = X(n\omega_0)/T_0$ from the Fourier transform of one period.

Worked Example 15.9 — Half-wave rectifier via the one-period transform

Recompute the coefficients of Worked Example 15.5 ($V_m = 1$, $\omega_0 = 1$) using $X_1(s)$.

Solution. One period is $\sin t$ on $[0,\pi]$ and 0 on $(\pi, 2\pi)$. Writing $\sin t\,[u(t) - u(t-\pi)] = \sin t\,u(t) + \sin(t-\pi)u(t-\pi)$ and using the shift theorem of Ch. 14,

$$X_1(s) = \frac{1+e^{-\pi s}}{s^2+1}.$$

For $n \neq \pm 1$: $c_n = \frac{1}{2\pi}\cdot\frac{1+(-1)^n}{1-n^2}$, which is 0 for odd n and $\frac{1}{\pi(1-n^2)}$ for even n. Then $a_n = 2\,\mathrm{Re}\,c_n = -\frac{2}{\pi(n^2-1)}$ — as before. For $n=0$, $c_0 = 1/\pi$. ✓

For $n = 1$, $X_1(s)$ has a removable $0/0$ at $s = j$: write $X_1(s) = \frac{1+e^{-\pi s}}{(s-j)(s+j)}$ and apply l'Hôpital to the numerator over $(s-j)$: $\lim_{s\to j}\frac{-\pi e^{-\pi s}}{1}\cdot\frac{1}{s+j} = \frac{-\pi e^{-j\pi}}{2j} = \frac{\pi}{2j}$. So $c_1 = \frac{1}{2\pi}\cdot\frac{\pi}{2j} = -\frac{j}{4}$ and $b_1 = -2\,\mathrm{Im}\,c_1 = \frac{1}{2}$. ✓

Lesson. The one-period transform is often the quickest route when x_1 is built from standard Laplace pairs and shifts; the special harmonics where the denominator vanishes need a limit.

Worked Example 15.10 — Reading coefficients off without integration

Find the complex Fourier coefficients of $x(t) = \cos^3\omega_0 t + 2\sin 2\omega_0 t$, and of the sawtooth $x(t) = t$ on $(-\pi, \pi)$ from its known sine series.

Solution. (a) By the power-reduction formula of Ch. 3, $\cos^3\theta = \frac{3}{4}\cos\theta + \frac{1}{4}\cos 3\theta$. So

$$x(t) = \tfrac{3}{8}\big(e^{j\omega_0 t} + e^{-j\omega_0 t}\big) + \tfrac{1}{8}\big(e^{j3\omega_0 t} + e^{-j3\omega_0 t}\big) + \tfrac{1}{j}\big(e^{j2\omega_0 t} - e^{-j2\omega_0 t}\big).$$

Hence $c_{\pm 1} = \frac{3}{8}$, $c_{\pm 3} = \frac{1}{8}$, $c_2 = -j$, $c_{-2} = j$, all others zero. Uniqueness of coefficients (§15.6) guarantees that *any* way of writing x as a finite sum of harmonics gives its Fourier series.

(b) From Worked Example 15.1, $a_n = 0$, $b_n = 2(-1)^{n+1}/n$. Then $c_n = \frac{a_n - jb_n}{2} = -j\frac{(-1)^{n+1}}{n} = j\frac{(-1)^n}{n}$ for $n \neq 0$, $c_0 = 0$. Check conjugate symmetry: $c_{-n} = j\frac{(-1)^n}{-n} = -j\frac{(-1)^n}{n} = c_n^*$. ✓

Pitfall — factor-of-two errors between the forms

The most frequent mistake in this whole subject is mixing the $1/T_0$ of c_n with the $2/T_0$ of a_n, b_n, or reading $|c_n|$ as the amplitude of the nth harmonic. For a real signal the amplitude of the harmonic at $n\omega_0$ is $A_n = 2|c_n|$; only the DC term has $A_0 = c_0$ with no factor 2. A quick sanity test: $\cos\omega_0 t$ has $c_{\pm 1} = \frac{1}{2}$ and $a_1 = 1$.

Engineering note — phasors are positive-frequency coefficients

In AC circuit analysis a voltage $V_m\cos(\omega t+\phi)$ is represented by the phasor $\mathbf{V}=V_m e^{j\phi}$ (peak) or $\frac{V_m}{\sqrt{2}}e^{j\phi}$ (RMS). In Fourier language $\mathbf{V}_{\text{peak}}=2c_1$. A multi-harmonic waveform is then a *list* of phasors $\mathbf{V}_n=2c_n$, one per harmonic, each analysed with the impedances $Z(jn\omega_0)$ at its own frequency — the method of §15.9.

15.4 Even and odd functions

Motivation

Most of the integrals in §15.2 turned out to be zero, and in every case the zero could have been predicted from a symmetry of the waveform. Recognising symmetry before integrating halves (or quarters) the work, provides an independent check on the answer, and — more importantly for engineers — links the *shape* of a waveform to its *harmonic content*: a symmetrical inverter output has no even harmonics, a balanced push–pull amplifier cancels even-order distortion, and an asymmetrical clipping characteristic generates a second harmonic and a DC shift. This section collects the symmetry rules and the half-range expansions that follow from them.

Even and odd functions

Even and odd functions

A function is **even** if $x(-t)=x(t)$ for all t (graph symmetric about the vertical axis) and **odd** if $x(-t)=-x(t)$ (graph symmetric under a 180° rotation about the origin). An odd function defined at 0 has $x(0)=0$.

The rules for products follow the rules for signs: even × even and odd × odd are even; even × odd is odd. For integrals over a symmetric interval,

$$\int_{-a}^{a} x_{\text{even}}(t)\,dt = 2\int_{0}^{a} x_{\text{even}}(t)\,dt, \qquad \int_{-a}^{a} x_{\text{odd}}(t)\,dt = 0.$$

Since $\cos n\omega_0 t$ is even and $\sin n\omega_0 t$ is odd, applying these rules to the Euler formulas over $(-T_0/2, T_0/2)$ gives:

Fourier series of even and odd functions

If x is even, $b_n=0$ for all n and x has a **Fourier cosine series**

$$x(t)\sim \frac{a_0}{2}+\sum_{n=1}^{\infty} a_n\cos n\omega_0 t, \qquad a_n=\frac{4}{T_0}\int_0^{T_0/2} x(t)\cos n\omega_0 t\,dt.$$

If x is odd, $a_n=0$ for all n (including a_0) and x has a **Fourier sine series**

$$x(t)\sim \sum_{n=1}^{\infty} b_n\sin n\omega_0 t, \qquad b_n=\frac{4}{T_0}\int_0^{T_0/2} x(t)\sin n\omega_0 t\,dt.$$

In complex form: x real and even $\iff$ c_n real and even in n; x real and odd $\iff$ c_n purely imaginary and odd in n.

Even/odd decomposition. Every function splits uniquely into even and odd parts,

$$x(t)=x_e(t)+x_o(t), \qquad x_e(t)=\frac{x(t)+x(-t)}{2}, \qquad x_o(t)=\frac{x(t)-x(-t)}{2}.$$

The cosine terms of the Fourier series of x are exactly the Fourier series of x_e, and the sine terms are the series of x_o. For real x: $x_e \leftrightarrow \operatorname{Re}\{c_n\}$ and $x_o \leftrightarrow j\operatorname{Im}\{c_n\}$.

Symmetry is a property of the waveform *and the chosen time origin*. The square wave of Worked Example 15.3 is odd about $t=0$; shifting the origin by $T_0/4$ makes it even, and its series becomes a cosine series with the same amplitudes (§15.7). Many waveforms have **hidden symmetry**: they become even or odd after subtracting their mean value (Worked Example 15.4) or after a shift of origin. Look for both before integrating.

Half-wave symmetry

Half-wave (odd-harmonic) symmetry

A T_0-periodic function has **half-wave symmetry** if

$$x\!\left(t+\frac{T_0}{2}\right)=-x(t) \qquad \text{for all } t.$$

Graphically, the second half-cycle is the first half-cycle shifted by $T_0/2$ and turned upside down. The mean value is necessarily zero.

Half-wave symmetry ⇔ odd harmonics only

x has half-wave symmetry if and only if $c_n=0$ (equivalently $a_n=b_n=0$) for every even n, including $n=0$. In that case, for odd n,

$$a_n=\frac{4}{T_0}\int_0^{T_0/2} x(t)\cos n\omega_0 t\,dt, \qquad b_n=\frac{4}{T_0}\int_0^{T_0/2} x(t)\sin n\omega_0 t\,dt.$$

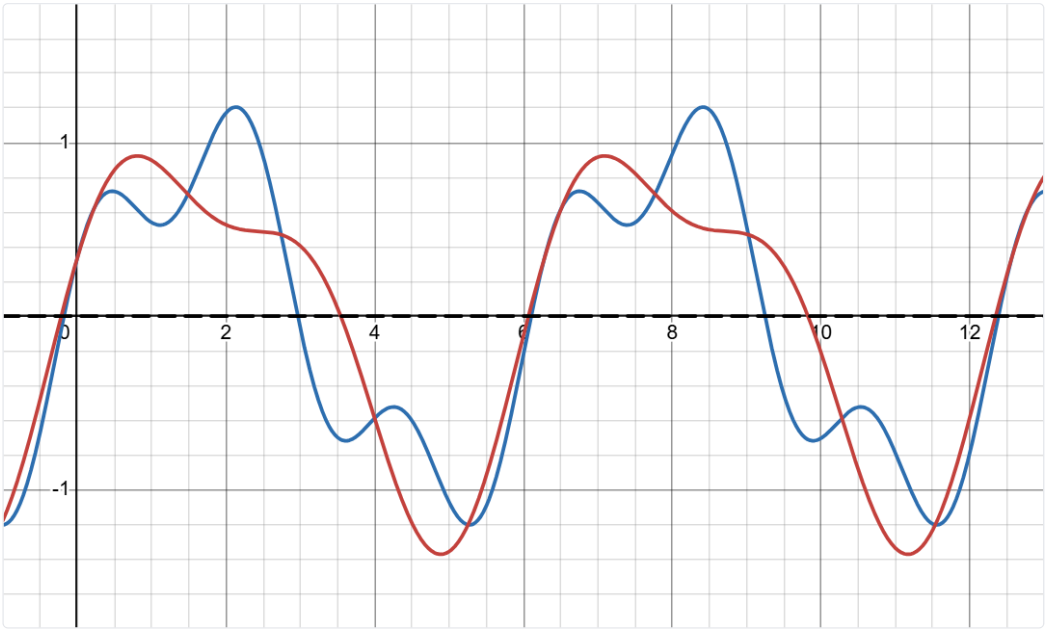
Proof. Split the coefficient integral into two half-periods and substitute $t = t' + T_0/2$ in the second:

$$c_n = \frac{1}{T_0}\int_0^{T_0/2} x(t)e^{-jn\omega_0 t}dt + \frac{1}{T_0}\int_0^{T_0/2} x\Big(t' + \frac{T_0}{2}\Big)e^{-jn\omega_0 t'}e^{-jn\pi}\,dt'.$$

With half-wave symmetry the second integrand is $-x(t')e^{-jn\omega_0 t'}(-1)^n$, so $c_n = \big(1-(-1)^n\big)\frac{1}{T_0}\int_0^{T_0/2} x(t)e^{-jn\omega_0 t}dt$: zero for even n and twice the half-period integral for odd n. Conversely, if only odd harmonics are present then each term satisfies $e^{jn\omega_0(t+T_0/2)} = (-1)^n e^{jn\omega_0 t} = -e^{jn\omega_0 t}$, so the sum does too. □

The physical reading is powerful: **any system that treats positive and negative half-cycles identically (apart from sign) produces only odd harmonics.** A symmetric clipper, a push–pull output stage, a transformer core with a symmetric B–H loop and a full-bridge inverter switched symmetrically all have half-wave-symmetric outputs when driven by a sinusoid. Even harmonics are the signature of *asymmetry*: a half-wave rectifier, a single-ended amplifier stage with a square-law term, or DC magnetisation of a transformer core.

The opposite condition $x(t + T_0/2) = +x(t)$ simply means that $T_0/2$ is a period; then only even harmonics of ω_0 appear — they are all the harmonics of the true fundamental $2\omega_0$. The full-wave rectified sine $|\sin \omega_0 t|$ is the standard example.



Blue: fundamental plus a third harmonic — each negative half-cycle is the inverted copy of the preceding positive one (half-wave symmetry). Red: fundamental plus a second harmonic — the positive and negative half-cycles have different shapes, and no shift-and-invert maps one onto the other.

Quarter-wave symmetry

Quarter-wave symmetry

A function has **quarter-wave symmetry** if it has half-wave symmetry *and* is either even or odd (about a suitable origin).

Then only odd harmonics are present and, in addition, only cosines (even case) or only sines (odd case). Combining the two reductions, one quarter-period suffices:

Quarter-wave formulas

If x is odd with half-wave symmetry, then $a_n = 0$ for all n, $b_n = 0$ for even n, and for odd n

$$b_n = \frac{8}{T_0}\int_0^{T_0/4} x(t)\sin n\omega_0 t\,dt.$$

If x is even with half-wave symmetry, then $b_n = 0$ for all n, $a_n = 0$ for even n, and for odd n

$$a_n = \frac{8}{T_0}\int_0^{T_0/4} x(t)\cos n\omega_0 t\,dt.$$

Proof sketch (odd case). Odd symmetry gives $b_n = \frac{4}{T_0}\int_0^{T_0/2} x\sin n\omega_0 t\,dt$. Odd plus half-wave implies $x(T_0/2 - t) = -x(t - T_0/2) = x(t)$: on $(0, T_0/2)$ the graph is symmetric about $t = T_0/4$. For odd n, $\sin n\omega_0(T_0/2 - t) = \sin(n\pi - n\omega_0 t) = \sin n\omega_0 t$, so the integrand is symmetric about $T_0/4$ too and the half-period integral is twice the quarter-period integral. □

Symmetry	Condition	Harmonics present	Coefficient integral
none	—	all, a_n and b_n	full period, $2/T_0$
even	$x(-t) = x(t)$	DC and cosines	half period, $4/T_0$
odd	$x(-t) = -x(t)$	sines only, no DC	half period, $4/T_0$
half-wave	$x(t + T_0/2) = -x(t)$	odd n only, no DC	half period, $4/T_0$

Symmetry	Condition	Harmonics present	Coefficient integral
even quarter-wave	even and half-wave	cosines, odd n	quarter period, $8/T_0$
odd quarter-wave	odd and half-wave	sines, odd n	quarter period, $8/T_0$

Worked Example 15.11 — Triangle wave by the quarter-wave formula

The triangle wave of peak A and period T_0 has $x(t) = A\big(1 - 4|t|/T_0\big)$ on $(-T_0/2, T_0/2)$. Find its Fourier series using symmetry.

Solution. Symmetry: x is even; and $x(t + T_0/2) = -x(t)$ (check at $t = 0$: $x(T_0/2) = A(1-2) = -A = -x(0)$; the graph is a straight line falling from A to $-A$ on $(0, T_0/2)$ and rising back on $(T_0/2, T_0)$). So it is even quarter-wave: only a_n with odd n.

Let $\theta = \omega_0 t$; on $(0, \pi/2)$, $x = A(1 - 2\theta/\pi)$. For odd n,

$$a_n = \frac{8}{T_0}\int_0^{T_0/4} x\cos n\omega_0 t\,dt = \frac{4}{\pi}\int_0^{\pi/2} A\Big(1-\frac{2\theta}{\pi}\Big)\cos n\theta\,d\theta.$$

By parts, $\int_0^{\pi/2}(1-\frac{2\theta}{\pi})\cos n\theta\,d\theta = \big[(1-\frac{2\theta}{\pi})\frac{\sin n\theta}{n}\big]_0^{\pi/2} + \frac{2}{\pi n}\int_0^{\pi/2}\sin n\theta\,d\theta = 0 + \frac{2}{\pi n^2}\Big(1-\cos\frac{n\pi}{2}\Big) = \frac{2}{\pi n^2}$ for odd n. Hence

$$a_n = \frac{8A}{\pi^2 n^2}\ (n \text{ odd}), \qquad x(t) = \frac{8A}{\pi^2}\Big(\cos\omega_0 t + \frac{1}{9}\cos 3\omega_0 t + \frac{1}{25}\cos 5\omega_0 t + \cdots\Big).$$

Checks. At $t = 0$: $\frac{8A}{\pi^2}\sum_{\text{odd}}\frac{1}{n^2} = \frac{8A}{\pi^2}\cdot\frac{\pi^2}{8} = A$. ✓ The coefficients fall as $1/n^2$ — the triangle is continuous with jumps only in its slope (§15.7). The fundamental, $0.811A$, is already very close to the waveform: the third harmonic is only $1/9$ of it.

Worked Example 15.12 — Quasi-square (120° conduction) waveform

A three-phase six-step inverter produces a line-to-line voltage that is $+V$ for $30^\circ < \theta < 150^\circ$, $-V$ for $210^\circ < \theta < 330^\circ$ and zero otherwise ($\theta = \omega_0 t$). Find its harmonic content.

Solution. With the origin as given, x is odd ($x(-\theta) = -x(\theta)$, since $-V$ occupies $(-150^\circ, -30^\circ) \equiv (210^\circ, 330^\circ)$) and half-wave symmetric. So only sine terms with odd n; use the quarter-wave formula with $T_0 = 2\pi$:

$$b_n = \frac{8}{2\pi}\int_0^{\pi/2} x\sin n\theta\,d\theta = \frac{4V}{\pi}\int_{\pi/6}^{\pi/2}\sin n\theta\,d\theta = \frac{4V}{n\pi}\Big[\cos\frac{n\pi}{6} - \cos\frac{n\pi}{2}\Big] = \frac{4V}{n\pi}\cos\frac{n\pi}{6} \quad (n \text{ odd}).$$

- $n = 3, 9, 15, \ldots$ (triplen): $\cos(n\pi/6) = \cos(k\pi/2)$ with odd k, which is 0. **No triplen harmonics** — the line-to-line voltage is the difference of two phase waveforms displaced by 120°, and triplens are in phase in all three phases (Ch. 3, §3.7), so they cancel.
- $n = 6k \pm 1$: $|\cos(n\pi/6)| = \sqrt{3}/2$, so $|b_n| = \frac{2\sqrt{3}V}{n\pi} = \frac{1.1027V}{n}$.

Numbers: $b_1 = 1.1027V$, $b_5 = -0.2205V$, $b_7 = -0.1575V$, $b_{11} = 0.1002V$, $b_{13} = 0.0848V$ — each exactly $1/n$ of the fundamental. **Check:** direct quadrature over a full period reproduces all five values to 10^{-12}. ✓ The distortion of this waveform is computed with Parseval's theorem in §15.8.

Half-range expansions

In boundary-value problems (Ch. 13) a function is given only on a half-interval $0 < x < L$ — the initial temperature along a rod, the initial displacement of a string, the voltage on one face of a box — and we are free to extend it to $(-L, 0)$ however suits the boundary conditions. Two extensions are standard.

Half-range expansions

Let f be given on $(0, L)$.

The **even extension** ($f(-x) = f(x)$, period $2L$) yields the **half-range cosine series**

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi x}{L}, \qquad a_n = \frac{2}{L}\int_0^L f(x)\cos\frac{n\pi x}{L}\,dx.$$

The **odd extension** ($f(-x) = -f(x)$, period $2L$) yields the **half-range sine series**

$$f(x) \sim \sum_{n=1}^{\infty} b_n \sin\frac{n\pi x}{L}, \qquad b_n = \frac{2}{L}\int_0^L f(x)\sin\frac{n\pi x}{L}\,dx.$$

Both series represent f on $(0, L)$; they differ outside it and at the endpoints. The sine series vanishes at $x = 0$ and $x = L$, suiting Dirichlet conditions $u = 0$ at the ends; the cosine series has zero slope there (each term does), suiting Neumann (insulated) conditions. Note the harmonics are π/L apart, half the spacing $2\pi/L$ of an ordinary series of period L — the extension doubles the period.

Worked Example 15.13 — Sine and cosine expansions of a ramp

Expand $f(x) = x$ on $(0, L)$ in (a) a half-range sine series and (b) a half-range cosine series, and compare them.

Solution. (a) Odd extension is the sawtooth x on $(-L, L)$. By parts,

$$b_n = \frac{2}{L}\int_0^L x \sin\frac{n\pi x}{L}\,dx = \frac{2}{L}\Big[-\frac{Lx}{n\pi}\cos\frac{n\pi x}{L}\Big]_0^L + \frac{2}{L}\cdot\frac{L}{n\pi}\int_0^L \cos\frac{n\pi x}{L}dx = \frac{2L(-1)^{n+1}}{n\pi}.$$

So $x = \frac{2L}{\pi}\big(\sin\frac{\pi x}{L} - \frac{1}{2}\sin\frac{2\pi x}{L} + \frac{1}{3}\sin\frac{3\pi x}{L} - \cdots\big)$ on $(0, L)$.

(b) Even extension is the triangle $|x|$ on $(-L, L)$. $a_0 = \frac{2}{L}\int_0^L x\,dx = L$, and for $n \geq 1$

$$a_n = \frac{2}{L}\int_0^L x\cos\frac{n\pi x}{L}\,dx = \frac{2}{L}\Big[\frac{L^2}{n^2\pi^2}\cos\frac{n\pi x}{L}\Big]_0^L = \frac{2L}{n^2\pi^2}\big((-1)^n - 1\big),$$

i.e. $-\frac{4L}{n^2\pi^2}$ for odd n, 0 for even n. So $x = \frac{L}{2} - \frac{4L}{\pi^2}\big(\cos\frac{\pi x}{L} + \frac{1}{9}\cos\frac{3\pi x}{L} + \cdots\big)$ on $(0, L)$.

Comparison. The odd extension jumps from L to $-L$ at $x = L$, so the sine series converges slowly ($1/n$), shows a Gibbs overshoot near $x = L$ (§15.6) and converges to 0 (the midpoint) at $x = L$. The even extension is continuous, so the cosine series converges uniformly with $1/n^2$ coefficients and gives the right value L at $x = L$. With $L = 1$ and three non-zero harmonics each (sine: $n = 1, 2, 3$; cosine: $n = 1, 3, 5$): at $x = 0.9$ the sine series gives 0.556 and the cosine series 0.912 (exact 0.9); at $x = 1$ the sine series gives 0, the cosine series 0.967 (exact 1). Choose the extension that is continuous whenever the boundary conditions leave you the choice.

Worked Example 15.14 — Even and odd parts from the coefficients

For the periodic exponential of Worked Example 15.8 (e^{-t} on $[0, 2\pi)$), find the Fourier series of its even part without further integration.

Solution. $c_n = \frac{0.158858}{1+jn} = 0.158858\frac{1-jn}{1+n^2}$. The even part has coefficients $\mathrm{Re}\{c_n\} = \frac{0.158858}{1+n^2}$, so

$$x_e(t) = 0.158858\Big(1 + 2\sum_{n=1}^{\infty}\frac{\cos nt}{1+n^2}\Big),$$

and the odd part is $x_o(t) = 0.317716\sum_{n\geq 1}\frac{n\sin nt}{1+n^2}$. Directly, $x_e(t) = \frac{1}{2}\big(e^{-t} + e^{-(2\pi - t)}\big)$ on $(0, 2\pi)$, which is $\frac{1}{2}(1 + e^{-2\pi}) = 0.500934$ at $t = 0^+$. **Check:** summing the cosine series at $t = 0$ to 200 000 terms gives $0.158858\,(1 + 2 \times 1.076674) = 0.50093$. ✓ (The continuous even part's series has $1/n^2$ coefficients; the odd part carries the jump and has $1/n$ coefficients.)

Pitfall — symmetry depends on the origin, and on the whole period

A waveform that "looks odd" over the half-period you sketched may not be odd as a periodic function; check $x(-t) = -x(t)$ using the periodic extension (for instance $x(-t) = x(T_0 - t)$). And half-wave symmetry is a statement about shifting by exactly $T_0/2$: a waveform whose negative half-cycle is a *mirror image in time* (rather than a shifted copy) of the positive one need not have half-wave symmetry.

Engineering note — even-order distortion and push–pull cancellation

If an amplifier's transfer characteristic is $y = a_1x + a_2x^2 + a_3x^3$, a push–pull pair of matched devices produces $y_1 - y_2$ with y_2 the same characteristic driven by $-x$: the even-order terms cancel, leaving $2a_1x + 2a_3x^3$, an odd function of x. An odd function of a half-wave-symmetric input is half-wave symmetric, so the output has no even harmonics. Measured second-harmonic distortion in such an amplifier is therefore a direct measure of device mismatch.

15.5 Harmonics

Motivation

Once a waveform has been decomposed, engineers stop thinking about the waveform and start thinking about its **harmonics**: "the 5th harmonic current exceeds the limit", "the third harmonic of the oscillator is 40 dB down", "the triplen harmonics overload the neutral". This section fixes the vocabulary, defines line spectra precisely in their one-sided and two-sided forms, and records how simple operations on a waveform — shifting it, compressing it in time, scaling it — move and reshape its spectrum.

Vocabulary

Harmonics

For a periodic signal with fundamental angular frequency ω_0:

- the **fundamental** (first harmonic) is the component at ω_0;
- the n**th harmonic** is the component $A_n\cos(n\omega_0 t + \phi_n)$ at $n\omega_0$, $n \geq 2$;
- the **DC component** is $A_0 = c_0$, sometimes called the zeroth harmonic;
- **odd** and **even** harmonics have odd and even n; harmonics with n a multiple of 3 are called **triplen** harmonics.

Three related terms are often confused:

- An **overtone** (acoustics) is any component above the fundamental; the first overtone is the *second* harmonic.

- An **interharmonic** is a component at a frequency that is *not* an integer multiple of the supply frequency, for instance a 175 Hz component in a 50 Hz system produced by a cycloconverter or an arc furnace. A signal containing interharmonics is periodic only with a longer period (if the frequency ratio is rational) or not at all (Ch. 3, §3.4), and its Fourier series with respect to 50 Hz does not exist.
- A **subharmonic** has a frequency ω_0/m below the fundamental. Its presence means the true period is mT_0; relative to that longer period, it is simply a low-order harmonic.

One-sided and two-sided line spectra

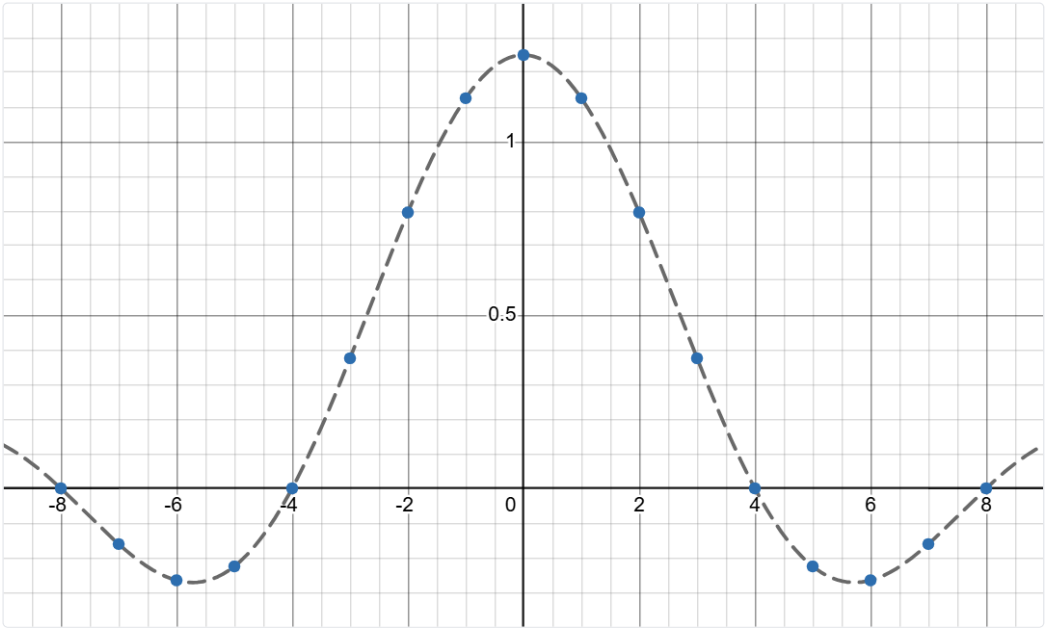
For a real signal there are two equivalent ways to draw the spectrum.

> **Line spectra**
>
> The **one-sided amplitude and phase spectra** are the sets of lines A_n and ϕ_n at frequencies $n\omega_0 \geq 0$, from the form $x(t) = A_0 + \sum_{n\geq 1} A_n \cos(n\omega_0 t + \phi_n)$.
>
> The **two-sided magnitude and phase spectra** are the lines $|c_n|$ and $\angle c_n$ at $n\omega_0$ for all $n \in \mathbb{Z}$, from $x(t) = \sum_n c_n e^{jn\omega_0 t}$.
>
> The **power spectrum** consists of lines $|c_n|^2$ (two-sided) or A_0^2, $A_n^2/2$ (one-sided), whose sum is the mean-square value (Parseval, §15.8).

The conversion is: one-sided line height $= 2\times$ two-sided height for $n \geq 1$; equal for DC. For phase, the one-sided spectrum is the right half of the two-sided one, and the left half is its negative. Lines with $A_n = 0$ have undefined phase and are omitted from phase plots.

Phase conventions matter when lines have negative real coefficients. A coefficient $a_n = -0.2$ can be plotted as amplitude -0.2 with phase 0, or (properly, since amplitudes are non-negative) as $A_n = 0.2$ with phase $\pm 180^\circ$. Both appear in textbooks; the non-negative-amplitude convention is the one used by instruments.

Amplitudes are frequently given in decibels relative to the fundamental, $20\log_{10}(A_n/A_1)$ dBc, or relative to a reference level such as 1 µV (dBµV, used in EMC). Because harmonics often span several orders of magnitude, a log scale is the only practical display.



Two-sided line spectrum $c_n = 1.25\,\mathrm{sinc}(n/4)$ of the 5 V, 25%-duty pulse train of Worked Example 15.7 (blue points, horizontal axis = harmonic number n), lying on the continuous sinc envelope (dashed). Every 4th harmonic is missing. Plotted as magnitudes, lines 5–7 would be drawn upwards with phase 180°.

How operations on the waveform change the spectrum

Let $x(t) \leftrightarrow c_n$ denote that x has coefficients c_n with fundamental ω_0. Each of the following follows in one line from the coefficient integral.

Amplitude scaling and adding a constant. $Kx(t) + B \leftrightarrow Kc_n + B\,\delta_{n0}$: scaling scales every line; adding a DC level changes only the $n = 0$ line.

Time shift. $x(t - t_0) \leftrightarrow c_n e^{-jn\omega_0 t_0}$. Substitute $t' = t - t_0$ in $\frac{1}{T_0}\int x(t - t_0)e^{-jn\omega_0 t}dt$. The magnitudes are unchanged; each phase shifts by $-n\omega_0 t_0$, **linearly in frequency**. A delay is invisible on an amplitude spectrum and is a straight line of slope $-t_0$ on a phase-versus-frequency plot.

Time scaling. $x(at)$, $a > 0$, has period T_0/a and fundamental $a\omega_0$, with the *same* coefficients c_n: compressing a waveform in time moves every line outward in frequency in proportion, without changing heights. A 1 kHz square wave and a 1 MHz square wave have identical coefficient lists.

Time reversal. $x(-t) \leftrightarrow c_{-n}$; for real x this is c_n^*, which negates every phase.

Harmonic content versus shape. The amplitude spectrum alone does not determine the waveform: phases matter. Ch. 3 showed a fundamental plus a third harmonic in phase and in anti-phase, with identical A_n, giving a flattened and a peaked waveform. Peak value, crest factor and the risk of clipping depend on phases; power and RMS do not (§15.8).

Worked Example 15.15 — One- and two-sided spectra of a rectified sine

Draw up the one-sided amplitude and phase spectrum and the two-sided spectrum of the half-wave rectified 230 V (RMS), 50 Hz mains, $V_m = 325.3$ V, up to the 6th harmonic, with amplitudes also in dBc.

Solution. From Worked Example 15.5, $x = V_m\left[\frac{1}{\pi} + \frac{1}{2}\sin\omega_0 t - \frac{2}{\pi}\sum_k \frac{\cos 2k\omega_0 t}{4k^2-1}\right]$. Convert to cosine-reference amplitude–phase form: $\frac{1}{2}\sin\omega_0 t = \frac{1}{2}\cos(\omega_0 t - 90^\circ)$, and $-\frac{2}{\pi(4k^2-1)}\cos 2k\omega_0 t = \frac{2}{\pi(4k^2-1)}\cos(2k\omega_0 t + 180^\circ)$.

n	f (Hz)	A_n/V_m	A_n (V)	ϕ_n	dBc	c_n (V)
0	0	0.3183	103.5	—	—	103.5
1	50	0.5000	162.6	-90°	0	$81.3\angle -90^\circ$
2	100	0.2122	69.0	180°	-7.4	$34.5\angle 180^\circ$
3, 5	150, 250	0	0	—	—	0
4	200	0.0424	13.8	180°	-21.4	$6.9\angle 180^\circ$
6	300	0.0182	5.9	180°	-28.8	$3.0\angle 180^\circ$

The two-sided spectrum has $c_{-n} = c_n^*$: lines of the same heights at $-50, -100, \ldots$ Hz, with phases $+90^\circ$ at -50 Hz and 180° (equivalently -180°) at the even negative harmonics.

Check. $A_2/A_1 = 0.2122/0.5 = 0.4244$, $20\log_{10} 0.4244 = -7.44$ dB. ✓ $A_6/V_m = 2/(35\pi) = 0.01819$. ✓

Worked Example 15.16 — Spectrum of a delayed pulse train

The pulse train of Worked Example 15.7 is delayed so that each pulse starts, rather than is centred, at $t = 0$: $x(t) = 5$ V for $0 < t < 1$ μs, period 4 μs. Find the new coefficients.

Solution. The new waveform is the old one delayed by $t_0 = \tau/2 = 0.5$ μs. By the shift rule,

$$c_n^{\text{new}} = 1.25\,\text{sinc}(n/4)\,e^{-jn\omega_0 t_0} = 1.25\,\text{sinc}(n/4)\,e^{-jn\pi/4},$$

since $\omega_0 t_0 = 2\pi \cdot 0.5/4 = \pi/4$. Magnitudes are unchanged; the phase of harmonic n is $-45^\circ n$ (plus 180° where sinc is negative). For example $c_1 = 1.1254\angle -45^\circ$, $c_2 = 0.7958\angle -90^\circ$, $c_3 = 0.3751\angle -135^\circ$.

Check by direct integration. $c_1 = \frac{5}{4\,\mu\text{s}}\int_0^{1\,\mu\text{s}} e^{-j\omega_0 t}dt = \frac{5}{4}\cdot\frac{1-e^{-j\pi/2}}{j\pi/2} = \frac{5}{4}\cdot\frac{1+j}{j\pi/2} = \frac{5}{2\pi}(1-j) = 0.7958 - 0.7958j = 1.1254\angle -45^\circ$. ✓

Harmonics in three-phase systems

In a balanced three-phase system the phase-b and phase-c waveforms are the phase-a waveform delayed by $T_0/3$ and $2T_0/3$. By the shift rule the nth harmonic of phase b is that of phase a shifted by $-n \cdot 120^\circ$. Hence:

- $n = 1, 4, 7, \ldots$ ($n = 3k+1$): shifts $-120^\circ, -240^\circ$ — **positive sequence** a-b-c, like the fundamental;
- $n = 2, 5, 8, \ldots$ ($n = 3k+2$): shifts $-240^\circ \equiv +120^\circ$ — **negative sequence** a-c-b, producing reverse torque in induction motors;
- $n = 3, 6, 9, \ldots$ (triplen): shifts that are multiples of 360° — **zero sequence**, in phase in all three lines.

Zero-sequence currents add in the neutral instead of cancelling, and cancel in any line-to-line quantity (Worked Example 15.12).

Worked Example 15.17 — Neutral current from triplen harmonics

Each phase of a balanced four-wire system supplies a nonlinear load drawing $I_1 = 10$ A (RMS) at 50 Hz and a third harmonic $I_3 = 3$ A (RMS), plus negligible higher harmonics. Find the RMS neutral current, and compare with the phase current.

Solution. The neutral current is $i_a + i_b + i_c$. Fundamental components are 120° apart and sum to zero; third-harmonic components are in phase and add: $i_N = 3 \times \sqrt{2}\,(3)\cos(3\omega_0 t + \phi_3)$. So $I_N = 9$ A RMS at 150 Hz.

The phase RMS current is $\sqrt{10^2 + 3^2} = 10.44$ A (orthogonality, §15.8). The neutral carries 86% of the phase current even though the load is perfectly balanced — the reason modern installations with many switch-mode loads size the neutral conductor at or above the phase conductors.

Pitfall — "the 3rd harmonic" of what?

Harmonic numbers refer to a stated fundamental. A 12-pulse rectifier's characteristic harmonics are $12k \pm 1$ of the *supply* frequency; the ripple of a full-wave rectifier is at the 2nd harmonic of the supply but the *fundamental* of the ripple waveform. When writing or reading a specification, say which fundamental the harmonic numbers refer to.

Engineering note — harmonic limits

Power-quality standards (e.g. IEC 61000-3-2 for equipment currents, IEEE 519 for system voltages and currents) specify limits harmonic by harmonic, typically up to the 40th or 50th, and in total via THD (§15.8). Measurement standards (IEC 61000-4-7) prescribe exactly the Fourier analysis of this chapter applied to a 10-cycle (50 Hz) or 12-cycle (60 Hz) window, which gives 5 Hz resolution and therefore also resolves interharmonics.

15.6 Convergence

Motivation

So far we have computed coefficients and written "$\sim$". An engineer is entitled to ask whether the sum really *is* the function. The question is not pedantic. The square-wave series converges to 0 at the jumps, where the square wave is $\pm A$ — so at some points the answer is "no". Its partial sums overshoot by 9% near every jump no matter how many terms are taken — so even "approximately equal" needs qualification. And the identities we want to use — Parseval's theorem, term-by-term differentiation, the LTI response formula — each require a particular kind of convergence. This section states the three standard modes of convergence (pointwise, uniform, mean-square), the conditions under which each holds, and the Gibbs phenomenon that marks the boundary between them.

Dirichlet conditions and pointwise convergence

Dirichlet conditions

A T_0-periodic function x satisfies the **Dirichlet conditions** if, over one period,

1. x is absolutely integrable: $\int_{T_0} |x(t)|\,dt < \infty$;
2. x has a finite number of maxima and minima;
3. x has a finite number of discontinuities, each of them finite (a jump).

Conditions 2 and 3 together say that x is of bounded variation on a period. A simpler sufficient set used in many texts is that x be **piecewise smooth**: the period splits into finitely many intervals on each of which x and x' are continuous and have finite one-sided limits at the ends.

Dirichlet's theorem (pointwise convergence)

If x satisfies the Dirichlet conditions, its Fourier series converges at every t, and

$$\lim_{N\to\infty} S_N(t) = \frac{x(t^+) + x(t^-)}{2},$$

where $x(t^\pm)$ are the one-sided limits. At points of continuity the series converges to $x(t)$; at a jump it converges to the **midpoint** of the jump.

All waveforms met in practice — square, triangle, sawtooth, rectified sines, pulse trains, PWM signals, exponential segments — satisfy the Dirichlet conditions. Functions that violate them are mathematical constructions such as $\sin(1/t)$ near $t = 0$ (infinitely many extrema) or $\tan t$ (not integrable).

Proof sketch. The argument has three steps and is worth knowing because the same kernel reappears in the DFT and in antenna-array theory.

Step 1 — the Dirichlet kernel. Use $T_0 = 2\pi$. Substituting the coefficient integral into the partial sum and exchanging the finite sum with the integral,

$$S_N(t) = \sum_{n=-N}^{N} \left[\frac{1}{2\pi}\int_{-\pi}^{\pi} x(s) e^{-jns} ds\right] e^{jnt} = \frac{1}{2\pi}\int_{-\pi}^{\pi} x(s)\, D_N(t-s)\, ds,$$

where, summing the geometric series as in Ch. 3, §3.6,

$$D_N(u) = \sum_{n=-N}^{N} e^{jnu} = \frac{\sin\left((N+\tfrac{1}{2})u\right)}{\sin(u/2)}.$$

D_N is even, 2π-periodic, has $\frac{1}{2\pi}\int_{-\pi}^{\pi} D_N(u)\,du = 1$ (only the $n=0$ term survives), and has a tall central peak of height $2N+1$ and width about $2\pi/(N+\tfrac{1}{2})$ flanked by oscillating side lobes. So $S_N(t)$ is a weighted local average of x near t, with weights that become more sharply concentrated as N grows — but that also oscillate in sign.

Step 2 — split at the point. Changing variable to $u = t - s$ and using evenness of D_N,

$$S_N(t) - \frac{x(t^+)+x(t^-)}{2} = \frac{1}{2\pi}\int_0^{\pi} \Big[x(t+u) - x(t^+) + x(t-u) - x(t^-)\Big] D_N(u)\,du,$$

because $\frac{1}{2\pi}\int_0^{\pi} D_N = \frac{1}{2}$.

Step 3 — Riemann–Lebesgue. Write $D_N(u) = \sin\left((N+\tfrac{1}{2})u\right)/\sin(u/2)$. The function $g(u) = \big[x(t+u) - x(t^+) + x(t-u) - x(t^-)\big]/\sin(u/2)$ is integrable on $(0,\pi)$: near $u=0$ the bracket is $O(u)$ when x is piecewise smooth (one-sided derivatives exist), cancelling the $u/2$ in the denominator. The **Riemann–Lebesgue lemma** states that $\int_a^b g(u)\sin(\lambda u)\,du \to 0$ as $\lambda\to\infty$ for every integrable g — rapid oscillation averages any fixed integrable function to zero (proof: true for step functions by direct integration; approximate g by step functions). With $\lambda = N + \tfrac{1}{2}$ the right-hand side tends to zero. $\square$

The Riemann–Lebesgue lemma applied to the coefficient integral itself gives the companion fact $c_n \to 0$ as $|n|\to\infty$ for every absolutely integrable x.

Uniqueness

If two piecewise-continuous T_0-periodic functions have the same Fourier coefficients, they are equal at every point where both are continuous. In particular, a function whose coefficients are all zero vanishes except at isolated points.

Uniqueness is what justifies reading off coefficients from any convenient expansion (Worked Example 15.10) and solving for unknown waveforms by matching harmonics.

Uniform convergence

Pointwise convergence can be very uneven: near a jump, the number of terms needed for a given accuracy grows without bound as t approaches the jump. **Uniform convergence** means $\max_t |S_N(t) - x(t)| \to 0$: a single N achieves a given accuracy everywhere.

Uniform convergence

(a) If $\sum_n |c_n| < \infty$, the Fourier series converges absolutely and uniformly to a continuous function (Weierstrass M-test), which equals x wherever x is continuous.

(b) If x is continuous everywhere (including across the period boundary, $x(t_1 + T_0) = x(t_1)$) and piecewise smooth, then $\sum |c_n| < \infty$ and the series converges uniformly to x.

(c) If x has a jump, the convergence cannot be uniform near that jump: a uniform limit of continuous functions is continuous.

Proof of (b). Let x' have coefficients d_n. Integrating by parts, with the boundary terms cancelling by continuity and periodicity, $d_n = jn\omega_0 c_n$ (§15.7). By Cauchy–Schwarz and Bessel's inequality applied to x',

$$\sum_{n\neq 0} |c_n| = \sum_{n\neq 0} \frac{|d_n|}{|n|\omega_0} \le \frac{1}{\omega_0}\Big(\sum_{n\neq 0}\frac{1}{n^2}\Big)^{1/2}\Big(\sum_{n}|d_n|^2\Big)^{1/2} \le \frac{1}{\omega_0}\sqrt{\frac{\pi^2}{3}}\,\|x'\| < \infty. \quad \square$$

So the triangle wave, the full-wave rectified sine and the trapezoid converge uniformly; the square wave, sawtooth, pulse trains and the periodic exponential of Worked Example 15.8 do not.

Mean-square convergence

The weakest but most useful mode is convergence in the RMS sense, $\|x - S_N\| \to 0$. It ignores what happens at individual points and measures only the power of the error — exactly the quantity that matters for energy, noise and distortion.

Mean-square convergence (completeness)

If $\frac{1}{T_0}\int_{T_0}|x|^2 dt < \infty$, then

$$\lim_{N\to\infty} \frac{1}{T_0}\int_{T_0} \big|x(t) - S_N(t)\big|^2 dt = 0.$$

Conversely (Riesz–Fischer), every sequence with $\sum |c_n|^2 < \infty$ is the coefficient sequence of some square-integrable x.

Proof sketch. By the best-approximation theorem of §15.1, $\|x - S_N\| \le \|x - p\|$ for every trigonometric polynomial p of degree N. So it suffices that trigonometric polynomials can approximate x arbitrarily well in RMS. First approximate x in RMS by a continuous periodic function g (possible for any square-integrable x). Then approximate g *uniformly* — hence also in RMS — by the **Fejér (Cesàro) means**

$$\sigma_N(t) = \frac{S_0(t) + S_1(t) + \cdots + S_N(t)}{N+1} = \frac{1}{2\pi}\int_{-\pi}^{\pi} g(t-u)F_N(u)\,du, \qquad F_N(u) = \frac{1}{N+1}\Big[\frac{\sin\big((N+1)u/2\big)}{\sin(u/2)}\Big]^2.$$

The **Fejér kernel** F_N is *non-negative*, has unit mean, and concentrates at $u = 0$; a positive averaging kernel of this kind converges uniformly for continuous g (Fejér's theorem). The triangle inequality completes the argument. $\square$

Mean-square convergence is what makes Parseval's theorem (§15.8) true for *every* finite-power periodic signal, square wave included, and it is the natural setting for the Fourier-series solutions of PDEs in Ch. 13.

Mode	Measure of error	Holds when	Square wave?
pointwise	$\lvert x(t) - S_N(t)\rvert$ at each t	Dirichlet conditions (limit = midpoint at jumps)	yes, with midpoint at jumps
uniform	$\max_t \lvert x - S_N\rvert$	x continuous and piecewise smooth, or $\sum\lvert c_n\rvert < \infty$	no
mean-square	$\|x - S_N\|$	x square-integrable	yes

Pitfall — continuity alone is not enough

There exist continuous periodic functions whose Fourier series diverge at a point (du Bois-Reymond, 1873). Continuity guarantees mean-square convergence and Fejér-mean convergence, but pointwise convergence of the ordinary partial sums needs something more — piecewise smoothness, bounded variation (Dirichlet), or a Hölder condition. For engineering waveforms this is never an issue, but it explains why theorems are stated with conditions that look fussier than "continuous".

Worked Example 15.18 — Using pointwise convergence to sum series

Use the convergence theorem to evaluate (a) $1 - \frac{1}{3} + \frac{1}{5} - \cdots$, (b) $\sum_{k\ge1} \frac{1}{(2k-1)^2}$, (c) the value of the sawtooth series $\sum_{n\ge1} \frac{2(-1)^{n+1}}{n} \sin nt$ at $t = \pi$.

Solution. (a) The square wave ($A = 1$, $\omega_0 = 1$) is continuous at $t = \pi/2$ with value 1: $\frac{4}{\pi}\left(\sin\frac{\pi}{2} + \frac{1}{3}\sin\frac{3\pi}{2} + \frac{1}{5}\sin\frac{5\pi}{2} + \cdots\right) = \frac{4}{\pi}\left(1 - \frac{1}{3} + \frac{1}{5} - \cdots\right) = 1$, so the sum is $\pi/4$.

(b) The triangle wave of Worked Example 15.11 is continuous at $t = 0$ with value A: $\frac{8A}{\pi^2}\sum_{\text{odd } n}\frac{1}{n^2} = A$, so $\sum \frac{1}{(2k-1)^2} = \frac{\pi^2}{8} = 1.2337$.

(c) Every term contains $\sin n\pi = 0$, so the series sums to 0. The sawtooth jumps from π (left limit) to $-\pi$ (right limit) at $t = \pi$; the midpoint is 0. ✓ This is not an approximation failure: the series *converges*, to the midpoint.

Check. $4\sum_{k=0}^{10^6-1}\frac{(-1)^k}{2k+1} = 3.1415917$ (error 10^{-6}, the slow $1/N$ convergence of an alternating series). The first three terms of (b) give $1 + \frac{1}{9} + \frac{1}{25} = 1.1511$; the exact $\pi^2/8 = 1.2337$. ✓

The Gibbs phenomenon

Near a jump, the partial sums of a Fourier series overshoot. As N grows, the overshoot moves closer to the jump and becomes narrower, but its height does **not** decrease. This is the **Gibbs phenomenon** (observed by Michelson's harmonic analyser in 1898, explained by Gibbs in 1899, and in fact found by Wilbraham in 1848).

Derivation for the square wave. Take $A = 1$, $\omega_0 = 1$ and let $S_M(t) = \frac{4}{\pi}\sum_{k=1}^{M}\frac{\sin(2k-1)t}{2k-1}$ contain the first M odd harmonics (highest harmonic $N = 2M - 1$). Differentiate and sum the cosines with the formula of Ch. 3, §3.6:

$$S_M'(t) = \frac{4}{\pi}\sum_{k=1}^{M}\cos(2k-1)t = \frac{2}{\pi}\cdot\frac{\sin 2Mt}{\sin t}.$$

The first zero of S_M' to the right of the jump at $t = 0$ is at $t^* = \pi/(2M)$, the location of the first and largest maximum. Since $S_M(0) = 0$,

$$S_M(t^*) = \frac{2}{\pi}\int_0^{\pi/(2M)}\frac{\sin 2Mt}{\sin t}\,dt.$$

Substitute $u = 2Mt$: $S_M(t^*) = \frac{2}{\pi}\int_0^{\pi}\frac{\sin u}{2M\sin(u/2M)}\,du$. As $M \to \infty$, $2M\sin(u/2M) \to u$, so

$$\lim_{M\to\infty} S_M(t^*) = \frac{2}{\pi}\int_0^{\pi}\frac{\sin u}{u}\,du = \frac{2}{\pi}\,\mathrm{Si}(\pi) = 1.178980,$$

where $\mathrm{Si}(z) = \int_0^z \frac{\sin u}{u}du$ is the sine integral, $\mathrm{Si}(\pi) = 1.851937$.

Gibbs phenomenon

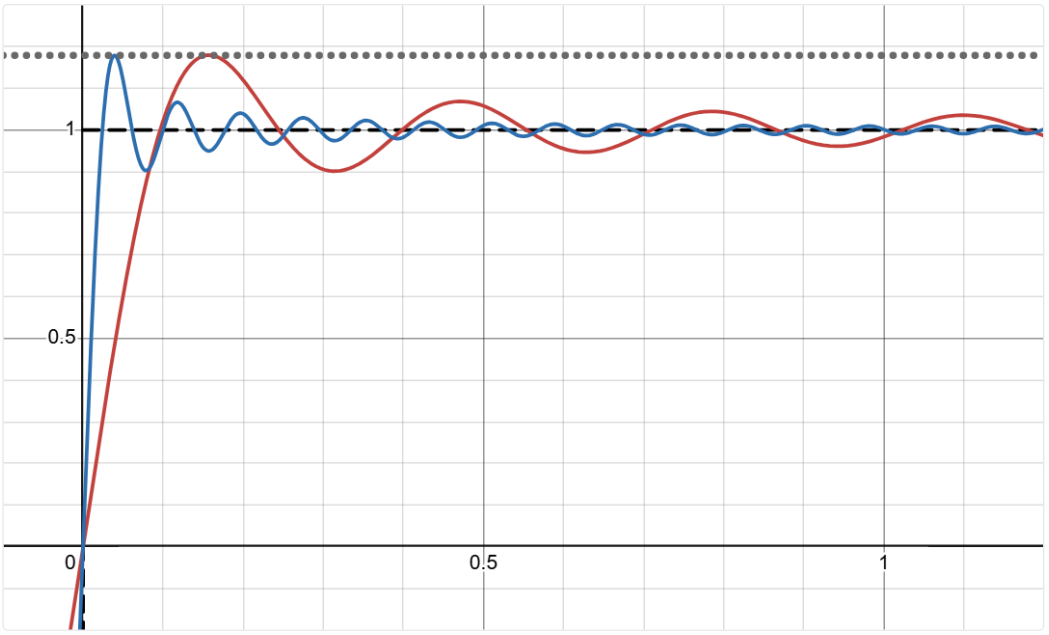
Let x be piecewise smooth with a jump of size $J = x(t_0^+) - x(t_0^-)$ at t_0. Then near t_0 the partial sums S_N overshoot $x(t_0^+)$ (and undershoot $x(t_0^-)$) by an amount tending to

$$J\left(\frac{1}{\pi}\,\mathrm{Si}(\pi) - \frac{1}{2}\right) = 0.08949\,J,$$

i.e. about **8.95% of the jump**, on each side. The overshoot occurs at a distance of order $T_0/(2N)$ from the jump, which tends to zero; its height does not.

The general case reduces to the square wave: subtract from x a multiple of a sawtooth or square wave with the same jump at t_0; the remainder is continuous near t_0 and its series converges uniformly there, so it contributes no overshoot.

Gibbs is fully consistent with the convergence theorems. At each *fixed* $t \neq t_0$ the overshoot eventually moves past t, so pointwise convergence holds; the overshoot region has width $O(1/N)$, so its contribution to the mean-square error tends to zero; but the *maximum* error never falls below about $0.09J$, so convergence is not uniform.



Zoom on the jump of the unit square wave at $t = 0$: partial sums with harmonics up to 19 (red) and up to 79 (blue). Increasing N squeezes the overshoot towards the jump but both peaks reach the same height, about 1.179 (dotted).

Worked Example 15.19 — Size and position of the Gibbs overshoot

A 1 kHz, ± 5 V square wave is passed through an ideal (brick-wall) low-pass filter that keeps harmonics up to 19 kHz and removes the rest. Find the peak output voltage and the time after each edge at which it occurs. Repeat for a 79 kHz cutoff.

Solution. The filter output is the partial sum with harmonics $1, 3, \dots, 19$, i.e. $M = 10$ odd harmonics. The peak is at $\theta^* = \pi/(2M) = \pi/20$ after the edge, i.e. $t^* = \frac{\theta^*}{\omega_0} = \frac{\pi/20}{2\pi \times 1000} = 25\ \mu$s. Its value for finite M: numerically $S_{10}(\pi/20) = 1.17981$, so the peak is $5 \times 1.1798 = 5.90$ V — an overshoot of 0.90 V, which is 9.0% of the 10 V jump.

With harmonics to 79 kHz ($M = 40$): $t^* = \frac{1}{4 \cdot 40 \cdot 1000} = 6.25\ \mu$s, and $S_{40}(t^*) = 1.17903$, peak 5.895 V. The overshoot has moved four times closer to the edge but is essentially the same size; the limit is $5 \times 1.17898 = 5.8949$ V.

Check. A brute-force search of S_M on a fine grid gives maxima 1.17981 at $t = 0.15708$ ($M = 10$) and 1.17903 at $t = 0.03927$ ($M = 40$), matching $\pi/20$ and $\pi/80$. ✓

Taming Gibbs: smoothing the partial sums

When a truncated series must be *used* — to synthesise a waveform, to design an FIR filter by truncating an ideal response, to display a reconstructed signal — the overshoot is usually unwanted. It comes from the abrupt truncation of the coefficients at $n = N$; tapering them removes it at the cost of a slower edge.

- **Fejér (Cesàro) means**: multiply c_n by $1 - \frac{|n|}{N+1}$. Since the Fejér kernel is non-negative, σ_N never leaves the range $[\min x, \max x]$: no overshoot at all, but a noticeably rounded edge.
- **Lanczos σ-factors**: multiply c_n by $\operatorname{sinc}\big(n/(N+1)\big)$. Equivalent to averaging S_N over a window one period of the highest harmonic wide; it reduces the overshoot to about 1% of the jump while keeping a reasonably sharp edge.

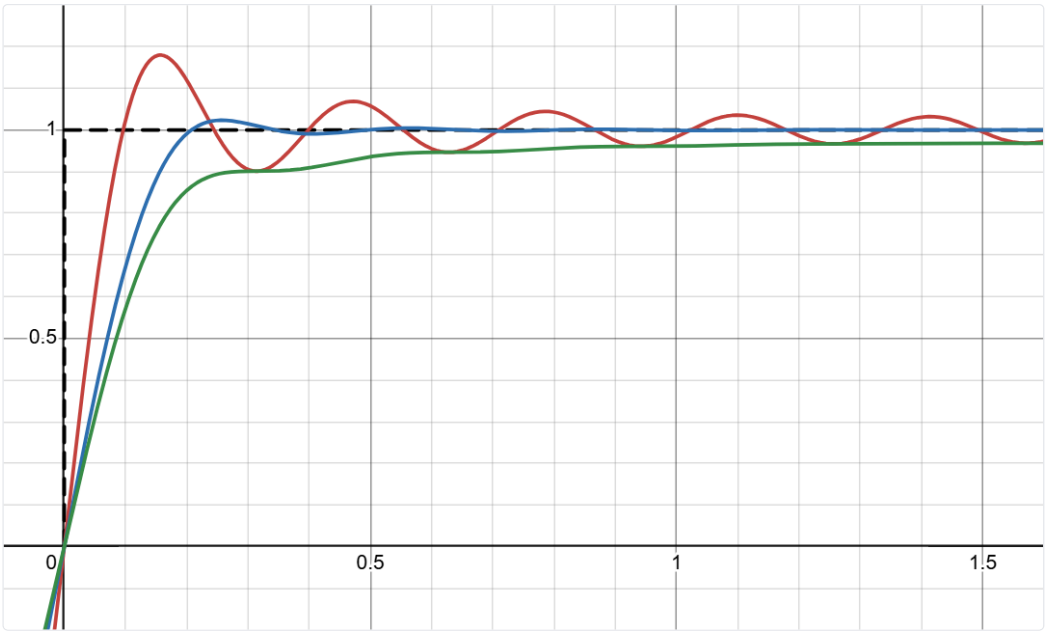
Worked Example 15.20 — Smoothed partial sums of a square wave

For the unit square wave with harmonics up to $N = 19$, compare the maximum of the raw partial sum with the Lanczos-smoothed and Fejér-smoothed sums.

Solution. The weights on harmonic n are 1, $\operatorname{sinc}(n/20)$ and $1 - n/20$ respectively; e.g. for $n = 19$ they are 1, 0.0524 and 0.05. Evaluating on a fine grid over a half-period, and measuring the 10%–90% rise time from the edge:

Partial sum	Maximum	Overshoot	10–90% rise (rad)
raw S_{19}	1.1798	18.0% of A (9.0% of jump)	0.074
Lanczos σ	1.0235	2.3% of A	0.143
Fejér	0.9682 (at $t = \pi/2$)	none	0.263

The Fejér sum stays below 1 everywhere, as the positivity of its kernel guarantees, but at the price of a rise time three and a half times that of the raw sum; the Lanczos sum is the usual compromise.



Harmonics up to 19 of the unit square wave: raw partial sum (red) with the Gibbs overshoot, Lanczos-smoothed (blue) with almost no overshoot, Fejér-smoothed (green) with none but a slower rise.

Engineering note — Gibbs in hardware

Any system with a sharp cutoff in frequency rings at sharp edges in time. The step response of an ideal low-pass filter overshoots by 8.95% for the same mathematical reason (Ch. 16); steep anti-aliasing filters and brick-wall digital filters show this ringing on square-wave tests; and FIR filters designed by truncating an ideal impulse response have ripples of about 9% (−21 dB) near the band edge whatever their length. Tapered windows (Hann, Hamming, Kaiser) are the filter-design version of the σ-factors above: they trade transition sharpness for suppressed ripple.

Pitfall — more terms do not remove the overshoot

It is tempting to blame a 9% overshoot in a reconstructed waveform on "too few harmonics". Adding harmonics only narrows it. If the overshoot matters, taper the coefficients (or, in hardware, use a filter with a gentler roll-off) — do not just raise N.

15.7 Fourier coefficients

Motivation

The Euler integrals of §15.2 always work, but integrating by parts over several pieces is slow and error-prone. Experienced analysts rarely do it. Instead they (i) build new series from known ones using the *properties* of the coefficient map — shifting, scaling, differentiating, integrating, multiplying; (ii) use the **jump method**, which reads the coefficients of any piecewise-polynomial waveform straight off the sizes and positions of its discontinuities; and (iii) compute coefficients numerically with the FFT when no formula is available, knowing exactly what error that introduces. This section develops all three, together with the result that ties them together: **the smoothness of a waveform determines how fast its coefficients decay**.

Properties of the coefficients

Write $x \leftrightarrow c_n$ and $y \leftrightarrow d_n$ for two signals with the same period T_0.

Fourier-coefficient properties

Operation	Signal	Coefficients
linearity	$\alpha x + \beta y$	$\alpha c_n + \beta d_n$
time shift	$x(t - t_0)$	$e^{-jn\omega_0 t_0} c_n$
frequency shift (modulation by a harmonic)	$e^{jm\omega_0 t} x(t)$	c_{n-m}
time reversal	$x(-t)$	c_{-n}
conjugation	$x^*(t)$	c^*_{-n}
time scaling ($a > 0$; period T_0/a)	$x(at)$	c_n (at frequencies $na\omega_0$)
differentiation	$x'(t)$	$jn\omega_0\, c_n$
integration (requires $c_0 = 0$)	$\int^t x(\tau)\, d\tau$	$\frac{c_n}{jn\omega_0}$ ($n \neq 0$), plus a constant
periodic convolution	$\frac{1}{T_0}\int_{T_0} x(\tau) y(t-\tau)\, d\tau$	$c_n d_n$
multiplication	$x(t)\, y(t)$	$\sum_{m=-\infty}^{\infty} c_m d_{n-m}$

Proofs. The shift, modulation, reversal and conjugation rules are substitutions in the coefficient integral. *Differentiation:* if x is continuous and periodic with piecewise-continuous x', integrate by parts:

$$\frac{1}{T_0}\int_{T_0} x'(t) e^{-jn\omega_0 t}\, dt = \frac{1}{T_0}\Big[x(t) e^{-jn\omega_0 t}\Big]_{t_1}^{t_1+T_0} + \frac{jn\omega_0}{T_0}\int_{T_0} x(t) e^{-jn\omega_0 t}\, dt = jn\omega_0 c_n,$$

the boundary term vanishing by periodicity. *Integration* is differentiation read backwards; the condition $c_0 = 0$ is needed because the integral of a signal with non-zero mean grows linearly and is not periodic. *Periodic convolution:* substitute the series of y and use orthogonality,

$$\frac{1}{T_0}\int_{T_0} x(\tau) \sum_m d_m e^{jm\omega_0(t-\tau)}\, d\tau = \sum_m d_m e^{jm\omega_0 t}\, \frac{1}{T_0}\int_{T_0} x(\tau) e^{-jm\omega_0 \tau}\, d\tau = \sum_m c_m d_m e^{jm\omega_0 t}.$$

Multiplication: the product of $\sum_m c_m e^{jm\omega_0 t}$ and $\sum_k d_k e^{jk\omega_0 t}$ collects, at frequency $n\omega_0$, all pairs with $m + k = n$. □

The last two rules are the duality at the heart of Fourier analysis: **convolution in time ↔ multiplication of coefficients; multiplication in time ↔ (discrete) convolution of coefficients.** Chapter 16 states the same pair for the transform. The multiplication rule explains why a nonlinearity spreads the spectrum: squaring a signal convolves its line spectrum with itself, producing lines at all sums and differences of the original frequencies (Ch. 3, §3.7).

Term-by-term differentiation and integration

(a) **Integration is always allowed.** If x is piecewise continuous, the Fourier series of x — whether or not it converges — may be integrated term by term between any limits, and the resulting series converges (uniformly) to the integral of x.

(b) **Differentiation needs care.** If x is continuous, periodic and piecewise smooth (in particular, no jumps, *including at the period boundary*), then the term-by-term derivative of its series is the Fourier series of x', converging to $\frac{1}{2}[x'(t^+) + x'(t^-)]$ wherever x' satisfies the Dirichlet conditions.

Integration improves convergence (coefficients gain a factor $1/n$); differentiation worsens it (coefficients gain a factor n). Differentiating the square-wave series term by term gives $\frac{4A\omega_0}{\pi}\sum_{\text{odd}} \cos n\omega_0 t$, which diverges at every t — the derivative of a function with jumps contains impulses, which no ordinary function represents. The jump method below makes this precise.

Worked Example 15.21 — Triangle wave from the square wave by integration

Obtain the series of the triangle wave of Worked Example 15.11 by integrating the square wave, and confirm the result.

Solution. Let $s(t)$ be the square wave of amplitude A (Worked Example 15.3), $s(t) = \frac{4A}{\pi}\sum_{\text{odd } n} \frac{\sin n\omega_0 t}{n}$. Its integral from 0 is a triangle: $\int_0^t s = At$ for $0 < t < T_0/2$, rising from 0 to $AT_0/2$, then falling back — a triangle of peak-to-peak $AT_0/2$ with mean $AT_0/4$. Integrating term by term:

$$\int_0^t s(\tau)\, d\tau = \frac{4A}{\pi\omega_0}\sum_{\text{odd } n}\frac{1-\cos n\omega_0 t}{n^2} = \frac{4A}{\pi\omega_0}\cdot\frac{\pi^2}{8} - \frac{4A}{\pi\omega_0}\sum_{\text{odd } n}\frac{\cos n\omega_0 t}{n^2},$$

using $\sum_{\text{odd}} 1/n^2 = \pi^2/8$ (Worked Example 15.18). The constant is $\frac{A\pi}{2\omega_0} = \frac{AT_0}{4}$, the mean. ✓ The triangle of Worked Example 15.11 has peak B at $t = 0$ and is $B - \frac{4B}{AT_0}\int_0^t s$ (scaled and inverted so that it falls from B to $-B$). Its cosine coefficients are therefore $\frac{4B}{AT_0}\cdot\frac{4A}{\pi\omega_0 n^2} = \frac{16B}{\pi T_0\omega_0 n^2} = \frac{8B}{\pi^2 n^2}$ for odd n. ✓

Smoothness and the rate of decay

Decay of Fourier coefficients

Let x be T_0-periodic and let $k \geq 0$ be an integer.

- If x has a jump discontinuity, then c_n decays like $1/n$ and no faster (as $|n| \to \infty$, typically $|c_n| \approx C/|n|$ along a subsequence).
- If $x, x', \ldots, x^{(k-1)}$ are continuous (as periodic functions) and $x^{(k)}$ is piecewise smooth with jumps, then $|c_n| = O\big(1/|n|^{k+1}\big)$, with the leading term set by the jumps of $x^{(k)}$.
- If x has continuous derivatives of every order, c_n decays faster than any power of $1/n$; if x is analytic (Ch. 18), c_n decays exponentially.

Proof. Apply the differentiation rule k times: $c_n = \frac{\text{coefficient of } x^{(k)}}{(jn\omega_0)^k}$, and the coefficients of the piecewise-smooth $x^{(k)}$ with jumps are $O(1/n)$ by one more integration by parts (the boundary terms at the jumps give exactly $1/n$; the remaining integral is $o(1/n)$ by Riemann–Lebesgue). □

Waveform	Smoothness	Decay of $\lvert c_n\rvert$	Coefficients
square, sawtooth, pulse train	jumps in x	$1/n$	square: $b_n = 4A/(n\pi)$
triangle, trapezoid, rectified sine	x continuous, jumps in x'	$1/n^2$	triangle: $a_n = 8A/(\pi^2 n^2)$
parabolic wave t^2 on $(-\pi,\pi)$	x continuous, jumps in x' at $\pm\pi$	$1/n^2$	$a_n = 4(-1)^n/n^2$
cubic wave $t(\pi^2 - t^2)$ on $(-\pi,\pi)$	x, x' continuous, jumps in x''	$1/n^3$	$b_n = 12(-1)^{n+1}/n^3$
$e^{\cos t}$	analytic	exponential	$c_n = I_n(1)$ (modified Bessel)

(The parabolic wave is continuous at $\pm\pi$ but its slope jumps from 2π to -2π there. The cubic wave vanishes at $\pm\pi$ and has slope $-2\pi^2$ at both ends, so it and its derivative are continuous; its second derivative $-6t$ jumps from -6π to 6π.)

This theorem has direct engineering consequences. **Edges cost bandwidth.** A clock with ideal vertical edges has harmonics falling at only 20 dB/decade, so its spectrum extends far beyond the clock frequency and causes interference; rounding the edges (finite rise time) turns the decay into 40 dB/decade above a corner frequency of about $1/(\pi t_r)$. Conversely, the number of harmonics needed to reproduce a waveform to a given accuracy can be estimated before computing anything.

The jump method

For piecewise-polynomial waveforms, the coefficients can be read off from the jumps. Suppose that over one period x is piecewise smooth with jumps $J_i = x(t_i^+) - x(t_i^-)$ at points t_i. Then, in the generalised sense of Ch. 14 and Ch. 16, x' is its ordinary (piecewise) derivative x'_{reg} plus impulses:

$$x'(t) = x'_{\text{reg}}(t) + \sum_i J_i\,\delta_{T_0}(t - t_i), \qquad \delta_{T_0}(t) = \sum_k \delta(t - kT_0),$$

and the periodic impulse train has coefficients $\frac{1}{T_0}\int_{T_0}\delta(t-t_i)e^{-jn\omega_0 t}dt = \frac{1}{T_0}e^{-jn\omega_0 t_i}$. Applying $c_n(x') = jn\omega_0 c_n(x)$:

Jump formula

For $n \neq 0$,

$$c_n = \frac{1}{jn\omega_0}\Big[\frac{1}{T_0}\sum_i J_i\,e^{-jn\omega_0 t_i} + c_n(x'_{\text{reg}})\Big] = \frac{1}{jn\omega_0 T_0}\sum_i J_i e^{-jn\omega_0 t_i} + \frac{1}{jn\omega_0}\,c_n(x'_{\text{reg}}).$$

Apply the same formula to x'_{reg}, using the jumps J_i' of the derivative, and so on. For a piecewise-polynomial waveform the process terminates, because some derivative is piecewise constant and its regular derivative is zero.

The same result follows without impulses by integrating by parts piece by piece; the impulses simply keep the bookkeeping clean.

Worked Example 15.22 — Square wave and sawtooth by the jump method

(a) The square wave of Worked Example 15.3 has jumps $+2A$ at $t = 0$ and $-2A$ at $t = T_0/2$, and is constant in between ($x'_{\text{reg}} = 0$). So

$$c_n = \frac{1}{jn\omega_0 T_0}\big(2A - 2Ae^{-jn\pi}\big) = \frac{2A\,(1-(-1)^n)}{j2\pi n} = \frac{2A}{j\pi n}\ (n \text{ odd}),\ 0\ (n \text{ even}),$$

and $b_n = -2\,\mathrm{Im}\,c_n = \frac{4A}{\pi n}$. ✓

(b) The sawtooth $x = t$ on $(-\pi,\pi)$ has one jump, -2π at $t = \pi$, and $x'_{\text{reg}} = 1$, whose coefficients vanish for $n \neq 0$. So $c_n = \frac{1}{jn\cdot 2\pi}(-2\pi)e^{-jn\pi} = \frac{-(-1)^n}{jn} = \frac{j(-1)^n}{n}$ — Worked Example 15.10(b) in one line. ✓

Worked Example 15.23 — Trapezoidal wave

A trapezoidal voltage (a square wave with finite rise time) rises linearly from 0 to A over $0 < \theta < \alpha$, stays at A until $\theta = \pi - \alpha$, falls linearly to 0 at $\theta = \pi$, and repeats with opposite sign on $(\pi, 2\pi)$ ($\theta = \omega_0 t$, $0 < \alpha \le \pi/2$). Find its coefficients by the jump method and evaluate them for $\alpha = \pi/6$ (30° ramps).

Solution. x is continuous, so it has no jumps. Its derivative (with respect to θ, so $\omega_0 = 1$, $T_0 = 2\pi$) is piecewise constant: $+A/\alpha$ on $(-\alpha, \alpha)$, $-A/\alpha$ on $(\pi - \alpha, \pi + \alpha)$, and 0 elsewhere. The jumps of x' are

position	$-\alpha$	α	$\pi - \alpha$	$\pi + \alpha$
jump J_i'	$+A/\alpha$	$-A/\alpha$	$-A/\alpha$	$+A/\alpha$

and $x''_{\text{reg}} = 0$. Applying the jump formula twice (x has no jumps, so only the second level contributes):

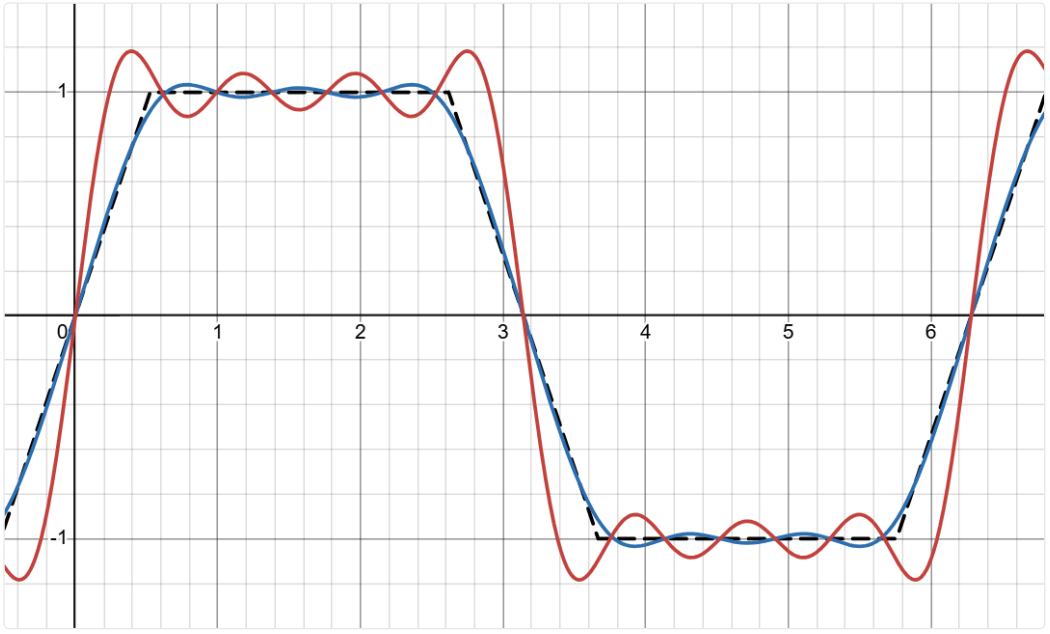
$$c_n = \frac{1}{jn} \cdot \frac{1}{jn\,2\pi} \sum_i J_i' e^{-jn\theta_i} = -\frac{A}{2\pi\alpha n^2}\left[e^{jn\alpha} - e^{-jn\alpha} - e^{-jn(\pi-\alpha)} + e^{-jn(\pi+\alpha)}\right].$$

The bracket is $2j\sin n\alpha - (-1)^n\left(e^{jn\alpha} - e^{-jn\alpha}\right) = 2j\sin n\alpha\,\left(1 - (-1)^n\right)$. So $c_n = 0$ for even n (half-wave symmetry ✓) and, for odd n, $c_n = -\frac{2jA\sin n\alpha}{\pi\alpha n^2}$, giving

$$b_n = -2\,\mathrm{Im}\,c_n = \frac{4A}{\pi\alpha} \cdot \frac{\sin n\alpha}{n^2} \quad (n \text{ odd}), \qquad a_n = 0.$$

With $\alpha = \pi/6$: $b_1 = 1.2159A$, $b_3 = 0.2702A$, $b_5 = 0.0486A$, $b_7 = -0.0248A$.

Checks. (i) Numerical quadrature reproduces all four values to 10^{-12}. ✓ (ii) As $\alpha \to 0$, $\frac{\sin n\alpha}{\alpha} \to n$ and $b_n \to \frac{4A}{\pi n}$ — the square wave. ✓ (iii) The ramps make the waveform continuous, so $b_n = O(1/n^2)$: compared with the square wave ($b_7 = 0.1819A$) the 7th harmonic is 7 times smaller. Writing $b_n = \frac{4A}{\pi n}\,\mathrm{sinc}\left(\frac{n\alpha}{\pi}\right)$ shows the trapezoid's spectrum as the square-wave spectrum multiplied by a sinc envelope whose first null is at $n = \pi/\alpha$: the frequency-domain picture of "finite rise time costs less bandwidth".



The trapezoidal wave with 30° ramps (dashed) and its partial sum through the 7th harmonic (blue), against the square-wave partial sum through the 7th harmonic (red). With the same four harmonics the trapezoid is reproduced closely because its coefficients fall as $1/n^2$.

Computing coefficients numerically: the DFT

When x is known only by samples — a measured current, a simulation output — the coefficient integral is approximated by a sum. Take N equally spaced samples $x_k = x(kT_0/N)$, $k = 0, \dots, N-1$, and apply the rectangle (equivalently, for periodic integrands, trapezoidal) rule:

$$\tilde{c}_n = \frac{1}{N}\sum_{k=0}^{N-1} x_k\, e^{-j2\pi nk/N}.$$

This is the **discrete Fourier transform** divided by N, computed by the FFT in $O(N \log N)$ operations (Ch. 24). Its error is described exactly by the following identity.

Aliasing of Fourier coefficients

If the Fourier series of x converges at the sample points (with the midpoint value used at any jump), then

$$\tilde{c}_n = \sum_{m=-\infty}^{\infty} c_{n+mN}.$$

In particular $\tilde{c}_n = c_n$ exactly for $|n| < N/2$ if x is band-limited with $c_n = 0$ for $|n| \ge N/2$.

Proof. Substitute $x_k = \sum_m c_m e^{j2\pi mk/N}$ and use $\frac{1}{N}\sum_{k=0}^{N-1} e^{j2\pi(m-n)k/N} = 1$ if $m \equiv n \pmod N$ and 0 otherwise (a finite geometric series). $\square$

Each computed coefficient is contaminated by every true coefficient whose index differs from it by a multiple of N — the coefficient-domain version of aliasing (Ch. 16, §16.8). The error therefore depends on how fast c_n decays: for a smooth periodic waveform it is tiny even for modest N (the trapezoidal rule is *spectrally accurate* for smooth periodic integrands), while for a waveform with jumps it is in general $O(1/N)$ (improving to $O(1/N^2)$ for a fixed harmonic when every jump falls on a sample instant and the midpoint value is used there).

Worked Example 15.24 — FFT estimate of sawtooth coefficients

Sample the sawtooth $x = t$ on $(-\pi, \pi)$ at $N = 16$ points $t_k = -\pi + 2\pi k/N$ (using the midpoint value 0 at the jump, $k = 0$), compute $\tilde{c}_1$ and $\tilde{c}_4$, and compare with the exact $c_n = j(-1)^n/n$. Repeat with $N = 64$.

Solution. Exact: $c_1 = -j$, $c_4 = 0.25j$. The FFT (with the phase factor $e^{jn\pi}$ that accounts for the samples starting at $-\pi$ instead of 0) gives

N	$\tilde{c}_1$	error	$\tilde{c}_4$	error
16	$-0.98712j$	1.3%	$0.19635j$	21%
64	$-0.99920j$	0.08%	$0.24678j$	1.3%

Explanation by the aliasing identity. For $N = 16$, $\tilde{c}_4 = \sum_m c_{4+16m} = j\sum_m \frac{1}{4+16m} = \frac{j}{16}\sum_m \frac{1}{m+\frac{1}{4}} = \frac{j}{16}\,\pi\cot\frac{\pi}{4} = \frac{j\pi}{16} = 0.19635j$, using the partial-fraction expansion $\pi\cot\pi z = \sum_m \frac{1}{z+m}$ (summed symmetrically; Ch. 18). ✓ For $N = 64$ the same formula gives $\frac{\pi}{64}\cot\frac{\pi}{16} = 0.24678$. ✓ The harmonic at a quarter of the sampling rate is badly aliased because the $1/n$ tail of the sawtooth is heavy, and the error of a fixed harmonic falls only like $1/N^2$ (here $21\% \to 1.3\%$ for a fourfold increase in N). A practical rule: to measure harmonics up to $n_{\max}$ to within a few per cent when the waveform has jumps, sample at many times $2n_{\max}$ per period, or low-pass filter before sampling.

Pitfall — coefficients of a waveform with jumps need an explicit midpoint

When sampling a waveform that has a jump exactly at a sample instant, use the midpoint value. Using either one-sided value adds a spurious constant $\pm J/(2N)$ to *every* $\tilde{c}_n$ — an error that does not decrease with n and that looks like a broadband noise floor.

Worked Example 15.25 — Full-wave rectified sine by the multiplication rule

Obtain the Fourier series of $|\sin t|$ from those of $\sin t$ and the square wave $s(t) = \operatorname{sgn}(\sin t)$, using $|\sin t| = \sin t \cdot s(t)$.

Solution. $\sin t$ has $d_1 = \frac{1}{2j}$, $d_{-1} = -\frac{1}{2j}$. The square wave ($A = 1$) has $c_m = \frac{2}{j\pi m}$ for odd m, zero for even m. By the multiplication rule the product has

$$e_n = \sum_m c_m d_{n-m} = c_{n-1}d_1 + c_{n+1}d_{-1} = \frac{1}{2j}\left(c_{n-1} - c_{n+1}\right).$$

For odd n, $n \pm 1$ is even, so $e_n = 0$. For even n, $e_n = \frac{1}{2j}\cdot\frac{2}{j\pi}\left(\frac{1}{n-1} - \frac{1}{n+1}\right) = -\frac{1}{\pi}\cdot\frac{2}{n^2-1}$. In particular $e_0 = \frac{2}{\pi}$. Then $a_n = 2e_n$:

$$|\sin t| = \frac{2}{\pi} - \frac{4}{\pi}\sum_{k=1}^{\infty}\frac{\cos 2kt}{4k^2-1}.$$

Check. This is twice the even-harmonic part of Worked Example 15.5, as $|\sin t| = 2\times$(half-wave rectified) $- \sin t$ requires. ✓ At $t = 0$: $\sum_k \frac{1}{4k^2-1} = \frac{1}{2}$ (telescoping), so the series gives $\frac{2}{\pi} - \frac{4}{\pi}\cdot\frac{1}{2} = 0 = |\sin 0|$. ✓

Engineering note — estimating EMI from edge rates

The jump formula applied to a trapezoidal clock of period T_0, duty 50% and rise time t_r gives the familiar EMC envelope: flat to f_0, -20 dB/decade from f_0 to about $1/(\pi\tau)$ (with τ the pulse width), and -40 dB/decade above $1/(\pi t_r)$. Slowing edges from 1 ns to 5 ns moves the second corner from 318 MHz to 64 MHz and reduces the harmonic amplitudes above 318 MHz by a factor 5 (14 dB), with a smaller reduction between the two corners — without changing the logic function.

15.8 Power and energy relationships

Motivation

The most frequently used result about Fourier series is not a formula for coefficients but a statement about *power*: the average power of a periodic signal equals the sum of the powers of its harmonics. It underlies every RMS meter calibrated for distorted waveforms, every harmonic-distortion specification, the definition of power factor in non-sinusoidal circuits, and the estimate of how much bandwidth a signal really needs. Chapter 3 (§3.7) proved it for a finite sum of harmonics; here we prove it for every finite-power periodic signal, extend it to products of two different signals (voltage and current), and use it to define and compute distortion measures.

Power signals

A periodic signal does not decay, so its total energy $\int_{-\infty}^{\infty}|x|^2dt$ is infinite. The meaningful quantities are the **energy per period** $E_{T_0} = \int_{T_0}|x|^2dt$ and the **average (normalised) power**

$$P_x = \frac{1}{T_0}\int_{T_0}|x(t)|^2dt = X_{\text{rms}}^2,$$

the power the signal would dissipate in a 1 Ω resistor if it were a voltage or a current. Periodic signals are **power signals**; the pulses and transients of Ch. 16, with finite total energy, are **energy signals**. For a voltage v across a resistance R the actual power is V_{rms}^2/R.

Parseval's theorem

Parseval's theorem for Fourier series

If x is T_0-periodic with $\int_{T_0}|x|^2dt<\infty$, then

$$\frac{1}{T_0}\int_{T_0}|x(t)|^2dt=\sum_{n=-\infty}^{\infty}|c_n|^2.$$

For real x in trigonometric and amplitude–phase form,

$$\frac{1}{T_0}\int_{T_0}x^2(t)\,dt=\left(\frac{a_0}{2}\right)^2+\frac{1}{2}\sum_{n=1}^{\infty}\left(a_n^2+b_n^2\right)=A_0^2+\sum_{n=1}^{\infty}\frac{A_n^2}{2}=A_0^2+\sum_{n=1}^{\infty}X_{n,\text{rms}}^2,$$

where $X_{n,\text{rms}}=A_n/\sqrt{2}$ is the RMS value of the nth harmonic.

Proof. By the best-approximation theorem (§15.1), $\|x-S_N\|^2=\|x\|^2-\sum_{|n|\le N}|c_n|^2$. By the mean-square convergence theorem (§15.6), the left side tends to 0 as $N\to\infty$. The trigonometric form follows from $|c_n|^2+|c_{-n}|^2=2|c_n|^2=\frac{A_n^2}{2}=\frac{a_n^2+b_n^2}{2}$ for real x. □

In words: **the mean-square value is the sum of the mean-square values of the harmonics, whatever their phases.** The harmonics are orthogonal "vectors" and Parseval's theorem is Pythagoras' theorem in the function space. It is a statement about power only: the peak value, the crest factor, and how hard an amplifier clips all depend on the phases.

The squared magnitudes $|c_n|^2$ form the **power spectrum**; the partial sums of Parseval give the **cumulative power**, and the error of a truncated series is

$$\|x-S_N\|^2=\sum_{|n|>N}|c_n|^2,$$

the power in the discarded harmonics. This is how one decides how many harmonics, or how much bandwidth, a waveform "needs".

Generalised Parseval (power of a product)

If $x\leftrightarrow c_n$ and $y\leftrightarrow d_n$ are T_0-periodic and square-integrable, then

$$\frac{1}{T_0}\int_{T_0}x(t)\,y^*(t)\,dt=\sum_{n=-\infty}^{\infty}c_n\,d_n^*.$$

For a real voltage $v=V_0+\sum_{n\ge1}V_n\cos(n\omega_0t+\alpha_n)$ and current $i=I_0+\sum_{n\ge1}I_n\cos(n\omega_0t+\beta_n)$ (peak amplitudes), the **average power** is

$$P=\frac{1}{T_0}\int_{T_0}v\,i\,dt=V_0I_0+\sum_{n=1}^{\infty}\frac{V_nI_n}{2}\cos(\alpha_n-\beta_n)=V_0I_0+\sum_{n=1}^{\infty}V_{n,\text{rms}}I_{n,\text{rms}}\cos\theta_n.$$

Proof. Apply Parseval to $x+y$ and $x+jy$ and use polarisation, or argue directly: the cross terms $c_md_n^*$ with $m\ne n$ average to zero by orthogonality. For the power formula take $c_n=\frac{V_n}{2}e^{j\alpha_n}$, $d_n=\frac{I_n}{2}e^{j\beta_n}$ and add the $\pm n$ terms: $2\,\text{Re}\{c_nd_n^*\}=\frac{V_nI_n}{2}\cos(\alpha_n-\beta_n)$. □

Only voltage and current harmonics of the same frequency transfer average power. A sinusoidal supply voltage delivers power only through the fundamental of the current; the current harmonics circulate without net energy transfer, yet they contribute to the RMS current and hence to losses in every conductor they pass through.

Distortion measures

Total harmonic distortion and related measures

For a periodic signal with zero mean, harmonic RMS values $X_1,X_2,\dots$ and total RMS value $X=\sqrt{\sum_{n\ge1}X_n^2}$:

- **THD (fundamental-referenced)**, the standard power-engineering definition (IEEE 519, IEC 61000):

$$\text{THD}_F=\frac{\sqrt{\sum_{n\ge2}X_n^2}}{X_1}=\sqrt{\frac{X^2}{X_1^2}-1};$$

- **THD (RMS-referenced)**, used by some audio analysers ("THD-R", or THD+N when noise is included):

$$\text{THD}_R=\frac{\sqrt{\sum_{n\ge2}X_n^2}}{X}=\frac{\text{THD}_F}{\sqrt{1+\text{THD}_F^2}};$$

- **distortion factor** (fundamental fraction) $X_1/X=1/\sqrt{1+\text{THD}_F^2}$;
- **crest factor** X_{peak}/X (equal to $\sqrt{2}$ for a sinusoid);
- **form factor** $X/\overline{|x|}$ (RMS over rectified mean; $\pi/(2\sqrt{2})=1.1107$ for a sinusoid).

THD_F can exceed 100%; THD_R cannot. For small distortion they agree. Parseval's theorem lets THD be computed from the total RMS value and the fundamental alone, without summing the infinite series of harmonics — the method used in all the examples below. A DC component, if present, is excluded from both numerator and denominator.

Power factor with a sinusoidal voltage. If v is a pure sinusoid at ω_0 and i is distorted, then $P = V I_1 \cos\theta_1$, with θ_1 the phase angle between v and the fundamental of i, while the apparent power is $S = VI$ (RMS values). Hence

$$\mathrm{PF} = \frac{P}{S} = \underbrace{\frac{I_1}{I}}_{\text{distortion factor}} \times \underbrace{\cos\theta_1}_{\text{displacement factor}} = \frac{\cos\theta_1}{\sqrt{1+\mathrm{THD}_i^2}}.$$

A load can have unity displacement factor and still a poor power factor.

Worked Example 15.26 — Summing series with Parseval

Use Parseval's theorem to evaluate $\sum_{n\ge1}\frac{1}{n^2}$, $\sum_{n\ge1}\frac{1}{n^4}$ and $\sum_{k\ge1}\frac{1}{(4k^2-1)^2}$.

Solution. (a) Sawtooth $x = t$ on $(-\pi,\pi)$: $b_n = \frac{2(-1)^{n+1}}{n}$, mean square $\frac{\pi^2}{3}$. Parseval: $\frac12\sum\frac{4}{n^2} = \frac{\pi^2}{3}$, so $\sum\frac{1}{n^2} = \frac{\pi^2}{6} = 1.644934$.

(b) Triangle of peak 1: $a_n = \frac{8}{\pi^2 n^2}$ (odd n), mean square $\frac13$ (a linear ramp from -1 to 1). Parseval: $\frac12\sum_{\text{odd}}\frac{64}{\pi^4 n^4} = \frac13$, so $\sum_{\text{odd}}\frac{1}{n^4} = \frac{\pi^4}{96}$. The even terms contribute $\sum\frac{1}{(2m)^4} = \frac{1}{16}\sum\frac{1}{m^4}$, so $\sum\frac{1}{n^4}\left(1-\frac{1}{16}\right) = \frac{\pi^4}{96}$ and $\sum\frac{1}{n^4} = \frac{\pi^4}{90} = 1.082323$.

(c) Full-wave rectified sine $|\sin t|$: mean square $\frac12$, DC $\frac{2}{\pi}$, $a_{2k} = -\frac{4}{\pi(4k^2-1)}$. Parseval: $\frac{4}{\pi^2} + \frac12\sum_k\frac{16}{\pi^2(4k^2-1)^2} = \frac12$, so $\sum_k\frac{1}{(4k^2-1)^2} = \frac{\pi^2}{16}\left(\frac12 - \frac{4}{\pi^2}\right) = \frac{\pi^2-8}{16} = 0.116850$.

Check. Direct summation to 10^6 terms: 1.644933, 1.082323, 0.116850. ✓

Worked Example 15.27 — THD of standard waveforms

Compute THD_F for the square wave, triangle, sawtooth, six-step (120° quasi-square) waveform of Worked Example 15.12 and the 30° trapezoid of Worked Example 15.23.

Solution. In each case compute the total mean square directly from the waveform and the fundamental's mean square $A_1^2/2$, then $\mathrm{THD}_F = \sqrt{X^2/X_1^2 - 1}$.

- *Square wave* ($\pm A$): $X^2 = A^2$; $X_1^2 = \frac12\left(\frac{4A}{\pi}\right)^2 = \frac{8A^2}{\pi^2}$. THD $= \sqrt{\frac{\pi^2}{8}-1} = 48.34\%$.
- *Triangle* (peak A): $X^2 = \frac{A^2}{3}$; $X_1^2 = \frac12\cdot\frac{64A^2}{\pi^4}$. THD $= \sqrt{\frac{\pi^4}{96}-1} = 12.12\%$.
- *Sawtooth* (t on $(-\pi,\pi)$): $X^2 = \frac{\pi^2}{3}$; $X_1^2 = \frac{2^2}{2} = 2$. THD $= \sqrt{\frac{\pi^2}{6}-1} = 80.31\%$.
- *Six-step*: the waveform is $\pm V$ for $\frac23$ of the period, so $X^2 = \frac23 V^2$; $b_1 = \frac{2\sqrt3}{\pi}V$, $X_1^2 = \frac{6}{\pi^2}V^2$. THD $= \sqrt{\frac{2\pi^2}{18}-1} = \sqrt{\frac{\pi^2}{9}-1} = 31.08\%$.
- *Trapezoid* ($\alpha = \pi/6$, peak A): each ramp of duration α contributes mean square $\frac13 A^2$, so over a half-period $X^2 = \frac{(\pi-2\alpha)A^2 + 2\alpha A^2/3}{\pi} = \left(1-\frac{4\alpha}{3\pi}\right)A^2 = 0.7778A^2$; $X_1^2 = \frac12(1.2159A)^2 = 0.7392A^2$. THD $= 22.86\%$.

Waveform	X/A	X_1/A	THD_F	THD_R	Crest factor
sine	0.7071	0.7071	0	0	1.414
triangle	0.5774	0.5732	12.1%	12.0%	1.732
trapezoid (30°)	0.8819	0.8597	22.9%	22.3%	1.134
six-step (120°)	0.8165	0.7797	31.1%	29.7%	1.225
square	1	0.9003	48.3%	43.5%	1
sawtooth ($A=\pi$)	0.5774	0.4502	80.3%	62.6%	1.732

Check. Summing $b_n^2/2$ numerically for the six-step wave over $n \le 20\,001$ and comparing with $\frac23 V^2$ reproduces 31.08%. ✓ Note how THD tracks smoothness (§15.7): jumps give large THD, kinks small THD.

Worked Example 15.28 — Power in a circuit with harmonics

A supply voltage $v(t) = 325\cos\omega_0 t + 30\cos 3\omega_0 t$ V drives a nonlinear load drawing $i(t) = 10\cos(\omega_0 t - 30^\circ) + 4\cos(3\omega_0 t - 60^\circ) + 2\cos(5\omega_0 t + 20^\circ)$ A. Find the RMS voltage and current, the average power, the apparent power, the power factor, and the voltage and current THD.

Solution. RMS values by Parseval: $V = \sqrt{\frac{325^2+30^2}{2}} = 230.79$ V, $\quad I = \sqrt{\frac{10^2+4^2+2^2}{2}} = \sqrt{60} = 7.746$ A.

Average power — only matching harmonics contribute; the 5th harmonic current has no 5th harmonic voltage to work against:

$$P = \frac{325\times10}{2}\cos 30^\circ + \frac{30\times4}{2}\cos 60^\circ + 0 = 1407.3 + 30.0 = 1437.3\ \text{W}.$$

Apparent power $S = VI = 1787.7$ VA; power factor $P/S = 0.804$.

$\mathrm{THD}_v = 30/325 = 9.23\%$; $\mathrm{THD}_i = \sqrt{4^2+2^2}/10 = 44.7\%$.

Interpretation. The fundamental displacement factor alone is $\cos 30^\circ = 0.866$; harmonics pull the true power factor down to 0.804. The third-harmonic product contributes a small amount of *real* power (30 W) because the supply itself is distorted.

Check. Averaging $v(t)\,i(t)$, v^2 and i^2 numerically over 10^6 points per period gives 1437.29 W, 230.787 V and 7.7460 A. ✓

Worked Example 15.29 — Power factor of a rectifier with a large DC inductor

A single-phase bridge rectifier with a large DC-side inductor draws a square-wave current of amplitude I_d in phase with the sinusoidal supply voltage $V\sqrt{2}\sin\omega_0 t$. Find the power factor.

Solution. The current is a square wave $\pm I_d$, so $I = I_d$ and $I_1 = \frac{4I_d}{\pi\sqrt{2}} = 0.9003 I_d$, in phase with the voltage ($\cos\theta_1 = 1$). Hence

$$\mathrm{PF} = \frac{I_1}{I}\cos\theta_1 = \frac{2\sqrt{2}}{\pi} = 0.9003, \qquad \mathrm{THD}_i = 48.3\%.$$

The load is "resistive" as far as the fundamental is concerned, yet about 10% of the conductor capacity is wasted on harmonic current. Active power-factor-correction front ends shape the current into a sinusoid to push PF above 0.99.

Worked Example 15.30 — How much bandwidth does a pulse train need?

For the 5 V, 25%-duty pulse train of Worked Example 15.7, what fraction of the total power lies in the harmonics inside the first spectral null ($|f| < 1/\tau$)? How much power lies in the DC term? For the $\pm A$ square wave, what fraction lies in the fundamental, in harmonics up to the 5th, and up to the 19th?

Solution. Total power of the pulse train: the signal is 5 V for a quarter of the time, so $P = 25 \times \frac{1}{4} = 6.25\ \mathrm{V}^2$. Inside the null, $|n| \le 3$:

$$\sum_{n=-3}^{3} |c_n|^2 = 1.25^2 + 2\big(1.1254^2 + 0.7958^2 + 0.3751^2\big) = 1.5625 + 2(1.2665 + 0.6333 + 0.1407) = 5.6435\ \mathrm{V}^2,$$

which is 90.3% of the total. The DC term alone carries $1.5625/6.25 = 25\%$.

For the square wave, the fraction in the odd harmonics up to N is $\frac{8}{\pi^2}\sum_{\text{odd } n\le N}\frac{1}{n^2}$: fundamental 81.06%; up to the 5th 93.31%; up to the 19th 97.98%.

Interpretation. Power converges far faster than the waveform does pointwise: the 19-harmonic square-wave approximation of Worked Example 15.19 has a 9% overshoot at every edge, yet it carries 98% of the power. "Bandwidth containing 90% of the power" is a standard practical definition of signal bandwidth.

Pitfall — adding RMS values linearly, or adding powers of the same frequency

RMS values of *different* harmonics add in quadrature, never linearly: 10 A at 50 Hz plus 3 A at 150 Hz is 10.44 A, not 13 A. But two components at the *same* frequency (two sources' 5th-harmonic currents meeting at a bus, say) must first be combined as phasors — their cross term does not average to zero, and the result can be anywhere between the difference and the sum of their amplitudes.

Engineering note — "true RMS" meters

An average-responding meter measures $\overline{|x|}$ and multiplies by the sinusoidal form factor 1.1107. On a square wave ($\overline{|x|} = A$, true RMS A) it reads $1.11A$, 11% high; on the narrow, peaky current of a capacitor-input rectifier (crest factor 2.5–3) it typically reads 20–40% low. A true-RMS meter computes $\sqrt{\overline{x^2}}$ directly, which by Parseval is the quadrature sum of all harmonics within its bandwidth.

15.9 Applications to periodic signals

Motivation

Section 15.1 promised that Fourier series would turn the response of a linear system to a periodic input into a list of multiplications. This section keeps that promise, states the conditions under which it is valid, and works through the standard situations: filtering a switching waveform, resonance with one harmonic of a non-sinusoidal drive, and ripple at the output of a rectifier. The applications are motivation; the method is general and applies equally to mechanical, thermal and acoustic systems described by linear differential equations with constant coefficients.

Steady-state response of an LTI system

Periodic steady state of an LTI system

Let a stable LTI system with frequency response $H(j\omega)$ be driven by a T_0-periodic input $x(t) = \sum_n c_n e^{jn\omega_0 t}$ with finite power. Then the output, after transients have decayed, is the T_0-periodic signal

$$y_{\mathrm{ss}}(t) = \sum_{n=-\infty}^{\infty} c_n\, H(jn\omega_0)\, e^{jn\omega_0 t},$$

i.e. the output coefficients are $d_n = c_n H(jn\omega_0)$. For real signals, in amplitude–phase form,

$$y_{\mathrm{ss}}(t) = A_0 H(0) + \sum_{n=1}^{\infty} A_n\, |H(jn\omega_0)| \cos\big(n\omega_0 t + \phi_n + \angle H(jn\omega_0)\big).$$

The output power is $P_y = \sum_n |c_n|^2 |H(jn\omega_0)|^2$.

Justification. Stability means $\int |h| < \infty$, so $H(j\omega)$ is bounded and continuous. The periodic steady state is $y_{\mathrm{ss}}(t) = \int_{-\infty}^{\infty} h(\tau)x(t-\tau)\,d\tau$ (the input applied for all past time). Writing the infinite convolution as a periodic convolution with the periodised impulse response $h_{T_0}(t) = \sum_k h(t - kT_0)$, whose Fourier coefficients are $\frac{1}{T_0}H(jn\omega_0)$, the periodic-convolution property of §15.7 gives $d_n = T_0 \cdot c_n \cdot \frac{1}{T_0}H(jn\omega_0) = c_n H(jn\omega_0)$. Because H is bounded, $\sum |d_n|^2 \le \max|H|^2 \sum |c_n|^2 <$

∞, so the output series converges in mean square; if $H(j\omega)$ falls off at high frequency, it converges faster than the input series. □

The **complete** response to an input switched on at $t = 0$ is y_{ss} plus a transient determined by the system's poles and initial conditions; for a stable system the transient decays and $y \to y_{ss}$. The Laplace transform (Ch. 14) gives both together; the Fourier series gives the steady state directly, which is usually what is wanted. The procedure is:

1. Find the input's Fourier coefficients (§15.2–15.7), using symmetry and tables.
2. Find $H(j\omega)$ — from impedances, from the differential equation by $d/dt \to j\omega$, or from $H(s)$ at $s = j\omega$.
3. Multiply harmonic by harmonic: $d_n = c_n H(jn\omega_0)$.
4. Recombine, or — more often — stop at step 3 and read off amplitudes, power, THD or ripple directly.

For a linear constant-coefficient ODE $\sum_k \alpha_k y^{(k)} = \sum_m \beta_m x^{(m)}$ the frequency response is $H(j\omega) = \sum_m \beta_m (j\omega)^m / \sum_k \alpha_k (j\omega)^k$, because d/dt acts on $e^{jn\omega_0 t}$ as multiplication by $jn\omega_0$ (the differentiation rule of §15.7).

Worked Example 15.31 — RC low-pass filter driven by a square wave

A ± 1 V, 1 kHz square wave drives an RC low-pass filter with $RC = 159.15\ \mu$s (corner frequency $f_c = 1/(2\pi RC) = 1$ kHz). Find the first four non-zero output harmonics, the output RMS value and THD, and the peak output voltage. Check the peak against a time-domain solution.

Solution. *Input:* $x(t) = \frac{4}{\pi}\sum_{\text{odd}} \frac{\sin n\omega_0 t}{n}$ (Worked Example 15.3), $\omega_0 = 2\pi \times 1000$ rad/s.

Frequency response: $H(j\omega) = \frac{1}{1+j\omega RC}$; at the harmonics $\omega_0 RC = 1$, so $H(jn\omega_0) = \frac{1}{1+jn}$, $|H| = \frac{1}{\sqrt{1+n^2}}$, $\angle H = -\arctan n$.

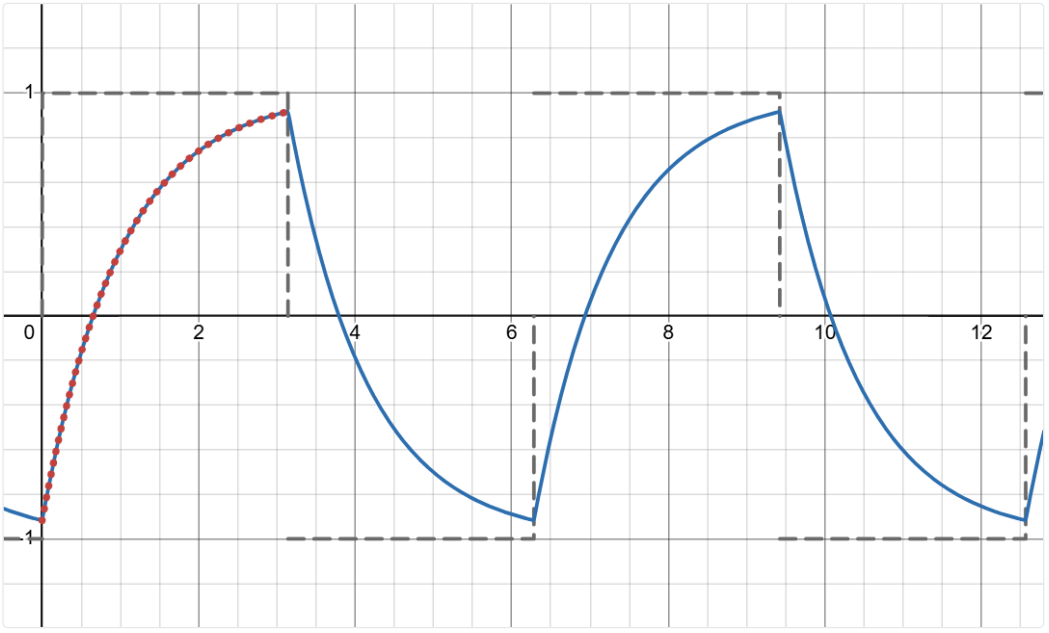
n	input amplitude $\frac{4}{n\pi}$ (V)	$\lvert H\rvert$	output amplitude (V)	phase shift
1	1.2732	0.7071	0.9003	$-45.0°$
3	0.4244	0.3162	0.1342	$-71.6°$
5	0.2546	0.1961	0.0499	$-78.7°$
7	0.1819	0.1414	0.0257	$-81.9°$

So $y_{ss}(t) = 0.9003\sin(\omega_0 t - 45°) + 0.1342\sin(3\omega_0 t - 71.6°) + 0.0499\sin(5\omega_0 t - 78.7°) + \cdots$ V. For large n the output amplitude is $\approx \frac{4}{\pi n^2}$: the filter converts the $1/n$ decay of the square wave into the $1/n^2$ decay of a waveform with kinks but no jumps — as it must, since a capacitor voltage cannot jump.

RMS and THD by Parseval: $Y^2 = \sum_{\text{odd}} \frac{1}{2}\left(\frac{4}{\pi n}\right)^2 \frac{1}{1+n^2} = 0.41612$ V² (summing to $n = 2\times 10^5$), so $Y = 0.6451$ V. The fundamental contributes $\frac{1}{2}(0.9003)^2 = 0.40528$ V², hence $\text{THD}_y = \sqrt{0.41612/0.40528 - 1} = 16.4\%$, down from 48.3% at the input.

Peak value: the output peaks at the end of each positive half-cycle, $t = T_0/2$. Summing the series there to 10^5 harmonics gives 0.91715 V.

Time-domain check. In the periodic steady state the capacitor charges towards $+1$ for half a period from $-V_p$ and must end at $+V_p$: $V_p = 1 - (1 + V_p)e^{-T_0/(2RC)}$, so $V_p = \frac{1-e^{-T_0/2RC}}{1+e^{-T_0/2RC}} = \tanh\frac{T_0}{4RC} = \tanh\frac{\pi}{2} = 0.917152$ V. ✓ The exact steady-state waveform $1 - (1 + V_p)e^{-t/RC}$ on the half-period also has RMS 0.64508 V. ✓



Steady-state RC filter output for a square-wave input with $\omega_0 RC = 1$ (horizontal axis $\omega_0 t$): the Fourier sum through the 119th harmonic (blue) and the exact exponential charging curve on the first half-period (red, dotted) coincide.

Worked Example 15.32 — Resonance with a harmonic

A lightly damped second-order system $\ddot{y} + 0.2\dot{y} + 25y = x(t)$ (natural frequency $\omega_n = 5$ rad/s, $\zeta = 0.02$) is driven by the unit square wave of period 2π s ($\omega_0 = 1$ rad/s). Find the steady-state response.

Solution. $H(j\omega) = \frac{1}{25 - \omega^2 + 0.2j\omega}$. With $b_n = \frac{4}{\pi n}$ for odd n:

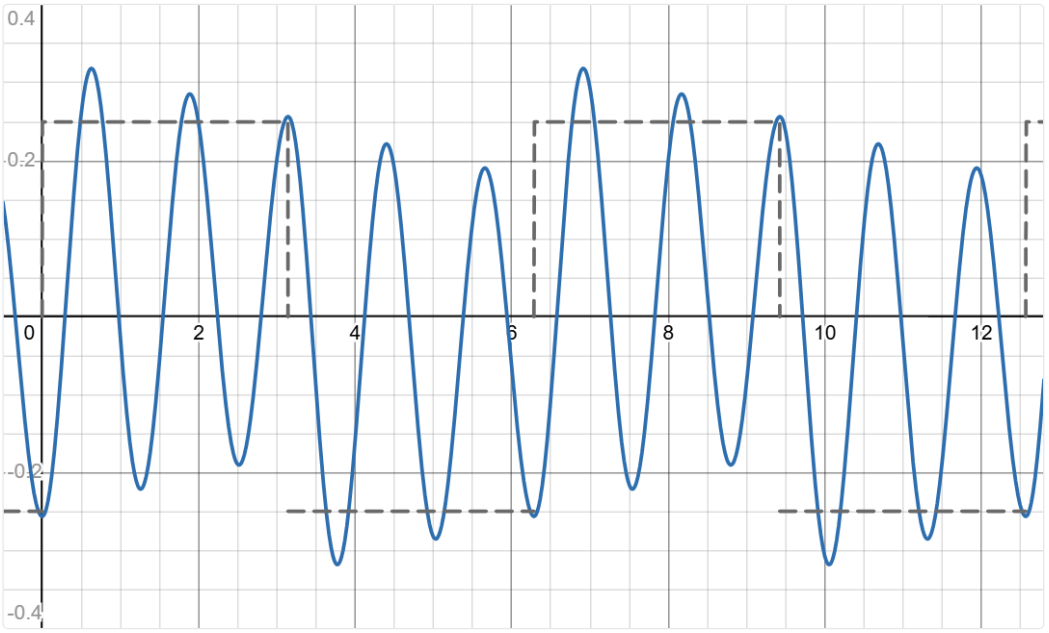
n	b_n	$\lvert H(jn)\rvert$	output amplitude	phase of H
1	1.2732	0.04167	0.05305	-0.5°
3	0.4244	0.06246	0.02651	-2.1°
5	0.2546	1.0000	**0.25465**	-90.0°
7	0.1819	0.04160	0.00757	-176.7°
9	0.1415	0.01786	0.00252	-178.2°

At $n = 5$ the denominator is $25 - 25 + 0.2j \cdot 5 = j$, so $H(j5) = -j$. The steady state is

$$y_{\text{ss}}(t) \approx 0.0531 \sin t + 0.0265 \sin 3t - 0.2546 \cos 5t + 0.0076 \sin(7t - 176.7^\circ) + \cdots .$$

The 5th harmonic, only 20% of the input fundamental, is amplified by the resonance until it dominates the output: Parseval gives the output mean square 0.034214, of which 94.8% is in the 5th harmonic. The system responds at *five times* the drive frequency.

The undamped limit. With no damping, $H(j5) = \frac{1}{25-25}$ is infinite: there is no periodic steady state, and the true solution contains a term growing like $t\cos 5t$ (Ch. 12, resonance). In general, $\ddot{y} + \omega_n^2 y = x(t)$ has a T_0-periodic solution if and only if the input has **no component at the resonant frequency**: either ω_n/ω_0 is not an integer, or $c_{\pm n} = 0$ for $n = \omega_n/\omega_0$. Had the system been tuned to $\omega_n = 4$, an even harmonic, the square wave (which has none) would drive it without resonance.



Steady-state response of $\ddot{y} + 0.2\dot{y} + 25y = x(t)$ to a unit square wave of period 2π (blue; the input is shown scaled by 0.25, dashed). The output oscillates five times per input period: the system has picked out the resonant 5th harmonic.

Worked Example 15.33 — Ripple at the output of a rectifier filter

A full-wave rectifier fed from 230 V RMS, 50 Hz ($V_m = 325$ V) supplies an LC filter: series inductor $L = 0.5$ H, shunt capacitor $C = 1000\ \mu$F, with a light load that can be neglected in computing the ripple. Assuming the rectifier output is the ideal $V_m|\sin \omega t|$, find the DC output and the ripple amplitudes.

Solution. *Input series* (Worked Example 15.25), fundamental of the rectified wave at 100 Hz:

$$v_r(t) = \frac{2V_m}{\pi} - \frac{4V_m}{\pi}\sum_{k\ge 1}\frac{\cos(2k\omega t)}{4k^2-1} = 206.90 - 137.93\cos(2\omega t) - 27.59\cos(4\omega t) - 11.82\cos(6\omega t) - \cdots \text{ V}.$$

Filter (unloaded LC divider): $H(j\Omega) = \frac{1/(j\Omega C)}{j\Omega L + 1/(j\Omega C)} = \frac{1}{1-\Omega^2 LC}$, with $H(0) = 1$.

component	Ω (rad/s)	input (V)	$\lvert H\rvert$	output (V)
DC	0	206.90	1	206.90
100 Hz	628.3	137.93	0.005092	0.702
200 Hz	1256.6	27.59	0.001268	0.035
300 Hz	1885.0	11.82	0.000563	0.0067

At 100 Hz, $\Omega^2 LC = 628.3^2 \times 0.5 \times 10^{-3} = 197.4$. The output is 206.9 V DC with 0.70 V of 100 Hz ripple; the RMS ripple by Parseval is 0.497 V, a ripple factor of 0.24%.

Interpretation. The second-order filter attenuates by $\approx 1/(\Omega^2 LC)$, i.e. 40 dB/decade, while the rectified-sine harmonics themselves fall as $1/k^2$; the combination makes the 200 Hz and higher ripple negligible. Only the lowest ripple harmonic matters for the design, which is why rectifier filter formulas are written for the 100 Hz (or 120 Hz) component alone.

Worked Example 15.34 — Choosing a filter from a harmonic specification

A 1 kHz, ± 1 V square wave is to be converted to a sine wave by a filter that leaves the fundamental essentially unchanged. What attenuation is needed at 3 kHz and above for the output THD to be below 1%? Is a second-order Butterworth low-pass with $f_c = 1$ kHz sufficient?

Solution. The output harmonics are $\frac{4}{\pi n}|H(jn\omega_0)|$. If the filter's attenuation grows at least as fast as n^2 beyond the 3rd harmonic, the 3rd dominates the THD and the requirement is approximately $\frac{1}{3}\frac{|H(j3\omega_0)|}{|H(j\omega_0)|} \le 0.01$, i.e. a relative attenuation of at least 33.3, or 30.5 dB, at 3 kHz.

A second-order Butterworth filter has $|H(j\omega)| = 1/\sqrt{1+(\omega/\omega_c)^4}$. With $f_c = 1$ kHz: $|H(j\omega_0)| = 0.7071$ and $|H(j3\omega_0)| = 1/\sqrt{82} = 0.1104$, a relative attenuation of only 6.4 (16 dB). Output harmonic ratios: 3rd $\frac{1}{3}\cdot\frac{0.1104}{0.7071} = 5.2\%$, 5th $\frac{1}{5}\cdot\frac{0.0400}{0.7071} = 1.1\%$, 7th 0.4%. THD $\approx \sqrt{5.2^2 + 1.1^2 + 0.4^2 + \cdots}\% \approx 5.3\%$. Not sufficient.

Moving the corner down to $f_c = 400$ Hz: $|H(j\omega_0)| = 1/\sqrt{1+2.5^4} = 0.1580$ and $|H(j3\omega_0)| = 1/\sqrt{1+7.5^4} = 0.01778$; 3rd-harmonic ratio $\frac{1}{3}\cdot\frac{0.01778}{0.1580} = 3.75\%$ — better, but the fundamental is now attenuated by 16 dB, and the filter still falls short. In fact, for this Butterworth filter the relative attenuation $|H(j\omega_0)|/|H(j3\omega_0)| = \sqrt{(1+81r^4)/(1+r^4)}$, $r = f_0/f_c$, can never exceed 9 (19 dB, the 40 dB/decade asymptote over a factor of 3 in frequency), whatever the corner. Since the 3rd harmonic needs 30.5 dB more attenuation than the fundamental, a much steeper roll-off is required: a filter of order 4 or more (e.g. a 4th-order Butterworth with $f_c \approx 1$ kHz gives a relative attenuation of $\sqrt{(1+3^8)/2} = 57.3$, 35 dB) or a band-pass tuned to 1 kHz.

Lesson. Harmonic specifications translate directly into frequency-response requirements at a few discrete frequencies — a far easier design problem than specifying a time-domain waveform.

Other uses of the same idea

- **Partial differential equations.** Separation of variables for the heat, wave and Laplace equations (Ch. 13) expands the initial or boundary data in a half-range sine or cosine series (§15.4); each spatial harmonic then evolves independently, decaying like $e^{-\kappa(n\pi/L)^2 t}$ in the heat equation. This is the same "multiply harmonic by harmonic" principle with the role of $H(jn\omega_0)$ played by the time evolution of each mode.
- **Periodic signals and the Fourier transform.** A periodic signal has the generalised Fourier transform $X(\omega) = 2\pi\sum_n c_n\delta(\omega - n\omega_0)$ — a train of impulses whose areas are the line-spectrum coefficients (Ch. 16, §16.6). The ideal impulse-train sampler of §16.8 is analysed exactly this way: its Fourier series has $c_n = 1/T_s$ for all n.
- **Nonlinear devices with periodic inputs.** A memoryless nonlinearity maps a periodic input to a periodic output with the same fundamental, whose coefficients follow from the multiplication rule (§15.7) or directly from the Euler integrals of the distorted waveform. Describing-function analysis of nonlinear feedback loops keeps only the fundamental of the output.
- **Switching power converters.** Their waveforms are piecewise linear (§15.7, jump method), and harmonic analysis of the switch-node voltage predicts ripple and EMI; the "average model" of a converter is exactly the $n = 0$ term.

Pitfall — the steady-state formula hides the transient and assumes stability

The harmonic-by-harmonic result describes only the periodic steady state of a *stable* system. It says nothing about the start-up transient (which can be much larger, e.g. inrush currents), and it fails outright for marginally stable systems driven at a resonance, as in the undamped case of Worked Example 15.32. Use the Laplace transform (Ch. 14) when the transient matters.

Engineering note — harmonic power flow studies

Power-system harmonic studies apply this section's theorem at network scale: each nonlinear load is modelled as a set of harmonic current sources I_n, the network as an impedance matrix $\mathbf{Z}(jn\omega_0)$ computed separately for each harmonic, and the harmonic bus voltages as $\mathbf{V}_n = \mathbf{Z}(jn\omega_0)\mathbf{I}_n$. Parallel resonances between line inductance and power-factor-correction capacitors, at which $|Z|$ peaks near the 5th or 7th harmonic, are the network-scale analogue of Worked Example 15.32.

Summary

Topic	Key formula or fact
Fundamental	T_0, $f_0 = 1/T_0$, $\omega_0 = 2\pi/T_0$; nth harmonic at $n\omega_0$
Inner product, RMS	$\langle x, y\rangle = \frac{1}{T_0}\int_{T_0} xy^*dt$; $\|x\| = X_{\text{rms}}$
Orthogonality	$\frac{1}{T_0}\int_{T_0} e^{jm\omega_0 t}e^{-jn\omega_0 t}dt = \delta_{mn}$
Trigonometric series	$x \sim \frac{a_0}{2} + \sum_{n\geq 1}(a_n\cos n\omega_0 t + b_n \sin n\omega_0 t)$
Euler formulas	$a_n = \frac{2}{T_0}\int_{T_0} x\cos n\omega_0 t\,dt$, $b_n = \frac{2}{T_0}\int_{T_0} x\sin n\omega_0 t\,dt$
Amplitude–phase form	$A_n = \sqrt{a_n^2+b_n^2}$, $\phi_n = \operatorname{atan2}(-b_n, a_n)$, $x \sim A_0 + \sum A_n\cos(n\omega_0 t + \phi_n)$
Complex series	$x \sim \sum_n c_n e^{jn\omega_0 t}$, $c_n = \frac{1}{T_0}\int_{T_0} xe^{-jn\omega_0 t}dt$
Conversions	$c_n = \frac{a_n - jb_n}{2}$, $c_0 = \frac{a_0}{2}$; real x: $c_{-n} = c_n^*$, $A_n = 2\|c_n\|$, $\phi_n = \angle c_n$
One-period transform	$c_n = \frac{1}{T_0}X_1(jn\omega_0)$, $X_1(s) = \int_0^{T_0} xe^{-st}dt$
Best approximation	S_N minimises $\|x-p\|$; error $= \|x\|^2 - \sum_{\|n\|\leq N}\|c_n\|^2$
Even / odd	even: cosines, $a_n = \frac{4}{T_0}\int_0^{T_0/2} x\cos$; odd: sines, $b_n = \frac{4}{T_0}\int_0^{T_0/2} x\sin$
Half-wave symmetry	$x(t+\frac{T_0}{2}) = -x(t) \iff$ odd harmonics only
Quarter-wave symmetry	odd n only, $\frac{8}{T_0}\int_0^{T_0/4}$
Half-range series on $(0, L)$	sine: $b_n = \frac{2}{L}\int_0^L f\sin\frac{n\pi x}{L}$; cosine: $a_n = \frac{2}{L}\int_0^L f\cos\frac{n\pi x}{L}$
Standard series	square $\frac{4A}{\pi}\sum_{\text{odd}}\frac{\sin n\omega_0 t}{n}$; triangle $\frac{8A}{\pi^2}\sum_{\text{odd}}\frac{\cos n\omega_0 t}{n^2}$; pulse train $c_n = AD\operatorname{sinc}(nD)$
Rectified sines	half-wave: $\frac{V_m}{\pi} + \frac{V_m}{2}\sin\omega_0 t - \frac{2V_m}{\pi}\sum_k\frac{\cos 2k\omega_0 t}{4k^2-1}$; full-wave: $\frac{2V_m}{\pi} - \frac{4V_m}{\pi}\sum_k\frac{\cos 2k\omega_0 t}{4k^2-1}$
Dirichlet theorem	$S_N(t) \to \frac{1}{2}[x(t^+) + x(t^-)]$
Uniform convergence	x continuous and piecewise smooth, or $\sum\|c_n\| < \infty$
Mean-square convergence	$\|x - S_N\| \to 0$ for every finite-power x
Gibbs phenomenon	overshoot $\to (\frac{1}{\pi}\operatorname{Si}(\pi) - \frac{1}{2})J = 0.0895J$; square wave peak $\frac{2}{\pi}\operatorname{Si}(\pi)A = 1.179A$
Shift, modulation	$x(t-t_0) \leftrightarrow e^{-jn\omega_0 t_0}c_n$; $e^{jm\omega_0 t}x \leftrightarrow c_{n-m}$
Differentiation, integration	$x' \leftrightarrow jn\omega_0 c_n$; $\int x \leftrightarrow c_n/(jn\omega_0)$ if $c_0 = 0$
Convolution, product	$\frac{1}{T_0}\int_{T_0} x(\tau)y(t-\tau)d\tau \leftrightarrow c_n d_n$; $xy \leftrightarrow \sum_m c_m d_{n-m}$
Jump formula	$c_n = \frac{1}{jn\omega_0 T_0}\sum_i J_i e^{-jn\omega_0 t_i} + \frac{1}{jn\omega_0}c_n(x'_{\text{reg}})$
Decay rate	jump in $x^{(k)}$, lower derivatives continuous $\Rightarrow \|c_n\| = O(1/n^{k+1})$
DFT aliasing	$\tilde{c}_n = \frac{1}{N}\sum_k x_k e^{-j2\pi nk/N} = \sum_m c_{n+mN}$
Parseval	$\frac{1}{T_0}\int_{T_0}\|x\|^2dt = \sum_n\|c_n\|^2 = A_0^2 + \sum_{n\geq 1}\frac{A_n^2}{2}$
Average power	$P = V_0 I_0 + \sum_{n\geq 1} V_{n,\text{rms}} I_{n,\text{rms}}\cos\theta_n$
THD	$\text{THD}_F = \sqrt{\sum_{n\geq 2} X_n^2}/X_1 = \sqrt{X^2/X_1^2 - 1}$; $\text{THD}_R = \text{THD}_F/\sqrt{1+\text{THD}_F^2}$
Power factor (sinusoidal v)	$\text{PF} = \frac{I_1}{I}\cos\theta_1 = \frac{\cos\theta_1}{\sqrt{1+\text{THD}_i^2}}$
THD values	square 48.3%, sawtooth 80.3%, six-step 31.1%, triangle 12.1%
LTI steady state	$d_n = c_n H(jn\omega_0)$; $P_y = \sum_n\|c_n\|^2\|H(jn\omega_0)\|^2$

Further reading

- E. Kreyszig, *Advanced Engineering Mathematics*, 10th ed., Wiley — Ch. 11: Fourier series, half-range expansions and the approximation viewpoint, with many worked examples.
- G. P. Tolstov, *Fourier Series*, Dover — a careful, readable account of convergence, the Dirichlet kernel, Gibbs phenomenon and term-by-term operations.
- A. V. Oppenheim, A. S. Willsky & S. H. Nawab, *Signals and Systems*, 2nd ed., Prentice Hall — Ch. 3: complex Fourier series, properties and the response of LTI systems to periodic inputs.
- J. Arrillaga & N. R. Watson, *Power System Harmonics*, 2nd ed., Wiley — harmonic sources, THD and power factor, and harmonic power-flow analysis.

Chapter 16 — Fourier Transforms

> **Prerequisites:** Ch. 2 (complex numbers), Ch. 5 (integral calculus), Ch. 14 (Laplace transforms), Ch. 15 (Fourier series and Fourier analysis) **Learning outcomes:** After this chapter you can
>
> - compute forward and inverse Fourier transforms of standard signals, including generalized transforms of impulses, steps, constants and sinusoids;
> - derive and apply the transform properties (linearity, shift, scaling, duality, differentiation, integration, modulation) so that most transforms are found without integrating;
> - use the convolution and multiplication theorems to analyse linear time-invariant systems and products of signals in the frequency domain;
> - interpret magnitude and phase spectra, frequency responses, group delay and bandwidth, including the time–bandwidth (uncertainty) trade-off;
> - analyse periodic and finite-length signals spectrally, including windowing, leakage and resolution;
> - compute energy and energy spectral density with Parseval's theorem, and state the mathematics of ideal sampling: impulse-train spectra, aliasing, the Nyquist condition and sinc reconstruction.

The Fourier series of Ch. 15 represents a *periodic* signal as a sum of harmonics at the discrete frequencies $k\omega_0$. Most signals an engineer meets are not periodic: a single radar pulse, a transient on a power line, a spoken word, a step applied to an amplifier, a bit of data on a cable. To describe such signals in terms of frequency we let the period grow without bound. The harmonics crowd together, their spacing ω_0 shrinks to $d\omega$, the sum becomes an integral, and we obtain the **Fourier transform**: a representation of an *aperiodic* signal as a continuum of complex exponentials $e^{j\omega t}$.

The Fourier transform sits between the two neighbouring chapters. It is the Laplace transform of Ch. 14 restricted to the imaginary axis $s = j\omega$ (whenever that axis lies in the region of convergence), and when a signal is sampled it becomes, after a change of variable, the discrete-time Fourier transform and the Z-transform of Ch. 17 on the unit circle. Compared with the Laplace transform it gives up the ability to handle growing signals directly, but gains a perfectly symmetric inverse, a clean physical meaning (the frequency content of the signal), and an energy theorem (Parseval) that makes "how much of the signal lies in this band?" a well-posed question.

Throughout this chapter $x(t)$ is a real or complex function of continuous time t (in seconds), ω is angular frequency in rad/s, and $f = \omega/2\pi$ is frequency in Hz. Wherever a voltage example helps, $x(t)$ is a voltage in volts.

16.1 Continuous Fourier transform

Motivation: from series to integral

A sinusoid $\cos(\omega t)$ passing through a linear circuit comes out as a sinusoid of the same frequency, scaled and phase-shifted; that is the whole content of phasor analysis. If we can write an arbitrary input as a superposition of sinusoids — or, more conveniently, of complex exponentials $e^{j\omega t}$ — then the response of any linear time-invariant circuit to that input is the same superposition of scaled exponentials. For periodic inputs, the Fourier series supplies the superposition. For aperiodic inputs we need a *continuous* superposition, and the weight given to each frequency is the Fourier transform.

Heuristic derivation. Let $x(t)$ be zero outside $|t| < T_0/2$, and let $\tilde{x}(t)$ be its periodic extension with period T_0 and fundamental $\omega_0 = 2\pi/T_0$. The exponential Fourier coefficients of $\tilde{x}$ (Ch. 15) are

$$c_k = \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} x(t)\, e^{-jk\omega_0 t}\, dt.$$

Define the function

$$X(\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt,$$

which, because x vanishes outside one period, gives $c_k = X(k\omega_0)/T_0$. The series becomes

$$\tilde{x}(t) = \sum_{k=-\infty}^{\infty} \frac{1}{T_0} X(k\omega_0)\, e^{jk\omega_0 t} = \frac{1}{2\pi}\sum_{k=-\infty}^{\infty} X(k\omega_0)\, e^{jk\omega_0 t}\, \omega_0.$$

Now let $T_0 \to \infty$. Then $\tilde{x} \to x$ (the neighbouring copies move off to infinity), the spacing ω_0 becomes the differential $d\omega$, the discrete frequencies $k\omega_0$ become a continuous variable ω, and the sum is a Riemann sum for

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)\, e^{j\omega t}\, d\omega.$$

Two facts come out of this argument and are worth keeping:

1. **The Fourier series coefficients of a periodic signal are samples of the transform of one period**, divided by the period: $c_k = X(k\omega_0)/T_0$.
2. As T_0 grows each individual coefficient c_k shrinks like $1/T_0$, but the *density* $T_0 c_k$ stays finite. The transform is that density — it is an amplitude *per unit frequency*, not an amplitude.

Definition

Fourier transform (angular-frequency convention)

For a signal $x(t)$, the Fourier transform is

$$X(\omega) = \mathcal{F}\{x(t)\} = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt,$$

where ω is angular frequency in rad/s. We write $x(t) \leftrightarrow X(\omega)$ and call x and X a **transform pair**.

$X(\omega)$ is in general complex even when $x(t)$ is real, and we write it in polar form $X(\omega) = |X(\omega)|e^{j\angle X(\omega)}$. If $x(t)$ is a voltage in V, then $X(\omega)$ has units V·s = V/Hz: it is a **spectral density**. Two immediate special values are useful checks on every computation:

$$X(0) = \int_{-\infty}^{\infty} x(t)\, dt \quad \text{(the area of } x\text{)}, \qquad x(0) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)\, d\omega \quad \text{(the area of } X\text{, divided by } 2\pi\text{)}.$$

The second follows from the inverse formula (Section 16.2) at $t = 0$.

Relation to the Laplace transform. The bilateral Laplace transform is $X_L(s) = \int x(t)e^{-st}dt$ with $s = \sigma + j\omega$. Setting $\sigma = 0$ gives the Fourier transform: $X(\omega) = X_L(j\omega)$, *provided the region of convergence (ROC) of X_L contains the imaginary axis*. For a causal signal this means all poles of $X_L(s)$ lie strictly in the left half-plane — the same condition as BIBO stability of an impulse response. If the ROC does not contain the $j\omega$-axis, the Fourier transform either does not exist or exists only as a generalized function (Section 16.2).

Existence

The integral defining $X(\omega)$ is improper on both ends, so we need conditions under which it makes sense.

Sufficient conditions for existence (Dirichlet conditions)

Suppose

1. $x(t)$ is **absolutely integrable**: $\int_{-\infty}^{\infty}|x(t)|\,dt < \infty$;
2. in every finite interval $x(t)$ has finitely many maxima and minima and finitely many finite discontinuities.

Then $X(\omega)$ exists for every ω, is continuous and bounded, $|X(\omega)| \le \int|x|\,dt$, and $X(\omega) \to 0$ as $|\omega| \to \infty$ (the **Riemann–Lebesgue lemma**).

Proof sketch. Since $|e^{-j\omega t}| = 1$, $|X(\omega)| \le \int|x(t)|dt < \infty$, which is the bound. Continuity follows because $|X(\omega+h) - X(\omega)| \le \int|x(t)|\,|e^{-jht} - 1|dt$, and the integrand tends to zero for each t while being dominated by $2|x(t)|$. For Riemann–Lebesgue, approximate x by a step function; for a single rectangle $\int_a^b e^{-j\omega t}dt = (e^{-j\omega a} - e^{-j\omega b})/(j\omega) \to 0$, and the approximation error can be made arbitrarily small uniformly in ω. □

There is a second, equally important class.

Finite-energy signals (Plancherel)

If $x(t)$ has finite energy, $\int_{-\infty}^{\infty}|x(t)|^2dt < \infty$, then the truncated transforms $X_T(\omega) = \int_{-T}^{T} x(t)e^{-j\omega t}dt$ converge **in mean square** to a function $X(\omega)$ as $T \to \infty$, the inverse formula holds in the same mean-square sense, and Parseval's theorem (Section 16.7) holds.

Absolute integrability and finite energy are different conditions. The sinc pulse $\sin(t)/t$ has finite energy but is not absolutely integrable (its envelope decays only as $1/|t|$); its transform exists in the mean-square sense and turns out to be a rectangle — discontinuous, which is consistent with the Dirichlet theorem *not* applying. Conversely $1/\sqrt{|t|}$ on $|t| < 1$ is absolutely integrable but has infinite energy.

Signals such as 1, $u(t)$, $\cos\omega_0 t$ and $\operatorname{sgn}(t)$ are neither absolutely integrable nor of finite energy. They are nevertheless indispensable — a DC supply, a switch closing, a carrier wave — and they have **generalized transforms** containing Dirac impulses $\delta(\omega)$, developed in Section 16.2.

Standard functions: rect, sinc, triangle, signum

We use the following standard shapes throughout the chapter.

Rectangle, sinc, triangle and signum

$$\operatorname{rect}(t) = \begin{cases} 1, & |t| < \tfrac{1}{2} \\ \tfrac{1}{2}, & |t| = \tfrac{1}{2} \\ 0, & |t| > \tfrac{1}{2} \end{cases}, \qquad \operatorname{sinc}(x) = \frac{\sin(\pi x)}{\pi x}, \quad \operatorname{sinc}(0) = 1,$$

$$\Lambda(t) = \max\bigl(0,\, 1 - |t|\bigr), \qquad \operatorname{sgn}(t) = \begin{cases} 1, & t > 0 \\ 0, & t = 0 \\ -1, & t < 0 \end{cases}.$$

So $\operatorname{rect}(t/T)$ is a unit-height pulse of width T centred on $t = 0$, and $\Lambda(t/T)$ is a unit-height triangle of base $2T$. This is the **normalized** sinc: its zeros are at the nonzero integers and $\int_{-\infty}^{\infty}\operatorname{sinc}(x)\,dx = 1$.

Common Pitfall — two sinc conventions

Mathematics texts and some circuits books define $\operatorname{sinc}(x) = \sin(x)/x$ (unnormalized, zeros at $x = k\pi$). This book uses $\sin(\pi x)/(\pi x)$, the convention of numerical libraries such as NumPy and MATLAB. Whenever a formula containing "sinc" is copied from another source, check which one it uses; the safest habit is to write the explicit ratio $2\sin(\omega T/2)/\omega$ when there is any doubt.

Worked examples

Worked Example 16.1 — the one-sided exponential

Find the transform of $x(t) = e^{-at}u(t)$, $a > 0$, and its magnitude and phase.

Step 1 — integrate. Only $t > 0$ contributes:

$$X(\omega) = \int_0^\infty e^{-at}e^{-j\omega t}\,dt = \int_0^\infty e^{-(a+j\omega)t}dt = \left[\frac{-e^{-(a+j\omega)t}}{a+j\omega}\right]_0^\infty = \frac{1}{a+j\omega}.$$

The upper limit vanishes because $|e^{-(a+j\omega)t}| = e^{-at} \to 0$; this is where $a > 0$ is used.

Step 2 — polar form. Multiply numerator and denominator by the conjugate, or read off directly:

$$|X(\omega)| = \frac{1}{\sqrt{a^2+\omega^2}}, \qquad \angle X(\omega) = -\arctan\frac{\omega}{a}.$$

Step 3 — interpret. $X(0) = 1/a$, which equals $\int_0^\infty e^{-at}dt$ ✓. At $\omega = a$ the magnitude is $1/(a\sqrt{2})$, i.e. $1/\sqrt{2}$ of its DC value (-3 dB), and the phase is $-45°$. For $\omega \gg a$, $|X| \approx 1/|\omega|$ and the phase approaches $-90°$. The result equals the Laplace transform $1/(s+a)$ at $s = j\omega$, which is legitimate because the ROC $\operatorname{Re}\{s\} > -a$ contains the $j\omega$-axis.

Numerical check ($a = 2$, $\omega = 2$): $X = 1/(2+2j) = 0.25 - 0.25j$, $|X| = 0.3536 = 1/(2\sqrt{2})$ ✓, $\angle X = -45°$ ✓.

Worked Example 16.2 — the rectangular pulse

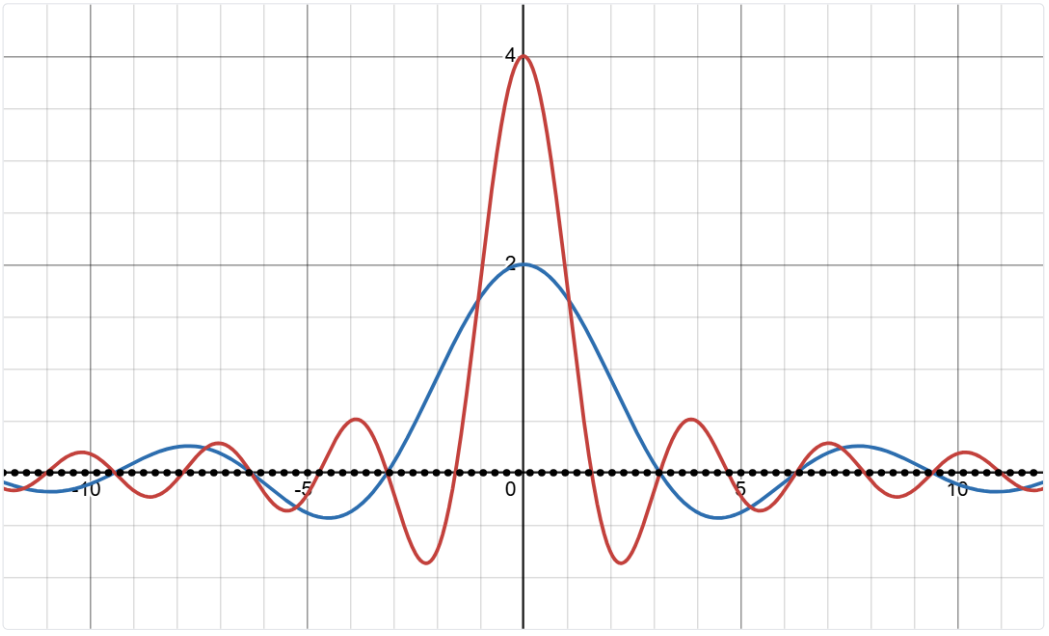
Find the transform of $x(t) = A\,\mathrm{rect}(t/T)$, a pulse of height A and width T.

$$X(\omega) = A\int_{-T/2}^{T/2} e^{-j\omega t}\,dt = A\,\frac{e^{j\omega T/2} - e^{-j\omega T/2}}{j\omega} = \frac{2A\sin(\omega T/2)}{\omega} = AT\,\operatorname{sinc}\left(\frac{\omega T}{2\pi}\right).$$

The last step writes $2\sin(\omega T/2)/\omega = T\sin(\pi u)/(\pi u)$ with $u = \omega T/2\pi$.

- $X(0) = AT$, the pulse area ✓.
- Zeros occur where $\omega T/2 = k\pi$, i.e. $\omega = 2\pi k/T$, $k = \pm 1, \pm 2, \ldots$ (in Hz: $f = k/T$).
- The main lobe runs from $-2\pi/T$ to $2\pi/T$: width $4\pi/T$ rad/s. **Halving the pulse width doubles the spectral width.**
- X is real because x is real and even; its sign alternates between lobes, so the phase alternates between 0 and $\pm\pi$.
- The envelope decays as $2A/|\omega|$ — slowly, because x has jump discontinuities.

For a 1 µs pulse ($T = 10^{-6}$ s) the first null is at $f = 1$ MHz: a rule of thumb used everywhere in pulse and digital-signal work is that a pulse of duration T needs a bandwidth of order $1/T$ Hz.



Spectra $T\,\operatorname{sinc}(\omega T/2\pi) = 2\sin(\omega T/2)/\omega$ of unit-height rectangular pulses of width $T = 2$ (blue) and $T = 4$ (red): the longer pulse has twice the peak and half the main-lobe width.

Worked Example 16.3 — the two-sided exponential

Find the transform of $x(t) = e^{-a|t|}$, $a > 0$.

Split the integral at $t = 0$:

$$X(\omega) = \int_{-\infty}^{0} e^{(a-j\omega)t}dt + \int_{0}^{\infty} e^{-(a+j\omega)t}dt = \frac{1}{a-j\omega} + \frac{1}{a+j\omega} = \frac{2a}{a^2+\omega^2}.$$

The result is real and even, as it must be for a real even signal, and positive for all ω. Its value at the origin is $X(0) = 2/a = \int e^{-a|t|}dt$ ✓. The decay is $1/\omega^2$, one power faster than for the one-sided exponential: $e^{-a|t|}$ is continuous (only its derivative jumps), whereas $e^{-at}u(t)$ itself jumps at $t = 0$. This shape is called a **Lorentzian**; it appears again as the power spectrum of a first-order Gauss–Markov process in Ch. 21.

Worked Example 16.4 — the Gaussian pulse

Find the transform of $x(t) = e^{-t^2/(2\sigma^2)}$, $\sigma > 0$.

Step 1 — complete the square in the exponent:

$$-\frac{t^2}{2\sigma^2} - j\omega t = -\frac{1}{2\sigma^2}\left(t + j\sigma^2\omega\right)^2 - \frac{\sigma^2\omega^2}{2}.$$

(Expand the square: $-(t^2 + 2j\sigma^2\omega t - \sigma^4\omega^2)/(2\sigma^2) = -t^2/(2\sigma^2) - j\omega t + \sigma^2\omega^2/2$; subtracting $\sigma^2\omega^2/2$ restores the original.)

Step 2 — shift the contour. Then

$$X(\omega) = e^{-\sigma^2\omega^2/2}\int_{-\infty}^{\infty} e^{-(t+j\sigma^2\omega)^2/(2\sigma^2)}\,dt.$$

The remaining integral is a Gaussian integral along a line parallel to the real axis. Because $e^{-z^2/(2\sigma^2)}$ is analytic and decays rapidly as $|\mathrm{Re}\, z| \to \infty$ in any horizontal strip, Cauchy's theorem (Ch. 18) lets us move the line back to the real axis without changing the value, giving $\int e^{-t^2/(2\sigma^2)}dt = \sigma\sqrt{2\pi}$.

$$e^{-t^2/(2\sigma^2)} \;\leftrightarrow\; \sigma\sqrt{2\pi}\, e^{-\sigma^2\omega^2/2}.$$

Interpretation. The transform of a Gaussian is a Gaussian. Its width in ω is $1/\sigma$: a pulse of time-width σ has frequency-width $1/\sigma$. With $\sigma = 1$ the pair is self-reciprocal up to the factor $\sqrt{2\pi}$. Because the Gaussian is infinitely smooth, its spectrum decays faster than any power of ω.

Numerical check ($\sigma = 0.5$, $\omega = 2$): formula gives $0.5\sqrt{2\pi}e^{-0.5} = 0.7602$; numerical integration of $\int e^{-2t^2}\cos(2t)\,dt$ (the sine part vanishes by oddness) also gives 0.7602 ✓.

Engineering Note — what the transform tells a circuit designer

The magnitude $|X(\omega)|$ shows which frequency bands the signal occupies, and therefore what bandwidth an amplifier, cable or filter must pass to reproduce it. The rectangle's $1/\omega$ tail is why sharp-edged digital signals radiate interference far above their clock rate; rounding the edges (Gaussian or raised-cosine shaping) converts that tail into a much faster decay. The phase $\angle X(\omega)$ records *when* each component occurs: a delay shows up purely in the phase (Section 16.3).

16.2 Inverse transform

Statement

The derivation in Section 16.1 already suggested how to recover $x(t)$ from $X(\omega)$. The precise statement is the **Fourier integral theorem**.

Inverse Fourier transform (Fourier integral theorem)

If $x(t)$ satisfies the Dirichlet conditions of Section 16.1, then

$$x(t) = \mathcal{F}^{-1}\{X(\omega)\} = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)\, e^{j\omega t}\, d\omega$$

at every point of continuity of x, where the integral is understood as the limit $\lim_{W\to\infty}\frac{1}{2\pi}\int_{-W}^{W}$. At a jump discontinuity the integral converges to the midpoint $\frac{1}{2}[x(t^-) + x(t^+)]$.

The inverse has exactly the same form as the forward transform except for the sign of the exponent and the factor $1/(2\pi)$. The factor appears because we integrate over angular frequency: $d\omega = 2\pi\, df$. In words, the inverse formula says that $x(t)$ is a superposition of complex exponentials $e^{j\omega t}$, the one at frequency ω having complex weight $X(\omega)\, d\omega/2\pi$.

Proof sketch. Insert the definition of X into the truncated inverse and exchange the order of integration (justified by absolute integrability of x and the finite range in ω):

$$x_W(t) = \frac{1}{2\pi}\int_{-W}^{W}\left[\int_{-\infty}^{\infty} x(\tau)e^{-j\omega\tau}d\tau\right]e^{j\omega t}d\omega = \int_{-\infty}^{\infty} x(\tau)\, D_W(t-\tau)\, d\tau, \qquad D_W(s) = \frac{\sin Ws}{\pi s}.$$

The kernel D_W is the continuous-time analogue of the Dirichlet kernel of Fourier series. For every W it has total area $\int D_W(s)\,ds = 1$, and as $W \to \infty$ it oscillates ever faster away from $s = 0$ while its central peak (height W/π, width about $2\pi/W$) grows. By the Riemann–Lebesgue lemma the contribution away from $s = 0$ tends to zero, so D_W acts in the limit like $\delta(s)$ and $x_W(t) \to x(t)$ at points of continuity. At a jump, half of the kernel's area lies on each side of $s = 0$

(it is even), so the limit is the average of the one-sided limits. □

The truncation x_W is exactly what an ideal low-pass filter of bandwidth W does to x, and near a jump it overshoots by about 9 % of the jump no matter how large W is — the **Gibbs phenomenon**, inherited unchanged from Fourier series.

The hertz convention

Many communications and optics texts (for example Bracewell) use frequency f in Hz instead of ω:

$$X_f(f) = \int_{-\infty}^{\infty} x(t)\, e^{-j2\pi f t}\, dt, \qquad x(t) = \int_{-\infty}^{\infty} X_f(f)\, e^{j2\pi f t}\, df.$$

The two are related by $X_f(f) = X(2\pi f)$. The Hz form is symmetric (no 2π outside either integral), which removes 2π factors from duality, the multiplication theorem and Parseval's theorem; the ω form matches the Laplace variable $s = \sigma + j\omega$ and circuit phasors. **This book uses ω throughout.** To convert a result from a Hz-convention source, substitute $f = \omega/2\pi$ in the transform and, in any integral over frequency, $df = d\omega/2\pi$. (A third, "unitary" convention puts $1/\sqrt{2\pi}$ in front of both integrals; it is common in physics and is not used here.)

Methods of inversion

1. **Table lookup plus properties.** This is the usual method. A short table of pairs (below) combined with the properties of Section 16.3 covers almost every signal met in practice.
2. **Partial fractions** for rational functions of $j\omega$, exactly as for Laplace transforms. Treat $j\omega$ as the variable, expand, and use

$$\frac{1}{(a + j\omega)^m} \;\leftrightarrow\; \frac{t^{m-1}}{(m-1)!}\, e^{-at} u(t), \qquad \operatorname{Re}\{a\} > 0.$$

If a term has $\operatorname{Re}\{a\} < 0$, the corresponding absolutely integrable time function is anticausal: $1/(a + j\omega) \leftrightarrow -e^{-at}u(-t)$ for $\operatorname{Re}\{a\} < 0$. A Fourier transform, unlike a unilateral Laplace transform, needs no separately stated ROC because the $j\omega$-axis *is* the ROC; the sign of $\operatorname{Re}\{a\}$ decides causal versus anticausal. 3. **Direct integration**, often by contour integration and residues (Ch. 18), which is how rational spectra with mixed left- and right-half-plane poles are inverted systematically.

Worked examples

Worked Example 16.5 — ideal low-pass spectrum

Let $X(\omega) = \operatorname{rect}(\omega/2W)$, i.e. $X = 1$ for $|\omega| < W$ and 0 otherwise. Find $x(t)$.

$$x(t) = \frac{1}{2\pi}\int_{-W}^{W} e^{j\omega t}\, d\omega = \frac{1}{2\pi}\cdot\frac{e^{jWt} - e^{-jWt}}{jt} = \frac{\sin(Wt)}{\pi t} = \frac{W}{\pi}\operatorname{sinc}\!\left(\frac{Wt}{\pi}\right).$$

Checks: $x(0) = W/\pi$, which equals $\frac{1}{2\pi}\int X\, d\omega = 2W/2\pi$ ✓. The zeros are at $t = k\pi/W$, $k \neq 0$.

If X is the frequency response of a filter, x is its impulse response. The ideal ("brick-wall") low-pass filter therefore has a sinc impulse response that extends over all $t < 0$: it responds before the impulse arrives. It is **non-causal** and hence not physically realisable — a fact read directly from the inverse transform. Realisable filters approximate the brick wall with a delayed, truncated sinc or with rational responses.

Worked Example 16.6 — partial fractions with distinct poles

Invert $X(\omega) = \dfrac{j\omega + 3}{(j\omega + 1)(j\omega + 2)}$.

Write $X = \dfrac{A}{1 + j\omega} + \dfrac{B}{2 + j\omega}$. By cover-up,

$$A = \left.\frac{j\omega + 3}{j\omega + 2}\right|_{j\omega = -1} = 2, \qquad B = \left.\frac{j\omega + 3}{j\omega + 1}\right|_{j\omega = -2} = -1.$$

Both poles are in the left half-plane ($a = 1, 2 > 0$), so both terms are causal:

$$x(t) = \left(2e^{-t} - e^{-2t}\right) u(t).$$

Check at $\omega = 0$: $X(0) = 3/2$ and $\int_0^\infty (2e^{-t} - e^{-2t})dt = 2 - \tfrac{1}{2} = \tfrac{3}{2}$ ✓. Check at $t = 0^+$: $x(0^+) = 1$, and the initial-value rule $x(0^+) = \lim_{\omega\to\infty} j\omega X(\omega) = 1$ ✓.

Worked Example 16.7 — complex poles, and a two-sided signal

(a) Invert $X(\omega) = \dfrac{1}{(j\omega)^2 + 2j\omega + 5}$.

Complete the square: $(j\omega)^2 + 2j\omega + 5 = (j\omega + 1)^2 + 4$. From the pair $e^{-at}\sin(bt)u(t) \leftrightarrow b/[(a + j\omega)^2 + b^2]$ (the Laplace pair at $s = j\omega$, valid since $a = 1 > 0$),

$$x(t) = \tfrac{1}{2}e^{-t}\sin(2t)\,u(t).$$

Check: $X(0) = 1/5$, and $\int_0^\infty \tfrac{1}{2}e^{-t}\sin 2t\,dt = \tfrac{1}{2}\cdot\tfrac{2}{1+4} = \tfrac{1}{5}$ ✓.

(b) Invert $Y(\omega) = \dfrac{6}{(2 + j\omega)(3 - j\omega)}$.

Partial fractions in $j\omega$: $Y = \dfrac{A}{2 + j\omega} + \dfrac{B}{3 - j\omega}$ with $A = 6/(3 + 2) = 6/5$ (set $j\omega = -2$) and $B = 6/(2 + 3) = 6/5$ (set $j\omega = 3$). The first term is causal. The second, $1/(3 - j\omega)$, is the transform of $e^{3t}u(-t)$ (replace $t \to -t$ in $e^{-3t}u(t) \leftrightarrow 1/(3 + j\omega)$, which replaces $\omega \to -\omega$). Hence

$$y(t) = \tfrac{6}{5}\left[e^{-2t}u(t) + e^{3t}u(-t)\right],$$

a two-sided signal that decays in both directions. Check: $Y(0) = 1$ and $\int y\,dt = \tfrac{6}{5}(\tfrac{1}{2} + \tfrac{1}{3}) = 1$ ✓. A unilateral Laplace table cannot produce this answer; the Fourier transform handles it naturally because its "ROC" is fixed to be the $j\omega$-axis.

Generalized transforms

Many important signals — constants, steps, sinusoids — are not absolutely integrable. They acquire transforms when we allow the Dirac impulse in the frequency domain. The cleanest route is to compute transforms of impulses directly and then use the inverse formula.

Impulse. By the sifting property, $\mathcal{F}\{\delta(t - t_0)\} = \int \delta(t - t_0)e^{-j\omega t}dt = e^{-j\omega t_0}$. In particular $\delta(t) \leftrightarrow 1$: an impulse contains all frequencies with equal weight, which is why an impulse response characterises an LTI system completely.

Complex exponential and constant. Apply the inverse formula to $X(\omega) = 2\pi\delta(\omega - \omega_0)$:

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} 2\pi\delta(\omega - \omega_0)\,e^{j\omega t}\,d\omega = e^{j\omega_0 t}.$$

Hence

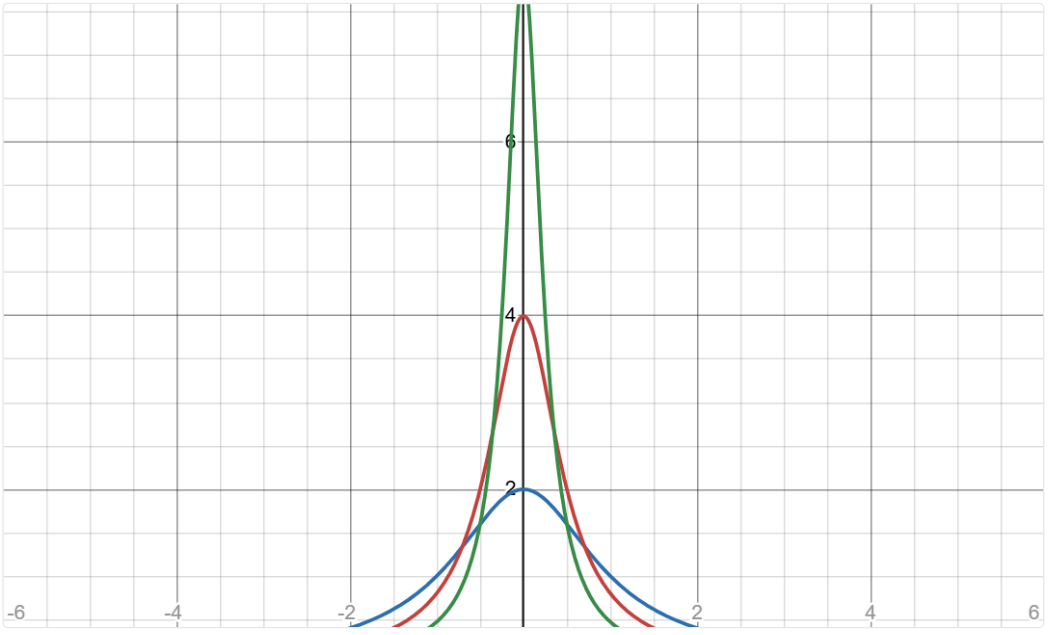
$$e^{j\omega_0 t} \leftrightarrow 2\pi\delta(\omega - \omega_0), \qquad 1 \leftrightarrow 2\pi\delta(\omega),$$

and by Euler's formula

$$\cos\omega_0 t \leftrightarrow \pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)], \qquad \sin\omega_0 t \leftrightarrow -j\pi[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)].$$

A pure tone has all its spectral content concentrated at $\pm\omega_0$, with impulse areas π (not $\tfrac{1}{2}$: the areas are 2π times the Fourier-series coefficients $\tfrac{1}{2}$).

A limiting argument. The same pair can be obtained without the inverse formula by viewing 1 as the limit of $e^{-a|t|}$ as $a \to 0^+$. Its transform $2a/(a^2 + \omega^2)$ has area $\int 2a/(a^2 + \omega^2)\,d\omega = 2\pi$ for every a, and as $a \to 0$ it becomes an ever taller, narrower peak at $\omega = 0$. Its limit is therefore $2\pi\delta(\omega)$ ✓.



The transforms $2a/(a^2 + \omega^2)$ of $e^{-a|t|}$ for $a = 1$ (blue), 0.5 (red) and 0.25 (green): every curve has area 2π, and as $a \to 0$ they converge to the impulse $2\pi\delta(\omega)$, the transform of the constant 1.

Signum. Write $\operatorname{sgn}(t) = \lim_{a\to 0^+}[e^{-at}u(t) - e^{at}u(-t)]$. The bracket has transform

$$\frac{1}{a+j\omega}-\frac{1}{a-j\omega}=\frac{-2j\omega}{a^2+\omega^2}\xrightarrow{a\to 0}\frac{-2j\omega}{\omega^2}=\frac{2}{j\omega}.$$

There is no impulse at $\omega=0$ because the bracket is odd, so its transform is odd and integrates to zero across the origin. Thus $\operatorname{sgn}(t)\leftrightarrow 2/(j\omega)$.

Unit step. Since $u(t)=\tfrac{1}{2}[1+\operatorname{sgn}(t)]$,

$$u(t)\leftrightarrow \pi\delta(\omega)+\frac{1}{j\omega}.$$

The impulse $\pi\delta(\omega)$ carries the average (DC) value $\tfrac{1}{2}$ of the step; the $1/(j\omega)$ term carries the jump.

Common Pitfall — substituting $s=j\omega$ outside the ROC

The Fourier transform of $u(t)$ is **not** obtained by putting $s=j\omega$ in $1/s$: the ROC of $\mathcal{L}\{u(t)\}$ is $\operatorname{Re}\{s\}>0$, which excludes the $j\omega$-axis. The extra $\pi\delta(\omega)$ accounts for the DC value $\tfrac{1}{2}$ of $u(t)$. Similarly $\mathcal{F}\{\cos(\omega_0 t)u(t)\}=\dfrac{j\omega}{\omega_0^2-\omega^2}+\dfrac{\pi}{2}[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)]$, not just $s/(s^2+\omega_0^2)$ at $s=j\omega$. Substitute $s=j\omega$ only when the ROC strictly contains the imaginary axis. When simple poles lie *on* the axis, each pole at $s=j\omega_p$ with residue r contributes an extra $\pi r\,\delta(\omega-\omega_p)$.

Worked Example 16.8 — a DC-plus-tone signal and a switched constant

(a) $x(t)=3+2\cos(5t)-\sin(5t)$. Term by term:

$$X(\omega)=6\pi\delta(\omega)+2\pi[\delta(\omega-5)+\delta(\omega+5)]+j\pi[\delta(\omega-5)-\delta(\omega+5)]=6\pi\delta(\omega)+(2+j)\pi\,\delta(\omega-5)+(2-j)\pi\,\delta(\omega+5).$$

The impulse weights at ±5 are complex conjugates, as they must be for a real signal. Dividing by 2π gives the Fourier-series coefficients $c_{\pm1}=1\pm j/2$, whose magnitude $\sqrt{5}/2$ is half the amplitude $\sqrt{2^2+1^2}=\sqrt{5}$ of the combined sinusoid $\sqrt{5}\cos(5t+26.6^\circ)$ ✓.

(b) $y(t)=V_0\,u(t-t_0)$, a supply switched on at t_0. By the time-shift property (Section 16.3),

$$Y(\omega)=V_0e^{-j\omega t_0}\left[\pi\delta(\omega)+\frac{1}{j\omega}\right]=\pi V_0\delta(\omega)+\frac{V_0e^{-j\omega t_0}}{j\omega},$$

where we used $e^{-j\omega t_0}\delta(\omega)=\delta(\omega)$ (the exponential equals 1 at $\omega=0$).

Table of standard pairs

$x(t)$	$X(\omega)$	Condition
$\delta(t)$	1	
$\delta(t-t_0)$	$e^{-j\omega t_0}$	
1	$2\pi\delta(\omega)$	
$u(t)$	$\pi\delta(\omega)+1/(j\omega)$	
$\operatorname{sgn}(t)$	$2/(j\omega)$	
$e^{-at}u(t)$	$1/(a+j\omega)$	$\operatorname{Re}\{a\}>0$
$te^{-at}u(t)$	$1/(a+j\omega)^2$	$\operatorname{Re}\{a\}>0$
$\dfrac{t^{m-1}}{(m-1)!}e^{-at}u(t)$	$1/(a+j\omega)^m$	$\operatorname{Re}\{a\}>0$
$e^{-at}\sin(bt)\,u(t)$	$b/[(a+j\omega)^2+b^2]$	$a>0$
$e^{-at}\cos(bt)\,u(t)$	$(a+j\omega)/[(a+j\omega)^2+b^2]$	$a>0$
$e^{-a\lvert t\rvert}$	$2a/(a^2+\omega^2)$	$a>0$
$1/(a^2+t^2)$	$(\pi/a)\,e^{-a\lvert\omega\rvert}$	$a>0$
$e^{-t^2/(2\sigma^2)}$	$\sigma\sqrt{2\pi}\,e^{-\sigma^2\omega^2/2}$	$\sigma>0$
$\operatorname{rect}(t/T)$	$T\operatorname{sinc}(\omega T/2\pi)=2\sin(\omega T/2)/\omega$	
$\Lambda(t/T)$	$T\operatorname{sinc}^2(\omega T/2\pi)$	
$\sin(Wt)/(\pi t)$	$\operatorname{rect}(\omega/2W)$	
$e^{j\omega_0 t}$	$2\pi\delta(\omega-\omega_0)$	
$\cos\omega_0 t$	$\pi[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)]$	
$\sin\omega_0 t$	$-j\pi[\delta(\omega-\omega_0)-\delta(\omega+\omega_0)]$	
$\sum_n\delta(t-nT)$	$\omega_s\sum_k\delta(\omega-k\omega_s)$	$\omega_s=2\pi/T$

16.3 Transform properties

Why properties matter

Evaluating $\int x(t)e^{-j\omega t}dt$ from scratch for every signal is slow and error-prone. In practice one memorises a dozen pairs and a dozen properties; together they generate hundreds of transforms by pure algebra. The properties are also where the physics lives: "a delay is a linear phase", "differentiation boosts high frequencies", "compressing a pulse spreads its spectrum", "multiplying by a carrier shifts the spectrum" are each one line of this section.

Throughout, $x(t) \leftrightarrow X(\omega)$ and $y(t) \leftrightarrow Y(\omega)$.

Property	Time domain	Frequency domain
Linearity	$ax(t) + by(t)$	$aX(\omega) + bY(\omega)$
Time shift	$x(t - t_0)$	$e^{-j\omega t_0}X(\omega)$
Frequency shift	$e^{j\omega_0 t}x(t)$	$X(\omega - \omega_0)$
Modulation	$x(t)\cos\omega_0 t$	$\tfrac{1}{2}[X(\omega-\omega_0) + X(\omega+\omega_0)]$
Scaling	$x(at)$, $a \neq 0$ real	$\dfrac{1}{\lvert a\rvert}X(\omega/a)$
Reversal	$x(-t)$	$X(-\omega)$
Conjugation	$x^*(t)$	$X^*(-\omega)$
Duality	$X(t)$	$2\pi x(-\omega)$
Differentiation in time	$d^n x/dt^n$	$(j\omega)^n X(\omega)$
Integration	$\int_{-\infty}^{t} x(\tau)d\tau$	$X(\omega)/(j\omega) + \pi X(0)\delta(\omega)$
Multiplication by t	$t^n\, x(t)$	$j^n\, d^n X/d\omega^n$
Convolution	$x * y$	$X(\omega)Y(\omega)$
Multiplication	$x(t)y(t)$	$\dfrac{1}{2\pi}(X * Y)(\omega)$
Parseval	$\int \lvert x\rvert^2 dt$	$\dfrac{1}{2\pi}\int \lvert X\rvert^2 d\omega$

The convolution, multiplication and Parseval rows are the subject of Sections 16.5 and 16.7. The others are derived now.

Derivations

Linearity is inherited from linearity of the integral.

Time shift. Substitute $\tau = t - t_0$:

$$\int x(t - t_0)e^{-j\omega t}dt = \int x(\tau)e^{-j\omega(\tau + t_0)}d\tau = e^{-j\omega t_0}X(\omega).$$

A delay multiplies the transform by a unit-magnitude factor: $\lvert X\rvert$ is unchanged and the phase acquires a term $-\omega t_0$, **linear in frequency**. Each frequency component is delayed by the same time t_0, which requires a phase lag proportional to its frequency.

Frequency shift. $\int e^{j\omega_0 t}x(t)e^{-j\omega t}dt = \int x(t)e^{-j(\omega-\omega_0)t}dt = X(\omega - \omega_0)$. Multiplying by a complex exponential slides the whole spectrum by ω_0. The **modulation** property follows by writing $\cos\omega_0 t = \tfrac{1}{2}(e^{j\omega_0 t} + e^{-j\omega_0 t})$; similarly $x(t)\sin\omega_0 t \leftrightarrow \tfrac{1}{2j}[X(\omega-\omega_0) - X(\omega+\omega_0)]$.

Scaling. For $a > 0$ substitute $\tau = at$, $dt = d\tau/a$:

$$\int x(at)e^{-j\omega t}dt = \frac{1}{a}\int x(\tau)e^{-j(\omega/a)\tau}d\tau = \frac{1}{a}X\left(\frac{\omega}{a}\right).$$

For $a < 0$ the substitution reverses the limits, producing a minus sign that turns $1/a$ into $1/\lvert a\rvert$. With $a = -1$ this gives the **reversal** property. Compressing in time ($\lvert a\rvert > 1$) expands the spectrum horizontally and lowers it by $1/\lvert a\rvert$; the factor keeps the area $\frac{1}{2\pi}\int X\,d\omega = x(0)$ unchanged, as it must be since $x(at)$ and $x(t)$ have the same value at $t = 0$.

Conjugation. $\int x^*(t)e^{-j\omega t}dt = \left[\int x(t)e^{j\omega t}dt\right]^* = X^*(-\omega)$. For real x this gives $X(-\omega) = X^*(\omega)$, the conjugate symmetry studied in Section 16.4.

Duality. Write the inverse formula with t replaced by $-t$ and multiply by 2π:

$$2\pi x(-t) = \int_{-\infty}^{\infty} X(\omega)e^{-j\omega t}d\omega.$$

Now rename the integration variable $\omega \to \tau$ and the free variable $t \to \omega$: $2\pi x(-\omega) = \int X(\tau)e^{-j\omega\tau}d\tau = \mathcal{F}\{X(t)\}$. So **if** $x(t) \leftrightarrow X(\omega)$ **then** $X(t) \leftrightarrow 2\pi x(-\omega)$. Every pair in the table therefore yields a second pair for free; the rectangle–sinc and sinc–rectangle pairs are duals.

Differentiation. Differentiate the inverse formula under the integral sign:

$$\frac{dx}{dt} = \frac{d}{dt}\frac{1}{2\pi}\int X(\omega)e^{j\omega t}d\omega = \frac{1}{2\pi}\int j\omega X(\omega)e^{j\omega t}d\omega,$$

so $dx/dt \leftrightarrow j\omega X(\omega)$; repeating gives $(j\omega)^n$. Differentiation multiplies the spectrum by $\lvert\omega\rvert$: it **emphasises high frequencies** and destroys DC. (Equivalently, integrate $\int \dot{x}e^{-j\omega t}dt$ by parts, the boundary terms vanishing because $x \to 0$ at $\pm\infty$.)

Multiplication by t. Differentiate the forward formula with respect to ω: $dX/d\omega = \int (-jt)x(t)e^{-j\omega t}dt$, so $t\,x(t) \leftrightarrow j\,dX/d\omega$. This is the dual of the differentiation property.

Integration. Let $y(t) = \int_{-\infty}^{t} x(\tau)d\tau$. Then $y = x * u$, and the convolution theorem (Section 16.5) with $u \leftrightarrow \pi\delta(\omega) + 1/(j\omega)$ gives

$$Y(\omega) = X(\omega)\left[\pi\delta(\omega) + \frac{1}{j\omega}\right] = \frac{X(\omega)}{j\omega} + \pi X(0)\delta(\omega).$$

The impulse term is present whenever x has nonzero area, because then $y(t) \to X(0)$ as $t \to \infty$ and y has a DC component $X(0)/2$ (it rises from 0 to $X(0)$).

Moment theorems. Setting $\omega = 0$ in the forward formula and in the t^n property gives

$$\int_{-\infty}^{\infty} t^n x(t)\,dt = j^n \left.\frac{d^n X}{d\omega^n}\right|_{\omega=0}.$$

The area, centroid and spread of a pulse can therefore be read from the value, slope and curvature of its transform at DC.

Worked examples

Worked Example 16.9 — shift and duality

Find the transforms of $x(t) = e^{-2(t-1)}u(t-1)$ and of $g(t) = \dfrac{1}{1+t^2}$.

(a) x is $e^{-2t}u(t)$ delayed by $t_0 = 1$. From $e^{-2t}u(t) \leftrightarrow 1/(2+j\omega)$ and the time-shift property:

$$X(\omega) = \frac{e^{-j\omega}}{2+j\omega}, \qquad |X| = \frac{1}{\sqrt{4+\omega^2}}, \qquad \angle X = -\omega - \arctan\frac{\omega}{2}.$$

The magnitude is that of the undelayed pulse; only the phase gained the linear term $-\omega$.

(b) From $e^{-|t|} \leftrightarrow 2/(1+\omega^2)$, duality gives $\dfrac{2}{1+t^2} \leftrightarrow 2\pi e^{-|-\omega|} = 2\pi e^{-|\omega|}$. Dividing by 2:

$$\frac{1}{1+t^2} \leftrightarrow \pi e^{-|\omega|}.$$

Check: $G(0) = \pi = \int_{-\infty}^{\infty} \frac{dt}{1+t^2} = [\arctan t]_{-\infty}^{\infty}$ ✓. Scaling then gives the general table entry $1/(a^2+t^2) \leftrightarrow (\pi/a)e^{-a|\omega|}$. Evaluating this integral directly requires residues (Ch. 18); duality did it in one line.

Worked Example 16.10 — combining scaling and shifting

(a) Find the transform of a unit pulse that starts at $t = 0$ and ends at $t = T$.

The pulse is rect $\big((t - T/2)/T\big)$, the centred pulse delayed by $T/2$:

$$X(\omega) = T\,\mathrm{sinc}\left(\frac{\omega T}{2\pi}\right)e^{-j\omega T/2} = \frac{1 - e^{-j\omega T}}{j\omega}.$$

(The second form is the direct integral $\int_0^T e^{-j\omega t}dt$; factoring $e^{-j\omega T/2}$ out of $1 - e^{-j\omega T}$ leaves $2j\sin(\omega T/2)$, confirming the first form.)

(b) Given $x(t) \leftrightarrow X(\omega)$, find the transform of $y(t) = x(2t - 3)$.

Write $y(t) = x\big(2(t - \tfrac{3}{2})\big)$. First scale: $x(2t) \leftrightarrow \tfrac{1}{2}X(\omega/2)$. Then delay *the scaled signal* by $\tfrac{3}{2}$:

$$Y(\omega) = \tfrac{1}{2}X\left(\frac{\omega}{2}\right)e^{-j3\omega/2}.$$

Alternatively shift first: $v(t) = x(t-3) \leftrightarrow e^{-j3\omega}X(\omega)$, and $y(t) = v(2t) \leftrightarrow \tfrac{1}{2}e^{-j3\omega/2}X(\omega/2)$ — the same answer, because scaling acts on *every* ω in the expression, including the one in the exponential.

Check with $x = \mathrm{rect}(t)$: $y = \mathrm{rect}(2t-3)$ is a pulse of width $\tfrac{1}{2}$ centred at $t = \tfrac{3}{2}$, so directly $Y = \tfrac{1}{2}\,\mathrm{sinc}(\omega/4\pi)e^{-j3\omega/2}$; the formula gives $\tfrac{1}{2}\,\mathrm{sinc}(\omega/4\pi)e^{-j3\omega/2}$ ✓.

Common Pitfall — the delay in $x(at - b)$

The delay of $x(at - b)$ is b/a, not b. Writing $e^{-j\omega b}$ instead of $e^{-j\omega b/a}$ is the single most common error with these two properties. Always rewrite the argument as $a(t - b/a)$ first.

Worked Example 16.11 — the triangle by differentiation

Find the transform of $x(t) = \Lambda(t/T)$ without integrating.

Step 1 — differentiate until only impulses remain. The triangle has slope $1/T$ on $(-T, 0)$ and $-1/T$ on $(0, T)$, so $\dot{x} = \frac{1}{T}[\text{rect on } (-T,0)] - \frac{1}{T}[\text{rect on } (0,T)]$. Differentiating again, each jump of the slope becomes an impulse:

$$\ddot{x}(t) = \frac{1}{T}\left[\delta(t+T) - 2\delta(t) + \delta(t-T)\right].$$

Step 2 — transform. By the shift property,

$$(j\omega)^2 X(\omega) = \frac{1}{T}\left[e^{j\omega T} - 2 + e^{-j\omega T}\right] = \frac{2\cos\omega T - 2}{T} = -\frac{4\sin^2(\omega T/2)}{T}.$$

Step 3 — divide. Since $(j\omega)^2 = -\omega^2$,

$$X(\omega) = \frac{4\sin^2(\omega T/2)}{T\omega^2} = T\,\mathrm{sinc}^2\!\left(\frac{\omega T}{2\pi}\right).$$

Division by ω^2 is safe here because x has finite area, so X has no impulse at the origin; the value $X(0) = T$ (by L'Hôpital or the sinc limit) equals the triangle's area $\frac{1}{2}\cdot 2T \cdot 1$ ✓. The method works for any **piecewise-polynomial** signal: differentiate until the result is a sum of impulses, transform, divide by $(j\omega)^n$, and fix the DC value by the area.

Worked Example 16.12 — multiplication by t

(a) From $e^{-at}u(t) \leftrightarrow (a + j\omega)^{-1}$,

$$te^{-at}u(t) \leftrightarrow j\frac{d}{d\omega}(a+j\omega)^{-1} = j\cdot(-1)(a+j\omega)^{-2}\cdot j = \frac{1}{(a+j\omega)^2},$$

confirming the table entry. Repeating gives the $t^{m-1}/(m-1)!$ pairs used for repeated poles.

(b) Find the transform of the "Gaussian monocycle" $x(t) = t\,e^{-t^2/2}$ (an ultra-wideband pulse shape).

From $e^{-t^2/2} \leftrightarrow \sqrt{2\pi}e^{-\omega^2/2}$:

$$X(\omega) = j\frac{d}{d\omega}\left(\sqrt{2\pi}e^{-\omega^2/2}\right) = -j\sqrt{2\pi}\,\omega\,e^{-\omega^2/2}.$$

The result is purely imaginary and odd, as it must be for a real odd signal, and it vanishes at DC (the monocycle has zero area). Its magnitude peaks where $\frac{d}{d\omega}(\omega e^{-\omega^2/2}) = 0$, i.e. at $\omega = 1$ rad/s, where $|X| = \sqrt{2\pi}e^{-1/2} = 1.520$. Equivalently, $x = -\frac{d}{dt}e^{-t^2/2}$, and the differentiation property gives $-j\omega\sqrt{2\pi}e^{-\omega^2/2}$ — the same answer by a second route ✓.

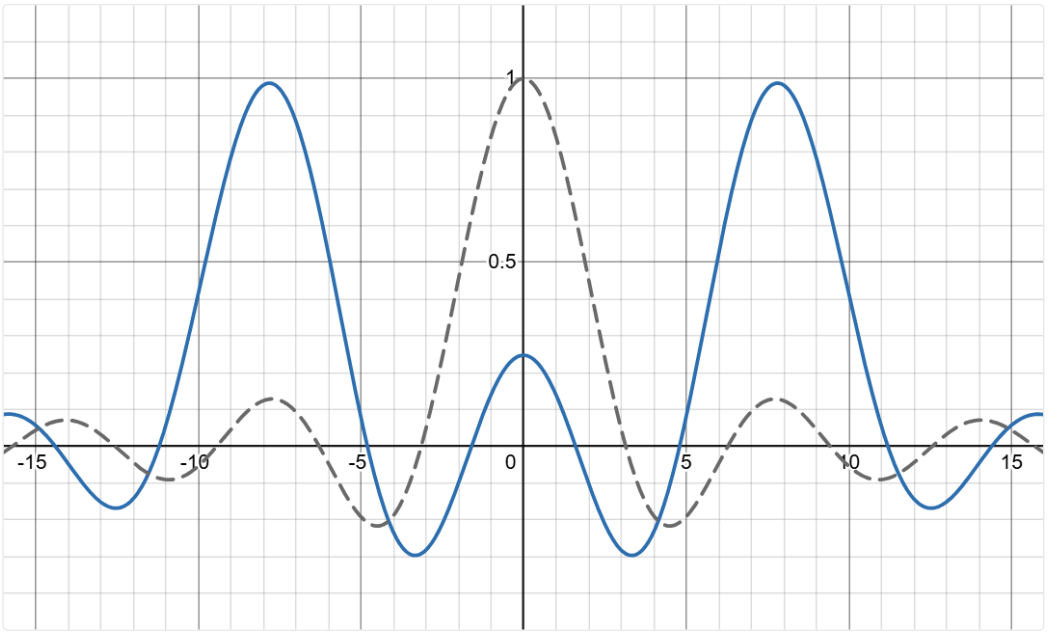
Worked Example 16.13 — an RF pulse

A radar transmitter emits $x(t) = A\,\mathrm{rect}(t/T)\cos(\omega_c t)$ with $\omega_c \gg 2\pi/T$. Find and sketch its spectrum.

By the modulation property applied to $A\,\mathrm{rect}(t/T) \leftrightarrow AT\,\mathrm{sinc}(\omega T/2\pi)$:

$$X(\omega) = \frac{AT}{2}\left[\mathrm{sinc}\!\left(\frac{(\omega-\omega_c)T}{2\pi}\right) + \mathrm{sinc}\!\left(\frac{(\omega+\omega_c)T}{2\pi}\right)\right].$$

The spectrum consists of two copies of the pulse spectrum, each of half the height, centred on $\pm\omega_c$. The main lobe around ω_c has null-to-null width $4\pi/T$ rad/s, or $2/T$ Hz. For $A = 1$ V, $T = 1$ µs, $f_c = 10$ MHz: peaks of 0.5 µV·s at ± 10 MHz, first nulls at 9 and 11 MHz. The two copies interfere slightly with each other through their sidelobes; the condition $\omega_c \gg 2\pi/T$ makes this negligible.



Modulation shifts the spectrum: the baseband pulse spectrum (grey dashed, $T = 2$, peak normalised to 1) is copied to $\pm\omega_c = \pm 8$ rad/s with half the height (blue).

Engineering Note — differentiation turns ODEs into algebra

For a circuit described by $a_2\ddot{y} + a_1\dot{y} + a_0 y = b_1\dot{x} + b_0 x$, the differentiation property gives $[a_2(j\omega)^2 + a_1 j\omega + a_0]Y = (b_1 j\omega + b_0)X$, so

$$Y(\omega) = H(\omega)X(\omega), \qquad H(\omega) = \frac{b_0 + jb_1\omega}{a_0 - a_2\omega^2 + ja_1\omega}.$$

This is the phasor method of AC circuit analysis, extended from single sinusoids to arbitrary spectra: an inductor's impedance $j\omega L$ and a capacitor's $1/(j\omega C)$ are the differentiation and integration properties applied to $v = L\,di/dt$ and $v = \frac{1}{C}\int i\,dt$. The Fourier method gives the zero-state response of a stable system; initial conditions and unstable systems are the territory of the Laplace transform (Ch. 14).

16.4 Frequency-domain representation

A signal and its transform are two complete descriptions of the same object. This section is about reading the frequency-domain description: what the magnitude and phase spectra mean, what symmetry a real signal forces on them, how an LTI system acts on them, and how "width" in one domain constrains width in the other.

Magnitude and phase spectra

The pair $|X(\omega)|$ (the **magnitude** or **amplitude spectrum**) and $\angle X(\omega)$ (the **phase spectrum**) is the frequency-domain representation of $x(t)$. Equivalently one may use the real and imaginary parts $X_R(\omega) = \mathrm{Re}\{X\}$, $X_I(\omega) = \mathrm{Im}\{X\}$.

- Magnitudes spanning many decades are plotted in decibels, $20\log_{10}|X(\omega)|$ (relative to a stated reference), against $\log\omega$ — the Bode format. A factor of 2 in magnitude is about 6 dB; a factor of 10 is 20 dB.
- Phase is defined only modulo 2π. Numerical routines return the **principal value** in $(-\pi, \pi]$, producing artificial jumps of 2π; **unwrapping** removes them by adding multiples of 2π to make the phase continuous wherever $|X| \neq 0$. Where X passes through zero and changes sign (the nulls of a sinc), the phase genuinely jumps by π.
- Phase matters as much as magnitude. Two signals with the same $|X|$ but different phases can look entirely different in time: an impulse and a "chirp" can share a flat magnitude spectrum.

Symmetry of real signals

Conjugate symmetry

If $x(t)$ is real then

$$X(-\omega) = X^*(\omega),$$

so $|X(\omega)|$ and $\mathrm{Re}\{X(\omega)\}$ are **even**, while $\angle X(\omega)$ and $\mathrm{Im}\{X(\omega)\}$ are **odd**. Moreover:

- x real and even $\Rightarrow$ X real and even;
- x real and odd $\Rightarrow$ X purely imaginary and odd;
- for any real x with even part $x_e(t) = \frac{1}{2}[x(t) + x(-t)]$ and odd part $x_o(t) = \frac{1}{2}[x(t) - x(-t)]$: $x_e \leftrightarrow \mathrm{Re}\{X\}$ and $x_o \leftrightarrow j\,\mathrm{Im}\{X\}$.

Proof. For real x, $X^*(\omega) = \int x(t)e^{+j\omega t}dt = X(-\omega)$. Writing $e^{-j\omega t} = \cos\omega t - j\sin\omega t$,

$$X(\omega) = \int x(t)\cos\omega t\,dt - j\int x(t)\sin\omega t\,dt.$$

The cosine integral sees only the even part of x (the odd part integrates to zero against an even function), and the sine integral sees only the odd part. This gives the even/odd statements; the cosine integral is even in ω, the sine integral odd. $\square$

Consequently, for real signals the negative-frequency half of the spectrum carries no new information. Engineers often plot only $\omega \geq 0$ (a **one-sided** spectrum), remembering to double energies when integrating over $\omega > 0$ only (Section 16.7). Negative frequencies are not a physical mystery: $\cos \omega_0 t$ is the sum of two counter-rotating phasors $e^{\pm j\omega_0 t}/2$, and the transform simply accounts for both.

Worked Example 16.14 — even and odd parts of the one-sided exponential

For $x(t) = e^{-at}u(t)$, $X = \dfrac{1}{a + j\omega} = \dfrac{a}{a^2+\omega^2} - j\dfrac{\omega}{a^2+\omega^2}$.

The even part of x is $x_e = \tfrac{1}{2}e^{-a|t|}$, whose transform is $\tfrac{1}{2}\cdot\dfrac{2a}{a^2+\omega^2} = \dfrac{a}{a^2+\omega^2} = \operatorname{Re}\{X\}$ ✓. The odd part is $x_o = \tfrac{1}{2}\operatorname{sgn}(t)e^{-a|t|}$, which must therefore have transform $j\operatorname{Im}\{X\} = -j\omega/(a^2+\omega^2)$. As $a \to 0$ this tends to $-j/\omega = 1/(j\omega)$, which is indeed the transform of $\tfrac{1}{2}\operatorname{sgn}(t)$ ✓.

This example shows a deeper fact: for a **causal** signal, $x_o(t) = \operatorname{sgn}(t)\, x_e(t)$, so the odd part is determined by the even part and hence $\operatorname{Im}\{X\}$ is determined by $\operatorname{Re}\{X\}$. (The precise relation is a Hilbert transform; for minimum-phase systems it also links the magnitude and phase of a frequency response.)

Frequency response of LTI systems

Complex exponentials are **eigenfunctions** of linear time-invariant (LTI) systems. If the impulse response is $h(t)$, the input $e^{j\omega t}$ produces

$$y(t) = \int_{-\infty}^{\infty} h(\tau)e^{j\omega(t-\tau)}d\tau = \left[\int h(\tau)e^{-j\omega\tau}d\tau\right] e^{j\omega t} = H(\omega)\, e^{j\omega t}, \qquad H(\omega) = \mathcal{F}\{h(t)\}.$$

The output is the input multiplied by a complex constant; the constant (the eigenvalue) is the **frequency response** $H(\omega)$. For a real system (h real), the input $\cos(\omega_0 t + \theta)$ — the real part of $e^{j\theta}e^{j\omega_0 t}$ — produces

$$y(t) = |H(\omega_0)|\cos\big(\omega_0 t + \theta + \angle H(\omega_0)\big).$$

The **magnitude response** $|H|$ is the gain and the **phase response** $\angle H$ the phase shift at each frequency. For a general input, superposition over all frequencies gives $Y(\omega) = H(\omega)X(\omega)$, proved formally as the convolution theorem in Section 16.5.

Worked Example 16.15 — RC low-pass filter

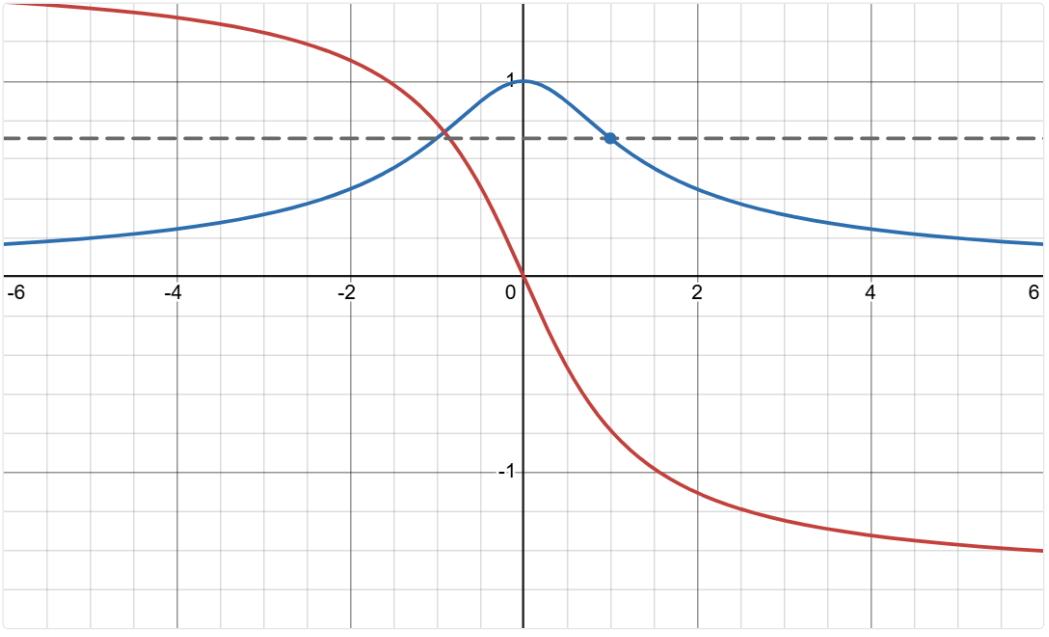
For a series R with shunt C, output across C: $RC\dot{y} + y = x$, so by the differentiation property

$$H(\omega) = \frac{1}{1 + j\omega RC}, \qquad h(t) = \frac{1}{RC}e^{-t/RC}u(t).$$

With $R = 1\ \text{k}\Omega$, $C = 1\ \mu\text{F}$: $RC = 1$ ms, so the cutoff is $\omega_c = 1/RC = 1000$ rad/s ($f_c \approx 159$ Hz).

For the input $\cos(2000t)$: $|H| = 1/\sqrt{1+2^2} = 0.447$ (-7.0 dB), $\angle H = -\arctan 2 = -63.4^\circ$, so in steady state $y(t) = 0.447\cos(2000t - 63.4^\circ)$.

For the input $\cos(100t)$: $|H| = 1/\sqrt{1.01} = 0.995$, $\angle H = -5.7^\circ$: frequencies well below cutoff pass almost unaltered. Far above cutoff $|H| \approx \omega_c/\omega$, a fall of 20 dB per decade.



Magnitude $|H| = 1/\sqrt{1+\omega^2}$ (blue, even) and phase $\angle H = -\arctan\omega$ in radians (red, odd) for $H = 1/(1+j\omega)$, with ω normalised to ω_c.

Worked Example 16.16 — a resonant second-order response

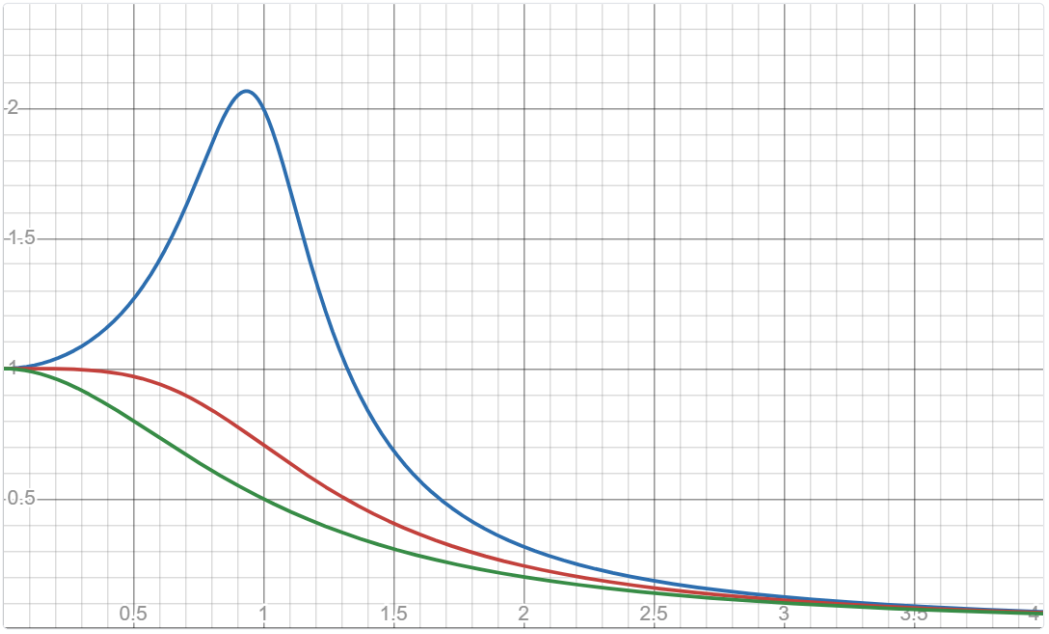
A series RLC circuit with output across C satisfies $LC\ddot{y} + RC\dot{y} + y = x$, so

$$H(\omega) = \frac{1}{1 - \omega^2 LC + j\omega RC} = \frac{1}{1 - (\omega/\omega_0)^2 + j(\omega/\omega_0)/Q}, \qquad \omega_0 = \frac{1}{\sqrt{LC}},\ Q = \frac{\omega_0 L}{R}.$$

Take $L = 10$ mH, $C = 1$ µF, $R = 50\ \Omega$: $\omega_0 = 1/\sqrt{10^{-8}} = 10^4$ rad/s ($f_0 = 1.59$ kHz) and $Q = 10^4 \cdot 0.01/50 = 2$.

ω	$\lvert H\rvert$	$\angle H$
$\omega_0/2 = 5000$	1.265	-18.4°
$\omega_0 = 10^4$	$2.000 = Q$	-90°
$2\omega_0 = 2\times 10^4$	0.316	-161.6°

(At $\omega_0/2$: denominator $= 0.75 + 0.25j$, magnitude 0.7906, so $\lvert H\rvert = 1.265$ and $\angle H = -\arctan(1/3)$. At $2\omega_0$: denominator $= -3 + j$, magnitude $\sqrt{10}$.) At resonance the real part of the denominator vanishes, the gain equals Q, and the phase passes through -90°. The magnitude response peaks near ω_0 and falls as $1/\omega^2$ (40 dB/decade) above it.



Magnitude of the second-order low-pass response against ω/ω_0 for $Q = 2$ (blue, resonant peak), $Q = 0.707$ (red, maximally flat) and $Q = 0.5$ (green, critically damped).

Distortionless transmission, phase delay and group delay

Distortionless system

A system is **distortionless** if every input emerges scaled and delayed but otherwise unchanged: $y(t) = Kx(t - t_d)$. By the time-shift property this is equivalent to

$$H(\omega) = Ke^{-j\omega t_d}: \qquad \lvert H(\omega)\rvert = K \ \text{(flat)}, \qquad \angle H(\omega) = -\omega t_d \ \text{(linear phase)}.$$

It suffices that these hold over the band occupied by the signal.

Departures from flat magnitude are **amplitude distortion**; departures from linear phase are **phase (delay) distortion**. Two delays characterise a non-ideal phase response $\phi(\omega) = \angle H(\omega)$:

$$\tau_p(\omega) = -\frac{\phi(\omega)}{\omega} \quad \text{(phase delay)}, \qquad \tau_g(\omega) = -\frac{d\phi}{d\omega} \quad \text{(group delay)}.$$

The phase delay is the delay of a single sinusoid at ω. The group delay is the delay of the **envelope** of a narrow-band signal centred at ω. To see why, let $x(t) = m(t)\cos\omega_c t$ with m slowly varying, so that X is concentrated near $\pm\omega_c$. Expand $\phi(\omega) \approx \phi(\omega_c) + \phi'(\omega_c)(\omega - \omega_c)$ near ω_c and take $\lvert H\rvert \approx \lvert H(\omega_c)\rvert$ there. Applying the shift and modulation properties to each term gives

$$y(t) \approx \lvert H(\omega_c)\rvert\, m\big(t - \tau_g(\omega_c)\big)\cos\big(\omega_c(t - \tau_p(\omega_c))\big).$$

The envelope travels at the group delay and the carrier at the phase delay. Both are equal (to t_d) for linear phase.

Worked Example 16.17 — group delay of an RC filter and of an all-pass section

(a) For $H = 1/(1 + j\omega RC)$, $\phi = -\arctan(\omega RC)$, so

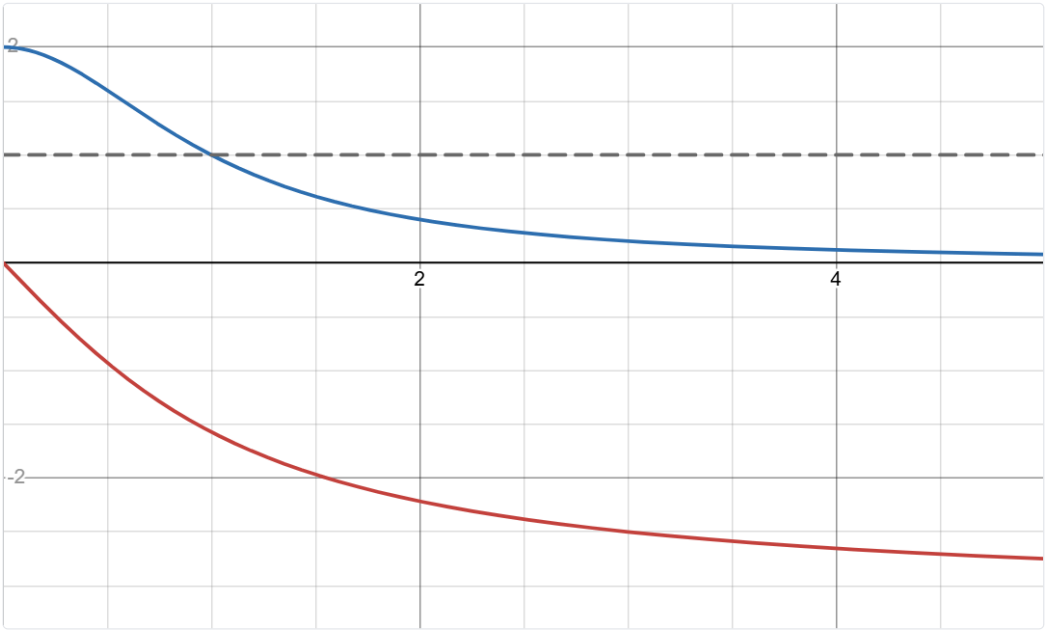
$$\tau_g(\omega) = \frac{RC}{1 + \omega^2 R^2 C^2}.$$

At DC the group delay is RC (1 ms for the values of Worked Example 16.15); at the cutoff it has halved to $RC/2$. A pulse whose spectrum lies well inside the passband is delayed by about RC with little distortion; a pulse that extends past cutoff suffers both amplitude and phase distortion.

(b) The first-order **all-pass** section $H(\omega) = \dfrac{a - j\omega}{a + j\omega}$, $a > 0$, has $|H| = 1$ for all ω — numerator and denominator are conjugates — so there is no amplitude distortion at all. Its phase is $\phi = -2\arctan(\omega/a)$ and

$$\tau_g(\omega) = \frac{2a}{a^2 + \omega^2},$$

which is $2/a$ at DC and falls towards zero at high frequency. With $a = 10^3$ s^{-1}: $\tau_g(0) = 2$ ms, $\tau_g(10^3) = 1$ ms, $\tau_g(10^4) = 19.8$ µs. A broadband pulse through this "flat" filter is dispersed, its low-frequency components arriving later than its high-frequency ones. All-pass sections are used to *equalise* group delay, i.e. to cancel the delay variation of another filter.



First-order all-pass with $a = 1$: flat magnitude 1 (grey dashed), phase $-2\arctan\omega$ in radians (red) and group delay $2/(1+\omega^2)$ in seconds (blue). Flat magnitude does not imply distortionless transmission.

Bandwidth and the time–bandwidth trade-off

Bandwidth measures. For a low-pass spectrum, common definitions are:

- the **3-dB bandwidth**, where $|X|$ first falls to $|X(0)|/\sqrt{2}$;
- the **null-to-null** or first-null bandwidth (for a rectangular pulse of width T: $2\pi/T$ rad/s one-sided);
- the **essential bandwidth**, containing a stated fraction (say 95 % or 99 %) of the energy (Section 16.7);
- the **RMS bandwidth** $\Delta\omega$ defined below.

For band-pass spectra the same measures are applied to the band around the centre frequency. No single definition is "correct"; a specification must say which one it uses.

The trade-off. The scaling property already implies that duration and bandwidth vary inversely: replacing $x(t)$ by $x(at)$ multiplies every width in time by $1/a$ and every width in frequency by a, leaving their product unchanged. The product depends only on the *shape*. How small can it be? With the root-mean-square widths

$$(\Delta t)^2 = \frac{\int t^2 |x(t)|^2 dt}{\int |x(t)|^2 dt}, \qquad (\Delta\omega)^2 = \frac{\int \omega^2 |X(\omega)|^2 d\omega}{\int |X(\omega)|^2 d\omega},$$

for a signal centred at $t = 0$ and $\omega = 0$:

Uncertainty principle (time–bandwidth inequality)

If $\sqrt{|t|}\,x(t) \to 0$ as $|t| \to \infty$ and the widths are finite, then

$$\Delta t\,\Delta\omega \;\geq\; \frac{1}{2},$$

with equality if and only if $x(t)$ is a Gaussian, $x(t) = Ce^{-t^2/(2\sigma^2)}$.

Proof sketch (real x). Let $E = \int x^2 dt$. By Parseval (Section 16.7) and $\dot{x} \leftrightarrow j\omega X$, $\int \omega^2 |X|^2 d\omega/2\pi = \int \dot{x}^2 dt$, so $(\Delta\omega)^2 = \int \dot{x}^2 dt / E$. The Cauchy–Schwarz inequality gives

$$\left(\int t\, x\, \dot{x}\, dt \right)^2 \le \int t^2 x^2 dt \int \dot{x}^2 dt = E^2 (\Delta t)^2 (\Delta\omega)^2.$$

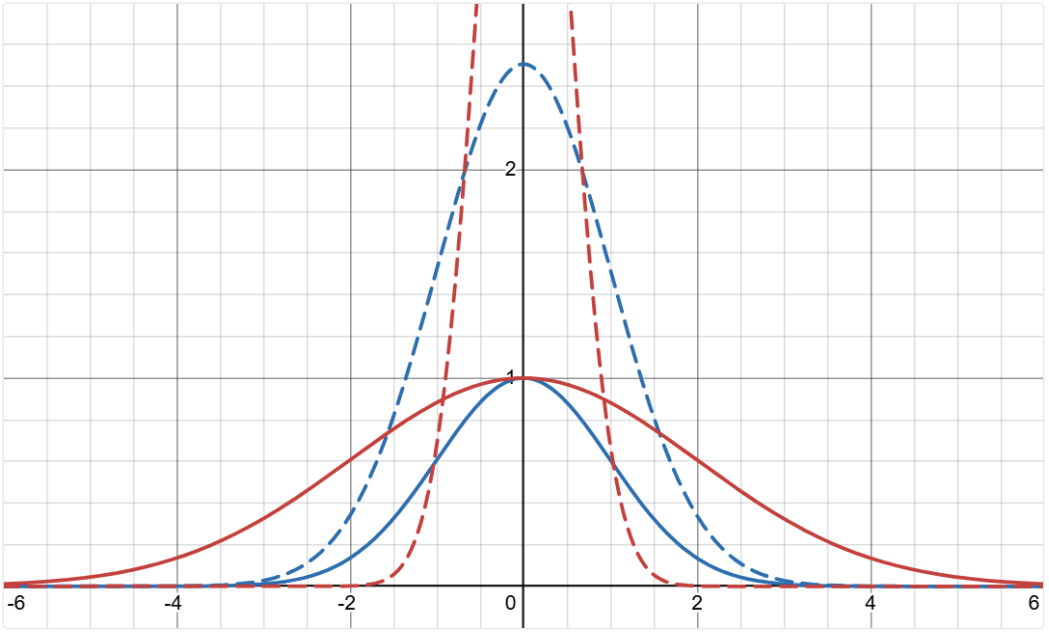
Integrating by parts, $\int t\, x\dot{x}\, dt = \int t\, \tfrac{1}{2} \tfrac{d}{dt}(x^2)\, dt = \left[\tfrac{1}{2} t x^2\right] - \tfrac{1}{2}\int x^2 dt = -E/2$, the boundary term vanishing by hypothesis. Hence $E^2/4 \le E^2 (\Delta t\, \Delta\omega)^2$. Equality in Cauchy–Schwarz requires $\dot{x} = -ctx$, whose solutions are Gaussians. □

Worked Example 16.18 — time–bandwidth products

(a) Gaussian. For $x = e^{-t^2/(2\sigma^2)}$, $x^2 = e^{-t^2/\sigma^2}$ is proportional to a normal density of variance $\sigma^2/2$, so $\Delta t = \sigma/\sqrt{2}$. Similarly $|X|^2 \propto e^{-\sigma^2\omega^2}$ has variance $1/(2\sigma^2)$, so $\Delta\omega = 1/(\sigma\sqrt{2})$. The product is $\tfrac{1}{2}$ exactly ✓.

(b) Two-sided exponential. For $x = e^{-|t|}$: $\int t^2 e^{-2|t|} dt = \tfrac{1}{2}$ and $\int e^{-2|t|} dt = 1$, so $(\Delta t)^2 = \tfrac{1}{2}$. In frequency, $(\Delta\omega)^2 = \int \dot{x}^2 dt / E = 1$ (because $\dot{x}^2 = e^{-2|t|}$). So $\Delta t\, \Delta\omega = 1/\sqrt{2} \approx 0.707 > \tfrac{1}{2}$ ✓.

(c) Rectangular pulse. $|X|^2 \propto \sin^2(\omega T/2)/\omega^2$, so $\omega^2 |X|^2$ does not decay and $\Delta\omega = \infty$: by the RMS measure a signal with jumps has infinite bandwidth, even though 90 % of its energy lies in the main lobe (Section 16.7). The RMS measure is sensitive to tails; the essential bandwidth is not.



Gaussian pulses (solid) and their transforms (dashed): $\sigma = 1$ (blue) gives a transform of the same width; $\sigma = 2$ (red) is twice as long in time and half as wide in frequency.

Smoothness and spectral decay

The examples so far display a pattern: the rectangle (jumps) has a spectrum decaying as $1/|\omega|$; the triangle and two-sided exponential (continuous, with jumps in the first derivative) decay as $1/\omega^2$; the Gaussian (infinitely smooth) decays faster than any power.

Smoothness and decay

If $x, \dot{x}, \ldots, x^{(n-1)}$ are continuous and absolutely integrable and $x^{(n)}$ is absolutely integrable (possibly with finitely many jumps), then $|X(\omega)| \le C/|\omega|^n$ for large $|\omega|$. If $x^{(n)}$ has jumps, $X(\omega)$ decays *exactly* as $1/|\omega|^{n+1}$.

Reason. $x^{(n)} \leftrightarrow (j\omega)^n X$, and the transform of an absolutely integrable function is bounded; if $x^{(n)}$ has a jump, its derivative contains an impulse whose transform does not decay, so $(j\omega)^{n+1} X$ stays bounded but nonzero at infinity.

Engineering Note — smooth edges, narrow spectra

The spectral occupancy of a digital waveform is set by its edges, not its bit rate alone. Trapezoidal edges (continuous signal, jumps in slope) give $1/\omega^2$ (40 dB/decade) decay beyond a corner frequency of roughly $1/(\pi t_r)$ for rise time t_r; raised-cosine edges give $1/\omega^3$. Electromagnetic-compatibility estimates and pulse-shaping filters in data links are applications of this theorem.

16.5 Convolution and multiplication

Motivation

The output of an LTI system with impulse response $h(t)$ to an input $x(t)$ is the **convolution**

$$y(t) = (x * h)(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau.$$

Each input value $x(\tau)\,d\tau$ launches a scaled, delayed copy of the impulse response, and the output is their sum. Convolution integrals are tedious to evaluate directly — the integrand changes form as t moves the flipped h across x — and the single most useful fact about the Fourier transform is that it turns convolution into multiplication. The dual statement, that multiplication in time becomes convolution in frequency, describes modulation, windowing and sampling.

The two theorems

Convolution theorem

If x and h have Fourier transforms X and H (for instance both absolutely integrable), and $y = x * h$, then

$$Y(\omega) = X(\omega)H(\omega).$$

Proof. By definition,

$$Y(\omega) = \int_{-\infty}^{\infty}\left[\int_{-\infty}^{\infty} x(\tau)h(t-\tau)\,d\tau\right]e^{-j\omega t}\,dt.$$

Exchange the order of integration (absolutely convergent when x, h are absolutely integrable) and in the inner integral put $t = \tau + \sigma$, $dt = d\sigma$:

$$Y(\omega) = \int x(\tau)\left[\int h(\sigma)e^{-j\omega(\tau+\sigma)}d\sigma\right]d\tau = \int x(\tau)e^{-j\omega\tau}d\tau \cdot \int h(\sigma)e^{-j\omega\sigma}d\sigma = X(\omega)H(\omega).\ \square$$

Algebraic properties of convolution follow at once from those of multiplication: $x * h = h * x$ (commutative), $(x * h) * g = x * (h * g)$ (associative — cascaded systems multiply their frequency responses), $x * (h + g) = x * h + x * g$ (parallel systems add), and $x * \delta = x$ (because $\mathcal{F}\{\delta\} = 1$). Also $x(t - t_1) * h(t - t_2) = y(t - t_1 - t_2)$, since delays multiply as $e^{-j\omega t_1}e^{-j\omega t_2}$.

Multiplication (modulation) theorem

$$x(t)\,y(t) \;\leftrightarrow\; \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\lambda)\,Y(\omega-\lambda)\,d\lambda = \frac{1}{2\pi}(X * Y)(\omega).$$

Proof. Write x via the inverse formula and compute the transform of the product:

$$\int\left[\frac{1}{2\pi}\int X(\lambda)e^{j\lambda t}d\lambda\right]y(t)e^{-j\omega t}dt = \frac{1}{2\pi}\int X(\lambda)\left[\int y(t)e^{-j(\omega-\lambda)t}dt\right]d\lambda = \frac{1}{2\pi}\int X(\lambda)Y(\omega-\lambda)d\lambda.\ \square$$

The $1/(2\pi)$ is the price of the angular-frequency convention; in the Hz convention it disappears. The two theorems are duals: **convolution in one domain is multiplication in the other.** The modulation property of Section 16.3 is the special case $y = \cos\omega_0 t$, for which Y is a pair of impulses and convolution with an impulse is a shift: $X(\omega) * \delta(\omega - \omega_0) = X(\omega - \omega_0)$.

Common Pitfall — the 2π in the multiplication theorem

Forgetting the $1/(2\pi)$ is the commonest error with this theorem. A quick test: take $y(t) = 1$. Then $x \cdot 1$ must transform to X, and indeed $\frac{1}{2\pi}X * 2\pi\delta = X$ ✓. Without the factor you would get $2\pi X$.

Worked examples

Worked Example 16.19 — convolution by transforms

Find $y = x * h$ for $x(t) = e^{-t}u(t)$ and $h(t) = e^{-2t}u(t)$.

$$Y(\omega) = \frac{1}{(1+j\omega)(2+j\omega)} = \frac{1}{1+j\omega} - \frac{1}{2+j\omega} \;\Rightarrow\; y(t) = \left(e^{-t} - e^{-2t}\right)u(t).$$

Direct check: for $t > 0$, $\int_0^t e^{-\tau}e^{-2(t-\tau)}d\tau = e^{-2t}\int_0^t e^{\tau}d\tau = e^{-2t}(e^t - 1) = e^{-t} - e^{-2t}$ ✓. Also $Y(0) = \tfrac{1}{2} = X(0)H(0) = 1 \cdot \tfrac{1}{2}$, consistent with the general rule that **areas multiply under convolution**.

Worked Example 16.20 — triangle from rectangles

Show that $\mathrm{rect}(t/T) * \mathrm{rect}(t/T) = T\,\Lambda(t/T)$ and deduce the transform of the triangle.

Time domain. The flipped rectangle $\mathrm{rect}((t-\tau)/T)$ overlaps $\mathrm{rect}(\tau/T)$ on an interval of length $T - |t|$ when $|t| < T$ and not at all otherwise. So the convolution is $\max(0, T - |t|) = T\Lambda(t/T)$, a triangle of height T and base $2T$.

Frequency domain. By the convolution theorem, $T\Lambda(t/T) \leftrightarrow \left[T\,\mathrm{sinc}(\omega T/2\pi)\right]^2$, hence

$$\Lambda(t/T) \leftrightarrow T\,\mathrm{sinc}^2(\omega T/2\pi),$$

confirming Worked Example 16.11 by a third route. The sinc² decays as $1/\omega^2$ rather than $1/\omega$ because the triangle is continuous while the rectangle has jumps — **convolution smooths, and smoother signals have faster-decaying spectra.** Each further convolution with the rectangle multiplies the spectrum by another sinc and adds one more order of smoothness; the n-fold convolution tends to a Gaussian shape (the central limit theorem of Ch. 19 in transform form).

Worked Example 16.21 — an RC filter driven by a pulse

A unit pulse of width T starting at $t = 0$, $x(t) = u(t) - u(t - T)$, drives the RC low-pass filter $H(\omega) = 1/(1 + j\omega\tau)$, $\tau = RC$. Find the output.

Step 1 — transforms. From Worked Example 16.10, $X(\omega) = (1 - e^{-j\omega T})/(j\omega)$. So

$$Y(\omega) = \left(1 - e^{-j\omega T}\right) G_0(\omega), \qquad G_0(\omega) = \frac{1}{j\omega(1 + j\omega\tau)}.$$

Step 2 — recognise the step response. The step response $g(t) = h * u$ has transform $H(\omega)[\pi\delta(\omega) + 1/(j\omega)] = \pi\delta(\omega) + G_0(\omega)$. In Y the factor $1 - e^{-j\omega T}$ vanishes at $\omega = 0$, so adding the impulse terms changes nothing: $Y = (1 - e^{-j\omega T})[\pi\delta + G_0] = G(\omega) - e^{-j\omega T}G(\omega)$, i.e.

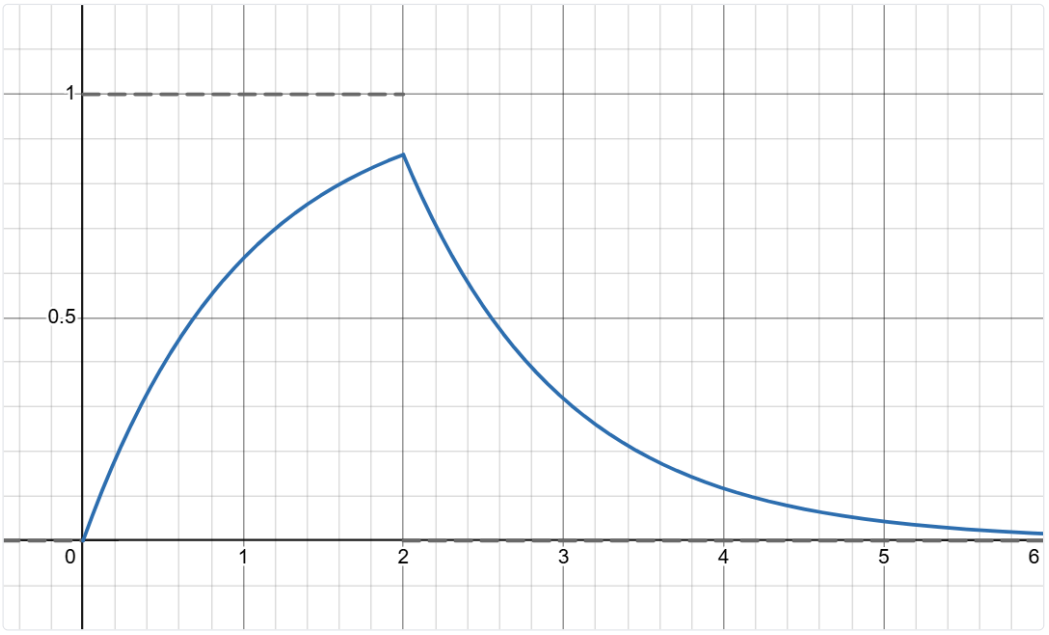
$$y(t) = g(t) - g(t - T).$$

Step 3 — find g. Partial fractions, $G_0 = \dfrac{1}{j\omega} - \dfrac{\tau}{1 + j\omega\tau}$, so $G = \pi\delta + \dfrac{1}{j\omega} - \dfrac{1}{1/\tau + j\omega}$, giving $g(t) = u(t) - e^{-t/\tau}u(t) = (1 - e^{-t/\tau})u(t)$.

Result.

$$y(t) = \left(1 - e^{-t/\tau}\right) u(t) - \left(1 - e^{-(t-T)/\tau}\right) u(t - T).$$

With $\tau = 1$ ms and $T = 2$ ms the output reaches $1 - e^{-2} = 0.865$ at the end of the pulse and then decays as $0.865\, e^{-(t-T)/\tau}$. Check: the output area is $X(0)H(0) = T \cdot 1 = 2$ ms, and integrating the result gives $\int_0^T (1 - e^{-t/\tau})dt + 0.865\tau = (T - \tau + \tau e^{-2}) + 0.865\tau = T$ ✓.



Rectangular input of width $T = 2\tau$ (grey dashed) and RC filter output (blue), time in units of τ: the output rises to 0.865 and decays exponentially — the low-pass filter has rounded the edges, removing the slow-decaying high-frequency content of the rectangle.

Worked Example 16.22 — the multiplication theorem

Find the transform of $x(t) = \left(\dfrac{\sin Wt}{\pi t}\right)^2$.

From Worked Example 16.5, $\sin(Wt)/(\pi t) \leftrightarrow \text{rect}(\omega/2W)$, a rectangle of width $2W$ in ω. By the multiplication theorem and Worked Example 16.20 (with $T = 2W$, variable ω),

$$X(\omega) = \frac{1}{2\pi}\,\text{rect}\left(\frac{\omega}{2W}\right) * \text{rect}\left(\frac{\omega}{2W}\right) = \frac{2W}{2\pi}\,\Lambda\left(\frac{\omega}{2W}\right) = \frac{W}{\pi}\Lambda\left(\frac{\omega}{2W}\right).$$

The squared sinc occupies **twice** the bandwidth of the sinc ($|\omega| < 2W$): squaring a signal doubles its bandwidth, which is why nonlinear devices generate new frequencies.

Check by $X(0) = \int x\,dt$: $\int \sin^2(Wt)/(\pi^2 t^2)\,dt = \pi W/\pi^2 = W/\pi$ (using $\int \sin^2(u)/u^2 du = \pi$) ✓.

Worked Example 16.23 — modulation and coherent demodulation

A message $m(t)$ band-limited to $|\omega| < \omega_M$ is transmitted as $s(t) = m(t)\cos\omega_c t$ with $\omega_c > \omega_M$ (double-sideband suppressed-carrier modulation). At the receiver it is multiplied again by $\cos\omega_c t$. Show that a low-pass filter recovers m.

Transmitter. $S(\omega) = \tfrac{1}{2}[M(\omega-\omega_c) + M(\omega+\omega_c)]$: two half-height copies of M centred on $\pm\omega_c$, occupying $\omega_c - \omega_M < |\omega| < \omega_c + \omega_M$.

Receiver. $r(t) = s(t)\cos\omega_c t = m(t)\cos^2\omega_c t = \tfrac{1}{2}m(t) + \tfrac{1}{2}m(t)\cos 2\omega_c t$, so

$$R(\omega) = \tfrac{1}{2}M(\omega) + \tfrac{1}{4}\left[M(\omega-2\omega_c) + M(\omega+2\omega_c)\right].$$

The baseband term occupies $|\omega| < \omega_M$ and the other two occupy $2\omega_c - \omega_M < |\omega| < 2\omega_c + \omega_M$. Since $\omega_c > \omega_M$, the gap $\omega_M < |\omega| < 2\omega_c - \omega_M$ separates them, and an ideal low-pass filter with gain 2 and any cutoff in that gap outputs exactly $m(t)$.

If the receiver's oscillator has a phase error, $\cos(\omega_c t + \theta)$, the baseband term becomes $\tfrac{1}{2}m(t)\cos\theta$: the output fades as $\theta \to 90^\circ$. This is why coherent receivers need carrier synchronisation — a conclusion reached entirely by the multiplication theorem.

Engineering Note — where the two theorems appear

- *Filtering* is multiplication of spectra: cascaded amplifier stages multiply gains, and in dB they add.
- *Mixing* (multiplying by a local oscillator) shifts spectra; superheterodyne receivers, frequency synthesisers and lock-in amplifiers are all the modulation property.
- *Windowing* a measured record (Section 16.6) and *sampling* (Section 16.8) are both multiplications in time and hence convolutions in frequency.
- Fast convolution in digital filters computes $x * h$ as $\mathcal{F}^{-1}\{XH\}$ with the FFT (Ch. 24), because a product is far cheaper than a convolution sum.

16.6 Spectral analysis

Spectral analysis asks: *which frequencies does a signal contain, and how strongly?* The Fourier transform answers it exactly for idealised signals defined for all time. A measured signal, however, is only ever observed for a finite time, and that limitation is itself a multiplication in time, so it changes the spectrum we see in a precisely calculable way. This section unifies periodic and aperiodic signals under one transform and then quantifies the effect of finite observation: resolution, leakage and windows.

Transforms of periodic signals

A periodic signal with Fourier series $x(t) = \sum_k c_k e^{jk\omega_0 t}$ (Ch. 15) has, term by term from $e^{jk\omega_0 t} \leftrightarrow 2\pi\delta(\omega - k\omega_0)$, the generalized transform

$$X(\omega) = 2\pi \sum_{k=-\infty}^{\infty} c_k\, \delta(\omega - k\omega_0),$$

a **line spectrum** of impulses at the harmonics, with areas $2\pi c_k$.

There is a second, illuminating derivation. Let $x_1(t)$ be one period of x (zero outside $|t| < T_0/2$), with transform $X_1(\omega)$. Then x is x_1 convolved with an impulse train, $x = x_1 * \sum_n \delta(t - nT_0)$, and the impulse-train pair (proved in Section 16.8) gives

$$X(\omega) = X_1(\omega)\cdot\omega_0\sum_k \delta(\omega - k\omega_0) = 2\pi\sum_k \frac{X_1(k\omega_0)}{T_0}\,\delta(\omega - k\omega_0).$$

Comparing, $c_k = X_1(k\omega_0)/T_0$ — the result found heuristically in Section 16.1, now as a theorem. **Periodicity in time is sampling in frequency.** The single-period spectrum $X_1(\omega)$ is the *envelope* of the line spectrum.

Aperiodic signals have continuous spectra; periodic ones have discrete spectra; a signal with both parts (a pulse plus a sinusoid, or a periodic signal plus a transient) has a mixed spectrum, and the transform handles all three with the same formulas.

Worked Example 16.24 — a rectangular pulse train

A clock signal consists of unit pulses of width $\tau = 1$ ms repeating every $T_0 = 4$ ms (duty cycle 25 %). Find its line spectrum.

One period is $x_1 = \mathrm{rect}(t/\tau)$, with $X_1(\omega) = \tau\,\mathrm{sinc}(\omega\tau/2\pi)$. With $\omega_0 = 2\pi/T_0$,

$$c_k = \frac{X_1(k\omega_0)}{T_0} = \frac{\tau}{T_0}\,\mathrm{sinc}\!\left(\frac{k\tau}{T_0}\right) = 0.25\,\mathrm{sinc}(0.25k).$$

k	0	±1	±2	±3	±4	±5
c_k	0.2500	0.2251	0.1592	0.0750	0	-0.0450

The DC value $c_0 = 0.25$ is the average of the waveform ✓. Every fourth harmonic vanishes because it falls on a null of the envelope ($k\tau/T_0$ an integer). The transform is $X(\omega) = 2\pi\sum_k c_k\delta(\omega - 2\pi k/T_0)$: lines every 250 Hz under a sinc envelope whose first null is at $1/\tau = 1$ kHz. Making the pulses narrower (smaller duty cycle) spreads the envelope and makes the lines denser relative to it; making the period longer packs the lines closer together under the same envelope shape — and in the limit $T_0 \to \infty$ they merge into the continuous spectrum of a single pulse.

Finite observation: windowing and leakage

Observing $x(t)$ only for $|t| < T/2$ is multiplication by a **window** $w(t) = \mathrm{rect}(t/T)$. By the multiplication theorem the spectrum we actually compute is

$$X_w(\omega) = \mathcal{F}\{x(t)w(t)\} = \frac{1}{2\pi}(X * W)(\omega), \qquad W(\omega) = T\,\mathrm{sinc}\left(\frac{\omega T}{2\pi}\right).$$

The true spectrum is blurred by the window's transform. A single spectral line, $2\pi c\,\delta(\omega-\omega_1)$, becomes $c\,W(\omega-\omega_1)$: an impulse is smeared into a sinc.

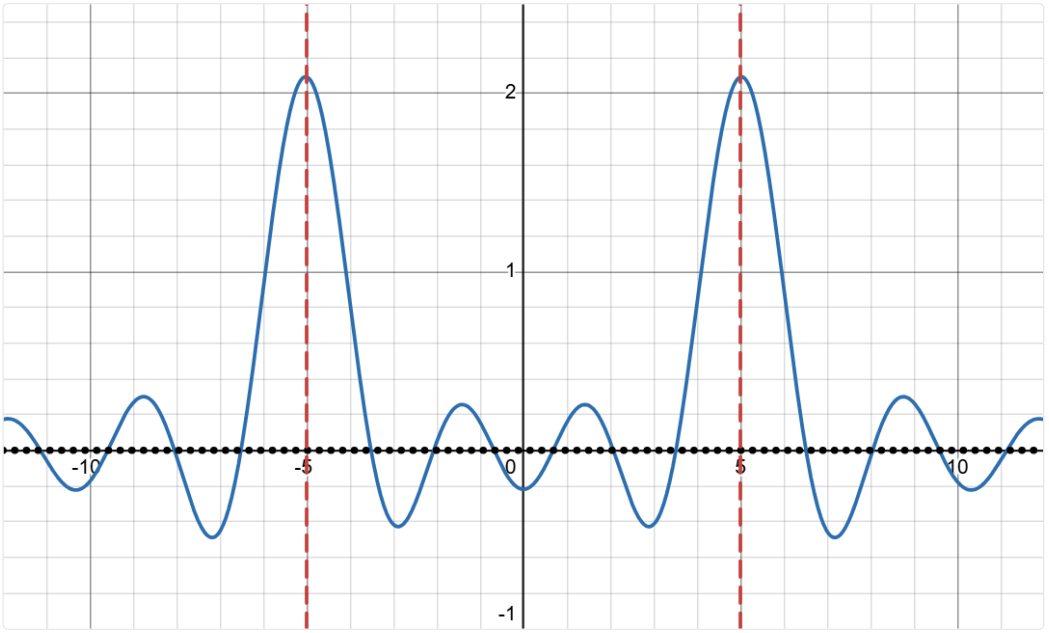
Worked Example 16.25 — a truncated cosine

Find the spectrum of $x(t) = \cos(\omega_0 t)\,\mathrm{rect}(t/T)$.

With $\cos\omega_0 t \leftrightarrow \pi[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)]$ and convolution with an impulse being a shift,

$$X_w(\omega) = \frac{T}{2}\left[\mathrm{sinc}\left(\frac{(\omega-\omega_0)T}{2\pi}\right) + \mathrm{sinc}\left(\frac{(\omega+\omega_0)T}{2\pi}\right)\right] = \frac{\sin\left((\omega-\omega_0)T/2\right)}{\omega-\omega_0} + \frac{\sin\left((\omega+\omega_0)T/2\right)}{\omega+\omega_0}.$$

Instead of two impulses we see two sinc lobes of peak $T/2$ (growing with observation time, as an impulse is approached) and main-lobe width $4\pi/T$, plus **sidelobes** whose largest is 0.217 of the peak (-13.3 dB), located about 1.43 bins ($1.43\times 2\pi/T$ rad/s) from the centre. This spreading of energy away from the true frequency is called **spectral leakage**.



Spectrum of $\cos(5t)$ observed for $T = 4$ s: the ideal lines at $\omega = \pm 5$ (red dashed) become sinc lobes of width $4\pi/T \approx 3.1$ rad/s with sidelobe leakage.

Two consequences govern every practical spectrum analysis.

- **Resolution.** Two sinusoids closer than about one main-lobe half-width, $2\pi/T$ rad/s (i.e. $1/T$ Hz), merge into a single lobe. Finer resolution requires a longer record: **frequency resolution is the reciprocal of observation time.** This is the time–bandwidth trade-off of Section 16.4 applied to measurement.
- **Leakage.** Sidelobes of a strong component can mask a weak component nearby. The rectangular window's sidelobes decay only as $1/|\omega|$ because the window has jumps.

Tapered windows

Leakage is reduced by replacing the abrupt rectangle with a window that goes smoothly to zero at its ends. By the smoothness theorem of Section 16.4, a smoother window has faster-decaying sidelobes. The standard example is the **Hann (raised-cosine) window**

$$w_H(t) = \tfrac{1}{2}\left[1+\cos\frac{2\pi t}{T}\right]\mathrm{rect}\left(\frac{t}{T}\right).$$

Its transform follows from the modulation property: the rectangle contributes $\frac{1}{2}W(\omega)$ and the cosine term contributes two shifted half-height copies $\frac{1}{4}W(\omega \mp 2\pi/T)$. With $u = \omega T/2\pi$,

$$W_H(\omega) = \frac{T}{2}\,\mathrm{sinc}(u) + \frac{T}{4}\left[\mathrm{sinc}(u-1)+\mathrm{sinc}(u+1)\right] = \frac{T}{2}\,\frac{\mathrm{sinc}(u)}{1-u^2}.$$

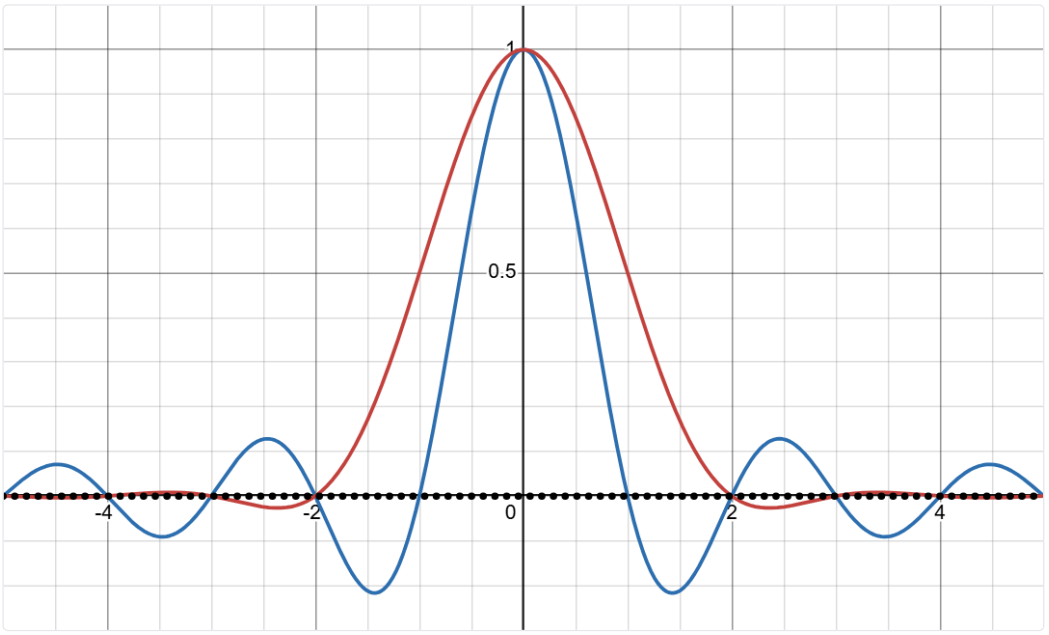
The three shifted sincs are arranged so that their sidelobes largely cancel. Compared with the rectangle:

- the first null moves from $u=1$ to $u=2$ (the main lobe is **twice as wide**);
- the highest sidelobe falls from -13.3 dB to -31.5 dB;
- the sidelobes decay as $1/|u|^3$ instead of $1/|u|$ (the Hann window is continuous with a continuous first derivative);
- the peak is $T/2$ instead of T (the window's area is halved).

Window	First null (u)	3-dB half-width (u)	Highest sidelobe	Sidelobe decay
Rectangular	1	0.44	-13.3 dB	$1/\lvert u\rvert$
Hann	2	0.72	-31.5 dB	$1/\lvert u\rvert^3$

Window	First null (u)	3-dB half-width (u)	Highest sidelobe	Sidelobe decay
Hamming ($0.54 + 0.46\cos$)	2	0.65	-42.7 dB	$1/\lvert u\rvert$
Blackman	3	0.82	-58.1 dB	$1/\lvert u\rvert^3$

The trade-off is universal: **lower sidelobes cost a wider main lobe**, i.e. better dynamic range costs resolution, because a window cannot be both short in time and narrow in frequency.



Normalised window transforms against $u = \omega T/2\pi$: rectangular $\operatorname{sinc}(u)$ (blue) and Hann $\operatorname{sinc}(u)/(1-u^2)$ (red). The Hann main lobe is twice as wide, but its sidelobes are almost invisible on this scale.

Worked Example 16.26 — choosing a record length and window

A signal contains a strong tone at 1000 Hz and a weak tone, 40 dB below it, at 1200 Hz. The analysis record is $T = 0.1$ s long. Can the weak tone be seen (a) with a rectangular window, (b) with a Hann window?

The frequency offset is $\Delta f = 200$ Hz, i.e. $u = \Delta f\, T = 20$ bins from the strong tone.

(a) Rectangular. The sidelobe envelope is $|\operatorname{sinc}(u)| \le 1/(\pi u)$; at $u = 20$ this is $1/(20\pi) = 0.0159$, i.e. -36.0 dB. The leakage from the strong tone at 1200 Hz is up to -36 dB, *above* the weak tone at -40 dB. The weak tone is masked.

(b) Hann. The envelope is $1/(\pi u(u^2-1))$; at $u = 20$ this is 4.0×10^{-5}, i.e. -88.0 dB. The weak tone stands about 48 dB above the leakage floor and is clearly visible. Its main lobe (null-to-null $4/T = 40$ Hz) is far narrower than the 200 Hz separation, so the wider Hann lobe costs nothing here.

(c) Resolution check. If instead the second tone were at 1015 Hz with equal amplitude, $u = 1.5$. With the rectangle the two main lobes (half-width 1 bin) are just separable; with Hann (half-width 2 bins) they merge. Either lengthen the record to $T \ge 0.2$ s, or accept the rectangle's leakage. Resolution and dynamic range compete, and the record length is the only free lunch.

Common Pitfall — reading peak heights from windowed spectra

The height of a windowed line depends on the window: it is $c\,W(0)$, i.e. proportional to the window's area (T for rectangular, $T/2$ for Hann). Amplitude estimates must be divided by the window's area (its "coherent gain"), and the frequency of a tone lying between computed frequency points is underestimated in height ("scalloping"). These are consequences of $X_w = \frac{1}{2\pi} X * W$, not instrument faults.

Time-varying spectra

The Fourier transform integrates over all time, so it cannot say *when* a frequency occurred: a chirp and a time-reversed chirp have the same magnitude spectrum. Sliding a short window along the signal and transforming each segment gives the **short-time Fourier transform**

$$X(\tau, \omega) = \int_{-\infty}^{\infty} x(t)\, w(t-\tau)\, e^{-j\omega t}\, dt,$$

whose magnitude squared (the spectrogram) is the standard picture of speech, radar returns and vibration. Its time and frequency resolutions are bounded by the window's duration and bandwidth, and the uncertainty principle forbids making both arbitrarily fine.

Engineering Note — from continuous theory to the FFT

A spectrum analyser's resolution bandwidth, and an FFT's bin spacing $1/T$ Hz, are both manifestations of $W(\omega)$. The discrete computation (DFT/FFT) evaluates samples of X_w, and its windowing behaviour is exactly the continuous theory above; the algorithms themselves are treated in Ch. 24.

16.7 Energy spectral density

Energy signals and power signals

How much energy does a pulse deliver, and how is that energy distributed over frequency? These questions decide how much bandwidth a channel must have to pass "most" of a pulse, how much noise a filter admits, and how much interference a transmitter puts into a neighbouring band.

Energy and power of a signal

The **energy** of $x(t)$ is

$$E = \int_{-\infty}^{\infty} |x(t)|^2 \, dt.$$

If x is a voltage across (or a current through) a $1\ \Omega$ resistor, E is the energy dissipated in joules; for a real load R divide (or multiply) by R. The **average power** is $P = \lim_{T\to\infty} \frac{1}{T}\int_{-T/2}^{T/2} |x(t)|^2 dt$. A signal with $0 < E < \infty$ is an **energy signal** (and has $P = 0$); a signal with $0 < P < \infty$ is a **power signal** (and has $E = \infty$).

Pulses and transients are energy signals; sinusoids, periodic waveforms, and stationary noise are power signals. This section treats energy signals; the power-signal theory is outlined at the end and developed for random signals in Ch. 21.

Parseval's theorem

Parseval's (Rayleigh's) energy theorem

If x has finite energy, then

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} |X(\omega)|^2 d\omega.$$

More generally, for two finite-energy signals,

$$\int_{-\infty}^{\infty} x(t)\, y^*(t)\, dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)\, Y^*(\omega)\, d\omega.$$

Proof. Substitute the inverse formula for x and exchange the order of integration:

$$\int x(t) y^*(t)\, dt = \int \left[\frac{1}{2\pi}\int X(\omega) e^{j\omega t} d\omega\right] y^*(t)\, dt = \frac{1}{2\pi}\int X(\omega)\left[\int y(t) e^{-j\omega t} dt\right]^* d\omega = \frac{1}{2\pi}\int X(\omega) Y^*(\omega)\, d\omega.$$

Setting $y = x$ gives the energy theorem. (Alternatively, apply the multiplication theorem to $x(t)x^*(t)$, using $x^* \leftrightarrow X^*(-\omega)$, and evaluate the result at $\omega = 0$.) $\square$

The general form says the Fourier transform preserves inner products (up to the factor 2π): it is a *rotation* of function space, in the language of Ch. 15's orthogonal expansions. In particular, two signals are orthogonal in time if and only if their transforms are orthogonal in frequency; signals occupying disjoint frequency bands are automatically orthogonal, which is the mathematical basis of frequency-division multiplexing.

Energy spectral density

Energy spectral density

The **energy spectral density (ESD)** of x is

$$\Psi_x(\omega) = |X(\omega)|^2.$$

By Parseval, $\Psi_x(\omega)\, d\omega/2\pi$ is the energy contained in the frequency band $d\omega$ (equivalently, $\Psi_x\, df$ in a band df Hz).

Units: for a voltage, $|X|^2$ is in $\text{V}^2\cdot\text{s}^2 = (\text{V}^2\cdot\text{s})/\text{Hz}$, i.e. joules (into $1\ \Omega$) per hertz. The ESD discards phase, so many different signals share the same ESD; a pulse and its delayed or time-reversed version, for example.

For **real** x the ESD is even, so the energy in the band $\omega_1 < |\omega| < \omega_2$ — counting both the positive and negative frequencies that make up the real signal components — is

$$E_{[\omega_1,\omega_2]} = \frac{1}{2\pi}\cdot 2\int_{\omega_1}^{\omega_2} |X(\omega)|^2 d\omega = \frac{1}{\pi}\int_{\omega_1}^{\omega_2} |X(\omega)|^2 d\omega.$$

ESD through an LTI system. Since $Y = HX$,

$$\Psi_y(\omega) = |H(\omega)|^2\, \Psi_x(\omega), \qquad E_y = \frac{1}{2\pi}\int |H(\omega)|^2 |X(\omega)|^2 d\omega.$$

The output ESD depends only on the **magnitude** response; phase distortion reshapes a pulse but does not change its energy or the energy in any band. The quantity $|H|^2$ is the **power gain**, and an ideal band-pass filter with $|H|^2 = 1$ on a band and 0 elsewhere measures the energy in that band — which is how a spectrum analyser physically measures the ESD.

Worked examples

Worked Example 16.27 — energy and essential bandwidth of an exponential pulse

For $x(t) = e^{-at}u(t)$: directly, $E = \int_0^\infty e^{-2at}dt = \dfrac{1}{2a}$. By Parseval,

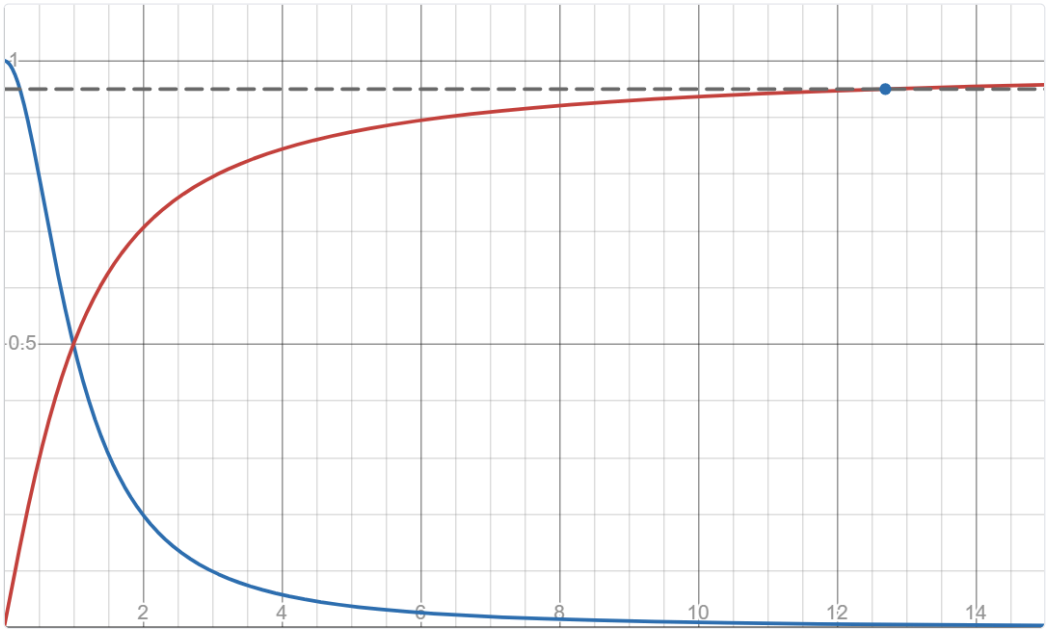
$$E = \frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{d\omega}{a^2+\omega^2} = \frac{1}{2\pi}\cdot\frac{\pi}{a} = \frac{1}{2a}\ \checkmark$$

The energy in $|\omega| < W$ is $\frac{1}{\pi}\int_0^W \frac{d\omega}{a^2+\omega^2} = \frac{1}{\pi a}\arctan\frac{W}{a}$, a fraction

$$\frac{E_W}{E} = \frac{2}{\pi}\arctan\frac{W}{a}.$$

At the 3-dB frequency $W = a$ this is exactly 50 %. For 95 %: $\arctan(W/a) = 0.95\pi/2$, so $W = a\tan(85.5^\circ) \approx 12.7a$. For 99 %, $W = a\tan(89.1^\circ) \approx 63.7a$. The slow $1/\omega^2$ decay of the ESD (caused by the jump at $t = 0$) makes the essential bandwidth very much larger than the 3-dB bandwidth.

With $a = 10^4$ s^{-1} (a 100 µs time constant), the 95 % essential bandwidth is 1.27×10^5 rad/s, about 20 kHz.



For $x = e^{-at}u(t)$ with ω in units of a: normalised ESD $a^2|X|^2 = 1/(1+\omega^2)$ (blue) and cumulative energy fraction $\frac{2}{\pi}\arctan\omega$ (red).

Worked Example 16.28 — the rectangular pulse and a sinc integral

(a) Use Parseval to evaluate $\displaystyle\int_{-\infty}^{\infty}\frac{\sin^2 u}{u^2}\,du$.

For $x = \mathrm{rect}(t)$, $E = 1$ and $X = 2\sin(\omega/2)/\omega$. Parseval gives $1 = \frac{1}{2\pi}\int\frac{4\sin^2(\omega/2)}{\omega^2}d\omega$. Put $u = \omega/2$, $d\omega = 2du$: $1 = \frac{1}{2\pi}\int\frac{4\sin^2 u}{4u^2}2\,du$, so

$$\int_{-\infty}^{\infty}\frac{\sin^2 u}{u^2}\,du = \pi.$$

(b) What fraction of the energy of a rectangular pulse of width T lies within its main lobe $|\omega| < 2\pi/T$?

$$\frac{E_{\text{main}}}{E} = \frac{1}{T}\cdot\frac{1}{\pi}\int_0^{2\pi/T}\frac{4\sin^2(\omega T/2)}{\omega^2}d\omega = \frac{2}{\pi}\int_0^{\pi}\frac{\sin^2 v}{v^2}dv \approx 0.903,$$

after substituting $v = \omega T/2$. The integral has no elementary closed form; numerical quadrature gives $\int_0^\pi \sin^2 v/v^2\,dv = 1.4182$, hence the fraction 0.903. So **about 90 % of the energy of a rectangular pulse lies in its main lobe**, even though its RMS bandwidth is infinite (Worked Example 16.18). A channel of bandwidth $1/T$ Hz passes 90 % of the energy of a pulse of duration T; it is the source of the "bandwidth ≈ 1/duration" rule.

Worked Example 16.29 — energy at the output of a filter

The pulse $x(t) = e^{-at}u(t)$ drives an RC low-pass filter $H = b/(b + j\omega)$, $b = 1/RC$. Find the output energy by Parseval and check it in the time domain.

Frequency domain. $|H|^2|X|^2 = \dfrac{b^2}{(b^2+\omega^2)(a^2+\omega^2)}$. With the partial-fraction identity

$$\frac{1}{(a^2+\omega^2)(b^2+\omega^2)} = \frac{1}{b^2-a^2}\left[\frac{1}{a^2+\omega^2} - \frac{1}{b^2+\omega^2}\right]$$

and $\int d\omega/(c^2+\omega^2) = \pi/c$, the integral is $\dfrac{\pi}{b^2-a^2}\left(\dfrac{1}{a}-\dfrac{1}{b}\right) = \dfrac{\pi}{ab(a+b)}$. Hence

$$E_y = \frac{1}{2\pi}\cdot b^2 \cdot \frac{\pi}{ab(a+b)} = \frac{b}{2a(a+b)}.$$

Time domain. For $a \neq b$, $y = x * h = \dfrac{b}{b-a}\left(e^{-at} - e^{-bt}\right)u(t)$, and

$$E_y = \frac{b^2}{(b-a)^2}\left[\frac{1}{2a} - \frac{2}{a+b} + \frac{1}{2b}\right] = \frac{b^2}{(b-a)^2}\cdot\frac{(b-a)^2}{2ab(a+b)} = \frac{b}{2a(a+b)}\ \checkmark$$

Numbers. $a = 500\ \text{s}^{-1}$, $b = 1000\ \text{s}^{-1}$: $E_x = 1/(2a) = 1.000\times10^{-3}$, $E_y = 1000/(2\cdot500\cdot1500) = 6.67\times10^{-4}$ (V²·s). The filter passes $E_y/E_x = b/(a+b) = 2/3$ of the input energy. As $b \to \infty$ (no filtering) the ratio tends to 1, and as $b \to 0$ to 0, as expected.

Worked Example 16.30 — essential bandwidth of a Gaussian pulse

For $x = e^{-t^2/(2\sigma^2)}$, $|X|^2 = 2\pi\sigma^2 e^{-\sigma^2\omega^2}$. The fraction of energy in $|\omega| < W$ is

$$\frac{E_W}{E} = \frac{\int_{-W}^{W} e^{-\sigma^2\omega^2}d\omega}{\int_{-\infty}^{\infty} e^{-\sigma^2\omega^2}d\omega} = \operatorname{erf}(\sigma W).$$

For 95 %, $\sigma W = \operatorname{erf}^{-1}(0.95) = 1.386$, so $W = 1.386/\sigma$. Compare the exponential pulse (Worked Example 16.27), whose 95 % bandwidth is 12.7 times its 3-dB bandwidth: the Gaussian's 3-dB bandwidth (where $e^{-\sigma^2\omega^2/2} = 1/\sqrt{2}$) is $\sqrt{\ln 2}/\sigma = 0.833/\sigma$, only a factor 1.66 below its 95 % bandwidth. Smooth pulses have compact spectra.

Autocorrelation and the energy Wiener–Khinchin relation

Deterministic autocorrelation

The autocorrelation of a finite-energy signal is

$$R_x(\tau) = \int_{-\infty}^{\infty} x(t+\tau)\,x^*(t)\,dt.$$

It measures the similarity of x with a copy of itself shifted by τ. For real x it is real and even, and $|R_x(\tau)| \le R_x(0) = E$ by the Cauchy–Schwarz inequality.

Substituting $s = -t$ shows $R_x(\tau) = x(\tau) * x^*(-\tau)$. Since $x^*(-t) \leftrightarrow X^*(\omega)$, the convolution theorem gives

Autocorrelation–ESD pair

$$R_x(\tau) \;\leftrightarrow\; |X(\omega)|^2 = \Psi_x(\omega), \qquad R_x(0) = E = \frac{1}{2\pi}\int \Psi_x(\omega)\,d\omega.$$

This is the energy-signal form of the **Wiener–Khinchin theorem**: the ESD is the Fourier transform of the autocorrelation. Because Ψ_x carries no phase information, neither does R_x; any two signals with the same magnitude spectrum have the same autocorrelation. Similarly the **cross-correlation** $R_{xy}(\tau) = \int x(t+\tau)y^*(t)\,dt$ transforms to the **cross-energy spectral density** $X(\omega)Y^*(\omega)$.

Worked Example 16.31 — autocorrelation of an exponential pulse

For $x = e^{-at}u(t)$ and $\tau \ge 0$: $R_x(\tau) = \int_0^\infty e^{-a(t+\tau)}e^{-at}dt = e^{-a\tau}/(2a)$. By evenness $R_x(\tau) = e^{-a|\tau|}/(2a)$. Its transform is $\dfrac{1}{2a}\cdot\dfrac{2a}{a^2+\omega^2} = \dfrac{1}{a^2+\omega^2} = |X(\omega)|^2$ ✓, and $R_x(0) = 1/(2a) = E$ ✓.

Power signals

For a power signal the ESD is infinite, and we work with power per unit bandwidth instead. Truncate the signal to $x_T(t) = x(t)\,\text{rect}(t/T)$, compute its ESD $|X_T(\omega)|^2$, divide by the duration and let $T \to \infty$:

$$S_x(\omega) = \lim_{T\to\infty}\frac{|X_T(\omega)|^2}{T}, \qquad P = \frac{1}{2\pi}\int_{-\infty}^{\infty} S_x(\omega)\,d\omega.$$

This is the **power spectral density**. For a periodic signal it reduces to Parseval's theorem for Fourier series, $P = \sum_k |c_k|^2$, with $S_x(\omega) = 2\pi\sum_k |c_k|^2\delta(\omega - k\omega_0)$. For random signals, where the limit must be combined with an expectation, the PSD $S_{XX}(\omega)$ and its relation to the autocorrelation $R_{XX}(\tau)$ are the subject of Ch. 21.

Common Pitfall — one-sided versus two-sided spectra

Integrating $|X|^2$ over $\omega > 0$ only and forgetting the factor 2 halves every energy figure. Conversely, many instruments display a *one-sided* density that already includes the factor 2. Before comparing numbers, establish whether a density is per rad/s or per Hz (they differ by 2π) and one- or two-sided (they differ by 2).

Engineering Note — matched filtering

The generalized Parseval theorem underlies the **matched filter**: among all filters, the one with $H(\omega) = X^*(\omega)e^{-j\omega t_0}$ maximises the output peak relative to white noise, and its output at t_0 is $\frac{1}{2\pi}\int|X|^2 d\omega = E$, the pulse energy. Its impulse response $x(t_0 - t)$ makes the filter output the autocorrelation $R_x(t - t_0)$. Radar range resolution and digital-receiver detection both rest on these two facts.

16.8 Sampling-related mathematical foundations

Every digital system — a data converter, a software radio, a digital oscilloscope, a control loop running on a microcontroller — replaces a continuous signal $x(t)$ by its samples $x(nT)$. Whether anything is lost, and if so what, is a question about Fourier transforms. This section gives the mathematics of ideal (instantaneous) sampling: the spectrum of an impulse train, the spectral replication caused by sampling, the Nyquist condition under which the replicas do not overlap, aliasing when they do, and the interpolation formula that reconstructs $x(t)$. Converter hardware is outside the scope of this book; the discrete-time transforms that take over once the samples exist are the subject of Ch. 17.

Notation: T is the sampling interval, $f_s = 1/T$ the sampling rate in Hz and $\omega_s = 2\pi/T$ in rad/s.

The impulse train and its transform

Impulse train (Dirac comb)

For $p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$,

$$P(\omega) = \omega_s \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_s), \qquad \omega_s = \frac{2\pi}{T}.$$

An impulse train in time, of spacing T and unit areas, transforms to an impulse train in frequency, of spacing $\omega_s = 2\pi/T$ and areas ω_s.

Proof. $p(t)$ is periodic with period T. Its Fourier series coefficients are

$$c_k = \frac{1}{T}\int_{-T/2}^{T/2} \delta(t) e^{-jk\omega_s t} dt = \frac{1}{T} \quad \text{for every } k,$$

so $p(t) = \frac{1}{T}\sum_k e^{jk\omega_s t}$. Transforming term by term with $e^{jk\omega_s t} \leftrightarrow 2\pi\delta(\omega - k\omega_s)$ gives $P(\omega) = \frac{2\pi}{T}\sum_k \delta(\omega - k\omega_s)$. □

The comb is (up to scale) its own transform, and its spacings are reciprocal: **dense sampling in time means widely spaced lines in frequency.** The series $p(t) = \frac{1}{T}\sum_k e^{jk\omega_s t}$ does not converge in the ordinary sense; it is an identity between generalized functions, meaning that both sides give the same result when integrated against any smooth, rapidly decaying test function.

Poisson summation formula

For a sufficiently smooth and rapidly decaying $x(t)$ with transform $X(\omega)$,

$$\sum_{n=-\infty}^{\infty} x(nT)\, e^{-j\omega nT} = \frac{1}{T}\sum_{k=-\infty}^{\infty} X(\omega - k\omega_s),$$

and in particular (at $\omega = 0$) $T\sum_n x(nT) = \sum_k X(k\omega_s)$.

This is the impulse-train theorem integrated against x, and it is the analytic heart of everything that follows: the left side is computed from samples, the right side from the continuous spectrum.

Ideal sampling and spectral replication

Ideal (impulse) sampling of $x(t)$ at interval T is modelled as multiplication by the comb:

$$x_s(t) = x(t)\, p(t) = \sum_{n=-\infty}^{\infty} x(nT)\, \delta(t - nT).$$

The product keeps only the values $x(nT)$, each as the area of an impulse. By the multiplication theorem,

$$X_s(\omega) = \frac{1}{2\pi} X(\omega) * \omega_s \sum_k \delta(\omega - k\omega_s) = \frac{1}{T}\sum_{k=-\infty}^{\infty} X(\omega - k\omega_s).$$

Spectrum of a sampled signal

$$X_s(\omega) = \frac{1}{T}\sum_{k=-\infty}^{\infty} X(\omega - k\omega_s) = \sum_{n=-\infty}^{\infty} x(nT)\, e^{-j\omega nT}.$$

Sampling replicates the spectrum at every multiple of ω_s, scaled by $1/T$. The sampled spectrum is periodic in ω with period ω_s.

The second form follows by transforming x_s term by term ($\delta(t-nT) \leftrightarrow e^{-j\omega nT}$); the equality of the two forms is the Poisson summation formula. The second form is the bridge to discrete time: with the normalised frequency $\Omega = \omega T$ (radians per sample), $X_s = \sum_n x[n]e^{-j\Omega n}$ is the **discrete-time Fourier transform** of the sequence $x[n] = x(nT)$, and with $z = e^{sT}$ (so $z = e^{j\Omega}$ on the $j\omega$-axis) it is the Z-transform of Ch. 17 evaluated on the unit circle.

Periodicity in frequency is the dual of Section 16.6's statement that periodicity in time gives a line spectrum: **sampling in one domain is periodic repetition in the other.**

Worked Example 16.32 — the sampled spectrum of a two-sided exponential

Sample $x(t) = e^{-a|t|}$ (transform $2a/(a^2+\omega^2)$, not band-limited) at interval T. Find $X_s(\omega)$ in closed form and compare TX_s with X.

From the samples. With $r = e^{-aT} < 1$, $x(nT) = r^{|n|}$, and two geometric series give

$$X_s(\omega) = \sum_{n=0}^{\infty} r^n e^{-j\omega nT} + \sum_{n=1}^{\infty} r^n e^{j\omega nT} = \frac{1}{1 - re^{-j\omega T}} + \frac{re^{j\omega T}}{1 - re^{j\omega T}} = \frac{1 - r^2}{1 - 2r\cos\omega T + r^2}.$$

From the replication formula the same function must equal $\frac{1}{T}\sum_k \dfrac{2a}{a^2 + (\omega - k\omega_s)^2}$. With $a = 1$, $T = 0.5$ s ($r = 0.6065$) and $\omega = 1.3$ rad/s, the closed form gives 1.57173; summing the replicas numerically for $|k| \le 20000$ gives 1.57173 ✓.

Aliasing error. Ideally $TX_s = X$ in the band $|\omega| < \omega_s/2$. Here:

ω	$X(\omega)$	$TX_s(\omega)$
0	2.000	$T\frac{1+r}{1-r} = 2.041$
$\omega_s/2 = 2\pi$ rad/s	0.0494	$T\frac{1-r}{1+r} = 0.1225$

Near DC the error is 2 %, but at the folding frequency the sampled spectrum is about 2.5 times the true one: the tails of the replicas centred at $\pm\omega_s$ have folded onto the band edge. A signal whose spectrum decays slowly suffers aliasing at any finite sampling rate.

Nyquist condition and reconstruction

If $X(\omega) = 0$ for $|\omega| > \omega_M$, the replicas $X(\omega - k\omega_s)$ occupy the intervals $|\omega - k\omega_s| < \omega_M$. Adjacent replicas do not overlap provided $\omega_s - \omega_M > \omega_M$, i.e. $\omega_s > 2\omega_M$. In that case the baseband replica ($k = 0$) can be isolated by a low-pass filter, and $x(t)$ recovered exactly.

Sampling theorem (Shannon–Nyquist)

If $x(t)$ is band-limited, $X(\omega) = 0$ for $|\omega| > \omega_M$, and

$$\omega_s > 2\omega_M \qquad (\text{equivalently } f_s > 2f_M),$$

then $x(t)$ is uniquely determined by its samples $x(nT)$ and

$$x(t) = \sum_{n=-\infty}^{\infty} x(nT)\,\operatorname{sinc}\!\left(\frac{t - nT}{T}\right) = \sum_{n=-\infty}^{\infty} x(nT)\,\frac{\sin\big(\pi(t-nT)/T\big)}{\pi(t-nT)/T}.$$

$2\omega_M$ (or $2f_M$) is the **Nyquist rate**; $\omega_s/2$ (or $f_s/2$) is the **Nyquist (folding) frequency**.

Proof. Because the replicas do not overlap, $X(\omega) = T\,X_s(\omega)$ for $|\omega| < \omega_s/2$ and $X(\omega) = 0$ for $\omega_M < |\omega|$. So multiplying X_s by the ideal low-pass filter

$$H_r(\omega) = T\,\operatorname{rect}\!\left(\frac{\omega}{\omega_s}\right) \qquad (\text{gain } T,\ \text{cutoff } \omega_s/2)$$

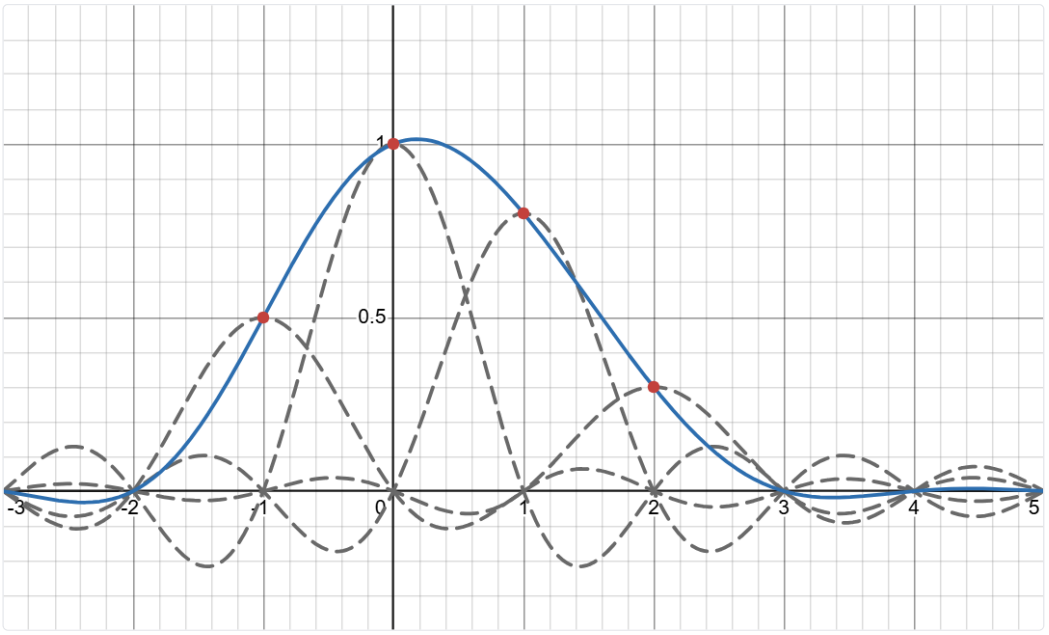
returns exactly X: $X(\omega) = H_r(\omega)X_s(\omega)$. By Worked Example 16.5 with $W = \omega_s/2 = \pi/T$, the impulse response of H_r is

$$h_r(t) = T \cdot \frac{\sin(\pi t/T)}{\pi t} = \operatorname{sinc}\!\left(\frac{t}{T}\right).$$

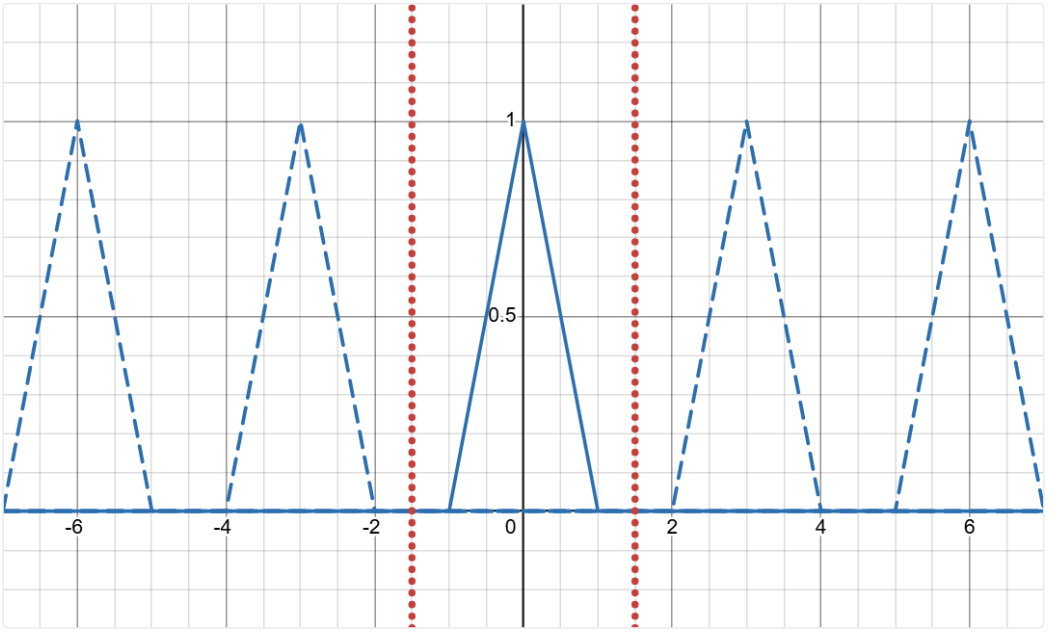
In the time domain, $x = h_r * x_s = h_r * \sum_n x(nT)\delta(t-nT) = \sum_n x(nT)h_r(t-nT)$, which is the interpolation formula. □

Two features of the interpolation formula deserve attention.

- **Interpolation property.** $\operatorname{sinc}((t-nT)/T)$ equals 1 at $t = nT$ and 0 at every other sample instant $t = mT$, $m \neq n$. So the formula passes exactly through every sample, and between samples each sinc contributes in proportion to its sample value.
- **Orthogonality.** The shifted sincs $\operatorname{sinc}((t-nT)/T)$ are mutually orthogonal, each with energy T. By the generalized Parseval theorem, with transforms $T\,\operatorname{rect}(\omega/\omega_s)e^{-j\omega nT}$, $\int \operatorname{sinc}\left(\frac{t-nT}{T}\right)\operatorname{sinc}\left(\frac{t-mT}{T}\right)dt = \frac{1}{2\pi}\int_{-\pi/T}^{\pi/T} T^2 e^{-j\omega(n-m)T}d\omega = T\,\delta_{nm}$, where δ_{nm} is the Kronecker delta. The sampling theorem is therefore an orthogonal expansion, like a Fourier series, in which the coefficients are simply the sample values: $x(nT) = \frac{1}{T}\int x(t)\operatorname{sinc}((t-nT)/T)\,dt$. It follows that $E = T\sum_n |x(nT)|^2$ for a band-limited signal sampled above the Nyquist rate.



Sinc interpolation with $T = 1$: four samples (red), the weighted sincs $x(nT)\,\mathrm{sinc}(t - n)$ (grey dashed), and their sum (blue), which passes through every sample because each sinc vanishes at the other sample instants.



Replicated spectrum $TX_s(\omega)$ for a triangular X with $\omega_M = 1$ and $\omega_s = 3 > 2\omega_M$: the replicas are separated by guard bands, so a low-pass filter with cutoff $\omega_s/2 = 1.5$ (red) recovers X exactly.

Worked Example 16.33 — Nyquist rates of derived signals

Let $x(t) = \dfrac{\sin(200\pi t)}{\pi t}$, band-limited to $\omega_M = 200\pi$ rad/s ($f_M = 100$ Hz). Find the Nyquist rate of each signal.

(a) $x(t)$ itself: $f_s > 2f_M = 200$ Hz.

(b) $x^2(t)$: by Worked Example 16.22 squaring doubles the bandwidth to $f_M = 200$ Hz, so $f_s > 400$ Hz.

(c) $x(t)\cos(1000\pi t)$: modulation moves the spectrum to $400 < |f| < 600$ Hz. Treated as a low-pass signal with highest frequency 600 Hz, $f_s > 1200$ Hz. (Band-pass sampling theory allows lower rates for such signals by exploiting the empty band below 400 Hz; the condition is that no replica lands on the occupied band.)

(d) $x(t) * y(t)$ with y band-limited to 50 Hz: the product $X(\omega)Y(\omega)$ vanishes outside the *narrower* band, so $f_s > 100$ Hz.

(e) $x(t) + x(t - t_0)$: delay does not change the band, so $f_s > 200$ Hz.

Multiplication widens bands (bandwidths add, by convolution of spectra); filtering and delay do not.

Worked Example 16.34 — reconstruction from samples, and truncation

The tone $x(t) = \cos(6\pi t)$ ($f_0 = 3$ Hz) is sampled at $f_s = 10$ Hz ($T = 0.1$ s), above its Nyquist rate of 6 Hz. Reconstruct $x(0.05)$, midway between two samples.

The samples are $x(nT) = \cos(0.6\pi n)$ and

$$x(0.05) = \sum_n \cos(0.6\pi n)\,\mathrm{sinc}(0.5 - n).$$

The exact value is $\cos(0.3\pi) = 0.587785$. Summing over $|n| \le N$:

N	5	50	500	5000
partial sum	0.584465	0.587725	0.587785	0.587785

The sum converges to the correct value, but slowly — the sinc tails decay only as $1/|n|$, so even 11 samples leave an error of 3×10^{-3}. Practical interpolators therefore use a faster-decaying kernel, which in turn requires sampling somewhat above the Nyquist rate so that a gentler filter edge fits into the guard band.

Aliasing

If $\omega_s < 2\omega_M$, adjacent replicas overlap. In the baseband $|\omega| < \omega_s/2$, $TX_s(\omega) = X(\omega) + X(\omega - \omega_s) + X(\omega + \omega_s) + \cdots$, and the extra terms are the high-frequency parts of X **folded** about $\pm\omega_s/2$. The folded content is indistinguishable from genuine low-frequency content: no processing of the samples can separate them. This is **aliasing**.

For a single sinusoid the effect is explicit. A tone of frequency f_0 sampled at f_s produces the same samples as one of frequency

$$f_a = |f_0 - k f_s|, \qquad k = \text{the integer nearest } f_0/f_s,$$

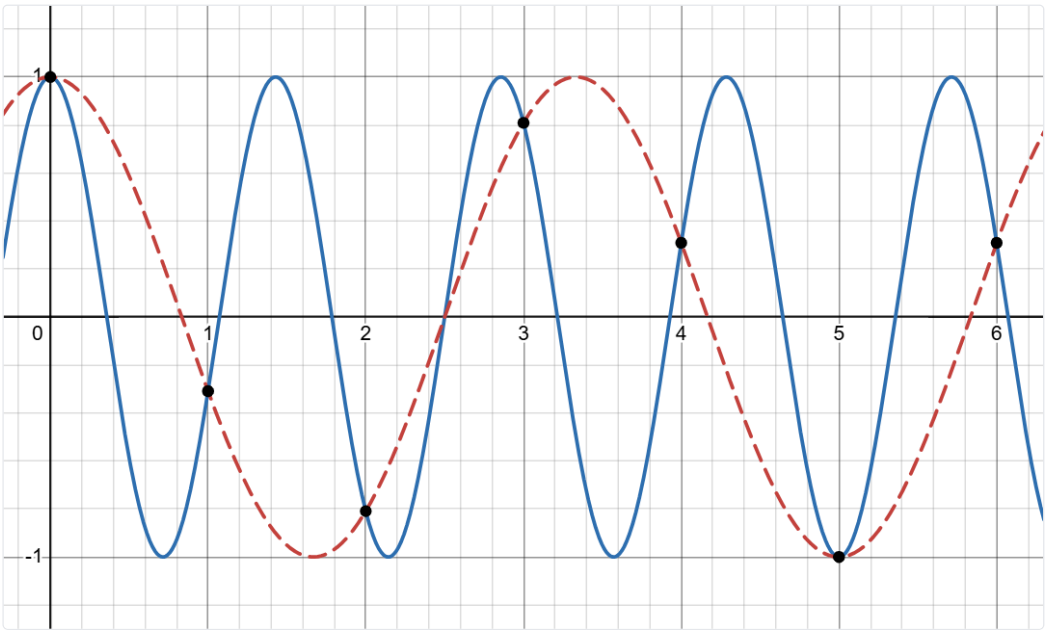
which lies in $[0, f_s/2]$. Indeed $\cos(2\pi f_0 nT + \theta) = \cos(2\pi(f_0 - kf_s)nT + \theta)$ because $2\pi k f_s nT = 2\pi kn$. When $f_0 - kf_s < 0$, writing the result in terms of the positive frequency f_a reverses the sign of the phase: a sine aliases to a *negated* sine.

Worked Example 16.35 — aliasing of a tone

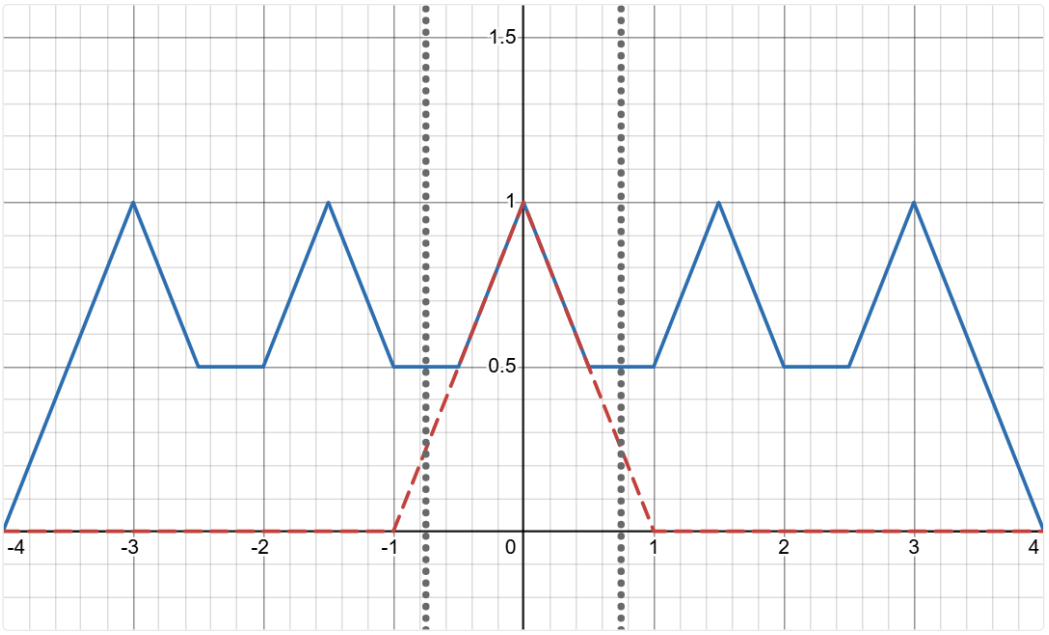
A 7 kHz tone is sampled at $f_s = 10$ kHz. Here $f_0/f_s = 0.7$, so $k = 1$ and $f_a = |7 - 10| = 3$ kHz.

The samples are $\cos(2\pi \cdot 0.7n)$ and $\cos(2\pi \cdot 0.3n)$, which are identical since $0.7n = n - 0.3n$ and cosine is even: both sequences are $1,\ -0.309,\ -0.809,\ 0.809,\ 0.309,\ -1, \ldots$ For a sine, $\sin(2\pi \cdot 0.7n) = -\sin(2\pi \cdot 0.3n)$: the alias has inverted phase.

To avoid this, the signal must be band-limited to below $f_s/2 = 5$ kHz *before* sampling, or f_s must exceed 14 kHz. A tone at 13 kHz would alias to $|13 - 10| = 3$ kHz as well, and one at 17 kHz to $|17 - 20| = 3$ kHz: every frequency $kf_s \pm 3$ kHz lands on the same alias.



Aliasing in time, with t measured in sample periods: the 7 kHz tone (blue) and the 3 kHz tone (red dashed) pass through exactly the same samples (black) at $f_s = 10$ kHz.



Undersampling ($\omega_M = 1, \omega_s = 1.5 < 2\omega_M$): the sum of overlapping replicas (blue) no longer matches the original triangle (red dashed) inside $|\omega| < \omega_s/2$ — aliasing.

> **Common Pitfall — sampling exactly at twice the highest frequency**
>
> Sampling at $f_s = 2f_M$ is not enough for a sinusoid at f_M: $\sin(2\pi f_M nT) = \sin(\pi n) = 0$ for all n, so the tone is invisible, and $\cos(\pi n + \theta) = \cos\theta\,(-1)^n$ cannot distinguish amplitude from phase. The inequality in the sampling theorem is strict. (For a signal whose spectrum contains no impulse at $\pm\omega_M$, equality is harmless, but nothing is gained by relying on it.)

Non-ideal reconstruction: the zero-order hold

The interpolation formula needs the ideal, non-causal sinc kernel. The simplest causal alternative holds each sample constant for one interval. Mathematically this is the convolution of x_s with a unit pulse on $[0, T)$:

$$x_{\text{ZOH}}(t) = \sum_n x(nT)\,\text{rect}\left(\frac{t - nT - T/2}{T}\right) \;\leftrightarrow\; X_s(\omega)\cdot T\,\text{sinc}\left(\frac{\omega T}{2\pi}\right) e^{-j\omega T/2}.$$

Compared with the ideal reconstruction filter (flat gain T in the baseband), the hold introduces (i) a magnitude **droop** $\text{sinc}(\omega T/2\pi)$, which at the folding frequency is $\text{sinc}(\tfrac{1}{2}) = 2/\pi = 0.637$ (-3.9 dB); (ii) a pure delay of $T/2$ (linear phase); and (iii) incomplete suppression of the replicas, attenuated only by the sinc sidelobes. Each is read directly from the convolution theorem.

> **Engineering Note — anti-aliasing and oversampling**
>
> Real systems place an analog anti-aliasing low-pass filter before the sampler and oversample (for example $f_s = 2.5f_M$, as in audio at 44.1 kHz for a 20 kHz band) to leave a guard band for a realisable filter roll-off. Choosing the filter order and the sampling rate is a design application of the replication formula $X_s = \frac{1}{T}\sum_k X(\omega - k\omega_s)$: the attenuation the filter provides at $f_s - f_M$ is the attenuation of the worst alias that can land in band.

Summary

Concept	Formula
Forward transform	$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$
Inverse transform	$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)e^{j\omega t}d\omega$ (midpoint at jumps)
Hz convention	$X_f(f) = X(2\pi f)$, $x(t) = \int X_f(f)e^{j2\pi ft}df$
Relation to Laplace	$X(\omega) = X_L(j\omega)$ if the ROC contains the $j\omega$-axis
DC values	$X(0) = \int x\,dt$, $x(0) = \frac{1}{2\pi}\int X\,d\omega$
Exponential	$e^{-at}u(t) \leftrightarrow 1/(a + j\omega)$; $e^{-a\lvert t\rvert} \leftrightarrow 2a/(a^2 + \omega^2)$
Rectangle / sinc	$\text{rect}(t/T) \leftrightarrow T\,\text{sinc}(\omega T/2\pi)$; $\sin(Wt)/(\pi t) \leftrightarrow \text{rect}(\omega/2W)$
Triangle	$\Lambda(t/T) \leftrightarrow T\,\text{sinc}^2(\omega T/2\pi)$
Gaussian	$e^{-t^2/2\sigma^2} \leftrightarrow \sigma\sqrt{2\pi}e^{-\sigma^2\omega^2/2}$
Generalized pairs	$\delta \leftrightarrow 1$, $1 \leftrightarrow 2\pi\delta(\omega)$, $u \leftrightarrow \pi\delta(\omega) + 1/(j\omega)$, $\text{sgn} \leftrightarrow 2/(j\omega)$
Sinusoids	$\cos\omega_0 t \leftrightarrow \pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$
Time shift / frequency shift	$x(t - t_0) \leftrightarrow e^{-j\omega t_0}X$; $e^{j\omega_0 t}x \leftrightarrow X(\omega - \omega_0)$
Modulation	$x\cos\omega_0 t \leftrightarrow \frac{1}{2}[X(\omega - \omega_0) + X(\omega + \omega_0)]$

Concept	Formula
Scaling	$x(at) \leftrightarrow X(\omega/a)/\lvert a\rvert$
Duality	$X(t) \leftrightarrow 2\pi x(-\omega)$
Differentiation / multiplication by t	$\dot{x} \leftrightarrow j\omega X$; $tx \leftrightarrow j\,dX/d\omega$
Integration	$\int_{-\infty}^{t} x\,d\tau \leftrightarrow X/(j\omega) + \pi X(0)\delta(\omega)$
Convolution / multiplication	$x * h \leftrightarrow XH$; $xy \leftrightarrow \frac{1}{2\pi} X * Y$
Real-signal symmetry	$X(-\omega) = X^*(\omega)$; $\lvert X\rvert$ even, $\angle X$ odd
Frequency response	$e^{j\omega t} \to H(\omega)e^{j\omega t}$, $H = \mathcal{F}\{h\}$
Distortionless system / group delay	$H = Ke^{-j\omega t_d}$; $\tau_g = -d\angle H/d\omega$
Uncertainty principle	$\Delta t\,\Delta\omega \geq \frac{1}{2}$, equality for the Gaussian
Periodic signal	$X = 2\pi\sum_k c_k\delta(\omega - k\omega_0)$, $c_k = X_1(k\omega_0)/T_0$
Windowing	$X_w = \frac{1}{2\pi} X * W$; resolution $\approx 1/T$ Hz
Hann window	$W_H = \frac{T}{2}\,\mathrm{sinc}(u)/(1-u^2)$, $u = \omega T/2\pi$
Parseval	$\int\lvert x\rvert^2 dt = \frac{1}{2\pi}\int\lvert X\rvert^2 d\omega$; $\int xy^*dt = \frac{1}{2\pi}\int XY^*d\omega$
ESD	$\Psi_x = \lvert X\rvert^2 \leftrightarrow R_x(\tau)$; $\Psi_y = \lvert H\rvert^2\Psi_x$
Band energy (real x)	$E_{[\omega_1,\omega_2]} = \frac{1}{\pi}\int_{\omega_1}^{\omega_2}\lvert X\rvert^2 d\omega$
Impulse train	$\sum_n \delta(t - nT) \leftrightarrow \omega_s\sum_k \delta(\omega - k\omega_s)$
Sampled spectrum	$X_s(\omega) = \frac{1}{T}\sum_k X(\omega - k\omega_s) = \sum_n x(nT)e^{-j\omega nT}$
Nyquist condition	$\omega_s > 2\omega_M$ ($f_s > 2f_M$)
Interpolation	$x(t) = \sum_n x(nT)\,\mathrm{sinc}((t-nT)/T)$
Alias frequency	$f_a = \lvert f_0 - kf_s\rvert$, k nearest integer to f_0/f_s

Further reading

- A. V. Oppenheim, A. S. Willsky & S. H. Nawab, *Signals and Systems*, 2nd ed., Prentice Hall — Chs. 4 (continuous-time Fourier transform), 6 (time and frequency characterisation) and 7 (sampling); uses the angular-frequency convention of this chapter.
- R. N. Bracewell, *The Fourier Transform and Its Applications*, 3rd ed., McGraw-Hill — the classic treatment in the Hz convention, with extensive pictorial dictionaries of pairs and a thorough account of the impulse train and sampling.
- B. P. Lathi & R. Green, *Signal Processing and Linear Systems*, 2nd ed., Oxford University Press — energy spectral density, essential bandwidth and filtering, with many circuit examples.
- A. Papoulis, *The Fourier Integral and Its Applications*, McGraw-Hill — rigorous yet engineering-oriented treatment of existence, generalized functions, windows and the uncertainty principle.

Chapter 17 — Z-Transforms

> **Prerequisites:** Ch. 2 (complex numbers), Ch. 4–5 (sequences, series and integral calculus), Ch. 14 (Laplace transforms), Ch. 16 (Fourier transforms and sampling) **Learning outcomes:** After this chapter you can
>
> - describe discrete-time signals as sequences, manipulate them with shifts, folds and convolution sums, and classify them (energy/power, periodic, causal);
> - compute bilateral and unilateral Z-transforms of standard sequences and apply the transform properties instead of summing series;
> - determine and sketch the region of convergence (ROC) for right-sided, left-sided, two-sided and finite sequences, and read causality and stability from it;
> - invert Z-transforms by partial fractions, power-series (long) division and the residue theorem;
> - solve linear constant-coefficient difference equations with initial conditions using the unilateral transform;
> - represent discrete-time LTI systems by their system function $H(z)$, pole–zero plots and block diagrams, and analyse their stability and frequency response $H(e^{j\omega})$, relating them to continuous-time systems through $z = e^{sT}$.

The Laplace transform of Ch. 14 turns linear differential equations into algebra; the Fourier transform of Ch. 16 describes signals as superpositions of sinusoids. Digital electronics, however, does not deal in functions of a continuous variable t. An analogue-to-digital converter delivers a list of numbers $x[0], x[1], x[2], \ldots$, one per clock tick; a digital filter, a microcontroller control loop or a sigma-delta modulator computes each new output from a handful of past inputs and outputs. The governing equations are *difference* equations rather than differential equations, and the natural transform is the **Z-transform**, which converts a sequence into a function of a complex variable z by using the sequence values as the coefficients of a power series in z^{-1}.

Everything in this chapter has a continuous-time twin: the Z-transform plays the role of the Laplace transform, the unit circle $|z| = 1$ plays the role of the imaginary axis $s = j\omega$, the inside of the unit circle plays the role of the left half-plane, and the region of convergence is an annulus instead of a vertical strip. The dictionary between the two worlds is the single relation $z = e^{sT}$, derived in §17.7. Because $X(z)$ is a Laurent series, the deeper theory — why the inverse transform is a contour integral, why the ROC is an annulus, why residues compute the inverse — belongs to complex analysis and is completed in Ch. 18. Here we state what we need and use it.

17.1 Sequences and discrete-time functions

Why sequences?

A continuous-time signal $x(t)$ sampled every T seconds gives the numbers $x(nT)$, $n \in \mathbb{Z}$. Once in a processor the sampling interval is no longer visible: the processor sees only the index n. It is therefore convenient to define discrete-time signals abstractly, as functions on the integers, and to carry T along only when we need to translate back to seconds and hertz.

Discrete-time signal (sequence)

A **discrete-time signal** is a function $x : \mathbb{Z} \to \mathbb{C}$, written $x[n]$ with square brackets. The integer n is the **sample index**. When $x[n] = x_a(nT)$ comes from sampling an analogue signal $x_a(t)$, T is the **sampling interval** and $f_s = 1/T$ the **sampling rate**.

Square brackets are a deliberate visual cue: $x[n]$ is defined only at integers, and $x[1.5]$ is meaningless (not zero). A finite sequence is often listed with an arrow, or here with a subscript marker, under the $n = 0$ entry:

$$x[n] = \{\, 2,\ \underset{\uparrow}{1},\ -1,\ 3 \,\} \quad \Longleftrightarrow \quad x[-1] = 2,\ x[0] = 1,\ x[1] = -1,\ x[2] = 3,$$

with $x[n] = 0$ for all other n.

Elementary sequences

Unit impulse, unit step and unit ramp

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases} \qquad u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases} \qquad r[n] = n\,u[n].$$

Unlike the Dirac delta of continuous time, the **Kronecker** (unit) impulse $\delta[n]$ is an ordinary function: it has the value 1 at one point. There is no limiting process, no distribution theory and no problem with $\delta[n]^2 = \delta[n]$. The three sequences are linked by differences and running sums, the discrete analogues of derivatives and integrals:

$$\delta[n] = u[n] - u[n-1], \qquad u[n] = \sum_{k=-\infty}^{n} \delta[k] = \sum_{k=0}^{\infty} \delta[n-k], \qquad r[n+1] - r[n] = u[n].$$

The sifting property is the discrete statement that any sequence is a weighted sum of shifted impulses:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k].$$

This identity, trivial as it looks, is the foundation of the convolution sum below.

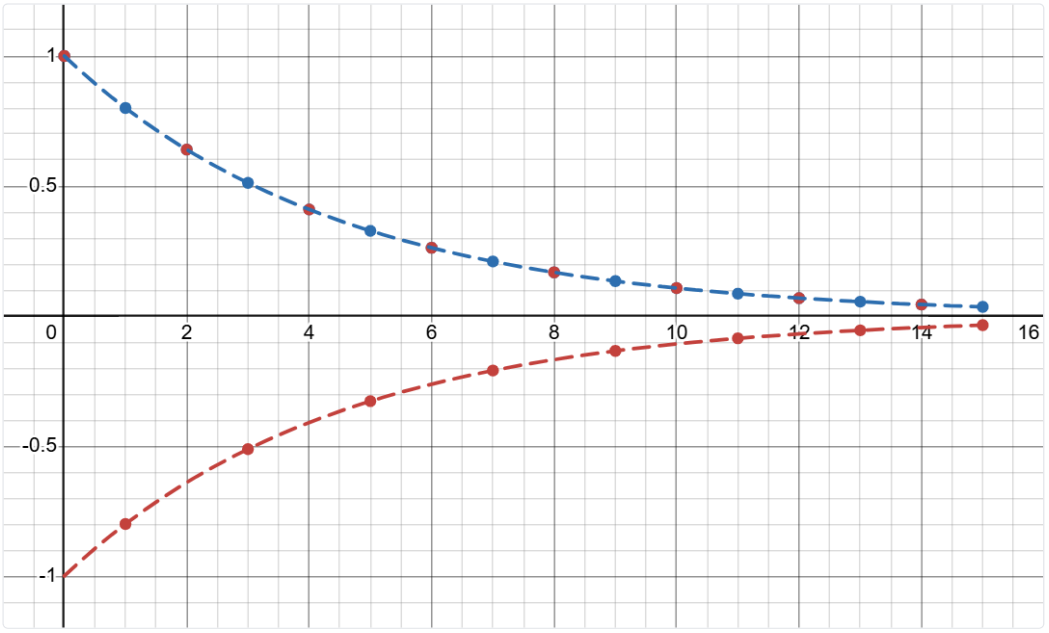
Exponential and sinusoidal sequences

The **complex exponential sequence** is $x[n] = A\,\alpha^n$ with $A, \alpha \in \mathbb{C}$. Writing $\alpha = re^{j\omega_0}$,

$$A\alpha^n = A\,r^n e^{j\omega_0 n} = A\,r^n\,(\cos\omega_0 n + j\sin\omega_0 n)\,.$$

For $r = 1$ this is a **discrete-time phasor** of frequency ω_0 (radians per sample); its real part $\cos(\omega_0 n + \theta)$ is a **sinusoidal sequence**.

The modulus $r = |\alpha|$ controls growth: $r < 1$ gives a decaying envelope, $r = 1$ a constant envelope, $r > 1$ a growing one. The angle ω_0 controls oscillation. For real α, $0 < \alpha < 1$ gives monotone decay, and $-1 < \alpha < 0$ gives decay with alternating sign — something a real continuous exponential e^{at} can never do, because $e^{aT} > 0$. A negative real pole is the discrete-time signature of oscillation at exactly half the sampling rate, $\omega_0 = \pi$.



The sequences $0.8^n u[n]$ (blue) and $(-0.8)^n u[n]$ (red) sampled at integers; both decay inside the envelope $\pm 0.8^n$ (dashed), but the negative base alternates in sign — oscillation at $\omega_0 = \pi$.

Discrete-time frequency is periodic

Two features of discrete-time sinusoids have no continuous-time counterpart, and both follow from the fact that n is an integer.

Frequency is only defined modulo 2π

For every integer k,

$$e^{j(\omega_0 + 2\pi k)n} = e^{j\omega_0 n}\,e^{j2\pi k n} = e^{j\omega_0 n}.$$

Hence the sequences with frequencies ω_0 and $\omega_0 + 2\pi k$ are *identical*. All distinct discrete-time frequencies lie in an interval of length 2π, conventionally $-\pi < \omega \le \pi$ or $0 \le \omega < 2\pi$.

In consequence, "high frequency" in discrete time means ω near $\pm\pi$ (the sequence alternates sign as fast as possible), and "low frequency" means ω near 0 or near any multiple of 2π. The sampled version of $\cos(\Omega t)$ with sampling interval T is $\cos(\Omega T n)$, so the discrete frequency is

$$\omega = \Omega T = 2\pi\frac{f}{f_s} \quad \text{rad/sample},$$

and the periodicity of ω is precisely the aliasing of §16.8: analogue frequencies f and $f + k f_s$ produce the same samples.

Periodicity of a discrete sinusoid

The sequence $e^{j\omega_0 n}$ is periodic with period N (a positive integer) if and only if $\omega_0 N = 2\pi m$ for some integer m, i.e. if and only if $\omega_0/2\pi = m/N$ is **rational**. The fundamental period is the smallest such N, obtained with m/N in lowest terms.

Proof. $e^{j\omega_0(n+N)} = e^{j\omega_0 n}$ for all n iff $e^{j\omega_0 N} = 1$ iff $\omega_0 N \in 2\pi\mathbb{Z}$. □

So $\cos(0.3\pi n)$ has $\omega_0/2\pi = 3/20$ and fundamental period $N = 20$ (it completes $m = 3$ cycles of the underlying cosine in 20 samples), whereas $\cos(n)$ — frequency 1 rad/sample — is *not periodic at all*, because $1/2\pi$ is irrational, even though its samples lie on a perfectly periodic continuous curve.

Operations on sequences

Operation	Definition	Remarks
Shift (delay by $n_0 > 0$)	$y[n] = x[n - n_0]$	moves the sequence n_0 samples right

Operation	Definition	Remarks
Advance	$y[n] = x[n + n_0]$	moves left; non-causal if used in real time
Fold (time reversal)	$y[n] = x[-n]$	mirror about $n = 0$
Decimation (downsampling by M)	$y[n] = x[Mn]$	keeps every Mth sample; information is lost
Expansion (upsampling by L)	$y[n] = x[n/L]$ if $L \mid n$, else 0	inserts $L - 1$ zeros between samples
First difference	$\nabla x[n] = x[n] - x[n-1]$	analogue of dx/dt (backward)
Accumulation	$y[n] = \sum_{k=-\infty}^{n} x[k]$	analogue of $\int_{-\infty}^{t} x\,d\tau$

When shift and fold are combined, the order matters: $x[-n+2] = x[-(n-2)]$ is $x[-n]$ shifted right by 2 — equivalently, $x[n]$ advanced by 2 and then folded. The safe method is to substitute: to sketch $y[n] = x[2-n]$, tabulate n, compute $2 - n$, and read off x.

Classes of sequences

- **Causal** (right-sided from 0): $x[n] = 0$ for $n < 0$. **Anticausal**: $x[n] = 0$ for $n > 0$.
- **Right-sided**: $x[n] = 0$ for $n < N_1$ for some N_1. **Left-sided**: $x[n] = 0$ for $n > N_2$. **Two-sided**: neither. **Finite-length**: both right- and left-sided.
- **Even**: $x[-n] = x[n]$; **odd**: $x[-n] = -x[n]$. Every sequence splits uniquely as $x_e[n] = \tfrac{1}{2}(x[n] + x[-n])$ plus $x_o[n] = \tfrac{1}{2}(x[n] - x[-n])$.
- **Energy**: $E_x = \sum_n |x[n]|^2$. **Average power**: $P_x = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} |x[n]|^2$. An **energy signal** has $0 < E_x < \infty$ (so $P_x = 0$); a **power signal** has $0 < P_x < \infty$ (so $E_x = \infty$).
- **Absolutely summable**: $\sum_n |x[n]| < \infty$ (written $x \in \ell^1$). **Bounded**: $|x[n]| \le B$ for all n.

The right/left/two-sided classification looks bookkeeping-level here, but in §17.3 it will determine the *shape* of the region of convergence.

Worked Example 17.1 — Energy and power of standard sequences

Classify (a) $x_1[n] = (0.5)^n u[n]$, (b) $x_2[n] = u[n]$, (c) $x_3[n] = \cos(0.3\pi n)$, (d) $x_4[n] = n\,u[n]$.

(a) $E = \sum_{n=0}^{\infty} 0.25^n = \dfrac{1}{1-0.25} = \dfrac{4}{3}$. Finite, so an energy signal; $P = 0$.

(b) $E = \sum_{n\ge 0} 1 = \infty$. For power, $\sum_{n=-N}^{N} u[n]^2 = N + 1$, so $P = \lim (N+1)/(2N+1) = \tfrac{1}{2}$. A power signal.

(c) $\cos^2\theta = \tfrac{1}{2}(1 + \cos 2\theta)$, and over a full period of 20 samples $\sum \cos(0.6\pi n) = 0$, so the average of $\cos^2(0.3\pi n)$ over one period is exactly $\tfrac{1}{2}$. Hence $P = \tfrac{1}{2}$: a power signal (as for any sinusoid of unit amplitude).

(d) $\sum_{n=0}^{N} n^2 \approx N^3/3$, so $P = \lim N^3/(3(2N+1)) = \infty$. Neither an energy nor a power signal.

Check: numerically, $\sum_{n=0}^{50} 0.25^n = 1.33333$ and $\frac{1}{40}\sum_{n=0}^{39} \cos^2(0.3\pi n) = 0.5000$.

Discrete-time systems and the convolution sum

A **discrete-time system** maps an input sequence $x[n]$ to an output sequence $y[n] = \mathcal{T}\{x[n]\}$. The properties of Ch. 14 carry over verbatim:

- **Linear**: $\mathcal{T}\{ax_1 + bx_2\} = a\mathcal{T}\{x_1\} + b\mathcal{T}\{x_2\}$.
- **Time-invariant** (shift-invariant): if $x[n] \mapsto y[n]$, then $x[n - n_0] \mapsto y[n - n_0]$ for every integer n_0.
- **Causal**: $y[n]$ depends only on $x[k]$ with $k \le n$.
- **BIBO stable**: every bounded input produces a bounded output.

For a linear time-invariant (LTI) system, define the **impulse response** $h[n] = \mathcal{T}\{\delta[n]\}$. Applying $\mathcal{T}$ to the sifting identity, linearity moves $\mathcal{T}$ inside the sum and time invariance turns $\mathcal{T}\{\delta[n-k]\}$ into $h[n-k]$:

Convolution sum

The output of an LTI system with impulse response $h[n]$ is

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k] = \sum_{k=-\infty}^{\infty} h[k]\,x[n-k] \;\equiv\; (x * h)[n].$$

Discrete convolution is commutative, associative and distributive over addition, and $\delta[n]$ is its identity: $x * \delta = x$, $x * \delta[n-n_0] = x[n-n_0]$.

Derivation. $y[n] = \mathcal{T}\{\sum_k x[k]\delta[n-k]\} = \sum_k x[k]\,\mathcal{T}\{\delta[n-k]\} = \sum_k x[k]h[n-k]$. The substitution $m = n - k$ gives the second form. □

Unlike the convolution integral, the convolution sum can be evaluated exactly by hand for short sequences. Two practical methods:

1. **Flip-and-slide.** Fold $h[k]$ to $h[-k]$, shift it to $h[n-k]$, multiply pointwise with $x[k]$, and add.
2. **Polynomial multiplication.** For finite sequences starting at $n = 0$, y has the coefficients of the product of the polynomials $\sum x[n]w^n$ and $\sum h[n]w^n$. This is not a coincidence: it is the convolution theorem of the Z-transform in disguise (§17.2).

Worked Example 17.2 — Convolution of finite sequences

Compute $y = x * h$ for $x[n] = \{\underset{\uparrow}{1}, 2, 3\}$ and $h[n] = \{\underset{\uparrow}{1}, -1, 2\}$.

Polynomial method. $(1 + 2w + 3w^2)(1 - w + 2w^2) = 1 + (2-1)w + (3 - 2 + 2)w^2 + (-3+4)w^3 + 6w^4$, so

$$y[n] = \{\underset{\uparrow}{1},\ 1,\ 3,\ 1,\ 6\}.$$

Check by flip-and-slide at $n = 2$: $y[2] = x[0]h[2] + x[1]h[1] + x[2]h[0] = 2 - 2 + 3 = 3$. ✓

Length rule. A length-L_1 sequence convolved with a length-L_2 sequence has length $L_1 + L_2 - 1 = 5$; if the first starts at N_1 and the second at N_2, the result starts at $N_1 + N_2$.

Worked Example 17.3 — Step response of a first-order system by convolution

An LTI system has $h[n] = a^n u[n]$ with $a \neq 1$. Find its step response $s[n] = (u * h)[n]$.

For $n < 0$, every term of $\sum_k h[k]u[n-k]$ has either $k < 0$ (so $h[k] = 0$) or $n - k < 0$ (so $u[n-k] = 0$); thus $s[n] = 0$. For $n \geq 0$ the product is non-zero only for $0 \leq k \leq n$:

$$s[n] = \sum_{k=0}^{n} a^k = \frac{1 - a^{n+1}}{1 - a}, \qquad n \geq 0.$$

If $|a| < 1$, $s[n] \to 1/(1-a)$, the **DC gain**. With $a = 0.5$: $s[0..4] = 1,\ 1.5,\ 1.75,\ 1.875,\ 1.9375 \to 2$.

Check: direct recursion $s[n] = 0.5\, s[n-1] + 1$ from $s[-1] = 0$ gives the same values.

Engineering Note — Sampled circuits are sequences

When an ADC samples the capacitor voltage of an RC circuit with time constant τ every T seconds, the free response $v(t) = V_0 e^{-t/\tau}$ becomes $v[n] = V_0 \left(e^{-T/\tau}\right)^n$: a geometric sequence with base $a = e^{-T/\tau}$. The ratio between consecutive samples, not the time constant, is what the processor sees. This is our first glimpse of $z = e^{sT}$: the continuous pole $s = -1/\tau$ becomes the discrete pole $a = e^{-T/\tau}$.

Common Pitfall — "Sampling a periodic signal gives a periodic sequence"

Only if f/f_s is rational. A 1 kHz tone sampled at $f_s = 44.1$ kHz gives $\omega_0/2\pi = 10/441$, periodic with $N = 441$ samples; a tone at $1000\sqrt{2}$ Hz sampled at the same rate gives a sequence that never repeats exactly. Algorithms that assume an integer number of samples per period (e.g. some RMS meters) then show leakage errors.

17.2 Definition and properties

Motivation: eigenfunctions of LTI systems

Feed the geometric sequence $x[n] = z^n$, with z any non-zero complex number, into an LTI system with impulse response $h[n]$. The convolution sum gives

$$y[n] = \sum_k h[k]\, z^{n-k} = z^n \sum_k h[k]\, z^{-k} = H(z)\, z^n,$$

provided the sum converges. The output is the input multiplied by a complex constant $H(z)$: geometric sequences are **eigenfunctions** of every LTI system, and $H(z)$ is the corresponding eigenvalue. Exactly as e^{st} led to the Laplace transform in continuous time, z^n leads to the Z-transform. Decomposing an input into geometric sequences and multiplying each by $H(z)$ is the whole strategy of this chapter.

Definition

Bilateral Z-transform

The (bilateral, or two-sided) **Z-transform** of a sequence $x[n]$ is the function of the complex variable z

$$X(z) = \mathcal{Z}\{x[n]\} = \sum_{n=-\infty}^{\infty} x[n]\, z^{-n},$$

defined on the set of z for which the series converges absolutely. That set is the **region of convergence** (ROC). We write $x[n] \overset{\mathcal{Z}}{\longleftrightarrow} X(z)$, ROC $= R_x$.

Unilateral Z-transform

The **unilateral** (one-sided) Z-transform is

$$\mathcal{X}(z) = \mathcal{Z}_u\{x[n]\} = \sum_{n=0}^{\infty} x[n]\, z^{-n}.$$

It ignores $x[n]$ for $n < 0$. For causal sequences the two transforms coincide. The unilateral form is the tool for difference equations with initial conditions (§17.5), just as the unilateral Laplace transform is for differential equations.

Writing $z = re^{j\omega}$,

$$X(re^{j\omega}) = \sum_{n} \left(x[n]\, r^{-n}\right) e^{-j\omega n},$$

so the Z-transform on the circle of radius r is the discrete-time Fourier transform (DTFT) of the exponentially weighted sequence $x[n]r^{-n}$. The weighting r^{-n} plays the role of the factor $e^{-\sigma t}$ in the Laplace transform: choosing r large tames sequences that grow as $n \to +\infty$; choosing r small tames those that grow as $n \to -\infty$. On the unit circle, $r = 1$, the Z-transform *is* the DTFT:

$$X(e^{j\omega}) = \sum_{n} x[n]\, e^{-j\omega n} \quad \text{(if the unit circle lies in the ROC)}.$$

The notation $X(e^{j\omega})$ — rather than $X(\omega)$ — is standard and deliberate: it reminds us that the DTFT is $X(z)$ evaluated at $z = e^{j\omega}$ and that it is 2π-periodic in ω.

The fundamental pair: the geometric sequence

Worked Example 17.4 — Transform of $a^n u[n]$

Let $x[n] = a^n u[n]$. Then

$$X(z) = \sum_{n=0}^{\infty} a^n z^{-n} = \sum_{n=0}^{\infty} \left(az^{-1}\right)^n = \frac{1}{1 - az^{-1}} = \frac{z}{z - a},$$

by the geometric series, which converges absolutely if and only if $|az^{-1}| < 1$, i.e. $|z| > |a|$.

ROC: the exterior of the circle of radius $|a|$. $X(z)$ has a **zero** at $z = 0$ and a **pole** at $z = a$; the pole lies on the inner boundary of the ROC.

Check ($a = 0.5$, $z = 2$): $\sum_{n=0}^{60} (0.25)^n = 1.33333 = 2/(2 - 0.5)$. ✓

Worked Example 17.5 — Transform of $-a^n u[-n-1]$

Let $x[n] = -a^n u[-n-1]$, which is non-zero only for $n \le -1$. Substituting $m = -n$,

$$X(z) = -\sum_{n=-\infty}^{-1} a^n z^{-n} = -\sum_{m=1}^{\infty} \left(a^{-1}z\right)^m = -\frac{a^{-1}z}{1 - a^{-1}z} = \frac{z}{z - a},$$

converging if and only if $|a^{-1}z| < 1$, i.e. $|z| < |a|$.

The same algebraic expression $z/(z-a)$ as in Example 17.4, but with ROC the *interior* of the circle $|z| = |a|$.

These two examples carry the most important lesson of the chapter: **a Z-transform is the pair (formula, ROC)**. The formula alone does not determine the sequence. In practice the ROC is often implied by context — causal signals in real-time processing have exterior ROCs — but whenever two-sided or non-causal sequences are possible it must be stated.

Table of standard pairs

Each pair below follows from the geometric series and the properties derived next.

$x[n]$	$X(z)$	ROC
$\delta[n]$	1	all z
$\delta[n - n_0]$, $n_0 > 0$	z^{-n_0}	$z \neq 0$
$\delta[n + n_0]$, $n_0 > 0$	z^{n_0}	all finite z ($z \neq \infty$)
$u[n]$	$\dfrac{1}{1 - z^{-1}} = \dfrac{z}{z - 1}$	$\lvert z\rvert > 1$
$-u[-n-1]$	$\dfrac{z}{z - 1}$	$\lvert z\rvert < 1$
$a^n u[n]$	$\dfrac{z}{z - a}$	$\lvert z\rvert > \lvert a\rvert$
$-a^n u[-n-1]$	$\dfrac{z}{z - a}$	$\lvert z\rvert < \lvert a\rvert$
$n\, u[n]$	$\dfrac{z}{(z-1)^2}$	$\lvert z\rvert > 1$
$n\, a^n u[n]$	$\dfrac{az}{(z-a)^2}$	$\lvert z\rvert > \lvert a\rvert$
$\binom{n+m-1}{m-1} a^n u[n]$	$\dfrac{z^m}{(z-a)^m}$	$\lvert z\rvert > \lvert a\rvert$
$\cos(\omega_0 n)\, u[n]$	$\dfrac{z(z - \cos\omega_0)}{z^2 - 2z\cos\omega_0 + 1}$	$\lvert z\rvert > 1$
$\sin(\omega_0 n)\, u[n]$	$\dfrac{z \sin\omega_0}{z^2 - 2z\cos\omega_0 + 1}$	$\lvert z\rvert > 1$
$r^n \cos(\omega_0 n)\, u[n]$	$\dfrac{z(z - r\cos\omega_0)}{z^2 - 2rz\cos\omega_0 + r^2}$	$\lvert z\rvert > r$
$r^n \sin(\omega_0 n)\, u[n]$	$\dfrac{zr \sin\omega_0}{z^2 - 2rz\cos\omega_0 + r^2}$	$\lvert z\rvert > r$
a^n for $0 \le n \le N-1$, else 0	$\dfrac{1 - a^N z^{-N}}{1 - az^{-1}}$	$z \neq 0$

Engineering Note — Write transforms in z^{-1} for implementation, in z for algebra

DSP texts and filter-design software (for example the coefficient vectors `b`, `a` passed to MATLAB's `filter` or SciPy's `lfilter`) write rational transforms as ratios of polynomials in z^{-1}, because z^{-1} is a one-sample delay that can be built from a register. Pole–zero and partial-fraction work is cleaner in positive powers of z. Converting is just multiplying numerator and denominator by the same power of z — but keep track of the poles or zeros at $z = 0$ that this creates or removes.

Linearity

Linearity

If $x_1 \leftrightarrow X_1$ (ROC R_1) and $x_2 \leftrightarrow X_2$ (ROC R_2), then

$$a\,x_1[n] + b\,x_2[n] \;\longleftrightarrow\; a\,X_1(z) + b\,X_2(z), \qquad \text{ROC} \supseteq R_1 \cap R_2.$$

The ROC can be larger than the intersection when poles cancel. For example $a^n u[n] - a^n u[n-1] = \delta[n]$: each term has ROC $|z| > |a|$, but the difference $\frac{z}{z-a} - \frac{a}{z-a} = 1$ converges everywhere.

Linearity immediately gives the sinusoid pairs. By Euler, $\cos(\omega_0 n)u[n] = \frac{1}{2}\left(e^{j\omega_0 n} + e^{-j\omega_0 n}\right)u[n]$, so with $a = e^{\pm j\omega_0}$:

$$\frac{1}{2}\left(\frac{z}{z - e^{j\omega_0}} + \frac{z}{z - e^{-j\omega_0}}\right) = \frac{z}{2}\cdot\frac{2z - 2\cos\omega_0}{z^2 - 2z\cos\omega_0 + 1} = \frac{z(z - \cos\omega_0)}{z^2 - 2z\cos\omega_0 + 1},$$

with ROC $|z| > 1$. The two poles $e^{\pm j\omega_0}$ sit *on* the unit circle at angle $\pm\omega_0$: a pole on the unit circle at angle ω_0 is an undamped oscillation at frequency ω_0.

Time shifting

Time shift (bilateral)

$$x[n - n_0] \;\longleftrightarrow\; z^{-n_0}X(z),$$

with the same ROC as X, except possibly for the addition or removal of $z = 0$ or $z = \infty$.

Proof. $\sum_n x[n-n_0]z^{-n} = \sum_m x[m]z^{-(m+n_0)} = z^{-n_0}X(z)$ with $m = n - n_0$. □

So a delay of one sample is multiplication by z^{-1} — the **unit-delay operator** in a block diagram. For the *unilateral* transform the shift rule must account for samples that cross $n = 0$; it is derived in §17.5 where it is needed.

Multiplication by an exponential (scaling in z)

Exponential multiplication

$$a^n x[n] \;\longleftrightarrow\; X\!\left(\frac{z}{a}\right), \qquad \text{ROC} = |a|R_x.$$

Every pole and zero p of X moves to ap: the pole–zero pattern is scaled by $|a|$ and rotated by $\angle a$.

Proof. $\sum_n a^n x[n] z^{-n} = \sum_n x[n](z/a)^{-n}$. □

With $a = e^{j\omega_0}$ this is **modulation**: $e^{j\omega_0 n}x[n] \leftrightarrow X(e^{-j\omega_0}z)$, a rotation of the pole–zero plot by ω_0, i.e. a frequency shift of the DTFT. With $a = r$ real it gives the damped sinusoid pairs in the table from the undamped ones: replace z by z/r in the $\cos(\omega_0 n)u[n]$ pair and simplify.

Differentiation in z

Multiplication by n

$$n\,x[n] \;\longleftrightarrow\; -z\frac{dX(z)}{dz}, \qquad \text{same ROC (up to } 0, \infty).$$

Proof. Inside its ROC a power series may be differentiated term by term (Ch. 18): $\frac{dX}{dz} = \sum_n (-n)x[n]z^{-n-1}$; multiply by $-z$. □

Applied to $a^n u[n]$:

$$-z\frac{d}{dz}\frac{z}{z-a} = -z\cdot\frac{(z-a) - z}{(z-a)^2} = \frac{az}{(z-a)^2},$$

the $na^n u[n]$ entry in the table. Applying the rule repeatedly produces $n^2 a^n u[n]$, and so on, and hence the double, triple, … pole pairs needed in partial-fraction inversion.

Time reversal and conjugation

Time reversal and conjugation

$$x[-n] \;\longleftrightarrow\; X(1/z), \quad \text{ROC} = 1/R_x; \qquad x^*[n] \;\longleftrightarrow\; X^*(z^*), \quad \text{ROC} = R_x.$$

Reversal maps a pole at p to a pole at $1/p$ and turns an exterior ROC $|z| > r$ into an interior one $|z| < 1/r$. The conjugation property implies that for **real** sequences $X(z^*) = X^*(z)$: poles and zeros are real or occur in complex-conjugate pairs, and on the unit circle $X(e^{-j\omega}) = X^*(e^{j\omega})$ — the magnitude response is even in ω and the phase is odd.

Convolution

Convolution theorem

$$(x * h)[n] \;\longleftrightarrow\; X(z)\,H(z), \qquad \text{ROC} \supseteq R_x \cap R_h.$$

Proof. Take $z \in R_x \cap R_h$, so both series converge absolutely and the order of summation may be exchanged:

$$\sum_{n}\sum_{k} x[k]h[n-k]z^{-n} = \sum_{k} x[k]z^{-k}\sum_{n} h[n-k]z^{-(n-k)} = X(z)H(z). \;\square$$

The polynomial-multiplication trick of Example 17.2 is this theorem with $w = z^{-1}$. More important, the theorem turns the output of an LTI system into a product, $Y(z) = H(z)X(z)$, which is the starting point of §17.6.

Accumulation and first difference

Because $\sum_{k=-\infty}^{n} x[k] = (x * u)[n]$ and $\mathcal{Z}\{u\} = z/(z-1)$:

$$\sum_{k=-\infty}^{n} x[k] \;\longleftrightarrow\; \frac{z}{z-1}X(z) = \frac{1}{1-z^{-1}}X(z), \qquad x[n] - x[n-1] \;\longleftrightarrow\; (1-z^{-1})X(z).$$

The accumulator has a pole at $z = 1$ — the discrete analogue of the integrator's pole at $s = 0$ — and the first difference has a zero there, like the differentiator's zero at $s = 0$.

Parseval's relation

For two sequences whose ROCs both contain the unit circle, the convolution theorem applied to $x[n]$ and $y^*[-n]$, evaluated at $n = 0$, gives

$$\sum_{n=-\infty}^{\infty} x[n]\,y^*[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\omega})\,Y^*(e^{j\omega})\,d\omega, \qquad E_x = \sum_{n}|x[n]|^2 = \frac{1}{2\pi}\int_{-\pi}^{\pi}|X(e^{j\omega})|^2 d\omega.$$

It is the discrete-time counterpart of the energy theorem of §16.7; the only difference is that the frequency integral runs over one period, 2π, rather than over the whole real line. For $x[n] = 0.5^n u[n]$, $|X(e^{j\omega})|^2 = 1/(1.25 - \cos\omega)$ and the integral evaluates to $\frac{1}{2\pi}\cdot\frac{2\pi}{\sqrt{1.25^2-1}} = \frac{1}{0.75} = \frac{4}{3}$, matching Example 17.1(a).

Initial and final value theorems

Initial value theorem

If $x[n]$ is causal, then

$$x[0] = \lim_{z\to\infty} X(z).$$

More generally $x[1] = \lim_{z\to\infty} z\,(X(z) - x[0])$, and so on.

Proof. $X(z) = x[0] + x[1]z^{-1} + x[2]z^{-2} + \cdots$; every term but the first vanishes as $z \to \infty$. $\square$

Final value theorem

If $x[n]$ is causal and $(z-1)X(z)$ has all its poles strictly inside the unit circle, then

$$\lim_{n\to\infty} x[n] = \lim_{z\to 1}(z-1)X(z) = \lim_{z\to 1}\left(1 - z^{-1}\right)X(z).$$

Proof sketch. $\mathcal{Z}\{x[n] - x[n-1]\} = (1-z^{-1})X(z)$. Summing the differences, $x[N] = \sum_{n=0}^{N}(x[n] - x[n-1])$; letting $N \to \infty$, $\lim x[N] = \lim_{z\to 1}\sum_{n\ge 0}(x[n] - x[n-1])z^{-n}$, the interchange of limits being justified by the pole condition, which guarantees the differences decay geometrically. $\square$

The hypothesis is essential: $X(z) = z/(z+1)$ (the sequence $(-1)^n u[n]$) gives $\lim_{z\to 1}(z-1)X(z) = 0$, yet $(-1)^n$ has no limit. The pole of $(z-1)X(z)$ at $z = -1$ lies on the unit circle, so the theorem does not apply.

Summary table of properties

Property	Sequence	Transform	ROC
Linearity	$ax_1 + bx_2$	$aX_1 + bX_2$	$\supseteq R_1 \cap R_2$
Time shift	$x[n - n_0]$	$z^{-n_0}X(z)$	R_x (except $0, \infty$)
Exponential scaling	$a^n x[n]$	$X(z/a)$	$\lvert a\rvert R_x$
Modulation	$e^{j\omega_0 n}x[n]$	$X(e^{-j\omega_0}z)$	R_x
Multiplication by n	$nx[n]$	$-z\,X'(z)$	R_x
Time reversal	$x[-n]$	$X(1/z)$	$1/R_x$

Property	Sequence	Transform	ROC
Conjugation	$x^*[n]$	$X^*(z^*)$	R_x
Convolution	$x * h$	XH	$\supseteq R_x \cap R_h$
Accumulation	$\sum_{k\le n} x[k]$	$X/(1-z^{-1})$	$\supseteq R_x \cap \{\lvert z\rvert > 1\}$
First difference	$x[n]-x[n-1]$	$(1-z^{-1})X$	$\supseteq R_x \cap \{\lvert z\rvert > 0\}$
Initial value (causal)	$x[0]$	$\lim_{z\to\infty} X(z)$	—
Final value (causal)	$x[\infty]$	$\lim_{z\to 1}(z-1)X(z)$	poles of $(z-1)X$ inside $\lvert z\rvert = 1$

Worked examples

Worked Example 17.6 — A finite sequence

Find $X(z)$ for $x[n] = \{2, \underset{\uparrow}{1}, -1, 3\}$.

Directly from the definition:

$$X(z) = 2z + 1 - z^{-1} + 3z^{-2} = \frac{2z^3 + z^2 - z + 3}{z^2}.$$

ROC. The positive power $2z$ diverges as $z \to \infty$, and the negative powers diverge at $z = 0$, so the ROC is $0 < \lvert z\rvert < \infty$. A finite sequence always has an ROC of the entire plane, except possibly 0 and/or ∞.

Check: at $z = 1$, $X(1) = 2 + 1 - 1 + 3 = 5 = \sum_n x[n]$, as it must be: the Z-transform at $z = 1$ is the sum of the sequence.

Worked Example 17.7 — A damped sinusoid via scaling

Find the Z-transform of $x[n] = (0.8)^n \cos(\pi n/3)\, u[n]$.

Step 1. Start from the cosine pair with $\cos(\pi/3) = \frac{1}{2}$:

$$\cos\left(\tfrac{\pi}{3}n\right) u[n] \longleftrightarrow \frac{z(z-0.5)}{z^2 - z + 1}, \quad \lvert z\rvert > 1.$$

Step 2. Multiply by 0.8^n, i.e. replace z by $z/0.8$:

$$X(z) = \frac{(z/0.8)(z/0.8 - 0.5)}{(z/0.8)^2 - z/0.8 + 1} = \frac{z(z-0.4)}{z^2 - 0.8z + 0.64}, \quad \lvert z\rvert > 0.8.$$

Poles at $z = 0.4 \pm j0.4\sqrt{3} = 0.8e^{\pm j\pi/3}$: radius 0.8 (the damping), angle $\pm\pi/3$ (the oscillation frequency). **Zeros** at 0 and 0.4.

Check at $z = 2$: the formula gives $2(1.6)/(4 - 1.6 + 0.64) = 3.2/3.04 = 1.052632$; summing $\sum_{n=0}^{80} 0.8^n \cos(\pi n/3)\, 2^{-n}$ numerically gives 1.052632. ✓

Worked Example 17.8 — Using several properties together

Find $X(z)$ for $x[n] = n\,(0.5)^{n-2} u[n-2]$.

Step 1. Let $g[n] = (0.5)^{n-2}u[n-2]$, the sequence $0.5^n u[n]$ delayed by 2. Then $G(z) = z^{-2} \cdot \dfrac{z}{z-0.5} = \dfrac{1}{z(z-0.5)}$, $\lvert z\rvert > 0.5$.

Step 2. $x[n] = n\,g[n]$, so

$$X(z) = -z\,\frac{d}{dz}\left[\frac{1}{z^2 - 0.5z}\right] = -z \cdot \frac{-(2z - 0.5)}{(z^2 - 0.5z)^2} = \frac{2z - 0.5}{z(z-0.5)^2}, \quad \lvert z\rvert > 0.5.$$

Check by leading coefficients. $x[0] = x[1] = 0$, $x[2] = 2$, $x[3] = 3(0.5) = 1.5$. Expanding $X(z)$ for large z: $X = 2z^{-2}\left(1 - \frac{0.25}{z}\right)\left(1 + \frac{1}{z} + \cdots\right) = 2z^{-2} + 1.5z^{-3} + \cdots$ ✓.

Worked Example 17.9 — Initial and final values without inversion

A causal sequence has $Y(z) = \dfrac{z^2}{(z-1)(z-0.5)}$. Find $y[0]$ and $\lim_{n\to\infty} y[n]$.

Initial value. $\lim_{z\to\infty} \dfrac{z^2}{z^2 - 1.5z + 0.5} = 1$, so $y[0] = 1$.

Final value. $(z-1)Y(z) = \dfrac{z^2}{z - 0.5}$ has its only pole at 0.5, inside the unit circle, so the theorem applies: $\lim_{z\to 1} \dfrac{z^2}{z-0.5} = \dfrac{1}{0.5} = 2$.

This Y is $\dfrac{z}{z-1} \cdot \dfrac{z}{z-0.5}$, the step response of $h[n] = 0.5^n u[n]$ (Example 17.3), whose values $1, 1.5, 1.75, \cdots \to 2$ confirm both results.

Common Pitfall — Dropping the ROC

Writing "$\mathcal{Z}\{x\} = z/(z-a)$" without an ROC is incomplete: Examples 17.4 and 17.5 show two different sequences with this transform. Tables in most texts list only causal sequences and silently assume the exterior ROC; when you invert, you must know whether that assumption is right for your problem.

17.3 Region of convergence

Why the ROC deserves its own section

Section 17.2 showed that $z/(z-a)$ is the transform of two different sequences. The ROC is what tells them apart, and it carries physical information: whether the sequence is right- or left-sided, whether the system it describes is causal, and whether it is stable. In continuous time the Laplace ROC is a vertical strip $\sigma_1 < \operatorname{Re} s < \sigma_2$; in discrete time it is an **annulus** $r_1 < |z| < r_2$, because the convergence of $\sum x[n]z^{-n}$ depends only on $|z|$.

The ROC is an annulus

Absolute convergence of $\sum_n x[n]z^{-n}$ means

$$\sum_{n=-\infty}^{\infty} |x[n]|\, r^{-n} < \infty, \qquad r = |z|.$$

The angle of z does not appear, so if one point on a circle centred at the origin is in the ROC, the whole circle is. Split the sum into the **causal part** $n \geq 0$ and the **anticausal part** $n < 0$:

$$\underbrace{\sum_{n=0}^{\infty} |x[n]|\, r^{-n}}_{\text{converges for } r > r_1} + \underbrace{\sum_{m=1}^{\infty} |x[-m]|\, r^{m}}_{\text{converges for } r < r_2}.$$

The first is a power series in $1/r$; by the root test (Ch. 4) it converges for $1/r < 1/r_1$, i.e. $r > r_1$, where

$$r_1 = \limsup_{n\to\infty} |x[n]|^{1/n}.$$

The second is a power series in r and converges for $r < r_2$, where $1/r_2 = \limsup_{m\to\infty} |x[-m]|^{1/m}$. Both must converge, so:

Shape of the region of convergence

The ROC of a Z-transform is an annulus centred at the origin,

$$R_x = \{\, z : \; r_1 < |z| < r_2 \,\}, \qquad 0 \leq r_1,\; r_2 \leq \infty,$$

possibly including one or both boundary circles, and possibly empty (when $r_1 \geq r_2$). The rate of growth of $x[n]$ as $n \to +\infty$ sets the inner radius r_1; the rate of growth as $n \to -\infty$ sets the outer radius r_2.

On the open annulus $X(z)$ is a convergent Laurent series and is therefore analytic (Ch. 18). An analytic function has no poles, which gives the next property.

Properties of the ROC

Properties of the ROC

1. **No poles inside.** The ROC contains no poles of $X(z)$. For rational $X(z)$ the ROC is bounded by poles (or extends to 0 or ∞).
2. **Finite-length sequences:** the ROC is the whole z-plane, except possibly $z = 0$ (if $x[n] \neq 0$ for some $n > 0$) and $z = \infty$ (if $x[n] \neq 0$ for some $n < 0$).
3. **Right-sided sequences:** the ROC is the **exterior** of a circle, $|z| > r_1$ — for rational X, outside the outermost pole. It includes $z = \infty$ if and only if the sequence is causal.
4. **Left-sided sequences:** the ROC is the **interior** of a circle, $|z| < r_2$ — for rational X, inside the innermost (non-zero) pole. It includes $z = 0$ if and only if the sequence is anticausal ($x[n] = 0$ for $n > 0$).
5. **Two-sided sequences:** the ROC is a **ring** $r_1 < |z| < r_2$ between two consecutive pole radii, or empty.
6. **Connectedness:** the ROC is a single connected region.

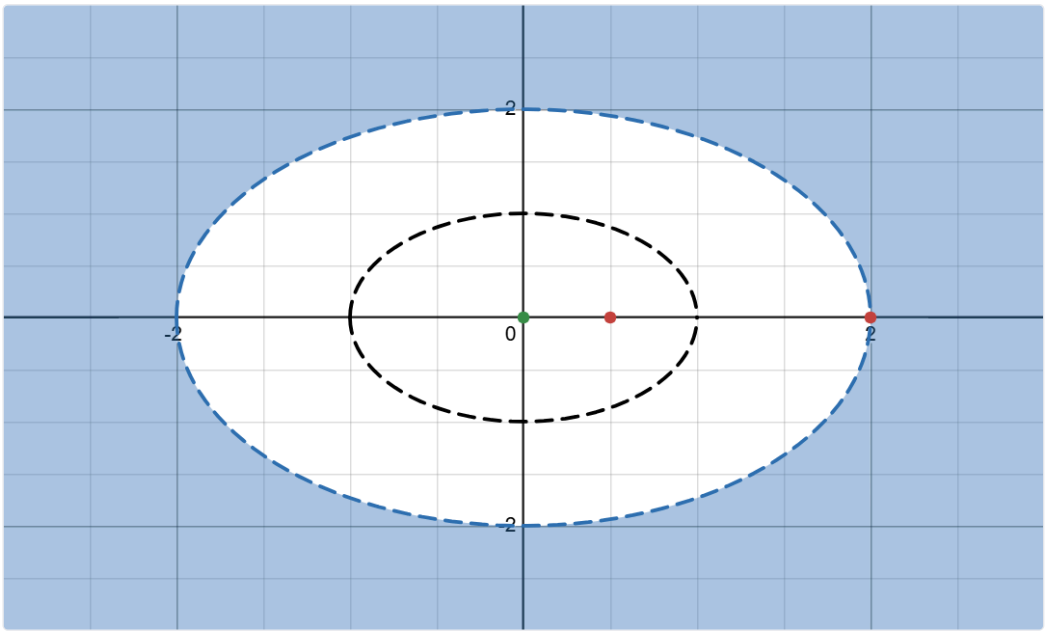
Proof of 3. If $x[n] = 0$ for $n < N_1$, the anticausal part has finitely many terms, so $r_2 = \infty$ (if $N_1 \geq 0$, even $z = \infty$ is allowed, because every term $x[n]z^{-n}$ with $n \geq 0$ stays finite as $z \to \infty$). Only r_1 is constrained: the ROC is $|z| > r_1$. Since there can be no poles in it, every pole has $|p| \leq r_1$, and the boundary must pass through the outermost pole. Property 4 is the mirror image. □

ROC sketches

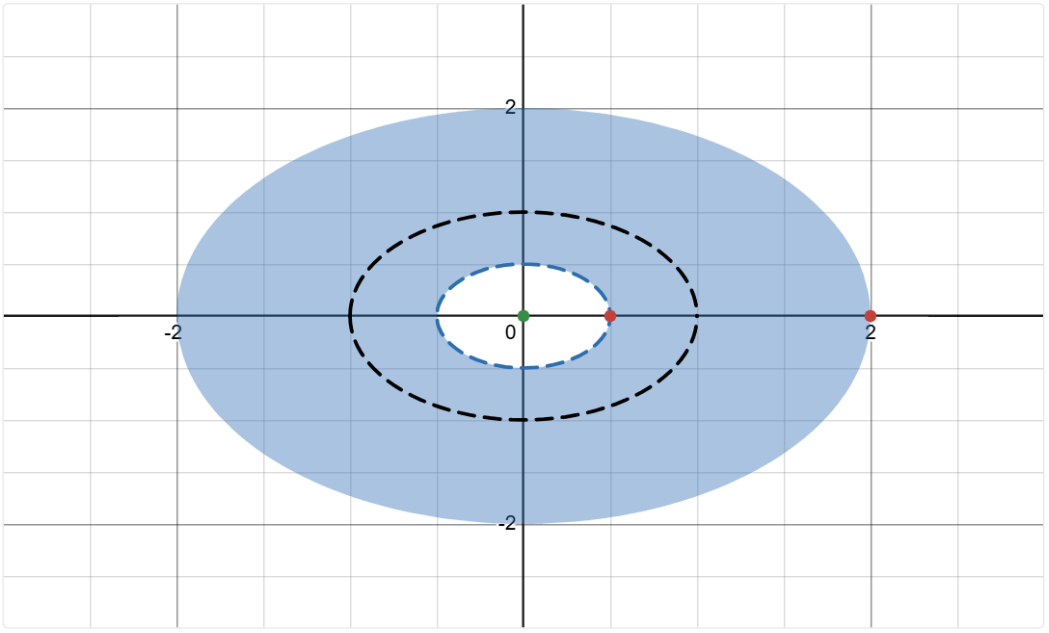
A sketch of the ROC consists of the pole–zero plot (poles $\times$, zeros $\circ$), the unit circle (for reference) and the shaded convergence region. Consider the rational function with poles at $z = 0.5$ and $z = 2$,

$$X(z) = \frac{z}{(z-0.5)(z-2)}.$$

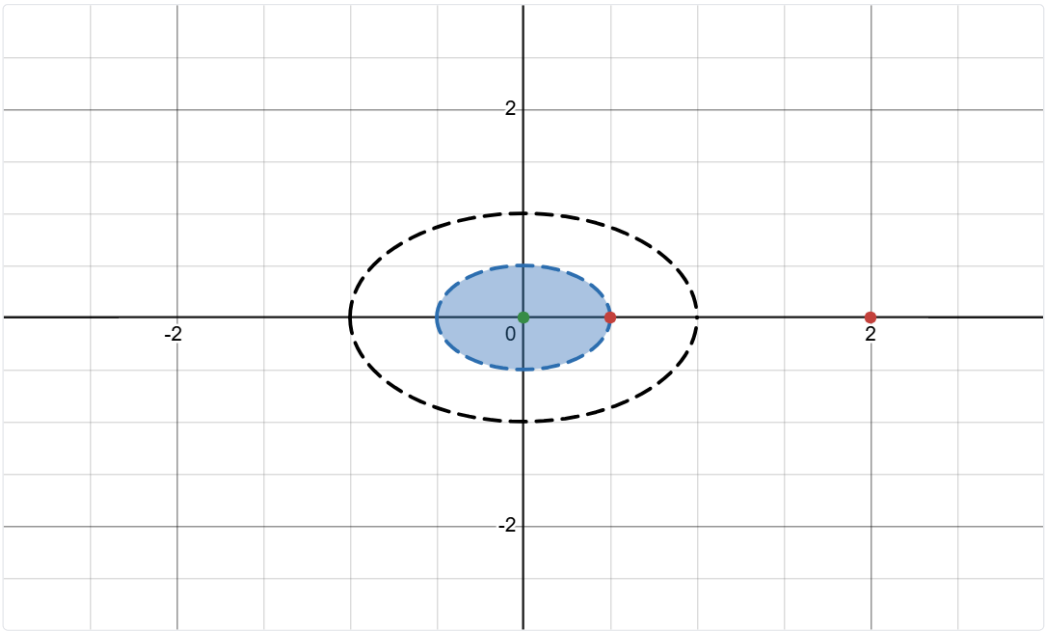
Its ROC must be an annulus bounded by pole radii, so there are exactly three possibilities.



ROC $|z| > 2$ (shaded): the right-sided (here causal) sequence $x[n] = \frac{2}{3}\left(2^n - 0.5^n\right) u[n]$. The ROC lies outside the outermost pole and excludes the unit circle (dashed), so the sequence grows without bound.



ROC $0.5 < |z| < 2$ (shaded ring): the two-sided sequence $x[n] = -\frac{2}{3}\,0.5^n u[n] - \frac{2}{3}\,2^n u[-n-1]$. The ring contains the unit circle, so this sequence is absolutely summable and has a DTFT.



ROC $|z| < 0.5$ (shaded disc): the left-sided sequence $x[n] = \frac{2}{3}\left(0.5^n - 2^n\right) u[-n-1]$, which blows up as $n \to -\infty$.

The three sequences are derived in Example 17.12 below; the point here is that one rational function has three inverse transforms, selected by the ROC.

Causality and stability read from the ROC

When $X(z) = H(z)$ is the transform of an impulse response, two system properties translate into ROC statements.

Causality and stability from the ROC

Let $h[n]$ have transform $H(z)$ with ROC R_h.

- The system is **causal** ($h[n] = 0$ for $n < 0$) if and only if R_h is the exterior of a circle *including* $z = \infty$. For rational $H(z)$: the ROC is outside the outermost pole, *and* the numerator degree does not exceed the denominator degree (in positive powers of z).
- The system is **BIBO stable** ($\sum |h[n]| < \infty$) if and only if R_h **includes the unit circle** $|z| = 1$.
- Consequently, a causal rational system is stable if and only if **all its poles lie strictly inside the unit circle**.

Proof of the stability statement. On $|z| = 1$, $\sum_n |h[n] z^{-n}| = \sum_n |h[n]|$. So the unit circle is in the ROC precisely when h is absolutely summable, which (§17.7) is precisely BIBO stability. □

The degree condition in the causality statement is what excludes $z = \infty$ from being a pole: $H(z) = z^2/(z - 0.5)$ has ROC $|z| > 0.5$ as a right-sided sequence, but it equals $z + 0.5 + \frac{0.25}{z - 0.5}$, whose polynomial part z is an advance $\delta[n+1]$. The sequence starts at $n = -1$: right-sided, but not causal.

Engineering Note — The unit circle is the frequency axis

Every real-time digital filter is causal, so its ROC is the exterior of its outermost pole. Stability then requires that exterior to contain the unit circle, i.e. every pole inside $|z| < 1$. This single picture — poles inside the circle — is the discrete-time version of "poles in the left half-plane". The distance of the outermost pole from the circle is the stability margin: a pole at radius 0.999 in a fixed-point implementation may cross the circle after coefficient quantisation.

Worked examples

Worked Example 17.10 — ROC of a sum of right-sided sequences

Find $X(z)$ and its ROC for $x[n] = (0.5)^n u[n] + \left(-\frac{1}{3}\right)^n u[n]$.

Step 1. Each term is right-sided:

$$(0.5)^n u[n] \leftrightarrow \frac{z}{z - 0.5}, \ |z| > 0.5; \qquad \left(-\tfrac{1}{3}\right)^n u[n] \leftrightarrow \frac{z}{z + \frac{1}{3}}, \ |z| > \tfrac{1}{3}.$$

Step 2. Add, over a common denominator:

$$X(z) = \frac{z(z + \frac{1}{3}) + z(z - 0.5)}{(z - 0.5)(z + \frac{1}{3})} = \frac{2z\left(z - \frac{1}{12}\right)}{(z - 0.5)(z + \frac{1}{3})}.$$

Step 3. ROC is the intersection, $|z| > 0.5$: outside the outermost pole, as it must be for a right-sided sequence. It contains the unit circle and $z = \infty$: the sequence is causal and absolutely summable.

Check: $X(\infty) = 2 = x[0] = 1 + 1$ ✓ (initial value theorem).

Worked Example 17.11 — The two-sided exponential $a^{|n|}$

Find the Z-transform of $x[n] = a^{|n|}$ for real a, and state when it exists.

Step 1. Split. $a^{|n|} = a^n u[n] + a^{-n} u[-n-1]$.

Step 2. Causal part. $a^n u[n] \leftrightarrow \dfrac{z}{z-a}$, ROC $|z| > |a|$.

Step 3. Anticausal part. $a^{-n} u[-n-1] = (1/a)^n u[-n-1]$. By Example 17.5 with base $1/a$, $-(1/a)^n u[-n-1] \leftrightarrow \dfrac{z}{z-1/a}$ for $|z| < 1/|a|$; hence $(1/a)^n u[-n-1] \leftrightarrow -\dfrac{z}{z-1/a}$.

Step 4. Combine.

$$X(z) = \frac{z}{z-a} - \frac{z}{z-1/a} = \frac{(a-1/a)\,z}{(z-a)(z-1/a)}, \qquad |a| < |z| < \frac{1}{|a|}.$$

Existence. The ring is non-empty only if $|a| < 1/|a|$, i.e. $|a| < 1$. For $|a| \geq 1$ the sequence grows (or fails to decay) in both directions and has **no** Z-transform. When it exists the ring always contains the unit circle, as it must for an absolutely summable sequence.

Check ($a = 0.5$, $z = 1$): $\sum_n 0.5^{|n|} = 1 + 2\sum_{n\geq 1} 0.5^n = 3$; the formula gives $(-1.5)(1)/((0.5)(-1)) = 3$ ✓.

Worked Example 17.12 — One formula, three sequences

For $X(z) = \dfrac{z}{(z-0.5)(z-2)}$, find the sequence for each possible ROC and say which is causal and which is stable.

Step 1. Partial fractions of $X(z)/z$ (method of §17.4):

$$\frac{X(z)}{z} = \frac{1}{(z-0.5)(z-2)} = \frac{-2/3}{z-0.5} + \frac{2/3}{z-2} \quad\Longrightarrow\quad X(z) = -\frac{2}{3}\frac{z}{z-0.5} + \frac{2}{3}\frac{z}{z-2}.$$

Step 2. Assign each pole's term according to the ROC. A pole *inside* the inner boundary of the ROC (the ROC lies outside it) gives a right-sided term $p^n u[n]$; a pole *outside* the outer boundary gives a left-sided term $-p^n u[-n-1]$.

ROC	pole 0.5	pole 2	$x[n]$	causal	stable
$\|z\| > 2$	right	right	$\frac{2}{3}\left(2^n - 0.5^n\right) u[n]$	yes	no
$0.5 < \|z\| < 2$	right	left	$-\frac{2}{3}\,0.5^n u[n] - \frac{2}{3}\,2^n u[-n-1]$	no	yes
$\|z\| < 0.5$	left	left	$\frac{2}{3}\left(0.5^n - 2^n\right) u[-n-1]$	no	no

Step 3. Interpret. No ROC makes this system both causal and stable, because one pole is outside the unit circle. The stable choice is two-sided and equals $-\frac{2}{3}\,0.5^{|n|}$ — compare Example 17.11 with $a = 0.5$, where X was -1.5 times this one.

Check (middle ROC): evaluating the inverse contour integral (§17.4) numerically on $|z| = 1$ gives $x[0] = -0.6667$, $x[1] = x[-1] = -0.3333$, $x[3] = x[-3] = -0.0833$, matching $-\frac{2}{3}(0.5)^{|n|}$.

Worked Example 17.13 — An empty ROC

Does $x[n] = 2^n u[n] - (0.5)^n u[-n-1]$ have a Z-transform?

The causal part needs $|z| > 2$; the anticausal part needs $|z| < 0.5$. The intersection is empty, so **no Z-transform exists** — even though the formal sum of the two rational expressions, $\frac{z}{z-2} + \frac{z}{z-0.5}$, can be written down. Likewise $x[n] = 1$ for all n (or $\cos\omega_0 n$ for all n) has no bilateral Z-transform: the causal part needs $|z| > 1$, the anticausal part $|z| < 1$. Such sequences are handled by the DTFT with impulses (as in §16.4) or, for $n \geq 0$ only, by the unilateral transform.

Common Pitfall — Choosing the ROC from the formula alone

Students often pick "$|z| >$ largest pole" automatically. That is right only if the sequence is known to be right-sided (e.g. a causal system). If instead you are told the system is *stable*, the ROC must contain the unit circle — which, with poles on both sides of it, produces a two-sided impulse response. Read the problem's physical constraints first, then choose the ROC.

17.4 Inverse Z-transform

The inversion integral

Recovering $x[n]$ from $X(z)$ is the step that turns an algebraic answer into a sequence you can plot or implement. The general formula is a contour integral, and every practical method is a shortcut for evaluating it.

Inverse Z-transform

If $X(z)$ has ROC R_x and C is any counter-clockwise circle $|z| = r$ lying in R_x, then

$$x[n] = \frac{1}{2\pi j} \oint_C X(z)\, z^{\,n-1}\, dz, \qquad n \in \mathbb{Z}.$$

Derivation. Substitute the series for X and integrate term by term (legitimate on a circle inside the ROC, where convergence is uniform):

$$\frac{1}{2\pi j} \oint_C \sum_k x[k] z^{-k} z^{n-1}\, dz = \sum_k x[k]\, \frac{1}{2\pi j} \oint_C z^{\,n-k-1}\, dz.$$

With $z = re^{j\theta}$, $dz = jz\, d\theta$, the inner integral is $\frac{r^{n-k}}{2\pi} \int_0^{2\pi} e^{j(n-k)\theta} d\theta = \delta[n-k]$. Only the $k = n$ term survives, leaving $x[n]$. $\square$

On the unit circle ($r = 1$, $z = e^{j\omega}$, $dz = jz\, d\omega$) the formula becomes the inverse DTFT,

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega})\, e^{j\omega n}\, d\omega,$$

the discrete-time counterpart of the inverse Fourier integral of §16.2. The Z-transform and its inverse are simply the Laurent-series expansion of X in the annulus R_x and the formula for its coefficients — which is why the choice of annulus (the ROC) changes the answer.

There are four practical methods:

1. **Inspection** using the table of pairs and properties;
2. **Partial-fraction expansion**, for rational $X(z)$ — the workhorse;
3. **Power-series expansion**, including **long division** — gives numerical values sample by sample and handles non-rational X;
4. **Residues** — evaluates the inversion integral directly; elegant for closed-form answers, especially with repeated poles.

Method 1: partial fractions

For a rational $X(z) = N(z)/D(z)$, the aim is to write X as a sum of terms $\frac{z}{z-p}$ (and $\frac{z}{(z-p)^m}$ for repeated poles), each of which inverts by the table. Because the table entries carry a factor z in the numerator, the cleanest procedure expands $X(z)/z$, not $X(z)$.

Partial-fraction procedure (expansion of $X(z)/z$)

1. Write $X(z)$ in positive powers of z and factor the denominator: poles $p_1, \dots, p_K$.
2. Form $F(z) = X(z)/z$. If F is improper (numerator degree $\geq$ denominator degree), first divide out the polynomial part.
3. Expand $F(z) = \sum_i \dfrac{A_i}{z - p_i}$ (+ repeated-pole terms), with **cover-up** residues for simple poles

$$A_i = (z - p_i) F(z)\big|_{z = p_i},$$

and for a pole of order m at p, the coefficients of $\frac{1}{(z-p)^k}$,

$$C_k = \frac{1}{(m-k)!} \frac{d^{\,m-k}}{dz^{\,m-k}} \Big[(z-p)^m F(z) \Big]\Big|_{z=p}, \quad k = 1, \dots, m.$$

4. Multiply back by z: $X(z) = \sum_i A_i \dfrac{z}{z - p_i} + \cdots$.
5. Invert each term using the ROC: for $|z| > |p|$, $\frac{z}{z-p} \to p^n u[n]$; for $|z| < |p|$, $\frac{z}{z-p} \to -p^n u[-n-1]$. Repeated poles use $\dfrac{z}{(z-p)^2} \to n\, p^{\,n-1} u[n]$ and $\dfrac{z}{(z-p)^3} \to \frac{n(n-1)}{2} p^{\,n-2} u[n]$ (right-sided case).

Dividing by z first has two advantages: a pole at $z = 0$, if created, simply produces a $\delta[n]$ term; and no separate polynomial division is needed when the numerator and denominator of X have equal degree — the common case for causal systems.

An equivalent route, preferred in DSP texts, expands X in powers of $w = z^{-1}$: $X = \sum_i A_i/(1 - p_i z^{-1})$ with $A_i = (1 - p_i z^{-1}) X\big|_{z = p_i}$. Both give the same coefficients when X is proper in z^{-1}.

Worked Example 17.14 — Distinct real poles

Find the causal sequence with $X(z) = \dfrac{z}{(z - 0.5)(z - 0.25)}$.

Step 1. $F(z) = \dfrac{X}{z} = \dfrac{1}{(z - 0.5)(z - 0.25)}$.

Step 2. Cover-up. $A_1 = \dfrac{1}{0.5 - 0.25} = 4$, $A_2 = \dfrac{1}{0.25 - 0.5} = -4$.

Step 3. $X(z) = 4\dfrac{z}{z - 0.5} - 4\dfrac{z}{z - 0.25}$; causal means $|z| > 0.5$, so both terms are right-sided:

$$x[n] = 4\left[(0.5)^n - (0.25)^n\right] u[n].$$

Check: $x[0] = 0$ (and $X(\infty) = 0$ ✓); $x[1] = 4(0.25) = 1$, $x[2] = 4(0.25 - 0.0625) = 0.75$. Running the recursion $x[n] = 0.75x[n-1] - 0.125x[n-2] + \delta[n-1]$ gives $0,\ 1,\ 0.75,\ 0.4375,\ 0.2344, \dots$, identical to the formula.

Worked Example 17.15 — A repeated pole

Find the causal inverse of $X(z) = \dfrac{z^2}{(z-1)(z-0.5)^2}$.

Step 1. $F(z) = \dfrac{z}{(z-1)(z-0.5)^2} = \dfrac{A}{z-1} + \dfrac{B}{z-0.5} + \dfrac{C}{(z-0.5)^2}$.

Step 2. Simple pole. $A = \left.\dfrac{z}{(z-0.5)^2}\right|_{z=1} = \dfrac{1}{0.25} = 4$.

Step 3. Double pole. With $G(z) = (z-0.5)^2 F(z) = \dfrac{z}{z-1}$:

$$C = G(0.5) = \frac{0.5}{-0.5} = -1, \qquad B = G'(0.5) = \left.\frac{-1}{(z-1)^2}\right|_{0.5} = -4.$$

Step 4. $X(z) = 4\dfrac{z}{z-1} - 4\dfrac{z}{z-0.5} - \dfrac{z}{(z-0.5)^2}$, and with $\dfrac{z}{(z-0.5)^2} \leftrightarrow n(0.5)^{n-1}u[n] = 2n(0.5)^n u[n]$:

$$x[n] = \left[4 - 4(0.5)^n - 2n(0.5)^n\right] u[n].$$

Check: $x[0..5] = 0,\ 1,\ 2,\ 2.75,\ 3.25,\ 3.5625$ from both the formula and a direct recursion; $x[n] \to 4$, which the final value theorem confirms: $\lim_{z\to 1}(z-1)X = 1/0.25 = 4$.

Worked Example 17.16 — Complex-conjugate poles

Find the causal inverse of $X(z) = \dfrac{z}{z^2 - z + 0.5}$.

Step 1. Poles. $z = \frac{1}{2} \pm \frac{1}{2}j = \frac{1}{\sqrt{2}}e^{\pm j\pi/4}$: radius $r = 1/\sqrt{2} \approx 0.707$, angle $\omega_0 = \pi/4$.

Step 2. Match the damped-sine pair $\dfrac{zr\sin\omega_0}{z^2 - 2rz\cos\omega_0 + r^2}$. Here $2r\cos\omega_0 = 2 \cdot \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = 1$ ✓ and $r^2 = 0.5$ ✓, while $r\sin\omega_0 = 0.5$. So $X(z) = 2 \times \dfrac{0.5\,z}{z^2 - z + 0.5}$ and

$$x[n] = 2\left(\tfrac{1}{\sqrt{2}}\right)^n \sin\left(\tfrac{\pi n}{4}\right) u[n].$$

Alternative (complex partial fractions). $F = X/z = \dfrac{1}{(z-p)(z-p^*)}$, $A = \dfrac{1}{p - p^*} = \dfrac{1}{j} = -j$, and the conjugate pole has $A^* = j$. Then $x[n] = -jp^n + jp^{*n} = 2\,\mathrm{Im}\{p^n\} = 2r^n\sin(\omega_0 n)$, the same answer. In general a conjugate pair with residues A, A^* contributes $2|A|r^n\cos(\omega_0 n + \angle A)\,u[n]$.

Check: $x[0..7] = 0,\ 1,\ 1,\ 0.5,\ 0,\ -0.25,\ -0.25,\ -0.125$, agreeing with the recursion $x[n] = x[n-1] - 0.5x[n-2] + \delta[n-1]$.

Worked Example 17.17 — Equal-degree numerator and a pole at the origin

Find the causal inverse of $X(z) = \dfrac{z^2 + 1}{z^2 - 0.25}$.

Step 1. $F(z) = \dfrac{z^2+1}{z(z-0.5)(z+0.5)}$: dividing by z creates a pole at 0, and F is now proper.

Step 2. Cover-up. $A_0 = \dfrac{1}{-0.25} = -4$, $A_{0.5} = \dfrac{1.25}{0.5 \cdot 1} = 2.5$, $A_{-0.5} = \dfrac{1.25}{(-0.5)(-1)} = 2.5$.

Step 3. $X(z) = -4 + 2.5\dfrac{z}{z-0.5} + 2.5\dfrac{z}{z+0.5}$, so

$$x[n] = -4\,\delta[n] + 2.5\left[(0.5)^n + (-0.5)^n\right] u[n].$$

The two exponentials cancel at odd n and add at even n: $x = 1,\ 0,\ 1.25,\ 0,\ 0.3125,\ 0, \ldots$.

Check: in powers of z^{-1}, $X = \dfrac{1 + z^{-2}}{1 - 0.25z^{-2}} = (1 + z^{-2})(1 + 0.25z^{-2} + 0.0625z^{-4} + \cdots) = 1 + 1.25z^{-2} + 0.3125z^{-4} + \cdots$ ✓.

Method 2: power series and long division

Since $X(z) = \sum x[n]z^{-n}$, any method that expands X as a power series in the ROC reads off $x[n]$ as the coefficients. For rational X the expansion is obtained by **polynomial long division**:

- For a **right-sided** (causal) ROC $|z| > r_1$, expand in *descending* powers of z, i.e. ascending powers of z^{-1}: arrange numerator and denominator in decreasing powers of z and divide.
- For a **left-sided** ROC $|z| < r_2$, expand in *ascending* powers of z: arrange both polynomials in increasing powers of z and divide.

Long division does not normally produce a closed form, but it is foolproof for the first few samples, it is how a computer "inverts" (it is the recursion of the difference equation), and it is the method of choice for checking a partial-fraction answer.

Worked Example 17.18 — Causal long division

Find the first four samples of the causal sequence with $X(z) = \dfrac{z}{z^2 - 0.75z + 0.125}$ (Example 17.14).

Divide z by $z^2 - 0.75z + 0.125$ in descending powers:

Step	Quotient term	Remainder after subtraction
$z \div z^2$	z^{-1}	$z - (z - 0.75 + 0.125z^{-1}) = 0.75 - 0.125z^{-1}$
$0.75 \div z^2$	$0.75z^{-2}$	$-0.125z^{-1} - (-0.5625z^{-1} + 0.09375z^{-2}) = 0.4375z^{-1} - 0.09375z^{-2}$
$0.4375z^{-1} \div z^2$	$0.4375z^{-3}$	$\dots$

So $X = 0 + z^{-1} + 0.75z^{-2} + 0.4375z^{-3} + \cdots$, giving $x[0..3] = 0,\ 1,\ 0.75,\ 0.4375$ — exactly the partial-fraction result.

Worked Example 17.19 — Anticausal long division

Expand $X(z) = \dfrac{z}{z - a}$ with ROC $|z| < |a|$.

Arrange in ascending powers of z: $\dfrac{z}{-a + z}$. Dividing, the first quotient term is $z/(-a) = -a^{-1}z$; the remainder is $z - (-a^{-1}z)(-a + z) = a^{-1}z^2$; the next quotient term is $-a^{-2}z^2$, and so on:

$$X(z) = -a^{-1}z - a^{-2}z^2 - a^{-3}z^3 - \cdots = \sum_{n=-\infty}^{-1} (-a^n)\, z^{-n}.$$

Hence $x[n] = -a^n u[-n-1]$, in agreement with Example 17.5.

Non-rational transforms are inverted by known series. The logarithm is the classic case.

Worked Example 17.20 — A logarithmic transform

Find the right-sided $x[n]$ with $X(z) = \ln\left(1 + az^{-1}\right)$, $|z| > |a|$.

Use $\ln(1+w) = \sum_{n=1}^{\infty} \dfrac{(-1)^{n+1}}{n} w^n$ for $|w| < 1$ with $w = az^{-1}$ (valid exactly when $|z| > |a|$):

$$X(z) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} a^n}{n} z^{-n} \quad \Longrightarrow \quad x[n] = \frac{(-1)^{n+1} a^n}{n}\, u[n-1].$$

Cross-check via the differentiation property: $-z\,X'(z) = \dfrac{az^{-1}}{1 + az^{-1}} = \dfrac{a}{z + a}$, whose inverse is $a(-a)^{n-1}u[n-1]$; dividing by n reproduces $x[n]$. Sequences of this type appear in cepstral analysis of speech and echo signals.

Method 3: residues

When $X(z)$ is rational, the integrand $X(z)z^{n-1}$ of the inversion integral is analytic everywhere except at finitely many poles, and the residue theorem (Ch. 18) evaluates the integral as a sum.

Residue formula for the inverse Z-transform

$$x[n] = \sum_{\text{poles } p_k \text{ inside } C} \operatorname{Res}_{z=p_k} \left[X(z)\, z^{n-1} \right],$$

where C is a circle in the ROC. For a pole of order m at p,

$$\operatorname{Res}_{z=p} F(z) = \frac{1}{(m-1)!} \frac{d^{m-1}}{dz^{m-1}} \Big[(z-p)^m F(z) \Big] \Big|_{z=p}; \qquad m = 1: \ \operatorname{Res} = (z-p)F(z)|_{z=p}.$$

Two practical remarks:

- For a **causal** X and $n \geq 0$, the poles inside C are just the poles of X; but for $n < 0$ the factor z^{n-1} adds a pole of order $1 - n$ at the origin. Rather than computing those awkward residues, use the fact that the sum of all residues, including the one at infinity, is zero: $x[n] = -\sum(\text{residues outside } C)$. For a causal X there are no poles outside C and, for $n < 0$, no residue at infinity, so $x[n] = 0$ — confirming causality.
- Even for $n \geq 0$, a factor z^{n-1} with $n = 0$ creates a pole at $z = 0$ if $X(0) \neq 0$. That is the $n = 0$ case requiring care; the $X(z)/z$ partial-fraction method handles it automatically.

Worked Example 17.21 — Residues for simple poles

Invert $X(z) = \dfrac{z}{(z-0.5)(z-0.25)}$, $|z| > 0.5$, by residues.

For $n \geq 0$, $X(z)z^{n-1} = \dfrac{z^n}{(z-0.5)(z-0.25)}$ has poles only at 0.5 and 0.25 (no pole at the origin, because the factor z in X cancels the z^{-1}):

$$\text{Res}_{0.5} = \frac{0.5^n}{0.25} = 4(0.5)^n, \qquad \text{Res}_{0.25} = \frac{0.25^n}{-0.25} = -4(0.25)^n.$$

Sum: $x[n] = 4(0.5^n - 0.25^n)$ for $n \geq 0$. For $n < 0$ there are no poles outside C (taken just larger than 0.5), so $x[n] = 0$. Same as Example 17.14.

Worked Example 17.22 — Residues for a repeated pole, and the $n = 0$ subtlety

(a) Invert $X(z) = \dfrac{z}{(z-a)^2}$, $|z| > |a|$.

For $n \geq 0$, $Xz^{n-1} = \dfrac{z^n}{(z-a)^2}$ has a double pole at a:

$$x[n] = \left.\frac{d}{dz} z^n\right|_{z=a} = n\,a^{n-1}, \qquad n \geq 0,$$

confirming the table entry $z/(z-a)^2 \leftrightarrow na^{n-1}u[n]$ with no differentiation of transforms needed.

(b) Invert $X(z) = \dfrac{1}{z-0.5}$, $|z| > 0.5$. Here $Xz^{n-1} = \dfrac{z^{n-1}}{z-0.5}$.

- $n \geq 1$: only the pole at 0.5; $x[n] = 0.5^{n-1}$.
- $n = 0$: $\dfrac{1}{z(z-0.5)}$ has poles at 0 and 0.5, with residues $\dfrac{1}{-0.5} = -2$ and $\dfrac{1}{0.5} = 2$. Sum: $x[0] = 0$.

So $x[n] = 0.5^{n-1}u[n-1]$ — the delayed geometric sequence, as the shift property predicts ($z^{-1} \cdot \frac{z}{z-0.5}$). Forgetting the pole at the origin would give the wrong value $x[0] = 2$.

Comparing the methods

Method	Gives	Best for	Weakness
Inspection / table	closed form	standard shapes, combined with properties	needs recognition
Partial fractions	closed form	any rational X; two-sided ROCs	repeated and complex poles need care
Long division	numbers, sample by sample	checking, first few samples, computer implementation	no closed form; errors accumulate by hand
Power series	closed form or numbers	non-rational X (logs, exponentials)	requires a known series
Residues	closed form	repeated poles; theory	pole at origin for small n

Common Pitfall — Expanding $X(z)$ instead of $X(z)/z$

Expanding $X(z) = \frac{z}{(z-0.5)(z-0.25)}$ directly gives $\frac{2}{z-0.5} - \frac{1}{z-0.25}$, whose terms are $\frac{1}{z-p} = z^{-1}\frac{z}{z-p} \leftrightarrow p^{n-1}u[n-1]$ — correct, but only if you remember the delay. Inverting $\frac{2}{z-0.5}$ as "$2(0.5)^n$" is a classic one-sample error. Dividing by z first avoids it.

Engineering Note — Numerical inversion

In software, a rational $X(z)$ in powers of z^{-1} is inverted by running the difference equation it defines with an impulse input (MATLAB `impz`, SciPy `signal.dimpulse` or `lfilter(b, a, impulse)`), and partial fractions are computed by `residuez`. Root-finding for high-order denominators is ill-conditioned; for orders above about ten, factor the filter into second-order sections rather than computing all poles of the full polynomial.

17.5 Difference equations

Where difference equations come from

A **linear constant-coefficient difference equation** (LCCDE) relates the present output of a discrete-time system to past outputs and present and past inputs:

$$\sum_{k=0}^{N} a_k\, y[n-k] = \sum_{k=0}^{M} b_k\, x[n-k], \qquad a_0 \neq 0.$$

The **order** is N, the number of past outputs the system remembers. Difference equations arise in three ways that an electrical engineer meets constantly:

- **By design**: a digital filter *is* its difference equation; the coefficients a_k, b_k are what the designer chooses and the processor stores.
- **By discretisation**: replacing derivatives in a differential equation by finite differences (Ch. 22) — for instance a forward-Euler model of an RC circuit — produces an LCCDE.
- **By counting**: recurrences in sampled-data control, compound-interest-like growth, ladder networks of identical sections and combinatorial counts (Fibonacci) are naturally indexed by an integer.

Normalising so that $a_0 = 1$, the equation can be solved for $y[n]$:

$$y[n] = -\sum_{k=1}^{N} a_k\, y[n-k] + \sum_{k=0}^{M} b_k\, x[n-k].$$

This is a **recursion**: given N initial values $y[-1], \dots, y[-N]$ and the input, it computes $y[0], y[1], \dots$ one after another. Unlike a differential equation, a difference equation can therefore always be "solved" numerically with nothing more than multiplication and addition. What the Z-transform supplies is the **closed form** — the formula valid for every n at once — and with it insight into the modes, stability and steady state.

The unilateral shift theorem

The bilateral shift rule $x[n-1] \leftrightarrow z^{-1}X(z)$ assumes knowledge of the whole sequence. In an initial-value problem we know $y[n]$ for $n \geq 0$ only through the equation, plus a few stored values $y[-1], y[-2], \dots$. The unilateral transform keeps exactly this information.

Unilateral time-shift (delay and advance)

With $\mathcal{Y}(z) = \sum_{n\geq 0} y[n] z^{-n}$:

$$\begin{aligned}
y[n-1] &\longleftrightarrow z^{-1}\mathcal{Y}(z) + y[-1],\\
y[n-2] &\longleftrightarrow z^{-2}\mathcal{Y}(z) + z^{-1}y[-1] + y[-2],\\
y[n-k] &\longleftrightarrow z^{-k}\mathcal{Y}(z) + \sum_{m=1}^{k} y[-m]\, z^{-(k-m)},\\
y[n+1] &\longleftrightarrow z\,\mathcal{Y}(z) - z\,y[0],\\
y[n+2] &\longleftrightarrow z^{2}\mathcal{Y}(z) - z^{2}y[0] - z\,y[1].
\end{aligned}$$

Derivation for one delay. $\sum_{n\geq 0} y[n-1]z^{-n} = y[-1] + \sum_{n\geq 1} y[n-1]z^{-n} = y[-1] + z^{-1}\sum_{m\geq 0} y[m]z^{-m}$. *For one advance:* $\sum_{n\geq 0} y[n+1]z^{-n} = z\sum_{m\geq 1} y[m]z^{-m} = z(\mathcal{Y} - y[0])$. The others follow by repeating. □

The parallel with the Laplace derivative rule $\mathcal{L}\{\dot y\} = sY - y(0^-)$ is exact: delays bring in *past* values ($y[-1]$, $y[-2]$), advances bring in *initial* values ($y[0]$, $y[1]$). Use the delay form when the equation is written with $y[n-k]$ and the data are $y[-1], y[-2], \dots$; use the advance form when it is written with $y[n+k]$ and the data are $y[0], y[1], \dots$.

For a **causal input** applied at $n = 0$, $x[-1] = x[-2] = \cdots = 0$, so the input terms shift without extra terms.

The solution procedure

Solving an LCCDE with the unilateral Z-transform

1. Take the unilateral transform of every term, using the shift theorem to bring in the initial conditions.
2. Solve the resulting **algebraic** equation for $\mathcal{Y}(z)$.
3. Invert by partial fractions of $\mathcal{Y}(z)/z$ (ROC: exterior of the outermost pole, since y is considered for $n \geq 0$).
4. Check $y[0]$ (and $y[1]$) against a direct run of the recursion.

Step 2 always produces the structure

$$\mathcal{Y}(z) = \underbrace{\frac{B(z)}{A(z)}\,\mathcal{X}(z)}_{\text{zero-state response}} + \underbrace{\frac{C(z)}{A(z)}}_{\text{zero-input response}}, \qquad A(z) = \sum_{k=0}^{N} a_k z^{-k}, \quad B(z) = \sum_{k=0}^{M} b_k z^{-k},$$

where $C(z)$ is a polynomial built from the initial conditions. The **zero-state response** (ZSR) is what the input produces from rest; the **zero-input response** (ZIR) is what the stored initial conditions produce with no input. Both share the denominator $A(z)$, whose roots — the **characteristic roots** or **natural modes** — are the same as those of the classical characteristic equation $\lambda^N + a_1\lambda^{N-1} + \cdots + a_N = 0$. The ratio B/A is the system function $H(z)$ of §17.6.

A second, equally useful split is into **natural** (transient) and **forced** (steady-state) parts: the natural response collects the terms from the poles of H, the forced response those from the poles of $\mathcal{X}$.

Worked examples

Worked Example 17.23 — First order with a step input

Solve $y[n] - 0.5\,y[n-1] = u[n]$ with $y[-1] = 1$.

Step 1. Transform. $\mathcal{Y} - 0.5\left(z^{-1}\mathcal{Y} + y[-1]\right) = \dfrac{z}{z-1}$.

Step 2. Solve. $\mathcal{Y}\left(1 - 0.5z^{-1}\right) = 0.5 + \dfrac{z}{z-1}$, so

$$\mathcal{Y}(z) = \underbrace{\frac{0.5z}{z-0.5}}_{\text{ZIR}} + \underbrace{\frac{z^2}{(z-1)(z-0.5)}}_{\text{ZSR}}.$$

Step 3. Invert. The ZIR is $0.5(0.5)^n = (0.5)^{n+1}$. For the ZSR, $\dfrac{z}{(z-1)(z-0.5)} = \dfrac{2}{z-1} - \dfrac{1}{z-0.5}$, so ZSR $= 2 - (0.5)^n$. Adding,

$$y[n] = 2 - (0.5)^n + (0.5)^{n+1} = 2 - (0.5)^{n+1}, \qquad n \geq 0.$$

Step 4. Check. Recursion: $y[0] = 0.5(1) + 1 = 1.5$, $y[1] = 0.5(1.5) + 1 = 1.75$, $y[2] = 1.875$; formula: $2 - 0.5 = 1.5$, $2 - 0.25 = 1.75$, $2 - 0.125 = 1.875$ ✓. The forced response is the constant $2 = H(1)$, the DC gain; the natural response $-(0.5)^{n+1}$ decays.

Worked Example 17.24 — Second order: ZIR, ZSR, natural and forced parts

Solve $y[n] - 0.75\,y[n-1] + 0.125\,y[n-2] = u[n]$ with $y[-1] = 1$, $y[-2] = 2$.

Step 1. Transform.

$$\mathcal{Y} - 0.75\left(z^{-1}\mathcal{Y} + 1\right) + 0.125\left(z^{-2}\mathcal{Y} + z^{-1}\cdot 1 + 2\right) = \frac{z}{z-1}.$$

Step 2. Collect. The initial-condition terms give $0.75 - 0.125z^{-1} - 0.25 = 0.5 - 0.125z^{-1}$, so

$$\mathcal{Y}\left(1 - 0.75z^{-1} + 0.125z^{-2}\right) = \frac{z}{z-1} + 0.5 - 0.125z^{-1}.$$

With $1 - 0.75z^{-1} + 0.125z^{-2} = (1 - 0.5z^{-1})(1 - 0.25z^{-1})$:

$$\mathcal{Y}_{\text{zi}} = \frac{0.5(1 - 0.25z^{-1})}{(1 - 0.5z^{-1})(1 - 0.25z^{-1})} = \frac{0.5z}{z - 0.5}, \qquad \mathcal{Y}_{\text{zs}} = \frac{z^3}{(z-1)(z-0.5)(z-0.25)}.$$

The initial conditions happen to cancel the 0.25 mode in the ZIR.

Step 3. Invert. ZIR: $0.5(0.5)^n$. ZSR: expanding $\mathcal{Y}_{\text{zs}}/z$ by cover-up,

$$A_1 = \frac{1}{0.5 \cdot 0.75} = \frac{8}{3}, \quad A_{0.5} = \frac{0.25}{(-0.5)(0.25)} = -2, \quad A_{0.25} = \frac{0.0625}{(-0.75)(-0.25)} = \frac{1}{3},$$

so ZSR $= \frac{8}{3} - 2(0.5)^n + \frac{1}{3}(0.25)^n$. Total:

$$y[n] = \underbrace{\tfrac{8}{3}}_{\text{forced}} \underbrace{- 1.5(0.5)^n + \tfrac{1}{3}(0.25)^n}_{\text{natural}}, \qquad n \geq 0.$$

Step 4. Check. Formula: $y[0] = 2.6667 - 1.5 + 0.3333 = 1.5$; $y[1] = 2.6667 - 0.75 + 0.0833 = 2.0$; $y[2] = 2.6667 - 0.375 + 0.0208 = 2.3125$. Recursion: $y[0] = 0.75(1) - 0.125(2) + 1 = 1.5$, $y[1] = 0.75(1.5) - 0.125(1) + 1 = 2.0$, $y[2] = 0.75(2) - 0.125(1.5) + 1 = 2.3125$ ✓. The steady state $8/3 = 1/(1 - 0.75 + 0.125) = H(1)$.

Worked Example 17.25 — The Fibonacci recurrence (advance form)

Solve $y[n+2] = y[n+1] + y[n]$ with $y[0] = 0$, $y[1] = 1$.

Step 1. Advance rule. $\left(z^2\mathcal{Y} - z^2 y[0] - z y[1]\right) = \left(z\mathcal{Y} - z y[0]\right) + \mathcal{Y}$, i.e. $z^2\mathcal{Y} - z = z\mathcal{Y} + \mathcal{Y}$.

Step 2. $\mathcal{Y}(z) = \dfrac{z}{z^2 - z - 1} = \dfrac{z}{(z-\varphi)(z-\psi)}$, with $\varphi = \dfrac{1+\sqrt{5}}{2} \approx 1.618$ and $\psi = \dfrac{1-\sqrt{5}}{2} \approx -0.618$.

Step 3. $\dfrac{\mathcal{Y}}{z} = \dfrac{1}{\varphi - \psi}\left(\dfrac{1}{z-\varphi} - \dfrac{1}{z-\psi}\right)$ with $\varphi - \psi = \sqrt{5}$, so

$$y[n] = \frac{\varphi^n - \psi^n}{\sqrt{5}} \qquad \text{(Binet's formula)}.$$

Check: $y[10] = (122.992 - 0.00813)/2.23607 = 55.000$ ✓. One characteristic root lies outside the unit circle, so the sequence grows geometrically, by the factor φ per step: as a "system" this recursion is unstable.

Worked Example 17.26 — Input at a natural frequency (discrete resonance)

Solve $y[n] - 0.5\,y[n-1] = (0.5)^n u[n]$ from rest.

$\mathcal{Y}(1 - 0.5z^{-1}) = \dfrac{z}{z-0.5}$, so $\mathcal{Y} = \dfrac{z^2}{(z-0.5)^2}$, a **double** pole: the input mode coincides with the natural mode. From the table ($m=2$): $\dfrac{z^2}{(z-a)^2} \leftrightarrow (n+1)a^n u[n]$, so

$$y[n] = (n+1)(0.5)^n, \qquad n \geq 0.$$

The factor $(n+1)$ is the discrete counterpart of the te^{-at} term in a continuous system driven at its own pole. *Check:* $y[0]=1$, $y[1] = 0.5(1)+0.5 = 1$, $y[2] = 0.5(1)+0.25 = 0.75$; formula $1,\ 1,\ 0.75$ ✓.

Worked Example 17.27 — Forward-Euler model of an RC low-pass circuit

An RC low-pass filter obeys $\tau\,\dot{v} = x - v$, $\tau = RC$. Approximating $\dot{v}(nT) \approx \left(v[n+1] - v[n]\right)/T$ gives

$$v[n+1] = \left(1 - \frac{T}{\tau}\right) v[n] + \frac{T}{\tau}\, x[n].$$

Find the step response for $\tau = 1$ ms, $T = 0.25$ ms, $v[0] = 0$, and compare with the exact samples $1 - e^{-nT/\tau}$.

Step 1. With $\alpha = T/\tau = 0.25$ and $x[n] = u[n]$, the advance rule gives $z\mathcal{V} - zv[0] = 0.75\mathcal{V} + 0.25\dfrac{z}{z-1}$, so

$$\mathcal{V}(z) = \frac{0.25z}{(z-1)(z-0.75)}.$$

Step 2. $\mathcal{V}/z = \dfrac{1}{z-1} - \dfrac{1}{z-0.75}$, so $v[n] = 1 - (0.75)^n$.

Step 3. Compare. The exact samples are $1 - (e^{-0.25})^n = 1 - (0.7788)^n$. At $n = 4$ ($t = \tau$): Euler gives 0.6836, exact 0.6321. The discrete pole 0.75 approximates the exact $e^{-T/\tau} = 0.7788$ — a first-order Taylor approximation of $z = e^{sT}$ (§17.7).

Step 4. Numerical stability. The pole $1 - T/\tau$ lies inside the unit circle only if $0 < T/\tau < 2$. With $T > 2\tau$ the simulation diverges although the circuit is perfectly stable — the origin of the step-size limits studied in Ch. 22.

Classical solution, for comparison

The method of undetermined coefficients works for difference equations exactly as for differential equations: the homogeneous solution is $\sum c_i \lambda_i^n$ over the characteristic roots λ_i (with $n^k\lambda^n$ for repeated roots), a particular solution is guessed in the form of the input, and the constants are fitted to the initial conditions. For Example 17.24: characteristic roots $0.5, 0.25$; particular solution $y_p = K$ with $K(1 - 0.75 + 0.125) = 1$, $K = 8/3$; then $y = \frac{8}{3} + c_1(0.5)^n + c_2(0.25)^n$ fitted to $y[0] = 1.5$, $y[1] = 2$ gives $c_1 = -1.5$, $c_2 = \frac{1}{3}$. The Z-transform does the same work but builds the initial conditions in automatically and handles resonant inputs (Example 17.26) without a separate rule.

Common Pitfall — Using $y[0]$ where the delay rule needs $y[-1]$

The delay rule needs the values *before* the input is applied ($y[-1], y[-2]$); the advance rule needs $y[0], y[1]$. If a problem gives $y[0]$ for an equation written in delays, first back-solve the recursion for $y[-1]$ (with the known $x[0]$) or rewrite the equation in advance form. Mixing them silently shifts the answer by one sample.

Engineering Note — Initial conditions are register contents

In an implemented IIR filter, $y[-1]$ and $y[-2]$ are literally the numbers sitting in the delay registers when processing starts. Clearing them (zero state) at power-up gives the zero-state response; a filter that is switched between coefficient sets on the fly carries its old register contents as initial conditions, and the resulting zero-input transient is the "click" heard when audio equaliser settings change abruptly.

17.6 System-function representation

The system function

For an LTI system with impulse response $h[n]$, the convolution theorem gives $Y(z) = H(z)X(z)$. The transform of the impulse response therefore characterises the system completely, in the same way the transfer function $H(s)$ does in continuous time.

System function (transfer function)

The **system function** of a discrete-time LTI system is

$$H(z) = \mathcal{Z}\{h[n]\} = \frac{Y(z)}{X(z)} \quad \text{(zero initial state)},$$

together with its ROC. Evaluated on the unit circle, $H(e^{j\omega})$ is the **frequency response**.

For a system described by the LCCDE $\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k]$, transforming with zero initial conditions (bilateral shift rule) gives

$$H(z) = \frac{\displaystyle\sum_{k=0}^{M} b_k z^{-k}}{\displaystyle\sum_{k=0}^{N} a_k z^{-k}} = \frac{b_0 + b_1 z^{-1} + \cdots + b_M z^{-M}}{a_0 + a_1 z^{-1} + \cdots + a_N z^{-N}}.$$

The coefficients of the difference equation are the coefficients of the numerator and denominator polynomials, read off directly. Going the other way, any rational $H(z)$ written in powers of z^{-1} is a difference equation: cross-multiply $Y \cdot A(z) = X \cdot B(z)$ and interpret each z^{-k} as a k-sample delay.

An LCCDE alone does not fix the ROC. If the system is known to be causal (the usual case), the ROC is the exterior of the outermost pole; if it is known to be stable, the ROC contains the unit circle. These two pieces of information, plus $H(z)$, determine $h[n]$ uniquely.

Pole–zero (factored) form

Factoring numerator and denominator,

$$H(z) = b_0\, z^{N-M}\, \frac{\prod_{k=1}^{M}(z - q_k)}{\prod_{k=1}^{N}(z - p_k)} \qquad (a_0 = 1),$$

with **zeros** q_k, **poles** p_k, and $|N - M|$ extra zeros or poles at $z = 0$. The **pole–zero plot** — poles marked $\times$, zeros $\circ$, with the unit circle drawn for reference — together with the gain b_0 is a complete description. It shows at a glance:

- **stability** of a causal system: all poles inside the unit circle;
- **modes** of the natural response: each pole $p = re^{j\theta}$ contributes r^n-weighted oscillation at θ rad/sample;
- **frequency selectivity**: poles near the unit circle at angle θ produce gain peaks near $\omega = \theta$; zeros on or near the circle produce nulls (§17.7).

Worked Example 17.28 — From difference equation to pole–zero plot

A causal system obeys $y[n] = 0.9\,y[n-1] - 0.81\,y[n-2] + x[n] - x[n-2]$. Find $H(z)$, its poles and zeros, and its ROC.

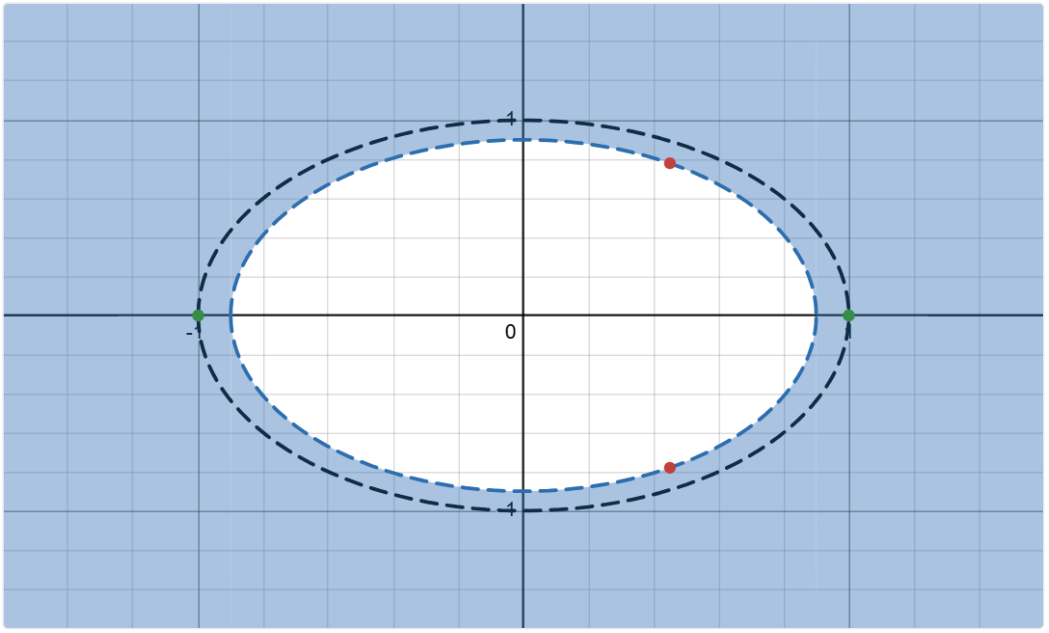
Step 1. Rearranged, $y[n] - 0.9y[n-1] + 0.81y[n-2] = x[n] - x[n-2]$, so

$$H(z) = \frac{1 - z^{-2}}{1 - 0.9z^{-1} + 0.81z^{-2}} = \frac{z^2 - 1}{z^2 - 0.9z + 0.81}.$$

Step 2. Zeros: $z^2 = 1$, i.e. $z = \pm 1$ — on the unit circle at $\omega = 0$ and $\omega = \pi$.

Step 3. Poles: $z = \dfrac{0.9 \pm \sqrt{0.81 - 3.24}}{2} = 0.45 \pm j0.7794 = 0.9e^{\pm j\pi/3}$ (check: $2r\cos\theta = 2(0.9)(0.5) = 0.9$ ✓, $r^2 = 0.81$ ✓).

Step 4. ROC. Causal $\Rightarrow |z| > 0.9$, which contains the unit circle: the system is stable. The zeros block DC and $\omega = \pi$; the poles near the circle at $\pm 60°$ boost $\omega \approx \pi/3$. This is a **second-order band-pass resonator**; its magnitude response is drawn in §17.7.



Pole–zero plot of $H(z) = (z^2 - 1)/(z^2 - 0.9z + 0.81)$ with the causal ROC $|z| > 0.9$ shaded. Poles (red) at radius 0.9 and angle $\pm\pi/3$; zeros (green) on the unit circle at ± 1.

FIR and IIR systems

FIR and IIR

A system is **finite impulse response (FIR)** if $h[n]$ has finitely many non-zero samples. For a causal FIR system $H(z) = \sum_{k=0}^{M} b_k z^{-k}$: all poles are at $z = 0$, so an FIR system is always stable. Its difference equation has no feedback: $y[n] = \sum_k b_k x[n-k]$, and $h[n] = b_n$.

A system is **infinite impulse response (IIR)** if $h[n]$ has infinitely many non-zero samples — for rational H, when there is at least one pole not cancelled by a zero away from the origin. IIR systems need feedback ($a_k \neq 0$ for some $k \geq 1$) and may be unstable.

Worked Example 17.29 — The moving-average filter: FIR with a recursive form

The 4-point moving average is $y[n] = \frac{1}{4}\left(x[n] + x[n-1] + x[n-2] + x[n-3]\right)$.

Step 1. $H(z) = \frac{1}{4}\left(1 + z^{-1} + z^{-2} + z^{-3}\right)$. By the geometric sum,

$$H(z) = \frac{1}{4}\,\frac{1 - z^{-4}}{1 - z^{-1}} = \frac{1}{4}\,\frac{z^4 - 1}{z^3(z - 1)}.$$

Step 2. Zeros: the fourth roots of unity $1, j, -1, -j$; **poles:** three at $z = 0$ and one at $z = 1$. The pole at 1 cancels the zero at 1, leaving zeros at $\omega = \pm\pi/2$ and π on the unit circle — frequencies that are completely rejected.

Step 3. Recursive implementation. The ratio form corresponds to the difference equation $y[n] = y[n-1] + \frac{1}{4}\left(x[n] - x[n-4]\right)$: one addition and one subtraction per sample regardless of length (the **running-sum** or CIC structure), instead of M additions. Its impulse response is still finite, because the pole is cancelled.

Check: $|H(e^{j0})| = 1$ (unit DC gain), $|H(e^{j\pi/2})| = 0$, $|H(e^{j\pi/4})| = 0.653$.

From impulse response to system function, and back

Worked Example 17.30 — Identifying a system from its impulse response

A causal system has $h[n] = \left[2(0.5)^n - (0.25)^n\right] u[n]$. Find $H(z)$ and the difference equation.

Step 1.

$$H(z) = \frac{2z}{z - 0.5} - \frac{z}{z - 0.25} = \frac{z\left[2(z - 0.25) - (z - 0.5)\right]}{(z - 0.5)(z - 0.25)} = \frac{z^2}{z^2 - 0.75z + 0.125}.$$

Step 2. In powers of z^{-1}: $H = \dfrac{1}{1 - 0.75z^{-1} + 0.125z^{-2}}$, so

$$y[n] = 0.75\,y[n-1] - 0.125\,y[n-2] + x[n].$$

Check: $h[0] = 2 - 1 = 1$ and the recursion gives $h[0] = 1$; $h[1] = 1 - 0.25 = 0.75$ and the recursion gives $0.75(1) = 0.75$ ✓. This is the system of Example 17.24, whose step response had steady state $H(1) = 1/0.375 = 8/3$.

Block diagrams and realisations

Three primitives build every LCCDE: the **adder**, the **multiplier** (gain) and the **unit delay** z^{-1}. Because the same $H(z)$ can be wired in many ways, the choice of **realisation** (structure) matters for memory, arithmetic and — above all in fixed-point hardware — sensitivity to coefficient rounding.

Direct forms

Write $H(z) = B(z) \cdot \dfrac{1}{A(z)}$.

- **Direct form I** implements $B(z)$ (a tapped delay line on x, M delays) followed by $1/A(z)$ (a tapped delay line on y with feedback, N delays): $M + N$ delays.
- **Direct form II** reverses the order, $\dfrac{1}{A(z)}$ then $B(z)$. Both sections then delay the same intermediate signal $w[n]$, so the delay lines merge: $w[n] = x[n] - \sum_{k=1}^{N} a_k w[n-k]$, $y[n] = \sum_{k=0}^{M} b_k w[n-k]$ — only $\max(M, N)$ delays, the minimum possible (**canonical**).
- **Transposed direct form II** reverses every arrow and swaps inputs and outputs (transposition preserves H); it is often preferred in floating point because the adders see smaller dynamic range.

Writing the DF-II equations for Example 17.28 ($b = [1, 0, -1]$, $a = [1, -0.9, 0.81]$):

$$w[n] = x[n] + 0.9\,w[n-1] - 0.81\,w[n-2], \qquad y[n] = w[n] - w[n-2],$$

two delays (storing $w[n-1]$, $w[n-2]$), three multiplications and three additions per sample.

Cascade and parallel forms

- **Cascade (series) form:** factor $H(z) = G\prod_k H_k(z)$ into first- and second-order sections ("biquads"), each with real coefficients — complex-conjugate pole and zero pairs are kept together. This is the standard implementation of IIR filters of order above two.
- **Parallel form:** a partial-fraction expansion in z^{-1}, $H(z) = C + \sum_k \dfrac{\beta_{0k} + \beta_{1k}z^{-1}}{1 + \alpha_{1k}z^{-1} + \alpha_{2k}z^{-2}}$, sums the outputs of independent sections.

The reason for cascades is numerical: the roots of a high-degree polynomial are extremely sensitive to its coefficients (Ch. 22). A 10th-order narrow-band filter stored as one denominator polynomial may become unstable after rounding its coefficients to 16 bits, while the same filter as five biquads, each with its own well-conditioned quadratic, stays stable.

Interconnections

The algebra of system functions mirrors that of Laplace transfer functions:

Connection	System function
Cascade H_1 then H_2	$H_1(z)H_2(z)$ (order irrelevant for LTI)
Parallel	$H_1(z) + H_2(z)$
Negative feedback, forward G, feedback F	$\dfrac{G(z)}{1 + G(z)F(z)}$
Inverse system	$H^{-1}(z) = 1/H(z)$

Worked Example 17.31 — A digital feedback loop

A digital integrating controller with a one-sample computation delay has forward path $G(z) = \dfrac{Kz^{-1}}{1 - z^{-1}} = \dfrac{K}{z - 1}$ in a unity negative-feedback loop. Find the closed-loop system function, the range of K for stability, and the step response for $K = 0.5$.

Step 1.

$$T(z) = \frac{G}{1 + G} = \frac{K/(z-1)}{1 + K/(z-1)} = \frac{K}{z - 1 + K} = \frac{Kz^{-1}}{1 - (1-K)z^{-1}}.$$

Step 2. Stability. The single closed-loop pole is $z = 1 - K$; stability requires $|1 - K| < 1$, i.e. $0 < K < 2$. The loop is unstable for too much gain, as a sampled control loop always is — the computation delay limits the achievable bandwidth.

Step 3. Step response ($K = 0.5$, pole 0.5): $Y = \dfrac{0.5}{z - 0.5}\cdot\dfrac{z}{z-1}$, $Y/z = \dfrac{1}{z-1} - \dfrac{1}{z-0.5}$, so $y[n] = [1 - (0.5)^n]\,u[n]$, with zero steady-state error ($T(1) = 1$) thanks to the integrator.

Check: simulated outputs $0,\ 0.5,\ 0.75,\ 0.875,\ 0.9375$ ✓. With $K = 1$ the pole is at 0 and the output reaches 1 in one step and stays there (a **deadbeat** response, impossible in continuous time); with $K = 1.5$ the pole is at -0.5 and the response overshoots with alternating sign: $0,\ 1.5,\ 0.75,\ 1.125,\ 0.9375, \ldots$.

Inverse systems and minimum phase

The inverse of $H(z)$ has the zeros of H as its poles. For the inverse to be **both causal and stable**, those poles — the zeros of H — must lie inside the unit circle.

Minimum-phase system

A causal, stable rational system is **minimum phase** if all its zeros are also strictly inside the unit circle. Exactly the minimum-phase systems have causal, stable inverses.

For example $H(z) = \dfrac{1 - 0.5z^{-1}}{1 - 0.8z^{-1}}$ (zero 0.5, pole 0.8) is minimum phase; its inverse $\dfrac{1 - 0.8z^{-1}}{1 - 0.5z^{-1}}$ is causal and stable. By contrast $H(z) = 1 - 2z^{-1}$ (zero at 2) has the inverse $1/(1 - 2z^{-1})$, which is either causal and unstable (ROC $|z| > 2$) or stable and anticausal (ROC $|z| < 2$). Channel equalisers in communication receivers face exactly this choice, and solve it by allowing a delay.

Common Pitfall — Pole–zero cancellation hides instability

Cancelling a factor $(z - p)$ between numerator and denominator is legitimate algebra, but if $|p| \geq 1$ and the cancellation occurs between two separately built subsystems (e.g. an unstable plant and a controller zero placed on its pole), the internal signal still contains the growing mode p^n. Any mismatch — and there always is one in hardware — lets it appear at the output. Cancel unstable poles on paper only.

Engineering Note — Reading a filter specification sheet

Coefficients exported by filter-design tools come either as `(b, a)` polynomials, as `(z, p, k)` zeros, poles and gain, or as a matrix of second-order sections `sos`, one row $[b_0, b_1, b_2, 1, a_1, a_2]$ per biquad. All three describe the same $H(z)$; the `sos` form is the one to implement.

17.7 Discrete-time system analysis

This section collects the analysis questions that the system function answers: Is the system stable? How does it respond to sinusoids? What is its frequency response, and how can it be read from the pole–zero plot? And how does all of this relate to the continuous-time system that a digital system usually imitates or controls?

BIBO stability

BIBO stability of LTI systems

A discrete-time LTI system is bounded-input bounded-output stable if and only if its impulse response is absolutely summable:

$$S = \sum_{n=-\infty}^{\infty} |h[n]| < \infty.$$

Equivalently, the ROC of $H(z)$ contains the unit circle. For a **causal** system with rational $H(z)$: stable $\iff$ every pole satisfies $|p_k| < 1$.

Proof. **Sufficiency.** If $|x[n]| \le B$ then $|y[n]| = |\sum_k h[k]x[n-k]| \le B\sum_k |h[k]| = BS$. **Necessity.** Suppose $S = \infty$. For real h, choose the bounded input $x[k] = \operatorname{sgn}(h[-k])$ (with $\operatorname{sgn} 0 = 0$). Then $y[0] = \sum_k h[k]x[-k] = \sum_k |h[k]| = \infty$, so a bounded input produces an unbounded output. **Pole form.** A causal rational $h[n]$ is a finite sum of terms $c\,n^m p^n u[n]$. Each is absolutely summable iff $|p| < 1$ (the ratio test: $|(n+1)^m p^{n+1}|/|n^m p^n| \to |p|$). $\square$

Three cases are worth naming for causal systems:

Pole location	Natural response	Classification
all $\lvert p\rvert < 1$	decays geometrically	**stable**
simple poles on $\lvert z\rvert = 1$, none outside	bounded, non-decaying (constant, $(-1)^n$, sinusoid)	**marginally stable** — *not* BIBO stable
any $\lvert p\rvert > 1$, or repeated poles on $\lvert z\rvert = 1$	grows	**unstable**

The accumulator $H(z) = 1/(1 - z^{-1})$ is the canonical marginal case: its impulse response $u[n]$ is bounded, but the bounded step input produces the unbounded ramp $(n+1)u[n]$ — the double pole of $Y = z^2/(z-1)^2$ on the unit circle.

Stability tests without factoring

For high-order denominators, root finding is avoidable. Write the denominator in positive powers as $D(z) = z^N + a_1 z^{N-1} + \cdots + a_N$.

Necessary conditions (any order)

If all roots of the real monic polynomial $D(z)$ lie strictly inside the unit circle, then

$$D(1) > 0, \qquad (-1)^N D(-1) > 0, \qquad |a_N| < 1.$$

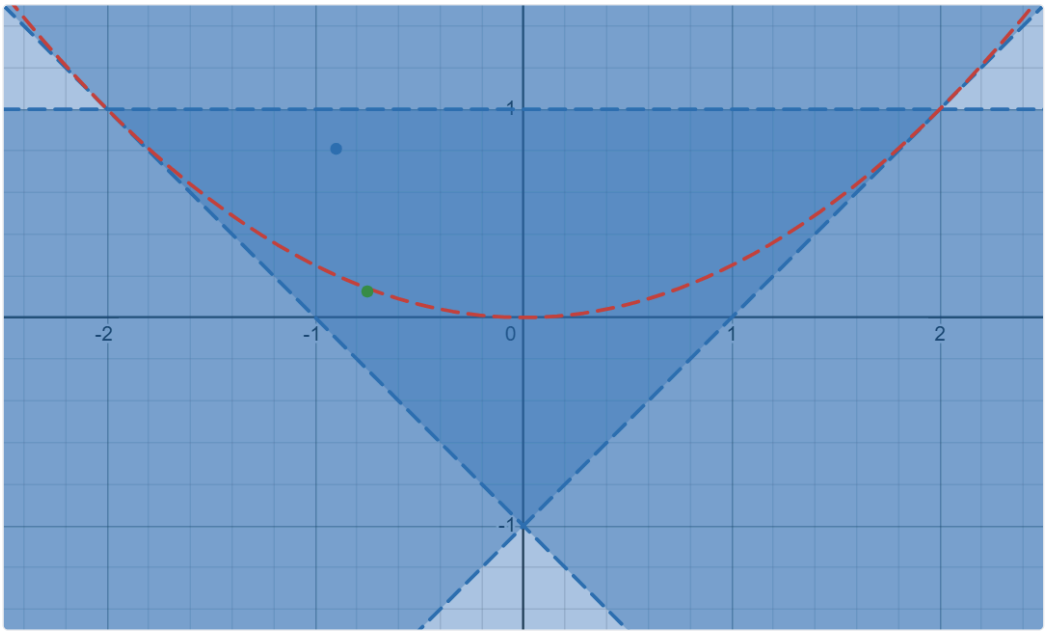
Reason. $D(1) = \prod(1 - p_k)$; for real roots in $(-1, 1)$ every factor is positive, and conjugate pairs contribute $|1 - p|^2 > 0$. Likewise $(-1)^N D(-1) = \prod(1 + p_k) > 0$. And $|a_N| = \prod |p_k| < 1$. $\square$

Second-order stability triangle

The polynomial $z^2 + a_1 z + a_2$ has both roots strictly inside the unit circle if and only if

$$|a_2| < 1 \qquad \text{and} \qquad |a_1| < 1 + a_2.$$

For $N = 2$ the necessary conditions above are also sufficient. The roots are complex for $a_2 > a_1^2/4$, in which case their radius is $\sqrt{a_2}$.



The stability triangle in the (a_1, a_2) plane for $z^2 + a_1 z + a_2$: the region where all three shadings overlap is stable. Above the dashed parabola $a_2 = a_1^2/4$ the poles are complex with radius $\sqrt{a_2}$; below it they are real. Both systems marked are stable.

For third and higher order, the **Jury** (or **Schur–Cohn**) table extends the test by a recursive construction; for a cubic $z^3 + a_1 z^2 + a_2 z + a_3$ it adds to the three necessary conditions the single extra inequality

$$\left|a_3^2 - 1\right| > \left|a_3 a_1 - a_2\right| .$$

The **bilinear map** $z = (1 + w)/(1 - w)$ sends the unit disc onto the left half w-plane, so applying the Routh–Hurwitz test of Ch. 14 to the transformed polynomial is an alternative route.

Worked Example 17.32 — Testing cubics

(a) $D(z) = z^3 - 1.2z^2 + 0.5z - 0.1$. Conditions: $D(1) = 1 - 1.2 + 0.5 - 0.1 = 0.2 > 0$ ✓; $-D(-1) = -(-1 - 1.2 - 0.5 - 0.1) = 2.8 > 0$ ✓; $|a_3| = 0.1 < 1$ ✓; $|0.01 - 1| = 0.99 > |0.12 - 0.5| = 0.38$ ✓. **Stable.** *Check:* the roots are 0.682 and $0.259 \pm j0.282$ (radius 0.383).

(b) $D(z) = z^3 - 1.5z^2 + 1.1z - 0.6$. $D(1) = 1 - 1.5 + 1.1 - 0.6 = 0$: the first condition fails with equality, so there is a root *on* the unit circle at $z = 1$. Dividing it out, $D = (z - 1)(z^2 - 0.5z + 0.6)$, whose quadratic factor has roots of radius $\sqrt{0.6} = 0.775$. **Marginally stable** (a simple pole at $z = 1$ makes the system an accumulator in disguise).

Frequency response

Sinusoidal steady state

Let a stable LTI system with real impulse response have frequency response $H(e^{j\omega}) = |H(e^{j\omega})| e^{j\angle H(e^{j\omega})}$. Then:

1. The eternal input $x[n] = e^{j\omega_0 n}$ produces exactly $y[n] = H(e^{j\omega_0})\, e^{j\omega_0 n}$.
2. The eternal input $x[n] = A\cos(\omega_0 n + \phi)$ produces

$$y[n] = A\left|H(e^{j\omega_0})\right| \cos\!\left(\omega_0 n + \phi + \angle H(e^{j\omega_0})\right) .$$

3. A causal input $A\cos(\omega_0 n + \phi)u[n]$ produces this response plus a transient made of the system's natural modes, which decays because the poles are inside the unit circle.

Derivation. Statement 1 is the eigenfunction property of §17.2 with $z = e^{j\omega_0}$ — the sum converges because the unit circle is in the ROC. For 2, write the cosine as $\frac{A}{2}\left(e^{j\phi}e^{j\omega_0 n} + e^{-j\phi}e^{-j\omega_0 n}\right)$ and use $H(e^{-j\omega_0}) = H^*(e^{j\omega_0})$ for real h. For 3, the partial-fraction expansion of $Y(z) = H(z)X(z)$ separates the terms with the input's poles $e^{\pm j\omega_0}$ (the steady state) from those with the poles of H (the transient). □

Properties of $H(e^{j\omega})$ that follow from §17.2:

- **Periodicity:** $H(e^{j(\omega + 2\pi)}) = H(e^{j\omega})$; only $-\pi < \omega \le \pi$ needs plotting.
- **Symmetry** (real h): $|H|$ is even and $\angle H$ is odd, so $0 \le \omega \le \pi$ suffices. In physical units $\omega = \pi$ corresponds to the Nyquist frequency $f_s/2$.
- **DC and Nyquist gains:** $H(e^{j0}) = H(1) = \sum h[n]$ and $H(e^{j\pi}) = H(-1) = \sum (-1)^n h[n]$.
- **Group delay:** $\tau_g(\omega) = -\dfrac{d\angle H(e^{j\omega})}{d\omega}$, in samples. A **linear-phase** FIR filter with symmetric coefficients $b_k = b_{M-k}$ has $\angle H = -\omega M/2$ (plus jumps of π at zeros), i.e. a constant delay of $M/2$ samples for all frequencies.

Worked Example 17.33 — First-order low-pass: the frequency response

The "exponential smoother" $y[n] = a\,y[n-1] + (1-a)\,x[n]$, $0 < a < 1$, has $H(z) = \dfrac{1-a}{1-az^{-1}}$.

Step 1. Magnitude. On $z = e^{j\omega}$: $|1 - a e^{-j\omega}|^2 = (1 - a\cos\omega)^2 + a^2\sin^2\omega = 1 - 2a\cos\omega + a^2$, so

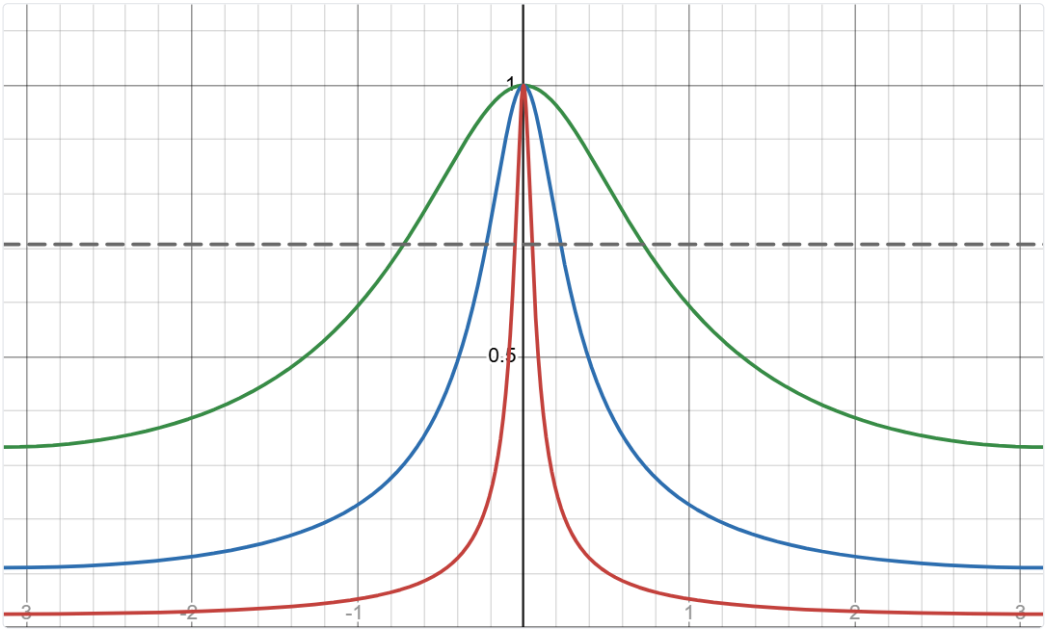
$$|H(e^{j\omega})| = \frac{1-a}{\sqrt{1 - 2a\cos\omega + a^2}}, \qquad \angle H(e^{j\omega}) = -\arctan\frac{a\sin\omega}{1 - a\cos\omega}.$$

Step 2. Values for $a = 0.8$. DC: $|H| = 0.2/0.2 = 1$. Nyquist ($\omega = \pi$): $0.2/1.8 = 0.111$, i.e. -19.1 dB.

Step 3. 3-dB frequency. Setting $|H|^2 = \tfrac{1}{2}$: $1 - 2a\cos\omega_c + a^2 = 2(1-a)^2$, so

$$\cos\omega_c = 1 - \frac{(1-a)^2}{2a} = 1 - \frac{0.04}{1.6} = 0.975, \qquad \omega_c = 0.2241 \text{ rad/sample}.$$

At $f_s = 10$ kHz this is $f_c = \omega_c f_s / 2\pi = 357$ Hz.



Magnitude responses $|H(e^{j\omega})| = (1-a)/\sqrt{1-2a\cos\omega+a^2}$ of the first-order smoother for $a = 0.5$ (green), 0.8 (blue) and 0.95 (red), plotted over one period $-\pi < \omega \le \pi$. As the pole approaches $z = 1$ the pass-band narrows; the dashed line is the -3 dB level.

Geometric evaluation from the pole–zero plot

Write H in factored form and set $z = e^{j\omega}$:

$$|H(e^{j\omega})| = |b_0|\,\frac{\prod_k |e^{j\omega} - q_k|}{\prod_k |e^{j\omega} - p_k|}, \qquad \angle H(e^{j\omega}) = \angle b_0 + (N-M)\omega + \sum_k \angle\left(e^{j\omega} - q_k\right) - \sum_k \angle\left(e^{j\omega} - p_k\right).$$

Each factor $e^{j\omega} - c$ is the vector from the pole or zero c to the point $e^{j\omega}$ on the unit circle. As ω sweeps from 0 to π, the point travels along the upper half of the circle, and:

- the **magnitude** is (gain) × (product of zero-vector lengths) / (product of pole-vector lengths);
- passing close to a **pole** makes a denominator length small: a **peak**, sharper the closer the pole is to the circle;
- passing through a **zero** on the circle makes the magnitude exactly **zero**;
- poles and zeros at the origin have vectors of length 1 for all ω: they affect only the phase (a pure delay).

For a pole at radius r close to 1, the 3-dB bandwidth of the peak it creates is approximately

$$\Delta\omega_{3\,\text{dB}} \approx 2(1-r) \quad \text{rad/sample},$$

because near the pole the unit circle is almost straight, and the pole-vector length grows by $\sqrt{2}$ when the point has moved a distance $1-r$ either side of the closest approach. This rule makes pole placement a design tool.

Worked Example 17.34 — The resonator of Example 17.28

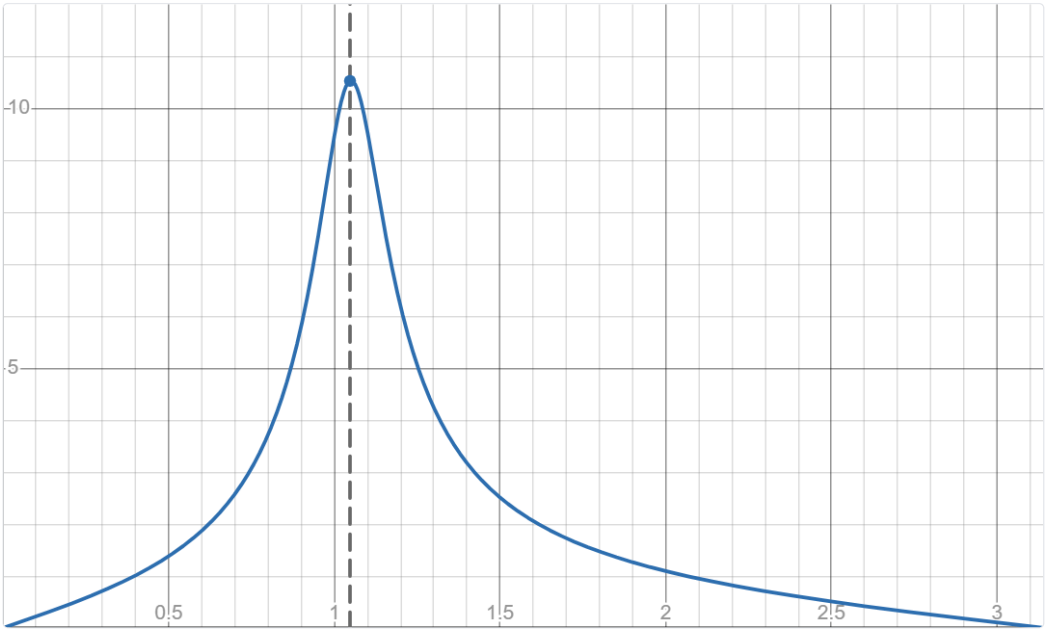
Analyse $H(z) = \dfrac{z^2 - 1}{z^2 - 0.9z + 0.81}$, poles $0.9e^{\pm j\pi/3}$, zeros ± 1.

Step 1. Geometric reading. At $\omega = 0$ and $\omega = \pi$ the point $e^{j\omega}$ sits on a zero: $|H| = 0$. Near $\omega = \pi/3$ it passes within $1 - r = 0.1$ of a pole: a peak, of approximate width $2(0.1) = 0.2$ rad.

Step 2. Peak value. At $\omega = \pi/3$: the zero vectors have lengths $|e^{j\pi/3} - 1| = 1$ and $|e^{j\pi/3} + 1| = \sqrt{3}$; the near pole vector has length 0.1; the far pole vector is $|e^{j\pi/3} - 0.9e^{-j\pi/3}| = |0.05 + j1.6454| = 1.6462$. So

$$|H(e^{j\pi/3})| = \frac{1 \cdot \sqrt{3}}{0.1 \times 1.6462} = 10.52.$$

Step 3. Steady-state response. For $x[n] = \cos(\pi n/3)$, a numerical evaluation gives $H(e^{j\pi/3}) = 10.52\, e^{j0.030}$, so in steady state $y[n] = 10.52\cos(\pi n/3 + 0.030)$. The exact maximum of $|H|$ is 10.53, at $\omega = 0.334\pi$, very slightly above $\pi/3$ (the far pole and the zeros tilt the peak).



Magnitude response of the resonator $H(z) = (1 - z^{-2})/(1 - 0.9z^{-1} + 0.81z^{-2})$ for $0 \le \omega \le \pi$: zeros at $z = \pm 1$ force nulls at DC and at Nyquist; the poles at $0.9e^{\pm j\pi/3}$ produce a peak at $\omega \approx \pi/3$ with bandwidth ≈ 0.2 rad.

Worked Example 17.35 — Designing a 50 Hz notch filter by pole–zero placement

Remove 50 Hz mains hum from an ECG sampled at $f_s = 1$ kHz, with a notch about 15 Hz wide.

Step 1. Notch frequency. $\omega_0 = 2\pi \cdot 50/1000 = 0.1\pi$ rad/sample.

Step 2. Zeros on the circle at $e^{\pm j\omega_0}$ give exact nulls: numerator $1 - 2\cos\omega_0 z^{-1} + z^{-2}$.

Step 3. Poles just inside, at the same angle, $re^{\pm j\omega_0}$, restore unit gain away from ω_0 (each pole nearly cancels the effect of its zero except near the notch). Bandwidth $2(1 - r) \approx 2\pi \cdot 15/1000 = 0.094$ rad gives $r \approx 0.95$:

$$H(z) = \frac{1 - 1.9021z^{-1} + z^{-2}}{1 - 1.8070z^{-1} + 0.9025z^{-2}}.$$

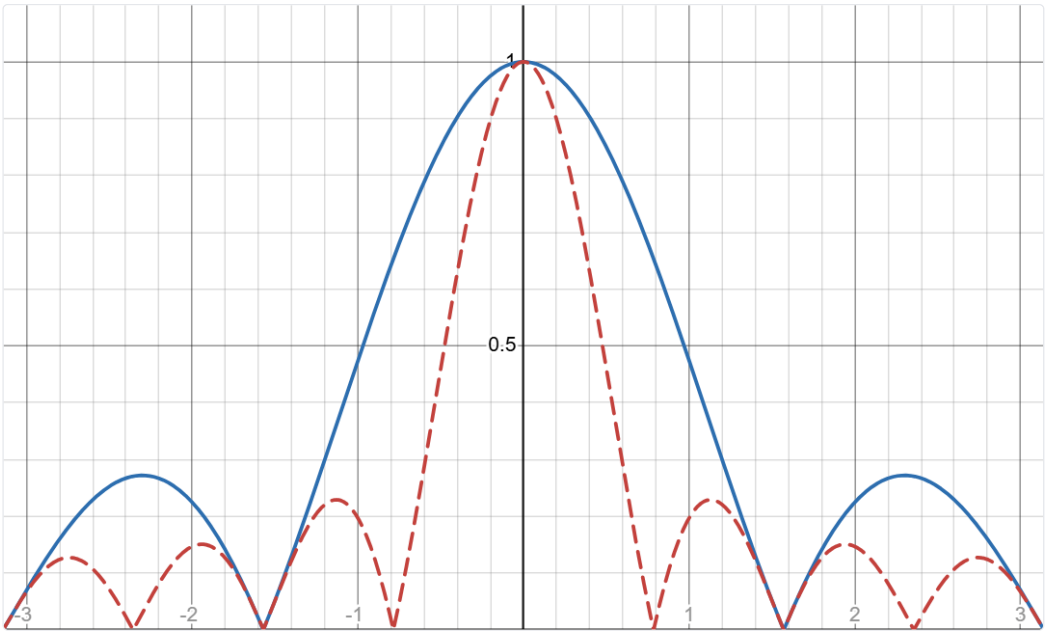
Step 4. Check. Numerically, $|H| = 0$ at 50 Hz, 1.025 at DC and 1.052 at Nyquist (scale by $1/1.025$ if exact unit DC gain is wanted), and the -3 dB notch width is 0.0936 rad $= 14.9$ Hz. The poles have radius 0.95: stable.

Worked Example 17.36 — Moving average and linear phase

For the 4-point moving average of Example 17.29,

$$H(e^{j\omega}) = \frac{1}{4}\,\frac{1 - e^{-j4\omega}}{1 - e^{-j\omega}} = \frac{1}{4}\,\frac{e^{-j2\omega}\,(2j\sin 2\omega)}{e^{-j\omega/2}\,\left(2j\sin\frac{\omega}{2}\right)} = e^{-j3\omega/2}\,\frac{\sin 2\omega}{4\sin(\omega/2)}.$$

The magnitude is the **Dirichlet kernel** $\left|\dfrac{\sin 2\omega}{4\sin(\omega/2)}\right|$, with zeros at $\omega = \pi/2, \pi$; the phase is -1.5ω (plus π jumps where the real factor changes sign) — **linear phase**, a constant group delay of $M/2 = 1.5$ samples, because the coefficients are symmetric.



Magnitude responses of the 4-point (blue) and 8-point (red, dashed) moving averages: crude low-pass filters with nulls at multiples of $2\pi/M$. Doubling the length halves the main-lobe width.

All-pass systems

The geometric picture shows how to build a filter with *flat* magnitude: place each zero at the **conjugate reciprocal** $1/p^*$ of a pole p. Then for $|z| = 1$,

$$H_{\text{ap}}(z) = \frac{z^{-1} - p^*}{1 - pz^{-1}}, \qquad |H_{\text{ap}}(e^{j\omega})| = \frac{|e^{-j\omega} - p^*|}{|1 - pe^{-j\omega}|} = \frac{|e^{-j\omega}|\,|1 - p^* e^{j\omega}|}{|1 - pe^{-j\omega}|} = 1,$$

since the last numerator is the conjugate of the denominator. All-pass sections change only the phase; they are used as phase equalisers, as building blocks of fractional delays, and in the theorem that **every rational stable system factors as (minimum phase) × (all-pass)**: reflecting a zero from outside the circle to its conjugate reciprocal inside leaves $|H|$ unchanged and reduces the phase lag, hence "minimum phase".

Relation to the Laplace transform: $z = e^{sT}$

Suppose $x[n] = x_a(nT)$ are samples of a continuous signal. Represent the sampled signal, as in §16.8, by the impulse train

$$x_s(t) = \sum_n x_a(nT)\,\delta(t - nT).$$

Its Laplace transform, using $\mathcal{L}\{\delta(t - nT)\} = e^{-snT}$, is

$$X_s(s) = \sum_n x[n]\,e^{-snT} = X(z)\big|_{z = e^{sT}}.$$

The s-to-z map

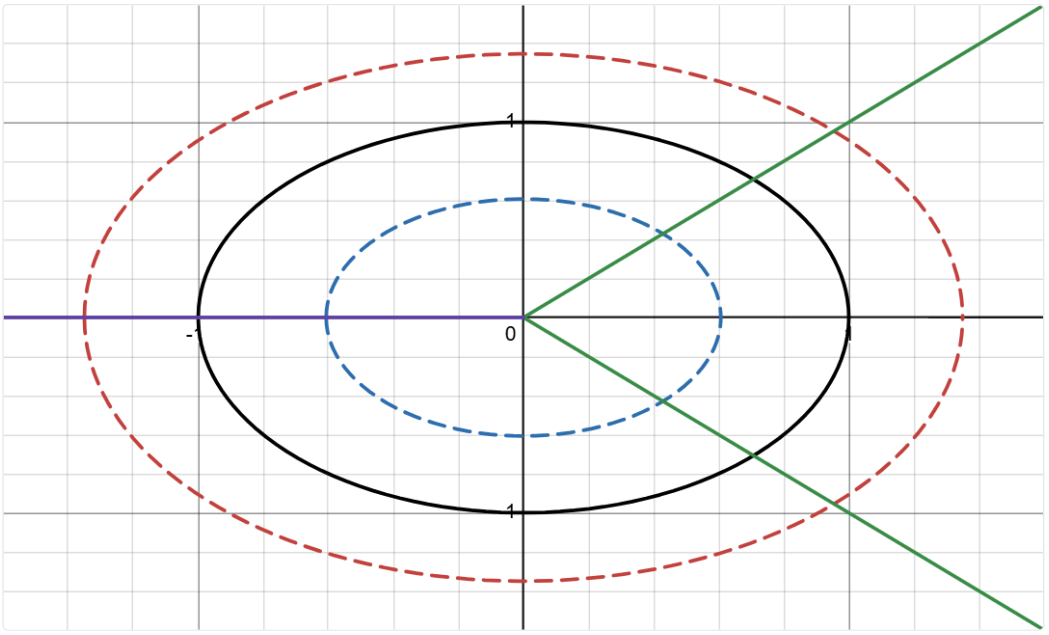
The Z-transform of the samples equals the Laplace transform of the impulse-sampled signal under the substitution

$$z = e^{sT}, \qquad s = \sigma + j\Omega \implies |z| = e^{\sigma T}, \quad \angle z = \Omega T.$$

In particular:

- the imaginary axis $\sigma = 0$ maps onto the unit circle, with digital frequency $\omega = \Omega T$;
- the open left half-plane $\sigma < 0$ maps into the open unit disc $|z| < 1$; the right half-plane maps outside it;
- a continuous pole at $s = -\alpha \pm j\Omega_d$ appears in the sampled signal as a pole at $z = e^{-\alpha T} e^{\pm j\Omega_d T}$;
- the map is **not one-to-one**: s and $s + jk\,2\pi/T$ give the same z. Each horizontal strip of height $\Omega_s = 2\pi/T$ covers the whole z-plane once.

The last statement is aliasing seen from the s-plane: frequencies Ω and $\Omega + k\Omega_s$ are indistinguishable after sampling, and so are poles that differ by $jk\Omega_s$. The **primary strip** $-\pi/T < \Omega \le \pi/T$ is the set of continuous frequencies that sampling represents faithfully, and its edges $\Omega = \pm\pi/T$ map to the negative real axis $z < 0$ — the Nyquist frequency.



Images under $z = e^{sT}$: the $j\Omega$ axis maps to the unit circle (black); the vertical lines $\sigma T = -0.5$ and $\sigma T = +0.3$ map to circles of radius $e^{-0.5} = 0.607$ (blue) and $e^{0.3} = 1.35$ (red); the horizontal lines $\Omega T = \pm\pi/4$ map to rays at $\pm 45^\circ$ (green); $\Omega T = \pm\pi$ maps to the negative real axis (purple).

Useful landmarks of the map: constant damping $\sigma \leftrightarrow$ circles; constant frequency $\Omega \leftrightarrow$ rays; the point $s = 0$ (DC) $\leftrightarrow z = 1$; $s = -\infty \leftrightarrow z = 0$. Fast continuous poles (large αT) crowd towards the origin; slow ones (small αT) crowd towards $z = 1$, which is why high sampling rates push all poles of a digital controller into a small, numerically delicate neighbourhood of $z = 1$.

Worked Example 17.37 — Mapping a damped oscillation

A continuous-time system has poles $s = -200 \pm j1000$ s⁻¹ and is sampled at $f_s = 4$ kHz ($T = 0.25$ ms). Where are the poles of the sampled system, and what are its digital damping and frequency?

Step 1. $\sigma T = -200 \times 0.25 \times 10^{-3} = -0.05$, $\Omega T = 1000 \times 0.25 \times 10^{-3} = 0.25$ rad.

Step 2. $z = e^{-0.05} e^{\pm j0.25} = 0.9512\, e^{\pm j0.25} = 0.9217 \pm j0.2353$.

Step 3. The pole radius $0.951 < 1$ confirms stability (left half-plane → inside the circle). The digital oscillation frequency 0.25 rad/sample corresponds to $0.25\, f_s/2\pi = 159$ Hz $= 1000/2\pi$ Hz, as it must. If the same system were sampled at only $f_s = 150$ Hz, the pole angle $\Omega T = 6.667$ rad would wrap to $6.667 - 2\pi = 0.384$ rad — an alias at 9.2 Hz.

Discretising continuous-time systems

The substitution $z = e^{sT}$ cannot be applied directly to a rational $H(s)$ — it would produce a non-rational $H(z)$. Practical discretisation methods are rational approximations that preserve different features.

Impulse invariance. Sample the analogue impulse response: $h[n] = T\, h_a(nT)$. For $H_a(s) = \sum_k \dfrac{A_k}{s - s_k}$, $h_a(t) = \sum_k A_k e^{s_k t} u(t)$, so

$$H(z) = \sum_k \frac{T A_k}{1 - e^{s_k T} z^{-1}}.$$

Each analogue pole s_k maps exactly to $e^{s_k T}$, but the frequency response is the aliased sum $H(e^{j\omega}) = \sum_k H_a\!\left(j\tfrac{\omega + 2\pi k}{T}\right)$, so the method suits only band-limited (low-pass, band-pass) prototypes.

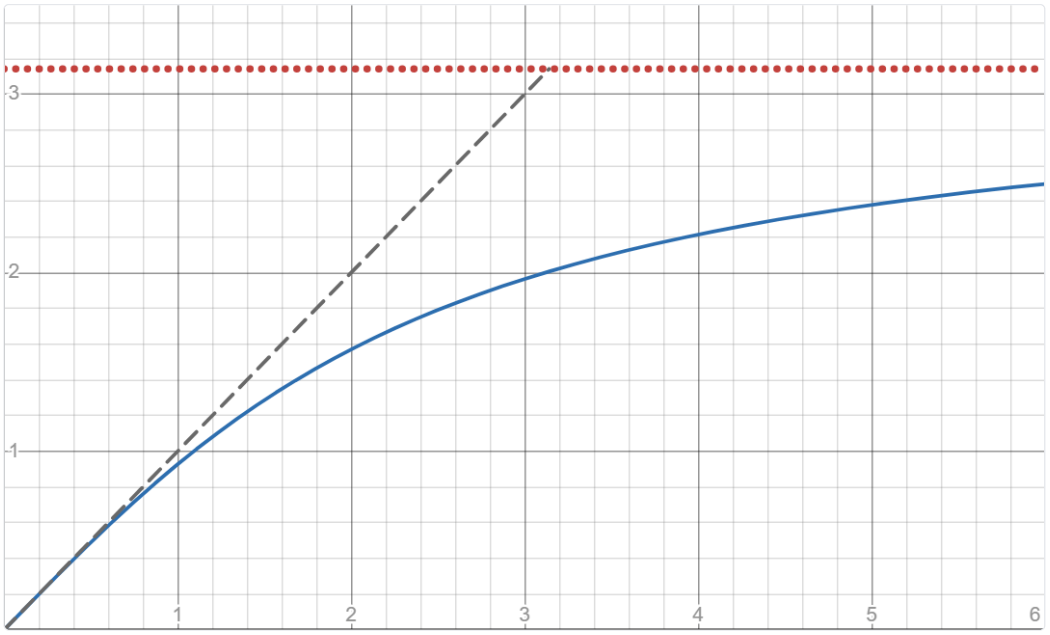
Bilinear transform. Replace s by the first-order rational approximation of $s = \frac{1}{T}\ln z$ (from the trapezoidal rule of integration):

$$s = \frac{2}{T}\,\frac{1 - z^{-1}}{1 + z^{-1}}, \qquad z = \frac{1 + sT/2}{1 - sT/2}.$$

It maps the left half-plane *exactly* onto the unit disc (so stable analogue filters always give stable digital ones) and the whole $j\Omega$ axis onto the unit circle *once* — no aliasing. The price is **frequency warping**: with $z = e^{j\omega}$,

$$s = \frac{2}{T}\,\frac{1 - e^{-j\omega}}{1 + e^{-j\omega}} = j\,\frac{2}{T}\tan\frac{\omega}{2} \quad\Longrightarrow\quad \Omega = \frac{2}{T}\tan\frac{\omega}{2}, \qquad \omega = 2\arctan\frac{\Omega T}{2}.$$

Critical frequencies are therefore **pre-warped**: to place a digital cutoff at ω_c, design the analogue prototype with cutoff $\Omega_c = \frac{2}{T}\tan\frac{\omega_c}{2}$.



Bilinear frequency warping, $\omega = 2\arctan(\Omega T/2)$ (blue) against the ideal $\omega = \Omega T$ (dashed): the two agree at low frequency, but the whole infinite analogue axis is compressed into $0 \le \omega < \pi$ (red dotted line).

Worked Example 17.38 — Impulse invariance versus bilinear for an RC low-pass

Discretise $H_a(s) = \dfrac{1}{1+s\tau}$ with $\tau = 1$ ms and $T = 0.25$ ms.

Impulse invariance. $h_a(t) = \frac{1}{\tau}e^{-t/\tau}u(t)$, so $h[n] = \frac{T}{\tau}\left(e^{-T/\tau}\right)^n u[n]$ and

$$H_{\text{ii}}(z) = \frac{0.25}{1 - 0.7788 z^{-1}}.$$

The pole $e^{-T/\tau} = 0.7788$ is exact (compare the Euler pole 0.75 of Example 17.27), but the DC gain is $0.25/(1-0.7788) = 1.130$, not 1: the analogue response is not band-limited, so aliased spectral copies add to the gain. In practice one rescales by $1/1.130$.

Bilinear. With $\frac{2}{T} = 8000\ \text{s}^{-1}$ and $\tau = 10^{-3}$ s: $1 + s\tau = 1 + 8\,\dfrac{1-z^{-1}}{1+z^{-1}} = \dfrac{9 - 7z^{-1}}{1+z^{-1}}$, so

$$H_{\text{bl}}(z) = \frac{1+z^{-1}}{9-7z^{-1}} = \frac{0.1111\,(1+z^{-1})}{1 - 0.7778 z^{-1}}.$$

DC gain $H(1) = 2/2 = 1$ exactly; $H(-1) = 0$ — the Nyquist frequency is a transmission zero, the image of $s = \infty$.

Worked Example 17.39 — Bilinear design with pre-warping

Design a first-order digital low-pass with 3-dB cutoff $f_c = 1$ kHz at $f_s = 8$ kHz.

Step 1. Digital cutoff. $\omega_c = 2\pi f_c / f_s = \pi/4$.

Step 2. Pre-warp. $\Omega_c = \dfrac{2}{T}\tan\dfrac{\omega_c}{2} = 16000\tan(\pi/8) = 6627$ rad/s (the naive value would be $2\pi \cdot 1000 = 6283$ rad/s). Then $\Omega_c T = 2\tan(\pi/8) = 0.8284$.

Step 3. Prototype $H_a(s) = \dfrac{\Omega_c}{s + \Omega_c}$. Substituting the bilinear map and multiplying through by $T(1+z^{-1})$:

$$H(z) = \frac{\Omega_c T\,(1+z^{-1})}{(2 + \Omega_c T) + (\Omega_c T - 2)z^{-1}} = \frac{0.2929\,(1+z^{-1})}{1 - 0.4142 z^{-1}}.$$

Step 4. Check. $|H(e^{j\pi/4})| = 0.7071$ exactly, i.e. -3 dB at 1 kHz ✓; $H(1) = 0.5858/0.5858 = 1$ ✓. Without pre-warping the cutoff would land at $\omega = 2\arctan(6283/16000) = 0.748$ rad, i.e. 952 Hz.

Common Pitfall — Confusing rad/sample with rad/s

ω (digital, rad/sample, periodic with period 2π) and Ω (analogue, rad/s) differ by the factor T. A digital filter "with cutoff 0.2" is meaningless until f_s is known, and design software variously normalises frequency to π rad/sample, to $f_s/2$ ("1.0 = Nyquist") or to f_s. Always convert with $\omega = 2\pi f/f_s$ and check that $\omega_c < \pi$.

Engineering Note — What the unit-circle picture buys a designer

Every specification of a digital filter translates into a pole–zero picture: stopband frequencies become zeros on the circle, passband peaks become poles near the circle at the corresponding angle, the sharpness of a feature becomes the distance $1-r$, and stability is the single constraint $r<1$. The magnitude plot is then the shadow of these poles and zeros on the circle. Sophisticated design methods (Butterworth, Chebyshev, elliptic prototypes transformed bilinearly; Parks–McClellan FIR design) are simply systematic ways of choosing those positions.

Summary

Concept	Formula
Discrete frequency	$\omega=\Omega T=2\pi f/f_s$ rad/sample; $e^{j(\omega+2\pi)n}=e^{j\omega n}$
Convolution sum	$y[n]=\sum_k x[k]h[n-k]$
Bilateral Z-transform	$X(z)=\sum_{n=-\infty}^{\infty}x[n]z^{-n}$, with ROC
Unilateral Z-transform	$\mathcal{X}(z)=\sum_{n=0}^{\infty}x[n]z^{-n}$
Geometric pairs	$a^nu[n]\leftrightarrow\frac{z}{z-a}$, $\lvert z\rvert>\lvert a\rvert$; $-a^nu[-n-1]\leftrightarrow\frac{z}{z-a}$, $\lvert z\rvert<\lvert a\rvert$
Repeated pole	$na^nu[n]\leftrightarrow\frac{az}{(z-a)^2}$
Damped cosine	$r^n\cos(\omega_0 n)u[n]\leftrightarrow\frac{z(z-r\cos\omega_0)}{z^2-2rz\cos\omega_0+r^2}$
Shift / scaling / multiply by n	$z^{-n_0}X(z)$; $a^nx[n]\leftrightarrow X(z/a)$; $nx[n]\leftrightarrow -zX'(z)$
Reversal / convolution	$x[-n]\leftrightarrow X(1/z)$; $x*h\leftrightarrow XH$
Initial / final value	$x[0]=\lim_{z\to\infty}X$; $x[\infty]=\lim_{z\to1}(z-1)X$
Parseval	$\sum\lvert x[n]\rvert^2=\frac{1}{2\pi}\int_{-\pi}^{\pi}\lvert X(e^{j\omega})\rvert^2d\omega$
ROC shapes	right-sided: $\lvert z\rvert>r_1$; left-sided: $\lvert z\rvert<r_2$; two-sided: $r_1<\lvert z\rvert<r_2$; finite: all z (except $0,\infty$)
Inversion integral	$x[n]=\frac{1}{2\pi j}\oint_C X(z)z^{n-1}dz=\sum\text{Res}\left[Xz^{n-1}\right]$
Partial fractions	expand $X(z)/z$; $A_i=(z-p_i)X(z)/z\,\lvert_{p_i}$
Unilateral delay	$y[n-1]\leftrightarrow z^{-1}\mathcal{Y}+y[-1]$; $y[n-2]\leftrightarrow z^{-2}\mathcal{Y}+z^{-1}y[-1]+y[-2]$
Unilateral advance	$y[n+1]\leftrightarrow z\mathcal{Y}-zy[0]$
System function	$H(z)=\dfrac{\sum b_kz^{-k}}{\sum a_kz^{-k}}=\dfrac{Y}{X}$
Feedback	$\dfrac{G}{1+GF}$
Stability (causal)	all poles $\lvert p_k\rvert<1\iff\sum\lvert h[n]\rvert<\infty$
Second-order test	$z^2+a_1z+a_2$: $\lvert a_2\rvert<1$, $\lvert a_1\rvert<1+a_2$
Frequency response	$H(e^{j\omega})$; $\cos\omega_0n\mapsto\lvert H\rvert\cos(\omega_0n+\angle H)$
Pole bandwidth	$\Delta\omega_{3\,\text{dB}}\approx2(1-r)$
s–z map	$z=e^{sT}$: LHP $\to\lvert z\rvert<1$, $j\Omega$ axis $\to$ unit circle
Impulse invariance	$\frac{A}{s-s_k}\to\frac{TA}{1-e^{s_kT}z^{-1}}$
Bilinear transform	$s=\frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}$, $\Omega=\frac{2}{T}\tan\frac{\omega}{2}$

Further reading

- A. V. Oppenheim & R. W. Schafer, *Discrete-Time Signal Processing*, 3rd ed., Pearson — Chs. 2–5: sequences, the Z-transform, ROC, system functions, structures and the bilinear transform; the standard reference.
- A. V. Oppenheim, A. S. Willsky & S. H. Nawab, *Signals and Systems*, 2nd ed., Prentice Hall — Ch. 10 develops the Z-transform in parallel with the Laplace transform of Ch. 9.
- J. G. Proakis & D. G. Manolakis, *Digital Signal Processing: Principles, Algorithms and Applications*, 4th ed., Pearson — many worked inversions and the unilateral transform for difference equations.
- K. Ogata, *Discrete-Time Control Systems*, 2nd ed., Prentice Hall — the s–z mapping, Jury stability test and sampled-data feedback loops.

Chapter 18 — Complex Analysis

> **Prerequisites:** Ch. 2 (complex numbers), Ch. 5 (integral calculus), Ch. 6 (multivariable calculus), Ch. 7 (vector calculus: Green's theorem), Ch. 14 (Laplace transforms), Ch. 17 (Z-transforms) **Learning outcomes:** After this chapter you can
>
> - treat a function $w = f(z)$ as a mapping of the plane, compute its real and imaginary parts, and picture what it does to grid lines (conformal mapping);
> - test a function for analyticity with the Cauchy–Riemann equations and construct harmonic conjugates;
> - parametrize contours, evaluate contour integrals, and bound them with the ML inequality;
> - apply Cauchy's integral theorem and integral formula, including contour deformation;
> - expand functions in Taylor and Laurent series on discs and annuli, and classify isolated singularities;
> - compute residues and use the residue theorem to evaluate real integrals (rational, trigonometric, Fourier-type via Jordan's lemma), to invert Laplace and Z-transforms, and to count zeros and poles with the argument principle — the mathematics behind the Nyquist stability criterion.

Every transform in Part V lives in the complex plane. The Laplace variable $s = \sigma + j\omega$ of Ch. 14, the Fourier frequency on the line $s = j\omega$ of Ch. 16, and the z of Ch. 17 are complex numbers, and the functions we manipulate — transfer functions $H(s)$, spectra $X(\omega)$, system functions $H(z)$ — are complex functions of a complex variable. Until now we have treated them algebraically: partial fractions, tables, pole–zero plots. This chapter supplies the analysis underneath. It explains *why* partial-fraction coefficients are what they are (they are residues), *why* the region of convergence of a Z-transform is an annulus (it is the annulus of a Laurent series), *why* the inverse Laplace transform is an integral along a vertical line (the Bromwich integral), and *why* counting encirclements of -1 by a Nyquist plot reveals closed-loop poles (the argument principle).

The central surprise of complex analysis is how rigid differentiable complex functions are. A real function can be differentiable once and nowhere twice; a complex function that is differentiable once in a region is differentiable infinitely often, equals its Taylor series, and is completely determined inside a closed curve by its values on the curve. That rigidity is what makes contour integration such a powerful computational tool.

Throughout, $z = x + jy = re^{j\theta}$ denotes the independent variable and $w = f(z) = u(x,y) + jv(x,y)$ the dependent one.

18.1 Complex functions

Why an engineer needs complex functions

A transfer function $H(s) = 1/(s^2 + 2\zeta\omega_n s + \omega_n^2)$ assigns to every complex frequency s a complex number: a gain and a phase. A Bode plot is a picture of H restricted to the imaginary axis; a root locus is a picture of where $1 + KH(s) = 0$; a Nyquist plot is the *image* of the imaginary axis under H. All three are statements about a function from $\mathbb{C}$ to $\mathbb{C}$. To use them fluently we must understand such functions as objects in their own right — in particular as *mappings* that carry curves in one plane to curves in another.

Complex function

Let $D \subseteq \mathbb{C}$. A **complex function** $f : D \to \mathbb{C}$ assigns to each $z \in D$ a complex number $w = f(z)$. Writing $z = x + jy$,

$$f(z) = u(x,y) + j\,v(x,y),$$

where $u = \operatorname{Re}\{f\}$ and $v = \operatorname{Im}\{f\}$ are real-valued functions of two real variables. In polar form $z = re^{j\theta}$ we may also write $f = u(r,\theta) + jv(r,\theta)$.

So a complex function is *a pair of real functions of two variables*. That viewpoint is honest but hides the structure; the payoff of complex analysis comes from treating z as a single variable.

Domains and regions. We will need a few words about sets in the plane.

- An **open disc** $|z - z_0| < \rho$ is a *neighbourhood* of z_0; the **punctured disc** $0 < |z - z_0| < \rho$ omits the centre.
- A set is **open** if every point has a neighbourhood inside the set. It is **connected** if any two points can be joined by a polygonal path inside it.
- A **domain** is an open connected set. A **region** is a domain possibly with some or all of its boundary points.
- A domain is **simply connected** if every closed curve in it can be shrunk continuously to a point without leaving it — informally, it has no holes. The disc $|z| < 1$ is simply connected; the annulus $1 < |z| < 2$ and the punctured plane $z \neq 0$ are not.

These distinctions are not pedantry: Cauchy's theorem (Section 18.5) holds on simply connected domains, and the Laurent series (Section 18.6) is the tool for annuli.

Elementary functions

Polynomials and rational functions. $P(z) = a_0 + a_1 z + \cdots + a_n z^n$ is defined everywhere. A rational function $R(z) = P(z)/Q(z)$ is defined except at zeros of Q — the *poles* of every transfer function you have met.

The exponential. Defined by

$$e^z = e^{x+jy} = e^x(\cos y + j \sin y),$$

so $|e^z| = e^x$ and $\angle e^z = y$. It agrees with the real exponential on the real axis and with Euler's formula on the imaginary axis. Key properties:

- $e^{z_1+z_2} = e^{z_1}e^{z_2}$, and $e^z \neq 0$ for every z;
- e^z is **periodic with period** $2\pi j$: $e^{z+2\pi j} = e^z$. The horizontal strip $-\pi < y \leq \pi$ already produces every nonzero value once.

Trigonometric and hyperbolic functions. Defined from the exponential,

$$\cos z = \frac{e^{jz}+e^{-jz}}{2}, \qquad \sin z = \frac{e^{jz}-e^{-jz}}{2j}, \qquad \cosh z = \frac{e^{z}+e^{-z}}{2}, \qquad \sinh z = \frac{e^{z}-e^{-z}}{2}.$$

Hence $\cos(jy) = \cosh y$ and $\sin(jy) = j\sinh y$, and

$$\cos(x+jy) = \cos x\cosh y - j\sin x\sinh y, \qquad \sin(x+jy) = \sin x\cosh y + j\cos x\sinh y.$$

A striking consequence: $|\sin z|^2 = \sin^2 x + \sinh^2 y$, which grows without bound as $|y|\to\infty$. **Complex sine and cosine are not bounded** — unlike their real versions. (Liouville's theorem in Section 18.5 will say no non-constant entire function can be bounded.)

The logarithm — the first multivalued function. We want $w = \log z$ to solve $e^w = z$. Write $z = re^{j\theta}$ and $w = u + jv$: then $e^u e^{jv} = re^{j\theta}$ gives $u = \ln r$ and $v = \theta + 2\pi k$. So

$$\log z = \ln|z| + j\arg z = \ln r + j(\theta + 2\pi k), \qquad k\in\mathbb{Z}.$$

Every nonzero z has infinitely many logarithms, differing by integer multiples of $2\pi j$. To get a genuine (single-valued) function we must choose.

Principal logarithm and branch cut

The **principal argument** $\operatorname{Arg} z \in (-\pi,\pi]$ and the **principal logarithm**

$$\operatorname{Log} z = \ln|z| + j\operatorname{Arg} z, \qquad z\neq 0,$$

define a single-valued function. It is continuous everywhere except on the negative real axis $\{x\le 0,\ y=0\}$, where Arg jumps by 2π. That excluded ray is a **branch cut**; $z=0$, around which the values wind, is a **branch point**. Any other choice of a ray from 0 and a 2π-wide interval of angles gives another **branch** of log.

Complex powers. For complex α, define $z^\alpha = e^{\alpha\log z}$; the **principal value** uses Log. If $\alpha = n$ is an integer this is single-valued; if $\alpha = p/q$ is rational in lowest terms there are exactly q distinct values (the q-th roots of Ch. 2); otherwise there are infinitely many. For example $z^{1/2}$ has two branches, $\pm\sqrt{r}\,e^{j\theta/2}$, which swap as z circles the origin.

Worked Example 18.1 — real and imaginary parts

Find u and v for (a) $f(z)=z^2$, (b) $f(z)=1/z$, (c) $f(z)=e^z$, and evaluate each at $z = 1+j$.

(a) $(x+jy)^2 = x^2 - y^2 + j\,2xy$, so $u = x^2-y^2$, $v = 2xy$. At $1+j$: $u=0$, $v=2$, i.e. $f = 2j$.

(b) Multiply by the conjugate:

$$\frac{1}{z} = \frac{\bar z}{|z|^2} = \frac{x-jy}{x^2+y^2}, \qquad u = \frac{x}{x^2+y^2}, \quad v = \frac{-y}{x^2+y^2}.$$

At $1+j$: $u = \tfrac12$, $v = -\tfrac12$, i.e. $f = \tfrac12 - \tfrac12 j$. Check: $(1+j)(\tfrac12 - \tfrac12 j) = \tfrac12 + \tfrac12 = 1$. ✓

(c) $u = e^x\cos y$, $v = e^x\sin y$. At $1+j$: $e\cos 1 = 1.4687$, $e\sin 1 = 2.2874$.

Worked Example 18.2 — logarithms and powers

Find (a) $\operatorname{Log}(-1-j)$, (b) all values of $\log(-1)$, (c) the principal value of j^j.

(a) $|-1-j| = \sqrt{2}$ and $-1-j$ lies in the third quadrant, so $\operatorname{Arg}(-1-j) = -3\pi/4$ (not $5\pi/4$, which lies outside $(-\pi,\pi]$). Hence $\operatorname{Log}(-1-j) = \tfrac12\ln 2 - j\tfrac{3\pi}{4} = 0.3466 - 2.3562j$.

(b) $|-1|=1$, $\arg(-1) = \pi + 2\pi k$, so $\log(-1) = j(2k+1)\pi$. The principal value is $j\pi$ — the logarithm of a negative real number is perfectly well defined in $\mathbb{C}$.

(c) $j^j = e^{j\operatorname{Log} j} = e^{j\cdot j\pi/2} = e^{-\pi/2} = 0.20788$. A complex number raised to a complex power can be real. The other values are $e^{-\pi/2 - 2\pi k}$.

Common Pitfall — "laws" that fail for multivalued functions

$\operatorname{Log}(z_1z_2) = \operatorname{Log}z_1 + \operatorname{Log}z_2$ is **false** in general. With $z_1 = z_2 = -1$: the left side is $\operatorname{Log}1 = 0$, the right side is $2\pi j$. The identity holds only modulo $2\pi j$. Likewise $(z_1z_2)^{1/2} \neq z_1^{1/2}z_2^{1/2}$ for principal branches in general, and $\sqrt{z^2}\neq z$. In software, `log`, `sqrt` and `angle` return principal values, so a phase computed as `angle(H)` jumps by 2π when the true phase crosses $\pm\pi$ — the reason phase plots must be *unwrapped*.

Functions as mappings

A real function $y = f(x)$ is pictured by its graph in the (x,y) plane. A complex function would need four real dimensions for its graph, so instead we draw **two planes**: the z-plane (coordinates x, y) and the w-plane (coordinates u, v), and we watch where f sends curves. A good test pattern is the rectangular grid $x =$ const, $y =$ const, or the polar grid $r =$ const, $\theta =$ const.

The map $w = z^2$. In polar form $w = r^2e^{j2\theta}$: moduli are squared and angles doubled. The first quadrant $0\le\theta\le\pi/2$ opens out to the whole upper half-plane. On the rectangular grid:

- the vertical line $x = a$ ($a\neq 0$) becomes $u = a^2 - y^2$, $v = 2ay$. Eliminating $y = v/(2a)$:

$$u = a^2 - \frac{v^2}{4a^2},$$

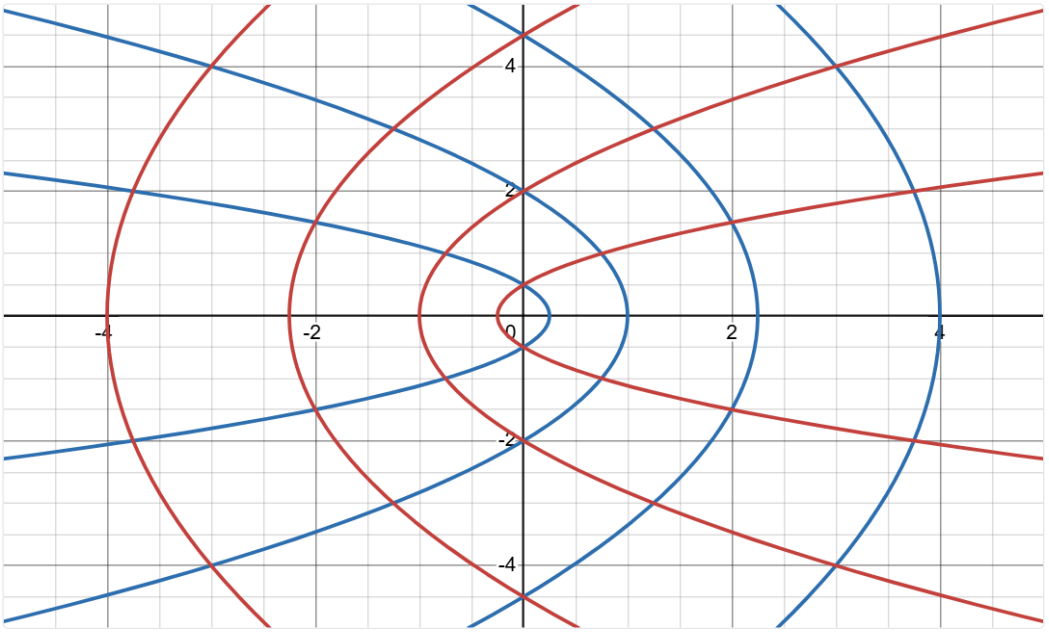
a parabola opening to the left with focus at the origin;

- the horizontal line $y = b$ ($b \neq 0$) becomes $u = x^2 - b^2$, $v = 2bx$, i.e.

$$u = \frac{v^2}{4b^2} - b^2,$$

a parabola opening to the right, also with focus at the origin.

The two families cross at right angles — exactly as the original grid lines did. That is not a coincidence: it is **conformality**, developed in Section 18.3.



Images under $w = z^2$ of the vertical lines $x = 0.5, 1, 1.5, 2$ (blue, left-opening parabolas) and horizontal lines $y = 0.5, 1, 1.5, 2$ (red, right-opening parabolas), drawn in the $w = u + jv$ plane; the two families of confocal parabolas meet at right angles.

The map $w = 1/z$ (inversion). In polar form $w = (1/r)e^{-j\theta}$: the modulus is inverted and the angle negated (reflection in the real axis). Points inside the unit circle go outside, and vice versa; the origin and the "point at infinity" swap. Using $u = x/(x^2 + y^2)$, $v = -y/(x^2 + y^2)$ and the inverse relations $x = u/(u^2 + v^2)$, $y = -v/(u^2 + v^2)$:

- the vertical line $x = a$ becomes $u/(u^2 + v^2) = a$, i.e.

$$\left(u - \frac{1}{2a}\right)^2 + v^2 = \frac{1}{4a^2},$$

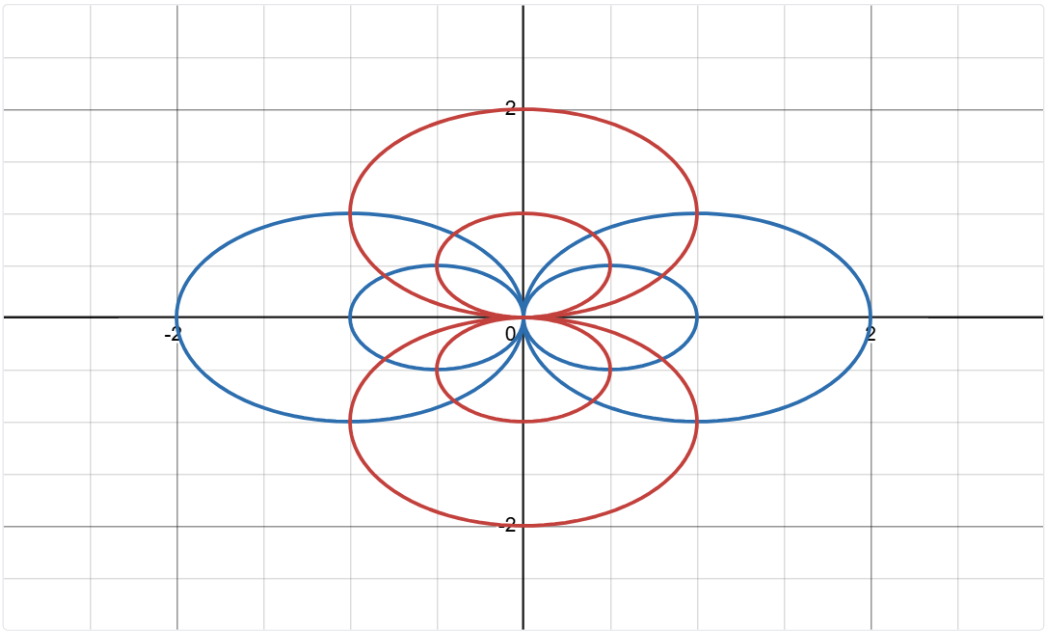
a circle through the origin with centre on the real axis;

- the horizontal line $y = b$ becomes

$$u^2 + \left(v + \frac{1}{2b}\right)^2 = \frac{1}{4b^2},$$

a circle through the origin with centre on the imaginary axis.

Lines map to circles through the origin (a line is a "circle through ∞", and $\infty \mapsto 0$). In general, **inversion maps circles-and-lines to circles-and-lines**, a property shared by all **Möbius (bilinear) transformations** $w = (az + b)/(cz + d)$, $ad - bc \neq 0$. The Smith chart is the Möbius map $\Gamma = (z_L - 1)/(z_L + 1)$ applied to the grid of normalized resistance and reactance; its circles are these circles.



Images under $w = 1/z$ of the lines $x = \pm 0.5, \pm 1$ (blue circles centred on the real axis) and $y = \pm 0.5, \pm 1$ (red circles centred on the imaginary axis); every circle passes through $w = 0$, the image of $z = \infty$, and blue meets red at right angles.

The map $w = e^z$. With $u + jv = e^x e^{jy}$: the vertical line $x = a$ maps to the circle $|w| = e^a$, traversed once each time y advances by 2π; the horizontal line $y = b$ maps to the ray $\angle w = b$. The exponential turns the Cartesian grid into the polar grid. Its inverse, Log, does the reverse: it straightens circles and rays into vertical and horizontal lines. That is exactly why the s-plane and z-plane of sampled systems are related by $z = e^{sT}$ (Ch. 17): the imaginary axis $s = j\omega$ wraps onto the unit circle, the left half-plane maps into the unit disc, and the strip $-\pi/T < \omega \le \pi/T$ is the primary strip that maps onto the whole z-plane once — the complex-analysis statement of aliasing.

Worked Example 18.3 — image of a region under e^z and z^2

(a) Find the image of the rectangle $0 \le x \le 1, 0 \le y \le \pi/2$ under $w = e^z$. (b) Find the image of the half-strip $0 < x < 1, y > 0$ under $w = z^2$.

(a) $|w| = e^x$ ranges over $[1, e]$ and $\angle w = y$ over $[0, \pi/2]$. The image is the quarter-annulus $1 \le |w| \le e, 0 \le \angle w \le \pi/2$. Its area is $\frac{1}{4}\pi(e^2 - 1) = 5.0180$, compared with $\pi/2 = 1.5708$ for the rectangle. The local area magnification is $|f'(z)|^2 = e^{2x}$, and indeed $\int_0^1 \int_0^{\pi/2} e^{2x}\, dy\, dx = \frac{\pi}{2} \cdot \frac{e^2-1}{2} = 5.0180$. ✓

(b) The boundary $x = 0, y > 0$ maps to $u = -y^2 \le 0, v = 0$: the negative real axis. The boundary $x = 1$ maps to the parabola $u = 1 - v^2/4$ with $v = 2y > 0$, the upper half of the left-opening parabola. The boundary segment $y = 0, 0 < x < 1$ maps to $0 < u < 1, v = 0$. The image is the region in the upper half-plane lying inside (to the left of) the parabola $u = 1 - v^2/4$ — test the interior point $z = 0.5 + 0.5j$: $w = 0.5j$, and $0 < 1 - 0.25/4$. ✓

Engineering Note — what a Nyquist plot really is

A Nyquist plot draws the curve $\{L(j\omega) : -\infty < \omega < \infty\}$ in the w-plane: the **image of the imaginary axis** under the loop transfer function $L(s)$. Similarly, a polar plot of an impedance $Z(j\omega)$ is the image of the frequency axis. Everything in this section — lines becoming circles, grids staying orthogonal — is what you are seeing in those plots. Section 18.9 uses the argument principle to read stability from such an image.

18.2 Analyticity

Why differentiability is special in the plane

In one real variable, x can approach x_0 only from the left or the right. In the complex plane, z can approach z_0 along infinitely many directions and along spirals. Demanding that a difference quotient have the *same* limit along every one of these paths is a far stronger condition than real differentiability, and the functions that meet it — analytic functions — inherit remarkable properties. The transfer functions, impedances and transforms of circuit theory are analytic away from their poles; this is why their behaviour on the $j\omega$-axis is tied so tightly to their pole–zero pattern.

Limits and continuity

Limit and continuity

$\lim_{z \to z_0} f(z) = L$ means: for every $\varepsilon > 0$ there is $\delta > 0$ such that $|f(z) - L| < \varepsilon$ whenever $0 < |z - z_0| < \delta$. The function is **continuous** at z_0 if $f(z_0)$ is defined and $\lim_{z \to z_0} f(z) = f(z_0)$.

Because $|z - z_0| < \delta$ is a disc, the limit must be the same however z approaches z_0. Writing $f = u + jv$ and $L = a + jb$, the complex limit exists if and only if $u \to a$ and $v \to b$ as two-variable limits. Sums, products, quotients (with nonzero denominator) and compositions of continuous functions are continuous; polynomials are continuous everywhere, rational functions wherever the denominator is nonzero.

Limits involving infinity. We say $f(z) \to \infty$ as $z \to z_0$ if $|f(z)| \to \infty$, regardless of direction. Adding a single point ∞ to $\mathbb{C}$ gives the **extended complex plane** (the Riemann sphere). Behaviour "at infinity" is studied by substituting $z = 1/\zeta$ and examining $\zeta \to 0$: for instance $H(s) = 1/(s+1)$ has $H(1/\zeta) = \zeta/(1+\zeta) \to 0$, a zero at infinity — the "strictly proper" property of transfer functions.

Worked Example 18.4 — a limit that does not exist

Does $\lim_{z\to 0} \dfrac{\bar{z}}{z}$ exist?

Along the real axis, $z = x$: $\bar{z}/z = x/x = 1$. Along the imaginary axis, $z = jy$: $\bar{z}/z = -jy/(jy) = -1$. Along the line $y = x$, $z = x(1+j)$: $\bar{z}/z = (1-j)/(1+j) = -j$. In polar form $\bar{z}/z = e^{-2j\theta}$, which takes every value on the unit circle depending on the direction of approach. The limit does not exist.

The complex derivative

Complex derivative

f is **differentiable** at z_0 if

$$f'(z_0) = \lim_{\Delta z \to 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}$$

exists (as a complex limit, independent of how $\Delta z \to 0$).

The algebra is identical to the real case, so all the familiar rules carry over: linearity, the product rule, the quotient rule, the chain rule, and $\dfrac{d}{dz} z^n = n z^{n-1}$ for integer n (with $z \neq 0$ when $n < 0$). For example, from $(z+\Delta z)^2 - z^2 = 2z\Delta z + (\Delta z)^2$ we get $(z^2)' = \lim(2z + \Delta z) = 2z$, independent of direction. Likewise $(e^z)' = e^z$, $(\sin z)' = \cos z$, $(\cos z)' = -\sin z$, and on its cut plane $(\operatorname{Log} z)' = 1/z$.

Differentiable implies continuous

If $f'(z_0)$ exists then f is continuous at z_0, since $f(z_0+\Delta z) - f(z_0) = \dfrac{f(z_0+\Delta z) - f(z_0)}{\Delta z} \cdot \Delta z \to f'(z_0) \cdot 0 = 0$.

The converse fails dramatically: there are continuous functions, like $\bar{z}$, that are differentiable nowhere.

Worked Example 18.5 — $\bar{z}$ and $|z|^2$

(a) Show $f(z) = \bar{z}$ is nowhere differentiable. (b) Show $g(z) = |z|^2$ is differentiable only at $z = 0$.

(a) $\dfrac{\overline{z+\Delta z} - \bar{z}}{\Delta z} = \dfrac{\overline{\Delta z}}{\Delta z}$, which by Example 18.4 has no limit as $\Delta z \to 0$. So $\bar{z}$ — a perfectly smooth map of the plane (a reflection) — has no complex derivative anywhere.

(b) $|z+\Delta z|^2 - |z|^2 = (z+\Delta z)(\bar{z} + \overline{\Delta z}) - z\bar{z} = \bar{z}\Delta z + z\overline{\Delta z} + \Delta z \overline{\Delta z}$. Dividing,

$$\frac{\Delta g}{\Delta z} = \bar{z} + z\frac{\overline{\Delta z}}{\Delta z} + \overline{\Delta z}.$$

The last term $\to 0$. The middle term has no limit unless $z = 0$. So $g'(0) = 0$ and $g'(z)$ does not exist for $z \neq 0$. Note that $g = x^2 + y^2$ is infinitely differentiable as a real function of (x, y) — real smoothness is not enough.

The lesson: a function that genuinely depends on $\bar{z}$ is not complex-differentiable. Analytic functions are, informally, functions of z alone. (One can make this precise with the Wirtinger derivative $\partial f/\partial \bar{z} = \frac{1}{2}(f_x + j f_y)$; the Cauchy–Riemann equations of Section 18.3 say exactly that it vanishes.)

Analytic functions

Differentiability at a single point, as in $|z|^2$ at 0, is a curiosity. What matters is differentiability throughout a neighbourhood.

Analytic (holomorphic) function

f is **analytic at** z_0 if it is differentiable at every point of some neighbourhood $|z - z_0| < \rho$. It is **analytic in a domain** D if it is analytic at every point of D. A function analytic in the whole finite plane is **entire**. A point where f fails to be analytic, but every neighbourhood of which contains points of analyticity, is a **singular point** (singularity) of f.

Examples:

- Polynomials, e^z, $\sin z$, $\cos z$, $\sinh z$, $\cosh z$ are entire.
- A rational function P/Q (in lowest terms) is analytic except at the zeros of Q, which are its singular points (poles).
- $\operatorname{Log} z$ is analytic in the plane cut along the negative real axis, $\mathbb{C} \setminus (-\infty, 0]$. Points on the cut are singular for this branch, though another branch is analytic there; $z = 0$ is singular for every branch.
- $\tan z = \sin z / \cos z$ is analytic except where $\cos z = 0$; since $|\cos z|^2 = \cos^2 x + \sinh^2 y$, these are exactly $z = \pi/2 + k\pi$ on the real axis — the complex extension creates no new singularities.
- $\bar{z}$, $\operatorname{Re} z$, $|z|$ are analytic nowhere.

Algebra of analytic functions

Sums, products and compositions of analytic functions are analytic; a quotient f/g is analytic wherever $g \neq 0$. If f is analytic and $f'(z) = 0$ throughout a domain, f is constant there.

Worked Example 18.6 — locating singular points

Where is $f(z) = \dfrac{z^2+1}{z^3+4z}$ analytic, and what is $f'(z)$?

The denominator is $z(z^2+4) = z(z-2j)(z+2j)$. The numerator $(z-j)(z+j)$ shares no factor with it, so f is analytic on $\mathbb{C} \setminus \{0, 2j, -2j\}$. By the quotient rule,

$$f'(z) = \frac{2z(z^3+4z) - (z^2+1)(3z^2+4)}{(z^3+4z)^2} = \frac{-z^4+z^2-4}{(z^3+4z)^2}.$$

Check at $z = 1$: the formula gives $-4/25 = -0.16$; a central difference quotient $[f(1.0001) - f(0.9999)]/0.0002$ also gives -0.16000. Because f is analytic at 1, the difference quotient taken in the imaginary direction, $[f(1+0.0001j) - f(1-0.0001j)]/(0.0002j)$, gives the same value -0.16000. ✓

Worked Example 18.7 — a real function explained by complex singularities

The real function $g(x) = 1/(1+x^2)$ is infinitely differentiable on all of $\mathbb{R}$, yet its Maclaurin series $1 - x^2 + x^4 - \dots$ converges only for $|x| < 1$. Why?

Nothing on the real axis explains the radius 1. Extend to $g(z) = 1/(1+z^2) = 1/[(z-j)(z+j)]$: it has singular points at $z = \pm j$, at distance 1 from the origin. We will prove in Section 18.6 that a Taylor series about z_0 converges in the largest disc about z_0 that avoids all singularities — here $|z| < 1$, whose intersection with the real line is $(-1, 1)$. The real series "feels" poles it cannot see. Expanding instead about $x_0 = 1$, the nearest singularities are at distance $|1 - j| = \sqrt{2}$, so that series converges for $|x - 1| < \sqrt{2} = 1.414$.

What analyticity buys

The results proved later in the chapter, previewed here, show how much the single hypothesis "analytic in D" delivers:

1. **Harmonic parts** (Section 18.3): u and v satisfy Laplace's equation $\nabla^2 u = \nabla^2 v = 0$.
2. **Conformality** (Section 18.3): wherever $f' \neq 0$, f preserves angles between curves.
3. **Path independence** (Section 18.5): $\oint_C f\,dz = 0$ around any closed curve in a simply connected D.
4. **Infinite differentiability** (Section 18.5): f' is analytic too, hence f'', f''', … all exist.
5. **Power series** (Section 18.6): f equals its Taylor series in the largest disc about z_0 contained in D.
6. **Identity theorem**: if two functions analytic in D agree on a segment of curve (or any set with a limit point in D), they agree on all of D. This underlies *analytic continuation*: the Laplace transform $1/(s+a)$, computed as an integral only for $\operatorname{Re} s > -a$, has a unique analytic extension to $\mathbb{C} \setminus \{-a\}$, and that is the function whose "poles" we speak of.

Engineering Note — causality, stability and analyticity

The Laplace transform $H(s) = \int_0^\infty h(t)e^{-st}dt$ of a causal impulse response is analytic throughout its region of convergence, a right half-plane $\operatorname{Re} s > \sigma_0$ (Ch. 14). A stable causal system has $\sigma_0 < 0$, so H is analytic on and to the right of the $j\omega$-axis. Analyticity in the right half-plane is therefore the mathematical signature of a *causal, stable* system, and it is what forces relationships between gain and phase (Bode's gain–phase relation, the Kramers–Kronig relations): you cannot specify $|H(j\omega)|$ and $\angle H(j\omega)$ independently.

Common Pitfall — "analytic at a point"

A function can be differentiable at a point without being analytic there ($|z|^2$ at 0). Analyticity is a neighbourhood property. When a theorem says "analytic inside and on C", it means analytic on some open set containing the curve and its interior.

18.3 Cauchy–Riemann equations

Motivation

The definition of the derivative is hard to check directly. We want a practical test in terms of u and v — something we can compute with ordinary partial derivatives. The test, the Cauchy–Riemann equations, does more than decide differentiability: it reveals that the real and imaginary parts of an analytic function are *harmonic* (they solve Laplace's equation, the equation of electrostatic potential, steady heat flow and ideal fluid flow) and that analytic maps preserve angles. Those two facts turn analytic functions into a toolkit for two-dimensional field problems.

Derivation

Suppose $f = u + jv$ is differentiable at $z_0 = x_0 + jy_0$. The limit defining $f'(z_0)$ must be the same along every path, in particular along the two coordinate directions.

Horizontal approach, $\Delta z = \Delta x$ real:

$$f'(z_0) = \lim_{\Delta x \to 0} \frac{u(x_0+\Delta x, y_0) - u(x_0, y_0)}{\Delta x} + j\frac{v(x_0+\Delta x, y_0) - v(x_0, y_0)}{\Delta x} = u_x + jv_x.$$

Vertical approach, $\Delta z = j\Delta y$:

$$f'(z_0) = \lim_{\Delta y \to 0} \frac{[u(x_0, y_0+\Delta y) - u] + j[v(x_0, y_0+\Delta y) - v]}{j\Delta y} = \frac{1}{j}(u_y + jv_y) = v_y - ju_y.$$

Equating real and imaginary parts of the two expressions gives the equations.

Cauchy–Riemann equations (necessary condition)

If $f = u + jv$ is differentiable at z_0, then the first partial derivatives of u and v exist at (x_0, y_0) and satisfy

$$u_x = v_y, \qquad u_y = -v_x.$$

Moreover $f'(z_0) = u_x + jv_x = v_y - ju_y$.

The equations are necessary but, on their own, not quite sufficient (there are pathological functions satisfying them at a point without being differentiable). A mild smoothness hypothesis closes the gap.

Sufficient condition

Let u and v have continuous first partial derivatives in a neighbourhood of z_0, and let them satisfy the Cauchy–Riemann equations at z_0. Then $f'(z_0)$ exists. If this holds at every point of a domain D, f is analytic in D.

Proof sketch. Continuity of the partials makes u and v differentiable as real functions, so $\Delta u = u_x\Delta x + u_y\Delta y + \varepsilon_1|\Delta z|$ and $\Delta v = v_x\Delta x + v_y\Delta y + \varepsilon_2|\Delta z|$ with $\varepsilon_{1,2} \to 0$. Substituting $u_y = -v_x$, $v_y = u_x$:

$$\Delta f = \Delta u + j\Delta v = (u_x + jv_x)(\Delta x + j\Delta y) + (\varepsilon_1 + j\varepsilon_2)|\Delta z|,$$

so $\Delta f/\Delta z \to u_x + jv_x$ because $|\Delta z|/\Delta z$ has modulus 1. $\square$

Matrix viewpoint. As a map of $\mathbb{R}^2$, f has Jacobian $\begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix}$. Under Cauchy–Riemann it becomes $\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$ with $a = u_x$, $b = v_x$ — a rotation by $\angle f'$ times a scaling by $|f'|$. That is precisely the matrix of multiplication by the complex number $f' = a + jb$. A complex-differentiable function is one whose local linear approximation is *multiplication by a complex number*, not an arbitrary 2×2 matrix. The Jacobian determinant is $a^2 + b^2 = |f'|^2$, the local area magnification used in Example 18.3.

Polar form. With $z = re^{j\theta}$ and $u(r,\theta)$, $v(r,\theta)$, the chain rule converts the equations to

$$u_r = \frac{1}{r}v_\theta, \qquad v_r = -\frac{1}{r}u_\theta, \qquad f'(z) = e^{-j\theta}(u_r + jv_r), \qquad r > 0.$$

These are convenient for functions built from r and θ, such as powers and logarithms.

Worked Example 18.8 — applying the test

Test (a) $f = z^2$, (b) $f = e^z$, (c) $f = \bar{z}$, (d) $f = x^2 + jy^2$, (e) $f = \operatorname{Log} z$.

(a) $u = x^2 - y^2$, $v = 2xy$: $u_x = 2x = v_y$, $u_y = -2y = -v_x$. ✓ everywhere; $f' = u_x + jv_x = 2x + 2jy = 2z$.

(b) $u = e^x\cos y$, $v = e^x \sin y$: $u_x = e^x\cos y = v_y$; $u_y = -e^x \sin y = -v_x$. ✓ everywhere, $f' = e^x\cos y + je^x \sin y = e^z$. The exponential is its own derivative in $\mathbb{C}$ too.

(c) $u = x$, $v = -y$: $u_x = 1$ but $v_y = -1$. ✗ everywhere — confirming Example 18.5.

(d) $u = x^2$, $v = y^2$: $u_x = 2x$, $v_y = 2y$, so the first equation needs $x = y$; the second, $0 = 0$, always holds. The partials are continuous, so f is differentiable exactly on the line $y = x$, with $f' = 2x$ there. A line has no interior, so f is **analytic nowhere**.

(e) $u = \ln r$, $v = \theta$ (principal branch, $-\pi < \theta < \pi$): $u_r = 1/r = \frac{1}{r}v_\theta$ and $v_r = 0 = -\frac{1}{r}u_\theta$. ✓ on the cut plane, $f' = e^{-j\theta}\cdot\frac{1}{r} = 1/z$.

Harmonic functions

Real and imaginary parts are harmonic

If $f = u + jv$ is analytic in D (and u, v have continuous second partials — automatically true, as Section 18.5 shows), then

$$\nabla^2 u = u_{xx} + u_{yy} = 0, \qquad \nabla^2 v = v_{xx} + v_{yy} = 0 \quad \text{in } D.$$

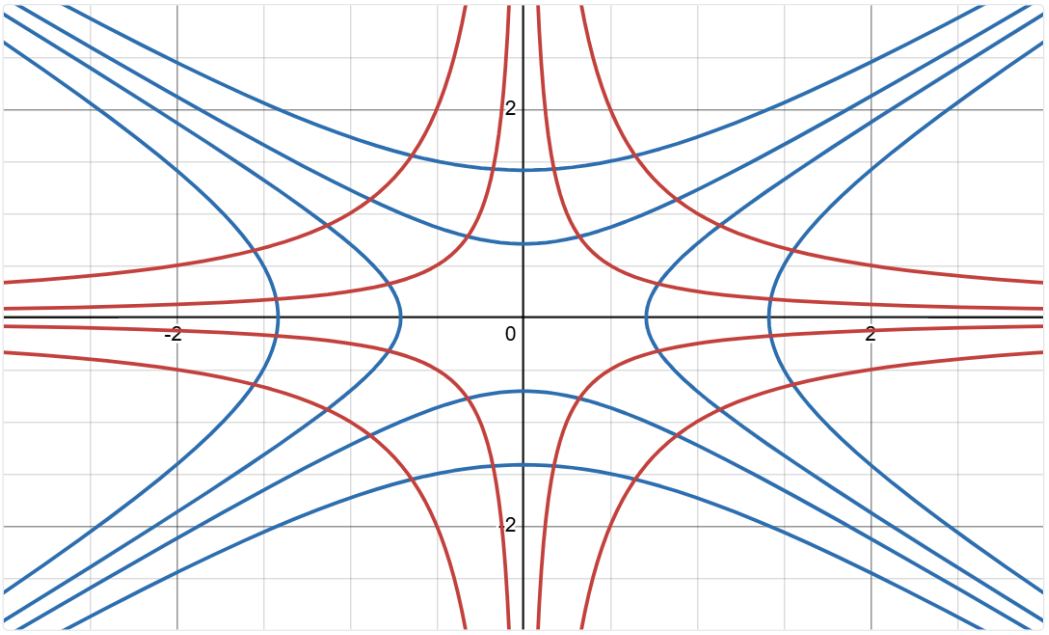
Proof. Differentiate $u_x = v_y$ with respect to x and $u_y = -v_x$ with respect to y: $u_{xx} = v_{yx}$ and $u_{yy} = -v_{xy}$. Adding, and using equality of mixed partials, $u_{xx} + u_{yy} = 0$. The same argument works for v. $\square$

A function satisfying Laplace's equation is **harmonic**. If u and v are harmonic and satisfy Cauchy–Riemann together, v is a **harmonic conjugate** of u. Every harmonic u on a simply connected domain has a harmonic conjugate, unique up to an additive constant, found by integrating the Cauchy–Riemann equations.

Orthogonality of level curves. The gradients are $\nabla u = (u_x, u_y)$ and $\nabla v = (v_x, v_y) = (-u_y, u_x)$, so

$$\nabla u \cdot \nabla v = -u_x u_y + u_y u_x = 0.$$

Where $f' \neq 0$, the curves $u = \text{const}$ and $v = \text{const}$ cross at right angles. In electrostatics, if u is the potential then $v = \text{const}$ are the field lines (flux lines); in fluid flow they are the equipotentials and streamlines.



Level curves of $u = \mathrm{Re}\{z^2\} = x^2 - y^2$ (blue) and $v = \mathrm{Im}\{z^2\} = 2xy$ (red): two families of hyperbolas that cross at right angles everywhere except at the critical point $z = 0$, where $f'(0) = 0$. They are the equipotentials and field lines near a right-angled conducting corner.

Worked Example 18.9 — constructing harmonic conjugates

(a) Show $u = x^2 - y^2 + 2x + 1$ is harmonic, find a harmonic conjugate v and the analytic f. (b) Repeat for $u = e^{-y}\cos x$. (c) Is $u = x^2 + y^2$ the real part of an analytic function?

(a) $u_{xx} + u_{yy} = 2 - 2 = 0$. ✓ From $v_y = u_x = 2x + 2$, integrate in y: $v = 2xy + 2y + h(x)$. Then $v_x = 2y + h'(x)$ must equal $-u_y = 2y$, so $h' = 0$ and $h = C$. Thus $v = 2xy + 2y + C$ and

$$f = u + jv = (x^2 - y^2 + 2x + 1) + j(2xy + 2y) + jC = (z + 1)^2 + jC.$$

Check at $z = j$: $(1 + j)^2 = 2j$, while $u(0, 1) = -1 + 1 = 0$ and $v(0, 1) = 2$. ✓

(b) $u_{xx} = -e^{-y}\cos x$, $u_{yy} = e^{-y}\cos x$; sum 0. ✓ From $v_y = u_x = -e^{-y}\sin x$: $v = e^{-y}\sin x + h(x)$. Then $v_x = e^{-y}\cos x + h'$ must equal $-u_y = e^{-y}\cos x$, so $h = C$. Hence $f = e^{-y}(\cos x + j\sin x) = e^{-y+jx} = e^{jz}$ (plus jC).

(c) $\nabla^2 u = 2 + 2 = 4 \neq 0$. Not harmonic, so it cannot be the real part of any analytic function.

A shortcut (Milne-Thomson). Since $f' = u_x - ju_y$, set $y = 0$ and $x = z$: $f'(z) = u_x(z, 0) - ju_y(z, 0)$, then integrate. In (a): $u_x(z, 0) = 2z + 2$, $u_y(z, 0) = 0$, so $f' = 2z + 2$ and $f = z^2 + 2z + c$ with $c = u(0, 0) = 1$ (up to jC). The method relies on the identity theorem: an analytic function is fixed by its values on the real axis.

Conformal mapping

Here is the geometric heart of the Cauchy–Riemann equations. Let a smooth curve $z(t)$ pass through $z_0 = z(t_0)$ with tangent vector $z'(t_0) \neq 0$. Its image $w(t) = f(z(t))$ has tangent, by the chain rule,

$$w'(t_0) = f'(z_0)\, z'(t_0).$$

Multiplying by the fixed complex number $f'(z_0) = |f'(z_0)|e^{j\alpha}$ rotates *every* tangent direction at z_0 by the same angle $\alpha = \angle f'(z_0)$ and stretches it by the same factor $|f'(z_0)|$. Two curves crossing at z_0 at angle β therefore have images crossing at angle β too.

Conformality of analytic maps

If f is analytic at z_0 and $f'(z_0) \neq 0$, then f is **conformal** at z_0: it preserves both the magnitude and the sense of the angle between any two smooth curves through z_0. Infinitesimally small figures near z_0 are rotated by $\angle f'(z_0)$ and scaled by $|f'(z_0)|$, so small squares map to (nearly) small squares, with area scaled by $|f'(z_0)|^2$.

At a **critical point**, where $f'(z_0) = 0$ and the first nonvanishing derivative is $f^{(m)}(z_0)$, angles are multiplied by m.

This is what the two graphs of Section 18.1 showed: the orthogonal grid stayed orthogonal under z^2 and $1/z$. At $z = 0$, $(z^2)' = 0$ with $m = 2$: the right angle between the positive real and imaginary axes opens into the straight angle between the positive and negative real w-axes.

Why conformal maps solve field problems. If $\Phi(u, v)$ is harmonic in the w-plane and $w = f(z)$ is analytic, then $\phi(x, y) = \Phi(u(x, y), v(x, y))$ is harmonic in the z-plane. (Locally $\Phi = \mathrm{Re}\, F(w)$ for analytic F, and $F(f(z))$ is analytic.) Conductor boundaries map to boundaries, and the boundary values carry over. So a hard geometry in the z-plane can be mapped to an easy one — parallel plates, a half-plane — solved there, and mapped back. The *complex potential* $\Omega = \phi + j\psi$ packages potential ϕ and flux function ψ into one analytic function.

Worked Example 18.10 — coaxial-line capacitance by conformal mapping

A coaxial line has inner conductor radius a at potential V_0 and outer radius b at 0, filled with dielectric ε. Find the capacitance per unit length by mapping to a parallel-plate geometry.

Use $w = \operatorname{Log} z = \ln r + j\theta$. The annulus $a < r < b$ maps to the rectangle $\ln a < u < \ln b$, $-\pi < v \le \pi$: the circle $r = a$ becomes the line $u = \ln a$ and $r = b$ the line $u = \ln b$. In the w-plane we have a parallel-plate capacitor of gap $d = \ln(b/a)$ and width 2π (per unit depth), whose edges $v = \pm\pi$ are glued together — so there are no fringing fields. The potential is linear in u:

$$\Phi = V_0 \frac{\ln b - u}{\ln(b/a)} \quad \Longrightarrow \quad \phi(r) = V_0 \frac{\ln(b/r)}{\ln(b/a)}.$$

A conformal map preserves capacitance (charge and potential are both preserved, since flux through corresponding curves is the same), so $C' = \varepsilon \cdot \dfrac{\text{width}}{\text{gap}} = \dfrac{2\pi\varepsilon}{\ln(b/a)}$.

Numerically, with $a = 0.5$ mm, $b = 1.75$ mm and $\varepsilon_r = 2.25$ (polyethylene): $C' = 2\pi(8.854 \times 10^{-12})(2.25)/\ln 3.5 = 99.9$ pF/m — the familiar value for a 50 Ω cable. Dimensions: ε in F/m divided by a pure number gives F/m. ✓

Engineering Note — where conformal maps appear

Microstrip and coplanar-waveguide impedance formulas come from Schwarz–Christoffel conformal maps (which send polygons to half-planes); the Smith chart is the Möbius map $\Gamma = (z_L - 1)/(z_L + 1)$; the bilinear transform $s = \frac{2}{T}\frac{z-1}{z+1}$ used to discretize analog filters is a conformal Möbius map of the left half s-plane onto the unit z-disc, which is why it preserves stability and why it warps frequency ($\omega_a = \frac{2}{T}\tan\frac{\omega_d T}{2}$).

Common Pitfall — checking only one equation, or only at one point

Both Cauchy–Riemann equations must hold, and they must hold throughout an open set for analyticity. $f = x^2 + jy^2$ satisfies them on a line and is analytic nowhere. Conversely, a function whose partials are discontinuous can satisfy the equations at a point and still fail to be differentiable there; quote the sufficiency theorem with its continuity hypothesis.

18.4 Contour integration

Motivation

The inverse Laplace transform is an integral along a vertical line in the s-plane; the inverse Z-transform is an integral around a circle in the z-plane; the Fourier integral is an integral along the $j\omega$-axis. These are integrals of complex functions along curves — *contour integrals*. This section defines them, shows how to compute them from a parametrization, and establishes the two workhorses: the antiderivative rule and the ML bound. The next sections show that for analytic integrands the curve can be deformed almost at will, which is where the power lies.

Curves and contours

Arcs, contours and orientation

A **smooth arc** is a curve $z(t) = x(t) + jy(t)$, $a \le t \le b$, with $z'(t)$ continuous and nonzero. A **contour** (piecewise smooth path) is a finite chain of smooth arcs joined end to end. A contour is **closed** if $z(b) = z(a)$, and **simple** if it does not cross itself (apart from the endpoints of a closed one). A simple closed contour is **positively oriented** if it is traversed counter-clockwise — the interior lies to the left of the direction of travel.

Standard parametrizations:

Curve	Parametrization
segment from z_1 to z_2	$z(t) = z_1 + (z_2 - z_1)t,\ 0 \le t \le 1$
circle $\lvert z - z_0\rvert = \rho$, counter-clockwise	$z(\theta) = z_0 + \rho e^{j\theta},\ 0 \le \theta \le 2\pi$
upper semicircle of radius R	$z(\theta) = Re^{j\theta},\ 0 \le \theta \le \pi$
vertical line $\operatorname{Re} z = \gamma$, upward	$z(\omega) = \gamma + j\omega,\ -\infty < \omega < \infty$

The last is the Bromwich contour of the inverse Laplace transform (Section 18.9). On a circle, $dz = j\rho e^{j\theta} d\theta = j(z - z_0)\, d\theta$, an identity used constantly.

Definition and evaluation

Contour integral

Let C be a contour $z(t)$, $a \le t \le b$, and f continuous on C. Then

$$\int_C f(z)\, dz = \int_a^b f\big(z(t)\big)\, z'(t)\, dt.$$

It is the limit of Riemann sums $\sum f(\zeta_k)\,\Delta z_k$ with $\Delta z_k = z_k - z_{k-1}$, and its value does not depend on which parametrization of C (with the same direction) is used. For a closed contour we write $\oint_C$.

Writing $f = u + jv$ and $dz = dx + j\,dy$ shows how the contour integral relates to the real line integrals of Ch. 7:

$$\int_C f\,dz = \int_C (u\,dx - v\,dy) + j\int_C (v\,dx + u\,dy).$$

This decomposition is what connects Cauchy's theorem to Green's theorem in the next section.

Properties

For contours C, C_1, C_2 and constants α, β:

1. Linearity: $\int_C (\alpha f + \beta g)\,dz = \alpha\int_C f\,dz + \beta\int_C g\,dz$.
2. Reversal: $\int_{-C} f\,dz = -\int_C f\,dz$, where $-C$ is C traversed backwards.
3. Additivity: $\int_{C_1+C_2} f\,dz = \int_{C_1} f\,dz + \int_{C_2} f\,dz$ for C_2 starting where C_1 ends.
4. Modulus bound: $\left|\int_C f\,dz\right| \le \int_C |f(z)|\,|dz|$, with $|dz| = |z'(t)|dt$ the element of arc length.

The fundamental integral

Worked Example 18.11 — integer powers around a circle

Let C be the circle $|z - z_0| = \rho$, counter-clockwise. Evaluate $\oint_C (z - z_0)^n\,dz$ for every integer n.

Parametrize $z = z_0 + \rho e^{j\theta}$, $dz = j\rho e^{j\theta} d\theta$:

$$\oint_C (z - z_0)^n\,dz = \int_0^{2\pi} \rho^n e^{jn\theta}\, j\rho e^{j\theta}\,d\theta = j\rho^{n+1}\int_0^{2\pi} e^{j(n+1)\theta}\,d\theta.$$

If $n \neq -1$, the integral of $e^{j(n+1)\theta}$ over a full period is 0. If $n = -1$, the integrand is 1 and the result is $2\pi j$:

$$\oint_C (z - z_0)^n\,dz = \begin{cases} 2\pi j, & n = -1 \\ 0, & n \neq -1. \end{cases}$$

The answer is independent of the radius ρ. This single computation is the seed of the whole residue theory: when a function is expanded in powers of $(z - z_0)$, only the $(z - z_0)^{-1}$ term survives integration around z_0.

Path dependence and the antiderivative rule

Worked Example 18.12 — z^2 versus $\bar{z}$ along two paths

Integrate from 0 to $1 + j$ along (i) the straight segment C_1, and (ii) the path C_2 that runs along the real axis to 1 and then straight up to $1 + j$. Do this for $f = z^2$ and $f = \bar{z}$.

$f = z^2$, **path (i):** $z = (1 + j)t$, $dz = (1 + j)dt$:

$$\int_{C_1} z^2 dz = \int_0^1 (1 + j)^3 t^2\,dt = \frac{(1 + j)^3}{3} = \frac{-2 + 2j}{3}.$$

$f = z^2$, **path (ii):** on $[0, 1]$, $\int_0^1 x^2 dx = \frac{1}{3}$. On the vertical leg $z = 1 + jy$, $dz = j\,dy$: $\int_0^1 (1 + jy)^2 j\,dy = j\int_0^1 (1 - y^2 + 2jy)\,dy = j\left(\frac{2}{3} + j\right) = -1 + \frac{2}{3}j$. Total $\frac{1}{3} - 1 + \frac{2}{3}j = \frac{-2+2j}{3}$ — the same.

$f = \bar{z}$, **path (i):** $\bar{z} = (1 - j)t$: $\int_0^1 (1 - j)t(1 + j)\,dt = 2 \cdot \frac{1}{2} = 1$.

$f = \bar{z}$, **path (ii):** $\int_0^1 x\,dx = \frac{1}{2}$, then $\int_0^1 (1 - jy)j\,dy = \int_0^1 (j + y)\,dy = j + \frac{1}{2}$. Total $1 + j$ — **different**.

The analytic integrand gives a path-independent answer; the non-analytic one does not. The difference, $(1 + j) - 1 = j$, equals $\oint \bar{z}\,dz$ around the closed triangle $C_2 - C_1$, and in general $\oint_C \bar{z}\,dz = 2j \times$ (area enclosed) — here the triangle has area $\frac{1}{2}$. ✓ (Around the unit circle this gives $2\pi j$, matching the direct computation $\int_0^{2\pi} e^{-j\theta} j e^{j\theta} d\theta = 2\pi j$.)

Antiderivative (fundamental theorem for contour integrals)

If f is continuous in a domain D and has an antiderivative F there ($F' = f$ throughout D), then for every contour C in D from z_1 to z_2,

$$\int_C f(z)\,dz = F(z_2) - F(z_1),$$

independent of the path. In particular $\oint_C f\,dz = 0$ for every closed contour in D.

Proof. $\frac{d}{dt}F(z(t)) = F'(z(t))z'(t) = f(z(t))z'(t)$, so the integral is $[F(z(t))]_a^b$. □

So $\int_C z^2 dz = [z^3/3]_0^{1+j}$ immediately. Conversely, Example 18.11 shows $1/(z - z_0)$ has **no** antiderivative on any domain containing a circle around z_0: its "antiderivative" $\log(z - z_0)$ is multivalued, and going once around changes it by $2\pi j$.

Worked Example 18.13 — $1/z$ and the branch of the logarithm

Compute $\int dz/z$ from $-j$ to j along (i) the right half of the unit circle and (ii) the left half.

(i) The right half lies in the cut plane $\mathbb{C} \setminus (-\infty, 0]$, where Log is an antiderivative: $\operatorname{Log} j - \operatorname{Log}(-j) = j\frac{\pi}{2} - (-j\frac{\pi}{2}) = j\pi$.

(ii) The left half crosses the negative real axis, so use a branch cut along the positive real axis instead, with $\arg z \in (0, 2\pi)$: then $\arg j = \pi/2$ and $\arg(-j) = 3\pi/2$, giving $j\pi/2 - j3\pi/2 = -j\pi$.

Directly: path (ii) is $z = e^{j\theta}$ with θ from $3\pi/2$ down to $\pi/2$, so $\int j\,d\theta = j(\pi/2 - 3\pi/2) = -j\pi$. ✓ The two paths differ by $2\pi j$: going up the right side and back down the left encircles the origin once counter-clockwise, $\oint dz/z = j\pi - (-j\pi) = 2\pi j$.

The ML inequality

Most contour arguments end by showing that some piece of the contour contributes nothing in a limit. The tool is a simple bound.

ML inequality (estimation lemma)

If $|f(z)| \le M$ for all z on a contour C of length L, then

$$\left| \int_C f(z)\,dz \right| \le ML.$$

Proof. $\left|\int_C f\,dz\right| \le \int_C |f||dz| \le M\int_C |dz| = ML$. □

To bound $|f|$ from above we need lower bounds on denominators, usually from the reverse triangle inequality $|z_1 + z_2| \ge \big||z_1| - |z_2|\big|$.

Worked Example 18.14 — bounding integrals

(a) Let C_R be the upper semicircle $|z| = R > 1$. Show $\int_{C_R} \frac{dz}{z^2+1} \to 0$ as $R \to \infty$. (b) Bound $\left|\oint_{|z|=2} \frac{e^z}{z^2+1} dz\right|$ and compare with the exact value.

(a) On C_R, $|z^2 + 1| \ge |z|^2 - 1 = R^2 - 1$, so $M = 1/(R^2-1)$; $L = \pi R$. Hence $\left|\int_{C_R}\right| \le \frac{\pi R}{R^2-1} \to 0$. For $R = 10$ the bound is 0.0317. The general rule: if $|f| \le K/R^k$ on C_R with $k > 1$, the integral over C_R vanishes as $R \to \infty$.

(b) On $|z| = 2$: $|e^z| = e^x \le e^2$ and $|z^2 + 1| \ge 4 - 1 = 3$, so $M = e^2/3$ and $L = 4\pi$: the bound is $4\pi e^2/3 = 30.95$. The exact value (Section 18.8) is $2\pi j \sin 1$, of modulus 5.287. The ML bound is crude — its job is to prove that something tends to zero, not to estimate it accurately.

Common Pitfall — forgetting $z'(t)$ or the orientation

$\int_C f\,dz$ is not $\int f\,dt$: the factor $z'(t)$ (on circles, $j\rho e^{j\theta}$) carries both the length element and the direction. A clockwise circle gives the negative of the counter-clockwise result, so $\oint dz/z = -2\pi j$ clockwise. Also, $\int_C f\,dz$ is not the same as the arc-length integral $\int_C f|dz|$; only the latter is used in bounds.

Engineering Note — contour integrals you have already met

The inverse Z-transform $x[n] = \frac{1}{2\pi j}\oint_C z^{n-1}X(z)\,dz$ (Ch. 17) is a contour integral around a circle in the ROC; on the unit circle $z = e^{j\Omega}$, $dz = jz\,d\Omega$ and it collapses to the inverse DTFT $\frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\Omega})e^{j\Omega n}d\Omega$. The inverse Fourier transform $\frac{1}{2\pi}\int X(\omega)e^{j\omega t}d\omega$ is $\frac{1}{2\pi j}\int_{j\mathbb{R}} X(s)e^{st}ds$ along the imaginary axis. Seeing these as contour integrals is what lets us evaluate them by residues.

18.5 Cauchy's integral theorem

Motivation

Example 18.12 showed an analytic integrand giving the same answer along two different paths. Cauchy's theorem says this is always so on a domain without holes: the integral of an analytic function around any closed curve is zero. Its corollary, the Cauchy integral formula, says something even stronger — the values of an analytic function *inside* a closed curve are completely determined by its values *on* the curve. These two results convert integration problems into the bookkeeping of singular points, and they deliver, almost for free, the infinite differentiability of analytic functions, Liouville's theorem, the fundamental theorem of algebra and the maximum-modulus principle.

The theorem

Cauchy's integral theorem (Cauchy–Goursat)

Let f be analytic in a simply connected domain D. Then for every closed contour C in D,

$$\oint_C f(z)\,dz = 0.$$

Equivalently: if f is analytic inside and on a simple closed contour C, then $\oint_C f\,dz = 0$.

Proof sketch (assuming f' continuous). Let C be simple and positively oriented, enclosing a region R. Split the integral as in Section 18.4 and apply Green's theorem (Ch. 7), $\oint_C (P\,dx + Q\,dy) = \iint_R (Q_x - P_y)\,dA$:

$$\oint_C f\,dz = \oint_C (u\,dx - v\,dy) + j\oint_C (v\,dx + u\,dy) = \iint_R (-v_x - u_y)\,dA + j\iint_R (u_x - v_y)\,dA.$$

Both integrands vanish identically by the Cauchy–Riemann equations. □

Goursat's refinement removes the assumption that f' is continuous: one subdivides a triangle repeatedly, uses the fact that f is approximated by $f(z_0) + f'(z_0)(z - z_0)$ near a limit point z_0, and shows the integral is smaller than any ε. The distinction matters theoretically — it is why "analytic" alone suffices in every later theorem — but not computationally.

Consequences on a simply connected domain

If f is analytic in a simply connected domain D:

1. $\int_{z_1}^{z_2} f\,dz$ is independent of the path in D;
2. $F(z) = \int_{z_0}^{z} f(\zeta)\,d\zeta$ is analytic in D with $F' = f$ — every analytic function has an antiderivative there.

Deformation of contours

On a domain with holes, closed curves around a hole need not give zero — $\oint dz/z = 2\pi j$ around the origin. But the answer depends only on *which holes are encircled*, not on the shape of the curve.

Deformation principle

Let C and C_1 be positively oriented simple closed contours with C_1 inside C, and let f be analytic in the closed region between them (it may have singularities inside C_1). Then

$$\oint_C f\,dz = \oint_{C_1} f\,dz.$$

More generally, if C encloses disjoint contours $C_1, \dots, C_n$ and f is analytic in the region inside C and outside every C_k, then $\oint_C f\,dz = \sum_{k=1}^{n} \oint_{C_k} f\,dz$.

Proof sketch. Join C to C_1 by a cross-cut. The contour that runs around C counter-clockwise, in along the cut, around C_1 clockwise and back out along the cut bounds a simply connected region where f is analytic, so its integral is 0. The two traversals of the cut cancel, leaving $\oint_C - \oint_{C_1} = 0$. □

The practical upshot: **shrink the contour onto the singularities.** Any contour around z_0 can be replaced by a tiny circle around z_0. Combined with Example 18.11, for any positively oriented simple closed contour C,

$$\oint_C \frac{dz}{(z-a)^n} = \begin{cases} 2\pi j, & n = 1 \text{ and } a \text{ inside } C \\ 0, & \text{otherwise } (n \geq 1 \text{ integer},\ a \notin C). \end{cases}$$

Worked Example 18.15 — partial fractions plus deformation

Evaluate $\oint_C \dfrac{dz}{z^2 - 1}$ for (a) $C : |z| = 2$, (b) $C : |z - 1| = 1$, (c) $C : |z| = \tfrac{1}{2}$.

Partial fractions: $\dfrac{1}{z^2-1} = \dfrac{1}{2}\left(\dfrac{1}{z-1} - \dfrac{1}{z+1}\right)$.

(a) Both ± 1 are inside: $\tfrac{1}{2}(2\pi j - 2\pi j) = 0$. (The integral is zero even though the integrand is not analytic inside — Cauchy's theorem is sufficient, not necessary.)

(b) Only $z = 1$ is inside ($|-1 - 1| = 2 > 1$): $\tfrac{1}{2} \cdot 2\pi j = \pi j$.

(c) Neither is inside; f is analytic on and inside C: 0.

The Cauchy integral formula

Cauchy integral formula

Let f be analytic inside and on a positively oriented simple closed contour C, and let z_0 be inside C. Then

$$f(z_0) = \frac{1}{2\pi j}\oint_C \frac{f(z)}{z - z_0}\,dz.$$

Proof. The integrand $f(z)/(z - z_0)$ is analytic except at z_0, so by deformation we may replace C by a small circle $C_\rho : |z - z_0| = \rho$. Write

$$\oint_{C_\rho} \frac{f(z)}{z - z_0}dz = f(z_0)\oint_{C_\rho} \frac{dz}{z - z_0} + \oint_{C_\rho} \frac{f(z) - f(z_0)}{z - z_0}dz = 2\pi j\, f(z_0) + I_\rho.$$

By continuity, $|f(z) - f(z_0)| < \varepsilon$ on C_ρ for small ρ, so by ML, $|I_\rho| \leq (\varepsilon/\rho)(2\pi\rho) = 2\pi\varepsilon$. Since the left side does not depend on ρ, $I_\rho = 0$. □

This is astonishing from a real-variable viewpoint: boundary values determine the interior completely. Taking C to be a circle $z = z_0 + \rho e^{j\theta}$ gives the **mean-value property**

$$f(z_0) = \frac{1}{2\pi}\int_0^{2\pi} f(z_0 + \rho e^{j\theta})\,d\theta,$$

and its real part says that a harmonic function (an electrostatic potential in a charge-free region) equals its average over any circle centred at the point.

Cauchy integral formula for derivatives

Under the same hypotheses f has derivatives of all orders at z_0, and

$$f^{(n)}(z_0) = \frac{n!}{2\pi j}\oint_C \frac{f(z)}{(z-z_0)^{n+1}}\,dz, \qquad n = 0, 1, 2, \ldots$$

Proof sketch. Differentiate the integral formula with respect to z_0 under the integral sign: $\frac{d}{dz_0}(z-z_0)^{-1} = (z-z_0)^{-2}$, $\frac{d^2}{dz_0^2}(z-z_0)^{-1} = 2(z-z_0)^{-3}$, and so on. Justifying the interchange is a uniform-convergence argument on C, where $|z-z_0|$ is bounded below. $\square$

Corollary. If f is analytic in D, so are f', f'', ... Hence u and v have continuous partial derivatives of all orders — the assumption made in proving harmonicity in Section 18.3 was automatically satisfied.

Read in reverse, the derivative formula is an *integration* rule, the form in which we use it:

$$\oint_C \frac{f(z)}{(z-z_0)^{n+1}}dz = \frac{2\pi j}{n!}f^{(n)}(z_0).$$

Worked Example 18.16 — direct use of the integral formula

(a) $\displaystyle\oint_{|z|=2}\frac{e^z}{z-1}dz$. (b) $\displaystyle\oint_{|z-j|=1}\frac{dz}{z^2+1}$. (c) $\displaystyle\oint_{|z|=3}\frac{e^z}{z^2+1}dz$.

(a) $f = e^z$ is entire and $z_0 = 1$ is inside: $2\pi j\, e^1 = 17.079j$.

(b) Only $z = j$ is inside the circle $|z-j| = 1$ ($-j$ is at distance 2). Write $\dfrac{1}{z^2+1} = \dfrac{1/(z+j)}{z-j}$, with $f(z) = 1/(z+j)$ analytic inside. Result: $2\pi j \cdot \dfrac{1}{2j} = \pi$.

(c) Both $\pm j$ are inside. Surround each by a small circle (deformation principle) and apply the formula to each:

$$\oint = 2\pi j\left[\frac{e^j}{2j} + \frac{e^{-j}}{-2j}\right] = 2\pi j\cdot\frac{e^j - e^{-j}}{2j} = 2\pi j\sin 1 = 5.2871j.$$

All three values were confirmed by numerical quadrature of the parametrized integrals.

Worked Example 18.17 — higher-order formula

(a) $\displaystyle\oint_{|z|=2}\frac{e^{2z}}{(z-1)^2}dz$. (b) $\displaystyle\oint_{|z|=2}\frac{\cos z}{(z-j)^3}dz$. (c) $\displaystyle\oint_{|z|=1}\frac{\sin z}{z^4}dz$.

(a) $n = 1$, $f = e^{2z}$, $f'(1) = 2e^2$: result $2\pi j\cdot 2e^2 = 4\pi e^2 j = 92.854j$.

(b) $n = 2$, $f = \cos z$, $f''(z) = -\cos z$, $f''(j) = -\cos j = -\cosh 1$: result $\dfrac{2\pi j}{2!}(-\cosh 1) = -\pi j\cosh 1 = -4.8477j$.

(c) $n = 3$, $f = \sin z$, $f'''(0) = -\cos 0 = -1$: result $\dfrac{2\pi j}{3!}(-1) = -\dfrac{\pi j}{3} = -1.0472j$. Check with the series of Section 18.6: $\sin z/z^4 = z^{-3} - \frac{1}{6}z^{-1} + \ldots$, and only the z^{-1} term contributes, giving $2\pi j(-\frac{1}{6})$. ✓

Consequences: estimates, Liouville, and the maximum principle

Cauchy's estimate. If $|f| \le M$ on the circle $|z-z_0| = \rho$, the derivative formula and ML give

$$|f^{(n)}(z_0)| \le \frac{n!}{2\pi}\cdot\frac{M}{\rho^{n+1}}\cdot 2\pi\rho = \frac{n!\,M}{\rho^n}.$$

Liouville's theorem

A bounded entire function is constant.

Proof. If $|f| \le M$ everywhere, Cauchy's estimate with $n = 1$ gives $|f'(z_0)| \le M/\rho$ for every ρ; let $\rho \to \infty$. So $f' \equiv 0$. $\square$

That is why $\sin z$ must be unbounded in $\mathbb{C}$. It also gives a two-line proof of the **fundamental theorem of algebra**: if a non-constant polynomial P had no zero, $1/P$ would be entire and (since $|P| \to \infty$) bounded, hence constant — contradiction. So every polynomial of degree $n \ge 1$ has a root, and by induction exactly n roots counted with multiplicity. Every characteristic polynomial in this book therefore factors completely over $\mathbb{C}$.

Maximum-modulus principle

If f is analytic and non-constant in a bounded domain D and continuous on its closure, then $|f|$ attains its maximum on the boundary of D, never at an interior point.

Proof idea. By the mean-value property, $|f(z_0)| \le$ the average of $|f|$ over any small circle about z_0; if $|f(z_0)|$ were a maximum, $|f|$ would have to equal it on the whole circle, and then (by the identity theorem) everywhere. $\square$

Engineering Note — why the H_∞ norm is measured on the $j\omega$-axis

For a stable transfer function $H(s)$, analytic in the closed right half-plane and bounded at infinity, the maximum-modulus principle (applied on large half-discs) shows $\sup_{\mathrm{Re}\, s\ge 0}|H(s)| = \sup_\omega |H(j\omega)|$. The peak gain over all exponentially growing inputs equals the peak of the Bode magnitude plot. This is the H_∞ norm of robust control, and it is computable from frequency-response data alone.

Common Pitfall — hypotheses of Cauchy's theorem

$\oint_C f\,dz = 0$ requires analyticity at **every** point inside C, not just on C. $\oint_{|z|=1} e^z/z^2\,dz = 2\pi j \neq 0$ even though e^z/z^2 is perfectly well behaved on the circle. Equally, a singularity *on* the contour makes the integral improper; it must be treated as a principal value with an indentation (Section 18.8), not ignored.

18.6 Taylor and Laurent series

Motivation

Power series are how analytic functions are computed, approximated and classified. A Taylor series describes f near a point where it is analytic. A Laurent series — a power series allowed to have negative powers — describes f near, and around, its singularities, and it is valid in an **annulus** rather than a disc. Engineers already use Laurent series under another name: the Z-transform $X(z) = \sum_n x[n] z^{-n}$ *is* a Laurent series in z, and its region of convergence is the annulus of that series. The coefficient of $(z - z_0)^{-1}$ in a Laurent series is the *residue*, the quantity on which Sections 18.7–18.9 are built.

Power series

Convergence of power series

For every power series $\sum_{n=0}^{\infty} a_n (z - z_0)^n$ there is a **radius of convergence** $R \in [0, \infty]$ such that the series converges absolutely for $|z - z_0| < R$ and diverges for $|z - z_0| > R$. It is given by

$$\frac{1}{R} = \limsup_{n\to\infty} |a_n|^{1/n} \qquad \left(\text{or } R = \lim_{n\to\infty} \left|\frac{a_n}{a_{n+1}}\right| \text{ when this limit exists}\right).$$

Inside the disc of convergence the sum is analytic and may be differentiated or integrated term by term, the new series having the same radius.

The region of convergence is always a disc: there is no such thing as a power series converging on an interval but not on the corresponding disc. On the circle $|z - z_0| = R$ itself anything can happen, point by point.

Taylor's theorem

Taylor's theorem

Let f be analytic in the disc $|z - z_0| < R$. Then for every z in the disc

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n, \qquad a_n = \frac{f^{(n)}(z_0)}{n!} = \frac{1}{2\pi j} \oint_C \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta,$$

where C is any circle $|\zeta - z_0| = \rho < R$. The series converges in the **largest open disc about z_0 in which f is analytic**: its radius is the distance from z_0 to the nearest singular point of f.

Proof. Fix z with $|z - z_0| < \rho < R$ and use the Cauchy formula on the circle C of radius ρ. For ζ on C, $|(z - z_0)/(\zeta - z_0)| < 1$, so the Cauchy kernel expands as a geometric series:

$$\frac{1}{\zeta - z} = \frac{1}{(\zeta - z_0)\left(1 - \dfrac{z - z_0}{\zeta - z_0}\right)} = \sum_{n=0}^{\infty} \frac{(z - z_0)^n}{(\zeta - z_0)^{n+1}}.$$

The convergence is uniform on C, so we may integrate term by term:

$$f(z) = \frac{1}{2\pi j} \oint_C \frac{f(\zeta)}{\zeta - z} d\zeta = \sum_{n=0}^{\infty} \left[\frac{1}{2\pi j} \oint_C \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta\right] (z - z_0)^n,$$

and the bracket is $f^{(n)}(z_0)/n!$ by the derivative formula. □

This is the payoff of complex differentiability: *analytic* and *equal to a convergent power series* are the same property. Standard Maclaurin series (all valid in $\mathbb{C}$ with the stated radius):

Function	Series	Radius
$\dfrac{1}{1 - z}$	$\sum_{n\ge 0} z^n$	1
e^z	$\sum_{n\ge 0} \dfrac{z^n}{n!}$	∞
$\sin z$	$\sum_{n\ge 0} \dfrac{(-1)^n z^{2n+1}}{(2n+1)!}$	∞
$\cos z$	$\sum_{n\ge 0} \dfrac{(-1)^n z^{2n}}{(2n)!}$	∞

Function	Series	Radius
$\operatorname{Log}(1+z)$	$\sum_{n\ge1}\dfrac{(-1)^{n+1}z^n}{n}$	1
$(1+z)^\alpha$ (principal)	$\sum_{n\ge0}\binom{\alpha}{n}z^n$	1 (unless $\alpha\in\mathbb{N}$)

Worked Example 18.18 — Taylor series and their radii

(a) Expand $f(z)=1/z$ about $z_0=1+j$ and give the radius. (b) Find the first three nonzero terms of $\tan z$ about 0 and the radius. (c) Expand $g(z)=\dfrac{1}{1+z^2}$ about $z_0=1$ to second order.

(a) Write $z=z_0+(z-z_0)$: $\dfrac{1}{z}=\dfrac{1}{z_0}\cdot\dfrac{1}{1+(z-z_0)/z_0}=\sum_{n\ge0}\dfrac{(-1)^n(z-z_0)^n}{z_0^{n+1}}$, valid for $|z-z_0|<|z_0|=\sqrt{2}$. The only singularity is $z=0$, at distance $\sqrt{2}$, as the theorem predicts.

(b) Divide series: $\tan z=\dfrac{z-z^3/6+z^5/120-\cdots}{1-z^2/2+z^4/24-\cdots}$. Set $\tan z=z+bz^3+cz^5$; multiplying by the denominator and matching: z^3: $b-\tfrac12=-\tfrac16\Rightarrow b=\tfrac13$; z^5: $c-\tfrac{b}{2}+\tfrac{1}{24}=\tfrac{1}{120}\Rightarrow c=\tfrac{2}{15}$. So $\tan z=z+\tfrac13z^3+\tfrac{2}{15}z^5+\cdots$. The nearest zeros of $\cos z$ are $\pm\pi/2$, so $R=\pi/2$. Check at $z=0.3$: $0.3+0.009+0.000324=0.309324$, vs $\tan 0.3=0.309336$.

(c) $g(1)=\tfrac12$; $g'=-2z/(1+z^2)^2$, $g'(1)=-\tfrac12$; $g''=(6z^2-2)/(1+z^2)^3$, $g''(1)=\tfrac12$. So $g\approx\tfrac12-\tfrac12(z-1)+\tfrac14(z-1)^2$, with radius $|1\mp j|=\sqrt{2}$ (Example 18.7).

Zeros. If f is analytic at z_0, $f(z_0)=0$ and not all derivatives vanish there, the Taylor series begins $f=a_m(z-z_0)^m+\cdots$ with $a_m\neq0$: z_0 is a **zero of order** m, and $f(z)=(z-z_0)^m g(z)$ with g analytic and $g(z_0)\neq0$. Since $g\neq0$ near z_0, **zeros of a non-constant analytic function are isolated** — the root of the identity theorem.

Laurent series

Near a singularity a Taylor series is impossible, but an expansion in both positive and negative powers works.

Laurent's theorem

Let f be analytic in the annulus $A:\ r_1<|z-z_0|<r_2$ (with $0\le r_1<r_2\le\infty$). Then for every z in A

$$f(z)=\sum_{n=-\infty}^{\infty}a_n(z-z_0)^n=\underbrace{\sum_{n=0}^{\infty}a_n(z-z_0)^n}_{\text{analytic (regular) part}}+\underbrace{\sum_{n=1}^{\infty}\frac{a_{-n}}{(z-z_0)^n}}_{\text{principal part}},$$

with

$$a_n=\frac{1}{2\pi j}\oint_C\frac{f(\zeta)}{(\zeta-z_0)^{n+1}}d\zeta,\qquad n\in\mathbb{Z},$$

C being any positively oriented circle $|\zeta-z_0|=\rho$ with $r_1<\rho<r_2$. The expansion in a given annulus is **unique**.

Proof sketch. For z in the annulus take circles C_2 (radius just below r_2) and C_1 (just above r_1) on either side of z. Cauchy's formula on the region between them gives $f(z)=\frac{1}{2\pi j}\oint_{C_2}\frac{f(\zeta)}{\zeta-z}d\zeta-\frac{1}{2\pi j}\oint_{C_1}\frac{f(\zeta)}{\zeta-z}d\zeta$. On C_2, $|z-z_0|<|\zeta-z_0|$, and the kernel expands in nonnegative powers exactly as in Taylor's proof. On C_1, $|\zeta-z_0|<|z-z_0|$, so instead

$$-\frac{1}{\zeta-z}=\frac{1}{(z-z_0)\left(1-\dfrac{\zeta-z_0}{z-z_0}\right)}=\sum_{n=1}^{\infty}\frac{(\zeta-z_0)^{n-1}}{(z-z_0)^n},$$

which produces the negative powers. Both circles can be deformed to any circle C in the annulus. □

Note that the coefficient formula with $n=-1$ reads

$$a_{-1}=\frac{1}{2\pi j}\oint_C f(\zeta)\,d\zeta:$$

the integral of f around the annulus is $2\pi j$ times the coefficient of $(z-z_0)^{-1}$. That is the content of Example 18.11 seen from the other side, and it is the definition of the residue in Section 18.7.

In practice, nobody computes Laurent coefficients from the integral formula. Because the expansion is unique, *any* legitimate manipulation — geometric series, known Taylor series, substitution, multiplication, partial fractions — that produces a series valid in the annulus gives *the* Laurent series. The essential tool is the geometric series used in two ways:

$$\frac{1}{1-w}=\sum_{n\ge0}w^n\ \ (|w|<1),\qquad \frac{1}{1-w}=-\frac{1}{w}\cdot\frac{1}{1-1/w}=-\sum_{n\ge1}w^{-n}\ \ (|w|>1).$$

Worked Example 18.19 — one function, four expansions

Expand $f(z) = \dfrac{1}{(z-1)(z-2)}$ in (a) $|z| < 1$, (b) $1 < |z| < 2$, (c) $|z| > 2$, (d) $0 < |z-1| < 1$.

Partial fractions: $f = \dfrac{1}{z-2} - \dfrac{1}{z-1}$. The singularities 1 and 2 cut the plane about $z_0 = 0$ into three regions; in each we expand each fraction in the convergent direction.

(a) $|z| < 1$ (so also $|z/2| < 1$):

$$\frac{1}{z-2} = -\frac{1}{2}\cdot\frac{1}{1-z/2} = -\sum_{n\ge 0}\frac{z^n}{2^{n+1}}, \qquad -\frac{1}{z-1} = \frac{1}{1-z} = \sum_{n\ge 0} z^n,$$

$$f = \sum_{n\ge 0}\left(1 - 2^{-(n+1)}\right) z^n = \tfrac{1}{2} + \tfrac{3}{4}z + \tfrac{7}{8}z^2 + \cdots \quad \text{(a Taylor series)}.$$

(b) $1 < |z| < 2$: keep the first expansion; for the second use $|1/z| < 1$:

$$-\frac{1}{z-1} = -\frac{1}{z}\cdot\frac{1}{1-1/z} = -\sum_{n\ge 1} z^{-n}, \qquad f = -\sum_{n\ge 0}\frac{z^n}{2^{n+1}} - \sum_{n\ge 1}\frac{1}{z^n}.$$

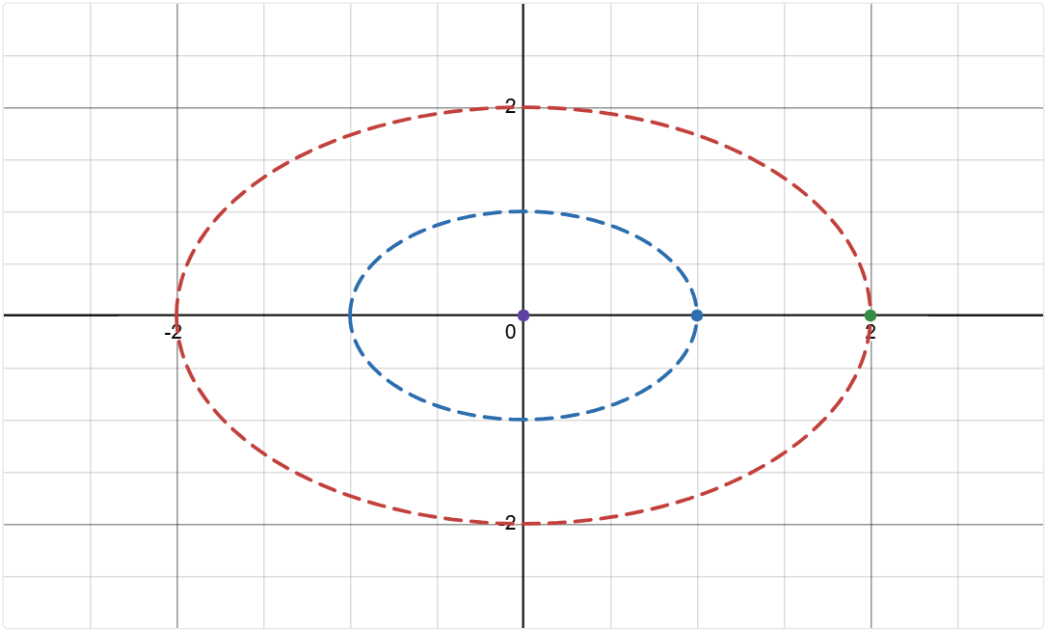
Check at $z = 1.5$: the first sum is $-\frac{1}{2}/(1-0.75) = -2$, the second $-\frac{2/3}{1-2/3} = -2$; total -4, and indeed $f(1.5) = 1/(0.5\cdot(-0.5)) = -4$. ✓

(c) $|z| > 2$: now $\dfrac{1}{z-2} = \dfrac{1}{z}\cdot\dfrac{1}{1-2/z} = \sum_{n\ge 1} 2^{n-1} z^{-n}$, so

$$f = \sum_{n\ge 1}\left(2^{n-1} - 1\right) z^{-n} = \frac{1}{z^2} + \frac{3}{z^3} + \frac{7}{z^4} + \cdots.$$

(d) About $z_0 = 1$, put $w = z - 1$: $f = \dfrac{1}{w(w-1)} = -\dfrac{1}{w}\cdot\dfrac{1}{1-w} = -\dfrac{1}{w} - 1 - w - w^2 - \cdots$ for $0 < |w| < 1$. The coefficient of $(z-1)^{-1}$ is -1.

Notice that in (c) the coefficient of z^{-1} is $2^0 - 1 = 0$: by the coefficient formula $\oint_{|z|=3} f\,dz = 0$, which agrees with Example 18.15-style reasoning ($\frac{1}{z-2}$ and $-\frac{1}{z-1}$ contribute $+2\pi j$ and $-2\pi j$). In (b) the coefficient of z^{-1} is -1, so $\oint_{|z|=1.5} f\,dz = -2\pi j$: only the pole at 1 is enclosed.



Regions of validity for the expansions of $1/((z-1)(z-2))$ about 0: a Taylor series inside the blue circle $|z| = 1$, a two-sided Laurent series in the annulus between the circles, and a series in negative powers only outside the red circle $|z| = 2$. Every boundary circle passes through a pole.

Engineering Note — the Z-transform ROC is a Laurent annulus

Put $X(z) = \sum_n x[n] z^{-n}$: the sequence values are the Laurent coefficients with $a_{-n} = x[n]$. Example 18.19 is therefore a complete catalogue of the three sequences sharing $X(z) = 1/((z-1)(z-2))$ (Ch. 17): region (c) gives a **right-sided (causal)** sequence $x[n] = 2^{n-1} - 1$ for $n \ge 1$; region (a) gives a **left-sided (anti-causal)** sequence; region (b) a **two-sided** one. The ROC is always an annulus bounded by poles, and the algebraic expression $X(z)$ alone does not determine the signal — the annulus does. A system is stable when its ROC contains the unit circle.

Isolated singularities

Classification of isolated singularities

z_0 is an **isolated singularity** of f if f is analytic in a punctured disc $0 < |z - z_0| < \rho$. Let $\sum a_n (z - z_0)^n$ be the Laurent series there. Then z_0 is

- a **removable singularity** if the principal part is zero ($a_{-n} = 0$ for all $n \geq 1$);
- a **pole of order** m if the principal part has finitely many terms with $a_{-m} \neq 0$ the last: $f = \dfrac{a_{-m}}{(z-z_0)^m} + \cdots + \dfrac{a_{-1}}{z - z_0} + \sum_{n \geq 0} a_n (z - z_0)^n$. A pole of order 1 is a **simple pole**;
- an **essential singularity** if the principal part has infinitely many nonzero terms.

Practical tests, which avoid computing series:

Type	Behaviour as $z \to z_0$	Test
removable	f bounded, has a finite limit	$\lim (z - z_0) f(z) = 0$
pole of order m	$\lvert f \rvert \to \infty$	$(z - z_0)^m f(z) \to$ finite nonzero limit
essential	no limit, finite or infinite	neither of the above

For a quotient $f = g/h$ with g, h analytic: if h has a zero of order m at z_0 and $g(z_0) \neq 0$, then z_0 is a pole of order m. More generally, if g has a zero of order k and h of order m, the result is a pole of order $m - k$ if $m > k$, otherwise a removable singularity.

Near an essential singularity the behaviour is wild: by the Casorati–Weierstrass theorem, f comes arbitrarily close to *every* complex value in every punctured neighbourhood (Picard's theorem says it actually takes every value, with at most one exception, infinitely often).

Worked Example 18.20 — classifying singularities by Laurent series

Classify the singularity at $z = 0$ and find the coefficient a_{-1} for (a) $\dfrac{\sin z}{z}$, (b) $\dfrac{1 - \cos z}{z^4}$, (c) $\dfrac{e^z}{z^3}$, (d) $e^{1/z}$, (e) $z^2 \sin(1/z)$.

(a) $\dfrac{\sin z}{z} = 1 - \dfrac{z^2}{6} + \dfrac{z^4}{120} - \cdots$: removable (define the value 1 at 0 — the sinc function is entire). $a_{-1} = 0$.

(b) $1 - \cos z = \dfrac{z^2}{2} - \dfrac{z^4}{24} + \dfrac{z^6}{720} - \cdots$, so $\dfrac{1 - \cos z}{z^4} = \dfrac{1}{2z^2} - \dfrac{1}{24} + \dfrac{z^2}{720} - \cdots$: a pole of order 2 (numerator zero of order 2, denominator of order 4). $a_{-1} = 0$.

(c) $\dfrac{e^z}{z^3} = \dfrac{1}{z^3} + \dfrac{1}{z^2} + \dfrac{1}{2z} + \dfrac{1}{6} + \cdots$: pole of order 3, $a_{-1} = \tfrac{1}{2}$.

(d) Substitute $1/z$ into the exponential series: $e^{1/z} = 1 + \dfrac{1}{z} + \dfrac{1}{2!z^2} + \dfrac{1}{3!z^3} + \cdots$, valid for $|z| > 0$. Infinitely many negative powers: essential. $a_{-1} = 1$. Along the positive real axis $e^{1/x} \to \infty$; along the negative real axis it $\to 0$; along the imaginary axis $|e^{1/(jy)}| = 1$.

(e) $z^2 \left(\dfrac{1}{z} - \dfrac{1}{6z^3} + \dfrac{1}{120 z^5} - \cdots \right) = z - \dfrac{1}{6z} + \dfrac{1}{120 z^3} - \cdots$: essential, $a_{-1} = -\tfrac{1}{6}$.

Non-isolated singularities. Branch points such as $z = 0$ for $\operatorname{Log} z$ or $z^{1/2}$ are not isolated singularities of any single-valued branch (the whole cut is singular), and have no Laurent expansion about them. The function $1/\sin(1/z)$ has poles at $z = 1/(k\pi)$ accumulating at 0, so 0 is a non-isolated singularity. Rational functions — and so every lumped-element transfer function — have only poles.

Common Pitfall — the "right" Laurent series

A function has a *different* Laurent series in each annulus. Asking for "the Laurent series of f about z_0" is meaningless until the annulus is specified. Residues are read from the series in the punctured disc $0 < |z - z_0| < \rho$ around the singularity itself — region (d) of Example 18.19, not region (b), even though both contain negative powers.

18.7 Residues

Motivation

When a function is integrated around a small circle enclosing one isolated singularity, every term of its Laurent series integrates to zero except one: the $(z - z_0)^{-1}$ term, which contributes $2\pi j \, a_{-1}$. That single coefficient is all the contour "sees" of the singularity. It deserves a name and a toolkit for computing it quickly — ideally without finding the whole Laurent series. The same numbers are, not by coincidence, the coefficients of partial-fraction expansions that you have computed by the cover-up rule in Ch. 14.

Residue

Let z_0 be an isolated singularity of f, with Laurent series $\sum_n a_n (z - z_0)^n$ in $0 < |z - z_0| < \rho$. The **residue** of f at z_0 is

$$\operatorname{Res}_{z = z_0} f(z) = a_{-1} = \frac{1}{2\pi j} \oint_C f(z)\, dz,$$

where C is any positively oriented simple closed contour around z_0 inside the punctured disc.

At a removable singularity the residue is 0. At an essential singularity it must be read from the Laurent series. At poles — the common case — there are formulas.

Computing residues at poles

Residue formulas

Simple pole.

$$\operatorname{Res}_{z=z_0} f = \lim_{z\to z_0} (z-z_0) f(z).$$

Quotient with a simple zero in the denominator. If $f = g/h$ with g, h analytic at z_0, $g(z_0) \neq 0$, $h(z_0) = 0$, $h'(z_0) \neq 0$, then

$$\operatorname{Res}_{z=z_0} \frac{g}{h} = \frac{g(z_0)}{h'(z_0)}.$$

Pole of order m.

$$\operatorname{Res}_{z=z_0} f = \frac{1}{(m-1)!} \lim_{z\to z_0} \frac{d^{m-1}}{dz^{m-1}} \Big[(z-z_0)^m f(z) \Big].$$

Derivation. At a pole of order m, $(z-z_0)^m f(z) = a_{-m} + a_{-m+1}(z-z_0) + \cdots + a_{-1}(z-z_0)^{m-1} + \ldots$ is analytic; a_{-1} is its Taylor coefficient of order $m-1$, i.e. $\frac{1}{(m-1)!}$ times its $(m-1)$-th derivative at z_0. For $m = 1$ this is the simple-pole formula. For the quotient rule, $h(z) = h'(z_0)(z-z_0) + \ldots$, so $(z-z_0)g/h \to g(z_0)/h'(z_0)$. $\square$

The quotient rule g/h' is the fastest route whenever the denominator is not already factored — for instance at the roots of z^4+1 or of $\sin z$.

Worked Example 18.21 — simple poles

(a) $f = \dfrac{ze^z}{z^2-1}$ at $z = \pm 1$. (b) $f = \dfrac{1}{z^4+1}$ at $z_1 = e^{j\pi/4}$. (c) $f = \cot z$ at $z = k\pi$. (d) $f = \dfrac{e^{jz}}{z^2+a^2}$ $(a>0)$ at $z = ja$.

(a) At $z=1$: $\lim (z-1)\dfrac{ze^z}{(z-1)(z+1)} = \dfrac{e}{2} = 1.3591$. At $z=-1$: $\dfrac{(-1)e^{-1}}{-2} = \dfrac{1}{2e} = 0.1839$.

(b) Quotient rule: $\dfrac{1}{4z_1^3} = \dfrac{z_1}{4z_1^4} = -\dfrac{z_1}{4} = -\dfrac{e^{j\pi/4}}{4} = -0.17678 - 0.17678j$ (using $z_1^4 = -1$).

(c) $\cot z = \cos z / \sin z$, $\sin(k\pi) = 0$, $(\sin z)' = \cos z$: residue $\cos(k\pi)/\cos(k\pi) = 1$ at every integer multiple of π. (This is why $\pi \cot \pi z$ is used to sum series $\sum f(n)$: it has residue 1 at every integer.)

(d) $z^2 + a^2 = (z-ja)(z+ja)$: residue $\dfrac{e^{j(ja)}}{2ja} = \dfrac{e^{-a}}{2ja}$. We will need this in Section 18.8.

Worked Example 18.22 — higher-order poles

(a) $f = \dfrac{e^{2z}}{(z-1)^2}$ at $z=1$. (b) $f = \dfrac{1}{(z^2+1)^2}$ at $z = j$. (c) $f = \dfrac{1}{z^3(z-1)}$ at $z = 0$ and $z = 1$.

(a) $m=2$: $\dfrac{d}{dz} e^{2z}\Big|_{z=1} = 2e^2 = 14.778$ — the same number as Example 18.17(a), as it must be.

(b) $(z^2+1)^2 = (z-j)^2(z+j)^2$, so $m=2$ and we differentiate $(z+j)^{-2}$: $-2(z+j)^{-3}$ at $z=j$ gives $-2/(2j)^3 = -2/(-8j) = \dfrac{1}{4j} = -\dfrac{j}{4}$.

(c) At $z=0$ $(m=3)$: $\dfrac{1}{2!}\dfrac{d^2}{dz^2}\left(\dfrac{1}{z-1}\right)\Big|_0 = \dfrac{1}{2}\cdot\dfrac{2}{(z-1)^3}\Big|_0 = -1$. Alternatively, from the series $-\dfrac{1}{z^3}\cdot\dfrac{1}{1-z} = -z^{-3} - z^{-2} - z^{-1} - \cdots$, $a_{-1} = -1$. At $z=1$ (simple): $1/1^3 = 1$. The two residues sum to 0 — see the residue at infinity below.

Residues and partial fractions

For a proper rational function $F(s) = N(s)/D(s)$ with simple poles p_k, the partial-fraction expansion $F = \sum_k A_k/(s-p_k)$ has

$$A_k = \lim_{s\to p_k} (s-p_k)F(s) = \operatorname{Res}_{s=p_k} F(s),$$

because $A_k/(s-p_k)$ is exactly the principal part of F at p_k. In general, **a proper rational function equals the sum of its principal parts at all its poles** (the difference would be an entire function tending to zero at infinity, hence zero by Liouville). The Heaviside cover-up rule is the simple-pole residue formula. At a repeated pole of order m, the coefficient of $(s-p)^{-k}$ is

$$A_k = \frac{1}{(m-k)!}\frac{d^{m-k}}{ds^{m-k}}\Big[(s-p)^m F(s)\Big]_{s=p},$$

of which the residue is the case $k=1$.

Worked Example 18.23 — partial fractions as principal parts

Expand $F(s) = \dfrac{s+3}{(s+1)(s+2)^2}$.

At $s=-1$ (simple): $A = \left.\dfrac{s+3}{(s+2)^2}\right|_{-1} = \dfrac{2}{1} = 2$.

At $s=-2$ (order 2), with $G(s) = (s+2)^2 F = \dfrac{s+3}{s+1}$: the $(s+2)^{-2}$ coefficient is $G(-2) = \dfrac{1}{-1} = -1$; the $(s+2)^{-1}$ coefficient (the residue) is $G'(-2) = \left.\dfrac{(s+1)-(s+3)}{(s+1)^2}\right|_{-2} = \dfrac{-2}{1} = -2$.

$$F(s) = \frac{2}{s+1} - \frac{2}{s+2} - \frac{1}{(s+2)^2}.$$

Check at $s=0$: $F(0) = 3/4$, and $2 - 1 - \tfrac{1}{4} = \tfrac{3}{4}$. ✓ Also, the residues 2 and -2 sum to zero, consistent with $F \sim 1/s^2$ at infinity (no $1/s$ term), as the next result explains.

The residue at infinity

Residue at infinity

If f is analytic for $|z| > R$, define

$$\operatorname{Res}_{z=\infty} f = -\frac{1}{2\pi j}\oint_{|z|=\rho} f(z)\,dz \quad (\rho > R) = -c_{-1},$$

where c_{-1} is the coefficient of $1/z$ in the Laurent expansion of f valid for $|z| > R$. Equivalently, $\operatorname{Res}_{z=\infty} f(z) = -\operatorname{Res}_{\zeta=0}\left[\dfrac{1}{\zeta^2} f\left(\dfrac{1}{\zeta}\right)\right]$.

Sum of all residues

If f has finitely many singularities in the extended plane, the sum of its residues at all of them, including ∞, is zero.

This is useful when many poles lie inside a contour and few outside: $\oint_C f\,dz = 2\pi j \sum_{\text{inside}} \operatorname{Res} = -2\pi j \left[\sum_{\text{outside}} \operatorname{Res} + \operatorname{Res}_\infty\right]$. For a rational $f = P/Q$ with $\deg Q \geq \deg P + 2$, $f = O(1/z^2)$ at infinity, the $1/z$ coefficient vanishes, and **the finite residues sum to zero** — as in Examples 18.22(c) and 18.23.

Worked Example 18.24 — many poles inside, use infinity

Evaluate $\displaystyle\oint_{|z|=2} \frac{z^{15}}{(z^2+1)^4(z^4+2)^3}dz$.

The denominator has twenty roots, all with modulus 1 or $2^{1/4} = 1.189$, all inside the contour. Computing twenty residues (with poles of orders 4 and 3!) is hopeless. Instead: the degree of the denominator is $8 + 12 = 20$, so for large $|z|$, $f = \dfrac{z^{15}}{z^{20}(1+\cdots)} = z^{-5}(1+\cdots)$ has no z^{-1} term: $c_{-1} = 0$. Hence the integral is $2\pi j\, c_{-1} = 0$.

Contrast $\displaystyle\oint_{|z|=2} \frac{z^3}{z^4+2}dz$: here $f = z^{-1}(1+2z^{-4})^{-1} = z^{-1} - 2z^{-5} + \cdots$, so $c_{-1} = 1$ and the integral is $2\pi j$ — confirming that the four simple poles each have residue $z_k^3/(4z_k^3) = \tfrac{1}{4}$.

Engineering Note — residues are the modal amplitudes

In the inverse Laplace transform of $F(s)$ with simple poles p_k, the time function is $\sum_k \operatorname{Res}_{s=p_k} F \cdot e^{p_k t}$ (Section 18.9). Each residue is the amplitude with which the natural mode $e^{p_k t}$ is excited. A small residue at a pole means that mode is weakly excited or weakly observed — the near cancellation of a pole by a nearby zero shows up as a tiny residue, which is why such a pole can be dropped in model reduction.

Common Pitfall — using the simple-pole formula at a double pole

$\lim (z - z_0) f(z)$ is infinite at a pole of order 2 or more; it is not "the residue". Determine the order first (factor the denominator, or count the order of the zero). Equally, $g(z_0)/h'(z_0)$ is valid only when h has a *simple* zero; at $z_0 = 0$ for $f = 1/(z\sin z)$, $h = z\sin z$ has a double zero and the residue is 0 (since $1/(z\sin z) = z^{-2} + \tfrac{1}{6} + \cdots$), not $1/h'(0)$, which is undefined.

18.8 Residue theorem

The theorem

Residue theorem

Let C be a positively oriented simple closed contour, and let f be analytic inside and on C except at finitely many isolated singularities $z_1, \dots, z_n$ inside C. Then

$$\oint_C f(z)\,dz = 2\pi j \sum_{k=1}^{n} \operatorname{Res}_{z=z_k} f(z).$$

Proof. Surround each z_k by a small circle C_k inside C, the circles disjoint. By the deformation principle (Section 18.5), $\oint_C = \sum_k \oint_{C_k}$. On C_k, integrate the Laurent series of f about z_k term by term: by Example 18.11 every term but $a_{-1}^{(k)}/(z-z_k)$ gives zero, and that one gives $2\pi j\, a_{-1}^{(k)}$. $\square$

Cauchy's theorem (no singularities, sum empty) and Cauchy's integral formula (one simple pole of $f(z)/(z-z_0)$ with residue $f(z_0)$) are special cases. The theorem reduces *any* contour integral of a function with isolated singularities to local algebra at the singular points.

Worked Example 18.25 — closed contour integrals

(a) $\displaystyle\oint_{|z|=2} \frac{ze^z}{z^2-1}dz$. (b) $\displaystyle\oint_{|z|=2} \tan z\,dz$. (c) $\displaystyle\oint_{|z|=1} e^{1/z}\,dz$.

(a) Both poles ± 1 are inside; by Example 18.21(a), $\oint = 2\pi j\left(\frac{e}{2}+\frac{1}{2e}\right) = 2\pi j \cosh 1 = 9.6955j$.

(b) Inside $|z|=2$, $\cos z = 0$ at $\pm\pi/2$ ($1.571 < 2$; the next, $3\pi/2$, is outside). Residues: $\sin z/(-\sin z) = -1$ at each. So $\oint = 2\pi j(-2) = -4\pi j = -12.566j$.

(c) The only singularity is the essential one at 0, with residue 1 (Example 18.20(d)): $\oint = 2\pi j$. No residue *formula* applies; the Laurent series does.

Each was confirmed by direct numerical quadrature on the parametrized circle.

The rest of this section is about the most celebrated use of the theorem: evaluating **real** definite integrals that resist real-variable methods. The strategy in every case is:

1. choose a complex function whose real or imaginary part, or whose restriction to the real axis, is the integrand;
2. choose a closed contour, part of which is the real interval of interest;
3. evaluate the closed integral by residues;
4. show that the extra parts of the contour contribute zero (or something computable) in a limit — this is where ML and Jordan's lemma come in.

Type I — trigonometric integrals over a full period

To evaluate $I = \displaystyle\int_0^{2\pi} R(\cos\theta, \sin\theta)\,d\theta$ with R rational, put $z = e^{j\theta}$. As θ runs over $[0, 2\pi]$, z runs once around the unit circle, and

$$\cos\theta = \frac{z+z^{-1}}{2}, \qquad \sin\theta = \frac{z-z^{-1}}{2j}, \qquad d\theta = \frac{dz}{jz}.$$

The integral becomes $\oint_{|z|=1} (\text{rational function of } z)\,dz$, evaluated by the residues **inside the unit circle**.

Worked Example 18.26 — $\displaystyle\int_0^{2\pi} \frac{d\theta}{2+\cos\theta}$

Substitute:

$$I = \oint_{|z|=1} \frac{1}{2+\frac{1}{2}(z+z^{-1})}\cdot\frac{dz}{jz} = \oint \frac{2\,dz}{j(z^2+4z+1)}.$$

The roots of z^2+4z+1 are $z = -2\pm\sqrt{3}$: $\alpha = -2+\sqrt{3} = -0.268$ (inside) and $\beta = -2-\sqrt{3} = -3.732$ (outside). Residue at α: $\dfrac{2}{j(\alpha-\beta)} = \dfrac{2}{j\,2\sqrt{3}} = \dfrac{1}{j\sqrt{3}}$. Hence

$$I = 2\pi j\cdot\frac{1}{j\sqrt{3}} = \frac{2\pi}{\sqrt{3}} = 3.6276.$$

Numerical quadrature: 3.62760. ✓ The same steps give the general result $\displaystyle\int_0^{2\pi}\frac{d\theta}{a+b\cos\theta} = \frac{2\pi}{\sqrt{a^2-b^2}}$ for $a > |b|$, and the same with $\sin\theta$ in place of $\cos\theta$ (e.g. $\int_0^{2\pi} d\theta/(5-4\sin\theta) = 2\pi/3$).

Worked Example 18.27 — a cosine-weighted integral

Evaluate $I = \displaystyle\int_0^{2\pi} \frac{\cos 2\theta}{5 - 4\cos\theta}\,d\theta$.

It is easier to compute $J = \int_0^{2\pi} \frac{e^{j2\theta}}{5-4\cos\theta}\,d\theta$ and take $I = \operatorname{Re} J$ (the denominator is real). With $e^{j2\theta} = z^2$ and $5 - 4\cos\theta = 5 - 2(z + z^{-1}) = -\dfrac{2z^2 - 5z + 2}{z} = -\dfrac{(2z-1)(z-2)}{z}$:

$$J = \oint_{|z|=1} \frac{z^2 \cdot z}{-(2z-1)(z-2)} \cdot \frac{dz}{jz} = \oint \frac{z^2\,dz}{-j(2z-1)(z-2)}.$$

Only $z = \frac{1}{2}$ is inside. Writing $2z - 1 = 2(z - \frac{1}{2})$, the residue is $\dfrac{(1/2)^2}{-j \cdot 2 \cdot (\frac{1}{2} - 2)} = \dfrac{1/4}{3j} = \dfrac{1}{12j}$. So $J = 2\pi j/(12j) = \pi/6$, real, and $I = \pi/6 = 0.5236$. ✓ (numerically 0.523599).

Generalization: $\displaystyle\int_0^{2\pi} \frac{\cos n\theta}{1 - 2a\cos\theta + a^2}\,d\theta = \frac{2\pi a^n}{1 - a^2}$ for $|a| < 1$. This is the Fourier-coefficient formula for the Poisson kernel, and in DSP terms it is the statement that $|1 - ae^{-j\Omega}|^{-2}$ has inverse DTFT $a^{|n|}/(1 - a^2)$.

Worked Example 18.28 — a double pole at the origin

Evaluate $I = \displaystyle\int_0^{2\pi} \frac{\sin^2\theta}{5 + 4\cos\theta}\,d\theta$.

Now $\sin^2\theta = -\dfrac{(z - z^{-1})^2}{4} = -\dfrac{(z^2-1)^2}{4z^2}$ and $5 + 4\cos\theta = \dfrac{(2z+1)(z+2)}{z}$, so

$$I = \oint_{|z|=1} f(z)\,dz, \qquad f(z) = -\frac{(z^2-1)^2}{4j\,z^2(2z+1)(z+2)}.$$

Inside the unit circle: a double pole at $z = 0$ and a simple pole at $z = -\frac{1}{2}$.

At $z = -\frac{1}{2}$: $\text{Res} = -\dfrac{(1/4 - 1)^2}{4j \cdot \frac{1}{4} \cdot 2 \cdot \frac{3}{2}} = -\dfrac{9/16}{3j} = -\dfrac{3}{16j}$.

At $z = 0$: let $g(z) = z^2 f(z) = -\dfrac{1}{4j} \cdot \dfrac{(z^2-1)^2}{2z^2 + 5z + 2}$. With $N = (z^2-1)^2$ ($N(0) = 1$, $N'(0) = 0$) and $D = 2z^2 + 5z + 2$ ($D(0) = 2$, $D'(0) = 5$): $(N/D)'(0) = \dfrac{0 \cdot 2 - 1 \cdot 5}{4} = -\dfrac{5}{4}$, so $\text{Res} = g'(0) = \dfrac{5}{16j}$.

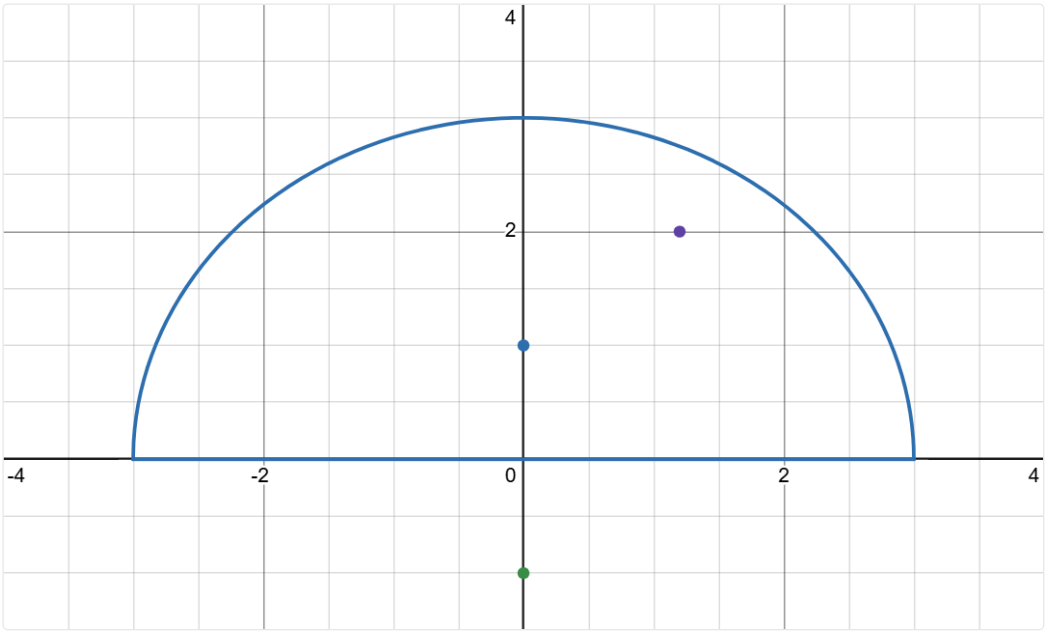
Sum $= \dfrac{2}{16j} = \dfrac{1}{8j}$, and $I = 2\pi j \cdot \dfrac{1}{8j} = \dfrac{\pi}{4} = 0.7854$. ✓

Common Pitfall — the pole on the wrong side

Only poles **inside** $|z| = 1$ count. For a denominator $a + b\cos\theta$ the two roots are reciprocals of each other (their product is 1), so exactly one is inside when $a > |b|$. If you find both inside or both outside, there is an algebra error. If $a \le |b|$ a root lies *on* the circle and the real integral diverges.

Type II — improper integrals of rational functions

To evaluate $\displaystyle\int_{-\infty}^{\infty} \frac{P(x)}{Q(x)}\,dx$, close the real segment $[-R, R]$ with the upper semicircle $C_R:\ z = Re^{j\theta}, 0 \le \theta \le \pi$.



The standard contour for Type II and III integrals: the real segment $[-R, R]$ closed by the upper semicircle C_R (here $R = 3$). As $R \to \infty$ it encloses every upper-half-plane pole; lower-half-plane poles never count.

Rational integrals over the real line

Let P and Q be polynomials with $\deg Q \geq \deg P + 2$ and $Q(x) \neq 0$ for real x. Then

$$\int_{-\infty}^{\infty} \frac{P(x)}{Q(x)} dx = 2\pi j \sum_{\operatorname{Im} z_k > 0} \operatorname{Res}_{z=z_k} \frac{P(z)}{Q(z)}.$$

Proof. For R large enough, the closed contour $[-R, R] + C_R$ encloses all upper-half-plane poles, so $\int_{-R}^{R} P/Q\,dx + \int_{C_R} P/Q\,dz = 2\pi j \sum \operatorname{Res}$. The degree condition gives $|P/Q| \leq K/R^2$ on C_R, so by ML $\left|\int_{C_R}\right| \leq \pi R \cdot K/R^2 \to 0$. (The degree condition also guarantees the improper integral converges, so the symmetric limit $\int_{-R}^{R}$ is the genuine integral.) $\square$

One could equally close in the lower half-plane, traversed clockwise, giving $-2\pi j \sum_{\operatorname{Im} z_k < 0} \operatorname{Res}$; for rational functions both give the same answer.

Worked Example 18.29 — three rational integrals

(a) $\displaystyle\int_{-\infty}^{\infty} \frac{dx}{(x^2+1)^2}$. (b) $\displaystyle\int_{-\infty}^{\infty} \frac{x^2\,dx}{(x^2+1)(x^2+4)}$. (c) $\displaystyle\int_{-\infty}^{\infty} \frac{dx}{x^4+1}$.

(a) The only upper-half-plane pole is $z = j$, of order 2, with residue $-j/4$ (Example 18.22(b)). Integral $= 2\pi j(-j/4) = \pi/2 = 1.5708$. ✓

(b) Degree condition: $4 \geq 2 + 2$. ✓ Upper poles j and $2j$, both simple:

$$\operatorname{Res}_j = \frac{j^2}{2j\,(j^2+4)} = \frac{-1}{6j}, \qquad \operatorname{Res}_{2j} = \frac{(2j)^2}{((2j)^2+1)\,4j} = \frac{-4}{-12j} = \frac{1}{3j}.$$

Sum $= \dfrac{1}{6j}$; integral $= 2\pi j/(6j) = \pi/3 = 1.0472$. ✓

(c) The roots of $z^4 = -1$ are $e^{j\pi/4}, e^{j3\pi/4}, e^{j5\pi/4}, e^{j7\pi/4}$; the upper two are $z_1 = e^{j\pi/4}$ and $z_2 = e^{j3\pi/4}$. By the quotient rule (Example 18.21(b)) the residues are $-z_k/4$:

$$\int = 2\pi j \cdot \left(-\frac{z_1 + z_2}{4}\right) = -\frac{\pi j}{2} \cdot j\sqrt{2} = \frac{\pi}{\sqrt{2}} = 2.2214,$$

using $z_1 + z_2 = \frac{1+j}{\sqrt{2}} + \frac{-1+j}{\sqrt{2}} = j\sqrt{2}$. ✓ (The same method gives $\int dx/(x^6+1) = 2\pi/3$.)

Type III — Fourier-type integrals and Jordan's lemma

Integrals such as $\displaystyle\int_{-\infty}^{\infty} \frac{\cos \omega x}{x^2 + a^2} dx$ are Fourier transforms of rational functions (Ch. 16). The trick is to integrate $R(z)e^{j\omega z}$ instead of $R(z)\cos \omega z$: on the upper half-plane, $|e^{j\omega z}| = e^{-\omega y} \leq 1$ for $\omega > 0$, whereas $\cos \omega z$ grows exponentially there. Then take real or imaginary parts.

Here the degree condition can be relaxed by one: the oscillation of $e^{j\omega z}$ is enough to kill the semicircle when $R(z)$ merely tends to zero.

Jordan's lemma

Let $\omega > 0$, and suppose $|R(z)| \leq M_R$ on the upper semicircle C_R with $M_R \to 0$ as $R \to \infty$. Then

$$\lim_{R \to \infty} \int_{C_R} R(z)\, e^{j\omega z}\, dz = 0.$$

Proof. On C_R, $z = Re^{j\theta}$ and $|e^{j\omega z}| = e^{-\omega R \sin\theta}$, so

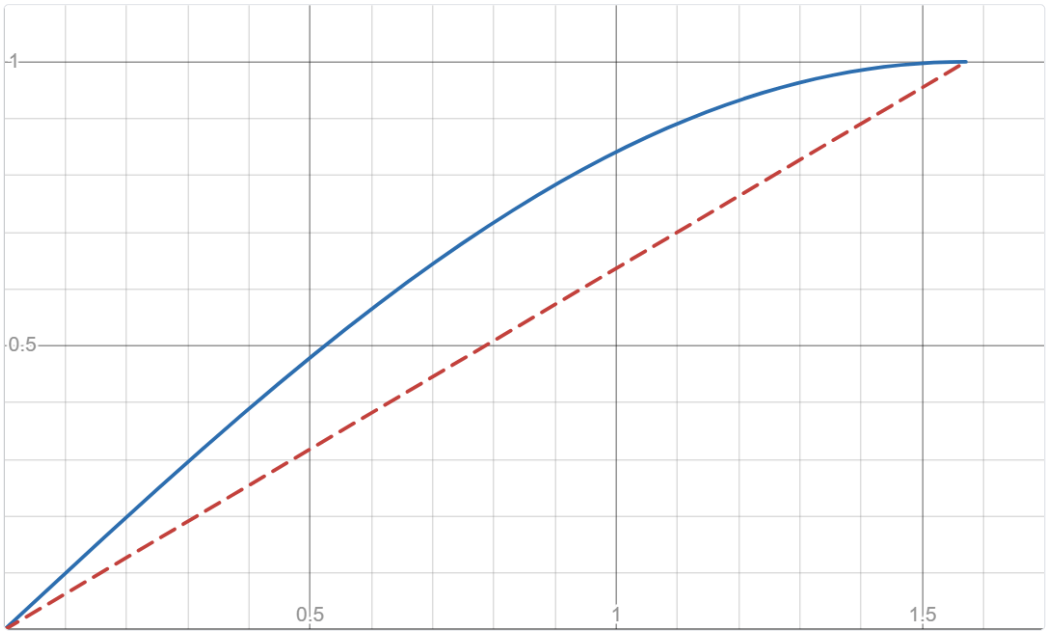
$$\left|\int_{C_R} R\, e^{j\omega z} dz\right| \leq M_R R \int_0^{\pi} e^{-\omega R \sin\theta} d\theta = 2M_R R \int_0^{\pi/2} e^{-\omega R \sin\theta} d\theta.$$

On $[0, \pi/2]$ the concave function $\sin\theta$ lies above its chord: $\sin\theta \geq 2\theta/\pi$ (**Jordan's inequality**). Therefore

$$\int_0^{\pi/2} e^{-\omega R \sin\theta} d\theta \leq \int_0^{\pi/2} e^{-2\omega R\theta/\pi} d\theta = \frac{\pi}{2\omega R}\left(1 - e^{-\omega R}\right) < \frac{\pi}{2\omega R},$$

and the whole bound is $< \pi M_R/\omega \to 0$. $\square$

The ML inequality alone would give only $\pi R M_R$, which need not vanish if $M_R \sim 1/R$; Jordan's lemma recovers the missing factor of R by noting that $e^{-\omega R \sin\theta}$ is tiny except near $\theta = 0$ and $\theta = \pi$.



Jordan's inequality: on $0 \le \theta \le \pi/2$ the sine curve (blue) lies above its chord $2\theta/\pi$ (red, dashed), so $e^{-\omega R \sin\theta} \le e^{-2\omega R\theta/\pi}$ — the bound that makes the semicircle vanish.

Fourier-type integrals

If $R = P/Q$ with $\deg Q \ge \deg P + 1$, no real poles, and $\omega > 0$, then

$$\int_{-\infty}^{\infty} R(x)\, e^{j\omega x} dx = 2\pi j \sum_{\operatorname{Im} z_k > 0} \operatorname{Res}_{z=z_k} \left[R(z) e^{j\omega z} \right],$$

and $\int R\cos\omega x\,dx$, $\int R\sin\omega x\,dx$ are its real and imaginary parts when R is real on the real axis. For $\omega < 0$ close in the **lower** half-plane, clockwise: $-2\pi j \sum_{\operatorname{Im} z_k < 0} \operatorname{Res}$.

Worked Example 18.30 — the Lorentzian and its Fourier transform

Evaluate $\displaystyle\int_{-\infty}^{\infty} \frac{\cos\omega x}{x^2 + a^2} dx$ for $a > 0$ and all real ω; then give the numerical value for $a = 1$, $\omega = 3$.

For $\omega > 0$: the only upper pole of $e^{j\omega z}/(z^2 + a^2)$ is ja, with residue $e^{-\omega a}/(2ja)$ (Example 18.21(d)). So

$$\int_{-\infty}^{\infty} \frac{e^{j\omega x}}{x^2 + a^2} dx = 2\pi j \cdot \frac{e^{-a\omega}}{2ja} = \frac{\pi}{a} e^{-a\omega}.$$

This is real, so the cosine integral equals it and the sine integral is zero (as oddness requires). The integrand is even in ω, so for all ω:

$$\int_{-\infty}^{\infty} \frac{\cos\omega x}{x^2 + a^2} dx = \frac{\pi}{a} e^{-a|\omega|}.$$

With $a = 1, \omega = 3$: $\pi e^{-3} = 0.15641$; numerical quadrature gives 0.156411. ✓ In transform language this is the pair $\dfrac{1}{t^2 + a^2} \leftrightarrow \dfrac{\pi}{a} e^{-a|\omega|}$, the dual of $e^{-a|t|} \leftrightarrow \dfrac{2a}{\omega^2 + a^2}$ in Ch. 16. The kink at $\omega = 0$ in the answer is the residue method's way of switching half-planes.

Worked Example 18.31 — a case needing Jordan's lemma

Evaluate $\displaystyle\int_{-\infty}^{\infty} \frac{x \sin x}{x^2 + 4} dx$.

Here $R(z) = z/(z^2 + 4)$ decays only like $1/R$, so ML fails and Jordan's lemma is essential. Consider $\displaystyle\int \frac{z e^{jz}}{z^2 + 4} dz$. Upper pole $2j$, residue $\dfrac{2j\, e^{j(2j)}}{2 \cdot 2j} = \dfrac{e^{-2}}{2}$. So

$$\int_{-\infty}^{\infty} \frac{x e^{jx}}{x^2 + 4} dx = 2\pi j \cdot \frac{e^{-2}}{2} = j\pi e^{-2}.$$

The imaginary part is the required integral: $\pi e^{-2} = 0.42517$ (numerically 0.425168 ✓). The real part, $\int x\cos x/(x^2 + 4)\,dx = 0$, is consistent with odd symmetry. (Strictly, this integral converges only conditionally; the contour argument produces it as $\lim_{R\to\infty} \int_{-R}^{R}$.)

Worked Example 18.32 — double pole, Fourier type

Evaluate $\displaystyle\int_{-\infty}^{\infty} \frac{\cos 2x}{(x^2+1)^2}\,dx$.

Take $f = e^{j2z}/(z^2+1)^2$, double pole at j. With $g = e^{j2z}/(z+j)^2$:

$$g'(z) = e^{j2z}\left[\frac{2j}{(z+j)^2} - \frac{2}{(z+j)^3}\right], \qquad g'(j) = e^{-2}\left[\frac{2j}{-4} - \frac{2}{-8j}\right] = e^{-2}\left[-\frac{j}{2} - \frac{j}{4}\right] = -\frac{3j}{4}e^{-2}.$$

Integral $= 2\pi j \cdot (-\frac{3j}{4}e^{-2}) = \dfrac{3\pi}{2e^2} = 0.63775$. ✓ (quadrature 0.637752).

Type IV — poles on the real axis: indentation and principal values

If the integrand has a simple pole on the real axis, the real integral does not exist in the ordinary sense, but its **Cauchy principal value** may:

$$\mathrm{PV}\int_{-\infty}^{\infty} f(x)\,dx = \lim_{\varepsilon\to 0^+}\left[\int_{-\infty}^{x_0-\varepsilon} + \int_{x_0+\varepsilon}^{\infty}\right] f(x)\,dx.$$

To compute it, detour around x_0 on a small semicircle γ_ε of radius ε in the upper half-plane.

Indentation (half-residue) lemma

If f has a simple pole at x_0 and γ_ε is the upper semicircle $z = x_0 + \varepsilon e^{j\theta}$ traversed clockwise (θ from π to 0), then

$$\lim_{\varepsilon\to 0}\int_{\gamma_\varepsilon} f(z)\,dz = -\pi j\,\mathrm{Res}_{z=x_0} f.$$

More generally an arc of angle α contributes αj times the residue, with sign set by the direction.

Proof. Near x_0, $f = \dfrac{a_{-1}}{z-x_0} + g(z)$ with g analytic. The g part is bounded and the arc length $\pi\varepsilon \to 0$. The principal-part integral is $a_{-1}\int_{\pi}^{0} \frac{j\varepsilon e^{j\theta}}{\varepsilon e^{j\theta}}\,d\theta = -\pi j\,a_{-1}$ exactly. □

Consequently, with the upper semicircle closed as before, a simple real-axis pole counts as **half** a residue:

$$\mathrm{PV}\int_{-\infty}^{\infty} f\,dx = 2\pi j \sum_{\mathrm{Im}\,z_k>0} \mathrm{Res}\,f + \pi j \sum_{\text{real simple poles}} \mathrm{Res}\,f.$$

Worked Example 18.33 — the Dirichlet integral $\int_0^\infty \frac{\sin x}{x}dx$

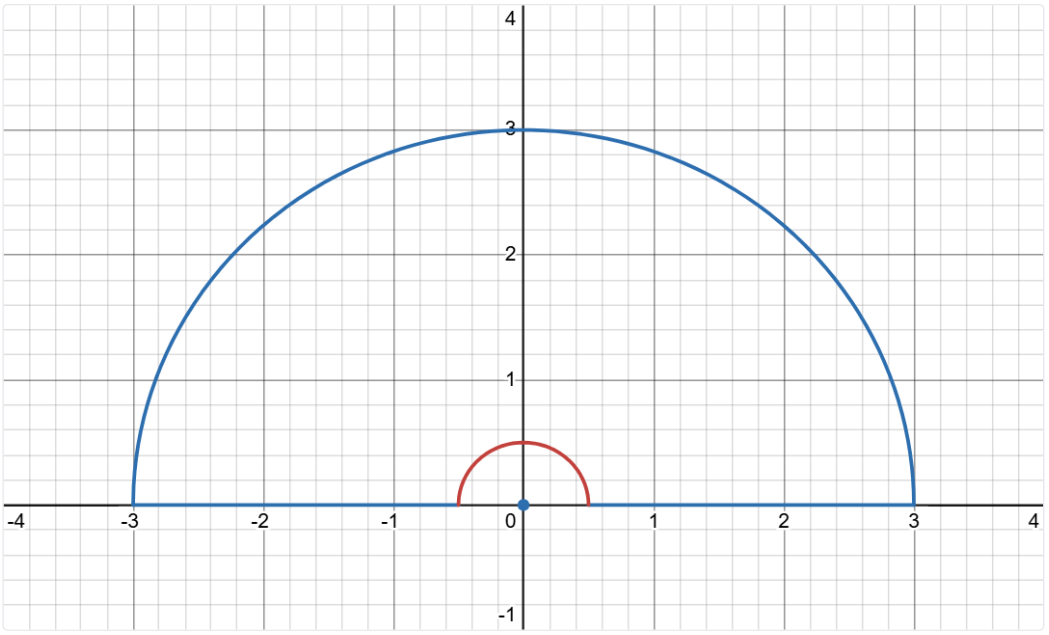
Consider $f(z) = e^{jz}/z$, which has a simple pole at 0 with residue 1 and no other singularities. Use the indented contour: $[-R, -\varepsilon]$, the small clockwise semicircle γ_ε, $[\varepsilon, R]$, and the large semicircle C_R. Nothing is enclosed, so the total is 0:

$$\left[\int_{-R}^{-\varepsilon} + \int_{\varepsilon}^{R}\right]\frac{e^{jx}}{x}dx + \int_{\gamma_\varepsilon} f\,dz + \int_{C_R} f\,dz = 0.$$

By Jordan's lemma ($M_R = 1/R$), $\int_{C_R} \to 0$. By the indentation lemma, $\int_{\gamma_\varepsilon} \to -\pi j$. Hence $\mathrm{PV}\int_{-\infty}^{\infty}\frac{e^{jx}}{x}dx = \pi j$. The real part, $\mathrm{PV}\int \cos x/x\,dx$, is 0 (odd integrand); the imaginary part is an ordinary convergent integral (the singularity of $\sin x/x$ is removable):

$$\int_{-\infty}^{\infty}\frac{\sin x}{x}dx = \pi, \qquad \int_0^{\infty}\frac{\sin x}{x}dx = \frac{\pi}{2}.$$

This is the area under the unnormalized sinc — the integral behind the Fourier transform of the rectangular pulse, and the limiting value $\mathrm{Si}(\infty) = \pi/2$ of the sine integral that governs the Gibbs overshoot in Ch. 15. The answer π agrees with numerical quadrature to all printed digits.



The indented contour for Example 18.33: the small red semicircle detours above the pole at the origin, so the pole lies outside the closed contour, and in the limit the detour contributes $-\pi j$ times the residue.

Branch cuts (keyhole contours), briefly. Integrals such as $\int_0^\infty \frac{x^{a-1}}{1+x}dx$ $(0<a<1)$ use a multivalued integrand z^{a-1} with its cut along the positive real axis. The *keyhole* contour runs out along the top of the cut, around a large circle, back along the bottom of the cut (where z^{a-1} has gained a factor $e^{j2\pi(a-1)}$) and around a small circle at the origin. The two cut integrals combine to $(1-e^{j2\pi a})I$; the residue at $z=-1=e^{j\pi}$ is $e^{j\pi(a-1)}$; solving gives

$$\int_0^\infty \frac{x^{a-1}}{1+x}dx = \frac{\pi}{\sin \pi a}.$$

For $a=\frac{1}{2}$ this is π; for $a=\frac{2}{3}$ it is $2\pi/\sqrt{3}=3.6276$, as numerical quadrature confirms.

Common Pitfall — closing on the wrong side

For $e^{j\omega z}$ with $\omega>0$, close in the **upper** half-plane, where $|e^{j\omega z}|=e^{-\omega y}$ decays. Closing in the lower half-plane makes the arc integral blow up and gives a wrong (often complex) answer. In inverse-transform applications this is exactly the choice that produces causality, as the next section shows. And never integrate $\cos \omega z$ directly over a semicircle: it contains both $e^{j\omega z}$ and $e^{-j\omega z}$, one of which grows on each side.

Engineering Note — why these integrals matter

Type I integrals are averages over one period of a periodic quantity (mean power in a phase-modulated signal, the response of a ring resonator over a revolution). Type II integrals give the total noise power at a filter output: for white noise into $H(s)$, the output variance is proportional to $\frac{1}{2\pi}\int |H(j\omega)|^2 d\omega$, a rational integral solved by residues (the basis of tabulated "I_n" integrals in control texts). Type III integrals are inverse Fourier transforms; Type IV, with its half-residues, reappears as the $\pi\delta(\omega)$ terms in the transforms of the step and the signum.

18.9 Applications to engineering analysis

This section collects the places where the machinery of Sections 18.5–18.8 underpins the transform methods of Part V. The theme is the same throughout: an engineering quantity is a contour integral, the contour can be closed or deformed, and the answer is a sum of residues or a count of zeros and poles.

The Bromwich inversion integral for the Laplace transform

Ch. 14 inverted Laplace transforms with tables and partial fractions. The general inversion formula is a contour integral.

Bromwich (complex inversion) integral

Let $F(s)=\mathcal{L}\{f(t)\}$ converge for $\operatorname{Re} s>\sigma_0$, and let $\gamma>\sigma_0$. Then, at points where f is continuous,

$$f(t)=\frac{1}{2\pi j}\int_{\gamma-j\infty}^{\gamma+j\infty} F(s)\,e^{st}\,ds = \frac{1}{2\pi j}\lim_{W\to\infty}\int_{\gamma-jW}^{\gamma+jW} F(s)e^{st}ds,$$

the path being the vertical line $\operatorname{Re} s=\gamma$ (the **Bromwich contour**), which lies to the right of every singularity of F. At a jump of f the integral gives the midpoint value.

Derivation from Fourier inversion. For fixed γ in the region of convergence, $g(t)=f(t)e^{-\gamma t}u(t)$ is absolutely integrable, and its Fourier transform is $G(\omega)=\int_0^\infty f(t)e^{-(\gamma+j\omega)t}dt = F(\gamma+j\omega)$. The Fourier inversion formula of Ch. 16 gives

$$f(t)e^{-\gamma t}=\frac{1}{2\pi}\int_{-\infty}^{\infty} F(\gamma+j\omega)e^{j\omega t}d\omega.$$

Multiply by $e^{\gamma t}$ and substitute $s = \gamma + j\omega$, $ds = j\,d\omega$: the result is the Bromwich integral. □ So the Laplace inversion formula is nothing more than Fourier inversion along a line shifted into the region of convergence. The value of γ does not matter, as long as the line stays right of all singularities — by Cauchy's theorem the integrand is analytic between any two such lines.

Evaluation by residues. Suppose F is rational and strictly proper (or, more generally, $|F(s)| \to 0$ uniformly as $|s| \to \infty$ in the left region). Close the line with a large arc Γ_R to the left.

- For $t > 0$: $|e^{st}| = e^{t\,\mathrm{Re}\,s}$ is small where $\mathrm{Re}\,s$ is very negative. A rotated form of Jordan's lemma (substitute $s = jz$; the left half-plane becomes the upper half-plane and $e^{st} = e^{jtz}$) shows $\int_{\Gamma_R} F(s)e^{st}ds \to 0$. The closed contour encloses **all** the poles of F.
- For $t < 0$: close to the **right** instead, where e^{st} now decays. No singularities lie right of the Bromwich line, so $f(t) = 0$ — **causality emerges from analyticity in the right half-plane**.

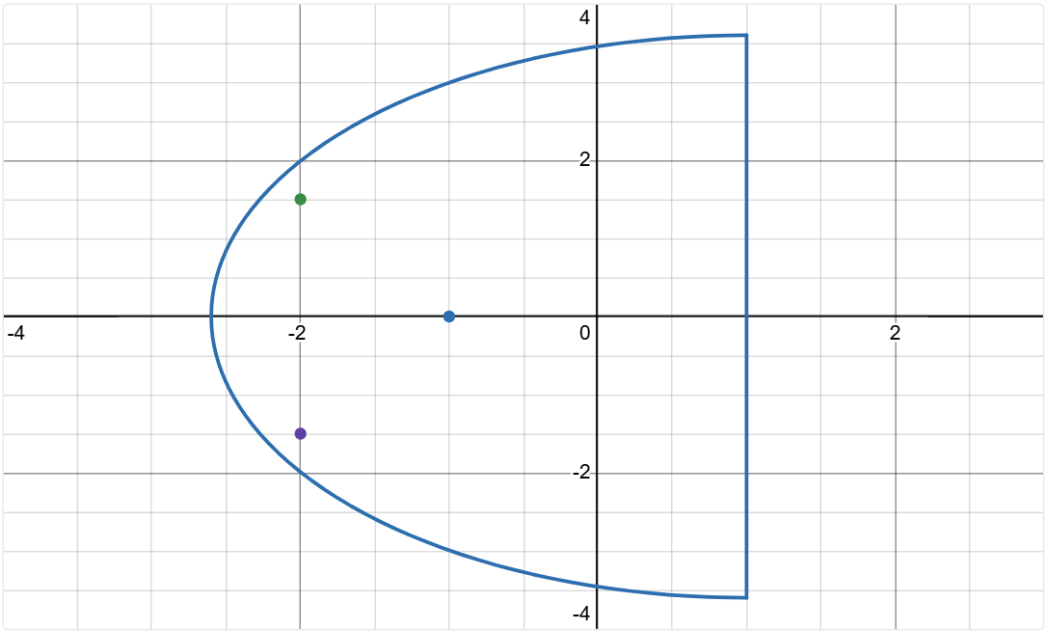
Inversion by residues

If $F(s)$ is rational and strictly proper, with poles $p_1, \dots, p_n$, then

$$f(t) = \sum_{k=1}^{n} \mathrm{Res}_{s=p_k}\left[F(s)e^{st}\right], \qquad t > 0,$$

and $f(t) = 0$ for $t < 0$. At a simple pole p the term is $\left[(s-p)F(s)\right]_{s=p} e^{pt}$; at a pole of order m it is $\dfrac{1}{(m-1)!}\dfrac{d^{m-1}}{ds^{m-1}}\left[(s-p)^m F(s)e^{st}\right]_{s=p}$, which produces the $t^k e^{pt}$ terms automatically.

This one formula replaces the table look-up and the partial-fraction step together: the derivative of e^{st} with respect to s is te^{st}, which is where the polynomial-times-exponential terms of repeated poles come from.



The Bromwich contour: the vertical line $\mathrm{Re}\,s = \gamma$ (here $\gamma = 1$) to the right of all poles, closed for $t > 0$ by a large arc to the left, which by Jordan's lemma contributes nothing as its radius grows; the enclosed residues of $F(s)e^{st}$ give $f(t)$.

Worked Example 18.34 — simple and repeated real poles

Invert (a) $F(s) = \dfrac{1}{s(s+1)^2}$ and (b) $F(s) = \dfrac{s+3}{(s+1)(s+2)^2}$.

(a) At $s = 0$ (simple): $\left[\dfrac{e^{st}}{(s+1)^2}\right]_{s=0} = 1$. At $s = -1$ (order 2):

$$\frac{d}{ds}\left[\frac{e^{st}}{s}\right]_{s=-1} = \left[\frac{te^{st}}{s} - \frac{e^{st}}{s^2}\right]_{s=-1} = -te^{-t} - e^{-t}.$$

So $f(t) = 1 - e^{-t} - te^{-t}$ for $t \geq 0$. Checks: $f(0) = 0$ (consistent with the initial-value theorem, $\lim sF = 0$); $f(\infty) = 1 = \lim_{s\to 0} sF$; at $t = 2$, $f = 1 - 3e^{-2} = 0.59399$, matching a numerical Bromwich inversion. ✓

(b) At $s = -1$: $\left[\dfrac{(s+3)e^{st}}{(s+2)^2}\right]_{-1} = 2e^{-t}$. At $s = -2$ (order 2), with $G(s) = \dfrac{(s+3)e^{st}}{s+1}$:

$$G'(s) = \frac{\left[e^{st} + t(s+3)e^{st}\right](s+1) - (s+3)e^{st}}{(s+1)^2}, \qquad G'(-2) = \frac{(1+t)e^{-2t}(-1) - e^{-2t}}{1} = -(2+t)e^{-2t}.$$

So $f(t) = 2e^{-t} - 2e^{-2t} - te^{-2t}$ — the time-domain image of the partial fractions of Example 18.23. At $t = 1.3$: 0.29996, matching numerical inversion. ✓

Worked Example 18.35 — a complex-conjugate pair

Invert $F(s) = \dfrac{1}{s^2 + 2s + 5}$.

Poles $p, \bar{p} = -1 \pm 2j$. At $p = -1 + 2j$: $\dfrac{e^{pt}}{p - \bar{p}} = \dfrac{e^{(-1+2j)t}}{4j}$. The residue at $\bar{p}$ is the conjugate (because F is real on the real axis), so

$$f(t) = 2\,\mathrm{Re}\left\{\frac{e^{(-1+2j)t}}{4j}\right\} = \frac{e^{-t}}{2}\,\mathrm{Re}\{-je^{j2t}\} = \frac{1}{2}e^{-t}\sin 2t.$$

In general a conjugate pair contributes $2\,\mathrm{Re}\{\mathrm{Res}_p\}$ — a damped sinusoid whose amplitude and phase are the modulus and angle of the residue. At $t = 1.3$: 0.070245. ✓

Common Pitfall — when the arc does not vanish

The residue-sum formula needs $F(s) \to 0$ on the left arc. It fails for an improper F (e.g. $F = s/(s+1) = 1 - 1/(s+1)$, whose constant part is the impulse $\delta(t)$ and must be split off first), and it must be adapted for a delay $e^{-sT}F(s)$: then $e^{s(t-T)}$ decays on the left arc only for $t > T$, so the residues give $f(t-T)$ for $t > T$ and closing right gives 0 for $t < T$ — the shift theorem recovered. Functions with branch points, such as $F = 1/\sqrt{s}$ (inverse $1/\sqrt{\pi t}$), need a contour that wraps around the branch cut on the negative real axis rather than a residue sum.

Inverse Z-transform by residues

The same idea inverts Z-transforms (Ch. 17). Since $X(z) = \sum_n x[n]z^{-n}$ is a Laurent series converging in the ROC annulus, the Laurent coefficient formula gives

$$x[n] = \frac{1}{2\pi j}\oint_C X(z)\,z^{n-1}\,dz = \sum_{\text{poles inside } C} \mathrm{Res}\left[X(z)z^{n-1}\right],$$

where C is a counter-clockwise circle in the ROC. For a causal sequence (ROC outside the outermost pole) C encloses all poles of X — but watch for an extra pole of z^{n-1} at $z = 0$ when n is small.

Worked Example 18.36 — causal inverse Z-transform

Find $x[n]$ for $X(z) = \dfrac{z}{(z - 0.5)(z - 1)}$, $|z| > 1$.

$X(z)z^{n-1} = \dfrac{z^n}{(z-0.5)(z-1)}$, which has no pole at $z = 0$ for $n \geq 0$. Residues: at $z = 1$, $\dfrac{1}{0.5} = 2$; at $z = 0.5$, $\dfrac{0.5^n}{-0.5} = -2(0.5)^n$. So

$$x[n] = \big(2 - 2(0.5)^n\big)u[n] = 0,\ 1,\ 1.5,\ 1.75,\ 1.875, \ldots$$

Check: $\sum_n x[n]z^{-n}$ at $z = 2$ sums to 1.3333, equal to $X(2) = 2/(1.5 \cdot 1)$. ✓ For $n < 0$ the integrand acquires a pole at $z = 0$ of order $-n$ whose residue cancels the others exactly, giving $x[n] = 0$ — causality again, provable more easily via the residue at infinity.

The argument principle

The final group of applications counts zeros and poles without finding them.

Argument principle

Let f be analytic inside and on a positively oriented simple closed contour C except for finitely many poles inside C, and let f have no zeros or poles on C. Then

$$\frac{1}{2\pi j}\oint_C \frac{f'(z)}{f(z)}dz = Z - P,$$

where Z and P are the numbers of zeros and poles of f inside C, each counted with multiplicity. Equivalently, $Z - P = \dfrac{1}{2\pi}\Delta_C \arg f(z)$: the net number of times the image curve $f(C)$ winds counter-clockwise around the origin.

Proof. The logarithmic derivative f'/f is analytic except at zeros and poles of f. Near a zero of order m, $f = (z - z_0)^m g$ with $g(z_0) \neq 0$, so $f'/f = \dfrac{m}{z - z_0} + \dfrac{g'}{g}$: a simple pole with residue m. Near a pole of order m, $f = (z - z_0)^{-m} g$, and the residue of f'/f is $-m$. The residue theorem gives $Z - P$. For the second form, note $f'/f = \frac{d}{dz}\log f$, so $\oint_C f'/f\,dz$ is the change of $\log f = \ln|f| + j\arg f$ around C; $\ln|f|$ returns to its starting value and $\arg f$ changes by $2\pi(Z - P)$. □

The second form is the useful one: **draw the image of C under f and count how many times it encircles the origin.** That is a graphical computation — and exactly what a Nyquist plot is.

Rouché's theorem

If f and g are analytic inside and on C and $|g(z)| < |f(z)|$ on C, then f and $f + g$ have the same number of zeros inside C.

(Intuition: on C, $f + g = f(1 + g/f)$ with $|g/f| < 1$, so $1 + g/f$ stays in the disc $|w - 1| < 1$ and cannot wind around 0; the winding of $f + g$ equals that of f. This is the "dog on a leash" argument.)

Worked Example 18.37 — locating polynomial roots

How many roots of $p(z) = z^5 + 3z + 1$ lie in $|z| < 1$, and how many in $1 < |z| < 2$?

On $|z| = 1$: take $f = 3z$, $g = z^5 + 1$. Then $|g| \le 2 < 3 = |f|$, so p has as many zeros in the unit disc as $3z$: **one**.

On $|z| = 2$: take $f = z^5$, $g = 3z + 1$. Then $|g| \le 7 < 32 = |f|$, so p has five zeros in $|z| < 2$.

Hence four roots lie in the annulus $1 < |z| < 2$. Numerically the roots have moduli 0.332, 1.263 (twice) and 1.374 (twice). ✓ The integral $\frac{1}{2\pi j}\oint_{|z|=1} p'/p\,dz$ evaluated by quadrature gives 1.0000. ✓ For a discrete-time characteristic polynomial, the same count on the unit circle decides stability: all roots must lie inside.

The Nyquist stability criterion — the idea

A feedback loop with loop transfer function $L(s)$ (forward path times feedback path) has closed-loop transfer function $T(s) = \dfrac{G(s)}{1 + L(s)}$. Its poles are the **zeros of** $1 + L(s)$. The closed loop is stable when $1 + L$ has no zeros in the closed right half-plane. Computing those zeros means factoring a polynomial; Nyquist's insight is to *count* them with the argument principle using only the frequency response $L(j\omega)$, which can be measured.

1. **The contour.** Take the *Nyquist contour* D: up the imaginary axis from $-jR$ to jR, then around the large semicircle $Re^{j\theta}$ from $\theta = \pi/2$ to $-\pi/2$. It encloses the right half-plane and is traversed **clockwise**. (If L has poles on the $j\omega$-axis, indent around them with small semicircles into the left half-plane.)
2. **Apply the argument principle to $F(s) = 1 + L(s)$.** Inside D, let Z = number of zeros of $1 + L$ (closed-loop RHP poles, unknown) and P = number of poles of $1 + L$, which are the poles of L (open-loop RHP poles, known from the model). Because D is clockwise, the image $F(D)$ encircles the origin $N = Z - P$ times **clockwise**.
3. **Shift the origin.** $F(D)$ encircling 0 is the same as $L(D) = F(D) - 1$ encircling -1. For strictly proper L, the large semicircle maps to the single point $L(\infty) = 0$, so $L(D)$ is just the Nyquist plot $L(j\omega)$, $-\infty < \omega < \infty$.

Nyquist criterion

Let N be the number of clockwise encirclements of the point -1 by the Nyquist plot $L(j\omega)$, $-\infty < \omega < \infty$, and P the number of open-loop poles of L in the right half-plane. The number of closed-loop poles in the right half-plane is

$$Z = N + P.$$

The closed loop is stable if and only if $Z = 0$, i.e. the plot encircles -1 exactly P times **counter-clockwise**. If L is open-loop stable ($P = 0$), stability requires no encirclement of -1.

Worked Example 18.38 — critical gain from the Nyquist plot

Let $L(s) = \dfrac{K}{(s+1)^3}$ with $K > 0$. Find the range of K for closed-loop stability.

All open-loop poles are at -1, so $P = 0$ and we need no encirclement of -1. On the imaginary axis,

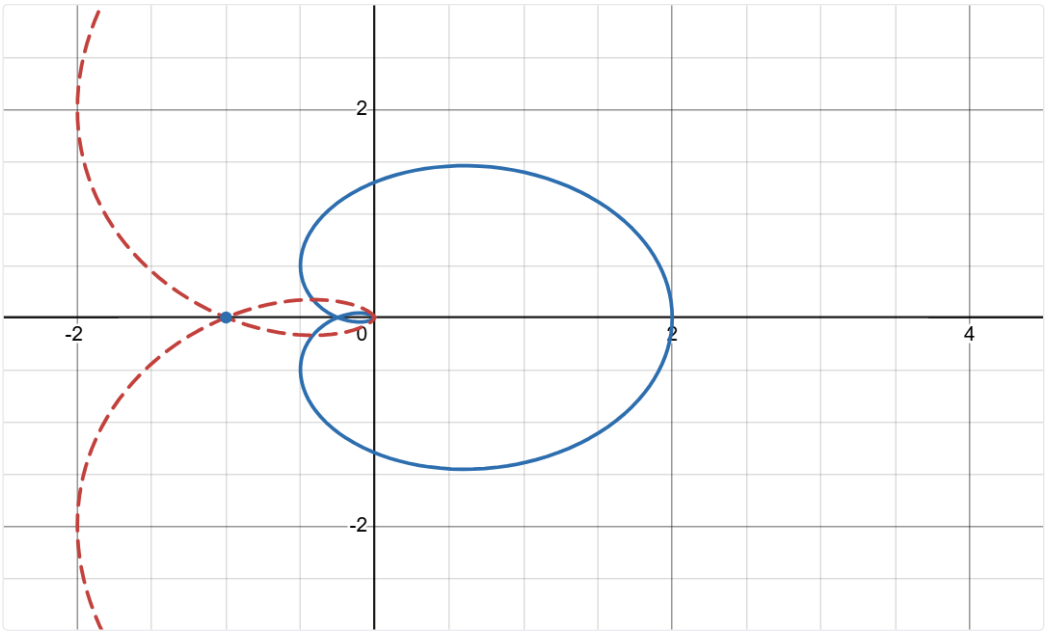
$$L(j\omega) = \frac{K}{(1 + j\omega)^3} = \frac{K}{(1+\omega^2)^{3/2}} e^{-j3\arctan\omega}.$$

At $\omega = 0$ the plot starts at K on the positive real axis; as $\omega \to \infty$ the phase tends to $-3\pi/2$ and the magnitude to 0. It crosses the negative real axis when the phase is $-\pi$: $3\arctan\omega = \pi$, $\omega_{180} = \tan(\pi/3) = \sqrt{3}$. There $|L| = K/(1+3)^{3/2} = K/8$, so the crossing point is $-K/8$.

- If $K < 8$, the crossing lies between -1 and 0: no encirclement, $Z = 0$, **stable**. The gain margin is $8/K$.
- If $K > 8$, the crossing lies left of -1; the plot (with its mirror image for $\omega < 0$) encircles -1 twice clockwise, so $Z = 2$: two closed-loop poles in the right half-plane.
- At $K = 8$ the plot passes through -1: closed-loop poles on the $j\omega$-axis at $\pm j\sqrt{3}$.

Check by algebra: the closed-loop characteristic polynomial is $(s+1)^3 + K = s^3 + 3s^2 + 3s + 1 + K$, and the Routh condition $3 \cdot 3 > 1 + K$ gives $K < 8$. With $K = 8$ its roots are -3 and $\pm j\sqrt{3} = \pm 1.732j$. ✓

Writing $L(j\omega)$ in polar form, $r = |L|$ and $\theta = \angle L = -3\arctan\omega$, we have $1/\sqrt{1+\omega^2} = \cos(\theta/3)$, so the whole Nyquist plot is the polar curve $r = K\cos^3(\theta/3)$, $|\theta| \le 3\pi/2$.



Nyquist plots of $L(s) = K/(s+1)^3$: for $K = 2$ (blue) the curve crosses the negative real axis at -0.25 and does not encircle -1 (stable); for $K = 8$ (red, dashed; only the part near the origin is visible) the crossing is exactly at -1, the edge of instability. Larger K scales the curve outward until it encircles -1.

Engineering Note — why Nyquist beats root-finding

The criterion needs only $L(j\omega)$, which can be **measured** with a network analyser on real hardware, including pure delays e^{-sT} (which make $1 + L$ non-polynomial and defeat Routh's test). Gain and phase margins are read directly as distances of the plot from -1. The mathematics underneath is entirely the argument principle: closed-loop poles are counted by winding numbers.

Common Pitfall — orientation and the open-loop count

Two sign conventions collide here: the Nyquist contour is traversed clockwise, and textbooks count encirclements clockwise as positive, giving $Z = N + P$. If you count counter-clockwise encirclements as positive the formula becomes $Z = P - N_{\text{ccw}}$. Also, P counts **open-loop RHP poles**, not zeros; forgetting an unstable open-loop pole (e.g. an inverted pendulum or a magnetic levitator) leads to the wrong conclusion even when the plot is drawn correctly.

Gain–phase relations (Kramers–Kronig), briefly

Let $H(\omega)$ be the frequency response of a causal, stable system, considered as a function of complex ω; causality makes it analytic in the lower half ω-plane for the $e^{j\omega t}$ convention ($s = j\omega$ in the right half s-plane), and suppose $H \to 0$ as $|\omega| \to \infty$. Integrate $H(\zeta)/(\zeta - \omega)$ around the real axis indented at $\zeta = \omega$ and closed in the half-plane of analyticity. There are no enclosed singularities, the large arc vanishes, and the indentation contributes πj times the residue $H(\omega)$ — exactly the Type IV computation of Example 18.33. The result is

$$H(\omega) = \frac{j}{\pi}\,\mathrm{PV}\int_{-\infty}^{\infty}\frac{H(\zeta)}{\zeta-\omega}d\zeta \quad\Longrightarrow\quad \operatorname{Re}H(\omega) = -\frac{1}{\pi}\,\mathrm{PV}\int_{-\infty}^{\infty}\frac{\operatorname{Im}H(\zeta)}{\zeta-\omega}d\zeta.$$

(The closed contour runs clockwise, and the small indentation passes below $\zeta = \omega$, so it turns counter-clockwise about the pole and contributes $+\pi j H(\omega)$; the PV integral plus this term is zero.) Check with $H = 1/(1 + j\omega)$: $\operatorname{Im}H = -\zeta/(1+\zeta^2)$, and the right side at $\omega = 0$ is $\frac{1}{\pi}\int_{-\infty}^{\infty}\frac{d\zeta}{1+\zeta^2} = 1 = \operatorname{Re}H(0)$. ✓ The real and imaginary parts of a causal frequency response are **Hilbert transforms** of each other: specifying one determines the other. Applied to $\log H = \ln|H| + j\angle H$ for minimum-phase systems (no RHP zeros, so $\log H$ is analytic too), this is Bode's gain–phase relation, the reason a slope of -20 dB/decade goes with about -90° of phase.

Summary

Topic	Key formula or statement
Complex function	$f(z) = u(x,y) + jv(x,y)$
Exponential, logarithm	$e^z = e^x(\cos y + j\sin y)$; $\operatorname{Log} z = \ln\lvert z\rvert + j\operatorname{Arg} z$, $\operatorname{Arg}\in(-\pi,\pi]$
Complex power	$z^\alpha = e^{\alpha \log z}$ (principal value uses Log)
Derivative	$f'(z_0) = \lim_{\Delta z\to 0}[f(z_0+\Delta z) - f(z_0)]/\Delta z$, direction-independent
Cauchy–Riemann	$u_x = v_y,\ u_y = -v_x$; $f' = u_x + jv_x$
Polar C–R	$u_r = v_\theta/r,\ v_r = -u_\theta/r$
Harmonic parts	$\nabla^2 u = \nabla^2 v = 0$; level curves of u, v orthogonal
Conformality	angles preserved where $f' \neq 0$; local scale $\lvert f'\rvert$, area scale $\lvert f'\rvert^2$
Contour integral	$\int_C f\,dz = \int_a^b f(z(t))z'(t)\,dt$
Fundamental integral	$\oint_{\lvert z-z_0\rvert=\rho}(z-z_0)^n dz = 2\pi j$ if $n=-1$, else 0

Topic	Key formula or statement
ML inequality	$\left\vert\int_C f\,dz\right\vert \le ML$
Cauchy's theorem	$\oint_C f\,dz = 0$ for f analytic inside and on C
Cauchy integral formula	$f^{(n)}(z_0) = \dfrac{n!}{2\pi j}\oint_C \dfrac{f(z)}{(z-z_0)^{n+1}}dz$
Cauchy estimate, Liouville	$\vert f^{(n)}(z_0)\vert \le n!M/\rho^n$; bounded entire $\Rightarrow$ constant
Taylor series	$a_n = f^{(n)}(z_0)/n!$; radius = distance to nearest singularity
Laurent series	$f = \sum_{n=-\infty}^{\infty} a_n (z-z_0)^n$ in $r_1 < \vert z-z_0\vert < r_2$; $a_n = \frac{1}{2\pi j}\oint \frac{f}{(z-z_0)^{n+1}}dz$
Singularities	removable / pole of order m / essential, by the principal part
Residue, simple pole	$\lim_{z\to z_0}(z-z_0)f(z) = g(z_0)/h'(z_0)$
Residue, order m	$\frac{1}{(m-1)!}\lim \frac{d^{m-1}}{dz^{m-1}}[(z-z_0)^m f]$
Residue theorem	$\oint_C f\,dz = 2\pi j \sum_{\text{inside}} \operatorname{Res} f$
Trig integrals	$z = e^{j\theta}$, $\cos\theta = \frac{z+z^{-1}}{2}$, $\sin\theta = \frac{z-z^{-1}}{2j}$, $d\theta = \frac{dz}{jz}$; residues in $\vert z\vert < 1$
Rational integrals	$\int_{-\infty}^{\infty} \frac{P}{Q}dx = 2\pi j \sum_{\operatorname{Im} z_k>0} \operatorname{Res}$, $\deg Q \ge \deg P + 2$
Jordan's lemma	$\int_{C_R} R(z)e^{j\omega z}dz \to 0$ ($\omega > 0$, $R\to 0$ on C_R)
Fourier-type integrals	$\int_{-\infty}^{\infty} R(x)e^{j\omega x}dx = 2\pi j \sum_{\operatorname{Im} z_k>0} \operatorname{Res}[Re^{j\omega z}]$, $\omega > 0$
Indentation	simple real pole contributes πj Res to the PV
Bromwich integral	$f(t) = \frac{1}{2\pi j}\int_{\gamma-j\infty}^{\gamma+j\infty} F(s)e^{st}ds = \sum \operatorname{Res}[F(s)e^{st}]$, $t > 0$
Inverse Z-transform	$x[n] = \frac{1}{2\pi j}\oint_C X(z)z^{n-1}dz = \sum_{\text{inside}} \operatorname{Res}[X(z)z^{n-1}]$
Argument principle	$\frac{1}{2\pi j}\oint_C \frac{f'}{f}dz = Z - P = \frac{1}{2\pi}\Delta_C \arg f$
Nyquist criterion	$Z = N + P$ (clockwise encirclements of -1); stable iff $Z = 0$

Further reading

- E. Kreyszig, *Advanced Engineering Mathematics*, 10th ed., Wiley, 2011 — Part D (complex analysis), with conformal mapping and potential theory.
- J. W. Brown and R. V. Churchill, *Complex Variables and Applications*, 9th ed., McGraw-Hill, 2014 — the standard undergraduate text; thorough on branches, residues and real integrals.
- E. B. Saff and A. D. Snider, *Fundamentals of Complex Analysis with Applications to Engineering, Science, and Mathematics*, 3rd ed., Pearson, 2003 — engineering-oriented, including the argument principle and transform inversion.
- T. Needham, *Visual Complex Analysis*, Oxford University Press, 1997 — geometric intuition for mappings, conformality and winding numbers.

Part VI — Probability, Statistics, and Stochastic Mathematics

Chapter 19 — Probability

> **Prerequisites:** Ch. 1 (sets, functions, sums), Ch. 5 (integral calculus), Ch. 6 (multiple integrals), Ch. 16 (Fourier transforms, for the characteristic function)
> **Learning outcomes:** After this chapter you can …
>
> - build a probability model — sample space, events, probability measure — from the axioms, and count outcomes with permutations and combinations;
> - compute conditional probabilities, apply the law of total probability, and invert conditioning with Bayes' theorem (binary channels, fault detectors, MAP decisions);
> - decide whether events are independent and compute the reliability of series, parallel, k-out-of-n and bridge systems;
> - describe discrete, continuous and mixed random variables by their PMF, PDF and CDF, and derive the distribution of a function of a random variable;
> - compute expectations, variances and higher moments, use the Markov and Chebyshev bounds, and obtain moments from the moment-generating and characteristic functions;
> - set up a joint distribution of two random variables, find marginals and conditionals, and compute the distribution of a sum of independent random variables.

Every measurement an engineer makes is uncertain. The voltage across a resistor fluctuates because electrons are in thermal motion; a bit sent over a wireless link is sometimes received in error; a batch of capacitors has values scattered around the nominal; a power-supply component fails after an unpredictable number of hours; packets arrive at a router at irregular instants. None of these can be predicted exactly, but all of them are *regular in the aggregate*: the fraction of bits in error over a long run settles to a stable number, the histogram of capacitor values has a repeatable shape, the average lifetime of a large population of components is well defined. **Probability theory** is the mathematics of that aggregate regularity. It lets us replace "we cannot know" by quantitative statements — "the bit-error rate is 10^{-6}", "the system survives five years with probability 0.98", "the noise exceeds the threshold 0.3 % of the time" — on which designs can be based.

This chapter builds the theory from the ground up. Sections 19.1–19.4 work with *events*: the axioms, counting, conditioning, Bayes' theorem and independence. Sections 19.5–19.8 introduce *random variables* — numerical outcomes — and the three functions that describe them (PMF, PDF, CDF), together with the most important families of distributions and the first ideas of joint distributions. Section 19.9 condenses a distribution into a few numbers: the mean, the variance, the higher moments, and the transforms (moment-generating and characteristic functions) that generate them. Ch. 20 turns the theory around — from a model to data, and from data back to a model — and catalogues the common distributions in more depth; Ch. 21 lets the random variable depend on time.

Throughout, outcomes are denoted ζ, events by capital letters $A, B, \ldots$, random variables by capital letters $X, Y, \ldots$ and their values by the corresponding lowercase letters $x, y, \ldots$.

19.1 Sample spaces and events

Motivation

Before we can attach numbers to uncertainty we must say precisely *what* is uncertain. "The probability that the receiver makes an error" is meaningless until we specify the experiment (one bit? one packet? one hour of operation?), list what can happen, and agree which collections of outcomes we want to talk about. This bookkeeping is the sample space and its events. Getting it right removes most of the paradoxes and wrong answers that plague informal probability arguments.

Experiments, outcomes and sample spaces

Random experiment, outcome, sample space

A **random experiment** is a procedure whose result cannot be predicted with certainty but whose set of possible results is known in advance. Each possible result is an **outcome** ζ. The set of all outcomes is the **sample space** Ω (some texts write S).

The outcomes must be **mutually exclusive** (exactly one occurs on each trial) and **collectively exhaustive** (one of them always occurs). How finely to describe the outcomes is a modelling choice: for a single coin toss $\Omega = \{H, T\}$ suffices, but if you care about where the coin lands you need a richer description.

Sample spaces come in three kinds:

Kind	Example experiment	Sample space
Finite	Send a 3-bit word and record which bits are in error	$\Omega = \{000, 001, \ldots, 111\}$, 8 outcomes (1 = bit in error)
Countably infinite	Retransmit a packet until it is received correctly; record the number of attempts	$\Omega = \{1, 2, 3, \ldots\}$
Uncountable (continuum)	Measure the lifetime of a capacitor in hours	$\Omega = [0, \infty)$
Uncountable, multidimensional	Sample the in-phase and quadrature noise voltages of a receiver	$\Omega = \mathbb{R}^2$

Events and set operations

Event

An **event** is a subset $A \subseteq \Omega$ of outcomes. We say the event A **occurs** on a trial if the outcome ζ of that trial satisfies $\zeta \in A$. The whole space Ω is the **certain event**; the empty set $\varnothing$ is the **impossible event**; a single outcome $\{\zeta\}$ is an **elementary event**.

Because events are sets, the algebra of sets (Ch. 1) becomes the logic of events:

Set notation	Read as	Event occurs when …
$A \cup B$	A or B (union)	at least one of A, B occurs
$A \cap B$ (also AB)	A and B (intersection)	both occur
A^c (also $\bar{A}$)	not A (complement)	A does not occur
$A \setminus B = A \cap B^c$	A but not B	A occurs and B does not
$A \subseteq B$	A implies B	whenever A occurs, B occurs
$A \cap B = \varnothing$	A, B mutually exclusive (disjoint)	they never occur together

The rules that matter most in calculation are the **distributive laws** and **De Morgan's laws**:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C), \qquad (A \cup B)^c = A^c \cap B^c, \qquad (A \cap B)^c = A^c \cup B^c.$$

De Morgan's laws extend to any number of events: "not (at least one occurs)" is "none occurs", $\left(\bigcup_i A_i\right)^c = \bigcap_i A_i^c$. This is the single most useful identity in reliability and error analysis, because "at least one component fails" is usually hard to compute directly while "no component fails" is easy.

A collection of events $B_1, B_2, \dots, B_n$ is a **partition** of Ω if the events are pairwise disjoint and their union is Ω. Exactly one event of a partition occurs on each trial. Partitions are the scaffolding of the law of total probability (Section 19.2).

Worked Example 19.1 — Events in a 3-bit word

A 3-bit word is transmitted; outcome $\zeta = b_1 b_2 b_3$ with $b_i = 1$ if bit i is received in error. Let A = "bit 1 is in error", B = "exactly one bit is in error", C = "the word is received correctly".

Step 1 — list the sets. $A = \{100, 101, 110, 111\}$, $B = \{100, 010, 001\}$, $C = \{000\}$.

Step 2 — combine. $A \cap B = \{100\}$ ("only bit 1 is wrong"); $A \cup B = \{100, 101, 110, 111, 010, 001\}$; C^c = "at least one error" $= \Omega \setminus \{000\}$, which contains 7 outcomes.

Step 3 — check disjointness. $B \cap C = \varnothing$: a word cannot have exactly one error and no error. $A \cap C = \varnothing$ as well. The events "0 errors", "1 error", "2 errors", "3 errors" form a partition of Ω into sets of sizes $1, 3, 3, 1$.

Which subsets are events? (σ-algebras)

For finite and countable Ω every subset can be an event. For a continuum such as $\Omega = [0, 1]$ it turns out to be impossible to assign a probability to *every* subset consistently with the axioms below (a classical result of measure theory). We therefore restrict attention to a family $\mathcal{F}$ of subsets, the **events**, that is closed under the operations we need.

σ-algebra (event field)

A family $\mathcal{F}$ of subsets of Ω is a **σ-algebra** if

1. $\Omega \in \mathcal{F}$;
2. $A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}$;
3. $A_1, A_2, \dots \in \mathcal{F} \Rightarrow \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$.

By De Morgan's laws it is then also closed under countable intersections. On $\mathbb{R}$ the standard choice is the **Borel** σ-algebra, the smallest one containing all intervals; it contains every set an engineer will ever write down.

For practical work you may treat "event" as "any subset you can describe". The σ-algebra is there to guarantee that limits of events — "the error count *eventually* stays below 5", "the voltage *ever* exceeds 1 V" — are themselves events.

The axioms of probability

Probability was put on a rigorous footing by Kolmogorov (1933), who took the following three properties as axioms and derived everything else from them.

Probability space (Kolmogorov axioms)

A **probability space** is a triple $(\Omega, \mathcal{F}, P)$ where P assigns to every event $A \in \mathcal{F}$ a real number $P(A)$ such that

1. **Non-negativity:** $P(A) \geq 0$;
2. **Normalization:** $P(\Omega) = 1$;
3. **Countable additivity:** if $A_1, A_2, \dots$ are pairwise disjoint, then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

The axioms do not say *what* number to assign to an event; they only constrain the assignment to be internally consistent. The numbers come from modelling: symmetry ("all 36 outcomes of two dice are equally likely"), physics (Boltzmann statistics of thermal noise), or measurement (a long-run relative frequency of failures). The **relative-frequency interpretation** — if the experiment is repeated N times and A occurs N_A times, then $N_A/N \to P(A)$ as $N \to \infty$ — is what connects the axioms to the laboratory, and it is a *theorem* of the theory (the law of large numbers, Section 19.9 and Ch. 20) rather than a definition.

Consequences of the axioms

Every rule of probability you will use follows from the three axioms by short arguments.

Elementary properties

For events A, B in a probability space:

1. $P(\varnothing) = 0$.
2. $P(A^c) = 1 - P(A)$.
3. If $A \subseteq B$ then $P(A) \le P(B)$ (monotonicity); in particular $0 \le P(A) \le 1$.
4. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ (addition rule).
5. $P(A \cup B) \le P(A) + P(B)$, and more generally $P\left(\bigcup_i A_i\right) \le \sum_i P(A_i)$ (union, or Boole, bound).

Proof sketch. (1) Take $A_1 = \Omega$ and $A_i = \varnothing$ for $i \ge 2$; additivity gives $P(\Omega) = P(\Omega) + \sum_{i \ge 2} P(\varnothing)$, forcing $P(\varnothing) = 0$. (2) A and A^c are disjoint with union Ω, so $P(A) + P(A^c) = 1$. (3) Write $B = A \cup (B \setminus A)$, a disjoint union, so $P(B) = P(A) + P(B \setminus A) \ge P(A)$. (4) Split $A \cup B$ into the three disjoint pieces $A \setminus B$, $A \cap B$, $B \setminus A$; also $P(A) = P(A \setminus B) + P(A \cap B)$ and similarly for B. Adding the last two and subtracting gives the result: the intersection has been counted twice. (5) follows from (4) because $P(A \cap B) \ge 0$, and by induction for finitely many events. $\square$

The addition rule generalizes to the **inclusion–exclusion principle**:

$$P\left(\bigcup_{i=1}^{n} A_i\right) = \sum_i P(A_i) - \sum_{i<j} P(A_i A_j) + \sum_{i<j<k} P(A_i A_j A_k) - \cdots + (-1)^{n+1} P(A_1 A_2 \cdots A_n).$$

For three events, $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(AB) - P(AC) - P(BC) + P(ABC)$. Truncating the series after the first sum gives the union bound (an upper bound); truncating after the second sum gives a lower bound (the **Bonferroni inequalities** alternate in this way).

A final consequence is **continuity of probability**: if $A_1 \subseteq A_2 \subseteq \cdots$ then $P\left(\bigcup_n A_n\right) = \lim_n P(A_n)$, and if $A_1 \supseteq A_2 \supseteq \cdots$ then $P\left(\bigcap_n A_n\right) = \lim_n P(A_n)$. This is what will make the CDF right-continuous in Section 19.8.

Worked Example 19.2 — Two test stations (addition rule)

Boards from a production line pass through an in-circuit test (ICT) and a functional test (FT). Records show $P(\text{fails ICT}) = 0.05$, $P(\text{fails FT}) = 0.04$ and $P(\text{fails both}) = 0.01$. Find the probability that a board fails at least one test, fails exactly one test, and passes both.

Step 1 — at least one. Let I and F be the failure events. By the addition rule,

$$P(I \cup F) = 0.05 + 0.04 - 0.01 = 0.08.$$

Step 2 — exactly one. "Exactly one" is $(I \cup F) \setminus (I \cap F)$, so its probability is $0.08 - 0.01 = 0.07$. Check: $P(I \setminus F) = 0.05 - 0.01 = 0.04$ and $P(F \setminus I) = 0.04 - 0.01 = 0.03$; total 0.07. ✓

Step 3 — passes both. By De Morgan, "passes both" is $I^c \cap F^c = (I \cup F)^c$, probability $1 - 0.08 = 0.92$.

The union bound would have given $P(I \cup F) \le 0.09$: close, because the overlap is small. When failures are rare, the union bound is nearly exact — the basis of many quick error-rate estimates.

Equally likely outcomes and counting

When Ω is finite with N outcomes and symmetry makes them equally likely, additivity forces each outcome to have probability $1/N$, and

$$P(A) = \frac{|A|}{|\Omega|} = \frac{\text{number of outcomes in } A}{\text{total number of outcomes}}.$$

Probability then reduces to **counting**, and counting large sets needs a few systematic tools — the subject of combinatorics.

Multiplication (product) principle

If a procedure consists of k stages, and stage i can be carried out in n_i ways regardless of how the earlier stages were done, then the whole procedure can be carried out in $n_1 n_2 \cdots n_k$ ways.

The four basic counts follow by choosing k items from n distinct items, distinguishing whether order matters and whether repetition is allowed:

	Order matters	Order does not matter
With replacement	n^k	$\binom{n+k-1}{k}$
Without replacement	$P(n,k) = \dfrac{n!}{(n-k)!}$	$\binom{n}{k} = \dfrac{n!}{k!\,(n-k)!}$

Derivations. Ordered with replacement: k stages, n choices each, giving n^k (the number of k-symbol words over an n-letter alphabet; 2^k binary words of length k). Ordered without replacement: n choices for the first item, $n-1$ for the second, ..., $n-k+1$ for the k-th, product $n!/(n-k)!$; with $k = n$ this is $n!$, the number of **permutations** of n items. Unordered without replacement: each unordered set of k items corresponds to exactly $k!$ ordered lists, so the number of **combinations** is $P(n,k)/k! = \binom{n}{k}$. Unordered with replacement ("stars and bars"): a multiset of size k from n types is a row of k stars separated into n bins by $n-1$ bars; choosing the positions of the k stars among $n+k-1$ symbols gives $\binom{n+k-1}{k}$.

The binomial coefficient obeys $\binom{n}{k} = \binom{n}{n-k}$, Pascal's rule $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$, and the **binomial theorem** $(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k}$, which will normalize the binomial distribution in Section 19.6.

Multinomial coefficient

The number of ways to split n distinct items into groups of sizes $k_1, \dots, k_r$ with $k_1 + \cdots + k_r = n$ (equivalently, the number of distinct arrangements of a word with k_i copies of letter i) is

$$\binom{n}{k_1, k_2, \dots, k_r} = \frac{n!}{k_1!\, k_2! \cdots k_r!}.$$

For example, the number of distinct orders in which 4 identical resistors, 3 identical capacitors and 2 identical inductors can be placed in 9 slots is $9!/(4!\,3!\,2!) = 1260$.

For large n, factorials are evaluated with **Stirling's approximation** $n! \approx \sqrt{2\pi n}\,(n/e)^n$, which is already within $1\,\%$ for $n = 10$.

Worked Example 19.3 — Two dice

Two fair dice are rolled. Find the probability that the sum is 7, and the probability that at least one die shows a 6.

Step 1 — sample space. Ordered pairs (d_1, d_2), $|\Omega| = 6^2 = 36$, all equally likely. (Using *unordered* pairs, 21 outcomes, would be wrong: $\{3,4\}$ can occur in two ways but $\{3,3\}$ in only one, so unordered outcomes are not equally likely.)

Step 2 — sum 7. $A = \{(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)\}$, so $P(A) = 6/36 = 1/6$.

Step 3 — at least one 6. The complement is "no 6", which has $5 \times 5 = 25$ outcomes. Hence $P = 1 - 25/36 = 11/36 \approx 0.306$. Inclusion–exclusion agrees: $1/6 + 1/6 - 1/36 = 11/36$. ✓

Worked Example 19.4 — Sampling inspection (hypergeometric counting)

A reel of 50 integrated circuits contains 5 defective ones. An inspector draws 4 at random without replacement. Find the probability that (a) none is defective, (b) exactly one is defective, (c) at least one is defective.

Step 1 — sample space. Unordered samples of 4 from 50: $\binom{50}{4} = 230\,300$, all equally likely.

Step 2 — none defective. All 4 must come from the 45 good parts: $\binom{45}{4} = 148\,995$. So

$$P(\text{none}) = \frac{\binom{45}{4}}{\binom{50}{4}} = \frac{148\,995}{230\,300} \approx 0.6470.$$

Step 3 — exactly one. Choose 1 of the 5 defectives and 3 of the 45 good parts (product principle):

$$P(\text{one}) = \frac{\binom{5}{1}\binom{45}{3}}{\binom{50}{4}} = \frac{5 \times 14\,190}{230\,300} \approx 0.3081.$$

Step 4 — at least one. $1 - 0.6470 = 0.3530$.

In general, drawing n from a population of N containing K "marked" items, the probability of exactly k marked items is $\binom{K}{k}\binom{N-K}{n-k}/\binom{N}{n}$ — the **hypergeometric** distribution. Sampling inspection plans in quality control are designed with it (Ch. 20).

Worked Example 19.5 — Address collisions (the birthday problem)

Each of n wireless sensor nodes picks a 16-bit address uniformly at random ($M = 2^{16} = 65\,536$ possibilities), independently of the others. What is the probability that at least two nodes pick the same address, for $n = 100$ and $n = 300$?

Step 1 — complement. "No collision" means the n addresses are all different. The number of ordered assignments with no repeat is $M(M-1)\cdots(M-n+1)$ out of M^n equally likely assignments. So

$$P(\text{no collision}) = \prod_{k=0}^{n-1} \left(1 - \frac{k}{M}\right).$$

Step 2 — approximation. For $k \ll M$, $1 - k/M \approx e^{-k/M}$, so the product is approximately $\exp\!\left(-\sum_{k=0}^{n-1} k/M\right) = \exp\!\left(-\dfrac{n(n-1)}{2M}\right)$.

Step 3 — numbers. For $n = 100$: exact $P(\text{collision}) = 0.0728$, approximation $1 - e^{-9900/131072} = 0.0727$. For $n = 300$: exact 0.4961, approximation 0.4956.

With only 300 nodes — less than half a percent of the address space — a collision is a coin flip. The same calculation with $M = 365$ gives the famous result that 23 people share a birthday with probability 0.507. Collisions become likely once $n \approx \sqrt{M}$, which is why hash functions and random identifiers need about twice as many bits as one would naively expect.

Common Pitfall — "equally likely" must be justified

The formula $P(A) = |A|/|\Omega|$ is valid only when the outcomes really are equally likely. Choosing a description in which they are not (unordered dice, "number of errors" as the outcome of a 3-bit word, "sum of the dice" as the outcome) and then counting gives wrong answers. When in doubt, describe outcomes at the finest level the physics makes symmetric — ordered tuples of individual trials — and count there.

Engineering Note — modelling first

Almost every probability error in engineering reports is a modelling error, not an arithmetic one. Before computing, write down (1) the experiment, (2) the sample space, (3) the events of interest as subsets, (4) the source of the probability numbers (symmetry, data sheet, measurement). The axioms then do the rest mechanically.

19.2 Conditional probability

Motivation

Information changes probabilities. The probability that a bit was received in error is 10^{-3} on average, but if the automatic gain control has just flagged a deep fade it may be 10^{-1}. The probability that a power module fails in the next month is small, but much larger if it has already run hot for a year. Engineering decisions are almost always made *after* some partial information has arrived — a sensor reading, a test result, a received symbol — and conditional probability is the tool that updates the model accordingly.

Definition

Suppose we learn that event B has occurred. Outcomes outside B are now impossible, and the outcomes inside B should keep their *relative* likelihoods. The natural rule is therefore to restrict to B and rescale so that B has total probability 1.

Conditional probability

For events A, B with $P(B) > 0$, the **conditional probability of A given B** is

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

In the relative-frequency picture: among the N_B trials in which B occurred, the fraction in which A also occurred is $N_{AB}/N_B = (N_{AB}/N)/(N_B/N) \to P(AB)/P(B)$. For equally likely outcomes, $P(A \mid B) = |A \cap B|/|B|$: count inside the **reduced sample space** B.

Conditional probability is a probability

For fixed B with $P(B) > 0$, the map $A \mapsto P(A \mid B)$ satisfies all three Kolmogorov axioms on $(\Omega, \mathcal{F})$. Consequently every rule of Section 19.1 holds conditionally: $P(A^c \mid B) = 1 - P(A \mid B)$, $P(A_1 \cup A_2 \mid B) = P(A_1 \mid B) + P(A_2 \mid B) - P(A_1 A_2 \mid B)$, and so on.

Proof. Non-negativity is clear. $P(\Omega \mid B) = P(B)/P(B) = 1$. If the A_i are disjoint, so are the $A_i \cap B$, and $P\left(\bigcup A_i \mid B\right) = P\left(\bigcup (A_i \cap B)\right)/P(B) = \sum P(A_i \cap B)/P(B) = \sum P(A_i \mid B)$. $\square$

Worked Example 19.6 — Reduced sample space

Two fair dice are rolled. Given that the first die shows 6, what is the probability that the sum is at least 10? Compare with the unconditional probability.

Step 1 — condition. $B = \{(6,1), \dots, (6,6)\}$, six equally likely outcomes.

Step 2 — count inside B. The sum is at least 10 for $(6,4), (6,5), (6,6)$: $P(A \mid B) = 3/6 = 1/2$.

Step 3 — unconditional. Sums 10, 11, 12 occur in $3 + 2 + 1 = 6$ of 36 outcomes: $P(A) = 1/6$.

Knowing the first die is a 6 triples the probability. Check with the formula: $P(A \cap B) = 3/36$, $P(B) = 6/36$, ratio $1/2$. ✓

The multiplication rule and the chain rule

Rearranging the definition gives the **multiplication rule**

$$P(A \cap B) = P(A \mid B)\, P(B) = P(B \mid A)\, P(A),$$

which is how joint probabilities are usually *computed* in sequential experiments: the probability of the first stage times the conditional probability of the second stage given the first. Applying it repeatedly yields the chain rule.

Chain rule

If $P(A_1 A_2 \cdots A_{n-1}) > 0$,

$$P(A_1 A_2 \cdots A_n) = P(A_1)\, P(A_2 \mid A_1)\, P(A_3 \mid A_1 A_2) \cdots P(A_n \mid A_1 \cdots A_{n-1}).$$

Proof. Write each conditional probability as a ratio; the product telescopes: $P(A_1) \cdot \frac{P(A_1 A_2)}{P(A_1)} \cdot \frac{P(A_1 A_2 A_3)}{P(A_1 A_2)} \cdots = P(A_1 \cdots A_n)$. $\square$

Sequential experiments are conveniently drawn as **probability trees**: each branch is labelled with a conditional probability given the path so far, and the probability of a complete path is the product of its branch labels. The labels on the branches leaving any node sum to 1, and the probabilities of all complete paths sum to 1 — two checks worth making every time.

Worked Example 19.7 — Drawing without replacement (chain rule)

A tray holds 10 MOSFETs, 3 of which are counterfeit. Parts are drawn one at a time without replacement. Find (a) the probability that the first two drawn are both counterfeit, and (b) the probability that the first counterfeit appears on the third draw.

(a) Let C_i = "draw i is counterfeit". Then

$$P(C_1 C_2) = P(C_1)P(C_2 \mid C_1) = \frac{3}{10}\cdot\frac{2}{9} = \frac{1}{15} \approx 0.0667.$$

Counting check: $\binom{3}{2}/\binom{10}{2} = 3/45 = 1/15$. ✓

(b) The event is $C_1^c C_2^c C_3$. By the chain rule,

$$P(C_1^c C_2^c C_3) = \frac{7}{10}\cdot\frac{6}{9}\cdot\frac{3}{8} = \frac{126}{720} = 0.175.$$

Each factor is a conditional probability computed in the *current* tray: after two genuine parts are removed, 8 parts remain, 3 of them counterfeit.

The law of total probability

Often $P(A)$ is hard to find directly but easy once we know which of several mutually exclusive "situations" holds. The law of total probability averages the conditional answers, weighting each situation by its probability.

Law of total probability

If $B_1, \dots, B_n$ is a partition of Ω with $P(B_i) > 0$ for all i, then for any event A

$$P(A) = \sum_{i=1}^{n} P(A \mid B_i)\,P(B_i).$$

The same holds for a countably infinite partition.

Proof. Because the B_i partition Ω, the sets $A \cap B_i$ are disjoint and their union is A. By additivity $P(A) = \sum_i P(A \cap B_i)$, and each term equals $P(A \mid B_i)P(B_i)$ by the multiplication rule. □

The simplest and most used case is the two-set partition $\{B, B^c\}$:

$$P(A) = P(A \mid B)P(B) + P(A \mid B^c)\big(1 - P(B)\big).$$

Geometrically, $P(A)$ is a weighted average of the conditional probabilities $P(A \mid B_i)$, so it always lies between the smallest and the largest of them.

Worked Example 19.8 — Three production lines

A plant makes the same voltage regulator on three lines. Line 1 produces 50 % of the output with a defect rate of 1 %, line 2 produces 30 % with 2 % defective, and line 3 produces 20 % with 3 % defective. What fraction of all regulators is defective?

Step 1 — partition. L_1, L_2, L_3 (which line made the part), with $P(L_1) = 0.5$, $P(L_2) = 0.3$, $P(L_3) = 0.2$; these sum to 1. ✓

Step 2 — conditionals. $P(D \mid L_1) = 0.01$, $P(D \mid L_2) = 0.02$, $P(D \mid L_3) = 0.03$.

Step 3 — total probability.

$$P(D) = 0.01(0.5) + 0.02(0.3) + 0.03(0.2) = 0.005 + 0.006 + 0.006 = 0.017.$$

So 1.7 % of regulators are defective — between the best line's 1 % and the worst's 3 %, as it must be. Section 19.3 asks the reverse question: given a defective part, which line made it?

Worked Example 19.9 — A channel with fades (Gilbert–Elliott model)

A radio link is in a "good" state 90 % of the time, with bit-error probability 10^{-3}, and in a "bad" (faded) state 10 % of the time, with bit-error probability 10^{-1}. Find the average bit-error probability, and the probability that two consecutive bits (sent within the same state, the state lasting much longer than a bit) are both in error. Assume errors are independent *given the state*.

Step 1 — average error probability. With E = "bit in error", G and G^c the states:

$$P(E) = 10^{-3}(0.9) + 10^{-1}(0.1) = 0.0009 + 0.01 = 0.0109.$$

The bad state, present only 10 % of the time, contributes over 90 % of the errors.

Step 2 — two errors. Conditionally on the state, two errors have probability p^2:

$$P(E_1E_2) = (10^{-3})^2(0.9) + (10^{-1})^2(0.1) = 9\times10^{-7} + 10^{-3} \approx 1.0009\times10^{-3}.$$

Step 3 — compare with a memoryless channel. If bits were independent with the average error rate, $P(E_1E_2)$ would be $0.0109^2 = 1.19\times10^{-4}$ — more than eight times too small. Moreover $P(E_2 \mid E_1) = 1.0009\times10^{-3}/0.0109 \approx 0.092$: once one error is seen, the next bit is almost 10 times more likely than average to be wrong. Errors on fading channels come in **bursts**, which is why such links use interleaving before the error-correcting code.

Common Pitfall — the conditioning bar is not symmetric

$P(A \mid B)$ and $P(B \mid A)$ are different numbers, related by $P(A \mid B)P(B) = P(B \mid A)P(A)$. "Most failed units had overheated" ($P(\text{hot} \mid \text{fail})$ large) does not mean "most units that overheat fail" ($P(\text{fail} \mid \text{hot})$ may be small). Confusing the two — sometimes called the *prosecutor's fallacy* — is the most common reasoning error in fault analysis. Bayes' theorem (Section 19.3) converts one into the other.

Common Pitfall — averaging conditionals with the wrong weights

The law of total probability weights $P(A \mid B_i)$ by $P(B_i)$, not equally. Averaging the three defect rates in Example 19.8 without weights gives 2 %, not the correct 1.7 %. And the partition must be genuinely exhaustive and disjoint; overlapping "situations" double-count.

Engineering Note — conditioning as a design tool

The law of total probability is the formal version of "divide into cases". It appears as the averaging of error rates over fading states, of failure rates over operating temperatures, of delay over network loads. When a direct calculation looks hard, ask: *which unknown, if I knew it, would make the problem easy?* Condition on it, then average.

19.3 Bayes' theorem

Motivation

Physical models usually run from cause to effect. A data sheet tells us the probability that a detector raises an alarm *given* that a fault is present; a channel model tells us the probability of receiving a 1 *given* that a 0 was sent. But the engineer's question runs the other way: the alarm has sounded — is there a fault? A 1 has been received — what was sent? Bayes' theorem inverts the conditioning, turning cause-to-effect models into effect-to-cause inferences. It is the mathematical core of detection, diagnosis, classification and estimation.

Statement and derivation

Bayes' theorem

Let $B_1, \dots, B_n$ be a partition of Ω with $P(B_i) > 0$, and let A be an event with $P(A) > 0$. Then

$$P(B_k \mid A) = \frac{P(A \mid B_k)\,P(B_k)}{\displaystyle\sum_{i=1}^{n} P(A \mid B_i)\,P(B_i)}.$$

For a two-set partition $\{B, B^c\}$:

$$P(B \mid A) = \frac{P(A \mid B)P(B)}{P(A \mid B)P(B) + P(A \mid B^c)P(B^c)}.$$

Derivation. The multiplication rule gives $P(B_k \cap A)$ in two ways: $P(B_k \mid A)P(A) = P(A \mid B_k)P(B_k)$. Divide by $P(A)$ and expand $P(A)$ with the law of total probability. □

The terms have standard names, which carry over unchanged to estimation theory (Ch. 20):

Term	Name	Meaning
$P(B_k)$	**prior**	belief in hypothesis B_k before the observation
$P(A \mid B_k)$	**likelihood**	how probable the observation is if B_k is true
$P(A)$	**evidence** (normalizer)	total probability of the observation
$P(B_k \mid A)$	**posterior**	updated belief after observing A

In words: *posterior is proportional to likelihood times prior*, and the denominator merely makes the posteriors sum to 1. This suggests a practical recipe: compute the products $P(A \mid B_i)P(B_i)$ for every hypothesis, then divide each by their sum.

Worked Example 19.10 — Which line made the defective part?

Continue Example 19.8. A regulator selected at random is found defective. Find the probability that it came from each line.

Step 1 — joint products. $P(D \mid L_i)P(L_i)$: line 1, 0.005; line 2, 0.006; line 3, 0.006. Sum $= P(D) = 0.017$.

Step 2 — normalize.

$$P(L_1 \mid D) = \frac{0.005}{0.017} = 0.294, \qquad P(L_2 \mid D) = \frac{0.006}{0.017} = 0.353, \qquad P(L_3 \mid D) = \frac{0.006}{0.017} = 0.353.$$

Check: the posteriors sum to 1. ✓

Line 1 makes half the parts but only 29 % of the defectives; line 3 makes a fifth of the parts but 35 % of the defectives. The prior (production share) and the likelihood (defect rate) pull in opposite directions, and Bayes' theorem weighs them exactly.

The binary channel

The simplest communication model has a binary input $X \in \{0, 1\}$ and a binary output $Y \in \{0, 1\}$. It is specified by the **prior** probabilities $P(X = 0) = p_0$, $P(X = 1) = p_1 = 1 - p_0$, and the **transition probabilities** $P(Y = y \mid X = x)$. In the **binary symmetric channel** (BSC) both kinds of error have the same probability ε: $P(Y = 1 \mid X = 0) = P(Y = 0 \mid X = 1) = \varepsilon$. In an asymmetric channel (optical on–off keying, where a "1" can be lost but noise rarely creates one, is a typical example) the two error probabilities differ.

Worked Example 19.11 — Binary asymmetric channel

A source sends 0 with probability 0.6 and 1 with probability 0.4. The channel has $P(Y = 1 \mid X = 0) = 0.05$ and $P(Y = 0 \mid X = 1) = 0.10$. Find (a) the output probabilities, (b) the posterior probabilities of the input given each output, (c) the error probability if the receiver simply decides $\hat{X} = Y$.

(a) Total probability.

$$P(Y = 1) = 0.05(0.6) + 0.90(0.4) = 0.03 + 0.36 = 0.39, \qquad P(Y = 0) = 0.95(0.6) + 0.10(0.4) = 0.57 + 0.04 = 0.61.$$

(b) Bayes.

$$P(X = 1 \mid Y = 1) = \frac{0.36}{0.39} = 0.923, \qquad P(X = 0 \mid Y = 0) = \frac{0.57}{0.61} = 0.934.$$

So a received 1 is wrong with probability 0.077 and a received 0 with probability 0.066.

(c) Error probability. An error occurs when $(X, Y) = (0, 1)$ or $(1, 0)$:

$$P_e = P(Y = 1 \mid X = 0)p_0 + P(Y = 0 \mid X = 1)p_1 = 0.05(0.6) + 0.10(0.4) = 0.07.$$

Check: $P_e = P(Y = 1)(1 - 0.923) + P(Y = 0)(1 - 0.934) = 0.39(0.0769) + 0.61(0.0656) = 0.030 + 0.040 = 0.070$. ✓

MAP and ML decisions

A receiver that observes $Y = y$ must decide which input was sent. The decision that minimizes the probability of error is the one that picks the input with the **largest posterior**.

MAP decision rule

Given an observation A and hypotheses $B_1, \dots, B_n$, the rule "decide B_k where k maximizes $P(B_k \mid A)$" — equivalently, maximizes $P(A \mid B_k)P(B_k)$ — minimizes the probability of a wrong decision. It is called the **maximum a posteriori (MAP)** rule. When all priors are equal, it reduces to the **maximum-likelihood (ML)** rule: maximize $P(A \mid B_k)$.

Proof sketch. For any decision rule, the probability of a *correct* decision is $\sum_{\text{observations } a} P(\text{decided hypothesis is true} \mid a)P(a)$. Each term is maximized separately by choosing, for each a, the hypothesis with the largest posterior. Since the denominators $P(a)$ are common to all hypotheses, comparing posteriors is the same as comparing the products likelihood $\times$ prior. □

For two hypotheses the MAP rule is a **likelihood-ratio test**:

$$\Lambda(a) = \frac{P(a \mid B_1)}{P(a \mid B_0)} \underset{B_0}{\overset{B_1}{\gtrless}} \frac{P(B_0)}{P(B_1)},$$

i.e. decide B_1 if the likelihood ratio exceeds the prior odds against it. Section 19.7 applies exactly this test to a signal in Gaussian noise.

Worked Example 19.12 — When MAP ignores the channel

In the channel of Example 19.11, MAP gives: for $Y=0$, compare $0.95(0.6)=0.57$ with $0.10(0.4)=0.04$ — decide 0; for $Y=1$, compare $0.05(0.6)=0.03$ with $0.90(0.4)=0.36$ — decide 1. So $\hat{X}=Y$ is already MAP, and $P_e=0.07$ is the minimum possible.

Now suppose 1s are rare: $p_1=0.03$, $p_0=0.97$, same channel.

Step 1 — $Y=1$ received. Compare $P(Y=1\mid 0)p_0=0.05(0.97)=0.0485$ with $P(Y=1\mid 1)p_1=0.90(0.03)=0.027$. The first is larger: **decide 0 even though 1 was received.**

Step 2 — $Y=0$ received. $0.95(0.97)$ vastly exceeds $0.10(0.03)$: decide 0.

Step 3 — error probabilities. The MAP receiver always outputs 0 and is wrong only when a 1 was sent: $P_e=0.03$. The "trust the channel" receiver $\hat{X}=Y$ has $P_e=0.05(0.97)+0.10(0.03)=0.0515$.

With strongly unequal priors, a noisy observation may carry less information than the prior. In practice one would not build such a receiver — it never detects a 1 — which shows that "minimum error probability" is not always the right criterion. Detection theory therefore also uses rules that weight the two errors by different *costs* (Bayes risk) or fix the false-alarm rate (Neyman–Pearson).

Base rates and fault detection

Bayes' theorem explains why detectors for rare events produce mostly false alarms, a fact that surprises almost everyone the first time.

Worked Example 19.13 — Insulation fault detector

An online partial-discharge monitor is fitted to a population of cable joints, of which $0.2\,\%$ have an incipient insulation fault. The monitor detects a real fault with probability 0.99 (sensitivity) and raises a false alarm on a healthy joint with probability 0.02. (a) An alarm is raised on a joint. What is the probability that it is faulty? (b) A second, independent monitor of the same specification also alarms on that joint. What now?

Step 1 — set up. F = faulty, A = alarm. Prior $P(F)=0.002$. Likelihoods $P(A\mid F)=0.99$, $P(A\mid F^c)=0.02$.

Step 2 — Bayes.

$$P(F\mid A)=\frac{0.99(0.002)}{0.99(0.002)+0.02(0.998)}=\frac{0.00198}{0.00198+0.01996}=0.0902.$$

Despite a "99 % accurate" detector, only about **9 %** of alarms are real. Out of 100 000 joints, 200 are faulty and 198 of those alarm; but 2% of the 99 800 healthy joints — 1996 — also alarm.

Step 3 — odds form for the second alarm. Write Bayes in odds: posterior odds = likelihood ratio $\times$ prior odds, where the likelihood ratio of an alarm is $\Lambda=0.99/0.02=49.5$. After the first alarm the odds are $0.0902/0.9098=0.0992$. A second alarm, conditionally independent given the joint's state, multiplies the odds again by 49.5:

$$\text{odds}=0.0992\times 49.5=4.91,\qquad P(F\mid A_1A_2)=\frac{4.91}{1+4.91}=0.831.$$

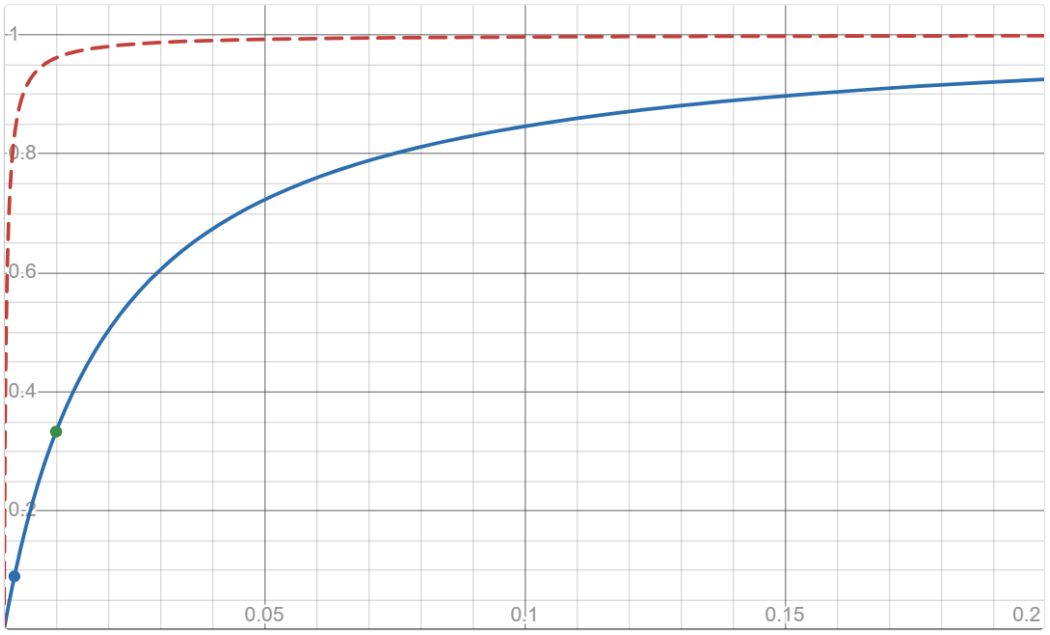
Two independent alarms move the probability from $0.2\,\%$ to $83\,\%$. Confirmation by an independent test is the standard engineering cure for the base-rate problem.

The **odds form** used in Step 3 is worth stating separately because it makes sequential updating trivial:

$$\underbrace{\frac{P(B\mid A)}{P(B^c\mid A)}}_{\text{posterior odds}}=\underbrace{\frac{P(A\mid B)}{P(A\mid B^c)}}_{\text{likelihood ratio}}\times\underbrace{\frac{P(B)}{P(B^c)}}_{\text{prior odds}}.$$

With independent observations the likelihood ratios multiply; taking logarithms, the evidence *adds*, which is why practical detectors accumulate **log-likelihood ratios**.

The dependence of the posterior on the prior is shown below for the detector of Example 19.13.



Posterior probability of a fault after one alarm (blue) and after two independent alarms (red, dashed) as a function of the prior fault probability x, for sensitivity 0.99 and false-alarm rate 0.02. For rare faults the single-alarm posterior collapses towards zero; a second alarm restores confidence.

Common Pitfall — base-rate neglect

A test's sensitivity $P(A \mid F)$ and specificity $1 - P(A \mid F^c)$ describe the *test*; the probability that an alarm is real, $P(F \mid A)$, depends also on the *population* through the prior $P(F)$. Quoting "99 % accurate" without the base rate says almost nothing about how alarms should be treated. The same detector deployed on a population with $10\,\%$ faulty units has $P(F \mid A) = 0.85$.

Engineering Note — Bayes everywhere

The binary channel and the fault detector are the same mathematics. So are spam filters, radar target detection, built-in self-test diagnostics, and the decoding of error-correcting codes (where the "hypotheses" are the possible codewords). In every case: write the prior, write the likelihood of the observation under each hypothesis, multiply, normalize, and decide by the largest posterior or by a cost-weighted threshold.

19.4 Independence

Motivation

Most probability models of real systems are built from parts: the bits of a word, the components of a circuit, the successive packets on a link, the samples of a noise waveform. The model becomes tractable when the parts do not influence each other, because then the probability of a compound event is simply the product of the probabilities of its pieces. Independence is that "no influence" assumption made precise. It is the most powerful simplification in applied probability — and the most frequently abused one, so knowing exactly what it means, and when it fails, matters as much as using it.

Independence of two events

Intuitively, A and B are independent if learning that B occurred does not change the probability of A: $P(A \mid B) = P(A)$. Multiplying by $P(B)$ gives a symmetric form that also makes sense when $P(B) = 0$, and this is taken as the definition.

Independent events

Events A and B are **independent** if

$$P(A \cap B) = P(A)\,P(B).$$

If $P(B) > 0$ this is equivalent to $P(A \mid B) = P(A)$; if $P(A) > 0$, to $P(B \mid A) = P(B)$.

Complements of independent events

If A and B are independent, so are A and B^c, A^c and B, and A^c and B^c.

Proof. $P(A \cap B^c) = P(A) - P(A \cap B) = P(A) - P(A)P(B) = P(A)\big(1 - P(B)\big) = P(A)P(B^c)$. The other cases follow by applying this twice. □

Common Pitfall — independent is not the same as disjoint

Disjoint events with non-zero probabilities are never independent: if $A \cap B = \varnothing$ then $P(AB) = 0 \neq P(A)P(B)$. Indeed, disjoint events are strongly *dependent* — knowing that A occurred tells you for certain that B did not. "Mutually exclusive" is a property of the sets; "independent" is a property of the probability measure.

Independence of several events

For three or more events, pairwise independence is not enough.

Mutual independence

Events $A_1, \dots, A_n$ are **(mutually) independent** if for *every* subcollection $A_{i_1}, \dots, A_{i_k}$ with $2 \le k \le n$ and distinct indices,

$$P(A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}) = P(A_{i_1})P(A_{i_2}) \cdots P(A_{i_k}).$$

For n events this is $2^n - n - 1$ conditions. An infinite family is independent if every finite subfamily is.

Worked Example 19.14 — Pairwise but not mutually independent (parity bit)

Two fair, independent data bits b_1, b_2 are sent with an even-parity bit $b_3 = b_1 \oplus b_2$. Let A_i = "$b_i = 1$". Are A_1, A_2, A_3 independent?

Step 1 — marginals. The four equally likely words are $000, 011, 101, 110$. Each bit equals 1 in exactly two of them: $P(A_i) = 1/2$.

Step 2 — pairs. $A_1 \cap A_2 = \{110\}$, $A_1 \cap A_3 = \{101\}$, $A_2 \cap A_3 = \{011\}$, each with probability $1/4 = P(A_i)P(A_j)$. So the events are **pairwise independent**.

Step 3 — triple. $A_1 \cap A_2 \cap A_3 = \varnothing$ (no valid word is 111), so $P = 0 \neq 1/8$. The events are **not mutually independent**: any two bits determine the third.

This is exactly why a parity bit detects errors: the three bits are not free.

Conditional independence

Events A and B are **conditionally independent given** C if $P(AB \mid C) = P(A \mid C)P(B \mid C)$. Conditional independence neither implies nor is implied by independence. The fading channel of Example 19.9 is the standard illustration: given the channel state, bit errors are independent; *unconditionally* they are strongly dependent, because an error is evidence that the channel is in the bad state, which makes the next error more likely. The second alarm in Example 19.13 was assumed conditionally independent given the joint's state — not independent outright, since both alarms respond to the same fault.

Repeated independent trials

The most important use of independence is to build the probability law of n repeated, independent trials of the same experiment. Suppose each trial results in "success" with probability p and "failure" with probability $q = 1 - p$ (a **Bernoulli trial**). By independence, any particular sequence with k successes and $n - k$ failures has probability $p^k q^{n-k}$, and there are $\binom{n}{k}$ such sequences (choose the positions of the successes). Therefore:

Binomial probability law

In n independent Bernoulli trials with success probability p, the probability of exactly k successes is

$$P(k \text{ successes}) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, 1, \dots, n.$$

These probabilities sum to $(p + q)^n = 1$ by the binomial theorem.

Worked Example 19.15 — Bit errors and a Hamming code

A binary channel has independent bit errors with probability $p = 0.01$. (a) What is the probability that an uncoded 4-bit data word is received correctly? (b) The 4 data bits are instead sent as a 7-bit Hamming codeword, which corrects any single error. What is the probability that the decoded word is wrong?

(a) All 4 bits correct: $(0.99)^4 = 0.9606$; word error probability $1 - 0.9606 = 0.0394$.

(b) The decoder fails only if 2 or more of the 7 bits are wrong:

$$P(\text{fail}) = 1 - \left[\binom{7}{0}(0.99)^7 + \binom{7}{1}(0.01)(0.99)^6\right] = 1 - [0.93207 + 0.06590] = 0.00203.$$

The code reduces the word-error probability by a factor of about 19, at the price of sending 7 bits instead of 4. The dominant failure term is $\binom{7}{2}p^2q^5 \approx 0.00199$, so $P(\text{fail}) \approx 21p^2$: a single-error-correcting code turns an error rate proportional to p into one proportional to p^2. This scaling argument is the essence of coding gain.

Reliability of systems

Reliability R is the probability that a component or system performs its function (over a stated time under stated conditions). If components fail independently, the reliability of a system follows from its structure by the rules of this section. Let component i work with probability p_i and fail with probability $q_i = 1 - p_i$.

Series system. The system works only if *every* component works (a chain: fuse, converter, regulator, load). By independence,

$$R_{\text{series}} = p_1 p_2 \cdots p_n = \prod_{i=1}^{n} p_i.$$

A series system is less reliable than its weakest component, and reliability decays geometrically with the number of parts: 100 parts of reliability 0.999 give only 0.905.

Parallel (redundant) system. The system works if *at least one* component works. By De Morgan, it fails only if all fail:

$$R_{\text{parallel}} = 1 - q_1 q_2 \cdots q_n = 1 - \prod_{i=1}^{n} (1 - p_i).$$

A parallel system is more reliable than its best component. Two units of 0.9 in parallel give 0.99; three give 0.999 — each additional unit adds "one nine".

k-out-of-n system. The system works if at least k of n identical, independent components work. By the binomial law,

$$R_{k/n} = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}.$$

Series is n-out-of-n, parallel is 1-out-of-n. The important special case 2-out-of-3 is **triple modular redundancy (TMR)** with a majority voter: $R_{\text{TMR}} = 3p^2 - 2p^3$ (for a perfect voter).

Series–parallel networks are reduced step by step: replace each parallel group by an equivalent component with the parallel formula, each series chain by the series formula, and repeat — much as series and parallel resistors are combined. Networks that cannot be reduced this way (the **bridge** is the classic one) are handled by **conditioning on a key component** with the law of total probability:

$$R = R(\text{system} \mid \text{component } k \text{ works})\, p_k + R(\text{system} \mid \text{component } k \text{ fails})\, q_k,$$

where in each case the network simplifies: a working component is replaced by a short circuit, a failed one by an open circuit.

Worked Example 19.16 — A redundant power path

A board is powered through an input fuse ($p = 0.999$), then two identical DC–DC converters in parallel (each $p = 0.95$), then a supervisory controller ($p = 0.99$). Find the system reliability, and compare with a single converter.

Step 1 — reduce the parallel pair. $R_{\text{conv}} = 1 - (0.05)^2 = 0.9975$.

Step 2 — series chain. $R = 0.999 \times 0.9975 \times 0.99 = 0.98654$.

Step 3 — without redundancy. $R = 0.999 \times 0.95 \times 0.99 = 0.93955$.

The second converter cuts the failure probability from $6.0\,\%$ to $1.35\,\%$. The fuse and controller now dominate: the remaining unreliability is about $0.1\,\% + 0.25\,\% + 1\,\%$. Further redundancy should go where the largest series term is — here the controller.

Worked Example 19.17 — 2-out-of-3 voting

Three channels of a safety sensor each work with probability 0.95, independently; a perfect voter outputs the majority. Find the system reliability.

Step 1 — binomial. At least 2 of 3 must work:

$$R = \binom{3}{2}(0.95)^2(0.05) + \binom{3}{3}(0.95)^3 = 0.135375 + 0.857375 = 0.99275.$$

Step 2 — check with the formula. $3p^2 - 2p^3 = 3(0.9025) - 2(0.857375) = 2.7075 - 1.71475 = 0.99275$. ✓

TMR improves on a single channel (0.95) — but only because $p > 1/2$. Solving $3p^2 - 2p^3 = p$ gives $p = 1/2$: if each channel is more likely to fail than to work, majority voting makes things *worse*.

Worked Example 19.18 — The bridge network

Five links connect a source S to a terminal T: link 1 from S to node u, link 2 from u to T, link 3 from S to node v, link 4 from v to T, and a bridging link 5 between u and v. Each works independently with probability $p = 0.9$. Find the probability that S and T are connected.

Step 1 — condition on link 5 working. Nodes u and v merge. The network becomes $(1 \parallel 3)$ in series with $(2 \parallel 4)$:

$$R_1 = \left[1 - (0.1)^2\right]^2 = 0.99^2 = 0.9801.$$

Step 2 — condition on link 5 failed. The network is two series paths in parallel, (1–2) ∥ (3–4):

$$R_0 = 1 - (1 - 0.81)^2 = 1 - 0.0361 = 0.9639.$$

Step 3 — total probability.

$$R = 0.9(0.9801) + 0.1(0.9639) = 0.88209 + 0.09639 = 0.97848.$$

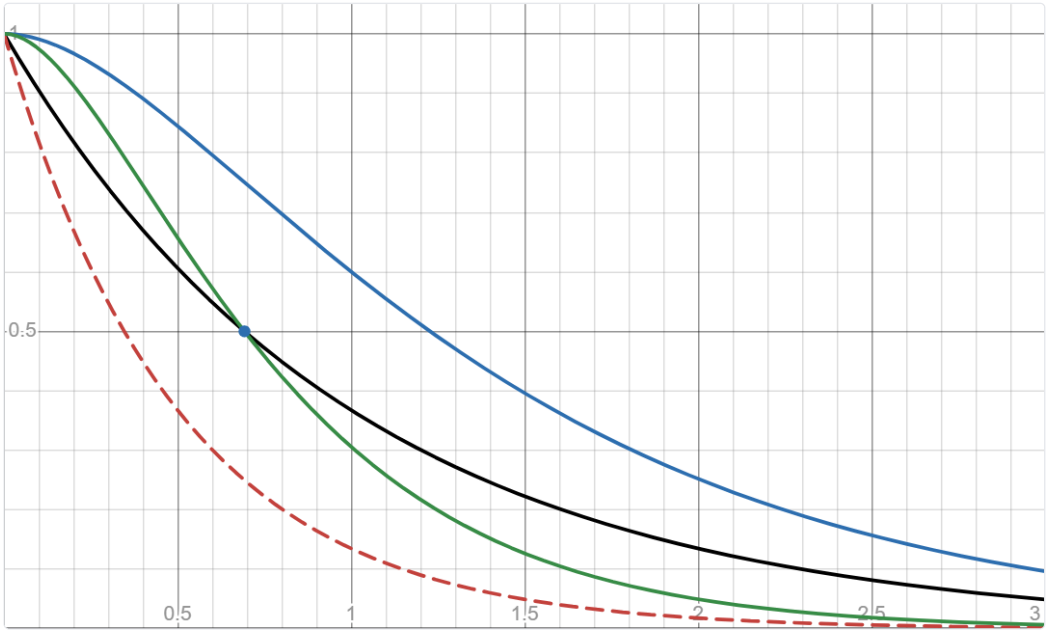
A brute-force check — summing the probabilities of all $2^5 = 32$ working/failed patterns for which a path exists — gives the same 0.97848. Conditioning reduced 32 cases to 2.

Reliability as a function of time

If a component's lifetime T has the exponential distribution (Section 19.6) with constant failure rate λ, its reliability at time t is $R(t) = P(T > t) = e^{-\lambda t}$. Substituting $p = e^{-\lambda t}$ into the structure formulas gives the reliability of a system as a function of time:

$$R_{\text{series}}(t) = e^{-n\lambda t}, \qquad R_{\text{par},2}(t) = 2e^{-\lambda t} - e^{-2\lambda t}, \qquad R_{\text{TMR}}(t) = 3e^{-2\lambda t} - 2e^{-3\lambda t}.$$

A series system of exponential components is again exponential, with failure rate equal to the sum of the component rates — the basis of the "parts count" reliability prediction method.



Reliability versus normalized time λt for a single unit (black), two units in series (red, dashed), two in parallel (blue) and 2-out-of-3 TMR (green). TMR is better than a single unit early in life but worse after $\lambda t = \ln 2$, when each channel's reliability drops below $1/2$; it suits short missions with repair or reset.

Common Pitfall — common-cause failures

The product formulas assume *independent* failures. Two "redundant" converters fed from the same overheated enclosure, two sensors calibrated with the same faulty reference, or three TMR channels running the same buggy software are not independent. A common cause that fails all channels at once with probability 10^{-3} caps the system reliability at about 0.999 no matter how much redundancy is added. Reliability engineers model this explicitly (the β-factor model) or diversify the redundant channels.

Engineering Note — where independence comes from

Independence is a *physical* assumption, justified by separation of mechanisms: different thermal noise sources in different resistors, bits separated by more than the channel's coherence time, components that share no stress. Write down why you believe it before using the product rule, and check it against data when you can — Ch. 20 gives tests for exactly this.

19.5 Random variables

Motivation

Outcomes of experiments are often not numbers: a received word, the state of a network, a sequence of heads and tails. Engineering questions, however, are almost always numerical — *how many* bits are wrong, *what* voltage was measured, *how long* until failure, *how much* power is in the noise. A random variable is the device that attaches a number to each outcome. Once we have numbers we can add, average, threshold and filter them, and the full machinery of calculus becomes available. From here on, almost all of probability is phrased in terms of random variables, and the underlying sample space fades into the background.

Definition

Random variable

A **random variable** (RV) on a probability space $(\Omega, \mathcal{F}, P)$ is a function $X : \Omega \to \mathbb{R}$ that assigns a real number $X(\zeta)$ to each outcome ζ, such that for every real x the set

$$\{X \le x\} = \{\zeta \in \Omega : X(\zeta) \le x\}$$

is an event (belongs to $\mathcal{F}$). The set of values X can take is its **range** S_X.

The technical condition (called *measurability*) guarantees that we can ask for $P(X \le x)$; it is automatically satisfied by every function an engineer writes down, and we shall not mention it again. Note the language: a random variable is neither random nor a variable — it is a deterministic *function*. The randomness lives in which outcome ζ occurs.

Once $\{X \le x\}$ is an event for every x, so are all the sets built from them by countable unions, intersections and complements, for instance

$$\{a < X \le b\} = \{X \le b\} \setminus \{X \le a\}, \qquad \{X = a\} = \bigcap_{n=1}^{\infty} \{a - 1/n < X \le a\}, \qquad \{X > a\} = \{X \le a\}^c.$$

So every question of the form "does X fall in the set B?" has a probability $P(X \in B)$. The assignment $B \mapsto P(X \in B)$ is itself a probability measure on the real line, called the **distribution** (or **law**) of X. Two random variables defined on completely different sample spaces can have the same distribution; for most purposes they are then interchangeable.

Worked Example 19.19 — Number of bit errors

A 3-bit word is sent over a channel with independent bit errors of probability $p = 0.1$. Let X = number of bits in error. Find the probability of each value of X.

Step 1 — the function. $\Omega = \{000, 001, \dots, 111\}$ (1 = error) and $X(\zeta)$ is the number of 1s in ζ: $X(000) = 0$, $X(010) = 1$, $X(111) = 3$, and so on. $S_X = \{0, 1, 2, 3\}$.

Step 2 — outcome probabilities. By independence, an outcome with k ones has probability $0.1^k\, 0.9^{3-k}$.

Step 3 — collect outcomes with the same value of X.

$$P(X=0) = 0.9^3 = 0.729, \qquad P(X=1) = 3(0.1)(0.9)^2 = 0.243,$$
$$P(X=2) = 3(0.1)^2(0.9) = 0.027, \quad P(X=3) = 0.1^3 = 0.001.$$

The probabilities sum to 1. ✓ Eight outcomes have been condensed into four numbers — the information relevant to the question "how many errors?".

Types of random variable

Discrete, continuous and mixed random variables

- X is **discrete** if its range S_X is finite or countably infinite: X takes isolated values $x_1, x_2, \dots$ with probabilities $P(X = x_k) > 0$ that sum to 1. (Counts, symbols, bits, quantizer outputs.)
- X is **continuous** if $P(X = x) = 0$ for every single value x and probabilities of intervals are given by integrals of a density: $P(a < X \le b) = \int_a^b f_X(x)\, dx$. (Voltages, times, phases, positions.)
- X is **mixed** if it has both: some isolated values with positive probability, and a continuous part spread over intervals. (A clipped voltage, the waiting time of a packet that is either sent at once or queued.)

Every random variable is completely described by its **cumulative distribution function** $F_X(x) = P(X \le x)$ (Section 19.8). For discrete RVs it is more convenient to list the probabilities of the individual values, the **probability mass function** (PMF); for continuous RVs, the derivative of the CDF, the **probability density function** (PDF). Both are developed in Section 19.7.

Common Pitfall — "probability zero" is not "impossible"

For a continuous RV every individual value has probability zero: the thermal-noise voltage is exactly $1.000\ldots$ mV with probability 0, yet some value always occurs. Zero-probability events can happen; they are just too "thin" to carry probability. Consequently, for a continuous RV, $P(a < X < b) = P(a \le X \le b)$ — endpoints do not matter. For discrete and mixed RVs they matter a great deal.

Indicator random variables

The simplest random variable takes only the values 0 and 1.

Indicator random variable

For an event A, the **indicator** of A is

$$I_A(\zeta) = \begin{cases} 1, & \zeta \in A \\ 0, & \zeta \notin A. \end{cases}$$

It satisfies $P(I_A = 1) = P(A)$, $I_{A^c} = 1 - I_A$, $I_{A \cap B} = I_A I_B$, and $I_{A \cup B} = I_A + I_B - I_A I_B$.

Indicators turn events into numbers and set operations into arithmetic. Their great use is in **counting**: the number of events among $A_1, \dots, A_n$ that occur is $X = I_{A_1} + \cdots + I_{A_n}$. In Example 19.19, X is the sum of three indicators "bit i is wrong". Section 19.9 shows that this decomposition makes the mean of X trivial to compute even when the events are dependent.

Functions of a random variable

If X is a random variable and g is an ordinary real function, then $Y = g(X)$ — the function $\zeta \mapsto g(X(\zeta))$ — is again a random variable. This is the probabilistic version of passing a signal through a memoryless system: a diode, a square-law detector, a limiter, a quantizer, an amplifier. The central question is: **given the distribution of X, what is the distribution of Y?**

The principle is always the same: an event about Y is an event about X,

$$P(Y \in B) = P\big(g(X) \in B\big) = P\big(X \in g^{-1}(B)\big), \qquad g^{-1}(B) = \{x : g(x) \in B\}.$$

For a discrete X this simply means collecting probability: every value y of Y gets the total probability of the x-values that map to it,

$$P(Y = y) = \sum_{x:\, g(x)=y} P(X = x).$$

For continuous X the same principle leads to the CDF method and the density-transformation formula of Sections 19.7 and 19.8.

Worked Example 19.20 — A squarer acting on a discrete signal

A 5-level signal X takes the values $-2, -1, 0, 1, 2$ V with probabilities $0.1, 0.2, 0.4, 0.2, 0.1$. It drives a square-law power detector whose output is $Y = X^2$ (in V^2). Find the PMF of Y.

Step 1 — range of Y. $g(x) = x^2$ maps $\{-2, -1, 0, 1, 2\}$ onto $\{0, 1, 4\}$.

Step 2 — collect preimages.

$$P(Y = 0) = P(X = 0) = 0.4, \qquad P(Y = 1) = P(X = -1) + P(X = 1) = 0.4, \qquad P(Y = 4) = P(X = -2) + P(X = 2) = 0.2.$$

Step 3 — check. $0.4 + 0.4 + 0.2 = 1$. ✓ Because g is not one-to-one, information is lost: from $Y = 1$ we cannot tell whether X was $+1$ or -1.

Worked Example 19.21 — A limiter produces a mixed random variable

A voltage X is equally likely to be anywhere in $(-2, 2)$ V, i.e. $P(a < X < b) = (b - a)/4$ for $-2 \le a < b \le 2$. It passes through an ideal limiter with output $Y = X$ for $|X| \le 1$, $Y = +1$ for $X > 1$ and $Y = -1$ for $X < -1$. Describe the distribution of Y.

Step 1 — clipped values. $P(Y = 1) = P(X > 1) = (2 - 1)/4 = 0.25$; likewise $P(Y = -1) = 0.25$. These are point masses.

Step 2 — the linear region. For $-1 < a < b < 1$, $P(a < Y < b) = P(a < X < b) = (b - a)/4$: probability is spread uniformly over $(-1, 1)$ with density $1/4$ per volt, total 0.5.

Step 3 — check. $0.25 + 0.25 + 0.5 = 1$. ✓

Y is **mixed**: it is neither discrete (it takes a continuum of values) nor continuous ($P(Y = 1) > 0$). Saturating amplifiers, ADCs driven beyond full scale, and rectifiers all produce mixed outputs; Section 19.8 draws their staircase-plus-ramp CDFs, and Section 19.7 represents the point masses with Dirac impulses.

Engineering Note — random variables are measurements

A good habit is to read "random variable" as "the reading on an instrument attached to the experiment". Different instruments on the same experiment give different RVs on the same sample space (X = number of errors, Y = position of the first error, Z = parity of the received word); their joint behaviour is the subject of the joint distributions introduced in Section 19.6. And a memoryless circuit applied to a reading gives a new reading, $Y = g(X)$, whose distribution is fixed by that of X.

19.6 Probability distributions

Motivation

A handful of distributions account for the overwhelming majority of probability models in electrical engineering. Each arises from a simple mechanism: counting successes in independent trials (binomial), waiting for the first success (geometric), counting rare events in a long interval (Poisson), complete ignorance over an interval (uniform), memoryless lifetimes (exponential), the sum of many small effects (Gaussian), the envelope of a two-dimensional Gaussian (Rayleigh). Recognizing the mechanism lets you write down the distribution without collecting data. This section introduces these key families and the first ideas of **joint** distributions of two random variables; Ch. 20 extends the catalogue (chi-square, Student t, lognormal, Weibull, ...) and shows how to fit and test distributions against data.

Distribution of a random variable

The **distribution** of X is the rule that gives $P(X \in B)$ for every event $B \subseteq \mathbb{R}$. In practice it is specified by a PMF $p_X(x_k) = P(X = x_k)$ for a discrete RV, by a PDF $f_X(x)$ with $P(X \in B) = \int_B f_X(x)\,dx$ for a continuous RV, or in all cases by the CDF $F_X(x) = P(X \le x)$. A **family** of distributions is a formula with adjustable **parameters** (n, p for the binomial; μ, σ for the Gaussian).

Discrete families

Bernoulli distribution

$X \sim \text{Bernoulli}(p)$: $P(X = 1) = p$, $P(X = 0) = 1 - p$. The indicator of any event with probability p; a single bit with error probability p.

Binomial distribution

$X \sim \text{Bin}(n, p)$: the number of successes in n independent Bernoulli(p) trials,

$$p_X(k) = \binom{n}{k} p^k (1 - p)^{n-k}, \qquad k = 0, 1, \ldots, n.$$

This is the law derived in Section 19.4; Example 19.19 was $\text{Bin}(3, 0.1)$. The PMF rises to a peak near $k \approx np$ and then falls; the ratio of successive terms, $p_X(k+1)/p_X(k) = \frac{(n-k)p}{(k+1)(1-p)}$, exceeds 1 exactly when $k < (n+1)p - 1$, which locates the mode at $\lfloor (n+1)p \rfloor$.

Geometric distribution

$N \sim \text{Geom}(p)$: the number of independent trials up to and including the first success,

$$p_N(k) = (1-p)^{k-1}p, \qquad k = 1, 2, 3, \ldots$$

with tail $P(N > k) = (1-p)^k$ (the first k trials all fail).

The geometric distribution is the only **memoryless** discrete distribution: $P(N > m + k \mid N > m) = (1-p)^{m+k}/(1-p)^m = P(N > k)$. Having already failed m times does not make success any closer.

Poisson distribution

$X \sim \text{Poisson}(\lambda)$, $\lambda > 0$:

$$p_X(k) = e^{-\lambda}\frac{\lambda^k}{k!}, \qquad k = 0, 1, 2, \ldots$$

The probabilities sum to $e^{-\lambda}e^{\lambda} = 1$ by the exponential series.

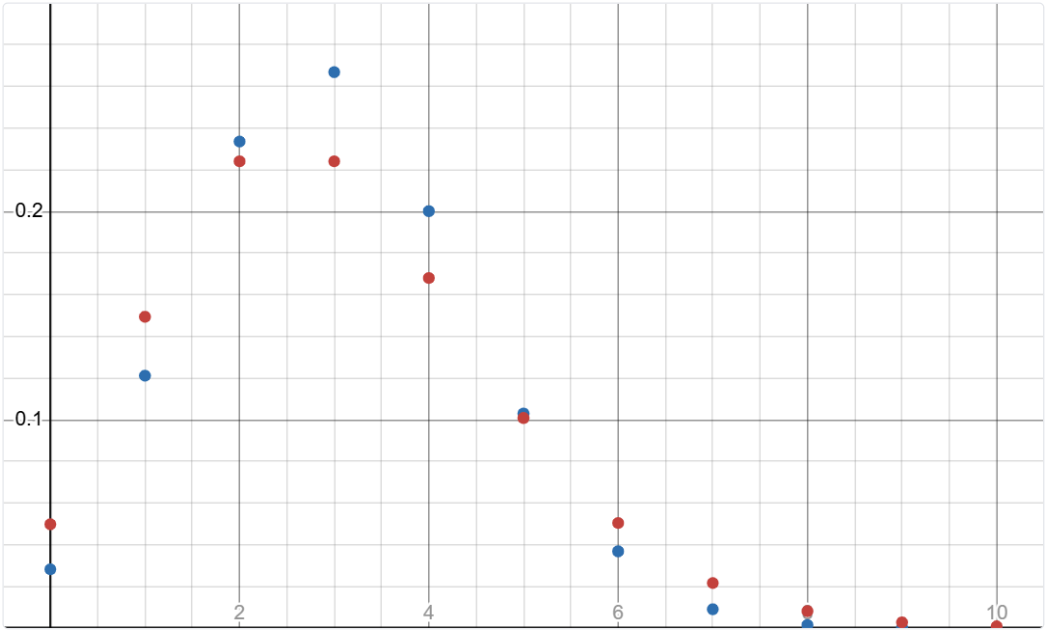
Poisson limit of the binomial

If $n \to \infty$ and $p \to 0$ with $np = \lambda$ fixed, then $\binom{n}{k}p^k(1-p)^{n-k} \to e^{-\lambda}\lambda^k/k!$ for each fixed k.

Proof. Write $p = \lambda/n$:

$$\binom{n}{k}\left(\frac{\lambda}{n}\right)^k\left(1-\frac{\lambda}{n}\right)^{n-k} = \frac{\lambda^k}{k!}\cdot\underbrace{\frac{n(n-1)\cdots(n-k+1)}{n^k}}_{\to 1}\cdot\underbrace{\left(1-\frac{\lambda}{n}\right)^{n}}_{\to e^{-\lambda}}\cdot\underbrace{\left(1-\frac{\lambda}{n}\right)^{-k}}_{\to 1}. \qquad \square$$

This is the **law of rare events**: whenever a large number of independent opportunities each produce an event with small probability, the count is approximately Poisson with λ = expected number of events. Photon arrivals at a photodiode, electrons crossing a barrier (shot noise), packet arrivals from many independent users, defects on a wafer, and bit errors in a long frame all follow this pattern. If events occur "at random" at an average **rate** r per second, the count in an interval of length τ is $\text{Poisson}(r\tau)$ — the **Poisson process** of Ch. 21.



PMFs of $\text{Bin}(10, 0.3)$ *(blue) and* $\text{Poisson}(3)$ *(red), both with mean 3. The Poisson PMF is slightly wider and has a tail beyond* $k = 10$*; with only* $n = 10$ *trials and* $p = 0.3$ *the Poisson approximation is rough, but it becomes excellent as* n *grows with* np *fixed.*

Worked Example 19.22 — Errors in a frame (Poisson approximation)

A 1000-bit frame is sent over a channel with independent bit errors, $p = 0.002$. Find the probability that the frame is error-free and the probability that it has 3 or more errors (more than a double-error-correcting code can handle).

Step 1 — exact model. $X \sim \text{Bin}(1000, 0.002)$. $P(X = 0) = 0.998^{1000} = 0.13506$.

Step 2 — Poisson model. $\lambda = np = 2$. $P(X = 0) \approx e^{-2} = 0.13534$ — within $0.2\,\%$.

Step 3 — three or more errors.

$$P(X \geq 3) = 1 - e^{-2}\left(1 + 2 + \frac{2^2}{2!}\right) = 1 - 5e^{-2} = 0.32332.$$

The exact binomial value is 0.32332 to five decimals. ✓

One frame in three would be uncorrectable; the design must lower p, shorten the frame or use a stronger code.

Worked Example 19.23 — Retransmissions (geometric)

An ARQ protocol resends a packet until it is acknowledged; each attempt succeeds independently with probability 0.8. Find the probability that more than 3 attempts are needed, and the probability that exactly 2 attempts are needed given that the first failed.

Step 1 — tail. $P(N > 3) = (0.2)^3 = 0.008$.

Step 2 — memorylessness. $P(N = 2 \mid N > 1) = P(N = 1) = 0.8$ by the memoryless property. Directly: $P(N = 2)/P(N > 1) = (0.2)(0.8)/0.2 = 0.8$. ✓

Continuous families

Uniform distribution

$X \sim \mathcal{U}(a, b)$: $f_X(x) = \dfrac{1}{b - a}$ for $a < x < b$ and 0 otherwise. Probability is proportional to length: $P(c < X < d) = (d - c)/(b - a)$ for $a \leq c < d \leq b$.

The uniform distribution models the phase of an oscillator that is not synchronized to anything ($\Theta \sim \mathcal{U}(0, 2\pi)$), and the rounding error of a fine quantizer with step Δ ($e \sim \mathcal{U}(-\Delta/2, \Delta/2)$, Section 19.9).

Exponential distribution

$T \sim \text{Exp}(\lambda)$, rate $\lambda > 0$:

$$f_T(t) = \lambda e^{-\lambda t}\ (t \geq 0), \qquad P(T > t) = e^{-\lambda t}\ (t \geq 0).$$

Memoryless property

$T \sim \text{Exp}(\lambda)$ satisfies $P(T > s + t \mid T > s) = P(T > t)$ for all $s, t \geq 0$, and it is the only continuous distribution on $[0, \infty)$ with this property.

Proof of the property. $P(T > s + t \mid T > s) = e^{-\lambda(s+t)}/e^{-\lambda s} = e^{-\lambda t}$. (Uniqueness: $G(t) = P(T > t)$ must satisfy $G(s + t) = G(s)G(t)$, whose only monotone solutions are exponentials.) □

A component with exponential lifetime does not age: a unit that has survived 10 000 hours is as good as new. This is realistic for electronic parts during their "useful life" — after infant mortality and before wear-out — where failures are caused by random external stresses rather than accumulated damage. The exponential is also the distribution of the time between events of a Poisson process: if events occur at rate r, the waiting time T to the first one satisfies $P(T > t) = P(\text{no events in } [0, t]) = e^{-rt}$.

Gaussian (normal) distribution

$X \sim \mathcal{N}(\mu, \sigma^2)$, with $\mu \in \mathbb{R}$, $\sigma > 0$:

$$f_X(x) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right), \qquad -\infty < x < \infty.$$

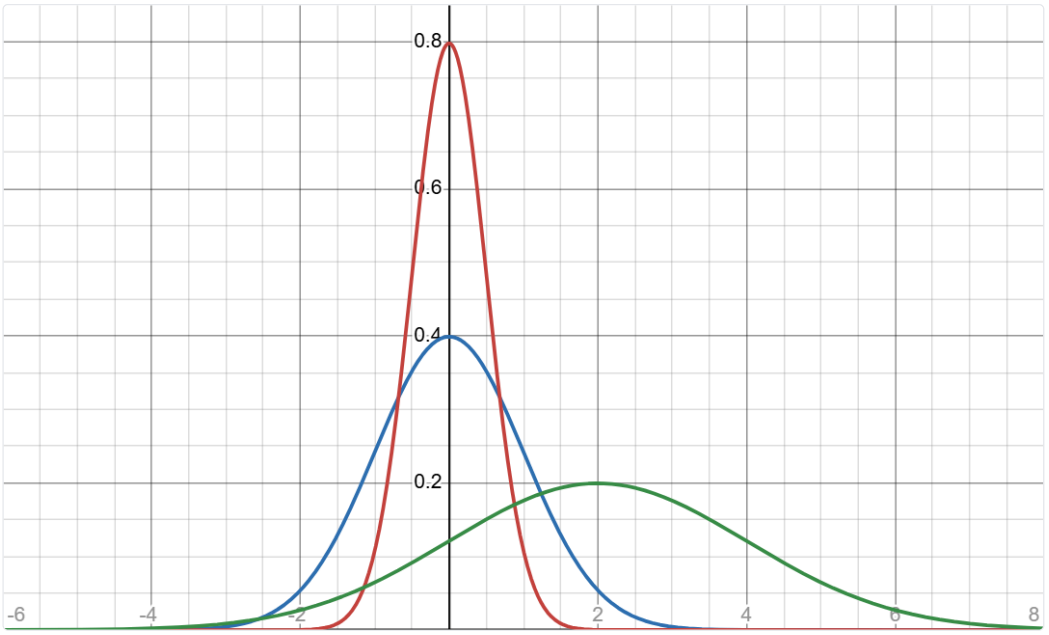
The **standard** Gaussian is $Z \sim \mathcal{N}(0, 1)$, with PDF $\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$, CDF $\Phi(z)$ and tail $Q(z) = 1 - \Phi(z)$.

Normalization. With $I = \int_{-\infty}^{\infty} e^{-z^2/2} dz$, convert I^2 to a double integral over the plane and use polar coordinates (Ch. 6):

$$I^2 = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(u^2+v^2)/2}\, du\, dv = \int_0^{2\pi}\int_0^{\infty} e^{-r^2/2}\, r\, dr\, d\phi = 2\pi,$$

so $I = \sqrt{2\pi}$, and the substitution $z = (x - \mu)/\sigma$ handles general μ, σ.

The Gaussian is symmetric about μ, has its inflection points at $\mu \pm \sigma$, and its tails fall off faster than any exponential. Its pre-eminence comes from the **central limit theorem** (Ch. 20): the sum of many small independent contributions is approximately Gaussian whatever the individual distributions. Thermal noise (the sum of the effects of enormous numbers of electrons), aggregate interference, and manufacturing tolerances that result from many small process variations are all modelled as Gaussian. Gaussian probabilities are computed from tables of Φ or Q (Section 19.8).



Gaussian PDFs: $\mathcal{N}(0,1)$ (blue), $\mathcal{N}(0,0.25)$ (red) and $\mathcal{N}(2,4)$ (green). The mean μ shifts the curve; the standard deviation σ scales its width by σ and its height by $1/\sigma$, keeping the area equal to 1.

Rayleigh distribution

$R \sim \text{Rayleigh}(\sigma)$:

$$f_R(r) = \frac{r}{\sigma^2} e^{-r^2/(2\sigma^2)} \ (r \ge 0), \qquad P(R > r) = e^{-r^2/(2\sigma^2)}.$$

It is the distribution of the envelope $R = \sqrt{X^2 + Y^2}$ when X, Y are independent $\mathcal{N}(0, \sigma^2)$ — the amplitude of narrowband Gaussian noise, or of a radio signal arriving over many scattered paths with no direct line of sight. Its mean-square value is $E[R^2] = 2\sigma^2$.

Worked Example 19.24 — Component lifetime and deep fades

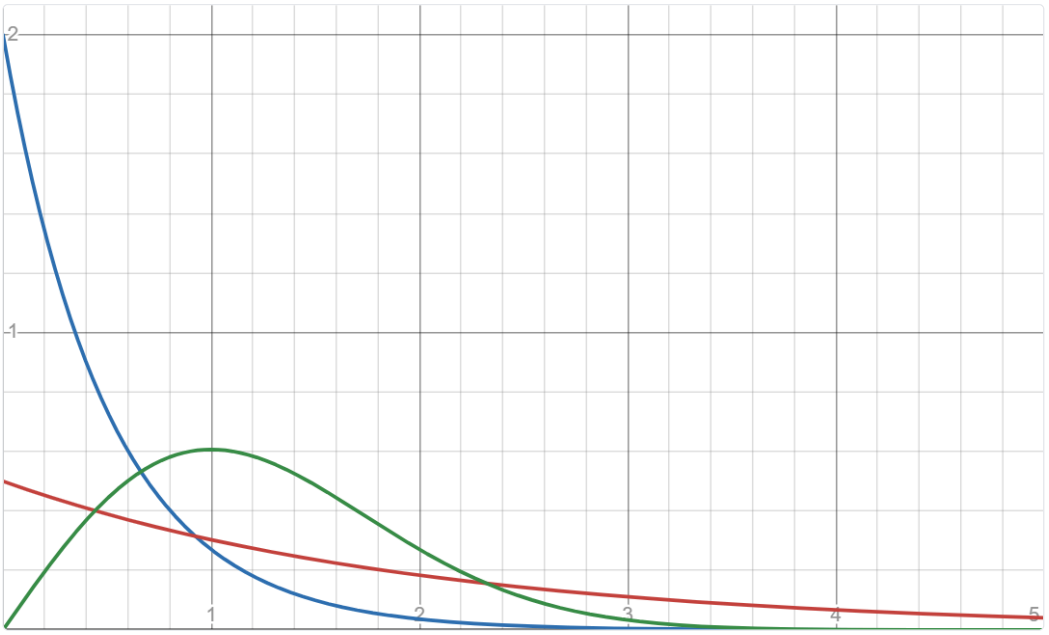
(a) A power MOSFET has an exponential lifetime with failure rate $\lambda = 2 \times 10^{-5}$ per hour. Find the probability that it survives one year of continuous operation (8760 h), and the probability that it survives a second year given that it survived the first. (b) A Rayleigh-fading signal has mean-square amplitude $E[R^2] = 2\sigma^2$. Find the probability that the amplitude is more than 20 dB below its RMS value.

(a) $P(T > 8760) = e^{-2\times10^{-5}\times 8760} = e^{-0.1752} = 0.839$. By memorylessness, the conditional probability of surviving the second year is also 0.839, and of surviving both years $0.839^2 = 0.704$.

(b) 20 dB below RMS in amplitude means $R < 0.1\sqrt{2\sigma^2}$, i.e. $R^2 < 0.01 \times 2\sigma^2$:

$$P(R < 0.1\,R_{\text{rms}}) = 1 - e^{-0.02\sigma^2/(2\sigma^2)} = 1 - e^{-0.01} = 0.00995.$$

About 1 % of the time the signal is in a 20 dB fade; by the same formula, 10 % of the time it is in a 10 dB fade ($1 - e^{-0.1} = 0.095$). Each extra 10 dB of fade margin buys a factor of 10 in outage probability — the classic Rayleigh rule of thumb.



Exponential PDFs with $\lambda = 2$ (blue) and $\lambda = 0.5$ (red), both maximal at $t = 0$: the most likely lifetime is "immediately", though the median is $\ln 2/\lambda$. The Rayleigh PDF with $\sigma = 1$ (green) vanishes at 0, peaks at $r = \sigma$ and has a Gaussian-like tail.

Joint distributions: a first look

Engineering problems usually involve several random quantities at once: the transmitted and received bits, the in-phase and quadrature noise components, the lifetimes of two redundant units. Their *individual* distributions do not say how they interact; that information is in the **joint distribution**. We introduce the essentials for two random variables; correlation and covariance as data-analysis tools are developed in Ch. 20, and families of RVs indexed by time in Ch. 21.

Joint PMF, joint PDF, joint CDF

For two random variables X, Y on the same probability space:

- the **joint CDF** is $F_{XY}(x, y) = P(X \le x,\ Y \le y)$;
- if both are discrete, the **joint PMF** is $p_{XY}(x_i, y_k) = P(X = x_i,\ Y = y_k)$, with $\sum_i \sum_k p_{XY}(x_i, y_k) = 1$;
- they are **jointly continuous** with **joint PDF** $f_{XY}(x, y) \ge 0$ if, for every region D of the plane,

$$P\big((X, Y) \in D\big) = \iint_D f_{XY}(x, y)\, dx\, dy, \qquad \iint_{\mathbb{R}^2} f_{XY}\, dx\, dy = 1.$$

Then $f_{XY}(x, y) = \dfrac{\partial^2 F_{XY}}{\partial x\, \partial y}$.

From the joint distribution we recover each variable alone — the **marginal** distributions — by summing or integrating out the other variable (law of total probability):

$$p_X(x_i) = \sum_k p_{XY}(x_i, y_k), \qquad f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y)\, dy.$$

The **conditional** distribution of Y given $X = x$ is the joint divided by the marginal, exactly as in Section 19.2:

$$p_{Y|X}(y_k \mid x_i) = \frac{p_{XY}(x_i, y_k)}{p_X(x_i)}, \qquad f_{Y|X}(y \mid x) = \frac{f_{XY}(x, y)}{f_X(x)} \quad (f_X(x) > 0).$$

Independent random variables

X and Y are **independent** if $P(X \in A,\ Y \in B) = P(X \in A)P(Y \in B)$ for all events A, B. Equivalently, the joint CDF (or joint PMF, or joint PDF) factors into the product of the marginals:

$$F_{XY}(x, y) = F_X(x) F_Y(y) \quad \text{for all } x, y, \qquad \text{or} \qquad f_{XY}(x, y) = f_X(x) f_Y(y).$$

The marginals determine the joint distribution only in the independent case; in general, many different joint distributions share the same marginals.

Worked Example 19.25 — Joint PMF of a binary channel

For the channel of Example 19.11 ($p_0 = 0.6$, $P(Y=1 \mid X=0) = 0.05$, $P(Y=0 \mid X=1) = 0.10$), tabulate the joint PMF of input X and output Y, find the marginals, and decide whether X and Y are independent.

Step 1 — joint PMF by the multiplication rule $p_{XY}(x,y) = P(Y=y \mid X=x)P(X=x)$:

	$Y=0$	$Y=1$	p_X
$X=0$	0.57	0.03	0.60
$X=1$	0.04	0.36	0.40
p_Y	0.61	0.39	1

Step 2 — marginals are the row and column sums (written in the margins — hence the name).

Step 3 — independence test. $p_X(0)p_Y(0) = 0.6 \times 0.61 = 0.366 \neq 0.57$. Not independent — fortunately, since a channel whose output is independent of its input carries no information.

Worked Example 19.26 — A joint PDF on a triangle

X and Y have joint PDF $f_{XY}(x,y) = 2$ on the triangle $0 < y < x < 1$ and 0 elsewhere. (For example, X and Y could be the larger and smaller of two independent $\mathcal{U}(0,1)$ readings.) Find the marginals, the conditional PDF of Y given $X = x$, and $P(X + Y < 1)$.

Step 1 — check normalization. The triangle has area $1/2$; $2 \times \frac{1}{2} = 1$. ✓

Step 2 — marginals.

$$f_X(x) = \int_0^x 2\,dy = 2x \quad (0 < x < 1), \qquad f_Y(y) = \int_y^1 2\,dx = 2(1-y) \quad (0 < y < 1).$$

Step 3 — conditional. $f_{Y|X}(y \mid x) = 2/(2x) = 1/x$ for $0 < y < x$: given $X = x$, Y is uniform on $(0, x)$.

Step 4 — a probability. The region $\{y < x,\ x + y < 1,\ y > 0\}$ is the triangle with vertices $(0,0)$, $(1,0)$, $(\frac{1}{2}, \frac{1}{2})$, of area $\frac{1}{4}$. So $P(X+Y<1) = 2 \times \frac{1}{4} = \frac{1}{2}$.

X and Y are not independent: $f_X(x)f_Y(y) = 4x(1-y)$ is not equal to 2, and in any case the support (a triangle, not a rectangle) already rules out independence.

Sums of independent random variables

The distribution of a sum $Z = X + Y$ arises constantly: signal plus noise, total delay of two stages, combined tolerance of two resistors in series, total lifetime of a unit and its cold spare. For independent X, Y, the event $\{Z \le z\}$ is the region $x + y \le z$, and

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_X(x) f_Y(y)\,dy\,dx.$$

Differentiating with respect to z gives the central result.

Distribution of a sum of independent RVs

If X and Y are independent with PDFs f_X, f_Y, then $Z = X + Y$ has PDF

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x)\, f_Y(z-x)\,dx = (f_X * f_Y)(z),$$

the **convolution** of the two densities. For independent integer-valued RVs, $p_Z(n) = \sum_k p_X(k)\, p_Y(n-k)$, the discrete convolution.

This is the same convolution that gives the output of an LTI system (Ch. 16, Ch. 17), and it can be computed the same way — by transforms. The transform of a density is its characteristic function (Section 19.9), and convolution becomes multiplication.

Worked Example 19.27 — Sums of uniform and exponential RVs

(a) Two independent errors $X, Y \sim \mathcal{U}(0,1)$ add. Find the PDF of $Z = X + Y$. (b) A unit with $\text{Exp}(\lambda)$ lifetime T_1 has a cold spare, switched in perfectly on failure, with independent $\text{Exp}(\lambda)$ lifetime T_2. Find the reliability of the pair, and evaluate it for $\lambda = 2 \times 10^{-5}\ \text{h}^{-1}$ at one year.

(a) $f_X(x) f_Y(z-x) = 1$ when $0 < x < 1$ and $0 < z - x < 1$, i.e. $\max(0, z-1) < x < \min(1, z)$. The length of that interval is z for $0 < z \le 1$ and $2 - z$ for $1 < z < 2$:

$$f_Z(z) = \begin{cases} z, & 0 < z \le 1 \\ 2 - z, & 1 < z < 2 \\ 0, & \text{otherwise} \end{cases}$$

— a **triangle**, the convolution of two rectangles. Adding more uniforms smooths the shape further towards a Gaussian (central limit theorem).

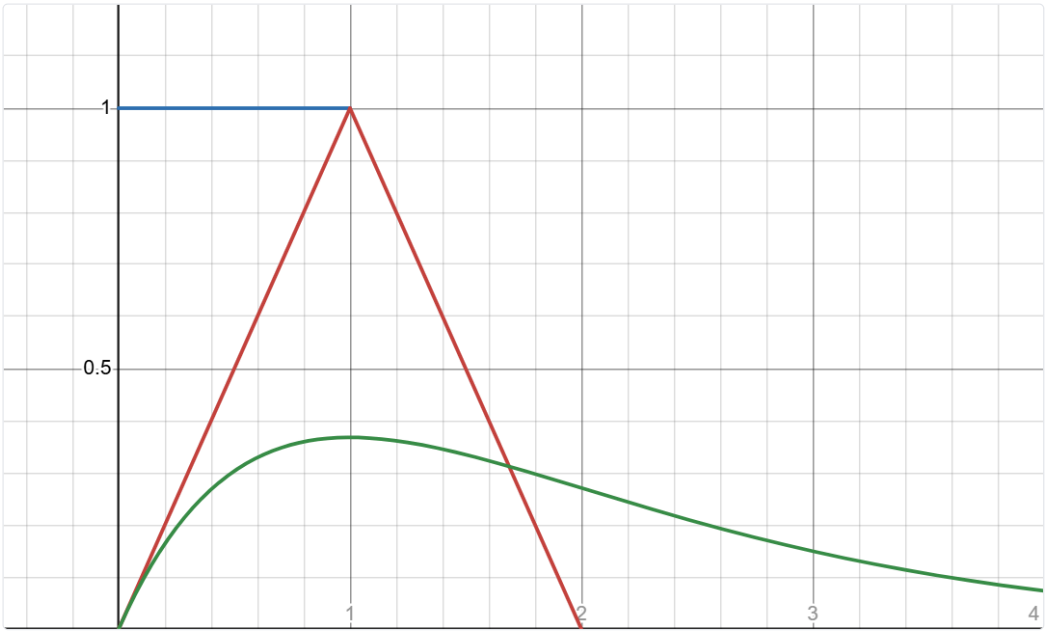
(b) For $z > 0$,

$$f_Z(z) = \int_0^z \lambda e^{-\lambda x}\, \lambda e^{-\lambda(z-x)}\, dx = \lambda^2 e^{-\lambda z} \int_0^z dx = \lambda^2 z e^{-\lambda z},$$

the **Erlang-2** (gamma) density. The reliability is

$$R(t) = P(Z > t) = \int_t^\infty \lambda^2 z e^{-\lambda z} dz = (1 + \lambda t) e^{-\lambda t}.$$

With $\lambda t = 0.1752$: $R = 1.1752\, e^{-0.1752} = 0.9863$, against 0.8393 for a single unit and $1 - (1 - 0.8393)^2 = 0.9742$ for two units running in *active* parallel. A cold spare beats a hot one because it does not age while waiting (assuming a perfect switch).



Convolution in action: the $\mathcal{U}(0,1)$ density (blue) convolved with itself gives the triangle (red); two independent $\text{Exp}(1)$ lifetimes add to the Erlang-2 density ze^{-z} (green), which, unlike the exponential, is zero at the origin — two failures rarely happen in quick succession.

Common Pitfall — marginals do not determine the joint

Knowing that X and Y are each $\mathcal{N}(0,1)$ does not determine $P(X + Y > 2)$: if $Y = X$ the answer is $Q(1) = 0.159$; if $Y = -X$ it is 0; if they are independent it is $Q(\sqrt{2}) = 0.079$. Any computation involving two random variables needs their joint distribution, and assuming independence is a modelling decision that must be justified.

Engineering Note — independence makes joint models cheap

Specifying a general joint PMF of n binary variables needs $2^n - 1$ numbers; under independence, n numbers suffice. This exponential saving is why independence (or conditional independence given a few hidden variables, as in the fading channel) underlies almost every tractable model of noise, errors and failures.

19.7 Probability density and mass functions

Motivation

The PMF and PDF are the working descriptions of a random variable: they are what you sketch, fit to histograms, multiply by likelihoods and integrate against. The PMF is simple — a list of probabilities. The PDF needs more care, because its values are *not* probabilities, it can exceed 1, it can contain impulses, and it transforms in a non-obvious way when the variable is passed through a nonlinearity. This section states the properties precisely, shows how to treat discrete,

continuous and mixed variables in one notation, and derives the transformation rule that gives the density at the output of a memoryless system.

The probability mass function

PMF

The **probability mass function** of a discrete random variable X with range $S_X = \{x_1, x_2, \dots\}$ is

$$p_X(x) = P(X = x).$$

It satisfies $p_X(x) \geq 0$, $p_X(x) = 0$ for $x \notin S_X$, and $\sum_k p_X(x_k) = 1$. For any set B, $P(X \in B) = \sum_{x_k \in B} p_X(x_k)$.

Any non-negative sequence summing to 1 is a valid PMF, and constructing a model often means choosing such a sequence from a family and fixing its parameters.

The probability density function

PDF

A random variable X is **continuous** with **probability density function** f_X if $f_X(x) \geq 0$, $\int_{-\infty}^{\infty} f_X(x)\,dx = 1$, and for all $a < b$

$$P(a < X \leq b) = \int_a^b f_X(x)\,dx.$$

The name "density" is literal. For a small interval of width Δx around a point where f_X is continuous,

$$P(x < X \leq x + \Delta x) \approx f_X(x)\,\Delta x,$$

so $f_X(x)$ is probability *per unit of* x. If X is a voltage, f_X has units of V^{-1}. This has three consequences that trip up newcomers:

1. $f_X(x)$ **is not a probability.** It can exceed 1: the $\mathcal{U}(0, 0.1)$ density equals 10 on its support.
2. **Only areas are probabilities.** $P(X = x) = \int_x^x f_X = 0$ for every x.
3. **Densities change under a change of units.** If $Y = 1000X$ (the same voltage in mV), then $f_Y(y) = f_X(y/1000)/1000$, so that $f_Y(y)\,dy = f_X(x)\,dx$ (see the transformation rule below).

A histogram of measured values, scaled so that its total area is 1 (bar height = fraction of samples in the bin divided by bin width), approximates the PDF; that is how densities are estimated from data in Ch. 20.

Useful single-number descriptions read off a PDF or PMF are the **mode** (the location of the maximum — the "most likely" value), the **median** m (with $P(X \leq m) = 1/2$, Section 19.8), and the **mean** (the balance point, Section 19.9). For a symmetric unimodal density like the Gaussian, all three coincide; for a skewed one like the exponential they differ (mode 0, median $\ln 2/\lambda$, mean $1/\lambda$).

Worked Example 19.28 — Normalizing a density

The normalized position X of a laser spot on a detector has PDF $f_X(x) = c\,x(1 - x)$ for $0 < x < 1$ and 0 otherwise. Find c, $P(X > 0.5)$ and $P(0.25 < X < 0.75)$.

Step 1 — normalize. $\int_0^1 x(1-x)\,dx = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$, so $c = 6$.

Step 2 — by symmetry about $x = \frac{1}{2}$, $P(X > 0.5) = 0.5$.

Step 3 — integrate. With antiderivative $G(x) = 3x^2 - 2x^3$,

$$P(0.25 < X < 0.75) = G(0.75) - G(0.25) = (1.6875 - 0.84375) - (0.1875 - 0.03125) = 0.84375 - 0.15625 = 0.6875.$$

The maximum value of the density is $f_X(0.5) = 1.5$ — larger than 1, which is perfectly legitimate.

Discrete and mixed variables as densities: Dirac impulses

Using the Dirac delta $\delta(x)$ (Ch. 16), a discrete random variable can be given a *generalized* density

$$f_X(x) = \sum_k p_X(x_k)\,\delta(x - x_k),$$

because $\int_a^b f_X(x)\,dx$ then picks up exactly the masses $p_X(x_k)$ with $a < x_k \leq b$. A mixed random variable has a density that is the sum of an ordinary part and impulses. The limiter output of Example 19.21, for instance, has

$$f_Y(y) = 0.25\,\delta(y + 1) + 0.25\,\delta(y - 1) + 0.25\ \mathrm{rect}(y/2),$$

where $\mathrm{rect}(y/2) = 1$ for $|y| < 1$. This unified notation lets every formula of this chapter — expectations, transformations, convolutions — be written once for all three types. The CDF (Section 19.8) achieves the same unification without generalized functions.

Conditional densities given an event

If B is an event with $P(B) > 0$, the **conditional PDF** of X given B is the derivative of the conditional CDF $P(X \leq x \mid B)$. When $B = \{X \in D\}$ is itself defined by X, the result is the original density restricted to D and renormalized:

$$f_{X|B}(x) = \begin{cases} \dfrac{f_X(x)}{P(X \in D)}, & x \in D \\ 0, & x \notin D. \end{cases}$$

For an exponential lifetime and $B = \{T > s\}$ this gives $f_{T|B}(t) = \lambda e^{-\lambda(t-s)}$ for $t > s$: the same exponential shape shifted to start at s — memorylessness again.

Bayes' theorem with densities

When the observation is continuous, the event $\{X = x\}$ has probability zero and Section 19.3's formula cannot be applied literally. Conditioning on the small interval $\{x < X \le x + \Delta x\}$ and letting $\Delta x \to 0$, the factors Δx cancel, leaving

$$P(H_i \mid X = x) = \frac{f_{X|H_i}(x)\,P(H_i)}{\sum_j f_{X|H_j}(x)\,P(H_j)},$$

i.e. Bayes' theorem with *conditional densities* playing the role of likelihoods. The MAP rule compares $f_{X|H_i}(x)P(H_i)$.

Worked Example 19.29 — MAP threshold for a binary signal in Gaussian noise

A receiver observes $X = s + N$, where $s = -A$ (symbol 0) with probability p_0 or $s = +A$ (symbol 1) with probability p_1, and $N \sim \mathcal{N}(0, \sigma^2)$ is independent noise. Find the MAP decision threshold, and evaluate it and the error probability for $A = 1$ V, $\sigma = 0.5$ V, $p_0 = 0.75$, $p_1 = 0.25$.

Step 1 — likelihoods. Given symbol 1, $X \sim \mathcal{N}(A, \sigma^2)$; given symbol 0, $X \sim \mathcal{N}(-A, \sigma^2)$.

Step 2 — likelihood-ratio test. Decide 1 when $p_1 f_{X|1}(x) > p_0 f_{X|0}(x)$:

$$\frac{f_{X|1}(x)}{f_{X|0}(x)} = \exp\!\left(\frac{(x+A)^2 - (x-A)^2}{2\sigma^2}\right) = \exp\!\left(\frac{2Ax}{\sigma^2}\right) > \frac{p_0}{p_1}.$$

Taking logarithms: decide 1 when $x > x_{\text{th}}$, with

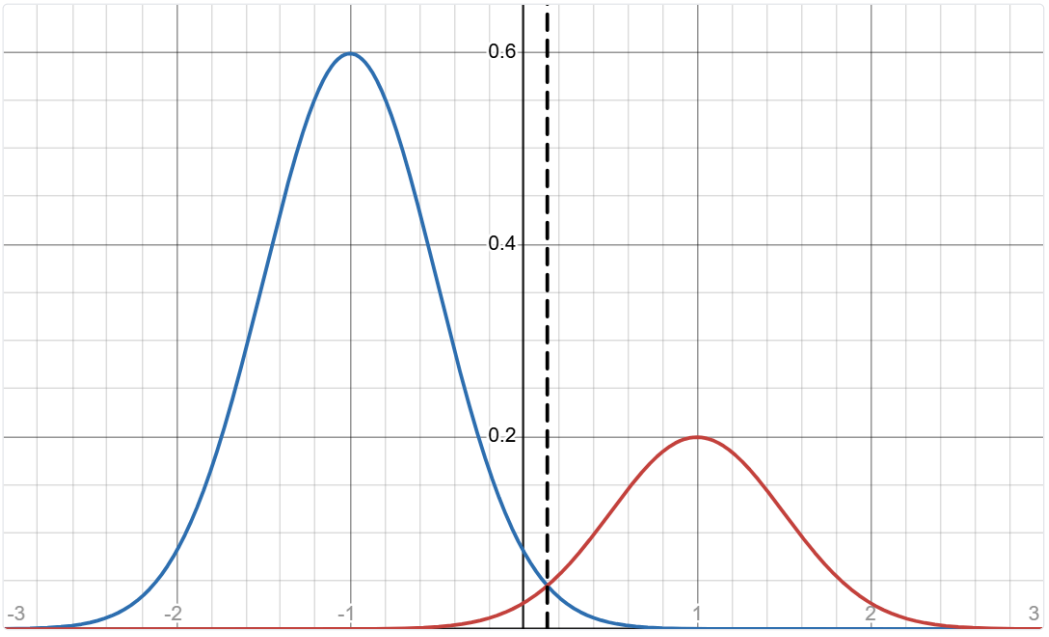
$$x_{\text{th}} = \frac{\sigma^2}{2A}\ln\frac{p_0}{p_1}.$$

Step 3 — numbers. $x_{\text{th}} = \dfrac{0.25}{2}\ln 3 = 0.1373$ V. The threshold moves *towards the less likely symbol*, so that more ambiguous observations are attributed to the more likely one.

Step 4 — error probability. An error occurs if symbol 0 is sent and $X > x_{\text{th}}$, or symbol 1 is sent and $X < x_{\text{th}}$:

$$P_e = p_0\,Q\!\left(\frac{x_{\text{th}} + A}{\sigma}\right) + p_1\,Q\!\left(\frac{A - x_{\text{th}}}{\sigma}\right) = 0.75\,Q(2.275) + 0.25\,Q(1.725) = 0.01916.$$

With the "natural" threshold 0 (the ML rule), $P_e = Q(A/\sigma) = Q(2) = 0.02275$. The MAP threshold reduces the error probability by about 16 %. With equal priors, $\ln(p_0/p_1) = 0$ and the two rules coincide.



Prior-weighted likelihoods $p_0 f_{X|0}(x)$ (blue) and $p_1 f_{X|1}(x)$ (red) for Example 19.29. The MAP threshold (dashed) is where the two curves cross, at $x = 0.137$ V rather than midway at 0; the error probability is the area under the lower curve on each side of the threshold.

Functions of a random variable: the density transformation

We now find the PDF of $Y = g(X)$ when X is continuous. Start with the simplest case.

Transformation by a monotone function

Let X have PDF f_X and let g be differentiable and strictly monotone (increasing or decreasing) on the range of X, with inverse $x = g^{-1}(y)$. Then $Y = g(X)$ has PDF

$$f_Y(y) = f_X\big(g^{-1}(y)\big)\left|\frac{dx}{dy}\right| = \left.\frac{f_X(x)}{|g'(x)|}\right|_{x=g^{-1}(y)}.$$

Derivation. For increasing g, $\{Y \le y\} = \{X \le g^{-1}(y)\}$, so $F_Y(y) = F_X(g^{-1}(y))$; differentiate by the chain rule to get $f_X(g^{-1}(y))\,\frac{d}{dy}g^{-1}(y)$. For decreasing g, $\{Y \le y\} = \{X \ge g^{-1}(y)\}$, so $F_Y(y) = 1 - F_X(g^{-1}(y))$, whose derivative carries a minus sign that the absolute value absorbs. Physically: probability is conserved, $f_Y(y)|dy| = f_X(x)|dx|$ — a steep part of g stretches a small dx into a large dy and thins the density. □

Transformation by a general function

If the equation $y = g(x)$ has finitely many real roots $x_1(y), \dots, x_m(y)$, at each of which $g'(x_i) \neq 0$, then

$$f_Y(y) = \sum_{i=1}^{m} \frac{f_X(x_i)}{|g'(x_i)|},$$

and $f_Y(y) = 0$ if there are no roots. Intervals on which g is constant produce point masses (impulses) in f_Y, as in the limiter.

Reason. The event $\{y < Y \le y + dy\}$ is the union of disjoint small intervals of x, one around each root, of widths $dy/|g'(x_i)|$; add their probabilities.

Worked Example 19.30 — Linear transformation of a Gaussian

A sensor output $X \sim \mathcal{N}(\mu, \sigma^2)$ is amplified and offset: $Y = aX + b$ with $a \neq 0$. Find the distribution of Y, and the density at the output of a gain-10, offset $+0.5$ V stage when $X \sim \mathcal{N}(0.1,\ 0.02^2)$ V.

Step 1 — inverse. $x = (y - b)/a$, $|dx/dy| = 1/|a|$.

Step 2 — substitute.

$$f_Y(y) = \frac{1}{|a|}\cdot\frac{1}{\sqrt{2\pi}\sigma}\exp\left(-\frac{\left(\frac{y-b}{a} - \mu\right)^2}{2\sigma^2}\right) = \frac{1}{\sqrt{2\pi}\,|a|\sigma}\exp\left(-\frac{(y - a\mu - b)^2}{2a^2\sigma^2}\right).$$

So $Y \sim \mathcal{N}(a\mu + b,\ a^2\sigma^2)$: **a linear function of a Gaussian is Gaussian.**

Step 3 — numbers. $Y \sim \mathcal{N}(10(0.1) + 0.5,\ (10 \times 0.02)^2) = \mathcal{N}(1.5,\ 0.2^2)$ V. In particular $Z = (X - \mu)/\sigma$ is $\mathcal{N}(0, 1)$ — **standardization**, which reduces every Gaussian probability to the standard tables.

Worked Example 19.31 — Square-law detector

Gaussian noise $X \sim \mathcal{N}(0, \sigma^2)$ drives a square-law detector, $Y = X^2$. Find f_Y and the probability that the detector output exceeds $4\sigma^2$.

Step 1 — roots. For $y > 0$, $x^2 = y$ has roots $x_{1,2} = \pm\sqrt{y}$, with $|g'(x)| = 2|x| = 2\sqrt{y}$. For $y < 0$ there are no roots.

Step 2 — add contributions.

$$f_Y(y) = \frac{f_X(\sqrt{y}) + f_X(-\sqrt{y})}{2\sqrt{y}} = \frac{1}{\sqrt{2\pi y}\,\sigma}\,e^{-y/(2\sigma^2)}, \qquad y > 0.$$

This is the **chi-square distribution with one degree of freedom** (scaled by σ^2). It is infinite at $y = 0$ — yet integrable, since $y^{-1/2}$ has a finite integral near 0.

Step 3 — tail probability. Easier through X than through f_Y: $P(Y > 4\sigma^2) = P(|X| > 2\sigma) = 2Q(2) = 0.0455$.

Worked Example 19.32 — A sinusoid with random phase

An oscillator's output sampled at a random instant is $Y = A\cos\Theta$, with $\Theta \sim \mathcal{U}(0, 2\pi)$. Find the PDF of Y and the fraction of time the sinusoid spends within half its amplitude of zero.

Step 1 — roots. For $|y| < A$, $A\cos\theta = y$ has two roots in $(0, 2\pi)$, $\theta_1 = \arccos(y/A)$ and $\theta_2 = 2\pi - \theta_1$. At each, $|g'(\theta)| = A|\sin\theta| = \sqrt{A^2 - y^2}$.

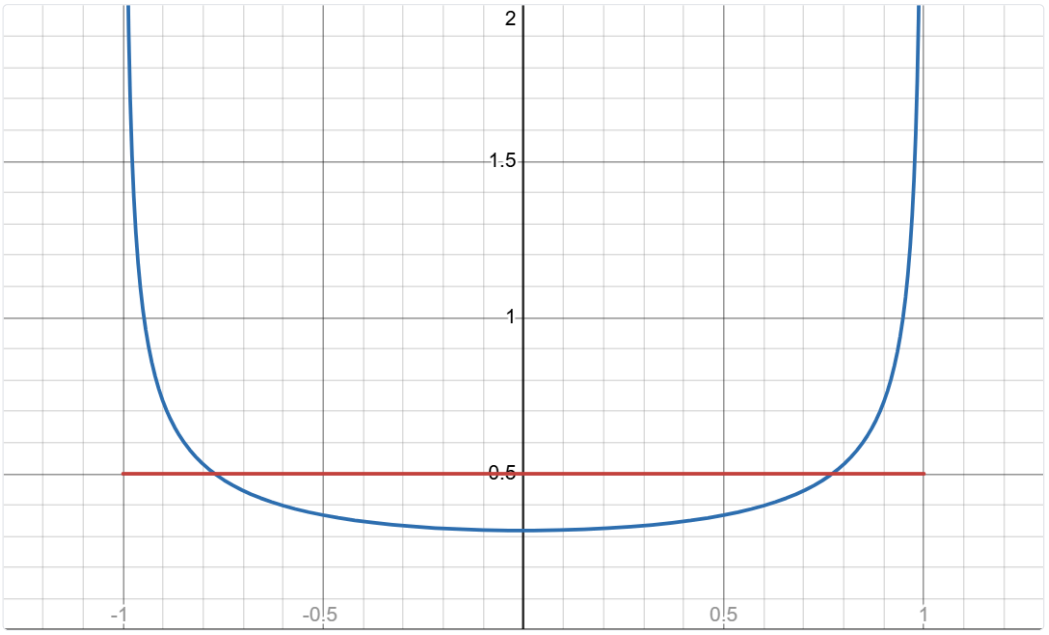
Step 2 — transform. $f_\Theta = 1/(2\pi)$ on $(0, 2\pi)$, so

$$f_Y(y) = 2\cdot\frac{1/(2\pi)}{\sqrt{A^2 - y^2}} = \frac{1}{\pi\sqrt{A^2 - y^2}}, \qquad |y| < A.$$

This is the **arcsine** density. It is smallest at $y = 0$ and infinite (but integrable) at $y = \pm A$: a sinusoid moves fastest through zero and lingers near its peaks.

Step 3 — probability. $P(|Y| < A/2) = \int_{-A/2}^{A/2} \frac{dy}{\pi\sqrt{A^2 - y^2}} = \frac{2}{\pi}\arcsin\frac{1}{2} = \frac{1}{3}$.

Only a third of the time is spent in the middle half of the range. This is why a histogram of ADC codes for a full-scale sine input (the "code density test") is bathtub-shaped, and why departures from the arcsine shape reveal the converter's non-linearity.



Arcsine density of $\cos\Theta$ with uniform phase (blue), compared with the uniform density on $(-1,1)$ (red). The sinusoid spends most of its time near its extremes; a triangle wave of the same amplitude would give the flat red density.

Common Pitfall — forgetting the Jacobian

The most common error in transforming densities is to substitute $x = g^{-1}(y)$ into f_X and stop, omitting the factor $|dx/dy|$. The result is not even normalized. A quick check: integrate your f_Y — it must give 1. A second check: the units of f_Y must be the reciprocal of the units of Y.

Engineering Note — memoryless nonlinearities

The transformation rule is the probabilistic counterpart of a transfer characteristic. Diodes, square-law and envelope detectors, logarithmic amplifiers, companders, limiters and quantizers all map an input distribution into an output distribution by exactly this rule. Designers use it in reverse, too: a compander is chosen so that the output of a speech signal with a peaked (Laplacian-like) amplitude distribution is spread more evenly across the quantizer levels.

19.8 Cumulative distributions

Motivation

The PMF works only for discrete variables and the PDF only for continuous ones (unless we admit impulses). The **cumulative distribution function** works for every random variable, discrete, continuous or mixed, without any generalized functions. It is also the function that answers the questions engineers most often ask — "what fraction of resistors are below 990 Ω?", "what is the probability that the component has failed by time t?", "what threshold is exceeded only 0.1 % of the time?" — and it is the natural tool for deriving distributions of functions of random variables, extremes of several variables, and for generating random numbers in simulation.

Definition and properties

Cumulative distribution function (CDF)

The **CDF** of a random variable X is

$$F_X(x) = P(X \le x), \qquad -\infty < x < \infty.$$

Properties of the CDF

Every CDF satisfies:

1. $0 \le F_X(x) \le 1$;
2. F_X is **non-decreasing**: $a < b \Rightarrow F_X(a) \le F_X(b)$;
3. $\lim_{x\to-\infty} F_X(x) = 0$ and $\lim_{x\to+\infty} F_X(x) = 1$;
4. F_X is **right-continuous**: $\lim_{h\downarrow 0} F_X(x+h) = F_X(x)$;
5. $P(a < X \le b) = F_X(b) - F_X(a)$;
6. $P(X = a) = F_X(a) - F_X(a^-)$, the size of the jump at a, where $F_X(a^-) = \lim_{h\downarrow 0} F_X(a-h)$.

Conversely, any function with properties 2–4 is the CDF of some random variable.

Proof sketch. (2) $\{X \le a\} \subseteq \{X \le b\}$, then monotonicity of P. (5) $\{X \le b\}$ is the disjoint union of $\{X \le a\}$ and $\{a < X \le b\}$. (3) and (4) follow from continuity of probability (Section 19.1): the events $\{X \le -n\}$ decrease to $\varnothing$, the events $\{X \le n\}$ increase to Ω, and $\{X \le x + 1/n\}$ decrease to $\{X \le x\}$. (6) The events $\{a - 1/n < X \le a\}$ decrease to $\{X = a\}$, and their probabilities are $F_X(a) - F_X(a - 1/n)$. $\square$

Property 6 is the key to reading a CDF. Wherever F_X is continuous, $P(X = x) = 0$; wherever it jumps, the jump is a point mass. Hence:

Type of RV	Shape of the CDF	Link to PMF/PDF
Discrete	staircase, flat between jumps	$F_X(x) = \sum_{x_k \le x} p_X(x_k)$; jumps $= p_X(x_k)$
Continuous	continuous (no jumps)	$F_X(x) = \int_{-\infty}^{x} f_X(u)\,du$, $f_X(x) = F_X'(x)$
Mixed	ramps plus jumps	jumps = point masses; slope = continuous density

For a continuous RV the relation $f_X = F_X'$ holds wherever F_X is differentiable; for a discrete or mixed RV it holds in the generalized sense, the derivative of a jump of height p being $p\,\delta(x - x_k)$.

Common Pitfall — $<$ versus $\le$

With the convention $F_X(x) = P(X \le x)$, the probability of an interval depends on whether its endpoints are included when X has point masses there: $P(a \le X \le b) = F_X(b) - F_X(a^-)$, $P(a < X < b) = F_X(b^-) - F_X(a)$. For a discrete count, $P(X < 3) = F_X(2)$, not $F_X(3)$. For continuous variables the distinction disappears.

Worked Example 19.33 — Staircase CDF of an error count

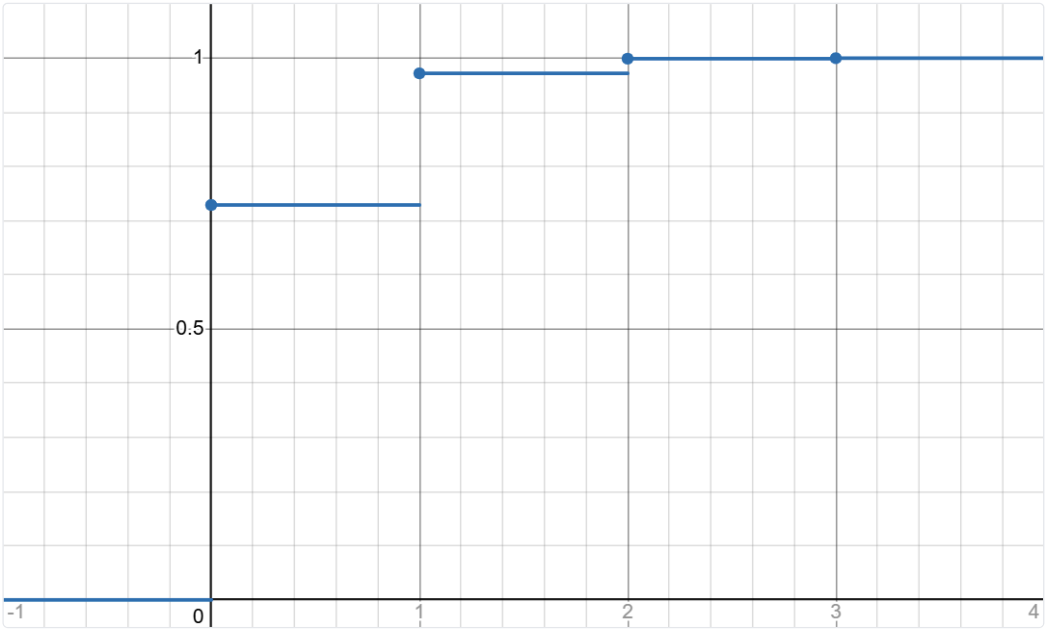
For the number of errors $X \sim \text{Bin}(3, 0.1)$ of Example 19.19, write $F_X(x)$, and use it to find $P(1 \le X \le 2)$ and $P(X > 1)$.

Step 1 — accumulate the PMF $(0.729, 0.243, 0.027, 0.001)$:

$$F_X(x) = \begin{cases} 0, & x < 0 \\ 0.729, & 0 \le x < 1 \\ 0.972, & 1 \le x < 2 \\ 0.999, & 2 \le x < 3 \\ 1, & x \ge 3. \end{cases}$$

Step 2 — interval with point masses. $P(1 \le X \le 2) = F_X(2) - F_X(1^-) = 0.999 - 0.729 = 0.270$. Check: $0.243 + 0.027 = 0.270$. ✓

Step 3 — tail. $P(X > 1) = 1 - F_X(1) = 1 - 0.972 = 0.028$.



Staircase CDF of $\text{Bin}(3, 0.1)$. *Each jump equals the PMF value at that point; the filled dots mark the right-continuity convention — the CDF takes the upper value at each jump.*

The Gaussian CDF and the Q-function

The Gaussian CDF has no closed form; it is tabulated, or computed from the **error function** $\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$, available in every numerical library:

$$\Phi(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-u^2/2}\,du = \frac{1}{2}\left[1 + \text{erf}\left(\frac{z}{\sqrt{2}}\right)\right], \qquad Q(z) = 1 - \Phi(z) = \frac{1}{2}\,\text{erfc}\left(\frac{z}{\sqrt{2}}\right).$$

By symmetry $\Phi(-z) = 1 - \Phi(z)$ and $Q(-z) = 1 - Q(z)$. For $X \sim \mathcal{N}(\mu, \sigma^2)$, standardization (Example 19.30) gives

$$F_X(x) = \Phi\left(\frac{x - \mu}{\sigma}\right), \qquad P(X > x) = Q\left(\frac{x - \mu}{\sigma}\right).$$

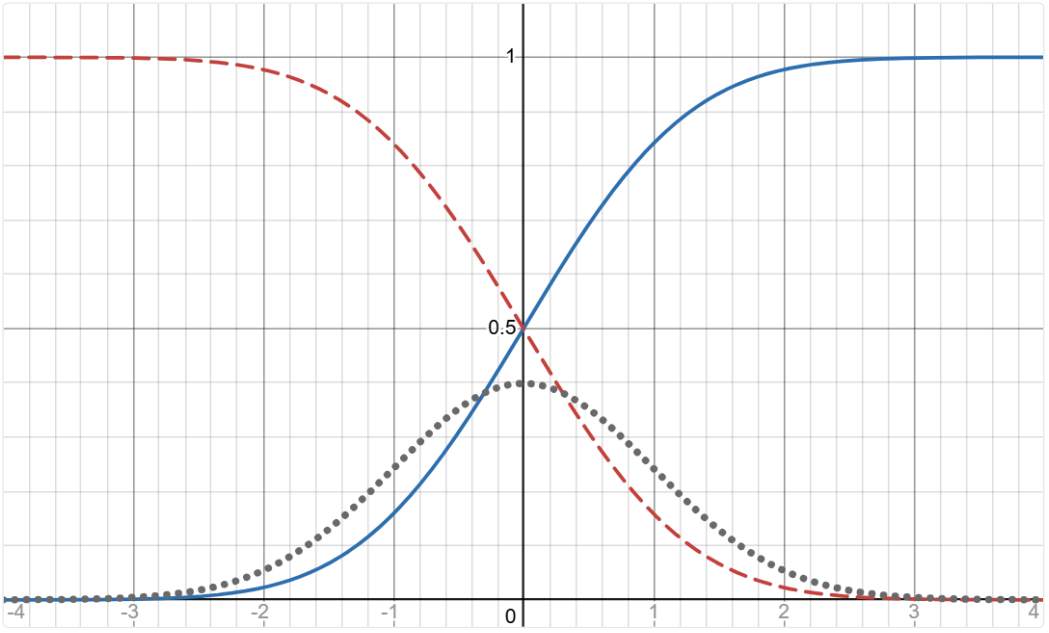
Communications engineers use Q because error probabilities are tail probabilities; statisticians tabulate Φ. A short table covers most needs:

z	$\Phi(z)$	$Q(z)$
0	0.5000	0.5000
1	0.8413	0.1587
1.645	0.9500	0.0500
1.960	0.9750	0.0250
2	0.9772	0.02275
2.326	0.9900	0.0100
2.576	0.9950	0.0050
3	0.99865	1.350×10^{-3}
4	—	3.167×10^{-5}
5	—	2.867×10^{-7}
6	—	9.866×10^{-10}

The probabilities of falling within 1, 2, 3 standard deviations of the mean are 0.6827, 0.9545 and 0.9973 — the "68–95–99.7 rule". For large z the bound

$$\frac{z}{1+z^2}\,\frac{e^{-z^2/2}}{\sqrt{2\pi}} < Q(z) < \frac{1}{z}\,\frac{e^{-z^2/2}}{\sqrt{2\pi}} \qquad (z>0)$$

shows that Q falls off essentially like $e^{-z^2/2}$; the simpler Chernoff bound $Q(z) \le \tfrac12 e^{-z^2/2}$ is often enough for design estimates.



Standard Gaussian CDF $\Phi(z)$ (blue), Q-function $1-\Phi(z)$ (red, dashed) and the PDF $\varphi(z)$ (grey, dotted). The CDF's slope is the PDF: steepest at $z=0$, where the density is largest.

Worked Example 19.34 — Resistor tolerance

A batch of nominal 1 kΩ resistors has values $R \sim \mathcal{N}(1000,\ 10^2)$ Ω. (a) What fraction lies between 990 and 1020 Ω? (b) What fraction lies outside a $\pm 2\,\%$ tolerance band? (c) What symmetric band contains 99.9 % of the batch?

(a) Standardize: $z_1 = (990-1000)/10 = -1$, $z_2 = (1020-1000)/10 = 2$.

$$P(990<R<1020) = \Phi(2) - \Phi(-1) = 0.97725 - 0.15866 = 0.8186.$$

(b) $\pm 2\,\%$ is $\pm 20\ \Omega = \pm 2\sigma$: $P(|R-1000|>20) = 2Q(2) = 0.0455$, i.e. 4.55 % of parts fail a 2 % specification.

(c) We need $2Q(z) = 0.001$, $Q(z) = 0.0005$, $z = 3.291$: the band is $1000 \pm 32.9\ \Omega$, i.e. about $\pm 3.3\,\%$.

Quantiles

Quantile, median, percentile

For $0<u<1$, the **u-quantile** of X is

$$x_u = F_X^{-1}(u) = \inf\{x : F_X(x) \ge u\},$$

the smallest x at which the CDF reaches u. The **median** is $x_{0.5}$; the **quartiles** are $x_{0.25}, x_{0.75}$; the $100u$-th **percentile** is x_u.

For a continuous, strictly increasing CDF, F_X^{-1} is the ordinary inverse. The "inf" in the definition handles flat parts and jumps. Quantiles are how specifications are written: "the 99th-percentile latency is 4 ms", "the B_{10} life of a bearing (10 % failed)", "the 1 % outage fade depth". For the exponential distribution, solving $1 - e^{-\lambda t} = u$ gives $t_u = -\ln(1-u)/\lambda$; the median life is $\ln 2/\lambda$, about 69 % of the mean life $1/\lambda$.

Reliability function and hazard rate

For a lifetime $T \geq 0$, the complement of the CDF is the **reliability** (or **survival**) function, and its relative rate of decrease is the **hazard rate**:

$$R(t) = P(T > t) = 1 - F_T(t), \qquad h(t) = \frac{f_T(t)}{R(t)} = -\frac{d}{dt}\ln R(t).$$

The hazard rate is the conditional failure density of a unit that has survived to t: $P(t < T \leq t + dt \mid T > t) \approx h(t)\,dt$. Integrating, $R(t) = \exp\!\left(-\int_0^t h(u)\,du\right)$. A constant hazard $h(t) = \lambda$ gives the exponential distribution — the mathematical statement of memorylessness. An increasing hazard means wear-out (electrolytic capacitors, fans, electromigration); a decreasing hazard means infant mortality (latent manufacturing defects, removed by burn-in). The sequence decreasing–constant–increasing over a product's life is the well-known **bathtub curve**; the Weibull family that models each phase is covered in Ch. 20.

The CDF method for functions of random variables

The most reliable way to find the distribution of $Y = g(X)$ — always valid, including for non-monotone g, flat regions, and mixed outputs — is to compute its CDF directly from the definition:

$$F_Y(y) = P(g(X) \leq y) = P\big(X \in \{x : g(x) \leq y\}\big),$$

and then differentiate if a density is wanted. The density-transformation formulas of Section 19.7 are shortcuts derived this way.

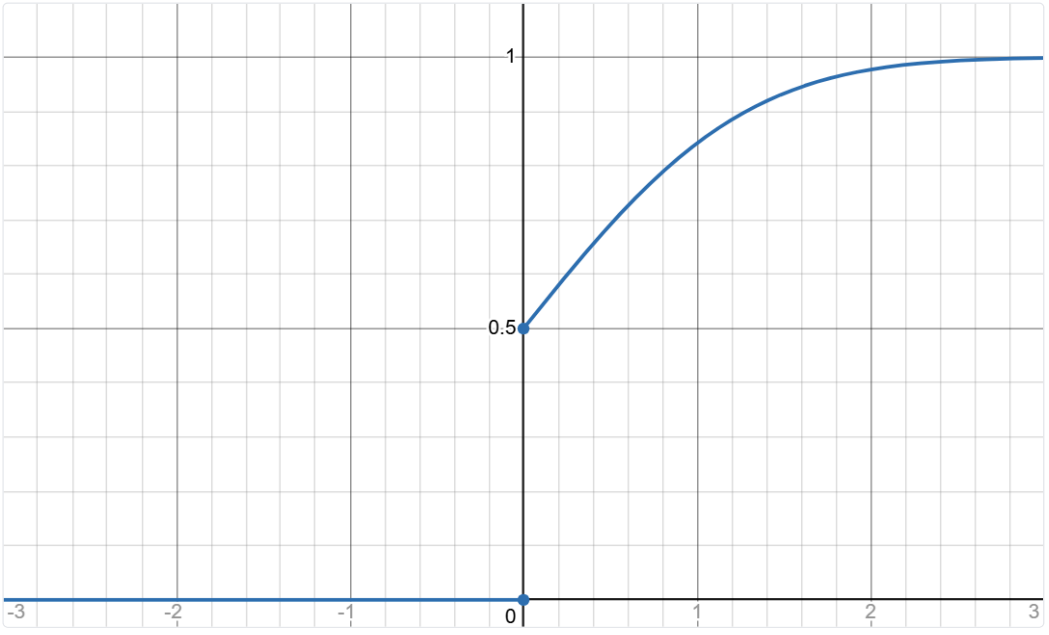
Worked Example 19.35 — Half-wave rectified noise (a mixed CDF)

Noise $X \sim \mathcal{N}(0,1)$ V passes through an ideal half-wave rectifier, $Y = \max(X, 0)$. Find F_Y and describe the distribution.

Step 1 — $y < 0$. $Y \geq 0$ always, so $F_Y(y) = 0$.

Step 2 — $y \geq 0$. $\{Y \leq y\} = \{X \leq y\}$ (if $X \leq 0$ then $Y = 0 \leq y$; if $X > 0$ then $Y = X$). So $F_Y(y) = \Phi(y)$.

Step 3 — interpret. F_Y jumps from 0 to $\Phi(0) = \tfrac{1}{2}$ at $y = 0$: a point mass $P(Y = 0) = \tfrac{1}{2}$ (all negative inputs map to 0). For $y > 0$ the CDF follows Φ, with density $\varphi(y)$. In generalized-density form, $f_Y(y) = \tfrac{1}{2}\delta(y) + \varphi(y)u(y)$.



CDF of half-wave rectified Gaussian noise: zero for $y < 0$, a jump of $1/2$ at $y = 0$ (the point mass of all clipped negative inputs), then the Gaussian CDF. A mixed random variable needs no impulses in this description.

Extremes: the minimum and the maximum

The CDF method is particularly efficient for the largest and smallest of several independent random variables, which appear as the lifetimes of parallel and series systems and as peak values of noise.

Distribution of the maximum and minimum

Let $X_1, \ldots, X_n$ be independent with CDFs $F_1, \ldots, F_n$. Then

$$F_{\max}(x) = P(\max_i X_i \leq x) = \prod_{i=1}^{n} F_i(x), \qquad P(\min_i X_i > x) = \prod_{i=1}^{n} \big(1 - F_i(x)\big).$$

Proof. The maximum is at most x exactly when every $X_i \le x$; the minimum exceeds x exactly when every $X_i > x$. Independence turns the intersections into products. □

For lifetimes these are the reliability formulas of Section 19.4 in CDF language: a series system fails at the *first* component failure, $T_{\text{sys}} = \min_i T_i$, so $R_{\text{sys}}(t) = \prod R_i(t)$; a parallel system fails at the *last*, $T_{\text{sys}} = \max_i T_i$, so $F_{\text{sys}}(t) = \prod F_i(t)$. With exponential components, $\min_i T_i \sim \text{Exp}(\lambda_1 + \cdots + \lambda_n)$.

Worked Example 19.36 — Peaks of noise

A comparator sees 1000 independent samples of Gaussian noise $X_i \sim \mathcal{N}(0, \sigma^2)$ per frame. Its threshold is set at 3σ, "because noise exceeds 3σ only 0.13 % of the time". Find the probability of at least one false trigger per frame, and the threshold needed to make it 1 %.

Step 1 — maximum. $P(\max_i X_i \le 3\sigma) = \Phi(3)^{1000} = 0.998650^{1000} = 0.2590$.

Step 2 — false trigger. $P(\text{at least one}) = 1 - 0.2590 = 0.741$. Almost three frames in four trigger falsely!

Step 3 — redesign. We need $\Phi(z)^{1000} = 0.99$, i.e. $\Phi(z) = 0.99^{1/1000} = 0.99998995$, so $Q(z) = 1.005 \times 10^{-5}$ and $z \approx 4.26$. The threshold must be about 4.3σ.

When many independent chances are taken, a rare event per trial becomes a likely event per frame (compare the union bound $1000 \times 0.00135 = 1.35$, which is useless here because it exceeds 1). This is the "look-elsewhere" effect that governs radar CFAR thresholds and test-limit settings on automatic test equipment.

Generating random variables: the inverse-transform method

The quantile function also runs the other way. If $U \sim \mathcal{U}(0,1)$ and F is any CDF, then

$$X = F^{-1}(U) \quad \text{has CDF } F,$$

because $P(F^{-1}(U) \le x) = P(U \le F(x)) = F(x)$. Conversely, if X is continuous then $F_X(X) \sim \mathcal{U}(0,1)$ (the **probability integral transform**). This is how simulations turn a uniform random-number generator into samples from any distribution: for $\text{Exp}(\lambda)$, $X = -\ln(1-U)/\lambda$; for Rayleigh(σ), $X = \sigma\sqrt{-2\ln(1-U)}$; for a discrete distribution, choose the smallest x_k with $F(x_k) \ge U$. Monte Carlo methods built on this idea are developed in Ch. 24.

Worked Example 19.37 — Simulating component lifetimes

A reliability simulation needs lifetimes with $\text{Exp}(\lambda)$, $\lambda = 2 \times 10^{-5}\ \text{h}^{-1}$. The uniform generator returns $u = 0.5$ and $u = 0.95$. What lifetimes are produced, and what do they mean?

Step 1 — formula. $t = -\ln(1-u)/\lambda$.

Step 2 — numbers. $u = 0.5$: $t = \ln 2/(2\times10^{-5}) = 34\,657$ h, the median life. $u = 0.95$: $t = -\ln(0.05)/(2\times10^{-5}) = 149\,787$ h, the 95th-percentile life.

Step 3 — check. $F(34\,657) = 1 - e^{-0.6931} = 0.5$. ✓ Each uniform sample is mapped to the lifetime with that cumulative probability.

Engineering Note — read specifications as CDF points

Nearly every probabilistic specification is a point on a CDF or its complement: a bit-error rate is $P(\text{error})$, an outage probability is $F_{\text{SNR}}(\text{threshold})$, a "six-sigma" process puts the specification limit where $Q(z)$ is tiny, an MTTF is a mean but a warranty period is a quantile. Plotting the empirical CDF of measured data (Ch. 20) is the quickest way to compare a batch against such a specification.

19.9 Expectation and variance

Motivation

A full distribution is often more information than a design needs. An engineer wants to know the DC level of a noisy signal, its RMS fluctuation, the average number of retransmissions, the mean time to failure, the noise power at the output of an ADC. These are all **expectations** — probability-weighted averages — of the random variable or of a function of it. The mean locates the distribution, the variance measures its spread, higher moments describe its shape, and the moment-generating and characteristic functions package all the moments into a single transform that turns the convolution of densities into multiplication. This section develops these tools and the inequalities that make them useful even when the distribution itself is unknown.

Expected value

Expectation (mean)

The **expected value** (mean) of X is

$$E[X] = \mu_X = \begin{cases} \displaystyle\sum_k x_k\, p_X(x_k), & X \text{ discrete}, \\[2mm] \displaystyle\int_{-\infty}^{\infty} x\, f_X(x)\, dx, & X \text{ continuous}, \end{cases}$$

provided the sum or integral converges absolutely ($\sum |x_k| p_X(x_k) < \infty$ or $\int |x| f_X\, dx < \infty$). Otherwise the mean does not exist.

The mean is the **centre of mass** of the probability distribution: if unit mass is distributed along the x-axis according to f_X, $E[X]$ is the balance point. Its link with experiment is the **law of large numbers**: the average $\frac{1}{N}\sum_{i=1}^{N} X_i$ of N independent repetitions converges to $E[X]$ as $N \to \infty$ (Ch. 20). Using the generalized density of Section 19.7, the two formulas are one: $E[X] = \int x f_X(x)\,dx$ with impulses turning the integral into a sum.

Common Pitfall — the mean need not exist

The **Cauchy** density $f_X(x) = \dfrac{1}{\pi(1+x^2)}$ — the distribution of $\tan\Theta$ for uniform Θ, and of the ratio of two independent zero-mean Gaussians — is symmetric about 0, yet $\int |x| f_X\,dx = \infty$, so $E[X]$ does not exist. Averages of Cauchy samples do not settle down as N grows: the average of N independent Cauchy variables is again standard Cauchy. Heavy-tailed quantities (ratios of noisy measurements, impulsive interference) can defeat averaging; always check that the mean exists before relying on it.

Expectation of a function of a random variable

To find $E[g(X)]$ we could first find the distribution of $Y = g(X)$ (Sections 19.7–19.8) and then its mean. Fortunately that is unnecessary.

Expectation of a function (law of the unconscious statistician)

$$E[g(X)] = \sum_{k} g(x_k)\,p_X(x_k) \qquad \text{or} \qquad E[g(X)] = \int_{-\infty}^{\infty} g(x)\,f_X(x)\,dx.$$

For two random variables, $E[g(X,Y)] = \iint g(x,y) f_{XY}(x,y)\,dx\,dy$ (or the double sum).

Proof (discrete case). $E[Y] = \sum_y y\,P(Y=y) = \sum_y y \sum_{x:g(x)=y} p_X(x) = \sum_x g(x) p_X(x)$, by regrouping the terms of an absolutely convergent sum. The continuous case follows by approximating with discrete variables, or directly from the transformation formula for monotone g. □

In Example 19.20, $E[Y] = E[X^2] = 4(0.1) + 1(0.2) + 0 + 1(0.2) + 4(0.1) = 1.2$, the same as from the PMF of Y: $0(0.4) + 1(0.4) + 4(0.2) = 1.2$. ✓

Linearity

Linearity of expectation

For any random variables X, Y (on the same space, with finite means) and constants a, b, c:

$$E[aX + bY + c] = a\,E[X] + b\,E[Y] + c.$$

By induction, $E\left[\sum_i X_i\right] = \sum_i E[X_i]$. **No independence is required.**

Proof. By the two-variable form of the previous theorem, $E[aX+bY] = \iint (ax+by) f_{XY}\,dx\,dy = a\int x f_X\,dx + b\int y f_Y\,dy$, using the marginal formulas of Section 19.6. □

Linearity combined with indicator variables is one of the most effective tricks in applied probability. Since $E[I_A] = 1\cdot P(A) + 0\cdot P(A^c) = P(A)$, the expected number of events that occur among $A_1, \dots, A_n$ is

$$E\left[\sum_i I_{A_i}\right] = \sum_i P(A_i),$$

however the events depend on each other.

Worked Example 19.38 — Means by indicators

(a) Find the mean of $X \sim \text{Bin}(n,p)$. (b) In Example 19.5, find the expected number of *pairs* of nodes that share an address, for $n = 300$ nodes and $M = 65\,536$ addresses.

(a) Write $X = I_1 + \cdots + I_n$, I_i the indicator of success on trial i. Then $E[X] = \sum_i P(\text{success on } i) = np$. (The direct route, $\sum_k k\binom{n}{k}p^k q^{n-k}$, needs an index shift and the binomial theorem — and gives the same answer.)

(b) For each of the $\binom{300}{2} = 44\,850$ pairs, let I_{ij} indicate that nodes i and j collide; $P(I_{ij}=1) = 1/M$. The indicators are dependent, but linearity does not care:

$$E[\text{number of colliding pairs}] = \binom{300}{2}\frac{1}{65\,536} = 0.684.$$

Compare: $P(\text{at least one collision}) = 0.496$. Since $P(N \ge 1) \le E[N]$ for any count N (Markov's inequality below), the expected number of collisions is a quick upper bound on the collision probability, and an accurate one when collisions are rare.

Mean of a non-negative variable and mean time to failure

For a non-negative random variable there is a useful alternative formula in terms of the tail of the distribution.

Tail formula

If $T \geq 0$, then

$$E[T] = \int_0^\infty P(T > t)\,dt = \int_0^\infty \big(1 - F_T(t)\big)\,dt.$$

For a non-negative integer-valued N, $E[N] = \sum_{k=0}^{\infty} P(N > k)$.

Proof. Write $T = \int_0^\infty I_{\{T>t\}}\,dt$ (the integrand is 1 for $t < T$ and 0 afterwards) and take expectations inside the integral: $E[T] = \int_0^\infty P(T > t)\,dt$. □

In reliability this is the **mean time to failure** $\mathrm{MTTF} = \int_0^\infty R(t)\,dt$ — the area under the reliability curve.

Worked Example 19.39 — MTTF of redundant configurations

Components have exponential lifetimes with $\lambda = 2 \times 10^{-5}\ \mathrm{h}^{-1}$. Find the MTTF of a single unit, two units in active parallel, TMR (2-out-of-3), and a unit with a perfectly switched cold spare.

Step 1 — single unit. $\int_0^\infty e^{-\lambda t}dt = 1/\lambda = 50\,000$ h.

Step 2 — active parallel. $\int_0^\infty (2e^{-\lambda t} - e^{-2\lambda t})\,dt = \frac{2}{\lambda} - \frac{1}{2\lambda} = \frac{3}{2\lambda} = 75\,000$ h.

Step 3 — TMR. $\int_0^\infty (3e^{-2\lambda t} - 2e^{-3\lambda t})\,dt = \frac{3}{2\lambda} - \frac{2}{3\lambda} = \frac{5}{6\lambda} = 41\,667$ h.

Step 4 — cold standby. $\int_0^\infty (1+\lambda t)e^{-\lambda t}\,dt = \frac{1}{\lambda} + \frac{1}{\lambda} = \frac{2}{\lambda} = 100\,000$ h. (Directly: $E[T_1 + T_2] = E[T_1] + E[T_2]$ by linearity.)

TMR has a *shorter* mean life than a single unit, although it is more reliable for $\lambda t < \ln 2$ (graph in Section 19.4). The MTTF is a poor figure of merit for systems intended for short missions; quote $R(t_{\text{mission}})$ instead.

Variance and standard deviation

Variance, standard deviation

The **variance** of X is the mean-square deviation from the mean,

$$\mathrm{Var}(X) = \sigma_X^2 = E\big[(X - \mu_X)^2\big],$$

and the **standard deviation** is $\sigma_X = \sqrt{\mathrm{Var}(X)}$, which has the same units as X.

Expanding the square and using linearity gives the computational form

$$\mathrm{Var}(X) = E[X^2] - 2\mu_X E[X] + \mu_X^2 = E[X^2] - \mu_X^2.$$

Properties of variance

1. $\mathrm{Var}(X) \geq 0$, with equality if and only if X is constant with probability 1.
2. $\mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X)$: a shift does not change spread; a scale multiplies the standard deviation by $|a|$.
3. If X, Y are independent, $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$; more generally (below) a covariance term appears.

Proof of 2. $aX + b - E[aX + b] = a(X - \mu_X)$, so $\mathrm{Var}(aX + b) = E[a^2(X - \mu_X)^2] = a^2\,\mathrm{Var}(X)$. □

The decomposition $E[X^2] = \mu_X^2 + \sigma_X^2$ has a direct electrical meaning. If X is a voltage across 1 Ω, $E[X^2]$ is the average (total) power, μ_X^2 is the power of the DC component and σ_X^2 is the power of the fluctuating (AC) component; σ_X is the AC RMS value read by a true-RMS meter with AC coupling.

Worked Example 19.40 — Quantization noise

An N-bit ADC with full-scale range $\pm V_{\mathrm{FS}}$ has step $\Delta = 2V_{\mathrm{FS}}/2^N$. For a busy input, the rounding error is modelled as $e \sim \mathcal{U}(-\Delta/2, \Delta/2)$. Find the mean and variance of e, and the signal-to-quantization-noise ratio (SQNR) for a full-scale sine wave, evaluated for $N = 8$, $V_{\mathrm{FS}} = 1$ V.

Step 1 — mean. By symmetry $E[e] = 0$.

Step 2 — variance.

$$\mathrm{Var}(e) = E[e^2] = \frac{1}{\Delta}\int_{-\Delta/2}^{\Delta/2} x^2\,dx = \frac{1}{\Delta}\cdot\frac{2}{3}\left(\frac{\Delta}{2}\right)^3 = \frac{\Delta^2}{12}.$$

The RMS quantization noise is $\Delta/\sqrt{12} \approx 0.289\Delta$.

Step 3 — SQNR. A full-scale sine has power $V_{\mathrm{FS}}^2/2$. With $\Delta = 2V_{\mathrm{FS}}/2^N$,

$$\mathrm{SQNR} = \frac{V_{\mathrm{FS}}^2/2}{\Delta^2/12} = \frac{3}{2}\,2^{2N}, \qquad 10\log_{10}\mathrm{SQNR} = 6.02N + 1.76\ \mathrm{dB}.$$

Step 4 — numbers. $\Delta = 2/256 = 7.8125$ mV, $\mathrm{Var}(e) = 5.086 \times 10^{-6}\ \mathrm{V}^2$, RMS noise 2.255 mV, $\mathrm{SQNR} = 10\log_{10}(0.5/5.086 \times 10^{-6}) = 49.93$ dB, matching $6.02(8) + 1.76 = 49.92$ dB. Every extra bit buys about 6 dB.

Means and variances of the key distributions

Distribution	Mean	Variance	Derivation hint
Bernoulli(p)	p	$p(1-p)$	$E[X^2]=E[X]=p$
Bin(n,p)	np	$np(1-p)$	sum of n independent Bernoullis
Geom(p)	$1/p$	$(1-p)/p^2$	tail formula: $\sum_k (1-p)^k = 1/p$
Poisson(λ)	λ	λ	MGF below, or $E[X(X-1)]=\lambda^2$
$\mathcal{U}(a,b)$	$(a+b)/2$	$(b-a)^2/12$	Example 19.40 shifted and scaled
Exp(λ)	$1/\lambda$	$1/\lambda^2$	$E[T^n]=n!/\lambda^n$
$\mathcal{N}(\mu,\sigma^2)$	μ	σ^2	standardize; $E[Z^2]=1$ by parts
Rayleigh(σ)	$\sigma\sqrt{\pi/2}$	$(2-\pi/2)\sigma^2$	$E[R^2]=2\sigma^2$

Two derivations. For $Z\sim\mathcal{N}(0,1)$, $E[Z]=0$ by symmetry and, integrating by parts with $u=z$, $dv=z\varphi(z)\,dz$ (so $v=-\varphi(z)$): $E[Z^2]=\int z^2\varphi(z)\,dz = [-z\varphi(z)]_{-\infty}^{\infty}+\int\varphi(z)\,dz = 1$. Then $X=\mu+\sigma Z$ has mean μ and variance σ^2, which justifies the names of the parameters. For $T\sim\text{Exp}(\lambda)$, repeated integration by parts gives $E[T^n]=\int_0^\infty t^n\lambda e^{-\lambda t}dt = n!/\lambda^n$, so $E[T]=1/\lambda$, $E[T^2]=2/\lambda^2$ and $\text{Var}(T)=1/\lambda^2$: the standard deviation of an exponential lifetime equals its mean.

Sums, covariance and independence

For two random variables, expanding $E[(X+Y-\mu_X-\mu_Y)^2]$ gives

$$\text{Var}(X+Y)=\text{Var}(X)+\text{Var}(Y)+2\,\text{Cov}(X,Y),\qquad \text{Cov}(X,Y)=E[(X-\mu_X)(Y-\mu_Y)]=E[XY]-\mu_X\mu_Y.$$

Expectation of a product of independent RVs

If X and Y are independent, then $E[XY]=E[X]E[Y]$, more generally $E[g(X)h(Y)]=E[g(X)]E[h(Y)]$. Hence $\text{Cov}(X,Y)=0$ and variances of independent RVs add:

$$\text{Var}\left(\sum_{i=1}^{n}X_i\right)=\sum_{i=1}^{n}\text{Var}(X_i)\qquad (X_i \text{ independent}).$$

Proof. $E[g(X)h(Y)]=\iint g(x)h(y)f_X(x)f_Y(y)\,dx\,dy=\int g f_X\,dx\int h f_Y\,dy$, the double integral factoring because the joint density does. □

The converse is false: zero covariance (**uncorrelated**) does not imply independence. For $X\sim\mathcal{N}(0,1)$ and $Y=X^2$, $\text{Cov}(X,Y)=E[X^3]=0$, yet Y is a function of X. Covariance, the correlation coefficient and their estimation from data are developed in Ch. 20; autocorrelation of signals in Ch. 21.

Two consequences are used constantly. First, the **variance of an average**: if $X_1,\dots,X_N$ are independent with common variance σ^2, then $\bar X=\frac{1}{N}\sum X_i$ has $\text{Var}(\bar X)=\sigma^2/N$, so averaging N independent readings reduces the RMS noise by $\sqrt{N}$ (signal averaging gives 3 dB of SNR per doubling of N). Second, **tolerance stack-up**: independent tolerances add in quadrature, not linearly.

Worked Example 19.41 — DC plus noise, and resistors in series

(a) A sensor output is $V=2+N$ V with $N\sim\mathcal{N}(0,0.5^2)$. Find the mean, variance and mean-square value of V, and the average power it delivers to 1 Ω. (b) Two independent resistors, each with mean 1000 Ω and standard deviation 10 Ω, are connected in series. Find the mean and standard deviation of the total, absolute and relative.

(a) $E[V]=2$ V, $\text{Var}(V)=0.25$ V^2, $E[V^2]=2^2+0.25=4.25$ V^2. Power into 1 Ω: 4.25 W, of which 4 W is DC and 0.25 W is noise.

(b) Mean 2000 Ω; variance $10^2+10^2=200$ Ω^2; standard deviation $\sqrt{200}=14.14$ Ω. Relative spread: $14.14/2000=0.71\,\%$, compared with $1\,\%$ for each resistor. Independent errors partially cancel; a worst-case (linear) stack-up of ±20 Ω would be pessimistic.

Conditional expectation and the law of total expectation

Conditioning works for expectations exactly as for probabilities. For an event B, $E[X\mid B]$ is the mean computed from the conditional distribution of X given B. If $B_1,\dots,B_n$ partition Ω,

$$E[X]=\sum_{i=1}^{n}E[X\mid B_i]\,P(B_i),$$

the **law of total expectation**, proved by applying total probability inside the sum or integral that defines $E[X]$.

Worked Example 19.42 — Mean number of transmissions by conditioning

In the ARQ protocol of Example 19.23, each attempt succeeds with probability p. Find $E[N]$ without summing a series.

Step 1 — condition on the first attempt. If it succeeds (probability p), $N = 1$. If it fails (probability $1-p$), one attempt has been used and, by independence, the situation restarts: $E[N \mid \text{fail}] = 1 + E[N]$.

Step 2 — solve. $E[N] = p \cdot 1 + (1-p)(1 + E[N])$, so $pE[N] = 1$ and $E[N] = 1/p$.

Step 3 — numbers. With $p = 0.8$, $E[N] = 1.25$ attempts: the throughput of the link is 80 % of its raw rate. Check by the tail formula: $\sum_{k \ge 0} (0.2)^k = 1/0.8 = 1.25$. ✓

Inequalities: Markov and Chebyshev

Means and variances bound tail probabilities even when the distribution is unknown.

Markov and Chebyshev inequalities

Markov: if $X \ge 0$ and $a > 0$, then $P(X \ge a) \le \dfrac{E[X]}{a}$.

Chebyshev: for any X with mean μ and variance σ^2, and any $k > 0$,

$$P(|X - \mu| \ge k\sigma) \le \frac{1}{k^2}.$$

Proof. Markov: $X \ge a\, I_{\{X \ge a\}}$ for $X \ge 0$; take expectations, $E[X] \ge aP(X \ge a)$. Chebyshev: apply Markov to the non-negative variable $(X-\mu)^2$ with $a = k^2\sigma^2$. □

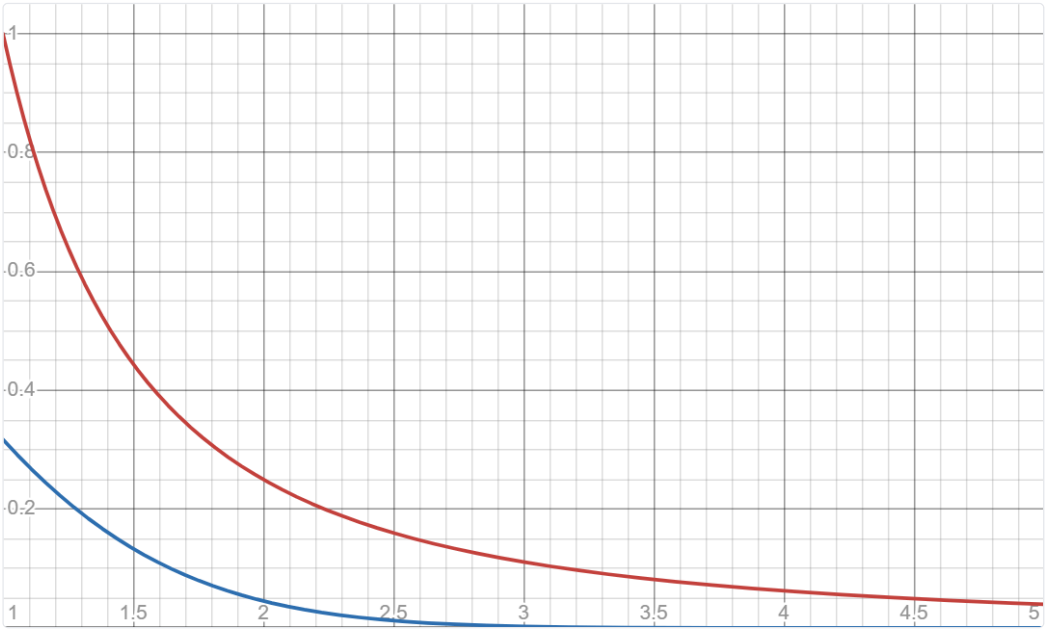
Worked Example 19.43 — Guaranteed yield without a distribution

A supplier guarantees only that a batch of resistors has mean 1000 Ω and standard deviation 10 Ω. What fraction can lie outside 1000 ± 30 Ω? Compare with the Gaussian model.

Step 1 — Chebyshev with $k = 3$. $P(|R - 1000| \ge 30) \le 1/9 = 0.111$.

Step 2 — Gaussian. $2Q(3) = 0.0027$.

Chebyshev's bound is distribution-free and therefore very conservative — about 40 times the Gaussian value here — but it is a *guarantee*. It is attained only by a three-point distribution, which real batches rarely resemble. Chebyshev's inequality also proves the weak law of large numbers in one line: $P(|\bar{X} - \mu| \ge \epsilon) \le \sigma^2/(N\epsilon^2) \to 0$.



Chebyshev bound $1/k^2$ (red) on $P(|X-\mu| \ge k\sigma)$, and the exact value $2Q(k)$ for a Gaussian (blue). The distribution-free bound decays only polynomially; Gaussian tails decay like $e^{-k^2/2}$.

Moments and the shape of a distribution

Moments

The n**-th moment** of X is $m_n = E[X^n]$ and the n**-th central moment** is $\mu_n = E[(X-\mu)^n]$. Thus $m_1 = \mu$, $\mu_2 = \sigma^2$. The normalized third and fourth central moments are the **skewness** and **kurtosis**:

$$\gamma_1 = \frac{\mu_3}{\sigma^3}, \qquad \beta_2 = \frac{\mu_4}{\sigma^4} \quad (\text{excess kurtosis } \beta_2 - 3).$$

Skewness measures asymmetry (positive when the right tail is longer); kurtosis measures the weight of the tails relative to a Gaussian, for which $\gamma_1 = 0$ and $\beta_2 = 3$. Central moments are found from ordinary moments by expanding the binomial: for instance $\mu_3 = m_3 - 3\mu m_2 + 2\mu^3$. For the exponential distribution, with $m_n = n!/\lambda^n$, this gives $\mu_3 = (6 - 6 + 2)/\lambda^3 = 2/\lambda^3$ and $\gamma_1 = 2$; similarly $\beta_2 = 9$ (excess kurtosis 6). Impulsive noise in power-line communications and switching transients is characterized by large excess kurtosis — its tails are much heavier than Gaussian, so Gaussian-based error-rate predictions are optimistic.

Knowing all the moments usually determines the distribution (it does, for example, whenever the moment-generating function below exists near 0). The transforms in the rest of the section encode the whole sequence of moments in one function.

The moment-generating function

Moment-generating function (MGF)

$$M_X(s) = E\left[e^{sX}\right] = \int_{-\infty}^{\infty} e^{sx} f_X(x)\,dx \quad \Big(\text{or } \sum_k e^{sx_k} p_X(x_k)\Big),$$

defined for the real s for which the expectation is finite. It always exists at $s = 0$, where $M_X(0) = 1$.

The MGF of a continuous RV is the two-sided Laplace transform (Ch. 14) of f_X evaluated at $-s$. Its name comes from the following property.

Properties of the MGF

If $M_X(s)$ is finite in an open interval around $s = 0$, then:

1. **Moments:** $E[X^n] = M_X^{(n)}(0) = \dfrac{d^n M_X}{ds^n}\Big|_{s=0}$, and $M_X(s) = \sum_{n=0}^{\infty} \dfrac{E[X^n]}{n!} s^n$.
2. **Uniqueness:** M_X determines the distribution of X.
3. **Linear transformation:** $M_{aX+b}(s) = e^{bs} M_X(as)$.
4. **Sums:** if X, Y are independent, $M_{X+Y}(s) = M_X(s) M_Y(s)$.

Proof of 1 and 4. Expand $e^{sX} = \sum_n (sX)^n/n!$ and take expectations term by term (justified by the finiteness of M_X near 0); the coefficient of $s^n/n!$ is $E[X^n]$. For 4, $E[e^{s(X+Y)}] = E[e^{sX}e^{sY}] = E[e^{sX}]E[e^{sY}]$ by the product theorem for independent variables. □

A related convenience is the **cumulant** function $\ln M_X(s)$: its first two derivatives at 0 are the mean and the variance directly, $(\ln M_X)'(0) = \mu$, $(\ln M_X)''(0) = \sigma^2$.

Worked Example 19.44 — MGFs of the exponential, Poisson and Gaussian

Find the MGFs and use them to obtain the means and variances.

Exponential. For $s < \lambda$: $M_T(s) = \int_0^\infty e^{st}\lambda e^{-\lambda t}dt = \dfrac{\lambda}{\lambda - s}$. Then $M'(s) = \lambda/(\lambda - s)^2$, $M''(s) = 2\lambda/(\lambda - s)^3$, so $E[T] = 1/\lambda$, $E[T^2] = 2/\lambda^2$, $\mathrm{Var}(T) = 1/\lambda^2$. ✓ (The MGF does not exist for $s \ge \lambda$ — the exponential tail is not light enough.)

Poisson. $M_X(s) = \sum_k e^{sk} e^{-\lambda}\lambda^k/k! = e^{-\lambda}\sum_k (\lambda e^s)^k/k! = \exp\big(\lambda(e^s - 1)\big)$. Its logarithm is $\lambda(e^s - 1)$, whose first and second derivatives at 0 are both λ: **mean = variance =** λ.

Gaussian. For $Z \sim \mathcal{N}(0,1)$, complete the square: $sz - z^2/2 = -\frac{1}{2}(z - s)^2 + \frac{s^2}{2}$, so $M_Z(s) = e^{s^2/2}\int \varphi(z - s)\,dz = e^{s^2/2}$. By property 3, $X = \mu + \sigma Z$ has

$$M_X(s) = \exp\big(\mu s + \tfrac{1}{2}\sigma^2 s^2\big)\,.$$

$\ln M_X = \mu s + \frac{1}{2}\sigma^2 s^2$, giving mean μ and variance σ^2 at once. Expanding $e^{s^2/2} = \sum s^{2k}/(2^k k!)$ also gives all the moments of Z: odd moments vanish and $E[Z^{2k}] = (2k)!/(2^k k!) = 1 \cdot 3 \cdot 5 \cdots (2k-1)$, so $E[Z^4] = 3$ (kurtosis 3).

Worked Example 19.45 — Closure of Poisson and Gaussian families under addition

(a) Packets from two independent sources arrive as Poisson(λ_1) and Poisson(λ_2) counts per slot. What is the distribution of the total? (b) A receiver input has independent thermal noise $N_1 \sim \mathcal{N}(0,1)$ and interference $N_2 \sim \mathcal{N}(0,4)$ (in $\mu\mathrm{V}^2$). What is the distribution of $N_1 + N_2$, and the probability that it exceeds 5 μV?

(a) $M(s) = e^{\lambda_1(e^s-1)}e^{\lambda_2(e^s-1)} = e^{(\lambda_1+\lambda_2)(e^s-1)}$, the MGF of Poisson($\lambda_1 + \lambda_2$). By uniqueness, merged independent Poisson streams are Poisson with the summed rate. (Doing this by discrete convolution requires the binomial theorem; the MGF makes it a one-liner.)

(b) $M(s) = e^{s^2/2}e^{4s^2/2} = e^{5s^2/2}$, so $N_1 + N_2 \sim \mathcal{N}(0,5)$. Then $P(N_1 + N_2 > 5) = Q(5/\sqrt{5}) = Q(2.236) = 0.0127$. Independent Gaussian noises add in *power* (variance), and the sum stays Gaussian — the property that makes Gaussian noise analysis of linear systems so clean (Ch. 21).

The characteristic function

The MGF may fail to exist (for the Cauchy distribution it is infinite for every $s \neq 0$). Replacing the real s by an imaginary $j\omega$ cures this, because $|e^{j\omega X}| = 1$.

Characteristic function

$$\Phi_X(\omega) = E\big[e^{j\omega X}\big] = \int_{-\infty}^{\infty} f_X(x)\, e^{j\omega x}\, dx, \qquad \omega \in \mathbb{R}.$$

It exists for every random variable, with $|\Phi_X(\omega)| \le \Phi_X(0) = 1$.

Comparing with the Fourier transform of Ch. 16, $\Phi_X(\omega)$ is the Fourier transform of the density with the sign of the exponent reversed: $\Phi_X(\omega) = \mathcal{F}\{f_X\}(-\omega)$. Every Fourier result transfers immediately:

- **Inversion:** $f_X(x) = \dfrac{1}{2\pi}\displaystyle\int_{-\infty}^{\infty} \Phi_X(\omega) e^{-j\omega x}\, d\omega$, so Φ_X determines the distribution.
- **Convolution theorem:** for independent X, Y, $\Phi_{X+Y}(\omega) = \Phi_X(\omega)\Phi_Y(\omega)$ — the transform-domain version of $f_{X+Y} = f_X * f_Y$ (Section 19.6).
- **Moments:** if $E[|X|^n] < \infty$, $E[X^n] = j^{-n}\Phi_X^{(n)}(0)$; smoothness of Φ_X at the origin reflects how many moments exist, just as smoothness of a signal reflects the decay of its spectrum.
- **Symmetry:** $\Phi_X(-\omega) = \Phi_X(\omega)^*$; Φ_X is real and even exactly when f_X is even.

Standard pairs: $\mathcal{N}(\mu,\sigma^2) \mapsto e^{j\mu\omega - \sigma^2\omega^2/2}$; $\mathrm{Exp}(\lambda) \mapsto \lambda/(\lambda - j\omega)$; $\mathcal{U}(-a,a) \mapsto \sin(a\omega)/(a\omega)$; Cauchy $\mapsto e^{-|\omega|}$ (not differentiable at 0, matching the non-existent mean); $\mathrm{Poisson}(\lambda) \mapsto \exp(\lambda(e^{j\omega} - 1))$.

For integer-valued random variables one also uses the **probability generating function** $G_N(z) = E[z^N] = \sum_k p_N(k) z^k$, which is the Z-transform of the PMF (Ch. 17) evaluated at z^{-1}; $G_N'(1) = E[N]$ and the PGF of a sum of independent counts is the product of the PGFs.

Worked Example 19.46 — Sum of independent uniform errors via characteristic functions

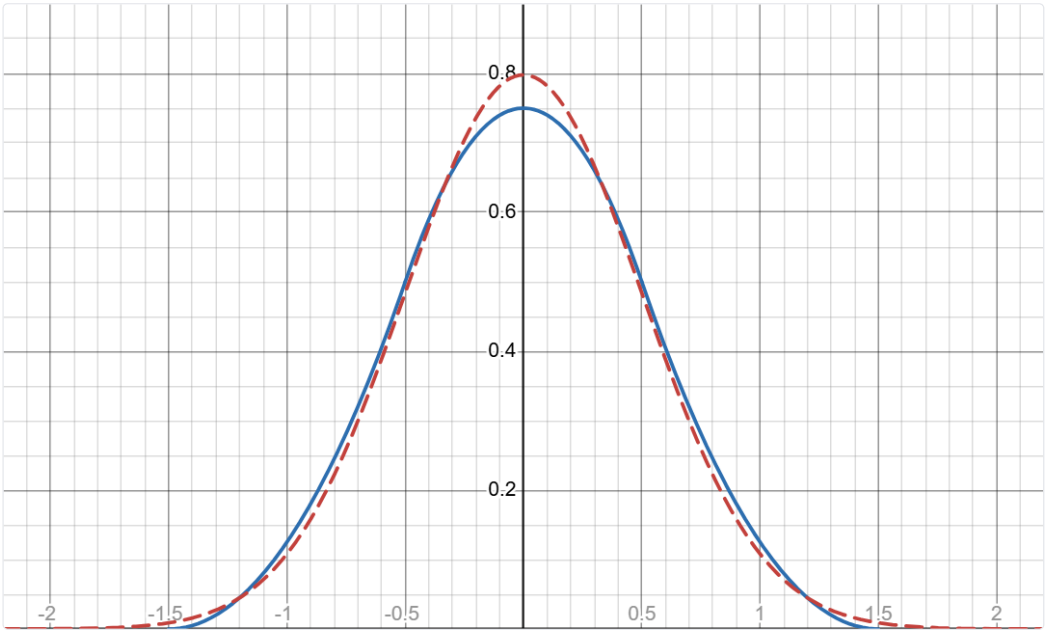
A measurement accumulates n independent rounding errors $e_i \sim \mathcal{U}(-\tfrac12, \tfrac12)$ (in units of one LSB). Find the characteristic function, mean and variance of the total error $S_n = e_1 + \cdots + e_n$, and compare its distribution with a Gaussian for $n = 3$.

Step 1 — one error. $\Phi_e(\omega) = \int_{-1/2}^{1/2} e^{j\omega x} dx = \dfrac{\sin(\omega/2)}{\omega/2}$.

Step 2 — the sum. By independence, $\Phi_{S_n}(\omega) = \left(\dfrac{\sin(\omega/2)}{\omega/2}\right)^n$ — the n-fold product, corresponding to the n-fold convolution of rectangles (the triangle of Example 19.27 for $n = 2$).

Step 3 — moments. Expand near $\omega = 0$: $\frac{\sin u}{u} = 1 - \frac{u^2}{6} + \cdots$ with $u = \omega/2$, so $\Phi_e(\omega) = 1 - \frac{\omega^2}{24} + \cdots$. Matching with $\Phi(\omega) = 1 + j\omega E[e] - \frac{\omega^2}{2}E[e^2] + \cdots$ gives $E[e] = 0$, $E[e^2] = 1/12$ (as in Example 19.40). Then $\Phi_{S_n}(\omega) = 1 - \frac{n\omega^2}{24} + \cdots$, so $\mathrm{Var}(S_n) = n/12$.

Step 4 — Gaussian comparison. For $n = 3$, $\mathrm{Var}(S_3) = 1/4$, $\sigma = 1/2$. The exact density (three convolved rectangles, a piecewise quadratic) has $f_{S_3}(0) = 3/4$; the Gaussian $\mathcal{N}(0, 1/4)$ has $f(0) = 1/(\sqrt{2\pi}\cdot\tfrac12) = 0.798$ — already within 6 %. Since $\left(\frac{\sin(\omega/2)}{\omega/2}\right)^n \approx \left(1 - \frac{\omega^2}{24}\right)^n \to$ a Gaussian shape after rescaling by $\sqrt{n}$, the characteristic function makes the central limit theorem plausible in a single line; Ch. 20 states it precisely.



Density of the sum of three independent $\mathcal{U}(-\frac12, \frac12)$ errors (blue, piecewise quadratic) and the Gaussian with the same variance $1/4$ (red, dashed). Three uniform terms are already nearly Gaussian; the difference lies mainly in the tails, which the uniform sum truncates at ± 1.5.

Common Pitfall — sign conventions for transforms

The characteristic function uses $e^{+j\omega x}$, the Fourier transform of this book uses $e^{-j\omega t}$, and the MGF uses the real exponent e^{sx}. Mixing conventions flips the sign of odd moments (and of the phase). When borrowing a Fourier pair for a characteristic function, replace ω by $-\omega$; for even densities nothing changes.

Engineering Note — moments as the designer's summary

Mean (DC level, bias), variance (AC power, noise power, tolerance), and the shape parameters (skewness, kurtosis) are the quantities that appear in specifications and link budgets. Second-order reasoning — means and variances, which add for independent contributions — handles the bulk of noise and tolerance analysis; the full distribution, usually Gaussian by the central limit theorem, is needed only at the final step, when the variance is converted into an error probability with $Q(\cdot)$.

Summary

Concept	Key formula
Axioms	$P(A) \geq 0$, $P(\Omega) = 1$, $P(\bigcup A_i) = \sum P(A_i)$ for disjoint A_i
Complement, addition rule	$P(A^c) = 1 - P(A)$, $P(A \cup B) = P(A) + P(B) - P(AB)$
Union bound	$P(\bigcup A_i) \leq \sum P(A_i)$
Counting	n^k, $n!/(n-k)!$, $\binom{n}{k} = \dfrac{n!}{k!(n-k)!}$, $\binom{n+k-1}{k}$
Conditional probability	$P(A \mid B) = P(AB)/P(B)$
Chain rule	$P(A_1 \cdots A_n) = P(A_1)P(A_2 \mid A_1) \cdots P(A_n \mid A_1 \cdots A_{n-1})$
Total probability	$P(A) = \sum_i P(A \mid B_i)P(B_i)$
Bayes' theorem	$P(B_k \mid A) = P(A \mid B_k)P(B_k) / \sum_i P(A \mid B_i)P(B_i)$
Odds form	posterior odds = likelihood ratio $\times$ prior odds
MAP threshold, $\pm A$ in $\mathcal{N}(0, \sigma^2)$	$x_{\text{th}} = \dfrac{\sigma^2}{2A} \ln \dfrac{p_0}{p_1}$
Independence	$P(AB) = P(A)P(B)$; for RVs $f_{XY} = f_X f_Y$
Series / parallel reliability	$\prod p_i$ / $1 - \prod(1 - p_i)$
k-out-of-n, TMR	$\sum_{i \geq k} \binom{n}{i} p^i (1-p)^{n-i}$; $3p^2 - 2p^3$
Binomial, geometric, Poisson	$\binom{n}{k} p^k q^{n-k}$; $q^{k-1}p$; $e^{-\lambda}\lambda^k/k!$
Uniform, exponential	$1/(b-a)$; $\lambda e^{-\lambda t}$, $P(T > t) = e^{-\lambda t}$
Gaussian	$\dfrac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2}$, $P(X > x) = Q\big((x-\mu)/\sigma\big)$
Rayleigh	$\dfrac{r}{\sigma^2} e^{-r^2/2\sigma^2}$, $P(R > r) = e^{-r^2/2\sigma^2}$
CDF	$F_X(x) = P(X \leq x)$, $P(a < X \leq b) = F_X(b) - F_X(a)$, $f_X = F_X'$
Marginal, conditional	$f_X = \int f_{XY}\, dy$, $f_{Y\mid X} = f_{XY}/f_X$
Sum of independent RVs	$f_{X+Y} = f_X * f_Y$
Density transformation	$f_Y(y) = \sum_i f_X(x_i)/\lvert g'(x_i)\rvert$, $g(x_i) = y$
Max / min of independent RVs	$F_{\max} = \prod F_i$, $P(\min > x) = \prod(1 - F_i)$
Hazard rate	$h(t) = f_T(t)/R(t)$, $R(t) = \exp\left(-\int_0^t h\right)$
Expectation, LOTUS	$E[g(X)] = \int g(x) f_X(x)\, dx$; $E[aX + bY] = aE[X] + bE[Y]$
Tail formula, MTTF	$E[T] = \int_0^\infty P(T > t)\, dt$
Variance	$\operatorname{Var}(X) = E[X^2] - \mu^2$, $\operatorname{Var}(aX + b) = a^2 \operatorname{Var}(X)$
Sum of variances	$\operatorname{Var}(X + Y) = \operatorname{Var} X + \operatorname{Var} Y + 2 \operatorname{Cov}(X, Y)$
Quantization noise	$\Delta^2/12$; SQNR $= 6.02N + 1.76$ dB
Markov, Chebyshev	$P(X \geq a) \leq E[X]/a$; $P(\lvert X - \mu\rvert \geq k\sigma) \leq 1/k^2$
MGF	$M_X(s) = E[e^{sX}]$, $E[X^n] = M_X^{(n)}(0)$, $M_{X+Y} = M_X M_Y$
Characteristic function	$\Phi_X(\omega) = E[e^{j\omega X}]$, $\Phi_{X+Y} = \Phi_X \Phi_Y$
Gaussian MGF	$\exp(\mu s + \sigma^2 s^2/2)$

Further reading

- A. Leon-Garcia, *Probability, Statistics, and Random Processes for Electrical Engineering*, 3rd ed., Pearson — the closest match to this chapter, with many communications and reliability examples.
- A. Papoulis and S. U. Pillai, *Probability, Random Variables, and Stochastic Processes*, 4th ed., McGraw-Hill — the standard graduate reference, especially for functions of random variables and characteristic functions.

- D. P. Bertsekas and J. N. Tsitsiklis, *Introduction to Probability*, 2nd ed., Athena Scientific — exceptionally clear on conditioning, Bayes' rule and expectation.
- S. M. Ross, *A First Course in Probability*, 10th ed., Pearson — combinatorics and classic problems in depth.

Chapter 20 — Statistical Analysis

> **Prerequisites:** Ch. 5 (Integral Calculus), Ch. 6 (Multivariable Calculus), Ch. 8 (Matrices and Linear Systems), Ch. 19 (Probability) **Learning outcomes:** After this chapter you can …
>
> - summarise a data set with location, spread and shape statistics, and choose robust statistics when outliers are present;
> - compute covariance, correlation coefficients and covariance matrices, and interpret them correctly;
> - recognise the standard distributions (Bernoulli, binomial, Poisson, uniform, exponential, Gaussian, Rayleigh, chi-square, t, F), their moments and the relations between them;
> - derive sampling distributions, apply the central limit theorem, and build estimators by the method of moments and maximum likelihood;
> - construct confidence intervals and carry out z, t, chi-square and F tests with correct interpretation of p-values and error probabilities;
> - fit and diagnose a linear regression, and evaluate and report measurement uncertainty in the GUM framework.

Probability theory (Ch. 19) starts from a model — a distribution with known parameters — and deduces what data should look like. **Statistics** runs the argument backwards: it starts from data and infers something about the model that produced them. An engineer who measures the noise floor of a receiver, calibrates a temperature sensor, qualifies a batch of capacitors or reports "4.702 kΩ ± 0.012 kΩ" is doing statistics, whether or not the word is used.

The chapter follows the natural order of a data analysis. We first describe data (Sections 20.1–20.3), then recall the distributions used as models (20.4). Sections 20.5–20.8 form the core of *inference*: how statistics computed from a sample are themselves random (sampling distributions), how to estimate parameters, how to attach intervals, and how to test claims. Section 20.9 fits relations between variables by regression, and Section 20.10 assembles everything into the internationally agreed method for evaluating measurement uncertainty (the *GUM*).

Throughout, $X_1, \dots, X_n$ denotes a random sample (random variables, uppercase) and $x_1, \dots, x_n$ the observed numbers (lowercase), following the notation of Ch. 19.

20.1 Descriptive statistics

Why describe before inferring

Before any model is fitted, a data set should be *looked at*. A single miswired channel, a saturated ADC, a unit error between millivolts and volts, or a bimodal population (two suppliers mixed in one reel) will invalidate every later calculation, and each is obvious from a summary or a plot but invisible in a single average. Descriptive statistics compress n numbers into a handful of meaningful ones and are the first defence against garbage-in, garbage-out.

Data, population and sample

A **population** is the complete set of items or outcomes about which we want to draw conclusions (all resistors produced by a process, all possible readings of a voltmeter on a fixed source). A **sample** is the finite subset $x_1, \dots, x_n$ actually observed; n is the **sample size**. A **statistic** is any function of the sample, $t = g(x_1, \dots, x_n)$, that does not depend on unknown parameters.

Data come in several types, and the type dictates the admissible summaries:

Type	Example	Meaningful operations
Nominal (categorical)	failure mode: open / short / drift	counting, mode
Ordinal	inspection grade A/B/C	ordering, median, ranks
Interval	temperature in °C	differences, mean, standard deviation
Ratio	resistance in Ω, time to failure	all of the above plus ratios, coefficient of variation

Averaging category codes ("open = 1, short = 2") is meaningless; quoting a coefficient of variation for a temperature in °C is meaningless because the zero is arbitrary.

Order statistics and the five-number summary

Sorting the sample gives the **order statistics** $x_{(1)} \le x_{(2)} \le \dots \le x_{(n)}$. Many robust summaries are built from them.

Quantiles, quartiles and the five-number summary

For $0 < p < 1$, a **p-quantile** q_p is a value with (approximately) a fraction p of the data below it and $1 - p$ above. The **median** is $q_{0.5}$; the **lower and upper quartiles** are $Q_1 = q_{0.25}$ and $Q_3 = q_{0.75}$. The **interquartile range** is $\mathrm{IQR} = Q_3 - Q_1$. The **five-number summary** is $\left(x_{(1)},\, Q_1,\, \tilde{x},\, Q_3,\, x_{(n)}\right)$, where $\tilde{x}$ is the median.

Several conventions exist for quartiles of a finite sample (software packages offer up to nine); for hand work we use the *median of halves*: Q_1 is the median of the lower half of the sorted data and Q_3 the median of the upper half (excluding the overall median when n is odd). Differences between conventions shrink as n grows and do not matter for the decisions quartiles are used for.

The **box plot** draws a box from Q_1 to Q_3 with a line at the median and whiskers to the most extreme points within the **fences**

$$Q_1 - 1.5\,\mathrm{IQR}, \qquad Q_3 + 1.5\,\mathrm{IQR}.$$

Points beyond the fences are flagged as potential outliers. For Gaussian data, about 0.7% of points fall outside the fences, so a flagged point is a reason to *investigate*, not automatically to delete.

Frequency tables and histograms

For larger n the data are grouped into K class intervals (bins) of width h, with **frequencies** f_k (counts), **relative frequencies** f_k/n and **frequency density** $f_k/(nh)$. A histogram plotted with frequency density has unit total area and is directly comparable with a probability density function $f_X(x)$ from Ch. 19 — this is the bridge between data and model.

The choice of bin width matters: too wide hides structure, too narrow shows noise. Two common rules are

$$K \approx 1 + \log_2 n \quad \text{(Sturges)}, \qquad h \approx 2\,\mathrm{IQR}\,n^{-1/3} \quad \text{(Freedman–Diaconis)}.$$

Sturges' rule assumes near-Gaussian data and undersmooths large samples; Freedman–Diaconis uses the IQR and is robust to outliers. The **empirical cumulative distribution function**

$$\hat{F}_n(x) = \frac{1}{n}\,\#\{i : x_i \le x\}$$

needs no binning at all: it is a staircase rising by $1/n$ at each data point, and by the Glivenko–Cantelli theorem it converges uniformly to the true CDF $F_X(x)$ as $n \to \infty$.

Shape: skewness and kurtosis

Location and spread (Section 20.2) are not the whole story. The *shape* of a distribution is summarised by standardised third and fourth moments. With $m_k = \frac{1}{n}\sum_i (x_i - \bar{x})^k$ the kth sample central moment,

$$g_1 = \frac{m_3}{m_2^{3/2}} \quad \text{(sample skewness)}, \qquad g_2 = \frac{m_4}{m_2^2} - 3 \quad \text{(sample excess kurtosis)}.$$

Skewness is zero for symmetric data, positive when the right tail is long (time-to-failure, power of noise, lognormal component values) and negative when the left tail is long. Excess kurtosis is zero for a Gaussian, positive for heavy-tailed data with more extreme values (impulsive noise, interference), and negative for flat, light-tailed data (a uniform distribution has $g_2 = -1.2$). Because they involve high powers of deviations, g_1 and g_2 are very sensitive to outliers and need n of at least several tens to be informative.

Worked Example 20.1 — Summarising a batch of resistors

Twelve resistors of nominal value 1 kΩ are measured (in Ω):

$$998,\ 1003,\ 1001,\ 995,\ 1007,\ 1002,\ 999,\ 1004,\ 1000,\ 1012,\ 997,\ 1001.$$

Find the five-number summary, the IQR, the fences, and comment on the shape.

Step 1 — sort. $995,\ 997,\ 998,\ 999,\ 1000,\ 1001,\ 1001,\ 1002,\ 1003,\ 1004,\ 1007,\ 1012$.

Step 2 — median. With $n = 12$ (even) the median is the average of the 6th and 7th values: $\tilde{x} = (1001 + 1001)/2 = 1001\ \Omega$.

Step 3 — quartiles (median of halves). Lower half $995, \dots, 1001$: $Q_1 = (998 + 999)/2 = 998.5\ \Omega$. Upper half $1001, \dots, 1012$: $Q_3 = (1003 + 1004)/2 = 1003.5\ \Omega$. Hence $\mathrm{IQR} = 5\ \Omega$.

Step 4 — five-number summary. $(995,\ 998.5,\ 1001,\ 1003.5,\ 1012)\ \Omega$; range $= 17\ \Omega$.

Step 5 — fences. $Q_1 - 1.5\,\mathrm{IQR} = 991\ \Omega$, $Q_3 + 1.5\,\mathrm{IQR} = 1011\ \Omega$. The value $1012\ \Omega$ lies beyond the upper fence and is flagged.

Step 6 — shape. Working with deviations $d_i = x_i - 1000$ keeps the arithmetic small: $\sum d_i = 19$, so $\bar{x} = 1001.583\ \Omega$. The central moments are $m_2 = 19.41\ \Omega^2$, $m_3 = 71.42\ \Omega^3$, $m_4 = 1267\ \Omega^4$, giving $g_1 = 71.42/19.41^{1.5} = 0.84$ and $g_2 = 1267/19.41^2 - 3 = 0.36$. The positive skewness is driven almost entirely by the flagged point. Before reporting statistics, one should check whether the $1012\ \Omega$ reading is a genuine part (perhaps from a different reel) or a contact-resistance error in the fixture.

Worked Example 20.2 — Statistics from a frequency table

The rise times of 40 logic buffers are grouped into 1 ns classes:

Class (ns)	4–5	5–6	6–7	7–8	8–9
Frequency f_k	3	9	15	10	3

Estimate the mean and standard deviation.

Step 1 — class marks. Represent each class by its midpoint $c_k = 4.5,\ 5.5,\ 6.5,\ 7.5,\ 8.5$ ns.

Step 2 — mean. $\bar{x} \approx \frac{1}{n}\sum_k f_k c_k = \frac{3(4.5)+9(5.5)+15(6.5)+10(7.5)+3(8.5)}{40} = \frac{261}{40} = 6.525$ ns.

Step 3 — variance. $s^2 \approx \frac{1}{n-1}\sum_k f_k (c_k - \bar{x})^2 = \frac{42.975}{39} = 1.102\ \mathrm{ns}^2$, so $s \approx 1.05$ ns.

Step 4 — grouping error. Replacing every value by its class mark adds an error that is roughly uniform over a class of width h; Sheppard's correction subtracts $h^2/12 = 0.083\ \mathrm{ns}^2$ from the variance, giving $s \approx 1.01$ ns. The correction is small when $h \ll s$ and is often ignored, but it shows why grouped statistics are approximate.

Engineering Note — Look at the raw data first

Production test logs routinely contain sentinel values (-999, 9.9×10^{37} "overload" from a DMM, zero for "no reading"). A five-number summary exposes them instantly because they appear as $x_{(1)}$ or $x_{(n)}$. Make computing the minimum and maximum the reflex first step of every analysis.

Common Pitfall — Deleting outliers to improve the statistics

Removing points only because they are inconvenient biases every later result (spread is underestimated, confidence intervals become too narrow). Delete a point only with a documented physical cause (fixture fault, logging error). Otherwise keep it and, if needed, use robust statistics (Section 20.2) whose value it cannot dominate.

20.2 Mean, median, variance, and standard deviation

Measures of location

Three numbers compete to describe "the typical value".

Sample mean, median and mode

- **Sample mean:** $\bar{x} = \dfrac{1}{n}\sum_{i=1}^{n} x_i$.
- **Sample median:** $\tilde{x} = x_{((n+1)/2)}$ for n odd, $\tilde{x} = \tfrac{1}{2}\big(x_{(n/2)} + x_{(n/2+1)}\big)$ for n even.
- **Mode:** the most frequent value (or, for continuous grouped data, the class with the highest density).

Each is the solution of a minimisation problem, which explains its behaviour:

$$\bar{x} = \arg\min_{c} \sum_{i=1}^{n} (x_i - c)^2, \qquad \tilde{x} = \arg\min_{c} \sum_{i=1}^{n} |x_i - c|.$$

Proof sketch. Differentiating $\sum (x_i - c)^2$ gives $-2\sum (x_i - c) = 0$, i.e. $c = \bar{x}$; the second derivative $2n > 0$ confirms a minimum. For $\sum |x_i - c|$ the derivative (where it exists) is $\#\{x_i < c\} - \#\{x_i > c\}$, which changes sign exactly when half the points lie on each side — at the median. $\square$

Squared loss punishes large deviations heavily, so a single wild point drags the mean; absolute loss treats all deviations linearly, so the median moves at most to the next data point. This is formalised by the **breakdown point**, the fraction of data that can be made arbitrarily bad before the statistic becomes arbitrarily bad: 0 (strictly $1/n$) for the mean, 50% for the median.

The mean has compensating virtues: it is linear ($\overline{ax + b} = a\bar{x} + b$), easy to update, and for Gaussian data it is the most efficient estimator of the centre (Section 20.6) — the median needs about $\pi/2 \approx 1.57$ times as many samples to achieve the same variance.

A useful compromise is the **trimmed mean**, which discards a fixed fraction (say 10%) from each end before averaging. It is common in sensor fusion and in reading-averaging routines in instrument firmware.

Weighted mean

When observations have different reliabilities, they should not count equally. If x_i has standard uncertainty u_i, the **inverse-variance weighted mean** is

$$\bar{x}_w = \frac{\sum_i w_i x_i}{\sum_i w_i}, \qquad w_i = \frac{1}{u_i^2}, \qquad u(\bar{x}_w) = \Big(\sum_i w_i\Big)^{-1/2}.$$

It is the minimum-variance unbiased linear combination of independent measurements of the same quantity: minimising $\operatorname{Var}\big(\sum a_i x_i\big) = \sum a_i^2 u_i^2$ subject to $\sum a_i = 1$ with a Lagrange multiplier (Ch. 6) gives $a_i \propto 1/u_i^2$.

Measures of spread

Sample variance and standard deviation

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n} (x_i - \bar{x})^2, \qquad s = \sqrt{s^2}.$$

The **range** is $x_{(n)} - x_{(1)}$; the **mean absolute deviation** is $\frac{1}{n}\sum |x_i - \bar{x}|$; the **median absolute deviation** is $\mathrm{MAD} = \operatorname{median}_i |x_i - \tilde{x}|$. The **coefficient of variation** (ratio data only) is $\mathrm{CV} = s/\bar{x}$.

The standard deviation s carries the units of the data and is the usual measure of spread. For Gaussian data, $1.4826\,\mathrm{MAD}$ and $\mathrm{IQR}/1.349$ are consistent estimates of σ that are robust to outliers.

Why divide by $n - 1$?

The divisor $n - 1$ (Bessel's correction) makes s^2 an *unbiased* estimator of the population variance σ^2. The derivation is short and worth knowing.

Unbiasedness of the sample variance

If $X_1, \dots, X_n$ are independent with common mean μ and variance σ^2, then $E[S^2] = \sigma^2$, where $S^2 = \frac{1}{n-1}\sum (X_i - \bar{X})^2$.

Proof. Write $X_i - \bar X = (X_i - \mu) - (\bar X - \mu)$ and expand:

$$\sum_{i=1}^{n}(X_i - \bar X)^2 = \sum_{i=1}^{n}(X_i - \mu)^2 - n(\bar X - \mu)^2,$$

because the cross term is $-2(\bar X - \mu)\sum(X_i - \mu) = -2n(\bar X - \mu)^2$. Taking expectations, $E[(X_i - \mu)^2] = \sigma^2$ and $E[(\bar X - \mu)^2] = \operatorname{Var}(\bar X) = \sigma^2/n$ (Ch. 19, variance of a sum of independent variables). Hence

$$E\Big[\sum (X_i - \bar X)^2\Big] = n\sigma^2 - n\cdot\frac{\sigma^2}{n} = (n-1)\sigma^2. \qquad \square$$

The intuition: deviations are measured from $\bar x$, which is itself fitted to the data and therefore sits closer to the points than the true μ does. One "degree of freedom" is used up in estimating μ — the n deviations satisfy one linear constraint $\sum(x_i - \bar x) = 0$, so only $n-1$ are free. The same counting gives the divisor $n - p$ when p parameters are fitted in regression (Section 20.9).

Note that s itself is *not* unbiased for σ: by Jensen's inequality $E[S] < \sigma$. For Gaussian data $E[S] = c_4\sigma$ with $c_4 = \sqrt{2/(n-1)}\,\Gamma(n/2)/\Gamma((n-1)/2)$, e.g. $c_4 = 0.798$ for $n = 2$ and 0.973 for $n = 10$. The bias is usually ignored except in quality-control charts.

Computational formulas and numerical stability

Expanding the square gives the "computational" formula

$$s^2 = \frac{1}{n-1}\Big(\sum x_i^2 - n\bar x^2\Big) = \frac{1}{n-1}\Big(\sum x_i^2 - \frac{(\sum x_i)^2}{n}\Big),$$

which needs only running sums. It is exact in real arithmetic but disastrous in floating point when $|\bar x| \gg s$: two large, nearly equal numbers are subtracted, and the relative rounding error of each (about 10^{-16} in double precision) is multiplied by $\bar x^2/s^2$ (catastrophic cancellation, Ch. 22).

The stable alternatives are (i) the **two-pass** algorithm (compute $\bar x$ first, then $\sum(x_i - \bar x)^2$), (ii) **shifting** the data by any value close to the mean before using the one-pass formula, or (iii) **Welford's online algorithm**, which updates mean and sum of squared deviations sample by sample:

$$\begin{aligned} \delta &= x_k - \bar x_{k-1}, \qquad \bar x_k = \bar x_{k-1} + \frac{\delta}{k},\\ M_k &= M_{k-1} + \delta\,(x_k - \bar x_k), \qquad s_k^2 = \frac{M_k}{k-1}. \end{aligned}$$

Welford's recursion is what a well-written DMM "statistics" mode or a streaming data logger should use.

Variance, RMS and power

For an electrical signal sampled as $v_1, \dots, v_n$, the mean is the DC component and the (population-divisor) variance is the AC power in a 1 Ω reference. Since

$$\frac{1}{n}\sum v_i^2 = \bar v^{\,2} + \frac{1}{n}\sum (v_i - \bar v)^2,$$

the squared RMS value equals the squared DC value plus the squared standard deviation:

$$V_{\text{rms}}^2 = V_{\text{DC}}^2 + \sigma_v^2.$$

A "true-RMS AC" meter that blocks DC reports σ_v; a DC-coupled true-RMS meter reports $\sqrt{V_{\text{DC}}^2 + \sigma_v^2}$. This identity is the statistical form of Parseval's theorem (Ch. 15) restricted to the zero-frequency term.

> **Worked Example 20.3 — Mean and standard deviation of ADC readings**
>
> Eight readings of a nominal 2.5 V reference are $2.47,\ 2.53,\ 2.51,\ 2.49,\ 2.55,\ 2.45,\ 2.50,\ 2.52$ V. Find $\bar x$, s, and the RMS value.
>
> **Step 1 — mean.** $\sum x_i = 20.02$, so $\bar x = 2.5025$ V.
>
> **Step 2 — deviations.** $x_i - \bar x = -0.0325,\ 0.0275,\ 0.0075,\ -0.0125,\ 0.0475,\ -0.0525,\ -0.0025,\ 0.0175$ (they sum to zero — a free arithmetic check).
>
> **Step 3 — sum of squares.** $\sum(x_i - \bar x)^2 = 0.00735$ V^2.
>
> **Step 4 — variance and standard deviation.** $s^2 = 0.00735/7 = 1.05 \times 10^{-3}$ V^2, $s = 32.4$ mV. (With divisor n one would get 30.3 mV; the difference is $\sqrt{8/7}$, i.e. about 7% for $n = 8$.)
>
> **Step 5 — RMS.** $\frac{1}{n}\sum x_i^2 = 50.1074/8 = 6.263425$ V^2, and indeed $\bar x^2 + \frac{1}{n}\sum(x_i - \bar x)^2 = 6.26250625 + 0.00091875 = 6.263425$. So $V_{\text{rms}} = 2.50268$ V, dominated by the DC value as expected.

> **Worked Example 20.4 — When the computational formula fails**
>
> A frequency counter logs four readings $10^9 + 4,\ 10^9 + 7,\ 10^9 + 13,\ 10^9 + 16$ (in Hz). The true sample variance is that of $4, 7, 13, 16$: $\bar d = 10$, $\sum(d_i - \bar d)^2 = 36 + 9 + 9 + 36 = 90$, $s^2 = 30$ Hz^2.
>
> Evaluating $\big(\sum x_i^2 - (\sum x_i)^2/n\big)/(n-1)$ in IEEE double precision returns -170.7 Hz^2 — a *negative* variance. Each $x_i^2 \approx 10^{18}$ is stored with an absolute rounding error of about $10^{18} \times 1.1 \times 10^{-16} \approx 100$, comparable to the 90 we are trying to recover. Subtracting 10^9 first (or using the two-pass or Welford algorithm) returns exactly 30 Hz^2.

Worked Example 20.5 — Combining three calibrations

Three laboratories report the value of a standard resistor as 10.12 ± 0.02, 10.05 ± 0.05 and $10.09 \pm 0.03\ \Omega$ (standard uncertainties). Find the weighted mean and its uncertainty.

Step 1 — weights. $w_i = 1/u_i^2 = 2500,\ 400,\ 1111.1\ \Omega^{-2}$; $\sum w_i = 4011.1$.

Step 2 — weighted mean. $\bar{x}_w = (2500 \times 10.12 + 400 \times 10.05 + 1111.1 \times 10.09)/4011.1 = 10.1047\ \Omega$.

Step 3 — uncertainty. $u(\bar{x}_w) = 1/\sqrt{4011.1} = 0.0158\ \Omega$ — smaller than the best single uncertainty, as it should be.

Step 4 — consistency check. $\chi^2 = \sum w_i (x_i - \bar{x}_w)^2 = 0.585 + 1.197 + 0.240 = 2.02$ with $n - 1 = 2$ degrees of freedom. A value near the degrees of freedom indicates the three results agree within their stated uncertainties (Section 20.8); a value much larger would suggest an unrecognised error in at least one laboratory.

Common Pitfall — Quoting s when you mean the uncertainty of the mean

The standard deviation s describes the scatter of *individual* readings. The uncertainty of their *average* is the standard error $s/\sqrt{n}$ (Section 20.5). In Example 20.3, the readings scatter by 32 mV but the mean is known to about $32/\sqrt{8} \approx 11$ mV. Confusing the two either overstates or understates the result by a factor $\sqrt{n}$.

Engineering Note — Robust statistics in firmware

Embedded systems reading noisy sensors often use a *median-of-k* filter (typically $k = 3$ or 5) to reject single-sample glitches from ESD or relay switching, followed by an averaging filter to reduce Gaussian noise. The median removes impulses that would corrupt a mean; the subsequent mean recovers the efficiency the median loses.

20.3 Covariance and correlation

Motivation

Engineering data are rarely one-dimensional. A sensor's output depends on temperature; two receiver channels share a local oscillator and therefore share its phase noise; the gain and offset of an ADC drift together with supply voltage. Quantifying how two variables vary *together* is needed to decide whether one predicts the other, to propagate uncertainties correctly when inputs are not independent (Section 20.10), and to design systems that cancel common-mode effects.

Population and sample covariance

Recall from Ch. 19 that for random variables X and Y with means μ_X, μ_Y,

$$\operatorname{Cov}(X,Y) = E\big[(X-\mu_X)(Y-\mu_Y)\big] = E[XY] - \mu_X\mu_Y, \qquad \rho_{XY} = \frac{\operatorname{Cov}(X,Y)}{\sigma_X \sigma_Y}.$$

Sample covariance and Pearson correlation coefficient

For paired data (x_i, y_i), $i = 1, \dots, n$, define the sums of squares and products

$$S_{xx} = \sum (x_i - \bar{x})^2, \qquad S_{yy} = \sum (y_i - \bar{y})^2, \qquad S_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y}).$$

The **sample covariance** is $s_{xy} = S_{xy}/(n-1)$ and the **sample (Pearson) correlation coefficient** is

$$r = \frac{s_{xy}}{s_x s_y} = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}.$$

The sign of s_{xy} says whether large x tends to go with large y (positive) or small y (negative). Its magnitude depends on units (Ω·°C, say), which is why the dimensionless r is preferred for describing the *strength* of association.

Bounds on the correlation coefficient

$-1 \le r \le 1$, with $|r| = 1$ if and only if all points lie exactly on a straight line $y = a + bx$ with $b \neq 0$ (sign of r = sign of b). The same holds for ρ_{XY}, with "exactly" replaced by "with probability one".

Proof. Let $\mathbf{u} = (x_i - \bar{x})_i$ and $\mathbf{v} = (y_i - \bar{y})_i$ be the centred data vectors in $\mathbb{R}^n$. Then $r = \langle \mathbf{u}, \mathbf{v} \rangle / (\|\mathbf{u}\|\|\mathbf{v}\|)$ is the cosine of the angle between them. The Cauchy–Schwarz inequality (Ch. 9) gives $|\langle \mathbf{u}, \mathbf{v} \rangle| \le \|\mathbf{u}\|\|\mathbf{v}\|$, with equality iff $\mathbf{v} = b\mathbf{u}$, i.e. $y_i - \bar{y} = b(x_i - \bar{x})$ for all i. $\square$

This geometric picture — **correlation is the cosine of the angle between centred data vectors** — is worth remembering: $r = 0$ means the centred vectors are orthogonal, and r^2 is the squared cosine, the fraction of $\|\mathbf{v}\|^2$ captured by projecting $\mathbf{v}$ onto $\mathbf{u}$. This is exactly the r^2 of simple linear regression (Section 20.9).

Further properties follow directly from the bilinearity of covariance:

- $\operatorname{Cov}(aX + b,\ cY + d) = ac \operatorname{Cov}(X,Y)$, so ρ is invariant to changes of units and origin (up to sign if $ac < 0$).
- $\operatorname{Var}(X \pm Y) = \operatorname{Var} X + \operatorname{Var} Y \pm 2 \operatorname{Cov}(X,Y)$.
- Independent $\Rightarrow$ uncorrelated; the converse fails in general but holds for jointly Gaussian variables (Ch. 19).

What correlation does not measure

Pearson's r measures **linear** association only. If X is symmetric about zero and $Y = X^2$, then Y is a deterministic function of X, yet $\operatorname{Cov}(X, Y) = E[X^3] - E[X]E[X^2] = 0$, so $\rho = 0$. Conversely, a strong r can be produced by a single extreme point, by mixing two groups with different means, or by a common cause (both x and y rise with ambient temperature). The four data sets of *Anscombe's quartet* share identical means, variances, $r = 0.816$ and regression lines while looking completely different when plotted. The rule is simple: never interpret r without a scatter plot.

Common Pitfall — Correlation is not causation, and not agreement

A correlation between solder-joint failures and production day of week may reflect a third variable (which operator was on shift). Separately, two instruments measuring the same voltage can have $r = 0.999$ while one reads consistently 2% high: correlation measures association, not agreement. For method comparison, analyse the *differences* $y_i - x_i$ (mean bias and its spread) instead.

Rank correlation

When the relation is monotonic but not linear, or the data contain outliers or are ordinal, replace values by their ranks.

Spearman's rank correlation

Let R_i and Q_i be the ranks of x_i among the x's and of y_i among the y's (average ranks for ties). **Spearman's** r_s is the Pearson correlation of the ranks. With no ties it simplifies to

$$r_s = 1 - \frac{6\sum_i d_i^2}{n(n^2-1)}, \qquad d_i = R_i - Q_i.$$

The shortcut follows from $\sum R_i = \sum Q_i = n(n+1)/2$ and $\sum R_i^2 = \sum Q_i^2 = n(n+1)(2n+1)/6$, which fix the means and variances of the ranks; expanding $\sum d_i^2 = \sum R_i^2 + \sum Q_i^2 - 2\sum R_i Q_i$ and substituting gives the formula. $r_s = \pm 1$ for any strictly monotonic relation, linear or not.

The covariance matrix

For a random vector $\mathbf{X} = (X_1, \dots, X_p)^{\mathsf{T}}$ with mean $\boldsymbol{\mu}$, the **covariance matrix** is

$$\boldsymbol{\Sigma} = E\big[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^{\mathsf{T}}\big], \qquad \Sigma_{ij} = \operatorname{Cov}(X_i, X_j).$$

Its sample counterpart, from n observation vectors $\mathbf{x}_k$, is $\mathbf{S} = \frac{1}{n-1}\sum_k (\mathbf{x}_k - \bar{\mathbf{x}})(\mathbf{x}_k - \bar{\mathbf{x}})^{\mathsf{T}}$.

Properties of the covariance matrix

1. $\boldsymbol{\Sigma}$ is symmetric and positive semidefinite: $\mathbf{a}^{\mathsf{T}}\boldsymbol{\Sigma}\mathbf{a} = \operatorname{Var}(\mathbf{a}^{\mathsf{T}}\mathbf{X}) \geq 0$ for every $\mathbf{a}$.
2. For a linear map $\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{b}$: $\boldsymbol{\Sigma}_Y = \mathbf{A}\boldsymbol{\Sigma}_X\mathbf{A}^{\mathsf{T}}$.
3. Its eigenvectors (Ch. 10) are orthogonal **principal axes**; the eigenvalues are the variances along those axes, and $\operatorname{tr}\boldsymbol{\Sigma} = \sum_i \sigma_i^2$ is the total variance.

Proof of 1 and 2. $\mathbf{a}^{\mathsf{T}}\mathbf{X}$ is a scalar random variable with mean $\mathbf{a}^{\mathsf{T}}\boldsymbol{\mu}$, so its variance is $E[(\mathbf{a}^{\mathsf{T}}(\mathbf{X} - \boldsymbol{\mu}))^2] = \mathbf{a}^{\mathsf{T}}E[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^{\mathsf{T}}]\mathbf{a}$, which is non-negative because it is a variance. For 2, $\mathbf{Y} - E\mathbf{Y} = \mathbf{A}(\mathbf{X} - \boldsymbol{\mu})$, and substituting into the definition gives $\mathbf{A}\boldsymbol{\Sigma}_X\mathbf{A}^{\mathsf{T}}$. Property 3 is the spectral theorem for real symmetric matrices. $\square$

Property 2 is the matrix form of uncertainty propagation used in Section 20.10. Property 3 underlies principal component analysis: rotating to the eigenvector basis produces *uncorrelated* components, and discarding those with small eigenvalues compresses the data with minimal loss of variance. The **correlation matrix** is $\mathbf{R} = \mathbf{D}^{-1}\boldsymbol{\Sigma}\mathbf{D}^{-1}$ with $\mathbf{D} = \operatorname{diag}(\sigma_1, \dots, \sigma_p)$; it has ones on the diagonal.

Worked Example 20.6 — Correlation of an RTD calibration

A platinum RTD is calibrated at six temperatures:

T (°C)	0	20	40	60	80	100
R (Ω)	100.0	107.9	115.5	123.2	130.9	138.5

Find s_{TR} and r.

Step 1 — means. $\bar{T} = 50\ ^\circ\text{C}$, $\bar{R} = 716/6 = 119.333\ \Omega$.

Step 2 — sums of squares. $S_{TT} = 2(50^2 + 30^2 + 10^2) = 7000\ (^\circ\text{C})^2$; $S_{RR} = 1035.29\ \Omega^2$.

Step 3 — sum of products. Because $\sum(T_i - \bar{T}) = 0$, $S_{TR} = \sum(T_i - \bar{T})R_i = -50(100.0) - 30(107.9) - 10(115.5) + 10(123.2) + 30(130.9) + 50(138.5) = 2692\ \Omega\,^\circ\text{C}$.

Step 4 — covariance and correlation. $s_{TR} = 2692/5 = 538.4\ \Omega\,^\circ\text{C}$; $r = 2692/\sqrt{7000 \times 1035.29} = 0.99999$.

Interpretation. r is so close to 1 that it no longer discriminates between "linear" and "slightly curved". The physically known curvature of platinum (Callendar–Van Dusen equation) is invisible in r; it shows up only in the residuals of a fit (Section 20.9). A high r is necessary but not sufficient for a good linear model.

Worked Example 20.7 — Rank correlation of capacitor ageing

Six electrolytic capacitors have equivalent series resistance (ESR, in mΩ) and measured remaining life (kh):

ESR	3.1	2.4	4.8	1.9	3.6	5.2
Life	12	15	7	18	9	8

Step 1 — rank each variable. ESR ranks: $3, 2, 5, 1, 4, 6$. Life ranks: $4, 5, 1, 6, 3, 2$.

Step 2 — rank differences. $d_i = -1, -3, 4, -5, 1, 4$; $\sum d_i^2 = 1 + 9 + 16 + 25 + 1 + 16 = 68$.

Step 3 — Spearman. $r_s = 1 - 6(68)/(6 \cdot 35) = 1 - 408/210 = -0.943$.

Step 4 — compare. Pearson's r on the raw values is -0.939. The two agree here because the relation is roughly linear; had one capacitor failed early with a huge ESR, r would be pulled strongly while r_s would change by at most one rank.

Worked Example 20.8 — Sum and difference of correlated noise

Two amplifier channels have output noise with $\sigma_1 = 2$ mV, $\sigma_2 = 3$ mV and correlation $\rho = 0.5$ (shared supply ripple). Find the noise of the sum and of the difference, and the principal axes.

Step 1 — covariance matrix. $\mathrm{Cov} = \rho\sigma_1\sigma_2 = 3\ \mathrm{mV}^2$, so

$$\boldsymbol{\Sigma} = \begin{pmatrix} 4 & 3 \\ 3 & 9 \end{pmatrix}\ \mathrm{mV}^2.$$

Step 2 — sum. $\mathbf{a} = (1, 1)^{\mathsf{T}}$: $\mathrm{Var} = \mathbf{a}^{\mathsf{T}}\boldsymbol{\Sigma}\mathbf{a} = 4 + 9 + 2(3) = 19\ \mathrm{mV}^2$, $\sigma_{1+2} = 4.36$ mV.

Step 3 — difference. $\mathbf{a} = (1, -1)^{\mathsf{T}}$: $\mathrm{Var} = 4 + 9 - 6 = 7\ \mathrm{mV}^2$, $\sigma_{1-2} = 2.65$ mV. A differential measurement partially cancels the common component — the statistical basis of common-mode rejection.

Step 4 — principal axes. The eigenvalues solve $\lambda^2 - 13\lambda + 27 = 0$: $\lambda = (13 \pm \sqrt{61})/2 = 10.41,\ 2.59\ \mathrm{mV}^2$. The unit-norm combination with the least variance is the eigenvector of $\lambda = 2.59$, approximately $(0.91, -0.42)^{\mathsf{T}}$, with $\sigma = 1.61$ mV. (For comparison, the normalised plain difference $(1, -1)^{\mathsf{T}}/\sqrt{2}$ has variance $3.5\ \mathrm{mV}^2$.) Weighting the quieter channel more heavily cancels more of the shared ripple.

Engineering Note — Covariance in multi-channel systems

In antenna arrays, the sample covariance matrix of the element signals is the starting point of adaptive beamforming and direction-of-arrival estimation: its large eigenvalues correspond to signals and its small ones to noise. In instrumentation, the off-diagonal terms decide whether averaging two channels helps (uncorrelated noise, $\sigma/\sqrt{2}$) or barely helps (strongly correlated noise).

20.4 Common distributions

Why a catalogue

Statistical inference needs models, and a surprisingly small set of distributions covers most engineering situations. Each arises from a specific mechanism — counting successes, counting rare events, waiting for the next event, adding many small effects, taking the magnitude of a 2-D Gaussian vector, summing squared Gaussian errors — and recognising the mechanism tells you which model to use. The last three entries (χ^2, t, F) are less physical models of data than *sampling distributions*: they describe statistics computed from Gaussian samples and are the workhorses of Sections 20.7–20.8.

General definitions (pmf, pdf, CDF, expectation, moment generating function $M_X(s) = E[e^{sX}]$) are in Ch. 19. For each distribution below we give the mechanism, the pmf or pdf, the mean and variance, and where it appears in EE.

Bernoulli distribution

Bernoulli(p)

$X \in \{0, 1\}$ with $P(X = 1) = p$, $P(X = 0) = 1 - p$. Then $E[X] = p$, $\mathrm{Var}(X) = p(1 - p)$, $M_X(s) = 1 - p + pe^s$.

The variance follows from $X^2 = X$: $E[X^2] = p$, so $\mathrm{Var}\,X = p - p^2$. It is maximal, $1/4$, at $p = 1/2$ — the worst case used for sample-size planning of proportions (Section 20.7). Models: one transmitted bit is in error or not; one device passes or fails test; one photon is detected or not.

Binomial distribution

Binomial(n, p)

The number of successes in n independent Bernoulli(p) trials:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \qquad k = 0, 1, \ldots, n,$$

with $E[X] = np$, $\mathrm{Var}(X) = np(1 - p)$, $M_X(s) = (1 - p + pe^s)^n$.

Derivation. Any particular sequence with k successes has probability $p^k(1-p)^{n-k}$, and there are $\binom{n}{k}$ such sequences. Writing $X = \sum_{i=1}^{n} B_i$ with independent Bernoulli B_i, linearity gives $E[X] = np$ and independence gives $\operatorname{Var} X = np(1-p)$; the MGF is the nth power of the Bernoulli MGF because MGFs of independent sums multiply. The pmf is unimodal with mode at $\lfloor (n+1)p \rfloor$.

Poisson distribution

Poisson(λ)

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \qquad k = 0, 1, 2, \ldots,$$

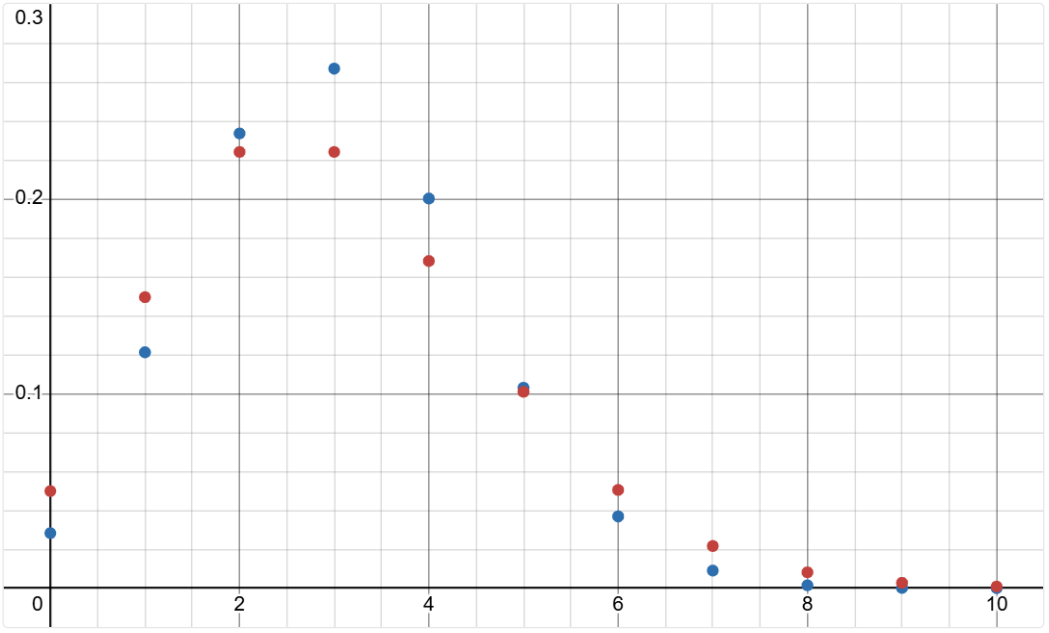
with $E[X] = \operatorname{Var}(X) = \lambda$ and $M_X(s) = \exp\big(\lambda(e^s - 1)\big)$.

Poisson limit of the binomial

If $n \to \infty$ and $p \to 0$ with $np = \lambda$ fixed, then $\binom{n}{k} p^k (1-p)^{n-k} \to \lambda^k e^{-\lambda}/k!$.

Proof sketch. $\binom{n}{k} p^k = \frac{n(n-1)\cdots(n-k+1)}{n^k} \frac{\lambda^k}{k!} \to \frac{\lambda^k}{k!}$, and $(1-\lambda/n)^{n-k} \to e^{-\lambda}$. Equivalently, the binomial MGF $(1 + p(e^s - 1))^n = (1 + \lambda(e^s - 1)/n)^n \to e^{\lambda(e^s - 1)}$. $\square$

So the Poisson law describes the count of *rare, independent* events in a fixed interval: electrons crossing a junction (shot noise), photons on a detector, radioactive decays, packet arrivals on a lightly loaded link, soft errors in a memory over a day. The equality of mean and variance is a useful diagnostic: count data whose variance greatly exceeds the mean are *overdispersed* (clustered events), and a plain Poisson model will underestimate their variability.



Binomial$(10, 0.3)$ pmf (blue) and Poisson(3) pmf (red): equal means, but the Poisson has variance 3 against the binomial's 2.1, so it is flatter with a longer right tail.

Uniform distribution

Uniform(a, b)

$$f_X(x) = \frac{1}{b-a} \quad (a \le x \le b), \qquad E[X] = \frac{a+b}{2}, \qquad \operatorname{Var}(X) = \frac{(b-a)^2}{12}.$$

Derivation of the variance. Centre at zero with half-width $c = (b-a)/2$: $\operatorname{Var} X = \frac{1}{2c}\int_{-c}^{c} x^2\,dx = \frac{c^2}{3} = \frac{(b-a)^2}{12}$.

The uniform law models complete ignorance within known bounds. Two cases matter enormously in EE. **Quantisation error** of an ideal ADC with step Δ is (under mild conditions) uniform on $[-\Delta/2, \Delta/2]$, with variance $\Delta^2/12$ — the source of the familiar $6.02N + 1.76$ dB SNR formula. **Type B uncertainty** from a manufacturer's "$\pm a$" specification is modelled as uniform on $[-a, a]$, giving standard uncertainty $a/\sqrt{3}$ (Section 20.10). The random phase of an unsynchronised oscillator is uniform on $[0, 2\pi)$.

Exponential distribution

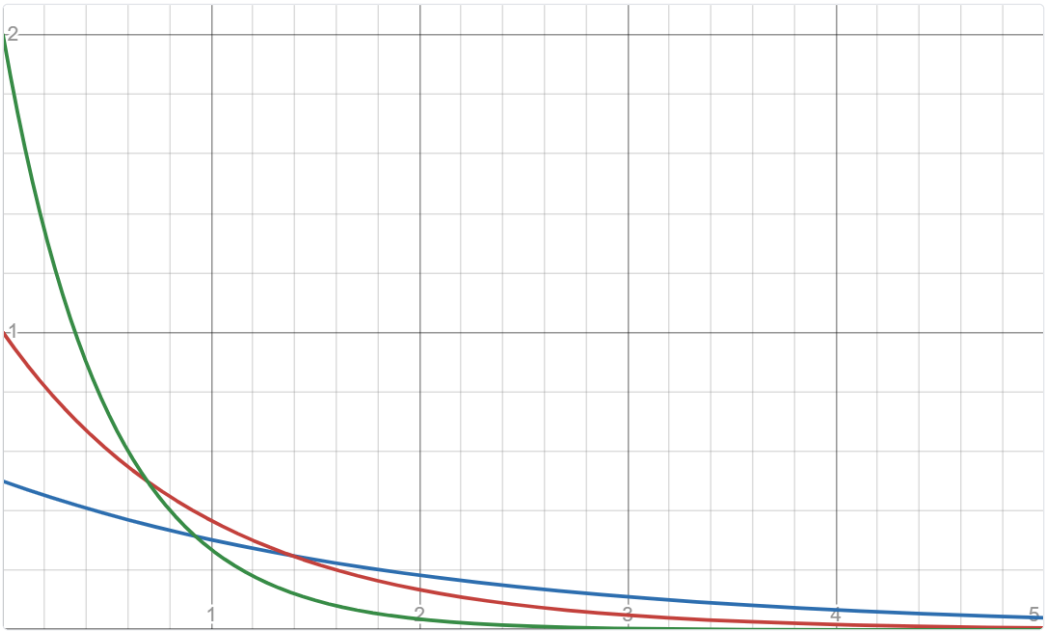
Exponential(λ)

$$f_X(x) = \lambda e^{-\lambda x} \ (x \ge 0), \qquad F_X(x) = 1 - e^{-\lambda x}, \qquad E[X] = \frac{1}{\lambda}, \qquad \operatorname{Var}(X) = \frac{1}{\lambda^2}.$$

It is the waiting time to the first event of a Poisson process of rate λ: $P(X > x) = P(\text{no events in } [0, x]) = e^{-\lambda x}$. It is the only continuous distribution with the **memoryless property**

$$P(X > s+t \mid X > s) = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = P(X > t).$$

In reliability engineering, λ is a constant **failure rate** (failures per hour, or FIT = failures per 10^9 device-hours) and $1/\lambda$ is the **MTBF** (mean time between failures). Memorylessness means a working part is "as good as new" — appropriate for the flat bottom of the bathtub curve (random failures), wrong for wear-out (electrolytic capacitors, fans, relays), which is modelled by the Weibull distribution instead.



Exponential pdfs for $\lambda = 0.5$ (blue), 1 (red) and 2 (green). The pdf starts at λ and decays with time constant $1/\lambda$ — the same shape as an RC discharge.

Gaussian (normal) distribution

Gaussian $\mathcal{N}(\mu, \sigma^2)$

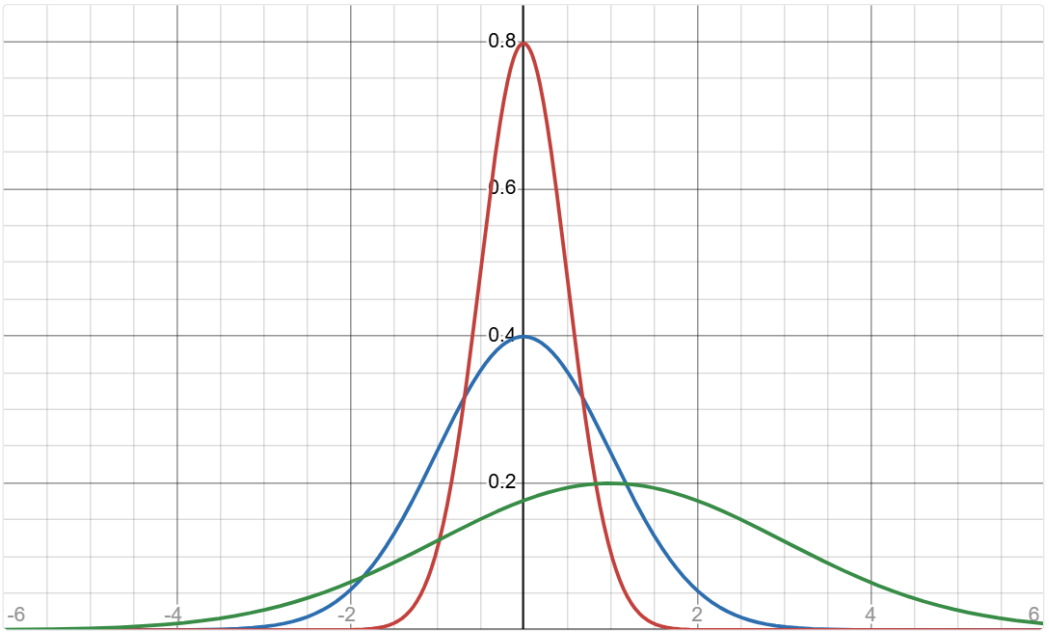
$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\!\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \qquad E[X] = \mu, \quad \operatorname{Var}(X) = \sigma^2, \quad M_X(s) = e^{\mu s + \sigma^2 s^2/2}.$$

The **standard** Gaussian $Z \sim \mathcal{N}(0,1)$ has CDF $\Phi(z)$; engineers usually use the tail function $Q(z) = 1 - \Phi(z) = P(Z > z)$.

Any Gaussian is standardised by $Z = (X-\mu)/\sigma$, so $P(X > x) = Q\big((x-\mu)/\sigma\big)$. Key values: $P(|Z| \le 1) = 0.6827$, $P(|Z| \le 2) = 0.9545$, $P(|Z| \le 3) = 0.9973$ (the "68–95–99.7 rule"); $Q(1.645) = 0.05$, $Q(1.960) = 0.025$, $Q(2.576) = 0.005$, $Q(3) = 1.35 \times 10^{-3}$. For large z, $Q(z) \approx \frac{1}{z\sqrt{2\pi}} e^{-z^2/2}$.

Three facts make the Gaussian central:

1. **Closure under linear operations.** Any linear combination of jointly Gaussian variables is Gaussian; the output of an LTI system driven by Gaussian noise is Gaussian (Ch. 21).
2. **The central limit theorem** (Section 20.5): sums of many small independent effects are approximately Gaussian, which is why thermal noise, aggregate manufacturing variation and many measurement errors are Gaussian.
3. **Maximum entropy.** Among all distributions with given mean and variance, the Gaussian has the largest entropy — it is the "least committal" model consistent with knowing only two moments.



Gaussian pdfs: $\mathcal{N}(0,1)$ (blue), $\mathcal{N}(0,0.25)$ (red) and $\mathcal{N}(1,4)$ (green). Halving σ doubles the peak height; the area stays 1.

Rayleigh distribution

Rayleigh(σ)

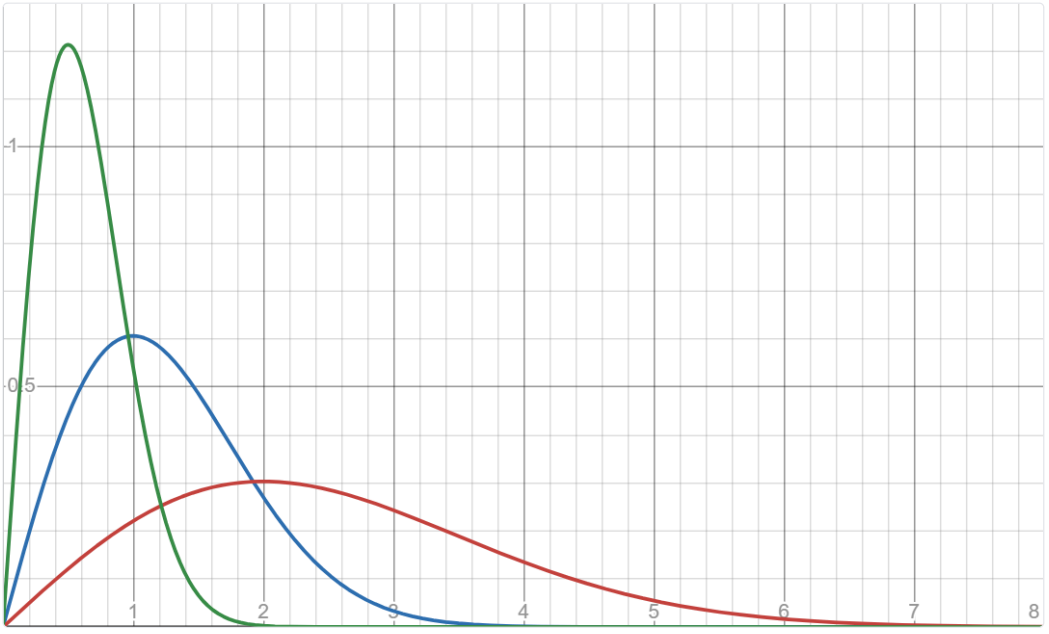
If X, Y are independent $\mathcal{N}(0,\sigma^2)$, then $R = \sqrt{X^2+Y^2}$ has

$$f_R(r) = \frac{r}{\sigma^2} e^{-r^2/(2\sigma^2)} \ (r \geq 0), \qquad P(R > r) = e^{-r^2/(2\sigma^2)},$$

$$E[R] = \sigma\sqrt{\pi/2} \approx 1.253\sigma, \qquad E[R^2] = 2\sigma^2, \qquad \mathrm{Var}(R) = \left(2 - \frac{\pi}{2}\right)\sigma^2 \approx 0.429\sigma^2.$$

Derivation. In polar coordinates (r,ϕ) the joint pdf $\frac{1}{2\pi\sigma^2} e^{-(x^2+y^2)/(2\sigma^2)}$ becomes, with the Jacobian r (Ch. 6), $\frac{r}{2\pi\sigma^2} e^{-r^2/(2\sigma^2)}$. Integrating out $\phi \in [0, 2\pi)$ gives f_R; the phase is uniform and independent of R. The moments follow from $\int_0^\infty r^2 e^{-r^2/(2\sigma^2)} dr = \sigma^3\sqrt{\pi/2}$.

The Rayleigh law is the **envelope of narrowband Gaussian noise** (in-phase and quadrature components are independent Gaussians) and the amplitude of a **multipath-faded** radio signal with no line-of-sight path. R^2 is exponential with mean $2\sigma^2$, so the *power* of a Rayleigh-faded signal is exponentially distributed. Adding a deterministic component gives the Rician distribution.



Rayleigh pdfs for $\sigma = 1$ (blue), $\sigma = 2$ (red) and $\sigma = 0.5$ (green). The mode is at $r = \sigma$; the long right tail is the occasional noise peak that triggers a false alarm.

Chi-square distribution

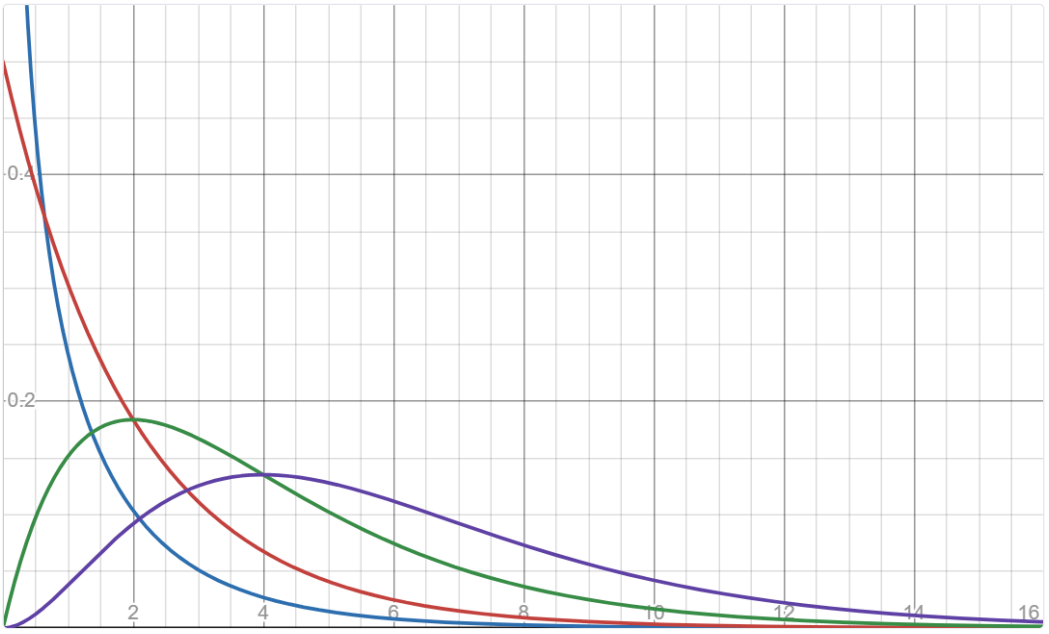
Chi-square χ^2_k

If $Z_1, \dots, Z_k$ are independent $\mathcal{N}(0,1)$, then $Q = \sum_{i=1}^{k} Z_i^2$ has the chi-square distribution with k **degrees of freedom**:

$$f_Q(q) = \frac{q^{k/2-1} e^{-q/2}}{2^{k/2}\Gamma(k/2)} \ (q > 0), \qquad E[Q] = k, \qquad \mathrm{Var}(Q) = 2k, \qquad M_Q(s) = (1-2s)^{-k/2}.$$

Derivation of the moments. For one term, $E[Z^2] = 1$ and $E[Z^4] = 3$, so $\mathrm{Var}(Z^2) = 2$; adding k independent terms gives mean k and variance $2k$. The MGF of Z^2 is $\int \frac{1}{\sqrt{2\pi}} e^{sz^2 - z^2/2} dz = (1-2s)^{-1/2}$, and the kth power gives M_Q, which is the MGF of a gamma distribution with shape $k/2$ and scale 2 — hence the pdf.

Special cases tie the catalogue together: χ^2_2 is exponential with mean 2, and $\sqrt{\chi^2_2} \cdot \sigma$ is Rayleigh. χ^2_k is skewed right for small k and approaches $\mathcal{N}(k, 2k)$ as $k \to \infty$ (CLT). In EE it describes **noise power estimates** (a sum of squared Gaussian samples), **energy detectors** in spectrum sensing, and, most importantly for this chapter, the sample variance of Gaussian data (Section 20.5) and goodness-of-fit statistics (Section 20.8).



Chi-square pdfs for $k = 1$ (blue), 2 (red), 4 (green) and 6 (purple). For $k \geq 3$ the mode is at $k - 2$; the distributions spread and become more symmetric as k grows.

Student's t distribution

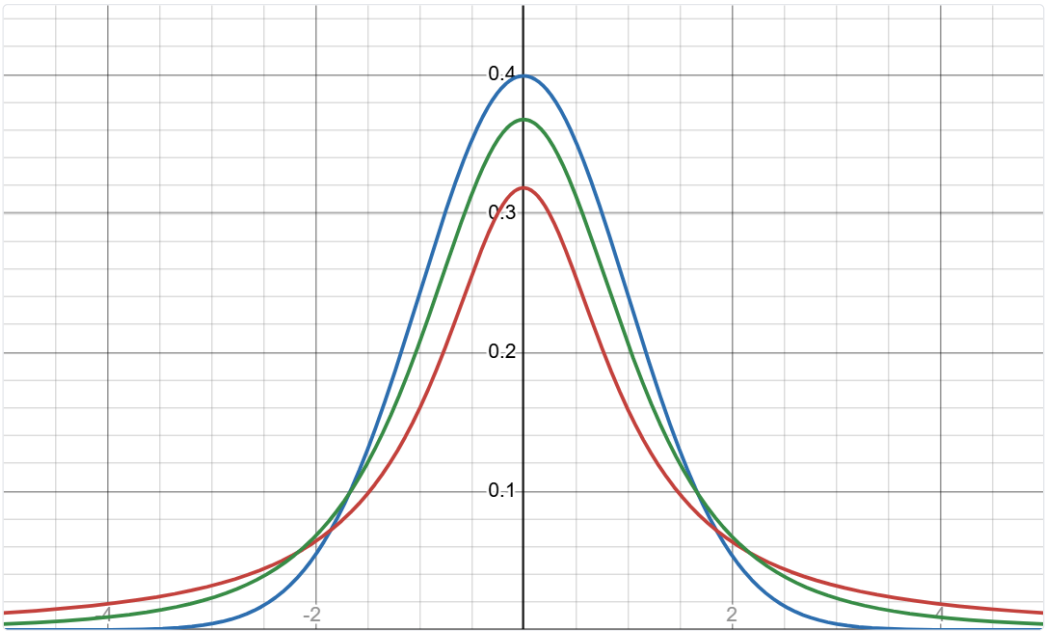
Student's t_ν

If $Z \sim \mathcal{N}(0,1)$ and $V \sim \chi^2_\nu$ are independent, then $T = Z / \sqrt{V/\nu}$ has the t distribution with ν degrees of freedom:

$$f_T(t) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\,\Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{t^2}{\nu}\right)^{-(\nu+1)/2},$$

with $E[T] = 0$ ($\nu > 1$) and $\mathrm{Var}(T) = \nu/(\nu-2)$ ($\nu > 2$).

The t distribution is what a standardised Gaussian becomes when the unknown σ is replaced by an estimate from ν degrees of freedom. The extra randomness of the estimate fattens the tails: with $\nu = 3$, $P(|T| > 3) = 0.058$, versus 0.0027 for the Gaussian — a factor of 21. As $\nu \to \infty$, $V/\nu \to 1$ and $t_\nu \to \mathcal{N}(0,1)$. The two-sided 95% critical value $t_{0.025,\nu}$ is 12.71 for $\nu = 1$, 3.18 for $\nu = 3$, 2.57 for $\nu = 5$, 2.23 for $\nu = 10$ and 2.04 for $\nu = 30$, against 1.96 for the Gaussian. The case $\nu = 1$ is the Cauchy distribution, which has no mean at all: averaging Cauchy samples does not reduce their spread.



Student t pdfs for $\nu = 1$ (red, Cauchy) and $\nu = 3$ (green) against $\mathcal{N}(0,1)$ (blue). Lower peaks, heavier tails: small samples must use wider intervals.

The F distribution

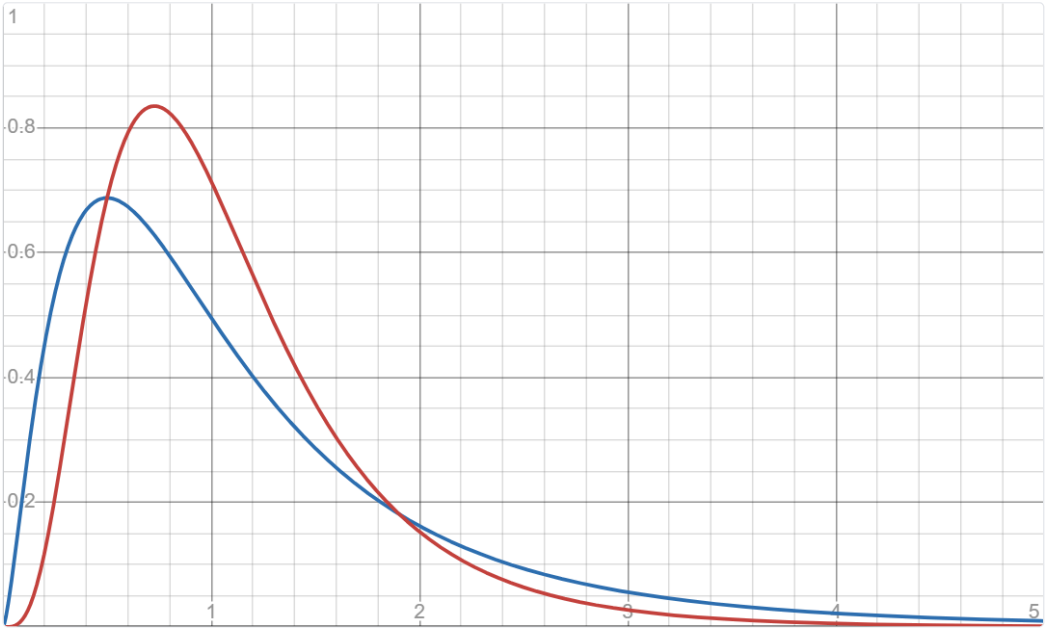
Fisher–Snedecor F_{d_1,d_2}

If $U \sim \chi^2_{d_1}$ and $V \sim \chi^2_{d_2}$ are independent, $F = \dfrac{U/d_1}{V/d_2}$ has pdf

$$f(x) = \frac{(d_1/d_2)^{d_1/2}}{B(d_1/2, d_2/2)}\, x^{d_1/2-1}\left(1 + \frac{d_1}{d_2}x\right)^{-(d_1+d_2)/2} \qquad (x > 0),$$

with $E[F] = d_2/(d_2 - 2)$ for $d_2 > 2$; B is the beta function.

F is the distribution of a **ratio of two independent variance estimates** from Gaussian data, normalised by their degrees of freedom; it is used to compare the noise of two instruments (Section 20.8) and to test the significance of regression models (Section 20.9). Useful identities: $1/F_{d_1,d_2} \sim F_{d_2,d_1}$, so lower critical values follow from upper ones, and $t_\nu^2 \sim F_{1,\nu}$.



F pdfs for $(d_1, d_2) = (5, 10)$ (blue) and $(10, 20)$ (red). Both concentrate near 1 — the value expected when two variance estimates agree — and tighten as the degrees of freedom grow.

Relations between the distributions

From	Operation	Gives
n independent Bernoulli(p)	sum	Binomial(n, p)

From	Operation	Gives
Binomial(n,p)	$n\to\infty$, $np=\lambda$	Poisson(λ)
Binomial, Poisson	large $np(1-p)$, large λ	Gaussian (CLT)
Poisson process rate λ	time to first event	Exponential(λ)
Two independent $\mathcal{N}(0,\sigma^2)$	magnitude $\sqrt{X^2+Y^2}$	Rayleigh(σ)
k independent $\mathcal{N}(0,1)$	sum of squares	χ^2_k
$\chi^2_2/2$	—	Exponential(1)
$\mathcal{N}(0,1)$ and χ^2_ν	$Z/\sqrt{V/\nu}$	t_ν
$\chi^2_{d_1}$ and $\chi^2_{d_2}$	ratio of normalised	F_{d_1,d_2}
Uniform$(0,1)$	$-\ln(U)/\lambda$	Exponential(λ)

The last line is the **inverse-transform method** for generating random numbers (Ch. 24): if F is a continuous CDF and U is uniform, $F^{-1}(U)$ has CDF F.

Worked Example 20.9 — Packet errors: binomial versus Poisson

A 100-bit packet is sent over a channel with independent bit errors of probability $p=0.01$. The error-correcting code fixes up to 2 bit errors. Find the probability that a packet is lost.

Step 1 — model. The number of bit errors is $X\sim\text{Binomial}(100,0.01)$; a packet is lost if $X\ge 3$.

Step 2 — exact. $P(X=0)=0.99^{100}=0.36603$; $P(X=1)=100(0.01)(0.99)^{99}=0.36973$; $P(X=2)=\binom{100}{2}(0.01)^2(0.99)^{98}=0.18486$. So $P(X\ge 3)=1-0.92063=0.0794$.

Step 3 — Poisson approximation. $\lambda=np=1$: $P(X\ge 3)\approx 1-e^{-1}(1+1+\tfrac12)=1-2.5e^{-1}=0.0803$.

Step 4 — comment. The approximation is within 1.2% (relative). Without the code, a packet would be lost whenever $X\ge 1$, with probability 0.634; the code reduces the loss rate by a factor of 8.

Worked Example 20.10 — Photon counting

A photodetector registers on average $\lambda=4$ photons per 1 ms gate. Find (a) the probability of an empty gate, (b) the probability of 8 or more counts, (c) the signal-to-noise ratio of a count.

(a) $P(X=0)=e^{-4}=0.0183$.

(b) $P(X\ge 8)=1-\sum_{k=0}^{7}e^{-4}4^k/k!=0.0511$. Doubling the mean count happens in 5% of gates purely by chance — the irreducible **shot noise** of light.

(c) Mean 4, standard deviation $\sqrt{4}=2$, so SNR $=\lambda/\sqrt{\lambda}=\sqrt{\lambda}=2$. To improve SNR by a factor of 10 one must collect 100 times more photons (longer gate or larger aperture): shot-noise-limited SNR grows only as the square root of signal.

Worked Example 20.11 — Reliability with exponential lifetimes

A power module has constant failure rate $\lambda=2\times10^{-5}\ \text{h}^{-1}$ (MTBF $=50\,000$ h).

(a) One-year survival. $P(T>8760\ \text{h})=e^{-8760/50000}=e^{-0.1752}=0.839$.

(b) Survival to the MTBF. $P(T>50\,000)=e^{-1}=0.368$: only 37% of units reach the MTBF — the MTBF is a mean, not a typical life.

(c) Memorylessness. Given a module has already run 3 years, the probability it survives one more year is still 0.839. Burn-in therefore does nothing for exponential lifetimes; it helps only when early failures (decreasing hazard rate) are present.

(d) Series system. Ten independent modules, all required: the system lifetime is the minimum of ten exponentials, which is exponential with rate 10λ, so $\text{MTBF}_{\text{sys}}=5000$ h.

Worked Example 20.12 — Gaussian yield and Rayleigh false alarms

(a) Yield. Resistor values are $\mathcal{N}(1000\ \Omega,\ (4\ \Omega)^2)$. The fraction inside a $\pm1\%$ tolerance band (990–1010 Ω) is $P(|Z|\le 10/4)=1-2Q(2.5)=1-2(0.00621)=0.9876$, i.e. 1.24% rejects.

(b) False alarms. A radar detector compares the envelope of receiver noise with a threshold. The I and Q noise components are each $\mathcal{N}(0,\sigma^2)$ with $\sigma=1$ mV, so the envelope is Rayleigh(1 mV). For a threshold of 3 mV, $P_{\text{FA}}=e^{-9/2}=0.0111$. To achieve $P_{\text{FA}}=10^{-6}$ the threshold must be $\sigma\sqrt{2\ln 10^6}=5.26$ mV. Note how slowly the threshold grows: a millionfold reduction in false alarms costs less than a doubling of threshold.

Common Pitfall — Assuming Gaussian tails far from the mean

Real component distributions are often truncated (parts outside tolerance were screened out) or have heavier tails than Gaussian. Predicting a 10^{-9} failure rate from a Gaussian fitted to 50 samples extrapolates far beyond the data; at 6σ the answer depends entirely on the assumed tail shape. Tail probabilities should be supported by data or by physics, not by the convenience of the Gaussian formula.

Engineering Note — Matching mechanism to model

Ask what generates the data. Counting events in a fixed window: binomial (fixed n) or Poisson (no natural n). Time between events: exponential (no ageing) or Weibull (ageing). Sum of many small effects: Gaussian. Magnitude of a 2-D Gaussian: Rayleigh. Product of many small factors (component values after several process steps): lognormal, since the logarithm is a sum.

20.5 Sampling

Statistics are random variables

A sample mean computed today from 16 readings will differ from one computed tomorrow from another 16. Before the data are taken, every statistic $T = g(X_1, \dots, X_n)$ is a random variable, and its distribution — the **sampling distribution** — tells us how far T is likely to be from the quantity it is meant to estimate. All of inference rests on this idea: a confidence interval is a statement about the sampling distribution of an estimator, and a hypothesis test compares an observed statistic with its sampling distribution under a hypothesis.

Random sample

$X_1, \dots, X_n$ is a **random sample** (i.i.d. sample) from a distribution F if the X_i are independent and each has CDF F. The **standard error** of a statistic is the standard deviation of its sampling distribution.

Independence is a physical assumption, not a mathematical gift. Readings taken faster than the correlation time of the noise (Ch. 21), units taken consecutively from one reel, or repeated measurements without re-seating a probe are *not* independent, and the formulas below then understate the true variability.

Sampling distribution of the mean

Mean and variance of the sample mean

If $X_1, \dots, X_n$ is a random sample from a distribution with mean μ and variance σ^2, then

$$E[\bar{X}] = \mu, \qquad \operatorname{Var}(\bar{X}) = \frac{\sigma^2}{n}, \qquad \operatorname{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

If the sample is from $\mathcal{N}(\mu, \sigma^2)$, then $\bar{X} \sim \mathcal{N}(\mu, \sigma^2/n)$ exactly.

Proof. Linearity of expectation gives $E[\bar{X}] = \frac{1}{n}\sum E[X_i] = \mu$. Independence lets variances add: $\operatorname{Var}(\sum X_i) = n\sigma^2$, and dividing by n^2 gives σ^2/n. A linear combination of independent Gaussians is Gaussian. □

The $1/\sqrt{n}$ **law** is the most important scaling in measurement: averaging four readings halves the noise, averaging a hundred reduces it tenfold. It is the statistical reason a lock-in amplifier, a signal averager on an oscilloscope, or a long integration time on a DMM improves resolution — and the reason improvement becomes expensive: each extra decimal digit costs a factor of 100 in measurement time.

If the readings are correlated with correlation coefficient ρ_k between samples k apart, the variance of the mean becomes

$$\operatorname{Var}(\bar{X}) = \frac{\sigma^2}{n}\left[1 + 2\sum_{k=1}^{n-1}\left(1 - \frac{k}{n}\right)\rho_k\right],$$

so positive correlation inflates the standard error. The ratio $n_{\text{eff}} = n/[\,\cdot\,]$ is the **effective number of independent samples**. For first-order (exponentially correlated) noise with $\rho_k = a^k$ and large n, $n_{\text{eff}} \approx n(1-a)/(1+a)$.

Finite populations. When sampling *without replacement* n items from a finite lot of N, the variance of the mean is multiplied by the finite-population correction $(N-n)/(N-1)$. It matters only when the sample is a sizeable fraction of the lot (more than about 5%).

The central limit theorem

The Gaussian result above needs Gaussian data. The CLT says that for *any* distribution with finite variance, the mean of a large sample is approximately Gaussian anyway.

Central limit theorem (Lindeberg–Lévy)

Let $X_1, X_2, \dots$ be i.i.d. with mean μ and finite variance $\sigma^2 > 0$. Then the standardised mean

$$Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} = \frac{\sum_{i=1}^{n} X_i - n\mu}{\sigma\sqrt{n}}$$

converges in distribution to $\mathcal{N}(0,1)$: $P(Z_n \le z) \to \Phi(z)$ for every z.

Proof sketch (via characteristic functions). Let $Y_i = (X_i - \mu)/\sigma$, so $E[Y_i] = 0$, $E[Y_i^2] = 1$, and let $\varphi(\omega) = E[e^{j\omega Y}]$ be their common characteristic function (the Fourier transform of the pdf, Ch. 16). A Taylor expansion about $\omega = 0$ gives

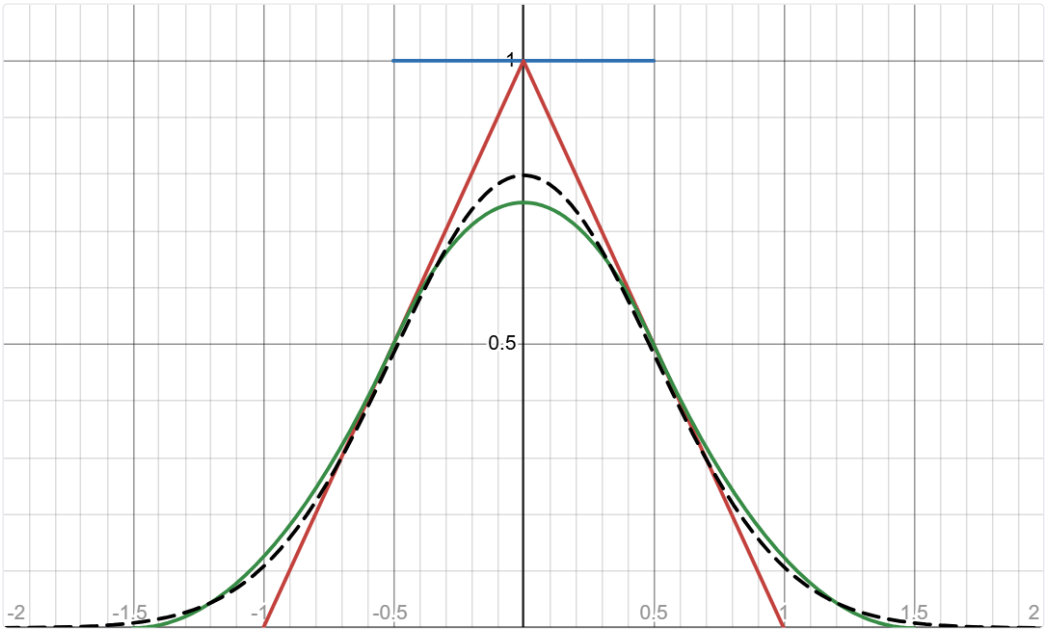
$$\varphi(\omega) = 1 + j\omega E[Y] - \frac{\omega^2}{2}E[Y^2] + o(\omega^2) = 1 - \frac{\omega^2}{2} + o(\omega^2).$$

Since $Z_n = \frac{1}{\sqrt{n}}\sum Y_i$ and characteristic functions of independent sums multiply,

$$\varphi_{Z_n}(\omega) = \Big[\varphi\Big(\frac{\omega}{\sqrt{n}}\Big)\Big]^n = \Big[1 - \frac{\omega^2}{2n} + o\Big(\frac{1}{n}\Big)\Big]^n \longrightarrow e^{-\omega^2/2},$$

which is the characteristic function of $\mathcal{N}(0,1)$. Lévy's continuity theorem converts convergence of characteristic functions into convergence of CDFs. □

In signal terms: the pdf of a sum is the *convolution* of the individual pdfs (Ch. 19), and repeated convolution of any reasonable pulse with itself tends to a Gaussian shape. The figure shows this for uniform pdfs, whose repeated convolutions (the Irwin–Hall densities) are piecewise polynomials.



Pdf of the centred sum of n independent Uniform$(0,1)$ variables: $n=1$ (blue), $n=2$ (red, triangle), $n=3$ (green, piecewise quadratic), against the Gaussian with the $n=3$ variance $3/12 = 0.25$ (dashed). Three uniforms are already close to Gaussian.

How large must n be? The **Berry–Esseen theorem** bounds the error: $\sup_z |P(Z_n \le z) - \Phi(z)| \le C\rho/(\sigma^3\sqrt{n})$ with $\rho = E|X-\mu|^3$ and $C < 0.48$. The convergence is fast for symmetric, light-tailed data (uniform: $n \approx 3$–5 suffices for the centre of the distribution), slow for skewed data (exponential, lognormal: n of 30–50), and fails for distributions without finite variance (Cauchy). The approximation is always worst in the far tails — the CLT is a statement about the *bulk* of the distribution.

The CLT also explains the **normal approximation to the binomial and Poisson**: a Binomial(n,p) count is a sum of n Bernoullis, so it is approximately $\mathcal{N}(np, np(1-p))$ when $np(1-p) \gtrsim 10$; a Poisson(λ) count is approximately $\mathcal{N}(\lambda,\lambda)$ for $\lambda \gtrsim 10$. A **continuity correction** (evaluating at $k \pm \tfrac{1}{2}$) improves the approximation for integer counts.

Sampling distribution of the variance

For Gaussian samples the sample variance has an exact distribution, and it is independent of the sample mean.

Sampling theorem for Gaussian samples

If $X_1, \dots, X_n$ is a random sample from $\mathcal{N}(\mu, \sigma^2)$, then

1. $\bar{X}$ and S^2 are independent;
2. $\dfrac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$;
3. $T = \dfrac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$.

Proof sketch. Choose an orthogonal matrix $\mathbf{Q}$ (Ch. 9) whose first row is $(1,\dots,1)/\sqrt{n}$, and set $\mathbf{Y} = \mathbf{Q}(\mathbf{X} - \mu\mathbf{1})/\sigma$. Because $\mathbf{Q}$ is orthogonal and the $(X_i-\mu)/\sigma$ are i.i.d. $\mathcal{N}(0,1)$, the Y_i are again i.i.d. $\mathcal{N}(0,1)$ (the covariance $\mathbf{QIQ}^{\mathsf{T}} = \mathbf{I}$). Now $Y_1 = \sqrt{n}(\bar{X}-\mu)/\sigma$, and since $\|\mathbf{Y}\|^2 = \|(\mathbf{X}-\mu\mathbf{1})/\sigma\|^2$,

$$\sum_{i=2}^{n} Y_i^2 = \frac{1}{\sigma^2}\Big[\sum_i (X_i-\mu)^2 - n(\bar{X}-\mu)^2\Big] = \frac{(n-1)S^2}{\sigma^2}.$$

So $\bar{X}$ depends only on Y_1 and S^2 only on $Y_2, \dots, Y_n$, giving independence (1), and $(n-1)S^2/\sigma^2$ is a sum of $n-1$ squared standard Gaussians, giving (2). Then $T = Y_1 / \sqrt{\chi^2_{n-1}/(n-1)}$, which is t_{n-1} by definition (3). □

The rotation argument shows geometrically where the "lost" degree of freedom goes: one dimension of the n-dimensional data space (the direction of $\mathbf{1}$) is used by the mean.

Consequences: $\operatorname{Var}(S^2) = 2\sigma^4/(n-1)$, so the relative standard uncertainty of S^2 is $\sqrt{2/(n-1)}$ and that of S is about $1/\sqrt{2(n-1)}$. With $n=10$ readings, s is uncertain by about 24%; with $n=50$, by 10%. Standard deviations estimated from a handful of readings are themselves very uncertain — which is precisely why the t distribution, rather than the Gaussian, is used for small samples.

Common Pitfall — Point 2 needs Gaussian data

The χ^2 result for S^2 is *not* protected by the CLT. For heavy-tailed data the variance of S^2 is $\sigma^4\left(\frac{2}{n-1}+\frac{\kappa}{n}\right)$ with κ the excess kurtosis, which can be much larger than the Gaussian value. Intervals and tests for variances (Sections 20.7–20.8) are therefore far less robust than those for means.

Two-sample statistics

For independent Gaussian samples of sizes n_1, n_2 with variances σ_1^2, σ_2^2:

$$\bar{X}_1 - \bar{X}_2 \sim \mathcal{N}\left(\mu_1-\mu_2,\ \frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}\right), \qquad \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F_{n_1-1,\,n_2-1}.$$

The second result follows directly from the definition of F as a ratio of independent normalised chi-squares.

Sample proportions

If $X \sim \text{Binomial}(n,p)$ counts defectives in a sample, the sample proportion $\hat{p} = X/n$ has $E[\hat{p}] = p$ and $\text{SE}(\hat{p}) = \sqrt{p(1-p)/n}$, and is approximately Gaussian for $np(1-p) \gtrsim 10$.

Worked Example 20.13 — How much averaging is enough?

An ADC channel has Gaussian reading noise with $\sigma = 5$ mV.

(a) For a 16-reading average, $\text{SE} = 5/\sqrt{16} = 1.25$ mV, and $P(|\bar{X}-\mu| > 2\text{ mV}) = 2Q(2/1.25) = 2Q(1.6) = 0.110$.

(b) To make the standard error at most 0.5 mV we need $\sigma/\sqrt{n} \le 0.5$, i.e. $n \ge (5/0.5)^2 = 100$ readings.

(c) If the readings are correlated with $\rho_k = 0.6^k$ (the noise is lowpass-filtered before sampling), the effective sample size is only $n_{\text{eff}} \approx n(0.4/1.6) = n/4$. One then needs about 400 readings — or, better, should sample more slowly so that successive readings become independent.

Worked Example 20.14 — CLT for skewed lifetimes

Component lifetimes are exponential with mean 1000 h (so $\sigma = 1000$ h). A life test runs 40 parts. Find the probability that the sample mean lifetime is below 800 h.

Step 1 — CLT. $\text{SE} = 1000/\sqrt{40} = 158.1$ h. $P(\bar{X} < 800) \approx \Phi\big((800-1000)/158.1\big) = \Phi(-1.265) = 0.103$.

Step 2 — exact. A sum of 40 independent exponentials is gamma distributed (shape 40, scale 1000), so $\bar{X}$ is gamma with shape 40 and scale 25; its CDF at 800 gives 0.0956.

Step 3 — comment. The CLT answer is about 8% high in relative terms because the exponential is strongly skewed; the true distribution of $\bar{X}$ is still slightly right-skewed at $n = 40$. For a quick engineering estimate the CLT is adequate; for an acceptance test with contractual consequences, use the exact distribution.

Worked Example 20.15 — Distribution of a noise-variance estimate

A noise generator is specified as $\sigma^2 = 4\ \text{mV}^2$. From $n = 10$ Gaussian samples, what is the probability that the estimate s^2 exceeds $6\ \text{mV}^2$, even though the generator meets specification?

Step 1 — pivot. $(n-1)S^2/\sigma^2 = 9S^2/4 \sim \chi^2_9$.

Step 2 — translate. $S^2 > 6$ is equivalent to $\chi^2_9 > 9(6)/4 = 13.5$.

Step 3 — evaluate. From chi-square tables or software, $P(\chi^2_9 > 13.5) = 0.141$.

Comment. In one test out of seven, a perfectly good generator would appear to be 50% noisier than specified. The central 95% range of χ^2_9 is $[2.70,\ 19.02]$, so s^2 ranges over $[1.20,\ 8.45]\ \text{mV}^2$ — a factor of seven. Ten samples are far too few to verify a variance.

Worked Example 20.16 — Sampling a production lot

A process has defect rate $p = 2\%$. In a sample of $n = 500$ parts, what is the probability that the observed defect rate is 3% or more (15 or more defectives)?

Step 1 — mean and SE. $E[X] = np = 10$, $\text{Var}\,X = np(1-p) = 9.8$, $\text{SD} = 3.13$.

Step 2 — normal approximation with continuity correction. $P(X \ge 15) \approx Q\big((14.5-10)/3.13\big) = Q(1.44) = 0.075$.

Step 3 — exact. The binomial tail $P(X \ge 15) = 0.081$. The approximation is serviceable; the remaining discrepancy reflects the right skew of a binomial with small p. (Without the continuity correction one would get $Q(1.60) = 0.055$, noticeably worse.)

Engineering Note — Dithering and the CLT

Adding a small random dither before quantisation and then averaging many readings lets an ADC resolve DC levels far finer than one LSB: the dither makes the quantisation errors of successive readings independent and zero-mean, so their average converges at the $1/\sqrt{n}$ rate. Without dither, a constant input always produces the same code and averaging gains nothing.

20.6 Estimation

The estimation task

A model has unknown parameters θ — the mean and standard deviation of a noise source, the failure rate of a component, the Rayleigh parameter of a fading channel, the slope of a sensor. An **estimator** $\hat\theta = g(X_1, \dots, X_n)$ is a rule that turns data into a value for θ; the number it produces for a particular data set is an **estimate**. Because $\hat\theta$ is a statistic, it has a sampling distribution, and "good" estimators are those whose sampling distribution is concentrated near the true θ.

Properties of estimators

Bias, variance, MSE, consistency, efficiency

- **Bias:** $b(\hat\theta) = E[\hat\theta] - \theta$. $\hat\theta$ is **unbiased** if $b = 0$ for all θ.
- **Mean squared error:** $\operatorname{MSE}(\hat\theta) = E\big[(\hat\theta - \theta)^2\big]$.
- **Consistency:** $\hat\theta_n \to \theta$ in probability as $n \to \infty$.
- **Efficiency:** among unbiased estimators, one with smaller variance is more efficient; the relative efficiency of $\hat\theta_1$ to $\hat\theta_2$ is $\operatorname{Var}\hat\theta_2 / \operatorname{Var}\hat\theta_1$.

Bias–variance decomposition

$$\operatorname{MSE}(\hat\theta) = \operatorname{Var}(\hat\theta) + b(\hat\theta)^2.$$

Proof. Add and subtract $E\hat\theta$: $E[(\hat\theta - E\hat\theta + E\hat\theta - \theta)^2] = E[(\hat\theta - E\hat\theta)^2] + 2(E\hat\theta - \theta)E[\hat\theta - E\hat\theta] + (E\hat\theta - \theta)^2$, and the middle expectation is zero. □

Unbiasedness is not sacred. A slightly biased estimator with much smaller variance can have lower MSE — for Gaussian data, the variance estimator with divisor $n+1$ has the smallest MSE of all estimators of the form $c\sum(X_i - \bar X)^2$, even though it is biased. A sufficient condition for consistency is that both bias and variance tend to zero; $\bar X$ is consistent for μ (by the law of large numbers, Ch. 19) and S^2 is consistent for σ^2.

The Cramér–Rao lower bound

How good can an unbiased estimator possibly be? For a model with pdf $f(x;\theta)$ that is smooth in θ, define the **score** $\partial_\theta \ln f(X;\theta)$ and the **Fisher information** per observation

$$I(\theta) = E\Big[\Big(\frac{\partial}{\partial\theta}\ln f(X;\theta)\Big)^2\Big] = -E\Big[\frac{\partial^2}{\partial\theta^2}\ln f(X;\theta)\Big].$$

Cramér–Rao lower bound (CRLB)

Under regularity conditions (support independent of θ, differentiation under the integral allowed), every unbiased estimator from an i.i.d. sample of size n satisfies

$$\operatorname{Var}(\hat\theta) \ge \frac{1}{n\,I(\theta)}.$$

An unbiased estimator attaining the bound is called **efficient**.

Proof sketch. Let $L(\theta) = \prod f(x_i;\theta)$ be the joint pdf and $U = \partial_\theta \ln L$ the total score. Differentiating $\int L\,d\mathbf{x} = 1$ gives $E[U] = 0$; differentiating $\int \hat\theta L\,d\mathbf{x} = \theta$ gives $E[\hat\theta U] = 1$, so $\operatorname{Cov}(\hat\theta, U) = 1$. The Cauchy–Schwarz inequality for covariances gives $1 \le \operatorname{Var}(\hat\theta)\operatorname{Var}(U)$, and $\operatorname{Var}(U) = nI(\theta)$ by independence. □

For a DC level θ in white Gaussian noise of variance σ^2, $\ln f = -\frac{(x-\theta)^2}{2\sigma^2} + \text{const}$, so $I = 1/\sigma^2$ and the bound is σ^2/n — exactly the variance of $\bar X$. The sample mean is efficient: no unbiased processing of the samples can do better. The CRLB is the standard benchmark in estimation theory for radar, communications and sensor arrays.

Method of moments

The oldest general recipe: equate sample moments to population moments and solve.

Method of moments (MoM)

If the model has p parameters and its first p moments are known functions $\mu'_k(\theta) = E[X^k]$, the MoM estimator solves

$$\frac{1}{n}\sum_{i=1}^{n} x_i^k = \mu'_k(\hat\theta), \qquad k = 1, \dots, p.$$

(Central moments may be used instead of raw moments.)

MoM is simple, needs only moment formulas, and gives consistent estimators (sample moments converge to population moments, and continuous functions preserve convergence). It is often a good *starting value* for an iterative maximum-likelihood search. Its weakness is that it can be inefficient and can produce impossible values (outside the parameter space or the data range).

Maximum likelihood

Likelihood and maximum likelihood estimator (MLE)

For observed data $x_1, \dots, x_n$, the **likelihood** is the joint pdf (or pmf) viewed as a function of the parameter:

$$L(\theta) = \prod_{i=1}^{n} f(x_i;\theta), \qquad \ell(\theta) = \ln L(\theta) = \sum_{i=1}^{n} \ln f(x_i;\theta).$$

The **MLE** $\hat\theta$ maximises L (equivalently ℓ). When the maximum is interior and ℓ is smooth, it solves the **likelihood equation** $\partial \ell / \partial \theta = 0$.

The MLE chooses the parameter value under which the observed data were most probable. It is the dominant estimation method in engineering because of its large-sample properties:

Asymptotic properties of the MLE

Under regularity conditions, as $n \to \infty$,

1. $\hat\theta$ is **consistent**;
2. $\hat\theta$ is **asymptotically Gaussian and efficient**: $\hat\theta \approx \mathcal{N}\big(\theta,\, 1/(n I(\theta))\big)$;
3. **invariance:** the MLE of $g(\theta)$ is $g(\hat\theta)$.

Point 2 provides standard errors almost for free: $\mathrm{SE}(\hat\theta) \approx 1/\sqrt{-\ell''(\hat\theta)}$, the inverse square root of the curvature of the log-likelihood at its peak (the **observed information**). A sharply peaked log-likelihood means the data pin θ down well. With several parameters, the inverse of the matrix of second derivatives (the negative Hessian of ℓ) approximates the covariance matrix of $\hat{\boldsymbol\theta}$. Point 3 means, for example, that the MLE of a standard deviation is the square root of the MLE of the variance, and the MLE of an MTBF is the reciprocal of the MLE of the failure rate.

MLE in Gaussian noise is least squares

If $x_i = s_i(\boldsymbol\theta) + w_i$ with w_i i.i.d. $\mathcal{N}(0,\sigma^2)$, then

$$\ell(\boldsymbol\theta) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}\big(x_i - s_i(\boldsymbol\theta)\big)^2,$$

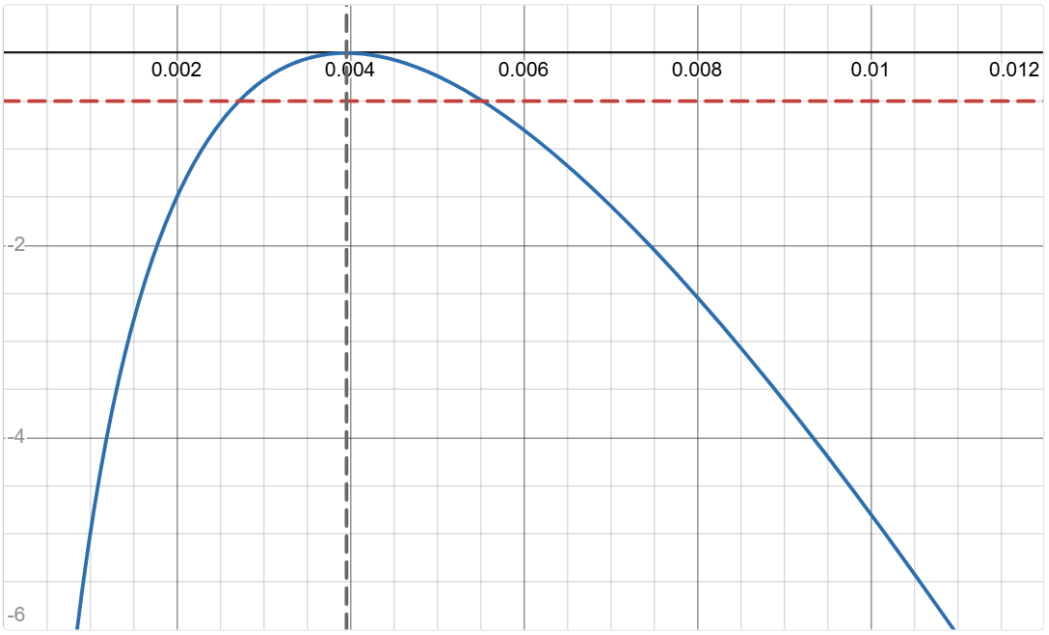
so maximising the likelihood over $\boldsymbol\theta$ is the same as minimising the sum of squared residuals.

This single observation justifies least-squares curve fitting (Section 20.9, Ch. 23) whenever the noise is Gaussian, and suggests weighted least squares (weights $1/\sigma_i^2$) when the noise variances differ.

Standard MLEs

Model	MLE	Notes
Bernoulli(p)	$\hat p = \bar x$ (proportion of successes)	unbiased
Poisson(λ)	$\hat\lambda = \bar x$	unbiased, efficient
Exponential(λ)	$\hat\lambda = 1/\bar x$	biased: $E\hat\lambda = n\lambda/(n-1)$
$\mathcal{N}(\mu,\sigma^2)$	$\hat\mu = \bar x$, $\hat\sigma^2 = \frac{1}{n}\sum(x_i-\bar x)^2$	$\hat\sigma^2$ biased by factor $(n-1)/n$
Uniform$(0,\theta)$	$\hat\theta = x_{(n)}$	biased low; support depends on θ
Rayleigh(σ)	$\hat\sigma^2 = \frac{1}{2n}\sum r_i^2$	unbiased for σ^2

Derivation for the Gaussian. $\ell(\mu,\sigma^2) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum(x_i-\mu)^2$. Setting $\partial\ell/\partial\mu = \frac{1}{\sigma^2}\sum(x_i-\mu) = 0$ gives $\hat\mu = \bar x$; setting $\partial\ell/\partial\sigma^2 = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum(x_i-\hat\mu)^2 = 0$ gives $\hat\sigma^2 = \frac{1}{n}\sum(x_i-\bar x)^2$. The MLE of the variance is biased because it does not account for the degree of freedom used by $\hat\mu$ — the MLE optimises fit, not unbiasedness. The bias vanishes as $n \to \infty$, consistent with the asymptotic theory.



Log-likelihood of the exponential failure rate in Example 20.18, shifted so that its maximum is 0. The MLE sits at the peak; the values of λ where ℓ has fallen by $\frac{1}{2}$ (red line) give an approximate 68% likelihood interval, which is visibly asymmetric.

Worked Example 20.17 — Uniform: method of moments versus maximum likelihood

Timing jitter is modelled as Uniform$(0,\theta)$ (in ns). Five observations: $0.8,\ 2.9,\ 1.7,\ 3.6,\ 2.2$. Estimate θ by both methods and compare their quality.

Step 1 — MoM. $E[X]=\theta/2$, so $\hat\theta_{\text{MoM}}=2\bar x=2(2.24)=4.48$ ns. It is unbiased, with $\operatorname{Var}=4\operatorname{Var}(X)/n=4\theta^2/(12n)=\theta^2/(3n)$.

Step 2 — MLE. $L(\theta)=\theta^{-n}$ if $\theta\ge x_{(n)}$, and 0 otherwise (some observation would be impossible). L decreases in θ, so it is maximised at the smallest admissible value: $\hat\theta_{\text{ML}}=x_{(n)}=3.6$ ns. The likelihood equation $\ell'=0$ has no solution — the maximum is at a boundary, and the regularity conditions of the CRLB fail.

Step 3 — bias correction. The maximum of n uniforms has $E[X_{(n)}]=n\theta/(n+1)$, so $\hat\theta_{\text{U}}=\frac{n+1}{n}x_{(n)}=4.32$ ns is unbiased, with variance $\theta^2/(n(n+2))$.

Step 4 — compare. For $n=5$: $\operatorname{Var}\hat\theta_{\text{MoM}}=\theta^2/15$, $\operatorname{Var}\hat\theta_{\text{U}}=\theta^2/35$. The order-statistic estimator is more than twice as efficient, and its advantage grows linearly with n (variance $\propto 1/n^2$ instead of $1/n$). Note also that MoM could produce an estimate *smaller* than the largest observation — a logically impossible value — for unlucky samples.

Worked Example 20.18 — Failure rate from a life test

Eight units run to failure at $120,\ 340,\ 75,\ 510,\ 260,\ 195,\ 430,\ 90$ h. Assume exponential lifetimes and find the MLE of λ, its standard error and the MLE of the MTBF.

Step 1 — log-likelihood. $\ell(\lambda)=\sum\left(\ln\lambda-\lambda t_i\right)=n\ln\lambda-\lambda\sum t_i=8\ln\lambda-2020\lambda$.

Step 2 — maximise. $\ell'(\lambda)=8/\lambda-2020=0$ gives $\hat\lambda=8/2020=3.96\times10^{-3}$ h^{-1}. $\ell''=-8/\lambda^2<0$ confirms a maximum.

Step 3 — standard error. Observed information $-\ell''(\hat\lambda)=n/\hat\lambda^2$, so $\text{SE}(\hat\lambda)\approx\hat\lambda/\sqrt{n}=1.40\times10^{-3}$ h^{-1} — a 35% relative uncertainty from eight failures.

Step 4 — invariance. $\widehat{\text{MTBF}}=1/\hat\lambda=\bar t=252.5$ h.

Comment. The relative uncertainty $1/\sqrt{n}$ depends only on the number of *failures*. Reliability demonstrations are expensive precisely because small failure rates produce few failures; Section 20.7 gives an exact interval using the chi-square distribution.

Worked Example 20.19 — Rayleigh parameter of a noise envelope

Envelope samples of narrowband receiver noise (mV): $1.2,\ 0.7,\ 2.1,\ 1.5,\ 0.9,\ 1.8$. Estimate σ by ML and by the method of moments.

Step 1 — log-likelihood. $\ell(\sigma^2)=\sum\ln r_i-n\ln\sigma^2-\frac{1}{2\sigma^2}\sum r_i^2$.

Step 2 — maximise. $\partial\ell/\partial\sigma^2=-\frac{n}{\sigma^2}+\frac{1}{2\sigma^4}\sum r_i^2=0\Rightarrow\hat\sigma^2=\frac{1}{2n}\sum r_i^2=\frac{12.64}{12}=1.053$ mV^2, so $\hat\sigma_{\text{ML}}=1.026$ mV (invariance).

Step 3 — method of moments. $E[R]=\sigma\sqrt{\pi/2}$, so $\hat\sigma_{\text{MoM}}=\bar r/1.2533=1.367/1.2533=1.090$ mV.

Step 4 — interpretation. The MLE uses $\sum r_i^2$, which is the estimated *noise power* ($E[R^2]=2\sigma^2$): it is the natural statistic, and $2n\hat\sigma^2/\sigma^2\sim\chi^2_{2n}$ exactly, which gives exact intervals. The two estimates differ by 6% — well within the sampling scatter for $n=6$.

Worked Example 20.20 — Cramér–Rao bound for a DC level

A DC voltage A is measured in white Gaussian noise of $\sigma = 2$ mV with $n = 25$ samples.

Step 1 — Fisher information. $\ln f(x;A) = -\frac{(x-A)^2}{2\sigma^2} + \text{const}$, $\partial_A^2 \ln f = -1/\sigma^2$, so $I(A) = 1/\sigma^2$.

Step 2 — bound. $\operatorname{Var}(\hat{A}) \geq \sigma^2/n = 4/25 = 0.16\ \text{mV}^2$, i.e. SD ≥ 0.4 mV.

Step 3 — efficiency of the mean. $\operatorname{Var}(\bar{X}) = \sigma^2/n$ equals the bound, so the sample mean is efficient. The median, with asymptotic variance $\pi\sigma^2/(2n) = 0.251\ \text{mV}^2$, reaches only 64% efficiency.

Step 4 — non-Gaussian noise. For Laplacian (double-exponential) noise with the same variance, the Fisher information doubles to $2/\sigma^2$ and the MLE becomes the *median*, which is then efficient while the mean is only 50% efficient. The best estimator depends on the noise distribution — a key lesson of estimation theory.

Common Pitfall — Maximum likelihood with few data

The attractive properties of the MLE are *asymptotic*. With small n it can be noticeably biased ($\hat{\sigma}^2$ with divisor n; $\hat{\lambda} = 1/\bar{t}$), and the Gaussian approximation to its distribution can be poor (the log-likelihood in Example 20.18 is visibly asymmetric). Quote small-sample results with exact intervals where available, or check by simulation (Ch. 24).

Engineering Note — Estimators inside equipment

A spectrum analyser's "noise marker" estimates power by averaging squared magnitudes (the Rayleigh MLE in disguise); its "video averaging" in log mode instead averages decibels, which is biased low by 2.51 dB for Gaussian noise — a correction built into good instruments. A GPS receiver computes position by iterative least squares, i.e. maximum likelihood under Gaussian range errors, and reports a dilution-of-precision that is the CRLB-like covariance of the fix.

20.7 Confidence intervals

From a point to an interval

A point estimate without an indication of its reliability is of limited use: "the noise floor is -112.3 dBm" means one thing if the estimate is good to 0.1 dB and quite another if it is good to 3 dB. A **confidence interval** reports a range of plausible parameter values together with a stated probability that the *procedure* captures the true value.

Confidence interval

An interval $[L, U]$ whose endpoints are statistics $L = g_1(X_1,\dots,X_n)$, $U = g_2(X_1,\dots,X_n)$ is a $100(1-\alpha)\%$ **confidence interval** for θ if

$$P(L \leq \theta \leq U) = 1 - \alpha \quad \text{for every value of } \theta.$$

$1-\alpha$ is the **confidence level** (coverage probability). One-sided bounds $(-\infty, U]$ or $[L, \infty)$ are defined similarly.

The probability statement concerns the random endpoints, not θ, which is fixed. Once the data are in and the interval is, say, $[2.475,\ 2.530]$ V, the true value either is or is not in it; the 95% describes the long-run success rate of the method. If many laboratories each compute a 95% interval from their own data, about 95% of the intervals will contain the true value.

The pivotal method

Most intervals in practice are built from a **pivot**: a function $Q(X_1,\dots,X_n;\theta)$ whose distribution is known and does not depend on θ. If $q_{\alpha/2}$ and $q_{1-\alpha/2}$ are quantiles of that distribution, then $P(q_{\alpha/2} \leq Q \leq q_{1-\alpha/2}) = 1-\alpha$, and solving the inequalities for θ yields the interval. The sampling theorems of Section 20.5 supply the pivots:

Parameter	Conditions	Pivot	Distribution
μ	Gaussian (or large n), σ known	$(\bar{X}-\mu)/(\sigma/\sqrt{n})$	$\mathcal{N}(0,1)$
μ	Gaussian, σ unknown	$(\bar{X}-\mu)/(S/\sqrt{n})$	t_{n-1}
σ^2	Gaussian	$(n-1)S^2/\sigma^2$	χ^2_{n-1}
σ_1^2/σ_2^2	two Gaussian samples	$(S_1^2/\sigma_1^2)/(S_2^2/\sigma_2^2)$	F_{n_1-1,n_2-1}
λ (exponential)	complete sample	$2\lambda\sum T_i$	χ^2_{2n}

Below, z_α, $t_{\alpha,\nu}$ and $\chi^2_{\alpha,\nu}$ denote **upper** α-points: $P(Z > z_\alpha) = \alpha$, and so on.

Interval for a mean

Confidence intervals for a mean

σ **known** (or n large, with s in place of σ):

$$\bar{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}.$$

σ **unknown, Gaussian data:**

$$\bar{x} \pm t_{\alpha/2,\,n-1}\frac{s}{\sqrt{n}}.$$

Derivation of the t interval. $P\big(-t_{\alpha/2,n-1} \le \frac{\bar{X}-\mu}{S/\sqrt{n}} \le t_{\alpha/2,n-1}\big) = 1-\alpha$; multiplying through by $S/\sqrt{n}$ and rearranging for μ gives $P\big(\bar{X} - t\,S/\sqrt{n} \le \mu \le \bar{X} + t\,S/\sqrt{n}\big) = 1-\alpha$. $\square$

The half-width $E = z_{\alpha/2}\sigma/\sqrt{n}$ is the **margin of error**. Inverting it gives the sample size needed for a target margin:

$$n = \left(\frac{z_{\alpha/2}\,\sigma}{E}\right)^2.$$

Because of the $1/\sqrt{n}$ law, halving the interval requires four times the data. The t interval is robust to moderate non-normality (thanks to the CLT) once n exceeds about 15–20, unless the data are strongly skewed or contain outliers.

Interval for a variance

Confidence interval for σ^2 (Gaussian data)

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2,\,n-1}} \;\le\; \sigma^2 \;\le\; \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,\,n-1}}.$$

Taking square roots gives the interval for σ.

The interval is not symmetric about s^2, because the chi-square distribution is skewed; the upper limit lies further from s^2 than the lower. Unlike the t interval, it is *not* robust to non-Gaussian data (Section 20.5).

Interval for a proportion

For x successes in n trials, $\hat{p} = x/n$. The textbook **Wald interval** $\hat{p} \pm z_{\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n}$ is simple but performs poorly when np is small: its actual coverage can fall well below the nominal level, and it can extend below zero. The **Wilson score interval** inverts the test statistic $(\hat{p}-p)/\sqrt{p(1-p)/n}$ exactly, solving a quadratic in p:

$$\frac{\hat{p} + \dfrac{z^2}{2n} \;\pm\; z\sqrt{\dfrac{\hat{p}(1-\hat{p})}{n} + \dfrac{z^2}{4n^2}}}{1 + \dfrac{z^2}{n}}, \qquad z = z_{\alpha/2}.$$

It is recommended for routine use. The exact (Clopper–Pearson) interval, computed from binomial tail probabilities, is conservative (coverage at least $1-\alpha$) and is preferred for contractual or safety-related claims.

Interval for a difference of means

For two independent samples, the **Welch interval** is

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2,\,\nu}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}, \qquad \nu = \frac{\left(s_1^2/n_1 + s_2^2/n_2\right)^2}{\dfrac{(s_1^2/n_1)^2}{n_1-1} + \dfrac{(s_2^2/n_2)^2}{n_2-1}},$$

where ν is the Welch–Satterthwaite effective degrees of freedom — the same formula reappears in the GUM (Section 20.10). If the two variances are known to be equal, the **pooled** variance $s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}$ with n_1+n_2-2 degrees of freedom may be used instead. For paired data (the same units measured twice), compute the differences d_i and use the one-sample t interval on them — pairing removes unit-to-unit variation and usually gives a much narrower interval.

Confidence, prediction and tolerance intervals

Three different questions are often confused:

Interval	Question	Gaussian form	As $n \to \infty$
Confidence for μ	where is the population mean?	$\bar{x} \pm t\,s/\sqrt{n}$	shrinks to a point
Prediction	where will the *next single* observation fall?	$\bar{x} \pm t\,s\sqrt{1+1/n}$	tends to $\mu \pm z\sigma$
Tolerance	where do $P\%$ of all units lie (with confidence γ)?	$\bar{x} \pm k\,s$ (tabulated k)	tends to $\mu \pm z\sigma$

The prediction interval follows because $X_{\text{new}} - \bar{X}$ has variance $\sigma^2(1 + 1/n)$. A datasheet "typical" range or a production limit is a tolerance-interval question, not a confidence-interval question.

Worked Example 20.21 — Interval for a reference voltage

Use the eight readings of Example 20.3: $\bar{x} = 2.5025$ V, $s = 32.40$ mV, $n = 8$.

(a) σ unknown. $\nu = 7$, $t_{0.025,7} = 2.365$. Half-width $= 2.365 \times 32.40/\sqrt{8} = 27.1$ mV. The 95% CI is $[2.4754,\ 2.5296]$ V.

(b) If $\sigma = 32.4$ mV were known from long experience with this ADC, the half-width would be $1.960 \times 32.4/\sqrt{8} = 22.5$ mV. The t interval is 21% wider: the price of estimating σ from only eight readings.

(c) Sample size. To obtain a 95% half-width of 5 mV (with $\sigma \approx 32.4$ mV): $n = (1.96 \times 32.4/5)^2 = 161.3$, so $n = 162$ readings.

(d) Prediction. A single future reading will lie in $2.5025 \pm 2.365 \times 32.40\sqrt{1 + 1/8} = 2.5025 \pm 0.0813$ V with 95% probability — three times wider than the CI for the mean.

Worked Example 20.22 — Interval for the noise standard deviation

From the same data, find a 95% CI for σ.

Step 1 — quantiles. For χ^2_7: $\chi^2_{0.975,7} = 1.690$, $\chi^2_{0.025,7} = 16.013$.

Step 2 — interval for σ^2. $(n-1)s^2 = 7 \times 1.05 \times 10^{-3} = 7.35 \times 10^{-3}$ V^2. The interval is $[7.35 \times 10^{-3}/16.013,\ 7.35 \times 10^{-3}/1.690] = [4.59 \times 10^{-4},\ 4.35 \times 10^{-3}]$ V^2.

Step 3 — interval for σ. $[21.4,\ 66.0]$ mV, around the point estimate 32.4 mV. The upper limit is double the estimate: a standard deviation from eight readings is known only to within a factor of about two.

Worked Example 20.23 — Defect rate: Wald, Wilson and exact

A sample of $n = 200$ boards contains $x = 6$ with solder defects. Find a 95% CI for the defect rate p.

Step 1 — estimate. $\hat{p} = 0.030$, $\sqrt{\hat{p}(1-\hat{p})/n} = 0.01206$.

Step 2 — Wald. $0.030 \pm 1.96(0.01206) = [0.0064,\ 0.0536]$.

Step 3 — Wilson. With $z^2/n = 0.01921$: centre $= (0.030 + 0.00960)/1.01921 = 0.0389$; half-width $= 1.96\sqrt{1.455 \times 10^{-4} + 2.40 \times 10^{-5}}/1.01921 = 0.0250$. Interval $[0.0138,\ 0.0639]$.

Step 4 — exact (Clopper–Pearson). $[0.0111,\ 0.0642]$.

Comment. With only six defects, the Wald interval is shifted too far to the left and is too short on the right; Wilson is close to the exact interval. Note the asymmetry: the data are compatible with a defect rate more than twice the point estimate.

Worked Example 20.24 — Exact interval for a failure rate, and zero-failure tests

(a) In Example 20.18, $n = 8$ failures in total time $\sum t_i = 2020$ h. The pivot $2\lambda \sum T_i \sim \chi^2_{16}$, with $\chi^2_{0.975,16} = 6.908$ and $\chi^2_{0.025,16} = 28.845$. Hence

$$\frac{6.908}{2 \times 2020} \le \lambda \le \frac{28.845}{2 \times 2020} \quad \Longrightarrow \quad 1.71 \times 10^{-3} \le \lambda \le 7.14 \times 10^{-3}\ \text{h}^{-1},$$

and the MTBF lies in $[140,\ 585]$ h with 95% confidence. The interval is strongly asymmetric about $\hat{\lambda} = 3.96 \times 10^{-3}$ — compare the Gaussian approximation $\hat{\lambda} \pm 1.96\,\text{SE} = [1.2,\ 6.7] \times 10^{-3}$.

(b) Zero failures. A supplier claims $\lambda \le 10^{-4}$ h^{-1}. How many device-hours T of testing with no failures demonstrate this at 90% confidence? For a time-terminated test with zero failures, the one-sided upper bound is $\lambda_U = \chi^2_{0.10,2}/(2T) = 4.605/(2T) = 2.303/T$. Setting $\lambda_U = 10^{-4}$ gives $T = 23\,026$ device-hours — e.g. 100 units for 230 h. (The same "rule of three" in another guise: with zero failures, the 95% upper bound is $3/T$.)

Common Pitfall — "There is a 95% probability that μ is in this interval"

In the frequentist framework used here, μ is not random, so this sentence is not strictly meaningful; the 95% is a property of the procedure. A related error is to judge two results "different" only if their 95% intervals do not overlap: two intervals can overlap substantially while the difference between the means is still significant. Compare means with an interval or test for the *difference* instead.

Engineering Note — Coverage factors in measurement reports

Calibration certificates quote an expanded uncertainty $U = ku_c$ with $k = 2$ "corresponding to a coverage probability of approximately 95%". This is the GUM version of a confidence interval: $k = 2$ replaces $z_{0.025} = 1.96$, and when the effective degrees of freedom are small, k is taken from the t distribution instead (Section 20.10).

20.8 Hypothesis testing

Decisions under uncertainty

Many engineering questions have yes/no answers: Does this batch meet the 1% tolerance? Did the firmware update change the mean current draw? Is the new supplier's capacitor noisier? Is this random number generator uniform? Is there a target in this radar cell? A **hypothesis test** is a rule for answering such questions from data, with controlled probabilities of the two possible mistakes.

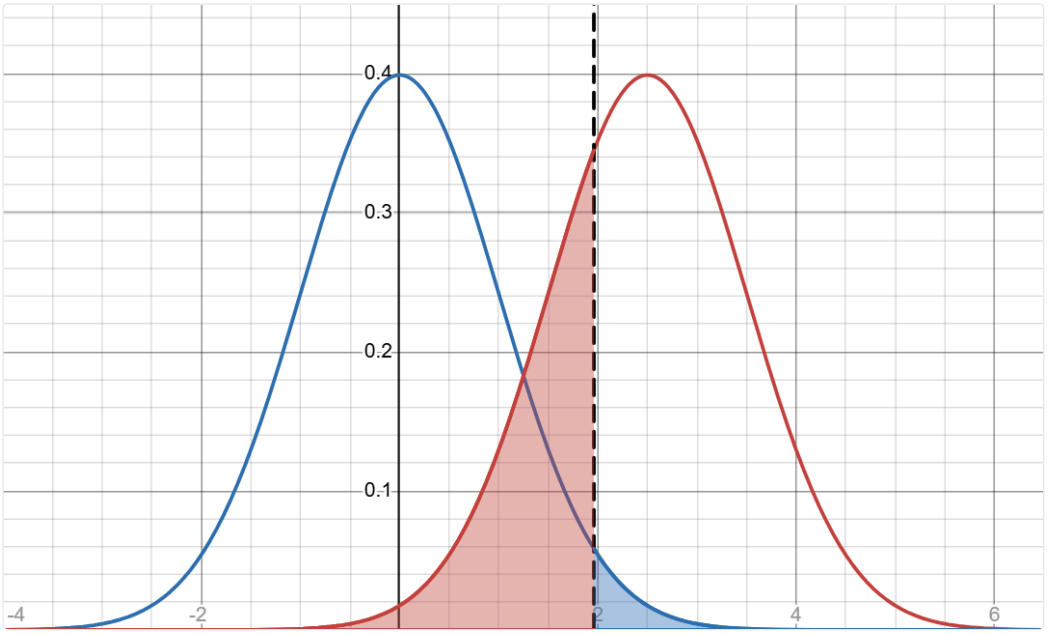
Hypotheses, test statistic, rejection region

- The **null hypothesis** H_0 is the default claim (no change, no effect, meets specification); the **alternative** H_1 is what we look for evidence of.
- A **test statistic** T is a statistic whose distribution under H_0 is known.
- The **rejection (critical) region** $\mathcal{R}$ is the set of values of T for which H_0 is rejected.
- A hypothesis such as $\mu = \mu_0$ that fixes the distribution completely is **simple**; one such as $\mu > \mu_0$ is **composite**.

	H_0 **true**	H_0 **false**
Reject H_0	**Type I error** (false alarm), probability α	correct (detection), probability $1-\beta$ = **power**
Do not reject H_0	correct	**Type II error** (miss), probability β

The **significance level** $\alpha = P(\text{reject } H_0 \mid H_0)$ is chosen in advance, conventionally 0.05 or 0.01. The **power** $1-\beta$ depends on how far the truth is from H_0, on n, and on α. For a fixed sample size, α and β trade against each other: moving the threshold to reduce false alarms increases misses. Only more data (or less noise) reduces both.

The radar vocabulary is exact, not metaphorical: α is the probability of false alarm P_{FA} and $1-\beta$ the probability of detection P_{D}; plotting P_{D} against P_{FA} as the threshold varies gives the **receiver operating characteristic** (ROC).



Distribution of the standardised statistic under H_0 (blue, mean 0) and under an alternative (red, mean 2.5), with threshold 1.96 (dashed). The blue area to the right is $\alpha = 0.025$; the red area to the left is $\beta = 0.295$. Moving the threshold right shrinks α and grows β.

The p-value

p-value

The **p-value** of an observed statistic t_{obs} is the probability, computed under H_0, of obtaining a value of T at least as extreme (in the direction of H_1) as t_{obs}:

$$p = P_{H_0}(T \geq t_{\text{obs}}) \ \text{(upper one-sided)}, \qquad p = P_{H_0}(|T| \geq |t_{\text{obs}}|) \ \text{(two-sided, symmetric } T).$$

Rejecting H_0 when $p \leq \alpha$ is equivalent to the fixed-α test.

The p-value measures how surprising the data would be *if H_0 were true*. It is not the probability that H_0 is true, and it is not the probability that the result is due to chance. A small p says "either H_0 is false or something unlikely happened"; a large p says the data are compatible with H_0 — which is not the same as evidence *for* H_0, particularly when n is small and the test has little power.

Duality of tests and confidence intervals

A two-sided level-α test of $H_0 : \theta = \theta_0$ rejects exactly when θ_0 lies outside the $100(1-\alpha)\%$ confidence interval built from the same pivot.

Proof. Both statements are the same inequality on the pivot: $|(\hat\theta - \theta_0)/\,\text{SE}| > c$ (reject) is equivalent to $\theta_0 \notin [\hat\theta - c\,\text{SE},\ \hat\theta + c\,\text{SE}]$. $\square$

The interval is usually more informative than the test result: it says not only *whether* the parameter differs from θ_0 but *by how much* it plausibly differs.

The general procedure

1. State H_0 and H_1 (one- or two-sided) *before* looking at the data.
2. Choose α, and ideally the sample size needed for adequate power at the smallest effect that matters.
3. Choose the test statistic and check its assumptions (independence, normality, equal variances).
4. Compute the statistic and the p-value (or compare with the critical value).
5. State the conclusion in the engineering context, with the size of the effect (a confidence interval), not just "significant".

Tests for means

One-sample z and t tests

For $H_0 : \mu = \mu_0$:

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim \mathcal{N}(0,1) \ \ (\sigma \text{ known}), \qquad T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t_{n-1} \ \ (\sigma \text{ unknown, Gaussian data}).$$

Reject at level α when $|Z| > z_{\alpha/2}$ (two-sided), $Z > z_\alpha$ ($H_1 : \mu > \mu_0$) or $Z < -z_\alpha$ ($H_1 : \mu < \mu_0$); likewise for T with $t_{\alpha/2,n-1}$ or $t_{\alpha,n-1}$.

Power and sample size. If the true mean is $\mu_0 + \delta$, the z statistic has mean $\delta\sqrt{n}/\sigma$, so for the one-sided test

$$1 - \beta = Q\Big(z_\alpha - \frac{\delta\sqrt{n}}{\sigma}\Big), \qquad n = \Big(\frac{(z_\alpha + z_\beta)\,\sigma}{\delta}\Big)^2,$$

and for a two-sided test z_α is replaced by $z_{\alpha/2}$ (neglecting the tiny opposite tail). The ratio δ/σ is the **effect size**; detecting an effect of a tenth of a standard deviation needs about a hundred times more data than detecting one of a full standard deviation.

Two-sample and paired tests

- **Welch t test** ($H_0 : \mu_1 = \mu_2$, independent samples, unequal variances): $T = \dfrac{\bar{X}_1 - \bar{X}_2}{\sqrt{S_1^2/n_1 + S_2^2/n_2}}$, approximately t_ν with the Welch–Satterthwaite ν of Section 20.7.
- **Pooled t test** (equal variances assumed): denominator $S_p\sqrt{1/n_1 + 1/n_2}$, exactly $t_{n_1+n_2-2}$.
- **Paired t test** (each unit measured under both conditions): one-sample t test on the differences $D_i = X_{1i} - X_{2i}$, with $n - 1$ degrees of freedom.

Chi-square tests

Test for a variance. For Gaussian data and $H_0 : \sigma^2 = \sigma_0^2$, the statistic $\chi^2 = (n-1)S^2/\sigma_0^2 \sim \chi^2_{n-1}$ under H_0. Large values indicate $\sigma^2 > \sigma_0^2$.

Goodness of fit. Data are classified into K categories with observed counts O_k; under H_0 the category probabilities are p_k, so the expected counts are $E_k = np_k$. **Pearson's statistic**

$$\chi^2 = \sum_{k=1}^{K} \frac{(O_k - E_k)^2}{E_k}$$

is approximately χ^2_{K-1-m} under H_0, where m is the number of parameters estimated from the data to compute the p_k. *Why chi-square:* each O_k is approximately Gaussian with variance close to E_k (Poisson-like counting), so each term is roughly a squared standard Gaussian; the constraint $\sum O_k = n$ removes one degree of freedom and each fitted parameter removes another. The approximation requires expected counts that are not too small — the usual rule is $E_k \geq 5$; merge sparse categories otherwise.

Independence in a contingency table. For an $r \times c$ table of counts O_{ij} with row totals R_i, column totals C_j and grand total n, the expected counts under independence are $E_{ij} = R_i C_j / n$, and $\sum_{i,j}(O_{ij} - E_{ij})^2/E_{ij}$ is approximately $\chi^2_{(r-1)(c-1)}$.

F test for equality of variances

For two independent Gaussian samples and $H_0 : \sigma_1^2 = \sigma_2^2$, $F = S_1^2/S_2^2 \sim F_{n_1-1,n_2-1}$. For a two-sided test, put the larger variance in the numerator and double the upper-tail probability. Like the chi-square variance test, it is sensitive to non-normality; Levene's test is a robust alternative.

The Neyman–Pearson lemma

Which test is *best*? For two simple hypotheses the answer is known exactly.

Neyman–Pearson lemma

For testing $H_0 : \theta = \theta_0$ against $H_1 : \theta = \theta_1$, the test that rejects H_0 when the **likelihood ratio**

$$\Lambda(\mathbf{x}) = \frac{L(\theta_1; \mathbf{x})}{L(\theta_0; \mathbf{x})} > \eta$$

(with η chosen so that the size is α) has the greatest power among all tests of size at most α.

For a Gaussian mean with known σ, $\ln \Lambda$ is an increasing linear function of $\bar{x}$ when $\mu_1 > \mu_0$, so the optimal test is "reject when $\bar{x}$ is large" — the z test. The likelihood-ratio test is the foundation of optimal detection in communications and radar: the matched filter followed by a threshold is the Neyman–Pearson detector for a known signal in white Gaussian noise.

Worked Example 20.25 — z test and power for a regulator

A 5.000 V regulator line has known output noise between units of $\sigma = 20$ mV. After a component change, $n = 25$ units give $\bar{x} = 5.009$ V. Has the mean shifted? Use $\alpha = 0.05$.

Step 1 — hypotheses. $H_0 : \mu = 5.000$, $H_1 : \mu \neq 5.000$ (two-sided: a shift either way matters).

Step 2 — statistic. $\text{SE} = 20/\sqrt{25} = 4$ mV; $z = 9/4 = 2.25$.

Step 3 — decision. $|z| > 1.96$; $p = 2Q(2.25) = 0.0244$. Reject H_0: the shift is statistically significant. The 95% CI for the shift is 9 ± 7.8 mV, i.e. $[1.2,\ 16.8]$ mV.

Step 4 — power. What is the probability that this test detects a true shift of $\delta = 10$ mV? The statistic then has mean $10/4 = 2.5$: power $= Q(1.96 - 2.5) + \Phi(-1.96 - 2.5) = 0.705$. A 10 mV shift would be missed 30% of the time.

Step 5 — sample size for 90% power. $n = \left((1.960 + 1.282) \times 20/10\right)^2 = 42.0$, so 43 units.

Worked Example 20.26 — One-sample t test and the danger of low power

For the eight reference readings of Example 20.3, test $H_0 : \mu = 2.500$ V.

$t = (2.5025 - 2.5000)/(0.03240/\sqrt{8}) = 0.0025/0.01146 = 0.218$ with 7 degrees of freedom; $p = 0.83$. There is no evidence of an offset. But the 95% CI is $[2.475,\ 2.530]$ V: offsets as large as ± 27 mV are entirely compatible with the data. "Not significant" here means "not enough data to tell", not "no offset".

Worked Example 20.27 — Paired versus unpaired comparison of drift

Six precision resistors are measured (kΩ) before and after a 1000 h burn-in:

Unit	1	2	3	4	5	6
Before	10.02	9.98	10.05	10.01	9.97	10.03
After	10.06	10.01	10.08	10.06	9.99	10.07

Step 1 — paired analysis. Differences $d_i = 0.04,\ 0.03,\ 0.03,\ 0.05,\ 0.02,\ 0.04$; $\bar{d} = 0.0350$, $s_d = 0.01049$. $t = 0.0350/(0.01049/\sqrt{6}) = 8.17$ with 5 degrees of freedom, $p = 0.00045$. The drift is unmistakable.

Step 2 — (wrong) unpaired analysis. Treating the columns as independent samples gives $t = 1.82$, $p = 0.10$ — "not significant".

Step 3 — explanation. Unit-to-unit variation (about 30 Ω) is much larger than the drift (35 Ω on average, but with spread only 10 Ω). Pairing subtracts each unit's own value and removes that variation. Always use the paired test when the same units are measured twice; the unpaired test is not only less powerful but answers a different question.

Worked Example 20.28 — Comparing two suppliers: Welch t and F tests

Capacitance (nF) of samples from two suppliers:

A: $47.2,\ 46.8,\ 47.5,\ 47.1,\ 46.9,\ 47.4,\ 47.0,\ 47.3$ ($n_1 = 8$, $\bar{x}_1 = 47.150$, $s_1 = 0.245$)

B: $46.1,\ 47.9,\ 46.5,\ 48.2,\ 45.8,\ 47.6,\ 46.9$ ($n_2 = 7$, $\bar{x}_2 = 47.000$, $s_2 = 0.924$)

Step 1 — means (Welch). $\text{SE} = \sqrt{0.245^2/8 + 0.924^2/7} = 0.360$; $t = 0.150/0.360 = 0.417$. Welch–Satterthwaite: $\nu = 6.74$. $p = 0.69$. No evidence that the means differ.

Step 2 — variances (F). $F = s_2^2/s_1^2 = 0.8533/0.0600 = 14.2$ with $(6, 7)$ degrees of freedom. The upper 2.5% point of $F_{6,7}$ is 5.12, so $H_0 : \sigma_1 = \sigma_2$ is rejected; the two-sided $p = 0.0026$.

Step 3 — conclusion. The suppliers agree on average, but supplier B's parts scatter almost four times as widely — exactly the difference that matters for a filter with tight tolerance. Note that the pooled t test would have been inappropriate here, since its equal-variance assumption is clearly violated; Welch's test does not need it.

Worked Example 20.29 — Chi-square tests for a bit generator and for shift dependence

(a) Goodness of fit. A hardware random-bit generator produces 400 two-bit symbols with counts 00: 112, 01: 91, 10: 85, 11: 112. Under H_0 (uniform) each expected count is 100:

$$\chi^2 = \frac{12^2 + 9^2 + 15^2 + 12^2}{100} = \frac{594}{100} = 5.94, \qquad \nu = 4 - 1 = 3.$$

The 5% critical value of χ^2_3 is 7.81; $p = 0.115$. No evidence of bias at this sample size (but a real generator would be tested with millions of symbols and a full test suite).

(b) Independence. Defective boards by shift: day 18 of 400, night 34 of 400. Under independence the expected defectives are 26 per shift and the expected good boards 374. Then

$$\chi^2 = \frac{8^2}{26} + \frac{8^2}{374} + \frac{8^2}{26} + \frac{8^2}{374} = 5.27, \qquad \nu = (2-1)(2-1) = 1,$$

exceeding 3.84; $p = 0.022$. The defect rate depends on shift (whether the cause is the shift itself or something correlated with it is a separate, engineering question).

Worked Example 20.30 — A borderline variance test

A sensor's noise is specified as $\sigma \le 2.0$ mV. Fifteen readings give $s = 2.6$ mV. Test $H_0 : \sigma = 2.0$ against $H_1 : \sigma > 2.0$ at $\alpha = 0.05$.

$\chi^2 = 14 \times 2.6^2 / 2.0^2 = 23.66$; the upper 5% point of χ^2_{14} is 23.68; $p = 0.0503$. Formally, H_0 is not rejected. Practically, a result this close to the threshold says the evidence is marginal; the sensible engineering response is to collect more readings, not to treat $p = 0.0503$ and $p = 0.0497$ as opposite conclusions.

Common Pitfall — Multiple testing and data dredging

Testing 20 independent parameters of a product at $\alpha = 0.05$ when nothing has changed produces at least one "significant" result with probability $1 - 0.95^{20} = 0.64$. When many tests are made, control the family-wise error, e.g. by the **Bonferroni** rule (use α/m for each of m tests), or the false-discovery rate. Choosing which hypothesis to test *after* seeing which looks interesting invalidates the stated α entirely.

Common Pitfall — Statistical versus practical significance

With 10^6 samples, a 0.01 mV offset in a 5 V supply can be "highly significant" ($p < 10^{-6}$) and completely irrelevant. With 5 samples, a 50 mV offset may be "not significant" and very relevant. Always report the estimated effect and its confidence interval alongside the p-value.

Engineering Note — Production test limits as hypothesis tests

A pass/fail limit on a test station is a hypothesis test performed on every unit. Measurement noise causes good units near the limit to fail (false rejects, Type I from the producer's view) and bad units to pass (escapes, Type II). **Guard-banding** moves the test limit inside the specification by a multiple of the measurement uncertainty to cut escapes at the cost of more false rejects — the α–β trade-off in hardware.

20.9 Regression

Fitting relations between variables

Calibration curves, sensor characterisation, extracting a time constant from a step response, finding a device's temperature coefficient, modelling battery capacity against cycle count: all of these fit a functional relation $y \approx g(x; \boldsymbol{\beta})$ to noisy data. **Regression analysis** estimates the parameters $\boldsymbol{\beta}$, quantifies their uncertainty, and — just as importantly — checks whether the assumed form of g is adequate. This section treats the statistical side; the optimisation side (nonlinear least squares, regularisation) is in Ch. 23.

The simple linear model

Simple linear regression model

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad i = 1, \dots, n,$$

where the x_i are known (not random, or measured without appreciable error) and the errors ε_i are independent with $E[\varepsilon_i] = 0$ and constant variance $\mathrm{Var}(\varepsilon_i) = \sigma^2$. For exact intervals and tests we further assume $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$.

Least-squares estimates

The **least-squares** estimates minimise the sum of squared vertical deviations

$$S(\beta_0, \beta_1) = \sum_{i=1}^{n} \big(y_i - \beta_0 - \beta_1 x_i\big)^2.$$

Setting both partial derivatives to zero (Ch. 6) gives the **normal equations**

$$\frac{\partial S}{\partial \beta_0} = -2\sum (y_i - \beta_0 - \beta_1 x_i) = 0, \qquad \frac{\partial S}{\partial \beta_1} = -2\sum x_i (y_i - \beta_0 - \beta_1 x_i) = 0.$$

Least-squares line

$$\hat\beta_1 = \frac{S_{xy}}{S_{xx}} = r\,\frac{s_y}{s_x}, \qquad \hat\beta_0 = \bar y - \hat\beta_1 \bar x.$$

The fitted line passes through $(\bar x, \bar y)$, and the residuals $e_i = y_i - \hat y_i$ satisfy $\sum e_i = 0$ and $\sum x_i e_i = 0$.

Derivation. The first normal equation gives $\bar y = \beta_0 + \beta_1 \bar x$, i.e. the line passes through the centroid. Substituting $\beta_0 = \bar y - \beta_1 \bar x$ into the second and using $\sum x_i (y_i - \bar y) = S_{xy}$, $\sum x_i (x_i - \bar x) = S_{xx}$ gives $S_{xy} - \beta_1 S_{xx} = 0$. The residual identities are the normal equations themselves. □

Under Gaussian errors, least squares is maximum likelihood (Section 20.6). Without the Gaussian assumption, the **Gauss–Markov theorem** guarantees that least squares gives the *best linear unbiased estimators* (minimum variance among all unbiased estimators that are linear in the y_i), provided the errors are uncorrelated with equal variance.

Sampling properties and inference

Since $\hat\beta_1 = \sum (x_i - \bar x) Y_i / S_{xx}$ is a linear combination of the independent Y_i,

$$E[\hat\beta_1] = \beta_1, \qquad \operatorname{Var}(\hat\beta_1) = \frac{\sigma^2}{S_{xx}}, \qquad \operatorname{Var}(\hat\beta_0) = \sigma^2\Big(\frac{1}{n} + \frac{\bar x^2}{S_{xx}}\Big), \qquad \operatorname{Cov}(\hat\beta_0, \hat\beta_1) = -\frac{\sigma^2 \bar x}{S_{xx}}.$$

The error variance is estimated from the residuals with $n - 2$ degrees of freedom (two parameters fitted):

$$s^2 = \frac{\mathrm{SSE}}{n-2} = \frac{1}{n-2}\sum_{i=1}^{n} e_i^2.$$

Inference for the regression line (Gaussian errors)

1. $\dfrac{\hat\beta_1 - \beta_1}{s/\sqrt{S_{xx}}} \sim t_{n-2}$; a $100(1-\alpha)\%$ CI for the slope is $\hat\beta_1 \pm t_{\alpha/2,n-2}\, s/\sqrt{S_{xx}}$.
2. CI for the **mean response** at x_0: $\hat y_0 \pm t_{\alpha/2,n-2}\, s\sqrt{\dfrac{1}{n} + \dfrac{(x_0 - \bar x)^2}{S_{xx}}}$.
3. **Prediction interval** for a new observation at x_0: $\hat y_0 \pm t_{\alpha/2,n-2}\, s\sqrt{1 + \dfrac{1}{n} + \dfrac{(x_0 - \bar x)^2}{S_{xx}}}$.

The term $(x_0 - \bar x)^2 / S_{xx}$ makes both bands narrowest at $\bar x$ and flare outwards: the line is pinned down best at the centre of the data, and uncertainty in the slope is amplified with distance. This is the quantitative argument against extrapolation. The formula $\operatorname{Var}(\hat\beta_1) = \sigma^2 / S_{xx}$ also guides experiment design: to measure a slope precisely, spread the x values as widely as the model remains valid — placing half the points at each end maximises S_{xx}, although it leaves no way to detect curvature.

Analysis of variance and r^2

Each observation's deviation from the mean splits into an explained part and a residual: $y_i - \bar y = (\hat y_i - \bar y) + (y_i - \hat y_i)$. Squaring and summing, the cross term vanishes (by the residual identities), giving the **ANOVA decomposition**

$$\underbrace{\sum (y_i - \bar y)^2}_{\mathrm{SST}} = \underbrace{\sum (\hat y_i - \bar y)^2}_{\mathrm{SSR}} + \underbrace{\sum (y_i - \hat y_i)^2}_{\mathrm{SSE}}.$$

Coefficient of determination

$$r^2 = \frac{\mathrm{SSR}}{\mathrm{SST}} = 1 - \frac{\mathrm{SSE}}{\mathrm{SST}},$$

the fraction of the variation in y explained by the fitted line. In simple linear regression it equals the square of the Pearson correlation r (Section 20.3). With p parameters, the **adjusted** $\bar r^2 = 1 - \dfrac{\mathrm{SSE}/(n-p)}{\mathrm{SST}/(n-1)}$ penalises extra parameters.

The overall significance of the regression is tested by $F = \dfrac{\mathrm{SSR}/1}{\mathrm{SSE}/(n-2)} \sim F_{1,n-2}$ under $H_0 : \beta_1 = 0$; in simple regression $F = t^2$ for the slope.

Residual analysis

The model assumptions — correct functional form, independent errors, constant variance, Gaussian errors — are checked by examining the residuals e_i, ideally the **standardised residuals** e_i / s. A good fit leaves residuals that look like structureless noise. Characteristic patterns diagnose specific failures:

Residual pattern	Plotted against	Diagnosis	Remedy
U-shape or arch	x or $\hat y$	missing curvature	add x^2 term, transform, physical model
Funnel (spread grows with x)	x or $\hat y$	non-constant variance	weighted least squares, log transform
Runs, waves	time/order	correlated errors, drift	model the drift, randomise run order
One very large $\lvert e_i/s\rvert$ (> 3)	—	outlier	check the point; robust fit
Curved normal-probability plot	normal quantiles	non-Gaussian errors	robust methods; intervals approximate

A point can also be **influential** without having a large residual: a point at extreme x (high **leverage** $h_{ii} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}}$) pulls the line towards itself. Cook's distance combines residual and leverage to measure how much the fit changes when a point is deleted.

Matrix form and multiple regression

With p parameters, write the model as $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, where the $n \times p$ **design matrix** $\mathbf{X}$ has a column of ones and one column per regressor (for a polynomial fit: $1,\ x_i,\ x_i^2, \dots$). Minimising $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2$ gives the normal equations and their solution (Ch. 8, Ch. 23):

$$\mathbf{X}^{\mathsf{T}}\mathbf{X}\,\hat{\boldsymbol{\beta}} = \mathbf{X}^{\mathsf{T}}\mathbf{y}, \qquad \hat{\boldsymbol{\beta}} = (\mathbf{X}^{\mathsf{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathsf{T}}\mathbf{y}, \qquad \operatorname{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}^{\mathsf{T}}\mathbf{X})^{-1}, \qquad s^2 = \frac{\mathrm{SSE}}{n-p}.$$

Geometrically, $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$ is the orthogonal projection of $\mathbf{y}$ onto the column space of $\mathbf{X}$, and the residual vector is orthogonal to every column — the matrix form of $\sum e_i = 0$, $\sum x_i e_i = 0$. The diagonal of the "hat" matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}^{\mathsf{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathsf{T}}$ gives the leverages. Numerically, solve by QR factorisation rather than by forming $\mathbf{X}^{\mathsf{T}}\mathbf{X}$, whose condition number is the square of that of $\mathbf{X}$ (Ch. 22).

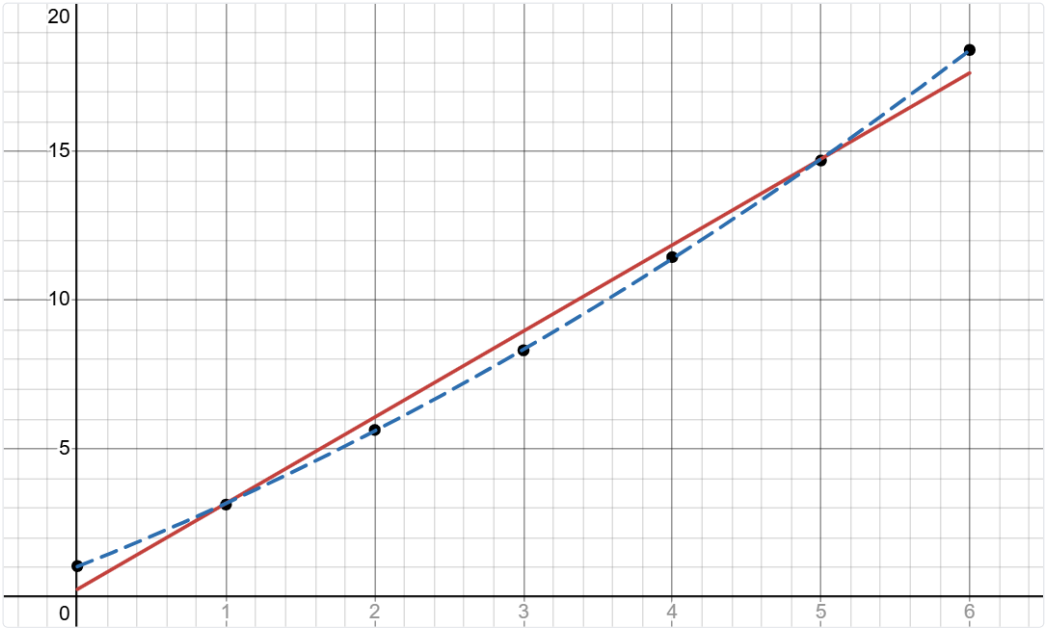
If the errors have different variances σ_i^2, **weighted least squares** minimises $\sum w_i(y_i - \mathbf{x}_i^{\mathsf{T}}\boldsymbol{\beta})^2$ with $w_i = 1/\sigma_i^2$, giving $\hat{\boldsymbol{\beta}} = (\mathbf{X}^{\mathsf{T}}\mathbf{W}\mathbf{X})^{-1}\mathbf{X}^{\mathsf{T}}\mathbf{W}\mathbf{y}$ — the regression version of the weighted mean of Section 20.2.

Linearising transformations

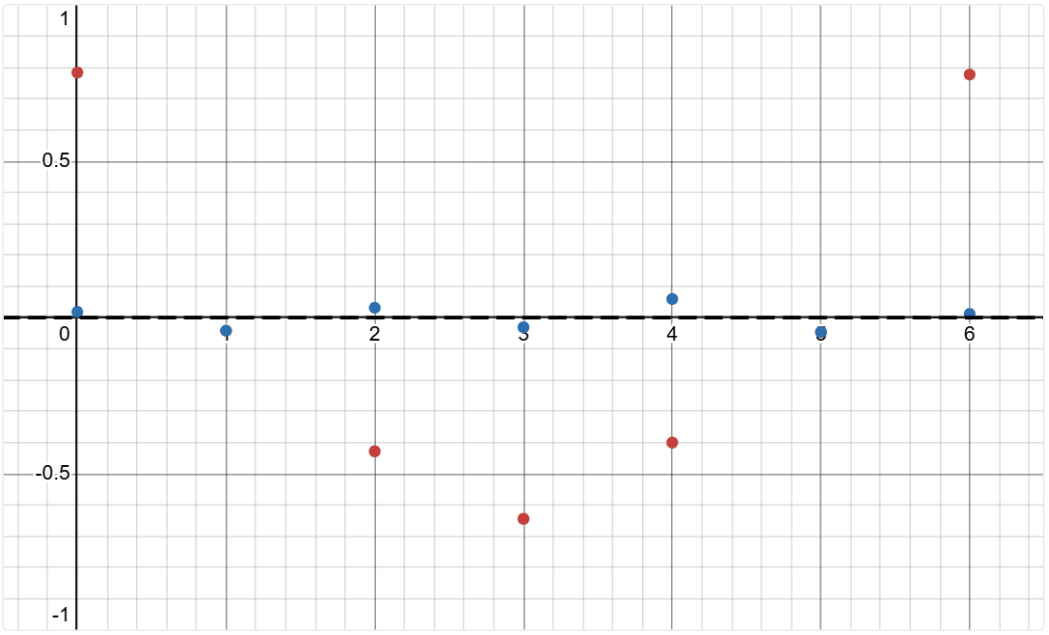
Many physical laws become linear after a transformation, and ordinary linear regression can then be used:

Model	Transformation	Linear form
$y = Ae^{-t/\tau}$ (RC discharge)	$\ln y$	$\ln y = \ln A - t/\tau$
$y = Ax^b$ (power law)	$\ln y,\ \ln x$	$\ln y = \ln A + b \ln x$
$I = I_s\left(e^{V/(nV_T)} - 1\right)$, $I \gg I_s$ (diode)	$\ln I$	$\ln I = \ln I_s + V/(nV_T)$
$R = R_0 e^{B(1/T - 1/T_0)}$ (NTC thermistor)	$\ln R,\ 1/T$	$\ln R = \text{const} + B/T$

A transformation also transforms the errors. If y has constant *absolute* noise σ, then $\ln y$ has noise $\approx \sigma/y$, which grows as y decays; the unweighted log fit then over-weights the small, noisy late values. Use weights $w_i \propto y_i^2$ in the log domain, or fit the nonlinear model directly (Ch. 23).



Data of Example 20.33 with the straight-line fit (red, $r^2 = 0.992$) and the quadratic fit (blue, dashed). Both look acceptable on this scale — the residual plot below shows the difference.



Residuals of the straight-line fit (red) form a clear U: the model is missing curvature. Residuals of the quadratic fit (blue) are small and patternless.

Worked Example 20.31 — Calibrating an RTD

Fit $R = \beta_0 + \beta_1 T$ to the data of Example 20.6 and give the sensitivity with a 95% CI.

Step 1 — slope and intercept. From Example 20.6, $S_{TT} = 7000$, $S_{TR} = 2692$, $\bar{T} = 50$, $\bar{R} = 119.333$. Hence $\hat{\beta}_1 = 2692/7000 = 0.384571\ \Omega/^\circ\text{C}$ and $\hat{\beta}_0 = 119.333 - 0.384571(50) = 100.105\ \Omega$.

Step 2 — residuals. $e_i = R_i - \hat{R}_i = -0.105,\ +0.104,\ +0.012,\ +0.021,\ +0.030,\ -0.062\ \Omega$ (they sum to zero). $\text{SSE} = 0.02705\ \Omega^2$.

Step 3 — error variance. $s^2 = 0.02705/4 = 6.76 \times 10^{-3}\ \Omega^2$, $s = 0.0822\ \Omega$.

Step 4 — standard errors. $\text{SE}(\hat{\beta}_1) = 0.0822/\sqrt{7000} = 9.83 \times 10^{-4}\ \Omega/^\circ\text{C}$; $\text{SE}(\hat{\beta}_0) = 0.0822\sqrt{1/6 + 2500/7000} = 0.0595\ \Omega$.

Step 5 — CI for the slope. $t_{0.025,4} = 2.776$: $\beta_1 \in 0.38457 \pm 0.00273 = [0.3818,\ 0.3873]\ \Omega/^\circ\text{C}$. Relative to $R_0 \approx 100\ \Omega$, this is a temperature coefficient of about $0.385\%/^\circ\text{C}$, consistent with the IEC value $0.00385\ /^\circ\text{C}$ for industrial platinum.

Step 6 — r^2 and the residual pattern. $r^2 = 1 - 0.02705/1035.29 = 0.99997$. The residuals are negative at both ends and positive in the middle — a faint arch consistent with the known negative quadratic term of platinum. A quadratic fit gives a curvature coefficient $-4.9 \times 10^{-5}\ \Omega/(^\circ\text{C})^2$, close to the Callendar–Van Dusen value -5.8×10^{-5} for a 100 Ω sensor. With six points and residuals near the noise level, the evidence for curvature is weak in isolation, but physics supports the quadratic model.

Worked Example 20.32 — Using the calibration: inverse prediction and bands

Using the fit of Example 20.31:

(a) Mean response at $T = 50\ ^\circ\text{C}$. $\hat{R} = 119.333\ \Omega$; half-width $2.776 \times 0.0822\sqrt{1/6} = 0.093\ \Omega$.

(b) Prediction for a single new reading at $50\ ^\circ\text{C}$. Half-width $2.776 \times 0.0822\sqrt{1 + 1/6} = 0.247\ \Omega$.

(c) Inverse prediction. The sensor reads $R = 120.00\ \Omega$. Solving the fitted line for T: $\hat{T} = (120.00 - 100.105)/0.384571 = 51.73\ ^\circ\text{C}$. Its uncertainty is roughly the prediction half-width converted to temperature: $0.247/0.3846 \approx 0.64\ ^\circ\text{C}$ (95%). This "calibrate y on x, then invert" procedure is the standard calibration method when the reference temperatures are known much better than the resistance readings.

(d) Extrapolation. At $T = 300\ ^\circ\text{C}$ the linear fit predicts $215.48\ \Omega$; the true Pt100 value is about $212.05\ \Omega$. The curvature invisible over 0–100 °C causes an error of $3.4\ \Omega$ ($9\ ^\circ\text{C}$) — far outside any band computed from the fit, because those bands assume the model is correct.

Worked Example 20.33 — Detecting curvature from the residuals

A sensor gives $y = 1.05,\ 3.12,\ 5.63,\ 8.31,\ 11.45,\ 14.70,\ 18.42$ at $x = 0, 1, \ldots, 6$.

Step 1 — straight line. $\bar{x} = 3$, $S_{xx} = 28$, $S_{xy} = 81.09$: $\hat{\beta}_1 = 2.8961$, $\hat{\beta}_0 = 0.2661$. SSE $= 1.982$, SST $= 236.82$, $r^2 = 0.9916$.

Step 2 — residuals. $0.78,\ -0.04,\ -0.43,\ -0.64,\ -0.40,\ -0.05,\ 0.78$: positive at the ends, negative in the middle. Despite $r^2 > 0.99$ the line is systematically wrong.

Step 3 — quadratic. With design matrix columns $(1,\ x,\ x^2)$, solving $\mathbf{X}^{\mathsf{T}}\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^{\mathsf{T}}\mathbf{y}$ gives $\hat{y} = 1.032 + 1.977x + 0.1532x^2$. Residuals are now all below 0.06 in magnitude with no pattern, and SSE $= 0.00987$ — two hundred times smaller.

Step 4 — is the improvement real? The partial F statistic for adding one parameter is $F = \frac{(1.982 - 0.00987)/1}{0.00987/4} = 799$ on $(1, 4)$ degrees of freedom, overwhelmingly significant. Adjusted $\bar{r}^2$ rises from 0.9900 to 0.99994.

Worked Example 20.34 — Time constant from an RC discharge

A capacitor discharges through a resistor; the voltage is sampled at $t = 0, 1, \ldots, 5$ ms: $5.02,\ 3.01,\ 1.86,\ 1.12,\ 0.67,\ 0.41$ V. Estimate τ.

Step 1 — linearise. $y_i = \ln v_i = 1.6134,\ 1.1019,\ 0.6206,\ 0.1133,\ -0.4005,\ -0.8916$.

Step 2 — fit. $\bar{t} = 2.5$, $S_{tt} = 17.5$, $S_{ty} = -8.7698$: slope $\hat{\beta}_1 = -0.50113$ ms^{-1}, intercept $\hat{\beta}_0 = 1.6124$.

Step 3 — parameters. $\hat{\tau} = -1/\hat{\beta}_1 = 1.9955$ ms, $\hat{V}_0 = e^{1.6124} = 5.015$ V.

Step 4 — uncertainty. $s = 0.00849$ (in ln units), $\mathrm{SE}(\hat{\beta}_1) = 0.00849/\sqrt{17.5} = 0.00203$. By first-order propagation (Section 20.10), $\mathrm{SE}(\hat{\tau}) = \mathrm{SE}(\hat{\beta}_1)/\hat{\beta}_1^2 = 0.0081$ ms. So $\tau = 1.996 \pm 0.022$ ms (95%, $t_{0.025,4} = 2.776$).

Step 5 — check. The residuals in the log domain $(+0.001,\ -0.009,\ +0.011,\ +0.004,\ -0.008,\ +0.002)$ show no trend, so a single exponential is adequate. If the noise were constant in volts, the later points (small v) would show larger log-residuals and a weighted fit would be preferable.

Common Pitfall — r^2 is not a test of model adequacy

A high r^2 says the line explains most of the variation *in this data set*; it says nothing about whether the functional form is right (Examples 20.31 and 20.33), whether it can be extrapolated, or whether the errors are independent. r^2 also depends on the range of x: the same physical relation sampled over a wider range gives a higher r^2. Judge fits by residual plots and by the size of s relative to the required accuracy.

Common Pitfall — Errors in x and regression dilution

Least squares assumes x is known exactly. If x is measured with noise of variance σ_u^2, the fitted slope is biased towards zero by the factor $S_{xx}/(S_{xx} + (n-1)\sigma_u^2)$ (**regression dilution**). When both variables are noisy and comparably so, use errors-in-variables methods (orthogonal/Deming regression) or make sure the reference variable is far more accurate than the response.

Engineering Note — Linearity specifications

An ADC's or sensor's "integral non-linearity" is the maximum residual from a fitted straight line, expressed in LSB or % of full scale. Datasheets specify whether the line is the *end-point* line (through the first and last codes) or the *best-fit* (least-squares) line; the best-fit figure is always smaller, sometimes by a factor of two, so the two are not comparable.

20.10 Experimental uncertainty

Why a common framework

A measurement result is incomplete without a statement of its uncertainty: without one, results cannot be compared with each other, with specifications, or with theory. Before 1993, laboratories used incompatible conventions ("random" and "systematic" errors combined linearly by some, in quadrature by others, at 1σ by some and 3σ by others). The *Guide to the Expression of Uncertainty in Measurement* (**GUM**, JCGM 100), issued jointly by BIPM, IEC, ISO and other bodies, established one method that is now used by every accredited calibration laboratory. It is the statistics of this chapter applied systematically to measurement.

Vocabulary

Measurand, error and uncertainty

- The **measurand** Y is the quantity intended to be measured (the resistance of *this* resistor at 23 °C).
- The **measurement model** $Y = f(X_1, \ldots, X_N)$ expresses the measurand in terms of **input quantities** X_i (readings, calibration corrections, environmental quantities).
- The **error** is the difference between the measured value and the true value; it is unknowable.
- The **standard uncertainty** $u(x_i)$ of an input estimate x_i is the standard deviation of the probability distribution that describes our knowledge of X_i.
- The **combined standard uncertainty** $u_c(y)$ is the standard uncertainty of the output estimate $y = f(x_1, \ldots, x_N)$.
- The **expanded uncertainty** $U = k\,u_c(y)$ defines an interval $y \pm U$ expected to contain a large fraction (the **coverage probability**, typically 95%) of the values that could reasonably be attributed to Y; k is the **coverage factor**.

Two further terms are often confused. **Accuracy** (closeness to the true value) is a qualitative concept; **precision** refers to the scatter of repeated readings (repeatability under the same conditions, reproducibility under changed conditions). Uncertainty is the quantitative statement that covers both. Known systematic effects are *corrected* — the GUM assumes the result has been corrected for all recognised significant systematic effects — and the uncertainty of each correction becomes an input to the budget.

Type A and Type B evaluation

The GUM classifies uncertainty components not by their physical nature ("random" vs "systematic") but by the **method used to evaluate them**.

Type A evaluation

Evaluation by statistical analysis of a series of observations. For n independent repeated readings $q_1, \dots, q_n$ of an input quantity, the estimate is $\bar{q}$ and

$$u(\bar{q}) = \frac{s(q)}{\sqrt{n}}, \qquad \nu = n - 1 \text{ degrees of freedom.}$$

This is the standard error of the mean of Section 20.5. If a well-characterised *pooled* standard deviation s_p from earlier extensive experiments is available, it can replace $s(q)$ (with its larger degrees of freedom), which is valuable when only a few readings are taken in the measurement itself.

Type B evaluation

Evaluation by any other means: manufacturer's specifications, calibration certificates, data from handbooks, resolution of a display, experience with similar instruments. The available information is converted to a standard deviation by assuming a probability distribution.

Information available	Assumed distribution	Standard uncertainty
Certificate: "U at $k = 2$" (or "95%")	Gaussian	$u = U/k$ (or $U/1.96$)
Limits $\pm a$, no other knowledge (specification, tolerance)	rectangular (uniform)	$u = a/\sqrt{3} \approx 0.577a$
Limits $\pm a$, values near centre more likely	triangular	$u = a/\sqrt{6} \approx 0.408a$
Quantity varies sinusoidally between $\pm a$ (e.g. RF mismatch, temperature cycling)	U-shaped (arcsine)	$u = a/\sqrt{2} \approx 0.707a$
Digital resolution δ (one count)	rectangular, half-width $\delta/2$	$u = \delta/\sqrt{12} \approx 0.289\delta$

The factors come from the variances of the distributions: uniform on $[-a, a]$ has variance $a^2/3$ (Section 20.4); the symmetric triangular distribution has variance $a^2/6$; the arcsine distribution of $a \sin\theta$ with θ uniform has variance $E[a^2 \sin^2\theta] = a^2/2$.

Type A and Type B uncertainties are both standard deviations and are combined in the same way. A Type B component with well-founded limits may be assigned effectively infinite degrees of freedom. When the limits themselves are uncertain, the GUM gives the approximation $\nu \approx \tfrac{1}{2}\big(\Delta u/u\big)^{-2}$, where $\Delta u/u$ is the relative uncertainty of the uncertainty (e.g. limits believed reliable to 25% give $\nu = 8$).

The law of propagation of uncertainty

Linearise the model about the input estimates (Ch. 4, Ch. 6):

$$Y \approx f(x_1, \dots, x_N) + \sum_{i=1}^{N} c_i\,(X_i - x_i), \qquad c_i = \left.\frac{\partial f}{\partial X_i}\right|_{\mathbf{x}}.$$

The c_i are **sensitivity coefficients**; they convert the uncertainty of each input into units of the output. Taking the variance of the linear approximation, with $\mathbf{c} = (c_1, \dots, c_N)^{\mathsf{T}}$ and the input covariance matrix $\boldsymbol{\Sigma}_x$, gives Section 20.3's rule $\operatorname{Var}(\mathbf{c}^{\mathsf{T}}\mathbf{X}) = \mathbf{c}^{\mathsf{T}}\boldsymbol{\Sigma}_x\mathbf{c}$:

Law of propagation of uncertainty (GUM)

$$u_c^2(y) = \sum_{i=1}^{N} c_i^2 u^2(x_i) + 2\sum_{i=1}^{N-1}\sum_{j=i+1}^{N} c_i c_j\, u(x_i, x_j),$$

where $u(x_i, x_j) = r(x_i, x_j)\, u(x_i)\, u(x_j)$ is the estimated covariance. For **uncorrelated** inputs,

$$u_c(y) = \sqrt{\sum_{i=1}^{N} \big(c_i\, u(x_i)\big)^2},$$

the root-sum-of-squares (RSS) of the individual contributions $u_i(y) = |c_i| u(x_i)$.

Two special cases are used constantly:

- **Sums and differences** $Y = \sum a_i X_i$: $c_i = a_i$, so $u_c^2 = \sum a_i^2 u^2(x_i)$ (plus covariance terms).
- **Products and quotients** $Y = A\, X_1^{p_1} X_2^{p_2} \cdots X_N^{p_N}$: $c_i = p_i y / x_i$, so the **relative** uncertainties combine:

$$\left(\frac{u_c(y)}{y}\right)^2 = \sum_{i=1}^{N} \left(p_i \frac{u(x_i)}{x_i}\right)^2.$$

The RSS combination has an important practical consequence: the largest contributions dominate. A component one third the size of the largest adds only about 5% to u_c. Effort spent reducing small components is wasted; the **uncertainty budget** shows where improvement is worthwhile.

Correlated inputs. If two inputs are fully correlated ($r = +1$, e.g. two voltages measured with the same uncalibrated meter on the same range), their contributions add *linearly*: $u_c = |c_1 u_1 + c_2 u_2|$. For a sum of two such readings this increases the uncertainty; for a difference or a ratio it can cancel it — the reason ratio and substitution methods (bridges, comparators, ratiometric ADC references) are so accurate.

Validity. The law of propagation is first order. It is accurate when the model is nearly linear over the range $x_i \pm u(x_i)$, i.e. when the second-order terms $\tfrac{1}{2}\sum \frac{\partial^2 f}{\partial X_i^2} u^2(x_i)$ are negligible compared with u_c. It fails, for example, for $Y = X^2$ when $x \approx 0$ (the linear term vanishes) or for strongly nonlinear functions such as decibel conversions of very noisy quantities. GUM Supplement 1 then prescribes the **Monte Carlo method**: draw M (typically 10^6) samples of each input from its assigned distribution, evaluate the model for each draw, and take the mean, standard deviation and the $2.5\%/97.5\%$ percentiles of the resulting output sample (Ch. 24).

Effective degrees of freedom and the coverage factor

If u_c is dominated by Type A components with few degrees of freedom, u_c is itself uncertain and $k = 2$ gives less than 95% coverage. The output is then treated as t-distributed with an effective number of degrees of freedom given by the **Welch–Satterthwaite formula**:

$$\nu_{\text{eff}} = \frac{u_c^4(y)}{\displaystyle\sum_{i=1}^{N} \frac{u_i^4(y)}{\nu_i}}, \qquad u_i(y) = |c_i| u(x_i).$$

It is obtained by matching the variance of the estimated u_c^2 to that of a scaled χ^2_ν variable: each u_i^2 estimated with ν_i degrees of freedom has $\operatorname{Var}(u_i^2) \approx 2u_i^4/\nu_i$ (Section 20.5), these add, and a χ^2_ν-based estimate of u_c^2 would have variance $2u_c^4/\nu$. The coverage factor is then $k = t_{0.025,\nu_{\text{eff}}}$ (truncating ν_{eff} down to an integer): for $\nu_{\text{eff}} \geq 30$, $k \approx 2$; for $\nu_{\text{eff}} = 5$, $k = 2.57$.

Reporting

The GUM recommends stating the result as, for example,

$$R = 4.9748\ \text{k}\Omega \pm 0.0045\ \text{k}\Omega,$$

"where the number following the symbol $\pm$ is the numerical value of an expanded uncertainty $U = k u_c$, with U determined from a combined standard uncertainty $u_c = 2.3\ \Omega$ and a coverage factor $k = 2$, corresponding to a coverage probability of approximately 95%." Uncertainties are rounded to at most two significant digits, and the result is rounded to the same decimal position. A compact equivalent with standard uncertainty is $R = 4.9748(23)\ \text{k}\Omega$, the digits in parentheses being u_c in units of the last digit.

Worked Example 20.35 — Type B evaluations from a datasheet and a certificate

(a) DMM specification. On the 10 V range a DMM is specified to $\pm(0.02\%$ of reading $+\ 5$ counts), with resolution 0.1 mV. For a reading of 4.9852 V: $a = 0.0002(4.9852) + 0.0005 = 1.497$ mV. With no further knowledge, rectangular: $u = 1.497/\sqrt{3} = 0.864$ mV.

(b) Resolution. One count $\delta = 0.1$ mV: $u = 0.1/\sqrt{12} = 0.029$ mV — negligible beside (a). (The DMM specification usually already includes resolution; it is listed separately here only to show its size.)

(c) Certificate. A standard resistor's certificate states $U = 0.8\ \mu\Omega/\Omega$ at $k = 2$. Then $u = 0.4\ \mu\Omega/\Omega$, with ν taken from the certificate or as large.

(d) RF mismatch. A power meter's mismatch uncertainty has limits ± 0.10 dB, arising from an unknown relative phase of reflections, so the U-shaped distribution applies: $u = 0.10/\sqrt{2} = 0.071$ dB — larger than the rectangular value 0.058 dB, because the error spends most of its time near the limits.

Worked Example 20.36 — Propagation for a product: power in a resistor

$P = V^2/R$ with $V = 12.00$ V, $u(V) = 0.02$ V and $R = 47.0\ \Omega$, $u(R) = 0.2\ \Omega$ (uncorrelated).

Step 1 — estimate. $P = 144/47 = 3.0638$ W.

Step 2 — relative uncertainties. $u(V)/V = 0.167\%$, $u(R)/R = 0.426\%$.

Step 3 — combine with exponents $p_V = 2$, $p_R = -1$.

$$\frac{u_c(P)}{P} = \sqrt{(2 \times 0.167\%)^2 + (0.426\%)^2} = \sqrt{0.111 + 0.181}\,\% = 0.541\%.$$

So $u_c(P) = 0.0166$ W and $U = 0.033$ W ($k = 2$): $P = 3.064 \pm 0.033$ W.

Step 4 — sensitivity coefficients (the same thing, explicitly). $c_V = 2V/R = 0.5106$ W/V, $c_R = -V^2/R^2 = -0.06519$ W/Ω; contributions 0.0102 W and 0.0130 W; RSS = 0.0166 W.

Step 5 — Monte Carlo check. Drawing 10^6 Gaussian samples of V and R and evaluating V^2/R gives mean 3.0639 W, standard deviation 0.0165 W and a 95% interval $[3.0316,\ 3.0964]$ W, confirming the linearised result: the model is nearly linear over $\pm u$.

Worked Example 20.37 — Correlated inputs: series and ratio

Two nominally equal 1 kΩ resistors are measured with the same ohmmeter, whose calibration uncertainty 0.5 Ω is common to both readings (correlation $r = 1$); random reading noise is negligible.

(a) Series combination $R_s = R_1 + R_2$: $c_1 = c_2 = 1$, so $u_c^2 = 0.25 + 0.25 + 2(1)(1)(0.5)(0.5) = 1.0\ \Omega^2$, $u_c = 1.0\ \Omega$. Ignoring the correlation would give $0.71\ \Omega$ — an underestimate.

(b) Ratio $Q = R_1/R_2$: write the calibration error as a common scale factor. With $c_1 = 1/R_2$ and $c_2 = -R_1/R_2^2$ (both $\approx \pm 10^{-3}\ \Omega^{-1}$), $u_c^2 \approx (10^{-3} \times 0.5)^2 + (10^{-3} \times 0.5)^2 - 2(10^{-3})(10^{-3})(0.25) = 0$. The common error cancels completely in the ratio. This is why a resistance bridge or a ratiometric measurement (sensor and ADC sharing one reference) can be far more accurate than the absolute accuracy of its parts.

Worked Example 20.38 — A complete uncertainty budget: $R = V/I$

A resistor is measured by passing a current and reading voltage and current with two DMMs. Ten simultaneous readings give $\bar{V} = 4.9852$ V with $s(V) = 1.2$ mV and $\bar{I} = 1.00210$ mA with $s(I) = 0.30\ \mu$A. The voltmeter is specified as in Example 20.35(a); the ammeter as $\pm(0.05\%$ of reading $+ 0.2\ \mu$A). Assume the two meters' errors are uncorrelated, as is the reading noise.

Step 1 — model and estimate. $R = V/I = 4.9852/1.00210 = 4.97475$ kΩ.

Step 2 — sensitivity coefficients. $c_V = 1/I = 0.99790$ kΩ/V; $c_I = -V/I^2 = -4.9643$ kΩ/mA.

Step 3 — input uncertainties.

- V, Type A: $1.2/\sqrt{10} = 0.379$ mV, $\nu = 9$.
- V, Type B (specification): 0.864 mV, $\nu = \infty$.
- I, Type A: $0.30/\sqrt{10} = 0.0949\ \mu$A, $\nu = 9$.
- I, Type B: $a = 0.0005(1.00210\text{ mA}) + 0.2\ \mu\text{A} = 0.701\ \mu$A; $u = 0.701/\sqrt{3} = 0.405\ \mu$A, $\nu = \infty$.

Step 4 — budget.

Source	$u(x_i)$	c_i	$\lvert u_i(R)\rvert$ (Ω)	ν_i
V repeatability (A)	0.379 mV	0.9979 kΩ/V	0.38	9
V meter spec (B, rect.)	0.864 mV	0.9979 kΩ/V	0.86	∞
I repeatability (A)	0.0949 μA	−4.964 kΩ/mA	0.47	9
I meter spec (B, rect.)	0.405 μA	−4.964 kΩ/mA	2.01	∞
Combined $u_c(R)$			**2.27**	$\nu_{\text{eff}} \approx 3400$

Step 5 — combine. $u_c = \sqrt{0.38^2 + 0.86^2 + 0.47^2 + 2.01^2} = 2.27\ \Omega$ (relative 0.046%).

Step 6 — effective degrees of freedom. $\nu_{\text{eff}} = 2.27^4/\left(0.38^4/9 + 0.47^4/9\right) \approx 3.4 \times 10^3$; Type B components dominate, so $k = 2$ is appropriate.

Step 7 — report. $R = 4.9748\ \text{k}\Omega \pm 0.0045\ \text{k}\Omega$ ($k = 2$, $\approx 95\%$).

Step 8 — read the budget. The ammeter specification contributes $2.01^2/2.27^2 = 78\%$ of the variance. Taking more readings (reducing the Type A terms) would barely help; a better ammeter, or measuring the current as the voltage across a calibrated standard resistor (turning the measurement into a ratio, Example 20.37), would.

Worked Example 20.39 — When few readings dominate: Welch–Satterthwaite

A frequency is measured four times; the Type A standard uncertainty of the mean is $u_A = 0.8$ Hz ($\nu = 3$). The counter's timebase contributes a Type B $u_B = 0.5$ Hz ($\nu = \infty$).

Step 1 — combine. $u_c = \sqrt{0.8^2 + 0.5^2} = 0.943$ Hz.

Step 2 — degrees of freedom. $\nu_{\text{eff}} = 0.943^4/(0.8^4/3) = 0.792/0.1365 = 5.8$, truncated to 5.

Step 3 — coverage factor. $k = t_{0.025,5} = 2.57$, so $U = 2.57 \times 0.943 = 2.4$ Hz. Using $k = 2$ would give 1.9 Hz and an actual coverage of only about 90%.

Step 4 — improvement. Taking 16 readings instead of 4 would halve u_A to 0.4 Hz (if s is unchanged) and raise ν to 15; then $u_c = 0.64$ Hz, $\nu_{\text{eff}} \approx 98$, $k \approx 1.98$ and $U \approx 1.3$ Hz — nearly half the previous value.

Common Pitfall — Adding uncertainties linearly, or double counting

Adding all contributions linearly ("worst case") overstates the uncertainty for independent inputs; RSS is correct unless the inputs are correlated. Conversely, listing the same physical effect twice — e.g. a DMM's resolution in addition to a specification that already includes it, or the scatter of readings that already reflects the temperature fluctuation you also list as Type B — inflates the budget. Each physical effect should appear once.

Common Pitfall — Confusing tolerance with uncertainty

A 1% resistor's tolerance is the limit of its deviation from nominal, a Type B input with $u = 1\%/\sqrt{3} = 0.58\%$ if nothing else is known. Once the resistor has been *measured*, its value is known to the measurement uncertainty, which is usually far smaller; the tolerance is then irrelevant. Using tolerance limits as if they were 1σ values, or as if they were 95% limits, are both common errors.

Engineering Note — Uncertainty and conformance decisions

Deciding whether a measured value conforms to a specification must account for the measurement uncertainty: a reading 0.3% inside a $\pm 1\%$ limit with $U = 0.5\%$ does not prove conformance. Standards such as ISO 14253-1 and ILAC-G8 formalise this with **guard bands** of width related to U — the measurement-uncertainty version of the production-test trade-off in Section 20.8. A common rule of thumb is a **test uncertainty ratio** of at least 4:1 between the specification tolerance and the measurement's expanded uncertainty.

Summary

Topic	Key formula or fact
Sample mean, variance	$\bar{x} = \frac{1}{n}\sum x_i,\ s^2 = \frac{1}{n-1}\sum (x_i - \bar{x})^2$ (unbiased)
Robust spread	$\mathrm{IQR} = Q_3 - Q_1$; fences $Q_1 - 1.5\,\mathrm{IQR}$, $Q_3 + 1.5\,\mathrm{IQR}$; $\sigma \approx 1.4826\,\mathrm{MAD}$
Weighted mean	$\bar{x}_w = \sum w_i x_i / \sum w_i$, $w_i = 1/u_i^2$, $u(\bar{x}_w) = (\sum w_i)^{-1/2}$
RMS and variance	$V_{\mathrm{rms}}^2 = V_{\mathrm{DC}}^2 + \sigma_v^2$
Correlation	$r = S_{xy}/\sqrt{S_{xx}S_{yy}}$, $-1 \le r \le 1$; Spearman $r_s = 1 - 6\sum d_i^2/(n(n^2-1))$
Covariance propagation	$\mathbf{\Sigma}_Y = \mathbf{A}\mathbf{\Sigma}_X\mathbf{A}^{\mathsf{T}}$; $\mathrm{Var}(X \pm Y) = \sigma_X^2 + \sigma_Y^2 \pm 2\,\mathrm{Cov}$
Binomial / Poisson	$np,\ np(1-p)\ /\ \lambda,\ \lambda$
Uniform / exponential	$\frac{a+b}{2},\ \frac{(b-a)^2}{12}\ /\ \frac{1}{\lambda},\ \frac{1}{\lambda^2}$; memoryless
Rayleigh	$E[R] = \sigma\sqrt{\pi/2}$, $E[R^2] = 2\sigma^2$, $P(R > r) = e^{-r^2/(2\sigma^2)}$
χ_k^2, t_ν, F_{d_1,d_2}	sum of k squared $\mathcal{N}(0,1)$; $Z/\sqrt{V/\nu}$; ratio of normalised χ^2
Sample mean	$E\bar{X} = \mu$, $\mathrm{SE} = \sigma/\sqrt{n}$; CLT: $\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \to \mathcal{N}(0,1)$
Gaussian sampling	$(n-1)S^2/\sigma^2 \sim \chi_{n-1}^2$; $\frac{\bar{X}-\mu}{S/\sqrt{n}} \sim t_{n-1}$; $\bar{X} \perp S^2$
Estimator quality	$\mathrm{MSE} = \mathrm{Var} + b^2$; CRLB $\mathrm{Var}\,\hat{\theta} \ge 1/(nI(\theta))$
MLE	maximise $\ell(\theta) = \sum \ln f(x_i;\theta)$; $\mathrm{SE} \approx 1/\sqrt{-\ell''(\hat{\theta})}$; Gaussian noise $\Rightarrow$ least squares
CI for mean	$\bar{x} \pm z_{\alpha/2}\sigma/\sqrt{n}$; $\bar{x} \pm t_{\alpha/2,n-1}s/\sqrt{n}$; $n = (z_{\alpha/2}\sigma/E)^2$
CI for variance	$\left[(n-1)s^2/\chi^2_{\alpha/2,n-1},\ (n-1)s^2/\chi^2_{1-\alpha/2,n-1}\right]$
Tests	z, t (one-sample, Welch, paired); $\chi^2 = \sum(O-E)^2/E$ with $K-1-m$ d.o.f.; $F = s_1^2/s_2^2$
Errors	$\alpha = P(\text{reject} \mid H_0)$, $\beta = P(\text{accept} \mid H_1)$; power $= 1-\beta$; $n = ((z_\alpha + z_\beta)\sigma/\delta)^2$
Regression	$\hat{\beta}_1 = S_{xy}/S_{xx}$, $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}$, $s^2 = \mathrm{SSE}/(n-2)$, $\mathrm{SE}(\hat{\beta}_1) = s/\sqrt{S_{xx}}$
r^2, matrix LS	$r^2 = 1 - \mathrm{SSE}/\mathrm{SST}$; $\hat{\boldsymbol{\beta}} = (\mathbf{X}^{\mathsf{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathsf{T}}\mathbf{y}$, $\mathrm{Cov} = \sigma^2(\mathbf{X}^{\mathsf{T}}\mathbf{X})^{-1}$
Type A / Type B	$u = s/\sqrt{n}$ ($\nu = n-1$); rectangular $a/\sqrt{3}$, triangular $a/\sqrt{6}$, U-shaped $a/\sqrt{2}$, resolution $\delta/\sqrt{12}$
Propagation (GUM)	$u_c^2 = \sum c_i^2 u^2(x_i) + 2\sum_{i<j} c_i c_j u(x_i,x_j)$, $c_i = \partial f/\partial x_i$
Products	$(u_c/y)^2 = \sum (p_i u(x_i)/x_i)^2$ for $y = A\prod x_i^{p_i}$
Expanded uncertainty	$U = k u_c$; $\nu_{\mathrm{eff}} = u_c^4/\sum(u_i^4/\nu_i)$; $k = t_{0.025,\nu_{\mathrm{eff}}}$ (≈ 2 for large ν)

Further reading

- D. C. Montgomery and G. C. Runger, *Applied Statistics and Probability for Engineers*, 7th ed., Wiley, 2018 — the standard engineering statistics text; thorough on intervals, tests and regression.
- A. Papoulis and S. U. Pillai, *Probability, Random Variables and Stochastic Processes*, 4th ed., McGraw-Hill, 2002 — distributions, estimation and the bridge to random processes (Ch. 21).
- S. M. Kay, *Fundamentals of Statistical Signal Processing, Vol. I: Estimation Theory*, Prentice Hall, 1993 — Cramér–Rao bounds and maximum likelihood with signal-processing examples.
- JCGM 100:2008, *Evaluation of Measurement Data — Guide to the Expression of Uncertainty in Measurement* (GUM), and JCGM 101:2008 (Supplement 1, Monte Carlo method), freely available from the BIPM — the authoritative source for Section 20.10.
- I. G. Hughes and T. P. A. Hase, *Measurements and their Uncertainties*, Oxford University Press, 2010 — a concise practical guide to error analysis and curve fitting.

Chapter 21 — Random Processes

> **Prerequisites:** Ch. 16 (Fourier Transforms), Ch. 17 (Z-Transforms), Ch. 19 (Probability), Ch. 20 (Statistical Analysis) **Learning outcomes:** After this chapter you can …
>
> - describe a random process as an ensemble of sample functions, distinguish ensemble averages from time averages, and compute mean, autocorrelation and autocovariance functions;
> - test a process for strict- and wide-sense stationarity and use the properties of autocorrelation and cross-correlation functions;
> - move between autocorrelation and power spectral density with the Wiener–Khinchin theorem, and compute the output mean, autocorrelation and PSD $S_{YY} = |H|^2 S_{XX}$ of an LTI system with random input, in continuous and discrete time (including AR(1) sequences);
> - work with Gaussian processes, white and band-limited white noise, and the noise-equivalent bandwidth of a filter;
> - decide when time averages may replace ensemble averages (ergodicity) and how long a record must be;
> - carry out standard noise calculations: thermal noise $4kTRB$, kT/C noise, SNR after filtering and averaging, and the matched filter.

A random variable (Ch. 19) assigns a number to each outcome of an experiment. Many engineering signals — the hiss at a receiver output, a speech waveform, the bit stream on a serial link, the drift of a gyroscope, the error of an ADC over time — are not single numbers but *functions of time* whose exact shape cannot be predicted in advance. Yet they are far from lawless: the noise at the output of a given amplifier always has about the same RMS value, its spectrum always has about the same shape, and a filter always reduces it by about the same amount. A **random process** is the mathematical object that captures this combination of unpredictable detail and predictable statistics. It assigns a whole waveform to each outcome of an experiment.

This chapter develops the **second-order theory** of random processes — means, correlation functions and power spectra. Second-order theory is remarkably economical: it is exactly what is preserved and transformed by linear systems, it completely characterizes Gaussian processes, and it answers the questions an engineer asks most often (What is the noise power? How is it distributed over frequency? How much does this filter remove? How long must I average?). We use the Fourier transform conventions of Ch. 16 ($X(\omega) = \int x(t) e^{-j\omega t} dt$) and the discrete-time transforms of Ch. 17 throughout, and the probability language of Ch. 19.

21.1 Random sequences and processes

Motivation

Suppose you connect an oscilloscope across a resistor at room temperature, set a large vertical gain, and press "single" a hundred times. Every trace is different — no two records of thermal noise ever coincide — but every trace has the same "look": the same typical amplitude, the same graininess. If you sample all hundred traces at the instant $t_1 = 3$ μs after the trigger, the hundred numbers you collect form a histogram that looks Gaussian with zero mean. If you instead take *one* trace and histogram its values along time, you get, for this kind of noise, the same Gaussian. These two procedures — averaging *across* records at a fixed time, and averaging *along* one record — are the two fundamental operations of this chapter, and the question of when they agree (ergodicity, §21.7) is one of its central themes.

Definition and the two viewpoints

Random process

Let $(\Omega, \mathcal{F}, P)$ be a probability space and T an index set. A **random process** (stochastic process) is a family of random variables $\{X(t),\ t \in T\}$ all defined on the same Ω. For each outcome $\zeta \in \Omega$ the function $t \mapsto X(t, \zeta)$ is a **sample function** (realization, sample path); the collection of all sample functions is the **ensemble**.

- $T \subseteq \mathbb{R}$ (an interval): **continuous-time** process $X(t)$.
- $T \subseteq \mathbb{Z}$: **discrete-time** process or **random sequence** $X[n]$.

Independently, the *values* may be continuous (a noise voltage) or discrete (a bit, a count).

A random process is thus a function of two variables, $X(t, \zeta)$, and there are two ways to slice it:

- **Fix t:** $X(t_1, \cdot)$ is an ordinary random variable with CDF $F_X(x; t_1) = P\{X(t_1) \le x\}$ and pdf $f_X(x; t_1) = \partial F_X / \partial x$. Everything in Ch. 19 applies to it.
- **Fix ζ:** $X(\cdot, \zeta_1)$ is a deterministic waveform — the single trace you see on the oscilloscope. Everything in Chs. 15–17 (Fourier series, transforms) applies to it, at least formally.

We follow the usual convention of suppressing ζ and writing $X(t)$; lowercase $x(t)$ denotes one particular sample function.

The four combinations of continuous/discrete time and continuous/discrete values all occur in practice:

	continuous values	discrete values
continuous time	thermal noise voltage $V_n(t)$	random telegraph signal ± 1; photon count $N(t)$
discrete time	sampled noise $V_n[n] = V_n(nT_s)$	data bits $b[n] \in \{0, 1\}$; quantized samples

Finite-dimensional distributions

A single random variable is described by its CDF. A random process involves infinitely many random variables, and their joint behaviour matters: the probability that a noise voltage exceeds a threshold at two instants 1 ns apart is very different from the probability for two instants 1 s apart.

Finite-dimensional distributions

The k**-th order distribution** of $X(t)$ is the joint CDF

$$F_X(x_1,\dots,x_k;\,t_1,\dots,t_k) = P\{X(t_1)\le x_1,\dots,X(t_k)\le x_k\},$$

with joint pdf $f_X(x_1,\dots,x_k;t_1,\dots,t_k)$. The process is **specified in distribution** when these are known for every k and every choice of times $t_1,\dots,t_k$.

Kolmogorov's extension theorem guarantees that any *consistent* family of finite-dimensional distributions (marginalizing the $(k+1)$-th order distribution over one variable gives the k-th order one, and permuting the times permutes the variables) defines a random process. In practice we almost never know all of these distributions. Two situations are tractable:

1. **The process is built explicitly** from a few random variables, e.g. $X(t) = A\cos(\omega_0 t + \Theta)$. Then all distributions follow from those of A and Θ.
2. **The process is Gaussian** (§21.5). Then the first two moments determine everything.

Otherwise we settle for the first two moments, which is what the rest of this section introduces.

Ensemble averages: mean, autocorrelation, autocovariance

Mean, autocorrelation, autocovariance

$$\mu_X(t) = E[X(t)] = \int_{-\infty}^{\infty} x\, f_X(x;t)\,dx$$

$$R_{XX}(t_1,t_2) = E[X(t_1)X(t_2)] = \iint x_1 x_2\, f_X(x_1,x_2;t_1,t_2)\,dx_1\,dx_2$$

$$C_{XX}(t_1,t_2) = E\big[(X(t_1)-\mu_X(t_1))(X(t_2)-\mu_X(t_2))\big] = R_{XX}(t_1,t_2) - \mu_X(t_1)\mu_X(t_2)$$

The **variance function** is $\sigma_X^2(t) = C_{XX}(t,t)$ and the **mean-square value** (average power at time t, in a 1 Ω normalization) is $E[X^2(t)] = R_{XX}(t,t)$. The **correlation coefficient** is $\rho_{XX}(t_1,t_2) = C_{XX}(t_1,t_2)/(\sigma_X(t_1)\sigma_X(t_2)) \in [-1,1]$. For a complex process, $R_{XX}(t_1,t_2) = E[X(t_1)X^*(t_2)]$ and $C_{XX}(t_1,t_2) = R_{XX}(t_1,t_2) - \mu_X(t_1)\mu_X^*(t_2)$.

These are **ensemble averages**: at each time they average over all outcomes ζ, i.e. vertically across the ensemble of traces. The mean function $\mu_X(t)$ is a deterministic function of time (for example, a DC offset that drifts as an amplifier warms up). The autocorrelation is a deterministic function of *two* times: it measures how strongly the value at t_1 is (linearly) related to the value at t_2.

Two facts follow immediately from the definition and are used constantly:

- **Symmetry:** $R_{XX}(t_1,t_2) = R_{XX}(t_2,t_1)$ for a real process (and $R_{XX}(t_1,t_2) = R_{XX}^*(t_2,t_1)$ for a complex one).
- **Non-negative definiteness:** for any times $t_1,\dots,t_n$ and real constants $a_1,\dots,a_n$,

$$\sum_{i=1}^{n}\sum_{k=1}^{n} a_i a_k\, R_{XX}(t_i,t_k) = E\Big[\Big(\sum_i a_i X(t_i)\Big)^2\Big] \ge 0.$$

So the matrix $[R_{XX}(t_i,t_k)]$ is a correlation matrix (Ch. 19) for every choice of times. Not every symmetric function of two variables can be an autocorrelation.

For two processes on the same probability space we also define the **cross-correlation** $R_{XY}(t_1,t_2) = E[X(t_1)Y(t_2)]$ and **cross-covariance** $C_{XY}(t_1,t_2) = R_{XY}(t_1,t_2) - \mu_X(t_1)\mu_Y(t_2)$; they are studied in §21.3.

Time averages

For a single sample function $x(t)$ we can also average *along time*:

Time averages of a sample function

$$\langle x\rangle = \lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T} x(t)\,dt, \qquad \mathcal{R}_x(\tau) = \langle x(t+\tau)x(t)\rangle = \lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T} x(t+\tau)x(t)\,dt.$$

For a sequence, $\langle x\rangle = \lim_{N\to\infty}\frac{1}{2N+1}\sum_{n=-N}^{N} x[n]$, and similarly for $\mathcal{R}_x[k]$.

The two kinds of average are conceptually very different:

	Ensemble average $E[\cdot]$	**Time average** $\langle\cdot\rangle$
averages over	outcomes ζ at fixed t	time t for fixed ζ
result	deterministic function of t	a number (or a function of τ) that may depend on ζ, i.e. a random variable
what you need	many independent records	one long record
what a DC voltmeter reads	—	$\langle x\rangle$ (over its averaging time)
what a true-RMS meter reads	—	$\sqrt{\langle x^2\rangle}$

Laboratory instruments compute time averages; theory is most easily done with ensemble averages. A process for which the two coincide (with probability 1) is called **ergodic** — §21.7 makes this precise and shows that it is a genuine assumption, not a tautology.

A catalogue of standard processes

Process	Construction	Typical EE use
Random-phase sinusoid	$X(t) = A\cos(\omega_0 t + \Theta)$, $\Theta \sim U[0, 2\pi)$	oscillator with unknown phase; carrier at a receiver
i.i.d. sequence	$X[n]$ independent, identically distributed	data symbols; ideal discrete white noise
Bernoulli / binary sequence	$X[n] \in \{-1, +1\}$ i.i.d., equally likely	random data bits, PRBS test patterns
Random walk	$S[n] = \sum_{k=1}^{n} X[k]$, $X[k]$ i.i.d. ± 1	phase-noise accumulation, drift, integrated error
Moving average MA(q)	$X[n] = \sum_{k=0}^{q} b_k W[n-k]$, W i.i.d.	FIR-filtered noise
Autoregressive AR(1)	$X[n] = aX[n-1] + W[n]$	IIR-filtered noise, sensor bias, channel fading (§21.4)
Random binary waveform	$X(t) = \sum_k a_k\, p(t - kT - D)$, $a_k = \pm A$, $D \sim U[0, T)$	NRZ data on a cable
Random telegraph signal	$X(t) = \pm 1$, sign flips at Poisson times of rate λ	burst (RTS) noise in MOSFETs, on/off traffic
Poisson counting process	$N(t)$ = number of events in $(0, t]$, rate λ	photon and electron arrivals, shot noise
Wiener process	limit of a scaled random walk (§21.5)	Brownian motion, oscillator phase noise
Gaussian process	all finite-dimensional distributions Gaussian	thermal noise, aggregate interference

A useful structural property shared by the random walk, the Poisson process and the Wiener process is:

Independent increments

$X(t)$ has **independent increments** if for any $t_1 < t_2 < \cdots < t_k$ the increments $X(t_2) - X(t_1),\ X(t_3) - X(t_2), \ldots, X(t_k) - X(t_{k-1})$ are mutually independent. It has **stationary increments** if the distribution of $X(t+s) - X(t)$ depends only on s.

For a process with independent increments starting from $X(0) = 0$, the autocovariance has the universal form $C_{XX}(t_1, t_2) = \sigma_X^2(\min(t_1, t_2))$: for $t_1 \le t_2$, write $X(t_2) = X(t_1) + [X(t_2) - X(t_1)]$; the bracket is independent of $X(t_1)$, so $\operatorname{Cov}(X(t_1), X(t_2)) = \operatorname{Var} X(t_1)$.

Worked examples

Worked Example 21.1 — Mean and autocorrelation of a random-phase sinusoid

Let $X(t) = A\cos(\omega_0 t + \Theta)$ with constants A, ω_0 and $\Theta \sim U[0, 2\pi)$.

Step 1 — mean. The pdf of Θ is $1/(2\pi)$ on $[0, 2\pi)$, so

$$\mu_X(t) = \frac{1}{2\pi}\int_0^{2\pi} A\cos(\omega_0 t + \theta)\, d\theta = 0 \quad \text{for every } t,$$

because a cosine integrates to zero over a full period of its argument.

Step 2 — autocorrelation. Using $\cos a \cos b = \tfrac{1}{2}[\cos(a-b) + \cos(a+b)]$,

$$R_{XX}(t_1, t_2) = \frac{A^2}{2} E\big[\cos\omega_0(t_1 - t_2)\big] + \frac{A^2}{2} E\big[\cos(\omega_0(t_1 + t_2) + 2\Theta)\big] = \frac{A^2}{2}\cos\omega_0(t_1 - t_2).$$

The first expectation is of a constant; the second is zero because 2Θ is uniform over two full periods.

Step 3 — interpretation. R_{XX} depends only on $t_1 - t_2$, not on absolute time — the hallmark of stationarity (§21.2). The power $R_{XX}(t, t) = A^2/2$ is the familiar mean-square value of a sinusoid of amplitude A.

Step 4 — first-order pdf. Since $\omega_0 t + \Theta \pmod{2\pi}$ is uniform, $X(t)$ has the arcsine pdf $f_X(x) = 1/(\pi\sqrt{A^2 - x^2})$ for $|x| < A$, the same for every t. A numerical check: $\int_{-A}^{A} x^2/(\pi\sqrt{A^2 - x^2})\, dx = A^2/2$. ✓

Step 5 — time averages. Take one sample function, $x(t) = A\cos(\omega_0 t + \theta_1)$. Its time average is 0 and its time autocorrelation is $\langle x(t+\tau)x(t)\rangle = \frac{A^2}{2}\cos\omega_0\tau$ — the same as the ensemble results, whatever θ_1 is. This process is ergodic (§21.7).

Worked Example 21.2 — Random walk: variance that grows

Let $S[0] = 0$ and $S[n] = \sum_{k=1}^{n} X[k]$ for $n \ge 1$, with $X[k]$ i.i.d., $P(X = \pm 1) = \tfrac{1}{2}$, so $E[X] = 0$ and $E[X^2] = 1$.

Mean. $\mu_S[n] = \sum_{k=1}^{n} E[X[k]] = 0$.

Autocorrelation. $R_{SS}[n_1, n_2] = E\Big[\sum_{k=1}^{n_1}\sum_{l=1}^{n_2} X[k]X[l]\Big] = \sum_k\sum_l E[X[k]X[l]]$. By independence $E[X[k]X[l]] = \delta[k-l]$, so only the terms with $k = l \le \min(n_1, n_2)$ survive:

$$R_{SS}[n_1, n_2] = \min(n_1, n_2).$$

This agrees with the independent-increments formula $C(t_1, t_2) = \sigma^2(\min(t_1, t_2))$.

Consequences. $\sigma_S^2[n] = n$: the ensemble spreads like $\sqrt{n}$. After $n = 10^4$ steps the typical excursion is about 100, and by the central limit theorem $P\{|S[10^4]| > 200\} \approx 2Q(2) \approx 0.046$. The correlation coefficient between $S[n]$ and $S[4n]$ is $n/\sqrt{n \cdot 4n} = 1/2$ for every n: the walk "remembers" its past indefinitely. Integrated noise — e.g. oscillator phase driven by white frequency noise, or the position estimate of an inertial navigator integrating noisy accelerometer readings — behaves the same way.

Worked Example 21.3 — The Poisson counting process

$N(t)$, $t \geq 0$, counts events in $(0, t]$; it has independent, stationary increments and $N(t+s) - N(t) \sim \text{Poisson}(\lambda s)$. Hence $\mu_N(t) = \lambda t$ and $\sigma_N^2(t) = \lambda t$.

Autocorrelation. For $t_1 \leq t_2$, by the independent-increments argument $C_{NN}(t_1, t_2) = \text{Var}\, N(t_1) = \lambda t_1$. So in general

$$C_{NN}(t_1, t_2) = \lambda \min(t_1, t_2), \qquad R_{NN}(t_1, t_2) = \lambda \min(t_1, t_2) + \lambda^2 t_1 t_2.$$

Numbers. A photodiode receives photons at $\lambda = 10^9$ s^{-1}. In a 1 μs window the expected count is $\lambda t = 1000$ with standard deviation $\sqrt{1000} = 31.6$, a relative fluctuation of 3.2%. This $1/\sqrt{\lambda t}$ fluctuation is **shot noise**; the corresponding photocurrent noise has one-sided PSD $2qI$ (§21.8). Note the mean grows with t, so $N(t)$ is not stationary — but its *increments over fixed-length windows* are.

Worked Example 21.4 — Sampling a process gives a random sequence

A continuous-time process $X(t)$ with $\mu_X = 0$ and $R_{XX}(t_1, t_2) = \sigma^2 e^{-\alpha|t_1 - t_2|}$ is sampled every T_s seconds: $X[n] = X(nT_s)$. Then

$$R_{XX}[n_1, n_2] = E[X(n_1 T_s) X(n_2 T_s)] = \sigma^2 e^{-\alpha T_s |n_1 - n_2|} = \sigma^2 a^{|n_1 - n_2|}, \qquad a = e^{-\alpha T_s}.$$

With $\alpha = 1000$ s^{-1} (correlation time 1 ms) and $T_s = 100$ μs, $a = e^{-0.1} = 0.905$: adjacent samples are 90% correlated, and samples are essentially uncorrelated ($\rho < 0.05$) only when $|n_1 - n_2| \geq 30$. Sampling at $T_s = 5$ ms instead gives $a = e^{-5} = 0.0067$: practically uncorrelated samples. This geometric autocorrelation is exactly that of the AR(1) sequence we study in §21.4 — sampling a first-order Gauss–Markov process produces an AR(1) sequence.

Common Pitfall — confusing the two averages

$E[X(t)]$ is a function of t computed across the ensemble; $\langle x \rangle$ is a number computed along one record. Writing "the mean of the noise" without saying which is ambiguous. For a process like $X(t) = A$ (a random constant, see §21.7), $E[X(t)] = E[A]$ but $\langle x \rangle = A$: they differ for every outcome except those with $A = E[A]$.

Engineering Note

Measurement noise, quantization error, channel noise, phase noise and interference are all modelled as random processes. What we need for design is rarely the full set of finite-dimensional distributions; it is the mean (DC offset), the power (RMS value) and *how that power is spread over time lag and frequency*. The autocorrelation (§21.3) and the power spectral density (§21.4) carry exactly this information, and linear systems transform them by simple rules.

21.2 Stationarity

Motivation

Most of the convenient machinery of signal analysis — transfer functions, spectra, filters designed once and used forever — rests on **time invariance**. For random signals the corresponding idea is that the *statistics* of the process do not change with time: the noise from a resistor in thermal equilibrium looks statistically the same today as tomorrow, even though no two sample functions agree. Stationarity is what makes the autocorrelation a function of one variable (the lag) instead of two, and that in turn is what allows the Fourier transform to enter (§21.4). Without it, the notion of a "noise spectrum" is not even well defined.

Strict-sense stationarity

Strict-sense stationarity (SSS)

$X(t)$ is **strict-sense stationary** if all its finite-dimensional distributions are invariant under a time shift: for every k, every set of times $t_1, \ldots, t_k$ and every shift c,

$$F_X(x_1, \ldots, x_k;\, t_1 + c, \ldots, t_k + c) = F_X(x_1, \ldots, x_k;\, t_1, \ldots, t_k).$$

If this holds only for $k \leq N$, the process is **stationary of order** N.

Consequences for an SSS process:

- **First order** ($k = 1$): $f_X(x; t) = f_X(x)$ does not depend on t. In particular $\mu_X(t) = \mu_X$ and $\sigma_X^2(t) = \sigma_X^2$ are constants.
- **Second order** ($k = 2$): $f_X(x_1, x_2; t_1, t_2)$ depends on the times only through $\tau = t_1 - t_2$ (choose $c = -t_2$). Hence $R_{XX}(t_1, t_2)$ is a function of $t_1 - t_2$ alone.

i.i.d. sequences are strictly stationary

If $X[n]$ are independent and identically distributed with CDF F, then $X[n]$ is SSS.

Proof. For any distinct $n_1, \ldots, n_k$ and shift c, independence gives $F_X(x_1, \ldots, x_k; n_1 + c, \ldots, n_k + c) = \prod_i F(x_i)$, which does not involve c. $\square$

A memoryless function of an SSS process, $Y(t) = g(X(t))$, is again SSS (the joint distribution of $Y(t_1 + c), \ldots, Y(t_k + c)$ is determined by that of $X(t_1 + c), \ldots, X(t_k + c)$). So is the output of any time-invariant (possibly nonlinear) system with memory driven by an SSS input, once start-up transients have died away.

SSS is a strong requirement and is hard to verify from data: it constrains distributions of every order. For linear-system analysis a much weaker condition, involving only the first two moments, suffices.

Wide-sense stationarity

Wide-sense stationarity (WSS)

$X(t)$ is **wide-sense stationary** (weakly stationary) if

1. $E[X^2(t)] < \infty$ for all t,
2. $\mu_X(t) = \mu_X$ is constant, and
3. $R_{XX}(t_1, t_2)$ depends only on $\tau = t_1 - t_2$.

We then write, with a slight abuse of notation,

$$R_{XX}(\tau) = E[X(t+\tau)X(t)], \qquad C_{XX}(\tau) = R_{XX}(\tau) - \mu_X^2, \qquad \sigma_X^2 = C_{XX}(0).$$

For a random sequence: $E[X[n]] = \mu_X$ and $R_{XX}[k] = E[X[n+k]X[n]]$ independent of n.

Joint wide-sense stationarity

$X(t)$ and $Y(t)$ are **jointly WSS** if each is WSS and $R_{XY}(t_1, t_2) = E[X(t_1)Y(t_2)]$ depends only on $t_1 - t_2$. We write $R_{XY}(\tau) = E[X(t+\tau)Y(t)]$.

Relations between the notions:

- **SSS with finite second moment** $\Rightarrow$ **WSS.** Shown above.
- **WSS** $\not\Rightarrow$ **SSS** in general (Example 21.8 below).
- **For Gaussian processes, WSS** $\Leftrightarrow$ **SSS** (§21.5), because the mean and autocovariance determine all finite-dimensional distributions.
- Each of X and Y being WSS does **not** make them jointly WSS. For example, with X WSS and $Y(t) = X(-t)$ (also WSS), $R_{XY}(t_1, t_2) = R_{XX}(t_1 + t_2)$, which depends on the sum of the times.

Engineering Note — why "WSS" is the working assumption

Every result about filters and spectra in this chapter needs only WSS. Since estimating higher-order distributions takes enormous amounts of data, practising engineers almost always mean "WSS" when they say "stationary". When a noise source is additionally Gaussian — the usual case for thermal noise and many aggregate interference sources — nothing is lost.

Testing for stationarity: worked examples

The procedure is always the same: compute $\mu_X(t)$ and $R_{XX}(t+\tau, t)$ and check whether t drops out.

Worked Example 21.5 — Random phase, random amplitude, random pair

(a) $X(t) = A\cos(\omega_0 t + \Theta)$, $\Theta \sim U[0, 2\pi)$: from Example 21.1, $\mu_X = 0$ and $R_{XX}(t+\tau, t) = \frac{A^2}{2}\cos\omega_0\tau$. **WSS.** (In fact SSS: shifting time by c is the same as shifting Θ by $\omega_0 c$, which leaves its uniform distribution modulo 2π unchanged.)

(b) $Y(t) = B\cos\omega_0 t$ with B a random variable, $E[B] = \mu_B$, $E[B^2] = m_2$:

$$\mu_Y(t) = \mu_B\cos\omega_0 t, \qquad R_{YY}(t+\tau, t) = m_2\cos\omega_0(t+\tau)\cos\omega_0 t = \frac{m_2}{2}\big[\cos\omega_0\tau + \cos\omega_0(2t+\tau)\big].$$

If $\mu_B \neq 0$ the mean depends on t; even with $\mu_B = 0$, R_{YY} depends on t through the second term. **Not WSS.** Physically: every sample function passes through zero at the same instants $t = (k + \tfrac{1}{2})\pi/\omega_0$, so the ensemble power $R_{YY}(t,t) = m_2\cos^2\omega_0 t$ is zero there.

(c) $Z(t) = B\cos\omega_0 t + C\sin\omega_0 t$ with B, C zero mean, uncorrelated, both of variance σ^2:

$$\begin{aligned} \mu_Z(t) &= 0, \\ R_{ZZ}(t_1, t_2) &= E[B^2]\cos\omega_0 t_1\cos\omega_0 t_2 + E[C^2]\sin\omega_0 t_1\sin\omega_0 t_2 + E[BC]\sin\omega_0(t_1+t_2) \\ &= \sigma^2\cos\omega_0(t_1 - t_2). \end{aligned}$$

WSS. The random phase is supplied by the random pair (B, C). If $E[B^2] \neq E[C^2]$ or $E[BC] \neq 0$, the t-dependence returns — both conditions are necessary. Z is WSS for *any* distribution of (B, C) with these moments, but it is SSS only if (B, C) is circularly symmetric (e.g. jointly Gaussian with equal variances).

Worked Example 21.6 — A phase that is not uniform enough

Let $X(t) = A\cos(\omega_0 t + \Theta)$ with $\Theta \sim U[0, \pi]$ (only half a cycle). Then

$$\mu_X(t) = \frac{A}{\pi}\int_0^{\pi}\cos(\omega_0 t + \theta)\,d\theta = \frac{A}{\pi}\big[\sin(\omega_0 t + \pi) - \sin\omega_0 t\big] = -\frac{2A}{\pi}\sin\omega_0 t.$$

The mean oscillates with amplitude $2A/\pi \approx 0.637A$, so the process is **not WSS**. Curiously, the autocorrelation is still $\frac{A^2}{2}\cos\omega_0(t_1 - t_2)$: the term $E[\cos(\omega_0(t_1 + t_2) + 2\Theta)]$ vanishes because 2Θ *is* uniform on $[0, 2\pi)$. Conditions 2 and 3 of the WSS definition are independent, and both must be checked. (The autocovariance, $R_{XX} - \mu_X(t_1)\mu_X(t_2)$, does depend on both times.)

Worked Example 21.7 — Random binary (NRZ) waveform

Let $X(t) = \sum_{k=-\infty}^{\infty} a_k\, p(t - kT - D)$, where $p(t) = 1$ for $0 \le t < T$ and 0 otherwise, the a_k are i.i.d. with $P(a_k = \pm A) = \tfrac{1}{2}$, and the timing offset $D \sim U[0, T)$ is independent of the a_k. This models a data stream whose clock phase is unknown to the observer.

Step 1 — mean. $E[X(t)] = \sum_k E[a_k]\, E[p(t - kT - D)] = 0$.

Step 2 — the product. $X(t)$ and $X(t+\tau)$ are each $\pm A$. If t and $t+\tau$ fall in the **same** bit interval, $X(t+\tau)X(t) = a_k^2 = A^2$. If they fall in **different** intervals, the product is $a_k a_l$ with $k \ne l$, which has zero mean. So

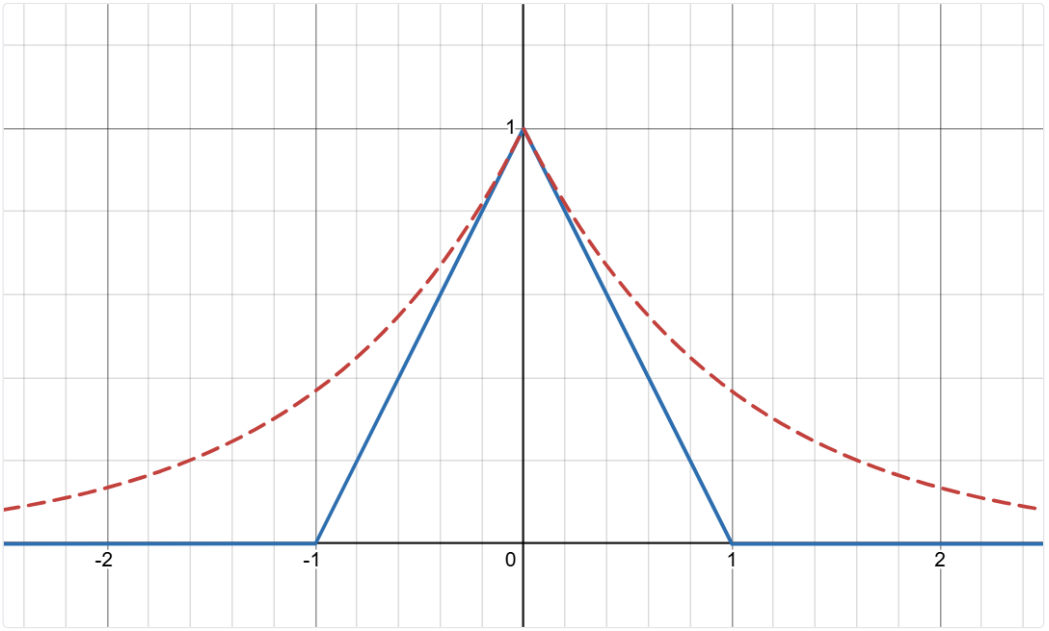
$$R_{XX}(t+\tau, t) = A^2\, P\{t \text{ and } t+\tau \text{ lie in the same bit}\}.$$

Step 3 — the probability. For $|\tau| \ge T$ it is 0. For $0 \le \tau < T$: the bit boundaries are at $kT + D$, and since D is uniform, the position of t within its bit, $U = (t - D) \bmod T$, is uniform on $[0, T)$. Both instants lie in the same bit iff $U + \tau < T$, which has probability $(T - \tau)/T$. By symmetry the same holds for $\tau < 0$ with $|\tau|$.

Result.

$$R_{XX}(\tau) = \begin{cases} A^2\left(1 - \dfrac{|\tau|}{T}\right), & |\tau| < T, \\ 0, & |\tau| \ge T. \end{cases}$$

The process is **WSS** with a triangular autocorrelation. Without the random offset D (a fixed, known clock) the probability in Step 3 depends on where t sits within its bit, and $R_{XX}(t+\tau, t)$ is periodic in t with period T — the process is then cyclostationary (below). The triangle reappears in §21.4 as the sinc^2 spectrum of NRZ data.



Triangular autocorrelation of the random NRZ waveform ($A = 1$, $T = 1$, blue) and the exponential autocorrelation $e^{-2\lambda|\tau|}$ of a random telegraph signal with the same average switching rate $\lambda = 1/(2T)$ (red, dashed). The NRZ signal is completely decorrelated after one bit; the telegraph signal, whose switching instants are not tied to a clock, decorrelates only gradually.

Worked Example 21.8 — WSS but not strictly stationary

Build a sequence of independent variables with $X[n] \sim \mathcal{N}(0, 1)$ for even n and $X[n] = \pm 1$ with probability $\tfrac{1}{2}$ each for odd n. Then $E[X[n]] = 0$ and $E[X^2[n]] = 1$ for all n, and $E[X[n+k]X[n]] = 0$ for $k \ne 0$ by independence, so

$$\mu_X = 0, \qquad R_{XX}[k] = \delta[k].$$

The sequence is **WSS** (indeed white, §21.6). But the first-order distribution alternates between a Gaussian and two point masses, so $P\{X[n] > 1.5\}$ equals $Q(1.5) \approx 0.067$ for even n and 0 for odd n: **not SSS**. A second-order description cannot see this difference — a reminder that "white" and "i.i.d." are not the same thing.

Cyclostationarity

Wide-sense cyclostationarity

$X(t)$ is **wide-sense cyclostationary** with period T_0 if

$$\mu_X(t + T_0) = \mu_X(t), \qquad R_{XX}(t_1 + T_0, t_2 + T_0) = R_{XX}(t_1, t_2) \quad \text{for all } t, t_1, t_2.$$

Digitally modulated signals $X(t) = \sum_k a_k\, p(t - kT_b)$ with a fixed clock, sampled-and-held noise, and noise in switched-capacitor and mixer circuits (whose operating point is periodically pumped by a clock or local oscillator) are all cyclostationary. The standard way of bringing them into the WSS framework is to randomize the time origin.

Stationarizing a cyclostationary process

If $X(t)$ is wide-sense cyclostationary with period T_0 and $D \sim U[0, T_0)$ is independent of X, then $Y(t) = X(t - D)$ is WSS with

$$\mu_Y = \frac{1}{T_0}\int_0^{T_0} \mu_X(t)\, dt, \qquad R_{YY}(\tau) = \bar{R}_{XX}(\tau) = \frac{1}{T_0}\int_0^{T_0} R_{XX}(t+\tau, t)\, dt.$$

Proof. Condition on $D = d$: $E[Y(t+\tau)Y(t) \mid D = d] = R_{XX}(t+\tau-d,\, t-d)$. Averaging over d uniform on one period gives $\frac{1}{T_0}\int_{t-T_0}^{t} R_{XX}(s+\tau, s)\, ds$ with $s = t - d$. Because $s \mapsto R_{XX}(s+\tau, s)$ is T_0-periodic, the integral over any interval of length T_0 is the same, so the result does not depend on t. The mean is handled identically. □

The **period-averaged autocorrelation** $\bar{R}_{XX}(\tau)$ is what a spectrum analyzer effectively measures for a cyclostationary signal, because the analyzer's sweep is not synchronized to the data clock. Example 21.7 is exactly this construction applied to NRZ data.

Non-stationary processes in practice

Many processes are not stationary but are close enough for practical purposes:

- **Asymptotically stationary:** a stable filter switched on at $t = 0$ with a WSS input produces an output whose statistics converge to stationary values as the transient decays (the AR(1) sequence started from $X[0] = 0$ in §21.4 is an example).
- **Locally stationary:** speech is commonly treated as stationary over 20–30 ms frames; a radio channel's fading statistics are stationary over distances of tens of wavelengths.
- **Stationary increments:** the random walk, Poisson process and Wiener process are not stationary, but their increments over fixed lags are. Oscillator phase noise and clock-jitter accumulation are analyzed this way (Allan variance).

Common Pitfall — stationarity belongs to the ensemble

Stationarity is a property of the *ensemble*, not of one record. A single sample function of a stationary process can look "non-stationary" over a short window (a burst of large values happens by chance), and a non-stationary process such as slow thermal drift can look stationary over a short window. Before invoking stationarity, state the time scale over which the assumption is meant to hold, and check that the record is long compared with the correlation time but short compared with the time scale of drift.

21.3 Autocorrelation and cross-correlation

Motivation

The autocorrelation function answers a simple question: *if I know the value of the signal now, how much do I know about its value τ seconds from now?* For thermal noise in a wideband amplifier the answer is "almost nothing after a few nanoseconds"; for the slowly wandering bias of a MEMS gyroscope, "a great deal, even after minutes". The **cross-correlation** asks the same question about two different signals, and it is the workhorse of synchronization, ranging, system identification and interference analysis: whenever an engineer needs to know whether, and with what delay, one signal is contained in another, a correlator is involved.

Throughout this section $X(t)$ and $Y(t)$ are real and (jointly) WSS unless stated otherwise.

Properties of the autocorrelation function

Properties of the autocorrelation function

1. **Power:** $R_{XX}(0) = E[X^2(t)] \ge 0$ is the average power (mean-square value).
2. **Even symmetry:** $R_{XX}(-\tau) = R_{XX}(\tau)$.
3. **Peak at the origin:** $|R_{XX}(\tau)| \le R_{XX}(0)$ for all τ.
4. **Non-negative definiteness:** for all n, times $t_1, \dots, t_n$ and real $a_1, \dots, a_n$,

$$\sum_{i=1}^{n}\sum_{k=1}^{n} a_i a_k\, R_{XX}(t_i - t_k) \ge 0.$$

Equivalently (Bochner's theorem, §21.4), its Fourier transform is non-negative. 5. **Periodicity:** if $R_{XX}(T_0) = R_{XX}(0)$ for some $T_0 \neq 0$, then R_{XX} is periodic with period T_0, and $X(t + T_0) = X(t)$ with probability 1 for every t. 6. **DC component:** if X has no periodic components and $\mu_X \neq 0$, then $R_{XX}(\tau) \to \mu_X^2$ as $|\tau| \to \infty$; i.e. $C_{XX}(\tau) \to 0$. 7. **Continuity:** if R_{XX} is continuous at $\tau = 0$, it is continuous everywhere.

Proofs.

1. Definition with $\tau = 0$.
2. $R_{XX}(-\tau) = E[X(t-\tau)X(t)]$; substitute $s = t - \tau$ and use stationarity: $E[X(s)X(s+\tau)] = R_{XX}(\tau)$.
3. Cauchy–Schwarz for random variables (Ch. 19): $\big(E[X(t+\tau)X(t)]\big)^2 \le E[X^2(t+\tau)]\,E[X^2(t)] = R_{XX}(0)^2$. Alternatively, $0 \le E[(X(t+\tau) \pm X(t))^2] = 2R_{XX}(0) \pm 2R_{XX}(\tau)$.
4. $0 \le E\big[(\sum_i a_i X(t_i))^2\big]$ expands to the double sum.

5. $E[(X(t+T_0)-X(t))^2] = 2R_{XX}(0) - 2R_{XX}(T_0) = 0$, so $X(t+T_0) = X(t)$ with probability 1. Then $R_{XX}(\tau+T_0) = E[X(t+\tau+T_0)X(t)] = E[X(t+\tau)X(t)] = R_{XX}(\tau)$.
6. Write $X = \mu_X + \tilde{X}$ with $\tilde{X}$ zero mean; $R_{XX} = \mu_X^2 + C_{XX}$, and "no periodic components" is the (informal) condition that $C_{XX}(\tau) \to 0$ — values far apart become uncorrelated.
7. $|R_{XX}(\tau+h) - R_{XX}(\tau)| = |E[(X(t+\tau+h) - X(t+\tau))X(t)]| \le \sqrt{E[(X(t+h)-X(t))^2]\,R_{XX}(0)} = \sqrt{2(R_{XX}(0) - R_{XX}(h))R_{XX}(0)}$, which tends to 0 as $h \to 0$ if R_{XX} is continuous at 0. $\square$

Property 7 has a physical reading: $E[(X(t+h)-X(t))^2] = 2[R_{XX}(0) - R_{XX}(h)]$. So R_{XX} continuous at 0 is exactly the statement that the process is **mean-square continuous** — its sample values do not jump on average. A large curvature of R_{XX} at the origin means a rapidly fluctuating process.

Reading an autocorrelation. A typical autocorrelation of a zero-mean noise process is a peaked, even function:

- its **height** $R_{XX}(0)$ is the power;
- its **width** is the correlation time — how long the process "remembers";
- its **shape at the origin** tells how smooth the sample functions are: a cusp ($e^{-\alpha|\tau|}$) means continuous but non-differentiable paths; a rounded top ($e^{-\alpha\tau^2}$, sinc) means smooth paths;
- a **constant floor** μ_X^2 at large lags is DC; a **persistent oscillation** is a sinusoidal component.

Correlation coefficient and correlation time

The normalized autocovariance

$$\rho_{XX}(\tau) = \frac{C_{XX}(\tau)}{C_{XX}(0)} \in [-1, 1]$$

is the **correlation coefficient function**. Several definitions of a single **correlation time** are in use; the two most common are

$$\tau_c = \int_0^\infty \rho_{XX}(\tau)\,d\tau \qquad \text{or} \qquad \tau_e : \ \rho_{XX}(\tau_e) = e^{-1}.$$

For $\rho = e^{-\alpha|\tau|}$ both give $1/\alpha$. Samples spaced much farther apart than τ_c are nearly uncorrelated; this is what makes long averages effective (§21.7).

Worked examples on autocorrelation

Worked Example 21.9 — Random telegraph signal

$X(t)$ takes values ± 1 and switches sign at the event times of a Poisson process of rate λ; $P(X(0) = 1) = \tfrac{1}{2}$, independent of the switching times.

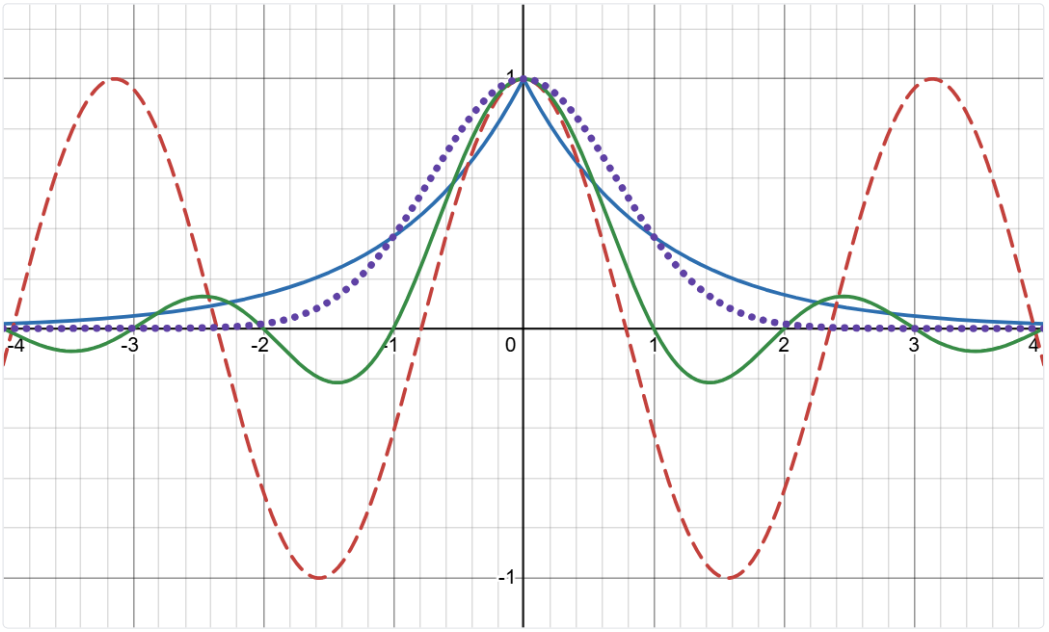
Step 1 — mean. By symmetry $P(X(t) = 1) = \tfrac{1}{2}$ for all t, so $\mu_X = 0$.

Step 2 — the product. $X(t+\tau)X(t) = +1$ if an even number of switches occurs in the interval between t and $t+\tau$ and -1 if odd. The number of switches N in an interval of length $|\tau|$ is Poisson with mean $\lambda|\tau|$.

Step 3 — sum the series.

$$R_{XX}(\tau) = P(N \text{ even}) - P(N \text{ odd}) = \sum_{k=0}^{\infty} (-1)^k \frac{(\lambda|\tau|)^k}{k!} e^{-\lambda|\tau|} = e^{-\lambda|\tau|} e^{-\lambda|\tau|} = e^{-2\lambda|\tau|}.$$

Interpretation. Power $R_{XX}(0) = 1$ (obvious, since $X^2 = 1$), correlation time $\tau_c = 1/(2\lambda)$. With $\lambda = 10^4$ switches per second (a typical MOSFET random-telegraph-noise trap), $\tau_c = 50\ \mu$s; values 200 µs apart have correlation $e^{-4} = 0.018$. The cusp at $\tau = 0$ reflects the jumps of the sample paths.



Four valid autocorrelation shapes: exponential (telegraph or RC-filtered noise, blue — note the cusp), cosine (random-phase sinusoid, red — never decays because the process is periodic), sinc (band-limited white noise, green — negative lobes are allowed) and Gaussian (smooth, differentiable noise, purple). All are even and peak at $\tau = 0$.

Worked Example 21.10 — Is this a valid autocorrelation?

Consider the rectangular function $g(\tau) = 1$ for $|\tau| \le 1$ and 0 otherwise. It is even, non-negative and peaks at 0 — properties 1–3 hold. Is it an autocorrelation?

Test property 4 with times $t_1 = 0,\ t_2 = 1,\ t_3 = 2$. The matrix $[g(t_i - t_k)]$ is

$$\mathbf{G} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \qquad \mathbf{a} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} : \quad \mathbf{a}^{\mathsf{T}}\mathbf{G}\mathbf{a} = 3 + 2(-1 - 1 + 0) = -1 < 0.$$

A "variance" of -1 is impossible, so g **is not an autocorrelation**. (The eigenvalues of $\mathbf{G}$ are 1 and $1 \pm \sqrt{2}$; one is negative.) In the frequency domain the same conclusion is immediate: the Fourier transform of g is $2\sin\omega/\omega$, which goes negative, violating $S_{XX} \ge 0$ (§21.4).

By contrast the triangle $1 - |\tau|$ (Example 21.7) has transform $\sin^2(\omega/2)/(\omega/2)^2 \ge 0$ and is valid. Being even and peaked is necessary but nowhere near sufficient.

Worked Example 21.11 — Moving-average (MA(1)) sequence

Let $W[n]$ be i.i.d. with zero mean and variance σ_W^2, and define the two-tap FIR output $X[n] = W[n] + bW[n-1]$.

Step 1 — mean: $E[X[n]] = 0$.

Step 2 — autocorrelation. $R_{XX}[k] = E\big[(W[n+k] + bW[n+k-1])(W[n] + bW[n-1])\big]$. Using $E[W[i]W[l]] = \sigma_W^2\delta[i-l]$, the four cross terms survive only when their indices match:

$$R_{XX}[0] = \sigma_W^2(1 + b^2), \qquad R_{XX}[\pm 1] = b\,\sigma_W^2, \qquad R_{XX}[k] = 0 \ \ (|k| \ge 2).$$

Step 3 — numbers. With $b = 0.5$, $\sigma_W^2 = 1$: $R_{XX}[0] = 1.25$, $R_{XX}[\pm 1] = 0.5$, $\rho_{XX}[1] = 0.4$. Notice $|\rho_{XX}[1]| = |b|/(1 + b^2) \le \tfrac{1}{2}$ for every b: an MA(1) sequence can never have lag-1 correlation exceeding 0.5 in magnitude. If a measured sequence has $\hat\rho[1] = 0.8$ and $\hat\rho[k] \approx 0$ for $k \ge 2$, it cannot be MA(1) — such constraints are exactly the non-negative-definiteness of property 4.

Generalization. For an FIR filter $X[n] = \sum_{m=0}^{q} b_m W[n-m]$ driven by white noise, $R_{XX}[k] = \sigma_W^2 \sum_m b_m b_{m+k}$ (the deterministic autocorrelation of the coefficient sequence), which vanishes for $|k| > q$.

Derivatives of a WSS process

Many physical quantities are derivatives of others (current in a capacitor, velocity from position), so it is useful to know the correlation of $X'(t)$. If $R_{XX}(\tau)$ is twice differentiable at $\tau = 0$, then $X(t)$ is **mean-square differentiable** and, interchanging derivative and expectation,

$$R_{X'X}(\tau) = E[X'(t+\tau)X(t)] = \frac{d}{d\tau}R_{XX}(\tau), \qquad R_{X'X'}(\tau) = E[X'(t+\tau)X'(t)] = -\frac{d^2}{d\tau^2}R_{XX}(\tau).$$

Derivation. $R_{X'X}(\tau) = \frac{\partial}{\partial t_1}E[X(t_1)X(t_2)]\big|_{t_1 = t+\tau, t_2 = t} = R'_{XX}(\tau)$. Then $R_{X'X'}(\tau) = \frac{\partial}{\partial t_2}R_{X'X}(t_1 - t_2) = -R''_{XX}(\tau)$.

Two consequences: since R_{XX} is even, $R'_{XX}(0) = 0$ and so $E[X'(t)X(t)] = 0$ — **a WSS process and its derivative are uncorrelated at the same instant**. And the derivative's power is $E[X'^2] = -R''_{XX}(0)$, which is finite only if R_{XX} is smooth at the origin.

Worked Example 21.12 — Smooth versus rough noise

(a) $R_{XX}(\tau) = \sigma^2 e^{-\alpha\tau^2}$. Then $R'_{XX} = -2\alpha\tau\sigma^2 e^{-\alpha\tau^2}$ and

$$R_{X'X'}(\tau) = -R''_{XX}(\tau) = 2\alpha\sigma^2(1 - 2\alpha\tau^2)e^{-\alpha\tau^2}, \qquad E[X'^2] = 2\alpha\sigma^2.$$

With $\sigma = 1$ mV and $\alpha = 10^8$ s^{-2}, the RMS slope is $\sqrt{2\times 10^8} \times 10^{-3} = 14.1$ V/s. This value is needed, for instance, to convert voltage noise into timing jitter at a comparator: $\sigma_t \approx \sigma_v/|\text{slope of the signal}|$, and the expected rate of zero crossings (Rice's formula, §21.5).

(b) $R_{XX}(\tau) = \sigma^2 e^{-\alpha|\tau|}$ has a cusp at 0: $R''_{XX}(0)$ does not exist (formally, $-R''_{XX}$ contains $2\alpha\sigma^2\delta(\tau)$). The process is mean-square continuous but **not differentiable** — the derivative would have infinite power. RC-filtered white noise and the telegraph signal are of this kind; their paths are rough at every scale down to where the idealized model breaks down.

Cross-correlation

Cross-correlation and cross-covariance

For jointly WSS X and Y:

$$R_{XY}(\tau) = E[X(t+\tau)Y(t)], \qquad C_{XY}(\tau) = R_{XY}(\tau) - \mu_X\mu_Y.$$

X and Y are **uncorrelated** if $C_{XY}(\tau) = 0$ for all τ, and **orthogonal** if $R_{XY}(\tau) = 0$ for all τ. Independent processes are uncorrelated; zero-mean uncorrelated processes are orthogonal.

The ordering convention matters: here the *first* subscript carries the time shift $+\tau$. Some texts use $E[X(t)Y(t+\tau)]$, which reverses the sign of τ. Check the convention before using a published formula.

Properties of the cross-correlation

1. $R_{XY}(\tau) = R_{YX}(-\tau)$ — **not** even in general; its peak may be anywhere.
2. $|R_{XY}(\tau)| \le \sqrt{R_{XX}(0) R_{YY}(0)}$.
3. $|R_{XY}(\tau)| \le \tfrac{1}{2}\big[R_{XX}(0) + R_{YY}(0)\big]$.
4. If $Z = X + Y$: $R_{ZZ}(\tau) = R_{XX}(\tau) + R_{YY}(\tau) + R_{XY}(\tau) + R_{YX}(\tau)$. For uncorrelated X, Y with at least one of them zero mean, $R_{ZZ} = R_{XX} + R_{YY}$: **powers of uncorrelated zero-mean processes add**.

Proofs. (1) Substitute $s = t + \tau$ in $E[X(t+\tau)Y(t)] = E[Y(s-\tau)X(s)] = R_{YX}(-\tau)$. (2) Cauchy–Schwarz. (3) The geometric mean is at most the arithmetic mean. (4) Expand $E[(X(t+\tau) + Y(t+\tau))(X(t) + Y(t))]$. □

Property 4 is the reason noise powers, not noise voltages, are added when independent sources are combined: two independent 1 mV-rms noise sources in series give $\sqrt{2}$ mV rms, not 2 mV.

Worked Example 21.13 — Signal plus independent noise

A WSS signal $S(t)$ with $R_{SS}(\tau) = 4e^{-100|\tau|}$ V^2 is corrupted by independent zero-mean noise $N(t)$ with $R_{NN}(\tau) = 0.25e^{-10^5|\tau|}$ V^2. The observed process is $X = S + N$.

Autocorrelation. By property 4 with $R_{SN} = R_{NS} = E[S]E[N] = 0$:

$$R_{XX}(\tau) = 4e^{-100|\tau|} + 0.25e^{-10^5|\tau|}.$$

Total power $R_{XX}(0) = 4.25$ V^2 (2.06 V rms); SNR $= 4/0.25 = 16$ (12.0 dB).

Separating them by lag. At $\tau = 50\ \mu$s the noise term is $0.25e^{-5} = 0.0017$ while the signal term is $4e^{-0.005} = 3.98$. A correlator evaluated at a lag longer than the noise correlation time but shorter than the signal's sees almost pure signal: the "spike" at the origin of a measured autocorrelation is the signature of wideband noise, and its height is the noise power. This is how noise power is often estimated in practice without switching the signal off.

Cross-correlation with the clean signal. $R_{XS}(\tau) = E[(S(t+\tau) + N(t+\tau))S(t)] = R_{SS}(\tau)$: independent noise drops out of a cross-correlation entirely (on average).

Worked Example 21.14 — Time-delay estimation

A transmitted signal $X(t)$ (WSS, zero mean, $R_{XX}(\tau) = e^{-1000|\tau|}$) arrives as $Y(t) = 0.5\,X(t - D) + N(t)$ with $D = 2$ ms and noise $N(t)$ zero mean and independent of X.

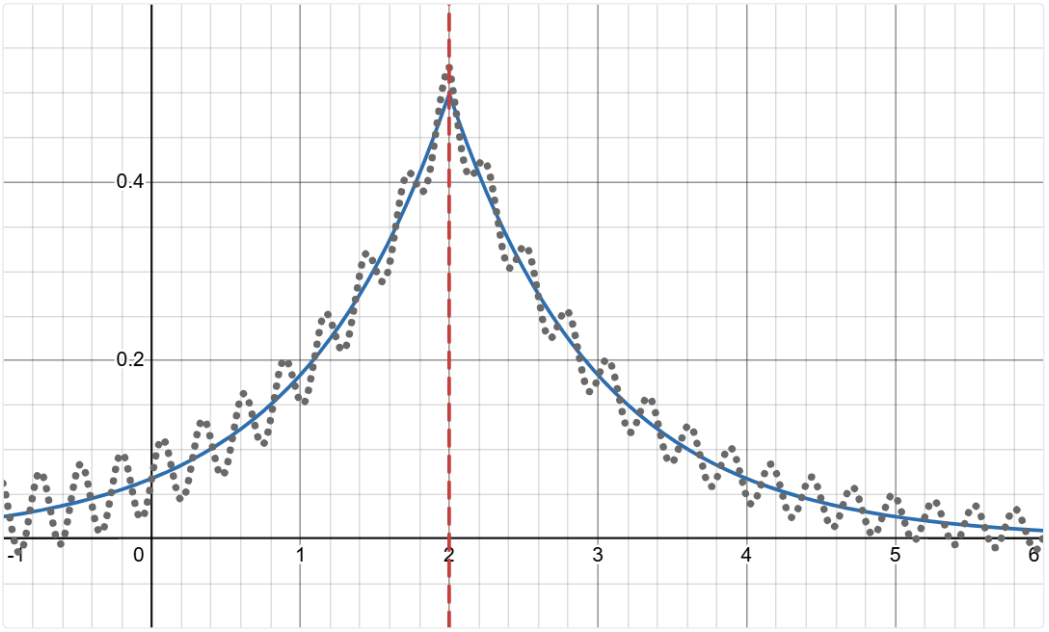
Step 1.

$$R_{YX}(\tau) = E[Y(t+\tau)X(t)] = 0.5\,E[X(t+\tau-D)X(t)] + E[N(t+\tau)]E[X(t)] = 0.5\,R_{XX}(\tau - D).$$

Step 2. So $R_{YX}(\tau) = 0.5\,e^{-1000|\tau - 0.002|}$, which peaks with value 0.5 at $\tau = D = 2$ ms. By property 1, $R_{XY}(\tau) = R_{YX}(-\tau)$ peaks at $\tau = -2$ ms.

Step 3 — resolution. The peak's half-width (where R_{YX} falls to $0.5\,e^{-1}$) is ± 1 ms. Two echoes closer than about one correlation time of X merge into one peak. Hence radar and GPS use signals with very **short correlation times** (wide bandwidth): a GPS C/A code with chip duration $T_c \approx 0.98\ \mu$s has a triangular autocorrelation of half-width T_c, corresponding to about 300 m of range, and receivers interpolate the peak to a small fraction of that.

Independent noise does not shift the peak; it only adds random fluctuations to the *estimated* correlation, whose size decreases with averaging time (§21.7).



Cross-correlation $R_{YX}(\tau)$ of Example 21.14 with τ in milliseconds (blue): the peak locates the delay $D = 2$ ms. The grey dotted curve suggests what a finite-length estimate looks like — fluctuations from the noise ride on top, but the peak location is preserved.

Engineering Note — correlators everywhere

In discrete time, $R_{XX}[k] = E[X[n+k]X[n]]$ and $R_{XY}[k] = E[X[n+k]Y[n]]$. Correlators in GNSS receivers (code phase search), preamble synchronizers in Wi-Fi and Ethernet PHYs, lock-in amplifiers (correlation with a reference sinusoid), time-domain reflectometers and acoustic source localizers all compute sample cross-correlations against a known reference and then look for the peak. The mathematics of this section — especially "independent noise drops out of R_{YX}" — is the reason they work.

Common Pitfall — correlation is not causation, and zero correlation is not independence

$R_{XY} \equiv 0$ says only that there is no *linear* relation at any lag. $Y(t) = X^2(t) - E[X^2]$ for a zero-mean Gaussian X is uncorrelated with X (odd moments of a Gaussian vanish) but completely determined by it. For jointly Gaussian processes, and only then, uncorrelated implies independent (§21.5).

21.4 Power spectral density

Motivation

Filters, amplifiers and channels are characterized by their frequency response, so the natural question about noise is *how its power is distributed over frequency*. For a deterministic energy signal $x(t)$, Parseval's theorem (Ch. 16) says $|X(\omega)|^2/(2\pi)$ is the energy per unit angular frequency. A sample function of a WSS process, however, has the same statistics forever, so its energy is infinite and its Fourier transform does not exist in the ordinary sense. The way out is to work with **power** per unit bandwidth, averaged over the ensemble. The remarkable fact — the Wiener–Khinchin theorem — is that this spectral power density is simply the Fourier transform of the autocorrelation function. Correlation in the time domain and power distribution in the frequency domain are two views of the same second-order information.

Definition via the truncated transform

Power spectral density

For a WSS process $X(t)$, let $X_T(t) = X(t)$ for $|t| \le T$ and 0 otherwise, with Fourier transform $X_T(\omega) = \int_{-T}^{T} X(t) e^{-j\omega t} dt$. The **power spectral density** (PSD) is

$$S_{XX}(\omega) = \lim_{T \to \infty} \frac{E\big[|X_T(\omega)|^2\big]}{2T}.$$

The quantity $|X_T(\omega)|^2/(2T)$ for one sample function is the **periodogram**: energy density of the truncated record divided by its duration, i.e. power density. The PSD is its ensemble average in the limit of a long record.

The Wiener–Khinchin theorem

Wiener–Khinchin theorem

If $X(t)$ is WSS and $\int_{-\infty}^{\infty} |\tau R_{XX}(\tau)|\, d\tau < \infty$ (so that the limit below may be taken; the result holds in the distributional sense under much weaker conditions), then the PSD and the autocorrelation are a Fourier-transform pair:

$$S_{XX}(\omega) = \int_{-\infty}^{\infty} R_{XX}(\tau)\, e^{-j\omega\tau}\, d\tau, \qquad R_{XX}(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XX}(\omega)\, e^{j\omega\tau}\, d\omega.$$

Proof. Write the squared magnitude as a double integral and take the expectation inside:

$$E\big[|X_T(\omega)|^2\big] = E\Big[\int_{-T}^{T}\!\!\int_{-T}^{T} X(t)X(s)e^{-j\omega(t-s)}\, dt\, ds\Big] = \int_{-T}^{T}\!\!\int_{-T}^{T} R_{XX}(t-s)e^{-j\omega(t-s)}\, dt\, ds.$$

The integrand depends only on $\tau = t - s$. In the (t, s) square of side $2T$, the line $t - s = \tau$ has length proportional to $2T - |\tau|$ for $|\tau| \le 2T$; changing variables to (τ, s) gives

$$E\big[|X_T(\omega)|^2\big] = \int_{-2T}^{2T} (2T - |\tau|)\, R_{XX}(\tau) e^{-j\omega\tau}\, d\tau.$$

Dividing by $2T$:

$$\frac{E|X_T(\omega)|^2}{2T} = \int_{-2T}^{2T} \Big(1 - \frac{|\tau|}{2T}\Big) R_{XX}(\tau) e^{-j\omega\tau}\, d\tau.$$

As $T \to \infty$ the triangular weight tends to 1 for each fixed τ, and the error term $\frac{1}{2T}\int |\tau R_{XX}(\tau)| d\tau \to 0$ under the stated condition. The limit is $\int R_{XX}(\tau) e^{-j\omega\tau} d\tau$. The inverse relation is the Fourier inversion theorem. $\square$

The triangular weight in the proof is worth remembering: a finite record of length $2T$ "sees" the autocorrelation through a triangular window, which is why spectral estimates from finite data are always smeared (§21.7).

A deep converse is **Bochner's theorem**: a continuous function $R(\tau)$ is non-negative definite (property 4 of §21.3) if and only if it is the inverse Fourier transform of a non-negative (possibly generalized) spectral measure. So a candidate autocorrelation is valid exactly when its transform is non-negative — the easiest test in practice (Example 21.10).

Properties of the PSD

Properties of the PSD

1. $S_{XX}(\omega)$ is **real** and **non-negative**: $S_{XX}(\omega) \geq 0$.
2. For real X, $S_{XX}(-\omega) = S_{XX}(\omega)$ (even).
3. **Total power:**
$$E[X^2(t)] = R_{XX}(0) = \frac{1}{2\pi}\int_{-\infty}^{\infty} S_{XX}(\omega)\,d\omega = \int_{-\infty}^{\infty} S_{XX}(2\pi f)\,df.$$
4. **Power in a band.** The power in the band $\omega_1 \leq |\omega| \leq \omega_2$ (both signs of frequency counted) is
$$P_{[\omega_1,\omega_2]} = \frac{1}{\pi}\int_{\omega_1}^{\omega_2} S_{XX}(\omega)\,d\omega.$$
5. **DC and periodic components.** A non-zero mean contributes $2\pi\mu_X^2\,\delta(\omega)$; a sinusoidal component of amplitude A at ω_0 contributes $\frac{\pi A^2}{2}[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)]$.
6. **Zero-frequency value:** $S_{XX}(0) = \int R_{XX}(\tau)\,d\tau$ — the area under the autocorrelation. For a zero-mean process this governs the variance of long time averages (§21.7).

Proofs. (1) Non-negativity is immediate from the definition as a limit of $E|X_T|^2/(2T) \geq 0$. Reality follows from R_{XX} being even: $S_{XX}(\omega) = \int R_{XX}(\tau)\cos\omega\tau\,d\tau$. (2) Cosine is even. (3) Set $\tau = 0$ in the inverse transform. (4) Integrate the PSD over $\omega_1 \leq |\omega| \leq \omega_2$ and use evenness. That this really is "the power passed by an ideal bandpass filter" follows from the LTI result $S_{YY} = |H|^2 S_{XX}$ below. (5) Transform of a constant and of a cosine. (6) Set $\omega = 0$. □

Common Pitfall — ω versus f, one-sided versus two-sided

With the angular-frequency convention used here, power is $\frac{1}{2\pi}\int S_{XX}(\omega)\,d\omega$. Expressed per hertz, $S_X(f) = S_{XX}(2\pi f)$ and power is $\int_{-\infty}^{\infty} S_X(f)\,df$, with units V^2/Hz. Instrument data sheets and spectrum analyzers usually quote the **one-sided** PSD $G_X(f) = 2S_X(f)$ for $f \geq 0$, so that power is $\int_0^{\infty} G_X(f)\,df$. Noise densities in $\mathrm{nV}/\sqrt{\mathrm{Hz}}$ are $\sqrt{G_X(f)}$. Mixing these conventions produces factor-of-2 or factor-of-2π errors; write the convention next to every PSD you quote.

Standard transform pairs

With $\alpha > 0$, sinc-type expressions written out explicitly:

$R_{XX}(\tau)$	$S_{XX}(\omega)$	**origin**
$\sigma^2 e^{-\alpha\lvert\tau\rvert}$	$\dfrac{2\alpha\sigma^2}{\alpha^2+\omega^2}$	RC-filtered white noise, telegraph signal
$\dfrac{A^2}{2}\cos\omega_0\tau$	$\dfrac{\pi A^2}{2}\big[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)\big]$	random-phase sinusoid
μ^2 (constant)	$2\pi\mu^2\delta(\omega)$	DC
$\dfrac{N_0}{2}\delta(\tau)$	$\dfrac{N_0}{2}$	white noise
$N_0 B\,\dfrac{\sin 2\pi B\tau}{2\pi B\tau}$	$\dfrac{N_0}{2}$ for $\lvert\omega\rvert < 2\pi B$, else 0	band-limited white noise
$\sigma^2\big(1-\lvert\tau\rvert/T\big)$ for $\lvert\tau\rvert \leq T$, else 0	$\sigma^2 T\,\dfrac{\sin^2(\omega T/2)}{(\omega T/2)^2}$	random NRZ data
$\sigma^2 e^{-\alpha\tau^2}$	$\sigma^2\sqrt{\pi/\alpha}\;e^{-\omega^2/(4\alpha)}$	smooth (Gaussian-shaped) noise
$\sigma^2 e^{-\alpha\lvert\tau\rvert}\cos\omega_0\tau$	$\alpha\sigma^2\Big[\dfrac{1}{\alpha^2+(\omega-\omega_0)^2}+\dfrac{1}{\alpha^2+(\omega+\omega_0)^2}\Big]$	narrowband (bandpass) noise

The last pair follows from the first by the modulation theorem: multiplying R_{XX} by $\cos\omega_0\tau$ shifts half the spectrum to $+\omega_0$ and half to $-\omega_0$.

Worked examples

Worked Example 21.15 — Exponential autocorrelation and the Lorentzian spectrum

Let $R_{XX}(\tau) = \sigma^2 e^{-\alpha|\tau|}$.

Step 1 — transform. Split the integral at $\tau = 0$:

$$S_{XX}(\omega) = \sigma^2\left[\int_0^\infty e^{-(\alpha+j\omega)\tau}d\tau + \int_0^\infty e^{-(\alpha-j\omega)\tau}d\tau\right] = \sigma^2\left[\frac{1}{\alpha+j\omega} + \frac{1}{\alpha-j\omega}\right] = \frac{2\alpha\sigma^2}{\alpha^2+\omega^2}.$$

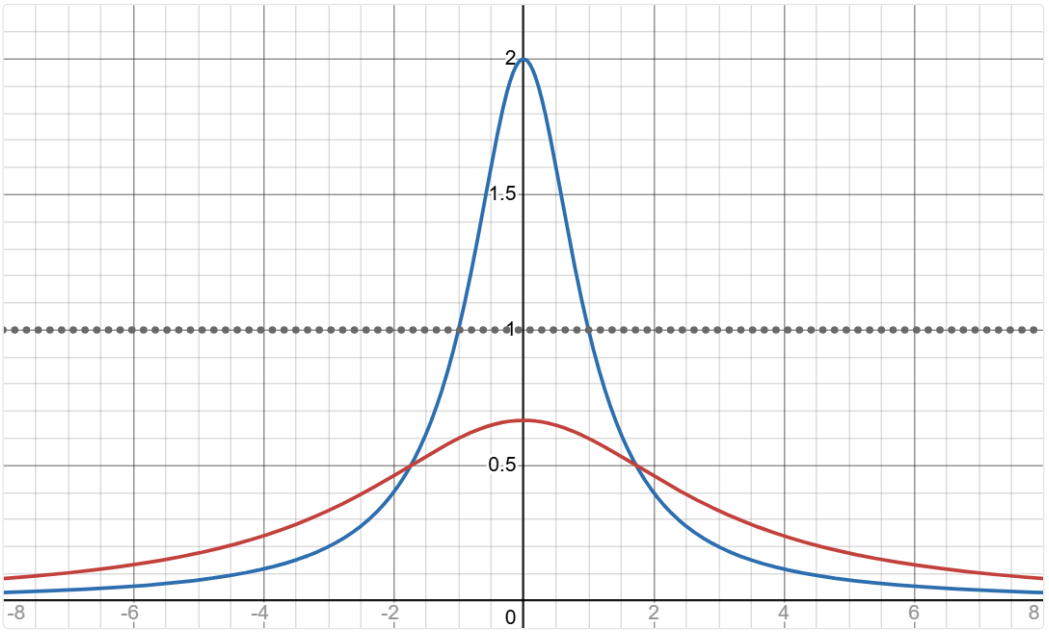
Step 2 — check the power. $\frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{2\alpha\sigma^2}{\alpha^2+\omega^2}d\omega = \frac{2\alpha\sigma^2}{2\pi}\cdot\frac{\pi}{\alpha} = \sigma^2 = R_{XX}(0)$. ✓

Step 3 — power in a band. By property 4, the power in $|\omega| \le \omega_1$ is

$$\frac{1}{\pi}\int_0^{\omega_1}\frac{2\alpha\sigma^2}{\alpha^2+\omega^2}d\omega = \frac{2\sigma^2}{\pi}\arctan\frac{\omega_1}{\alpha}.$$

With $\alpha = 1000\ \text{s}^{-1}$ and a 1 kHz bandwidth ($\omega_1 = 2\pi\times 1000$), the fraction is $\frac{2}{\pi}\arctan(6.283) = 0.900$: 90% of the power lies below 1 kHz. Half the power lies below $\omega = \alpha$ (159 Hz), which is also the half-power (−3 dB) point of the spectrum.

Step 4 — interpretation. Shorter correlation time (larger α) means a wider, flatter spectrum with the same total power: time scale and bandwidth trade off exactly as for deterministic signals (Ch. 16). The value at the origin, $S_{XX}(0) = 2\sigma^2/\alpha = \int R_{XX}d\tau$, confirms property 6.



Lorentzian PSDs $S_{XX}(\omega) = 2\alpha/(\alpha^2+\omega^2)$ for $\alpha = 1$ (blue) and $\alpha = 3$ (red), both with $\sigma^2 = 1$. The areas under the two curves are equal (same power); faster decorrelation spreads that power over a wider band. The grey line marks half the peak of the blue curve, reached at $\omega = \alpha = 1$.

Worked Example 21.16 — Spectrum of random NRZ data

From Example 21.7, $R_{XX}(\tau) = A^2(1 - |\tau|/T)$ for $|\tau| < T$.

Step 1 — transform. The triangle of half-width T and height 1 is the self-convolution of a rectangle of width T and height $1/\sqrt{T}$; its transform is therefore the square of the rectangle's transform:

$$S_{XX}(\omega) = A^2 T\,\frac{\sin^2(\omega T/2)}{(\omega T/2)^2}.$$

In hertz, $S_X(f) = A^2T\left[\sin(\pi fT)/(\pi fT)\right]^2$, with nulls at $f = k/T$, $k \neq 0$.

Step 2 — power in the main lobe. Total power is A^2. The fraction in $|f| < 1/T$ is

$$\int_{-1/T}^{1/T} T\left[\frac{\sin \pi fT}{\pi fT}\right]^2 df = \int_{-1}^{1}\left[\frac{\sin \pi u}{\pi u}\right]^2 du \approx 0.903.$$

So a 1 Gb/s NRZ stream ($T = 1$ ns) has 90.3% of its power below 1 GHz, while the remaining 9.7% in the sidelobes is what causes crosstalk and EMI — the motivation for pulse shaping.

Step 3 — check at DC. $S_{XX}(0) = A^2T = \int R_{XX}\,d\tau$ (area of the triangle $= A^2T$). ✓ There is no impulse at DC because the data are equally likely $\pm A$ (zero mean). With unipolar data $\{0, A\}$ the mean is $A/2$ and an impulse $2\pi(A/2)^2\delta(\omega)$ appears.

Worked Example 21.17 — Separating a sinusoid from noise in the spectrum

$X(t) = 2\cos(2\pi \cdot 50\, t + \Theta) + N(t)$, $\Theta \sim U[0, 2\pi)$, with N independent zero-mean noise of $R_{NN}(\tau) = 0.5e^{-200|\tau|}$. Then

$$R_{XX}(\tau) = 2\cos(100\pi\tau) + 0.5e^{-200|\tau|}, \qquad S_{XX}(\omega) = 2\pi\big[\delta(\omega - 100\pi) + \delta(\omega + 100\pi)\big] + \frac{200}{200^2 + \omega^2}.$$

The sinusoid's power 2 sits in two impulses of weight 2π each (check: $\frac{1}{2\pi} \cdot 2 \cdot 2\pi = 2$); the noise's power 0.5 is spread as a Lorentzian. In a measured spectrum with resolution bandwidth Δf, a line appears as a peak of height $\approx P_{\text{line}}/\Delta f$ while the noise floor stays at its density; halving Δf raises the line 3 dB above the floor. In the autocorrelation the same separation appears in the lag domain: the cosine persists forever, the noise dies out within about $5/200 = 25$ ms.

Cross-power spectral density

Cross-power spectral density

For jointly WSS X, Y:

$$S_{XY}(\omega) = \int_{-\infty}^{\infty} R_{XY}(\tau) e^{-j\omega\tau} d\tau, \qquad S_{YX}(\omega) = S_{XY}^*(\omega) \;\; (\text{real } X, Y).$$

S_{XY} is complex in general; its phase carries delay information. The **magnitude-squared coherence** is

$$\gamma_{XY}^2(\omega) = \frac{|S_{XY}(\omega)|^2}{S_{XX}(\omega) S_{YY}(\omega)} \in [0, 1],$$

which measures how linearly related X and Y are at each frequency (Cauchy–Schwarz in the frequency domain).

For the delayed-echo model of Example 21.14, $R_{YX}(\tau) = 0.5 R_{XX}(\tau - D)$ gives $S_{YX}(\omega) = 0.5\, e^{-j\omega D} S_{XX}(\omega)$: a linear phase of slope $-D$. Measuring that slope estimates the delay; this is the frequency-domain twin of peak-picking the cross-correlation.

LTI systems with random inputs

This is the single most-used result of the chapter. Let a WSS process $X(t)$ drive a stable LTI system with impulse response $h(t)$ and frequency response $H(\omega) = \int h(t) e^{-j\omega t} dt$, producing

$$Y(t) = \int_{-\infty}^{\infty} h(\alpha) X(t - \alpha)\, d\alpha.$$

Input–output relations for LTI systems

Y is WSS, jointly WSS with X, and

$$\begin{aligned} \mu_Y &= \mu_X\, H(0), \\ R_{YX}(\tau) &= h(\tau) * R_{XX}(\tau), & S_{YX}(\omega) &= H(\omega)\, S_{XX}(\omega), \\ R_{XY}(\tau) &= h(-\tau) * R_{XX}(\tau), & S_{XY}(\omega) &= H^*(\omega)\, S_{XX}(\omega), \\ R_{YY}(\tau) &= h(\tau) * h(-\tau) * R_{XX}(\tau), & S_{YY}(\omega) &= |H(\omega)|^2 S_{XX}(\omega). \end{aligned}$$

The output power is

$$E[Y^2] = R_{YY}(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} |H(\omega)|^2 S_{XX}(\omega)\, d\omega.$$

If X is Gaussian, so is Y (§21.5).

Derivation.

- **Mean.** $E[Y(t)] = \int h(\alpha) E[X(t-\alpha)] d\alpha = \mu_X \int h(\alpha) d\alpha = \mu_X H(0)$ — the DC gain applied to the DC value.
- **Cross-correlation.** $R_{YX}(\tau) = E[Y(t+\tau)X(t)] = \int h(\alpha) E[X(t+\tau-\alpha)X(t)]\, d\alpha = \int h(\alpha) R_{XX}(\tau - \alpha)\, d\alpha = (h * R_{XX})(\tau)$. Independent of t, so X, Y are jointly WSS.
- **Autocorrelation.** $R_{YY}(\tau) = E[Y(t+\tau)Y(t)] = \int h(\beta) E[Y(t+\tau)X(t-\beta)]\, d\beta = \int h(\beta) R_{YX}(\tau + \beta)\, d\beta$. With $\beta \to -\beta$ this is the convolution of R_{YX} with $h(-\tau)$.
- **Spectra.** Transform the convolutions, using $\mathcal{F}\{h(-\tau)\} = H(-\omega) = H^*(\omega)$ for real h. Then $S_{YY} = H\, H^*\, S_{XX} = |H|^2 S_{XX}$. □

Notice that only $|H|$ affects the output PSD: the **phase response of a filter has no effect on the output noise power or spectrum**, although it does affect the cross-spectrum S_{YX} and hence the shape of the noise waveforms.

More generally, if $Y_1 = h_1 * X$ and $Y_2 = h_2 * X$ are two filtered versions of the same input,

$$S_{Y_1 Y_2}(\omega) = H_1(\omega) H_2^*(\omega) S_{XX}(\omega).$$

If the passbands of H_1 and H_2 do not overlap, $S_{Y_1 Y_2} \equiv 0$: **outputs of non-overlapping filters are uncorrelated** — the basis of filter-bank spectrum analyzers and of treating different frequency bins of noise as independent.

Two immediate uses of the theorem:

- **System identification with white noise.** If $S_{XX} = N_0/2$ then $R_{YX}(\tau) = \frac{N_0}{2}h(\tau)$: cross-correlating the output with a white input measures the impulse response directly. Pseudo-random binary sequences (PRBS), whose autocorrelation is very nearly an impulse, are used for exactly this in acoustics and in-circuit test.
- **Shaping filters.** Any rational PSD $S(\omega) = |H(\omega)|^2 N_0/2$ can be generated by filtering white noise through H — the basis of noise simulation and of Kalman-filter noise models. Finding a stable, minimum-phase H from a given S is called **spectral factorization**.

Worked Example 21.18 — White noise through an RC lowpass

$H(\omega) = 1/(1 + j\omega RC)$ and $S_{XX} = N_0/2$ (white noise, §21.6).

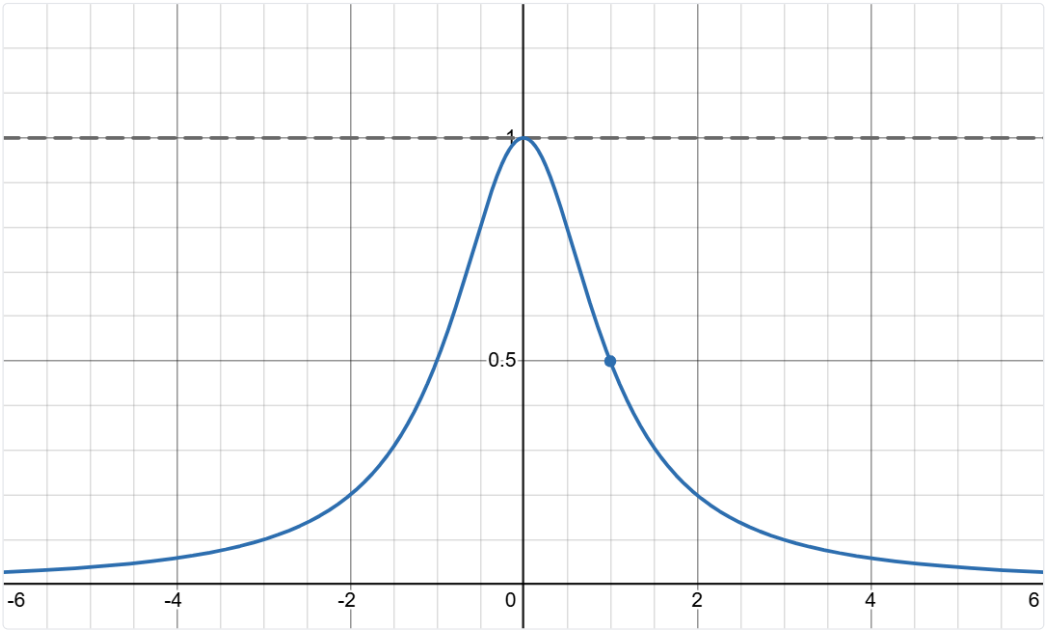
Step 1 — output PSD. $S_{YY} = |H|^2 S_{XX} = \dfrac{N_0/2}{1 + \omega^2 R^2 C^2}$. Writing $\alpha = 1/(RC)$, $S_{YY} = \dfrac{N_0}{2} \cdot \dfrac{\alpha^2}{\alpha^2 + \omega^2} = \dfrac{N_0 \alpha}{4} \cdot \dfrac{2\alpha}{\alpha^2 + \omega^2}$.

Step 2 — output autocorrelation. Matching with Example 21.15 ($\sigma^2 = N_0\alpha/4$):

$$R_{YY}(\tau) = \frac{N_0}{4RC}\, e^{-|\tau|/RC}, \qquad E[Y^2] = \frac{N_0}{4RC}.$$

Step 3 — cross-check in the time domain. $h(t) = \alpha e^{-\alpha t}u(t)$, so $R_{YY}(0) = \frac{N_0}{2}\int h^2 dt = \frac{N_0}{2} \cdot \frac{\alpha^2}{2\alpha} = \frac{N_0\alpha}{4}$. ✓ (For white input, $E[Y^2] = \frac{N_0}{2}\int h^2(t)\,dt$ — the output power is the input density times the filter's impulse-response energy.)

Step 4 — numbers. $R = 1$ kΩ, $C = 1$ μF ($RC = 1$ ms), $N_0/2 = 10^{-6}$ V²/Hz: $E[Y^2] = 2 \times 10^{-6}/(4 \times 10^{-3}) = 5 \times 10^{-4}$ V², i.e. 22.4 mV rms. The output samples decorrelate on the time scale RC: white noise has become exponentially correlated (Lorentzian) noise.



White input PSD (normalized to 1, dashed) and the output PSD of a first-order RC lowpass, $|H(\omega)|^2 = 1/(1 + \omega^2 R^2 C^2)$, plotted against ωRC. The filter turns white noise into Lorentzian, exponentially correlated noise.

Worked Example 21.19 — Coloured noise through a differentiator-like highpass

The noise of Example 21.15 ($\sigma^2 = 1$ V², $\alpha = 1000$ s^{-1}) is applied to a first-order highpass $H(\omega) = \dfrac{j\omega\tau_0}{1 + j\omega\tau_0}$ with $\tau_0 = 1$ ms (corner at $\omega = 1000$ rad/s, equal to α). Find the output power.

Step 1. $|H|^2 = \dfrac{\omega^2\tau_0^2}{1 + \omega^2\tau_0^2} = \dfrac{\omega^2}{\alpha^2 + \omega^2}$ (since $1/\tau_0 = \alpha$).

Step 2. $S_{YY} = \dfrac{\omega^2}{\alpha^2 + \omega^2} \cdot \dfrac{2\alpha\sigma^2}{\alpha^2 + \omega^2} = \dfrac{2\alpha\sigma^2\omega^2}{(\alpha^2 + \omega^2)^2}$.

Step 3. Using $\int_{-\infty}^{\infty} \frac{\omega^2}{(\alpha^2+\omega^2)^2} d\omega = \frac{\pi}{2\alpha}$,

$$E[Y^2] = \frac{1}{2\pi} \cdot 2\alpha\sigma^2 \cdot \frac{\pi}{2\alpha} = \frac{\sigma^2}{2} = 0.5 \text{ V}^2.$$

A highpass whose corner coincides with the noise's own corner removes exactly half the noise power. The complementary lowpass $1/(1 + j\omega\tau_0)$ passes $\frac{1}{2\pi}\int \frac{\alpha^2}{\alpha^2+\omega^2}\frac{2\alpha\sigma^2}{\alpha^2+\omega^2} d\omega = \sigma^2/2$ as well: the two outputs' powers add to the input power here because $|H_{LP}|^2 + |H_{HP}|^2 = 1$ — a power-complementary pair.

Discrete-time random sequences

All of the above has an exact discrete-time counterpart, which is what is actually computed in DSP systems.

Discrete-time Wiener–Khinchin and LTI relations

For a WSS sequence with $\sum_k |R_{XX}[k]| < \infty$,

$$S_{XX}(e^{j\Omega}) = \sum_{k=-\infty}^{\infty} R_{XX}[k]\, e^{-j\Omega k}, \qquad R_{XX}[k] = \frac{1}{2\pi}\int_{-\pi}^{\pi} S_{XX}(e^{j\Omega}) e^{j\Omega k}\, d\Omega,$$

a 2π-periodic, real, non-negative spectrum in the normalized frequency Ω (rad/sample). The power is $E[X^2[n]] = \frac{1}{2\pi}\int_{-\pi}^{\pi} S_{XX}(e^{j\Omega})d\Omega$. If $Y[n] = \sum_m h[m]X[n-m]$ with $H(e^{j\Omega}) = \sum_m h[m]e^{-j\Omega m}$, then

$$\mu_Y = \mu_X H(e^{j0}), \qquad S_{YX}(e^{j\Omega}) = H(e^{j\Omega})S_{XX}(e^{j\Omega}), \qquad S_{YY}(e^{j\Omega}) = |H(e^{j\Omega})|^2 S_{XX}(e^{j\Omega}).$$

In z-transform form, $S_{YY}(z) = H(z)H(z^{-1})S_{XX}(z)$.

The derivation is word-for-word the continuous one with sums for integrals. When a continuous process is sampled at rate $f_s = 1/T_s$, $\Omega = \omega T_s$, and the sampled PSD is the aliased sum $S_{X_d}(e^{j\Omega}) = \frac{1}{T_s}\sum_m S_{XX}\left(\frac{\Omega - 2\pi m}{T_s}\right)$; for a process band-limited below $f_s/2$ there is no aliasing.

The AR(1) sequence

First-order autoregressive (AR(1)) sequence

$$X[n] = aX[n-1] + W[n], \qquad |a| < 1,$$

where $W[n]$ is zero-mean white noise with variance σ_W^2 ("innovations"), with $W[n]$ uncorrelated with past values $X[n-1], X[n-2], \ldots$.

This is the discrete-time analogue of RC-filtered white noise and the simplest model of a correlated sequence. We derive its statistics in three independent ways.

(i) As a filtered white sequence. The recursion is an LTI filter with $H(z) = \dfrac{1}{1 - az^{-1}}$, $h[n] = a^n u[n]$. In steady state,

$$S_{XX}(e^{j\Omega}) = \frac{\sigma_W^2}{|1 - ae^{-j\Omega}|^2} = \frac{\sigma_W^2}{1 - 2a\cos\Omega + a^2},$$

and the variance is $\sigma_W^2 \sum_{n\ge 0} h^2[n] = \sigma_W^2 \sum a^{2n} = \dfrac{\sigma_W^2}{1-a^2}$.

(ii) Yule–Walker recursion. Multiply the recursion by $X[n-k]$ ($k \ge 1$) and take expectations. Since $W[n]$ is uncorrelated with the past, $E[W[n]X[n-k]] = 0$, giving

$$R_{XX}[k] = aR_{XX}[k-1] \quad (k \ge 1) \qquad \Longrightarrow \qquad R_{XX}[k] = R_{XX}[0]\, a^{|k|}.$$

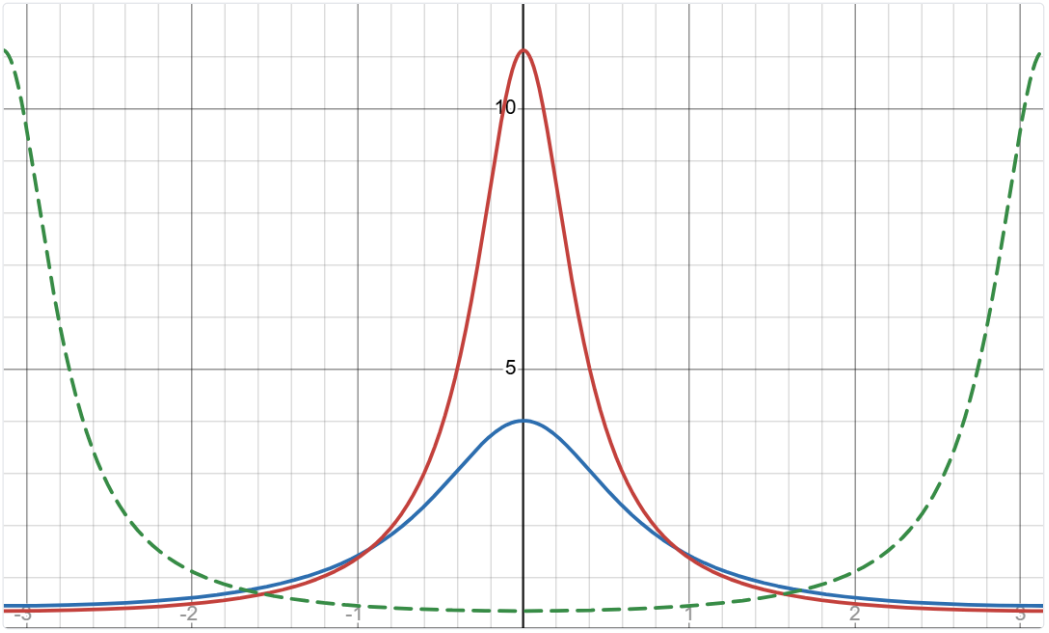
For $k = 0$: $R_{XX}[0] = E[(aX[n-1] + W[n])^2] = a^2 R_{XX}[0] + \sigma_W^2$, so $R_{XX}[0] = \sigma_W^2/(1-a^2)$. Hence

$$R_{XX}[k] = \frac{\sigma_W^2}{1-a^2}\, a^{|k|}.$$

(iii) Wiener–Khinchin check. Summing the two geometric series,

$$\sum_k \frac{\sigma_W^2}{1-a^2} a^{|k|} e^{-j\Omega k} = \frac{\sigma_W^2}{1-a^2}\left[\frac{1}{1-ae^{-j\Omega}} + \frac{1}{1-ae^{j\Omega}} - 1\right] = \frac{\sigma_W^2}{1 - 2a\cos\Omega + a^2}. \checkmark$$

The spectrum is lowpass for $0 < a < 1$ (peak $\sigma_W^2/(1-a)^2$ at $\Omega = 0$) and highpass for $-1 < a < 0$ (peak at $\Omega = \pi$, with alternating-sign correlation $R[k] \propto a^{|k|}$). As $a \to 1$ the correlation time $-1/\ln a \approx 1/(1-a)$ samples grows and the spectrum concentrates near DC; at $a = 1$ the recursion becomes the random walk of Example 21.2, which has no stationary state.



AR(1) power spectra $S_{XX}(e^{j\Omega}) = 1/(1 - 2a\cos\Omega + a^2)$ over one period $-\pi \le \Omega \le \pi$ ($\sigma_W^2 = 1$): $a = 0.5$ (blue) and $a = 0.7$ (red) are lowpass, sharper for larger a; $a = -0.7$ (green, dashed) is the mirror-image highpass spectrum.

Worked Example 21.20 — AR(1) statistics and start-up transient

Take $a = 0.9$, $\sigma_W^2 = 1$.

Step 1 — steady state. $\sigma_X^2 = 1/(1 - 0.81) = 5.263$; $\rho[k] = 0.9^{|k|}$, so $\rho[10] = 0.349$ and $\rho[22] = 0.098$. The spectrum ranges from $S(e^{j0}) = 1/(0.1)^2 = 100$ down to $S(e^{j\pi}) = 1/(1.9)^2 = 0.277$ — a 25.6 dB lowpass. Its half-power point solves $1 - 1.8\cos\Omega + 0.81 = 2(0.01)$, giving $\Omega_{3\text{dB}} = 0.105$ rad/sample $\approx 1 - a$.

Step 2 — power check by integration. $\frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{d\Omega}{1.81 - 1.8\cos\Omega} = \frac{1}{\sqrt{1.81^2 - 1.8^2}} = \frac{1}{\sqrt{0.0361}} = 5.263$. ✓ (Using $\frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{d\Omega}{A - B\cos\Omega} = 1/\sqrt{A^2 - B^2}$ for $A > |B|$.)

Step 3 — transient. If the recursion starts at $X[0] = 0$, then $X[n] = \sum_{m=0}^{n-1} a^m W[n-m]$ and

$$\operatorname{Var} X[n] = \sigma_W^2 \frac{1 - a^{2n}}{1 - a^2}.$$

After $n = 10$ steps the variance is 4.62 (88% of the steady state); after $n = 50$, 5.263 to four digits. The process is **asymptotically WSS**. In simulation, discard the first several correlation times of output ("burn-in"), or start from $X[0] \sim \mathcal{N}(0, \sigma_W^2/(1 - a^2))$, which makes the sequence stationary from the start.

Worked Example 21.21 — Fitting an AR(1) model to measured correlations

The bias of a temperature sensor, sampled once per second, has estimated autocorrelation $\hat{R}[0] = 2.0\ \text{K}^2$ and $\hat{R}[1] = 1.6\ \text{K}^2$. Fit an AR(1) model and predict the bias one and ten seconds ahead.

Step 1 — parameters (Yule–Walker). $\hat{a} = \hat{R}[1]/\hat{R}[0] = 0.8$ and $\hat{\sigma}_W^2 = \hat{R}[0](1 - \hat{a}^2) = 2.0 \times 0.36 = 0.72\ \text{K}^2$.

Step 2 — one-step prediction. Given the present value $X[n] = x$, the best linear predictor of $X[n+1]$ is $\hat{X}[n+1] = ax = 0.8x$, with error $W[n+1]$ of variance 0.72 — compared with 2.0 without using x. Knowledge of the present removes 64% ($= a^2$) of the uncertainty.

Step 3 — ten steps ahead. Iterating, $\hat{X}[n+10] = a^{10}x = 0.107x$, error variance $\sigma_X^2(1 - a^{20}) = 2.0(1 - 0.0115) = 1.977\ \text{K}^2$: after ten correlation times the present value is nearly useless for prediction.

Step 4 — continuous-time equivalent. Matching $a = e^{-\alpha T_s}$ with $T_s = 1$ s gives $\alpha = -\ln 0.8 = 0.223\ \text{s}^{-1}$, i.e. a Gauss–Markov process (§21.5) with correlation time 4.5 s. This is the standard bias model in a Kalman filter.

Engineering Note — why AR models are everywhere

AR(p) models $X[n] = \sum_{i=1}^{p} a_i X[n-i] + W[n]$ generalize the above and are fitted by solving the Yule–Walker equations, a Toeplitz linear system in $R_{XX}[0], \dots, R_{XX}[p]$ (solved efficiently by the Levinson–Durbin recursion). They underlie linear-predictive speech coding (LPC in mobile-phone codecs), high-resolution "parametric" spectrum estimation, channel-fading simulators and the process models of Kalman filters.

21.5 Gaussian processes

Motivation

Second-order theory tells us means, correlations and spectra, but an engineer often needs *probabilities*: How often will the noise exceed the decision threshold? What is the chance that a sensor error exceeds its specification? In general such questions need the full distribution. For one enormously important class — Gaussian processes — the mean and autocorrelation already determine every distribution, so everything computed in §§21.3–21.4 converts directly into probabilities. And Gaussian processes are not a mathematical convenience only: by the central limit theorem (Ch. 19), any signal formed as the sum of a very large number of small, independent contributions — the thermal agitation of 10^{20} electrons, the superposition of many weak interferers, the accumulated effect of many small manufacturing variations — is very nearly Gaussian.

Definition

Gaussian process

$X(t)$ is a **Gaussian process** if for every k and every set of times $t_1, \dots, t_k$ the random vector $\mathbf{X} = [X(t_1), \dots, X(t_k)]^{\mathsf{T}}$ is jointly Gaussian (Ch. 19):

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{(2\pi)^{k/2}(\det \mathbf{C})^{1/2}} \exp\Big(-\tfrac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^{\mathsf{T}} \mathbf{C}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \Big),$$

with $\mu_i = \mu_X(t_i)$ and $C_{ik} = C_{XX}(t_i, t_k)$. (If $\mathbf{C}$ is singular the vector is a degenerate Gaussian, defined through its characteristic function $\Phi_{\mathbf{X}}(\boldsymbol{\omega}) = \exp\big(j\boldsymbol{\omega}^{\mathsf{T}}\boldsymbol{\mu} - \tfrac{1}{2}\boldsymbol{\omega}^{\mathsf{T}}\mathbf{C}\boldsymbol{\omega}\big)$.) An equivalent definition: **every finite linear combination** $\sum_i a_i X(t_i)$ is a (possibly degenerate) Gaussian random variable.

Two processes X, Y are **jointly Gaussian** if every vector formed from samples of both is jointly Gaussian.

Common Pitfall — Gaussian marginals are not enough

A process whose first-order pdf $f_X(x;t)$ is Gaussian for every t need **not** be a Gaussian process. Example: let $Z \sim \mathcal{N}(0,1)$ and $S = \pm 1$ independent and equally likely; the pair (Z, SZ) has Gaussian marginals, but $Z + SZ$ equals 0 with probability $\frac{1}{2}$, which no Gaussian variable does. The joint Gaussian property must hold for all finite collections of samples.

Key properties

Properties of Gaussian processes

1. **Second-order specification:** a Gaussian process is completely specified, in distribution, by $\mu_X(t)$ and $C_{XX}(t_1, t_2)$. Conversely, for any function $\mu(t)$ and any symmetric non-negative-definite $C(t_1, t_2)$ there exists a Gaussian process with these moments.
2. **WSS ⇔ SSS** for Gaussian processes.
3. **Uncorrelated ⇒ independent:** jointly Gaussian samples (or processes) that are uncorrelated are independent.
4. **Closure under linear operations:** if X is Gaussian, then the output of any linear system — samples, sums, derivatives, integrals, LTI filtering — is Gaussian and jointly Gaussian with X.
5. **Linear conditional means:** for jointly Gaussian X and Y, $E[Y \mid X]$ is an affine function of X, so the best (minimum mean-square error) estimator of one Gaussian quantity from others is linear, and the error variance does not depend on the observed values.

Proof sketches.

1. The joint pdf above involves only $\boldsymbol{\mu}$ and $\mathbf{C}$. Existence follows from Kolmogorov's extension theorem, because the Gaussian distributions with these parameters are consistent (marginals of Gaussians are Gaussian with the corresponding sub-vector and sub-matrix).
2. If X is WSS, the mean vector and covariance matrix of $[X(t_1 + c), \ldots, X(t_k + c)]$ are $\mu_X \mathbf{1}$ and $[C_{XX}(t_i - t_k)]$, which do not involve c. By property 1 the joint distribution is therefore shift-invariant for every k: SSS.
3. If $\mathbf{C}$ is diagonal, the quadratic form in the exponent separates into a sum and the joint pdf factors into the product of the marginals.
4. A sample, sum or finite difference of Gaussian samples is a linear combination, hence Gaussian by the equivalent definition. An integral $\int h(\alpha) X(t-\alpha)\,d\alpha$ is a mean-square limit of Riemann sums, each Gaussian; mean-square limits of Gaussian variables are Gaussian (their characteristic functions $e^{j\omega\mu_n - \omega^2\sigma_n^2/2}$ converge to one of the same form). Derivatives are limits of difference quotients, likewise.
5. For jointly Gaussian (X, Y) with correlation coefficient ρ, the conditional distribution of Y given $X = x$ is $\mathcal{N}\big(\mu_Y + \rho\frac{\sigma_Y}{\sigma_X}(x - \mu_X),\ \sigma_Y^2(1-\rho^2)\big)$ (Ch. 19). □

Property 4 is what makes Gaussian noise analysis tractable: to find the distribution of a filtered Gaussian noise sample we need only its mean and variance, both of which follow from $\mu_Y = \mu_X H(0)$ and $S_{YY} = |H|^2 S_{XX}$ (§21.4). Non-Gaussian inputs, by contrast, generally produce outputs whose distributions are hard to compute — although narrowband filtering of almost any wideband noise tends to make the output *more* Gaussian (a central-limit effect: each output sample is a weighted sum of many nearly independent input samples).

Worked examples

Worked Example 21.22 — Probability of a large increment

$X(t)$ is a zero-mean Gaussian WSS process with $R_{XX}(\tau) = 4e^{-|\tau|}$ (units V², τ in seconds). Find $P\{X(1) - X(0) > 2\}$.

Step 1 — distribution of the increment. $D = X(1) - X(0)$ is a linear combination of jointly Gaussian samples, hence Gaussian, with mean 0 and

$$\operatorname{Var}(D) = R_{XX}(0) + R_{XX}(0) - 2R_{XX}(1) = 8 - 8e^{-1} = 5.057\ \text{V}^2, \qquad \sigma_D = 2.249\ \text{V}.$$

Step 2 — probability. $P\{D > 2\} = Q(2/2.249) = Q(0.889) \approx 0.187$, where $Q(x) = 1 - \Phi(x)$.

Step 3 — comparison. $P\{X(0) > 2\} = Q(2/2) = Q(1) \approx 0.159$. The correlation between $X(0)$ and $X(1)$ is $\rho = e^{-1} = 0.368$; if the samples were independent, $\operatorname{Var}(D)$ would be 8, and if perfectly correlated it would be 0. The general formula $\operatorname{Var}[X(t+\tau) - X(t)] = 2[R_{XX}(0) - R_{XX}(\tau)]$ — the **structure function** — is used to specify jitter and drift over a given time interval.

Worked Example 21.23 — Predicting a Gaussian process

Same process as Example 21.22. The value $X(0) = 3$ V is observed. Find the distribution of $X(0.5)$ and the probability that it still exceeds 3 V.

Step 1 — correlation. $\rho = R_{XX}(0.5)/R_{XX}(0) = e^{-0.5} = 0.607$.

Step 2 — conditional distribution (property 5 with $\mu = 0$, $\sigma_X = \sigma_Y = 2$):

$$X(0.5) \mid X(0) = 3 \ \sim\ \mathcal{N}\big(\rho \cdot 3,\ 4(1-\rho^2)\big) = \mathcal{N}(1.820,\ 2.528).$$

The conditional standard deviation is 1.590 V.

Step 3 — probability. $P\{X(0.5) > 3 \mid X(0) = 3\} = Q\big((3 - 1.820)/1.590\big) = Q(0.742) \approx 0.229$, compared with the unconditional $P\{X(0.5) > 3\} = Q(1.5) = 0.067$. Knowing that the noise is high now makes it more than three times as likely to be high half a second later.

Remark. Because the conditional variance $4(1-\rho^2)$ does not depend on the observed value, the prediction error of this optimal linear predictor is known in advance — the fact that makes Kalman-filter covariance computations possible before any data arrive. For the exponential autocorrelation, observing X at earlier times $t < 0$ in addition to $X(0)$ would not improve the prediction: this is the Markov property of the Gauss–Markov process (below).

Worked Example 21.24 — Zero-crossing rate (Rice's formula)

For a zero-mean stationary Gaussian process with twice-differentiable R_{XX}, the expected number of crossings of the level u per unit time (both directions) is **Rice's formula**

$$\nu_u = \frac{1}{\pi}\sqrt{\frac{-R_{XX}''(0)}{R_{XX}(0)}}\,\exp\!\Big(-\frac{u^2}{2R_{XX}(0)}\Big).$$

Sketch. A crossing occurs in $(t, t+dt)$ if $X(t)$ is within $|X'(t)|dt$ of u. Since $X(t)$ and $X'(t)$ are uncorrelated (§21.3), hence independent (Gaussian), the expected count is $dt\, f_X(u)\, E|X'| = dt\, f_X(u)\sqrt{2/\pi}\,\sigma_{X'}$. With $\sigma_{X'}^2 = -R_{XX}''(0)$ and $f_X(u) = e^{-u^2/2\sigma^2}/(\sqrt{2\pi}\sigma)$ the formula follows.

Check on a sinusoid. For $R_{XX} = \cos\omega_0\tau$, $\nu_0 = \omega_0/\pi = 2f_0$: two zero crossings per period. ✓

Band-limited noise. For flat noise on $|f| < B$, $R_{XX}(\tau) = \sigma^2\frac{\sin 2\pi B\tau}{2\pi B\tau}$ and $-R_{XX}''(0)/R_{XX}(0) = (2\pi B)^2/3$, so $\nu_0 = 2B/\sqrt{3}$. With $B = 1$ kHz: $\nu_0 = 1155$ crossings per second. Crossings of $u = 3\sigma$ (both directions) occur at $1155\,e^{-4.5} = 12.8$ per second, i.e. about 6.4 upward excursions per second. This is how false-trigger rates of comparators and false-alarm rates of threshold detectors are estimated.

Gauss–Markov and Wiener processes

Gauss–Markov (Ornstein–Uhlenbeck) process

The stationary Gaussian process with

$$R_{XX}(\tau) = \sigma^2 e^{-\alpha|\tau|}, \qquad S_{XX}(\omega) = \frac{2\alpha\sigma^2}{\alpha^2+\omega^2}$$

is the only stationary Gaussian process (up to trivial cases) that is also **Markov**: given $X(t)$, the future is independent of the past. It is the output of a first-order lowpass filter driven by white Gaussian noise, $\dot X = -\alpha X + W$, and its samples at spacing T_s form an AR(1) sequence with $a = e^{-\alpha T_s}$ (Example 21.4).

It is the standard model for slowly varying sensor biases and drifts (gyroscope bias, clock frequency error, the bias of Example 21.21) and for the velocity of a Brownian particle.

Wiener process (Brownian motion)

$W(t)$, $t \ge 0$, is the Gaussian process with $W(0) = 0$, independent stationary increments, $W(t+s) - W(t) \sim \mathcal{N}(0, \sigma^2 s)$. Equivalently, it is the zero-mean Gaussian process with

$$R_{WW}(t_1, t_2) = \sigma^2 \min(t_1, t_2).$$

The Wiener process is the continuous-time limit of the random walk: scale steps of size $\pm\sigma\sqrt{\Delta t}$ every Δt seconds; by the CLT, the walk at time t is approximately $\mathcal{N}(0, \sigma^2 t)$, and its autocorrelation $\sigma^2\Delta t\min(n_1, n_2) \to \sigma^2\min(t_1, t_2)$. It is not stationary ($\mathrm{Var}\,W(t) = \sigma^2 t$), its paths are continuous but nowhere differentiable, and — formally — its derivative is white Gaussian noise of PSD σ^2 (§21.6): $\frac{\partial^2}{\partial t_1 \partial t_2}\sigma^2\min(t_1,t_2) = \sigma^2\delta(t_1 - t_2)$. Free-running oscillator phase noise is modelled as a Wiener process, which is why phase error grows like $\sqrt{t}$ and why the phase-noise spectrum of an oscillator falls as $1/f^2$ away from the carrier.

Bandpass Gaussian noise

At the output of a narrowband IF or RF filter centred at ω_0, noise looks like a sinusoid at ω_0 with slowly wandering amplitude and phase. This is made precise by the quadrature representation.

In-phase/quadrature representation of narrowband noise

Let $N(t)$ be zero-mean, WSS, Gaussian with PSD concentrated in $|\omega \mp \omega_0| < W$, $W < \omega_0$. Then

$$N(t) = N_c(t)\cos\omega_0 t - N_s(t)\sin\omega_0 t = R(t)\cos\big(\omega_0 t + \Phi(t)\big),$$

where the in-phase and quadrature components N_c, N_s are jointly Gaussian, zero-mean, lowpass processes, each with the same power as N:

$$E[N_c^2] = E[N_s^2] = E[N^2] = \sigma^2, \qquad S_{N_cN_c}(\omega) = S_{N_sN_s}(\omega) = \big[S_{NN}(\omega-\omega_0) + S_{NN}(\omega+\omega_0)\big]_{|\omega|<W}.$$

If S_{NN} is symmetric about $\pm\omega_0$, N_c and N_s are independent. At any instant the **envelope** $R = \sqrt{N_c^2 + N_s^2}$ is **Rayleigh** distributed and the phase Φ is uniform on $[0, 2\pi)$, independent of R:

$$f_R(r) = \frac{r}{\sigma^2}e^{-r^2/(2\sigma^2)}\ (r \ge 0), \qquad P\{R > r\} = e^{-r^2/(2\sigma^2)}.$$

The Rayleigh result follows from transforming the pdf of two independent $\mathcal{N}(0,\sigma^2)$ variables to polar coordinates (Ch. 19): $f_{R\Phi}(r,\phi) = \frac{r}{2\pi\sigma^2}e^{-r^2/2\sigma^2}$, which factors.

Worked Example 21.25 — Envelope-detector false-alarm threshold

A radar receiver's IF noise is Gaussian narrowband noise with power σ^2. An envelope detector declares a target when $R(t)$ exceeds a threshold r_0. Choose r_0 so that the probability of a false alarm on any single sample is $P_{fa} = 10^{-6}$.

Step 1. $P\{R > r_0\} = e^{-r_0^2/(2\sigma^2)} = 10^{-6}$, so $r_0 = \sigma\sqrt{2\ln 10^6} = \sigma\sqrt{27.63} = 5.26\,\sigma$.

Step 2 — compare with a baseband threshold. For a real Gaussian sample, $P\{N > 5.26\sigma\} = Q(5.26) \approx 7.2 \times 10^{-8}$, about 14 times smaller. The envelope combines two independent Gaussian components, so large values are more probable; using the Gaussian tail here would make the false-alarm rate 14 times worse than designed.

Step 3 — in decibels. The threshold is $20\log_{10} 5.26 = 14.4$ dB above the RMS noise of each component, which sets the minimum detectable signal of the radar.

Engineering Note — where Gaussianity fails

The Gaussian model is excellent for thermal noise and usually for the sum of many independent interferers, but it fails for impulsive noise (lightning, ignition, switching transients), for a small number of dominant interferers, for quantization error of a slowly varying signal (closer to uniform), and for clipped or rectified signals. Heavy-tailed noise makes rare large errors far more likely than $Q(\cdot)$ predicts; robust detectors and median filters are the response. Always check a histogram, ideally on a logarithmic probability axis where the Gaussian tail is a parabola.

21.6 White noise

Motivation

Every receiver, amplifier and sensor adds noise, and in most cases that noise is spread almost uniformly over a frequency range far wider than the circuit that processes it. The thermal noise of a resistor is flat to within 1% up to about 100 GHz at room temperature; the circuits that see it have bandwidths of hertz to gigahertz. In such cases the detailed spectrum of the noise beyond the circuit's passband is irrelevant — only its *density* inside the passband matters. **White noise** is the idealization that makes this explicit: a flat PSD at all frequencies, characterized by one number. It is the "impulse" of random-signal analysis — unphysical on its own, indispensable as an input.

Definition and interpretation

White noise

A zero-mean WSS process $W(t)$ is **white** if its PSD is constant over all frequencies:

$$S_{WW}(\omega) = \frac{N_0}{2} \quad \Longleftrightarrow \quad R_{WW}(\tau) = \frac{N_0}{2}\,\delta(\tau).$$

If in addition W is Gaussian, it is **white Gaussian noise** (WGN). The factor $\frac{1}{2}$ is a convention that makes N_0 the **one-sided** PSD, in V^2/Hz (or W/Hz for available power).

Consequences of the definition:

- **Infinite power.** $R_{WW}(0) = \infty$; equivalently $\frac{1}{2\pi}\int \frac{N_0}{2}\,d\omega$ diverges. White noise is not a process with finite-variance samples; it is a generalized process that only makes sense after filtering or integration, just as $\delta(t)$ only makes sense under an integral.
- **Uncorrelated at every separation.** $R_{WW}(\tau) = 0$ for every $\tau \neq 0$, however small: the process has no memory at all.
- **Filtered white noise is well defined.** By §21.4, a filter h with finite energy turns it into a finite-power process with

$$S_{YY}(\omega) = \frac{N_0}{2}|H(\omega)|^2, \qquad R_{YY}(\tau) = \frac{N_0}{2}\int h(t+\tau)h(t)\,dt, \qquad E[Y^2] = \frac{N_0}{2}\int_{-\infty}^{\infty} h^2(t)\,dt.$$

The output autocorrelation is the *deterministic autocorrelation of the impulse response*: the filter "imprints" its own memory on the noise.

- **Integrated white noise is a Wiener process.** $\int_0^t W(s)\,ds$ has variance $\frac{N_0}{2}t$ and independent increments (§21.5).

White noise is useful precisely *because* physical noise is flat over a band far wider than any system that processes it, so only N_0 matters and the system's own bandwidth sets the output power.

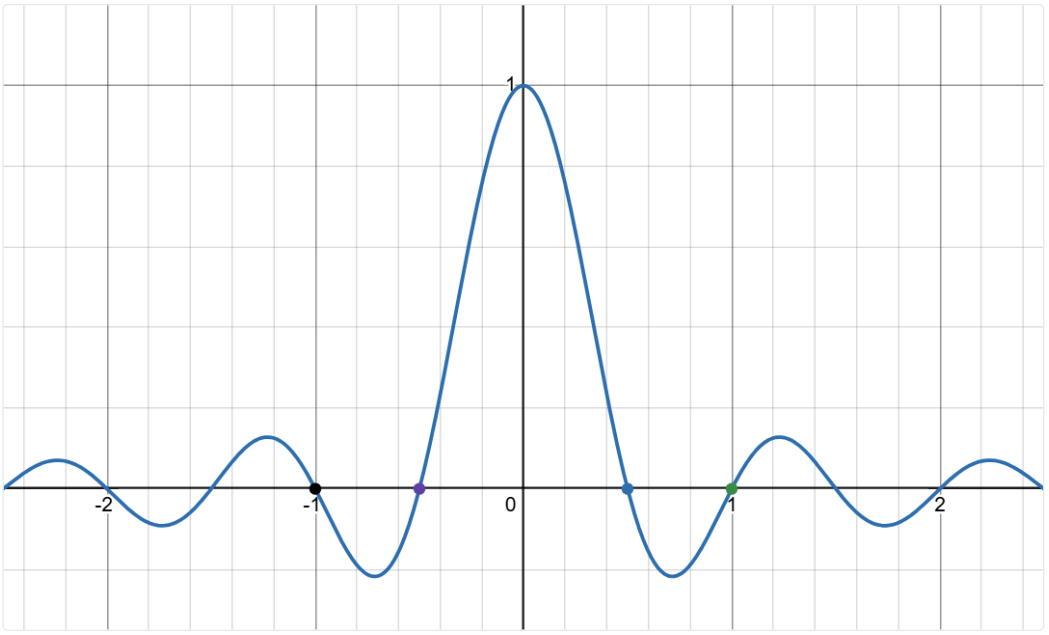
Band-limited white noise

Band-limited (lowpass) white noise

$$S_{XX}(\omega) = \begin{cases} N_0/2, & |\omega| < 2\pi B \\ 0, & \text{otherwise} \end{cases} \quad \Longrightarrow \quad R_{XX}(\tau) = N_0 B\,\frac{\sin(2\pi B\tau)}{2\pi B\tau}.$$

Derivation. $R_{XX}(\tau) = \frac{1}{2\pi}\int_{-2\pi B}^{2\pi B} \frac{N_0}{2} e^{j\omega\tau}\,d\omega = \frac{N_0}{2}\cdot\frac{1}{2\pi}\cdot\frac{2\sin(2\pi B\tau)}{\tau} = N_0 B\,\frac{\sin(2\pi B\tau)}{2\pi B\tau}$.

The power is $R_{XX}(0) = N_0 B$: **noise power = one-sided density × bandwidth**. This is the single most-used formula in link budgets.



Normalized autocorrelation of band-limited white noise with $B = 1$: zeros at $\tau = k/(2B)$. Samples taken at the Nyquist rate $2B$ are uncorrelated — and, if the noise is Gaussian, independent.

Similarly, **bandpass white noise** with density $N_0/2$ on $|f \mp f_0| < B/2$ has $R_{XX}(\tau) = N_0 B \frac{\sin \pi B\tau}{\pi B\tau} \cos 2\pi f_0 \tau$ and power $N_0 B$ — a sinc envelope on a carrier, consistent with the narrowband representation of §21.5.

Discrete-time white noise

Discrete-time white noise and sampling

A sequence $W[n]$ with $E[W[n]] = 0$ and $R_{WW}[k] = \sigma_W^2\,\delta[k]$ has the flat spectrum $S_{WW}(e^{j\Omega}) = \sigma_W^2$ for all Ω, and — unlike its continuous-time counterpart — finite power σ_W^2. Sampling band-limited white noise of bandwidth B at exactly $f_s = 2B$ produces discrete white noise with $\sigma_W^2 = N_0 B$.

Proof of the sampling statement. $R_{XX}[k] = R_{XX}(k/(2B)) = N_0 B \frac{\sin \pi k}{\pi k} = N_0 B\,\delta[k]$. □

Sampling at a higher rate $f_s > 2B$ gives correlated samples (sinc values at non-integer multiples of π); sampling unfiltered wideband noise at f_s aliases all of its power into $[-f_s/2, f_s/2]$. This is why an anti-aliasing filter is needed even when the *signal* is band-limited: without it, wideband noise is folded into the band of interest and the SNR degrades.

Worked Example 21.26 — Noise power in a receiver bandwidth

White noise with two-sided PSD $N_0/2 = 10^{-8}$ V^2/Hz passes through an ideal lowpass filter of bandwidth $B = 5$ kHz.

Step 1. $P = N_0 B = 2 \times 10^{-8} \times 5 \times 10^3 = 10^{-4}$ V^2, so $V_{\text{rms}} = 10$ mV.

Step 2 — scaling. Doubling the bandwidth doubles the noise power (+3 dB) and raises the RMS noise by $\sqrt{2}$ to 14.1 mV. Halving it gives 7.07 mV. In general $V_{\text{rms}} \propto \sqrt{B}$, which is why noise densities are quoted in V/$\sqrt{\text{Hz}}$: here $\sqrt{N_0} = \sqrt{2 \times 10^{-8}} = 141\ \mu\text{V}/\sqrt{\text{Hz}}$, and $141\ \mu\text{V} \times \sqrt{5000} = 10$ mV. ✓

Step 3 — samples. If this output is sampled at $f_s = 10$ kHz $= 2B$, the samples are uncorrelated with variance 10^{-4} V^2; if the noise is Gaussian, $P\{|v| > 30\ \text{mV}\} = 2Q(3) = 0.0027$ per sample, i.e. about 27 samples per second exceed ± 30 mV.

Noise-equivalent bandwidth

Real filters do not have brick-wall responses. To keep the convenient formula "power = density × bandwidth", we define the bandwidth that makes it exact.

Noise-equivalent bandwidth

For a lowpass filter with DC gain $H(0) \neq 0$, the **noise-equivalent bandwidth** is

$$B_N = \frac{1}{|H(0)|^2} \int_0^\infty |H(2\pi f)|^2\, df,$$

the width (in Hz) of an ideal rectangular filter of the same peak gain that passes the same white-noise power. For a bandpass filter, replace $|H(0)|$ by the peak gain $|H(2\pi f_0)|$ and integrate over $f > 0$.

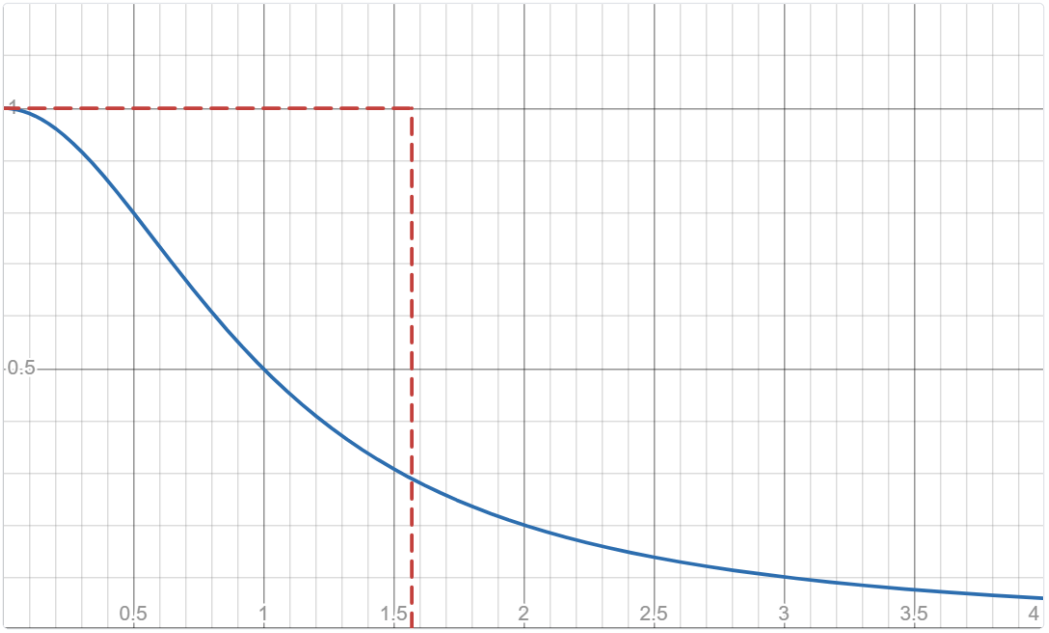
With this definition the output power of a filter driven by white noise of one-sided density N_0 is exactly

$$E[Y^2] = N_0\, B_N\, |H(0)|^2.$$

By Parseval's theorem there is an equivalent **time-domain formula**:

$$B_N = \frac{\int_{-\infty}^{\infty} h^2(t)\,dt}{2\left(\int_{-\infty}^{\infty} h(t)\,dt\right)^2},$$

since $\int_0^\infty |H(2\pi f)|^2 df = \tfrac{1}{2}\int_{-\infty}^{\infty}|H|^2 df = \tfrac{1}{2}\int h^2 dt$ and $H(0) = \int h\,dt$.



Noise-equivalent bandwidth of a first-order lowpass, with frequency normalized to $f_{3\text{dB}}$: the area under $|H|^2 = 1/(1+x^2)$ (blue) over $x>0$ is $\pi/2$, equal to the area of the red rectangle of height 1 and width $\pi/2 \approx 1.571$.

Worked Example 21.27 — Noise bandwidth of RC filters

(a) Single RC lowpass. $|H(2\pi f)|^2 = 1/(1+(2\pi f RC)^2)$, $H(0)=1$:

$$B_N = \int_0^\infty \frac{df}{1+(2\pi f RC)^2} = \frac{1}{2\pi RC}\cdot\frac{\pi}{2} = \frac{1}{4RC} = \frac{\pi}{2} f_{3\text{dB}},$$

where $f_{3\text{dB}} = 1/(2\pi RC)$. The time-domain formula agrees: $h = \frac{1}{RC}e^{-t/RC}u(t)$, $\int h^2 = \frac{1}{2RC}$, $\int h = 1$, so $B_N = \frac{1}{4RC}$. ✓ With $RC = 1$ ms, $f_{3\text{dB}} = 159$ Hz and $B_N = 250$ Hz: using $f_{3\text{dB}}$ in place of B_N would underestimate the noise power by 36% (1.96 dB).

(b) Two identical RC sections, buffered. $|H|^2 = 1/(1+(2\pi f RC)^2)^2$. With $u = 2\pi f RC$ and $\int_0^\infty \frac{du}{(1+u^2)^2} = \frac{\pi}{4}$,

$$B_N = \frac{1}{2\pi RC}\cdot\frac{\pi}{4} = \frac{1}{8RC}.$$

The 3 dB frequency of the cascade solves $(1+u^2)^2 = 2$, $u = \sqrt{\sqrt{2}-1} = 0.644$, so $f_{3\text{dB}} = 0.1024/RC$ and $B_N/f_{3\text{dB}} = 1.22$. Steeper skirts bring B_N closer to $f_{3\text{dB}}$.

The ratio $B_N/f_{3\text{dB}}$ for an n-th order Butterworth lowpass is $\dfrac{\pi/(2n)}{\sin(\pi/(2n))}$:

order n	1	2	3	4	∞
$B_N/f_{3\text{dB}}$	1.571	1.111	1.047	1.026	1

Worked Example 21.28 — Averaging as filtering

A **moving-average** (integrate-and-dump) filter outputs $Y(t) = \frac{1}{T}\int_{t-T}^{t} X(s)\,ds$, i.e. $h(t) = \frac{1}{T}$ on $[0,T]$. Then $\int h = 1$, $\int h^2 = 1/T$ and

$$B_N = \frac{1/T}{2\cdot 1^2} = \frac{1}{2T}.$$

Averaging white noise of density N_0 for T seconds gives output variance $N_0/(2T)$. A digital voltmeter integrating over one power-line cycle ($T = 20$ ms at 50 Hz) has $B_N = 25$ Hz; increasing the integration time to 10 cycles reduces the noise bandwidth to 2.5 Hz and the RMS noise by $\sqrt{10}$. The frequency response, $|H(2\pi f)| = |\sin \pi f T/(\pi f T)|$, also has nulls at multiples of $1/T$ — which is why integrating over exact line cycles rejects hum.

Worked Example 21.29 — Noise bandwidth of a tuned circuit

A second-order bandpass (e.g. a parallel RLC tank driven by a current, with voltage output) has

$$|H(2\pi f)|^2 = \frac{1}{1 + Q^2\left(f/f_0 - f_0/f\right)^2}$$

with peak gain 1 at f_0 and $B_{3\text{dB}} = f_0/Q$. The integral (by contour methods, Ch. 18, or numerically) is

$$B_N = \int_0^\infty |H|^2 df = \frac{\pi f_0}{2Q} = \frac{\pi}{2} B_{3\text{dB}}.$$

For a 10.7 MHz IF filter with $Q = 50$: $B_{3\text{dB}} = 214$ kHz, $B_N = 336$ kHz. A numerical check with $f_0 = 1$, $Q = 50$ gives 0.03138 against $\pi/100 = 0.03142$ (the small difference is the truncated tail). The same $\pi/2$ factor as the first-order lowpass appears because the bandpass response is the lowpass response mapped onto the frequency axis.

Coloured noise

Noise whose PSD is not flat is called **coloured**. The common types are named by their spectral slope:

Name	One-sided PSD	Autocorrelation behaviour	Where it occurs
white	$G(f) = N_0$	$\delta(\tau)$	thermal, shot noise
pink / flicker	$\propto 1/f$	very long memory, no finite variance over $0 < f < \infty$	MOSFETs, carbon resistors, oscillator close-in phase noise
brown / red	$\propto 1/f^2$	integral of white noise (Wiener)	random walk, free-running oscillator phase
Lorentzian	$\propto 1/(1 + (f/f_c)^2)$	$e^{-2\pi f_c \lvert\tau\rvert}$	RC-filtered white, RTS noise
blue / violet	$\propto f$, $\propto f^2$	derivative-like	differentiated noise, noise-shaped quantization

Worked Example 21.30 — Integrating a white-plus-flicker noise density

An op-amp has input voltage noise with one-sided density $G(f) = e_w^2\left(1 + f_c/f\right)$, white level $e_w = 10\ \text{nV}/\sqrt{\text{Hz}}$ and **1/f corner** $f_c = 100$ Hz. Find the RMS noise from $f_1 = 0.1$ Hz to $f_2 = 10$ kHz.

Step 1 — integrate.

$$\overline{v^2} = \int_{f_1}^{f_2} e_w^2\left(1 + \frac{f_c}{f}\right) df = e_w^2\left[(f_2 - f_1) + f_c \ln\frac{f_2}{f_1}\right].$$

Step 2 — numbers. $e_w^2 = 10^{-16}\ \text{V}^2/\text{Hz}$; $(f_2 - f_1) = 9999.9$ Hz; $f_c \ln(10^5) = 100 \times 11.51 = 1151$ Hz. Total $\overline{v^2} = 10^{-16} \times 11151 = 1.115 \times 10^{-12}\ \text{V}^2$, so $v_{\text{rms}} = 1.06\ \mu\text{V}$.

Step 3 — interpretation. The 1/f part contributes the equivalent of 1151 Hz of white bandwidth, i.e. 10% of the total power. Its contribution grows only logarithmically with the observation time ($f_1 \approx 1/T_{\text{obs}}$): each additional decade at low frequency adds $f_c \ln 10 = 230$ Hz. The variance does not converge as $f_1 \to 0$ — flicker noise has no well-defined stationary variance, a symptom of its very long memory.

Common Pitfall — "white" is not "Gaussian"

"White" refers to the *spectrum* (uncorrelated samples); "Gaussian" refers to the *amplitude distribution*. They are independent properties. A random ± 1 i.i.d. sequence is white but not Gaussian; the Gauss–Markov process is Gaussian but not white; thermal noise is both. Statements such as "white noise has a Gaussian distribution" or "Gaussian noise has a flat spectrum" are false in general.

Engineering Note — quantization noise as discrete white noise

An ADC with step Δ, digitizing a busy signal, adds an error that is well modelled as discrete white noise, uniformly distributed on $[-\Delta/2, \Delta/2]$, with variance $\Delta^2/12$ spread evenly over $|f| < f_s/2$. Its one-sided density is therefore $\Delta^2/(6 f_s)$: **oversampling** by a factor M (then filtering to the signal band) reduces the in-band quantization noise by M — 3 dB, or half a bit, per doubling of f_s. Noise-shaping ($\Sigma\Delta$) converters go further by making the quantization noise "blue", pushing it out of band.

21.7 Ergodicity

Motivation

Everything so far has been defined by ensemble averages — expectations over infinitely many hypothetical repetitions of the experiment. In the laboratory we usually have **one** long record: one amplifier, one data capture, one night of sensor logging. A spectrum analyzer, an RMS voltmeter and an oscilloscope's "average" mode all compute *time* averages over that single record. The property that justifies replacing $E[\cdot]$ by $\langle\cdot\rangle$ is **ergodicity**. It is not automatic, and the theory below tells us both *when* it holds and *how long* the record must be for the time average to be accurate — the practical question behind every measurement-time specification.

Time averages as random variables

For a finite record, define the time averages

$$\langle X\rangle_T = \frac{1}{2T}\int_{-T}^{T} X(t)\,dt, \qquad \hat{R}_T(\tau) = \frac{1}{2T}\int_{-T}^{T} X(t+\tau)X(t)\,dt.$$

Because they are computed from a random sample function, these are **random variables**. For a WSS process, interchanging expectation and integration,

$$E\big[\langle X\rangle_T\big] = \frac{1}{2T}\int_{-T}^{T} \mu_X\,dt = \mu_X,$$

so $\langle X\rangle_T$ is an **unbiased estimator** of μ_X (Ch. 20). The question is whether its variance shrinks to zero as T grows.

Ergodicity in the mean and in autocorrelation

A WSS process is **mean-ergodic** if $\langle X\rangle_T \to \mu_X$ in mean square as $T\to\infty$:

$$\lim_{T\to\infty} E\big[(\langle X\rangle_T - \mu_X)^2\big] = 0.$$

It is **autocorrelation-ergodic** if $\hat{R}_T(\tau) \to R_{XX}(\tau)$ in mean square for every τ, and **ergodic** (in the strict sense) if *every* time average of every measurable function of the process converges to the corresponding ensemble average. For a sequence, replace $\frac{1}{2T}\int_{-T}^{T}$ by $\frac{1}{N}\sum_{n=0}^{N-1}$.

The mean-ergodic theorem

Mean-ergodic theorem (Slutsky)

For a WSS process,

$$\operatorname{Var}\big(\langle X\rangle_T\big) = \frac{1}{2T}\int_{-2T}^{2T} C_{XX}(\tau)\Big(1-\frac{|\tau|}{2T}\Big)d\tau,$$

and X is mean-ergodic if and only if this tends to zero. Equivalently (Slutsky's condition), $\frac{1}{2T}\int_{-2T}^{2T} C_{XX}(\tau)\,d\tau \to 0$. A **sufficient** condition is $\int_{-\infty}^{\infty}|C_{XX}(\tau)|\,d\tau < \infty$, in which case, for large T,

$$\operatorname{Var}\big(\langle X\rangle_T\big) \approx \frac{1}{2T}\int_{-\infty}^{\infty} C_{XX}(\tau)\,d\tau = \frac{S_{CC}(0)}{2T},$$

where $S_{CC}(0)$ is the PSD of the zero-mean part $X-\mu_X$ at zero frequency. For a WSS sequence with N samples: $\operatorname{Var}(\bar{X}_N) = \frac{1}{N}\sum_{|k|<N}\big(1-\frac{|k|}{N}\big)C_{XX}[k] \approx \frac{1}{N}\sum_k C_{XX}[k]$.

Derivation.

$$\operatorname{Var}\big(\langle X\rangle_T\big) = E\Big[\Big(\frac{1}{2T}\int_{-T}^{T}(X(t)-\mu_X)dt\Big)^2\Big] = \frac{1}{4T^2}\int_{-T}^{T}\int_{-T}^{T} C_{XX}(t-s)\,dt\,ds.$$

This is the same double integral over a square that appeared in the proof of Wiener–Khinchin (§21.4), with $\omega=0$. Collapsing it along lines $t-s=\tau$ gives $\frac{1}{4T^2}\int_{-2T}^{2T}(2T-|\tau|)C_{XX}(\tau)\,d\tau$, which is the stated formula. If $\int|C_{XX}|<\infty$, the triangular weight tends to 1 and the integral is bounded, so the variance behaves like $(1/2T)\int C_{XX}$ and tends to zero. □

Two important interpretations:

- **Effective number of independent samples.** If X had independent samples every Δ seconds, a record of length $2T$ would contain $2T/\Delta$ of them and the variance of the mean would be $\sigma_X^2\Delta/(2T)$. Comparing with the formula, a correlated process behaves as if it had independent samples spaced $\Delta_{\text{eff}} = \int C_{XX}\,d\tau/\sigma_X^2 = 2\tau_c$ apart (twice the correlation time). The **effective sample size** is $N_{\text{eff}} = 2T/(2\tau_c)$.
- **Frequency-domain view.** Averaging over $2T$ is a lowpass filter of noise bandwidth $B_N = 1/(4T)$ (Example 21.28 with $T\to 2T$). The residual fluctuation is the noise power in that tiny band around DC, $2\cdot S_{CC}(0)\cdot B_N = S_{CC}(0)/(2T)$ — exactly the formula. Only the noise density at DC matters, which is why 1/f noise, whose density diverges at DC, defeats long averaging.

Worked Example 21.31 — How long must we average?

A sensor output is $X(t)=\mu+N(t)$, where N has $C_{NN}(\tau)=\sigma^2 e^{-\alpha|\tau|}$ with $\sigma = 1$ mV and $\alpha = 1000\ \text{s}^{-1}$. We estimate μ by averaging a record of total length $2T = 1$ s.

Step 1 — asymptotic formula.

$$\operatorname{Var}(\hat{\mu}) \approx \frac{1}{2T}\int_{-\infty}^{\infty}\sigma^2 e^{-\alpha|\tau|}d\tau = \frac{1}{2T}\cdot\frac{2\sigma^2}{\alpha} = \frac{2\times 10^{-6}}{1\times 1000} = 2\times 10^{-9}\ \text{V}^2,$$

so the standard error is $\sqrt{2\times10^{-9}} \approx 44.7\ \mu\text{V}$.

Step 2 — exact formula. Evaluating the triangular-weighted integral exactly with $L=2T$ gives $\frac{2\sigma^2}{\alpha L}\Big[1-\frac{1-e^{-\alpha L}}{\alpha L}\Big] = 2\times10^{-9}(1-0.001) = 1.998\times 10^{-9}\ \text{V}^2$: the asymptotic formula is excellent once $L \gg 1/\alpha$.

Step 3 — effective samples. $\tau_c = 1/\alpha = 1$ ms, so $N_{\text{eff}} = L/(2\tau_c) = 500$, and $\sigma/\sqrt{500} = 44.7\ \mu$V. ✓ Sampling the record at 1 MHz (a million samples) does not help: the samples are highly correlated, and the error is set by the 500 effectively independent ones.

Step 4 — design. To reach a standard error of 10 μV, we need $N_{\text{eff}} = (1\ \text{mV}/10\ \mu\text{V})^2 = 10^4$, i.e. $L = 2\tau_c N_{\text{eff}} = 20$ s of data.

Worked Example 21.32 — Sample mean of an AR(1) sequence

Take the AR(1) sequence of Example 21.20 ($a = 0.9$, $\sigma_W^2 = 1$, $\sigma_X^2 = 5.263$) and $N = 1000$ samples.

Step 1 — sum of covariances. $\sum_k C_{XX}[k] = \sigma_X^2 \sum_k a^{|k|} = \sigma_X^2 \frac{1+a}{1-a} = 5.263 \times 19 = 100$. (This equals $S_{XX}(e^{j0}) = \sigma_W^2/(1-a)^2 = 100$, as it must.)

Step 2 — variance of the mean. $\mathrm{Var}(\bar{X}_N) \approx 100/1000 = 0.1$; the exact finite-$N$ sum gives 0.0991. Standard error 0.316.

Step 3 — comparison with i.i.d. Treating the samples as independent would give $\sigma_X^2/N = 0.00526$ — **19 times too small**, an error bar $\sqrt{19} = 4.4$ times too optimistic. The effective sample size is $N_{\text{eff}} = N\frac{1-a}{1+a} = 52.6$.

This is a common error when applying the confidence intervals of Ch. 20 to time-series data: those formulas assume independent observations.

Non-ergodic processes

Worked Example 21.33 — Stationary processes that are not ergodic

(a) Random constant. Let $X(t) = A$ for all t, where $A \sim \mathcal{N}(0,1)$. Then $\mu_X = 0$ and $R_{XX}(\tau) = 1$: the process is SSS. But each sample function is a constant, so $\langle x \rangle_T = A$, which is not 0 in general. Here $C_{XX}(\tau) = 1$ does not decay, and the variance formula gives $\mathrm{Var}\langle X \rangle_T = 1 \not\to 0$. One record never reveals the ensemble mean. In the PSD this shows up as an impulse $2\pi\delta(\omega)$ in the spectrum of the zero-mean part — a random DC component.

(b) Random-phase sinusoid. $A\cos(\omega_0 t + \Theta)$ with fixed A *is* ergodic in mean and autocorrelation: every sample function has time average 0 and time autocorrelation $\frac{A^2}{2}\cos\omega_0\tau$ (Example 21.1, Step 5).

(c) Random amplitude and phase. $A\cos(\omega_0 t + \Theta)$ with *random* A (independent of Θ) is still mean-ergodic, but **not autocorrelation-ergodic**: the time autocorrelation of one record is $\frac{A^2}{2}\cos\omega_0\tau$, while the ensemble one is $\frac{E[A^2]}{2}\cos\omega_0\tau$. A power meter on one record reports that record's $A^2/2$.

(d) Random choice of source. A switch, set once at random, connects either a 1 mV-rms or a 10 mV-rms noise source to the output. The resulting process is stationary, with ensemble power $\frac{1}{2}(1 + 100) = 50.5$ mV^2, but every individual record has time-average power 1 or 100 mV^2 — never 50.5. Non-ergodicity typically means "the ensemble is a mixture of different sub-populations, and each record sees only one".

The general principle: a stationary process fails to be ergodic when some random quantity is fixed for the whole record (an amplitude, a DC offset, a choice of state). In engineering terms, ergodicity is a statement that **one device, observed for a long time, is representative of the population of devices**. This is plausible for the thermal noise of a resistor; it is false for, say, the offset voltage of op-amps, which varies from unit to unit but is constant in time for each unit. Unit-to-unit variation needs ensemble statistics (many devices), not long records.

Autocorrelation ergodicity

Autocorrelation-ergodicity is mean-ergodicity of the product process $Z_\tau(t) = X(t+\tau)X(t)$, whose autocovariance involves **fourth** moments of X. For a zero-mean Gaussian process, the Isserlis (Wick) moment theorem gives $E[X_1X_2X_3X_4] = R_{12}R_{34} + R_{13}R_{24} + R_{14}R_{23}$, and one finds

$$C_{Z_\tau Z_\tau}(s) = R_{XX}^2(s) + R_{XX}(s+\tau)R_{XX}(s-\tau).$$

Autocorrelation ergodicity of Gaussian processes

A zero-mean stationary Gaussian process with $\int_{-\infty}^{\infty} R_{XX}^2(\tau)\,d\tau < \infty$ (equivalently, finite-energy correlation, which holds whenever its spectrum has no impulses and is square-integrable) is autocorrelation-ergodic. More generally, a stationary Gaussian process is ergodic (in every sense) if and only if its spectrum has no discrete (impulsive) part.

So for Gaussian noise without spectral lines — the typical receiver-noise case — time averages of any kind converge to ensemble averages, which is why the ergodic assumption is so rarely questioned in noise measurements.

Estimating correlations and spectra from data

From N samples $x[0], \ldots, x[N-1]$ of a zero-mean sequence, the standard estimators are

$$\hat{R}_{XX}[k] = \frac{1}{N}\sum_{n=0}^{N-1-k} x[n+k]x[n] \quad (0 \le k < N), \qquad \hat{R}_{XX}[-k] = \hat{R}_{XX}[k],$$

$$\hat{S}_{XX}(e^{j\Omega}) = \frac{1}{N}\Big|\sum_{n=0}^{N-1} x[n]e^{-j\Omega n}\Big|^2 = \sum_{|k|<N} \hat{R}_{XX}[k]e^{-j\Omega k} \quad \text{(periodogram)}.$$

Properties (all resting on an ergodicity assumption):

- **Bias of $\hat{R}$.** $E[\hat{R}_{XX}[k]] = \big(1 - \frac{|k|}{N}\big)R_{XX}[k]$: biased (a triangular taper), but asymptotically unbiased, and guaranteed non-negative definite. Dividing by $N - |k|$ instead removes the bias but can produce invalid (non-positive-definite) estimates at large lags.
- **Bias of the periodogram.** $E[\hat{S}] = \frac{1}{2\pi}\big(S_{XX} * F_N\big)(e^{j\Omega})$, the true spectrum convolved with the Fejér kernel $F_N(\Omega) = \frac{1}{N}\frac{\sin^2(N\Omega/2)}{\sin^2(\Omega/2)}$: finite records smear spectral detail finer than about $2\pi/N$ rad/sample and leak power from strong peaks into their sidelobes. Windowing (Hann, Blackman) lowers the sidelobes at the expense of resolution.
- **Variance of the periodogram.** For Gaussian noise, at each frequency $\hat{S} \approx S \cdot \chi_2^2/2$ — an exponential random variable whose standard deviation equals its mean, **however large N is**. The raw periodogram is not a consistent estimator: more data gives more frequency points, not smoother ones.
- **Welch's method.** Split the record into K segments of length L, window each, compute K periodograms and average them. The variance drops by about K (more with 50% overlap), while the resolution coarsens to $\sim f_s/L$. This is what every FFT spectrum analyzer's "averaging" does.

Worked Example 21.34 — Planning a Welch spectral estimate

A noise signal is captured at $f_s = 100$ kHz for 1 s ($N = 10^5$ samples). We want a PSD estimate with about 100 Hz resolution and better than 10% relative standard deviation.

Step 1 — segment length. Resolution $\approx f_s/L$; choose $L = 1024$, giving $\Delta f = 97.7$ Hz.

Step 2 — number of segments. Non-overlapping: $K = \lfloor 10^5/1024 \rfloor = 97$. Relative standard deviation of the averaged estimate $\approx 1/\sqrt{K} = 1/\sqrt{97} = 10.2\%$ — just at the target. With 50% overlap and a Hann window, $K \approx 194$ segments whose effective independence is about 0.9, bringing the relative error to roughly 7.5%.

Step 3 — trade-off. Halving L to 512 doubles K (relative error $\div\sqrt{2}$) but doubles Δf to 195 Hz. Resolution × (relative error)^2 × record length is roughly constant — a time–bandwidth uncertainty relation for spectral estimation: to get both fine resolution and low variance, you need a long record.

Common Pitfall — ergodicity cannot be checked from one record

Ergodicity is a property relating one record to the ensemble; a single record, however long, cannot reveal whether other records would behave differently (Example 21.33(d)). It must be justified physically ("this noise is generated by the same mechanism in every realization, with no hidden parameters fixed per record") or checked with several independent records — for instance, several devices or several power-up cycles.

21.8 Applications to communications and measurement

Motivation

The tools of §§21.1–21.7 combine into a short, repeatable procedure that settles most noise questions in circuits, instruments and communication links: model each noise source by its PSD, carry it through the system with $|H|^2$, add the PSDs of independent sources, integrate to get power, and — if the noise is Gaussian — convert powers into probabilities with $Q(\cdot)$. This section applies that procedure to the physical noise sources an EE meets most often (thermal and shot noise), to the figures of merit used to specify them (noise bandwidth, SNR, noise figure), and to the central design question of detection: which filter makes a known signal most visible in white noise? The aim is to show the mathematics at work; the device physics and communication-system design belong to other courses.

Thermal (Johnson–Nyquist) noise

The random thermal motion of charge carriers in a resistor R at absolute temperature T produces a fluctuating open-circuit voltage. Because it is the sum of the effects of an enormous number of independent carrier collisions, each lasting about 10^{-14} s, it is to an excellent approximation zero-mean, **Gaussian** (central limit theorem) and **white** up to frequencies of order $kT/h \approx 6$ THz at 300 K.

Johnson–Nyquist formula

The open-circuit noise voltage of a resistor R in thermal equilibrium at temperature T has

$$G_V(f) = 4kTR \ \ \text{V}^2/\text{Hz (one-sided)}, \qquad S_V(f) = 2kTR \ \ \text{(two-sided, per Hz)},$$

where $k = 1.380649 \times 10^{-23}$ J/K is Boltzmann's constant. In a bandwidth B (or noise bandwidth B_N):

$$\overline{v_n^2} = 4kTRB.$$

Equivalently (Norton form), a noise current source of one-sided PSD $G_I = 4kT/R$ in parallel with a noiseless R.

Sketch of Nyquist's argument. Connect two equal resistors R by a lossless transmission line of characteristic impedance R and length L; everything is at temperature T. Each resistor delivers noise power into the line, matched, so it is absorbed by the other. Now short both ends: the line becomes a resonator with modes at $f_m = mc/(2L)$, i.e. $2L\,B/c$ modes in a band B. By the equipartition theorem each mode holds average energy kT, so the energy in band B is $kT \cdot 2LB/c$; the line delivers it in a transit time L/c in each direction, giving power kTB from each resistor into the other. A resistor with open-circuit noise voltage v_n delivers $\overline{v_n^2}/(4R)$ into a matched load, so $\overline{v_n^2}/(4R) = kTB$, i.e. $\overline{v_n^2} = 4kTRB$. □

Two corollaries are used constantly:

- **Available noise power** $P_n = kTB$, independent of R: $kT_0 = 4.00 \times 10^{-21}$ W/Hz at the reference temperature $T_0 = 290$ K, i.e. **−174 dBm/Hz**.
- **Combining resistors.** Independent noise sources add in power. Two resistors in series: $4kT(R_1 + R_2)B$ — the same as one resistor $R_1 + R_2$. In parallel: $4kT(R_1 \parallel R_2)B$. More generally, the noise of any passive network of resistors, capacitors and inductors at a uniform temperature is $4kT\,\mathrm{Re}\{Z(f)\}$ per hertz, where Z is the network's impedance (the fluctuation–dissipation theorem).

Worked Example 21.35 — Resistor noise in audio and RF bands

(a) Audio. $R = 10\ \text{k}\Omega$, $T = 300$ K, $B = 20$ kHz:

$$v_{n,\text{rms}} = \sqrt{4kTRB} = \sqrt{4 \times 1.381 \times 10^{-23} \times 300 \times 10^{4} \times 2 \times 10^{4}} = \sqrt{3.31 \times 10^{-12}} \approx 1.82\ \mu\text{V}.$$

The often-quoted figure $\sqrt{4kTR} \approx 12.9\ \text{nV}/\sqrt{\text{Hz}}$ for 10 kΩ is the same result per root-hertz ($12.9\ \text{nV} \times \sqrt{20000} = 1.82\ \mu\text{V}$ ✓).

(b) RF. $R = 50\ \Omega$, $T = 290$ K, $B = 1$ MHz: $\overline{v_n^2} = 8.0 \times 10^{-13}\ \text{V}^2$, $v_{n,\text{rms}} = 0.895\ \mu$V open-circuit. The available power is $kTB = 4.0 \times 10^{-15}\ \text{W} = -114$ dBm — the noise floor of a 1 MHz-wide receiver before any amplifier noise is added.

(c) Parallel combination. Two 1 kΩ resistors in parallel at 300 K, $B = 10$ kHz: equivalent 500 Ω, $v_{n,\text{rms}} = \sqrt{4kT \cdot 500 \cdot 10^{4}} = 0.288\ \mu$V. Check via Norton sources: each contributes current density $4kT/1000$; the two independent currents add in power to $8kT/1000$, and across 500 Ω give voltage density $8kT/1000 \times 500^2 = 4kT \times 500$. ✓

Worked Example 21.36 — kT/C noise

Model a resistor R (with its noise source) connected across a capacitor C. The capacitor voltage is the RC-lowpass output of white noise with one-sided density $4kTR$, i.e. $N_0/2 = 2kTR$ two-sided.

Step 1. By Example 21.18, $\overline{v_C^2} = \dfrac{N_0}{4RC} = \dfrac{4kTR}{4RC} = \dfrac{kT}{C}$.

Step 2 — the resistance cancels. A larger R produces more noise density but a proportionally narrower noise bandwidth $B_N = 1/(4RC)$. Equivalently, by equipartition, the capacitor's mean stored energy $\tfrac{1}{2}C\overline{v_C^2}$ equals $\tfrac{1}{2}kT$.

Step 3 — numbers. For $C = 1$ pF at 300 K, $v_{\text{rms}} = \sqrt{4.14 \times 10^{-21}/10^{-12}} \approx 64\ \mu$V. For a 12-bit ADC with 1 V full scale (LSB $= 244\ \mu$V), the quantization noise is $244/\sqrt{12} = 70\ \mu$V rms, so a 1 pF sampling capacitor already contributes noise comparable to the quantization error. This sets a floor on the noise of sample-and-hold circuits, switched-capacitor filters and CMOS image-sensor pixels, and is why low-noise ADC front ends use large sampling capacitors.

Shot noise

Current in a diode or photodetector consists of discrete charges q crossing a barrier at random (Poisson) times. Modelling the current as $I(t) = q\sum_k \delta(t - t_k)$ with Poisson arrivals of rate $\lambda = \bar{I}/q$ (Example 21.3), one finds that its fluctuation about the mean $\bar{I}$ is white with one-sided density

$$G_I(f) = 2q\bar{I} \qquad \Longrightarrow \qquad \overline{i_n^2} = 2q\bar{I}B.$$

Sketch. For a Poisson impulse train of rate λ, the autocovariance is $\lambda q^2\delta(\tau)$ (increments over disjoint intervals are independent), so the two-sided PSD of the fluctuation is $\lambda q^2 = q\bar{I}$, and the one-sided density is $2q\bar{I}$.

For example, a photodiode current of $\bar{I} = 1\ \mu$A in a 1 MHz bandwidth has $\sqrt{2 \times 1.602 \times 10^{-19} \times 10^{-6} \times 10^{6}} = 0.566$ nA rms of shot noise — a relative fluctuation of 5.7×10^{-4}, the $1/\sqrt{\lambda t}$ law of Example 21.3 in another form.

Signal-to-noise ratio, noise figure and averaging

Signal-to-noise ratio

For a signal component with power P_s and an additive noise component with power P_n at the same point, $\text{SNR} = P_s/P_n$, usually quoted as $10\log_{10}(\text{SNR})$ dB. If signal $s(t)$ (deterministic or random, with PSD S_{ss}) and independent noise with PSD S_{NN} pass through H,

$$\text{SNR}_{\text{out}} = \frac{\frac{1}{2\pi}\int |H|^2 S_{ss}\, d\omega}{\frac{1}{2\pi}\int |H|^2 S_{NN}\, d\omega}.$$

Noise factor and noise figure

For a two-port with available gain G, driven by a source at $T_0 = 290$ K, the **noise factor** is

$$F = \frac{\text{SNR}_{\text{in}}}{\text{SNR}_{\text{out}}} = 1 + \frac{T_e}{T_0} \;\geq 1,$$

where T_e is the **equivalent input noise temperature**: the amplifier's internal noise referred to its input as if it were extra thermal noise kT_eB. The **noise figure** is $NF = 10\log_{10} F$ dB.

For a cascade of stages with gains $G_1, G_2, \ldots$ and noise factors $F_1, F_2, \ldots$, refer each stage's added noise to the input of the chain by dividing by the gain preceding it. Because the noise contributions are independent, their powers add, and one obtains **Friis's formula**:

$$F = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \cdots$$

Worked Example 21.37 — Receiver noise floor and cascade

A receiver front end has an LNA with $NF_1 = 1.5$ dB ($F_1 = 1.413$) and gain 15 dB ($G_1 = 31.6$), followed by a mixer with $NF_2 = 10$ dB ($F_2 = 10$).

Step 1 — cascade. $F = 1.413 + \dfrac{10 - 1}{31.6} = 1.413 + 0.285 = 1.697$, i.e. $NF = 2.30$ dB. The noisy mixer costs only 0.8 dB because the LNA's gain divides its contribution.

Step 2 — sensitivity. The noise floor referred to the input in a noise bandwidth $B_N = 200$ kHz is $-174 + 10\log_{10}(2 \times 10^5) + NF = -174 + 53.0 + 2.3 = -118.7$ dBm. If the demodulator needs SNR $= 10$ dB, the minimum detectable signal is -108.7 dBm.

Step 3 — reversing the order. Put the mixer first; suppose it is passive with a conversion loss of 7 dB ($G = 0.2$). Then $F = 10 + \dfrac{1.413 - 1}{0.2} = 12.07$, i.e. $NF = 10.8$ dB — 8.5 dB worse, and the sensitivity drops to about -100 dBm. The first stage dominates; this is the mathematical reason the lowest-noise, highest-gain stage goes first.

SNR gain from averaging

If a deterministic waveform $s(t)$ is observed M times in independent zero-mean noise of variance σ^2 (per sample), and the M records are averaged point by point (synchronous or **coherent averaging**), the signal is unchanged while the noise variance becomes σ^2/M. The SNR improves by a factor M (10 dB per decade of M). If the noise in successive records is correlated with correlation coefficients ρ_{ik}, the improvement is $M^2/\sum_{i,k}\rho_{ik}$ instead.

Proof. The average of the noise samples has variance $\frac{1}{M^2}\sum_{i,k}\operatorname{Cov}(N_i, N_k) = \frac{\sigma^2}{M^2}\sum_{i,k}\rho_{ik}$, which is σ^2/M for uncorrelated records. □

This is how oscilloscopes, evoked-potential recorders and radar pulse integrators dig repetitive signals out of noise; Example 21.31 is the same principle for a constant signal.

The matched filter

A pulse $s(t)$ of known shape, nonzero only on $[0, T]$, is received in additive white noise of two-sided PSD $N_0/2$. It is passed through a filter h and the output is sampled at time T_0. Which h makes the pulse most visible — i.e. maximizes the output SNR at T_0?

At the sampling instant the signal output is $y_s(T_0) = \int h(\alpha)s(T_0 - \alpha)\,d\alpha$, and the noise output has variance $\frac{N_0}{2}\int h^2(\alpha)\,d\alpha$ (§21.6). So

$$\mathrm{SNR}(T_0) = \frac{\left(\int h(\alpha)s(T_0-\alpha)\,d\alpha\right)^2}{\frac{N_0}{2}\int h^2(\alpha)\,d\alpha}.$$

Matched filter

The output SNR at T_0 is maximized by

$$h_{\mathrm{opt}}(t) = c\,s(T_0 - t) \qquad (\text{any constant } c \neq 0), \qquad H_{\mathrm{opt}}(\omega) = c\,S^*(\omega)e^{-j\omega T_0},$$

and the maximum is

$$\mathrm{SNR}_{\max} = \frac{2E}{N_0}, \qquad E = \int s^2(t)\,dt,$$

which depends only on the pulse **energy**, not its shape. Choosing $T_0 = T$ makes h_{opt} causal.

Proof. By the Cauchy–Schwarz inequality for integrals,

$$\left(\int h(\alpha)s(T_0-\alpha)\,d\alpha\right)^2 \le \int h^2(\alpha)\,d\alpha \cdot \int s^2(T_0-\alpha)\,d\alpha = E\int h^2(\alpha)\,d\alpha,$$

with equality iff $h(\alpha) \propto s(T_0 - \alpha)$. Substituting, $\mathrm{SNR} \le E/(N_0/2) = 2E/N_0$. □

The filter's impulse response is the pulse **reversed in time**; equivalently, its output at T_0 is the **correlation** $\int r(t)s(t)\,dt$ of the received waveform with the known pulse — a correlator receiver. Its frequency response emphasizes the frequencies where the signal is strong ($|H| \propto |S|$) and aligns all their phases to peak simultaneously at T_0. For coloured noise of PSD S_{NN}, the same argument applied after a whitening filter gives $H_{\mathrm{opt}} \propto S^*(\omega)e^{-j\omega T_0}/S_{NN}(\omega)$.

Worked Example 21.38 — Matched filter for a rectangular pulse, and what a simple RC filter loses

Pulse. $s(t) = A$ on $[0, T]$, $A = 1$ mV, $T = 1$ ms; noise $N_0/2 = 10^{-10}$ V^2/Hz.

Step 1 — matched filter. $h(t) = s(T - t)$ is again a rectangle on $[0, T]$: the **integrate-and-dump** filter. $E = A^2 T = 10^{-9}$ V^{2}s and

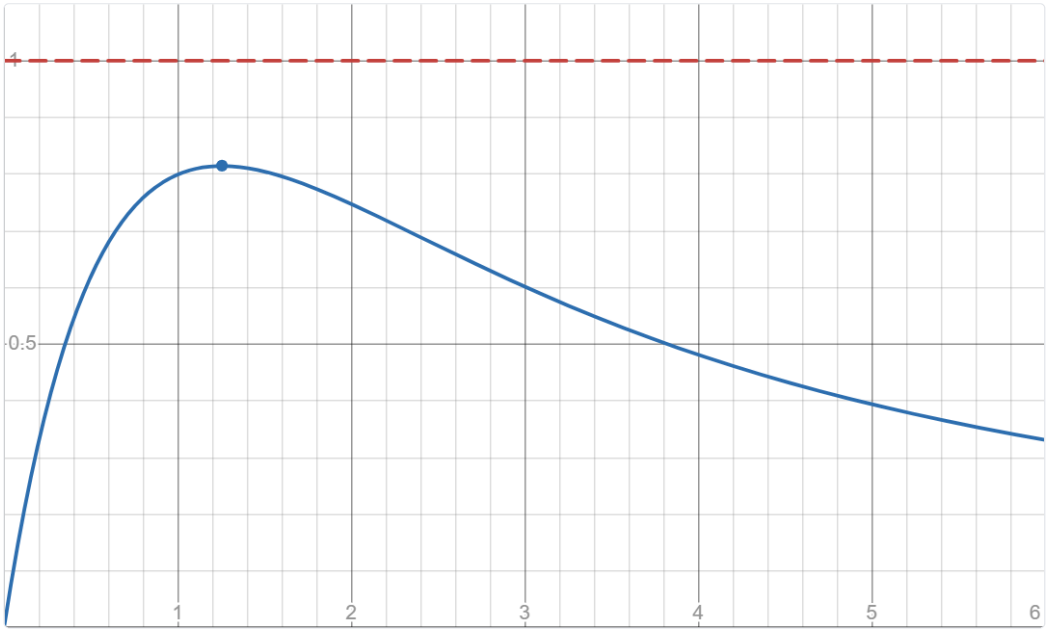
$$\mathrm{SNR}_{\max} = \frac{2E}{N_0} = \frac{2 \times 10^{-9}}{2 \times 10^{-10}} = 10 \ \ (10 \text{ dB}).$$

Step 2 — detection. For antipodal signalling ($\pm s(t)$) with a threshold at zero and Gaussian noise, the error probability is $P_e = Q\big(\sqrt{\mathrm{SNR}_{\max}}\big) = Q\big(\sqrt{2E/N_0}\big) = Q(3.16) = 7.8 \times 10^{-4}$. Raising E/N_0 to 9.6 dB gives the textbook $P_e \approx 10^{-5}$.

Step 3 — RC lowpass instead. Use $H = 1/(1 + j\omega RC)$ and sample at T. The signal output is $A(1 - e^{-T/RC})$; the noise variance is $N_0/(4RC)$ (Example 21.18). With $x = T/RC$,

$$\mathrm{SNR}_{RC} = \frac{A^2(1 - e^{-x})^2}{N_0/(4RC)} = \frac{2E}{N_0} \cdot \frac{2(1 - e^{-x})^2}{x}.$$

The factor $g(x) = 2(1 - e^{-x})^2/x$ is maximized numerically at $x = 1.26$ (i.e. $RC = 0.80\,T$), where $g = 0.815$: the best RC filter is only 0.89 dB worse than the matched filter. Too narrow a filter ($x \to 0$) cuts signal; too wide ($x \to \infty$) admits noise.



SNR of an RC lowpass relative to the matched filter for a rectangular pulse, against $x = T/RC$ (blue). The matched filter (red, dashed) is the upper bound set by Cauchy–Schwarz; the RC filter comes within 0.89 dB of it at $T/RC \approx 1.26$.

Engineering Note — estimation filters

The matched filter maximizes SNR for *detecting* a known pulse. When instead a random signal $S(t)$ must be *estimated* from $X = S + N$, the filter minimizing $E[(S - \hat{S})^2]$ is the (non-causal) **Wiener filter** $H(\omega) = S_{SX}(\omega)/S_{XX}(\omega) = S_{SS}/(S_{SS} + S_{NN})$ for independent S, N: it passes frequencies where the signal dominates and attenuates those where the noise does. Its recursive, time-domain counterpart for state-space models is the Kalman filter. Both are built entirely from the correlation functions and spectra of this chapter.

Engineering Note — the noise-calculation workflow

Nearly every noise calculation in circuits and communications follows the same steps: (1) model each noise source by its PSD (thermal $4kTR$, shot $2qI$, amplifier e_n^2, i_n^2, flicker $\propto 1/f$); (2) transfer each to the output with its own $|H_i(\omega)|^2$; (3) add the PSDs of independent sources; (4) integrate (or multiply by a noise bandwidth) to get power; (5) refer to the input by dividing by the signal gain if an input-referred figure is wanted; (6) if Gaussian, convert to probabilities with $Q(\cdot)$; (7) if the result comes from a measurement, check the averaging time with the ergodic formulas of §21.7.

Common Pitfall — adding RMS values

Independent noise sources add **in power** (mean-square), never in RMS: 1 μV and 1 μV rms of independent noise give 1.41 μV, not 2 μV. Only fully correlated sources (the same noise reaching the output by two paths) add in amplitude — and then with their relative phase, possibly cancelling. Also remember that the PSD conventions (one-sided G, two-sided S, per hertz or per rad/s) must be the same for all terms before adding.

Summary

Concept	Key formula
Mean, autocorrelation	$\mu_X(t) = E[X(t)]$, $R_{XX}(t_1, t_2) = E[X(t_1)X(t_2)]$

Concept	Key formula
Autocovariance	$C_{XX}(t_1,t_2) = R_{XX}(t_1,t_2) - \mu_X(t_1)\mu_X(t_2)$
Time averages	$\langle x\rangle = \lim \frac{1}{2T}\int_{-T}^{T} x\,dt$, $\langle x(t+\tau)x(t)\rangle$
Independent increments	$C_{XX}(t_1,t_2) = \sigma_X^2(\min(t_1,t_2))$; random walk $\min(n_1,n_2)$; Poisson $\lambda\min(t_1,t_2)$
WSS	μ_X constant, $R_{XX}(\tau) = E[X(t+\tau)X(t)]$
Cyclostationary → WSS	$\bar{R}_{XX}(\tau) = \frac{1}{T_0}\int_0^{T_0} R_{XX}(t+\tau,t)\,dt$
Autocorrelation properties	$R_{XX}(0) = E[X^2]$, even, $\lvert R_{XX}(\tau)\rvert \le R_{XX}(0)$, non-negative definite
Derivative	$R_{X'X'}(\tau) = -R''_{XX}(\tau)$; $E[X'X] = 0$
Cross-correlation	$R_{XY}(\tau) = E[X(t+\tau)Y(t)] = R_{YX}(-\tau)$
Wiener–Khinchin	$S_{XX}(\omega) = \int R_{XX}(\tau)e^{-j\omega\tau}d\tau$
Power	$E[X^2] = \frac{1}{2\pi}\int S_{XX}(\omega)\,d\omega = \int S_X(f)\,df = \int_0^\infty G_X(f)\,df$
Exponential pair	$\sigma^2 e^{-\alpha\lvert\tau\rvert} \leftrightarrow 2\alpha\sigma^2/(\alpha^2+\omega^2)$
NRZ data	$A^2(1-\lvert\tau\rvert/T) \leftrightarrow A^2T\,\sin^2(\omega T/2)/(\omega T/2)^2$
LTI output	$\mu_Y = \mu_X H(0)$, $S_{YX} = HS_{XX}$, $S_{YY} = \lvert H\rvert^2 S_{XX}$
Discrete WK, LTI	$S_{XX}(e^{j\Omega}) = \sum_k R_{XX}[k]e^{-j\Omega k}$; $S_{YY} = \lvert H(e^{j\Omega})\rvert^2 S_{XX}$
AR(1)	$R_{XX}[k] = \frac{\sigma_W^2}{1-a^2}a^{\lvert k\rvert}$, $S_{XX} = \frac{\sigma_W^2}{1-2a\cos\Omega+a^2}$
Gaussian conditional	$X(t+\tau)\mid X(t)=x \sim \mathcal{N}(\rho x,\ \sigma^2(1-\rho^2))$ (zero mean)
Rice's formula	$\nu_u = \frac{1}{\pi}\sqrt{-R''(0)/R(0)}\,e^{-u^2/2R(0)}$
Narrowband envelope	Rayleigh: $P\{R>r\} = e^{-r^2/2\sigma^2}$
White noise	$S = N_0/2$, $R = \frac{N_0}{2}\delta(\tau)$; filtered: $E[Y^2] = \frac{N_0}{2}\int h^2 dt$
Band-limited white noise	$R = N_0 B\,\frac{\sin 2\pi B\tau}{2\pi B\tau}$, power $N_0 B$
Noise-equivalent bandwidth	$B_N = \frac{1}{\lvert H(0)\rvert^2}\int_0^\infty \lvert H(2\pi f)\rvert^2 df$; RC: $1/(4RC)$; averager: $1/(2T)$
Variance of time average	$\mathrm{Var}\langle X\rangle_T \approx \frac{1}{2T}\int C_{XX}(\tau)d\tau$; sequence: $\frac{1}{N}\sum_k C_{XX}[k]$
Thermal noise	$\overline{v_n^2} = 4kTRB$; available kTB (−174 dBm/Hz); on a capacitor kT/C
Shot noise	$\overline{i_n^2} = 2q\bar{I}B$
Friis	$F = F_1 + (F_2-1)/G_1 + \cdots$
Matched filter	$h(t) = s(T_0 - t)$, $\mathrm{SNR}_{\max} = 2E/N_0$

Further reading

- A. Papoulis and S. U. Pillai, *Probability, Random Variables, and Stochastic Processes*, 4th ed., McGraw-Hill — the standard reference for stationarity, ergodicity and spectral theory.
- A. Leon-Garcia, *Probability, Statistics, and Random Processes for Electrical Engineering*, 3rd ed., Pearson — gentler, with many EE examples, including AR models and estimation.
- P. Z. Peebles, *Probability, Random Variables, and Random Signal Principles*, 4th ed., McGraw-Hill — concise treatment of noise in linear systems, noise bandwidth and noise figure.
- R. G. Gallager, *Stochastic Processes: Theory for Applications*, Cambridge University Press — rigorous treatment of Gaussian and Poisson processes.

Part VII — Numerical and Computational Mathematics

Chapter 22 — Numerical Methods

> **Prerequisites:** Ch. 4 (Differential Calculus — Taylor's theorem), Ch. 5 (Integral Calculus), Ch. 8 (Matrices and Linear Systems), Ch. 10 (Eigenvalues and Eigenvectors), Ch. 11–13 (ODEs and PDEs) **Learning outcomes:** After this chapter you can …
>
> - explain how IEEE-754 floating-point numbers are stored, bound the rounding error of an arithmetic operation, and recognise catastrophic cancellation;
> - separate truncation error from rounding error, and distinguish the *conditioning* of a problem from the *stability* of an algorithm;
> - solve nonlinear equations by bisection, Newton–Raphson, secant and fixed-point iteration, and predict their order of convergence and failure modes;
> - build interpolating polynomials (Lagrange, Newton) and cubic splines, explain the Runge phenomenon and cure it with Chebyshev nodes, and compute least-squares polynomial approximations;
> - derive finite-difference and quadrature rules with their error orders, and accelerate them by Richardson extrapolation;
> - integrate ODEs with Euler and Runge–Kutta methods, recognise stiffness, and discretise the heat, wave and Laplace equations with step sizes that respect stability (CFL) limits;
> - state the consistency–stability–convergence relationship (Lax equivalence) and carry out a von Neumann stability analysis.

Almost every quantitative result an engineer produces today passes through a computer. SPICE solves the nonlinear equations of a circuit by Newton's method; an electromagnetic field solver discretises Maxwell's equations on a grid; a control engineer simulates a plant by integrating ODEs with an adaptive Runge–Kutta method; a DSP engineer worries about whether a 16-bit accumulator will overflow. In each case the answer on the screen is *not* the exact mathematical answer — it is an approximation, and its quality depends on choices that the user often leaves at default values without understanding them.

This chapter is about those choices. The guiding questions are always the same three:

1. **What is computed?** (the algorithm, i.e. the discrete approximation to a continuous problem);
2. **How wrong is it?** (truncation error from the discretisation, rounding error from finite-precision arithmetic, and how both propagate);
3. **How does the error behave as we refine?** (order of convergence, stability, cost).

Throughout, small Python snippets show the idea in executable form; every number quoted in a worked example was checked by running such code in IEEE-754 double precision.

22.1 Floating-point arithmetic

Why an engineer must care

Real numbers have infinitely many digits; computer words have 16, 32 or 64 bits. Every real number that enters a computation is therefore replaced by a nearby *representable* number, and every arithmetic operation on representable numbers is followed by a rounding back to the representable set. Usually the resulting errors are around 10^{-16} relative and harmless. Occasionally they are amplified into total nonsense — a subtraction of nearly equal numbers, a recursion that multiplies errors by n at every step, a matrix that is nearly singular. To tell the two situations apart you need a precise model of floating-point arithmetic.

Floating-point number systems

Floating-point number system

A (normalised, binary) floating-point number system $\mathbb{F}(p, e_{\min}, e_{\max})$ consists of zero and all numbers of the form

$$x = \pm\,(1.d_1 d_2 \ldots d_{p-1})_2 \times 2^{e}, \qquad d_i \in \{0,1\}, \quad e_{\min} \le e \le e_{\max}.$$

The integer p is the **precision** (number of significant bits, including the leading 1), $(1.d_1 \ldots d_{p-1})_2$ is the **significand** (or mantissa) and e the **exponent**.

Two features follow at once.

- **The numbers are not evenly spaced.** Between 2^{e} and 2^{e+1} there are exactly 2^{p-1} equally spaced numbers with spacing 2^{e-p+1}. Each time you cross a power of two the spacing doubles. Relative spacing, however, is roughly constant: this is the whole point of floating point.
- **There is a largest and a smallest positive normalised number**, $x_{\max} = (2 - 2^{1-p})\,2^{e_{\max}}$ and $x_{\min} = 2^{e_{\min}}$. Results larger than $x_{\max}$ **overflow**; results smaller than $x_{\min}$ **underflow**.

Machine epsilon, unit roundoff, ulp

- **Machine epsilon** $\varepsilon_M = 2^{1-p}$ is the distance from 1 to the next larger floating-point number.
- **Unit roundoff** $u = \tfrac{1}{2}\varepsilon_M = 2^{-p}$ is the maximum relative error committed when a real number in the normal range is rounded to nearest.
- The **ulp** (unit in the last place) of x is the spacing of floating-point numbers near x: $\operatorname{ulp}(x) = 2^{e-p+1}$ when $2^{e} \le |x| < 2^{e+1}$.

Be aware that some texts (and MATLAB's `eps`) call ε_M "machine epsilon" while others reserve that name for u. The factor of two rarely matters, but always check which one a bound uses.

The IEEE-754 standard

Since 1985 essentially all hardware has implemented the IEEE-754 standard (revised 2008 and 2019). The formats an EE meets most often are:

Format	Total bits	Sign / exponent / fraction bits	Precision p	$\varepsilon_M = 2^{1-p}$	Largest normal	Smallest normal
binary16 (half)	16	1 / 5 / 10	11	9.77×10^{-4}	65504	6.10×10^{-5}
binary32 (single, `float`)	32	1 / 8 / 23	24	1.19×10^{-7}	3.40×10^{38}	1.18×10^{-38}
binary64 (double)	64	1 / 11 / 52	53	2.22×10^{-16}	1.80×10^{308}	2.23×10^{-308}

So single precision carries about 7 significant decimal digits and double precision about 16.

Encoding. A binary64 word is laid out as

$$\underbrace{s}_{1\text{ bit}}\ \underbrace{E}_{11\text{ bits}}\ \underbrace{f}_{52\text{ bits}}, \qquad x = (-1)^s\,(1.f)_2 \times 2^{E-1023} \quad (1 \le E \le 2046).$$

The exponent is stored with a **bias** of 1023 (127 for single), so that the stored field E is an unsigned integer. The leading 1 of the significand is *not* stored ("hidden bit"), which is why 52 stored fraction bits give $p = 53$. The two extreme exponent fields are reserved:

Stored E	f	Meaning
0	0	± 0 (signed zero)
0	$\neq 0$	**subnormal**: $(-1)^s (0.f)_2 \times 2^{-1022}$
$1 \ldots 2046$	any	normal number
2047	0	$\pm\infty$
2047	$\neq 0$	NaN (Not a Number)

Subnormals fill the gap between 0 and $x_{\min}$ evenly ("gradual underflow"); the smallest positive subnormal double is $2^{-1074} \approx 4.94 \times 10^{-324}$. Infinities and NaNs make arithmetic *closed*: $1/0 = +\infty$, $0/0 = \text{NaN}$, $\sqrt{-1} = \text{NaN}$, and any comparison involving NaN is false (so `x != x` is the classic NaN test).

Worked Example 22.1 — Encoding a number in binary64

Encode $x = 5.75$.

Step 1 — binary expansion. $5 = 101_2$ and $0.75 = 0.11_2$, so $x = 101.11_2$.

Step 2 — normalise. $101.11_2 = 1.0111_2 \times 2^2$, so $s = 0$, $e = 2$, fraction $f = 0111\,000\ldots0$ (52 bits).

Step 3 — bias the exponent. $E = 2 + 1023 = 1025 = 100\,0000\,0001_2$.

Step 4 — assemble. 0 10000000001 0111 0000 … Grouping in fours: 0100 0000 0001 0111 0000 …, i.e. the hexadecimal word `0x4017000000000000`.

Check. In Python, `struct.pack('>d', 5.75).hex()` returns `'4017000000000000'`. Because 5.75 has a finite binary expansion, it is stored *exactly*.

Worked Example 22.2 — Why $0.1 + 0.2 \neq 0.3$

In binary, $0.1 = 0.0\overline{0011}_2$ repeats forever, so it must be rounded. The stored double nearest to 0.1 is

$$\text{fl}(0.1) = 0.1000000000000000055511151231257827\ldots$$

(hex `0x3FB9999999999A` — note the final `A`: the repeating `9` pattern rounded up). The absolute error is 5.55×10^{-18}, so the relative error is $5.55 \times 10^{-18}/0.1 \approx 5.6 \times 10^{-17} < u = 1.11 \times 10^{-16}$, as promised.

Adding $\text{fl}(0.1) + \text{fl}(0.2)$ and rounding gives `0.30000000000000004`, which is one ulp above $\text{fl}(0.3)$. Likewise, adding 0.1 ten times in a loop gives `0.9999999999999999`. Neither result is a bug; each is the correctly rounded result of operations on numbers that were never exactly 0.1, 0.2, 0.3.

Lesson. Never test floating-point results for exact equality. Use a tolerance, e.g. `abs(a-b) <= rtol*max(abs(a),abs(b)) + atol`.

Rounding and the standard model

IEEE-754 requires the five basic operations $+, -, \times, \div, \sqrt{}$ to be **correctly rounded**: the result is the exact real result, rounded to the representable set according to the current rounding mode (default: *round to nearest, ties to even*). This single requirement gives the most useful model in numerical analysis.

Standard model of floating-point arithmetic

Let $\text{fl}(\cdot)$ denote rounding to nearest, and let $u = 2^{-p}$. If x lies in the normal range, then

$$\text{fl}(x) = x(1+\delta), \qquad |\delta| \le u.$$

Consequently, for floating-point numbers a, b and $\circ \in \{+, -, \times, \div\}$ (no overflow or underflow),

$$\text{fl}(a \circ b) = (a \circ b)(1+\delta), \qquad |\delta| \le u.$$

Proof sketch. Suppose $2^e \le x < 2^{e+1}$. Representable numbers in this interval are spaced $\Delta = 2^{e-p+1}$ apart, so the nearest one is within $\Delta/2 = 2^{e-p}$ of x. Dividing by $x \ge 2^e$ gives a relative error at most $2^{-p} = u$. Correct rounding means the computed $a \circ b$ is just fl applied to the exact result. $\square$

The model says that *each individual operation* is accurate to about 16 digits in double precision. Large errors can only arise from the way errors from many operations accumulate or are amplified, which is the subject of Section 22.2.

Consequences: non-associativity and summation error

Floating-point addition is commutative but **not associative**. In double precision, $(10^{16} + 1) - 10^{16} = 0$ because $10^{16} + 1$ rounds to 10^{16} (the ulp at 10^{16} is 2), whereas $10^{16} + (1 - 10^{16}) = 1$. The order of operations matters.

For a sum $s_n = x_1 + \cdots + x_n$ computed left to right, repeated application of the standard model gives

$$|\hat{s}_n - s_n| \;\le\; (n-1)\,u \sum_{i=1}^{n} |x_i| + O(u^2).$$

The bound grows linearly in n; in practice, with randomly signed rounding errors, the error typically grows like $\sqrt{n}\,u$. For a million-term sum in single precision ($u \approx 6 \times 10^{-8}$) the worst-case bound is already of order 6×10^{-2} relative — a serious problem for, say, accumulating energy in a long FFT or a power meter. Two cures:

- **Pairwise summation** (add in a binary tree): error bound $O(u \log_2 n)$. NumPy's `sum` does this.
- **Compensated (Kahan) summation**: carry a correction term that recaptures the low-order bits lost in each addition; the error bound becomes $2u \sum |x_i| + O(nu^2)$, essentially independent of n.

```
def kahan_sum(xs):
    s, c = 0.0, 0.0           # running sum and compensation
    for x in xs:
        y = x - c             # subtract the error from last time
        t = s + y             # low bits of y may be lost here ...
        c = (t - s) - y       # ... and are recovered algebraically
        s = t
    return s
```

Catastrophic cancellation

Cancellation

When two nearly equal numbers $a \approx b$ that already carry relative errors are subtracted, the leading digits cancel and the relative error of $a - b$ is magnified by the factor $|a|/|a - b|$. The subtraction itself is exact (Sterbenz's lemma: if $b/2 \le a \le 2b$ then $a - b$ is exactly representable); it merely *exposes* the errors that were already present.

Worked Example 22.3 — Evaluating $1 - \cos x$ for small x

Compute $g(x) = (1 - \cos x)/x^2$ at $x = 1.2 \times 10^{-5}$. The exact value is very close to $\tfrac{1}{2}$ (Taylor: $\tfrac{1}{2} - x^2/24$).

Naive formula. $\cos x = 1 - 7.2 \times 10^{-11} + \ldots$ is computed with absolute error up to about $u \approx 1.1 \times 10^{-16}$. The subtraction $1 - \cos x \approx 7.2 \times 10^{-11}$ then has relative error up to $1.1 \times 10^{-16} / 7.2 \times 10^{-11} \approx 1.5 \times 10^{-6}$. In double precision the naive formula returns 0.4999997330 — only six correct digits.

Rewritten formula. Use the identity $1 - \cos x = 2\sin^2(x/2)$, which involves no subtraction. This returns 0.49999999999400, correct to full precision.

At $x = 10^{-8}$ the naive formula gives $1 - \cos x = 0$ exactly (because $\cos x$ rounds to 1), a 100 % error, while $2\sin^2(x/2) = 5.0 \times 10^{-17}$ is correct.

Worked Example 22.4 — A stable quadratic formula

Solve $x^2 - 10^8 x + 1 = 0$. The roots are $x_1 \approx 10^8$ and $x_2 \approx 10^{-8}$.

Textbook formula. $x_2 = \left(10^8 - \sqrt{10^{16} - 4}\right)/2$. Here $\sqrt{10^{16} - 4}$ rounds to a number within one ulp ($\approx 1.5 \times 10^{-8}$) of 10^8, and subtracting it from 10^8 cancels all the good digits. Double precision returns $x_2 = 7.45 \times 10^{-9}$, a 25 % error.

Stable approach. Compute the large-magnitude root without cancellation, $x_1 = \left(-b + \operatorname{sign}(-b)\sqrt{b^2 - 4ac}\right)/(2a) = 10^8$, then use Vieta's relation $x_1 x_2 = c/a$: $x_2 = c/(a x_1) = 10^{-8}$, correct to full precision.

The general rule: **rearrange formulas so that nearly equal quantities that carry error are never subtracted.**

Overflow, underflow and scaling

Intermediate results can overflow even when the final answer is modest. Computing the magnitude of a phasor $|a + jb| = \sqrt{a^2 + b^2}$ with $a = b = 10^{200}$ overflows at a^2, although the answer 1.41×10^{200} is representable. Library functions such as `hypot(a,b)` scale first: with $m = \max(|a|, |b|)$, $\sqrt{a^2 + b^2} = m\sqrt{(a/m)^2 + (b/m)^2}$. Similarly, probabilities of long sequences (Ch. 19) and likelihoods underflow, which is why they are accumulated as logarithms.

Common Pitfall — Comparing floats and counting with floats

Loops such as `t = 0.0; while t != 1.0: t += 0.1` never terminate, because t steps over 1.0. Loop over an integer counter and compute $t = n\,h$ from it. The same applies to time stepping in simulations: computing $t_n = nh$ avoids the drift that $t \mathrel{+}= h$ accumulates.

Engineering Note — Fixed point versus floating point in DSP

Many low-cost DSPs and FPGAs use **fixed-point** arithmetic, e.g. the Q15 format, a 16-bit two's-complement integer k representing $k/2^{15} \in [-1, 1)$. The spacing is uniform ($2^{-15} \approx 3.05 \times 10^{-5}$), so the *absolute* error is constant but the *relative* error of small signals is large, and sums can overflow. Designers therefore scale filter coefficients and signals carefully and analyse quantisation noise statistically (each rounding is modelled as additive uniform noise of variance $\Delta^2/12$ — see Ch. 19–21). Floating point removes most scaling headaches at the cost of silicon area and power; half precision (binary16) and bfloat16 are now common in machine-learning accelerators, where $\varepsilon_M \approx 10^{-3}$ is acceptable.

22.2 Numerical error

Measuring error

Absolute and relative error

If $\hat{x}$ approximates x, the **absolute error** is $|\hat{x} - x|$ and, for $x \neq 0$, the **relative error** is

$$\rho(\hat{x}) = \frac{|\hat{x} - x|}{|x|}.$$

For vectors replace $|\cdot|$ by a norm. $\hat{x}$ has about k **correct significant digits** when $\rho \lesssim 0.5 \times 10^{-k}$.

Relative error is dimensionless and scale-invariant, which is why it is the natural measure for floating-point results ($\rho \leq u$ after a single rounding). Absolute error is the right measure near zero, where relative error is meaningless: a computed voltage of 3×10^{-17} V where the exact answer is 0 has an infinite relative error but is perfectly acceptable.

Sources of error

Any numerical answer is contaminated by four distinct kinds of error, and it pays to keep them apart:

1. **Modelling error** — the equations idealise reality (an ideal diode, a lumped-element line, a linearised amplifier). Not a numerical issue, but often the largest error (Ch. 25).
2. **Data error** — component tolerances, measurement noise, constants known to finite accuracy.
3. **Truncation (discretisation) error** — the error of replacing a limiting process by a finite one: truncating a Taylor series, replacing a derivative by a difference quotient, an integral by a finite sum, an infinite iteration by a finite number of steps. It exists even in exact arithmetic and typically behaves like Ch^p for a step h.
4. **Rounding error** — from finite-precision arithmetic (Section 22.1). It exists even for an exact algorithm and typically behaves like $C'u/h^q$ when it is amplified by small steps.

Worked Example 22.5 — Truncation error of a Taylor series

How many terms of $e^x = \sum_{k} x^k/k!$ are needed to compute e with error below 10^{-6}?

Step 1 — remainder. By Taylor's theorem with Lagrange remainder (Ch. 4), truncating after the x^n term leaves

$$R_n(1) = \frac{e^{\xi}}{(n+1)!}, \qquad 0 < \xi < 1, \qquad \text{so} \quad 0 < R_n(1) < \frac{e}{(n+1)!}.$$

Step 2 — solve the inequality. We need $(n+1)! > e \times 10^6 \approx 2.72 \times 10^6$. Since $9! = 362\,880$ and $10! = 3\,628\,800$, $n+1 = 10$, i.e. $n = 9$ (ten terms, $k = 0, \dots, 9$).

Check. $\sum_{k=0}^{9} 1/k! = 2.718281526$, and $e - 2.718281526 = 3.03 \times 10^{-7} < 10^{-6}$. The truncation error is a property of the *formula*; rounding contributes only $\sim 10^{-16}$ here.

Order notation

We write $E(h) = O(h^p)$ as $h \to 0$ if $|E(h)| \leq Ch^p$ for some constant C and all sufficiently small h. A method with error $O(h^p)$ is said to be **of order** p. The order tells you how the error responds to refinement: halving h divides the error by about 2^p. Conversely, from errors E_1, E_2 at steps h_1, h_2 you can *measure* the order:

$$p \approx \frac{\ln(E_1/E_2)}{\ln(h_1/h_2)}.$$

This "observed order" test is the single most useful debugging tool for numerical code: if a method that should be second order shows $p \approx 1$, there is a bug (often in a boundary condition).

For iterations $x_k \to x^*$ the analogous notion is the order of convergence q: $|x_{k+1} - x^*| \approx C|x_k - x^*|^q$ (Section 22.3).

Propagation of errors

If $y = f(x_1, \dots, x_n)$ and each input carries a small error Δx_i, a first-order Taylor expansion (Ch. 6) gives

$$\Delta y \approx \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}\, \Delta x_i.$$

Bounding term by term gives the **worst-case** estimate $|\Delta y| \le \sum_i |\partial f/\partial x_i|\,|\Delta x_i|$. If the errors are independent random variables with standard deviations σ_i, the variance formula (Ch. 19–20) gives the **root-sum-square** estimate $\sigma_y^2 \approx \sum_i (\partial f/\partial x_i)^2 \sigma_i^2$. For products and quotients it is convenient to work with relative errors: if $y = x_1^{a_1} x_2^{a_2} \cdots$, then $\Delta y / y \approx \sum_i a_i \, \Delta x_i / x_i$.

Worked Example 22.6 — Error in a computed power

A resistor $R = 50\ \Omega \pm 2\%$ carries a measured voltage $V = 10\ \text{V} \pm 1\%$. Estimate the error in $P = V^2/R$.

Nominal value. $P = 100/50 = 2\ \text{W}$.

Relative sensitivities. $P = V^2 R^{-1}$, so $\Delta P/P \approx 2\,\Delta V/V - \Delta R/R$.

Worst case. $2(1\%) + 1(2\%) = 4\%$, i.e. $P = 2.00 \pm 0.08\ \text{W}$.

Statistical (independent tolerances treated as standard deviations). $\sqrt{(2\cdot 1\%)^2 + (2\%)^2} = 2.83\%$, i.e. $\pm 0.057\ \text{W}$.

The squared voltage doubles its relative error — the exponent in a power law is exactly the condition number defined next.

Conditioning of a problem

Error propagation describes how the *problem* responds to perturbations of its data, independently of any algorithm. This intrinsic sensitivity is captured by a condition number.

Condition number of a scalar function

The (relative) condition number of evaluating a differentiable f at x is

$$\kappa_f(x) = \lim_{\varepsilon \to 0} \sup_{|\Delta x| \le \varepsilon |x|} \frac{|f(x+\Delta x) - f(x)|/|f(x)|}{|\Delta x|/|x|} = \left| \frac{x\, f'(x)}{f(x)} \right|.$$

The problem is **well-conditioned** if κ is modest and **ill-conditioned** if κ is large. A relative data error ρ produces a relative result error of about $\kappa \rho$.

Worked Example 22.7 — Condition numbers of elementary operations

Problem	$\kappa_f(x)$	Comment
$f(x) = \sqrt{x}$	$\tfrac{1}{2}$	always well-conditioned
$f(x) = x^n$	$\lvert n \rvert$	squaring doubles relative error (Example 22.6)
$f(x) = e^x$	$\lvert x \rvert$	ill-conditioned for large $\lvert x \rvert$: e^{700} is sensitive
$f(x) = \ln x$	$1/\lvert \ln x \rvert$	ill-conditioned near $x = 1$
$f(x) = x - 1$	$\lvert x/(x-1) \rvert$	at $x = 1.0001$: $\kappa = 10\,001$
$f(x) = \tan x$	$\lvert x/(\sin x \cos x) \rvert$	ill-conditioned near $x = \pi/2$

The entry for $x - 1$ is cancellation (Section 22.1) seen as a *conditioning* statement: subtracting nearly equal numbers is an ill-conditioned problem, so no algorithm can save you if x itself is inexact.

For a linear system $\mathbf{A}\mathbf{x} = \mathbf{b}$ the corresponding quantity is the **matrix condition number** (Ch. 8)

$$\kappa(\mathbf{A}) = \|\mathbf{A}\|\,\|\mathbf{A}^{-1}\| = \frac{\sigma_{\max}}{\sigma_{\min}} \quad \text{(2-norm)},$$

and perturbation theory gives, for $\mathbf{A}(\mathbf{x}+\Delta\mathbf{x}) = \mathbf{b} + \Delta\mathbf{b}$,

$$\frac{\|\Delta\mathbf{x}\|}{\|\mathbf{x}\|} \le \kappa(\mathbf{A})\, \frac{\|\Delta\mathbf{b}\|}{\|\mathbf{b}\|}.$$

Derivation. $\Delta\mathbf{x} = \mathbf{A}^{-1}\Delta\mathbf{b}$ gives $\|\Delta\mathbf{x}\| \le \|\mathbf{A}^{-1}\|\|\Delta\mathbf{b}\|$, and $\mathbf{b} = \mathbf{A}\mathbf{x}$ gives $\|\mathbf{b}\| \le \|\mathbf{A}\|\|\mathbf{x}\|$, i.e. $1/\|\mathbf{x}\| \le \|\mathbf{A}\|/\|\mathbf{b}\|$. Multiply the two inequalities. $\square$

A rule of thumb follows: **solving a linear system in floating point loses about $\log_{10}\kappa(\mathbf{A})$ decimal digits.**

Worked Example 22.8 — An ill-conditioned circuit matrix

Consider

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1.0001 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 2 \\ 2.0001 \end{bmatrix} \Rightarrow \mathbf{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Such matrices arise when two nodes are joined by a very small resistance compared with everything else, so their equations are nearly identical.

Condition number. $\kappa_2(\mathbf{A}) \approx 4.0 \times 10^4$.

Perturb the data. Change b_2 by 10^{-4} to 2.0002 — a relative change $\|\Delta\mathbf{b}\|/\|\mathbf{b}\| = 10^{-4}/2.83 = 3.5 \times 10^{-5}$. The solution becomes $\mathbf{x} = (0, 2)^{\mathsf{T}}$: a relative change of $\|(-1, 1)\|/\|(1, 1)\| = 100\%$.

Compare with the bound. $\kappa \cdot 3.5 \times 10^{-5} \approx 1.4$, so the observed amplification ($\approx 2.8 \times 10^4$) is close to the worst case. Nothing is wrong with the solver — the *problem* is ill-conditioned.

Hilbert matrices $H_{ij} = 1/(i + j - 1)$, which appear in least-squares polynomial fitting (Section 22.6), are a notorious family: $\kappa_2(\mathbf{H}_3) \approx 524$, $\kappa_2(\mathbf{H}_5) \approx 4.8 \times 10^5$, $\kappa_2(\mathbf{H}_8) \approx 1.5 \times 10^{10}$, $\kappa_2(\mathbf{H}_{12}) \approx 1.6 \times 10^{16}$ — at $n = 12$ double precision has no correct digits left.

Stability of an algorithm

Conditioning belongs to the problem; **stability** belongs to the algorithm. Let f be the exact map from data to result and $\hat{f}$ the algorithm executed in floating point.

Forward error, backward error, backward stability

- The **forward error** is $|\hat{f}(x) - f(x)|$ (relative: divided by $|f(x)|$).
- The **backward error** is the smallest $|\Delta x|/|x|$ such that $\hat{f}(x) = f(x + \Delta x)$: how much would the data have to change for the computed answer to be *exact*?
- An algorithm is **backward stable** if its backward error is $O(u)$ for all inputs. It is (forward) **unstable** if it can produce forward errors much larger than $\kappa_f(x)\,u$.

These ideas combine in the most important inequality in numerical analysis:

$$\text{forward error} \lesssim \text{condition number} \times \text{backward error}.$$

A backward-stable algorithm therefore delivers the best accuracy the problem allows: about $\kappa\,u$. Gaussian elimination with partial pivoting, Householder QR and the standard algorithms for $\sqrt{}$, sin, the FFT, etc. are backward stable (or nearly so). When a computed answer is bad, ask first: *is the problem ill-conditioned, or is my algorithm unstable?* The cures are completely different — reformulate the problem (or get better data) in the first case, change the algorithm in the second.

Worked Example 22.9 — An unstable recurrence and its stable reversal

Let $I_n = \int_0^1 x^n e^{x-1}\,dx$. Integration by parts gives the exact recurrence

$$I_n = 1 - n\,I_{n-1}, \qquad I_0 = 1 - e^{-1} \approx 0.632121.$$

Since $0 < x^n e^{x-1} \le x^n$, we know $0 < I_n < 1/(n+1)$, so $I_n \to 0$ slowly.

Forward recurrence in double precision.

n	0	1	5	10	15	17	18	19	20
$\hat{I}_n$	0.632121	0.367879	0.145533	0.083877	0.059034	0.057192	−0.029454	1.55962	−30.1924

By $n = 17$ the value has started to *increase* and at $n = 18$ it is negative, both impossible.

Explanation. An error ϵ_0 in I_0 (about $u \cdot I_0 \approx 7 \times 10^{-17}$ from rounding) propagates as $\epsilon_n = -n\,\epsilon_{n-1}$, so $|\epsilon_n| = n!\,|\epsilon_0|$. With $20! \approx 2.4 \times 10^{18}$, $|\epsilon_{20}| \sim 10^2$ — consistent in order of magnitude with the observed -30. The *problem* (computing I_{20}) is perfectly well conditioned; the *algorithm* is unstable.

Stable reversal. Run the recurrence backwards, $I_{n-1} = (1 - I_n)/n$. Now errors are *divided* by n at every step. Starting from the crude guess $I_{30} = 0$ (error $< 1/31$) and recurring down, the error at $n = 20$ is reduced by $30!/20! \approx 1.1 \times 10^{14}$ and is below 10^{-15}. This gives $I_{20} = 0.0455449$, $I_{10} = 0.0838771$, and $I_0 = 0.6321205588285577$, which agrees with $1 - e^{-1}$ to all 16 digits.

Common Pitfall — Blaming the computer for an ill-conditioned problem

Using higher precision (quad, arbitrary precision) cures *instability* only by brute force and does nothing for a problem whose *data* are uncertain. If the entries of $\mathbf{b}$ in Example 22.8 are known only to 10^{-4}, the solution is unknown to 100 % no matter how the arithmetic is done. Conversely, a large residual from a backward-stable solver is a sign of a bug, not of ill-conditioning: backward-stable methods always produce small residuals $\|\mathbf{b} - \mathbf{A}\hat{\mathbf{x}}\| \approx u\|\mathbf{A}\|\|\hat{\mathbf{x}}\|$ even when $\hat{\mathbf{x}}$ is inaccurate.

Engineering Note — Ill-conditioning in practice

Ill-conditioned problems are common in EE: polynomial root-finding from coefficients (Wilkinson showed that changing the x^{19} coefficient of $\prod_{k=1}^{20}(x-k)$ by 2^{-23} moves some roots far into the complex plane), high-order IIR filters implemented in direct form (pole positions are ill-conditioned functions of the polynomial coefficients — hence the universal use of cascaded second-order sections), deconvolution and inverse problems (which need regularisation), and nodal analysis with resistances spanning many decades. Recognising ill-conditioning — and reformulating to avoid it — is a design skill.

22.3 Root-finding methods

The problem

Given a continuous $f:\mathbb{R}\to\mathbb{R}$, find x^* with $f(x^*)=0$. Nonlinear equations without closed-form solutions appear constantly in engineering: the operating point of a diode or transistor circuit, the resonant frequency of a distributed structure ($\tan\beta\ell=\dots$), the propagation constant of a waveguide mode, the time at which a transient crosses a threshold, Kepler-type equations in orbital mechanics, the break frequency at which a gain equals -3 dB. All practical methods are **iterative**: they generate a sequence $x_0, x_1, x_2, \dots$ that (one hopes) converges to x^*.

Two questions organise the subject: **Does the iteration converge** (from which starting points)? And **how fast**?

Existence and bracketing

Intermediate value theorem (root form)

If f is continuous on $[a,b]$ and $f(a)f(b)<0$, then f has at least one root in (a,b).

An interval with a sign change is called a **bracket**. Bracketing methods (bisection, regula falsi, Brent) keep a bracket at every step and therefore *cannot fail* to converge to a root; open methods (Newton, secant, fixed-point) use only the current iterates, converge faster near the root, but can diverge from a poor start. Robust library routines (`scipy.optimize.brentq`, MATLAB `fzero`) combine both: open steps when they behave, bisection steps when they do not.

Order of convergence

Order of convergence

Let $x_k\to x^*$ and $e_k=x_k-x^*$. The sequence converges with **order** $q\ge 1$ and **asymptotic constant** C if

$$\lim_{k\to\infty}\frac{|e_{k+1}|}{|e_k|^q}=C, \qquad 0<C<\infty,$$

with $C<1$ required when $q=1$. The cases $q=1$, $1<q<2$ and $q=2$ are called **linear**, **superlinear** and **quadratic** convergence.

The practical meaning: with linear convergence the number of correct digits grows by a *fixed amount* $-\log_{10}C$ per iteration; with order q the number of correct digits is *multiplied by* q per iteration. Quadratic convergence roughly doubles the digits each step, so from 2 correct digits one reaches full double precision in about three steps.

Fixed-point iteration

Rewrite $f(x)=0$ in the equivalent form $x=g(x)$ and iterate

$$x_{k+1}=g(x_k).$$

A **fixed point** of g is a point with $g(x^*)=x^*$. There are many ways to rewrite one equation: for $x^2-2=0$ one could use $g(x)=2/x$, $g(x)=x-(x^2-2)/3$, or $g(x)=\tfrac12(x+2/x)$ — and they behave very differently.

Contraction mapping (Banach fixed-point) theorem, scalar form

Suppose g maps the closed interval $I=[a,b]$ into itself and is a contraction there:

$$|g(x)-g(y)|\le L|x-y| \quad \text{for all } x,y\in I, \qquad 0\le L<1$$

(sufficient: g differentiable with $|g'(x)|\le L<1$ on I). Then

1. g has exactly one fixed point $x^*\in I$;
2. for every $x_0\in I$ the iteration $x_{k+1}=g(x_k)$ converges to x^*;
3. the error satisfies $|x_k-x^*|\le L^k|x_0-x^*|$ and the computable bound $|x_k-x^*|\le \dfrac{L}{1-L}|x_k-x_{k-1}|$.

Proof sketch. $|x_{k+1}-x_k|=|g(x_k)-g(x_{k-1})|\le L|x_k-x_{k-1}|$, so the step lengths decrease geometrically and $\sum|x_{k+1}-x_k|$ converges; hence (x_k) is Cauchy and has a limit $x^*\in I$ (I is closed). Continuity of g gives $g(x^*)=x^*$. If y^* were a second fixed point, $|x^*-y^*|=|g(x^*)-g(y^*)|\le L|x^*-y^*|$ forces $x^*=y^*$. Finally $|x_k-x^*|=|g(x_{k-1})-g(x^*)|\le L|x_{k-1}-x^*|$, and the a posteriori bound follows from $|x_{k-1}-x^*|\le|x_{k-1}-x_k|+|x_k-x^*|$. $\square$

Local behaviour. Near the fixed point, Taylor expansion gives

$$e_{k+1}=g(x^*+e_k)-g(x^*)=g'(x^*)\,e_k+\tfrac12 g''(\xi_k)e_k^2.$$

Hence:

- if $0 < |g'(x^*)| < 1$: **linear** convergence with $C = |g'(x^*)|$; if $g'(x^*) < 0$ the iterates alternate sides of x^*, if $g'(x^*) > 0$ they approach monotonically;
- if $|g'(x^*)| > 1$: the fixed point **repels** and the iteration diverges from it;
- if $g'(x^*) = 0$ and $g''(x^*) \neq 0$: **quadratic** convergence. More generally, if $g'(x^*) = \cdots = g^{(q-1)}(x^*) = 0 \neq g^{(q)}(x^*)$, the order is q. Newton's method (Section 22.4) is precisely the choice of g that makes $g'(x^*) = 0$.

Worked Example 22.10 — Solving $x = \cos x$

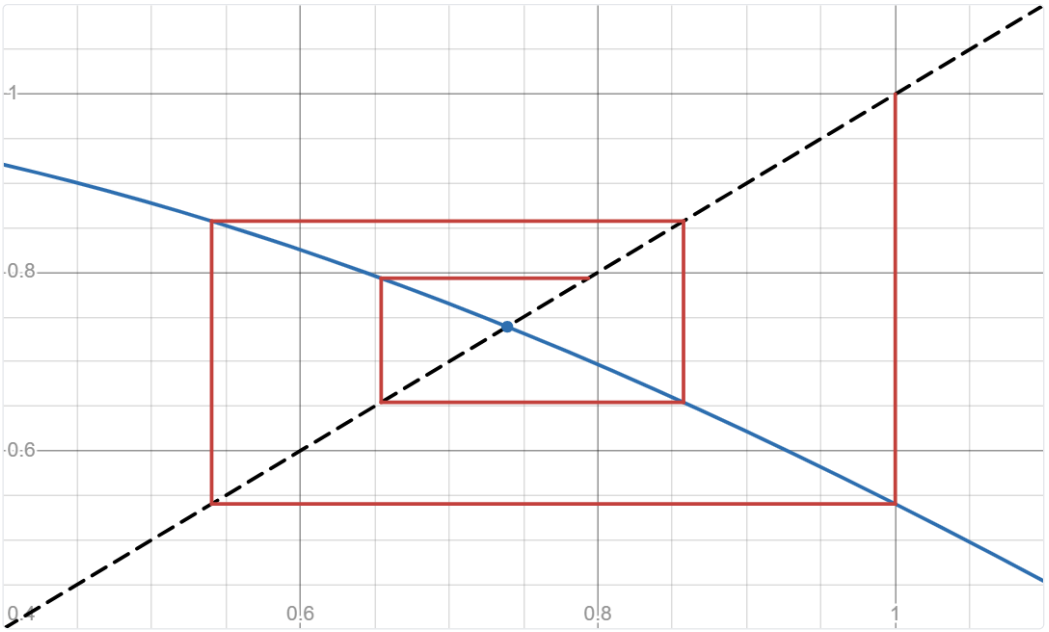
Step 1 — check the theorem. Take $I = [0, 1]$. Then cos maps I into $[\cos 1, 1] = [0.540, 1] \subset I$, and $|g'(x)| = |\sin x| \leq \sin 1 = 0.841 < 1$ on I. So there is a unique fixed point and the iteration converges from any $x_0 \in I$.

Step 2 — iterate from $x_0 = 1$.

k	1	2	3	4	5	6	10
x_k	0.540302	0.857553	0.654290	0.793480	0.701369	0.763960	0.744237

Step 3 — rate. The fixed point is $x^* = 0.7390851332$ and $g'(x^*) = -\sin x^* = -0.6736$. The iterates alternate about x^* (negative g') and each error is about 0.67 times the previous one: only $-\log_{10} 0.674 \approx 0.17$ digits per iteration, so about 90 iterations are needed for full double precision. Linear convergence with a constant this close to 1 is slow — the motivation for Newton's method.

The **cobweb diagram** makes the geometry visible: from (x_k, x_k) on the line $y = x$, move vertically to the curve $y = g(x)$, then horizontally back to $y = x$; the abscissa is x_{k+1}. A slope g' between -1 and 0 produces an inward spiral.



Cobweb diagram for $x_{k+1} = \cos x_k$ from $x_0 = 1$: because $-1 < g'(x^*) < 0$ the path spirals inward to the fixed point $x^* \approx 0.739$.

Worked Example 22.11 — Three rearrangements of $x^2 = 2$

$g(x)$	$g'(x^*)$ at $x^* = \sqrt{2}$	Behaviour from $x_0 = 1$
$2/x$	-1	$1, 2, 1, 2, \ldots$ — oscillates forever
$x - (x^2 - 2)/3$	$1 - 2\sqrt{2}/3 = 0.0572$	$1.3333, 1.4074, 1.41381, 1.414190, 1.4142122$ — linear, gains ≈ 1.24 digits per step
$\frac{1}{2}(x + 2/x)$	0	$1.5, 1.41667, 1.4142157, 1.41421356237469, 1.414213562373095$ — quadratic

The third rearrangement is Newton's method (Heron's ancient square-root algorithm): its errors are $8.6 \times 10^{-2}, 2.5 \times 10^{-3}, 2.1 \times 10^{-6}, 1.6 \times 10^{-12}$ and then machine precision — digits double each step. The choice of g is everything.

The secant method

Newton's method (next section) needs f'. If the derivative is unavailable or expensive — f may be the output of a circuit simulation — replace it by the slope of the chord through the last two iterates:

$$x_{k+1} = x_k - f(x_k)\,\frac{x_k - x_{k-1}}{f(x_k) - f(x_{k-1})}.$$

Geometrically, x_{k+1} is where the secant line through $(x_{k-1}, f(x_{k-1}))$ and $(x_k, f(x_k))$ crosses the axis. Only one *new* function evaluation is needed per step.

Convergence of the secant method

If $f \in C^2$ near a simple root x^* ($f'(x^*) \neq 0$) and x_0, x_1 are sufficiently close to x^*, the secant iterates converge with order

$$q = \varphi = \frac{1+\sqrt{5}}{2} \approx 1.618, \qquad \text{and} \quad e_{k+1} \approx \frac{f''(x^*)}{2f'(x^*)}\, e_k\, e_{k-1}.$$

Derivation sketch. Expanding f about x^* to second order in the secant formula gives $e_{k+1} \approx M e_k e_{k-1}$ with $M = f''(x^*)/(2f'(x^*))$. Try $|e_{k+1}| \approx A|e_k|^q$; then $|e_{k-1}| \approx (|e_k|/A)^{1/q}$ and substitution gives $A|e_k|^q \approx MA^{-1/q}|e_k|^{1+1/q}$. Matching exponents, $q = 1 + 1/q$, i.e. $q^2 - q - 1 = 0$, whose positive root is the golden ratio. □

Per function evaluation the secant method is actually more efficient than Newton when f' costs as much as f: two secant steps (two evaluations) multiply the digits by $\varphi^2 = 2.618 > 2$, whereas one Newton step (two evaluations: f and f') multiplies them by 2.

Worked Example 22.12 — Secant method for $x^3 - x - 2 = 0$

$f(1) = -2$, $f(2) = 4$, so $[1, 2]$ is a bracket. Start with $x_0 = 1$, $x_1 = 2$.

Step 1. $x_2 = 2 - 4 \cdot \dfrac{2-1}{4-(-2)} = 2 - \dfrac{4}{6} = 1.333333.$

Subsequent steps (root $x^* = 1.5213797068$):

k	2	3	4	5	6	7	8
x_k	1.333333	1.462687	1.531169	1.520926	1.521376	1.5213797080	1.5213797068
$\lvert e_k \rvert$	1.9×10^{-1}	5.9×10^{-2}	9.8×10^{-3}	4.5×10^{-4}	3.4×10^{-6}	1.2×10^{-9}	3×10^{-15}

Order check. $\ln(1.2 \times 10^{-9}/3.4 \times 10^{-6}) / \ln(3.4 \times 10^{-6}/4.5 \times 10^{-4}) = (-7.95)/(-4.89) = 1.63 \approx \varphi$. Note that the secant method does not maintain a bracket: x_2 and x_3 both lie below the root, and x_4 overshoots it.

Regula falsi (false position) uses the same chord formula but keeps the two points that bracket the root. It always converges, but typically only linearly, because one endpoint tends to get "stuck". The Illinois modification (halve the stored function value at a stuck endpoint) and Brent's method repair this.

Stopping criteria

An iteration needs a rule for stopping. Common tests, usually combined:

- **step size:** $|x_{k+1} - x_k| \le \text{atol} + \text{rtol}\,|x_{k+1}|$;
- **residual:** $|f(x_k)| \le \tau$;
- **bracket width** (bracketing methods): $b_k - a_k \le \text{tol}$;
- **iteration cap** — always, to guard against non-convergence.

Neither the step test nor the residual test alone is reliable. For a function with a very flat zero ($f'(x^*)$ small), the residual can be tiny far from the root; for a steep function, the residual can be large even at the floating-point number closest to the root. The **attainable accuracy** of any root-finder is limited by conditioning: if f can only be evaluated to absolute accuracy δ, the root is determined only to about $\delta/|f'(x^*)|$ (the "condition number of a root" is $1/|f'(x^*)|$), and much worse for a multiple root.

Polynomial roots

For polynomials, special methods exist. The most robust general approach — used by `numpy.roots` and MATLAB `roots` — is to form the **companion matrix** whose characteristic polynomial is $p(x)$ and compute its eigenvalues by the QR algorithm (Ch. 10). Once one root r is found, **deflation** divides $p(x)$ by $(x - r)$ (synthetic division) to obtain a lower-degree polynomial for the remaining roots; deflation should proceed from the smallest-magnitude roots to limit error growth. Remember Wilkinson's warning (Section 22.2): roots are often ill-conditioned functions of the coefficients, so if you have the polynomial in factored or state-space form, do not multiply it out.

Common Pitfall — Multiple and clustered roots

At a root of multiplicity $m > 1$, f touches the axis without (for even m) changing sign, so bracketing methods may not even see it, and f is so flat that rounding errors of size δ smear the root over an interval of width about $(\delta\, m!/|f^{(m)}(x^*)|)^{1/m}$ — in double precision a double root is typically determined only to about 8 digits. Newton and secant methods slow to linear convergence there (Section 22.4).

Engineering Note — Root-finding inside simulators

SPICE's DC operating-point analysis is a multidimensional root-finding problem $\mathbf{F}(\mathbf{v}) = \mathbf{0}$ (Kirchhoff's current law at every node). It uses Newton's method with damping, "source stepping" (ramping supplies from zero) and "gmin stepping" (adding small conductances to ground) — all techniques to bring the starting point inside the region of convergence. The message "no convergence in DC analysis" is a root-finder giving up.

22.4 Bisection and Newton–Raphson methods

These two methods are the extremes of the root-finding spectrum: bisection is slow but guaranteed, Newton–Raphson is fast but needs a good start. Every serious root-finder is built from one or both.

The bisection method

Given a bracket $[a_0, b_0]$ with $f(a_0)f(b_0) < 0$, repeat for $k = 0, 1, 2, \dots$:

1. $m_k = \tfrac{1}{2}(a_k + b_k)$;
2. if $f(a_k)f(m_k) < 0$ set $[a_{k+1}, b_{k+1}] = [a_k, m_k]$, otherwise $[a_{k+1}, b_{k+1}] = [m_k, b_k]$.

The bracket halves every step and always contains a root.

Bisection error bound

If f is continuous on $[a_0, b_0]$ with $f(a_0)f(b_0) < 0$, the midpoints satisfy

$$|m_k - x^*| \le \frac{b_0 - a_0}{2^{k+1}}$$

for some root $x^* \in [a_0, b_0]$. Hence an absolute accuracy τ is guaranteed after

$$k \ \ge\ \log_2 \frac{b_0 - a_0}{\tau} - 1 \quad \text{iterations.}$$

Proof. By induction $b_k - a_k = (b_0 - a_0)/2^k$ and $[a_k, b_k]$ contains a root by the intermediate value theorem; the midpoint is within half the width of every point of the bracket. $\square$

The error bound halves each step, so bisection behaves like a linearly convergent method with $C = \tfrac{1}{2}$: one binary digit, or $\log_{10} 2 \approx 0.3$ decimal digits, per function evaluation. Its virtues: it needs only continuity and sign evaluations, the iteration count is known *in advance*, and it is immune to round-off misbehaviour (as long as the sign of f is computed correctly). In floating point compute the midpoint as `a + (b - a)/2` and test signs with `copysign` or `sign(fa) != sign(fm)` rather than `fa*fm < 0`, which can underflow to zero.

```
def bisect(f, a, b, tol=1e-12, maxit=200):
    fa = f(a)
    for _ in range(maxit):
        m = a + (b - a) / 2
        fm = f(m)
        if fm == 0 or (b - a) / 2 < tol:
            return m
        if (fa < 0) != (fm < 0):   # sign change in [a, m]
            b = m
        else:
            a, fa = m, fm
    return a + (b - a) / 2
```

Worked Example 22.13 — Bisection for $x^3 - x - 2 = 0$

Bracket. $f(1) = -2 < 0$, $f(2) = 4 > 0$.

k	a_k	b_k	m_k	$f(m_k)$
0	1	2	1.5	-0.125
1	1.5	2	1.75	1.609
2	1.5	1.75	1.625	0.666
3	1.5	1.625	1.5625	0.252
4	1.5	1.5625	1.53125	0.0591
5	1.5	1.53125	1.515625	-0.0341
6	1.515625	1.53125	1.5234375	0.0123
7	1.515625	1.5234375	1.51953125	-0.0110

Error check. After 8 midpoints, $|m_7 - x^*| = |1.51953 - 1.52138| = 1.8 \times 10^{-3} \le 1/2^8 = 3.9 \times 10^{-3}$. ✓

Iterations for $\tau = 10^{-6}$. $\log_2(1/10^{-6}) - 1 = 19.93 - 1$, so $k = 19$, i.e. 20 midpoint evaluations; for $\tau = 10^{-12}$ about 40. Compare Newton below.

Newton–Raphson: derivation

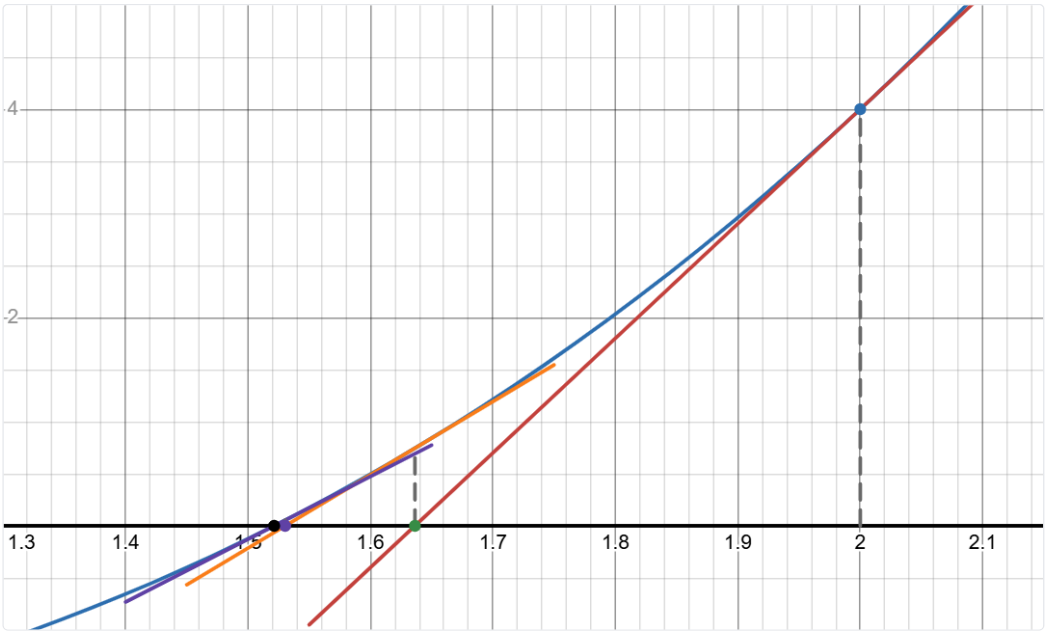
Replace f near the current iterate by its tangent line (first-order Taylor polynomial, Ch. 4):

$$f(x) \approx f(x_k) + f'(x_k)(x - x_k).$$

Setting the linear model to zero gives the next iterate

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}.$$

Equivalently, Newton is fixed-point iteration with $g(x) = x - f(x)/f'(x)$; differentiating, $g'(x) = f(x)f''(x)/f'(x)^2$, which **vanishes at a simple root**. By Section 22.3 this already suggests quadratic convergence.



Newton–Raphson for $f(x) = x^3 - x - 2$ from $x_0 = 2$: each tangent line (red, orange, purple) meets the axis at the next iterate, $x_1 = 1.6364$, $x_2 = 1.5304$, $x_3 = 1.52144$; the tangents become ever better models as the iterates close in on the root at 1.52138.

Quadratic convergence

Local quadratic convergence of Newton's method

Let $f \in C^2$ on an interval containing a simple root x^* ($f(x^*) = 0$, $f'(x^*) \neq 0$). Then there is $\delta > 0$ such that for every x_0 with $|x_0 - x^*| < \delta$ the Newton iterates converge to x^* and

$$\lim_{k \to \infty} \frac{e_{k+1}}{e_k^2} = \frac{f''(x^*)}{2f'(x^*)}.$$

Proof. Taylor's theorem about x_k evaluated at x^*:

$$0 = f(x^*) = f(x_k) + f'(x_k)(x^* - x_k) + \tfrac{1}{2} f''(\xi_k)(x^* - x_k)^2.$$

Divide by $f'(x_k)$ and rearrange using the Newton formula $x_{k+1} = x_k - f(x_k)/f'(x_k)$:

$$x_{k+1} - x^* = \frac{f''(\xi_k)}{2f'(x_k)} (x_k - x^*)^2, \qquad \text{i.e.} \quad e_{k+1} = \frac{f''(\xi_k)}{2f'(x_k)} e_k^2.$$

On a small interval around x^*, $|f'| \geq m > 0$ and $|f''| \leq M$, so $|e_{k+1}| \leq \frac{M}{2m} |e_k|^2$. If $\frac{M}{2m}|e_0| < 1$, then $|e_{k+1}| \leq \left(\frac{M}{2m}|e_0|\right) |e_k|$ and the errors decrease to zero; as $x_k \to x^*$ also $\xi_k \to x^*$, giving the limit. $\square$

The constant $M/(2m)$ explains *when* Newton is happy: small curvature f'' and steep slope f'. The radius $\delta \approx 2m/M$ of guaranteed convergence can be very small for strongly curved functions — the exponential of a diode being the prime example. (The Newton–Kantorovich theorem gives a checkable sufficient condition based on $f(x_0)$, $f'(x_0)$ and a Lipschitz bound on f'.)

Worked Example 22.14 — Newton for $x^3 - x - 2 = 0$

$f'(x) = 3x^2 - 1$. Start from the bisection midpoint $x_0 = 1.5$.

Step 1. $f(1.5) = 3.375 - 1.5 - 2 = -0.125$, $f'(1.5) = 5.75$, $x_1 = 1.5 + 0.125/5.75 = 1.5217391304$.

Further steps.

k	x_k	$\lvert e_k \rvert$	$\lvert e_k \rvert / \lvert e_{k-1} \rvert^2$
0	1.5	2.14×10^{-2}	—
1	1.521739130434783	3.59×10^{-4}	0.79
2	1.521379805964786	9.9×10^{-8}	0.77
3	1.521379706804575	7.5×10^{-15}	0.77
4	1.521379706804568	$< 10^{-16}$	—

Check against theory. $f''(x^*)/(2f'(x^*)) = 6(1.52138)/(2 \cdot 5.9438) = 0.768$, matching the last column. The number of correct digits goes $2 \to 3 \to 7 \to 14 \to 16$: four Newton steps versus about fifty bisection steps for the same accuracy.

Newton's method in circuit analysis

Worked Example 22.15 — Diode operating point

A 5 V source drives a 1 kΩ resistor in series with a silicon diode, $I_D = I_S(e^{V/V_T} - 1)$ with $I_S = 10^{-14}$ A and $V_T = 25.852$ mV (300 K). Find the diode voltage V.

Step 1 — KCL. Current through the resistor equals diode current:

$$F(V) = \frac{5 - V}{1000} - I_S\left(e^{V/V_T} - 1\right) = 0, \qquad F'(V) = -\frac{1}{1000} - \frac{I_S}{V_T}e^{V/V_T}.$$

Step 2 — good start. Engineering judgement says $V \approx 0.7$ V.

k	V_k (V)	$F(V_k)$ (A)
1	0.693518214	-1.7×10^{-4}
2	0.692561670	-3.0×10^{-6}
3	0.692543639	-1.0×10^{-9}
4	0.692543633	$\approx 10^{-16}$

So $V = 0.69254$ V and $I_D = (5 - 0.69254)/1000 = 4.307$ mA. Quadratic convergence is evident in the residuals.

Step 3 — bad start. From $V_0 = 0$ the diode term is negligible, the tangent is essentially the resistor's line, and $V_1 = 4.99999999807$ V. At 5 V the exponential is astronomically steep ($e^{5/0.0259} \approx 10^{84}$), so each subsequent Newton step reduces V by only about V_T: $4.9741, 4.9483, 4.9224, \ldots$ It would take about 165 iterations to crawl down to 0.7 V — and with a slightly larger overshoot e^{V/V_T} would overflow to ∞.

Cure. SPICE-type simulators **limit** the change in junction voltage per iteration (e.g. to a few V_T on a logarithmic scale, the "pnjlim" algorithm) or **damp** the step, $V_{k+1} = V_k - \lambda F/F'$ with $\lambda \le 1$ chosen so that $|F|$ decreases.

When Newton fails

Newton's method is only *locally* convergent. The classic failure modes are:

1. **Zero or tiny derivative.** If $f'(x_k) \approx 0$ the step is huge (the tangent is nearly horizontal).
2. **Cycling.** For $f(x) = x^3 - 2x + 2$ and $x_0 = 0$: $x_1 = 0 - 2/(-2) = 1$, $x_2 = 1 - 1/1 = 0$, and the iterates alternate $0, 1, 0, 1, \ldots$ forever, although a real root exists at $x^* \approx -1.7693$.
3. **Divergence.** For $f(x) = \arctan x$ (root 0), Newton converges only for $|x_0| < 1.3917$; from $x_0 = 1.5$ the iterates oscillate with growing magnitude ($-1.69, 2.32, -5.11, \ldots$).
4. **Convergence to an unwanted root** — the basins of attraction of different roots can be intricately interleaved (in the complex plane they form fractals).
5. **Overflow** of f itself in the first steps (Example 22.15).

Safeguarding. Robust implementations combine Newton with a bracket: if the Newton step would leave the current bracket or fails to reduce $|f|$ sufficiently, take a bisection step instead. This "Newton–bisection hybrid" (and Brent's method, which does the same with inverse quadratic interpolation) inherits bisection's guarantee and Newton's speed.

Multiple roots

At a root of multiplicity m, $f(x) = (x - x^*)^m h(x)$ with $h(x^*) \neq 0$. Then $g(x) = x - f/f'$ satisfies $g'(x^*) = 1 - 1/m$, so Newton converges only **linearly** with constant $(m-1)/m$ — for a double root, the error just halves each step. Two remedies:

- **Modified Newton**, if m is known: $x_{k+1} = x_k - m\,f(x_k)/f'(x_k)$ restores quadratic convergence (now $g'(x^*) = 0$).
- Apply Newton to $\mu(x) = f(x)/f'(x)$, which has a *simple* root at x^* whatever m is (requires f'').

Worked Example 22.16 — A double root

$f(x) = (x-1)^2(x+2)$ has a double root at 1. From $x_0 = 2$:

k	1	2	3	4	5
Newton e_k	0.556	0.298	0.155	0.0796	0.0403
Modified ($m=2$) e_k	0.111	1.95×10^{-3}	6.3×10^{-7}	6.7×10^{-14}	—

Plain Newton's errors approach the ratio $\frac{1}{2}$ predicted by $(m-1)/m$; the modified iteration is quadratic again. A practical symptom of a multiple root: Newton suddenly converges linearly with ratio $\approx 1/2$ or $2/3$.

Newton's method for systems

For $\mathbf{F} : \mathbb{R}^n \to \mathbb{R}^n$, the tangent line becomes the tangent *plane* given by the **Jacobian** $\mathbf{J}(\mathbf{x}) = [\partial F_i/\partial x_j]$ (Ch. 6):

$$\mathbf{F}(\mathbf{x}) \approx \mathbf{F}(\mathbf{x}_k) + \mathbf{J}(\mathbf{x}_k)(\mathbf{x} - \mathbf{x}_k) \quad \Longrightarrow \quad \mathbf{J}(\mathbf{x}_k)\,\Delta\mathbf{x}_k = -\mathbf{F}(\mathbf{x}_k), \qquad \mathbf{x}_{k+1} = \mathbf{x}_k + \Delta\mathbf{x}_k.$$

Each step solves a *linear* system (never form $\mathbf{J}^{-1}$). Convergence is again quadratic near a root at which $\mathbf{J}$ is nonsingular. This is exactly the inner loop of nonlinear circuit simulation, power-flow analysis in power systems (the Newton–Raphson load flow), and implicit ODE solvers (Section 22.9). When $\mathbf{J}$ is expensive, **quasi-Newton** methods (Broyden) update an approximation to it — a theme of Ch. 23.

```
import numpy as np
def newton_system(F, J, x, tol=1e-12, maxit=50):
    for _ in range(maxit):
        dx = np.linalg.solve(J(x), -F(x))
        x = x + dx
        if np.linalg.norm(dx) <= tol * (1 + np.linalg.norm(x)):
            break
    return x
```

Worked Example 22.17 — A 2×2 nonlinear system

Solve $x^2 + y^2 = 4$, $xy = 1$ (intersection of a circle and a hyperbola) starting from $(x_0, y_0) = (2, 0.5)$.

Set-up. $\mathbf{F} = \begin{bmatrix} x^2 + y^2 - 4 \\ xy - 1 \end{bmatrix}$, $\mathbf{J} = \begin{bmatrix} 2x & 2y \\ y & x \end{bmatrix}$.

Step 1. $\mathbf{F}(2, 0.5) = (0.25, 0)^{\mathsf{T}}$, $\mathbf{J} = \begin{bmatrix} 4 & 1 \\ 0.5 & 2 \end{bmatrix}$. Solving $\mathbf{J}\Delta = -\mathbf{F}$: $\det \mathbf{J} = 7.5$, $\Delta = \frac{1}{7.5}\begin{bmatrix} 2 & -1 \\ -0.5 & 4 \end{bmatrix}\begin{bmatrix} -0.25 \\ 0 \end{bmatrix} = (-0.06667, 0.01667)^{\mathsf{T}}$, so $(x_1, y_1) = (1.93333, 0.51667)$.

Further steps. $\|\mathbf{F}\|$: $0.25 \to 4.9 \times 10^{-3} \to 3.4 \times 10^{-6} \to 2.5 \times 10^{-12} \to 4 \times 10^{-16}$. The solution is $(1.9318517, 0.5176381)$.

Check. Exactly, $x^2 + y^2 = 4$ and $xy = 1$ give $(x + y)^2 = 6$, $(x - y)^2 = 2$, so $x = (\sqrt{6} + \sqrt{2})/2 = 1.9318517$, $y = (\sqrt{6} - \sqrt{2})/2 = 0.5176381$. ✓

Common Pitfall — Trusting a converged Newton iteration blindly

Newton converging tells you that you found *a* root — not necessarily the physical one. A transistor circuit with positive feedback (a latch, a Schmitt trigger) has several DC operating points; which one Newton finds depends on the starting guess. Always sanity-check the answer (signs of currents, device regions) and, for multi-stable circuits, perform a transient or continuation analysis.

Engineering Note — Newton in hardware

Reciprocal and square-root units in many processors and DSPs start from a small lookup-table estimate and refine it with one or two Newton steps, using only multiplications: for $1/a$, apply Newton to $f(x) = 1/x - a$ to get $x_{k+1} = x_k(2 - a x_k)$; for $1/\sqrt{a}$, $x_{k+1} = \tfrac{1}{2}x_k(3 - a x_k^2)$ (the iteration behind the famous "fast inverse square root"). An 8-bit table estimate becomes 16 bits after one step and 32 bits after two — quadratic convergence in silicon.

22.5 Interpolation

Motivation

Interpolation constructs a function that passes *exactly* through given data points. Engineers use it to read between the entries of a datasheet table (a transistor's h_{FE} versus I_C, a thermistor's resistance versus temperature), to implement function generators by table lookup, to resample a signal at a new rate or a fractional delay, and to draw smooth curves through simulation output. Equally important, interpolation is the *theoretical engine* behind most of this chapter: numerical differentiation (Section 22.7) differentiates an interpolant, and quadrature (Section 22.8) integrates one. The error of those methods is inherited from the interpolation error formula derived here.

Existence and uniqueness

Polynomial interpolation theorem

Given $n + 1$ distinct nodes $x_0, \dots, x_n$ and values $y_0, \dots, y_n$, there is exactly one polynomial p_n of degree at most n with $p_n(x_i) = y_i$ for $i = 0, \dots, n$.

Proof. Writing $p_n(x) = \sum_{k=0}^{n} a_k x^k$, the interpolation conditions form the linear system $\mathbf{V}\mathbf{a} = \mathbf{y}$ with the **Vandermonde matrix** $V_{ik} = x_i^k$, whose determinant is $\prod_{i<j}(x_j - x_i) \neq 0$ for distinct nodes. (Alternatively: if p and q both interpolate, $p - q$ has degree $\le n$ and $n + 1$ roots, hence is zero; existence follows from the Lagrange construction below.) □

Solving the Vandermonde system is a poor way to *compute* the interpolant: $\mathbf{V}$ is notoriously ill-conditioned (for 20 equally spaced nodes on $[0, 1]$, $\kappa(\mathbf{V}) \approx 10^{16}$). The Lagrange and Newton forms below represent the same unique polynomial in better-conditioned bases.

Lagrange form

Lagrange basis polynomials

For distinct nodes $x_0, \dots, x_n$,

$$\ell_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^{n} \frac{x - x_j}{x_i - x_j}, \qquad \text{so that} \quad \ell_i(x_k) = \delta_{ik},$$

and the interpolating polynomial is

$$p_n(x) = \sum_{i=0}^{n} y_i \, \ell_i(x).$$

Each ℓ_i is a "cardinal" function: 1 at its own node, 0 at all others. The Lagrange form is ideal for theory (quadrature weights are $\int \ell_i$, differentiation weights are ℓ_i'). For computation, the algebraically equivalent **barycentric formula**

$$p_n(x) = \frac{\displaystyle\sum_{i=0}^{n} \frac{w_i}{x - x_i}\, y_i}{\displaystyle\sum_{i=0}^{n} \frac{w_i}{x - x_i}}, \qquad w_i = \frac{1}{\prod_{j \neq i}(x_i - x_j)},$$

costs $O(n)$ per evaluation once the weights are known and is numerically stable; it is what modern software uses.

Newton's divided-difference form

Divided differences

$$f[x_i] = f(x_i), \qquad f[x_i, \dots, x_{i+k}] = \frac{f[x_{i+1}, \dots, x_{i+k}] - f[x_i, \dots, x_{i+k-1}]}{x_{i+k} - x_i}.$$

Newton form of the interpolant

$$p_n(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1) + \cdots + f[x_0, \dots, x_n] \prod_{j=0}^{n-1} (x - x_j).$$

Proof sketch. Let p_k interpolate at $x_0, \dots, x_k$. Then $p_{k+1} - p_k$ has degree $\le k+1$ and vanishes at $x_0, \dots, x_k$, so $p_{k+1} = p_k + c_{k+1} \prod_{j=0}^{k}(x - x_j)$, and c_{k+1} is the leading coefficient of p_{k+1}. Comparing the leading coefficients of two interpolants on overlapping node sets yields the recursive definition of the divided difference. $\square$

Two practical advantages: adding a new data point adds one term without recomputing the others, and the form is evaluated cheaply by nested multiplication (Horner's scheme). Divided differences also approximate derivatives: if $f \in C^k$, then $f[x_0, \dots, x_k] = f^{(k)}(\xi)/k!$ for some ξ in the span of the nodes.

Worked Example 22.18 — Cubic through four points, two ways

Data: $(0,1), (1,3), (2,2), (3,5)$. Find p_3 and evaluate it at $x = 1.5$.

Divided-difference table.

x_i	$f[x_i]$	first	second	third
0	1			
		$(3-1)/1 = 2$		
1	3		$(-1-2)/2 = -1.5$	
		$(2-3)/1 = -1$		$(2-(-1.5))/3 = 7/6$
2	2		$(3-(-1))/2 = 2$	
		$(5-2)/1 = 3$		
3	5			

Newton form. The top diagonal gives

$$p_3(x) = 1 + 2x - 1.5\,x(x-1) + \tfrac{7}{6}\,x(x-1)(x-2).$$

Evaluate at 1.5. $1 + 3 - 1.5(0.75) + \frac{7}{6}(1.5)(0.5)(-0.5) = 4 - 1.125 - 0.4375 = 2.4375$.

Lagrange check. With $\ell_0(1.5) = \frac{(0.5)(-0.5)(-1.5)}{(-1)(-2)(-3)} = -0.0625$, $\ell_1(1.5) = \frac{(1.5)(-0.5)(-1.5)}{(1)(-1)(-2)} = 0.5625$, $\ell_2(1.5) = \frac{(1.5)(0.5)(-1.5)}{(2)(1)(-1)} = 0.5625$, $\ell_3(1.5) = \frac{(1.5)(0.5)(-0.5)}{(3)(2)(1)} = -0.0625$ (they sum to 1, as they must): $p_3(1.5) = 1(-0.0625) + 3(0.5625) + 2(0.5625) + 5(-0.0625) = 2.4375$. ✓

Expanded, $p_3(x) = \frac{7}{6}x^3 - 5x^2 + \frac{35}{6}x + 1$, which matches `numpy.polyfit` with degree 3.

The interpolation error

Interpolation error formula

Let $f \in C^{n+1}[a,b]$ and let p_n interpolate f at distinct nodes $x_0, \dots, x_n \in [a,b]$. For every $x \in [a,b]$ there is $\xi = \xi(x) \in (a,b)$ with

$$f(x) - p_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}\,\omega_{n+1}(x), \qquad \omega_{n+1}(x) = \prod_{i=0}^{n}(x - x_i).$$

Proof. Fix x not a node and define $\phi(t) = f(t) - p_n(t) - \dfrac{f(x) - p_n(x)}{\omega_{n+1}(x)}\,\omega_{n+1}(t)$. Then ϕ vanishes at the $n+2$ points $x_0, \dots, x_n, x$. By Rolle's theorem applied repeatedly, $\phi^{(n+1)}$ has a zero ξ. Since $p_n^{(n+1)} = 0$ and $\omega_{n+1}^{(n+1)} = (n+1)!$, $0 = f^{(n+1)}(\xi) - (n+1)!\,(f(x) - p_n(x))/\omega_{n+1}(x)$. $\square$

The formula separates two influences: the **smoothness of** f (through $f^{(n+1)}$) and the **placement of the nodes** (through ω_{n+1}). The second factor is ours to choose.

Worked Example 22.19 — Quadratic interpolation of $\sin x$

Interpolate $\sin x$ at $0,\ \pi/4,\ \pi/2$ and estimate the error at $x = \pi/8$.

Divided differences. $f[x_0, x_1] = 0.70711/0.78540 = 0.90032$, $f[x_1, x_2] = (1 - 0.70711)/0.78540 = 0.37292$, $f[x_0, x_1, x_2] = (0.37292 - 0.90032)/1.5708 = -0.33575$.

Interpolant. $p_2(x) = 0.90032x - 0.33575\,x(x - \pi/4)$; at $\pi/8 = 0.392699$: $p_2 = 0.35355 + 0.05178 = 0.40533$.

Actual error. $\sin(\pi/8) = 0.38268$, so $|f - p_2| = 0.02265$.

Bound. $|f'''| = |\cos| \le 1$, so $|f - p_2| \le \frac{1}{3!}|(\pi/8)(\pi/8 - \pi/4)(\pi/8 - \pi/2)| = \frac{1}{6}(0.3927)(0.3927)(1.1781) = 0.0303$. The bound holds and is not wildly pessimistic.

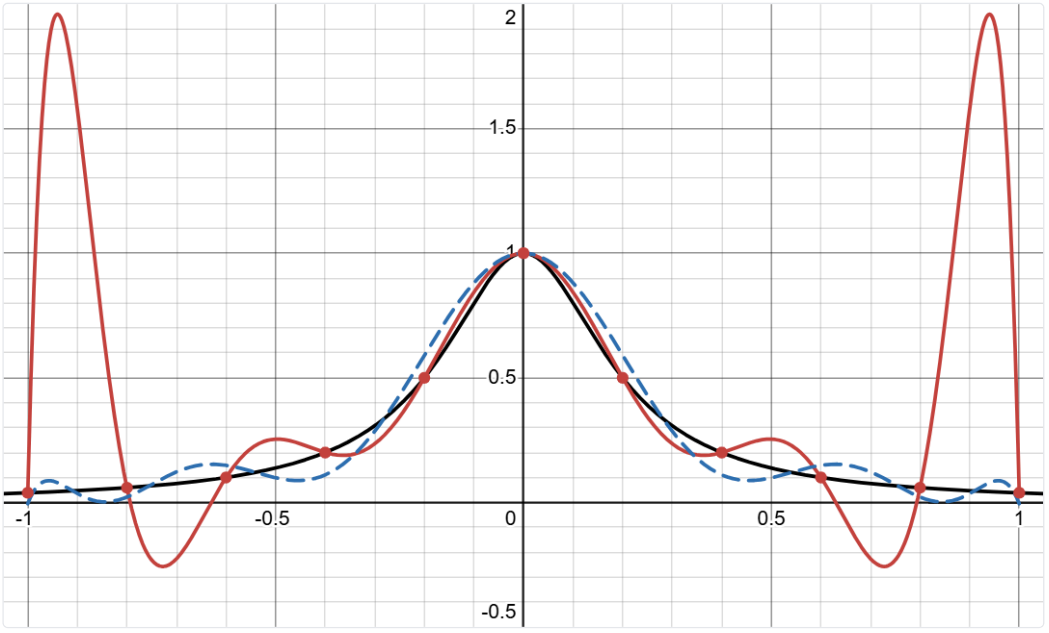
The Runge phenomenon

It is tempting to think that more nodes always give a better fit. For **equally spaced** nodes this is false, even for perfectly smooth functions. Runge's classic example is

$$f(x) = \frac{1}{1 + 25x^2}, \qquad x \in [-1, 1].$$

n (degree)	4	8	12	16	20
$\max\|f - p_n\|$, equispaced nodes	0.438	1.045	3.66	14.4	59.8
$\max\|f - p_n\|$, Chebyshev nodes	0.402	0.171	0.069	0.033	0.015

With equispaced nodes the error near the ends of the interval *grows exponentially* with n; with Chebyshev nodes (defined below) it decreases steadily.



Runge phenomenon: $f(x) = 1/(1 + 25x^2)$ (black), its degree-10 interpolant at 11 equally spaced nodes (red, nodes marked) overshooting to ≈ 1.96 near $x = \pm 0.94$ (an error of 1.92), and the degree-10 interpolant at Chebyshev nodes (blue dashed) staying within 0.11 of f everywhere.

Why it happens. Two factors in the error formula conspire. First, $|\omega_{n+1}(x)|$ for equispaced nodes is much larger near the ends of the interval than in the middle (by a factor that grows like 2^n). Second, f has complex poles at $x = \pm j/5$, close to the real interval, so its derivatives grow like $(n+1)!\,5^{n+1}$ — fast enough to beat the $(n+1)!$ in the denominator. Complex analysis (Ch. 18) makes this precise: polynomial interpolants at equispaced nodes converge only inside a certain lens-shaped region of the complex plane, and if a singularity of f lies inside that region, interpolation diverges near the endpoints.

Chebyshev nodes

Since $f^{(n+1)}$ is beyond our control, minimise the other factor: choose nodes that make $\max_{[-1,1]}|\omega_{n+1}|$ as small as possible.

Chebyshev polynomials and nodes

The Chebyshev polynomials are $T_k(x) = \cos(k \arccos x)$ on $[-1, 1]$, equivalently $T_0 = 1$, $T_1 = x$, $T_{k+1} = 2xT_k - T_{k-1}$. The **Chebyshev nodes** (of the first kind) for degree n are the $n + 1$ zeros of T_{n+1}:

$$x_i = \cos\left(\frac{(2i+1)\pi}{2(n+1)}\right), \qquad i = 0, \dots, n.$$

On a general interval $[a, b]$ use $\frac{a+b}{2} + \frac{b-a}{2}x_i$.

The nodes are projections onto the axis of equally spaced points on a semicircle, so they *cluster near the endpoints* — exactly where equispaced interpolation goes wrong.

Minimax property of Chebyshev nodes

Among all monic polynomials of degree $n+1$, $2^{-n}T_{n+1}(x)$ has the smallest maximum magnitude on $[-1,1]$, namely 2^{-n}. Consequently, for Chebyshev nodes $\omega_{n+1} = 2^{-n}T_{n+1}$ and

$$\max_{[-1,1]} |f - p_n| \le \frac{\max |f^{(n+1)}|}{2^n (n+1)!}.$$

Proof sketch. $2^{-n}T_{n+1}$ is monic and attains $\pm 2^{-n}$ alternately at the $n+2$ points $\cos(k\pi/(n+1))$. If a monic q of the same degree had smaller maximum, $2^{-n}T_{n+1} - q$ (degree $\le n$) would alternate in sign at those $n+2$ points and so have $n+1$ zeros — hence be identically zero, a contradiction. □

A finer measure of node quality is the **Lebesgue constant** $\Lambda_n = \max_x \sum_i |\ell_i(x)|$, which bounds how much interpolation can amplify errors in the data and relates the interpolation error to the best possible: $\|f - p_n\|_\infty \le (1 + \Lambda_n)\, \|f - p_n^*\|_\infty$, where p_n^* is the best (minimax) approximation. For equispaced nodes $\Lambda_n \sim 2^{n+1}/(e\, n \ln n)$ (exponential growth); for Chebyshev nodes $\Lambda_n \le \frac{2}{\pi}\ln(n+1) + 1$ (logarithmic growth, essentially optimal). With Chebyshev nodes, interpolation of any Lipschitz-continuous function converges, and for analytic functions it converges geometrically. This is the basis of the "Chebfun" philosophy of computing with functions.

Piecewise interpolation and splines

The other cure for Runge's phenomenon is to **keep the degree low and subdivide**: interpolate by a low-degree polynomial on each subinterval between nodes.

Piecewise linear interpolation (what a plotting program does when it "connects the dots") on nodes with spacing h has error

$$\max |f - s| \le \frac{h^2}{8} \max |f''|,$$

from the error formula with $n = 1$ and $\max_{[x_i, x_{i+1}]} |(x - x_i)(x - x_{i+1})| = h^2/4$. It converges for any C^2 function, but the interpolant has corners.

Cubic spline

A **cubic spline** on nodes $a = x_0 < x_1 < \cdots < x_n = b$ is a function S that is a cubic polynomial on each $[x_i, x_{i+1}]$ and has continuous first and second derivatives on $[a, b]$. An **interpolating** cubic spline also satisfies $S(x_i) = y_i$. Two extra conditions are required, typically:

- **natural**: $S''(a) = S''(b) = 0$;
- **clamped**: $S'(a)$, $S'(b)$ prescribed;
- **not-a-knot**: S''' continuous at x_1 and x_{n-1} (MATLAB/SciPy default).

Counting. n cubics have $4n$ coefficients. Interpolation at both ends of each piece gives $2n$ conditions, continuity of S' and S'' at the $n-1$ interior nodes gives $2n - 2$; two end conditions complete the count.

Derivation (moment form). Let $M_i = S''(x_i)$ and $h_i = x_{i+1} - x_i$. Since S'' is linear on each piece, integrating twice and imposing $S(x_i) = y_i$, $S(x_{i+1}) = y_{i+1}$ gives on $[x_i, x_{i+1}]$:

$$S(x) = M_i \frac{(x_{i+1} - x)^3}{6h_i} + M_{i+1} \frac{(x - x_i)^3}{6h_i} + \left(\frac{y_i}{h_i} - \frac{M_i h_i}{6}\right)(x_{i+1} - x) + \left(\frac{y_{i+1}}{h_i} - \frac{M_{i+1} h_i}{6}\right)(x - x_i).$$

Requiring S' to be continuous at each interior node x_i yields the **tridiagonal** system

$$h_{i-1} M_{i-1} + 2(h_{i-1} + h_i) M_i + h_i M_{i+1} = 6\left(\frac{y_{i+1} - y_i}{h_i} - \frac{y_i - y_{i-1}}{h_{i-1}}\right), \qquad i = 1, \dots, n-1,$$

which is strictly diagonally dominant, hence uniquely solvable in $O(n)$ operations by the Thomas algorithm (Gaussian elimination without pivoting on a tridiagonal matrix).

Properties of cubic splines

1. **Accuracy.** For $f \in C^4$ and the clamped (or not-a-knot) spline with maximum spacing h: $\|f - S\|_\infty \le \frac{5}{384} h^4 \|f^{(4)}\|_\infty$, with $O(h^3)$ for S' and $O(h^2)$ for S''.
2. **Minimum curvature.** Among all C^2 functions interpolating the data (with the same end conditions, natural case), the natural cubic spline minimises $\int_a^b (g'')^2\, dx$ — the bending energy of a thin elastic beam, which is where the name "spline" (a draughtsman's flexible strip) comes from.
3. **Locality.** Changing one data value changes S everywhere, but the influence decays geometrically (by a factor of about $2 - \sqrt{3} \approx 0.27$ per interval for uniform nodes).

Worked Example 22.20 — A natural cubic spline

Data: $(0,0),(1,1),(2,0),(3,1)$, so $h=1$ and $M_0=M_3=0$.

Step 1 — system for M_1, M_2. With $h_i=1$ the equations read $M_{i-1}+4M_i+M_{i+1}=6(y_{i+1}-2y_i+y_{i-1})$:

$$\begin{aligned} 4M_1+M_2 &= 6(0-2+0) = -12,\\ M_1+4M_2 &= 6(1-0+1) = 12, \end{aligned}$$

giving $M_1=-4$, $M_2=4$.

Step 2 — pieces. On $[0,1]$: $S_0(x) = -4\frac{x^3}{6} + \left(1+\frac{4}{6}\right)x = -\frac{2}{3}x^3 + \frac{5}{3}x$. On $[1,2]$: $S_1(x) = -\frac{2}{3}(2-x)^3 + \frac{2}{3}(x-1)^3 + \frac{5}{3}(2-x) - \frac{2}{3}(x-1)$. On $[2,3]$, by the symmetry $y_i \mapsto 1-y_{3-i}$, $S_2(x) = 1 - S_0(3-x)$.

Step 3 — check smoothness at $x=1$. $S_0'(1) = -2+\frac{5}{3} = -\frac{1}{3}$; $S_1'(1) = 2(2-1)^2 + 0 - \frac{5}{3} - \frac{2}{3} = -\frac{1}{3}$. ✓ Second derivatives: $S_0''(1) = -4 = M_1 = S_1''(1)$. ✓ And $S_1(1.5) = 0.5$, as symmetry requires.

Hermite interpolation

If derivatives are known as well as values (for example position *and* velocity, or a circuit waveform and its slope), **Hermite interpolation** matches both. Matching f and f' at $n+1$ nodes determines a unique polynomial of degree $\le 2n+1$, with error $f^{(2n+2)}(\xi)\,\omega_{n+1}(x)^2/(2n+2)!$. The piecewise cubic Hermite interpolant (matching f, f' at both ends of every interval) underlies "PCHIP", a shape-preserving interpolant that avoids the overshoots a cubic spline can produce for monotone data.

Common Pitfall — Extrapolation and high-degree fits

The interpolation error formula contains $\omega_{n+1}(x)$, which grows like $|x|^{n+1}$ outside the node interval. **Extrapolating** an interpolating polynomial beyond the data is therefore dangerous, and the higher the degree the worse. Likewise, never fit a single polynomial of degree 15 through 16 equally spaced measurements; use a spline, or least squares with a low degree (Section 22.6).

Engineering Note — Interpolation in signal processing

A DAC with a zero-order hold performs piecewise-constant interpolation; a first-order hold is piecewise linear. The ideal band-limited reconstruction of the sampling theorem (Ch. 16) is interpolation with the sinc kernel, which is the limit of Lagrange interpolation on infinitely many equally spaced nodes. Practical **fractional-delay** filters and sample-rate converters use Lagrange interpolation over 4–8 neighbouring samples, often in the efficient **Farrow** structure, whose polynomial coefficients are exactly the Newton/Lagrange coefficients of this section.

22.6 Polynomial approximation

Interpolation versus approximation

Interpolation forces the approximant through every data point. That is the wrong goal when the data are **noisy** (fitting a polynomial through every noisy sample fits the noise), when there are **more data than parameters** (a thousand measurements of a sensor characteristic that is nearly quadratic), or when we want the **best possible accuracy** for a given degree (a function-evaluation routine in a math library or an FPGA). Approximation instead minimises a measure of the error in a chosen norm:

- the **uniform (minimax, L^∞) norm** $\|f-p\|_\infty = \max_{x\in[a,b]}|f(x)-p(x)|$ — the worst-case error, used for library functions and lookup tables;
- the **least-squares (L^2) norm** $\|f-p\|_2 = \left(\int_a^b w(x)\,(f-p)^2\,dx\right)^{1/2}$ or, for data, $\left(\sum_i w_i (y_i - p(x_i))^2\right)^{1/2}$ — the energy of the error, used for data fitting because it is optimal for Gaussian noise (Ch. 20) and leads to *linear* equations.

That polynomials are a sensible choice at all is guaranteed by a classical theorem.

Weierstrass approximation theorem

If f is continuous on $[a,b]$, then for every $\varepsilon>0$ there is a polynomial p with $\|f-p\|_\infty<\varepsilon$.

A constructive proof uses the **Bernstein polynomials** $B_n f(x) = \sum_{k=0}^{n} f(k/n)\binom{n}{k}x^k(1-x)^{n-k}$ on $[0,1]$, which converge uniformly to f — but only slowly ($O(1/n)$ for smooth f). The theorem guarantees existence; the methods below achieve good accuracy at low degree.

Taylor polynomials: good locally, poor globally

A Taylor polynomial (Ch. 4) is exact at one point with all derivatives up to order n and its error, $f^{(n+1)}(\xi)(x-x_0)^{n+1}/(n+1)!$, grows rapidly away from x_0. For e^x on $[0,1]$, the degree-1 Taylor polynomial $1+x$ has maximum error $e-2 = 0.718$ at $x=1$; we shall see that other degree-1 polynomials do seven times better. Taylor polynomials are the right tool for analysis near a point (linearisation, small-signal models) and the wrong tool for approximation over an interval.

Best uniform (minimax) approximation

Chebyshev equioscillation theorem

Let $f \in C[a,b]$. A polynomial p^* of degree $\le n$ is the best uniform approximation to f if and only if there are $n+2$ points $a \le t_0 < t_1 < \cdots < t_{n+1} \le b$ at which the error $e = f - p^*$ attains its maximum magnitude with alternating signs:

$$e(t_i) = \pm(-1)^i \|f - p^*\|_\infty.$$

The best approximation exists and is unique.

The theorem is the same alternation argument that proved the minimax property of T_{n+1} in Section 22.5 (which is the special case "best approximation of x^{n+1} by lower degree"). In practice the minimax polynomial is computed by the **Remez exchange algorithm**, which iteratively adjusts a set of $n+2$ reference points until the error equioscillates. The same equioscillation principle defines **equiripple (Parks–McClellan) FIR filters** and **elliptic (Cauer) analog filters** — optimal filter design is minimax approximation.

Worked Example 22.21 — Best straight line for e^x on $[0,1]$

Seek $p^*(x) = mx + c$ with $n+2 = 3$ alternation points. Because e^x is convex, the error $e^x - mx - c$ has its extreme values at the endpoints $0, 1$ and at one interior point ξ where $e' = 0$, i.e. $e^\xi = m$.

Equioscillation conditions. $e(0) = +E$, $e(\xi) = -E$, $e(1) = +E$:

$$1 - c = E, \qquad e^\xi - m\xi - c = -E, \qquad e - m - c = E.$$

Solve. Subtract the first from the third: $m = e - 1 = 1.718282$. Then $\xi = \ln m = 0.541325$. Adding the first two equations: $1 + m - m\xi - 2c = 0$, so $c = \tfrac{1}{2}(1 + m - m\xi) = 0.894067$, and $E = 1 - c = 0.105933$.

Result. $p^*(x) = 1.718282x + 0.894067$ with $\|e^x - p^*\|_\infty = 0.1059$ — versus 0.718 for the Taylor line $1 + x$. A numerical scan of $|e^x - p^*(x)|$ on $[0,1]$ confirms the maximum 0.10593 at $x = 0,\ 0.5413,\ 1$.

Discrete least squares

Given data (x_i, y_i), $i = 1, \dots, N$, and a model $p(x) = \sum_{k=0}^{n} a_k \phi_k(x)$ that is *linear in the parameters* (for polynomials $\phi_k = x^k$, but sinusoids, exponentials with known rates, or any fixed basis work identically), least squares chooses $\mathbf{a}$ to minimise

$$S(\mathbf{a}) = \sum_{i=1}^{N} \big(y_i - p(x_i)\big)^2 = \|\mathbf{y} - \mathbf{A}\mathbf{a}\|_2^2, \qquad A_{ik} = \phi_k(x_i).$$

Setting $\nabla S = \mathbf{0}$ (Ch. 6, Ch. 23) gives the **normal equations**

$$\mathbf{A}^{\mathsf{T}}\mathbf{A}\,\mathbf{a} = \mathbf{A}^{\mathsf{T}}\mathbf{y},$$

which have a unique solution when the columns of $\mathbf{A}$ are linearly independent (for polynomials: at least $n+1$ distinct x_i). Geometrically, $\mathbf{A}\mathbf{a}$ is the orthogonal projection of $\mathbf{y}$ onto the column space of $\mathbf{A}$: the residual $\mathbf{r} = \mathbf{y} - \mathbf{A}\mathbf{a}$ is orthogonal to every column, $\mathbf{A}^{\mathsf{T}}\mathbf{r} = \mathbf{0}$ (Ch. 9).

Worked Example 22.22 — Fitting a resistor's I–V line

Measured voltage–current pairs (V in volts, I in mA): $(0, 0.1), (1, 1.9), (2, 4.2), (3, 5.8), (4, 8.1)$. Fit $I = a_0 + a_1 V$.

Normal equations. $\sum 1 = 5$, $\sum V_i = 10$, $\sum V_i^2 = 30$, $\sum I_i = 20.1$, $\sum V_i I_i = 0 + 1.9 + 8.4 + 17.4 + 32.4 = 60.1$:

$$\begin{bmatrix} 5 & 10 \\ 10 & 30 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 20.1 \\ 60.1 \end{bmatrix}.$$

Solve. Determinant $150 - 100 = 50$; $a_1 = (5 \cdot 60.1 - 10 \cdot 20.1)/50 = (300.5 - 201)/50 = 1.99$ mA/V, $a_0 = (20.1 - 10a_1)/5 = 0.04$ mA.

Interpretation. Conductance 1.99 mS, i.e. $R = 1/1.99$ mS $= 503\ \Omega$; offset 0.04 mA (instrument offset). Residuals $0.06, -0.13, 0.18, -0.21, 0.10$ sum to zero (as orthogonality to the constant column requires) and their sum of squares is 0.107, giving an estimated noise standard deviation $\sqrt{0.107/(5-2)} = 0.19$ mA (Ch. 20).

Numerical caution. Forming $\mathbf{A}^{\mathsf{T}}\mathbf{A}$ *squares* the condition number: $\kappa(\mathbf{A}^{\mathsf{T}}\mathbf{A}) = \kappa(\mathbf{A})^2$. For the quadratic fit to the data above, $\kappa(\mathbf{A}) = 27$ but $\kappa(\mathbf{A}^{\mathsf{T}}\mathbf{A}) = 735$; for high-degree monomial fits the squaring is catastrophic. Stable practice is to solve the least-squares problem by a **QR factorisation** $\mathbf{A} = \mathbf{Q}\mathbf{R}$ (then $\mathbf{R}\mathbf{a} = \mathbf{Q}^{\mathsf{T}}\mathbf{y}$) or the SVD, which is what `numpy.linalg.lstsq` and MATLAB's backslash do; and to **centre and scale** the abscissae (use $(x - \bar{x})/s$) before fitting polynomials.

Continuous least squares and orthogonal polynomials

To approximate a *function* f on $[a,b]$ in the L^2 sense with $p(x) = \sum_k a_k x^k$, minimise $\int_a^b (f - p)^2 dx$. The normal equations become

$$\sum_{k=0}^{n} \left(\int_a^b x^{j+k} dx \right) a_k = \int_a^b x^j f(x)\, dx, \qquad j = 0, \dots, n.$$

On $[0,1]$ the matrix is the **Hilbert matrix** $H_{jk} = 1/(j+k+1)$ — which we met in Section 22.2 as the archetype of ill-conditioning. The monomials are a terrible basis: x^{10} and x^{11} look almost alike on $[0,1]$.

The cure is an **orthogonal basis**. If $\langle \phi_j, \phi_k \rangle = \int_a^b w\, \phi_j \phi_k\, dx = 0$ for $j \ne k$, the normal-equation matrix is diagonal and the coefficients decouple:

$$a_k = \frac{\langle f, \phi_k \rangle}{\langle \phi_k, \phi_k \rangle}.$$

This is exactly the Fourier-coefficient formula of Ch. 15 with polynomials in place of sinusoids — and, as there, adding a term does not change the previous coefficients, and the error decreases monotonically as terms are added (Bessel's inequality, Parseval).

Family	Interval	Weight $w(x)$	Recurrence	Norm $\langle \phi_k, \phi_k \rangle$
Legendre P_k	$[-1, 1]$	1	$(k+1)P_{k+1} = (2k+1)xP_k - kP_{k-1}$	$2/(2k+1)$
Chebyshev T_k	$[-1, 1]$	$(1-x^2)^{-1/2}$	$T_{k+1} = 2xT_k - T_{k-1}$	π $(k=0)$, $\pi/2$ $(k \geq 1)$
Laguerre L_k	$[0, \infty)$	e^{-x}	$(k+1)L_{k+1} = (2k+1-x)L_k - kL_{k-1}$	1
Hermite H_k	$(-\infty, \infty)$	e^{-x^2}	$H_{k+1} = 2xH_k - 2kH_{k-1}$	$\sqrt{\pi}\, 2^k k!$

The first Legendre polynomials are $P_0 = 1$, $P_1 = x$, $P_2 = \frac{1}{2}(3x^2 - 1)$, $P_3 = \frac{1}{2}(5x^3 - 3x)$. Any family of orthogonal polynomials can be generated from the monomials by Gram–Schmidt (Ch. 9), and every such family satisfies a three-term recurrence.

Worked Example 22.23 — Least-squares straight line for e^x on $[0, 1]$

Normal equations (monomial basis).

$$\begin{bmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} \int_0^1 e^x dx \\ \int_0^1 x e^x dx \end{bmatrix} = \begin{bmatrix} e - 1 \\ 1 \end{bmatrix}.$$

Solve. Determinant $\frac{1}{3} - \frac{1}{4} = \frac{1}{12}$; $a_0 = 12\left(\frac{e-1}{3} - \frac{1}{2}\right) = 4e - 10 = 0.873127$, $a_1 = 12\left(1 - \frac{e-1}{2}\right) = 18 - 6e = 1.690309$.

Compare with the minimax line of Example 22.21.

Approximation	$\|e^x - p\|_\infty$	$\|e^x - p\|_2$
Taylor $1 + x$	0.718	—
Least squares $0.8731 + 1.6903x$	0.155	**0.0628**
Minimax $0.8941 + 1.7183x$	**0.106**	0.0723

Each is optimal in its own norm, and they are close to each other — a general phenomenon for smooth functions.

Worked Example 22.24 — Legendre expansion of e^x on $[-1, 1]$

Coefficients. $a_k = \frac{2k+1}{2} \int_{-1}^{1} e^x P_k(x)\, dx$:

$$a_0 = \tfrac{1}{2}(e - e^{-1}) = 1.175201, \quad a_1 = \tfrac{3}{2} \cdot 2e^{-1} = 1.103638, \quad a_2 = \tfrac{5}{2}\,(e - 7e^{-1}) = 0.357814.$$

(The integrals follow by repeated integration by parts: $\int_{-1}^{1} xe^x dx = 2/e$, $\int_{-1}^{1} x^2 e^x dx = e - 5/e$.)

Quadratic approximation. $p_2 = a_0 + a_1 x + a_2 \cdot \frac{1}{2}(3x^2 - 1) = 0.996294 + 1.103638x + 0.536722x^2$.

Accuracy. $\max_{[-1,1]} |e^x - p_2| = 0.0816$, compared with 0.218 for the Taylor quadratic $1 + x + x^2/2$. Adding $a_3 P_3$ would not change a_0, a_1, a_2.

Chebyshev series and economisation

Expanding in Chebyshev polynomials (weight $(1-x^2)^{-1/2}$) gives a series $f = \sum c_k T_k$ whose truncation is **near-minimax**: since $|T_k| \leq 1$, the truncation error is at most $\sum_{k>n} |c_k|$, and because the first omitted term $c_{n+1}T_{n+1}$ usually dominates and equioscillates $n+2$ times, the truncated series is very close to the true minimax polynomial. Interpolating at Chebyshev nodes (Section 22.5) gives almost the same polynomial and is easier to compute (by FFT).

Economisation turns a Taylor polynomial into a better approximation of lower degree: express the highest power through Chebyshev polynomials and drop the T_n term, whose magnitude is known exactly.

Worked Example 22.25 — Economising the cubic Taylor polynomial of e^x

Step 1. $e^x \approx 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3$ on $[-1, 1]$ (max error 0.0516).

Step 2. From $T_3 = 4x^3 - 3x$: $x^3 = \frac{1}{4}(T_3 + 3x)$, so $\frac{1}{6}x^3 = \frac{1}{24}T_3 + \frac{1}{8}x$.

Step 3. Drop $\frac{1}{24}T_3$ (error at most $1/24 = 0.0417$): $e^x \approx 1 + 1.125x + 0.5x^2$.

Accuracy. A numerical scan gives $\max|e^x - (1 + 1.125x + 0.5x^2)| = 0.093$, close to the predicted $0.0516 + 0.0417$ and far better than the Taylor quadratic (0.218). For comparison, interpolating at the three Chebyshev nodes gives $1 + 1.12977x + 0.53204x^2$ with error 0.056.

Choosing the degree

In data fitting, the residual always decreases as the degree increases, until at degree $N - 1$ the fit interpolates and the residual is zero — while the polynomial oscillates wildly between data points. This is **overfitting**: the extra parameters model the noise. Choose the degree by physics (a resistor is linear; a thermistor's $\ln R$ is nearly linear in $1/T$), by checking that residuals look like white noise, or by statistical criteria (adjusted R^2, cross-validation, the F-test of Ch. 20). Weighted least squares (weights $w_i = 1/\sigma_i^2$) is appropriate when measurements have unequal uncertainties.

Common Pitfall — Fitting in the wrong variables

Least squares minimises errors in the variable you fit. Linearising an exponential model $y = Ae^{bx}$ by fitting $\ln y$ versus x is convenient but changes the error weighting — small y values (often the noisiest) get magnified influence. If the noise is additive in y, either weight the log-fit by y_i^2 or fit the nonlinear model directly (Gauss–Newton, Ch. 23).

Engineering Note — Polynomial approximation in hardware and firmware

The `sin`, `exp` and `log` routines in a math library (or an FPGA's CORDIC alternative) typically reduce the argument to a small interval and then evaluate a **minimax** polynomial or rational function of modest degree, coefficients precomputed by the Remez algorithm. Sensor linearisation in firmware — thermocouple voltage to temperature (the NIST ITS-90 polynomials are degree 8–10), strain-gauge calibration, RTD Callendar–Van Dusen curves — uses **least-squares** polynomial fits, evaluated by Horner's scheme to minimise multiplications and rounding error.

22.7 Numerical differentiation

Motivation

Derivatives are needed numerically whenever a function is known only through samples (a digitised waveform, a measured C–V curve whose slope gives doping profile), or through a black-box program (the Jacobian for Newton's method, sensitivities $\partial V_{\text{out}}/\partial R_k$ of a simulated circuit), and — most importantly — as the building block for solving differential equations on grids (Sections 22.9–22.10). Numerical differentiation is easy to do and easy to do badly: it is the one basic numerical operation in which **truncation and rounding errors pull in opposite directions**.

Finite-difference formulas from Taylor series

Let f be smooth and $h > 0$ a step. Taylor expansions about x give

$$f(x+h) = f + hf' + \tfrac{h^2}{2}f'' + \tfrac{h^3}{6}f''' + \tfrac{h^4}{24}f^{(4)} + \cdots$$
$$f(x-h) = f - hf' + \tfrac{h^2}{2}f'' - \tfrac{h^3}{6}f''' + \tfrac{h^4}{24}f^{(4)} - \cdots$$

(all derivatives evaluated at x). Rearranging the first line:

$$\textbf{forward difference:}\quad D_+ f(x) = \frac{f(x+h) - f(x)}{h} = f'(x) + \frac{h}{2}f''(\xi),$$

so the truncation error is $O(h)$ — first order. The **backward difference** $D_- f = (f(x) - f(x-h))/h$ is likewise first order. Subtracting the two expansions cancels the even terms:

$$\textbf{central difference:}\quad D_0 f(x) = \frac{f(x+h) - f(x-h)}{2h} = f'(x) + \frac{h^2}{6}f'''(\xi),$$

which is **second order**. Adding them cancels the odd terms and gives the standard second-derivative formula:

$$D_+ D_- f(x) = \frac{f(x+h) - 2f(x) + f(x-h)}{h^2} = f''(x) + \frac{h^2}{12}f^{(4)}(\xi).$$

The general pattern: a symmetric stencil kills alternate error terms, gaining one order for free.

Order of a difference approximation

A difference formula $D_h f$ approximates $f^{(m)}(x)$ with **order** p if $D_h f - f^{(m)}(x) = Ch^p f^{(m+p)}(\xi)$ with $C \neq 0$ for all smooth f. The quantity $D_h f - f^{(m)}(x)$ is the **truncation error**; C is its error constant.

Approximation	Formula	Truncation error
f' forward	$\frac{f_1 - f_0}{h}$	$\frac{h}{2}f''$
f' backward	$\frac{f_0 - f_{-1}}{h}$	$-\frac{h}{2}f''$
f' central	$\frac{f_1 - f_{-1}}{2h}$	$\frac{h^2}{6}f'''$
f' one-sided, 2nd order	$\frac{-3f_0 + 4f_1 - f_2}{2h}$	$-\frac{h^2}{3}f'''$
f' central, 4th order	$\frac{-f_2 + 8f_1 - 8f_{-1} + f_{-2}}{12h}$	$-\frac{h^4}{30}f^{(5)}$
f'' central	$\frac{f_1 - 2f_0 + f_{-1}}{h^2}$	$\frac{h^2}{12}f^{(4)}$
f'' central, 4th order	$\frac{-f_2 + 16f_1 - 30f_0 + 16f_{-1} - f_{-2}}{12h^2}$	$-\frac{h^4}{90}f^{(6)}$
f''' central	$\frac{f_2 - 2f_1 + 2f_{-1} - f_{-2}}{2h^3}$	$\frac{h^2}{4}f^{(5)}$

Here $f_k = f(x + kh)$. The one-sided second-order formula is what one uses at a boundary, where no point to the left exists (Neumann boundary conditions in Section 22.10).

Method of undetermined coefficients. Any stencil can be derived systematically. To approximate $f'(0)$ from f_0, f_1, f_2, write $f'(0) \approx (af_0 + bf_1 + cf_2)/h$, expand each f_k in Taylor series, and demand that the coefficients of f, f', f'' be $0, 1, 0$:

$$a+b+c=0, \qquad b+2c=1, \qquad \tfrac{1}{2}b+2c=0 \quad\Longrightarrow\quad (a,b,c)=\left(-\tfrac{3}{2},\ 2,\ -\tfrac{1}{2}\right),$$

which is the one-sided formula in the table. Equivalently, one can differentiate the interpolating polynomial through the stencil points (Section 22.5): the weights are $\ell_i'(x)$. That viewpoint generalises to unequally spaced points.

Worked Example 22.26 — Derivatives of e^x at $x=0$

Exact values: $f'(0)=f''(0)=1$.

h	forward	error	central	error	D_+D_- for f''	error
0.1	1.0517092	5.2×10^{-2}	1.0016675	1.67×10^{-3}	1.0008336	8.3×10^{-4}
0.05	1.0254219	2.5×10^{-2}	1.0004167	4.17×10^{-4}	1.0002084	2.1×10^{-4}

Order check. Halving h halves the forward error (ratio 2.0, order 1) and quarters the central and second-difference errors (ratio 4.0, order 2).

Error constants. Predicted forward error $\frac{h}{2}f''=0.05$ ✓; central $\frac{h^2}{6}f'''=1.67\times10^{-3}$ ✓; second difference $\frac{h^2}{12}f^{(4)}=8.3\times10^{-4}$ ✓.

Higher order. At $h=0.1$ the one-sided second-order formula gives an error of -3.6×10^{-3} (predicted $-\frac{h^2}{3}=-3.3\times10^{-3}$) — much better than the forward difference's 5.2×10^{-2} with the same points available on one side — and the five-point fourth-order formula gives -3.3×10^{-6} (predicted $-h^4/30=-3.3\times10^{-6}$).

Richardson extrapolation

If an approximation has an error expansion in known powers of h,

$$D(h)=L+c_1h^p+c_2h^q+\cdots \qquad (q>p),$$

then combining two step sizes eliminates the leading error term:

$$D(h)-L\approx c_1h^p, \quad D(h/2)-L\approx c_1(h/2)^p \quad\Longrightarrow\quad L\approx R(h)=\frac{2^pD(h/2)-D(h)}{2^p-1},$$

with error $O(h^q)$. For the central difference, $p=2$ and the expansion contains only even powers, so $R(h)=\frac{4D_0(h/2)-D_0(h)}{3}$ is **fourth order**. Repeating the process builds a triangular table (the same idea gives Romberg integration in Section 22.8). Richardson extrapolation also provides an **error estimate** for free: $D(h/2)-L\approx\frac{D(h/2)-D(h)}{2^p-1}$.

Worked Example 22.27 — Richardson for the derivative of e^x

From Example 22.26, $D_0(0.1)=1.00166750$ and $D_0(0.05)=1.00041672$.

Extrapolate. $R=\dfrac{4(1.00041672)-1.00166750}{3}=0.99999979$, error -2.1×10^{-7} — an improvement by a factor of 2000 over $D_0(0.05)$, using no new function values.

Error estimate. $(D_0(0.05)-D_0(0.1))/3=-4.17\times10^{-4}$, so $D_0(0.05)\approx L+4.17\times10^{-4}$: the estimate matches the actual error of $D_0(0.05)$ to three digits.

(Algebraically $R(h)$ equals the five-point formula with step $h/2$, whose predicted error $-(0.05)^4/30=-2.1\times10^{-7}$ agrees.)

Rounding error: the trade-off

The formulas above assume exact function values. In floating point, $f(x\pm h)$ carries an absolute error of about $u|f|$ (more if f is computed by a long program). For the forward difference these errors are divided by h, giving a total error

$$E(h)\approx \underbrace{\frac{h}{2}|f''|}_{\text{truncation}}+\underbrace{\frac{2u|f|}{h}}_{\text{rounding}}.$$

The first term falls with h, the second rises. Minimising over h ($E'(h)=0$):

$$h_{\text{opt}}=2\sqrt{\frac{u|f|}{|f''|}}\sim\sqrt{u}\approx10^{-8}, \qquad E_{\min}=2\sqrt{u|f||f''|}\sim\sqrt{u}\approx10^{-8}.$$

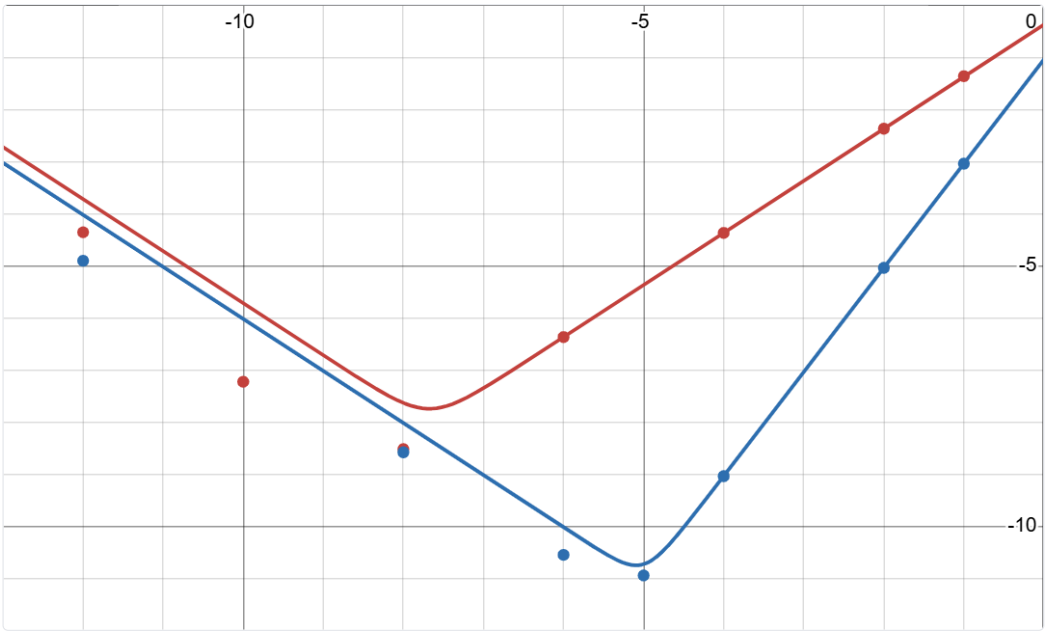
So in double precision **a forward difference can never deliver more than about 8 correct digits.** For the central difference, $E(h)\approx\frac{h^2}{6}|f'''|+u|f|/h$, giving $h_{\text{opt}}=(3u|f|/|f'''|)^{1/3}\sim u^{1/3}\approx5\times10^{-6}$ and $E_{\min}\sim u^{2/3}\approx10^{-11}$. Higher-order formulas push the optimum to larger h and smaller minimum error, but every formula for an m-th derivative suffers rounding amplification like u/h^m.

Worked Example 22.28 — The error "V" for $\frac{d}{dx}\sin x$ at $x = 1$

Exact: $\cos 1 = 0.540302306$.

h	10^{-1}	10^{-2}	10^{-4}	10^{-5}	10^{-6}	10^{-8}	10^{-10}	10^{-12}
forward error	4.3×10^{-2}	4.2×10^{-3}	4.2×10^{-5}	4.2×10^{-6}	4.2×10^{-7}	3.0×10^{-9}	5.8×10^{-8}	4.3×10^{-5}
central error	9.0×10^{-4}	9.0×10^{-6}	9.0×10^{-10}	1.1×10^{-11}	2.8×10^{-11}	2.6×10^{-9}	5.8×10^{-8}	1.2×10^{-5}

Reading the table. Both errors first fall at their theoretical rates (one decade per decade of h for forward, two for central). The forward error bottoms out near $h = 10^{-8}$ at about 3×10^{-9} (theory: $h_{\text{opt}} = 2\sqrt{u} = 2.1\times10^{-8}$, $E_{\min} \approx \sqrt{u}\cdot 2\sin 1 \approx 2\times10^{-8}$; the observed value is smaller because rounding errors partially cancel). The central error bottoms out near $h = 10^{-5}$ at about 10^{-11} (theory: $h_{\text{opt}} = (3u\tan 1)^{1/3} = 8\times10^{-6}$). For smaller h both curves *rise* along the same line, slope -1, because rounding dominates.



Total error of the forward (red) and central (blue) difference for $(\sin x)'$ at $x = 1$, on log–log axes ($x = \log_{10} h$, $y = \log_{10}$ error): model curves $\frac{h}{2}|f''| + 2u|f|/h$ and $\frac{h^2}{6}|f'''| + u|f|/h$ with measured errors as dots. Truncation dominates on the right (slopes 1 and 2), rounding on the left (slope -1).

The complex-step derivative

Engineers comfortable with j have a neat escape from the rounding trade-off when f is analytic and implemented in complex arithmetic. Expand with an *imaginary* step:

$$f(x + jh) = f(x) + jhf'(x) - \tfrac{h^2}{2}f''(x) - j\tfrac{h^3}{6}f'''(x) + \cdots \quad\Longrightarrow\quad f'(x) = \frac{\operatorname{Im}\{f(x+jh)\}}{h} + \frac{h^2}{6}f'''(x) + \cdots$$

There is **no subtraction**, hence no cancellation, so h can be taken absurdly small: with $h = 10^{-20}$ the truncation error $\sim 10^{-40}$ is invisible and the result for $(\sin x)'$ at $x = 1$ is correct to the last bit (computed error exactly 0). The method is widely used for accurate sensitivities in optimisation codes; it requires that the code for f be "complex-safe" (no `abs`, no comparisons on real parts that break analyticity).

Differentiating noisy data

Measured data contain noise of size σ that does not decrease with h. The rounding analysis then applies with $u|f|$ replaced by σ: a central difference of samples spaced h apart has noise standard deviation $\sigma/(\sqrt{2}\,h)$, and a second difference $\sqrt{6}\,\sigma/h^2$. Differentiation is a **high-pass** operation — its frequency response is $j\omega$ (Ch. 16) — so it amplifies high-frequency noise. Practical remedies smooth before (or while) differentiating: fit a local low-degree least-squares polynomial and differentiate it (the **Savitzky–Golay** filter), differentiate a smoothing spline, or use a band-limited differentiator FIR filter.

Common Pitfall — "Smaller h is always more accurate"

It is not. Below the optimum step, every halving of h *doubles* the rounding error (for a first derivative) or quadruples it (for a second). A finite-difference Jacobian in a Newton solver with $h = 10^{-14}|x|$ can be pure noise and stall the iteration. A sound default is $h = \sqrt{u}\,\max(|x|, x_{\text{typ}})$ for forward differences and $h = u^{1/3}\max(|x|, x_{\text{typ}})$ for central differences, with h rounded so that $x + h$ is exactly representable (`h = (x + h) - x`).

Engineering Note — Differentiators in practice

An oscilloscope's "derivative" math channel, a PID controller's D term and a digital tachometer all implement numerical differentiation of noisy samples. That is why PID implementations filter the derivative term (a first-order low-pass on D, "derivative kick" avoidance) and why rotational speed is often estimated by counting encoder pulses over a longer window (which is a long-baseline difference, trading bandwidth for noise) rather than by differentiating successive samples.

22.8 Numerical integration

Motivation

Definite integrals without elementary antiderivatives are everywhere: the error function in bit-error-rate calculations (Ch. 19), the RMS value of a measured waveform, the energy of a pulse, Fourier coefficients of sampled signals (Ch. 15), the inductance of a loop from Neumann's formula, radiation patterns of antennas, and integrals of tabulated data. **Quadrature** (the classical name for numerical integration) approximates

$$I(f) = \int_a^b f(x)\,dx \qquad \text{by} \qquad Q(f) = \sum_{i=0}^{n} w_i f(x_i),$$

a weighted sum of function values at **nodes** x_i with **weights** w_i. All the methods of this section are of this form; they differ in how nodes and weights are chosen.

> **Degree of exactness**
>
> A quadrature rule has **degree of exactness** (or degree of precision) d if $Q(p) = I(p)$ for every polynomial p of degree $\le d$, but not for some polynomial of degree $d+1$.

Since Q and I are both linear, it suffices to check the monomials $1, x, \dots, x^d$. A rule obtained by integrating the interpolating polynomial through $n+1$ nodes — an **interpolatory rule**, with weights $w_i = \int_a^b \ell_i(x)\,dx$ — has degree at least n.

Newton–Cotes rules

Interpolatory rules on **equally spaced** nodes are called Newton–Cotes rules.

Trapezoidal rule (linear interpolation through the endpoints):

$$\int_a^b f\,dx \approx \frac{b-a}{2}\big(f(a)+f(b)\big), \qquad E = -\frac{(b-a)^3}{12} f''(\xi).$$

Derivation of the error. By the interpolation error formula (Section 22.5), $f - p_1 = \tfrac{1}{2}f''(\xi_x)(x-a)(x-b)$. Since $(x-a)(x-b) \le 0$ on $[a,b]$ does not change sign, the integral mean-value theorem gives $\int_a^b (f-p_1)\,dx = \tfrac{1}{2}f''(\xi)\int_a^b (x-a)(x-b)\,dx = \tfrac{1}{2}f''(\xi)\left(-\tfrac{(b-a)^3}{6}\right)$. $\square$

Midpoint rule $\int_a^b f \approx (b-a) f\big(\tfrac{a+b}{2}\big)$ has error $+\tfrac{(b-a)^3}{24}f''(\xi)$ — half the trapezoidal error and of opposite sign, a fact exploited by Simpson's rule.

Simpson's rule (quadratic interpolation through a, $m = \tfrac{a+b}{2}$, b; $h = \tfrac{b-a}{2}$):

$$\int_a^b f\,dx \approx \frac{h}{3}\big(f(a) + 4f(m) + f(b)\big), \qquad E = -\frac{h^5}{90} f^{(4)}(\xi).$$

Although it interpolates with a quadratic, Simpson's rule integrates **cubics exactly** (degree of exactness 3): by symmetry about m, the rule integrates $(x-m)^3$ exactly (both sides are zero). This "bonus degree" occurs for every symmetric rule with an odd number of points. Simpson's rule can also be seen as the combination $S = \tfrac{2}{3}M + \tfrac{1}{3}T$ of midpoint and trapezoid, which cancels their f'' errors.

Higher-order Newton–Cotes rules exist (Simpson's 3/8 rule, Boole's rule), but beyond about 8 points their weights become negative and large — the integration counterpart of the Runge phenomenon — so in practice one uses low-order rules **composed** over many subintervals.

Composite rules

Divide $[a,b]$ into n subintervals of width $h = (b-a)/n$, nodes $x_i = a + ih$, and apply the basic rule on each:

$$T_n = h\left[\tfrac{1}{2}f_0 + f_1 + f_2 + \cdots + f_{n-1} + \tfrac{1}{2}f_n\right], \qquad E_T = -\frac{(b-a)h^2}{12} f''(\xi),$$

$$S_n = \frac{h}{3}\left[f_0 + 4f_1 + 2f_2 + 4f_3 + \cdots + 2f_{n-2} + 4f_{n-1} + f_n\right] \ (n \text{ even}), \qquad E_S = -\frac{(b-a)h^4}{180} f^{(4)}(\xi).$$

The composite error follows by summing the n local errors $-\tfrac{h^3}{12}f''(\xi_i)$ and replacing the average $\tfrac{1}{n}\sum f''(\xi_i)$ by $f''(\xi)$ (intermediate value theorem). Each local error is one power of h higher than the global error: $n \propto 1/h$ local errors add up. So the composite trapezoidal rule is **second order** and composite Simpson **fourth order**.

```
import numpy as np
def trapezoid(f, a, b, n):
    x = np.linspace(a, b, n + 1); y = f(x)
    return (b - a) / n * (y.sum() - 0.5 * (y[0] + y[-1]))
def simpson(f, a, b, n):                    # n must be even
    x = np.linspace(a, b, n + 1); y = f(x); h = (b - a) / n
    return h / 3 * (y[0] + y[-1] + 4 * y[1:-1:2].sum() + 2 * y[2:-1:2].sum())
```

Worked Example 22.29 — $\int_0^1 e^{-x^2}\,dx$ by trapezoid and Simpson

The exact value is $\frac{\sqrt{\pi}}{2}\operatorname{erf}(1) = 0.746824132812$.

Trapezoid, $n = 4$ ($h = 0.25$). f values: $1,\ 0.939413,\ 0.778801,\ 0.569783,\ 0.367879$. $T_4 = 0.25\,[0.5 + 0.939413 + 0.778801 + 0.569783 + 0.183940] = 0.742984$.

Simpson, $n = 4$. $S_4 = \frac{0.25}{3}\,[1 + 4(0.939413) + 2(0.778801) + 4(0.569783) + 0.367879] = 0.746855$.

n	T_n	error	S_n	error
2	0.731370	-1.55×10^{-2}	0.747180	3.6×10^{-4}
4	0.742984	-3.84×10^{-3}	0.746855	3.1×10^{-5}
8	0.745866	-9.59×10^{-4}	0.746826	2.0×10^{-6}

Orders. Trapezoid errors fall by 4.0 per halving (h^2); Simpson by 11.4 and 15.7, approaching 16 (h^4).

Error prediction. The Euler–Maclaurin formula below predicts $T_n - I \approx \frac{h^2}{12}\big(f'(1) - f'(0)\big) = \frac{h^2}{12}(-2e^{-1})$, i.e. -3.832×10^{-3} for $n = 4$, matching the observed -3.840×10^{-3}.

The Euler–Maclaurin formula and Romberg integration

Euler–Maclaurin formula

If $f \in C^{2m+2}[a, b]$, the composite trapezoidal rule satisfies

$$T_n - \int_a^b f\,dx = \sum_{k=1}^{m} \frac{B_{2k} h^{2k}}{(2k)!}\left(f^{(2k-1)}(b) - f^{(2k-1)}(a)\right) + O(h^{2m+2}),$$

where $B_2 = \frac{1}{6}$, $B_4 = -\frac{1}{30}$, $B_6 = \frac{1}{42}$, … are Bernoulli numbers. In particular the error has an expansion in **even** powers of h: $T_n = I + c_1 h^2 + c_2 h^4 + c_3 h^6 + \cdots$.

Two important consequences:

1. **Romberg integration.** Since the error expansion is in powers of h^2, Richardson extrapolation (Section 22.7) applies repeatedly. With $R_{k,0} = T_{2^k}$,

$$R_{k,m} = R_{k,m-1} + \frac{R_{k,m-1} - R_{k-1,m-1}}{4^m - 1}.$$

The column $R_{k,1}$ is exactly composite Simpson; $R_{k,2}$ is composite Boole (order 6); and so on.

2. **Periodic functions.** If f is smooth and periodic with period $b - a$, all the derivative differences $f^{(2k-1)}(b) - f^{(2k-1)}(a)$ vanish, so the trapezoidal error is smaller than *every* power of h. For analytic periodic functions the error decays **geometrically** in n. The humble trapezoidal rule is the best possible method for integrating smooth periodic functions over a full period.

Worked Example 22.30 — Romberg table for $\int_0^1 e^{-x^2}\,dx$

k	$n = 2^k$	$R_{k,0}$ (trapezoid)	$R_{k,1}$ (Simpson)	$R_{k,2}$	$R_{k,3}$
0	1	0.6839397			
1	2	0.7313703	0.7471804		
2	4	0.7429841	0.7468554	0.7468337	
3	8	0.7458656	0.7468261	0.74682417	0.74682402

For example $R_{1,1} = 0.7313703 + (0.7313703 - 0.6839397)/3 = 0.7471804$. With 9 function evaluations, $R_{3,2}$ is correct to 4×10^{-8}, versus 10^{-3} for T_8 alone. The last entry $R_{3,3}$ (-1.1×10^{-7}) is slightly worse than $R_{3,2}$: extrapolation assumes the asymptotic regime, and the crude $R_{0,0}$ (one interval!) contaminates the far corner of the table. Practical Romberg codes stop when successive diagonal entries agree to tolerance.

Worked Example 22.31 — Spectral accuracy for a periodic integrand

$I = \int_0^{2\pi} \frac{dx}{2 + \cos x} = \frac{2\pi}{\sqrt{3}} = 3.6275987285$ (residue calculus, Ch. 18). For a periodic function the trapezoidal rule with n points is simply $\frac{2\pi}{n}\sum_{i=0}^{n-1} f(2\pi i/n)$:

n	2	4	8	16
T_n	4.18879	3.66519	3.62779	3.6275987336
error	5.6×10^{-1}	3.8×10^{-2}	1.9×10^{-4}	5.1×10^{-9}

Each doubling of n *squares* the error, roughly — geometric convergence (the error behaves like $(2 - \sqrt{3})^n \approx 0.268^n$, set by the location of the integrand's poles at $\cos x = -2$). A fourth-order method would only gain a factor of 16 per doubling.

Gaussian quadrature

Newton–Cotes rules fix the nodes and choose $n+1$ weights, achieving degree n (or $n+1$). If the **nodes are free too**, there are $2n+2$ parameters and one can hope for degree $2n+1$. Gauss showed this is achievable, and optimal.

Gauss–Legendre quadrature

For each $n \geq 1$ there is a unique n-point rule

$$\int_{-1}^{1} f(x)\,dx \approx \sum_{i=1}^{n} w_i f(x_i)$$

with degree of exactness $2n-1$; no n-point rule has higher degree. Its nodes are the zeros of the Legendre polynomial P_n, all real, distinct and in $(-1,1)$, and its weights $w_i = \int_{-1}^{1} \ell_i(x)\,dx = \dfrac{2}{(1-x_i^2)[P_n'(x_i)]^2}$ are all positive. The error is

$$E_n = \frac{2^{2n+1}(n!)^4}{(2n+1)[(2n)!]^3}\, f^{(2n)}(\xi).$$

Proof sketch of exactness. Let p have degree $\leq 2n-1$. Divide by P_n: $p = qP_n + r$ with $\deg q, \deg r \leq n-1$. Then $\int_{-1}^{1} p = \int qP_n + \int r = 0 + \int r$, because P_n is orthogonal to all polynomials of lower degree (Section 22.6). At the nodes $P_n(x_i) = 0$, so $\sum w_i p(x_i) = \sum w_i r(x_i) = \int r$, the last equality because the rule is interpolatory with n nodes and $\deg r \leq n-1$. *Optimality:* $p = \prod (x - x_i)^2$ has degree $2n$, is positive, and is integrated to 0 by the rule. $\square$

n	nodes x_i	weights w_i	exact for degree
1	0	2	1
2	$\pm 1/\sqrt{3} = \pm 0.5773503$	$1,\ 1$	3
3	$0,\ \pm\sqrt{3/5} = \pm 0.7745967$	$8/9,\ 5/9,\ 5/9$	5
4	$\pm 0.3399810,\ \pm 0.8611363$	$0.6521452,\ 0.3478548$	7

Deriving the 2-point rule directly. Require $w_1 f(x_1) + w_2 f(x_2)$ to integrate $1, x, x^2, x^3$ exactly on $[-1,1]$: $w_1 + w_2 = 2$, $w_1 x_1 + w_2 x_2 = 0$, $w_1 x_1^2 + w_2 x_2^2 = \tfrac{2}{3}$, $w_1 x_1^3 + w_2 x_2^3 = 0$. Symmetry suggests $x_2 = -x_1$, $w_1 = w_2 = 1$; then $2x_1^2 = \tfrac{2}{3}$ gives $x_{1,2} = \pm 1/\sqrt{3}$.

General interval. Map $x \in [-1,1]$ to $t = \frac{a+b}{2} + \frac{b-a}{2}x$:

$$\int_a^b g(t)\,dt \approx \frac{b-a}{2} \sum_{i=1}^{n} w_i\, g\!\left(\frac{a+b}{2} + \frac{b-a}{2} x_i\right).$$

Worked Example 22.32 — Gauss–Legendre for $\int_0^1 e^{-x^2}dx$ and $\int_0^{\pi} \sin x\,dx$

2-point rule on $[0,1]$. Nodes $t = \frac{1}{2} \pm \frac{1}{2\sqrt{3}} = 0.211325,\ 0.788675$; weights $\frac{1}{2}$ each. $Q_2 = \frac{1}{2}\left(e^{-0.044658} + e^{-0.622008}\right) = \frac{1}{2}(0.956324 + 0.536865) = 0.746595$, error -2.3×10^{-4}.

3- and 4-point rules. $Q_3 = 0.7468146$ (error -9.5×10^{-6}), $Q_4 = 0.74682447$ (error 3.4×10^{-7}).

Comparison at equal cost. Q_4 with 4 evaluations beats Simpson S_8 with 9 evaluations (2.0×10^{-6}).

Second integral (exact value 2). Trapezoid $T_4 = 1.896119$, Simpson $S_4 = 2.004560$, Gauss $Q_2 = 1.935820$, $Q_3 = 2.001389$. Three function values in the right places do better than five equally spaced ones.

Other Gaussian rules. The same construction with a weight function $w(x)$ uses the corresponding orthogonal polynomials (Section 22.6): **Gauss–Chebyshev** for $\int_{-1}^{1} f(x)(1-x^2)^{-1/2}dx$ (nodes are Chebyshev nodes, equal weights π/n); **Gauss–Laguerre** for $\int_0^{\infty} f(x)e^{-x}dx$; **Gauss–Hermite** for $\int_{-\infty}^{\infty} f(x)e^{-x^2}dx$ — the natural rule for expectations over Gaussian noise (Ch. 19), e.g. average bit-error rate over a Gaussian-distributed parameter. Variants that include endpoints as nodes (Gauss–Lobatto, Gauss–Radau) underlie spectral-element field solvers and some implicit Runge–Kutta methods.

Adaptive quadrature and difficult integrands

A fixed composite rule wastes evaluations where f is smooth and under-resolves where it is not. **Adaptive** quadrature estimates the error on each subinterval — e.g. by comparing Simpson on $[a,b]$ with Simpson on its two halves, $|S_{\text{halves}} - S_{\text{whole}}|/15$ being an error estimate by Richardson's argument — and recursively bisects only the subintervals that fail the tolerance. Library routines (QUADPACK's `qags`, behind `scipy.integrate.quad`; MATLAB `integral`) use adaptive Gauss–Kronrod pairs, which reuse the Gauss nodes in a higher-order Kronrod extension so the error estimate costs few extra evaluations.

Worked Example 22.33 — An endpoint singularity

$\int_0^1 \sqrt{x}\,dx = \frac{2}{3}$. The integrand is continuous but f' is infinite at 0, so the error constants of the composite rules are infinite and convergence degrades: composite Simpson gives errors -1.3×10^{-3}, -1.6×10^{-4}, -2.0×10^{-5} for $n = 16, 64, 256$ — order $h^{1.5}$ instead of h^4. An adaptive Simpson code with tolerance 10^{-8} concentrates its subintervals near $x = 0$ and reaches an error of 2.6×10^{-11} with 317 evaluations. Better still is to remove the singularity analytically — here the substitution $x = t^2$ gives $\int_0^1 2t^2\,dt$, which Simpson integrates *exactly*.

Infinite intervals and singularities. Transform them away when possible: substitutions ($x = t^2$, $x = \tan t$, $x = t/(1-t)$), subtracting off the singular part and integrating it analytically, or Gaussian rules with the singular weight built in. For oscillatory integrands such as Fourier integrals, specialised (Filon, Levin) methods or the FFT are preferable to general-purpose quadrature.

Common Pitfall — Quadrature on data you cannot refine

When f is a table of measurements, you cannot choose the nodes, so Gaussian quadrature is unavailable, and the "error formulas" involve derivatives of an unknown function. Use composite trapezoid or Simpson (or integrate a spline fit), estimate the error by comparing the rule on all points with the rule on every second point, and remember that noise in the data integrates to a random error of order $\sigma\sqrt{b-a}\sqrt{h}$ — integration, unlike differentiation, *averages* noise down.

Engineering Note — The FFT is a trapezoidal rule

The DFT coefficient $X_k = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}$ is, up to the factor T/N, the trapezoidal rule for the Fourier-series coefficient $\frac{1}{T}\int_0^T x(t)e^{-jk\omega_0 t}\,dt$ of a periodic signal. Example 22.31 explains why FFT-based spectral analysis of smooth periodic signals is so accurate — and why a *non*-periodic record (a window that does not contain an integer number of periods) loses this accuracy and produces leakage: the endpoint terms of Euler–Maclaurin no longer cancel. RMS meters, power analysers and energy meters integrate sampled $v(t)i(t)$ with exactly these rules.

22.9 Numerical solution of ODEs

The initial-value problem

Most ODEs that describe real systems — nonlinear circuits, motors with saturation, power converters, oscillators, chemical kinetics in batteries — have no closed-form solution. We consider the **initial-value problem (IVP)**

$$\mathbf{y}'(t) = \mathbf{f}(t, \mathbf{y}), \qquad \mathbf{y}(t_0) = \mathbf{y}_0,$$

where $\mathbf{y} \in \mathbb{R}^m$ is the state vector (Ch. 11–12). If $\mathbf{f}$ is continuous and **Lipschitz** in $\mathbf{y}$, $\|\mathbf{f}(t,\mathbf{y}) - \mathbf{f}(t,\mathbf{z})\| \le L\|\mathbf{y}-\mathbf{z}\|$, the solution exists and is unique (Picard–Lindelöf). Higher-order equations are reduced to first-order systems: the series RLC equation $L\ddot{q} + R\dot{q} + q/C = v(t)$ becomes, with $\mathbf{y} = (q, i)$, $i = \dot{q}$,

$$\frac{d}{dt}\begin{bmatrix} q \\ i \end{bmatrix} = \begin{bmatrix} i \\ \big(v(t) - Ri - q/C\big)/L \end{bmatrix}.$$

A numerical method produces approximations $\mathbf{y}_n \approx \mathbf{y}(t_n)$ on a grid $t_n = t_0 + nh$ (or with variable steps h_n).

Euler's method

Replace the derivative at t_n by a forward difference (Section 22.7):

$$\mathbf{y}_{n+1} = \mathbf{y}_n + h\,\mathbf{f}(t_n, \mathbf{y}_n).$$

Geometrically, follow the tangent of the solution curve through $(t_n, \mathbf{y}_n)$ for a time h. Each step is cheap — one evaluation of $\mathbf{f}$ — but, as we shall see, inaccurate.

Local and global truncation error

- The **local truncation error** τ_{n+1} is the error made in one step starting from the exact solution: for Euler, $\tau_{n+1} = y(t_{n+1}) - y(t_n) - hf(t_n, y(t_n)) = \frac{h^2}{2}y''(\xi_n) = O(h^2)$.
- The **global error** is $e_n = y(t_n) - y_n$, the accumulated error at a fixed time t_n.
- A one-step method is **of order** p if its local truncation error is $O(h^{p+1})$; then (under a Lipschitz condition) its global error is $O(h^p)$.

Convergence of Euler's method

Let f be Lipschitz in y with constant L and let $|y''| \le M$ on $[t_0, T]$. Then the Euler iterates satisfy

$$|e_n| \le \frac{Mh}{2L}\left(e^{L(t_n - t_0)} - 1\right), \qquad t_0 \le t_n \le T,$$

so the global error is $O(h)$: Euler's method is first order.

Proof. Subtract the Euler step from the exact relation $y(t_{n+1}) = y(t_n) + hf(t_n, y(t_n)) + \tau_{n+1}$:

$$e_{n+1} = e_n + h\big[f(t_n, y(t_n)) - f(t_n, y_n)\big] + \tau_{n+1} \;\Rightarrow\; |e_{n+1}| \le (1 + hL)|e_n| + \tfrac{M}{2}h^2.$$

With $e_0 = 0$, induction gives $|e_n| \le \frac{M}{2}h^2 \sum_{k=0}^{n-1}(1+hL)^k = \frac{Mh}{2L}\big((1+hL)^n - 1\big)$, and $(1+hL)^n \le e^{nhL}$. $\square$

The proof displays the general mechanism, which we shall meet again in Section 22.11: **global error = (local errors, $O(h^{p+1})$ each) × (number of steps, $O(1/h)$) × (an amplification factor that stays bounded — stability)**.

Worked Example 22.34 — Charging an RC circuit with Euler's method

A capacitor charges through R from a 5 V step, $\tau = RC = 1$ ms: $\dot v = (5 - v)/\tau$, $v(0) = 0$, exact $v = 5(1 - e^{-t/\tau})$. Use $h = 0.2$ ms.

Step 1. $v_1 = 0 + 0.2 \cdot (5 - 0)/1 = 1.0$ V (exact 0.906).

Step 2. $v_2 = 1 + 0.2(5 - 1) = 1.8$ V (exact 1.648).

t (ms)	0.2	0.4	0.6	0.8	1.0
Euler v_n (V)	1.000	1.800	2.440	2.952	3.362
exact (V)	0.906	1.648	2.256	2.753	3.161

Interpretation. Here $v_n = 5\big(1 - (1 - h/\tau)^n\big)$, so Euler replaces $e^{-t/\tau}$ by $(1 - h/\tau)^{t/h} = 0.8^5 = 0.328$ at $t = \tau$ instead of 0.368 — a 6 % error in the voltage at one time constant. Halving h halves the error (first order). Notice also that $0 < 1 - h/\tau < 1$ is needed for the numerical solution to decay monotonically like the true one; this is a *stability* requirement, discussed below.

Runge–Kutta methods

Euler uses the slope only at the start of the step. Runge–Kutta (RK) methods sample the slope at several points within the step and combine the samples so that the Taylor expansion of the numerical solution matches the true one to higher order.

Heun's method (improved Euler, explicit trapezoid), order 2:

$$\mathbf{k}_1 = \mathbf{f}(t_n, \mathbf{y}_n), \quad \mathbf{k}_2 = \mathbf{f}(t_n + h, \mathbf{y}_n + h\mathbf{k}_1), \quad \mathbf{y}_{n+1} = \mathbf{y}_n + \tfrac{h}{2}(\mathbf{k}_1 + \mathbf{k}_2).$$

Explicit midpoint method, also order 2: $\mathbf{k}_2 = \mathbf{f}(t_n + \tfrac{h}{2}, \mathbf{y}_n + \tfrac{h}{2}\mathbf{k}_1)$, $\mathbf{y}_{n+1} = \mathbf{y}_n + h\mathbf{k}_2$.

Why order 2. For scalar $y' = f(t, y)$, the exact solution satisfies $y(t + h) = y + hf + \tfrac{h^2}{2}(f_t + f_y f) + O(h^3)$. Expanding Heun's $k_2 = f + h(f_t + f_y f) + O(h^2)$ gives $y_{n+1} = y + hf + \tfrac{h^2}{2}(f_t + f_y f) + O(h^3)$ — the two expansions agree through h^2, so the local error is $O(h^3)$.

The classical fourth-order Runge–Kutta method (RK4):

$$\begin{aligned} \mathbf{k}_1 &= \mathbf{f}(t_n,\ \mathbf{y}_n), & \mathbf{k}_2 &= \mathbf{f}\big(t_n + \tfrac{h}{2},\ \mathbf{y}_n + \tfrac{h}{2}\mathbf{k}_1\big), \\ \mathbf{k}_3 &= \mathbf{f}\big(t_n + \tfrac{h}{2},\ \mathbf{y}_n + \tfrac{h}{2}\mathbf{k}_2\big), & \mathbf{k}_4 &= \mathbf{f}(t_n + h,\ \mathbf{y}_n + h\mathbf{k}_3), \\ \mathbf{y}_{n+1} &= \mathbf{y}_n + \tfrac{h}{6}\left(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4\right). \end{aligned}$$

When $\mathbf{f}$ depends on t only, RK4 reduces to Simpson's rule (weights $\tfrac{1}{6}, \tfrac{4}{6}, \tfrac{1}{6}$, with the two midpoint samples sharing the middle weight). Four function evaluations per step buy fourth-order accuracy: halving h reduces the error by 16 for only twice the work.

```
def rk4_step(f, t, y, h):
    k1 = f(t, y)
    k2 = f(t + h/2, y + h/2 * k1)
    k3 = f(t + h/2, y + h/2 * k2)
    k4 = f(t + h,   y + h   * k3)
    return y + h/6 * (k1 + 2*k2 + 2*k3 + k4)
```

Worked Example 22.35 — One RK4 step, and observed orders

Solve $y' = y - t^2 + 1$, $y(0) = 0.5$, whose exact solution is $y = (t + 1)^2 - \tfrac{1}{2}e^t$.

One step with $h = 0.2$. $k_1 = f(0, 0.5) = 1.5$; $k_2 = f(0.1,\ 0.5 + 0.1 \cdot 1.5) = f(0.1, 0.65) = 0.65 - 0.01 + 1 = 1.64$; $k_3 = f(0.1,\ 0.5 + 0.1 \cdot 1.64) = f(0.1, 0.664) = 1.654$; $k_4 = f(0.2,\ 0.5 + 0.2 \cdot 1.654) = f(0.2, 0.8308) = 0.8308 - 0.04 + 1 = 1.7908$; $y_1 = 0.5 + \tfrac{0.2}{6}(1.5 + 3.28 + 3.308 + 1.7908) = 0.5 + 0.329293 = 0.829293$. Exact: $y(0.2) = 1.44 - 0.5e^{0.2} = 0.829299$. The one-step error is 5.3×10^{-6}.

Global errors at $t = 2$ (exact $y(2) = 5.305472$):

h	Euler	Heun	RK4
0.2	-0.440	-7.24×10^{-2}	-1.09×10^{-4}
0.1	-0.242	-1.89×10^{-2}	-6.99×10^{-6}
0.05	-0.127	-4.82×10^{-3}	-4.42×10^{-7}
ratio per halving	$\to 2$ (order 1)	3.9 (order 2)	15.8 (order 4)

To reach an error of 10^{-6}, Euler would need $h \approx 4 \times 10^{-7}$ (about five million steps and evaluations); RK4 needs $h \approx 0.06$ — about 33 steps, 130 evaluations.

Step-size control

A fixed step is wasteful: transients need small steps, quiescent intervals large ones. Production integrators (MATLAB `ode45`, SciPy `solve_ivp` with `RK45`) use **embedded Runge–Kutta pairs**, such as Dormand–Prince 5(4): two RK formulas of orders p and $p - 1$ share the same stage evaluations, and their difference estimates the local error err of the lower-order one. The step is accepted if $\text{err} \le \text{tol}$ and the next step is chosen as

$$h_{\text{new}} = h \cdot \min\left(f_{\max},\ \max\left(f_{\min},\ 0.9\left(\frac{\text{tol}}{\text{err}}\right)^{1/p}\right)\right),$$

where the exponent follows from err $\propto h^p$ and the safety factor 0.9 and limits $f_{\min}$, $f_{\max}$ (e.g. 0.2, 5) prevent oscillation. The tolerance is usually mixed, $\text{tol} = \text{atol} + \text{rtol} \cdot |y|$. Note that the controller bounds the *local* error per step; the global error is typically of the same order as the tolerance for well-behaved problems, but is not guaranteed by it.

Multistep methods

Instead of sampling within the step, **linear multistep methods** reuse past values. The second-order **Adams–Bashforth** method,

$$\mathbf{y}_{n+1} = \mathbf{y}_n + \frac{h}{2}\big(3\mathbf{f}_n - \mathbf{f}_{n-1}\big),$$

integrates the linear interpolant of the last two slopes and costs only *one* new evaluation per step. Implicit **Adams–Moulton** methods and the **backward differentiation formulas (BDF)**, e.g. BDF2 $\frac{3}{2}\mathbf{y}_{n+1} - 2\mathbf{y}_n + \frac{1}{2}\mathbf{y}_{n-1} = h\mathbf{f}_{n+1}$, are the workhorses for stiff problems (below). Multistep methods need a starting procedure (a few RK steps) and are more awkward to use with variable steps.

Absolute stability: the test equation

Convergence is a statement about $h \to 0$. In practice h is finite, and a method can produce wildly wrong, exponentially growing answers for a decaying problem if h is too large. The analysis uses the scalar **test equation**

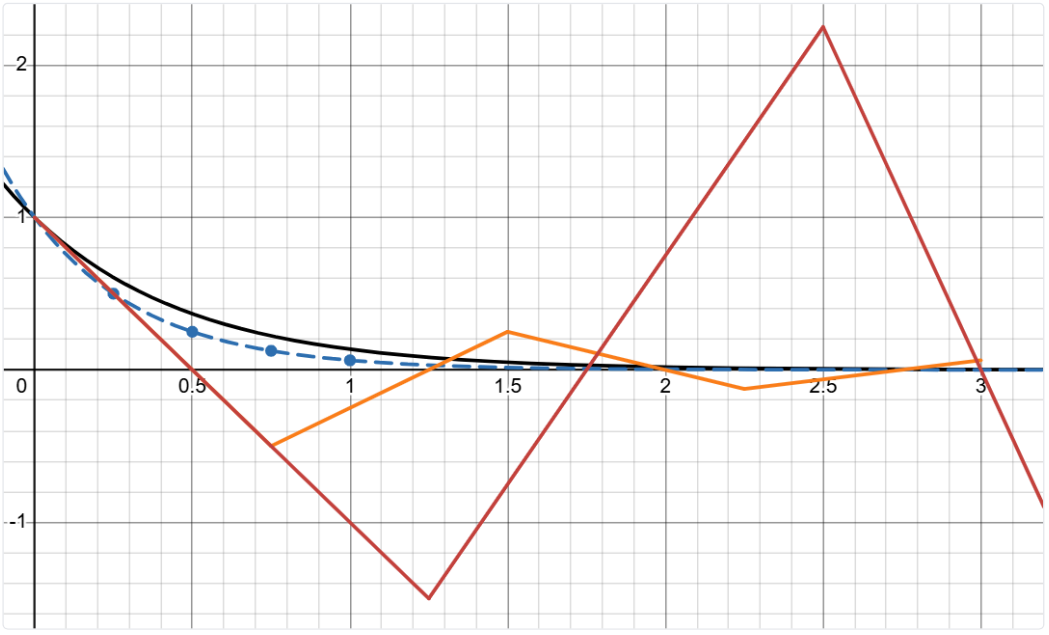
$$y' = \lambda y, \qquad \lambda \in \mathbb{C},\ \operatorname{Re}\lambda < 0,$$

whose solution decays. (It models each eigen-mode of a linear system $\mathbf{y}' = \mathbf{A}\mathbf{y}$, with λ an eigenvalue of $\mathbf{A}$ — Ch. 10.) Applied to it, every RK method gives $y_{n+1} = R(z)\,y_n$ with $z = h\lambda$, where R is the **stability function**:

Method	$R(z)$	Real stability interval
Euler (explicit)	$1 + z$	$-2 \le z \le 0$
Heun / midpoint (RK2)	$1 + z + \frac{z^2}{2}$	$-2 \le z \le 0$
RK4	$1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24}$	$-2.785 \le z \le 0$
backward (implicit) Euler	$\dfrac{1}{1 - z}$	all $z \le 0$
trapezoidal (Crank–Nicolson)	$\dfrac{1 + z/2}{1 - z/2}$	all $z \le 0$

> **Region of absolute stability; A-stability**
>
> The **region of absolute stability** of a method is $\{z \in \mathbb{C} : |R(z)| \le 1\}$. A method is **A-stable** if this region contains the entire left half-plane $\operatorname{Re} z \le 0$, so that the numerical solution decays whenever the true one does, for every step size.

For explicit Euler the region is the disk $|1 + z| \le 1$ of radius 1 centred at $z = -1$. For real $\lambda < 0$ the condition is $h \le 2/|\lambda|$; for monotone (non-oscillating) decay, $h \le 1/|\lambda|$. Explicit methods always have bounded stability regions (their R is a polynomial, which is unbounded); the implicit Euler and trapezoidal rules are A-stable.



Euler's method on $y' = -2y$, $y(0) = 1$ (exact e^{-2t}, black). With $h = 0.25$ ($z = -0.5$, blue) the numerical solution 0.5^n decays monotonically but too fast; with $h = 0.75$ ($z = -1.5$, orange) it oscillates in sign while decaying ($|R| = 0.5$); with $h = 1.25$ ($z = -2.5$, red) $|R| = 1.5 > 1$ and it grows without bound — outside the stability region.

Stiffness

Stiff problem (working definition)

An ODE system is **stiff** when it contains modes with very different time scales — eigenvalues of the Jacobian $\partial \mathbf{f}/\partial \mathbf{y}$ with $\operatorname{Re}\lambda < 0$ and $\max|\operatorname{Re}\lambda| \gg \min|\operatorname{Re}\lambda|$ — such that the step size of an explicit method is limited by **stability** for the fast, already-decayed modes rather than by **accuracy** for the slow mode of interest. The ratio $\max|\operatorname{Re}\lambda|/\min|\operatorname{Re}\lambda|$ is the stiffness ratio.

Stiffness is endemic in electronics: a circuit with a 1 ns parasitic time constant and a 1 ms thermal or bias time constant has stiffness ratio 10^6; an explicit method would need a million steps per millisecond merely to remain stable, long after the fast transient has died.

Implicit methods. Backward Euler,

$$\mathbf{y}_{n+1} = \mathbf{y}_n + h\,\mathbf{f}(t_{n+1}, \mathbf{y}_{n+1}),$$

requires solving a (generally nonlinear) equation for $\mathbf{y}_{n+1}$ at every step — by Newton's method (Section 22.4) with Jacobian $\mathbf{I} - h\,\partial\mathbf{f}/\partial\mathbf{y}$ — but is A-stable, so h can be chosen for accuracy alone. Higher-order stiff solvers (BDF methods in `ode15s` and `solve_ivp(method='BDF')`, implicit RK methods such as Radau IIA) follow the same pattern. The **Dahlquist barrier** limits what is possible: no explicit linear multistep method is A-stable, and an A-stable linear multistep method has order at most 2 (the trapezoidal rule has the smallest error constant among them).

Worked Example 22.36 — A stiff scalar problem

$y' = -1000\,(y - \cos t)$, $y(0) = 0$. After a fast initial transient (e^{-1000t}, time constant 1 ms) the solution follows $y \approx \cos t$ closely. Exact value at $t = 0.1$: $y(0.1) = 0.9951030$. Here $\lambda = -1000$, so explicit Euler is stable only for $h \le 2/1000 = 0.002$.

h	steps to $t = 0.1$	explicit Euler $y(0.1)$	implicit Euler $y(0.1)$
0.001	100	0.9951035	0.9951025
0.0021	48	-96.0	0.9950226
0.01	10	-3.5×10^9	0.9950980

Implicit Euler step. Because f is linear in y, the implicit equation $y_{n+1} = y_n - 1000h\,(y_{n+1} - \cos t_{n+1})$ is solved in closed form: $y_{n+1} = \dfrac{y_n + 1000h\cos t_{n+1}}{1 + 1000h}$.

Lesson. Just 5% beyond the stability limit ($h = 0.0021$, $z = -2.1$, $|R| = 1.1$) explicit Euler explodes; implicit Euler gives a 0.0005 % accurate answer with ten steps. The interesting solution ($\cos t$) varies on a time scale of seconds; only the stability of the method forced explicit Euler to millisecond steps.

Common Pitfall — Energy drift in oscillators

For an undamped LC oscillator, $\ddot{x} + x = 0$, the eigenvalues are $\pm j$ — on the imaginary axis, which lies *outside* the stability region of explicit Euler except at $z = 0$. Euler multiplies the "energy" $x^2 + v^2$ by exactly $1 + h^2$ every step: with $h = 0.1$, after one period (62 steps) the amplitude has grown by a factor $\sqrt{1.853} = 1.36$. Backward Euler errs the other way, damping a lossless circuit. The trapezoidal rule ($|R(j\omega h)| = 1$ exactly) conserves amplitude, and *symplectic* methods (symplectic Euler, Störmer–Verlet/leapfrog) conserve a nearby energy for long times — the methods of choice for lossless systems such as orbits and particle accelerators.

Engineering Note — How SPICE integrates

SPICE's transient analysis uses implicit methods throughout — the **trapezoidal rule** by default and **Gear (BDF) methods** of order 1–2 as alternatives — with Newton iteration at every time point and step control based on a local-truncation-error estimate. The trapezoidal rule's tendency to produce a spurious point-to-point ringing on stiff modes (since $R(z) \to -1$ as $z \to -\infty$: the fast mode is not damped, just sign-flipped) is the reason SPICE offers `method=gear`, whose $R(z) \to 0$ ("L-stability") damps fast modes strongly. When a simulation shows ringing at the time-step frequency, switch the integration method before blaming the circuit.

22.10 Numerical solution of PDEs

Grids and notation

The three model PDEs of Ch. 13 — the **heat (diffusion)** equation, the **wave** equation and the **Laplace/Poisson** equation — are prototypes of the parabolic, hyperbolic and elliptic classes, and each class demands a different numerical strategy. EE instances abound: diffusion of dopants and heat in a chip, the telegrapher's equations on a transmission line, Maxwell's equations in an FDTD simulator, the potential in an electrostatic lens or around a microstrip.

The **finite-difference method** replaces the continuous domain by a grid $x_i = i\,\Delta x$, $t_n = n\,\Delta t$, the unknown function by grid values $u_i^n \approx u(x_i, t_n)$, and every derivative by a difference quotient from Section 22.7. Throughout we use

$$\delta_x^2 u_i^n = u_{i+1}^n - 2u_i^n + u_{i-1}^n, \qquad \frac{\delta_x^2 u_i^n}{\Delta x^2} = u_{xx}(x_i, t_n) + O(\Delta x^2).$$

The heat equation: explicit scheme (FTCS)

For $u_t = \alpha u_{xx}$ on $0 < x < L$ with boundary values $u(0,t)$, $u(L,t)$ and initial data $u(x,0)$, use a **f**orward difference in **t**ime and a **c**entral difference in **s**pace:

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = \alpha\,\frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2} \quad\Longleftrightarrow\quad u_i^{n+1} = u_i^n + r\left(u_{i+1}^n - 2u_i^n + u_{i-1}^n\right), \qquad r = \frac{\alpha\,\Delta t}{\Delta x^2}.$$

The dimensionless **mesh ratio** r controls everything. The scheme is explicit — each new value comes directly from three old ones — and its truncation error is $O(\Delta t) + O(\Delta x^2)$.

Stability by Fourier modes. Substitute a single grid Fourier mode $u_i^n = G^n e^{j\theta i}$ ($\theta = k\Delta x$) into the scheme:

$$G = 1 + r\left(e^{j\theta} - 2 + e^{-j\theta}\right) = 1 - 2r(1 - \cos\theta) = 1 - 4r\sin^2\frac{\theta}{2}.$$

The true solution mode decays, so we need $|G| \le 1$ for all θ. The worst case is the sawtooth mode $\theta = \pi$ ($u_i \propto (-1)^i$), where $G = 1 - 4r$, and $|1 - 4r| \le 1$ requires

$$\boxed{r = \frac{\alpha\,\Delta t}{\Delta x^2} \le \frac{1}{2}}.$$

This is a severe restriction: **halving Δx forces Δt to be quartered**, so refining the grid by 2 costs 8 times the work in one dimension. (Section 22.11 turns this calculation into the general von Neumann method.)

Worked Example 22.37 — One FTCS step

$u_t = u_{xx}$ on $[0, 1]$, $u = 0$ at both ends, $u(x, 0) = \sin \pi x$; exact solution $u = e^{-\pi^2 t}\sin \pi x$. Take $\Delta x = 0.25$ and $r = 0.25$, so $\Delta t = r\Delta x^2 = 0.015625$.

Initial values. $u^0 = (0,\ 0.707107,\ 1,\ 0.707107,\ 0)$.

Step. $u_1^1 = 0.707107 + 0.25(1 - 2\cdot 0.707107 + 0) = 0.603553$; $u_2^1 = 1 + 0.25(0.707107 - 2 + 0.707107) = 0.853553$; $u_3^1 = u_1^1$ by symmetry.

Compare. Exact: $e^{-\pi^2 \cdot 0.015625} = 0.857090$, so $u(0.25, \Delta t) = 0.606054$, $u(0.5, \Delta t) = 0.857090$. The numerical mode is multiplied by $G = 1 - 4(0.25)\sin^2(\pi/8) = 0.853553$ per step versus the exact factor 0.857090 — an error of 0.4 % per step from the coarse grid.

Worked Example 22.38 — Watching FTCS go unstable

Same problem with $\Delta x = 0.05$ (21 points), run to $t \approx 0.1$, with a tiny perturbation $10^{-6}(-1)^i$ (think rounding noise) added to the initial data.

r	steps	$\max_i \lvert u_i - u_{\text{exact}}\rvert$
0.50	80	1.5×10^{-3} (truncation error only)
0.60	67	3.9×10^{3}

At $r = 0.6$ the sawtooth mode has $G = 1 - 2.4 = -1.4$; after 67 steps it has grown by $1.4^{67} \approx 6 \times 10^9$, turning the 10^{-6} perturbation into thousands. Instability amplifies *whatever* high-frequency content is present — rounding error suffices — so an unstable scheme is useless however small the perturbation.

Implicit schemes: BTCS and Crank–Nicolson

Evaluating the spatial difference at the *new* time level gives the **backward-time (BTCS) scheme**

$$-r\,u_{i-1}^{n+1} + (1 + 2r)\,u_i^{n+1} - r\,u_{i+1}^{n+1} = u_i^n,$$

a **tridiagonal** linear system per step (solved in $O(N)$ by the Thomas algorithm, as for splines). Its amplification factor $G = 1/(1 + 4r\sin^2(\theta/2))$ satisfies $0 < G \le 1$ for *every* r: the scheme is **unconditionally stable**, though still only first order in time.

Averaging the explicit and implicit schemes gives **Crank–Nicolson**:

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = \frac{\alpha}{2}\left(\frac{\delta_x^2 u_i^{n+1}}{\Delta x^2} + \frac{\delta_x^2 u_i^n}{\Delta x^2}\right), \qquad G = \frac{1 - 2r\sin^2(\theta/2)}{1 + 2r\sin^2(\theta/2)},$$

second order in both Δt and Δx and unconditionally stable ($|G| \le 1$ always). It is the trapezoidal rule of Section 22.9 in disguise, and it inherits the trapezoidal rule's weakness: for large r the high-frequency modes have $G \to -1$ (at $r = 5$, $\theta = \pi$: $G = -9/11 = -0.82$), so sharp initial data (a step in temperature, a switched voltage) produce slowly decaying point-to-point oscillations. A few initial BTCS steps ("Rannacher smoothing") suppress them.

The method-of-lines view

Discretising *only in space* turns the PDE into a system of ODEs, $\dot{\mathbf{u}} = \frac{\alpha}{\Delta x^2}\mathbf{T}\mathbf{u}$, where $\mathbf{T} = \text{tridiag}(1, -2, 1)$. The eigenvalues of $\mathbf{T}$ (Ch. 10) are $-4\sin^2\left(\frac{k\pi}{2(N+1)}\right)$, so the ODE eigenvalues range from about $-\alpha\pi^2/L^2$ (the slow fundamental mode) to about $-4\alpha/\Delta x^2$ (the fast grid-scale mode). The system is therefore **stiff**, with stiffness ratio $\sim (2L/(\pi\Delta x))^2$, growing as the grid is refined. FTCS is explicit Euler on this system and its stability limit $\Delta t \cdot 4\alpha/\Delta x^2 \le 2$ is exactly $r \le \frac{1}{2}$; BTCS and Crank–Nicolson are implicit Euler and the trapezoidal rule. The method of lines thus lets any stiff ODE solver from Section 22.9 serve as a PDE solver — a common and powerful approach.

The wave equation and the CFL condition

For $u_{tt} = c^2 u_{xx}$, central differences in both variables give the explicit **leapfrog** scheme

$$u_i^{n+1} = 2u_i^n - u_i^{n-1} + C^2\left(u_{i+1}^n - 2u_i^n + u_{i-1}^n\right), \qquad C = \frac{c\,\Delta t}{\Delta x},$$

second order in both variables. C is the **Courant number**. The first step needs u^{-1}; with initial velocity $u_t(x, 0) = g(x)$, the central-difference condition $u_i^1 - u_i^{-1} = 2\Delta t\, g_i$ gives $u_i^1 = u_i^0 + \Delta t\, g_i + \frac{C^2}{2}\delta_x^2 u_i^0$.

Courant–Friedrichs–Lewy (CFL) condition

A necessary condition for the convergence of an explicit difference scheme for a hyperbolic equation is that the **numerical domain of dependence contain the physical domain of dependence**. For the leapfrog scheme for $u_{tt} = c^2 u_{xx}$ (and for upwind schemes for $u_t + c u_x = 0$) this is

$$C = \frac{c\,\Delta t}{\Delta x} \le 1,$$

and for leapfrog it is also sufficient for stability.

Reasoning. The exact solution at (x_i, t_n) depends on initial data in $[x_i - ct_n,\ x_i + ct_n]$ (d'Alembert, Ch. 13). The scheme's value u_i^n depends only on initial grid values in $[x_i - n\Delta x,\ x_i + n\Delta x]$ since information moves one cell per step. If $c\Delta t > \Delta x$ the numerical interval is smaller than the physical one, and changing the initial data in the gap changes the true solution but not the numerical one — so the scheme cannot converge. Von Neumann analysis confirms it: the amplification factors satisfy $G^2 - 2\big(1 - 2C^2\sin^2\tfrac{\theta}{2}\big)G + 1 = 0$, whose roots have $|G| = 1$ if $C \le 1$, while for $C > 1$ one root has $|G| > 1$ at $\theta = \pi$. $\square$

Worked Example 22.39 — The CFL limit in action

A plucked string: $c = 1$, $u = \sin \pi x$ at $t = 0$, $u_t = 0$, fixed ends, $\Delta x = 0.05$, run to $t = 2$ (one full period); exact $u = \cos(\pi t)\sin(\pi x)$. A 10^{-6} sawtooth perturbation is added to the initial data.

Courant number C	steps	max error at $t \approx 2$
1.00	40	1.0×10^{-6} (only the perturbation)
0.50	80	1.2×10^{-5}
1.05	38	7.2×10^{3}

Why $C = 1$ is exact. With $C = 1$ the scheme becomes $u_i^{n+1} = u_{i+1}^n + u_{i-1}^n - u_i^{n-1}$, which is satisfied exactly by any $F(x - ct) + G(x + ct)$ on the grid: the "magic time step" reproduces d'Alembert's solution. For $C < 1$ the scheme is stable but **dispersive** (different wavelengths travel at slightly wrong, different speeds). For $C = 1.05$, only 5 % over the limit, the sawtooth mode grows explosively.

First-order hyperbolic equations. For the advection equation $u_t + a\,u_x = 0$ ($a > 0$, a signal travelling along a lossless line), the natural explicit scheme takes the spatial difference **upwind** (from the side the information comes from):

$$u_i^{n+1} = u_i^n - C\left(u_i^n - u_{i-1}^n\right), \qquad C = \frac{a\Delta t}{\Delta x} \le 1.$$

It is stable for $0 \le C \le 1$, first order, and numerically diffusive (it smears sharp fronts). The tempting FTCS version with a central difference $u_{i+1}^n - u_{i-1}^n$ is **unconditionally unstable** ($|G|^2 = 1 + C^2\sin^2\theta > 1$) — a scheme that is perfectly consistent but useless, illustrating that consistency is not enough (Section 22.11). Second-order schemes (Lax–Wendroff, leapfrog) reduce diffusion at the cost of dispersive wiggles near discontinuities.

Engineering Note — FDTD and the Courant limit

The finite-difference time-domain (FDTD, Yee) method for Maxwell's equations is a staggered leapfrog scheme for $\mathbf{E}$ and $\mathbf{H}$. Its stability limit in 3-D on a cubic grid is $c\,\Delta t \le \Delta x/\sqrt{3}$ — the CFL condition with the 3-D domain of dependence. One consequence: halving the cell size to resolve a fine feature (a thin trace, a narrow slot) halves the allowable time step for the *whole* simulation, so run time scales like $(\Delta x)^{-4}$. This is why FDTD codes offer sub-gridding and why frequency-domain (FEM, MoM) solvers compete for electrically small fine structure.

Laplace's and Poisson's equations

Elliptic problems have no time variable: the potential in a region is determined all at once by its boundary values. For Poisson's equation $\nabla^2 V = -\rho/\varepsilon$ on a square grid with spacing h, the **five-point stencil**

$$\frac{V_{i+1,j} + V_{i-1,j} + V_{i,j+1} + V_{i,j-1} - 4V_{i,j}}{h^2} = -\frac{\rho_{i,j}}{\varepsilon}$$

has truncation error $O(h^2)$. For Laplace's equation ($\rho = 0$) it says each interior value is the **average of its four neighbours** — the discrete mean-value property, which also yields a discrete maximum principle (no interior extrema), mirroring the continuous theory of Ch. 13. Writing one equation per interior node gives a large, sparse, symmetric positive-definite linear system $\mathbf{A}\mathbf{V} = \mathbf{b}$, with boundary values moved into $\mathbf{b}$.

Worked Example 22.40 — Potential in a square trough

A square region has its top side at 100 V and the other three sides grounded. Use $h = \frac{1}{4}$ of the side, giving a 3×3 array of interior nodes.

Equations. For each interior node, $4V_{i,j} - (\text{sum of neighbours}) = (\text{sum of neighbouring boundary values})$. For example the top-left node: $4V_{1,1} - V_{1,2} - V_{2,1} = 100$; the centre: $4V_{2,2} - V_{1,2} - V_{3,2} - V_{2,1} - V_{2,3} = 0$.

Solution (by symmetry 6 unknowns, solved directly):

row	left	centre	right
top ($y = \frac{3}{4}$)	42.857	52.679	42.857
middle ($y = \frac{1}{2}$)	18.750	25.000	18.750
bottom ($y = \frac{1}{4}$)	7.143	9.821	7.143

Check. The centre value 25 V is exact even on this coarse grid: by superposition, the four problems with one side at 100 V sum to a box entirely at 100 V, so each contributes 25 V at the centre — and the discrete problem has the same symmetry. The Fourier-series solution (Ch. 13) gives 25.000 as well. Each value is also the average of its neighbours, e.g. $(100 + 18.75 + 52.679 + 0)/4 = 42.857$ for the top-left node.

Iterative solution. Real grids have 10^4–10^8 nodes, and direct elimination fills in the sparse matrix. The classical iterations sweep the grid repeatedly:

- **Jacobi:** replace every $V_{i,j}$ by the average of its neighbours' *old* values.
- **Gauss–Seidel:** use new values as soon as they are available (sweeping in order); converges twice as fast as Jacobi for this problem.
- **Successive over-relaxation (SOR):** $V_{i,j} \leftarrow (1-\omega)V_{i,j} + \omega \cdot (\text{Gauss–Seidel value})$ with $1 < \omega < 2$; for the square with N intervals per side the optimal factor is $\omega_{\text{opt}} = 2/(1 + \sin(\pi/N))$.

For the model problem the Jacobi iteration matrix has spectral radius $\cos(\pi h) \approx 1 - \frac{\pi^2 h^2}{2}$, so the number of iterations to reduce the error by a fixed factor grows like h^{-2} (Gauss–Seidel: same order, half the count); optimal SOR's radius $\approx 1 - 2\pi h$ makes it grow only like h^{-1}. In Example 22.40, reducing the error below 10^{-6} from a zero start takes 51 Jacobi, 25 Gauss–Seidel and 12 SOR sweeps ($\omega_{\text{opt}} = 2/(1+\sin(\pi/4)) = 1.172$). Modern solvers use preconditioned conjugate gradients and multigrid, whose cost is nearly proportional to the number of unknowns (Ch. 24).

```
import numpy as np
def laplace_sor(V, fixed, omega=1.8, tol=1e-8, maxit=10000):
    """V: 2-D array with boundary values set; fixed: boolean mask of Dirichlet nodes."""
    for it in range(maxit):
        delta = 0.0
        for i in range(1, V.shape[0] - 1):
            for j in range(1, V.shape[1] - 1):
                if fixed[i, j]:
                    continue
                gs = 0.25 * (V[i+1, j] + V[i-1, j] + V[i, j+1] + V[i, j-1])
                new = (1 - omega) * V[i, j] + omega * gs
                delta = max(delta, abs(new - V[i, j]))
                V[i, j] = new
        if delta < tol:
            return V, it + 1
    return V, maxit
```

Boundary conditions. Dirichlet values (fixed potentials, electrode voltages) enter $\mathbf{b}$ directly. A Neumann condition $\partial V/\partial n = g$ (a symmetry plane, an insulating boundary) is imposed with a **ghost point** outside the domain: at $x = 0$, the central difference $(V_{1,j} - V_{-1,j})/(2h) = g$ eliminates $V_{-1,j}$ from the stencil while keeping second-order accuracy. Curved boundaries and material interfaces (dielectric steps, where $\varepsilon\,\partial V/\partial n$ must be continuous) need modified stencils; this is where finite-element methods, which handle unstructured meshes naturally, usually take over.

Common Pitfall — Accuracy near corners and edges

The $O(h^2)$ error estimate assumes the solution is smooth. Near re-entrant corners and electrode edges, fields are singular (e.g. $E \propto r^{-1/3}$ near a 90° re-entrant corner, $r^{-1/2}$ near a thin conductor's edge) and convergence degrades sharply. Quantities computed from derivatives — fields, capacitance per unit length from surface charge — suffer most. Refine locally near singularities, or extract capacitance from a variational (energy) expression, which is less sensitive.

22.11 Stability and convergence

Three properties of a discretisation

Every section of this chapter has met the same pattern: a discretisation that looks accurate (small truncation error) may still produce garbage (Examples 22.9, 22.36, 22.38, 22.39). This final section isolates the concepts that explain the pattern and states the theorems that make them precise. We phrase them for a linear time-dependent problem $u_t = \mathcal{L}u$ (with $\mathcal{L}$ a spatial differential operator) discretised by a one-step scheme

$$\mathbf{u}^{n+1} = \mathbf{C}(\Delta t)\,\mathbf{u}^{n},$$

where $\mathbf{u}^n$ is the vector of grid values and $\mathbf{C}$ the **solution (amplification) operator** of the scheme, with the grid spacing Δx tied to Δt by a fixed rule (e.g. r or C held constant). The same ideas apply to ODE solvers ($\mathbf{C} = R(h\mathbf{A})$ for $\mathbf{y}' = \mathbf{A}\mathbf{y}$) and to elliptic problems.

Consistency, stability, convergence

- **Consistency.** The scheme is consistent if its **local truncation error** — the residual $\boldsymbol{\tau}^n = \left(\mathbf{u}_{\text{exact}}^{n+1} - \mathbf{C}\,\mathbf{u}_{\text{exact}}^{n}\right)/\Delta t$ obtained by inserting the exact solution — tends to zero as $\Delta t, \Delta x \to 0$. It is consistent **of order** (p, q) if $\|\boldsymbol{\tau}\| = O(\Delta t^p + \Delta x^q)$.
- **Stability.** The scheme is stable (in the sense of Lax–Richtmyer) if for every $T > 0$ there is a constant K_T, independent of Δt and Δx, such that

$$\|\mathbf{C}(\Delta t)^n\| \le K_T \qquad \text{for all } n \text{ with } n\Delta t \le T.$$

- **Convergence.** The scheme is convergent if, for every initial condition (in a suitable class), $\max_{n\Delta t \le T} \|\mathbf{u}^n - \mathbf{u}_{\text{exact}}(t_n)\| \to 0$ as $\Delta t, \Delta x \to 0$.

Consistency is a local, *one-step* property and is checked by Taylor expansion. Stability is a global property of *many steps*: errors introduced at any time (truncation, rounding, data) must not be amplified without bound. Convergence is what we actually want, but it is hard to verify directly, because it involves the unknown exact solution.

The Lax equivalence theorem

Lax equivalence theorem (Lax–Richtmyer, 1956)

For a consistent finite-difference approximation to a **well-posed linear** initial-value problem,

$$\text{stability} \iff \text{convergence}.$$

Moreover, a stable scheme that is consistent of order (p, q) converges with the same order: $\|\mathbf{e}^n\| = O(\Delta t^p + \Delta x^q)$.

Proof of the important direction (stability + consistency ⇒ convergence). Let $\mathbf{e}^n = \mathbf{u}_{\text{exact}}^n - \mathbf{u}^n$. By the definition of $\boldsymbol{\tau}$,

$$\mathbf{u}_{\text{exact}}^{n+1} = \mathbf{C}\,\mathbf{u}_{\text{exact}}^{n} + \Delta t\,\boldsymbol{\tau}^n, \qquad \mathbf{u}^{n+1} = \mathbf{C}\,\mathbf{u}^n \quad\Longrightarrow\quad \mathbf{e}^{n+1} = \mathbf{C}\,\mathbf{e}^n + \Delta t\,\boldsymbol{\tau}^n.$$

Unrolling with $\mathbf{e}^0 = \mathbf{0}$:

$$\mathbf{e}^n = \Delta t \sum_{k=0}^{n-1} \mathbf{C}^{\,n-1-k}\boldsymbol{\tau}^k \quad\Longrightarrow\quad \|\mathbf{e}^n\| \le \Delta t \sum_{k=0}^{n-1} K_T \|\boldsymbol{\tau}^k\| \le K_T\, T \max_k \|\boldsymbol{\tau}^k\|.$$

Consistency makes the right side tend to zero, at the rate of the truncation error. □

This is the same "local error × number of steps × bounded amplification" argument as for Euler's method in Section 22.9 — now in its general form. The converse (convergence ⇒ stability) uses the uniform boundedness principle of functional analysis.

The practical value of the theorem is enormous: **to prove that a linear scheme converges, check consistency (Taylor series) and stability (usually von Neumann analysis); you never need the exact solution.** It also explains the failures we saw — FTCS for the heat equation with $r > \tfrac{1}{2}$, leapfrog with $C > 1$, central-space FTCS for advection — all consistent, all unstable, therefore all non-convergent.

Common Pitfall — The scope of the Lax theorem

The theorem is for **linear** problems. For nonlinear PDEs (shock-forming conservation laws, nonlinear diffusion) stability of the linearised scheme is necessary but not sufficient, and convergence may be to a wrong (non-physical) weak solution unless the scheme is also *conservative* and satisfies an *entropy* condition (Lax–Wendroff theorem). Also, "stable" is relative to a norm: a scheme can be stable in ℓ^2 but produce overshoots in max-norm.

Von Neumann (Fourier) stability analysis

For linear constant-coefficient schemes on uniform grids (with periodic boundaries, or away from boundaries), Fourier modes $u_i^n = G^n e^{j\theta i}$ diagonalise the solution operator, and $\|\mathbf{C}^n\|_2 = \max_\theta |G(\theta)|^n$ by Parseval's theorem (Ch. 15–16).

Von Neumann stability condition

A one-step, constant-coefficient scheme with amplification factor $G(\theta, \Delta t)$ is stable in ℓ^2 if and only if

$$|G(\theta, \Delta t)| \le 1 + K\Delta t \qquad \text{for all } \theta \in [-\pi, \pi]$$

and some constant K independent of $\Delta t, \Delta x$. For problems whose exact solutions do not grow, the practical form is $|G| \le 1$.

The allowance $1 + K\Delta t$ permits physically growing solutions (e.g. $u_t = u_{xx} + u$): then $|G|^n \le e^{KT}$, still bounded. For multi-level schemes (leapfrog), one analyses the matrix amplification factor or the characteristic equation for G, requiring its roots to satisfy $|G| \le 1$ (and simple roots on the unit circle).

Recipe.

1. Substitute $u_i^n = G^n e^{j\theta i}$ into the homogeneous scheme.
2. Divide by $G^n e^{j\theta i}$ and solve for $G(\theta)$.
3. Find the conditions on the mesh ratios under which $\max_\theta |G| \le 1$.

Worked Example 22.41 — Von Neumann analysis of the upwind scheme

Scheme: $u_i^{n+1} = u_i^n - C(u_i^n - u_{i-1}^n)$ for $u_t + au_x = 0$, $C = a\Delta t/\Delta x$.

Step 1–2. $G = 1 - C\left(1 - e^{-j\theta}\right) = (1 - C) + Ce^{-j\theta}$.

Step 3. $|G|^2 = (1 - C + C\cos\theta)^2 + C^2\sin^2\theta = 1 - 2C(1 - C)(1 - \cos\theta)$. For $0 \le C \le 1$, $C(1 - C) \ge 0$ and $|G| \le 1$ for all θ: **stable**. For $C > 1$ (or $C < 0$, i.e. "downwind"), $C(1 - C) < 0$ and $|G| > 1$ for $\theta \ne 0$: **unstable**. The von Neumann condition reproduces the CFL condition.

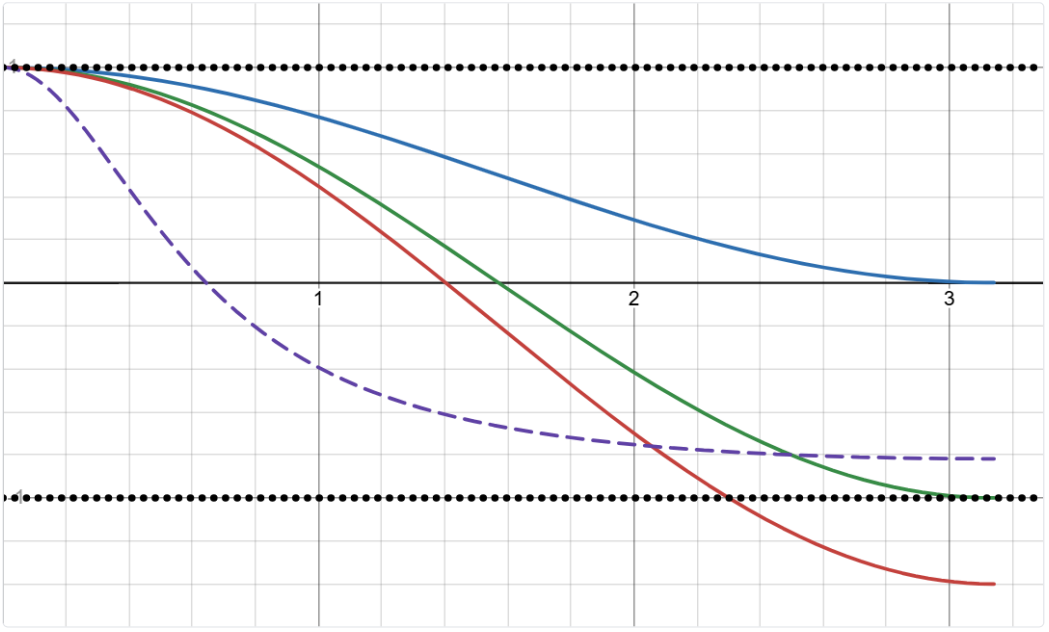
Numerical check ($C = 0.5$): $|G|^2 = 1,\ 0.854,\ 0.5,\ 0.146,\ 0$ at $\theta = 0,\ \pi/4,\ \pi/2,\ 3\pi/4,\ \pi$. The short-wavelength modes are strongly damped — the numerical diffusion of the upwind scheme.

Worked Example 22.42 — Stability of the heat-equation schemes compared

From Section 22.10, with $s = \sin^2(\theta/2) \in [0, 1]$:

Scheme	$G(\theta)$	Condition	Order
FTCS (explicit)	$1 - 4rs$	$r \le \frac{1}{2}$	$(1, 2)$
BTCS (implicit)	$\dfrac{1}{1 + 4rs}$	none (unconditional)	$(1, 2)$
Crank–Nicolson	$\dfrac{1 - 2rs}{1 + 2rs}$	none (unconditional)	$(2, 2)$

The exact amplification of mode θ over one step is $e^{-\alpha k^2 \Delta t} = e^{-r\theta^2}$; all three schemes match it for small θ (well-resolved modes), and differ only in how they treat grid-scale modes. Only FTCS can *amplify* them.



FTCS amplification factor $G(\theta) = 1 - 4r\sin^2(\theta/2)$ versus $\theta \in [0, \pi]$ for $r = 0.25$ (blue), 0.5 (green, just touching -1 at $\theta = \pi$) and 0.6 (red, dropping below -1: unstable), with Crank–Nicolson at $r = 5$ (purple dashed), which stays inside $[-1, 1]$ but approaches -1 for grid-scale modes.

Stability for ODE methods: zero-stability and absolute stability

For ODE solvers two different stability notions are needed, and it is important not to confuse them.

Zero-stability concerns $h \to 0$ on a *fixed* interval and is the ODE counterpart of Lax–Richtmyer stability. For a linear multistep method $\sum_{k=0}^{s} \alpha_k y_{n+k} = h\sum_{k=0}^{s} \beta_k f_{n+k}$, it is governed by the **first characteristic polynomial** $\rho(\zeta) = \sum \alpha_k \zeta^k$.

Dahlquist equivalence theorem

A linear multistep method is convergent if and only if it is **consistent** ($\rho(1) = 0$, $\rho'(1) = \sum \beta_k$) and **zero-stable**: all roots of $\rho(\zeta)$ satisfy $|\zeta| \le 1$, and roots on the unit circle are simple (the **root condition**). All one-step methods (Euler, RK) are automatically zero-stable.

Worked Example 22.43 — A high-order method that does not converge

The explicit two-step method

$$y_{n+2} + 4y_{n+1} - 5y_n = h\,(4f_{n+1} + 2f_n)$$

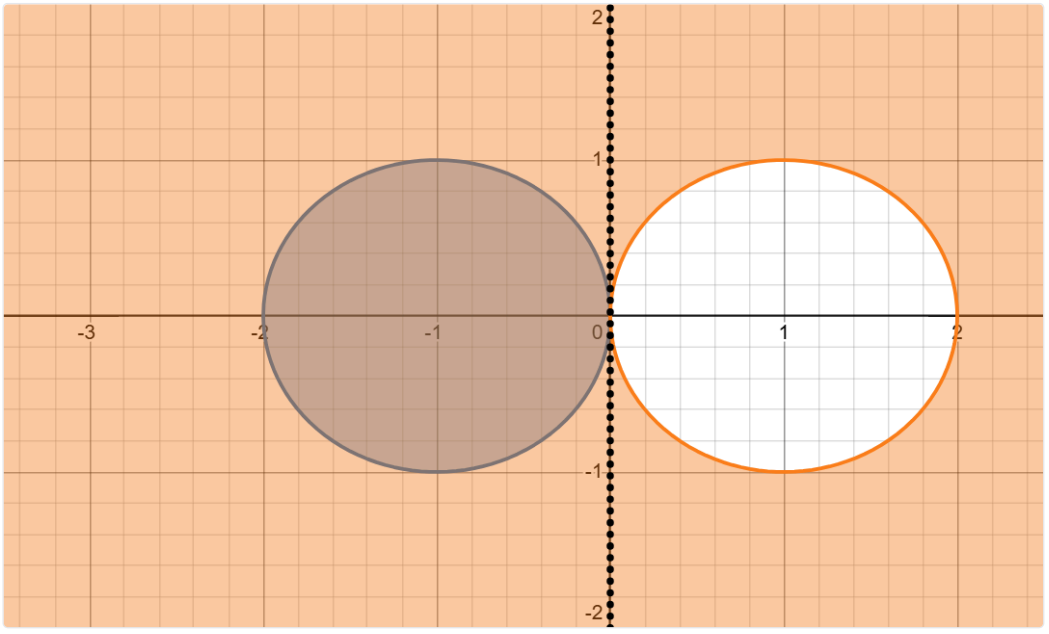
has the highest possible order (3) for its form — its truncation error is tiny. But $\rho(\zeta) = \zeta^2 + 4\zeta - 5 = (\zeta - 1)(\zeta + 5)$ has the root -5: the root condition fails.

Experiment on $y' = -y$, $y(0) = 1$, with exact starting values, integrating to $t = 1$ (exact $e^{-1} = 0.368$):

h	**0.1**	**0.05**	**0.025**
y_N	-6.68	-4.65×10^{6}	-2.88×10^{19}

The results get *worse* as h decreases: the parasitic solution $(-5)^n$ is excited by rounding and truncation errors and grows faster when there are more steps. Consistency with high order, yet no convergence — the Dahlquist (and Lax) theorems in action.

Absolute stability (Section 22.9) concerns a *fixed* step $h > 0$ applied to a decaying problem, and requires $h\lambda$ to lie in the stability region for every eigenvalue λ of the Jacobian. A method can be zero-stable and convergent but useless at practical step sizes on stiff problems because its absolute stability region is small.



Absolute stability regions in the complex $z = h\lambda$ plane: explicit Euler, $|1 + z| \le 1$ (blue disk centred at -1); implicit Euler, $|1 - z| \ge 1$ (orange, everything outside the disk centred at $+1$), which contains the whole left half-plane — A-stability. The imaginary axis (dotted) is where undamped oscillators live, outside the explicit Euler region.

Dissipation, dispersion and the modified equation

Stability guarantees that errors stay bounded; it says nothing about their *character*. Writing $G = |G|e^{-j\omega_{\text{num}}\Delta t}$ for a wave-like problem and comparing with the exact factor $e^{-j\omega\Delta t}$:

- $|G| < 1$ where the exact mode is undamped is **numerical dissipation** (amplitude error): sharp features are smeared;
- $\omega_{\text{num}} \neq \omega$ is **numerical dispersion** (phase error): different wavelengths travel at wrong and different speeds, producing trailing or leading wiggles.

The **modified equation** makes these effects quantitative: it is the PDE that the scheme solves *more accurately* than the original one, obtained by Taylor-expanding the scheme and eliminating time derivatives. For the upwind scheme,

$$u_t + a u_x = \frac{a\Delta x}{2}(1 - C)\,u_{xx} + O(\Delta x^2),$$

so upwinding is, to leading order, the advection equation plus an artificial diffusion with coefficient $\frac{a\Delta x}{2}(1 - C)$ — positive (hence stable) exactly when $0 \le C \le 1$, and zero at $C = 1$, the exact "magic" step. For leapfrog and Lax–Wendroff the leading error term is u_{xxx}, a dispersive term, which explains their oscillations near fronts. In FDTD simulations numerical dispersion makes waves travel slightly slower than c, with an error depending on direction and on the number of cells per wavelength — the reason for the rule of thumb of at least 10–20 cells per wavelength.

Verifying convergence in practice

The theorems tell us what *should* happen; verification checks that our code does it.

1. **Grid-refinement study.** Run on grids $h, h/2, h/4$; with a known exact solution, compute the observed order $p \approx \log_2(E_h/E_{h/2})$ (Section 22.2). Without one, use three grids: $p \approx \log_2 \dfrac{u_h - u_{h/2}}{u_{h/2} - u_{h/4}}$.
2. **Method of manufactured solutions.** Pick any smooth $u_{\text{m}}(x,t)$, compute the source term $f = (u_{\text{m}})_t - \mathcal{L}u_{\text{m}}$ symbolically, add f to the code, and check that the numerical solution converges to u_{m} at the design order. This tests every term, including boundary conditions.

3. **Richardson extrapolation** of the finest results to estimate the remaining discretisation error (as in Sections 22.7–22.8).
4. **Conservation checks.** Total charge, energy or mass that the PDE conserves should be conserved by the scheme to the expected accuracy.

> **Engineering Note — Reading a simulator's warnings**
>
> "Time step too small", "timestep reduced below minimum", "Courant number exceeded", "solution diverging" — these messages are the stability theory of this section surfacing in commercial tools. An explicit electromagnetic or thermal solver chooses its time step from the CFL or $r \le \frac{1}{2}$ condition applied to the *smallest* cell; an implicit solver chooses it for accuracy but may fail when its Newton iteration does not converge. Before trusting a simulation, perform at least one refinement study: halve the mesh size (and time step) and check that the quantity you care about changes by roughly the amount the method's order predicts.

Summary

Topic	Key formula or result	Notes
Floating point	$x = \pm(1.f)_2 \times 2^e$; $\mathrm{fl}(a \circ b) = (a \circ b)(1+\delta)$, $\lvert\delta\rvert \le u$	double: $u = 2^{-53} \approx 1.1 \times 10^{-16}$, $\varepsilon_M = 2^{-52}$
Cancellation	relative error of $a-b$ amplified by $\lvert a\rvert/\lvert a-b\rvert$	rewrite formulas (e.g. $1-\cos x = 2\sin^2\frac{x}{2}$)
Condition number	$\kappa_f = \lvert x f'(x)/f(x)\rvert$; $\kappa(\mathbf{A}) = \lVert\mathbf{A}\rVert\lVert\mathbf{A}^{-1}\rVert$	lose $\approx \log_{10}\kappa$ digits
Stability (algorithm)	forward error $\lesssim \kappa\times$ backward error	backward stable $\Rightarrow$ error $\approx \kappa u$
Observed order	$p \approx \ln(E_1/E_2)/\ln(h_1/h_2)$	the key debugging test
Bisection	$\lvert m_k - x^*\rvert \le (b-a)/2^{k+1}$	guaranteed, 1 bit per step
Fixed point	$x_{k+1} = g(x_k)$; converges if $\lvert g'\rvert \le L < 1$; $e_{k+1} \approx g'(x^*)e_k$	linear unless $g'(x^*) = 0$
Newton–Raphson	$x_{k+1} = x_k - f(x_k)/f'(x_k)$; $e_{k+1} \approx \frac{f''}{2f'}e_k^2$	quadratic; local; linear at multiple roots
Newton (systems)	$\mathbf{J}(\mathbf{x}_k)\Delta\mathbf{x} = -\mathbf{F}(\mathbf{x}_k)$	solve, never invert
Secant	$x_{k+1} = x_k - f_k\frac{x_k - x_{k-1}}{f_k - f_{k-1}}$	order $\varphi = 1.618$
Lagrange interpolation	$p_n = \sum y_i\ell_i$, $\ell_i = \prod_{j\ne i}\frac{x-x_j}{x_i-x_j}$	unique, degree $\le n$
Interpolation error	$f - p_n = \frac{f^{(n+1)}(\xi)}{(n+1)!}\prod(x-x_i)$	Runge phenomenon for equispaced nodes
Chebyshev nodes	$x_i = \cos\frac{(2i+1)\pi}{2(n+1)}$; $\max\lvert\omega_{n+1}\rvert = 2^{-n}$	near-optimal; $\Lambda_n \sim \frac{2}{\pi}\ln n$
Cubic spline	tridiagonal system for $M_i = S''(x_i)$; error $\frac{5}{384}h^4\lVert f^{(4)}\rVert$	minimum curvature
Least squares	$\mathbf{A}^{\mathsf{T}}\mathbf{A}\mathbf{a} = \mathbf{A}^{\mathsf{T}}\mathbf{y}$ (solve by QR)	orthogonal polynomials decouple coefficients
Minimax	error equioscillates at $n+2$ points	Remez; equiripple filters
Forward difference	$\frac{f(x+h)-f(x)}{h} = f' + \frac{h}{2}f''$	$h_{\text{opt}} \sim \sqrt{u}$
Central difference	$\frac{f(x+h)-f(x-h)}{2h} = f' + \frac{h^2}{6}f'''$	$h_{\text{opt}} \sim u^{1/3}$
Second difference	$\frac{f_1 - 2f_0 + f_{-1}}{h^2} = f'' + \frac{h^2}{12}f^{(4)}$	
Richardson	$R = \frac{2^p D(h/2) - D(h)}{2^p - 1}$	also gives error estimate
Composite trapezoid	$T_n = h[\frac{1}{2}f_0 + f_1 + \cdots + \frac{1}{2}f_n]$, $E = -\frac{(b-a)h^2}{12}f''$	spectral for periodic f
Composite Simpson	$S_n = \frac{h}{3}[f_0 + 4f_1 + 2f_2 + \cdots + f_n]$, $E = -\frac{(b-a)h^4}{180}f^{(4)}$	exact for cubics
Gauss–Legendre	n nodes = zeros of P_n; exact for degree $2n-1$	2-pt: $\pm 1/\sqrt{3}$, weights 1
Euler	$\mathbf{y}_{n+1} = \mathbf{y}_n + h\mathbf{f}(t_n, \mathbf{y}_n)$	order 1; stable for $\lvert 1 + h\lambda\rvert \le 1$
RK4	$\mathbf{y}_{n+1} = \mathbf{y}_n + \frac{h}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4)$	order 4; real stability limit $h\lambda \ge -2.785$
Implicit Euler / trapezoid	$R(z) = \frac{1}{1-z}$, $\frac{1+z/2}{1-z/2}$	A-stable; for stiff problems
Heat equation FTCS	$u_i^{n+1} = u_i^n + r\,\delta_x^2 u_i^n$, $r = \frac{\alpha\Delta t}{\Delta x^2} \le \frac{1}{2}$	BTCS, Crank–Nicolson unconditionally stable
Wave equation leapfrog	$u_i^{n+1} = 2u_i^n - u_i^{n-1} + C^2\delta_x^2 u_i^n$, $C = \frac{c\Delta t}{\Delta x} \le 1$	CFL condition
Laplace 5-point	$V_{i,j} = \frac{1}{4}(V_{i+1,j} + V_{i-1,j} + V_{i,j+1} + V_{i,j-1})$	Jacobi, Gauss–Seidel, SOR $\omega_{\text{opt}} = \frac{2}{1+\sin(\pi/N)}$
Lax equivalence	consistent + stable $\iff$ convergent (linear, well-posed)	order of convergence = order of consistency
Von Neumann	$u_i^n = G^n e^{ji\theta}$; need $\lvert G(\theta)\rvert \le 1 + K\Delta t$	Fourier modes diagonalise the scheme
Dahlquist	consistent + root condition $\iff$ convergent (multistep)	A-stable LMM has order ≤ 2

Further reading

- R. L. Burden, J. D. Faires and A. M. Burden, *Numerical Analysis*, 10th ed., Cengage, 2016 — a thorough, example-driven introduction to every topic of this chapter.
- N. J. Higham, *Accuracy and Stability of Numerical Algorithms*, 2nd ed., SIAM, 2002 — the definitive reference on floating-point arithmetic, conditioning and backward error analysis.
- L. N. Trefethen, *Approximation Theory and Approximation Practice*, extended ed., SIAM, 2019 — Chebyshev interpolation, the Runge phenomenon and Gaussian quadrature from a modern viewpoint.

- R. J. LeVeque, *Finite Difference Methods for Ordinary and Partial Differential Equations*, SIAM, 2007 — stability regions, stiffness, the CFL condition, von Neumann analysis and the Lax equivalence theorem.
- E. Hairer, S. P. Nørsett and G. Wanner, *Solving Ordinary Differential Equations I: Nonstiff Problems*, and E. Hairer and G. Wanner, *II: Stiff and Differential-Algebraic Problems*, Springer, 1993/1996 — the standard references on Runge–Kutta and multistep methods.

Chapter 23 — Optimization

> **Prerequisites:** Ch. 4 (single-variable extrema, §4.10), Ch. 6 (gradient, Hessian, constrained extrema and Lagrange multipliers, §6.10–6.11), Ch. 8–Ch. 10 (linear systems, least squares, positive-definite matrices, SVD), Ch. 22 (Newton's method for roots, convergence order). **Learning outcomes:** After this chapter you can
>
> - turn an engineering design question into a precise optimization problem: decision variables, objective, constraints, and scaling;
> - solve small linear programs graphically and by the simplex method, and read the dual variables as shadow prices;
> - state and apply the first- and second-order optimality conditions, including the Karush–Kuhn–Tucker (KKT) conditions for inequality constraints;
> - run and analyse gradient descent, Newton's method and the BFGS quasi-Newton method, choosing step sizes by line search;
> - recognise convex sets, functions and problems, and explain why convexity turns "a local minimum" into "the global minimum";
> - solve linear and nonlinear least-squares problems, with and without ridge (Tikhonov) regularization, and handle trade-offs between competing objectives with Pareto fronts, weighted sums and the ε-constraint method.

Optimization is the mathematics of *choosing well*. Every design an engineer signs off is the answer to an implicit optimization problem: the resistor values that meet a cutoff frequency at least cost, the generator outputs that meet a load at least fuel, the filter taps that best approximate an ideal response, the parameters of a model that best explain measured data. Chapter 4 handled the one-variable case with "set the derivative to zero", and Chapter 6 extended that to several variables and to equality constraints with Lagrange multipliers. This chapter takes the subject to the point where it becomes a working tool: many variables, inequality constraints, problems with no closed-form solution, and iterative algorithms that a computer runs for you — together with the theory that tells you whether the number the computer returns can be trusted.

Throughout, the standard problem is written

$$\begin{aligned} \min_{\mathbf{x}\in\mathbb{R}^n} \quad & f(\mathbf{x}) \\ \text{subject to} \quad & g_i(\mathbf{x}) \le 0, \quad i = 1,\dots,m, \\ & h_k(\mathbf{x}) = 0, \quad k = 1,\dots,p. \end{aligned}$$

Maximization of F is minimization of $f = -F$, and a constraint $g(\mathbf{x}) \ge 0$ is written $-g(\mathbf{x}) \le 0$, so this single template covers everything.

23.1 Objective functions

Motivation

The hardest step in optimization is usually not the algorithm; it is deciding *what you are optimizing*. "Design a good low-pass filter" is not a mathematical problem. "Choose R and C to minimise board area subject to a cutoff of $1\ \text{kHz} \pm 1\,\%$ and an output noise below $0.1\ \mu\text{V}$" is. The objective function encodes what "good" means, and a poorly chosen objective produces a design that is optimal for the wrong question — a perfect answer to a misstatement. This section sets up the vocabulary and shows how engineering goals become functions.

Definitions

Decision variables, objective, optimum

The **decision variables** $\mathbf{x} = (x_1, \dots, x_n)$ are the quantities the designer is free to choose. The **objective function** (cost, loss, or merit function) $f : D \subseteq \mathbb{R}^n \to \mathbb{R}$ assigns a single real number to each candidate design. A point $\mathbf{x}^*$ is a

- **global minimiser** if $f(\mathbf{x}^*) \le f(\mathbf{x})$ for every feasible $\mathbf{x}$;
- **local minimiser** if there is a $\delta > 0$ with $f(\mathbf{x}^*) \le f(\mathbf{x})$ for all feasible $\mathbf{x}$ with $\|\mathbf{x} - \mathbf{x}^*\| < \delta$;
- **strict** (local or global) minimiser if the inequality is strict for $\mathbf{x} \ne \mathbf{x}^*$.

The value $f^* = f(\mathbf{x}^*)$ is the **optimal value**. If the infimum of f is not attained (e.g. $f(x) = e^{-x}$ on $\mathbb{R}$) the problem has an optimal value $f^* = 0$ but no minimiser.

The distinction between local and global is the central difficulty of nonlinear optimization. Every practical algorithm in this chapter is *local*: it looks at f and its derivatives near the current point and moves downhill. Whether the point where it stops is the global answer depends on the global shape of f — which is why §23.7 on convexity matters so much.

Weierstrass extreme value theorem

If f is continuous on a nonempty, closed and bounded (compact) set $S \subset \mathbb{R}^n$, then f attains both a minimum and a maximum on S.

A useful variant for unbounded domains: if f is continuous on $\mathbb{R}^n$ and **coercive** — $f(\mathbf{x}) \to \infty$ whenever $\|\mathbf{x}\| \to \infty$ — then f has a global minimiser. (Proof sketch: pick any $\mathbf{x}_0$; coercivity means the sublevel set $\{\mathbf{x} : f(\mathbf{x}) \le f(\mathbf{x}_0)\}$ is bounded, it is closed by continuity, so Weierstrass applies to it, and nothing outside it can do better.) Quadratic losses $\|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2 + \lambda\|\mathbf{x}\|^2$ with $\lambda > 0$ are coercive; this is one quiet reason regularization (§23.8) is popular.

Kinds of objective

Objective type	Form	Typical EE instance
Linear	$\mathbf{c}^{\mathsf{T}}\mathbf{x}$	fuel cost of generators with linear cost curves; total component cost
Quadratic	$\tfrac{1}{2}\mathbf{x}^{\mathsf{T}}\mathbf{Q}\mathbf{x} + \mathbf{c}^{\mathsf{T}}\mathbf{x}$	ohmic loss $\sum R_k I_k^2$; energy $\tfrac{1}{2}CV^2$; squared error

Objective type	Form	Typical EE instance
Sum of squares	$\sum_i r_i(\mathbf{x})^2$	curve fitting, filter design in the L^2 sense
Minimax	$\max_i \lvert r_i(\mathbf{x})\rvert$	equiripple (Chebyshev) filter design; worst-case tolerance
Ratio / log	$\log(\text{SNR})$, $\eta = P_{\text{out}}/P_{\text{in}}$	efficiency, noise figure, capacity $\log_2(1+\text{SNR})$
Expected value	$E[\ell(\mathbf{x}, \boldsymbol{\xi})]$	mean-square error of an estimator with random noise $\boldsymbol{\xi}$

The **norm** you put on an error vector $\mathbf{r}$ is a modelling choice with real consequences. The L^2 norm $\sum r_i^2$ is smooth and leads to linear equations (§23.8), but a few large errors dominate it. The L^1 norm $\sum \lvert r_i\rvert$ is robust to outliers and can be written as a linear program. The L^∞ norm $\max \lvert r_i\rvert$ controls the worst case, which is what a specification mask demands; it, too, becomes a linear program after an epigraph trick (§23.2).

Scaling and units

An objective must be a single number, so all its terms must share units. When the natural terms do not — dollars of capital and kilowatt-hours of loss, ohms and farads — you convert them to a common currency (cost per year) or nondimensionalise them by reference values. This is not cosmetic: gradient-based algorithms (§23.5) move in the direction of steepest descent *as measured in the units you chose*, and a variable measured in farads (10^{-9}) next to one measured in ohms (10^{4}) gives a problem that is numerically ill-conditioned by a factor of 10^{13}. Rescale every variable so that its plausible range is of order one before handing it to software.

Worked Example 23.1 — maximum power transfer as an optimization problem

A source with open-circuit voltage V_s and internal resistance R_s drives a load R_L. Choose $R_L > 0$ to maximise the power delivered.

Step 1 (variables and objective): the decision variable is R_L; the load current is $I = V_s/(R_s + R_L)$, so

$$P(R_L) = I^2 R_L = \frac{V_s^2 R_L}{(R_s + R_L)^2}.$$

Units: $\mathrm{V}^2\,\Omega/\Omega^2 = \mathrm{V}^2/\Omega = \mathrm{W}$. ✓

Step 2 (stationary point): by the quotient rule

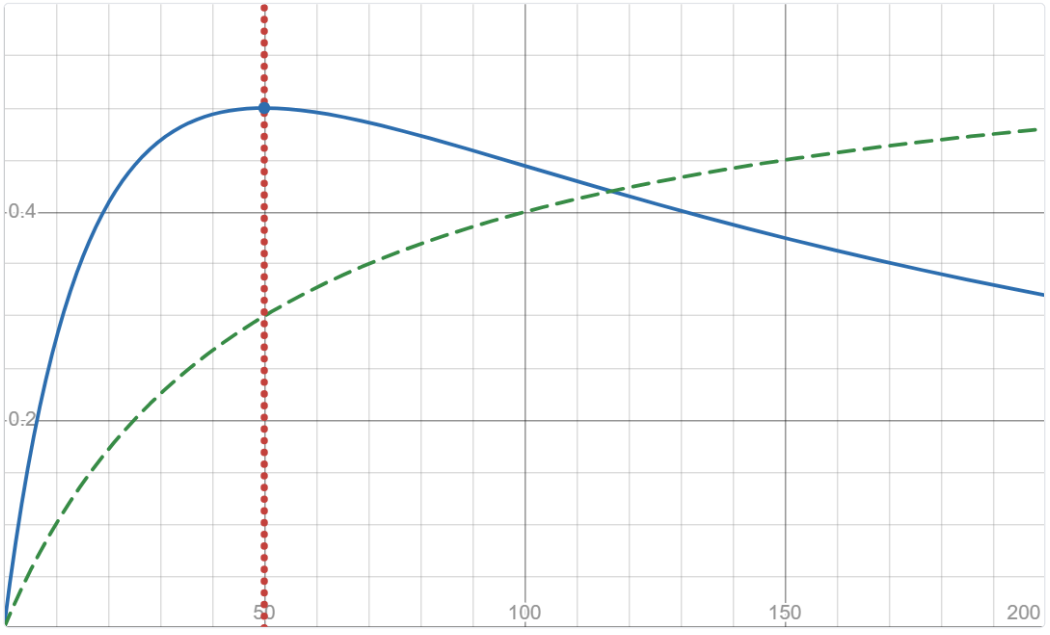
$$P'(R_L) = V_s^2 \frac{(R_s + R_L)^2 - 2R_L(R_s + R_L)}{(R_s + R_L)^4} = V_s^2 \frac{R_s - R_L}{(R_s + R_L)^3},$$

which vanishes only at $R_L = R_s$.

Step 3 (classify): $P' > 0$ for $R_L < R_s$ and $P' < 0$ for $R_L > R_s$, so $R_L = R_s$ is the global maximiser on $(0, \infty)$, with $P_{\max} = V_s^2/(4R_s)$.

Step 4 (check): $V_s = 10$ V, $R_s = 50\ \Omega$: $P(50) = 100 \cdot 50/100^2 = 0.5$ W $= V_s^2/(4R_s)$ ✓; $P(40) = 4000/8100 = 0.494$ W and $P(60) = 6000/12100 = 0.496$ W, both smaller ✓.

Step 5 (question the objective): at $R_L = R_s$ the efficiency is only $50\,\%$. A power utility, whose objective is efficiency, chooses $R_L \gg R_s$. Same circuit, different objective, opposite design.



Load power $P(R_L)$ (blue, watts) for $V_s = 10$ V, $R_s = 50\ \Omega$; the dashed green curve is efficiency $R_L/(R_s + R_L)$ scaled by 0.6. The two objectives peak in different places.

Worked Example 23.2 — Kelvin's law for conductor sizing

A feeder carries a steady current I for T hours per year. A conductor of cross-section A has annualised capital cost $c_1 A$ (cost per unit area per year) and resistance $\rho \ell / A$, so its yearly loss cost is $c_2 I^2 \rho \ell T / A$ with c_2 the price of energy. Find the economic cross-section.

Step 1 (objective): collect the constants $k = c_2 I^2 \rho \ell T$; the yearly cost is

$$f(A) = c_1 A + \frac{k}{A}, \qquad A > 0.$$

Step 2 (stationary point): $f'(A) = c_1 - k/A^2 = 0 \Rightarrow A^* = \sqrt{k/c_1}$.

Step 3 (classify): $f''(A) = 2k/A^3 > 0$ for all $A > 0$, so f is convex on its domain and A^* is the global minimiser. At the optimum $c_1 A^* = k/A^* = \sqrt{c_1 k}$: **the annual capital cost equals the annual loss cost** (Kelvin's law, 1881).

Step 4 (numbers): $I = 200$ A, $\ell = 1$ km, $\rho = 1.72 \times 10^{-8}\ \Omega\,\mathrm{m}$, $T = 8760$ h, energy 0.10 per kWh, capital $c_1 = 0.02$ per mm^2 per metre per year, i.e. 20 per mm^2 per year for the whole kilometre. Then the loss cost is

$$\frac{k}{A} = \frac{(200)^2 (1.72 \times 10^{-8})(1000)}{A\,[\mathrm{m}^2]}\ \mathrm{W} \times 8760\ \mathrm{h} \times \frac{0.10}{1000\ \mathrm{Wh}}.$$

With A in mm^2 ($1\ \mathrm{mm}^2 = 10^{-6}\ \mathrm{m}^2$): resistive loss $= 688/A$ kW, energy $= 6.03 \times 10^6 / A$ kWh, cost $= 6.03 \times 10^5 / A$. So $k = 6.03 \times 10^5$ and $A^* = \sqrt{6.03 \times 10^5 / 20} = 174$ mm^2. Check: capital $20 \times 174 = 3.47 \times 10^3$, loss $6.03 \times 10^5 / 174 = 3.47 \times 10^3$ ✓. One would then pick the next standard size (185 mm^2); the cost curve is flat near the optimum (f rises only 0.2 %), so rounding costs almost nothing.

That last remark is general and worth stating: near a smooth minimum, $f(\mathbf{x}^* + \boldsymbol{\delta}) \approx f^* + \tfrac{1}{2}\boldsymbol{\delta}^{\mathsf{T}} \nabla^2 f(\mathbf{x}^*) \boldsymbol{\delta}$, so a *first-order* error in the design costs only a *second-order* penalty in the objective. Optima are forgiving; this is why engineers can round to preferred values.

Worked Example 23.3 — writing a filter-design objective

We want a linear-phase FIR filter whose amplitude response $A(\omega; \mathbf{h}) = h_0 + 2\sum_{k=1}^{M} h_k \cos k\omega$ approximates an ideal low-pass $D(\omega)$ (1 in the passband $[0, \omega_p]$, 0 in the stopband $[\omega_s, \pi]$). Formulate three candidate objectives.

Step 1 (variables): $\mathbf{h} = (h_0, \dots, h_M) \in \mathbb{R}^{M+1}$. A is *linear* in $\mathbf{h}$: $A(\omega; \mathbf{h}) = \mathbf{a}(\omega)^{\mathsf{T}} \mathbf{h}$ with $\mathbf{a}(\omega) = (1, 2\cos\omega, \dots, 2\cos M\omega)$.

Step 2 (grid): sample $\omega_1, \dots, \omega_N$ in the bands (not in the transition band, which is left free) and form the error $e_i(\mathbf{h}) = W(\omega_i)\,[\mathbf{a}(\omega_i)^{\mathsf{T}}\mathbf{h} - D(\omega_i)]$ with a weight W (larger in the stopband if attenuation matters more).

Step 3 (objectives):

- $f_2(\mathbf{h}) = \sum_i e_i^2$: least squares — a quadratic, solved by one linear system (§23.8);
- $f_\infty(\mathbf{h}) = \max_i |e_i|$: minimax — the equiripple design; non-smooth, but an LP (§23.2–23.3);
- $f_1(\mathbf{h}) = \sum_i |e_i|$: least absolute error — also an LP.

All three are **convex** in $\mathbf{h}$ (§23.7), so each has a single global optimum that any correct algorithm finds. The choice between them is an engineering judgement about what the specification sheet actually penalises.

Common Pitfall — optimizing a proxy

Objectives are often proxies: "minimise component count" for cost, "minimise mean-square error" for audible distortion. An optimizer exploits every weakness of the proxy. If the optimal design looks strange — a resistor of 0.3 mΩ, a filter with huge gain in the transition band — the objective or the constraints are incomplete, not the algorithm.

Engineering Note — sensitivity of the optimal value

Once you have $\mathbf{x}^*(p)$ for a parameter p (a price, a load, a tolerance), the **envelope theorem** gives the sensitivity cheaply: $\dfrac{df^*}{dp} = \dfrac{\partial f}{\partial p}\Big|_{\mathbf{x}^*}$ — the indirect effect through $\mathbf{x}^*$ vanishes because $\nabla_{\mathbf{x}} f = \mathbf{0}$ there. In Example 23.2, $f^* = 2\sqrt{c_1 k}$ and indeed $\partial f / \partial c_1 = A^* = \sqrt{k/c_1} = df^*/dc_1$. With constraints the same role is played by the Lagrange multipliers (§23.4).

23.2 Constraints

Motivation

Unconstrained optimization is the exception in engineering. Resistances are positive, currents are bounded by fuse ratings, generator outputs lie between a technical minimum and a nameplate maximum, and the specification sheet is a list of inequalities: gain at least 40 dB, ripple at most 0.5 dB, power at most 2 W. Constraints change the character of the problem: the optimum is very often *on the boundary* of what is allowed, where the derivative of the objective is not zero at all. Learning to describe constraints precisely — and to see which ones actually matter — is the core of practical optimization.

Definitions

Feasible set, active constraints

For the standard problem with inequality constraints $g_i(\mathbf{x}) \le 0$ and equality constraints $h_k(\mathbf{x}) = 0$, the **feasible set** is

$$S = \{\mathbf{x} \in \mathbb{R}^n : \; g_i(\mathbf{x}) \le 0 \;\; (i = 1, \dots, m), \;\; h_k(\mathbf{x}) = 0 \;\; (k = 1, \dots, p)\}.$$

A point in S is **feasible**. At a feasible $\mathbf{x}$ the inequality g_i is **active** (binding, tight) if $g_i(\mathbf{x}) = 0$ and **inactive** if $g_i(\mathbf{x}) < 0$. The **active set** is $\mathcal{A}(\mathbf{x}) = \{i : g_i(\mathbf{x}) = 0\}$. Equality constraints are always active. If $S = \varnothing$ the problem is **infeasible**; if f is unbounded below on S it is **unbounded**.

Inactive constraints have no local influence: if $g_i(\mathbf{x}^*) < 0$, a small neighbourhood of $\mathbf{x}^*$ satisfies $g_i < 0$ too, so $\mathbf{x}^*$ would still be a local minimiser with g_i deleted. The difficulty of inequality-constrained optimization is that you do not know in advance *which* constraints will be active; the KKT conditions of §23.4 are, in effect, a systematic way to guess and verify the active set.

Common constraint types:

Type	Form	Example
Bound (box)	$\ell_j \le x_j \le u_j$	$P_{\min} \le P_g \le P_{\max}$ for a generator
Linear inequality	$\mathbf{a}_i^{\mathsf{T}}\mathbf{x} \le b_i$	total current from a supply $\le$ rating
Linear equality	$\mathbf{A}\mathbf{x} = \mathbf{b}$	Kirchhoff's current law at every node; power balance
Nonlinear	$g(\mathbf{x}) \le 0$	$RC = \tau$; $\lvert H(j\omega_s)\rvert \le 10^{-2}$; $\sqrt{kT/C} \le v_{\max}$
Integer / discrete	$x_j \in \mathbb{Z}$ or $x_j \in \{\text{E12 values}\}$	number of parallel cells; preferred resistor values

Discrete constraints make a problem combinatorial and are outside the scope of this chapter; the standard engineering practice is to solve the continuous relaxation and then round to the nearest available values, checking the specification afterwards (the forgiving-optimum argument of §23.1 says the penalty is second order).

Handling constraints: four techniques

1. Elimination. An equality constraint that can be solved for one variable removes that variable. For $\min f(x, y)$ subject to $h(x, y) = 0$ with $y = \varphi(x)$, minimise $F(x) = f(x, \varphi(x))$. This is the cleanest method when it applies, and it is exactly what Chapter 6 did before introducing multipliers (§6.10).

2. Change of variables. Simple inequality constraints can often be removed by a substitution: $x > 0$ becomes $x = e^u$ with $u \in \mathbb{R}$; $0 \le x \le 1$ becomes $x = \sin^2 u$ or $x = 1/(1 + e^{-u})$. The resulting problem is unconstrained. For positive physical quantities (resistances, capacitances, frequencies) the logarithmic substitution is particularly natural because tolerances and preferred-value series are logarithmic anyway; in §23.10 it turns a component-sizing problem into a convex one.

3. Slack variables. An inequality $g_i(\mathbf{x}) \le 0$ becomes an equality $g_i(\mathbf{x}) + s_i = 0$ with a new variable $s_i \ge 0$ (the **slack**). Linear programming uses this to put all constraints in equality form (§23.3). The slack is zero exactly when the constraint is active.

4. Penalty and barrier functions. Replace the constrained problem by a sequence of unconstrained ones:

- an *exterior penalty* adds $\mu\big(\sum_k h_k(\mathbf{x})^2 + \sum_i \max(0, g_i(\mathbf{x}))^2\big)$, with $\mu \to \infty$; iterates approach S from outside;
- an *interior (logarithmic) barrier* adds $-\frac{1}{t}\sum_i \ln(-g_i(\mathbf{x}))$, which is finite only strictly inside S; as $t \to \infty$ the minimiser approaches the constrained optimum from inside. Barrier methods with Newton steps are the **interior-point methods** that modern LP and convex solvers use.

The epigraph trick

A non-smooth objective such as $\max_i r_i(\mathbf{x})$ can be made smooth by adding one variable t:

$$\min_{\mathbf{x}} \; \max_i r_i(\mathbf{x}) \qquad \Longleftrightarrow \qquad \min_{\mathbf{x}, t} \; t \quad \text{subject to} \quad r_i(\mathbf{x}) \le t, \;\; i = 1, \dots, N.$$

At the optimum of the right-hand problem t is pushed down until it touches the largest r_i, so the two problems have the same solutions. With r_i linear in $\mathbf{x}$, minimax design becomes a linear program. Similarly $|r| \le t$ is the pair of linear inequalities $-t \le r \le t$, which turns L^1 and L^∞ fitting into LPs.

Worked Example 23.4 — a voltage divider: feasibility first

A divider R_1 (top), R_2 (bottom) from $V_{\text{in}} = 5$ V must give $V_{\text{out}} = 2$ V (ratio $\alpha = R_2/(R_1+R_2) = 0.4$), draw at most $I_{\max} = 100\ \mu$A, and have Thevenin output resistance $R_1 \parallel R_2 \le 10$ kΩ. Find the feasible designs.

Step 1 (eliminate the equality): with total resistance $R_T = R_1 + R_2$, the ratio gives $R_2 = \alpha R_T$, $R_1 = (1-\alpha)R_T$, so there is one free variable R_T.

Step 2 (rewrite the inequalities):

- current: $V_{\text{in}}/R_T \le I_{\max} \Rightarrow R_T \ge 5/10^{-4} = 50$ kΩ;
- output resistance: $R_1 R_2/R_T = \alpha(1-\alpha)R_T = 0.24 R_T \le 10$ kΩ $\Rightarrow R_T \le 41.7$ kΩ.

Step 3 (feasibility): $50 \le R_T \le 41.7$ is impossible: **the specification is infeasible.** No algorithm can fix that; the specification must change. (This is why solvers report "infeasible" as a legitimate outcome rather than an error.)

Step 4 (relax): allow $I_{\max} = 150\ \mu$A. Then $R_T \ge 33.3$ kΩ and the feasible set is the interval $[33.3,\ 41.7]$ kΩ.

Step 5 (optimise): to minimise quiescent power V_{in}^2/R_T, take the largest feasible $R_T = 41.7$ kΩ: $R_1 = 25.0$ kΩ, $R_2 = 16.7$ kΩ. The output-resistance constraint is **active**, the current constraint is **inactive**. Check: $5/41.7\text{k} = 120\ \mu\text{A} \le 150$ ✓; $25 \parallel 16.7 = 10.0$ kΩ ✓; ratio $16.7/41.7 = 0.40$ ✓.

Worked Example 23.5 — quadratic penalty converging to a constrained optimum

Minimise $f(x,y) = x^2 + y^2$ subject to $x + y = 1$ using the penalty function $P_\mu(x,y) = x^2 + y^2 + \mu(x+y-1)^2$.

Step 1 (exact answer, for reference): by elimination ($y = 1-x$), $F(x) = x^2 + (1-x)^2$, $F'(x) = 4x - 2 = 0$, so $(x^*, y^*) = (\tfrac12, \tfrac12)$, $f^* = \tfrac12$.

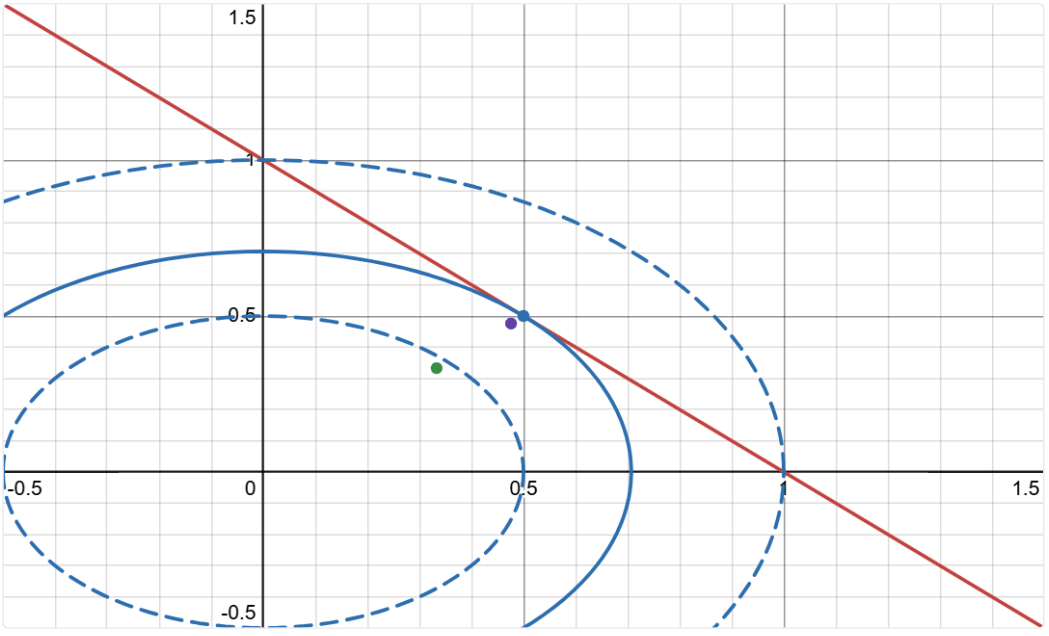
Step 2 (penalised problem): $\nabla P_\mu = \mathbf{0}$ gives $2x + 2\mu(x+y-1) = 0$ and $2y + 2\mu(x+y-1) = 0$. Subtracting, $x = y$; then $2x + 2\mu(2x-1) = 0$, so

$$x_\mu = y_\mu = \frac{\mu}{1+2\mu}.$$

Step 3 (behaviour): $\mu = 1$: $x_\mu = 0.333$; $\mu = 10$: 0.476; $\mu = 100$: 0.4975; $\mu = 10^4$: 0.499975. The constraint violation is $1 - 2x_\mu = 1/(1+2\mu)$, which goes to zero only as $\mu \to \infty$.

Step 4 (multiplier for free): the quantity $2\mu(x_\mu + y_\mu - 1) = -2\mu/(1+2\mu) \to -1$. Comparing with the Lagrange condition $\nabla f + \lambda \nabla h = \mathbf{0}$, i.e. $2x + \lambda = 0$ at $x = \tfrac12$, gives $\lambda^* = -1$: the penalty term's gradient converges to the multiplier term.

Step 5 (the catch): the Hessian of P_μ is $\begin{pmatrix} 2+2\mu & 2\mu \\ 2\mu & 2+2\mu \end{pmatrix}$ with eigenvalues 2 and $2+4\mu$; its condition number $1+2\mu$ grows without bound. Large penalties make the unconstrained subproblems ill-conditioned — the reason practical codes use *augmented Lagrangian* or barrier methods instead of a pure penalty.



Contours of $x^2 + y^2$ (blue) and the constraint line (red). The constrained optimum is where a contour is tangent to the line; penalty solutions approach it from the infeasible side as μ grows.

Worked Example 23.6 — minimax fitting as an LP

Fit a constant c to three measurements $1.0,\ 1.4,\ 2.0$ so that the worst-case error is as small as possible.

Step 1 (epigraph form): minimise t over (c, t) subject to $-t \le c - 1.0 \le t$, $-t \le c - 1.4 \le t$, $-t \le c - 2.0 \le t$ — six linear inequalities.

Step 2 (which constraints bind): only the extremes matter: $c - 1.0 \le t$ and $2.0 - c \le t$. Adding, $1.0 \le 2t$, so $t \ge 0.5$, with equality when $c - 1.0 = 2.0 - c$, i.e. $c = 1.5$.

Step 3 (compare): the minimax constant is the **midrange** 1.5 (worst error 0.5), the least-squares constant is the mean 1.467 (worst error 0.533), and the L^1 constant is the median 1.4 (worst error 0.6). Three norms, three answers; each is correct for its objective.

Common Pitfall — strict inequalities

Constraints in optimization are always non-strict ($\le$, not $<$). A feasible set defined by $x > 0$ is open, and $\min x$ over it has no minimiser. If a physical quantity must be strictly positive, impose $x \ge \varepsilon$ with a small physically meaningful ε (say $1\ \Omega$) or use the substitution $x = e^u$.

Engineering Note — constraint qualification in plain words

The optimality theory of §23.4 needs the active constraints to be "independent" at the optimum (their gradients linearly independent, or the feasible set having an interior). Redundant constraints — writing the same power-balance equation twice, or a bound that duplicates another — can break this and make solvers report spurious failures. Keep the constraint list minimal and non-redundant.

23.3 Linear programming

Motivation

When the objective and every constraint are linear, the problem is a **linear program** (LP). Linearity sounds restrictive, but LPs model a remarkable range of engineering decisions: dispatching generators with piecewise-linear cost curves, routing power or data flows through a network, allocating bandwidth, scheduling production, and — through the epigraph trick of §23.2 — minimax and L^1 fitting. LPs with millions of variables are solved routinely. More importantly for us, LP is the cleanest setting in which to see three ideas that run through all of optimization: the optimum lives on the boundary, an algorithm can walk along that boundary efficiently, and every problem has a **dual** whose variables are prices.

Standard and canonical forms

Linear program

A linear program in **inequality (canonical) form** is

$$\max_{\mathbf{x}} \ \mathbf{c}^{\mathsf{T}}\mathbf{x} \quad \text{subject to} \quad \mathbf{A}\mathbf{x} \le \mathbf{b},\ \ \mathbf{x} \ge \mathbf{0},$$

and in **standard (equality) form**

$$\min_{\mathbf{x}} \ \mathbf{c}^{\mathsf{T}}\mathbf{x} \quad \text{subject to} \quad \mathbf{A}\mathbf{x} = \mathbf{b},\ \ \mathbf{x} \ge \mathbf{0},$$

with $\mathbf{A} \in \mathbb{R}^{m \times n}$ and, in standard form, $\operatorname{rank} \mathbf{A} = m < n$. Vector inequalities hold componentwise.

Any LP can be brought to either form by these moves: maximise $\mathbf{c}^{\mathsf{T}}\mathbf{x} \leftrightarrow$ minimise $-\mathbf{c}^{\mathsf{T}}\mathbf{x}$; turn $\mathbf{a}^{\mathsf{T}}\mathbf{x} \le b$ into $\mathbf{a}^{\mathsf{T}}\mathbf{x} + s = b$ with slack $s \ge 0$ (or $\mathbf{a}^{\mathsf{T}}\mathbf{x} \ge b$ into $\mathbf{a}^{\mathsf{T}}\mathbf{x} - s = b$ with surplus $s \ge 0$); replace a free variable x_j by $x_j^+ - x_j^-$ with both parts nonnegative.

Geometry: the optimum is at a vertex

The feasible set of an LP, $S = \{\mathbf{x} : \mathbf{A}\mathbf{x} \le \mathbf{b},\ \mathbf{x} \ge \mathbf{0}\}$, is an intersection of half-spaces — a **convex polyhedron** (a polygon in two dimensions). Its "corners" are **vertices** (extreme points): points of S that are not the midpoint of any two other points of S. Algebraically, in standard form a vertex is a **basic feasible solution**: choose m linearly independent columns of $\mathbf{A}$ (the *basis* $\mathbf{B}$), set the other $n - m$ variables to zero, and solve $\mathbf{B}\mathbf{x}_B = \mathbf{b}$; if $\mathbf{x}_B \ge \mathbf{0}$ the point is a vertex.

Fundamental theorem of linear programming

For an LP in standard form:

1. if the feasible set is nonempty, it has at least one basic feasible solution (vertex);
2. if the LP has an optimal solution, it has an optimal solution at a vertex.

Consequently an LP is either infeasible, unbounded, or attains its optimum at one of finitely many vertices (at most $\binom{n}{m}$ of them).

Proof sketch of part 2. Let $\mathbf{x}^*$ be optimal but not a vertex. Then $\mathbf{x}^* = \tfrac{1}{2}(\mathbf{y} + \mathbf{z})$ for distinct feasible $\mathbf{y}, \mathbf{z}$, so $\mathbf{d} = \mathbf{y} - \mathbf{x}^*$ is a direction along which one can move both ways and stay feasible for a while. Since $\mathbf{c}^{\mathsf{T}}(\mathbf{x}^* \pm t\mathbf{d}) = \mathbf{c}^{\mathsf{T}}\mathbf{x}^* \pm t\,\mathbf{c}^{\mathsf{T}}\mathbf{d}$ cannot drop below the optimum in either direction, $\mathbf{c}^{\mathsf{T}}\mathbf{d} = 0$. Move along $\mathbf{d}$ until one more variable hits zero; the objective is unchanged and the new point has strictly more zero components. Repeating at most n times ends at a vertex with the same optimal value. $\square$

The intuition: the objective $\mathbf{c}^{\mathsf{T}}\mathbf{x} = z$ is a family of parallel lines (planes). Slide the line in the improving direction $\mathbf{c}$ until it is about to leave S; the last contact is a vertex, or a whole edge if the line happens to be parallel to it.

The graphical method

For two variables:

1. draw each constraint boundary and shade the feasible side;
2. identify the feasible polygon and its vertices (intersections of pairs of active boundaries);
3. draw one objective contour $\mathbf{c}^{\mathsf{T}}\mathbf{x} = z_0$ and the direction $\mathbf{c}$ of improvement;
4. slide the contour in that direction to the last vertex touched — or, equivalently, evaluate the objective at every vertex and take the best.

Worked Example 23.7 — a production LP solved graphically

A small plant assembles two power-supply modules. Module 1 earns 4 units of profit, module 2 earns 3. Per batch, module 1 uses 1 hour of test time, 2 hours of assembly and 1 kg of copper; module 2 uses 1 hour of test, 1 hour of assembly and 3 kg of copper. Available per week: 8 test hours, 12 assembly hours, 18 kg copper. Maximise profit.

Step 1 (formulate): with x_1, x_2 batches,

$$\max\ z = 4x_1 + 3x_2 \quad \text{s.t.} \quad x_1 + x_2 \le 8,\ \ 2x_1 + x_2 \le 12,\ \ x_1 + 3x_2 \le 18,\ \ x_1, x_2 \ge 0.$$

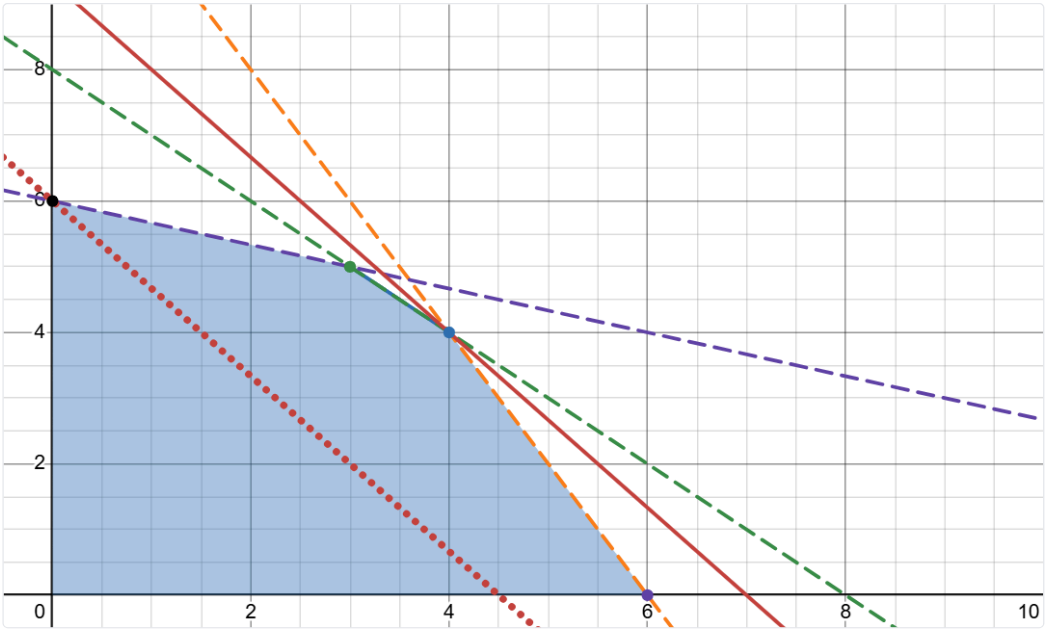
Step 2 (vertices): intersect pairs of boundaries and keep the feasible points:

Vertex	Active constraints	(x_1, x_2)	z
A	$x_1 = 0,\ x_2 = 0$	$(0, 0)$	0
B	$x_2 = 0$, assembly	$(6, 0)$	24
C	assembly, test	$(4, 4)$	**28**
D	test, copper	$(3, 5)$	27
E	copper, $x_1 = 0$	$(0, 6)$	18

(For C: subtract $x_1 + x_2 = 8$ from $2x_1 + x_2 = 12$ to get $x_1 = 4$, then $x_2 = 4$. For D: subtract $x_1 + x_2 = 8$ from $x_1 + 3x_2 = 18$ to get $x_2 = 5$, $x_1 = 3$.)

Step 3 (optimum): $x^* = (4, 4)$, $z^* = 28$. At C the test and assembly constraints are active; copper is inactive with slack $18 - (4 + 12) = 2$ kg.

Step 4 (contour check): the objective gradient $\mathbf{c} = (4, 3)$ has slope $-4/3$ for its contours. At C the two active edges have slopes -1 (test) and -2 (assembly); since $-2 < -4/3 < -1$, the contour $4x_1 + 3x_2 = 28$ touches the polygon only at C. ✓



Feasible region of Example 23.7 (blue) bounded by the test (green), assembly (orange) and copper (purple) constraints. The objective contour $z = 18$ (dotted) slides in the direction $(4, 3)$ until $z = 28$ (solid red) just touches vertex C.

Worked Example 23.8 — a minimisation LP, an unbounded region, and alternative optima

A utility must buy blocks from two contracts. A block from contract A supplies 1 MW of energy and 1 MW of reserve at cost 2; a block from B supplies 1 MW of energy and 3 MW of reserve at cost 3. It needs at least 4 MW of energy and 6 MW of reserve, and contract A is limited to 5 blocks.

Step 1 (formulate): $\min\ 2x + 3y$ s.t. $x + y \geq 4$, $x + 3y \geq 6$, $x \leq 5$, $x, y \geq 0$.

Step 2 (region): the feasible region is *unbounded* (any large y is allowed), but the objective increases in every feasible direction of escape, so a minimum exists. Its vertices are $(0, 4)$, $(3, 1)$ [from $x + y = 4$, $x + 3y = 6$], and $(5, \tfrac{1}{3})$ [from $x = 5$, $x + 3y = 6$].

Step 3 (evaluate): $2x + 3y$ gives 12, **9**, 11. Optimum $(3, 1)$ with cost 9: both the energy and reserve constraints are active.

Step 4 (alternative optima): change B's price so that the cost is $x + 3y$. Now the contours are parallel to the reserve edge $x + 3y = 6$; vertices give 12, 6, 6, and **every** point on the segment from $(3, 1)$ to $(5, \tfrac{1}{3})$ is optimal with cost 6. A solver will report one vertex; the existence of others is signalled by a zero reduced cost for a nonbasic variable in the final simplex tableau.

The simplex method

The fundamental theorem suggests an algorithm: start at a vertex, move along an edge to an adjacent vertex with a better objective, and stop when no edge improves. This is Dantzig's **simplex method** (1947). Adjacent vertices differ by exchanging one basic variable for one nonbasic variable, which is a single Gauss–Jordan pivot (Ch. 8).

Consider the canonical maximisation problem with $\mathbf{b} \geq \mathbf{0}$. Adding slacks $\mathbf{s}$ gives $\mathbf{A}\mathbf{x} + \mathbf{s} = \mathbf{b}$, and the point $\mathbf{x} = \mathbf{0},\ \mathbf{s} = \mathbf{b}$ is an immediate basic feasible solution (the slacks form the basis). Write the objective as a row $z - \mathbf{c}^{\mathsf{T}}\mathbf{x} = 0$. The **tableau** stores the constraint rows and this objective row.

Simplex iteration (maximisation, tableau form)

1. **Optimality test.** If every entry of the objective row (the **reduced costs**, here $-\bar{c}_j$) is ≥ 0, stop: the current vertex is optimal.
2. **Entering variable.** Otherwise choose a column q with a negative entry (Dantzig's rule: the most negative). Increasing x_q from zero increases z.
3. **Ratio test (leaving variable).** For each row i with positive entry $a_{iq} > 0$, compute b_i / a_{iq}. The smallest ratio identifies the row p whose basic variable first reaches zero as x_q grows. If no $a_{iq} > 0$, the LP is **unbounded**.
4. **Pivot.** Divide row p by a_{pq}, then subtract multiples of it from every other row (including the objective row) to zero column q. Return to step 1.

Why it works: in the current basis every basic variable is expressed in terms of the nonbasic ones; the objective row says how z changes per unit increase of each nonbasic variable. The ratio test increases the entering variable as far as possible *without making any basic variable negative*, so feasibility is preserved and z never decreases. With finitely many bases and strict improvement at each step (absent degeneracy), the method terminates.

Worked Example 23.9 — simplex on Example 23.7

Initial tableau (basis s_1, s_2, s_3; vertex A):

Basis	x_1	x_2	s_1	s_2	s_3	RHS
s_1	1	1	1	0	0	8
s_2	2	1	0	1	0	12
s_3	1	3	0	0	1	18
z	-4	-3	0	0	0	0

Iteration 1. Most negative reduced cost: x_1 (-4). Ratios: $8/1 = 8$, $12/2 = \mathbf{6}$, $18/1 = 18$; s_2 leaves. Pivot on the entry 2:

Basis	x_1	x_2	s_1	s_2	s_3	RHS
s_1	0	0.5	1	-0.5	0	2
x_1	1	0.5	0	0.5	0	6
s_3	0	2.5	0	-0.5	1	12
z	0	-1	0	2	0	24

This is vertex B $= (6, 0)$, $z = 24$.

Iteration 2. x_2 has reduced cost -1. Ratios: $2/0.5 = \mathbf{4}$, $6/0.5 = 12$, $12/2.5 = 4.8$; s_1 leaves. Pivot on 0.5:

Basis	x_1	x_2	s_1	s_2	s_3	RHS
x_2	0	1	2	-1	0	4
x_1	1	0	-1	1	0	4
s_3	0	0	-5	2	1	2
z	0	0	2	1	0	28

Optimality. All reduced costs are ≥ 0: optimum $x_1 = 4$, $x_2 = 4$, $s_3 = 2$ (spare copper), $z^* = 28$ — vertex C, as found graphically. The path was A → B → C: two pivots instead of examining all five vertices.

Bonus reading. The objective-row entries under the slack columns, $(2, 1, 0)$, are the **shadow prices** of test time, assembly time and copper — the dual solution, explained next.

Starting when the origin is infeasible. If some constraints are $\geq$ or equalities, $\mathbf{x} = \mathbf{0}$ is not feasible and there is no obvious initial basis. The **two-phase method** first minimises the sum of artificial variables added to those rows (phase I); a zero optimum gives a feasible vertex for phase II, a positive optimum proves the LP infeasible. The **big-M method** does both at once by charging each artificial variable a large cost M.

Degeneracy and cycling. If a ratio test ties, some basic variable becomes zero and the next pivot may not improve z. In rare constructed cases the method can then cycle forever; Bland's rule (always choose the lowest-indexed eligible variable) provably prevents this.

Complexity. The simplex method can take exponentially many pivots on contrived examples (Klee–Minty), but on practical problems it typically needs a small multiple of m pivots. **Interior-point methods** (Karmarkar 1984 and successors), which follow a smooth path through the interior using barrier functions (§23.2) and Newton steps (§23.6), have polynomial worst-case complexity and are preferred for very large sparse problems.

Duality

Every LP has a companion LP, its **dual**, obtained by asking: *what is the best upper bound on the primal objective that can be proved by combining the constraints?*

Take Example 23.7. Multiply the three constraints by nonnegative numbers y_1, y_2, y_3 and add:

$$(y_1 + 2y_2 + y_3)\,x_1 + (y_1 + y_2 + 3y_3)\,x_2 \;\leq\; 8y_1 + 12y_2 + 18y_3.$$

If the coefficients on the left are at least the profit coefficients, $y_1 + 2y_2 + y_3 \geq 4$ and $y_1 + y_2 + 3y_3 \geq 3$, then (because $x_1, x_2 \geq 0$) the left side is at least $4x_1 + 3x_2 = z$. So every such $\mathbf{y}$ gives the bound $z \leq 8y_1 + 12y_2 + 18y_3$. The tightest bound is the dual LP:

$$\min\; 8y_1 + 12y_2 + 18y_3 \quad \text{s.t.} \quad y_1 + 2y_2 + y_3 \geq 4,\;\; y_1 + y_2 + 3y_3 \geq 3,\;\; \mathbf{y} \geq \mathbf{0}.$$

Primal–dual pair

$$\text{(P)}\;\; \max\, \mathbf{c}^{\mathsf{T}}\mathbf{x} \;\text{ s.t. }\; \mathbf{A}\mathbf{x} \leq \mathbf{b},\; \mathbf{x} \geq \mathbf{0} \qquad\qquad \text{(D)}\;\; \min\, \mathbf{b}^{\mathsf{T}}\mathbf{y} \;\text{ s.t. }\; \mathbf{A}^{\mathsf{T}}\mathbf{y} \geq \mathbf{c},\; \mathbf{y} \geq \mathbf{0}.$$

Each primal constraint corresponds to a dual variable and each primal variable to a dual constraint. The dual of the dual is the primal.

LP duality

1. **Weak duality.** For any primal-feasible $\mathbf{x}$ and dual-feasible $\mathbf{y}$: $\mathbf{c}^{\mathsf{T}}\mathbf{x} \le \mathbf{b}^{\mathsf{T}}\mathbf{y}$.
2. **Strong duality.** If either problem has a finite optimum, so does the other, and $\mathbf{c}^{\mathsf{T}}\mathbf{x}^{*} = \mathbf{b}^{\mathsf{T}}\mathbf{y}^{*}$.
3. **Complementary slackness.** Feasible $\mathbf{x}$ and $\mathbf{y}$ are both optimal if and only if
$$y_i\,(b_i - \mathbf{a}_i^{\mathsf{T}}\mathbf{x}) = 0 \ \text{ for every } i, \qquad x_j\,\big((\mathbf{A}^{\mathsf{T}}\mathbf{y})_j - c_j\big) = 0 \ \text{ for every } j.$$
4. If the primal is unbounded the dual is infeasible, and vice versa (both can be infeasible).

Proof of weak duality and complementary slackness. For feasible $\mathbf{x}, \mathbf{y}$,
$$\mathbf{c}^{\mathsf{T}}\mathbf{x} \;\le\; (\mathbf{A}^{\mathsf{T}}\mathbf{y})^{\mathsf{T}}\mathbf{x} \;=\; \mathbf{y}^{\mathsf{T}}\mathbf{A}\mathbf{x} \;\le\; \mathbf{y}^{\mathsf{T}}\mathbf{b},$$
the first inequality using $\mathbf{x} \ge \mathbf{0}$ and $\mathbf{A}^{\mathsf{T}}\mathbf{y} \ge \mathbf{c}$, the second using $\mathbf{y} \ge \mathbf{0}$ and $\mathbf{A}\mathbf{x} \le \mathbf{b}$. Equality $\mathbf{c}^{\mathsf{T}}\mathbf{x} = \mathbf{b}^{\mathsf{T}}\mathbf{y}$ — which by weak duality certifies that both are optimal — holds exactly when both inequalities are equalities, i.e. when each product of a nonnegative variable and a nonnegative slack vanishes. Strong duality is deeper (it follows from Farkas' lemma, or from the fact that the simplex method terminates with a dual-feasible objective row); we state it without proof. □

Complementary slackness says: **a constraint with spare capacity has zero price, and a resource with positive price is fully used.**

Worked Example 23.10 — solving the dual by complementary slackness

Use the primal optimum $\mathbf{x}^{*} = (4, 4)$ of Example 23.7 to find the dual optimum.

Step 1 (inactive primal constraint): copper has slack 2, so $y_3 = 0$.

Step 2 (positive primal variables): $x_1, x_2 > 0$, so both dual constraints are tight: $y_1 + 2y_2 = 4$ and $y_1 + y_2 = 3$. Hence $y_2 = 1$, $y_1 = 2$.

Step 3 (verify): $\mathbf{y}^{*} = (2, 1, 0) \ge \mathbf{0}$ ✓ (dual feasible); dual objective $8(2) + 12(1) + 18(0) = 28 = z^{*}$ ✓. These are exactly the slack-column entries of the final simplex tableau.

Step 4 (interpret): one more test hour is worth 2 units of profit, one more assembly hour 1 unit, more copper nothing. Check with a re-solve: with 13 assembly hours the optimum moves to $(5, 3)$ and $z = 29 = 28 + 1$ ✓. With 8.4 test hours it moves to $(3.6, 4.8)$ and $z = 28.8 = 28 + 2(0.4)$ ✓.

Step 5 (range of validity): shadow prices are valid only while the optimal basis stays the same. Adding Δ test hours changes the basic variables by Δ times the s_1 column of the final tableau, $(2, -1, -5)$: copper slack becomes $2 - 5\Delta$, which hits zero at $\Delta = 0.4$. Beyond 8.4 test hours copper becomes binding and extra test time is worth nothing: with 8.5 hours the optimum is still 28.8.

Engineering Note — locational marginal prices

In an electricity market the operator solves an LP (or QP) that dispatches generators at minimum cost subject to power balance at each bus and line flow limits. The dual variable of the balance constraint at bus k is the **locational marginal price**: the cost of serving one more MW at that bus. When a transmission line is congested its limit constraint is active, its dual variable is positive, and prices on either side of the line differ — a direct, monetised reading of complementary slackness.

Common Pitfall — forgetting nonnegativity or sign conventions

Most LP software assumes $\mathbf{x} \ge \mathbf{0}$ by default (SciPy's `linprog` does). A variable that can be negative — a current direction, a voltage relative to ground — must be declared free, or the solver will silently clamp it at zero and return a wrong "optimum". Equally, solvers differ in the sign they report for dual values (marginals); check against a one-unit perturbation as in Example 23.10.

A minimal Python check of Examples 23.7 and 23.10 (`linprog` minimises, so the profit is negated):

```
from scipy.optimize import linprog
res = linprog(c=[-4, -3],
              A_ub=[[1, 1], [2, 1], [1, 3]], b_ub=[8, 12, 18],
              bounds=[(0, None), (0, None)])
print(res.x, -res.fun)          # [4. 4.] 28.0
print(-res.ineqlin.marginals)   # shadow prices [2. 1. 0.]
```

23.4 Nonlinear optimization

Motivation

Most engineering objectives are nonlinear: losses go as I^2R, gain as a ratio of polynomials in frequency, efficiency as a ratio of powers, model error as a sum of squared exponentials. Before choosing an algorithm we need to know what a solution *looks like* — which equations it satisfies and how to tell a minimum from a saddle. Those **optimality conditions** do double duty: for small problems they are solved by hand to give the answer, and for large ones they are the stopping test that every algorithm checks. This section collects them, first without constraints, then with equality constraints (Lagrange, already met in Ch. 6 §6.11), and finally with inequality constraints (Karush–Kuhn–Tucker).

Unconstrained optimality conditions

Let $f:\mathbb{R}^n\to\mathbb{R}$ be twice continuously differentiable, with gradient $\nabla f(\mathbf{x})$ and Hessian $\nabla^2 f(\mathbf{x})=\left[\partial^2 f/\partial x_i\partial x_j\right]$ (Ch. 6). Everything rests on the second-order Taylor expansion

$$f(\mathbf{x}^*+\mathbf{d})=f(\mathbf{x}^*)+\nabla f(\mathbf{x}^*)^{\mathsf T}\mathbf{d}+\tfrac12\,\mathbf{d}^{\mathsf T}\nabla^2 f(\mathbf{x}^*)\,\mathbf{d}+o(\|\mathbf{d}\|^2).$$

Optimality conditions without constraints

1. **First-order necessary condition (FONC).** If $\mathbf{x}^*$ is a local minimiser, then $\nabla f(\mathbf{x}^*)=\mathbf{0}$ ($\mathbf{x}^*$ is a **stationary point**).
2. **Second-order necessary condition (SONC).** If $\mathbf{x}^*$ is a local minimiser, then $\nabla^2 f(\mathbf{x}^*)$ is positive semidefinite.
3. **Second-order sufficient condition (SOSC).** If $\nabla f(\mathbf{x}^*)=\mathbf{0}$ and $\nabla^2 f(\mathbf{x}^*)$ is positive definite, then $\mathbf{x}^*$ is a strict local minimiser.

Proof. (1) If $\mathbf{g}=\nabla f(\mathbf{x}^*)\neq\mathbf{0}$, take $\mathbf{d}=-t\mathbf{g}$: $f(\mathbf{x}^*-t\mathbf{g})=f(\mathbf{x}^*)-t\|\mathbf{g}\|^2+O(t^2)<f(\mathbf{x}^*)$ for small $t>0$, contradicting minimality. (2) With $\nabla f=\mathbf{0}$, if some $\mathbf{d}$ had $\mathbf{d}^{\mathsf T}\nabla^2 f\,\mathbf{d}<0$, then $f(\mathbf{x}^*+t\mathbf{d})-f(\mathbf{x}^*)=\tfrac12 t^2\mathbf{d}^{\mathsf T}\nabla^2 f\,\mathbf{d}+o(t^2)<0$ for small t. (3) If $\nabla^2 f(\mathbf{x}^*)\succ 0$ with smallest eigenvalue $\lambda_{\min}>0$, then $\tfrac12\mathbf{d}^{\mathsf T}\nabla^2 f\,\mathbf{d}\geq\tfrac12\lambda_{\min}\|\mathbf{d}\|^2$, which beats the $o(\|\mathbf{d}\|^2)$ remainder for $\mathbf{d}$ small enough, so $f(\mathbf{x}^*+\mathbf{d})>f(\mathbf{x}^*)$ for small $\mathbf{d}\neq\mathbf{0}$. $\square$

Definiteness is checked by eigenvalues or, for small matrices, by leading principal minors (Ch. 10 §10.6). At a stationary point:

Hessian	Conclusion
positive definite (all $\lambda_i>0$)	strict local minimum
negative definite (all $\lambda_i<0$)	strict local maximum
indefinite (eigenvalues of both signs)	saddle point
semidefinite and singular	inconclusive — look at higher-order terms

Worked Example 23.11 — stationary points and their type

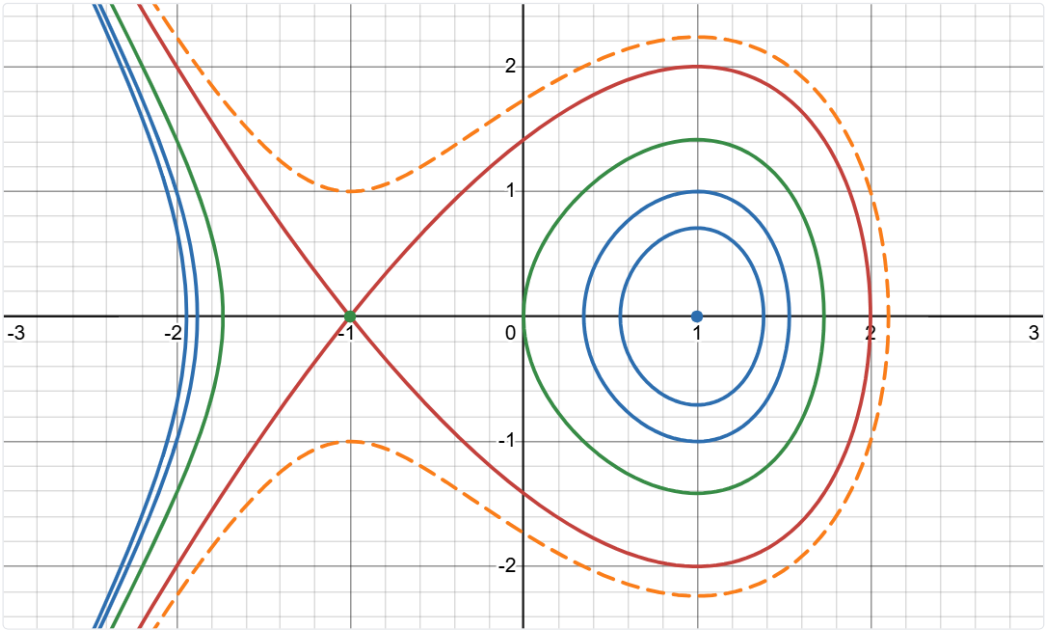
Classify the stationary points of $f(x,y)=x^3-3x+y^2$.

Step 1 (gradient): $\nabla f=(3x^2-3,\;2y)=\mathbf{0}$ gives $x=\pm1$, $y=0$.

Step 2 (Hessian): $\nabla^2 f=\begin{pmatrix}6x & 0\\ 0 & 2\end{pmatrix}$.

- At $(1,0)$: $\operatorname{diag}(6,2)\succ 0\to$ strict local minimum, $f=-2$.
- At $(-1,0)$: $\operatorname{diag}(-6,2)$ indefinite $\to$ saddle, $f=2$.

Step 3 (global?): along $y=0$, $f(x,0)=x^3-3x\to-\infty$ as $x\to-\infty$. The local minimum is **not** global; f is unbounded below. A descent algorithm started at $(-3,0)$ never finds $(1,0)$ — it runs off to $-\infty$.



Level curves of $f=x^3-3x+y^2$. Closed curves surround the local minimum at $(1,0)$; the red level $f=2$ crosses itself at the saddle $(-1,0)$, and to the left the function falls without bound.

Common Pitfall — a semidefinite Hessian proves nothing

$f_1=x^4+y^2$, $f_2=x^3+y^2$ and $f_3=-x^4+y^2$ all have $\nabla f=\mathbf{0}$ and Hessian $\operatorname{diag}(0,2)$ at the origin. The origin is a strict minimum of f_1, a degenerate saddle of f_2 and a saddle of f_3. When the Hessian is singular, the second-order test is silent; examine f along the null direction (here the x-axis).

Equality constraints: the Lagrange conditions

For $\min f(\mathbf{x})$ subject to $h_k(\mathbf{x}) = 0$ ($k = 1, \dots, p$), Chapter 6 showed that at a constrained extremum the gradient of f must be a linear combination of the constraint gradients, provided those gradients are linearly independent (Ch. 6 §6.11). With the **Lagrangian**

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) = f(\mathbf{x}) + \sum_{k=1}^{p} \lambda_k h_k(\mathbf{x}),$$

the conditions are $\nabla_{\mathbf{x}} \mathcal{L} = \mathbf{0}$ and $\nabla_{\boldsymbol{\lambda}} \mathcal{L} = \mathbf{h}(\mathbf{x}) = \mathbf{0}$: $n + p$ equations in $n + p$ unknowns. The geometric reason: any feasible direction $\mathbf{d}$ is tangent to every constraint surface, $\nabla h_k^{\mathsf{T}} \mathbf{d} = 0$, and at a minimum f cannot decrease along such directions, so $\nabla f \perp \mathbf{d}$ for all tangent $\mathbf{d}$ — that is, ∇f lies in the span of the ∇h_k. (Chapter 6 writes $\nabla f = \lambda \nabla h$; with our sign convention $\nabla f + \lambda \nabla h = \mathbf{0}$ the multiplier simply changes sign.)

The multiplier is a **sensitivity**: if the constraint is $h(\mathbf{x}) = b$ rather than 0, then $\dfrac{df^*}{db} = -\lambda^*$ in our convention. In Example 23.5, relaxing $x + y = 1$ to $x + y = b$ gives $f^* = b^2/2$ and $df^*/db = 1 = -\lambda^*$ at $b = 1$ ✓.

Inequality constraints: the KKT conditions

Now add inequalities $g_i(\mathbf{x}) \le 0$. At a candidate $\mathbf{x}^*$ only the active ones matter. For an active g_i, a direction $\mathbf{d}$ keeps us feasible to first order if $\nabla g_i^{\mathsf{T}} \mathbf{d} \le 0$ (we may move *into* the feasible side, not just along the boundary). At a minimum, no feasible direction may decrease f:

$$\nabla g_i(\mathbf{x}^*)^{\mathsf{T}} \mathbf{d} \le 0 \ \text{ for all active } i \quad \Longrightarrow \quad \nabla f(\mathbf{x}^*)^{\mathsf{T}} \mathbf{d} \ge 0.$$

By Farkas' lemma (the same theorem of alternatives that underlies LP duality), this implication holds exactly when $-\nabla f(\mathbf{x}^*)$ is a **nonnegative** combination of the active constraint gradients: $-\nabla f = \sum_{i \in \mathcal{A}} \mu_i \nabla g_i$ with $\mu_i \ge 0$. Geometrically: the downhill direction $-\nabla f$ points straight out of the feasible set, into the cone spanned by the outward normals of the active walls. If it pointed anywhere else, there would be a feasible downhill direction.

Setting $\mu_i = 0$ for inactive constraints and including equality constraints gives the central theorem of constrained optimization.

Karush–Kuhn–Tucker (KKT) conditions

Let $\mathbf{x}^*$ be a local minimiser of f subject to $g_i(\mathbf{x}) \le 0$ ($i = 1, \dots, m$) and $h_k(\mathbf{x}) = 0$ ($k = 1, \dots, p$), all continuously differentiable, and suppose a **constraint qualification** holds at $\mathbf{x}^*$ — for instance LICQ: the gradients of the active inequalities and of all equalities are linearly independent. Then there exist multipliers μ_i and λ_k such that

$$\begin{aligned}
&\text{stationarity:} && \nabla f(\mathbf{x}^*) + \sum_{i=1}^{m} \mu_i \nabla g_i(\mathbf{x}^*) + \sum_{k=1}^{p} \lambda_k \nabla h_k(\mathbf{x}^*) = \mathbf{0}, \\
&\text{primal feasibility:} && g_i(\mathbf{x}^*) \le 0, \quad h_k(\mathbf{x}^*) = 0, \\
&\text{dual feasibility:} && \mu_i \ge 0, \\
&\text{complementary slackness:} && \mu_i \, g_i(\mathbf{x}^*) = 0 \quad \text{for every } i.
\end{aligned}$$

If the problem is **convex** (f and all g_i convex, all h_k affine), the KKT conditions are also **sufficient**: any KKT point is a global minimiser (§23.7).

Remarks:

- **Complementary slackness** encodes the active-set logic: either a constraint is active ($g_i = 0$) or its multiplier is zero. It is the nonlinear version of the LP statement "spare capacity has zero price".
- **The sign of μ_i matters**, unlike λ_k. A negative multiplier on an active inequality says the objective would improve by moving *into* the interior, so the constraint should not be active — the guess of active set was wrong.
- **Sensitivity.** If $g_i(\mathbf{x}) \le b_i$, then $\partial f^*/\partial b_i = -\mu_i^*$: relaxing a constraint by one unit lowers the optimal cost by about μ_i^*. The multipliers are shadow prices, exactly as in LP; for an LP, the KKT conditions *are* primal feasibility, dual feasibility and complementary slackness.
- **Second-order conditions** for constrained problems test the Hessian of the Lagrangian, $\nabla^2_{\mathbf{xx}} \mathcal{L}$, only on directions tangent to the active constraints (the bordered-Hessian test of Ch. 6 is the two-variable special case).

A recipe for small problems

1. Write all constraints as $g_i \le 0$, $h_k = 0$ and form $\mathcal{L} = f + \sum \mu_i g_i + \sum \lambda_k h_k$.
2. Solve the unconstrained problem; if its minimiser is feasible and the problem is convex, you are done (all $\mu_i = 0$).
3. Otherwise, guess an active set $\mathcal{A}$. Solve stationarity together with $g_i = 0$ ($i \in \mathcal{A}$), $h_k = 0$, and $\mu_i = 0$ ($i \notin \mathcal{A}$).
4. Accept the guess if all inactive constraints are satisfied and all $\mu_i \ge 0$; otherwise try another active set. With m inequalities there are at most 2^m guesses — fine by hand for $m \le 3$; algorithms (active-set and interior-point methods) organise this search for large m.

Worked Example 23.12 — projection onto a half-plane

Minimise $f = (x - 2)^2 + (y - 1)^2$ subject to $g_1 = x + y - 2 \le 0$ and $g_2 = -x \le 0$.

Step 1 (unconstrained): the minimiser $(2, 1)$ has $g_1 = 1 > 0$: infeasible, so g_1 must be active.

Step 2 (guess $\mathcal{A} = \{1\}$): stationarity $\big(2(x - 2) + \mu_1,\ 2(y - 1) + \mu_1\big) = \mathbf{0}$ gives $x - 2 = y - 1$, i.e. $x = y + 1$. With $x + y = 2$: $y = \tfrac{1}{2}$, $x = \tfrac{3}{2}$, and $\mu_1 = -2(x - 2) = 1$.

Step 3 (verify): $\mu_1 = 1 \ge 0$ ✓; inactive $g_2 = -\tfrac{3}{2} < 0$ ✓ with $\mu_2 = 0$. KKT point $(1.5, 0.5)$, $f^* = 0.5$. The problem is convex, so this is the global minimum: the closest point to $(2, 1)$ in the region.

Step 4 (sensitivity): relax to $x + y \le 2 + \varepsilon$; the projection becomes $(1.5 + \varepsilon/2,\ 0.5 + \varepsilon/2)$ and $f^* = 2\big(\tfrac{1 - \varepsilon}{2}\big)^2 = \tfrac{1}{2}(1 - \varepsilon)^2$, so $df^*/d\varepsilon = -1 = -\mu_1$ at $\varepsilon = 0$ ✓.

Worked Example 23.13 — the minimum-norm point of a half-plane

Minimise $f = x^2 + y^2$ subject to $x + 2y \geq 2$.

Step 1 (standard form): $g = 2 - x - 2y \leq 0$. The unconstrained minimiser $(0, 0)$ has $g = 2 > 0$, infeasible.

Step 2 (active): stationarity $(2x - \mu,\ 2y - 2\mu) = \mathbf{0}$ gives $x = \mu/2$, $y = \mu$. Then $x + 2y = \frac{5}{2}\mu = 2 \Rightarrow \mu = \frac{4}{5}$, $(x, y) = (\frac{2}{5}, \frac{4}{5})$.

Step 3 (check): $\mu = 0.8 \geq 0$ ✓, $f^* = \frac{4}{25} + \frac{16}{25} = \frac{4}{5}$. Geometric check: the distance from the origin to the line $x + 2y = 2$ is $2/\sqrt{5}$, whose square is $\frac{4}{5}$ ✓. And $-\nabla f = (-0.8, -1.6) = \mu \nabla g = 0.8(-1, -2)$: the downhill direction points out of the feasible set along the outward normal, as the theory says.

Worked Example 23.14 — KKT with case analysis

Minimise $f = (x - 3)^2 + (y - 2)^2$ subject to $g_1 = x^2 + y^2 - 5 \leq 0$ (a disc) and $g_2 = x + 2y - 4 \leq 0$ (a half-plane).

Step 1 (unconstrained): $(3, 2)$ violates both ($13 > 5$, $7 > 4$).

Step 2 (only g_1 active): the closest point of the circle to $(3, 2)$ is $(3, 2)\sqrt{5}/\sqrt{13} = (1.861, 1.240)$. But $g_2 = 1.861 + 2.481 - 4 = 0.342 > 0$: infeasible, reject.

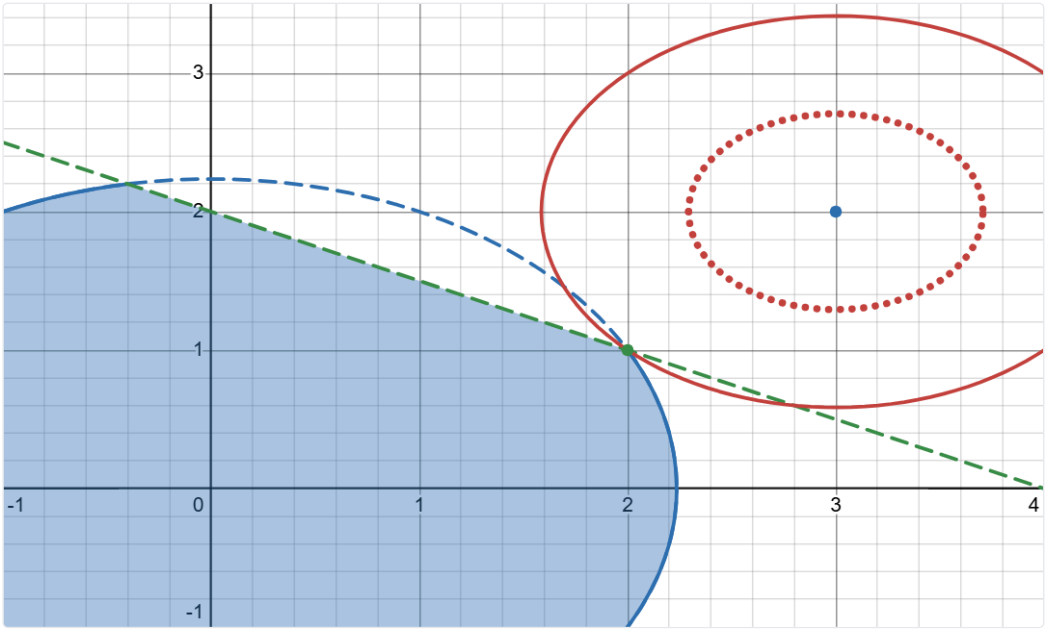
Step 3 (only g_2 active): the projection onto the line $x + 2y = 4$ is $(3, 2) - \frac{3+4-4}{5}(1, 2) = (2.4, 0.8)$. But $g_1 = 5.76 + 0.64 - 5 = 1.4 > 0$: infeasible, reject.

Step 4 (both active): $x = 4 - 2y$ in $x^2 + y^2 = 5$ gives $5y^2 - 16y + 11 = 0$, $y = 1$ or $y = \frac{11}{5}$. Take $(2, 1)$ (the other root $(-0.4, 2.2)$ is farther from $(3, 2)$; one can check its multipliers are not both nonnegative). Stationarity at $(2, 1)$:

$$\begin{pmatrix} -2 \\ -2 \end{pmatrix} + \mu_1 \begin{pmatrix} 4 \\ 2 \end{pmatrix} + \mu_2 \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \mathbf{0} \ \Rightarrow\ 4\mu_1 + \mu_2 = 2,\ \ 2\mu_1 + 2\mu_2 = 2 \ \Rightarrow\ \mu_1 = \tfrac{1}{3},\ \mu_2 = \tfrac{2}{3}.$$

Step 5 (accept): both multipliers are positive ✓, so $(2, 1)$ is a KKT point with $f^* = 2$. The problem is convex (disc and half-plane are convex sets, f is convex), so it is the global minimum. A numerical solver (`scipy.optimize.minimize` with SLSQP) returns $(2.000, 1.000)$, $f = 2.000$ ✓.

Step 6 (sensitivity): enlarging the disc to radius^2 5.1 lowers f^* by about $\mu_1 \times 0.1 = 0.033$; the re-solved value is 1.968, a drop of 0.032 ✓.



Example 23.14: the feasible set (blue) is the part of the disc below the green line. The smallest contour of f (red) that touches it does so at the corner $(2, 1)$, where both constraints are active.

Engineering Note — water-filling

Allocating total power P across parallel channels with noise levels N_i to maximise capacity, $\max \sum_i \log(1 + p_i/N_i)$ s.t. $\sum p_i = P$, $p_i \geq 0$, is a convex problem whose KKT conditions give $p_i = \max(0,\ \nu - N_i)$ for a common "water level" ν. Channels whose noise exceeds the level get no power — their nonnegativity constraints are active. This one-line KKT result underlies power allocation in OFDM and MIMO systems.

23.5 Gradient-based methods

Motivation

The optimality conditions are equations, but for realistic problems — a 64-tap filter, a 20-parameter device model, a neural-network equaliser with 10^5 weights — they cannot be solved in closed form. Instead we **iterate**: start from a guess $\mathbf{x}_0$ and generate $\mathbf{x}_1, \mathbf{x}_2, \ldots$ with f decreasing at each step. The simplest and most widely used idea is to walk downhill along the negative gradient. Gradient descent is cheap (one gradient per step, no matrices), works in any dimension, and is the engine of adaptive filtering and machine learning. It is also slow on badly scaled problems, and understanding exactly why is the key to every faster method.

Descent directions and the generic iteration

All methods in this section and the next have the form

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{p}_k,$$

with a **search direction** $\mathbf{p}_k$ and a **step size** (learning rate) $\alpha_k > 0$.

Descent direction

$\mathbf{p}$ is a descent direction for f at $\mathbf{x}$ if $\nabla f(\mathbf{x})^{\mathsf T}\mathbf{p} < 0$. Then $f(\mathbf{x}+\alpha\mathbf{p}) < f(\mathbf{x})$ for all sufficiently small $\alpha > 0$, because $\dfrac{d}{d\alpha} f(\mathbf{x}+\alpha\mathbf{p})\Big|_{\alpha=0} = \nabla f(\mathbf{x})^{\mathsf T}\mathbf{p} < 0$.

Among unit vectors $\mathbf{p}$, the directional derivative $\nabla f^{\mathsf T}\mathbf{p} = \|\nabla f\|\cos\theta$ is most negative for $\theta = \pi$, i.e. $\mathbf{p} = -\nabla f/\|\nabla f\|$ (Ch. 6 §6.5). This gives **steepest descent** (gradient descent):

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha_k \nabla f(\mathbf{x}_k).$$

The negative gradient is perpendicular to the level curve through $\mathbf{x}_k$; each step leaves the contour at right angles. "Steepest" is relative to the Euclidean norm, i.e. to the units the variables are measured in — rescaling the variables changes the direction, which is the root of both the method's weakness and its cure (§23.6).

Fixed step sizes: when does it converge?

Suppose ∇f is **Lipschitz continuous** with constant L: $\|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\| \le L\|\mathbf{x}-\mathbf{y}\|$ (for a C^2 function, L bounds the largest Hessian eigenvalue in magnitude). Then f lies below a quadratic upper bound,

$$f(\mathbf{y}) \le f(\mathbf{x}) + \nabla f(\mathbf{x})^{\mathsf T}(\mathbf{y}-\mathbf{x}) + \frac{L}{2}\|\mathbf{y}-\mathbf{x}\|^2,$$

and substituting $\mathbf{y} = \mathbf{x} - \alpha\nabla f(\mathbf{x})$ yields the **descent lemma**

$$f(\mathbf{x}_{k+1}) \le f(\mathbf{x}_k) - \alpha\Big(1 - \frac{L\alpha}{2}\Big)\|\nabla f(\mathbf{x}_k)\|^2.$$

Convergence of gradient descent with a fixed step

If f is bounded below and ∇f is L-Lipschitz, then for any fixed $0 < \alpha < 2/L$ the iterates satisfy $f(\mathbf{x}_{k+1}) < f(\mathbf{x}_k)$ whenever $\nabla f(\mathbf{x}_k) \neq \mathbf{0}$, and $\nabla f(\mathbf{x}_k) \to \mathbf{0}$. The choice $\alpha = 1/L$ maximises the guaranteed decrease, $f(\mathbf{x}_{k+1}) \le f(\mathbf{x}_k) - \frac{1}{2L}\|\nabla f(\mathbf{x}_k)\|^2$.

Proof sketch. The bracket $1 - L\alpha/2$ is positive for $\alpha < 2/L$, so f decreases by a positive multiple of $\|\nabla f\|^2$. Summing the inequality from $k=0$ to $N-1$ gives $c\sum_k \|\nabla f(\mathbf{x}_k)\|^2 \le f(\mathbf{x}_0) - f(\mathbf{x}_N) \le f(\mathbf{x}_0) - f_{\inf} < \infty$, so the gradient norms are square-summable and tend to zero. □

Note what is *not* claimed: convergence to a global minimum. Gradient descent finds a stationary point; convexity (§23.7) is what makes that point the answer.

The quadratic model problem

Everything about gradient descent can be seen on $f(\mathbf{x}) = \tfrac12 \mathbf{x}^{\mathsf T}\mathbf{A}\mathbf{x} - \mathbf{b}^{\mathsf T}\mathbf{x}$ with $\mathbf{A}$ symmetric positive definite, eigenvalues $0 < m = \lambda_{\min} \le \cdots \le \lambda_{\max} = L$. The minimiser solves $\mathbf{A}\mathbf{x}^* = \mathbf{b}$, and the gradient is $\mathbf{A}\mathbf{x} - \mathbf{b} = \mathbf{A}(\mathbf{x} - \mathbf{x}^*)$. The error $\mathbf{e}_k = \mathbf{x}_k - \mathbf{x}^*$ obeys

$$\mathbf{e}_{k+1} = (\mathbf{I} - \alpha\mathbf{A})\,\mathbf{e}_k.$$

Expanding $\mathbf{e}_0$ in the eigenvectors $\mathbf{v}_i$ of $\mathbf{A}$ (Ch. 10), each component is multiplied by $1 - \alpha\lambda_i$ per step. Hence:

- convergence for every start $\iff$ $|1-\alpha\lambda_i| < 1$ for all i $\iff$ $0 < \alpha < 2/L$;
- the slowest component decays like $\max_i |1-\alpha\lambda_i|^k$;
- minimising $\max(|1-\alpha m|, |1-\alpha L|)$ gives the **best fixed step** $\alpha^* = \dfrac{2}{m+L}$ and the **convergence factor**

$$\rho = \frac{L-m}{L+m} = \frac{\kappa - 1}{\kappa + 1}, \qquad \kappa = \frac{L}{m} = \operatorname{cond}(\mathbf{A}).$$

The **condition number** κ governs everything. To gain one decimal digit of accuracy needs about $\ln 10 / \ln(1/\rho) \approx 1.15\,\kappa$ iterations when κ is large. A problem with $\kappa = 10^4$ needs tens of thousands of gradient steps — while Newton's method (§23.6) would take one.

Worked Example 23.15 — fixed step on a one-dimensional quadratic

$f(x) = 2x^2$, so $f'(x) = 4x$, $L = 4$, $x^* = 0$, starting at $x_0 = 1$.

Step 1 (iteration): $x_{k+1} = x_k - 4\alpha x_k = (1-4\alpha)x_k$, so $x_k = (1-4\alpha)^k$.

Step 2 (cases):

- $\alpha = 0.1$: factor 0.6, monotone convergence; $x_{10} = 0.006$.
- $\alpha = 0.25 = 1/L$: factor 0, exact in one step (in one dimension $1/L$ is Newton's step).
- $\alpha = 0.4$: factor -0.6, convergence with sign flips (overshooting).
- $\alpha = 0.5 = 2/L$: factor -1, $x_k = (-1)^k$ oscillates forever.
- $\alpha = 0.6$: factor -1.4, divergence.

Step 3 (lesson): the stability limit $\alpha < 2/L$ is exactly the condition for a first-order discrete-time system $x_{k+1} = (1-\alpha L)x_k$ to have its pole inside the unit circle (Ch. 17). Gradient descent *is* a discrete-time feedback loop.

Worked Example 23.16 — step size on an anisotropic bowl

$f(x,y) = x^2 + 5y^2$, i.e. $\mathbf{A} = \operatorname{diag}(2, 10)$, $m = 2$, $L = 10$, $\kappa = 5$. Start at $(5, 1)$, $f_0 = 30$.

Step 1 (per-coordinate factors): $x_{k+1} = (1 - 2\alpha)x_k$ and $y_{k+1} = (1 - 10\alpha)y_k$.

Step 2 (stability): need $|1 - 10\alpha| < 1$, i.e. $\alpha < 0.2 = 2/L$. The *stiff* direction y (large curvature) sets the limit, while the *soft* direction x (small curvature) sets the speed.

Step 3 (compare after 20 steps):

α	x factor	y factor	$f(\mathbf{x}_{20})$
0.05	0.9	0.5	0.37
0.10	0.8	0	3.3×10^{-3}
1/6 (optimal)	$2/3$	$-2/3$	2.7×10^{-6}
0.18	0.64	-0.8	6.7×10^{-4}
0.21	0.58	-1.1	226 (diverging)

For the optimal step both factors have magnitude $\rho = (5-1)/(5+1) = 2/3$, so f falls by $\rho^2 = 4/9$ per step: $30(4/9)^{20} = 2.7 \times 10^{-6}$.

Step 4 (lesson): the best step is a compromise dictated by the two extreme curvatures. If the problem were rescaled so that $\kappa = 1$ (substitute $y = u/\sqrt{5}$), a single step would suffice.

Line search

Rather than a fixed α, choose α_k at each step by searching along the ray $\phi(\alpha) = f(\mathbf{x}_k + \alpha \mathbf{p}_k)$.

Exact line search minimises ϕ. For the quadratic, $\phi(\alpha) = f(\mathbf{x}_k) - \alpha \mathbf{g}_k^{\mathsf T}\mathbf{g}_k + \tfrac{1}{2}\alpha^2 \mathbf{g}_k^{\mathsf T}\mathbf{A}\mathbf{g}_k$ along $\mathbf{p}_k = -\mathbf{g}_k$, so

$$\alpha_k = \frac{\mathbf{g}_k^{\mathsf T}\mathbf{g}_k}{\mathbf{g}_k^{\mathsf T}\mathbf{A}\mathbf{g}_k}.$$

At the exact minimiser along the line, $\phi'(\alpha_k) = \nabla f(\mathbf{x}_{k+1})^{\mathsf T}\mathbf{p}_k = 0$: **the new gradient is orthogonal to the old search direction**. For steepest descent this means consecutive steps are at right angles — the notorious **zig-zag**. The classical bound (Kantorovich) is

$$f(\mathbf{x}_{k+1}) - f^\ast \le \left(\frac{\kappa - 1}{\kappa + 1}\right)^2 \big(f(\mathbf{x}_k) - f^\ast\big),$$

the same rate as the optimal fixed step: exact line search removes the need to know m and L, but does not beat the condition number.

Inexact line search is what real codes do, because each evaluation of f may be expensive. The aim is just "enough" decrease:

Armijo (sufficient decrease) and Wolfe conditions

For a descent direction $\mathbf{p}_k$ and constants $0 < c_1 < c_2 < 1$ (typically $c_1 = 10^{-4}$, $c_2 = 0.9$), a step α satisfies

- the **Armijo condition** if $f(\mathbf{x}_k + \alpha\mathbf{p}_k) \le f(\mathbf{x}_k) + c_1 \alpha \nabla f(\mathbf{x}_k)^{\mathsf T}\mathbf{p}_k$;
- the **curvature condition** if $\nabla f(\mathbf{x}_k + \alpha\mathbf{p}_k)^{\mathsf T}\mathbf{p}_k \ge c_2 \nabla f(\mathbf{x}_k)^{\mathsf T}\mathbf{p}_k$.

Both together are the **Wolfe conditions**. Armijo forbids steps that are too long (too little decrease for their length); the curvature condition forbids steps that are too short (the slope has not flattened enough).

Backtracking implements Armijo simply: try $\alpha = \alpha_0$ (say 1); while the Armijo condition fails, set $\alpha \leftarrow \beta\alpha$ (say $\beta = \tfrac{1}{2}$). Because $\mathbf{p}_k$ is a descent direction, a small enough α always passes, so the loop terminates.

Worked Example 23.17 — exact line search and the zig-zag

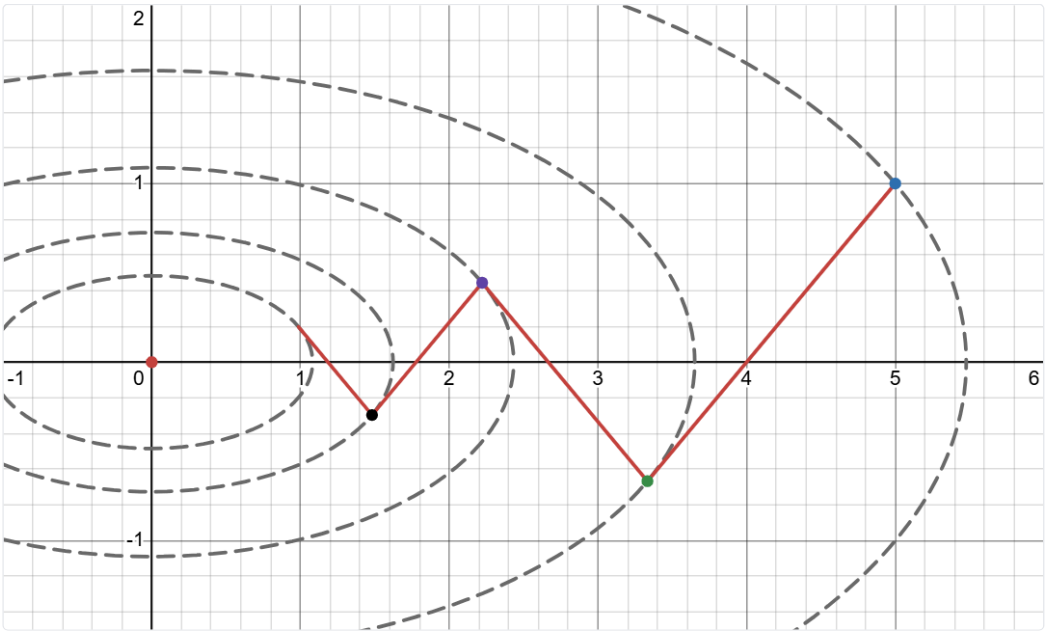
Apply steepest descent with exact line search to $f = x^2 + 5y^2$ from $(5, 1)$.

Step 1: $\mathbf{g}_0 = (2x, 10y) = (10, 10)$. $\mathbf{g}_0^{\mathsf T}\mathbf{g}_0 = 200$, $\mathbf{g}_0^{\mathsf T}\mathbf{A}\mathbf{g}_0 = 2(100) + 10(100) = 1200$, so $\alpha_0 = \tfrac{1}{6}$ and $\mathbf{x}_1 = (5, 1) - \tfrac{1}{6}(10, 10) = (\tfrac{10}{3}, -\tfrac{2}{3})$, $f_1 = \tfrac{100}{9} + \tfrac{20}{9} = 13.33$.

Step 2: $\mathbf{g}_1 = (\tfrac{20}{3}, -\tfrac{20}{3})$. Check orthogonality: $\mathbf{g}_1^{\mathsf T}\mathbf{g}_0 = \tfrac{200}{3} - \tfrac{200}{3} = 0$ ✓. Again $\alpha_1 = \tfrac{1}{6}$, giving $\mathbf{x}_2 = (\tfrac{20}{9}, \tfrac{4}{9})$, $f_2 = 5.93$.

Step 3 (pattern): $\mathbf{x}_3 = (\tfrac{40}{27}, -\tfrac{8}{27})$, $\mathbf{x}_4 = (\tfrac{80}{81}, \tfrac{16}{81})$, and $f_k = 30(\tfrac{4}{9})^k$: $30,\ 13.3,\ 5.93,\ 2.63,\ 1.17$. Each step multiplies f by exactly $(\tfrac{\kappa-1}{\kappa+1})^2 = \tfrac{4}{9}$: this starting point is the worst case of the Kantorovich bound.

Step 4 (picture): each step goes along $\pm 45^\circ$ and turns 90°; the path bounces across the narrow valley.



Steepest descent with exact line search on $f = x^2 + 5y^2$ (contours dashed). Every step leaves a contour at right angles and is orthogonal to the previous step, producing the characteristic zig-zag toward the minimum.

Worked Example 23.18 — backtracking on the Rosenbrock function

The Rosenbrock function $f(x, y) = (1 - x)^2 + 100(y - x^2)^2$ has its minimum $f = 0$ at $(1, 1)$ at the bottom of a long, curved, narrow valley. Take one backtracking step from the standard start $(-1.2, 1)$ with $\alpha_0 = 1$, $\beta = \tfrac{1}{2}$, $c_1 = 10^{-4}$.

Step 1 (gradient): $\partial f/\partial x = -2(1 - x) - 400x(y - x^2)$ and $\partial f/\partial y = 200(y - x^2)$. At $(-1.2, 1)$: $y - x^2 = -0.44$, so $\nabla f = (-4.4 - 211.2,\ -88) = (-215.6,\ -88)$, $\|\nabla f\|^2 = 54\,227$, $f_0 = 24.2$.

Step 2 (backtrack): trial f values for $\alpha = 1, \tfrac{1}{2}, \tfrac{1}{4}, \ldots$ are 2.1×10^{11}, 1.3×10^{10}, …, 543.5 ($\alpha = 2^{-6}$), 211.4, 149.6, 35.1 ($\alpha = 2^{-9}$) — all rejected; the Armijo bound $24.2 - 10^{-4}\alpha\,(54\,227)$ is about 24.2.

Step 3 (accept): $\alpha = 2^{-10} \approx 9.77 \times 10^{-4}$ gives $(-0.989,\ 1.086)$ with $f = 5.10 < 24.2$ ✓.

Step 4 (the long run): continuing with backtracking, gradient descent needs about 1.4×10^4 iterations to reach $\|\nabla f\| < 10^{-6}$. The Hessian at the minimum is $\begin{pmatrix} 802 & -400 \\ -400 & 200 \end{pmatrix}$, with condition number about 2500. Newton's method (§23.6) gets there in 6 iterations.

Beyond plain gradient descent

- **Momentum (heavy ball).** $\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k) + \beta(\mathbf{x}_k - \mathbf{x}_{k-1})$. The extra term is a low-pass filter on the step sequence that damps the zig-zag. On quadratics, optimally tuned α, β improve the convergence factor from $(\kappa - 1)/(\kappa + 1)$ to $(\sqrt{\kappa} - 1)/(\sqrt{\kappa} + 1)$, so the iteration count scales with $\sqrt{\kappa}$ instead of κ: for $\kappa = 10^4$, roughly 100 times fewer iterations. Nesterov's accelerated gradient achieves this rate for general smooth convex functions.
- **Conjugate gradient (CG).** For quadratics, choosing each direction to be $\mathbf{A}$-conjugate to the previous ones ($\mathbf{p}_i^{\mathsf{T}} \mathbf{A} \mathbf{p}_j = 0$) cures the zig-zag: CG reaches the exact minimiser in at most n steps and is the standard iterative solver for large sparse SPD systems (Ch. 24). Nonlinear variants (Fletcher–Reeves, Polak–Ribière) extend it to general f.
- **Stochastic gradient.** When $f = \frac{1}{N}\sum_i f_i$ is an average over many data points, use the gradient of one (or a few) f_i per step: noisy but N times cheaper.

Engineering Note — the LMS adaptive filter is stochastic gradient descent

An adaptive FIR filter with weights $\mathbf{w}$ and input vector $\mathbf{u}_n$ should minimise the mean-square error $E[e_n^2]$, $e_n = d_n - \mathbf{w}^{\mathsf{T}}\mathbf{u}_n$. The true gradient is $-2E[e_n \mathbf{u}_n]$; replacing the expectation by its instantaneous value gives Widrow's **least-mean-squares** update $\mathbf{w}_{n+1} = \mathbf{w}_n + \mu\, e_n \mathbf{u}_n$. The stability limit $\alpha < 2/L$ becomes the classical LMS bound $0 < \mu < 2/\lambda_{\max}(\mathbf{R})$, where $\mathbf{R} = E[\mathbf{u}_n \mathbf{u}_n^{\mathsf{T}}]$ is the input autocorrelation matrix (Ch. 21), and the convergence speed is governed by its eigenvalue spread — exactly the condition number story above.

Common Pitfall — stopping tests

"Stop when f stops changing" can halt in a flat region far from a minimum; "stop when $\|\nabla f\| < \varepsilon$" depends on the units of f and $\mathbf{x}$. Robust codes combine a *relative* gradient test, $\|\nabla f(\mathbf{x}_k)\| \le \varepsilon \max(1, |f(\mathbf{x}_k)|)$, with a small-step test and an iteration limit, and report which one fired.

A minimal implementation:

```
import numpy as np

def grad_descent(f, grad, x0, alpha0=1.0, beta=0.5, c1=1e-4, tol=1e-6, maxit=100000):
    x = np.asarray(x0, float)
    for k in range(maxit):
        g = grad(x)
        if np.linalg.norm(g) < tol:
            break
        a = alpha0
        while f(x - a*g) > f(x) - c1*a*(g @ g):   # Armijo backtracking
            a *= beta
        x = x - a*g
    return x, k
```

23.6 Newton and quasi-Newton methods

Motivation

Gradient descent uses only the slope; it does not know how curved the function is, so it cannot tell a gentle valley from a steep one and must use a step that is safe for the steepest direction. Newton's method also uses the **curvature** (the Hessian): it fits a quadratic bowl to f at the current point and jumps straight to the bottom of the bowl. Near a minimum this converges *quadratically* — the number of correct digits doubles every iteration — and is immune to bad scaling. The price is computing and factoring an $n \times n$ Hessian at every step. **Quasi-Newton** methods, BFGS above all, get most of the benefit while using only gradients, by learning the curvature from the history of steps. BFGS is the default unconstrained method in most scientific software, including SciPy's `minimize`.

Newton's method

At $\mathbf{x}_k$, with $\mathbf{g}_k = \nabla f(\mathbf{x}_k)$ and $\mathbf{H}_k = \nabla^2 f(\mathbf{x}_k)$, the quadratic model is

$$m_k(\mathbf{p}) = f(\mathbf{x}_k) + \mathbf{g}_k^{\mathsf T}\mathbf{p} + \tfrac{1}{2}\mathbf{p}^{\mathsf T}\mathbf{H}_k\mathbf{p}.$$

If $\mathbf{H}_k \succ 0$, the model has a unique minimiser where $\nabla m_k = \mathbf{g}_k + \mathbf{H}_k\mathbf{p} = \mathbf{0}$:

Newton step

$$\mathbf{H}_k\,\mathbf{p}_k^{\mathrm N} = -\mathbf{g}_k, \qquad \mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{p}_k^{\mathrm N},$$

with $\alpha_k = 1$ for the *pure* Newton method. The linear system is solved (never by forming $\mathbf{H}_k^{-1}$) by Cholesky factorisation, which also tests $\mathbf{H}_k \succ 0$ for free: the factorisation fails if it is not (Ch. 10 §10.6).

Two other views of the same iteration are useful:

- **Root-finding view.** Minimising f means solving $\nabla f(\mathbf{x}) = \mathbf{0}$. Newton's method for the nonlinear system $\mathbf{F}(\mathbf{x}) = \mathbf{0}$ uses the Jacobian of $\mathbf{F}$ (Ch. 6 §6.6, Ch. 22); with $\mathbf{F} = \nabla f$ the Jacobian is the Hessian, and we recover the Newton step. In one dimension: $x_{k+1} = x_k - f'(x_k)/f''(x_k)$.
- **Rescaling view.** If $\mathbf{H} \succ 0$, change variables to $\mathbf{u} = \mathbf{H}^{1/2}\mathbf{x}$. In $\mathbf{u}$ the local quadratic model is a perfectly round bowl ($\kappa = 1$) and steepest descent with unit step goes straight to its centre. Translating back gives the Newton step. Newton's method is **affine invariant**: a linear change of variables $\mathbf{x} = \mathbf{T}\mathbf{z}$ does not change the iterates (mapped back), so units and scaling do not matter — in sharp contrast to gradient descent.

A **descent property**: if $\mathbf{H}_k \succ 0$ then $\mathbf{g}_k^{\mathsf T}\mathbf{p}_k^{\mathrm N} = -\mathbf{g}_k^{\mathsf T}\mathbf{H}_k^{-1}\mathbf{g}_k < 0$, so the Newton direction is a descent direction and a line search can be applied. If $\mathbf{H}_k$ is indefinite, the "Newton step" heads for the stationary point of an indefinite quadratic — a saddle — and may go uphill.

Local quadratic convergence of Newton's method

Let $f \in C^2$ with Lipschitz-continuous Hessian near a point $\mathbf{x}^\ast$ where $\nabla f(\mathbf{x}^\ast) = \mathbf{0}$ and $\nabla^2 f(\mathbf{x}^\ast) \succ 0$. Then for $\mathbf{x}_0$ close enough to $\mathbf{x}^\ast$, pure Newton iterates are well defined, converge to $\mathbf{x}^\ast$, and satisfy

$$\|\mathbf{x}_{k+1} - \mathbf{x}^\ast\| \le C\,\|\mathbf{x}_k - \mathbf{x}^\ast\|^2$$

for a constant C (roughly $\tfrac{1}{2}\times$ Hessian Lipschitz constant $\times \|\nabla^2 f(\mathbf{x}^\ast)^{-1}\|$).

Proof sketch (one dimension). Write $e_k = x_k - x^\ast$. Taylor's theorem gives $0 = f'(x^\ast) = f'(x_k) - f''(x_k)e_k + \tfrac{1}{2}f'''(\xi)e_k^2$. Divide by $f''(x_k)$ and rearrange: $e_k - f'(x_k)/f''(x_k) = \frac{f'''(\xi)}{2f''(x_k)}e_k^2$. The left side is e_{k+1}. Hence $e_{k+1} \approx \dfrac{f'''(x^\ast)}{2f''(x^\ast)}e_k^2$. $\square$

Quadratic convergence is spectacular *once you are close*; far away the pure method may diverge, cycle or head for a maximum. The theorem says nothing about how close is close enough.

Worked Example 23.19 — quadratic convergence in one dimension

Minimise $f(x) = e^x - 2x$ (minimiser $x^* = \ln 2 = 0.693147$) from $x_0 = 0$.

Step 1 (iteration): $f'(x) = e^x - 2$, $f''(x) = e^x$, so $x_{k+1} = x_k - (e^{x_k} - 2)/e^{x_k} = x_k - 1 + 2e^{-x_k}$.

Step 2 (iterates):

k	x_k	error $e_k = x_k - \ln 2$	e_k / e_{k-1}^2
0	0	-0.693	—
1	1.000000	3.07×10^{-1}	0.64
2	0.735759	4.26×10^{-2}	0.45
3	0.694042	8.95×10^{-4}	0.49
4	0.6931476	4.00×10^{-7}	0.50
5	0.69314718056	8.0×10^{-14}	0.50

Step 3 (check the constant): the theory predicts $e_{k+1}/e_k^2 \to f'''(x^*)/(2f''(x^*)) = e^{x^*}/(2e^{x^*}) = \tfrac{1}{2}$ ✓. The number of correct digits goes 1, 1, 3, 6, 13: doubling each step.

Worked Example 23.20 — one Newton step solves a quadratic

For $f(\mathbf{x}) = \tfrac{1}{2}\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} - \mathbf{b}^{\mathsf{T}}\mathbf{x}$ with $\mathbf{A} \succ 0$, the Hessian is $\mathbf{A}$ everywhere and $\mathbf{p}^{\mathrm{N}} = -\mathbf{A}^{-1}(\mathbf{A}\mathbf{x}_0 - \mathbf{b}) = \mathbf{x}^* - \mathbf{x}_0$. One step lands exactly on $\mathbf{x}^*$ from any start, whatever κ. On $f = x^2 + 5y^2$ from $(5, 1)$: $\mathbf{g} = (10, 10)$, $\mathbf{H} = \operatorname{diag}(2, 10)$, $\mathbf{p} = -(5, 1)$, $\mathbf{x}_1 = (0, 0)$ ✓ — compared with the endless zig-zag of Example 23.17.

Worked Example 23.21 — pure Newton on the Rosenbrock function

From $(-1.2, 1)$, using the Hessian

$$\nabla^2 f = \begin{pmatrix} 2 - 400(y - x^2) + 800x^2 & -400x \\ -400x & 200 \end{pmatrix}:$$

k	(x_k, y_k)	$f(\mathbf{x}_k)$
0	$(-1.2000,\ 1.0000)$	24.2
1	$(-1.1753,\ 1.3807)$	4.73
2	$(0.7631,\ -3.1750)$	**1411.8**
3	$(0.7634,\ 0.5828)$	0.0560
4	$(0.999995,\ 0.9440)$	0.313
5	$(0.9999957,\ 0.9999914)$	1.9×10^{-11}
6	$(1,\ 1)$	3×10^{-20}

Observations: (i) six iterations versus about fourteen thousand for gradient descent (Example 23.18); (ii) the pure method is not a descent method — at step 2 the objective jumps from 4.7 to 1412 because the quadratic model is trusted far outside its region of validity; it happens to recover here, but on other problems such a jump diverges. (iii) The last two steps show the quadratic convergence. A line search (damped Newton) would have rejected the full step at iteration 2 and kept f monotone.

Making Newton's method robust

- **Damped Newton.** Use the Newton direction with a backtracking line search starting from $\alpha = 1$ (§23.5). Far from the solution, short steps keep f decreasing; near the solution $\alpha = 1$ is accepted and quadratic convergence is retained.
- **Hessian modification.** If $\mathbf{H}_k$ is not positive definite, replace it by $\mathbf{H}_k + \tau\mathbf{I}$ with $\tau > 0$ large enough that Cholesky succeeds. For small τ this is nearly Newton; for large τ it is a short gradient step $-\mathbf{g}/\tau$.
- **Trust region.** Minimise the model $m_k(\mathbf{p})$ only over $\|\mathbf{p}\| \le \Delta_k$, then compare the actual decrease with the predicted decrease. If they agree, accept and enlarge Δ; if not, shrink Δ. The KKT conditions of this subproblem give $(\mathbf{H}_k + \tau\mathbf{I})\mathbf{p} = -\mathbf{g}_k$ with $\tau \ge 0$ the multiplier of the radius constraint — so Hessian modification and trust regions are the same idea. For least squares this is the **Levenberg–Marquardt** method (§23.8).

Cost. Each Newton step needs the Hessian ($n(n+1)/2$ second derivatives) and a Cholesky factorisation ($\approx n^3/3$ flops). For n in the hundreds this is cheap; for $n = 10^5$ it is prohibitive unless the Hessian is sparse — which motivates quasi-Newton and, for very large problems, limited-memory variants.

Quasi-Newton methods and BFGS

The idea: maintain an approximation $\mathbf{B}_k \approx \nabla^2 f(\mathbf{x}_k)$ (or its inverse $\mathbf{H}_k \approx \nabla^2 f(\mathbf{x}_k)^{-1}$), take steps $\mathbf{p}_k = -\mathbf{H}_k\mathbf{g}_k$, and after each step update the approximation using only the gradient change.

With $\mathbf{s}_k = \mathbf{x}_{k+1} - \mathbf{x}_k$ and $\mathbf{y}_k = \mathbf{g}_{k+1} - \mathbf{g}_k$, a first-order expansion of the gradient gives $\mathbf{y}_k \approx \nabla^2 f\, \mathbf{s}_k$ — exactly for a quadratic. So we require the new approximation to satisfy the

Secant (quasi-Newton) condition

$$\mathbf{B}_{k+1}\mathbf{s}_k = \mathbf{y}_k \qquad \Longleftrightarrow \qquad \mathbf{H}_{k+1}\mathbf{y}_k = \mathbf{s}_k.$$

This is n equations for the $n(n+1)/2$ entries of a symmetric matrix, so there is freedom. The Broyden–Fletcher–Goldfarb–Shanno (BFGS) update chooses the symmetric $\mathbf{H}_{k+1}$ satisfying the secant condition that is *closest* to $\mathbf{H}_k$ in a suitably weighted Frobenius norm. The result is a rank-two correction:

BFGS update

With $\rho_k = 1/(\mathbf{y}_k^{\mathsf T}\mathbf{s}_k)$,

$$\mathbf{H}_{k+1} = \left(\mathbf{I} - \rho_k \mathbf{s}_k \mathbf{y}_k^{\mathsf T}\right)\mathbf{H}_k\left(\mathbf{I} - \rho_k \mathbf{y}_k \mathbf{s}_k^{\mathsf T}\right) + \rho_k \mathbf{s}_k \mathbf{s}_k^{\mathsf T}.$$

Properties:

1. $\mathbf{H}_{k+1}$ is symmetric and satisfies $\mathbf{H}_{k+1}\mathbf{y}_k = \mathbf{s}_k$;
2. if $\mathbf{H}_k \succ 0$ and the **curvature condition** $\mathbf{y}_k^{\mathsf T}\mathbf{s}_k > 0$ holds, then $\mathbf{H}_{k+1} \succ 0$ — so every BFGS direction is a descent direction;
3. on a strictly convex quadratic with exact line searches, BFGS terminates in at most n steps with $\mathbf{H}_n = \mathbf{A}^{-1}$;
4. near a minimiser with $\nabla^2 f \succ 0$, BFGS with a Wolfe line search converges **superlinearly**: $\|\mathbf{x}_{k+1} - \mathbf{x}^*\| / \|\mathbf{x}_k - \mathbf{x}^*\| \to 0$.

Why property 1 holds. Multiply by $\mathbf{y}_k$: the right factor gives $(\mathbf{I} - \rho_k \mathbf{y}_k \mathbf{s}_k^{\mathsf T})\mathbf{y}_k = \mathbf{y}_k - \rho_k \mathbf{y}_k(\mathbf{s}_k^{\mathsf T}\mathbf{y}_k) = \mathbf{0}$, so only the last term survives: $\rho_k \mathbf{s}_k(\mathbf{s}_k^{\mathsf T}\mathbf{y}_k) = \mathbf{s}_k$ ✓. *Why property 2 holds.* For any $\mathbf{z} \neq \mathbf{0}$, put $\mathbf{w} = \mathbf{z} - \rho_k \mathbf{y}_k(\mathbf{s}_k^{\mathsf T}\mathbf{z})$; then $\mathbf{z}^{\mathsf T}\mathbf{H}_{k+1}\mathbf{z} = \mathbf{w}^{\mathsf T}\mathbf{H}_k\mathbf{w} + \rho_k(\mathbf{s}_k^{\mathsf T}\mathbf{z})^2$. Both terms are ≥ 0, and they cannot both vanish (if $\mathbf{s}_k^{\mathsf T}\mathbf{z} = 0$ then $\mathbf{w} = \mathbf{z} \neq \mathbf{0}$). The Wolfe curvature condition of §23.5 guarantees $\mathbf{y}_k^{\mathsf T}\mathbf{s}_k > 0$, which is why BFGS is always paired with a Wolfe line search.

Related methods: **DFP** is the same formula with the roles of $\mathbf{s}$ and $\mathbf{y}$ (and $\mathbf{B}$ and $\mathbf{H}$) swapped, and is less robust in practice; **SR1** is a symmetric rank-one update that need not stay positive definite (useful inside trust regions); **L-BFGS** stores only the last $m \approx 5$–20 pairs $(\mathbf{s}_i, \mathbf{y}_i)$ and applies $\mathbf{H}_k$ implicitly, using $O(mn)$ memory — the method of choice for large smooth problems.

Worked Example 23.22 — BFGS by hand on a quadratic

Minimise $f = x^2 + 5y^2$ ($\mathbf{A} = \operatorname{diag}(2, 10)$) from $\mathbf{x}_0 = (5, 1)$ with $\mathbf{H}_0 = \mathbf{I}$ and exact line search.

Step 1 (first step = steepest descent): $\mathbf{g}_0 = (10, 10)$, $\mathbf{p}_0 = -(10, 10)$, $\alpha_0 = \tfrac{1}{6}$ (as in Example 23.17), $\mathbf{x}_1 = (\tfrac{10}{3}, -\tfrac{2}{3})$, $\mathbf{g}_1 = (\tfrac{20}{3}, -\tfrac{20}{3})$.

Step 2 (update data): $\mathbf{s}_0 = (-\tfrac{5}{3}, -\tfrac{5}{3})$, $\mathbf{y}_0 = \mathbf{g}_1 - \mathbf{g}_0 = (-\tfrac{10}{3}, -\tfrac{50}{3})$, $\mathbf{y}_0^{\mathsf T}\mathbf{s}_0 = \tfrac{50}{9} + \tfrac{250}{9} = \tfrac{100}{3} > 0$ ✓, $\rho_0 = 0.03$.

Step 3 (BFGS update): evaluating the formula gives

$$\mathbf{H}_1 = \frac{1}{36}\begin{pmatrix} 53 & -7 \\ -7 & 5 \end{pmatrix}.$$

Secant check: $\mathbf{H}_1\mathbf{y}_0 = \tfrac{1}{36}\big(53(-\tfrac{10}{3}) + (-7)(-\tfrac{50}{3}),\ (-7)(-\tfrac{10}{3}) + 5(-\tfrac{50}{3})\big) = \tfrac{1}{36}(-60, -60) = (-\tfrac{5}{3}, -\tfrac{5}{3}) = \mathbf{s}_0$ ✓. Positive definite: trace > 0, determinant $(265 - 49)/36^2 > 0$ ✓.

Step 4 (second step): $\mathbf{p}_1 = -\mathbf{H}_1\mathbf{g}_1 = -\tfrac{20}{3} \cdot \tfrac{1}{36}(53 + 7,\ -7 - 5) = (-\tfrac{100}{9},\ \tfrac{20}{9})$. Exact line search: $\alpha_1 = -\mathbf{g}_1^{\mathsf T}\mathbf{p}_1 / (\mathbf{p}_1^{\mathsf T}\mathbf{A}\mathbf{p}_1) = 0.3$, so $\mathbf{x}_2 = (\tfrac{10}{3} - \tfrac{10}{3},\ -\tfrac{2}{3} + \tfrac{2}{3}) = (0, 0)$ — **the exact minimiser after $n = 2$ steps** ✓.

Step 5 (learned curvature): one more update gives $\mathbf{H}_2 = \operatorname{diag}(0.5,\ 0.1) = \mathbf{A}^{-1}$ exactly: BFGS has learned the inverse Hessian from two gradient differences.

Comparing the methods

On the Rosenbrock function from $(-1.2, 1)$ (tolerance about 10^{-5} on the gradient):

Method	Information used	Iterations	Function evaluations	Cost per iteration
Gradient descent + backtracking	$f, \nabla f$	$\approx 14\,000$	many more	$O(n)$
Nelder–Mead simplex (derivative-free)	f	85	159	$O(n)$
Nonlinear conjugate gradient	$f, \nabla f$	36	78	$O(n)$
BFGS	$f, \nabla f$	32	39	$O(n^2)$
Newton (pure)	$f, \nabla f, \nabla^2 f$	6	6	$O(n^3)$

(SciPy 1.18 defaults; exact counts depend on implementation and tolerances.) The rule of thumb: use Newton when Hessians are cheap and n is modest; BFGS when only gradients are available; L-BFGS or CG when n is large; derivative-free methods only when gradients are unavailable and n is small.

Common Pitfall — wrong gradients

Quasi-Newton methods trust the gradient you supply. A sign error or a missing factor of 2 produces failed line searches, "precision loss" warnings, or convergence to a non-stationary point. Always check an analytic gradient against central finite differences, $[f(\mathbf{x} + h\mathbf{e}_i) - f(\mathbf{x} - h\mathbf{e}_i)]/(2h)$ with $h \approx 10^{-6}|x_i|$, before optimising (`scipy.optimize.check_grad` does this).

Engineering Note — circuit simulators run Newton's method

SPICE's DC operating-point analysis solves the nonlinear nodal equations with Newton–Raphson and the device Jacobian; its convergence aids ("gmin stepping", "source stepping", step limiting on junction voltages) are exactly the damping and continuation ideas above. When a simulation reports "no convergence", the remedies are the optimiser's remedies: a better starting point, smaller steps, or a regularised Jacobian.

```
import numpy as np
from scipy.optimize import minimize

def f(v):    x, y = v; return (1 - x)**2 + 100*(y - x**2)**2
def grad(v): x, y = v; return np.array([-2*(1 - x) - 400*x*(y - x**2), 200*(y - x**2)])

res = minimize(f, x0=[-1.2, 1.0], jac=grad, method="BFGS")
print(res.x, res.nit, res.nfev)     # approx [1. 1.], 32, 39
```

23.7 Convex optimization fundamentals

Motivation

Sections 23.5–23.6 delivered algorithms that find *local* minima — points where no small move helps. For a general nonlinear problem that is all anyone can promise: a local method has no way to know whether a better valley exists somewhere else. There is, however, a large and very useful class of problems for which this worry disappears entirely. For **convex** problems every local minimum is global, the KKT conditions are sufficient as well as necessary, duality gives a certificate of optimality, and reliable algorithms find the solution in polynomial time. Least squares, linear programming, minimax and L^1 fitting, many filter-design problems, power allocation, economic dispatch with convex costs, and (after a change of variables) many component-sizing problems are all convex. The practical motto, due to Boyd and Vandenberghe, is that the great watershed in optimization is not between linear and nonlinear problems but between **convex and nonconvex** ones. Recognising convexity — or reformulating a problem until it is convex — is the most valuable modelling skill in this chapter.

Convex sets

Convex set

A set $C \subseteq \mathbb{R}^n$ is **convex** if for all $\mathbf{x}, \mathbf{y} \in C$ and all $\theta \in [0,1]$,

$$\theta \mathbf{x} + (1-\theta)\mathbf{y} \in C.$$

That is, the line segment joining any two points of C lies in C.

Standard examples:

- **hyperplanes** $\{\mathbf{x} : \mathbf{a}^{\mathsf T}\mathbf{x} = b\}$ and **half-spaces** $\{\mathbf{x} : \mathbf{a}^{\mathsf T}\mathbf{x} \le b\}$;
- **polyhedra** $\{\mathbf{x} : \mathbf{A}\mathbf{x} \le \mathbf{b},\ \mathbf{C}\mathbf{x} = \mathbf{d}\}$ (LP feasible sets);
- **norm balls** $\{\mathbf{x} : \|\mathbf{x} - \mathbf{x}_c\| \le r\}$ for any norm, and **ellipsoids** $\{\mathbf{x} : (\mathbf{x}-\mathbf{x}_c)^{\mathsf T}\mathbf{P}^{-1}(\mathbf{x}-\mathbf{x}_c) \le 1\}$, $\mathbf{P} \succ 0$;
- the **positive semidefinite cone** $\{\mathbf{X} = \mathbf{X}^{\mathsf T} : \mathbf{X} \succeq 0\}$;
- sublevel sets $\{\mathbf{x} : f(\mathbf{x}) \le c\}$ of a convex function f (shown below).

Non-examples: an annulus $1 \le x^2 + y^2 \le 4$ (the segment between opposite points passes through the hole); the set $\{(x,y) : xy \ge 1\}$ including both branches; any finite set of two or more points (e.g. E12 resistor values — which is why discrete design is hard).

Operations that preserve convexity of sets

Intersections (of any number) of convex sets are convex. Images and inverse images of convex sets under affine maps $\mathbf{x} \mapsto \mathbf{T}\mathbf{x} + \mathbf{t}$ are convex.

Proof for intersections. If $\mathbf{x}, \mathbf{y}$ lie in every C_i, so does each point of the segment between them, since each C_i is convex. $\square$

This is why a feasible set defined by many convex constraints — $g_i(\mathbf{x}) \le 0$ with each g_i convex, and affine equalities — is convex: it is an intersection of convex sublevel sets and hyperplanes. Note that a *nonlinear equality* $h(\mathbf{x}) = 0$ generally destroys convexity (a circle $x^2 + y^2 = 1$ is not convex, though the disc $x^2 + y^2 \le 1$ is).

Convex functions

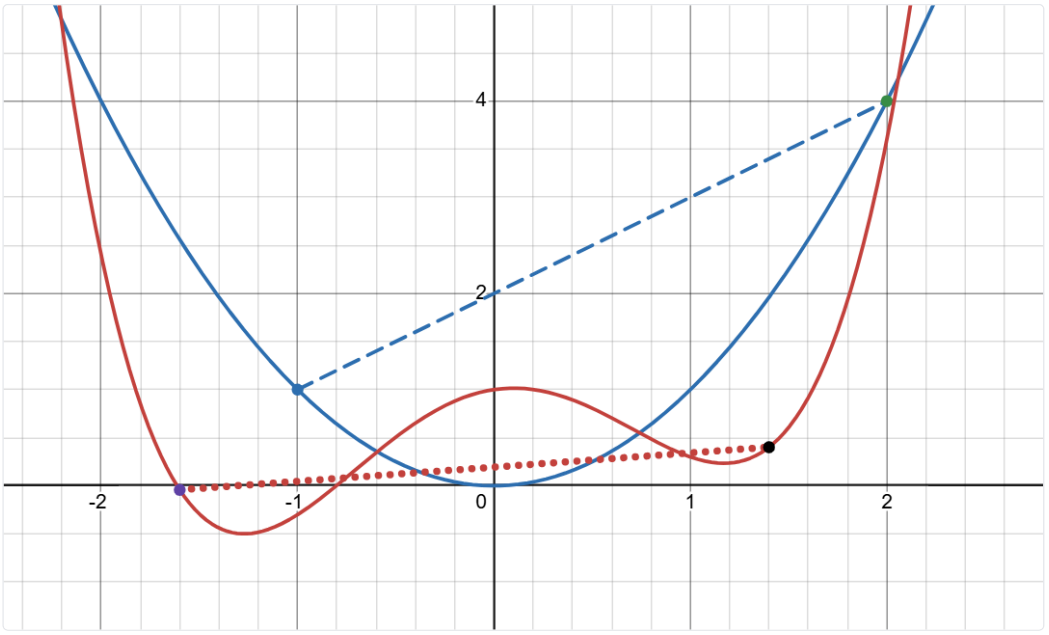
Convex function

$f : C \to \mathbb{R}$ on a convex set C is **convex** if for all $\mathbf{x}, \mathbf{y} \in C$ and $\theta \in [0,1]$,

$$f\big(\theta \mathbf{x} + (1-\theta)\mathbf{y}\big) \le \theta f(\mathbf{x}) + (1-\theta) f(\mathbf{y}),$$

i.e. the chord lies on or above the graph. It is **strictly convex** if the inequality is strict for $\mathbf{x} \ne \mathbf{y}$, $\theta \in (0,1)$, and **strongly convex** with parameter $m > 0$ if $f(\mathbf{x}) - \tfrac{m}{2}\|\mathbf{x}\|^2$ is convex. f is **concave** if $-f$ is convex.

Equivalently, f is convex iff its **epigraph** $\{(\mathbf{x}, t) : f(\mathbf{x}) \le t\}$ is a convex set — this is how the epigraph trick of §23.2 preserves convexity. The defining inequality extends by induction to **Jensen's inequality**: $f\big(\sum \theta_i \mathbf{x}_i\big) \le \sum \theta_i f(\mathbf{x}_i)$ for weights $\theta_i \ge 0$ summing to 1, and in probabilistic form $f(E[X]) \le E[f(X)]$ (Ch. 19).



Blue: $f = x^2$ is convex — every chord (dashed) lies above the graph. Red: a quartic with two local minima is not convex — the dotted chord between points near its two wells passes below the hump in the middle.

For differentiable functions there are two much more practical tests.

First- and second-order characterisations

Let C be open and convex.

1. **First order.** A differentiable f is convex on C iff for all $\mathbf{x}, \mathbf{y} \in C$

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^{\mathsf{T}}(\mathbf{y} - \mathbf{x}),$$

i.e. every tangent plane is a global under-estimator. 2. **Second order.** A twice-differentiable f is convex on C iff $\nabla^2 f(\mathbf{x}) \succeq 0$ for all $\mathbf{x} \in C$. If $\nabla^2 f(\mathbf{x}) \succ 0$ everywhere, f is strictly convex; if $\nabla^2 f(\mathbf{x}) \succeq m\mathbf{I}$, it is m-strongly convex.

Proof sketch. (1, $\Rightarrow$) From the definition, $f(\mathbf{x} + \theta(\mathbf{y} - \mathbf{x})) \leq f(\mathbf{x}) + \theta(f(\mathbf{y}) - f(\mathbf{x}))$; rearrange to $\frac{f(\mathbf{x}+\theta(\mathbf{y}-\mathbf{x})) - f(\mathbf{x})}{\theta} \leq f(\mathbf{y}) - f(\mathbf{x})$ and let $\theta \to 0^+$: the left side tends to $\nabla f(\mathbf{x})^{\mathsf{T}}(\mathbf{y} - \mathbf{x})$. (1, $\Leftarrow$) Apply the inequality at $\mathbf{z} = \theta\mathbf{x} + (1-\theta)\mathbf{y}$ toward both $\mathbf{x}$ and $\mathbf{y}$, multiply by θ and $1-\theta$, add; the gradient terms cancel. (2) Restrict f to a line, $\varphi(t) = f(\mathbf{x} + t\mathbf{d})$; f is convex iff every such φ is, and a one-variable C^2 function is convex iff $\varphi''(t) = \mathbf{d}^{\mathsf{T}}\nabla^2 f\,\mathbf{d} \geq 0$. $\square$

The first-order condition is the heart of the matter. If $\nabla f(\mathbf{x}^*) = \mathbf{0}$, it says $f(\mathbf{y}) \geq f(\mathbf{x}^*)$ for **every** $\mathbf{y}$: a stationary point of a convex function is a global minimiser. Local information certifies a global fact.

A toolkit for recognising convex functions

Building blocks (convex on the stated domain):

Function	Domain	Note
$\mathbf{a}^{\mathsf{T}}\mathbf{x} + b$	$\mathbb{R}^n$	affine: both convex and concave
$\frac{1}{2}\mathbf{x}^{\mathsf{T}}\mathbf{Q}\mathbf{x}$	$\mathbb{R}^n$	iff $\mathbf{Q} \succeq 0$
$\lVert\mathbf{x}\rVert$ (any norm)	$\mathbb{R}^n$	triangle inequality
e^{ax}	$\mathbb{R}$	
x^p	$x > 0$	for $p \geq 1$ or $p \leq 0$; concave for $0 \leq p \leq 1$
$-\ln x$, $x \ln x$, $1/x$	$x > 0$	
$\log \sum_i e^{x_i}$	$\mathbb{R}^n$	smooth approximation of $\max_i x_i$
$\max_i (\mathbf{a}_i^{\mathsf{T}}\mathbf{x} + b_i)$	$\mathbb{R}^n$	piecewise-linear

Operations that preserve convexity of functions:

1. nonnegative weighted sums $\sum w_i f_i$, $w_i \geq 0$;
2. composition with an affine map, $f(\mathbf{A}\mathbf{x} + \mathbf{b})$ — so $\lVert\mathbf{A}\mathbf{x} - \mathbf{b}\rVert$ and $\lVert\mathbf{A}\mathbf{x} - \mathbf{b}\rVert^2$ are convex;
3. pointwise maximum (or supremum) of convex functions — so $\max_i |r_i(\mathbf{x})|$ with affine r_i is convex;
4. composition $h(g(\mathbf{x}))$ with h convex and nondecreasing and g convex (e.g. $e^{g(\mathbf{x})}$).

A product or ratio of convex functions is in general *not* convex (xy is not; x^2/y for $y > 0$ happens to be).

Worked Example 23.23 — testing convexity with the Hessian

Decide which are convex on $\mathbb{R}^2$: (a) $f = x^2 + xy + y^2$; (b) $f = xy$; (c) $f = x^2 + 4xy + y^2$; (d) $f = e^{x+y} + x^2$.

(a) $\nabla^2 f = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, eigenvalues $1, 3 > 0$: strictly (indeed 1-strongly) convex. ✓

(b) $\nabla^2 f = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, eigenvalues ± 1: indefinite, not convex (a saddle surface). Along $y = -x$, $f = -x^2$ is concave.

(c) $\nabla^2 f = \begin{pmatrix} 2 & 4 \\ 4 & 2 \end{pmatrix}$, eigenvalues 6 and -2: not convex, even though both "diagonal" terms are convex. Along $y = -x$: $f = x^2 - 4x^2 + x^2 = -2x^2$.

(d) $\nabla^2 f = e^{x+y}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}$: a sum of two PSD matrices, so PSD — convex. (Or by the toolkit: e^u composed with the affine $u = x + y$, plus x^2.) Neither term alone is positive definite, but the sum has determinant $(e^{x+y} + 2)e^{x+y} - e^{2(x+y)} = 2e^{x+y} > 0$ and positive trace, so it is positive definite: f is strictly convex.

Worked Example 23.24 — convexity in engineering variables

(a) The conductor-cost function $f(A) = c_1 A + k/A$ of Example 23.2 has $f''(A) = 2k/A^3 > 0$ on $A > 0$: strictly convex, so the stationary point found there is the unique global optimum.

(b) The set of capacitances meeting a kT/C noise limit, $\{C > 0 : \sqrt{kT/C} \le v_{\max}\} = \{C \ge kT/v_{\max}^2\}$, is a half-line: convex.

(c) The power-transfer objective $P(R_L) = V_s^2 R_L/(R_s + R_L)^2$ of Example 23.1 is *not* concave on $(0, \infty)$: P'' changes sign at $R_L = 2R_s$ (check: $P'' \propto 2(R_L - 2R_s)/(R_s + R_L)^4$). It is, however, **unimodal**, so the stationary point is still the global maximum. Convexity is sufficient for "local = global", not necessary.

(d) The set $\{(x, y) : x > 0,\ y > 0,\ xy \ge 1\}$ (the region above one branch of a hyperbola) is convex: it is $\{-\ln x - \ln y \le 0\}$, a sublevel set of a convex function.

Convex optimization problems

Convex optimization problem

$$\min\ f(\mathbf{x}) \quad \text{s.t.} \quad g_i(\mathbf{x}) \le 0\ (i = 1, \dots, m), \qquad \mathbf{A}\mathbf{x} = \mathbf{b},$$

with f and every g_i convex. (Equality constraints must be affine.)

Local equals global

For a convex optimization problem:

1. every local minimiser is a global minimiser;
2. the set of global minimisers is convex (a point, a segment, a face…);
3. if f is strictly convex, there is at most one minimiser;
4. a feasible $\mathbf{x}^*$ is optimal iff $\nabla f(\mathbf{x}^*)^{\mathsf{T}}(\mathbf{y} - \mathbf{x}^*) \ge 0$ for all feasible $\mathbf{y}$ (for differentiable f), and under Slater's condition iff it satisfies the KKT conditions.

Proof of 1. Suppose $\mathbf{x}^*$ is a local minimiser but some feasible $\mathbf{y}$ has $f(\mathbf{y}) < f(\mathbf{x}^*)$. The feasible set is convex, so $\mathbf{z}_\theta = \theta\mathbf{y} + (1-\theta)\mathbf{x}^*$ is feasible, and by convexity $f(\mathbf{z}_\theta) \le \theta f(\mathbf{y}) + (1-\theta) f(\mathbf{x}^*) < f(\mathbf{x}^*)$ for every $\theta \in (0, 1]$. As $\theta \to 0$, $\mathbf{z}_\theta$ comes arbitrarily close to $\mathbf{x}^*$ with a smaller value, contradicting local minimality. □

Proof of 3. If $\mathbf{x}_1 \ne \mathbf{x}_2$ were both minimisers with value f^*, strict convexity would give $f(\tfrac12\mathbf{x}_1 + \tfrac12\mathbf{x}_2) < f^*$ at a feasible point. □

Condition 4 says: at the optimum, no feasible direction goes downhill — the convex-set version of "the gradient is zero". In the interior of the feasible set it reduces to $\nabla f(\mathbf{x}^*) = \mathbf{0}$.

Lagrangian duality

LP duality (§23.3) is a special case of a construction that works for any problem. With the Lagrangian $\mathcal{L}(\mathbf{x}, \boldsymbol{\mu}, \boldsymbol{\lambda}) = f(\mathbf{x}) + \sum \mu_i g_i(\mathbf{x}) + \sum \lambda_k h_k(\mathbf{x})$, define the **dual function**

$$q(\boldsymbol{\mu}, \boldsymbol{\lambda}) = \inf_{\mathbf{x}}\ \mathcal{L}(\mathbf{x}, \boldsymbol{\mu}, \boldsymbol{\lambda}).$$

For $\boldsymbol{\mu} \ge \mathbf{0}$ and any feasible $\mathbf{x}$, $\mathcal{L}(\mathbf{x}, \boldsymbol{\mu}, \boldsymbol{\lambda}) \le f(\mathbf{x})$ (the added terms are ≤ 0), so $q(\boldsymbol{\mu}, \boldsymbol{\lambda}) \le f^*$: every dual-feasible point gives a lower bound. The **dual problem** is to find the best bound, $\max_{\boldsymbol{\mu} \ge 0, \boldsymbol{\lambda}} q$. As an infimum of affine functions of $(\boldsymbol{\mu}, \boldsymbol{\lambda})$, q is always concave — the dual is a convex problem even when the primal is not.

Weak and strong duality

Weak duality $q^* \le f^*$ always holds. **Strong duality** $q^* = f^*$ holds for convex problems that satisfy **Slater's condition**: there is a strictly feasible point, $g_i(\mathbf{x}_0) < 0$ for all i (affine inequalities need only be satisfied), with $\mathbf{A}\mathbf{x}_0 = \mathbf{b}$. The gap $f(\mathbf{x}) - q(\boldsymbol{\mu}, \boldsymbol{\lambda}) \ge 0$ between any primal-feasible and dual-feasible pair bounds the suboptimality of $\mathbf{x}$ — a **certificate** that solvers use as their stopping test.

Worked Example 23.25 — the dual of a small QP

Primal (Example 23.13): $\min x^2 + y^2$ s.t. $2 - x - 2y \le 0$, with $f^\ast = \tfrac{4}{5}$ at $(\tfrac{2}{5}, \tfrac{4}{5})$.

Step 1 (Lagrangian): $\mathcal{L} = x^2 + y^2 + \mu(2 - x - 2y)$.

Step 2 (minimise over x, y): $\partial\mathcal{L}/\partial x = 2x - \mu = 0$, $\partial\mathcal{L}/\partial y = 2y - 2\mu = 0$, so $x = \mu/2$, $y = \mu$. Substituting,

$$q(\mu) = \frac{\mu^2}{4} + \mu^2 + 2\mu - \frac{\mu^2}{2} - 2\mu^2 = 2\mu - \frac{5}{4}\mu^2.$$

Step 3 (dual problem): maximise over $\mu \ge 0$: $q'(\mu) = 2 - \tfrac{5}{2}\mu = 0 \Rightarrow \mu^\ast = \tfrac{4}{5}$, $q^\ast = \tfrac{8}{5} - \tfrac{4}{5} = \tfrac{4}{5}$.

Step 4 (check): $q^\ast = f^\ast = 0.8$ — strong duality, as Slater predicts (the point $(2, 2)$ is strictly feasible). The dual optimum $\mu^\ast = 0.8$ is the KKT multiplier of Example 23.13. Any μ gives a bound: $\mu = 0.5$ gives $q = 0.6875 \le 0.8$, certifying that no feasible point beats 0.6875.

The standard convex problem classes

Class	Objective	Constraints	Typical EE use
LP	linear	linear	dispatch, flows, minimax / L^1 fitting
QP	convex quadratic	linear	least squares with bounds, MPC, portfolio-style allocation
SOCP	linear	$\|\mathbf{A}_i\mathbf{x} + \mathbf{b}_i\| \le \mathbf{c}_i^\mathsf{T}\mathbf{x} + d_i$	robust beamforming, FIR design with magnitude bounds
SDP	linear	$\sum x_i \mathbf{F}_i \preceq \mathbf{G}$	Lyapunov stability, LMI control design, relaxations
GP (after log)	posynomial	posynomial ≤ 1, monomial $= 1$	component and circuit sizing (§23.10)

Each class contains the one above it. Interior-point methods solve all of them to high accuracy in a few tens of Newton steps, nearly independent of problem size; modelling tools (CVX, CVXPY) let you write the problem in mathematical form and verify convexity automatically through the composition rules above ("disciplined convex programming").

Common Pitfall — "convex" in the wrong variables

Convexity depends on the parametrisation. The gain-bandwidth product of an amplifier, a monomial like $c\,W^{0.5}L^{-1}I^{0.5}$ in transistor width, length and current, is neither convex nor concave in (W, L, I), but its logarithm is *linear* in $(\ln W, \ln L, \ln I)$. Before concluding a problem is hopelessly nonconvex, try logarithmic variables, epigraph variables, or a different but equivalent objective (e.g. minimise $\|\cdot\|^2$ instead of a ratio).

Engineering Note — what to do with a nonconvex problem

Many real problems (filter design with free phase, AC optimal power flow, neural-network training) are nonconvex. Practical strategies: (i) find a convex reformulation or relaxation (AC power flow → SDP or DC power flow); (ii) run a local method from many starting points and keep the best; (iii) use the convex solution of a simplified model as the starting point for the full model; (iv) use the dual bound to judge how far from optimal a local solution might be.

23.8 Least-squares optimization

Motivation

Least squares is the most-used optimization problem in engineering. Fitting a calibration line to sensor readings, estimating a resistance from noisy voltage and current samples, identifying the parameters of a transfer function from a measured frequency response, designing an FIR filter, solving an overdetermined system of channel equations, training a linear equaliser — all are "minimise a sum of squared errors". It is so prevalent for good reasons: the problem is convex, its solution is given by a linear system, it is the maximum-likelihood estimator under Gaussian noise (Ch. 20), and it extends naturally to weighted, regularised and nonlinear versions. Chapter 9 (§9.11) treated least squares geometrically as orthogonal projection; here we treat it as an optimization problem, add regularization, and handle models that are nonlinear in their parameters.

Linear least squares and the normal equations

Given $\mathbf{A} \in \mathbb{R}^{N \times n}$ ($N \ge n$ measurements, n unknowns) and data $\mathbf{b} \in \mathbb{R}^N$, the system $\mathbf{A}\mathbf{x} = \mathbf{b}$ generally has no solution. We minimise the squared residual norm

$$f(\mathbf{x}) = \|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2 = \mathbf{x}^\mathsf{T}\mathbf{A}^\mathsf{T}\mathbf{A}\mathbf{x} - 2\mathbf{b}^\mathsf{T}\mathbf{A}\mathbf{x} + \mathbf{b}^\mathsf{T}\mathbf{b}.$$

This is a quadratic with gradient $\nabla f = 2(\mathbf{A}^\mathsf{T}\mathbf{A}\mathbf{x} - \mathbf{A}^\mathsf{T}\mathbf{b})$ and Hessian $2\mathbf{A}^\mathsf{T}\mathbf{A} \succeq 0$, so it is convex and every stationary point is a global minimiser (§23.7).

Normal equations

$\mathbf{x}^\ast$ minimises $\|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2$ if and only if

$$\mathbf{A}^\mathsf{T}\mathbf{A}\,\mathbf{x}^\ast = \mathbf{A}^\mathsf{T}\mathbf{b}.$$

The minimiser is unique iff $\mathbf{A}$ has full column rank ($\operatorname{rank}\mathbf{A} = n$), in which case $\mathbf{A}^\mathsf{T}\mathbf{A} \succ 0$ and $\mathbf{x}^\ast = (\mathbf{A}^\mathsf{T}\mathbf{A})^{-1}\mathbf{A}^\mathsf{T}\mathbf{b}$. The optimal residual $\mathbf{r}^\ast = \mathbf{b} - \mathbf{A}\mathbf{x}^\ast$ is orthogonal to every column of $\mathbf{A}$: $\mathbf{A}^\mathsf{T}\mathbf{r}^\ast = \mathbf{0}$.

Proof. Setting $\nabla f = \mathbf{0}$ gives the normal equations; convexity makes them sufficient. For uniqueness, $\mathbf{z}^\mathsf{T}\mathbf{A}^\mathsf{T}\mathbf{A}\mathbf{z} = \|\mathbf{A}\mathbf{z}\|^2$ vanishes only for $\mathbf{z} \in \operatorname{null}\mathbf{A}$, which is $\{\mathbf{0}\}$ exactly when the columns are independent. The orthogonality statement is the normal equations rewritten. □

The orthogonality $\mathbf{A}^{\mathsf{T}}\mathbf{r}^{*} = \mathbf{0}$ is the geometry of Chapter 9: $\mathbf{A}\mathbf{x}^{*}$ is the orthogonal projection of $\mathbf{b}$ onto the column space of $\mathbf{A}$, and the residual is what is left over, perpendicular to that space. It is also a cheap correctness check on any fit.

Worked Example 23.26 — fitting a resistor's V–I line

A resistor with a small offset in the measurement chain gives the readings

I (mA)	1	2	3	4	5
V (V)	2.3	4.4	6.8	8.9	11.1

Fit $V = RI + V_0$.

Step 1 (matrix form): unknowns $\mathbf{x} = (R, V_0)$; row i of $\mathbf{A}$ is $(I_i, 1)$, $\mathbf{b} = \mathbf{V}$.

Step 2 (normal equations):

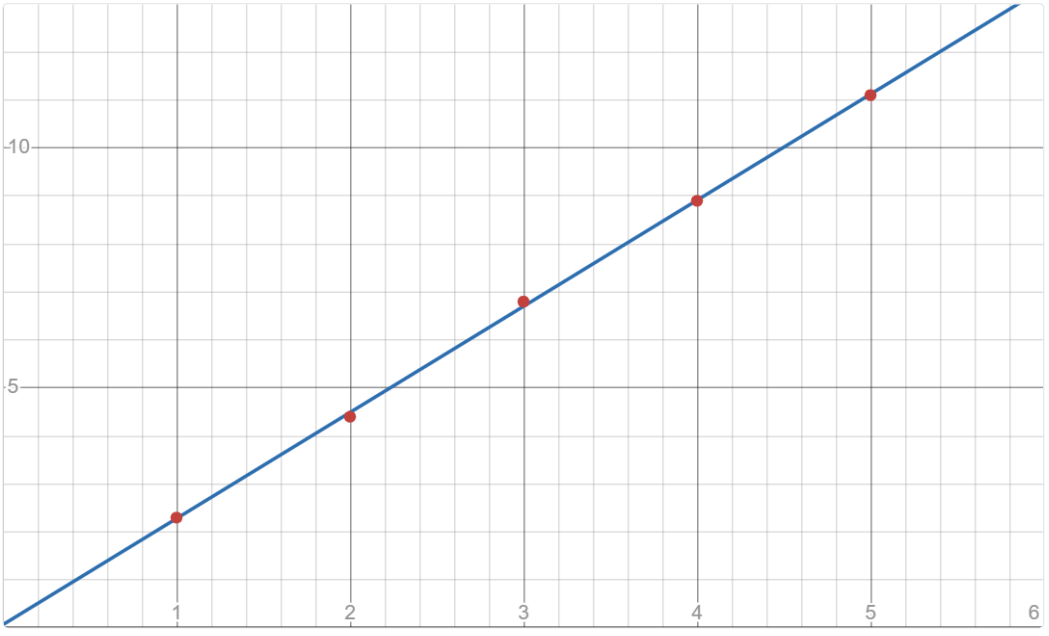
$$\mathbf{A}^{\mathsf{T}}\mathbf{A} = \begin{pmatrix} \sum I_i^2 & \sum I_i \\ \sum I_i & N \end{pmatrix} = \begin{pmatrix} 55 & 15 \\ 15 & 5 \end{pmatrix}, \qquad \mathbf{A}^{\mathsf{T}}\mathbf{b} = \begin{pmatrix} \sum I_i V_i \\ \sum V_i \end{pmatrix} = \begin{pmatrix} 122.6 \\ 33.5 \end{pmatrix}.$$

Step 3 (solve): determinant $55 \cdot 5 - 15^2 = 50$;

$$R = \frac{5(122.6) - 15(33.5)}{50} = 2.21, \qquad V_0 = \frac{55(33.5) - 15(122.6)}{50} = 0.07.$$

Units: R in V/mA $=$ kΩ, so $R = 2.21$ kΩ and $V_0 = 70$ mV.

Step 4 (check): residuals $V_i - (2.21 I_i + 0.07) = (0.02,\ -0.09,\ 0.10,\ -0.01,\ -0.02)$; $\sum r_i = 0$ ✓ and $\sum I_i r_i = 0.02 - 0.18 + 0.30 - 0.04 - 0.10 = 0$ ✓ (orthogonal to both columns). Sum of squares 0.019 V^2, so the rms residual is $\sqrt{0.019/3} = 80$ mV (dividing by $N - n = 3$ degrees of freedom).



Least-squares line $V = 2.21I + 0.07$ through the five measurements (current in mA, voltage in V). The fit minimises the sum of squared vertical distances.

Solving least squares reliably

The normal equations are perfect for derivations and for small, well-conditioned problems, but they have a numerical flaw: $\operatorname{cond}(\mathbf{A}^{\mathsf{T}}\mathbf{A}) = \operatorname{cond}(\mathbf{A})^2$. If $\mathbf{A}$ has condition number 10^6, forming $\mathbf{A}^{\mathsf{T}}\mathbf{A}$ in double precision (≈ 16 digits) leaves about 4 correct digits. Standard practice:

- **QR factorisation.** Write $\mathbf{A} = \mathbf{Q}\mathbf{R}$ with $\mathbf{Q}$ having orthonormal columns and $\mathbf{R}$ upper triangular (Gram–Schmidt or Householder, Ch. 9 §9.10). Since $\|\mathbf{A}\mathbf{x} - \mathbf{b}\|$ is unchanged by orthogonal transformations, the problem reduces to the triangular system $\mathbf{R}\mathbf{x} = \mathbf{Q}^{\mathsf{T}}\mathbf{b}$, whose sensitivity is governed by $\operatorname{cond}(\mathbf{A})$, not its square.
- **SVD.** With $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\mathsf{T}}$ (Ch. 10 §10.7), $\mathbf{x}^{*} = \sum_i \frac{\mathbf{u}_i^{\mathsf{T}}\mathbf{b}}{\sigma_i}\mathbf{v}_i$. Small singular values amplify the data component $\mathbf{u}_i^{\mathsf{T}}\mathbf{b}$ — including its noise — by $1/\sigma_i$. The SVD also gives the minimum-norm solution when $\mathbf{A}$ is rank deficient (the pseudoinverse $\mathbf{A}^{+}\mathbf{b}$), and it makes regularization transparent (below).

NumPy's `np.linalg.lstsq` uses the SVD; MATLAB's backslash uses QR.

Weighted least squares

If measurement i has noise variance σ_i^2, it deserves weight $w_i = 1/\sigma_i^2$:

$$\min_{\mathbf{x}} \sum_i w_i(\mathbf{a}_i^{\mathsf{T}}\mathbf{x} - b_i)^2 = (\mathbf{A}\mathbf{x} - \mathbf{b})^{\mathsf{T}}\mathbf{W}(\mathbf{A}\mathbf{x} - \mathbf{b}), \qquad \mathbf{A}^{\mathsf{T}}\mathbf{W}\mathbf{A}\,\mathbf{x}^{*} = \mathbf{A}^{\mathsf{T}}\mathbf{W}\mathbf{b}.$$

More generally, with noise covariance $\boldsymbol{\Sigma}$, take $\mathbf{W} = \boldsymbol{\Sigma}^{-1}$. By the Gauss–Markov theorem this gives the minimum-variance linear unbiased estimator, with covariance $(\mathbf{A}^{\mathsf{T}}\mathbf{W}\mathbf{A})^{-1}$ — the formula that error bars on fitted parameters come from. Weighted LS is ordinary LS on the "whitened" data $\mathbf{W}^{1/2}\mathbf{A}$, $\mathbf{W}^{1/2}\mathbf{b}$.

Equality-constrained least squares

To minimise $\|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2$ subject to $\mathbf{C}\mathbf{x} = \mathbf{d}$ (e.g. a filter with exactly unit DC gain, a fit forced through a known point), form the Lagrangian $\|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2 + \boldsymbol{\lambda}^{\mathsf{T}}(\mathbf{C}\mathbf{x} - \mathbf{d})$. Its stationarity conditions are the linear **KKT system**

$$\begin{pmatrix} 2\mathbf{A}^{\mathsf{T}}\mathbf{A} & \mathbf{C}^{\mathsf{T}} \\ \mathbf{C} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{x} \\ \boldsymbol{\lambda} \end{pmatrix} = \begin{pmatrix} 2\mathbf{A}^{\mathsf{T}}\mathbf{b} \\ \mathbf{d} \end{pmatrix}.$$

Because the problem is convex with affine constraints, any solution of this system is the global constrained optimum.

Regularization: ridge (Tikhonov) least squares

When $\mathbf{A}$ is ill-conditioned — nearly collinear regressors, a high-order polynomial fit, deconvolution, an inverse problem in imaging or field reconstruction — the least-squares solution is extremely sensitive to noise. The cure is to add a penalty on the size of $\mathbf{x}$:

Ridge (Tikhonov) regression

For $\lambda > 0$,

$$\mathbf{x}_\lambda = \arg\min_{\mathbf{x}} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2 + \lambda\|\mathbf{x}\|^2 \quad \Longleftrightarrow \quad (\mathbf{A}^{\mathsf{T}}\mathbf{A} + \lambda\mathbf{I})\,\mathbf{x}_\lambda = \mathbf{A}^{\mathsf{T}}\mathbf{b}.$$

More generally $\lambda\|\mathbf{L}\mathbf{x}\|^2$ with a matrix $\mathbf{L}$ (e.g. a difference operator, to penalise roughness rather than size).

Derivation. The gradient of the objective is $2(\mathbf{A}^{\mathsf{T}}\mathbf{A} + \lambda\mathbf{I})\mathbf{x} - 2\mathbf{A}^{\mathsf{T}}\mathbf{b}$. Since $\mathbf{A}^{\mathsf{T}}\mathbf{A} + \lambda\mathbf{I}$ has eigenvalues $\sigma_i^2 + \lambda \geq \lambda > 0$, it is always invertible — even if $\mathbf{A}$ is rank deficient — and the objective is 2λ-strongly convex, so $\mathbf{x}_\lambda$ exists and is unique. Equivalently, ridge is ordinary LS on the augmented system $\begin{pmatrix} \mathbf{A} \\ \sqrt{\lambda}\,\mathbf{I} \end{pmatrix}\mathbf{x} \approx \begin{pmatrix} \mathbf{b} \\ \mathbf{0} \end{pmatrix}$, which is how it is computed stably.

SVD view. Substituting $\mathbf{A} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^{\mathsf{T}}$,

$$\mathbf{x}_\lambda = \sum_i \underbrace{\frac{\sigma_i^2}{\sigma_i^2 + \lambda}}_{\text{filter factor } \phi_i} \frac{\mathbf{u}_i^{\mathsf{T}}\mathbf{b}}{\sigma_i}\,\mathbf{v}_i.$$

Components with $\sigma_i^2 \gg \lambda$ pass almost unchanged ($\phi_i \approx 1$); components with $\sigma_i^2 \ll \lambda$ — the ones that amplify noise — are suppressed ($\phi_i \approx \sigma_i^2/\lambda$). Ridge is a smooth low-pass filter in "singular-value space"; truncated SVD is the brick-wall version.

Trade-off. As λ grows, the residual $\|\mathbf{A}\mathbf{x}_\lambda - \mathbf{b}\|$ increases and the norm $\|\mathbf{x}_\lambda\|$ decreases monotonically: a bias–variance trade-off, and in fact a two-objective problem whose weighted-sum solutions trace a Pareto front (§23.9). The parameter is chosen by cross-validation, by the "corner" of the log–log plot of norm versus residual (the **L-curve**), or by the discrepancy principle (choose λ so that the residual matches the known noise level). Statistically, ridge is the MAP estimate when $\mathbf{x}$ has a Gaussian prior with variance $\sigma_{\text{noise}}^2/\lambda$.

Worked Example 23.27 — ridge stabilises a nearly collinear fit

Fit $\mathbf{x} = (x_1, x_2)$ to

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1.01 \\ 1 & 0.99 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 2 \\ 2.03 \\ 1.96 \end{pmatrix}.$$

The data were generated from $\mathbf{x}_{\text{true}} = (1, 1)$ with perturbations of order 0.02.

Step 1 (diagnose): singular values $\sigma_1 = 2.449$, $\sigma_2 = 0.0100$; $\operatorname{cond}(\mathbf{A}) = 245$ and $\operatorname{cond}(\mathbf{A}^{\mathsf{T}}\mathbf{A}) \approx 6 \times 10^4$. The second column is almost the first.

Step 2 (plain LS): $\mathbf{x}^* = (-1.503,\ 3.500)$ with residual norm 0.0041. The fit is excellent and the parameters are nonsense: the small noise has been amplified by $1/\sigma_2 = 100$ along $\mathbf{v}_2 \approx (1, -1)/\sqrt{2}$.

Step 3 (ridge):

λ	$\mathbf{x}_\lambda$	$\|\mathbf{A}\mathbf{x}_\lambda - \mathbf{b}\|$	$\phi_2 = \sigma_2^2/(\sigma_2^2 + \lambda)$
0	$(-1.503,\ 3.500)$	0.0041	1
10^{-3}	$(0.771,\ 1.226)$	0.0324	0.091
10^{-2}	$(0.972,\ 1.021)$	0.0357	0.0099
10^{-1}	$(0.979,\ 0.985)$	0.0669	0.001
1	$(0.855,\ 0.856)$	0.495	10^{-4}

Step 4 (interpret): a tiny $\lambda = 0.01$ (compared with $\sigma_1^2 = 6$) removes the unstable component and gives parameters within 3 % of the truth at the cost of a residual comparable to the noise level ($\|\mathbf{b} - \mathbf{A}\mathbf{x}_{\text{true}}\| = 0.028$). Too large a λ ($= 1$) starts shrinking the well-determined component as well — bias.

Nonlinear least squares

Many models are nonlinear in their parameters: an exponential decay $v(t) = a e^{-bt}$, a resonance $|H(j\omega)| = K/\sqrt{(1-\omega^2/\omega_0^2)^2 + (\omega/(Q\omega_0))^2}$, a diode law $I = I_s(e^{V/(nV_T)} - 1)$. With residuals $r_i(\mathbf{p}) = y_i - m(t_i;\mathbf{p})$ we minimise

$$F(\mathbf{p}) = \tfrac{1}{2}\sum_{i=1}^{N} r_i(\mathbf{p})^2 = \tfrac{1}{2}\|\mathbf{r}(\mathbf{p})\|^2.$$

With the residual Jacobian $\mathbf{J} = \partial \mathbf{r}/\partial \mathbf{p}$ ($N \times n$), the gradient and Hessian are

$$\nabla F = \mathbf{J}^{\mathsf T}\mathbf{r}, \qquad \nabla^2 F = \mathbf{J}^{\mathsf T}\mathbf{J} + \sum_i r_i \nabla^2 r_i.$$

The second Hessian term is small when the residuals are small (a good fit) or the model is nearly linear. Dropping it gives an approximation that needs only first derivatives and is automatically positive semidefinite.

Gauss–Newton and Levenberg–Marquardt

Gauss–Newton: solve the linear least-squares problem for the step,

$$(\mathbf{J}_k^{\mathsf T}\mathbf{J}_k)\,\boldsymbol{\delta}_k = -\mathbf{J}_k^{\mathsf T}\mathbf{r}_k, \qquad \mathbf{p}_{k+1} = \mathbf{p}_k + \boldsymbol{\delta}_k,$$

i.e. linearise the model about $\mathbf{p}_k$ and fit the linearised model exactly by least squares. **Levenberg–Marquardt:** $(\mathbf{J}_k^{\mathsf T}\mathbf{J}_k + \tau_k \mathbf{I})\boldsymbol{\delta}_k = -\mathbf{J}_k^{\mathsf T}\mathbf{r}_k$, with τ_k decreased after a successful step and increased after a failed one — a trust-region Gauss–Newton method (§23.6) that behaves like gradient descent far from the solution and like Gauss–Newton near it.

(If $\mathbf{r} = \mathbf{y} - \mathbf{m}$, then $\mathbf{J} = -\partial\mathbf{m}/\partial\mathbf{p}$; with the model Jacobian $\mathbf{J}_m = \partial\mathbf{m}/\partial\mathbf{p}$ the Gauss–Newton step reads $\mathbf{J}_m^{\mathsf T}\mathbf{J}_m\boldsymbol{\delta} = \mathbf{J}_m^{\mathsf T}\mathbf{r}$.) Gauss–Newton converges quadratically for zero-residual problems and linearly (often fast) for small-residual ones; it can fail on large-residual or badly initialised problems, which is when LM's damping matters. A good starting point is essential, and is often available from a linearising transformation.

Worked Example 23.28 — fitting an RC discharge by Gauss–Newton

A capacitor discharging through a resistor is sampled:

t (ms)	0	1	2	3	4
v (V)	5.0	3.0	1.9	1.1	0.7

Fit $v = a e^{-bt}$ (with $b = 1/\tau$).

Step 1 (starting guess by linearisation): $\ln v = \ln a - bt$ is linear. A straight-line LS fit of $\ln v_i$ against t_i gives $b = 0.4936\ \text{ms}^{-1}$, $\ln a = 1.6048$, $a = 4.977\ \text{V}$. (This minimises errors in $\ln v$, which over-weights the small late samples, so it is a starting point rather than the answer.)

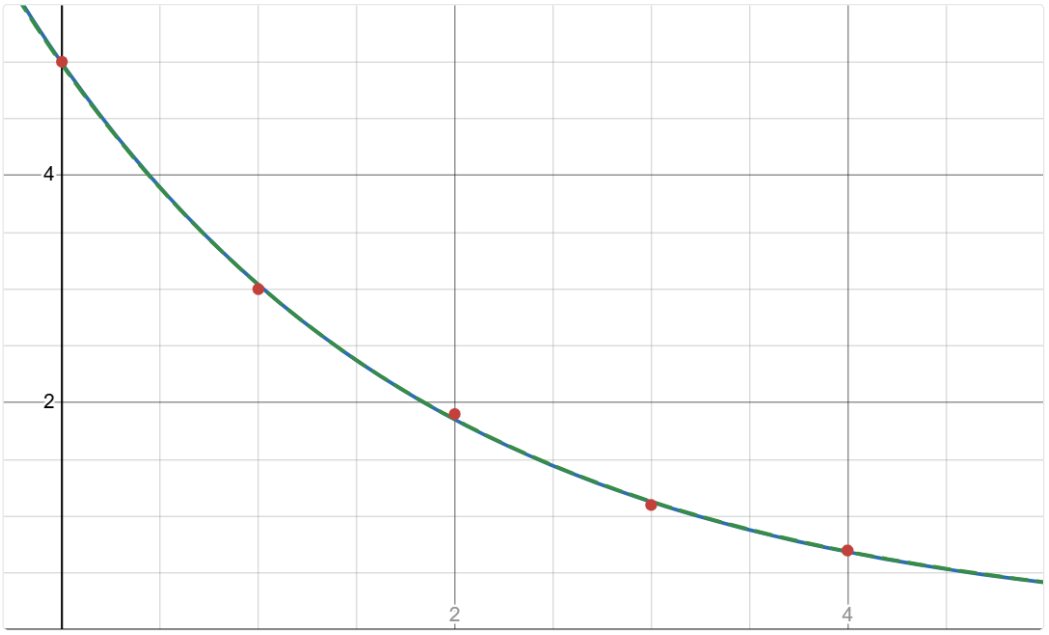
Step 2 (Jacobian): the model Jacobian has rows $\left(\partial m/\partial a,\ \partial m/\partial b\right) = \left(e^{-bt_i},\ -a t_i e^{-bt_i}\right)$.

Step 3 (iterate from a deliberately poor start $(a,b) = (4,\ 0.2)$):

k	a (V)	b (ms^{-1})	$\sum r_i^2$ (V^2)
0	4.0000	0.2000	4.09
1	4.8385	0.4319	0.142
2	4.9844	0.4920	0.00540
3	4.98933	0.495605	0.00496029
4	4.989372	0.4956304	0.00496027

Step 4 (result): $a = 4.989\ \text{V}$, $b = 0.4956\ \text{ms}^{-1}$, so $\tau = 1/b = 2.018\ \text{ms}$; residual rms $\sqrt{0.00496/3} = 41\ \text{mV}$. Starting from the log-linear guess, Gauss–Newton converges in two iterations.

Step 5 (check): at the solution $\mathbf{J}_m^{\mathsf T}\mathbf{r} = \mathbf{0}$ to machine precision — the nonlinear normal equations hold ✓.



RC discharge data (red), the nonlinear least-squares fit $4.989e^{-0.4956t}$ (blue) and the log-linear starting fit (green dashed); time in ms, voltage in V. The two curves nearly coincide here because the noise is small.

Common Pitfall — fitting in transformed coordinates

Fitting $\ln y$ instead of y, or $1/y$ instead of y, changes the error weighting: a 0.05 V error on a 0.7 V sample is a $7\,\%$ error in $\ln v$, but on a 5 V sample only $1\,\%$. Linearising transforms give excellent starting values; the final fit should minimise errors in the quantity that was actually measured with known noise (or use weights $w_i \propto y_i^2$ to compensate).

Engineering Note — system identification

Fitting a transfer function $H(s) = \frac{b_0 + b_1 s + \dots}{1 + a_1 s + \dots}$ to measured frequency-response data is nonlinear in the a_k. Levy's method multiplies through by the denominator to obtain a *linear* least-squares problem (an equation-error formulation); Sanathanan–Koerner iteration reweights it to approximate the true output-error criterion, and the result seeds a Levenberg–Marquardt refinement. Vector fitting, used for passive macromodels of interconnects, refines this idea with pole relocation.

```
import numpy as np
from scipy.optimize import least_squares

t = np.array([0, 1, 2, 3, 4.0]);  v = np.array([5.0, 3.0, 1.9, 1.1, 0.7])
res = least_squares(lambda p: v - p[0]*np.exp(-p[1]*t), x0=[4.0, 0.2], method="lm")
print(res.x)                                  # approx [4.989, 0.4956]

A = np.array([[1, 1], [1, 1.01], [1, 0.99]]);  b = np.array([2, 2.03, 1.96]);  lam = 1e-2
x_ridge = np.linalg.lstsq(np.vstack([A, np.sqrt(lam)*np.eye(2)]),
                          np.concatenate([b, np.zeros(2)]), rcond=None)[0]
print(x_ridge)                                # approx [0.972, 1.021]
```

23.9 Multi-objective optimization

Motivation

Real design specifications rarely reduce to one number. A low-noise amplifier should have low noise *and* low power consumption; a filter should have a sharp transition *and* low passband ripple *and* few taps; a power system should run at low cost *and* low emissions *and* high reliability; a regularised fit should have small residual *and* small parameter norm (§23.8). These goals conflict: improving one makes another worse. There is then no single "optimal" design but a family of best compromises, and the engineer's job is to *see* that family and choose from it. Multi-objective optimization provides the vocabulary (dominance, Pareto optimality) and the standard techniques (weighted sums, ε-constraints) that turn a vector of objectives back into ordinary single-objective problems.

Dominance and Pareto optimality

Consider $\min_{\mathbf{x} \in \mathcal{S}} \; \mathbf{F}(\mathbf{x}) = \big(f_1(\mathbf{x}), \dots, f_q(\mathbf{x})\big)$.

Pareto optimality

- A feasible $\mathbf{x}$ **dominates** $\mathbf{y}$ if $f_i(\mathbf{x}) \le f_i(\mathbf{y})$ for all i and $f_j(\mathbf{x}) < f_j(\mathbf{y})$ for at least one j: $\mathbf{x}$ is at least as good in every objective and strictly better in one.
- $\mathbf{x}^* \in S$ is **Pareto optimal** (efficient, non-dominated) if no feasible point dominates it. It is **weakly Pareto optimal** if no feasible point is strictly better in *every* objective.
- The **Pareto set** is the set of Pareto-optimal decisions; its image $\{\mathbf{F}(\mathbf{x}^*)\}$ in objective space is the **Pareto front** (trade-off curve).
- The **ideal (utopia) point** $\mathbf{F}^{\mathrm{id}}$ has components $f_i^{\mathrm{id}} = \min_S f_i$, each objective minimised on its own. It is usually not attainable.

At a Pareto-optimal point you cannot improve any objective without worsening another. A design that is *not* Pareto optimal is simply wasteful: some other design is better in at least one respect and worse in none. The first, uncontroversial, step of any trade-off study is therefore to discard dominated designs; only the choice *along* the front involves preferences.

The slope of a smooth two-objective front, $df_2/df_1 < 0$, is the **marginal rate of trade-off**: how many units of f_2 must be given up per unit of f_1 gained. Near the ends of the front this rate is often extreme — the last few percent of improvement in one objective is bought at great cost in the other — and a "knee" in the middle marks the region where a balanced compromise lives.

The weighted-sum method

Combine the objectives with nonnegative weights:

$$\min_{\mathbf{x}\in S} \sum_{i=1}^{q} w_i f_i(\mathbf{x}), \qquad w_i \ge 0, \;\; \sum w_i = 1.$$

Weighted sums and the Pareto front

1. If all $w_i > 0$, every minimiser of the weighted sum is Pareto optimal. (With some $w_i = 0$, it is at least weakly Pareto optimal.)
2. If the problem is **convex** (convex S and convex f_i), then for every Pareto-optimal $\mathbf{x}^*$ there are weights $w_i \ge 0$, not all zero, for which $\mathbf{x}^*$ minimises the weighted sum.
3. For nonconvex problems, Pareto-optimal points on non-convex parts of the front cannot be found by any choice of weights.

Proof of 1. If some $\mathbf{y}$ dominated a weighted-sum minimiser $\mathbf{x}^*$, then $\sum w_i f_i(\mathbf{y}) < \sum w_i f_i(\mathbf{x}^*)$ (strict because one f_j is strictly smaller and $w_j > 0$), contradicting minimality. □

Why 2 and 3 hold (two objectives). In the (f_1, f_2) plane the weighted sum $w_1 f_1 + w_2 f_2 = c$ is a straight line with slope $-w_1/w_2$. Minimising c slides this line toward the origin until it last touches the set of attainable objective vectors. The touching point is a point where the line **supports** the attainable set. For a convex problem the attainable set (plus everything dominated by it) is convex, and every boundary point of a convex set has a supporting line — so every front point is reachable. A part of the front that bulges *away* from the origin cannot be touched by a supporting line from below-left; the sliding line always jumps past it to the ends.

At a weighted-sum optimum on a smooth front, the line is tangent to the front, so the **front slope equals** $-w_1/w_2$: the weights *are* the trade-off rate one is willing to accept.

Common Pitfall — weights depend on units

$w_1 f_1 + w_2 f_2$ with f_1 in milliwatts and f_2 in $\mathrm{nV}^2/\mathrm{Hz}$ means nothing until the units are fixed; changing f_1 to watts changes the effective weight by 1000. Normalise each objective first, for example $\tilde{f}_i = (f_i - f_i^{\mathrm{id}})/(f_i^{\mathrm{nad}} - f_i^{\mathrm{id}})$ using the ideal point and the **nadir point** (worst value of f_i over the Pareto set), so that each normalised objective runs from 0 to 1 along the front. Even then, equal weights do not guarantee "equal importance"; sweep the weights and look at the front.

The ε-constraint method

Keep one objective and turn the others into constraints:

$$\min_{\mathbf{x}\in S} \; f_1(\mathbf{x}) \quad \text{subject to} \quad f_i(\mathbf{x}) \le \varepsilon_i, \;\; i = 2, \dots, q.$$

ε-constraint method

1. Every solution of the ε-constraint problem is weakly Pareto optimal; if it is unique, it is Pareto optimal.
2. Conversely, every Pareto-optimal $\mathbf{x}^*$ solves the ε-constraint problem with $\varepsilon_i = f_i(\mathbf{x}^*)$ — **convex or not**.
3. The KKT multiplier μ_i of the constraint $f_i \le \varepsilon_i$ gives the local trade-off rate: $\partial f_1^*/\partial \varepsilon_i = -\mu_i$.

Proof of 2. If some feasible $\mathbf{y}$ had $f_i(\mathbf{y}) \le f_i(\mathbf{x}^*)$ for $i \ge 2$ and $f_1(\mathbf{y}) < f_1(\mathbf{x}^*)$, it would dominate $\mathbf{x}^*$. □

The ε-constraint form is also how specifications are usually written ("minimise power subject to noise $\le$ spec"), so it connects directly to the design process: sweeping ε traces the front, and the multiplier tells the designer what each tightening of the specification costs. Its drawbacks are that it needs sensible ranges for ε_i and that each point requires a constrained solve.

Other scalarisations worth knowing: **goal programming / compromise programming** minimises the distance $\|\mathbf{F}(\mathbf{x}) - \mathbf{F}^{\mathrm{id}}\|$ (weighted, in some norm) to the ideal point; the **Chebyshev (min–max) scalarisation** $\min \max_i w_i(f_i - f_i^{\mathrm{id}})$ can reach every Pareto point even on nonconvex fronts. Population-based evolutionary algorithms (NSGA-II and relatives) approximate a whole front at once for difficult nonconvex problems.

Worked Example 23.29 — a convex front traced two ways

Objectives $f_1(x) = x^2$ and $f_2(x) = (x-2)^2$ over $x \in \mathbb{R}$ (two targets 0 and 2 that cannot both be met).

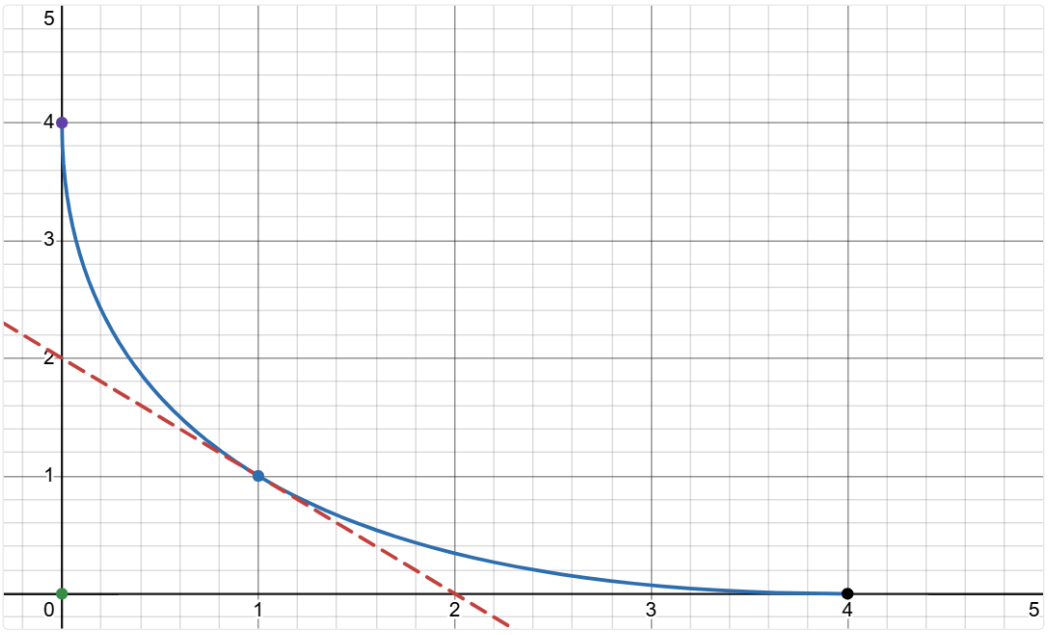
Step 1 (Pareto set): for $x < 0$, moving toward 0 improves both; for $x > 2$, moving toward 2 improves both. For $0 \le x \le 2$, increasing x helps f_2 and hurts f_1. Pareto set: $[0, 2]$. Ideal point $(0, 0)$, not attainable.

Step 2 (front): on $[0, 2]$, $x = \sqrt{f_1}$, so the front is $f_2 = (\sqrt{f_1} - 2)^2$, $0 \le f_1 \le 4$ — a convex curve from $(0, 4)$ to $(4, 0)$.

Step 3 (weighted sum): minimise $wx^2 + (1-w)(x-2)^2$: derivative $2wx + 2(1-w)(x-2) = 0$ gives $x^* = 2(1-w)$. As w runs from 1 to 0, x^* sweeps the whole Pareto set — every front point is reached, as the convex theory promises. For $w = \frac{1}{2}$: $x^* = 1$, $(f_1, f_2) = (1, 1)$.

Step 4 (ε-constraint): minimise x^2 s.t. $(x-2)^2 \le \varepsilon$. For $0 < \varepsilon < 4$ the constraint is active: $x^* = 2 - \sqrt{\varepsilon}$, $f_1^* = (2 - \sqrt{\varepsilon})^2$. With $\varepsilon = 1$: $x^* = 1$, $f_1^* = 1$ — the same point as $w = \frac{1}{2}$.

Step 5 (trade-off rate): $df_1^*/d\varepsilon = -(2 - \sqrt{\varepsilon})/\sqrt{\varepsilon} = -1$ at $\varepsilon = 1$, so $\mu = 1$. From KKT directly: $2x + \mu \cdot 2(x-2) = 0$ at $x = 1$ gives $\mu = 1$ ✓. And the weighted-sum tangent slope $-w/(1-w) = -1$ ✓: the multiplier and the weight ratio carry the same information.



Pareto front of $f_1 = x^2$, $f_2 = (x-2)^2$ (blue). The weighted-sum line for $w = \frac{1}{2}$ (red dashed, slope -1) is tangent to the front at $(1, 1)$; sweeping w rolls the tangent line along the whole convex front.

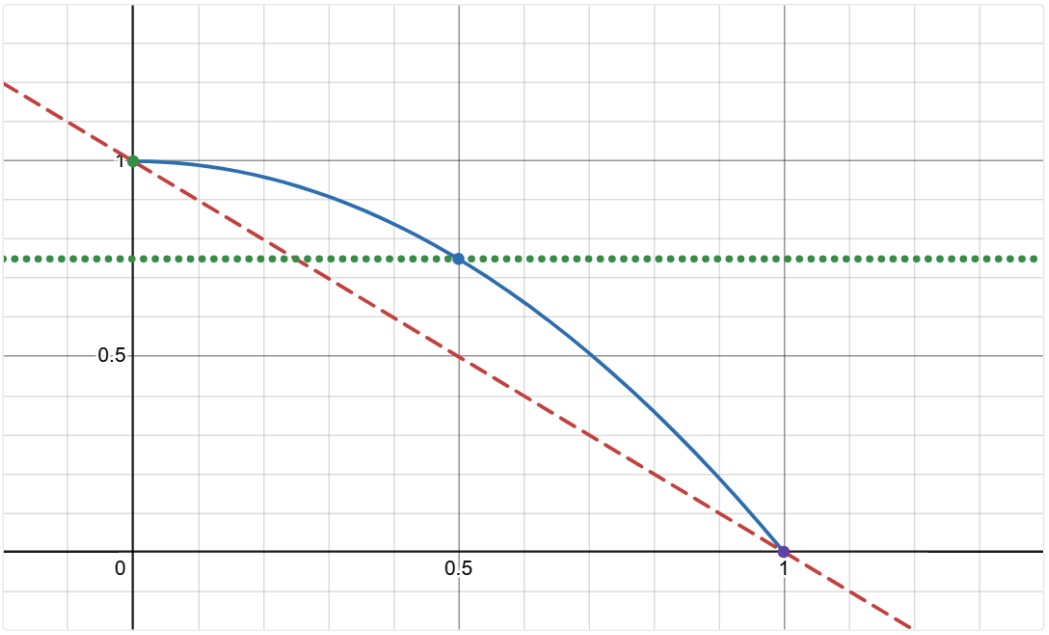
Worked Example 23.30 — a nonconvex front defeats the weighted sum

Objectives $f_1(x) = x$ and $f_2(x) = 1 - x^2$ on $S = [0, 1]$.

Step 1 (front): f_1 increases and f_2 decreases along S, so every $x \in [0, 1]$ is Pareto optimal and the front is $f_2 = 1 - f_1^2$, bulging *away* from the ideal point $(0, 0)$.

Step 2 (weighted sum): $\phi(x) = wx + (1-w)(1-x^2)$ has $\phi'' = -2(1-w) \le 0$: it is concave in x, so its minimum over $[0, 1]$ is at an endpoint. Comparing $\phi(0) = 1 - w$ and $\phi(1) = w$: the minimiser is $x = 1$ for $w < \frac{1}{2}$ and $x = 0$ for $w > \frac{1}{2}$. **Only the two end points of the front are ever found**, no matter how finely w is swept.

Step 3 (ε-constraint): minimise x s.t. $1 - x^2 \le \varepsilon$: $x^* = \sqrt{1 - \varepsilon}$. For $\varepsilon = 0.75$, $x^* = 0.5$ and $(f_1, f_2) = (0.5, 0.75)$, an interior front point unreachable by weights. Every front point is obtained as ε sweeps $[0, 1]$ ✓.



A nonconvex (concave-shaped) front $f_2 = 1 - f_1^2$. The equal-weight line (red dashed) touches only the end points; the ε-constraint $f_2 \le 0.75$ (green dotted) picks out the interior point $(0.5, 0.75)$.

Worked Example 23.31 — power versus noise in an amplifier stage

For a bipolar input stage biased at collector current I (mA) from a 3.3 V supply, model the power as $f_1 = 3.3I$ (mW) and the input voltage-noise density (shot-noise dominated) as $f_2 = 4/I$ (in nV^2/Hz), with $0.5 \le I \le 10$ mA.

Step 1 (front): f_1 rises and f_2 falls with I, so every feasible I is Pareto optimal. Eliminating I: $f_1 f_2 = 13.2$, a hyperbola from $(1.65, 8)$ to $(33, 0.4)$ — a convex front.

Step 2 (ε-constraint, the specification form): minimise $3.3I$ subject to $4/I \le 2$ (noise spec $\sqrt{2} = 1.41$ $\text{nV}/\sqrt{\text{Hz}}$). The constraint gives $I \ge 2$, so $I^* = 2$ mA and $f_1^* = 6.6$ mW.

Step 3 (multiplier): stationarity $3.3 + \mu(-4/I^2) = 0$ at $I = 2$ gives $\mu = 3.3$ mW per nV^2/Hz. Tightening the spec to $f_2 \le 1.9$ should cost about $0.1\mu = 0.33$ mW; the exact re-solve gives $I = 2.105$ mA, $f_1 = 6.95$ mW, an increase of 0.35 mW ✓.

Step 4 (weighted sum, normalised): with weights on the raw objectives, $w(3.3I) + (1-w)(4/I)$ is minimised at $I^* = \sqrt{4(1-w)/(3.3w)}$; $w = \frac{1}{2}$ gives $I^* = 1.10$ mA, $(f_1, f_2) = (3.63, 3.63)$ — a point that depends entirely on the arbitrary choice of units. Expressing the same requirement as a noise specification (Step 2) is usually more meaningful to the designer than choosing a weight.

Engineering Note — trade-off curves in design reviews

Presenting a Pareto front instead of a single "optimal" design changes design conversations: it makes explicit what each specification costs, reveals knees where a small concession buys a large gain, and exposes dominated legacy designs. Typical fronts in EE: noise figure versus power in LNAs, settling time versus overshoot in control loops, filter order versus transition width, conversion efficiency versus power density in converters, and residual versus solution norm (the L-curve) in regularised estimation.

23.10 Applications to engineering design

Motivation

The previous sections developed methods; this one shows how they fit together on problems that look like real design work. The pattern is always the same: (1) choose decision variables and put them in sensible units or logarithms; (2) write the objective and the specifications as functions; (3) recognise the problem class — LP, QP, convex, least squares, nonconvex; (4) solve with the matching method; (5) read the multipliers as sensitivities and check the design against the original, un-idealised requirements (discrete component values, tolerances, unmodelled effects). The three applications below — generator dispatch, component sizing, and FIR filter design — each exercise a different part of the chapter. They are motivation for the mathematics; the engineering details belong in power-systems, circuits and DSP texts.

Application 1: economic dispatch of generators

Generators $i = 1, \dots, G$ must together supply a demand D (MW). Each has a convex fuel-cost curve, commonly quadratic, $C_i(P_i) = a_i P_i^2 + b_i P_i + c_i$ (\$/\text{h}\$, with $a_i > 0$), and operating limits $P_i^{\min} \le P_i \le P_i^{\max}$. Losses are neglected here.

$$\min_{\mathbf{P}} \sum_{i=1}^{G} C_i(P_i) \quad \text{s.t.} \quad \sum_{i=1}^{G} P_i = D, \qquad P_i^{\min} \le P_i \le P_i^{\max}.$$

This is a convex QP: a convex separable objective, one affine equality and box constraints, so the KKT conditions are necessary and sufficient (§23.4, §23.7).

Without limits. The Lagrangian $\mathcal{L} = \sum C_i(P_i) + \lambda\big(D - \sum P_i\big)$ gives $\partial\mathcal{L}/\partial P_i = C_i'(P_i) - \lambda = 0$:

Equal incremental cost criterion

At the optimal unconstrained dispatch, every generator operates at the same **incremental cost** $dC_i/dP_i = \lambda$. For quadratic costs $P_i = (\lambda - b_i)/(2a_i)$, and the power balance gives

$$\lambda = \frac{D + \sum_i b_i/(2a_i)}{\sum_i 1/(2a_i)}.$$

The multiplier λ ($\$/\text{MWh}$) is the **system marginal cost**: $dC^*/dD = \lambda$.

The reason is the forgiving-optimum argument in reverse: if one unit had a lower incremental cost than another, shifting a little output from the expensive unit to the cheap one would lower total cost while keeping the balance.

With limits. With multipliers $\underline{\mu}_i, \overline{\mu}_i \ge 0$ for the lower and upper bounds, stationarity reads $C_i'(P_i) - \lambda - \underline{\mu}_i + \overline{\mu}_i = 0$. Complementary slackness then gives the textbook rule:

$$C_i'(P_i) = \lambda \text{ if } P_i^{\min} < P_i < P_i^{\max}, \qquad C_i'(P_i) \le \lambda \text{ if } P_i = P_i^{\max}, \qquad C_i'(P_i) \ge \lambda \text{ if } P_i = P_i^{\min}.$$

A unit at its maximum would *like* to produce more (its incremental cost is below the system price); the gap $\overline{\mu}_i = \lambda - C_i'(P_i^{\max})$ is the value of one more MW of its capacity.

Worked Example 23.32 — dispatching three units

Units with $a=(0.005,\ 0.008,\ 0.010)\ \$/\text{MW}^2\text{h}$, $b=(6.0,\ 6.5,\ 7.0)\ \$/\text{MWh}$ (fixed costs c_i do not affect the dispatch) supply $D = 700$ MW.

Step 1 (no limits): $\sum 1/(2a_i) = 100 + 62.5 + 50 = 212.5$ MW per $\$/\text{MWh}$, and $\sum b_i/(2a_i) = 600 + 406.25 + 350 = 1356.25$ MW. So

$$\lambda = \frac{700 + 1356.25}{212.5} = 9.676\ \$/\text{MWh}, \qquad P = \left(\frac{9.676 - 6}{0.01},\ \frac{9.676 - 6.5}{0.016},\ \frac{9.676 - 7}{0.02}\right) = (367.6,\ 198.5,\ 133.8)\ \text{MW}.$$

Check: sum $= 700.0$ ✓; incremental costs $2a_iP_i + b_i$ all equal 9.676 ✓. Variable cost $\sum(a_iP_i^2+b_iP_i)=5603.3\ \$/\text{h}$.

Step 2 (add the limit $P_1 \le 300$ MW): unit 1 violates it, so guess it is active: $P_1 = 300$ and units 2 and 3 share the remaining 400 MW at a common λ:

$$\lambda = \frac{400 + 406.25 + 350}{62.5 + 50} = 10.278\ \$/\text{MWh}, \qquad P_2 = 236.1,\ \ P_3 = 163.9\ \text{MW}.$$

Step 3 (verify KKT): unit 1's incremental cost at 300 MW is $2(0.005)(300) + 6 = 9.0 \le \lambda$ ✓, so $\overline\mu_1=10.278-9.0=1.278\ \$/\text{MWh}\ge0$ ✓. The limit raises the cost to $5646.5\ \$/\text{h}$ (+43.2) and raises the system marginal cost from 9.68 to 10.28 $\$/\text{MWh}$.

Step 4 (interpret): $\overline{\mu}_1$ says one more MW of capacity on unit 1 would save about $1.28 per hour — the information a planner needs to decide whether an uprate is worth it.

Worked Example 23.33 — linear costs give merit-order dispatch

If the same three units have *linear* costs $6.0,\ 6.5,\ 7.0\ \$/\text{MWh}$, each limited to 300 MW, and $D = 700$ MW, the problem is an LP. By the vertex theorem (§23.3) the optimum loads units fully in order of cost: $P = (300,\ 300,\ 100)$ MW, cost $1800+1950+700=4450\ \$/\text{h}$. The dual variable of the balance constraint is $\lambda=7.0\ \$/\text{MWh}$, the cost of the **marginal unit** (the only one not at a limit), and the capacity constraints of units 1 and 2 have shadow prices $7.0 - 6.0 = 1.0$ and $7.0-6.5=0.5\ \$/\text{MWh}$, by complementary slackness.

Application 2: sizing components — log variables and geometric programming

Component values are positive and span decades, and many circuit relationships are **monomials** $c\,x_1^{a_1}x_2^{a_2}\cdots$ (time constants RC, cutoff $1/(2\pi RC)$, kT/C noise, gain g_mR). Sums of monomials with positive coefficients are **posynomials**. A **geometric program** (GP) minimises a posynomial subject to posynomial ≤ 1 and monomial $= 1$ constraints. It is not convex as written, but in logarithmic variables $u_j = \ln x_j$:

- a monomial becomes $e^{\mathbf{a}^{\mathsf T}\mathbf{u} + \ln c}$, and the constraint "monomial $= 1$" becomes *linear*, $\mathbf{a}^{\mathsf T}\mathbf{u} + \ln c = 0$;
- a posynomial becomes a sum of exponentials of affine functions, which is convex; taking its logarithm gives the convex log-sum-exp form.

So every GP is a convex problem in disguise, with all the guarantees of §23.7. GP is used for sizing transistors in op-amps, wires in clock trees and gates in digital paths.

Worked Example 23.34 — sizing an RC low-pass filter

Choose R and C for a first-order low-pass with time constant $\tau = RC = 1$ ms (cutoff 159 Hz). Board area is modelled as $J = R/R_0 + C/C_0$ with $R_0 = 1\ \text{k}\Omega$ and $C_0 = 10$ nF (one unit of area per kilohm and per 10 nF). The integrated output noise $\sqrt{kT/C}$ at $T = 300$ K must not exceed 0.1 μV.

Step 1 (constraints in physical form): $kT = 4.14 \times 10^{-21}$ J, so $\sqrt{kT/C} \le 10^{-7}$ V $\iff C \ge kT/10^{-14} = 414$ nF.

Step 2 (log variables): let $u = \ln(R/R_0)$, $v = \ln(C/C_0)$. Then $RC = \tau$ becomes $u + v = \ln\big(\tau/(R_0C_0)\big) = \ln 100 = 4.605$, the noise constraint becomes $v \ge \ln 41.4 = 3.724$, and the objective is $J = e^u + e^v$ — convex. The problem is a convex program with one linear equality and one linear inequality.

Step 3 (ignore the noise constraint first): by symmetry of $e^u + e^v$ on the line $u + v = 4.605$, the optimum is $u = v = 2.303$: $R = 10\ \text{k}\Omega$, $C = 100$ nF, $J = 20$. But $C = 100$ nF violates $C \ge 414$ nF, so the noise constraint must be active.

Step 4 (active noise constraint): $v = 3.724$, $u = 4.605 - 3.724 = 0.881$: $C = 414$ nF, $R = \tau/C = 2.41\ \text{k}\Omega$, $J = 2.41 + 41.4 = 43.8$.

Step 5 (KKT check and sensitivity): with $\mathcal{L} = e^u + e^v + \lambda(u + v - 4.605) + \mu(3.724 - v)$: $\partial_u\mathcal{L} = e^u + \lambda = 0 \Rightarrow \lambda = -2.41$; $\partial_v\mathcal{L} = e^v + \lambda - \mu = 0 \Rightarrow \mu = 41.4 - 2.41 = 39.0 \ge 0$ ✓. In log variables μ is the cost per unit of $\ln C_{\min}$: a 1 % looser noise budget on $C_{\min}$ saves about 0.39 area units (re-solving with $C_{\min} = 410$ nF gives $J = 43.44$, a saving of 0.40 ✓). The noise specification is what drives the design.

Step 6 (round to preferred values): the nearest E12 capacitor not below 414 nF is 470 nF; then $R = \tau/C = 2.13\ \text{k}\Omega$, and the nearest E96 resistor 2.15 kΩ gives $\tau = 1.011$ ms (+1.1 %). The area becomes $2.15 + 47 = 49.2$ — 12 % above the continuous optimum, mostly because the E12 capacitor series is coarse. Rounding is a discrete optimisation; the continuous solution tells you where to look.

Application 3: FIR filter design by least squares and by linear programming

A linear-phase FIR low-pass filter of length $2M + 1$ has amplitude response $A(\omega) = h_0 + 2\sum_{k=1}^{M} h_k \cos k\omega$ (Example 23.3). Design means choosing $\mathbf{h}$ to approximate the ideal response $D(\omega)$.

Continuous least squares. Minimise $E(\mathbf{h}) = \int_0^\pi \big(A(\omega) - D(\omega)\big)^2 d\omega$. Because the functions $1, \cos\omega, \dots, \cos M\omega$ are orthogonal on $[0, \pi]$ — $\int_0^\pi \cos k\omega \cos l\omega\, d\omega = \frac{\pi}{2}\delta_{kl}$ for $k, l \ge 1$ and $\int_0^\pi 1\, d\omega = \pi$ — the normal equations are *diagonal*, and each coefficient is found independently:

$$h_0 = \frac{1}{\pi}\int_0^\pi D(\omega)\, d\omega, \qquad h_k = \frac{1}{\pi}\int_0^\pi D(\omega)\cos k\omega\, d\omega.$$

These are the Fourier-series coefficients of D (Ch. 15): **truncating a Fourier series is the least-squares approximation** by a trigonometric polynomial of that order. For an ideal low-pass with cutoff ω_c, $h_0 = \omega_c/\pi$ and $h_k = \sin(k\omega_c)/(k\pi)$ — the windowed-sinc design with a rectangular window.

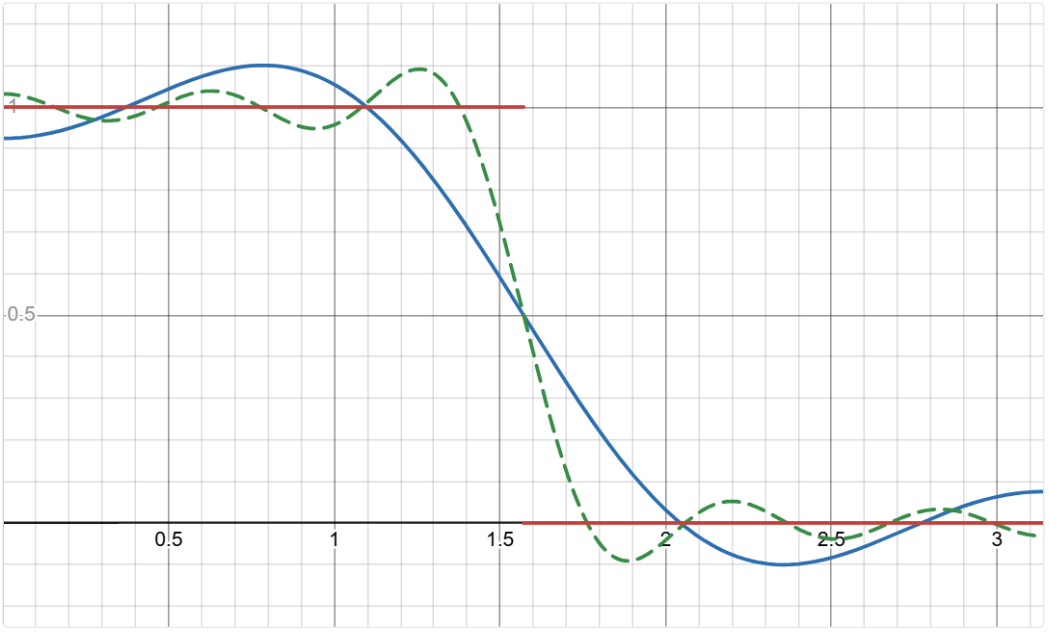
Worked Example 23.35 — a 7-tap least-squares half-band filter

Take $\omega_c = \pi/2$ and $M = 3$.

Step 1 (coefficients): $h_0 = \frac{1}{2}$, $h_1 = \sin(\pi/2)/\pi = 0.3183$, $h_2 = \sin\pi/(2\pi) = 0$, $h_3 = \sin(3\pi/2)/(3\pi) = -0.1061$. The impulse response is $(h_3, h_2, h_1, h_0, h_1, h_2, h_3) = (-0.106,\ 0,\ 0.318,\ 0.5,\ 0.318,\ 0,\ -0.106)$.

Step 2 (response): $A(\omega) = 0.5 + 0.6366\cos\omega - 0.2122\cos 3\omega$. Check $A(0) = 0.924$, $A(\pi/2) = 0.5$ (half amplitude at the cutoff, by symmetry), $A(\pi) = 0.076$.

Step 3 (error): the minimum integrated squared error is 0.078; the peak of A is 1.100 at $\omega = \pi/4$ — a 10 % overshoot. Raising M to 10 lowers the integrated error to 0.032 but the overshoot stays near 9 %: the **Gibbs phenomenon**. Least squares minimises average error, not peak error.



Least-squares approximations of an ideal half-band low-pass (red) by 7-tap (blue) and 21-tap (green dashed) linear-phase FIR filters, against normalised frequency ω in rad/sample. More taps sharpen the transition, but the overshoot near the edge (Gibbs) does not shrink.

Improving the formulation. Two changes to the *objective* do more than adding taps:

1. **Leave a transition band free.** Fit only on $[0, \omega_p] \cup [\omega_s, \pi]$ and solve the resulting discrete weighted least-squares problem on a frequency grid (§23.8). The jump disappears from the target, and with it most of the Gibbs overshoot.
2. **Change the norm.** Minimising the *peak* weighted error $\max_i |A(\omega_i) - D(\omega_i)|$ is an LP by the epigraph trick (§23.2); its solution is the **equiripple** (minimax) filter that the Parks–McClellan algorithm computes by a specialised exchange method.

Worked Example 23.36 — least squares versus minimax for a 21-tap filter

Specification: passband $[0, 0.4\pi]$, stopband $[0.6\pi, \pi]$, $M = 10$ (21 taps), 200 grid points in each band, equal weights.

Step 1 (least squares): `numpy.linalg.lstsq` on the 400×11 system gives a peak error of 0.0268, i.e. stopband peak $20 \log_{10} 0.0268 = -31.4$ dB.

Step 2 (minimax LP): minimise t over $(\mathbf{h}, t)$ subject to $-t \le A(\omega_i; \mathbf{h}) - D(\omega_i) \le t$ for all 400 grid points — 12 variables, 800 linear inequalities. `linprog` returns $t = 0.0114$: stopband peak -38.9 dB, about 7.5 dB better in the worst case, with the same number of taps.

Step 3 (moral): the minimax filter has larger *average* error and smaller *peak* error than the least-squares one. Which is "optimal" depends on whether the specification is an energy criterion or a mask — the point of §23.1.

A design-optimization checklist

1. **Variables.** What can actually be chosen? Use units of order one, or logarithms for positive quantities spanning decades.
2. **Objective.** What does "better" mean? Is it a proxy? If several goals conflict, plan a trade-off study (§23.9) rather than inventing weights.
3. **Constraints.** List every specification as $g \le 0$ or $h = 0$. Check feasibility first (Example 23.4). Remove redundant constraints.
4. **Classify.** Linear → LP; convex quadratic → QP; sum of squares → least squares; monomials → GP in log variables; otherwise check convexity with the toolkit of §23.7.
5. **Solve.** Use the matching solver. For nonconvex problems use several starting points, and seed them from a convex simplification.
6. **Verify.** Check KKT residuals, the sign of multipliers, and the duality gap. Perturb the data and re-solve: is the answer stable?
7. **Interpret.** Read multipliers as the price of each specification. Round to available values, re-check the specifications, and simulate the rounded design.

Common Pitfall — optimising the model, not the product

An optimizer drives the design to wherever the model is most flattering — often the edge of its validity (a transistor model used outside its fitted bias range, a linear cost curve far beyond the tested load). Add constraints that keep the design inside the model's domain of validity, and verify the final design with a higher-fidelity model or measurement.

Engineering Note — where these problems appear in tools

Optimal power flow and unit commitment in grid operation (LP/QP/MILP), automatic matching-network and filter synthesis in RF tools (nonlinear least squares and minimax), parameter extraction for device models (Levenberg–Marquardt), model predictive control running a QP every sample period, and analogue-circuit sizing by geometric programming are all direct applications of this chapter's methods.

Summary

Topic	Key result	Formula
Standard problem	minimise subject to inequalities and equalities	$\min f(\mathbf{x})$ s.t. $g_i(\mathbf{x}) \le 0,\ h_k(\mathbf{x}) = 0$
Existence	continuous f on a compact set, or coercive f	Weierstrass
Epigraph trick	minimax / L^1 to LP	$\min t$ s.t. $r_i(\mathbf{x}) \le t$
LP primal–dual	weak/strong duality	$\max \mathbf{c}^\mathsf{T}\mathbf{x},\ \mathbf{A}\mathbf{x} \le \mathbf{b},\ \mathbf{x} \ge \mathbf{0} \ \leftrightarrow\ \min \mathbf{b}^\mathsf{T}\mathbf{y},\ \mathbf{A}^\mathsf{T}\mathbf{y} \ge \mathbf{c},\ \mathbf{y} \ge \mathbf{0}$
Complementary slackness	spare capacity has zero price	$y_i(b_i - \mathbf{a}_i^\mathsf{T}\mathbf{x}) = 0,\ \mu_i g_i(\mathbf{x}) = 0$
Unconstrained optimality	FONC, SOSC	$\nabla f(\mathbf{x}^*) = \mathbf{0},\ \nabla^2 f(\mathbf{x}^*) \succ 0$
KKT stationarity	constrained optimality	$\nabla f + \sum \mu_i \nabla g_i + \sum \lambda_k \nabla h_k = \mathbf{0},\ \mu_i \ge 0$
Multiplier as sensitivity	shadow price	$\partial f^*/\partial b_i = -\mu_i$ for $g_i \le b_i$
Gradient descent	stable step, quadratic rate	$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha \nabla f,\ 0 < \alpha < 2/L,\ \rho = \dfrac{\kappa - 1}{\kappa + 1}$
Exact line search (quadratic)	step length	$\alpha_k = \dfrac{\mathbf{g}_k^\mathsf{T}\mathbf{g}_k}{\mathbf{g}_k^\mathsf{T}\mathbf{A}\mathbf{g}_k}$
Armijo condition	sufficient decrease	$f(\mathbf{x} + \alpha\mathbf{p}) \le f(\mathbf{x}) + c_1 \alpha \nabla f^\mathsf{T}\mathbf{p}$
Newton step	quadratic convergence	$\nabla^2 f(\mathbf{x}_k)\,\mathbf{p} = -\nabla f(\mathbf{x}_k)$
Secant condition	quasi-Newton	$\mathbf{H}_{k+1}\mathbf{y}_k = \mathbf{s}_k$
BFGS update	inverse Hessian	$\mathbf{H}_{k+1} = (\mathbf{I} - \rho \mathbf{s}\mathbf{y}^\mathsf{T})\mathbf{H}_k(\mathbf{I} - \rho \mathbf{y}\mathbf{s}^\mathsf{T}) + \rho \mathbf{s}\mathbf{s}^\mathsf{T},\ \rho = 1/\mathbf{y}^\mathsf{T}\mathbf{s}$
Convex function	chord above graph; tangent below	$f(\mathbf{y}) \ge f(\mathbf{x}) + \nabla f(\mathbf{x})^\mathsf{T}(\mathbf{y} - \mathbf{x});\ \nabla^2 f \succeq 0$
Convex problem	local = global; KKT sufficient	Slater ⇒ strong duality

Topic	Key result	Formula
Normal equations	linear least squares	$\mathbf{A}^{\mathsf{T}}\mathbf{A}\mathbf{x} = \mathbf{A}^{\mathsf{T}}\mathbf{b}$, $\mathbf{A}^{\mathsf{T}}\mathbf{r}^{*} = \mathbf{0}$
Ridge regression	regularised LS	$(\mathbf{A}^{\mathsf{T}}\mathbf{A} + \lambda\mathbf{I})\mathbf{x} = \mathbf{A}^{\mathsf{T}}\mathbf{b}$, filter factors $\sigma_i^2/(\sigma_i^2+\lambda)$
Gauss–Newton	nonlinear LS	$\mathbf{J}^{\mathsf{T}}\mathbf{J}\boldsymbol{\delta} = -\mathbf{J}^{\mathsf{T}}\mathbf{r}$ (LM: $+\tau\mathbf{I}$)
Weighted sum	Pareto point (convex fronts only)	$\min \sum w_i f_i$, front slope $= -w_1/w_2$
ε-constraint	any Pareto point	$\min f_1$ s.t. $f_i \le \varepsilon_i$, $\partial f_1^{*}/\partial \varepsilon_i = -\mu_i$
Economic dispatch	equal incremental cost	$C_i'(P_i) = \lambda$ for units within limits

Further reading

- S. Boyd and L. Vandenberghe, *Convex Optimization*, Cambridge University Press — the standard text on convex sets, functions, duality and problem classes, with many engineering applications (freely available from the authors).
- J. Nocedal and S. J. Wright, *Numerical Optimization*, 2nd ed., Springer — line searches, trust regions, Newton, quasi-Newton (BFGS, L-BFGS), nonlinear least squares and constrained algorithms in depth.
- D. G. Luenberger and Y. Ye, *Linear and Nonlinear Programming*, Springer — simplex method, LP duality, and the convergence theory of gradient and Newton methods.
- A. J. Wood, B. F. Wollenberg and G. B. Sheblé, *Power Generation, Operation, and Control*, Wiley — economic dispatch, optimal power flow and their optimization formulations.

Chapter 24 — Computational Mathematics

> **Prerequisites:** Ch. 8 (elimination, LU, norms, condition number, sparse storage, Jacobi and Gauss–Seidel), Ch. 9 (least squares, QR), Ch. 10 (eigenvalues, SVD), Ch. 19–Ch. 20 (random variables, variance, estimation), Ch. 22 (floating point, numerical error, Newton's method). **Learning outcomes:** After this chapter you can
>
> - estimate the cost of an algorithm in flops and memory traffic, read and use big-O notation, and restructure a computation (reordering, vectorization) to change its cost;
> - decide when a symbolic computation is worth doing, recognise expression swell, and combine symbolic derivation with numerical evaluation;
> - compute QR and SVD factorisations, count their cost, and use backward error and the condition number to judge whether a computed answer can be trusted;
> - choose and tune iterative solvers (SOR, conjugate gradients, GMRES) and preconditioners, and predict their convergence from spectral information;
> - store and factor sparse matrices efficiently, controlling fill-in by reordering;
> - design Monte Carlo experiments with quantified error and variance reduction, fit nonlinear models by Gauss–Newton and Levenberg–Marquardt, propagate uncertainty by linearization (GUM), Monte Carlo and adjoint sensitivities, and make every such experiment reproducible.

Chapters 8 and 22 taught the *building blocks* of numerical computation: how a floating-point number is stored, how Gaussian elimination works, how Newton's method converges, how a system is ill-conditioned. This chapter is about *putting them together at scale*. A post-layout transistor-level simulation solves a sparse linear system with a million unknowns thousands of times; a finite-element model of a motor or an antenna produces systems with tens of millions; a yield study runs the same simulation ten thousand times with randomly perturbed component values; a device model is calibrated by fitting twenty parameters to a few thousand measured points. In every case the questions are the same. How much work will this take, and how does the work grow with problem size? Can I trust the answer, and how much of the error comes from the data, the model, the algorithm and the arithmetic? Could someone else — or I, in six months — get the same answer again?

The chapter therefore mixes three kinds of material: **cost** (Sections 24.1, 24.5), **algorithms with guaranteed accuracy** (24.2–24.4, 24.7) and **statistical computation** (24.6, 24.8), finishing with the discipline that ties them together, **reproducibility** (24.9). Short Python fragments using NumPy, SciPy and SymPy illustrate each idea; they are written to be read, not just run, and every number quoted in a worked example has been checked by such a script.

24.1 Algorithmic thinking

Motivation

Two programs that compute the same mathematical quantity can differ in running time by a factor of a million. The difference rarely comes from clever micro-optimisation; it comes from the *algorithm* — the sequence of operations chosen — and from how that sequence interacts with the hardware. A discrete Fourier transform of 2^{20} samples computed from the definition takes about 10^{12} complex multiply–adds; the FFT takes about 10^8 flops. Multiplying three matrices in the wrong order can cost a thousand times more than in the right order. Algorithmic thinking is the habit of asking, *before* writing code, "how does the work grow with n?" and "is there an equivalent formulation that does less?"

Measuring cost

Flop, cost model

A **flop** is one floating-point addition, subtraction, multiplication or division. The **arithmetic cost** of an algorithm is the number of flops it performs as a function of the problem size n (number of unknowns, samples, grid points). The **memory cost** is the number of words it stores; the **communication cost** is the number of words moved between levels of the memory hierarchy (main memory, cache, registers, or between processors).

Flop counts are the classical measure and remain the right first estimate. On modern hardware, however, moving a number from main memory costs far more time than an arithmetic operation on it — typically 10–100 flop-times. An algorithm whose ratio of flops to memory words (its **arithmetic intensity**) is high, like dense matrix–matrix multiplication ($2n^3$ flops on $3n^2$ words), runs near peak speed; one with intensity $O(1)$, like a sparse matrix–vector product (2 nnz flops on about nnz words), is limited by memory bandwidth no matter how fast the processor is.

Asymptotic notation

We usually care about how cost grows for large n, not about exact counts.

Big-O, big-Omega, big-Theta

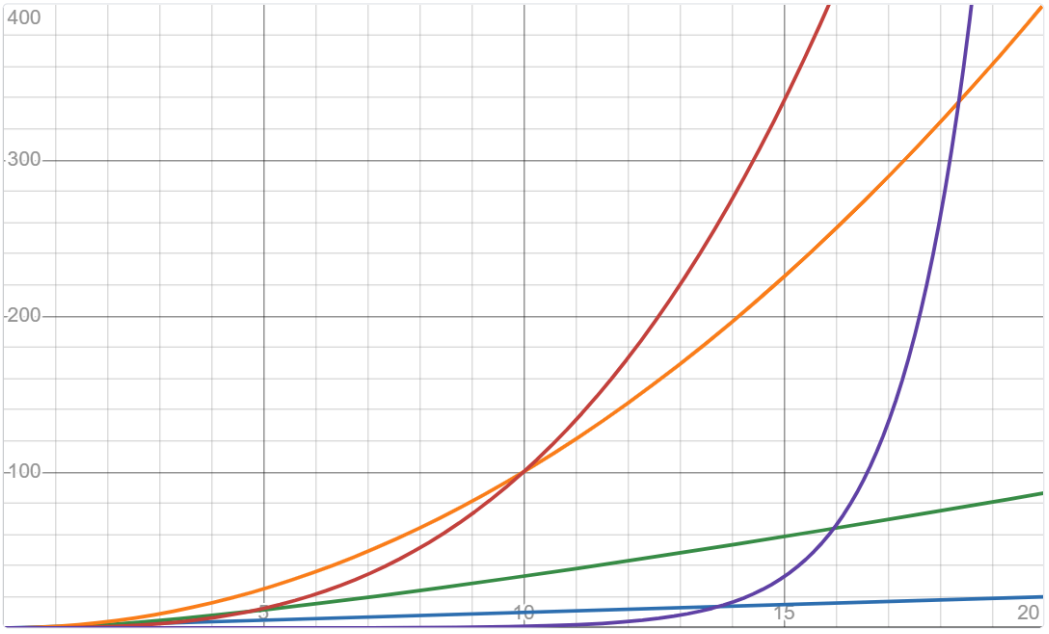
For positive functions $f(n)$, $g(n)$:

- $f = O(g)$ if there are constants $c > 0$, n_0 with $f(n) \le c\,g(n)$ for all $n \ge n_0$ (grows *no faster* than g);
- $f = \Omega(g)$ if $f(n) \ge c\,g(n)$ for $n \ge n_0$ (grows *at least as fast*);
- $f = \Theta(g)$ if both hold (grows *at the same rate*).

We also write $f \sim g$ if $f(n)/g(n) \to 1$; this keeps the leading constant, as in "LU costs $\sim \tfrac{2}{3}n^3$ flops".

Big-O hides constants, so it answers *scaling* questions ("if I double n, how much longer?") rather than *absolute* ones ("which of two $O(n^3)$ algorithms is faster?"). For the latter we keep the leading term with its constant, $\sim c\,n^p$. Typical growth classes in numerical work are:

Class	Example
$O(1)$	one evaluation of a closed-form formula
$O(\log n)$	bisection to fixed absolute accuracy on an interval of length n; binary search
$O(n)$	vector addition, inner product, tridiagonal (Thomas) solve, one sparse matrix–vector product
$O(n \log n)$	FFT, sorting, nested-dissection fill for 2-D grids
$O(n^2)$	dense matrix–vector product, triangular solve, direct DFT
$O(n^3)$	dense LU, QR, SVD, eigenvalues, matrix–matrix product
$O(2^n)$, $O(n!)$	exhaustive search over subsets or orderings; cofactor-expansion determinant



Growth of n (blue), $n \log_2 n$ (green), n^2 (orange), $n^3/10$ (red) and $2^n/1000$ (purple). The constants are deliberately chosen so that the "worse" algorithms start out faster: for small n any of them may win, but beyond a crossover the exponent (or the exponential) always dominates.

Empirical scaling exponent

If a cost behaves as $T(n) \approx c\, n^p$, then measurements at two sizes give

$$p \approx \frac{\ln\big(T(n_2)/T(n_1)\big)}{\ln(n_2/n_1)},$$

the slope of T against n on log–log axes. Predicting the time at a new size is then $T(n_3) \approx T(n_2)\,(n_3/n_2)^p$.

This is the single most useful back-of-envelope tool in computational work: time a code at two modest sizes, compute p, and you know whether the production run will take minutes or months — and whether the code really has the complexity you think it has.

Recurrences and divide-and-conquer

Many fast algorithms split a problem of size n into a subproblems of size n/b and combine the results with $O(n^d)$ extra work, giving the recurrence $T(n) = a\,T(n/b) + O(n^d)$.

Master theorem (simplified)

For $T(n) = a\,T(n/b) + \Theta(n^d)$ with $a \ge 1$, $b > 1$:

$$T(n) = \begin{cases} \Theta(n^d) & \text{if } d > \log_b a, \\ \Theta(n^d \log n) & \text{if } d = \log_b a, \\ \Theta(n^{\log_b a}) & \text{if } d < \log_b a. \end{cases}$$

Sketch. Unrolling the recursion gives $\log_b n$ levels; level ℓ has a^ℓ subproblems each costing $(n/b^\ell)^d$, i.e. total $n^d (a/b^d)^\ell$ at that level. This is a geometric series in ℓ with ratio a/b^d: if the ratio is below 1 the top level dominates, if it equals 1 all $\log_b n$ levels contribute equally, and if it exceeds 1 the $a^{\log_b n} = n^{\log_b a}$ leaves dominate. $\square$

The radix-2 FFT has $a = b = 2$, $d = 1$, so $T = \Theta(n \log n)$. Strassen's matrix multiplication replaces eight half-size products by seven, $a = 7$, $b = 2$, $d = 2$, giving $\Theta(n^{\log_2 7}) = \Theta(n^{2.807})$.

Worked Example 24.1 — Polynomial evaluation: naive versus Horner

Count the flops needed to evaluate $p(x) = a_0 + a_1 x + \cdots + a_n x^n$ (a) by computing each power x^k from scratch, (b) by accumulating powers, (c) by Horner's rule $p = a_0 + x(a_1 + x(a_2 + \cdots + x\,a_n))$.

(a) The term $a_k x^k$ needs k multiplications ($k-1$ for the power and one for the coefficient), so the total is $\sum_{k=1}^{n} k = \tfrac{1}{2}n(n+1)$ multiplications plus n additions: $\Theta(n^2)$.

(b) Keeping $x^k = x \cdot x^{k-1}$ costs one multiplication per power and one per coefficient: $2n$ multiplications and n additions, $3n$ flops.

(c) Horner performs one multiplication and one addition per coefficient: $2n$ flops — optimal, since any method must at least touch every coefficient.

Check, $n = 3$, $x = 2$, $p = 1 - 3x + 0x^2 + 2x^3$**.** Horner: $b_3 = 2$, $b_2 = 0 + 2 \cdot 2 = 4$, $b_1 = -3 + 2 \cdot 4 = 5$, $b_0 = 1 + 2 \cdot 5 = 11$; directly $1 - 6 + 0 + 16 = 11$ ✓. For a degree-20 filter polynomial the counts are 230, 60 and 40 flops. Horner is also the most accurate of the three in floating point, and its intermediate values b_k are the coefficients of the deflated polynomial $p(x)/(x - x_0)$ — two benefits for the price of one loop.

Worked Example 24.2 — Predicting a run time from two measurements

A dense solver takes 0.21 s for $n = 2000$ and 1.62 s for $n = 4000$. Estimate the scaling exponent and the time for $n = 20\,000$. Would $n = 10^6$ (a modest post-layout netlist) be feasible?

Exponent. $p = \ln(1.62/0.21)/\ln 2 = \ln 7.714/0.6931 = 2.043/0.6931 \approx 2.95$ — consistent with an $O(n^3)$ LU factorisation.

Prediction. $T(20\,000) \approx 1.62 \times (20\,000/4000)^{2.95} = 1.62 \times 5^{2.95} = 1.62 \times 115.5 \approx 187$ s, about three minutes.

Extrapolation to 10^6**.** $T \approx 1.62 \times (250)^3 \approx 2.5 \times 10^7$ s, about ten months — and the matrix alone would need $8n^2 = 8 \times 10^{12}$ bytes. Dense methods are out; the problem must be solved by exploiting sparsity (Section 24.5), where the cost for circuit matrices is typically close to $O(n^{1.1\text{–}1.5})$.

Reading. The measured $p = 2.95$ rather than exactly 3 is normal: at moderate n lower-order terms and cache effects still contribute. Always measure at sizes large enough that the leading term dominates, and treat the prediction as an order of magnitude.

Worked Example 24.3 — Associativity changes the cost

Let $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times n}$ and $\mathbf{x} \in \mathbb{R}^n$. Compare the cost of $(\mathbf{AB})\mathbf{x}$ and $\mathbf{A}(\mathbf{Bx})$ for $n = 1000$. Then consider $\mathbf{y} = (\mathbf{I} + \mathbf{u}\mathbf{v}^{\mathsf{T}})\mathbf{x}$.

Products. A matrix–matrix product costs $2n^3$ flops (n^2 entries, each an inner product of length n: n multiplications and $n-1$ additions); a matrix–vector product costs $2n^2$.

- $(\mathbf{AB})\mathbf{x}$: $2n^3 + 2n^2 \approx 2.002 \times 10^9$ flops.
- $\mathbf{A}(\mathbf{Bx})$: $2n^2 + 2n^2 = 4 \times 10^6$ flops.

The results are mathematically identical; the second is 500 times cheaper, and the ratio grows as $n/2$.

Rank-one update. Forming $\mathbf{I} + \mathbf{u}\mathbf{v}^{\mathsf{T}}$ explicitly and multiplying costs n^2 (outer product) $+ 2n^2$ (product) flops and n^2 words of memory. Writing instead $\mathbf{y} = \mathbf{x} + \mathbf{u}(\mathbf{v}^{\mathsf{T}}\mathbf{x})$ costs $2n$ (inner product) $+ 2n$ (scaled add) $= 4n$ flops and no extra storage. For $n = 10^6$ the explicit matrix would not even fit in memory.

Rule. Never form a matrix you only need to multiply by. The same rule appears as "never form $\mathbf{A}^{-1}$" (Ch. 8), "never form $\mathbf{A}^{\mathsf{T}}\mathbf{A}$" (Section 24.3) and "never form the dense Jacobian" (Section 24.8).

Vectorization and the memory hierarchy

Interpreted languages such as Python or MATLAB execute an explicit loop body at perhaps 10^7 iterations per second; the same operation expressed as a whole-array operation runs in compiled, cache-friendly library code at 10^9–10^{11} flops per second. **Vectorization** means expressing an algorithm in terms of whole-array operations — elementwise arithmetic, reductions, matrix products — so that the inner loops run inside optimised libraries (BLAS, LAPACK, FFTW).

```
import numpy as np
rng = np.random.default_rng(0)
t = np.linspace(0.0, 1e-3, 1_000_000)          # 1 ms, 1 M samples
# loop version: ~1 s in CPython
v = np.empty_like(t)
for k in range(t.size):
    v[k] = 5.0 * (1.0 - np.exp(-t[k] / 1e-4))
# vectorized version: ~10 ms
v = 5.0 * (1.0 - np.exp(-t / 1e-4))
```

The libraries are themselves organised in three levels. **Level-1 BLAS** (vector–vector, $O(n)$ work on $O(n)$ data), **level-2** (matrix–vector, $O(n^2)$ on $O(n^2)$) and **level-3** (matrix–matrix, $O(n^3)$ on $O(n^2)$). Only level-3 operations have enough arithmetic intensity to reuse data in cache, so high-performance LU, QR and Cholesky are written as **blocked algorithms**: the matrix is partitioned into $b \times b$ tiles (with b of order 100) and most of the work is recast as tile–tile products. This is why `numpy.linalg.solve` on a 4000×4000 matrix runs at a large fraction of the processor's peak, while a hand-written triple loop runs a thousand times slower *with the same flop count*.

Worked Example 24.4 — Direct DFT versus FFT, and a vectorized DFT

A spectrum is needed for a record of $N = 2^{20} \approx 1.05 \times 10^6$ samples. Compare the costs of the direct DFT $X_k = \sum_{n=0}^{N-1} x_n e^{-j2\pi kn/N}$ and the radix-2 FFT.

Direct DFT. N outputs, each a sum of N complex products: $N^2 \approx 1.1 \times 10^{12}$ complex multiply–adds, about 8×10^{12} real flops (a complex multiply–add is 4 real multiplications and 4 additions).

FFT. About $5N \log_2 N = 5 \times 2^{20} \times 20 \approx 1.05 \times 10^8$ real flops.

Ratio. $8N^2/(5N \log_2 N) = 8N/(5 \log_2 N) \approx 8.4 \times 10^4$. At 10^{10} flops per second the DFT takes about 13 minutes, the FFT about 10 ms.

Vectorization is not a substitute for a better algorithm. The DFT can be vectorized as a matrix–vector product, `X = np.exp(-2j*np.pi*np.outer(k, n)/N) @ x`, which is fast for $N = 1024$ — but the matrix has N^2 entries (16 TB in complex double precision for $N = 2^{20}$). Vectorization reduces the constant; only the FFT reduces the exponent.

Common Pitfall

Big-O comparisons hold only *asymptotically*. Insertion sort ($O(n^2)$) beats merge sort for $n \lesssim 20$, which is why library sorts switch to it for small subarrays; Strassen's algorithm only wins for n in the thousands and is less accurate; a sparse solver can be slower than a dense one for a 50×50 matrix. Conversely, a "fast" algorithm whose constant is small but whose exponent is large will lose eventually. Measure at realistic sizes, and report both the exponent and the constant.

Engineering Note

SPICE-class simulators spend most of their time in two places: evaluating device models (an $O(n)$ loop over transistors, vectorized and parallelized in modern simulators) and solving the sparse linear system at each Newton iteration. For large digital blocks the solve dominates; for analog blocks with complex transistor models the evaluation dominates. Profiling to find out *which* — before optimising anything — is itself part of algorithmic thinking. The same holds for an FPGA or DSP implementation: a filter whose coefficients are evaluated by Horner's rule needs n multipliers rather than $n(n+1)/2$, and an FFT-based convolution beats direct convolution once the filter length exceeds roughly 64 taps.

Accuracy is part of the algorithm

Cost is not the only property of an algorithm. Two algorithms with the same flop count can deliver very different accuracy: the textbook variance formula $\frac{1}{N}\sum x_i^2 - \bar{x}^2$ and the two-pass formula $\frac{1}{N}\sum (x_i - \bar{x})^2$ both cost $O(N)$, but the first suffers catastrophic cancellation when the mean is large compared with the spread (Ch. 22). An algorithm is judged by the triple **(cost, accuracy, robustness)** — and the next sections develop the tools (backward error, condition number, convergence rate) for the second and third.

Worked Example 24.5 — One-pass variance: cost versus accuracy

Ten resistor readings near 10 kΩ are recorded as $x_i = 10\,000 + \delta_i$ Ω with $\delta_i \in \{0.1, 0.2, \ldots, 1.0\}$. Compute the variance in single precision ($\varepsilon \approx 6 \times 10^{-8}$) by the one-pass and two-pass formulas.

Exact. The δ_i are $0.1k$, $k = 1, \ldots, 10$, with mean 0.55 and population variance $0.01 \times \mathrm{Var}(k) = 0.01 \times 8.25 = 0.0825\ \Omega^2$.

One-pass. $\frac{1}{N}\sum x_i^2 \approx 1.0000110 \times 10^8$; each x_i^2 is about 10^8, and single precision represents numbers near 10^8 only to the nearest multiple of 8. The difference $\frac{1}{N}\sum x_i^2 - \bar{x}^2$ is therefore a difference of two numbers each uncertain by several units, whose true value is 0.0825: the result is pure rounding noise. Run in NumPy `float32`, it returns -8.0 — a *negative* variance.

Two-pass. Subtracting $\bar{x} \approx 10\,000.55$ first leaves residuals between -0.45 and 0.45 with small relative error; the result is 0.08254, which agrees to five digits with the variance of the stored `float32` values computed in double precision. (It differs from 0.0825 in the fourth digit only because $10\,000.1$ etc. are themselves stored with errors up to 5×10^{-4}.)

Welford's algorithm obtains the two-pass accuracy in one pass by updating the mean and the sum of squared deviations recursively, $\bar{x}_k = \bar{x}_{k-1} + (x_k - \bar{x}_{k-1})/k$, $M_k = M_{k-1} + (x_k - \bar{x}_{k-1})(x_k - \bar{x}_k)$ — the method of choice for streaming data from an ADC.

24.2 Symbolic versus numerical computation

Motivation

A computer can manipulate mathematics in two quite different ways. A **computer algebra system** (CAS) such as SymPy, Mathematica or Maple manipulates *expressions*: it can differentiate $\ln(1 + e^{x/V_T})$, factor $s^3 + 2s^2 + 2s + 1$, or solve a nodal system with symbolic R and C and return a transfer function. A **numerical** library such as NumPy, SciPy or LAPACK manipulates *floating-point numbers*: it evaluates, factorises and integrates, quickly and approximately. Engineers need both, and the skill is knowing which to use for which part of a problem. Symbolic results give insight (which component controls the pole?), exactness (no rounding) and formulas that can be reused; numerical results give speed and scale. Each fails in characteristic ways, and the best workflows combine them.

Two models of computation

Symbolic and numerical computation

Symbolic computation represents mathematical objects exactly — integers of arbitrary length, rationals, algebraic numbers such as $\sqrt{2}$, symbols, and expression trees built from them — and transforms them by rewriting rules (expansion, simplification, differentiation, Gröbner bases, symbolic integration). **Numerical computation** represents real numbers by fixed-precision floating-point approximations (Ch. 22) and evaluates algorithms whose every step is a rounded arithmetic operation. **Exact numeric** (arbitrary-precision rational or integer) arithmetic and **arbitrary-precision floating point** (e.g. `mpmath` with 50 digits) are intermediate options.

The trade-off is between three resources:

	Symbolic / exact	Floating point
Cost per operation	grows with the *size* of the operands (digits, terms)	constant (one machine instruction)
Error	none (or explicit)	rounding, $\lesssim \varepsilon \approx 1.1 \times 10^{-16}$ per operation in double precision
Output	formula valid for all parameter values	number valid for one parameter value
Scale	tens of unknowns	millions of unknowns

Expression swell

The central failure mode of symbolic computation is that intermediate and final expressions can grow without bound even when the inputs are small.

Expression swell

Expression swell is the growth in size (number of terms, nesting depth, or number of digits) of the expressions produced during a symbolic or exact computation, so that time and memory grow much faster than the operation count of the corresponding floating-point algorithm.

Three sources recur.

1. **Combinatorial growth.** The determinant of a general symbolic $n \times n$ matrix has $n!$ terms in expanded form: 24 for $n = 4$, 3 628 800 for $n = 10$. Floating-point LU computes the numerical value in $\tfrac{2}{3}n^3 \approx 670$ flops for $n = 10$.
2. **Digit growth.** Exact rational arithmetic roughly doubles the number of digits at each multiplication. Gaussian elimination on an integer matrix, done naively over the rationals, can produce entries whose length grows exponentially with n (fraction-free Bareiss elimination keeps the growth polynomial, but still far from $O(1)$ per operation).
3. **Closed-form blow-up.** Formulas that exist in principle can be useless in practice. A quartic has closed-form roots, but SymPy's expression for a single root of $x^4 + x + 1$ already contains 72 operations of nested radicals; by the Abel–Ruffini theorem degree five and above have no general radical formula at all.

Worked Example 24.6 — Digit growth in exact Newton iteration

Apply Newton's method $x_{k+1} = \tfrac{1}{2}(x_k + 2/x_k)$ for $\sqrt{2}$ from $x_0 = 1$ in exact rational arithmetic, and compare with floating point.

Iterates. $x_1 = \tfrac{3}{2}$, $x_2 = \tfrac{17}{12}$, $x_3 = \tfrac{577}{408}$, $x_4 = \tfrac{665\,857}{470\,832}$, …

Errors and sizes.

k	1	2	3	4	5	6	7
digits in numerator	1	2	3	6	12	25	49
error $x_k - \sqrt{2}$	8.6×10^{-2}	2.5×10^{-3}	2.1×10^{-6}	1.6×10^{-12}	$\approx 10^{-24}$	$\approx 10^{-48}$	$\approx 10^{-97}$

Quadratic convergence doubles the number of correct digits each step; in exact arithmetic it *also* doubles the number of digits stored, so the cost of each step doubles too. In double precision the iteration reaches full accuracy at $k = 5$ and every step costs the same three flops.

Reading. Exactness was not buying anything: after step 5 the extra digits exceed any physical meaning. When the answer is going to be used as a number, use floating point (or arbitrary precision with a *fixed* number of digits) and control the error analytically.

Worked Example 24.7 — Exact versus floating-point inversion of Hilbert matrices

The Hilbert matrix $\mathbf{H}_n$, $h_{ij} = 1/(i + j - 1)$, is the classic ill-conditioned matrix of Ch. 8. Compare exact rational and floating-point inversion.

Exact. $\mathbf{H}_n^{-1}$ has *integer* entries, computable exactly in SymPy. Their size grows rapidly: the largest entry is 192 for $n = 3$, 179 200 for $n = 5$, 4.25×10^9 for $n = 8$ and 3.48×10^{12} for $n = 10$.

Floating point. The 2-norm condition numbers are $\kappa_2(\mathbf{H}_5) = 4.8 \times 10^5$, $\kappa_2(\mathbf{H}_8) = 1.5 \times 10^{10}$, $\kappa_2(\mathbf{H}_{10}) = 1.6 \times 10^{13}$. The worst-case rule of thumb "relative error $\lesssim \kappa\varepsilon$" allows about 2×10^{-6} for $n = 8$, 2×10^{-3} for $n = 10$ and order 1 for $n = 12$ ($\kappa_2 \approx 2 \times 10^{16}$); the measured relative errors of `numpy.linalg.inv` against the exact inverse are 2.1×10^{-9}, 8.8×10^{-5} and 3.1×10^{-2} — better than the bound, as is usual, but growing at the same exponential rate.

Which wins? If the matrix entries are *exact data* (as in a proof, or a combinatorial identity), exact arithmetic is the right tool for small n. If the entries come from measurements with three significant figures, exact inversion is meaningless: the problem itself amplifies data errors by κ, and no arithmetic can recover what the data do not contain. The cure is reformulation (an orthogonal polynomial basis instead of monomials, Section 24.3), not more digits.

What symbolic computation does well

- **Deriving formulas once, evaluating many times.** A symbolic transfer function, a Jacobian for Newton's method, or a sensitivity $\partial y/\partial p$ derived symbolically and then compiled to numerical code (SymPy `lambdify`, code generation to C) is exact in form and fast in evaluation.
- **Insight into parameter dependence.** A symbolic pole location shows which component moves it; a series expansion shows the leading behaviour.
- **Removing cancellation before it happens.** Rewriting an expression analytically (e.g. $1-\cos x = 2\sin^2(x/2)$) eliminates catastrophic cancellation that no amount of numerical care can fix afterwards.
- **Verification.** Symbolic results provide exact references against which numerical code is tested (Section 24.9).

What numerical computation does well

- Problems with **no closed form**: roots of degree-5 polynomials, nonlinear ODEs, most integrals of measured or tabulated data.
- **Large problems**: 10^6 unknowns, where any symbolic representation is hopeless.
- Problems whose **data are inexact anyway**: measurements, tolerances, fitted models.
- Problems whose answer is wanted **only as a number** at specific parameter values.

Worked Example 24.8 — Cancellation removed symbolically

Evaluate $f(x) = \dfrac{1-\cos x}{x^2}$ at $x = 10^{-4}$ and $x = 10^{-8}$ in double precision. Fix it with a symbolic rewrite.

Direct evaluation. At $x = 10^{-8}$, $\cos x = 1 - 5\times10^{-17}$ rounds to exactly 1 in double precision (the spacing of doubles just below 1 is 1.1×10^{-16}), so the numerator is 0 and $f = 0$. The true value is $\tfrac12$: 100% error. At $x = 10^{-4}$ the direct formula gives $0.499\,999\,996\,96$, relative error 6×10^{-9} — eight digits lost to cancellation.

Symbolic rewrite. $1-\cos x = 2\sin^2(x/2)$, so $f(x) = \dfrac{2\sin^2(x/2)}{x^2} = \dfrac12\left(\dfrac{\sin(x/2)}{x/2}\right)^2$. There is no subtraction; the evaluation gives 0.5 at $x = 10^{-8}$ and $0.499\,999\,999\,583$ at $x = 10^{-4}$, correct to all digits shown.

Series check. SymPy's `series((1-cos(x))/x**2, x, 0, 6)` returns $\tfrac12 - \tfrac{x^2}{24} + \tfrac{x^4}{720} + O(x^6)$; at $x = 10^{-4}$, $\tfrac12 - 4.17\times10^{-10} = 0.499\,999\,999\,583$ ✓.

Lesson. The symbolic step costs nothing at run time and removes the error at its source. This is the typical successful division of labour: algebra *before* the numbers.

Worked Example 24.9 — A hybrid workflow: symbolic transfer function, numerical evaluation

A two-section RC ladder (series R, shunt C, series R, shunt C) is driven by v_{in}. Derive $H(s) = V_2/V_{\text{in}}$ symbolically, then evaluate $|H(j\omega)|$ numerically over 1000 frequencies.

Symbolic nodal analysis. With node voltages V_1 (first capacitor) and V_2 (output) and $V_{\text{in}} = 1$,

$$\begin{bmatrix} \frac{2}{R}+sC & -\frac{1}{R} \\ -\frac{1}{R} & \frac{1}{R}+sC \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{R} \\ 0 \end{bmatrix}.$$

```
import sympy as sp, numpy as np
s, R, C = sp.symbols('s R C', positive=True)
G = sp.Matrix([[2/R + s*C, -1/R], [-1/R, 1/R + s*C]])
V = G.LUsolve(sp.Matrix([1/R, 0]))
H = sp.factor(sp.simplify(V[1]))      # 1/(C**2*R**2*s**2 + 3*C*R*s + 1)
Hnum = sp.lambdify((s, R, C), H, 'numpy')
w = np.logspace(1, 6, 1000)
mag = np.abs(Hnum(1j*w, 1e3, 100e-9))  # vectorized over w
```

Result. $H(s) = \dfrac{1}{R^2C^2s^2 + 3RCs + 1}$. The symbolic form tells us immediately that the poles are at $s = -\dfrac{3\pm\sqrt{5}}{2RC}$, i.e. $-0.382/RC$ and $-2.618/RC$ (the same golden-ratio numbers that appear in infinite ladders): the loading of the first section by the second splits the double pole of two buffered sections apart by a factor $2.618^2 \approx 6.85$.

Numerical check. With $R = 1\ \text{k}\Omega$, $C = 100\ \text{nF}$ ($RC = 10^{-4}$ s), at $\omega = 10^4$ rad/s: $sRC = j$, so $H = 1/(-1+3j+1) = 1/(3j)$, $|H| = 1/3$ and the phase is -90° — what `Hnum(1j*1e4, 1e3, 100e-9)` returns.

Why hybrid? The symbolic solve of a 2×2 system is instant and gives insight; repeating it for a 40-node filter would produce an enormous rational function that is slower to evaluate, and less accurate, than a numerical sparse solve at each frequency. The rule: go symbolic for small subsystems whose *structure* you want to understand; go numerical for the full system.

Choosing between them

Engineering Note — a decision checklist

1. **Is a formula the deliverable** (a design equation, a sensitivity expression, a model to hand to someone else)? Use symbolic computation, on the smallest subsystem that captures the effect.
2. **Are there more than about ten coupled unknowns, or nonlinear equations without closed-form solution?** Use numerical methods.
3. **Will the expression be evaluated many times?** Derive symbolically, simplify, then generate numerical code (`lambdify` , `codegen`) — or use automatic differentiation (Section 24.8), which gives exact derivatives without symbolic expression swell.
4. **Is the computation losing digits to cancellation?** Rewrite symbolically first.
5. **Are the data exact integers or rationals and the size small** (combinatorics, coding theory, exact filter coefficients)? Exact arithmetic is appropriate.
6. **Is the problem ill-conditioned?** Neither exact arithmetic nor more digits fixes ill-conditioned *data*; reformulate.

Common Pitfall — floats leaking into symbolic code

In SymPy, `sp.Rational(1, 10)` is exactly $\frac{1}{10}$, but `0.1` is the binary float $0.1000000000000000055\ldots$ and turns the whole expression into floating-point arithmetic, silently losing exactness (and `sp.nsimplify` may not recover the intended rational). Similarly, `sp.sqrt(2)` is exact but `np.sqrt(2)` is not. Conversely, never call `simplify` inside a loop over thousands of parameter values: simplification is heuristic and can take seconds; derive once, `lambdify` , then loop numerically.

Common Pitfall — symbolic solutions with hidden assumptions

A CAS returns *generic* answers. Solving $ax = b$ symbolically gives $x = b/a$, valid only for $a \neq 0$; the symbolic inverse of a matrix ignores the parameter values at which the determinant vanishes. When a symbolic result will be evaluated across a parameter range, check the denominators (SymPy `fraction` , `solve(..., dict=True)` with assumptions) or verify numerically at the extremes of the range.

24.3 Numerical linear algebra

Motivation

Almost every large computation in engineering ends in linear algebra: Newton's method for a nonlinear circuit solves a linear system per iteration; a least-squares fit factorises a tall matrix; a modal analysis computes eigenvalues; a model-order reduction or a beamformer computes an SVD. Chapter 8 developed Gaussian elimination, LU and Cholesky factorisation, norms and the condition number; Chapter 9 introduced Gram–Schmidt and QR for least squares; Chapter 10 introduced the SVD. This section asks the computational questions those chapters postponed: *how* are QR and the SVD computed in floating point, what do they cost, and in what precise sense are the answers "correct"?

Forward error, backward error and stability

Let f be the exact map from data to answer (for a linear system, $\mathbf{b} \mapsto \mathbf{A}^{-1}\mathbf{b}$) and $\hat{f}$ the computed version.

Forward error, backward error

The **forward error** of a computed answer $\hat{\mathbf{y}} = \hat{f}(\mathbf{x})$ is $\|\hat{\mathbf{y}} - f(\mathbf{x})\|$ (absolute) or $\|\hat{\mathbf{y}} - f(\mathbf{x})\|/\|f(\mathbf{x})\|$ (relative). The **backward error** is the size of the smallest perturbation of the *data* that makes the computed answer exact:

$$\eta(\hat{\mathbf{y}}) = \min\left\{ \frac{\|\Delta\mathbf{x}\|}{\|\mathbf{x}\|} \;:\; \hat{\mathbf{y}} = f(\mathbf{x} + \Delta\mathbf{x}) \right\}.$$

An algorithm is **backward stable** if for every input its backward error is $O(\varepsilon)$ — that is, it computes the exact answer to a slightly perturbed problem, with the perturbation comparable to rounding the data.

The two are linked by the condition number κ of the *problem*:

Rule of thumb: forward error ≤ condition × backward error

To first order,

$$\frac{\|\hat{\mathbf{y}} - \mathbf{y}\|}{\|\mathbf{y}\|} \lesssim \kappa \cdot \eta.$$

In particular, a backward-stable algorithm applied to a problem with condition number κ delivers a relative forward error of about $\kappa\varepsilon$: roughly $\log_{10}\kappa$ significant digits are lost, and no algorithm can do systematically better with data known only to precision ε.

This separates responsibilities cleanly. **Conditioning** is a property of the problem — the circuit, the measurement geometry, the choice of basis — and is the engineer's business. **Stability** is a property of the algorithm and is the library writer's business. A backward-stable algorithm is "as good as the data deserve".

For linear systems the backward error has a computable formula (Rigal–Gaches): for an approximate solution $\hat{\mathbf{x}}$ with residual $\mathbf{r} = \mathbf{b} - \mathbf{A}\hat{\mathbf{x}}$, the smallest normwise perturbation $\Delta\mathbf{A}$ with $(\mathbf{A} + \Delta\mathbf{A})\hat{\mathbf{x}} = \mathbf{b}$ has

$$\eta_A(\hat{\mathbf{x}}) = \frac{\|\mathbf{r}\|}{\|\mathbf{A}\|\,\|\hat{\mathbf{x}}\|}.$$

A *small residual* therefore certifies a *small backward error* — but, as Ch. 8 showed, not a small forward error.

Worked Example 24.10 — Small residual, large error

$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1.0001 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 2 \\ 2.0001 \end{bmatrix}$, exact solution $\mathbf{x} = (1, 1)^{\mathsf{T}}$. A candidate answer is $\hat{\mathbf{x}} = (2, 0)^{\mathsf{T}}$. Compute its residual, backward error and forward error in the ∞-norm, and check the rule of thumb.

Residual. $\mathbf{A}\hat{\mathbf{x}} = (2, 2)^{\mathsf{T}}$, so $\mathbf{r} = (0,\ 0.0001)^{\mathsf{T}}$, $\|\mathbf{r}\|_\infty = 10^{-4}$.

Backward error. $\|\mathbf{A}\|_\infty = 2.0001$, $\|\hat{\mathbf{x}}\|_\infty = 2$: $\eta = 10^{-4}/(2.0001 \times 2) = 2.5 \times 10^{-5}$. The wrong answer is the *exact* answer to a problem whose matrix differs from $\mathbf{A}$ in the fifth significant digit.

Forward error. $\hat{\mathbf{x}} - \mathbf{x} = (1, -1)^{\mathsf{T}}$, relative error $1/1 = 100\%$.

Condition number. $\mathbf{A}^{-1} = 10^4 \begin{bmatrix} 1.0001 & -1 \\ -1 & 1 \end{bmatrix}$, $\|\mathbf{A}^{-1}\|_\infty = 2.0001 \times 10^4$, so $\kappa_\infty = 4.0004 \times 10^4$. Then $\kappa\eta = 4.0 \times 10^4 \times 2.5 \times 10^{-5} = 1.0$ ✓ — the bound is attained.

Reading. If the matrix entries are resistor values known to 0.01%, both $(1, 1)$ and $(2, 0)$ are equally valid answers. The singular values $\sigma_1 = 2.00005$, $\sigma_2 = 5.0 \times 10^{-5}$ show why: the direction $\mathbf{v}_2 \approx (1, -1)/\sqrt{2}$ is almost in the null space.

Householder QR

Gram–Schmidt (Ch. 9) builds $\mathbf{Q}$ column by column by subtracting projections. In floating point, *classical* Gram–Schmidt loses orthogonality in proportion to $\kappa(\mathbf{A})^2\varepsilon$ and *modified* Gram–Schmidt in proportion to $\kappa(\mathbf{A})\varepsilon$. Production codes instead apply a sequence of orthogonal transformations to $\mathbf{A}$ that introduce zeros below the diagonal — orthogonal *triangularisation* rather than triangular *orthogonalisation*.

Householder reflector

For a nonzero vector $\mathbf{v} \in \mathbb{R}^m$, the **Householder reflector**

$$\mathbf{H} = \mathbf{I} - 2\frac{\mathbf{v}\mathbf{v}^{\mathsf{T}}}{\mathbf{v}^{\mathsf{T}}\mathbf{v}}$$

is symmetric and orthogonal ($\mathbf{H}^{\mathsf{T}}\mathbf{H} = \mathbf{H}^2 = \mathbf{I}$); it reflects vectors in the hyperplane orthogonal to $\mathbf{v}$. Given $\mathbf{a} \neq \mathbf{0}$, the choice

$$\mathbf{v} = \mathbf{a} + \operatorname{sign}(a_1)\|\mathbf{a}\|_2\,\mathbf{e}_1$$

gives $\mathbf{H}\mathbf{a} = -\operatorname{sign}(a_1)\|\mathbf{a}\|_2\,\mathbf{e}_1$: all entries but the first are zeroed.

Derivation. $\mathbf{H}\mathbf{a} = \mathbf{a} - 2\mathbf{v}(\mathbf{v}^{\mathsf{T}}\mathbf{a})/(\mathbf{v}^{\mathsf{T}}\mathbf{v})$. With $\mathbf{v} = \mathbf{a} + \alpha\mathbf{e}_1$, $\alpha = \operatorname{sign}(a_1)\|\mathbf{a}\|$, we have $\mathbf{v}^{\mathsf{T}}\mathbf{a} = \|\mathbf{a}\|^2 + \alpha a_1$ and $\mathbf{v}^{\mathsf{T}}\mathbf{v} = \|\mathbf{a}\|^2 + 2\alpha a_1 + \alpha^2 = 2(\|\mathbf{a}\|^2 + \alpha a_1)$, so $2\mathbf{v}^{\mathsf{T}}\mathbf{a}/\mathbf{v}^{\mathsf{T}}\mathbf{v} = 1$ and $\mathbf{H}\mathbf{a} = \mathbf{a} - \mathbf{v} = -\alpha\mathbf{e}_1$. The sign choice makes $|v_1| = |a_1| + \|\mathbf{a}\|$ a sum of same-signed quantities, avoiding cancellation when $\mathbf{a}$ is nearly parallel to $\mathbf{e}_1$. □

Applying reflectors $\mathbf{H}_1, \dots, \mathbf{H}_n$ to zero columns $1, \dots, n$ below the diagonal in turn gives $\mathbf{H}_n \cdots \mathbf{H}_1\mathbf{A} = \mathbf{R}$, i.e. $\mathbf{A} = \mathbf{Q}\mathbf{R}$ with $\mathbf{Q} = \mathbf{H}_1 \cdots \mathbf{H}_n$. The reflectors are never formed as matrices: $\mathbf{H}\mathbf{x} = \mathbf{x} - \beta\mathbf{v}(\mathbf{v}^{\mathsf{T}}\mathbf{x})$ with $\beta = 2/\mathbf{v}^{\mathsf{T}}\mathbf{v}$ costs $4m$ flops (Worked Example 24.3 again), and $\mathbf{Q}$ is stored implicitly as the list of vectors $\mathbf{v}_k$.

Cost and stability of Householder QR

For $\mathbf{A} \in \mathbb{R}^{m \times n}$, $m \geq n$, Householder QR costs $\sim 2mn^2 - \tfrac{2}{3}n^3$ flops ($\tfrac{4}{3}n^3$ for square $\mathbf{A}$, twice LU). It is backward stable: the computed $\hat{\mathbf{R}}$ is the exact triangular factor of $\mathbf{A} + \Delta\mathbf{A}$ with $\|\Delta\mathbf{A}\| \leq c\,mn\,\varepsilon\|\mathbf{A}\|$, and the computed $\hat{\mathbf{Q}}$ is orthogonal to working precision, $\|\hat{\mathbf{Q}}^{\mathsf{T}}\hat{\mathbf{Q}} - \mathbf{I}\| = O(\varepsilon)$, *independently of* $\kappa(\mathbf{A})$.

Cost sketch. Step k applies a reflector of length $m - k + 1$ to the remaining $n - k + 1$ columns, at $4(m - k + 1)(n - k + 1)$ flops. Summing, $\sum_{k=1}^{n} 4(m - k)(n - k) \approx 4\int_0^n (m - t)(n - t)\,dt = 4(mn^2 - \tfrac{1}{2}(m + n)n^2 + \tfrac{1}{3}n^3) = 2mn^2 - \tfrac{2}{3}n^3$. □

Givens rotations $\begin{bmatrix} c & s \\ -s & c \end{bmatrix}$ with $c^2 + s^2 = 1$ zero one entry at a time. They cost more for dense matrices (about $3mn^2 - n^3$) but are ideal when only a few entries need zeroing — sparse or banded matrices, or *updating* a QR factorisation when one new measurement row arrives (recursive least squares).

Worked Example 24.11 — Householder QR by hand

Compute the QR factorisation of $\mathbf{A} = \begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix}$ with one Householder reflector.

Reflector. $\mathbf{a} = (3,4)^{\mathsf{T}}$, $\|\mathbf{a}\| = 5$, $\operatorname{sign}(a_1) = +1$, so $\mathbf{v} = (3+5,\ 4)^{\mathsf{T}} = (8,4)^{\mathsf{T}}$, $\mathbf{v}^{\mathsf{T}}\mathbf{v} = 80$, $\beta = 2/80 = 0.025$.

First column. $\mathbf{H}\mathbf{a} = -5\mathbf{e}_1 = (-5,0)^{\mathsf{T}}$ by construction. (Check: $\mathbf{v}^{\mathsf{T}}\mathbf{a} = 24 + 16 = 40$, $\mathbf{a} - 0.025 \times 40\,\mathbf{v} = (3,4) - (8,4) = (-5,0)$ ✓.)

Second column. $\mathbf{x} = (1,2)^{\mathsf{T}}$: $\mathbf{v}^{\mathsf{T}}\mathbf{x} = 8 + 8 = 16$, $\mathbf{H}\mathbf{x} = (1,2) - 0.025 \times 16\,(8,4) = (1 - 3.2,\ 2 - 1.6) = (-2.2,\ 0.4)$.

Result.

$$\mathbf{R} = \begin{bmatrix} -5 & -2.2 \\ 0 & 0.4 \end{bmatrix}, \qquad \mathbf{Q} = \mathbf{H} = \mathbf{I} - 0.025\,\mathbf{v}\mathbf{v}^{\mathsf{T}} = \begin{bmatrix} -0.6 & -0.8 \\ -0.8 & 0.6 \end{bmatrix}.$$

Checks. $\mathbf{QR}$: first row $(-0.6)(-5) = 3$, $(-0.6)(-2.2) + (-0.8)(0.4) = 1.32 - 0.32 = 1$ ✓; second row 4, $1.76 + 0.24 = 2$ ✓. $\det \mathbf{A} = 2$, $\det \mathbf{Q} \det \mathbf{R} = (-1)(-2) = 2$ ✓ (a reflector has determinant -1). `numpy.linalg.qr` returns exactly these factors; the negative diagonal of $\mathbf{R}$ is the Householder sign convention, not an error.

Worked Example 24.12 — Loss of orthogonality: Gram–Schmidt versus Householder

The Läuchli matrix with $\epsilon = 10^{-8}$,

$$\mathbf{L} = \begin{bmatrix} 1 & 1 & 1 \\ \epsilon & 0 & 0 \\ 0 & \epsilon & 0 \\ 0 & 0 & \epsilon \end{bmatrix},$$

has nearly parallel columns ($\kappa_2 \approx 1.7 \times 10^{8}$). Orthogonalise its columns by classical Gram–Schmidt (CGS), modified Gram–Schmidt (MGS) and Householder QR, and measure the loss of orthogonality $\|\hat{\mathbf{Q}}^{\mathsf{T}}\hat{\mathbf{Q}} - \mathbf{I}\|_F$ (Frobenius norm).

CGS by hand. $1 + \epsilon^2$ rounds to 1, so $\mathbf{q}_1 = (1,\epsilon,0,0)$. Then $\mathbf{a}_2 - (\mathbf{q}_1^{\mathsf{T}}\mathbf{a}_2)\mathbf{q}_1 = (0,-\epsilon,\epsilon,0)$, $\mathbf{q}_2 = (0,-1,1,0)/\sqrt{2}$. For $\mathbf{a}_3$ CGS projects the *original* column on each $\mathbf{q}_i$: $\mathbf{q}_1^{\mathsf{T}}\mathbf{a}_3 = 1$, $\mathbf{q}_2^{\mathsf{T}}\mathbf{a}_3 = 0$, giving $(0,-\epsilon,0,\epsilon)$, $\mathbf{q}_3 = (0,-1,0,1)/\sqrt{2}$. But $\mathbf{q}_2^{\mathsf{T}}\mathbf{q}_3 = \tfrac{1}{2}$ — the "orthonormal" vectors are at 60°!

MGS. Projects the *updated* vector: after removing $\mathbf{q}_1$, $\mathbf{a}_3$ becomes $(0,-\epsilon,0,\epsilon)$, and removing its component along $\mathbf{q}_2$ ($= \epsilon/\sqrt{2}$) gives $(0,-\epsilon/2,-\epsilon/2,\epsilon)$, orthogonal to $\mathbf{q}_2$.

Measured (NumPy, double precision):

Method	$\|\hat{\mathbf{Q}}^{\mathsf{T}}\hat{\mathbf{Q}} - \mathbf{I}\|_F$
classical Gram–Schmidt	0.71
modified Gram–Schmidt	1.2×10^{-8}
Householder	3.5×10^{-16}

MGS loses orthogonality like $\kappa\varepsilon$ ($1.7 \times 10^{8} \times 1.1 \times 10^{-16} \approx 2 \times 10^{-8}$ ✓); Householder is orthogonal to machine precision regardless of κ.

Least squares: normal equations versus QR

For the overdetermined problem $\min\|\mathbf{A}\mathbf{x} - \mathbf{b}\|_2$ (Ch. 9), the normal equations $\mathbf{A}^{\mathsf{T}}\mathbf{A}\mathbf{x} = \mathbf{A}^{\mathsf{T}}\mathbf{b}$ solved by Cholesky cost $\sim mn^2 + \tfrac{1}{3}n^3$ flops, about half of QR. But

$$\kappa_2(\mathbf{A}^{\mathsf{T}}\mathbf{A}) = \kappa_2(\mathbf{A})^2,$$

because the singular values of $\mathbf{A}^{\mathsf{T}}\mathbf{A}$ are the squares of those of $\mathbf{A}$. Forming $\mathbf{A}^{\mathsf{T}}\mathbf{A}$ in floating point squares the condition number *before* any solving begins: if $\kappa(\mathbf{A}) = 10^{8}$, the normal equations have $\kappa = 10^{16}$ and no digits survive.

The QR route avoids this: with $\mathbf{A} = \mathbf{Q}\mathbf{R}$ (reduced, $\mathbf{Q} \in \mathbb{R}^{m\times n}$),

$$\|\mathbf{A}\mathbf{x} - \mathbf{b}\|_2^2 = \|\mathbf{R}\mathbf{x} - \mathbf{Q}^{\mathsf{T}}\mathbf{b}\|_2^2 + \|(\mathbf{I} - \mathbf{Q}\mathbf{Q}^{\mathsf{T}})\mathbf{b}\|_2^2,$$

so the solution is $\mathbf{R}\hat{\mathbf{x}} = \mathbf{Q}^{\mathsf{T}}\mathbf{b}$, a triangular solve whose sensitivity is governed by $\kappa(\mathbf{A})$ (plus a term $\kappa^2\|\mathbf{r}\|$ that only matters when the residual is large, and is intrinsic to the problem).

Worked Example 24.13 — A line fit by QR

Fit $V = x_1 + x_2 I$ to $(I, V) = (1, 2.1), (2, 3.9), (3, 6.2), (4, 7.8)$ by QR.

Factorisation. For $\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \end{bmatrix}^{\mathsf{T}}$, Householder QR gives

$$\mathbf{R} = \begin{bmatrix} -2 & -5 \\ 0 & -2.2361 \end{bmatrix}, \qquad \mathbf{Q}^{\mathsf{T}}\mathbf{b} = \begin{bmatrix} -10 \\ -4.3380 \end{bmatrix}.$$

(The first column of $\mathbf{Q}$ is $-\tfrac{1}{2}(1,1,1,1)$, so $(\mathbf{Q}^{\mathsf{T}}\mathbf{b})_1 = -\tfrac{1}{2}\sum V_i = -10$; the second is $\propto (-3,-1,1,3)$, the centred I values.)

Back substitution. $x_2 = -4.3380/-2.2361 = 1.940$; $x_1 = (-10 + 5 \times 1.940)/(-2) = 0.150$.

Check against the closed form. $\bar{I} = 2.5$, $\bar{V} = 5$, $\sum (I-\bar{I})(V-\bar{V}) = (-1.5)(-2.9) + (-0.5)(-1.1) + (0.5)(1.2) + (1.5)(2.8) = 9.7$, $\sum (I-\bar{I})^2 = 5$, slope 1.94 ✓, intercept $5 - 1.94 \times 2.5 = 0.15$ ✓.

Conditioning. $\kappa_2(\mathbf{A}) = 7.47$, $\kappa_2(\mathbf{A}^{\mathsf{T}}\mathbf{A}) = 55.8 = 7.47^2$ ✓. Harmless here; the next example is not.

Worked Example 24.14 — Polynomial fitting: when the normal equations fail

Fit a degree-d polynomial in the monomial basis to $m = 50$ equally spaced points on $[0, 1]$, with data generated from the polynomial whose coefficients are all 1 (so the exact answer is known). Compare normal equations and QR.

degree d	$\kappa_2(\mathbf{V})$	rel. error, normal equations	rel. error, QR
6	2.0×10^{4}	2.7×10^{-9}	5.3×10^{-13}
10	2.0×10^{7}	1.0×10^{-3}	7.1×10^{-11}
14	2.3×10^{10}	6.0 (no correct digits)	1.4×10^{-6}

Reading. Each column follows the $\kappa\varepsilon$ rule: QR errors track $\kappa(\mathbf{V})\varepsilon$, normal-equation errors track $\kappa(\mathbf{V})^2\varepsilon$ (for $d = 14$, $\kappa^2\varepsilon \approx 6 \times 10^4$: hopeless). Even QR is losing ten digits at $d = 14$, which is a property of the *problem*: the monomials t^{13} and t^{14} are nearly parallel on $[0, 1]$. The real cure is a better basis — Chebyshev or Legendre polynomials on $[-1, 1]$, for which κ grows only slowly with d — exactly the reformulation advice of Worked Example 24.7.

The SVD computationally

The SVD $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\mathsf{T}}$ (Ch. 10) is computed in two phases. **Bidiagonalisation**: Householder reflectors applied alternately from the left and the right reduce $\mathbf{A}$ to upper bidiagonal form $\mathbf{B} = \mathbf{U}_1^{\mathsf{T}}\mathbf{A}\mathbf{V}_1$ (Golub–Kahan), at $\sim 4mn^2 - \frac{4}{3}n^3$ flops. **Diagonalisation**: an implicitly shifted QR iteration (or a divide-and-conquer method) drives the superdiagonal of $\mathbf{B}$ to zero, at $O(n^2)$ flops per sweep for the values. Forming $\mathbf{U}$ and $\mathbf{V}$ explicitly multiplies the cost several times.

Never compute the SVD from the eigenvalues of $\mathbf{A}^{\mathsf{T}}\mathbf{A}$: as with the normal equations this squares the condition number, and small singular values below $\sqrt{\varepsilon}\,\sigma_1$ are lost completely.

Accuracy of computed singular values

The computed singular values $\hat{\sigma}_i$ of a backward-stable SVD satisfy $|\hat{\sigma}_i - \sigma_i| \le c\,\varepsilon\,\sigma_1$. Singular values are perfectly conditioned in the *absolute* sense (Weyl's inequality: $|\sigma_i(\mathbf{A}+\mathbf{E}) - \sigma_i(\mathbf{A})| \le \|\mathbf{E}\|_2$), so large singular values are computed to high relative accuracy and singular values below $\varepsilon\sigma_1$ are indistinguishable from zero.

This makes the SVD the reliable tool for **numerical rank**: the number of singular values above a tolerance τ, typically $\tau = \max(m,n)\,\varepsilon\,\sigma_1$ for exact data, or the noise level $\|\mathbf{E}\|$ for measured data. The **truncated SVD** solution of a least-squares problem keeps only the k components with $\sigma_i > \tau$:

$$\hat{\mathbf{x}}_k = \sum_{i=1}^{k} \frac{\mathbf{u}_i^{\mathsf{T}}\mathbf{b}}{\sigma_i}\mathbf{v}_i.$$

Discarding the small σ_i removes the components that would divide noise by a tiny number — a form of *regularisation*.

Worked Example 24.15 — Truncated SVD stabilises a near-singular solve

Return to $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1.0001 \end{bmatrix}$. The right-hand side is measured as $\mathbf{b} = (2,\ 2.0002)^{\mathsf{T}}$ — the true $(2, 2.0001)$ plus an error of 10^{-4} in the second entry. Compare the exact solve with the rank-1 truncated SVD.

Exact solve. $x_1 + x_2 = 2$, $x_1 + 1.0001x_2 = 2.0002$ give $0.0001x_2 = 0.0002$: $\mathbf{x} = (0, 2)^{\mathsf{T}}$, far from the true $(1, 1)$.

SVD. $\sigma_1 = 2.00005$, $\sigma_2 = 4.99988 \times 10^{-5}$; $\mathbf{v}_1 \approx (0.70709,\ 0.70712)$, $\mathbf{v}_2 \approx (-0.70712,\ 0.70709)$ (up to sign). The coefficients are $\mathbf{u}_1^{\mathsf{T}}\mathbf{b} = 2.82857$ and $\mathbf{u}_2^{\mathsf{T}}\mathbf{b} = 7.07 \times 10^{-5}$ (signs as returned by NumPy).

- Component 1 contributes $(2.82857/2.00005)\mathbf{v}_1 = (1.0000,\ 1.0000)$.
- Component 2 contributes $(7.07 \times 10^{-5}/5.0 \times 10^{-5})\mathbf{v}_2 \approx 1.414\,\mathbf{v}_2 = (-1,\ 1)$ — almost entirely the amplified measurement error.

Truncated. Keeping $k = 1$: $\hat{\mathbf{x}}_1 = (1.0000,\ 1.00005)$, within 5×10^{-5} of the truth.

Reading. With data error of order 10^{-4}, any singular value below about 10^{-4} cannot be trusted; here $\sigma_2 = 5 \times 10^{-5}$ is below that level and is discarded. Choosing the truncation level is a *modelling* decision based on the noise, not a numerical one.

Cost summary

Task	Algorithm	Flops (leading term)	Stable?
Solve $\mathbf{A}\mathbf{x}=\mathbf{b}$, general	LU, partial pivoting	$\frac{2}{3}n^3$	in practice (growth factor)
Solve, SPD	Cholesky	$\frac{1}{3}n^3$	yes
Solve, extra right-hand side	triangular solves with stored factors	$2n^2$	yes
Least squares $m \times n$	normal equations + Cholesky	$mn^2+\frac{1}{3}n^3$	no: κ^2
Least squares $m \times n$	Householder QR	$2mn^2-\frac{2}{3}n^3$	yes
Singular values only	Golub–Kahan + QR iteration	$4mn^2-\frac{4}{3}n^3$	yes
Full SVD, square	with $\mathbf{U},\mathbf{V}$	$\approx 21n^3$	yes
Symmetric eigenvalues	tridiagonalisation + QR/divide & conquer	$\frac{4}{3}n^3$ ($\approx 9n^3$ with vectors)	yes
Nonsymmetric eigenvalues	Hessenberg + shifted QR	$\approx 10n^3$ ($\approx 25n^3$ with vectors)	yes (but eigenvalues may be ill-conditioned)
Determinant	from LU	$\frac{2}{3}n^3$	yes (but beware overflow; use $\log\lvert\det\rvert$)

(Approximate counts from Golub and Van Loan; constants for the iterative phases depend on the matrix.)

Gaussian elimination with partial pivoting is backward stable in practice

GEPP computes $\hat{\mathbf{x}}$ with $(\mathbf{A}+\Delta\mathbf{A})\hat{\mathbf{x}}=\mathbf{b}$, $\|\Delta\mathbf{A}\|_\infty \le c\,n^3\rho_n\varepsilon\|\mathbf{A}\|_\infty$, where the **growth factor** $\rho_n=\max_{i,j,k}|a_{ij}^{(k)}|/\max_{i,j}|a_{ij}|$ measures how large the entries become during elimination. The worst case is $\rho_n=2^{n-1}$ — attained by Wilkinson's matrix (1 on the diagonal and in the last column, -1 below the diagonal), for which $n=20$ gives $\rho=2^{19}=524\,288$ — but matrices arising in practice almost never show growth beyond $\rho\approx 10$.

Eigenvalues: a computational note

Eigenvalues are roots of the characteristic polynomial, but **no one computes them that way**: polynomial roots are violently ill-conditioned functions of the coefficients (Wilkinson's polynomial $\prod_{k=1}^{20}(x-k)$: changing one coefficient by 2^{-23} moves roots into the complex plane). Instead the **QR algorithm** iterates $\mathbf{A}_k=\mathbf{Q}_k\mathbf{R}_k$, $\mathbf{A}_{k+1}=\mathbf{R}_k\mathbf{Q}_k$ (an orthogonal similarity) on a Hessenberg or tridiagonal form with shifts, converging to triangular (Schur) form. For one dominant eigenvalue of a huge sparse matrix the **power method** $\mathbf{v}_{k+1}=\mathbf{A}\mathbf{v}_k/\|\mathbf{A}\mathbf{v}_k\|$ suffices, converging at the rate $|\lambda_2/\lambda_1|$ per step; Lanczos and Arnoldi methods (the eigenvalue cousins of the CG and GMRES solvers of Section 24.4) find a few extreme eigenvalues of matrices with millions of rows.

Worked Example 24.16 — Power iteration

Apply the power method with Rayleigh quotient $\lambda^{(k)}=\mathbf{v}_k^{\mathsf T}\mathbf{A}\mathbf{v}_k$ to $\mathbf{A}=\begin{bmatrix}2&1\\1&3\end{bmatrix}$ from $\mathbf{v}_0=(1,0)$.

Iterates. The Rayleigh quotients (using the current normalised vector) are $2,\ 3,\ 3.5,\ 3.6,\ 3.6154,\ldots$, and the vectors $(0.894,0.447)$, $(0.707,0.707)$, $(0.6,0.8)$, $(0.555,0.832)$, $(0.537,0.844)$.

Exact. $\lambda=(5\pm\sqrt{5})/2=3.6180,\ 1.3820$; dominant eigenvector $\propto(0.526,0.851)$.

Rate. $|\lambda_2/\lambda_1|=0.382$: the vector error shrinks by 0.382 per step and, for a symmetric matrix, the Rayleigh-quotient error by $0.382^2=0.146$. The errors $0.118, 0.018, 0.0027$ in the last three quotients fall by about 0.15 per step ✓.

Common Pitfall

"The residual is tiny, so the answer is right" is false for ill-conditioned problems (Worked Example 24.10), and "the answer changed in the fifth digit when I switched libraries, so one of them is buggy" is usually false too — both may be backward stable answers to a problem with $\kappa\approx 10^{11}$. Before blaming an algorithm, estimate κ (`numpy.linalg.cond`, or the cheap LAPACK estimate from an existing factorisation) and compare the observed discrepancy with $\kappa\varepsilon$.

Engineering Note

Scaling matters. A modified nodal analysis matrix mixing conductances in siemens (10^{-3}) with unit entries for voltage sources, or a state-space model mixing millivolts and kiloamps, can have an enormous condition number that is purely an artefact of units. **Equilibration** — scaling rows and columns by diagonal matrices so that entries are of comparable size, $\mathbf{D}_r\mathbf{A}\mathbf{D}_c$ — often reduces κ by many orders of magnitude at negligible cost, and LAPACK's expert drivers (`dgesvx`) do it automatically. It is the matrix version of choosing sensible units, the topic of nondimensionalisation in Ch. 25.

24.4 Iterative solvers

Motivation

A direct solver (LU, Cholesky) delivers the solution of $\mathbf{A}\mathbf{x}=\mathbf{b}$ in a fixed number of operations. For a 3-D finite-element model of a transformer with $n=10^7$ unknowns, however, even the best fill-reducing ordering (Section 24.5) leaves factors with 10^{10} or more nonzeros — too much memory and too many flops. **Iterative methods** never factor $\mathbf{A}$. They need only the ability to compute $\mathbf{A}\mathbf{v}$ for a given vector $\mathbf{v}$ — $O(\text{nnz})$ work and no extra storage — and improve an approximate solution step by step. Their cost is (number of iterations) $\times$ (cost per iteration), and the whole art lies in making the first factor small. Chapter 8 introduced the Jacobi and Gauss–Seidel iterations and their convergence criterion; here we develop the general theory, over-relaxation, the Krylov methods (conjugate gradients and GMRES) that dominate modern practice, and preconditioning.

Stationary iterations: the general picture

Split $\mathbf{A} = \mathbf{M} - \mathbf{N}$ with $\mathbf{M}$ easy to invert. Then $\mathbf{A}\mathbf{x} = \mathbf{b}$ is equivalent to $\mathbf{M}\mathbf{x} = \mathbf{N}\mathbf{x} + \mathbf{b}$, suggesting

$$\mathbf{x}^{(k+1)} = \mathbf{M}^{-1}\big(\mathbf{N}\mathbf{x}^{(k)} + \mathbf{b}\big) = \mathbf{x}^{(k)} + \mathbf{M}^{-1}\mathbf{r}^{(k)}, \qquad \mathbf{r}^{(k)} = \mathbf{b} - \mathbf{A}\mathbf{x}^{(k)}.$$

The second form shows the structure: *correct the current iterate by an approximate solve of the residual equation*. With $\mathbf{A} = \mathbf{D} + \mathbf{L}_A + \mathbf{U}_A$ (diagonal, strictly lower, strictly upper), Jacobi is $\mathbf{M} = \mathbf{D}$ and Gauss–Seidel is $\mathbf{M} = \mathbf{D} + \mathbf{L}_A$.

Convergence of stationary iterations

The error $\mathbf{e}^{(k)} = \mathbf{x}^{(k)} - \mathbf{x}$ satisfies $\mathbf{e}^{(k+1)} = \mathbf{T}\mathbf{e}^{(k)}$ with **iteration matrix** $\mathbf{T} = \mathbf{M}^{-1}\mathbf{N} = \mathbf{I} - \mathbf{M}^{-1}\mathbf{A}$. The iteration converges for every $\mathbf{x}^{(0)}$ if and only if the spectral radius $\rho(\mathbf{T}) < 1$, and asymptotically

$$\|\mathbf{e}^{(k)}\| \approx C\,\rho(\mathbf{T})^k, \qquad k_\epsilon \approx \frac{\ln \epsilon}{\ln \rho(\mathbf{T})}$$

iterations reduce the error by a factor ϵ. The quantity $R = -\ln \rho(\mathbf{T})$ is the **asymptotic rate of convergence**.

(Proof in Ch. 8: $\mathbf{T}^k \to \mathbf{0}$ iff $\rho(\mathbf{T}) < 1$, which follows from the Jordan form of Ch. 10.) When ρ is close to 1, $R \approx 1 - \rho$ and the iteration count is $\approx \ln(1/\epsilon)/(1-\rho)$: halving $1 - \rho$ doubles the work.

Two caveats are worth stating. First, $\rho(\mathbf{T}) < 1$ controls the *asymptotic* rate; for non-normal $\mathbf{T}$ the error can grow for many steps before decaying. Second, the iteration stops on the residual, not the error: since $\mathbf{e} = -\mathbf{A}^{-1}\mathbf{r}$,

$$\frac{\|\mathbf{e}^{(k)}\|}{\|\mathbf{x}\|} \le \kappa(\mathbf{A})\,\frac{\|\mathbf{r}^{(k)}\|}{\|\mathbf{b}\|},$$

so a relative-residual tolerance of 10^{-8} guarantees only $10^{-8}\kappa$ in the error.

The model problem

The standard test case is the discrete Laplacian. In one dimension, $-u'' = f$ on $(0,1)$ with n interior points and spacing $h = 1/(n+1)$ gives the tridiagonal matrix $\mathbf{A} = \operatorname{tridiag}(-1, 2, -1)$ — also the nodal matrix of a uniform resistive ladder with grounded ends. Its eigenvalues are

$$\lambda_k = 2 - 2\cos(k\pi h) = 4\sin^2\!\Big(\frac{k\pi h}{2}\Big), \qquad k = 1, \dots, n,$$

with eigenvectors $\sin(k\pi i h)$. The Jacobi iteration matrix $\mathbf{I} - \tfrac{1}{2}\mathbf{A}$ has eigenvalues $\cos(k\pi h)$, so

$$\rho_J = \cos(\pi h) \approx 1 - \tfrac{1}{2}\pi^2 h^2.$$

The convergence factor approaches 1 as the grid is refined: the number of Jacobi iterations grows like $h^{-2} \propto n^2$. The slowest modes are the *smooth* ones (k small) — the iteration damps oscillatory errors quickly and smooth errors very slowly, an observation that is the starting point of **multigrid** methods.

Successive over-relaxation (SOR)

Gauss–Seidel moves each unknown to the value that satisfies its own equation. Over-relaxation moves it *further* in the same direction by a factor ω:

$$x_i^{(k+1)} = (1-\omega)x_i^{(k)} + \frac{\omega}{a_{ii}}\Big(b_i - \sum_{j<i} a_{ij}x_j^{(k+1)} - \sum_{j>i} a_{ij}x_j^{(k)}\Big).$$

In matrix form $\mathbf{M}_\omega = \frac{1}{\omega}\mathbf{D} + \mathbf{L}_A$ and

$$\mathbf{T}_\omega = (\mathbf{D} + \omega\mathbf{L}_A)^{-1}\big((1-\omega)\mathbf{D} - \omega\mathbf{U}_A\big).$$

$\omega = 1$ is Gauss–Seidel; $\omega > 1$ is over-relaxation, $\omega < 1$ under-relaxation.

SOR convergence (Kahan, Ostrowski–Reich, Young)

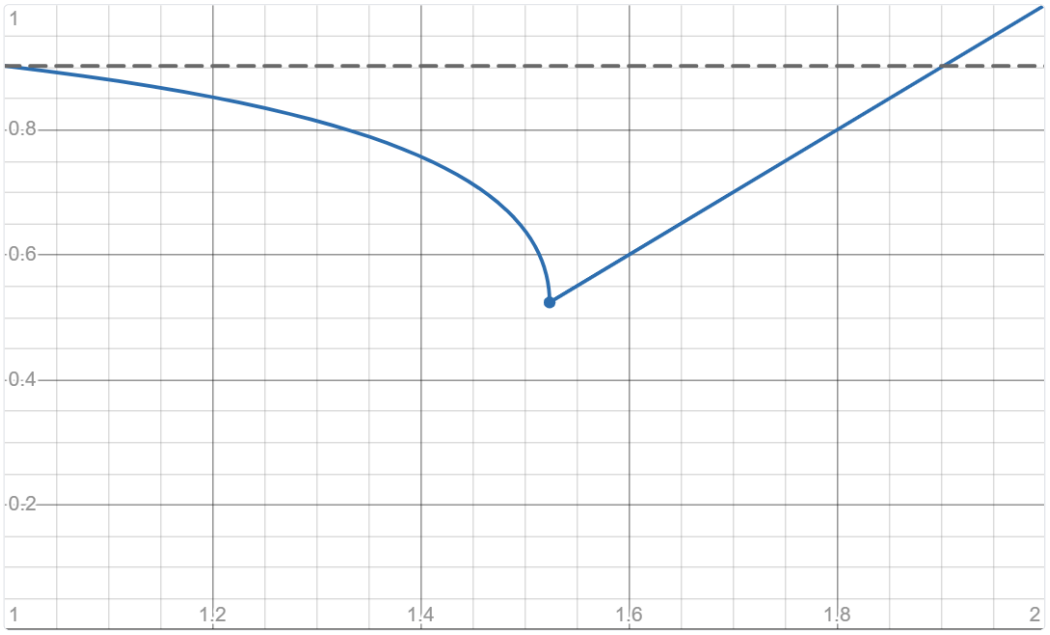
1. (**Kahan**) $\rho(\mathbf{T}_\omega) \ge |\omega - 1|$, so SOR can converge only for $0 < \omega < 2$.
2. (**Ostrowski–Reich**) If $\mathbf{A}$ is symmetric positive definite, SOR converges for every $0 < \omega < 2$.
3. (**Young**) If $\mathbf{A}$ is *consistently ordered* (true for tridiagonal matrices and for the 5-point Laplacian in natural or red–black ordering) and the Jacobi eigenvalues are real with $\rho_J < 1$, then the optimal relaxation factor and the resulting spectral radius are

$$\omega_{\text{opt}} = \frac{2}{1 + \sqrt{1 - \rho_J^2}}, \qquad \rho(\mathbf{T}_{\omega_{\text{opt}}}) = \omega_{\text{opt}} - 1.$$

Moreover $\rho(\mathbf{T}_1) = \rho_J^2$: Gauss–Seidel converges exactly twice as fast as Jacobi.

Sketch of 1. $\det \mathbf{T}_\omega = \det(\mathbf{D} + \omega\mathbf{L}_A)^{-1}\det((1-\omega)\mathbf{D} - \omega\mathbf{U}_A) = \det(\mathbf{D})^{-1}(1-\omega)^n \det \mathbf{D} = (1-\omega)^n$ because both factors are triangular. The determinant is the product of the eigenvalues, so at least one has modulus $\ge |1-\omega|$. $\square$

Sketch of 3. For consistently ordered matrices, each Jacobi eigenvalue μ and SOR eigenvalue λ are related by Young's equation $(\lambda + \omega - 1)^2 = \lambda\omega^2\mu^2$. Solving this quadratic in $\sqrt{\lambda}$ gives $\sqrt{\lambda} = \tfrac{1}{2}\big(\omega\mu \pm \sqrt{\omega^2\mu^2 - 4(\omega-1)}\big)$. For ω below the value making the discriminant zero for $\mu = \rho_J$, the largest $|\lambda|$ decreases with ω; above it the roots are complex with $|\lambda| = \omega - 1$, increasing with ω. The minimum is where the discriminant vanishes, $\omega^2\rho_J^2 = 4(\omega-1)$, whose root in $(1,2)$ is ω_{opt}. $\square$



Spectral radius of SOR against ω for a consistently ordered matrix with $\rho_J = 0.95$. At $\omega = 1$ (Gauss–Seidel) $\rho = 0.9025$ (dashed); the minimum $\rho = 0.524$ at $\omega_{\text{opt}} = 1.524$ is a sharp corner. Overestimating ω costs little (slope 1 on the right); underestimating it costs a lot (infinite slope on the left).

Worked Example 24.17 — SOR on the three-node network

Apply SOR with $\omega = 1.2$ to the nodal system of Ch. 8, Worked Example 8.39:

$$\begin{bmatrix} 4 & -2 & -1 \\ -2 & 4 & -1 \\ -1 & -1 & 4 \end{bmatrix} \mathbf{v} = \begin{bmatrix} 8 \\ 2 \\ 1 \end{bmatrix}, \qquad \mathbf{v} = (4, 3, 2)^{\mathsf{T}},$$

from $\mathbf{v}^{(0)} = \mathbf{0}$, and compare spectral radii.

Iteration 1.

- v_1: Gauss–Seidel value $\tfrac{1}{4}(8) = 2$; SOR $v_1 = -0.2 \cdot 0 + 1.2 \cdot 2 = 2.4$.
- v_2: GS value $\tfrac{1}{4}(2 + 2 \cdot 2.4 + 0) = 1.7$; $v_2 = 1.2 \times 1.7 = 2.04$.
- v_3: GS value $\tfrac{1}{4}(1 + 2.4 + 2.04) = 1.36$; $v_3 = 1.2 \times 1.36 = 1.632$.

Iterations 2 and 3. $\mathbf{v}^{(2)} = (3.6336,\ 2.8618,\ 1.9222)$, $\mathbf{v}^{(3)} = (3.9670,\ 2.9845,\ 2.0010)$. Maximum errors: 1.6, 0.366, 0.033 — against 2.0, 0.969, 0.456 for Gauss–Seidel.

Spectral radii (computed from $\mathbf{T}_\omega$):

ω	1.0	1.1	1.16	1.2	1.3	1.5
$\rho(\mathbf{T}_\omega)$	0.472	0.350	0.230	0.271	0.367	0.555

The optimum is near $\omega = 1.16$, where the iterations needed for 10^{-6} fall from 18 (GS) to 9.5. Young's formula with $\rho_J = 0.683$ would predict $\omega_{\text{opt}} = 1.155$; it applies only approximately because this matrix is not consistently ordered, but it is a good starting estimate.

Worked Example 24.18 — The model problem: Jacobi, Gauss–Seidel and optimal SOR

For the 1-D model problem with $n = 50$ interior points, how many iterations does each method need to reduce the error by 10^{-6}? Repeat for $n = 100$.

$n = 50$. $h = 1/51$, $\rho_J = \cos(\pi/51) = 0.998103$. Then

- Jacobi: $k = \ln 10^{-6} / \ln 0.998103 \approx 7277$;
- Gauss–Seidel: $\rho = \rho_J^2 = 0.996210$, $k \approx 3639$;
- SOR: $\omega_{\text{opt}} = 2/(1 + \sqrt{1 - \rho_J^2}) = 1.884$, $\rho = 0.884$, $k \approx 112$.

$n = 100$. $\rho_J = 0.999516$: Jacobi 28 554, Gauss–Seidel 14 277, SOR ($\omega_{\text{opt}} = 1.940$) 222.

Scaling. Doubling n multiplies Jacobi/GS work per unit accuracy by 4 (since $1 - \rho \propto h^2$) but optimal SOR only by 2 (since $1 - \rho_{\text{SOR}} \approx 2\pi h$). Optimal SOR turns an $O(n^2)$ iteration count into $O(n)$ — the same improvement, we shall see, that conjugate gradients achieve *without* needing to know ρ_J.

Krylov subspace methods

Stationary methods use only the latest residual. **Krylov methods** use the whole history: after k steps from $\mathbf{x}_0$ (residual $\mathbf{r}_0$) they choose the best approximation from the affine space $\mathbf{x}_0 + \mathcal{K}_k$, where

$$\mathcal{K}_k(\mathbf{A}, \mathbf{r}_0) = \operatorname{span}\{\mathbf{r}_0,\ \mathbf{A}\mathbf{r}_0,\ \mathbf{A}^2\mathbf{r}_0, \dots, \mathbf{A}^{k-1}\mathbf{r}_0\}$$

is the **Krylov subspace**. Every vector in it is $p(\mathbf{A})\mathbf{r}_0$ for a polynomial p of degree $< k$, so the error after k steps has the form $\mathbf{e}_k = q_k(\mathbf{A})\mathbf{e}_0$ with q_k a polynomial of degree k and $q_k(0) = 1$. Krylov methods differ in *which* q_k they choose — that is, in what "best" means.

Conjugate gradients

For SPD $\mathbf{A}$, solving $\mathbf{A}\mathbf{x} = \mathbf{b}$ is equivalent to minimising the quadratic "energy"

$$\phi(\mathbf{x}) = \tfrac{1}{2}\mathbf{x}^{\mathsf T}\mathbf{A}\mathbf{x} - \mathbf{b}^{\mathsf T}\mathbf{x}, \qquad \nabla\phi = \mathbf{A}\mathbf{x} - \mathbf{b} = -\mathbf{r}.$$

(For a resistive network, ϕ is half the dissipated power minus the power delivered by the sources, and the solution minimises it — Maxwell's minimum-heat theorem.) Steepest descent searches along $\mathbf{r}$ each time, which zigzags when the level sets are elongated. CG instead searches along directions $\mathbf{p}_0, \mathbf{p}_1, \dots$ that are **$\mathbf{A}$-conjugate**, $\mathbf{p}_i^{\mathsf T}\mathbf{A}\mathbf{p}_j = 0$ for $i \neq j$, so that minimising along a new direction never spoils the minimisation along the old ones.

Conjugate gradient algorithm

Given SPD $\mathbf{A}$, $\mathbf{b}$, $\mathbf{x}_0$: set $\mathbf{r}_0 = \mathbf{b} - \mathbf{A}\mathbf{x}_0$, $\mathbf{p}_0 = \mathbf{r}_0$. For $k = 0, 1, 2, \dots$:

$$\begin{aligned} \alpha_k &= \frac{\mathbf{r}_k^{\mathsf T}\mathbf{r}_k}{\mathbf{p}_k^{\mathsf T}\mathbf{A}\mathbf{p}_k}, & \mathbf{x}_{k+1} &= \mathbf{x}_k + \alpha_k \mathbf{p}_k, & \mathbf{r}_{k+1} &= \mathbf{r}_k - \alpha_k \mathbf{A}\mathbf{p}_k, \\ \beta_k &= \frac{\mathbf{r}_{k+1}^{\mathsf T}\mathbf{r}_{k+1}}{\mathbf{r}_k^{\mathsf T}\mathbf{r}_k}, & \mathbf{p}_{k+1} &= \mathbf{r}_{k+1} + \beta_k \mathbf{p}_k. \end{aligned}$$

Each iteration costs one matrix–vector product, two inner products and three vector updates — about $2\,\mathrm{nnz} + 10n$ flops — and stores four vectors.

Properties of CG

In exact arithmetic:

1. The residuals are mutually orthogonal and the directions mutually $\mathbf{A}$-conjugate.
2. $\mathbf{x}_k$ minimises the energy norm of the error $\|\mathbf{e}\|_{\mathbf{A}} = \sqrt{\mathbf{e}^{\mathsf T}\mathbf{A}\mathbf{e}}$ over $\mathbf{x}_0 + \mathcal{K}_k$; equivalently
$$\|\mathbf{e}_k\|_{\mathbf{A}} = \min_{q \in \mathcal{P}_k,\, q(0)=1} \|q(\mathbf{A})\mathbf{e}_0\|_{\mathbf{A}} \le \min_{q} \max_{\lambda \in \sigma(\mathbf{A})} |q(\lambda)|\, \|\mathbf{e}_0\|_{\mathbf{A}}.$$
3. CG terminates with the exact solution in at most n steps — in fact in at most d steps, where d is the number of distinct eigenvalues of $\mathbf{A}$.
4. With $\kappa = \lambda_{\max}/\lambda_{\min}$,
$$\frac{\|\mathbf{e}_k\|_{\mathbf{A}}}{\|\mathbf{e}_0\|_{\mathbf{A}}} \le 2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^{k}.$$

Sketch. Property 2 follows because $\phi(\mathbf{x}) - \phi(\mathbf{x}^\ast) = \tfrac12\|\mathbf{x} - \mathbf{x}^\ast\|_{\mathbf{A}}^2$ and conjugacy makes the minimisation over the growing Krylov space separable, one direction at a time. Expanding $\mathbf{e}_0$ in eigenvectors, $\|q(\mathbf{A})\mathbf{e}_0\|_{\mathbf{A}}^2 = \sum_i \lambda_i q(\lambda_i)^2 c_i^2 \le \max_i q(\lambda_i)^2\, \|\mathbf{e}_0\|_{\mathbf{A}}^2$. Property 3: the polynomial $q(\lambda) = \prod_{i=1}^{d}(1 - \lambda/\lambda_i)$ vanishes on the spectrum. Property 4: choose q to be the Chebyshev polynomial on $[\lambda_{\min}, \lambda_{\max}]$ scaled so $q(0) = 1$; its maximum on the interval is $1/T_k\!\left(\frac{\kappa+1}{\kappa-1}\right) \le 2\big(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\big)^k$. $\square$

The bound gives an iteration count of about $k \approx \tfrac12\sqrt{\kappa}\ln(2/\epsilon)$, compared with $\tfrac12\kappa\ln(1/\epsilon)$ for steepest descent. For the 2-D model problem $\kappa \propto h^{-2}$, so CG needs $O(h^{-1}) = O(\sqrt{n})$ iterations — the same as optimal SOR, with no parameter to tune.

Worked Example 24.19 — Conjugate gradients by hand

Solve $\begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix}\mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ by CG from $\mathbf{x}_0 = \mathbf{0}$.

Step 0. $\mathbf{r}_0 = \mathbf{p}_0 = (1, 2)$, $\mathbf{r}_0^{\mathsf T}\mathbf{r}_0 = 5$, $\mathbf{A}\mathbf{p}_0 = (6, 7)$, $\mathbf{p}_0^{\mathsf T}\mathbf{A}\mathbf{p}_0 = 6 + 14 = 20$, $\alpha_0 = 0.25$. $\mathbf{x}_1 = (0.25,\ 0.5)$, $\mathbf{r}_1 = (1, 2) - 0.25(6, 7) = (-0.5,\ 0.25)$, $\mathbf{r}_1^{\mathsf T}\mathbf{r}_1 = 0.3125$, $\beta_0 = 0.3125/5 = 0.0625$, $\mathbf{p}_1 = (-0.5, 0.25) + 0.0625(1, 2) = (-0.4375,\ 0.375)$.

Step 1. $\mathbf{A}\mathbf{p}_1 = (-1.75 + 0.375,\ -0.4375 + 1.125) = (-1.375,\ 0.6875)$, $\mathbf{p}_1^{\mathsf T}\mathbf{A}\mathbf{p}_1 = 0.6016 + 0.2578 = 0.859375$, $\alpha_1 = 0.3125/0.859375 = 0.363636 = \tfrac{4}{11}$. $\mathbf{x}_2 = (0.25, 0.5) + \tfrac{4}{11}(-0.4375, 0.375) = (0.090909,\ 0.636364) = (\tfrac{1}{11}, \tfrac{7}{11})$, and $\mathbf{r}_2 = \mathbf{0}$.

Checks. $4/11 + 7/11 = 1$ ✓, $1/11 + 21/11 = 2$ ✓. Termination in $n = 2$ steps ✓. Orthogonality: $\mathbf{r}_0^{\mathsf T}\mathbf{r}_1 = -0.5 + 0.5 = 0$ ✓; conjugacy: $\mathbf{p}_0^{\mathsf T}\mathbf{A}\mathbf{p}_1 = (1, 2)\cdot(-1.375, 0.6875) = 0$ ✓.

Worked Example 24.20 — Iteration counts from the condition number

An SPD system from a capacitance extraction has $\kappa = 10^4$. Estimate the iterations for a 10^{-6} reduction of the energy-norm error with steepest descent and with CG.

Steepest descent. Error factor $\le \big(\frac{\kappa-1}{\kappa+1}\big)^k \approx e^{-2k/\kappa}$: $k \approx \frac{\kappa}{2}\ln 10^6 = 5000 \times 13.8 \approx 69\,000$.

CG. $k \approx \frac{\sqrt{\kappa}}{2}\ln(2 \times 10^6) = 50 \times 14.5 \approx 725$.

Reading. CG's dependence on $\sqrt{\kappa}$ rather than κ is worth a factor of 100 here. And the bound is pessimistic: if the eigenvalues are clustered (a few outliers and the rest close together), CG "removes" each outlier in about one iteration and then converges at the rate set by the cluster. That is exactly the effect a good preconditioner aims for.

GMRES

For nonsymmetric or indefinite matrices — modified nodal analysis with voltage sources, convection–diffusion models, integral-equation formulations of electromagnetic scattering — there is no energy to minimise. **GMRES** (generalised minimal residual) chooses $\mathbf{x}_k \in \mathbf{x}_0 + \mathcal{K}_k$ to minimise the residual norm $\|\mathbf{b} - \mathbf{A}\mathbf{x}_k\|_2$.

Arnoldi process and GMRES

The **Arnoldi process** builds an orthonormal basis $\mathbf{q}_1, \dots, \mathbf{q}_k$ of $\mathcal{K}_k$ by modified Gram–Schmidt: $\mathbf{q}_1 = \mathbf{r}_0/\beta$ with $\beta = \|\mathbf{r}_0\|$, and for each j, $\mathbf{w} = \mathbf{A}\mathbf{q}_j$ is orthogonalised against $\mathbf{q}_1, \dots, \mathbf{q}_j$ (coefficients h_{ij}) and normalised to give $\mathbf{q}_{j+1}$ ($h_{j+1,j} = \|\mathbf{w}\|$). In matrix form $\mathbf{A}\mathbf{Q}_k = \mathbf{Q}_{k+1}\tilde{\mathbf{H}}_k$ with $\tilde{\mathbf{H}}_k$ a $(k+1) \times k$ upper Hessenberg matrix. Writing $\mathbf{x}_k = \mathbf{x}_0 + \mathbf{Q}_k\mathbf{y}$,

$$\|\mathbf{b} - \mathbf{A}\mathbf{x}_k\|_2 = \|\beta\mathbf{e}_1 - \tilde{\mathbf{H}}_k\mathbf{y}\|_2,$$

a small $(k+1) \times k$ least-squares problem solved by Givens rotations. **GMRES** returns the minimising $\mathbf{y}$.

GMRES convergence

GMRES residuals are non-increasing and satisfy

$$\frac{\|\mathbf{r}_k\|}{\|\mathbf{r}_0\|} = \min_{q \in \mathcal{P}_k,\, q(0)=1} \frac{\|q(\mathbf{A})\mathbf{r}_0\|}{\|\mathbf{r}_0\|} \le \kappa(\mathbf{V}) \min_{q} \max_{\lambda \in \sigma(\mathbf{A})} |q(\lambda)|$$

for diagonalisable $\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$. GMRES terminates in at most d steps if $\mathbf{A}$ has a minimal polynomial of degree d. Convergence is fast when the eigenvalues are clustered away from the origin and $\mathbf{V}$ is well conditioned.

The price of GMRES is that every step must orthogonalise against *all* previous basis vectors: step k costs $O(kn)$ flops and the method stores k vectors. In practice it is **restarted** every m steps (GMRES(m), with m typically 20–100), which bounds the memory but can slow or even stall convergence. Short-recurrence alternatives for nonsymmetric systems — BiCGSTAB, QMR, IDR(s) — have fixed cost per step but less predictable convergence.

Worked Example 24.21 — GMRES and the minimal polynomial

A 50×50 nonsymmetric matrix $\mathbf{A} = \mathbf{S}\mathbf{\Lambda}\mathbf{S}^{-1}$ is constructed with random $\mathbf{S}$ and eigenvalues 1 (25 times) and 3 (25 times). Predict, and then observe, the GMRES iteration count.

Prediction. $(\mathbf{A} - \mathbf{I})(\mathbf{A} - 3\mathbf{I}) = \mathbf{S}(\mathbf{\Lambda} - \mathbf{I})(\mathbf{\Lambda} - 3\mathbf{I})\mathbf{S}^{-1} = \mathbf{0}$, so the polynomial $q(\lambda) = (1-\lambda)(1-\lambda/3)$, with $q(0) = 1$, annihilates $\mathbf{A}$. GMRES must converge in 2 steps regardless of n.

Observation. SciPy's `gmres` reports relative residuals 0.988 after one step and 1.6×10^{-13} after two ✓.

Contrast. For a 1024-unknown convection–diffusion matrix (32×32 grid) with scattered eigenvalues, full GMRES needed 74 iterations for a 10^{-8} residual reduction while GMRES(20) needed 229 — restarting throws away the subspace that was making progress. This is why preconditioning, which clusters the spectrum, matters even more for GMRES than for CG.

Preconditioning

The convergence of every Krylov method depends on the spectrum of $\mathbf{A}$. **Preconditioning** replaces the system by an equivalent one with a better spectrum:

$$\text{left: } \mathbf{P}^{-1}\mathbf{A}\mathbf{x} = \mathbf{P}^{-1}\mathbf{b}, \qquad \text{right: } \mathbf{A}\mathbf{P}^{-1}\mathbf{y} = \mathbf{b},\ \ \mathbf{x} = \mathbf{P}^{-1}\mathbf{y},$$

where the preconditioner $\mathbf{P}$ satisfies two competing requirements: $\mathbf{P} \approx \mathbf{A}$ (so $\mathbf{P}^{-1}\mathbf{A} \approx \mathbf{I}$, eigenvalues clustered near 1) and $\mathbf{P}\mathbf{z} = \mathbf{r}$ cheap to solve. $\mathbf{P} = \mathbf{I}$ is cheap but useless; $\mathbf{P} = \mathbf{A}$ is perfect but as hard as the original problem. For CG, $\mathbf{P}$ must be SPD, and the preconditioned algorithm (PCG) is CG in the $\mathbf{P}$-inner product: the only change is that each iteration solves $\mathbf{P}\mathbf{z}_k = \mathbf{r}_k$ and uses $\mathbf{z}_k$ in place of $\mathbf{r}_k$ in the direction update and in α_k, β_k (via $\mathbf{r}_k^{\mathsf{T}}\mathbf{z}_k$).

Common preconditioners, in rough order of cost and power:

Preconditioner	$\mathbf{P}$	Comment
Jacobi (diagonal)	$\mathbf{D}$	removes bad *scaling*; free; weak otherwise
SSOR	$(\mathbf{D} + \omega\mathbf{L}_A)\mathbf{D}^{-1}(\mathbf{D} + \omega\mathbf{U}_A)$	symmetric version of SOR; no setup
Incomplete Cholesky IC(0) / incomplete LU ILU(0)	$\tilde{\mathbf{L}}\tilde{\mathbf{U}}$ with the sparsity of $\mathbf{A}$	elimination that discards all fill; workhorse of circuit and FEM codes
ILUT(τ, p)	incomplete LU keeping fill above threshold τ, at most p per row	tunable between ILU(0) and full LU
Multigrid / algebraic multigrid (AMG)	one V-cycle	$O(n)$ work and iteration count independent of h for elliptic problems
Domain decomposition	local solves on subdomains plus a coarse problem	parallel

Worked Example 24.22 — Diagonal scaling as a preconditioner

$\mathbf{A} = \begin{bmatrix} 10^4 & 1 \\ 1 & 1 \end{bmatrix}$ (a stiff spring coupled to a soft one, or a 10 kΩ and a 1 Ω branch sharing a node). Compare $\kappa_2(\mathbf{A})$ with that of the symmetrically Jacobi-preconditioned matrix $\mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}$.

Unpreconditioned. Eigenvalues $\approx 10^4$ and ≈ 0.9999: $\kappa_2 \approx 1.0 \times 10^4$.

Scaled. $\mathbf{D}^{-1/2} = \mathrm{diag}(10^{-2}, 1)$ gives $\begin{bmatrix} 1 & 0.01 \\ 0.01 & 1 \end{bmatrix}$, eigenvalues 1 ± 0.01, $\kappa_2 = 1.01/0.99 = 1.020$.

Reading. CG on the scaled system converges in 2 steps anyway ($n = 2$), but the iteration-count bound drops from about $\frac{1}{2}\sqrt{10^4}\ln(2/\epsilon)$ to essentially 1 step per digit. For large systems diagonal scaling never hurts and fixes purely unit-induced ill-conditioning; it cannot fix the *geometric* ill-conditioning of the model problem, whose diagonal is constant.

Worked Example 24.23 — Incomplete Cholesky on the 2-D Poisson problem

The 5-point Laplacian on a $k \times k$ grid ($n = k^2$ unknowns) is solved by CG to a relative residual of 10^{-8}, with and without an IC(0) preconditioner. Observed iteration counts:

grid	n	$\kappa(\mathbf{A})$	CG	PCG with IC(0)
16×16	256	116	28	17
32×32	1024	441	59	29
64×64	4096	1712	119	—

Reading. Without preconditioning the count doubles each time k doubles, tracking $\sqrt{\kappa} \propto k$ ($\sqrt{441} = 21$, $\sqrt{1712} = 41$); the bound $\frac{1}{2}\sqrt{\kappa}\ln(2/\epsilon) \approx 200$ for $k = 32$ is pessimistic by a factor of 3, as usual. IC(0) halves the count but does not change its growth rate (its preconditioned condition number still grows like h^{-2}, with a smaller constant). Each PCG iteration costs about twice a CG iteration (two triangular solves), so the net gain is modest here; it is much larger for problems with variable coefficients (materials with different permittivities or conductivities), where IC also corrects the scaling. Only multigrid makes the count independent of k.

Common Pitfall — wrong method for the matrix

CG requires $\mathbf{A}$ (and the preconditioner) to be symmetric positive definite. Applied to a nonsymmetric MNA matrix, or with a nonsymmetric ILU preconditioner, it may stall or diverge without any error message — in a test with an ILU preconditioner on the Poisson problem SciPy's `cg` ran to its iteration limit and returned garbage, while the same preconditioner inside GMRES worked. Check symmetry ($\|\mathbf{A} - \mathbf{A}^{\mathsf{T}}\|$) before choosing CG, use symmetric preconditioners (Jacobi, SSOR, IC) with it, and always check the returned convergence flag (`info` in SciPy).

Common Pitfall — stopping on the wrong quantity

A relative residual of 10^{-6} does not mean six correct digits: the error bound is κ times larger. Conversely, demanding 10^{-14} in a problem whose data are known to 10^{-3} wastes iterations. Choose the tolerance from the accuracy actually needed in the *output quantity* — often a single node voltage or a capacitance — and, if possible, monitor that quantity directly.

Engineering Note

In IC design, power-grid analysis (IR drop) solves SPD systems with 10^8 unknowns by PCG with algebraic multigrid or incomplete Cholesky preconditioners; full-wave EM solvers using integral equations (method of moments) solve dense complex systems by GMRES with matrix–vector products accelerated by the fast multipole method; transient circuit simulation, where each Newton step needs only a modest improvement on the previous solution, uses the previous LU factors as a preconditioner for a few GMRES steps. The common thread: the matrix–vector product and the preconditioner are designed together with the physics.

```
import numpy as np, scipy.sparse as sp, scipy.sparse.linalg as spla
k = 64
T = sp.diags([-1.0, 2.0, -1.0], [-1, 0, 1], shape=(k, k))
A = (sp.kron(sp.eye(k), T) + sp.kron(T, sp.eye(k))).tocsr()   # 2-D Laplacian
b = np.ones(k * k)
its = []
x, info = spla.cg(A, b, rtol=1e-8, callback=lambda xk: its.append(1))
print(info, len(its))          # 0 (converged), 119 iterations
```

24.5 Sparse computation

Motivation

The matrices of engineering are overwhelmingly **sparse**: a node in a circuit connects to a handful of others, a finite-element basis function overlaps only its neighbours, a bus in a power grid connects to a few lines. Chapter 8 introduced sparse storage (COO and CSR), fill-in and the effect of elimination order through the star-network example, and the Thomas algorithm for tridiagonal systems. This section makes those ideas quantitative: how the storage formats support the kernels that iterative and direct solvers need, how fill-in is predicted from the graph before any arithmetic is done, how reordering algorithms work, and how the cost of sparse factorisation scales with the geometry of the problem.

Storage formats and their kernels

CSR and CSC

For an $m \times n$ matrix with nnz nonzeros, **compressed sparse row** (CSR) stores three arrays: `data` (the nonzero values, row by row), `indices` (the column index of each value) and `indptr` (length $m+1$; row i occupies positions `indptr[i]` to `indptr[i+1]-1`). **Compressed sparse column** (CSC) stores the same information column by column: `indices` then holds row indices and `indptr` has length $n+1$. Both need $2\,\mathrm{nnz} + (\text{dimension} + 1)$ words.

The choice between them is dictated by the access pattern of the kernel:

Kernel	Natural format	Why
$\mathbf{y} = \mathbf{A}\mathbf{x}$	CSR	each y_i is an inner product of row i with $\mathbf{x}$
$\mathbf{y} = \mathbf{A}^{\mathsf{T}}\mathbf{x}$	CSC (or CSR of $\mathbf{A}$ read as CSC of $\mathbf{A}^{\mathsf{T}}$)	column access
left-looking LU, Cholesky, column pivoting	CSC	algorithms proceed column by column
row slicing, row-wise assembly	CSR	
assembly by stamping	COO	append (row, col, value) triplets; duplicates summed on conversion
ladder/1-D problems	banded / diagonal (DIA)	contiguous diagonals

The CSR matrix–vector product is the most executed loop in scientific computing:

```
def csr_matvec(data, indices, indptr, x):
    y = [0.0] * (len(indptr) - 1)
    for i in range(len(y)):
        s = 0.0
        for p in range(indptr[i], indptr[i + 1]):
            s += data[p] * x[indices[p]]      # indirect access to x
        y[i] = s
    return y
```

It performs $2\,\mathrm{nnz}$ flops and reads nnz values, nnz indices and (irregularly) entries of $\mathbf{x}$: an arithmetic intensity well below one flop per word, so its speed is set by memory bandwidth (Section 24.1). Orderings that cluster nonzeros near the diagonal improve cache reuse of $\mathbf{x}$ — a second, purely performance-related reason to reorder.

Worked Example 24.24 — CSR and CSC of a nonsymmetric matrix, and a product

Store

$$\mathbf{M} = \begin{bmatrix} 4 & 0 & -1 & 0 \\ 1 & 3 & 0 & -2 \\ 0 & 0 & 5 & 0 \\ 0 & -2 & 0 & 6 \end{bmatrix}$$

in CSR and CSC (0-based), and compute $\mathbf{M}\mathbf{x}$ for $\mathbf{x} = (1,2,3,4)^{\mathsf{T}}$ from the CSR arrays.

CSR. `data` $= [4,-1,\ 1,3,-2,\ 5,\ -2,6]$, `indices` $= [0,2,\ 0,1,3,\ 2,\ 1,3]$, `indptr` $= [0,2,5,6,8]$.

CSC. Reading down the columns: column 0 has 4 (row 0), 1 (row 1); column 1 has 3 (row 1), -2 (row 3); column 2 has -1 (row 0), 5 (row 2); column 3 has -2 (row 1), 6 (row 3). So `data` $= [4,1,\ 3,-2,\ -1,5,\ -2,6]$, `indices` $= [0,1,\ 1,3,\ 0,2,\ 1,3]$, `indptr` $= [0,2,4,6,8]$.

Product. Row 0: positions 0–1, $4\cdot x_0 + (-1)\cdot x_2 = 4 - 3 = 1$. Row 1: positions 2–4, $1\cdot 1 + 3\cdot 2 + (-2)\cdot 4 = -1$. Row 2: position 5, $5\cdot 3 = 15$. Row 3: positions 6–7, $-2\cdot 2 + 6\cdot 4 = 20$. So $\mathbf{M}\mathbf{x} = (1,-1,15,20)^{\mathsf{T}}$ — 16 flops for 8 nonzeros.

Check. A dense product gives the same result with 28 flops (16 multiplications, 12 additions), half of them on zeros. For a 10^6-node circuit with 5 nonzeros per row the ratio is $10^6/5 = 200\,000$.

The graph of a sparse matrix and the elimination game

Adjacency graph and fill

The **graph** $G(\mathbf{A})$ of a structurally symmetric $n \times n$ matrix has vertices $1,\dots,n$ and an edge $\{i,j\}$ whenever $a_{ij} \neq 0$, $i \neq j$. Eliminating vertex k (one step of Gaussian elimination) removes k and adds an edge between every pair of its remaining neighbours that were not already adjacent; these new edges are the **fill-in**. The **filled graph** $G^{+}(\mathbf{A})$ is $G(\mathbf{A})$ plus all fill edges; its edges are exactly the nonzeros of $\mathbf{L} + \mathbf{L}^{\mathsf{T}}$ (barring numerical cancellation).

This purely combinatorial description has three consequences that sparse direct solvers exploit.

1. **Symbolic factorisation.** The pattern of $\mathbf{L}$ can be computed from the graph alone, before any floating-point work, in time roughly proportional to $\mathrm{nnz}(\mathbf{L})$. Memory is allocated once.
2. **Reuse.** In Newton iterations, frequency sweeps, transient time steps and Monte Carlo runs, the *values* of the matrix change but its *pattern* does not. Ordering and symbolic factorisation are done once; only the numeric factorisation is repeated. SPICE has worked this way since the 1970s.
3. **The elimination tree.** Define the parent of column j as the row index of the first off-diagonal nonzero in column j of $\mathbf{L}$. The resulting tree encodes all dependencies: columns in disjoint subtrees can be factored independently, in parallel. Groups of consecutive columns with identical pattern (**supernodes**) are processed as dense blocks with level-3 BLAS, recovering the cache efficiency of dense code.

Reordering

The amount of fill depends on the elimination order, i.e. on the symmetric permutation $\mathbf{P}\mathbf{A}\mathbf{P}^{\mathsf{T}}$. Finding the order that minimises fill is NP-hard, so heuristics are used.

- **Bandwidth and profile reduction: (reverse) Cuthill–McKee.** Number the vertices by breadth-first search from a peripheral vertex, visiting neighbours in order of increasing degree; reversing the order (RCM) usually reduces the profile further. Elimination on a banded matrix creates fill only inside the band, so the fill is at most $n\times$(bandwidth).
- **Minimum degree** (and its fast approximation AMD): at each step eliminate the vertex with fewest neighbours in the current graph, since eliminating a degree-d vertex creates at most $d(d-1)/2$ fill edges. This is the greedy rule of the star example in Ch. 8 — leaves before hubs.
- **Nested dissection.** Find a small vertex **separator** S that splits the graph into two halves A and B with no edges between them; number A, then B, then S last; recurse on A and B. Since A and B are not connected, no fill ever links them; fill is confined to the separator blocks.
- **Column orderings for unsymmetric LU** (COLAMD) order the columns to limit fill in the factors of $\mathbf{A}^{\mathsf{T}}\mathbf{A}$'s pattern, leaving row pivoting free for numerical stability.

Fill and work for model grids

For the 5-point (2-D) and 7-point (3-D) Laplacians on grids with n unknowns, Cholesky factorisation costs, to leading order in n:

Ordering	2-D: $\operatorname{nnz}(\mathbf{L})$	2-D: flops	3-D: $\operatorname{nnz}(\mathbf{L})$	3-D: flops
natural (banded)	$n^{3/2}$	n^2	$n^{5/3}$	$n^{7/3}$
nested dissection	$n\log n$	$n^{3/2}$	$n^{4/3}$	n^2

Nested dissection is asymptotically optimal for these grids (George, 1973; Lipton–Rose–Tarjan).

Sketch, 2-D. A $k\times k$ grid ($n=k^2$) has a separator of k vertices (a grid line). After the two halves are eliminated the separator block is dense, $k\times k$, costing $O(k^3)=O(n^{3/2})$ flops to factor; the recursion $W(n)=4W(n/4)+O(n^{3/2})$ (splitting in both directions) gives $W=O(n^{3/2})$ by the master theorem (Section 24.1, $a=4$, $b=4$, $d=3/2>1$). The separators at level ℓ contribute 4^{ℓ} dense blocks of size $(k/2^{\ell})^2$, i.e. $O(n)$ fill per level over $\log n$ levels: $O(n\log n)$. With natural ordering the bandwidth is $k=\sqrt{n}$, the band fills completely ($n\cdot k=n^{3/2}$ entries) and the banded Cholesky costs $nk^2=n^2$. $\square$

In 3-D the separators are planes of $n^{2/3}$ vertices, and their dense blocks cost $(n^{2/3})^3=n^2$ — this is why 3-D field problems are the domain of iterative methods while 2-D problems and circuits (whose graphs have small separators) are routinely solved directly.

Worked Example 24.25 — Reverse Cuthill–McKee on a badly numbered chain

Six components form a chain whose nodes, in physical order along the chain, were numbered $0,3,5,1,4,2$ by a netlister. The nodal matrix has 2 on the diagonal and -1 for each link. Find its bandwidth, then renumber by RCM.

Original. The links are $\{0,3\},\{3,5\},\{5,1\},\{1,4\},\{4,2\}$, so the nonzero off-diagonals are in positions with $|i-j|=3,2,4,3,2$. The bandwidth is 4; banded elimination would treat the matrix as having up to 9 diagonals.

RCM. Start from a peripheral vertex (degree 1): vertex 2 or vertex 0. BFS from 0 visits $0,3,5,1,4,2$ — the physical order. Reversed: $2,4,1,5,3,0$. (SciPy's `reverse_cuthill_mckee` returns exactly this permutation.)

Result. In the new numbering consecutive chain nodes have consecutive numbers: the matrix is tridiagonal, bandwidth 1, and can be solved by the Thomas algorithm in $8n$ flops with zero fill.

Reading. For a chain this is obvious by eye; for a 10^5-node mesh it is not, and RCM does it in $O(\operatorname{nnz})$ time.

Worked Example 24.26 — Measured fill on a 2-D grid

The 2-D Laplacian on $k\times k$ grids is factored by SciPy's SuperLU with three column orderings. Nonzeros in $\mathbf{L}+\mathbf{U}$:

k	n	$\operatorname{nnz}(\mathbf{A})$	natural	COLAMD	minimum degree on $\mathbf{A}+\mathbf{A}^{\mathsf{T}}$
30	900	4 380	54 058	30 338	20 196
60	3 600	17 760	432 118	179 720	108 430
100	10 000	49 600	2 000 198	645 750	371 346

Check against theory. Natural ordering fills the band: $\operatorname{nnz}(\mathbf{L}+\mathbf{U})\approx 2nk=2k^3$, i.e. 54 000, 432 000, 2 000 000 ✓ — growing as $n^{3/2}$ (factor 8 per doubling of k). Minimum degree grows by a factor of 5.4 from $k=30$ to $k=60$, close to the $n\log n$ prediction $4\times\log(3600)/\log(900)\approx 4.8$.

Consequence. At $k=100$ the minimum-degree factors need 5.4 times less memory than natural ordering, and the flop savings are larger still (the flop count scales as the sum of squared column counts). At $k=1000$ ($n=10^6$) natural ordering would need 2×10^9 nonzeros (16 GB) and $\sim 10^{12}$ flops, nested dissection about $\frac{31}{4}n\log_2 n\approx 1.5\times 10^8$ nonzeros (1.2 GB) and $\frac{829}{84}n^{3/2}\approx 10^{10}$ flops (George's constants for the model grid).

Engineering Note — the circuit matrix is special

Circuit matrices differ from FEM matrices in ways that matter for solver choice: they are very sparse (often under 5 nonzeros per row), have a few dense rows and columns (supply rails, clock nets), are unsymmetric when controlled sources and voltage sources are present, and have small separators (circuits are "almost planar" locally). The KLU solver, designed for them, first permutes to **block triangular form** (each strongly connected component of the graph becomes a diagonal block, so only those blocks need factoring), then orders each block with AMD and factors it with partial pivoting. Power-grid analysis matrices, by contrast, are SPD and mesh-like, and are solved with sparse Cholesky (for $n \lesssim 10^7$) or PCG with multigrid.

Common Pitfall — accidental densification

Several innocent operations silently produce dense matrices: `A.toarray()` or `A.todense()` "for debugging"; `scipy.sparse.linalg.inv(A)` (the inverse of a sparse matrix is almost always dense — a node voltage depends on every source); `A + c` with a scalar $c \neq 0$; `numpy.linalg.solve(A.toarray(), b)` when a sparse solve was intended; or building a sparse matrix by element insertion into CSR (`A[i, j] = v` in a loop, quadratic cost — assemble in COO and convert once). Also beware of explicit zeros: a stamped matrix may store entries that cancel to exactly zero; call `eliminate_zeros()` only if the pattern will not change later (otherwise the stored zero keeps the symbolic factorisation valid).

Common Pitfall — ordering versus pivoting

For SPD matrices any symmetric ordering is numerically stable, so the ordering can be chosen purely for sparsity. For unsymmetric or indefinite matrices (MNA with voltage sources has zeros on the diagonal), the pivot sequence must also guard against small pivots. Sparse LU uses **threshold pivoting**: accept the sparsity-preferred pivot a_{kk} if $|a_{kk}| \geq u \max_i |a_{ik}|$ with u around 0.1 (or 0.001 in circuit simulators), otherwise pivot. Setting $u = 0$ for speed can give catastrophic growth.

Direct or iterative?

Putting Sections 24.4 and 24.5 together gives a practical rule:

Situation	Usual choice
circuits, 1-D and 2-D problems up to $\sim 10^6$–10^7 unknowns; many right-hand sides; need robustness	sparse direct (LU, Cholesky) with fill-reducing ordering
3-D field problems, $> 10^6$ unknowns	preconditioned Krylov (PCG for SPD, GMRES/BiCGSTAB otherwise), multigrid
matrix not available explicitly (only $\mathbf{A}\mathbf{v}$, e.g. fast multipole)	Krylov
sequence of slowly varying systems	direct factors of one reused as preconditioner for the others
low accuracy acceptable, cheap iterations	Krylov with loose tolerance

24.6 Monte Carlo methods

Motivation

Some quantities are expectations that are hard to compute any other way: the probability that a filter built from 5% components meets its specification, the bit-error rate of a receiver with a nonlinear equaliser, the average power absorbed by a randomly oriented antenna, an integral over a ten-dimensional parameter space. **Monte Carlo methods** estimate such quantities by *simulation*: draw random samples of the inputs, evaluate the quantity for each, and average. They are slow to converge — the error falls only as $1/\sqrt{N}$ — but that rate does not depend on the dimension of the problem or the smoothness of the function, and the method needs nothing more than the ability to evaluate the model. For high-dimensional problems, discontinuous pass/fail criteria, and black-box simulators, Monte Carlo is often the only practical method. Random variables, expectation, variance and the generation of samples from a given distribution (inverse transform) were developed in Ch. 19; sampling distributions and confidence intervals in Ch. 20.

The basic estimator

Monte Carlo estimator

To estimate $\mu = E[f(\mathbf{X})]$, where $\mathbf{X}$ is a random vector with known distribution, draw independent samples $\mathbf{X}_1, \dots, \mathbf{X}_N$ and form

$$\hat{\mu}_N = \frac{1}{N}\sum_{i=1}^{N} f(\mathbf{X}_i), \qquad s_N^2 = \frac{1}{N-1}\sum_{i=1}^{N} \bigl(f(\mathbf{X}_i) - \hat{\mu}_N\bigr)^2.$$

An integral $\int_\Omega g(\mathbf{x})\,d\mathbf{x}$ over a region of volume V is the special case $\mu = V\,E[g(\mathbf{X})]$ with $\mathbf{X}$ uniform on Ω. A probability $P(A)$ is the special case $f = \mathbf{1}_A$, the indicator of the event.

Convergence of Monte Carlo

If $\sigma^2 = \mathrm{Var}(f(\mathbf{X})) < \infty$, then

1. $\hat{\mu}_N$ is unbiased, $E[\hat{\mu}_N] = \mu$, with $\mathrm{Var}(\hat{\mu}_N) = \sigma^2/N$;
2. $\hat{\mu}_N \to \mu$ almost surely (strong law of large numbers);
3. $\sqrt{N}(\hat{\mu}_N - \mu)/\sigma \to \mathcal{N}(0,1)$ in distribution (central limit theorem), so for large N

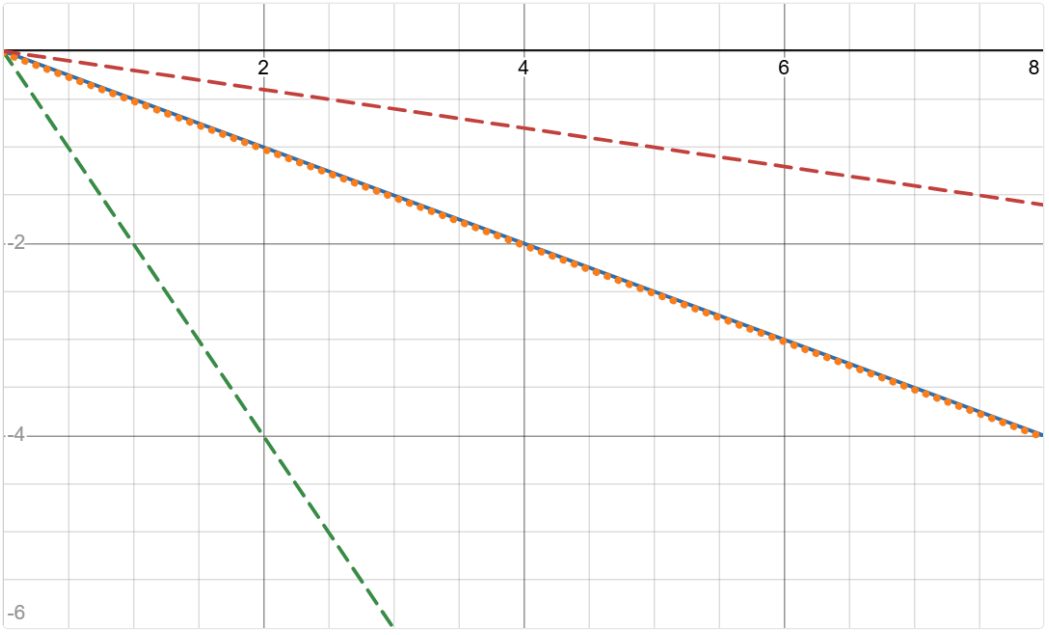
$$\mu \in \left[\hat{\mu}_N - z\frac{s_N}{\sqrt{N}},\ \hat{\mu}_N + z\frac{s_N}{\sqrt{N}}\right]$$

with confidence about 95% for $z = 1.96$ and 99.7% for $z = 3$.

The quantity $s_N/\sqrt{N}$ is the **standard error** of the estimate.

Proof of 1. Linearity of expectation gives $E[\hat{\mu}_N] = \frac{1}{N}\sum E[f(\mathbf{X}_i)] = \mu$; independence gives $\mathrm{Var}(\hat{\mu}_N) = \frac{1}{N^2}\sum \mathrm{Var}(f(\mathbf{X}_i)) = \sigma^2/N$. □

Three features stand out. The error decreases as $N^{-1/2}$: **each additional correct digit costs 100 times more samples**. The constant is σ, the standard deviation of the *integrand* — so reducing σ (variance reduction) is as good as increasing N. And neither the dimension d of $\mathbf{X}$ nor the smoothness of f appears. Compare a product trapezoidal rule in d dimensions, whose error is $O(h^2) = O(N^{-2/d})$ for N points: it beats Monte Carlo for $d < 4$, ties at $d = 4$, and loses for $d > 4$.



Log–log error versus number of function evaluations ($x = \log_{10} N$, $y = \log_{10}$ error, constants set to 1): Monte Carlo (blue, slope $-\frac{1}{2}$ in any dimension) against the trapezoidal rule in 1 dimension (green, slope -2), 4 dimensions (orange, slope $-\frac{1}{2}$, coinciding with Monte Carlo) and 10 dimensions (red, slope -0.2). In high dimension Monte Carlo wins.

Worked Example 24.27 — A one-dimensional integral and its error bar

Estimate $I = \int_0^1 e^x\,dx = e - 1 = 1.718\,282$ by Monte Carlo with $N = 10^4$ uniform samples, and predict the standard error.

Variance of the integrand. With $U \sim \mathcal{U}(0,1)$, $E[e^{2U}] = \frac{1}{2}(e^2 - 1) = 3.19453$, so $\sigma^2 = 3.19453 - (e-1)^2 = 0.242036$, $\sigma = 0.49197$.

Predicted standard error. $\sigma/\sqrt{N} = 0.49197/100 = 0.00492$.

One run (NumPy `default_rng(2024)`): $\hat{I} = 1.71993$, $s_N/\sqrt{N} = 0.00493$. The actual error 0.00165 is 0.33 standard errors — consistent with the CLT.

Reading. Three correct digits from 10^4 evaluations; the trapezoidal rule gets six digits with 40 points (error 9×10^{-5} with $n = 40$, Section 24.9). In one dimension Monte Carlo is a poor integrator; its value lies elsewhere.

Variance reduction

Since the error is $\sigma/\sqrt{N}$, any transformation that keeps the mean but reduces the variance of the averaged quantity by a factor r is equivalent to multiplying N by r. Four standard techniques:

Variance-reduction techniques

- **Antithetic variates.** Average f over pairs of negatively correlated samples, e.g. $\tfrac12\big(f(U)+f(1-U)\big)$. The variance of a pair mean is $\tfrac12\sigma^2(1+\rho)$, where $\rho=\operatorname{Corr}(f(U),f(1-U))$; it helps when f is monotone.
- **Control variates.** If $h(\mathbf{X})$ has *known* mean ν and is correlated with f, estimate $E[f-c(h-\nu)]$. The optimal coefficient and the resulting variance are

$$c^{*}=\frac{\operatorname{Cov}(f,h)}{\operatorname{Var}(h)},\qquad \operatorname{Var}\big(f-c^{*}(h-\nu)\big)=\sigma^2(1-\rho_{fh}^2).$$

- **Stratified sampling.** Partition the domain into strata, sample each stratum separately (proportionally to its probability) and combine. The between-strata component of the variance is eliminated.
- **Importance sampling.** Sample from a different density q that puts more mass where $|f|p$ is large, and reweight: $\mu=E_p[f(\mathbf{X})]=E_q\big[f(\mathbf{X})\,w(\mathbf{X})\big]$ with likelihood ratio $w=p/q$.

Derivation of the control-variate optimum. $\operatorname{Var}(f-ch)=\sigma_f^2-2c\operatorname{Cov}(f,h)+c^2\sigma_h^2$, a quadratic in c minimised at $c^{*}=\operatorname{Cov}(f,h)/\sigma_h^2$, where it equals $\sigma_f^2-\operatorname{Cov}(f,h)^2/\sigma_h^2=\sigma_f^2(1-\rho^2)$. A correlation of 0.99 reduces the variance 50-fold. □

Worked Example 24.28 — Three variance reductions on the same integral

Apply antithetic variates, a control variate and stratification to $I=\int_0^1 e^{x}\,dx$, at a cost of 10^4 evaluations of e^x each.

Antithetic. $\operatorname{Cov}(e^{U},e^{1-U})=E[e]-(e-1)^2=2.71828-2.95249=-0.23421$, so a pair mean has variance $\tfrac12(0.24204-0.23421)=0.003912$. Two independent samples averaged would have variance $0.24204/2=0.12102$: a reduction by a factor 30.9 for the same work. With 5000 pairs: $\hat I=1.71789\pm0.00088$.

Control variate $h(U)=U$ (mean $\tfrac12$, variance $\tfrac{1}{12}$). $\operatorname{Cov}(e^{U},U)=E[Ue^{U}]-\tfrac12(e-1)=1-0.85914=0.14086$, so $c^{*}=0.14086\times12=1.6903$ and the residual variance is $0.24204-0.14086^2\times12=0.003940$: a factor 61. With 10^4 samples: $\hat I=1.718462\pm0.00063$.

Stratified, one sample in each of 10^4 equal strata. $\hat I=1.718\,281\,80$, error 3×10^{-8} — the estimator is now essentially a randomised midpoint rule, whose error for smooth f falls as $N^{-3/2}$.

Reading. Variance reduction exploits knowledge of the integrand (monotonicity, a correlated function with known mean, smoothness). The cost is a little analysis; the gain can be orders of magnitude.

Rare events and importance sampling

Estimating a small probability p with plain Monte Carlo is expensive: the indicator has variance $p(1-p)$, so the *relative* standard error is

$$\frac{\sqrt{p(1-p)/N}}{p}\approx\frac{1}{\sqrt{Np}},$$

and a 10% relative error needs $N\approx100/p$ samples — about 100 *observed* events.

Worked Example 24.29 — A tail probability and a bit-error rate

(a) Estimate $p=P(X>4)$ for $X\sim\mathcal{N}(0,1)$ (exact $p=Q(4)=3.167\times10^{-5}$) with $N=10^4$ samples, plainly and by importance sampling from $\mathcal{N}(4,1)$. (b) How many simulated bits does a plain Monte Carlo BER measurement at $P_e=10^{-9}$ need?

(a) Plain. Relative standard error $\approx1/\sqrt{Np}=1/\sqrt{0.317}=1.78$ — with 10^4 samples we expect to see 0.3 events; the estimate is usually exactly zero. A 10% error would need 3.2×10^6 samples.

Importance sampling. Draw $Y_i\sim\mathcal{N}(4,1)$; the likelihood ratio is $w(y)=\dfrac{e^{-y^2/2}}{e^{-(y-4)^2/2}}=e^{-4y+8}$. The estimator $\frac1N\sum\mathbf{1}\{Y_i>4\}w(Y_i)$ is unbiased. Its second moment is $E_q[\mathbf{1}w^2]=e^{16}Q(8)=4.53\times10^{-9}$, giving a relative standard error $\sqrt{(4.53\times10^{-9}-p^2)/N}/p=2.1\%$ at $N=10^4$. One run: $\hat p=3.12\times10^{-5}\pm0.07\times10^{-5}$ ✓ — about 7000 times fewer samples than plain Monte Carlo for the same accuracy.

(b) BER. $N\approx100/P_e=10^{11}$ bits for a 10% error. At 10^6 simulated bits per second that is 28 hours per point on the BER curve. Importance sampling (biasing the noise variance or mean towards the decision boundary) or semi-analytic methods (simulate the noiseless waveform, compute the error probability analytically for each symbol) cut this to seconds.

Common Pitfall — importance sampling can make things worse

The variance of an importance-sampling estimator is $E_q[f^2w^2]-\mu^2$, which can be *infinite* if q has lighter tails than fp (e.g. sampling a heavy-tailed p from a Gaussian q). The estimator then converges erratically, with occasional enormous weights, and the sample standard error is meaningless. Always choose q with tails at least as heavy as p where $f\neq0$, and inspect the distribution of the weights; the **effective sample size** $(\sum w_i)^2/\sum w_i^2$ should not be tiny compared with N.

Quasi-Monte Carlo

Pseudo-random points cluster and leave gaps by chance. **Low-discrepancy sequences** (Halton, Sobol', lattice rules) fill the cube more evenly by design. For integrands of bounded variation the Koksma–Hlawka inequality bounds the error by the variation times the discrepancy of the point set, $O\big((\log N)^d/N\big)$ — asymptotically almost $1/N$ instead of $1/\sqrt{N}$. The points are deterministic, so error bars are obtained by **randomised QMC** (random shifts or scrambling,

repeated a few times). For smooth integrands in moderate dimension (up to tens), QMC routinely gains one to two orders of magnitude; for discontinuous indicators (pass/fail) the gain is smaller. SciPy provides `scipy.stats.qmc.Sobol`.

Monte Carlo tolerance analysis of a circuit

The most common engineering use of Monte Carlo is **statistical tolerance analysis**: components are manufactured with random deviations from their nominal values; what is the distribution of a circuit's performance, and what fraction of units (the **yield**) meets specification?

Tolerance analysis

Let a performance measure $y = g(p_1, \dots, p_m)$ depend on component values p_k with nominal values p_k^0 and tolerances $\pm t_k p_k^0$. **Worst-case analysis** finds the extreme values of y over the tolerance box. **Statistical analysis** models each p_k as a random variable (commonly uniform on the tolerance band for precision parts, or Gaussian with 3σ equal to the tolerance), and estimates the distribution of y; the **yield** is $Y = P(y \in [y_{\min}, y_{\max}])$.

The Monte Carlo procedure is:

1. Choose distributions (and correlations, e.g. resistors in the same thin-film network track each other) for all component values.
2. For $i = 1, \dots, N$: draw a component set, evaluate $y_i = g(\mathbf{p}_i)$ (a formula or a full circuit simulation).
3. Report $\hat{\mu}_y$, s_y, percentiles and the yield estimate $\hat{Y} = \frac{1}{N}\#\{y_i \text{ in spec}\}$ with standard error $\sqrt{\hat{Y}(1-\hat{Y})/N}$.

Worked Example 24.30 — Gain spread of a non-inverting amplifier

A non-inverting op-amp stage has gain $G = 1 + R_2/R_1$ with $R_1 = 1\ \text{k}\Omega$ and $R_2 = 9\ \text{k}\Omega$, both $\pm 1\%$ resistors modelled as independent and uniform on their tolerance bands. The specification is $G = 10 \pm 1\%$. Find the worst case, the standard deviation (by linearisation and by Monte Carlo) and the yield.

Worst case. $G_{\max} = 1 + 9 \times 1.01/0.99 = 10.1818$, $G_{\min} = 1 + 9 \times 0.99/1.01 = 9.8218$: $\pm 1.8\%$. Worst-case design says the spec cannot be guaranteed.

Linearisation. The relative sensitivities are $S^G_{R_2} = \dfrac{R_2}{G}\dfrac{\partial G}{\partial R_2} = \dfrac{R_2/R_1}{G} = 0.9$ and $S^G_{R_1} = -0.9$. A uniform $\pm 1\%$ deviation has standard deviation $\sigma_r = 0.01/\sqrt{3} = 0.5774\%$, so

$$\frac{\sigma_G}{G} \approx \sqrt{0.9^2 + 0.9^2}\,\sigma_r = 1.273 \times 0.5774\% = 0.735\%, \qquad \sigma_G \approx 0.0735.$$

Monte Carlo, $N = 10^5$:

```
rng = np.random.default_rng(12345)
N = 100_000
R1 = 1e3 * (1 + 0.01 * rng.uniform(-1, 1, N))
R2 = 9e3 * (1 + 0.01 * rng.uniform(-1, 1, N))
G = 1 + R2 / R1
yield_ = np.mean(np.abs(G - 10) <= 0.1)
```

Results: mean 10.0001, standard deviation 0.0738 (linearisation ✓), range $[9.8225,\ 10.1809]$ (inside the worst case, as it must be), yield within $\pm 1\%$: $\hat{Y} = 80.0\%$ (standard error 0.13%); within $\pm 1.5\%$: 97.2%.

Why not just use the Gaussian approximation? Treating G as Gaussian with $\sigma = 0.0735$ predicts a $\pm 1\%$ yield of $2\Phi(0.1/0.0735) - 1 = 82.6\%$. The true distribution of the sum of two uniform deviations is *triangular*, with thinner tails and more mass in the shoulders; the Gaussian overestimates the yield by 2.6 points. Monte Carlo makes no such assumption.

How many samples? With only $N = 1000$ samples, three seeds gave 78.6%, 79.3% and 80.2%, consistent with the standard error $\sqrt{0.8 \times 0.2/1000} = 1.3\%$. To quote yield to $\pm 0.5\%$ (one standard error) needs $N = 0.8 \times 0.2/0.005^2 = 6400$ runs.

Worked Example 24.31 — A nonlinear performance: RC cutoff frequency

An RC low-pass filter with $R = 10\ \text{k}\Omega \pm 5\%$ and $C = 10\ \text{nF} \pm 10\%$ (both uniform) has nominal $f_c = 1/(2\pi RC) = 1591.5$ Hz. The spec is f_c within $\pm 10\%$ of nominal. Compare linearised and Monte Carlo results.

Linearised. $S^{f_c}_R = S^{f_c}_C = -1$, so $\sigma_{f_c}/f_c = \sqrt{(0.05/\sqrt{3})^2 + (0.10/\sqrt{3})^2} = 6.45\%$, $\sigma_{f_c} = 102.7$ Hz, mean predicted 1591.5 Hz.

Monte Carlo ($N = 10^5$). Mean 1598.4 Hz, standard deviation 103.6 Hz, 0.5% and 99.5% percentiles 1396.7 and 1832.8 Hz, i.e. -12.2% and $+15.2\%$; yield within $\pm 10\%$: 87.4%.

Reading. Because $f_c \propto 1/(RC)$ is convex, the mean is *shifted upward* by 0.43% (second-order effect: $E[1/X] \approx \frac{1}{\mu}(1 + \sigma^2/\mu^2)$, and $(0.0645)^2 \approx 0.42\%$ ✓), and the distribution is skewed towards high frequencies. The linearised standard deviation is still accurate to 1%, but the asymmetric tails — which are what the yield depends on — are visible only in the simulation. Section 24.8 compares the two approaches systematically.

Engineering Note

SPICE simulators implement this directly (`.MC` analysis in many dialects; Monte Carlo with `mismatch` and `process` statistical models in foundry PDKs). Foundry models separate **global (process)** variation, common to all devices on a die, from **local (mismatch)** variation, independent per device and scaling as $1/\sqrt{\text{area}}$ (Pelgrom's law); a correct Monte Carlo draws one global sample per run and independent local samples per device. Because each run is a full circuit simulation, the techniques of this section — fewer runs through variance reduction, QMC, or importance sampling towards the failure region for high-sigma SRAM yield (failure probabilities of 10^{-6}–10^{-9} per cell) — translate directly into days of compute saved.

Common Pitfall — reporting Monte Carlo results without error bars

A Monte Carlo yield of "93.4%" from 200 runs has a standard error of 1.8%; the next digit is noise, and comparing it with "92.1%" from another design proves nothing. Always report the standard error (or a confidence interval), and when comparing two designs use **common random numbers** — the same random component deviations for both — so that the comparison is not swamped by sampling noise.

24.7 Parameter estimation

Motivation

Models contain parameters that must be learned from measurements: the time constant and final value of a charging capacitor, the saturation current and ideality factor of a diode, the ESR and inductance of a capacitor from an impedance sweep, the poles and zeros of a loudspeaker from its frequency response, the twenty parameters of a transistor compact model. When the model is linear in the parameters, the problem is linear least squares (Ch. 9), solved in one step by QR (Section 24.3). Most physical models are *nonlinear* in at least some parameters — τ in $e^{-t/\tau}$, n in $e^{V/(nV_T)}$, a resonance frequency — and need an iterative method. This section develops the Gauss–Newton and Levenberg–Marquardt algorithms, which exploit the least-squares structure far better than a general optimiser (Ch. 23), and shows how to attach uncertainties to the fitted parameters. The statistical justification — least squares is maximum likelihood under Gaussian noise — is in Ch. 20.

Problem statement

Nonlinear least squares

Given data (t_i, y_i), $i = 1, \dots, m$, and a model $y = \phi(t; \boldsymbol{\theta})$ with parameters $\boldsymbol{\theta} \in \mathbb{R}^n$ ($m > n$), define the **residuals** $r_i(\boldsymbol{\theta}) = y_i - \phi(t_i; \boldsymbol{\theta})$ and the cost

$$S(\boldsymbol{\theta}) = \tfrac{1}{2} \sum_{i=1}^{m} w_i \, r_i(\boldsymbol{\theta})^2 = \tfrac{1}{2} \, \mathbf{r}^{\mathsf{T}} \mathbf{W} \mathbf{r},$$

with weights $w_i = 1/\sigma_i^2$ (inverse noise variances; $\mathbf{W} = \mathbf{I}$ if the noise is uniform). The **Jacobian** $\mathbf{J} \in \mathbb{R}^{m \times n}$ of the model has entries $J_{ik} = \partial \phi(t_i; \boldsymbol{\theta}) / \partial \theta_k$, so that $\partial \mathbf{r} / \partial \boldsymbol{\theta} = -\mathbf{J}$.

The gradient and Hessian of S (with $\mathbf{W} = \mathbf{I}$) are

$$\nabla S = -\mathbf{J}^{\mathsf{T}} \mathbf{r}, \qquad \nabla^2 S = \mathbf{J}^{\mathsf{T}} \mathbf{J} - \sum_{i=1}^{m} r_i \nabla^2 \phi(t_i; \boldsymbol{\theta}).$$

The first Hessian term needs only first derivatives; the second is small when the residuals are small or the model is nearly linear. This structure is what the specialised methods exploit.

Gauss–Newton

Linearise the model about the current estimate $\boldsymbol{\theta}_k$: $\phi(t_i; \boldsymbol{\theta}_k + \boldsymbol{\delta}) \approx \phi(t_i; \boldsymbol{\theta}_k) + \mathbf{J}_i \boldsymbol{\delta}$. The residual becomes $\mathbf{r} - \mathbf{J}\boldsymbol{\delta}$, and minimising $\|\mathbf{r} - \mathbf{J}\boldsymbol{\delta}\|^2$ is a *linear* least-squares problem.

Gauss–Newton iteration

$$\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k + \boldsymbol{\delta}_k, \qquad \boldsymbol{\delta}_k = \arg\min_{\boldsymbol{\delta}} \|\mathbf{r}_k - \mathbf{J}_k \boldsymbol{\delta}\|_2, \quad \text{i.e.} \quad (\mathbf{J}_k^{\mathsf{T}} \mathbf{J}_k) \boldsymbol{\delta}_k = \mathbf{J}_k^{\mathsf{T}} \mathbf{r}_k.$$

The step should be computed by QR of $\mathbf{J}_k$, not by forming $\mathbf{J}_k^{\mathsf{T}} \mathbf{J}_k$ (Section 24.3), whenever $\mathbf{J}$ is ill-conditioned.

Gauss–Newton is Newton's method for $\nabla S = \mathbf{0}$ with the second Hessian term dropped. Consequently:

Local convergence of Gauss–Newton

Near a minimiser $\boldsymbol{\theta}^*$ with $\mathbf{J}(\boldsymbol{\theta}^*)$ of full rank, Gauss–Newton converges

- **quadratically** if the residuals at the solution are zero (the model fits exactly);
- **linearly** otherwise, with rate approximately $\rho\big((\mathbf{J}^{\mathsf{T}}\mathbf{J})^{-1} \sum_i r_i \nabla^2 \phi_i\big)$ evaluated at $\boldsymbol{\theta}^*$ — fast when the residuals are small relative to the model's curvature, and possibly divergent when they are large.

Far from the solution it may diverge: the linearised model can be a poor approximation, and the full step can overshoot.

Worked Example 24.32 — One Gauss–Newton step by hand

Fit $v(t) = a\,e^{-bt}$ to $(t, v) = (0, 5.0),\ (1, 3.1),\ (2, 1.8)$ starting from $(a, b) = (5,\ 0.5)$.

Model, residuals, Jacobian. $\phi = (5,\ 5e^{-0.5},\ 5e^{-1}) = (5,\ 3.03265,\ 1.83940)$; $\mathbf{r} = (0,\ 0.06735,\ -0.03940)$; $S_0 = \sum r_i^2 = 0.006088$ (we quote $\sum r_i^2$, i.e. $2S$). With $\partial\phi/\partial a = e^{-bt}$ and $\partial\phi/\partial b = -a\,t\,e^{-bt}$,

$$\mathbf{J} = \begin{bmatrix} 1 & 0 \\ 0.60653 & -3.03265 \\ 0.36788 & -3.67879 \end{bmatrix}.$$

Normal equations of the step.

$$\mathbf{J}^{\mathsf T}\mathbf{J} = \begin{bmatrix} 1.50321 & -3.19275 \\ -3.19275 & 22.73051 \end{bmatrix}, \qquad \mathbf{J}^{\mathsf T}\mathbf{r} = \begin{bmatrix} 0.02635 \\ -0.05930 \end{bmatrix}.$$

The determinant is $1.50321 \times 22.73051 - 3.19275^2 = 23.975$, so

$$\boldsymbol{\delta} = \frac{1}{23.975}\begin{bmatrix} 22.73051 & 3.19275 \\ 3.19275 & 1.50321 \end{bmatrix}\begin{bmatrix} 0.02635 \\ -0.05930 \end{bmatrix} = \begin{bmatrix} 0.01709 \\ -0.00021 \end{bmatrix}.$$

New estimate. $(a, b) = (5.01709,\ 0.49979)$, $\sum r_i^2 = 0.005625$. A library solver (`scipy.optimize.least_squares`) returns the same values to five digits and the same cost: one step has reached the minimum, because the starting point was close and the residuals small.

Contrast with the log-linear fit. Fitting $\ln v = \ln a - bt$ by linear least squares gives $(a, b) = (5.055,\ 0.511)$ — different, because the log transform reweights the data (small values gain influence). It is an excellent *starting point*, not the answer to the original problem (Ch. 9).

Levenberg–Marquardt

To make Gauss–Newton robust far from the solution, Levenberg (1944) and Marquardt (1963) added a damping term:

Levenberg–Marquardt step

$$\left(\mathbf{J}_k^{\mathsf T}\mathbf{J}_k + \lambda_k \mathbf{D}_k\right)\boldsymbol{\delta}_k = \mathbf{J}_k^{\mathsf T}\mathbf{r}_k,$$

with damping parameter $\lambda_k \ge 0$ and a positive diagonal scaling $\mathbf{D}_k$ (Levenberg: $\mathbf{D} = \mathbf{I}$; Marquardt: $\mathbf{D} = \operatorname{diag}(\mathbf{J}^{\mathsf T}\mathbf{J})$, which makes the method invariant to the units of the parameters). A typical update: if the step reduces S, accept it and decrease λ (e.g. $\lambda \leftarrow \lambda/10$); otherwise reject it, increase λ ($\lambda \leftarrow 10\lambda$) and recompute.

The damping interpolates between two methods:

- $\lambda \to 0$: the Gauss–Newton step (fast near the solution);
- $\lambda \to \infty$: $\boldsymbol{\delta} \approx \lambda^{-1}\mathbf{D}^{-1}\mathbf{J}^{\mathsf T}\mathbf{r} = -\lambda^{-1}\mathbf{D}^{-1}\nabla S$, a short (scaled) steepest-descent step, which always reduces S for λ large enough.

Equivalently, the LM step minimises the linearised cost $\|\mathbf{r} - \mathbf{J}\boldsymbol{\delta}\|^2$ subject to $\|\mathbf{D}^{1/2}\boldsymbol{\delta}\| \le \Delta$ for some trust radius Δ that decreases as λ increases: LM is a **trust-region** method, and modern implementations (MINPACK, `scipy.optimize.least_squares` with `method='lm'` or `'trf'`) adjust Δ directly by comparing the actual decrease of S with the decrease predicted by the linear model. The LM system is again best solved as an augmented least-squares problem $\min \left\| \begin{bmatrix} \mathbf{J} \\ \sqrt{\lambda}\,\mathbf{D}^{1/2} \end{bmatrix}\boldsymbol{\delta} - \begin{bmatrix} \mathbf{r} \\ \mathbf{0} \end{bmatrix} \right\|$ by QR.

Worked Example 24.33 — Fitting an RC charging curve: Gauss–Newton diverges, Levenberg–Marquardt converges

A capacitor charging through a resistor is sampled every 0.25 ms:

t (ms)	0.25	0.50	0.75	1.00	1.25	1.50	1.75	2.00
v (V)	1.342	2.330	3.036	3.550	3.943	4.213	4.440	4.616

(The data were synthesised from $V = 5$ V, $\tau = 0.8$ ms with Gaussian noise of $\sigma = 20$ mV.) Fit $v = V(1 - e^{-t/\tau})$ from the poor initial guess $V = 2$ V, $\tau = 5$ ms.

Jacobian. $\partial\phi/\partial V = 1 - e^{-t/\tau}$, $\partial\phi/\partial\tau = -V\,\dfrac{t}{\tau^2}e^{-t/\tau}$.

Gauss–Newton. The initial cost is $\sum r_i^2 = 80.3$ V^2. The first GN step jumps to $V = -42.4$ V, $\tau = -167$ ms — the linearisation about a nearly straight initial curve (with $\tau = 5$ ms the model is almost linear in t over the data) wildly extrapolates — and the iteration then diverges to $|V| \sim 10^{15}$.

Levenberg–Marquardt (Marquardt scaling, $\lambda_0 = 10^{-2}$, factor 10):

iter	V (V)	τ (ms)	$\sum r_i^2$ (V^2)	λ after step
0	2.000	5.000	80.27	10^{-2}
1	3.197	1.466	28.67	1
2	4.358	0.5445	0.4349	10^{-1}
3	4.644	0.6725	0.09904	10^{-2}
4	4.939	0.7815	0.004695	10^{-3}
5	5.0127	0.80667	0.0013139	10^{-4}
9	5.015855	0.807757	0.0013084	10^{-7}

In the first iteration three trial steps were rejected (λ raised to 10) before a short, safe step was accepted; thereafter λ falls by ten each iteration and the method becomes Gauss–Newton, converging rapidly. From the better start $(4\text{ V},\ 0.3\text{ ms})$ plain Gauss–Newton also converges, in about 6 iterations.

Reading. The final residual $\sqrt{0.0013084/6} = 14.8$ mV (dividing by $m - n = 6$ degrees of freedom) is consistent with the 20 mV noise. Fitted values: $V = 5.016$ V, $\tau = 0.808$ ms. How good are they? That needs the next subsection.

Uncertainty of the fitted parameters

Near the solution the model is approximately linear, $\mathbf{y} \approx \boldsymbol{\phi}(\boldsymbol{\theta}^*) + \mathbf{J}(\boldsymbol{\theta} - \boldsymbol{\theta}^*) + \text{noise}$, so the linear least-squares theory applies with $\mathbf{J}$ in place of the design matrix.

Asymptotic covariance of least-squares estimates

If the measurement errors are independent with variance σ^2 and the model is correct, the estimate $\hat{\boldsymbol{\theta}}$ has approximately

$$\operatorname{Cov}(\hat{\boldsymbol{\theta}}) \approx \sigma^2 (\mathbf{J}^{\mathsf{T}}\mathbf{J})^{-1}, \qquad \hat{\sigma}^2 = \frac{\sum_i r_i(\hat{\boldsymbol{\theta}})^2}{m - n},$$

with $\mathbf{J}$ evaluated at $\hat{\boldsymbol{\theta}}$. With weights $\mathbf{W} = \operatorname{diag}(1/\sigma_i^2)$ the covariance is $(\mathbf{J}^{\mathsf{T}}\mathbf{W}\mathbf{J})^{-1}$. Standard errors are the square roots of the diagonal; the **correlation** between $\hat{\theta}_j$ and $\hat{\theta}_k$ is $C_{jk}/\sqrt{C_{jj}C_{kk}}$. For small samples, approximate 95% intervals use Student's t_{m-n} quantile (Ch. 20).

Sketch. Linearising, $\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^* \approx (\mathbf{J}^{\mathsf{T}}\mathbf{J})^{-1}\mathbf{J}^{\mathsf{T}}\boldsymbol{\epsilon}$, whose covariance is $(\mathbf{J}^{\mathsf{T}}\mathbf{J})^{-1}\mathbf{J}^{\mathsf{T}}(\sigma^2\mathbf{I})\mathbf{J}(\mathbf{J}^{\mathsf{T}}\mathbf{J})^{-1} = \sigma^2(\mathbf{J}^{\mathsf{T}}\mathbf{J})^{-1}$. This is the linear error propagation of Section 24.8 applied to the map from data to estimate. □

Worked Example 24.34 — Error bars and correlation for the RC fit

Continue Worked Example 24.33.

Covariance. With $\hat{\sigma} = 14.8$ mV and $\mathbf{J}$ at the solution, $\hat{\sigma}^2(\mathbf{J}^{\mathsf{T}}\mathbf{J})^{-1}$ gives standard errors $u(V) = 0.020$ V and $u(\tau) = 0.0077$ ms, and a correlation coefficient $+0.93$.

Intervals. With $t_{6,0.975} = 2.447$: $V = 5.016 \pm 0.049$ V, $\tau = 0.808 \pm 0.019$ ms (95%). Both contain the true values 5 V and 0.8 ms ✓.

The correlation matters. The data stop at $t = 2$ ms $= 2.5\tau$, before the curve has settled; a larger final value with a slower time constant fits almost as well as a smaller one with a faster time constant. The individual error bars understate how well the *combination* is determined, and any quantity derived from both (e.g. the initial slope V/τ) must be propagated with the full covariance (Section 24.8). Extending the record to 5τ would decorrelate them.

Scaling. In SI units the columns of $\mathbf{J}$ differ by a factor of order 10^3–10^4 (V per V versus V per second), and $\kappa(\mathbf{J}^{\mathsf{T}}\mathbf{J}) = 5.1 \times 10^7$. Measuring τ in milliseconds gives $\kappa = 65$. The answer is the same in exact arithmetic, but a well-scaled problem is kinder to every algorithm — Marquardt's diagonal scaling achieves the same effect automatically.

Identifiability and good formulations

Identifiability

A parameter vector is **(locally) identifiable** from an experiment if distinct parameter values near $\boldsymbol{\theta}^*$ produce distinct model outputs, i.e. $\mathbf{J}(\boldsymbol{\theta}^*)$ has full column rank. It is **practically identifiable** if, in addition, $\mathbf{J}^\mathsf{T}\mathbf{J}$ is well enough conditioned that the noise does not produce huge parameter uncertainties.

Practical non-identifiability is common and shows up as near-singular $\mathbf{J}^\mathsf{T}\mathbf{J}$, correlations near ± 1 and enormous standard errors. The remedies are experimental (measure over a wider range, excite the relevant dynamics) or mathematical (reparametrise, fix a parameter, add prior information as regularisation).

Worked Example 24.35 — Diode parameters: correlation by design

The forward diode law $I = I_s e^{V/(nV_T)}$ becomes linear after taking logarithms: $\ln I = \ln I_s + \dfrac{1}{nV_T} V$, with unknowns $\theta_1 = \ln I_s$ and $\theta_2 = 1/(nV_T)$. Measurements are taken at 11 equally spaced voltages from 0.5 V to 0.7 V. Find the correlation between the estimates, and fix it.

Correlation. The design matrix has columns $\mathbf{1}$ and $\mathbf{V}$. With equal noise, $\mathrm{Cov} \propto (\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}$, and the correlation is $-\bar{V}/\sqrt{\overline{V^2}} = -0.6/\sqrt{0.36 + 0.004} = -0.994$. The intercept $\ln I_s$ is an extrapolation to $V = 0$, far outside the data; any error in the slope swings it strongly.

Reparametrisation. Write $\ln I = \ln I_0 + \theta_2 (V - 0.6)$, where I_0 is the current at the centre of the data. The columns $\mathbf{1}$ and $\mathbf{V} - \bar{V}$ are orthogonal, the correlation is exactly 0, and $\ln I_0$ is determined to the noise level divided by $\sqrt{11}$. $I_s = I_0 e^{-0.6\theta_2}$ can be computed afterwards — with its large but now honestly propagated uncertainty.

Reading. Parametrise models by quantities the data actually measure: a value at the centre of the data rather than an extrapolated intercept, log of scale parameters that span decades (like $I_s \sim 10^{-14}$ A), time constants rather than poles when the data are in the time domain.

Common Pitfall — local minima and bad starting values

Nonlinear least squares can have several local minima — fitting a sum of two exponentials, a resonance frequency, or a phase is notorious — and every method here finds only a local one. Get starting values from physics (the final value of a step response, the -3 dB point of a response), from a linearising transformation (Worked Example 24.32), or by a coarse grid search over the nonlinear parameters; and for models linear in some parameters, eliminate those analytically (**variable projection**: for each trial τ, the best V is a one-line linear least-squares solve), leaving a lower-dimensional search.

Engineering Note

Device-model extraction (BSIM transistor models, IBIS buffer models, battery equivalent circuits) is large-scale weighted nonlinear least squares, usually with LM and carefully staged parameter subsets so that each stage is identifiable. Impedance spectroscopy fits equivalent circuits to complex data by stacking real and imaginary residuals, $\mathbf{r} = [\mathrm{Re}\{\mathbf{Z}_\mathrm{meas} - \mathbf{Z}_\mathrm{model}\},\ \mathrm{Im}\{\cdots\}]$, often weighted by $1/|Z|$ so that every decade of frequency counts equally. Vector fitting, used to build rational macromodels of interconnects from S-parameter data, is an iterated *linear* least-squares method that relocates poles — an example of choosing a formulation that avoids nonlinear search altogether.

24.8 Sensitivity and uncertainty propagation

Motivation

Every computed result inherits uncertainty from its inputs: component tolerances, measured parameters, fitted model coefficients, environmental conditions. Two related questions follow. **Sensitivity analysis** asks how much the output changes per unit change of each input — which resistor sets the gain error, which permittivity dominates the resonance shift — and therefore where to spend money on tighter tolerances. **Uncertainty propagation** asks what the resulting uncertainty of the output is, so that a measurement or a simulation can be reported as "$y = \hat{y} \pm U$". Both depend on derivatives of the output with respect to the inputs, and computing those derivatives accurately and cheaply for large models is itself a computational problem. This section treats derivative computation (finite differences, complex step, automatic differentiation and the adjoint method), then linearised propagation in the framework of the *Guide to the Expression of Uncertainty in Measurement* (GUM), then Monte Carlo propagation and the conditions under which the two disagree.

Sensitivities

Sensitivity

For an output $y = f(p_1, \dots, p_m)$, the **absolute sensitivity** to p_k is $\partial y / \partial p_k$ and the **relative (normalised, Bode) sensitivity** is

$$S_{p_k}^{y} = \frac{p_k}{y}\frac{\partial y}{\partial p_k} = \frac{\partial \ln |y|}{\partial \ln |p_k|},$$

the percentage change in y per percent change in p_k. The vector of absolute sensitivities is the gradient $\nabla_{\mathbf{p}} y$; for several outputs it is the **sensitivity matrix** (Jacobian) $\mathbf{S} = \partial \mathbf{y} / \partial \mathbf{p}$.

Relative sensitivities are dimensionless and obey simple rules: $S_p^{y_1 y_2} = S_p^{y_1} + S_p^{y_2}$, $S_p^{1/y} = -S_p^{y}$, $S_p^{y^n} = n S_p^{y}$. For an RC cutoff $f_c = 1/(2\pi RC)$, $S_R^{f_c} = S_C^{f_c} = -1$; for an inverting amplifier $G = -R_2/R_1$, $S_{R_2}^{G} = 1$, $S_{R_1}^{G} = -1$.

Computing derivatives numerically

Finite differences. The forward difference $\big(f(x+h)-f(x)\big)/h$ has truncation error $\tfrac{1}{2}hf''$ and rounding error about $2\varepsilon|f|/h$ (Ch. 22). Balancing them gives $h_{\text{opt}} \approx \sqrt{\varepsilon}|x| \approx 10^{-8}|x|$ and a best relative accuracy of about $\sqrt{\varepsilon} \approx 10^{-8}$ — half the digits. The central difference $\big(f(x+h)-f(x-h)\big)/(2h)$ has truncation error $\tfrac{1}{6}h^2 f'''$, giving $h_{\text{opt}} \approx \varepsilon^{1/3}$ and accuracy about $\varepsilon^{2/3} \approx 10^{-11}$. Each gradient of m parameters costs m (forward) or $2m$ (central) extra evaluations of f.

Complex step. If f is real-analytic and implemented with operations that extend to complex arguments, Taylor expansion gives $f(x+jh) = f(x) + jhf'(x) - \tfrac{1}{2}h^2 f''(x) - \tfrac{1}{6}jh^3 f'''(x) + \cdots$, so

$$f'(x) = \frac{\operatorname{Im}\{f(x+jh)\}}{h} + O(h^2).$$

There is **no subtraction**, hence no cancellation: h can be made as small as 10^{-20} or 10^{-100} and the derivative is accurate to machine precision. The cost is one complex evaluation per parameter. Circuit and EM codes that already use complex arithmetic for phasors can often be differentiated this way with almost no change — but functions such as `abs`, `max` or comparisons must be redefined for complex arguments, and the code must not take the conjugate of the perturbed variable.

Worked Example 24.36 — Finite differences versus complex step

Differentiate $f(x) = \dfrac{e^{x}}{\sqrt{\sin^3 x + \cos^3 x}}$ at $x = 1.5$ (exact $f'(1.5) = 4.053\,427\,894$). Relative errors:

h	forward difference	central difference	complex step
10^{-2}	1.2×10^{-2}	1.3×10^{-4}	1.3×10^{-4}
10^{-4}	1.2×10^{-4}	1.3×10^{-8}	1.3×10^{-8}
10^{-6}	1.2×10^{-6}	1.4×10^{-10}	1.3×10^{-12}
10^{-8}	2.5×10^{-8}	8.4×10^{-9}	$< 10^{-16}$
10^{-10}	3.5×10^{-6}	2.0×10^{-7}	$< 10^{-16}$
10^{-12}	2.7×10^{-4}	5.4×10^{-5}	$< 10^{-16}$
10^{-20}	(meaningless)	(meaningless)	2.2×10^{-16}

Reading. Finite differences reach their best accuracy at an intermediate h (10^{-8} forward, about 10^{-6} central) and get *worse* as h decreases further — the V-shaped error curve of Ch. 22. The complex step improves monotonically and saturates at machine precision. For both large and small h, the central difference and complex step share the same $O(h^2)$ truncation error (identical first two rows).

Automatic differentiation (AD). A program is a composition of elementary operations whose derivatives are known. AD applies the chain rule to the *program*, operation by operation, producing derivatives exact to rounding error with no step size at all. **Forward mode** propagates, alongside each intermediate value v, its derivative $\dot v = \partial v/\partial p$ with respect to *one* input (dual numbers $v + \dot v\epsilon$ with $\epsilon^2 = 0$; the complex step is a close relative); a full gradient needs m passes. **Reverse mode** first evaluates the program and records it, then propagates *adjoints* $\bar v = \partial y/\partial v$ backwards from *one* output; a full gradient with respect to all m inputs costs a small constant multiple (typically 2–5) of one function evaluation, independent of m. Reverse-mode AD is "backpropagation" in neural networks and is what JAX, PyTorch and similar tools implement.

The adjoint method

The reverse-mode idea, applied to a model defined by a linear system, is the **adjoint method**. It is the key to sensitivity analysis of large circuits and field models, where there may be thousands of parameters but only a few outputs.

Let the state $\mathbf{x} \in \mathbb{R}^n$ be defined by $\mathbf{A}(\mathbf{p})\mathbf{x} = \mathbf{b}(\mathbf{p})$ and the output be $y = \mathbf{c}^{\mathsf T}\mathbf{x}$ (a node voltage, a port current). Differentiating the system with respect to p_k:

$$\mathbf{A}\frac{\partial \mathbf{x}}{\partial p_k} = \frac{\partial \mathbf{b}}{\partial p_k} - \frac{\partial \mathbf{A}}{\partial p_k}\mathbf{x}.$$

The **direct** method solves this for $\partial\mathbf{x}/\partial p_k$ — one solve per parameter — and forms $\partial y/\partial p_k = \mathbf{c}^{\mathsf T}\partial\mathbf{x}/\partial p_k$. The **adjoint** method notices that only the combination $\mathbf{c}^{\mathsf T}\mathbf{A}^{-1}$ is needed.

Adjoint sensitivity formula

Let $\boldsymbol{\lambda}$ solve the **adjoint system** $\mathbf{A}^{\mathsf T}\boldsymbol{\lambda} = \mathbf{c}$. Then for every parameter p_k

$$\frac{\partial y}{\partial p_k} = \boldsymbol{\lambda}^{\mathsf T}\Big(\frac{\partial \mathbf{b}}{\partial p_k} - \frac{\partial \mathbf{A}}{\partial p_k}\mathbf{x}\Big).$$

All m sensitivities of one output cost **one** extra solve (with the transposed matrix, whose LU factors are those of $\mathbf{A}$ transposed) plus m cheap inner products — instead of m solves.

Proof. $\dfrac{\partial y}{\partial p_k} = \mathbf{c}^{\mathsf T}\dfrac{\partial \mathbf{x}}{\partial p_k} = \mathbf{c}^{\mathsf T}\mathbf{A}^{-1}\Big(\dfrac{\partial \mathbf{b}}{\partial p_k} - \dfrac{\partial \mathbf{A}}{\partial p_k}\mathbf{x}\Big)$, and $\mathbf{c}^{\mathsf T}\mathbf{A}^{-1} = (\mathbf{A}^{-\mathsf T}\mathbf{c})^{\mathsf T} = \boldsymbol{\lambda}^{\mathsf T}$. $\square$

The rule of thumb: **direct** sensitivities when there are few parameters and many outputs; **adjoint** sensitivities when there are many parameters and few outputs (the common case in design optimisation and tolerance analysis).

For a conductance g connected between nodes a and b of a nodal matrix, $\partial\mathbf{G}/\partial g = (\mathbf{e}_a - \mathbf{e}_b)(\mathbf{e}_a - \mathbf{e}_b)^{\mathsf T}$ (the stamp of a unit conductance), so

$$\frac{\partial y}{\partial g} = -(\lambda_a - \lambda_b)(v_a - v_b)\,:$$

the product of the branch voltage in the original circuit and the branch voltage in the adjoint circuit. This is the classical **adjoint-network** result derived from Tellegen's theorem (Director and Rohrer, 1969). For a reciprocal (symmetric) network the adjoint network is the original network, driven by a unit current source at the output node.

Worked Example 24.37 — Sensitivities of a node voltage to every conductance with one extra solve

The three-node network of Worked Example 24.17 has conductances (in siemens) $g_{12}=2$, $g_{13}=1$, $g_{23}=1$ between nodes and $g_1=1$, $g_2=1$, $g_3=2$ from each node to ground, and source currents $\mathbf{b}=(8,2,1)$ A, giving $\mathbf{G}=\begin{bmatrix}4&-2&-1\\-2&4&-1\\-1&-1&4\end{bmatrix}$ and $\mathbf{v}=(4,3,2)$ V. Find $\partial v_3/\partial g$ for all six conductances and $\partial v_3/\partial b_k$ for all three sources.

Adjoint solve. $\mathbf{G}^{\mathsf T}\boldsymbol{\lambda}=\mathbf{e}_3$ ($\mathbf{G}$ is symmetric). Check that $\boldsymbol{\lambda}=(\tfrac16,\tfrac16,\tfrac13)$: row 1, $\tfrac46-\tfrac26-\tfrac13=0$ ✓; row 2 similarly 0 ✓; row 3, $-\tfrac16-\tfrac16+\tfrac43=1$ ✓.

Source sensitivities. $\partial\mathbf{b}/\partial b_k=\mathbf{e}_k$, so $\partial v_3/\partial b_k=\lambda_k$: $(\tfrac16,\tfrac16,\tfrac13)\ \Omega$ — these are transfer resistances, by reciprocity.

Conductance sensitivities $-(\lambda_a-\lambda_b)(v_a-v_b)$ (for a grounded element, $\lambda_b=v_b=0$):

element	$\lambda_a-\lambda_b$	v_a-v_b (V)	$\partial v_3/\partial g$ (V/S)	relative $S_g^{v_3}$
g_{12}	0	1	0	0
g_{13}	$-\tfrac16$	2	$+0.3333$	$+0.167$
g_{23}	$-\tfrac16$	1	$+0.1667$	$+0.083$
g_1	$\tfrac16$	4	-0.6667	-0.333
g_2	$\tfrac16$	3	-0.5000	-0.250
g_3	$\tfrac13$	2	-0.6667	-0.667

Check by finite difference. Increasing g_{13} by 10^{-6} and re-solving changes v_3 by 3.3333×10^{-7} ✓; every row agrees to six digits.

Reading. One factorisation, one extra (transposed) solve, and six inner products give every sensitivity; the direct method would need six solves, and for a circuit with 10^4 elements the adjoint saves a factor of 10^4. The table also explains the physics: g_{12} does not affect v_3 at all because nodes 1 and 2 are at the same potential in the adjoint circuit, and the shunt g_3 at the output node has the largest relative influence.

Linearised uncertainty propagation (GUM)

Standard uncertainty and the GUM law of propagation

The **standard uncertainty** $u(x)$ of an input estimate x is its estimated standard deviation — evaluated statistically from repeated observations (**Type A**) or from other information such as a datasheet tolerance or calibration certificate (**Type B**; a uniform interval $\pm a$ gives $u=a/\sqrt{3}$). For a measurement model $Y=f(X_1,\dots,X_n)$ with estimates x_i, the **combined standard uncertainty** of $y=f(x_1,\dots,x_n)$ is, to first order,

$$u_c^2(y)=\sum_{i=1}^{n}c_i^2u^2(x_i)+2\sum_{i<j}c_ic_j\,u(x_i,x_j),\qquad c_i=\left.\frac{\partial f}{\partial x_i}\right|_{\mathbf{x}},$$

where c_i are the **sensitivity coefficients** and $u(x_i,x_j)$ the covariances. In matrix form, for a vector output, $\mathbf{U}_y=\mathbf{J}\mathbf{U}_x\mathbf{J}^{\mathsf T}$. The **expanded uncertainty** is $U=k\,u_c(y)$ with **coverage factor** k, usually $k=2$ (about 95% coverage for a Gaussian output).

Derivation. Expand f to first order, $Y-y\approx\sum_i c_i(X_i-x_i)$; the variance of a linear combination is $\sum_{i,j}c_ic_j\,\mathrm{Cov}(X_i,X_j)$. $\square$

For products and quotients of powers, $y=\prod x_i^{a_i}$ with uncorrelated inputs, the rule takes the convenient relative form $\big(u_c(y)/y\big)^2=\sum_i a_i^2\big(u(x_i)/x_i\big)^2$ — the relative sensitivities are just the exponents.

Worked Example 24.38 — Power from voltage and resistance

The power dissipated in a resistor is computed as $P=V^2/R$ from $V=5.000$ V with $u(V)=0.020$ V and $R=100.0\ \Omega$ with $u(R)=0.50\ \Omega$ (independent). Find $u_c(P)$ and the expanded uncertainty.

Estimate. $P=25/100=0.2500$ W.

Relative form. Exponents 2 and -1: $\big(u_c/P\big)^2=(2\times0.004)^2+(0.005)^2=6.4\times10^{-5}+2.5\times10^{-5}=8.9\times10^{-5}$, so $u_c/P=0.943\%$ and $u_c(P)=2.36$ mW.

Budget. The voltage contributes $64/89=72\%$ of the variance despite its smaller relative uncertainty, because of the square. A better voltmeter is the cheaper improvement.

Result. $P=(250.0\pm4.7)$ mW with $k=2$.

Monte Carlo check (10^5 Gaussian draws): mean 250.00 mW, standard deviation 2.35 mW ✓. The model is nearly linear over $\pm1\%$, so linearisation is excellent.

Worked Example 24.39 — Correlated inputs: a quantity derived from fitted parameters

From the RC fit of Worked Example 24.34, compute the initial slope $s = V/\tau$ and its uncertainty, with and without the correlation $\rho = 0.931$.

Estimate. $s = 5.0159/0.80776 \times 10^{-3} = 6210$ V/s.

Sensitivities (relative). $s = V\tau^{-1}$: exponents $+1$ and -1. Relative uncertainties $u(V)/V = 0.398\%$, $u(\tau)/\tau = 0.949\%$.

Ignoring correlation. $\sqrt{0.398^2 + 0.949^2}\% = 1.03\%$, $u(s) = 64$ V/s.

With correlation. The cross term is $2 \times (+1)(-1) \times \rho \times 0.398\% \times 0.949\%$:

$$\left(\frac{u(s)}{s}\right)^2 = 0.398^2 + 0.949^2 - 2(0.931)(0.398)(0.949) = 0.1584 + 0.9006 - 0.7033 = 0.3557\ (\%)^2,$$

so $u(s)/s = 0.596\%$, $u(s) = 37$ V/s.

Reading. V and τ err in the *same* direction, so their ratio — which the early data points measure directly — is much better determined than either. Dropping the covariance overstates the uncertainty by 73% here; with $\rho < 0$ it would understate it. When propagating from fitted parameters, always use the full covariance matrix $\mathbf{J}_s \mathbf{C} \mathbf{J}_s^{\mathsf{T}}$.

Monte Carlo propagation and the limits of linearisation

The **GUM Supplement 1** approach replaces the first-order formula by simulation: assign a probability distribution to each input (Gaussian, uniform, triangular, or a joint distribution for correlated inputs), draw N samples (typically 10^5–10^6), evaluate the model for each, and summarise the output distribution by its mean, standard deviation and a *coverage interval* taken from the empirical quantiles (e.g. the 2.5% and 97.5% percentiles). It is exactly the tolerance analysis of Section 24.6, with measurement uncertainties in place of tolerances.

Linearisation and Monte Carlo agree when the model is approximately linear over the region where the input distributions have most of their mass, i.e. when second-order terms $\tfrac{1}{2} f'' u^2$ are small compared with $f' u$. They disagree when:

1. **The model is strongly nonlinear over the uncertainty range** (large relative uncertainties; exponentials, ratios with uncertain denominators near zero).
2. **The first derivative vanishes** at the estimate (an extremum: maximum power transfer, the peak of a resonance, a balanced bridge), so the linear formula predicts zero uncertainty.
3. **The output distribution is far from Gaussian** (bounded, skewed or bimodal), so that $y \pm 2u_c$ is not a 95% interval.
4. **The quantity of interest is a probability or a quantile** (yield, coverage) rather than a standard deviation.

For case 2 the GUM provides a second-order correction: for a single Gaussian input, $u_c^2 = c_1^2 u^2 + \tfrac{1}{2}\left(f''\right)^2 u^4$.

Worked Example 24.40 — Matched load: where linearisation predicts zero

A source with $R_s = 50\ \Omega$ drives a load R_L; the delivered power relative to the maximum is $\eta = \dfrac{4R_s R_L}{(R_s + R_L)^2}$. The load is nominally 50 Ω with a 5% Gaussian standard uncertainty. Find the uncertainty of η by GUM (first and second order) and by Monte Carlo.

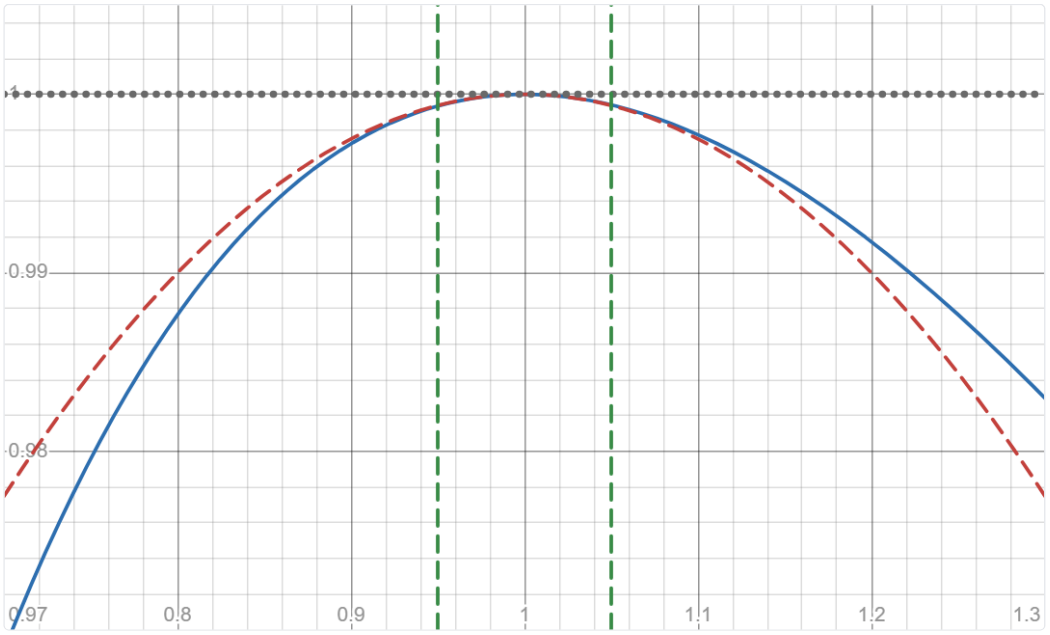
First order. With $x = R_L/R_s$, $\eta = 4x/(1+x)^2$, $d\eta/dx = 4(1-x)/(1+x)^3 = 0$ at $x = 1$. The linear law gives $u_c(\eta) = 0$: "the power is exactly 100%".

Second order. Expanding about $x = 1 + e$: $\eta = \dfrac{1+e}{(1+e/2)^2} \approx (1+e)(1 - e + \tfrac{3}{4}e^2) = 1 - \tfrac{1}{4}e^2$. So $\eta'' = -\tfrac{1}{2}$, and with $e \sim \mathcal{N}(0, 0.05^2)$:

$$E[\eta] \approx 1 - \tfrac{1}{4}(0.05)^2 = 0.999375, \qquad u(\eta) = \tfrac{1}{4}\sqrt{\mathrm{Var}(e^2)} = \tfrac{1}{4}\sqrt{2}(0.05)^2 = 0.000884.$$

Monte Carlo (10^5 samples). Mean 0.999373, standard deviation 0.000896, median 0.99972, 95% coverage interval $[0.99685,\ 0.99999]$.

Reading. The second-order GUM matches the mean and standard deviation, but the distribution is one-sided (it can never exceed 1) and strongly skewed: $\eta \pm 2u$ would include impossible values above 1. The Monte Carlo coverage interval is the honest answer. Mismatch is always a *loss*, never a gain — a qualitative fact the linear method cannot express.



Relative delivered power $\eta = 4x/(1+x)^2$ against $x = R_L/R_s$ (blue) with its quadratic approximation $1-(x-1)^2/4$ (red, dashed). The green lines mark ± 1 standard uncertainty of the load. At the maximum the tangent is horizontal, so a first-order propagation sees no uncertainty at all; the curvature maps a symmetric input spread into a one-sided output spread below 1.

	Linearised (GUM)	Monte Carlo (GUM-S1)
Cost	one evaluation plus m derivatives (one adjoint solve for many inputs)	10^4–10^6 evaluations
Nonlinearity	first order (second order with extra effort)	exact for the assumed input distributions
Output distribution	assumed Gaussian (or Student t)	estimated
Correlated inputs	covariance terms	joint sampling
Insight	uncertainty budget: contribution of each input	needs extra runs (one-at-a-time or variance-based indices)
Best for	small uncertainties, many inputs, design iterations	large uncertainties, extrema, yield and coverage questions, validation of the linear result

Engineering Note — global sensitivity

Derivatives describe *local* sensitivity at the nominal point. When inputs vary over wide ranges, **variance-based (Sobol') indices** quantify how much of $\mathrm{Var}(y)$ each input explains on its own ($S_i = \mathrm{Var}(E[y \mid X_i])/\mathrm{Var}(y)$) and through interactions. They are estimated by structured Monte Carlo designs and are the standard way to decide which of fifty process parameters actually matter for a circuit's yield. For a linear model with independent inputs they reduce to the normalised terms $c_i^2 u_i^2 / u_c^2$ of the GUM uncertainty budget.

Common Pitfall — tolerance is not standard uncertainty

A "$\pm 1\%$" resistor tolerance is a bound, not a standard deviation. Treating it as $u = 1\%$ overstates the uncertainty by $\sqrt{3}$ for a uniform distribution, and the true distribution of a purchased batch is often neither uniform nor Gaussian (parts near nominal may have been sorted into a tighter tolerance grade, leaving a bimodal distribution). State the assumed distribution explicitly, and where it matters, measure a sample.

24.9 Reproducible numerical experiments

Motivation

A numerical result is only useful if it can be trusted, and it can only be trusted if it can be checked. Six months after a tolerance study, someone asks why the yield was quoted as 87% when a re-run gives 85%. A colleague's filter-design script produces different coefficients on a new laptop. A regression test fails after a library upgrade, and nobody knows whether the old or the new answer is right. Each of these is a failure of **reproducibility**, and each has a mathematical component (random-number streams, floating-point non-associativity, ill-conditioning) as well as an organisational one (versions, environments, records). This section treats computations as *experiments* that must be designed, recorded and tested like laboratory measurements.

Levels of reproducibility

- **Repeatability (bitwise reproducibility):** the same code, data, parameters and environment produce bit-for-bit identical output.
- **Reproducibility:** a re-run by someone else, possibly on different hardware or library versions, produces results that agree *within a stated tolerance* justified by the numerical and statistical error analysis.
- **Replicability:** an independent implementation of the same method, or a new experiment, leads to the same scientific or engineering conclusion.

Bitwise repeatability is a debugging tool; reproducibility within tolerance is the engineering requirement.

Sources of irreproducibility

Source	Mechanism	Control
random numbers	different seeds, shared global generator, parallel streams overlapping	explicit seeded generators, independent spawned streams, seeds recorded
floating-point order	$(a+b)+c \neq a+(b+c)$; parallel reductions, multithreaded BLAS, vectorised loops, fused multiply–add, GPU atomics change the order	accept and bound the variation; deterministic modes where needed; compensated summation
library versions	changed algorithms, defaults (tolerances, orderings, pivoting thresholds), bug fixes	pinned versions, environment files, containers
ill-conditioning	tiny rounding differences amplified by κ	estimate κ; set test tolerances accordingly; reformulate
data and parameters	edited input files, manual steps, undocumented options	version control for code *and* inputs; configuration files; scripted pipelines
hardware and OS	different CPU instruction sets, math-library implementations of exp, sin	record platform; tolerance-based comparison

Floating-point non-associativity

Floating-point addition is commutative but not associative: each operation rounds, and the rounding depends on the operands' magnitudes (Ch. 22). Any change in the *order* of a reduction can therefore change the last few bits of the result — and multithreaded libraries choose the order dynamically, depending on the number of threads and on how the data are split.

Worked Example 24.41 — The order of a sum

(a) Compare $(0.1+0.2)+0.3$ with $0.1+(0.2+0.3)$, and $(10^{16}+1)-10^{16}$ with $(10^{16}-10^{16})+1$. (b) Sum 10^6 copies of 0.1 by a naive loop, by NumPy's `sum`, and by `math.fsum`. (c) Sum 10^6 standard-normal samples in forward, reversed and sorted order.

(a) $(0.1+0.2)+0.3 = 0.6000000000000001$ but $0.1+(0.2+0.3) = 0.6$. $(10^{16}+1)-10^{16} = 0$ (the spacing of doubles near 10^{16} is 2, so $10^{16}+1$ rounds back to 10^{16}), but $(10^{16}-10^{16})+1 = 1$.

(b) Naive loop: 100 000.000 001 332 88 (relative error 1.3×10^{-11}: each addition rounds, and errors accumulate roughly as $n\varepsilon$ in the worst case). NumPy `sum`: 100 000.000 000 000 03 — NumPy uses **pairwise summation**, whose error grows like $\varepsilon\log_2 n$. `math.fsum`: exactly 100 000.0, the correctly rounded sum of the stored values.

(c) Forward: 998.570 649 438 621 3; reversed: 998.570 649 438 621 4; sorted ascending: 998.570 649 438 654 1; `fsum`: 998.570 649 438 621 3. The results differ in the 12th–14th significant digit.

Reading. None of these is a bug. A test that compares such results with `==` will fail on another machine, or with a different thread count. Results that must agree should be compared with a tolerance derived from the error analysis — here about $n\varepsilon\max|x_i|/|\sum x_i|$, far larger than ε when there is cancellation. When bitwise repeatability matters (regression debugging, audit), fix the thread count, use deterministic library modes, or use compensated (Kahan) or exact summation in the few places that need it.

Random-number streams

Pseudo-random number generators (PRNGs) are deterministic: the same seed gives the same sequence. Reproducible Monte Carlo therefore requires that every random draw comes from a generator whose seed is chosen deliberately and recorded.

Seeded generators and independent streams

A **seed** is the initial state from which a PRNG's sequence is generated. A **stream** is the sequence produced by one generator. For parallel or multi-component simulations, each worker or component needs its own **statistically independent stream**, obtained not by seeds $1, 2, 3, \ldots$ chosen ad hoc but by a mechanism designed to avoid overlap and correlation — in NumPy, `SeedSequence.spawn`, which derives child seeds by hashing; in other libraries, jump-ahead or counter-based generators (Philox, Threefry).

```
import numpy as np
root = np.random.SeedSequence(20241021)          # record this number
children = root.spawn(8)                          # one per worker / per component group
rngs = [np.random.default_rng(s) for s in children]
# worker k draws only from rngs[k]; results are independent of scheduling
R1 = 1e3 * (1 + 0.01 * rngs[0].uniform(-1, 1, 10_000))
```

Three rules follow. Never use the legacy global generator (`np.random.seed`, `np.random.rand`): any library call that draws from it silently shifts every later draw. Never seed from the clock in production code (do it, if at all, once — and *log* the seed). And structure the draws so that adding a new random component does not change existing ones (give each component its own stream), so that a design change can be compared with the old design using **common random numbers** (Section 24.6).

Versions and environments

The result of a computation is a function of the code, its inputs and the environment — interpreter, libraries, compiler flags, hardware. All three must be pinned and recorded.

- **Version control** (git) for code, configuration and small input data. Every reported result should carry the commit hash of the code that produced it; a "dirty" working tree (uncommitted changes) should be flagged in the output.
- **Environment specification.** A `requirements.txt` with exact versions (`numpy==2.1.3`), a conda `environment.yml` or lock file, or a container image. Library defaults change: tolerances of iterative solvers, the default ordering of a sparse LU, the algorithm behind a random-number distribution. A pinned environment can be rebuilt; "latest" cannot.
- **Metadata with every output.** Write alongside each result file the parameters, the seeds, the commit hash, the library versions and the platform. This costs a few lines:

```
import json, platform, subprocess, numpy, scipy
meta = {
    "commit": subprocess.run(["git", "rev-parse", "HEAD"], capture_output=True, text=True).stdout.strip(),
    "numpy": numpy.__version__, "scipy": scipy.__version__,
    "python": platform.python_version(), "machine": platform.machine(),
    "seed": 20241021, "N": 100_000, "tolerance_model": "uniform, 1%",
}
json.dump(meta, open("results/mc_gain_meta.json", "w"), indent=2)
```

- **Scripted pipelines.** Figures and tables should be generated by scripts from raw data with a single command, not by interactive sessions or edits in a spreadsheet. A notebook is acceptable if it is run top to bottom from a clean kernel before results are recorded.

Testing numerical code

Numerical code fails silently: a wrong sign in a Jacobian slows Newton's method rather than crashing it; an off-by-one in a stencil changes the convergence order from 2 to 1 while still giving "reasonable" answers. Tests must therefore check *mathematical* properties, with tolerances justified by error analysis.

1. **Exact or analytic references.** Compare against cases with known solutions — a symbolic result (Section 24.2), a closed-form circuit response, a problem constructed so that the answer is known (data generated from known parameters, as in Worked Example 24.33).
2. **Convergence-order tests.** For a method of order p, the error behaves as Ch^p; halving h should divide it by 2^p. The **observed order** $p_{\text{obs}} = \log_2\big(e(h)/e(h/2)\big)$ is a far more sensitive check than the error itself.
3. **Method of manufactured solutions** (for ODE/PDE solvers): choose an arbitrary smooth solution $u^\ast$, substitute it into the equation to compute the source term that makes it exact, run the solver with that source, and verify both the error and its convergence order.
4. **Invariants and properties.** Kirchhoff's current law at every node (residual of $\mathbf{G}\mathbf{v} - \mathbf{i}$), symmetry and positive definiteness of matrices that must have them, reciprocity ($\mathbf{Z} = \mathbf{Z}^{\mathsf{T}}$ for passive networks), conservation of energy or charge, orthogonality of computed $\mathbf{Q}$. Property-based testing frameworks generate many random inputs and check such properties automatically.
5. **Regression tests.** Store reference outputs and compare future runs against them with a tolerance — catching unintended changes, not establishing correctness.
6. **Statistical tests for stochastic code.** A Monte Carlo estimate is a random variable; test that it lies within z standard errors of the known answer.

Worked Example 24.42 — An observed-order test for the trapezoidal rule

Test a trapezoidal-rule implementation on $\int_0^1 e^x\,dx = e - 1$ with $n = 10, 20, 40$ subintervals.

Errors. $T_{10} - I = 1.4317 \times 10^{-3}$, $T_{20} - I = 3.5796 \times 10^{-4}$, $T_{40} - I = 8.9493 \times 10^{-5}$.

Observed order. $\log_2(1.4317/0.35796) = \log_2(3.9996) = 1.9998$, and $\log_2(3.5796/0.89493) = 2.0000$ ✓ — the expected $p = 2$.

Leading constant. The Euler–Maclaurin formula predicts $T_n - I \approx \frac{h^2}{12}\big(f'(1) - f'(0)\big) = \frac{h^2}{12}(e - 1)$, i.e. 1.4319×10^{-3} for $h = 0.1$ ✓.

As a test. `assert abs(p_obs - 2) < 0.05`. A typical bug — weighting both end points by 1 instead of $\frac{1}{2}$ — gives an error of order h and $p_{\text{obs}} \approx 1$, which this test catches immediately even though T_{40} would still be correct to two digits.

Worked Example 24.43 — Setting tolerances for deterministic and stochastic tests

(a) A regression test checks the solution of a linear system with $\kappa(\mathbf{A}) \approx 10^6$ computed by LU. What relative tolerance is appropriate? (b) A Monte Carlo yield estimator is tested on a case with known yield $Y = 0.80$, using $N = 10^4$ samples. What acceptance band gives a false-failure probability below 10^{-4}?

(a) A backward-stable solver guarantees a relative error of about $\kappa\varepsilon \approx 10^6 \times 1.1 \times 10^{-16} \approx 10^{-10}$; differences between platforms or thread counts can be of this size. A tolerance of `rtol=1e-8` (a safety factor of 100) is appropriate; `rtol=1e-15` would fail spuriously, and `rtol=1e-3` would hide real bugs.

(b) The standard error is $\sqrt{0.8 \times 0.2/10^4} = 0.004$. Accepting $|\hat{Y} - 0.80| \le 4 \times 0.004 = 0.016$ has false-failure probability $2Q(4) = 6.3 \times 10^{-5}$ (a 3-standard-error band would fail spuriously 0.27% of the time — about once in every 370 runs of the test suite). With a fixed seed the test is also bitwise repeatable, but the band should still be set statistically, so that the test survives a legitimate change of seed or generator.

Rule. Tolerances come from error analysis: $\kappa\varepsilon$ for deterministic linear algebra, the truncation error Ch^p for discretisations, $z \cdot \mathrm{SE}$ for Monte Carlo. A tolerance chosen by trial and error ("1e-6 seemed to work") is a test that has not been designed.

```
import numpy as np
from numpy.testing import assert_allclose

def test_mc_gain_yield():
    rng = np.random.default_rng(7)
    N = 10_000
    R1 = 1e3 * (1 + 0.01 * rng.uniform(-1, 1, N))
    R2 = 9e3 * (1 + 0.01 * rng.uniform(-1, 1, N))
    G = 1 + R2 / R1
    Y = np.mean(np.abs(G - 10) <= 0.1)
    se = np.sqrt(Y * (1 - Y) / N)
    assert abs(Y - 0.80) < 4 * se          # reference 0.80 from a 10^6-sample run

def test_kcl_residual(G, v, i):
    assert_allclose(G @ v, i, rtol=1e-10, atol=1e-12)
```

Common Pitfall — the reference answer is also a computation

A regression test that compares a new result with an old one only shows that nothing *changed*. If the reference was produced by the same buggy code, the test enshrines the bug. Independent references — analytic results, a different algorithm (sparse versus dense solve, Monte Carlo versus linearisation), a published benchmark, or a measurement — are what establish correctness. Keep at least one independent check for every major result.

Common Pitfall — "it worked on my machine"

An analysis run in an interactive session with variables left over from earlier cells, a file edited by hand, a parameter changed and changed back, or an unrecorded library upgrade cannot be reproduced even by its author. The cure is mundane: scripts under version control, a pinned environment, a recorded seed, and a single command that regenerates every number in the report.

Engineering Note

In circuit design, reproducibility has an extra layer: the simulator version, the model files (PDK release), the simulator options (`reltol` , `abstol` , integration method, Gmin stepping) and the netlist all affect the numbers. Sign-off flows record all of them, and the same Monte Carlo seed is used for "before" and "after" runs of an engineering change so that differences reflect the design, not the sampling. Measurement laboratories accredited to ISO/IEC 17025 must document their uncertainty calculations (Section 24.8) and the software that performs them, which is validated like any other instrument.

Summary

Topic	Key formula or fact
Asymptotic cost	$f = O(g)$: $f \le cg$ for large n; scaling exponent $p \approx \ln(T_2/T_1)/\ln(n_2/n_1)$
Master theorem	$T = aT(n/b) + n^d$: n^d, $n^d \log n$ or $n^{\log_b a}$ as $d \gtrless \log_b a$
Typical costs	matrix–vector $2n^2$; matrix–matrix $2n^3$; LU $\tfrac{2}{3}n^3$; Cholesky $\tfrac{1}{3}n^3$; FFT $5N\log_2 N$; sparse matvec $2\,\mathrm{nnz}$
Expression swell	symbolic det has $n!$ terms; exact Newton doubles digits per step
Backward error (linear system)	$\eta = \|\mathbf{r}\|/(\|\mathbf{A}\|\|\hat{\mathbf{x}}\|)$; forward error $\lesssim \kappa\eta$
Householder reflector	$\mathbf{H} = \mathbf{I} - 2\mathbf{v}\mathbf{v}^{\mathsf{T}}/\mathbf{v}^{\mathsf{T}}\mathbf{v}$, $\mathbf{v} = \mathbf{a} + \operatorname{sign}(a_1)\|\mathbf{a}\|\mathbf{e}_1$
QR cost	$2mn^2 - \frac{2}{3}n^3$; backward stable; $\|\hat{\mathbf{Q}}^{\mathsf{T}}\hat{\mathbf{Q}} - \mathbf{I}\| = O(\varepsilon)$
Normal equations	$\kappa(\mathbf{A}^{\mathsf{T}}\mathbf{A}) = \kappa(\mathbf{A})^2$ — use QR or SVD
Truncated SVD	$\hat{\mathbf{x}}_k = \sum_{i\le k}(\mathbf{u}_i^{\mathsf{T}}\mathbf{b}/\sigma_i)\mathbf{v}_i$
Stationary iteration	$\mathbf{e}^{(k+1)} = \mathbf{T}\mathbf{e}^{(k)}$; converges iff $\rho(\mathbf{T}) < 1$; $k_\epsilon \approx \ln\epsilon/\ln\rho$
Model problem	$\rho_J = \cos\pi h$; $\rho_{GS} = \rho_J^2$
Optimal SOR	$\omega_{\text{opt}} = 2/(1+\sqrt{1-\rho_J^2})$, $\rho = \omega_{\text{opt}} - 1$
CG	$\|\mathbf{e}_k\|_{\mathbf{A}} \le 2\big(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\big)^k\|\mathbf{e}_0\|_{\mathbf{A}}$; $k \approx \frac{1}{2}\sqrt{\kappa}\ln(2/\epsilon)$
GMRES	minimises $\|\mathbf{r}_k\|_2$ over $\mathcal{K}_k$; exact in $\deg(\text{minimal polynomial})$ steps
Preconditioning	solve $\mathbf{P}^{-1}\mathbf{A}\mathbf{x} = \mathbf{P}^{-1}\mathbf{b}$ with $\mathbf{P} \approx \mathbf{A}$, $\mathbf{P}\mathbf{z} = \mathbf{r}$ cheap
Sparse storage	CSR/CSC: $2\,\mathrm{nnz} + n + 1$ words
Fill, 2-D grid	natural: $n^{3/2}$ fill, n^2 flops; nested dissection: $n\log n$ fill, $n^{3/2}$ flops
Monte Carlo	$\hat{\mu}_N = \frac{1}{N}\sum f(\mathbf{X}_i)$, $\mathrm{SE} = s_N/\sqrt{N}$
Control variate	$c^* = \mathrm{Cov}(f,h)/\mathrm{Var}(h)$; variance $\times(1-\rho^2)$
Importance sampling	$E_p[f] = E_q[f\,p/q]$
Rare event	relative SE $\approx 1/\sqrt{Np}$; $N \approx 100/p$ for 10%
Yield	$\hat{Y} = \#\text{pass}/N$, $\mathrm{SE} = \sqrt{\hat{Y}(1-\hat{Y})/N}$
Gauss–Newton	$(\mathbf{J}^{\mathsf{T}}\mathbf{J})\boldsymbol{\delta} = \mathbf{J}^{\mathsf{T}}\mathbf{r}$
Levenberg–Marquardt	$(\mathbf{J}^{\mathsf{T}}\mathbf{J} + \lambda\,\mathrm{diag}(\mathbf{J}^{\mathsf{T}}\mathbf{J}))\boldsymbol{\delta} = \mathbf{J}^{\mathsf{T}}\mathbf{r}$

Topic	Key formula or fact
Parameter covariance	$\hat{\sigma}^2(\mathbf{J}^\mathsf{T}\mathbf{J})^{-1}$, $\hat{\sigma}^2 = \sum r_i^2/(m-n)$
Relative sensitivity	$S_p^y = (p/y)\,\partial y/\partial p$
Complex step	$f'(x) \approx \mathrm{Im}\{f(x+jh)\}/h$, no cancellation
Adjoint sensitivity	$\mathbf{A}^\mathsf{T}\boldsymbol{\lambda} = \mathbf{c}$; $\partial y/\partial p = \boldsymbol{\lambda}^\mathsf{T}(\partial_p \mathbf{b} - \partial_p \mathbf{A}\,\mathbf{x})$; branch: $-(\lambda_a - \lambda_b)(v_a - v_b)$
GUM propagation	$u_c^2 = \sum c_i^2 u_i^2 + 2\sum_{i<j} c_i c_j u_{ij}$; $\mathbf{U}_y = \mathbf{J}\mathbf{U}_x\mathbf{J}^\mathsf{T}$; $U = k u_c$
Uniform tolerance	$\pm a \Rightarrow u = a/\sqrt{3}$
Observed order	$p_{\text{obs}} = \log_2\left(e(h)/e(h/2)\right)$
Test tolerances	$\kappa\varepsilon$ (linear algebra), Ch^p (discretisation), $z \cdot \mathrm{SE}$ (Monte Carlo)

Further reading

- L. N. Trefethen and D. Bau III, *Numerical Linear Algebra*, 25th anniversary ed., SIAM, 2022 — QR, SVD, conditioning, backward stability, Arnoldi, GMRES and CG, in short, beautifully argued lectures.
- G. H. Golub and C. F. Van Loan, *Matrix Computations*, 4th ed., Johns Hopkins University Press, 2013 — the reference for algorithms, flop counts and implementation details.
- Y. Saad, *Iterative Methods for Sparse Linear Systems*, 2nd ed., SIAM, 2003 — Krylov methods, preconditioning and sparse storage; T. A. Davis, *Direct Methods for Sparse Linear Systems*, SIAM, 2006, is the companion for sparse factorisation and ordering.
- N. J. Higham, *Accuracy and Stability of Numerical Algorithms*, 2nd ed., SIAM, 2002 — backward error analysis, summation, and testing numerical software.
- A. B. Owen, *Monte Carlo Theory, Methods and Examples*, online text, 2013– — variance reduction, importance sampling and quasi-Monte Carlo; with JCGM 100:2008 (*GUM*) and JCGM 101:2008 (*GUM Supplement 1*) for uncertainty propagation, freely available from the BIPM.

Chapter 25 — Mathematical Modeling

> **Prerequisites:** Ch. 1 (Mathematical Language and Algebra), Ch. 4 (Differential Calculus), Ch. 6 (Multivariable Calculus), Ch. 8 (Matrices and Linear Systems), Ch. 11–12 (Ordinary and Linear Differential Equations), Ch. 14 (Laplace Transforms), Ch. 20 (Statistical Analysis), Ch. 22–23 (Numerical Methods, Optimization) **Learning outcomes:** After this chapter you can …
>
> - run the modeling cycle deliberately: state the purpose, choose the system boundary, write conservation laws and constitutive relations, and decide what to neglect;
> - separate state variables, parameters, inputs, disturbances and outputs, and put a lumped model into state-space form;
> - apply the Buckingham Pi theorem to reduce the number of variables in a problem and to obtain scaling laws without solving any equation;
> - nondimensionalize a model (RLC circuit, thermal network, heat equation, nonlinear oscillator) so that its behaviour is governed by a minimal set of dimensionless groups such as ζ, ω_n, Biot and Fourier numbers;
> - linearize a nonlinear model about an operating point, compute small-signal parameters, and estimate the amplitude range over which the linear model is trustworthy;
> - validate a model with residual analysis, goodness-of-fit measures and cross-validation, identify its parameters from step responses or by least squares, test identifiability, and choose between analytical and numerical solution.

Every chapter of this book has taken a mathematical problem as given: *solve this differential equation*, *invert this transform*, *minimize this function*. In engineering practice the problem is almost never given. Someone hands you a circuit board that oscillates when it should not, a sensor whose reading drifts with temperature, a transmission line whose losses are higher than the datasheet suggests, and the first and hardest step is to decide **which mathematics describes it**. That step is *mathematical modeling*.

Modeling is a craft as much as a science, but it is not mysterious. It has a recognisable cycle (Section 25.1), a vocabulary (25.2), a set of tools that extract information *before* any equation is solved (dimensional analysis and scaling, 25.3–25.4), a standard way to make hard problems tractable (linearization, 25.5), a discipline of stating and checking assumptions (25.6), and a way to confront the model with data — validating its structure (25.7) and estimating its numbers (25.8). The chapter ends with the choice every modeller faces between closed-form and computed solutions (25.9), and a short synthesis that ties the methods of the whole book together.

Because this is the final chapter, it leans on almost everything before it. Cross-references are given where a technique is used; the emphasis here is on *when* and *why* to use each one, not on re-deriving it.

25.1 Model formulation

What a model is, and what it is for

> **Mathematical model**
>
> A **mathematical model** of a system is a set of mathematical relations — algebraic equations, differential or difference equations, probability distributions, inequalities — together with a statement of the variables they relate, the assumptions under which they hold, and the purpose for which they are intended.

The last clause is the one most often forgotten. A model is a tool built for a job, and "Is the model correct?" is the wrong question; the right one is "Is the model *adequate for this purpose* over *this range of conditions*?" The same resistor is

- an ideal $v = Ri$ for a DC bias calculation,
- $v = R(T)\,i$ with $R(T) = R_0\,[1 + \alpha(T - T_0)]$ for a precision current shunt,
- R in series with a lead inductance L_s and in parallel with a pad capacitance C_p for a 2 GHz matching network,
- R plus a Johnson noise source with power spectral density $4kTR$ for a low-noise amplifier front end (Ch. 21).

None of these is "the" model of a resistor. Each is the simplest model that answers a particular question. George Box's often-quoted remark that all models are wrong but some are useful expresses exactly this point.

> **Engineering Note — the purpose fixes the fidelity**
>
> Before writing an equation, write one sentence: *"This model must predict ____ to within ____ for inputs in the range ____."* For example: "predict the junction temperature of Q3 to within 5 °C for dissipations from 0 to 25 W, on time scales longer than 1 s." That sentence decides the boundary, the time scale, the variables to keep and the accuracy needed from every parameter. Models built without it tend to be either too crude to answer the question or too elaborate to be understood, calibrated or trusted.

The modeling cycle

Modeling is iterative. A widely used description has six stages:

1. **Problem statement and purpose.** What question must the model answer? What accuracy, what operating range, what time scale?
2. **Conceptualization.** Choose the system boundary, identify the inputs crossing it and the outputs of interest, and decide which physical effects matter. Sketch an idealized system (a circuit diagram, a lumped thermal network, a free-body diagram).
3. **Mathematical formulation.** Write the governing equations: conservation laws plus constitutive relations, with initial and boundary conditions.
4. **Analysis and solution.** Simplify (dimensional analysis, scaling, linearization), then solve — analytically or numerically.
5. **Validation.** Compare predictions with independent measurements. Are the residuals consistent with measurement noise? Does the model work on data not used to build it?

6. **Use and refinement.** Use the model for design or prediction; when it fails, return to stage 2 or 3 with what the failure taught.

$$\text{purpose} \;\to\; \text{concept} \;\to\; \text{equations} \;\to\; \text{solution} \;\to\; \text{validation} \;\to\; \text{use} \;\circlearrowleft$$

The loop back is essential. The first model is rarely the last; but a *documented* first model, with explicit assumptions, is the only thing that can be systematically improved.

Building blocks: conservation laws and constitutive relations

Almost every physical model in electrical engineering is assembled from two kinds of equation.

Balance (conservation) law and constitutive relation

A **balance law** states that the rate of change of an extensive quantity inside a boundary equals its net inflow plus its internal production:

$$\frac{d}{dt}(\text{amount inside}) = (\text{rate in}) - (\text{rate out}) + (\text{rate generated}).$$

A **constitutive relation** characterises a particular material or component: it links the *flow* of the conserved quantity to the *effort* (potential difference) that drives it, or links the stored amount to the effort.

Balance laws are universal and exact (within classical physics); constitutive relations are empirical, approximate and valid over limited ranges. That asymmetry tells you where model error usually lives: in the constitutive relations and in what was left out, rather than in the conservation laws themselves.

Domain	Conserved quantity	Effort (potential)	Flow	Storage relation	Dissipation relation
Electrical	charge q	voltage v	current i	$q = Cv$	$v = Ri$
Magnetic (flux linkage)	flux linkage λ	—	—	$\lambda = Li$	—
Thermal	heat Q	temperature T	heat flow P	$Q = C_{\text{th}}T$	$\Delta T = R_{\text{th}}P$
Mechanical (translation)	momentum p	velocity v	force F	$p = mv$	$F = bv$
Mechanical (rotation)	angular momentum	angular velocity ω	torque τ	$J\omega$	$\tau = b\omega$
Fluid	volume V	pressure p	volume flow Q	$V = C_f\,p$	$p = R_f\,Q$

Kirchhoff's current law is charge conservation at a node; Kirchhoff's voltage law is energy conservation around a loop (in the quasi-static, lumped regime of Section 25.6). The *analogy* across domains is not decorative: it lets an electrical engineer write the thermal model of a heat sink or the mechanical model of a motor shaft as an equivalent circuit, and reuse every circuit-analysis technique of Ch. 12 and Ch. 14.

Levels of description

Models can be classified along several independent axes. Choosing a position on each axis is part of formulation.

Axis	Choices	Mathematics that results
Spatial	lumped / distributed	ODEs (Ch. 11–12) / PDEs (Ch. 13)
Time	static / dynamic; continuous / discrete	algebraic / differential / difference equations (Ch. 17)
Linearity	linear / nonlinear	superposition, transforms (Ch. 14–17) / iteration, simulation (Ch. 22)
Time dependence of parameters	time-invariant / time-varying	constant-coefficient / variable-coefficient equations
Certainty	deterministic / stochastic	ODEs / random processes (Ch. 21)
Origin	white-box (physics) / grey-box / black-box (data)	first-principles equations / fitted structure / regression, ARX, neural nets

A **white-box** model is derived entirely from physical laws with parameters known from geometry and material data. A **black-box** model is a flexible input–output relation fitted to data with no physical interpretation of its coefficients. Most useful engineering models are **grey-box**: the structure comes from physics (a first-order thermal lag, a second-order mechanical resonance), and the parameter values come from measurement (Section 25.8).

Worked Example 25.1 — An RC charging model from first principles

Problem. A sensor node draws its supply from a capacitor $C = 100\ \mu\text{F}$ charged through a resistor $R = 1\ \text{k}\Omega$ from a source $V_s = 5\ \text{V}$ that is switched on at $t = 0$. The load draws a constant current $I_L = 1\ \text{mA}$. Formulate a model for the capacitor voltage $v(t)$ and find its steady state and time constant.

Step 1 — purpose. Predict $v(t)$ to a few percent over the first second after switch-on, to check the node can start when $v > 3\ \text{V}$.

Step 2 — concept. Boundary: the capacitor node. Inputs: V_s (switched), I_L. Neglect capacitor leakage and equivalent series resistance (justified later if the leakage current is $\ll I_L$).

Step 3 — equations. Charge balance at the node (KCL):

$$C\frac{dv}{dt} = \frac{V_s - v}{R} - I_L, \qquad v(0) = 0.$$

Rearranged:

$$RC\,\frac{dv}{dt} + v = V_s - RI_L.$$

Step 4 — solution. This is the standard first-order linear equation of Ch. 11 with time constant $\tau = RC = 10^3 \times 10^{-4} = 0.1\ \text{s}$ and steady state $v_\infty = V_s - RI_L = 5 - 1 = 4\ \text{V}$:

$$v(t) = v_\infty\left(1 - e^{-t/\tau}\right) = 4\left(1 - e^{-t/0.1}\right)\ \text{V}.$$

Step 5 — use. The node reaches 3 V when $1 - e^{-t/\tau} = 0.75$, i.e. $t = \tau \ln 4 = 0.139\ \text{s}$.

Dimensional check. $[RC] = (\text{V/A})(\text{A s/V}) = \text{s}$ and $[RI_L] = \text{V}$, so every term has the right units.

The example is elementary on purpose: notice that the mathematics (Step 4) occupied one line, while purpose, boundary and assumptions occupied the rest. That proportion is typical of real modeling.

Worked Example 25.2 — A lumped thermal model of a power transistor

Problem. A MOSFET dissipating $P = 20\ \text{W}$ is mounted on a heat sink in ambient air at $T_a = 40\ ^\circ\text{C}$. Datasheets give the junction-to-case resistance $R_{jc} = 1.5\ \text{K/W}$, the interface (thermal pad) resistance $R_{cs} = 0.5\ \text{K/W}$ and the sink-to-ambient resistance $R_{sa} = 2.0\ \text{K/W}$. The heat sink (aluminium, $m = 0.25\ \text{kg}$, $c_p = 900\ \text{J/(kg K)}$) has much larger heat capacity than the transistor package. Formulate a model for the junction temperature after power is applied.

Concept. Heat flows junction → case → sink → ambient through three series resistances. Heat is *stored* mainly in the sink, with thermal capacitance $C_s = mc_p = 0.25 \times 900 = 225\ \text{J/K}$. Neglect the package's own capacitance (it charges in seconds, while the sink takes minutes).

Equations. Energy balance on the sink, with sink temperature T_s:

$$C_s\frac{dT_s}{dt} = P - \frac{T_s - T_a}{R_{sa}},$$

and, because the package stores negligible heat, the full P flows through $R_{jc} + R_{cs}$ at every instant:

$$T_j = T_s + (R_{jc} + R_{cs})P.$$

Solution. With $T_s(0) = T_a$, the sink equation is first order with $\tau = R_{sa}C_s = 2.0 \times 225 = 450\ \text{s}$ and

$$T_s(t) = T_a + PR_{sa}\left(1 - e^{-t/\tau}\right).$$

Hence

$$T_j(t) = T_a + (R_{jc} + R_{cs})P + R_{sa}P\left(1 - e^{-t/\tau}\right) = 40 + 40 + 40\left(1 - e^{-t/450}\right)\ ^\circ\text{C}.$$

At switch-on the junction jumps (on the model's time scale) to 80 °C, and it approaches 120 °C with a 7.5-minute time constant.

What the model omits. The instantaneous jump is an artefact of neglecting the package capacitance; on a millisecond time scale the junction rises smoothly. That is acceptable for the stated purpose (steady state and minutes-scale warm-up) but would be wrong for a 10 ms overload pulse — there the package capacitance is the *only* thing that matters. Same device, different purpose, different model.

Worked Example 25.3 — An electromechanical model: the DC motor

Problem. Formulate a model for a permanent-magnet DC motor driven by voltage $v_a(t)$ and loaded by torque $\tau_L(t)$, suitable for designing a speed controller.

Concept. Two coupled domains. Electrical: armature resistance R_a, inductance L_a, and back-EMF proportional to speed. Mechanical: rotor and load inertia J, viscous friction b. Coupling: the torque constant K_t (torque per ampere) and back-EMF constant K_e (volts per rad/s). Energy conservation across the air gap (electrical power $e\,i$ converted equals mechanical power $\tau\omega$) forces $K_t = K_e \equiv K$ in SI units.

Equations. KVL round the armature and Newton's law for rotation:

$$L_a \frac{di}{dt} = v_a - R_a i - K\omega,$$
$$J \frac{d\omega}{dt} = K i - b\omega - \tau_L.$$

If shaft position θ is also wanted, add $\dot{\theta} = \omega$.

Check. Units: $[K\omega] = (\mathrm{V\,s/rad})(\mathrm{rad/s}) = \mathrm{V}$; $[Ki] = (\mathrm{N\,m/A})(\mathrm{A}) = \mathrm{N\,m}$; and $\mathrm{V\,s} = \mathrm{N\,m/A}$ since $1\ \mathrm{V\,A} = 1\ \mathrm{W} = 1\ \mathrm{N\,m/s}$. Steady state with $\tau_L = 0$: $\omega_\infty = K v_a / (R_a b + K^2)$, which is the familiar no-load speed when b is small, $\omega_\infty \approx v_a / K$.

This model is linear only because friction was taken viscous. Real gearboxes add Coulomb friction $\tau_c\,\mathrm{sgn}(\omega)$, which is nonlinear and discontinuous at $\omega = 0$ — the reason speed controllers behave poorly near standstill. Whether to include it depends, once again, on the purpose.

Common Pitfall — modeling the device instead of the question

A frequent failure is to start from "let us model everything about the motor": winding temperature, magnetic saturation, brush arcing, cogging torque, shaft flexibility. The result has dozens of parameters, most unknown, and no-one can tell which effects drive the answer. Start with the simplest model that could possibly answer the question; add an effect only when validation (Section 25.7) shows the simple model fails *in a way that effect explains*.

25.2 State variables and parameters

Five kinds of quantity

Once a model is written down, every symbol in it plays one of a small number of roles. Getting the roles straight is what turns a page of equations into something that can be simulated, identified and controlled.

Roles of quantities in a model

- **State variables** $\mathbf{x}(t) \in \mathbb{R}^n$: the minimal set of quantities whose present values, together with future inputs, determine all future behaviour. They carry the system's *memory*.
- **Inputs** $\mathbf{u}(t)$: externally imposed signals that the modeller (or a controller) chooses — a drive voltage, a commanded current, a heater power.
- **Disturbances** $\mathbf{w}(t)$: external signals that act on the system but are *not* chosen — ambient temperature, load torque, supply ripple, noise.
- **Parameters** $\mathbf{p}$: quantities that characterise the system and are constant (or vary much more slowly than the states) over the time scale of interest — R, L, C, J, K, R_{th}.
- **Outputs** $\mathbf{y}(t)$: quantities that are measured or are of interest; functions of the states and inputs.

The general lumped, continuous-time model is

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}, \mathbf{w}; \mathbf{p}, t), \qquad \mathbf{y} = \mathbf{h}(\mathbf{x}, \mathbf{u}; \mathbf{p}, t), \qquad \mathbf{x}(t_0) = \mathbf{x}_0.$$

Constants that are *universal* — π, the Boltzmann constant k, the elementary charge q — are sometimes called **constants** to distinguish them from parameters, which belong to a particular device.

The distinction between a parameter and a state is a matter of *time scale*, not of physics. The temperature of a resistor is a parameter in a circuit simulation lasting a microsecond, because it cannot change appreciably in that time; it is a state in a thermal simulation lasting an hour. The same goes for battery state of charge, capacitor ageing, or a slowly drifting oscillator frequency. A useful rule:

Engineering Note — parameter or state?

If a quantity changes by less than the required model accuracy over the time window of interest, treat it as a parameter. If it changes by more, it must be a state with its own equation, or the model must be re-identified as it drifts (adaptive or gain-scheduled models).

The distinction between an input and a disturbance is a matter of *control*. Ambient temperature is a disturbance for an amplifier, but an input for the environmental chamber used to test it.

Why states: the state-space form

The state-space form $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u})$ is the standard because

- it is first order, so every numerical ODE solver of Ch. 22 accepts it directly;
- its linear version $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$ is analysed by the eigenvalue methods of Ch. 10 (modes, stability) and the matrix exponential;
- the initial condition $\mathbf{x}(t_0)$ is exactly the information needed — no more, no less.

For linear time-invariant (LTI) systems,

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}, \qquad \mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u},$$

with solution

$$\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}(0) + \int_0^t e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau)\,d\tau$$

and transfer function (Ch. 14), for zero initial state,

$$\mathbf{G}(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}.$$

The poles of $\mathbf{G}(s)$ are among the eigenvalues of $\mathbf{A}$; they coincide with all of them when the realization is minimal (controllable and observable).

Choosing state variables

Natural state variables for lumped physical systems

In a lumped network of ideal elements, a natural choice of states is one *energy variable per independent energy-storage element*: the capacitor voltages (or charges) and inductor currents (or flux linkages) in a circuit; the masses' velocities and springs' deflections in a mechanism; the node temperatures of thermal capacitances. The stored energy is then a quadratic form in the states, e.g. $E = \tfrac{1}{2}\sum C_k v_k^2 + \tfrac{1}{2}\sum L_m i_m^2$.

Why it works (sketch). The constitutive relations of storage elements are the only ones involving derivatives ($i = C\,dv/dt$, $v = L\,di/dt$). Resistive elements and Kirchhoff's laws are algebraic. Solving the algebraic equations for the capacitor currents and inductor voltages in terms of the capacitor voltages, inductor currents and sources gives exactly $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u})$.

The word *independent* matters. Two capacitors directly in parallel share one voltage and contribute one state; a loop consisting only of capacitors and voltage sources, or a cut-set consisting only of inductors and current sources, removes a state. Such configurations are called **degenerate**; the order of the network is

$$n = n_C + n_L - n_{\text{C-loops}} - n_{\text{L-cutsets}},$$

where $n_{\text{C-loops}}$ counts independent loops of capacitors (and voltage sources only) and $n_{\text{L-cutsets}}$ counts independent cut-sets of inductors (and current sources only).

States are not unique. Any invertible linear transformation $\mathbf{z} = \mathbf{T}\mathbf{x}$ gives an equally valid state vector, with

$$\dot{\mathbf{z}} = \mathbf{T}\mathbf{A}\mathbf{T}^{-1}\mathbf{z} + \mathbf{T}\mathbf{B}\mathbf{u}, \qquad \mathbf{y} = \mathbf{C}\mathbf{T}^{-1}\mathbf{z} + \mathbf{D}\mathbf{u}.$$

The eigenvalues ($\mathbf{T}\mathbf{A}\mathbf{T}^{-1}$ is similar to $\mathbf{A}$, Ch. 10) and the transfer function are unchanged: $\mathbf{C}\mathbf{T}^{-1}(s\mathbf{I} - \mathbf{T}\mathbf{A}\mathbf{T}^{-1})^{-1}\mathbf{T}\mathbf{B} = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}$. What changes is interpretability and numerical conditioning — physical states with similar magnitudes are usually the best-conditioned choice.

Worked Example 25.4 — State-space model of the series RLC circuit

Problem. A series RLC circuit with $R = 10\ \Omega$, $L = 1$ mH, $C = 1\ \mu$F is driven by $v_s(t)$. The output is the capacitor voltage. Write the state-space model, find the eigenvalues and interpret them.

States. One capacitor and one inductor, no degeneracy: $n = 2$, $\mathbf{x} = [v_C,\ i]^{\mathsf{T}}$.

Equations. Capacitor: $C\dot{v}_C = i$. KVL: $L\dot{i} = v_s - Ri - v_C$. Hence

$$\frac{d}{dt}\begin{bmatrix} v_C \\ i \end{bmatrix} = \begin{bmatrix} 0 & 1/C \\ -1/L & -R/L \end{bmatrix}\begin{bmatrix} v_C \\ i \end{bmatrix} + \begin{bmatrix} 0 \\ 1/L \end{bmatrix} v_s, \qquad y = \begin{bmatrix} 1 & 0 \end{bmatrix}\mathbf{x}.$$

Eigenvalues. $\det(\lambda\mathbf{I} - \mathbf{A}) = \lambda^2 + (R/L)\lambda + 1/(LC) = 0$. Numerically $R/L = 10^4\ \text{s}^{-1}$ and $1/(LC) = 10^9\ \text{s}^{-2}$, so

$$\lambda = -5000 \pm j\sqrt{10^9 - 2.5\times 10^7} = -5000 \pm j\,31\,225\ \text{s}^{-1}.$$

The real part gives an envelope decay time $1/5000 = 0.2$ ms; the imaginary part gives a ringing frequency $31\,225/(2\pi) = 4.97$ kHz.

Transfer function. $G(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} = \dfrac{1/(LC)}{s^2 + (R/L)s + 1/(LC)}$, the standard second-order low-pass form whose dimensionless parameters are extracted in Section 25.4.

Worked Example 25.5 — Counting states in a degenerate circuit

Problem. A voltage source v_s feeds a resistor R_1 into node A. Node A has capacitor C_1 to ground and capacitor C_2 to node B; node B has C_3 to ground and R_2 to ground. How many states? Now suppose R_1 is removed so the source connects directly to node A. How many states?

Case 1. Three capacitors. Is there a loop of capacitors only? The loop ground–C_1–A–C_2–B–C_3–ground contains only capacitors, so $n_{\text{C-loops}} = 1$ and $n = 3 - 1 = 2$. Indeed $v_{C_1} = v_{C_2} + v_{C_3}$ (KVL round the loop), so only two capacitor voltages are independent; choose $\mathbf{x} = [v_{C_1}, v_{C_3}]^{\mathsf{T}}$.

Case 2. With the source wired straight to node A, $v_{C_1} = v_s$ is now forced: the loop source–C_1 is a second loop of capacitors and voltage sources only. So $n = 3 - 2 = 1$, with state v_{C_3} (and $v_{C_2} = v_s - v_{C_3}$).

Consequence. In case 2 a step in v_s produces an *impulse* of current into C_1 and C_2 — the ideal model predicts infinite current. A circuit simulator will either fail or insert a tiny series resistance. The degeneracy is a symptom of an idealization (zero source resistance) that is invalid on short time scales, and is a reminder that assumptions must be checked (Section 25.6).

Worked Example 25.6 — Changing state coordinates for the DC motor

Problem. For the DC motor of Example 25.3 with $\tau_L = 0$, the states $\mathbf{x} = [i,\ \omega]^{\mathsf{T}}$ give

$$\mathbf{A} = \begin{bmatrix} -R_a/L_a & -K/L_a \\ K/J & -b/J \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1/L_a \\ 0 \end{bmatrix}.$$

Use instead the momentum-like states $\mathbf{z} = [\lambda,\ h]^{\mathsf{T}} = [L_a i,\ J\omega]^{\mathsf{T}}$ (flux linkage and angular momentum). Find the new matrices and confirm the eigenvalues are unchanged. Use $R_a = 2\ \Omega$, $L_a = 5$ mH, $K = 0.05$ V s/rad, $J = 2 \times 10^{-5}$ kg m^2, $b = 10^{-5}$ N m s/rad.

Transformation. $\mathbf{T} = \operatorname{diag}(L_a, J)$, so

$$\mathbf{T}\mathbf{A}\mathbf{T}^{-1} = \begin{bmatrix} -R_a/L_a & -K/J \\ K/L_a & -b/J \end{bmatrix}, \qquad \mathbf{T}\mathbf{B} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

In the new coordinates the input enters with unit gain ($\dot\lambda = v_a - \dots$), which is the energy-based (port-Hamiltonian) form preferred in some control designs.

Eigenvalues. Trace $= -R_a/L_a - b/J = -400 - 0.5 = -400.5\ \text{s}^{-1}$; determinant $= (R_a b + K^2)/(L_a J) = (2 \times 10^{-5} + 2.5 \times 10^{-3})/(10^{-7}) = 25\,200\ \text{s}^{-2}$. Both matrices share trace and determinant, so both give

$$\lambda^2 + 400.5\lambda + 25\,200 = 0 \ \Rightarrow\ \lambda = \frac{-400.5 \pm \sqrt{400.5^2 - 4(25\,200)}}{2} = \frac{-400.5 \pm 244.1}{2},$$

that is $\lambda_1 \approx -78.2\ \text{s}^{-1}$ and $\lambda_2 \approx -322.3\ \text{s}^{-1}$. The slow mode (time constant 12.8 ms) is essentially mechanical, related to $JR_a/K^2 = 16$ ms; the fast mode (3.1 ms) is essentially electrical, related to $L_a/R_a = 2.5$ ms. Because the domains are coupled through K, neither eigenvalue equals the naive single-domain estimate — a small but genuine lesson about decoupled "back-of-envelope" models.

Common Pitfall — treating a derivative of an input as a state

In a model like $\ddot y + a\dot y + by = c\dot u + du$, it is tempting to take $\mathbf{x} = [y, \dot y]^{\mathsf{T}}$. Then $\ddot y$ contains $\dot u$, and the "state equation" needs the derivative of the input — which does not exist for a step input. The fix is a proper realization (e.g. controllable canonical form, Ch. 14) in which the states are internal variables and u enters without differentiation; the output is then $y = \mathbf{C}\mathbf{x}$ with $\mathbf{C} = [d,\ c]$.

Algebraic variables and DAEs

Many models contain variables that obey *algebraic* rather than differential equations: node voltages with no capacitor, currents through resistors, a junction temperature determined by an instantaneous heat flow (Example 25.2). The full model is then a **differential-algebraic equation** (DAE)

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{z}, \mathbf{u}), \qquad \mathbf{0} = \mathbf{g}(\mathbf{x}, \mathbf{z}, \mathbf{u}),$$

with differential states $\mathbf{x}$ and algebraic variables $\mathbf{z}$. If $\partial \mathbf{g}/\partial \mathbf{z}$ is invertible (an *index-1* DAE), the implicit function theorem (Ch. 6) lets us eliminate $\mathbf{z}$ locally and recover an ODE. Circuit simulators of the SPICE family solve the DAE form directly using modified nodal analysis, because eliminating every algebraic variable by hand would destroy the sparsity of the network equations.

Discrete-time states

Sampled-data systems and digital filters have states that evolve by a difference equation,

$$\mathbf{x}[n+1] = \mathbf{A}_d\mathbf{x}[n] + \mathbf{B}_d\mathbf{u}[n], \qquad \mathbf{y}[n] = \mathbf{C}\mathbf{x}[n] + \mathbf{D}\mathbf{u}[n].$$

For a continuous plant whose input is held constant between samples (zero-order hold) at period T, the exact discretization is $\mathbf{A}_d = e^{\mathbf{A}T}$ and $\mathbf{B}_d = \int_0^T e^{\mathbf{A}\sigma}\,d\sigma\,\mathbf{B}$; the eigenvalues map as $z = e^{sT}$ (Ch. 17). The states of a digital filter are its delay-line contents; the parameters are its coefficients.

25.3 Dimensional analysis

Motivation: information for free

Before solving a single equation — sometimes before even knowing the equations — one can deduce the *form* of a physical relation from the requirement that it be dimensionally consistent. Dimensional analysis

- detects errors (an equation that adds volts to amperes is wrong, full stop);
- reduces the number of variables an experiment or simulation must sweep, often dramatically;
- yields **scaling laws** ("double the length, quadruple the radiated power") that guide design before detailed analysis;
- organises results into dimensionless groups that are universal across sizes and materials.

Chapter 1 introduced dimensional consistency as an error check. This section turns it into a method.

Dimensions, units and homogeneity

Dimension, base dimensions, dimensional formula

A **dimension** is a physical kind of quantity (length, mass, time…); a **unit** is a particular standard amount of it (metre, kilogram, second). The SI uses seven **base dimensions**; for electrical engineering four or five suffice: mass M, length L, time T, current I, and temperature Θ. Every derived quantity Q has a **dimensional formula**

$$[Q] = \mathrm{M}^{a_1}\mathrm{L}^{a_2}\mathrm{T}^{a_3}\mathrm{I}^{a_4}\Theta^{a_5},$$

and the exponent vector $(a_1, \dots, a_5)$ is its **dimension vector**. A quantity is **dimensionless** if all exponents are zero.

Some dimension vectors used constantly in EE:

Quantity	SI unit	M	L	T	I	Θ
Voltage V	V	1	2	−3	−1	0
Resistance R	Ω	1	2	−3	−2	0
Capacitance C	F	−1	−2	4	2	0
Inductance L	H	1	2	−2	−2	0
Permittivity ε	F/m	−1	−3	4	2	0
Permeability μ	H/m	1	1	−2	−2	0
Conductivity σ	S/m	−1	−3	3	2	0
Power P	W	1	2	−3	0	0
Thermal resistance R_{th}	K/W	−1	−2	3	0	1
Thermal conductivity k	W/(m K)	1	1	−3	0	−1
Heat-transfer coefficient h	W/(m² K)	1	0	−3	0	−1

Principle of dimensional homogeneity

A physically meaningful equation must have the same dimensions in every additive term. Consequently, the arguments of transcendental functions (exp, ln, sin, …) must be dimensionless, and any relation among physical quantities can be rearranged so that it relates dimensionless combinations only.

The first sentence is the error check of Ch. 1. The second is the seed of the Pi theorem: a law of nature cannot depend on whether we measure in metres or feet, so it must be expressible in quantities that do not change when units change — dimensionless groups.

The Buckingham Pi theorem

Buckingham Pi theorem

Suppose a physical relation involves n quantities $Q_1, \dots, Q_n$,

$$f(Q_1, Q_2, \dots, Q_n) = 0,$$

and is dimensionally homogeneous. Let $\mathbf{D}$ be the $k \times n$ **dimension matrix** whose column i is the dimension vector of Q_i, and let $r = \operatorname{rank}\mathbf{D}$. Then the relation is equivalent to one among $n - r$ independent dimensionless products

$$F(\Pi_1, \Pi_2, \dots, \Pi_{n-r}) = 0, \qquad \Pi_m = Q_1^{a_{1m}} Q_2^{a_{2m}} \cdots Q_n^{a_{nm}},$$

where the exponent vectors $\mathbf{a}_m = (a_{1m}, \dots, a_{nm})^{\mathsf{T}}$ form a basis of the null space of $\mathbf{D}$.

Proof sketch. A product $\Pi = \prod Q_i^{a_i}$ has dimension vector $\mathbf{D}\mathbf{a}$, so it is dimensionless exactly when $\mathbf{D}\mathbf{a} = \mathbf{0}$. By the rank–nullity theorem (Ch. 9) the null space has dimension $n - r$, so there are exactly $n - r$ independent dimensionless products; any other dimensionless product is a product of powers of these. Now pick r of the quantities whose dimension vectors are linearly independent (the **repeating variables**). Changing the size of each base unit rescales every Q_i by a factor, and because the repeating variables span the dimension space, units can be chosen so that each repeating variable has numerical value 1. In those units the relation $f = 0$ depends only on the remaining $n - r$ quantities, whose numerical values are now precisely the Π groups. Since the physical relation cannot depend on the choice of units, it must be a relation among the Π's alone. $\square$

The theorem's power lies in the count: a relation among n quantities becomes one among $n - r$ groups. A function of one variable needs one curve to describe it; a function of zero variables is just a constant.

Procedure — applying the Pi theorem

1. **List** every quantity that could influence the result, including the result itself. Omitting a relevant quantity is the main way the method fails.
2. **Write** the dimension matrix $\mathbf{D}$ and find its rank r.
3. **Choose** r repeating variables with independent dimensions, preferably those that are "given" (geometry, material constants), not the unknown.
4. **Form** each remaining quantity into a Π by multiplying by powers of the repeating variables and solving the linear equations for the exponents that make it dimensionless. (Equivalently: compute a null-space basis of $\mathbf{D}$.)
5. **Write** the result as $\Pi_1 = \Phi(\Pi_2, \dots, \Pi_{n-r})$ with the unknown in Π_1, and determine Φ by theory, simulation or experiment.

Worked Example 25.7 — The period of a pendulum

Problem. A simple pendulum has bob mass m, length ℓ, release angle θ_0, in gravity g. Find the form of the period T_p.

List. $T_p, m, \ell, g, \theta_0$: $n = 5$.

Dimension matrix (rows M, L, T):

$$\mathbf{D} = \begin{array}{c|ccccc} & T_p & m & \ell & g & \theta_0 \\ \hline \mathrm{M} & 0 & 1 & 0 & 0 & 0 \\ \mathrm{L} & 0 & 0 & 1 & 1 & 0 \\ \mathrm{T} & 1 & 0 & 0 & -2 & 0 \end{array}$$

The columns of m, ℓ, g are independent, so $r = 3$ and there are $5 - 3 = 2$ groups.

Groups. θ_0 (an angle, radians, is already dimensionless): $\Pi_2 = \theta_0$. For T_p, try $\Pi_1 = T_p\, m^a \ell^b g^c$: M: $a = 0$; L: $b + c = 0$; T: $1 - 2c = 0$. So $c = \tfrac{1}{2}$, $b = -\tfrac{1}{2}$:

$$\Pi_1 = T_p \sqrt{\frac{g}{\ell}}, \qquad T_p = \sqrt{\frac{\ell}{g}}\, \Phi(\theta_0).$$

Conclusions without solving anything. (i) The period is independent of the mass — no dimensionless group can contain m, because m is the only quantity carrying M. (ii) It scales as $\sqrt{\ell}$: a pendulum four times longer swings half as fast. (iii) Only one curve $\Phi(\theta_0)$ need be measured. Solving the ODE (Section 25.5) gives $\Phi(0) = 2\pi$ and $\Phi(\theta_0) = 4K(\sin(\theta_0/2))$ in terms of the complete elliptic integral K; dimensional analysis cannot find the constant 2π, but it found everything else.

Worked Example 25.8 — Dimensionless groups of the RLC circuit

Problem. The free response of a series RLC circuit is characterised by a frequency ω (any characteristic frequency). Which dimensionless groups govern it?

List. ω, L, C, R: $n = 4$. **Dimension matrix** (rows M, L, T, I):

$$\mathbf{D} = \begin{array}{c|cccc} & \omega & L & C & R \\ \hline \mathrm{M} & 0 & 1 & -1 & 1 \\ \mathrm{L} & 0 & 2 & -2 & 2 \\ \mathrm{T} & -1 & -2 & 4 & -3 \\ \mathrm{I} & 0 & -2 & 2 & -2 \end{array}$$

Rows M, L and I are proportional (each component appears only through the combination $\mathrm{ML^2I^{-2}}$), so $r = 2$ and there are $4 - 2 = 2$ groups even though four base dimensions appear. This is why the rank, not the number of base dimensions, must be used.

Null space. Solving $\mathbf{D}\mathbf{a} = \mathbf{0}$ gives basis vectors $(2, 1, 1, 0)$ and $(0, -1, 1, 2)$ for (ω, L, C, R) — the second equivalent, after taking a square root, to $(0, -\tfrac{1}{2}, \tfrac{1}{2}, 1)$. Hence

$$\Pi_1 = \omega^2 LC \;\to\; \omega\sqrt{LC}, \qquad \Pi_2 = R\sqrt{\frac{C}{L}}.$$

Any characteristic frequency is therefore $\omega = \Phi\!\left(R\sqrt{C/L}\right)/\sqrt{LC}$. Section 25.4 shows these are exactly $\omega_n = 1/\sqrt{LC}$ and $\zeta = \tfrac{1}{2}R\sqrt{C/L}$; the damped frequency $\omega_d = \omega_n\sqrt{1-\zeta^2}$ is one instance of the general form.

Worked Example 25.9 — Skin depth from dimensions alone

Problem. At angular frequency ω, current in a conductor of conductivity σ and permeability μ concentrates in a surface layer of thickness δ. Assuming δ depends only on ω, μ, σ, find its form and estimate the scaling with frequency.

List and matrix. $n = 4$: $\delta, \omega, \mu, \sigma$. Rows M, L, T, I:

$$\mathbf{D} = \begin{array}{c|cccc} & \delta & \omega & \mu & \sigma \\ \hline \mathrm{M} & 0 & 0 & 1 & -1 \\ \mathrm{L} & 1 & 0 & 1 & -3 \\ \mathrm{T} & 0 & -1 & -2 & 3 \\ \mathrm{I} & 0 & 0 & -2 & 2 \end{array}$$

Rank $r = 3$, so a single group exists. Try $\delta^2 \omega^a \mu^b \sigma^c$. M: $b - c = 0$; I: $-2b + 2c = 0$; L: $2 + b - 3c = 0 \Rightarrow c = 1, b = 1$; T: $-a - 2b + 3c = 0 \Rightarrow a = 1$. So

$$\Pi = \delta^2 \omega \mu \sigma = \text{const} \quad\Longrightarrow\quad \delta = \frac{c_0}{\sqrt{\omega\mu\sigma}}.$$

With one group and no other, Φ is a pure constant. Solving the diffusion equation for the field in the conductor (Ch. 13) gives $c_0 = \sqrt{2}$, i.e. $\delta = \sqrt{2/(\omega\mu\sigma)}$.

Numbers. For copper, $\sigma = 5.8 \times 10^7$ S/m, $\mu = \mu_0$: $\delta = 8.53$ mm at 60 Hz, 66 μm at 1 MHz, 2.09 μm at 1 GHz — every factor of 100 in frequency shrinks δ by 10, exactly as $\delta \propto \omega^{-1/2}$ demands.

Worked Example 25.10 — Radiated power of a short dipole (two groups)

Problem. A short wire antenna of length ℓ carries a uniform sinusoidal current of amplitude I_0 at angular frequency ω in free space (μ_0, ε_0). Find the form of the time-averaged radiated power P.

List. $P, I_0, \ell, \omega, \mu_0, \varepsilon_0$: $n = 6$. **Matrix** (rows M, L, T, I):

$$\mathbf{D} = \begin{array}{c|cccccc} & P & I_0 & \ell & \omega & \mu_0 & \varepsilon_0 \\ \hline \mathrm{M} & 1 & 0 & 0 & 0 & 1 & -1 \\ \mathrm{L} & 2 & 0 & 1 & 0 & 1 & -3 \\ \mathrm{T} & -3 & 0 & 0 & -1 & -2 & 4 \\ \mathrm{I} & 0 & 1 & 0 & 0 & -2 & 2 \end{array}$$

Rank $r = 4$, so $6 - 4 = 2$ groups.

Null space. A basis is $\mathbf{a}_1 = (-1, 2, 1, 1, 1, 0)$ and $\mathbf{a}_2 = (1, -2, 1, 1, 0, 1)$, i.e.

$$\Pi_a = \frac{I_0^2\, \ell\, \omega\, \mu_0}{P}, \qquad \Pi_b = \frac{P\, \ell\, \omega\, \varepsilon_0}{I_0^2}.$$

These are correct but not illuminating. Any products of powers are equally valid groups, so recombine: $\Pi_a \Pi_b = \omega^2 \ell^2 \mu_0 \varepsilon_0 = (k\ell)^2$ with $k = \omega\sqrt{\mu_0 \varepsilon_0} = 2\pi/\lambda$, and $\sqrt{\Pi_b / \Pi_a} = (P/I_0^2)\sqrt{\varepsilon_0/\mu_0} = P/(\eta_0 I_0^2)$, with $\eta_0 = \sqrt{\mu_0/\varepsilon_0} \approx 377\ \Omega$. Therefore

$$P = \eta_0 I_0^2\, \Phi(k\ell).$$

Physics fills in Φ. For a short dipole ($k\ell \ll 1$) the field solution gives $\Phi = (k\ell)^2/(12\pi)$, i.e. $P = \eta_0 I_0^2 (k\ell)^2/(12\pi) = 40\pi^2 (\ell/\lambda)^2 I_0^2$ watts. For $\ell = 1$ m, $f = 10$ MHz ($\lambda = 30$ m), $I_0 = 1$ A: $P = 40\pi^2/900 = 0.439$ W. Dimensional analysis alone already told us the answer is $\eta_0 I_0^2$ times a function of electrical length, and that a geometrically similar antenna scaled together with the wavelength radiates the same power for the same current.

Choice of groups. The lesson of the recombination step: the null space is unique, its basis is not. Choose groups that have recognised physical meanings (electrical length $k\ell$, power normalised to $\eta_0 I_0^2$).

Worked Example 25.11 — Forced-air cooling of a heat sink: Nusselt, Reynolds and Prandtl

Problem. The convective heat-transfer coefficient h of a heat-sink fin in an air stream depends on air speed V, fin length ℓ, and the air's thermal conductivity k, density ρ, viscosity μ and specific heat c_p. Reduce the problem.

List. $n = 7$; base dimensions M, L, T, Θ, and the dimension matrix has rank $r = 4$. So there are $7 - 4 = 3$ groups. A null-space computation gives the basis (after recombination):

$$\mathrm{Nu} = \frac{h\ell}{k}, \qquad \mathrm{Re} = \frac{\rho V \ell}{\mu}, \qquad \mathrm{Pr} = \frac{\mu c_p}{k}.$$

Check one: $[h\ell/k] = (\mathrm{MT}^{-3}\Theta^{-1})(\mathrm{L})/(\mathrm{MLT}^{-3}\Theta^{-1}) = 1$.

Result. $\mathrm{Nu} = \Phi(\mathrm{Re}, \mathrm{Pr})$. A seven-variable problem has become a two-variable one; for air $\mathrm{Pr} \approx 0.71$ is essentially fixed, so a *single curve* $\mathrm{Nu}(\mathrm{Re})$ — measured once in a wind tunnel on any convenient size of fin — serves every fin and every fan speed. Laminar flat-plate theory gives $\mathrm{Nu} = 0.664\, \mathrm{Re}^{1/2} \mathrm{Pr}^{1/3}$, from which $h \propto \sqrt{V/\ell}$: doubling the fan speed raises h by only $\sqrt{2}$.

Common Pitfall — the missing variable

Dimensional analysis is only as good as the list of variables. Leave out g in the pendulum and no group containing T_p can be formed — the method then "proves" that the period is meaningless. Worse, leave out a variable whose dimensions happen to be expressible through the others (say, the wire radius a in the dipole problem, which has the same dimension as ℓ) and the method silently returns a wrong, incomplete answer: the true result would contain the additional group a/ℓ. Physical insight decides what goes in the list; the algebra only processes it.

Common Pitfall — dimensionless is not the same as negligible

Angles, ratios such as a/ℓ, and numbers like Pr are dimensionless but may matter enormously. Dimensional analysis never tells you which groups can be dropped; that comes from scaling (Section 25.4) or experiment.

Engineering Note — scaling laws in electronics

Dennard's MOSFET scaling rules are dimensional reasoning applied to a device: scale all lengths and the supply voltage by $1/\kappa$ and doping by κ, so that every electric field — and therefore every dimensionless group in the device equations — is unchanged. Then current scales as $1/\kappa$, capacitance as $1/\kappa$, gate delay CV/I as $1/\kappa$, and power density stays constant. When threshold-voltage and leakage effects (which introduce the thermal voltage kT/q, a quantity that does *not* scale) became important, the rules broke down — a dimensional argument explaining why the "free lunch" of scaling ended.

25.4 Scaling and nondimensionalization

From dimensionless groups to dimensionless equations

Dimensional analysis (Section 25.3) works on a *list of variables* and tells us which combinations can appear. **Nondimensionalization** works on the *equations themselves*: every dependent and independent variable is divided by a characteristic scale, and the equation is rewritten in the new variables. The parameters then collapse into a handful of dimensionless groups, each of which can be read as a ratio of competing effects. When the scales are chosen well, the size of each group tells us which terms dominate and which can be neglected.

Characteristic scale, scaled variable

For a variable q with characteristic magnitude q_c (a constant built from the model's parameters, inputs or geometry), the **scaled variable** is $q^* = q/q_c$. For time, $t^* = t/t_c$, and derivatives transform by the chain rule:

$$\frac{dq}{dt} = \frac{q_c}{t_c}\frac{dq^*}{dt^*}, \qquad \frac{d^2q}{dt^2} = \frac{q_c}{t_c^2}\frac{d^2q^*}{dt^{*2}}.$$

A scaling is **good** when, for the behaviour of interest, the scaled variables and their derivatives are of order one, so that the size of each dimensionless coefficient directly measures the importance of its term.

The procedure is mechanical:

1. Identify dependent variables, independent variables and parameters.
2. Introduce scaled variables with *unknown* scales q_c, t_c, x_c.
3. Substitute and divide through by the coefficient of one chosen term (usually the one that must always be present).
4. Choose the scales to make as many coefficients as possible equal to 1.
5. The coefficients that remain are the governing dimensionless groups.

The number of groups that survive equals the number predicted by the Pi theorem for the quantities in the equation — the two methods are consistent, but scaling also tells you *where* in the equation each group sits.

The series RLC circuit: ω_n and ζ

The capacitor voltage of a series RLC circuit driven by $v_s(t)$ obeys

$$LC\frac{d^2v}{dt^2} + RC\frac{dv}{dt} + v = v_s(t).$$

Let $v = V_0\, y$ (with V_0 the size of the input, e.g. the step height) and $t = t_c\,\tau$:

$$\frac{LC}{t_c^2}\,y'' + \frac{RC}{t_c}\,y' + y = u(\tau), \qquad u = v_s/V_0,$$

with primes denoting $d/d\tau$. Choosing t_c so that the first coefficient is 1 gives $t_c = \sqrt{LC}$, and the middle coefficient becomes $RC/\sqrt{LC} = R\sqrt{C/L}$. Writing it as 2ζ:

Standard form of a second-order system

With $\omega_n = 1/\sqrt{LC}$, $\tau = \omega_n t$ and

$$\zeta = \frac{R}{2}\sqrt{\frac{C}{L}} = \frac{1}{2Q},$$

the series RLC circuit becomes

$$y'' + 2\zeta y' + y = u(\tau).$$

Every series RLC circuit — every mass–spring–damper, every second-order servo — with the same ζ has the *same* scaled response; ω_n only stretches the time axis and V_0 only stretches the voltage axis.

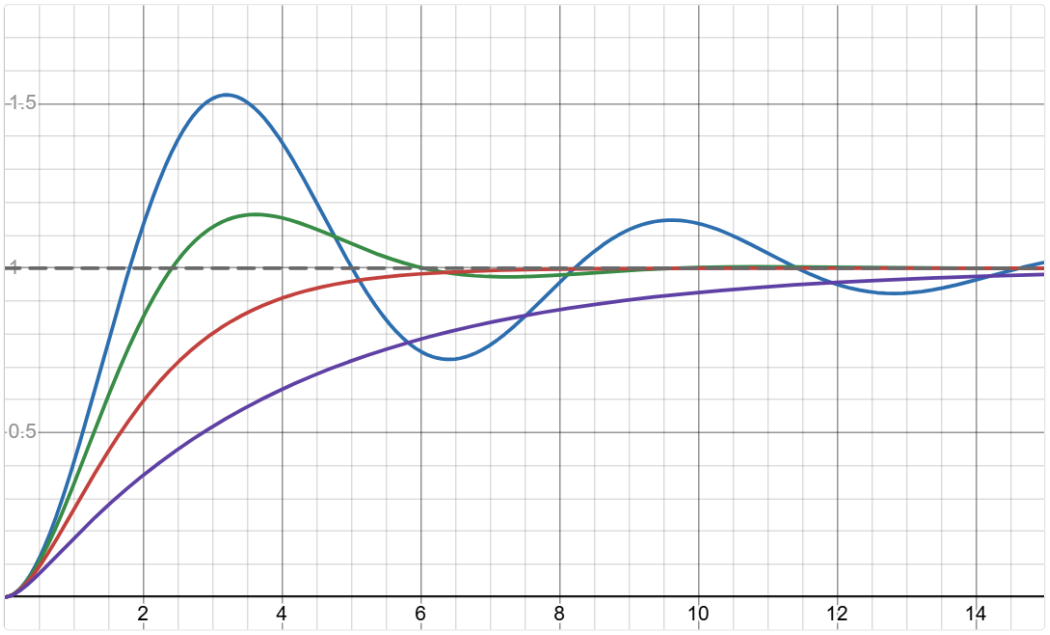
Three parameters (R, L, C) have become one (ζ) plus two scales. Equivalently, the physical step response is

$$v(t) = V_0\, S_\zeta(\omega_n t),$$

where S_ζ is the universal unit-step response of the standard form. For $0 < \zeta < 1$,

$$S_\zeta(\tau) = 1 - e^{-\zeta\tau}\left[\cos\!\left(\sqrt{1-\zeta^2}\,\tau\right) + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\!\left(\sqrt{1-\zeta^2}\,\tau\right)\right],$$

for $\zeta = 1$, $S_1(\tau) = 1 - (1+\tau)e^{-\tau}$, and for $\zeta > 1$ it is the sum of two decaying real exponentials with rates $\zeta \pm \sqrt{\zeta^2 - 1}$.



Universal step responses $S_\zeta(\tau)$ of $y''+2\zeta y'+y=u$ for $\zeta=0.2$ (blue), 0.5 (green), 1 (red, critically damped) and 2 (purple, overdamped); the horizontal axis is $\tau=\omega_n t$.

The dimensionless figures of merit follow directly: percent overshoot $\mathrm{PO}=100\,e^{-\pi\zeta/\sqrt{1-\zeta^2}}$, peak time $\tau_p=\pi/\sqrt{1-\zeta^2}$, and 2 % settling time $\tau_s\approx 4/\zeta$. Each is a function of ζ only; physical times are recovered by dividing by ω_n.

Worked Example 25.12 — Reading the RLC circuit of Example 25.4 in scaled form

Problem. For $R=10\ \Omega$, $L=1$ mH, $C=1\ \mu$F, find ω_n, ζ, Q, overshoot, peak time and settling time of the capacitor voltage after a 1 V step. What resistance gives critical damping?

Scales. $\omega_n=1/\sqrt{10^{-3}\times 10^{-6}}=1/\sqrt{10^{-9}}=31\,623$ rad/s ($f_n=5.03$ kHz). $\zeta=\frac{10}{2}\sqrt{10^{-6}/10^{-3}}=5\times 0.031\,62=0.158$; $Q=1/(2\zeta)=3.16$.

Figures of merit. $\sqrt{1-\zeta^2}=0.9874$.

$$\mathrm{PO}=100\exp\!\left(-\frac{\pi\times 0.158}{0.9874}\right)=100\,e^{-0.503}=60.5\,\%.$$

Peak time $t_p=\pi/(\omega_n\sqrt{1-\zeta^2})=\pi/31\,225=100.6\ \mu$s; settling time $t_s\approx 4/(\zeta\omega_n)=4/5000=0.8$ ms.

Critical damping. $\zeta=1$ requires $R=2\sqrt{L/C}=2\sqrt{1000}=63.2\ \Omega$. The quantity $\sqrt{L/C}=31.6\ \Omega$ is the **characteristic impedance** of the resonator — the natural resistance scale of the circuit — and ζ is simply R measured in units of $2\sqrt{L/C}$.

Check against Example 25.4. There the eigenvalues were $-5000\pm j\,31\,225$: real part $-\zeta\omega_n=-0.158\times 31\,623=-5000$ ✓, imaginary part $\omega_n\sqrt{1-\zeta^2}=31\,225$ ✓.

Thermal models: lumped scaling

The sink equation of Example 25.2, $C_s\dot T_s=P-(T_s-T_a)/R_{sa}$, scales with temperature-rise scale $\Delta T_c=PR_{sa}$ and time scale $t_c=R_{sa}C_s$. Setting $\theta=(T_s-T_a)/(P_0R_{sa})$ and $\tau=t/(R_{sa}C_s)$,

$$\frac{d\theta}{d\tau}+\theta=p(\tau),\qquad p=P/P_0.$$

There is *no* dimensionless parameter left: every first-order lag is the same system, and its only "parameters" are the two scales. This is why one plot of $1-e^{-\tau}$ serves RC filters, thermometers and heat sinks alike.

Distributed thermal models: Biot and Fourier numbers

The lumped model assumes the body is at a uniform temperature. Whether that is reasonable is itself answered by scaling. Consider a plate of half-thickness ℓ, conductivity k, diffusivity $\alpha=k/(\rho c_p)$, initially at T_0, suddenly exposed on both faces to a fluid at T_∞ with heat-transfer coefficient h. By symmetry, on $0\le x\le \ell$:

$$\frac{\partial T}{\partial t}=\alpha\frac{\partial^2 T}{\partial x^2},\qquad \frac{\partial T}{\partial x}(0,t)=0,\qquad -k\frac{\partial T}{\partial x}(\ell,t)=h\,\big(T(\ell,t)-T_\infty\big),\qquad T(x,0)=T_0.$$

Scale with $\xi=x/\ell$, $\theta=(T-T_\infty)/(T_0-T_\infty)$ and an as-yet-free time scale t_c:

$$\frac{\ell^2}{\alpha t_c}\frac{\partial\theta}{\partial\tau}=\frac{\partial^2\theta}{\partial\xi^2},\qquad \frac{\partial\theta}{\partial\xi}(1,\tau)=-\frac{h\ell}{k}\,\theta(1,\tau).$$

Choosing $t_c=\ell^2/\alpha$ (the diffusion time) makes the scaled time the **Fourier number** and leaves a single group in the boundary condition, the **Biot number**:

$$\mathrm{Fo}=\frac{\alpha t}{\ell^2},\qquad \mathrm{Bi}=\frac{h\ell}{k}=\frac{\ell/k}{1/h}=\frac{\text{internal conduction resistance}}{\text{surface convection resistance}}.$$

The lumped-capacitance criterion

If $\mathrm{Bi} \ll 1$ (in practice $\mathrm{Bi} < 0.1$), the temperature inside the body is uniform to within about 5 % of the overall temperature difference, and the distributed model reduces to the lumped first-order model

$$\frac{d\theta}{dt} = -\frac{hA}{\rho c_p V}\,\theta, \qquad \tau_{\text{lumped}} = \frac{\rho c_p V}{hA} = \frac{\ell^2}{\alpha\,\mathrm{Bi}} \quad (\text{plate: } V/A = \ell).$$

Sketch. With $\mathrm{Bi} \to 0$ the boundary condition says $\partial\theta/\partial\xi \approx 0$ at both faces, so to leading order θ is independent of ξ. Integrating the PDE over $0 \le \xi \le 1$ and using the boundary conditions gives $d\bar\theta/d\mathrm{Fo} = -\mathrm{Bi}\,\theta(1) \approx -\mathrm{Bi}\,\bar\theta$, i.e. decay on the slow time $\mathrm{Fo} \sim 1/\mathrm{Bi}$, which is the lumped model. The internal profile relaxes on the fast time $\mathrm{Fo} \sim 1$; when $\mathrm{Bi} \ll 1$ these two time scales are well separated. □

Worked Example 25.13 — Is the heat sink really lumped?

Problem. The heat sink of Example 25.2 has a 5 mm thick aluminium base ($k = 200$ W/(m K), $\alpha = 8.4 \times 10^{-5}$ m²/s), cooled by natural convection, $h \approx 10$ W/(m²K) (effective, per unit fin area, $h_{\text{eff}} \approx 100$ W/(m²K) referred to the base area). A second design uses a 1.6 mm FR-4 board as the "sink" ($k = 0.3$ W/(m K)). Compute Biot numbers and diffusion times.

Aluminium. Heated from one side, the relevant length is the full thickness $\ell = 5$ mm:

$$\mathrm{Bi} = \frac{100 \times 0.005}{200} = 2.5 \times 10^{-3} \ll 0.1,$$

and the diffusion time $\ell^2/\alpha = 2.5 \times 10^{-5}/8.4 \times 10^{-5} = 0.30$ s, versus a lumped time constant of 450 s. Lumped model: excellent.

FR-4. $\mathrm{Bi} = 100 \times 0.0016/0.3 = 0.53$. The lumped model is *not* valid: the board surface far from the component runs much cooler than the copper under it, and the heat spreads laterally over a distance comparable to the board thickness times $1/\sqrt{\mathrm{Bi}}$. A distributed (PDE or finite-element) model, or at least a multi-node RC ladder, is required.

Moral. The *same equation* (first-order lag) is a superb model in one case and a poor one in the other; the dimensionless number tells which, before any simulation is run.

Time-scale separation and small parameters

A good scaling often exposes a **small parameter** $\varepsilon \ll 1$ multiplying one term. Neglecting that term gives a **reduced model**; the conditions under which this is legitimate are the subject of perturbation theory.

- If ε multiplies a term that is *not* the highest derivative, setting $\varepsilon = 0$ is a **regular** perturbation: the solution changes smoothly, by $O(\varepsilon)$.
- If ε multiplies the *highest* derivative, setting $\varepsilon = 0$ lowers the order of the equation and some initial conditions can no longer be satisfied: a **singular** perturbation. The neglected dynamics act in a fast **boundary layer** of duration $O(\varepsilon)$ in scaled time, after which the reduced model is accurate.

The RLC circuit with very small inductance is the canonical example. Scaling time on the *slow* scale $t_c = RC$ instead of $\sqrt{LC}$:

$$\varepsilon\, y'' + y' + y = u, \qquad \varepsilon = \frac{L/R}{RC} = \frac{L}{R^2 C} = \frac{1}{4\zeta^2}.$$

When $\zeta \gg 1$, $\varepsilon \ll 1$ and the circuit behaves as an RC circuit with time constant RC, except for a brief initial interval of order L/R during which the inductor current builds up. The two time constants are approximately RC and L/R, and their ratio is ε — a direct measure of the **stiffness** that will matter for numerical solution (Section 25.9).

Worked Example 25.14 — A nonlinear oscillator reduced to one parameter

Problem. A negative-resistance LC oscillator (for example, a tunnel-diode or cross-coupled-pair oscillator) is modelled by an inductor L and capacitor C in parallel with a nonlinear element drawing current $i_N(v) = -av + bv^3$ ($a > 0$ is the small-signal negative conductance, $b > 0$ models gain compression). Nondimensionalize and predict amplitude and frequency for $L = 10\ \mu\text{H}$, $C = 1\ \text{nF}$, $a = 1\ \text{mS}$, $b = 1\ \text{mA/V}^3$.

Equations. KCL at the top node with inductor current i_L: $C\dot{v} + i_L + i_N(v) = 0$, $L\,\dot{i}_L = v$. Differentiate the first and substitute the second:

$$C\ddot{v} - a\left(1 - \frac{3b}{a}v^2\right)\dot{v} + \frac{v}{L} = 0.$$

Scaling. Let $t = \sqrt{LC}\,\tau$ and $v = V_c\,x$. Multiplying by L:

$$x'' - a\sqrt{\frac{L}{C}}\left(1 - \frac{3bV_c^2}{a}x^2\right)x' + x = 0.$$

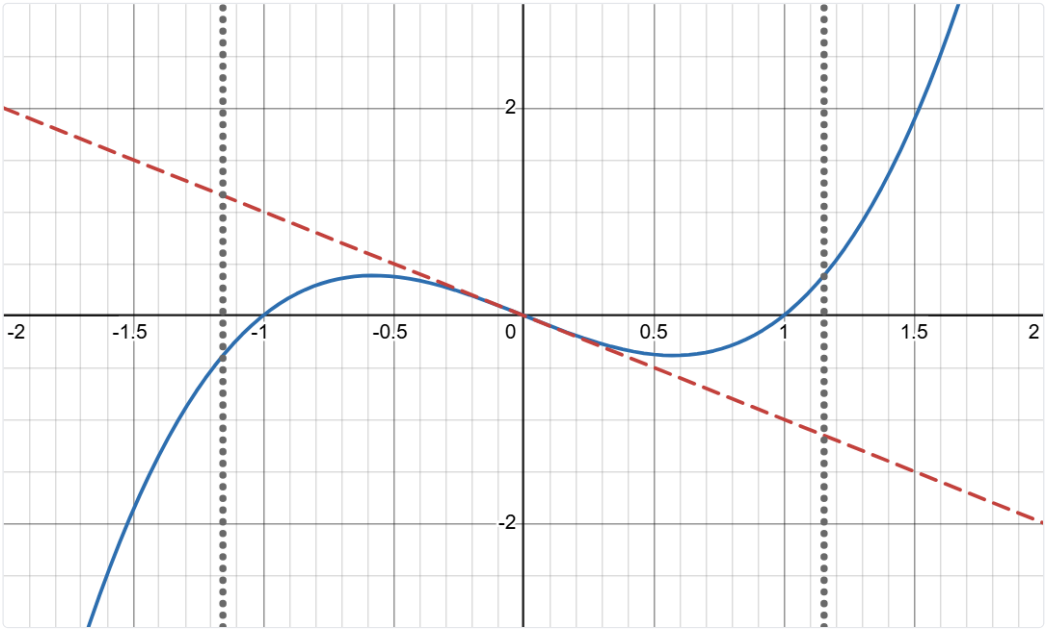
Choose $V_c = \sqrt{a/(3b)}$ to make the bracket $1 - x^2$. The result is the **Van der Pol equation**

$$x'' - \mu\,(1 - x^2)\,x' + x = 0, \qquad \mu = a\sqrt{\frac{L}{C}}.$$

Five physical parameters (L, C, a, b) plus time and voltage units have collapsed into *one* number μ, the ratio of the negative conductance to the tank's characteristic admittance $\sqrt{C/L}$.

Numbers. $\sqrt{L/C} = \sqrt{10^{-5}/10^{-9}} = 100\ \Omega$, so $\mu = 10^{-3} \times 100 = 0.1$. $V_c = \sqrt{10^{-3}/(3 \times 10^{-3})} = 0.577$ V. For small μ the Van der Pol limit cycle is nearly sinusoidal with amplitude $x \approx 2$ and scaled angular frequency ≈ 1; numerical integration gives amplitude 2.000 and period $1.0006 \times 2\pi$ for $\mu = 0.1$. Hence $v(t) \approx 1.155\cos(\omega_0 t)$ V with $f_0 = 1/(2\pi\sqrt{LC}) = 1.59$ MHz.

The parameter's meaning. For $\mu = 1$ the scaled amplitude is 2.01 and the period $1.06 \times 2\pi$; for $\mu = 5$ the oscillation becomes a *relaxation* oscillation with period $1.85 \times 2\pi$ and sharp transitions — a square-ish rather than sinusoidal waveform. A designer wanting a clean sine wave therefore wants $\mu \ll 1$, i.e. just enough negative conductance to sustain oscillation; this design rule is invisible in the five-parameter form and obvious in the one-parameter form.



Nonlinear element current $i_N(v) = -av + bv^3$ in mA against v in volts ($a = 1$ mS, $b = 1$ mA/V³): the small-signal slope $-a$ (red dashed) supplies energy near $v = 0$, the cubic term absorbs it at large swing, and the dotted lines mark the predicted oscillation amplitude $2V_c = 1.155$ V.

Common Pitfall — scaling with the wrong scale

A scaling is a *hypothesis* about which balance dominates. Scaling the RLC circuit with $t_c = \sqrt{LC}$ when $\zeta = 50$ produces a coefficient $2\zeta = 100$ in front of y' and hides the fact that the response has two widely separated time scales; scaling with $t_c = RC$ instead reveals $\varepsilon = 1/(4\zeta^2) = 10^{-4}$. If a scaled solution produces derivatives far from order one, the chosen scales were wrong for that regime — rescale.

Engineering Note — normalized design charts

Filter tables (Butterworth, Chebyshev), Smith charts, universal resonance curves and heat-sink design charts are all products of nondimensionalization. A Butterworth table lists poles for $\omega_c = 1$ rad/s and $R = 1\ \Omega$; *frequency scaling* and *impedance scaling* then produce any physical design. Recognising that a design problem is a scaled version of a tabulated one is often the fastest route to an answer.

25.5 Linearization

Why linearize

Almost every real device is nonlinear: diodes and transistors are exponential or square-law, magnetic cores saturate, pendulums follow $\sin\theta$, power is quadratic in voltage. Yet the overwhelming majority of engineering analysis — transfer functions, Bode plots, phasors, eigenvalue stability, Kalman filters, linear control — is linear. The bridge is **linearization**: replacing a nonlinear model by its first-order Taylor approximation about an **operating point**, and treating the deviations from that point as small signals. Small-signal analysis of amplifiers, the "AC analysis" of every circuit simulator, and the stability analysis of every power converter and control loop rest on it.

The price is a **regime of validity**: the linear model is accurate only for deviations small enough that the neglected second-order terms are negligible. An honest linear model states how small "small" is.

Operating points and the Jacobian

Operating point and linearized model

For the model $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u})$, $\mathbf{y} = \mathbf{h}(\mathbf{x}, \mathbf{u})$, an **operating point** (equilibrium) is a pair $(\bar{\mathbf{x}}, \bar{\mathbf{u}})$ with $\mathbf{f}(\bar{\mathbf{x}}, \bar{\mathbf{u}}) = \mathbf{0}$; the corresponding output is $\bar{\mathbf{y}} = \mathbf{h}(\bar{\mathbf{x}}, \bar{\mathbf{u}})$. Writing $\mathbf{x} = \bar{\mathbf{x}} + \delta\mathbf{x}$, $\mathbf{u} = \bar{\mathbf{u}} + \delta\mathbf{u}$, $\mathbf{y} = \bar{\mathbf{y}} + \delta\mathbf{y}$, the **linearized (small-signal) model** is

$$\delta\dot{\mathbf{x}} = \mathbf{A}\,\delta\mathbf{x} + \mathbf{B}\,\delta\mathbf{u}, \qquad \delta\mathbf{y} = \mathbf{C}\,\delta\mathbf{x} + \mathbf{D}\,\delta\mathbf{u},$$

with Jacobian matrices evaluated at the operating point:

$$\mathbf{A} = \left.\frac{\partial \mathbf{f}}{\partial \mathbf{x}}\right|_{\bar{\mathbf{x}},\bar{\mathbf{u}}}, \quad \mathbf{B} = \left.\frac{\partial \mathbf{f}}{\partial \mathbf{u}}\right|_{\bar{\mathbf{x}},\bar{\mathbf{u}}}, \quad \mathbf{C} = \left.\frac{\partial \mathbf{h}}{\partial \mathbf{x}}\right|_{\bar{\mathbf{x}},\bar{\mathbf{u}}}, \quad \mathbf{D} = \left.\frac{\partial \mathbf{h}}{\partial \mathbf{u}}\right|_{\bar{\mathbf{x}},\bar{\mathbf{u}}}.$$

Derivation. Taylor's theorem in several variables (Ch. 6) gives

$$\mathbf{f}(\bar{\mathbf{x}} + \delta\mathbf{x}, \bar{\mathbf{u}} + \delta\mathbf{u}) = \underbrace{\mathbf{f}(\bar{\mathbf{x}}, \bar{\mathbf{u}})}_{=\mathbf{0}} + \mathbf{A}\,\delta\mathbf{x} + \mathbf{B}\,\delta\mathbf{u} + \mathbf{r}(\delta\mathbf{x}, \delta\mathbf{u}),$$

and since $\dot{\bar{\mathbf{x}}} = \mathbf{0}$, $\delta\dot{\mathbf{x}} = \dot{\mathbf{x}}$. Dropping the remainder $\mathbf{r}$ gives the linear model. If $\mathbf{f}$ is twice continuously differentiable, the remainder is bounded by

$$\|\mathbf{r}\| \le \tfrac{1}{2} M \left(\|\delta\mathbf{x}\| + \|\delta\mathbf{u}\|\right)^2,$$

where M bounds the second derivatives of $\mathbf{f}$ near the operating point. The linear terms are of size $\|\mathbf{A}\|\|\delta\mathbf{x}\|$, so the *relative* error of linearization grows in proportion to the signal amplitude.

Linearization and local stability (Lyapunov's indirect method)

Let $\bar{\mathbf{x}}$ be an equilibrium of $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$ with $\mathbf{f}$ continuously differentiable and Jacobian $\mathbf{A}$.

1. If every eigenvalue of $\mathbf{A}$ has negative real part, $\bar{\mathbf{x}}$ is locally asymptotically stable for the nonlinear system.
2. If some eigenvalue has positive real part, $\bar{\mathbf{x}}$ is unstable for the nonlinear system.
3. If no eigenvalue has positive real part but some lie on the imaginary axis, linearization is **inconclusive**; the nonlinear terms decide.

Moreover (Hartman–Grobman), when no eigenvalue lies on the imaginary axis (a *hyperbolic* equilibrium), the nonlinear trajectories near $\bar{\mathbf{x}}$ are a continuous deformation of the linear ones.

The theorem is the formal licence for small-signal stability analysis — and case 3 is its fine print.

Small-signal model of a diode

Worked Example 25.15 — The diode small-signal resistance and its range of validity

Problem. A silicon diode obeys the Shockley equation $i = I_S\left(e^{v/(nV_T)} - 1\right)$ with $n = 1$ and $V_T = kT/q = 25.85$ mV at 300 K. It is biased at $I_D = 1$ mA. Find the small-signal resistance and determine how large a voltage excursion the linear model tolerates for 2 %, 10 % and 20 % accuracy in the current change.

Linearization. At the bias point $\bar{v} = V_D$, with $I_D \gg I_S$,

$$g_d = \left.\frac{di}{dv}\right|_{V_D} = \frac{I_S}{nV_T}e^{V_D/(nV_T)} \approx \frac{I_D}{nV_T}, \qquad r_d = \frac{1}{g_d} = \frac{nV_T}{I_D} = \frac{25.85\ \text{mV}}{1\ \text{mA}} = 25.85\ \Omega.$$

Remarkably, r_d depends only on the bias current, not on I_S — an unknown, process-dependent parameter has dropped out.

Exact versus linear. The exact current change for a voltage step δv is $\delta i = I_D\left(e^{\delta v/V_T} - 1\right)$; the linear prediction is $\delta i_{\text{lin}} = \delta v / r_d = I_D\,\delta v/V_T$. With $x = \delta v/V_T$, the relative error is

$$\frac{\delta i_{\text{lin}} - \delta i}{\delta i} = \frac{x}{e^{x}-1} - 1 \approx -\frac{x}{2} \quad (|x| \ll 1).$$

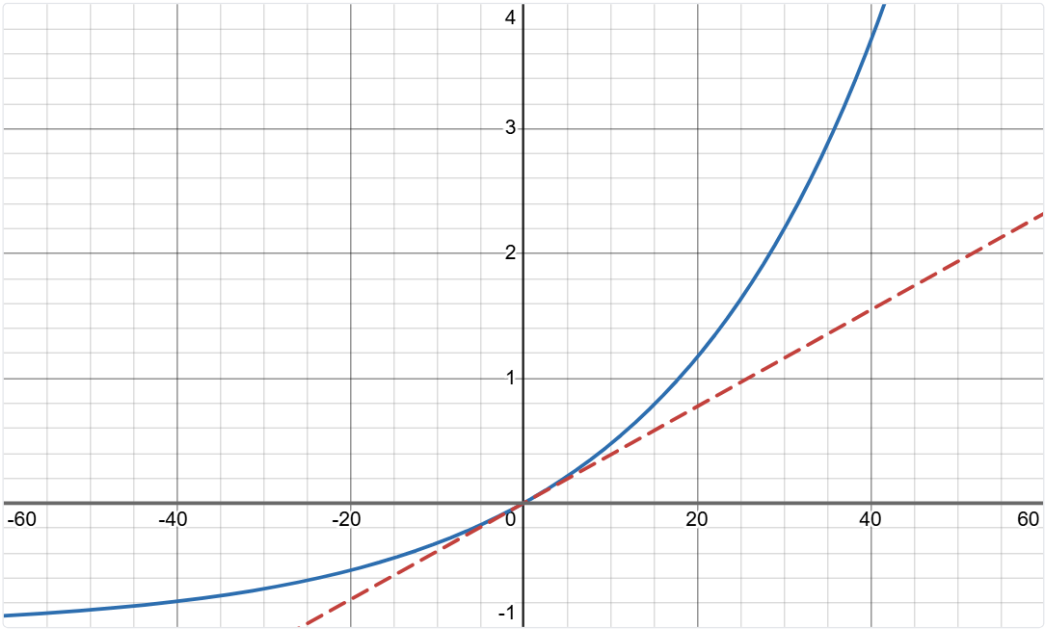
δv	exact δi (µA)	linear δi (µA)	error
$+1$ mV	39.44	38.68	−1.9 %
$+5$ mV	213.4	193.4	−9.4 %
$+10$ mV	472.3	386.8	−18.1 %
-10 mV	−320.8	−386.8	+20.6 %
$+20$ mV	1167.6	773.6	−33.7 %

Regime of validity. Since the error is about $|\delta v|/(2V_T)$, an accuracy e requires $|\delta v| \lesssim 2eV_T$: about 1 mV for 2 %, 5 mV for 10 %, 10 mV for 20 %. The often-quoted rule "small signal means $|v_d| \ll V_T$" is this calculation in words. Note the asymmetry between ± 10 mV: the linear model *underestimates* forward excursions and *overestimates* reverse ones, which is the origin of even-order distortion.

The second-order Taylor term gives that distortion quantitatively. For $\delta v = \hat{V}\cos\omega t$,

$$\delta i \approx I_D\left[\frac{\hat{V}}{V_T}\cos\omega t + \frac{1}{2}\left(\frac{\hat{V}}{V_T}\right)^2\cos^2\omega t\right] = I_D\left[\frac{\hat{V}}{V_T}\cos\omega t + \frac{\hat{V}^2}{4V_T^2} + \frac{\hat{V}^2}{4V_T^2}\cos 2\omega t\right],$$

so the second-harmonic distortion is $\text{HD}_2 \approx \hat{V}/(4V_T)$: 0.97 % at $\hat{V} = 1$ mV and 4.8 % at 5 mV (the exact Bessel-function result, $I_2(\hat{V}/V_T)/I_1(\hat{V}/V_T)$, agrees to three digits). The constant term $\hat{V}^2/(4V_T^2)$ is a shift of the operating point — **rectification** — and is exactly what the linear model misses entirely.



Diode current change $\delta i/I_D = e^{\delta v/V_T} - 1$ (blue) and its tangent-line linearization $\delta v/V_T$ (red dashed), with δv in mV; they agree only within a few millivolts of the operating point.

The nonlinear pendulum (and every oscillator with a $\sin$)

The pendulum $\ddot{\theta} + \omega_0^2 \sin\theta = 0$, $\omega_0^2 = g/\ell$, is the prototype for a whole class of EE models: the swing equation of a synchronous generator ($M\ddot{\delta} + D\dot{\delta} = P_m - P_{\max}\sin\delta$), the phase-locked loop with a sinusoidal phase detector, and the Josephson junction all have the same sin nonlinearity.

With $\mathbf{x} = [\theta,\ \dot{\theta}]^{\mathsf{T}}$, $\mathbf{f} = [\dot{\theta},\ -\omega_0^2\sin\theta]^{\mathsf{T}}$, equilibria satisfy $\sin\bar{\theta} = 0$: $\bar{\theta} = 0$ (hanging) and $\bar{\theta} = \pi$ (inverted). The Jacobian is

$$\mathbf{A}(\bar\theta) = \begin{bmatrix} 0 & 1 \\ -\omega_0^2 \cos\bar\theta & 0 \end{bmatrix}.$$

Worked Example 25.16 — Linearizing the pendulum about both equilibria

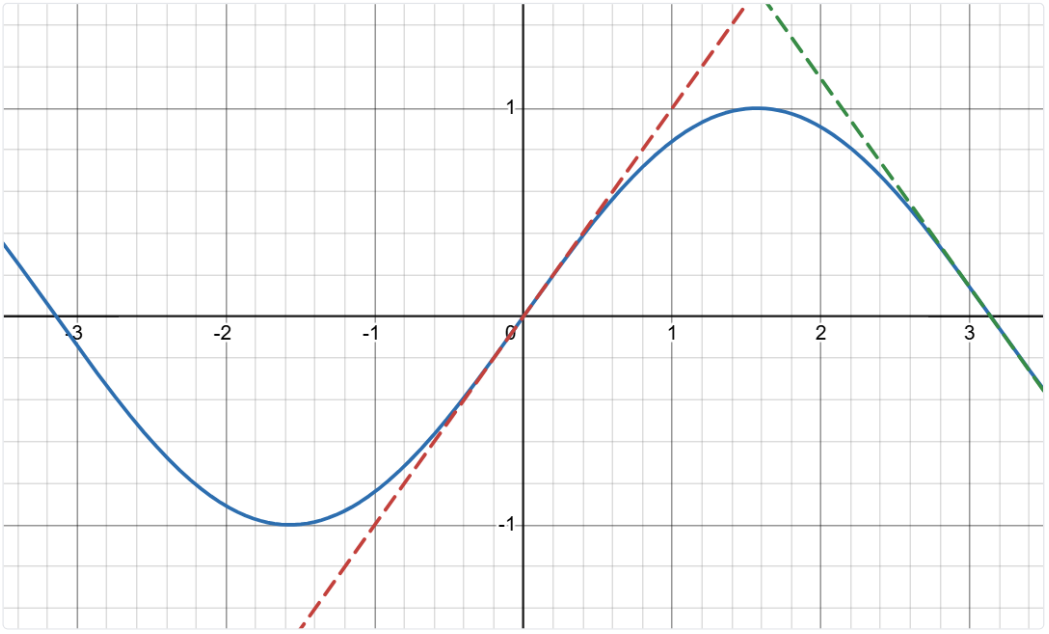
Hanging, $\bar\theta = 0$. $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -\omega_0^2 & 0 \end{bmatrix}$, eigenvalues $\pm j\omega_0$. Linearization predicts a centre (undamped oscillation at ω_0) — but by the theorem this is case 3, *inconclusive*. Here energy conservation, $E = \tfrac{1}{2}\dot\theta^2 + \omega_0^2(1 - \cos\theta)$, settles it: level curves of E near the bottom are closed, so the nonlinear system also has a centre.

Inverted, $\bar\theta = \pi$. $\cos\pi = -1$, so $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ \omega_0^2 & 0 \end{bmatrix}$, eigenvalues $\pm\omega_0$: a saddle, hyperbolic, so the nonlinear equilibrium is genuinely unstable, with small deviations growing as $e^{\omega_0 t}$. For $\ell = 1$ m, $\omega_0 = 3.13$ s^{-1}: an error doubles every $\ln 2/\omega_0 = 0.22$ s, which sets the bandwidth an inverted-pendulum balancing controller must have.

How good is the hanging linearization? The linear model predicts a period $T_0 = 2\pi/\omega_0$ independent of amplitude. The exact period is $T = (2/\pi)\,K(\sin(\theta_0/2))\,T_0$, with K the complete elliptic integral of the first kind; its series is $T/T_0 = 1 + \theta_0^2/16 + \dots$

Amplitude θ_0	exact T/T_0	$1 + \theta_0^2/16$	linear-model error
10°	1.0019	1.0019	0.19 %
20°	1.0077	1.0076	0.77 %
30°	1.0174	1.0171	1.7 %
45°	1.0400	1.0386	4.0 %
90°	1.1803	1.1542	18 %
150°	1.7622	1.4284	76 %

The "small-angle approximation" is good to 1 % up to about 23° and fails completely as $\theta_0 \to 180^\circ$, where the period diverges.



The pendulum nonlinearity $\sin\theta$ (blue) and its linearizations about $\bar\theta = 0$ (red, slope $+1$) and $\bar\theta = \pi$ (green, slope -1): the sign of the slope decides between oscillation and exponential divergence.

A multivariable example: magnetic levitation

Worked Example 25.17 — Linearizing an electromagnet suspension

Problem. A steel ball of mass m hangs below an electromagnet. With gap x (measured downward from the magnet face) and coil current i, the magnetic force is approximately $k\,i^2/x^2$ upward, so

$$m\ddot{x} = mg - k\frac{i^2}{x^2}.$$

For $m = 0.05$ kg, $k = 10^{-4}$ N m^2/A^2, find the current that holds the ball at $x_0 = 10$ mm, linearize, and classify the equilibrium.

Operating point. $mg = ki_0^2/x_0^2 \Rightarrow i_0 = x_0\sqrt{mg/k} = 0.01\sqrt{0.4905/10^{-4}} = 0.700$ A.

Jacobian. With $\mathbf{x} = [x,\ \dot{x}]^{\mathsf{T}}$ and input i, $f_2 = g - (k/m)\,i^2/x^2$:

$$\frac{\partial f_2}{\partial x} = \frac{2ki_0^2}{mx_0^3} = \frac{2g}{x_0}, \qquad \frac{\partial f_2}{\partial i} = -\frac{2ki_0}{mx_0^2} = -\frac{2g}{i_0},$$

where the equilibrium condition was used to simplify. So

$$\delta\ddot{x} = \frac{2g}{x_0}\,\delta x - \frac{2g}{i_0}\,\delta i = 1962\,\delta x - 28.0\,\delta i.$$

Classification. Open loop ($\delta i = 0$), $\lambda = \pm\sqrt{2g/x_0} = \pm 44.3$ s^{-1}: an unstable saddle with time constant $1/44.3 = 23$ ms. The linearized model, with transfer function $\delta X/\delta I = -28.0/(s^2 - 1962)$, is the starting point for designing the stabilizing feedback loop.

Validity. The second-order term in δx is $\tfrac{1}{2}(\partial^2 f_2/\partial x^2)\delta x^2 = -(3g/x_0^2)\,\delta x^2$; compared with the linear term $(2g/x_0)\delta x$, the ratio is $1.5\,|\delta x|/x_0$. For 10 % accuracy the ball must stay within about ± 0.7 mm of the set point — the linear controller design is valid only if the controller keeps it there.

Beyond equilibria

Two extensions are worth knowing by name:

- **Linearization about a trajectory.** If $\bar{\mathbf{x}}(t)$ is a nominal solution (e.g. a switching converter's periodic steady state), the deviations obey a *time-varying* linear system $\delta\dot{\mathbf{x}} = \mathbf{A}(t)\delta\mathbf{x} + \mathbf{B}(t)\delta\mathbf{u}$, with Jacobians evaluated along $\bar{\mathbf{x}}(t)$. For periodic trajectories this leads to Floquet theory and to state-space averaging in power electronics.
- **Quasi-linearization (describing functions).** For a sinusoidal signal of amplitude A through a static nonlinearity, replace the nonlinearity by the gain of its fundamental Fourier component (Ch. 15), $N(A)$. This amplitude-dependent "linearization" predicts limit cycles, as in the Van der Pol oscillator, where it gives amplitude $x = 2$ exactly.

Common Pitfall — linearizing what is not differentiable

Saturation, dead zones, ideal switches, Coulomb friction $\tau_c\,\mathrm{sgn}(\omega)$ and absolute-value detectors have no derivative at the critical point. A Jacobian taken "on one side" describes only that side, and small-signal analysis at the corner is meaningless: an amplifier biased exactly at the edge of saturation has *no* well-defined small-signal gain. Either bias away from the corner or use describing-function, piecewise-linear or simulation methods.

Engineering Note — what SPICE "AC analysis" actually does

A circuit simulator's AC analysis first solves the nonlinear DC operating-point equations by Newton's method (Ch. 22), then replaces every nonlinear device by its Jacobian — g_m, r_π, r_o, junction capacitances evaluated at the bias point — and solves the resulting *linear* phasor equations at each frequency. An AC analysis therefore reports a gain of, say, 40 dB for a 10 V input signal without complaint; it is the user's responsibility to remember that the result is valid only for signals small enough to keep the operating point fixed.

25.6 Model assumptions

Assumptions are part of the model

Every model in this chapter so far rested on assumptions: the capacitor had no leakage (Example 25.1), the package stored no heat (25.2), friction was viscous (25.3), the source had zero resistance (25.5), the diode signal was small (25.15). An assumption is not a weakness to be hidden; it is the *specification* of the model's domain of validity. A model without stated assumptions cannot be checked, cannot be trusted outside the conditions it was tested in, and cannot be improved systematically.

Assumption and regime of validity

A **modeling assumption** is a statement that some effect, quantity or interaction is negligible, constant, ideal or of a particular form. Its **regime of validity** is the set of operating conditions — usually expressible as an inequality on a dimensionless group — under which the error it introduces is below the accuracy required by the model's purpose.

The key move is the second sentence: turning a qualitative assumption ("the wire is short") into a quantitative inequality ("$\ell < \lambda/10$"). Sections 25.3–25.5 provide the tools; this section collects the standard assumptions of EE modeling and the inequalities that police them.

A catalogue of standard assumptions

Assumption	What it buys	Regime of validity (typical)	When it fails
Lumped (quasi-static) circuit	ODEs instead of PDEs; Kirchhoff's laws	size $\ell \ll \lambda$, say $\ell < \lambda/10$; or edge rise time $t_r > 6\,t_{pd}$	long cables, fast digital edges, RF
Lumped thermal body	one temperature per body	$\mathrm{Bi} = h\ell/k < 0.1$	thin, poorly conducting boards (Ex. 25.13)
Linearity	superposition, transforms, eigenvalues	deviations small vs. nonlinearity scale, e.g. $\lvert v_d\rvert \ll V_T$	large signals, saturation, clipping
Time invariance	constant coefficients, transfer functions	parameters drift less than accuracy over the window	warm-up, ageing, battery discharge
Ideal op-amp	virtual short, infinite input impedance	loop gain $\lvert A\beta\rvert \gg 1$, i.e. $f \ll f_{\mathrm{GBW}}\beta$	high frequency, high closed-loop gain
Ideal capacitor	$Z = 1/(j\omega C)$	f well below self-resonance $1/(2\pi\sqrt{L_{\mathrm{ESL}}C})$	decoupling at hundreds of MHz
Constant resistance	R fixed	$\alpha\,\Delta T \ll$ accuracy	windings, filaments, current shunts
Deterministic	no noise terms	signal $\gg$ noise (SNR large)	low-level sensing, receivers
Neglect of a time constant	lower-order model	that time constant $\ll$ time scale of interest (Sec. 25.4)	stiff, closely spaced modes
Continuum / Ohm's law	$\mathbf{J} = \sigma\mathbf{E}$	dimensions $\gg$ mean free path; fields moderate	nanoscale devices, breakdown

Making an assumption quantitative

The pattern is always the same: identify the neglected term, estimate its size relative to the retained terms using the scales of Section 25.4, and require the ratio to be below the tolerated error.

Worked Example 25.18 — Is a PCB trace a lumped wire or a transmission line?

Problem. A 10 cm trace on FR-4 (effective relative permittivity $\varepsilon_{\mathrm{eff}} \approx 4$) carries (a) a 100 MHz sinusoid, (b) a digital signal with 0.5 ns rise time. Is the lumped assumption valid?

Propagation. $v_p = c/\sqrt{\varepsilon_{\mathrm{eff}}} = 1.5 \times 10^8$ m/s, so the one-way delay is $t_{pd} = 0.1/1.5 \times 10^8 = 0.67$ ns.

(a) Sinusoid. $\lambda = v_p/f = 1.5$ m, $\ell/\lambda = 0.067 < 0.1$. The phase shift along the trace is $360^\circ \times 0.067 = 24^\circ$ — borderline. For a rough amplitude estimate the lumped model is acceptable; for a phase-critical path it is not.

(b) Digital edge. The relevant comparison is rise time versus delay. With $t_r = 0.5$ ns $< t_{pd} = 0.67$ ns, the edge has fully risen at the driver before it reaches the load: reflections will be seen as distinct steps. Equivalently, the signal's "knee" frequency $0.35/t_r = 700$ MHz has wavelength 0.21 m, and $\ell/\lambda \approx 0.5$. The trace must be modelled as a transmission line (Ch. 13, telegrapher's equations) and terminated.

Moral. "Is the circuit lumped?" has no answer until the *signal* is specified. The assumption belongs to the combination of system and input.

Worked Example 25.19 — When is an op-amp ideal?

Problem. A non-inverting amplifier with ideal gain $1/\beta = 100$ uses an op-amp with DC gain $A_0 = 10^5$ and a single pole giving gain–bandwidth product $f_{\mathrm{GBW}} = 1$ MHz (pole at 10 Hz). Over what frequency range is the ideal-op-amp model (gain exactly 100) accurate to 1 %? To 3 dB?

Model. The closed-loop gain is

$$G = \frac{A}{1 + A\beta} = \frac{1}{\beta}\cdot\frac{1}{1 + 1/(A\beta)}, \qquad A(f) = \frac{A_0}{1 + jf/10\ \mathrm{Hz}}.$$

The ideal model sets $1/(A\beta) = 0$; the neglected term is the reciprocal of the loop gain.

DC. $A_0\beta = 1000$, error $\approx 0.1\,\%$: excellent.

Frequency dependence. For $f \gg 10$ Hz, $A \approx -jf_{\mathrm{GBW}}/f$, so $1/(A\beta) \approx jf/(f_{\mathrm{GBW}}\beta) = jf/10$ kHz, and

$$|G| \approx \frac{100}{\sqrt{1 + (f/10\ \mathrm{kHz})^2}}.$$

A 1 % magnitude error requires $(f/10\ \mathrm{kHz})^2 < 0.02$, i.e. $f < 1.4$ kHz. The 3 dB point is $f = f_{\mathrm{GBW}}\beta = 10$ kHz. A direct computation gives $|G| = 99.41$ at 1 kHz, 89.4 at 5 kHz and 70.7 (−3.0 dB) at 10 kHz, confirming the estimate.

Moral. The "ideal op-amp" assumption is not a property of the op-amp; it is a condition, $|A\beta| \gg 1$, on the op-amp *in this circuit at this frequency*. An op-amp that is ideal for an audio buffer is far from ideal at gain 100 in the same band.

Worked Example 25.20 — Hidden parasitics: the capacitor that becomes an inductor

Problem. A 100 nF ceramic decoupling capacitor has equivalent series inductance $L_{\text{ESL}} = 1$ nH and equivalent series resistance $R_{\text{ESR}} = 10$ mΩ. Over what frequency range is it "a capacitor"?

Model. $Z(\omega) = R_{\text{ESR}} + j\left(\omega L_{\text{ESL}} - \dfrac{1}{\omega C}\right)$, a series RLC. Its self-resonant frequency is

$$f_{\text{SRF}} = \frac{1}{2\pi\sqrt{L_{\text{ESL}}C}} = \frac{1}{2\pi\sqrt{10^{-9}\times 10^{-7}}} = 15.9\ \text{MHz}.$$

At 1 MHz, $|Z| = 1.585\ \Omega$ versus the ideal $1/(\omega C) = 1.592\ \Omega$ — the ideal model is within 0.4 %. At 15.9 MHz the reactances cancel and $|Z| = R_{\text{ESR}} = 10\ \text{m}\Omega$. At 100 MHz, $|Z| = 0.61\ \Omega$ and the part is *inductive*: an ideal-capacitor model would predict 0.016 Ω, wrong by a factor of 38.

Regime. The ideal model holds for $f \ll f_{\text{SRF}}$; the inequality comes from the ratio of the neglected to the retained reactance, $\omega^2 L_{\text{ESL}} C = (f/f_{\text{SRF}})^2$.

Checking assumptions a posteriori

Many assumptions can only be checked after the model has been solved, because the size of the neglected term depends on the solution. The discipline is:

1. Solve the simplified model.
2. Use the solution to *evaluate* every neglected term.
3. Confirm each is small compared with the retained terms, by the margin the purpose requires.
4. If not, restore the term and re-solve.

Worked Example 25.21 — A posteriori check of a constant-resistance assumption

Problem. In the DC motor model of Example 25.3, $R_a = 2\ \Omega$ was taken constant. The copper winding has temperature coefficient $\alpha = 0.003\,93\ \text{K}^{-1}$. A thermal model of the motor shows that at rated load the winding settles 40 K above the temperature at which R_a was measured. The speed model must be accurate to 5 %. Is the assumption acceptable?

Evaluate the neglected effect. $R_a(T) = R_a(1 + \alpha\Delta T) = 2(1 + 0.157) = 2.31\ \Omega$ — a 15.7 % change.

Effect on the output. At load torque τ_L the steady speed (with b small) is $\omega \approx v_a/K - R_a\tau_L/K^2$. The *speed droop* term is proportional to R_a, so it is underestimated by 15.7 %. If the droop is 10 % of no-load speed, the speed error is 1.6 %: acceptable. If the motor runs at stall-like loads where droop is 50 %, the error is 7.9 %: not acceptable, and R_a must become temperature-dependent — making temperature a *state* coupled to an electrical model (Section 25.2).

Moral. Whether an assumption is valid depends on how sensitive the *output of interest* is to it, which is a question about the model's sensitivity, not only about the size of the neglected effect.

Engineering Note — write the assumptions down

Professional models (a thermal design report, a link budget, a SPICE netlist submitted for review) should begin with a numbered list of assumptions, each with its validity criterion and, ideally, the computed margin ("lumped: $\ell/\lambda = 0.03$ at the highest frequency of interest"). When the design changes — a faster edge, a hotter enclosure — the list tells you immediately which parts of the model must be revisited.

Common Pitfall — validity by habit

Assumptions inherited from a textbook or a previous project ("op-amps are ideal", "wires are short", "noise is white", "the load is resistive") are the most dangerous, because they are never questioned. The circuit that worked at 1 MHz does not work at 100 MHz for precisely this reason. Each assumption should be re-justified for every new operating range.

Conservative and non-conservative assumptions

Some assumptions err in a known direction. Neglecting a heat path *over*-predicts temperature (conservative for reliability); neglecting a parasitic capacitance may *under*-predict oscillation risk (non-conservative). When an assumption cannot be justified precisely, choosing the conservative direction lets the model still give a safe answer — a bound rather than an estimate. Worst-case and corner analysis (setting each uncertain parameter at the extreme that hurts most) is the systematic version of this idea; Monte Carlo analysis (Ch. 24) is the statistical version.

25.7 Model validation

Verification, calibration, validation

Three activities are often confused:

Verification, calibration and validation

- **Verification** asks *"Are we solving the equations right?"* — is the code or derivation a correct solution of the stated mathematical model? (Checked against analytical solutions, convergence tests, conservation checks.)
- **Calibration** (parameter identification, Section 25.8) asks *"What parameter values make the model match these data?"*
- **Validation** asks *"Are we solving the right equations?"* — does the calibrated model reproduce the behaviour of the real system, to the accuracy required, on data that were **not** used for calibration?

A model can be perfectly verified and perfectly calibrated and still be invalid: a first-order model fitted by least squares to a second-order system will be solved exactly and fitted optimally, and still be wrong. Validation is the step that catches structural error, and it requires data the model has not seen.

Residuals

Residual

Given measurements y_k ($k = 1, \dots, N$) and model predictions $\hat{y}_k$ at the same conditions, the **residuals** are

$$e_k = y_k - \hat{y}_k.$$

If the model is structurally correct and the measurements contain independent zero-mean noise of standard deviation σ, the residuals should look like that noise: zero mean, standard deviation $\approx \sigma$, no trend against time, input or predicted value, and no correlation between neighbours.

Residual analysis is the single most informative validation tool, because *structure in the residuals points to the missing physics*. A bowed residual plot means a missing curvature (a quadratic term, a saturation); residuals growing with $\hat{y}$ mean multiplicative rather than additive error; residuals that oscillate in time mean an unmodelled mode; a step in the residuals means an unmodelled event.

Goodness-of-fit measures

For a model with p fitted parameters and N data points, with $\bar{y} = \frac{1}{N}\sum y_k$:

Measure	Definition	Interpretation
Sum of squared errors	$\mathrm{SSE} = \sum e_k^2$	raw misfit; depends on units and N
Root-mean-square error	$\mathrm{RMSE} = \sqrt{\mathrm{SSE}/N}$	typical error, in units of y
Residual standard error	$s = \sqrt{\mathrm{SSE}/(N-p)}$	unbiased estimate of σ (Ch. 20)
Coefficient of determination	$R^2 = 1 - \mathrm{SSE}/\mathrm{SST},\ \mathrm{SST} = \sum (y_k - \bar{y})^2$	fraction of variance "explained"
Adjusted R^2	$1 - \dfrac{\mathrm{SSE}/(N-p)}{\mathrm{SST}/(N-1)}$	penalises extra parameters
Chi-square	$\chi^2 = \sum (e_k/\sigma_k)^2$	with known noise σ_k
Reduced chi-square	$\chi^2_\nu = \chi^2/\nu,\ \nu = N - p$	≈ 1 for an adequate model
Normalized fit	$100\left(1 - \dfrac{\lVert \mathbf{y} - \hat{\mathbf{y}} \rVert}{\lVert \mathbf{y} - \bar{y}\mathbf{1} \rVert}\right)\%$	used in system identification

Reduced chi-square test

If the model is correct, the noise is Gaussian with known standard deviations σ_k, and p linear parameters are fitted, then χ^2 follows a chi-square distribution with $\nu = N - p$ degrees of freedom (Ch. 20). Hence $E[\chi^2_\nu] = 1$ and $\mathrm{Var}(\chi^2_\nu) = 2/\nu$. A value of χ^2_ν far above $1 + 2\sqrt{2/\nu}$ indicates model inadequacy (or underestimated noise); a value far below 1 indicates overestimated noise or overfitting.

Common Pitfall — a high R^2 does not validate a model

R^2 measures how much of the *variation in the data* the model reproduces. When the data span a wide range (a sensor calibrated from 0 to 400 °C), almost any monotonic model achieves $R^2 > 0.999$, while its errors can still be fifty times the measurement noise. R^2 cannot detect structural error; residual plots and χ^2_ν can.

Worked Example 25.22 — Validating a linear RTD model

Problem. A platinum resistance thermometer is calibrated at nine temperatures; the resistance meter has standard uncertainty $\sigma = 0.02\ \Omega$.

T (°C)	0	50	100	150	200	250	300	350	400
R (Ω)	100.04	119.41	138.51	157.29	175.85	194.09	212.05	229.70	247.09

Fit (a) a linear model $R = c_0 + c_1 T$ and (b) a quadratic model $R = c_0 + c_1 T + c_2 T^2$. Validate each.

(a) Linear. Least squares (Section 25.8) gives $c_0 = 101.362\ \Omega$, $c_1 = 0.367\,65\ \Omega/^\circ\mathrm{C}$. Then

$$\mathrm{SSE} = 6.242\ \Omega^2, \quad R^2 = 0.999\,69, \quad \mathrm{RMSE} = 0.833\ \Omega, \quad \chi^2_\nu = \frac{6.242/0.02^2}{7} = 2229.$$

R^2 looks superb. But the residuals are

$$-1.32,\ -0.34,\ +0.38,\ +0.78,\ +0.96,\ +0.82,\ +0.39,\ -0.34,\ -1.33\ \ \Omega,$$

a symmetric bow, up to sixty-five times the noise level, and $\chi^2_\nu \approx 2229$ is absurdly larger than 1. The linear model is **invalid** at this accuracy: it is missing curvature. As a temperature sensor it would err by up to $1.33/0.368 \approx 3.6\ ^\circ\mathrm{C}$.

(b) Quadratic. $c_0 = 100.034\ \Omega$, $c_1 = 0.390\,43\ \Omega/^\circ\mathrm{C}$, $c_2 = -5.694 \times 10^{-5}\ \Omega/^\circ\mathrm{C}^2$, with

$$\mathrm{SSE} = 1.15 \times 10^{-3}\ \Omega^2, \quad \mathrm{RMSE} = 0.011\ \Omega, \quad \chi^2_\nu = \frac{2.87}{6} = 0.48.$$

Residuals $(+0.006, -0.003, +0.003, -0.026, +0.009, +0.009, +0.013, -0.008, -0.004\ \Omega)$ show no pattern and are at the noise level. With $\nu = 6$, the 95 % range of χ^2_ν is roughly 0.21 to 2.41, so 0.48 is consistent with a correct model (the data were also rounded to 0.01 Ω, which slightly reduces the apparent noise).

Physics check. The fitted $c_1/c_0 = 3.903 \times 10^{-3}$ and $c_2/c_0 = -5.69 \times 10^{-7}$ per °C² are close to the Callendar–Van Dusen standard coefficients $A = 3.9083 \times 10^{-3}$, $B = -5.775 \times 10^{-7}$ — independent confirmation that the quadratic structure is physical, not just a better curve.

Residual tests for dynamic models

For time-series data, residuals must also be **uncorrelated in time**. The sample autocorrelation of the residuals,

$$\hat{r}(k) = \frac{\sum_{n=1}^{N-k} (e_n - \bar{e})(e_{n+k} - \bar{e})}{\sum_{n=1}^{N} (e_n - \bar{e})^2},$$

is approximately $\mathcal{N}(0, 1/N)$ for each $k \geq 1$ if the residuals are white (Ch. 21). Values outside $\pm 1.96/\sqrt{N}$ at more than about 5 % of lags indicate unmodelled dynamics. A complementary test is the cross-correlation between residuals and past inputs, which should also vanish if the model has captured the input–output dynamics.

Worked Example 25.23 — Detecting a missing time constant

Problem. A sensor's step response is sampled at 251 points over 5 s, with noise of standard deviation 0.01. Unknown to the modeller, the true response is second order with time constants 1.0 s and 0.2 s. Fit a first-order model $K(1 - e^{-t/\tau})$ and a second-order model, and compare residual autocorrelations.

First-order fit. Nonlinear least squares gives $K = 1.037$, $\tau = 1.334$ s, RMSE $= 0.030$ — only three times the noise, and a plot of model against data "looks good". But the residual autocorrelation is

$$\hat{r}(1) = 0.88, \quad \hat{r}(2) = 0.88, \quad \hat{r}(5) = 0.84, \quad \hat{r}(10) = 0.75,$$

against a whiteness bound $1.96/\sqrt{251} = 0.124$. The residuals are a slow, smooth curve — the signature of a missing mode. Note also that the fitted τ (1.33 s) is not either true time constant: an incorrect structure biases the parameters.

Second-order fit. $K = 0.999$, $\tau_1 = 0.991$ s, $\tau_2 = 0.205$ s, RMSE $= 0.0096$ (matching the noise), $\hat{r}(1) = -0.06$, $\hat{r}(2) = -0.04$, $\hat{r}(5) = -0.08$, $\hat{r}(10) = 0.02$: all within the bound. The model is accepted, and its parameters are now close to the true ones.

Overfitting and the need for independent data

Adding parameters always reduces SSE on the data used for fitting: a polynomial of degree $N - 1$ passes through all N points exactly. A model that fits the noise rather than the system is **overfitted**: it reproduces the calibration data beautifully and predicts new data badly. The expected prediction error of a model decomposes (for squared error) as

$$E\left[(y_{\text{new}} - \hat{y})^2\right] = \underbrace{\sigma^2}_{\text{irreducible noise}} + \underbrace{\left(\text{bias}\right)^2}_{\text{structural error}} + \underbrace{\text{variance}}_{\text{parameter uncertainty}},$$

the **bias–variance trade-off**. Too simple a model has large bias; too complex a model has large variance because its many parameters are poorly determined by finite data. The goal is the minimum of the sum, and it can only be found by testing on data that were not used for fitting.

Cross-validation

- **Hold-out validation:** split the data into a *training* set (for fitting) and a *validation* set (for evaluating prediction error).
- **k-fold cross-validation:** split the data into k parts; fit k times, each time leaving one part out and computing the prediction error on it; average the k errors.
- **Leave-one-out cross-validation (LOOCV):** $k = N$. For linear least squares it needs no refitting: with hat matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}^{\mathsf{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathsf{T}}$ and leverages h_{kk}, the leave-one-out residual is $e_k/(1 - h_{kk})$, and the **PRESS** statistic is

$$\mathrm{PRESS} = \sum_{k=1}^{N} \left(\frac{e_k}{1 - h_{kk}} \right)^2 .$$

Why the shortcut holds (sketch). Removing observation k changes the normal-equation matrix by a rank-one term $-\mathbf{x}_k\mathbf{x}_k^{\mathsf{T}}$; the Sherman–Morrison formula for the inverse of a rank-one update gives the new prediction at $\mathbf{x}_k$ in closed form, and simplifying yields $y_k - \hat{y}_{k,(-k)} = e_k/(1 - h_{kk})$. Points with high leverage (h_{kk} near 1, typically at the ends of the range) have their leave-one-out errors amplified the most, which is exactly where overfitted polynomials misbehave.

Worked Example 25.24 — Choosing the polynomial order by cross-validation

Problem. For the RTD data of Example 25.22, compare degrees 1, 2 and 3 by LOOCV.

Computation (PRESS via leverages):

Degree	p	Training RMSE (Ω)	LOOCV RMSE $\sqrt{\mathrm{PRESS}/N}$ (Ω)	AICc
1	2	0.833	1.192	— (structurally inadequate)
2	3	0.0113	**0.0161**	−69.9
3	4	0.0097	0.0175	−65.5

The training RMSE keeps falling as parameters are added, as it always must. The leave-one-out error is smallest for the quadratic: the cubic term reduces the training error by fitting noise, and that "improvement" does not survive prediction of withheld points. The small-sample Akaike criterion, $\mathrm{AICc} = N\ln(\mathrm{SSE}/N) + 2p + 2p(p+1)/(N-p-1)$, agrees. Both select the physically motivated quadratic model.

Common Pitfall — cross-validating time series by random shuffling

For dynamic data, neighbouring samples are correlated. Randomly assigning samples to folds lets the model "see" the neighbours of every validation point, and the cross-validation error is optimistically biased. Use **blocked** folds (contiguous segments of time), or better, validate on an entirely separate experiment with a different input signal.

Engineering Note — validate against the purpose

A model is validated *for a purpose*. A thermal model intended to predict peak junction temperature should be validated on the peak, under the worst-case load profile, not on the average temperature under a gentle one. Choose validation experiments that stress the conditions where the model will be used, and especially those where the assumptions of Section 25.6 are closest to failing.

25.8 Parameter identification

The identification problem

A grey-box model has a structure from physics and parameters from data. **Parameter identification** (estimation, calibration, fitting) is the inverse problem: given input–output measurements, find the parameter vector $\mathbf{p}$ that makes the model reproduce them.

Output-error parameter estimation

Given data $\{(u_k, y_k)\}_{k=1}^{N}$ and a model that predicts $\hat{y}_k(\mathbf{p})$, the **least-squares estimate** is

$$\hat{\mathbf{p}} = \arg\min_{\mathbf{p}} J(\mathbf{p}), \qquad J(\mathbf{p}) = \sum_{k=1}^{N} w_k \big(y_k - \hat{y}_k(\mathbf{p})\big)^2,$$

with weights $w_k = 1/\sigma_k^2$ when measurement uncertainties differ. If the errors are independent Gaussian, this is also the maximum-likelihood estimate (Ch. 20).

Three families of method are used in practice, in increasing generality: graphical/characteristic-point methods (read parameters off a step response), linear least squares (when the model is linear in the parameters) and nonlinear least squares (everything else). All of them must be followed by an identifiability check: are the data actually capable of determining the parameters?

Identification from a step response

The step response is the most common identification experiment in EE: switch on a heater, a motor voltage, a reference input, and record the output. Standard model structures have parameters that can be read from characteristic features.

First-order step response

For $\tau\dot{y} + y = Ku$ with a step of size Δu at $t = 0$ from equilibrium, $y(t) - y(0) = K\Delta u\left(1 - e^{-t/\tau}\right)$. Hence

- **gain:** $K = \Delta y_\infty / \Delta u$;
- **time constant:** τ is the time to reach $1 - e^{-1} = 63.2\,\%$ of the final change; equivalently, the initial slope is $K\Delta u/\tau$;
- **log-linear form:** $\ln\left(1 - \dfrac{y(t) - y(0)}{\Delta y_\infty}\right) = -\dfrac{t}{\tau}$, a straight line through the origin with slope $-1/\tau$.

The log-linear form is preferable to reading a single point, because it uses all the data and its straightness is itself a validation test: curvature at early times reveals a second, faster time constant or a dead time. For a first-order-plus-dead-time (FOPDT) model $Ke^{-\theta s}/(\tau s + 1)$, a classic two-point method uses the times t_{28} and t_{63} at which the response reaches 28.3 % and 63.2 %: $\tau = 1.5\,(t_{63} - t_{28})$, $\theta = t_{63} - \tau$ (these follow from $1 - e^{-1/3} = 0.283$ and $1 - e^{-1} = 0.632$).

Worked Example 25.25 — Identifying a first-order system from step data

Problem. The supply node of Example 25.1 is tested: after switch-on, its voltage settles at $v_\infty = 4.00$ V, and samples are

t (s)	0.02	0.05	0.10	0.15	0.20	0.30
v (V)	0.743	1.571	2.524	3.081	3.459	3.798

Estimate τ.

Single point. $0.632 \times 4.00 = 2.528$ V is reached at about $t = 0.10$ s, so $\tau \approx 0.10$ s.

Log-linear fit. Compute $z = \ln(1 - v/4.00)$: $-0.2055, -0.4988, -0.9970, -1.4708, -2.0006, -2.9858$. The model $z = -t/\tau$ is a line through the origin, so the least-squares slope is

$$\hat{m} = \frac{\sum t_k z_k}{\sum t_k^2} = -9.947\ \text{s}^{-1}, \qquad \hat{\tau} = -1/\hat{m} = 0.1005\ \text{s}.$$

The points lie on the line without systematic curvature (e.g. $z/t = -10.3, -9.98, -9.97, -9.81, -10.0, -9.95$), supporting the first-order structure. Combined with $R = 1\ \text{k}\Omega$ known, $C = \tau/R = 100.5\ \mu\text{F}$.

Caution. The log transform amplifies noise near the end of the transient, where $1 - v/v_\infty$ is small; points beyond about 3τ should be down-weighted or omitted, and an error in v_∞ bends the whole line. Nonlinear least squares on the untransformed data avoids both problems.

Underdamped second-order step response

For $y'' + 2\zeta\omega_n y' + \omega_n^2 y = \omega_n^2 K u$ with $0 < \zeta < 1$, the fractional overshoot M_p and the peak time t_p satisfy

$$M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}}, \qquad t_p = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}.$$

Inverting,

$$\zeta = \frac{-\ln M_p}{\sqrt{\pi^2 + \ln^2 M_p}}, \qquad \omega_n = \frac{\pi}{t_p\sqrt{1-\zeta^2}}.$$

For a free (or impulse) response with successive peaks x_k, x_{k+1} one period apart, the **logarithmic decrement** $\delta = \ln(x_k/x_{k+1}) = 2\pi\zeta/\sqrt{1-\zeta^2}$ gives $\zeta = \delta/\sqrt{4\pi^2 + \delta^2}$; using peaks m periods apart, $\delta = \frac{1}{m}\ln(x_k/x_{k+m})$, is more accurate.

Worked Example 25.26 — Identifying an RLC network from overshoot and peak time

Problem. A filter section modelled as a series RLC circuit with known $C = 1\ \mu\text{F}$ shows, in its capacitor-voltage step response, 25 % overshoot at $t_p = 1.2$ ms. Identify ζ, ω_n, L and R.

Damping. $\ln 0.25 = -1.3863$:

$$\zeta = \frac{1.3863}{\sqrt{9.8696 + 1.9218}} = \frac{1.3863}{3.4339} = 0.404.$$

Natural frequency. $\omega_d = \pi/t_p = 2618$ rad/s; $\sqrt{1-\zeta^2} = 0.9149$, so $\omega_n = 2862$ rad/s ($f_n = 455$ Hz).

Element values. From $\omega_n = 1/\sqrt{LC}$: $L = 1/(\omega_n^2 C) = 1/(8.19 \times 10^6 \times 10^{-6}) = 0.122$ H. From $\zeta = \frac{R}{2}\sqrt{C/L}$: $R = 2\zeta\sqrt{L/C} = 2(0.404)(349.5) = 282\ \Omega$.

Identifiability note. Without knowing C, the step response determines only ω_n and ζ — two numbers — and not the three element values. We return to this below.

Linear least squares

If the model output is linear in the parameters, $\hat{y}_k = \boldsymbol{\varphi}_k^{\mathsf{T}}\mathbf{p}$ (the **regressors** $\boldsymbol{\varphi}_k$ may be nonlinear functions of the inputs), stacking gives $\hat{\mathbf{y}} = \mathbf{X}\mathbf{p}$ with $\mathbf{X}$ the $N \times p$ regression matrix.

Least-squares estimate and its covariance

The minimizer of $\|\mathbf{y} - \mathbf{X}\mathbf{p}\|^2$ satisfies the **normal equations** $\mathbf{X}^{\mathsf{T}}\mathbf{X}\hat{\mathbf{p}} = \mathbf{X}^{\mathsf{T}}\mathbf{y}$; if $\mathbf{X}$ has full column rank,

$$\hat{\mathbf{p}} = (\mathbf{X}^{\mathsf{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathsf{T}}\mathbf{y}.$$

If $\mathbf{y} = \mathbf{X}\mathbf{p}_0 + \boldsymbol{\epsilon}$ with $E[\boldsymbol{\epsilon}] = \mathbf{0}$ and $\mathrm{Cov}(\boldsymbol{\epsilon}) = \sigma^2\mathbf{I}$, then $\hat{\mathbf{p}}$ is unbiased and

$$\mathrm{Cov}(\hat{\mathbf{p}}) = \sigma^2(\mathbf{X}^{\mathsf{T}}\mathbf{X})^{-1},$$

with σ^2 estimated by $s^2 = \mathrm{SSE}/(N-p)$. (Gauss–Markov: no other linear unbiased estimator has smaller variance.)

Derivation. $J = (\mathbf{y}-\mathbf{X}\mathbf{p})^{\mathsf{T}}(\mathbf{y}-\mathbf{X}\mathbf{p})$ has gradient $-2\mathbf{X}^{\mathsf{T}}(\mathbf{y}-\mathbf{X}\mathbf{p})$ (Ch. 6); setting it to zero gives the normal equations, and the Hessian $2\mathbf{X}^{\mathsf{T}}\mathbf{X}$ is positive definite under full rank, so the stationary point is the global minimum (Ch. 23). Substituting $\mathbf{y}$: $\hat{\mathbf{p}} - \mathbf{p}_0 = (\mathbf{X}^{\mathsf{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathsf{T}}\boldsymbol{\epsilon}$, whose mean is zero and whose covariance is $(\mathbf{X}^{\mathsf{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathsf{T}}(\sigma^2\mathbf{I})\mathbf{X}(\mathbf{X}^{\mathsf{T}}\mathbf{X})^{-1} = \sigma^2(\mathbf{X}^{\mathsf{T}}\mathbf{X})^{-1}$. $\square$

Numerically, the normal equations square the condition number of $\mathbf{X}$; QR factorization or the SVD (Ch. 22, Ch. 24) should be used for anything but small, well-conditioned problems.

Many nonlinear models become linear in transformed parameters. The diode equation, the Arrhenius law, power laws $y = ax^b$ and exponential decays all linearize under a logarithm. The transformation also transforms the noise: additive noise of constant variance on y becomes noise of variance σ^2/y^2 on $\ln y$, so an unweighted log-domain fit over-weights small values. Conversely, *multiplicative* (percentage) noise becomes additive with constant variance in the log domain — then the log fit is exactly right.

Worked Example 25.27 — Extracting diode parameters by linear least squares

Problem. A small-signal diode is measured at 300 K ($V_T = 25.85$ mV); the ammeter error is about 2 % of reading.

V (V)	0.40	0.45	0.50	0.55	0.60	0.65
I (A)	1.744×10^{-5}	5.186×10^{-5}	1.653×10^{-4}	4.737×10^{-4}	1.441×10^{-3}	4.479×10^{-3}

Estimate I_S and n with standard errors.

Linearize the model. For $I \gg I_S$, $\ln I = \ln I_S + V/(nV_T)$: linear in $p_1 = \ln I_S$ and $p_2 = 1/(nV_T)$, with regressors $\boldsymbol{\varphi}_k = [1,\ V_k]^{\mathsf{T}}$. Because the noise is a percentage of reading, it is additive with constant variance on $\ln I$, so an unweighted fit is appropriate.

Normal equations. $\sum 1 = 6$, $\sum V = 3.15$, $\sum V^2 = 1.6975$, giving $\mathbf{X}^{\mathsf{T}}\mathbf{X} = \begin{bmatrix} 6 & 3.15 \\ 3.15 & 1.6975 \end{bmatrix}$. Solving with $\mathbf{X}^{\mathsf{T}}\mathbf{y}$ from $y_k = \ln I_k$:

$$\hat{p}_1 = -19.820, \qquad \hat{p}_2 = 22.153\ \mathrm{V}^{-1}.$$

Parameters. $\hat{n} = 1/(\hat{p}_2 V_T) = 1/(22.153 \times 0.025\,85) = 1.746$; $\hat{I}_S = e^{-19.820} = 2.47\times10^{-9}$ A.

Uncertainty. Residual standard error $s = 0.0237$ (i.e. 2.4 % — consistent with the stated 2 % meter error). $\mathrm{Cov}(\hat{\mathbf{p}}) = s^2(\mathbf{X}^{\mathsf{T}}\mathbf{X})^{-1}$ gives standard errors 0.060 for $\hat{p}_1$ and 0.113 V^{-1} for $\hat{p}_2$. Propagating (Ch. 20): $\sigma_n = \sigma_{p_2}/(p_2^2 V_T) = 0.009$, and since $\sigma_{\ln I_S} = 0.060$, I_S is known to about 6 %. Result: $n = 1.746 \pm 0.009$, $I_S = (2.47 \pm 0.15)$ nA.

Correlation. The correlation coefficient between $\hat{p}_1$ and $\hat{p}_2$ is -0.987. The intercept $\ln I_S$ is an extrapolation to $V = 0$, far outside the data (0.40–0.65 V); any tilt of the line moves the intercept. This strong correlation is a mild form of the identifiability problem discussed next. Re-parameterizing around the data centre, $\ln I = \ln I(V_0) + (V - V_0)/(nV_T)$ with $V_0 = 0.525$ V, makes the two estimates uncorrelated.

Validation. At higher currents the series resistance R_s makes the curve bend below the straight line; residuals trending negative at the top of the range would be the signal to add R_s to the model ($V = nV_T\ln(I/I_S) + IR_s$, no longer linear in the parameters).

Nonlinear least squares

When the model is not linear in its parameters — a step response in τ, a resonance in ω_n and ζ, a diode with series resistance — $J(\mathbf{p})$ is minimized iteratively by the **Gauss–Newton** or **Levenberg–Marquardt** methods of Ch. 23. Linearizing the model about the current estimate $\mathbf{p}^{(m)}$,

$$\hat{\mathbf{y}}(\mathbf{p}) \approx \hat{\mathbf{y}}(\mathbf{p}^{(m)}) + \mathbf{S}\,\Delta\mathbf{p}, \qquad S_{ki} = \left.\frac{\partial \hat{y}_k}{\partial p_i}\right|_{\mathbf{p}^{(m)}},$$

turns each step into a linear least-squares problem for $\Delta\mathbf{p}$:

$$(\mathbf{S}^{\mathsf{T}}\mathbf{S} + \lambda\mathbf{I})\,\Delta\mathbf{p} = \mathbf{S}^{\mathsf{T}}\big(\mathbf{y} - \hat{\mathbf{y}}(\mathbf{p}^{(m)})\big),$$

with $\lambda = 0$ for Gauss–Newton and $\lambda > 0$ adapted for robustness in Levenberg–Marquardt. The **sensitivity matrix** $\mathbf{S}$ plays the role of $\mathbf{X}$; at convergence, $\mathrm{Cov}(\hat{\mathbf{p}}) \approx \sigma^2(\mathbf{S}^{\mathsf{T}}\mathbf{S})^{-1}$. Good starting values — often from the graphical methods above — are essential, because J may have several local minima.

Identifiability

Structural and practical identifiability

A parameter is **structurally identifiable** if, with perfect noise-free data from the chosen experiment, it can be determined uniquely (globally) or up to a finite set of values (locally). It is **practically identifiable** if it can be determined with acceptable precision from the finite, noisy data actually available.

Structural identifiability is a property of the *model and experiment*, independent of the data quality. The test for a linear model: compute the transfer function from the measured input to the measured output; its coefficients are all the data can ever reveal. Each parameter is identifiable only if it can be recovered from those coefficients.

Worked Example 25.28 — Which RC parameters can a step test identify?

Problem. In the circuit of Example 25.1, a voltage step drives R and C, and only the capacitor voltage is measured. Can R and C both be identified? What if the source current is measured too?

Voltage only. The transfer function is $V_C(s)/V_s(s) = 1/(RCs+1)$. Its single coefficient is RC. Every pair (R, C) with the same product gives *identical* data: R and C are individually **not structurally identifiable**; only $\tau = RC$ is. No amount of data or cleverness changes this.

Current as well. $I(s)/V_s(s) = Cs/(RCs+1)$. The initial current after the step is V_s/R, giving R; combined with τ, $C = \tau/R$. Both are now identifiable. The remedy for structural non-identifiability is a *different experiment* — another sensor, another input — not a better optimizer.

The same reasoning explains Example 25.26: the capacitor-voltage step response of a series RLC circuit has transfer function $1/(LCs^2 + RCs + 1)$, i.e. two coefficients for three unknowns, so one element value must be known independently.

Practical identifiability is diagnosed from the **Fisher information matrix** $\mathbf{F} = \mathbf{S}^{\mathsf{T}}\mathbf{S}/\sigma^2$, whose inverse approximates the covariance of the estimates (the Cramér–Rao bound). Warning signs are a large condition number of $\mathbf{F}$, correlation coefficients between parameter estimates near ± 1, and standard errors comparable to the parameters themselves.

Worked Example 25.29 — Practical identifiability depends on the experiment duration

Problem. A first-order step response $y = K(1 - e^{-t/\tau})$ with true $K = 1$, $\tau = 1$ s is sampled at 50 equally spaced points over a window $[0, W]$, with noise $\sigma = 0.01$. Compute the standard errors of $\hat{K}$ and $\hat{\tau}$ and their correlation for $W = 0.2$ s, 1 s and 5 s.

Sensitivities. $\partial y/\partial K = 1 - e^{-t/\tau}$, $\partial y/\partial \tau = -(Kt/\tau^2)e^{-t/\tau}$. Form $\mathbf{S}$ (50 × 2), $\mathbf{F} = \mathbf{S}^{\mathsf{T}}\mathbf{S}/\sigma^2$, and invert:

Window W	$\sigma_{\hat{K}}$	$\sigma_{\hat{\tau}}$ (s)	corr$(\hat{K}, \hat{\tau})$	cond$(\mathbf{F})$
0.2τ	0.63	0.68	0.9998	9 600
1τ	0.028	0.041	0.993	320
5τ	0.0026	0.0097	0.75	36

Interpretation. For $t \ll \tau$, $y \approx Kt/\tau$: the early response depends only on the *ratio* K/τ, so the two parameters are almost perfectly correlated and individually undetermined (63 % and 68 % uncertainty despite 1 % noise!). Recording until the response has settled separates them, since the final value fixes K alone. The model is structurally identifiable in all three cases; practically, only the long experiment identifies it.

Design rule. Plan identification experiments so that each parameter has a distinct, strong effect on the measured output somewhere in the record: run step tests to at least 4–5τ, excite all frequencies of interest (Ch. 21: a persistently exciting input, e.g. a pseudo-random binary sequence), and measure the variables that break structural ambiguities.

Discrete-time (ARX) identification

For sampled data, a powerful shortcut is to fit a difference equation directly. The first-order ARX model

$$y[n] = a\,y[n-1] + b\,u[n-1] + \epsilon[n]$$

is *linear* in (a, b), so linear least squares applies with regressors $\boldsymbol{\varphi}_n = [y[n-1],\ u[n-1]]^{\mathsf{T}}$. The continuous parameters follow from the ZOH relations $a = e^{-T/\tau}$, $b = K(1-a)$: $\hat{\tau} = -T/\ln \hat{a}$, $\hat{K} = \hat{b}/(1-\hat{a})$. Because past noisy outputs appear as regressors, the estimate is biased when the measurement noise is not white in the equation-error sense; instrumental-variable or output-error methods remove the bias. Higher-order ARX, ARMAX and subspace methods generalize the idea and are the workhorses of black- and grey-box system identification.

Common Pitfall — fitting parameters that the model cannot carry

If the residuals of Section 25.7 show structure, fitted parameters are biased in ways that cannot be fixed by more data (Example 25.23: the first-order fit gave $\tau = 1.33$ s, matching neither true time constant). Parameters are only meaningful once the structure has been validated; identification and validation must iterate together.

25.9 Analytical versus numerical solutions

Two ways to answer the same question

Once a model is formulated, scaled, simplified and calibrated, it must be solved. There are two broad routes. An **analytical** (closed-form) solution expresses the answer as a formula in the parameters — $v(t) = v_\infty(1 - e^{-t/RC})$, $\delta = \sqrt{2/(\omega\mu\sigma)}$, $r_d = nV_T/I_D$. A **numerical** solution computes the answer for specific parameter values by an algorithm — an ODE integrator, a finite-element solver, Newton iteration, a Monte Carlo simulation (Ch. 22, Ch. 24). Neither is universally better. The choice is a modeling decision, made with the purpose in mind.

Criterion	Analytical	Numerical
Scope	linear, low-order, simple geometry, special nonlinearities	essentially unrestricted: nonlinear, high-order, complex geometry

Criterion	Analytical	Numerical
Parameter dependence	explicit — sensitivities, scaling laws and design rules read off directly	one parameter set per run; trends need sweeps
Insight	high: shows *why* and which groups matter	low unless combined with scaling and visualization
Accuracy	exact for the model (but the model may be oversimplified to permit it)	approximate: discretization, round-off, convergence tolerances
Cost	cheap to evaluate once found; finding it may be impossible	cheap to set up; can be expensive to run (stiff, 3-D, many runs)
Failure modes	algebra errors; hidden assumptions made "to get a formula"	instability, non-convergence, under-resolved meshes, silently wrong answers
Verification	check limits, dimensions, special cases	compare with analytical cases, refine steps, check conservation

The most effective practice combines both: use analysis to understand structure, find scales, derive limiting cases and check codes; use computation to handle the full model.

When a closed form exists — and when it does not

Closed forms are available for linear constant-coefficient ODEs and PDEs on simple domains (Chs. 11–16), for some nonlinear first-order equations (separable, Bernoulli, Riccati with a known solution), and for special nonlinear problems whose solutions are **special functions** (elliptic integrals for the pendulum, Bessel functions for cylindrical problems, the Lambert W function for diode circuits). Outside these islands — nonlinear circuits of order two or more, coupled multi-physics, irregular geometry — numerical methods are the only route.

Between the two lie **semi-analytical** methods: perturbation expansions in a small parameter (Section 25.4), averaging, describing functions (Section 25.5), and series or transform solutions evaluated numerically. They often give the best of both worlds — formulas that are accurate in a stated regime.

Worked Example 25.30 — A diode circuit: special function versus root-finding

Problem. A source $V_s = 5$ V drives the diode of Example 25.27 ($I_S = 2.47$ nA, $n = 1.746$) through $R = 1$ kΩ. Find the operating point (a) numerically, (b) in closed form.

Equation. KCL: $\dfrac{V_s - v}{R} = I_S\left(e^{v/(nV_T)} - 1\right)$, a transcendental equation with no solution in elementary functions.

(a) Numerical. Bracketing on $[0, 1]$ V and applying Brent's method (or Newton's method, Ch. 22) gives $v = 0.6492$ V, $I = 4.351$ mA in a few iterations.

(b) Closed form. Rearranging to the form $we^{w} = z$ gives, with $a = V_s + RI_S$,

$$v = a - nV_T\, W\!\left(\frac{RI_S}{nV_T}\, e^{a/(nV_T)}\right),$$

where W is the Lambert W function (the inverse of we^{w}). Evaluating gives $v = 0.6492$ V, identical.

Which is better? For one operating point, either. The closed form, however, shows the structure at a glance — v depends on $RI_S/(nV_T)$ and $V_s/(nV_T)$ only, two dimensionless groups — and can be differentiated to obtain sensitivities in closed form. In practice W is itself evaluated by a Newton iteration inside a library, so the "analytical" answer is numerical one level down. The real distinction is not formula versus algorithm, but whether the parameter dependence is *explicit*.

Numerical solutions must be verified

A numerical solution is an approximation whose error must be controlled. Analytical solutions of simplified cases are the main verification tool, and analytical insight (eigenvalues, time scales) predicts the step sizes a numerical method needs.

Stability of forward Euler for a linear system

Applied to $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ with step h, forward Euler gives $\mathbf{x}_{k+1} = (\mathbf{I} + h\mathbf{A})\mathbf{x}_k$, which is stable if and only if $|1 + h\lambda_i| < 1$ for every eigenvalue λ_i of $\mathbf{A}$ (for diagonalizable $\mathbf{A}$; Ch. 22). For $\lambda = -\sigma \pm j\omega_d$, this requires

$$h < \frac{2\sigma}{\sigma^2 + \omega_d^2} = \frac{2\zeta}{\omega_n}.$$

Worked Example 25.31 — Verifying an integrator on the RLC circuit

Problem. Simulate the step response of the RLC circuit of Example 25.4 ($\omega_n = 31\,623$ rad/s, $\zeta = 0.158$) to $t = 1$ ms by forward Euler and by fourth-order Runge–Kutta, and compare with the exact solution $v_C(1\text{ ms}) = 0.993\,59$ V for a 1 V step.

Predicted stability limit. $h_{\max} = 2\zeta/\omega_n = 2(0.158)/31\,623 = 10.0\ \mu$s — even though the ringing period is 201 μs.

Results.

Method	h (μs)	$v_C(1\text{ ms})$ (V)	error (V)
Forward Euler	1	0.9892	4.4×10^{-3}
Forward Euler	2	0.9821	1.1×10^{-2}
Forward Euler	5	0.9208	7.3×10^{-2}
Forward Euler	10	0.0039	≈ 1 (no decay: at the stability limit)
Forward Euler	12	−1.698	diverging
RK4	10	0.993 59	3.4×10^{-6}

Interpretation. The Euler error roughly doubles when h doubles (first-order convergence) until the stability limit predicted analytically is reached, where the answer becomes meaningless. RK4 at $h = 10\ \mu$s is a thousand times more accurate than Euler at 1 μs. Without the exact solution to compare against, the $h = 5\ \mu$s Euler answer (7 % low) might well have been accepted. The habit to cultivate: *before trusting a simulation of the full model, run it on a special case with a known answer*, and *refine the step until the answer stops changing*.

Stiffness: when analysis tells you which solver to use

Stiff system

A system is **stiff** when its eigenvalues (time constants) span many orders of magnitude and the behaviour of interest is governed by the slow ones. Explicit methods are then forced, by stability rather than accuracy, to use steps comparable to the *fastest* time constant over a horizon set by the *slowest*.

Worked Example 25.32 — A stiff RLC circuit

Problem. A series circuit has $R = 1$ kΩ, a parasitic inductance $L = 1\ \mu$H, and $C = 1\ \mu$F. Characterise its time scales and choose a solver for a 5 ms simulation.

Scaling (Section 25.4). $\zeta = \frac{R}{2}\sqrt{C/L} = 500$, so $\varepsilon = 1/(4\zeta^2) = 10^{-6}$. The eigenvalues are the roots of $\lambda^2 + 10^9\lambda + 10^{12} = 0$: $\lambda_1 \approx -10^3$ s^{-1} ($RC = 1$ ms) and $\lambda_2 \approx -10^9$ s^{-1} ($L/R = 1$ ns). The stiffness ratio is 10^6.

Explicit solver. Forward Euler needs $h < 2/10^9 = 2$ ns for stability, i.e. at least 2.5×10^6 steps for 5 ms — all spent resolving a nanosecond transient that ended long ago.

Implicit solver. Backward Euler, the trapezoidal rule or BDF methods (Ch. 22) are stable for any step on this problem; accuracy on the slow mode requires only $h \sim 10$–$50\ \mu$s, about 100–500 steps. Circuit simulators use implicit methods for precisely this reason.

Or remove the stiffness analytically. Since $\varepsilon \ll 1$ and the purpose concerns the millisecond behaviour, the reduced model $RC\dot{v} + v = v_s$ (Section 25.4) is accurate to about ε after the first few nanoseconds, and has a closed-form solution. Analysis has not only chosen the solver; it has made the solver unnecessary.

Common Pitfall — believing the simulation because it is detailed

A detailed numerical model inspires confidence it may not deserve. Mesh too coarse, step too long, tolerance too loose, a unit typed as mV instead of V, a boundary condition on the wrong face — all produce smooth, plausible, wrong results. The analytical checks of this chapter (dimensions, limiting cases, scaling laws, conservation, special-case solutions) are the defence. A simulation that has never been compared with anything is a hypothesis, not a result.

Engineering Note — a practical decision rule

1. Always derive what can be derived cheaply: equilibria, scales, dimensionless groups, linearized eigenvalues, limiting cases.
2. If the question can be answered from those, stop — the formula is the best deliverable.
3. Otherwise, solve numerically, using the analytical results to choose the method and step, to verify the code, and to interpret the output in terms of the dimensionless groups.
4. Present numerical results in scaled form wherever possible, so that one set of curves covers a family of designs.

Synthesis: the book as a modeling toolkit

This chapter has been, in effect, a map of how the rest of the book is used. Every method in the earlier chapters enters the modeling cycle at a particular stage:

Stage of modeling	Methods from this book
Formulating relations	algebra, functions and dimensional consistency (Ch. 1); complex numbers and phasors (Ch. 2–3); derivatives as rates (Ch. 4); integrals as accumulation (Ch. 5); vector calculus for fields and balance laws (Ch. 7)

Stage of modeling	Methods from this book
Organizing the equations	matrices and linear systems (Ch. 8); vector spaces, null spaces and rank — the heart of the Pi theorem (Ch. 9); ODEs and PDEs as the language of dynamics and distributed systems (Ch. 11–13)
Simplifying	Taylor expansion and Jacobians for linearization (Ch. 4, 6); eigenvalues for modes, stability and time-scale separation (Ch. 10); scaling and dimensional analysis (Ch. 1, 25)
Solving	Laplace, Fourier and Z-transforms for linear time-invariant models (Ch. 14–17); complex analysis for inversion and stability (Ch. 18); numerical methods, optimization and computational mathematics when closed forms run out (Ch. 22–24)
Confronting data	probability, statistics and random processes for noise, residuals, estimation, identification and uncertainty (Ch. 19–21); least squares and optimization for calibration (Ch. 20, 23)

Three ideas recur throughout. **Linearity** — superposition, transforms, eigen-decomposition — is the great simplifier, and linearization is how nonlinear reality is brought within its reach. **Dimensionless thinking** — scales, groups, regimes of validity — tells us which effects matter and when a simplification is legitimate. **Comparison with data**, through residuals, likelihoods and cross-validation, is what turns a mathematical exercise into an engineering model. The mathematics of this book is not a collection of separate techniques but a single toolkit, and modeling is the craft of choosing the right tool for each stage of the question.

Summary

Topic	Key formula or rule
Modeling cycle	purpose → concept → equations → solution → validation → use, iterated
Balance law	$\frac{d}{dt}(\text{stored}) = \text{in} - \text{out} + \text{generated}$; constitutive relations close the system
State-space model	$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}; \mathbf{p}),\ \mathbf{y} = \mathbf{h}(\mathbf{x}, \mathbf{u}; \mathbf{p})$
LTI transfer function	$\mathbf{G}(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}$; invariant under $\mathbf{z} = \mathbf{T}\mathbf{x}$
Circuit order	$n = n_C + n_L - n_{\text{C-loops}} - n_{\text{L-cutsets}}$
Buckingham Pi	n quantities, dimension matrix rank $r \Rightarrow n - r$ groups; exponents span null$(\mathbf{D})$
Skin depth	$\delta = \sqrt{2/(\omega\mu\sigma)}$ (form from dimensions alone)
Short dipole	$P = \eta_0 I_0^2\, \Phi(k\ell),\ \Phi = (k\ell)^2/(12\pi)$ for $k\ell \ll 1$
Convection	$\mathrm{Nu} = h\ell/k = \Phi(\mathrm{Re}, \mathrm{Pr}),\ \mathrm{Re} = \rho V \ell/\mu,\ \mathrm{Pr} = \mu c_p/k$
Second-order standard form	$y'' + 2\zeta y' + y = u,\ \tau = \omega_n t,\ \omega_n = 1/\sqrt{LC},\ \zeta = \frac{R}{2}\sqrt{C/L} = 1/(2Q)$
Step-response metrics	$M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}},\ t_p = \pi/(\omega_n\sqrt{1-\zeta^2}),\ t_s \approx 4/(\zeta\omega_n)$
Biot and Fourier numbers	$\mathrm{Bi} = h\ell/k$ (lumped if < 0.1), $\mathrm{Fo} = \alpha t/\ell^2$
Singular perturbation (RLC)	$\varepsilon y'' + y' + y = u,\ \varepsilon = L/(R^2C) = 1/(4\zeta^2)$
Van der Pol scaling	$x'' - \mu(1 - x^2)x' + x = 0,\ \mu = a\sqrt{L/C},\ v = x\sqrt{a/(3b)}$
Linearization	$\mathbf{A} = \partial\mathbf{f}/\partial\mathbf{x},\ \mathbf{B} = \partial\mathbf{f}/\partial\mathbf{u}$ at $(\bar{\mathbf{x}}, \bar{\mathbf{u}})$; stability from eigenvalues unless on the $j\omega$-axis
Diode small signal	$r_d = nV_T/I_D$; relative error $\approx \lvert\delta v\rvert/(2nV_T)$; $\mathrm{HD}_2 \approx \hat{V}/(4nV_T)$
Pendulum	$T/T_0 = \frac{2}{\pi}K(\sin\frac{\theta_0}{2}) \approx 1 + \theta_0^2/16$
Lumped-circuit validity	$\ell < \lambda/10$, or $t_r \gtrsim 6\, t_{pd}$
Ideal op-amp validity	$\lvert A\beta\rvert \gg 1$; closed-loop bandwidth $\approx f_{\mathrm{GBW}}\beta$
Capacitor self-resonance	$f_{\mathrm{SRF}} = 1/(2\pi\sqrt{L_{\mathrm{ESL}}C})$
Goodness of fit	$R^2 = 1 - \mathrm{SSE}/\mathrm{SST},\ s = \sqrt{\mathrm{SSE}/(N-p)},\ \chi^2_\nu = \sum(e_k/\sigma_k)^2/(N-p) \approx 1$
Residual whiteness	$\lvert\hat{r}(k)\rvert < 1.96/\sqrt{N}$ for most lags
LOOCV (linear LS)	$\mathrm{PRESS} = \sum\big(e_k/(1-h_{kk})\big)^2,\ \mathbf{H} = \mathbf{X}(\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}\mathbf{X}^\mathsf{T}$
First-order identification	$K = \Delta y_\infty/\Delta u$; τ at 63.2 %; $\ln(1 - \Delta y/\Delta y_\infty) = -t/\tau$
Second-order identification	$\zeta = -\ln M_p/\sqrt{\pi^2 + \ln^2 M_p};\ \omega_n = \pi/(t_p\sqrt{1-\zeta^2});\ \zeta = \delta/\sqrt{4\pi^2 + \delta^2}$
Linear least squares	$\hat{\mathbf{p}} = (\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}\mathbf{X}^\mathsf{T}\mathbf{y},\ \mathrm{Cov}(\hat{\mathbf{p}}) = \sigma^2(\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}$
Nonlinear LS step	$(\mathbf{S}^\mathsf{T}\mathbf{S} + \lambda\mathbf{I})\Delta\mathbf{p} = \mathbf{S}^\mathsf{T}(\mathbf{y} - \hat{\mathbf{y}})$
Identifiability	structural: recoverable from transfer-function coefficients; practical: $\mathbf{F} = \mathbf{S}^\mathsf{T}\mathbf{S}/\sigma^2$ well conditioned
ARX first order	$y[n] = ay[n-1] + bu[n-1];\ \tau = -T/\ln a,\ K = b/(1-a)$
Forward Euler stability	$\lvert 1 + h\lambda\rvert < 1$; for $\lambda = -\zeta\omega_n \pm j\omega_d$: $h < 2\zeta/\omega_n$
Stiffness ratio	$\lvert\lambda_{\max}\rvert/\lvert\lambda_{\min}\rvert \gg 1 \Rightarrow$ implicit solver or reduced model

Further reading

- E. A. Bender, *An Introduction to Mathematical Modeling*, Dover, 2000 — short, readable treatment of the modeling cycle, dimensional analysis and model simplification.
- C. C. Lin and L. A. Segel, *Mathematics Applied to Deterministic Problems in the Natural Sciences*, SIAM Classics, 1988 — the standard reference on scaling, nondimensionalization and perturbation methods.

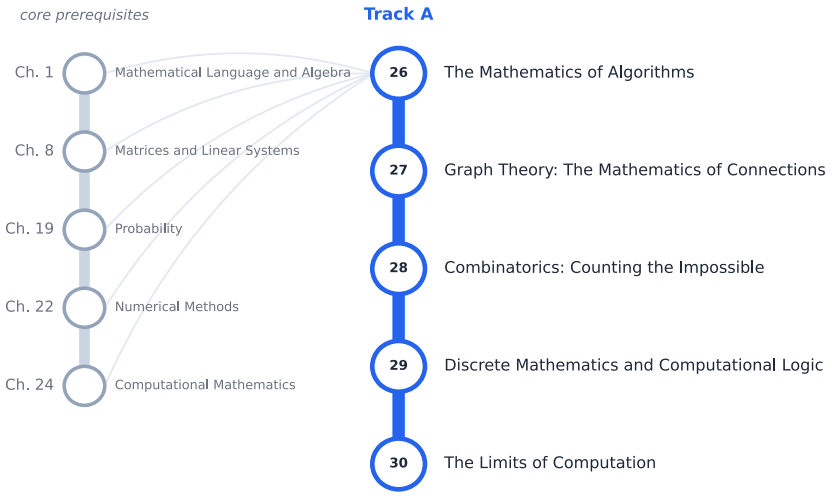
- G. I. Barenblatt, *Scaling*, Cambridge University Press, 2003 — dimensional analysis and similarity in depth, including the Pi theorem and its limitations.
- L. Ljung, *System Identification: Theory for the User*, 2nd ed., Prentice Hall, 1999 — parameter estimation, model validation, identifiability and ARX/output-error methods for dynamic systems.
- K. J. Åström and R. M. Murray, *Feedback Systems: An Introduction for Scientists and Engineers*, 2nd ed., Princeton University Press, 2021 — state-space modeling, linearization and the use of models in control (freely available online).

Level 2 — Specializations

Tracks A–I. Each track has its own prerequisite graph; read the ones you need, in any order.

Track A — The Mathematics Behind Algorithms

> **Level 2 — Specializations.** How to make programs faster, prove they work, and understand the limits of computation.



Suggested reading order for Track A. The grey stations on the left are the core chapters it leans on.

Before you start

This track assumes you are comfortable with these core chapters:

- Ch. 1 — Mathematical Language and Algebra
- Ch. 8 — Matrices and Linear Systems
- Ch. 19 — Probability
- Ch. 22 — Numerical Methods
- Ch. 24 — Computational Mathematics

Chapters

Ch.	Chapter	What it explores
26	The Mathematics of Algorithms	Asymptotic analysis, Big-O/Theta/Omega, recurrences, amortized analysis, algorithmic lower bounds, and complexity trade-offs.
27	Graph Theory: The Mathematics of Connections	Paths, trees, flows, matchings, connectivity, graph coloring, planar graphs, network centrality, spectral graph theory.
28	Combinatorics: Counting the Impossible	Permutations, combinations, inclusion–exclusion, recurrence relations, generating functions, pigeonhole principle, Ramsey theory, extremal combinatorics.
29	Discrete Mathematics and Computational Logic	Propositional and predicate logic, induction, relations, sets, discrete structures, Boolean algebra, invariants, proof techniques.
30	The Limits of Computation	Automata, formal languages, Turing machines, decidability, computability, reductions, P versus NP, and undecidable problems.

Every chapter follows the same arc: **the hook** (a puzzle or a failure), **the big idea**, **the mathematics**, **the algorithm**, **the experiment** you can run, **the challenge**, and **loose threads**: the questions the chapter deliberately leaves open.

Chapter 26 — The Mathematics of Algorithms

> **Level 2 · Track A — The Mathematics Behind Algorithms Builds on:** Ch. 1, Ch. 17, Ch. 24 · **Leads to:** Ch. 27, Ch. 28, Ch. 30 · **Related:** Ch. 52, Ch. 32 **Reading time:** ~25 min

26.1 The hook

Two engineers are asked to write the same routine. It takes a list of n items (sensors to switch on, jobs to schedule, wires to route) and picks the best combination.

Engineer X writes the obvious thing: try every subset of the n items, score each one, keep the best. There are 2^n subsets, and each one is scored in about one elementary step. Engineer Y writes something cleverer, full of tables and bookkeeping, that needs about $50\,n^2$ elementary steps.

They race the two programs on the test suite, where n runs from 5 to 12. X wins every single test. At $n = 10$, X takes $2^{10} = 1024$ steps and Y takes $50 \times 100 = 5000$: the "naive" code is almost five times faster. Y's code is longer, harder to read, and slower. The team ships X.

Six months later a customer runs it with $n = 40$. On a machine doing 10^9 steps per second, X needs $2^{40} \approx 1.1 \times 10^{12}$ steps, about **18 minutes**. Y would have needed $50 \times 1600 = 80{,}000$ steps: 80 microseconds. A second customer tries $n = 60$. X now needs $2^{60} \approx 1.15 \times 10^{18}$ steps, about **36 years**. Y: 180,000 steps, under a fifth of a millisecond.

Nothing was wrong with the test suite. Every measurement was correct. The machine did not get slower. And yet a program that was five times *faster* became trillions of times *slower*, just because the input grew from 10 to 60.

> **The challenge of this chapter:** Why can an algorithm that is faster on small inputs become terrible on large ones? And can you tell, *before* the customer does, which one it will be?

The answer is not "computers are mysterious". It is a theorem, and a short one. By the end of this chapter you will be able to prove it, predict exactly where the crossover happens, and show that no amount of hardware or clever coding can rescue the exponential.

26.2 The big idea

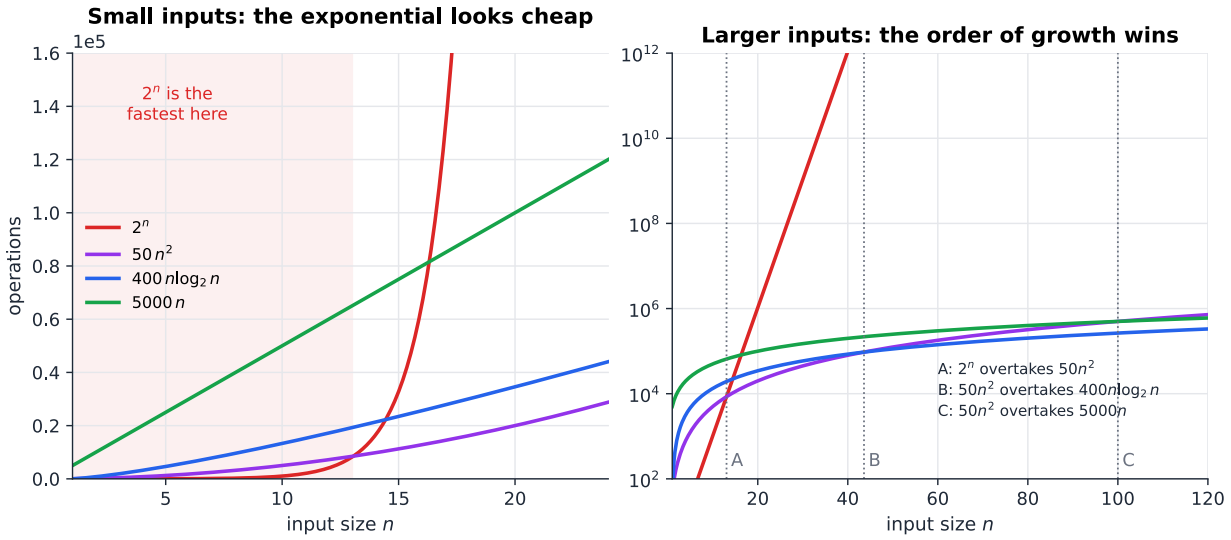
Forget seconds. An algorithm's running time is a **function of the input size**, $T(n)$, and what matters for large inputs is the *shape* of that function, not its value at any particular n.

Here is the most useful single experiment in algorithm analysis: **double the input and see what happens to the cost.**

Cost $T(n)$	$T(2n)$	Doubling the input…
n	$2n$	doubles the cost
$n \log_2 n$	$2n \log_2 n + 2n$	slightly more than doubles it
n^2	$4n^2$	quadruples it
n^3	$8n^3$	multiplies it by 8
2^n	$2^{2n} = (2^n)^2$	**squares** it

That last row is the whole story of the hook. For a polynomial cost, doubling n multiplies the cost by a *fixed* factor (2^k for n^k). For an exponential cost, doubling n *squares* the cost. A thousand steps becomes a million; a million becomes a trillion. No fixed factor can keep up with squaring.

An analogy: a polynomial algorithm is a salary with a raise; an exponential algorithm is compound interest. Over the first few years the big salary looks better. Over a lifetime compound interest wins, and the only question is *when*.



Costs with very different constants. On small inputs (left) the exponential is the cheapest of all four; on a logarithmic scale (right) it is a straight line climbing away from every polynomial. The polynomials reorder too: crossovers A, B, C are where a slower-growing cost finally pays off its larger constant.

Look at the right panel carefully, because it hides a second lesson. The polynomial curves cross each other too. $5000\,n$ has a large constant and is the *worst* of the three polynomials until $n = 100$, after which it beats $50\,n^2$ forever. Every pair of growth rates has a crossover point. Below it the constants decide; above it the **order of growth** decides, and it decides permanently.

So the answer to the challenge question is already visible: small inputs measure constants, large inputs measure growth rates, and the two can disagree. The mathematics will tell us *how far* the crossover can be pushed, and the answer is shockingly small.

26.3 The mathematics

Asymptotic notation, in one paragraph

The definitions of O, Ω and Θ are in Ch. 24, §24.1: $f = O(g)$ means $f(n) \le c\,g(n)$ for all large n; Ω is the reverse inequality; Θ is both. We add one tool and one symbol.

Little-o and the limit test

$f = o(g)$ ("f grows strictly slower than g") means $f(n)/g(n) \to 0$ as $n \to \infty$. If $\lim_{n\to\infty} f(n)/g(n) = L$ exists, then

- $L = 0$ gives $f = o(g)$, hence $f = O(g)$ but not $\Theta(g)$;
- $0 < L < \infty$ gives $f = \Theta(g)$;
- $L = \infty$ gives $g = o(f)$.

The limit test turns "which algorithm scales better?" into a calculus question, and L'Hôpital's rule (Ch. 4) often answers it in one line. For example, $\ln n / n^{0.01} \to 0$: even the feeblest power of n eventually beats a logarithm.

Result 1: every exponential beats every polynomial

Polynomials lose to exponentials

For every fixed $k \ge 0$ and every base $b > 1$,

$$\lim_{n\to\infty} \frac{n^k}{b^n} = 0.$$

Consequently, for any constants $c_1, c_2 > 0$ there is an N such that $c_1\,n^k < c_2\,b^n$ for all $n > N$.

Proof. Let $a_n = n^k / b^n$. The ratio of consecutive terms is

$$\frac{a_{n+1}}{a_n} = \left(1 + \frac{1}{n}\right)^k \frac{1}{b} \;\longrightarrow\; \frac{1}{b} < 1.$$

Pick any q with $1/b < q < 1$. Because the ratio converges to $1/b$, there is an M with $a_{n+1} \le q\,a_n$ for all $n \ge M$. Then $a_n \le a_M\,q^{n-M}$ for $n \ge M$, and $q^{n-M} \to 0$ because $0 < q < 1$. So $a_n \to 0$.

For the second claim, $c_1 n^k / (c_2 b^n) = (c_1/c_2)\,a_n \to 0$, so it is eventually below 1. ∎

That proves the hook's disaster was inevitable. But it does not yet say *when*, and that is where the real surprise is.

How far can a constant push the crossover?

Suppose the exponential algorithm is made c times cheaper than a quadratic one: compare 2^n against $c\,n^2$. Take $\log_2$ of both sides. The exponential is still the cheaper one exactly when

$$n \;<\; \log_2 c + 2\log_2 n.$$

The constant enters only through $\log_2 c$. Every factor of two you squeeze out of the exponential algorithm (a faster chip, better code, a parallel cluster of two machines) buys you **one** more input item before the crossover. A factor of a million buys about $\log_2 10^6 \approx 20$.

With the hook's numbers, $c = 50$ gives a crossover at $n \approx 13.1$. Make the exponential code a million times faster ($c = 5 \times 10^7$) and the crossover moves to $n \approx 35.9$. Twenty-three more items, bought with a millionfold speed-up. Go to $c = 5 \times 10^9$ and you reach $n \approx 43$. The customer with $n = 60$ is out of reach of every computer that will ever be built.

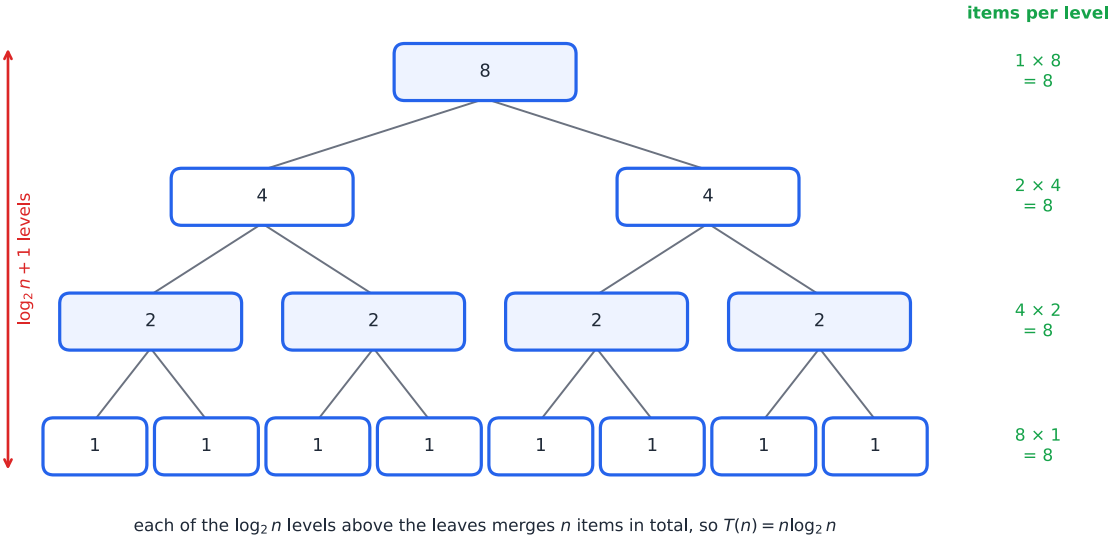
> **The engineering rule**
>
> Hardware improvements multiply your speed, which **adds** to the feasible n of an exponential algorithm and **multiplies** the feasible n of a polynomial one. Moore's law makes an $O(n^2)$ code handle $\sqrt{2}$ times larger inputs every doubling; it makes an $O(2^n)$ code handle exactly one more item.

Result 2: solving a recurrence with a recursion tree

Divide-and-conquer algorithms are analysed by **recurrences**. Merge sort splits a list in half, sorts each half recursively, and merges the two sorted halves in at most n comparisons:

$$T(n) = 2\,T(n/2) + n, \qquad T(1) = 0.$$

A recurrence is a difference equation in disguise. Ch. 17 solves linear difference equations with the Z-transform; here the index *halves* instead of decreasing by one, so we draw the recursion instead.



The recursion tree of merge sort for $n = 8$. Each level splits the work into more, smaller pieces, but the total per level stays n. The height is the number of halvings, $\log_2 n$.

> **Merge-sort recurrence**
>
> For $n = 2^m$, the recurrence $T(n) = 2T(n/2) + n$, $T(1) = 0$, has the exact solution $T(n) = n\log_2 n$.

Proof by induction on m. For $m = 0$: $T(1) = 0 = 1 \cdot \log_2 1$. Assume $T(2^m) = 2^m m$. Then

$$T(2^{m+1}) = 2\,T(2^m) + 2^{m+1} = 2 \cdot 2^m m + 2^{m+1} = 2^{m+1}(m+1),$$

which is $n\log_2 n$ for $n = 2^{m+1}$. ■

The tree shows *why*: at depth i there are 2^i subproblems of size $n/2^i$, so each level costs $2^i \cdot n/2^i = n$, and there are $\log_2 n$ levels that do merging. The general version, $T(n) = a\,T(n/b) + \Theta(n^d)$, is the master theorem stated in Ch. 24, §24.1. The recursion tree explains its three cases in one sentence: the work per level forms a geometric series with ratio a/b^d. If the ratio is below 1 the root dominates, $\Theta(n^d)$; if it equals 1 every level counts equally and you pay a $\log n$ factor; if it is above 1 the leaves dominate, $\Theta(n^{\log_b a})$.

> **Why the exponent of matrix multiplication is not 3**
>
> Multiplying two $n \times n$ matrices by splitting into 2×2 blocks needs 8 block products: $T(n) = 8T(n/2) + \Theta(n^2)$, so $\Theta(n^{\log_2 8}) = \Theta(n^3)$. Strassen found a way with 7 block products. The ratio $7/2^2 > 1$ puts us in the leaf-dominated case: $\Theta(n^{\log_2 7}) = \Theta(n^{2.807})$. Saving one multiplication out of eight at the top changes the exponent, because the saving compounds at every level.

? The exponent nobody knows

Strassen's trick was improved again and again, and as of 2024 the best known exponent for matrix multiplication is about 2.371. The obvious lower bound is 2 (you have to read n^2 entries). Is the true exponent ω equal to 2? Nobody knows. It is one of the great open problems of algorithm theory, and how mathematicians attack problems like it is the subject of Ch. 98.

Result 3: amortized analysis — the cheap average of an expensive worst case

A dynamic array (Python's `list`, C++'s `vector`) stores items in a fixed block of memory. When the block is full, the next append allocates a block **twice as large** and copies everything over. A single append can therefore cost n copies. Is appending $O(n)$?

Doubling makes appends $O(1)$ amortized

Starting from capacity 1 and doubling when full, any sequence of n appends performs fewer than $3n$ elementary writes in total. The **amortized** cost per append is therefore $O(1)$, even though a single append can cost $\Theta(n)$.

Proof. Each append writes its own item once: n writes. Copies happen when the size reaches $1, 2, 4, \dots, 2^k$ where $2^k < n$, costing $1 + 2 + 4 + \cdots + 2^k = 2^{k+1} - 1 < 2n$ copies. Total: fewer than $n + 2n = 3n$. ∎

Amortization is a statement about *sequences* of operations, not single ones. Here is the trap in the other direction: grow the block by a fixed $+100$ slots instead of doubling, and the copies cost $100 + 200 + \cdots \approx n^2/200$, which is $\Theta(n^2)$ in total. Both versions are instant in a unit test with 1000 items. Only one survives ten million.

The proof above simply added the costs (the **aggregate method**). The more powerful **potential method** stores "prepaid credit" Φ in the data structure: each cheap append pays 3 units, uses 1, and banks 2 against the next expensive copy. It is the same accounting idea an engineer uses for energy in a capacitor: cheap steps charge it, the expensive step discharges it, and the books balance over any window.

Result 4: a lower bound — no sorting algorithm can beat $n \log n$

Big-O answers "how fast is *this* algorithm?". A **lower bound** answers a far bolder question: "how fast can *any* algorithm possibly be?". Lower bounds are rare and precious, because they speak about programs nobody has written yet.

The comparison-sorting lower bound

Any algorithm that sorts n distinct items using only pairwise comparisons ("is $x_i < x_j$?") needs at least $\lceil \log_2 n! \rceil$ comparisons in the worst case, and $\log_2 n! = \Theta(n \log n)$.

Proof. Model any such algorithm as a **decision tree**: each internal node is a comparison, each branch is its outcome, and each leaf is a final answer ("the sorted order is $x_3 < x_1 < x_2$"). Different input orderings must reach different leaves, otherwise the algorithm gives the same answer to two inputs that need different answers. There are $n!$ orderings, so the tree has at least $n!$ leaves. A binary tree of height h has at most 2^h leaves, so $2^h \geq n!$, that is $h \geq \log_2 n!$. The height is the worst-case number of comparisons.

For the growth rate: the largest $n/2$ factors of $n!$ are each at least $n/2$, so $n! \geq (n/2)^{n/2}$ and $\log_2 n! \geq \frac{n}{2} \log_2 \frac{n}{2} = \Omega(n \log n)$. Also $n! \leq n^n$ gives $\log_2 n! \leq n \log_2 n$. ∎

For $n = 10$ the bound is $\lceil \log_2 3{,}628{,}800 \rceil = \lceil 21.79 \rceil = 22$ comparisons. For $n = 10^6$ it is about 1.85×10^7, while merge sort uses at most $n \log_2 n \approx 1.99 \times 10^7$. Merge sort is within 8% of the best that is *logically possible*. That is the power of a lower bound: it tells you when to stop optimizing.

Lower bounds have fine print

The theorem is about the *comparison model*. Sort n integers that fit in a machine word and radix sort runs in $O(n)$, because it looks at digits instead of comparing. A lower bound is only as strong as the model it is proved in, and many famous "impossibility" results are really statements about one model of computation.

Trade-offs: what big-O does not tell you

Big-O picks the winner as $n \to \infty$. Engineers live at finite n, and three trade-offs decide real choices:

- **Time against memory.** A lookup table turns a computation into $O(1)$ time at the cost of memory; memoization (Ch. 52) turns some exponential recursions into polynomial ones by storing every sub-answer.
- **Worst case against typical case.** Quicksort is $\Theta(n^2)$ in the worst case and roughly $1.39\, n \log_2 n$ comparisons on average; it is usually faster than merge sort anyway.
- **Constants against growth.** Library sorts switch to insertion sort ($\Theta(n^2)$) for sub-lists below about 16–32 items, exactly because of the left panel of the first figure.

? Where does the 1.39 come from?

Quicksort with a random pivot makes about $2n \ln n \approx 1.39\, n \log_2 n$ comparisons on average. The worst case is quadratic, so the average needs a probability argument over all $n!$ orderings, and the constant $2 \ln 2 \approx 1.39$ comes out of a harmonic sum. That argument, counting random arrangements, is developed in Ch. 32.

26.4 The algorithm

Analysis becomes real when you can *count*. Here is merge sort instrumented to return the number of comparisons it makes, the quantity the lower bound talks about:

```
def merge_sort(a):
    """Return (sorted copy of a, number of comparisons)."""
    if len(a) <= 1:
        return list(a), 0
    mid = len(a) // 2
    left, cl = merge_sort(a[:mid])
    right, cr = merge_sort(a[mid:])
    out, i, j, comps = [], 0, 0, cl + cr
    while i < len(left) and j < len(right):
        comps += 1
        if left[i] <= right[j]:
            out.append(left[i]); i += 1
        else:
            out.append(right[j]); j += 1
    out += left[i:] + right[j:]          # one side is exhausted: no comparisons needed
    return out, comps
```

On a random permutation of 1000 items this makes about 8700 comparisons, below the worst-case $n \log_2 n \approx 9966$ and above the information-theoretic floor $\log_2 1000! \approx 8530$.

Complexity. Time $\Theta(n \log n)$ in every case (best, average, worst), by the recursion tree. Extra memory $\Theta(n)$ for the merged lists. Recursion depth $\lceil \log_2 n \rceil$, so about 20 for a million items: no risk of stack overflow.

How to analyse any algorithm in four moves.

1. *Choose the cost unit.* Comparisons, arithmetic operations, memory reads, disk seeks, messages over a network. The right unit is the one that dominates the wall-clock time.
2. *Write the cost as a function of n.* Loops become sums ($\sum_{i=1}^{n} i = n(n+1)/2$ for a triangular double loop); recursion becomes a recurrence.
3. *Solve or bound it.* Sums by closed forms or integrals (Ch. 5); recurrences by recursion trees, the master theorem, or generating functions (Ch. 28).
4. *Check the constant at realistic sizes.* Measure. Big-O says who wins eventually, not who wins at *your* n.

Failure cases worth knowing.

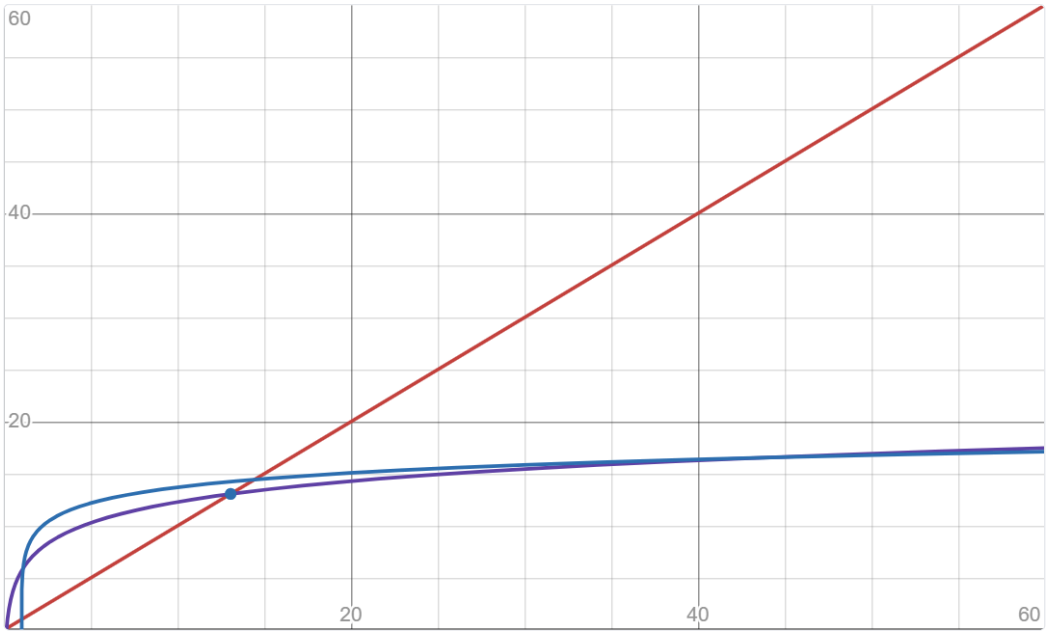
- *Quicksort on sorted input.* With "first element as pivot", an already-sorted list splits into sizes 0 and $n-1$ every time: $T(n) = T(n-1) + n = \Theta(n^2)$, and recursion depth n. Random pivots fix the expected cost, not the worst case.
- *Hash tables under attack.* A hash table is $O(1)$ per lookup *on average*. An adversary who knows the hash function can make every key collide, turning a web server's $O(n)$ request parsing into $O(n^2)$. This is a real denial-of-service attack, and the reason many languages randomize their hash seeds.
- *The hidden factor in "linear".* String concatenation in a loop (`s = s + piece`) copies the whole string every time: $\Theta(n^2)$ characters copied, the same trap as growing an array by $+100$.

> **Measuring is not analysing**
>
> Timing a program at $n = 10, 20, 40$ and fitting a curve can tell you the exponent of a polynomial (log–log slope), but it cannot distinguish n^{10} from 2^n over a short range, and it cannot see a worst case that your test data never triggers. Measurement estimates constants; analysis guarantees growth.

26.5 The experiment

Where is the crossover? (drag the constant)

Plotting $\log_2$ of the cost turns the exponential 2^n into the straight line $y = n$ and turns $c\,n^2$ into $\log_2 c + 2\log_2 n$, a curve that bends over and flattens:



Vertical axis: $\log_2(\text{cost})$. *Red is* 2^n, *purple is* $50\,n^2$, *blue is* $400\,n\log_2 n$. *The red line crosses each curve once and never comes back.*

Change the **50** in the purple curve to **50000000** (a millionfold handicap for the exponential algorithm). The purple curve rises by about 20 units, and the crossover slides only from $n \approx 13$ to $n \approx 36$. Then change the **2** in front of the second term to **3** (a cubic algorithm) and see how little that changes the picture: the logarithm of any polynomial is a gently bending curve, and the logarithm of an exponential is a straight line that always gets out from under it.

Insertion sort against merge sort (run it)

Insertion sort is $\Theta(n^2)$ but each step is tiny; merge sort is $\Theta(n \log n)$ but each comparison carries the overhead of recursion and list copying. The script counts comparisons for both and charges merge sort a configurable overhead per comparison.

```
import numpy as np
import matplotlib.pyplot as plt

def insertion_cost(a):              # comparisons made by insertion sort
    a, c = list(a), 0
    for i in range(1, len(a)):
        j = i
        while j > 0:
            c += 1
            if a[j - 1] <= a[j]:
                break
            a[j - 1], a[j] = a[j], a[j - 1]
            j -= 1
    return c

def merge_cost(a):                  # comparisons made by merge sort
    if len(a) <= 1:
        return list(a), 0
    m = len(a) // 2
    (L, cl), (R, cr) = merge_cost(a[:m]), merge_cost(a[m:])
    out, i, j, c = [], 0, 0, cl + cr
    while i < len(L) and j < len(R):
        c += 1
        if L[i] <= R[j]: out.append(L[i]); i += 1
        else:            out.append(R[j]); j += 1
    return out + L[i:] + R[j:], c

OVERHEAD = 4.0                      # KNOB: extra cost per merge-sort comparison
rng = np.random.default_rng(0)
ns = np.arange(2, 201, 2)           # KNOB: range of input sizes
ins = [np.mean([insertion_cost(rng.permutation(n)) for _ in range(20)]) for n in ns]
mer = [OVERHEAD * np.mean([merge_cost(list(rng.permutation(n)))[1] for _ in range(20)]) for n in ns]
cross = ns[np.argmax(np.array(ins) > np.array(mer))]
print("insertion sort stops winning at n =", cross)
plt.plot(ns, ins, label="insertion sort"); plt.plot(ns, mer, label=f"merge sort x {OVERHEAD}")
plt.axvline(cross, ls=":"); plt.xlabel("n"); plt.ylabel("cost"); plt.legend(); plt.show()
```

With `OVERHEAD = 4.0` the crossover lands at about $n = 78$; with `1.0` it is about 4, with `2.0` about 28. Try `8.0` and widen `ns` to 600: the crossover keeps moving right, but it always exists. Then replace `rng.permutation(n)` with `np.arange(n)` (already sorted input) and watch insertion sort become *linear*: the "worst" algorithm is the best one on nearly-sorted data, which is why it survives inside every production sort.

26.6 The challenge

1. **Intuition.** Algorithm P costs n^{100} steps and algorithm E costs 1.01^n steps. Theorem 1 says E eventually loses. Without computing anything exactly, explain why "eventually" here means an astronomically large n, and argue which one you would actually deploy for $n \le 10^6$.
2. **Derivation.** Solve the recurrence $T(n) = 3\,T(n/2) + n$, $T(1) = 1$, for $n = 2^m$ exactly, by summing the recursion tree level by level. Show that the answer is $\Theta(n^{\log_2 3})$ and identify which level of the tree carries most of the work. (This recurrence is Karatsuba's fast multiplication of n-digit numbers.)
3. **Implementation.** Write a dynamic array class with a `growth` parameter: capacity is multiplied by `growth` when full. Count total element copies for 10^6 appends with growth factors 1.1, 1.5, 2 and 4, and also record the peak wasted memory (capacity minus size). Plot copies against waste. Which factor would you choose, and why do real libraries use something between 1.5 and 2?
4. **Open investigation.** The comparison lower bound $\lceil \log_2 n! \rceil$ is not always achievable. For $n = 12$ it equals 29, yet the minimum worst-case number of comparisons needed to sort 12 items is known to be 30. Write a program that searches for optimal comparison strategies for small n ($n \le 6$ is feasible) and compare with $\lceil \log_2 n! \rceil$. The exact optimum is still unknown for many n above 15: what patterns can you find?

26.7 Loose threads

This chapter answered its challenge: an algorithm that is fast on small inputs becomes terrible on large ones when its *order of growth* is worse, because constants only shift the crossover by a logarithmic amount, while growth rates compound. We proved that every exponential beats every polynomial, solved a recurrence exactly, showed that amortization tames occasional expensive steps, and proved that no comparison sort can beat $n \log n$.

But we only ever counted *steps*. Several questions were raised and left on the table.

> **? Is the exponential ever necessary?**
>
> Engineer X used 2^n steps because they tried every subset. For many subset problems (the knapsack problem, scheduling, circuit satisfiability) nobody knows any algorithm that is polynomial in the worst case, and nobody can prove one does not exist. Our sorting lower bound worked because the comparison model is simple; for general programs, proving that a problem *needs* exponential time is the P versus NP question, the subject of Ch. 30.

The other threads run through the rest of this track and beyond:

- **Recurrences that don't halve.** Fibonacci-style recurrences $T(n) = T(n-1) + T(n-2)$ grow like φ^n with $\varphi \approx 1.618$. Where that golden ratio comes from, and how generating functions solve such recurrences mechanically, is in Ch. 28.
- **From exponential to polynomial by remembering.** Many exponential recursions collapse to polynomial time when you store sub-answers: Ch. 52.
- **Algorithms on networks.** The most-used algorithms on Earth (route planning, matching, page ranking) run on graphs, and their analysis needs the language of Ch. 27.
- **Proving the code is right, not just fast.** A fast wrong answer is worthless. Loop invariants and induction, the tools that turn "it passed the tests" into "it works for every input", are in Ch. 29.

Further reading

- T. H. Cormen, C. E. Leiserson, R. L. Rivest, C. Stein, *Introduction to Algorithms*, 4th ed., MIT Press, 2022. The standard reference for asymptotics, recurrences, amortized analysis and lower bounds.
- J. Kleinberg, É. Tardos, *Algorithm Design*, Pearson, 2005. Strong on intuition and on why problems are hard.
- R. Sedgewick, P. Flajolet, *An Introduction to the Analysis of Algorithms*, 2nd ed., Addison-Wesley, 2013. Exact counting, recurrences and generating functions.
- D. E. Knuth, *The Art of Computer Programming*, Vol. 3: *Sorting and Searching*, 2nd ed., Addison-Wesley, 1998. The definitive treatment of comparison counts, including minimum-comparison sorting.

Chapter 27 — Graph Theory: The Mathematics of Connections

> **Level 2 · Track A — The Mathematics Behind Algorithms Builds on:** Ch. 8, Ch. 10, Ch. 26 · **Leads to:** Ch. 28, Ch. 87, Ch. 90 · **Related:** Ch. 34, Ch. 48, Ch. 52 **Reading time:** ~25 min

27.1 The hook

It is 6 p.m. on a Friday at a ride-hailing company. Three engineers have three problems.

The router must send a car from the depot to a pickup across a city of 200,000 intersections. The number of distinct routes is astronomically large: on a modest 20×20 grid of streets there are already $\binom{40}{20} \approx 1.4 \times 10^{11}$ shortest-looking staircase routes alone, and the real city is not a grid. Yet the app must answer in milliseconds, for millions of requests an hour.

The matcher has 1000 free drivers and 1000 waiting riders. Each driver can reach only some riders in time. She wants as many riders served as possible. The number of ways to pair 1000 drivers with 1000 riders is $1000! \approx 10^{2567}$, a number with 2568 digits. Trying them all is not slow; it is meaningless.

The analyst has a year of trip data: who rides from where to where. Management suspects the city has "hidden neighbourhoods", clusters of places that trade riders mostly among themselves, which do not match any map. Nobody has told her how many clusters there are, or where.

> **The challenge of this chapter:** How would you find the best route, match thousands of people, or detect hidden communities?

All three problems have the same shape: things, and connections between things. Strip away the cars, the people and the streets, and what is left is a drawing of dots and lines. That drawing is a **graph**, and the surprise of this chapter is that one drawing supports three completely different kinds of mathematics. The router will be saved by a greedy argument, the matcher by an alternating path, and the analyst by an eigenvector.

27.2 The big idea

A graph forgets everything except **who is connected to whom** (and sometimes, how strongly). A street map becomes intersections (vertices) and road segments (edges) with travel times as weights. A dating app becomes two columns of people with lines for "would accept". A friendship network becomes dots with lines for "talks to".

Once the problem is a graph, three big ideas do most of the work.

Grow outward, never look back. To find shortest routes from the depot, don't enumerate routes. Instead, spread out from the depot like a ripple in a pond, settling each intersection at the moment the ripple reaches it. A place reached first by the ripple can never be reached sooner by a longer detour. That is Dijkstra's algorithm, and it touches each road only once.

Improve by flipping. To enlarge a matching, look for a path that zig-zags between unmatched and matched edges, starting and ending at people nobody has served yet. Flip every edge along it, matched to unmatched and back, and the matching grows by one. When no such path exists, the matching is provably the largest possible.

Let the network vibrate. Encode the graph as a matrix and compute its eigenvectors. A graph with two loosely tied communities "vibrates" like two masses joined by a weak spring: the slowest non-trivial mode moves one community up while the other moves down. The sign of that eigenvector reveals the communities, with no labels and no guesswork about where they are.

The analogy that ties these together: a graph is a mechanical structure. Shortest paths are the strings that go taut when you pull two knots apart; matchings are how you pair up the bolts; communities are the parts that wobble together.

27.3 The mathematics

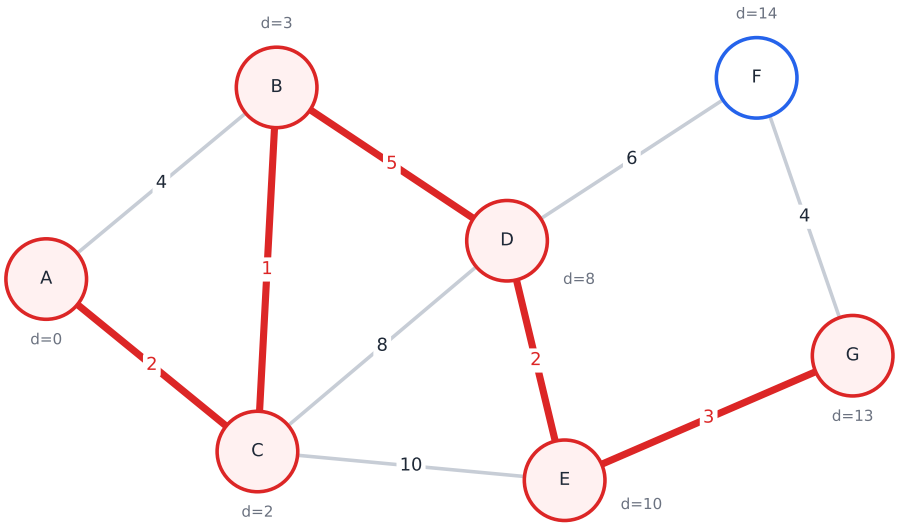
Vocabulary in five lines

> **Graphs**
>
> A **graph** $G = (V, E)$ is a set V of vertices and a set E of edges, each edge joining two vertices. The **degree** $\deg v$ is the number of edges at v. A **path** is a sequence of distinct vertices with consecutive ones adjacent; a **cycle** is a path that returns to its start. G is **connected** if every pair of vertices is joined by a path. A **tree** is a connected graph with no cycles. In a **weighted** graph each edge e carries a length $w(e) \geq 0$, and the length of a path is the sum of its edge lengths.

Two facts come up constantly. First, the **handshake lemma**: $\sum_v \deg v = 2|E|$, because every edge has two ends and is counted once at each. (So the number of odd-degree vertices is always even.) Second, every tree on n vertices has exactly $n - 1$ edges: remove a leaf and its edge, and induct.

Graphs are also matrices. The **adjacency matrix** $\mathbf{A}$ has $A_{ij} = 1$ if i and j are adjacent, else 0; the **degree matrix** $\mathbf{D}$ is diagonal with $D_{ii} = \deg i$. Both are symmetric, which will matter at the end.

Result 1: shortest paths — why the greedy ripple is correct



shortest A to G: A - C - B - D - E - G (length 13)

A small road network. The direct road A–B (length 4) loses to the detour A–C–B (length 3), and the shortest route to G (length 13) passes through four intermediate intersections. Each label d is the distance Dijkstra's algorithm assigns.

Dijkstra's algorithm keeps a tentative distance $d(v)$ for every vertex, starting at $d(s) = 0$ for the source and ∞ elsewhere, and a set S of **settled** vertices, initially empty. Repeatedly:

1. pick the unsettled vertex u with the smallest $d(u)$ and settle it;
2. for every edge (u, v), **relax** it: $d(v) \leftarrow \min\{d(v),\ d(u) + w(u, v)\}$.

On the network above the vertices settle in the order A (0), C (2), B (3), D (8), E (10), G (13), F (14). Notice that B is first given tentative distance 4 by the direct road, then lowered to 3 when C is settled. Once settled, a distance never changes again. That is the claim we must prove.

> **Correctness of Dijkstra's algorithm**
>
> If all edge weights are non-negative, then when a vertex u is settled, $d(u)$ equals the true shortest-path distance $\delta(s, u)$.

Proof. Every tentative value $d(v)$ is the length of some actual path (relaxation only ever writes path lengths), so $d(v) \geq \delta(s, v)$ always. Suppose, for contradiction, that u is the **first** vertex settled with $d(u) > \delta(s, u)$. Take a true shortest path P from s to u. It starts inside S (at s) and ends outside, so it has a first edge (x, y) with $x \in S$ and $y \notin S$.

- x was settled before u, so (by the choice of u) $d(x) = \delta(s, x)$, and when x was settled the edge (x, y) was relaxed. Hence $d(y) \leq \delta(s, x) + w(x, y) = \delta(s, y)$, the last step because P's prefix up to y is a shortest path. So $d(y) = \delta(s, y)$.
- Weights are non-negative, so the rest of P from y to u has length ≥ 0: $\delta(s, y) \leq \delta(s, u)$.

Putting it together, $d(y) = \delta(s, y) \leq \delta(s, u) < d(u)$. But the algorithm chose u because it had the *smallest* tentative distance among unsettled vertices, and y is unsettled with a smaller one. Contradiction. ■

Look at where the proof used "non-negative": the remainder of the path cannot be negative, so a detour can never come back cheaper. That single word is load-bearing.

> **? The clue we're holding back**
>
> Put a single negative weight on one road (a toll refund, a downhill battery recharge for an electric car) and Dijkstra can return a wrong answer with full confidence. The fix, the Bellman–Ford algorithm, relaxes every edge $|V| - 1$ times and costs $O(|V|\,|E|)$ instead of roughly $O(|E| \log |V|)$. Why $|V| - 1$ rounds are exactly enough is a dynamic-programming argument about the number of edges in a path: Ch. 52.

Flows and matchings

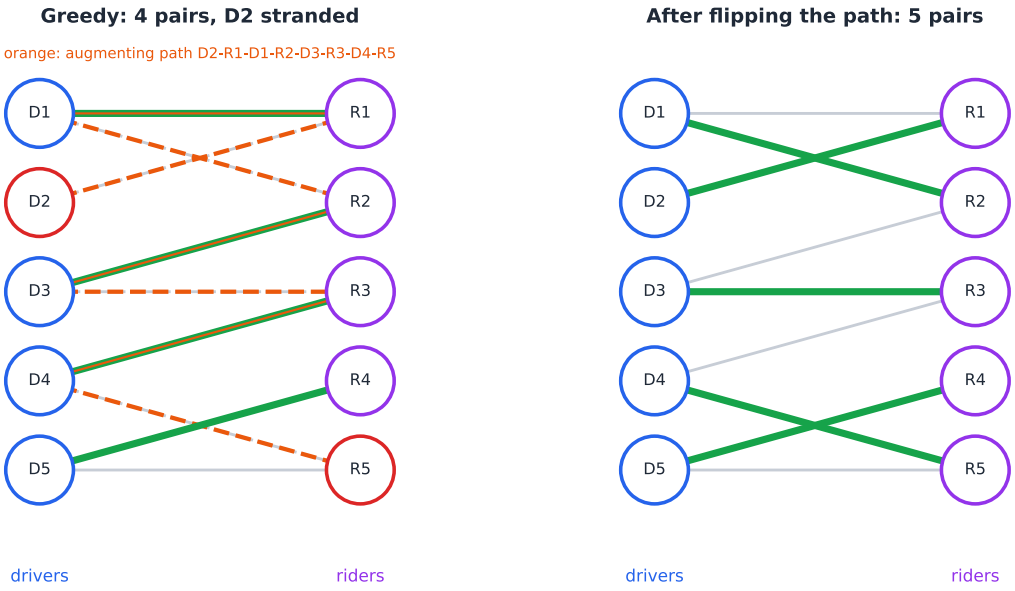
A **flow network** has a source s, a sink t, and a capacity $c(e) \geq 0$ on each directed edge. A flow assigns $f(e)$ with $0 \leq f(e) \leq c(e)$, conserved at every vertex except s and t (in equals out, Kirchhoff's current law from circuit theory). A **cut** splits the vertices into a side containing s and a side containing t; its capacity is the total capacity of edges crossing from the s-side to the t-side.

Any flow must cross every cut, so the value of any flow is at most the capacity of any cut. The astonishing theorem is that the bound is tight.

> **Max-flow min-cut (stated)**
>
> In any flow network, the maximum value of an s–t flow equals the minimum capacity of an s–t cut.

The algorithmic idea is the same "improve by flipping" from §27.2: while some path from s to t has spare capacity (counting the option of *cancelling* existing flow on reverse edges), push more flow along it. When no such path exists, the vertices still reachable from s form a cut whose capacity equals the current flow, and both are optimal at once. Max-flow min-cut is a linear program and its dual, two problems with the same optimum; why that always happens is the theme of Ch. 48.

Matching is flow in disguise. Put drivers on the left and riders on the right, an edge for every feasible pairing, a source joined to every driver and every rider joined to a sink, all with capacity 1. A maximum flow is a maximum matching.



*Greedy matching gets stuck at four pairs: D2's only rider R1 is taken. The orange **augmenting path** alternates unmatched and matched edges between two free vertices; flipping it swaps three matched edges for four and serves everyone.*

When can *every* driver be matched? An obvious obstacle: if some group of k drivers can, between them, reach fewer than k riders, someone is left out. Write $N(W)$ for the set of riders adjacent to at least one driver in W.

Hall's marriage theorem

A bipartite graph with left side L has a matching that covers every vertex of L if and only if $|N(W)| \geq |W|$ for every subset $W \subseteq L$.

Proof of necessity. If a matching covers L, then the drivers in W are matched to $|W|$ *distinct* riders, all of them in $N(W)$. So $|N(W)| \geq |W|$. ■

Sufficiency, the deep direction, says the obvious obstacle is the *only* obstacle: if no small group is starved, the augmenting-path algorithm never gets stuck. It follows from max-flow min-cut (a minimum cut that is smaller than $|L|$ can be read off as a starved group W). The matcher's problem with 1000 drivers is solved not by trying 10^{2567} pairings but by at most 1000 augmenting-path searches, each $O(|E|)$.

Colouring and planarity

Assign each vertex a colour so that adjacent vertices differ. The minimum number of colours is the **chromatic number** $\chi(G)$. Colouring is scheduling in disguise: exams are vertices, an edge joins two exams sharing a student, colours are time slots. Radio transmitters that interfere need different frequencies; registers in a compiler that are live at the same time need different physical registers.

A graph is **planar** if it can be drawn in the plane with no crossing edges, like a circuit on a single-layer board. A drawing divides the plane into F regions (faces, counting the outside).

Euler's formula and its consequence

For a connected planar drawing, $V - E + F = 2$. If moreover $V \geq 3$, then $E \leq 3V - 6$.

Proof sketch of Euler's formula. A tree has $E = V - 1$ and one face: $V - (V - 1) + 1 = 2$. Every other connected planar graph is a tree plus extra edges, and each extra edge closes a cycle, splitting one face into two: E and F both rise by 1 and $V - E + F$ is unchanged.

Proof of the inequality. Every face is bounded by at least 3 edges (no loops or double edges, and $V \geq 3$), and every edge borders at most 2 faces. Counting edge–face incidences two ways gives $3F \leq 2E$. Substitute $F = 2 - V + E$ from Euler's formula: $3(2 - V + E) \leq 2E$, that is $E \leq 3V - 6$. ■

The inequality instantly proves an impossibility. The complete graph K_5 (five vertices, all joined) has $V = 5$ and $E = 10$, but $3V - 6 = 9$. So K_5 **cannot be drawn without a crossing**, however clever you are. No search, no trial and error: one line of counting rules out infinitely many drawings at once. (The same argument with faces of length at least 4 shows the "three houses, three utilities" graph $K_{3,3}$ is not planar either.)

? Four colours, and a proof no human has checked

Every planar map can be coloured with at most **four** colours so that neighbouring regions differ. The inequality $E \le 3V - 6$ quickly shows that five colours always suffice. The step from five to four resisted mathematicians for 124 years, and when it was finally proved in 1976 the proof required a computer to check nearly two thousand configurations. What it means to trust such a proof, and how modern proof assistants have since verified it, is part of Ch. 98 and Ch. 67.

Planar graphs are the gentle case. For a general graph, deciding whether three colours suffice is NP-complete: no known algorithm beats exponential time in the worst case. Why that classification is so damning is Ch. 30's story.

Centrality: who matters in a network?

There is no single answer, which is why there are several centrality measures.

- **Degree centrality**: how many connections. Simple, local, easy to fool.
- **Betweenness centrality**: the fraction of shortest paths between other pairs that pass through v. A bridge between two communities has low degree but huge betweenness, and it is the vertex whose failure splits the network.
- **Eigenvector centrality and PageRank**: you are important if important vertices point to you. That circular definition is an eigenvector equation, $\mathbf{A}^{\mathsf{T}}\mathbf{x} = \lambda \mathbf{x}$, and PageRank adds a small chance of jumping to a random page so the eigenvector is unique.

Engineering note

In a power grid, betweenness identifies transmission lines whose loss reroutes the most power flow; in a chip's interconnect, it flags wires that are on many critical paths. Degree alone would miss both.

Result 2: the Laplacian sees connectivity

Define the **graph Laplacian** $\mathbf{L} = \mathbf{D} - \mathbf{A}$. It is the discrete cousin of ∇^2 (Ch. 13) and the nodal admittance matrix of a resistor network in which every edge is a 1 S conductance (Ch. 8). Its key property is a quadratic form.

The Laplacian quadratic form

For every vector $\mathbf{x} \in \mathbb{R}^V$,

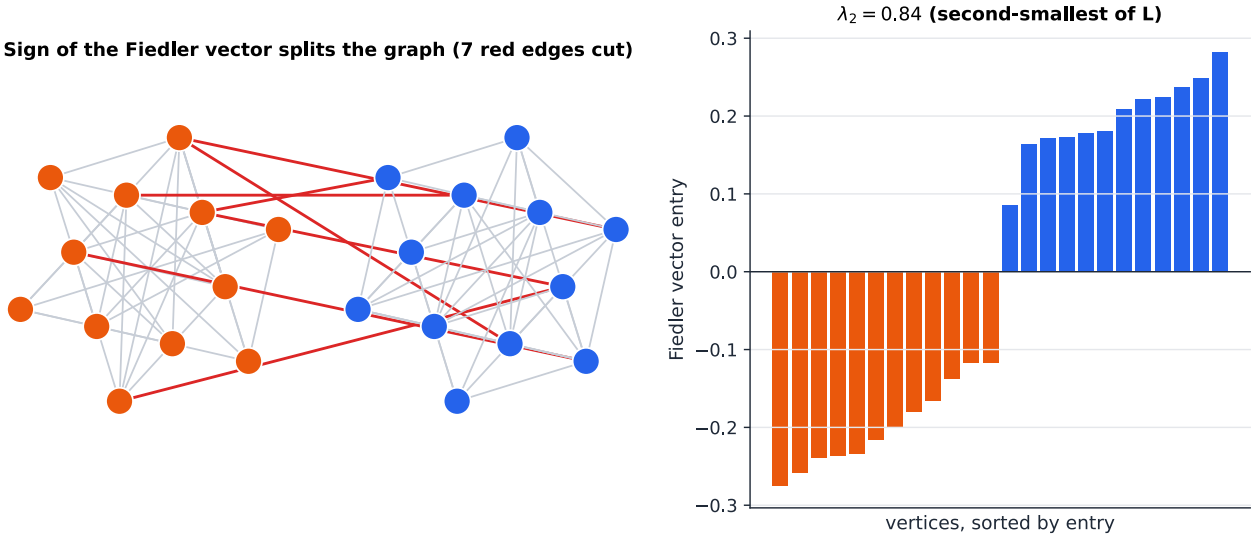
$$\mathbf{x}^{\mathsf{T}} \mathbf{L}\, \mathbf{x} = \sum_{\{i,j\} \in E} (x_i - x_j)^2 .$$

Hence $\mathbf{L}$ is positive semidefinite, and **the number of zero eigenvalues of $\mathbf{L}$ equals the number of connected components of G.**

Proof. $\mathbf{x}^{\mathsf{T}}\mathbf{D}\mathbf{x} = \sum_i \deg(i)\, x_i^2$, and by the handshake idea this equals $\sum_{\{i,j\}\in E}(x_i^2 + x_j^2)$: each edge contributes the squares of its two ends. Also $\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} = \sum_{i,j} A_{ij} x_i x_j = \sum_{\{i,j\}\in E} 2 x_i x_j$. Subtracting, each edge gives $x_i^2 - 2x_i x_j + x_j^2 = (x_i - x_j)^2$.

A sum of squares is ≥ 0, so $\mathbf{L}$ is positive semidefinite and its eigenvalues are real and non-negative (Ch. 10, §10.5). Now $\mathbf{L}\mathbf{x} = \mathbf{0}$ implies $\mathbf{x}^{\mathsf{T}}\mathbf{L}\mathbf{x} = 0$, so $x_i = x_j$ across every edge, so $\mathbf{x}$ is constant on each connected component. Conversely, any vector constant on each component satisfies $\mathbf{L}\mathbf{x} = \mathbf{0}$ (each row of $\mathbf{L}$ sums to zero). The null space is therefore spanned by the indicator vectors of the k components, which are linearly independent: its dimension is k. Since $\mathbf{L}$ is symmetric, the multiplicity of eigenvalue 0 equals the dimension of the null space. ∎

So a matrix computation can *count the pieces* of a network. Now push the idea one step. If the graph is connected, the smallest eigenvalue $\lambda_1 = 0$ belongs to the constant vector, and the second-smallest, $\lambda_2 > 0$, is small exactly when the graph is *almost* disconnected. Its eigenvector, the **Fiedler vector**, minimizes $\sum_E (x_i - x_j)^2$ among unit vectors orthogonal to the constants: it tries to give neighbours equal values while being forced to take both signs. The cheapest way to do that is to be positive on one tightly-knit community, negative on the other, and pay the penalty only on the few edges between them.



The analyst's trick on a small example. Colouring each vertex by the sign of the second eigenvector of $\mathbf{L}$ recovers the two planted communities exactly; only the 7 red edges run between them. The algorithm was never told that there were communities.

> **? Why does one eigenvalue measure a bottleneck?**
>
> We showed that $\lambda_2 = 0$ exactly when the graph falls apart. The quantitative version, **Cheeger's inequality**, says λ_2 is small *if and only if* the graph has a cut that removes few edges relative to the sizes of the pieces, and it gives explicit bounds in both directions. The proof needs the variational theory of eigenvalues and the analogy with vibrating membranes, which is developed in Ch. 90.

27.4 The algorithm

Dijkstra with a priority queue

```
import heapq

def dijkstra(adj, source):
    """adj[u] = list of (v, weight) with weight >= 0. Returns (dist, prev)."""
    dist, prev = {source: 0}, {}
    heap = [(0, source)]                 # (tentative distance, vertex)
    while heap:
        d, u = heapq.heappop(heap)
        if d > dist[u]:                  # stale entry: u was already settled cheaper
            continue
        for v, w in adj[u]:
            nd = d + w
            if nd < dist.get(v, float("inf")):
                dist[v], prev[v] = nd, u   # relax edge (u, v)
                heapq.heappush(heap, (nd, v))
    return dist, prev
```

Follow `prev` backwards from the target to read off the route. Each edge is relaxed at most once per settled endpoint and each heap operation costs $O(\log |V|)$, so the running time is $O((|V| + |E|)\log |V|)$. Road networks have $|E| \approx 2.5|V|$, so 200,000 intersections cost a few million elementary steps: milliseconds.

Failure cases. A negative edge weight breaks the proof (and the answer). Floating-point weights that are "almost equal" can make two routes tie and flip between runs, so production routers break ties deterministically. And for a whole continent, even milliseconds per query is too slow: real map services *preprocess* the graph (contraction hierarchies, landmark heuristics such as A*) so that each query explores only a thin corridor instead of a disc around the source.

Maximum bipartite matching by augmenting paths

```
def max_matching(adj, left):
    """adj[d] = riders driver d can reach. Returns dict rider -> driver."""
    match = {}                           # rider -> driver

    def try_augment(d, seen):
        for r in adj[d]:
            if r in seen:
                continue
            seen.add(r)
            # r is free, or r's current driver can be re-routed elsewhere
            if r not in match or try_augment(match[r], seen):
                match[r] = d             # flip the edges along the path
                return True
        return False

    for d in left:
        try_augment(d, set())
    return match
```

Each call to `try_augment` searches for one augmenting path by depth-first search, in $O(|E|)$; there are $|L|$ calls, so the total is $O(|L|\,|E|)$. The Hopcroft–Karp algorithm finds many shortest augmenting paths at once and improves this to $O(\sqrt{|V|}\,|E|)$. On 1000 drivers each reaching 50 riders that is about 50,000 edges, solved in well under a second, compared with the 10^{2567} pairings of brute force.

Failure cases. The recursion depth can reach $|L|$; for very large graphs, write the search iteratively. And this finds *a* largest matching, not the *cheapest* one. If each pairing has a cost (pickup time), you need the weighted version, the assignment problem, solved by the Hungarian algorithm in $O(n^3)$ or as a linear program (Ch. 23, §23.3).

Spectral community detection

1. Build $\mathbf{L} = \mathbf{D} - \mathbf{A}$ (sparse, if the graph is large).
2. Compute the eigenvector $\mathbf{v}_2$ of the second-smallest eigenvalue.
3. Split the vertices by the sign of $\mathbf{v}_2$ (or, better, try every threshold along the sorted entries and keep the cut with the best ratio of cut edges to community sizes).

A dense eigen-solve costs $O(|V|^3)$, fine for a few thousand vertices. For a million, use a sparse iterative method such as Lanczos (Ch. 24, §24.4), which needs only matrix–vector products. For k communities, take the k smallest eigenvectors as coordinates and cluster the points.

Failure cases. One isolated vertex creates an extra zero eigenvalue and the Fiedler vector happily "discovers" a community of size 1. Very unequal degrees distort the split, which is why practitioners often use the *normalized* Laplacian $\mathbf{I} - \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}$ instead.

27.5 The experiment

The script plants two communities of 100 people, links people in the same community with probability `P_IN` and across communities with probability `P_OUT`, shuffles the vertices to hide the structure, and asks the Fiedler vector to find it.

```
import numpy as np
import matplotlib.pyplot as plt

N_PER = 100          # KNOB: people per community
P_IN = 0.10          # KNOB: chance two people in the SAME community are linked
P_OUT = 0.02         # KNOB: chance two people in DIFFERENT communities are linked
SEED = 1

rng = np.random.default_rng(SEED)
truth = np.repeat([0, 1], N_PER)
N = truth.size
P = np.where(truth[:, None] == truth[None, :], P_IN, P_OUT)
A = np.triu(rng.random((N, N)) < P, 1).astype(float)
A = A + A.T                                  # undirected adjacency matrix
L = np.diag(A.sum(axis=1)) - A               # graph Laplacian L = D - A
vals, vecs = np.linalg.eigh(L)               # symmetric: real eigenvalues, ascending
fiedler = vecs[:, 1]
guess = (fiedler > 0).astype(int)
acc = max(np.mean(guess == truth), np.mean(guess != truth))
print(f"lambda_2 = {vals[1]:.3f}, communities recovered: {100 * acc:.1f}%")

perm = rng.permutation(N)                    # hide the structure, then reveal it
fig, ax = plt.subplots(1, 3, figsize=(12, 4))
ax[0].imshow(A[np.ix_(perm, perm)], cmap="Greys"); ax[0].set_title("adjacency, shuffled")
order = np.argsort(fiedler[perm])
ax[1].imshow(A[np.ix_(perm[order], perm[order])], cmap="Greys"); ax[1].set_title("sorted by Fiedler vector")
ax[2].plot(vals[:10], "o-"); ax[2].set_title("smallest eigenvalues of L")
plt.tight_layout(); plt.show()
```

With the default knobs the Fiedler vector recovers about 95% of the labels, and the middle panel turns a shapeless grey square into two dark blocks. Now turn the knobs:

- Lower `P_IN` to 0.08, 0.07, 0.06 and run several seeds. Recovery becomes erratic, sometimes above 90%, sometimes near 55%, even though each person still has more than twice as many friends inside their community as outside.
- Lower `P_IN` to 0.03. Now some people have no links at all, λ_2 drops to 0, and the method collapses to a coin toss (about 50%): it has found an isolated vertex, not a community. Count the zero eigenvalues in the right panel and compare with the theorem of §27.3.
- Set `P_OUT = 0` to make the communities disconnected. Two eigenvalues are exactly 0, and the split is perfect.

Somewhere between "easy" and "impossible" there is a boundary, and it is sharper than you might expect.

27.6 The challenge

1. **Intuition.** In the road network figure, which single road could you remove to increase the distance from A to G the most, and by how much? Answer by inspection first, then explain why the roads *not* on the shortest path can be removed for free. What does that say about the union of all shortest paths from A (it is a famous kind of graph)?
2. **Derivation.** A **star** on n vertices has one centre joined to $n-1$ leaves. Write down its Laplacian and find all its eigenvalues by hand (hint: try vectors that are zero at the centre and sum to zero on the leaves). Show that $\lambda_2 = 1$ for every n, and explain why a star, despite being connected, is "fragile".
3. **Implementation.** Extend the matching code to the **stable marriage** problem: each driver ranks riders and each rider ranks drivers. Implement the Gale–Shapley proposal algorithm, prove (on paper) that it always terminates after at most n^2 proposals, and test on random preferences how often the driver-proposing and rider-proposing versions give different answers as n grows.
4. **Open investigation.** In the experiment, fix `N_PER = 500` and `P_OUT = 0.01` and scan `P_IN` finely. Plot recovery (averaged over 10 seeds) against `P_IN`. Is there a threshold where recovery jumps from near 50% to well above it? How does the threshold move with `N_PER`? Try replacing $\mathbf{L}$ by the normalized Laplacian, or by $\mathbf{A}$ itself, and see whether the threshold moves.

27.7 Loose threads

The three engineers have their answers. The router uses a greedy ripple, proved correct by a first-failure argument that leans entirely on non-negative weights. The matcher uses augmenting paths, and Hall's theorem tells her exactly when a perfect match is impossible. The analyst uses the Laplacian, whose zero eigenvalues count the pieces of the network and whose second eigenvector cuts it along its weakest seam.

What we did not tell you:

- **Negative weights** need a different algorithm and a different cost, and the reason lives in dynamic programming (Ch. 52).
- **The four-colour theorem** is true, but its proof is a story about computers and trust (Ch. 98).
- **Cheeger's inequality**, the exact link between λ_2 and bottlenecks, needs spectral theory (Ch. 90).

- **Detectability.** The threshold you met in the experiment is real. For large random networks with planted communities, there is a sharp limit below which *no* algorithm can beat random guessing, and it was located only in the 2010s, with ideas borrowed from statistical physics. Real networks, with hubs, small worlds and power-law degrees, are the subject of Ch. 87.
- **PageRank** is the stationary distribution of a random walk on the web graph; why that walk settles down at all is Ch. 34.
- **Hard graph problems.** Shortest paths and matchings are easy; colouring with three colours, the travelling-salesman tour and the longest path are not known to be. Where that line falls is Ch. 30, and how to attack the hard side anyway is Ch. 49. And counting the graphs themselves (how many trees, how many colourings) is Ch. 28.

Further reading

- D. B. West, *Introduction to Graph Theory*, 2nd ed., Prentice Hall, 2001. Rigorous and complete: matchings, connectivity, colouring, planarity.
- R. Diestel, *Graph Theory*, 5th ed., Springer, 2017. The standard graduate text, freely available online in a preview edition.
- J. Kleinberg, É. Tardos, *Algorithm Design*, Pearson, 2005. Chapters on shortest paths, network flow and matching, with real applications.
- D. A. Spielman, *Spectral and Algebraic Graph Theory*, lecture notes, Yale University. The Laplacian, Cheeger's inequality and graph partitioning, from first principles.

Chapter 28 — Combinatorics: Counting the Impossible

> **Level 2 · Track A — The Mathematics Behind Algorithms Builds on:** Ch. 1, Ch. 17, Ch. 19, Ch. 26 · **Leads to:** Ch. 29, Ch. 32, Ch. 57 · **Related:** Ch. 27, Ch. 33, Ch. 18 **Reading time:** ~25 min

28.1 The hook

Pick up an ordinary deck of 52 cards and shuffle it well. The order you are holding has, almost certainly, **never existed before in the history of the universe**, and will never exist again.

That sounds like mysticism. It is arithmetic. The number of orderings of 52 cards is

$$52! = 52 \times 51 \times 50 \times \cdots \times 2 \times 1 \approx 8.07 \times 10^{67}.$$

If every one of the roughly eight billion people alive had shuffled a deck once per second since the Big Bang, about 4×10^{17} seconds ago, they would have produced around 3×10^{27} orderings: a fraction 4×10^{-41} of the possibilities.

The deck is small change. The number of atoms in the observable universe is usually estimated at about 10^{80}. A Go board has 361 points, each empty, black or white, so at most $3^{361} \approx 1.7 \times 10^{172}$ configurations; the number of *legal* positions was computed exactly in 2016 and is about 2.08×10^{170}. That is more than 10^{90} positions **for every atom in the universe**. And the simplest example of all: a memory of just 266 bits (34 bytes, shorter than this sentence) has $2^{266} \approx 1.2 \times 10^{80}$ possible contents, already more than there are atoms.

> **The challenge of this chapter:** How can a problem have more possible configurations than atoms in the observable universe?

And if the space is that large, how can we say *anything* about it, how many configurations have a property, whether some pattern must appear, without ever looking at them? That is what combinatorics is: the mathematics of counting things you could never list.

28.2 The big idea

The answer to the challenge is one rule, applied over and over:

> **If a choice is made in stages, the counts multiply.**

A configuration of a big system is a sequence of small choices. 266 bits are 266 two-way choices: $2 \times 2 \times \cdots \times 2 = 2^{266}$. A deck is 52 choices of "which card next" with one fewer option each time. Physical systems add their parts, $1 + 1 + \ldots$; configuration spaces multiply them, $2 \times 2 \times \ldots$. Adding builds sizes like the number of atoms. Multiplying builds sizes like the number of states of those atoms. That is why a modest number of parts outruns the universe: Ch. 26 showed that every exponential eventually beats every polynomial, and here the exponential is not a running time but a count.

So listing is hopeless. Combinatorics replaces listing with four tricks of reasoning:

1. **Multiply** stages (the product rule), and **divide** out orderings you don't care about.
2. **Biject:** count a set by matching it one-to-one with a set that is easier to count.
3. **Correct overcounts** systematically: inclusion–exclusion.
4. **Guarantee without searching:** the pigeonhole principle and Ramsey theory prove that a pattern *must* exist in every configuration, even in spaces too large to search.

The analogy: you cannot count the grains of sand on a beach one by one, but you can measure the beach's volume and one grain's volume and divide. Combinatorics is the art of finding the right "volume" to measure.

28.3 The mathematics

Permutations and combinations

Counting arrangements and selections

The number of **permutations** (orderings) of n distinct objects is $n! = n(n-1)\cdots 1$, with $0! = 1$. The number of ordered selections of k out of n is $n(n-1)\cdots(n-k+1) = \dfrac{n!}{(n-k)!}$. The number of **combinations**, unordered selections of k out of n, is the binomial coefficient

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}.$$

The combination count comes from the "divide out" trick: each unordered k-set appears $k!$ times among the ordered selections. A lottery that draws 6 numbers from 49 has $\binom{49}{6} = 13{,}983{,}816$ outcomes; a 20×20 street grid has $\binom{40}{20} \approx 1.38 \times 10^{11}$ shortest staircase routes from one corner to the other (choose which 20 of the 40 steps go east).

How big is $n!$? **Stirling's approximation** answers it:

$$n! \approx \sqrt{2\pi n}\left(\frac{n}{e}\right)^n,$$

with relative error about $1/(12n)$. For $n = 52$ it gives 8.05×10^{67}, within 0.2% of the truth. So $\log n! \approx n \log n$: factorials grow faster than any exponential b^n, because the "base" n/e keeps increasing.

Result 1: Pascal's rule, by a story rather than algebra

Pascal's rule

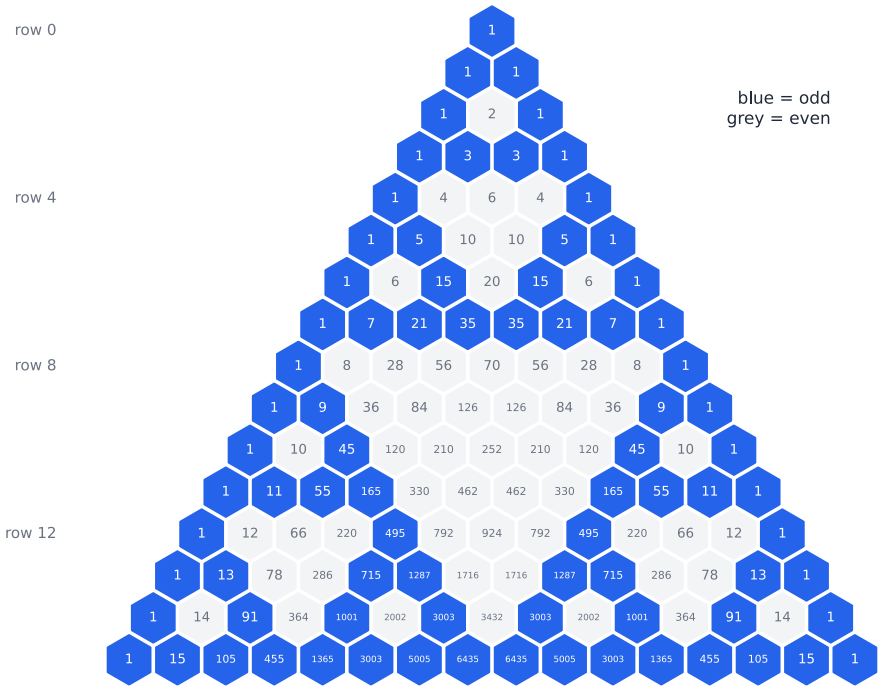
For $1 \le k \le n-1$,

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

Proof (by double counting). Count the k-member committees chosen from n people, one of whom is Ada. Every committee either includes Ada or not. Those that include her are completed by choosing $k-1$ of the other $n-1$ people: $\binom{n-1}{k-1}$ ways. Those that exclude her choose all k from the other $n-1$: $\binom{n-1}{k}$ ways. The two cases do not overlap and cover everything, so the counts add. ■

No factorials were harmed. This is the combinatorial style of proof: find two honest ways to count the same set, and equate them. Applied to $(1+x)^n = (1+x)(1+x)\cdots(1+x)$, where the coefficient of x^k counts ways to pick the x from k of the n factors, the same idea gives the **binomial theorem** $(1+x)^n = \sum_k \binom{n}{k} x^k$.

Pascal's rule builds the famous triangle: each entry is the sum of the two above it. Now do something strange with it: colour each entry by whether it is odd or even.



Pascal's triangle with odd entries in blue. A fractal appears: three copies of the top half of the triangle, with a grey hole in the middle, each copy made of three smaller copies, and so on. Rows 1, 3, 7 and 15 are entirely odd.

Why should a rule about adding numbers produce a fractal? Taking parity turns Pascal's rule into $a \oplus b$ (addition mod 2, the XOR gate), and the triangle becomes a cellular automaton: each cell is the XOR of its two parents. That much explains *that* there is a self-similar pattern. It does not explain *which* entries are odd.

? The odd ones

$\binom{n}{k}$ is odd exactly when every binary digit of k is at most the corresponding binary digit of n, so row n contains exactly 2^s odd entries, where s is the number of 1s in n written in binary. (Row 12 = 1100_2 has 4 odd entries; check the picture.) The reason is Lucas's theorem on binomial coefficients modulo a prime, and it lives in Ch. 57.

Result 2: inclusion–exclusion and the hat-check problem

Adding the sizes of overlapping sets overcounts the overlaps. For two sets, $|A \cup B| = |A| + |B| - |A \cap B|$, which is the same formula as $P(A \cup B)$ in Ch. 19, §19.1. For many sets the corrections alternate.

Inclusion–exclusion

For finite sets $A_1, \dots, A_n$,

$$\left|\bigcup_{i=1}^{n} A_i\right| = \sum_i |A_i| - \sum_{i<j} |A_i \cap A_j| + \sum_{i<j<k} |A_i \cap A_j \cap A_k| - \cdots + (-1)^{n+1} |A_1 \cap \cdots \cap A_n|.$$

Proof. Take any element x that lies in exactly $m \geq 1$ of the sets. It is counted $\binom{m}{1}$ times in the first sum, $\binom{m}{2}$ times in the second, and so on, so its net count on the right is

$$\binom{m}{1} - \binom{m}{2} + \binom{m}{3} - \cdots = 1 - \sum_{j=0}^{m} (-1)^j \binom{m}{j} = 1 - (1-1)^m = 1,$$

by the binomial theorem. An element in none of the sets is counted 0 times on both sides. So both sides count every element exactly once. ∎

The hat-check problem. n people check their hats, and the hats are handed back in random order. What is the chance that *nobody* gets their own hat? A permutation with no fixed point is a **derangement**. Let A_i be the permutations that fix person i. Any k specified people can be fixed in $(n-k)!$ ways, and there are $\binom{n}{k}$ ways to specify them, so inclusion–exclusion gives the number of permutations fixing *somebody*, and subtracting from $n!$:

$$D_n = \sum_{k=0}^{n} (-1)^k \binom{n}{k} (n-k)! = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!}.$$

The probability is $D_n/n! = \sum_{k=0}^{n} (-1)^k/k!$, the partial sum of the series for e^{-1}. So the answer converges to $1/e \approx 0.3679$, and it does so astonishingly fast: for $n = 10$ it is $0.367879\ldots$, already correct to six decimal places, and $D_{10} = 1{,}334{,}961$. Remarkably, the answer barely depends on whether there are 10 guests or 10 million.

Result 3: recurrences and generating functions

Many counts are easier to describe by a rule than by a formula. How many ways t_n are there to tile a $1 \times n$ strip with squares (1×1) and dominoes (1×2)? Look at the last tile: it is a square (leaving a strip of length $n-1$) or a domino (leaving $n-2$). So

$$t_n = t_{n-1} + t_{n-2}, \qquad t_0 = 1,\ t_1 = 1,$$

the Fibonacci recurrence: $1, 1, 2, 3, 5, 8, 13, \ldots$. Recurrences like this are exactly the difference equations of Ch. 17, §17.5, and combinatorics has its own name for the Z-transform: the **generating function**, the power series whose coefficients are the counts.

Closed form by generating function

The tiling numbers have generating function $T(x) = \sum_{n \geq 0} t_n x^n = \dfrac{1}{1 - x - x^2}$, and

$$t_n = \frac{\varphi^{n+1} - \psi^{n+1}}{\sqrt{5}}, \qquad \varphi = \frac{1+\sqrt{5}}{2} \approx 1.618, \quad \psi = \frac{1-\sqrt{5}}{2} \approx -0.618.$$

Proof. Multiply the recurrence by x^n and sum over $n \geq 2$: $T(x) - 1 - x = x\,(T(x) - 1) + x^2 T(x)$. Solving, $T(x)(1 - x - x^2) = 1$. The denominator factors as $(1 - \varphi x)(1 - \psi x)$, because $\varphi + \psi = 1$ and $\varphi\psi = -1$. Partial fractions give

$$\frac{1}{(1-\varphi x)(1-\psi x)} = \frac{1}{\varphi - \psi}\left(\frac{\varphi}{1-\varphi x} - \frac{\psi}{1-\psi x}\right),$$

and expanding each term as a geometric series, the coefficient of x^n is $(\varphi^{n+1} - \psi^{n+1})/(\varphi - \psi)$, with $\varphi - \psi = \sqrt{5}$. ∎

Since $|\psi| < 1$, the second term dies out and t_n is the nearest integer to $\varphi^{n+1}/\sqrt{5}$: tilings grow like 1.618^n, an exponential, which is the configuration explosion of the hook in miniature. The growth rate is set by the root of the denominator closest to the origin, $x = 1/\varphi$, exactly like the dominant pole of a discrete-time system.

Generating functions do far more than solve recurrences. A product of generating functions counts combined choices; that is how one counts **partitions** of an integer (ways to write n as a sum of positive integers, ignoring order). The generating function is $\prod_{k \geq 1} \frac{1}{1 - x^k}$, and $p(100) = 190{,}569{,}292$.

? A formula with π in it, for a problem with no circles

The number of partitions grows like $p(n) \sim \dfrac{1}{4n\sqrt{3}} \exp\!\left(\pi\sqrt{2n/3}\right)$. A question about adding whole numbers has an answer involving π, e and $\sqrt{3}$. Hardy and Ramanujan found it by treating the generating function as a function of a *complex* variable and studying how it blows up near the unit circle. That method of reading growth from singularities is built on the tools of Ch. 18, and it is the gateway to analytic combinatorics.

Result 4: the pigeonhole principle and the party problem

The pigeonhole principle

If more than k objects are placed into k boxes, some box contains at least two objects. More generally, if N objects go into k boxes, some box contains at least $\lceil N/k \rceil$.

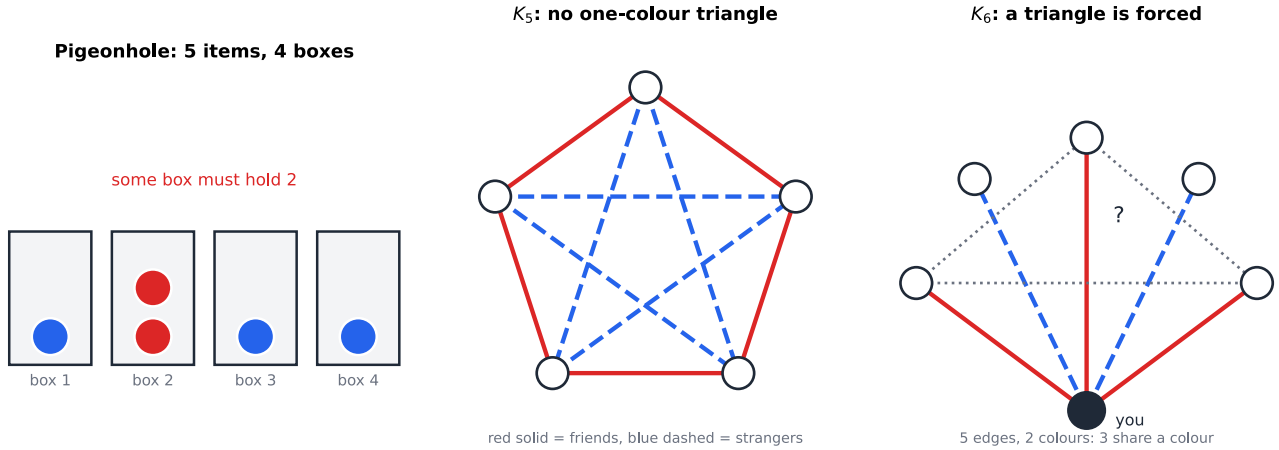
Proof. If every box had at most $\lceil N/k \rceil - 1 < N/k$ objects, the total would be less than $k \cdot N/k = N$. ∎

It is the most obvious statement in this book and one of the most powerful, because it proves existence without construction.

Why no compressor can shrink every file

There are 2^n files of exactly n bits, but only $2^0 + 2^1 + \cdots + 2^{n-1} = 2^n - 1$ files shorter than n bits. A lossless compressor must map different files to different outputs, so by pigeonhole at least one n-bit file cannot get shorter. Every compressor that shrinks some files must enlarge others. How much compression is possible *on average* is the entropy bound of Ch. 33.

Now the party problem. At any party of six people, either some three all know each other, or some three are mutual strangers. Model people as vertices of the complete graph K_6, colour each edge red (acquainted) or blue (strangers). The claim is that every red/blue colouring of K_6 contains a **monochromatic triangle**.



Pigeonhole (left) powers Ramsey's theorem. Five guests are not enough: the pentagon/pentagram colouring of K_5 (middle) has no single-colour triangle. With six, any vertex has five edges, so three share a colour; whatever colour the dotted edges get, a one-colour triangle appears (right).

The party theorem: $R(3,3) = 6$

Every 2-colouring of the edges of K_6 contains a monochromatic triangle, and K_5 has a 2-colouring with none.

Proof. Fix a vertex v of K_6. It has 5 edges and there are 2 colours, so by pigeonhole at least $\lceil 5/2 \rceil = 3$ of them share a colour, say red, going to a, b, c. If any of the edges ab, bc, ca is red, it forms a red triangle with v. If none is red, then a, b, c form a blue triangle. Either way a monochromatic triangle exists. For K_5, colour the five outer edges of a pentagon red and the five diagonals blue: the red edges form a 5-cycle and the blue edges form another 5-cycle (the pentagram), and a 5-cycle contains no triangle. ■

The **Ramsey number** $R(s,t)$ is the smallest N such that every red/blue colouring of K_N contains a red K_s or a blue K_t. Ramsey's theorem says it always exists: complete disorder is impossible, and any large enough structure contains an orderly piece. The slogan of the field: in a big enough system, patterns are not a coincidence but a certainty.

But the numbers are brutal. $R(3,3) = 6$; $R(4,4) = 18$. For $R(5,5)$, the best we know is $43 \le R(5,5) \le 46$ (the upper bound was lowered to 46 only in 2024). Why can't we just check? Settling whether 43 suffices means examining 2-colourings of K_{43}, and there are $2^{\binom{43}{2}} = 2^{903} \approx 10^{272}$ of them. The configuration explosion of the hook is the very thing that keeps a five-person party question unsolved.

? Five friends, and a problem that defeats every computer

What is $R(5,5)$? Symmetry and clever pruning got the bounds to $43 \le R(5,5) \le 46$, but no known method closes the gap. For $R(6,6)$ the gap is far wider: at least 102, and an upper bound still in the 160s. It is a genuine open problem; how mathematicians chip away at problems like this (and why a famous quip says we would be wiser to fight an alien army than to compute $R(6,6)$) is in Ch. 98.

Extremal combinatorics: how much can you have without a pattern?

Ramsey asks when a pattern is forced. **Extremal combinatorics** asks the quantitative version: how large can a structure be while still avoiding the pattern? The first result of the subject is a gem.

Mantel's theorem

A graph on n vertices with no triangle has at most $n^2/4$ edges, and the complete bipartite graph with two halves of size $n/2$ achieves this.

Proof. If uv is an edge, u and v have no common neighbour (that would make a triangle), so $\deg u + \deg v \le n$. Summing over all edges, each vertex w appears in $\deg w$ edges, so $\sum_w (\deg w)^2 \le n|E|$. By the Cauchy–Schwarz inequality, $\sum_w (\deg w)^2 \ge \frac{1}{n}\left(\sum_w \deg w\right)^2 = \frac{4|E|^2}{n}$ (handshake lemma, Ch. 27). Combining, $4|E|^2/n \le n|E|$, so $|E| \le n^2/4$. ■

So a network of 100 nodes with more than 2500 links *must* contain a triangle, whatever its wiring. The extremal example, two halves with every link between them, is exactly the "two communities that only talk across" picture: the extremal answer has structure.

28.4 The algorithm

Combinatorics changes how you compute in two opposite ways: it tells you how to compute huge counts *exactly* without listing anything, and it tells you when a search is hopeless.

Counting without listing: dynamic programming over a recurrence

The partition numbers satisfy no simple formula, but the generating function $\prod_k 1/(1-x^k)$ translates directly into a loop: multiply in one factor $1/(1-x^k) = 1 + x^k + x^{2k} + \cdots$ at a time.

```
def partitions(N):
    """p[n] = number of partitions of n, for n = 0..N."""
    p = [1] + [0] * N                # the series for the empty product: 1
    for k in range(1, N + 1):        # multiply by 1/(1 - x^k)
        for n in range(k, N + 1):
            p[n] += p[n - k]         # allow any number of parts equal to k
    return p

print(partitions(100)[100])         # 190569292
```

That is $O(N^2)$ additions to compute a number that counts 190 million objects, none of which is ever constructed. The same pattern (a recurrence evaluated bottom-up, from small cases to large) counts lattice paths, tilings, binary trees and coin-change combinations; it is the counting cousin of dynamic programming (Ch. 52).

Binomial coefficients without overflow

Computing $\binom{n}{k}$ as `factorial(n) // (factorial(k) * factorial(n - k))` builds enormous intermediate numbers. In fixed-width arithmetic it overflows long before the answer does: $\binom{67}{33}$ fits in 64 bits but $67!$ needs more than 300 bits. The fix multiplies and divides alternately,

$$\binom{n}{k} = \prod_{i=1}^{k} \frac{n-k+i}{i},$$

where after step i the running product is itself the integer $\binom{n-k+i}{i}$, so every division is exact. Use $k \leftarrow \min(k, n-k)$ to halve the work. In floating point, work with logarithms instead: `math.lgamma(n + 1)` gives $\ln n!$ without ever forming $n!$, which is how probabilities like $\binom{1000}{500}2^{-1000}$ are computed safely.

Exhaustive search, and how to survive it

Sometimes you must actually enumerate (to find a colouring of K_5 with no monochromatic triangle, or to test a conjecture on small cases). Three techniques push the frontier of what is feasible:

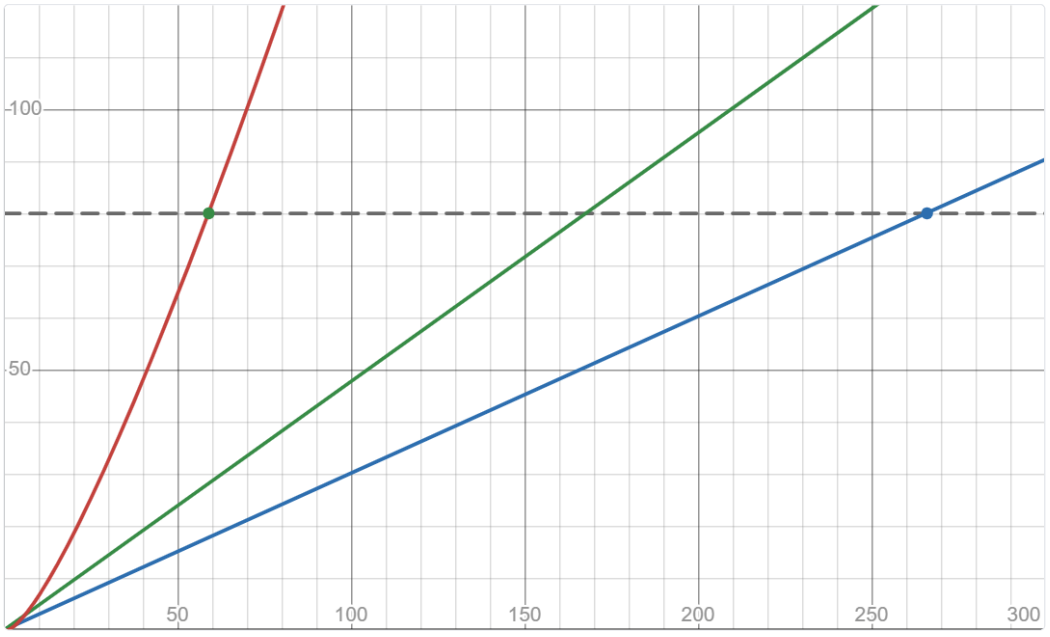
- **Backtracking**: build a configuration choice by choice and abandon a partial one the moment it is doomed. Colouring edges of K_N one at a time and stopping at the first one-colour triangle prunes almost all of the $2^{\binom{N}{2}}$ colourings.
- **Symmetry breaking**: relabelling the vertices or swapping colours gives an equivalent colouring, so search only one representative per class, saving up to a factor $2 \cdot N!$.
- **Reduction to SAT**: encode "no monochromatic K_s" as a Boolean formula and hand it to an industrial satisfiability solver. Several Ramsey-type results were settled this way, with machine-checkable certificates.

Even so, the counts in this chapter are a warning: every exponential search hits a wall, and the wall's position moves only logarithmically with your hardware (Ch. 26). Whether some cleverer method could always avoid the search is the question of Ch. 30.

28.5 The experiment

Where does a configuration space pass the universe?

Plot $\log_{10}$ of each count against the number of parts n. The horizontal line at 80 is the number of atoms in the observable universe.



Blue: 2^n (bits). Green: 3^n (Go points). Red: $n!$ via Stirling (orderings of a deck). The grey line is 10^{80}.

The blue line crosses the universe at $n \approx 266$, the green at $n \approx 168$, and the red at only $n \approx 59$: a deck of 59 cards has more orderings than the universe has atoms. Change the **3** in the green curve to **10** (decimal digits) and then to **26** (letters): how long a password beats the universe? Then add the line `y=x^{2}\frac{\ln(2)}{\ln(10)}` : that is the number of black-and-white images of $n \times n$ pixels, and it leaves the screen almost immediately.

The party and the hats (run it)

The script does two things. It checks the party theorem by brute force over every colouring of K_N, and it simulates the hat-check problem against the inclusion–exclusion formula.

```
import itertools
import numpy as np
import matplotlib.pyplot as plt

# Part 1: party problem by brute force. Is there a 2-colouring of K_N with no one-colour triangle?
N = 5                                  # KNOB: try 5, then 6 (7 takes a few minutes)
edges = list(itertools.combinations(range(N), 2))
index = {e: i for i, e in enumerate(edges)}
tris = [(index[(a, b)], index[(a, c)], index[(b, c)]) for a, b, c in itertools.combinations(range(N), 3)]
good = 0
for colours in itertools.product((0, 1), repeat=len(edges)):
    if all(not (colours[x] == colours[y] == colours[z]) for x, y, z in tris):
        good += 1
print(f"K_{N}: {2 ** len(edges)} colourings, {good} with no one-colour triangle")

# Part 2: the hat-check problem. Return n hats at random; chance nobody gets their own?
rng = np.random.default_rng(0)
TRIALS = 20000                         # KNOB: more trials, less noise
ns = np.arange(1, 13)
sim = [np.mean([not np.any(rng.permutation(n) == np.arange(n)) for _ in range(TRIALS)]) for n in ns]
exact = [sum((-1) ** k / np.prod(np.arange(1, k + 1)) for k in range(n + 1)) for n in ns]
plt.plot(ns, sim, "o", label="simulation")
plt.plot(ns, exact, "-", label="inclusion-exclusion")
plt.axhline(np.exp(-1), ls=":", label="1/e")
plt.xlabel("number of hats n"); plt.ylabel("P(no one gets own hat)"); plt.legend(); plt.show()
```

With `N = 5` it reports 1024 colourings, of which exactly 12 avoid a one-colour triangle. Each of them is a pentagon/pentagram picture like the figure, with the vertices relabelled. With `N = 6` it reports 32,768 colourings and **zero** good ones: the party theorem, verified by exhaustion. Try `N = 7` and notice that the run time grows by a factor of more than 64 for one extra guest: you are watching the configuration explosion from the inside. Can you make it faster by stopping each colouring at its first bad triangle?

In the second part, the probability settles onto $1/e$ by about $n = 5$. Change `TRIALS` to 200 and see how much of the wiggle in the dots is just sampling noise.

28.6 The challenge

1. **Intuition.** In a room of 23 people the chance that two share a birthday is just over one half, yet the chance that someone shares *your* birthday is only about 6%. Explain, by counting *pairs* rather than people, why these two numbers are so different, without computing either one exactly.
2. **Derivation.** Count the number of surjections (onto functions) from a set of n elements to a set of k elements using inclusion–exclusion, with A_i the functions that miss element i. Check your formula for $n = 4$, $k = 2$ by listing, and use it to show that $\sum_{j=0}^{k} (-1)^j \binom{k}{j} (k-j)^n = 0$ whenever $n < k$.

3. **Implementation.** Write a backtracking search that 2-colours the edges of K_N one at a time and backtracks as soon as a red K_3 or a blue K_4 appears. Use it to find a colouring of K_8 with neither, and estimate how many partial colourings your search visits compared with the full 2^{28}. (The fact $R(3,4)=9$ says K_9 has no such colouring; can your code confirm it in reasonable time?)
4. **Open investigation.** Colour each edge of K_N red or blue **at random**. Estimate by simulation the probability that there is no monochromatic K_4, for N from 6 to 17. How does it fall off as N approaches $R(4,4)=18$? Then compute the *expected number* of monochromatic K_4s exactly by counting. What does the comparison suggest about how good random colourings are at avoiding patterns, and how far could the idea be pushed for larger k?

28.7 Loose threads

The challenge is answered. A problem has more configurations than the universe has atoms because configurations **multiply** while matter **adds**: 266 binary choices, 168 three-way choices or 59 cards in order are enough. And combinatorics lets us reason about such spaces without listing them: counting by multiplication and bijection, correcting overcounts by inclusion–exclusion, solving recurrences with generating functions, and forcing patterns to exist with the pigeonhole principle and Ramsey's theorem.

What we left on the table:

- **Lucas's theorem**, the arithmetic behind the fractal in Pascal's triangle, waits in Ch. 57.
- **Analytic combinatorics**: the π in the partition asymptotics comes from complex analysis (Ch. 18), and the same method turns any generating function into its growth rate.
- **Ramsey numbers**: $R(5,5)$ is unknown, and the frontier of such problems is in Ch. 98. The challenge's fourth problem hints at the most surprising tool for it.
- **The probabilistic method.** In 1947 Erdős proved that $R(k,k)>2^{k/2}$ without exhibiting a single good colouring: he showed that a *random* colouring works with positive probability. It is a proof of existence by counting random worlds, the subject of Ch. 32. Nobody knows how to *construct* colourings nearly as good.
- **Counting as logic.** Every proof in this chapter quietly used induction or double counting. What makes such arguments airtight, and how to prove a statement for *all* n at once, is Ch. 29.

Further reading

- R. A. Brualdi, *Introductory Combinatorics*, 5th ed., Pearson, 2010. A gentle, complete first course: counting, inclusion–exclusion, recurrences, generating functions, pigeonhole and Ramsey.
- H. S. Wilf, *generatingfunctionology*, 3rd ed., A K Peters, 2006. The classic on generating functions, freely available from the author.
- P. Flajolet, R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009. How singularities of generating functions determine growth rates.
- N. Alon, J. H. Spencer, *The Probabilistic Method*, 4th ed., Wiley, 2016. Erdős's random-colouring idea and everything it grew into.

Chapter 29 — Discrete Mathematics and Computational Logic

> **Level 2 · Track A — The Mathematics Behind Algorithms Builds on:** Ch. 1, Ch. 26 · **Leads to:** Ch. 30, Ch. 65, Ch. 67 · **Related:** Ch. 28, Ch. 63, Ch. 79 **Reading time:** ~25 min

29.1 The hook

Binary search is the first "clever" algorithm most programmers meet. You have a sorted array, you look at the middle element, and you throw away the half that cannot contain your target. Ten lines of code. Everyone has written it.

Almost everyone has written it wrong. In his book *Programming Pearls*, Jon Bentley reports giving the problem to professional programmers in courses. Given a couple of hours, about ninety percent of them handed in code with a bug. Knuth notes that binary search was first published in 1946, but the first version that worked for every array size did not appear until 1962.

Then there is the bug nobody saw. In 2006 Joshua Bloch, who had written the binary search in Java's standard library, reported that it had been broken for about nine years. The bug was in this line, which also appears in Bentley's own carefully argued version:

```
mid = (low + high) // 2
```

With 32-bit integers, `low + high` overflows once the array has more than about $2^{30} \approx 1.07 \times 10^9$ elements. The sum wraps round to a negative number, `mid` becomes negative, and the program crashes or reads garbage. When the code was written, arrays that large were rare. By 2006 they were routine.

So the code had been read by experts, published in a classic book, shipped to millions of machines and tested, and it was still wrong. Testing could not have saved it. Two 32-bit indices give $2^{64} \approx 1.8 \times 10^{19}$ input pairs; at a billion tests per second, trying them all takes about 585 years. Edsger Dijkstra's remark is the one to keep in mind: testing can show that bugs are present, but never that they are absent.

That leaves the challenge of this chapter:

> **How do you prove that a program must work for every valid input?**

You cannot run infinitely many cases. But you can write a *finite* argument that covers them all. That idea, a finite proof standing in for an infinite set of tests, is the core of discrete mathematics, and it is what this chapter builds.

29.2 The big idea

An electrical engineer analysing a circuit never simulates every possible voltage waveform. She writes Kirchhoff's current law at each node: *the current in equals the current out, at every instant*. That one statement holds whatever the inputs do, and every later step of the analysis leans on it. It is a **conserved quantity**.

Programs have conserved quantities too. They are called **invariants**. Look at binary search again. At every moment the array splits into three zones (Figure 29.3, in §29.3):

- everything to the left of `lo` is known to be smaller than the target,
- everything from `hi` onwards is known to be larger,
- the target, if it is anywhere, is in the window between them.

Each step of the loop shrinks the window, *and the three statements stay true*. When the window is empty the invariant tells you the answer: the target is not there. You never had to try a single input. You proved one statement about one step, and an induction argument carried it through every step of every run.

To make that rigorous you need a small toolkit:

1. a language precise enough to state what "correct" means (**logic**),
2. objects to state it about (**sets, relations, functions**),
3. a way to push a claim from one step to the next, forever (**induction**),
4. a guarantee that the loop actually stops (**well-founded orders**).

Underneath all of it sits a very small algebra of two values, true and false, which turns out to be the same algebra your laptop's transistors compute. We start there.

29.3 The mathematics

Propositional logic and Boolean algebra

A **proposition** is a statement that is either true or false. We combine propositions p, q with connectives: $\neg p$ (not), $p \wedge q$ (and), $p \vee q$ (or), $p \oplus q$ (exclusive or), $p \Rightarrow q$ (implies). Ch. 1 introduces the symbols and the contrapositive in §1.10; here we treat them as *functions*. Write 1 for true and 0 for false. Then each connective is a function from $\{0,1\}^k$ to $\{0,1\}$, and a truth table is just its table of values (Figure 29.1).

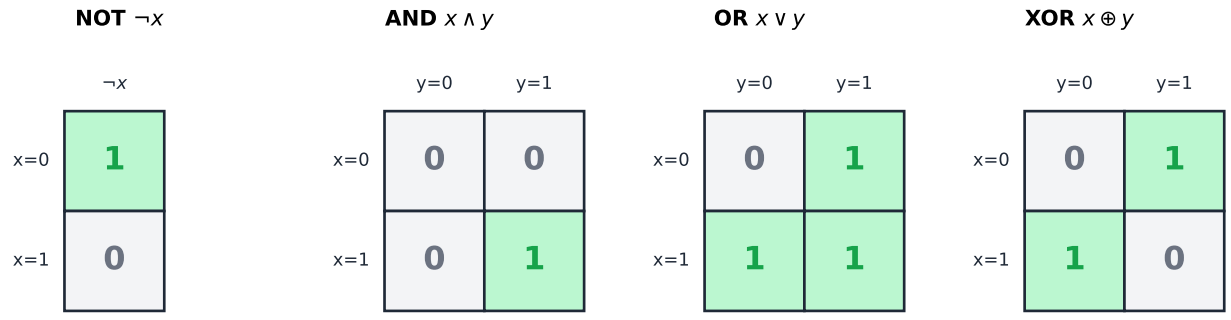


Figure 29.1 — The four basic connectives as grids. Green cells are 1 (true). AND has a single green cell, OR a single grey one, and XOR is green exactly where the inputs differ.

Boolean function, formula, equivalence

A **Boolean function** of n variables is any map $f : \{0,1\}^n \to \{0,1\}$. A **formula** is an expression built from variables with $\neg, \wedge, \vee$. Two formulas are **equivalent**, $\varphi \equiv \psi$, if they have the same truth table. A formula is a **tautology** if it is 1 under every assignment, and **satisfiable** if it is 1 under at least one.

There are 2^n rows in a truth table on n variables, and each row can be 0 or 1, so there are 2^{2^n} Boolean functions: 16 for $n = 2$, 256 for $n = 3$, and 2^{64} for $n = 6$. The algebra obeys a short list of laws, each checkable from the grids:

$$\begin{aligned} &p \wedge q \equiv q \wedge p, && p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r),\\ &p \vee q \equiv q \vee p, && p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r),\\ &\neg(p \wedge q) \equiv \neg p \vee \neg q, && \neg(p \vee q) \equiv \neg p \wedge \neg q,\\ &p \vee \neg p \equiv 1, && p \wedge \neg p \equiv 0. \end{aligned}$$

The third line is **De Morgan's laws**. Notice the second distributive law, $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$: it fails for ordinary numbers ($2 + 3 \cdot 4 \neq 5 \cdot 6$) but holds here. Boolean algebra is not arithmetic in disguise. Its closest arithmetic relative is XOR, which is addition modulo 2: $p \oplus q = p + q \bmod 2$. Pair XOR with AND as multiplication and you have the two-element field $\mathbb{F}_2$, the arithmetic behind error-correcting codes in Ch. 79.

How many connectives do we actually need? Here is the first result worth proving.

Theorem 29.1 — Every Boolean function has a formula

Every Boolean function $f : \{0,1\}^n \to \{0,1\}$ is equivalent to a formula using only $\neg, \wedge, \vee$. Consequently the single connective NAND, $p \uparrow q := \neg(p \wedge q)$, can express every Boolean function.

Proof. If f is never 1, take the formula $x_1 \wedge \neg x_1$. Otherwise, for each input $\mathbf{a} = (a_1, \dots, a_n)$ with $f(\mathbf{a}) = 1$ build the **minterm**

$$m_{\mathbf{a}} = \ell_1 \wedge \ell_2 \wedge \cdots \wedge \ell_n, \qquad \ell_i = \begin{cases} x_i & a_i = 1,\\ \neg x_i & a_i = 0. \end{cases}$$

By construction $m_{\mathbf{a}}$ is 1 on input $\mathbf{a}$ and on no other input: on any $\mathbf{b} \neq \mathbf{a}$ some coordinate differs, and the matching literal is 0. Now take the OR of all these minterms,

$$\varphi = \bigvee_{\mathbf{a}\,:\, f(\mathbf{a})=1} m_{\mathbf{a}}.$$

On an input where $f = 1$, its own minterm is 1, so $\varphi = 1$. On an input where $f = 0$, every minterm is 0, so $\varphi = 0$. Hence $\varphi \equiv f$. This is the **disjunctive normal form**.

For NAND it is enough to build $\neg$, $\wedge$ and $\vee$ from it:

$$\neg p \equiv p \uparrow p, \qquad p \wedge q \equiv (p \uparrow q) \uparrow (p \uparrow q), \qquad p \vee q \equiv (p \uparrow p) \uparrow (q \uparrow q).$$

The first holds because $p \wedge p = p$. The second negates a NAND. The third is De Morgan: $\neg(\neg p \wedge \neg q) \equiv p \vee q$. Substitute these into the DNF and only NAND remains. $\square$

This is why a chip factory can build an entire processor from one kind of gate. It also hides a trap. The DNF of f can have up to 2^n minterms; the parity function $x_1 \oplus \cdots \oplus x_n$ needs all 2^{n-1} of its minterms in DNF. "Expressible" and "expressible *cheaply*" are very different statements.

Satisfiable is not the same as easy to check

Deciding whether a formula is a tautology is the same as deciding whether its negation is unsatisfiable. The obvious method fills in all 2^n rows of the truth table. For $n = 300$ variables, a typical size for a small hardware-verification problem, that is more rows than there are atoms in the observable universe.

Predicate logic: saying "for every input"

Propositional logic cannot say "for every input". For that we need variables that range over a domain, **predicates** about them, and the quantifiers $\forall$ and $\exists$ (their order and negation rules are in §1.10). With them, a specification becomes a precise sentence. "The array $a[0..n)$ is sorted" is

$$\forall i \, \forall j \, \big(0 \le i < j < n \Rightarrow a[i] \le a[j]\big),$$

and "search returns r correctly" is

$$\big(r \geq 0 \wedge a[r] = t\big) \;\vee\; \big(r = -1 \wedge \forall i\,(0 \leq i < n \Rightarrow a[i] \neq t)\big).$$

Many real bugs are wrong specifications, not wrong code. If t appears three times, the predicate above accepts any of its indices; a caller who assumed "the first one" has a bug no proof will catch.

Sets, relations and orders

A **relation** R on a set S is a subset of $S \times S$; we write $x\,R\,y$ for $(x, y) \in R$. Three families matter constantly in computing.

An **equivalence relation** is reflexive ($x\,R\,x$), symmetric ($x\,R\,y \Rightarrow y\,R\,x$) and transitive ($x\,R\,y \wedge y\,R\,z \Rightarrow x\,R\,z$). Every equivalence relation chops S into disjoint **classes**, and every partition comes from one. "Same remainder mod 3" splits the integers into three classes; Ch. 30 builds a machine whose states *are* those classes.

A **partial order** $\preceq$ is reflexive, transitive and antisymmetric ($x \preceq y \wedge y \preceq x \Rightarrow x = y$). Not every pair need be comparable: neither of 6 and 10 divides the other. We draw a partial order as a **Hasse diagram**, with an edge from x up to y when y covers x (Figure 29.2).

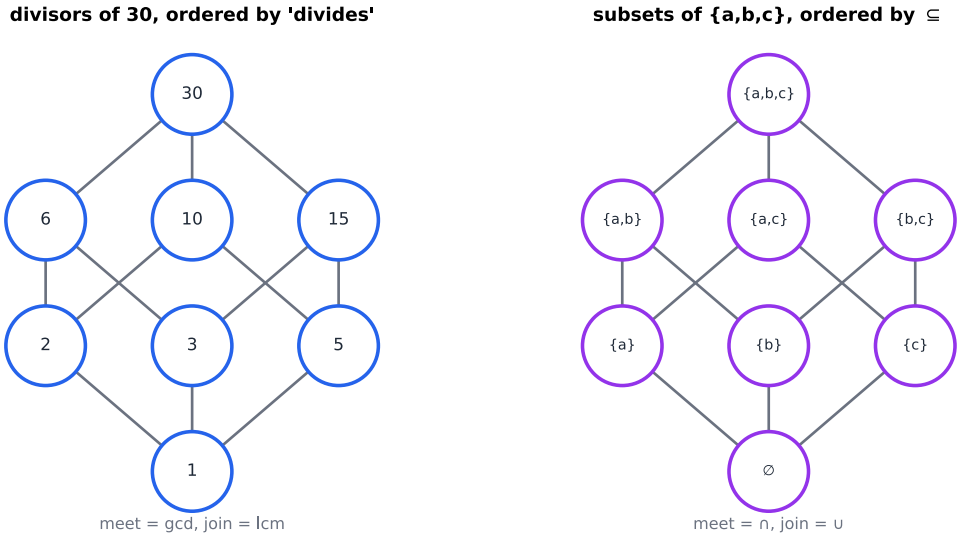


Figure 29.2 — Two partial orders with the same shape: the divisors of 30 under "divides" and the subsets of $\{a, b, c\}$ under $\subseteq$. Both are cubes, because $30 = 2 \cdot 3 \cdot 5$ and each divisor picks a subset of the three primes.

In both orders every pair has a greatest lower bound (gcd, or intersection) and a least upper bound (lcm, or union). Such an order is a **lattice**. In the cube on the right, the meet is $\wedge$, the join is $\vee$, and complement is $\neg$: the subsets of an n-element set form a Boolean algebra, the same algebra as the truth tables. Hardware, set theory and number theory meet in one picture.

A partial order is **well-founded** if it has no infinite descending chain $x_0 \succ x_1 \succ x_2 \succ \cdots$. The natural numbers under $\leq$ are the standard example, and they are the reason loops stop.

Induction, three ways

The **principle of induction** says: if $P(0)$ holds and $P(k) \Rightarrow P(k+1)$ for every k, then $P(n)$ holds for every $n \in \mathbb{N}$. Ch. 1 gives the standard worked example in §1.10. Three variants do most of the work in computer science.

- **Strong induction:** assume $P(0), \ldots, P(k)$ all hold and deduce $P(k+1)$. This is the natural form for divide-and-conquer, where a problem of size $k+1$ calls itself on sizes like $\lfloor (k+1)/2 \rfloor$.
- **Well-ordering:** every non-empty subset of $\mathbb{N}$ has a least element. To prove $P(n)$ for all n, suppose not, take the *smallest* counterexample, and derive a contradiction by finding a smaller one. This is equivalent to induction, and it is often the easier way to think.
- **Structural induction:** to prove something about every formula, tree or list, prove it for the atoms and show that each constructor preserves it. The proof of Theorem 29.1 is essentially a structural induction in disguise: we built the formula from literals up.

Loop invariants: induction over time

Now the central tool. Consider a loop

```
INIT
while GUARD:
    BODY
```

Theorem 29.2 — The loop-invariant principle

Let I be a predicate on the program state and V an integer-valued function of the state. Suppose

1. **Initialisation:** I holds after INIT.
2. **Maintenance:** if I and GUARD hold before BODY, then I holds after it.
3. **Variant:** whenever I and GUARD hold, $V \geq 0$, and each execution of BODY strictly decreases V.

Then the loop terminates, and on exit $I \wedge \neg$GUARD holds.

Proof. Let s_k be the state after k executions of BODY (if the loop runs that long). We show by induction on k that $I(s_k)$ holds. The base case $k = 0$ is initialisation. If $I(s_k)$ holds and the loop runs again, GUARD held in s_k, so maintenance gives $I(s_{k+1})$. That proves the claim for every k the loop reaches.

For termination, suppose the loop ran forever. Then GUARD and I hold in every s_k, so the variant gives $V(s_0) > V(s_1) > V(s_2) > \cdots$, an infinite strictly decreasing sequence of non-negative integers. By well-ordering the set $\{V(s_k)\}$ has a least element $V(s_m)$, yet $V(s_{m+1}) < V(s_m)$ is in the set: contradiction. So the loop stops after some K iterations. It stopped because GUARD failed, and $I(s_K)$ holds by the induction. □

The theorem converts one infinite question (does this work on all inputs?) into three finite ones, each about a *single* step. Let us cash it in.

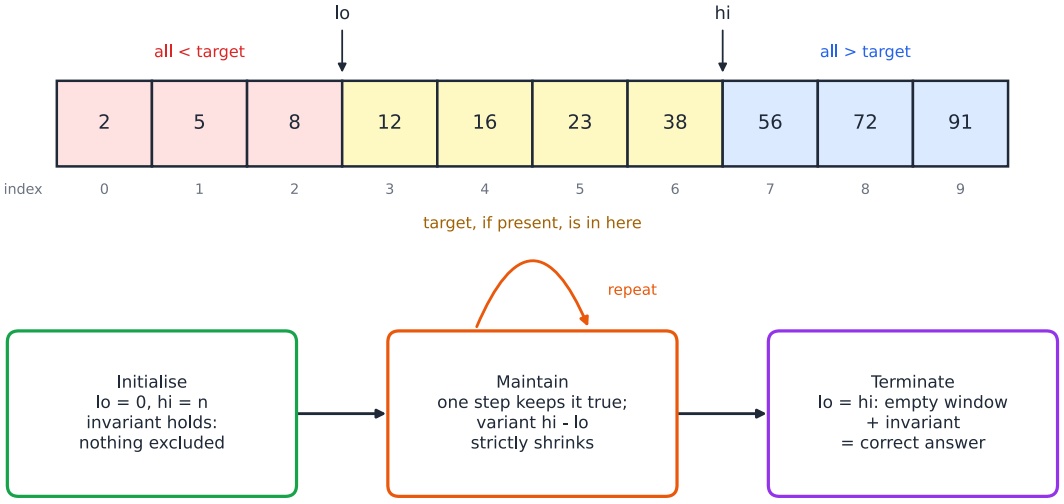


Figure 29.3 — Binary search mid-run, searching for 38 or anything else. Red cells are proved smaller than the target, blue cells proved larger; the yellow window is all that is left. The three boxes are the three hypotheses of Theorem 29.2.

Theorem 29.3 — Binary search is correct

Let $a[0..n)$ be sorted. The loop `lo, hi = 0, n; while lo < hi: mid = lo + (hi-lo)//2; if a[mid] < t: lo = mid+1 elif a[mid] > t: hi = mid else: return mid`, followed by `return -1`, always terminates and meets the specification of the previous subsection. With exact integers, every index it touches lies in $[0, n)$.

Proof. Take the invariant

$$I: \quad 0 \leq lo \leq hi \leq n \ \wedge\ \forall i < lo:\ a[i] < t \ \wedge\ \forall i \geq hi\,(i < n):\ a[i] > t,$$

and the variant $V = hi - lo$.

Initialisation. With $lo = 0$, $hi = n$ both quantified parts are about empty ranges, so they hold vacuously.

Maintenance. Suppose I holds and $lo < hi$. Then $mid = lo + \lfloor (hi - lo)/2 \rfloor$ satisfies $lo \leq mid < hi \leq n$, so $a[mid]$ is a valid read. If $a[mid] < t$, sortedness gives $a[i] \leq a[mid] < t$ for every $i \leq mid$, so setting $lo = mid + 1$ keeps the left clause, and $lo \leq hi$ still holds since $mid < hi$. If $a[mid] > t$, then $a[i] \geq a[mid] > t$ for every $i \geq mid$, so $hi = mid$ keeps the right clause. If $a[mid] = t$ we return a correct index.

Variant. $V \geq 1$ while the guard holds. The new window is $[mid + 1, hi)$ or $[lo, mid)$, both strictly shorter than $[lo, hi)$ because $lo \leq mid < hi$.

Exit. By Theorem 29.2 the loop stops. If it stops without returning, $lo = hi$, and the invariant says every element is either $< t$ (index below lo) or $> t$ (index at or above hi). So t is absent and -1 is correct. □

Where was the 2006 bug? Look at the phrase "with exact integers". The proof uses $lo \leq mid < hi$, which is true in $\mathbb{Z}$. In 32-bit arithmetic, $(lo + hi)/2$ can be negative, so the step "$a[mid]$ is a valid read" is false. The proof was never wrong. The *model* was: it assumed integers that machines do not have. Writing $lo + (hi - lo)/2$ keeps every intermediate value between lo and hi, so the proof goes through for machine integers too. A careful proof would have demanded exactly that check; a test suite of arrays under a billion elements never could.

? The invariant nobody gave you

Every proof in this section started with someone *handing* us the invariant. Finding it was the creative step. Could a program read any loop and produce its invariant, or decide whether a given program meets its specification at all? Ch. 30 proves that no algorithm can do this in general, and Ch. 67 shows what verification tools do instead.

A toolbox of proof techniques

Ch. 1 lists direct proof, contrapositive, contradiction and induction (§1.10). Computing adds a few more habits:

Technique	Pattern	Where it shows up
Invariant	a property preserved by every step	loops, protocols, puzzles (the 15-puzzle's parity)
Variant / potential	a quantity in a well-founded order that must drop	termination, amortised analysis (Ch. 26)
Construction	prove $\exists x$ by building x	DNF in Theorem 29.1, Euclid's coefficients
Counterexample	refute $\forall x\, P(x)$ with one x	the 2006 overflow input
Pigeonhole	$n+1$ objects in n boxes	hashing collisions (Ch. 28)
Diagonalisation	build an object that differs from every item in a list	the halting problem (Ch. 30)

Each of these is a finite argument about infinitely many cases. That is the whole trick.

? Is every true statement provable?

Our proofs worked inside predicate logic. Is there a fixed set of rules that proves *every* true sentence of predicate logic? For pure logic, yes, and that is Gödel's completeness theorem. For arithmetic on $\mathbb{N}$, the very structure induction lives in, the answer flips to no. Why the same logician got opposite answers is the story of Ch. 65 and Ch. 68.

29.4 The algorithm

Writing the invariant into the code

A proof is most useful when it lives next to the code. Here is binary search with its invariant and variant written as runtime checks. The checks are expensive (the invariant check is $O(n)$ per step, which ruins the $O(\log n)$ running time), so you switch them off in production, but they turn the proof into something a test can falsify.

```
def binary_search(a, t):
    lo, hi = 0, len(a)
    while lo < hi:
        assert all(x < t for x in a[:lo]) and all(x > t for x in a[hi:])  # invariant I
        v = hi - lo                                                       # variant V
        mid = lo + (hi - lo) // 2          # never leaves [lo, hi), even in 32-bit
        if a[mid] < t:
            lo = mid + 1
        elif a[mid] > t:
            hi = mid
        else:
            return mid
        assert hi - lo < v                                                # V decreased
    return -1
```

The window halves (or better) each step, so after k steps $V \le n/2^k$, and the loop runs at most $\lfloor \log_2 n \rfloor + 1$ times: 30 iterations for a billion elements.

A second example: Euclid's algorithm

The oldest non-trivial algorithm has the cleanest invariant.

```
def gcd(a, b):            # a, b >= 0, not both zero
    while b != 0:
        a, b = b, a % b   # invariant: gcd(a, b) never changes; variant: b
    return a
```

Invariant. Any common divisor of a and b divides $a \bmod b = a - qb$, and any common divisor of b and $a \bmod b$ divides $a = qb + (a \bmod b)$. So the pairs (a, b) and $(b, a \bmod b)$ have exactly the same common divisors and the same gcd. *Variant.* $0 \le a \bmod b < b$, so b strictly decreases in $\mathbb{N}$. *Exit.* When $b = 0$, $\gcd(a, 0) = a$. By Theorem 29.2, the function returns gcd of the original inputs. For $\gcd(1071, 462)$ the states are $(1071, 462) \to (462, 147) \to (147, 21) \to (21, 0)$, and the answer is 21.

The variant gives termination but a weak bound (b steps). A sharper look shows that b at least halves every two steps, so the loop runs $O(\log b)$ times; the worst case is consecutive Fibonacci numbers, which ties this chapter back to recurrences in Ch. 28.

Brute-force logic, and why it does not scale

Theorem 29.1 suggests an algorithm for checking any propositional claim: enumerate the truth table, `all(f(*bits) for bits in itertools.product([False, True], repeat=n))`. This is correct and costs 2^n evaluations. Doubling the variable count squares the work. For the specifications of real programs, which talk about 32- or 64-bit integers, it is hopeless: a single 64-bit variable is already 64 Boolean variables. Industrial tools (SAT solvers) do something far cleverer and routinely handle millions of variables in practice, yet nobody knows an algorithm that is guaranteed to be fast on *every* formula.

> **? The price of a truth table**
>
> Checking that one given assignment satisfies a formula takes time linear in the formula's length. *Finding* one seems to need exponential time. Is that gap real, or have we just not been clever enough? This is the P versus NP question, and the satisfiability problem sits at its very centre: see Ch. 30.

Failure cases

- **Wrong model.** The proof assumes mathematical integers, exact reals, or memory that never runs out; the machine has none of these. Overflow, rounding (Ch. 22) and aliasing break proofs that were valid on paper.
- **Wrong specification.** The proof shows the code meets the predicate you wrote, not the one you meant.
- **Too-weak invariant.** "$0 \le lo \le hi \le n$" is preserved by binary search, but it says nothing about where t is, so it proves nothing useful. A good invariant is strong enough to imply the goal at exit and weak enough to be true at the start.

29.5 The experiment

Proof and exhaustive testing are not enemies. On *small* inputs you can test everything, and a planted bug almost always shows up on a tiny case. The script below checks binary search, with its invariant asserted, on every sorted array of length up to `N_MAX` with entries from `0..VALUES-1`, for every target. Then it reproduces the 2006 overflow with 32-bit numbers.

```
import itertools
import numpy as np

N_MAX = 6        # knob: longest array tested
VALUES = 4       # knob: entries drawn from 0..VALUES-1
BUGGY = False    # knob: True plants an off-by-one bug

def search(a, t):
    lo, hi = 0, len(a)
    while lo < hi:
        # invariant: a[:lo] < t  and  a[hi:] > t
        assert all(x < t for x in a[:lo]) and all(x > t for x in a[hi:])
        mid = lo + (hi - lo) // 2
        if a[mid] < t:
            lo = mid + 1
        elif a[mid] > t:
            hi = mid if not BUGGY else mid - 1
        else:
            return mid
    return -1

cases = 0
for n in range(N_MAX + 1):
    for a in itertools.combinations_with_replacement(range(VALUES), n):
        for t in range(-1, VALUES + 1):
            r = search(list(a), t)
            ok = (r == -1 and t not in a) or (r >= 0 and a[r] == t)
            assert ok, f"wrong answer on a={a}, t={t}"
            cases += 1
print("all", cases, "cases correct")

# the 2006 bug: (lo + hi) // 2 in 32-bit arithmetic
lo, hi = np.int32(1_500_000_000), np.int32(2_000_000_000)
with np.errstate(over="ignore"):
    print("(lo+hi)//2   =", (lo + hi) // np.int32(2))
print("lo+(hi-lo)//2 =", lo + (hi - lo) // np.int32(2))
```

With the default knobs it reports `all 1260 cases correct`, then prints a midpoint of $-397{,}483{,}648$ for the naive formula and $1{,}750{,}000{,}000$ for the safe one.

Things to try:

1. Set `BUGGY = True`. The very first failure is on the array `(0, 1)` with target `0`, an input of length two. Notice *which* assertion fires: not the invariant check but the final answer check, because the loop exits right after the bad step, before the invariant is checked again. Where should a second invariant check go so that it catches the bug itself?
2. Change `mid = lo + (hi - lo) // 2` to `mid = lo + (hi - lo + 1) // 2` (round up instead of down). Does it still pass? Now also change `lo = mid + 1` to `lo = mid`. What goes wrong, and which hypothesis of Theorem 29.2 failed? The **small-scope hypothesis** behind this experiment says that most bugs have a small counterexample. It is a heuristic, not a theorem: the overflow bug has no small counterexample at all, because it needs over a billion elements. That is exactly the case where only a proof helps.

29.6 The challenge

No answers are given. Work them out, then compare with a friend.

1. **Intuition.** A loop starts with $x = 0$, $y = 1$ and repeats `x, y = y, x + y`. Without running it, find a quantity built from x and y that stays the same up to sign on every step (hint: try $y^2 - xy - x^2$). What does your invariant tell you about consecutive Fibonacci numbers, and why does it imply that they are coprime?
2. **Derivation.** Prove that $\{\neg, \vee\}$ is functionally complete but $\{\wedge, \vee\}$ is not. For the second part, find a property that every $\{\wedge, \vee\}$ formula has (structural induction) and that $\neg x$ lacks.
3. **Implementation.** Write insertion sort with an explicit invariant ("the prefix `a[:i]` is sorted and is a permutation of the original prefix") and an explicit variant, both as `assert` statements. Test it exhaustively on every list of length at most 7 over three values, then plant three different bugs and see how small the smallest counterexample is for each.
4. **Open investigation.** The *Collatz loop* `while n != 1: n = n // 2 if n % 2 == 0 else 3 * n + 1` has been checked to terminate for every start below $2^{68} \approx 3 \times 10^{20}$, and nobody can prove it terminates for all n. Starting from 27 it takes 111 steps and climbs to 9232. Try to build a variant function: a quantity that decreases on every step, or on every *block* of steps. Experiment with candidates such as $\log n$ averaged over many steps. Where does each attempt break?

29.7 Loose threads

This chapter answered its challenge. You prove a program works for every valid input by writing a precise specification in predicate logic, finding an invariant, and using induction over the steps of the computation, with a variant in a well-founded order to show that the steps run out. Binary search, the program that fooled experts for decades, fell to a one-page argument, and the argument even pointed at the line where machine integers break the mathematics.

But we cheated in three places, and each one opens a door.

- **We were handed the invariants.** Is finding them a matter of skill, or is there a mechanical method? The first mystery above points to Ch. 30, which proves a startling negative answer for the general question "does this program meet its specification?", and to Ch. 67, where tools such as Hoare logic and proof assistants make the best of it.
- **We checked logic by brute force.** Truth tables grow like 2^n. Whether satisfiability can be decided in polynomial time is a million-dollar open problem, and Ch. 30 explains why so many other problems rise or fall with it.
- **We trusted our proof rules.** Induction, the quantifier rules and the Boolean laws were taken on faith. Are they consistent? Are they enough? Ch. 65 formalises them, and Ch. 68 shows that for arithmetic they can never be enough.

There is one further crack. Theorem 29.2 needed a variant, and the Collatz loop has none that anyone can find. Some loops may stop for every input with no proof that they do. Whether that is a temporary embarrassment or a permanent feature of mathematics is the question the next chapter takes head-on.

Further reading

- K. H. Rosen, *Discrete Mathematics and Its Applications*, 8th ed., McGraw-Hill, 2019. Broad coverage of logic, sets, relations and induction, with many exercises.
- E. Lehman, F. T. Leighton and A. R. Meyer, *Mathematics for Computer Science*, MIT OpenCourseWare. Free, lively, and strong on proofs, invariants and well-ordering.
- J. Bentley, *Programming Pearls*, 2nd ed., Addison-Wesley, 2000. The binary search chapter and the "writing correct programs" essays.
- D. Gries, *The Science of Programming*, Springer, 1981. The classic on developing programs and their invariants together.

Chapter 30 — The Limits of Computation

> **Level 2 · Track A — The Mathematics Behind Algorithms Builds on:** Ch. 26, Ch. 28, Ch. 29, Ch. 24 · **Leads to:** Ch. 65, Ch. 68, Ch. 77 · **Related:** Ch. 49, Ch. 64, Ch. 86 **Reading time:** ~25 min

30.1 The hook

Here is a five-line program.

```
def collatz(n):
    while n != 1:
        n = n // 2 if n % 2 == 0 else 3 * n + 1
    return "done"
```

Does it return for every positive integer n? Start at 27 and it wanders for 111 steps, climbing as high as 9232 before falling to 1. Computers have checked every starting value below $2^{68} \approx 3 \times 10^{20}$, and every one of them reaches 1. Yet nobody can prove that all of them do. The question has been open since 1937. The mathematician Paul Erdős said that mathematics was not yet ready for problems like it.

Now imagine you are the engineer in charge of a code-review tool. Your manager wants a feature: *flag every function that can loop forever*. You would not need to understand Collatz yourself. Your tool would just say "halts" or "loops", and Collatz would be settled. So would a frightening number of other open problems, because many famous conjectures can be written as "this search program never finds a counterexample". Goldbach's conjecture is one: write a loop that tests $4, 6, 8, \ldots$ for being a sum of two primes and stops at the first failure.

In 1928 David Hilbert asked for something even bigger, the **Entscheidungsproblem**: a mechanical procedure that decides, for any mathematical statement, whether it is provable. If it existed, mathematics would become computation.

In 1936 Alan Turing, aged 23, answered: no such procedure exists. Not "we haven't found one yet". It *cannot* exist. To prove that, he first had to say exactly what a "mechanical procedure" is, and in doing so he invented the modern computer on paper.

> **Are there problems that no computer program can ever solve?**

By the end of this chapter you will have a proof that the answer is yes, a sense of how *many* such problems there are, and a map of a second, subtler frontier: problems that can be solved, but perhaps never quickly.

30.2 The big idea

There are two ideas, and both fit in a paragraph.

Idea 1: programs are data. Every program is a finite string of characters. Strings can be listed: first all strings of length 1, then length 2, and so on. So the programs can be numbered $P_1, P_2, P_3, \ldots$. A *problem*, on the other hand, is any yes/no question about inputs, which amounts to any subset of the set of all strings. There are as many such subsets as there are real numbers, which is strictly more than there are whole numbers (Ch. 64 proves Cantor's theorem behind this). There are not enough programs to go round. Almost every problem has no program at all.

Idea 2: a program that feeds on itself. The counting argument says that unsolvable problems exist but does not name one. To name one, Turing used the oldest trick in logic, the liar's paradox ("this sentence is false"). Because programs are data, a program can be given *its own text* as input. Suppose some program claims to predict the behaviour of every program. Build a contrarian that asks the predictor about itself and then does the opposite. The predictor must be wrong about the contrarian. That is the halting problem, and §30.3 makes it rigorous.

The second half of the chapter asks a finer question. Among problems that *can* be solved, some seem to need exponential time. Ch. 29 left one hanging: checking a proposed satisfying assignment of a formula is fast, but finding one appears hard. Is "easy to check" really different from "easy to solve"? That is the P versus NP question, the most famous open problem in computer science.

A useful picture is a ladder of machines. Each rung adds memory and can recognise more:

Machine	Memory	Recognises	Example it can't handle
finite automaton	a fixed number of states	regular languages	$\{0^n 1^n\}$
pushdown automaton	one stack	context-free languages	$\{a^n b^n c^n\}$
Turing machine	unbounded tape	recursively enumerable languages	the complement of HALT

The top rung is as high as the ladder goes. Every computer ever built, and every programming language, sits at exactly that rung or below.

30.3 The mathematics

Strings, languages and finite automata

An **alphabet** Σ is a finite set of symbols, such as $\{0, 1\}$. A **string** is a finite sequence of symbols, Σ^* is the set of all strings (including the empty string ε), and a **language** is any subset $L \subseteq \Sigma^*$. Every yes/no problem is a language: the language of binary numerals divisible by 3, of satisfiable formulas, of programs that halt.

Deterministic finite automaton (DFA)

A DFA is $M = (Q, \Sigma, \delta, q_0, F)$: a finite set of states Q, an alphabet Σ, a transition function $\delta : Q \times \Sigma \to Q$, a start state q_0 and a set of accepting states $F \subseteq Q$. It reads a string one symbol at a time, moving $q \mapsto \delta(q, c)$, and **accepts** if it ends in F. The language it accepts is $L(M)$; languages accepted by some DFA are **regular**.

The state set is the machine's entire memory, so a DFA is the right model for anything with bounded memory: a vending machine, a protocol handshake, a lexer, a digital circuit with flip-flops. Figure 30.1 shows a three-state DFA that reads a binary numeral from the most significant bit and accepts exactly the multiples of 3. Its states are the three equivalence classes of "same remainder mod 3" from Ch. 29. Reading bit b turns a value v into $2v + b$, so the remainder r becomes $(2r + b) \bmod 3$.

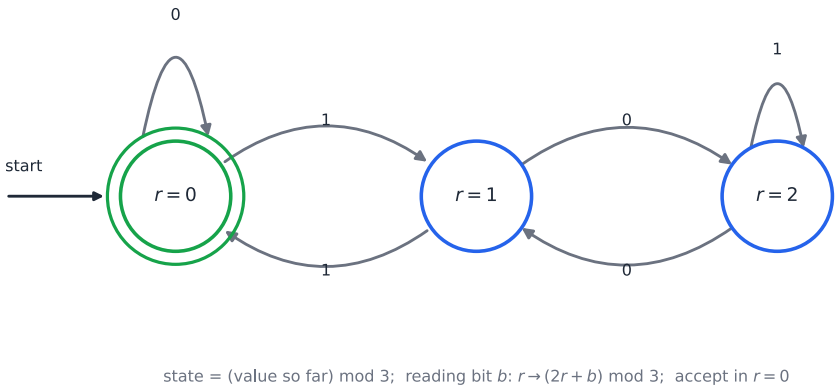


Figure 30.1 — A DFA for "divisible by 3". The double ring marks the accepting state. Trace `110` *(six):* $0 \to 1 \to 0 \to 0$*, accept. Trace* `111` *(seven):* $0 \to 1 \to 0 \to 1$*, reject.*

How much can a finite memory do? Not much, and the proof is the pigeonhole principle from Ch. 28.

Theorem 30.1 — Counting needs unbounded memory

The language $L = \{0^n 1^n : n \geq 0\}$ (some zeros followed by the *same number* of ones) is not regular.

Proof. Suppose a DFA M with k states accepts L. Feed it the $k + 1$ strings $\varepsilon, 0, 00, \ldots, 0^k$ and record the state after each. There are $k + 1$ strings and only k states, so two of them, say 0^i and 0^j with $i < j$, leave M in the same state q. From q, the machine's future depends only on the rest of the input. Append 1^i: the string $0^i 1^i \in L$ must be accepted, so reading 1^i from q ends in an accepting state. But $0^j 1^i$ also reaches q and then reads the same 1^i, so it is accepted too, though $j \neq i$ means $0^j 1^i \notin L$. Contradiction. □

The same argument proves that no DFA checks balanced parentheses, which is why compilers need a stack (a pushdown automaton, the next rung) to parse code. Its general form, the **pumping lemma**, is a standard tool for proving languages non-regular.

Turing machines

Turing's model gives a finite control unlimited scratch space.

Turing machine

A Turing machine has a finite set of states, including a start state and a halting state, and a tape of cells extending without bound, each holding a symbol (blank, written $\lrcorner$, by default). A head sits over one cell. At each step the machine reads the symbol under the head and, according to a finite table $\delta(q, s) = (q', s', D)$, switches to state q', writes s', and moves one cell left or right ($D \in \{L, R\}$). It **halts** if it reaches the halting state. On some inputs it may run forever.

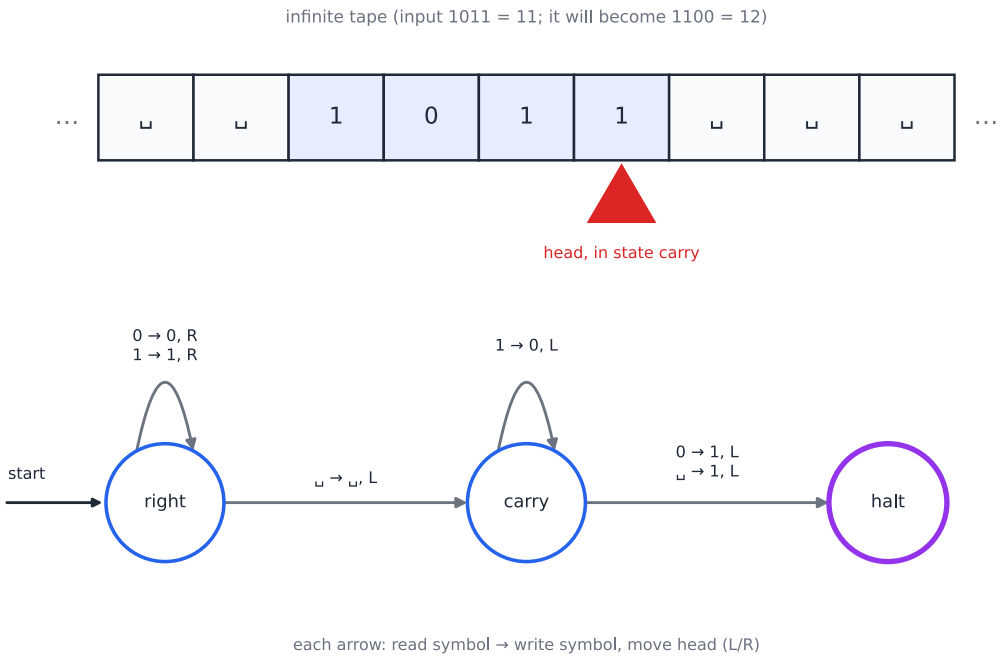


Figure 30.2 — A three-state Turing machine that adds 1 to a binary number. In state right *it runs to the end of the input; in state* carry *it walks back, turning trailing 1s into 0s, until it can write a single 1. The head is shown mid-carry on input 1011.*

This looks absurdly primitive, but every algorithm anyone has written down can be compiled into such a table. Two facts make the model central:

- **Universality.** There is a single Turing machine U that takes as input the description $\langle M \rangle$ of any machine M plus an input w, and simulates M on w. This is a stored-program computer, described nine years before the first one was built.
- **The Church–Turing thesis.** Every procedure that a human or a physical machine can carry out step by step can be done by a Turing machine. This is not a theorem (it relates a mathematical object to an informal idea), but every alternative model ever proposed, from lambda calculus to cellular automata to the programming language you use, has turned out to be equivalent.

Decidable, recognisable and the halting problem

> **Decidable and recognisable**
>
> A language L is **decidable** if some Turing machine halts on every input, accepting exactly the strings in L. It is **recognisable** if some machine accepts exactly the strings in L, but may run forever on strings outside it.

The difference matters. To recognise "programs that halt", just run the program and say yes if it stops. If it never stops, you never answer. The question is whether something cleverer always answers.

> **Theorem 30.2 — The halting problem is undecidable (Turing, 1936)**
>
> The language $\mathrm{HALT} = \{\langle M, w\rangle : M \text{ halts on input } w\}$ is recognisable but not decidable.

Proof. It is recognisable: the universal machine simulates M on w and accepts if the simulation halts.

Suppose, for contradiction, that a machine H decides it: $H(\langle M\rangle, w)$ always halts, answering *yes* if M halts on w and *no* otherwise. Build a new machine D that, on input $\langle M\rangle$:

1. runs $H(\langle M\rangle, \langle M\rangle)$, asking whether M halts when fed its own description;
2. if H says *yes*, loops forever; if H says *no*, halts.

D is a legitimate machine because H always halts. Now run D on its own description $\langle D\rangle$.

- If D halts on $\langle D\rangle$, then $H(\langle D\rangle, \langle D\rangle)$ said *yes*, so by step 2 D loops forever. Contradiction.
- If D loops forever on $\langle D\rangle$, then H said *no*, so by step 2 D halts. Contradiction.

Either way we reach a contradiction, so no such H exists. □

This is **diagonalisation**. List all machines down the side of a table and all inputs across the top, and write in each cell whether that machine halts on that input. D is built to disagree with the diagonal: on input $\langle M_i\rangle$ it does the opposite of what M_i does on $\langle M_i\rangle$. So D differs from every row, yet D is supposed to be a row. It is Cantor's proof that the reals are uncountable, turned into a statement about programs.

What the theorem does and does not say

It does not say you can never tell whether a *particular* program halts: for `while True: pass` you obviously can. It says no single algorithm is right about *every* program. A practical tool can answer "halts", "loops" or "don't know" and be useful, as long as it is allowed to say "don't know".

Reductions: undecidability spreads

Once you have one unsolvable problem, you can find many by **reduction**. Problem A reduces to problem B, written $A \le B$, if a computable function f turns every instance x of A into an instance $f(x)$ of B with the same answer. Then a solver for B would solve A, so if A is undecidable, so is B.

For example, "does program P ever print `hello`?" is undecidable. Given $\langle M, w\rangle$, build the program that silently simulates M on w (with all output suppressed) and then prints `hello`. It prints `hello` exactly when M halts on w. A `hello`-detector would therefore decide HALT.

Pushing this idea to its limit gives a theorem that should worry anyone who writes analysis tools.

Rice's theorem (1953)

Let S be any property of the *function a program computes* (its input-output behaviour, not its text), such that some programs have it and some do not. Then "does program P have property S?" is undecidable.

"Does this program ever divide by zero?", "does it compute the same function as that one?", "does it meet its specification?" are all undecidable in general. This is why the first mystery of Ch. 29 can have no complete mechanical solution.

? The machine that could not prove itself

Turing's diagonal trick looks a lot like the sentence "this statement is not provable". Five years earlier, Kurt Gödel had used that sentence to show that any consistent formal system rich enough for arithmetic leaves some true statements unprovable. Are the two results the same theorem in different clothes, and can one be derived from the other? The story is in Ch. 65 and Ch. 68.

Efficient computation: P and NP

Decidable is not the same as practical. A problem whose best algorithm takes 2^n steps is decidable, but at $n = 300$ it is as good as unsolvable. Complexity theory sorts decidable problems by how their cost grows (the growth-rate language is from Ch. 26).

P and NP

A decision problem is in **P** if some algorithm solves it in time $O(n^k)$ for a fixed k, where n is the input length. It is in **NP** if every *yes*-instance has a **certificate** of polynomial length that a polynomial-time **verifier** can check: the verifier accepts (x, c) for some c exactly when x is a *yes*-instance.

Satisfiability (SAT: does this formula have an assignment making it true?) is in NP, because the certificate is the assignment and checking it takes one pass over the formula. Sorting, shortest paths (Ch. 27) and, since 2002, primality testing are in P. Every problem in P is in NP (the verifier can ignore the certificate and solve the problem), so P $\subseteq$ NP. Whether the inclusion is strict is the open question.

Polynomial-time reductions, $A \le_p B$, are reductions computed in polynomial time. A problem B is **NP-hard** if every problem in NP reduces to it, and **NP-complete** if it is also in NP. An NP-complete problem is a hardest problem in NP: a polynomial algorithm for *one* of them gives a polynomial algorithm for *all* of NP.

Cook–Levin theorem (1971–73)

SAT is NP-complete.

The idea of the proof is to write the entire run of a polynomial-time verifier, tape cell by tape cell and step by step, as a giant but polynomial-size Boolean formula that is satisfiable exactly when some certificate makes the verifier accept. We state it without proof. Once SAT was known to be hard, Richard Karp showed in 1972 that 21 classic problems were NP-complete too, by chains of reductions. Here is one link of such a chain, proved in full.

Let $G = (V, E)$ be a graph. A set $S \subseteq V$ is **independent** if no edge joins two vertices of S; it is a **vertex cover** if every edge has at least one endpoint in S; it is a **clique** if every two vertices of S are joined by an edge.

Theorem 30.3 — Three faces of one problem

For any graph $G = (V, E)$ with $|V| = n$ and any $S \subseteq V$:

1. S is independent in G if and only if $V \setminus S$ is a vertex cover of G;
2. S is independent in G if and only if S is a clique in the complement graph $\bar{G}$ (same vertices, an edge exactly where G has none).

Hence "does G have an independent set of size $\ge k$?", "does G have a vertex cover of size $\le n - k$?" and "does $\bar{G}$ have a clique of size $\ge k$?" have the same answer, and each problem reduces to the others in polynomial time.

Proof. (1) Suppose S is independent and take any edge uv. Not both of u, v lie in S, so at least one lies in $V \setminus S$: the complement covers every edge. Conversely, if $V \setminus S$ covers every edge, no edge can have both endpoints in S, so S is independent.

(2) Two distinct vertices $u, v \in S$ are non-adjacent in G exactly when they are adjacent in $\bar{G}$. So "no pair in S is adjacent in G" and "every pair in S is adjacent in $\bar{G}$" are the same statement.

The reductions are the maps $(G, k) \mapsto (G, n - k)$ and $(G, k) \mapsto (\bar{G}, k)$. The first is free; the second writes down $\binom{n}{2}$ adjacency bits, which takes $O(n^2)$ time. $\square$

A classic reduction sends 3-SAT to Independent Set (by a gadget that turns each clause into a triangle); combined with Theorem 30.3 this makes Clique and Vertex Cover NP-complete as well. Thousands of problems are now known to be NP-complete: scheduling, timetabling, circuit layout, protein folding models, Sudoku on $n^2 \times n^2$ grids, and the integer programs of Ch. 49. They stand or fall together.

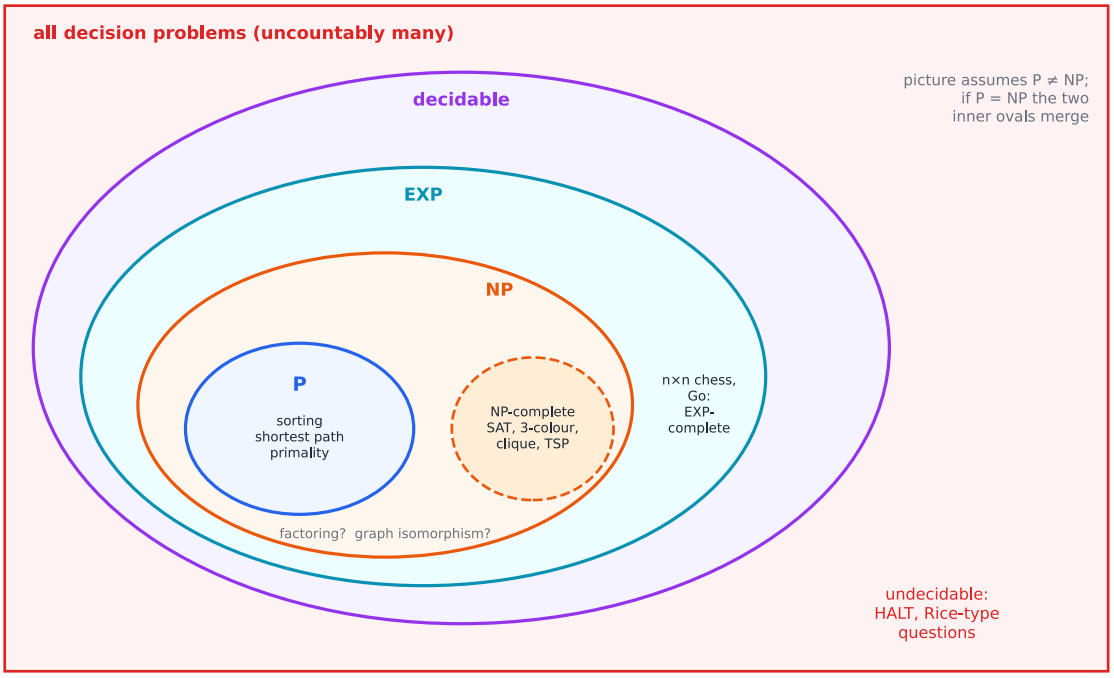


Figure 30.3 — The landscape as most researchers believe it to be. We know $\mathrm{P} \subseteq \mathrm{NP} \subseteq \mathrm{EXP}$ *and, by a diagonal argument,* $\mathrm{P} \neq \mathrm{EXP}$. *Every other boundary in the picture inside the decidable region is unproved.*

> **? The million-dollar inclusion**
>
> Is $\mathrm{P} = \mathrm{NP}$? The Clay Mathematics Institute has offered US$1,000,000 for an answer since 2000. Strangely, much of modern life is built on betting that the answer is no: if every efficiently checkable problem were efficiently solvable, the one-way functions behind online banking could not exist. Why cryptography needs hardness, and which hardness it needs, is Ch. 77.

30.4 The algorithm

Simulating machines

A DFA is a loop over the input with one table lookup per symbol: linear time and constant memory, which is why regular expressions (compiled to automata) are fast. A Turing machine simulator is barely longer. The tape is a dictionary, so it grows in both directions on demand. Notice the `max_steps` argument: a simulator *must* have one, and Theorem 30.2 says that no clever replacement for it exists.

```
def run_tm(rules, tape, state="right", head=0, halt="halt", max_steps=10_000):
    tape = dict(enumerate(tape))
    for step in range(max_steps):
        if state == halt:
            return "".join(tape[i] for i in sorted(tape)).strip("_"), step
        state, write, move = rules[state, tape.get(head, "_")]
        tape[head] = write
        head += 1 if move == "R" else -1
    return None, max_steps          # gave up: maybe it never halts

inc = {("right", "0"): ("right", "0", "R"), ("right", "1"): ("right", "1", "R"),
       ("right", "_"): ("carry", "_", "L"),
       ("carry", "1"): ("carry", "0", "L"), ("carry", "0"): ("halt", "1", "L"),
       ("carry", "_"): ("halt", "1", "L")}
print(run_tm(inc, "1011"), run_tm(inc, "111"))   # ('1100', 8) ('1000', 8)
```

Verifying versus searching

Here is the asymmetry at the heart of NP, in code. A formula in conjunctive normal form is a list of clauses; literal `3` means x_3 and `-3` means $\neg x_3$.

```
from itertools import product

def verify(clauses, bits):                    # O(total formula length)
    return all(any(bits[abs(l) - 1] == (l > 0) for l in c) for c in clauses)

def brute_sat(clauses, n):                    # O(2^n * formula length)
    for bits in product([False, True], repeat=n):
        if verify(clauses, bits):
            return bits
    return None

F = [(1, 2, 3), (-1, -2), (-1, -3), (-2, -3), (-1, 2), (1, -3)]
print(brute_sat(F, 3))                        # (False, True, False)
```

`verify` reads the formula once. `brute_sat` may call it 2^n times. Real SAT solvers (DPLL and its descendant, conflict-driven clause learning) prune the search brilliantly and solve industrial instances with millions of variables, but on carefully built hard families every known algorithm still takes exponential time.

The program you cannot write

Theorem 30.2, written as Python, is short enough to memorise.

```
def halts(program_source, input_data):
    ...                               # suppose someone hands you this, always correct

def contrarian(source):
    if halts(source, source):
        while True:
            pass                      # loop forever
    return "halted"

# contrarian(<source code of contrarian>) can neither halt nor loop.
```

Whatever body you write for `halts`, `contrarian` run on its own source code proves it wrong. In practice, analysis tools accept this and choose a side: **sound** tools (which never miss a real bug but raise false alarms) or **complete** ones (which never raise false alarms but miss bugs). Rice's theorem says you cannot have both and also always answer.

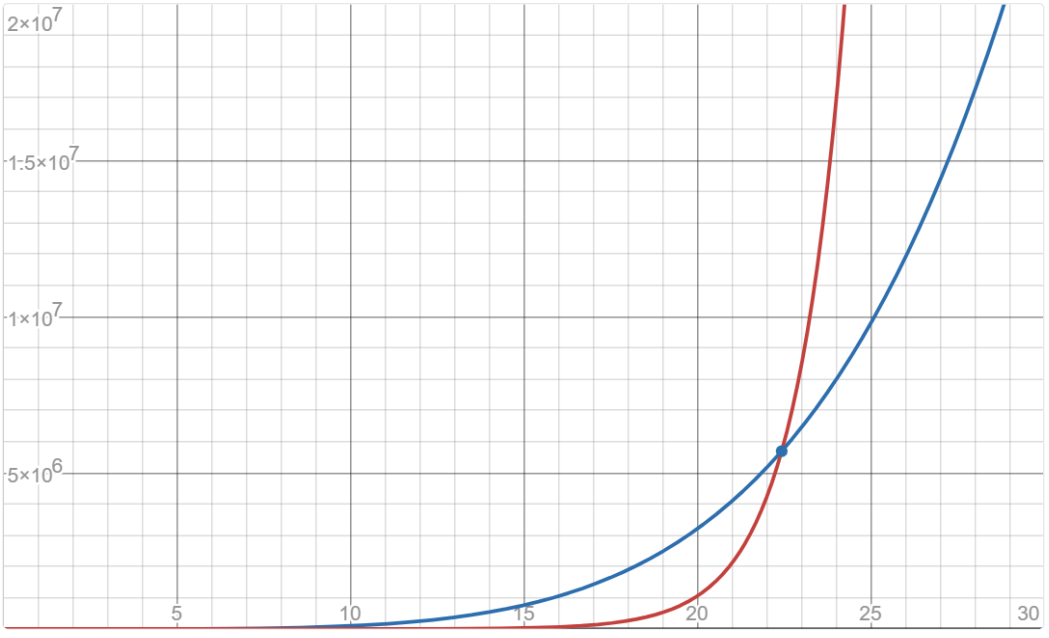
Failure cases

- **Confusing worst case with typical case.** NP-completeness is a worst-case statement. Many instances of NP-complete problems are easy, and SAT solvers exploit that every day.
- **Forgetting the encoding.** "Polynomial" means polynomial in the *length of the input*. Trial division tests whether N is prime in about $\sqrt{N}$ steps, which looks polynomial, but the input is $d \approx \log_{10} N$ digits long, so $\sqrt{N} \approx 10^{d/2}$ is exponential in the input length.

30.5 The experiment

Polynomial versus exponential

Every polynomial is eventually beaten by every exponential, but "eventually" can be far away. The graph compares n^5 with 2^n.



An algorithm costing 2^n beats one costing n^5 until $n \approx 22.4$; after that it loses, and the gap explodes.

Change the exponent 5 to 10: the crossover moves out to $n \approx 58.8$ (widen to `right=70` and `top=10^{18}` to see it, and move the label point). This is the honest caveat of complexity theory. An n^{10} algorithm is "efficient" in theory and useless in practice, while a 1.001^n algorithm is "inefficient" and fine for any n you will ever meet. The theory is about the shape of growth, not about this afternoon's run time.

The cliff in random 3-SAT

Generate random 3-SAT formulas with n variables and m clauses, each clause three distinct variables with random signs. With few clauses, almost every formula is satisfiable. With many, almost none is. The surprise is how sharply one regime gives way to the other. The script below brute-forces every formula over all 2^n assignments at once, using numpy.

```
import numpy as np
import matplotlib.pyplot as plt

N = 12            # knob: number of variables (each +1 doubles the work)
TRIALS = 200      # knob: random formulas per ratio
RATIOS = np.arange(1.0, 8.01, 0.25)   # clauses per variable, m/n
rng = np.random.default_rng(1)

# every assignment of N Boolean variables, as rows of a 2^N x N table
A = ((np.arange(2**N)[:, None] >> np.arange(N)) & 1).astype(bool)

def satisfiable(m):
    """Random 3-SAT with m clauses; brute force over all 2^N assignments."""
    var = np.array([rng.choice(N, 3, replace=False) for _ in range(m)])
    neg = rng.random((m, 3)) < 0.5
    alive = np.ones(2**N, dtype=bool)
    for v, s in zip(var, neg):
        alive &= (A[:, v] ^ s).any(axis=1)   # clause = OR of 3 literals
        if not alive.any():
            return False
    return True

frac = [np.mean([satisfiable(int(round(r * N))) for _ in range(TRIALS)]) for r in RATIOS]
for r, f in zip(RATIOS, frac):
    print(f"m/n = {r:4.2f}   P(satisfiable) = {f:.2f}")

plt.plot(RATIOS, frac, "o-")
plt.axvline(4.27, ls="--", color="grey")
plt.xlabel("clauses per variable  m/n")
plt.ylabel("fraction satisfiable")
plt.title(f"random 3-SAT, n = {N}")
plt.show()
```

With the default knobs it runs in a few seconds. Every formula is satisfiable up to $m/n = 3$, the fraction is 0.74 at $m/n = 4.25$, it crosses one half near $m/n \approx 4.8$, and it is down to about 0.04 by $m/n = 7$. The dashed line marks 4.27, where the transition is believed to sit for very large n.

Things to try:

1. Run with `N = 8`, then `N = 14`. The drop gets steeper and slides toward the dashed line as N grows. Estimate where it crosses one half for each N.
2. Time each call of `satisfiable` and plot the average time against m/n. The cost is not monotone: under-constrained formulas are satisfied at once, over-constrained ones die after a few clauses, and the hardest ones cluster near the cliff.
3. Switch to 2-SAT (two variables per clause). Where is the cliff now? Unlike 3-SAT, 2-SAT is in P, and its threshold can be computed exactly.

30.6 The challenge

No answers are given.

1. **Intuition.** A program has 1 KB of memory and no other storage. Explain why "does it halt on this input?" *is* decidable for it, and estimate how many steps a decider might need. Why does this not contradict Theorem 30.2?
2. **Derivation.** Use a pigeonhole argument in the style of Theorem 30.1 to prove that the language of palindromes over $\{0, 1\}$ is not regular. Then design a DFA with 5 states that accepts binary numerals divisible by 5, and prove that no DFA with fewer states can do it.
3. **Implementation.** The **busy beaver** $\mathrm{BB}(n)$ is the largest number of steps any halting n-state, two-symbol Turing machine takes when started on a blank tape. Extend `run_tm` to enumerate every 2-state machine (each table entry is a choice of write symbol, move and next state or halt) and find $\mathrm{BB}(2)$ by experiment. Why can no program compute $\mathrm{BB}(n)$ for every n?
4. **Open investigation.** In 2024 a large online collaboration proved, with a computer-checked proof, that $\mathrm{BB}(5) = 47{,}176{,}870$. Some 6-state machines are known to behave like Collatz-type iterations. Write a simulator for a 6-state machine of your own design that mimics the Collatz map on unary or binary numbers, and investigate: what would proving that your machine halts actually require?

30.7 Loose threads

The challenge question has an answer, and it is yes. Some problems no program can ever solve, whatever the hardware, the language or the century. The halting problem is the first, proved by making a program argue with itself, and Rice's theorem spreads the damage to every non-trivial question about what programs do. A simple count shows that these are not rare exceptions. Almost every problem is unsolvable, because there are only countably many programs.

Then the chapter drew a second frontier inside the solvable world, between P and NP, and left it exactly where mathematics leaves it: unresolved. That is not a gap in this book. It is a gap in human knowledge.

What we did not tell you:

- **Why logic breaks in the same place.** The diagonal trick that kills the halting problem also limits proof itself. The first mystery points to Ch. 65 and Ch. 68.
- **What hardness is good for.** If P were equal to NP, modern cryptography would collapse; it is built on problems believed hard on average, which is a stronger demand than NP-hardness. The second mystery points to Ch. 77.
- **Whether coins help.** Every algorithm in this chapter was deterministic.

> **? Does randomness buy speed?**
>
> Some problems have fast randomised algorithms that flip coins and are wrong with probability below 2^{-100}, while no equally fast deterministic algorithm is known. Primality testing was one of them for 25 years until a deterministic method appeared in 2002. Is randomness ever *really* necessary for efficiency, or can it always be faked? Many researchers believe the second, and the reason has to do with pseudorandom generators: see Ch. 81.

Simple rules running for a long time also turn up in nature and in games: Conway's Game of Life can simulate a universal Turing machine, so even "will this pattern ever die out?" is undecidable. That lead is followed in Ch. 86.

Further reading

- M. Sipser, *Introduction to the Theory of Computation*, 3rd ed., Cengage, 2012. The standard text: automata, computability and complexity, with very clear proofs.
- C. Moore and S. Mertens, *The Nature of Computation*, Oxford University Press, 2011. Wide-ranging and readable, including the physics of the 3-SAT phase transition.
- S. Aaronson, *Quantum Computing Since Democritus*, Cambridge University Press, 2013. Witty essays on computability, complexity and what they say about the world.

Track B — Probability, Uncertainty, and Information

> **Level 2 — Specializations.** The mathematics of randomness, prediction, communication, and decision-making under uncertainty.

Suggested reading order for Track B. The grey stations on the left are the core chapters it leans on.

Before you start

This track assumes you are comfortable with these core chapters:

- Ch. 19 — Probability
- Ch. 20 — Statistical Analysis
- Ch. 21 — Random Processes
- Ch. 24 — Computational Mathematics

Chapters

Ch.	Chapter	What it explores
31	Probability: The Science of Uncertainty	Conditional probability, Bayes' theorem, independence, inequalities, limit theorems.
32	The Art of Counting Random Worlds	Advanced combinatorics, random variables, distributions, tail bounds.
33	Information Theory: Measuring the Unknown	Entropy, mutual information, KL divergence, coding, compression.
34	Random Walks and Markov Worlds	Markov chains, stationary distributions, mixing times, hitting times.
35	Bayesian Thinking and Statistical Inference	Bayesian inference, likelihood, estimators, hypothesis testing, model selection.
36	Stochastic Processes: When Randomness Evolves	Poisson processes, Brownian motion, martingales, stochastic calculus.
37	Causal Inference: Correlation Is Not Destiny	Causal graphs, interventions, counterfactuals, confounding, experimental design.
38	Monte Carlo: Computing with Randomness	Sampling, simulation, variance reduction, Markov chain Monte Carlo.

Every chapter follows the same arc: **the hook** (a puzzle or a failure), **the big idea**, **the mathematics**, **the algorithm**, **the experiment** you can run, **the challenge**, and **loose threads**: the questions the chapter deliberately leaves open.

Chapter 31 — Probability: The Science of Uncertainty

> **Level 2 · Track B — Probability, Uncertainty, and Information Builds on:** Ch. 19, Ch. 20 · **Leads to:** Ch. 32, Ch. 35, Ch. 38 · **Related:** Ch. 33, Ch. 97 **Reading time:** ~25 min

31.1 The hook

In September 1990 a reader sent a puzzle to Marilyn vos Savant's column in *Parade* magazine. You are on a game show facing three closed doors. Behind one is a car; behind the other two, goats. You pick door 1. The host, who knows where the car is, opens door 3 and shows you a goat. He then offers you the chance to switch to door 2. Should you?

Vos Savant said yes: switching wins two times in three. The magazine reportedly received around ten thousand letters in reply, many from people with doctorates, a large share of them telling her she was wrong. Two doors are left, they said, so it must be fifty-fifty. Paul Erdős, one of the most prolific mathematicians of the century, is said to have stayed unconvinced until he watched a computer simulation.

Now change one detail. The host does *not* know where the car is. He opens one of the two doors you did not pick at random, and it happens to show a goat. Everything you can see is identical: you chose door 1, door 3 is open, a goat is standing there, door 2 is closed. Yet now switching wins exactly half the time.

Same doors, same goat, same view, different answer. If that bothers you, good. It means you have found the central fact of this chapter: **a probability is not a property of the world you see. It depends on how you came to see it.** The answer to "should I switch?" depends on a question that seems to have nothing to do with doors: *what was the host's rule?*

The second half of the chapter asks a matching question about long runs. A casino does not know whether the next spin of its roulette wheel will win or lose, yet it can predict its takings for the year to within a fraction of a percent. Somewhere between one spin and a million, uncertainty turns into near-certainty. Where does that happen, how fast, and can it fail? It can, and you will watch it fail in Figure 31.2.

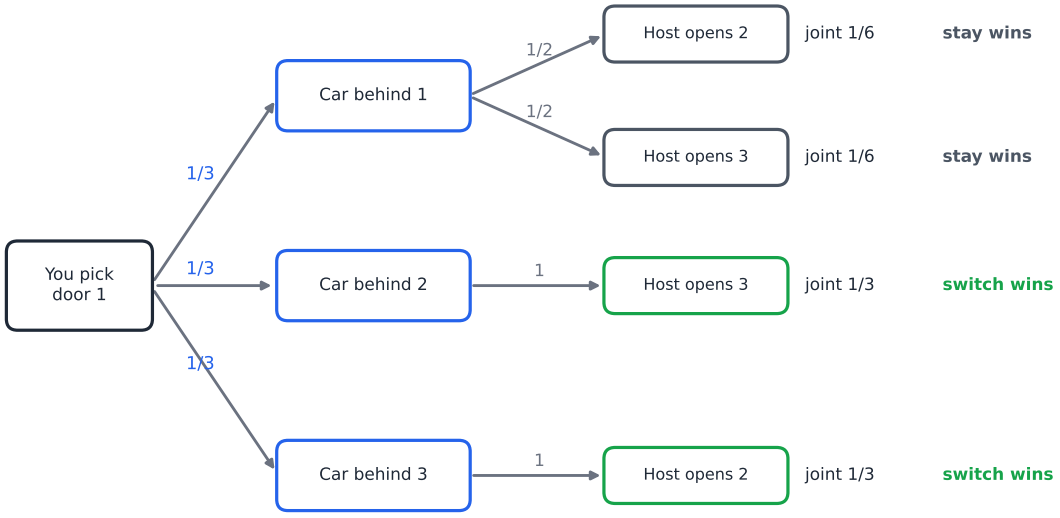
31.2 The big idea

Think of probability as **bookkeeping for ignorance**. Before the host acts you have a ledger: one third of your belief on each door. Every observation is a new entry. Conditioning on an event does not change the world. It deletes the parts of your ledger that the observation rules out and rescales what remains, so that it sums to one again.

The catch is that an observation is an event in the *full* story, host's behaviour included, and not just in the picture of the doors. "Door 3 shows a goat" is one event when the host chose door 3 on purpose because it hid a goat. It is a different event when he picked door 3 by chance and got lucky. Each version deletes a different part of the ledger:

- **Knowing host.** If the car is behind door 2, he is *forced* to open door 3. If the car is behind door 1, he opens door 3 only half the time. So seeing door 3 opened is twice as likely in the "car behind 2" world. That is evidence, and it pushes belief towards door 2.
- **Ignorant host.** He opens door 3 half the time whatever is behind door 1 or door 2, so the opening alone tells you nothing. Only the goat matters, and the goat rules out door 3 alone, leaving doors 1 and 2 equal.

The probability tree in Figure 31.1 makes this mechanical. Every path from the root to a leaf is one complete story, and its probability is the product of the branch probabilities. Conditioning means keeping the leaves that agree with what you saw and renormalising.



Given 'host opens 3': stay wins with 1/6, switch wins with 1/3 -> P(switch wins) = (1/3)/(1/6 + 1/3) = 2/3

Figure 31.1 — The knowing-host game as a tree. The leaves consistent with "host opens door 3" carry joint probability 1/6 (you were right first time) and 1/3 (the car is behind door 2). Renormalised, switching wins 2/3.

The analogy for the second half. A single coin toss is like one voter in an election: unpredictable. A million tosses are like the national result, which polls predict well. The law of large numbers says the average of many independent random quantities settles down to its expectation. The central limit theorem says how the remaining wobble is shaped (a bell curve) and how big it is (it shrinks like $1/\sqrt{n}$). Both are covered in the core (§20.5, with proofs in Appendix E.4–E.5). This chapter pushes on what the core left soft. In what sense does the average "settle"? What guarantees it? What breaks it?

31.3 The mathematics

Conditional probability, the law of total probability and Bayes' theorem are developed in §19.2 and §19.3. Independence is in §19.4, and Markov's and Chebyshev's inequalities are in §19.9. We use all of them freely.

Result 1: the host's rule decides the answer

Generalise the host. Let $C \in \{1, 2, 3\}$ be the car's door, uniform. You pick door 1. If the car is behind door 1, the host can open either goat door. Say he opens door 3 with probability q and door 2 with probability $1 - q$. If the car is behind door 2 or 3, he must open the other goat door. Let O_3 be the event "host opens door 3".

Monty Hall with a biased host

For the knowing host with preference $q \in [0, 1]$,

$$P(C = 2 \mid O_3) = \frac{1}{1+q}, \qquad P(C = 1 \mid O_3) = \frac{q}{1+q}.$$

For the ignorant host, who opens door 2 or 3 at random, $P(C = 2 \mid O_3,\ \text{goat}) = \tfrac{1}{2}$.

Proof. By the law of total probability,

$$P(O_3) = P(O_3 \mid C = 1)P(C = 1) + P(O_3 \mid C = 2)P(C = 2) + P(O_3 \mid C = 3)P(C = 3) = \frac{q}{3} + \frac{1}{3} + 0.$$

Bayes' theorem gives

$$P(C = 2 \mid O_3) = \frac{P(O_3 \mid C = 2)P(C = 2)}{P(O_3)} = \frac{1/3}{(q+1)/3} = \frac{1}{1+q}.$$

For the ignorant host, $P(O_3 \mid C = c) = \tfrac{1}{2}$ for every c. The goat is seen exactly when $C \neq 3$. So $P(O_3 \cap \{\text{goat}\} \cap \{C = c\}) = \tfrac{1}{6}$ for $c = 1, 2$ and 0 for $c = 3$. Renormalising, doors 1 and 2 get $\tfrac{1}{2}$ each. □

With $q = \tfrac{1}{2}$, the usual host, switching wins $\tfrac{2}{3}$. Even in the worst case, $q = 1$ (a host who always opens door 3 when he can), switching wins $\tfrac{1}{2}$. It is never worse than staying. If he opens door 2 instead, the mirror-image formula is $1/(2 - q)$, which is also at least $\tfrac{1}{2}$. So "always switch" is a dominant strategy, whatever the host's taste in doors. And the ignorant-host answer is not a special value of q at all: it is a different event in a different sample space.

Common Pitfall — conditioning on what you saw instead of what happened

"Two doors remain, so it's 50–50" quietly assumes the ignorant-host model. The data ("door 3 is open with a goat") cannot settle which model is true. Only the *protocol* can. The same trap appears in the boy–girl paradox, in survivorship bias, and in every observational study where the way the data were collected depends on the outcome (Ch. 37).

Independence is more fragile than it looks

Events A, B, C can be independent in pairs without being independent together. Toss two fair bits X, Y and let $Z = X \oplus Y$ (exclusive-or). Every pair is independent: for instance $P(X = 1, Z = 1) = P(X = 1, Y = 0) = \tfrac{1}{4} = P(X = 1)P(Z = 1)$. But any two of them determine the third, so $P(X = 1, Y = 1, Z = 1) = 0 \neq \tfrac{1}{8}$. This is not a curiosity. It is the principle behind the one-time pad and RAID-5 parity: one parity bit tells you nothing on its own, yet together with the other bits it tells you everything. The definition of mutual independence in §19.4 needs the product rule for *every* subfamily for exactly this reason.

Independence can also appear or disappear when you condition. Two independent fair dice are no longer independent once you are told their sum is 7. Two strongly correlated alarm signals can become independent once you know whether there really is a fire. Conditioning is the single most powerful operation in probability, and the most dangerous.

Result 2: Jensen's inequality

Markov and Chebyshev turn means and variances into tail bounds. The next inequality is the one that information theory (Ch. 33) is built on.

Jensen's inequality

If φ is convex and X and $\varphi(X)$ have finite expectations, then

$$\varphi\big(E[X]\big) \le E\big[\varphi(X)\big].$$

For concave φ the inequality reverses. If φ is strictly convex, equality holds only when X is almost surely constant.

Proof. Let $\mu = E[X]$. A convex function lies above each of its supporting lines. So there is a slope m (the derivative at μ if it exists, otherwise any subgradient) such that

$$\varphi(x) \ge \varphi(\mu) + m(x - \mu) \quad \text{for all } x.$$

Put $x = X$ and take expectations. Linearity gives $E[\varphi(X)] \geq \varphi(\mu) + m(E[X] - \mu) = \varphi(\mu)$. If φ is strictly convex, the inequality between the curve and the line is strict for $x \neq \mu$. Equality in expectation then forces $X = \mu$ with probability one. □

Choosing φ gives a family of classics:

$\varphi(x)$	Jensen says	Reading
x^2	$E[X]^2 \leq E[X^2]$	variance is non-negative
e^x	$e^{E[X]} \leq E[e^X]$	the MGF is never below e^{μ} (the seed of Chernoff bounds, Ch. 32)
$\log x$ (concave)	$E[\log X] \leq \log E[X]$	geometric mean $\leq$ arithmetic mean; entropy $\leq \log$(alphabet size)
$1/x$ on $x > 0$	$1/E[X] \leq E[1/X]$	average travel time at random speeds is worse than the time at the average speed

A second workhorse is **Cauchy–Schwarz**, $|E[XY]| \leq \sqrt{E[X^2]E[Y^2]}$. It follows because $E[(X - tY)^2] \geq 0$ for every real t, so this quadratic in t has a non-positive discriminant. Applied to the centred variables, it shows that the correlation coefficient of §20.3 always lies in $[-1, 1]$.

Result 3: averages settle, almost surely

The weak law of large numbers (Appendix E.4) says that for each fixed $\varepsilon > 0$,

$$P\big(|\bar{X}_n - \mu| > \varepsilon\big) \to 0, \qquad \bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

Read it carefully. It is a statement about each *single* n: at a large time, a large deviation is unlikely. It does not say that any particular run of the experiment eventually stays close to μ. A run could, in principle, wander away rarely but infinitely often. The **strong law** rules that out. It is a statement about whole sample paths:

$$P\Big(\lim_{n\to\infty} \bar{X}_n = \mu\Big) = 1.$$

The tool that turns "rarely" into "only finitely often" is short enough to prove in full.

First Borel–Cantelli lemma

If $A_1, A_2, \ldots$ are events with $\sum_n P(A_n) < \infty$, then with probability one only finitely many of the A_n occur.

Proof. "Infinitely many A_n occur" is the event $\bigcap_N \bigcup_{n \geq N} A_n$. For every N it is contained in $\bigcup_{n \geq N} A_n$, so by the union bound

$$P(\text{infinitely many } A_n) \leq \sum_{n \geq N} P(A_n) \xrightarrow[N\to\infty]{} 0,$$

because the tail of a convergent series tends to zero. □

Now the strong law, under an extra assumption that keeps the proof to a few lines.

Strong law of large numbers (finite fourth moment)

Let $X_1, X_2, \ldots$ be i.i.d. with mean μ and $E[(X_1 - \mu)^4] = \kappa < \infty$. Then $\bar{X}_n \to \mu$ with probability one.

Proof. Put $Y_i = X_i - \mu$, so $E[Y_i] = 0$ and $E[Y_i^2] = \sigma^2$. Expand $E[(Y_1 + \cdots + Y_n)^4]$ as a sum of terms $E[Y_iY_jY_kY_l]$. By independence and $E[Y_i] = 0$, any term containing an index exactly once vanishes. What survives is the n terms $E[Y_i^4] = \kappa$ and the terms $E[Y_i^2Y_j^2] = \sigma^4$ with $i \neq j$. There are $\binom{4}{2} = 6$ ways to choose which two of the four positions hold i, and $\binom{n}{2}$ unordered pairs $\{i, j\}$, giving $3n(n-1)$ such terms. Hence

$$E\Big[\Big(\sum_{i=1}^{n} Y_i\Big)^4\Big] = n\kappa + 3n(n-1)\sigma^4 \leq Kn^2, \qquad K = \kappa + 3\sigma^4.$$

Markov's inequality applied to the fourth power gives, for any $\varepsilon > 0$,

$$P\big(|\bar{X}_n - \mu| > \varepsilon\big) = P\Big(\Big(\sum Y_i\Big)^4 > n^4\varepsilon^4\Big) \leq \frac{Kn^2}{n^4\varepsilon^4} = \frac{K}{\varepsilon^4 n^2}.$$

The right side is summable in n. By Borel–Cantelli, with probability one $|\bar{X}_n - \mu| > \varepsilon$ happens only finitely often. Apply this with $\varepsilon = 1/m$ for $m = 1, 2, \ldots$. A countable union of probability-zero events has probability zero, so with probability one, for every m, $|\bar{X}_n - \mu| \leq 1/m$ for all large n. That is $\bar{X}_n \to \mu$. □

Notice why Chebyshev alone was not enough. It gives a bound of order $1/n$, and $\sum 1/n$ diverges, so Borel–Cantelli cannot use it. Raising the power from 2 to 4 buys a $1/n^2$ bound, and that is summable. Exponential bounds, which buy e^{-cn}, are the business of Ch. 32.

? The strong law with nothing but a mean

Our proof needed a finite fourth moment. Kolmogorov's strong law needs only $E|X| < \infty$: no variance, no fourth moment, nothing else. The proof truncates the variables, controls what was cut off, and needs a rigorous account of what "with probability one" means for an infinite sequence of coin tosses. That rebuilding of probability on measure theory is the subject of Ch. 97.

When averages refuse to settle

Every one of these laws needs $E|X| < \infty$. The standard **Cauchy** distribution, with density $f(x) = \dfrac{1}{\pi(1+x^2)}$, is symmetric about zero, yet $\int |x| f(x)\,dx$ diverges logarithmically. It has no mean. It is not exotic: it is the distribution of $\tan\Theta$ for a uniformly random angle Θ, so it describes where a randomly aimed beam from a lighthouse hits the shore. Its characteristic function is $e^{-|\omega|}$, so the average of n independent copies has characteristic function $\left(e^{-|\omega|/n}\right)^n = e^{-|\omega|}$. **The average of n Cauchy samples is again standard Cauchy.** Averaging a million of them is no better than taking one.

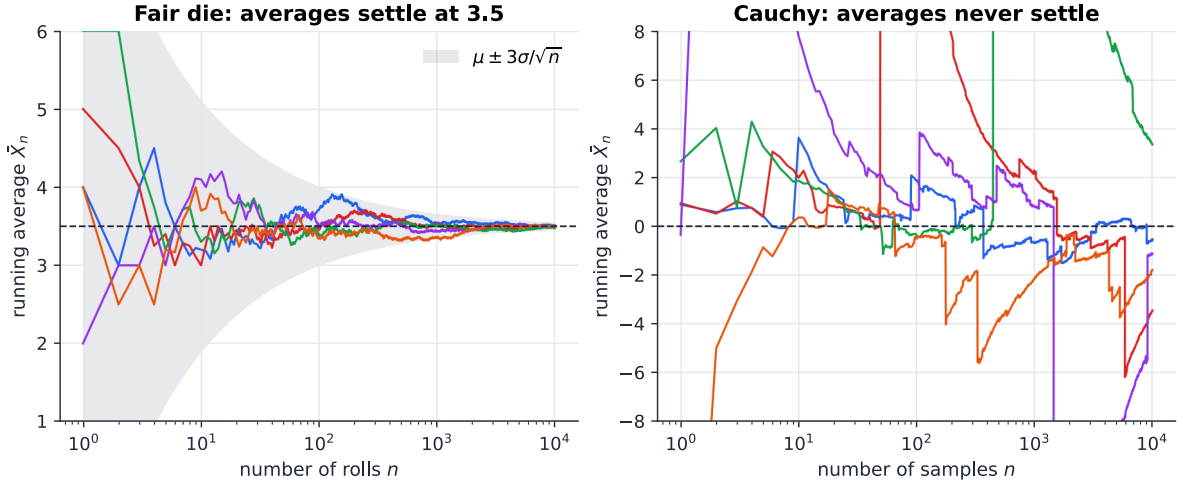


Figure 31.2 — Left: five runs of the running average of a fair die, closing in on 3.5 inside the $\pm 3\sigma/\sqrt{n}$ envelope. Right: five runs of the running average of standard Cauchy samples. Each time a single huge sample arrives, the average jumps, and it never settles.

The central limit theorem (§20.5) completes the finite-variance story. The wobble $\bar{X}_n - \mu$ is approximately $\mathcal{N}(0, \sigma^2/n)$, which is the grey envelope in Figure 31.2. It is a statement about *shape*: it tells you the distribution of the error at typical scales, $\pm$ a few $\sigma/\sqrt{n}$. It does not reliably tell you about errors far out in the tails. That distinction will matter a great deal in the next chapter.

31.4 The algorithm

The law of large numbers is an algorithm in disguise. **To estimate a probability, simulate and count.** If I_k is the indicator that trial k produced the event, then $\hat{p}_n = \frac{1}{n}\sum I_k$ converges to p. The questions an engineer must answer are *how many trials*, and *how do I accumulate the answer without numerical trouble*.

```
import numpy as np

def estimate(trial, n, rng):
    """Monte Carlo estimate of P(event) with a CLT confidence half-width."""
    mean, m2 = 0.0, 0.0                     # Welford's running mean and sum of squares
    for k in range(1, n + 1):
        x = float(trial(rng))               # 1.0 if the event happened, else 0.0
        d = x - mean
        mean += d / k
        m2 += d * (x - mean)
    var = m2 / (n - 1)
    return mean, 1.96 * np.sqrt(var / n)    # ~95 % interval: mean +- half-width
```

Welford's update avoids the cancellation of the naive formula $\frac{1}{n}\sum x^2 - \bar{x}^2$, which can lose every significant digit when the variance is tiny compared with the mean squared. Each step costs $O(1)$ time and $O(1)$ memory, so the estimator can run on a stream that never ends.

How many trials? To get $|\hat{p} - p| \le \varepsilon$ with failure probability at most δ:

- *Chebyshev* (a guarantee): $\mathrm{Var}(\hat{p}) \le \frac{1}{4n}$, so $n \ge \dfrac{1}{4\delta\varepsilon^2}$. For $\varepsilon = 0.01$ and $\delta = 0.05$ that is 50 000 trials.
- *CLT* (an approximation): $n \approx \dfrac{z^2}{4\varepsilon^2}$ with $z = 1.96$, about 9 604 trials, five times fewer.

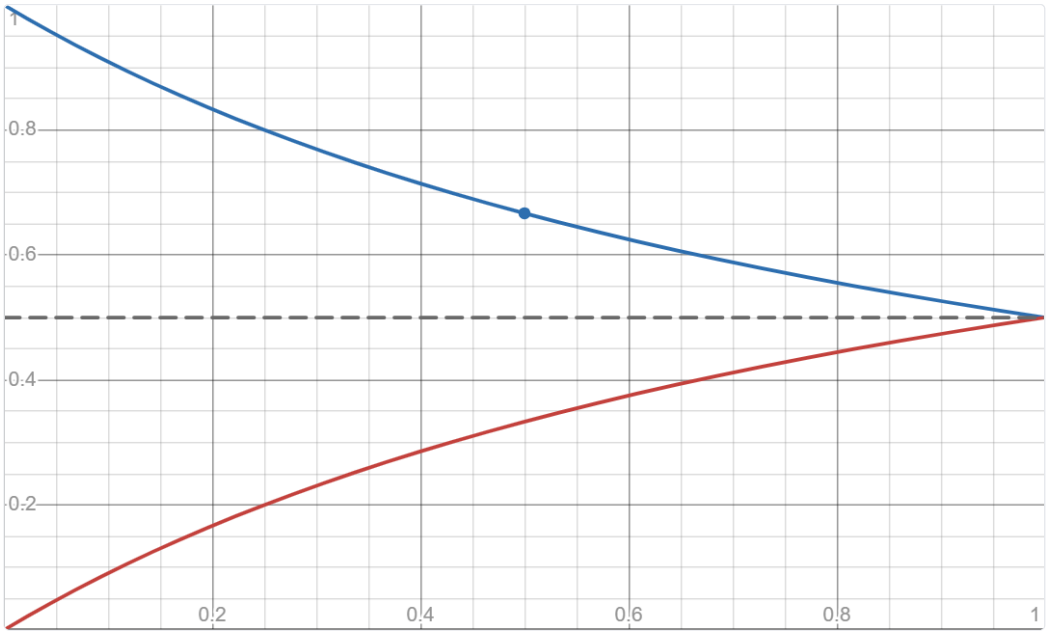
The gap between a rigorous bound and an approximate one is a recurring theme. In Ch. 32 the Chernoff–Hoeffding bound closes most of it, giving a guarantee with $\log(1/\delta)$ in place of $1/\delta$.

Failure cases.

1. *Rare events.* The relative error of $\hat{p}$ is about $1/\sqrt{np}$. To estimate $p = 10^{-6}$ to $\pm 10\,\%$ at 95 % confidence you need roughly 4×10^8 trials. Bit-error-rate testing at 10^{-12} is impossible by brute force. That is why importance sampling exists (Ch. 38).
2. *Heavy tails.* If the quantity being averaged has infinite variance, the $1/\sqrt{n}$ rate is gone. If it has no mean, as in Figure 31.2, there is nothing to converge to.
3. *Hidden dependence.* A poor random-number generator, or trials that share state, quietly break the "independent" in i.i.d. The estimate then converges to the wrong answer with great confidence.

31.5 The experiment

First, see the host-bias formula $1/(1+q)$ from Result 1. The horizontal axis is q, the probability that the knowing host opens door 3 when he has a choice.



Probability that switching wins (blue) and that staying wins (red) as the host's door preference q varies. The curves meet only at $q = 1$. Add the line `y=\frac{1}{2-x}` (switching wins when the host opened door 2 instead) and check that it too never drops below the dashed line at $1/2$. Then change the fixed point to the q you think a real host might have.

Now simulate the game itself. The knobs are at the top.

```
import numpy as np

DOORS = 3           # try 10 or 100: the host opens all but one other door
HOST_KNOWS = True   # False: host opens doors at random (games showing the car are discarded)
Q = 0.5             # knowing host, car behind your door: chance he leaves the lowest-numbered other door shut
TRIALS = 100_000
rng = np.random.default_rng(1)

stay = switch = kept = 0
for _ in range(TRIALS):
    car, pick = rng.integers(DOORS), 0
    others = [d for d in range(DOORS) if d != pick]
    if HOST_KNOWS:
        if car == pick:
            closed = others[0] if rng.random() < Q else rng.choice(others[1:])
        else:
            closed = car                    # he must keep the car hidden
    else:
        closed = rng.choice(others)         # the door left shut, chosen blindly
        if car not in (pick, closed):       # he revealed the car: game void
            continue
    kept += 1
    stay += (car == pick)
    switch += (car == closed)

print(f"games kept: {kept}")
print(f"P(win | stay)   = {stay / kept:.4f}")
print(f"P(win | switch) = {switch / kept:.4f}")
print(f"theory, usual knowing host: switch wins {(DOORS - 1) / DOORS:.4f}")
```

Things to try. Set `DOORS = 100` with `HOST_KNOWS = True`: switching wins about 99 % of the time, and the puzzle stops feeling paradoxical. Set `HOST_KNOWS = False` and watch both answers fall to $1/2$, while `games kept` drops to about $2/\text{DOORS}$ of the trials: most games are thrown away because the blind host exposed the car. Then plot the running estimate of `switch / kept` against the number of games on log axes. Does its wobble shrink like $1/\sqrt{n}$?

31.6 The challenge

1. **Intuition.** A test for a disease affecting 1 in 1000 people is 99 % sensitive and has a 5 % false-positive rate. Before computing anything, guess the probability that a person who tests positive is ill. Then explain, using only the idea of "deleting the impossible parts of the ledger", why the true answer is so much smaller than most people guess. What changes if the person was tested *because* they had symptoms?
2. **Derivation.** In the n-door knowing-host game, the host opens k goat doors ($1 \le k \le n-2$) chosen uniformly from those available, and you then switch to a uniformly random closed door other than yours. Derive the probability that switching wins, and show that it is always larger than $1/n$.
3. **Implementation.** Write a function that takes any finite probability tree (nested lists of `(probability, label, subtree)`) and an observed event, given as a predicate on root-to-leaf paths, and returns the posterior distribution over leaves. Use it to reproduce Result 1 for both hosts and the boy–girl paradox ("at least one is a boy" versus "the elder is a boy").

4. **Open investigation.** For the running average of a symmetric distribution with tails $P(|X| > x) \sim x^{-\alpha}$, explore by simulation for which α the average seems to converge, at what rate, and to what *shape* of fluctuation. What happens near $\alpha = 1$ and near $\alpha = 2$? Can you guess a scaling law $n^{\beta(\alpha)}$ that makes the fluctuations look the same at every n?

31.7 Loose threads

This chapter treated probability as a calculus of belief updates and long-run averages. In doing so it took several things on trust.

We assumed the host's rule was known. In real problems it never is. A doctor does not know the exact false-positive rate, and a radar does not know the clutter statistics. The honest response is to put a probability distribution on the unknown rule itself and let the data update *that*.

> **? What if you do not know the host's rule?**
>
> Suppose you play the game many times against the same host and do not know his q. Each game you see which door he opens. Can you learn q, and how should your decision to switch change as you learn? Treating q itself as random, with a prior that the data sharpen into a posterior, is Bayesian inference, the subject of Ch. 35.

We proved the strong law with a fourth moment, and the general proof was left for later. We also saw that Chebyshev gives deviation probabilities falling only like $1/n$, while the true probabilities in a coin-tossing experiment fall far faster. For 1000 fair tosses, Chebyshev bounds the chance that the fraction of heads is outside $[0.45, 0.55]$ by 0.1. The truth is about 0.0014.

> **? Exponentially unlikely**
>
> The probability that the average of n coin tosses is off by more than ε does not fall like $1/n$. It falls like e^{-cn}, and the constant c has an exact value that depends on ε. Where does the exponential come from, and what is c? Chernoff's method in Ch. 32 gives the bound. The identity of c, a quantity that measures how *distinguishable* two coins are, waits in Ch. 33.

Finally, the Cauchy distribution showed that averaging can fail outright. Between "finite variance, Gaussian fluctuations" and "no mean, no convergence" lies a whole zoo of heavy-tailed limits, the stable laws. They govern network traffic bursts, financial returns and impulsive noise. Their story continues with processes that evolve in time in Ch. 36, and the computational consequences of rare events in Ch. 38.

Further reading

- J. K. Blitzstein and J. Hwang, *Introduction to Probability*, 2nd ed., CRC Press, 2019. Conditioning, Monty Hall variants and the inequalities, with excellent intuition.
- G. Grimmett and D. Stirzaker, *Probability and Random Processes*, 4th ed., Oxford University Press, 2020. The laws of large numbers, Borel–Cantelli and modes of convergence, with full proofs.
- W. Feller, *An Introduction to Probability Theory and Its Applications*, Vol. 1, 3rd ed., Wiley, 1968. The classic; its chapters on fluctuations in coin tossing are still startling.
- J. S. Rosenthal, *Struck by Lightning: The Curious World of Probabilities*, HarperCollins, 2006. A light, accurate tour of everyday probability paradoxes.

Chapter 32 — The Art of Counting Random Worlds

> **Level 2 · Track B — Probability, Uncertainty, and Information Builds on:** Ch. 19, Ch. 31 · **Leads to:** Ch. 33, Ch. 34, Ch. 38 · **Related:** Ch. 28, Ch. 26 **Reading time:** ~25 min

32.1 The hook

A load balancer sends each of 1000 incoming requests to one of 1000 servers, chosen uniformly at random and independently. It is the simplest possible policy, with no coordination and no state. Three questions decide whether it is good enough.

1. **How many servers sit idle?** One might guess "hardly any, since there is one request per server". The answer is about 368, more than a third of the fleet.
2. **How busy is the busiest server?** The average load is exactly 1. The maximum is usually 5 or 6, and that maximum sets your latency tail.
3. **How many requests until every server has received at least one?** Not 1000, and not 2000. It takes about 7485 on average.

Now make one small change. For each request, pick *two* random servers and send it to the less loaded of the pair. The busiest server's load drops from about 5 or 6 to 3, almost every time. One extra random probe per request is enough to tame the maximum.

Each of these numbers is a property of an astronomically large set of equally likely "worlds": 1000^{1000} ways to assign the requests. Nobody can list them. Yet the answers come out in a few lines of algebra. This chapter is about how: counting random worlds *without enumerating them*, and bounding the rare worlds where things go badly wrong. It ends with a piece of magic. We will prove that certain objects exist by showing that a random one works, without ever constructing one.

32.2 The big idea

Two ideas do almost all the work.

Idea 1: break a count into indicators and add expectations. To find the expected number of idle servers, do not attempt the distribution of that number. It is horrible, because the servers' states are dependent. Instead, attach to each server a little switch I_j that reads 1 if server j is idle. The count is $\sum_j I_j$, and expectation is linear *whether or not the switches are independent*. So you need only $P(I_j = 1)$ for one server, multiplied by 1000. Linearity of expectation (§19.9) is the most underrated theorem in probability. It ignores dependence completely.

Idea 2: the average is not enough; bound the tails. "On average 7485 requests" does not tell an engineer how often it takes 10 000. The tail of the distribution is where systems fail: the server that overflows, the hash bucket that degrades to a linked list, the noise spike that flips a bit. Chapter 19 gave Markov's and Chebyshev's inequalities, which bound tails from the mean and variance alone. They are honest but weak. For sums of many independent pieces, a cleverer use of Markov's inequality gives bounds that decay *exponentially*. That is the **Chernoff bound**, and it is the reason randomized systems can be made reliable.

The analogy. Think of the random world as a city of 1000^{1000} houses and a property as a colour painted on some of them. Linearity of expectation lets you count the colour by asking one typical house at a time. Tail bounds tell you that the strange houses, the ones far from typical, are so rare that a random visitor essentially never finds one. The probabilistic method turns this round: if the strange houses are rare enough, then *most* houses are normal, and in particular a normal house *exists*.

32.3 The mathematics

Counting rules (permutations, combinations, inclusion–exclusion) are in §19.1 and treated in depth in Ch. 28. The binomial, geometric and Poisson families are in §19.6. Here we use them as tools.

Indicators at work

Idle servers. Throw m balls into n bins. Server j is idle when every ball misses it, so $P(I_j = 1) = (1 - 1/n)^m$ and

$$E[\#\text{idle}] = n\Big(1 - \frac{1}{n}\Big)^m \approx ne^{-m/n}.$$

With $m = n = 1000$ this is 367.7, close to $n/e = 367.9$. More generally, the load of one bin is $\text{Binomial}(m, 1/n)$, which for large n is close to $\text{Poisson}(m/n)$. With $m = n$, about 36.8 % of bins hold 0 balls, 36.8 % hold 1, 18.4 % hold 2, 6.1 % hold 3, and 1.5 % hold 4. This is the **Poisson paradigm**: many rare, weakly dependent events produce Poisson counts.

Fixed points. Shuffle a deck of n cards. How many cards end up where they started? Let I_k indicate that card k is fixed, so $P(I_k = 1) = 1/n$ and $E[\#\text{fixed}] = n \cdot \frac{1}{n} = 1$. The answer is one, for every n, even though the events are dependent (if $n - 1$ cards are fixed, so is the last). The chance of *no* fixed point, a derangement, tends to $\sum_k (-1)^k/k! = 1/e \approx 0.368$. Inclusion–exclusion gives that (Ch. 28); the Poisson paradigm predicts it at once.

Result 1: the coupon collector

Coupon collector

Draw uniformly from n coupon types until every type has appeared. The number of draws T satisfies

$$E[T] = nH_n = n\Big(1 + \frac{1}{2} + \cdots + \frac{1}{n}\Big) = n\ln n + \gamma n + O(1),$$

where $\gamma \approx 0.5772$, and for every $c > 0$

$$P\big(T > n\ln n + cn\big) \le e^{-c}.$$

Proof of the mean. Split the collection into stages. Stage i starts when you hold $i-1$ distinct types and ends when you get a new one. During stage i each draw is new with probability $p_i = (n-i+1)/n$, independently of the past. So the length T_i of stage i is geometric with mean $1/p_i$ (§19.9). By linearity,

$$E[T] = \sum_{i=1}^{n} \frac{n}{n-i+1} = n\sum_{k=1}^{n}\frac{1}{k} = nH_n.$$

Proof of the tail. $T > t$ means some type is still missing after t draws. Type j is missing with probability $(1-1/n)^t \le e^{-t/n}$, using $1-x \le e^{-x}$. By the union bound over the n types,

$$P(T > t) \le n e^{-t/n}.$$

Put $t = n\ln n + cn$: the bound becomes $n \cdot e^{-\ln n - c} = e^{-c}$. $\square$

For $n = 1000$: $E[T] = 7485.5$, and $P(T > 9908) \le e^{-3} \approx 0.05$. The stages are independent, so their variances add: $\mathrm{Var}(T) = \sum (1-p_i)/p_i^2 \le n^2 \sum 1/k^2 < \pi^2 n^2/6$. That gives a standard deviation below 1283. The collector's time is concentrated in a window of width $O(n)$ around $n\ln n$. The last few coupons are the expensive ones: the final stage alone takes n draws on average.

The birthday bound, made general

The birthday calculation of Example 19.5 gives the exact chance that k random labels from M are all distinct, $\prod_{i<k}(1 - i/M)$. Apply $1 - x \le e^{-x}$ to each factor:

$$P(\text{no collision}) \le \exp\Big(-\sum_{i=0}^{k-1}\frac{i}{M}\Big) = \exp\Big(-\frac{k(k-1)}{2M}\Big).$$

This bound is also an excellent approximation while $k \ll M$. It says collisions become likely once k is of order $\sqrt{2M\ln 2} \approx 1.18\sqrt{M}$. For $M = 365$ that is 22.5; for 64-bit random identifiers, about 5×10^9. A hash with b output bits gives only about $b/2$ bits of collision resistance. That is why SHA-256, and not a 128-bit hash, is used where collisions must be infeasible (Ch. 77).

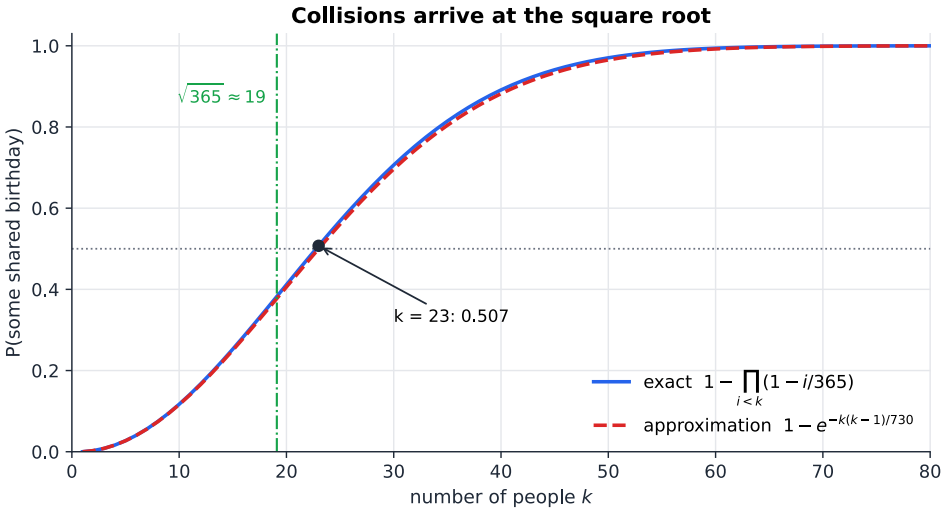


Figure 32.1 — Probability of a shared birthday among k people, exact (blue) and the exponential approximation (red, dashed). The curve passes $1/2$ at $k = 23$, just after $\sqrt{365} \approx 19$: collisions arrive at the square root of the space size.

Result 2: the Chernoff bound

Markov's inequality, $P(Y \ge a) \le E[Y]/a$ for $Y \ge 0$, is weak on its own. The trick is to apply it to e^{sX} rather than to X. For any $s > 0$, $X \ge a$ if and only if $e^{sX} \ge e^{sa}$, so

$$P(X \ge a) \le e^{-sa}\,E\big[e^{sX}\big] = e^{-sa}M_X(s),$$

with the moment-generating function of §19.9. The bound holds for every $s > 0$, so we may choose the best one. When X is a sum of independent pieces, the MGF *factors*, and that is where the exponential decay comes from.

Chernoff bound for sums of independent Bernoulli variables

Let $X = \sum_{i=1}^{n} X_i$ with independent $X_i \sim \text{Bernoulli}(p_i)$ and $\mu = E[X] = \sum p_i$. For every $\delta > 0$,

$$P\big(X \geq (1+\delta)\mu\big) \leq \left(\frac{e^{\delta}}{(1+\delta)^{1+\delta}}\right)^{\mu},$$

and for $0 < \delta \leq 1$ this implies $P\big(X \geq (1+\delta)\mu\big) \leq e^{-\delta^2\mu/3}$.

Proof. Each factor of the MGF is $E[e^{sX_i}] = 1 - p_i + p_i e^s = 1 + p_i(e^s - 1) \leq \exp\big(p_i(e^s-1)\big)$, using $1 + x \leq e^x$. By independence,

$$M_X(s) = \prod_{i=1}^{n} E\big[e^{sX_i}\big] \leq \exp\big(\mu(e^s - 1)\big).$$

Hence, for every $s > 0$,

$$P\big(X \geq (1+\delta)\mu\big) \leq \exp\big(\mu(e^s-1) - s(1+\delta)\mu\big).$$

The exponent is minimised where its derivative $\mu e^s - (1+\delta)\mu$ vanishes, at $s = \ln(1+\delta) > 0$. Substituting gives $\exp\big(\mu\delta - \mu(1+\delta)\ln(1+\delta)\big)$, which is the first bound. For the second, we need $(1+\delta)\ln(1+\delta) \geq \delta + \delta^2/3$ on $[0,1]$. Let $g(\delta) = (1+\delta)\ln(1+\delta) - \delta - \delta^2/3$. Then $g(0) = 0$ and $g'(\delta) = \ln(1+\delta) - \frac{2}{3}\delta$. This is concave, zero at 0, and positive at $\delta = 1$ ($\ln 2 - \frac{2}{3} > 0$), so $g' \geq 0$ on $[0,1]$ and therefore $g \geq 0$. $\square$

The lower tail has a matching bound, $P(X \leq (1-\delta)\mu) \leq e^{-\delta^2\mu/2}$. A close cousin, **Hoeffding's inequality**, handles any independent variables confined to intervals $[a_i, b_i]$:

$$P\big(X - E[X] \geq t\big) \leq \exp\!\left(-\frac{2t^2}{\sum_i (b_i - a_i)^2}\right).$$

How much better is this than Chebyshev? Toss 100 fair coins and ask for at least 75 heads.

Bound	Value for $P(X \geq 75)$
Markov, μ/a	0.67
Chebyshev, $\sigma^2/(a-\mu)^2$	0.04
Chernoff, $\delta = \frac{1}{2}$, $e^{-\delta^2\mu/3}$	0.016
Chernoff, $\left(e^{\delta}/(1+\delta)^{1+\delta}\right)^{\mu}$	0.0045
Hoeffding, $e^{-2t^2/n}$ with $t = 25$	3.7×10^{-6}
Chernoff with the exact Bernoulli MGF (no $1+x \leq e^x$ step)	2.1×10^{-6}
True value	2.8×10^{-7}

The crude Chernoff forms are built for small p_i, where $1 + x \leq e^x$ is nearly tight. For fair coins it is better to keep the exact factor $(\frac{1}{2} + \frac{1}{2}e^s)^n$ and optimise s directly. That is the bottom bound row, and the blue curve in Figure 32.2. On a log scale, it runs parallel to the true tail: it has the right *exponential rate*, missing only a polynomial prefactor. Chebyshev and Markov fall hopelessly behind.

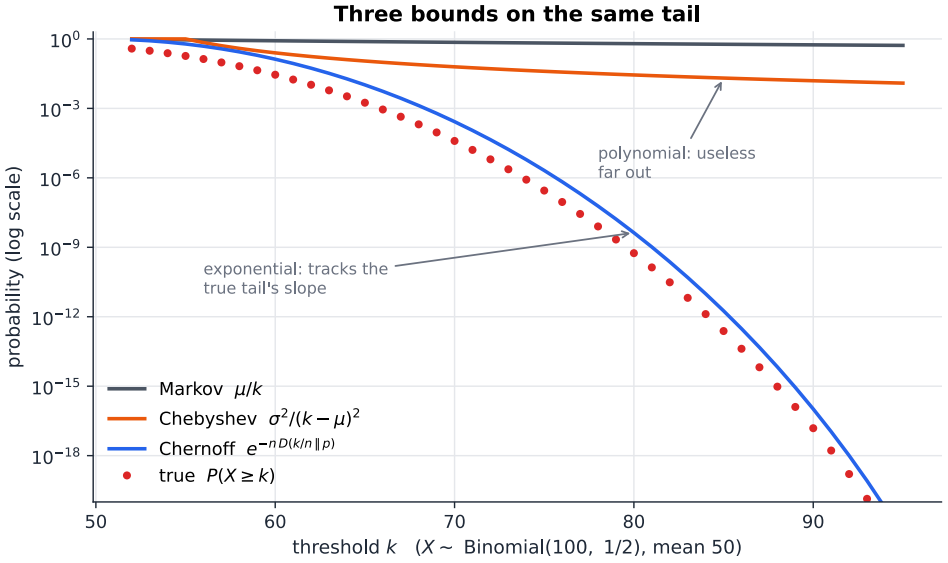


Figure 32.2 — Upper bounds on $P(X \geq k)$ for $X \sim \text{Binomial}(100, \frac{1}{2})$. Markov and Chebyshev decay polynomially. The optimised Chernoff bound decays at the same exponential rate as the truth (red dots).

Max load. The Chernoff bound answers the hook's second question. A bin's load is a sum of n Bernoulli$(1/n)$ variables with $\mu = 1$. For large $1 + \delta = k$ the bound is roughly $(e/k)^k$. A union bound over n bins shows that the maximum load exceeds k with small probability once $(e/k)^k \ll 1/n$. Solving gives $k \approx \dfrac{\ln n}{\ln \ln n}$, so the maximum grows, but extraordinarily slowly. A matching lower bound shows that this is the true order.

Result 3: the probabilistic method

Colour each edge of the complete graph K_N red or blue. A **monochromatic** K_k is a set of k vertices whose $\binom{k}{2}$ connecting edges all have the same colour. Ramsey's theorem (Ch. 28) says that for N large enough, every colouring has one. The Ramsey number $R(k,k)$ is the least such N. How large must N be?

> **Erdős's lower bound (1947)**
>
> If $\binom{N}{k}2^{1-\binom{k}{2}} < 1$, then some 2-colouring of K_N has no monochromatic K_k, so $R(k,k) > N$. In particular $R(k,k) > 2^{k/2}$ for all $k \geq 3$.

Proof. Colour every edge independently red or blue with probability $\frac{1}{2}$ each. A fixed set S of k vertices is monochromatic with probability $2 \cdot 2^{-\binom{k}{2}}$: all red or all blue. By the union bound over the $\binom{N}{k}$ choices of S,

$$P(\text{some monochromatic } K_k) \leq \binom{N}{k} 2^{1-\binom{k}{2}} < 1.$$

So with positive probability the random colouring has *no* monochromatic K_k. A positive-probability event cannot be empty, so such a colouring exists. For the explicit bound, take $N = \lfloor 2^{k/2} \rfloor$ and use $\binom{N}{k} \leq N^k/k!$:

$$\binom{N}{k} 2^{1-\binom{k}{2}} \leq \frac{2^{k^2/2}}{k!} 2^{1-k(k-1)/2} = \frac{2^{1+k/2}}{k!} < 1 \quad (k \geq 3). \qquad \square$$

Read that proof again. It proves that a colouring of K_{32} with no monochromatic K_{10} exists, and the sharper condition reaches K_{100}. Yet it gives no way to find one other than "try random colourings". Nobody knows an explicit construction that comes close. This is the **probabilistic method**: to show that an object with a property exists, show that a random object has the property with positive probability.

> **? The exponential gap in Ramsey theory**
>
> The probabilistic method gives $R(k,k) > 2^{k/2} = (\sqrt{2})^k$. A simple counting argument (Erdős–Szekeres, 1935) gives $R(k,k) < 4^k$. For almost ninety years neither base could be improved. In 2023 the upper base was finally pushed below 4, by a tiny amount. Is the truth $(\sqrt{2})^k$, 4^k, or something in between? Even $R(5,5)$ is known only to lie between 43 and 46. This is a famous **open problem**. Ch. 28 describes the battlefield, and Ch. 98 discusses how such problems are attacked.

32.4 The algorithm

The probabilistic method has an algorithmic twin: if a random choice is good with probability p, then trying repeatedly finds a good one in $1/p$ attempts on average. Here is the idea on a classic optimisation problem, **MAX-CUT**: split the vertices of a graph into two sides so that as many edges as possible cross.

Put each vertex on a random side. Each edge crosses with probability $\frac{1}{2}$, so by linearity the expected cut is $m/2$ for a graph with m edges. Hence a cut of at least $m/2$ *exists*. No cut can exceed m, so this is a $\frac{1}{2}$-approximation. **Derandomisation by conditional expectations** turns the coin flips into decisions. Place vertices one at a time, each on the side that keeps the conditional expected cut at least $m/2$:

```
def greedy_cut(n, edges):
    """Deterministic cut with at least half the edges, via conditional expectations."""
    side = [None] * n
    nbrs = [[] for _ in range(n)]
    for u, v in edges:
        nbrs[u].append(v); nbrs[v].append(u)
    for v in range(n):
        # Edges to still-unplaced vertices cross with prob. 1/2 either way, so only
        # the already-placed neighbours matter: join the side that cuts more of them.
        on0 = sum(1 for w in nbrs[v] if side[w] == 0)
        on1 = sum(1 for w in nbrs[v] if side[w] == 1)
        side[v] = 1 if on0 >= on1 else 0
    cut = sum(1 for u, v in edges if side[u] != side[v])
    return side, cut
```

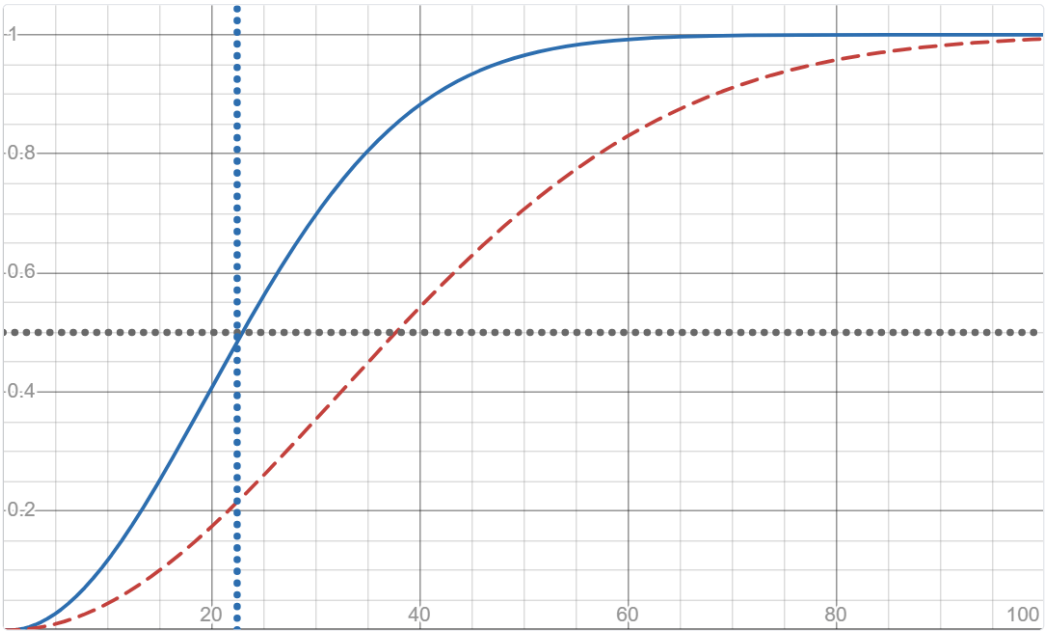
The running time is $O(n+m)$. The guarantee is cut $\geq m/2$ always, with no probability left. The invariant is that the conditional expectation, over the vertices not yet placed, never drops below $m/2$. At the end nothing is random any more, so the actual cut is at least $m/2$.

Load balancing in practice. Random assignment is $O(1)$ per request and needs no shared state, but the $\ln n / \ln \ln n$ maximum load grows with the fleet. The *two-choice* rule costs one extra probe and cuts the maximum to $\ln \ln n / \ln 2 + O(1)$. That is an exponential improvement, and real systems from Nginx to distributed caches use it.

Failure cases. Union bounds become useless when the events are many and not rare (the bound exceeds 1). Chernoff bounds need *independence*, or at least something close to it: a fault that hits all servers at once is not a sum of independent indicators. Linearity of expectation never fails, but an expectation can be dominated by rare catastrophic worlds. The *expected* load of a bin is always 1, yet that says nothing about the maximum.

32.5 The experiment

The birthday approximation, with the space size M as a knob:



Collision probability for k random labels drawn from $M = 365$ (blue) and $M = 1000$ (red). The dotted vertical line marks the 50 % point $\sqrt{2M \ln 2}$. Change 365 to 10000: how far does the curve move? Then check that the 50 % point moves like $\sqrt{M}$ and not like M.

Now simulate balls into bins, with one and two choices, and the coupon collector:

```
import numpy as np

N = 1000          # servers (bins) and requests (balls); try 100, 10_000, 100_000
TRIALS = 200
CHOICES = 2       # 1 = purely random, 2 = power of two choices, try 3
rng = np.random.default_rng(7)

def max_load(n, d):
    load = np.zeros(n, dtype=int)
    for _ in range(n):
        cand = rng.integers(0, n, d)              # d random candidate servers
        load[cand[np.argmin(load[cand])]] += 1    # send to the least loaded
    return load.max(), np.sum(load == 0)

res = np.array([max_load(N, CHOICES) for _ in range(TRIALS)])
print(f"{CHOICES} choice(s): max load histogram {np.bincount(res[:, 0])}")
print(f"mean idle servers {res[:, 1].mean():.1f}   (one-choice theory n/e = {N / np.e:.1f})")

def coupon(n):
    seen, t = np.zeros(n, bool), 0
    while not seen.all():
        seen[rng.integers(0, n, 4 * n)] = True    # draw in blocks for speed...
        t += 4 * n
    return t                                       # ...so t overshoots by < 4n

H = np.sum(1 / np.arange(1, N + 1))
print(f"coupon collector: coarse simulation {np.mean([coupon(N) for _ in range(20)]):.0f}, theory n*H_n = {N * H:.0f}")
print(f"one-choice max-load scale ln n/ln ln n = {np.log(N) / np.log(np.log(N)):.2f}")
```

With `CHOICES = 1` and `N = 1000` you should see maximum loads of 4 to 8, mostly 5 or 6. With `CHOICES = 2` nearly every trial gives 3. Push `N` to 100 000 (be patient) and see how little the two-choice maximum moves. The `coupon` function is deliberately coarse. Rewrite it to count draws one at a time, compare its average with nH_n, and plot the histogram of $(T - n \ln n)/n$. Its shape does not depend on n. What is it?

32.6 The challenge

1. **Intuition.** Linearity of expectation says the expected number of fixed points of a random permutation is 1 for every n. Without computing, explain why the *variance* of that number should also be close to 1 for large n, and why that fits the Poisson paradigm.
2. **Derivation.** Using the Chernoff bound, show that if $m = n \ln n$ balls are thrown into n bins, then with probability tending to 1 every bin holds between $\frac{1}{2} \ln n$ and $3 \ln n$ balls. Explain why "many balls per bin" behaves so differently from "one ball per bin".
3. **Implementation.** Write a randomised algorithm that, given N and k, searches for a 2-colouring of K_N with no monochromatic K_k by sampling random colourings and repairing violations. Find a colouring of K_{17} with no monochromatic K_4. What happens when you try K_{18}, and why?
4. **Open investigation.** In the two-choice process, what if the second choice is not independent but correlated with the first, for instance a neighbour in a ring of servers? Measure the maximum load as a function of n. Does the $\ln \ln n$ behaviour survive? Which kinds of correlation destroy it?

32.7 Loose threads

We proved that the Chernoff bound has the right exponential *rate*, but we left one thing unnamed. When we kept the exact MGF of a fair coin and optimised, we got $P(X \geq an) \leq e^{-nD}$ for a number D that depends on a and on the coin's bias. For $a = 0.75$ and a fair coin, $D \approx 0.131$ nats per toss. That number is not an accident of the algebra.

> **? What is the exponent?**
>
> The optimised Chernoff exponent for n tosses of a p-coin, asked to show a fraction a of heads, is
>
> $$D = a \ln \frac{a}{p} + (1-a) \ln \frac{1-a}{1-p}.$$
>
> It is zero when $a = p$, it is never negative, and it is not symmetric in a and p. It measures how hard it is to mistake a p-coin for an a-coin. It is a *distance between probability distributions*, and it is the central quantity of Ch. 33.

The coupon collector is a random walk on the complete graph: each draw "visits" a uniformly random type, and T is the time to visit everything. On a ring, a grid or a social network, the walk can only move to neighbours, and the time to visit everything, the **cover time**, can be far larger.

> **? Covering a graph**
>
> A random walk on a cycle of n nodes takes about $n^2/2$ steps to visit every node, compared with $n \ln n$ for the complete graph. A walk on a two-dimensional grid (a torus) of n nodes takes about $\frac{1}{\pi} n (\ln n)^2$. Why do these exponents and logarithms appear, and how is the cover time linked to the eigenvalues of the graph? The answers live in Ch. 34.

We also stated, without proof, that two random choices cut the maximum load from $\ln n / \ln \ln n$ to $\ln \ln n / \ln 2$. The proof follows the fraction of bins with load at least i: that fraction roughly *squares* at each level, which gives a doubly exponential decay. For practical versions, such as the one in the open investigation above, only partial answers are known. Finally, the probabilistic method only ever gave us existence. Turning "a random object works" into "here is one" is sometimes easy (conditional expectations), sometimes very hard (explicit Ramsey graphs), and sometimes equivalent to deep questions about whether randomness helps computation at all (Ch. 81).

Further reading

- N. Alon and J. H. Spencer, *The Probabilistic Method*, 4th ed., Wiley, 2016. The bible of the subject, from Ramsey lower bounds to the Lovász local lemma.
- M. Mitzenmacher and E. Upfal, *Probability and Computing*, 2nd ed., Cambridge University Press, 2017. Balls into bins, Chernoff bounds and the power of two choices, written for computer scientists.
- R. Motwani and P. Raghavan, *Randomized Algorithms*, Cambridge University Press, 1995. Derandomisation, tail bounds and randomized data structures.
- P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009. Where exact counting meets asymptotics, for readers who want precise constants.

Chapter 33 — Information Theory: Measuring the Unknown

> **Level 2 · Track B — Probability, Uncertainty, and Information Builds on:** Ch. 19, Ch. 21, Ch. 31, Ch. 32 · **Leads to:** Ch. 80, Ch. 35, Ch. 79 · **Related:** Ch. 30, Ch. 34 **Reading time:** ~25 min

33.1 The hook

Think of a whole number between 1 and 1 000 000. I will find it with twenty yes/no questions. Each answer halves the remaining candidates, and $2^{20} = 1\,048\,576$ is just over a million. Twenty questions is also the best possible: nineteen answers can only tell apart $2^{19} = 524\,288$ cases, so some numbers would share the same answers. So "a number up to a million" contains about twenty *bits* of uncertainty.

Now change the game. I roll a loaded six-sided die whose faces come up with probabilities $0.35, 0.20, 0.15, 0.12, 0.10, 0.08$. You must find the face by yes/no questions, and we play thousands of rounds. Six outcomes need three questions if you treat them equally. But you can do better *on average* by asking about the likely faces first. How much better? Is there a floor that no questioning strategy can beat?

There is, and it is the same floor that limits every compression program ever written. Zip a file of English text and it typically shrinks to about a third of its size. Zip the zipped file and almost nothing happens. Zip a file of truly random bytes and it gets slightly *bigger*. Something in the data is being used up, and once it is gone no clever algorithm can squeeze further. In 1948 Claude Shannon gave that something a name, a formula and a unit. He also proved two theorems that have governed every modem, disk drive and phone since: one for how far data can be compressed, and one for how fast it can be sent through noise.

This chapter measures the unknown. By the end you will know the exact floor for the loaded die (2.396 questions per roll), a simple algorithm that gets within 0.06 of it, and why the rest of the gap cannot be closed one roll at a time.

33.2 The big idea

Information is surprise. Learning that an event of probability p happened tells you $\log_2(1/p)$ bits. A fair coin landing heads is 1 bit. A one-in-a-million event is about 20 bits, exactly the twenty questions above. An event you were sure of tells you nothing. The logarithm is forced on us by one requirement: for *independent* events, surprises should add, $\log \frac{1}{pq} = \log \frac{1}{p} + \log \frac{1}{q}$.

Entropy is average surprise. The entropy of a random variable is the surprise you expect before you look. It has three readings that turn out to be the same number:

1. the average number of yes/no questions an optimal player needs to identify the outcome;
2. the average number of bits per symbol an optimal compressor needs;
3. how *spread out* the distribution is: zero for a certainty, maximal for a uniform distribution.

Mutual information is shared surprise. If Y is a noisy measurement of X, then seeing Y removes some of your uncertainty about X. The amount removed is the information Y carries about X. It is symmetric: X tells you exactly as much about Y as Y tells you about X. Figure 33.2 draws this as a Venn diagram.

KL divergence is the price of being wrong. If the world follows p but you designed your code (or your bets, or your model) for q, you waste $D(p\|q)$ extra bits per symbol. That one quantity turns out to be the exponent in the Chernoff bound of Ch. 32, and it measures how quickly an experiment can tell p from q.

The analogy. Entropy is to information what energy is to physics: a conserved currency that can be moved around, converted between forms and lost, but never created by processing. Shannon borrowed the word from thermodynamics, reportedly on von Neumann's advice. The resemblance is not only in the name.

33.3 The mathematics

Logarithms are base 2 (bits) unless written ln (nats; 1 nat $= 1/\ln 2 \approx 1.443$ bits). By convention $0 \log 0 = 0$.

Entropy, joint and conditional entropy

For a discrete random variable X with PMF $p(x)$,

$$H(X) = -\sum_{x} p(x) \log p(x) = E\Big[\log \frac{1}{p(X)}\Big].$$

For a pair, $H(X,Y) = -\sum_{x,y} p(x,y) \log p(x,y)$. The conditional entropy is $H(Y \mid X) = \sum_x p(x) H(Y \mid X = x)$, the average remaining uncertainty about Y once X is known.

The **chain rule** follows by taking logarithms of $p(x,y) = p(x)p(y \mid x)$ and averaging:

$$H(X,Y) = H(X) + H(Y \mid X).$$

The uncertainty of a pair is the uncertainty of the first, plus what remains about the second once the first is known.

For a coin with $P(\text{heads}) = p$, the entropy is the **binary entropy function**

$$H_b(p) = -p \log p - (1-p) \log (1-p),$$

plotted in Figure 33.1. It is 1 bit at $p = \frac{1}{2}$, 0 at $p = 0$ or 1, and $H_b(0.11) \approx 0.5$: a coin that lands heads 11 % of the time carries half a bit per toss.

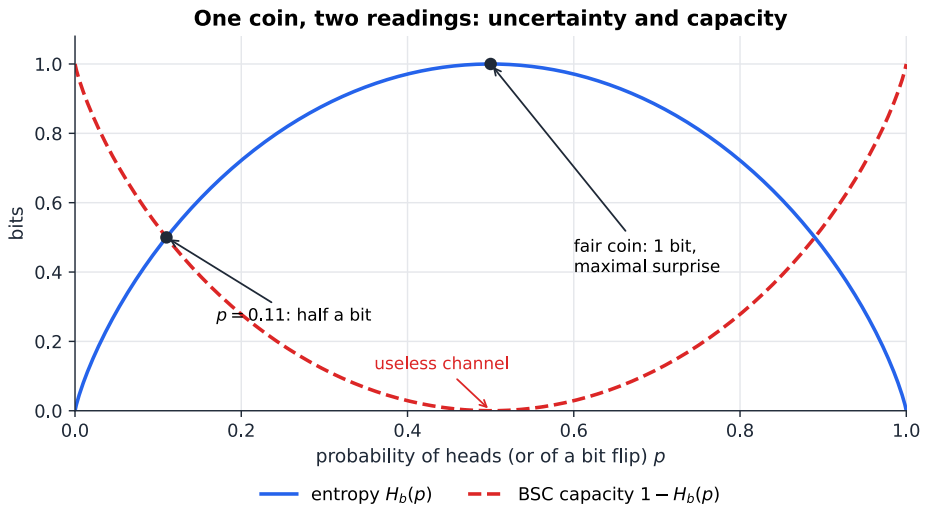


Figure 33.1 — The binary entropy $H_b(p)$ (blue) and, dashed, $1 - H_b(p)$: the capacity of a channel that flips each bit with probability p. A channel that flips half its bits is useless; one that flips all of them is perfect (just invert).

Result 1: Gibbs' inequality and the maximum of entropy

Kullback–Leibler divergence

For PMFs p and q on the same alphabet, with $q(x) > 0$ wherever $p(x) > 0$,

$$D(p\|q) = \sum_x p(x) \log \frac{p(x)}{q(x)}.$$

Gibbs' inequality

$D(p\|q) \geq 0$, with equality if and only if $p = q$.

Proof. Take expectations under p over the support of p. The logarithm is strictly concave, so Jensen's inequality (§31.3) gives

$$-D(p\|q) = E_p\left[\log \frac{q(X)}{p(X)}\right] \leq \log E_p\left[\frac{q(X)}{p(X)}\right] = \log \sum_{x:\, p(x)>0} q(x) \leq \log 1 = 0.$$

Equality in Jensen requires $q(X)/p(X)$ to be constant on the support of p, and equality in the last step requires that support to carry all of q's mass. Together these force $q = p$. □

Two consequences follow at once.

- **Entropy is at most** $\log(\text{alphabet size})$**.** Take q uniform on n symbols. Then $0 \leq D(p\|q) = \log n - H(p)$, so $H(X) \leq \log n$, with equality only for the uniform distribution. The fair die is the hardest die to guess.
- **Mutual information is non-negative.** Define

$$I(X;Y) = D\big(p_{XY} \,\|\, p_X p_Y\big) = \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p(x)p(y)} \;\geq 0,$$

with equality exactly when X and Y are independent. Expanding the logarithm gives the Venn-diagram identities of Figure 33.2:

$$I(X;Y) = H(X) - H(X \mid Y) = H(Y) - H(Y \mid X) = H(X) + H(Y) - H(X,Y).$$

Since $I \geq 0$, we get $H(X \mid Y) \leq H(X)$: *on average, learning something never increases uncertainty*. A single surprising observation can leave you more confused. On average it cannot.

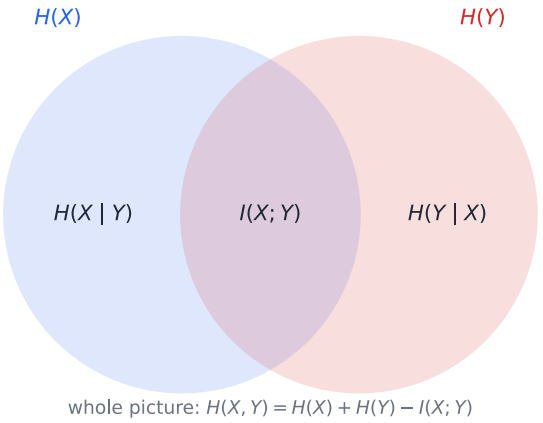


Figure 33.2 — How the entropies of two variables fit together. The overlap is the mutual information. Each circle minus the overlap is a conditional entropy. The union is the joint entropy. (For three or more variables the picture breaks: the central region can be negative.)

Worked Example 33.1 — A noisy binary link

A bit X, equally likely 0 or 1, is sent through a **binary symmetric channel** that flips it with probability $\varepsilon = 0.11$. How much information does the received bit Y carry about X?

Step 1. Y is also uniform, so $H(Y) = 1$ bit.

Step 2. Given X, the output is the input flipped with probability 0.11, so $H(Y \mid X) = H_b(0.11) = 0.500$ bit.

Step 3. $I(X;Y) = H(Y) - H(Y \mid X) = 1 - 0.500 = 0.500$ bit.

Each channel use delivers half a bit, even though 89 % of the bits arrive intact. The other half bit is eaten by our not knowing *which* bits flipped. Shannon showed that $1 - H_b(\varepsilon)$ is the channel's **capacity**: the largest rate at which bits can be sent with an error probability as small as you like. That claim is far from obvious, and we hold it back for later.

Result 2: prefix codes, Kraft, and the source coding bound

A **prefix code** assigns each symbol a binary codeword so that no codeword is the beginning of another. A stream of codewords can then be decoded left to right with no separators: read bits until they match a codeword, output its symbol, and repeat. Every prefix code is a binary tree whose leaves are the symbols (Figure 33.3).

Kraft inequality

Codeword lengths $\ell_1, \dots, \ell_n$ are achievable by a binary prefix code if and only if

$$\sum_{i=1}^{n} 2^{-\ell_i} \le 1.$$

Proof. Read a codeword $c = c_1 c_2 \dots c_\ell$ as the binary fraction $0.c_1 c_2 \dots c_\ell$ and attach to it the interval $[0.c_1 \dots c_\ell,\ 0.c_1 \dots c_\ell + 2^{-\ell})$. These are exactly the numbers in $[0, 1)$ whose binary expansion begins with c. If no codeword is a prefix of another, the intervals are disjoint. They all lie in $[0, 1)$, so their lengths $2^{-\ell_i}$ sum to at most 1. Conversely, given lengths satisfying the inequality, sort them so that $\ell_1 \le \ell_2 \le \dots$ and lay intervals of length $2^{-\ell_i}$ end to end from 0. Each starts at a multiple of its own length (all earlier lengths are larger powers of $\tfrac{1}{2}$), so its left endpoint has an exact ℓ_i-bit binary expansion. Take that expansion as the codeword. Disjoint intervals mean no prefixes. □

Source coding bound (Shannon, 1948)

Every prefix code for X has expected length $L = \sum p_i \ell_i \ge H(X)$. There is a prefix code with $L < H(X) + 1$.

Proof. Lower bound. Let $Z = \sum_j 2^{-\ell_j} \le 1$ and define the PMF $r_i = 2^{-\ell_i}/Z$. Then $\ell_i = \log \frac{1}{r_i} - \log Z$, and

$$L - H(X) = \sum_i p_i \ell_i + \sum_i p_i \log p_i = \sum_i p_i \log \frac{p_i}{r_i} - \log Z = D(p \| r) + \log \frac{1}{Z} \ge 0,$$

because both terms are non-negative (Gibbs, and $Z \le 1$).

Upper bound. Choose $\ell_i = \lceil \log(1/p_i) \rceil$. Then $2^{-\ell_i} \le p_i$, so $\sum 2^{-\ell_i} \le 1$, and Kraft supplies a prefix code with these lengths. Since $\ell_i < \log(1/p_i) + 1$, we get $L < H(X) + 1$. □

The proof says more than the statement. Equality $L = H$ needs $p_i = 2^{-\ell_i}$ exactly, so every probability must be a power of $\tfrac{1}{2}$. And $L - H$ is a KL divergence: *the redundancy of a code is the divergence between the true distribution and the distribution the code implicitly assumes*.

For the loaded die of the hook, $H = 2.396$ bits. A fixed 3-bit code wastes 0.60 bits per roll. The Shannon code with lengths $\lceil \log(1/p_i) \rceil$ averages 2.95. The Huffman code of the next section achieves 2.45.

The Chernoff exponent, named

Ch. 32 left a number unexplained. For $X \sim \text{Binomial}(n, p)$ and $a > p$, keep the exact MGF in the Chernoff bound:

$$P(X \geq an) \leq \min_{s>0} e^{-san}\left(1 - p + pe^{s}\right)^{n}.$$

Setting the derivative of the exponent to zero gives $e^{s} = \dfrac{a(1-p)}{p(1-a)}$, which exceeds 1 when $a > p$. Substituting, $1 - p + pe^{s} = \dfrac{1-p}{1-a}$, and the bound becomes

$$P(X \geq an) \leq \exp\!\left(-n\left[a \ln \frac{a}{p} + (1-a)\ln \frac{1-a}{1-p}\right]\right) = e^{-n\,D(a\|p)},$$

with D in nats, where $D(a\|p)$ is the divergence between a Bernoulli(a) and a Bernoulli(p) coin. The chance that a p-coin *impersonates* an a-coin for n tosses decays at a rate equal to the divergence of the impostor from the truth. For $a = 0.75$ and $p = 0.5$ the rate is 0.131 nats per toss, the slope of the blue curve in Figure 32.2. The bound is tight in the exponent: Cramér's theorem (and its generalisation, Sanov's theorem) says no better exponent is possible.

Divergence is not symmetric. $D(0.5\|0.9) = 0.737$ bits but $D(0.9\|0.5) = 0.531$ bits. If the coin is really fair, the balanced stretches it produces quickly rule out a 0.9-coin. If it really is a 0.9-coin, ruling out "fair" takes longer. So D is not a distance in the geometric sense, and that asymmetry has consequences in statistics, machine learning and hypothesis testing (Ch. 35).

> **? The capacity claim**
>
> Worked Example 33.1 said a channel flipping 11 % of bits can carry half a bit per use with an error rate *as small as you like*. Repetition codes do not come close: to drive errors to zero they send the rate to zero. Shannon's proof of the noisy-channel coding theorem picks a code *at random*, the probabilistic method of Ch. 32 again, and shows that its average error vanishes. Why does a random code work, and why did it take until the 1990s (turbo and LDPC codes) to build practical codes that come close to capacity? See Ch. 80.

33.4 The algorithm

Huffman's algorithm (1952). Repeatedly merge the two least probable symbols into a single node whose probability is their sum, until one node remains. The merge history is the code tree. Label each left branch 0 and each right branch 1.

```
import heapq, itertools

def huffman(probs):
    """probs: dict symbol -> probability. Returns dict symbol -> codeword string."""
    tie = itertools.count()                     # breaks ties so tuples always compare
    heap = [(p, next(tie), {s: ""}) for s, p in probs.items()]
    heapq.heapify(heap)
    if len(heap) == 1:
        return {s: "0" for s in probs}
    while len(heap) > 1:
        p0, _, c0 = heapq.heappop(heap)         # the two least likely subtrees
        p1, _, c1 = heapq.heappop(heap)
        merged = {s: "0" + w for s, w in c0.items()}
        merged.update({s: "1" + w for s, w in c1.items()})
        heapq.heappush(heap, (p0 + p1, next(tie), merged))
    return heap[0][2]

die = {"A": .35, "B": .20, "C": .15, "D": .12, "E": .10, "F": .08}
code = huffman(die)
print(code, sum(die[s] * len(w) for s, w in code.items()))   # average length 2.45
```

For the loaded die the merges are $F + E = 0.18$, $D + C = 0.27$, $0.18 + B = 0.38$, $0.27 + A = 0.62$, and finally 1.00. The likely faces A and B get 2-bit codewords and the rest get 3 bits. The average is 2.45 bits per roll, only 0.054 above the entropy.

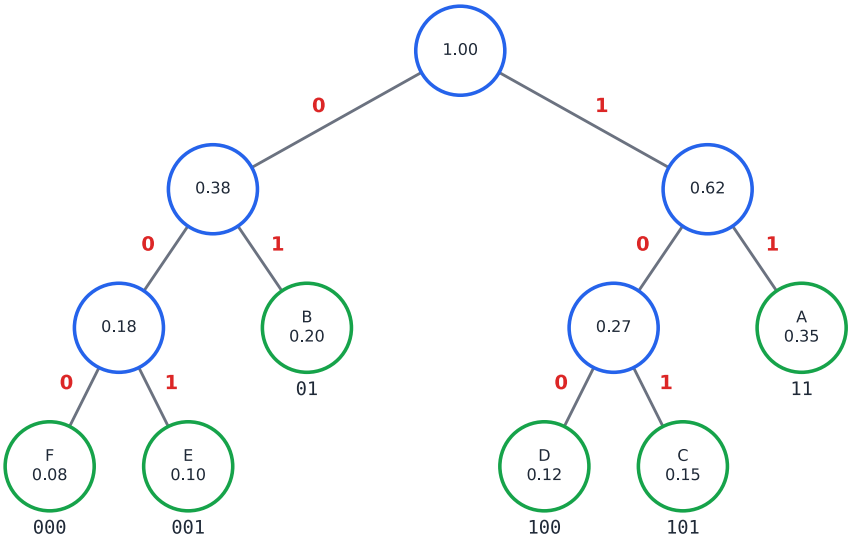


Figure 33.3 — The Huffman tree for the loaded die. Read a codeword from the root down: F is 000, A is 11. Likely symbols sit near the root.

Complexity. With a binary heap, each of the $n - 1$ merges costs $O(\log n)$, for $O(n \log n)$ in total. Decoding walks the tree one bit at a time, or uses a lookup table indexed by the next few bits.

Optimality. Among all prefix codes for a given distribution, Huffman's has the smallest expected length. The proof is an exchange argument: in some optimal tree, the two least likely symbols are siblings at the deepest level, so merging them first loses nothing. You are asked to complete it in the challenge.

Failure cases.

1. *Skewed sources.* A binary source with $P(1) = 0.01$ has entropy $H_b(0.01) = 0.081$ bits. But Huffman on single symbols must spend at least 1 bit per symbol, over twelve times the floor. The cure is to code *blocks* of k symbols, which brings the overhead down to below $1/k$ bits per symbol, or to use arithmetic coding, which effectively allows fractional codeword lengths.
2. *Wrong model.* Huffman is optimal for the probabilities you give it. Coding with q when the truth is p costs $H(p) + D(p\|q)$ bits per symbol on average, which is the KL divergence again as a price. For the loaded die, the ideal code for a fair die (an average of $\log_2 6 = 2.585$ bits) wastes $D(p\|\text{uniform}) = 0.189$ bits per roll.
3. *Ignoring memory.* Letter frequencies alone give English about 4.1 bits per letter. But letters are not independent: after "q" comes "u". Shannon's 1951 experiments with human predictors estimated the true rate of English at around 1 bit per letter. Huffman on single letters cannot see that. Context-modelling compressors can.

33.5 The experiment

The binary entropy and the divergence from a fair coin, with fixed numbers you can edit:

Blue: $H_b(x)$. Red: $D(x\|0.5)$, the cost of believing a fair coin when the true bias is x. Green (dashed): $D(0.5\|x)$, the reverse. The red and green curves differ: KL divergence is not symmetric. Replace every 0.5 by 0.9 to see the divergences from a biased coin, and find where each one is zero.

Now measure the entropy of real text and compare three compressors: the entropy floor for independent letters, a Huffman code, and zlib, which exploits repetition and context.

```
import zlib, heapq, itertools
from collections import Counter
import numpy as np

TEXT = open(__file__).read() * 1     # try: a novel from a text file, or "ab" * 5000
ORDER = 1                            # 1 = letters; 2 = pairs of letters as symbols

symbols = [TEXT[i:i + ORDER] for i in range(0, len(TEXT) - ORDER + 1, ORDER)]
counts = Counter(symbols)
p = np.array(list(counts.values()), float) / len(symbols)
H = -np.sum(p * np.log2(p)) / ORDER                    # bits per character

tie = itertools.count()
heap = [(c, next(tie), 0) for c in counts.values()]    # (weight, tiebreak, cost so far)
heapq.heapify(heap); total = 0
while len(heap) > 1:                                   # total bits = sum of merged weights
    a, _, _ = heapq.heappop(heap); b, _, _ = heapq.heappop(heap)
    total += a + b; heapq.heappush(heap, (a + b, next(tie), 0))
huff = total / len(TEXT)

z = 8 * len(zlib.compress(TEXT.encode(), 9)) / len(TEXT)
print(f"characters {len(TEXT)}, distinct symbols {len(counts)}")
print(f"entropy (order {ORDER})  {H:.3f} bits/char")
print(f"Huffman              {huff:.3f} bits/char")
print(f"zlib                 {z:.3f} bits/char   (raw ASCII: 8)")
```

The script compresses its own source code by default. Then try `TEXT = "ab" * 5000`: the letter entropy is a full 1 bit per character, yet zlib needs almost nothing, because the string is perfectly predictable. Try `ORDER = 2` on a long English text and watch the per-character entropy fall. Then try random text, `"".join(np.random.choice(list("abcdefgh"), 10000))`, where zlib cannot beat 3 bits per character. What does that tell you about the difference between "the entropy of the letters" and "the entropy of the text"?

33.6 The challenge

1. **Intuition.** A biased coin with $P(\text{heads}) = 0.9$ has entropy 0.469 bits. Explain in words, without formulas, how a sequence of 1000 such tosses could be stored in about 470 bits, and why any scheme that stores *every* possible sequence in 470 bits must fail on some sequences.
2. **Derivation.** Prove that among all distributions on $\{0, 1, 2, \dots\}$ with mean μ, the geometric distribution has the largest entropy. Use Gibbs' inequality with a well-chosen q, and find the maximum entropy as a function of μ.
3. **Implementation.** Complete the exchange-argument proof that Huffman codes are optimal. Then implement **canonical Huffman** coding, where only the codeword *lengths* are transmitted, together with an encoder and decoder that round-trip a file. Measure the header overhead as a fraction of the file size.
4. **Open investigation.** Estimate the entropy rate of English in bits per character as the context length k grows ($k = 0, 1, 2, \dots$), using frequency counts from a large text, and plot the estimate against k. Why does it eventually start to *fall too fast*, below any plausible true value? How would you design an estimator that does not fool itself? (Modern language models give the best-known estimates, and they still disagree.)

33.7 Loose threads

Entropy as defined here is an average over a probability distribution. It says nothing about a *single* string. The string 0101 . . . 01 and a string of fair coin flips of the same length have the same letter frequencies and the same order-0 entropy. Yet one has a ten-character description ("01, 500 times") and the other has no description much shorter than itself.

> **? The information in one object**
>
> The *Kolmogorov complexity* of a string is the length of the shortest program that prints it. It captures exactly the intuition above, and for typical outputs of a random source it matches entropy. But it is **uncomputable**: no algorithm can find the shortest program for every string. Why not, and what does that say about the limits of compression and of science itself? The answer is a diagonal argument from Ch. 30.

We proved $H \le L < H + 1$ one symbol at a time, and we noted that coding blocks drives the overhead to zero. Behind that remark is a much stronger fact. Long sequences from a source with entropy H almost all lie in a "typical set" of about 2^{nH} sequences, each with probability close to 2^{-nH}. This is the **asymptotic equipartition property**, a law of large numbers for surprise. The same idea makes random channel codes work, which is why the capacity mystery above lives in Ch. 80.

Everything here was discrete. For continuous variables, $-\int f \log f$ (the *differential* entropy) behaves strangely. It can be negative, and it changes when you change units. Yet differences of it, such as mutual information, remain perfectly meaningful. The Gaussian maximises differential entropy for a given variance (§20.4), and that one fact leads to $C = B \log_2(1 + \text{SNR})$, the most famous formula in communications.

> **? Choosing beliefs by maximising ignorance**
>
> If all you know about a quantity is its mean, which distribution should you assume? Gibbs' inequality suggests an answer: the one with *maximum entropy*, the least committal belief consistent with the facts. Is that a principle of logic, a convenient habit, or a disguised prior? The debate, and when maximum entropy agrees with Bayes' rule, is taken up in Ch. 35.

Further reading

- T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed., Wiley, 2006. The standard text: entropy, AEP, source and channel coding, large deviations.
- D. J. C. MacKay, *Information Theory, Inference, and Learning Algorithms*, Cambridge University Press, 2003. Free online; connects compression, coding and Bayesian inference with exceptional clarity.
- C. E. Shannon, "A Mathematical Theory of Communication," *Bell System Technical Journal*, 27, 1948. Still readable, and still surprising.
- J. R. Pierce, *An Introduction to Information Theory: Symbols, Signals and Noise*, 2nd ed., Dover, 1980. A gentle, non-technical companion.

Chapter 34 — Random Walks and Markov Worlds

> **Level 2 · Track B — Probability, Uncertainty, and Information Builds on:** Ch. 10, Ch. 19, Ch. 21 · **Leads to:** Ch. 36, Ch. 38, Ch. 53 · **Related:** Ch. 27, Ch. 87 **Reading time:** ~30 min

34.1 The hook

A man leaves a bar and walks home along a long straight street. He is too drunk to remember where he is going, so at every corner he tosses a mental coin and steps one block left or one block right. Will he ever get back to the bar's door?

Yes, with probability one. He will return, and return again, infinitely often. But here is the first twist: the *expected* time until his first return is infinite. He is certain to come back, and on average he takes forever to do it.

Now let the drunk loose in a city grid, where at every corner he picks one of four directions. He still returns with probability one. Finally, give him wings. A drunk bird flapping randomly through a three-dimensional lattice of air returns to its starting point with probability only about 0.34. Two-thirds of the time it is lost forever. The mathematician Shizuo Kakutani summarized it in one line: a drunk man will find his way home, but a drunk bird may get lost forever.

Why should the number of dimensions decide between "certainly comes back" and "probably never does"? Nothing in the rule changed: one random step at a time, no memory of the past.

Hold that question. Here are two more that look unrelated.

The shuffle. You riffle-shuffle a deck of 52 cards. How many shuffles until the deck is "random"? Three shuffles leave it almost perfectly ordered in a statistical sense. Four is no better. Then, between five and nine shuffles, the deck's distance from random collapses from nearly total to nearly nothing, and the folklore answer "seven" sits in the middle of the fall. Not a gradual improvement: a cliff.

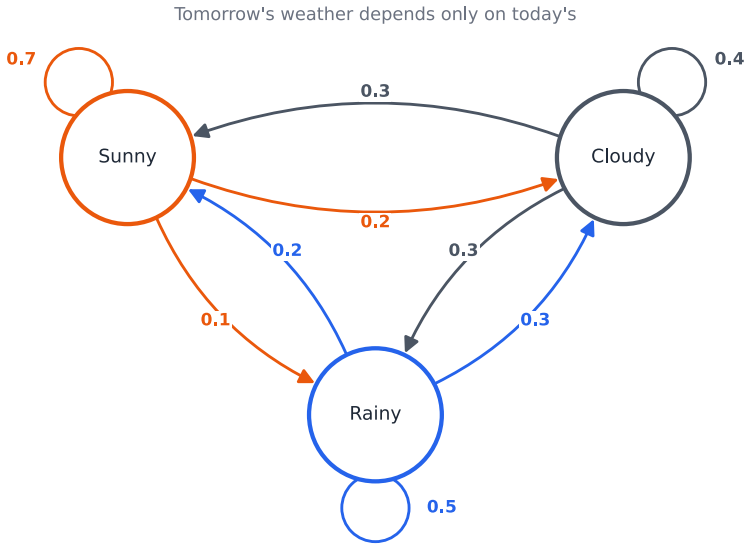
The search engine. In 1998 two students ranked the web by imagining a bored surfer clicking links at random, forever, and asking what fraction of time the surfer spends on each page. That fraction, PageRank, is a number that the surfer's starting page has no influence on at all.

All three stories are about the same object: a system that jumps between states with probabilities that depend only on where it is now. This chapter builds that object, proves the theorem that makes PageRank work, computes when gamblers go broke, and leaves the drunk bird and the seven shuffles hanging, because their full answers need tools the next chapters supply.

34.2 The big idea

Forget the past; keep the present. A Markov chain is a random process with amnesia. To predict tomorrow you need today, and nothing earlier. That single restriction turns an impossibly rich object (a whole random history) into something you can write on a napkin: a table of probabilities "from state i, go to state j with probability P_{ij}".

Here is the napkin for a toy weather model.



The weather chain. Each arrow carries the probability of tomorrow's state given today's; the arrows leaving any state sum to 1.

Run the chain for a long time and two things happen.

1. **The start is forgotten.** Whether today is Sunny or Rainy, the probability that it is Sunny 30 days from now is the same number. The chain "mixes".
2. **The long-run fractions settle.** Over a year, the fraction of Sunny days converges to a fixed value, the *stationary distribution*. That fraction is also the probability of Sunny on a single day far in the future.

The analogy that carries the whole chapter is **ink in water**. Put a drop of ink (all the probability) on one state. Each step, the ink in every state splits along the arrows in the proportions P_{ij}. Ink is never created or destroyed: probability is conserved. After enough steps the ink is spread in a fixed pattern that the flow no longer changes. That pattern is the stationary distribution, and how fast the ink spreads is the mixing time.

Three questions organise everything:

- **Where does the ink end up?** (stationary distributions)
- **How fast does it get there?** (mixing times)
- **When does a single particle first reach a target?** (hitting times)

The drunk on the street is the same story with infinitely many states, one per street corner. There the ink never settles: it spreads thinner and thinner forever. That is why the drunk always returns yet takes infinite expected time, and the escape of the bird is the same spreading, faster in more dimensions.

34.3 The mathematics

Chains and their matrices

Markov chain

A sequence of random variables $X_0, X_1, X_2, \dots$ taking values in a countable state space S is a (time-homogeneous) **Markov chain** with **transition matrix** $\mathbf{P}$ if

$$P(X_{n+1} = j \mid X_n = i,\ X_{n-1} = i_{n-1}, \dots, X_0 = i_0) = P_{ij}$$

for all n and all histories. The matrix is **row-stochastic**: $P_{ij} \geq 0$ and $\sum_j P_{ij} = 1$ for each row i.

For the weather chain, with states ordered (Sunny, Cloudy, Rainy),

$$\mathbf{P} = \begin{bmatrix} 0.7 & 0.2 & 0.1 \\ 0.3 & 0.4 & 0.3 \\ 0.2 & 0.3 & 0.5 \end{bmatrix}.$$

Write the distribution of X_n as a row vector $\boldsymbol{\mu}_n$, with $(\boldsymbol{\mu}_n)_j = P(X_n = j)$. Conditioning on the state at time n (the law of total probability from Ch. 19) gives

$$(\boldsymbol{\mu}_{n+1})_j = \sum_i (\boldsymbol{\mu}_n)_i P_{ij} \quad \Longrightarrow \quad \boldsymbol{\mu}_{n+1} = \boldsymbol{\mu}_n \mathbf{P}, \qquad \boldsymbol{\mu}_n = \boldsymbol{\mu}_0 \mathbf{P}^n.$$

The entry $(\mathbf{P}^n)_{ij}$ is the probability of going from i to j in exactly n steps. Everything about the chain's distribution is a question about matrix powers, which Ch. 10 §10.4 already taught you to compute by eigen-decomposition. (Ch. 10 writes probability vectors as columns and uses the transposed, column-stochastic matrix; the two conventions are the same mathematics. Probabilists use rows.)

Stationary distributions

Stationary distribution

A probability row vector $\boldsymbol{\pi}$ ($\pi_j \geq 0$, $\sum_j \pi_j = 1$) is **stationary** for $\mathbf{P}$ if

$$\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}.$$

If $X_0 \sim \boldsymbol{\pi}$ then $X_n \sim \boldsymbol{\pi}$ for every n.

So $\boldsymbol{\pi}$ is a left eigenvector of $\mathbf{P}$ with eigenvalue 1. Such an eigenvalue always exists: $\mathbf{P}\mathbf{1} = \mathbf{1}$ because rows sum to one, and a matrix and its transpose share eigenvalues.

For the weather chain, $\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}$ is three linear equations, one redundant. Taking the first two with the normalisation:

$$\begin{aligned} 0.7\pi_S + 0.3\pi_C + 0.2\pi_R &= \pi_S, \\ 0.2\pi_S + 0.4\pi_C + 0.3\pi_R &= \pi_C, \\ \pi_S + \pi_C + \pi_R &= 1, \end{aligned} \qquad \Longrightarrow \qquad \boldsymbol{\pi} = \left(\tfrac{21}{46},\ \tfrac{13}{46},\ \tfrac{12}{46}\right) \approx (0.457,\ 0.283,\ 0.261).$$

In the long run it is sunny 45.7 % of days. The remaining eigenvalues of $\mathbf{P}$ are $(3 \pm \sqrt{3})/10 \approx 0.473$ and 0.127; we will see in a moment that the larger of these controls the speed.

Two things can go wrong, and the convergence theorem must rule both out.

- **Reducible chains.** If the state space splits into pieces that cannot reach each other, there is one stationary distribution per closed piece, and where you end up depends on where you started. A chain is **irreducible** if every state can reach every other.
- **Periodic chains.** The two-state chain that swaps deterministically, $\mathbf{P} = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$, has stationary distribution $(\tfrac{1}{2}, \tfrac{1}{2})$, yet started in state 1 it is in state 1 at even times and state 2 at odd times. $\boldsymbol{\mu}_n$ oscillates forever. A chain is **aperiodic** if returns to each state are not locked to multiples of some period $d > 1$. Its eigenvalue -1 is the culprit: it has modulus one, so it never decays.

The convergence theorem, with a proof

To measure how far a distribution is from equilibrium we use the **total variation distance**

$$d_{TV}(\boldsymbol{\mu}, \boldsymbol{\nu}) = \tfrac{1}{2} \sum_j |\mu_j - \nu_j| = \max_{A \subseteq S} |\mu(A) - \nu(A)|,$$

the largest disagreement the two distributions can have about the probability of any event. It lies between 0 and 1.

The general theorem says: a finite, irreducible, aperiodic chain has a unique stationary distribution and $\boldsymbol{\mu}_0\mathbf{P}^n \to \boldsymbol{\pi}$ from every start. We prove a version with a clean, quantitative argument due to Doeblin and Dobrushin. The key quantity is the **contraction coefficient**

$$\delta(\mathbf{P}) = \max_{i,k} d_{TV}\big(\mathbf{P}_{i\cdot},\ \mathbf{P}_{k\cdot}\big),$$

the largest total-variation distance between two rows of $\mathbf{P}$. It measures how much the chain's next step still "remembers" which state it came from.

Contraction theorem

For any two probability vectors $\boldsymbol{\mu}, \boldsymbol{\nu}$,

$$d_{TV}(\boldsymbol{\mu}\mathbf{P}, \boldsymbol{\nu}\mathbf{P}) \le \delta(\mathbf{P})\, d_{TV}(\boldsymbol{\mu}, \boldsymbol{\nu}).$$

If $\delta(\mathbf{P}) < 1$ (in particular, if every entry of $\mathbf{P}$ is at least $\varepsilon > 0$ on an N-state space, which gives $\delta \le 1 - N\varepsilon$), then $\mathbf{P}$ has exactly one stationary distribution $\boldsymbol{\pi}$, and for every start

$$d_{TV}(\boldsymbol{\mu}_0\mathbf{P}^n, \boldsymbol{\pi}) \le \delta(\mathbf{P})^n.$$

Proof. **Step 1: split the difference.** Let $a = d_{TV}(\boldsymbol{\mu}, \boldsymbol{\nu})$; if $a = 0$ there is nothing to prove. The vector $\boldsymbol{\mu} - \boldsymbol{\nu}$ sums to zero, so its positive part and negative part each have total mass a. Normalise them: $\boldsymbol{\mu} - \boldsymbol{\nu} = a(\boldsymbol{\alpha} - \boldsymbol{\beta})$ with $\boldsymbol{\alpha}, \boldsymbol{\beta}$ probability vectors.

Step 2: push through one step. By linearity $\boldsymbol{\mu}\mathbf{P} - \boldsymbol{\nu}\mathbf{P} = a(\boldsymbol{\alpha}\mathbf{P} - \boldsymbol{\beta}\mathbf{P})$, so $d_{TV}(\boldsymbol{\mu}\mathbf{P}, \boldsymbol{\nu}\mathbf{P}) = a\, d_{TV}(\boldsymbol{\alpha}\mathbf{P}, \boldsymbol{\beta}\mathbf{P})$.

Step 3: bound by rows. Since $\sum_i \alpha_i = \sum_k \beta_k = 1$,

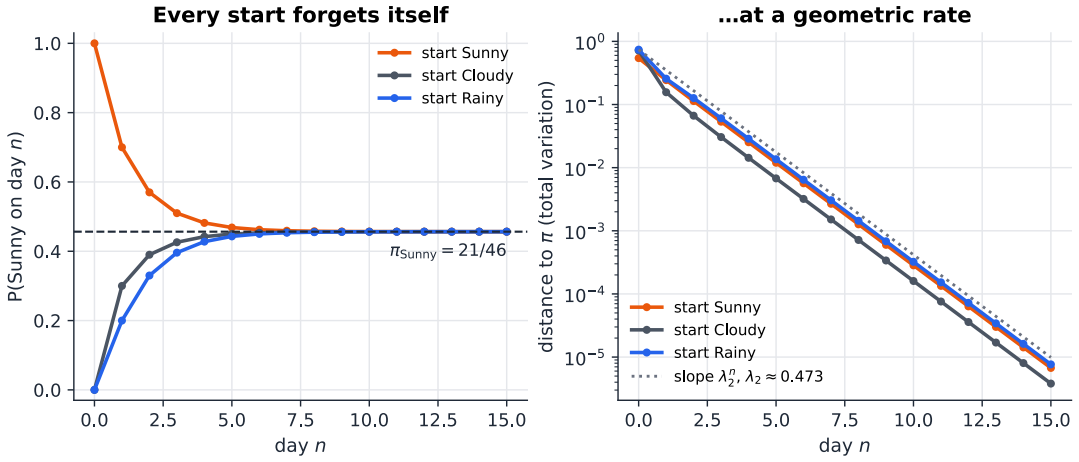
$$\boldsymbol{\alpha}\mathbf{P} - \boldsymbol{\beta}\mathbf{P} = \sum_i \alpha_i \mathbf{P}_{i\cdot} - \sum_k \beta_k \mathbf{P}_{k\cdot} = \sum_{i,k} \alpha_i \beta_k \big(\mathbf{P}_{i\cdot} - \mathbf{P}_{k\cdot}\big).$$

Take half the ℓ_1 norm and use the triangle inequality: $d_{TV}(\boldsymbol{\alpha}\mathbf{P}, \boldsymbol{\beta}\mathbf{P}) \le \sum_{i,k} \alpha_i\beta_k\, d_{TV}(\mathbf{P}_{i\cdot}, \mathbf{P}_{k\cdot}) \le \delta(\mathbf{P})$. This proves the inequality.

Step 4: the bound on δ. For two rows, $d_{TV}(\mathbf{p}, \mathbf{q}) = 1 - \sum_j \min(p_j, q_j)$ (because $|p - q| = p + q - 2\min(p, q)$). If every entry is at least ε, each minimum is at least ε and $\delta \le 1 - N\varepsilon$.

Step 5: existence, uniqueness, convergence. If $\delta < 1$ the map $\boldsymbol{\mu} \mapsto \boldsymbol{\mu}\mathbf{P}$ is a contraction on the closed set of probability vectors, so the sequence $\boldsymbol{\mu}_0\mathbf{P}^n$ is Cauchy and converges to a fixed point (the contraction-mapping argument of Ch. 22). Two fixed points $\boldsymbol{\pi}, \boldsymbol{\pi}'$ would satisfy $d_{TV}(\boldsymbol{\pi}, \boldsymbol{\pi}') \le \delta\, d_{TV}(\boldsymbol{\pi}, \boldsymbol{\pi}')$, forcing equality. Finally $d_{TV}(\boldsymbol{\mu}_0\mathbf{P}^n, \boldsymbol{\pi}) = d_{TV}(\boldsymbol{\mu}_0\mathbf{P}^n, \boldsymbol{\pi}\mathbf{P}^n) \le \delta^n d_{TV}(\boldsymbol{\mu}_0, \boldsymbol{\pi}) \le \delta^n$. ∎

For the weather chain the smallest entry is 0.1, giving the crude guarantee $\delta \le 1 - 3(0.1) = 0.7$; computing the row distances directly gives $\delta = 0.5$. Either way the distance to equilibrium at least halves every day. The truth is a little better still:



Left: the probability of a sunny day n days ahead, from three different starting days. Right: total-variation distance to $\boldsymbol{\pi}$ on a log scale; the lines are straight with slope set by the second eigenvalue $\lambda_2 = (3 + \sqrt{3})/10 \approx 0.473$.

What about chains where some entries are zero, like a random walk on a sparse graph? If the chain is irreducible and aperiodic, then some power $\mathbf{P}^m$ has all entries positive (a fact from number theory about the return times), and you apply the theorem to $\mathbf{P}^m$. This is the general convergence theorem.

The straight lines in the figure come from the eigen-decomposition. If $\mathbf{P}$ is diagonalisable with eigenvalues $1 = \lambda_1 > |\lambda_2| \ge \cdots$, then $\boldsymbol{\mu}_0\mathbf{P}^n = \boldsymbol{\pi} + \sum_{k \ge 2} c_k \lambda_k^n \mathbf{v}_k$, and the distance decays like $|\lambda_2|^n$. The quantity $1 - |\lambda_2|$ is the **spectral gap**. Big gap, fast mixing.

Reversibility: the shortcut to $\boldsymbol{\pi}$

Solving $\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}$ for a million-state chain is expensive. For an important class of chains there is a shortcut.

Detailed balance

If a probability vector $\boldsymbol{\pi}$ satisfies $\pi_i P_{ij} = \pi_j P_{ji}$ for all i, j, then $\boldsymbol{\pi}$ is stationary.

Proof. Sum the detailed-balance equation over i: $\sum_i \pi_i P_{ij} = \pi_j \sum_i P_{ji} = \pi_j$. ∎

Detailed balance says that in equilibrium the *flow* of probability from i to j equals the flow back, edge by edge. Such chains are called **reversible**: a film of the chain in equilibrium looks statistically the same run backwards.

The most important reversible chain is the **random walk on a graph**: from vertex v, move to a uniformly chosen neighbour, so $P_{vw} = 1/\deg(v)$ for each edge vw. Try $\pi_v = \deg(v)/(2|E|)$: then $\pi_v P_{vw} = 1/(2|E|) = \pi_w P_{wv}$. Detailed balance holds, so a random walker on any connected graph spends time at each vertex in proportion to its degree. Well-connected vertices are visited more, a first hint of why random walks can rank things.

Hitting times and the gambler's ruin

Now follow one particle instead of the ink. The **hitting time** of a set A is $\tau_A = \min\{n \geq 0 : X_n \in A\}$. The workhorse for computing anything about τ_A is **first-step analysis**: condition on the first move, and the Markov property hands you a linear system.

Gambler's ruin

A gambler bets one unit on a fair coin each round, starting with k units, and stops on reaching 0 (ruin) or N (target). Then

$$P(\text{reach } N \text{ before } 0) = \frac{k}{N}, \qquad E[\text{number of bets}] = k(N-k).$$

Proof. Let h_k be the probability of reaching N first from fortune k. After one bet the fortune is $k \pm 1$ with probability $\frac{1}{2}$ each, and the future is a fresh game from there, so

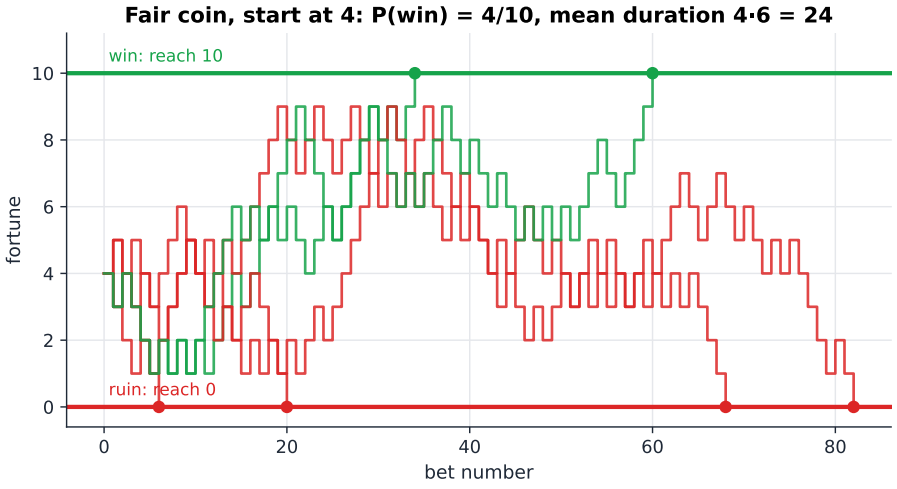
$$h_k = \tfrac{1}{2} h_{k+1} + \tfrac{1}{2} h_{k-1}, \qquad h_0 = 0,\ h_N = 1.$$

Rearranged, $h_{k+1} - h_k = h_k - h_{k-1}$: the differences are constant, so h_k is linear, and the boundary values give $h_k = k/N$.

For the duration, let m_k be the expected number of bets from k. One bet is always made, then the game continues from $k \pm 1$:

$$m_k = 1 + \tfrac{1}{2} m_{k+1} + \tfrac{1}{2} m_{k-1}, \qquad m_0 = m_N = 0.$$

Now the second difference is constant: $m_{k+1} - 2m_k + m_{k-1} = -2$. A quadratic with second difference -2 is $-k^2 + bk + c$; the boundary conditions give $c = 0$, $b = N$, so $m_k = k(N-k)$. ∎



Six games starting from 4 units with target 10. On average 4 in 10 games are won, and a game lasts 24 bets. Individual games vary enormously in length.

Look at the duration formula again. Start in the middle, $k = N/2$, and the expected game length is $N^2/4$: a random walk needs about N^2 steps to travel distance N. Distance grows like the *square root* of time. This $\sqrt{n}$ law is the fingerprint of every diffusive process, and it reappears as the envelope of Brownian motion in Ch. 36.

The same algebra with a biased coin (win probability p, $q = 1-p$, $r = q/p \neq 1$) gives

$$P(\text{reach } N \text{ before } 0) = \frac{1 - r^k}{1 - r^N}.$$

At a roulette-like $p = 0.49$, a gambler who starts with 50 and wants to double to 100 succeeds only about 12 % of the time, against 50 % for a fair coin. A 2 % edge, compounded over the long journey a random walk takes, becomes crushing.

Two more facts about hitting times, stated without proof:

- **Kac's formula.** For an irreducible chain with stationary $\boldsymbol{\pi}$, the expected time to return to state i, starting from i, is $1/\pi_i$. In the weather chain a rainy day recurs on average every $46/12 \approx 3.83$ days. First-step analysis also gives the expected wait for rain starting from a sunny day: $20/3 \approx 6.67$ days.
- **Recurrence on the infinite line.** The fair walk on $\mathbb{Z}$ has no stationary distribution (the ink spreads forever). Letting $N \to \infty$ in the gambler's ruin, the chance of escaping to $+N$ before returning to 0 from $k = 1$ is $1/N \to 0$, so the walker returns with probability one. Yet the expected return time is infinite: Kac's formula with every $\pi_i = 0$ gives $1/0$. The hook's first paradox is just this.

? The drunk bird

On $\mathbb{Z}^d$ the simple random walk returns to its start with probability one for $d = 1, 2$ (Pólya, 1921) but only with probability about 0.3405 for $d = 3$. The gambler's-ruin trick only works in one dimension. The real proof turns the question into whether the sum $\sum_n P(X_{2n} = 0)$ diverges, and computing $P(X_{2n} = 0) \sim c\, n^{-d/2}$ needs Fourier analysis on the lattice (Ch. 16) or, in the continuum, the Green's function of the Laplacian, which behaves like $1/r$ in three dimensions (Ch. 89). Why should the dimension of space, and not the coin, decide whether you get home?

Mixing times

The convergence theorem says *that* the ink spreads. The practical question is *how long* to wait.

Mixing time

$$t_{\text{mix}}(\varepsilon) = \min\left\{ n : \max_{i \in S} d_{TV}\left(\mathbf{P}^n_{i,\cdot}, \boldsymbol{\pi}\right) \le \varepsilon \right\},$$

the number of steps after which, from the *worst* starting state, the chain is within ε of equilibrium. By convention $t_{\text{mix}} = t_{\text{mix}}(1/4)$.

For the weather chain, after one day the worst start (Rainy) is still at distance 0.257; after two days every start is below 0.127. So $t_{\text{mix}} = 2$ days. Weather forgets fast.

For reversible chains the spectral gap gives two-sided bounds, roughly

$$\frac{1}{\text{gap}} - 1 \;\lesssim\; t_{\text{mix}} \;\lesssim\; \frac{1}{\text{gap}} \ln \frac{1}{\pi_{\min}},$$

up to constants. The logarithm on the right matters: a chain on $52!$ states can still mix in a handful of steps if its gap is not tiny.

? Seven shuffles and the cliff

Bayer and Diaconis (1992) computed the exact distance to uniform for riffle-shuffling 52 cards: 1.000 after four shuffles, then $0.924,\ 0.614,\ 0.334,\ 0.167,\ 0.085$ after five to nine. For n cards the drop happens near $\tfrac{3}{2}\log_2 n$ shuffles and gets sharper as n grows. This sudden collapse is the **cutoff phenomenon**. Eigenvalues alone predict smooth geometric decay, so they cannot explain a cliff. Which chains have cutoff, and why, is a major open problem: Peres's "product condition" is necessary, fails to be sufficient in general, and has been proven sufficient only for special families such as birth-and-death chains. Ch. 38 shows why the answer matters every time someone runs a Markov chain Monte Carlo sampler and asks when to stop.

34.4 The algorithm

PageRank by power iteration

The random surfer of the hook follows a link chosen uniformly from the current page. Two defects stop this from being a well-behaved chain: some pages have no outgoing links (the surfer is stuck), and the web is not irreducible (some regions cannot be left). The fix is to add **teleportation**: with probability d (the damping factor, classically 0.85) follow a random link; with probability $1 - d$ jump to a page chosen uniformly from all n. The new transition matrix is

$$\mathbf{G} = d\,\mathbf{P} + (1 - d)\,\frac{1}{n}\mathbf{1}\mathbf{1}^{\mathsf{T}},$$

and every entry of $\mathbf{G}$ is at least $(1 - d)/n > 0$. Every entry positive means the contraction theorem of §34.3 applies directly: a unique stationary distribution exists, and iteration converges to it from any start. In fact the same proof, applied with the rank-one teleport term, gives $\delta(\mathbf{G}) \le d$, so the error shrinks by at least 0.85 per sweep, independent of the size of the web.

```
import numpy as np

def pagerank(links, n, d=0.85, tol=1e-10, max_iter=1000):
    """links: dict page -> list of pages it links to. Returns the stationary vector."""
    x = np.full(n, 1.0 / n)
    for it in range(max_iter):
        new = np.full(n, (1 - d) / n)          # teleport: every page gets (1-d)/n
        dangling = 0.0
        for i in range(n):
            out = links.get(i, [])
            if out:
                share = d * x[i] / len(out)
                for j in out:
                    new[j] += share
            else:
                dangling += d * x[i]            # page with no links: jump anywhere
        new += dangling / n
        if np.abs(new - x).sum() < tol:
            return new, it + 1
        x = new
    return x, max_iter

print(pagerank({0: [1, 2], 1: [2], 2: [0], 3: [2]}, 4))
```

On the four-page web A→{B, C}, B→C, C→A, D→C, it converges in 47 sweeps to

$$\boldsymbol{\pi} \approx (0.373,\ 0.196,\ 0.394,\ 0.0375).$$

Page C wins: three pages point to it. Page A comes second although only one page links to it, because that one page is C, the most important page. Importance is inherited from who links to you, which is exactly what $\boldsymbol{\pi} = \boldsymbol{\pi}\mathbf{G}$ says. Page D, which nobody links to, gets only the teleport share $(1-d)/n = 0.0375$.

Cost. Each sweep touches every link once, so it costs $O(n + |E|)$, never forming the dense matrix $\mathbf{G}$. To reach error ε you need about $\ln \varepsilon / \ln d$ sweeps: about 140 for $\varepsilon = 10^{-10}$ in the worst case, usually far fewer. Compare a direct solve of $\boldsymbol{\pi}(\mathbf{I} - \mathbf{G}) = \mathbf{0}$ at $O(n^3)$ for a dense matrix: hopeless for billions of pages.

Stability. Each sweep maps probability vectors to probability vectors, so errors cannot grow, and nothing is ever divided by a small number. Round-off drift in the total mass can be removed by renormalising $\mathbf{x}$ to sum to 1 after each sweep.

Failure cases

- **Periodic chains.** Without teleportation, power iteration on a bipartite graph oscillates forever, exactly like the two-state swap. The cure is either teleportation or a **lazy** chain $\frac{1}{2}(\mathbf{I} + \mathbf{P})$, which has the same $\boldsymbol{\pi}$ and no negative eigenvalues.
- **Small spectral gap.** If $d \to 1$ the guarantee $\delta \le d$ degrades, and on webs with weakly connected clusters $|\lambda_2|$ approaches 1: convergence becomes painfully slow, and the answer becomes hypersensitive to small changes in the links. This is one reason $d = 0.85$ rather than 0.99.
- **Simulating instead of iterating.** Estimating $\boldsymbol{\pi}$ by running a single surfer and counting visits also works (the ergodic theorem, the Markov-chain cousin of Ch. 21 §21.7), but the error falls like $1/\sqrt{\text{steps}}$, inflated by the correlation between successive states. The slower the chain mixes, the more steps each "independent" sample costs.

34.5 The experiment

Part 1: the weather in your terminal

The script below simulates the weather chain for 2000 days, compares the running fraction of sunny days with $\pi_S = 21/46$, and plots the distance to equilibrium.

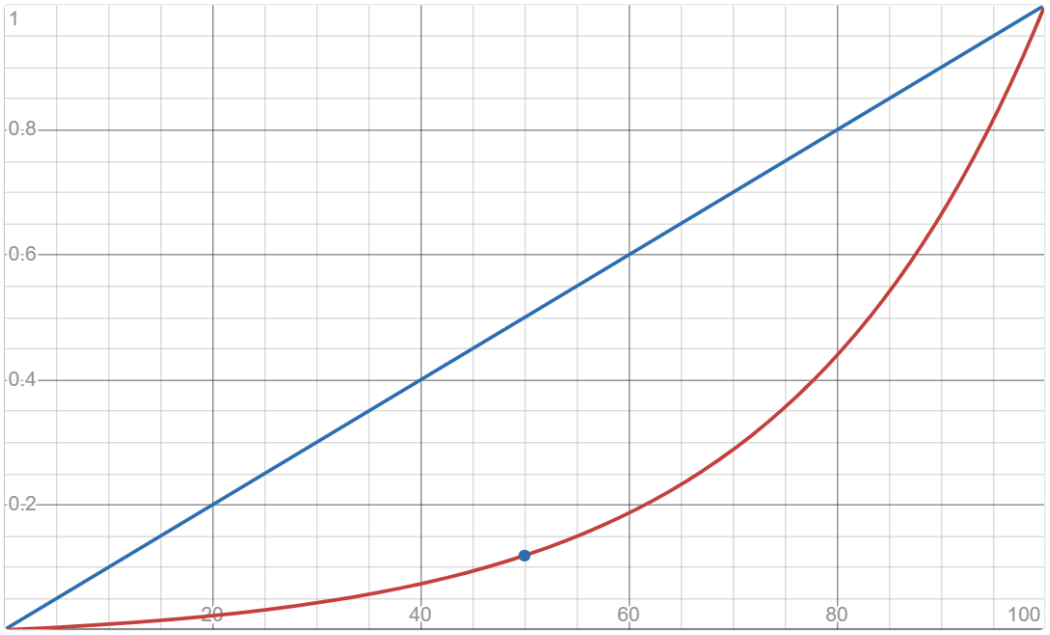
```
import numpy as np
import matplotlib.pyplot as plt

# ---- knobs ---------------------------------------------------------------
P = np.array([[0.7, 0.2, 0.1],      # Sunny  -> (Sunny, Cloudy, Rainy)
              [0.3, 0.4, 0.3],      # Cloudy -> ...
              [0.2, 0.3, 0.5]])     # Rainy  -> ...
start = 2          # 0 = Sunny, 1 = Cloudy, 2 = Rainy
days = 2000        # length of one simulated history
seed = 1
# --------------------------------------------------------------------------

w, V = np.linalg.eig(P.T)                    # left eigenvectors of P
pi = np.real(V[:, np.argmin(abs(w - 1))]); pi /= pi.sum()
lam = sorted(abs(w))[-2]
print("stationary pi  :", np.round(pi, 4))
print("second |eigval|:", round(lam, 4))

rng = np.random.default_rng(seed)
x, path = start, []
for _ in range(days):
    path.append(x)
    x = rng.choice(3, p=P[x])                # one Markov step
path = np.array(path)
running = np.cumsum(path == 0) / np.arange(1, days + 1)
print("simulated fraction Sunny:", round(running[-1], 4))

mu, tv = np.eye(3)[start], []
for n in range(30):
    tv.append(0.5 * abs(mu - pi).sum()); mu = mu @ P

fig, (a, b) = plt.subplots(1, 2, figsize=(10, 3.5))
a.plot(running); a.axhline(pi[0], ls="--", c="k")
a.set(xscale="log", xlabel="day", ylabel="running fraction Sunny")
b.semilogy(tv, "o-"); b.semilogy(tv[0] * lam ** np.arange(30), "k:")
b.set(xlabel="day n", ylabel="TV distance to pi")
plt.tight_layout(); plt.show()
```

With the knobs as given it prints $\boldsymbol{\pi} \approx (0.4565, 0.2826, 0.2609)$, $|\lambda_2| \approx 0.4732$, and a simulated sunny fraction of about 0.47. Things to change:

1. **Make the weather sticky.** Set the diagonal to 0.98 and spread 0.01 to each other state. The stationary distribution becomes uniform, but $|\lambda_2|$ jumps to 0.97. Watch the running average wander for hundreds of days before it settles. Same $\boldsymbol{\pi}$, very different mixing.
2. **Break aperiodicity.** Use the cyclic chain Sunny→Cloudy→Rainy→Sunny with probability 1. The distance curve stops decaying. Which eigenvalues have modulus 1 now?
3. **Break irreducibility.** Make Rainy absorbing (row $[0, 0, 1]$). What is $\boldsymbol{\pi}$, and does the running fraction of sunny days still converge to it?
4. Increase `days` by factors of 10 and check that the simulation error shrinks roughly like $1/\sqrt{\text{days}}$.

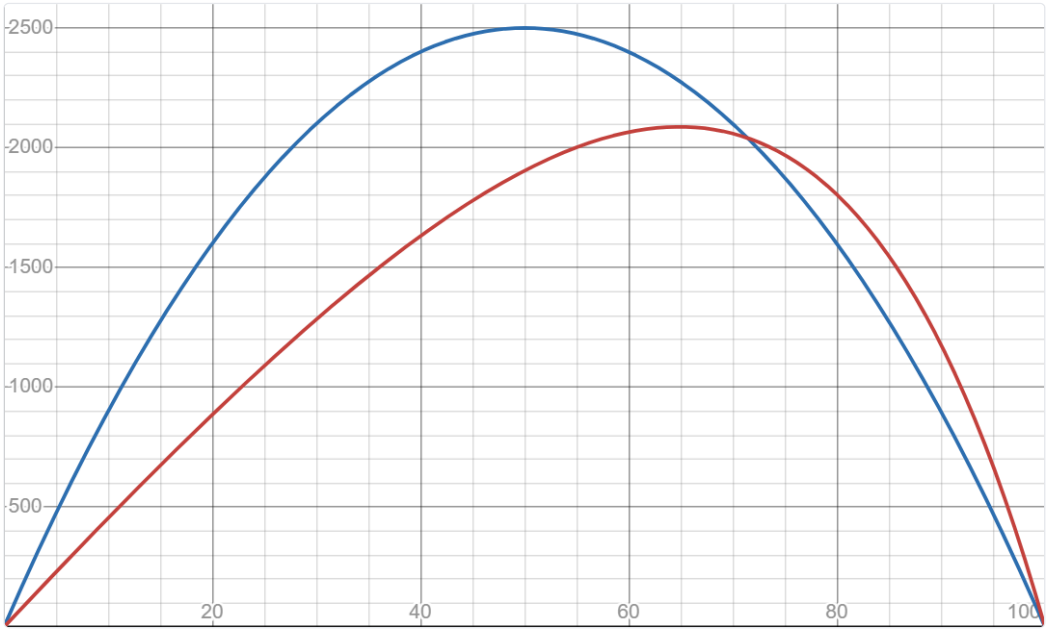
Part 2: the casino's edge

The probability of reaching 100 before 0, as a function of the starting fortune x, for a fair coin (blue) and a coin with $p = 0.49$ (red):



Doubling 50 to 100: a fair coin succeeds half the time; a 2 % edge cuts it to 12 %.

Change $0.51/0.49$ to $0.52/0.48$, or change the target 100 (in both places, and in the fair line) to 1000. How small must the edge be before the red curve starts to look like the blue line? The expected game lengths tell the same story: from 50, the fair game lasts $50 \cdot 50 = 2500$ bets on average, the $p = 0.49$ game about 1904 bets. The house edge does not just tilt the outcome; it ends the game sooner.



Expected number of bets before the game ends, fair (blue) and $p = 0.49$ (red). The red formula is $k/(q-p) - \frac{N}{q-p} \cdot \frac{1-r^k}{1-r^N}$; the 0.02 is $q - p$.

34.6 The challenge

1. **Intuition.** A random walk on a connected graph has stationary distribution proportional to degree. A random walk on a *directed* graph does not, in general. Construct a three-vertex directed graph where the vertex with the most incoming links is *not* the most visited, and explain in words what the walker is doing.
2. **Derivation.** For the biased gambler's ruin, derive $h_k = (1 - r^k)/(1 - r^N)$ from the recurrence $h_k = p\,h_{k+1} + q\,h_{k-1}$. Then let $N \to \infty$ and find the probability that a gambler with a favourable coin ($p > \tfrac{1}{2}$) starting with k units is *never* ruined.
3. **Implementation.** Write a function that estimates $t_{\text{mix}}(1/4)$ for any transition matrix by computing $\max_i d_{TV}(\mathbf{P}^n_{i\cdot}, \boldsymbol{\pi})$ for increasing n. Apply it to the lazy random walk on a cycle of n vertices for $n = 10, 20, 40, 80$ and fit the growth law. Is it closer to n, n^2 or $n \ln n$?
4. **Open investigation.** Take the four-page web of §34.4 and add one page that links only to itself and to A. How does each page's rank change as d goes from 0.5 to 0.99? Can a page owner raise their own rank by adding outgoing links, removing them, or creating a ring of pages that link to each other? Search for the strongest manipulation you can find with, say, three extra pages. (No closed-form answer is known for the general problem of optimal "link farms".)

34.7 Loose threads

This chapter worked almost entirely with finite chains in discrete time, where matrices and eigenvalues do the heavy lifting. Several doors were left shut on purpose.

> **? Can you build a chain for any target?**
>
> Here everything ran forward: given $\mathbf{P}$, find $\boldsymbol{\pi}$. Turn it around. Suppose you want samples from a distribution $\boldsymbol{\pi}$ on a space far too large to enumerate, known only up to a normalising constant: every configuration of a million-spin magnet, or every value of a model's parameters given data. Detailed balance hints that you could *design* a chain whose stationary distribution is exactly $\boldsymbol{\pi}$, using only ratios π_j/π_i, and then just run it. That recipe, Metropolis–Hastings, is one of the most important algorithms of the twentieth century. How it works, and how to tell when it has run long enough, is in Ch. 38. The posterior distributions of Ch. 35 are its main customers.

Other threads:

- **Continuous time and continuous space.** Shrink the step and the step size of the random walk together, with steps of size $\sqrt{\Delta t}$ every Δt, and the walk becomes Brownian motion. The gambler's-ruin answers survive the limit. Why they survive, and the strange calculus that results, is Ch. 36.
- **Chains with choices.** If, in each state, you may *choose* the transition matrix and you collect rewards, you have a Markov decision process. The stationary-distribution machinery becomes Bellman's equation: Ch. 52 and Ch. 53.
- **Random walks as a lens on networks.** Hitting times between vertices behave like effective resistances in an electrical network with a resistor on every edge, a link that runs both ways and is developed with graph Laplacians in Ch. 27 and Ch. 87.
- **The entropy of a chain.** A stationary chain produces information at a definite rate, $-\sum_i \pi_i \sum_j P_{ij} \log_2 P_{ij}$ bits per step. Why that is the right number for compressing its output is a question for Ch. 33.

Further reading

- D. A. Levin and Y. Peres, *Markov Chains and Mixing Times*, 2nd ed., AMS, 2017. The modern reference on mixing, coupling and cutoff; the first chapters are very readable.
- J. R. Norris, *Markov Chains*, Cambridge University Press, 1997. Clean treatment of hitting times, recurrence and continuous-time chains.
- P. G. Doyle and J. L. Snell, *Random Walks and Electric Networks*, MAA, 1984 (free online). Pólya's theorem through resistor networks.
- D. Bayer and P. Diaconis, "Trailing the dovetail shuffle to its lair," *Annals of Applied Probability* 2 (1992), 294–313. The seven-shuffles paper.

Chapter 35 — Bayesian Thinking and Statistical Inference

> **Level 2 · Track B — Probability, Uncertainty, and Information Builds on:** Ch. 19, Ch. 20 · **Leads to:** Ch. 37, Ch. 38, Ch. 91 · **Related:** Ch. 33, Ch. 41 **Reading time:** ~30 min

35.1 The hook

The perfect fab line. A new production line has made its first 20 chips, and all 20 pass the test. Your manager wants a number for the datasheet: the defect rate. The textbook estimator from Ch. 20, the maximum-likelihood estimate, is $0/20 = 0$. Zero defects. Would you print "defect rate: 0 %" and promise every customer a perfect part? Nobody would. But if not zero, then what, and on what grounds?

The sunrise. Pierre-Simon Laplace asked a version of the same question in 1814. Suppose all you know is that the sun has risen every day for 5000 years, about 1,826,213 days. What is the probability that it rises tomorrow? He gave an exact number, and the method behind it is the one that rescues the fab line.

The almost-significant coin. A colleague tosses a coin 100 times and gets 60 heads. The two-sided p-value for "the coin is fair" is 0.057: just short of the conventional 0.05, "nearly significant", the sort of result that gets written up as "a trend toward bias". A second analyst looks at the same 100 tosses and reports that the data favour the coin being fair, by a small margin. Same data, opposite leaning. One of them must have made a mistake?

Neither did. They answered different questions. This chapter is about the difference: what it means to *learn* a parameter from data, how prior knowledge enters (it always does, openly or not), and why the most common number in science, the p-value, is not the probability that a hypothesis is true. By the end you will compute the fab line's defect rate, Laplace's sunrise, and the Bayes factor for the coin, and you will see the coin dispute become a full-blown paradox as the sample grows.

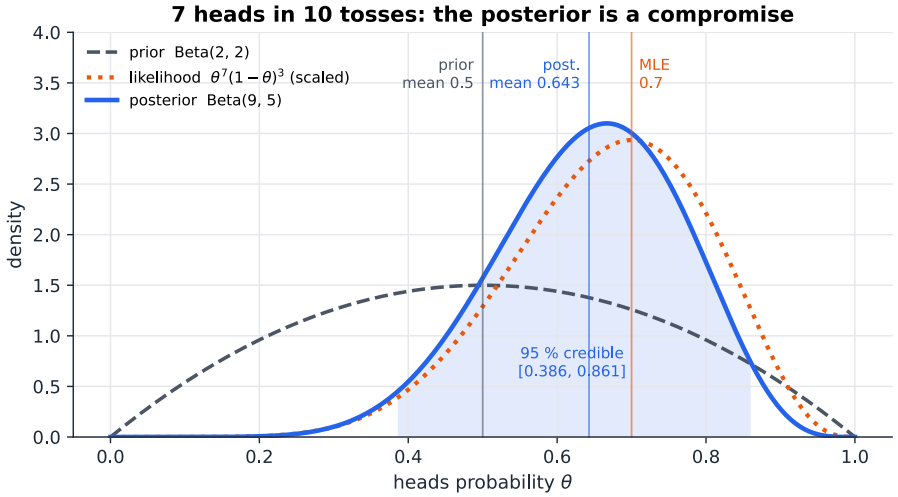
35.2 The big idea

Treat the unknown as random. In Ch. 19 §19.3, Bayes' theorem updated the probability of an *event*: is the patient ill, is the bit a 1? Bayesian inference applies the same rule to a *parameter*: the heads probability θ of a coin, the defect rate of a line, the resistance of a sensor. Before data, your knowledge of θ is a distribution, the **prior**. Data arrive. The rule

$$\text{posterior} \;\propto\; \text{likelihood} \times \text{prior}$$

turns the prior into the **posterior**: everything you now know about θ, as a full distribution rather than a single number.

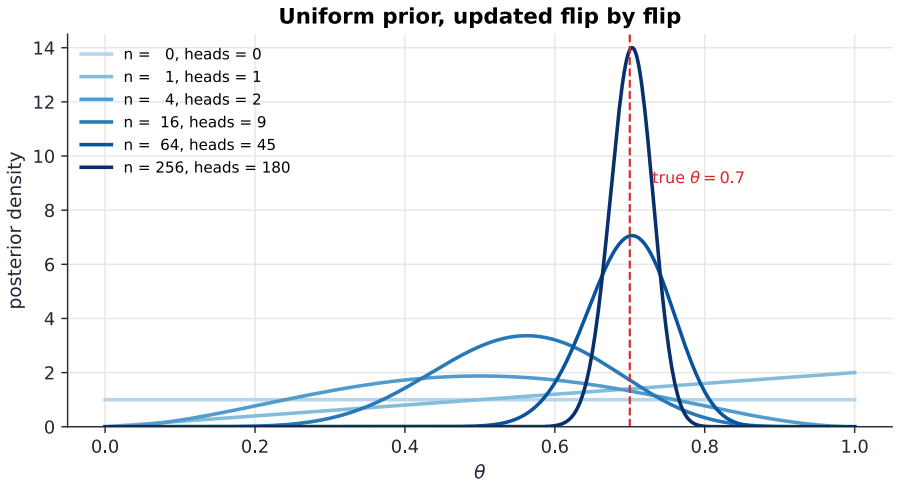
The analogy is a **detective's suspect board**. Before evidence, each suspect has some plausibility. Each clue raises the plausibility of the suspects it fits and lowers the others, in proportion to how well it fits (the likelihood). The detective never jumps to "the butler did it, probability 1" on one clue, and never refuses to update. With enough clues, detectives with different initial hunches end up agreeing.



Seven heads in ten tosses. A mild prior centred at 0.5 (grey) meets the likelihood (orange, peaking at the MLE 0.7). The posterior (blue) sits between them, narrower than either, with its central 95 % shaded.

Three consequences make the method worth learning.

1. **Uncertainty comes for free.** The posterior's width says how much you know. Twenty perfect chips do not prove a zero defect rate; they rule out large rates and leave small ones plausible.
2. **Learning is sequential.** Today's posterior is tomorrow's prior. You can update after every chip, every toss, every radar return, and the result is the same as updating once with all the data.
3. **Data eventually win.** As the sample grows, the likelihood becomes sharp and the prior's influence fades.



A flat prior updated by tosses of a coin with true $\theta = 0.7$. After 256 tosses the posterior is a narrow spike near the truth.

The price is that you must say what you believed before seeing data. Critics call that subjective. Bayesians reply that every method makes such assumptions; this one writes them down where they can be argued with.

35.3 The mathematics

Posterior, likelihood and evidence

Let data D have probability (or density) $p(D \mid \theta)$ given a parameter θ. Read as a function of θ with D fixed, this is the **likelihood** $L(\theta) = p(D \mid \theta)$ of Ch. 20 §20.6. With a prior density $p(\theta)$, Bayes' theorem for densities gives

$$p(\theta \mid D) = \frac{p(D \mid \theta)\, p(\theta)}{p(D)}, \qquad p(D) = \int p(D \mid \theta)\, p(\theta)\, d\theta.$$

The denominator $p(D)$, the **evidence** or **marginal likelihood**, does not depend on θ: it only normalises the posterior. It will matter enormously when we compare models.

The coin: conjugacy

Model n tosses as independent Bernoulli(θ) trials with k heads. The likelihood is $L(\theta) = \binom{n}{k}\theta^k(1-\theta)^{n-k}$. For the prior choose the **Beta distribution**

$$p(\theta) = \frac{\theta^{a-1}(1-\theta)^{b-1}}{B(a,b)}, \quad 0 < \theta < 1, \qquad B(a,b) = \int_0^1 t^{a-1}(1-t)^{b-1}dt = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}.$$

Its mean is $a/(a+b)$. The parameters behave like pseudo-counts: $a-1$ imagined heads and $b-1$ imagined tails. Beta$(1,1)$ is uniform, Beta$(2,2)$ is a gentle hump at 0.5, Beta$(50,50)$ is a confident belief that the coin is nearly fair.

Beta–binomial conjugacy

With prior $\theta \sim \text{Beta}(a,b)$ and k heads in n tosses,

$$\theta \mid D \;\sim\; \text{Beta}(a+k,\; b+n-k),$$

and the posterior mean is a weighted average of the prior mean and the MLE $\hat\theta = k/n$:

$$E[\theta \mid D] = \frac{a+k}{a+b+n} = \underbrace{\frac{a+b}{a+b+n}}_{\text{weight on prior}} \cdot \frac{a}{a+b} + \underbrace{\frac{n}{a+b+n}}_{\text{weight on data}} \cdot \frac{k}{n}.$$

Proof. Multiply likelihood and prior and drop every factor free of θ:

$$p(\theta \mid D) \propto \theta^k(1-\theta)^{n-k} \cdot \theta^{a-1}(1-\theta)^{b-1} = \theta^{(a+k)-1}(1-\theta)^{(b+n-k)-1}.$$

This is the kernel of Beta$(a+k, b+n-k)$, and a density is determined by its kernel, since the constant is fixed by integrating to 1. The mean of a Beta(α,β) is $\alpha/(\alpha+\beta)$; the weighted-average form follows by writing $a+k = (a+b)\frac{a}{a+b} + n\frac{k}{n}$. ∎

A prior is **conjugate** to a likelihood when the posterior stays in the prior's family. Updating then reduces to bookkeeping: add heads to a, tails to b. The weight on the data, $n/(a+b+n)$, tends to 1: the prior counts as $a+b$ observations' worth of opinion, and real observations outnumber it eventually.

For the figure, Beta$(2,2)$ and 7 heads in 10 give Beta$(9,5)$: posterior mean $9/14 \approx 0.643$, between the prior mean 0.5 and the MLE 0.7. The central 95 % **credible interval** is $[0.386,\ 0.861]$, and $P(\theta > 0.5 \mid D) \approx 0.867$.

Prediction and the rule of succession

The question you usually care about is not "what is θ" but "what happens next". The **posterior predictive** probability averages over everything θ could be.

Predictive probability and the rule of succession

Under the model above, the probability that toss $n+1$ is a head is the posterior mean:

$$P(\text{head on toss } n+1 \mid D) = \int_0^1 \theta\, p(\theta \mid D)\, d\theta = \frac{a+k}{a+b+n}.$$

With a uniform prior ($a = b = 1$) this is **Laplace's rule of succession**, $\dfrac{k+1}{n+2}$.

Proof. Condition on θ: given θ, the next toss is a head with probability θ, independently of the past. By the law of total probability over the posterior, $P(\text{head} \mid D) = E[\theta \mid D]$, and conjugacy gives the value. ∎

Now the hook's first two puzzles fall.

- **Sunrise.** $k = n = 1{,}826{,}213$: $P(\text{sunrise}) = \frac{n+1}{n+2} \approx 1 - 5.5 \times 10^{-7}$. Laplace himself added that anyone who understands the physics of the solar system would be far more confident: the uniform prior encodes ignorance, and he was not ignorant.
- **Fab line.** $k = 0$ defects in $n = 20$ with a uniform prior on the defect rate: the posterior is $\text{Beta}(1, 21)$, the chance that the next chip is defective is $1/22 \approx 4.5\,\%$, and the 95 % upper credible bound on the rate is $1 - 0.05^{1/21} \approx 13.3\,\%$. That is an honest datasheet line: "no failures in 20; rate below about 13 % with 95 % credibility." Engineers know its frequentist cousin as the **rule of three**: with zero failures in n trials, the 95 % upper confidence bound is about $3/n$ (exactly $1 - 0.05^{1/20} \approx 13.9\,\%$ here).

Three estimators, three losses

If you must report one number, which one? The posterior offers several.

- **MLE**, $\arg\max_\theta L(\theta)$: ignores the prior. For the coin, k/n.
- **MAP** (maximum a posteriori), $\arg\max_\theta p(\theta \mid D)$: the posterior's peak. For $\text{Beta}(\alpha, \beta)$ with $\alpha, \beta > 1$ it is $(\alpha - 1)/(\alpha + \beta - 2)$; for $\text{Beta}(9, 5)$, $8/12 \approx 0.667$. With a uniform prior MAP equals MLE.
- **Posterior mean**, $E[\theta \mid D]$: 0.643 for $\text{Beta}(9, 5)$.

Which is "best" depends on what an error costs you, and decision theory makes this precise.

The posterior mean minimises expected squared error

For any estimate c computed from the data,

$$E\big[(\theta - c)^2 \mid D\big] = \operatorname{Var}(\theta \mid D) + \big(E[\theta \mid D] - c\big)^2,$$

so the posterior expected squared loss is minimised exactly by $c = E[\theta \mid D]$.

Proof. Write $m = E[\theta \mid D]$ and expand $(\theta - c)^2 = (\theta - m)^2 + 2(\theta - m)(m - c) + (m - c)^2$. Take the posterior expectation: the first term gives the posterior variance, the middle term vanishes because $E[\theta - m \mid D] = 0$ and $m - c$ is a constant given D, and the last term is constant. ∎

The same argument applied to absolute error $|\theta - c|$ picks out the posterior **median**, and a 0–1 loss on small intervals picks out the MAP. There is no loss-free notion of the best estimate. Notice also what the theorem says about the MLE: it is not the minimum-MSE answer once you believe anything about θ. Estimators that pull the MLE toward a prior value are called **shrinkage** estimators, and they can beat the MLE on mean squared error even under frequentist rules, a fact that shocked statisticians when Stein proved a version of it in 1956.

Credible intervals versus confidence intervals

A 95 % **credible interval** contains θ with posterior probability 0.95: a statement about θ, given the data you saw. A 95 % **confidence interval** (Ch. 20 §20.7) is produced by a procedure that captures the true θ in 95 % of repeated experiments: a statement about the procedure, not about this particular interval. In large samples with mild priors the two nearly coincide numerically, which is why they are so often confused. In small samples, and especially at boundaries like zero defects, they can differ substantially.

Testing hypotheses: the Bayes factor

To compare two hypotheses H_0 and H_1, apply Bayes' theorem to the hypotheses themselves:

$$\underbrace{\frac{P(H_0 \mid D)}{P(H_1 \mid D)}}_{\text{posterior odds}} = \underbrace{\frac{p(D \mid H_0)}{p(D \mid H_1)}}_{\text{Bayes factor } B_{01}} \times \underbrace{\frac{P(H_0)}{P(H_1)}}_{\text{prior odds}}.$$

The **Bayes factor** is the part supplied by the data. Each $p(D \mid H)$ is a marginal likelihood: the probability of the data averaged over that hypothesis's prior on its parameters.

For the 60-heads coin, take $H_0 : \theta = \frac{1}{2}$ and $H_1 : \theta \sim \text{Uniform}(0, 1)$, "the coin is biased, by an amount I cannot predict". Then

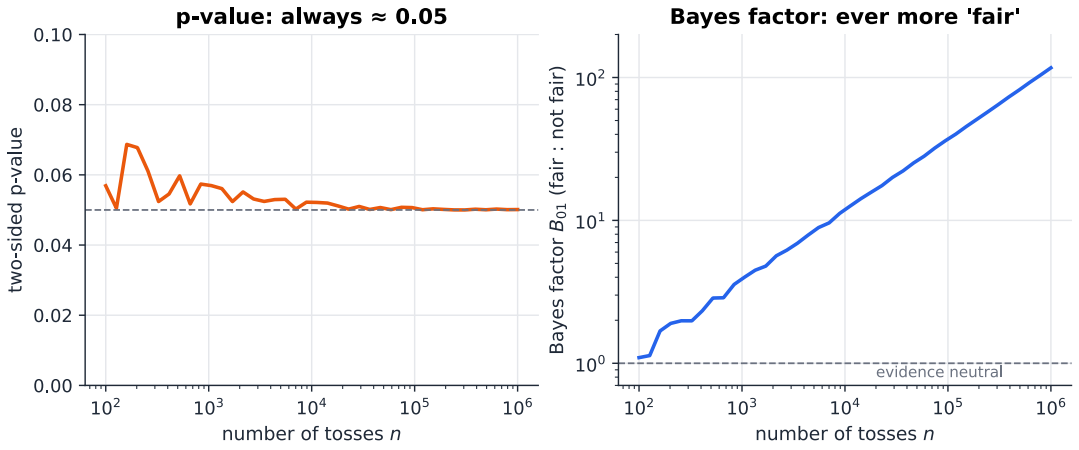
$$p(D \mid H_0) = \binom{100}{60} 2^{-100} \approx 0.01084,$$

and under H_1, integrating the binomial against the uniform prior with the Beta integral,

$$p(D \mid H_1) = \int_0^1 \binom{n}{k} \theta^k (1-\theta)^{n-k} d\theta = \binom{n}{k} \frac{k!\,(n-k)!}{(n+1)!} = \frac{1}{n+1} = \frac{1}{101} \approx 0.00990.$$

So $B_{01} \approx 0.01084 \times 101 \approx 1.10$. The data favour "fair" by about 10 %: essentially no evidence either way. Meanwhile the exact two-sided p-value is 0.057. The second analyst of the hook was right, and so was the first. The p-value asks "how surprising are data this extreme *if* H_0 is true?" The Bayes factor asks "how much better does H_0 predict the data than H_1 does?" Sixty heads is mildly surprising under a fair coin, but it is *also* unimpressive under a vague H_1, which spreads its bets over every θ from 0 to 1 and so predicts any particular count poorly.

Now hold the p-value fixed near 0.05 and let the sample grow, by choosing $k \approx n/2 + 0.98\sqrt{n}$ heads (a z-score of 1.96). The p-value stays at 0.05 forever. The Bayes factor does not.



Every experiment on the left is "just significant at 5 %". On the right, the same data favour the fair coin more and more strongly: $B_{01} \approx 1.1$ *at* $n = 100$, 3.7 *at* $n = 1000$, 117 *at* $n = 10^6$.

This is **Lindley's paradox** (1957). The reason is the $\sqrt{n}$ in the z-score: an excess of $0.98\sqrt{n}$ heads is a *shrinking fraction* $0.98/\sqrt{n}$ of the tosses, so the observed frequency closes in on $\frac{1}{2}$, which is exactly what H_0 predicts and what a uniform H_1 finds unremarkable. A small p-value in a huge sample can signal a tiny real effect, a tiny bias in the experiment, or nothing, and the Bayes factor rate of growth, roughly $\sqrt{n}$, measures how differently the two philosophies weigh that.

> **Common Pitfall — the p-value is not $P(H_0 \mid D)$**
>
> "$p = 0.04$, so there is a 4 % chance the null hypothesis is true" is wrong. A p-value is computed assuming H_0 is true; it cannot also be the probability that H_0 is true. Converting it into $P(H_0 \mid D)$ needs a prior and an alternative, as above. The Neyman–Pearson framework of Ch. 20 §20.8 controls error *rates* over repeated tests; it never claims to say how likely a single hypothesis is.

Model selection and the automatic Occam's razor

The same machinery compares whole models. Given models $M_1, M_2, \ldots$ with their own parameters, the evidence $p(D \mid M) = \int p(D \mid \theta, M) p(\theta \mid M)\, d\theta$ rewards fit but punishes flexibility. A model with many parameters spreads its prior probability over a huge space of possible datasets, so it assigns relatively little probability to the one you observed, unless it needed that flexibility. Simple models that predict sharply and are right win. This is Occam's razor, derived rather than assumed.

For large n and d parameters, a Laplace approximation of the evidence integral gives the **Bayesian information criterion**

$$\ln p(D \mid M) \approx \ln L(\hat{\theta}) - \frac{d}{2} \ln n,$$

stated here without proof: log-likelihood at the best fit, minus a penalty of $\frac{1}{2} \ln n$ per parameter. Compare polynomial fits of degree 1 and 5 to 50 noisy points and BIC will usually prefer degree 1 unless the curvature is real, exactly the overfitting guard that Ch. 20 §20.9 achieves with residual analysis.

> **? Where should the prior come from?**
>
> Everything here assumed a prior, and the Bayes factor is notoriously sensitive to it: replace the uniform H_1 by Beta(50, 50) ("biased, but only slightly") and B_{01} for 60 heads drops from 1.10 to 0.52, so the same data now favour "biased" by two to one. Is there an "objective" prior that represents ignorance? The uniform prior is not it: uniform in θ is not uniform in θ^2 or in the log-odds, so "ignorance" depends on the parametrisation. Harold Jeffreys proposed $p(\theta) \propto \sqrt{I(\theta)}$, built from the Fisher information, which gives the same answer in every parametrisation. For the coin it is Beta$(\frac{1}{2}, \frac{1}{2})$. Why the square root of Fisher information, and what geometry makes it invariant, is the doorway to information geometry (Ch. 59, with Fisher information and KL divergence from Ch. 33). Whether any prior can truly represent ignorance is still argued.

35.4 The algorithm

Conjugate priors are a gift that the real world rarely gives. Change the coin to a logistic-regression model of chip failure versus temperature, and the posterior has no closed form. The universal fallback in one or two dimensions is the **grid posterior**: evaluate prior times likelihood on a grid of θ values and normalise numerically.

```
GRID-POSTERIOR(data, log_prior, log_lik, grid θ_1..θ_G with spacing Δ)
  for g = 1..G:
      s_g ← log_prior(θ_g) + log_lik(data | θ_g)      # work in logs
  s_max ← max_g s_g
  for g = 1..G:
      w_g ← exp(s_g − s_max)                          # largest weight is exactly 1
  Z ← Δ · Σ_g w_g
  return posterior density w_g / Z,  log-evidence  s_max + ln Z
```

Numerical stability. The likelihood of 10 000 tosses is a product of 10 000 numbers below one, around 10^{-3000}, which underflows to zero in double precision (smallest positive double about 10^{-308}). Two rules fix this: add log-likelihoods instead of multiplying likelihoods, and subtract the maximum before exponentiating (the **log-sum-exp** trick). The returned log-evidence is exact up to the grid's quadrature error, which is what Bayes factors need.

Cost. One likelihood evaluation per grid point: $O(Gn)$ for n data points. That is trivial in one dimension. In d dimensions a grid with G points per axis has G^d points: 100 points per axis and $d = 10$ parameters is 10^{20} evaluations. This **curse of dimensionality** is why grids die beyond three or four parameters and why the rest of modern Bayesian computation exists.

Streaming updates. When a conjugate prior is available, the opposite extreme holds: the update is $O(1)$ per observation. A monitoring system tracking a defect rate keeps two counters (a, b) and increments one per chip, never storing the data.

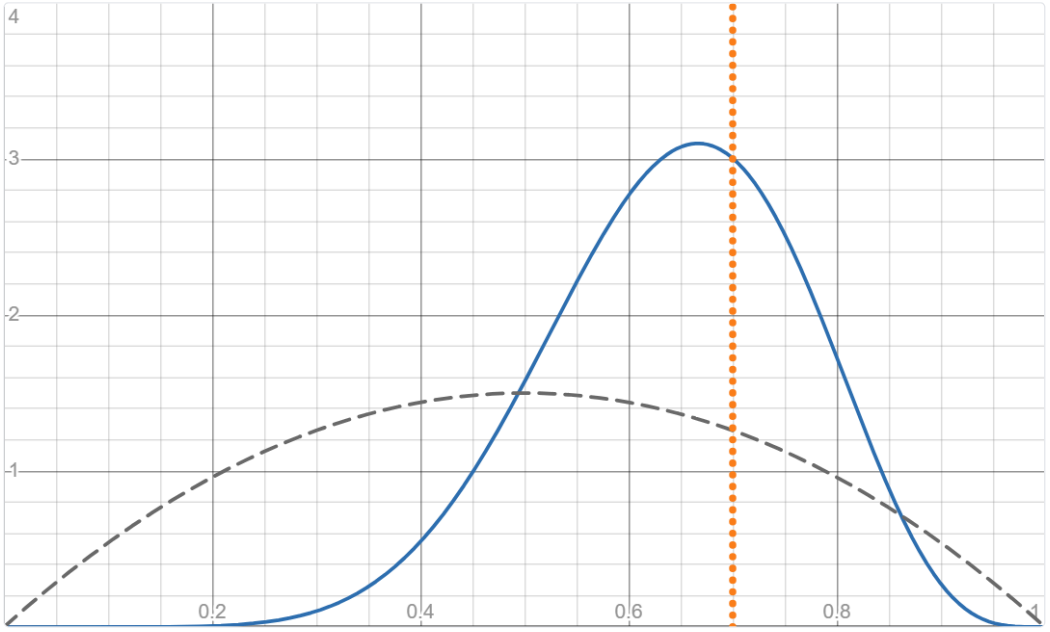
Failure cases.

- *Improper priors in Bayes factors.* A "flat prior on the whole real line" is fine for estimation (the posterior is usually proper) but makes the evidence $p(D \mid H_1)$ defined only up to an arbitrary constant. Bayes factors computed with improper priors are meaningless.
- *Grids that miss the mass.* With lots of data the posterior is a spike of width about $1/\sqrt{n}$. A grid too coarse to resolve it, or one placed where the spike is not, returns confident nonsense. Always check the posterior visibly fits inside the grid.
- *Prior–data conflict.* If a confident prior and the data disagree sharply, the posterior can land between them, where neither has support. That is a symptom worth reporting, not hiding.

35.5 The experiment

Part 1: drag the posterior

The posterior $\mathrm{Beta}(9, 5)$ for 7 heads in 10 tosses under a $\mathrm{Beta}(2, 2)$ prior, with the prior in grey:



Posterior (blue), prior (grey) and the MLE 0.7 *(orange). The blue curve is* $\mathrm{Beta}(a + k,\ b + n - k)$ *with* $a = b = 2$, $k = 7$, $n = 10$.

Edit the numbers. The two blue parameters are $a + k$ and $b + n - k$; the grey ones are a and b. Try:

1. **More data, same frequency.** Replace $9, 5$ by $72, 32$ (70 heads in 100) and change `top` to 10. The posterior narrows like $1/\sqrt{n}$ and the prior stops mattering.
2. **A stubborn prior.** Set the prior to $\mathrm{Beta}(30, 30)$ (grey 30, 30) and the posterior to $\mathrm{Beta}(37, 33)$. Seven heads in ten barely move it. How many tosses at 70 % does it take before the posterior mean passes 0.65?
3. **Zero defects.** Set the posterior to $\mathrm{Beta}(1, 21)$ and the prior to $\mathrm{Beta}(1, 1)$. This is the fab line. Most of the mass sits below 0.1, none of it at exactly zero.

Part 2: a grid posterior you can break

```
import numpy as np
import matplotlib.pyplot as plt

# ---- knobs ----------------------------------------------------------------
true_theta = 0.7      # the coin's real heads probability (hidden from the analyst)
n_flips    = 30       # how many tosses we observe
a, b       = 2, 2     # Beta(a, b) prior; try (1,1), (20,20), (1,10)
seed       = 0
# ---------------------------------------------------------------------------

rng = np.random.default_rng(seed)
data = rng.random(n_flips) < true_theta
k = int(data.sum())

theta = np.linspace(0.0005, 0.9995, 2000)          # grid over the parameter
dth = theta[1] - theta[0]
log_prior = (a - 1) * np.log(theta) + (b - 1) * np.log(1 - theta)
log_lik = k * np.log(theta) + (n_flips - k) * np.log(1 - theta)
log_post = log_prior + log_lik
post = np.exp(log_post - log_post.max())            # subtract max: no underflow
post /= post.sum() * dth                            # normalise numerically

mean = (theta * post).sum() * dth
cdf = np.cumsum(post) * dth
lo, hi = theta[np.searchsorted(cdf, 0.025)], theta[np.searchsorted(cdf, 0.975)]
print(f"heads {k}/{n_flips}   MLE {k/n_flips:.3f}   MAP {theta[post.argmax()]:.3f}")
print(f"posterior mean {mean:.3f}  (exact {(a+k)/(a+b+n_flips):.3f})")
print(f"95% credible interval [{lo:.3f}, {hi:.3f}]")
print(f"P(theta > 0.5 | data) = {post[theta > 0.5].sum()*dth:.3f}")

prior = np.exp(log_prior - log_prior.max()); prior /= prior.sum() * dth
plt.plot(theta, prior, "--", label="prior")
plt.plot(theta, post, label="posterior")
plt.axvline(true_theta, color="k", ls=":", label="true theta")
plt.xlabel("theta"); plt.ylabel("density"); plt.legend(); plt.show()
```

With the default knobs it reports 20 heads in 30, posterior mean 0.647 (matching the exact conjugate answer), a 95 % credible interval of about $[0.48,\ 0.80]$, and $P(\theta > 0.5 \mid D) \approx 0.96$. Things to try:

1. Set `n_flips = 100000` and remove the `- log_post.max()`. Watch the posterior turn into `nan`: that is underflow, live.
2. Replace the Beta prior by a *bimodal* one, say `log_prior = np.log(np.exp(-((theta-0.2)/0.05)**2) + np.exp(-((theta-0.8)/0.05)**2))`. No conjugate formula exists now, but the grid does not care. How much data does it take for one hump to win?
3. Run 1000 seeds and count how often the 95 % credible interval contains `true_theta`. Is it 95 %? Does the answer depend on the prior?

35.6 The challenge

1. **Intuition.** Two engineers estimate the same defect rate. One updates after every chip; the other waits for the whole batch of 500 and updates once. Explain without algebra why they must end with the same posterior, and name a situation in which a real analysis could break this equivalence.
2. **Derivation.** Show that the Gamma distribution is a conjugate prior for the rate λ of a Poisson count: with prior $\text{Gamma}(\alpha, \beta)$ (density $\propto \lambda^{\alpha-1} e^{-\beta\lambda}$) and counts $x_1, \dots, x_n$, find the posterior and write its mean as a weighted average of the prior mean and the sample mean. Interpret α and β as pseudo-data.
3. **Implementation.** Extend the grid code to two parameters: a Gaussian with unknown mean μ and standard deviation σ, observed through 20 noisy voltage readings. Plot the joint posterior as a contour map, then the marginal posterior of μ. Measure how runtime grows when you go from 100×100 to 400×400 grid points, and predict what three parameters would cost.
4. **Open investigation.** Lindley's paradox used a uniform alternative. For a fixed p-value of 0.05, how does the growth of B_{01} with n depend on the prior under H_1? Try $\text{Beta}(c, c)$ for several c, and a prior that shrinks with n. Is there a prior that makes the Bayes factor and the p-value agree at every sample size, and would you be willing to use it? (Statisticians still disagree about what the right reconciliation is.)

35.7 Loose threads

The chapter stayed with one or two parameters and friendly priors on purpose. The frontier starts exactly where it stopped.

> **? The integral you cannot do**
>
> Every posterior here was either conjugate or small enough for a grid. A model with 50 parameters has a posterior whose normalising integral $p(D) = \int p(D \mid \theta) p(\theta)\, d\theta$ is a 50-dimensional integral that no grid and no formula can touch. Yet modern science computes such posteriors routinely. The trick is to stop computing the posterior and start *sampling* from it, using a Markov chain built so that its stationary distribution is the posterior (Ch. 34 supplies the convergence theorem). The normalising constant cancels out of every step. How to build that chain, and why the evidence $p(D)$ remains hard even after the posterior becomes easy, is Ch. 38.

? What if the model is wrong?

As $n \to \infty$ the posterior concentrates at the true parameter and becomes Gaussian with width $1/\sqrt{nI(\theta)}$, so Bayesian and frequentist intervals agree. This is the Bernstein–von Mises theorem, and it needs conditions, including that the true distribution is in the model family. If the coin's tosses are secretly correlated, the posterior still concentrates, confidently, on the parameter that makes the wrong model least wrong in the KL sense of Ch. 33. The precise statement, its failures in infinite-dimensional models, and the measure-theoretic machinery needed even to state it are in Ch. 97.

Other threads:

- **Inference is not intervention.** A posterior that says "chips made on Tuesday fail more" tells you nothing about what would happen if you moved production off Tuesdays. Learning causes from data needs extra structure: Ch. 37.
- **Inverse problems.** Deblurring an image or reconstructing a CT scan is Bayesian inference with millions of parameters, where the prior is what makes the problem solvable at all: Ch. 91.
- **Beliefs about other people's beliefs.** When the parameter you are inferring is another player's private information and they know you are inferring it, Bayes' rule becomes a strategic tool: Ch. 41.
- **Filtering.** Updating the belief about a *moving* state, one measurement at a time, with the posterior as the next prior, is the Kalman filter when everything is Gaussian. It is Bayes' rule running on a Markov chain.

Further reading

- A. Gelman, J. B. Carlin, H. S. Stern, D. B. Dunson, A. Vehtari and D. B. Rubin, *Bayesian Data Analysis*, 3rd ed., CRC Press, 2013. The standard applied reference.
- E. T. Jaynes, *Probability Theory: The Logic of Science*, Cambridge University Press, 2003. A passionate argument that probability is extended logic.
- D. J. C. MacKay, *Information Theory, Inference, and Learning Algorithms*, Cambridge University Press, 2003 (free online). Chapter 28 on Occam's razor and model comparison is a classic.
- R. McElreath, *Statistical Rethinking*, 2nd ed., CRC Press, 2020. Gentle, code-first, and honest about priors.

Chapter 36 — Stochastic Processes: When Randomness Evolves

> **Level 2 · Track B — Probability, Uncertainty, and Information Builds on:** Ch. 19, Ch. 21, Ch. 34 · **Leads to:** Ch. 38, Ch. 89, Ch. 97 · **Related:** Ch. 13, Ch. 54, Ch. 92 **Reading time:** ~30 min

36.1 The hook

The bus that is always late. The timetable says buses come "every 10 minutes on average". They are not scheduled; they arrive at random, independently of each other, at an average rate of one per 10 minutes. You walk to the stop at a random moment. How long do you expect to wait?

The obvious answer is 5 minutes: on average you land halfway through a 10-minute gap. The correct answer is 10 minutes. And if you ask how long the gap you landed in lasts, from the bus before you to the bus after you, the answer is 20 minutes, twice the average gap. Every passenger, every day, experiences gaps twice as long as the timetable's average. No one is lying, and no buses are behaving strangely.

The integral that will not behave. Let $W(t)$ be a random signal that wanders like the pen of a seismograph, the Brownian motion of Ch. 21 §21.5. Ordinary calculus says $\int_0^t W\,dW = \frac{1}{2}W(t)^2$, by the same rule that gives $\int x\,dx = \frac{1}{2}x^2$. Compute the integral numerically, as a limit of Riemann sums, the way engineers compute every integral. You get

$$\int_0^t W\,dW = \tfrac{1}{2}W(t)^2 - \tfrac{1}{2}t.$$

An extra term, $-t/2$, appears out of nowhere, and it does not shrink as the grid gets finer. Worse, if you evaluate the Riemann sum at the midpoint of each interval instead of the left end, you get the classical answer back. For ordinary functions the evaluation point never matters in the limit. Here it changes the answer.

The investment that grows and shrinks. A stock model widely used in finance says the expected value of an investment grows by 65 % over ten years. In the same model, about 59 % of investors lose money over those ten years. Both statements are correct.

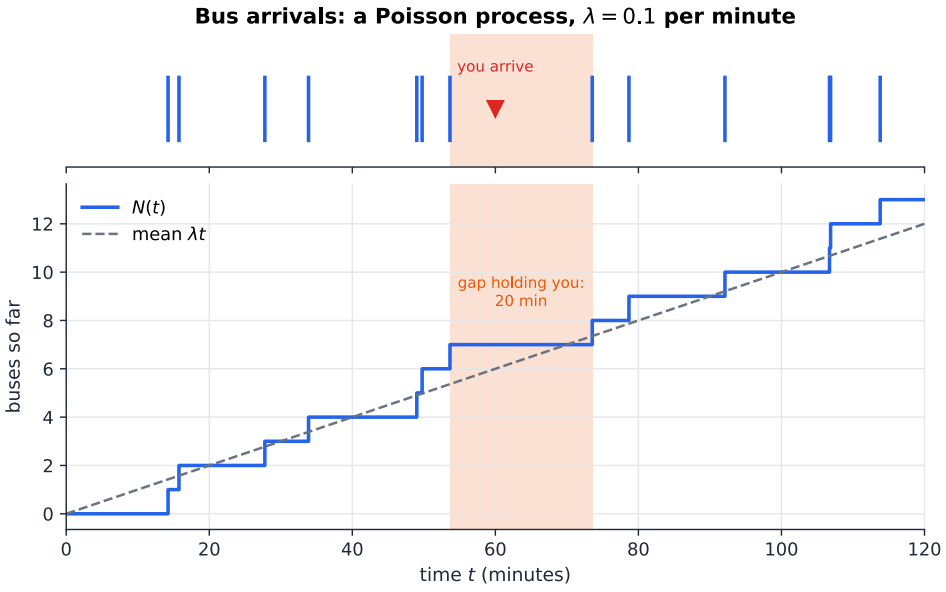
All three surprises come from one fact: when randomness accumulates in time, its *fluctuations* have rules of their own, and those rules are not the rules of smooth functions. This chapter builds the two fundamental random processes, the Poisson process (random events) and Brownian motion (random wandering), introduces the martingale (the mathematical fair game), and opens the door to stochastic calculus. It resolves the bus, the integral and the investment. It leaves some deeper doors shut.

36.2 The big idea

Randomness that adds up. A random variable is one roll of a die. A stochastic process is a whole history of rolls, indexed by time, where what happens later can build on what happened earlier. Core Ch. 21 studied processes through their averages: means, autocorrelations, spectra. This chapter studies their *paths*: what does one realisation actually look like, when does it first cross a line, how do you integrate against it?

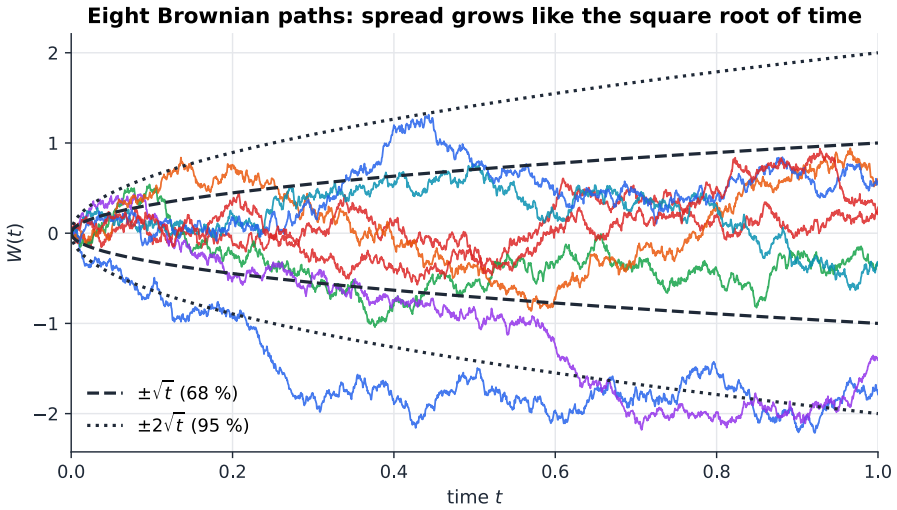
Two building blocks cover a surprising amount of the world.

- **Counting random events: the Poisson process.** Photons hitting a detector, packets arriving at a router, decays of radioactive atoms, buses. Each tiny time slot has a tiny independent chance of an event. Paths are staircases that jump by one at random times.
- **Accumulating random nudges: Brownian motion.** A pollen grain kicked by water molecules, the phase of a free-running oscillator, the integrated error of a gyroscope. Each instant adds an independent, tiny, symmetric nudge. Paths are continuous but jagged at every scale.



Top: arrival times of a Poisson process with rate 0.1 *per minute. Bottom: the count* $N(t)$ *against its mean* λt*. A passenger arriving at* $t = 60$ *falls into the longest gap in sight: long gaps are easier to land in.*

The analogy for both is **a drunk with a stopwatch**. In Ch. 34 the drunk took one step per tick. Now let the ticks get faster and the steps smaller. If each step is a rare "jump" that happens with small probability, you get the Poisson process. If every tick produces a step of size $\sqrt{\Delta t}$, you get Brownian motion. That square root is the whole secret: in a time Δt a Brownian path moves about $\sqrt{\Delta t}$, which is *much larger* than Δt when Δt is small. The path's velocity, displacement over time, is about $1/\sqrt{\Delta t}$ and blows up. The path has no velocity anywhere. That is why calculus has to be rebuilt.



Eight sample paths of standard Brownian motion. The dashed and dotted curves are $\pm\sqrt{t}$ and $\pm 2\sqrt{t}$: at each time about 68 % and 95 % of paths lie inside them.

The third idea is a lens rather than a process. A **martingale** is a fair game: a process whose expected future value, given everything you know now, is its present value. Martingales are the reason gambling systems cannot beat a fair casino, and they turn hard questions about random paths (when, where, how likely) into one line of algebra.

36.3 The mathematics

The Poisson process

Poisson process

A counting process $N(t)$, $t \geq 0$, with $N(0) = 0$, is a **Poisson process of rate** λ if

1. it has independent increments: counts in disjoint time intervals are independent;
2. it has stationary increments: the distribution of $N(t+s) - N(t)$ depends only on s;
3. in a short interval of length h, $P(\text{one event}) = \lambda h + o(h)$ and $P(\text{two or more}) = o(h)$.

Condition 3 says events are rare and arrive one at a time. From these three local rules a global formula follows.

Poisson counts

$$P\big(N(t) = k\big) = \frac{(\lambda t)^k}{k!} e^{-\lambda t}, \qquad k = 0, 1, 2, \ldots$$

Proof. Let $p_k(t) = P(N(t) = k)$. To have k events at time $t + h$, either there were k at time t and none in $(t, t+h]$, or $k-1$ and exactly one, or fewer and at least two. By independent increments and rule 3,

$$p_k(t+h) = p_k(t)(1 - \lambda h) + p_{k-1}(t)\,\lambda h + o(h).$$

Subtract $p_k(t)$, divide by h and let $h \to 0$:

$$p_0'(t) = -\lambda p_0(t), \qquad p_k'(t) = -\lambda p_k(t) + \lambda p_{k-1}(t) \quad (k \geq 1),$$

with $p_0(0) = 1$, $p_k(0) = 0$. The first equation gives $p_0(t) = e^{-\lambda t}$. For the rest, substitute $p_k(t) = q_k(t) e^{-\lambda t}$: the system becomes $q_k' = \lambda q_{k-1}$ with $q_0 = 1$, $q_k(0) = 0$, so $q_1 = \lambda t$, $q_2 = (\lambda t)^2/2$, and by induction $q_k = (\lambda t)^k / k!$. ∎

This is a system of linear ODEs of the type solved in Ch. 11, and the answer is the Poisson distribution of Ch. 19 §19.6. Two consequences follow at once.

- **Gaps are exponential.** The time T_1 to the first event satisfies $P(T_1 > t) = P(N(t) = 0) = e^{-\lambda t}$: exponential with mean $1/\lambda$. By independent, stationary increments, every later gap is the same and independent (Ch. 20 §20.4).
- **No memory.** $P(T > s + t \mid T > s) = e^{-\lambda(s+t)} / e^{-\lambda s} = e^{-\lambda t}$. Having already waited s minutes tells you nothing about the remaining wait.

The bus paradox, resolved

Memorylessness answers the first half of the hook immediately. You arrive at a random time. Whatever happened before, the time until the next bus is exponential with mean $1/\lambda = 10$ minutes. Looking backwards, the time since the last bus is (for large t) also exponential with mean 10 minutes, by the same argument run in reverse. The gap you are in averages $10 + 10 = 20$ minutes.

Why is that not a contradiction with an average gap of 10 minutes? Because you do not pick a gap at random; you pick a *moment* at random, and a gap twice as long contains twice as many moments. This is **length-biased sampling**. If gaps have density $f(x)$, the gap containing a random moment has density proportional to $x f(x)$:

$$f_{\text{seen}}(x) = \frac{x f(x)}{E[X]}, \qquad E[X_{\text{seen}}] = \frac{E[X^2]}{E[X]} = E[X] + \frac{\operatorname{Var}(X)}{E[X]}.$$

For exponential gaps $E[X^2] = 2/\lambda^2$, so $E[X_{\text{seen}}] = 2/\lambda = 20$ minutes. The formula shows the paradox is general: the gap you experience exceeds the average gap by $\operatorname{Var}(X)/E[X]$. Only perfectly regular buses ($\operatorname{Var} = 0$) give the naive answer. The same effect is why your friends have more friends than you on average, and why sampled classes look more crowded than the average class size.

Two more facts, stated without proof, make Poisson processes easy to work with. **Superposition:** merging independent Poisson streams of rates λ_1 and λ_2 gives a Poisson stream of rate $\lambda_1 + \lambda_2$. **Thinning:** keeping each event independently with probability p gives a Poisson stream of rate $p\lambda$. Together they explain why aggregated traffic from many independent sources looks Poisson even when no single source is.

Brownian motion: the square-root law

Ch. 21 §21.5 defined the Wiener process. We use the standard version, $\sigma = 1$: $W(0) = 0$, independent increments, $W(t+s) - W(t) \sim \mathcal{N}(0, s)$, continuous paths. Two properties drive everything.

Scaling. For any $c > 0$, the process $\frac{1}{\sqrt{c}} W(ct)$ is again a standard Brownian motion (its increments are Gaussian with variance $cs/c = s$). Zoom in on a Brownian path by a factor c in time and $\sqrt{c}$ in space and it looks statistically identical. The path is a random fractal, and $\operatorname{Var} W(t) = t$ means it typically sits within $\pm\sqrt{t}$ of zero, the envelope in the figure above. This is the continuum version of the gambler's-ruin observation in Ch. 34 that a walk needs N^2 steps to travel N.

Quadratic variation. Chop $[0, t]$ into n intervals of length $\Delta t = t/n$ with increments ΔW_k.

Quadratic variation of Brownian motion

$$Q_n = \sum_{k=1}^{n} (\Delta W_k)^2 \;\longrightarrow\; t \qquad \text{in mean square as } n \to \infty.$$

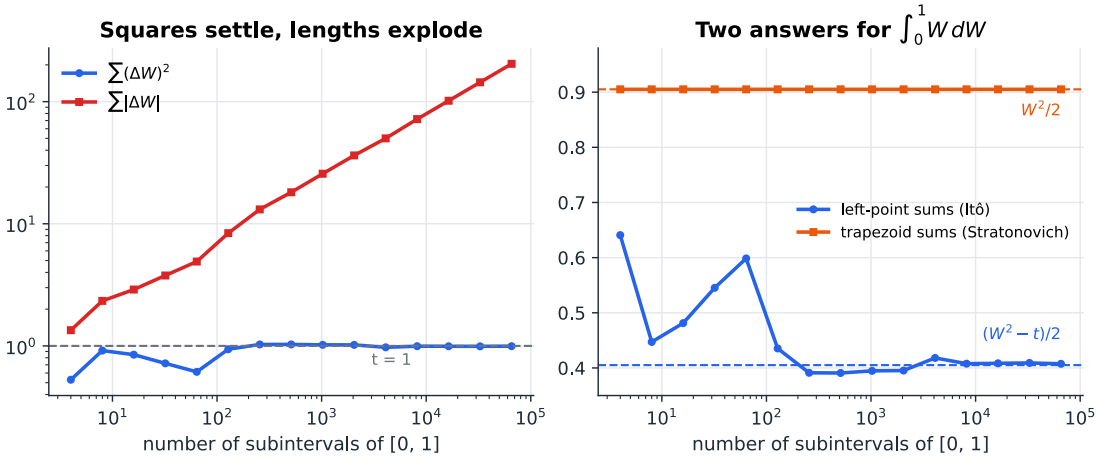
Proof. Each $\Delta W_k \sim \mathcal{N}(0, \Delta t)$, so $E[(\Delta W_k)^2] = \Delta t$ and, using $E[Z^4] = 3$ for a standard Gaussian Z, $\operatorname{Var}\left((\Delta W_k)^2\right) = 3\Delta t^2 - \Delta t^2 = 2\Delta t^2$. The increments are independent, so

$$E[Q_n] = n\,\Delta t = t, \qquad \operatorname{Var}(Q_n) = n \cdot 2\Delta t^2 = \frac{2t^2}{n} \to 0.$$

Hence $E[(Q_n - t)^2] = \operatorname{Var}(Q_n) \to 0$. ∎

For any differentiable function f, the same sum $\sum (\Delta f_k)^2 \approx \sum f'(t_k)^2 \Delta t^2 \le \max|f'|^2\, t\, \Delta t \to 0$. Smooth functions have zero quadratic variation; Brownian motion has quadratic variation exactly t, *deterministically*, even though every increment is random. The squares of the random kicks add up to a clock.

There is a sting in the tail. Since $Q_n \le \max_k |\Delta W_k| \cdot \sum_k |\Delta W_k|$, and $\max_k |\Delta W_k| \to 0$ by continuity while $Q_n \to t > 0$, the total length $\sum_k |\Delta W_k|$ must go to infinity. A Brownian path has **infinite length** on every interval. In fact (Paley, Wiener and Zygmund, 1933) it is differentiable at no point at all.



Left: on one path over $[0, 1]$, the sum of squared increments settles at $t = 1$ while the sum of absolute increments grows without bound. Right: left-point Riemann sums of $\int_0^1 W\, dW$ converge to $\frac{1}{2}(W^2 - t)$; trapezoid sums give $\frac{1}{2}W^2$.

Martingales: the mathematics of fair games

Martingale

A process $M(t)$ with $E|M(t)| < \infty$ is a **martingale** with respect to the information $\mathcal{F}_s$ available at time s (the history of the process up to s) if

$$E\big[M(t) \mid \mathcal{F}_s\big] = M(s) \qquad \text{for all } s \le t.$$

The best prediction of the future value is the present value. Examples:

- $W(t)$: given the past, $W(t) = W(s) + [W(t) - W(s)]$ and the bracket is independent with mean zero.
- $N(t) - \lambda t$: the Poisson count minus its expected growth, the "surprise" part of the arrivals.
- $W(t)^2 - t$, which deserves a proof because it is the engine of what follows.

$W(t)^2 - t$ is a martingale

For $s \le t$, $E\big[W(t)^2 - t \mid \mathcal{F}_s\big] = W(s)^2 - s$.

Proof. Write $W(t) = W(s) + D$ with $D = W(t) - W(s) \sim \mathcal{N}(0, t-s)$ independent of $\mathcal{F}_s$. Then $W(t)^2 = W(s)^2 + 2W(s)D + D^2$. Conditionally on $\mathcal{F}_s$, $W(s)$ is known, $E[D] = 0$ and $E[D^2] = t - s$, so $E[W(t)^2 \mid \mathcal{F}_s] = W(s)^2 + t - s$. Subtract t. ∎

The power of martingales comes from a theorem we state without proof.

Optional stopping (one version)

If M is a martingale, τ is a stopping time (a random time whose occurrence by time t is decided by the history up to t), and $M(t \wedge \tau)$ stays bounded, then $E[M(\tau)] = E[M(0)]$.

Now watch it redo Ch. 34's gambler's ruin, in continuous time and in two lines. Let Brownian motion start at 0 and stop at the first time τ it hits $-a$ or $+b$. Let $p = P(\text{hit } b \text{ first})$. Since W is a martingale,

$$0 = E[W(\tau)] = p\,b - (1-p)\,a \quad\Longrightarrow\quad p = \frac{a}{a+b}.$$

Since $W^2 - t$ is a martingale (with a limiting argument for the unbounded t),

$$0 = E[W(\tau)^2 - \tau] = p\,b^2 + (1-p)\,a^2 - E[\tau] \quad\Longrightarrow\quad E[\tau] = ab.$$

These are exactly k/N and $k(N-k)$ from the discrete game with $a = k$, $b = N - k$, now with no recurrences at all.

? The gambler who always wins

Bet 1 on a fair coin. If you lose, double the bet; keep doubling until you win once, then stop. You are certain to stop, and when you do you are up exactly 1. Your fortune is a martingale, it started at 0, and at the stopping time it equals 1, so $E[M(\tau)] = 1 \neq 0$. Optional stopping fails. The boundedness condition was violated (your losses before the win are unbounded, and you need infinite credit). Exactly which conditions make "you cannot beat a fair game" true, and the measure-theoretic notion of conditional expectation needed to state them precisely, are the business of Ch. 97.

Stochastic integrals: where the extra term comes from

We want to make sense of $\int_0^t H(s)\,dW(s)$: the accumulated effect of random kicks dW scaled by a time-varying gain H. A Brownian path has infinite length, so the classical Riemann–Stieltjes integral does not exist. Itô's idea was to *choose* the left endpoint and define

$$\int_0^t H\,dW = \lim_{n\to\infty} \sum_{k=0}^{n-1} H(t_k)\big(W(t_{k+1}) - W(t_k)\big),$$

the limit taken in mean square. The left endpoint is not arbitrary: it means the gain is decided *before* the kick arrives, which is how any causal system, a controller, a trader, a filter, actually operates.

The Itô integral of W

$$\int_0^t W\,dW = \tfrac{1}{2}W(t)^2 - \tfrac{1}{2}t.$$

Proof. Write $W_k = W(t_k)$ and $\Delta W_k = W_{k+1} - W_k$. The algebraic identity $b^2 - a^2 = 2a(b-a) + (b-a)^2$ with $a = W_k$, $b = W_{k+1}$ gives

$$W_k\,\Delta W_k = \tfrac{1}{2}\big(W_{k+1}^2 - W_k^2\big) - \tfrac{1}{2}(\Delta W_k)^2.$$

Sum over k. The first part telescopes to $\frac{1}{2}\big(W(t)^2 - W(0)^2\big) = \frac{1}{2}W(t)^2$. The second part is $-\frac{1}{2}Q_n$, and $Q_n \to t$ in mean square by the quadratic-variation theorem. ∎

The proof shows exactly where ordinary calculus breaks. For a smooth function the $(\Delta)^2$ terms add up to zero and you get $\frac{1}{2}W^2$. For Brownian motion they add up to t. Evaluate at the midpoint (or average the endpoints, the trapezoid rule) and the cross term is gone: $\frac{1}{2}(W_k + W_{k+1})\Delta W_k = \frac{1}{2}(W_{k+1}^2 - W_k^2)$ exactly, which telescopes to $\frac{1}{2}W^2$. That is the **Stratonovich** integral. The two differ by half the quadratic variation, and the right-hand panel of the figure above shows the two Riemann sums converging to the two answers on one simulated path.

The Itô choice buys one beautiful property: an Itô integral is a martingale with mean zero. Indeed $E[\frac{1}{2}W(t)^2 - \frac{1}{2}t] = \frac{1}{2}t - \frac{1}{2}t = 0$. The classical answer $\frac{1}{2}W^2$ has mean $t/2$, so it would have the system gaining on average from kicks it could not anticipate.

Itô's lemma: the new chain rule

The same bookkeeping, applied to any smooth function $f(W)$, gives the fundamental theorem of stochastic calculus. Taylor-expand one step:

$$\Delta f = f'(W)\,\Delta W + \tfrac{1}{2}f''(W)(\Delta W)^2 + \cdots.$$

In ordinary calculus the second term is negligible. Here $(\Delta W)^2$ sums to Δt, so it survives, while higher terms ($|\Delta W|^3 \sim \Delta t^{3/2}$) vanish. The mnemonic is $(dW)^2 = dt$, $dt\,dW = 0$, $(dt)^2 = 0$.

Itô's lemma (stated)

If X satisfies $dX = a\,dt + b\,dW$ and $f(t,x)$ is twice continuously differentiable, then

$$df(t,X) = \Big(\frac{\partial f}{\partial t} + a\frac{\partial f}{\partial x} + \frac{1}{2}b^2\frac{\partial^2 f}{\partial x^2}\Big)dt + b\frac{\partial f}{\partial x}\,dW.$$

The extra $\frac{1}{2}b^2 f''$ is the Itô correction. With $f = x^2$ and $X = W$ ($a = 0, b = 1$) it reproduces $d(W^2) = 2W\,dW + dt$, the theorem above in differential form.

The investment paradox, resolved

The standard model for a price, a population, or any quantity whose random fluctuations are proportional to its size is **geometric Brownian motion**:

$$dS = \mu S\,dt + \sigma S\,dW, \qquad S(0) = S_0.$$

Naively, divide by S and integrate: $\ln S$ should grow like $\mu t + \sigma W$. Itô's lemma with $f = \ln x$ ($f' = 1/x$, $f'' = -1/x^2$, $a = \mu S$, $b = \sigma S$) says otherwise:

$$d(\ln S) = \Big(\mu - \tfrac{1}{2}\sigma^2\Big)dt + \sigma\,dW \quad\Longrightarrow\quad S(t) = S_0\exp\Big(\big(\mu - \tfrac{1}{2}\sigma^2\big)t + \sigma W(t)\Big).$$

Two different growth rates now coexist. The **mean** grows at rate μ: $E[S(t)] = S_0 e^{\mu t}$, using $E[e^{\sigma W}] = e^{\sigma^2 t/2}$. The **typical path** (the median, and the long-run growth of almost every path) grows at the smaller rate $\mu - \frac{1}{2}\sigma^2$.

With the hook's numbers, $\mu = 0.05$ per year and $\sigma = 0.4$, over $t = 10$ years: $E[S]/S_0 = e^{0.5} \approx 1.65$, a 65 % expected gain. But $\mu - \frac{1}{2}\sigma^2 = -0.03$, so the median is $e^{-0.3} \approx 0.74$ and

$$P\big(S(10) < S_0\big) = P\Big(W(10) < \frac{0.3}{0.4}\Big) = \Phi\Big(\frac{0.3}{0.4\sqrt{10}}\Big) \approx 0.594,$$

so about 59 % to 60 % of paths lose money. The mean is dragged up by a few enormous winners. Volatility is not just risk around a trend: it *lowers* the trend that a typical path follows, by exactly the Itô correction $\frac{1}{2}\sigma^2$. Engineers meet the same effect as the gap between average power and the power of the average in log-normal fading channels.

36.4 The algorithm

Simulating a Poisson process

The exponential-gap property gives an exact $O(\text{number of events})$ algorithm: draw $E_1, E_2, \dots \sim \text{Exponential}(\lambda)$ and set $T_k = E_1 + \cdots + E_k$ until $T_k > T$. An exponential sample is $-\ln(U)/\lambda$ with U uniform on $(0, 1]$.

When the rate varies in time, $\lambda(t) \le \lambda_{\max}$, use **thinning** (Lewis–Shedler, 1979):

```
THINNING(λ(·), λ_max, T)
  t ← 0, events ← []
  loop:
      t ← t − ln(U₁)/λ_max           # next candidate of a rate-λ_max process
      if t > T: return events
      if U₂ ≤ λ(t)/λ_max: append t   # keep it with probability λ(t)/λ_max
```

It is exact, by the thinning property of §36.3. Its efficiency is the average of $\lambda(t)/\lambda_{\max}$: a sharply peaked rate wastes most candidates.

Simulating Brownian motion and SDEs

On a grid $t_k = k\,\Delta t$, Brownian motion is exact at the grid points: $W_{k+1} = W_k + \sqrt{\Delta t}\,Z_k$ with $Z_k \sim \mathcal{N}(0,1)$ independent. For a stochastic differential equation $dX = a(X)\,dt + b(X)\,dW$, the analogue of Euler's method (Ch. 22) is **Euler–Maruyama**:

$$X_{k+1} = X_k + a(X_k)\,\Delta t + b(X_k)\sqrt{\Delta t}\,Z_k.$$

Note the left-point evaluation of b: this scheme converges to the *Itô* solution. Its accuracy has two different orders.

- **Strong order $\frac{1}{2}$**: the error on each individual path, $E|X_N - X(T)|$, shrinks like $\Delta t^{1/2}$. Halving the error needs four times the steps.
- **Weak order 1**: the error in expectations, $|E[g(X_N)] - E[g(X(T))]|$, shrinks like Δt.

If you only need averages (a price, a mean error), the cheaper weak order is what counts, and Monte Carlo sampling error, which falls like $1/\sqrt{\text{paths}}$, usually dominates.

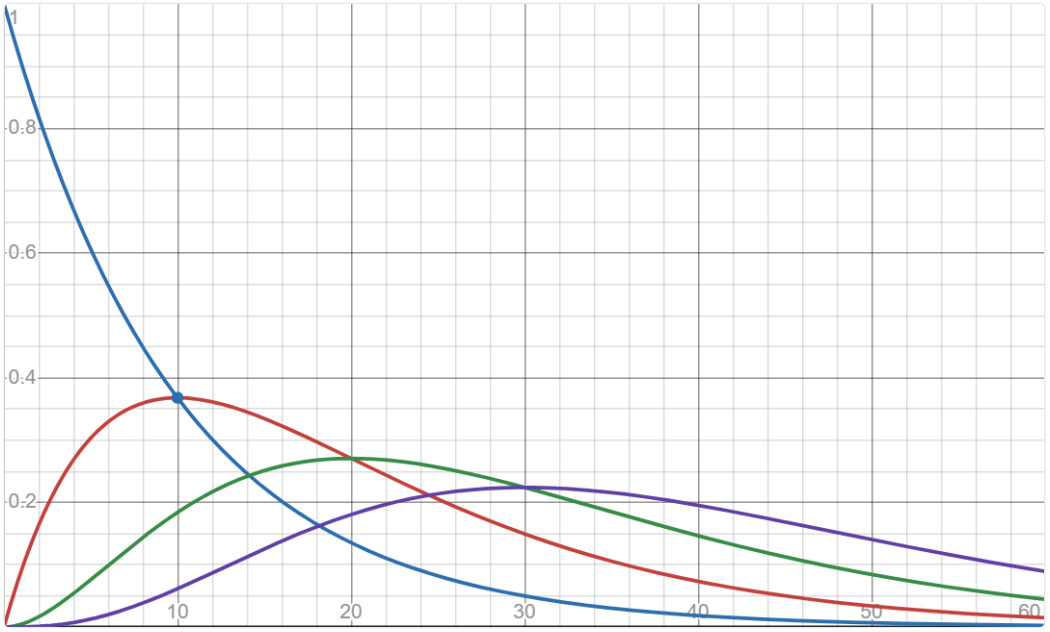
Failure cases.

- *Wrong calculus.* Evaluate b at the midpoint, or use a deterministic ODE solver with the noise treated as an ordinary input signal, and you silently solve the Stratonovich equation instead. For geometric Brownian motion that shifts the growth rate by $\frac{1}{2}\sigma^2$.
- *Positivity.* A geometric Brownian motion is always positive, but an Euler–Maruyama step $S(1 + \mu\Delta t + \sigma\sqrt{\Delta t}Z)$ goes negative whenever $Z < -(1 + \mu\Delta t)/(\sigma\sqrt{\Delta t})$. With coarse steps and high volatility this happens. Simulate $\ln S$ instead, where the scheme is exact.
- *Shared randomness.* To compare two step sizes on the same path, the coarse increments must be sums of the fine ones. Drawing fresh random numbers for each resolution compares different paths and hides the convergence.

36.5 The experiment

Part 1: waiting for the bus

The probability of exactly k buses in the first t minutes, for $k = 0, 1, 2, 3$ and rate $\lambda = 0.1$ per minute:



$P(N(t) = k)$ for $k = 0$ (blue) to 3 (red, green, purple). Each curve for $k \geq 1$ peaks at $t = k/\lambda$, where the expected count equals k.

At $t = 10$ the blue curve reads $e^{-1} \approx 0.368$: more than a third of the time you see *no* bus in a full average gap. Change every 0.1 to 0.2 (a bus every 5 minutes on average) and confirm the peaks move to half the times. Then add the curve for $k = 4$, `y=\frac{(0.1x)^{4}}{24}e^{-0.1x}`, and check where it peaks.

Part 2: the investment paradox in 20 000 paths

The script simulates geometric Brownian motion by Euler–Maruyama and by the exact Itô solution, and checks the Itô integral of the hook along the way.

```
import numpy as np
import matplotlib.pyplot as plt

# ---- knobs --------------------------------------------------------------
mu, sigma = 0.05, 0.4     # drift and volatility of dS = mu S dt + sigma S dW
T, steps  = 10.0, 250     # horizon (years) and Euler–Maruyama steps
paths     = 20000         # number of simulated paths
seed      = 36
# -------------------------------------------------------------------------

rng = np.random.default_rng(seed)
dt = T / steps
dW = rng.normal(0.0, np.sqrt(dt), (paths, steps))

S = np.ones(paths)                        # Euler–Maruyama, S(0) = 1
for k in range(steps):
    S = S + mu * S * dt + sigma * S * dW[:, k]
W_T = dW.sum(axis=1)
S_exact = np.exp((mu - 0.5 * sigma**2) * T + sigma * W_T)   # Ito's formula

print(f"mean   S(T): EM {S.mean():.3f}  exact {S_exact.mean():.3f}  theory {np.exp(mu*T):.3f}")
print(f"median S(T): EM {np.median(S):.3f}  exact {np.median(S_exact):.3f}"
      f"  theory {np.exp((mu-0.5*sigma**2)*T):.3f}")
print(f"fraction of paths that lost money: {(S_exact < 1).mean():.3f}")
print(f"EM paths below zero: {(S < 0).sum()}")

W = np.concatenate([np.zeros((paths, 1)), dW.cumsum(axis=1)], axis=1)
ito = (W[:, :-1] * dW).sum(axis=1)        # left-point sums for int W dW
print(f"mean Ito sum {ito.mean():+.3f}   mean (W^2 - T)/2 {((W_T**2 - T)/2).mean():+.3f}")

plt.hist(np.log(S_exact), bins=80, density=True, alpha=0.6, label="log S(T) exact")
plt.axvline(mu * T, color="r", label="log of mean")
plt.axvline((mu - 0.5 * sigma**2) * T, color="k", ls="--", label="mean of log")
plt.xlabel("log S(T)"); plt.legend(); plt.show()
```

With the default knobs it prints a mean near 1.63 (theory 1.65), a median near 0.73 (theory 0.74), and a loss fraction near 0.60. The mean of the Itô sums is close to zero, as a martingale's must be. Things to try:

1. **Crank the volatility.** Set `sigma = 1.0` and `steps = 10`. Count the Euler–Maruyama paths that go negative. Then raise `steps` and watch them disappear.

2. **Find the break-even volatility.** For $\mu = 0.05$, what σ makes exactly half the paths lose money? Predict it from $\mu - \frac{1}{2}\sigma^2 = 0$ first.
3. **Switch calculus.** Replace the left-point sum by the trapezoid sum `0.5*(W[:, :-1] + W[:, 1:]) * dW`. What is its mean now, and why is it $T/2$?
4. **Watch the mean misbehave.** Increase `T` to 50 with `sigma = 0.4`. The sample mean of `S_exact` starts to undershoot the theory badly. Why does estimating $E[S]$ from samples get harder as the distribution becomes more skewed?

36.6 The challenge

1. **Intuition.** Buses on a real route tend to bunch: a late bus picks up more passengers, slows down, and the next bus catches it up. Does bunching make the passenger's experienced gap longer or shorter than the Poisson value of $2/\lambda$? Argue from the formula $E[X] + \mathrm{Var}(X)/E[X]$ without computing anything.
2. **Derivation.** Let $N(t)$ be a Poisson process of rate λ. Show that $(N(t) - \lambda t)^2 - \lambda t$ is a martingale. Then use Itô's lemma to find $d\big(e^{\theta W(t) - \theta^2 t/2}\big)$ and show that this exponential is a martingale for every real θ.
3. **Implementation.** Implement Euler–Maruyama for the Ornstein–Uhlenbeck process $dX = -\alpha X\,dt + \sigma\,dW$ of Ch. 21. Measure the strong error at $T = 1$ against the exact solution driven by the same Brownian increments, for $\Delta t = 2^{-4}, \dots, 2^{-12}$, and confirm the order $\frac{1}{2}$ on a log-log plot. Do you actually see order $\frac{1}{2}$, or something better, and why might this particular equation be special?
4. **Open investigation.** Brownian motion started at the centre of a square first hits the boundary at a random point. Simulate that exit point and estimate its distribution along the edges. Then put a "temperature" g on the boundary (say 1 on one edge, 0 on the others), and estimate $E[g(\text{exit point})]$ for many starting points. Compare the resulting map with the steady-state heat distribution from Ch. 13 §13.6. How does the Monte Carlo error scale with the number of paths, and how badly does the time-step bias near the boundary spoil it?

36.7 Loose threads

The last challenge hints at the deepest connection this chapter has kept off the page.

> **? Brownian motion knows the heat equation**
>
> Run Brownian paths from a point x and average a function of where they are at time t: $u(t, x) = E[f(x + W(t))]$. This average satisfies the heat equation $\partial u/\partial t = \frac{1}{2}\,\partial^2 u/\partial x^2$ of Ch. 13, exactly. With killing, drift and boundaries added, every linear parabolic PDE and every Laplace boundary-value problem has a "random walker" solution, the Feynman–Kac formula. Why should a diffusing particle and a diffusing temperature obey the same law, and what does that mean for solving PDEs in a thousand dimensions where grids are hopeless? The answer starts in Ch. 89.

> **? Itô or Stratonovich?**
>
> Mathematics gave two consistent calculi that disagree by $\frac{1}{2}$ of the quadratic variation. Which one does nature use? A physical noise source is never truly white: it has a tiny correlation time. Smooth the noise, solve the ordinary differential equation, then let the correlation time go to zero, and the answer converges to the *Stratonovich* solution (the Wong–Zakai theorem). Yet finance, filtering and biology routinely use Itô. When is each correct, and what goes wrong when a model is built with the wrong one? The physics of noise-driven systems is taken up in Ch. 92.

Other threads:

- **Why the paths exist at all.** Defining a process with continuous but nowhere-differentiable paths, and making "conditional expectation given the past" precise, needs measure theory. The construction of Brownian motion, filtrations and the full martingale theory are in Ch. 97.
- **Simulation as a tool.** Every expectation in this chapter can be estimated by simulating paths. Doing that efficiently, with variance reduction and multilevel tricks that exploit the strong and weak orders, is Ch. 38.
- **Deciding under evolving randomness.** Choosing when to stop, invest or switch while a stochastic process unfolds is optimal stopping and stochastic control: Ch. 51 and Ch. 54.
- **Back to the walk.** Poisson jumps plus Brownian wandering, plus Markov chains in continuous time, cover nearly every stochastic model used in engineering. The discrete-time roots are in Ch. 34.

Further reading

- S. M. Ross, *Introduction to Probability Models*, 12th ed., Academic Press, 2019. Poisson processes, renewal theory and the inspection paradox at an engineering level.
- B. Øksendal, *Stochastic Differential Equations*, 6th ed., Springer, 2003. The standard first text on Itô calculus.
- D. J. Higham, "An algorithmic introduction to numerical simulation of stochastic differential equations," *SIAM Review* 43 (2001), 525–546. Short, practical, with code.
- P. Mörters and Y. Peres, *Brownian Motion*, Cambridge University Press, 2010. The paths themselves, in depth.

Chapter 37 — Causal Inference: Correlation Is Not Destiny

> **Level 2 · Track B — Probability, Uncertainty, and Information Builds on:** Ch. 19 (conditional probability, independence), Ch. 20 (correlation, two-sample statistics) · **Leads to:** Ch. 38, Ch. 53 · **Related:** Ch. 35, Ch. 27, Ch. 48 **Reading time:** ~30 min

37.1 The hook

In 1986 a group of urologists in London compared two ways of removing kidney stones: open surgery (call it treatment A) and a newer keyhole procedure (treatment B). They had 700 patients, 350 on each treatment. The headline numbers were these:

	Treatment A (open)	Treatment B (keyhole)
All patients	273/350 = **78%** success	289/350 = **83%** success

Keyhole surgery wins, clearly. It is also cheaper and less painful. Case closed?

Now split the same 700 patients by the size of their stone:

	Treatment A (open)	Treatment B (keyhole)
Small stones	81/87 = **93%**	234/270 = **87%**
Large stones	192/263 = **73%**	55/80 = **69%**
All patients	273/350 = **78%**	289/350 = **83%**

Treatment A is better for small stones. Treatment A is better for large stones. Treatment B is better overall. No arithmetic error is hiding anywhere; you can check every fraction. This is **Simpson's paradox**, and it is not a curiosity. It turns up in admissions data, drug trials and A/B tests every week.

So which treatment should *you* choose if you walk in with a kidney stone of unknown size? Most people say A, because it wins in every group. Now consider a second story with *exactly the same numbers*. Replace "stone size" by "blood pressure measured one week after the treatment": low versus high. Treatment A is better among patients who ended up with low blood pressure, and better among those who ended up with high blood pressure, and worse overall. Which do you choose now?

This time the right answer is B, and the aggregate table is the one to trust. Same numbers, opposite answers.

That should be disturbing. It means that **no amount of staring at the data can tell you whether to aggregate or split**. The answer depends on something that is not in the table: *how the variables cause one another*. Stone size influences which treatment a doctor picks and also how well any treatment works. Post-treatment blood pressure is something the treatment itself changes. One is a confounder, the other is a mediator, and the difference lives entirely in the arrows.

This chapter is about those arrows: how to draw them, how to compute with them, and how to answer the question every engineer, doctor and policymaker actually cares about. Not "what do I see when X is large?" but "what *happens* if I *make* X large?"

37.2 The big idea

Seeing versus doing

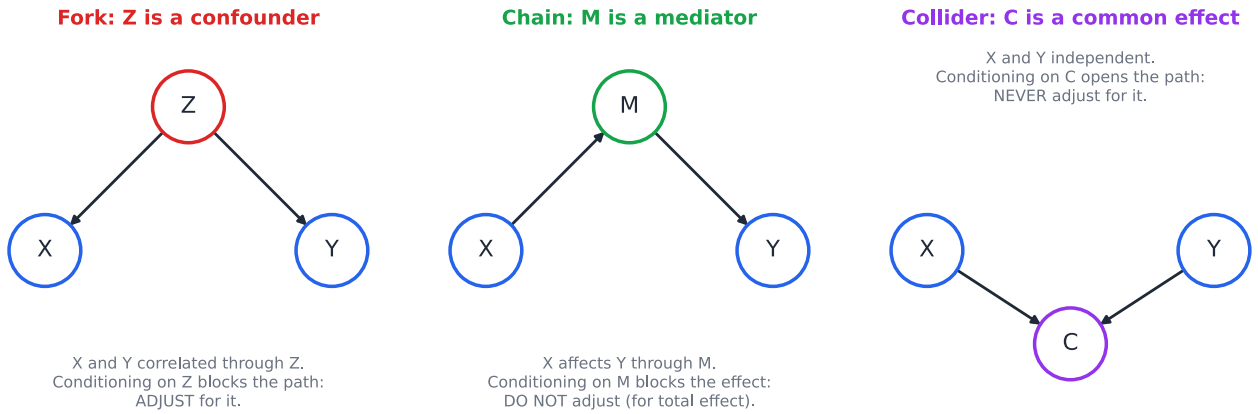
A barometer's needle falls before a storm. The correlation is excellent, and you can use it to predict rain. But if you grab the needle and push it down, no storm arrives. Observing a low reading and *forcing* a low reading are different operations, and probability theory as taught in Ch. 19 has symbols only for the first.

Conditional probability $P(\text{storm} \mid \text{needle low})$ answers: *among the days on which the needle happened to be low, how often did a storm follow?* Those days were selected by nature, and nature lowered the needle *because* the air pressure was dropping. The question we want, $P(\text{storm} \mid do(\text{needle low}))$, asks: *if I reach into the world and set the needle, overriding whatever normally sets it, how often does a storm follow?* The new symbol $do(\cdot)$ is the whole subject in four letters.

Draw the arrows

The tool that makes *do* precise is a **causal graph**: a directed acyclic graph (DAG) with one node per variable and an arrow $A \to B$ whenever A is one of the things that directly determines B.

Three tiny graphs, built from three variables, explain almost everything.

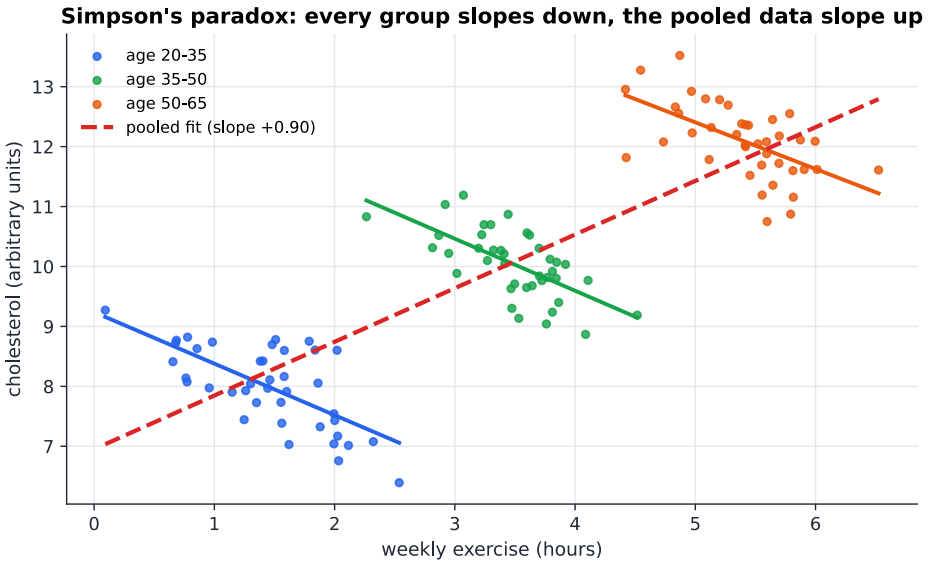


The three junctions. In a fork the middle node is a common cause; in a chain it passes an effect along; in a collider it is a common effect. Conditioning on the middle node closes the first two paths and opens the third.

- **Fork** ($X \leftarrow Z \rightarrow Y$). Stone size Z pushes doctors toward open surgery X and independently lowers success Y. X and Y become correlated even if X did nothing. Compare like with like, within each level of Z, and the spurious part disappears. Z is a **confounder**: adjust for it.
- **Chain** ($X \rightarrow M \rightarrow Y$). Treatment X changes blood pressure M, which changes recovery Y. If you compare only patients with the same M you have frozen the very channel through which X works, and you will under-count its effect. M is a **mediator**: for the total effect, leave it alone.
- **Collider** ($X \rightarrow C \leftarrow Y$). Talent X and good looks Y are independent in the population, but Hollywood casts people who have enough of one or the other, C. Among actors, the talented ones seem less attractive. Nothing about talent affects looks; you created the correlation by looking only at $C = 1$. This is **Berkson's paradox**, and colliders are the reason that "control for everything" is terrible advice.

Simpson's paradox, drawn

Here is the same phenomenon with continuous variables. Within each age group, more exercise goes with lower cholesterol. But older people both exercise more (in this invented population, retirees with time on their hands) and have higher cholesterol. Pool the groups and the slope flips sign.

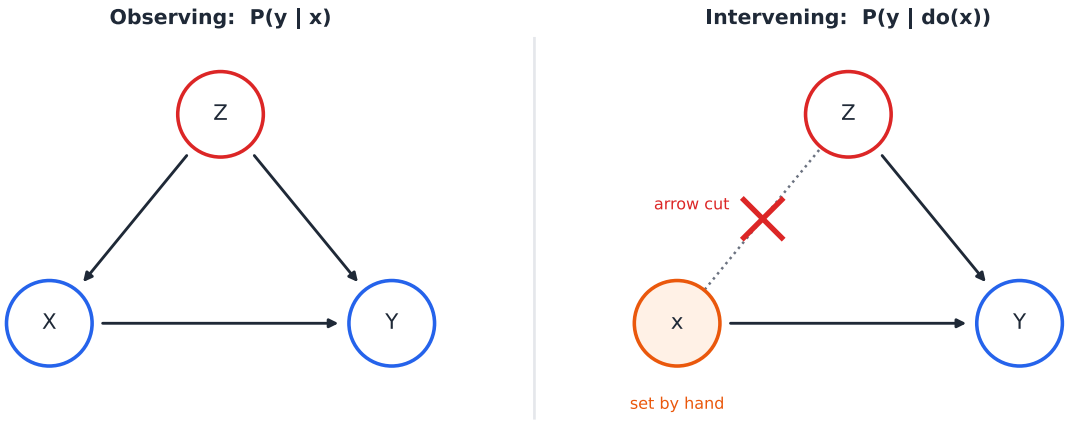


Simpson's paradox in a scatter plot: the within-group slopes are about -0.8, the pooled slope is about $+0.9$. Age is a fork, so the within-group lines tell the causal story.

Whether the coloured lines or the red dashed line is "right" is not a statistical question. If age causes both exercise and cholesterol, the coloured lines are right. If (absurdly) exercise caused aging, age would be a mediator and the dashed line would be the total effect. The data are identical in both worlds.

The surgery

How do we compute $P(y \mid do(x))$? The answer is beautifully mechanical. *Intervening on X means cutting every arrow that points into X*, because whatever used to determine X no longer does: you do. Everything else in the world keeps working as before.



Graph surgery. Observing X keeps the arrow $Z \to X$, so seeing X tells you something about Z. Setting X cuts that arrow; Z keeps its natural distribution.

In the mutilated graph, X carries no information about Z. So the only route by which x can change Y is the direct arrow. That is what we want to measure, and it is what a randomized trial measures, because a coin flip is a way of performing the surgery in the real world.

37.3 The mathematics

Structural causal models

Structural causal model (SCM)

An SCM consists of exogenous "noise" variables $U_1, \dots, U_n$, mutually independent with distribution $P(\mathbf{u})$, and endogenous variables $V_1, \dots, V_n$ each determined by a **structural equation**

$$V_i = f_i\big(\mathrm{pa}(V_i),\, U_i\big),$$

where $\mathrm{pa}(V_i)$ are the **parents** of V_i in a directed acyclic graph G.

Each structural equation is an *assignment*, not an algebraic identity: $Y := 2X + U_Y$ says "nature computes Y from X", and you may not rearrange it into an equation for X. Because the graph is acyclic and the noises are independent, the joint distribution factorizes along the graph:

$$P(v_1, \dots, v_n) = \prod_{i=1}^{n} P\big(v_i \mid \mathrm{pa}(v_i)\big).$$

This is the chain rule of Ch. 19 with most conditioning variables dropped, which is what the graph's missing arrows promise.

Intervention

The intervention $do(X = x)$ replaces the structural equation for X by the constant assignment $X := x$ and leaves every other equation unchanged. The **interventional distribution** $P(\mathbf{v} \mid do(x))$ is the distribution of the modified model. Graphically: delete all arrows into X.

Since only one factor changes, the modified model factorizes as the original one with X's factor removed. This is the **truncated factorization**:

$$P(\mathbf{v} \mid do(x)) = \prod_{i:\, V_i \neq X} P\big(v_i \mid \mathrm{pa}(v_i)\big)\Big|_{X = x}.$$

Result 1: the adjustment formula

Take the kidney-stone graph: $Z \to X$, $Z \to Y$, $X \to Y$. The observational joint is $P(z)P(x \mid z)P(y \mid x, z)$.

Adjustment for a confounder

If Z is the only common cause of X and Y (graph $Z \to X \to Y$, $Z \to Y$), then

$$P(y \mid do(x)) = \sum_{z} P(y \mid x, z)\, P(z).$$

Proof. By truncated factorization, the post-intervention joint of (Z, Y) is $P(z)\,P(y \mid x, z)$: the factor $P(x \mid z)$ is gone and x is fixed. Summing over z gives the result. ■

Compare it with plain conditioning, which by the law of total probability is

$$P(y \mid x) = \sum_{z} P(y \mid x, z)\, P(z \mid x).$$

The two formulas differ in one symbol: $P(z)$ versus $P(z \mid x)$. Conditioning weights each stratum by how common it is *among the people who got x*; intervention weights it by how common it is *in the population*. When X and Z are dependent, which is what makes Z a confounder, the weights differ.

The kidney stones, settled

Small stones are $357/700 = 0.51$ of patients, large ones 0.49. Adjusting:

$$P(\text{success} \mid do(A)) = 0.931 \times 0.51 + 0.730 \times 0.49 = 0.833,$$
$$P(\text{success} \mid do(B)) = 0.867 \times 0.51 + 0.688 \times 0.49 = 0.779.$$

Treatment A is better by about 5 percentage points. The raw comparison went the other way because doctors gave A to 77% of the large-stone (hard) cases but only 24% of the small-stone (easy) cases. A was handed the difficult patients.

For the blood-pressure story the graph is $X \to M \to Y$ (possibly with $X \to Y$ too). There are no arrows into X, so surgery changes nothing and $P(y \mid do(x)) = P(y \mid x)$: the aggregate table is the causal answer. Splitting by M would compute something else, a "controlled direct effect", which is a legitimate quantity but not the one a patient choosing a treatment needs.

Result 2: why randomization works

Randomization identifies the causal effect

If X is assigned by an independent random device R (so its structural equation is $X := f_X(R)$ with R independent of everything else, and X has no other parents), then

$$P(y \mid do(x)) = P(y \mid x).$$

Proof. Write $\mathbf{V}'$ for all variables other than X and R. Because R is a parent of X only, the factorization of the joint is

$$P(r, x, \mathbf{v}') = P(r)\, P(x \mid r) \prod_{V_i \in \mathbf{V}'} P\big(v_i \mid \mathrm{pa}(v_i)\big),$$

and no factor in the product mentions r. Summing over r gives $P(x) \prod_i P(v_i \mid \mathrm{pa}(v_i))$, and dividing by $P(x)$,

$$P(\mathbf{v}' \mid x) = \prod_{V_i \in \mathbf{V}'} P\big(v_i \mid \mathrm{pa}(v_i)\big)\Big|_{X = x}.$$

That is exactly the truncated factorization for $P(\mathbf{v}' \mid do(x))$. Summing both sides over everything except Y gives the claim. ■

The proof is short, but the idea is one of the great inventions of the twentieth century. R. A. Fisher's randomized field trials in the 1920s did not need to know what the confounders were. The coin cuts *all* incoming arrows at once, including the ones nobody thought of. That is why randomized controlled trials (RCTs) are the gold standard, and why "we controlled for age, sex and income" is always weaker: it cuts only the arrows you remembered.

Result 3: conditioning on a collider manufactures correlation

Collider bias, Gaussian case

Let $X, Y \sim \mathcal{N}(0,1)$ be independent and $C = X + Y + U$ with $U \sim \mathcal{N}(0,1)$ independent. Then, conditional on C,

$$\mathrm{Corr}(X, Y \mid C) = -\tfrac{1}{2}.$$

Proof. The vector (X, Y, C) is jointly Gaussian with covariance matrix

$$\mathbf{\Sigma} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 3 \end{pmatrix}.$$

For jointly Gaussian vectors, the conditional covariance of (X, Y) given C is the Schur complement $\mathbf{\Sigma}_{11} - \mathbf{\Sigma}_{12}\mathbf{\Sigma}_{22}^{-1}\mathbf{\Sigma}_{21}$ (the same formula that drives the Kalman filter mentioned in Ch. 21):

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \frac{1}{3}\begin{pmatrix} 1 \\ 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \end{pmatrix} = \begin{pmatrix} 2/3 & -1/3 \\ -1/3 & 2/3 \end{pmatrix}.$$

The conditional correlation is $(-1/3)/(2/3) = -1/2$, and it does not depend on the value of C. ■

Two independent causes become *strongly* anticorrelated once you look within a slice of their common effect. If you knew C exactly with no noise ($U = 0$), the correlation would be -1: knowing X would pin down Y.

The back-door criterion

The three junctions combine into one rule for general graphs. A **path** between X and Y is any sequence of edges, ignoring direction. It is **blocked** by a set $\mathbf{Z}$ if it contains a fork or chain whose middle node is in $\mathbf{Z}$, or a collider whose middle node is *not* in $\mathbf{Z}$ (and has no descendant in $\mathbf{Z}$). A **back-door path** is a path from X to Y that starts with an arrow *into* X.

Back-door adjustment (Pearl, 1993)

If a set $\mathbf{Z}$ contains no descendant of X and blocks every back-door path from X to Y, then

$$P(y \mid do(x)) = \sum_{\mathbf{z}} P(y \mid x, \mathbf{z})\, P(\mathbf{z}).$$

Result 1 is the special case of a single fork. The rule tells you, by inspecting the graph alone, which variables to put in your regression or stratification. It also tells you when *no* set of measured variables works, for instance when the confounder is unmeasured. Then other tricks are needed: the **front-door criterion** (adjust through a fully observed mediator), or an **instrumental variable** (something that nudges X but touches Y only through X, like a lottery for school places).

? Can the arrows be read off the data?

Everything above assumed someone handed you the graph. With two variables, $X \to Y$ and $Y \to X$ produce the same family of joint distributions, so data alone cannot orient that edge. Yet with three or more variables, colliders leave a fingerprint (independent parents that become dependent when you condition on the child) and some arrows *can* be discovered from passive observation. How much of a causal graph is learnable, and with how much data, is an active research area built on conditional-independence tests; its information-theoretic core, conditional mutual information, is in Ch. 33.

The ladder of causation

Judea Pearl organizes causal questions into three rungs.

1. **Association**, $P(y \mid x)$: *what does seeing x tell me?* Statistics and machine learning live here.
2. **Intervention**, $P(y \mid do(x))$: *what if I do x?* Experiments, policies, control.
3. **Counterfactuals**, $P(y_{x'} \mid x, y)$: *I did x and saw y; what would have happened had I done x'?* Regret, blame, explanation, legal liability.

Each rung needs strictly more knowledge than the one below. The adjustment formula climbs from rung 1 to rung 2 using only the graph. Rung 3 needs the structural equations themselves, because it reasons about *this particular individual*, whose noise terms have already been revealed.

Counterfactual (abduction, action, prediction)

To compute what Y would have been had X been x', given evidence $\mathbf{e}$:

1. **Abduction.** Use $\mathbf{e}$ to infer the exogenous variables $\mathbf{U}$ (or their posterior $P(\mathbf{u} \mid \mathbf{e})$).
2. **Action.** Apply $do(X = x')$.
3. **Prediction.** Run the modified model with the inferred $\mathbf{U}$ and read off Y.

A linear counterfactual

Let $Z := U_Z$, $X := Z + U_X$, $Y := 2X + 3Z + U_Y$. We observe one unit with $(Z, X, Y) = (1, 2, 9)$. Abduction: $U_Z = 1$, $U_X = 1$, $U_Y = 9 - 4 - 3 = 2$. Action: set $X := 0$. Prediction: $Y = 0 + 3 \cdot 1 + 2 = 5$. Had this unit received $X = 0$, its outcome would have been 5, not 9.

In a linear model, the counterfactual difference $9 - 5 = 4 = 2 \times 2$ is the same for every unit. In nonlinear models it is not, and the population distribution of counterfactuals is usually *not* identifiable even from perfect experimental data.

? The questions no experiment can answer

"Did the drug *cause* this patient's recovery?" is a rung-3 question: the probability that she would not have recovered without it. Even an infinitely large RCT gives only bounds on it. Remarkably, the sharpest bounds come from a *linear program* over the unknown response types, and the bounds' tightness is governed by LP duality. That machinery is in Ch. 48.

Experimental design in one page

If you *can* intervene, do. A randomized experiment estimates the **average treatment effect**

$$\text{ATE} = E[Y \mid do(X = 1)] - E[Y \mid do(X = 0)]$$

by the difference of sample means $\bar{Y}_1 - \bar{Y}_0$, an unbiased estimator by Result 2. Its variance, from the two-sample results of Ch. 20, is $\sigma_1^2/n_1 + \sigma_0^2/n_0$. For a binary outcome near 50% success, $\sigma^2 \approx 0.25$, and detecting a 10-point improvement with the usual 5% significance and 80% power needs about

$$n = \frac{2\,(z_{0.975} + z_{0.8})^2\,\sigma^2}{\Delta^2} = \frac{2\,(1.96 + 0.84)^2 (0.25)}{0.1^2} \approx 392$$

patients per arm. Two design ideas sharpen this:

- **Blocking** (stratified randomization): randomize *within* levels of a known prognostic variable such as stone size. The estimator stays unbiased and its variance drops, because between-block variation no longer enters, exactly the stratification trick from Monte Carlo in Ch. 24.
- **Blinding**: neither patient nor assessor knows the arm, so expectations cannot open a new path $X \to$ belief $\to Y$.

37.4 The algorithm

Suppose the graph is drawn and the back-door criterion has given you an adjustment set $\mathbf{Z}$. Two classical estimators turn observational records $(x_i, \mathbf{z}_i, y_i)$, $i = 1, \dots, N$, into an estimate of the ATE.

Stratification (direct adjustment). Estimate the adjustment formula term by term:

```
def ate_stratified(x, z, y):
    """x: 0/1 treatment, z: hashable stratum label, y: outcome (numpy arrays)."""
    ate = 0.0
    for s in np.unique(z):
        m = (z == s)
        if x[m].all() or not x[m].any():
            raise ValueError(f"stratum {s} has only one arm: positivity fails")
        ate += (y[m & (x == 1)].mean() - y[m & (x == 0)].mean()) * m.mean()
    return ate
```

Inverse propensity weighting (IPW). Let $e(\mathbf{z}) = P(X = 1 \mid \mathbf{z})$ be the **propensity score**. Then

$$E[Y \mid do(X = 1)] = E\left[\frac{X\,Y}{e(\mathbf{Z})}\right], \qquad E[Y \mid do(X = 0)] = E\left[\frac{(1 - X)\,Y}{1 - e(\mathbf{Z})}\right].$$

To see the first, condition on $\mathbf{Z}$: $E[XY \mid \mathbf{z}] = P(X = 1 \mid \mathbf{z})\,E[Y \mid X = 1, \mathbf{z}] = e(\mathbf{z})\,E[Y \mid X = 1, \mathbf{z}]$. Divide by $e(\mathbf{z})$ and average over $P(\mathbf{z})$, and you have the adjustment formula. IPW builds a *pseudo-population* in which every treated patient counts $1/e$ times: a patient who received treatment despite having only a 10% chance of it stands in for ten patients like her. In that pseudo-population treatment is independent of $\mathbf{Z}$, just as in a trial.

Cost and scaling. Stratification costs $O(N)$ after grouping, but the number of strata grows as k^d for d covariates with k levels each. With ten binary covariates there are already 1024 cells, most of them nearly empty. Rosenbaum and Rubin's theorem (1983) rescues us: if $\mathbf{Z}$ is a valid adjustment set, so is the single number $e(\mathbf{Z})$. You can fit e by logistic regression and then stratify on it, match on it, or weight by it. One dimension instead of d.

Failure cases. Each of these breaks the estimate silently, with no error message.

1. **Positivity violations.** If some stratum contains only treated (or only untreated) units, the adjustment formula needs a quantity the data never show. In IPW this appears as $e \approx 0$ or 1 and enormous weights.
2. **Unmeasured confounding.** No estimator can adjust for a variable you did not record.
3. **Bad controls.** Adjusting for a mediator removes part of the effect; adjusting for a collider (or a descendant of one) *creates* bias. "Add every available covariate to the regression" can make the answer worse.

> **? Why do the weights explode?**
>
> IPW is an importance-sampling estimator in disguise: you have samples from the world as it is (doctors choosing), and you reweight them by a likelihood ratio to estimate an average in a world that does not exist (a coin choosing). When a propensity approaches 0 the estimator remains unbiased, yet its variance can grow without bound and a single patient can dominate the answer. Why that happens, and how to diagnose it with an *effective sample size*, is the subject of importance sampling in Ch. 38. The same reweighting lets a reinforcement-learning agent evaluate a policy it has never run, in Ch. 53.

37.5 The experiment

The script below builds a small world with a known answer. Severity Z makes doctors more likely to treat ($Z \to X$) and makes recovery less likely ($Z \to Y$). The treatment truly raises recovery probability by 10 points. Run it, then try to break each estimator.

```
import numpy as np
rng = np.random.default_rng(1)

N      = 100_000   # patients
EFFECT = 0.10      # true causal effect of treatment on recovery probability
CONF   = 0.60      # how strongly severity drives treatment choice (0 = none)
HARM   = 0.30      # how much severity lowers recovery

def world(randomize):
    Z = rng.random(N) < 0.5                        # severe case?
    if randomize:
        X = rng.random(N) < 0.5                    # coin flip: do(X)
    else:
        X = rng.random(N) < 0.5 + CONF * (Z - 0.5) # doctors treat the severe
    pY = 0.5 + EFFECT * X - HARM * Z
    Y = rng.random(N) < pY
    return Z, X, Y

Z, X, Y = world(randomize=False)
naive = Y[X].mean() - Y[~X].mean()

adj = 0.0                                          # back-door adjustment
for z in (False, True):
    s = Z == z
    adj += (Y[s & X].mean() - Y[s & ~X].mean()) * s.mean()

e = 0.5 + CONF * (Z - 0.5)                         # propensity score P(X=1|Z)
ipw = np.mean(X * Y / e) - np.mean((~X) * Y / (1 - e))

Zr, Xr, Yr = world(randomize=True)
rct = Yr[Xr].mean() - Yr[~Xr].mean()

print(f"true effect      {EFFECT:+.3f}")
print(f"naive difference {naive:+.3f}")
print(f"adjusted for Z   {adj:+.3f}")
print(f"IPW estimate     {ipw:+.3f}")
print(f"randomized trial {rct:+.3f}")
```

With the settings shown, one run prints a naive difference of about -0.085 (its expected value is exactly -0.08): a treatment that helps every patient *looks harmful*, because doctors aim it at the patients most likely to do badly. The adjusted and IPW estimates come out near $+0.09$ and the randomized trial near $+0.10$, each within sampling error of the truth.

Things to change:

- Set `CONF = 0`. Doctors now ignore severity, the arrow $Z \to X$ disappears, and the naive estimate becomes correct. Confounding needs *both* arrows of the fork.
- Set `CONF = 0.99`. Propensities become 0.005 and 0.995. Change the seed a few times and watch the IPW estimate wander far more than the stratified one. Then set `CONF = 1.0` and see the stratified estimator fail on an empty arm: positivity in action.
- Add a collider. Inside the randomized branch, compute `C = Xr | (Yr & (rng.random(N) < 0.3))` (a unit enters a follow-up study if treated, or sometimes if recovered) and recompute `rct` using only units with `C` true. A perfect experiment, ruined by whom you chose to look at.

37.6 The challenge

1. **Intuition.** A study finds that hospitalized patients with disease A rarely have disease B, and concludes that A protects against B. Draw a causal graph in which A and B are independent in the population yet negatively associated among hospital patients. Which junction is responsible, and what single change to the study design would remove the effect?

2. **Derivation.** For the front-door graph $X \to M \to Y$ with an unobserved confounder U pointing into both X and Y (and no other arrows), use truncated factorization to show that

$$P(y \mid do(x)) = \sum_{m} P(m \mid x) \sum_{x'} P(y \mid m, x')\, P(x').$$

Explain in words why M must be fully observed and must intercept every directed path from X to Y.

3. **Implementation.** Extend the experiment to three binary confounders Z_1, Z_2, Z_3, each pushing both treatment and outcome. Implement (a) stratification over all eight cells and (b) IPW with a propensity fitted by logistic regression via a few iterations of Newton's method (using only numpy). Compare their errors over 200 repetitions at $N = 500$ and $N = 50\,000$. At what point does stratification begin to suffer from empty cells?

4. **Open investigation.** In a linear-Gaussian SCM, the arrow direction between two variables cannot be learned from data. It can be learned if the noise is *non-Gaussian*, because only in the causal direction is the residual independent of the input. Simulate $Y := X + U$ with X and U uniform, regress both ways, and measure how dependent the residual is on the regressor (for example by the correlation between their squares). How does your ability to tell the directions apart degrade as the noise becomes more Gaussian, say a sum of k uniforms with growing k? There is no clean formula for the sample size you need; find out empirically.

37.7 Loose threads

This chapter gave you the grammar of causation and quietly skipped several hard parts.

- **Where do the graphs come from?** We assumed the arrows. In practice they come from domain knowledge, from experiments, and partly from data. How far data alone can go is the first mystery of 37.3; its toolkit of conditional-independence tests and conditional mutual information is in Ch. 33.
- **Is the back-door criterion the whole story?** No. Pearl's **do-calculus** is a set of three rewriting rules that turn a *do*-expression into an ordinary probability whenever that is possible at all, and it is known to be *complete*: if the rules fail, the effect is not identifiable from the graph. The completeness proof is well beyond this chapter.
- **What about rung 3?** Counterfactual quantities like the probability that a treatment was *necessary* are generally only partially identified. The second mystery of 37.3 points to the linear programs that compute their sharpest bounds, in Ch. 48.
- **Why reweighting is fragile.** The mystery of 37.4: IPW is importance sampling, and it degrades badly when propensities approach 0 or 1. The diagnosis is in Ch. 38, and the reinforcement-learning version (off-policy evaluation) in Ch. 53.

The kidney-stone table will never look the same again. Every pair of correlated numbers you meet now comes with a question attached: *which arrows made this?*

Further reading

- Judea Pearl and Dana Mackenzie, *The Book of Why* (Basic Books, 2018). The ladder of causation and the history of the field, without equations.
- Judea Pearl, Madelyn Glymour and Nicholas P. Jewell, *Causal Inference in Statistics: A Primer* (Wiley, 2016). Short and exact; the best next step after this chapter.
- Miguel A. Hernán and James M. Robins, *Causal Inference: What If* (Chapman & Hall/CRC, 2020). Potential outcomes, IPW and the practice of observational studies.
- Jonas Peters, Dominik Janzing and Bernhard Schölkopf, *Elements of Causal Inference* (MIT Press, 2017). Causal discovery and the non-Gaussian ideas behind challenge 4.

Chapter 38 — Monte Carlo: Computing with Randomness

> **Level 2 · Track B — Probability, Uncertainty, and Information Builds on:** Ch. 24 (§24.6 Monte Carlo methods), Ch. 19, Ch. 34 (stationary distributions) · **Leads to:** Ch. 35, Ch. 71, Ch. 90 · **Related:** Ch. 37, Ch. 81, Ch. 92 **Reading time:** ~30 min

38.1 The hook

In 1946 the mathematician Stanisław Ulam was recovering from an illness and playing a lot of solitaire. He wanted to know the chance that a randomly shuffled deck could be played out to the end. He tried the combinatorics and got nowhere: too many cases. Then a lazier, better idea arrived. *Just play a hundred games and count the wins.* He mentioned it to John von Neumann, who saw at once that the same trick would answer questions about neutrons diffusing through the core of a bomb, questions nobody could solve on paper. They needed a code name for a secret method that ran on chance. Nicholas Metropolis suggested the casino where Ulam's uncle liked to gamble: **Monte Carlo**.

You have met the basic idea already in Ch. 24, §24.6: an expectation $\mu = E[f(\mathbf{X})]$ is estimated by an average of f over random samples, the error falls as $\sigma/\sqrt{N}$ whatever the dimension, and variance-reduction tricks (antithetic pairs, control variates, stratification, importance sampling) shrink σ. That chapter treated Monte Carlo as an *integrator*. This one is about the hard part it took for granted.

Every Monte Carlo estimate begins with "draw $\mathbf{X}_1, \dots, \mathbf{X}_N$ from p". But what if you cannot draw from p?

Here is the situation that changed science. In 1953 Metropolis, the Rosenbluths and the Tellers wanted the average energy of a liquid: a few hundred hard disks in a box, in thermal equilibrium. Statistical mechanics says the probability of a configuration $\mathbf{x}$ (all the positions together, a point in a space of hundreds of dimensions) is

$$p(\mathbf{x}) = \frac{e^{-E(\mathbf{x})/kT}}{Z}, \qquad Z = \int e^{-E(\mathbf{x})/kT}\, d\mathbf{x}.$$

The energy $E(\mathbf{x})$ is easy to compute for any configuration. The normalizing constant Z is an integral over hundreds of dimensions: computing it is *harder* than the problem you started with. And without Z you do not even know p, only something proportional to it.

Modern Bayesian statistics lives in exactly this trap. A posterior (Ch. 35) is likelihood times prior divided by the evidence, and the evidence is an integral nobody can do. You can evaluate the shape of the distribution at any point you like, but you cannot normalize it, cannot invert its CDF, and in a hundred dimensions cannot even find where most of its mass lies.

So here is the question this chapter answers:

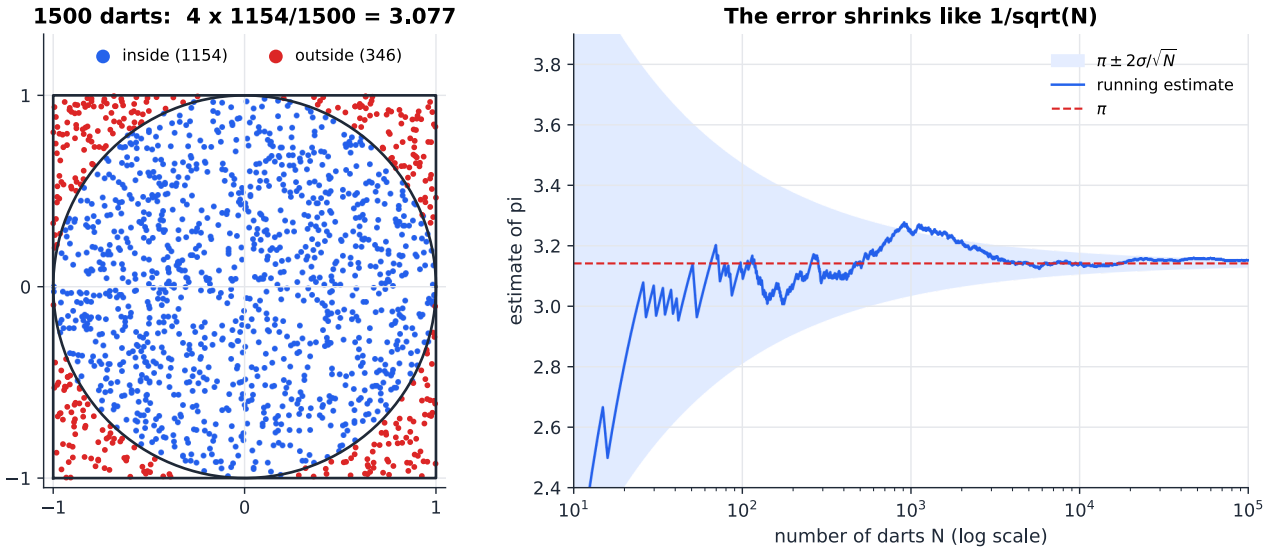
How do you draw samples from a probability distribution that you can evaluate only up to an unknown constant, in a space with hundreds of dimensions?

The answer, found on a 1950s computer called MANIAC, is a random walk with one carefully designed rule. It is one of the most cited algorithms of the twentieth century.

38.2 The big idea

Darts first

Throw darts uniformly at the square $[-1, 1]^2$. The fraction that land inside the unit circle estimates the area ratio $\pi/4$.



Estimating π with darts. Each dart is a Bernoulli trial with success probability $\pi/4$, so the estimate $4 \times (\text{hits}/N)$ has standard deviation $1.64/\sqrt{N}$: two correct digits need about 10^4 darts, three need about 10^6.

This works because we *can* sample the square uniformly, and the circle sits comfortably inside it, so most darts are useful. Two things go wrong when we try the same trick on a hard distribution.

The target hides. In a 100-dimensional space, the mass of a typical distribution is concentrated in a region that is an absurdly small fraction of any box you could draw around it. The volume of the unit ball in d dimensions, as a fraction of its enclosing cube, is $\pi^{d/2}/\big(2^d\,\Gamma(d/2+1)\big)$: 79% for $d=2$, 0.25% for $d=10$, and about 2×10^{-70} for $d=100$. Uniform darts almost never land anywhere interesting.

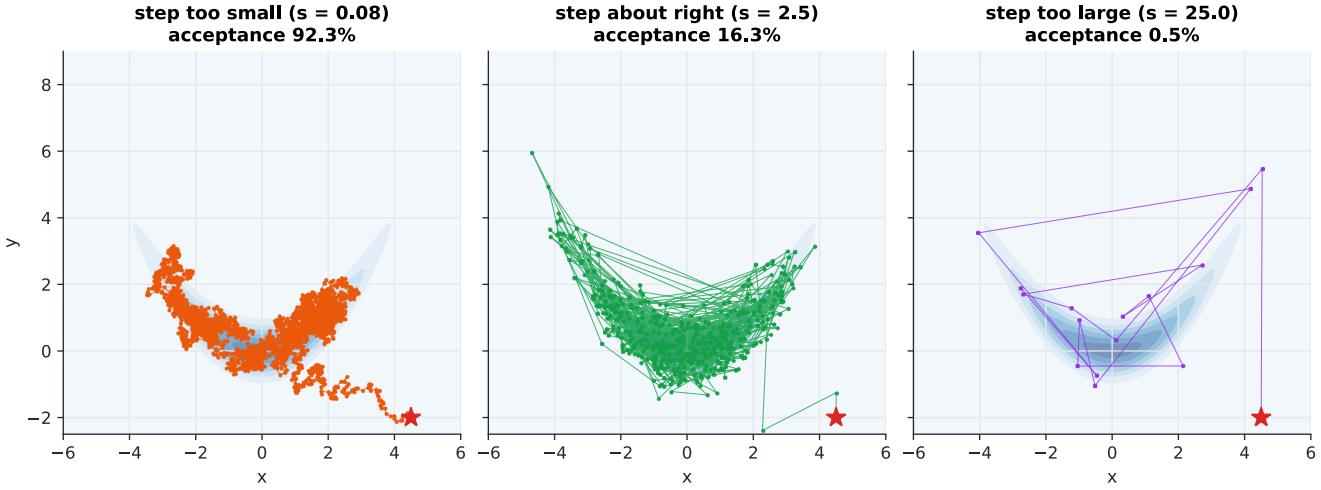
The target is unnormalized. We can compute $\tilde{p}(\mathbf{x}) = Z\,p(\mathbf{x})$ but not Z.

The hiker in the fog

Imagine a hiker on a foggy mountain range who can read only an altimeter showing the local "height" $\tilde{p}(\mathbf{x})$. She wants to spend time at each place in proportion to its height, so that a log of where she has been is a sample from p. She cannot see the landscape. She follows one rule:

> Propose a random step. If it goes uphill, take it. If it goes downhill, take it only with probability (new height)/(old height); otherwise stay put and record the current position again.

She never needs the total volume of the mountain range, only *ratios* of heights, and in a ratio the unknown Z cancels. She drifts uphill toward the mass, but keeps occasionally stepping down, so she does not just sit on the summit; she explores the slopes in exactly the right proportions. This is the **Metropolis algorithm**. Each recorded position depends on the previous one, so her log is a **Markov chain**, the object of Ch. 34, and we will design that chain so that its stationary distribution is p.



Three Metropolis chains of 3000 steps on the same banana-shaped density, all starting at the red star. Tiny steps are almost always accepted but crawl; huge steps are almost always rejected, so the chain freezes for hundreds of steps at a time; a moderate step explores the whole density.

Notice that the best-looking chain accepts only about one proposal in six. A high acceptance rate is not the goal; *exploration* is.

38.3 The mathematics

Sampling when you can: inverse transform and rejection

If X has a CDF F you can invert, then $F^{-1}(U)$ with U uniform has CDF F (inverse-transform sampling, Ch. 19). That covers the exponential, the Rayleigh, the Cauchy. For everything else, the first general tool is rejection.

Rejection sampling

Let $\tilde{p}\ge 0$ be an unnormalized target with $\int \tilde{p} = Z$, and let q be a density you can sample, with $\tilde{p}(\mathbf{x}) \le M\,q(\mathbf{x})$ for all $\mathbf{x}$. Repeat: draw $\mathbf{Y}\sim q$ and $U\sim\mathcal{U}(0,1)$; accept $\mathbf{Y}$ if $U \le \tilde{p}(\mathbf{Y})/\big(Mq(\mathbf{Y})\big)$. Then accepted values have density exactly $p=\tilde{p}/Z$, and each trial is accepted with probability Z/M.

Proof. For any set A,

$$P(\mathbf{Y}\in A,\ \text{accept}) = \int_A q(\mathbf{y})\,\frac{\tilde{p}(\mathbf{y})}{Mq(\mathbf{y})}\,d\mathbf{y} = \frac{1}{M}\int_A \tilde{p}(\mathbf{y})\,d\mathbf{y}.$$

Taking A to be the whole space gives $P(\text{accept}) = Z/M$. Dividing, $P(\mathbf{Y}\in A \mid \text{accept}) = \frac{1}{Z}\int_A \tilde{p}$, which is p. ∎

Geometrically, you throw darts uniformly under the curve Mq and keep those that fall under $\tilde{p}$: the π experiment with a smarter box. For a standard half-normal target, $\tilde{p}(x) = e^{-x^2/2}$ on $x\ge0$, with an Exp(1) proposal the best envelope constant gives an acceptance rate of $\sqrt{\pi/(2e)} = 0.760$. Very good.

Now the bad news. Take the target $\mathcal{N}(\mathbf{0},\mathbf{I}_d)$ and a proposal only slightly wider, $\mathcal{N}(\mathbf{0},\sigma^2\mathbf{I}_d)$. The ratio p/q is largest at the origin, where it equals σ^d, so the acceptance rate is σ^{-d}. With $\sigma=1.2$ and $d=100$: $1.2^{-100}\approx 1.2\times10^{-8}$. One accepted sample per hundred million tries, for a proposal that is "only" 20% too wide. **Rejection sampling dies exponentially with dimension.**

Importance sampling, revisited

Ch. 24 introduced importance sampling: draw from q, weight by $w = p/q$. When p is known only up to Z, use the **self-normalized** estimator

$$\hat{\mu} = \frac{\sum_{i=1}^{N} \tilde{w}_i \, f(\mathbf{X}_i)}{\sum_{i=1}^{N} \tilde{w}_i}, \qquad \tilde{w}_i = \frac{\tilde{p}(\mathbf{X}_i)}{q(\mathbf{X}_i)}, \quad \mathbf{X}_i \sim q.$$

Both numerator and denominator are Monte Carlo averages: divided by N, the numerator tends to $Z\mu$ and the denominator to Z. So Z cancels and $\hat{\mu} \to \mu$. The price is a small bias of order $1/N$ (a ratio of unbiased estimators is not unbiased), which is usually negligible.

The deeper problem is the one the effective sample size $\mathrm{ESS} = (\sum w_i)^2 / \sum w_i^2$ from Ch. 24 diagnoses. For large N, $\mathrm{ESS}/N \approx (E_q[w])^2 / E_q[w^2]$. In the same Gaussian example (target $\mathcal{N}(\mathbf{0}, \mathbf{I}_d)$, proposal $\mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_d)$), a direct Gaussian integral gives $E_q[w^2] = \left(\sigma^2 / \sqrt{2\sigma^2 - 1}\right)^d$. With $\sigma = 1.2$:

Dimension d	1	10	100
ESS/N	0.95	0.61	0.0074

At $d = 100$, a million weighted samples are worth about 7400 independent ones, and the situation gets exponentially worse from there. A handful of samples carry nearly all the weight. This is the same degeneracy that makes inverse propensity weighting fragile in Ch. 37: a patient with a tiny propensity is a sample with a huge weight.

Both independent-sampling methods fail for the same reason: they propose blindly, from a fixed q that knows nothing about where the target's mass is. The fix is to let each proposal depend on where you are now.

Markov chains that sample

A Markov chain on a continuous space is described by a **transition kernel** $K(\mathbf{x} \to \mathbf{y})$: the density of the next state $\mathbf{y}$ given the current state $\mathbf{x}$. A density π is **stationary** for K if a chain started in π stays in π:

$$\int \pi(\mathbf{x}) \, K(\mathbf{x} \to \mathbf{y}) \, d\mathbf{x} = \pi(\mathbf{y}) \quad \text{for all } \mathbf{y}.$$

(We write π for the target in this section, as is traditional in MCMC; it has nothing to do with 3.14159.) Checking this integral equation directly is hard. A local condition is much easier.

Detailed balance implies stationarity

If $\pi(\mathbf{x}) \, K(\mathbf{x} \to \mathbf{y}) = \pi(\mathbf{y}) \, K(\mathbf{y} \to \mathbf{x})$ for all $\mathbf{x}, \mathbf{y}$, then π is stationary for K.

Proof. Integrate both sides over $\mathbf{x}$:

$$\int \pi(\mathbf{x}) K(\mathbf{x} \to \mathbf{y}) \, d\mathbf{x} = \pi(\mathbf{y}) \int K(\mathbf{y} \to \mathbf{x}) \, d\mathbf{x} = \pi(\mathbf{y}),$$

because $K(\mathbf{y} \to \cdot)$ is a probability density and integrates to 1. ∎

Detailed balance says the *flow* of probability from $\mathbf{x}$ to $\mathbf{y}$ equals the flow back. A chain in equilibrium, filmed and played backwards, looks statistically the same; such chains are called **reversible**. Stationarity only requires the net flow into each state to balance; detailed balance asks for balance across every single pair, which is stronger but checkable one pair at a time.

Metropolis–Hastings

Now design a kernel with detailed balance for a target we know only as $\tilde{\pi} = Z\pi$. Start from any proposal density $q(\mathbf{y} \mid \mathbf{x})$, such as a Gaussian step around $\mathbf{x}$, and accept the proposal with probability

$$\alpha(\mathbf{x}, \mathbf{y}) = \min\left(1, \; \frac{\tilde{\pi}(\mathbf{y}) \, q(\mathbf{x} \mid \mathbf{y})}{\tilde{\pi}(\mathbf{x}) \, q(\mathbf{y} \mid \mathbf{x})}\right).$$

If the proposal is rejected, the chain stays at $\mathbf{x}$ (and $\mathbf{x}$ is recorded again: this repetition is essential). Only the ratio $\tilde{\pi}(\mathbf{y})/\tilde{\pi}(\mathbf{x}) = \pi(\mathbf{y})/\pi(\mathbf{x})$ appears, so **the normalizing constant is never needed**. For a symmetric proposal, $q(\mathbf{x} \mid \mathbf{y}) = q(\mathbf{y} \mid \mathbf{x})$, the q's cancel and we recover the hiker's rule: that is Metropolis's original 1953 algorithm, and Hastings generalized it to asymmetric proposals in 1970.

Metropolis–Hastings satisfies detailed balance

The Metropolis–Hastings kernel $K(\mathbf{x} \to \mathbf{y}) = q(\mathbf{y} \mid \mathbf{x}) \, \alpha(\mathbf{x}, \mathbf{y})$ for $\mathbf{y} \neq \mathbf{x}$, plus a point mass at $\mathbf{x}$ for rejections, satisfies detailed balance with respect to π. Hence π is stationary.

Proof. The point mass at $\mathbf{x}$ only ever connects $\mathbf{x}$ to itself, and detailed balance is trivial when $\mathbf{y} = \mathbf{x}$. For $\mathbf{y} \neq \mathbf{x}$, write $a = \pi(\mathbf{x}) q(\mathbf{y} \mid \mathbf{x})$ and $b = \pi(\mathbf{y}) q(\mathbf{x} \mid \mathbf{y})$, so that $\alpha(\mathbf{x}, \mathbf{y}) = \min(1, b/a)$ and $\alpha(\mathbf{y}, \mathbf{x}) = \min(1, a/b)$. Then

$$\begin{aligned} \pi(\mathbf{x}) K(\mathbf{x} \to \mathbf{y}) &= a \, \min(1, b/a) = \min(a, b), \\ \pi(\mathbf{y}) K(\mathbf{y} \to \mathbf{x}) &= b \, \min(1, a/b) = \min(a, b). \end{aligned}$$

The two flows are equal. ∎

The proof is three lines, and it explains the strange-looking formula. The acceptance probability is the *largest* one that equalizes the two flows: whichever direction is "overused" by the proposal gets throttled down to match the other.

Stationarity is not yet enough; the chain must also *reach* π from wherever it starts, and its time averages must converge. The relevant result, stated here without proof, extends the ergodic theorem for finite chains in Ch. 34:

Ergodic theorem for MCMC (stated)

If the chain is **irreducible** (from any start it can reach any set of positive π-probability) and **aperiodic**, and π is stationary, then for any f with $E_\pi|f| < \infty$ and almost any starting point,

$$\frac{1}{N}\sum_{t=1}^{N} f(\mathbf{X}_t) \longrightarrow E_\pi[f(\mathbf{X})] \quad \text{as } N \to \infty.$$

A Gaussian random-walk proposal on a target that is positive everywhere satisfies both conditions. So the algorithm is *correct*. Whether it is *fast* is another matter entirely.

The price of dependence

Successive MCMC samples are correlated, so N of them carry less information than N independent draws. Let ρ_k be the lag-k autocorrelation of the sequence $f(\mathbf{X}_t)$ in equilibrium. A calculation like the one for the variance of a sample mean of a stationary process (Ch. 21) gives, for large N,

$$\operatorname{Var}\left(\frac{1}{N}\sum_{t=1}^{N} f(\mathbf{X}_t)\right) \approx \frac{\sigma_f^2}{N}\,\tau, \qquad \tau = 1 + 2\sum_{k=1}^{\infty}\rho_k.$$

The **integrated autocorrelation time** τ is the number of chain steps that are worth one independent sample, and N/τ is the effective sample size of the chain. For a chain whose correlations decay geometrically, $\rho_k = \rho^k$, the sum is $\tau = (1+\rho)/(1-\rho)$: with $\rho = 0.9$, $\tau = 19$; with $\rho = 0.99$, $\tau = 199$.

This is where the step size bites. Tiny steps are accepted, but successive states are nearly identical, $\rho \approx 1$, and τ is huge. Huge steps are rejected, the chain repeats itself, $\rho \approx 1$ again. In between lies a sweet spot. For a random-walk Metropolis sampler on a d-dimensional Gaussian-like target, Roberts, Gelman and Gilks (1997) showed that as $d \to \infty$ the optimal proposal scale is about $2.38/\sqrt{d}$ times the target's standard deviation, which gives an acceptance rate of **0.234**. In low dimensions the optimum is higher (about 0.44 in one dimension). These are rules of thumb, not laws; the experiment below finds a different optimum for a curved target.

? How long is long enough?

The ergodic theorem promises convergence but not a speed. How many steps until a chain started at the red star has "forgotten" its start? For a finite chain the answer is governed by the gap between the largest and second-largest eigenvalue of its transition matrix (Ch. 34). For a sampler on a continuous space the transition kernel is an *operator*, its spectrum can be continuous, and bounding its gap is a genuinely hard problem in operator theory. The tools are in Ch. 90; for most real posteriors, nobody knows the answer.

Gibbs sampling in one paragraph

If you can sample each coordinate from its conditional distribution given all the others, $\pi(x_i \mid \mathbf{x}_{-i})$, then cycling through the coordinates and replacing each with a fresh conditional draw is a Metropolis–Hastings chain whose acceptance probability is exactly 1 (put $q = \pi(x_i \mid \mathbf{x}_{-i})$ into α and the ratio equals 1). That is **Gibbs sampling**, the workhorse behind early Bayesian software and image restoration. It shines when conditionals are simple and fails when coordinates are strongly correlated, because moving one axis at a time along a narrow diagonal ridge is like walking down a corridor sideways.

38.4 The algorithm

Random-walk Metropolis in its entirety:

```
def metropolis(log_target, x0, step, n, rng):
    x = np.asarray(x0, float); lp = log_target(x)
    chain = np.empty((n, x.size)); accepted = 0
    for t in range(n):
        y = x + step * rng.normal(size=x.size)      # symmetric Gaussian proposal
        lp_y = log_target(y)
        if np.log(rng.random()) < lp_y - lp:        # accept with prob min(1, ratio)
            x, lp = y, lp_y; accepted += 1
        chain[t] = x                                # record x even after a rejection
    return chain, accepted / n
```

Four engineering details matter more than they look.

1. **Work in logarithms.** In a hundred dimensions, $\tilde\pi(\mathbf{x})$ can be 10^{-300} and underflow to zero, making every ratio $0/0$. Compare $\log U$ with $\log\tilde\pi(\mathbf{y}) - \log\tilde\pi(\mathbf{x})$ instead. The `min(1, ·)` is implicit: if the difference is positive, the test always passes.
2. **Record rejections.** A common bug is to append only accepted states. That silently samples the wrong distribution, overweighting regions where moves are easy.
3. **Cost.** One target evaluation per step, $O(d)$ arithmetic for the proposal. The real cost is the number of steps, which is $N_{\text{eff}} \times \tau$, and τ for random-walk Metropolis typically grows like d (the step must shrink like $1/\sqrt{d}$, and a diffusion needs (distance/step)^2 steps to cross the target).
4. **Burn-in and diagnostics.** Early samples remember the starting point; discard them. Then check convergence, imperfectly: run several chains from dispersed starting points and compare their within-chain and between-chain variances (the Gelman–Rubin statistic $\hat R$, which should be close to 1), and estimate τ from the autocorrelations.

Failure cases.

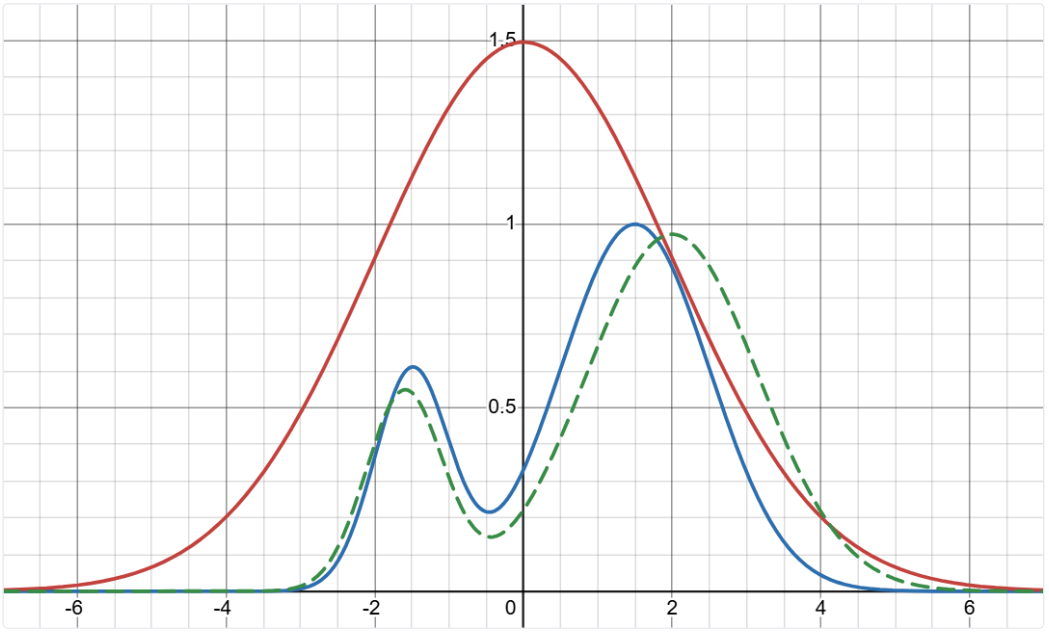
- **Multimodality.** If the target has two peaks separated by a valley of near-zero density, a random walk may never cross it. Every diagnostic computed from a single chain will look perfect, and the answer will be wrong by a factor of two. Several chains from different starts are the only cheap defence, and they are not a proof.
- **Bad scaling.** If the target is narrow in one direction and wide in another, an isotropic step that suits the narrow direction crawls along the wide one. The fix is to rescale the proposal by an estimate of the target's covariance (adaptive Metropolis).
- **Heavy tails and sharp ridges.** Curved, funnel-shaped posteriors (common in hierarchical Bayesian models) have regions where no single step size works.

> **? The trick of adding momentum**
>
> Random-walk Metropolis diffuses, so it needs about d steps to cross a d-dimensional target. **Hamiltonian Monte Carlo** gives each sample an imaginary momentum, lets it slide frictionlessly over the landscape $-\log\tilde{\pi}$ for a while, and then applies an accept/reject test. It can make long, nearly always accepted moves, and the number of steps it needs grows only about like $d^{1/4}$. Why does it work? Because Hamilton's equations preserve volume in phase space and conserve energy, so the Metropolis ratio is almost exactly 1; and the integrator that keeps those properties under discretization is not the obvious one. The physics, including symplectic structure, is in Ch. 92.

38.5 The experiment

Rejection sampling: fit the envelope

The blue curve below is an unnormalized bimodal target $\tilde{p}(x)$, with area $Z \approx 3.259$. The red curve is the envelope $M\,q(x)$ with $q = \mathcal{N}(0, 2^2)$ and $M = 7.5$. Rejection sampling is valid only if red lies above blue everywhere, and its acceptance rate is Z/M, about 43% here.



Rejection sampling with a Gaussian envelope. The dashed green curve is the acceptance probability $\tilde{p}/(Mq)$; it must stay below 1.

Change the `7.5` (in both the red and green lines) to `5` and find where the envelope dips below the target: there the sampler would produce too few samples, silently. The smallest valid M is about 7.29, set by the point $x = 2$. Then change the proposal width: replace `/8` by `/2` and `2\sqrt{2\pi}` by `\sqrt{2\pi}` (standard deviation 1). Push M as high as you like; the green curve still crosses 1 far out on the right, because the target's tail $e^{-(x-1.5)^2/2}$ eventually beats $e^{-x^2/2}$. A proposal with lighter tails than the target can never be fixed by a bigger constant.

Metropolis on the banana

The script samples the banana density of the figure in 38.2, whose true means are $E[x^2] = 4$ and $E[y] = 1$. It reports the acceptance rate and an effective sample size computed from the autocorrelation of y.

```
import numpy as np
rng = np.random.default_rng(0)

STEP   = 1.0       # proposal step size: try 0.1, 1.0, 3.0, 10.0
N      = 50_000    # chain length
BURN   = 1_000     # samples discarded at the start
START  = (4.5, -2.0)

def log_target(x, y):
    """Unnormalized log density of a curved 'banana'. True E[y] = 1."""
    return -x**2 / 8 - (y - 0.25 * x**2)**2 / (2 * 0.5**2)

def metropolis(n, step, start):
    chain = np.empty((n, 2))
    cur = np.array(start, float); lp = log_target(*cur); acc = 0
    for i in range(n):
        prop = cur + step * rng.normal(size=2)      # symmetric proposal
        lp_prop = log_target(*prop)
        if np.log(rng.random()) < lp_prop - lp:     # accept w.p. min(1, ratio)
            cur, lp = prop, lp_prop; acc += 1
        chain[i] = cur
    return chain, acc / n

def ess(v, max_lag=2000):
    """Effective sample size from the autocorrelation of one coordinate."""
    v = v - v.mean(); n = len(v); tau = 1.0
    for k in range(1, max_lag):
        rho = np.dot(v[:-k], v[k:]) / np.dot(v, v)
        if rho < 0.05: break
        tau += 2 * rho
    return n / tau

chain, acc = metropolis(N, STEP, START)
y = chain[BURN:, 1]
print(f"acceptance rate  {acc:.2f}")
print(f"estimate of E[y] {y.mean():.3f}   (true 1.000)")
print(f"estimate of E[x^2] {np.mean(chain[BURN:, 0]**2):.3f} (true 4.000)")
print(f"effective sample size for y: {ess(y):.0f} out of {len(y)}")
```

With `STEP = 1.0` one run accepts 40% of proposals and estimates $E[y] \approx 1.05$ with an effective sample size of roughly 500 out of 49 000: each independent-equivalent sample costs about 100 steps. Things to try:

- Sweep `STEP` through 0.1, 1.0, 2.5, 5.0, 10.0 and tabulate acceptance against ESS. On this curved target the best ESS comes at steps between about 2.5 and 5, with acceptance rates of only 6–16%, well below the textbook 0.234. Why might a two-dimensional banana prefer bold steps?
- With `STEP = 0.1` the acceptance rate is above 90% and the ESS collapses to a few dozen. Plot `chain[:, 1]` against step number to *see* the slow wander.
- Set `BURN = 0` and `N = 2000`, and compare estimates from `START = (4.5, -2.0)` with `START = (0, 0)`. How many steps does the start keep influencing the answer?
- Replace the banana by two well-separated Gaussian bumps centred at $(\pm 4, 0)$ and run from one of them. Look at the fraction of time spent on each side. Then run five chains from different starts.

38.6 The challenge

1. **Intuition.** A colleague runs Metropolis, keeps only the accepted proposals ("the rejected ones were never real samples") and finds the histogram too spread out compared with the known answer. Explain, without equations, which regions of the target are over-represented by this mistake, and why.

2. **Derivation.** For the independence sampler, a Metropolis–Hastings chain whose proposal $q(\mathbf{y})$ ignores the current state, write down the acceptance probability in terms of the importance weights $w(\mathbf{x}) = \tilde{\pi}(\mathbf{x})/q(\mathbf{x})$. Then show that if $w \le M$ everywhere, every step accepts with probability at least Z/M, and relate this to the acceptance rate of rejection sampling with the same envelope.

3. **Implementation.** Write a Gibbs sampler for the bivariate Gaussian with unit variances and correlation r, using the conditionals $x \mid y \sim \mathcal{N}(ry,\, 1 - r^2)$ and $y \mid x \sim \mathcal{N}(rx,\, 1 - r^2)$. Measure the lag-1 autocorrelation of the x-chain for $r = 0.5, 0.9, 0.99$ and compare it with r^2. Then implement random-walk Metropolis on the same target and find, for $r = 0.99$, the step size that maximizes ESS per target evaluation. Which method wins?

4. **Open investigation.** Take the bimodal target $\tilde{\pi}(x) = e^{-(x-a)^2/2} + e^{-(x+a)^2/2}$ and a Gaussian random-walk proposal with step 1. Measure, by simulation, the average number of steps before a chain started at $+a$ first crosses $x = 0$, for $a = 1, 2, 3, 4, 5$. How does it grow with a? Try to explain the growth with a barrier-height argument, and then design a modification (tempering, larger occasional jumps, or anything you invent) that makes the crossing time grow polynomially instead. There is no universal recipe for multimodal targets in high dimension.

38.7 Loose threads

The hook asked how to sample from a distribution known only up to a constant, in hundreds of dimensions. Metropolis–Hastings answers it: build a Markov chain whose only contact with the target is through ratios, prove detailed balance in three lines, and let the ergodic theorem turn the chain's time averages into expectations. What the chapter did not answer is how *well* this works.

- **Speed.** The first mystery: we proved the chain converges, but not how fast. Mixing-time bounds for continuous-space chains come from the spectral gap of an operator, in Ch. 90. The finite-state version is in Ch. 34.

- **Better proposals.** The second mystery: Hamiltonian Monte Carlo replaces blind diffusion by physics, and why its energy-conserving, volume-preserving dynamics are exactly what the acceptance test needs is a story about symplectic geometry, in Ch. 92.
- **Where does the randomness come from?** Every line of code above calls `rng.random()`, which is a deterministic program. Why a deterministic sequence can stand in for true chance, and how to tell when it cannot, is the subject of Ch. 81.
- **The constant we threw away.** MCMC never computes Z, yet in Bayesian model comparison (Ch. 35) Z *is* the answer: the evidence for a model. Estimating it requires further machinery (thermodynamic integration, nested sampling, annealed importance sampling).
- **Light transport.** Path tracing in computer graphics estimates an integral over all light paths in a scene. The same importance sampling, and even Metropolis itself (Metropolis light transport), turn up in Ch. 71.

? Is randomness necessary at all?

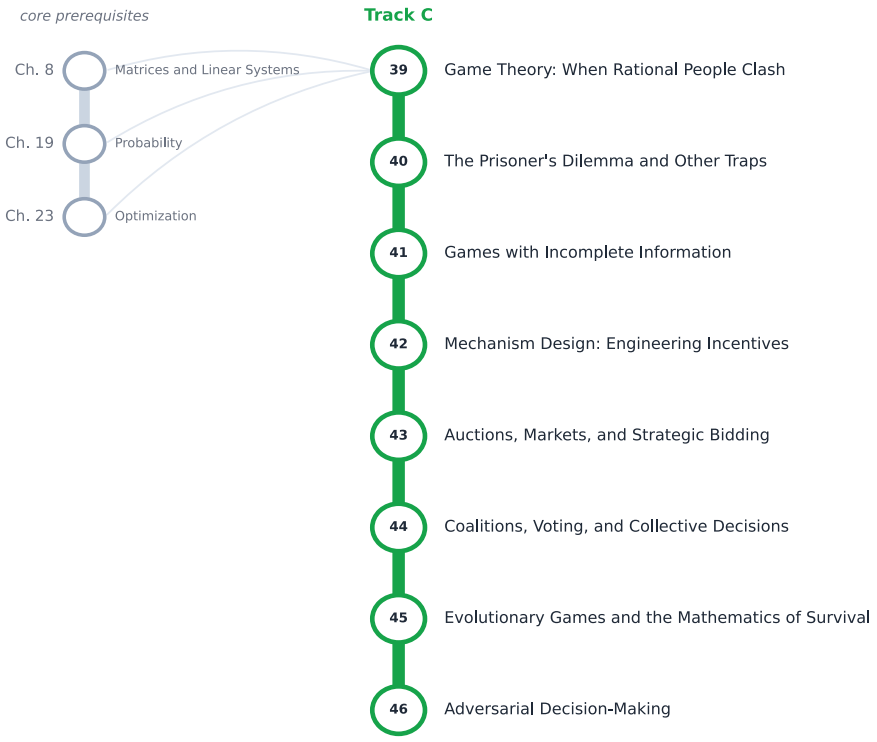
Quasi-Monte Carlo (Ch. 24) replaced random points by carefully spread deterministic ones and gained almost a factor of $\sqrt{N}$ in accuracy. Can a Markov chain be "derandomized" the same way? Can every randomized algorithm be replaced by a deterministic one with only polynomial slowdown? The second question is the famous open problem whether $\mathsf{BPP} = \mathsf{P}$, and the evidence for and against it runs through Ch. 81.

Further reading

- Christian P. Robert and George Casella, *Monte Carlo Statistical Methods*, 2nd ed. (Springer, 2004). The standard reference for rejection, importance sampling and MCMC with proofs.
- Art B. Owen, *Monte Carlo Theory, Methods and Examples* (online book, 2013). Clear on variance reduction, importance sampling and quasi-Monte Carlo.
- David J. C. MacKay, *Information Theory, Inference, and Learning Algorithms* (Cambridge University Press, 2003), Chapters 29–30. The most intuitive introduction to Metropolis, Gibbs and Hamiltonian Monte Carlo.
- Steve Brooks, Andrew Gelman, Galin Jones and Xiao-Li Meng (eds.), *Handbook of Markov Chain Monte Carlo* (Chapman & Hall/CRC, 2011). Includes Radford Neal's classic chapter on Hamiltonian dynamics.

Track C — Game Theory and Strategic Intelligence

> **Level 2 — Specializations.** What happens when the thing you are optimizing against can think back.



Suggested reading order for Track C. The grey stations on the left are the core chapters it leans on.

Before you start

This track assumes you are comfortable with these core chapters:

- Ch. 8 — Matrices and Linear Systems
- Ch. 19 — Probability
- Ch. 23 — Optimization

Chapters

Ch.	Chapter	What it explores
39	Game Theory: When Rational People Clash	Normal-form games, dominant strategies, Nash equilibrium.
40	The Prisoner's Dilemma and Other Traps	Cooperation, repeated games, social dilemmas, evolutionary stability.
41	Games with Incomplete Information	Bayesian games, beliefs, signaling, screening.
42	Mechanism Design: Engineering Incentives	Social choice, incentive compatibility, truthful mechanisms.
43	Auctions, Markets, and Strategic Bidding	Auction formats, revenue, allocation, market design.
44	Coalitions, Voting, and Collective Decisions	Cooperative games, Shapley value, voting systems, bargaining.
45	Evolutionary Games and the Mathematics of Survival	Replicator dynamics, evolutionary stable strategies, population games.
46	Adversarial Decision-Making	Minimax, zero-sum games, regret, robust strategies.

Every chapter follows the same arc: **the hook** (a puzzle or a failure), **the big idea**, **the mathematics**, **the algorithm**, **the experiment** you can run, **the challenge**, and **loose threads**: the questions the chapter deliberately leaves open.

Chapter 39 — Game Theory: When Rational People Clash

> **Level 2 · Track C — Game Theory and Strategic Intelligence Builds on:** Ch. 8, Ch. 19, Ch. 23 · **Leads to:** Ch. 40, Ch. 41, Ch. 46 · **Related:** Ch. 30, Ch. 48, Ch. 58 **Reading time:** ~25 min

39.1 The hook

Here is a game you can play with a lecture hall, a mailing list, or a newspaper's readership. Everybody writes down a whole number from 0 to 100. The winner is whoever comes closest to **two thirds of the average** of all the numbers. That's the whole game.

Take a minute and pick your number before reading on.

Most people start like this: "If everyone picks at random, the average is about 50, so I should say 33." A sharper player thinks one step further: "But everyone else will reason the same way and say 33, so I should say 22." The next player says 15, then 10, and the staircase keeps going down. Where does it stop?

A cold argument says: at zero. Two thirds of any average of numbers between 0 and 100 can be at most $66.\overline{6}$, so a guess above 67 is never better than a guess of 67, whatever anyone else does. A player who knows nobody guesses above 67 knows the target is at most $\frac{2}{3} \cdot 66.\overline{6} \approx 44.4$. Then 29.6, 19.8, 13.2, and so on. Each round of "I know that you know that I know..." shrinks the upper bound by another factor of $\frac{2}{3}$, and the only number that survives every round is 0.

Now the twist. When this game is played with real people, the winner almost never picked 0. Averages land somewhere in the twenties or thirties, and the person who guessed 0, the "perfectly rational" one, loses badly. Being right about rationality is not the same as being right about people.

Keep that tension in mind, because it runs through this whole track. Once you start optimizing against something that can think back, the best move stops being a property of the problem and becomes a property of what everyone believes about everyone else. In Ch. 23 the objective function sat still while you climbed it. Here it moves, because the other players are climbing too.

A second example, which we will come back to: a penalty kick in football. The kicker shoots left or right, and the keeper has to dive left or right before seeing the ball. If the kicker always goes left, a keeper who has done any homework always dives left. If the keeper always dives left, the kicker shoots right. Neither of them can be predictable, and yet each has to *choose*. What does a rational choice even mean when being predictable is exactly what loses?

By the end of this chapter you will be able to compute the answer to that question, down to the percentage. You will also see why the answer is stranger than it first looks.

39.2 The big idea

The basic move of game theory is to stop asking "what is the best action?" and ask "what is the best action **given what the others do**?" That conditional best choice is a **best response**. The central idea of the subject is then almost too simple to state:

> A situation is stable when every player is already playing a best response to everyone else.

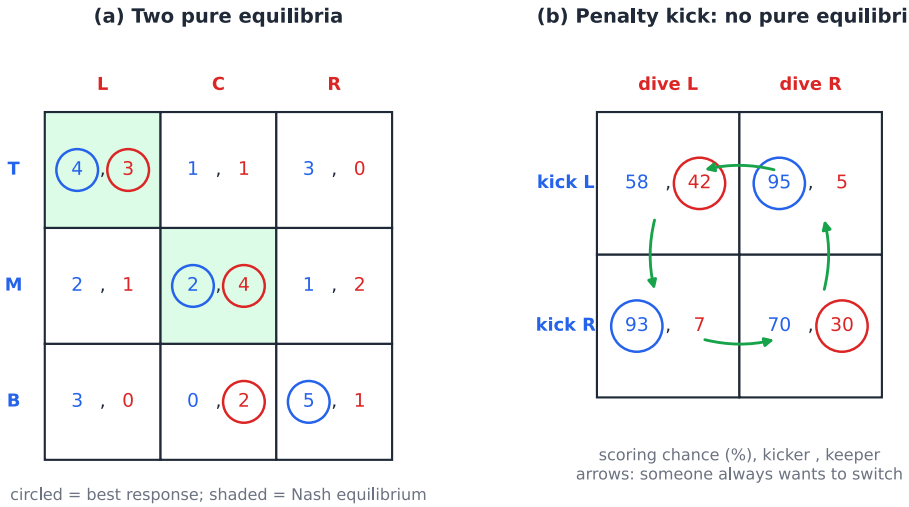
Nobody, looking back, wishes they had done something different, given what the others actually did. That's a **Nash equilibrium**: a profile of choices with no regrets.

An analogy helps. Picture a crowded beach on a hot afternoon. Everyone has spread a towel somewhere, and each person picked their spot given where the others already were: not too close to the loud group, not too far from the water. Once everyone is settled, nobody wants to get up and move. That doesn't make the arrangement *good* (maybe everyone would be happier if the loud group moved), but it is *stable*. Equilibrium is a statement about stability, not about how good the outcome is. Ch. 40 builds an entire chapter on how far apart those two ideas can be.

There are two ways to find stable outcomes, and they correspond to the two examples in the hook.

Elimination. Some choices are bad *no matter what* the others do. Throw them away. Once they're gone, other choices that used to be reasonable may become bad no matter what, so throw those away too. The 2/3 game is pure elimination: the staircase 100, 67, 44, 30, ... is a chain of "nobody sensible would ever..." arguments.

Mixing. In the penalty kick, nothing can be eliminated. Every pure action is a best response to *something*, and every pure action is beaten by *something*. The way out is to choose at random, with probabilities tuned so precisely that the opponent gains nothing from knowing them. The kicker cannot hide the probabilities, only the outcome of the coin flip. That turns out to be enough.



Best responses marked by circles: blue for the row player (largest in its column), red for the column player (largest in its row). A cell with both circles is a pure Nash equilibrium. In the penalty kick the circles never meet; the arrows show that from any cell, one of the players wants to switch.

39.3 The mathematics

Games in normal form

Normal-form game

A **game in normal form** consists of

- a finite set of **players** $N = \{1, \dots, n\}$;
- for each player i a set of **pure strategies** S_i;
- for each player a **payoff function** $u_i : S_1 \times \cdots \times S_n \to \mathbb{R}$.

A **strategy profile** is $s = (s_1, \dots, s_n)$. We write s_{-i} for everyone's choices except player i's, so $s = (s_i, s_{-i})$.

With two players and finite strategy sets, the whole game fits into two matrices. Let player 1 (the **row player**) have strategies indexed by $i = 1, \dots, m$ and player 2 (the **column player**) by $j = 1, \dots, n$. The payoffs are a_{ij} and b_{ij}, giving the **bimatrix** $(\mathbf{A}, \mathbf{B})$ that you see in the figure: each cell lists "row payoff, column payoff". If $\mathbf{A} + \mathbf{B} = \mathbf{0}$, what one player wins the other loses: a **zero-sum** game. The penalty kick, measured in scoring probability, is zero-sum after a shift (the keeper's payoff is $1-$ the kicker's), and that's all zero-sum requires.

Payoffs are utilities, not money. The only thing that matters about them is how a player ranks outcomes and lotteries over outcomes. That last part is a real assumption (von Neumann–Morgenstern expected utility), and we use it every time we average payoffs over randomness.

Dominance

Strict dominance

Strategy s_i **strictly dominates** s_i' if $u_i(s_i, s_{-i}) > u_i(s_i', s_{-i})$ for **every** s_{-i}. A strategy that is strictly dominated by some other strategy is never a best response, so a rational player never uses it.

In the left matrix of the figure, compare the column player's options C and R. Column C pays $(1, 4, 2)$ against rows (T, M, B); column R pays $(0, 2, 1)$. C is better in every row, so R is strictly dominated and the column player will never play it. Once R is gone, look at the row player: row T now pays $(4, 1)$ against (L, C), and row B pays $(3, 0)$. With R deleted, T strictly dominates B. Two rounds of elimination shrink the 3×3 game to a 2×2 core:

	L	C
T	$(4, 3)$	$(1, 1)$
M	$(2, 1)$	$(2, 4)$

This is **iterated elimination of strictly dominated strategies** (IESDS), the same procedure as the $2/3$ staircase. It's not free, though. The first round only needs the column player to be rational. The second round needs the row player to *know* the column player is rational. A third round would need the column player to know that the row player knows it, and so on. IESDS to depth k needs rationality to be "known" k levels deep. The $2/3$ game in a real lecture hall shows what happens when the hierarchy breaks off after one or two levels.

Nash equilibrium

Best response and Nash equilibrium

The **best-response set** of player i to s_{-i} is

$$\mathrm{BR}_i(s_{-i}) = \arg\max_{s_i \in S_i} u_i(s_i, s_{-i}).$$

A profile s^* is a **(pure) Nash equilibrium** if $s_i^* \in \mathrm{BR}_i(s_{-i}^*)$ for every player i, that is,

$$u_i(s_i^*, s_{-i}^*) \;\geq\; u_i(s_i, s_{-i}^*) \quad \text{for all } i \text{ and all } s_i \in S_i.$$

In a bimatrix game this is a finger exercise. Circle the largest a_{ij} in each column (the row player's best response to column j) and the largest b_{ij} in each row (the column player's best response to row i). Every cell with two circles is a pure equilibrium. The left game has two: (T, L) with payoffs $(4, 3)$ and (M, C) with payoffs $(2, 4)$. The players disagree about which one they prefer, and nothing in the definition says which one will happen. That *equilibrium selection* problem comes back again and again.

How do dominance and equilibrium fit together? The next result says elimination is safe: it never throws away a stable outcome.

IESDS preserves Nash equilibria

If s^* is a Nash equilibrium of a finite game, then no strategy s_i^* in it is ever deleted by iterated elimination of strictly dominated strategies.

Proof. Suppose the claim fails, and look at the **first** round in which some equilibrium strategy, say s_i^*, is deleted. It is deleted because some strategy t_i that is still present strictly dominates it against every profile of opponents' strategies that is still present. Since this is the first round to delete any part of s^*, all of s_{-i}^* is still present. Dominance then gives

$$u_i(t_i, s_{-i}^*) > u_i(s_i^*, s_{-i}^*),$$

which says s_i^* is not a best response to s_{-i}^*. That contradicts s^* being an equilibrium. ∎

So if IESDS leaves only one profile, that profile is the unique Nash equilibrium. (The argument shows every equilibrium survives, so at most one exists; the surviving profile is an equilibrium because every strategy that could beat it has been deleted for good reason. Checking that last step carefully is a nice exercise.) In the $2/3$ game, IESDS (with weak dominance in the first round) leaves only "everyone says 0".

When no pure equilibrium exists: mixed strategies

The penalty kick has no cell with two circles. Let's write it in scoring probabilities, with rows the kicker's choice and columns the keeper's dive:

$$\mathbf{S} = \begin{pmatrix} 0.58 & 0.95 \\ 0.93 & 0.70 \end{pmatrix}.$$

The numbers are illustrative, but they're in the range that studies of professional penalties report. Shooting to the side the keeper dives to still scores sometimes, and shooting to the open side almost always scores. The kicker maximizes $\mathbf{S}$, and the keeper minimizes it.

Mixed strategy

A **mixed strategy** for player i is a probability distribution σ_i over S_i. For a two-player game, row mixes are vectors $\mathbf{x} \in \Delta_m = \{\mathbf{x} \geq 0,\ \sum_i x_i = 1\}$ and column mixes are $\mathbf{y} \in \Delta_n$. The expected payoffs are

$$u_1(\mathbf{x}, \mathbf{y}) = \mathbf{x}^{\mathsf{T}} \mathbf{A} \mathbf{y}, \qquad u_2(\mathbf{x}, \mathbf{y}) = \mathbf{x}^{\mathsf{T}} \mathbf{B} \mathbf{y}.$$

A **mixed Nash equilibrium** is a pair $(\mathbf{x}^*, \mathbf{y}^*)$ in which each mix is a best response to the other.

Expected values here are the ordinary expectations of Ch. 19, with the players' coin flips independent. The bilinear form $\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{y}$ is all the linear algebra we need. The key structural fact about mixed equilibria is short to prove and it does all the work in practice.

Indifference principle (support lemma)

Let $(\mathbf{x}^*, \mathbf{y}^*)$ be a mixed Nash equilibrium, and let $v_1 = u_1(\mathbf{x}^*, \mathbf{y}^*)$. Then every pure strategy i with $x_i^* > 0$ earns exactly v_1 against $\mathbf{y}^*$:

$$(\mathbf{A}\mathbf{y}^*)_i = v_1 \quad \text{whenever } x_i^* > 0, \qquad (\mathbf{A}\mathbf{y}^*)_k \leq v_1 \text{ for all } k.$$

The same holds for player 2.

Proof. Write $w_k = (\mathbf{A}\mathbf{y}^*)_k$ for the payoff of pure row k against $\mathbf{y}^*$, and $w_{\max} = \max_k w_k$. Playing the pure row with the largest w_k is one of player 1's options, and since $\mathbf{x}^*$ is a best response, $v_1 \geq w_{\max}$. On the other hand,

$$v_1 = \sum_k x_k^* w_k \leq \sum_k x_k^* w_{\max} = w_{\max}.$$

So $v_1 = w_{\max}$, and the inequality in the middle holds with equality. That means $\sum_k x_k^*(w_{\max} - w_k) = 0$. Each term is a nonnegative weight times a nonnegative gap, so each term is zero: wherever $x_k^* > 0$ we must have $w_k = w_{\max} = v_1$. ∎

Read that theorem backwards and it becomes a recipe, along with the surprising core of mixed play. **In a mixed equilibrium, you randomize so that the *other* player is indifferent.** The keeper's probabilities aren't chosen to make the keeper happy. They're chosen to leave the kicker with nothing to exploit.

Solving the penalty kick

Let the kicker shoot left with probability p and the keeper dive left with probability q. Neither pure profile is stable, so both must mix, and by the indifference principle each must make the other indifferent.

Keeper's q makes the kicker indifferent between shooting left and right:

$$0.58q + 0.95(1-q) = 0.93q + 0.70(1-q) \implies 0.25 = 0.60q \implies q = \tfrac{5}{12} \approx 0.417.$$

Kicker's p makes the keeper indifferent between diving left and right:

$$0.58p + 0.93(1-p) = 0.95p + 0.70(1-p) \implies 0.23 = 0.60p \implies p = \tfrac{23}{60} \approx 0.383.$$

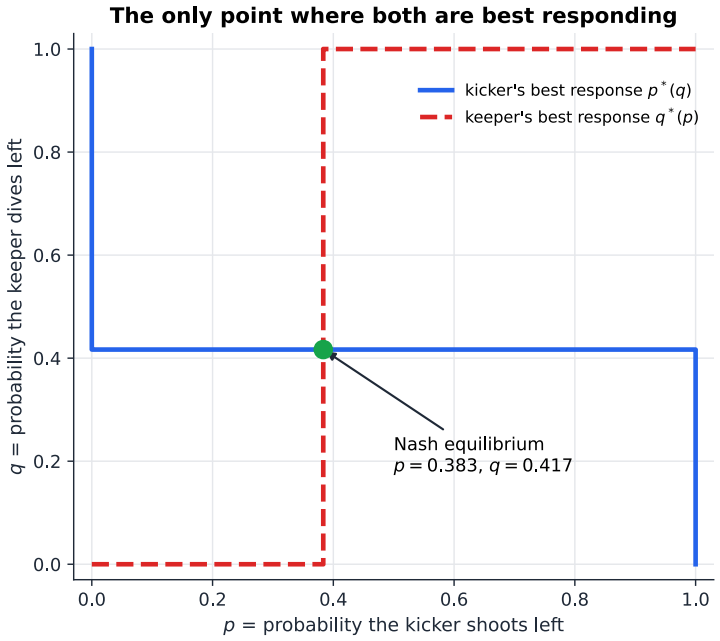
The scoring probability at equilibrium is $0.95 - 0.37 \cdot \frac{5}{12} \approx 0.796$, the same whichever way the kicker shoots, as the theorem promised.

For any 2×2 game with no pure equilibrium, the same two lines of algebra give a closed form. For the row player's payoffs a_{ij}, the column mix that makes rows indifferent is

$$q^* = \frac{a_{22} - a_{12}}{a_{11} - a_{12} - a_{21} + a_{22}},$$

and symmetrically for p^* using b_{ij}. Notice something odd: **the keeper's equilibrium probability depends only on the kicker's payoffs.** If the kicker gets better at shooting left (raise a_{11}), the kicker's own mix doesn't change at all. The keeper's does. People find this hard to believe the first time, and the experiment section lets you test it.

The best-response curves make it visible. The kicker's best response is a step function of q: shoot left if $q < 5/12$, right if $q > 5/12$, anything at all if $q = 5/12$. The keeper's best response is a step function of p. The two staircases cross at one point only.



Each curve is one player's best response to the other's mix. The only point lying on both is the mixed Nash equilibrium $(p, q) = (23/60,\ 5/12)$.

The left game of the figure has a mixed equilibrium too, on top of its two pure ones. The same algebra on the 2×2 core gives $\mathbf{x}^* = (3/5,\ 2/5)$ over (T, M) and $\mathbf{y}^* = (1/3,\ 2/3)$ over (L, C), with payoffs $(2,\ 2.2)$. That is *worse for both players* than either pure equilibrium. The players randomize to stay unexploitable, and they sometimes miscoordinate as a result.

Does an equilibrium always exist?

The penalty kick had no pure equilibrium but did have a mixed one. Is that luck?

Nash (1950)

Every finite game, with finitely many players and finitely many pure strategies each, has at least one Nash equilibrium in mixed strategies.

We won't prove it here. The idea, though, is clean. Collect all players' mixes into one point σ of a product of simplices, a compact convex set. Define a continuous map f that nudges each player's mix toward strategies that do better than the current mix. A fixed point $f(\sigma) = \sigma$ is a profile where no nudge helps, which is exactly an equilibrium. Then a theorem from topology guarantees that the fixed point exists.

? A theorem that promises without delivering

Nash's proof uses Brouwer's fixed-point theorem: every continuous map of a disc (or simplex, or ball) into itself fixes some point. The proof tells you an equilibrium *exists* but gives you no way to find it, and nobody knows how to make it constructive at a cost that stays polynomial. Why must a continuous map of a disc fix a point, while a map of an annulus need not? The answer is topological, and it's waiting in Ch. 58.

For **zero-sum** games there is an older and sharper result. Von Neumann's minimax theorem (1928) says

$$\max_{\mathbf{x}\in\Delta_m}\ \min_{\mathbf{y}\in\Delta_n}\ \mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{y} \;=\; \min_{\mathbf{y}\in\Delta_n}\ \max_{\mathbf{x}\in\Delta_m}\ \mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{y}.$$

The number in the middle, 0.796 for the penalty kick, is the **value** of the game. In zero-sum games, equilibrium is the same thing as security: each side guarantees itself the value, however clever the opponent is. The general-sum game on the left has no such guarantee.

Fix $\mathbf{y}$ and the inner maximum is a linear program, and so is the outer minimization. That's no accident: the minimax theorem and linear-programming duality are two views of the same fact, which is why a zero-sum game can be solved with the simplex method of Ch. 23. The duality side of the story is Ch. 48, and the adversarial side is Ch. 46.

39.4 The algorithm

Pure equilibria. For an $m \times n$ bimatrix, compute the column maxima of $\mathbf{A}$ and the row maxima of $\mathbf{B}$, then check each cell. That costs $O(mn)$, which is optimal, since you have to look at every payoff anyway.

Mixed equilibria: support enumeration. The indifference principle turns equilibrium-finding into linear algebra *once you guess the supports*. Fix a set I of rows and a set J of columns that will get positive probability. For a nondegenerate game, $|I| = |J| = k$. Then solve two square linear systems:

- find $\mathbf{y}$ on J with $\sum_j y_j = 1$ that makes every row in I earn the same v_1;
- find $\mathbf{x}$ on I with $\sum_i x_i = 1$ that makes every column in J earn the same v_2.

Accept the pair if both mixes are nonnegative and no pure strategy *outside* the support does better than v_1 (resp. v_2).

```
import itertools
import numpy as np

def support_enumeration(A, B, tol=1e-9):
    """All Nash equilibria of a nondegenerate bimatrix game (A: row payoffs, B: column payoffs)."""
    m, n = A.shape
    found = []
    for k in range(1, min(m, n) + 1):
        for I in itertools.combinations(range(m), k):
            for J in itertools.combinations(range(n), k):
                # column mix y on J makes row indifferent on I; row mix x on I makes column indifferent on J
                My = np.block([[A[np.ix_(I, J)], -np.ones((k, 1))], [np.ones((1, k)), np.zeros((1, 1))]])
                Mx = np.block([[B[np.ix_(I, J)].T, -np.ones((k, 1))], [np.ones((1, k)), np.zeros((1, 1))]])
                rhs = np.r_[np.zeros(k), 1.0]
                try:
                    sy, sx = np.linalg.solve(My, rhs), np.linalg.solve(Mx, rhs)
                except np.linalg.LinAlgError:
                    continue
                x, y = np.zeros(m), np.zeros(n)
                x[list(I)], y[list(J)] = sx[:k], sy[:k]
                if x.min() < -tol or y.min() < -tol:
                    continue
                # no profitable deviation outside the support
                if (A @ y).max() <= sy[k] + tol and (x @ B).max() <= sx[k] + tol:
                    found.append((x.round(4), y.round(4)))
    return found

A = np.array([[4, 1, 3], [2, 2, 1], [3, 0, 5]])
B = np.array([[3, 1, 0], [1, 4, 2], [0, 2, 1]])
for x, y in support_enumeration(A, B):
    print(x, y)
```

On the 3×3 game of the figure it prints exactly three equilibria: (T, L), (M, C), and the mix $\mathbf{x} = (0.6, 0.4, 0)$, $\mathbf{y} = (0.333, 0.667, 0)$. No support involving B or R survives, which matches the dominance argument.

Complexity. For each k there are $\binom{m}{k}\binom{n}{k}$ support pairs, and each costs a $k \times k$ solve. Summed over k, that's $\binom{m+n}{m}$ support pairs, exponential in the game's size. Ten strategies each already means about 185,000 pairs. Smarter pivoting methods (Lemke–Howson, a cousin of the simplex method of Ch. 23) find *one* equilibrium much faster on typical games, but there are known games where they take exponentially many pivots.

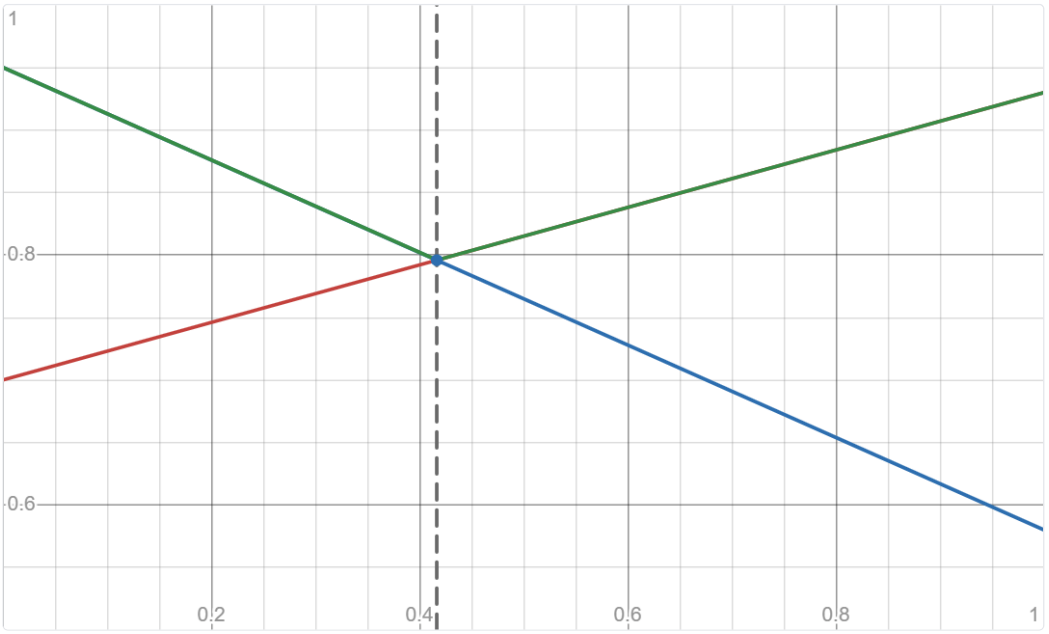
Failure cases.

- *Degenerate games* (ties in payoffs, or a mixed strategy with k-element support whose best responses number more than k) can have a whole continuum of equilibria. The square systems then become singular, and the code above silently skips the continuum.
- *Numerical tolerance.* An equilibrium probability of 10^{-12} is either a rounding error or a real but tiny weight. The test `x.min() < -tol` has to decide which, and the choice of `tol` changes the answer on badly scaled payoffs.
- *Zero-sum games* don't need any of this: they reduce to a single linear program and are solvable in polynomial time.

> **? Hard to find, easy to check**
>
> Checking a proposed equilibrium takes one matrix–vector product. Finding one in a general two-player game is **PPAD-complete** (Daskalakis–Goldberg–Papadimitriou, Chen–Deng, 2006): as hard as any problem whose answer is guaranteed by a "follow the path to its end" parity argument. Nobody believes it can be done in polynomial time, but nobody can prove that either. What PPAD is, and why "a solution is guaranteed to exist" doesn't make a problem easy, belongs to Ch. 30.

39.5 The experiment

Drag the indifference point. The graph shows the kicker's scoring probability as a function of the keeper's mix q (horizontal axis). The blue line is "shoot left", the red line is "shoot right", and the thick green curve is the kicker's best, the upper envelope. The keeper wants to sit where that envelope is lowest, which is the crossing point.



The kicker's scoring chance against a keeper who dives left with probability x; the keeper picks the lowest point of the green envelope, $x = 5/12$.

Change the **0.58** in both the blue and green lines to 0.70 (the kicker gets better at hitting a covered left corner). Where does the crossing move, and whose equilibrium mix just changed? Then change **0.95** to 0.85 and watch what happens when the crossing slides near $x = 0$ or $x = 1$. When the lines stop crossing inside $[0, 1]$, one shot dominates the other and the mixed equilibrium disappears.

Watch players learn. Equilibrium is a static notion. Do players who don't know any game theory ever reach it? In **fictitious play** (Brown, 1951), each player best-responds to the *empirical frequency* of the opponent's past actions, as if the opponent were a fixed coin.

```
import numpy as np
import matplotlib.pyplot as plt

# Scoring probability: rows = kicker (L, R), columns = keeper (L, R).
S = np.array([[0.58, 0.95],
              [0.93, 0.70]])          # KNOB: change these four numbers
rounds = 5000                          # KNOB: how long they learn
kick_counts = np.array([1.0, 0.0])     # KNOB: initial "memories"
dive_counts = np.array([0.0, 1.0])

p_hist, q_hist = [], []
for t in range(rounds):
    q = dive_counts / dive_counts.sum()       # kicker's belief about the keeper
    p = kick_counts / kick_counts.sum()       # keeper's belief about the kicker
    kick = np.argmax(S @ q)                   # kicker maximises scoring chance
    dive = np.argmin(p @ S)                   # keeper minimises it
    kick_counts[kick] += 1
    dive_counts[dive] += 1
    p_hist.append(kick_counts[0] / kick_counts.sum())
    q_hist.append(dive_counts[0] / dive_counts.sum())

print(f"long-run: kicker shoots left {p_hist[-1]:.3f}, keeper dives left {q_hist[-1]:.3f}")
fig, ax = plt.subplots(figsize=(7, 4))
ax.semilogx(p_hist, label="kicker: fraction left")
ax.semilogx(q_hist, label="keeper: fraction left")
ax.axhline(23/60, ls=":", c="grey"); ax.axhline(5/12, ls=":", c="grey")
ax.set_xlabel("round"); ax.set_ylabel("empirical frequency"); ax.legend()
plt.show()
```

After 5,000 rounds the frequencies come out at about 0.386 and 0.425, closing in on $23/60$ and $5/12$ slowly and with a sawtooth. Each player, round by round, is completely predictable (it's an `argmax`!), yet the *frequencies* approach the equilibrium mix. Things to try:

- Increase `rounds` to 10^5. How fast does the error shrink?
- Replace `S` with a non-zero-sum game by giving the keeper its own payoff matrix (and `argmax` for both). Try the coordination core of the left figure.
- Try Shapley's 3×3 game, where the row player gets $\left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{smallmatrix}\right)$ and the column player gets $\left(\begin{smallmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{smallmatrix}\right)$. Watch whether the frequencies settle down.

39.6 The challenge

1. **Intuition.** In the penalty kick, the kicker practises and her accuracy into a covered right corner rises from 0.70 to 0.80. Without calculating, decide which player's equilibrium mix changes and in which direction. Then explain in one paragraph why a player's improvement shows up in the *opponent's* behaviour.

2. **Derivation.** For the 2/3 game with n players choosing real numbers in $[0, 100]$, show that "everyone chooses 0" is the unique Nash equilibrium. Then change the rule to "closest to $\frac{2}{3}$ of the average **plus 10**", and find the new equilibrium and the number of elimination rounds it takes to get there to within 0.1.

3. **Implementation.** Extend `support_enumeration` to detect degenerate cases: report when a square system is singular *and* the game has a pure strategy that ties with the support's payoff. Test it on the game $\mathbf{A} = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 2 \end{smallmatrix}\right)$, $\mathbf{B} = \left(\begin{smallmatrix} 1 & 1 \\ 2 & 0 \end{smallmatrix}\right)$ and describe the full set of equilibria you find.

4. **Open investigation.** Generate random $n \times n$ bimatrix games with independent uniform payoffs and count their Nash equilibria (pure and mixed) for $n = 2, \dots, 8$. How does the average count grow with n? How often is it exactly one? Then compare with how often fictitious play converges on the same games. Is there a simple property of a random game that predicts when learning succeeds?

39.7 Loose threads

This chapter gave you the core vocabulary (strategies, payoffs, dominance, best responses, Nash equilibrium, mixing) and one genuinely useful tool, the indifference principle. Along the way it stepped around several holes.

- **Existence without construction.** Nash's theorem guarantees an equilibrium, but its proof is a fixed-point argument that doesn't tell you where the fixed point is. Why that argument works at all is topology (Ch. 58). Why turning it into an algorithm seems to need exponential time is complexity theory (Ch. 30).
- **Which equilibrium?** The left game has three equilibria. The theory as built here can't say which one players will reach. History, focal points, learning dynamics and evolution all have a say. Ch. 40 and Ch. 45 start to break the tie.
- **The 2/3 game was not "solved".** Real people stop after one or two levels of reasoning. The equilibrium predicts that *everyone* plays 0, and it fails as a forecast even though every individual step of the argument is valid. Models of bounded "level-k" reasoning take that failure seriously.
- **Everyone knew the payoffs.** The kicker knew the keeper's matrix and vice versa. Take that away, so that you don't know whether the other player is strong or weak, honest or bluffing, and you need beliefs about *types*. That's Ch. 41.
- **Mixed strategies are strange.** Does a professional kicker really flip a mental coin? One modern reading says the probabilities describe the *opponent's uncertainty* about you rather than your own randomness. That reading becomes precise once beliefs are on the table in Ch. 41.

> **? Why does learning sometimes fail?**
>
> Fictitious play converges to the mixed equilibrium in every two-player zero-sum game (Robinson, 1951). In Shapley's 3×3 game from the experiment it cycles forever, with longer and longer runs, and the frequencies never settle. What separates games where simple learning finds equilibrium from games where it can't is a story about dynamics on the simplex, told in Ch. 45 and Ch. 46.

Further reading

- M. J. Osborne, *An Introduction to Game Theory*, Oxford University Press, 2004. The gentlest rigorous start; Chapters 2–4 cover this chapter.
- R. B. Myerson, *Game Theory: Analysis of Conflict*, Harvard University Press, 1991. Deeper foundations, including utility theory and refinements.
- J. Nash, "Non-cooperative games", *Annals of Mathematics* 54 (1951). The original, and short.
- N. Nisan, T. Roughgarden, É. Tardos, V. Vazirani (eds.), *Algorithmic Game Theory*, Cambridge University Press, 2007. Chapters 2–3 on computing equilibria and PPAD.

Chapter 40 — The Prisoner's Dilemma and Other Traps

> **Level 2 · Track C — Game Theory and Strategic Intelligence Builds on:** Ch. 39, Ch. 10, Ch. 19 · **Leads to:** Ch. 41, Ch. 45 · **Related:** Ch. 34, Ch. 46, Ch. 52 **Reading time:** ~25 min

40.1 The hook

Your laptop and your neighbour's phone share one Wi-Fi channel. When both want to transmit, the 802.11 protocol tells each of them to wait a random number of slots first, drawn from a *contention window*, and to double that window after every collision. It's a polite protocol: everyone backs off, collisions stay rare, and the channel stays busy with useful traffic.

Now suppose you edit your driver so that it always uses the smallest possible window. Against polite neighbours you win nearly every race for the channel, and your throughput jumps. This isn't hypothetical; papers in the early 2000s measured exactly this kind of "greedy" MAC misbehaviour. So why doesn't everyone do it? If *everyone* does it, every slot is a collision, the doubling rule never gets a chance to work, and the whole channel collapses. Each of you is better off cheating whatever the others do, and all of you together are worse off when everyone cheats.

That structure has a name, and a famous story. Two suspects are held in separate cells. Each is offered a deal: testify against the other and walk free, unless the other also testifies, in which case both get a reduced sentence. If both stay silent, the police can only convict them of something minor. Whatever your partner does, testifying shortens *your* sentence. So both testify, and both serve far longer than if they had kept quiet.

Overfishing, arms races, price wars, antibiotic overuse, doping in sport, climate negotiations, and every group project where someone coasts: the same pattern appears again and again. Individual rationality, applied carefully and correctly, leads to collective disaster.

Yet the real world is full of cooperation. Wi-Fi mostly works. Neighbouring fishing villages sometimes manage their stocks for centuries. In 1980 the political scientist Robert Axelrod invited game theorists to submit computer programs to play the dilemma against each other, two hundred rounds a match. The winner was the shortest program submitted, four lines long, written by Anatol Rapoport: *cooperate first, then do whatever the other player did last time*. Axelrod ran the tournament again with 62 entrants who knew the result. Tit-for-tat won again.

How can a strategy that never beats any opponent head-to-head (it can at best tie) win the whole tournament? And how can cooperation survive when defection is always individually better? The answer lies in one variable the one-shot story hides: **the future**.

40.2 The big idea

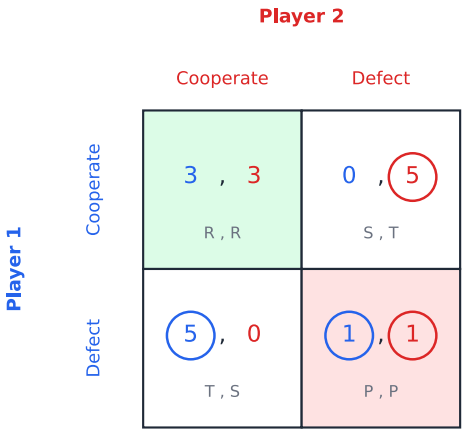
The prisoner's dilemma is a trap because the game is played **once**. Defecting has no consequences beyond the current payoff. Repeat the game, so the same two players meet again and again, and today's choice becomes a message about tomorrow. A defection can be punished, and cooperation can be rewarded. Whether that's enough depends on how much the players care about what comes next.

Think of a restaurant on a highway versus one in your own neighbourhood. The highway diner can serve bad food to travellers it will never see again. The neighbourhood café can't, because its customers come back, or don't. Same incentives, same people. The only difference is the **shadow of the future**. Game theorists measure it with a single number, the discount factor $\delta \in [0, 1)$: the weight of tomorrow's payoff relative to today's, or equivalently the probability that there will be a tomorrow at all.

The big idea of this chapter in one line:

> Cooperation is sustainable when the long-run gain from continued cooperation outweighs the one-time gain from betrayal, and that comparison has an exact threshold in δ.

The second big idea is about *populations* rather than pairs. Instead of asking what a rational player should do, ask which behaviours **survive** when strategies reproduce in proportion to their payoffs. A strategy that can't be invaded by a small group of mutants is *evolutionarily stable*. Remarkably, the answers from rationality and from evolution often coincide, and the places where they don't are where the most interesting biology and engineering happen.



Why it is a trap

T > R	temptation beats reward
P > S	punishment beats being the sucker
R > P	mutual cooperation beats mutual defection
green	what both would choose together
red	the only Nash equilibrium
circles	best responses: Defect, every time

The canonical numbers $T = 5$, $R = 3$, $P = 1$, $S = 0$. Every best-response circle falls on Defect, so mutual defection (red) is the only equilibrium, though both players prefer mutual cooperation (green).

40.3 The mathematics

The one-shot dilemma

Prisoner's dilemma

A symmetric 2×2 game with actions Cooperate (C) and Defect (D) and row-player payoffs

	C	D
C	R	S
D	T	P

is a **prisoner's dilemma** if $T > R > P > S$ (Temptation, Reward, Punishment, Sucker). For repeated play one also asks $2R > T + S$, so that taking turns exploiting each other is worse than steady cooperation.

The analysis takes one line. Against C, defecting gives $T > R$; against D, defecting gives $P > S$. So **D strictly dominates C**, and by the result of Ch. 39 the unique Nash equilibrium is (D, D) with payoffs (P, P). But (C, C) gives (R, R) with $R > P$ for both. The equilibrium is **Pareto-dominated**: some other outcome is better for everybody.

That's the precise sense of "trap". Nash equilibrium is a statement about stability, and stability has nothing to do with efficiency.

Other traps in the same family

Changing the order of T, R, P, S gives a small zoo of social dilemmas, each with its own personality.

- **Stag hunt** ($R > T \geq P > S$): hunting the stag together pays most, but hunting a hare alone is safe. Both (C, C) and (D, D) are equilibria. The trap is *trust*: cooperation is stable, but risky if you doubt your partner.
- **Chicken / Hawk–Dove** ($T > R > S > P$): two drivers head at each other. Swerving is humiliating, but crashing is worst. The pure equilibria are "one swerves, one doesn't", and there's a mixed equilibrium in which crashes happen with positive probability.
- **Public goods** (n players): each player can contribute 1 to a pot that is multiplied by r (with $1 < r < n$) and shared equally. A unit you contribute returns only $r/n < 1$ to you, so not contributing is strictly dominant, even though full contribution pays everyone $r > 1$ instead of 0. With $n = 4$ and $r = 1.6$, each contributed unit costs you 1 and returns you 0.4. This is the n-player prisoner's dilemma, and it's the mathematics behind "the tragedy of the commons".

Finitely repeated: the future doesn't help

Repeat the dilemma a known, finite number of times K, adding up the payoffs. Surely the shadow of the future helps now?

Unravelling

In the prisoner's dilemma repeated a commonly known finite number of times K, every subgame-perfect equilibrium has both players defecting in every round.

Proof. We go by backward induction on the number of rounds left. A subgame-perfect equilibrium must be an equilibrium of every subgame, starting from any history.

Last round. Whatever happened before, round K is a one-shot prisoner's dilemma, since no future follows to reward or punish anything. D strictly dominates C, so both defect, whatever the history.

Inductive step. Suppose that from round $k + 1$ onward both players defect regardless of history. Then nothing a player does in round k changes what happens later, so round k is strategically a one-shot game plus a constant. Again D strictly dominates C, so both defect in round k, whatever the history. By induction, both defect in every round. ■

The "known" in "known finite" is doing all the work. If the horizon is uncertain, so that after each round the game continues with probability δ, there is no last round to start the unravelling from.

Infinitely repeated: thresholds for cooperation

Now let the game continue forever, with discounted payoffs

$$U_i = \sum_{t=0}^{\infty} \delta^t \, u_i(a^t), \qquad 0 \le \delta < 1.$$

It helps to compare **average** payoffs $(1-\delta)U_i$, which are on the same scale as one round: steady cooperation has average R.

A strategy in the repeated game is a rule mapping every history to an action, an enormous object. Two simple ones:

- **Grim trigger**: cooperate until the opponent defects once, then defect forever.
- **Tit-for-tat (TFT)**: cooperate in round 0, then copy the opponent's previous move.

When grim trigger sustains cooperation

Both players using grim trigger is a Nash equilibrium of the infinitely repeated prisoner's dilemma if and only if

$$\delta \;\ge\; \delta^{*} = \frac{T-R}{T-P}.$$

Proof. Against grim trigger, the opponent's behaviour depends only on whether you have ever defected. Before your first defection, cooperating forever gives

$$U_{\text{coop}} = R + \delta R + \delta^2 R + \cdots = \frac{R}{1-\delta}.$$

Suppose instead you first defect in some round t_0. Discounting is time-consistent, so it is enough to compare from t_0 onward. In round t_0 you get T. From then on the opponent defects forever, and your best reply to permanent defection is to defect too, earning P per round. So the best deviation gives

$$U_{\text{dev}} = T + \delta P + \delta^2 P + \cdots = T + \frac{\delta P}{1-\delta}.$$

Cooperation is a best response if and only if $U_{\text{coop}} \ge U_{\text{dev}}$. Multiply by $1-\delta > 0$:

$$R \;\ge\; (1-\delta)T + \delta P \iff \delta(T-P) \;\ge\; T - R \iff \delta \;\ge\; \frac{T-R}{T-P}.$$

By symmetry the same holds for the other player. ∎

With the canonical numbers, $\delta^{*} = (5-3)/(5-1) = \tfrac{1}{2}$. If there's at least a 50% chance of meeting again, the threat of permanent punishment keeps both players honest.

Tit-for-tat is more forgiving, which makes it harder to check. Against TFT, any deviation that eventually returns to cooperation gets punished only once. Two deviations matter: defecting forever (average payoff $(1-\delta)T + \delta P$, same as above) and **alternating** D, C, D, C, ..., which against TFT produces the outcome sequence $(D,C),(C,D),(D,C),\ldots$ with payoffs $T, S, T, S, \ldots$. Its value is $(T+\delta S)/(1-\delta^2)$, and comparing with $R/(1-\delta)$ gives $R(1+\delta) \ge T + \delta S$. Both together give the condition

$$\delta \;\ge\; \max\left\{ \frac{T-R}{T-P},\; \frac{T-R}{R-S} \right\} = \max\left\{ \tfrac{1}{2},\; \tfrac{2}{3} \right\} = \tfrac{2}{3}$$

for the canonical payoffs. This is Axelrod's condition for TFT to be "collectively stable". Forgiveness has a price: a slightly longer shadow of the future is needed.

? Why is it enough to check one deviation at a time?

In both proofs we quietly compared cooperation against a *handful* of deviations, while a repeated-game strategy can deviate in infinitely many, history-dependent ways. The justification is the **one-shot deviation principle**: with discounting, a strategy profile is subgame perfect if and only if no player gains by deviating in a single period and then returning to the plan. It is Bellman's principle of optimality in disguise, and it is proved in Ch. 52.

Too many equilibria

Here's the uncomfortable sequel. Once δ is large, grim trigger can sustain not only (C,C) forever but almost *any* pattern: cooperate on even rounds only, or let one player exploit the other two rounds out of three. The **folk theorem** (so called because everyone knew it before anyone published it) says that, for δ close enough to 1, every feasible average payoff pair that gives each player strictly more than their punishment payoff P is a subgame-perfect equilibrium outcome.

Repetition fixes the problem of *no* good equilibria by creating *infinitely many* equilibria. The theory explains how cooperation is possible. It doesn't explain why it happens, or which form it takes.

Evolutionary stability

Change the question. Suppose a large population plays the stage game in random pairs, and strategies that earn more spread (by imitation, learning, or reproduction). Write $u(\sigma,\tau)$ for the payoff of strategy σ against τ.

Evolutionarily stable strategy (Maynard Smith and Price, 1973)

A strategy σ is an **ESS** if, for every mutant strategy $\mu \neq \sigma$, either

1. $u(\sigma, \sigma) > u(\mu, \sigma)$, or
2. $u(\sigma, \sigma) = u(\mu, \sigma)$ and $u(\sigma, \mu) > u(\mu, \mu)$.

The intuition: a small fraction ε of mutants appears. They mostly meet residents. Condition 1 says the residents do strictly better in those meetings. If there is a tie there, condition 2 says the residents do better in the rare meetings with mutants. Either way, the mutants shrink.

Every ESS is a symmetric Nash equilibrium (condition 1 or 2 implies $u(\sigma, \sigma) \geq u(\mu, \sigma)$), but not every equilibrium is an ESS. In the one-shot prisoner's dilemma, ALL-D is an ESS: $u(D, D) = P > S = u(C, D)$.

The **Hawk–Dove** game is the classic example where evolution predicts a mix. Two animals contest a resource worth V. Hawks fight; doves posture and retreat. A fight costs the loser C, and we assume $C > V$. The row player's payoffs are

	H	D
H	$\frac{V-C}{2}$	V
D	0	$\frac{V}{2}$

The Hawk–Dove ESS

If $C > V > 0$, the strategy "play Hawk with probability $p^* = V/C$" is the unique ESS of the Hawk–Dove game.

Proof. Let $u(x, y)$ be the expected payoff of a player who plays Hawk with probability x against one who plays Hawk with probability y:

$$u(x, y) = xy\,\frac{V - C}{2} + x(1 - y)V + (1 - x)(1 - y)\,\frac{V}{2}.$$

Equilibrium. Against $y = p^*$, Hawk earns $p^*\frac{V-C}{2} + (1 - p^*)V$ and Dove earns $(1 - p^*)\frac{V}{2}$. Their difference is $\frac{1}{2}\big(p^*(V - C) + (1 - p^*)V\big) = \frac{1}{2}(V - p^*C) = 0$. So every x earns the same against p^*, and condition 1 holds with equality for every mutant.

Stability. So we need condition 2 for every mutant $m \neq p^*$. Because u is linear in its first argument,

$$u(p^*, m) - u(m, m) = (p^* - m)\,\frac{\partial u}{\partial x}(\cdot, m) = (p^* - m) \cdot \frac{V - mC}{2} = \frac{C}{2}\,(p^* - m)^2 > 0,$$

using $V = p^*C$. Condition 2 holds, so p^* is an ESS. Pure Hawk isn't an ESS (Dove does better against it, $0 > \frac{V-C}{2}$), and pure Dove isn't either ($V > \frac{V}{2}$); a different mix $x \neq p^*$ isn't even a best reply to itself, so it can't be an ESS. ■

With $V = 2$ and $C = 10$, a fifth of contests are played by hawks. Notice how the indifference principle from Ch. 39 reappears, now holding a *population* in balance rather than a single player's coin.

What about the *repeated* dilemma? Is tit-for-tat an ESS? Here is where the story gets slippery.

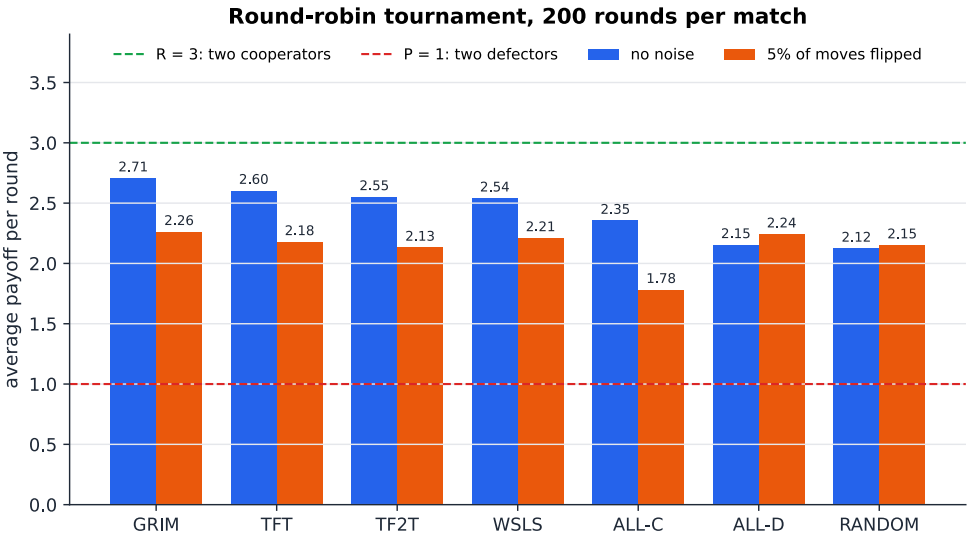
? Tit-for-tat can't hold the fort

In a population of TFT players, a mutant that always cooperates (ALL-C) earns exactly the same against TFT, and exactly the same against itself. Neither ESS condition holds, so ALL-C can drift in unopposed. Once there are enough ALL-C players, ALL-D can invade by exploiting them. Cooperation then collapses, and TFT can return. Populations may cycle between cooperation and defection forever. Whether these cycles are real, and what they look like, needs dynamics, not just stability conditions: the **replicator equation** of Ch. 45.

40.4 The algorithm

There are two computational questions, and they need different tools.

Simulation: who wins a tournament? A round-robin tournament of k strategies, each match L rounds long, costs $O(k^2 L)$ moves. Every strategy is a function from the pair of histories (mine, theirs) to an action. Deterministic strategies make it exact; noisy ones need repeated runs and averages, with the usual $1/\sqrt{N}$ Monte Carlo error.



A small Axelrod-style tournament (each strategy meets every other and its own twin, 200 rounds, averaged over 20 runs). Without noise the "nice" strategies (never defect first) take the top four places. With 5% of moves flipped by accident, every nice strategy loses ground, ALL-C is exploited, and ALL-D climbs to near the top.

Two lessons in that chart deserve to be stated carefully, because they're not what people expect.

1. **The winner depends on the field.** In this field GRIM edges out TFT, mostly because GRIM punishes RANDOM forever while TFT keeps forgiving it. In Axelrod's fields TFT won. No strategy is best against every population, since the best reply depends on who else is there. That's Ch. 39 all over again.
2. **Noise is brutal to reciprocity.** One accidental defection between two TFT players starts an echo: C/D, D/C, C/D, ... until another error breaks it. Two GRIM players never recover at all.

Exact analysis: memory-one strategies as Markov chains. If a strategy looks only at the last round, it is four numbers: the probabilities $(p_{CC}, p_{CD}, p_{DC}, p_{DD})$ of cooperating after each outcome (own move first). TFT is $(1, 0, 1, 0)$. *Win-stay, lose-shift* (WSLS, also called Pavlov) repeats its move after a good outcome (R or T) and switches after a bad one, giving $(1, 0, 0, 1)$. Two memory-one players drive a **Markov chain** on the four outcomes CC, CD, DC, DD. With execution errors the chain is irreducible, and the long-run payoff is the stationary distribution $\boldsymbol{\pi}$ (the left eigenvector of the transition matrix for eigenvalue 1, as in Ch. 10 and Ch. 34) dotted with the payoff vector.

```
import numpy as np

T, R, P, S = 5, 3, 1, 0

def long_run(p, q, eps=0.01):
    """Long-run payoffs of two memory-one strategies.
    p, q: probability of cooperating after the last outcome (CC, CD, DC, DD), seen from each player's side."""
    p = (1 - eps) * np.asarray(p) + eps * (1 - np.asarray(p))   # execution errors
    q = (1 - eps) * np.asarray(q) + eps * (1 - np.asarray(q))
    q = q[[0, 2, 1, 3]]                    # player 2 sees CD as DC and vice versa
    M = np.zeros((4, 4))
    for s in range(4):
        a, b = p[s], q[s]
        M[s] = [a * b, a * (1 - b), (1 - a) * b, (1 - a) * (1 - b)]
    w, V = np.linalg.eig(M.T)              # stationary distribution = left eigenvector for eigenvalue 1
    pi = np.real(V[:, np.argmin(abs(w - 1))]); pi /= pi.sum()
    return pi @ [R, S, T, P], pi @ [R, T, S, P]

TFT, WSLS, ALLD, GTFT = [1, 0, 1, 0], [1, 0, 0, 1], [0, 0, 0, 0], [1, 1/3, 1, 1/3]
for name, s in [("TFT", TFT), ("WSLS", WSLS), ("GTFT", GTFT), ("ALL-D", ALLD)]:
    print(f"{name:5s} vs itself: {long_run(s, s)[0]:.3f}   vs ALL-D: {long_run(s, ALLD)[0]:.3f}")
```

With a 1% error rate, two TFT players average only 2.250, exactly $(R + S + T + P)/4$, because errors leave them wandering uniformly over all four outcomes. Two WSLS players average 2.951, and two players of *generous* TFT (forgive a defection with probability $1/3$) average 2.970. The price of being nice shows up against ALL-D: TFT keeps 1.020, GTFT 0.693, WSLS only 0.535.

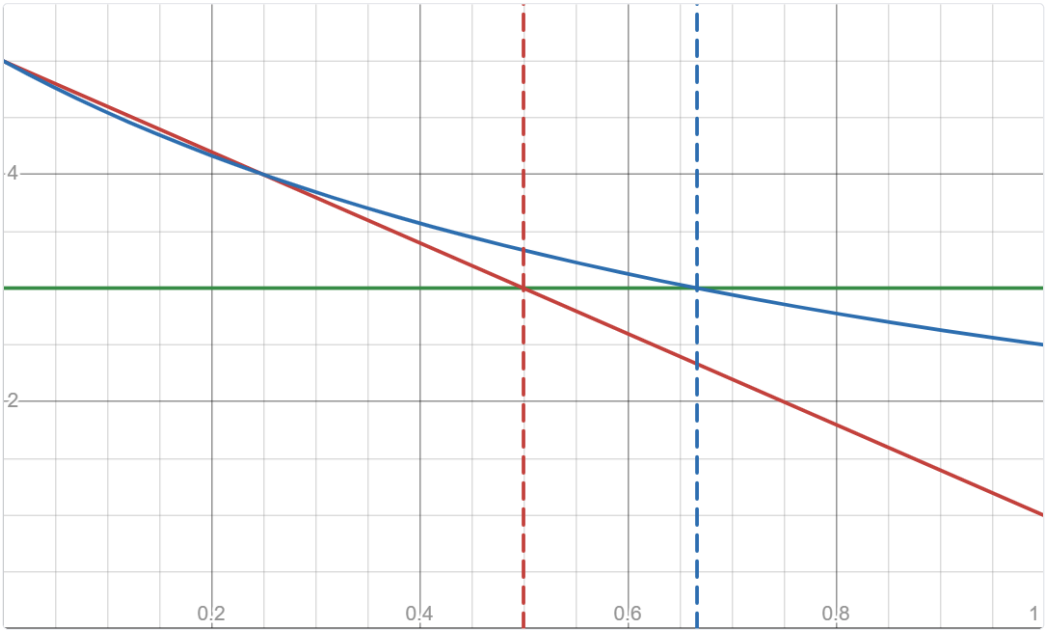
Cost and stability. The exact method costs one 4×4 eigenproblem per pair, independent of the match length, which is far cheaper than simulation. It does fail in one place. With `eps=0`, chains like TFT vs TFT are *reducible* (they get stuck in CC forever, or in an echo), the eigenvalue 1 is repeated, and "the" stationary distribution depends on the starting state. `np.linalg.eig` then returns an arbitrary vector in that eigenspace. Keep a small positive error rate, or handle the starting state explicitly.

? A strategy that dictates your score

In 2012, the physicist Freeman Dyson and computer scientist William Press found memory-one strategies that let one player **unilaterally** force a linear relation between the two long-run scores, for example "my surplus over P is always three times yours", whatever the opponent does. These **zero-determinant** strategies came out of a determinant identity hiding in exactly the Markov chain above. Are extortioners a real threat to cooperation, or do they eat themselves in a population? The adversarial side is in Ch. 46, and the evolutionary verdict in Ch. 45.

40.5 The experiment

Find the threshold. The graph shows average payoffs (per round) as functions of the discount factor $x = \delta$. The green line is steady cooperation, $R = 3$. The red curve is the best deviation against grim trigger, $(1-\delta)T + \delta P$. The blue curve is the alternating deviation against TFT, $T/(1+\delta)$ (with $S = 0$). Cooperation is safe wherever green lies above the others.



Grim trigger holds cooperation for $\delta \geq 1/2$ (red crossing); tit-for-tat needs $\delta \geq 2/3$ (blue crossing).

Change **5** (the temptation T) to 8 in both curves: how much longer must the shadow of the future be? Change the **1** in $1x$ (the punishment P) to 2: does harsher or milder punishment help grim trigger? Finally change the **0** in $0x$ (the sucker's payoff S) to 1 and watch the blue crossing move while the red one stays put.

Run your own tournament. This is the engine behind the bar chart, trimmed to fit.

```
import numpy as np

T, R, P, S = 5, 3, 1, 0                      # KNOB: payoffs (need T > R > P > S)
PAY = {('C', 'C'): (R, R), ('C', 'D'): (S, T), ('D', 'C'): (T, S), ('D', 'D'): (P, P)}
ROUNDS, NOISE = 200, 0.0                     # KNOB: game length, chance a move is flipped
rng = np.random.default_rng(1)

STRATS = {                                   # each sees (my history, their history)
    'ALL-C':  lambda me, op: 'C',
    'ALL-D':  lambda me, op: 'D',
    'TFT':    lambda me, op: op[-1] if op else 'C',
    'GRIM':   lambda me, op: 'D' if 'D' in op else 'C',
    'TF2T':   lambda me, op: 'D' if op[-2:] == ['D', 'D'] else 'C',
    'WSLS':   lambda me, op: 'C' if not me else (me[-1] if op[-1] == 'C' else ('D' if me[-1] == 'C' else 'C')),
    'RANDOM': lambda me, op: 'C' if rng.random() < 0.5 else 'D',
}

def play(f, g):
    a, b, sa, sb = [], [], 0, 0
    for _ in range(ROUNDS):
        x, y = f(a, b), g(b, a)
        if rng.random() < NOISE: x = 'D' if x == 'C' else 'C'
        if rng.random() < NOISE: y = 'D' if y == 'C' else 'C'
        pa, pb = PAY[(x, y)]
        a.append(x); b.append(y); sa += pa; sb += pb
    return sa / ROUNDS, sb / ROUNDS

names = list(STRATS)
score = {n: [] for n in names}
for i, n in enumerate(names):                # round robin, including a match against a twin
    for m in names[i:]:
        s1, s2 = play(STRATS[n], STRATS[m])
        score[n].append(s1)
        if m != n: score[m].append(s2)
for n in sorted(names, key=lambda n: -np.mean(score[n])):
    print(f"{n:7s} {np.mean(score[n]):.3f}")
```

Things to try:

- Set `NOISE = 0.05` and rerun with several seeds. How stable is the ranking?
- Remove `'RANDOM'`, or add three copies of `'ALL-D'`. Does TFT overtake GRIM?

- Write *generous TFT* (after the opponent defects, still cooperate with probability $1/3$) and *suspicious TFT* (defect first, then copy). Which of the two does noise hurt more?
- Make it **ecological**: after each tournament, give each strategy a population share proportional to its score and rerun with those shares. Which strategies go extinct first?

40.6 The challenge

1. **Intuition.** Two Wi-Fi stations repeatedly share a channel. One argues "I will always back off politely, whatever you do"; the other says "I will back off as long as you do, and jam you for ten seconds if you cheat". Which promise does more to keep the channel cooperative, and why does the *credibility* of the second promise matter?

2. **Derivation.** For the "k-period punishment" strategy (cooperate; after a defection, defect for exactly k rounds, then return to cooperation), find the smallest δ, as a function of k, T, R, P, at which mutual use of it is an equilibrium. Check that it tends to the grim-trigger threshold $\frac{T-R}{T-P}$ as $k \to \infty$.

3. **Implementation.** Using `long_run`, scan all memory-one strategies on a grid $p_s \in \{0, 0.25, 0.5, 0.75, 1\}$ (625 strategies). For each, find its payoff against itself and the best payoff any grid strategy can earn against it. Plot the strategies that are "nearly stable" (no invader gains more than 0.01) and describe what they have in common.

4. **Open investigation.** Run the ecological version of the tournament (strategies reproduce in proportion to score) with a mutation process that occasionally introduces random memory-one strategies. Over long runs, does the population settle, cycle between cooperation and defection, or wander? How does the answer change with the error rate? There is no known closed-form description of the long-run behaviour for general mutation processes.

40.7 Loose threads

This chapter turned the prisoner's dilemma from a paradox into a calculation: cooperation is a self-enforcing equilibrium once δ passes an exact threshold. It also left several doors ajar.

- **Infinitely many equilibria.** The folk theorem says repetition can sustain almost anything, including exploitation. Which equilibrium a society lands in is a question for history, learning and evolution, not for the definition of equilibrium (Ch. 45).
- **The proofs checked only a few deviations.** That was legitimate, but only because of the one-shot deviation principle, which is dynamic programming in disguise (Ch. 52).
- **We assumed everyone sees everything.** Every strategy here observed the opponent's actual move. Real Wi-Fi stations, firms and nations see only noisy signals: a collision might be bad luck or sabotage. With imperfect monitoring, punishment has to be triggered by *evidence*, and the analysis needs beliefs about what the other player is and did. That's the doorway into Ch. 41.
- **ESS is a static test.** It says which states resist invasion, not how a population moves between them, or whether it moves in circles. That's the replicator equation of Ch. 45.
- **Designing the game itself.** If the dilemma is the problem, why not change the payoffs? Protocol designers do exactly that: they add fines, reputations and verifiable commitments so that cooperation becomes the dominant strategy. Engineering the incentives instead of the players is Ch. 42.

Further reading

- R. Axelrod, *The Evolution of Cooperation*, Basic Books, 1984. The tournaments, told by the person who ran them.
- J. Maynard Smith, *Evolution and the Theory of Games*, Cambridge University Press, 1982. ESS from the source.
- G. J. Mailath and L. Samuelson, *Repeated Games and Reputations*, Oxford University Press, 2006. The folk theorems in full rigour.
- W. H. Press and F. J. Dyson, "Iterated Prisoner's Dilemma contains strategies that dominate any evolutionary opponent", *PNAS* 109 (2012).

Chapter 41 — Games with Incomplete Information

> **Level 2 · Track C — Game Theory and Strategic Intelligence Builds on:** Ch. 39, Ch. 19, Ch. 40 · **Leads to:** Ch. 42, Ch. 43 · **Related:** Ch. 35, Ch. 33, Ch. 96 **Reading time:** ~25 min

41.1 The hook

You want to buy a used car. There are good ones ("peaches") and bad ones ("lemons"), and on a test drive they look identical. The seller knows which one she has, because she has driven it for five years. You don't.

So you do the sensible thing and offer a price that reflects an *average* car. Now look at that offer from the sellers' side. The owner of a peach finds your average price insulting, because her car is worth more, and keeps it. The owner of a lemon is delighted to sell. The cars actually on offer at your price are therefore *worse than average*. Once you realize this, you lower your offer. More peach owners withdraw. The average falls again. In 1970, George Akerlof showed that this spiral can run all the way down: a market in which every buyer values every car more than its owner does, so that every trade would make both sides happier, can collapse to *no trade at all*. The paper was famously rejected by three journals as either trivial or wrong. It later earned a Nobel Prize.

The same spiral threatens health insurance (the sickest are keenest to buy), loans (the riskiest borrowers accept any rate), freelance marketplaces, and spectrum auctions. The problem isn't greed. It is **asymmetric information**: one side knows something the other doesn't, and everyone knows that.

Two escape routes suggest themselves, and they're opposites.

- The informed side can try to **signal**. A peach owner might offer a two-year warranty, something a lemon owner can't afford to promise. A job applicant may spend four years earning a degree that a weaker applicant would find too costly to get.
- The uninformed side can try to **screen**. The insurer offers a menu, a cheap policy with a big deductible and an expensive one with full cover, and lets customers sort *themselves* by what they choose.

Here's a puzzle to carry through the chapter. Why do so many scam emails announce, in the first line, that they come from a foreign prince with a fortune to share? Wouldn't a more plausible story catch more victims? Hold that thought. By the end you'll see the absurd opening line as a precisely tuned piece of strategy.

41.2 The big idea

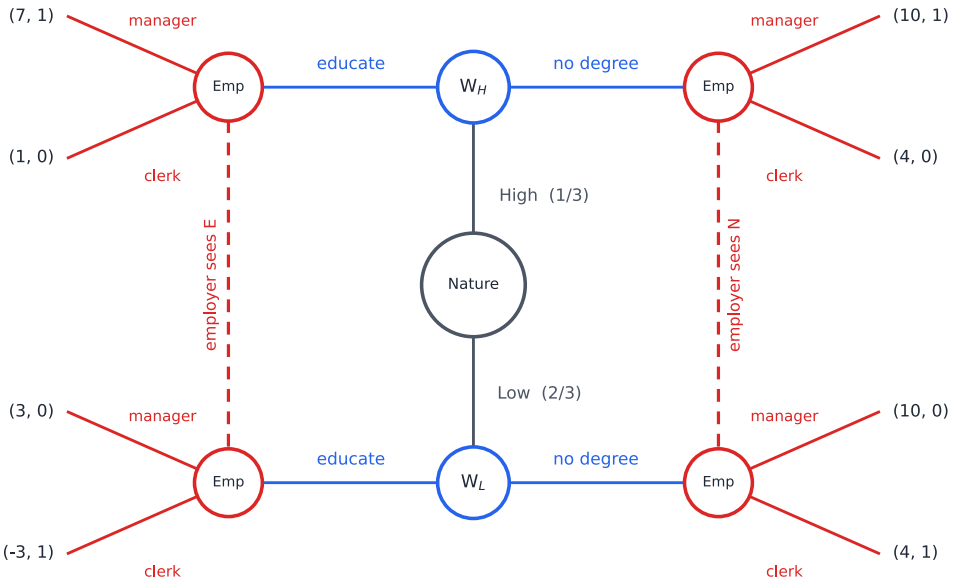
In Ch. 39 every player knew every payoff. Now some players have **private information**: their own cost, value, quality, or intention. The others can't see it, so they must form **beliefs** about it. And because beliefs drive choices, actions themselves become evidence. Whatever you do, others will read it.

The key move, due to John Harsanyi (1967), turns this mess into something we already know how to handle. Pretend that before the game starts, a neutral player called **Nature** rolls dice and deals each player a secret **type**: "peach" or "lemon", "able" or "weak", "high value" or "low value". The odds on the dice are common knowledge, even though each player sees only their own roll. Uncertainty about *who you're playing* becomes uncertainty about *what Nature did*, and probability theory (Ch. 19) knows how to deal with that.

From there, two ideas carry everything.

1. **A strategy is a plan for every type.** A seller's strategy isn't "sell" or "keep" but "if peach, keep; if lemon, sell". You must plan for all the types you could have been, because the *other* players' best responses depend on what you would have done in each case.
2. **Beliefs must be Bayesian where they can be.** When you see an action, update your beliefs about the actor's type by Bayes' rule. An analogy: a poker player who bets big is either strong or bluffing. A good opponent doesn't take the bet at face value; she asks how often a player of each type would make this bet, and updates.

Signaling is the art of choosing actions that are *too expensive to fake*. A signal works not because it is informative in itself (a degree in classics may teach nothing about software), but because it costs different types different amounts.



payoffs: (worker, employer). Manager job worth 10, clerk 4; degree costs 3 (High) or 7 (Low)

A job-market signaling game. Nature deals the worker a type; the worker chooses whether to get a degree; the employer, seeing only the degree (dashed information sets), assigns a job. Payoffs are (worker, employer).

41.3 The mathematics

The lemons spiral, exactly

Let car quality q be uniform on $[0, 1]$. An owner values her car at q; a buyer values it at kq with $k > 1$, so every trade would create value. Quality is private to the owner. Buyers are competitive, so the market price equals buyers' expected value of the cars actually on offer.

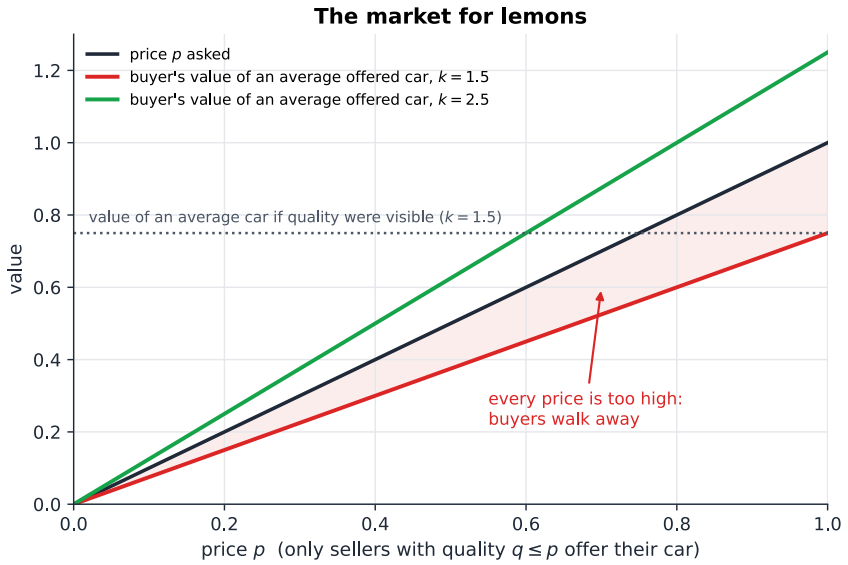
> **Market collapse (Akerlof)**
>
> If $1 < k < 2$, the only equilibrium price is $p = 0$: essentially no car is traded, even though every trade would benefit both sides.

Proof. At price $p \in (0, 1]$, exactly the owners with $q \le p$ sell. Conditional on $q \le p$, quality is uniform on $[0, p]$, so the expected quality on offer is $p/2$ and buyers value a random offered car at $kp/2$. An equilibrium needs buyers' value to equal the price:

$$\frac{kp}{2} = p \iff p\left(\frac{k}{2} - 1\right) = 0.$$

For $k < 2$ the bracket is negative, so the only solution is $p = 0$. At any positive price, $kp/2 < p$ and buyers refuse to pay it. (At $p > 1$ every car is offered, average quality is $1/2$, and $k/2 < 1 < p$, so no price above 1 works either.) ∎

Watch what makes the argument work: the price determines *who participates*, and who participates determines what the price should be. That feedback loop is the signature of **adverse selection**. For $k = 1.5$, a buyer who could see quality would pay 0.75 on average; without that information, the market pays nothing. Information is worth something to the market as a whole, a fact that Ch. 33 teaches you to measure in bits.



At price p only cars of quality at most p are offered, so buyers value the offered pool at $kp/2$. For $k = 1.5$ (red) that's always below the price and the market unravels; only $k \geq 2$ (green) keeps it alive.

Bayesian games

Bayesian game

A **Bayesian game** has

- players $i = 1, \dots, n$, each with an action set A_i;
- for each player a set of **types** Θ_i;
- a **common prior** π on $\Theta = \Theta_1 \times \cdots \times \Theta_n$;
- payoffs $u_i(a, \theta)$ that may depend on everyone's actions *and* types.

Player i learns only θ_i, and forms the posterior belief $\pi(\theta_{-i} \mid \theta_i)$ by Bayes' rule. A **strategy** is a function $s_i : \Theta_i \to A_i$, one action for every possible type.

Bayesian Nash equilibrium (BNE)

A strategy profile s^* is a **Bayesian Nash equilibrium** if every type of every player maximizes expected payoff given its beliefs:

$$s_i^*(\theta_i) \in \arg\max_{a_i \in A_i} \; \mathbb{E}_{\theta_{-i} \sim \pi(\cdot \mid \theta_i)} \left[u_i\big(a_i,\; s_{-i}^*(\theta_{-i}),\; \theta\big) \right] \quad \text{for all } i,\ \theta_i.$$

Harsanyi's insight is that a BNE is simply a Nash equilibrium of a bigger ordinary game in which the "players" are the type-contingent strategies, and Nature is just a chance move. Nash's existence theorem from Ch. 39 therefore applies, for finite type and action sets.

The cleanest worked example has a continuum of types.

A first-price sealed-bid auction

Two bidders each have a private value v_i, independent and uniform on $[0, 1]$. Each submits a sealed bid; the higher bid wins and **pays its own bid**. A winner with value v who bids b gets $v - b$; a loser gets 0.

Guess a symmetric equilibrium in which each bids a fixed fraction of value, $b(v) = \beta v$. **Check** that it is a best response. Bidder 1 with value v bidding b wins when $\beta v_2 < b$, which has probability $\min(b/\beta,\ 1)$. For $b \leq \beta$ the expected payoff is

$$\Pi(b) = (v - b)\,\frac{b}{\beta}.$$

Setting $\Pi'(b) = (v - 2b)/\beta = 0$ gives $b = v/2$, a maximum since $\Pi'' < 0$. (Bids above β win for sure but pay more than necessary.) The best response to "the other bids βv" is to bid $v/2$, *whatever* β is, as long as $\beta \geq 1/2$. Consistency requires $\beta = 1/2$.

So $b(v) = v/2$ is a BNE: **shade your bid to half your value.** The seller's expected revenue is the expected winning bid, $\mathbb{E}[\max(v_1, v_2)]/2 = \frac{2}{3} \cdot \frac{1}{2} = \frac{1}{3}$.

Bid shading is the defining feature of incomplete information. You'd never bid half your value if you knew the other bid; you'd bid a hair above it. Not knowing, you trade off the probability of winning against the profit when you win.

? The coincidence that isn't one

Change the rule: highest bid wins but pays the **second**-highest bid. Now bidding your true value is a dominant strategy (prove it in one line!), and the seller's expected revenue is $\mathbb{E}[\min(v_1, v_2)] = \frac{1}{3}$, *exactly* the same as in the first-price auction. Very different rules, very different bidding, identical revenue. That isn't luck. The **revenue equivalence theorem** in Ch. 43 says when and why it must happen.

Signaling games and perfect Bayesian equilibrium

In the tree above, the worker moves first and the employer second, after watching. Plain BNE is too weak a concept here: it allows the employer to make threats at information sets that are never reached, like "if you educate, I'll make you a clerk anyway". We need beliefs at *every* information set, and actions that are optimal given those beliefs.

Perfect Bayesian equilibrium (PBE)

In a signaling game, a **PBE** is a sender strategy $\sigma(\theta)$, a receiver strategy $\rho(m)$ for each message m, and beliefs $\mu(\theta \mid m)$, such that

1. each sender type's message maximizes its payoff given ρ;
2. after each message, $\rho(m)$ maximizes the receiver's expected payoff given $\mu(\cdot \mid m)$;
3. for messages sent with positive probability, μ is obtained by Bayes' rule (Ch. 19.3) from the prior and σ. For messages never sent, μ may be anything.

PBEs come in two broad kinds. In a **separating** equilibrium, different types send different messages, so the message reveals the type. In a **pooling** equilibrium, all types send the same message, and the receiver learns nothing.

Take the game in the figure: prior $P(H) = \frac{1}{3}$, a manager job worth 10 to the worker and a clerk job worth 4, a degree costing 3 for the high type and 7 for the low type, and an employer who scores 1 for putting each type in the right job.

Two very different equilibria

In the job-market game of the figure:

1. **Separating:** "High educates, Low doesn't; the employer makes graduates managers and the rest clerks" is a PBE.
2. **Pooling:** "Nobody educates; the employer makes everybody a clerk, and would also make a graduate a clerk" is a PBE, supported by the off-path belief $\mu(H \mid E) \le \frac{1}{2}$.

Proof. **Separating.** Bayes' rule gives $\mu(H \mid E) = 1$ and $\mu(H \mid N) = 0$, so the employer's replies (manager after E, clerk after N) are optimal. High's payoffs: educate $10 - 3 = 7$, don't 4, so High educates. Low's payoffs: educate $10 - 7 = 3$, don't 4, so Low doesn't. No one wants to deviate.

Pooling. On path, everyone sends N, so Bayes gives $\mu(H \mid N) = \frac{1}{3}$. The employer's expected score is $\frac{1}{3}$ for "manager" and $\frac{2}{3}$ for "clerk", so clerk is optimal. Message E is never sent, so condition 3 allows any belief there. With $\mu(H \mid E) \le \frac{1}{2}$, clerk is optimal after E as well. Then educating gives High $4 - 3 = 1 < 4$ and Low $4 - 7 < 4$, so neither deviates. ■

The two equilibria have exactly the same game, the same payoffs and the same rationality, and in one of them the able worker is recognized while in the other she is stuck as a clerk forever. The pooling equilibrium survives only because of an off-path belief: the employer "thinks" that a graduate is probably the low type, even though the low type would never want a degree. That belief is allowed but feels absurd.

The continuous version is Michael Spence's job-market model (1973). Let education level $e \ge 0$ cost e/θ for a worker of productivity $\theta \in \{1, 2\}$, with wage equal to expected productivity. Separation at a degree level e^* requires that Low doesn't mimic, $2 - e^* \le 1$, and that High is willing, $2 - e^*/2 \ge 1$. Together these give $1 \le e^* \le 2$: **a whole interval of separating equilibria**, each needing a different amount of costly, socially useless schooling.

? Which beliefs are reasonable?

PBE lets beliefs off the equilibrium path be anything, and so it supports pooling equilibria that rest on absurd suspicions, plus a continuum of separating ones. Game theorists responded with **refinements**, the most famous being the *Intuitive Criterion* of Cho and Kreps (1987), which says to ignore types that couldn't possibly gain from a deviation. Applied here, it kills the pooling equilibrium and selects the cheapest separating one, $e^* = 1$. Why that criterion, and whether any refinement is the "right" one, is a live debate you'll meet in Ch. 96.

Screening: let them sort themselves

In signaling the informed party moves first. In **screening**, the uninformed party moves first by designing a menu, and the informed party responds by choosing from it. An insurer can't see who is risky, so it offers two contracts: full cover at a high premium, and partial cover (a large deductible) at a low premium. Safe drivers, who expect few claims, find the deductible cheap and take the low premium. Risky drivers pay for full cover. The menu is built so that each type *prefers* the contract meant for it. That self-selection requirement is called **incentive compatibility**.

Two features of good screening menus are universal, and we'll only state them here.

- The "best" type (the one the designer most wants to identify) gets an efficient contract, but has to be left some surplus, an **information rent**, or it would pretend to be another type.
- The other types get *distorted* contracts (a deductible larger than efficient), deliberately made worse so that they're unattractive to mimic.

Now the scam email. The scammer's real cost isn't sending emails, which is nearly free. It's the time spent on each reply, the back-and-forth, before it turns out the target isn't going to pay. The absurd story is a **screen**. Anyone sceptical enough to eventually back out is filtered out by the first line, and the replies come mostly from people who will go all the way. As computer scientist Cormac Herley observed in 2012, a more believable email would generate more replies and *less* profit. The implausibility is the design.

? Is a menu always enough?

Screening, auctions, voting rules, and tax schedules all ask the same question: what outcomes can a designer achieve when everyone else holds private information? It looks hopeless, since the set of possible mechanisms is enormous. Yet one theorem lets you restrict attention to mechanisms that simply *ask* each player for their type and make honesty optimal. That theorem, the **revelation principle**, opens Ch. 42.

41.4 The algorithm

Finite Bayesian games → finite normal-form games. If player i has $|\Theta_i|$ types and $|A_i|$ actions, it has $|A_i|^{|\Theta_i|}$ pure strategies. List them, compute each profile's *ex ante* expected payoff by summing over the prior, and every tool from Ch. 39 applies. This is correct but expensive, since the strategy count is exponential in the number of types. In practice one checks best responses type by type instead, which costs only $|\Theta_i| \cdot |A_i|$ evaluations per player.

Pure PBE of a signaling game. The search has three nested loops:

1. Enumerate the sender's pure strategies $\sigma : \Theta \to M$ (here $2^2 = 4$ of them).
2. For each message, compute the receiver's belief: Bayes' rule if the message is used, otherwise scan a grid of off-path beliefs.
3. Compute the receiver's best reply to each belief, then check that no sender type wants to switch message.

```
import itertools
import numpy as np

prior = {"H": 1/3, "L": 2/3}                # Nature's odds of each worker type
cost = {"H": 3, "L": 7}                     # cost of education by type
wage = {"M": 10, "C": 4}                    # worker's value of each job
fit = {("H", "M"): 1, ("L", "C"): 1}        # employer gets 1 for a correct match, else 0

def worker(t, e, job): return wage[job] - (cost[t] if e == "E" else 0)
def best_job(mu_H):                         # employer's reply given belief P(H | signal)
    vM, vC = mu_H * fit.get(("H", "M"), 0), (1 - mu_H) * fit.get(("L", "C"), 0)
    return "M" if vM > vC else "C"

for sH, sL in itertools.product("EN", repeat=2):     # the sender's pure strategies
    sigma = {"H": sH, "L": sL}
    for off in np.linspace(0, 1, 11):                # belief at an unused signal
        beliefs, reply = {}, {}
        for e in "EN":
            senders = [t for t in "HL" if sigma[t] == e]
            if senders:                              # on path: Bayes' rule
                mass = sum(prior[t] for t in senders)
                beliefs[e] = prior["H"] / mass if "H" in senders else 0.0
            else:                                    # off path: any belief allowed
                beliefs[e] = off
            reply[e] = best_job(beliefs[e])
        ok = all(worker(t, sigma[t], reply[sigma[t]]) >= worker(t, d, reply[d])
                 for t in "HL" for d in "EN")
        if ok:
            print(f"H->{sH}, L->{sL}  beliefs P(H|E)={beliefs['E']:.1f}, P(H|N)={beliefs['N']:.1f}"
                  f"  employer: E->{reply['E']}, N->{reply['N']}")
        if sH != sL: break                           # separating: no off-path signal to scan
```

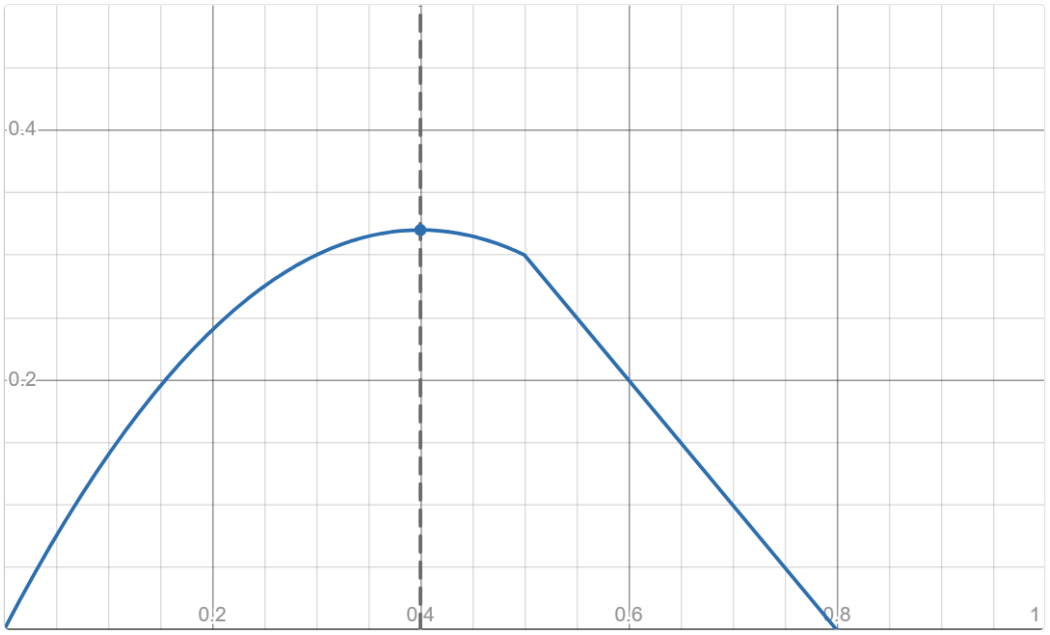
It prints the separating equilibrium once, and the pooling-on-N equilibrium for every grid belief $\mu(H \mid E) \in \{0, 0.1, \ldots, 0.5\}$ (at exactly 0.5 the employer is indifferent, and the code's tie-break picks clerk). Pooling on E and the "reverse" separation (Low educates, High doesn't) never appear.

Cost and failure cases.

- With $|\Theta|$ types and $|M|$ messages there are $|M|^{|\Theta|}$ sender strategies, each with a grid of off-path beliefs. That's fine for toy games and hopeless for realistic ones.
- *Mixed* (semi-separating) equilibria, where a type randomizes so that the receiver's posterior lands exactly on an indifference point, are invisible to a pure search. They need the indifference principle of Ch. 39 applied to posteriors.
- *Grids can lie.* The pooling equilibrium holds for $\mu(H \mid E) \le \frac{1}{2}$, a closed interval. A grid that skipped 0.5 would still find it, but a game whose equilibrium needs a belief in a tiny interval could slip between the grid points. Solve the receiver's indifference condition exactly when you can.
- *Ties.* Indifference is common at equilibrium, and the tie-breaking rule inside `best_job` silently selects among equilibria. Change `>` to `>=` and watch the output change.

41.5 The experiment

Shade your bid. In the first-price auction, a bidder with value $v = 0.8$ who bids x against an opponent bidding half her value wins with probability $\min(2x, 1)$, so her expected profit is $(0.8 - x)\min(2x, 1)$.



Expected profit of a bidder with value 0.8; the peak sits at exactly half her value.

Change **0.8** to other values in both places and confirm the peak always sits at $v/2$. Then change the **2** in $\min(2x, 1)$ to **3**: the opponent now bids a third of her value, so you win more easily. Does your best bid change? Try it with value 0.5 and with value 0.8, and explain why the two answers differ.

Watch a market die. This script runs the lemons spiral with real cars: each week, only owners whose car is worth less than the price sell, and buyers reset the price to what the offered cars are worth.

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(0)
cars = 10_000
quality = rng.uniform(0, 1, cars)     # KNOB: try rng.beta(2, 5, cars) or rng.uniform(0.5, 1, cars)
k = 1.5                               # KNOB: buyers value a car at k * quality (k > 1: trade is efficient)
price = k * quality.mean()            # buyers start by paying for the average car
history = [price]

for week in range(25):
    offered = quality[quality <= price]          # only owners of worse-than-price cars sell
    if offered.size == 0:
        break
    price = k * offered.mean()                   # buyers pay what the offered pool is worth
    history.append(price)
    share = offered.size / cars
    print(f"week {week:2d}: price {price:.3f}, cars on sale {share:6.1%}")

plt.plot(history, "o-")
plt.xlabel("week"); plt.ylabel("market price")
plt.title(f"Lemons market, k = {k}")
plt.show()
```

With $k = 1.5$ the price starts at 0.75 and loses a quarter of its value every week: 0.56, 0.42, 0.31, ..., with only about 6% of cars still on sale by week 9. Things to try:

- Set `k = 2.5`. Does the market survive, and at what price?
- Keep `k = 1.5` but use `rng.uniform(0.5, 1, cars)`, so no car is truly terrible. Does a *floor* on quality rescue the market?
- Add a **warranty**: suppose owners of cars with `quality > 0.7` can certify them at a cost of `0.05`, and certified cars sell in a separate market. Which owners certify, and what happens to the uncertified market?

41.6 The challenge

1. **Intuition.** A hospital adopts a policy that every applicant for a nursing position must complete an unpaid two-week trial shift. Explain how this could function as a signal, a screen, or neither, depending on who bears the cost. What would have to be true about applicants' outside options for it to separate types?

2. **Derivation.** Generalize the first-price auction to n bidders with independent values uniform on $[0, 1]$. Guess $b(v) = \beta v$, derive $\beta = \frac{n-1}{n}$, and compute the seller's expected revenue. Then show that a second-price auction with the same bidders gives the same revenue.

3. **Implementation.** Extend the PBE search to **three** worker types (low, medium, high, with education costs 7, 5, 3), three messages (no degree, bachelor's, master's), and three jobs. Report every pure PBE, classify each as separating, pooling, or partially pooling, and count how many depend on off-path beliefs.

4. **Open investigation.** In the lemons simulation, let a fraction ϕ of buyers be "experts" who can see quality and pay kq for exactly the cars they want, while the rest trade blind. How large must ϕ be to stop the unravelling at $k = 1.5$? Explore how the answer depends on the shape of the quality distribution. Is there a distribution for which no amount of expertise short of $\phi = 1$ saves the market?

41.7 Loose threads

This chapter added one ingredient to game theory, private information, and the world got stranger: markets that should exist can vanish, useless degrees can be rational, and absurd emails can be optimal. Several threads were left hanging on purpose.

- **Revenue equivalence.** Two auctions with completely different bidding behaviour raised exactly $\frac{1}{3}$. Whether that's a theorem or a coincidence, and what breaks it (risk aversion, correlated values, asymmetric bidders), is Ch. 43.
- **Too many equilibria, again.** Signaling games produce pooling and a continuum of separating equilibria, all of them PBE. The refinements that try to pick out the reasonable ones, and the arguments about whether they should, lead into Ch. 96.
- **Designing the menu.** We stated the shape of an optimal screening menu (rent at the top, distortion below) without deriving it. The derivation, and the revelation principle that makes it possible, is Ch. 42.
- **Where do common priors come from?** Harsanyi's trick assumes all players agree on Nature's dice. Drop that assumption and you face beliefs about beliefs about beliefs, an infinite hierarchy that ties into the Bayesian updating of Ch. 35.
- **Learning the type over time.** In Ch. 40 cooperation rested on repeated interaction. Combine repetition with private types and you get **reputation**: a player who might be a committed cooperator can sustain trust even in a finitely repeated dilemma, breaking the unravelling argument. How much a small doubt can achieve is one of the most surprising results in the theory of repeated games.

Further reading

- G. A. Akerlof, "The Market for 'Lemons': Quality Uncertainty and the Market Mechanism", *Quarterly Journal of Economics* 84 (1970).
- M. Spence, "Job Market Signaling", *Quarterly Journal of Economics* 87 (1973).
- R. Gibbons, *Game Theory for Applied Economists*, Princeton University Press, 1992. Chapters 3–4 are the clearest short treatment of Bayesian and perfect Bayesian equilibrium.
- V. Krishna, *Auction Theory*, 2nd ed., Academic Press, 2010. First-price auctions and bid shading in depth.

Chapter 42 — Mechanism Design: Engineering Incentives

> **Level 2 · Track C — Game Theory and Strategic Intelligence Builds on:** Ch. 23, Ch. 19, Ch. 39, Ch. 41 · **Leads to:** Ch. 43, Ch. 44, Ch. 96 · **Related:** Ch. 49, Ch. 30 **Reading time:** ~25 min

42.1 The hook

Two women stand before King Solomon. Each claims the same baby. Solomon cannot see into their hearts; he can only listen to what they say, and both say the same thing. So he announces a rule: the child will be cut in two and each woman given half. One woman says "do it". The other says "give her the child, only let it live". Solomon hands the baby to the second woman.

Strip away the drama and you have an engineering problem. The king holds **private information he does not have**: who the real mother is. The people who do have it will say whatever serves them. Solomon's trick was not to ask a better question but to design a **rule** under which the two women's *answers* would differ, because their *preferences* differ. That is the whole subject of this chapter in one story, and it is also a warning: Solomon's rule only works once. A second false mother who has heard the story simply says "let it live" too.

Here is a modern, less bloody version. Three households on a river want a footbridge. It costs 300. The town clerk proposes the obvious rule: "Tell me what the bridge is worth to you. If the total is at least 300, we build it and split the cost equally, 100 each." The true values are 250, 50 and 40. Honestly reported, the total is 340, the bridge goes up, and household 3, who values it at 40, pays 100 for the privilege. Household 3 is no fool. If negative reports are allowed, it reports -1000 and the bridge dies. If they are not, household 1 has the opposite temptation: in a slightly different world where the others valued it less, writing "1,000,000" costs nothing and guarantees the bridge. The clerk's rule asks for the truth and rewards lying.

And it gets worse before it gets better. Suppose the town abandons money and simply votes. Three council members rank three bridge designs X, Y, Z:

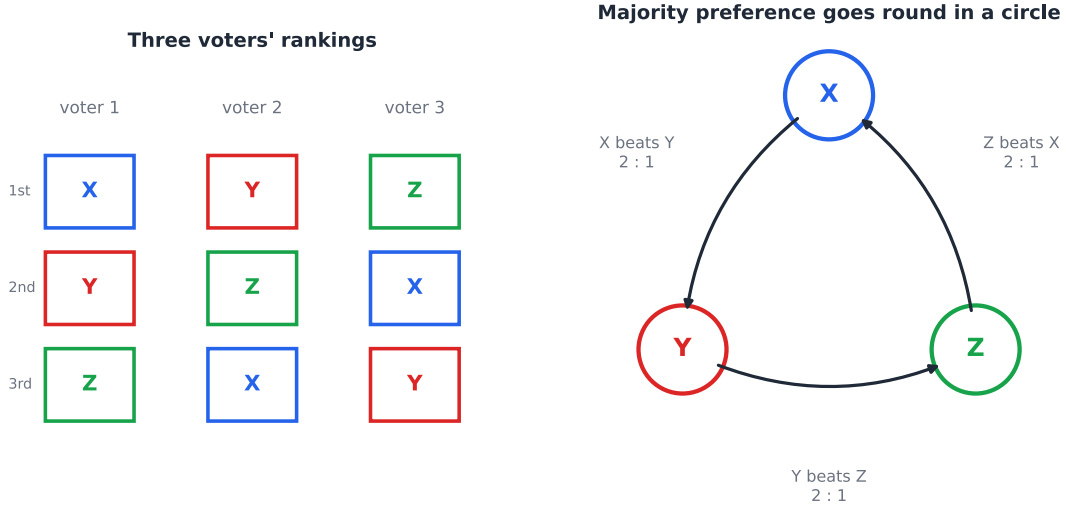


Figure 42.1 — The Condorcet cycle. Every voter is perfectly rational, yet "the majority prefers" is not a ranking at all.

Each member's ranking is perfectly sensible. Yet a majority prefers X to Y, a majority prefers Y to Z, and a majority prefers Z to X. Whoever controls the *order* of pairwise votes controls the winner: put Y against Z first, then the survivor against X, and X wins; reorder the agenda and Y or Z wins instead. The group has no "will" to discover.

So the question this chapter chases is simple to ask and surprisingly deep to answer:

> **Can we write down rules, for votes, for auctions, for bridges, under which telling the truth is every participant's selfish best move?**

We will find a remarkable "yes" when money can change hands, a precise reason why, and the first glimpse of a theorem that says "no" when it cannot.

42.2 The big idea

Ordinary game theory (Ch. 39) takes the game as given and asks what rational players will do. **Mechanism design runs game theory backwards.** You start from the outcome you *want*, say "build the bridge exactly when it is worth more than it costs", and you design the game whose equilibrium produces that outcome, even though every player is selfish and holds private information.

That is why it is sometimes called *reverse game theory*, and why engineers should feel at home in it. A control engineer designs a feedback law so that the closed loop does what she wants whatever the plant's internal state. A mechanism designer designs rules so that the "closed loop" of selfish agents does what she wants whatever their private preferences.

Picture a mechanism as a box:

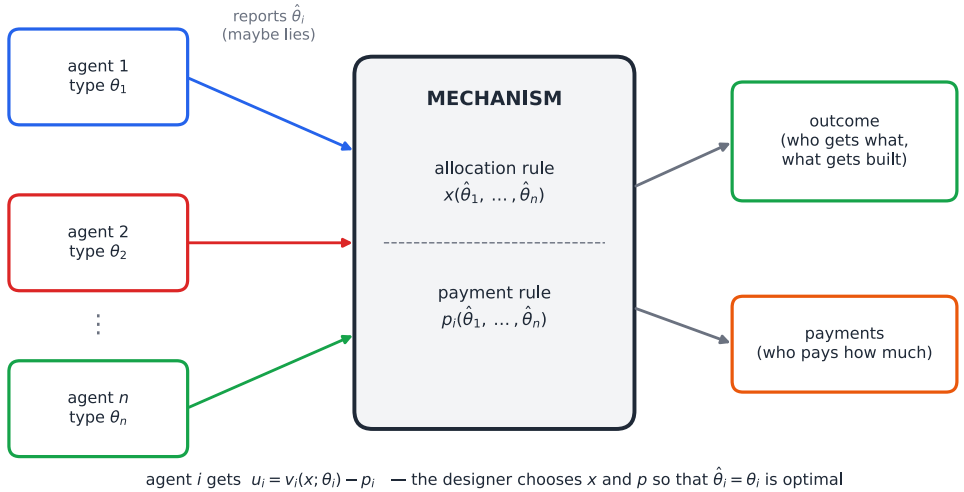


Figure 42.2 — A mechanism. Agents hold private types, send reports that may be lies, and the box turns reports into an outcome and a payment for each agent.

Three ingredients matter.

1. **Types.** Each agent i privately knows a type θ_i: how much the bridge is worth to her, how she ranks the designs, what a radio licence would earn her. This is exactly the setting of Bayesian games in Ch. 41.
2. **Reports.** The agents send messages. In the simplest mechanisms, the message *is* a claimed type $\hat{\theta}_i$.
3. **Rules.** The box applies an **allocation rule** (what happens) and a **payment rule** (who pays what).

The magic word is **incentive compatibility**: the rules are built so that reporting $\hat{\theta}_i = \theta_i$ is optimal for each agent. The analogy to hold on to is a well-designed tax form. A bad form invites you to shade the numbers; a good one is constructed so that shading cannot help you, and you stop bothering to think strategically at all. That last point is the real prize. A mechanism with a dominant truthful strategy is *robust*: it works without the designer knowing anyone's beliefs, and without participants needing to out-think each other.

The other big idea is a disappointment that turns into a design principle. Without money, the Condorcet cycle is not a curiosity but a symptom: as we will state below, essentially every voting rule over three or more options can be manipulated. With money, a single clever trick fixes the bridge problem. The trick is to **charge each agent for the harm her report does to everyone else.** Once you are billed for your externality, your private interest and society's interest line up, and lying stops paying.

42.3 The mathematics

Social choice functions

Let $N = \{1, \ldots, n\}$ be agents and O a set of possible outcomes (build or not; who gets which item). Agent i has a type $\theta_i \in \Theta_i$, and $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_n)$.

Social choice function and mechanism

A **social choice function** $f : \Theta_1 \times \cdots \times \Theta_n \to O$ says which outcome the designer *wants* for each profile of true types. A **mechanism** $(M_1, \ldots, M_n, g)$ gives each agent a message space M_i and an outcome rule $g : M_1 \times \cdots \times M_n \to O$. It is **direct** if $M_i = \Theta_i$: you simply announce a type.

The designer never observes $\boldsymbol{\theta}$. She observes messages, and the messages are chosen strategically.

When there is no money, agents only have preferences over O. Arrow's work in 1951 and the Gibbard–Satterthwaite theorem of the 1970s say how little you can then do:

Arrow's impossibility theorem (statement)

With at least three alternatives, no rule that turns every profile of individual rankings into a social ranking can satisfy all of: **unanimity** (if everyone prefers a to b, so does society), **independence of irrelevant alternatives** (society's verdict on a versus b depends only on individual verdicts on a versus b), and **non-dictatorship**.

Gibbard–Satterthwaite (statement)

With at least three possible winners and every ranking allowed, any voting rule that is **strategy-proof** (no voter ever gains by misreporting her ranking) and **onto** (every candidate can win) is a **dictatorship**.

The Condorcet cycle of Figure 42.1 is the smallest witness that majority rule breaks independence-plus-transitivity. The full proofs are combinatorial gems, and we are not going to give them here.

? Why must it be a dictator?

Gibbard–Satterthwaite does not merely say "manipulation is possible". It says that the *only* escape is to hand all power to one voter. How can two innocent-looking axioms force something as specific as a dictator? The proof hunts down a "pivotal voter" whose ranking flips society's choice, and shows that voter controls everything. The full argument, and its kinship with Arrow's theorem, are in Ch. 96.

Adding money: quasilinear utility

Now let money move. In the **quasilinear** model, an outcome is a pair (decision $x \in O$, payments $p_1, \dots, p_n$), and agent i's utility is

$$u_i = v_i(x; \theta_i) - p_i.$$

Money enters linearly and is worth the same to everyone. That assumption is strong (rich and poor do not value a euro equally) but it is what makes the theory tick.

Dominant-strategy incentive compatibility (DSIC)

A direct mechanism $(x(\cdot), p(\cdot))$ is **DSIC** (or **strategy-proof**) if for every agent i, every true type θ_i, every false report $\hat{\theta}_i$, and every report profile $\hat{\boldsymbol{\theta}}_{-i}$ of the others,

$$v_i\big(x(\theta_i, \hat{\boldsymbol{\theta}}_{-i}); \theta_i\big) - p_i(\theta_i, \hat{\boldsymbol{\theta}}_{-i}) \;\geq\; v_i\big(x(\hat{\theta}_i, \hat{\boldsymbol{\theta}}_{-i}); \theta_i\big) - p_i(\hat{\theta}_i, \hat{\boldsymbol{\theta}}_{-i}).$$

Truth is a dominant strategy: it is optimal *whatever the others do*, truthful or not.

A weaker notion, **Bayesian incentive compatibility** (BIC), asks only that truth be optimal *in expectation*, assuming the others are truthful and types are drawn from known distributions, i.e. that truth-telling is a Bayes–Nash equilibrium in the sense of Ch. 41. The expectations are the ordinary ones of Ch. 19.

Result 1: the revelation principle

Why study only direct mechanisms, where you just announce your type? Real mechanisms have rounds, clocks, bids and counter-bids. The first central result says you lose nothing.

Revelation principle (dominant-strategy version)

If some mechanism (M, g) has a dominant-strategy equilibrium $s_i^* : \Theta_i \to M_i$ that implements f, meaning $g(s_1^*(\theta_1), \dots, s_n^*(\theta_n)) = f(\boldsymbol{\theta})$ for all $\boldsymbol{\theta}$, then the direct mechanism $g^{\text{dir}}(\hat{\boldsymbol{\theta}}) = g(s_1^*(\hat{\theta}_1), \dots, s_n^*(\hat{\theta}_n))$ implements f and is DSIC.

Proof. Build a "proxy box": agent i tells the proxy her type $\hat{\theta}_i$, and the proxy plays $s_i^*(\hat{\theta}_i)$ on her behalf in the original mechanism. Suppose agent i with true type θ_i considers lying, reporting $\hat{\theta}_i$, when the others report $\hat{\boldsymbol{\theta}}_{-i}$. Her outcome is then $g(s_i^*(\hat{\theta}_i), \mathbf{s}^*_{-i}(\hat{\boldsymbol{\theta}}_{-i}))$. But $s_i^*(\hat{\theta}_i)$ is just *some* message $m_i \in M_i$, and since $s_i^*(\theta_i)$ is dominant in the original mechanism, it does at least as well as any m_i against *any* messages of the others, in particular against $\mathbf{s}^*_{-i}(\hat{\boldsymbol{\theta}}_{-i})$. So reporting θ_i is at least as good as reporting $\hat{\theta}_i$. Truth is dominant, and with truthful reports the outcome is $g(\mathbf{s}^*(\boldsymbol{\theta})) = f(\boldsymbol{\theta})$. ∎

The proof is almost insultingly short, and that is the point: anything clever an agent could do, the proxy already does for her. The same argument with "Bayes–Nash" in place of "dominant" gives the Bayesian revelation principle. The practical moral is double-edged. The principle lets theorists search only over truthful direct mechanisms. It does *not* tell engineers to build direct mechanisms: asking a bidder to report a value for every one of 2^{50} bundles of licences is hopeless, which is why real designs are indirect (Ch. 43).

Result 2: VCG makes truth dominant

Now the trick promised in 42.2. Let the designer's goal be **efficiency**: choose the decision maximising total value,

$$x^*(\hat{\boldsymbol{\theta}}) \in \arg\max_{x \in O} \sum_{j=1}^{n} v_j(x; \hat{\theta}_j).$$

Vickrey–Clarke–Groves (VCG) mechanism with the Clarke pivot

Given reports $\hat{\boldsymbol{\theta}}$: choose $x^*(\hat{\boldsymbol{\theta}})$ as above, and charge agent i

$$p_i(\hat{\boldsymbol{\theta}}) \;=\; \underbrace{\max_{x \in O} \sum_{j \neq i} v_j(x; \hat{\theta}_j)}_{\text{others' best welfare without } i} \;-\; \underbrace{\sum_{j \neq i} v_j\big(x^*(\hat{\boldsymbol{\theta}}); \hat{\theta}_j\big)}_{\text{others' welfare with } i}.$$

Agent i pays exactly the **externality** she imposes: how much worse off her presence makes everyone else.

VCG is DSIC

In the quasilinear model, the VCG mechanism is dominant-strategy incentive compatible.

Proof. Fix agent i, her true type θ_i, and arbitrary reports $\hat{\boldsymbol{\theta}}_{-i}$ of the others. Write $h_i = \max_x \sum_{j \neq i} v_j(x; \hat{\theta}_j)$; it does not depend on anything agent i says. If i reports $\hat{\theta}_i$, leading to decision $\hat{x} = x^*(\hat{\theta}_i, \hat{\boldsymbol{\theta}}_{-i})$, her utility is

$$u_i(\hat{\theta}_i) = v_i(\hat{x}; \theta_i) - h_i + \sum_{j \neq i} v_j(\hat{x}; \hat{\theta}_j) = \Big[v_i(\hat{x}; \theta_i) + \sum_{j \neq i} v_j(\hat{x}; \hat{\theta}_j)\Big] - h_i.$$

The bracket is the total welfare *as measured with her true value and the others' reports*, evaluated at $\hat{x}$. Agent i can influence only which $\hat{x}$ is chosen, and h_i is a constant to her. The bracket is maximised over all of O by choosing $\hat{x} = \arg\max_x [v_i(x;\theta_i) + \sum_{j\neq i} v_j(x;\hat{\theta}_j)]$, and that is precisely the decision the mechanism picks when she reports $\hat{\theta}_i = \theta_i$. So no report beats the truth, whatever the others report. ■

Notice what the proof did: VCG aligned each agent's objective with the designer's objective, *up to a constant she cannot affect*. This is the same move as a well-chosen Lagrangian or penalty term in optimization (Ch. 23): change the objective so that the local optimiser's best response is the global optimum.

Two side facts, easy to check. Payments are never negative, because the others' best welfare without i is at least their welfare at x^*. And truthful agents never regret participating (**individual rationality**) when values are non-negative and "nothing happens to you" is always available, because $u_i = \max_x \sum_j v_j - \max_x \sum_{j\neq i} v_j \geq 0$.

The footbridge, done right

Cost 300, split 100 each if built. Net values (value minus share) are $150,\ -50,\ -60$, summing to $+40$, so VCG builds. Write w_i for net value; "not build" gives everyone 0.

- Household 1: the others' net sum is -110. Without household 1 they would not build and get 0; with her they get -110. She pays $0 - (-110) = 110$ on top of her 100 share, so 210 in all. Her utility is $250 - 210 = 40$.
- Households 2 and 3: without household 2 the others' sum is $90 > 0$, so the bridge is built anyway and her presence changes nothing for them: she pays no extra. Same for household 3 (others' sum 100).

Now let household 2 exaggerate her dislike, reporting net -200. The total becomes -110, the bridge is cancelled, and she is now pivotal: the others lose 90, which she pays. Her utility drops from -50 to -90. Lying cost her 40.

Household 1's extra 110 is called the **Clarke tax**. Only **pivotal** agents, those whose report flips the decision, pay it.

? Where does the money go?

In the bridge example VCG collects 110 in Clarke tax that no one needs. If you hand it back to the households, you change their incentives and truth-telling can collapse. If you burn it, you waste welfare. Is there *any* rule that is efficient, truthful, voluntary, and exactly balances the budget? For two people trading a single good, the Myerson–Satterthwaite theorem says no, and the proof is a beautiful integration argument. It waits for you in Ch. 96.

Result 3: one-dimensional types and monotonicity

VCG targets efficiency. What can a DSIC mechanism do in general? For the most common case, **single-parameter** agents, the answer is clean. Each agent has one number $v_i \geq 0$ (value per unit of "stuff"), the mechanism gives her an amount $x_i(\mathbf{b}) \in [0,1]$ (a probability of winning, a share of bandwidth) based on bids $\mathbf{b}$, and her utility is $v_i x_i(\mathbf{b}) - p_i(\mathbf{b})$. Fix the others' bids and drop the subscript: write $x(b)$ and $p(b)$ as functions of her own bid.

Monotonicity is necessary

If (x,p) is DSIC, then $x(b)$ is non-decreasing in b, and for any $z < y$,

$$z\,[x(y) - x(z)] \;\leq\; p(y) - p(z) \;\leq\; y\,[x(y) - x(z)].$$

Proof. Truth for an agent of value z means she does not want to pretend to be y; truth for an agent of value y means she does not want to pretend to be z:

$$z\,x(z) - p(z) \geq z\,x(y) - p(y),$$
$$y\,x(y) - p(y) \geq y\,x(z) - p(z).$$

Rearranging the first gives the left inequality of the sandwich, the second gives the right one. The two ends of the sandwich can only be ordered if $(y-z)[x(y) - x(z)] \geq 0$, and since $y > z$ this forces $x(y) \geq x(z)$. ■

Divide the sandwich by $y - z$ and let $y \to z$: wherever x is differentiable, $p'(z) = z\,x'(z)$. Integrating, with $p(0) = 0$,

$$p(b) \;=\; b\,x(b) - \int_0^b x(z)\,dz.$$

This is **Myerson's lemma**: in single-parameter settings, *every* monotone allocation rule can be made truthful, and the payment is then *forced* (the converse, that this payment makes any monotone rule DSIC, follows from the same picture). Geometrically, the payment is the area to the *left* of the allocation curve, between it and the vertical axis, up to height $x(b)$.

Try it on one item and a "highest bid wins" allocation. If the best rival bid is B, then $x(b) = 0$ for $b < B$ and $x(b) = 1$ for $b > B$. The formula gives $p(b) = b - (b - B) = B$. **The winner pays the second-highest bid.** We have just *derived* the Vickrey second-price auction, rather than guessed it. It is also VCG for a single item, as you can check from the definition.

42.4 The algorithm

VCG is a recipe you can code for any finite outcome set. It needs one welfare maximisation with everyone, and one more without each agent: $n + 1$ optimisations in all.

```
def vcg(outcomes, values):
    """outcomes: list of decisions; values[i](x) = agent i's reported value for x."""
    n = len(values)
    welfare = lambda x, skip=None: sum(v(x) for j, v in enumerate(values) if j != skip)
    x_star = max(outcomes, key=welfare)                       # efficient decision
    pay = []
    for i in range(n):
        best_without_i = max(welfare(x, skip=i) for x in outcomes)
        pay.append(best_without_i - welfare(x_star, skip=i))    # externality of i
    return x_star, pay

# footbridge: outcome 1 = build, 0 = don't; values are net of the 100 share
net = [150, -50, -60]
print(vcg([0, 1], [lambda x, w=w: w * x for w in net]))       # (1, [110, 0, 0])
```

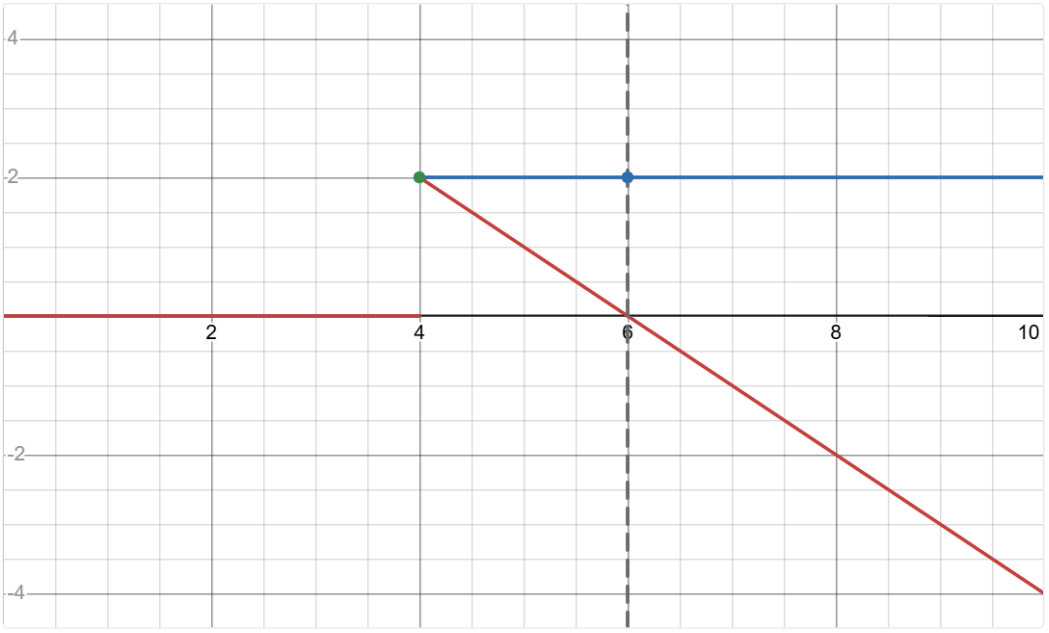
Complexity. The cost is $(n+1)$ times the cost of maximising welfare. For a yes/no decision that is trivial. For allocating m distinct items to bidders with values on *bundles*, the outcome set has up to $(n+1)^m$ elements, and welfare maximisation is the NP-hard **winner determination problem**, a weighted set-packing integer program (Ch. 49). That is a deep tension: truthfulness was proved assuming the mechanism finds the *exact* optimum.

Failure cases worth knowing:

- **Approximation breaks truth.** Replace the argmax by a heuristic and the VCG proof collapses: an agent may now gain by lying to steer the heuristic toward an outcome it would not find on its own. Designing mechanisms that are both truthful and computationally efficient is the field of *algorithmic mechanism design*, which leans on the limits in Ch. 30.
- **Collusion.** DSIC is about single agents. Two losing bidders can sometimes jointly lie and both win (you will meet a spectacular case in Ch. 43).
- **Numerics.** Payments are *differences* of large optima. With floating-point values of similar size, a pivotal agent's tax can be swamped by rounding; use exact arithmetic (integers, fractions) for money. The same catastrophic-cancellation warning appears in Ch. 22.
- **False-name bids.** VCG assumes each agent has one identity. On the internet, one person can bid as many, and VCG is not robust to that.

42.5 The experiment

A. Feel the incentives. The graph plots the utility of a bidder whose true value is 6 against a rival whose bid is 4, as a function of her own bid x. Blue is the second-price auction (win and pay 4 if $x > 4$). Red is a first-price auction (win and pay x if $x > 4$).



Blue never rises above its value at the truthful bid $x = 6$; red rewards shading down toward 4 and punishes bids above 6.

Change the rival's bid from 4 to 7: in the four domain restrictions replace 4 by 7, and replace the blue line $y = 2$ by $y = -1$ (value 6 minus price 7); the red line $y = 6 - x$ is unchanged. Now the truthful bid 6 loses, and the blue line shows that bidding above 7 would win at a loss. Truth is still optimal. Then try to find *any* rival bid for which a blue bid other than 6 does strictly better. You can't, and the proof in 42.3 says why.

B. Try to cheat VCG. This script runs the Clarke-pivot footbridge and lets household `i` try every report on a grid. Change `true_net`, the cost share inside it, and `i`.

```
import numpy as np

def clarke(reports):
    """Build iff total net report > 0. Returns (decision, Clarke taxes)."""
    reports = np.asarray(reports, dtype=float)
    build = reports.sum() > 0
    taxes = []
    for k in range(len(reports)):
        others = reports.sum() - reports[k]
        pivotal = (others > 0) != build
        taxes.append(abs(others) if pivotal else 0.0)
    return build, np.array(taxes)

true_net = np.array([150.0, -50.0, -60.0])   # value minus 100 cost share
i = 1                                        # the household trying to cheat

def utility(report):
    r = true_net.copy(); r[i] = report
    build, tax = clarke(r)
    return true_net[i] * build - tax[i]

grid = np.linspace(-500, 500, 2001)
u = np.array([utility(r) for r in grid])
print(f"truthful report {true_net[i]:+.0f}: utility {utility(true_net[i]):.1f}")
print(f"best report on grid: {grid[u.argmax()]:+.0f}, utility {u.max():.1f}")
print(f"best lie strictly better than truth? {u.max() > utility(true_net[i]) + 1e-9}")

# The naive clerk's rule: build iff sum > 0, no taxes. Lying pays here.
naive = lambda rep: true_net[i] * ((true_net.sum() - true_net[i] + rep) > 0) + 0.0
print("naive rule, truth:", naive(true_net[i]), " best lie:", max(naive(r) for r in grid))
```

With the numbers given, household 2 cannot beat its truthful utility of -50 under VCG, but under the naive rule a lie lifts it from -50 to 0. Now set `true_net = [30, -20, -5]` and look at the total tax collected. Then randomise `true_net` a thousand times and plot the histogram of total Clarke tax: that money is the price of truth.

42.6 The challenge

1. **Intuition.** Solomon's mechanism stops working once the false mother knows it. Explain, in the language of 42.3, which incentive constraint fails. Then propose a version of Solomon's test that uses *money* (each woman may pay to claim the child) and argue informally which woman would pay more, and what could still go wrong.
2. **Derivation.** For a single item, show directly from the VCG definition that the winner's payment equals the second-highest reported value and losers pay nothing. Then take an allocation rule that gives one unit to the highest bidder *only if* her bid exceeds a reserve r, and use Myerson's formula $p(b) = b\,x(b) - \int_0^b x(z)\,dz$ to find what she pays.
3. **Implementation.** Extend the `vcg` function to k identical items among n bidders who each want at most one. Verify numerically that the winners all pay the same price, identify that price, and check with random lies that no bidder ever gains.
4. **Open investigation.** For three voters and three candidates there are $6^3 = 216$ ranking profiles. Write a program that, for plurality with alphabetical tie-breaking, Borda, and instant-runoff voting (Ch. 44), counts the fraction of profiles in which some single voter can profitably misreport. How does the fraction grow with the number of voters and candidates, and does any non-dictatorial rule come close to zero? (No one has a clean formula for most rules.)

42.7 Loose threads

This chapter promised truthfulness, and with money it delivered: VCG makes honesty a dominant strategy whenever the designer's goal is total welfare and utilities are quasilinear. But look at what we swept aside.

- **We only maximised welfare.** A seller does not want efficiency; she wants revenue. The Clarke tax in the bridge example was an accident, not an objective. What mechanism extracts the *most* money from bidders whose values she can only guess at?

> **? The seller's mechanism**
>
> Myerson's lemma pins down payments for any monotone allocation. So the expected revenue of a truthful mechanism is a function of the allocation rule alone. Which allocation maximises it? The answer involves a strange quantity, the "virtual value" $v - \frac{1-F(v)}{f(v)}$, and sometimes tells the seller to refuse to sell to the only bidder present. That surprising rule, and why it beats every auction format you know, is the business of Ch. 43.

- **We never proved the impossibility theorems.** Arrow and Gibbard–Satterthwaite were stated, not proved; their proofs are in Ch. 96.
- **Budget balance** was left as a mystery above.
- **Groups were ignored.** DSIC protects against one liar, not coalitions; cooperative behaviour is the subject of Ch. 44.
- **Computation was waved at.** When the welfare optimum is NP-hard, can a truthful mechanism get close to it? For some problems, provably not without losing truthfulness; for others, this is open.

And one deeper unease. Every result here assumed agents are perfectly rational expected-utility maximisers who understand the rules. Real people in a second-price auction sometimes overbid anyway. A "strategy-proof" mechanism that people do not *believe* is strategy-proof may behave like a manipulable one. Designing for real minds, rather than ideal ones, is where much of the current research lives.

Further reading

- T. Börgers, *An Introduction to the Theory of Mechanism Design*, Oxford University Press, 2015. A compact, rigorous introduction covering both dominant-strategy and Bayesian design.
- N. Nisan, T. Roughgarden, É. Tardos, V. Vazirani (eds.), *Algorithmic Game Theory*, Cambridge University Press, 2007. Chapters 9–12 treat mechanism design with an eye on computation.
- T. Roughgarden, *Twenty Lectures on Algorithmic Game Theory*, Cambridge University Press, 2016. Lectures 2–4 derive Myerson's lemma and VCG in the single-parameter style used here.
- A. Mas-Colell, M. Whinston, J. Green, *Microeconomic Theory*, Oxford University Press, 1995. Chapters 21–23 for Arrow, Gibbard–Satterthwaite and the classical theory.

Chapter 43 — Auctions, Markets, and Strategic Bidding

> **Level 2 · Track C — Game Theory and Strategic Intelligence Builds on:** Ch. 19, Ch. 5, Ch. 41, Ch. 42 · **Leads to:** Ch. 46, Ch. 96 · **Related:** Ch. 48, Ch. 49, Ch. 38 **Reading time:** ~25 min

43.1 The hook

In the year 2000, several European governments sold licences for the third generation of mobile phone networks, the radio spectrum that every smartphone now lives on. The licences were broadly similar, the bidders were often the very same telecom giants, and the auctions happened within months of each other. The outcomes were not similar at all.

The United Kingdom raised about £22.5 billion, roughly 2.5% of national income, in an auction that ran for seven weeks and 150 rounds. The Netherlands, a few months later, raised about €2.7 billion, a small fraction of what had been forecast, with about the same number of licences as there were strong incumbent firms ready to bid. In Switzerland, bidders withdrew or merged before the auction began, four licences went to four bidders at close to the reserve price, and the government collected about CHF 205 million, a tiny fraction of the British take per head of population. The economist Paul Klemperer, who helped design the British auction, summed up the lesson: in auctions, *details matter*.

Go back ten years and the details bite harder. In 1990 New Zealand sold spectrum licences with a **second-price** sealed-bid auction: the highest bidder wins and pays the second-highest bid. Economists love this format; we derived it in Ch. 42 as the unique truthful way to sell one item. The public did not love it. As John McMillan later reported, in one licence the winning bid was NZ$100,000 and the price paid was NZ$6, because the only other bid was NZ$6. Headlines asked why the government was giving away the airwaves.

So which is it? Is the second-price auction a mathematical triumph (truthful, efficient, simple) or a revenue disaster? Why did "the same" auction raise, per head of population, more than 25 times as much in one country as in another? And what is a seller supposed to do when she does not even know what the bidders think the item is worth?

By the end of this chapter you will be able to compute, for a clean model, exactly what each format earns. And you will meet a theorem so strong it feels like a trick: in that clean model, a whole family of very different-looking auctions earns *exactly the same* expected revenue. The real-world disasters are all stories about where that model breaks.

43.2 The big idea

There are four classic formats for selling one item.

Format	How it runs	Winner pays
English (ascending)	Price rises; bidders drop out; last one standing wins	The price at which the second-to-last bidder quit
Dutch (descending)	Price falls from high; first to shout "mine" wins	The price at which she shouted
First-price sealed	Everyone submits one sealed bid; highest wins	Her own bid
Second-price sealed (Vickrey)	Same, but…	The second-highest bid

Two pairs collapse at once. A Dutch auction is **strategically identical** to a first-price sealed auction: in both, your only decision is the price at which you commit, and you pay that price if it is the highest. And when bidders know their own values (the **private values** setting), an English auction works like a second-price auction: you stay in until the price reaches your value, and the winner pays roughly where the runner-up dropped out.

So really there are two worlds, and they push bidders in opposite directions.

- In the **second-price** world, your bid controls only *whether* you win, not *what you pay*. So you should bid exactly your value: bidding higher only adds wins at a loss, bidding lower only loses wins you wanted (Ch. 42 has the picture).
- In the **first-price** world, your bid is your price. Bid your value and you win nothing (zero profit); bid lower and you keep a margin, at the risk of losing. So everyone **shades**: they bid below their values, trading off a fatter margin against a thinner chance of winning.

The analogy is the difference between a fixed-price menu and haggling. In the first, you only need to know what you want. In the second, you need a theory of everybody else.

Here is the surprise to keep in mind. In the second-price auction the seller collects the *second*-highest value. In the first-price auction she collects the highest *bid*, which is shaded. One number is a lower value, the other a shaded higher value. They could be ordered either way. Section 43.3 shows they are equal on average, and not by coincidence.

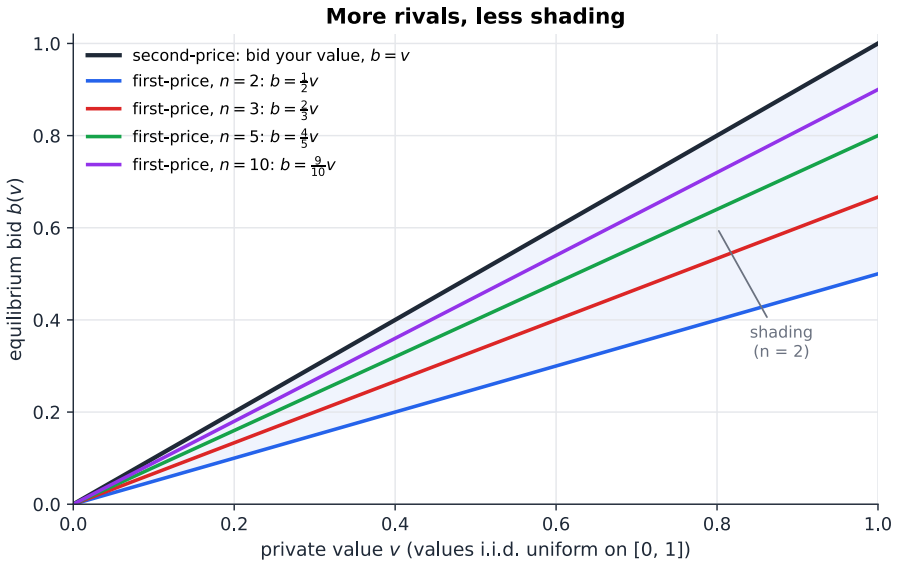


Figure 43.1 — Equilibrium bidding with values uniform on $[0,1]$. Second-price bidders bid their value; first-price bidders shade, and the shading shrinks as rivals multiply.

43.3 The mathematics

The model

There are n bidders. Bidder i has a private value v_i, and the values are **independent and identically distributed** with a continuous distribution function F and density f on $[0, \bar{v}]$ (the *independent private values*, or IPV, model). Each bidder knows her own value and only the distribution of the others', which makes this a Bayesian game in the sense of Ch. 41. Utility is quasilinear: win at price p and get $v_i - p$; lose and get 0. For concrete numbers we use the uniform case $F(v) = v$ on $[0,1]$.

One fact from probability (Ch. 19) does most of the work. The maximum of k independent uniforms has CDF $P(\max \le t) = t^k$, so its density is kt^{k-1} and its mean is $\int_0^1 t \cdot kt^{k-1}\,dt = \frac{k}{k+1}$. For the second-highest of n uniforms, the mean is $\frac{n-1}{n+1}$.

Result 1: equilibrium shading in the first-price auction

First-price equilibrium, uniform values

With n bidders and values i.i.d. uniform on $[0,1]$, the strategy $\beta(v) = \frac{n-1}{n}\,v$ played by everyone is a symmetric Bayes–Nash equilibrium of the first-price auction.

Proof. Guess that the other $n-1$ bidders use $\beta(v) = c\,v$ for some $c \in (0,1]$, and look for bidder 1's best bid b when her value is v. She wins if every rival bids below b, i.e. if every rival value is below b/c. For $b \le c$, independence gives

$$P(\text{win}) = \left(\frac{b}{c}\right)^{n-1}, \qquad \pi(b) = (v-b)\left(\frac{b}{c}\right)^{n-1}.$$

Differentiate and set to zero (the constant $c^{-(n-1)}$ drops out):

$$-b^{n-1} + (n-1)(v-b)\,b^{n-2} = 0 \quad\Longrightarrow\quad b = \frac{n-1}{n}\,v.$$

The second derivative is negative at this point, so it is a maximum, and bids above c win with certainty but cost more, so they are worse. Her best response is linear with slope $\frac{n-1}{n}$. For a *symmetric* equilibrium her best response must be the strategy we assumed for the others, so $c = \frac{n-1}{n}$. Since $\frac{n-1}{n}v \le c$ for every $v \le 1$, the condition $b \le c$ we used holds. ∎

With two bidders you bid half your value. With ten, you bid 90%. Competition squeezes out the shading, which is Figure 43.1.

There is a pretty interpretation. If Y is the highest of the other $n-1$ values, then $E[Y \mid Y < v] = \frac{n-1}{n}v$ for uniforms. You bid what you expect the strongest rival's value to be, *given that you beat it*. That formula, $\beta(v) = E[Y \mid Y < v]$, is the general first-price equilibrium for any distribution F.

Result 2: revenue equivalence

Expected revenue in the second-price auction is the expected second-highest value: $\frac{n-1}{n+1}$. In the first-price auction it is the expected highest bid, $\frac{n-1}{n} \cdot E[\max_i v_i] = \frac{n-1}{n} \cdot \frac{n}{n+1} = \frac{n-1}{n+1}$. **Identical.** For $n = 2$, both give $\frac{1}{3}$.

Coincidence? Ch. 42 already has the reason. Apply Myerson's lemma to the *expected* allocation and payment. Let $x(v)$ be the probability that a bidder of value v wins and $m(v)$ her expected payment, both in equilibrium. Truth-telling to the "direct" version of any equilibrium (the revelation principle) gives the same sandwich argument as in Ch. 42, now in expectation, and hence

$$m(v) = v\,x(v) - \int_0^v x(z)\,dz + m(0).$$

Revenue equivalence

In the IPV model, any two auctions in which, in equilibrium, (i) the bidder with the highest value always wins, and (ii) a bidder with value 0 pays nothing in expectation, give every bidder the same expected payment $m(v)$, and hence give the seller the same expected revenue.

Proof. Under (i), a bidder of value v wins exactly when all $n-1$ others have lower values, so $x(v) = F(v)^{n-1}$, the same in both auctions. Under (ii), $m(0) = 0$. The formula above then determines $m(v)$ completely from x, so it is the same in both auctions. Expected revenue is $n\,E[m(v_i)]$, which is therefore the same too. ∎

Check it in the uniform case: $x(v) = v^{n-1}$, so $m(v) = v^n - \frac{v^n}{n} = \frac{n-1}{n}v^n$, and $n\,E[m] = (n-1)\int_0^1 v^n\,dv = \frac{n-1}{n+1}$. It also covers stranger formats: an **all-pay auction**, where every bidder pays her bid win or lose (a model of lobbying, patent races, or the effort spent on a job application), earns the same expected revenue too.

What revenue equivalence does not say

It says *expected* revenue is equal. The distributions differ: the second-price price is more variable, which is exactly how New Zealand ended up selling a licence for NZ$6 when there was only one serious bidder. A risk-averse seller, or a government facing headlines, is not indifferent.

Result 3: the optimal auction

The mystery in Ch. 42 asked which mechanism earns the seller the most. Revenue equivalence says all "efficient" formats tie, so to do better the seller must sometimes **not** give the item to the highest bidder. The cleanest lever is a **reserve price** r: no sale unless someone bids at least r.

Start with one bidder. With a posted price r, she buys if $v \ge r$, so revenue is $r\,(1 - F(r)) = r(1-r)$ for uniform values. Maximising, $r^* = \frac{1}{2}$ and revenue is $\frac{1}{4}$. With no reserve, one bidder in a second-price auction pays 0. So the seller *should* risk not selling at all.

For n bidders the same calculation becomes a short identity. Take the expected payment $m(v)$ above, with $m(0) = 0$, average it over $v \sim F$, and swap the order of integration in the double integral:

$$\int_0^{\bar v}\!\!\int_0^{v} x(z)\,dz\,f(v)\,dv = \int_0^{\bar v} x(z)\,\big(1 - F(z)\big)\,dz.$$

That gives

$$E\big[m(v)\big] = \int_0^{\bar v} x(v)\left[v - \frac{1-F(v)}{f(v)}\right] f(v)\,dv = E\big[\varphi(v)\,x(v)\big], \qquad \varphi(v) = v - \frac{1-F(v)}{f(v)}.$$

Myerson's optimal auction (regular case)

Expected revenue of any truthful mechanism equals the expected **virtual value** of the winner, $E\big[\sum_i \varphi(v_i)\,x_i\big]$. If φ is increasing, revenue is maximised by giving the item to the bidder with the highest $\varphi(v_i)$, provided it is positive, and to no one otherwise. For i.i.d. bidders this is a second-price auction with reserve r^* solving $\varphi(r^*) = 0$.

Proof. Summing over bidders, expected revenue is $E[\sum_i \varphi(v_i)x_i(\mathbf{v})]$. For each value profile, the expression $\sum_i \varphi(v_i)x_i$ with $\sum_i x_i \le 1$ is largest when all weight goes to the largest $\varphi(v_i)$ if that is positive, and nothing is sold if all are negative. If φ is increasing, this allocation is monotone in each bid, so by Myerson's lemma it can be made truthful, and the payment it forces is "the larger of the reserve and the second-highest bid". Maximising pointwise maximises the expectation. ∎

For uniform values $\varphi(v) = v - (1-v) = 2v - 1$, so $r^* = \frac{1}{2}$, **whatever the number of bidders**. With two bidders, the reserve lifts expected revenue from $\frac{1}{3} \approx 0.333$ to $\frac{5}{12} \approx 0.417$, a 25% gain, at the cost of sometimes not selling to a bidder who valued the item. Optimal revenue and efficiency pull in different directions.

? When values are not private

Everything above assumed you know what the item is worth *to you*. Spectrum, oil leases and takeover targets are worth roughly the same to everyone, and nobody knows how much. Then winning is bad news: you win exactly when you were the most optimistic, and the naive bidder systematically overpays. This **winner's curse** breaks revenue equivalence, and the English auction, which lets you watch rivals quit, now beats the sealed formats by "linking" your price to their information. The *linkage principle* and the affiliated-values model that makes it precise are in Ch. 96.

Many items: packages and markets

The 3G auctions sold many licences at once, and values were **not additive**. A firm that wins one regional licence may need the neighbouring one to build a viable network: the pair is worth more than the sum of its parts (complements). Figure 43.2 shows a toy version.

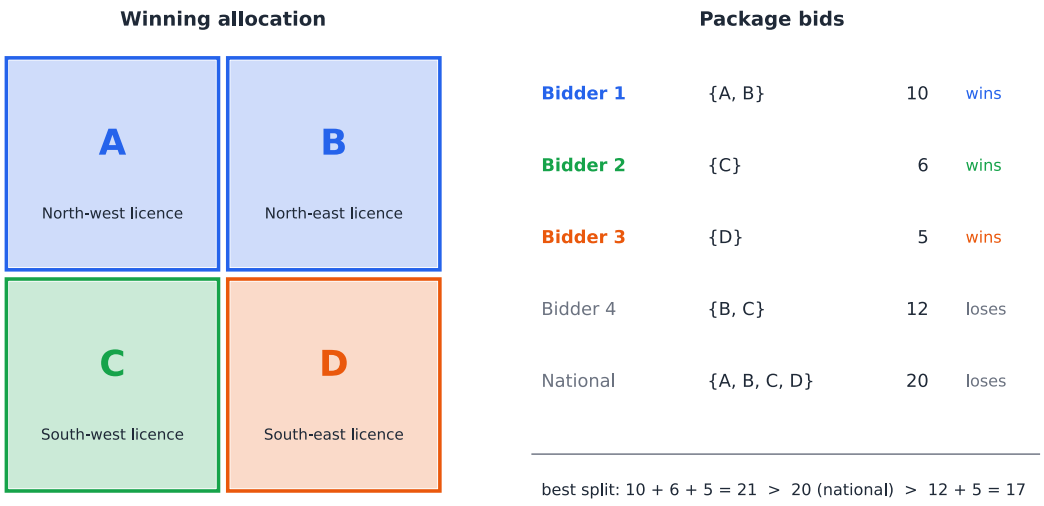


Figure 43.2 — A package auction. Bids are on bundles; the auctioneer picks the set of non-overlapping bids with the largest total.

Picking the winners is the **winner determination problem**: choose a set of pairwise disjoint packages with maximum total bid. Here the three local bidders together (21) beat the national bidder (20). Now compute VCG payments as in Ch. 42. Without bidder 1, the best allocation is the national bid, 20; with her, the others get $6 + 5 = 11$; so she pays 9. Similarly bidder 2 pays $20 - 15 = 5$ and bidder 3 pays $20 - 16 = 4$. **Total revenue: 18**, although a single bidder was offering 20 for the lot. Worse, if bidders 2 and 3 were one firm, the incentives would change again. Truthful does not mean sensible.

Two design ideas tame this. **Simultaneous ascending (clock) auctions** raise prices on all licences together round by round, so bidders discover which packages are affordable; the UK 3G sale was of this family. And the theory of **markets** says that when goods are substitutes, a set of item prices exists at which every bidder demands exactly what she is allocated, a **Walrasian equilibrium**, which an ascending clock can find.

Not all markets have prices. Doctors and hospitals, students and schools, kidney donors and patients: here money cannot or should not clear the market. Gale and Shapley showed in 1962 that in a two-sided matching market, a **stable matching** (no doctor and hospital who would both rather be with each other than with their assigned partners) always exists, and the next section gives the algorithm that finds it.

43.4 The algorithm

Deferred acceptance (Gale–Shapley). One side proposes; the other side holds its best offer so far and rejects the rest, until no one is left to propose.

```
def deferred_acceptance(prop_prefs, recv_prefs):
    """prop_prefs[p] = list of receivers, best first; recv_prefs[r] likewise.
    Returns dict receiver -> proposer for a proposer-optimal stable matching."""
    rank = {r: {p: k for k, p in enumerate(lst)} for r, lst in recv_prefs.items()}
    nxt = {p: 0 for p in prop_prefs}             # next receiver p will try
    held = {}                                    # receiver -> proposer held
    free = list(prop_prefs)
    while free:
        p = free.pop()
        if nxt[p] >= len(prop_prefs[p]):
            continue                             # p has been rejected everywhere
        r = prop_prefs[p][nxt[p]]; nxt[p] += 1
        if p not in rank[r]:
            free.append(p); continue             # r finds p unacceptable
        if r not in held:
            held[r] = p
        elif rank[r][p] < rank[r][held[r]]:
            free.append(held[r]); held[r] = p    # r trades up
        else:
            free.append(p)                       # r rejects p
    return held
```

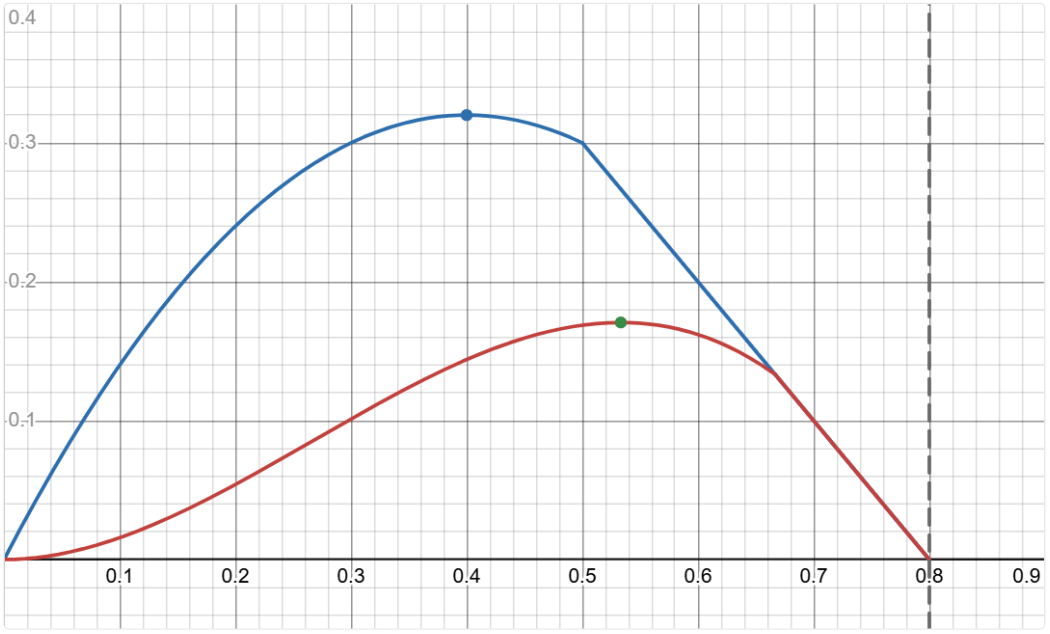
Why it stops, and why the result is stable. Each proposer proposes to each receiver at most once, so with n on each side there are at most n^2 proposals: $O(n^2)$ time. A receiver's held partner only ever improves. Now suppose p and r would both prefer each other to their final partners. Then p proposed to r before reaching her final partner (she ranks r higher), and r rejected her, at that time or later, for someone r prefers. Since r only trades up, r's final partner beats p, contradicting the assumption. So no blocking pair exists.

The auction side. Monte Carlo (Ch. 38) is the workhorse for comparing formats: draw values, apply the equilibrium strategies, average revenue. Its error falls like $1/\sqrt{\text{samples}}$, so to separate $\frac{1}{3}$ from $\frac{5}{12}$ reliably you need thousands of draws, and to see a 1% revenue difference you need hundreds of thousands. For package auctions, winner determination is a weighted set-packing integer program (Ch. 49); brute force over subsets of bids, as used to check the numbers of Figure 43.2, is exponential and fine only for toy cases.

Failure cases. Deferred acceptance is strategy-proof for the *proposing* side only; receivers can sometimes gain by misreporting. With couples who must be placed together (a real issue in medical residency matching), a stable matching may not exist at all. Clock auctions can suffer the **exposure problem**: a bidder who chases a complementary pair can end up holding only one licence at a price above its stand-alone value.

43.5 The experiment

A. Find your best first-price bid. Your value is 0.8. Blue: one rival who bids half her (uniform) value, so you win with probability $2x$. Red: two such rivals bidding two-thirds of their values, so you win with probability $(1.5x)^2$. The curves are your expected profit at bid x.



The peaks sit at $\frac{1}{2}\cdot 0.8 = 0.4$ and $\frac{2}{3}\cdot 0.8 \approx 0.533$: exactly the equilibrium bids of Result 1.

Change the value 0.8 to 0.5 (in all four curves) and confirm that the peaks move to 0.25 and 0.333. Then make the one rival more aggressive: replace $2x$ by $1.25x$ (she bids 80% of her value, for $x < 0.8$). Does your best bid rise or fall? Is it still a best response to bid half your value?

B. Race the formats. Draw values, play the equilibrium strategies, compare revenues. Change `n`, `reserve`, and the value distribution.

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(0)
n, trials, reserve = 2, 200_000, 0.5
v = rng.random((trials, n))                # i.i.d. uniform private values
s = np.sort(v, axis=1)
top, second = s[:, -1], s[:, -2]

first_price  = (n - 1) / n * top           # highest equilibrium bid
second_price = second                      # truthful bids: pay the runner-up
with_reserve = np.where(top < reserve, 0.0, np.maximum(second, reserve))

for name, rev in [("first-price", first_price), ("second-price", second_price),
                  (f"second-price, reserve {reserve}", with_reserve)]:
    se = rev.std() / np.sqrt(trials)
    print(f"{name:28s} mean {rev.mean():.4f} +/- {1.96*se:.4f}   std {rev.std():.3f}")
print("theory (no reserve):", (n - 1) / (n + 1))

plt.hist(first_price, bins=80, alpha=0.6, density=True, label="first-price revenue")
plt.hist(second_price, bins=80, alpha=0.6, density=True, label="second-price revenue")
plt.xlabel("seller revenue"); plt.legend(); plt.show()
```

Same mean, different spread: the histogram shows why a risk-averse seller cares about the format. Now replace `rng.random` with `rng.beta(2, 5, ...)` (values usually low) and note that the first-price line `(n-1)/n * top` is no longer the equilibrium. What breaks first, the code or the theorem?

43.6 The challenge

1. **Intuition.** Explain without equations why the optimal reserve price for uniform values is $\frac{1}{2}$ *no matter how many bidders there are*, even though with 100 bidders the reserve almost never binds. Then explain why governments are often told to set reserves well above what this theory suggests.
2. **Derivation.** Let two bidders have values on $[0, 1]$ with CDF $F(v) = v^2$ (strong bidders are common). Use revenue equivalence to find the first-price equilibrium bid $\beta(v)$, the seller's expected revenue, and the optimal reserve price from the virtual value $\varphi(v)$. Is φ increasing here?
3. **Implementation.** Write a brute-force winner-determination solver for package bids and a VCG payment routine. Reproduce the payments $9, 5, 4$ for Figure 43.2. Then find a package bid that bidders 2 and 3, acting together, could *add* (a fake shill bid) to lower their total payment.
4. **Open investigation.** Simulate a repeated first-price auction in which each of several bidders adapts her shading factor by a simple learning rule (for example, nudge the factor toward whatever would have earned more last round). Do the factors converge to $\frac{n-1}{n}$? Does the answer change with the learning rate, with noise, or with asymmetric bidders? Online ad exchanges, which moved from second-price to first-price auctions around 2019, face exactly this question.

43.7 Loose threads

We answered the questions of the hook in a clean world: independent private values, risk-neutral bidders, one item. In that world the second-price auction is truthful and the first-price auction is not, yet they earn the same, and the best a seller can do is a second-price auction with a well-chosen reserve. The real failures came from where that world ends: too few serious bidders (Switzerland, the Netherlands, the NZ$6 licence), common values, packages, and collusion.

? Prices are shadows of a linear program

In a market where buyers each want one item and sellers each have one, the efficient assignment is a linear program, and something odd happens: its **dual variables** are exactly a set of market-clearing prices, and the set of all such prices forms a lattice with a buyer-best and a seller-best corner. Why does the dual of an assignment problem know about competitive equilibrium? That bridge between optimisation and markets runs through LP duality in Ch. 48.

? Bidding against a machine that learns

A real bidder on an ad exchange does not know F, faces millions of auctions, and competes with algorithms that are themselves adapting. Is there a strategy whose total profit is guaranteed to be nearly as good as the best fixed shading factor in hindsight, whatever the opponents do? Strategies with vanishing **regret** do exactly that; how they work, and what equilibrium they converge to, is the subject of Ch. 46.

We also left untouched:

- **Asymmetric bidders.** When one bidder is known to be strong, the first-price equilibrium has no closed form and revenue equivalence fails. Handicapping the strong bidder can *raise* revenue.
- **Many-to-one matching with couples** and the strategic behaviour of receiving sides.
- **Why the VCG revenue in Figure 43.2 was "too low"**: the set of payments that no coalition can object to is called the *core*, and modern spectrum auctions use core-selecting payment rules. The theory of the core belongs to cooperative games, which is where Ch. 44 begins.

Further reading

- V. Krishna, *Auction Theory*, 2nd ed., Academic Press, 2010. The standard graduate text; Chapters 2–5 cover everything in 43.3 for general distributions.
- P. Klemperer, *Auctions: Theory and Practice*, Princeton University Press, 2004. Includes the account of the European 3G auctions and why their designs mattered.
- P. Milgrom, *Putting Auction Theory to Work*, Cambridge University Press, 2004. Package auctions, clock auctions and the linkage principle from one of the designers of the US spectrum auctions.
- A. E. Roth, *Who Gets What — and Why*, Houghton Mifflin Harcourt, 2015. A readable account of matching markets, from medical residencies to kidney exchange.
- J. McMillan, "Selling spectrum rights", *Journal of Economic Perspectives* 8(3), 1994. The New Zealand story and early lessons in spectrum auction design.

Chapter 44 — Coalitions, Voting, and Collective Decisions

> **Level 2 · Track C — Game Theory and Strategic Intelligence Builds on:** Ch. 23, Ch. 39, Ch. 42, Ch. 43 · **Leads to:** Ch. 45, Ch. 96 · **Related:** Ch. 28, Ch. 38, Ch. 48 **Reading time:** ~25 min

44.1 The hook

When the European Economic Community was founded in 1958, its six members wrote a voting rule for the Council of Ministers. On many questions each country cast a weighted vote: France, West Germany and Italy 4 each, Belgium and the Netherlands 2 each, Luxembourg 1. There were 17 votes in all, and a proposal passed with 12. Luxembourg, the smallest member, had one vote out of seventeen, about 6% of the total. A fair-looking rule.

Here is the strange part. List every possible coalition of countries voting "yes", and ask in how many of them Luxembourg's vote *made the difference* between passing and failing. The answer is **none**. Not one. Luxembourg's vote could never change the outcome of a weighted vote, in any configuration of the other five. The country had a vote, but it had **no power**.

That puzzle is about *voting* power, but the same question hides in every joint venture. Three energy firms build a shared microgrid. Alone, firm A could earn 10 (in millions per year), firm B 20, and firm C nothing: it owns storage but no generation. Any two together earn more (A and B 40, A and C 50, B and C 60) and all three together earn 90. Firm C is worthless on its own and essential in partnership. How should they split the 90? Equally? In proportion to what each could earn alone (which gives C nothing)? Something else?

And the question hides in elections too. A hundred voters rank three candidates:

Voters	1st	2nd	3rd
40	A	C	B
35	B	C	A
20	C	B	A
5	C	A	B

Under plurality (most first choices), A wins. Under ranked-choice instant runoff, B wins. Under the Borda count, and in every head-to-head contest, C wins. Same voters, same ballots, three different "winners".

So the chapter asks: **when a group acts together, who really holds power, who deserves what, and whose choice is "the group's choice"?** Mathematics cannot settle what is fair. What it can do, remarkably, is pin down exactly which fairness rules are even possible.

44.2 The big idea

The previous chapters were *non-cooperative*: each player chose a strategy alone, and we asked what equilibrium emerged. Now we zoom out. **Cooperative game theory** does not model how players negotiate. It assumes groups can sign binding agreements, and asks only: what can each coalition achieve, and how should the proceeds be split?

The central object is a single function. For each coalition S of players, $v(S)$ is the total value that S can guarantee on its own. Everything else, the alliances, the haggling, the threats to walk away, is compressed into those numbers.

The key intuition is **marginal contribution**. Imagine the players arrive at a room one at a time, in random order. Each, on arrival, adds something to the coalition already there. Firm C arriving first adds 0. Arriving after A, it adds $50 - 10 = 40$. Arriving last, it adds $90 - 40 = 50$. Your fair share, says Lloyd Shapley, is your **average marginal contribution over all arrival orders**. That turns out to be the only rule satisfying a handful of very mild fairness axioms.

The same idea measures **power**: in a voting game, a country's "contribution" is 1 if its arrival turns a losing coalition into a winning one, and 0 otherwise. Luxembourg's average contribution is zero because its single vote never tips any total across the line.

Voting rules are a different kind of aggregation: not dividing money but combining rankings. Here the big idea is negative, and you have already met it in Ch. 42: Arrow's theorem says no ranking rule is perfect. So the practical question becomes "which imperfection do we prefer?", and the ranked-choice example shows the choice has teeth.

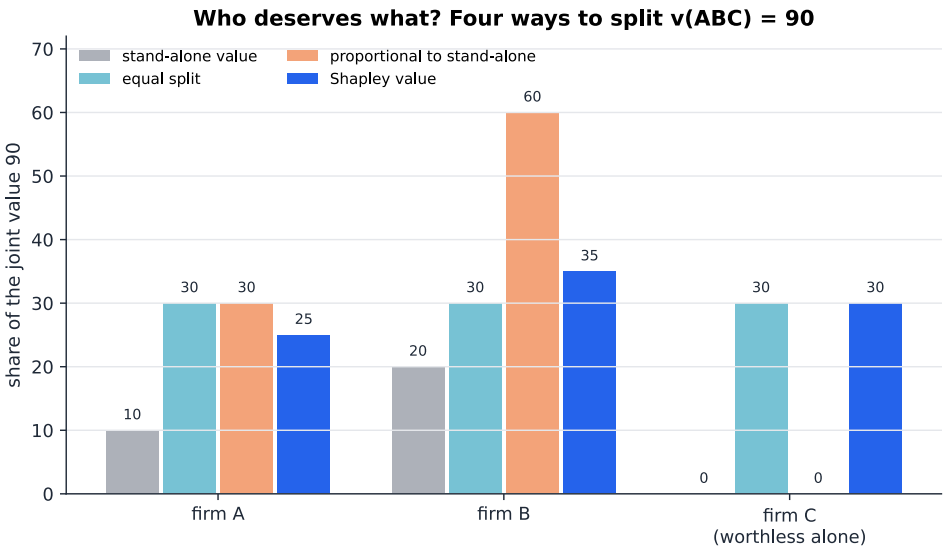


Figure 44.1 — Four ways to split the microgrid's 90. The Shapley value (25, 35, 30) rewards C for what it enables, not for what it earns alone.

44.3 The mathematics

Coalitional games and the core

Transferable-utility (TU) game

A set of players $N = \{1, \dots, n\}$ and a **characteristic function** $v : 2^N \to \mathbb{R}$ with $v(\varnothing) = 0$. The number $v(S)$ is the total payoff coalition S can secure and divide freely among its members. The game is **superadditive** if $v(S \cup T) \geq v(S) + v(T)$ for disjoint S, T.

An **allocation** is a vector $\mathbf{x} = (x_1, \dots, x_n)$ with $\sum_i x_i = v(N)$: the grand coalition's value is shared out completely.

The core

The **core** is the set of allocations that no coalition can object to:

$$\sum_{i \in S} x_i \;\geq\; v(S) \quad \text{for every } S \subseteq N.$$

If some S received less than $v(S)$, its members could walk away together and do better.

For the microgrid, the Shapley allocation $(25, 35, 30)$ is in the core: each firm gets more than it earns alone, and the pairs get $60 \geq 40$, $55 \geq 50$, $65 \geq 60$. But the core can be empty, and the smallest example is worth proving.

The three-player majority game has an empty core

Let $v(S) = 1$ if $|S| \geq 2$ and $v(S) = 0$ otherwise (any two of three can take the whole prize). No allocation lies in the core.

Proof. Suppose $\mathbf{x}$ were in the core. The three two-player constraints say $x_1 + x_2 \geq 1$, $x_1 + x_3 \geq 1$, $x_2 + x_3 \geq 1$. Adding them, $2(x_1 + x_2 + x_3) \geq 3$. But $x_1 + x_2 + x_3 = v(N) = 1$, so $2 \geq 3$, a contradiction. ∎

The proof is a hint of something bigger: it multiplies the coalition constraints by weights ($\frac{1}{2}$ each) that "cover" every player exactly once and finds them demanding more than exists.

? Exactly when is the core non-empty?

The majority game failed because a balanced family of coalitions, each weighted by $\frac{1}{2}$, together demanded $\frac{3}{2}$ of a prize of size 1. Is that the *only* way a core can be empty? The Bondareva–Shapley theorem says yes: the core is non-empty precisely when no balanced weighting of coalitions demands more than $v(N)$. The proof is a two-line consequence of a deeper fact about linear programs and their duals. Find that fact, and the theorem, in Ch. 48.

The Shapley value

Let π be an ordering of the players and P_i^π the set of players who come before i in π. Player i's marginal contribution in that order is $v(P_i^\pi \cup \{i\}) - v(P_i^\pi)$.

Shapley value

$$\phi_i(v) = \frac{1}{n!} \sum_{\pi} \Big[v\big(P_i^\pi \cup \{i\}\big) - v\big(P_i^\pi\big) \Big] \;=\; \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(n - |S| - 1)!}{n!} \Big[v(S \cup \{i\}) - v(S) \Big].$$

The two formulas agree, and the Shapley value is efficient

(a) The permutation average equals the subset sum above. (b) $\sum_i \phi_i(v) = v(N)$.

Proof. (a) Group the $n!$ orderings by the set S of players who precede i. An ordering has predecessors exactly S when it lists the $|S|$ members of S in any order, then i, then the remaining $n - |S| - 1$ players in any order. That is $|S|!\,(n - |S| - 1)!$ orderings, each contributing the same marginal term $v(S \cup \{i\}) - v(S)$. Dividing by $n!$ gives the subset formula. The counting is the standard one of Ch. 28.

(b) Fix one ordering $\pi = (i_1, i_2, \dots, i_n)$. The marginal contributions along it telescope:

$$\big[v(\{i_1\}) - v(\varnothing)\big] + \big[v(\{i_1, i_2\}) - v(\{i_1\})\big] + \cdots + \big[v(N) - v(N \setminus \{i_n\})\big] = v(N) - v(\varnothing) = v(N).$$

So the contributions of all players sum to $v(N)$ in *every* ordering. Averaging over orderings preserves this. ■

Here is the microgrid in full. Each row is an arrival order; entries are marginal contributions.

Order	A	B	C
A, B, C	10	30	50
A, C, B	10	40	40
B, A, C	20	20	50
B, C, A	30	20	40
C, A, B	50	40	0
C, B, A	30	60	0
Average	**25**	**35**	**30**

Each row sums to 90, as the proof promised. Shapley characterised his value by four axioms: **efficiency** (shares add to $v(N)$), **symmetry** (interchangeable players get equal shares), **null player** (a player who never adds anything gets nothing), and **additivity** (the value of the sum of two games is the sum of the values). He proved that **exactly one** rule satisfies all four.

Four innocent axioms, and no freedom left at all. Yet additivity is the axiom economists argue about most: why should fairness in a *sum* of two unrelated games be the sum of the fair shares? The uniqueness proof, and the values you get by dropping additivity, are in Ch. 96.

Power in weighted voting

A **weighted voting game** $[q;\ w_1, \dots, w_n]$ has $v(S) = 1$ if $\sum_{i \in S} w_i \geq q$ and 0 otherwise. The Shapley value of such a game is the **Shapley–Shubik power index**: the probability that player i is **pivotal**, the one whose arrival first pushes the total to q. A cousin, the **Banzhaf index**, counts the coalitions in which i is a **swing** (the coalition wins with i and loses without), instead of orderings.

For the 1958 Council, $[12;\ 4, 4, 4, 2, 2, 1]$, enumeration of all 720 orderings gives Shapley–Shubik indices $\frac{7}{30}$ for each large country, $\frac{3}{20}$ for Belgium and the Netherlands, and 0 for Luxembourg. Banzhaf counts are 10, 10, 10, 6, 6, 0 swings. Luxembourg's zero has a one-line proof.

Why Luxembourg never swings

Luxembourg is a swing in coalition $S \ni$ Lu exactly when the others in S have weight W with $W < 12$ but $W + 1 \geq 12$, i.e. $W = 11$. Every other country's weight is even ($4, 4, 4, 2, 2$), so W is always even and never 11. ■

Notice that the weights are a terrible guide to power: Luxembourg had $\frac{1}{17} \approx 5.9\%$ of the votes and 0% of the power, while Belgium had $\frac{2}{17} \approx 11.8\%$ of the votes and 15% of the (Shapley–Shubik) power.

Voting rules

With rankings instead of money, a **voting rule** maps a profile of rankings to a winner. Four common ones:

- **Plurality:** most first choices wins.
- **Borda count:** with m candidates, a candidate gets $m - 1$ points for each first place, $m - 2$ for each second, and so on.
- **Instant-runoff (ranked-choice):** repeatedly eliminate the candidate with fewest first choices, moving each of its ballots to its next surviving choice, until someone has a majority.
- **Condorcet:** the candidate who beats every other head to head, if one exists (Figure 42.1 shows it need not).

The hook's profile separates them all. Plurality: A 40, B 35, C 25, so A. Borda (2, 1, 0 points): $\text{A} = 2 \cdot 40 + 5 = 85$, $\text{B} = 2 \cdot 35 + 20 = 90$, $\text{C} = 40 + 35 + 2 \cdot 25 = 125$, so C. Head to head, C beats A by 60 to 40 and B by 65 to 35: C is the Condorcet winner. And instant runoff eliminates C first:

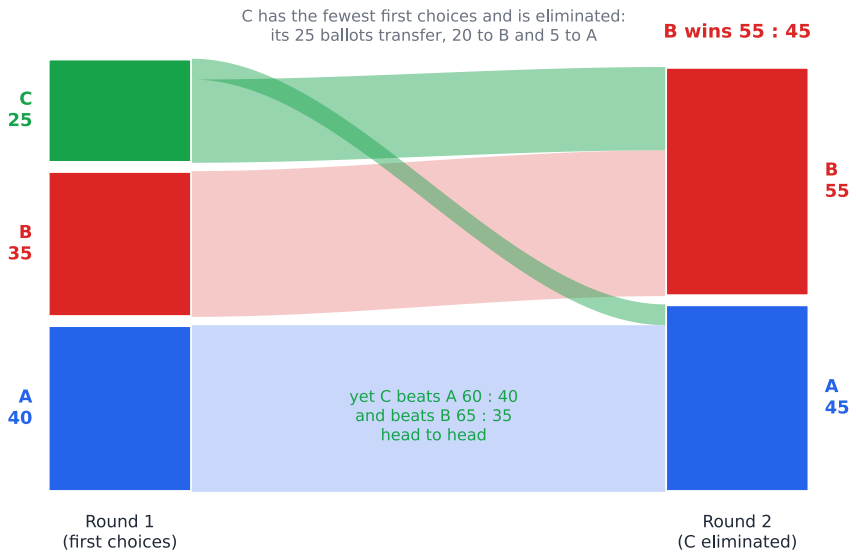


Figure 44.2 — Instant runoff on the hook's ballots. The candidate everyone would accept as a compromise is the first one thrown out.

This is the "squeezed centre" effect: a broadly acceptable candidate who is few voters' *first* choice can be eliminated before her broad support is ever counted. Arrow's theorem (Ch. 42) guarantees every rule has some such flaw. For **two** candidates, by contrast, May's theorem says simple majority is the only rule that treats voters equally, treats candidates equally, and responds positively to extra support. All the trouble starts at three.

Bargaining

Two players can split a surplus if they agree; if they don't, they get disagreement payoffs d_1, d_2. Nash (1950) proposed four axioms (efficiency, symmetry, invariance to rescaling each player's utility, and independence of irrelevant alternatives) and proved they single out the allocation that maximises the **Nash product**

$$(u_1 - d_1)(u_2 - d_2) \quad \text{over feasible agreements with } u_1 \ge d_1,\ u_2 \ge d_2.$$

Splitting a unit surplus

Feasible agreements are $u_1 = x$, $u_2 = 1 - x$. Maximise $g(x) = (x - d_1)(1 - x - d_2)$. Then $g'(x) = (1 - x - d_2) - (x - d_1) = 0$ gives

$$x^* = \frac{1 + d_1 - d_2}{2}, \qquad 1 - x^* = \frac{1 - d_1 + d_2}{2},$$

and $g'' = -2 < 0$ confirms a maximum. Each player gets her disagreement payoff plus **half the surplus** $1 - d_1 - d_2$. A better outside option is worth exactly half its value at the table.

Compare this with the two-player Shapley value: $\phi_1 = \tfrac{1}{2}v(\{1\}) + \tfrac{1}{2}\big[v(N) - v(\{2\})\big] = v(\{1\}) + \tfrac{1}{2}\big[v(N) - v(\{1\}) - v(\{2\})\big]$. With $d_i = v(\{i\})$ it is the *same* split. Two very different philosophies, averaging marginal contributions and maximising a product, agree whenever only two players are involved.

? Where does the Nash product come from?

Nash's solution falls out of axioms, not out of any model of people actually haggling. Yet if two players alternate offers, and each round of delay shrinks the pie a little, the unique equilibrium of that very non-cooperative game converges to the Nash bargaining solution as the delay shrinks to zero. Why should an axiom about fairness and a game about impatience give the same answer? Rubinstein's alternating-offers model is worked out in Ch. 96.

44.4 The algorithm

Exact Shapley values cost $n!$ orderings or, with the subset formula, 2^{n-1} coalitions per player: $O(n\,2^n)$ evaluations of v. That is fine for $n \le 20$ and hopeless beyond. Two escapes:

1. **Monte Carlo over orderings** (Ch. 38). Sample random permutations and average marginal contributions. Each sample is an unbiased estimate, so the error falls like $1/\sqrt{m}$ for m samples. This is how "SHAP" explanations of machine-learning models approximate the Shapley value of each input feature, with "players" being features and $v(S)$ the model's prediction using only the features in S.
2. **Dynamic programming for weighted voting.** Player i swings in S when the other members' weight W satisfies $q - w_i \le W \le q - 1$. So count the subsets of the *other* players by total weight, a subset-sum count:

```
def banzhaf_swings(weights, q):
    """Number of coalitions in which each player is a swing. O(n^2 * total weight)."""
    total = sum(weights)
    swings = []
    for i, wi in enumerate(weights):
        count = [1] + [0] * total            # count[W] = #subsets of others with weight W
        for j, wj in enumerate(weights):
            if j != i:
                for W in range(total, wj - 1, -1):
                    count[W] += count[W - wj]
        swings.append(sum(count[W] for W in range(max(q - wi, 0), q)))
    return swings

print(banzhaf_swings([4, 4, 4, 2, 2, 1], 12))   # [10, 10, 10, 6, 6, 0]
```

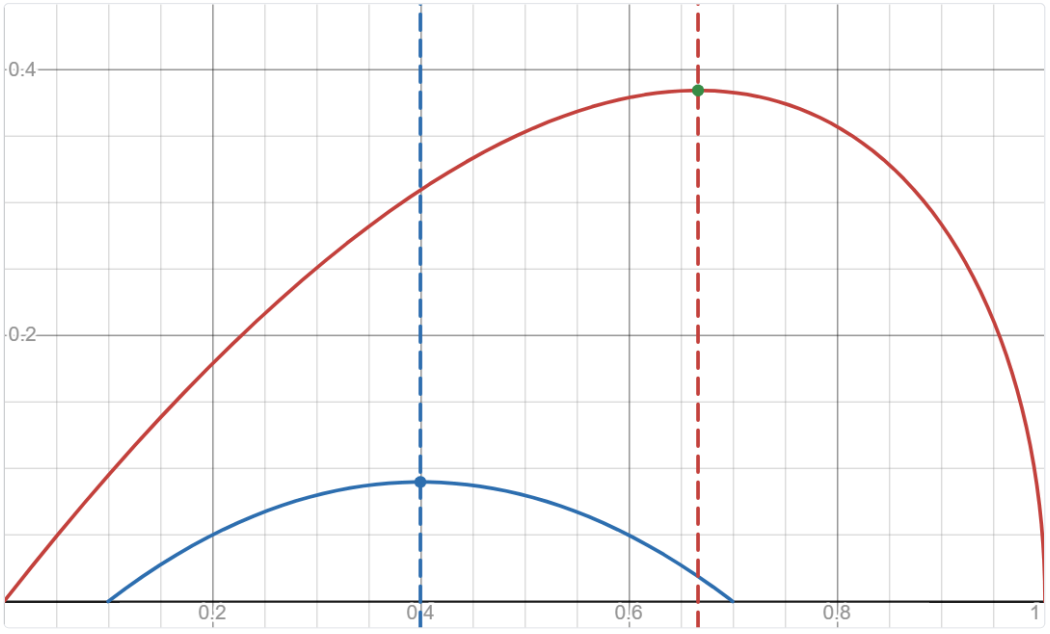
The running time is $O(n^2 \sum w)$: polynomial in the weights' *size*, exponential in their number of digits, so "pseudo-polynomial". With large integer weights (populations, say), computing exact power indices is NP-hard, a fact that connects this chapter to Ch. 30. A similar DP, also tracking coalition size, gives the Shapley–Shubik index.

Instant runoff is a loop: tally first surviving choices, stop if someone has a majority, else drop the lowest and repeat. It runs in $O(m \cdot B)$ for m candidates and B ballots, but it hides a numerical trap: **ties for last place**. Different tie-breaking rules can change the winner, and real election laws spend pages on them.

Failure cases. The core may be empty, so "find a core allocation" can have no answer; Shapley values always exist but may lie outside the core and then invite defection. Monte Carlo Shapley estimates for players with tiny values can be swamped by noise from large ones; stratify by position in the ordering to reduce variance.

44.5 The experiment

A. Bargain with a risk-averse partner. Blue is the Nash product for splitting 1 with disagreement points $d_1 = 0.1$, $d_2 = 0.3$. Red is player 1 bargaining with a *risk-averse* player 2 whose utility for money y is $\sqrt{y}$, and disagreement points 0.



Blue peaks at $x = \frac{1+0.1-0.3}{2} = 0.4$. Red peaks at $x = \frac{2}{3}$: the risk-averse partner, who fears disagreement more, concedes more.

Change $d_1 = 0.1$ to 0.3 in the blue curve (and its domain) and predict the new peak before you look. Then replace $\sqrt{1-x}$ by $(1-x)^{0.25}$ (even more risk-averse): how much more does player 1 get?

B. Measure power, split value, count votes. Change `v`, the voting weights and quota, and the ballots.

```
import itertools
import numpy as np

def shapley(players, v):
    phi = dict.fromkeys(players, 0.0)
    orders = list(itertools.permutations(players))
    for order in orders:
        seen = frozenset()
        for p in order:
            phi[p] += (v(seen | {p}) - v(seen)) / len(orders)
            seen = seen | {p}
    return {p: round(x, 4) for p, x in phi.items()}

grid = {"": 0, "A": 10, "B": 20, "C": 0, "AB": 40, "AC": 50, "BC": 60, "ABC": 90}
print("microgrid:", shapley("ABC", lambda S: grid["".join(sorted(S))]))

w = {"Fr": 4, "De": 4, "It": 4, "Be": 2, "Nl": 2, "Lu": 1}; q = 12
print("1958 power:", shapley(list(w), lambda S: float(sum(w[p] for p in S) >= q)))

ballots = [(40, "ACB"), (35, "BCA"), (20, "CBA"), (5, "CAB")]
def irv(ballots, alive):
    while True:
        tally = {c: sum(n for n, r in ballots if [x for x in r if x in alive][0] == c) for c in alive}
        top = max(tally, key=tally.get)
        if 2 * tally[top] > sum(tally.values()): return top, tally
        alive = alive - {min(tally, key=tally.get)}
borda = {c: sum(n * (2 - r.index(c)) for n, r in ballots) for c in "ABC"}
print("IRV:", irv(ballots, set("ABC")), " Borda:", borda)
```

Now try to give Luxembourg power. What is the smallest change to one weight, or to the quota, that makes its index positive? Then replace the microgrid's $v(\{A,B\}) = 40$ by 70 and check whether the Shapley value is still in the core.

44.6 The challenge

1. **Intuition.** In the "glove game", player L owns a left glove and players R1 and R2 each own a right glove; a pair is worth 1, a single glove nothing. Before computing anything, guess how the Shapley value splits the 1, and what the core contains. Explain why the two answers disagree so strongly, and which one you find fairer.
2. **Derivation.** Prove that for any weighted voting game, a player whose weight is 0 has Shapley–Shubik index 0, and that two players with equal weights have equal indices. Then find weights and a quota for three players in which the player with the *largest* weight has the *same* power as the smallest.
3. **Implementation.** Write a Monte Carlo Shapley estimator that samples m random orderings. For a 15-player weighted voting game with random weights, plot the error against the exact (DP) index as a function of m on log–log axes, and check the slope is $-\tfrac{1}{2}$.
4. **Open investigation.** Draw 10,001 voters whose rankings of 3 candidates are independent and uniformly random over the 6 orders ("impartial culture"). Estimate by simulation how often there is no Condorcet winner, and how often plurality, Borda and instant runoff each fail to elect the Condorcet winner when one exists. How do these frequencies change with 4, 5, 6 candidates, and with voters whose preferences are correlated along a left–right axis? Which rule would you recommend, and on what evidence?

44.7 Loose threads

We found three ways a group can "decide": dividing value (the core, the Shapley value), measuring power (Shapley–Shubik and Banzhaf), and aggregating rankings (plurality, Borda, runoff, Condorcet). Each gave crisp mathematics and an uncomfortable surprise: an empty core, a member with no power, a compromise candidate eliminated first.

What we did not tell you:

- **Why the Shapley value is unique**, and what changes if you drop additivity, is in Ch. 96; so is **Rubinstein's bargaining game**, which turns Nash's axioms into a story about impatience.
- **When the core exists** is a linear-programming duality fact, held over for Ch. 48.
- **How coalitions are actually formed**, who approaches whom and why some alliances are stable, was compressed into the single function v. Non-cooperative models of coalition formation are an active research area.

And one thread runs forward. Throughout this chapter we assumed players are *rational*: they calculate shares, compare outside options, and vote strategically or sincerely by choice. Yet in laboratory bargaining games, people reject "unfair" offers that would leave them better off, and they offer close to 50–50 when they could keep far more.

> **? Fairness from nowhere**
>
> If a rational responder should accept any positive offer, why do human beings across many cultures reject offers below about a quarter of the pie, and why do proposers rarely make them? One answer is that fairness norms are *strategies that survived*: in a population that plays bargaining games over and over, the rules people follow are shaped by which ones prosper. How a population can evolve toward fair splits without anyone choosing them is the story of Ch. 45.

Further reading

- M. Maschler, E. Solan, S. Zamir, *Game Theory*, Cambridge University Press, 2013. Chapters 15–20 cover TU games, the core, the Shapley value and bargaining with full proofs.

- A. E. Roth (ed.), *The Shapley Value: Essays in Honor of Lloyd S. Shapley*, Cambridge University Press, 1988. Includes the original paper and the power-index literature.
- A. D. Taylor, A. M. Pacelli, *Mathematics and Politics*, 2nd ed., Springer, 2008. An accessible tour of voting rules, power indices and Arrow's theorem.
- M. J. Osborne, A. Rubinstein, *Bargaining and Markets*, Academic Press, 1990. Nash's axiomatic solution and its strategic foundations.

Chapter 45 — Evolutionary Games and the Mathematics of Survival

> **Level 2 · Track C — Game Theory and Strategic Intelligence Builds on:** Ch. 11, Ch. 10, Ch. 39 · **Leads to:** Ch. 46, Ch. 83, Ch. 84 · **Related:** Ch. 40, Ch. 34, Ch. 53 **Reading time:** ~25 min

45.1 The hook

On the grassy hillsides of California's Coast Range lives a small lizard, *Uta stansburiana*, the side-blotched lizard. The males come in three throat colours, and each colour is a way of life.

- **Orange** males are big, aggressive and territorial. They hold large patches of ground with many females.
- **Blue** males are smaller. Each guards a single female closely and cooperates with blue neighbours.
- **Yellow** males look like females. They hold no territory at all; they sneak into other males' patches and mate on the sly.

Here is the trouble. Orange beats blue: a single blue mate-guarder cannot hold out against a bully with a large territory. Yellow beats orange: the bully's territory is too large to watch, and the sneakers slip in. Blue beats yellow: a male who never takes his eyes off his one female cannot be cuckolded.

That is rock, paper, scissors, played not by people but by genes. Field studies in the 1990s found that the three morphs do not settle down. The common colour of one breeding season becomes rarer later, its nemesis rises, and the cycle turns over the course of a few years. Nobody in this game is thinking. No lizard has a strategy in the sense of Ch. 39; each simply *is* a strategy, fixed at birth. And yet the population behaves as if it were solving a game.

There is an older puzzle too. Why do so many animals fight by ritual? Stags roar and walk side by side before they ever lock antlers. Snakes wrestle without biting. If victory brings a female or a territory, why doesn't every individual fight all-out? A population of polite ritual fighters looks like an open invitation to one ruthless mutant.

Both puzzles have the same mathematical shape. We drop the idea of a rational player and replace it with a **population**: a crowd of individuals, each hard-wired to one strategy, meeting at random, with payoff measured as reproductive success. Strategies that score above the average spread. Strategies that score below it fade. The question is no longer "what should I do?" but "what survives?"

The answers are surprisingly sharp. Selection can find a Nash equilibrium without anyone computing one. It can also cycle forever, or shatter the population against the edges of the simplex, depending on one inequality between two numbers. By the end of this chapter you will know which inequality.

45.2 The big idea

Picture a large population as a bag of coloured marbles: a fraction x_1 play strategy 1, a fraction x_2 play strategy 2, and so on. The vector $\mathbf{x} = (x_1, \dots, x_n)$ has non-negative entries summing to 1. It lives on the **simplex**, a line segment for two strategies, a triangle for three.

Every generation, individuals pair off at random. An individual of type i meets type k with probability x_k, so its expected payoff is

$$f_i(\mathbf{x}) = \sum_k a_{ik}\, x_k = (\mathbf{A}\mathbf{x})_i,$$

where $\mathbf{A}$ is the payoff matrix. The average payoff in the population is $\bar{f} = \mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x}$.

Now the one idea the chapter turns on. Payoff is fitness: the number of offspring. A type that does better than average grows *as a share of the population*. A type that does worse shrinks. Its growth rate is proportional to how far it is above the average:

$$\frac{\dot{x}_i}{x_i} = f_i(\mathbf{x}) - \bar{f}(\mathbf{x}).$$

That is the **replicator equation**. It is the "rich get richer, relative to the average" rule, and it has three features worth noticing before any algebra.

1. **Nobody optimises.** There is no argmax anywhere. Selection is a blind ratchet that only compares each type to the current average.
2. **The environment is the population.** Your fitness depends on who else is around, so the target moves as you climb towards it. That is what makes this a *game* and not ordinary optimisation.
3. **Extinct types stay extinct.** If $x_i = 0$ then $\dot{x}_i = 0$. Replicator dynamics cannot invent a strategy; only mutation can. So we will have to ask what happens when a rare mutant appears: can it invade?

That last question gives the second idea of the chapter, due to John Maynard Smith and George Price (1973). A strategy is **evolutionarily stable** if, once the whole population uses it, no small band of mutants playing anything else can invade. Think of an immune system rather than a chess player: the question is not whether the strategy is clever, but whether it can repel every intruder.

The analogy to hold on to is a thermostat. Hawks do well when hawks are rare (they bully the doves). Hawks do badly when hawks are common (they keep meeting each other and getting injured). So the hawk share is pushed up when it is low and down when it is high. Like a thermostat, the population settles at the share where the push is zero. That share turns out to be a mixed Nash equilibrium, discovered by a process that knows nothing about Nash.

45.3 The mathematics

The replicator equation and the simplex

Replicator dynamics

For a symmetric game with $n \times n$ payoff matrix $\mathbf{A}$, the replicator dynamics on the simplex $\Delta_n = \{\mathbf{x} \in \mathbb{R}^n : x_i \geq 0,\ \sum_i x_i = 1\}$ are

$$\dot{x}_i = x_i\big[(\mathbf{A}\mathbf{x})_i - \mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x}\big], \qquad i = 1, \dots, n.$$

Two quick checks show the definition makes sense. Summing over i gives $\sum_i \dot{x}_i = \bar{f} - \bar{f}\sum_i x_i = 0$ on the simplex, so shares keep summing to 1. And each face $x_i = 0$ is invariant, so the simplex is never left. This is a system of nonlinear ODEs of the kind studied in Section 11.9; we only need its rest points and their stability.

Adding a constant c to a column of $\mathbf{A}$ adds $c\,x_k$ to every f_i and to $\bar{f}$ alike, so it leaves the dynamics unchanged. Only payoff *differences* within a column matter. Keep this in mind: it will tell us which numbers in the rock–paper–scissors matrix actually drive the outcome.

Rest points are (almost) Nash equilibria

Folk theorem of evolutionary game theory (first part)

If $\mathbf{p}$ is a symmetric Nash equilibrium of the game, then $\mathbf{p}$ is a rest point of the replicator dynamics. Conversely, an interior rest point is a Nash equilibrium, and any rest point that is the limit of an interior trajectory is a Nash equilibrium.

Proof of the first two claims. $\mathbf{p}$ is a Nash equilibrium when every strategy in its support earns the same payoff and no strategy earns more (Ch. 39 gives this as the indifference principle). So for each i, either $p_i = 0$ or $(\mathbf{A}\mathbf{p})_i = \bar{f}$, and in both cases $\dot{x}_i = 0$. Conversely, at an interior rest point every $x_i > 0$, so $(\mathbf{A}\mathbf{x})_i = \bar{f}$ for all i. All pure strategies earn the same, so none is a profitable deviation. ■

The word "almost" in the heading matters. Every pure strategy is a rest point (a population of all-hawks stays all-hawks, because nothing else exists to grow), even when it is a terrible strategy. Rest points are cheap. *Stable* rest points are what we want.

Hawk and dove

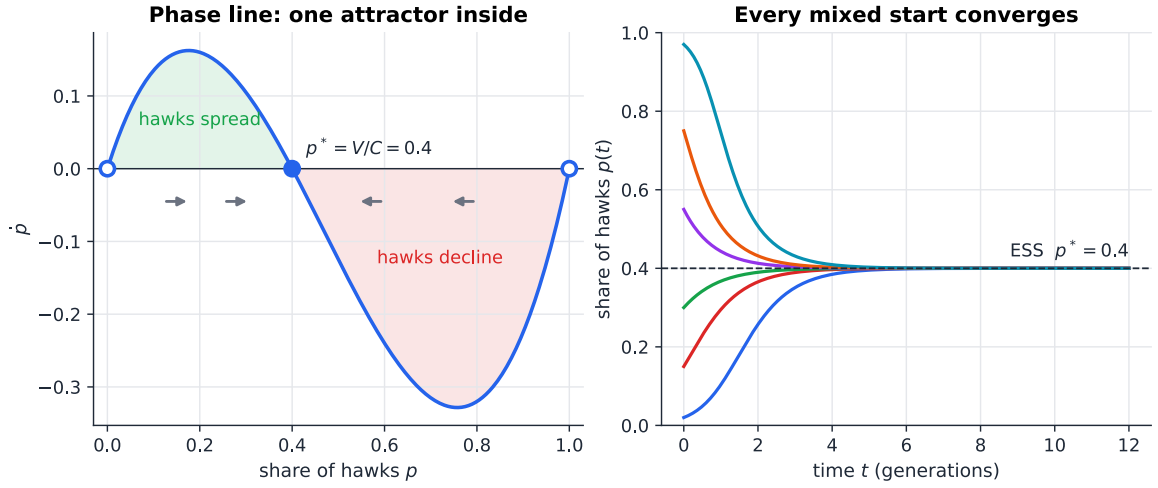
Two animals contest a resource worth V. A hawk escalates until injured or the opponent retreats; an injury costs C. A dove displays and retreats if the other escalates. Two doves share. With $C > V > 0$:

$$\mathbf{A} = \begin{pmatrix} \dfrac{V - C}{2} & V \\[2mm] 0 & \dfrac{V}{2} \end{pmatrix} \qquad \begin{matrix} \text{row 1: hawk} \\ \text{row 2: dove} \end{matrix}$$

Let p be the hawk share. Then $f_H = p\frac{V-C}{2} + (1-p)V$ and $f_D = (1-p)\frac{V}{2}$, so $f_H - f_D = \frac{V - Cp}{2}$. With two strategies, $f_H - \bar{f} = (1-p)(f_H - f_D)$, and the replicator equation collapses to a single ODE:

$$\dot{p} = \frac{p(1-p)(V - Cp)}{2}.$$

The rest points are $p = 0$, $p = 1$ and $p^* = V/C$. The derivative of the right-hand side at p^* is $-\frac{1}{2}p^*(1-p^*)C < 0$, so p^* is a stable node; at $p = 0$ the derivative is $V/2 > 0$ and at $p = 1$ it is $(C - V)/2 > 0$, so both pure states are unstable. For $V = 4$, $C = 10$ the attractor is $p^* = 0.4$ and the local rate is -1.2 per generation: a disturbance decays like $e^{-1.2t}$.



Left: the phase line $\dot{p}$ against the hawk share p for $V = 4$, $C = 10$; hawks grow below $p^* = 0.4$ and shrink above it. Right: six populations started anywhere strictly between 0 and 1 all converge to 40% hawks.

This answers the ritual-fighting puzzle in one line. All-dove is not stable: a hawk among doves earns $V = 4$ against the doves' $V/2 = 2$. All-hawk is not stable either: when injuries are costly ($C > V$), a dove among hawks earns 0 while hawks earn $(V - C)/2 = -3$. The population settles at a mixture, with the hawk share rising as the prize grows and falling as injuries get worse. Restraint is not kindness. It is what is left when escalation becomes too expensive.

Notice something unsettling. At $p^* = 0.4$ every individual earns 1.2. In an all-dove population every individual would earn 2. Selection drove the population to a state that is *worse for everyone* than one it cannot stay in. Evolution maximises nothing; this is the prisoner's dilemma of Ch. 40 wearing fur.

Evolutionary stability

Evolutionarily stable strategy (ESS)

A (possibly mixed) strategy $\mathbf{p}$ is an ESS if for every other strategy $\mathbf{q} \neq \mathbf{p}$:

1. $\mathbf{q}^{\mathsf{T}}\mathbf{A}\mathbf{p} \leq \mathbf{p}^{\mathsf{T}}\mathbf{A}\mathbf{p}$ (a Nash equilibrium: mutants do no better against the residents), and
2. if $\mathbf{q}^{\mathsf{T}}\mathbf{A}\mathbf{p} = \mathbf{p}^{\mathsf{T}}\mathbf{A}\mathbf{p}$, then $\mathbf{p}^{\mathsf{T}}\mathbf{A}\mathbf{q} > \mathbf{q}^{\mathsf{T}}\mathbf{A}\mathbf{q}$ (residents beat mutants in the rare mutant–mutant encounters).

Condition 2 is the tie-breaker: if a mutant is as good as the resident against residents, the resident must win against the mutant. An equivalent form (a standard result we quote rather than prove) is **local superiority**: $\mathbf{p}$ is an ESS if and only if

$$\mathbf{p}^{\mathsf{T}}\mathbf{A}\mathbf{x} > \mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} \quad \text{for all } \mathbf{x} \neq \mathbf{p} \text{ in a neighbourhood of } \mathbf{p} \text{ in } \Delta_n.$$

In words: near $\mathbf{p}$, the strategy $\mathbf{p}$ does better against the current population than the population does against itself.

For hawk–dove with $\mathbf{p} = (p^*, 1-p^*)$ and $\mathbf{x} = (p, 1-p)$, a short computation gives $\mathbf{p}^{\mathsf{T}}\mathbf{A}\mathbf{x} - \mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} = \frac{C}{2}(p - p^*)^2 > 0$, so V/C hawks is an ESS. (Check it yourself: the difference is $(p^* - p)(f_H - f_D)$ evaluated at $\mathbf{x}$.)

The central theorem: an ESS is an attractor

Why should a static "can't be invaded" condition say anything about the dynamics? Because of a clever Lyapunov function borrowed from information theory.

ESS implies asymptotic stability (Taylor and Jonker; Hofbauer, Schuster and Sigmund)

If $\mathbf{p}$ is an ESS, then $\mathbf{p}$ is an asymptotically stable rest point of the replicator dynamics.

Proof. Let S be the support of $\mathbf{p}$ (the strategies it uses) and consider, for $\mathbf{x}$ with $x_i > 0$ on S, the relative entropy

$$H(\mathbf{x}) = \sum_{i \in S} p_i \ln \frac{p_i}{x_i}.$$

This is the Kullback–Leibler divergence of Ch. 33: $H \geq 0$ with equality only at $\mathbf{x} = \mathbf{p}$ (Gibbs' inequality), so H is a candidate Lyapunov function. Differentiate along a trajectory, using $\dot{x}_i/x_i = (\mathbf{A}\mathbf{x})_i - \bar{f}$:

$$\begin{aligned} \dot{H} &= -\sum_{i \in S} p_i \frac{\dot{x}_i}{x_i} = -\sum_{i \in S} p_i \big[(\mathbf{A}\mathbf{x})_i - \mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x}\big] \\ &= -\big[\mathbf{p}^{\mathsf{T}}\mathbf{A}\mathbf{x} - \mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x}\big]. \end{aligned}$$

The last step uses $\sum_{i \in S} p_i = 1$. By local superiority, the bracket is strictly positive for $\mathbf{x} \neq \mathbf{p}$ near $\mathbf{p}$, so $\dot{H} < 0$ there. A function that is zero at $\mathbf{p}$, positive nearby, and strictly decreasing along trajectories forces nearby trajectories into $\mathbf{p}$: that is Lyapunov's theorem, the nonlinear cousin of the eigenvalue test in Section 10.9. ∎

The proof is worth rereading for its meaning. Selection steadily reduces the information-theoretic "surprise" between the population and the ESS. The population is, in a precise sense, *learning* $\mathbf{p}$. The converse fails: there are asymptotically stable rest points that are not ESSs, so the static notion and the dynamic one are not the same thing.

? Selection as learning

The relative entropy H decreasing along selection looks exactly like a Bayesian posterior concentrating on the truth, and the replicator equation looks exactly like the multiplicative-weights update you will meet in Ch. 46. Is natural selection literally an inference algorithm? The formal bridge between replicator dynamics and Bayesian updating, and its limits, is a thread picked up in Ch. 35 and Ch. 53.

Rock, paper, scissors: one inequality decides everything

Back to the lizards. Use a generalised rock–paper–scissors matrix in which a win pays $a > 0$ and a loss costs $b > 0$:

$$\mathbf{A} = \begin{pmatrix} 0 & -b & a \\ a & 0 & -b \\ -b & a & 0 \end{pmatrix} \qquad \text{(rows: Rock, Paper, Scissors)}.$$

By symmetry the unique interior rest point is the centre $\mathbf{p} = (\tfrac{1}{3}, \tfrac{1}{3}, \tfrac{1}{3})$. Every row and column sums to $a - b$. Write $\mathbf{x} = \mathbf{p} + \mathbf{d}$ with $\sum_i d_i = 0$. Then:

- $\mathbf{p}^{\mathsf{T}}\mathbf{A}\mathbf{x} = \frac{1}{3}\sum_k x_k (\text{column sum}) = \frac{a-b}{3}$;
- $\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} = (a-b)(x_1x_2 + x_2x_3 + x_3x_1)$, because each pair $\{i,k\}$ contributes a one way and $-b$ the other;
- $x_1x_2 + x_2x_3 + x_3x_1 = \frac{1}{2}\big(1 - \|\mathbf{x}\|^2\big) = \frac{1}{3} - \frac{1}{2}\|\mathbf{d}\|^2$.

Putting these together gives an exact identity, valid on the whole simplex, not only near the centre:

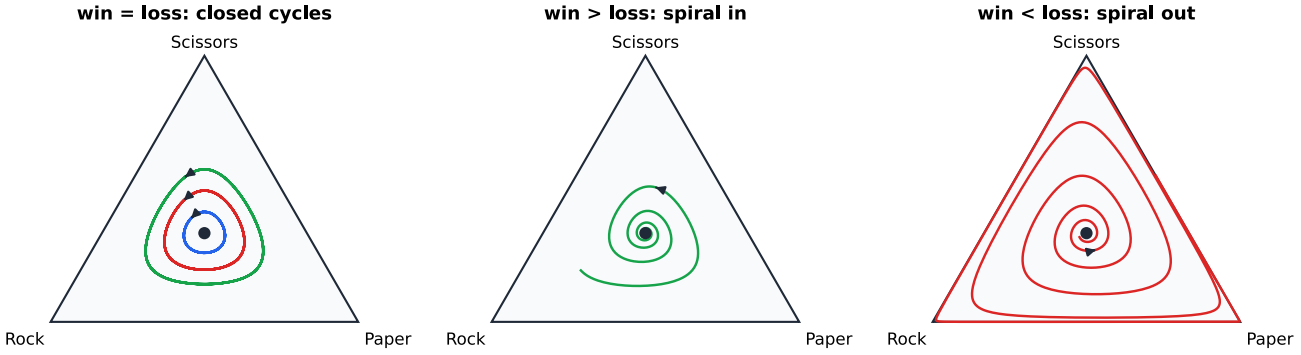
$$\mathbf{p}^{\mathsf{T}}\mathbf{A}\mathbf{x} - \mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} = \frac{a-b}{2}\,\|\mathbf{x}-\mathbf{p}\|^2.$$

Feed it into the Lyapunov computation, where now $H = -\ln 3 - \frac{1}{3}\ln(x_1x_2x_3)$:

$$\dot{H} = -\frac{a-b}{2}\,\|\mathbf{x}-\mathbf{p}\|^2.$$

Three regimes follow at once.

Case	Sign of $\dot{H}$	What happens	ESS?
$a > b$ (wins pay more than losses cost)	negative	every interior orbit spirals into the centre	yes
$a = b$ (zero-sum)	zero	$x_1x_2x_3$ is conserved; closed cycles forever	no
$a < b$ (losses hurt more)	positive	orbits spiral out towards the boundary	no



Replicator orbits on the strategy triangle. Left, $a = b = 1$: each orbit is a level curve of $x_1x_2x_3$ and cycles forever. Centre, $a = 1.5$, $b = 1$: the centre is an ESS and attracts. Right, $a = 1$, $b = 1.5$: the orbit winds outwards and hugs the edges, so each strategy in turn nearly goes extinct.

In the zero-sum case the cycle period near the centre follows from linearisation. The Jacobian at $\mathbf{p}$ restricted to the simplex has eigenvalues $\pm j/\sqrt{3}$ (check it with the eigenvalue tools of Ch. 10), so small cycles have period $2\pi\sqrt{3} \approx 10.9$ generations. Purely imaginary eigenvalues usually mean linearisation tells you nothing about stability; here the conserved quantity rescues us and proves the orbits really are closed.

The right-hand panel is the dramatic one. In a population where losing is worse than winning is good, the orbit approaches the boundary of the triangle: long stretches in which almost everyone plays one strategy, then a sudden takeover by its nemesis. In a finite population, one of those near-extinctions will eventually become a real one, and the cycle ends with a single survivor. Which survivor? That depends on luck, and luck is not in the replicator equation at all.

> **? When the deterministic model lies**
>
> The replicator equation treats the population as infinite, so a share of 10^{-30} still counts as alive. Real populations have N individuals, and a share below $1/N$ is extinction. How the deterministic cycles of the right-hand panel turn into a random walk that is absorbed at a corner, and how long that takes, is a question about Markov chains with absorbing states: see Ch. 34 and Ch. 36.

Population games beyond pairs

Nothing above required pairwise encounters. In a **population game** each strategy i has a fitness $f_i(\mathbf{x})$ that is any smooth function of the population state: drivers choosing routes, where your travel time depends on how many others chose the same road, or packets choosing paths in a network. The replicator equation $\dot{x}_i = x_i(f_i - \bar{f})$ still makes sense, and the folk theorem still holds. When the fitnesses come from a **potential**, $f_i = \partial\Phi/\partial x_i$, the dynamics climb Φ like a gradient flow (Section 23.5) and cannot cycle. Hawk–dove has a potential; rock–paper–scissors does not, which is exactly why it can cycle.

45.4 The algorithm

Simulating replicator dynamics takes a few lines of code, and it is easy to get wrong in a way that looks like biology.

```
REPLICATOR(A, x0, h, steps):
    x ← x0                                  # a point on the simplex
    for k = 1 .. steps:
        fit  ← A x                          # payoff of each pure strategy
        avg  ← xᵀ fit                       # population average
        x    ← x + h · x ⊙ (fit − avg)      # forward Euler step
        x    ← max(x, 0); x ← x / sum(x)    # guard against round-off
    return x
```

Each step costs one matrix–vector product, $O(n^2)$ for n strategies. The simplex is preserved exactly in exact arithmetic (the update sums to zero), and the clipping line only cleans up floating-point dust.

The failure case. Run this on zero-sum rock–paper–scissors ($a = b$), where the true orbits are closed. Forward Euler spirals *outwards*. The reason is the stability analysis of ODE solvers from Ch. 22 in miniature: near the centre the system behaves like $\dot{\mathbf{z}} = \mathbf{J}\mathbf{z}$ with eigenvalues $\pm j\omega$, $\omega = 1/\sqrt{3}$. One Euler step multiplies a mode by $1 + jh\omega$, whose magnitude is $\sqrt{1 + h^2\omega^2} > 1$. Over one unit of time the amplitude grows by roughly $e^{h\omega^2/2} = e^{h/6}$. With $h = 0.1$ and 200

generations that is a factor of about $e^{3.3} \approx 28$ in amplitude near the centre, and the orbit is flung against the edges of the triangle. A student who sees this plot and concludes "rock–paper–scissors populations are unstable" has discovered a property of Euler's method, not of lizards.

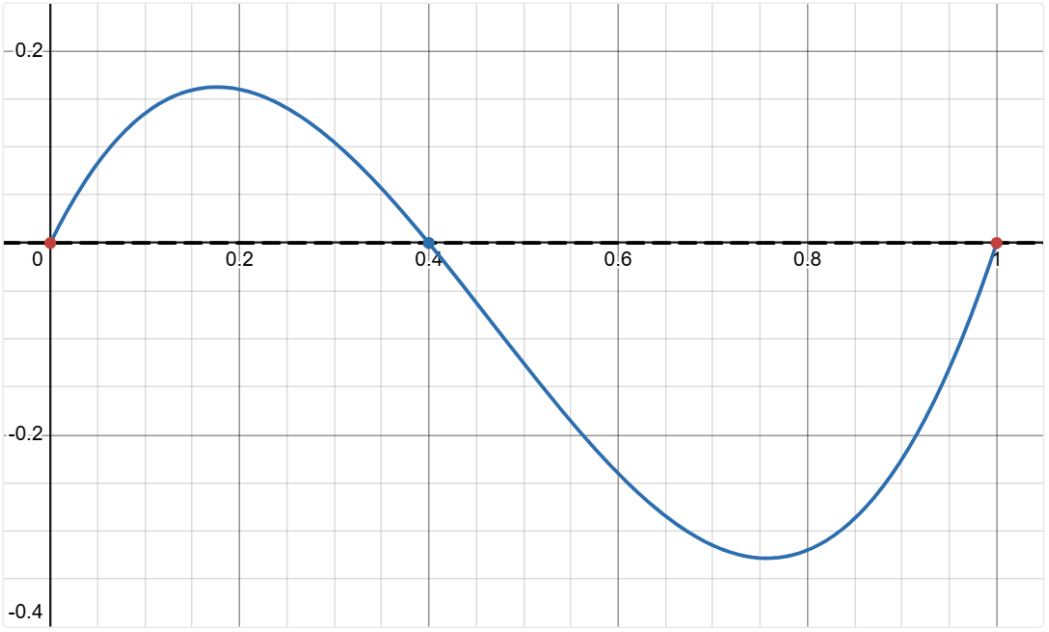
The fixes are standard:

- Use a higher-order integrator (RK4, or an adaptive solver). The drift does not vanish, but it drops from $O(h)$ to $O(h^4)$ per unit time.
- Monitor a conserved quantity. In the zero-sum case $x_1 x_2 x_3$ must stay constant; its drift is a free error meter.
- Beware the popular **discrete-time replicator** $x_i' = x_i \, \dfrac{c + f_i}{c + \bar{f}}$ with a background fitness $c > 0$. It is a perfectly good model of non-overlapping generations, but it is a *different* dynamical system. For zero-sum rock–paper–scissors it spirals outwards to the boundary, so its long-run behaviour disagrees with the continuous model.

Finding ESSs. For small games, enumerate supports. For each subset S of strategies, solve the linear indifference system $(\mathbf{A}\mathbf{p})_i = (\mathbf{A}\mathbf{p})_k$ for $i, k \in S$ with $\sum_{i \in S} p_i = 1$ (Ch. 8), keep solutions with $\mathbf{p} \geq 0$ that satisfy the Nash condition, then test the second ESS condition. That is 2^n subsets, fine for $n \leq 15$ and hopeless beyond. Deciding whether a game has an ESS at all is known to be computationally hard in general, so there is no shortcut hiding in the background.

45.5 The experiment

First, the hawk–dove phase line. The curve is $\dot{p} = p(1 - p)(V - Cp)/2$ with $V = 4$ and $C = 10$; the filled point is the ESS.



The phase line for hawk–dove. Change the 4 (the prize V) and both 10s (the injury cost C) and watch the ESS slide to V/C; set V above C and the interior root leaves the interval, so all-hawk wins.

Now the cycle. The script below integrates generalised rock–paper–scissors and plots the three shares and the product $x_1 x_2 x_3$ on a log scale.

```
import numpy as np
import matplotlib.pyplot as plt

# --- knobs ---------------------------------------------------------------
a, b = 1.0, 1.0             # win payoff and loss cost: try a=1.5 or b=1.5
h = 0.01                    # time step: try 0.1 and watch the cycle break
T = 200.0                   # total time in generations
x = np.array([0.5, 0.3, 0.2])   # starting shares of Rock, Paper, Scissors
method = "rk4"              # "euler" or "rk4"
# -------------------------------------------------------------------------

A = np.array([[0, -b, a], [a, 0, -b], [-b, a, 0]])

def f(x):
    fit = A @ x
    return x * (fit - x @ fit)

steps = int(T / h)
xs = np.empty((steps, 3))
for k in range(steps):
    xs[k] = x
    if method == "euler":
        x = x + h * f(x)
    else:
        k1 = f(x); k2 = f(x + h/2*k1); k3 = f(x + h/2*k2); k4 = f(x + h*k3)
        x = x + h/6 * (k1 + 2*k2 + 2*k3 + k4)

t = np.arange(steps) * h
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(10, 4))
for i, name in enumerate(["Rock", "Paper", "Scissors"]):
    ax1.plot(t, xs[:, i], label=name)
ax1.set_xlabel("generations"); ax1.set_ylabel("share"); ax1.legend()
ax2.semilogy(t, xs.prod(axis=1))
ax2.set_xlabel("generations"); ax2.set_ylabel("x1*x2*x3 (conserved if a=b)")
print("final shares:", xs[-1].round(4), " product:", xs[-1].prod())
plt.tight_layout(); plt.show()
```

Things to try, in order:

1. Run it as is. The product starts at $0.5 \times 0.3 \times 0.2 = 0.03$ and RK4 keeps it at 0.03 to about ten decimal places. The shares cycle forever.
2. Switch `method = "euler"`. With `h = 0.01` the product drifts down to about 0.025 in 200 generations; with `h = 0.1` it collapses to a few times 10^{-6} and Paper ends with over 99% of the population. Same model, different integrator, opposite conclusion.
3. Back on RK4, set `a = 1.5`. The product climbs towards its maximum $1/27 \approx 0.037$: the centre is an ESS. Then try `b = 1.5`: the product plunges towards zero and the population lives on the edges.
4. Starting share of `1e-12` for Scissors: how long until Scissors "comes back"? What would a real population of a million lizards do in that time?

45.6 The challenge

1. **Intuition.** In hawk–dove, the ESS hawk share V/C does not depend on how many individuals there are or how often they meet. Explain in words why a larger prize leads to *more* fighting at equilibrium, and why everyone at the ESS earns less than in an all-dove world, yet all-dove cannot last.
2. **Derivation.** For the two-strategy coordination game $\mathbf{A} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ find all rest points of the replicator dynamics, classify their stability, and determine which are ESSs. Then show directly from the definition that the interior Nash equilibrium is *not* an ESS.
3. **Implementation.** Write the support-enumeration ESS finder sketched in 45.4 for $n \times n$ matrices. Test it on hawk–dove, on the three regimes of rock–paper–scissors, and on a 4×4 random matrix. Then compare: for 1,000 random 3×3 games, what fraction have an ESS, and in what fraction does a random interior start converge to one under RK4?
4. **Open investigation.** Add a fourth strategy to rock–paper–scissors, for example a "Loner" who opts out of contests for a fixed payoff σ. Explore numerically how the long-run behaviour changes as σ varies: do you see rest points, cycles, or something irregular? Is there a value of σ at which the qualitative behaviour changes abruptly? (There is no single right answer; the shape of what you find is the point.)

45.7 Loose threads

This chapter took a deliberately narrow view: infinite populations, random pairing, payoffs that never change, and no mutation beyond the occasional invader. Every one of those assumptions hides a story.

We showed that the centre of rock–paper–scissors either attracts, repels, or is ringed by closed orbits. With three strategies that is the whole menu, because a flow on a two-dimensional triangle cannot do anything wilder than settle or cycle (the Poincaré–Bendixson theorem). With four strategies, the triangle becomes a tetrahedron, and that safety net disappears.

> **? Can selection be chaotic?**
>
> With four or more strategies the replicator equation is a three-dimensional flow, and nothing forbids a strange attractor. There are explicit four-strategy games whose replicator orbits never settle and never repeat, with sensitive dependence on the initial mix of strategies. What makes a flow chaotic, and how you would recognise it in one of these population games, is the subject of Ch. 84; the change of variables that turns any replicator system into a Lotka–Volterra predator–prey model, and lets you import its theory wholesale, waits in Ch. 83.

The second loose thread is the mystery of finite populations already raised in 45.3: drift, absorption and fixation probabilities. The third is the word "learning". The replicator update multiplies each share by a factor that grows with its payoff. Keep that picture in mind as you turn the page: in Ch. 46 the same multiplicative rule reappears, run not by genes but by an algorithm facing an adversary, and there it comes with a guarantee that no amount of bad luck can break.

We have also said nothing about where the payoff matrix comes from. In the lizards, the matrix is not fixed: it depends on density, on the season, on the mix of females. Allowing the game itself to evolve in response to the population is one of the active frontiers where evolutionary game theory meets ecology.

Further reading

- J. Maynard Smith, *Evolution and the Theory of Games*, Cambridge University Press, 1982. The founding text, readable and full of animal examples.
- J. Hofbauer and K. Sigmund, *Evolutionary Games and Population Dynamics*, Cambridge University Press, 1998. The standard mathematical treatment of replicator dynamics, ESS and Lotka–Volterra equivalence.
- M. A. Nowak, *Evolutionary Dynamics: Exploring the Equations of Life*, Harvard University Press, 2006. Finite populations, spatial games and cooperation, with a light touch.
- W. H. Sandholm, *Population Games and Evolutionary Dynamics*, MIT Press, 2010. Population games, potential games and the many dynamics beyond replicator.

Chapter 46 — Adversarial Decision-Making

> **Level 2 · Track C — Game Theory and Strategic Intelligence Builds on:** Ch. 23, Ch. 19, Ch. 39 · **Leads to:** Ch. 48, Ch. 52, Ch. 53, Ch. 54 · **Related:** Ch. 45, Ch. 77, Ch. 30 **Reading time:** ~25 min

46.1 The hook

A penalty kick takes less time than the goalkeeper needs to react. The keeper must commit to a side before the ball is struck, and the kicker must choose a side knowing the keeper is guessing. Suppose a kicker scores with these probabilities:

	keeper dives left	keeper dives right
kick left	0.60	0.95
kick right	0.90	0.70

Shooting to the left is strong if the keeper goes the wrong way, but this keeper is very good at saving left-side kicks. What should the kicker do?

Here is a strange offer. The keeper's coach has hacked the kicker's phone and read the kicker's game plan. Before the kick, the keeper will know exactly how the kicker intends to decide. Should the kicker panic?

If the plan is "always kick right", yes: the keeper dives right and the scoring chance drops to 0.70. If the plan is "always kick left", the chance drops to 0.60. Any fixed choice can be punished. But suppose the plan is: *roll a die-like device and kick left with probability exactly* $4/11$. The keeper reads it, thinks hard, and discovers that it does not matter which way they dive. The scoring chance is 0.7909 either way. The leak has cost the kicker nothing.

That is the miracle of this chapter, and it is not a trick of this particular table. In any two-player game where one side's gain is the other's loss, there is a randomised strategy you can publish to the world and still guarantee the best result you could ever have hoped for against a perfect opponent. John von Neumann proved it in 1928, and he later said that until it was proved he saw nothing worth publishing in game theory.

The same idea runs a long way past football. A chess program assumes the opponent always finds the best reply. A spam filter assumes the spammer adapts to every rule. A portfolio manager follows a panel of expert advisers, some of whom are wrong in ways designed to fool them. A robust controller is tuned for the worst disturbance, not the average one. In each case the question is the same:

Can you make decisions that are guaranteed good, against an opponent who knows exactly how you make them?

The answer is yes, in a precise and surprising sense. And the guarantee comes from an algorithm so simple it fits on two lines.

46.2 The big idea

Ordinary optimisation (Ch. 23) treats the world as fixed: the objective function sits still while you climb it. Probability (Ch. 19) treats the world as random but indifferent: the dice do not care what you chose. Adversarial thinking takes the third view. Whatever you do, assume the world responds in the way that hurts you most.

That sounds paranoid, and sometimes it is. But it buys something priceless: a **guarantee**. If your plan is good against the worst case, it is good against every case. You never need a model of your opponent.

Three ideas make the guarantee useful rather than hopeless.

Pessimism has a value. For each plan, imagine the opponent's best reply and record what you get. Pick the plan whose worst case is best. That number, the **maxmin** value, is what you can secure no matter what. The opponent can do the same from their side, and secure the **minmax**. The gap between the two numbers is the territory the players are actually fighting over.

Randomness is armour. If you are predictable, an adversary who knows your plan exploits it. A coin flip the adversary can *read* but cannot *predict* closes the gap. The opponent knows the probabilities but not the outcome, and against the right probabilities every reply is equally bad for them.

Regret replaces the impossible benchmark. In repeated play you usually cannot beat an adaptive adversary in absolute terms, because they can make every round costly. But you can compare yourself to a humbler target: the best *single* fixed action you could have chosen with hindsight. The shortfall against that target is your **regret**. A remarkable family of algorithms guarantees that regret grows more slowly than the number of rounds, against *any* sequence of losses, chosen by *anyone*. Average regret per round goes to zero. Nobody can stop it.

The surprise at the end of 46.3 is that the third idea proves the first two. Two players who each just try to keep their regret small end up, on average, playing von Neumann's equilibrium. Learning finds the minimax solution without ever being told it exists.

46.3 The mathematics

Zero-sum games and their two values

> **Two-player zero-sum game**
>
> Player 1 (rows) chooses $i \in \{1, \dots, m\}$, player 2 (columns) chooses $k \in \{1, \dots, n\}$, and player 1 receives a_{ik} while player 2 receives $-a_{ik}$. Mixed strategies are probability vectors $\mathbf{x} \in \Delta_m$, $\mathbf{y} \in \Delta_n$, and the expected payoff to player 1 is $\mathbf{x}^{\mathsf{T}} \mathbf{A} \mathbf{y}$.

Player 1 can guarantee

$$\underline{v} = \max_{\mathbf{x} \in \Delta_m} \min_{\mathbf{y} \in \Delta_n} \mathbf{x}^{\mathsf{T}} \mathbf{A} \mathbf{y}$$

(choose $\mathbf{x}$ first, then let the opponent react), and player 2 can hold player 1 down to

$$\overline{v} = \min_{\mathbf{y} \in \Delta_n} \max_{\mathbf{x} \in \Delta_m} \mathbf{x}^{\mathsf{T}} \mathbf{A} \mathbf{y}.$$

For fixed $\mathbf{x}$ the inner minimum of a linear function of $\mathbf{y}$ is attained at a pure strategy, so $\min_{\mathbf{y}} \mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{y} = \min_k (\mathbf{x}^{\mathsf{T}}\mathbf{A})_k$. The opponent never needs to randomise in *reply*. Randomisation matters only for whoever moves first.

Weak minimax inequality

For every matrix, $\underline{v} \le \overline{v}$. Moving second never hurts.

Proof. Fix any $\mathbf{x}$ and $\mathbf{y}$. Then

$$\min_{\mathbf{y}'} \mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{y}' \;\le\; \mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{y} \;\le\; \max_{\mathbf{x}'} \mathbf{x}'^{\mathsf{T}}\mathbf{A}\mathbf{y}.$$

The left side depends only on $\mathbf{x}$ and the right side only on $\mathbf{y}$, and the inequality holds for every pair. So the largest possible left side is at most the smallest possible right side: $\max_{\mathbf{x}} \min_{\mathbf{y}'} \le \min_{\mathbf{y}} \max_{\mathbf{x}'}$. $\blacksquare$

With **pure** strategies only, the inequality is often strict. In matching pennies ($a_{ik} = +1$ if the choices match, -1 otherwise) the pure maxmin is -1 and the pure minmax is $+1$: whoever must reveal a pure choice first loses. Mixing at $\tfrac{1}{2}$ each brings both values to 0.

The penalty kick, solved

Let the kicker kick left with probability q. Against a keeper diving left the kicker scores $0.60q + 0.90(1-q) = 0.90 - 0.30q$; against a keeper diving right, $0.95q + 0.70(1-q) = 0.70 + 0.25q$. The keeper picks the smaller of the two, so the kicker maximises the lower envelope of two lines. One line falls and one rises, so the maximum sits where they cross:

$$0.90 - 0.30q = 0.70 + 0.25q \quad\Longrightarrow\quad q^{*} = \frac{0.20}{0.55} = \frac{4}{11} \approx 0.364, \qquad v = 0.90 - 0.30 \cdot \tfrac{4}{11} \approx 0.7909.$$

The same calculation from the keeper's side, diving left with probability r, gives $0.95 - 0.35r = 0.70 + 0.20r$, so $r^{*} = 5/11 \approx 0.455$ and again $v \approx 0.7909$. Both pessimists arrive at the same number. The kicker can guarantee at least 0.7909; the keeper can guarantee at most 0.7909. Here $\underline{v} = \overline{v}$, and neither side benefits from knowing the other's plan.

Notice the counter-intuitive detail: the kicker's left side is the *weaker* option against a correct guess, yet the kicker still uses it about a third of the time. The point of kicking left is not to score on the left. It is to stop the keeper from leaning right.

Von Neumann's theorem and its linear program

Minimax theorem (von Neumann, 1928)

For every finite matrix $\mathbf{A}$,

$$\max_{\mathbf{x} \in \Delta_m} \min_{\mathbf{y} \in \Delta_n} \mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{y} \;=\; \min_{\mathbf{y} \in \Delta_n} \max_{\mathbf{x} \in \Delta_m} \mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{y} \;=\; v.$$

The common value v is the **value of the game**; any pair of optimisers $(\mathbf{x}^{*}, \mathbf{y}^{*})$ is a Nash equilibrium, and every Nash equilibrium is such a pair.

The row player's problem is a linear program of the kind solved in Section 23.3:

$$\begin{aligned} \text{maximise} \quad & t \\ \text{subject to} \quad & (\mathbf{x}^{\mathsf{T}}\mathbf{A})_k \ge t \quad (k = 1, \dots, n), \\ & \textstyle\sum_i x_i = 1, \quad \mathbf{x} \ge \mathbf{0}. \end{aligned}$$

The column player's problem is another LP, and the two are *dual* to each other. Strong duality, the deep theorem of Ch. 48, says their optimal values coincide, which is the minimax theorem. That route is short but leans on machinery we have not built. Instead we will prove the theorem with nothing more than an inequality about exponentials, and the proof will hand us an algorithm as a bonus.

Regret and the Hedge algorithm

The setting is **online learning with experts**. There are n actions ("experts"). In each round $t = 1, \dots, T$:

1. you choose a probability vector $\mathbf{p}_t \in \Delta_n$;
2. an adversary, who may know everything about your algorithm and the past, chooses a loss vector $\boldsymbol{\ell}_t \in [0,1]^n$;
3. you suffer expected loss $\mathbf{p}_t^{\mathsf{T}}\boldsymbol{\ell}_t$ and see the whole vector $\boldsymbol{\ell}_t$.

Regret

$$R_T = \sum_{t=1}^{T} \mathbf{p}_t^{\mathsf{T}}\boldsymbol{\ell}_t \;-\; \min_{i} \sum_{t=1}^{T} \ell_{t,i}.$$

Your total loss, minus the total loss of the best single action in hindsight.

The obvious algorithm, **Follow the Leader** (put all your weight on the action with the smallest loss so far), is a disaster. With two actions, let the first losses be $(\frac{1}{2}, 0)$ and then alternate $(0,1), (1,0), (0,1), \ldots$. The leader always switches to the action about to lose. FTL loses 1 almost every round, while each fixed action loses only about half the time, so $R_T \approx T/2$: linear regret, the worst possible order. The flaw is that FTL is deterministic and jumps; the adversary reads it and punishes each jump.

Hedge (also called multiplicative weights or exponential weights) softens the jump. Keep a weight per action, start at $w_{1,i} = 1$, and after each round multiply:

$$w_{t+1,i} = w_{t,i}\, e^{-\eta\, \ell_{t,i}}, \qquad \mathbf{p}_t = \frac{\mathbf{w}_t}{\sum_i w_{t,i}}.$$

Actions that lose are penalised exponentially but never killed. Compare the replicator equation of Ch. 45: shares multiplied by a factor depending on payoff. The resemblance is not a coincidence.

Regret bound for Hedge

For any adversary and any $\eta > 0$,

$$R_T \le \frac{\ln n}{\eta} + \frac{\eta T}{2}.$$

With $\eta = \sqrt{2 \ln n / T}$ this gives $R_T \le \sqrt{2T \ln n}$, so the average regret $R_T/T \le \sqrt{2 \ln n / T} \to 0$.

Proof. Track the total weight $W_t = \sum_i w_{t,i}$, with $W_1 = n$. Two estimates squeeze $\ln W_{T+1}$.

Upper bound. For $z \ge 0$, $e^{-z} \le 1 - z + z^2/2$. With $z = \eta \ell_{t,i} \in [0, \eta]$:

$$\frac{W_{t+1}}{W_t} = \sum_i p_{t,i}\, e^{-\eta \ell_{t,i}} \le 1 - \eta\, \mathbf{p}_t^{\mathsf{T}} \boldsymbol{\ell}_t + \frac{\eta^2}{2} \sum_i p_{t,i} \ell_{t,i}^2 \le 1 - \eta\, \mathbf{p}_t^{\mathsf{T}} \boldsymbol{\ell}_t + \frac{\eta^2}{2}.$$

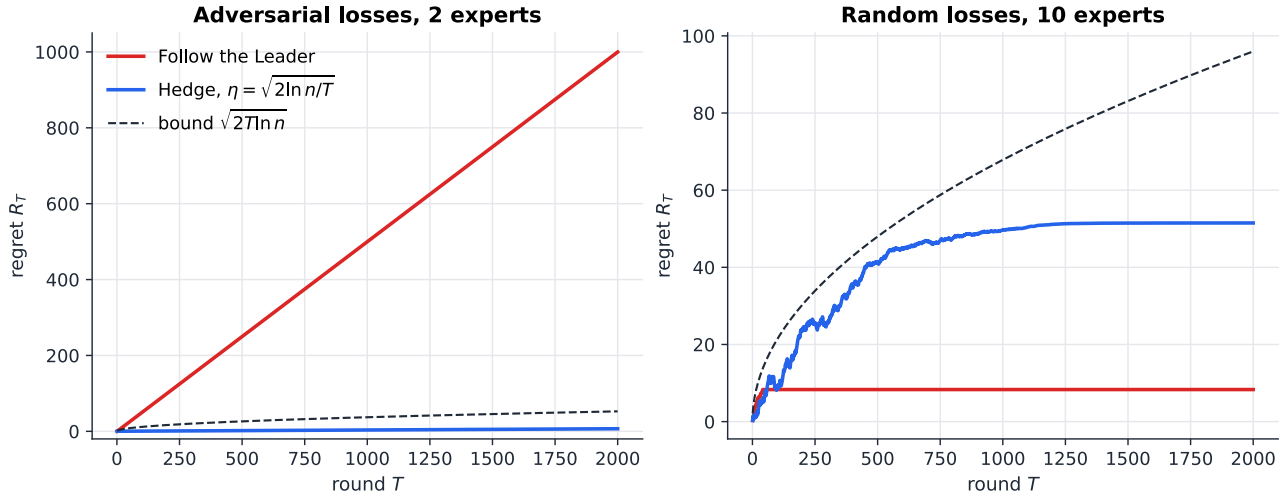
Since $\ln(1+u) \le u$, taking logs and summing over t:

$$\ln \frac{W_{T+1}}{W_1} \le -\eta \sum_{t=1}^{T} \mathbf{p}_t^{\mathsf{T}} \boldsymbol{\ell}_t + \frac{\eta^2 T}{2}.$$

Lower bound. The total weight is at least the weight of the best action i^*: $W_{T+1} \ge w_{T+1,i^*} = e^{-\eta L^*}$, where $L^* = \min_i \sum_t \ell_{t,i}$. So $\ln(W_{T+1}/W_1) \ge -\eta L^* - \ln n$.

Combine. $-\eta L^* - \ln n \le -\eta \sum_t \mathbf{p}_t^{\mathsf{T}} \boldsymbol{\ell}_t + \eta^2 T/2$. Divide by η and rearrange: $\sum_t \mathbf{p}_t^{\mathsf{T}} \boldsymbol{\ell}_t - L^* \le \ln n / \eta + \eta T/2$. Minimising the right side over η gives $\eta = \sqrt{2 \ln n / T}$ and the bound $\sqrt{2T \ln n}$. ∎

Read the proof again and notice what it never assumed. Nothing about where the losses come from. Nothing about randomness, stationarity, or the adversary's computing power. The only facts used are that losses lie in $[0,1]$ and that the adversary must commit to $\boldsymbol{\ell}_t$ without seeing your coin flip. The number of actions enters only through $\ln n$: a million experts cost you about $\sqrt{2 \cdot 13.8\, T} \approx 5.3\sqrt{T}$, against $2.1\sqrt{T}$ for ten. A hundred thousand times more advisers, two and a half times the regret.



Left: on the adversarial sequence that alternates losses between two actions, Follow the Leader's regret grows linearly (about 1000 after 2000 rounds) while Hedge stays near 7, well under its guarantee $\sqrt{2T \ln 2} \approx 53$. Right: on random losses with ten actions, one slightly better, FTL actually beats Hedge (regret about 8 against about 51); Hedge pays an insurance premium for its worst-case guarantee.

The right-hand panel is a warning against reading guarantees as predictions. When the world is benign, a greedy algorithm can win; Hedge, tuned for the worst case, explores more than it needs to. Insurance always costs something on a sunny day.

? Is $\sqrt{T \ln n}$ the end of the road?

The bound grows like $\sqrt{T}$. Could a smarter algorithm guarantee regret growing like $\ln T$, or even a constant, against every adversary? The answer is no, and the proof is a probabilistic surprise: let the adversary choose losses by *fair coin flips*, so no algorithm can do better than $T/2$ on average, while the best of n independent random walks beats $T/2$ by about $\sqrt{(T/2) \ln n}$. Why the maximum of n random walks behaves like that is a question about fluctuations of random walks, answered in Ch. 34.

The minimax theorem from regret

Now the payoff. Rescale $\mathbf{A}$ so its entries lie in $[0, 1]$ (adding a constant and scaling changes nothing essential). Let both players run a no-regret algorithm such as Hedge against each other for T rounds:

- the row player, who wants a large $\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{y}$, feeds Hedge the losses $\boldsymbol{\ell}_t = \mathbf{1} - \mathbf{A}\mathbf{y}_t$;
- the column player, who wants it small, feeds Hedge the losses $\boldsymbol{\ell}_t = \mathbf{A}^{\mathsf{T}}\mathbf{x}_t$.

Write ε_T for the larger of the two average regrets, $\bar{v} = \frac{1}{T}\sum_t \mathbf{x}_t^{\mathsf{T}}\mathbf{A}\mathbf{y}_t$ for the average payoff, and $\bar{\mathbf{x}}, \bar{\mathbf{y}}$ for the average strategies.

> **No regret implies minimax**
>
> $\overline{v} \le \underline{v} + 2\varepsilon_T$. Since $\varepsilon_T \to 0$ for Hedge, $\overline{v} = \underline{v}$, and $(\bar{\mathbf{x}}, \bar{\mathbf{y}})$ is a $2\varepsilon_T$-approximate equilibrium.

Proof. The row player's regret bound says that no fixed row does much better than what was actually earned: $\max_i \frac{1}{T}\sum_t (\mathbf{A}\mathbf{y}_t)_i \le \bar{v} + \varepsilon_T$. The left side is $\max_{\mathbf{x}} \mathbf{x}^{\mathsf{T}}\mathbf{A}\bar{\mathbf{y}}$. Symmetrically, the column player's bound gives $\min_{\mathbf{y}} \bar{\mathbf{x}}^{\mathsf{T}}\mathbf{A}\mathbf{y} \ge \bar{v} - \varepsilon_T$. Chain them:

$$\overline{v} = \min_{\mathbf{y}} \max_{\mathbf{x}} \mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{y} \le \max_{\mathbf{x}} \mathbf{x}^{\mathsf{T}}\mathbf{A}\bar{\mathbf{y}} \le \bar{v} + \varepsilon_T \le \min_{\mathbf{y}} \bar{\mathbf{x}}^{\mathsf{T}}\mathbf{A}\mathbf{y} + 2\varepsilon_T \le \max_{\mathbf{x}} \min_{\mathbf{y}} \mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{y} + 2\varepsilon_T = \underline{v} + 2\varepsilon_T.$$

Let $T \to \infty$ and combine with the weak inequality $\underline{v} \le \overline{v}$. $\blacksquare$

Pause on how strange this is. Neither player knows the game is zero-sum, or that an equilibrium exists, or what the other is doing. Each merely refuses to have regrets. The theorem of the existence of a value falls out of an exponential inequality.

> **? What happens when the game is not zero-sum?**
>
> Run the same two Hedge players on a general-sum game, the prisoner's dilemma of Ch. 40 or a 3×3 coordination game, and the averages no longer need to approach a Nash equilibrium. They converge to a weaker object, a *coarse correlated equilibrium*. Is there any fast learning rule that finds a Nash equilibrium of a general game? The evidence says no: computing one is complete for a complexity class called PPAD, believed intractable. Where that hardness comes from is a story from Ch. 30.

Robust decisions: when the adversary is Nature

Sometimes the "opponent" is not a mind but uncertainty itself: an unknown demand, an unmodelled disturbance, a parameter you could not measure. Abraham Wald proposed in 1945 to treat Nature as an adversary and choose the action with the best worst case, the **maxmin** rule. Leonard Savage objected that this is too timid and proposed **minimax regret**: for each scenario compute how much you lose relative to the best action for that scenario, and minimise the largest such regret.

> **Sizing a plant**
>
> Profit (in millions) for three capacities under three demand scenarios:
>
> | capacity | low | medium | high | worst case | worst regret |
> |---|---|---|---|---|---|
> | small | 4 | 5 | 5 | **4** | 7 |
> | medium | 2 | 7 | 8 | 2 | **4** |
> | large | −2 | 5 | 12 | −2 | 6 |
>
> The best profits per scenario are $4, 7, 12$, so the regret of *medium* in the three scenarios is $2, 0, 4$. Wald's maxmin chooses *small*, guaranteeing 4. Savage's minimax regret chooses *medium*, never more than 4 below the best possible outcome. Two worst-case philosophies, two different plants.

Neither rule is "right". Maxmin protects the level of the outcome; minimax regret protects against looking foolish in hindsight, the same benchmark Hedge uses. Robust optimisation (Ch. 54) turns these ideas into tractable programs where the scenarios form a continuous uncertainty set, and robust control does the same for a disturbance signal of bounded energy.

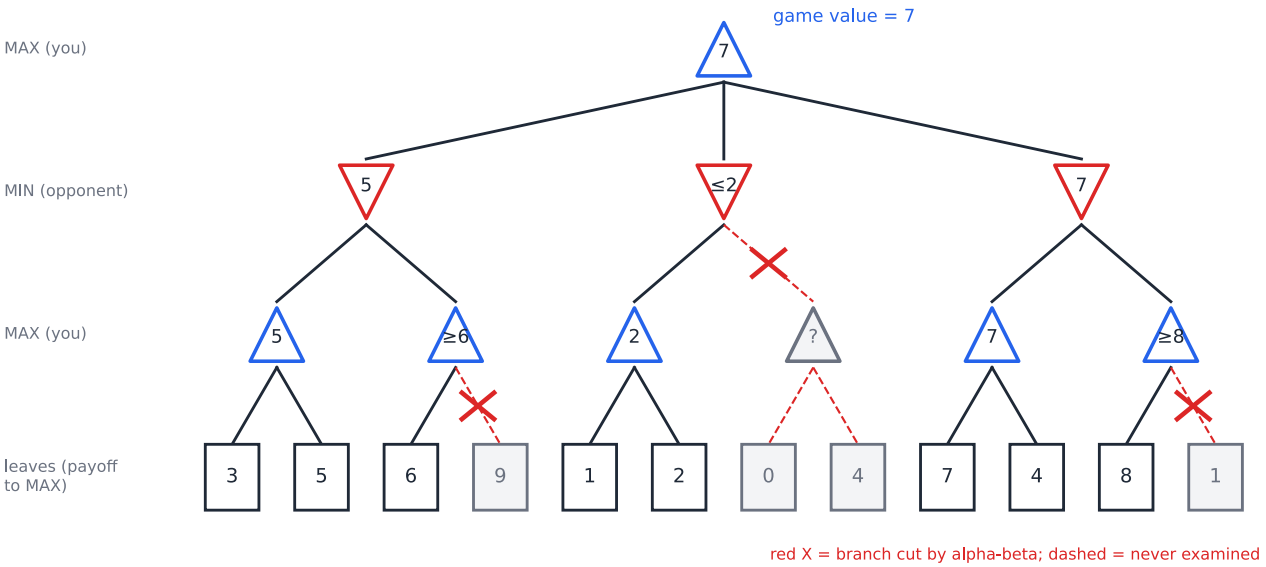
46.4 The algorithm

Minimax search with alpha–beta pruning

In a game of perfect information such as chess or Go, there is no need to randomise: the players alternate, every move is visible, and the value of a position is defined by backing up values from the leaves of the game tree. MAX nodes take the maximum of their children, MIN nodes the minimum. This backward recursion is dynamic programming on a tree (Ch. 52), and it proves a fact von Neumann and Zermelo knew early on: every finite two-player game of perfect information has a value achieved by pure strategies.

The trouble is size. A tree with branching factor b and depth d has b^d leaves. **Alpha–beta pruning** computes exactly the same root value while skipping branches that provably cannot matter. It carries two numbers down the tree: α, the best value MAX is already guaranteed somewhere above, and β, the best value MIN is already guaranteed.

```
ALPHABETA(node, α, β, maximizing):
    if node is a leaf (or depth limit reached):
        return EVAL(node)
    if maximizing:
        v ← −∞
        for each child c:
            v ← max(v, ALPHABETA(c, α, β, false))
            α ← max(α, v)
            if α ≥ β: break          # MIN above will never let play reach here
        return v
    else:
        v ← +∞
        for each child c:
            v ← min(v, ALPHABETA(c, α, β, true))
            β ← min(β, v)
            if α ≥ β: break          # MAX above already has something better
        return v
```



A three-ply game tree. Alpha–beta examines 8 of the 12 leaves and still returns the exact value 7. In the left subtree, once MAX sees a 6, the MIN node above (which already holds 5) will never allow it, so the 9 is never read. In the middle subtree, MIN finds a reply worth only 2, below the 5 MAX already has at the root, so the whole second branch is skipped. Values marked $\geq$ or $\leq$ are bounds, not exact values.

The correctness argument is a short induction: a pruned branch could only change a node's value in a direction that some ancestor would reject anyway. The efficiency depends entirely on **move ordering**. If the best move is always searched first, Knuth and Moore (1975) showed that alpha–beta examines about $b^{\lceil d/2 \rceil} + b^{\lfloor d/2 \rfloor} - 1$ leaves instead of b^d. For chess-like numbers, $b = 35$ and $d = 10$, that is roughly 1.05×10^8 positions instead of 2.76×10^{15}, a saving of a factor of about 26 million. In effect the search can go twice as deep for the same effort. With the worst ordering, alpha–beta prunes nothing.

Failure cases. Real programs cannot reach the leaves, so they stop at a depth limit and call a heuristic `EVAL`. Two famous pathologies follow.

- *The horizon effect.* A program facing an unavoidable loss can make delaying moves that push the loss just past its search depth, and so believe it has escaped.
- *Error amplification.* Minimax trusts its evaluation function absolutely. If `EVAL` is slightly wrong in a consistent direction, the search actively steers towards the positions where it is most wrong, because those look best.

Hedge in practice

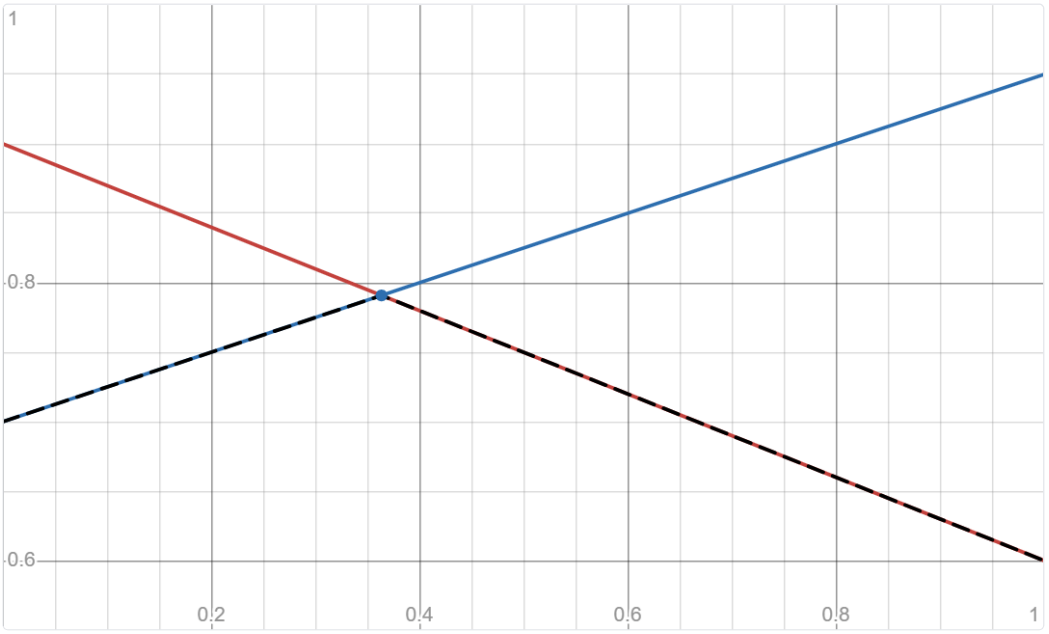
```
def hedge(losses, eta):
    """losses: T x n array in [0, 1]. Returns the T x n array of strategies played."""
    T, n = losses.shape
    cum = np.zeros(n)
    P = np.empty((T, n))
    for t in range(T):
        w = np.exp(-eta * (cum - cum.min()))   # shift: avoids underflow
        P[t] = w / w.sum()
        cum += losses[t]
    return P
```

Each round costs $O(n)$. Two numerical details matter. Subtracting the minimum cumulative loss before exponentiating changes nothing mathematically (it cancels in the normalisation) but prevents every weight from underflowing to zero after a few thousand rounds. And the tuned rate $\eta = \sqrt{2 \ln n / T}$ needs T in advance; if you do not know the horizon, the *doubling trick* (restart with $T = 1, 2, 4, 8, \ldots$) or a decreasing rate $\eta_t \propto \sqrt{\ln n / t}$ keeps the $O(\sqrt{T \ln n})$ guarantee at the price of a constant factor.

Solving a matrix game. For an exact answer, solve the LP of 46.3 with any LP solver: polynomial time in m and n. For huge games where even writing down $\mathbf{A}$ is costly, self-play Hedge finds an ε-equilibrium in $O(\ln n/\varepsilon^2)$ rounds, and it only needs to evaluate $\mathbf{A}\mathbf{y}$ and $\mathbf{A}^{\mathsf{T}}\mathbf{x}$. This is the engine behind modern poker programs, which run regret-minimising algorithms on games far too large to tabulate.

46.5 The experiment

First, the kicker's view of the penalty. The two lines are the scoring probabilities against each dive, as functions of the probability q (on the x-axis) of kicking left; the keeper always picks the lower line.



The kicker maximises the lower envelope (dashed). Change the four scoring probabilities 0.9, 0.3, 0.7, 0.25 (they come from the table in 46.1) and watch the optimal mix move; make one line lie entirely above the other and the optimum jumps to a pure strategy.

Next, let two Hedge learners play the penalty game against each other and watch them discover von Neumann's solution.

```
import numpy as np
import matplotlib.pyplot as plt

# --- knobs ---------------------------------------------------------------
A = np.array([[0.60, 0.95],     # P(goal): rows = kicker L/R,
              [0.90, 0.70]])    #          cols = keeper dives L/R
T = 5000                        # rounds of play
eta = np.sqrt(2 * np.log(2) / T)  # learning rate: try 10x larger or smaller
# -------------------------------------------------------------------------

Lk, Lg = np.zeros(2), np.zeros(2)   # cumulative losses: kicker, goalkeeper
avg_k, avg_g, gap = np.zeros(2), np.zeros(2), []
for t in range(1, T + 1):
    p = np.exp(-eta * (Lk - Lk.min())); p /= p.sum()   # kicker's mix
    q = np.exp(-eta * (Lg - Lg.min())); q /= q.sum()   # keeper's mix
    Lk += 1 - A @ q          # kicker loses when no goal
    Lg += A.T @ p            # keeper loses when goal
    avg_k += (p - avg_k) / t
    avg_g += (q - avg_g) / t
    # duality gap of the averages: best reply value minus worst case value
    gap.append((A @ avg_g).max() - (avg_k @ A).min())

print("kicker avg P(left) :", avg_k[0].round(4), " theory 4/11 =", round(4/11, 4))
print("keeper avg P(left) :", avg_g[0].round(4), " theory 5/11 =", round(5/11, 4))
print("value bracket      :", (avg_k @ A).min().round(4), (A @ avg_g).max().round(4))
plt.loglog(np.arange(1, T + 1), gap, label="duality gap of average play")
plt.loglog(np.arange(1, T + 1), 2*np.sqrt(2*np.log(2)*T)/np.arange(1, T + 1), "--",
           label="2 x regret bound / t (tuned for t = T)")
plt.xlabel("round t"); plt.legend(); plt.show()
```

With the default knobs the averages come out near 0.347 and 0.467 (theory: 0.364 and 0.455), and the value is bracketed between about 0.787 and 0.793, enclosing the true 0.7909. The final duality gap, about 0.007, sits well below the guaranteed $2\varepsilon_T \approx 0.033$. Things to try:

1. Multiply `eta` by 10. The averages still converge, but look at the *current* strategies `p` and `q` (print them every 500 rounds): they swing around the equilibrium like the rock–paper–scissors orbits of Ch. 45 and never settle. Only the averages converge.
2. Divide `eta` by 10. The learners are now too timid; after 5000 rounds they have barely left the uniform mix.
3. Replace `A` by rock–paper–scissors rescaled to $[0,1]$, for example $\frac{1}{2}(\mathbf{A}_{\text{RPS}}+1)$, and check the averages approach $(\frac{1}{3},\frac{1}{3},\frac{1}{3})$.
4. Make the keeper play Follow the Leader instead of Hedge. Who now exploits whom?

46.6 The challenge

1. **Intuition.** In the penalty table, suppose the kicker becomes *better* at scoring left when the keeper guesses wrong (raise 0.95 to 0.99). Does the kicker now kick left more often or less often at equilibrium? Predict before computing, and explain the answer in terms of what each player is trying to prevent.
2. **Derivation.** Prove that for any 2×2 zero-sum game without a pure saddle point, the row player's optimal probability on the first row is $q^* = \dfrac{a_{22} - a_{21}}{a_{11} - a_{12} - a_{21} + a_{22}}$, and derive the value. Then show that Follow the Leader has regret at least $T/2 - 1$ on the alternating sequence of 46.3.
3. **Implementation.** Implement tic-tac-toe with plain minimax and with alpha–beta. Count the leaves each examines from the empty board, then try three move orderings (fixed, random, centre-and-corners first) and compare the counts with the Knuth–Moore estimate.
4. **Open investigation.** Hedge with a fixed η has $O(\sqrt{T})$ regret against adversaries but, as the right panel of the regret figure shows, can be beaten by Follow the Leader when losses are random. Design an algorithm that does about as well as FTL on benign data and as well as Hedge on adversarial data, and test it on both sequences of the figure. How much must it give up on each side? (Algorithms of this "best of both worlds" type exist; how close you can get with a simple design is for you to find out.)

46.7 Loose threads

This chapter gave you a guarantee and an algorithm, and it hid how far both reach.

We solved perfect-information games by searching a tree with a hand-made evaluation function, and we solved matrix games by letting regret minimisers play each other. Modern game-playing systems merge the two ideas: they search a tree, but the evaluation and the move ordering are learned, and the learning happens by playing against copies of themselves.

> **? Learning to evaluate by playing yourself**
>
> A Go board has more legal positions than there are atoms in the observable universe, so no evaluation function can be written by hand and no tree can be searched to the end. Programs trained purely by self-play nevertheless surpassed every human player. What exactly is being learned, why self-play does not simply cycle like the over-eager Hedge learners of 46.5, and what replaces alpha–beta in those systems are questions taken up in Ch. 53, building on the Bellman recursion of Ch. 52.

We also leaned on one hidden assumption throughout: that you see the whole loss vector after each round. A slot machine, an advertising slot or a network route shows you only the loss of the action you took. That is the *bandit* problem, where you must pay to explore, and the regret guarantee changes shape in an interesting way.

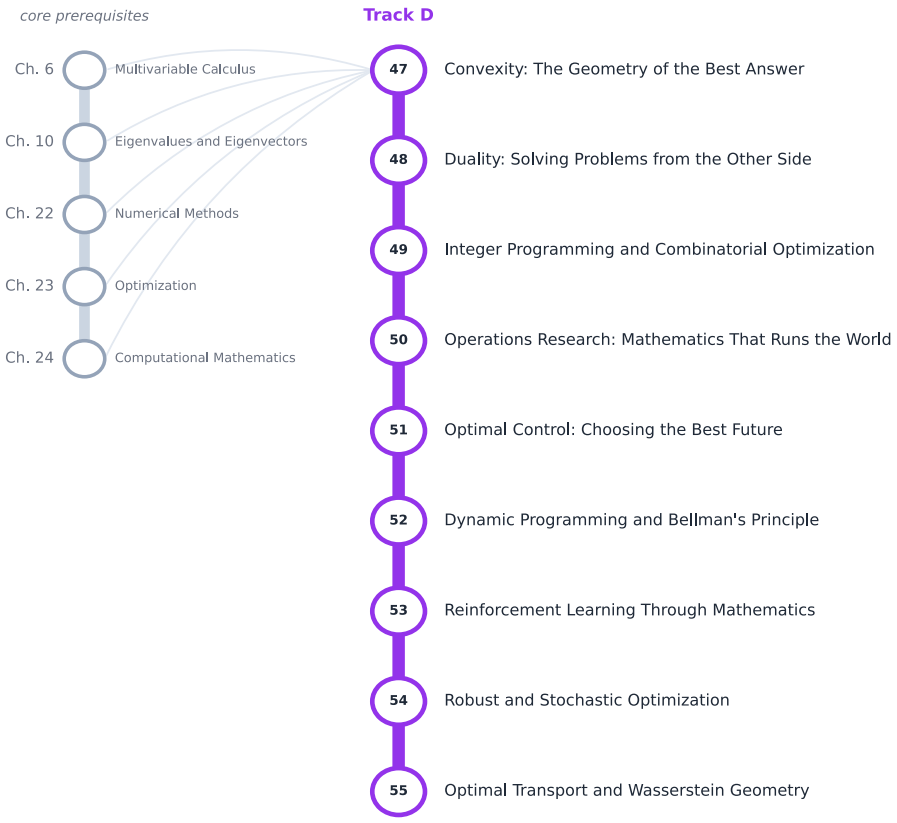
Finally, minimax thinking has a cousin that is far older than game theory: the cryptographer's assumption that the enemy knows the system. Kerckhoffs' principle says a cipher must stay secure even when everything except the key is public, which is exactly the spirit of publishing your mixed strategy. In Ch. 77 the adversary returns, with unlimited patience and bounded computing power, and the guarantees take yet another form.

Further reading

- J. von Neumann and O. Morgenstern, *Theory of Games and Economic Behavior*, Princeton University Press, 1944. The founding book; the minimax theorem in its original setting.
- N. Cesa-Bianchi and G. Lugosi, *Prediction, Learning, and Games*, Cambridge University Press, 2006. The definitive treatment of regret, experts and the minimax theorem via learning.
- S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed., Pearson, 2020. Clear chapters on adversarial search and alpha–beta.
- A. Ben-Tal, L. El Ghaoui and A. Nemirovski, *Robust Optimization*, Princeton University Press, 2009. Worst-case decision making as tractable optimisation.

Track D — Optimization and Operations Research

> **Level 2 — Specializations.** From introductory engineering tools to the advanced frameworks that schedule, route and control the world.



Suggested reading order for Track D. The grey stations on the left are the core chapters it leans on.

Before you start

This track assumes you are comfortable with these core chapters:

- Ch. 6 — Multivariable Calculus
- Ch. 10 — Eigenvalues and Eigenvectors
- Ch. 22 — Numerical Methods
- Ch. 23 — Optimization
- Ch. 24 — Computational Mathematics

Chapters

Ch.	Chapter	What it explores
47	Convexity: The Geometry of the Best Answer	Convex sets/functions, separation theorems, convex optimization.
48	Duality: Solving Problems from the Other Side	Lagrangian duality, KKT conditions, sensitivity analysis.
49	Integer Programming and Combinatorial Optimization	Integer models, branch-and-bound, cutting planes, relaxations.
50	Operations Research: Mathematics That Runs the World	Scheduling, inventory, logistics, resource allocation.
51	Optimal Control: Choosing the Best Future	Dynamic optimization, Pontryagin's principle, control constraints.
52	Dynamic Programming and Bellman's Principle	Bellman equations, optimal policies, state-space methods.
53	Reinforcement Learning Through Mathematics	Markov decision processes, value functions, policy optimization.
54	Robust and Stochastic Optimization	Uncertain parameters, chance constraints, worst-case optimization.
55	Optimal Transport and Wasserstein Geometry	Transport plans, Wasserstein distances, distribution matching.

Every chapter follows the same arc: **the hook** (a puzzle or a failure), **the big idea**, **the mathematics**, **the algorithm**, **the experiment** you can run, **the challenge**, and **loose threads**: the questions the chapter deliberately leaves open.

Chapter 47 — Convexity: The Geometry of the Best Answer

> **Level 2 · Track D — Optimization and Operations Research Builds on:** Ch. 6, Ch. 10, Ch. 23 · **Leads to:** Ch. 48, Ch. 49, Ch. 54 · **Related:** Ch. 39, Ch. 61, Ch. 75 **Reading time:** ~30 min

47.1 The hook

In 1958 Frank Rosenblatt built a machine that learned. The Mark I Perceptron looked at a grid of photocells, took a weighted sum of their outputs, and fired if the sum crossed a threshold. Each time it guessed wrong, it nudged its weights toward the right answer. Newspapers predicted thinking machines within a decade.

Eleven years later Marvin Minsky and Seymour Papert pointed out something the machine could never learn. Put four points on a square:

point	$(0,0)$	$(1,1)$	$(1,0)$	$(0,1)$
label	A	A	B	B

This is the exclusive-or pattern: label B when exactly one coordinate is 1. A perceptron draws one straight line and puts A on one side and B on the other. Try it. Every line you draw through that square leaves at least one point on the wrong side. Run the perceptron's learning rule on these four points and the weights never settle: they cycle forever.

That raises a question that matters well beyond 1969:

> **How do you *prove* that no line works?**

There are infinitely many lines, so you can't try them all. "I tried for an hour and failed" doesn't count as a proof. What you want is a **certificate**: a short piece of evidence that anyone can check in ten seconds and that settles the matter for good.

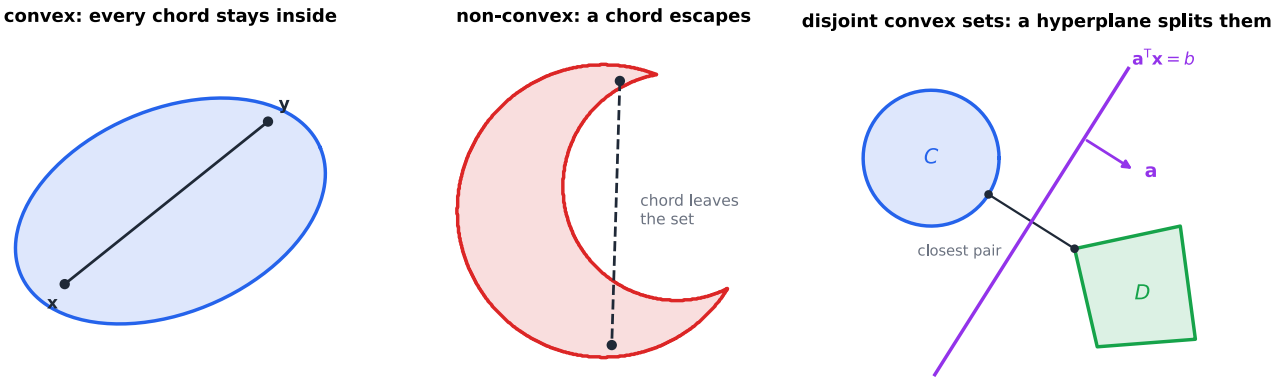
The same question turns up all over engineering. A filter specification asks for passband ripple under 0.1 dB, stopband attenuation over 80 dB and only 20 taps; is that achievable at all? A power network must carry a set of flows without overloading any line; is there any feasible dispatch? An optimizer reports a cost of 41.7; is that the best possible, or did it stop in a ditch? In every case "yes" is easy to certify: show the design. "No" and "this is optimal" are much harder. They are claims about *every* possible design.

This chapter shows that for one family of shapes, the **convex** ones, the hard claims have short proofs. A wall is the proof: a single hyperplane. By the end you will be able to write down the ten-second certificate for XOR. You will also see why the same geometry makes "local" answers into "global" ones.

47.2 The big idea

The core chapter defined convex sets and functions and proved the payoff: for a convex problem every local minimum is global (§23.7). Here is the intuition behind that result, and the idea this whole track grows from.

A convex set has no hidden corners. Stand anywhere inside it and you can see every other point along a straight line of sight that never leaves the set. A crescent moon fails this test: from one horn you can't see the other.



Left: in a convex set every chord stays inside. Centre: in a crescent a chord escapes. Right: two disjoint convex sets can always be split by a hyperplane, and the shortest segment between them points along its normal $\mathbf{a}$.

A convex set has two descriptions. You can describe it from the *inside*, as all the mixtures (convex combinations) of its points. Or you can describe it from the *outside*, as the intersection of all the half-spaces that contain it, like a gemstone defined by the flat cuts of its facets. For a closed convex set the two descriptions agree. Most of optimization theory is a conversation between them:

- an inside description lets you *exhibit* things: here is a feasible design, here is a point in both sets;
- an outside description lets you *rule things out*: here is a wall, and everything you want lies on the far side of it.

The certificate the hook asked for is an outside description. If two point clouds can't be separated, it's because their convex "shadows" overlap, and a single shared point proves it. If they can be separated, the line itself is the proof.

An analogy: walking downhill in fog. You are on a mountainside in thick fog and can only feel the slope under your feet. On a general mountain, reaching a spot where every direction goes up proves nothing. There may be a deeper valley over the ridge. On a convex bowl that spot *is* the bottom, because the ground under your feet already tells you about the whole bowl. The tangent plane you feel is a global floor (Figure 47.2). Local information is global information. That one sentence is why convex optimization is reliable and fast, and why so much of applied mathematics is spent turning problems into convex ones.

47.3 The mathematics

We work in $\mathbb{R}^n$ with the Euclidean inner product. Convex sets, convex functions, the first- and second-order tests, and "local implies global" are in §23.7 and are not repeated here. We need one more definition.

Convex combination and convex hull

A **convex combination** of points $\mathbf{x}_1, \dots, \mathbf{x}_k$ is $\sum_i \theta_i \mathbf{x}_i$ with $\theta_i \ge 0$ and $\sum_i \theta_i = 1$. The **convex hull** $\operatorname{conv} S$ of a set S is the set of all convex combinations of points of S. It is the smallest convex set containing S.

For a finite set of points in the plane, the convex hull is the shape a rubber band makes when you stretch it around them (computing it fast is a classic of Ch. 75). **Carathéodory's theorem** says that in $\mathbb{R}^n$ every point of $\operatorname{conv} S$ is a convex combination of at most $n+1$ points of S: in the plane, every point of the hull lies in some triangle with corners in S.

The closest point

Everything in this chapter flows from one elementary fact.

Theorem 47.1 — Projection onto a closed convex set

Let $C \subseteq \mathbb{R}^n$ be nonempty, closed and convex, and let $\mathbf{z} \in \mathbb{R}^n$. There is exactly one point $\mathbf{p} = P_C(\mathbf{z}) \in C$ closest to $\mathbf{z}$. It is characterised by the **obtuse-angle condition**

$$(\mathbf{z} - \mathbf{p})^{\mathsf{T}} (\mathbf{y} - \mathbf{p}) \le 0 \qquad \text{for all } \mathbf{y} \in C.$$

Proof. **Existence.** Pick any $\mathbf{y}_0 \in C$ and let $r = \|\mathbf{z} - \mathbf{y}_0\|$. Any closer point lies in $C \cap \{\mathbf{y} : \|\mathbf{y} - \mathbf{z}\| \le r\}$, which is closed and bounded, hence compact. The continuous function $\mathbf{y} \mapsto \|\mathbf{y} - \mathbf{z}\|$ attains its minimum there.

Characterisation. Let $\mathbf{p}$ be a closest point and $\mathbf{y} \in C$. By convexity $\mathbf{p} + t(\mathbf{y} - \mathbf{p}) \in C$ for $t \in [0,1]$, so

$$\varphi(t) = \|\mathbf{z} - \mathbf{p} - t(\mathbf{y} - \mathbf{p})\|^2 = \|\mathbf{z} - \mathbf{p}\|^2 - 2t(\mathbf{z} - \mathbf{p})^{\mathsf{T}}(\mathbf{y} - \mathbf{p}) + t^2 \|\mathbf{y} - \mathbf{p}\|^2$$

has its minimum on $[0,1]$ at $t = 0$. Then $\varphi'(0) = -2(\mathbf{z} - \mathbf{p})^{\mathsf{T}}(\mathbf{y} - \mathbf{p}) \ge 0$, which is the stated inequality. Conversely, if $\mathbf{p} \in C$ satisfies the inequality, then for any $\mathbf{y} \in C$, $\varphi(1) = \|\mathbf{z} - \mathbf{y}\|^2 \ge \|\mathbf{z} - \mathbf{p}\|^2 + \|\mathbf{y} - \mathbf{p}\|^2$, so $\mathbf{p}$ is closest.

Uniqueness. If $\mathbf{p}_1, \mathbf{p}_2$ are both closest, apply the inequality at $\mathbf{p}_1$ with $\mathbf{y} = \mathbf{p}_2$ and at $\mathbf{p}_2$ with $\mathbf{y} = \mathbf{p}_1$, then add: $(\mathbf{p}_2 - \mathbf{p}_1)^{\mathsf{T}}(\mathbf{p}_2 - \mathbf{p}_1) \le 0$. Hence $\mathbf{p}_1 = \mathbf{p}_2$. $\square$

In words: the line from $\mathbf{z}$ to its projection makes an obtuse (or right) angle with every direction that stays in C. For the unit disc and $\mathbf{z} = (2, 2)$, $P_C(\mathbf{z}) = (1/\sqrt{2}, 1/\sqrt{2}) \approx (0.707, 0.707)$. Every chord of the disc starting at that point heads away from $\mathbf{z}$. Convexity was essential. For the crescent of Figure 47.1, a point in the "bite" has *two* equally close points on the inner arc, and the obtuse-angle condition fails.

Separation

Theorem 47.2 — Separating a point from a convex set

Let C be nonempty, closed and convex, and $\mathbf{z} \notin C$. Then there are $\mathbf{a} \neq \mathbf{0}$ and b with

$$\mathbf{a}^{\mathsf{T}}\mathbf{y} \le b < \mathbf{a}^{\mathsf{T}}\mathbf{z} \qquad \text{for all } \mathbf{y} \in C.$$

Proof. Let $\mathbf{p} = P_C(\mathbf{z})$ and $\mathbf{a} = \mathbf{z} - \mathbf{p} \neq \mathbf{0}$ (since $\mathbf{z} \notin C$). The obtuse-angle condition reads $\mathbf{a}^{\mathsf{T}}\mathbf{y} \le \mathbf{a}^{\mathsf{T}}\mathbf{p}$ for all $\mathbf{y} \in C$. Put $b = \mathbf{a}^{\mathsf{T}}\mathbf{p}$. Then $\mathbf{a}^{\mathsf{T}}\mathbf{z} - b = \mathbf{a}^{\mathsf{T}}(\mathbf{z} - \mathbf{p}) = \|\mathbf{a}\|^2 > 0$. $\square$

The proof is constructive: the normal of the wall is the vector from the closest point to the outsider. One immediate corollary is the "outside description" promised in §47.2. **A closed convex set equals the intersection of all closed half-spaces containing it**, because any point outside is cut off by one of them.

Theorem 47.3 — Separating two convex sets

Let C and D be disjoint nonempty convex sets with C closed and D compact. Then some hyperplane strictly separates them: there are $\mathbf{a} \neq \mathbf{0}$, b with $\mathbf{a}^{\mathsf{T}}\mathbf{x} < b < \mathbf{a}^{\mathsf{T}}\mathbf{y}$ for all $\mathbf{x} \in C$, $\mathbf{y} \in D$.

Proof. The difference set $E = D - C = \{\mathbf{y} - \mathbf{x}\}$ is convex (it is the image of the convex set $D \times C$ under a linear map). It is closed because a compact set plus a closed set is closed, and it doesn't contain $\mathbf{0}$ because the sets are disjoint. Theorem 47.2 separates $\mathbf{0}$ from E. With $\mathbf{w} = -P_E(\mathbf{0})$ it gives $\mathbf{w}^{\mathsf{T}}(\mathbf{y} - \mathbf{x}) \ge \|\mathbf{w}\|^2 > 0$ for all pairs. Hence $\inf_D \mathbf{w}^{\mathsf{T}}\mathbf{y} \ge \sup_C \mathbf{w}^{\mathsf{T}}\mathbf{x} + \|\mathbf{w}\|^2$, and any b strictly between the two numbers works with $\mathbf{a} = \mathbf{w}$. $\square$

The vector $\mathbf{w}$ is exactly the shortest segment from C to D, the one drawn in the right panel of Figure 47.1. Two relatives are worth stating without proof:

- **Separation without compactness.** Any two disjoint convex sets can be separated *weakly* ($\mathbf{a}^\mathsf{T}\mathbf{x} \le b \le \mathbf{a}^\mathsf{T}\mathbf{y}$). Strictness can fail: the region above $y = e^x$ and the region below $y = 0$ are disjoint, but the gap between them shrinks to zero.
- **Supporting hyperplanes.** At every boundary point $\mathbf{x}_0$ of a convex set C there is a hyperplane through $\mathbf{x}_0$ with all of C on one side. A convex set can be "touched" from outside at every point of its surface. A crescent cannot: no line touches the inner arc while keeping the whole moon on one side.

The certificate for XOR

Now we can answer the hook.

Theorem 47.4 — Linear separability

Two finite point sets $A, B \subset \mathbb{R}^n$ can be strictly separated by a hyperplane if and only if $\operatorname{conv} A \cap \operatorname{conv} B = \varnothing$.

Proof. ($\Leftarrow$) Convex hulls of finite sets are compact and convex, so Theorem 47.3 applies. ($\Rightarrow$) Suppose $\mathbf{a}^\mathsf{T}\mathbf{x} < b$ on A and $\mathbf{a}^\mathsf{T}\mathbf{y} > b$ on B, and some point $\mathbf{m}$ lies in both hulls. Writing $\mathbf{m} = \sum \theta_i \mathbf{x}_i$ with $\mathbf{x}_i \in A$ gives $\mathbf{a}^\mathsf{T}\mathbf{m} = \sum \theta_i \mathbf{a}^\mathsf{T}\mathbf{x}_i < b$. Writing it as a combination of points of B gives $\mathbf{a}^\mathsf{T}\mathbf{m} > b$. Contradiction. $\square$

For XOR the certificate is one point, $\mathbf{m} = (\tfrac12, \tfrac12)$:

$$\tfrac12(0,0) + \tfrac12(1,1) = (\tfrac12, \tfrac12) = \tfrac12(1,0) + \tfrac12(0,1).$$

The two diagonals of the square cross. Any line that put both A points strictly on one side would have to put their midpoint there too. The B points would force that same midpoint onto the other side. That is the ten-second proof, and it rules out all infinitely many lines at once. The "yes" answers have short proofs too: for $A = \{(0,0),(1,0)\}$ and $B = \{(0,1),(1,1)\}$ the line $y = \tfrac12$ is its own certificate.

Engineering Note: support vector machines

When two classes are separable, there are many separating hyperplanes. The proof of Theorem 47.3 picks a special one: the perpendicular bisector of the shortest segment between the two hulls. That hyperplane has the widest possible margin, and finding it is exactly the **hard-margin support vector machine**, a convex quadratic program. The few points that touch the margin are the "support vectors".

Farkas' lemma: infeasibility has a certificate

The same idea certifies that a system of linear equations has **no nonnegative solution**. This is the version that powers linear programming. The core chapter uses it to justify the KKT conditions (§23.4) and the strong duality of LP (§23.3) without proving it. Here is the proof.

Theorem 47.5 — Farkas' lemma

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$. Exactly one of the following holds:

1. there is $\mathbf{x} \ge \mathbf{0}$ with $\mathbf{A}\mathbf{x} = \mathbf{b}$;
2. there is $\mathbf{y}$ with $\mathbf{A}^\mathsf{T}\mathbf{y} \ge \mathbf{0}$ and $\mathbf{b}^\mathsf{T}\mathbf{y} < 0$.

Proof. Both cannot hold: otherwise $0 \le (\mathbf{A}^\mathsf{T}\mathbf{y})^\mathsf{T}\mathbf{x} = \mathbf{y}^\mathsf{T}\mathbf{A}\mathbf{x} = \mathbf{y}^\mathsf{T}\mathbf{b} < 0$. Now suppose (1) fails. The cone $K = \{\mathbf{A}\mathbf{x} : \mathbf{x} \ge \mathbf{0}\}$, all nonnegative combinations of the columns of $\mathbf{A}$, is convex, and as a finitely generated cone it is closed (a standard fact we take on trust). Since $\mathbf{b} \notin K$, Theorem 47.2 gives $\mathbf{a}$ and β with $\mathbf{a}^\mathsf{T}\mathbf{k} \le \beta < \mathbf{a}^\mathsf{T}\mathbf{b}$ for all $\mathbf{k} \in K$. Because $\mathbf{0} \in K$, $\beta \ge 0$. Because K is a cone, $t\,\mathbf{k} \in K$ for every $t > 0$, so $\mathbf{a}^\mathsf{T}\mathbf{k} \le 0$ for all $\mathbf{k} \in K$; otherwise scaling would push it past β. Applying this to each column $\mathbf{A}\mathbf{e}_j$ gives $\mathbf{A}^\mathsf{T}\mathbf{a} \le \mathbf{0}$. Put $\mathbf{y} = -\mathbf{a}$: then $\mathbf{A}^\mathsf{T}\mathbf{y} \ge \mathbf{0}$ and $\mathbf{b}^\mathsf{T}\mathbf{y} = -\mathbf{a}^\mathsf{T}\mathbf{b} < -\beta \le 0$. $\square$

Worked Example 47.1: reading an infeasibility certificate

Is there $\mathbf{x} \ge \mathbf{0}$ with

$$x_1 + x_2 + x_3 = 1, \qquad x_1 - x_2 + 3x_3 = 4\ ?$$

Geometry. The columns $(1,1)$, $(1,-1)$, $(1,3)$ point at angles $45°$, $-45°$ and $71.6°$. The target $\mathbf{b} = (1,4)$ points at $76.0°$, just outside the cone the columns sweep out.

Certificate. Take $\mathbf{y} = (3,-1)$. Then $\mathbf{A}^\mathsf{T}\mathbf{y} = (3-1,\ 3+1,\ 3-3) = (2,4,0) \ge \mathbf{0}$ and $\mathbf{b}^\mathsf{T}\mathbf{y} = 3 - 4 = -1 < 0$.

What it means. Three times the first equation minus the second gives $2x_1 + 4x_2 + 0x_3 = -1$. The left side is nonnegative for every $\mathbf{x} \ge \mathbf{0}$, and the right side is negative. No solution exists. You don't have to trust any solver: one subtraction settles it.

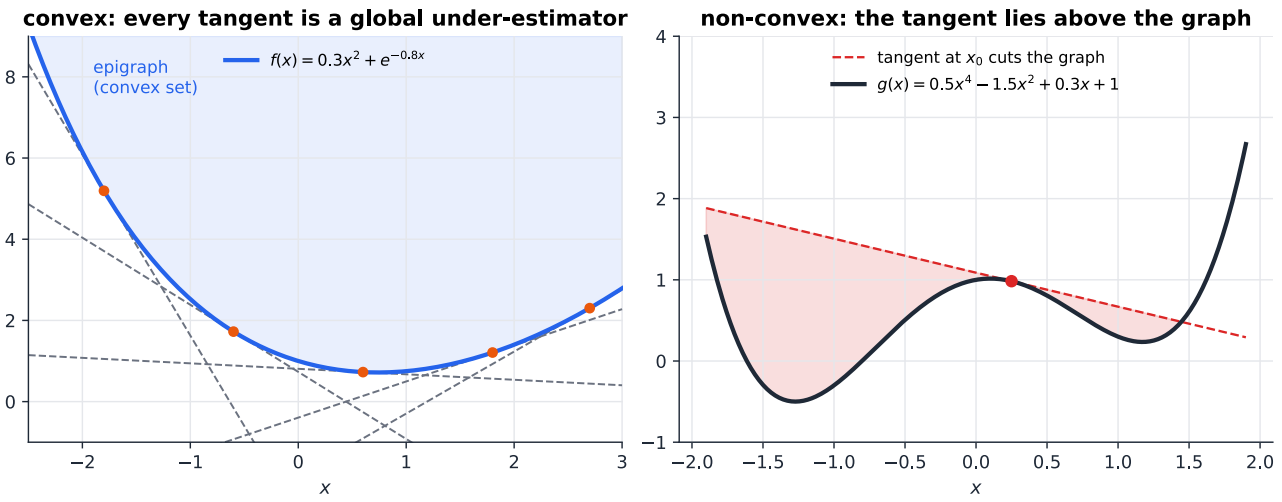
The hook's two questions now have one answer. **For convex problems, "no" always has a short certificate, and the certificate is a hyperplane.** In linear programming the hyperplane's coefficients $\mathbf{y}$ will turn out to be *prices*, but that story belongs to the next chapter.

Convex functions from the outside

Separation also explains the geometry of convex functions. The core chapter proved that a differentiable convex f lies above each of its tangents, $f(\mathbf{y}) \ge f(\mathbf{x}) + \nabla f(\mathbf{x})^\mathsf{T}(\mathbf{y} - \mathbf{x})$. Equality holds at $\mathbf{y} = \mathbf{x}$, so

$$f(\mathbf{y}) = \max_{\mathbf{x}} \Big[f(\mathbf{x}) + \nabla f(\mathbf{x})^\mathsf{T}(\mathbf{y} - \mathbf{x}) \Big].$$

A convex function is the upper envelope of its tangent planes. This is the outside description of its epigraph: the region above the graph is cut out by the half-spaces above the tangents.



Left: every tangent of $f(x) = 0.3x^2 + e^{-0.8x}$ is a global floor, and f is their upper envelope. Right: for a non-convex function the tangent at a hump overestimates the function on both sides (shaded), so local slope information says nothing reliable about distant points.

For non-smooth functions the tangent is replaced by any supporting line, which exists by the supporting-hyperplane theorem applied to the epigraph.

Subgradient

$\mathbf{g}$ is a **subgradient** of a convex f at $\mathbf{x}$ if $f(\mathbf{y}) \geq f(\mathbf{x}) + \mathbf{g}^{\mathsf{T}}(\mathbf{y} - \mathbf{x})$ for all $\mathbf{y}$. The set of all of them is the **subdifferential** $\partial f(\mathbf{x})$.

For $f(x) = |x|$, $\partial f(0) = [-1, 1]$: every line through the origin with slope between -1 and 1 stays under the V. The optimality condition becomes $\mathbf{0} \in \partial f(\mathbf{x}^*)$, and its proof is one line: if $\mathbf{0}$ is a subgradient, then $f(\mathbf{y}) \geq f(\mathbf{x}^*) + \mathbf{0}$ for all $\mathbf{y}$. This is how one handles the L^1 penalties of sparse signal recovery, the hinge loss of SVMs and the max-functions of minimax design, none of which are differentiable at their optimum.

Constraints: the normal cone

For a convex problem $\min_{\mathbf{x} \in C} f(\mathbf{x})$ the core chapter's optimality test (§23.7, "local equals global", item 4) says that no feasible direction goes downhill. In the language of this chapter:

Normal cone

$N_C(\mathbf{x}) = \{\mathbf{g} : \mathbf{g}^{\mathsf{T}}(\mathbf{y} - \mathbf{x}) \leq 0 \text{ for all } \mathbf{y} \in C\}$, the outward normals of all supporting hyperplanes of C at $\mathbf{x}$.

The optimality condition is then $-\nabla f(\mathbf{x}^*) \in N_C(\mathbf{x}^*)$: the downhill direction must point straight out through a supporting wall. Compare Theorem 47.1. The projection of $\mathbf{z}$ is the point $\mathbf{p}$ with $\mathbf{z} - \mathbf{p} \in N_C(\mathbf{p})$. In the interior the normal cone is $\{\mathbf{0}\}$ and we recover $\nabla f = \mathbf{0}$. At a corner of a polyhedron, Farkas' lemma says the normal cone is spanned by the normals of the active constraints, and the condition becomes the KKT equations of §23.4 with nonnegative multipliers. KKT is the normal-cone condition with the cone written out in coordinates.

Common Pitfall: convex objective, non-convex set

All of this needs *both* a convex function and a convex feasible set. Minimising the convex $\|\mathbf{x} - \mathbf{z}\|$ over a non-convex set (integer points, a circle, "at most k nonzero entries") can have many local minima, and the obtuse-angle condition no longer identifies the answer. Ch. 49 is about exactly this kind of trouble.

? The wall is a price tag

In Farkas' lemma the separating vector $\mathbf{y}$ combined the constraints into a contradiction. For a feasible LP, a similar vector combines them into a *bound* on the objective, and at the optimum the bound is tight. The coefficients of that wall turn out to be the marginal value of each resource, the shadow prices of §23.3. Why must the best bound equal the best design, with no gap? And why does this stop being true for non-convex problems? Separation is the hidden engine, and Ch. 48 shows how.

47.4 The algorithm

Separation theorems prove that certificates exist. Algorithms have to find them. Two simple algorithms do most of the work, and both are built from the projection of Theorem 47.1.

Projected gradient descent

To minimise a smooth convex f over a closed convex set C, take an ordinary gradient step (§23.5) and then snap back to the nearest feasible point:

```
input: f, its gradient, projection P_C, start x0 in C, step alpha
for k = 0, 1, 2, ...
    y      = x_k − alpha * grad f(x_k)        # unconstrained step
    x_{k+1} = P_C(y)                          # back to the feasible set
    stop when ||x_{k+1} − x_k|| is tiny
```

The fixed points are exactly the optimal points. $\mathbf{x} = P_C(\mathbf{x} - \alpha\nabla f(\mathbf{x}))$ says, by Theorem 47.1, that $-\alpha\nabla f(\mathbf{x}) \in N_C(\mathbf{x})$, which is the normal-cone condition. Two facts make the iteration trustworthy.

- **Projection never stretches distances:** $\|P_C(\mathbf{u}) - P_C(\mathbf{v})\| \le \|\mathbf{u} - \mathbf{v}\|$. To see it, add the obtuse-angle inequalities at the two projections and apply Cauchy–Schwarz. The snap-back step can never undo the progress of the gradient step.
- **Rates.** If ∇f is L-Lipschitz and $\alpha = 1/L$, then $f(\mathbf{x}_k) - f^* \le L\|\mathbf{x}_0 - \mathbf{x}^*\|^2/(2k)$: error $O(1/k)$, the same as unconstrained gradient descent. If f is also m-strongly convex, the distance to $\mathbf{x}^*$ shrinks by a factor of at least $1 - m/L$ per step. That is linear convergence, with the condition number $\kappa = L/m$ of §23.5 setting the speed.

Cost per step is one gradient plus one projection. Projections are cheap for boxes (clip each coordinate), balls (rescale), half-spaces (one formula) and the probability simplex (a sort, $O(n \log n)$). For a general polyhedron the projection is itself a quadratic program, and other methods (interior-point, §23.2) take over.

Alternating projections: finding a point, or a wall

To find a point in $C \cap D$, bounce between the two sets:

```
x = any starting point
repeat:  p = P_C(x);  x = P_D(p)
```

This is von Neumann's method of **alternating projections** (called POCS, projection onto convex sets, in signal processing). If the sets intersect, the iterates converge to a point in the intersection. If they are disjoint and one is compact, they converge to a **closest pair** $(\mathbf{p}, \mathbf{x})$. The vector $\mathbf{x} - \mathbf{p}$ is then the normal of a separating hyperplane through the midpoint (proof of Theorem 47.3). The same loop answers "feasible?" either way: it either exhibits a common point or hands you the certificate wall.

The perceptron, revisited

Rosenblatt's rule is a cousin of these methods. With labels $s_i = \pm 1$ and augmented points $\tilde{\mathbf{x}}_i = (\mathbf{x}_i, 1)$:

```
w = 0
repeat: pick any i with s_i * w.x_i <= 0  (a misclassified point)
        w = w + s_i * x_i
until no point is misclassified
```

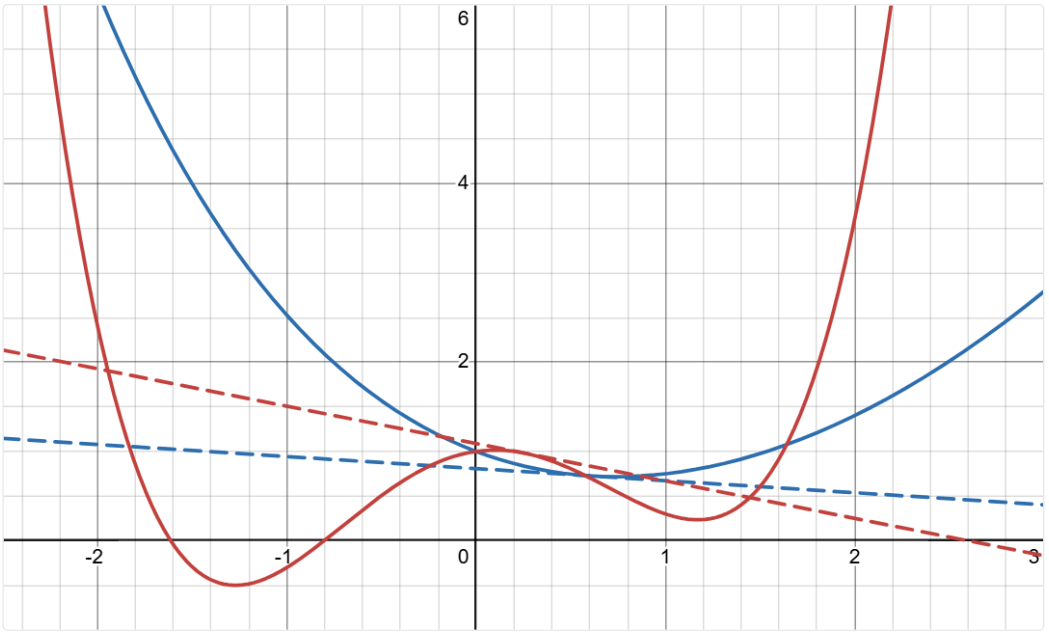
Novikoff's theorem (1962): if some unit vector separates the data with margin $\gamma > 0$ and every $\|\tilde{\mathbf{x}}_i\| \le R$, the rule stops after at most $(R/\gamma)^2$ updates. The margin γ is a quantitative version of "the convex hulls are disjoint": the further apart the hulls, the faster the learning. On XOR the hulls meet at $(\tfrac{1}{2}, \tfrac{1}{2})$, $\gamma = 0$, and the loop never ends.

Failure cases

- **Non-convex sets** (sparse vectors, integer points, rank constraints). Projection may not be unique, alternating projections can get stuck in cycles, and none of the guarantees above hold.
- **Sets that meet at a sliver.** If C and D touch at a very small angle, alternating projections creep toward the intersection, with each step shrinking the error only by about the cosine of that angle.
- **Expensive projections.** If P_C costs as much as the original problem, these methods offer nothing. For that case, ellipsoid and cutting-plane methods need only a *separation oracle*: a routine that, given a point outside C, returns one wall that cuts it off.

47.5 The experiment

Drag a tangent. The first two lines are $f(x) = 0.3x^2 + e^{-0.8x}$ and its tangent at $x_0 = 0.6$. Replace every `0.6` in the second line with another value (try -1.5, 0, 2.5): the tangent always stays under the graph. Then do the same with the tangent to the double-well g at $x_0 = 0.25$ (fourth line; try 1.2 and -1.3). For some x_0 the tangent pokes above the graph.



Blue: a convex function and one tangent, always below. Red: a non-convex function whose tangents can cross the graph. Change the x_0 values and watch which family keeps its promise.

Find a point or a wall. The script below runs alternating projections between the unit disc C and the half-plane $D = \{x + 2y \geq b\}$. With $b = 3$ the sets are disjoint. It prints the distance $3/\sqrt{5} - 1 \approx 0.341641$ and a separator, which after scaling is $x + 2y \approx 2.618$, halfway between the disc's tangent line $x + 2y = \sqrt{5}$ and D's edge $x + 2y = 3$. With $b = 1.5$ the distance drops to 0 and the last iterate is a common point.

```
import numpy as np
import matplotlib.pyplot as plt

# --- knobs ---------------------------------------------------------------
c, r = np.array([0.0, 0.0]), 1.0          # disc C: ||x - c|| <= r
a, b = np.array([1.0, 2.0]), 3.0          # half-plane D: a.x >= b  (try b = 1.5)
x = np.array([-2.0, 2.5])                 # starting point
iters = 30

def proj_disc(z):                         # closest point of C to z
    d = z - c; n = np.linalg.norm(d)
    return z if n <= r else c + r * d / n

def proj_half(z):                         # closest point of D to z
    s = a @ z - b
    return z if s >= 0 else z - s * a / (a @ a)

path = [x]
for k in range(iters):
    p = proj_disc(x); x = proj_half(p)
    path += [p, x]
path = np.array(path)
gap = np.linalg.norm(x - p)
print(f"distance between the sets ~ {gap:.6f}")
if gap > 1e-6:                            # build the separating hyperplane
    w = x - p; beta = w @ (x + p) / 2
    print(f"separator: {w[0]:.4f} x + {w[1]:.4f} y = {beta:.4f}")

t = np.linspace(0, 2 * np.pi, 200)
plt.fill(c[0] + r * np.cos(t), c[1] + r * np.sin(t), alpha=0.2)
xs = np.linspace(-3, 3, 2)
plt.plot(xs, (b - a[0] * xs) / a[1], "g", label="boundary of D")
plt.plot(path[:, 0], path[:, 1], "o-", ms=3, color="k", label="alternating projections")
plt.axis("equal"); plt.legend(); plt.show()
```

Things to try: move the disc centre `c` toward the line and watch the number of useful iterations grow as the sets begin to touch. Replace `proj_disc` with a projection onto a *ring* $\{0.5 \leq \|\mathbf{x}\| \leq 1\}$ (non-convex) and look for starting points where the method stalls far from the true closest pair.

47.6 The challenge

1. **Intuition.** Sketch two non-convex sets in the plane that are disjoint but cannot be separated by any line. Then explain in one sentence which step of the proof of Theorem 47.3 fails for them.
2. **Derivation.** Prove that projection onto a closed convex set is non-expansive, $\|P_C(\mathbf{u}) - P_C(\mathbf{v})\| \leq \|\mathbf{u} - \mathbf{v}\|$, using only the obtuse-angle condition of Theorem 47.1. Then derive the explicit projection onto the box $\{\mathbf{l} \leq \mathbf{x} \leq \mathbf{u}\}$ and check that it satisfies the condition.

3. **Implementation.** Write a function that takes two finite point sets in $\mathbb{R}^2$ and returns either a separating line or a *certificate of inseparability*: weights $\boldsymbol{\theta} \geq 0$, $\boldsymbol{\phi} \geq 0$, each summing to 1, with $\sum \theta_i \mathbf{a}_i = \sum \phi_j \mathbf{b}_j$. (Hint: Farkas' lemma tells you which linear feasibility problem to pose.) Test it on XOR and on two well-separated clusters.
4. **Open investigation.** Alternating projections between two convex sets that touch at a single point can converge very slowly. Build families of such pairs (two tangent discs, a disc and a tangent line, a parabola's epigraph and a line) and measure how the error after k steps decays: $1/k$, $1/\sqrt{k}$, or something else? Is there a geometric quantity at the touching point that predicts the rate?

47.7 Loose threads

This chapter built the geometry and left three doors closed.

? A hyperplane in a space of signals

Everything here lived in $\mathbb{R}^n$. Engineers constantly optimise over *functions*: pulse shapes, filter responses, control signals. Those live in infinite-dimensional spaces, where closed bounded sets need not be compact and the existence step of Theorem 47.1 can fail. Does every convex set of signals still have a separating wall? The answer is the Hahn–Banach theorem, one of the pillars of Ch. 61, and it changes what "a hyperplane" means.

? Why do non-convex problems so often behave?

Training a neural network is wildly non-convex: millions of variables, symmetries, saddle points everywhere. Yet plain gradient descent, with no certificate of anything, routinely finds solutions that generalise well. Nobody has a complete explanation of why these landscapes are so benign. It remains an open problem at the meeting point of optimization and learning theory, and Ch. 53 meets the same puzzle in policy optimization.

The third thread is the price-tag mystery of §47.3: the certificate vector $\mathbf{y}$ is about to become the dual variable, and the whole of Ch. 48 is about what it measures. Two further pointers: von Neumann's minimax theorem for zero-sum games (Ch. 39) is a separation theorem in disguise, and when the data of a convex problem are uncertain, the feasible set itself becomes an intersection of infinitely many half-spaces (Ch. 54).

Further reading

- S. Boyd and L. Vandenberghe, *Convex Optimization*, Cambridge University Press, 2004. Chapters 2–3 on sets and functions, with the separation theorems in §2.5. Freely available from the authors.
- R. T. Rockafellar, *Convex Analysis*, Princeton University Press, 1970. The classical foundation, including subgradients and support functions.
- J.-B. Hiriart-Urruty and C. Lemaréchal, *Fundamentals of Convex Analysis*, Springer, 2001. A gentler and very geometric route through the same material.
- D. P. Bertsekas, *Convex Optimization Theory*, Athena Scientific, 2009. Proofs organised around separation, and a natural bridge to duality.

Chapter 48 — Duality: Solving Problems from the Other Side

> **Level 2 · Track D — Optimization and Operations Research Builds on:** Ch. 6, Ch. 23, Ch. 47 · **Leads to:** Ch. 49, Ch. 51, Ch. 55 · **Related:** Ch. 39, Ch. 43, Ch. 50 **Reading time:** ~30 min

48.1 The hook

Every five minutes, somewhere in a control room, an operator has to decide how much electricity each of several hundred power stations should generate. Demand must be met exactly. No transmission line may overheat. The total cost must be as low as possible. Each station has its own fuel curve, ramp limits and quirks, and most of those details are private. A coal plant does not want to publish its true costs. A wind farm doesn't know its output until the wind blows.

One way to run this is central planning: collect every detail, solve one enormous problem, and send each station an order. Modern electricity markets mostly do something stranger. They broadcast **a price**, one number per location. Each station looks at the price, decides on its own how much it wants to produce, and offers that. If the total is too low, the price goes up; if it's too high, the price goes down. When the market clears, the stations' selfish choices add up to the minimum-cost dispatch, exactly what the central planner would have ordered.

That should seem too good to be true. Why should prices, which carry so much less information than the full problem, be enough to coordinate the optimum? And there's a second, sharper puzzle hidden in it. In some markets, after the price is set, the operator has to make extra side payments called "uplift" to certain stations. Without those payments, the stations would lose money by following the "optimal" schedule. So prices sometimes *fail*. What is the mathematical difference between a world where prices work and one where they don't?

This chapter answers both questions with a single construction, the **Lagrangian dual**. It turns every minimisation problem into a maximisation over prices. The two problems look at the same truth from opposite sides. For convex problems they meet exactly. For non-convex ones a gap opens, and the uplift payment is that gap, measured in money.

48.2 The big idea

Picture two people working on the same problem.

The **designer** stands inside the feasible region. Every feasible design she tries has some cost, and that cost is an *upper bound* on the best possible cost: the optimum can't be worse than something she has already built. Her bounds come down as she finds better designs.

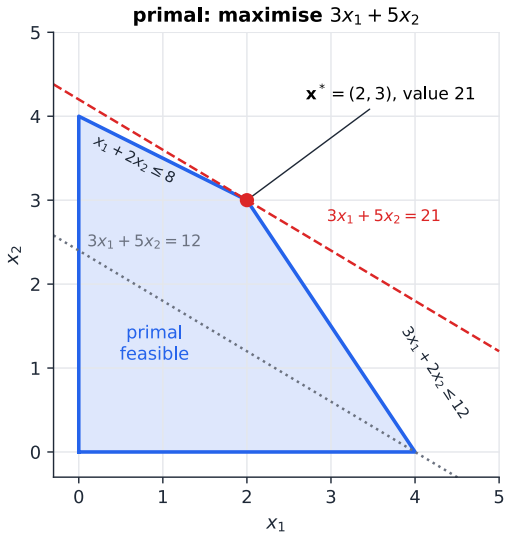
The **price-setter** never builds anything. She attaches a price to each constraint, which you can think of as a fine for each unit of violation. Then she asks: if constraints could be broken freely at these prices, what is the cheapest total of cost plus fines? Because any truly feasible design pays no fines, and at nonnegative prices can only collect credits for unused slack, that relaxed minimum is a *lower bound* on the true optimum. Her bounds go up as she tunes the prices.

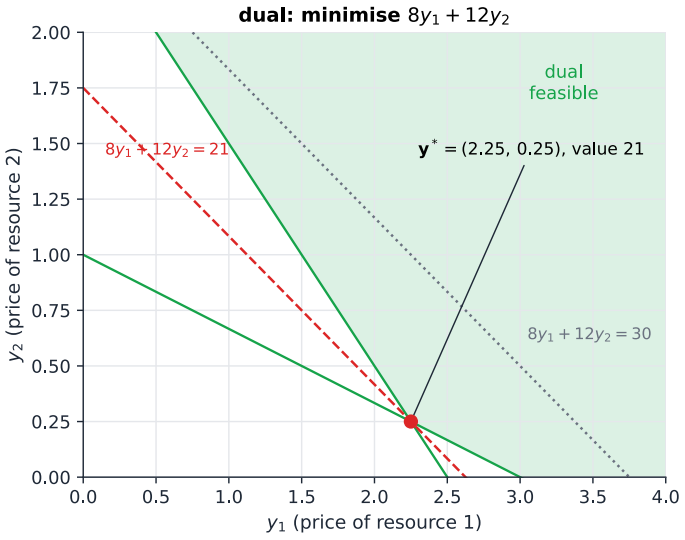
The designer's numbers come down from above and the price-setter's come up from below. **Weak duality** says the floor can never pass through the ceiling. **Strong duality** says that, for convex problems, the two meet. When they meet you hold two numbers that agree, and the agreement is the certificate of optimality that Ch. 47 promised. Nobody has to search further.

Here is the smallest example. A workshop makes two products with profits 3 and 5 per unit. It has 8 hours of machine time (product 1 needs 1 hour per unit, product 2 needs 2) and 12 kg of material (3 kg and 2 kg). The primal problem is

$$\max\ 3x_1 + 5x_2 \quad \text{s.t.} \quad x_1 + 2x_2 \le 8, \quad 3x_1 + 2x_2 \le 12, \quad x_1, x_2 \ge 0.$$

Suppose a buyer offers to rent the whole workshop, paying y_1 per machine hour and y_2 per kilogram. You'll accept only if the rent for the resources in each product is at least that product's profit: $y_1 + 3y_2 \ge 3$ and $2y_1 + 2y_2 \ge 5$. The buyer wants to pay as little as possible, $\min 8y_1 + 12y_2$. That is the **dual** problem, and it is the same "combine the constraints into a bound" trick as Farkas' lemma in §47.3, now aimed at the objective instead of at a contradiction.





Two problems, one number. The best production plan $(2, 3)$ earns 21 (left). The cheapest fair rent $(2.25, 0.25)$ also costs 21 (right). Every other feasible plan earns less than 21 and every other fair rent costs more: the dotted lines show a plan worth 12 and a rent worth 30.

The two answers are both 21. A machine hour is worth 2.25 at the margin and a kilogram 0.25. Those are the **shadow prices** of §23.3, now read as the price-setter's whole strategy. Give the workshop one more machine hour and its best profit rises to 23.25; one more kilogram, and it rises to 21.25. The prices tell you the value of resources you don't yet have.

48.3 The mathematics

The Lagrangian, dual function and the statements of weak and strong duality are in §23.7, and the KKT conditions are in §23.4 (with their equality-only ancestor, Lagrange multipliers, in §6.11). Here we prove the results and see *why* they hold.

Setup

The primal problem is

$$p^* = \min_{\mathbf{x}} \; f(\mathbf{x}) \quad \text{s.t.} \quad g_i(\mathbf{x}) \le 0, \; i = 1, \dots, m.$$

(Equality constraints add free-sign multipliers and change nothing essential. We leave them out to keep the notation light.) The **Lagrangian** is $\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) = f(\mathbf{x}) + \boldsymbol{\lambda}^{\mathsf{T}} \mathbf{g}(\mathbf{x})$, the **dual function** is $q(\boldsymbol{\lambda}) = \inf_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda})$, and the **dual problem** is $d^* = \max_{\boldsymbol{\lambda} \ge 0} q(\boldsymbol{\lambda})$. Note that the infimum is over *all* $\mathbf{x}$, feasible or not. That freedom is what lets the dual decompose.

Theorem 48.1 — Weak duality

q is concave, and $q(\boldsymbol{\lambda}) \le f(\mathbf{x})$ for every $\boldsymbol{\lambda} \ge \mathbf{0}$ and every feasible $\mathbf{x}$. Hence $d^* \le p^*$, whatever f and g_i are.

Proof. For fixed $\mathbf{x}$, $\boldsymbol{\lambda} \mapsto \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda})$ is affine. An infimum of affine functions is concave, because its hypograph is an intersection of half-spaces, which is the outside description of Ch. 47. For feasible $\mathbf{x}$ and $\boldsymbol{\lambda} \ge \mathbf{0}$ each term $\lambda_i g_i(\mathbf{x}) \le 0$, so $q(\boldsymbol{\lambda}) \le \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) \le f(\mathbf{x})$. $\square$

The dual is *always* a concave maximisation, even when the primal is a nightmare. That is the first reason duality is useful. You can always get a bound by solving an easy problem.

The picture: supporting lines

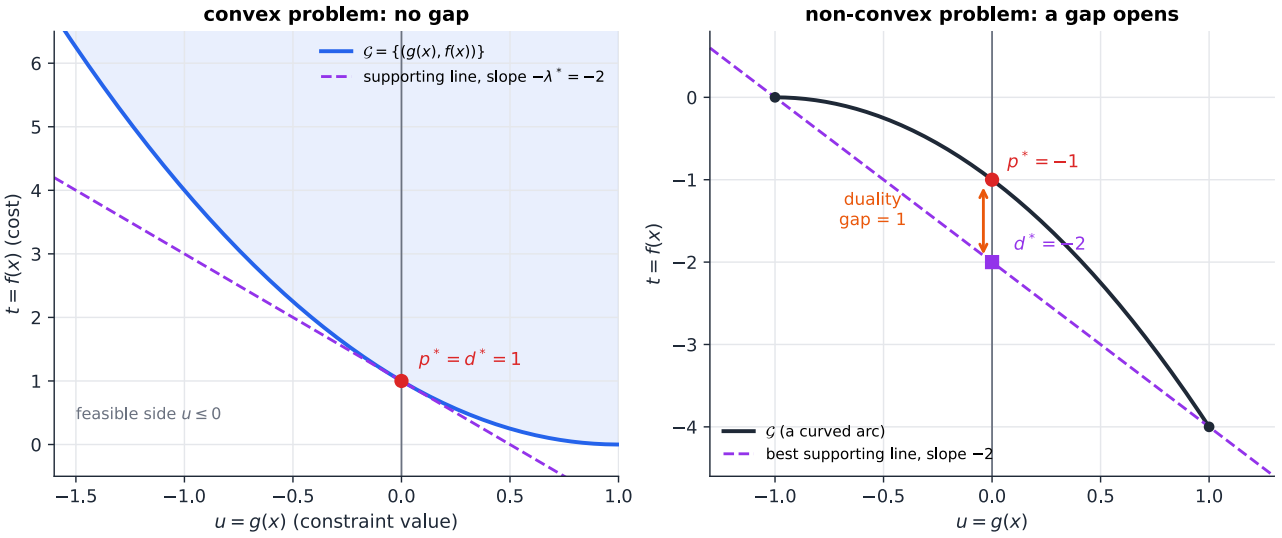
Take a single constraint and collect every pair of (constraint value, cost) that any $\mathbf{x}$ can achieve:

$$\mathcal{G} = \{(u, t) : \; u = g(\mathbf{x}), \; t = f(\mathbf{x}) \text{ for some } \mathbf{x}\} \subset \mathbb{R}^2.$$

The primal optimum is the lowest point of $\mathcal{G}$ to the left of the vertical axis: $p^* = \inf\{t : (u, t) \in \mathcal{G}, \; u \le 0\}$. The dual function is

$$q(\lambda) = \inf_{(u,t) \in \mathcal{G}} (t + \lambda u).$$

The lines $t + \lambda u = c$ have slope $-\lambda$. Push such a line up from below until it touches $\mathcal{G}$: its intercept with the axis $u = 0$ is exactly $q(\lambda)$. **The dual problem tilts a supporting line to make its intercept as high as possible.** Weak duality is now visible. A line lying under all of $\mathcal{G}$ with nonpositive slope must pass under the lowest feasible point too.



The geometry of duality. Left: $\min x^2$ s.t. $1 - x \le 0$. The set $\mathcal{G}$ is convex from below, a tilted line touches it at $(0, 1)$, and $p^* = d^* = 1$. Right: $\min -x^2$ s.t. $x - 1 \le 0$, $0 \le x \le 2$. The arc bends the wrong way, no line can reach $(0, -1)$ without cutting the arc, and the best intercept is -2. The vertical distance is the duality gap.

Worked Example 48.1: a duality gap, computed

Primal: $\min_{0\le x\le 2} -x^2$ subject to $x \le 1$. Feasible $x \in [0,1]$, so $p^* = -1$ at $x = 1$.

Dual function. $q(\lambda) = \min_{0\le x\le 2} \big(-x^2 + \lambda(x-1)\big)$. For fixed λ the bracket is concave in x, so its minimum over the interval is at an endpoint: $x = 0$ gives $-\lambda$, and $x = 2$ gives $\lambda - 4$. Hence $q(\lambda) = \min(-\lambda,\ \lambda - 4)$.

Dual problem. The two pieces cross at $\lambda^* = 2$, so $d^* = q(2) = -2$.

Gap. $p^* - d^* = -1 - (-2) = 1 > 0$. The Lagrangian minimiser at $\lambda = 2$ is $x = 0$ or $x = 2$, and neither is the primal optimum $x = 1$. The price can't "talk" the problem into choosing the middle point, because the middle point is never the cheapest at any price.

This is the electricity-market puzzle in miniature. When a generator has a start-up cost or a minimum stable output, its cost curve is non-convex. The price that clears the market makes the planner's schedule unprofitable for that generator, which on its own would rather sit at an "endpoint" (off, or full). The uplift payment covers the gap.

? Where else does the gap open?

Integer constraints such as "build the plant or don't" and "route the truck or don't" make $\mathcal{G}$ a scattered cloud of points, and the gap of Worked Example 48.1 becomes the normal state of affairs. Yet the dual bound is still the most valuable number an integer solver has, and clever relaxations can shrink the gap to zero. How the bound is used to prune billions of possibilities, and how cuts close the gap, is the story of Ch. 49.

Strong duality: separation does the work

Theorem 48.2 — Strong duality under Slater's condition

Suppose f and every g_i are convex, p^* is finite, and there is a strictly feasible point $\mathbf{x}_0$ with $g_i(\mathbf{x}_0) < 0$ for all i (**Slater's condition**). Then $d^* = p^*$, and the dual optimum is attained.

Proof. Use the "upper-right shadow" of $\mathcal{G}$ instead of $\mathcal{G}$ itself:

$$\mathcal{A} = \{(\mathbf{u}, t) \in \mathbb{R}^m \times \mathbb{R} :\ \exists\, \mathbf{x} \text{ with } \mathbf{g}(\mathbf{x}) \le \mathbf{u},\ f(\mathbf{x}) \le t\}.$$

Step 1: $\mathcal{A}$ is convex. If $(\mathbf{u}_1, t_1)$ and $(\mathbf{u}_2, t_2)$ are witnessed by $\mathbf{x}_1$ and $\mathbf{x}_2$, then for $\theta \in [0,1]$ the point $\theta\mathbf{x}_1 + (1-\theta)\mathbf{x}_2$ witnesses the mixture, because convexity gives $g_i(\theta\mathbf{x}_1 + (1-\theta)\mathbf{x}_2) \le \theta g_i(\mathbf{x}_1) + (1-\theta) g_i(\mathbf{x}_2) \le \theta u_{1i} + (1-\theta)u_{2i}$, and the same for f. This is the one place convexity is used.

Step 2: separate. The set $\mathcal{B} = \{(\mathbf{0}, t) : t < p^*\}$ is convex and disjoint from $\mathcal{A}$, since a point in both would be a feasible $\mathbf{x}$ with $f(\mathbf{x}) < p^*$. By the separation theorem for disjoint convex sets (§47.3, the non-strict version) there is $(\boldsymbol{\lambda}, \mu) \neq (\mathbf{0}, 0)$ with

$$\boldsymbol{\lambda}^{\mathsf{T}}\mathbf{u} + \mu t \ \ge\ \mu s \qquad \text{for all } (\mathbf{u}, t) \in \mathcal{A},\ s < p^*.$$

Step 3: signs. $\mathcal{A}$ is closed under increasing any coordinate. If some λ_i or μ were negative, sending that coordinate to $+\infty$ would push the left side to $-\infty$ and break the inequality. So $\boldsymbol{\lambda} \ge \mathbf{0}$ and $\mu \ge 0$. Letting $s \uparrow p^*$ gives $\boldsymbol{\lambda}^{\mathsf{T}}\mathbf{u} + \mu t \ge \mu p^*$ on $\mathcal{A}$.

Step 4: the line is not vertical. If $\mu = 0$, then $\boldsymbol{\lambda} \neq \mathbf{0}$, $\boldsymbol{\lambda} \ge \mathbf{0}$, and $\boldsymbol{\lambda}^{\mathsf{T}}\mathbf{u} \ge 0$ on $\mathcal{A}$. But Slater's point gives $(\mathbf{g}(\mathbf{x}_0), f(\mathbf{x}_0)) \in \mathcal{A}$ with $\boldsymbol{\lambda}^{\mathsf{T}}\mathbf{g}(\mathbf{x}_0) < 0$. Contradiction. So $\mu > 0$.

Step 5: read off the bound. Divide by μ and put $\boldsymbol{\lambda}^* = \boldsymbol{\lambda}/\mu \ge \mathbf{0}$. Each $\mathbf{x}$ gives the point $(\mathbf{g}(\mathbf{x}), f(\mathbf{x})) \in \mathcal{A}$, so $f(\mathbf{x}) + \boldsymbol{\lambda}^{*\mathsf{T}}\mathbf{g}(\mathbf{x}) \ge p^*$ for every $\mathbf{x}$, i.e. $q(\boldsymbol{\lambda}^*) \ge p^*$. With weak duality, $q(\boldsymbol{\lambda}^*) = d^* = p^*$. $\square$

The proof is a picture. Convexity makes $\mathcal{A}$ a convex blob, separation gives a supporting line at the optimum, and Slater's condition keeps that line from standing vertical, where it would have no slope to read a price from. The optimal multipliers *are* the slope of the separating hyperplane of Ch. 47, as that chapter's first mystery promised.

Common Pitfall: Slater is sufficient, not necessary

Without a strictly feasible point strong duality *can* fail even for convex problems. A classic is $\min e^{-x_2}$ subject to $x_1^2/x_2 \le 0$ with $x_2 > 0$: the feasible set is $x_1 = 0$, $p^* = 1$, yet $d^* = 0$. For linear constraints the condition relaxes to plain feasibility, which is why LP strong duality (§23.3) needs no Slater point.

KKT falls out

Theorem 48.3 — Zero gap forces the KKT conditions

If $\mathbf{x}^*$ is primal optimal, $\boldsymbol{\lambda}^*$ is dual optimal and $p^* = d^*$, then

1. $\mathbf{x}^*$ minimises $\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}^*)$ over all $\mathbf{x}$ (so $\nabla f(\mathbf{x}^*) + \sum \lambda_i^* \nabla g_i(\mathbf{x}^*) = \mathbf{0}$ when everything is differentiable);
2. $\lambda_i^* g_i(\mathbf{x}^*) = 0$ for every i (**complementary slackness**).

Proof. Chain the definitions:

$$f(\mathbf{x}^*) = q(\boldsymbol{\lambda}^*) = \inf_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}^*) \le \mathcal{L}(\mathbf{x}^*, \boldsymbol{\lambda}^*) = f(\mathbf{x}^*) + \sum_i \lambda_i^* g_i(\mathbf{x}^*) \le f(\mathbf{x}^*).$$

The two ends are equal, so both inequalities are equalities. The first equality says $\mathbf{x}^*$ attains the infimum: claim 1, and stationarity follows because an unconstrained minimiser of a differentiable function has zero gradient. The second says $\sum_i \lambda_i^* g_i(\mathbf{x}^*) = 0$. Every term is ≤ 0, so every term is 0: claim 2. $\square$

Read complementary slackness as an economist would. **A resource that isn't used up has price zero; a resource with a positive price is used up exactly.** In the workshop both resources are exhausted at $(2, 3)$ ($2+6=8$, $6+6=12$), and both have positive prices.

The chain also shows that $(\mathbf{x}^*, \boldsymbol{\lambda}^*)$ is a **saddle point**: $\mathcal{L}(\mathbf{x}^*, \boldsymbol{\lambda}) \le \mathcal{L}(\mathbf{x}^*, \boldsymbol{\lambda}^*) \le \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}^*)$. Strong duality says $\min_{\mathbf{x}} \max_{\boldsymbol{\lambda}\ge 0} \mathcal{L} = \max_{\boldsymbol{\lambda}\ge 0} \min_{\mathbf{x}} \mathcal{L}$. It's a two-player game between designer and price-setter, and it doesn't matter who moves first. That is the same exchange of min and max as von Neumann's minimax theorem for zero-sum games (Ch. 39, Ch. 46).

Sensitivity: what a price predicts

Perturb the constraints to $\mathbf{g}(\mathbf{x}) \le \mathbf{u}$ and call the new optimal value $p(\mathbf{u})$. Then $p(\mathbf{0}) = p^*$. Loosening ($\mathbf{u} > \mathbf{0}$) can only help, and tightening can only hurt.

Theorem 48.4 — Global sensitivity inequality

If strong duality holds with dual optimum $\boldsymbol{\lambda}^*$, then for **every** $\mathbf{u}$

$$p(\mathbf{u}) \;\ge\; p^* - \boldsymbol{\lambda}^{*\mathsf{T}}\mathbf{u}.$$

If p is differentiable at $\mathbf{0}$, then $\nabla p(\mathbf{0}) = -\boldsymbol{\lambda}^*$.

Proof. Let $\mathbf{x}$ satisfy $\mathbf{g}(\mathbf{x}) \le \mathbf{u}$. Since $\boldsymbol{\lambda}^* \ge \mathbf{0}$,

$$f(\mathbf{x}) = \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}^*) - \boldsymbol{\lambda}^{*\mathsf{T}}\mathbf{g}(\mathbf{x}) \;\ge\; q(\boldsymbol{\lambda}^*) - \boldsymbol{\lambda}^{*\mathsf{T}}\mathbf{u} = p^* - \boldsymbol{\lambda}^{*\mathsf{T}}\mathbf{u}.$$

Take the infimum over such $\mathbf{x}$. For the derivative, the inequality says the affine function $p^* - \boldsymbol{\lambda}^{*\mathsf{T}}\mathbf{u}$ touches p at $\mathbf{0}$ from below, so $-\boldsymbol{\lambda}^*$ is a subgradient of p there. A differentiable function's only subgradient is its gradient. $\square$

The inequality is one-sided, and that is the practical lesson:

- **Large λ_i^*, tighten constraint i:** the optimal cost is *guaranteed* to rise steeply, at least by λ_i^* per unit. Never tighten an expensive constraint casually.
- **Large λ_i^*, loosen constraint i:** the theorem promises nothing beyond a first-order estimate. The gain could be smaller than λ_i^* per unit, because another constraint may become active.
- **$\lambda_i^* = 0$:** small changes in either direction don't matter to first order. That doesn't make the constraint useless at large changes.

Check on the workshop, a *maximisation*, so the signs flip. One extra machine hour raises the profit from 21 to 23.25, exactly $y_1^* = 2.25$. One extra kilogram raises it to 21.25, exactly $y_2^* = 0.25$. LP values are piecewise linear in $\mathbf{u}$, so the prediction is exact until the optimal vertex changes. For the convex example $\min x^2$ s.t. $x \ge b$, the value $p(b) = b^2$ has slope $2b$, and the multiplier at $b = 1$ is $\lambda^* = 2$.

Engineering Note: locational marginal prices

In a power-market dispatch problem, the multiplier of the power-balance constraint at bus k is the cost of serving one more megawatt there, the **locational marginal price**. Theorem 48.4 is why it is a fair price, and complementary slackness is why prices differ across a congested line but agree across an uncongested one. Planners use these multipliers to decide where a new line or a battery is worth most.

48.4 The algorithm

The hook's market isn't just an analogy. It is an algorithm called **dual decomposition** (or **dual ascent**), and it is how very large problems that share a few constraints get solved in practice.

Why prices decompose problems

Suppose n agents each choose $\mathbf{x}_k$ with private cost f_k, and they share one resource: $\sum_k \mathbf{a}_k^{\mathsf{T}}\mathbf{x}_k \le c$. The Lagrangian

$$\mathcal{L} = \sum_k f_k(\mathbf{x}_k) + \lambda\Big(\sum_k \mathbf{a}_k^{\mathsf{T}}\mathbf{x}_k - c\Big) = \sum_k \Big[f_k(\mathbf{x}_k) + \lambda\,\mathbf{a}_k^{\mathsf{T}}\mathbf{x}_k\Big] - \lambda c$$

is a **sum of separate problems**: once λ is fixed, agent k minimises its own bracket and nobody needs to know anyone else's f_k. The only shared information is the price. That is why markets, network congestion control and distributed machine learning can all be organised around prices.

The price update is a gradient step

How should the price move? There's a one-line answer.

Proposition: the constraint violation is a supergradient of the dual

If $\mathbf{x}(\boldsymbol{\lambda})$ minimises $\mathcal{L}(\cdot, \boldsymbol{\lambda})$, then for every $\boldsymbol{\lambda}'$, $q(\boldsymbol{\lambda}') \le q(\boldsymbol{\lambda}) + (\boldsymbol{\lambda}' - \boldsymbol{\lambda})^{\mathsf{T}}\mathbf{g}(\mathbf{x}(\boldsymbol{\lambda}))$.

Proof. $q(\boldsymbol{\lambda}') \le \mathcal{L}(\mathbf{x}(\boldsymbol{\lambda}), \boldsymbol{\lambda}') = \mathcal{L}(\mathbf{x}(\boldsymbol{\lambda}), \boldsymbol{\lambda}) + (\boldsymbol{\lambda}' - \boldsymbol{\lambda})^{\mathsf{T}}\mathbf{g}(\mathbf{x}(\boldsymbol{\lambda}))$, and the first term is $q(\boldsymbol{\lambda})$. $\square$

So to climb the concave dual, move the price in the direction of the constraint violation, and never let it go negative:

```
lambda = initial prices (>= 0)
repeat:
    each agent k:   x_k = argmin  f_k(x_k) + lambda . (a_k x_k)      # in parallel, privately
    violation       = sum_k a_k x_k - c
    lambda          = max(0, lambda + alpha * violation)              # raise price if over-demanded
until violation is small and the bounds agree
```

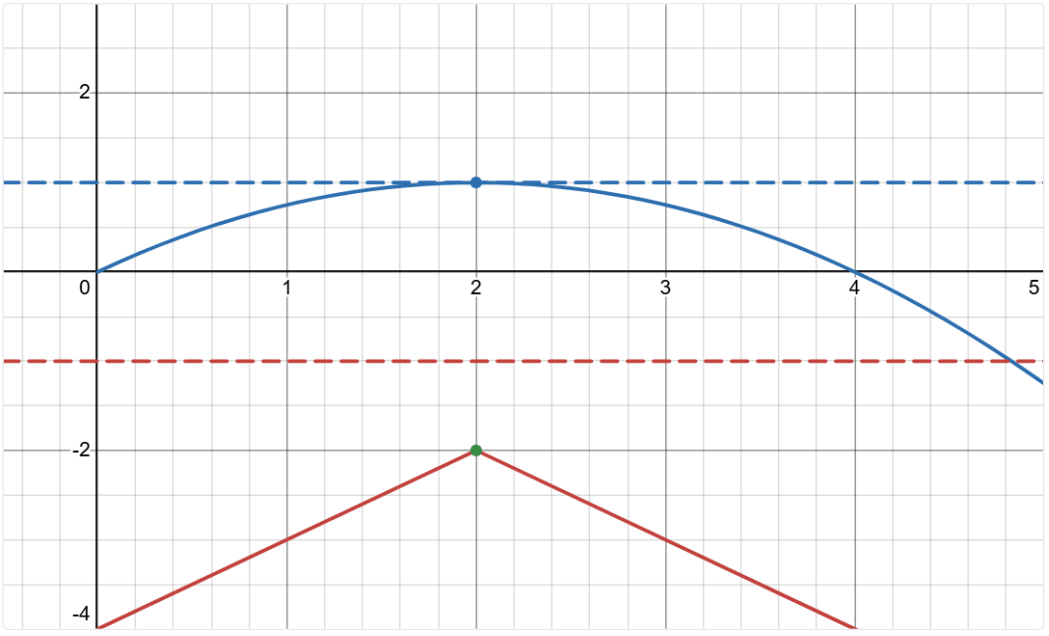
Convergence. If every f_k is strictly convex, q is differentiable and a small fixed step α converges. If the f_k are merely convex (linear costs, for example), q has kinks, and you need diminishing steps such as $\alpha_k \propto 1/\sqrt{k}$. Even then the error falls only like $O(1/\sqrt{k})$, which is slow. Each iteration hands you a certificate for free: the dual value $q(\boldsymbol{\lambda}_k)$ bounds the optimum from one side, and any feasible repair of $\mathbf{x}(\boldsymbol{\lambda}_k)$ bounds it from the other.

Failure cases.

- **Linear costs make agents jumpy.** If f_k is linear, the minimiser $\mathbf{x}(\lambda)$ leaps from one extreme to the other as λ crosses a threshold, and the total demand oscillates around c without settling. The standard cure is to add a quadratic penalty $\frac{\rho}{2}\|\text{violation}\|^2$ to the Lagrangian (the **augmented Lagrangian**, and its distributed form **ADMM**). That buys smoothness at the price of coupling.
- **Primal infeasibility.** The dual can converge while $\mathbf{x}(\boldsymbol{\lambda}_k)$ remains slightly infeasible. Engineers report averaged iterates or project the final answer onto the feasible set.
- **Non-convexity.** Worked Example 48.1 shows the limit. With a gap, no price makes the agents' selfish choices add up to the optimum, and the loop can cycle between "endpoints" forever.

48.5 The experiment

Tilt the dual. The blue curve is the dual function $q(\lambda) = b\lambda - \lambda^2/4$ of $\min x^2$ s.t. $x \geq b$ with $b = 1$. The blue dashed line is the primal value $p^\star = b^2 = 1$. Change both `1` s in the first line and the dashed level to a new b (try $b = 1.5$: the peak moves to $\lambda^\star = 2b = 3$ and the level to 2.25). The peak always touches the primal value. That is strong duality, and its location is the sensitivity $dp^\star/db = 2b$. The red curve is the dual $\min(-\lambda, \lambda - 4)$ of Worked Example 48.1, and the red dashed line is its primal value -1. The peak at -2 never reaches the line.



Dual functions plotted against the price λ (horizontal axis). The convex problem's dual (blue) rises to meet its primal value; the non-convex one (red) stops a full unit short.

Run a market for power. The script allocates a total power P across channels with noise levels N_k to maximise capacity $\sum_k \ln(1 + p_k/N_k)$, the water-filling problem of §23.4. It never solves the problem directly. It announces a price λ, lets each channel buy $p_k = \max(0, 1/\lambda - N_k)$, and adjusts the price. Each round it records two bounds: the dual value (an upper bound, since this is a maximisation) and the capacity of a rescaled feasible design (a lower bound).

```
import numpy as np
import matplotlib.pyplot as plt

# --- knobs ---------------------------------------------------------------
N = np.array([0.2, 0.5, 1.0, 1.6])   # noise level of each channel
P = 2.0                              # total power budget
lam, alpha, iters = 2.0, 0.3, 40     # starting price, price step, rounds

def best_response(lam):              # each channel maximises log(1+p/N) - lam*p
    return np.maximum(0.0, 1.0 / lam - N)

def utility(p):
    return np.sum(np.log(1 + p / N))

upper, lower, prices = [], [], []
for k in range(iters):
    p = best_response(lam)
    upper.append(utility(p) - lam * p.sum() + lam * P)    # dual value g(lam)
    feas = p * P / p.sum() if p.sum() > 0 else np.full_like(N, P / len(N))
    lower.append(utility(feas))                           # a feasible design
    prices.append(lam)
    lam = max(1e-6, lam + alpha * (p.sum() - P))          # raise price if over budget

print("final price  :", round(lam, 4), " water level 1/price:", round(1 / lam, 4))
print("final powers :", np.round(best_response(lam), 4))
print("bounds       :", round(lower[-1], 5), "<= optimum <=", round(upper[-1], 5))

fig, ax = plt.subplots(1, 2, figsize=(10, 3.5))
ax[0].plot(upper, label="dual bound g(lambda)"); ax[0].plot(lower, label="feasible design")
ax[0].set_xlabel("round"); ax[0].set_ylabel("capacity (nats)"); ax[0].legend()
ax[1].plot(prices); ax[1].set_xlabel("round"); ax[1].set_ylabel("price lambda")
plt.tight_layout(); plt.show()
```

With the default knobs the price settles at $\lambda^* \approx 0.8108$ (water level $1/\lambda^* = 1.2333$). The powers come out as $(1.0333,\ 0.7333,\ 0.2333,\ 0)$: the noisiest channel gets nothing, its nonnegativity constraint is active. Both bounds agree at 2.93175 nats. Things to try: raise `alpha` to 0.6 and the price keeps oscillating without settling (at 1.5 it runs away entirely); set `P = 0.2` so only one channel is worth buying; replace the utility with a non-concave one, say $\sqrt{p}$ capped at a threshold, and see whether the two bounds still meet.

48.6 The challenge

1. **Intuition.** A constraint in your design has multiplier $\lambda^* = 0$. A colleague proposes deleting it "because it doesn't matter". Describe, with a sketch of $\mathcal{G}$ or of $p(u)$, a situation in which deleting it changes the optimum completely.
2. **Derivation.** Derive the dual of the minimum-norm problem $\min \tfrac{1}{2}\|\mathbf{x}\|^2$ s.t. $\mathbf{A}\mathbf{x} = \mathbf{b}$, solve the dual in closed form, and show that strong duality recovers $\mathbf{x}^* = \mathbf{A}^{\mathsf{T}}(\mathbf{A}\mathbf{A}^{\mathsf{T}})^{-1}\mathbf{b}$. Then do the same with $\mathbf{x} \geq \mathbf{0}$ added, and explain why the closed form disappears.
3. **Implementation.** Extend the water-filling script to two shared budgets, a total power limit and a per-group limit on channels 1–2. Use two prices updated simultaneously. Plot both prices and verify complementary slackness numerically by printing each price next to the slack of its constraint.
4. **Open investigation.** Take a market with a few non-convex agents, each of which either produces nothing or at least a minimum output $m_k > 0$, and run price adjustment. Measure the duality gap as you add more and more small agents. Does the gap per agent shrink? Find, by experiment, how the gap scales with the number of agents and with the size of the non-convexity (this is related to the Shapley–Folkman lemma, but the rate in your model is for you to discover).

48.7 Loose threads

Duality was presented here for a handful of variables. Each generalisation opens a new door.

> **? A price for every instant**
>
> In optimal control the decision is a whole trajectory $\mathbf{u}(t)$ and the constraint is a differential equation $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u})$ that must hold at *every* instant. Attach a multiplier to each instant and you get a multiplier *function* $\mathbf{p}(t)$, which itself obeys a differential equation running backward in time. Why backward? And what does this "costate" price? Pontryagin's principle in Ch. 51 answers both.

> **? The dual of moving dirt**
>
> Moving a pile of sand into a hole of the same shape at least total cost is a linear program with infinitely many variables, one for each pair of source and destination points. Its dual looks for a *price field* $\varphi(\mathbf{x})$ that a hauling company could charge. The Kantorovich duality that links them, and the strange "conjugate" operation that turns one price field into another, wait in Ch. 55.

The third thread is §48.3's mystery: the gap of non-convex problems, and how integer programming exploits the dual bound anyway (Ch. 49). When the constraints themselves are uncertain, the dual of a worst-case problem becomes a problem about probability distributions (Ch. 54). For the economics of prices that fail, see auctions and markets in Ch. 43.

Further reading

- S. Boyd and L. Vandenberghe, *Convex Optimization*, Cambridge University Press, 2004. Chapter 5: Lagrange duality, its geometric interpretation, and sensitivity analysis.
- D. P. Bertsekas, *Nonlinear Programming*, 3rd ed., Athena Scientific, 2016. Duality, dual methods and the augmented Lagrangian, with many distributed examples.
- D. G. Luenberger and Y. Ye, *Linear and Nonlinear Programming*, 4th ed., Springer, 2016. LP duality, shadow prices and sensitivity from the operations-research side.
- S. Boyd, N. Parikh, E. Chu, B. Peleato and J. Eckstein, "Distributed Optimization and Statistical Learning via the Alternating Direction Method of Multipliers", *Foundations and Trends in Machine Learning* 3(1), 2011. Dual decomposition and ADMM in practice.

Chapter 49 — Integer Programming and Combinatorial Optimization

> **Level 2 · Track D — Optimization and Operations Research Builds on:** Ch. 8, Ch. 23, Ch. 47, Ch. 48 · **Leads to:** Ch. 50, Ch. 52 · **Related:** Ch. 27, Ch. 28, Ch. 30 **Reading time:** ~30 min

49.1 The hook

A data-centre team is filling a hall with server racks. There are two designs. A type-X rack delivers 5 units of compute and draws 5 kW; a type-Y rack delivers 8 units and draws 9 kW. The hall has 6 rack positions and a 45 kW power feed. How many of each should they install?

An engineer who knows linear programming (§23.3) writes it down in a minute:

$$\max\ 5x + 8y \quad \text{s.t.} \quad x + y \le 6, \quad 5x + 9y \le 45, \quad x, y \ge 0,$$

and the solver replies at once: $x = 2.25$, $y = 3.75$, compute 41.25.

You can't install a quarter of a rack. The obvious fix is to round. Rounding to the nearest integers gives $(2, 4)$, which draws $10 + 36 = 46$ kW and trips the breaker. Rounding down to $(2, 3)$ is safe but gives only 34, 18% below the LP's promise. Hunting around nearby, $(3, 3)$ gives 39. The true optimum turns out to be $(0, 5)$: five type-Y racks, compute 40, and *zero* of the type the LP wanted 2.25 of. The best integer answer isn't a rounding of the LP answer at all. It sits on the far side of the feasible region.

With two variables you can list all 25 feasible integer points and check. Real problems aren't like that. An airline assigning crews to flights, a chip designer placing logic cells, a utility deciding which power stations to switch on tomorrow: each has thousands or millions of yes/no decisions. One hundred binary choices already give $2^{100} \approx 1.27 \times 10^{30}$ combinations. At a billion checks per second that would take about 4×10^{13} years, thousands of times the age of the universe.

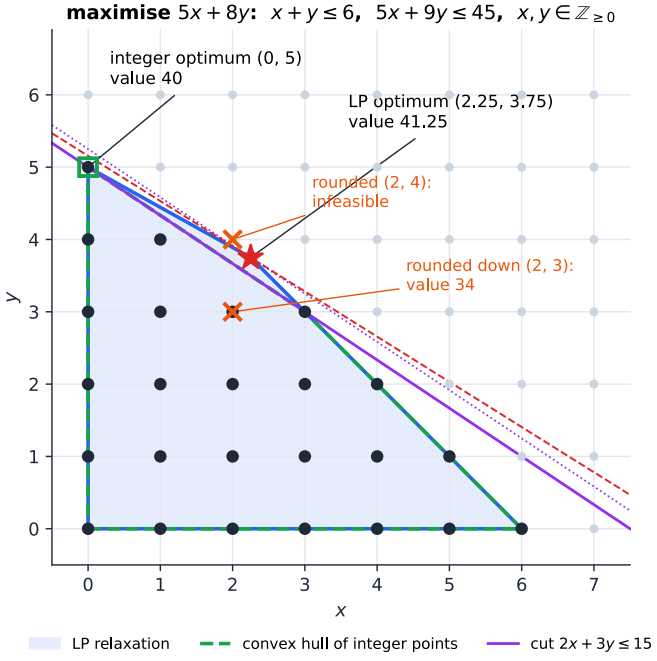
Yet commercial solvers settle problems like these to proven optimality every day, often in minutes. They don't round, and they don't enumerate. This chapter is about the trick: **use the easy continuous problem as a bound, and cut the hopeless search tree down with it.**

49.2 The big idea

The integer points sit inside the LP's polygon like seeds in a fruit. The LP relaxation, which drops the integrality requirement, is a *convex* problem (Ch. 47): easy to solve, and it always over-promises. Its value is an upper bound on what any integer design can achieve. Integer programming runs on two ideas built on that bound.

Idea 1: divide and bound. Split the problem into pieces, "$x \le 2$" and "$x \ge 3$", which together contain every integer point and remove the fractional LP solution in between. Solve each piece's LP. If a piece's bound is already worse than a design you hold in hand, the whole piece, with every integer point inside it, can be thrown away without being looked at. That is **branch-and-bound**. Its power comes entirely from how much the bounds let you skip.

Idea 2: tighten the fruit. Between the integer points and the LP polygon there's wasted space. If you could shrink the polygon to the **convex hull** of the integer points, the tightest convex shape containing them, its corners would *be* integer points, and a single LP would solve the integer problem. You can't compute the hull directly; it may have astronomically many facets. But you can add a few of its walls, **cutting planes**, exactly where the LP solution is cheating.



The rack problem. The LP relaxation (blue) has a fractional corner worth 41.25. The convex hull of the integer points (green dashed) is smaller. One extra wall, the cut $2x + 3y \le 15$ (purple), slices off the fractional corner, and on the trimmed polygon the LP optimum is the integer point $(0, 5)$.

A useful analogy is detective work. Branching is splitting the suspects into groups and interrogating each group's alibi. A bound is an alibi so strong that the whole group goes home at once. A cut is a new fact that rules out an impossible story ("nobody was in two rooms at the same time") without naming a suspect. Real solvers do both, constantly. That is **branch-and-cut**.

49.3 The mathematics

Modelling with integers

The step from "continuous" to "integer" is mostly about what you can *say*. Binary variables $z \in \{0,1\}$ turn logic into linear inequalities.

Requirement	Constraint
choose at most one of items 1–4	$z_1 + z_2 + z_3 + z_4 \le 1$
if you build A you must build B	$z_A \le z_B$
power p only if the station is on, $0 \le p \le P_{\max}$	$0 \le p \le P_{\max} z$
fixed start-up cost F plus running cost c	cost $Fz + cp$ with $p \le P_{\max} z$
either $\mathbf{a}^{\mathsf T}\mathbf{x} \le b_1$ or $\mathbf{c}^{\mathsf T}\mathbf{x} \le b_2$	$\mathbf{a}^{\mathsf T}\mathbf{x} \le b_1 + Mz$, $\mathbf{c}^{\mathsf T}\mathbf{x} \le b_2 + M(1-z)$

The last row uses a **big-M**, a constant large enough that the constraint switches off when its binary says so. The fixed-charge row is exactly the start-up cost that broke the electricity market's prices in §48.3. Non-convexity that defeats prices can often be *modelled* exactly with a few binaries.

The classic test problem is the **0–1 knapsack**: $\max \sum v_i z_i$ s.t. $\sum w_i z_i \le W$, $z_i \in \{0,1\}$. Pack items of value v_i and weight w_i into a bag of capacity W. It looks innocent. It is NP-hard.

> **Common Pitfall: big-M too big**
>
> Any M large enough gives a *correct* model, but not an equally good one. In the relaxation, z may take the value $p/P_{\max}$, so a huge M lets z be tiny and almost cancels the fixed cost. The LP bound becomes weak, and branch-and-bound must work far harder. Choose the smallest M that is valid. Physical limits ($P_{\max}$ itself) are usually the right choice.

Relaxations bound, and hulls are exact

Write the integer program as $z_{\mathrm{IP}} = \max\{\mathbf{c}^{\mathsf T}\mathbf{x} : \mathbf{x} \in S\}$ with $S = P \cap \mathbb{Z}^n$, where $P = \{\mathbf{x} : \mathbf{A}\mathbf{x} \le \mathbf{b}\}$ is the LP polyhedron, and let $z_{\mathrm{LP}} = \max\{\mathbf{c}^{\mathsf T}\mathbf{x} : \mathbf{x} \in P\}$.

> **Theorem 49.1 — The relaxation bound and the convex hull**
>
> 1. $z_{\mathrm{IP}} \le z_{\mathrm{LP}}$.
> 2. If S is finite and nonempty, $z_{\mathrm{IP}} = \max\{\mathbf{c}^{\mathsf T}\mathbf{x} : \mathbf{x} \in \operatorname{conv} S\}$, and the maximum is attained at a point of S.

Proof. (1) $S \subseteq P$, and maximising over a larger set can't give less. (2) Every $\mathbf{x} \in \operatorname{conv} S$ is a convex combination $\sum \theta_k \mathbf{s}_k$ of points of S (§47.3), so $\mathbf{c}^{\mathsf T}\mathbf{x} = \sum \theta_k \mathbf{c}^{\mathsf T}\mathbf{s}_k \le \max_k \mathbf{c}^{\mathsf T}\mathbf{s}_k \le z_{\mathrm{IP}}$. A linear function can't do better on a mixture than on the best ingredient. The reverse inequality holds because $S \subseteq \operatorname{conv} S$. □

So integer programming *is* linear programming over the right polygon, $\operatorname{conv} S$. The difficulty is entirely in describing that polygon. For the racks, $\operatorname{conv} S$ has corners $(0,0)$, $(6,0)$, $(3,3)$, $(0,5)$ and the inequality description

$$x \ge 0, \qquad y \ge 0, \qquad x + y \le 6, \qquad 2x + 3y \le 15.$$

The **integrality gap** $z_{\mathrm{LP}} - z_{\mathrm{IP}} = 41.25 - 40 = 1.25$ measures how much the relaxation over-promises. It's the integer-programming cousin of the duality gap in Ch. 48.

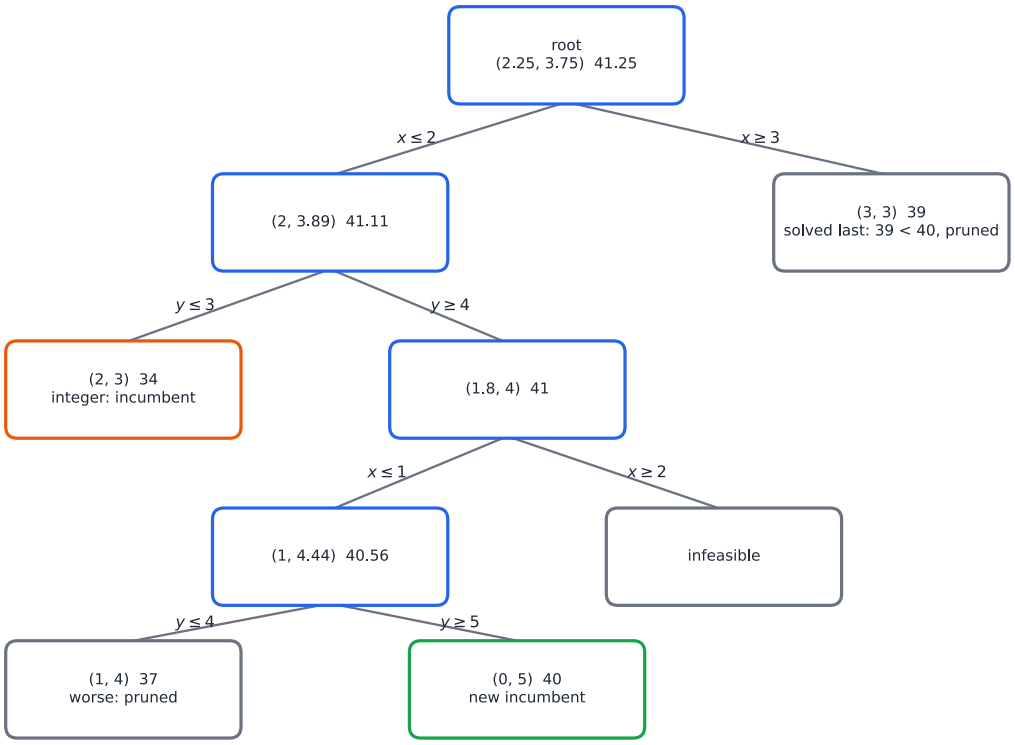
Branch-and-bound

Branch-and-bound explores a tree. Each node is the original problem plus some extra bounds on variables, and its LP value $\bar z$ is an upper bound for every integer point in that node. The best integer solution found so far, the **incumbent**, has value z°.

> **Theorem 49.2 — Pruning never loses the optimum**
>
> Suppose the integer points of a node are split into two children by $x_j \le \lfloor \bar x_j \rfloor$ and $x_j \ge \lfloor \bar x_j \rfloor + 1$, and a node is discarded only if (a) its LP is infeasible, (b) its LP solution is integer (it then updates the incumbent), or (c) $\bar z \le z^\circ$. Then when no open nodes remain, the incumbent is optimal.

Proof. Every integer point of a node lies in exactly one child, because no integer lies strictly between $\lfloor \bar x_j \rfloor$ and $\lfloor \bar x_j \rfloor + 1$. So the leaves of the tree always partition S. Take an optimal $\mathbf{x}^\star$ and follow it down the tree to the leaf that contains it. That leaf was closed for one of three reasons. (a) is impossible, since $\mathbf{x}^\star$ is feasible for it. Under (b), the leaf's LP optimum is an integer point of value at least $\mathbf{c}^{\mathsf T}\mathbf{x}^\star$, and it entered the incumbent. Under (c), $\mathbf{c}^{\mathsf T}\mathbf{x}^\star \le \bar z \le z^\circ$. In each case the final incumbent is at least as good as $\mathbf{x}^\star$. The tree is finite when the variables are bounded, because each branch shrinks an integer range. □

Here is the whole search for the racks, branching on the more fractional variable and exploring the "$\le$" child first.



Branch-and-bound on the rack problem. Each box shows the node's LP solution and its value, an upper bound for that branch. Depth-first search finds the incumbent 34, improves it to 40 at $(0, 5)$, and then closes the right branch without branching on it: that branch's integer optimum is only 39.

Nine LPs settle the problem, and the final one is closed by its bound. In larger problems the bound does far more work: whole subtrees with billions of leaves vanish in a single comparison.

Cutting planes

Theorem 49.3 — Chvátal–Gomory cuts are valid

Let $\mathbf{A}\mathbf{x} \le \mathbf{b}$, $\mathbf{x} \ge \mathbf{0}$ define P and let $\mathbf{u} \ge \mathbf{0}$. Then every integer point of P satisfies

$$\sum_j \lfloor (\mathbf{u}^{\mathsf{T}}\mathbf{A})_j \rfloor x_j \;\le\; \lfloor \mathbf{u}^{\mathsf{T}}\mathbf{b} \rfloor.$$

Proof. Combining the rows with weights $\mathbf{u} \ge \mathbf{0}$ gives the valid inequality $\sum_j (\mathbf{u}^{\mathsf{T}}\mathbf{A})_j x_j \le \mathbf{u}^{\mathsf{T}}\mathbf{b}$, the same "combine constraints" move as in Farkas' lemma and LP duality. Because $\mathbf{x} \ge \mathbf{0}$, rounding the coefficients *down* can only shrink the left side, so $\sum_j \lfloor (\mathbf{u}^{\mathsf{T}}\mathbf{A})_j \rfloor x_j \le \mathbf{u}^{\mathsf{T}}\mathbf{b}$ still holds. For integer $\mathbf{x}$ the left side is now an integer, and an integer at most $\mathbf{u}^{\mathsf{T}}\mathbf{b}$ is at most $\lfloor \mathbf{u}^{\mathsf{T}}\mathbf{b} \rfloor$. $\square$

The last step is the only place integrality enters, and it's where fractional points get cut off. For the racks take $\mathbf{u} = (\tfrac{3}{4}, \tfrac{1}{4})$:

$$\tfrac{3}{4}(x+y) + \tfrac{1}{4}(5x+9y) \le \tfrac{3}{4}\cdot 6 + \tfrac{1}{4}\cdot 45 \quad\Longrightarrow\quad 2x + 3y \le 15.75 \quad\Longrightarrow\quad 2x + 3y \le 15.$$

The LP optimum has $2(2.25) + 3(3.75) = 15.75$, so it violates the cut by 0.75. Re-solving with the cut added gives $(0, 5)$, value 40, the integer optimum, in a single round. The cut is exactly the missing facet of the hull.

That was lucky. In general Gomory (1958) showed how to read a violated cut directly from the final simplex tableau. Chvátal (1973) proved that repeating the round-down operation finitely many times always reaches $\operatorname{conv} S$, but the number of rounds can be enormous. In practice cuts and branching work together: cut until progress stalls, then branch.

When the LP is already integral

Some integer programs need no integer machinery at all.

Total unimodularity

A matrix $\mathbf{A}$ is **totally unimodular** (TU) if every square submatrix has determinant 0, $+1$ or -1.

Theorem 49.4 — TU matrices give integer vertices

If $\mathbf{A}$ is TU and $\mathbf{b}$ is an integer vector, every vertex of $P = \{\mathbf{x} : \mathbf{A}\mathbf{x} \le \mathbf{b},\ \mathbf{x} \ge \mathbf{0}\}$ is integer. Hence the LP relaxation solves the integer program.

Proof. Stacking $\mathbf{A}$ on $-\mathbf{I}$ (for $\mathbf{x} \geq \mathbf{0}$) keeps the matrix TU, because expanding a determinant along the rows coming from $-\mathbf{I}$ leaves a $\pm$ determinant of a submatrix of $\mathbf{A}$. A vertex is the unique solution of $\mathbf{B}\mathbf{x} = \mathbf{b}'$ for some nonsingular square submatrix $\mathbf{B}$ of the stacked matrix and the matching integer entries $\mathbf{b}'$. By Cramer's rule (Ch. 8), $x_i = \det \mathbf{B}_i / \det \mathbf{B}$, where $\mathbf{B}_i$ is $\mathbf{B}$ with column i replaced by $\mathbf{b}'$. The numerator is an integer, and the denominator is ± 1. $\square$

Incidence matrices of directed graphs, and of bipartite graphs, are TU. That's why shortest paths, maximum flows and the assignment problem (n workers, n jobs) are "integer programs" that a plain LP, or a specialised network algorithm (Ch. 27), solves exactly in polynomial time. Recognising hidden network structure in an integer model is one of the most profitable skills in operations research.

Lagrangian relaxation

Duality (Ch. 48) gives a second kind of bound. Move the awkward constraints into the objective with prices $\boldsymbol{\lambda} \geq \mathbf{0}$ and keep the easy ones, integrality included. The dual function is then an upper bound on z_{IP} (for a maximisation), and a classical result of Geoffrion (1974) says the best such bound is *never worse* than the LP bound, and often strictly better. The gap of Worked Example 48.1 doesn't go away, but it can be pushed below the integrality gap. For the crew-scheduling and unit-commitment problems of Ch. 50, that is often the difference between a bound you can use and one you can't.

How hard is this, really?

Deciding whether a 0–1 knapsack can reach value V is NP-complete, and general integer programming is NP-hard. Unless P = NP, no algorithm solves every instance in time polynomial in the input size. Branch-and-bound offers no guarantee either: there are tiny-looking families on which it must explore exponentially many nodes (§49.4 and §49.5 below show one).

> **? Hard in theory, easy in practice?**
>
> Solvers routinely close industrial integer programs with millions of variables, yet the theory says the worst case is hopeless. Is that because real instances have hidden structure, or because P = NP after all? Nobody knows. The P versus NP question is open, and Ch. 30 explains what a proof either way would have to overcome. Even the practical success of branch-and-cut has no satisfying theoretical explanation.

49.4 The algorithm

Branch-and-bound, as code

```
input: max c.x  s.t.  A x <= b,  x >= 0,  x_j integer for j in J
incumbent = None, z_best = -infinity
open = [ root problem ]
while open is not empty:
    node = select(open)                    # depth-first, best-bound, or a mix
    solve the LP relaxation of node -> (x_bar, z_bar)
    if infeasible or z_bar <= z_best:  continue          # prune
    if x_bar_j is integer for all j in J:
        incumbent, z_best = x_bar, z_bar;  continue      # new incumbent
    j = choose a fractional variable       # branching rule
    open += [ node + {x_j <= floor(x_bar_j)},  node + {x_j >= floor(x_bar_j) + 1} ]
report incumbent, and the gap  (max bound over open nodes) - z_best
```

Three choices decide whether this finishes in a second or in a century.

- **Node selection.** *Depth-first* dives quickly to integer leaves, so it finds incumbents early and uses little memory. *Best-bound* always expands the node with the highest $\bar{z}$, so it proves optimality with the fewest nodes but can keep a huge open list. Modern solvers dive first and switch later.
- **Branching variable.** "Most fractional" is the textbook rule and is often poor. *Strong branching* tentatively solves both children's LPs for several candidates and picks the variable whose children improve the bound most. It's expensive per node, but it often cuts the tree by orders of magnitude.
- **Warm starts.** A child differs from its parent by one bound, so the parent's optimal basis is still dual-feasible. A few **dual simplex** pivots (§23.3) re-solve it, far cheaper than starting from scratch.

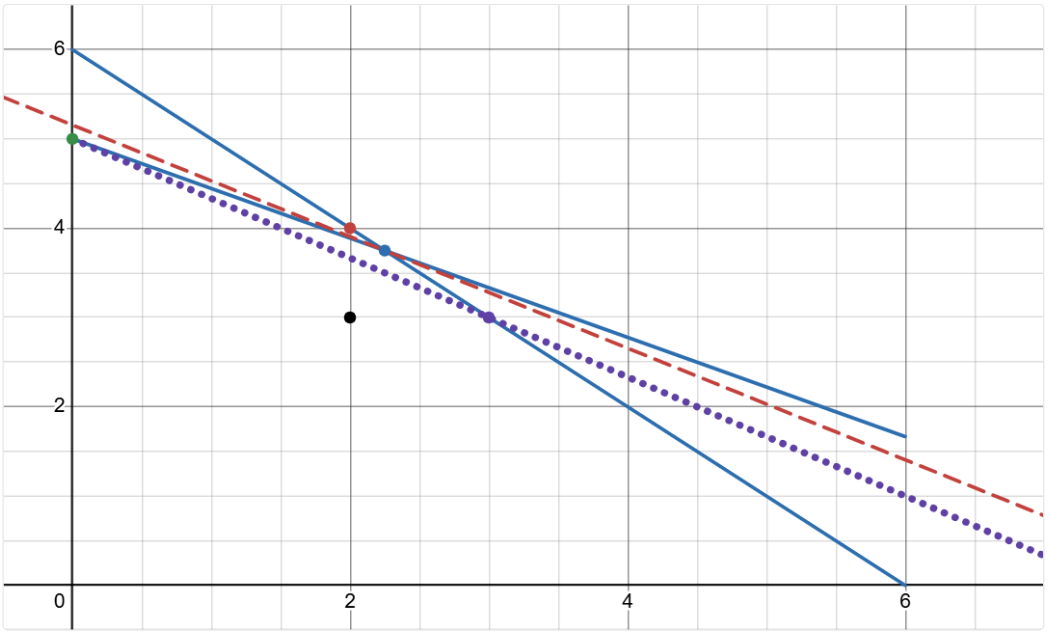
Anytime answers. At every moment the solver holds an incumbent (a lower bound) and the best open-node bound (an upper bound). The **optimality gap** $(\bar{z}_{\max} - z^{\circ})/|z^{\circ}|$ tells you how far from optimal the current design could be. Industrial users often stop at 1%: a guarantee that rounding never gave them.

Branch-and-cut inserts a cut-separation loop at each node: solve the LP, look for violated cuts (Gomory, knapsack covers, clique inequalities…), add them, re-solve, and branch only when the cuts stop helping. Add primal heuristics that try to round or repair LP solutions into good incumbents, and you have the skeleton of every commercial MIP solver.

Failure cases. The worst case is exponential, and it isn't exotic. Take $2z_1 + 2z_2 + \cdots + 2z_n = n$ with n odd and $z_i \in \{0, 1\}$. No solution exists (the left side is even), but setting every free variable to $\frac{1}{2}$-ish keeps the LP feasible until about half the variables are fixed. So plain branch-and-bound must explore at least about $2^{n/2}$ nodes before proving infeasibility. A single cut ($\sum z_i \leq \lfloor n/2 \rfloor$ from rounding) or a parity argument ends it at once, which shows why cuts matter. Other trouble spots are weak big-M formulations (§49.3), symmetry (identical machines make many branches equivalent copies of each other) and numerical tolerance ("integer" means within 10^{-6} or so, which can mislead with huge coefficients).

49.5 The experiment

Slide the objective. The graph draws the rack constraints and the integer points that matter. The dashed objective line $5x + 8y = 41.25$ touches the LP corner. Change `41.25` to `40`: the line drops to the integer optimum $(0, 5)$, and no integer point lies between the two lines. Now change the objective to $5x + 7y$ (edit both `8`s and recompute the level). Does the integer optimum stay at $(0, 5)$?



The rack problem in Desmos. Blue: the LP constraints. Purple dotted: the Chvátal–Gomory cut. Red dashed: an objective level line to slide.

Watch the tree explode. The script solves a 0–1 knapsack by depth-first branch-and-bound. The bound at each node is the knapsack's LP relaxation, which has a closed form: sort items by value per weight, take them greedily, and take a fraction of the first item that doesn't fit (Dantzig's bound). The script needs only numpy.

```
import numpy as np
import matplotlib.pyplot as plt

# --- knobs ---------------------------------------------------------------
n, seed = 30, 1                       # number of items, random instance
rng = np.random.default_rng(seed)
w = rng.integers(10, 100, n)          # weights   (try: v = w + 10 for a hard instance)
v = rng.integers(10, 100, n)          # values
cap = int(0.5 * w.sum())              # capacity

order = np.argsort(-v / w); v, w = v[order], w[order]   # best value per weight first

def lp_bound(k, room, val):           # LP relaxation of items k..n-1: greedy + one fraction
    for i in range(k, n):
        if w[i] <= room: room -= w[i]; val += v[i]
        else: return val + v[i] * room / w[i]
    return val

best, nodes, trace = 0, 0, []
stack = [(0, cap, 0)]                 # (next item, remaining room, value so far)
while stack:
    k, room, val = stack.pop(); nodes += 1
    best = max(best, val)
    if k == n or lp_bound(k, room, val) <= best:
        continue                      # leaf, or pruned: this branch cannot beat 'best'
    stack.append((k + 1, room, val))                    # skip item k
    if w[k] <= room:
        stack.append((k + 1, room - w[k], val + v[k]))   # take item k (explored first)
    trace.append(best)

print(f"optimum {best},  root LP bound {lp_bound(0, cap, 0):.2f},  nodes {nodes}"
      f"  (full tree would have {2**(n+1)-1:.3g})")
plt.plot(trace); plt.xlabel("nodes branched"); plt.ylabel("best value found")
plt.title("incumbent over the search"); plt.show()
```

With the default random instance ($n = 30$) the optimum is 1275 (checked against an exact dynamic program), the root bound is 1282.85, and the search visits only **59 nodes** out of a full tree of about 2.15×10^9. The bound is so tight that almost everything is pruned. Doubling to $n = 50$ still needs only 169 nodes.

Now make the instance nasty. Replace the three instance lines with `w = 2 * rng.integers(5, 50, n)`, `v = w.copy()` and `cap = int(0.5 * w.sum()) | 1`. Every weight is even and the capacity is odd, so the bag can never be filled exactly, and the LP bound always sits one unit above anything achievable. Nothing gets pruned early. The node counts become:

n	16	20	24
nodes visited	28,543	222,024	8,475,918

That's roughly an eight- to forty-fold increase for every four extra items: exponential growth, live on your laptop. (The $n = 24$ run can take a few minutes.) Things to try: switch the stack to a best-bound priority queue and compare node counts; add the parity cut "total weight $\leq$ cap -1" to `lp_bound` and watch the nasty instance collapse.

49.6 The challenge

1. **Intuition.** Draw a two-variable integer program whose LP relaxation is feasible and bounded but which has **no** integer feasible point at all. Then draw one where the LP optimum is far from *every* integer optimum, at least 10 units away.
2. **Derivation.** For the rack problem, derive the cut $2x + 3y \le 15$ as a Gomory fractional cut from the optimal simplex tableau of the LP relaxation (introduce slacks, find the row of a fractional basic variable, and take fractional parts). Check that you obtain the same inequality as the Chvátal–Gomory weights $\mathbf{u} = (\tfrac{3}{4}, \tfrac{1}{4})$.
3. **Implementation.** Write a general branch-and-bound for small integer programs in $\max\{\mathbf{c}^{\mathsf T}\mathbf{x} : \mathbf{A}\mathbf{x} \le \mathbf{b},\ \mathbf{x} \in \mathbb{Z}^n_{\ge 0}\}$ form, using `scipy.optimize.linprog` for the node LPs. Reproduce the tree of Figure 49.2. Then compare depth-first and best-bound node counts on random instances with $n = 10$ to 20.
4. **Open investigation.** For random knapsack instances (values and weights uniform on $[1, R]$), how does the expected number of branch-and-bound nodes grow with n? Polynomially? Does the answer change if values and weights are correlated, as in the `v = w + 10` variant? Plot node counts on log–log and log–linear axes and argue from your data which growth law fits.

49.7 Loose threads

Integer programming is where convexity ends and cleverness begins. Three of the deepest questions it raises are answered elsewhere.

> **? A different way to slice the knapsack**
>
> Branch-and-bound split the knapsack by *items*. There's another way to organise the same search, by *remaining capacity*: "what is the best value achievable with capacity c using the first k items?" That table has only nW entries, so it solves the knapsack in time proportional to n times the *number* W, which can be fast or hopeless depending on how big W is. Why doesn't this contradict NP-hardness, and what principle makes the table correct? Bellman's principle of optimality in Ch. 52.

> **? Models too big to write down**
>
> An airline's crew-scheduling model has one binary variable for every *legal pairing*, a sequence of flights a crew can fly in one trip. There are billions of them, far too many to list, let alone branch on. Yet such models are solved routinely by generating variables only when the dual prices of Ch. 48 say a new one would help. How does an LP solver "price out" a column it has never seen? That is column generation, in Ch. 50.

The third thread is the P versus NP mystery of §49.3 (Ch. 30). For counting how many integer points a polytope holds, and why that number grows like a polynomial when the polytope is scaled up, see the combinatorics of Ch. 28.

Further reading

- L. A. Wolsey, *Integer Programming*, 2nd ed., Wiley, 2020. The standard introduction: formulations, relaxations, branch-and-bound and cutting planes.
- M. Conforti, G. Cornuéjols and G. Zambelli, *Integer Programming*, Springer, 2014. Rigorous polyhedral theory, including total unimodularity and the Chvátal closure.
- A. Schrijver, *Theory of Linear and Integer Programming*, Wiley, 1986. The reference for the deep theory.
- F. S. Hillier and G. J. Lieberman, *Introduction to Operations Research*, McGraw-Hill (many editions). Accessible modelling examples, including the classic "rounding fails" illustration.

Chapter 50 — Operations Research: Mathematics That Runs the World

> **Level 2 · Track D — Optimization and Operations Research Builds on:** Ch. 6, Ch. 23, Ch. 25 · **Leads to:** Ch. 51, Ch. 52, Ch. 54 · **Related:** Ch. 27, Ch. 30, Ch. 36, Ch. 48, Ch. 49 **Reading time:** ~30 min

50.1 The hook

A small print shop takes five orders on a Monday morning. Every order goes through the same two machines: first the press prints it, then the binder folds, stitches and trims it. The binder cannot start a job until the press has finished that job, and each machine handles one job at a time. The times, in hours, are:

job	A	B	C	D	E
press a_j	4	7	2	6	3
binder b_j	5	2	6	3	7

The press has 22 hours of work and the binder 23. The only decision is the *order* in which the jobs go through. The manager does the obvious thing and runs them alphabetically. The last book leaves the binder at hour 29.

Is that good? There are $5! = 120$ possible orders. Try them all and the worst takes 35 hours and the best takes 25. The alphabetical order sits in between, and it wastes four hours of binder time. Over a year of Mondays that is a lot of overtime.

Now scale up. A real shop with 20 jobs has $20! \approx 2.4 \times 10^{18}$ orders. A computer checking a billion orders a second would need about 77 years. Yet in 1954 Selmer Johnson, working at the RAND Corporation, published a rule that finds a best order for *any* number of jobs on two machines with nothing more than a sort. It takes milliseconds.

> **How can a rule beat a search over 10^{18} possibilities without looking at them?**

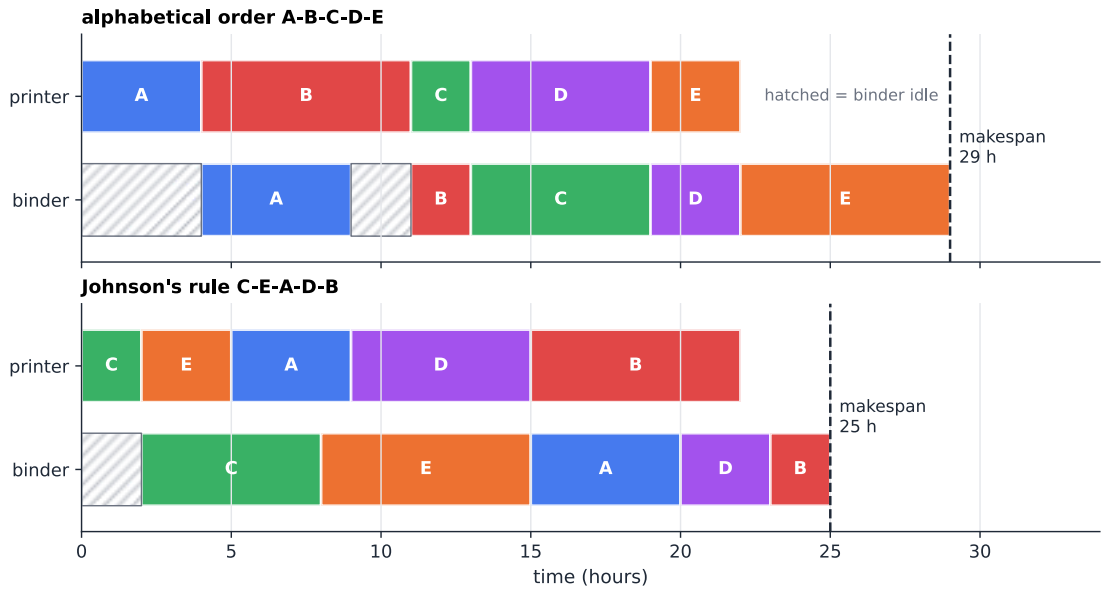
That question is the soul of operations research (OR). The field was born in the Second World War, when mathematicians were asked where to put radar stations, how big to make convoys, and how to pattern anti-submarine patrols. It now decides when your parcel leaves the depot, how many spare parts a power utility keeps, which aircraft flies which route, and how a mobile phone shares its transmit power across frequencies. This chapter shows you three of its classic results, each a small piece of mathematics with an enormous footprint: a scheduling rule, an inventory formula, and an allocation principle. Each one replaces a search with a *structural insight*.

50.2 The big idea

OR models have a common shape. You choose **decisions** (an order, a quantity, an allocation), subject to **constraints** (one job per machine, a budget, demand must be met), to minimise a **cost** or maximise a **benefit**. That is the optimization template of §23.1. What makes OR its own subject is the kind of structure the real world hands you:

- **Scheduling.** Decisions are orders or start times. The feasible set is a set of permutations: finite, huge, with no gradient. The trick is to find an *exchange argument*: show that swapping two neighbours in the wrong order never hurts. Then a sort is enough.
- **Inventory.** Decisions are how much and when to order. Two costs pull in opposite directions: ordering often is expensive in paperwork and trucks; ordering rarely means paying to store goods. The optimum sits where the two pulls balance, and the cost curve is surprisingly *flat* around it.
- **Logistics.** Decisions are flows on a network: so many tonnes from this factory to that warehouse. These are linear programs (§23.3), and their dual variables are prices (Ch. 48). When the decisions become yes/no (open this depot? send this truck along that road?), they turn into the integer programs of Ch. 49.
- **Resource allocation.** Decisions are how much of a scarce resource each activity gets. If returns diminish, the rule is *equalise the marginal returns*: the last dollar (or milliwatt) should buy the same benefit everywhere it goes.

Here is the analogy to keep in mind. A good OR model is like a well-cut key. The lock (the real system) is messy, but if you cut the key to match only the few pins that matter, it turns. Johnson's rule ignores almost everything about the shop and keeps one fact: which machine is the bottleneck for each job. That single fact is enough.



The same five jobs in two orders. Hatched blocks are hours the binder sits idle waiting for the press. Johnson's order starts with the job that frees the binder fastest and ends with the job that keeps it busy longest after the press stops.

Look at the picture. The binder can never start before the press has finished its first job, and it can never stop before it has done all 23 hours of its own work. So no schedule can finish before $\min_j a_j + \sum_j b_j = 2 + 23 = 25$ hours. Johnson's order hits that bound exactly, so it is optimal, and we have a *certificate*: a short argument anyone can check. The alphabetical order loses because it opens with job A (four hours on the press while the binder waits) and then job B, which is slow on the press and quick on the binder, starving the binder again.

50.3 The mathematics

Result 1: the makespan of a two-machine flow shop

Fix an order $\sigma = (\sigma_1, \dots, \sigma_n)$ of the jobs. Write $A_k = \sum_{i=1}^{k} a_{\sigma_i}$ for the time the press finishes the k-th job (the press never idles), and C_k for the time the binder finishes it. The binder can start job k only when both it is free and the press is done:

$$C_k = \max\left(C_{k-1},\ A_k\right) + b_{\sigma_k}, \qquad C_0 = 0.$$

Makespan formula

For the order σ, the time the last job leaves the binder is

$$C_{\max}(\sigma) = \max_{1 \le k \le n} \left(\sum_{i=1}^{k} a_{\sigma_i} + \sum_{i=k}^{n} b_{\sigma_i} \right).$$

Proof. Unroll the recursion. Since $\max(x, y) + c = \max(x + c, y + c)$,

$$\begin{aligned} C_n &= \max\left(C_{n-1} + b_{\sigma_n},\ A_n + b_{\sigma_n}\right) \\ &= \max\left(C_{n-2} + b_{\sigma_{n-1}} + b_{\sigma_n},\ A_{n-1} + b_{\sigma_{n-1}} + b_{\sigma_n},\ A_n + b_{\sigma_n}\right) \\ &= \cdots = \max_{1 \le k \le n} \left(A_k + \sum_{i=k}^{n} b_{\sigma_i} \right), \end{aligned}$$

where the final step uses $C_0 = 0$, so the last leftover term $C_0 + \sum_{i \ge 1} b_{\sigma_i}$ is dominated by the $k = 1$ term. ∎

The formula has a vivid reading. Term k is the length of a path through the Gantt chart: run along the press up to job k, drop down to the binder, then run along the binder to the end. The makespan is the longest such path, a **critical path**. Each job k is a candidate "hand-over point".

Result 2: the exchange argument and Johnson's rule

Now compare two orders that differ only by swapping neighbours: job i immediately before job j, or j before i. Every term of the makespan formula with the hand-over point elsewhere is unchanged, because it contains both jobs on the same machine or neither. Let P be the press time before the pair and S the binder time after it. The two terms that change are, for i first,

$$P + a_i + b_i + b_j + S \quad \text{and} \quad P + a_i + a_j + b_j + S,$$

whose maximum is $P + S + a_i + b_j + \max(b_i, a_j)$. For j first it is $P + S + a_j + b_i + \max(b_j, a_i)$. Subtract $P + S + a_i + a_j + b_i + b_j$ from both and use $\max(x, y) - x - y = -\min(x, y)$:

$$i \text{ before } j \text{ is no worse} \iff -\min(a_j, b_i) \le -\min(a_i, b_j) \iff \min(a_i, b_j) \le \min(a_j, b_i).$$

Johnson's rule (1954)

Put first the jobs with $a_j \le b_j$, in increasing order of a_j. Put after them the jobs with $a_j > b_j$, in decreasing order of b_j. The resulting order minimises the makespan on two machines.

Why does the rule satisfy the exchange condition for every neighbouring pair? Take two jobs from the first group with $a_i \le a_j$. Then $\min(a_i, b_j) \le a_i$, and $a_i \le \min(a_j, b_i)$ because $a_i \le a_j$ and $a_i \le b_i$. The second group is the mirror image, and for a first-group job i before a second-group job j, $\min(a_i, b_j)$ is at most $a_i \le b_i$ and at most $b_j < a_j$, so it is at most $\min(a_j, b_i)$. Starting from any optimal order, you can bubble-sort it into Johnson's order one neighbour swap at a time, and no swap ever increases the makespan. Ties need a little care; the challenge asks you to supply it.

For the print shop, the first group is $\{C, E, A\}$ ($a = 2, 3, 4$) and the second is $\{D, B\}$ ($b = 3, 2$), giving C-E-A-D-B with makespan 25. Eight of the 120 orders achieve 25; Johnson's rule finds one with a single sort, $O(n \log n)$ work.

? Add a third machine

Put a packing station after the binder and the exchange argument collapses: there are orders where no swap of neighbours helps, yet the order is far from optimal. In 1976 Garey, Johnson and Sethi proved that the three-machine flow shop is NP-hard, so no sort, and probably no polynomial algorithm of any kind, can solve it. What "NP-hard" really means, and why one extra machine crosses that line, is the subject of Ch. 30.

Result 3: the economic order quantity

A shop sells D units per year at a steady rate. Each order costs a fixed K (paperwork, delivery, set-up) no matter its size, and keeping one unit on the shelf for a year costs h (capital, space, spoilage). If you order Q units each time and the next delivery arrives just as the shelf empties, the stock follows a sawtooth from Q down to 0. The average stock is $Q/2$ and you place D/Q orders per year, so the annual cost is

$$C(Q) = \underbrace{\frac{KD}{Q}}_{\text{ordering}} + \underbrace{\frac{hQ}{2}}_{\text{holding}}.$$

Dimensions check: K is dollars per order, D/Q is orders per year, h is dollars per unit-year and $Q/2$ is units, so both terms are dollars per year.

EOQ and its flatness (Harris, 1913)

C is minimised at

$$Q^{*} = \sqrt{\frac{2KD}{h}}, \qquad C(Q^{*}) = \sqrt{2KDh},$$

and for any other order size $Q = rQ^{*}$,

$$\frac{C(rQ^{*})}{C(Q^{*})} = \frac{1}{2}\left(r + \frac{1}{r}\right).$$

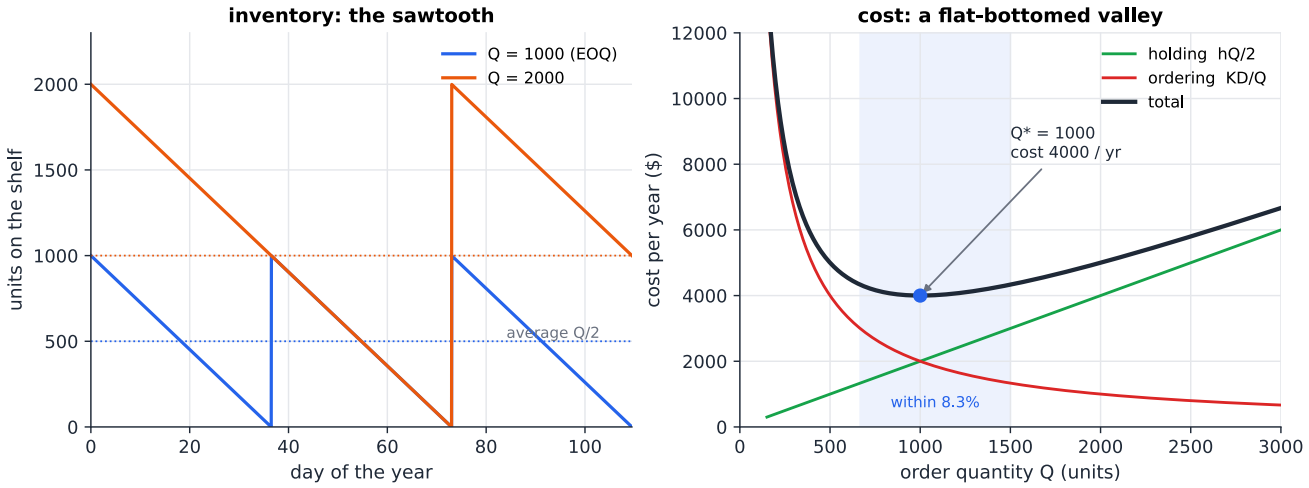
Proof. $C'(Q) = -KD/Q^2 + h/2$ vanishes at $Q^{*} = \sqrt{2KD/h}$, and $C''(Q) = 2KD/Q^3 > 0$, so this is the global minimum on $Q > 0$. Substituting, each term equals $\tfrac{1}{2}\sqrt{2KDh}$: at the optimum the two costs are *equal*. Writing $C^{*} = \sqrt{2KDh}$, we have $hQ^{*}/2 = KD/Q^{*} = C^{*}/2$, so

$$C(rQ^{*}) = \frac{KD}{rQ^{*}} + \frac{h\,rQ^{*}}{2} = \frac{1}{r}\cdot\frac{C^{*}}{2} + r\cdot\frac{C^{*}}{2}.$$

∎

The square root is a famous OR surprise. If demand doubles, you should *not* double the order size; it grows by only $\sqrt{2} \approx 1.41$. Big shops therefore carry relatively less stock than small ones, which is one quiet reason that scale pays.

The ratio formula is the real engineering gift. With $D = 10{,}000$ units/yr, $K = \$200$ and $h = \$4$/unit/yr, $Q^{*} = 1000$ units and $C^{*} = \$4000/yr$, $tenordersayear.Order1500atatimeinstead(r = 1.5\$)$ and the cost rises to $\$4333$, $only8.3r = 2\$)$ costs 25% more. A 50% mistake in Q costs single-digit percent. Because the error enters as $r + 1/r$, which is flat at $r = 1$, rough estimates of K and h are good enough. Estimate K as twice its true value and Q^{*} rises by only $\sqrt{2}$, costing 6%.



Left: stock on the shelf for two order sizes; doubling Q doubles the average stock. Right: the total cost is a flat-bottomed valley. Anywhere in the shaded band ($667 \le Q \le 1500$) you pay at most 8.3% over the optimum.

> **? What if customers don't behave?**
>
> The EOQ assumes demand flows at a perfectly known rate. Real demand is random, deliveries take time, and running out has a price. Then you must also choose a **safety stock**, and in the one-shot version (a newspaper seller who must decide this morning how many papers to buy) the optimal order is a *quantile* of the demand distribution, not its mean. Which quantile, and why a cost ratio picks it, is worked out in Ch. 54.

Resource allocation: equalise the margins

Suppose a transmitter has total power P to share among n frequency channels with noise levels $N_1, \dots, N_n$. Channel i given power p_i carries $\log(1 + p_i/N_i)$ units of data (Shannon's formula, Ch. 33). We want

$$\max_{p_i \ge 0} \sum_{i=1}^{n} \log\left(1 + \frac{p_i}{N_i}\right) \quad \text{subject to} \quad \sum_{i=1}^{n} p_i = P.$$

The returns are concave (each extra milliwatt buys less), so the KKT conditions of §23.4 are necessary and sufficient. Attach a multiplier ν to the budget (§6.11). At the optimum, every channel that gets power has the same marginal return, and channels left off would return less:

$$\frac{1}{N_i + p_i} = \nu \ \text{ if } p_i > 0, \qquad \frac{1}{N_i} \le \nu \ \text{ if } p_i = 0.$$

Writing $\mu = 1/\nu$, this is $p_i = \max(0,\ \mu - N_i)$: pour water of total volume P into a vessel whose floor has height N_i over channel i, and the water depth is the power. This is **water-filling**. With noise levels $1, 3, 6$ and $P = 6$, trying all three channels gives $\mu = 16/3 < 6$, so channel 3 is dry; the two others give $\mu = 5$ and powers $(4, 2, 0)$. Total rate: $\log_2 5 + \log_2(5/3) \approx 3.06$ bits per use, against 2.74 for an equal split.

The same principle runs budgets, crews and computing time: give the next unit of a resource to whichever activity values it most, and stop when the margins are level. The multiplier ν is the **shadow price** of the resource, the benefit of one more unit. That reading of multipliers as prices is the theme of Ch. 48.

Logistics and queues, in brief

Two more pillars deserve a sentence each. **Transportation problems** ship goods from supplies s_i to demands d_j at unit costs c_{ij}: minimise $\sum c_{ij} x_{ij}$ with $\sum_j x_{ij} \le s_i$, $\sum_i x_{ij} \ge d_j$, $x_{ij} \ge 0$. This is a linear program whose constraint matrix is so special that integer supplies and demands always admit an optimal flow in whole units, with no integer programming needed. **Queues** obey a law with almost no assumptions. If λ is the arrival rate, W the average time a customer spends in the system and L the average number present, then

$$L = \lambda W \qquad \text{(Little's law)}.$$

A wafer fab that starts 50 wafers a day and holds 1500 in progress has an average cycle time of $1500/50 = 30$ days. To cut that time you must cut work in progress, which is the mathematical core of "lean" manufacturing.

50.4 The algorithm

Johnson's rule is four lines. The makespan recursion is a few more:

```
def johnson(a, b):
    """Optimal two-machine flow-shop order. a, b: lists of times on machines 1, 2."""
    first = sorted([j for j in range(len(a)) if a[j] <= b[j]], key=lambda j: a[j])
    last  = sorted([j for j in range(len(a)) if a[j] >  b[j]], key=lambda j: -b[j])
    return first + last

def makespan2(seq, a, b):
    t1 = t2 = 0
    for j in seq:
        t1 += a[j]                  # press finishes job j
        t2 = max(t2, t1) + b[j]     # binder starts when free AND job is printed
    return t2
```

Complexity. Sorting costs $O(n \log n)$ and the makespan $O(n)$. The brute force it replaces costs $O(n \cdot n!)$.

EOQ is a closed form, but in practice you compute it with care. Real price lists give quantity discounts, so $C(Q)$ becomes piecewise; evaluate the EOQ on each price band, clip it to the band, and compare the few candidates. Production with a finite rate $R > D$ changes the average stock to $\frac{Q}{2}(1 - D/R)$, which replaces h by $h(1 - D/R)$ in the formula.

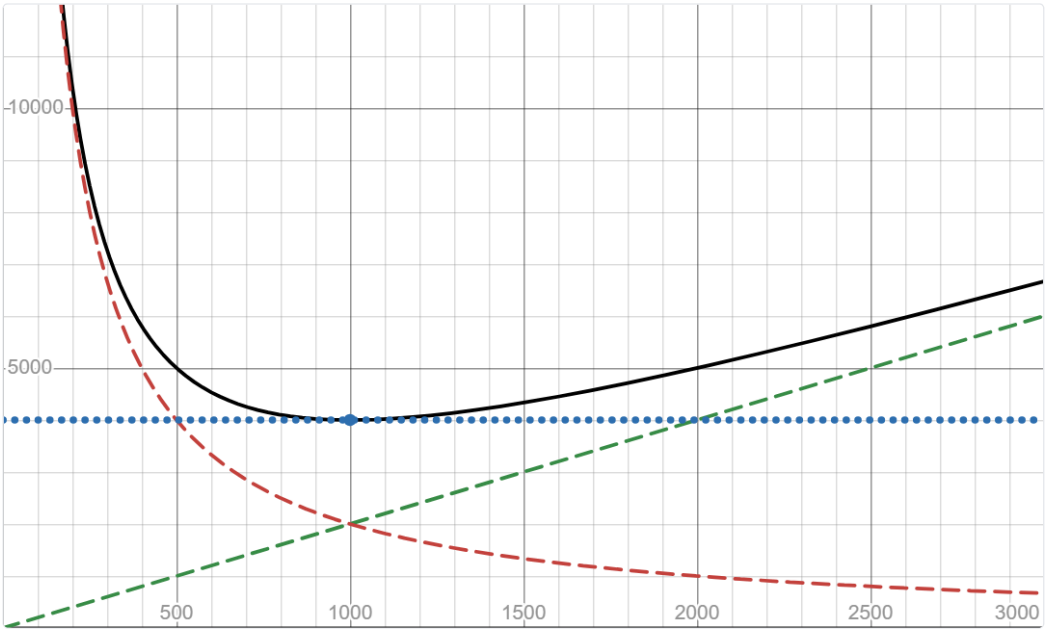
Water-filling needs the level μ. Sort the noise levels; for $m = n, n-1, \ldots$ try using the m quietest channels, set $\mu = (P + \sum_{i \le m} N_i)/m$, and stop at the first m with $\mu > N_m$. That is $O(n \log n)$. Bisection on μ also works and generalises to returns that have no closed form.

Failure cases. Each of these algorithms is exact *only for its model*. Johnson's rule says nothing about set-up times that depend on the previous job, about due dates, or about a third machine. EOQ breaks when demand is lumpy or random. Water-filling assumes the channel states are known; if they are estimated, the "optimal" allocation can pour power into a channel that is actually bad. The professional skill in OR is knowing which assumption is about to fail.

> **Engineering note**
>
> In practice, OR problems are rarely solved with a single elegant rule. They are written as integer or linear programs and handed to a solver. The elegant rules still matter. They give warm starts, lower bounds that certify a solver's answer, and a sanity check on its output.

50.5 The experiment

Part 1: feel the flat bottom. The graph shows the EOQ cost $C(Q) = KD/Q + hQ/2$ with $KD = 2{,}000{,}000$ and $h/2 = 2$. Change $h/2$ from 2 to 8 (storage becomes four times dearer). The minimum moves left, but by a factor of only 2, and the valley stays flat.



Total annual cost (black) as the sum of holding (green) and ordering (red). The minimum is where the two dashed curves cross.

Part 2: break Johnson's rule. The script below draws random 6-job instances, schedules them with Johnson's rule, and compares the result with the best of all 720 orders. Run it with `MACHINES = 2`: the rule is optimal every time. Then set `MACHINES = 3`. The script squashes the three machines into two virtual ones (machines 1+2 and 2+3) and applies Johnson's rule. With the seed given it is optimal in only 54% of trials, averages 2.6% above the best, and is 16.3% worse in the worst case. Try `N_JOBS = 8` (40,320 orders; slow but feasible) and other heuristics: sort by total time, or by the ratio a_j/c_j.

```
import itertools
import numpy as np

rng = np.random.default_rng(1)
N_JOBS, MACHINES, TRIALS = 6, 2, 200        # try MACHINES = 3

def makespan(seq, P):
    """P[j, m] = processing time of job j on machine m; jobs visit machines in order."""
    done = np.zeros(P.shape[1])
    for j in seq:
        done[0] += P[j, 0]
        for m in range(1, P.shape[1]):
            done[m] = max(done[m], done[m - 1]) + P[j, m]
    return done[-1]

def johnson(a, b):
    first = sorted([j for j in range(len(a)) if a[j] <= b[j]], key=lambda j: a[j])
    last = sorted([j for j in range(len(a)) if a[j] > b[j]], key=lambda j: -b[j])
    return first + last

gaps = []
for _ in range(TRIALS):
    P = rng.integers(1, 10, size=(N_JOBS, MACHINES))
    if MACHINES == 2:
        seq = johnson(P[:, 0], P[:, 1])
    else:   # squash three machines into two "virtual" ones and hope
        seq = johnson(P[:, 0] + P[:, 1], P[:, 1] + P[:, 2])
    best = min(makespan(s, P) for s in itertools.permutations(range(N_JOBS)))
    gaps.append(makespan(seq, P) / best - 1)

gaps = np.array(gaps)
print(f"Johnson-style order optimal in {np.mean(gaps == 0):.0%} of {TRIALS} trials")
print(f"average gap {gaps.mean():.1%}, worst gap {gaps.max():.1%}")
```

Watch what the numbers say. On two machines, structure beats search completely. On three, structure gives a good guess but no guarantee, and the only way to be sure is to search, exactly what the 1954 rule let us avoid.

50.6 The challenge

1. **Intuition.** A job takes 1 hour on the press and 9 on the binder. Explain in words, without formulas, why Johnson's rule wants it *first*, and why a job with 9 hours on the press and 1 on the binder belongs *last*. Then explain why the lower bound $\min_j a_j + \sum_j b_j$ is not always achievable.
2. **Derivation.** Finish the proof of Johnson's rule. Show that when several jobs tie (for example $a_i = b_i$, or two jobs with equal a), every tie-break produces the same makespan, and turn the "bubble-sort an optimal order" sketch into a careful induction. Then derive the EOQ with a finite production rate $R > D$ and show the optimal batch is $\sqrt{2KD/(h(1 - D/R))}$.
3. **Implementation.** Write a water-filling routine for n channels by bisection on μ. Test it on noise levels $1, 3, 6$ and power $P = 6$, and then plot total rate against P from 0 to 30. Mark the values of P at which each new channel switches on. What shape does the curve have between switch-on points?
4. **Open investigation.** For the three-machine flow shop, design the best polynomial-time heuristic you can, and measure its worst gap on random instances of 8–10 jobs against brute force. Can you find a family of instances on which your heuristic's ratio to the optimum keeps growing? No clean answer is known for what the best achievable guarantee is in practice; the literature (NEH, Taillard's benchmarks) is a good place to compare yourself.

50.7 Loose threads

This chapter solved three problems by *structure*: an exchange argument, a square-root balance, and level margins. Each worked because the model was static and certain. The threads that pull hardest are the ones that relax those two words.

> **? Why does Little's law need no assumptions?**
>
> $L = \lambda W$ holds for a supermarket queue, a router buffer, a hospital ward and a wafer fab, whatever the distributions of arrivals and service times, whatever the order in which customers are served. That universality looks too good to be true. The proof is a beautiful sample-path argument that compares two ways of measuring the same area under a curve, and it belongs to the theory of stochastic processes in Ch. 36.

Two more pointers. First, *time*. The inventory problem here was a single static choice of Q. If demand changes month by month, the decision becomes a sequence where each choice shapes the next state: that is optimal control in continuous time (Ch. 51) and dynamic programming in discrete stages (Ch. 52). The Wagner–Whitin algorithm for lot sizing with time-varying demand is a dynamic program. Second, *feedback*. Supply chains in which each stage orders from the next amplify small swings in demand into wild oscillations upstream, the "bullwhip effect". It is a stability phenomenon of a feedback loop, and the tools for taming it come from control (Ch. 51). When routes and depots become yes/no decisions, return to the branch-and-bound machinery of Ch. 49; and shortest-route ideas behind every parcel are in Ch. 27.

Further reading

- F. S. Hillier and G. J. Lieberman, *Introduction to Operations Research*, 11th ed., McGraw-Hill, 2021. The standard broad survey: LP, networks, inventory, queues.
- M. L. Pinedo, *Scheduling: Theory, Algorithms, and Systems*, 6th ed., Springer, 2022. Johnson's rule, flow shops, and the complexity map of scheduling.
- P. H. Zipkin, *Foundations of Inventory Management*, McGraw-Hill, 2000. EOQ and its stochastic descendants, carefully derived.

- S. Boyd and L. Vandenberghe, *Convex Optimization*, Cambridge University Press, 2004. §5.5 derives water-filling from the KKT conditions.

Chapter 51 — Optimal Control: Choosing the Best Future

> **Level 2 · Track D — Optimization and Operations Research Builds on:** Ch. 6, Ch. 11, Ch. 12, Ch. 23 · **Leads to:** Ch. 52, Ch. 53, Ch. 54 · **Related:** Ch. 50, Ch. 61, Ch. 83, Ch. 92 **Reading time:** ~30 min

51.1 The hook

A robotic cart sits at rest on a straight rail. It must stop, exactly, at a mark 100 m away, in the shortest possible time. Its motor can push it forward or backward with any acceleration up to 1 $\mathrm{m/s^2}$ in size, and you choose the acceleration at every instant. What should you do?

Ask a room of engineers and most will sketch something smooth: ease in, build speed, ease off, glide to a stop. It feels right. Sudden changes look crude, and smooth curves look like good engineering.

Here is the smoothest sensible plan. Let the acceleration fall linearly from $+a$ to $-a$ over the trip. Among all plans of a given duration, this is the one that uses the least control effort $\int u^2\,dt$ (you'll derive it in §51.3). To keep $|u| \le 1$ $\mathrm{m/s^2}$, the trip must last $\sqrt{600} \approx 24.5$ s.

Now the crude plan: floor it for 10 s, then brake as hard as possible for 10 s. The cart reaches 10 m/s at the midpoint and stops at exactly 100 m after **20 s**. The smooth plan is 22% slower, and no amount of polishing fixes that, because the problem's best answer is not smooth at all.

Two questions follow, and they are much deeper than the cart:

> **How do you prove that nothing beats 20 seconds?** And when the system is a satellite, a chemical reactor or an epidemic, where intuition fails, **how do you find the best plan at all?**

The set of possible plans is a set of *functions*: an infinite-dimensional space. You cannot list them, and you cannot set a gradient to zero in the usual way, because the best plan sits on the boundary of the allowed set, with $|u| = 1$ almost all the time.

In the mid-1950s Lev Pontryagin, a topologist who had been blind since the age of fourteen, turned to problems of fastest manoeuvres posed by engineers. With his students Boltyanskii, Gamkrelidze and Mishchenko he found the answer: a **maximum principle** that turns the search over functions into a pointwise choice at each instant, guided by a hidden companion trajectory called the **costate**. This chapter builds that principle from the Lagrange multipliers you already know.

51.2 The big idea

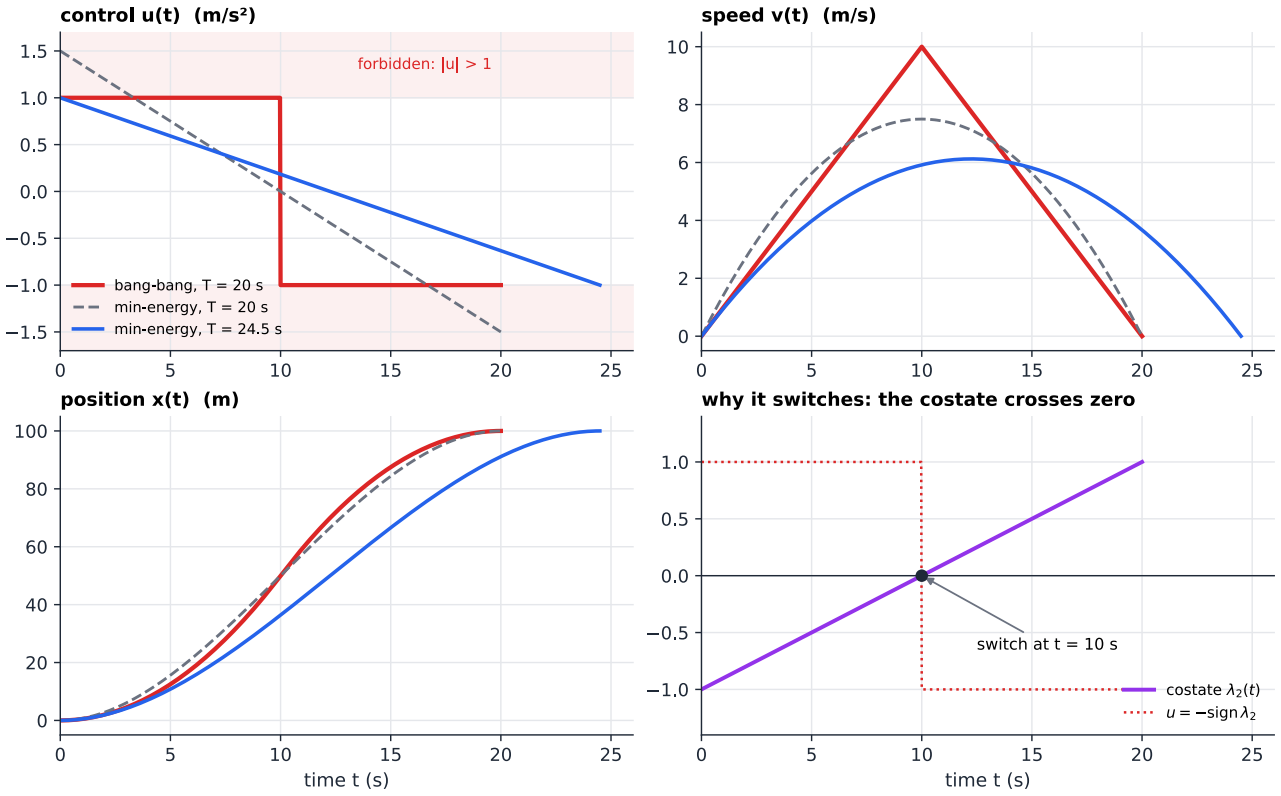
An optimal-control problem has three ingredients.

1. A **state** $\mathbf{x}(t)$ that summarises the system now (position and velocity; charge and current; susceptible and infected).
2. **Dynamics** $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}, t)$, driven by a **control** $\mathbf{u}(t)$ that you choose from an allowed set U. This is the state-variable form of §12.7.
3. A **cost** to minimise, such as total time, fuel used, or tracking error, usually $J = \varphi(\mathbf{x}(T)) + \int_0^T L(\mathbf{x}, \mathbf{u}, t)\,dt$.

The obstacle is that each choice of $\mathbf{u}$ at time t affects every later state. A good move now must account for its whole future. Here is the trick. Attach to each state variable a **price**, $\lambda_i(t)$: the marginal value, at time t, of having a little more of x_i. If you knew those prices you could judge each instant on its own. You would pick the $\mathbf{u}$ that minimises "running cost now plus the value of the state change it causes", and the future would be taken care of by the prices.

That is the whole idea of Pontryagin's principle. The prices are the costate. They are found by running a second differential equation *backwards* from the final time, because the value of the state at time t depends on what happens after t.

An economic analogy helps. A factory manager doesn't need to plan every future year to decide today's production; she needs today's price of the factory's stock of capital. Prices summarise the future. In the cart problem, the costate turns out to say "speed is valuable" early in the trip and "speed is a liability" late in the trip. The moment the price of speed crosses zero, the right action flips from full throttle to full brake.



Top left: the bang-bang control (red) stays on the limits; the minimum-energy plan for the same 20 s (dashed) needs 1.5 m/s², breaking the limit, and the feasible smooth plan (blue) takes 24.5 s. Bottom right: the costate λ_2 is a straight line, and the control is the negative of its sign.

Look at the speed panel. Distance is the area under the speed curve, so every plan must enclose exactly 100 m of area. The speed can rise no faster than slope 1 and must fall back to zero at the same rate. The tallest triangle you can fit under those two constraints is the bang-bang one. Every other plan wastes area by rising or falling more slowly than it could.

51.3 The mathematics

The necessary conditions, derived

Take the problem with a free final state and no constraint on $\mathbf{u}$:

$$\min_{\mathbf{u}(\cdot)} J = \varphi(\mathbf{x}(T)) + \int_0^T L(\mathbf{x},\mathbf{u},t)\,dt \quad \text{subject to} \quad \dot{\mathbf{x}} = \mathbf{f}(\mathbf{x},\mathbf{u},t),\ \ \mathbf{x}(0) = \mathbf{x}_0.$$

Treat the dynamics as a constraint at every instant, and give it a Lagrange multiplier function $\boldsymbol{\lambda}(t)$, exactly as in §6.11 but with one multiplier per time. Define the **Hamiltonian**

$$H(\mathbf{x},\mathbf{u},\boldsymbol{\lambda},t) = L(\mathbf{x},\mathbf{u},t) + \boldsymbol{\lambda}^{\mathsf{T}}\mathbf{f}(\mathbf{x},\mathbf{u},t).$$

Along any trajectory that obeys the dynamics, adding $\int \boldsymbol{\lambda}^{\mathsf{T}}(\mathbf{f} - \dot{\mathbf{x}})\,dt = 0$ changes nothing, so

$$J = \varphi(\mathbf{x}(T)) + \int_0^T \left(H - \boldsymbol{\lambda}^{\mathsf{T}}\dot{\mathbf{x}}\right) dt.$$

Integrate the last term by parts: $-\int_0^T \boldsymbol{\lambda}^{\mathsf{T}}\dot{\mathbf{x}}\,dt = -\boldsymbol{\lambda}(T)^{\mathsf{T}}\mathbf{x}(T) + \boldsymbol{\lambda}(0)^{\mathsf{T}}\mathbf{x}(0) + \int_0^T \dot{\boldsymbol{\lambda}}^{\mathsf{T}}\mathbf{x}\,dt$. Now perturb the control by $\delta\mathbf{u}(t)$. The state changes by $\delta\mathbf{x}(t)$, with $\delta\mathbf{x}(0) = \mathbf{0}$ because the start is fixed. To first order,

$$\delta J = \left(\nabla\varphi - \boldsymbol{\lambda}(T)\right)^{\mathsf{T}}\delta\mathbf{x}(T) + \int_0^T \left[\left(\frac{\partial H}{\partial \mathbf{x}} + \dot{\boldsymbol{\lambda}}\right)^{\mathsf{T}}\delta\mathbf{x} + \left(\frac{\partial H}{\partial \mathbf{u}}\right)^{\mathsf{T}}\delta\mathbf{u}\right] dt.$$

The state perturbation $\delta\mathbf{x}$ depends on $\delta\mathbf{u}$ in a complicated way, through the whole history. The multiplier lets us avoid computing it: *choose* $\boldsymbol{\lambda}$ so that its coefficients vanish.

First-order conditions for optimal control

If $\mathbf{u}^*$ is optimal (and $L, \mathbf{f}, \varphi$ are smooth), there is a costate $\boldsymbol{\lambda}(t)$ with

$$\dot{\mathbf{x}} = \frac{\partial H}{\partial \boldsymbol{\lambda}}, \qquad \dot{\boldsymbol{\lambda}} = -\frac{\partial H}{\partial \mathbf{x}}, \qquad \boldsymbol{\lambda}(T) = \nabla\varphi(\mathbf{x}(T)), \qquad \frac{\partial H}{\partial \mathbf{u}} = \mathbf{0}.$$

Proof. With $\boldsymbol{\lambda}$ defined by the backward equation $\dot{\boldsymbol{\lambda}} = -\partial H/\partial \mathbf{x}$ and the end condition $\boldsymbol{\lambda}(T) = \nabla\varphi$ (a linear ODE, so a solution exists), the first two pieces of δJ vanish, leaving $\delta J = \int_0^T (\partial H/\partial \mathbf{u})^{\mathsf{T}} \delta\mathbf{u}\, dt$. At a minimum this first variation must be zero for every perturbation $\delta\mathbf{u}$. Choosing $\delta\mathbf{u} = -\epsilon\, \partial H/\partial \mathbf{u}$ gives $\delta J = -\epsilon \int_0^T \|\partial H/\partial \mathbf{u}\|^2 dt$, which is negative unless $\partial H/\partial \mathbf{u} \equiv \mathbf{0}$. ∎

If the final state is fixed instead, $\delta\mathbf{x}(T) = \mathbf{0}$ and the end condition on $\boldsymbol{\lambda}$ disappears; the two boundary conditions on $\mathbf{x}$ take its place. Notice the structure: state forward from $t = 0$, costate backward from $t = T$, coupled through the control. This is a **two-point boundary-value problem** (§11.8), and that coupling is what makes optimal control hard to compute.

The minimum-energy cart

State x (position) and v (speed), dynamics $\dot{x} = v$, $\dot{v} = u$, cost $\int_0^T \tfrac{1}{2}u^2\, dt$, boundary conditions $x(0) = v(0) = v(T) = 0$, $x(T) = d$. The Hamiltonian is $H = \tfrac{1}{2}u^2 + \lambda_1 v + \lambda_2 u$, so

$$\dot{\lambda}_1 = -\frac{\partial H}{\partial x} = 0, \qquad \dot{\lambda}_2 = -\frac{\partial H}{\partial v} = -\lambda_1, \qquad \frac{\partial H}{\partial u} = u + \lambda_2 = 0.$$

So λ_1 is constant, λ_2 is linear in t, and $u = -\lambda_2$ is linear. Fitting the four boundary conditions gives $u(t) = \dfrac{6d}{T^2}\left(1 - \dfrac{2t}{T}\right)$, a cubic position profile and peak acceleration $6d/T^2$. With $d = 100$ m and $T = 20$ s the peak is $1.5\ \mathrm{m/s^2}$, over the limit. Keeping the peak at 1 needs $T = \sqrt{6d} \approx 24.5$ s.

Constraints: Pontryagin's principle

When the control is limited, $\mathbf{u}(t) \in U$, the optimum usually sits on the boundary of U, where $\partial H/\partial \mathbf{u}$ need not vanish. The fix is to replace "stationary" with "best":

Pontryagin's minimum principle (statement)

If $\mathbf{u}^*$ is optimal, there exist a constant $\lambda_0 \in \{0, 1\}$ and a costate $\boldsymbol{\lambda}(t)$, not both zero, such that with $H = \lambda_0 L + \boldsymbol{\lambda}^{\mathsf{T}}\mathbf{f}$ the state and costate equations above hold, and at almost every time

$$H\left(\mathbf{x}^*(t), \mathbf{u}^*(t), \boldsymbol{\lambda}(t), t\right) = \min_{\mathbf{u} \in U} H\left(\mathbf{x}^*(t), \mathbf{u}, \boldsymbol{\lambda}(t), t\right).$$

If the final time is free and the problem does not depend explicitly on t, then $H \equiv 0$ along the optimal trajectory.

(Pontryagin's group wrote it with the opposite sign as a *maximum* principle; it is the same statement.) The proof uses "needle" variations: change $\mathbf{u}$ to some other value in U on a tiny interval, and show that the cost cannot drop. It is not hard, but it is long; the first-order derivation above is the special case where the minimum over U is an interior stationary point. The constant λ_0 handles rare "abnormal" problems where the constraints alone pin down the answer; for the problems here $\lambda_0 = 1$.

Result: the minimum-time cart is bang-bang

Now minimise $T = \int_0^T 1\, dt$ with $\dot{x} = v$, $\dot{v} = u$, $|u| \le a$. The Hamiltonian is

$$H = 1 + \lambda_1 v + \lambda_2 u.$$

H is *linear* in u, so its minimum over $[-a, a]$ is at an end: $u = -a\ \operatorname{sign}(\lambda_2)$. The costate equations are the same as before: λ_1 constant, $\lambda_2(t) = c - \lambda_1 t$, a straight line. A straight line changes sign at most once. If λ_2 were identically zero, λ_1 would be too, and $H = 1 \neq 0$, which contradicts $H \equiv 0$. So:

Bang-bang structure

Every time-optimal control for the double integrator takes only the values $\pm a$ and switches at most once.

For the trip from rest at 0 to rest at d, the only such plan that works is "accelerate for t_s, brake for t_s", and $x(2t_s) = a t_s^2 = d$ gives $T = 2\sqrt{d/a}$. With $d = 100$ m and $a = 1\ \mathrm{m/s^2}$, $T = 20$ s. Using $H \equiv 0$ at $t = 0$, where $v = 0$ and $u = +a$, fixes $\lambda_2(0) = -1/a$, and the switch at $t_s = 10$ s makes $\lambda_2(t) = -1 + t/10$ for $a = 1$: the purple line in the figure.

Pontryagin's principle is only *necessary*. For the cart there is also a two-line proof of optimality that needs no costate, and it is worth seeing because it gives a certificate:

Proof that no plan beats $2\sqrt{d/a}$. Let any admissible plan reach rest at d at time T. Since $|\dot{v}| \le a$ and $v(0) = 0$, we have $v(t) \le at$. Since $v(T) = 0$, also $v(t) \le a(T - t)$. Then

$$d = \int_0^T v(t)\, dt \le \int_0^T a \min(t,\ T - t)\, dt = \frac{aT^2}{4},$$

so $T \ge 2\sqrt{d/a}$, with equality only when v follows the tent $a\min(t, T - t)$, the bang-bang plan. ∎

That is the speed panel's triangle argument made exact.

? What *is* the costate, really?

We introduced $\boldsymbol{\lambda}(t)$ as a bookkeeping device to make δJ simple, then called it a "price". Is it really one? Here is the claim: if $V(\mathbf{x}, t)$ is the best achievable cost starting from state $\mathbf{x}$ at time t, then along an optimal trajectory $\boldsymbol{\lambda}(t) = \nabla_{\mathbf{x}} V(\mathbf{x}^*(t), t)$, the sensitivity of the optimal future to the present state. Proving it requires a partial differential equation for V, the Hamilton–Jacobi–Bellman equation, which grows out of a completely different idea: Ch. 52.

From open loop to feedback: the switching curve

The plan above is **open loop**: a schedule $u(t)$ computed in advance. If a gust pushes the cart, the schedule is wrong. Better is a **feedback law** $u = k(x, v)$ that gives the optimal action from wherever you are. For the double integrator with target $(0, 0)$ we can find it by geometry.

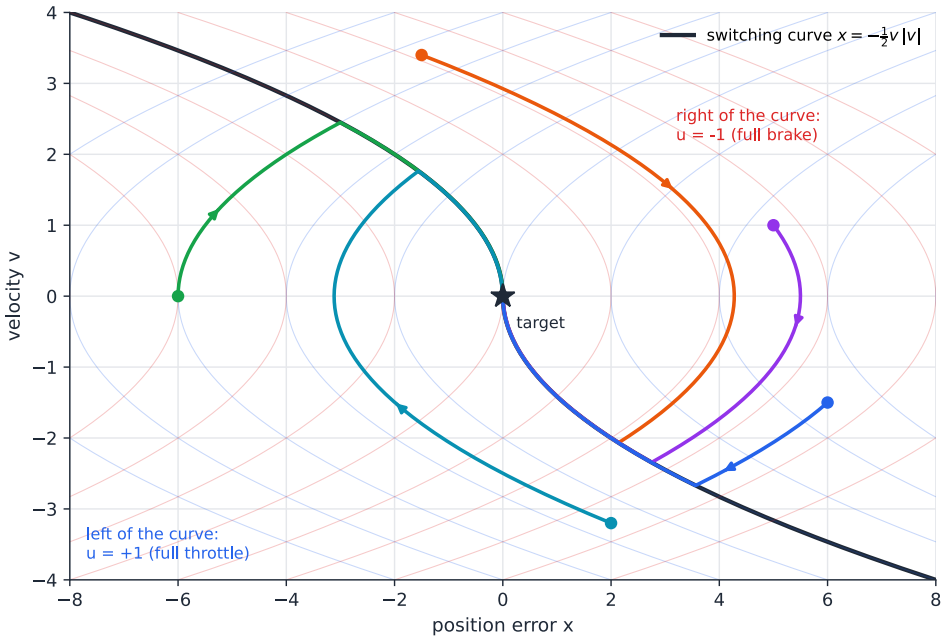
Under $u = +a$, the trajectories in the (x, v) plane are parabolas $x = v^2/(2a) + c$; under $u = -a$, they are $x = -v^2/(2a) + c$. Exactly one parabola of each family passes through the origin, and the halves that arrive there form the **switching curve**

$$x = -\frac{v|v|}{2a}.$$

From any point you ride a parabola to the curve, switch, and slide down the curve into the origin. With $s = x + v|v|/(2a)$, the optimal feedback is

$$u = -a\ \operatorname{sign}(s),$$

and the minimum time from (x_0, v_0) is $T^* = \left(\sigma v_0 + 2\sqrt{\sigma a x_0 + v_0^2/2}\right)/a$ with $\sigma = \operatorname{sign}(s_0)$.



Every optimal trajectory has at most one corner: a parabola of one family until it meets the black switching curve, then the curve itself into the origin. To the left of the curve the motor pushes forward at full power; to the right it brakes at full power.

Smooth costs, linear feedback

Bang-bang control comes from costs that are linear in u. When effort is penalised quadratically and the dynamics are linear, everything is smooth and the answer is a linear feedback. For $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$ and $J = \int_0^\infty (\mathbf{x}^\mathsf{T}\mathbf{Q}\mathbf{x} + \mathbf{u}^\mathsf{T}\mathbf{R}\mathbf{u})\,dt$, the **linear–quadratic regulator** (LQR) is

$$\mathbf{u} = -\mathbf{R}^{-1}\mathbf{B}^\mathsf{T}\mathbf{P}\mathbf{x}, \qquad \mathbf{A}^\mathsf{T}\mathbf{P} + \mathbf{P}\mathbf{A} - \mathbf{P}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^\mathsf{T}\mathbf{P} + \mathbf{Q} = \mathbf{0},$$

where the second equation, the **algebraic Riccati equation**, has a unique positive-definite solution under mild conditions. For the scalar system $\dot{x} = u$ with $J = \int (x^2 + u^2)\,dt$ it reads $P^2 = 1$, so $u = -x$. The costate is $\lambda = 2Px$: linear in the state, which is why the feedback is linear. The LQR is the workhorse of aerospace and process control, and its stability guarantees come from the eigenvalue analysis of §10.9.

51.4 The algorithm

Real problems rarely have closed forms. There are three families of numerical methods.

Indirect (shooting) methods solve Pontryagin's two-point boundary-value problem. Guess the unknown initial costate $\boldsymbol{\lambda}(0)$, integrate state and costate forward with the control from the minimum condition, and measure the miss at time T. Then adjust the guess with Newton's method (§22.4):

```
guess lam0
repeat
    integrate x' = H_lam, lam' = -H_x, with u = argmin_u H, from (x0, lam0) to T
    miss = boundary_conditions(x(T), lam(T))
    lam0 = lam0 - J_miss^{-1} * miss        # Jacobian by finite differences
until |miss| < tol
```

Shooting is very accurate when it converges. But it is fragile: the costate equations are often unstable forward in time, so small errors in $\boldsymbol{\lambda}(0)$ blow up, and bang-bang controls make the miss function discontinuous.

Direct methods ("discretise, then optimise") chop $[0, T]$ into N steps, treat $\mathbf{u}_0, \dots, \mathbf{u}_{N-1}$ as ordinary variables, impose the discretised dynamics as constraints, and give the whole thing to a nonlinear-programming solver (Ch. 23). Control bounds become simple box constraints. The solver's Lagrange multipliers for the dynamics constraints approximate the costate, which is the discrete echo of our derivation. This is what most practical software does, from rocket-landing

guidance to robot motion planning.

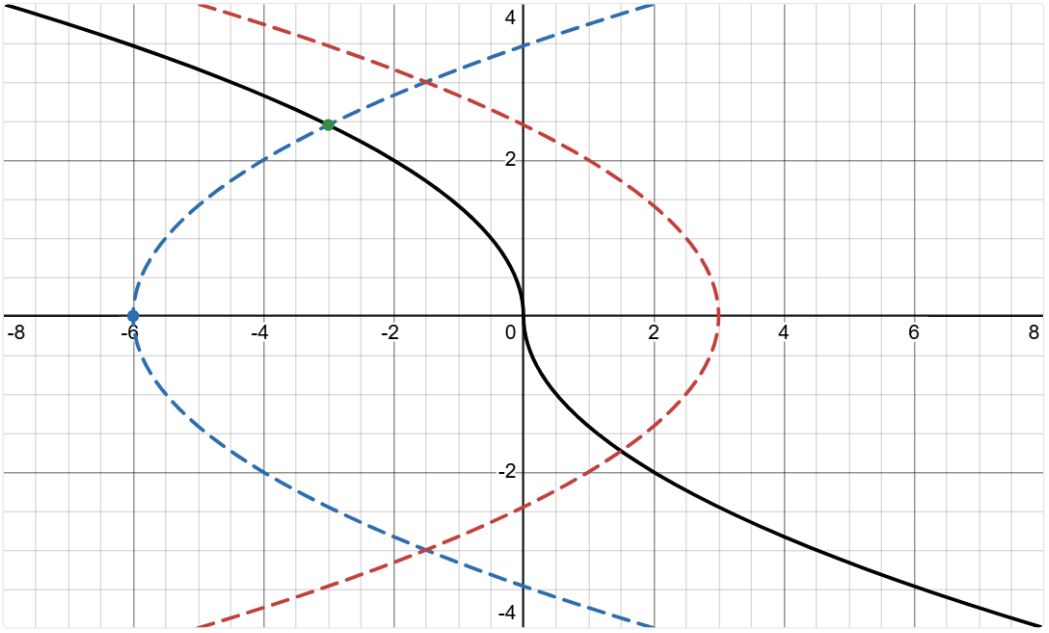
Dynamic programming computes the feedback law over a grid of states. It is robust and global, but its cost grows exponentially with the state dimension. That is the next chapter.

Failure cases. Bang-bang feedback implemented with a finite sampling time **chatters**: near the switching curve the sign of s flips back and forth every sample, wearing out relays and exciting unmodelled dynamics. Engineers soften it with a boundary layer, replacing $\operatorname{sign}(s)$ by a saturated linear function near $s = 0$. A second failure: *singular arcs*. If $\lambda_2 \equiv 0$ on an interval, the minimum condition says nothing about u, and you must differentiate the switching function repeatedly to recover it.

> **Engineering note**
>
> Model predictive control (MPC), standard in chemical plants and increasingly in power electronics and cars, solves a direct-method optimal-control problem over a short horizon *at every sampling instant*, applies only the first control move, measures the state again, and repeats. It turns open-loop optimisation into feedback by brute repetition.

51.5 The experiment

Part 1: the parabolas. In the phase plane (horizontal x, vertical y for velocity), the black curve is the switching curve for $a = 1$. The blue parabola is a full-throttle arc $x = y^2/2 + c$ with $c = -6$, starting from rest 6 m behind the target. Change c (try -2 and -10) and find where each arc meets the switching curve. That intersection is the switching point.



The full-throttle arc from $(-6, 0)$ meets the switching curve at $x = -3$, $v = \sqrt{6}$. The red dashed arc is a full-brake arc for comparison.

Part 2: fly the feedback. The script runs the optimal feedback $u = -a\ \operatorname{sign}(s)$ with a simple Euler integrator and compares the arrival time with the formula T^*. Things to change:

- `X0, V0`: try `5.0, 1.0` (theory 5.69 s) and `2.0, -3.2` (theory 6.73 s). In the second case the simulation reports **3 switches**, though the theory says one. Plot u against t and find where the extra switches happen.
- `DT = 0.1`: arrival slips from about 4.9 s to 6.0 s, with an extra switch. The controller overshoots the curve between samples.
- `U_MAX = 0.5`: the minimum time scales as $1/\sqrt{a}$ (here 6.93 s), not $1/a$.

```
import numpy as np
import matplotlib.pyplot as plt

X0, V0 = -6.0, 0.0      # start: 6 m behind the target, at rest
DT = 0.01               # time step (s); try 0.1 and 0.001
U_MAX = 1.0             # thrust limit (m/s^2); try 0.5
TOL = 0.05              # "arrived" when |x| and |v| are both below this

def feedback(x, v):
    s = x + v * abs(v) / (2 * U_MAX)        # signed distance to the switching curve
    return -U_MAX * np.sign(s) if s != 0 else -U_MAX * np.sign(v)

x, v, t, switches, u_old = X0, V0, 0.0, 0, None
xs, vs, us = [x], [v], []
while not (abs(x) < TOL and abs(v) < TOL) and t < 30:
    u = feedback(x, v)
    switches += (u_old is not None and u != u_old)
    u_old = u
    v += u * DT; x += v * DT; t += DT
    xs.append(x); vs.append(v); us.append(u)

s0 = X0 + V0 * abs(V0) / (2 * U_MAX)
T_star = (np.sign(s0) * V0 + 2 * np.sqrt(np.sign(s0) * U_MAX * X0 + V0**2 / 2)) / U_MAX
print(f"arrived at t = {t:.2f} s (theory {T_star:.2f} s) after {switches} switches")

fig, ax = plt.subplots(1, 2, figsize=(10, 4))
ax[0].plot(xs, vs); vv = np.linspace(-4, 4, 200)
ax[0].plot(-vv * np.abs(vv) / (2 * U_MAX), vv, "k--", label="switching curve")
ax[0].set_xlabel("x"); ax[0].set_ylabel("v"); ax[0].legend()
ax[1].step(np.arange(len(us)) * DT, us); ax[1].set_xlabel("t (s)"); ax[1].set_ylabel("u")
plt.show()
```

The default run prints an arrival at 4.85 s against the theoretical 4.90 s; the difference is the tolerance `TOL`, since the simulation stops just before the exact target.

51.6 The challenge

1. **Intuition.** For a cost $\int_0^T |u|\,dt$ (fuel, for a thruster whose consumption is proportional to thrust) with $|u| \le 1$ and a *fixed* trip time longer than the minimum, guess the shape of the optimal control. Explain why it should contain a phase where the motor is off, and why neither bang-bang nor smooth is the answer.
2. **Derivation.** Use the first-order conditions to solve the minimum-energy cart in full: find λ_1, $\lambda_2(0)$, $u(t)$, $v(t)$ and $x(t)$, and verify the cost $\int_0^T u^2\,dt = 12d^2/T^3$. Then derive the minimum-time formula $T^*(x_0, v_0)$ from the switching-curve construction.
3. **Implementation.** Write a direct method for the cart: $N = 50$ time steps, variables $u_0, \dots, u_{49}$, bounds $|u_k| \le 1$, and the end conditions as equality constraints. Minimise the trip time by bisection on T (is it feasible within T?). Feasibility for fixed T is a linear program. Compare your discrete answer with $T = 20$ s, and see what the LP's dual variables look like next to $\lambda_2(t)$.
4. **Open investigation.** Fuller's problem: minimise $\int_0^\infty x^2\,dt$ for $\dot{x} = v$, $\dot{v} = u$, $|u| \le 1$, from $(x_0, v_0) = (1, 0)$. Simulate candidate bang-bang feedbacks $u = -\operatorname{sign}(x + k\,v|v|)$ for a range of k, find the best k numerically, and count the switches. You should find something strange happening as the trajectory nears the origin. Which nonlinear systems show this behaviour, and why, is still an active research topic.

51.7 Loose threads

We found the best future by attaching a price to the state at every instant and then choosing greedily against those prices. Three things were left unsaid.

> **? Does a best plan even exist?**
>
> Everything in this chapter assumed an optimal control exists and then described it. In infinite-dimensional spaces that is not automatic: a sequence of ever-better controls can oscillate faster and faster and converge to nothing that is itself a control. Guaranteeing existence needs compactness of the right kind, usually convexity of the set of achievable velocities plus a weak form of convergence. The tools are those of Ch. 61, and Filippov's theorem is the classical answer.

> **? Steering through fog**
>
> Our cart knew its exact position and the exact dynamics. Real controllers see noisy measurements and face random disturbances. Remarkably, for linear systems with Gaussian noise and quadratic cost, the best you can do splits cleanly into two separate problems: estimate the state as well as possible, then apply the LQR as if the estimate were the truth. Why this "separation" holds there and fails almost everywhere else is a question for stochastic and robust optimization in Ch. 54, and for learning controllers that don't know the model at all in Ch. 53.

The costate mystery from §51.3 is resolved in the next chapter, which starts from a seemingly unrelated idea, that *the tail of an optimal path is optimal*, and arrives at the same Hamiltonian from the other side (Ch. 52). The pair of equations $\dot{\mathbf{x}} = \partial H/\partial \boldsymbol{\lambda}$, $\dot{\boldsymbol{\lambda}} = -\partial H/\partial \mathbf{x}$ is no accident either: they are Hamilton's equations of classical mechanics, and the principle of least action is an optimal-control problem posed by nature (Ch. 92). For the phase-plane geometry of the switching curve, and what happens to feedback systems near their equilibria, see Ch. 83.

Further reading

- D. E. Kirk, *Optimal Control Theory: An Introduction*, Dover, 2004 (reprint of the 1970 edition). The clearest first book: calculus of variations, Pontryagin, and dynamic programming side by side.
- D. Liberzon, *Calculus of Variations and Optimal Control Theory: A Concise Introduction*, Princeton University Press, 2012. A careful proof of the maximum principle with needle variations.
- A. E. Bryson and Y.-C. Ho, *Applied Optimal Control*, Hemisphere, 1975. The engineer's classic, rich in aerospace examples and numerical methods.
- J. T. Betts, *Practical Methods for Optimal Control Using Nonlinear Programming*, 3rd ed., SIAM, 2020. Direct methods as used in industry.

Chapter 52 — Dynamic Programming and Bellman's Principle

> **Level 2 · Track D — Optimization and Operations Research Builds on:** Ch. 19, Ch. 23, Ch. 27, Ch. 34 · **Leads to:** Ch. 53, Ch. 54 · **Related:** Ch. 40, Ch. 46, Ch. 50, Ch. 51, Ch. 89 **Reading time:** ~30 min

52.1 The hook

In 1950 Richard Bellman arrived at the RAND Corporation to work on multistage decision problems: how to schedule, stock, steer and fight when each decision changes the situation for the next. RAND was funded by the Air Force, and the Secretary of Defense of the day was said to turn red at the word *research*. So Bellman, by his own account, chose a name for his method that no congressman could object to. "Programming" meant planning, as in a military programme. "Dynamic" sounded impressive and, he joked, was impossible to use in a bad sense. The name was camouflage. The idea underneath it has since become one of the most widely used in all of applied mathematics.

Here is a problem of the kind he had in mind. A courier must get from depot S to customer T through three stages of relay points, choosing one point at each stage. Each leg has a cost (minutes, litres, risk). The legs are drawn in the figure in §52.2. A tempting strategy is **greedy**: at each point, take the cheapest leg out. From S the cheapest leg costs 2, to A1, then 6 to B2, 3 to C2 and 4 to T, for 15 in total. Not bad?

The best route costs 11. Greedy lost 36% by grabbing a cheap first step that led into an expensive neighbourhood.

You could list every route: $3 \times 3 \times 2 = 18$ of them. But a real network might have 20 stages with 10 points each, and then there are 10^{20} routes. Bellman's method solves that larger network with under 2000 additions, and it never looks at a single complete route.

> **How can you find the best path without ever comparing paths?**

The same question, in disguise, sits under other chapters of this book. Ch. 27 promised that the Bellman–Ford algorithm needs exactly $|V| - 1$ rounds to find shortest paths with negative edges. Ch. 40 claimed that to check a strategy in an infinitely repeated game you only need to test deviations *one period at a time*. Ch. 51 introduced a mysterious costate and called it a price. All three debts are paid in this chapter, by one principle.

52.2 The big idea

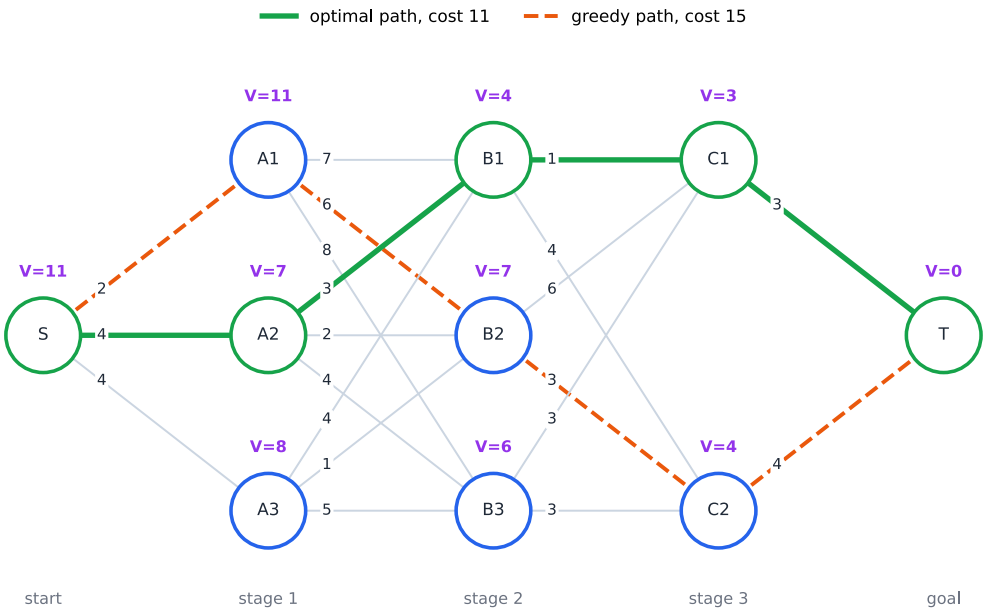
> **Bellman's principle of optimality**
>
> An optimal policy has the property that, whatever the initial state and first decision, the remaining decisions form an optimal policy for the state that results from the first decision.

In plain words: **every tail of an optimal path is itself optimal.** If the best route from S to T passes through B1, then its part from B1 to T must be the best route from B1 to T. If there were a better way from B1, you could splice it in and beat the "best" route.

That sounds almost too obvious to be useful. Its power comes from turning it around. Instead of asking "what is the best route from here?" for one starting point, ask it for *every* point, and start at the end. Write $V(\text{point})$ for the cheapest cost from that point to T, the **value** or **cost-to-go**. Then:

- $V(T) = 0$.
- For each point one stage back, $V(\text{point}) =$ min over legs of (leg cost $+ V$ of where the leg lands).
- Repeat, stage by stage, back to S.

Each point is processed once, and each leg is looked at once. The work grows with the number of *legs*, not the number of *routes*.



Solving backwards. Each purple number is the cheapest cost from that node to T, computed from the numbers one stage to its right. Greedy grabs the cheap first leg into A1, whose cost-to-go is 11; the green route pays 4 to reach A2, whose cost-to-go is only 7.

An analogy: imagine hiking in fog with signposts at every junction, each showing the distance to the valley along the best trail. At each junction, add each trail's length to the next signpost's number, and walk toward the smallest total. Dynamic programming is the art of writing those signposts. Once they exist, acting optimally is greedy, but greedy with respect to the *value*, not the immediate cost.

That last sentence is the deep lesson. Greedy failed because the immediate cost is a bad guide. Greedy with respect to V is perfect, because V has already folded the entire future into one number per state.

52.3 The mathematics

Finite-horizon dynamic programming

A **finite-horizon decision problem** has stages $k = 0, 1, \dots, N$, a set of states, an action set $A(s)$ in each state, a transition $s' = g_k(s, a)$ and a stage cost $c_k(s, a)$, with a terminal cost $c_N(s)$. A policy picks an action in every state at every stage.

Backward recursion

Define $V_N(s) = c_N(s)$ and, for $k = N-1, \dots, 0$,

$$V_k(s) = \min_{a \in A(s)} \left[c_k(s, a) + V_{k+1}\left(g_k(s, a)\right) \right].$$

Then $V_k(s)$ is the minimum total cost from state s at stage k, and choosing at each stage an action that attains the minimum is optimal.

Proof. By backward induction on k. At $k = N$ it is the definition. Suppose V_{k+1} is the true optimal cost-to-go. Any plan from (s, k) takes some first action a, paying $c_k(s, a)$, and then follows some plan from $s' = g_k(s, a)$ whose cost is at least $V_{k+1}(s')$. So every plan costs at least $c_k(s, a) + V_{k+1}(s') \geq V_k(s)$. Conversely, taking the minimising a and then an optimal plan from s' costs exactly $V_k(s)$. Hence $V_k(s)$ is the optimum, and it is attained by the greedy-in-V choice. ∎

With uncertainty, the next state is random with probabilities $P(s' \mid s, a)$, and the same proof goes through with an expectation (§19.9):

$$V_k(s) = \min_a \left[c_k(s, a) + \sum_{s'} P(s' \mid s, a)\, V_{k+1}(s') \right].$$

The key point is that **the state carries everything about the past that matters for the future**. That is the Markov property of Ch. 34. If the cost depended on the whole history, the recursion would fail unless you enlarged the state to hold that history.

The cost. With S states, A actions and N stages, the recursion takes $O(NSA)$ work (deterministic) or $O(NS^2A)$ (stochastic), against A^N plans for brute force. For the 20-stage, 10-point courier network that is $10 + 19 \times 100 + 10 = 1920$ leg evaluations instead of 10^{20} routes.

Result 1: why Bellman–Ford needs $|V| - 1$ rounds

Shortest paths on a general directed graph have no stages, but you can manufacture them by counting edges. For a source s, let $d_k(v)$ be the minimum weight of a walk from s to v that uses *at most* k edges ($d_0(s) = 0$, and $d_0(v) = \infty$ otherwise). The last edge of such a walk enters v from some u, and by the principle of optimality the rest must be a best walk to u with at most $k-1$ edges. So

$$d_k(v) = \min\left(d_{k-1}(v),\ \min_{(u,v) \in E} \left[d_{k-1}(u) + w(u, v) \right] \right).$$

One "round" of Bellman–Ford, relaxing every edge once, computes d_k from d_{k-1} in $O(|E|)$ time.

Bellman–Ford

If no cycle of negative total weight is reachable from s, then $d_{|V|-1}(v)$ is the shortest-path distance from s to every v. If $d_{|V|}(v) < d_{|V|-1}(v)$ for some v, a negative cycle is reachable.

Proof. Take any shortest walk from s to v. If it visits some vertex twice, it contains a cycle. That cycle has weight ≥ 0 by assumption, so cutting it out gives a walk that is no heavier and has fewer edges. Repeat until no vertex repeats. The result is a simple path, which visits at most $|V|$ vertices and so uses at most $|V|-1$ edges. Hence the minimum over walks with at most $|V|-1$ edges already equals the true distance, and further rounds cannot change anything. Conversely, if round $|V|$ still improves some $d(v)$, there is a cheapest walk with exactly $|V|$ edges, which must repeat a vertex; the cycle it contains must be negative, or removing it would contradict the improvement. ■

So the factor $|V|$ that Ch. 27 warned about is the **horizon** of a dynamic program: the longest a useful path can be. Dijkstra's algorithm avoids it by exploiting non-negative weights to settle vertices in order, which a negative edge can wreck.

The infinite horizon: Bellman's equation

Many problems have no natural end: a thermostat, an inventory system (Ch. 50), a robot that keeps working. Then we use a **Markov decision process** (MDP): states s, actions a, rewards $r(s,a)$, transition probabilities $P(s' \mid s,a)$, and a **discount factor** $0 \leq \gamma < 1$ that makes a reward k steps away worth γ^k of one now. (Rewards to be maximised are the same as costs to be minimised with a sign flip; this section uses rewards, as reinforcement learning does.) A stationary policy π picks action $\pi(s)$ in state s, and its value is

$$V^{\pi}(s) = E\left[\sum_{t=0}^{\infty} \gamma^t \, r(s_t, \pi(s_t)) \,\middle|\, s_0 = s\right].$$

The optimal value $V^*(s) = \sup_{\pi} V^{\pi}(s)$ satisfies **Bellman's optimality equation**, the stationary version of the backward recursion:

$$V^*(s) = \max_{a} \left[r(s,a) + \gamma \sum_{s'} P(s' \mid s,a)\, V^*(s') \right].$$

Call the right-hand side $(\mathcal{T}V)(s)$. The **Bellman operator** $\mathcal{T}$ maps a value table V to a new one. Bellman's equation says V^* is a fixed point, $V^* = \mathcal{T}V^*$. Two questions arise immediately. Is the fixed point unique? And can we find it?

The Bellman operator is a contraction

For any two value tables V, W,

$$\|\mathcal{T}V - \mathcal{T}W\|_\infty \leq \gamma \, \|V - W\|_\infty,$$

where $\|V\|_\infty = \max_s |V(s)|$. Consequently $\mathcal{T}$ has exactly one fixed point V^*, and **value iteration** $V_{k+1} = \mathcal{T}V_k$ converges to it from any start, with $\|V_k - V^*\|_\infty \leq \gamma^k \|V_0 - V^*\|_\infty$.

Proof. First a small fact: for any two functions f, g of a, $|\max_a f(a) - \max_a g(a)| \leq \max_a |f(a) - g(a)|$. (If $\max f \geq \max g$ and a^* maximises f, then $\max f - \max g \leq f(a^*) - g(a^*)$.) Apply it with $f(a) = r(s,a) + \gamma \sum_{s'} P(s' \mid s,a) V(s')$ and g the same with W:

$$|(\mathcal{T}V)(s) - (\mathcal{T}W)(s)| \leq \max_a \, \gamma \sum_{s'} P(s' \mid s,a)\, |V(s') - W(s')| \leq \gamma \|V - W\|_\infty,$$

because the probabilities sum to 1. Take the maximum over s. If V^* and W^* were two fixed points, $\|V^* - W^*\|_\infty \leq \gamma \|V^* - W^*\|_\infty$ would force them equal. For convergence, $\|V_k - V^*\|_\infty = \|\mathcal{T}V_{k-1} - \mathcal{T}V^*\|_\infty \leq \gamma \|V_{k-1} - V^*\|_\infty$, and induction finishes it. Existence follows because the iterates form a Cauchy sequence in a complete space, the Banach fixed-point theorem. ■

The discount factor is the *rate* of convergence. With $\gamma = 0.9$ the error shrinks a thousandfold in about 66 sweeps; with $\gamma = 0.99$ it takes about 687. Far-sighted problems are slow to solve.

Result 2: one-step deviations are enough

Here is the debt to Ch. 40. A policy can be beaten, in principle, by an elaborate alternative that deviates differently at every step forever. Do you have to check all of those?

One-step deviation principle (single decision-maker)

Let π be a stationary policy. If no **single** deviation improves it, that is, if for every state s and action a

$$V^{\pi}(s) \;\geq\; r(s,a) + \gamma \sum_{s'} P(s' \mid s,a)\, V^{\pi}(s'),$$

(deviate once to a, then return to π), then π is optimal: $V^{\pi} = V^*$.

Proof. The value of a policy satisfies its own Bellman equation, $V^{\pi}(s) = r(s,\pi(s)) + \gamma \sum_{s'} P(s' \mid s,\pi(s)) V^{\pi}(s')$: condition on the first step. So the right-hand side of the hypothesis *equals* $V^{\pi}(s)$ when $a = \pi(s)$, and it is at most $V^{\pi}(s)$ for every other a. The maximum over a is therefore $V^{\pi}(s)$, which says $\mathcal{T}V^{\pi} = V^{\pi}$. By the contraction theorem, $\mathcal{T}$ has only one fixed point, V^*. Hence $V^{\pi} = V^*$. ■

Discounting does the work. Any infinitely long, cunning deviation is worth only finitely many steps of deviation, up to an error of order γ^n, and a finite deviation can be undone one step at a time from its last step backwards, each step being a one-shot deviation that does not help. In a repeated game, each player faces exactly such an MDP once the other players' strategies are fixed, and this is the one-shot deviation principle that Ch. 40 used.

The same idea gives an algorithm. If π *can* be improved by a one-step deviation, switch to the improving action everywhere at once. The **policy improvement theorem** says the new greedy policy $\pi'(s) = \arg\max_a [r(s,a) + \gamma \sum_{s'} P V^{\pi}(s')]$ satisfies $V^{\pi'} \geq V^{\pi}$ in every state, with strict improvement somewhere unless π was already optimal. (Sketch: $\mathcal{T}_{\pi'} V^{\pi} = \mathcal{T} V^{\pi} \geq V^{\pi}$, and applying the monotone map $\mathcal{T}_{\pi'}$ repeatedly only increases the left side, which converges to $V^{\pi'}$.)

> **? How many improvements does it take?**
>
> Policy iteration, alternating "evaluate V^{π}" and "improve greedily", stops after finitely many steps, since there are finitely many policies and each step strictly improves. In practice it almost always finishes in a handful of iterations, even for millions of states. For a fixed discount factor, Ye proved in 2011 that it is strongly polynomial. Without that restriction, cleverly built examples force related variants to take exponentially many steps. An MDP is also a linear program, and whether linear programs can be solved in *strongly* polynomial time is one of Smale's problems for the twenty-first century, still open. The LP view of MDPs and its dual are in the spirit of Ch. 48.

Result 3 (sketch): where the costate came from

Return to continuous time, with dynamics $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u})$ and cost $\int L\, dt$ as in Ch. 51. Let $V(\mathbf{x}, t)$ be the optimal cost-to-go. Apply the principle of optimality over a tiny step dt: pay $L\, dt$, move to $\mathbf{x} + \mathbf{f}\, dt$, and continue optimally:

$$V(\mathbf{x}, t) = \min_{\mathbf{u}} \left[L(\mathbf{x}, \mathbf{u})\, dt + V(\mathbf{x} + \mathbf{f}\, dt,\ t + dt) \right].$$

Expand V to first order, cancel $V(\mathbf{x}, t)$ and divide by dt. This gives the **Hamilton–Jacobi–Bellman equation**

$$-\frac{\partial V}{\partial t} = \min_{\mathbf{u}} \left[L(\mathbf{x}, \mathbf{u}) + \nabla_{\mathbf{x}} V^{\mathsf{T}} \mathbf{f}(\mathbf{x}, \mathbf{u}) \right] = \min_{\mathbf{u}} H\left(\mathbf{x}, \mathbf{u}, \nabla_{\mathbf{x}} V\right).$$

The bracket is exactly Pontryagin's Hamiltonian with $\boldsymbol{\lambda}$ replaced by $\nabla_{\mathbf{x}} V$. Differentiate the HJB equation with respect to $\mathbf{x}$ along the optimal trajectory (the minimising $\mathbf{u}$ drops out by the envelope theorem), and you find that $\boldsymbol{\lambda}(t) = \nabla_{\mathbf{x}} V(\mathbf{x}^*(t), t)$ obeys $\dot{\boldsymbol{\lambda}} = -\partial H / \partial \mathbf{x}$, Ch. 51's costate equation. So the costate *is* a price. It is the marginal cost-to-go, the amount the optimal future changes per unit nudge of the present state.

This derivation quietly assumed V is differentiable. For the minimum-time cart of Ch. 51 it is not: V has a crease along the switching curve.

> **? A PDE whose solutions have corners**
>
> If the value function has kinks, in what sense does it "solve" the HJB equation? Classical derivatives don't exist on the crease, and many non-smooth functions satisfy the equation almost everywhere. Which one is the true value? The answer, found by Crandall and Lions in the 1980s, is the theory of **viscosity solutions**, one of the great ideas in modern PDE theory: Ch. 89.

State-space methods: the linear–quadratic case

Usually V must be stored as a table, one number per state. One famous exception is linear dynamics with quadratic costs, where V stays quadratic forever and the "table" is a matrix. Take the scalar system $x_{k+1} = x_k + u_k$ with stage cost $x_k^2 + u_k^2$. Guess $V_{k+1}(x) = P_{k+1} x^2$. Then

$$V_k(x) = \min_u \left[x^2 + u^2 + P_{k+1}(x + u)^2 \right],$$

minimised at $u = -\dfrac{P_{k+1}}{1 + P_{k+1}}\, x$, which gives $V_k(x) = P_k x^2$ with

$$P_k = 1 + \frac{P_{k+1}}{1 + P_{k+1}}.$$

This is a **Riccati recursion**, the discrete twin of the LQR equation in Ch. 51. Starting from $P_N = 0$, it produces $1, \tfrac{3}{2}, \tfrac{8}{5}, \tfrac{21}{13}, \ldots$, ratios of Fibonacci numbers, converging to the fixed point $P^2 - P - 1 = 0$, that is $P = \varphi = (1 + \sqrt{5})/2 \approx 1.618$. The optimal long-run feedback is $u = -x/\varphi \approx -0.618\, x$. The golden ratio, hiding in a regulator. For vector states, P_k becomes a matrix and the recursion costs $O(n^3)$ per step, even though the state space is a continuum.

Without such structure, a table over a d-dimensional state space with m grid points per axis has m^d entries. Bellman called this the **curse of dimensionality**. A modest robot arm with 12 state variables and 20 grid points each needs $20^{12} \approx 4 \times 10^{15}$ values.

52.4 The algorithm

Value iteration applies $\mathcal{T}$ until the table stops moving:

```
V = 0 for every state
repeat
    for each state s:
        V_new(s) = max over a of [ r(s,a) + gamma * sum_{s'} P(s'|s,a) V(s') ]
    delta = max_s |V_new(s) - V(s)|;   V = V_new
until delta < eps * (1 - gamma) / gamma
policy(s) = argmax over a of the same bracket
```

The stopping rule is not arbitrary. The contraction property implies $\|V_{k+1} - V^*\|_\infty \leq \frac{\gamma}{1-\gamma} \|V_{k+1} - V_k\|_\infty$, so the test guarantees the final table is within ε of V^*. Each sweep costs $O(S^2 A)$ in general, and $O(SA)$ when each action leads to only a few successor states.

Policy iteration alternates two steps: *evaluate* the current policy by solving the linear system $(\mathbf{I} - \gamma \mathbf{P}_\pi)\mathbf{v} = \mathbf{r}_\pi$ (Ch. 8), then *improve* greedily. Each iteration is expensive, $O(S^3)$ for a dense solve, but there are few of them.

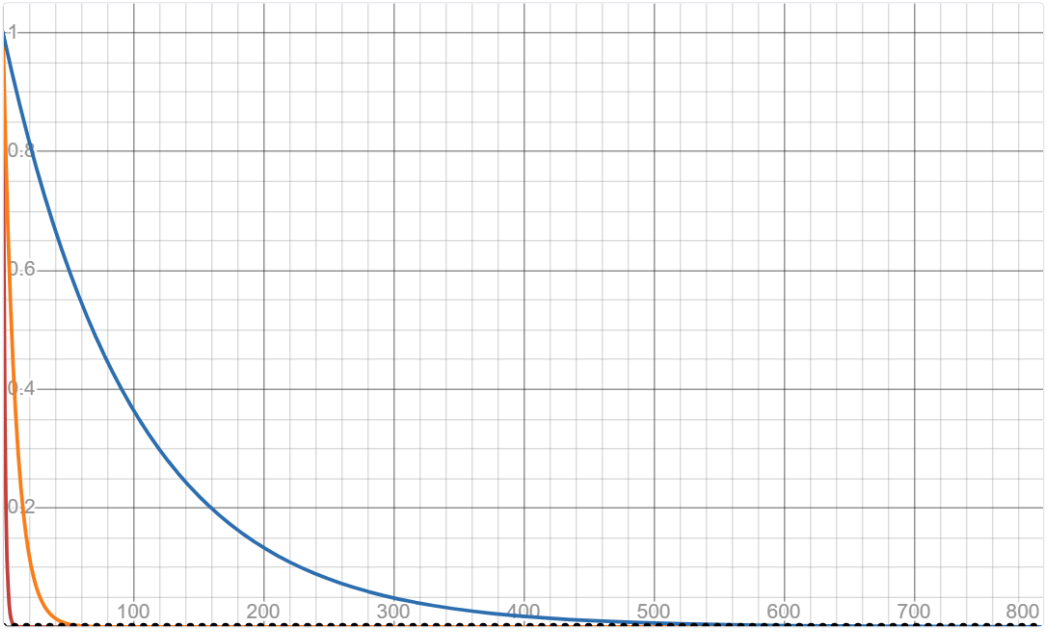
Numerical notes. $\mathbf{I} - \gamma \mathbf{P}_\pi$ is always invertible for $\gamma < 1$, because the eigenvalues of a stochastic matrix have magnitude at most 1 (Ch. 34). As $\gamma \to 1$, however, it becomes ill-conditioned, and value iteration slows to a crawl.

Failure cases. Dynamic programming needs (1) a *known model*, the probabilities and rewards; (2) a *small state space*, or special structure like LQR; and (3) the *Markov property*, a state that summarises the past. Break (3) and the recursion gives wrong answers with full confidence. A common fix is to enlarge the state, for example by adding the last few observations, which feeds the curse of dimensionality in (2).

> **Engineering note**
>
> Dynamic programming hides inside familiar engineering tools. The **Viterbi decoder** in every mobile phone and hard-disk read channel is a shortest-path DP over the states of a convolutional code (Ch. 79). Sequence alignment in genomics, speech recognition, and the dynamic time warping used to match signals are all DP over a grid.

52.5 The experiment

Part 1: the price of patience. Value iteration's error shrinks like γ^k. The three curves are $\gamma = 0.5$, 0.9 and 0.99; the dotted line is a factor of 1000. Read off where each curve crosses it (about 10, 66 and 687 sweeps). Then change 0.99 to 0.999 and see how far right the crossing moves.



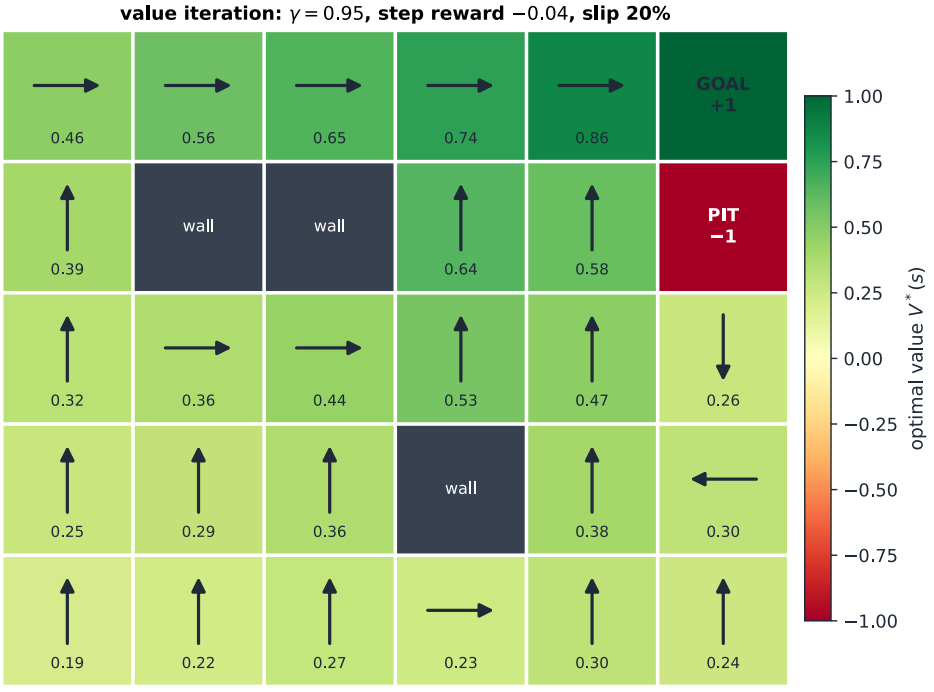
Error bound after x sweeps of value iteration, as a fraction of the starting error.

Part 2: a grid world. A robot lives on a small grid with walls (`#`), a goal worth $+1$ (`G`) and a pit worth -1 (`P`). Each move costs `STEP`. With probability `SLIP` the robot slides sideways instead of going where it intends. The script runs value iteration and prints the values and the greedy policy.

```
import numpy as np

GAMMA, SLIP, STEP = 0.95, 0.2, -0.04      # knobs: try STEP = -0.5, SLIP = 0.0, GAMMA = 0.5
GRID = ["....G",
        ".#.#P",
        "....."]                          # G = goal (+1), P = pit (-1), # = wall
R, C = len(GRID), len(GRID[0])
MOVES = {"^": (-1, 0), "v": (1, 0), "<": (0, -1), ">": (0, 1)}
SIDE = {"^": "<>", "v": "<>", "<": "^v", ">": "^v"}

def go(r, c, a):
    r2, c2 = r + MOVES[a][0], c + MOVES[a][1]
    ok = 0 <= r2 < R and 0 <= c2 < C and GRID[r2][c2] != "#"
    return (r2, c2) if ok else (r, c)

def q(V, r, c, a):                        # expected return of action a, then acting optimally
    nxt = [(1 - SLIP, a)] + [(SLIP / 2, b) for b in SIDE[a]]
    return STEP + GAMMA * sum(p * V[go(r, c, b)] for p, b in nxt)

V = np.zeros((R, C))
for k in range(1000):
    V_new = V.copy()
    for r in range(R):
        for c in range(C):
            cell = GRID[r][c]
            if cell in "GP#":
                V_new[r, c] = {"G": 1.0, "P": -1.0, "#": 0.0}[cell]
            else:
                V_new[r, c] = max(q(V, r, c, a) for a in MOVES)
    change = np.abs(V_new - V).max()
    V = V_new
    if change < 1e-8:
        break

print(f"converged after {k + 1} sweeps\n", np.round(V, 2))
for r in range(R):
    print(" ".join(GRID[r][c] if GRID[r][c] in "GP#" else
                   max(MOVES, key=lambda a: q(V, r, c, a)) for c in range(C)))
```

With the defaults it converges in 33 sweeps. Look at the bottom-right cell, directly below the pit: its policy is `<`, walking *away* from the goal to avoid sliding into the pit. Now try the knobs:

- `SLIP = 0.0`: convergence drops to 8 sweeps (the world is deterministic, and values settle as soon as the goal's information has travelled across the grid).
- `STEP = -0.5`: life is so painful that the bottom-right cell now chooses `^`, straight into the pit. A quick -1 beats a long walk at -0.5 a step. The robot has rationally chosen to end it.
- `GAMMA = 0.5`: the robot is so short-sighted that the goal is barely visible; most values are near zero, and the bottom-right cell bumps into the wall (`v`) rather than risk anything.



A larger grid world solved by value iteration with the default knobs. Values fall smoothly with distance from the goal. In the cell below the pit the arrow points away from the goal: with a 20% chance of slipping sideways, the safe detour is worth more than the short cut.

52.6 The challenge

1. **Intuition.** In the staged network, the cost-to-go of A1 is 11 and that of A3 is 8. Without recomputing anything, explain how the optimal route from S would change if the leg S→A3 dropped from 4 to 2. Then explain why changing a leg in stage 3 can change the optimal *first* decision, but changing a leg leaving S cannot change the cost-to-go of any node other than S.
2. **Derivation.** Prove the stopping rule: if $\|V_{k+1} - V_k\|_\infty \le \delta$, then $\|V_{k+1} - V^*\|_\infty \le \frac{\gamma}{1-\gamma}\delta$. Then complete the proof of the policy improvement theorem, using that $\mathcal{T}_{\pi'}$ is monotone (if $V \le W$ everywhere, then $\mathcal{T}_{\pi'}V \le \mathcal{T}_{\pi'}W$).
3. **Implementation.** Add policy iteration to the grid-world script: evaluate V^π by solving the linear system with `np.linalg.solve`, then improve greedily, until the policy stops changing. Count iterations for $\gamma = 0.5, 0.9, 0.99$ and compare with the number of value-iteration sweeps. Then implement Bellman–Ford with negative-cycle detection and test it on a graph with one negative edge.
4. **Open investigation.** Make the grid world big (say 50×50) with random walls and random pits. How does the number of policy-iteration steps grow with grid size and with γ? Can you construct a small MDP that forces policy iteration (with greedy switching in every state) to take many steps? Exponential lower bounds are known for some variants and not for others; see how close you can get.

52.7 Loose threads

Dynamic programming replaced a search over paths with a table of values, one number per state, filled in from the future backwards. It paid three debts: the $|V| - 1$ horizon of Bellman–Ford, the one-step deviation principle of repeated games, and the meaning of Ch. 51's costate. But everything rested on two quiet assumptions: we knew the model, and the table was small enough to write down.

> **? What if nobody tells you the probabilities?**
>
> Every Bellman update above needed $P(s' \mid s, a)$ and $r(s, a)$. A robot in a new building, a trading agent in a market, or a program learning to play Go knows neither. It only experiences one transition at a time. Can you still converge to V^* by averaging experience into the Bellman equation as you go? Remarkably, yes, under conditions on how you explore and how fast you average. That is Q-learning and the whole of Ch. 53.

The curse of dimensionality is the other loose thread, and it has no complete answer. Linear–quadratic structure defeats it exactly. Monte Carlo sampling (Ch. 38) defeats it for some problems. Neural networks that approximate V appear to defeat it in practice (Ch. 53), with few guarantees. When the transition probabilities themselves are uncertain, a robust version of the Bellman equation takes a worst case over a set of models (Ch. 54). When an opponent picks some of the transitions, the max becomes a max–min, and you are back in the game trees of Ch. 46.

Further reading

- R. Bellman, *Dynamic Programming*, Princeton University Press, 1957 (Dover reprint 2003). The original, still readable, full of examples.
- D. P. Bertsekas, *Dynamic Programming and Optimal Control*, Vols. I–II, 4th ed., Athena Scientific, 2017. The standard reference, from finite-horizon problems to approximate DP.
- M. L. Puterman, *Markov Decision Processes: Discrete Stochastic Dynamic Programming*, Wiley, 1994. Rigorous treatment of value iteration, policy iteration and the LP formulation.
- T. H. Cormen, C. E. Leiserson, R. L. Rivest and C. Stein, *Introduction to Algorithms*, 4th ed., MIT Press, 2022. Chapter on dynamic programming and the Bellman–Ford algorithm.

Chapter 53 — Reinforcement Learning Through Mathematics

> **Level 2 · Track D — Optimization and Operations Research Builds on:** Ch. 19, Ch. 23, Ch. 34, Ch. 52 · **Leads to:** Ch. 54, Ch. 61 · **Related:** Ch. 37, Ch. 38, Ch. 45, Ch. 46 **Reading time:** ~30 min

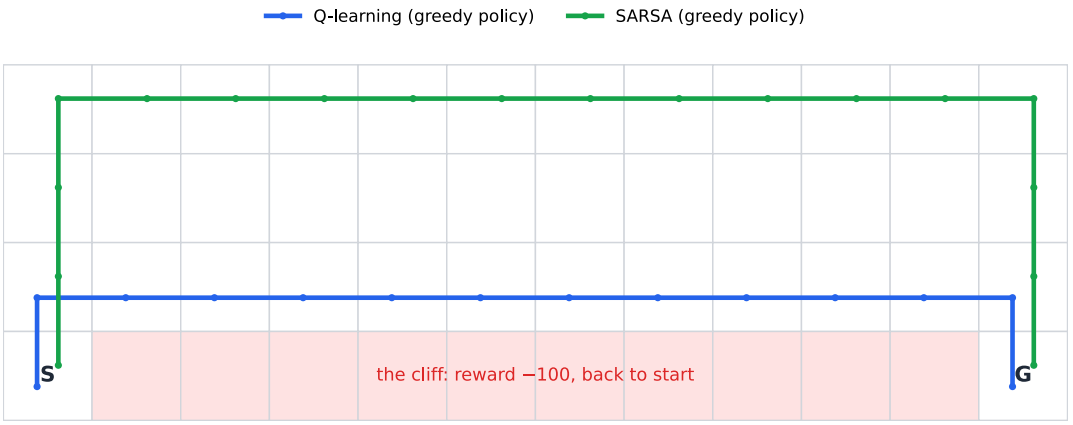
53.1 The hook

Picture a grid four squares tall and twelve wide. A walker starts in the bottom-left corner and wants to reach the bottom-right corner. Every step costs one point. Between the two corners, along the whole bottom row, runs a cliff. Step onto it and you lose a hundred points and are carried back to the start.

The walker has no map. Nobody tells it where the cliff is, or that there is a cliff at all. All it sees is the square it stands on and, after each step, a number: -1, or occasionally a shocking -100. From those numbers alone it must learn how to walk.

Now let two learning programs loose on this grid. Both try a random direction one time in ten and otherwise take the move that currently looks best. Both get the same 500 attempts. The first, **Q-learning**, ends up knowing the truly optimal route, the 13-step path that hugs the cliff edge. The second, **SARSA**, learns a timid detour up to the top row and back down, 17 steps long.

Then you look at the scores they actually earned during the last hundred attempts. The program that knows the best path averaged about -49 points per attempt. The timid one averaged about -26. The student who learned the right answer did almost twice as badly as the one who learned the wrong answer.



The same world, the same experience, two different conclusions. Q-learning's greedy policy (blue) takes the shortest route, one square from the edge. SARSA's (green) keeps three squares of safety margin. With 10% random moves, the shortest route sends the walker off the cliff about once every three attempts.

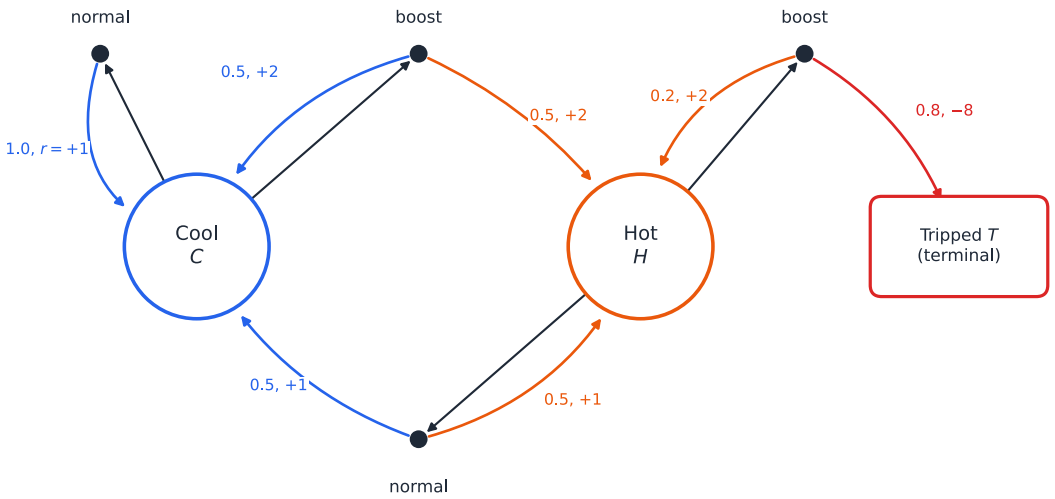
Q-learning has learned the value of walking the edge *if you never stumble*. But it does stumble, one step in ten, and on the edge a stumble is a fall. SARSA learned the value of its own clumsy behaviour, stumbles included, and concluded the edge is not worth it. Each program answered a different question correctly.

So which question should a learner be answering? And how can anything learn the value of a path at all from a stream of -1s and -100s, without ever seeing the map? The answer is a small amount of probability, one fixed-point theorem, and an idea so simple it is almost embarrassing: *guess, then correct your guess using your next guess.*

53.2 The big idea

Ch. 52 solved sequential decision problems when you know the rules: the transition probabilities and the rewards. Bellman's principle turned "plan the whole future" into "plan one step, then trust the value of where you land", and value iteration computed those values by sweeping over every state. Reinforcement learning asks what is left of that machinery when the rules are hidden and you can only *act and observe*.

Here is a small world we will use all chapter. An electric motor can run **cool** or **hot**. In each state the controller chooses **normal** or **boost**. Boost earns more (2 units of work per step against 1), but it heats the motor. A cool motor under boost becomes hot half the time. A hot motor under boost trips its thermal protection with probability 0.8, which costs a penalty and ends the run. Future rewards are discounted by $\gamma = 0.9$ per step.



The motor-drive decision process. Each black dot is a choice; the arrows leaving it are what Nature does next, with probability and reward. From Hot, boost pays $+2$ and then trips the motor 80% of the time, for a net reward of -8.

If you know this diagram, Ch. 52's value iteration gives the answer in a few lines: boost when cool, run normally when hot. The interesting case is when you don't know the diagram. You can only run the motor and watch.

Three ideas carry the whole subject.

1. Value is an average of the future. The value of "boost while cool" is the expected discounted sum of all rewards that follow. You cannot compute that expectation without the model, but you can *sample* it: try the action, see what happens. Averages of samples converge to expectations (Ch. 19).

2. Bootstrapping. Waiting for the end of a run is slow. Instead, after a single step you already have a better guess than before: the reward you just got plus your current estimate of where you landed. Move your old guess a little toward that one-step target. This is **temporal-difference learning**, and it is Bellman's equation turned into a correction rule. You are learning a guess from a guess. That such a thing converges at all is the mathematical heart of this chapter.

3. Exploration versus exploitation. If the motor controller always takes the action that currently looks best, it may never try the action that is actually best. A controller that starts with all estimates at zero, runs normal once and sees $+1$, will prefer normal forever and never discover that boost is better. It has to spend some of its time doing things that look worse, *to find out*. That cost is the -49 of the cliff walker.

An analogy: you move to a new city and want the best restaurant. Each meal is a noisy sample of a restaurant's quality (idea 1). You judge a neighbourhood partly by the restaurants you have already rated nearby (idea 2). And every night you must decide between your current favourite and a place you have never tried (idea 3).

53.3 The mathematics

Markov decision processes

Markov decision process (MDP)

An MDP is a tuple $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$: a set of states $\mathcal{S}$, a set of actions $\mathcal{A}$, transition probabilities $P(s' \mid s, a)$, an expected reward $r(s, a)$ for taking action a in state s, and a discount factor $0 \le \gamma < 1$. A **policy** $\pi(a \mid s)$ is a probability distribution over actions in each state. The agent's **return** from time t is

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \cdots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}.$$

The next state depends on the past only through the current state and action, as in the chains of Ch. 34. Fix a policy and the MDP *becomes* a Markov chain with transition matrix $P_\pi(s' \mid s) = \sum_a \pi(a \mid s) P(s' \mid s, a)$. A policy is a way of choosing which Markov chain you live in.

The discount has two readings. Economically, a reward now is worth more than a reward later. Probabilistically, γ is the chance the world continues for another step, so $1/(1-\gamma)$ is the **effective horizon**: 10 steps for $\gamma = 0.9$, 100 for $\gamma = 0.99$. Since $|R| \le R_{\max}$ implies $|G_t| \le R_{\max}/(1-\gamma)$, returns are always finite.

Value functions

The **state value** and the **action value** of a policy π are

$$V^\pi(s) = E_\pi\left[G_t \mid S_t = s\right], \qquad Q^\pi(s, a) = E_\pi\left[G_t \mid S_t = s,\ A_t = a\right].$$

The optimal action value is $Q^*(s, a) = \max_\pi Q^\pi(s, a)$.

Ch. 52 derives the Bellman equations and proves that the Bellman operator on state values is a contraction; the same argument, word for word, shows that the **Bellman optimality operator** on action values

$$(\mathcal{T}Q)(s,a) = r(s,a) + \gamma \sum_{s'} P(s' \mid s,a) \max_{a'} Q(s',a')$$

is a γ-contraction in the max-norm, $\|\mathcal{T}Q - \mathcal{T}Q'\|_\infty \le \gamma \|Q - Q'\|_\infty$. So Q^* is its unique fixed point, value iteration $Q_{k+1} = \mathcal{T}Q_k$ converges to it geometrically, and any policy that is greedy with respect to Q^* is optimal. The new problem is that $\mathcal{T}$ contains P, and P is exactly what we don't have.

The motor drive, solved with the model

Write C, H for cool and hot, n, b for normal and boost. Running value iteration from $Q = 0$ with $\gamma = 0.9$ converges to

$$Q^* = \begin{pmatrix} Q^*(C,n) & Q^*(C,b) \\ Q^*(H,n) & Q^*(H,b) \end{pmatrix} = \begin{pmatrix} 14.95 & 15.50 \\ 14.50 & -3.39 \end{pmatrix}.$$

Check one entry against the fixed-point equation: $Q^*(H,b) = 0.2\,(2 + 0.9 \times 14.50) + 0.8 \times (-8) = 0.4 + 2.61 - 6.4 = -3.39$. The optimal policy boosts when cool and runs normally when hot, with $V^*(C) = 15.5$. Always running normal gives only $V(C) = 10$, and always boosting gives $V(C) = -2.35$. Notice how close the two actions in state C are: 15.50 against 14.95. A learner estimating these from noisy samples has to resolve a gap of 0.55 in numbers around 15.

Learning a fixed point from samples

Look at $\mathcal{T}Q$ again. It is an expectation over the next state s'. If we take action a in state s and observe a reward r and a next state s', then the single number

$$y = r + \gamma \max_{a'} Q(s',a')$$

is an unbiased sample of $(\mathcal{T}Q)(s,a)$: its expected value, over the randomness of the environment, is exactly $(\mathcal{T}Q)(s,a)$. We cannot apply $\mathcal{T}$, but we can draw noisy samples of it, one transition at a time. **Q-learning** nudges the estimate toward each sample:

$$Q(s,a) \leftarrow Q(s,a) + \alpha \underbrace{\big[\, r + \gamma \max_{a'} Q(s',a') - Q(s,a) \,\big]}_{\text{temporal-difference error } \delta}.$$

The bracket δ is the surprise: how much better or worse the step went than the current estimate predicted. When $Q = Q^*$ the surprise is zero *on average*, which is all Bellman's equation says.

Why should such a rule converge? Strip away the noise first and look at what the update does on average.

Theorem 53.1 — The averaged Q-learning step is a contraction

Let $\mathcal{T}$ be a γ-contraction in $\|\cdot\|_\infty$ with fixed point Q^*, and let $0 < \alpha \le 1$. The averaged update $\mathcal{U}Q = (1-\alpha)Q + \alpha\,\mathcal{T}Q$ satisfies

$$\|\mathcal{U}Q - Q^*\|_\infty \le \big(1 - \alpha(1-\gamma)\big)\,\|Q - Q^*\|_\infty.$$

Hence iterating $\mathcal{U}$ converges to Q^* geometrically, at rate $1 - \alpha(1-\gamma) < 1$.

Proof. Because $Q^* = \mathcal{T}Q^*$, we can write $Q^* = (1-\alpha)Q^* + \alpha\,\mathcal{T}Q^*$. Subtract this from $\mathcal{U}Q$:

$$\mathcal{U}Q - Q^* = (1-\alpha)(Q - Q^*) + \alpha(\mathcal{T}Q - \mathcal{T}Q^*).$$

Take max-norms, use the triangle inequality, and then the contraction property:

$$\|\mathcal{U}Q - Q^*\|_\infty \le (1-\alpha)\|Q - Q^*\|_\infty + \alpha\gamma\|Q - Q^*\|_\infty = \big(1 - \alpha(1-\gamma)\big)\|Q - Q^*\|_\infty.$$

Since $\alpha > 0$ and $\gamma < 1$, the factor is strictly less than 1, and the Banach fixed-point argument of Ch. 52 applies. $\square$

Two practical lessons follow. Small α slows convergence: with $\alpha = 0.1$ and $\gamma = 0.9$ the factor is 0.99, so the error shrinks by only 1% per sweep. Long horizons hurt too: as $\gamma \to 1$ the factor approaches 1 for any fixed α.

Real Q-learning does not apply $\mathcal{U}$. It replaces $\mathcal{T}Q$ by one noisy sample and updates one entry at a time. The noise forces a compromise in the step size. A constant α keeps reacting to every new sample, so the estimate jitters forever around Q^*. A step size that shrinks too fast stops learning before it arrives. The classical result of stochastic approximation (Robbins and Monro, 1951), applied to Q-learning by Watkins and Dayan (1992) and Tsitsiklis (1994), says the right compromise is

$$\sum_t \alpha_t(s,a) = \infty, \qquad \sum_t \alpha_t(s,a)^2 < \infty$$

for every pair (s,a), for example $\alpha = 1/N(s,a)$ or $1/N(s,a)^{0.7}$, where N counts visits. The first sum lets the estimate travel any distance; the second makes the accumulated noise finite.

Convergence of tabular Q-learning (stated)

In a finite MDP, if every state–action pair is updated infinitely often and the step sizes satisfy the two conditions above, then $Q_t \to Q^*$ with probability 1, *whatever policy generated the data.*

The last clause is the cliff walker's secret. Because the target uses $\max_{a'}$, Q-learning learns the value of the *greedy* policy even while behaving with ε-random moves. It is **off-policy**: it learns about one policy while following another. SARSA replaces $\max_{a'} Q(s',a')$ by $Q(s',a')$ for the action a' it actually takes next, so it learns Q^π for its own exploring policy. It is **on-policy**.

Notice what the theorem assumes and nothing supplies: *every pair visited infinitely often*. Exploration is a separate decision.

Optimizing the policy directly

Value-based methods learn Q^* and act greedily. That breaks down when actions are continuous (what is $\max_{a'}$ over all motor voltages?) or when the best policy is genuinely random, as in the zero-sum games of Ch. 39. The alternative is to write down a parametrized policy $\pi_\theta(a \mid s)$, say a softmax over action scores, and do gradient ascent (§23.5) on the expected return

$$J(\theta) = E_{\tau\sim\pi_\theta}\left[R(\tau)\right], \qquad R(\tau) = \sum_{t=0}^{T-1} \gamma^t r_{t+1},$$

where $\tau = (s_0, a_0, r_1, s_1, a_1, \dots)$ is a whole trajectory. The obstacle is that θ appears inside the *distribution* of τ, not inside the thing being averaged, and that distribution involves the unknown dynamics P. It looks as if you would need to differentiate the environment.

Theorem 53.2 — Policy gradient (REINFORCE form)

For an episodic task of length T,

$$\nabla_\theta J(\theta) = E_{\tau\sim\pi_\theta}\left[R(\tau)\sum_{t=0}^{T-1}\nabla_\theta \log \pi_\theta(a_t \mid s_t)\right].$$

The right side contains no derivative of P, so it can be estimated from sampled trajectories alone.

Proof. The probability of a trajectory is a product of policy terms and environment terms:

$$p_\theta(\tau) = \rho_0(s_0)\prod_{t=0}^{T-1}\pi_\theta(a_t \mid s_t)\,P(s_{t+1} \mid s_t, a_t).$$

Differentiate under the sum (or integral) over trajectories and use the identity $\nabla p = p\,\nabla \log p$, valid wherever $p > 0$:

$$\nabla_\theta J = \sum_\tau R(\tau)\nabla_\theta p_\theta(\tau) = \sum_\tau p_\theta(\tau) R(\tau)\nabla_\theta \log p_\theta(\tau) = E_{\tau\sim\pi_\theta}\left[R(\tau)\nabla_\theta \log p_\theta(\tau)\right].$$

Now take the logarithm of the product:

$$\log p_\theta(\tau) = \log \rho_0(s_0) + \sum_t \log \pi_\theta(a_t \mid s_t) + \sum_t \log P(s_{t+1} \mid s_t, a_t).$$

The first and last terms do not depend on θ, so their gradients vanish. Only the policy terms survive. □

The **log-derivative trick** in the middle step is worth remembering on its own: it turns the gradient of an expectation into the expectation of something you can sample. The resulting estimator has a clean meaning. $\nabla \log \pi_\theta(a_t \mid s_t)$ points in the direction that makes the action you actually took more likely. Weight that direction by how good the whole trajectory turned out, average, and step. Good trajectories get reinforced, hence the name.

It is also noisy. The estimate multiplies a sum of T random vectors by a random total reward. Two standard repairs keep the expectation unchanged and shrink the variance: replace $R(\tau)$ in the term for time t by the reward-to-go from t (an action cannot influence rewards that came before it), and subtract a **baseline** $b(s_t)$, typically an estimate of $V^\pi(s_t)$. The subtraction is harmless because $E_{a\sim\pi_\theta}\left[\nabla \log \pi_\theta(a \mid s)\right] = \sum_a \nabla \pi_\theta(a \mid s) = \nabla 1 = 0$. With the baseline V^π, what multiplies $\nabla \log \pi$ is an estimate of $Q^\pi(s,a) - V^\pi(s)$, the **advantage** of the action over the policy's average. Learn V by temporal differences and the policy with this gradient, and you have an **actor–critic** method: the two halves of this section, working together.

? Climbing a mountain you cannot see

$J(\theta)$ is not concave in θ, not even for a softmax policy on a tiny MDP. Gradient ascent on a non-concave function should get stuck on local peaks. Yet for tabular softmax policies, with exact gradients, it provably reaches the *global* optimum, and in large neural-network policies it usually reaches something good. The tabular proof rests on a "gradient domination" inequality that has no analogue in general non-convex optimization. Why policy landscapes are so much kinder than they have any right to be is the same open puzzle Ch. 47 raised for neural networks, and for deep policies it remains unsolved.

Learning about one policy from another's data

Suppose you have a log of trajectories recorded under an old behaviour policy μ and want the value of a new policy π without running it, because running it on a real motor, patient or power grid could be dangerous. This is **off-policy evaluation**, and its basic tool is importance sampling (Ch. 38).

Theorem 53.3 — Importance-sampled return

If $\mu(a \mid s) > 0$ whenever $\pi(a \mid s) > 0$, then with $\rho(\tau) = \prod_{t=0}^{T-1} \pi(a_t \mid s_t)/\mu(a_t \mid s_t)$,

$$E_{\tau\sim\mu}\left[\rho(\tau)R(\tau)\right] = E_{\tau\sim\pi}\left[R(\tau)\right].$$

Proof. Write p_π and p_μ for the trajectory distributions. In their ratio, the initial-state and transition factors cancel, because both policies face the same environment, leaving $p_\pi(\tau)/p_\mu(\tau) = \rho(\tau)$. So $E_\mu[\rho R] = \sum_\tau p_\mu(\tau)\frac{p_\pi(\tau)}{p_\mu(\tau)}R(\tau) = E_\pi[R]$. (The support condition prevents division by zero.) □

The price, again, is variance. If π is deterministic and μ picks uniformly between two actions, each step contributes a factor π/μ that is 2 or 0, and $E_\mu[\rho^2] = 2^T$. Over a 20-step horizon the weight has variance $2^{20} - 1 \approx 1.05 \times 10^6$. This is the exploding inverse-propensity weight of Ch. 37, compounded once per step: off-policy evaluation is causal inference with a horizon.

53.4 The algorithm

Tabular Q-learning with ε-greedy exploration fits in a dozen lines.

```
initialise Q(s, a) = 0 for all s, a;  N(s, a) = 0
repeat for each episode:
    s ← start state
    while s is not terminal:
        with probability ε pick a at random, else a ← argmax_a Q(s, a)
        take a, observe reward r and next state s'
        N(s, a) ← N(s, a) + 1;   α ← 1 / N(s, a)^0.7
        y ← r                          if s' is terminal
            r + γ · max_a' Q(s', a')   otherwise
        Q(s, a) ← Q(s, a) + α (y − Q(s, a))
        s ← s'
```

For SARSA, choose the next action a' *before* the update and use $y = r + \gamma Q(s', a')$; then take a'.

Cost. Each step costs $O(|\mathcal{A}|)$ and memory is one number per state–action pair. The price is samples: the best known bounds for accuracy ϵ grow like $|\mathcal{S}||\mathcal{A}|/\big((1-\gamma)^3\epsilon^2\big)$, so going from $\gamma = 0.9$ to 0.99 can cost a factor of a thousand.

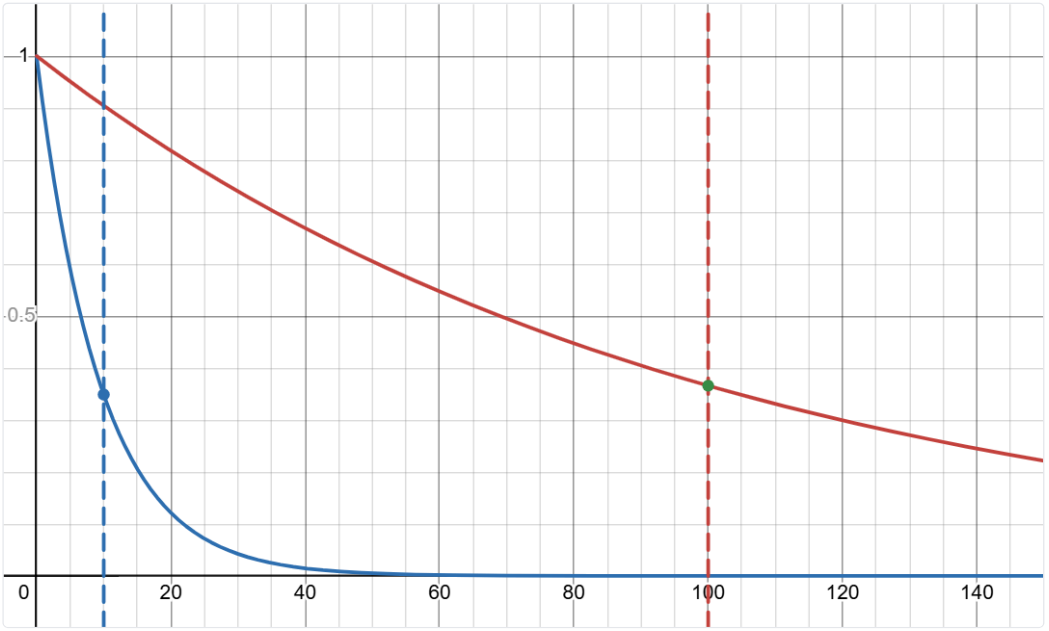
Failure modes worth knowing before you trust a learning curve.

- *No exploration, no learning.* With $\varepsilon = 0$ and all estimates starting at zero, the motor controller of 53.5 runs normal, sees $+1$, and never tries boost again. The theorem's "every pair infinitely often" is silently violated. In the cliff world the same $\varepsilon = 0$ learner explores anyway, because every reward is negative and so a zero initial estimate is *optimistic*: every untried action looks better than the disappointments already seen. Initial values are a hidden exploration policy.
- *Maximization bias.* The target $\max_{a'} Q(s', a')$ is the maximum of noisy estimates, and $E[\max] \geq \max E$. Early in learning Q-learning systematically overestimates. **Double Q-learning** keeps two tables, picks the maximizing action with one and evaluates it with the other, which removes the upward bias.
- *Function approximation.* Tables do not scale. Backgammon has about 10^{20} positions and Go about 10^{170}, so practical systems replace the table by a parametrized function $Q_{\mathbf{w}}(s, a)$, from a linear combination of features up to a deep network, and apply the same temporal-difference correction to $\mathbf{w}$. Theorem 53.1 then stops applying, because the update is no longer $\mathcal{T}$ but "apply $\mathcal{T}$, then squeeze the result back into the family of representable functions".

> **? The deadly triad**
>
> Combine three individually harmless ingredients: function approximation, bootstrapping (learning a guess from a guess) and off-policy data. The result can diverge to infinity, even with a *linear* $Q_{\mathbf{w}}$ and exact expected updates. A seven-state example due to Baird (1995) does it. Remove any one ingredient and the divergence disappears. The reason is that the projection onto representable functions is non-expansive only in a norm weighted by the *on-policy* state distribution, and the Bellman operator is a contraction only in the max-norm. When the two norms disagree, nothing forces their composition to contract. The geometry of projections in weighted function spaces is in Ch. 61. A complete theory of when deep Q-learning is stable is an open problem.

53.5 The experiment

First, a picture of what γ means. The weight a reward k steps in the future receives is γ^k. The weights add up to the effective horizon $1/(1-\gamma)$.



Discount weights γ^k for $\gamma = 0.9$ (blue) and 0.99 (red). At the effective horizon $k = 1/(1-\gamma)$ the weight has fallen to about e^{-1}. Change 0.99 to 0.999 and watch the horizon run off the screen; that is why long-horizon problems need so many more samples.

Now learn the motor drive from experience. The script below runs Q-learning for 50,000 steps with no access to the transition probabilities, and tracks the largest error against the Q^* computed in 53.3.

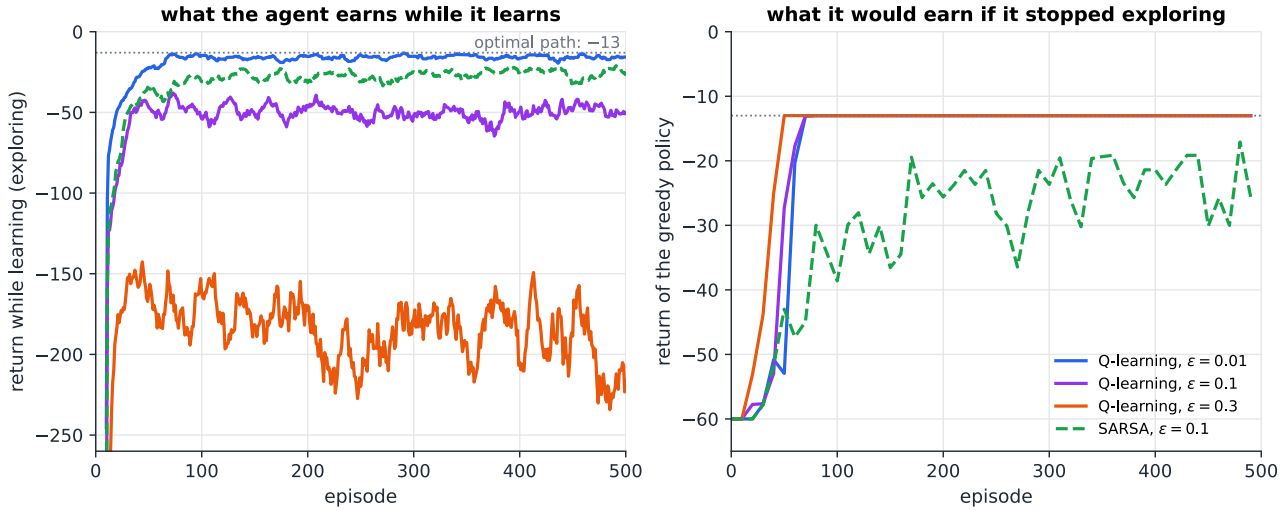
```
import numpy as np
import matplotlib.pyplot as plt

# --- knobs ---------------------------------------------------------------
eps, gamma, steps, seed = 0.1, 0.9, 50_000, 1
Q_star = np.array([[14.95, 15.5], [14.5, -3.39]])  # from value iteration
# -------------------------------------------------------------------------
rng = np.random.default_rng(seed)
C, H, T = 0, 1, 2                       # states; actions 0 = normal, 1 = boost
def env(s, a):
    if s == C and a == 0: return C, 1.0
    if s == C and a == 1: return (C if rng.random() < 0.5 else H), 2.0
    if s == H and a == 0: return (C if rng.random() < 0.5 else H), 1.0
    return (H, 2.0) if rng.random() < 0.2 else (T, -8.0)

Q, N = np.zeros((2, 2)), np.zeros((2, 2))
err, s = [], C
for t in range(steps):
    a = rng.integers(2) if rng.random() < eps else int(np.argmax(Q[s]))
    s2, r = env(s, a)
    N[s, a] += 1
    alpha = 1.0 / N[s, a] ** 0.7         # try 0.1 (constant) or 1/N
    target = r if s2 == T else r + gamma * Q[s2].max()
    Q[s, a] += alpha * (target - Q[s, a])
    s = C if s2 == T else s2             # a trip ends the episode
    err.append(np.abs(Q - Q_star).max())

print("learned Q:\n", Q.round(2), "\nvisits:\n", N.astype(int))
print("greedy policy (0=normal, 1=boost):", Q.argmax(1))
plt.semilogy(err); plt.xlabel("step"); plt.ylabel("max |Q - Q*|")
plt.show()
```

With the default knobs the learned policy is optimal (boost when cool, normal when hot) and the table is $\begin{pmatrix} 14.92 & 15.49 \\ 14.48 & -2.53 \end{pmatrix}$: three entries within 0.03 of Q^*, and one off by 0.86. That one is $Q(H, b)$, the disastrous action, which the agent tried only 1,247 times in 50,000 steps because it already knew enough to avoid it. Q-learning is accurate where accuracy matters for the decision and sloppy elsewhere.

Things to change:

- `eps = 0.0`: the agent never boosts, ends with $Q(C, n) = 10$ and the wrong policy, and its error stays at 15.5 forever.
- `eps = 0.01`: the policy still comes out right, but $Q(C, n)$ stalls near 12.1, because after the first few hundred steps the agent almost never runs normal while cool.
- `eps = 0.5`: every entry is within 0.08 of Q^*, at the price of behaving badly half the time.
- `gamma = 0.99` (and recompute `Q_star` with value iteration from Ch. 52): the same number of steps leaves a much larger error.

Finally, back to the cliff. The next figure shows what ε does there.



Average of 20 runs, $\alpha = 0.5$. Left: the reward actually collected while exploring. Over the last 100 episodes Q-learning earns about -16, -49 and -195 for $\varepsilon = 0.01$, 0.1 and 0.3, and SARSA with $\varepsilon = 0.1$ earns about -26. Right: the reward the current greedy policy would earn. Every Q-learning run has found the optimal -13 path within about 70 episodes; SARSA's greedy policy stays on its safer, longer detour.

The right panel measures what you *know*; the left, what you *earn while finding out*. That is the hook's paradox as a trade-off with numbers on it.

53.6 The challenge

1. **Intuition.** In the cliff world, suppose ε is slowly decreased to zero over many episodes. Explain why SARSA's learned path should then drift down towards the cliff edge and end up identical to Q-learning's. Then explain why, for any fixed $\varepsilon > 0$, it never does. Which of the two programs would you rather have controlling a real machine *during* training, and which *after*?
2. **Derivation.** For a fixed ε-greedy policy π, define the SARSA operator $(\mathcal{T}^{\pi}Q)(s,a) = r(s,a) + \gamma \sum_{s'} P(s' \mid s,a) \sum_{a'} \pi(a' \mid s') Q(s',a')$. Prove that it is a γ-contraction in the max-norm. Then solve $Q = \mathcal{T}^{\pi}Q$ exactly for the motor drive with $\varepsilon = 0.1$ around the optimal policy, as a 4×4 linear system, and compare the two entries of state C with those of Q^{*}.
3. **Implementation.** Change the experiment script into Double Q-learning: keep two tables, on each step update one of them at random, using it to choose $\arg\max_{a'}$ and the *other* to evaluate that action. Over 200 seeds, plot the average of $Q(C,b) - Q^{*}(C,b)$ against the step number for both algorithms during the first 2,000 steps, and measure the maximization bias you have removed.
4. **Open investigation.** In the cliff world, vary the cliff penalty (from -10 to -1000) and ε (from 0.01 to 0.3), and record which row SARSA's greedy path uses. Map the boundary in the (penalty, ε) plane where the safe detour takes over. Is there a simple formula for that boundary, perhaps in terms of the probability of surviving n random steps next to the edge? Does it change with grid width?

53.7 Loose threads

> **? How much exploration is enough?**
>
> ε-greedy explores blindly: it tries a known-terrible action as often as a promising untried one. Smarter schemes add an "optimism bonus" to rarely tried actions, or sample from a posterior over which action is best (Thompson sampling). For the simplest case, a single state with k actions (the multi-armed bandit), the best possible regret grows like $\log T$, and the constant in front is a sum of reciprocal KL divergences between the arms' reward distributions. Why it is exactly KL divergence, and why no algorithm can beat it, comes from how fast an experiment can distinguish two distributions: the ingredients are in Ch. 33, and the Bayesian side of Thompson sampling in Ch. 35.

Two promises from other chapters are partly kept here. Ch. 46 asked what a self-play Go program learns. The answer is a pair of functions in the sense of this chapter: a value network approximating V^{*} of the game from the current player's point of view, and a policy network trained to imitate the moves that the search itself ends up preferring. Tree search guided by both replaces alpha–beta. Why self-play does not simply cycle remains only partly understood, and is a live research question. Ch. 45 asked whether selection is learning. Take a one-state problem with softmax probabilities $\pi_i \propto e^{\theta_i}$ and raise each score at the rate of its reward, $\dot{\theta}_i = r_i$, which is the natural policy gradient for this case. Then $\dot{\pi}_i = \pi_i \left(r_i - \sum_j \pi_j r_j \right)$, which is exactly the replicator equation. Natural selection and natural policy gradient are the same dynamics.

Untouched here: continuous time, where Bellman's equation becomes the Hamilton–Jacobi–Bellman PDE of Ch. 51, and safety. A learner that explores a real system can damage it. Requiring that a learned policy keep its constraints *with high probability, whatever the true model turns out to be* is the subject of Ch. 54.

Further reading

- R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed., MIT Press, 2018. The standard introduction; temporal-difference learning, Q-learning versus SARSA, and policy gradients.
- D. P. Bertsekas and J. N. Tsitsiklis, *Neuro-Dynamic Programming*, Athena Scientific, 1996. The stochastic-approximation proofs of convergence, and the analysis of function approximation.
- Cs. Szepesvári, *Algorithms for Reinforcement Learning*, Morgan & Claypool, 2010. A short, mathematically careful survey.
- T. Lattimore and Cs. Szepesvári, *Bandit Algorithms*, Cambridge University Press, 2020. Exploration done properly, including the regret lower bounds.

Chapter 54 — Robust and Stochastic Optimization

> **Level 2 · Track D — Optimization and Operations Research Builds on:** Ch. 19, Ch. 23, Ch. 47, Ch. 48 · **Leads to:** Ch. 52, Ch. 55 · **Related:** Ch. 38, Ch. 46, Ch. 53, Ch. 75 **Reading time:** ~30 min

54.1 The hook

Here is a small production plan. Two products, x_1 and x_2 units, each worth one unit of profit. Machine A has 8 hours and needs 2 hours per unit of product 1 and 1 hour per unit of product 2. Machine B has 9 hours and needs 1 and 3. Maximize $x_1 + x_2$. The linear program of §23.3 answers in a moment: $x_1 = 3$, $x_2 = 2$, profit 5, both machines fully booked.

Now admit what everyone on the factory floor knows: the "2 hours" was measured, and measurements wobble. Suppose each of the four time coefficients is really its nominal value plus a small random error, independent, with mean zero. How often does the optimal plan overrun a machine?

Think before reading on. The errors are small, so surely the plan works almost always?

It fails **75% of the time**. And the answer does not depend on how small the errors are. Make them 10% of the coefficients or 0.001%, and still three plans in four overrun. The reason takes one line. The optimum sits exactly on both constraint boundaries, $2x_1 + x_2 = 8$ and $x_1 + 3x_2 = 9$. A symmetric error pushes each machine's actual load above or below its capacity with probability $\frac{1}{2}$ each. The chance that *both* land on the safe side is $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.

This is not a quirk of the example. The simplex method always returns a vertex, where as many constraints as there are variables hold with equality. Optimization by its nature drives you to the edge of the feasible set, and the edge is where any uncertainty hurts most. In 2000, Aharon Ben-Tal and Arkadi Nemirovski took the classical NETLIB collection of real-world linear programs, perturbed the uncertain coefficients by just 0.01%, and found that in a substantial fraction of the problems the nominal optimal solutions violated some constraints by far more than the perturbation, in several cases by more than the entire right-hand side. Solutions that were optimal to many decimal places were, in practice, nonsense.

So what should an optimizer do when it does not know the numbers? There are three classical answers, each with its own mathematics, and the surprise of this chapter is that for the simplest kind of uncertainty all three lead to *the same shape of constraint*, differing only in a single number.

54.2 The big idea

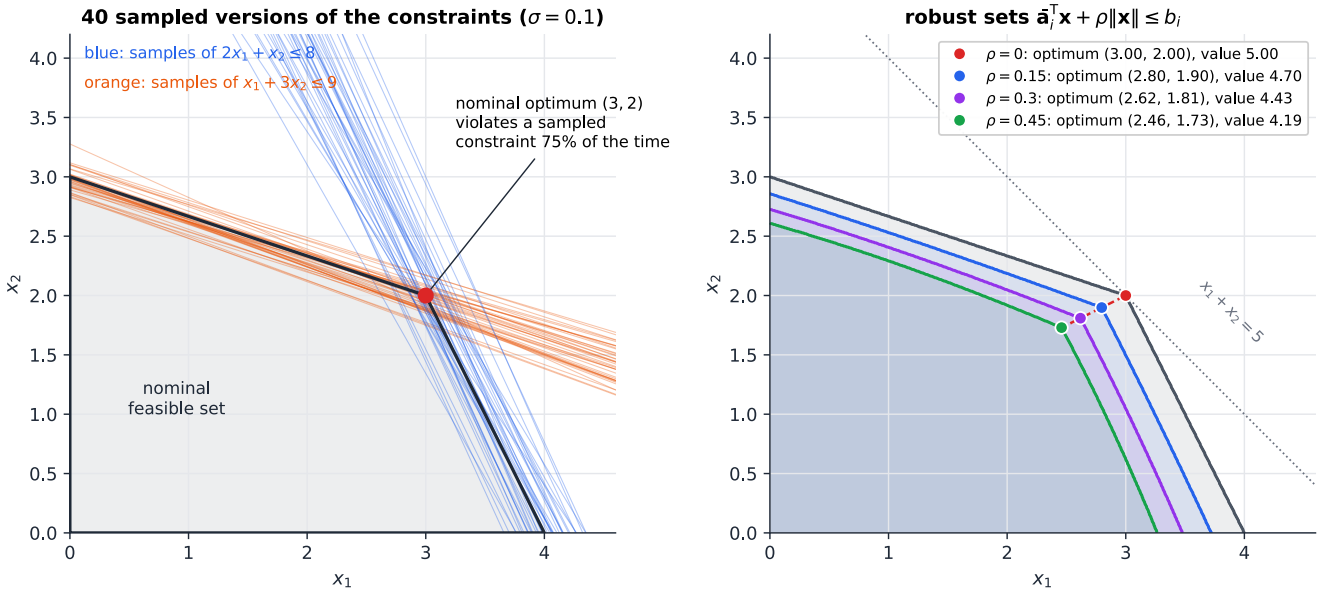
Imagine three engineers given the same uncertain plan.

The **gambler** says: the coefficients are random, so optimize the *expected* outcome, averaging over everything that might happen. This is **stochastic programming**. It is the right philosophy when you play the same game many times and the law of large numbers evens things out: a shop ordering stock every week, a grid operator committing generators every day.

The **actuary** says: I accept a small risk, but I want a guarantee on it. Each constraint must hold with probability at least $1 - \varepsilon$, say 95%. This is a **chance constraint**. It is the language of reliability engineering, of "loss of load probability" in power systems, and of service levels in supply chains.

The **paranoid** says: I don't trust any probability model. Give me a set $\mathcal{U}$ of coefficient values I believe are possible, and I will make sure the plan works for *every one of them*. This is **robust optimization**. Nature is treated as an adversary, exactly as in Wald's maxmin rule of Ch. 46, except that now the scenarios form a continuous set.

Geometrically, the paranoid's feasible set is an intersection: the plan must lie on the safe side of every constraint line that $\mathcal{U}$ allows. Each possible line is a half-plane, and the intersection of infinitely many half-planes is still convex (Ch. 47), but its boundary is no longer straight. Picture the feasible polygon shrinking and rounding its corners as the uncertainty set grows.



Left: forty random versions of each machine constraint. The nominal optimum $(3, 2)$ is outside roughly half of each bundle. Right: the robust feasible sets for ellipsoidal uncertainty of radius ρ. As ρ grows the set shrinks, the corner rounds off, and the best plan retreats inward from the danger, paying for safety with profit.

The cost of that retreat has a name, the **price of robustness**. Too little and the plan fails; too much and you leave money on the table for protection against things that will never happen. The whole craft is in choosing $\mathcal{U}$, or ε, so that the price is worth paying.

54.3 The mathematics

Write one uncertain linear constraint as $\mathbf{a}^{\mathsf{T}}\mathbf{x} \le b$, where $\mathbf{a} \in \mathbb{R}^n$ is not known exactly. Uncertainty in the objective can always be moved into a constraint (minimize t subject to $\mathbf{c}^{\mathsf{T}}\mathbf{x} \le t$), and uncertainty in b can be absorbed into $\mathbf{a}$ by appending a coordinate, so this one form covers everything. Different constraints are handled one at a time.

Robust counterparts

Robust counterpart

Given an uncertainty set $\mathcal{U} \subset \mathbb{R}^n$, the **robust counterpart** of $\mathbf{a}^{\mathsf{T}}\mathbf{x} \le b$ is the constraint

$$\mathbf{a}^{\mathsf{T}}\mathbf{x} \le b \quad \text{for all } \mathbf{a} \in \mathcal{U}, \qquad \text{equivalently} \qquad \max_{\mathbf{a} \in \mathcal{U}} \mathbf{a}^{\mathsf{T}}\mathbf{x} \le b.$$

As written this is infinitely many constraints. The trick that makes robust optimization practical is that, for well-chosen $\mathcal{U}$, the inner maximum has a closed form.

Theorem 54.1 — Ellipsoidal uncertainty gives a second-order cone constraint

Let $\mathcal{U} = \{\bar{\mathbf{a}} + \mathbf{P}\mathbf{u} : \|\mathbf{u}\|_2 \le 1\}$, an ellipsoid with centre $\bar{\mathbf{a}}$ and shape matrix $\mathbf{P}$. Then

$$\mathbf{a}^{\mathsf{T}}\mathbf{x} \le b \ \ \forall \mathbf{a} \in \mathcal{U} \quad \Longleftrightarrow \quad \bar{\mathbf{a}}^{\mathsf{T}}\mathbf{x} + \|\mathbf{P}^{\mathsf{T}}\mathbf{x}\|_2 \le b.$$

Proof. For any $\mathbf{a} = \bar{\mathbf{a}} + \mathbf{P}\mathbf{u}$ in $\mathcal{U}$,

$$\mathbf{a}^{\mathsf{T}}\mathbf{x} = \bar{\mathbf{a}}^{\mathsf{T}}\mathbf{x} + \mathbf{u}^{\mathsf{T}}(\mathbf{P}^{\mathsf{T}}\mathbf{x}) \le \bar{\mathbf{a}}^{\mathsf{T}}\mathbf{x} + \|\mathbf{u}\|_2 \|\mathbf{P}^{\mathsf{T}}\mathbf{x}\|_2 \le \bar{\mathbf{a}}^{\mathsf{T}}\mathbf{x} + \|\mathbf{P}^{\mathsf{T}}\mathbf{x}\|_2,$$

by Cauchy–Schwarz and $\|\mathbf{u}\|_2 \le 1$. So the right-hand condition implies the robust one. Conversely, if $\mathbf{P}^{\mathsf{T}}\mathbf{x} \neq \mathbf{0}$, the choice $\mathbf{u}^{*} = \mathbf{P}^{\mathsf{T}}\mathbf{x} / \|\mathbf{P}^{\mathsf{T}}\mathbf{x}\|_2$ lies in the unit ball and makes both inequalities equalities. That $\mathbf{a}^{*} = \bar{\mathbf{a}} + \mathbf{P}\mathbf{u}^{*}$ is in $\mathcal{U}$, so the robust constraint at $\mathbf{a}^{*}$ is exactly the right-hand condition. (If $\mathbf{P}^{\mathsf{T}}\mathbf{x} = \mathbf{0}$ both sides read $\bar{\mathbf{a}}^{\mathsf{T}}\mathbf{x} \le b$.) $\square$

Read the result as a recipe: **take the nominal constraint and add a safety margin proportional to how strongly $\mathbf{x}$ is exposed to the uncertainty.** The margin $\|\mathbf{P}^{\mathsf{T}}\mathbf{x}\|_2$ is a convex function of $\mathbf{x}$, so the robust problem is still convex: a second-order cone program, solvable by interior-point methods almost as fast as an LP. The proof also hands you the adversary's best move, $\mathbf{a}^{*}$, which will matter in 54.4.

In the factory example with $\mathbf{P} = \rho\mathbf{I}$ (every coefficient may move by up to ρ in Euclidean length), the robust constraints are $2x_1 + x_2 + \rho\|\mathbf{x}\|_2 \le 8$ and $x_1 + 3x_2 + \rho\|\mathbf{x}\|_2 \le 9$. The margin grows with $\|\mathbf{x}\|$, so the boundary curves inward, which is the rounding you saw in the figure. For $\rho = 0.15$ the robust optimum is $(2.80, 1.90)$ with profit 4.70; for $\rho = 0.3$ it is $(2.62, 1.81)$ with profit 4.43.

Other shapes of $\mathcal{U}$ give other norms. If each coefficient may move independently within $\pm\delta_j$ (a box), the same argument with Hölder's inequality gives $\bar{\mathbf{a}}^{\mathsf{T}}\mathbf{x} + \sum_j \delta_j |x_j| \le b$, which is linear after introducing auxiliary variables. In general a ball in one norm produces the *dual* norm in the margin. And for a polyhedral $\mathcal{U} = \{\mathbf{a} : \mathbf{D}\mathbf{a} \le \mathbf{d}\}$, the inner maximum is itself an LP, and replacing it by its dual from Ch. 48 turns the robust constraint into a finite set of linear constraints with new variables. Robust LPs with polyhedral uncertainty stay LPs.

Chance constraints

The actuary does not want *every* $\mathbf{a}$, only most of them. Suppose $\mathbf{a}$ is Gaussian, $\mathbf{a} \sim \mathcal{N}(\bar{\mathbf{a}}, \mathbf{\Sigma})$, and we demand

$$P\left(\mathbf{a}^{\mathsf{T}}\mathbf{x} \le b\right) \ge 1 - \varepsilon.$$

Write Φ for the standard normal CDF (§19.8) and $z_{1-\varepsilon} = \Phi^{-1}(1-\varepsilon)$ for its quantile: $z_{0.95} = 1.645$, $z_{0.99} = 2.326$.

Theorem 54.2 — Gaussian chance constraints are second-order cone constraints

If $\mathbf{a} \sim \mathcal{N}(\bar{\mathbf{a}}, \mathbf{\Sigma})$ then

$$P\left(\mathbf{a}^{\mathsf{T}}\mathbf{x} \le b\right) \ge 1 - \varepsilon \quad \Longleftrightarrow \quad \bar{\mathbf{a}}^{\mathsf{T}}\mathbf{x} + z_{1-\varepsilon}\,\|\mathbf{\Sigma}^{1/2}\mathbf{x}\|_2 \le b,$$

and the set of such $\mathbf{x}$ is convex whenever $\varepsilon \le \tfrac{1}{2}$.

Proof. For fixed $\mathbf{x}$, the load $L = \mathbf{a}^{\mathsf{T}}\mathbf{x}$ is a linear combination of jointly Gaussian variables, hence Gaussian, with mean $\bar{\mathbf{a}}^{\mathsf{T}}\mathbf{x}$ and variance $\mathbf{x}^{\mathsf{T}}\mathbf{\Sigma}\mathbf{x} = \|\mathbf{\Sigma}^{1/2}\mathbf{x}\|_2^2 = s^2$. If $s > 0$, standardize:

$$P(L \le b) = \Phi\left(\frac{b - \bar{\mathbf{a}}^{\mathsf{T}}\mathbf{x}}{s}\right) \ge 1 - \varepsilon \quad \Longleftrightarrow \quad \frac{b - \bar{\mathbf{a}}^{\mathsf{T}}\mathbf{x}}{s} \ge z_{1-\varepsilon},$$

because Φ is strictly increasing. Multiply by $s > 0$ to get the stated form. (If $s = 0$ the load is deterministic and both sides say $\bar{\mathbf{a}}^{\mathsf{T}}\mathbf{x} \le b$, assuming $\varepsilon < \tfrac{1}{2}$.) For convexity: when $\varepsilon \le \tfrac{1}{2}$, $z_{1-\varepsilon} \ge 0$, so the left side is a linear function plus a nonnegative multiple of a norm, which is convex, and a sublevel set of a convex function is convex. $\square$

When $\varepsilon > \tfrac{1}{2}$ the coefficient $z_{1-\varepsilon}$ is negative, the margin becomes a *bonus*, and the feasible set is generally not convex. Asking for less than a coin-flip's reliability breaks the geometry.

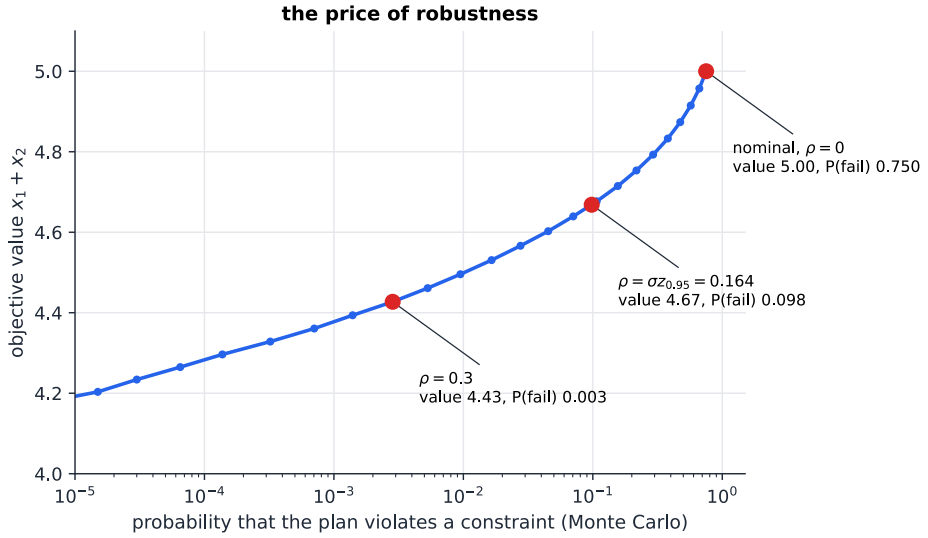
Put the two theorems side by side. The robust constraint with ellipsoid $\mathbf{P} = \rho\,\boldsymbol{\Sigma}^{1/2}$ and the Gaussian chance constraint are *the same inequality* when $\rho = z_{1-\varepsilon}$. A paranoid engineer who chooses an uncertainty set of radius 1.645 standard deviations has, without saying so, bought a 95% guarantee for Gaussian data. Even the distribution-free version fits the pattern: if you know only the mean and covariance of $\mathbf{a}$ and demand the guarantee for *every* distribution with those moments, the constraint becomes the same cone with radius $\sqrt{(1-\varepsilon)/\varepsilon}$, a consequence of Cantelli's one-sided Chebyshev inequality, which is tight.

guarantee	margin radius, $\varepsilon = 0.05$	$\varepsilon = 0.01$
robust over $\{\bar{\mathbf{a}} + \rho\boldsymbol{\Sigma}^{1/2}\mathbf{u} : \lVert\mathbf{u}\rVert \le 1\}$	ρ (your choice)	ρ
Gaussian chance constraint	$z_{0.95} = 1.645$	$z_{0.99} = 2.326$
any distribution with that mean and covariance	$\sqrt{19} = 4.359$	$\sqrt{99} = 9.950$

Ignorance about the shape of the distribution costs a factor of about 2.6 in the margin at 95%, and over 4 at 99%. That is a precise answer to "how much is a probability model worth?"

The factory, made 95% safe

Take every coefficient to have independent noise with $\sigma = 0.1$, so $\boldsymbol{\Sigma}^{1/2} = 0.1\,\mathbf{I}$, and ask each machine to be safe with probability 95%: $\rho = 0.1 \times 1.645 = 0.1645$. The robust optimum is $(2.779,\ 1.889)$ with profit 4.668, a price of 6.6%. A Monte Carlo check with 400,000 samples finds each constraint violated 5.0% of the time, as promised, and *at least one* violated 9.8% of the time, close to $1 - 0.95^2 = 0.0975$. Individual guarantees do not add up to a joint one; for that, split the risk budget, for example $\varepsilon/2$ per constraint.



The price of robustness for the factory, $\sigma = 0.1$. Each dot is the robust optimum for one ρ. Cutting the failure rate from 75% to 10% costs 6.6% of profit; from 10% to 0.3% costs another 5%. Each further factor of ten in reliability costs another 2–4%.

Stochastic programming: optimizing the average

The gambler's problem is $\min_{\mathbf{x}} E_{\xi}\left[F(\mathbf{x}, \xi)\right]$, where ξ is the random data. The classic example has one variable and teaches the central lesson.

A shop must order q units of a perishable product before seeing demand D, a random variable with CDF F. Each unit costs c and sells for $p > c$. Unsold units are worth nothing. Profit is $p\min(q, D) - cq$.

Theorem 54.3 — The newsvendor's critical fractile

If D has a continuous CDF F, the expected profit $\pi(q) = pE[\min(q, D)] - cq$ is concave and is maximized at

$$F(q^*) = \frac{p - c}{p} = \frac{c_u}{c_u + c_o},$$

where $c_u = p - c$ is the cost of ordering one unit too few and $c_o = c$ the cost of one unit too many.

Proof. Write $\min(q, D) = \int_0^q \mathbf{1}\{D > t\}\,dt$ for $D \ge 0$. Taking expectations, $E[\min(q, D)] = \int_0^q P(D > t)\,dt = \int_0^q \big(1 - F(t)\big)\,dt$. Hence

$$\pi'(q) = p\big(1 - F(q)\big) - c.$$

This is non-increasing in q because F is non-decreasing, so π is concave. Its zero is where $1 - F(q^*) = c/p$, that is $F(q^*) = (p - c)/p$. $\square$

The answer is a *quantile*, not an average. With $p = 5$, $c = 2$ and $D \sim \mathcal{N}(100, 20^2)$, the fractile is 0.6 and $q^* = 100 + 20 \times 0.253 = 105.1$: order a little more than average demand, because running out costs more than leftovers. With a thin margin, $c = 4.5$, the fractile drops to 0.1 and $q^* = 100 - 20 \times 1.282 = 74.4$. Ordering the mean demand of 100 then earns an expected profit of about 10.1 instead of 32.4. **Planning for the average scenario is not the same as planning for the average outcome**, and the difference can be most of your profit. The gap has a name in stochastic programming: the *value of the stochastic solution*.

Real problems rarely have a closed-form F. **Sample average approximation (SAA)** replaces the expectation with an average over N observed or simulated scenarios $\xi_1, \dots, \xi_N$ and solves $\min_{\mathbf{x}} \frac{1}{N}\sum_k F(\mathbf{x}, \xi_k)$. For the newsvendor, SAA's answer is simply the empirical 0.6-quantile of past demand. As $N \to \infty$ the SAA optimum converges to the true one, and its error shrinks like $1/\sqrt{N}$, the familiar Monte Carlo rate of Ch. 38.

Hedging against the model itself

Each approach needs something the engineer may not have: a distribution (gambler, actuary) or a believable set (paranoid). **Distributionally robust optimization** blends them. Take a set $\mathcal{P}$ of plausible *distributions*, typically everything close to the empirical distribution of your data, and minimize the worst expected cost over $\mathcal{P}$:

$$\min_{\mathbf{x}} \sup_{Q \in \mathcal{P}} E_{\xi \sim Q}\left[F(\mathbf{x}, \xi)\right].$$

The inner problem is an optimization over probability measures, an infinite-dimensional LP. Its dual, in the spirit of Ch. 48, is a problem over functions, and for the right $\mathcal{P}$ it collapses to something finite. Which $\mathcal{P}$? A ball defined by KL divergence (Ch. 33) contains only distributions on the same support as your data, so it can never imagine a demand larger than any you have seen. A ball measured in the distance of the next chapter has no such blindness.

> **? How far is one distribution from another?**
>
> For a Wasserstein ball of radius r around the empirical distribution and a cost that is L-Lipschitz in ξ, the worst-case expected cost is at most the sample average plus rL: a simple regularization term falls out of an infinite-dimensional sup. Why the Lipschitz constant appears, and what "distance between distributions" means so that this works, is the Kantorovich–Rubinstein duality of Ch. 55.

54.4 The algorithm

Two practical routes lead to a robust or chance-constrained solution.

Route 1: reformulate and call a cone solver. Theorems 54.1 and 54.2 turn each uncertain constraint into a second-order cone constraint. Hand the result to any interior-point SOCP solver. Cost is polynomial, typically a few dozen Newton steps, each dominated by a linear solve. This is how almost all production robust optimization is done.

Route 2: let the adversary propose cuts. When no closed form exists, or you want to see the mechanism, alternate between a planner and an adversary. The planner solves the problem against a *finite* list of scenarios. The adversary looks at the plan and finds the scenario that hurts it most. Add that scenario to the list and repeat.

```
scenarios ← { a_nominal }                    for each uncertain constraint
repeat:
    x ← solve LP: max cᵀx  s.t.  aᵀx ≤ b for every a in scenarios
    for each constraint:
        a* ← argmax over a in U of aᵀx        (ellipsoid: a* = ā + P Pᵀx / ‖Pᵀx‖)
        if a*ᵀx > b + tol: add a* to scenarios
until no constraint was violated
```

Each round adds one half-space that cuts off the current plan, so this is a **cutting-plane method**, the outside description of a convex set from Ch. 47 built up one face at a time. It converges to the robust optimum, though only approximately in finitely many steps when $\mathcal{U}$ is curved: an ellipsoid has infinitely many faces to discover. Its real strength is generality. The adversary step can be any oracle, even a simulation, as long as it can find a violating scenario when one exists.

A third route needs no model at all. The **scenario approach** samples N independent realizations of the uncertain data, imposes the constraint for every one of them, and solves the resulting ordinary LP. The theorem of Calafiore and Campi (2006), sharpened by Campi and Garatti (2008), says: for a convex problem with d decision variables, the probability that the scenario solution violates the constraint on a fresh sample with probability more than ε is at most

$$\sum_{i=0}^{d-1} \binom{N}{i} \varepsilon^i (1-\varepsilon)^{N-i},$$

whatever the distribution of the data. For the factory ($d = 2$), $\varepsilon = 0.05$ and a confidence level of $1 - 10^{-6}$, this needs $N = 326$ samples. The simpler sufficient condition $N \geq \frac{2}{\varepsilon}\left(\ln\frac{1}{\beta} + d\right)$ gives 633.

Failure modes.

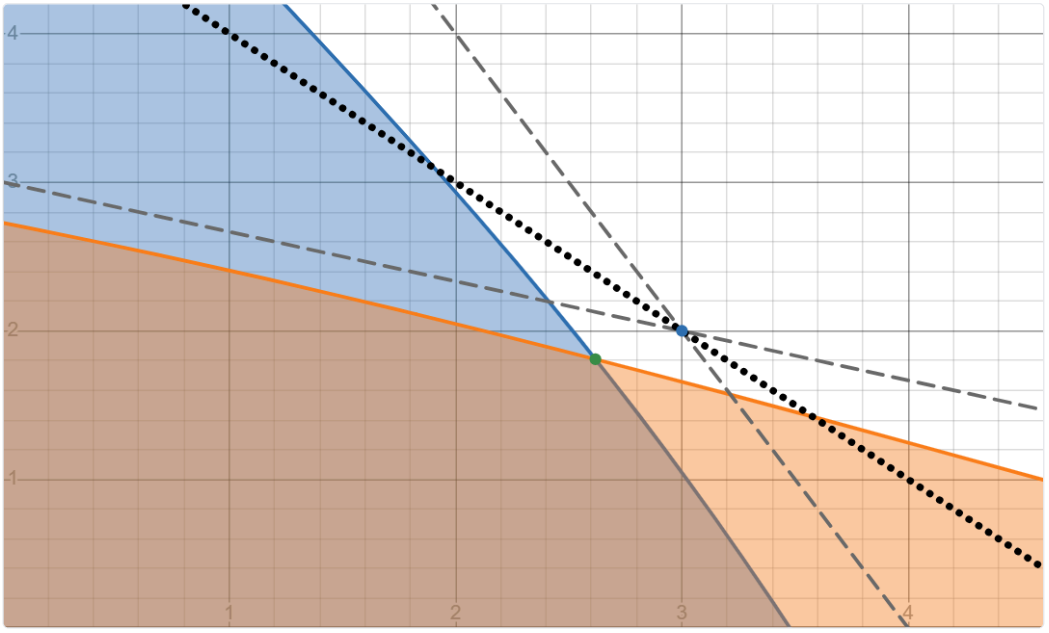
- *Over-conservatism.* A box uncertainty set lets *every* coefficient reach its worst value simultaneously, which with fifty uncertain coefficients is absurdly unlikely. Bertsimas and Sim's **budget of uncertainty** allows at most Γ coefficients per constraint to deviate. Then the violation probability decays like $e^{-\Gamma^2/(2n)}$ for n independent symmetric coefficients, so Γ of order $\sqrt{n}$ buys high reliability at a fraction of the box's price.
- *Joint guarantees.* Separate 95% guarantees on two constraints give about 90% jointly, as the example showed. Bonferroni splitting is safe but conservative; exact joint chance constraints are usually non-convex.
- *Non-convex chance constraints.* For a general (say discrete) distribution, $\{\mathbf{x} : P(\mathbf{a}^\mathsf{T}\mathbf{x} \leq b) \geq 1 - \varepsilon\}$ can be a union of disjoint pieces, and optimizing over it is NP-hard. The standard convex safe approximation replaces the probability by the **conditional value at risk**, the expected violation in the worst ε fraction of cases.
- *Bad uncertainty sets.* Robust optimization is only as good as $\mathcal{U}$. Too small and the guarantee is fiction; too large and the solution is useless or the problem becomes infeasible.

? A guarantee that knows nothing about the distribution

The scenario bound depends on the number of decision variables d and on nothing else: not on the distribution, not on the number of constraints, not on the geometry. How can a result about random constraints be so indifferent to their randomness? The key is that a convex program in d variables has at most d **support constraints**, scenarios whose removal would change the solution, a cousin of Helly's theorem on intersecting convex sets. The combinatorics of support constraints, and the LP-type problems where they rule, are in Ch. 75.

54.5 The experiment

First, see the robust feasible set shrink. The two shaded regions are the robust versions of the factory constraints with $\rho = 0.3$; the dashed lines are the nominal ones, and the dotted line is the objective level $x_1 + x_2 = 5$ through the nominal optimum.



Robust factory constraints with $\rho = 0.3$. The overlap of the two shadings is the robust feasible set; its best corner has moved from $(3, 2)$ to $(2.618, 1.809)$. Change both 0.3s to 0.1645 (95% per constraint for $\sigma = 0.1$ Gaussian noise) or to 1, and watch the set collapse towards the origin.

Now the stochastic side. The script below plays the newsvendor many times. Each trial sees only N past demands and orders either the SAA solution (the empirical quantile at the critical fractile) or the "plan for the average" quantity (the sample mean). A million fresh demands stand in for the true distribution to score each order.

```
import numpy as np
import matplotlib.pyplot as plt

# --- knobs ---------------------------------------------------------------
price, cost = 5.0, 2.0            # sell at 5, buy at 2, unsold stock is worth 0
N, trials, seed = 20, 2000, 0     # data points per trial, repeated trials
def demand(n):                    # try rng.lognormal(4.55, 0.4, n) for a skewed world
    return np.maximum(rng.normal(100, 20, n), 0)
# -------------------------------------------------------------------------
rng = np.random.default_rng(seed)
frac = (price - cost) / price                 # critical fractile c_u / (c_u + c_o)
big = demand(1_000_000)                       # stands in for "the true distribution"
profit = lambda q: (price * np.minimum(q, big) - cost * q).mean()
q_true = np.quantile(big, frac)

q_saa = np.array([np.quantile(demand(N), frac) for _ in range(trials)])
q_mean = np.array([demand(N).mean() for _ in range(trials)])   # "plan for the average"

print(f"critical fractile {frac:.2f}, true optimal order {q_true:.1f}, profit {profit(q_true):.2f}")
for name, q in [("SAA", q_saa), ("mean", q_mean)]:
    lost = profit(q_true) - np.array([profit(x) for x in q[:300]])
    print(f"{name:5s}: order {q.mean():6.1f} +- {q.std():4.1f}, expected profit lost {lost.mean():.2f}")

plt.hist(q_saa, 60, alpha=0.6, label="SAA order (empirical quantile)")
plt.hist(q_mean, 60, alpha=0.6, label="order = sample mean")
plt.axvline(q_true, color="k", ls="--", label="true optimum")
plt.xlabel("order quantity"); plt.legend(); plt.show()
```

With the default knobs the true optimum is to order 105.1 for an expected profit of 261.4. SAA with only 20 data points orders 104.8 ± 5.5 and loses 1.48 of profit on average; ordering the sample mean loses 1.99. Things to change:

- `N = 200` and `N = 2000`: SAA's loss falls to 0.14 and then 0.02, roughly by a factor of ten for each factor of ten in data, because the profit curve is flat at its peak and the loss goes like the *square* of an error that shrinks like $1/\sqrt{N}$. The mean-based order stays wrong, losing about 1.3 however much data it has: more data cannot fix the wrong question.
- `cost = 4.5`: a thin margin, fractile 0.1. The optimal order drops to 74.3 with profit 32.4. SAA with 20 points loses about 1.2; ordering the mean loses about 24, three quarters of the attainable profit.
- Switch `demand` to the lognormal line. Which estimator suffers more from the skew?
- Make the shop paranoid: order the *worst-case* optimal quantity over all demands between the smallest and largest of the N observations. What does maxmin choose, and what does it cost on average?

54.6 The challenge

1. **Intuition.** In the hook, the nominal optimum failed 75% of the time because two constraints were active. A three-variable LP typically has three active constraints at its optimum. What failure probability do you predict, and why must it be at least $\frac{1}{2}$ for *any* LP whose optimum lies on at least one uncertain constraint with symmetric noise? What property of the optimal solution would make it immune?
2. **Derivation.** Derive the robust counterpart of $\mathbf{a}^{\mathsf{T}}\mathbf{x} \le b$ for the box $|a_j - \bar{a}_j| \le \delta_j$, and rewrite it as a set of linear inequalities using auxiliary variables $t_j \ge |x_j|$. Then generalize: for $\mathcal{U} = \{\bar{\mathbf{a}} + \mathbf{u} : \|\mathbf{u}\| \le \rho\}$ in an arbitrary norm, show that the margin is $\rho\|\mathbf{x}\|_*$, where $\|\cdot\|_*$ is the dual norm.
3. **Implementation.** Implement the scenario approach for the factory with `scipy.optimize.linprog`: draw $N = 326$ perturbed copies of each constraint, solve, then estimate the true violation probability of the solution with 10^5 fresh samples. Repeat 1,000 times and plot the histogram of violation probabilities against $\varepsilon = 0.05$. How conservative is the Campi–Garatti bound in this example?
4. **Open investigation.** For a *joint* chance constraint on both factory machines, $P(\text{both safe}) \ge 0.95$, compare three strategies: Bonferroni (2.5% per machine), an optimized split of the risk budget between the machines, and the scenario approach with the same reliability. Now correlate the noise between coefficients. How does the best split depend on the correlation? Is there a rule for allocating risk among m constraints that is near-optimal in general?

54.7 Loose threads

The chapter treated decisions made once, here and now, before the uncertainty is revealed. Real plans unfold.

> **? Decisions that wait for information**
>
> A grid operator commits generators today, but adjusts their output tomorrow after seeing the wind. The second decision is not a number but a *function* of what will be observed, and optimizing over functions makes robust problems intractable in general. A popular escape restricts the recourse to be an affine function of the uncertainty, which turns the problem back into a cone program, at an unknown cost in optimality. When is that loss small, and what is the exact answer? The exact answer is a dynamic program over information states, Bellman's recursion of Ch. 52, with all the curse of dimensionality that implies. How much the affine shortcut gives up in general is still an open question.

The other threads lead outward. The Wasserstein ball of 54.3 is defined, and its duality proved, in Ch. 55. The reinforcement-learning agent of Ch. 53 that must explore without breaking anything faces a robust MDP, in which the transition probabilities themselves lie in an uncertainty set. Robust control, where the uncertainty is a disturbance signal of bounded energy, turns the worst case into an H_∞ norm. And the stochastic processes of Ch. 36 supply the scenario generators that SAA needs.

Further reading

- A. Ben-Tal, L. El Ghaoui and A. Nemirovski, *Robust Optimization*, Princeton University Press, 2009. The theory of robust counterparts, including the NETLIB study.
- D. Bertsimas and M. Sim, "The Price of Robustness", *Operations Research* 52(1), 2004. Budgets of uncertainty and their probabilistic guarantees.
- J. R. Birge and F. Louveaux, *Introduction to Stochastic Programming*, 2nd ed., Springer, 2011. Two-stage problems, recourse, and the value of the stochastic solution.
- A. Shapiro, D. Dentcheva and A. Ruszczyński, *Lectures on Stochastic Programming*, 3rd ed., SIAM, 2021. Sample average approximation, chance constraints and risk measures, rigorously.

Chapter 55 — Optimal Transport and Wasserstein Geometry

> **Level 2 · Track D — Optimization and Operations Research Builds on:** Ch. 19, Ch. 23, Ch. 33, Ch. 48 · **Leads to:** Ch. 59, Ch. 89, Ch. 97 · **Related:** Ch. 54, Ch. 61, Ch. 76 **Reading time:** ~30 min

55.1 The hook

Play a game of hot-and-cold. I hide a coin somewhere on a line, at position 0. You stand at position θ and want to find it, and after each step I tell you how far off you are. The game is easy because my answers change as you move: "warmer" tells you which way to go.

Now play it with a very strict referee. Instead of measuring distance, he compares *where you are* with *where the coin is* and reports how much the two pictures overlap. You are a single point at θ; the coin is a single point at 0. Unless $\theta = 0$ exactly, the overlap is zero. Move from $\theta = 5$ to $\theta = 0.01$ and his answer does not change at all. You are told you are wrong, but never *how* wrong, so you have no idea which way to step.

That strict referee is not a straw man. He is the Kullback–Leibler divergence of Ch. 33, and its symmetrized cousin, the Jensen–Shannon divergence, and the total-variation distance. All of them compare two probability distributions *point by point*, density against density. Put the coin's distribution δ_0 against yours, δ_θ, and for every $\theta \neq 0$ the KL divergence is $+\infty$, the Jensen–Shannon divergence is exactly $\ln 2$, and the total variation is exactly 1. These numbers have no slope. A learning algorithm that adjusts θ by gradient descent on any of them stands still.

This happened in practice. Around 2014–2017, generative adversarial networks were trained to make realistic images by, in effect, minimizing a Jensen–Shannon divergence between generated and real images. Real images occupy a thin, low-dimensional sliver of the space of all pixel arrays, and so, early in training, do the generator's. Two thin slivers in a huge space almost never overlap. Training stalled, or oscillated, or collapsed onto a few outputs. In 2017 Arjovsky, Chintala and Bottou proposed replacing the strict referee with one who *measures distance*. Their fix was a quantity first studied by a French engineer in 1781 who wanted to move dirt.

How do you measure the distance between two probability distributions so that it knows how far apart they are, not merely whether they overlap? And why should the answer be an optimization problem?

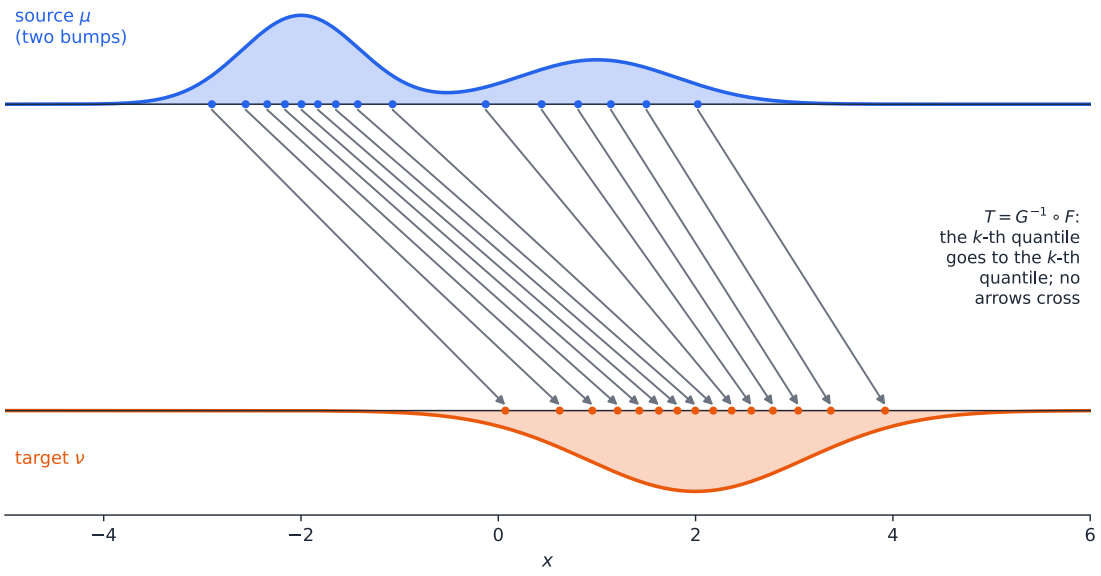
55.2 The big idea

Think of each distribution as a pile of sand of total mass 1. The first pile is where the probability is now; the second is where you want it. Moving a grain from x to y costs $c(x, y)$, say the distance $|x - y|$ or its square. The **optimal transport cost** is the least total effort needed to reshape the first pile into the second.

Gaspard Monge posed this in 1781 for building fortifications: the *déblais* (earth dug out of a trench) had to be carried to the *remblais* (embankments) as cheaply as possible. In 1942 Leonid Kantorovich, working on Soviet production planning, recast it as a linear program, the kind of problem he had invented a few years earlier, complete with its dual prices. (He shared the 1975 economics Nobel prize for that work.)

The cost of transport is a distance that "knows geometry". Move a pile a distance d without changing its shape and the transport cost is exactly d (with $c = |x - y|$): every grain travels d. Back to the coin: transporting δ_θ to δ_0 costs $|\theta|$, which shrinks smoothly as you approach and has slope ± 1 pointing the right way. Hot-and-cold works again.

In one dimension the optimal plan is something you would guess: **never let paths cross.** Sort both piles from left to right. Send the leftmost grain of the first pile to the leftmost position of the second, the median to the median, the 90th percentile to the 90th percentile.



The optimal transport map in one dimension sends each quantile of the source (blue, two bumps) to the same quantile of the target (orange, one bump). Grains from the left bump travel far; grains from the right bump travel little. No two arrows cross: crossing would always waste effort.

Notice what this picture is *not*. The point-by-point comparison would say these two densities differ a lot on the left and a little on the right. Transport says something more useful: *how* to turn one into the other, and how much work that takes. Distances between distributions become distances between *places*.

Two further consequences make optimal transport more than a better ruler. First, it gives a way to *interpolate* between distributions: let each grain travel only halfway along its arrow. The halfway pile between two narrow bumps is a narrow bump in the middle. Averaging the densities would instead give two half-height bumps, a picture of no physical situation at all. Second, the set of all distributions with this distance has a geometry of its own, with straight lines (those half-way piles), angles and curvature. Physics turns out to live there.

55.3 The mathematics

Two formulations

Let μ and ν be probability distributions on $\mathbb{R}^d$ and $c(x,y)\ge 0$ a cost.

Monge and Kantorovich problems

Monge: find a map T that pushes μ to ν (if $X\sim\mu$ then $T(X)\sim\nu$) and minimizes $E_{X\sim\mu}\,[c(X,T(X))]$.

Kantorovich: find a joint distribution π of a pair (X,Y), called a **coupling** or **transport plan**, whose marginals are $X\sim\mu$ and $Y\sim\nu$, and which minimizes $E_{(X,Y)\sim\pi}\,[c(X,Y)]$. The set of couplings is written $\Pi(\mu,\nu)$.

A map must send each grain to one place. A plan can split a grain. That matters: a point mass at 0 cannot be mapped onto two half-masses at ± 1 by any map, but a plan can split it. Kantorovich's relaxation always has a solution under mild conditions, and when an optimal map exists it is also an optimal plan.

For discrete distributions $\mu=\sum_i a_i\delta_{x_i}$ and $\nu=\sum_j b_j\delta_{y_j}$, a plan is a nonnegative matrix $\mathbf{P}$ with row sums $\mathbf{a}$ and column sums $\mathbf{b}$, and the problem is the linear program

$$\min_{\mathbf{P}\ge 0}\ \sum_{i,j} C_{ij}P_{ij}\quad\text{s.t.}\quad \sum_j P_{ij}=a_i,\quad \sum_i P_{ij}=b_j,\qquad C_{ij}=c(x_i,y_j).$$

This is the transportation problem of operations research (§23.3), with nm variables and $n+m$ constraints.

Wasserstein distance

For $p\ge 1$ and $c(x,y)=\|x-y\|^p$,

$$W_p(\mu,\nu)=\left(\min_{\pi\in\Pi(\mu,\nu)} E_\pi\left[\|X-Y\|^p\right]\right)^{1/p}.$$

W_p is a genuine metric on distributions with finite p-th moments: symmetric, zero only when $\mu=\nu$, and obeying the triangle inequality (glue an optimal plan from μ to ν to one from ν to ρ along their common marginal). Unlike KL, it is finite even when the supports are disjoint.

The one-dimensional solution

Theorem 55.1 — Monotone matching is optimal on the line

Let $x_1\le\cdots\le x_n$ and $y_1\le\cdots\le y_n$ be points on $\mathbb{R}$, each carrying mass $1/n$, and let $c(x,y)=h(x-y)$ with h convex. Among all one-to-one assignments σ, the total cost $\sum_i h(x_i-y_{\sigma(i)})$ is minimized by the identity, $\sigma(i)=i$.

Proof. First the key inequality, for two points and two targets. Take $x_1\le x_2$ and $y_1\le y_2$, and compare the uncrossed pairing with the crossed one. Set

$$\alpha=x_1-y_2,\qquad \beta=x_1-y_1,\qquad \gamma=x_2-y_2,\qquad \delta=x_2-y_1.$$

Then $\alpha\le\beta\le\delta$, $\alpha\le\gamma\le\delta$ and $\beta+\gamma=\alpha+\delta$. So $\beta=(1-t)\alpha+t\delta$ for some $t\in[0,1]$, and then $\gamma=\alpha+\delta-\beta=t\alpha+(1-t)\delta$. By convexity,

$$h(\beta)+h(\gamma)\le\big[(1-t)h(\alpha)+th(\delta)\big]+\big[th(\alpha)+(1-t)h(\delta)\big]=h(\alpha)+h(\delta).$$

The left side is the cost of the uncrossed pairing $(x_1\to y_1,\ x_2\to y_2)$, the right side that of the crossed one. Uncrossing never costs more.

Now take any assignment σ other than the identity. It has an *inversion*: indices $i<k$ with $\sigma(i)>\sigma(k)$, a pair of crossing arrows. Swap their targets. By the inequality the total cost does not increase, and the number of inversions strictly decreases. After finitely many swaps we reach the identity, at a cost no greater than that of σ. $\square$

Two remarks extend this. The LP above, with uniform weights and $n=m$, always has an optimal solution that is a permutation matrix: its feasible set is the Birkhoff polytope, whose vertices are exactly the permutation matrices, and an LP attains its optimum at a vertex. So restricting to assignments lost nothing. And for general distributions on $\mathbb{R}$ with CDFs F and G, the same sorting becomes "match quantiles": the optimal map is $T=G^{-1}\circ F$ and

$$W_p^p(\mu,\nu)=\int_0^1 \left|F^{-1}(u)-G^{-1}(u)\right|^p\,du.$$

In one dimension, optimal transport costs a sort: $O(n\log n)$.

Theorem 55.2 — Wasserstein distance between Gaussians on the line

$$W_2^2\left(\mathcal{N}(m_1,\sigma_1^2),\,\mathcal{N}(m_2,\sigma_2^2)\right)=(m_1-m_2)^2+(\sigma_1-\sigma_2)^2.$$

Proof. The quantile function of $\mathcal{N}(m,\sigma^2)$ is $m+\sigma\,\Phi^{-1}(u)$. If U is uniform on $(0,1)$ then $Z=\Phi^{-1}(U)$ is standard normal, so by the quantile formula

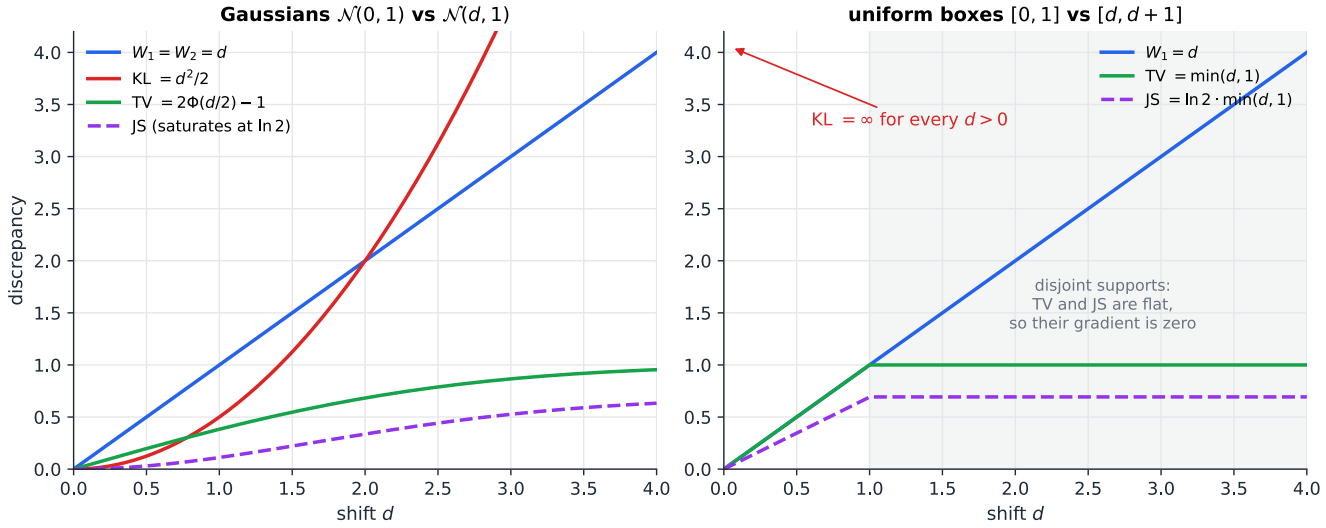
$$W_2^2 = E\left[\big((m_1-m_2)+(\sigma_1-\sigma_2)Z\big)^2\right] = (m_1-m_2)^2 + 2(m_1-m_2)(\sigma_1-\sigma_2)E[Z] + (\sigma_1-\sigma_2)^2E[Z^2],$$

and $E[Z]=0$, $E[Z^2]=1$. $\square$

Compare the Kullback–Leibler divergence between the same two Gaussians,

$$D\left(\mathcal{N}(m_1,\sigma_1^2)\,\|\,\mathcal{N}(m_2,\sigma_2^2)\right) = \ln\frac{\sigma_2}{\sigma_1} + \frac{\sigma_1^2+(m_1-m_2)^2}{2\sigma_2^2} - \frac{1}{2}.$$

Three differences jump out. W_2 is symmetric and KL is not. W_2 measures a shift in the units of x; KL measures it in units of the *width*, $(m_1-m_2)^2/(2\sigma^2)$ for equal widths, so as both bumps become narrow KL explodes while W_2 stays put. And as $\sigma\to 0$, W_2 becomes the plain distance between the means: the coin game. For $\mathcal{N}(0,1)$ against $\mathcal{N}(1,4)$, $W_2=\sqrt{2}=1.414$ while the KL divergence is 0.443 nats.



Four ways to compare a distribution with its shifted copy. With Gaussians (left) every measure increases with the shift, but TV and Jensen–Shannon flatten out quickly. With boxes (right) the supports stop overlapping at $d=1$; from then on TV and JS are constant, KL is infinite throughout, and only the Wasserstein distance still says how far there is to go.

Duality: the price of moving dirt

Ch. 48 promised a dual for this LP. Attach a price φ_i to each source constraint and ψ_j to each target constraint. Think of a haulage company that offers to take over the whole job. It charges $\varphi(x)$ per unit for picking sand up at x and $\psi(y)$ per unit for dropping it off at y, so your bill is $\sum_i a_i\varphi_i + \sum_j b_j\psi_j$, and it wants that bill as large as possible. You accept only prices that are never dearer than doing the job yourself, on every possible route:

$$\varphi(x)+\psi(y)\le c(x,y) \qquad \text{for all } x,y.$$

Theorem 55.3 — Kantorovich duality

$$\min_{\pi\in\Pi(\mu,\nu)} E_\pi[c(X,Y)] = \max_{\varphi(x)+\psi(y)\le c(x,y)} \Big(E_\mu[\varphi(X)] + E_\nu[\psi(Y)]\Big).$$

Proof of $\ge$ (weak duality). For any coupling π and any admissible prices, $(X,Y)\sim\pi$ has $X\sim\mu$ and $Y\sim\nu$, so

$$E_\mu[\varphi(X)] + E_\nu[\psi(Y)] = E_\pi\left[\varphi(X)+\psi(Y)\right] \le E_\pi\left[c(X,Y)\right].$$

Take the max over prices on the left and the min over plans on the right. $\square$

For discrete distributions equality is LP strong duality (Ch. 48); for general ones it needs a compactness argument we do not give. The optimal prices are not arbitrary. Given φ, the best admissible ψ is the largest one allowed,

$$\psi(y) = \varphi^c(y) = \inf_x \big(c(x,y)-\varphi(x)\big),$$

the c**-transform** of φ: the cheapest way to deliver to y, net of what you were paid at the pickup. For $c(x,y)=-x^{\mathsf T}y$ it is, up to signs, the convex conjugate of convex analysis. This is the "strange conjugate operation" Ch. 48 mentioned.

For the cost $c(x,y)=|x-y|$ the conjugation simplifies dramatically. Admissible prices can be taken to satisfy $\psi=-\varphi$ with φ **1-Lipschitz**, $|\varphi(x)-\varphi(y)|\le|x-y|$, giving the **Kantorovich–Rubinstein formula**

$$W_1(\mu,\nu) = \sup_{\mathrm{Lip}(f)\le 1} \Big(E_\mu[f(X)] - E_\nu[f(Y)]\Big).$$

The easy half is again one line: for any coupling and any 1-Lipschitz f, $E_\mu f - E_\nu f = E_\pi[f(X)-f(Y)] \le E_\pi|X-Y|$. This is the formula behind the Wasserstein GAN of 55.1. A second network, the "critic", searches over (approximately) 1-Lipschitz functions f for the one that best tells real data from generated data. The value it achieves estimates W_1, and its gradient tells the generator which way to move.

The same easy half settles a promise of Ch. 54. Let $\hat{P}$ be the empirical distribution of your data and ℓ an L-Lipschitz loss. If $W_1(Q, \hat{P}) \le r$, apply the formula to $f = \ell / L$:

$$E_Q[\ell] \le E_{\hat{P}}[\ell] + L\, W_1(Q, \hat{P}) \le E_{\hat{P}}[\ell] + Lr.$$

Robustness against every distribution in a Wasserstein ball costs at most a Lipschitz penalty, which is why "Wasserstein distributionally robust" learning so often reduces to ordinary learning with a regularizer. For many losses the bound is attained.

> **? Why is the optimal map a gradient?**
>
> In $\mathbb{R}^d$ with cost $\|x - y\|^2$ there is no sorting, yet Brenier's theorem (1991) says that, if μ has a density, the optimal plan is a map, and that map is the *gradient of a convex function*, $T = \nabla u$. In one dimension a gradient of a convex function is just an increasing function, so this is the "no crossing" rule in disguise. Pushing a density f forward to g by ∇u then forces $g(\nabla u(x))\, \det \nabla^2 u(x) = f(x)$, the **Monge–Ampère equation**, one of the most important fully nonlinear PDEs. Why it is so hard, and what regularity its solutions have, is a story for Ch. 89.

55.4 The algorithm

In one dimension, sort and match. In higher dimensions the transport LP has nm variables; with $n = m = 10^4$ points that is 10^8 unknowns, and exact network-flow solvers cost roughly $O(n^3)$. For years this made optimal transport a beautiful theory with few applications. What changed it, around 2013, was a small perturbation.

Entropic regularization. Add a penalty that rewards spreading the plan out:

$$\min_{\mathbf{P} \ge 0} \sum_{i,j} C_{ij} P_{ij} + \varepsilon \sum_{i,j} P_{ij} \left(\ln P_{ij} - 1\right) \quad \text{s.t.} \quad \mathbf{P}\mathbf{1} = \mathbf{a}, \quad \mathbf{P}^{\mathsf{T}}\mathbf{1} = \mathbf{b}.$$

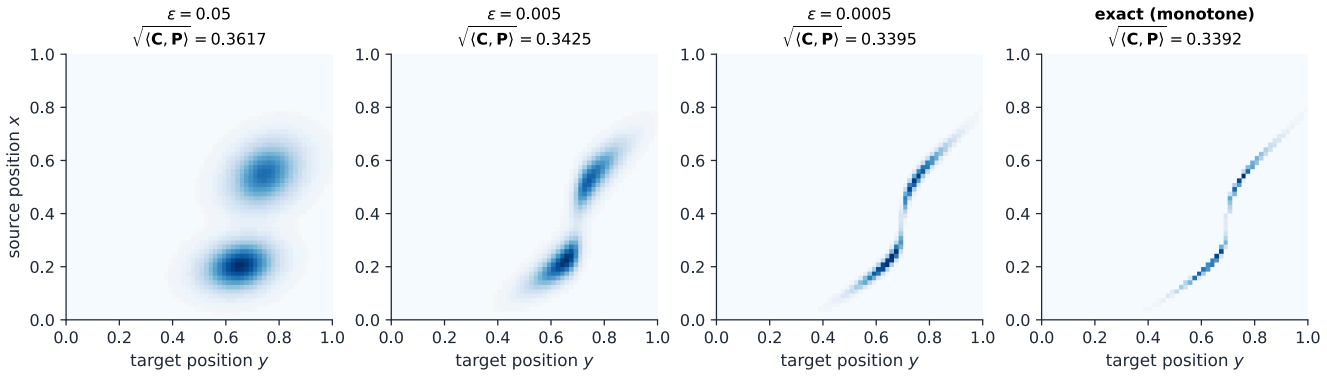
The penalty is strictly convex, so the solution is unique. Attach multipliers f_i and g_j to the two sets of constraints and set the derivative of the Lagrangian with respect to P_{ij} to zero:

$$C_{ij} + \varepsilon \ln P_{ij} - f_i - g_j = 0 \quad \Longrightarrow \quad P_{ij} = \underbrace{e^{f_i/\varepsilon}}_{u_i}\, \underbrace{e^{-C_{ij}/\varepsilon}}_{K_{ij}}\, \underbrace{e^{g_j/\varepsilon}}_{v_j}.$$

So the optimal plan has the form $\mathbf{P} = \operatorname{diag}(\mathbf{u})\, \mathbf{K}\, \operatorname{diag}(\mathbf{v})$ for a fixed **Gibbs kernel** $\mathbf{K} = e^{-\mathbf{C}/\varepsilon}$ (elementwise), and only the $n + m$ scaling factors are unknown. The row-sum condition reads $\mathbf{u} \odot (\mathbf{K}\mathbf{v}) = \mathbf{a}$, the column-sum condition $\mathbf{v} \odot (\mathbf{K}^{\mathsf{T}}\mathbf{u}) = \mathbf{b}$. Solving each one for its own unknown, alternately, gives **Sinkhorn's algorithm**:

```
K ← exp(−C / ε);   v ← 1
repeat until the marginals match:
    u ← a ./ (K v)          fix the row sums
    v ← b ./ (Kᵀ u)         fix the column sums
P ← diag(u) K diag(v)
```

Each sweep is two matrix–vector products, $O(nm)$ operations, perfectly suited to GPUs. The number of sweeps grows as ε shrinks, roughly like $1/\varepsilon$.



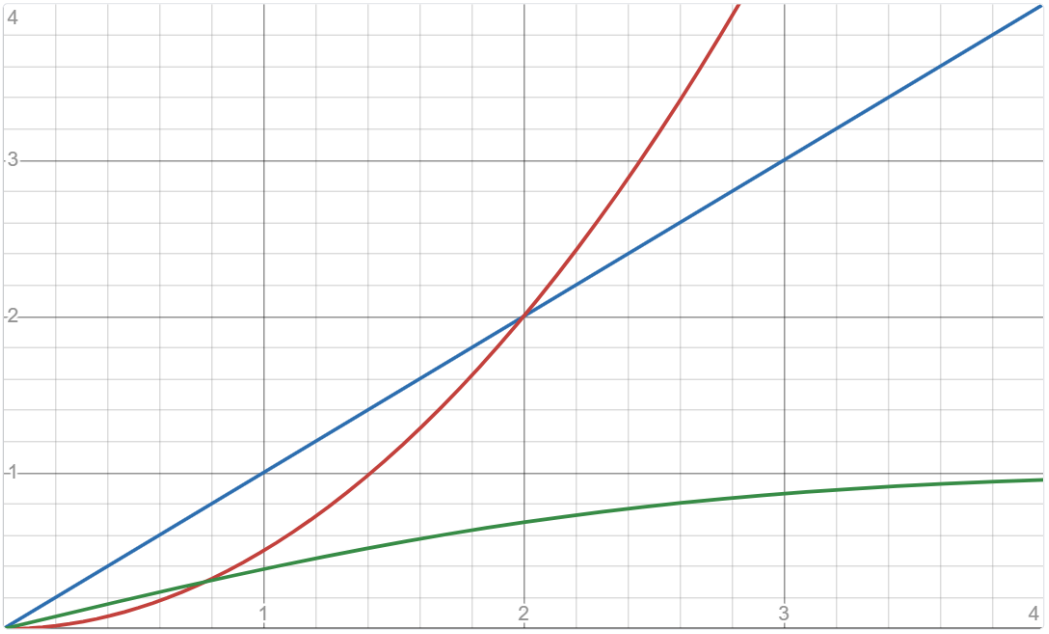
Entropic plans for the source and target of the first figure (now on $[0, 1]$). With $\varepsilon = 0.05$ the plan is a blur and its cost $\sqrt{\langle \mathbf{C}, \mathbf{P} \rangle} = 0.3617$ overestimates $W_2 = 0.3392$ by 7%. At $\varepsilon = 0.0005$ it is within 0.1% and has sharpened onto the monotone map: an increasing curve, steep where the source's gap between its two bumps is squeezed into a narrow stretch of the target.

Failure modes.

- *Bias.* For $\varepsilon > 0$ the plan is blurred and its cost overestimates the true one. A corrected quantity, the **Sinkhorn divergence** $S_\varepsilon(\mu, \nu) = \mathrm{OT}_\varepsilon(\mu, \nu) - \frac{1}{2}\mathrm{OT}_\varepsilon(\mu, \mu) - \frac{1}{2}\mathrm{OT}_\varepsilon(\nu, \nu)$, removes the self-blur and vanishes when $\mu = \nu$.
- *Underflow.* $e^{-C/\varepsilon}$ is below the smallest double ($\approx 10^{-308}$) once $C/\varepsilon > 708$. Then $\mathbf{K}\mathbf{v}$ has zeros, the division produces infinities, and the iteration returns `nan`. The cure is to iterate on $f = \varepsilon \ln u$ and $g = \varepsilon \ln v$ with a log-sum-exp, which costs a little speed and never overflows.
- *Slow convergence for small ε*, exactly when the answer is most accurate. Practical codes start with a large ε and decrease it, warm-starting each stage.
- *High dimension.* Estimating W_p from n samples in $\mathbb{R}^d$ converges like $n^{-1/d}$, so in dimension 100 more data barely helps. Regularization partly escapes this; see the challenge.

55.5 The experiment

First the shape of the comparison. The graph shows, against the shift d between two Gaussians of equal width $\sigma = 1$, the Wasserstein distance (blue), the KL divergence $d^2/(2\sigma^2)$ (red) and the total variation $\operatorname{erf}\left(d/(2\sqrt{2}\,\sigma)\right)$ (green).



Discrepancy between $\mathcal{N}(0,\sigma^2)$ and $\mathcal{N}(d,\sigma^2)$ as a function of d, with $\sigma = 1$. Change both `1`*s (the width) to* `0.3`*. The Wasserstein line does not move at all, the KL parabola shoots up eleven times faster, and the total variation jumps to 1 almost immediately: narrow distributions are "completely different" to point-by-point measures, but only d apart to transport.*

Now Sinkhorn against the exact answer. The script builds a two-bump source histogram and a shifted, widened target on a grid, runs Sinkhorn, and compares the result with the exact quantile formula of 55.3.

```
import numpy as np
import matplotlib.pyplot as plt

# --- knobs ---------------------------------------------------------------
n, eps, iters = 200, 0.01, 2000       # grid size, entropic blur, Sinkhorn sweeps
shift, width = 0.3, 0.05              # target = source moved right and widened
# -------------------------------------------------------------------------
x = np.linspace(0, 1, n)
bump = lambda m, s: np.exp(-0.5 * ((x - m) / s) ** 2)
a = bump(0.25, 0.05) + 0.6 * bump(0.45, 0.04); a /= a.sum()   # source histogram
b = bump(0.25 + shift, 0.05 + width);           b /= b.sum()   # target histogram
C = (x[:, None] - x[None, :]) ** 2                             # squared-distance cost

K = np.exp(-C / eps)                  # Gibbs kernel; underflows when eps is tiny
v = np.ones(n)
for k in range(iters):
    u = a / (K @ v)                   # fix the row sums
    v = b / (K.T @ u)                 # fix the column sums
P = u[:, None] * K * v[None, :]       # the entropic transport plan

# exact 1-D answer: match quantiles, W2^2 = integral of (F^-1 - G^-1)^2
q = np.linspace(0.0005, 0.9995, 4000)
Finv = np.interp(q, np.cumsum(a), x)
Ginv = np.interp(q, np.cumsum(b), x)
print("Sinkhorn  sqrt(<C,P>) =", np.sqrt((C * P).sum()))
print("exact W2 (quantiles)  =", np.sqrt(np.mean((Finv - Ginv) ** 2)))
print("row-sum error =", np.abs(P.sum(1) - a).max())
print("L2 distance between histograms =", np.linalg.norm(a - b))

fig, ax = plt.subplots(1, 2, figsize=(10, 4))
ax[0].plot(x, a, label="source"); ax[0].plot(x, b, label="target"); ax[0].legend()
ax[1].imshow(P, origin="lower", extent=[0, 1, 0, 1], cmap="Blues")
ax[1].set_xlabel("to y"); ax[1].set_ylabel("from x")
plt.show()
```

With the default knobs Sinkhorn reports 0.2452 against the exact $W_2 = 0.2367$, with the row sums matched to 10^{-17}. Things to change:

- `eps = 0.001`: the estimate tightens to 0.2378, and the plan in the right panel sharpens into a curve.
- `eps = 0.0001`: everything becomes `nan`. Look at `K`: most of its entries are exactly zero. This is the underflow of 55.4.
- `shift = 0.5`, then `0.7`: the exact W_2 grows from 0.237 to 0.434 to 0.588, tracking the shift, while the L^2 distance between the histograms barely moves (0.150, 0.176, 0.198) once the bumps have separated.
- `width = 0.0`: a pure shift of the second bump only. How does the plan change?

- Replace `C` by `np.abs(x[:, None] - x[None, :])` to compute W_1 instead. Does the plan still look like a monotone curve? (In one dimension many plans can be optimal for W_1; Sinkhorn picks the most spread-out one.)

55.6 The challenge

1. **Intuition.** Two narrow bumps, one at 0 and one at 1. Describe the mixture $\frac{1}{2}\mu + \frac{1}{2}\nu$ and the Wasserstein midpoint (each grain travels half-way). If μ and ν are images of a moving object at two instants, which one looks like the object at the middle instant? Give one engineering situation where the mixture is the right answer instead.
2. **Derivation.** Show from the quantile formula that, in one dimension, $W_1(\mu,\nu) = \int_{-\infty}^{\infty} |F(x) - G(x)|\,dx$, the area between the two CDFs. (Hint: draw both CDFs and compute the same area by horizontal and by vertical strips.) Use it to find W_1 between the uniform distributions on $[0,1]$ and on $[0,2]$, then between δ_0 and $\mathcal{N}(0,1)$.
3. **Implementation.** Rewrite the experiment's Sinkhorn loop in the log domain: iterate on $f = \varepsilon \ln u$ and $g = \varepsilon \ln v$ using a numerically stable log-sum-exp. Confirm that it works at $\varepsilon = 10^{-4}$, where the plain version fails. Then plot the number of sweeps needed to reach a row-sum error of 10^{-9} against ε on log–log axes. What power law do you see?
4. **Open investigation.** Draw two independent samples of size n from the *same* standard Gaussian in $\mathbb{R}^d$ and compute the exact W_2 between them as an assignment problem (`scipy.optimize.linear_sum_assignment`). Measure how W_2 shrinks with n for $d = 1, 2, 3, 5, 10$ and estimate the exponent. Now repeat with the Sinkhorn divergence at fixed ε. Does it escape the curse of dimensionality, and at what price in what it actually measures?

55.7 Loose threads

The chapter used optimal transport as a ruler. Its deeper life is as a geometry.

? Heat flow is the steepest descent of entropy

Equip the space of probability densities with the W_2 distance. Now ask: which way should a density move to *increase its entropy* as fast as possible, per unit of Wasserstein distance travelled? The answer, found by Jordan, Kinderlehrer and Otto in 1998, is the heat equation $\partial_t \rho = \nabla^2 \rho$. Diffusion is gradient descent on entropy, in this geometry and in no other. Many other PDEs turn out to be gradient flows of the same kind. Making sense of "gradient" on a space of distributions requires a Riemannian structure on it, the subject of Ch. 59.

? Why does Sinkhorn converge so reliably?

Alternately rescaling rows and columns of a positive matrix looks naive, yet it converges linearly from any start, at a rate that depends only on $\mathbf{K}$. The reason is that each rescaling is a contraction in Hilbert's projective metric, a way of measuring distance between positive vectors that ignores their overall scale, by the Birkhoff–Hopf theorem on positive linear maps. The geometry of cones and positive operators that makes this work is part of Ch. 61.

The first mystery of 55.3, Brenier's map and the Monge–Ampère equation, waits in Ch. 89. A rigorous account of couplings, of weak convergence and of why the duality holds beyond the discrete case needs measure theory (Ch. 97). Other doors open from here: colour transfer and image registration in Ch. 76, matching markets in economics, and the Wasserstein balls of Ch. 54, which this chapter has now made precise.

Further reading

- C. Villani, *Topics in Optimal Transportation*, American Mathematical Society, 2003. A readable introduction by one of the field's architects; his *Optimal Transport: Old and New* (Springer, 2009) is the encyclopaedia.
- F. Santambrogio, *Optimal Transport for Applied Mathematicians*, Birkhäuser, 2015. Duality, one-dimensional problems and gradient flows, at a pace suited to engineers.
- G. Peyré and M. Cuturi, *Computational Optimal Transport*, Foundations and Trends in Machine Learning 11(5–6), 2019. Sinkhorn, its variants and applications, with code.
- M. Arjovsky, S. Chintala and L. Bottou, "Wasserstein Generative Adversarial Networks", *Proceedings of ICML*, 2017. The hot-and-cold argument for machine learning.

Track E — The Abstract Mathematics Behind Everything

> **Level 2 — Specializations.** The common structures hiding beneath seemingly unrelated problems.

Suggested reading order for Track E. The grey stations on the left are the core chapters it leans on.

Before you start

This track assumes you are comfortable with these core chapters:

- Ch. 1 — Mathematical Language and Algebra
- Ch. 2 — Complex Numbers and Engineering Representation
- Ch. 4 — Differential Calculus
- Ch. 8 — Matrices and Linear Systems
- Ch. 9 — Vector Spaces and Linear Transformations
- Ch. 10 — Eigenvalues and Eigenvectors

Chapters

Ch.	Chapter	What it explores
56	Abstract Algebra: Symmetry as a Language	Groups, rings, fields, homomorphisms, quotient structures, group actions, and symmetry. Applications: cryptography, coding theory, geometry, physics.
57	Number Theory: The Hidden Arithmetic of the Digital World	Primes, modular arithmetic, congruences, Diophantine equations, finite fields, elliptic curves, computational number theory. Applications: cryptography, pseudorandom generators, error-correcting codes.
58	Topology: What Survives When Shapes Bend?	Continuity, compactness, connectedness, homeomorphisms, fundamental groups, homology, topological data analysis. Applications: geometry, robotics, data analysis, dynamical systems.
59	Differential Geometry: Mathematics on Curved Worlds	Manifolds, tangent spaces, differential forms, curvature, geodesics, tensors, Riemannian geometry. Applications: robotics, computer graphics, mechanics, relativity.
60	Real Analysis: Making Calculus Rigorous	Limits, continuity, sequences, series, metric spaces, completeness, measure, integration, convergence theorems. Applications: probability theory, optimization, mathematical proof.
61	Functional Analysis: Infinite-Dimensional Thinking	Normed and Banach spaces, Hilbert spaces, operators, dual spaces, spectral theory, weak convergence. Applications: quantum mechanics, PDEs, signal processing, inverse problems.
62	Category Theory: Mathematics About Mathematics	Categories, functors, natural transformations, universal properties, limits, adjunctions. Applications: type theory, programming language design, abstract algebra, compositional systems.

Ch.	Chapter	What it explores
99	Abstract Linear Algebra: Vectors Without Coordinates	Vector spaces over arbitrary fields, linear maps as the primary objects, dual spaces, quotients, direct sums, invariant subspaces, Jordan and rational canonical forms, multilinear forms. An independent advanced module bridging the core linear algebra chapters to functional analysis, representation theory and tensors.

Every chapter follows the same arc: **the hook** (a puzzle or a failure), **the big idea**, **the mathematics**, **the algorithm**, **the experiment** you can run, **the challenge**, and **loose threads**: the questions the chapter deliberately leaves open.

Chapter 56 — Abstract Algebra: Symmetry as a Language

> **Level 2 · Track E — The Abstract Mathematics Behind Everything Builds on:** Ch. 1, Ch. 2, Ch. 8, Ch. 9 · **Leads to:** Ch. 57, Ch. 62, Ch. 95, Ch. 99 · **Related:** Ch. 29, Ch. 70, Ch. 77, Ch. 79 **Reading time:** ~25 min

56.1 The hook

In the 1880s a small wooden toy swept across America and Europe. Fifteen numbered tiles sat in a 4-by-4 frame with one empty slot, and you slid tiles into the gap to put them in order. Puzzle columnist Sam Loyd stirred up the craze by publicising a cruel variant: start with the tiles in order except that **14 and 15 are swapped**, and restore the natural order. Loyd claimed to offer $1,000 for a solution. People missed work over it. Ship captains were said to have run aground.

Nobody ever collected, and nobody ever will. The 14–15 puzzle cannot be solved.

Think about what kind of claim that is. The frame can hold the 16 objects (15 tiles and a blank) in $16! = 20{,}922{,}789{,}888{,}000$ arrangements. You cannot try them all by hand, and "I tried for a month and failed" proves nothing. Yet there is a short, airtight argument that no sequence of slides, however long or clever, swaps two tiles and leaves everything else in place. Moreover, the same argument tells you that *exactly half* of all arrangements are reachable, 10,461,394,944,000 of them, and the other half are forever cut off.

The Rubik's cube has the same flavour. Its reachable positions number

$$43{,}252{,}003{,}274{,}489{,}856{,}000 = 2^{27}\,3^{14}\,5^{3}\,7^{2}\,11,$$

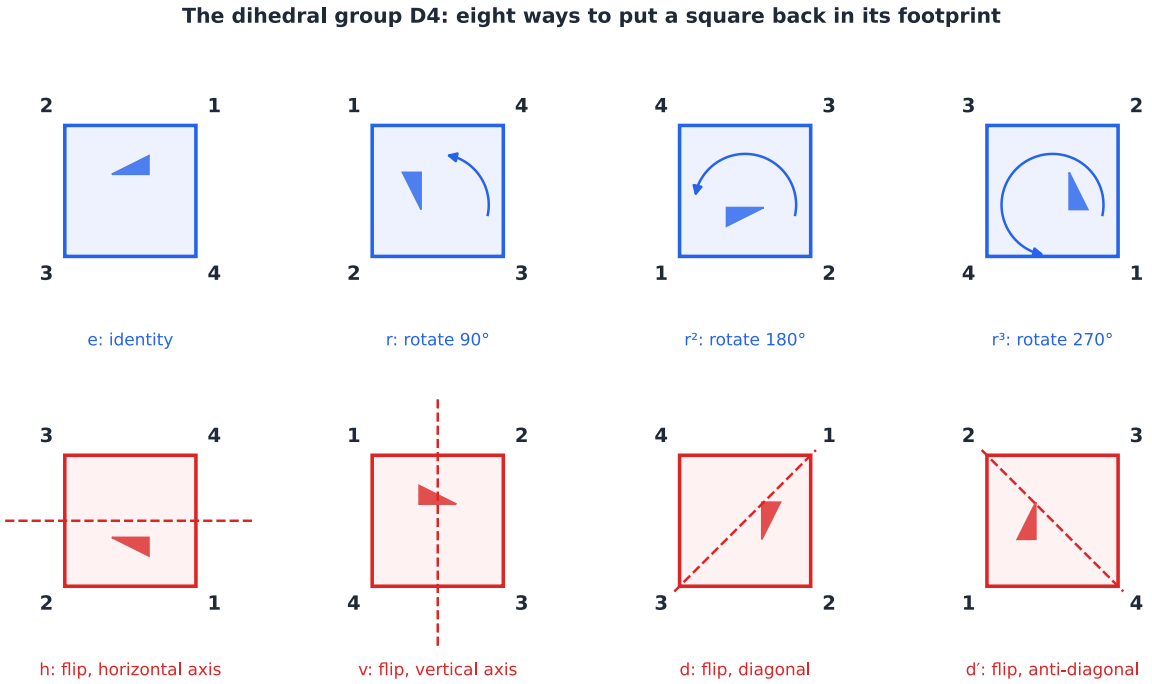
and you can know that exact number, and know that twisting a single corner in place is impossible, without ever building a list.

> **How can you prove that a move is impossible, or count an astronomically large set of states exactly, without searching?**

The answer is that the moves of a puzzle are not just a list. They *compose*, they can be *undone*, and those two facts force a rigid structure on everything the puzzle can do. That structure is called a **group**, and once you can see it you will find it inside crystals, codes, ciphers, quantum particles and the rotations of a spacecraft.

56.2 The big idea

Pick up a square tile and ask: in how many ways can I put it back into its square hole? You can leave it alone. You can turn it by 90°, 180° or 270°. You can flip it over along a horizontal, vertical or diagonal axis. That is eight ways, and there are no others.



The dihedral group D_4. Each panel shows where the corner labels 1–4 end up; the small flag shows orientation. Blue panels are rotations, red panels are reflections (dashed line = mirror).

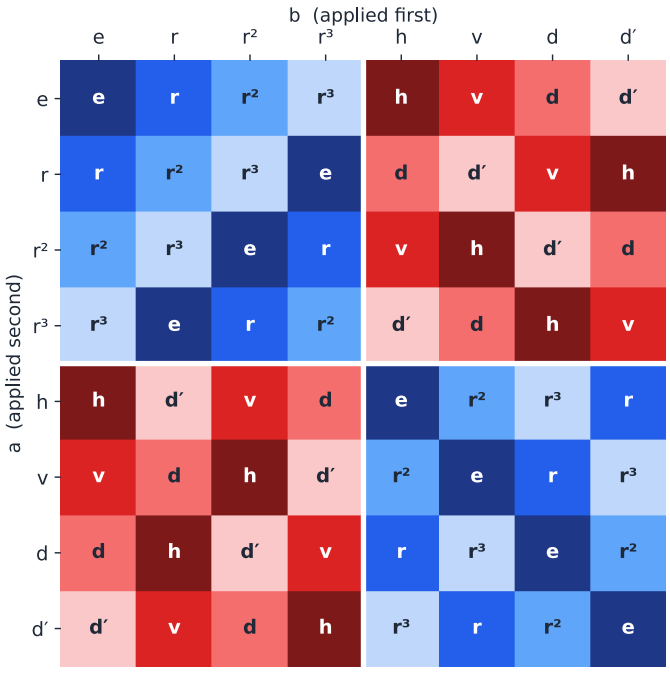
The eight motions are interesting not individually but as a system:

- **Doing one after another gives another one.** Rotate 90°, then flip horizontally: the result is one of the eight (a diagonal flip, as it turns out).
- **Doing nothing is one of them.**
- **Every one can be undone by one of them.** A 90° turn is undone by a 270° turn; every flip undoes itself.

Those three facts, plus the harmless observation that "first a, then b, then c" doesn't care how you bracket it, are the whole definition of a group. Notice what is *missing*: there is no mention of squares, angles or tiles. The same three facts hold for the twelve hours on a clock, for the invertible 3×3 matrices, for the rotations of a quantum spin, for the shuffles of a deck of cards. Abstract algebra is the decision to study the pattern and forget the costume.

Here is the payoff, stated as an analogy. A detective who knows that a suspect is 1.80 m tall can rule out everyone shorter without interviewing them. An algebraist who knows that some quantity is *preserved by every allowed move* (an **invariant**) can rule out every state where that quantity differs without exploring them. For the 15 puzzle the invariant is a single bit, the *parity* of a permutation, and it is enough to kill Loyd's challenge.

The whole Cayley table of D_4 fits in one picture:



Cayley table of D4: entry = a∘b. Blue = rotation, red = reflection

Row a, column b holds $a \circ b$ (do b first, then a). The four blocks tell a story: rotation∘rotation and reflection∘reflection are rotations; mixed products are reflections. That block pattern is a homomorphism onto $\{+1, -1\}$ in disguise. Also, the table is not symmetric: $r \circ h = d$ but $h \circ r = d'$.

Look at the table for a moment. Every row and every column contains each element exactly once (a **Latin square**), which is the "undo" axiom in visual form. The four colour blocks say that "is it a reflection?" behaves like a sign: two reflections make a rotation, just as two minus signs make a plus. Remembering only that sign throws away information yet respects composition. That is the first example of the most important idea in the chapter: a structure-preserving map, or **homomorphism**.

56.3 The mathematics

Groups

Group

A **group** $(G, *)$ is a set G with an operation $* : G \times G \to G$ such that

1. (associativity) $(a * b) * c = a * (b * c)$ for all a, b, c;
2. (identity) there is $e \in G$ with $e * a = a * e = a$ for all a;
3. (inverses) for every a there is a^{-1} with $a * a^{-1} = a^{-1} * a = e$.

The group is **abelian** if also $a * b = b * a$ for all a, b. The number of elements $|G|$ is its **order**.

Closure is built into "$* : G \times G \to G$". Usually we write ab for $a * b$. Some examples you already know:

Group	Elements	Operation	Order	Abelian?
$\mathbb{Z}$	integers	$+$	infinite	yes
$\mathbb{Z}_n$	$\{0, 1, \dots, n-1\}$	addition mod n	n	yes
μ_n	n-th roots of unity $e^{i2\pi k/n}$	complex multiplication	n	yes
D_n	symmetries of a regular n-gon	composition	$2n$	no, for $n \geq 3$
S_n	permutations of $\{1, \dots, n\}$	composition	$n!$	no, for $n \geq 3$
$GL_n(\mathbb{R})$	invertible $n \times n$ matrices	matrix product	infinite	no, for $n \geq 2$
$SO(3)$	rotations of 3-D space	composition	infinite	no

The roots of unity are covered in Ch. 2 §2.6; non-commuting matrix products are in Ch. 8 §8.2. Non-commutativity is the rule, not the exception: put on socks then shoes, versus shoes then socks.

Two small consequences of the axioms are used constantly. The identity is unique (if e and e' both work, $e = ee' = e'$), and so is each inverse. And $(ab)^{-1} = b^{-1}a^{-1}$, the socks-and-shoes rule: to undo "socks then shoes" you remove shoes first.

A **subgroup** $H \le G$ is a subset that is itself a group under the same operation. The rotations $\{e, r, r^2, r^3\}$ form a subgroup of D_4; so do $\{e, h\}$ and $\{e, r^2\}$. The **order of an element** g is the smallest $k \ge 1$ with $g^k = e$, and the powers $\langle g \rangle = \{e, g, g^2, \dots, g^{k-1}\}$ form the **cyclic subgroup** generated by g.

Lagrange's theorem: sizes are not free

Here is the first sign that the axioms are stronger than they look.

> **Lagrange's theorem**
>
> If G is finite and $H \le G$, then $|H|$ divides $|G|$. In particular the order of every element divides $|G|$.

Proof. For $g \in G$ define the **left coset** $gH = \{gh : h \in H\}$.

1. *Every coset has $|H|$ elements.* The map $h \mapsto gh$ is one-to-one, because $gh_1 = gh_2$ implies $h_1 = h_2$ after multiplying on the left by g^{-1}.
2. *Two cosets are equal or disjoint.* Suppose $x \in g_1H \cap g_2H$, so $x = g_1h_1 = g_2h_2$. Then $g_1 = g_2h_2h_1^{-1}$, and for any $h \in H$, $g_1h = g_2(h_2h_1^{-1}h) \in g_2H$ because H is closed. So $g_1H \subseteq g_2H$, and by symmetry they are equal.
3. *The cosets cover G,* since $g = ge \in gH$.

So G is cut into disjoint pieces of equal size $|H|$, and $|G| = [G : H]\,|H|$ where the **index** $[G : H]$ counts the pieces. For an element g, apply this to $H = \langle g \rangle$, whose size is the order of g. ∎

Immediately, a group of prime order p has no subgroups except $\{e\}$ and itself, so it is generated by any non-identity element and is cyclic. And $g^{|G|} = e$ for every g. Applied to the multiplicative group of non-zero residues mod p (order $p - 1$), this last fact is **Fermat's little theorem**, the engine of RSA. Ch. 57 takes it from there.

Homomorphisms, and the death of the 14–15 puzzle

> **Homomorphism, kernel, isomorphism**
>
> A map $\varphi : G \to H$ between groups is a **homomorphism** if $\varphi(ab) = \varphi(a)\varphi(b)$ for all a, b. Its **kernel** is $\ker \varphi = \{g : \varphi(g) = e_H\}$. A bijective homomorphism is an **isomorphism**, written $G \cong H$: the two groups are the same up to renaming.

Examples: $\det : GL_n(\mathbb{R}) \to \mathbb{R}^\times$, because $\det(\mathbf{AB}) = \det \mathbf{A} \det \mathbf{B}$. The exponential $x \mapsto e^x$ turns $(\mathbb{R}, +)$ into $(\mathbb{R}_{>0}, \times)$, which is why logarithm tables worked. The map $k \mapsto e^{j2\pi k/n}$ is an isomorphism $\mathbb{Z}_n \cong \mu_n$: adding hours on a clock *is* multiplying roots of unity.

The homomorphism that solves the hook is the **sign of a permutation**. A **transposition** swaps two items and fixes the rest; every permutation is a product of transpositions (sort a list by swaps). Count the **inversions** of a permutation σ, the pairs $i < j$ with $\sigma(i) > \sigma(j)$. One can check that composing with a single transposition changes the number of inversions by an odd amount, so the parity of "number of transpositions used" does not depend on how you write σ. That makes

$$\operatorname{sgn} : S_n \to \{+1, -1\}, \qquad \operatorname{sgn}(\sigma) = (-1)^{\#\text{transpositions}}$$

well defined, and it is a homomorphism: write σ with a swaps and τ with b swaps, and $\sigma\tau$ uses $a + b$. Its kernel, the **even** permutations, is the **alternating group** A_n, of order $n!/2$.

> **The 14–15 puzzle is unsolvable**
>
> No sequence of slides returns the blank to its corner with tiles 14 and 15 exchanged and all other tiles fixed.

Proof. Treat the blank as a 16th tile and colour the board like a chessboard. Each slide swaps the blank with a neighbouring tile, so it is a transposition of the 16 objects, and it moves the blank to a square of the opposite colour. If the blank ends where it started, it has made an even number of moves, so the overall permutation is a product of an even number of transpositions and has sign $+1$. The target configuration differs from the start by one transposition, $(14\ 15)$, of sign -1. Since sgn is well defined, no sequence of moves produces it. ∎

The full story is that the reachable configurations are exactly those whose sign matches the colour parity of the blank, which is why exactly half of the 16! arrangements are reachable. Proving the "every even one is reachable" half takes a constructive argument; the impossibility half took four lines.

Normal subgroups and quotients

The kernel of a homomorphism is not just any subgroup. If $k \in \ker \varphi$ then $\varphi(gkg^{-1}) = \varphi(g)\varphi(g)^{-1} = e$, so gkg^{-1} is in the kernel too. Subgroups with this property, $gNg^{-1} = N$ for every g, are called **normal**, written $N \trianglelefteq G$. For a normal subgroup the cosets themselves form a group, the **quotient** G/N, with $(aN)(bN) = (ab)N$.

The quotient is how algebra "forgets on purpose". $\mathbb{Z}/n\mathbb{Z}$ is the integers with multiples of n forgotten, which is clock arithmetic. $D_4/\{\text{rotations}\}$ has two elements, "rotation" and "reflection", and is exactly the block pattern of the Cayley table. The general principle is:

> **First isomorphism theorem**
>
> If $\varphi : G \to H$ is a homomorphism, then $G/\ker \varphi \cong \varphi(G)$.

So $S_n/A_n \cong \{\pm 1\}$, $GL_n(\mathbb{R})/SL_n(\mathbb{R}) \cong \mathbb{R}^\times$ (where SL_n is the determinant-one matrices), and $\mathbb{Z}/n\mathbb{Z} \cong \mu_n$. Every homomorphic image of a group is one of its quotients. We state this without proof; the proof checks that $g \ker \varphi \mapsto \varphi(g)$ is well defined and bijective.

> **? The atoms of symmetry**
>
> Integers factor into primes. Groups "factor" too: if N is normal in G, you can study N and G/N separately. Groups with no normal subgroups except $\{e\}$ and themselves are called **simple**, and they are the atoms. After about 10,000 journal pages written over half a century, mathematicians finished classifying every finite simple group: a few infinite families plus 26 "sporadic" exceptions, the largest being the Monster, with about 8×10^{53} elements. Why the Monster's numbers turn up in the coefficients of a function from 19th-century complex analysis ("monstrous moonshine") is a story about representations of groups as matrices: see Ch. 95.

Group actions and the orbit–stabilizer theorem

Groups are most useful when they *do* something. An **action** of G on a set X assigns to each g a map $x \mapsto g \cdot x$ with $e \cdot x = x$ and $g \cdot (h \cdot x) = (gh) \cdot x$. D_4 acts on the corners of the square, on colourings of the square, on the cells of a chessboard. The **orbit** of x is everything you can reach from it, $Gx = \{g \cdot x\}$. The **stabilizer** is what leaves it alone, $G_x = \{g : g \cdot x = x\}$, and it is a subgroup.

> **Orbit–stabilizer**
>
> For a finite group acting on a set, $|Gx| \cdot |G_x| = |G|$.

Proof. Send each coset gG_x to the point $g \cdot x$. This is well defined and one-to-one, because $g \cdot x = g' \cdot x \iff g^{-1}g' \cdot x = x \iff g^{-1}g' \in G_x \iff gG_x = g'G_x$. It is onto the orbit by definition. So the orbit has as many points as there are cosets of G_x, namely $|G|/|G_x|$ by Lagrange. ■

This is a counting machine. The rotation group of a cube acts on its 6 faces; one face can be sent to any face (orbit size 6), and the rotations fixing it are the 4 turns about its axis. So the cube has $6 \times 4 = 24$ rotational symmetries, found without listing them. The same trick gives 12 rotations for a tetrahedron and 60 for an icosahedron.

Counting *orbits* rather than group elements needs one more step. Count the pairs (g, x) with $g \cdot x = x$ in two ways: by x it is $\sum_x |G_x|$, and by g it is $\sum_g |\mathrm{Fix}(g)|$. Each orbit O contributes $\sum_{x \in O} |G|/|O| = |G|$ to the first sum. Hence

$$\#\text{orbits} = \frac{1}{|G|} \sum_{g \in G} |\mathrm{Fix}(g)|,$$

Burnside's lemma. How many genuinely different ways are there to colour the corners of a square with 3 colours, counting two as the same if a symmetry carries one to the other? The identity fixes all $3^4 = 81$ colourings. The quarter turns r, r^3 fix only the 3 one-colour ones; r^2 fixes $3^2 = 9$. The diagonal flips d, d' fix two corners and swap a pair, $3^3 = 27$ each; the edge flips h, v swap two pairs, $3^2 = 9$ each. The average is $(81 + 3 + 9 + 3 + 27 + 27 + 9 + 9)/8 = 168/8 = 21$.

Rings and fields: two operations at once

Groups have one operation. Numbers have two, and they interact through the distributive law.

> **Ring, field**
>
> A **ring** $(R, +, \cdot)$ is an abelian group under $+$, with an associative multiplication that distributes over addition: $a(b + c) = ab + ac$ and $(a + b)c = ac + bc$. A **field** is a commutative ring with $1 \neq 0$ in which every non-zero element has a multiplicative inverse.

$\mathbb{Z}$ is a ring but not a field (no $1/2$). $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ are fields. Polynomials with real coefficients form a ring, as do $n \times n$ matrices (a non-commutative one). The clock $\mathbb{Z}_n$ is a ring, and here algebra gets its first surprise: in $\mathbb{Z}_6$, $2 \cdot 3 = 0$. Non-zero things multiplying to zero are **zero divisors**, and a ring with them cannot be a field. $\mathbb{Z}_n$ is a field exactly when n is prime, a fact Ch. 57 proves with the extended Euclidean algorithm.

The ring analogue of a normal subgroup is an **ideal**, a subset closed under addition and under multiplication by *anything* in the ring, and the analogue of a quotient group is a **quotient ring**. Take real polynomials, forget multiples of $x^2 + 1$, and what is left is $\mathbb{R}[x]/(x^2 + 1)$. In that ring $x^2 = -1$. You have just built $\mathbb{C}$, with x playing the role of j, and without ever saying "imagine a square root of -1". Do the same with polynomials over $\mathbb{Z}_2$ modulo $x^8 + x^4 + x^3 + x + 1$ and you build the 256-element field in which the AES cipher does its arithmetic.

Where the structures live

Structure	Where engineers meet it	Chapter
cyclic group $\mathbb{Z}_p^\times$	Diffie–Hellman key exchange, discrete logs	Ch. 57, Ch. 77
elliptic-curve group	ECDSA signatures, TLS	Ch. 78
finite field $\mathbb{F}_{2^m}$, ideals in $\mathbb{F}_2[x]$	CRC checks, Reed–Solomon and cyclic codes	Ch. 79
$SO(3)$, unit quaternions	attitude control, 3-D graphics	Ch. 70
μ_n acting on sequences	the DFT is the "character table" of $\mathbb{Z}_n$	Ch. 16
space groups (230 of them)	crystallography, band structure	Ch. 92

> **? Symmetry that you can't see but can conserve**
>
> Every physical law you use is unchanged if you shift the clock by Δt, slide the lab by $\Delta \mathbf{x}$, or rotate it. Those are *continuous* groups, not finite tables. In 1918 Emmy Noether proved that each such continuous symmetry of a system's action forces a conserved quantity: time shifts give energy, space shifts give momentum, rotations give angular momentum. Why should a symmetry *produce* a number that never changes? The answer needs the calculus of variations and the infinitesimal version of a group: see Ch. 92.

56.4 The algorithm

A computer cannot hold "the set of all symmetries" as an idea, but it can hold a few **generators** and close them up. Represent each symmetry as a permutation, a tuple `p` where `p[i]` is where point `i` goes, and repeatedly multiply by the generators until nothing new appears.

```
def compose(p, q):            # (p o q)(i) = p[q[i]]: do q first, then p
    return tuple(p[i] for i in q)

def generate(gens):
    """All elements of the group generated by permutations gens (breadth-first closure)."""
    n = len(gens[0])
    e = tuple(range(n))
    group, frontier = {e}, [e]
    while frontier:
        new = []
        for g in frontier:
            for s in gens:
                h = compose(s, g)
                if h not in group:
                    group.add(h)
                    new.append(h)
        frontier = new
    return group

def count_orbits(group, k):
    """Burnside: number of k-colourings of the points, up to the group."""
    def cycles(p):
        seen, c = set(), 0
        for i in range(len(p)):
            if i not in seen:
                c += 1
                while i not in seen:
                    seen.add(i); i = p[i]
        return c
    return sum(k ** cycles(g) for g in group) // len(group)

r = (1, 2, 3, 0)              # quarter turn of the square's corners
s = (0, 3, 2, 1)              # flip across the diagonal through corner 0
D4 = generate([r, s])
print(len(D4), count_orbits(D4, 3))          # 8 21
S5 = generate([(1, 2, 3, 4, 0), (1, 0, 2, 3, 4)])
print(len(S5))                               # 120
```

Two details deserve comment.

Why closing under the generators is enough. In a finite group the inverse of s is a power of s (Lagrange says $s^{|G|} = e$, so $s^{-1} = s^{|G|-1}$), so products of generators already reach every inverse. The breadth-first search is just a graph search on the **Cayley graph**, whose vertices are group elements and whose edges are "multiply by a generator". Its cost is $O(|G| \cdot m \cdot n)$ for m generators acting on n points, plus memory for every element.

Why Burnside counts cycles. A colouring is fixed by g exactly when it is constant on each cycle of g, so $|\mathrm{Fix}(g)| = k^{c(g)}$ where $c(g)$ is the number of cycles. Running the code on the rotation group of a cube acting on its 6 faces gives 24 elements, 10 two-colourings and 57 three-colourings, numbers you can check against the orbit–stabilizer count above.

> **Common Pitfall: listing the group is the wrong plan for big groups**
>
> Breadth-first closure stores every element. The Rubik's cube group has about 4.3×10^{19} elements, far beyond any memory. Real computer-algebra systems use the **Schreier–Sims algorithm**, which stores a chain of stabilizers $G \geq G_{x_1} \geq G_{x_1 x_2} \geq \cdots$ and uses orbit–stabilizer at each level, so that $|G|$ comes out as a product of orbit lengths. It handles the cube in milliseconds. The cube's exact order in §56.1 drops out of such a computation.

The 15 puzzle also shows a second failure mode: if you search *configurations* rather than the group, you need an invariant to know when to stop. A solver that does not check the sign first will search all 10^{13} reachable states before admitting that the 14–15 position is not among them.

56.5 The experiment

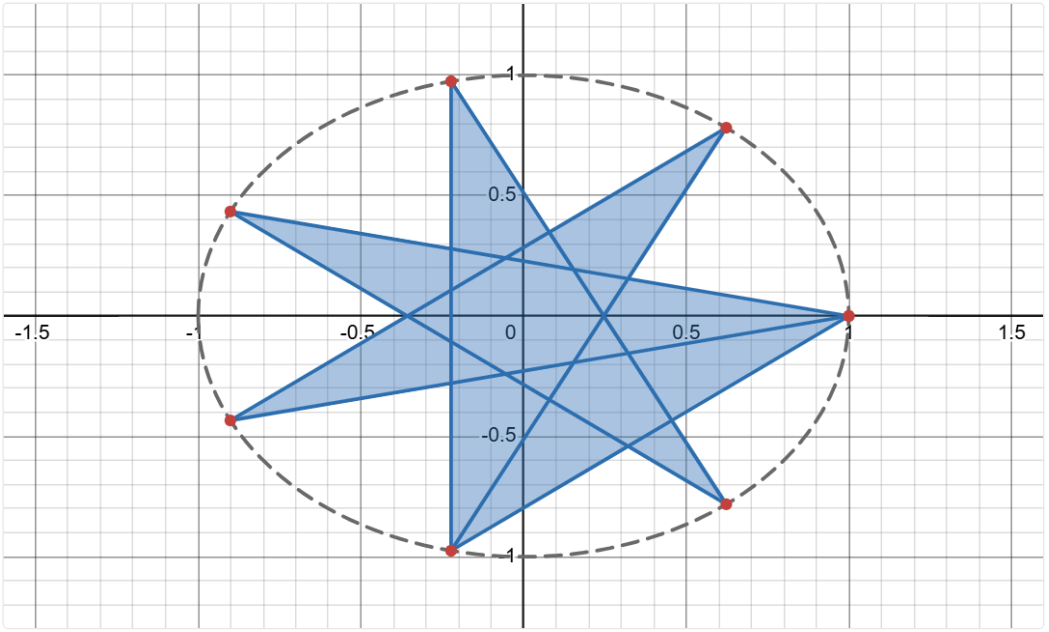
The script below builds the dihedral group D_n of a regular n-gon from scratch, checks that its Cayley table is a Latin square, tests Lagrange on every element, and counts necklaces with Burnside's lemma.

```
import numpy as np
import matplotlib.pyplot as plt

n = 6          # knob: symmetries of a regular n-gon (try 3, 4, 5, 12)
k = 2          # knob: number of bead colours for the necklace count

# D_n as permutations of the n corners: rotations r^a, reflections s r^a
rot = [tuple((i + a) % n for i in range(n)) for a in range(n)]
ref = [tuple((a - i) % n for i in range(n)) for a in range(n)]
G = rot + ref
index = {g: t for t, g in enumerate(G)}
comp = lambda p, q: tuple(p[i] for i in q)           # do q first, then p

T = np.array([[index[comp(a, b)] for b in G] for a in G])
assert all(sorted(row) == list(range(2 * n)) for row in T)   # Latin square

def order(g):
    e, h, m = tuple(range(n)), g, 1
    while h != e:
        h, m = comp(g, h), m + 1
    return m

orders = [order(g) for g in G]
print("element orders:", orders)
print("all divide |G| =", 2 * n, ":", all((2 * n) % o == 0 for o in orders))
print("abelian:", (T == T.T).all())

def cycles(p):
    seen, c = set(), 0
    for i in range(n):
        if i not in seen:
            c += 1
            while i not in seen:
                seen.add(i); i = p[i]
    return c

print("necklaces (rotations only):", sum(k ** cycles(g) for g in rot) // n)
print("bracelets (with flips):   ", sum(k ** cycles(g) for g in G) // (2 * n))
plt.imshow(T, cmap="tab20"); plt.title(f"Cayley table of D{n}"); plt.show()
```

With $n = 6, k = 2$ it reports element orders 1, 6, 3, 2, 3, 6 for the rotations and 2 for all six reflections, all dividing 12; the group is not abelian; and there are 14 two-colour necklaces of six beads but only 13 bracelets, because one pair of necklaces are mirror images. Things to try:

- Change `n` to 7, then 8. For a prime n every non-trivial rotation has order n; for $n = 8$ you see orders 2, 4 and 8. Why are there never orders like 3 when $n = 8$?
- Set `k = 3` and increase `n`. How fast does the bracelet count grow compared with $k^n/(2n)$, the naive "divide by the group size" guess? When is the naive guess exactly right?
- Change `ref` so that reflections are omitted from `G` and rerun the Latin-square check on `rot` alone. Is the rotation subgroup normal? (Test ghg^{-1}.)

For a geometric view of the cyclic group μ_7, the graph below joins the powers of ω^3 where $\omega = e^{j2\pi/7}$. Because 3 and 7 share no factor, the star visits every seventh root of unity before closing.



Powers of ω^3 in μ_7. In the blue line, change both 7s to 8 and both 3s to 2: the path now closes after 4 steps, tracing the subgroup of order $8/\gcd(8,2) = 4$. Try 8 and 3 next, then 12 and 4. The red dots are the seventh roots for reference.

56.6 The challenge

No answers are given.

1. **Intuition.** On a Rubik's cube, twisting a single corner cubie by 120° in place, leaving everything else solved, is impossible. Guess the invariant: assign each corner a twist of 0, 1 or 2, and ask what every face turn does to the *sum* of the eight twists. Why does a homomorphism to $\mathbb{Z}_3$ settle the question, and what is its kernel?
2. **Derivation.** Prove that every group of order 4 is isomorphic either to $\mathbb{Z}_4$ or to the Klein four-group $\mathbb{Z}_2 \times \mathbb{Z}_2$. (Use Lagrange to list the possible element orders.) Then show D_4 has exactly 10 subgroups and decide which of them are normal.
3. **Implementation.** Extend `generate` to the 15 puzzle restricted to a 2×3 board (5 tiles and a blank). Generate all reachable configurations by breadth-first search, confirm that exactly half of the $6! = 720$ arrangements are reached, and record the largest number of moves needed (the board's "God's number").
4. **Open investigation.** Burnside's lemma counts colourings up to symmetry. Choose a structure you care about (6-bit binary words up to cyclic shift, which is relevant to cyclic codes; chemical isomers of a ring molecule; seating plans at a round table) and compute counts for several sizes. Then look for a closed formula, and investigate what happens when you weight colours unequally (Pólya's enumeration theorem is the place to end up).

56.7 Loose threads

The hook asked how to prove impossibility without searching. The answer is an **invariant that is a homomorphism**: a map from the group of moves to a small group that respects composition. The sign $\operatorname{sgn} : S_{16} \to \{\pm 1\}$, combined with the colour of the blank's square, kills the 14–15 puzzle in four lines and simultaneously shows that half of all arrangements are unreachable. Exact counting comes from the same structure: Lagrange's theorem forces sizes to divide, orbit–stabilizer turns "how many symmetries?" into a product, and Burnside's lemma turns "how many different-looking things?" into an average.

What we did not tell you:

- **Why the symmetries of a polynomial's roots decide whether it can be solved.** There is a formula for the roots of quadratics, cubics and quartics, but provably none, using only arithmetic and radicals, for the general quintic. The reason is that S_5, the group permuting five roots, contains the simple group A_5 (order 60, which you generated in §56.4), and it is not abelian. That is Galois theory, which deserves a book of its own; Stewart's in the reading list is the friendliest route.
- **How groups become matrices.** Every finite group can be realised as a group of matrices, and the traces of those matrices (its **characters**) carry almost everything about it. The DFT is the character table of the cyclic group $\mathbb{Z}_n$, which is why convolution diagonalises. The first mystery points to Ch. 95.
- **What the common shape of all these definitions is.** Groups, rings and vector spaces each have substructures, structure-preserving maps, quotients and an isomorphism theorem. That family resemblance is not a coincidence; it is the subject of Ch. 62.

> **? A group drawn on a curve**
>
> Draw the curve $y^2 = x^3 - x + 1$. Through two points on it, draw the straight line; it meets the curve in exactly one more point. Reflect that point in the x-axis and call it the "sum". Astonishingly, this rule is associative and makes the curve's points into an abelian group, and the hardness of undoing repeated addition in that group protects most of the world's secure web connections. Why is a rule made of lines associative? See Ch. 57 for the arithmetic, and Ch. 78 for the proof and the cryptography.

Further reading

- I. Stewart, *Why Beauty Is Truth: A History of Symmetry*, Basic Books, 2007. A popular history from Babylonian equations to Galois and the Monster.
- J. A. Gallian, *Contemporary Abstract Algebra*, 10th ed., CRC Press, 2021. Gentle, example-rich undergraduate text with many applications.
- M. A. Armstrong, *Groups and Symmetry*, Springer, 1988. Short chapters, a geometric viewpoint, and Burnside counting done properly.
- I. Stewart, *Galois Theory*, 4th ed., CRC Press, 2015. Where the unsolvable quintic is finally explained.
- D. Joyner, *Adventures in Group Theory: Rubik's Cube, Merlin's Machine, and Other Mathematical Toys*, 2nd ed., Johns Hopkins University Press, 2008. Puzzles as a route into permutation groups.

Chapter 57 — Number Theory: The Hidden Arithmetic of the Digital World

> **Level 2 · Track E — The Abstract Mathematics Behind Everything Builds on:** Ch. 56, Ch. 1, Ch. 29 · **Leads to:** Ch. 77, Ch. 78, Ch. 79, Ch. 81 · **Related:** Ch. 26, Ch. 38, Ch. 18, Ch. 94 **Reading time:** ~25 min

57.1 The hook

The English mathematician G. H. Hardy wrote in 1940 that number theory was gloriously useless: nobody had found a warlike purpose for it, and he did not expect anyone to. Thirty-seven years later, three researchers at MIT turned the arithmetic of primes into the lock on the world's secrets.

Here is the puzzle they solved. You want to buy something from a website you have never visited. Every bit you send crosses networks owned by strangers, and an eavesdropper can copy all of it. You and the website have never met and share no secret password. Yet within a fraction of a second your browser and the server agree on a secret key that the eavesdropper, who saw *every message*, cannot work out.

That sounds impossible. If I tell you everything I tell my partner, how can my partner know something you don't?

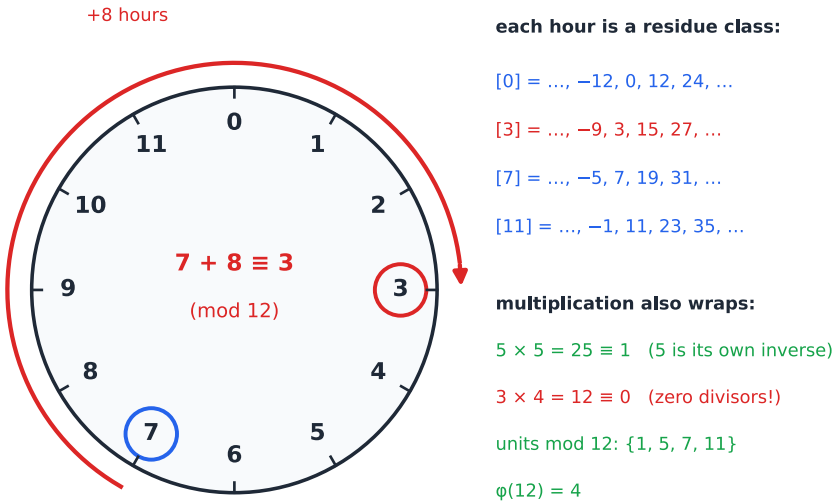
The trick is an asymmetry of effort that lives entirely in whole numbers. Multiply two 300-digit primes and a laptop finishes in microseconds. Hand someone only the 600-digit product and ask for the two primes back, and every computer on Earth working together would not finish before the Sun burns out, as far as anyone knows. Similarly, raising 5 to the 6th power and keeping the remainder after dividing by 23 is easy ($5^6 = 15625 \equiv 8$). Recovering the exponent 6 from the remainder 8 is, for large numbers, believed to be hopeless.

> **How can arithmetic on remainders make some calculations effortless one way and practically impossible the other way, and how do two strangers use that to share a secret in public?**

The answer needs four ideas that are each more than 200 years old: Euclid's algorithm, the clock arithmetic of Gauss, Fermat's little theorem and Euler's generalisation of it. You will see the full machine working by the end of §57.3, on numbers small enough to check by hand.

57.2 The big idea

Number theory studies the integers, but its modern power comes from looking at them through a keyhole: **remainders**. Fix a modulus n and identify any two integers that differ by a multiple of n. On a 12-hour clock, 7 o'clock plus 8 hours is 3 o'clock, and 15, 27 and -9 all "are" 3.

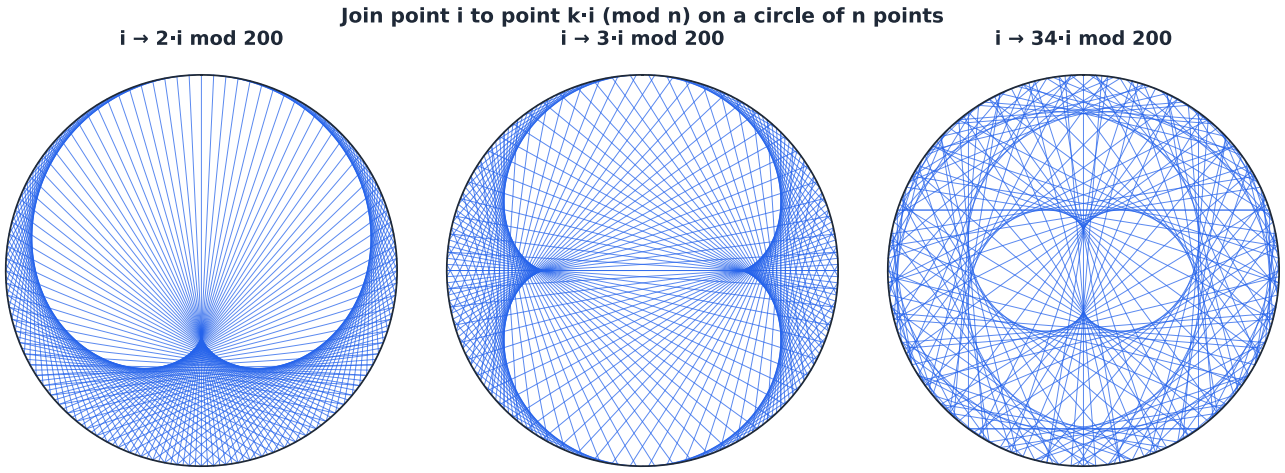


Arithmetic mod 12. Each hour stands for an entire class of integers. Addition wraps around; so does multiplication, which brings surprises: some numbers have inverses ($5 \times 5 \equiv 1$) and some multiply to zero ($3 \times 4 \equiv 0$).

Wrapping around destroys size. In ordinary arithmetic, 5^k grows steadily and you can recover k with a logarithm. Mod 23, the powers of 5 go

$$5,\ 2,\ 10,\ 4,\ 20,\ 8,\ 17,\ 16,\ 11,\ 9,\ 22,\ 18,\ 21,\ 13,\ 19,\ 3,\ 15,\ 6,\ 7,\ 12,\ 14,\ 1,\ \ldots$$

visiting all 22 non-zero residues in an order that looks scrambled. Computing forward is easy; going backward (the **discrete logarithm**) has no landmarks. The second picture shows how strange modular multiplication looks when drawn. Put n points on a circle and join point i to point $ki \bmod n$:



Multiplication mod 200 by 2, 3 and 34. Small multipliers trace smooth envelopes (a cardioid, a nephroid); larger ones give tangled, nearly structureless webs. The same operation can look orderly or chaotic depending on the numbers.

Here is the one-paragraph version of the public-key idea, using a paint analogy. Alice and Bob agree in public on a common colour. Each mixes in a secret colour of their own and sends the mixture across. Each then adds their own secret to what they received. Both end up with "common + Alice's + Bob's", while the eavesdropper holds only the two half-mixtures and cannot un-mix paint. In number theory, "mixing" is exponentiation mod a prime, and un-mixing is the discrete logarithm.

Underneath all of this sit the **primes**, the numbers with no divisors but 1 and themselves. They are the atoms of multiplication, they make clock arithmetic into a field, and their distribution is the deepest unsolved mystery in mathematics.

57.3 The mathematics

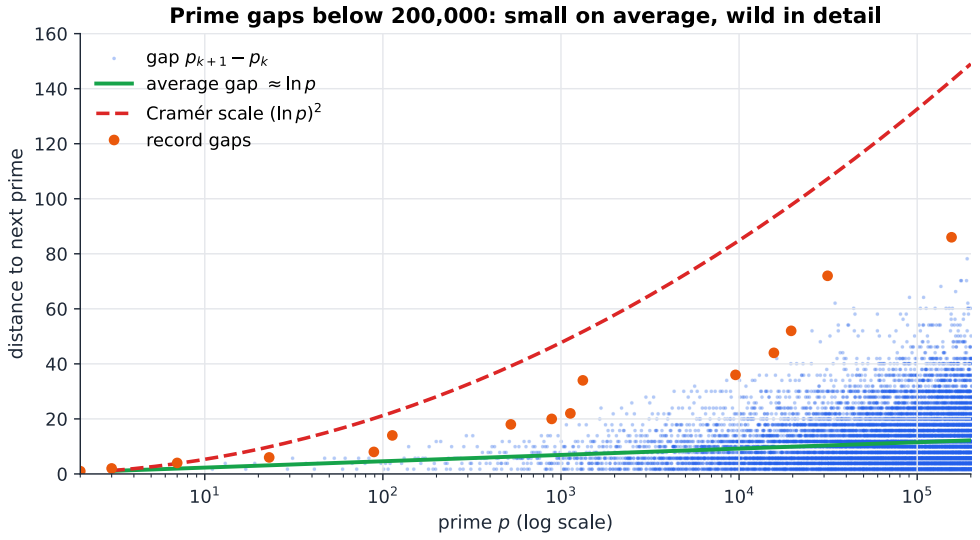
Primes and the fundamental theorem

Every integer $n \geq 2$ factors into primes, and the factorisation is unique up to order: $3120 = 2^4 \cdot 3 \cdot 5 \cdot 13$. (Existence is a short induction; uniqueness needs Euclid's lemma below.) The first great theorem about primes is also the oldest.

> **Euclid (c. 300 BC)**
>
> There are infinitely many primes.

Proof. Suppose $p_1, \dots, p_k$ were all of them. Let $N = p_1 p_2 \cdots p_k + 1$. Dividing N by any p_i leaves remainder 1, so no p_i divides N. But $N \geq 2$ has some prime factor, which is then a prime missing from the list. ■

How many are there? The **prime number theorem** (Hadamard and de la Vallée Poussin, 1896) says the number $\pi(x)$ of primes up to x satisfies $\pi(x) \sim x / \ln x$. There are 78,498 primes below 10^6, against $10^6 / \ln 10^6 \approx 72{,}382$. So near a large number p the average gap between primes is about $\ln p$, and a random 300-digit number is prime with probability roughly $1/\ln(10^{300}) \approx 1/691$. That is why finding big primes for keys is cheap: test a few hundred random odd candidates.



Gaps between consecutive primes below 200,000. The average gap tracks $\ln p$ (green), but individual gaps scatter wildly; record gaps (orange) stay below the $(\ln p)^2$ curve that a random model of the primes predicts. The longest here is 86, after 155,921.

Congruences and Euclid's algorithm

Congruence

$a \equiv b \pmod{n}$ means n divides $a - b$. The residue classes mod n form the ring $\mathbb{Z}_n = \{0, 1, \dots, n-1\}$ of Ch. 56, with addition and multiplication done mod n.

Congruences can be added and multiplied like equations, but **not divided** freely: $2 \cdot 3 \equiv 2 \cdot 9 \pmod{12}$, yet $3 \not\equiv 9$. Division by a is legal exactly when a has a multiplicative inverse mod n, and the tool for finding inverses is the oldest non-trivial algorithm still in daily use. Euclid's algorithm computes $\gcd(a, b)$ by repeatedly replacing (a, b) with $(b,\ a \bmod b)$; Ch. 29 §29.4 proves it correct with a loop invariant. Running it *backwards* gives more.

Bézout's identity

For integers a, b not both zero there are integers x, y with $ax + by = \gcd(a, b)$.

Proof. Let $S = \{ax + by > 0 : x, y \in \mathbb{Z}\}$ and let $d = ax_0 + by_0$ be its smallest element. Divide a by d: $a = qd + r$ with $0 \le r < d$. Then $r = a - qd = a(1 - qx_0) + b(-qy_0)$ is again a combination of a and b, and it is smaller than d, so it cannot be in S; hence $r = 0$ and d divides a. Likewise d divides b. Any common divisor of a and b divides $ax_0 + by_0 = d$. So d is the greatest common divisor. ■

The **extended Euclidean algorithm** finds x, y explicitly (§57.4). Two consequences matter everywhere:

1. *a is invertible mod n if and only if* $\gcd(a, n) = 1$. If $ax + ny = 1$ then $ax \equiv 1 \pmod{n}$, so x is the inverse. Conversely, if $ax \equiv 1$ then $ax - 1 = ny$ for some y, and any common divisor of a, n divides 1.
2. *$\mathbb{Z}_p$ is a field when p is prime*, since every $a \not\equiv 0$ is coprime to p. For composite n it is not: a factor d of n is a zero divisor. This also gives **Euclid's lemma** (if a prime divides ab it divides a or b), which proves uniqueness of factorisation.

The same identity solves the simplest **Diophantine equations**, equations where only integer solutions count. $ax + by = c$ has integer solutions exactly when $\gcd(a, b)$ divides c: with 6-litre and 10-litre jugs you can measure 4 litres, but never 5. Other Diophantine equations are wildly harder. $x^2 + y^2 = z^2$ has infinitely many solutions (3, 4, 5; 5, 12, 13; …), while $x^n + y^n = z^n$ has none in positive integers for $n \ge 3$, as Fermat claimed in 1637 and Andrew Wiles proved in 1994 using elliptic curves. Hilbert asked in 1900 for an algorithm that decides whether any given Diophantine equation has a solution; in 1970 Matiyasevich showed none exists, which is the halting problem of Ch. 30 wearing integer clothes.

The units, the invertible residues, form a group $\mathbb{Z}_n^\times$ under multiplication. Its size is **Euler's totient** $\varphi(n)$, the count of $1 \le a \le n$ with $\gcd(a, n) = 1$. For a prime, $\varphi(p) = p - 1$; for distinct primes, $\varphi(pq) = (p-1)(q-1)$.

Chinese remainder theorem

If $\gcd(m, n) = 1$, the map $x \mapsto (x \bmod m,\ x \bmod n)$ is a ring isomorphism $\mathbb{Z}_{mn} \cong \mathbb{Z}_m \times \mathbb{Z}_n$.

So a congruence mod 12 is the same information as one mod 3 plus one mod 4: "$x \equiv 2 \pmod 3$, $x \equiv 3 \pmod 4$" has the unique solution $x \equiv 11 \pmod{12}$. Engineers use this to split a big modular computation into small parallel ones (residue number systems, fast RSA decryption).

Fermat, Euler and the trapdoor

Euler's theorem

If $\gcd(a, n) = 1$, then $a^{\varphi(n)} \equiv 1 \pmod{n}$. For a prime p not dividing a, this is **Fermat's little theorem**, $a^{p-1} \equiv 1 \pmod{p}$.

Proof. Let $u_1, \dots, u_k$ be the $k = \varphi(n)$ units mod n. Multiplying each by the unit a gives $au_1, \dots, au_k$. These are again units, and they are distinct, because $au_i \equiv au_j$ implies $u_i \equiv u_j$ after multiplying by a^{-1}. So they are the same k units in a different order, and their products agree:

$$\prod_{i=1}^{k} (au_i) \equiv \prod_{i=1}^{k} u_i \pmod{n} \quad \Longrightarrow \quad a^k \prod u_i \equiv \prod u_i .$$

The product of units is a unit, so cancel it: $a^k \equiv 1$. ■

(Equivalently: $\mathbb{Z}_n^\times$ is a group of order $\varphi(n)$, and by Lagrange's theorem in Ch. 56 every element's order divides it.)

Now the machine. **RSA** (Rivest, Shamir, Adleman, 1977):

1. Choose primes p, q; publish $n = pq$. Compute $\varphi(n) = (p-1)(q-1)$ privately.
2. Choose a public exponent e with $\gcd(e, \varphi(n)) = 1$; compute the private $d = e^{-1} \bmod \varphi(n)$ by extended Euclid.
3. Encrypt a message $m < n$ as $c = m^e \bmod n$. Decrypt as $c^d \bmod n$.

Why it works. $ed = 1 + t\varphi(n)$ for some integer t, so when $\gcd(m, n) = 1$,

$$c^d \equiv m^{ed} = m \cdot \left(m^{\varphi(n)}\right)^t \equiv m \cdot 1^t = m \pmod{n}.$$

(The rare case $\gcd(m, n) > 1$ also works, by checking mod p and mod q separately and gluing with the CRT.)

RSA with toy numbers

Take $p = 61$, $q = 53$: $n = 3233$, $\varphi(n) = 3120$. With $e = 17$, extended Euclid gives $17 \cdot 2753 = 46801 = 15 \cdot 3120 + 1$, so $d = 2753$. The message $m = 65$ encrypts to $65^{17} \bmod 3233 = 2790$, and $2790^{2753} \bmod 3233 = 65$. Anyone can encrypt with $(n, e) = (3233, 17)$. To find d you need $\varphi(n)$, which means factoring 3233, trivial here and believed infeasible when n has 600 digits.

The secret-sharing trick of the hook is **Diffie–Hellman** (1976). Publish a prime p and a base g. Alice picks secret a and sends $A = g^a \bmod p$; Bob picks b and sends $B = g^b \bmod p$. Alice computes B^a, Bob computes A^b, and both get $g^{ab} \bmod p$. With $p = 23$, $g = 5$, $a = 6$, $b = 15$: $A = 8$, $B = 19$, and $19^6 \equiv 8^{15} \equiv 2 \pmod{23}$. The eavesdropper sees 23, 5, 8 and 19, and must solve a discrete logarithm to get the shared 2.

That answers the hook. Modular exponentiation is fast (§57.4), but its inverses, factoring and discrete logarithms, have no known efficient classical algorithm. The best general factoring method, the number field sieve, runs in roughly $\exp\big(1.92\,(\ln n)^{1/3}(\ln \ln n)^{2/3}\big)$ steps: sub-exponential, but still astronomically slow for 2048-bit n.

? A machine that factors

Every public key built on factoring or discrete logs shares one weakness: both problems reduce to *finding the period* of a function like $k \mapsto a^k \bmod n$. In 1994 Peter Shor showed that a quantum computer can find such periods efficiently, using a quantum version of the Fourier transform you met in Ch. 16. Why does period-finding crack factoring, and what are cryptographers replacing RSA with before such machines exist? See Ch. 77.

Finite fields

$\mathbb{Z}_p$ is a field with p elements. Are there others? Yes, exactly one (up to isomorphism) of each prime-power size p^k, and none of any other size. The field $\mathbb{F}_{p^k}$ is built like $\mathbb{C}$ in Ch. 56: take polynomials with coefficients mod p and reduce modulo an irreducible polynomial of degree k. Engineers live in $\mathbb{F}_{2^8}$, whose 256 elements are bytes: addition is XOR, and multiplication is polynomial multiplication reduced mod $x^8 + x^4 + x^3 + x + 1$. AES encryption, Reed–Solomon codes on QR codes and in deep-space links, and RAID-6 disk arrays all compute there. The multiplicative group of every finite field is **cyclic**, which is why a generator g like the 5 mod 23 above always exists.

Elliptic curves, in one paragraph

An **elliptic curve** over a field is the set of solutions of $y^2 = x^3 + ax + b$ (with $4a^3 + 27b^2 \neq 0$), plus one extra "point at infinity". Its points form an abelian group under a chord-and-tangent rule (the mystery at the end of Ch. 56). Over $\mathbb{F}_p$ the curve is a finite cloud of points: $y^2 = x^3 + 2x + 3$ over $\mathbb{F}_{97}$ has exactly 100 points. **Hasse's theorem** says the count N always satisfies $|N - (p+1)| \le 2\sqrt{p}$; here $|100 - 98| = 2 \le 19.7$. The discrete logarithm in these groups appears to be much harder than in $\mathbb{Z}_p^\times$, which is why a 256-bit elliptic-curve key offers security comparable to a 3072-bit RSA key.

? The shortcut that elliptic curves seem to lack

For $\mathbb{Z}_p^\times$ and for factoring there are sub-exponential "index calculus" attacks that exploit the fact that ordinary integers factor into small primes. For a well-chosen elliptic curve, the best known attack is generic and takes about $\sqrt{N}$ steps. Is there a hidden notion of "small" points on a curve that would let index calculus in? Nobody knows; if there were, much of the internet would need new locks. The current state of play is in Ch. 78.

Pseudorandomness and codes

Number theory also manufactures *fake* randomness. A **linear congruential generator** iterates $x_{k+1} = (a x_k + c) \bmod m$. Its outputs look random but sit on a few lattice planes when plotted in pairs or triples, and the Hull–Dobell theorem says the period reaches its maximum m exactly when $\gcd(c, m) = 1$, $a - 1$ is divisible by every prime factor of m, and $a - 1$ is divisible by 4 if m is. The Mersenne Twister, the default generator in NumPy's legacy interface and many languages, has period $2^{19937} - 1$, a Mersenne prime. None of these generators is safe for cryptography: Ch. 81 explains what "unpredictable" must mean. The same finite fields carry **error-correcting codes**: a Reed–Solomon code treats a block of data as the coefficients of a polynomial over $\mathbb{F}_{2^8}$ and transmits its values at extra points, so that a few corrupted bytes can be located and repaired (Ch. 79).

57.4 The algorithm

Three short routines carry nearly all of computational number theory: fast exponentiation, the extended Euclidean algorithm, and a randomised primality test.

```
import random

def power_mod(a, k, n):
    """a^k mod n by square-and-multiply: O(log k) multiplications."""
    result, a = 1, a % n
    while k > 0:
        if k & 1:                # this binary digit of k is 1
            result = result * a % n
        a = a * a % n            # a, a^2, a^4, a^8, ...
        k >>= 1
    return result

def egcd(a, b):
    """Return (g, x, y) with a*x + b*y = g = gcd(a, b)."""
    x0, y0, x1, y1 = 1, 0, 0, 1
    while b:
        q, a, b = a // b, b, a % b
        x0, x1 = x1, x0 - q * x1
        y0, y1 = y1, y0 - q * y1
    return a, x0, y0

def inverse_mod(a, n):
    g, x, _ = egcd(a, n)
    if g != 1:
        raise ValueError("not invertible")
    return x % n

def is_probable_prime(n, rounds=20):
    """Miller-Rabin: a composite survives one round with probability <= 1/4."""
    if n < 4:
        return n in (2, 3)
    d, s = n - 1, 0
    while d % 2 == 0:
        d, s = d // 2, s + 1
    for _ in range(rounds):
        x = power_mod(random.randrange(2, n - 1), d, n)
        if x in (1, n - 1):
            continue
        for _ in range(s - 1):
            x = x * x % n
            if x == n - 1:
                break
        else:
            return False         # a is a witness: n is certainly composite
    return True

print(power_mod(65, 17, 3233), inverse_mod(17, 3120))       # 2790 2753
print(is_probable_prime(561), is_probable_prime(2**127 - 1))  # False True
```

Cost. Square-and-multiply reads the exponent bit by bit, so $a^k \bmod n$ costs at most $2\log_2 k$ modular multiplications. For a 2048-bit RSA exponent that is about 4,000 multiplications instead of 2^{2048}. Euclid's algorithm takes $O(\log \min(a, b))$ division steps; its worst case is consecutive Fibonacci numbers (Lamé, 1844). These two facts are the "easy direction" of the hook.

Why Miller–Rabin and not plain Fermat. Fermat's theorem suggests testing whether $a^{n-1} \equiv 1 \pmod n$. Composites can pass: $341 = 11 \cdot 31$ fools base 2, and **Carmichael numbers** such as $561 = 3 \cdot 11 \cdot 17$ fool *every* base coprime to them. Miller–Rabin adds a second check. Since n prime makes $\mathbb{Z}_n$ a field, the only square roots of 1 are ± 1; the inner loop watches the chain $a^d, a^{2d}, a^{4d}, \ldots$ for a square root of 1 that is not ± 1. For an odd composite, at least three quarters of the bases expose it, so 20 rounds leave an error chance below $4^{-20} \approx 10^{-12}$. (A deterministic polynomial-time test exists, AKS from 2002, but it is far slower in practice.)

> **Common Pitfall: textbook RSA is not secure RSA**
>
> The toy RSA above is deterministic: the same message always encrypts to the same ciphertext, so an attacker can test guesses. Encrypting m^e with a tiny m and $e = 3$ may never wrap mod n, and an ordinary cube root recovers it. Real systems pad messages with structured randomness (OAEP), use $e = 65537$, and use constant-time arithmetic so that the *timing* of square-and-multiply doesn't leak the bits of d. The number theory is the easy part; Ch. 77 covers the rest.

57.5 The experiment

The script counts primes, hunts for composites that fool the Fermat test, and measures the period of a linear congruential generator, plotting successive outputs so you can see the lattice structure hidden inside "random" numbers.

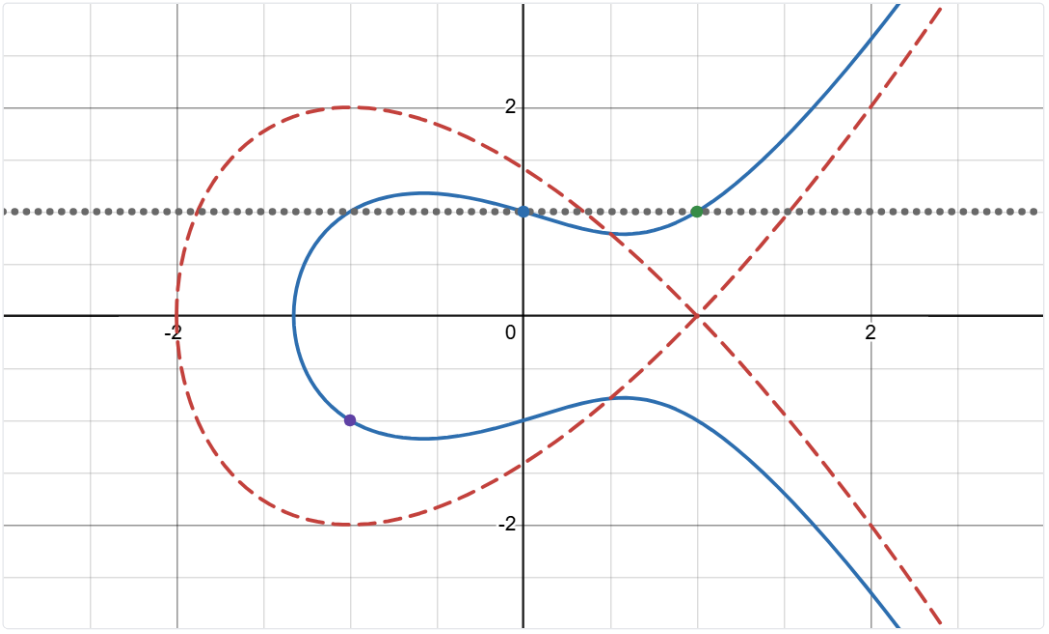
```
import numpy as np
import matplotlib.pyplot as plt

N = 100_000        # knob: search range for primes and pseudoprimes
BASES = [2]        # knob: Fermat-test bases, try [2, 3] or [2, 3, 5, 7]
a, c, m = 21, 7, 64  # knob: linear congruential generator x -> (a x + c) mod m

sieve = np.ones(N + 1, bool); sieve[:2] = False
for p in range(2, int(N ** 0.5) + 1):
    if sieve[p]:
        sieve[p * p::p] = False

def fermat(n):           # "probably prime" if b^(n-1) = 1 (mod n) for every base
    return all(pow(b, n - 1, n) == 1 for b in BASES if b % n)

liars = [n for n in range(3, N + 1, 2) if fermat(n) and not sieve[n]]
print("primes below N:", sieve.sum(), "   N/ln N =", round(N / np.log(N)))
print("composites that fool the test:", len(liars), liars[:8])

# period of the LCG from seed 0
x, seen = 0, {}
for step in range(m + 1):
    if x in seen:
        print("LCG period:", step - seen[x], "out of a possible", m)
        break
    seen[x] = step
    x = (a * x + c) % m

xs = [0]
for _ in range(2000):
    xs.append((a * xs[-1] + c) % m)
xs = np.array(xs) / m
plt.plot(xs[:-1], xs[1:], ".", ms=4)
plt.title(f"successive LCG outputs, a={a}, c={c}, m={m}")
plt.xlabel("$x_n/m$"); plt.ylabel("$x_{n+1}/m$"); plt.show()
```

With the settings shown it finds 9,592 primes below 10^5 (against $N/\ln N \approx 8{,}686$) and 78 odd composites that fool base 2, starting 341, 561, 645, 1105. With `BASES = [2, 3, 5, 7]` only three survive: 29341, 46657 and 75361, all Carmichael numbers. The LCG reaches its full period 64, and its 64 points fall on a handful of parallel lines.

Things to change:

- Set `a = 20`. The Hull–Dobell condition fails ($a - 1 = 19$ shares no factor with 64), and the generator collapses into a fixed point. Try `c = 8` with `a = 21`: now $\gcd(c, m) \neq 1$ and the period drops to 8.
- Set `a, c, m = 17, 1, 256`. Hull–Dobell holds, so the period is the full 256, yet the plotted pairs fall on just a few steep lines. A long period is not the same thing as randomness.
- Increase `N` to 10^6 and compare `liars` for base 2 with the Carmichael numbers you find with many bases. Do the Carmichael numbers get rarer or more common relative to the primes?

The elliptic curves of §57.3 also have a real picture. The graph shows a smooth curve and a singular one:



$y^2 = x^3 - x + 1$ (blue) with $P = (0, 1)$ and $Q = (1, 1)$. The line through them meets the curve again at $(-1, 1)$; reflecting gives $P + Q = (-1, -1)$. The red curve has $4a^3 + 27b^2 = 0$ and crosses itself at $(1, 0)$, where the group law breaks. Change $-3x + 2$ to $-3x + 3$ and the crossing opens into a smooth curve; change the blue curve's $+1$ to $+0.3$ and it splits into two pieces.

57.6 The challenge

No answers are given.

1. **Intuition.** Explain without symbols why the times-table picture for $i \mapsto 2i \bmod n$ approaches a cardioid as n grows, while $i \mapsto (n/2+1)\,i \bmod n$ (for even n) looks completely different. Which pictures are permutations of the points, and how can you tell from $\gcd(k, n)$?
2. **Derivation.** Prove that if $2^k + 1$ is prime then k is a power of 2, and that if $2^k - 1$ is prime then k is prime. Then show that every Carmichael number is odd and has at least three prime factors.
3. **Implementation.** Write a function that counts points on $y^2 = x^3 + ax + b$ over $\mathbb{F}_p$ for all a, b with $p = 101$, and plot the histogram of $N - (p+1)$. Check Hasse's bound, then implement the chord-and-tangent addition and find a point whose multiples visit the whole group.
4. **Open investigation.** Twin primes are pairs $p, p+2$ both prime. Count them below x for x up to 10^7 and compare with the Hardy–Littlewood prediction $2C_2\,x/(\ln x)^2$ with $C_2 \approx 0.6602$. Then look at the *last digits* of consecutive primes: is a prime ending in 1 more likely to be followed by one ending in 1, 3, 7 or 9? (This bias was only noticed in 2016.)

57.7 Loose threads

The hook asked how arithmetic can be easy one way and hard the other. Modular reduction throws away size, so the landmarks that make logarithms and factoring easy over the real numbers vanish. Euclid's algorithm and square-and-multiply keep the forward direction fast, Euler's theorem guarantees that the RSA exponents e and d undo each other, and Diffie–Hellman lets two strangers build g^{ab} in public. The hard direction, factoring or taking discrete logarithms, has resisted every classical attack for almost fifty years. That is a belief, not a theorem; no one has proved that these problems are hard.

What we did not tell you:

- **Why the primes are where they are.** The prime number theorem gives the average density, but the error term, how far $\pi(x)$ strays from its smooth approximation, is controlled by the zeros of a single complex function.
- **What a quantum computer changes.** The first mystery points to Ch. 77.
- **Whether elliptic curves really are harder.** The second mystery points to Ch. 78, and the geometry of such curves, which links them to Fermat's last theorem, is the territory of Ch. 94.

> **? A million dollars for the zeros**
>
> Euler noticed that $\sum_{n\ge 1} n^{-s} = \prod_{p\ \text{prime}} (1 - p^{-s})^{-1}$, which ties all integers to all primes. Riemann extended this **zeta function** to complex s using the analytic continuation of Ch. 18 and conjectured in 1859 that all its non-trivial zeros lie on the line $\operatorname{Re}\{s\} = \tfrac{1}{2}$. If true, the primes are as evenly spread as a random model allows: $\pi(x)$ differs from its smooth approximation by roughly $\sqrt{x}\ln x$. More than ten trillion zeros have been checked. Nobody has a proof; it is one of the Clay Millennium Problems, and how mathematicians attack such problems is the subject of Ch. 98.

Further reading

- J. H. Silverman, *A Friendly Introduction to Number Theory*, 4th ed., Pearson, 2012. Exactly what the title says, from primes to RSA and elliptic curves.
- M. du Sautoy, *The Music of the Primes*, HarperCollins, 2003. A popular history of the Riemann hypothesis.
- J. Hoffstein, J. Pipher and J. H. Silverman, *An Introduction to Mathematical Cryptography*, 2nd ed., Springer, 2014. The number theory of public-key systems, done carefully.
- R. Crandall and C. Pomerance, *Prime Numbers: A Computational Perspective*, 2nd ed., Springer, 2005. Sieves, primality tests and factoring algorithms in depth.
- D. E. Knuth, *The Art of Computer Programming*, Vol. 2: *Seminumerical Algorithms*, 3rd ed., Addison-Wesley, 1997. The classic analysis of congruential generators and the spectral test.

Chapter 58 — Topology: What Survives When Shapes Bend?

> **Level 2 · Track E — The Abstract Mathematics Behind Everything Builds on:** Ch. 4, Ch. 18, Ch. 56, Ch. 27 · **Leads to:** Ch. 59, Ch. 60, Ch. 62 · **Related:** Ch. 39, Ch. 55, Ch. 75, Ch. 83 **Reading time:** ~25 min

58.1 The hook

A factory robot has a single rotating joint: a shoulder that can spin all the way round. Its controller needs a **motion planner**, a rule that takes the current angle a and a target angle b and returns a path from one to the other. The engineers add one sensible requirement. The planner should be **continuous**: if you nudge the target slightly, the planned path should change only slightly. Nobody wants a robot that swings left for a target at 179.9° and right for one at 180.1°.

The obvious rule is "take the shorter way round". It is perfect, until the target is directly opposite the start. Then the two ways are equally short and the planner must pick one; just past that point it must suddenly switch to the other. Fine, say the engineers, always turn counter-clockwise. Now a target just *clockwise* of the start is reached by turning almost a full circle, while a target exactly at the start needs no motion at all. Again a jump.

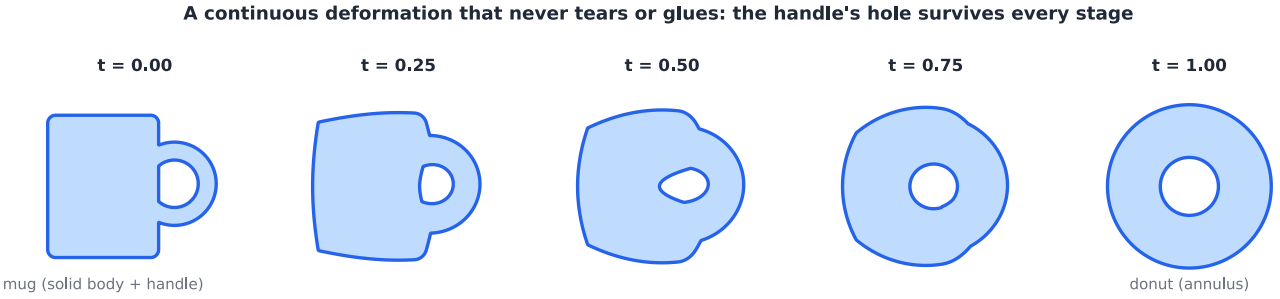
Every rule the team tries breaks somewhere. Two joints, a shoulder and an elbow, make things worse, and a drone's orientation in 3-D space is worse still.

> **Is there any continuous motion planner for a rotating joint, or is a discontinuity unavoidable? And if it is unavoidable, what property of the space of poses forces it?**

The answer has nothing to do with angles, lengths, motors or the particular formula you write. It depends only on the *shape* of the space of poses, in a sense of "shape" so loose that a coffee mug and a doughnut count as the same. That sense of shape is topology.

58.2 The big idea

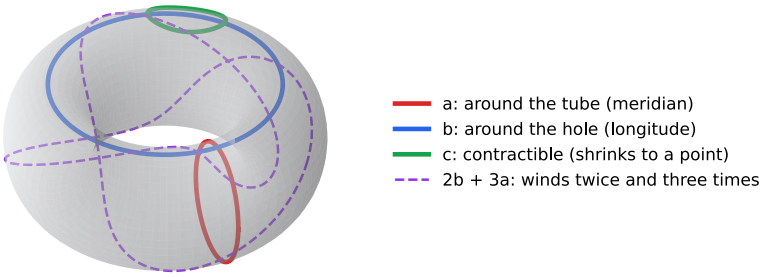
Geometry measures: lengths, angles, curvature. Topology asks what is left when you are allowed to **stretch, bend and squash, but never tear or glue**. A circle and a square are the same; so is a wobbly closed loop. A doughnut and a coffee mug are the same, because the mug's handle *is* the doughnut's hole and the cup part can be squashed into the handle's base.



A continuous deformation from a mug's silhouette to a donut's. Each stage is a blend of the two shapes' distance functions; the hole through the handle is never created or destroyed, only moved and widened.

So what *can't* be bent away? Holes. Pieces. Whether a loop can be shrunk to a point. Those are the **topological invariants**, and they are the whole point of the subject: if two spaces differ in an invariant, no amount of clever stretching turns one into the other.

Now think of the robot again. The set of all its poses is a space. One rotating joint gives a circle S^1: angles 0° and 360° are the same pose. Two joints give a pair of angles, each on a circle, which is a **torus**, the surface of a doughnut.



Configuration space of a two-joint arm. Loop a spins the elbow once, loop b spins the shoulder once, loop c wiggles a little and comes back. Only c can be shrunk to a point while staying on the surface. Any loop is, up to deformation, "m times b plus n times a".

Here is the intuition for the hook. On a circle, a loop that goes once round can't be shrunk to a point without leaving the circle. A continuous motion planner would let you *build* such a shrinking, as §58.3 shows, so it can't exist. The obstacle is a hole in the configuration space, and topology turns "there is a hole" into a theorem.

The same thinking applies to data. A cloud of sensor readings has no edges or surfaces, only points. But if you thicken every point into a small ball and watch which holes appear and disappear as the balls grow, you can measure the shape of the data. That is **topological data analysis**, and it closes the chapter.

58.3 The mathematics

Open sets and continuity

In Ch. 4 §4.1, f is continuous at x_0 if for every $\varepsilon > 0$ there is a $\delta > 0$ with $|x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \varepsilon$. Topology rewrites that definition without distances. A set $U \subseteq \mathbb{R}^n$ is **open** if around each of its points there is a small ball still inside U. Then, as you can check by unpacking the definitions, f is continuous everywhere exactly when **the preimage of every open set is open**.

That reformulation mentions only open sets, so it can be used on any set where we have *declared* which subsets are open.

> **Topological space, continuity, homeomorphism**
>
> A **topology** on a set X is a collection of subsets called open sets, containing $\varnothing$ and X, and closed under arbitrary unions and finite intersections. A map $f : X \to Y$ between topological spaces is **continuous** if $f^{-1}(V)$ is open for every open $V \subseteq Y$. A **homeomorphism** is a continuous bijection whose inverse is also continuous; X and Y are then **homeomorphic**, written $X \cong Y$.

The inverse must be continuous too. Wrapping the interval $[0, 2\pi)$ onto the circle by $t \mapsto (\cos t, \sin t)$ is a continuous bijection, but it is not a homeomorphism: points near $(1, 0)$ on the circle come from both ends of the interval, so the inverse tears the circle open. Indeed the two spaces are not homeomorphic, as we are about to prove.

Connectedness and the intermediate value theorem

> **Connected**
>
> A space X is **connected** if it cannot be written as the union of two disjoint non-empty open sets.

The real interval $[a, b]$ is connected. (Sketch: if $[a, b] = U \cup V$ with $a \in U$, let c be the supremum of the points t for which all of $[a, t]$ lies in U; openness of U and V makes c lie in neither, a contradiction. This is where completeness of $\mathbb{R}$ enters, the subject of Ch. 60.)

> **Continuous images of connected spaces are connected**
>
> If $f : X \to Y$ is continuous and X is connected, then $f(X)$ is connected.

Proof. Suppose $f(X) = U \cup V$ with U, V disjoint, non-empty and open in $f(X)$. Then $f^{-1}(U)$ and $f^{-1}(V)$ are open (by continuity), disjoint, non-empty, and cover X. That separates X, which is impossible. ■

Apply this to a continuous $f : [a, b] \to \mathbb{R}$. The image is a connected subset of $\mathbb{R}$, and the connected subsets of $\mathbb{R}$ are exactly the intervals. So if $f(a) < 0 < f(b)$, the value 0 lies in the image: **the intermediate value theorem** is a statement about connectedness. It is the guarantee behind bisection root-finding in Ch. 22 §22.4.

Connectedness is also a cheap invariant. Remove one point from an open interval and it falls into two pieces; remove any one point from a circle and it stays in one piece. A homeomorphism would carry "interval minus a point" to "circle minus a point", so **an interval and a circle are not homeomorphic**. Similarly, the letter X is not homeomorphic to the letter O (removing the centre of X leaves four pieces).

Compactness

Compact

A space is **compact** if every covering of it by open sets has a finite sub-covering. In $\mathbb{R}^n$ (Heine–Borel), the compact sets are exactly the closed and bounded ones.

Compactness is "finiteness in disguise". Its key property has the same one-line proof shape as connectedness: the continuous image of a compact space is compact (pull back an open cover, pick a finite sub-cover, push it forward). Since compact subsets of $\mathbb{R}$ are closed and bounded, a continuous real function on a compact space is bounded and **attains** its maximum and minimum. That is the extreme value theorem, the existence guarantee behind every "minimise the cost over a closed, bounded design region" problem in Ch. 23. Over the open interval $(0,1)$, $f(x) = 1/x$ has no maximum: the open interval is not compact. So $[0,1] \not\cong (0,1)$, even though both are connected.

Homotopy and the fundamental group

Connectedness counts pieces. To detect holes we need loops. Two continuous maps $f, g : X \to Y$ are **homotopic** if one can be deformed continuously into the other: there is a continuous $H : X \times [0,1] \to Y$ with $H(x,0) = f(x)$ and $H(x,1) = g(x)$. You have seen this idea before. Deforming a contour of integration without crossing a singularity, in Ch. 18 §18.5, is a homotopy, and Cauchy's theorem says the integral cannot tell the two contours apart.

Fundamental group

Fix a base point $x_0 \in X$. The **fundamental group** $\pi_1(X, x_0)$ is the set of loops starting and ending at x_0, with two loops identified if they are homotopic through loops based at x_0. The product of two loops is "do one, then the other". The identity is the constant loop, and the inverse runs a loop backwards.

The group axioms of Ch. 56 hold *up to homotopy*: for example, running a loop and then its reverse is not literally the constant loop, but it shrinks to it. A space is **simply connected** if π_1 is trivial, meaning every loop shrinks to a point. Some answers, stated without proof:

Space	π_1	What a loop "remembers"
$\mathbb{R}^n$, a disk, a ball	trivial	nothing
circle S^1	$\mathbb{Z}$	how many times it winds round
torus $S^1 \times S^1$	$\mathbb{Z} \times \mathbb{Z}$	windings (m, n) round each circle
sphere S^2	trivial	nothing: a rubber band slides off a ball
figure-eight	free group on two letters	the word in a, b, a^{-1}, b^{-1} it spells; $ab \neq ba$
plane minus a point	$\mathbb{Z}$	the winding number about the point

The entry for the circle is the deep one. The integer attached to a loop is its **winding number**. In the complex plane, the winding number of a closed curve γ about the origin is $\frac{1}{2\pi j}\oint_\gamma \frac{dz}{z}$, the integral behind the argument principle and the Nyquist stability criterion of Ch. 18 §18.9. Proving that it is a *complete* invariant, that loops with equal winding numbers are homotopic and loops with different ones are not, uses the lifting of loops on the circle to paths on the real line; we take it as given.

No continuous motion planner on a circle

There is no continuous rule s that assigns to each pair of angles $(a, b) \in S^1 \times S^1$ a path $s(a,b) : [0,1] \to S^1$ from a to b, depending continuously on (a, b).

Proof. Suppose s existed. Fix a starting pose a_0 and define $H(b,t) = s(a_0, b)(t)$ for $b \in S^1$, $t \in [0,1]$. By assumption H is continuous. At $t = 0$, $H(b,0) = a_0$ for every b: the constant map. At $t = 1$, $H(b,1) = b$: the identity map of the circle. So H is a homotopy from the constant map to the identity. Now trace the identity map once round the circle: it is the loop of winding number 1. The homotopy H deforms it continuously, through closed loops $b \mapsto H(b,t)$, into a constant loop, which has winding number 0. On the circle the winding number cannot change under such a deformation, even one that moves the base point, because it is an integer that varies continuously. So $1 = 0$, a contradiction. ■

That answers the hook. A discontinuity is unavoidable for any planner on a single rotating joint, and the reason is the hole in the circle, $\pi_1(S^1) \cong \mathbb{Z} \neq 0$. The best you can do is split the configuration pairs into pieces and use a different continuous rule on each. For the circle two pieces suffice: on pairs that are not exactly opposite, take the shorter way round; on opposite pairs, always turn counter-clockwise. Each rule is continuous on its own piece. The minimum number of pieces is a topological invariant of the space, Farber's **topological complexity**, and it grows with the number of independent holes. A two-joint arm on a torus needs three pieces.

? The loop test for spheres

In two dimensions the sphere is the only closed surface on which every loop shrinks to a point. In 1904 Henri Poincaré asked whether the same is true one dimension up: is every closed, simply connected 3-dimensional space homeomorphic to the 3-sphere? It took a century. Grigori Perelman proved it in 2003 by letting the *geometry* of the space flow like heat, smoothing curvature until the shape revealed itself, and then declined both the Fields Medal and a million-dollar prize. How can a curvature flow prove a purely topological fact? Curvature is the subject of Ch. 59.

Homology: counting holes with linear algebra

The fundamental group is powerful but hard to compute, and non-abelian groups are awkward to compare. **Homology** trades some information for computability: it counts holes of each dimension with ranks of matrices.

Build a space out of simple pieces: vertices, edges, filled triangles (a **simplicial complex**). Write C_k for the vector space, over $\mathbb{F}_2 = \{0, 1\}$ to keep signs out of it, spanned by the k-dimensional pieces. The **boundary map** $\partial_k : C_k \to C_{k-1}$ sends an edge to the sum of its two end vertices and a triangle to the sum of its three edges. The key identity is

$$\partial_{k-1} \circ \partial_k = 0,$$

because the boundary of a boundary is empty: each vertex of a triangle appears in exactly two of its edges, and $1 + 1 = 0$ in $\mathbb{F}_2$. So every boundary is a **cycle** (an element of $\ker \partial$), and the k-th homology counts cycles that are *not* boundaries:

$$H_k = \ker \partial_k / \operatorname{im} \partial_{k+1}, \qquad \beta_k = \dim H_k = \dim \ker \partial_k - \operatorname{rank} \partial_{k+1}.$$

This is the quotient construction of Ch. 56 applied to vector spaces, and the dimensions come from rank–nullity (Ch. 8 §8.5). The **Betti number** β_0 counts connected pieces, β_1 counts independent loops that bound nothing, and β_2 counts enclosed voids.

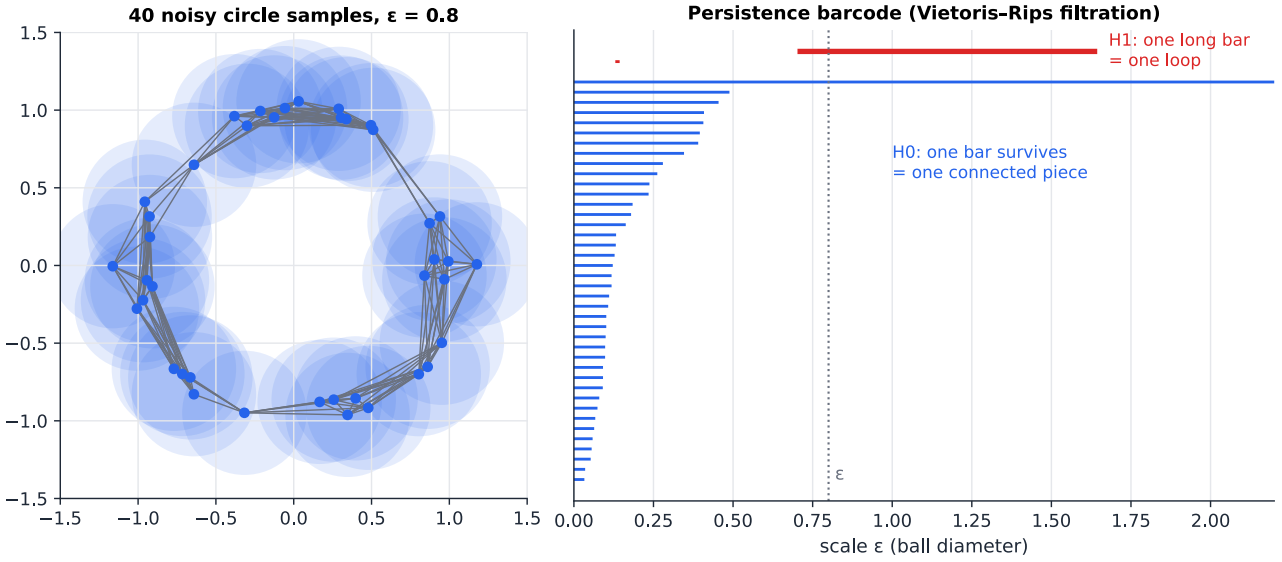
> **A hollow triangle versus a filled one**
>
> Take vertices u, v, w and edges uv, vw, wu. The boundary map ∂_1 is a 3×3 matrix over $\mathbb{F}_2$ in which every column has two 1s; its columns sum to zero, so its rank is 2. Then $\beta_0 = 3 - 2 = 1$ (one piece), and $\dim \ker \partial_1 = 3 - 2 = 1$. With no triangle filled in, $\operatorname{rank} \partial_2 = 0$, so $\beta_1 = 1$: one loop, the hole. Fill in the triangle uvw: now ∂_2 sends it to $uv + vw + wu$, which is exactly the cycle, so $\operatorname{rank} \partial_2 = 1$ and $\beta_1 = 0$. The hole is gone.

For the surfaces we have met, the answers are: circle $(\beta_0, \beta_1) = (1, 1)$; sphere $(\beta_0, \beta_1, \beta_2) = (1, 0, 1)$; torus $(1, 2, 1)$, one piece, two independent loops a and b, and one enclosed void. The alternating sum $\chi = \beta_0 - \beta_1 + \beta_2$ is the **Euler characteristic**, and a short rank–nullity computation shows it also equals $V - E + F$, counted on *any* triangulation. A cube has $8 - 12 + 6 = 2$, like the sphere; a torus has 0. That is why Euler's polyhedron formula $V - E + F = 2$ holds for every convex polyhedron: they are all homeomorphic to a sphere.

> **? Why you can't comb a hairy ball**
>
> Because $\chi(S^2) = 2 \neq 0$, every continuous field of tangent arrows on a sphere must vanish somewhere: at any moment there is a point on Earth where the horizontal wind speed is zero, and a smooth "hairy ball" always has a cowlick. The torus, with $\chi = 0$, can be combed flat. A cousin of this result, Brouwer's theorem that every continuous map of a disk to itself has a fixed point, is exactly what John Nash needed to prove that every finite game has an equilibrium. How does a statement about shapes guarantee a stable strategy? See Ch. 39.

From shapes to data: persistent homology

A point cloud has $\beta_0 =$ number of points and $\beta_1 = 0$: on its own it has no interesting homology. The fix is to look at it **at every scale**. For each $\varepsilon \geq 0$, the **Vietoris–Rips complex** R_ε has a vertex per data point, an edge between any two points closer than ε, and a filled triangle on any three points that are pairwise closer than ε. As ε grows, simplices are only ever added, so the complexes are nested: $R_\varepsilon \subseteq R_{\varepsilon'}$ for $\varepsilon < \varepsilon'$. A hole is **born** at the scale where its loop closes and **dies** when it gets filled in. Plotting every hole as a bar from birth to death gives a **persistence barcode**.



Left: 40 noisy samples of the unit circle, with edges drawn at scale $\varepsilon = 0.8$. Right: the barcode. Many short H_0 bars die quickly as nearby points merge; one infinite bar survives. In H_1 a single long bar, born near $\varepsilon \approx 0.70$ when the ring closes and dying near 1.64 when triangles fill the disk, says "this data has one loop". A tiny H_1 bar near 0.13 is noise.

The principle is that **long bars are features and short bars are noise**. It is used on protein structures, on the activity of neurons that encode an animal's head direction (the data really does lie on a circle), on sensor networks checking for coverage holes, and on time series embedded with delays to detect periodic behaviour in a dynamical system (Ch. 83).

58.4 The algorithm

Computing a barcode is linear algebra over $\mathbb{F}_2$, done in a clever order.

1. **Build the filtration.** List every simplex with the scale at which it appears: points at 0, an edge at its length, a triangle at its longest edge. Sort by scale, breaking ties so that every face comes before the simplices it bounds.
2. **Form the boundary matrix D**, with one column per simplex in that order. Column k holds the faces of simplex k (two vertices for an edge, three edges for a triangle).
3. **Reduce the columns left to right.** Let $\mathrm{low}(k)$ be the row of the lowest 1 in column k. While some earlier column has the same low, add it to column k (mod 2). Adding *earlier* columns only is what keeps the answer faithful to the order of the filtration.
4. **Read off the bars.** If reduced column k is non-zero with $\mathrm{low}(k) = i$, simplex i created a hole that simplex k killed: a bar from the scale of i to the scale of k, in dimension $\dim(i)$. A simplex that creates a hole and is never paired gives an infinite bar.

```
for k in 0 .. N−1:                      # columns in filtration order
    while column k is non-zero and low(k) == low(j) for some j < k:
        column k += column j  (mod 2)
    if column k is non-zero:
        record bar  [ scale(low(k)),  scale(k) )
```

Why does this work? Step 3 is Gaussian elimination restricted to adding columns from the left, so at every stage ε the reduced matrix still has the same rank structure as the boundary maps of R_ε. A zero column means the new simplex closed a cycle, so it gives birth to a homology class. A non-zero column means it filled one in, and the lowest-1 rule pairs it with the *youngest* class it kills, the "elder rule" that makes barcodes well defined.

Cost and failure cases. The reduction is cubic in the number of simplices in the worst case, and the number of simplices explodes: n points give up to $\binom{n}{3}$ triangles, about 1.7×10^8 for $n = 1000$. Practical software (Ripser, GUDHI) caps the maximum scale, computes cohomology instead (which reduces far fewer columns), and never stores the full matrix. Rips complexes are also sensitive to **outliers**: a single stray point can split a long bar in two, which is why robust variants weight points by local density. And barcodes depend on the metric you choose; Euclidean distance on raw features with different units can create holes that are artefacts of scaling.

58.5 The experiment

The script samples a noisy circle, builds the full Rips filtration up to triangles, runs the column reduction above, and plots the bars longer than 0.02.

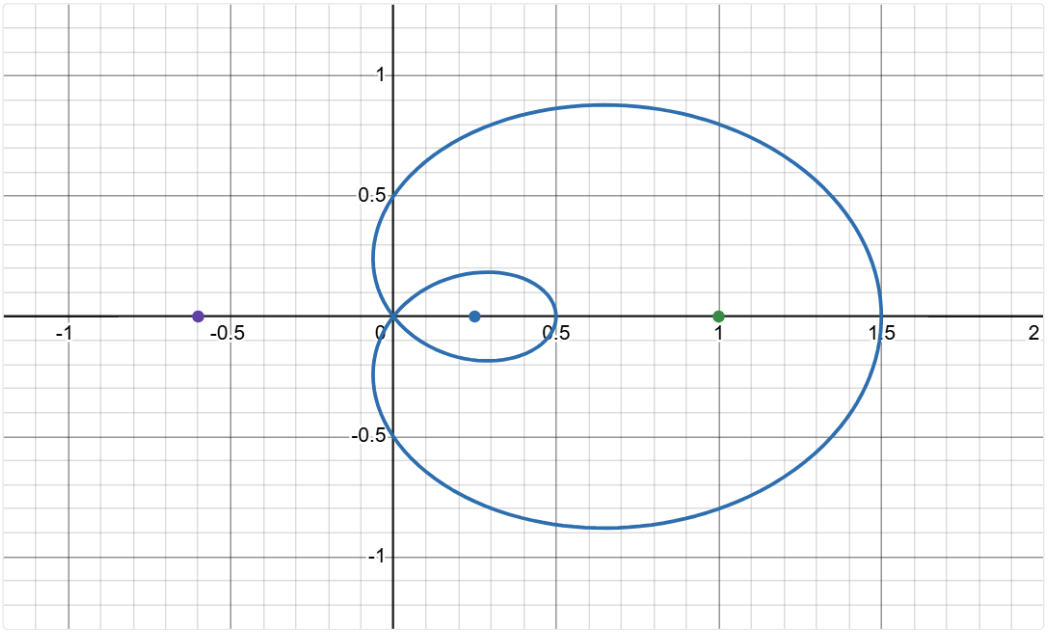
```
import numpy as np
import matplotlib.pyplot as plt
from itertools import combinations

n, noise, seed = 40, 0.08, 7   # knobs: number of points, noise level, random seed
two_loops = False               # knob: True samples a figure-eight instead of a circle
rng = np.random.default_rng(seed)
t = rng.uniform(0, 2 * np.pi, n)
P = np.c_[np.cos(t), np.sin(t)]
if two_loops:
    P[n // 2:, 0] += 2.2                 # second circle, touching the first
P += rng.normal(0, noise, P.shape)
D = np.linalg.norm(P[:, None] − P[None], axis=2)

# Vietoris–Rips filtration up to triangles: each simplex enters at its longest edge
S = [(0.0, (i,)) for i in range(n)]
S += [(D[i, j], (i, j)) for i, j in combinations(range(n), 2)]
S += [(max(D[i, j], D[i, k], D[j, k]), (i, j, k)) for i, j, k in combinations(range(n), 3)]
S.sort(key=lambda s: (s[0], len(s[1])))
index = {s: k for k, (_, s) in enumerate(S)}

cols, pivot, bars = [], {}, []
for k, (val, s) in enumerate(S):           # column reduction over GF(2)
    col = {index[f] for f in combinations(s, len(s) − 1)} if len(s) > 1 else set()
    while col and max(col) in pivot:
        col ^= cols[pivot[max(col)]]
    cols.append(col)
    if col:
        pivot[max(col)] = k
        birth = S[max(col)][0]
        bars.append((len(s) − 2, birth, val))   # dimension, birth, death

for dim in (0, 1):
    life = sorted((d − b for k, b, d in bars if k == dim), reverse=True)
    print(f"H{dim}: longest finite bars", np.round(life[:4], 3))
for k, b, d in bars:
    if d − b > 0.02:
        plt.plot([b, d], [k + rng.uniform(−0.3, 0.3)] * 2, color="C%d" % k, lw=2)
plt.yticks([0, 1], ["H0", "H1"]); plt.xlabel("scale"); plt.show()
```

With the settings shown, the longest finite H_1 bar has length 0.942 and the next one 0.014: one loop, unmistakably. (The infinite H_0 bar is not listed, since it never dies.) Things to try:

- Set `two_loops = True`. Two H_1 bars appear, but the second is short (0.32) because each circle has only 20 points. Raise `n` to 80 and both bars are long (about 1.09 and 0.92). Sampling density, not just shape, controls what you can see.

- Raise `noise` to 0.3. The H_1 bar shrinks to about 0.67 and a second bar of 0.23 appears. Where would you draw the line between feature and noise? Try several seeds before you decide.
- Replace the circle with points on a line segment (`P = np.c_[t, 0 * t]`). Every H_1 bar should vanish. Then sample a filled disk; why does it look like the segment, not like the circle?

The winding number of §58.3 is easy to explore by eye. The limaçon below winds twice round points inside its small loop, once round points in the big loop only, and not at all round points outside.



$r = 0.5 + \cos\theta$. Change 0.5 to 1.2 and the inner loop disappears (every inside point now has winding number 1); change it to 0 and the curve becomes a circle traced twice. Count the winding numbers in each case by tracing the curve with a finger.

58.6 The challenge

No answers are given.

1. **Intuition.** Classify the capital letters A to Z (in a plain sans-serif font, drawn as curves of zero thickness) up to homeomorphism. Which invariant separates the classes you cannot tell apart by counting holes? (Hint: remove a point.)
2. **Derivation.** Prove that every continuous map $f : [0,1] \to [0,1]$ has a fixed point, using only the intermediate value theorem. Then explain why the same argument says nothing about continuous maps of a circle to itself, and give a fixed-point-free example.
3. **Implementation.** Triangulate a torus as an $m \times m$ grid of squares, each cut into two triangles, with opposite edges identified. Build ∂_1 and ∂_2 over $\mathbb{F}_2$, compute their ranks, and verify $(\beta_0, \beta_1, \beta_2) = (1, 2, 1)$ and $V - E + F = 0$. Repeat with the identification that makes a Klein bottle; what changes, and what changes if you work over $\mathbb{R}$ instead of $\mathbb{F}_2$?
4. **Open investigation.** Record a periodic signal $x(t)$ (a sine wave, a heartbeat trace, a vowel sound) and embed it as points $(x(t), x(t+\tau))$ or in 3-D with a second delay. Compute barcodes as you vary τ and add noise. How well does the length of the longest H_1 bar work as a "periodicity score", and how does it compare with an autocorrelation peak (Ch. 21)?

58.7 Loose threads

The hook asked whether a robot joint can have a continuous motion planner. It cannot. Any such planner would continuously deform the loop that goes once round the circle into a point, and the winding number, the generator of $\pi_1(S^1) \cong \mathbb{Z}$, forbids that. The discontinuity is not a defect of any formula; it is forced by the hole in the configuration space. The same tools that prove this (connectedness, compactness, loops and homology) also gave the intermediate and extreme value theorems their real meaning, and, turned towards data, let us read the shape of a point cloud from its barcode.

What we did not tell you:

- **Why topology needs geometry to finish its biggest proofs.** The first mystery points to Ch. 59.
- **Why fixed points guarantee equilibria.** The second mystery points to Ch. 39.
- **What all these invariants have in common.** Every invariant in this chapter is a rule that turns spaces into algebraic objects (numbers, groups, vector spaces) and continuous maps into homomorphisms, compatibly with composition. That pattern is called a **functor**, and it is the founding example of Ch. 62.

? How far apart are two barcodes?

If you jiggle the data a little, does the barcode only jiggle a little? It does: the **stability theorem** of Cohen-Steiner, Edelsbrunner and Harer (2007) bounds the distance between barcodes by the size of the perturbation. But "distance between barcodes" needs a definition: it is a matching problem, pairing bars of one barcode with bars of the other at least cost, with unmatched bars sent to zero length. That is optimal transport in disguise, and the Wasserstein distance it produces is the subject of Ch. 55.

Further reading

- J. Stillwell, *Classical Topology and Combinatorial Group Theory*, 2nd ed., Springer, 1993. Surfaces, the fundamental group and its computation, with history.
- J. R. Munkres, *Topology*, 2nd ed., Prentice Hall, 2000. The standard first course in point-set and algebraic topology.
- D. S. Richeson, *Euler's Gem: The Polyhedron Formula and the Birth of Topology*, Princeton University Press, 2008. A popular account from $V - E + F = 2$ to the Poincaré conjecture.
- H. Edelsbrunner and J. Harer, *Computational Topology: An Introduction*, American Mathematical Society, 2010. Persistence and the reduction algorithm, from its inventors.
- M. Farber, *Invitation to Topological Robotics*, European Mathematical Society, 2008. Configuration spaces and topological complexity of motion planning.

Chapter 59 — Differential Geometry: Mathematics on Curved Worlds

> **Level 2 · Track E — The Abstract Mathematics Behind Everything Builds on:** Ch. 6, Ch. 7, Ch. 10 · **Leads to:** Ch. 93, Ch. 92, Ch. 70 · **Related:** Ch. 58, Ch. 55, Ch. 69 **Reading time:** ~30 min

59.1 The hook

New York and Madrid sit at almost the same latitude: 40.7° N and 40.4° N. Draw them on a classroom wall map and the obvious route is a straight line due east across the Atlantic, about 5,940 km along the 40th parallel.

No airline flies it. A flight leaving JFK for Madrid sets off on a heading of about 66°, well north of east. It climbs to about 46° N, passes south of Newfoundland, and then curves back down into Spain. That route is 5,770 km, roughly 170 km shorter than "straight east". On the wall map it looks like a detour. On a globe it is the straightest path there is.

So which line is straight? Your eyes, trained on flat paper, say the parallel. The globe says the arc that bulges toward the pole. The disagreement is not a quirk of map projections. It is a fact about the surface itself, and it can be detected without ever leaving it.

In the 1820s Carl Friedrich Gauss ran a land survey of the Kingdom of Hanover. As part of it he measured the angles of a huge triangle whose corners were three mountain tops, the Hohenhagen, the Brocken and the Inselsberg, with sides of roughly 69, 85 and 107 km. In a flat world the angles of every triangle add up to 180°. Gauss's added up to 180° plus about 15 seconds of arc. That is exactly what a triangle of that size on a sphere the size of the Earth should give. (Gauss probably also wondered whether *space itself* might be curved. His triangle was too small to tell, and the question waited a century for Einstein.)

Now push the idea to its extreme. Start at the North Pole. Walk due south to the equator. Turn left 90° and walk a quarter of the way around the equator. Turn left 90° again and walk north. You arrive back at the pole, having walked a triangle with **three right angles**: an angle sum of 270°.

Here is the question this chapter is built around:

> **How can a creature that lives *inside* a surface, and can only measure lengths and angles on it, discover that its world is curved, and by how much?**

The answer leads to manifolds, tangent spaces, geodesics and curvature: the language in which robots plan motions on curved configuration spaces, graphics engines measure distance across a mesh, and GPS satellites correct their clocks for the curvature of spacetime.

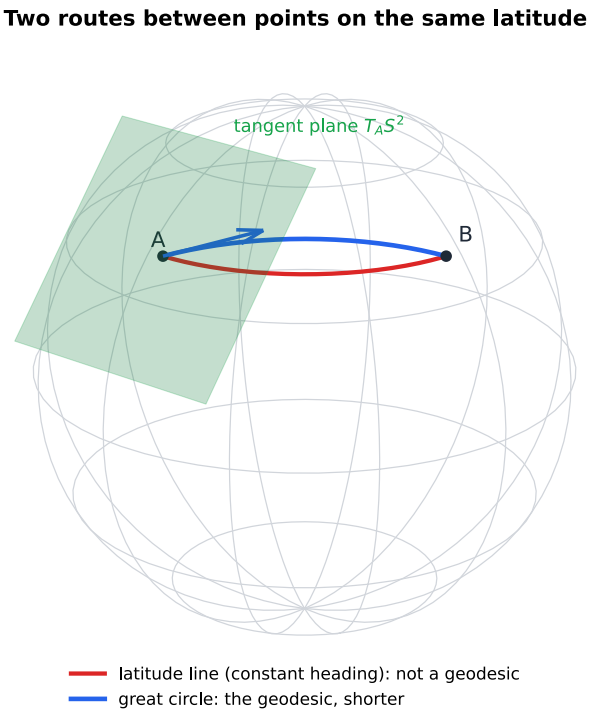
59.2 The big idea

Two ideas carry the whole subject.

Idea 1: curved worlds are flat up close. The Earth looks flat from your garden. Any smooth surface, looked at under a strong enough magnifier, looks like a piece of plane. A **manifold** is anything with this property: a space that, near each point, looks like ordinary $\mathbb{R}^n$. The flat approximation at a point p is the **tangent space** T_pM. Calculus happens there: velocities, gradients and linear approximations all live in tangent spaces, exactly as in §4.8.

The catch is that the flat pictures don't glue together into one flat picture. An atlas of the Earth needs many pages, and where two pages overlap they disagree about distances. That disagreement *is* the curvature.

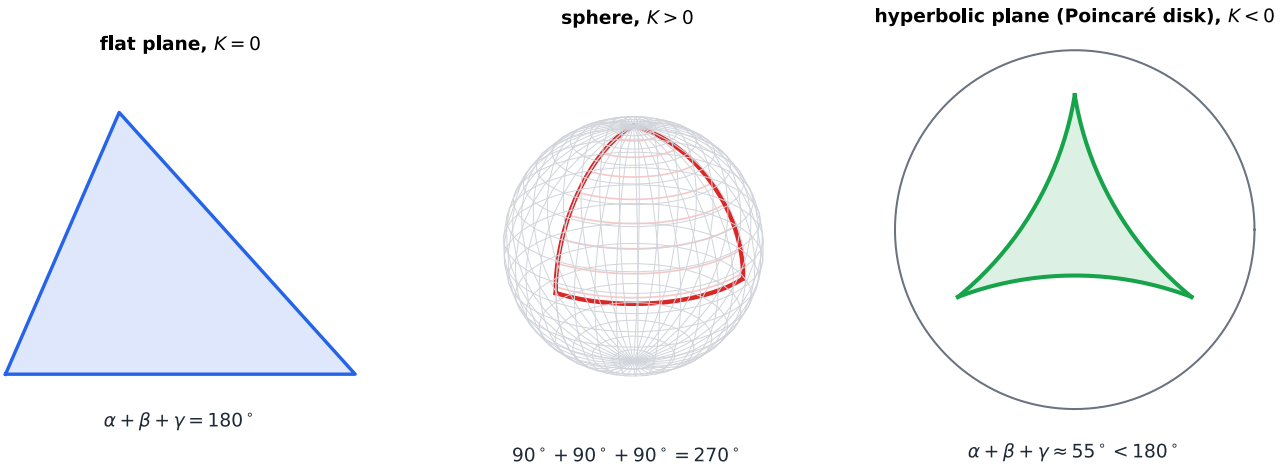
Idea 2: "straight" means "not turning, as seen from inside". An ant walking on a globe can't see that its path bends in three-dimensional space. All it can tell is whether it is steering left or right. A path along which the ant never steers is a **geodesic**. On a sphere the geodesics are the great circles: the equator and every circle cut by a plane through the centre. A latitude line other than the equator is *not* a geodesic. An ant walking along the 40th parallel has to keep steering slightly toward the pole, as you would when driving around a large roundabout.



Two routes from A to B on the same latitude. The great circle (blue) is the geodesic and is shorter. The latitude line (red) keeps a constant compass heading but is always turning, as seen from inside the surface. The green plane is the tangent space at A, where the geodesic's initial velocity lives.

How does an insider measure curvature? Through **triangles and circles**. Draw a triangle whose sides are geodesics and add up its angles:

- on a flat plane the sum is exactly 180°;
- on a sphere it is *more* than 180°, and the excess grows with the triangle's area;
- on a saddle-shaped surface it is *less* than 180°.



Angle sums reveal curvature. Flat: exactly 180°. Sphere: the octant triangle has three right angles. Hyperbolic plane, drawn in the Poincaré disk where geodesics are circular arcs meeting the boundary at right angles: this triangle's angles add up to only about 55°.

An analogy: a pizza slice. A flat slice of pizza can bend in one direction without stretching, which is why folding it lengthwise stiffens it: once it curves one way, it cannot also droop the other way without tearing. That rule, that the *product* of the two bendings is fixed by the dough's intrinsic geometry, is Gauss's great theorem in disguise. A flat sheet has curvature zero, so if it bends one way it must stay straight the other way. Curvature in this sense is something the dough itself knows, not something about how it is held in space.

59.3 The mathematics

Manifolds, charts and tangent spaces

Smooth manifold (informal)

An n-dimensional **manifold** M is a space covered by **charts**: one-to-one maps $\varphi : U \to \mathbb{R}^n$ from open pieces $U \subset M$ onto open sets of $\mathbb{R}^n$. Wherever two charts overlap, the **transition map** $\psi \circ \varphi^{-1}$ must be smooth (infinitely differentiable). The collection of charts is an **atlas**.

Examples you already know:

- The sphere S^2, charted by latitude and longitude (which fail at the poles, so you need at least two charts).
- The torus T^2: the configuration space of a robot arm with two revolute joints, one angle per joint, each angle wrapping around.
- The rotation group $SO(3)$: every possible orientation of a drone or a camera. It is a 3-dimensional manifold, but not a sphere, and no single chart covers it without singularities. Gimbal lock is a chart failing. That story belongs to Ch. 70.

The **tangent space** T_pM is the set of velocities $\dot\gamma(0)$ of smooth curves γ through $\gamma(0) = p$. For a surface sitting in $\mathbb{R}^3$ it is the familiar tangent plane. For the unit sphere it is $T_pS^2 = \{\mathbf{v} : \mathbf{v}\cdot\mathbf{p} = 0\}$, the vectors perpendicular to the radius. It is a vector space of dimension n in the sense of Ch. 9. The derivative of a map between manifolds is a linear map between tangent spaces. In coordinates it is the Jacobian of §6.6.

The metric: measuring from inside

To measure lengths we need an inner product on every tangent space, varying smoothly from point to point. In a chart with coordinates $x^1, \dots, x^n$ it is a symmetric positive-definite matrix $g_{ij}(x)$, the **metric tensor**:

$$ds^2 = \sum_{i,j} g_{ij}(x)\, dx^i\, dx^j, \qquad L[\gamma] = \int_a^b \sqrt{\sum_{i,j} g_{ij}\big(x(t)\big)\, \dot x^i \dot x^j}\; dt.$$

A manifold with a metric is a **Riemannian manifold**. On a sphere of radius R in colatitude θ (measured from the north pole) and longitude ϕ, which is the spherical system of §6.9 with the radius frozen:

$$ds^2 = R^2\left(d\theta^2 + \sin^2\theta\, d\phi^2\right), \qquad g = \begin{pmatrix} R^2 & 0 \\ 0 & R^2\sin^2\theta \end{pmatrix}.$$

The factor $\sin^2\theta$ says that a degree of longitude is long at the equator and shrinks to nothing at the poles. That is all the sphere's geometry, written in one matrix.

Why "tensor"? If you change coordinates $x \to y$ with Jacobian $J^i{}_a = \partial x^i/\partial y^a$, the metric must transform as $\tilde g_{ab} = \sum_{i,j} J^i{}_a J^j{}_b\, g_{ij}$, so that lengths come out the same in every chart. A **tensor** is any array of numbers that transforms by such a rule, one Jacobian factor per index. The rule guarantees that the geometric object, not its coordinate shadow, is what you are computing with. Ch. 93 builds the full algebra.

Geodesics

A geodesic is a curve that is locally the shortest path. Minimizing length directly is awkward because of the square root, and because length doesn't care how fast you traverse the curve. The standard trick is to minimize the **energy** $E[\gamma] = \frac12\int g_{ij}\dot x^i\dot x^j\, dt$ instead. Its minimizers are the shortest curves, traversed at constant speed.

Geodesic equation

A curve $x(t)$ is a critical point of the energy if and only if

$$\ddot x^k + \sum_{i,j} \Gamma^k_{ij}\, \dot x^i \dot x^j = 0, \qquad \Gamma^k_{ij} = \tfrac12 \sum_l g^{kl}\left(\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij}\right),$$

where g^{kl} is the inverse matrix of g_{kl}. The Γ^k_{ij} are the **Christoffel symbols**.

The equation says "acceleration, as seen from inside, is zero". The Γ terms are the corrections needed because the coordinate grid itself bends and stretches. The general derivation is the Euler–Lagrange equation applied to $\frac12 g_{ij}\dot x^i\dot x^j$. Here it is done explicitly for the sphere, where it answers the hook.

Derivation on the sphere. The Lagrangian is $\mathcal{L} = \frac12 R^2(\dot\theta^2 + \sin^2\theta\,\dot\phi^2)$. The Euler–Lagrange equation $\frac{d}{dt}\frac{\partial\mathcal{L}}{\partial\dot q} = \frac{\partial\mathcal{L}}{\partial q}$ for each coordinate gives:

$$\begin{aligned} \theta: &\quad R^2\ddot\theta = R^2\sin\theta\cos\theta\,\dot\phi^2 &&\Longrightarrow \quad \ddot\theta - \sin\theta\cos\theta\,\dot\phi^2 = 0,\\ \phi: &\quad \frac{d}{dt}\left(R^2\sin^2\theta\,\dot\phi\right) = 0 &&\Longrightarrow \quad \sin^2\theta\,\dot\phi = \text{const}. \end{aligned}$$

Reading off the Christoffel symbols: $\Gamma^\theta_{\phi\phi} = -\sin\theta\cos\theta$ and $\Gamma^\phi_{\theta\phi} = \Gamma^\phi_{\phi\theta} = \cot\theta$, all others zero.

Which circles of latitude are geodesics

The circle of colatitude θ_0, traversed at constant speed, is a geodesic of the sphere if and only if $\theta_0 = \pi/2$, the equator.

Proof. Put $\theta(t) = \theta_0$ and $\phi(t) = \omega t$ with $\omega \neq 0$. The ϕ equation holds, since $\sin^2\theta_0\,\omega$ is constant. In the θ equation $\ddot\theta = 0$, so it reduces to $\sin\theta_0\cos\theta_0\,\omega^2 = 0$. With $0 < \theta_0 < \pi$ we have $\sin\theta_0 \neq 0$, so the equation forces $\cos\theta_0 = 0$, which means $\theta_0 = \pi/2$. ∎

The leftover term $\sin\theta_0\cos\theta_0\,\omega^2$ is exactly the steering an ant on the 40th parallel must apply. Rotations of the sphere carry geodesics to geodesics, and every great circle is a rotated equator. So every great circle is a geodesic, and since a geodesic is fixed by its starting point and direction, there are no others.

The conserved quantity $\sin^2\theta\,\dot\phi$ is **Clairaut's relation**. For a unit-speed path it says $\sin\theta\,\sin a = \text{const}$, where a is the compass heading measured from north. Leaving New York (colatitude 49.29°) on heading 65.66°, the path is heading due east ($a = 90^\circ$) at the colatitude where $\sin\theta = \sin 49.29^\circ\,\sin 65.66^\circ = 0.6905$, which is $\theta = 43.68^\circ$, or latitude 46.3° N. That is the northernmost point of the flight in the hook.

Curvature from circles

Here is the cleanest way for an insider to measure curvature. Stand at a point, walk out a distance r along geodesics in every direction, and measure the circumference $C(r)$ of the circle you trace out.

On a sphere of radius R, centre the circle at the north pole. Walking a geodesic distance r takes you to colatitude $\theta = r/R$. That circle of latitude has Euclidean radius $R\sin(r/R)$, so

$$C(r) = 2\pi R \sin\frac{r}{R} = 2\pi r\left(1 - \frac{r^2}{6R^2} + O(r^4)\right).$$

The circle is *short*: less than $2\pi r$. The relative shortfall, at leading order, is $r^2/(6R^2)$. Turn this into a definition:

Gaussian curvature (Bertrand–Diguet–Puiseux)

$$K(p) = \lim_{r\to 0} \frac{3}{\pi}\,\frac{2\pi r - C(r)}{r^3}.$$

For the sphere, $2\pi r - C(r) \approx \pi r^3/(3R^2)$, so $K = 1/R^2$. A flat plane has $K = 0$. On the hyperbolic plane circles are *long*: $C(r) = 2\pi\sinh r$ for $K = -1$. For the Earth, a circle of geodesic radius 1,000 km has circumference about 25.8 km less than $2\pi \times 1000$ km, and a careful enough surveyor could measure that.

Everything in this definition is measured from inside the surface. That gives the punchline of Gauss's career:

Theorema Egregium (Gauss, 1827)

For a surface in $\mathbb{R}^3$, the Gaussian curvature equals the product of the two principal curvatures, $K = \kappa_1\kappa_2$, which describe how the surface bends in space. Yet K depends only on the metric g_{ij} and its derivatives. It is unchanged by any bending of the surface that preserves lengths.

This is why no flat map of the Earth can preserve all distances: the sphere has $K = 1/R^2$, paper has $K = 0$, and a distance-preserving map would have to preserve K. It is also the pizza rule: a flat slice has $K = \kappa_1\kappa_2 = 0$, so if you force $\kappa_1 \neq 0$ by folding it, then $\kappa_2 = 0$ and the tip can't droop.

? How can the ant compute $\kappa_1\kappa_2$ without seeing the third dimension?

Each principal curvature on its own depends on how the surface sits in space: a flat sheet and a rolled-up cylinder have different κ_1 but the same metric. Yet their product can be computed from g_{ij} alone. Gauss called this "remarkable", and the proof he gave is a page of brute calculation that hides the reason. The reason is the **Riemann curvature tensor**, built from derivatives of the Christoffel symbols. It measures how a vector fails to come back to itself when carried around a tiny loop. That tensor, and why it only exists because of the transformation law above, is the subject of Ch. 93.

Triangles: Girard's theorem

Now for the angle sums. On the sphere the result can be proved with nothing more than areas.

Girard's theorem

A geodesic triangle on a sphere of radius R with angles α, β, γ and area T satisfies

$$\alpha + \beta + \gamma = \pi + \frac{T}{R^2} = \pi + K\,T.$$

Proof. Two great circles meeting at angle α cut the sphere into four **lunes** (orange-segment shapes). The lune with angle α takes the fraction $\alpha/(2\pi)$ of the sphere's area, so its area is $\frac{\alpha}{2\pi}\cdot 4\pi R^2 = 2\alpha R^2$.

Extend the three sides of the triangle to full great circles. At vertex A they bound a lune containing the triangle, and its antipodal twin, which contains the antipodal triangle. Together the two have area $4\alpha R^2$. Do the same at B and C. These six lunes cover the whole sphere. Every point is covered exactly once, except points of the triangle and of its antipodal copy, which are each covered three times. So the six lunes overcount the sphere by $2T + 2T$:

$$4\alpha R^2 + 4\beta R^2 + 4\gamma R^2 = 4\pi R^2 + 4T.$$

Divide by $4R^2$. ∎

Check: the octant triangle of the hook has area $\frac{1}{8}\cdot 4\pi R^2 = \frac{\pi}{2}R^2$, so its angle sum is $\pi + \pi/2 = 3\pi/2 = 270^\circ$. Gauss's mountain triangle has area about 2,930 km². Dividing by $R^2 = 6371^2$ km² gives an excess of 7.2×10^{-5} rad, which is 14.9″.

For a general surface where K varies, the same statement holds with KT replaced by $\iint_T K\,dA$. This is the local **Gauss–Bonnet theorem**. Triangles on saddles, where $K < 0$, have angle deficits.

? Curvature that counts holes

Apply Gauss–Bonnet to a whole closed surface, triangulated into geodesic triangles, and something strange falls out. The total curvature $\iint_M K\,dA$ of *any* surface shaped like a sphere (a potato, a pear, a dented ball) is exactly 4π, and for any doughnut it is exactly 0. A number built from local bending comes out as a topological invariant, 2π times the Euler characteristic from Ch. 58. Why should integrating geometry produce topology? The bookkeeping uses $V - E + F$, but the reason runs much deeper. Its higher-dimensional heir, the Chern–Gauss–Bonnet theorem, and its cousins the Chern numbers of gauge fields, are waiting in Ch. 92.

Differential forms in one paragraph

Vector calculus has four "fundamental theorems": the ordinary one, Green's, Stokes' and the divergence theorem (§7.9–§7.11). On a manifold they become one. A **k-form** is a machine that eats k tangent vectors and returns a number, linearly and antisymmetrically. A 1-form such as df eats a velocity and returns a rate of change. A 2-form such as $dx \wedge dy$ eats two vectors and returns the signed area of the parallelogram they span. The **exterior derivative** d turns k-forms into

$(k+1)$-forms, and $d(d\omega) = 0$ packages both "curl of a gradient is zero" and "divergence of a curl is zero". Then, for any oriented k-manifold M with boundary ∂M and any $(k-1)$-form ω:

$$\int_M d\omega = \int_{\partial M} \omega.$$

This single line is the generalized Stokes theorem. Forms need no metric at all, which is why physicists write Maxwell's equations as $dF = 0$ and $d{\star}F = J$: the same equations in every coordinate system, flat or curved.

59.4 The algorithm

Shooting a geodesic. To fly a geodesic from a starting point in a given direction, integrate the geodesic equation as a first-order system in $(\theta, \phi, \dot\theta, \dot\phi)$. Fourth-order Runge–Kutta (§22.9) is ideal.

```
import numpy as np
R = 6371.0                                  # Earth radius, km

def rhs(s):
    th, ph, dth, dph = s                    # colatitude, longitude, and their rates
    return np.array([dth, dph,
                     np.sin(th) * np.cos(th) * dph**2,           # -Gamma^th_phph dph^2
                     -2 * np.cos(th) / np.sin(th) * dth * dph])  # -2 Gamma^ph_thph

def rk4(s, h, n):
    for _ in range(n):
        k1 = rhs(s); k2 = rhs(s + h/2*k1); k3 = rhs(s + h/2*k2); k4 = rhs(s + h*k3)
        s = s + h/6*(k1 + 2*k2 + 2*k3 + k4)
    return s

lat0, lon0, heading = 40.71, -74.01, 65.66   # New York, compass heading (deg)
th0, ph0, a = np.radians(90 - lat0), np.radians(lon0), np.radians(heading)
s0 = np.array([th0, ph0, -np.cos(a), np.sin(a) / np.sin(th0)])   # unit speed
L = 5768.6 / R                              # arc length to fly, radians
s = rk4(s0, L / 2000, 2000)
print("arrive at lat %.2f, lon %.2f" % (90 - np.degrees(s[0]), np.degrees(s[1])))
print("speed^2 drift:", s[2]**2 + np.sin(s[0])**2 * s[3]**2 - 1)
```

It lands at 40.42° N, 3.70° W, which is Madrid. The squared speed $\dot\theta^2 + \sin^2\theta\,\dot\phi^2$ stays equal to 1 to about 10^{-15}. This is a free accuracy check: geodesics have constant speed, so any drift is integration error.

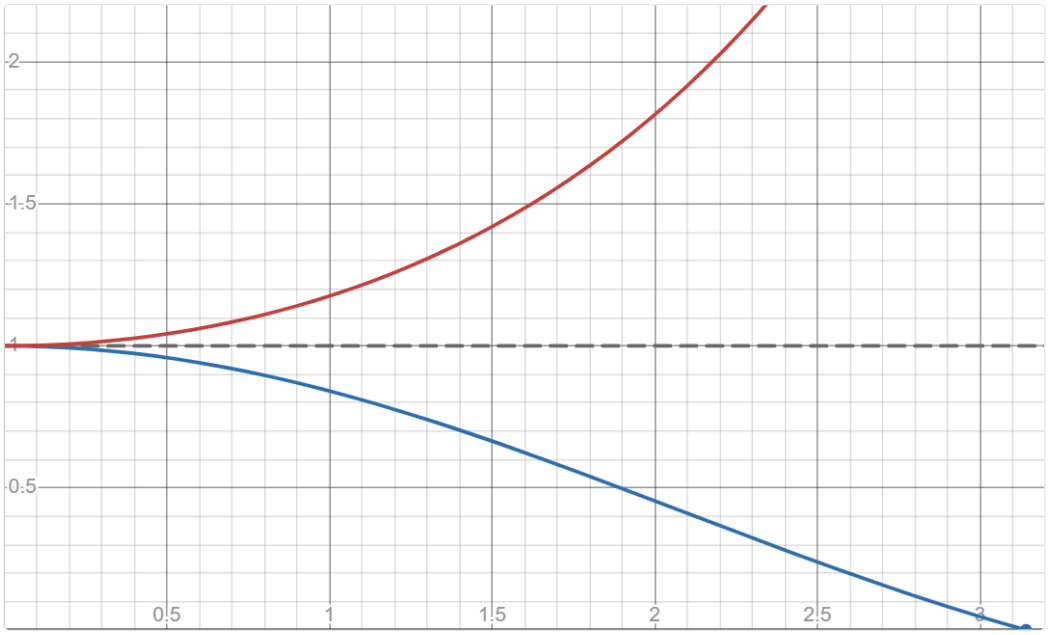
Cost and accuracy. Each step costs $O(n^2)$ Christoffel evaluations on an n-manifold ($O(n^3)$ terms in general, fewer when most vanish). RK4 has global error $O(h^4)$.

Failure cases worth knowing.

- *Chart singularities.* The term $\cot\theta$ blows up at the poles. That is a defect of the latitude–longitude chart, not of the sphere. A route over the pole needs a second chart, or an integration in the embedding $\mathbb{R}^3$ that projects back onto the sphere. Every serious geometric code switches charts or avoids them.
- *Boundary-value problems.* "Find the geodesic from A to B" is harder than "fly from A in direction v". The usual fix is shooting: adjust the initial heading with Newton's method or bisection until the endpoint hits B.
- *Non-uniqueness.* Between antipodal points there are infinitely many shortest geodesics. A geodesic that runs past such a **conjugate point** stops being the shortest path, even though it still satisfies the equation. Satisfying the geodesic equation means "critical", not "minimal".
- *Meshes.* On a triangle mesh in graphics, running Dijkstra's algorithm along edges overestimates geodesic distance by an error that does not shrink as the mesh is refined: up to about 8% on a square grid with diagonal links. Fast-marching and heat-method algorithms work on the faces instead.

59.5 The experiment

Circles tell curvature. The ratio $C(r)/(2\pi r)$ is plotted below for a unit sphere ($K = 1$, blue), the plane (grey) and the hyperbolic plane ($K = -1$, red). Change the sphere's radius by replacing $\sin(x)/x$ with $2\sin(x/2)/x$ (radius 2), and watch the curve flatten. Larger spheres look flatter for longer.



Circumference of a geodesic circle divided by $2\pi r$: below 1 on the sphere, exactly 1 on the plane, above 1 on the hyperbolic plane. The initial curvature of each graph is $-K/6$.

Angle sums grow with area. Run this script, which builds right-angled isosceles triangles at the north pole with legs of geodesic length d on the unit sphere, and compares the angle excess with the area:

```
import numpy as np

def angle(P, Q, S):
    """Angle at P of the spherical triangle PQS (unit vectors)."""
    t1 = Q - (P @ Q) * P; t2 = S - (P @ S) * P      # tangent directions at P
    return np.arccos(t1 @ t2 / np.linalg.norm(t1) / np.linalg.norm(t2))

def area(A, B, C):                                   # Van Oosterom–Strackee formula
    num = abs(A @ np.cross(B, C))
    den = 1 + A @ B + B @ C + C @ A
    return 2 * np.arctan2(num, den)

for d in [0.01, 0.1, 0.3, 0.6, 1.0, 1.4]:            # leg length, radians
    A = np.array([0, 0, 1.0])
    B = np.array([np.sin(d), 0, np.cos(d)])
    C = np.array([0, np.sin(d), np.cos(d)])
    s = angle(A, B, C) + angle(B, C, A) + angle(C, A, B)
    print(f"d={d:4.2f}  angle sum={np.degrees(s):8.3f} deg   "
          f"excess={s-np.pi:.5f}   area={area(A,B,C):.5f}")
```

The excess column and the area column agree to every printed digit: 180.003° at $d = 0.01$, 190.9° at $d = 0.6$, 250.7° at $d = 1.4$. Things to change:

- Set $d = \pi/2$ to recover the 270° triangle.
- Rotate `B` and `C` so the vertex angle at `A` is 60° rather than 90°.
- Replace the area formula with $d^2/2$, the flat answer, and see when the flat world's prediction becomes visibly wrong.

59.6 The challenge

1. **Intuition.** Carry a spear around the 270° octant triangle, always keeping it pointed in the same direction as seen from inside the surface: never rotating it relative to the path's turns. Show by a sketch that it comes back rotated by 90°. Then guess the rotation after a trip around a small triangle of area T on a sphere of radius R.
2. **Derivation.** For the surface of revolution $ds^2 = dr^2 + f(r)^2\,d\phi^2$, find the Christoffel symbols and the geodesic equations, and show that $K = -f''/f$. Check that $f = \sin r$ gives $K = 1$, $f = r$ gives $K = 0$, and $f = \sinh r$ gives $K = -1$. Then verify the circle formula $C(r) = 2\pi f(r)$ against the Bertrand–Diguet–Puiseux definition.
3. **Implementation.** Turn the RK4 geodesic code into a *boundary-value* solver. Given two airports, use Newton's method on the initial heading and arc length to hit the destination. Compare its answer with the closed-form great-circle distance $R\arccos(\sin\varphi_1\sin\varphi_2 + \cos\varphi_1\cos\varphi_2\cos\Delta\lambda)$, where φ is latitude and λ longitude, and make it survive routes that pass over a pole.
4. **Open investigation.** The Earth is not a sphere but an oblate ellipsoid, with polar radius 21 km less than the equatorial radius. Write the metric of an ellipsoid of revolution and integrate its geodesics. Do they close up after one circuit, the way great circles do? Explore numerically how much a geodesic "precesses" per orbit as a function of its maximum latitude, and look for a pattern.

59.7 Loose threads

This chapter measured curvature on surfaces, where one number K says everything. In three or more dimensions curvature depends on *which* 2-dimensional slice you look at, and a single number is no longer enough. That is why the Riemann tensor of the first mystery has $n^2(n^2-1)/12$ independent components: 1 on a surface, 6 in three dimensions, 20 in four.

> **? Matter tells spacetime how to curve**
>
> A GPS satellite's clock runs fast by about 45 microseconds per day because it sits higher in Earth's gravity. It runs slow by about 7 microseconds per day because of its orbital speed. Uncorrected, the net 38 µs per day would put your position off by about 11 km after a single day. Einstein's explanation is that gravity is not a force at all. Spacetime is a 4-dimensional manifold whose metric is curved by mass, and falling objects simply follow its geodesics. Which curvature does matter control, and why that one? The Einstein field equations, and the Riemannian geometry they need, are waiting in Ch. 92.

The spear of challenge problem 1 is the start of **holonomy**: the rotation a vector picks up when it is carried around a loop. It explains the Foucault pendulum and the way a falling cat turns over with zero angular momentum. It is also why interpolating rotations for a robot or a game camera is a geodesic problem on the curved manifold $SO(3)$ (Ch. 70). Spaces of probability distributions carry geometries too, and moving mass optimally across a space has its own Riemannian structure (Ch. 55).

Further reading

- M. P. do Carmo, *Differential Geometry of Curves and Surfaces*, 2nd ed., Dover, 2016. The classic route through the Theorema Egregium and Gauss–Bonnet, with superb exercises.
- T. Needham, *Visual Differential Geometry and Forms*, Princeton University Press, 2021. Geometry first and algebra second, with hundreds of figures. The best intuition-builder there is.
- J. M. Lee, *Introduction to Riemannian Manifolds*, 2nd ed., Springer, 2018. The modern, rigorous treatment of metrics, connections and curvature.
- K. Crane, *Discrete Differential Geometry: An Applied Introduction*, lecture notes, Carnegie Mellon University. Curvature, geodesics and forms on triangle meshes, as used in graphics. Freely available.

Chapter 60 — Real Analysis: Making Calculus Rigorous

> **Level 2 · Track E — The Abstract Mathematics Behind Everything Builds on:** Ch. 4, Ch. 5, Ch. 15 · **Leads to:** Ch. 61, Ch. 97, Ch. 64 · **Related:** Ch. 63, Ch. 31, Ch. 22 **Reading time:** ~30 min

60.1 The hook

Here is a proof that $1 = \frac{1}{2}$.

Start from a series every calculus student meets, the alternating harmonic series:

$$S = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \frac{1}{7} - \frac{1}{8} + \cdots = \ln 2 \approx 0.6931.$$

Now write the same terms in a different order: one positive term, then two negative ones.

$$S = \left(1 - \frac{1}{2}\right) - \frac{1}{4} + \left(\frac{1}{3} - \frac{1}{6}\right) - \frac{1}{8} + \left(\frac{1}{5} - \frac{1}{10}\right) - \frac{1}{12} + \cdots$$

Every term of the original series appears exactly once. Nothing has been added or removed. Now do the subtractions in brackets:

$$S = \frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \frac{1}{10} - \frac{1}{12} + \cdots = \frac{1}{2}\left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots\right) = \frac{1}{2}S.$$

So $S = \frac{1}{2}S$, and since $S = \ln 2 \neq 0$, dividing by S gives $1 = \frac{1}{2}$.

Addition is commutative. Every schoolchild knows that $a + b = b + a$. So where did it go wrong?

This isn't an isolated trick. In the nineteenth century calculus was the most successful tool in science, and it kept producing nonsense whenever it was pushed:

- In 1821 Augustin-Louis Cauchy published a "theorem" that a convergent sum of continuous functions is continuous. Fourier's series had already shown sums of smooth sine waves converging to a square wave with jumps in it (§15.6). Abel spotted the contradiction in 1826.
- Everyone "knew" that a continuous curve must be smooth at most of its points, apart from a few corners. In 1872 Karl Weierstrass presented a function that is continuous everywhere and differentiable *nowhere*. Charles Hermite wrote that he turned away "with fright and horror from this lamentable plague".
- Engineers and physicists swapped limits, sums, integrals and derivatives freely: $\frac{d}{dt}\int = \int \frac{d}{dt}$, $\lim \int = \int \lim$, $\sum \int = \int \sum$. Usually it worked. Sometimes it gave nonsense, and nobody could say in advance which.

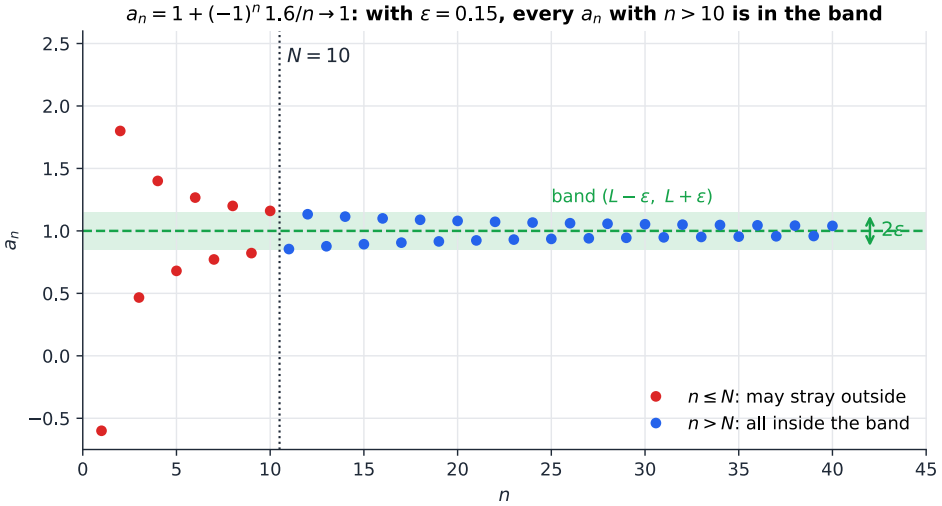
The central question of this chapter:

> **When can you trust the moves of calculus, such as reordering a sum, swapping a limit with an integral, or calling a limit "the answer", and how do you prove it?**

Real analysis is the answer the nineteenth century built. It is the reason the core chapters of this book can say "converges", "continuous" and "integrable" and mean something exact.

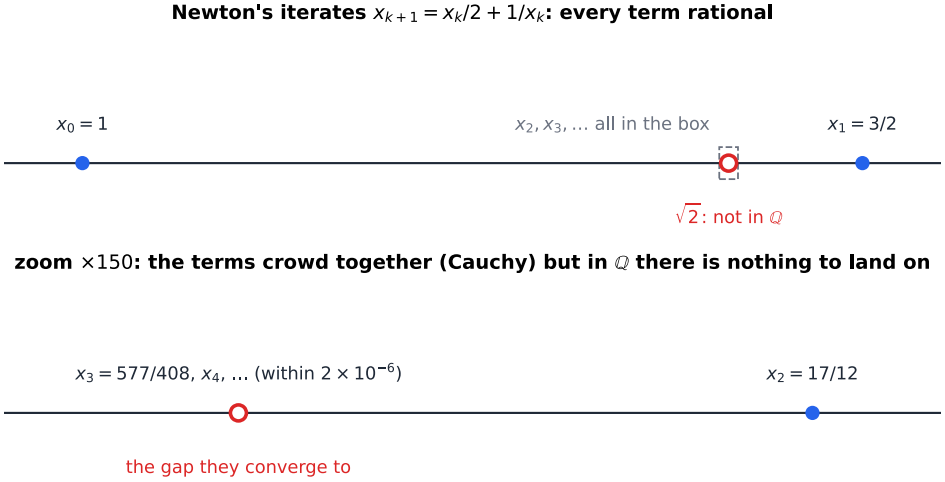
60.2 The big idea

Idea 1: a limit is a game against an adversary. "The sequence a_n approaches L" sounds like motion, but there is nothing moving. Analysis turns it into a challenge-and-response game. Your opponent names a tolerance $\varepsilon > 0$, as small as they like. You must reply with an index N such that *every* term after N lies within ε of L. If you can always reply, whatever ε they pick, then $a_n \to L$. The phrase "gets closer and closer" is gone, and so is the paradox.



The ε–N game. For the tolerance $\varepsilon = 0.15$ the reply $N = 10$ works: every term after it lies in the band. A smaller ε needs a later N, but there is always one.

Idea 2: the real numbers have no holes, and that is an axiom. The rationals $\mathbb{Q}$ are fine for arithmetic but useless for limits. You can write down a sequence of fractions whose terms crowd ever closer together, yet the point they crowd toward isn't a fraction. Calculus needs a number line where every such crowd has a landing spot. The reals $\mathbb{R}$ are, by definition, that number line. This property, **completeness**, is behind every existence theorem you have used: the intermediate value theorem, the maximum of a continuous function on $[a, b]$, the convergence of bisection (§22.4), and the existence of $\int_a^b f$.



A Cauchy sequence in $\mathbb{Q}$. Newton's iterates for $\sqrt{2}$ are all fractions. They bunch together ever more tightly, but the point they bunch toward is missing from $\mathbb{Q}$. In $\mathbb{R}$ it is there.

Idea 3: "infinitely many" comes in different strengths. The rearrangement paradox happens because the alternating harmonic series converges only through cancellation. Its positive terms alone add up to $+\infty$ and its negative terms alone to $-\infty$. Reordering lets you spend the two infinite budgets at different rates, and you can steer the total anywhere you like. Series whose absolute values add up to something finite are immune to this. Much of analysis is the art of telling the safe kind of infinity from the dangerous kind.

60.3 The mathematics

Sequences and the completeness axiom

Limit of a sequence

$a_n \to L$ means: for every $\varepsilon > 0$ there is an N such that $|a_n - L| < \varepsilon$ for all $n > N$.

(a_n) is a **Cauchy sequence** if for every $\varepsilon > 0$ there is an N such that $|a_m - a_n| < \varepsilon$ for all $m, n > N$.

A Cauchy sequence is one whose terms eventually crowd together. The definition doesn't mention any limit, which is the point: you can check it without knowing where the sequence is going. Every convergent sequence is Cauchy, by the triangle inequality with $\varepsilon/2$. The converse is what can fail.

A Cauchy sequence in $\mathbb{Q}$ with no limit in $\mathbb{Q}$

Let $x_0 = 1$ and $x_{k+1} = \dfrac{x_k}{2} + \dfrac{1}{x_k}$. Every x_k is rational, the sequence is Cauchy, and it has no rational limit.

Proof. By induction each x_k is a positive rational, since the recursion uses only field operations. A direct computation gives

$$x_{k+1}^2 - 2 = \left(\frac{x_k}{2} + \frac{1}{x_k}\right)^2 - 2 = \left(\frac{x_k}{2} - \frac{1}{x_k}\right)^2 = \left(\frac{x_k^2 - 2}{2x_k}\right)^2 \geq 0.$$

So $x_k^2 \geq 2$ for $k \geq 1$, and then $x_{k+1} - x_k = \dfrac{2 - x_k^2}{2x_k} \leq 0$. From $k = 1$ on, the sequence is decreasing and bounded below by 1. The terms are $\frac{3}{2}$, $\frac{17}{12}$, $\frac{577}{408}$, $\frac{665857}{470832}$, $\ldots$.

Next we show it is Cauchy, using only rational arithmetic. Write $e_k = x_k^2 - 2 \geq 0$. For $k \geq 1$ we have $x_k \geq 1$, so the identity gives $e_{k+1} = e_k^2/(4x_k^2) \leq e_k^2/4$. Starting from $e_1 = \frac{1}{4}$ this gives $e_k \leq 4^{1-2^k}$, which tends to zero extremely fast. Since $x_k - x_{k+1} = e_k/(2x_k) \leq e_k/2$, the tails satisfy $|x_m - x_n| \leq \sum_{k \geq n} e_k/2$ for $m > n$. That sum is as small as we like for large n, so the sequence is Cauchy.

Finally, suppose $x_k \to L \in \mathbb{Q}$. Then $L \geq 1$, and passing to the limit in the recursion gives $L = L/2 + 1/L$, so $L^2 = 2$. But no rational number squares to 2: if p/q in lowest terms did, then $p^2 = 2q^2$ would make p even, then q even, a contradiction. ∎

The errors are 8.6×10^{-2}, 2.5×10^{-3}, 2.1×10^{-6} and 1.6×10^{-12}. The number of correct digits roughly doubles at each step. This is Newton's quadratic convergence, running in a number system where the answer doesn't exist.

Completeness axiom

Every non-empty set of real numbers that is bounded above has a **least upper bound** (supremum) in $\mathbb{R}$.

This is the one axiom that separates $\mathbb{R}$ from $\mathbb{Q}$. The set $\{q \in \mathbb{Q} : q^2 < 2\}$ is bounded above in $\mathbb{Q}$ but has no least rational upper bound. In $\mathbb{R}$ its supremum is $\sqrt{2}$. Everything else follows from it, beginning with this.

Monotone convergence theorem

A sequence of reals that is increasing and bounded above converges, to its supremum.

Proof. Let $s = \sup_n a_n$, which exists by completeness. Given $\varepsilon > 0$, the number $s - \varepsilon$ is not an upper bound, so some $a_N > s - \varepsilon$. Because the sequence is increasing, $s - \varepsilon < a_N \le a_n \le s$ for every $n > N$, and so $|a_n - s| < \varepsilon$. ∎

Two more consequences, stated without proof:

- **Bolzano–Weierstrass:** every bounded sequence of reals has a convergent subsequence.
- **Cauchy criterion:** in $\mathbb{R}$, a sequence converges if and only if it is Cauchy.

From these, the familiar theorems of calculus follow: the intermediate value theorem, the extreme value theorem, and the integrability of continuous functions. Each is an "existence" statement, and each one fails over $\mathbb{Q}$. For example, $f(x) = x^2 - 2$ changes sign on $[1, 2]$ but has no rational root.

Series: why the hook went wrong

A series $\sum a_n$ converges if its partial sums do. It converges **absolutely** if $\sum |a_n| < \infty$, and **conditionally** if it converges but not absolutely.

Rearrangements (Dirichlet; Riemann)

If $\sum a_n$ converges absolutely, every rearrangement converges to the same sum. If it converges only conditionally, then for any target $T \in \mathbb{R}$ (or $\pm\infty$) some rearrangement converges to T.

Why Riemann's half is true. In a conditionally convergent series the positive terms and the negative terms each add up to an infinite total, while the terms themselves shrink to zero. The greedy recipe goes like this. Add positive terms until the partial sum passes T. Then add negative terms until it drops below T. Repeat. You can never get stuck, because each budget is infinite. The overshoot is never more than the last term used, and the terms tend to zero, so the partial sums close in on T.

The hook used one positive term for every two negative ones. In general, taking p positive terms for every q negative terms of the alternating harmonic series gives the sum $\ln 2 + \tfrac{1}{2}\ln(p/q)$. With $p = 1$, $q = 2$ that is $\ln 2 - \tfrac{1}{2}\ln 2 = \tfrac{1}{2}\ln 2$, the hook's result: about 0.3466 rather than 0.6931. So the error in the hook was never about arithmetic. It was the assumption that an infinite sum behaves like a finite one.

Continuity and the two kinds of convergence

The ε–δ definition of continuity is in §4.1. For sequences of *functions* there are two different notions of convergence, and Cauchy's 1821 mistake was to treat them as the same.

Pointwise and uniform convergence

$f_n \to f$ **pointwise** on D if $f_n(x) \to f(x)$ for each $x \in D$ separately: the N may depend on x.

$f_n \to f$ **uniformly** on D if $\sup_{x \in D} |f_n(x) - f(x)| \to 0$: one N works for every x at once.

The standard example is $f_n(x) = x^n$ on $[0, 1]$. It converges pointwise to 0 for $x < 1$ and to 1 at $x = 1$. Every f_n is continuous, but the limit has a jump. The convergence isn't uniform: for any n, the point $x = 2^{-1/n}$ has $f_n(x) = \tfrac{1}{2}$.

Uniform limits of continuous functions are continuous

If each f_n is continuous at x_0 and $f_n \to f$ uniformly, then f is continuous at x_0.

Proof (the ε/3 argument). Let $\varepsilon > 0$. Choose n with $\sup_x |f_n(x) - f(x)| < \varepsilon/3$. Since f_n is continuous at x_0, choose $\delta > 0$ with $|f_n(x) - f_n(x_0)| < \varepsilon/3$ whenever $|x - x_0| < \delta$. Then for such x:

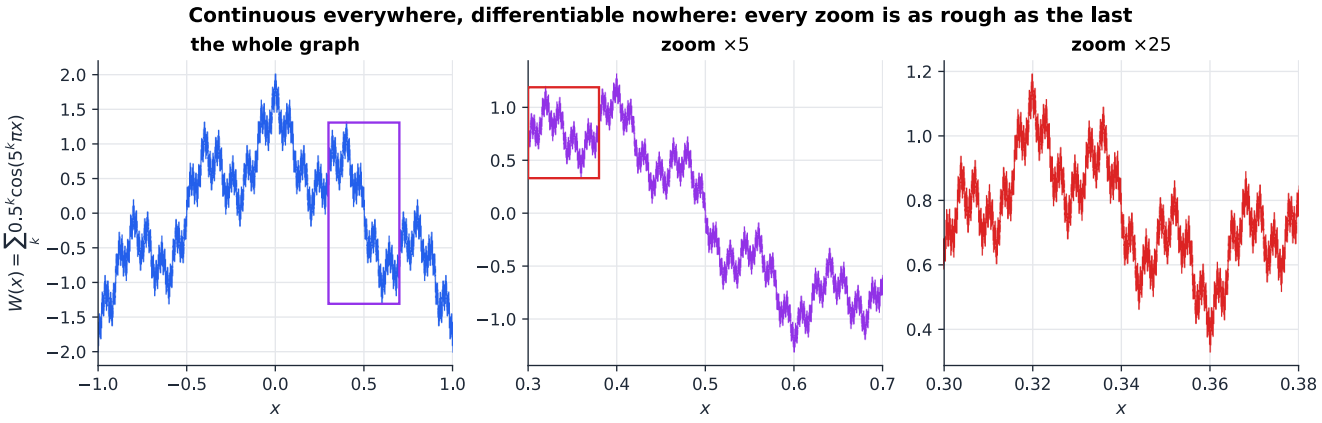
$$|f(x) - f(x_0)| \le \underbrace{|f(x) - f_n(x)|}_{<\varepsilon/3} + \underbrace{|f_n(x) - f_n(x_0)|}_{<\varepsilon/3} + \underbrace{|f_n(x_0) - f(x_0)|}_{<\varepsilon/3} < \varepsilon. \quad \blacksquare$$

The proof needs one n that works at x and at x_0 at the same time, and that is exactly what uniformity provides. A practical test is the **Weierstrass M-test**: if $|g_k(x)| \le M_k$ for all x and $\sum M_k < \infty$, then $\sum g_k$ converges uniformly.

Apply it to Weierstrass's monster:

$$W(x) = \sum_{k=0}^{\infty} a^k \cos\!\left(b^k \pi x\right), \qquad 0 < a < 1.$$

The terms are bounded by $M_k = a^k$, a convergent geometric series, so W is a uniform limit of continuous functions and is continuous. Yet when $ab \ge 1$ (Hardy, 1916; Weierstrass needed a stronger condition), W has no derivative at any point. Each new term adds wiggles that are b times faster but only a times smaller. Their slopes grow like $(ab)^k$, and no tangent line can survive.



Weierstrass's function with $a = 0.5$, $b = 5$. A smooth curve flattens into a straight line under magnification. This one never does: every zoom shows new corners.

? Monsters are the majority

Weierstrass's function looks like a pathological exception, built by hand to embarrass calculus. The truth is stranger. In a precise sense, *most* continuous functions are differentiable nowhere, and the smooth functions of engineering are a thin and exceptional set. Making "most" precise needs a way of measuring the size of sets inside an infinite-dimensional space of functions. That is the Baire category theorem, and it is one of the tools of Ch. 61.

Metric spaces: distance, abstracted

Every argument above used only one thing about numbers: the distance $|x - y|$ and its triangle inequality. Abstracting that one thing gives analysis a much bigger stage.

Metric space

A set X with a function $d : X \times X \to [0, \infty)$ such that $d(x, y) = 0 \iff x = y$, $d(x, y) = d(y, x)$, and $d(x, z) \le d(x, y) + d(y, z)$. It is **complete** if every Cauchy sequence converges to a point of X.

Examples:

- $\mathbb{R}^n$ with the Euclidean distance, or with the taxicab distance $\sum |x_i - y_i|$.
- Bit strings with the Hamming distance (Ch. 79).
- Continuous functions on $[0, 1]$ with the uniform distance $d_\infty(f, g) = \max_x |f(x) - g(x)|$. The theorem above says exactly that this space, $C[0, 1]$, is complete.
- The same functions with $d_1(f, g) = \int_0^1 |f - g| \, dx$. This space is *not* complete: continuous ramps steepening toward a step function form a Cauchy sequence whose limit is not continuous.

The payoff of the abstraction is that one theorem serves everywhere:

Banach fixed-point theorem (contraction mapping)

Let (X, d) be complete, and let $T : X \to X$ satisfy $d(Tx, Ty) \le q \, d(x, y)$ for some $q < 1$. Then T has exactly one fixed point x^*. The iteration $x_{n+1} = T x_n$ converges to it from any start, with

$$d(x_n, x^*) \le \frac{q^n}{1 - q} \, d(x_1, x_0), \qquad d(x_n, x^*) \le \frac{q}{1 - q} \, d(x_n, x_{n-1}).$$

The iterates form a Cauchy sequence, since their steps shrink geometrically. Completeness supplies the limit, and the contraction makes it unique. With X a space of functions, this one theorem proves that solutions of $\dot{y} = f(t, y)$ exist and are unique (Picard's theorem, behind Ch. 11). It also proves that value iteration in reinforcement learning converges (Ch. 52).

Measure and integration: Lebesgue's idea

The Riemann integral of Ch. 5 slices the *domain* into thin strips. It breaks down on functions that are too wild. The Dirichlet function, equal to 1 on rationals and 0 on irrationals, has every upper Riemann sum equal to 1 and every lower sum equal to 0 on $[0, 1]$, so it has no Riemann integral. Worse, limits of Riemann-integrable functions can fail to be Riemann integrable, so the Riemann integral is "incomplete" in the same sense as $\mathbb{Q}$.

Lebesgue (1902) sliced the *range* instead. To integrate f, ask for each height y how much of the domain has $f(x) > y$, and add up. This needs a way to measure the "length" of complicated sets.

Null set

$A \subset \mathbb{R}$ has **measure zero** if for every $\varepsilon > 0$ it can be covered by countably many intervals of total length less than ε.

The rationals have measure zero

$\mathbb{Q}$ is a null set, even though it is dense in $\mathbb{R}$.

Proof. List the rationals as $q_1, q_2, q_3, \ldots$, which is possible because they are countable (Ch. 64). Cover q_k by an interval of length $\varepsilon/2^k$. The total length is $\sum_k \varepsilon/2^k = \varepsilon$. ∎

So the Dirichlet function equals 0 except on a set of measure zero. Its Lebesgue integral is 0, which is surely the right answer. Lebesgue measure extends "length" to an enormous family of sets, the measurable ones, and the Lebesgue integral comes with the swap theorems the nineteenth century lacked:

Convergence theorems (stated)

Monotone convergence: if $0 \le f_1 \le f_2 \le \cdots$ and $f_n \to f$ pointwise, then $\int f_n \to \int f$.

Dominated convergence: if $f_n \to f$ pointwise and $|f_n| \le g$ for a single g with $\int g < \infty$, then $\int f_n \to \int f$.

The domination hypothesis is not decoration. Let $f_n = n$ on $(0, 1/n)$ and 0 elsewhere. Then $\int f_n = 1$ for every n, while $f_n(x) \to 0$ at every point. So $\lim \int f_n = 1 \neq 0 = \int \lim f_n$. The mass escapes into an ever-taller, ever-thinner spike, and no single integrable g can lie above all of them.

These theorems are why probability theory works. An expectation is a Lebesgue integral (§19.9). Swapping a derivative with an expectation, $\frac{d}{d\theta} E[f(X,\theta)] = E[\partial_\theta f]$, is the step behind every stochastic-gradient method (Ch. 54), and it is valid exactly when a domination condition holds.

? A set too wild to measure

Lebesgue measure assigns a length to a vast family of sets. Can it be extended to *every* subset of $\mathbb{R}$, keeping the two properties that make it "length": translating a set doesn't change its measure, and countable unions of disjoint pieces add up? In 1905 Giuseppe Vitali showed the answer is no, by building a set no consistent length can be assigned to. His construction uses the axiom of choice, and in some models of set theory *without* choice, every set of reals is measurable. Whether non-measurable sets "exist" turns out to depend on which axioms you accept. See Ch. 64 for choice and Ch. 97 for the measure theory.

60.4 The algorithm

Analysis is usually thought of as the opposite of computation, but its central proofs are algorithms with error certificates. The Banach theorem's bound turns fixed-point iteration into a method that tells you when to stop, and *guarantees* that its answer is correct to the stated tolerance.

```
import math

def fixed_point(T, x0, q, tol=1e-6, max_iter=10_000):
    """Iterate x <- T(x) for a contraction with constant q < 1.
    Stops when the a-posteriori bound q/(1-q)*|x_n - x_{n-1}| <= tol."""
    x = x0
    for n in range(1, max_iter + 1):
        x_new = T(x)
        bound = q / (1 - q) * abs(x_new - x)   # guaranteed |x_new - x*| <= bound
        x = x_new
        if bound <= tol:
            return x, n, bound
    raise RuntimeError("no convergence: is T really a contraction?")

q = math.sin(1.0)                   # |T'(x)| = |sin x| <= sin 1 on [0, 1]
x, n, b = fixed_point(math.cos, 0.0, q)
print(f"x* ~ {x:.10f} after {n} steps, certified error <= {b:.1e}")
print("a-priori estimate of steps:", math.ceil(math.log(1e-6 * (1 - q)) / math.log(q)))
```

For $T(x) = \cos x$ on $[0,1]$, the mean value theorem gives $|\cos x - \cos y| \le \sin(1)\,|x - y|$ with $q = \sin 1 \approx 0.841$. The cosine maps $[0,1]$ into itself, and $[0,1]$ is complete. The run stops after 40 steps at 0.7390850787, with a certified error of at most 7.2×10^{-7}. The true fixed point is 0.7390851332, so the actual error is 5×10^{-8}. The a-priori bound predicted 91 steps. It is pessimistic because near the fixed point the true contraction factor is $\sin(0.739) \approx 0.67$, not 0.84.

Cost and failure modes.

- Each step costs one evaluation of T. The error falls like q^n, which is linear convergence: about $\log_{10}(1/q)$ digits per step, so 0.075 digits per step here. Compare Newton's digit-doubling in §60.3.
- If q is close to 1, convergence is painfully slow. If $q \ge 1$, the theorem is silent: the iteration may still converge, cycle, or diverge.
- Estimating q wrongly, too small, voids the certificate. The bound is only as honest as the Lipschitz estimate behind it.
- Floating-point rounding means the computed iterates aren't exactly those of T. The certificate then holds only up to about $\epsilon_{\text{mach}}/(1-q)$ (§22.1).

The same skeleton, a monotone or contracting sequence plus completeness, gives bisection's certified bracket $[a_n, b_n]$ of width $(b-a)/2^n$ and Picard iteration for ODEs.

60.5 The experiment

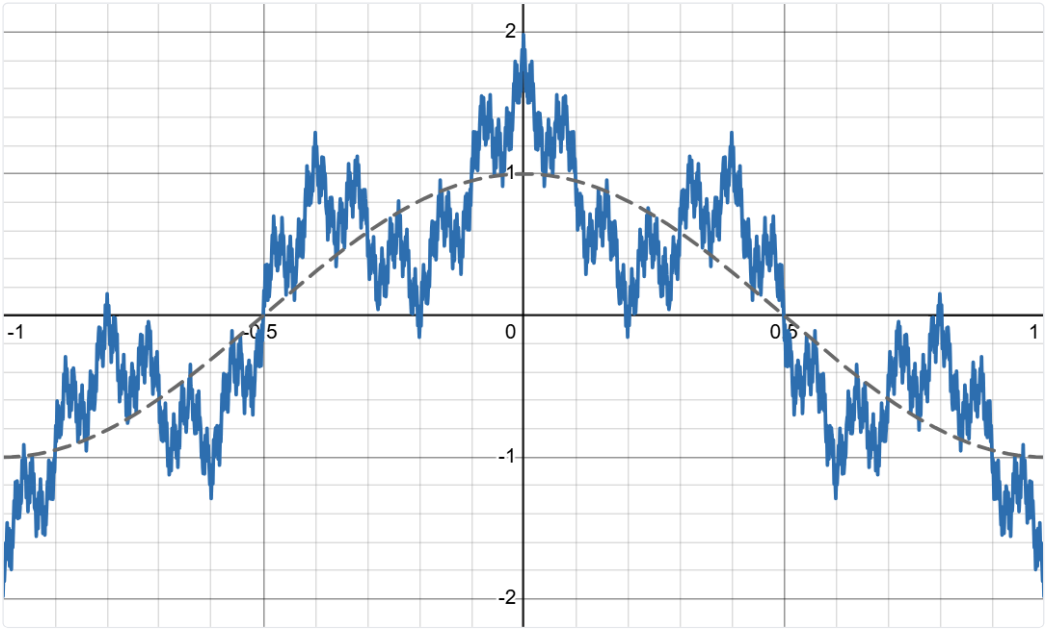
Steer a series anywhere. This script runs Riemann's greedy rearrangement of $1 - \frac{1}{2} + \frac{1}{3} - \cdots$ toward any target you choose:

```
import numpy as np

target = np.pi          # any real number you like: try -1, 0, 100
N = 200_000             # how many terms to use

pos = iter(1 / k for k in range(1, 10**9, 2))     # +1, +1/3, +1/5, ...
neg = iter(-1 / k for k in range(2, 10**9, 2))    # -1/2, -1/4, ...
s, partial, used_pos = 0.0, [], 0
for _ in range(N):
    if s <= target:
        s += next(pos); used_pos += 1
    else:
        s += next(neg)
    partial.append(s)

print(f"sum after {N} terms = {s:.6f}  (target {target:.6f})")
print(f"positive terms used: {used_pos}, negative: {N - used_pos}")
print(f"ratio pos/neg ~ {used_pos / (N - used_pos):.1f};  e^(2*target)/4 = {np.exp(2*target)/4:.1f}")
```

With target π the sum after 200,000 terms is 3.14138. The greedy rule spends about 134 positive terms for every negative one, matching $e^{2T}/4 = e^{2\pi}/4 \approx 133.9$, which is what the formula $\ln 2 + \frac{1}{2}\ln(p/q) = T$ predicts. Things to try:

- Set `target = 100`. How many terms would you need? (Hint: e^{200}.)
- Plot `partial` with matplotlib and watch it oscillate around the target.
- Replace the series with $\sum(-1)^{k+1}/k^2$, which converges absolutely, and show that no amount of greed can move its sum.

Zoom into a monster. Below are the first seven terms of Weierstrass's function with $a = 0.5$ and $b = 5$. Change the upper limit from 6 to 2, then back to 6, and watch the corners appear. Then change b from 5 to 2, which makes $ab = 1$, the borderline case. Then zoom in with Desmos's own controls.



Partial sums of Weierstrass's function. The dashed curve is the first term. Every extra term adds wiggles five times faster and only half as tall.

60.6 The challenge

1. **Intuition.** Explain, without formulas, why reordering the terms of $\sum(-1)^{k+1}/k^2$ can never change its sum, while reordering $\sum(-1)^{k+1}/k$ can produce anything. What is the one property of the terms that decides it?
2. **Derivation.** Prove that $f_n(x) = nx\,e^{-nx^2}$ converges pointwise to 0 on $[0, 1]$. Compute $\int_0^1 f_n\,dx$ exactly and its limit. Is $\lim \int = \int \lim$? Find $\sup_x f_n(x)$, and explain which hypothesis of the dominated convergence theorem fails.
3. **Implementation.** Write a function that takes a contraction T on $\mathbb{R}^2$, given as code, plus a Lipschitz constant q, and returns a certified fixed point. Use it on $T(x, y) = \left(\frac{1}{2}\cos y,\ \frac{1}{2}\sin x + 0.1\right)$. Then estimate q numerically by sampling $\|T(\mathbf{u}) - T(\mathbf{v})\|/\|\mathbf{u} - \mathbf{v}\|$, and explain why a sampled estimate can never be trusted for the certificate.
4. **Open investigation.** For the Weierstrass function with $a = 0.5$, estimate numerically how the oscillation $\max_{|t|\le h}|W(x + t) - W(x)|$ scales with h as $h \to 0$, for several values of b. You should find a power law h^{α}. Guess a formula for α in terms of a and b, then compare it with the box-counting dimension of the graph estimated from your plots. Does the relationship hold for every x, or only for most?

60.7 Loose threads

The ε–N game, completeness, and Lebesgue's integral are the foundations. This chapter showed you the ground floor and closed several doors without opening them.

? When does a Fourier series actually converge?

Ch. 15 promised that well-behaved periodic signals equal their Fourier series. In 1873 Paul du Bois-Reymond built a *continuous* function whose Fourier series diverges at a point. Later it was shown that for continuous functions divergence can even happen on a dense set. Then in 1966 Lennart Carleson proved that the Fourier series of every square-integrable function converges at *almost every* point, meaning everywhere except on a set of measure zero. The statement uses only this chapter's words, but the proof is among the hardest in twentieth-century analysis. The framework it lives in is Ch. 97.

Two more threads to follow. The space $C[0,1]$ with the distance d_1 was not complete. Completing it, the way $\mathbb{R}$ completes $\mathbb{Q}$, produces the Lebesgue space L^1, and that construction opens Ch. 61. The careful logic of ε–N proofs, with its nested "for every… there exists", is a subject in its own right in Ch. 63. And probability, rebuilt on measure by Kolmogorov in 1933, is the destination of Ch. 31 and Ch. 97.

Further reading

- S. Abbott, *Understanding Analysis*, 2nd ed., Springer, 2015. The friendliest rigorous introduction, built around exactly the paradoxes of this chapter.
- W. Rudin, *Principles of Mathematical Analysis*, 3rd ed., McGraw-Hill, 1976. Terse, complete, and the standard reference for metric spaces and uniform convergence.
- T. Tao, *Analysis I* and *Analysis II*, 4th ed., Springer (Hindustan Book Agency), 2022. Builds $\mathbb{R}$ from scratch, then reaches Lebesgue integration.
- D. Bressoud, *A Radical Approach to Real Analysis*, 2nd ed., MAA, 2007. Follows the history: Fourier, Cauchy's error, Abel and Weierstrass, in the order they happened.

Chapter 61 — Functional Analysis: Infinite-Dimensional Thinking

> **Level 2 · Track E — The Abstract Mathematics Behind Everything Builds on:** Ch. 9, Ch. 10, Ch. 15 · **Leads to:** Ch. 90, Ch. 91, Ch. 89 · **Related:** Ch. 60, Ch. 47, Ch. 92 **Reading time:** ~30 min

61.1 The hook

In the summer of 1925, a 23-year-old Werner Heisenberg, recovering from hay fever on the island of Heligoland, invented a new mechanics. In it, the position and momentum of an electron were no longer numbers but arrays. Max Born recognised the arrays as matrices, and with Pascual Jordan he found the rule at the centre of the theory: position X and momentum P don't commute. Instead,

$$XP - PX = j\hbar\mathbf{I}.$$

Everything quantum follows from this one line: the uncertainty principle, discrete energy levels, the stability of atoms.

Here is the strange part. The equation has **no solution in matrices**. Not 2×2, not 1000×1000, not any finite size. The proof takes one line, and you will see it in §61.3. Yet the theory worked, its predictions were confirmed to many decimal places, and transistors, lasers and MRI scanners are all built on it.

So what are X and P, if they can't be matrices?

The same puzzle shows up wherever engineers work with signals rather than lists of numbers:

- A signal $x(t)$ is a vector with one coordinate for every instant. Fourier series write it in a basis of sines (Ch. 15), but there are infinitely many basis vectors. Is "orthogonal projection" still the best approximation? Does every Cauchy sequence of signals have a limit?
- Differentiation is a linear map on signals. So is integration. Integration smooths, so a small change in the input gives a small change in the output. Differentiation does the opposite: a tiny high-frequency wiggle in the data becomes a huge error in the derivative. Why are two inverse operations so different?
- Deblurring a photo, or reconstructing a CT image, means inverting a linear map. With exact data the inverse exists. With real data, however good the instrument, the naive inverse returns garbage. Why?

The question of this chapter:

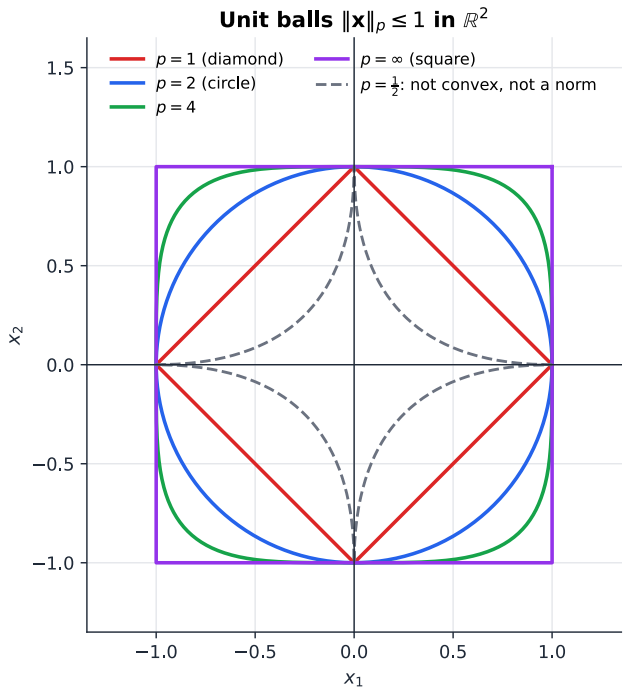
> **What survives of linear algebra when the dimension becomes infinite, and what new phenomena appear that no matrix could ever show?**

The subject that answers it is functional analysis. It was built in the 1900s–1930s by Hilbert, Riesz, Banach and von Neumann, and von Neumann used it to put Heisenberg's matrices on solid ground.

61.2 The big idea

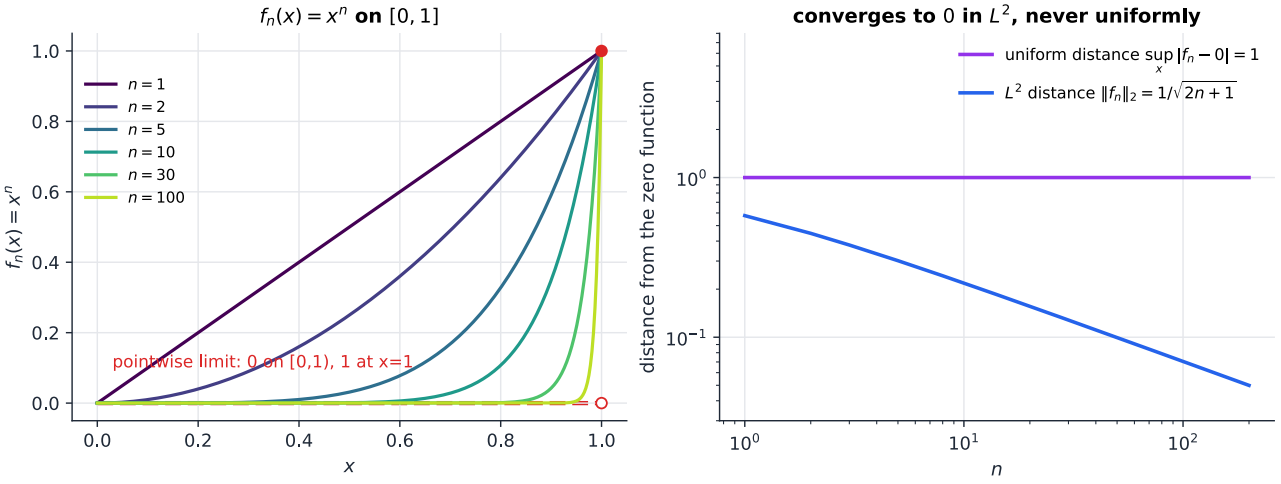
Functions are vectors. You can add two signals and scale a signal, so signals form a vector space (§9.1). Everything you know about linear maps, projections and eigenvectors has a candidate meaning for them. The catch is that no finite basis spans the space.

In infinite dimensions, "how you measure" changes "what converges". In $\mathbb{R}^n$ it doesn't matter whether you measure the size of a vector by the largest component, the sum of components, or the Euclidean length: a sequence that converges in one measure converges in all of them. The unit balls of these norms differ in shape (a square, a diamond, a disc), but each fits inside a scaled copy of the others.



The unit balls of $\|\mathbf{x}\|_p = (|x_1|^p + |x_2|^p)^{1/p}$. As p grows they swell from the diamond to the square. For $p < 1$ the "ball" is not convex and the triangle inequality fails, so it isn't a norm.

For functions that comfort disappears. Take $f_n(x) = x^n$ on $[0, 1]$. Measured by *energy*, the L^2 norm $\left(\int_0^1 |f|^2\right)^{1/2}$, the sequence shrinks to zero: $\|x^n\|_2 = 1/\sqrt{2n+1}$. Measured by *peak value*, the sup norm, it never shrinks at all, because $\max x^n = 1$ for every n. One norm says "converges to zero" and the other says "stays at distance 1 forever". Both are right. You have to choose, and the choice is physics: energy-type questions call for L^2, and questions about peak voltage call for the sup norm.



Same sequence, two verdicts. In L^2 the functions x^n converge to the zero function. In the sup norm they never get closer than 1. Gibbs' overshoot in Fourier series (§15.6) is the same phenomenon.

An analogy: an infinite orchestra. A finite-dimensional vector is a chord played by a fixed set of instruments. A signal is a chord played by infinitely many instruments, each a little higher in pitch. In finite dimensions you can't hide energy anywhere. In an infinite orchestra the energy can drift up to ever-higher instruments and *leak away* while the total stays constant. The sequence $\sin(nt)$ has fixed energy but moves to ever-higher pitch. Its overlap with any fixed signal fades to zero, yet it never converges to anything. Most of what is new in functional analysis is this escape to infinity: how to see it, how to forbid it, and how to use it.

61.3 The mathematics

Normed and Banach spaces

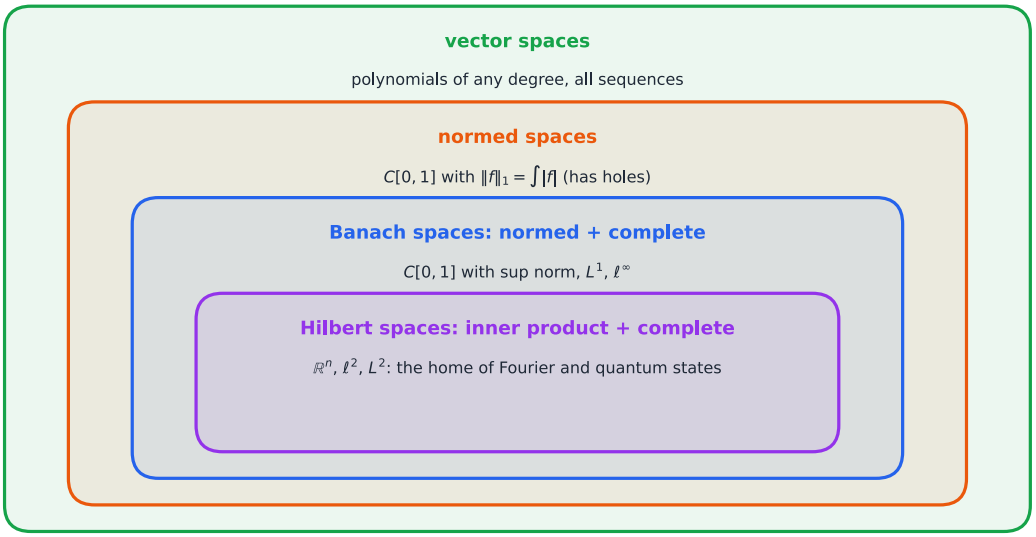
Normed space, Banach space

A **norm** on a vector space V is a function $\|\cdot\| : V \to [0, \infty)$ with $\|v\| = 0 \iff v = 0$, $\|\alpha v\| = |\alpha|\|v\|$, and $\|u + v\| \le \|u\| + \|v\|$. A normed space is a **Banach space** if it is complete for the metric $d(u, v) = \|u - v\|$, in the sense of Ch. 60: every Cauchy sequence converges.

The main examples:

- ℓ^p, the sequences with $\|x\|_p = (\sum_k |x_k|^p)^{1/p} < \infty$, for $1 \le p < \infty$.
- ℓ^∞, the bounded sequences with the sup norm.
- $C[a, b]$ with the sup norm.
- $L^p[a, b]$, the functions with $\int |f|^p < \infty$. Strictly, these are classes of functions that agree almost everywhere, and the integral must be Lebesgue's (§60.3).

Completeness is not automatic. $C[0, 1]$ with the L^1 norm is not complete, and its completion is L^1, just as $\mathbb{R}$ completes $\mathbb{Q}$. The fact that the L^p spaces *are* complete (the Riesz–Fischer theorem, in its general form) is the main reason for Lebesgue's integral. With Riemann's integral, limits of signals could fall out of the space.



The family tree. Each level adds structure: a length, then completeness, then angles. Most of signal processing and quantum mechanics lives in the innermost box.

In finite dimensions all norms are **equivalent**: for any two norms there are constants with $c\|v\|_a \le \|v\|_b \le C\|v\|_a$. In infinite dimensions they are not, as the x^n example shows. There $\|f_n\|_2 \to 0$ while $\|f_n\|_\infty = 1$, so no constant C can make $\|f\|_\infty \le C\|f\|_2$ hold.

Hilbert spaces: geometry with infinitely many axes

A **Hilbert space** is a Banach space whose norm comes from an inner product, $\|v\|^2 = \langle v, v\rangle$. Inner products, orthogonality and Gram–Schmidt are covered in §9.8–§9.10. The examples that matter are ℓ^2 and L^2, where $\langle f, g\rangle = \int f\,\overline{g}$. Among the ℓ^p spaces, only $p = 2$ comes from an inner product. The test is the parallelogram law $\|u + v\|^2 + \|u - v\|^2 = 2\|u\|^2 + 2\|v\|^2$, which the diamond and the square both fail.

Three facts carry over from finite dimensions, each needing completeness:

The Hilbert-space toolkit (stated)

Projection theorem. If M is a closed subspace of a Hilbert space H, every $f \in H$ has a unique closest point $Pf \in M$, and $f - Pf \perp M$. This is the infinite-dimensional version of least squares (§9.11).

Orthonormal bases. If $\{e_k\}$ is a complete orthonormal set, then $f = \sum_k \langle f, e_k\rangle e_k$, with convergence in norm, and **Parseval's identity** $\|f\|^2 = \sum_k |\langle f, e_k\rangle|^2$ holds. Fourier series are this theorem for $e_k = e^{jkt}/\sqrt{2\pi}$ (§15.8).

Riesz representation. Every continuous linear functional $\phi : H \to \mathbb{C}$ has the form $\phi(f) = \langle f, g\rangle$ for exactly one $g \in H$.

Now the first genuinely new phenomenon.

The unit ball of an infinite-dimensional Hilbert space is not compact

Let $\{e_n\}$ be an infinite orthonormal sequence. Then:

1. $\|e_m - e_n\| = \sqrt{2}$ for $m \ne n$, so no subsequence converges in norm.
2. For every fixed $f \in H$, $\langle f, e_n\rangle \to 0$.

Proof. (1) By Pythagoras, $\|e_m - e_n\|^2 = \|e_m\|^2 + \|e_n\|^2 = 2$, since $e_m \perp e_n$. A convergent subsequence would be Cauchy, but its terms all stay $\sqrt{2}$ apart.

(2) For any N, the projection of f onto $\operatorname{span}\{e_1, \dots, e_N\}$ is $\sum_{n \le N} \langle f, e_n\rangle e_n$. Its norm is at most $\|f\|$, which gives **Bessel's inequality** $\sum_{n=1}^{N} |\langle f, e_n\rangle|^2 \le \|f\|^2$. Since N is arbitrary, the series $\sum_n |\langle f, e_n\rangle|^2$ converges, so its terms tend to zero. ■

In $\mathbb{R}^n$, Bolzano–Weierstrass guarantees that every bounded sequence has a convergent subsequence. Part (1) says that in infinite dimensions this fails for the simplest bounded sequence there is. Part (2) says the sequence *does* go to zero in a weaker sense. This is the orchestra's leaking energy, made precise.

Weak convergence

$f_n \rightharpoonup f$ (**weakly**) if $\langle f_n, g\rangle \to \langle f, g\rangle$ for every $g \in H$. Norm convergence implies weak convergence, but not conversely: $e_n \rightharpoonup 0$ while $\|e_n\| = 1$.

In $L^2[0,1]$ the functions $\sqrt{2}\sin(n\pi x)$ form such a sequence. Their weak convergence to zero is the Riemann–Lebesgue lemma. Weak convergence is what rescues compactness: every bounded sequence in a Hilbert space has a *weakly* convergent subsequence. Existence proofs for PDEs and optimal controls work this way. Find a minimizing sequence, extract a weakly convergent subsequence, and show the limit is a solution.

Operators: bounded means continuous

A linear map between normed spaces is called an **operator**. Its size is the **operator norm** $\|T\| = \sup_{\|f\|\le 1}\|Tf\|$. For a matrix with the Euclidean norm this is the largest singular value (§10.7).

Bounded ⇔ continuous

A linear operator T between normed spaces is continuous if and only if it is **bounded**, meaning $\|Tf\| \le C\|f\|$ for some constant C.

Proof. If T is bounded, then $\|Tf - Tg\| = \|T(f-g)\| \le C\|f-g\|$, so T is Lipschitz and in particular continuous.

Conversely, suppose T is continuous at 0. Take $\varepsilon = 1$: there is a $\delta > 0$ such that $\|h\| \le \delta$ implies $\|Th\| \le 1$. For any $f \neq 0$, put $h = \delta f/\|f\|$, which has norm δ. By linearity,

$$\|Tf\| = \frac{\|f\|}{\delta}\,\|Th\| \le \frac{1}{\delta}\|f\|,$$

so $C = 1/\delta$ works. ∎

Every matrix is bounded. In infinite dimensions many natural operators are not. Differentiation $D = d/dx$ on smooth functions with the sup norm sends $f_n(x) = \sin(nx)$, which has $\|f_n\|_\infty = 1$, to $n\cos(nx)$, which has norm n. No constant C bounds it, so **differentiation is discontinuous**. That is the reason numerical differentiation amplifies noise (§22.7): it isn't sloppy coding, it's a theorem.

Integration is the opposite. The **Volterra operator** $(Vf)(x) = \int_0^x f(t)\,dt$ on $L^2[0,1]$ is bounded, with $\|V\| = 2/\pi \approx 0.637$. More than that, it is **compact**: it maps bounded sets to sets whose closure is compact, so it squeezes the infinite-dimensional unit ball into something essentially finite-dimensional. Its singular values are

$$\sigma_k = \frac{2}{(2k-1)\pi}, \qquad k = 1, 2, 3, \dots$$

which decay to zero. A compact operator always has singular values tending to zero. Now the inverse problem of the hook makes sense. Inverting V means dividing by the σ_k, and $1/\sigma_k \to \infty$. The inverse of a compact operator on an infinite-dimensional space is never bounded.

Dual spaces and Hahn–Banach

The **dual** V^* of a normed space is the space of its continuous linear functionals: the "measurements" you can make on its elements. Examples are evaluating a filter output at one instant, or taking an inner product with a template. Riesz says a Hilbert space is its own dual. For $1 < p < \infty$, the dual of ℓ^p is ℓ^q with $1/p + 1/q = 1$. That is why Hölder's inequality $|\sum x_k y_k| \le \|x\|_p\|y\|_q$ is everywhere.

Does a general Banach space have *enough* functionals to tell its points apart? In $\mathbb{R}^n$, separating a point from a convex set by a hyperplane was the engine of Ch. 47. Its infinite-dimensional heir is this theorem.

Hahn–Banach (stated)

A continuous linear functional defined on a subspace of a normed space extends to the whole space with the same norm. Geometric form: a closed convex set and a point outside it can always be strictly separated by a continuous linear functional, a "hyperplane" $\{\phi = c\}$.

So the walls of convex optimization survive in spaces of signals. That is why duality theory works for optimal control and for infinite linear programs. The proof needs no geometry of $\mathbb{R}^n$ at all. It uses Zorn's lemma, a form of the axiom of choice (Ch. 64).

Baire's theorem: why monsters win

The last pillar answers a question from Ch. 60.

Baire category theorem and uniform boundedness (stated)

A complete metric space is not a countable union of nowhere-dense sets. Consequence (Banach–Steinhaus): if a family of bounded operators on a Banach space is bounded at each point, $\sup_\alpha \|T_\alpha f\| < \infty$ for every f, then it is uniformly bounded, $\sup_\alpha \|T_\alpha\| < \infty$.

Banach applied this in 1931. In $C[0,1]$, the functions that have a derivative at even one point form a countable union of nowhere-dense sets, so "most" continuous functions are nowhere differentiable. Similarly, the Fourier partial-sum operators $f \mapsto S_N f(0)$ on continuous functions have norms (the Lebesgue constants) that grow like $\frac{4}{\pi^2}\ln N$. They are not uniformly bounded, so by Banach–Steinhaus some continuous function has a Fourier series that diverges at 0. In both cases the monster's existence is proved without ever building it.

Spectral theory: eigenvalues and beyond

For a symmetric matrix the spectral theorem gives an orthonormal eigenbasis (§10.5). The closest infinite-dimensional analogue is this.

Spectral theorem for compact self-adjoint operators (stated)

If T is compact and self-adjoint on a Hilbert space, then $Tf = \sum_k \lambda_k \langle f, e_k \rangle e_k$, where the e_k are orthonormal eigenvectors and the real eigenvalues $\lambda_k \to 0$.

Separation of variables for the heat and wave equations (§13.3) is this theorem in action. The inverse of $-d^2/dx^2$ with zero boundary values is a compact self-adjoint integral operator on $L^2[0,1]$ with eigenfunctions $\sin(k\pi x)$ and eigenvalues $1/(k\pi)^2$.

Without compactness, eigenvectors can vanish entirely. Define the **spectrum** $\sigma(T)$ as the set of λ for which $T - \lambda I$ has no bounded inverse. In finite dimensions that is exactly the set of eigenvalues. In infinite dimensions it can be much more:

- **Multiplication**, $(Mf)(x) = xf(x)$ on $L^2[0,1]$, is bounded and self-adjoint. It has *no eigenvalues at all*: $xf(x) = \lambda f(x)$ forces $f = 0$ almost everywhere. Yet its spectrum is the whole interval $[0,1]$. This is the position operator of quantum mechanics, and it has a *continuous spectrum*.
- **The right shift** $S(x_1, x_2, \dots) = (0, x_1, x_2, \dots)$ on ℓ^2 has no eigenvalues, preserves norms, and has spectrum equal to the closed unit disk.

Back to the hook:

Heisenberg's relation has no finite-dimensional solution

There are no $n \times n$ matrices X, P (for any $n \geq 1$) with $XP - PX = j\hbar \mathbf{I}$, $\hbar \neq 0$.

Proof. The trace satisfies $\operatorname{tr}(AB) = \operatorname{tr}(BA)$. Taking traces of both sides gives $\operatorname{tr}(XP) - \operatorname{tr}(PX) = 0$ on the left and $j\hbar \operatorname{tr} \mathbf{I} = j\hbar n \neq 0$ on the right. ∎

On $L^2(\mathbb{R})$, by contrast, take $(X\psi)(x) = x\psi(x)$ and $(P\psi)(x) = -j\hbar\,\psi'(x)$. Then

$$(XP - PX)\psi = -j\hbar\, x\psi' + j\hbar\,(x\psi)' = j\hbar\,\psi,$$

so the relation holds. The answer to the hook is that X and P are operators on an infinite-dimensional Hilbert space, and neither is a matrix. They pay a price for it: P is a differentiation and is unbounded, and on $L^2(\mathbb{R})$ the operator X is unbounded too.

? Could bounded operators have done the job?

The trace trick only rules out matrices. Infinite-dimensional operators have no trace in general. Maybe some clever pair of *bounded* operators on ℓ^2 satisfies $XP - PX = j\hbar I$, which would spare physics the problems of unbounded operators: domains, self-adjointness and convergence. The answer (Wintner 1947, Wielandt 1949) is no, and the proof is a short and beautiful spectral argument. It forces quantum theory into unbounded operators, and the Stone–von Neumann theorem then says the pair (X, P) is essentially unique. Both stories are in Ch. 92.

? Eigenvectors that aren't there

Physicists write position "eigenstates" $|x_0\rangle$ with $X|x_0\rangle = x_0|x_0\rangle$ and expand every state in them, as if M had a full set of eigenvectors. But M has *no* eigenvectors in L^2, and the would-be eigenvector is a Dirac delta, which isn't a function. What exactly replaces "a basis of eigenvectors" for an operator with continuous spectrum? The answer is spectral measures and projection-valued integrals, $T = \int \lambda\, dE(\lambda)$. That is the full spectral theorem of Ch. 90.

61.4 The algorithm

Seeing an ill-posed problem. Discretize the Volterra operator on n grid points, measure its singular values against the formula, and try to undo it on slightly noisy data:

```
import numpy as np

n = 200                                  # grid points on [0, 1]
h = 1.0 / n
x = (np.arange(n) + 0.5) * h             # midpoints
V = h * np.tril(np.ones((n, n)))         # (Vf)(x) = integral_0^x f(t) dt
V -= 0.5 * h * np.eye(n)                 # midpoint correction on the diagonal

U, s, Wt = np.linalg.svd(V)
k = np.arange(1, 6)
print("sigma_k  numeric:", np.round(s[:5], 4))
print("sigma_k  theory :", np.round(2 / ((2 * k - 1) * np.pi), 4))

f_true = np.cos(3 * np.pi * x)           # the signal we want to recover
rng = np.random.default_rng(0)
g = V @ f_true + 1e-3 * rng.standard_normal(n)   # measured, slightly noisy

f_naive = np.linalg.solve(V, g)          # exact inverse = differentiate the data
r = 20                                   # keep only r singular values
f_tsvd = Wt[:r].T @ ((U[:, :r].T @ g) / s[:r])
for name, f in [("naive", f_naive), ("TSVD r=20", f_tsvd)]:
    print(f"{name:10s} relative error = {np.linalg.norm(f - f_true) / np.linalg.norm(f_true):.3f}")
print("condition number of V:", round(s[0] / s[-1]))
```

The first five singular values match $2/((2k-1)\pi) = 0.6366,\ 0.2122,\ 0.1273,\ 0.0909,\ 0.0707$ to four digits. The matrix's condition number is about 6.5×10^4, and it grows without bound as the grid is refined, because the matrix is approximating an operator whose inverse is unbounded. With noise of only 10^{-3}, the naive solve returns a reconstruction with a relative error of 12, or 1,200%. **Truncated SVD**, which keeps only the 20 largest singular values, brings the error down to 14%.

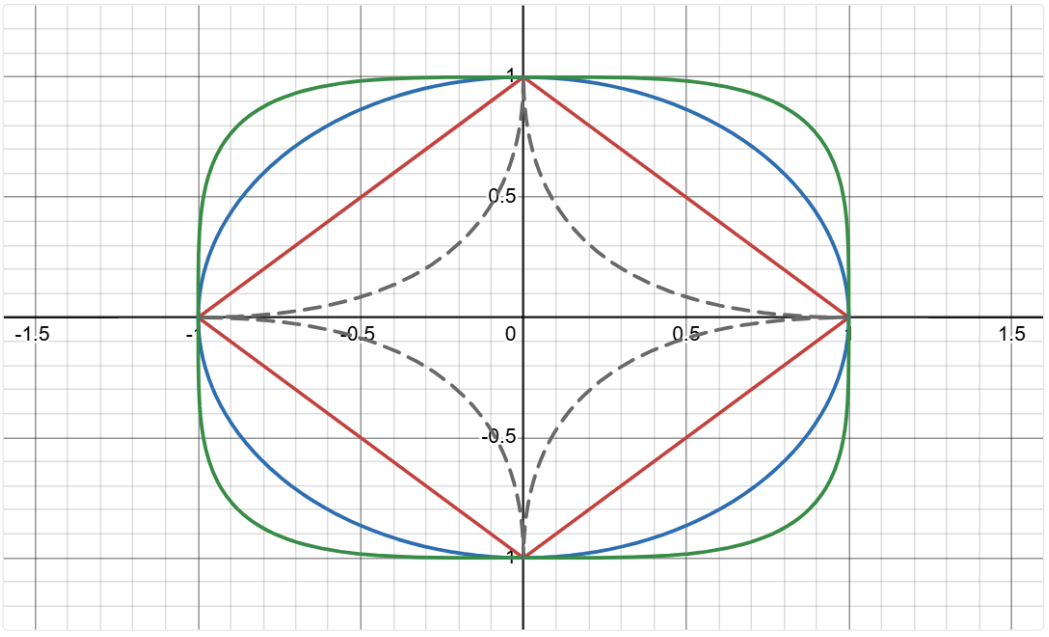
What the algorithm is doing. The data component along the k-th singular vector gets divided by σ_k. For small σ_k that component is almost pure noise, and dividing by a tiny number blows the noise up. Truncation trades a little bias (discarding fine detail) for a large cut in variance. **Tikhonov regularization** makes the same trade smoothly, replacing $1/\sigma_k$ by $\sigma_k/(\sigma_k^2 + \alpha)$.

Cost and failure modes.

- A full SVD costs $O(n^3)$. For large problems one uses iterative Krylov methods. Their iteration count itself acts as a regularizer, which is called *semi-convergence*: stop too late and the noise floods back in.
- Refining the grid does *not* fix ill-posedness. It makes the condition number worse, because a finer grid samples more of the operator's decaying spectrum.
- The best truncation level depends on the noise level and on how smooth the true signal is. Pick it from the same noisy data and you can fool yourself.

61.5 The experiment

Reshape the unit ball. Change the exponent 4 to 20 and watch the ball approach the square, which is the $p = \infty$ ball. Change 0.5 to 0.9 and then 1, and watch the star inflate to the diamond. It becomes convex exactly at $p = 1$.



Unit "balls" of $\|\cdot\|_p$ for $p = 1$ (red), 2 (blue), 4 (green) and $\frac{1}{2}$ (dashed grey, not a norm). Only $p = 2$ comes from an inner product.

Converging in energy but not in peak. This script measures the Fourier partial sums of a square wave in two norms:

```
import numpy as np

x = np.linspace(-np.pi, np.pi, 20_001)
dx = x[1] - x[0]
f = np.sign(x)                                  # square wave on one period

def partial_sum(N):
    """Fourier partial sum of the square wave, odd harmonics up to N."""
    k = np.arange(1, N + 1, 2)
    return (4 / np.pi) * (np.sin(np.outer(x, k)) / k).sum(axis=1)

print("   N    L2 error   sup error   max of S_N")
for N in [1, 5, 25, 125, 625]:
    e = partial_sum(N) - f
    l2 = np.sqrt(np.sum(e**2) * dx / (2 * np.pi))  # RMS = L2 norm / sqrt(2 pi)
    print(f"{N:4d}   {l2:8.4f}   {np.abs(e).max():8.4f}   {partial_sum(N).max():8.4f}")
```

The RMS error falls from 0.435 at $N = 1$ to 0.026 at $N = 625$, roughly halving each time N grows five-fold. The sup error stays pinned at 1.000, because a continuous partial sum can't follow a jump. The overshoot settles at 1.179, the Gibbs constant. Knobs to turn:

- Replace `np.sign(x)` by the continuous triangle wave `np.abs(x)`. Its partial sums use cosines, $\frac{\pi}{2} - \frac{4}{\pi}\sum_{k\ \text{odd}} \cos(kx)/k^2$. Watch the sup error go to zero too.
- Measure the L^1 error. Does it decay faster or slower than L^2?
- Change the exponent of `k` and see how the smoothness of a signal controls the decay rates in different norms.

61.6 The challenge

1. **Intuition.** In ℓ^2, explain without calculation why the sequence $e_1, e_2, e_3, \ldots$ "goes to zero" when measured against any fixed vector, but never gets close to zero. Then decide: does $(e_1 + \cdots + e_n)/n$ converge in norm? To what?

2. **Derivation.** Show that the Volterra operator's adjoint is $(V^*g)(x) = \int_x^1 g(t)\,dt$. Then show that the eigenfunctions of V^*V satisfy $-u'' = \lambda^{-1}u$, with $u(1) = 0$ and $u'(0) = 0$. Deduce the singular values $\sigma_k = 2/((2k-1)\pi)$, and so $\|V\| = 2/\pi$.
3. **Implementation.** Add Tikhonov regularization to the inverse-problem script. For noise levels $10^{-2}, 10^{-3}, 10^{-4}$, plot the reconstruction error against α on a log scale and find the best α for each. Then implement one rule that chooses α *without* knowing `f_true`, such as the discrepancy principle or the L-curve, and compare it with the optimum.
4. **Open investigation.** For the right shift S on ℓ^2, compute the eigenvalues of its $n \times n$ truncation (a nilpotent Jordan block), and then of slightly perturbed versions: add 10^{-10} in the corner entry $(1, n)$. Where do the eigenvalues go as n grows? Compare with the true spectrum, the closed unit disk, and look up "pseudospectra". Is there a rule that predicts which finite truncations of an infinite operator can be trusted?

61.7 Loose threads

Functional analysis turns "signal", "state" and "field" into points of a space, and "filter", "observable" and "PDE" into operators on it. This chapter set up the stage: norms, completeness, Hilbert geometry, bounded and compact operators, duality and the first spectral theorem. The plot is still to come.

> **? How much regularization is right?**
>
> Truncated SVD worked once we chose r, and Tikhonov works once we choose α, but the best choice depended on the unknown true signal. Is there a principled rule that is provably near-optimal with only the noisy data in hand? It depends on how fast the singular values decay compared with the signal's coefficients (the Picard condition). The Bayesian reading, where α is a ratio of noise variance to prior variance, gives a second answer that is in tension with the first. Both are in Ch. 91.

Further pointers. Sobolev spaces, which are Hilbert spaces of functions with square-integrable derivatives, turn PDEs into problems of projection and duality (Ch. 89). Compressed sensing exploits the pointed corners of the ℓ^1 ball from the figure, and convex duality supplies its certificates (Ch. 48). Category theory, next in this track, asks what all these "spaces with structure-preserving maps" have in common (Ch. 62).

Further reading

- E. Kreyszig, *Introductory Functional Analysis with Applications*, Wiley, 1978. The gentlest full introduction, written for readers coming from engineering mathematics.
- B. D. MacCluer, *Elementary Functional Analysis*, Springer, 2009. Short, modern and readable, with good coverage of spectral theory.
- M. Reed and B. Simon, *Methods of Modern Mathematical Physics I: Functional Analysis*, Academic Press, 1980. The standard bridge from Hilbert spaces to quantum mechanics.
- P. C. Hansen, *Discrete Inverse Problems: Insight and Algorithms*, SIAM, 2010. SVD, truncation and Tikhonov regularization in practice, with code.

Chapter 62 — Category Theory: Mathematics About Mathematics

> **Level 2 · Track E — The Abstract Mathematics Behind Everything Builds on:** Ch. 1, Ch. 9, Ch. 56 · **Leads to:** Ch. 66, Ch. 99, Ch. 95 · **Related:** Ch. 58, Ch. 29, Ch. 64, Ch. 93 **Reading time:** ~30 min

62.1 The hook

A colleague hands you a compiled library function. You can't read its source. All you know is its type signature:

```
mystery : List[A] -> List[A]      # for every type A
```

It works on lists of integers, strings or circuit netlists. You run it once, `mystery([1, 2, 4])`, and get `[4, 2, 1]`. Maybe it reverses. Maybe it does something stranger on longer lists. Now take any function `f`, say `even : Int -> Bool`. Is it true that

```
mystery(map(f, xs)) == map(f, mystery(xs))      # for every list xs?
```

In words: does the order of "transform the elements" and "apply `mystery`" matter? You are being asked about code you have never seen, and yet the type seems to know. A function that must work for *every* type `A` can't look inside the elements, so it can't compare, add or print them. Is that restriction enough to settle the question?

Here is a second puzzle that looks unrelated. Write down five constructions you know:

- the Cartesian product $A \times B$ of two sets,
- the greatest common divisor $\gcd(a, b)$ of two integers,
- the minimum $\min(a, b)$ of two real numbers,
- the logical conjunction "p AND q",
- a tuple type `(A, B)` in a programming language.

They come from set theory, number theory, analysis, logic and software. Yet they are *the same construction*, done in five different worlds: not similar, the same, in a sense precise enough to prove theorems about all five at once.

Both puzzles have one answer: stop looking at what things *are*, and look at how they *connect*. That shift of attention is **category theory**. It began in 1945, when Eilenberg and Mac Lane wanted to say precisely what "natural" meant, and grew into a language that runs through algebra, logic, programming-language design and composable systems. By the end of this chapter you will have proved the `mystery` claim for a whole family of functions, and seen why gcd, min, AND, tuples and Cartesian products are one idea.

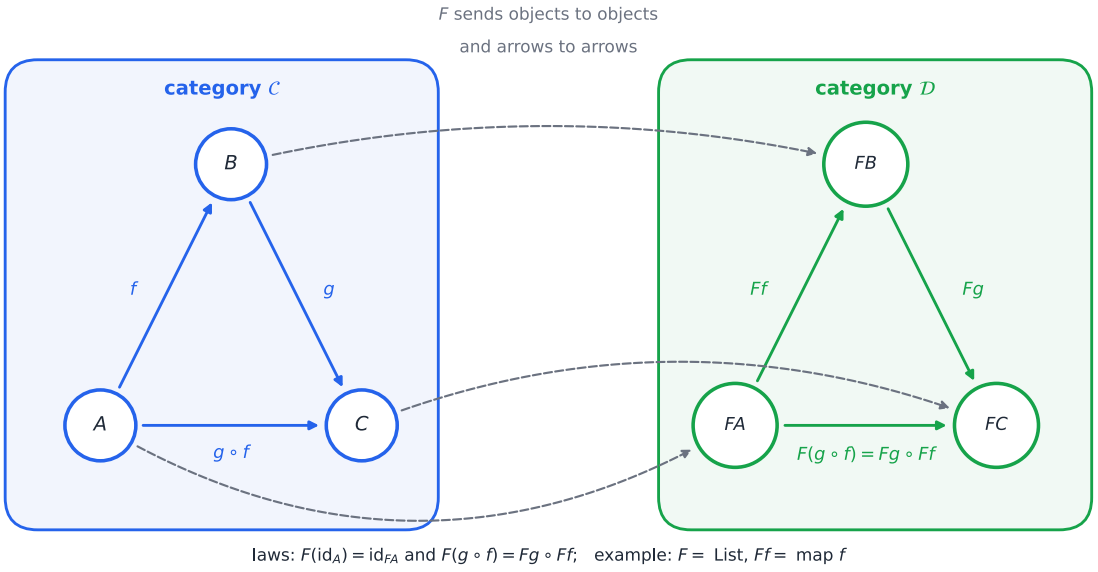
62.2 The big idea

Most of mathematics studies **objects**: numbers, sets, groups, vector spaces. Category theory studies the **arrows between them** and how arrows compose. It treats the inside of an object as off limits. Whatever you want to know about an object, you must learn by probing it with arrows.

An analogy: you could describe a person by height, address and blood type, or by their relationships, who they call and who calls them. The second description never opens the person up, and yet it largely determines them. Category theory pushes that view as far as it will go, which turns out to be very far.

Three layers build on each other.

1. A **category** is a world: objects, arrows between them, and a rule for composing arrows end to end. Sets with functions, vector spaces with linear maps (Ch. 9, §9.5), groups with homomorphisms (Ch. 56), and types with programs are all categories.
2. A **functor** is a map between worlds that respects composition. "Take the list of" is a functor: it sends a type `A` to the type `List[A]`, and a function `f` to the function `map f`.
3. A **natural transformation** is a map between functors that does the same thing "uniformly", without peeking inside. `reverse` is one: it converts lists into lists the same way for every element type.



A functor F copies the shape of one world into another: every object gets an image, every arrow gets an image, and composites go to composites. For $F = \text{List}$, the image of an arrow f is `map f`.

Above all three sits the idea that answers the second puzzle: the **universal property**. Instead of defining $A \times B$ by what is inside it (ordered pairs), define it by its relationships: the object with arrows to A and to B through which every other such pair of arrows passes in exactly one way. That never mentions elements, so it makes sense in any category: pairs in sets, the gcd among divisors, the minimum among reals.

62.3 The mathematics

Categories

Category

A **category** $\mathcal{C}$ consists of

- a collection of **objects** $A, B, C, \ldots$;
- for each pair of objects a set $\mathrm{Hom}(A, B)$ of **arrows** (morphisms) $f : A \to B$;
- a **composition** $g \circ f : A \to C$ for every $f : A \to B$ and $g : B \to C$;
- an **identity** arrow $\mathrm{id}_A : A \to A$ for every object,

such that composition is associative, $h \circ (g \circ f) = (h \circ g) \circ f$, and identities are neutral, $f \circ \mathrm{id}_A = f = \mathrm{id}_B \circ f$.

That is all. Nothing says arrows must be functions or objects must be sets. Some examples, from the familiar to the strange:

Category	Objects	Arrows $A \to B$	Composition
Set	sets	functions	usual composition
$\mathbf{Vect}_K$	vector spaces over a field K	linear maps	composition (matrix product)
Grp	groups	homomorphisms	composition
a poset $(P, \leq)$	elements of P	one arrow if $a \leq b$, none otherwise	transitivity
a monoid M	a single object $\star$	the elements of M	the monoid operation
Type	types of a language	(total) programs	piping one into the next

The poset and monoid examples show how little a category needs. In the divisibility poset there is an arrow $4 \to 12$ because 4 divides 12, and composition is the transitivity of divisibility. In a monoid all the information sits in how arrows compose.

An arrow $f : A \to B$ is an **isomorphism** if it has a two-sided inverse g, with $g \circ f = \mathrm{id}_A$ and $f \circ g = \mathrm{id}_B$: a bijection in **Set**, an invertible linear map in $\mathbf{Vect}_K$. Category theory never asks whether two objects are *equal*, only whether they are isomorphic, and *how*.

Every category has an **opposite** $\mathcal{C}^{\mathrm{op}}$ with every arrow reversed, so every theorem about all categories comes with a free dual theorem. Every definition below has a "co-" version for nothing.

Functors

Functor

A **functor** $F : \mathcal{C} \to \mathcal{D}$ assigns to each object A of $\mathcal{C}$ an object FA of $\mathcal{D}$, and to each arrow $f : A \to B$ an arrow $Ff : FA \to FB$, such that

$$F(\mathrm{id}_A) = \mathrm{id}_{FA}, \qquad F(g \circ f) = Fg \circ Ff.$$

The **List functor** sends a type A to $\mathrm{List}(A)$ and $f : A \to B$ to map f. The laws say that mapping the identity does nothing, and that mapping f then mapping g is the same as mapping $g \circ f$ once. Compilers use exactly this law to fuse two loops into one.

A **forgetful functor** $U : \mathbf{Vect}_K \to \mathbf{Set}$ forgets the linear structure. A **contravariant** functor reverses arrows: the dual space construction sends $T : V \to W$ to $T^* : W^* \to V^*$, the transpose in disguise, which is why $(\mathbf{AB})^{\mathsf{T}} = \mathbf{B}^{\mathsf{T}}\mathbf{A}^{\mathsf{T}}$ reverses the order (Ch. 99).

The engineer's favourite functor is in hiding. Lumped linear time-invariant systems compose by **cascading**, and the transfer function sends a cascade to a product, $H_{g \circ f}(s) = H_g(s)\,H_f(s)$, and a plain wire to $H = 1$ (Ch. 14). That is the functor law: block diagrams can be "multiplied out" because the Laplace transform is functorial.

Functors **preserve isomorphisms**: if f has inverse g, then $Fg \circ Ff = F(g \circ f) = F(\mathrm{id}) = \mathrm{id}$. Read backwards, if Ff is *not* an isomorphism, f wasn't either, so a functor into a simpler world can prove something impossible in a harder one.

? A functor that proves a theorem

Topologists attach a group to every shape and a group homomorphism to every continuous map, functorially. From that plus a one-line diagram chase it follows that every continuous map of a disc to itself has a fixed point (Brouwer's theorem), a statement with no visible algebra in it. Which functor does it, and why the diagram chase is unavoidable, is the business of the fundamental group in Ch. 58.

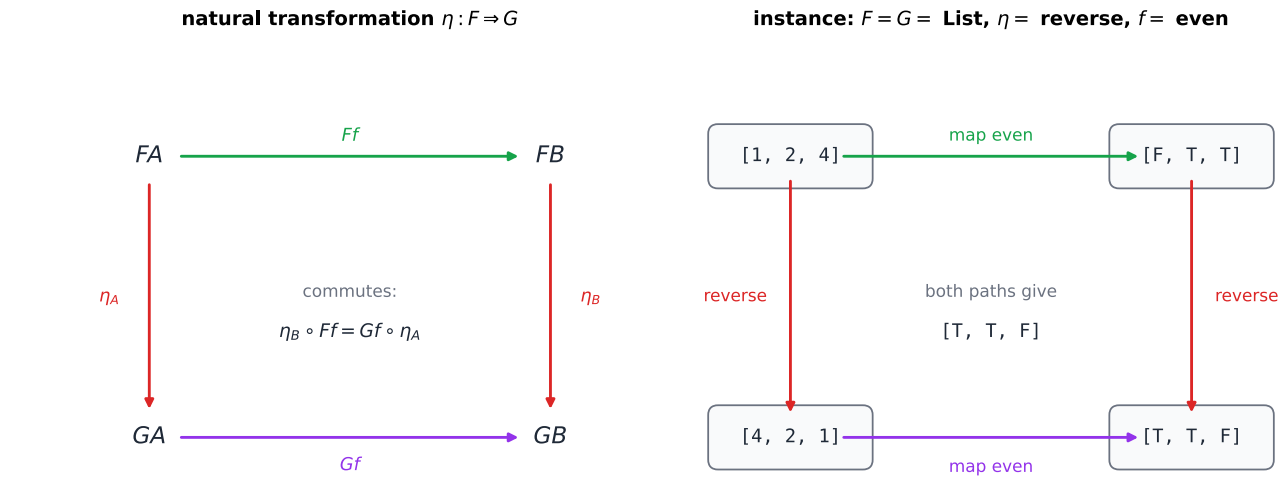
Natural transformations

Natural transformation

Given functors $F, G : \mathcal{C} \to \mathcal{D}$, a **natural transformation** $\eta : F \Rightarrow G$ is a family of arrows $\eta_A : FA \to GA$, one for each object A, such that for every arrow $f : A \to B$

$$\eta_B \circ Ff = Gf \circ \eta_A.$$

The equation is the **naturality square**: going across and then down equals going down and then across.



The naturality square. On the right, the hook's claim for `mystery = reverse` *and* `f = even`*: both routes through the square land on* `[T, T, F]`*.*

Natural transformations are what "uniform" or "canonical" means. The hook's claim is exactly that `mystery` is a natural transformation List ⇒ List. A function that can't look inside the elements can only shuffle, copy and drop them by *position*, and for that whole family we can prove it.

Position-only list functions are natural

Suppose that for each length n there is a list of indices $\sigma_n = (i_1, \ldots, i_m)$ with every $i_k \in \{1, \ldots, n\}$ (the length m may depend on n), and define, for every type A,

$$\eta_A(x_1, \ldots, x_n) = (x_{i_1}, \ldots, x_{i_m}).$$

Then η is a natural transformation List ⇒ List: for every $f : A \to B$, $\eta_B \circ \mathrm{map}\, f = \mathrm{map}\, f \circ \eta_A$.

Proof. Take $xs = (x_1, \ldots, x_n)$. Mapping first gives $(f(x_1), \ldots, f(x_n))$, which still has length n, so η_B uses the same index list σ_n and returns $(f(x_{i_1}), \ldots, f(x_{i_m}))$. Going the other way, $\eta_A(xs) = (x_{i_1}, \ldots, x_{i_m})$, and mapping f over it gives $(f(x_{i_1}), \ldots, f(x_{i_m}))$. The two lists agree entry by entry. The key step is that map f never changes the length, and η looks at nothing but the length. ■

Reverse ($\sigma_n = (n, \ldots, 1)$), "take the first two", "every other element" and "repeat each element twice" are all of this form, so all are natural. A sort is not: it chooses positions by *comparing contents*. With $f(x) = -x$, sorting $[1, 2]$ and negating gives $[-1, -2]$, but negating then sorting gives $[-2, -1]$. That is why `sort` needs an ordering on `A` in its type and can't have the plain signature `List[A] -> List[A]`.

The converse, that *every* function with the polymorphic type must be position-based, is a deeper fact called **parametricity**; §62.7 says where it is proved.

Naturality also separates constructions that merely exist from constructions that are canonical. A finite-dimensional space V is isomorphic both to its dual V^* and to its double dual V^{**}, but only one of these isomorphisms is natural.

? Two isomorphisms, one of them fake

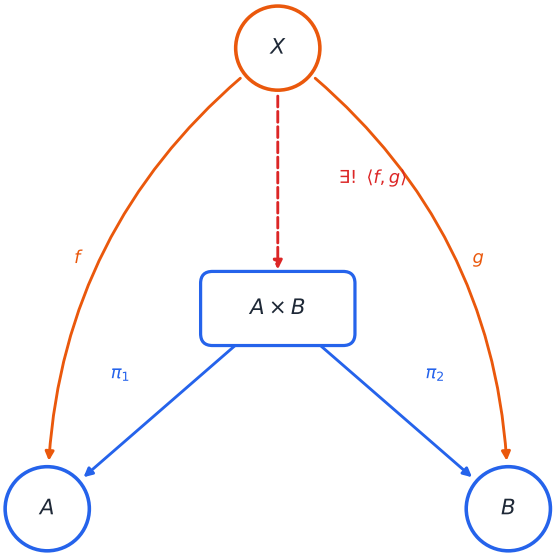
The map $V \to V^{**}$ that sends v to "evaluate at v" needs no choices and passes the naturality test for every linear map. Every isomorphism $V \to V^*$ requires picking a basis or an inner product, and *no* choice makes it natural. Why the double dual is canonical and the single dual is not, and why the double-dual map fails to be onto in infinite dimensions, is where Ch. 99 begins.

Universal properties

Now the second puzzle. Here is a definition of "product" that never mentions elements.

Product

A **product** of objects A and B is an object $A \times B$ with two arrows $\pi_1 : A \times B \to A$ and $\pi_2 : A \times B \to B$ such that for every object X and every pair $f : X \to A$, $g : X \to B$ there is **exactly one** arrow $\langle f, g \rangle : X \to A \times B$ with $\pi_1 \circ \langle f, g \rangle = f$ and $\pi_2 \circ \langle f, g \rangle = g$.



$\pi_1 \circ \langle f, g \rangle = f, \quad \pi_2 \circ \langle f, g \rangle = g$

category	product $A \times B$
Sets	pairs (a, b)
Vector spaces	direct sum $A \oplus B$
Divisibility of integers	gcd(a, b)
Real numbers with $\leq$	min(a, b)
Logic (implication)	a AND b
Program types	tuple (A, B)

one definition, six familiar constructions

$A \times B$ is the "best" object with arrows to both A and B: every rival X factors through it uniquely. Read in six categories, one diagram gives six familiar constructions.

Check it in a few worlds. In **sets**, the only function into the pairs that projects correctly is $x \mapsto (f(x), g(x))$. In a **poset** an arrow is a single fact "$x \leq y$", so for **divisibility** the definition says: p divides a and b, and every common divisor divides p. That is $\gcd(a, b)$. For **reals with** $\leq$ it is $\min(a, b)$; for **propositions ordered by implication**, $p \wedge q$. For **vector spaces** it is the direct sum $V \oplus W$, and $\langle \mathbf{F}, \mathbf{G} \rangle$ is the matrix $\mathbf{F}$ stacked on top of $\mathbf{G}$.

The definition does not promise that a product exists (the experiment in §62.5 builds a case where none does). But when one exists, it is essentially unique. The definition does not claim a product exists; it says what one must do. Sometimes none exists, and the experiment in §62.5 builds such a case. But when one exists, it is essentially unique, and this is the theorem that makes universal properties worth having.

Universal objects are unique up to unique isomorphism

If (P, π_1, π_2) and (Q, ρ_1, ρ_2) are both products of A and B, there is exactly one arrow $u : Q \to P$ compatible with the projections ($\pi_i \circ u = \rho_i$), and it is an isomorphism.

Proof. Because P is a product, the pair ρ_1, ρ_2 from Q factors through P by a unique $u : Q \to P$ with $\pi_i \circ u = \rho_i$. Because Q is a product, the pair π_1, π_2 factors through Q by a unique $v : P \to Q$ with $\rho_i \circ v = \pi_i$. Now look at $u \circ v : P \to P$. It satisfies $\pi_i \circ (u \circ v) = \rho_i \circ v = \pi_i$. But the identity id_P satisfies $\pi_i \circ \mathrm{id}_P = \pi_i$ too, and the universal property (with $X = P$, $f = \pi_1$, $g = \pi_2$) says there is only *one* arrow $P \to P$ with that property. Hence $u \circ v = \mathrm{id}_P$. The same argument in Q gives $v \circ u = \mathrm{id}_Q$. ■

Notice what the proof never used: elements, pairs, numbers. It works in every category at once: one proof, infinitely many theorems. It is also why mathematicians say "*the* product" although, as sets, $A \times B$ and $B \times A$ differ: they are uniquely isomorphic.

Turn every arrow around and you get the **coproduct** $A + B$: disjoint union in sets, lcm among divisors, max among reals, OR in logic, a **sum type** such as `Either[A, B]` in programming. In vector spaces the coproduct is the direct sum *again*, so $V \oplus W$ is both product and coproduct, a coincidence special to linear algebra that lets matrices be cut into blocks in both directions.

Limits, briefly

Products are the simplest **limits**: take a diagram of objects and arrows, and look for the best object with a compatible family of arrows into it.

- The limit of the empty diagram is a **terminal object**, with exactly one arrow into it from everywhere: a one-point set, the zero space, "true", the unit type.
- The limit of two parallel arrows $f, g : A \to B$ is the **equalizer**: $\{a : f(a) = g(a)\}$ in sets, the kernel $\ker(f - g)$ for linear maps (Ch. 9, §9.6). Solving an equation is computing an equalizer.
- The limit of $A \xrightarrow{f} C \xleftarrow{g} B$ is the **pullback** $\{(a, b) : f(a) = g(b)\}$, which database engineers call the **join** of two tables on a key.

Dually, **colimits** include initial objects, coequalizers (quotients, as in Ch. 56) and pushouts (gluing two shapes along an edge). "Solve", "intersect", "join" and "glue" are all limits or colimits.

Adjunctions

The last notion is the most powerful. For a set S, let $F(S)$ be the vector space with basis S: formal linear combinations of elements of S. Let $U(W)$ be the underlying set of a vector space W. The most-used fact of linear algebra is that **a linear map is determined by what it does to a basis, and any values on the basis are allowed** (Ch. 9, §9.5). In categorical language:

> **Free–forgetful adjunction**
>
> For every set S and vector space W there is a bijection, natural in both S and W,
>
> $$\mathrm{Hom}_{\mathbf{Vect}}\big(F(S),\, W\big) \;\cong\; \mathrm{Hom}_{\mathbf{Set}}\big(S,\, U(W)\big).$$

Proof. Send a linear map $T : F(S) \to W$ to its restriction $s \mapsto T(s)$, a function $S \to U(W)$. Conversely, send a function $\phi : S \to U(W)$ to the linear map $\hat{\phi}\big(\sum_k c_k s_k\big) = \sum_k c_k\, \phi(s_k)$; this is well defined because every element of $F(S)$ is a *unique* finite combination of basis elements. Restricting $\hat{\phi}$ recovers ϕ, and a linear map that agrees with T on a basis equals T everywhere, so the two constructions are inverse. Naturality (compatibility with maps of S and of W) is a one-line check. ∎

Whenever $\mathrm{Hom}_{\mathcal{D}}(FX, Y) \cong \mathrm{Hom}_{\mathcal{C}}(X, GY)$ naturally, F is **left adjoint** to G, written $F \dashv G$. The pattern is everywhere.

- **Free constructions.** Free groups, polynomial rings and the free monoid on an alphabet (all strings) are left adjoints of forgetful functors.
- **Currying.** $\mathrm{Hom}(A \times B, C) \cong \mathrm{Hom}(A, C^B)$: a function of two arguments is a function returning a function, so "$- \times B$" is left adjoint to "$B \to -$". Categories where this works (*cartesian closed*) are the semantics of typed functional languages.
- **Rounding.** Order the integers and reals by $\le$ and let $i : \mathbb{Z} \to \mathbb{R}$ be the inclusion. For every real x and integer n,

$$\lceil x \rceil \le n \iff x \le i(n),$$

so the ceiling function is the left adjoint of the inclusion (and the floor its right adjoint). Between posets, adjunctions are called **Galois connections**.

Adjoints obey general laws, the most useful being that **left adjoints preserve colimits and right adjoints preserve limits**. For the ceiling this predicts $\lceil \max(x, y) \rceil = \max(\lceil x \rceil, \lceil y \rceil)$, since max is a coproduct. For the free vector space it predicts $F(S \sqcup T) \cong F(S) \oplus F(T)$. Each "obvious" identity is one instance of a single theorem.

Where it pays off

Type theory and programming languages. Tuples, tagged unions, function types and the unit type are products, coproducts, exponentials and the terminal object, so their typing and pattern-matching rules follow from universal properties. Generic containers are functors, and polymorphic functions are natural transformations, which is why the hook could be answered from a type signature.

Abstract algebra. The first isomorphism theorem (Ch. 56) says a quotient has a universal property: every homomorphism that kills a normal subgroup factors uniquely through it. Free groups, polynomial rings and tensor products are defined, and proved unique, the same way.

Compositional systems. Series wiring is composition; parallel wiring is a second operation $\otimes$, and categories with both are **monoidal categories**. Their arrows are drawn as *string diagrams*, block diagrams with rigorous semantics, and signal-flow graphs, quantum circuits and electrical networks have all been treated this way.

62.4 The algorithm

Category theory is a language, not a numerical method, but it gives algorithms for one task: **checking laws**. A functor must satisfy two equations, a natural transformation one square per arrow. For finite categories these are finite checks; for infinite ones, such as all types, you test random instances, which is how property-based testing tools (QuickCheck and its descendants) catch broken `map` implementations.

Here is a naturality tester for list functions, using random arrows $f : \{0, \dots, 9\} \to \{0, \dots, 9\}$.

```
import random

def fmap(f, xs):                # the List functor on arrows
    return [f(x) for x in xs]

candidates = {
    "reverse":   lambda xs: xs[::-1],
    "first_two": lambda xs: xs[:2],
    "evens_pos": lambda xs: xs[::2],
    "doubled":   lambda xs: [x for x in xs for _ in (0, 1)],
    "sort":      lambda xs: sorted(xs),
    "dedupe":    lambda xs: list(dict.fromkeys(xs)),
}

def is_natural(eta, trials=2000, seed=1):
    rng = random.Random(seed)
    for _ in range(trials):
        xs = [rng.randrange(10) for _ in range(rng.randrange(7))]
        table = [rng.randrange(10) for _ in range(10)]
        f = lambda x: table[x]   # a random arrow int -> int
        if eta(fmap(f, xs)) != fmap(f, eta(xs)):
            return False, (xs, table)
    return True, None

for name, eta in candidates.items():
    ok, witness = is_natural(eta)
    print(f"{name:10s} natural: {ok}" + ("" if ok else f"   counterexample xs={witness[0]}"))
```

The first four pass all 2,000 trials; `sort` and `dedupe` fail, on `xs=[7, 7, 8, 3, 5, 3]` and `xs=[5, 8, 6, 8, 3]`. Each test costs $O(n)$.

Three cautions.

- **A test is not a proof.** Passing 2,000 random squares makes naturality plausible, not certain; a function could misbehave only on lists of length 1,000. The proof in §62.3 settles the position-based case.
- **`dedupe` is the instructive failure.** It looks harmless, but a non-injective f creates duplicates: deduplicating then mapping keeps them, mapping then deduplicating removes them. Equality testing is a peek inside the elements, and any peek breaks uniformity.
- **Finite categories can be checked exhaustively.** With a composition table, associativity costs $O(|\mathrm{Arr}|^3)$ checks. In a poset, testing whether a candidate is a product of two objects costs $O(|\mathrm{Obj}|)$ comparisons, as in the experiment below.

62.5 The experiment

Any poset is a category, and in a poset the universal property becomes a finite search: the product of a and b is a lower bound of both that sits above every other lower bound. The script searches for products among the divisors of N ordered by divisibility, then checks the answer against `gcd`.

```
import numpy as np

# A poset is a category: one arrow x -> y exactly when x <= y.
# KNOBS: the ground set and the order. Try N = 60, 360, or remove elements.
N = 60
removed = set()                # try {6}, or {1}
P = [d for d in range(1, N + 1) if N % d == 0 and d not in removed]
leq = lambda x, y: y % x == 0      # "x divides y"; try lambda x, y: x <= y

def product(a, b):
    """Universal property: p <= a, p <= b, and every such x satisfies x <= p."""
    cones = [x for x in P if leq(x, a) and leq(x, b)]
    best = [p for p in cones if all(leq(x, p) for x in cones)]
    return best[0] if best else None

def coproduct(a, b):
    cocones = [x for x in P if leq(a, x) and leq(b, x)]
    best = [c for c in cocones if all(leq(c, x) for x in cocones)]
    return best[0] if best else None

M = np.array([[product(a, b) or 0 for b in P] for a in P])
print("objects:", P)
print("product table (0 = no product exists):")
print(M)
missing = [(a, b) for a in P for b in P if product(a, b) is None]
print("pairs with no product:", missing[:8], "..." if len(missing) > 8 else "")
print("coproduct(4, 6) =", coproduct(4, 6) if 4 in P and 6 in P else "n/a")
print("gcd check:", all(product(a, b) == np.gcd(a, b) for a in P for b in P) if not removed else "skipped")
```

With $N = 60$ the twelve divisors $1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60$ give a full 12×12 product table, every entry equals the gcd, and `coproduct(4, 6)` returns $12 = \mathrm{lcm}(4, 6)$. Nobody told the program what a gcd is. Things to change:

1. **Delete an object.** Set `removed = {6}`. Now 12 and 30 have common lower bounds $1, 2, 3$, none above the other two, so their product **does not exist**. The universal property is a demand, and the world can refuse it.
2. **Delete the bottom.** Set `removed = {1}`. Now 2 and 3 have no common lower bound at all. How many pairs lose their product?
3. **Change the order.** Replace `leq` by `lambda x, y: x <= y`. The product becomes `min` and the coproduct `max`. Same code, different category.

4. **Grow the world.** Try $N = 360$ or $N = 2^{10}$. For a power of a prime the divisors form a chain, and every product is just the smaller element.

62.6 The challenge

1. **Intuition.** In the category with one object whose arrows are the integers under addition, what is an isomorphism? In the poset of subsets of $\{1, 2, 3\}$ ordered by inclusion, what are the product and coproduct of $\{1, 2\}$ and $\{2, 3\}$, and what are the terminal and initial objects? Why does a poset have at most one isomorphism between any two objects?
2. **Derivation.** Using only the Galois connection $\lceil x \rceil \le n \iff x \le n$, prove that $\lceil \max(x, y) \rceil = \max(\lceil x \rceil, \lceil y \rceil)$ for all real x, y. Then prove the general statement it illustrates: if $F \dashv G$ between posets, then F sends every join (least upper bound) that exists to a join.
3. **Implementation.** Store a finite category as Python dictionaries (arrows with sources and targets, plus a composition table). Write a checker for the identity and associativity laws, then a brute-force search for products of two objects, now checking uniqueness of the factorising arrow. Test it on two objects joined by two parallel non-identity arrows.
4. **Open investigation.** Principal component analysis (Ch. 10, §10.7) picks a "best" basis for a data cloud. Is it natural? For which linear maps applied to every data point (rotations, uniform scalings, axis scalings, shears) does "PCA, then transform" agree with "transform, then PCA"? Explore numerically, and decide what a category of data sets would have to be for PCA to become natural.

62.7 Loose threads

This chapter built the vocabulary and stopped short of two theorems. The **Yoneda lemma** says an object is determined, up to unique isomorphism, by the functor $\mathrm{Hom}(-, A)$ recording all arrows into it: §62.2's "relationships determine the person", made exact. And the **converse of the hook**, that the type `List[A] -> List[A]` *forces* a function to be position-based, is Reynolds's parametricity theorem ("theorems for free", in Wadler's slogan), proved for typed languages in Ch. 66.

> **? Where do monads come from?**
>
> Every adjunction $F \dashv G$ produces a functor $T = G \circ F$ with two extra natural transformations, $A \to TA$ and $TTA \to TA$. For the free vector space, T sends a set to the set of formal linear combinations of its elements. Functional programmers use exactly this structure, called a **monad**, to model exceptions, state, randomness and input/output inside pure code, and probability distributions form one too. Why one structure captures all side effects, and how it relates to proofs, is taken up with type theory in Ch. 66.

Group representations are functors from a one-object category into $\mathbf{Vect}_K$, with natural transformations as intertwining maps (Ch. 95). The tensor product is defined by a universal property for bilinear maps (Ch. 93). Whether categories can replace sets as a foundation (topos theory, univalent foundations) belongs with Ch. 64 and Ch. 68.

Further reading

- Saunders Mac Lane, *Categories for the Working Mathematician*, 2nd ed., Springer, 1998. The classic, by one of the founders; dense but definitive.
- Tom Leinster, *Basic Category Theory*, Cambridge University Press, 2014 (free on arXiv). The best short first course: adjunctions, representables, limits.
- Emily Riehl, *Category Theory in Context*, Dover, 2016. Examples drawn from all of mathematics.
- Brendan Fong and David Spivak, *An Invitation to Applied Category Theory: Seven Sketches in Compositionality*, Cambridge University Press, 2019. Databases, circuits, signal-flow graphs and resource theories.
- Bartosz Milewski, *Category Theory for Programmers*, 2019. Functors, naturality and monads in code.

Chapter 99 — Abstract Linear Algebra: Vectors Without Coordinates

Level 2 · Track E — The Abstract Mathematics Behind Everything Builds on: Ch. 8, Ch. 9, Ch. 10 · **Leads to:** Ch. 61, Ch. 93, Ch. 95 · **Related:** Ch. 56, Ch. 62, Ch. 79, Ch. 90 **Reading time:** ~30 min

99.1 The hook

A temperature sensor sits on a metal bar. Along the bar the temperature rises steadily, and you record two things at the sensor: a small **displacement** d (how far you moved it) and the **gradient** g (how fast temperature changes with position). Working in metres, you write $d = 2$ m and $g = 3$ K/m. The temperature change is $g\,d = 6$ K.

Your colleague works in centimetres. She writes $d = 200$ cm and $g = 0.03$ K/cm. The temperature change is still 6 K, as it must be. But look at what happened to the numbers. When you shrank the unit by a factor of 100, the displacement's component got **100 times bigger** and the gradient's component got **100 times smaller**. Both are "vectors" with one component. They respond to a change of coordinates in *opposite* ways.

In three dimensions the same thing happens with full matrices: under a change of basis $\mathbf{P}$, displacements transform with $\mathbf{P}^{-1}$ and gradients with $\mathbf{P}^{\mathsf{T}}$. Store both as columns of three numbers and the same code works in an orthonormal frame and silently breaks in a skewed one; this is why graphics code must transform surface normals, which are gradients, by the inverse transpose.

So what *is* a vector, if a list of numbers can lie about it? And here is a second, sharper question. Ch. 10 showed how to diagonalize a matrix by choosing a basis of eigenvectors. But

$$\mathbf{N} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \qquad \text{and} \qquad \mathbf{R} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

refuse. $\mathbf{N}$ has only one independent eigenvector, and $\mathbf{R}$, a quarter-turn of the plane, has no real eigenvectors at all. If diagonal is impossible, what is the *simplest* form to which every linear map can be brought, over any number system? The answer turns out to need almost no matrices. It needs the right way to think about vector spaces when nobody hands you coordinates.

99.2 The big idea

The core chapters taught linear algebra through matrices, which is how you compute. This chapter turns the picture round: the **linear map** is the real object, and a matrix is a *shadow* it casts once you pick a basis. Different bases cast different shadows ($\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$, Ch. 9), and the questions worth asking are about what is the same in all of them.

An analogy: a sentence can be written in English, Bengali or French, and a grammar of its *meaning* must work in all of them. Coordinate-free linear algebra is that grammar, with four new tools.

1. **The dual space.** Gradients are not vectors at all. They are *machines that eat a vector and return a number* (the temperature change along it). Such machines form their own vector space, the dual V^*, and that is why they transform the other way.
2. **The quotient.** Sometimes you want to *ignore* a subspace: voltages up to a common ground offset, signals up to a DC component, states up to an unobservable mode. The quotient V/U makes "equal up to something in U" into a new vector space.
3. **Invariant subspaces.** A map T that cannot be diagonalized can still be *chopped*: find subspaces that T maps into themselves, and study T on each piece.
4. **Multilinear forms.** Determinants, inner products and the quadratic forms of Ch. 10 are functions of several vectors at once, linear in each. Seen without coordinates, the determinant becomes a single number attached to a map, the factor by which it scales oriented volume.

The payoff is the answer to the hook: every linear map on a finite-dimensional space, over any field, splits into a direct sum of a few standard pieces. Over $\mathbb{C}$ the pieces are **Jordan blocks**; over any field they are **companion matrices**. Both lists are complete invariants: two maps are similar exactly when their lists agree.

99.3 The mathematics

Fields and spaces

Everything in Ch. 9 works with scalars from any **field** K: a number system where you can add, subtract, multiply and divide by non-zero elements (Ch. 56). Besides $\mathbb{R}$ and $\mathbb{C}$, three fields matter in practice.

- $\mathbb{F}_2 = \{0, 1\}$ with $1 + 1 = 0$. The space $\mathbb{F}_2^3$ has just 8 vectors, and there are $(8-1)(8-2)(8-4) = 168$ invertible 3×3 matrices over $\mathbb{F}_2$. Linear codes are subspaces of $\mathbb{F}_2^n$ (Ch. 79), and every theorem below applies to them.
- $\mathbb{Q}$, the rationals. $\mathbb{R}$ is a vector space over $\mathbb{Q}$ of *infinite* dimension, since $1, \sqrt{2}, \sqrt{3}, \pi, \ldots$ are linearly independent over $\mathbb{Q}$.
- $\mathbb{C}$, whose key property is that every polynomial has a root. That single fact is why Jordan form exists over $\mathbb{C}$ and not over $\mathbb{R}$.

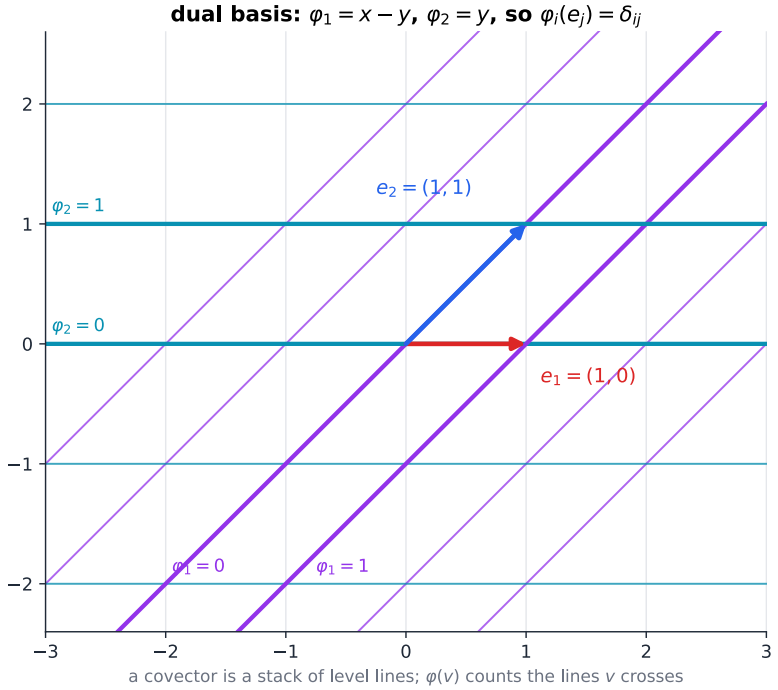
The linear maps $V \to W$ form a vector space $\mathrm{Hom}(V, W)$ of their own, of dimension $\dim V \cdot \dim W$.

Dual spaces

Dual space

The **dual** of V is $V^* = \mathrm{Hom}(V, K)$, the space of linear functionals $\varphi : V \to K$, also called **covectors**. If $e_1, \ldots, e_n$ is a basis of V, the **dual basis** $\varphi^1, \ldots, \varphi^n$ of V^* is defined by $\varphi^i(e_j) = \delta_{ij}$, so φ^i reads off the i-th coordinate.

A covector on the plane is best pictured as a family of parallel, equally spaced **level lines**: $\varphi(v)$ counts how many lines v crosses. The gradient of the hook is one: its value on a displacement is the temperature change. When the basis is not orthonormal, the dual basis does not point along the basis vectors at all.



The dual basis of $e_1 = (1, 0)$, $e_2 = (1, 1)$ is $\varphi^1 = x - y$, $\varphi^2 = y$. Each e_j crosses exactly one line of its own covector and lies along a level line of the other. Tilting e_2 would tilt the purple lines, not the teal ones.

Linear maps act on covectors *backwards*. Given $T : V \to W$, the **dual map** $T^* : W^* \to V^*$ is $T^*(\psi) = \psi \circ T$: to measure a vector of V with ψ, push it through T first. No coordinates are involved. But choose coordinates and something familiar appears.

The dual map is the transpose

Let $T : V \to W$ have matrix $\mathbf{A}$ with respect to bases (e_j) of V and (f_i) of W. Then T^* has matrix $\mathbf{A}^{\mathsf{T}}$ with respect to the dual bases (ψ^i) of W^* and (φ^j) of V^*.

Proof. The matrix of T is defined by $Te_j = \sum_i a_{ij} f_i$. To find the matrix of T^*, expand $T^*\psi^i$ in the basis φ^j; the coefficient of φ^j is $(T^*\psi^i)(e_j)$, because φ^j is the only dual basis vector that is non-zero on e_j. Compute it:

$$(T^*\psi^i)(e_j) = \psi^i(Te_j) = \psi^i\Big(\sum_k a_{kj} f_k\Big) = a_{ij}.$$

So $T^*\psi^i = \sum_j a_{ij}\varphi^j$: the entry in row j, column i of the matrix of T^* is a_{ij}. That is $\mathbf{A}^{\mathsf{T}}$. ∎

The transpose was never "flip the array". It is the matrix of the map that pulls measurements back, and $(ST)^* = T^*S^*$ reverses order because pulling back reverses order. Now the hook is explained. Under a change of basis by $\mathbf{P}$, coordinates of vectors change by $\mathbf{P}^{-1}$, so coordinates of covectors must change by $\mathbf{P}^{\mathsf{T}}$ to keep every pairing $\varphi(v)$ unchanged. Displacements are vectors, gradients are covectors, and an inner product is exactly the extra structure that lets you convert one into the other (Ch. 9, §9.8). Without an inner product the two kinds stay apart.

The double dual. Every $v \in V$ defines a functional on V^*, "evaluate at v": $\mathrm{ev}_v(\varphi) = \varphi(v)$. This gives a map $\iota : V \to V^{**}$ that uses no basis, no inner product, nothing.

Finite-dimensional spaces are their own double duals

If $\dim V = n < \infty$, then $\iota : V \to V^{**}$, $v \mapsto \mathrm{ev}_v$, is a linear isomorphism, and it is natural: $T^{**} \circ \iota_V = \iota_W \circ T$ for every linear $T : V \to W$.

Proof. Linearity: $\mathrm{ev}_{av+bw}(\varphi) = a\varphi(v) + b\varphi(w)$. Injectivity: if $v \neq 0$, extend v to a basis and let φ be the first dual basis vector; then $\mathrm{ev}_v(\varphi) = 1 \neq 0$, so the kernel is zero. Dimensions: the dual basis shows $\dim V^* = n$, hence $\dim V^{**} = n$, and an injective linear map between spaces of equal finite dimension is onto (rank–nullity, Ch. 9, §9.6). Naturality: for $\psi \in W^*$, $(T^{**}\,\mathrm{ev}_v)(\psi) = \mathrm{ev}_v(T^*\psi) = \psi(Tv) = \mathrm{ev}_{Tv}(\psi)$. ∎

Here is the point that Ch. 62 promised. V and V^* also have the same dimension, so they are isomorphic, but every isomorphism between them needs a choice (a basis, or an inner product), and none of them is natural. The double dual needs no choice. In the proof, the only place where finite dimension entered was the dimension count.

? When the double dual is bigger

In infinite dimensions $\iota : V \to V^{**}$ is still injective, but it can fail to be onto: the dual of the space of absolutely summable sequences is the space of bounded sequences, and the dual of *that* is far larger than where we started. Worse, the "algebraic" dual of a function space is so big that analysts throw most of it away and keep only the continuous functionals, among them the Dirac delta. Which functionals survive, and why the Dirac delta turns out to be a covector, is the business of Ch. 61.

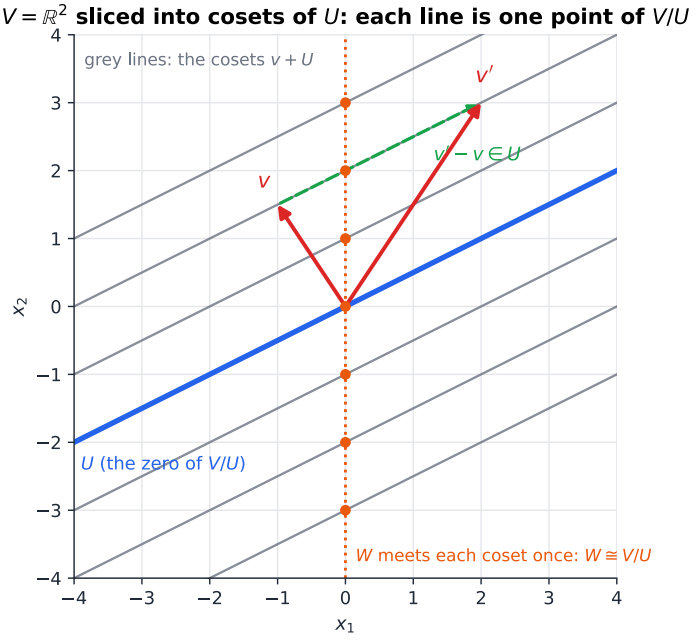
For a subspace $U \subseteq V$, the **annihilator** $U^0 = \{\varphi \in V^* : \varphi(u) = 0 \text{ for all } u \in U\}$ is the set of measurements blind to U, and $\dim U + \dim U^0 = \dim V$. A subspace of dimension k in $\mathbb{R}^n$ is cut out by exactly $n - k$ independent linear equations; the annihilator is the space of those equations.

Quotients and direct sums

Quotient space

For a subspace $U \subseteq V$, the **coset** of v is $v + U = \{v + u : u \in U\}$. The **quotient** V/U is the set of cosets, with $(v + U) + (w + U) = (v + w) + U$ and $c(v + U) = cv + U$. Its zero is U itself, and $v + U = v' + U$ exactly when $v' - v \in U$.

The operations are well defined because changing representatives changes the answer only by something in U. Geometrically, V/U collapses each translate of U to a single point.



V/U for a line U in $\mathbb{R}^2$: every grey line is one element. Any complement W meets each coset exactly once, which is why $V/U \cong W$, but the quotient itself does not depend on which W you pick.

Engineers already work in quotients. Node potentials are defined only up to a common constant, so the physical object lives in $\mathbb{R}^n/\operatorname{span}(1, \dots, 1)$, and "grounding a node" is choosing a complement (Ch. 9, §9.6). An indefinite integral is an element of a quotient by the constants.

First isomorphism theorem

For linear $T : V \to W$, the map $\bar{T} : V/\ker T \to \operatorname{im} T$, $\bar{T}(v + \ker T) = Tv$, is a well-defined linear isomorphism. In particular, if V is finite-dimensional, $\dim V = \dim \ker T + \dim \operatorname{im} T$.

Proof. Well defined: if $v + \ker T = v' + \ker T$ then $v' - v \in \ker T$, so $Tv' = Tv$. Linear: inherited from T. Onto $\operatorname{im} T$: by definition. One-to-one: $\bar{T}(v + \ker T) = 0$ means $Tv = 0$, so $v \in \ker T$, so $v + \ker T$ is the zero coset. For the dimension count, pick a basis $u_1, \dots, u_k$ of $U = \ker T$ and extend it to a basis $u_1, \dots, u_k, w_1, \dots, w_m$ of V. The cosets $w_1 + U, \dots, w_m + U$ span V/U (the u's vanish modulo U) and are independent (a relation $\sum c_i w_i \in U$ forces all $c_i = 0$ by independence of the full basis), so $\dim V/U = \dim V - \dim U$. Combine with $V/\ker T \cong \operatorname{im} T$. ∎

That is rank–nullity (Ch. 9) with the matrices removed: a linear map is "collapse the kernel, then embed isomorphically". The same theorem, word for word, holds for groups and rings (Ch. 56).

Direct sums. $V = U \oplus W$ means every v is *uniquely* $u + w$ with $u \in U$, $w \in W$; equivalently $U + W = V$ and $U \cap W = \{0\}$. Direct sums correspond exactly to **projections**: linear maps with $P^2 = P$. Given P, take $U = \operatorname{im} P$ and $W = \ker P$; given the decomposition, let $P(u + w) = u$. In a basis adapted to $U \oplus W$, a projection is $\operatorname{diag}(\mathbf{I}, \mathbf{0})$, and a map that preserves both pieces is **block diagonal**.

Invariant subspaces

A subspace U is **invariant** under $T : V \to V$ if $T(U) \subseteq U$. Eigenvector lines are the one-dimensional examples; $\ker T$, $\operatorname{im} T$ and $\ker p(T)$ for any polynomial p are others. An invariant subspace gives two smaller maps: the **restriction** $T|_U : U \to U$ and the **induced map** $\bar{T} : V/U \to V/U$, $\bar{T}(v + U) = Tv + U$ (well defined precisely because $T(U) \subseteq U$). In a basis that starts with a basis of U,

$$[T] = \begin{bmatrix} [T|_U] & * \\ \mathbf{0} & [\bar{T}] \end{bmatrix}, \qquad \chi_T(x) = \chi_{T|_U}(x)\,\chi_T(x),$$

since the determinant of a block-triangular matrix is the product of the diagonal blocks' determinants (Ch. 8, §8.3). This is the engine of every canonical form: find an invariant subspace, split off the restriction, and *recurse on the quotient*. Over $\mathbb{C}$ the engine never stalls, because every map has an eigenvalue and so a one-dimensional invariant subspace. Induction on dimension then gives **Schur's triangularization**: every operator on a finite-dimensional complex space has an upper-triangular matrix in a suitable basis. Over $\mathbb{R}$ the engine can stall: the quarter-turn $\mathbf{R}$ has no invariant line, so the smallest invariant pieces may be planes.

? Two maps at once

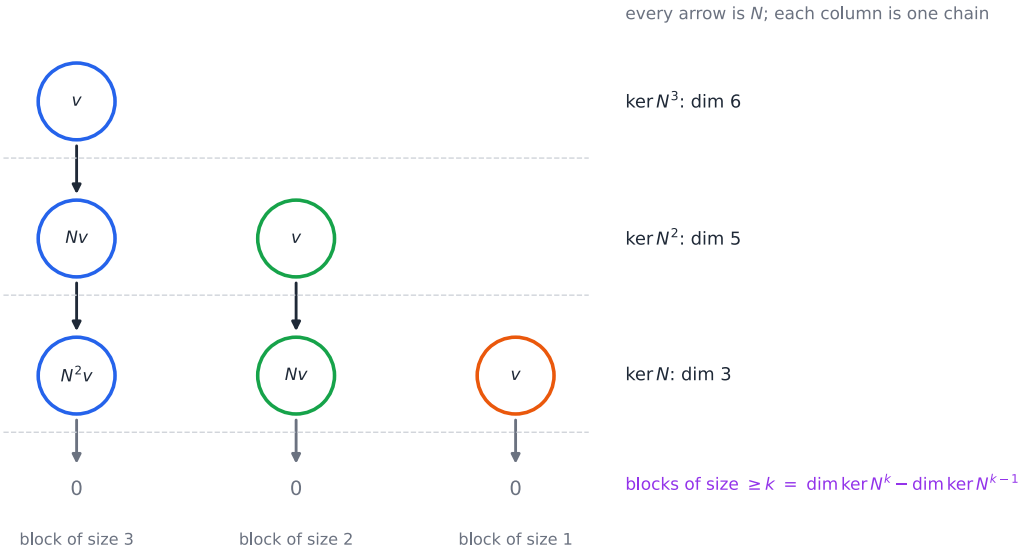
The recursion above classifies one map up to change of basis completely. Try two maps S, T on the same space, changed by the *same* basis $\mathbf{P}$ ($\mathbf{P}^{-1}\mathbf{S}\mathbf{P}$, $\mathbf{P}^{-1}\mathbf{T}\mathbf{P}$), and the problem changes character: there is no finite list of standard pieces, and a full classification of pairs would classify the modules of every finite-dimensional algebra at once. Mathematicians call such problems **wild**. Why pairs are wild while single maps are tame is one of the deep results of Ch. 95.

Jordan and rational canonical forms

Over $\mathbb{C}$ the characteristic polynomial factors as $\chi_T(x) = \prod_i (x - \lambda_i)^{m_i}$. The **generalized eigenspace** $V_\lambda = \ker(T - \lambda I)^m$ is invariant, and V is their direct sum (a consequence of Cayley–Hamilton, Ch. 10, §10.1, plus coprimality of the factors). On each V_λ, the map $N = T - \lambda I$ is **nilpotent**: some power is zero. So everything reduces to understanding nilpotent maps.

A nilpotent N moves vectors along **chains**: $v \mapsto Nv \mapsto N^2v \mapsto \cdots \mapsto 0$. The main structure theorem, which we state, says that V_λ has a basis consisting of disjoint chains. In the basis $N^{k-1}v, \ldots, Nv, v$ of one chain, N shifts each vector to the previous one, and its matrix is a **Jordan block**: zeros on the diagonal, ones just above it. Adding back λI:

$$J_k(\lambda) = \begin{bmatrix} \lambda & 1 & & \\ & \lambda & \ddots & \\ & & \ddots & 1 \\ & & & \lambda \end{bmatrix} \quad (k \times k).$$



A nilpotent map on a 6-dimensional space with chains of lengths 3, 2, 1. The bottom row spans $\ker N$*; the bottom two rows span* $\ker N^2$*; all three span* $\ker N^3$*. Counting dots row by row recovers the block sizes.*

The remarkable fact is that the chain lengths, and hence the Jordan form, are determined by numbers anyone can compute.

Counting Jordan blocks

Let N be nilpotent on V with a chain basis, and let $d_k = \dim \ker N^k$ ($d_0 = 0$). Then

$$\#\{\text{blocks of size } \geq k\} = d_k - d_{k-1}, \qquad \#\{\text{blocks of size exactly } k\} = 2d_k - d_{k-1} - d_{k+1}.$$

In particular the block sizes do not depend on which chain basis you found.

Proof. Number the vectors of a chain of length ℓ by their **height**: $N^{\ell-j}v$ has height j, so v has height ℓ and the last non-zero vector $N^{\ell-1}v$ has height 1. A basis vector of height j is sent to zero by N^k exactly when $j \leq k$, and N^k maps the basis vectors of height $j > k$ to *distinct* basis vectors of height $j - k$, which are independent. So $\ker N^k$ is spanned by the basis vectors of height at most k, and d_k counts them. Hence $d_k - d_{k-1}$ counts basis vectors of height exactly k. Each chain of length $\geq k$ has exactly one such vector, and shorter chains have none. That is the first formula; the second is the first at k minus the first at $k+1$. The d_k depend only on N, never on the basis, so neither do the block sizes. ∎

In the figure, $d_1, d_2, d_3 = 3, 5, 6$, so there are 3 blocks of size ≥ 1, 2 of size ≥ 2, 1 of size ≥ 3: sizes $3, 2, 1$. The hook's $\mathbf{N}$ is $J_2(0)$, the smallest map that no basis can diagonalize. The **Jordan form theorem** puts the pieces together: over $\mathbb{C}$, every operator is similar to a block-diagonal matrix of Jordan blocks, unique up to the order of the blocks, and two matrices are similar exactly when they have the same blocks.

Over a field where polynomials don't split, such as $\mathbb{R}$ or $\mathbb{Q}$, use a different chain. Pick v and follow $v, Tv, T^2v, \dots$ until the vectors become dependent. The span is invariant, and in the basis $v, Tv, \dots, T^{k-1}v$ the matrix of T is the **companion matrix** of the polynomial $x^k + c_{k-1}x^{k-1} + \cdots + c_0$ that expresses $T^k v$ in terms of the earlier vectors. The **rational canonical form** says every operator over every field is similar to a block-diagonal matrix of companion matrices $C(p_1), \dots, C(p_r)$ with $p_1 \mid p_2 \mid \cdots \mid p_r$, uniquely. For the quarter-turn $\mathbf{R}$, $v = e_1$ gives $Tv = e_2$ and $T^2v = -e_1$, so the polynomial is $x^2 + 1$ and $\mathbf{R}$ is already its own companion matrix. That is the honest answer to the hook over $\mathbb{R}$: the irreducible polynomial $x^2 + 1$, not a pair of imaginary eigenvalues, is the real invariant.

Multilinear forms

A **bilinear form** $B : V \times V \to K$ is linear in each slot. Inner products, the quadratic energy forms $\mathbf{x}^{\mathsf{T}}\mathbf{K}\mathbf{x}$ of Ch. 10, and the symplectic form of mechanics are examples. In a basis, $B(x, y) = \mathbf{x}^{\mathsf{T}}\mathbf{B}\mathbf{y}$, and under a change of basis $\mathbf{B} \mapsto \mathbf{P}^{\mathsf{T}}\mathbf{B}\mathbf{P}$, *not* $\mathbf{P}^{-1}\mathbf{B}\mathbf{P}$. Forms and operators are different objects that both happen to be stored as square arrays. A form eats two vectors; an operator eats one vector and returns a vector. They agree only when $\mathbf{P}^{\mathsf{T}} = \mathbf{P}^{-1}$, which is why the distinction is invisible in orthonormal coordinates. For real symmetric forms the invariant under congruence is not the eigenvalues but only their *signs*: the numbers of positive, negative and zero eigenvalues (**Sylvester's law of inertia**).

An n-linear form ω on an n-dimensional space is **alternating** if it vanishes whenever two arguments are equal (so it changes sign when two are swapped). Expanding each argument in a basis shows that ω is fixed by its single value $\omega(e_1, \dots, e_n)$, so the alternating n-forms form a **one-dimensional** space. Any operator T acts on it by $\omega \mapsto \omega(T\cdot, \dots, T\cdot)$, and a linear map on a one-dimensional space is multiplication by a scalar. That scalar *is* the determinant:

$$\omega(Tv_1, \dots, Tv_n) = (\det T)\, \omega(v_1, \dots, v_n) \quad \text{for all } \omega, v_1, \dots, v_n.$$

No matrix, no cofactor expansion, no basis. Multiplicativity is now one line: applying S after T multiplies by $\det T$, then by $\det S$, so $\det(ST) = \det S \det T$. And $\det T$ is visibly the factor by which T scales oriented volume, since ω measures oriented volume.

> **? One space for all bilinear maps**
>
> Bilinear maps $V \times W \to X$ are not linear maps on $V \times W$. Yet there is a single space $V \otimes W$ such that bilinear maps out of $V \times W$ are *exactly* linear maps out of $V \otimes W$, and the alternating forms above live in a quotient of $V \otimes \cdots \otimes V$. How that space is built, and why physicists' upper and lower indices are just vectors and covectors in disguise, is the subject of Ch. 93.

99.4 The algorithm

The counting theorem is an algorithm: to find the Jordan blocks for an eigenvalue λ, compute $d_k = \dim \ker(\mathbf{A} - \lambda\mathbf{I})^k = n - \operatorname{rank}(\mathbf{A} - \lambda\mathbf{I})^k$ until it stops growing, and read off the sizes. No chain basis is needed.

```
import numpy as np

def jordan_structure(A, lam, tol=1e-9):
    """Sizes of the Jordan blocks of A for eigenvalue lam, from kernel dimensions."""
    n = A.shape[0]
    N = A - lam * np.eye(n)
    d = [0]                                  # d[k] = dim ker N^k
    P = np.eye(n)
    while True:
        P = P @ N
        d.append(int(n - np.linalg.matrix_rank(P, tol=tol)))
        if d[-1] == d[-2]:
            break
    at_least = [d[k] - d[k - 1] for k in range(1, len(d))]   # blocks of size >= k
    sizes = []
    for k, m in enumerate(at_least, start=1):
        nxt = at_least[k] if k < len(at_least) else 0
        sizes += [k] * (m - nxt)                             # blocks of size exactly k
    return d[1:-1], sorted(sizes, reverse=True)

# A matrix with eigenvalue 2 only, disguised by a change of basis
J = np.diag([2.0] * 6) + np.diag([1, 1, 0, 1, 0], k=1)       # blocks 3, 2, 1
S = np.eye(6) + np.eye(6, k=1) + np.eye(6, k=-1)               # det S = 1
A = S @ J @ np.linalg.inv(S)
print(np.round(A, 3))
print("dim ker (A-2I)^k:", *jordan_structure(A, 2.0))
print("eigvals of A, distance from 2:", np.round(np.sort(abs(np.linalg.eigvals(A) - 2)), 8))

# the same computation after a tiny perturbation of size 1e-10
E = 1e-10 * np.random.default_rng(0).standard_normal((6, 6))
for tol in (1e-8, 1e-12):
    print(f"perturbed, tol={tol:g}:", jordan_structure(A + E, 2.0, tol))
```

The disguised matrix is an integer matrix with no visible structure, yet the script reports $d_k = 3, 5, 6$ and blocks $[3, 2, 1]$, the figure's chains exactly. Each rank costs one SVD, $O(n^3)$, and there are at most n powers, so the cost is $O(n^4)$ per eigenvalue.

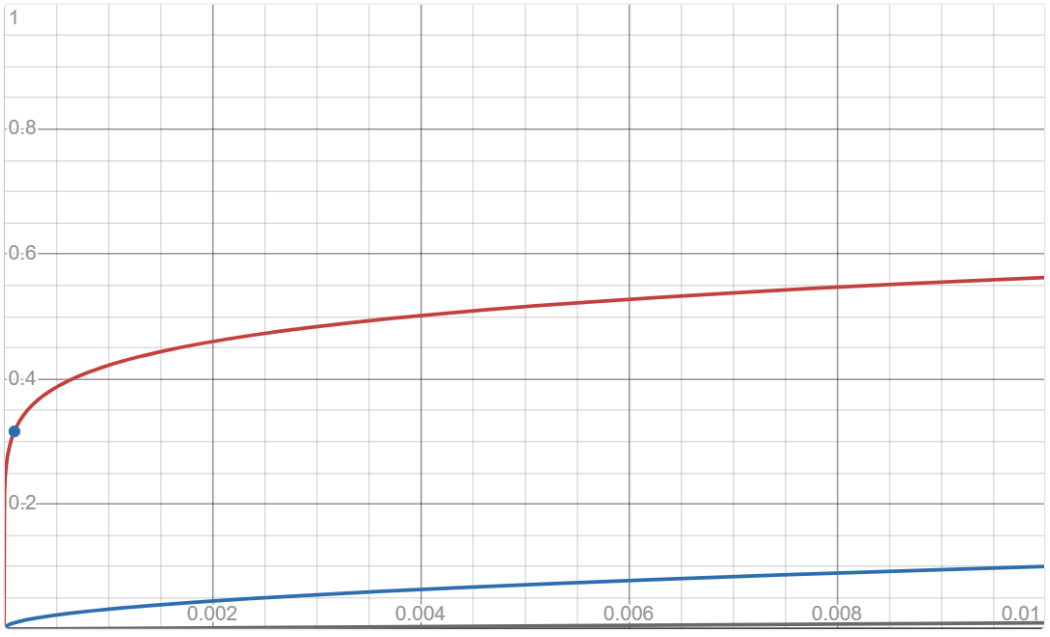
Now the warning. The same script asks `numpy` for the eigenvalues of the *exact* integer matrix $\mathbf{A}$, whose only eigenvalue is 2. It returns three of them about 3.3×10^{-6} away from 2. Rounding errors of size 10^{-16} have been amplified to 10^{-6}, roughly $(10^{-16})^{1/3}$, because a Jordan block of size 3 spreads a perturbation ε into eigenvalues at distance about $\varepsilon^{1/3}$. After a deliberate perturbation of size 10^{-10}, the rank test with tolerance 10^{-8} still finds $[3, 2, 1]$; with tolerance 10^{-12} it finds

no eigenvalue 2 at all, which is the exact truth for the perturbed matrix.

Both answers are correct. The Jordan form is a **discontinuous** function of the matrix: every non-diagonalizable matrix is a limit of diagonalizable ones. So numerical libraries compute the stable Schur form instead, and the Jordan form is asked for only in exact arithmetic over $\mathbb{Q}$ or $\mathbb{F}_p$. The tolerance is your claim about how much of the matrix is real.

99.5 The experiment

How sensitive is a Jordan block? Take the $n \times n$ block $J_n(0)$, whose only eigenvalue is 0, and put ε in the bottom-left corner. The characteristic polynomial becomes $x^n - \varepsilon$, so every eigenvalue jumps to distance $\varepsilon^{1/n}$ from 0. A diagonalizable matrix would move them by about ε.



Eigenvalue displacement against perturbation size ε: grey for a diagonalizable matrix, blue for a 2×2 Jordan block, red for an 8×8 block. Change the exponent $\frac{1}{8}$ to $\frac{1}{20}$ and watch the curve become a wall.

The script draws the eigenvalues in the complex plane. With the defaults it prints `max|lambda| = 1.000e-02` for $\varepsilon = 10^{-16}$: a perturbation at the level of rounding error moves the eigenvalues of an 8×8 block by 0.01, fourteen orders of magnitude more.

```
import numpy as np
import matplotlib.pyplot as plt

# KNOBS: block size n, perturbation sizes, and where the perturbation goes
n = 8
epsilons = [1e-16, 1e-12, 1e-8, 1e-4]
corner_only = True        # False: perturb every entry at random

J = np.diag(np.ones(n - 1), k=1)          # nilpotent Jordan block: only eigenvalue 0
rng = np.random.default_rng(1)
fig, ax = plt.subplots(figsize=(6, 6))
for eps in epsilons:
    E = np.zeros((n, n))
    if corner_only:
        E[-1, 0] = eps                    # the single entry that closes the chain
    else:
        E = eps * rng.standard_normal((n, n))
    lam = np.linalg.eigvals(J + E)
    r = np.abs(lam).max()
    print(f"eps={eps:.0e}  max|lambda|={r:.3e}  eps^(1/n)={eps ** (1 / n):.3e}")
    ax.plot(lam.real, lam.imag, "o", label=f"eps = {eps:.0e}")
ax.set_aspect("equal"); ax.grid(True); ax.legend()
ax.set_title(f"eigenvalues of a perturbed {n}x{n} Jordan block (exact answer: all 0)")
plt.show()
```

Things to try: set `n = 20` and see $\varepsilon = 10^{-16}$ produce eigenvalues of size about 0.16; set `corner_only = False` and check that random perturbations obey the same $\varepsilon^{1/n}$ law; replace `J` by a diagonal matrix and watch the effect collapse to size ε.

99.6 The challenge

1. **Intuition.** In $\mathbb{R}^3$, describe V/U when U is a line, and when U is a plane. Why is there a natural map $V \to V/U$ but no natural map $V/U \to V$? Which of the two covector families in the dual-basis figure moves if you replace e_1 by $(2, 0)$?
2. **Derivation.** Prove $\dim U + \dim U^0 = \dim V$, and show that $(\operatorname{im} T)^0 = \ker T^*$. Deduce, without matrices, that row rank equals column rank.

3. **Implementation.** Write `rational_form(A)` for integer matrices using exact `fractions.Fraction` arithmetic: grow cyclic chains $v, Av, A^2v, \dots$ until they become dependent, record the polynomial, and split off the invariant subspace. Test it on $\mathbf{R}$ (expect x^2+1) and on a 4×4 matrix with characteristic polynomial $(x^2+1)^2$.
4. **Open investigation.** For random $n \times n$ matrices $\mathbf{E}$ with independent Gaussian entries, how does the *distribution* of eigenvalues of $J_n(0) + \varepsilon \mathbf{E}$ look as n grows with ε fixed? Does it concentrate on a circle of radius $\varepsilon^{1/n}$, fill a disc, or do something else? Explore numerically and form a conjecture.

99.7 Loose threads

This chapter stated, rather than proved, the existence of chain bases and the rational canonical form. Both are one theorem in disguise: a vector space with an operator T is a module over the polynomial ring $K[x]$ (let x act as T), and the **structure theorem for finitely generated modules over a principal ideal domain** decomposes it into cyclic pieces. The same theorem classifies finite abelian groups, with $\mathbb{Z}$ in place of $K[x]$; its home is the ring theory of Ch. 56.

The numerical fragility of §99.4 also has a modern answer. Since eigenvalues of non-normal matrices are unreliable, analysts study **pseudospectra**, the sets where $\|(\mathbf{A} - z\mathbf{I})^{-1}\|$ is large, which are stable and explain transient growth in fluid flows and lasers (Ch. 90).

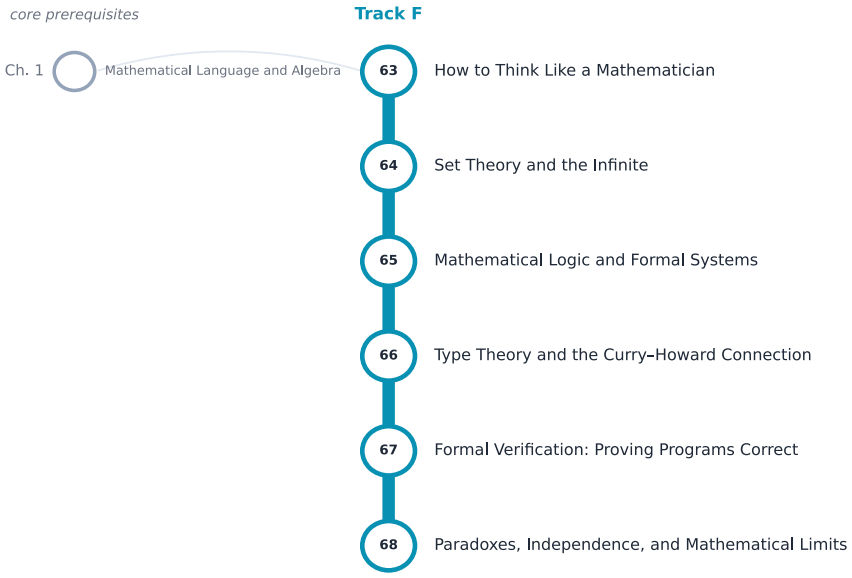
The three mysteries lead out of the track: continuous functionals in infinite dimensions (Ch. 61), the wildness of pairs of matrices (Ch. 95), and the tensor product (Ch. 93). Quotients by "functions that vanish almost everywhere", which turn functions into vectors of L^2, are waiting in measure theory (Ch. 97).

Further reading

- Sheldon Axler, *Linear Algebra Done Right*, 4th ed., Springer, 2024 (open access). Operators first, determinants last; the natural next book.
- Paul Halmos, *Finite-Dimensional Vector Spaces*, 2nd ed., Van Nostrand, 1958 (Springer reprint, 1974). The classic coordinate-free treatment, written as preparation for Hilbert space.
- Kenneth Hoffman and Ray Kunze, *Linear Algebra*, 2nd ed., Prentice Hall, 1971. Thorough on invariant subspaces and the rational and Jordan forms.
- Lloyd N. Trefethen and Mark Embree, *Spectra and Pseudospectra*, Princeton University Press, 2005. What happens to all of this on a computer.

Track F — Logic, Proofs, and the Foundations of Truth

> **Level 2 — Specializations.** For writing reliable algorithms, designing programming languages, and understanding why some truths cannot be proved.



Suggested reading order for Track F. The grey stations on the left are the core chapters it leans on.

Before you start

This track assumes you are comfortable with these core chapters:

- Ch. 1 — Mathematical Language and Algebra

Chapters

Ch.	Chapter	What it explores
63	How to Think Like a Mathematician	Direct proof, contradiction, induction, construction, counterexamples.
64	Set Theory and the Infinite	Cardinality, ordinals, countability, uncountability, axioms.
65	Mathematical Logic and Formal Systems	Predicate logic, completeness, soundness, compactness, incompleteness.
66	Type Theory and the Curry–Howard Connection	Types as propositions, proofs as programs, dependent types.
67	Formal Verification: Proving Programs Correct	Hoare logic, invariants, model checking, theorem provers.
68	Paradoxes, Independence, and Mathematical Limits	Russell's paradox, Gödel's incompleteness theorems, independence results.

Every chapter follows the same arc: **the hook** (a puzzle or a failure), **the big idea**, **the mathematics**, **the algorithm**, **the experiment** you can run, **the challenge**, and **loose threads**: the questions the chapter deliberately leaves open.

Chapter 63 — How to Think Like a Mathematician

> **Level 2 · Track F — Logic, Proofs, and the Foundations of Truth Builds on:** Ch. 1, Ch. 29 · **Leads to:** Ch. 64, Ch. 65, Ch. 67 · **Related:** Ch. 28, Ch. 57, Ch. 66 **Reading time:** ~25 min

63.1 The hook

Put two points on a circle and join them: the disc is cut into 2 regions. Three points, all chords drawn: 4 regions. Four points: 8. Five points: 16. Anyone who has seen the sequence $1, 2, 4, 8, 16$ knows what comes next. Draw six points (in general position, so that no three chords meet at one interior point), count carefully, and you get **31**.

Not 32. Thirty-one. The pattern was a liar, and it lied politely for five steps before it betrayed you.

It gets worse. The polynomial $n^2 + n + 41$, found by Euler, produces primes for $n = 0, 1, 2, \ldots, 39$: $41, 43, 47, 53, 61, \ldots, 1601$. Forty primes in a row. At $n = 40$ it gives $1681 = 41^2$. Euler also guessed that you need at least k k-th powers to add up to a k-th power. For two centuries nobody found an exception, until in 1966 a computer search turned up

$$27^5 + 84^5 + 110^5 + 133^5 = 144^5,$$

four fifth powers summing to a fifth power. And Fermat, perhaps the greatest pattern-spotter of all, claimed that every number $2^{2^m} + 1$ is prime. It is for $m = 0, 1, 2, 3, 4$ (giving $3, 5, 17, 257, 65537$). For $m = 5$, Euler found $2^{32} + 1 = 4294967297 = 641 \times 6700417$.

Now a puzzle with the opposite flavour. Take a $2^n \times 2^n$ chessboard and delete any one square you like. Can the rest always be tiled by L-shaped trominoes (three squares forming an L)? For $n = 1$ the board is 2×2 minus a square, which *is* an L. For $n = 2$ you can find a tiling by hand in a minute. For $n = 10$ the board has 1,048,576 squares and over a million choices of missing square, and the number of ways to place trominoes is beyond any computer. Testing is hopeless. Yet the answer is yes, for every n and every missing square, and by the end of §63.3 you will have seen why with your own eyes, in about ten lines.

So here is the question this chapter is about:

> **When a pattern has held a thousand times, how do you know it will hold the thousand-and-first time, and how do you decide which tool will tell you?**

63.2 The big idea

A **proof** is a finite argument that settles infinitely many cases at once. Testing checks cases one at a time and never finishes; a proof finds a reason that applies to all of them. The two activities are partners: testing suggests what might be true, and proof decides it. A single failed test kills a universal claim outright, while a million passed tests do not establish it.

A programmer's analogy works well. A theorem is a specification, a proof is an implementation that has been checked to meet it, and a counterexample is a failing test case. A failing test is decisive; a passing test suite is only evidence. (Ch. 29 takes this analogy literally and proves programs correct; Ch. 66 will show it is more than an analogy.)

The second idea is that the *shape of the claim* usually tells you which tool to reach for. Mathematicians are not geniuses who pull proofs out of the air; they read the logical form of the statement and pick from a small toolkit, the way an engineer reads a circuit and decides between nodal and mesh analysis.

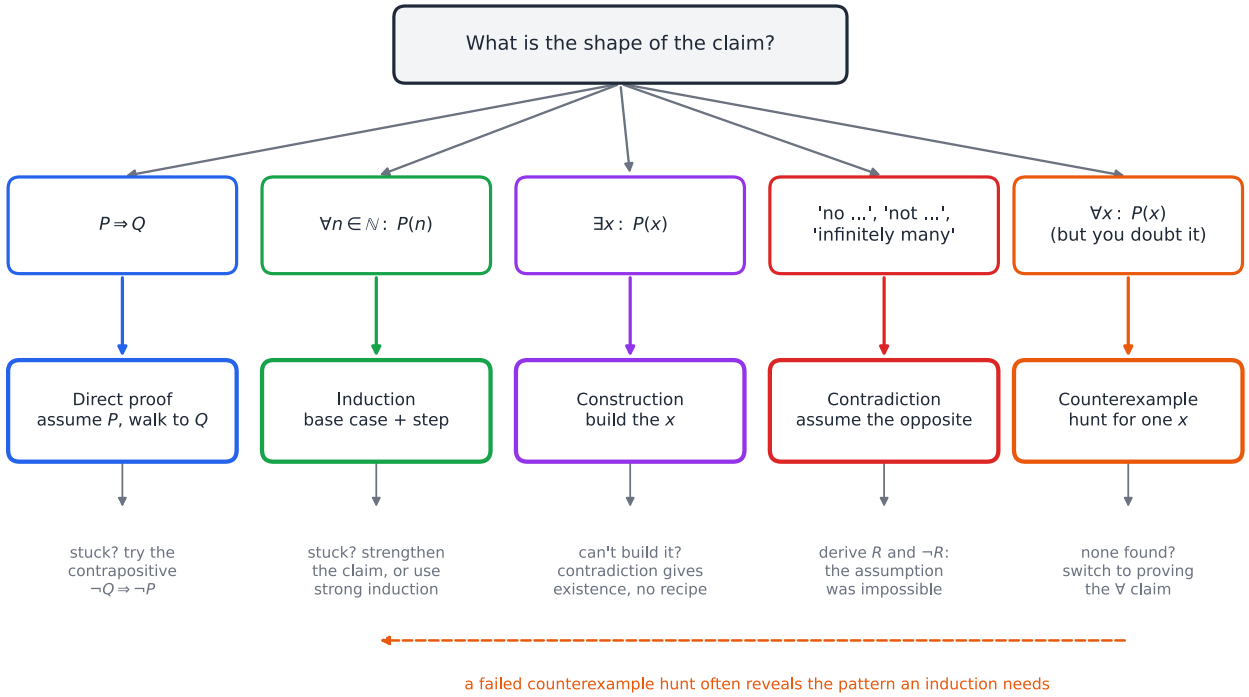


Figure 63.1 — Read the claim's logical shape, pick the matching strategy, and know the fallback when you get stuck. Hunting for counterexamples is not a separate activity: it is how you find out which box you are in.

The five boxes are the whole chapter:

- **Direct proof.** To prove $P \Rightarrow Q$, assume P and walk, step by valid step, to Q.
- **Contradiction.** To prove S, assume $\neg S$ and derive something absurd. Best for negative claims: "there is no...", "is irrational", "there are infinitely many...".
- **Induction.** To prove $P(n)$ for every natural number n, prove $P(0)$ and prove that $P(k)$ forces $P(k+1)$.
- **Construction.** To prove $\exists x\, P(x)$, build an x and check it.
- **Counterexample.** To refute $\forall x\, P(x)$, exhibit one x with $\neg P(x)$.

Ch. 1 lists the first four in §1.10, and Ch. 29 develops the variants of induction (strong induction, well-ordering, structural induction) in §29.3. This chapter is about *using* them: what each one feels like from the inside, where each one breaks, and how to choose.

63.3 The mathematics

We need one piece of vocabulary from logic, covered in Ch. 29's predicate-logic section: $P \Rightarrow Q$ is false only when P is true and Q false, and $\neg\forall x\, P(x)$ is equivalent to $\exists x\, \neg P(x)$. That last equivalence is the entire theory of counterexamples.

Direct proof and the contrapositive

> **Even, odd, divides**
>
> An integer n is **even** if $n = 2k$ for some integer k, and **odd** if $n = 2k + 1$. We write $a \mid b$ ("a divides b") if $b = ak$ for some integer k.

A direct proof unpacks definitions, manipulates, and repacks. Claim: *if $a \mid b$ and $b \mid c$ then $a \mid c$.* Assume $b = ak$ and $c = bl$. Then $c = a(kl)$, and kl is an integer, so $a \mid c$. □ Three lines, and nothing clever: the definitions did the work. Most proofs you will ever write look like this, and the skill is mostly in writing down the definitions precisely before you start.

Sometimes the forward walk has nowhere to go. Claim: *if n^2 is even then n is even.* Assume $n^2 = 2k$. Now what? Taking square roots leads nowhere. The **contrapositive** $\neg Q \Rightarrow \neg P$ is logically equivalent to $P \Rightarrow Q$ (check the four rows of a truth table), and here it is easy: if n is odd, $n = 2k + 1$, then $n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$ is odd. □ The lesson is general: when the hypothesis is weak and the conclusion is strong, flip the implication so that the strong statement becomes the hypothesis. This lemma is exactly the step used in the irrationality of $\sqrt{2}$ in §1.1.

Proof by contradiction

Contradiction is the most powerful and the most dangerous tool. You assume the claim is false, reason until you reach a statement and its negation, and conclude that the assumption was impossible. Its power is that you get a free hypothesis to work with. Its danger is that if you make an ordinary mistake along the way, you also "reach a contradiction", and the proof looks just as convincing.

Theorem 63.1 — Euclid: there are infinitely many primes

For every finite list of primes, there is a prime not on the list.

Proof. Suppose, for contradiction, that $p_1, p_2, \dots, p_m$ is a complete list of all primes. Let $N = p_1 p_2 \cdots p_m + 1$. Since $N \geq 3 > 1$, N has at least one prime factor q (the smallest divisor of N greater than 1 must be prime, because any proper factor of it would be a smaller divisor). By assumption q is one of the p_i. Then $q \mid p_1 \cdots p_m$ and $q \mid N$, so q divides their difference, $N - p_1 \cdots p_m = 1$. No prime divides 1. Contradiction, so no finite list contains all primes. □

A common misreading: "$p_1 \cdots p_m + 1$ is always prime." It is not. With the first six primes, $2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 + 1 = 30031 = 59 \times 509$. The proof never claimed N was prime, only that its prime factors were new. Read the proof again and notice that it can be rewritten *without* contradiction: given any list, N's smallest prime factor is a new prime, and that is a construction. Many proofs by contradiction hide a direct proof inside; finding it usually gives you more information.

Induction, and the art of strengthening

Ch. 29 states the principle and its variants. Figure 63.2 is the picture to keep in your head.

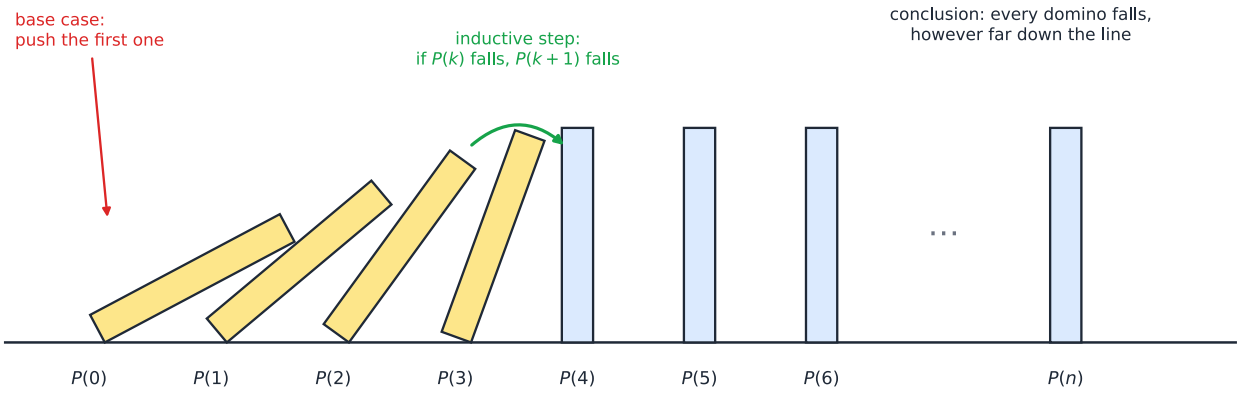


Figure 63.2 — Induction as a line of dominoes. The base case knocks over the first one; the inductive step guarantees that every falling domino knocks over the next. Neither half alone is enough: a perfect step with no push does nothing, and a push with one gap in the line stops the cascade.

The domino picture also explains the classic failure. "All horses are the same colour": in any set of $k+1$ horses, the first k share a colour and the last k share a colour, and these overlap, so all $k+1$ do. The step fails exactly from $k = 1$ to $k = 2$, where the "first one" and the "last one" do not overlap. One gap in the line of dominoes, and the whole cascade stops at the second domino.

Now the tromino puzzle from the hook. Here is the surprising lesson: the stronger claim is *easier* to prove.

Theorem 63.2 — Tromino tiling

For every $n \geq 1$ and every choice of one square to remove, the remaining $4^n - 1$ squares of a $2^n \times 2^n$ board can be tiled by L-trominoes.

Proof. By induction on n. Let $P(n)$ be the full statement, "for **every** missing square".

Base case $n = 1$. A 2×2 board minus any one square is itself an L-tromino.

Inductive step. Assume $P(k)$ and take a $2^{k+1} \times 2^{k+1}$ board with one square removed. Cut it into four $2^k \times 2^k$ quadrants. The missing square lies in one of them. Place a single tromino at the centre of the board so that it covers one square from each of the *other three* quadrants (the three central corners). Now each of the four quadrants is a $2^k \times 2^k$ board with exactly one square unavailable: the original hole in one, a square covered by the central tromino in the other three. By $P(k)$, applied four times, each quadrant can be tiled. Together with the central tromino, that tiles the whole board. □

Look at what made this work. If we had tried to prove only "the board minus a *corner* can be tiled", the inductive step would have failed: three of the four quadrants end up missing a *central* square, not a corner, and the weaker hypothesis would not apply to them. By asking for more (every missing square), we got a stronger hypothesis to use in the step. This is the single most useful trick in induction: **if the step does not go through, try proving something stronger.** It feels backwards, and it works constantly. The count also checks out: $4^n - 1$ is always divisible by 3 (it equals $1, 5, 21, 85, 341$ trominoes for $n = 1, \dots, 5$), as it must be.

The proof is also an algorithm. Follow it recursively and it *places* the trominoes; §63.4 turns it into code.

Construction: proving that something exists

To prove "there exist n consecutive composite numbers, for every n", build them. The numbers

$$(n+1)! + 2,\ (n+1)! + 3,\ \dots,\ (n+1)! + (n+1)$$

are n consecutive integers, and for each $2 \leq k \leq n+1$, k divides both $(n+1)!$ and k, so $k \mid (n+1)! + k$, which is therefore composite. For $n = 5$: $722, 723, 724, 725, 726$, divisible by $2, 3, 4, 5, 6$ respectively. So primes have arbitrarily large gaps, even though Theorem 63.1 says they never run out. The construction is crude (the gap after 113 already has length 13, far below 14!) but crude is fine: an existence proof only needs *one* witness.

Now a stranger existence proof.

Theorem 63.3 — An irrational power of an irrational can be rational

There exist irrational numbers a and b such that a^b is rational.

Proof. Consider $x = \sqrt{2}^{\sqrt{2}}$. Either x is rational or it is not. If it is, take $a = b = \sqrt{2}$, which is irrational by §1.1. If it is not, take $a = x$ and $b = \sqrt{2}$: then

$$a^b = \left(\sqrt{2}^{\sqrt{2}}\right)^{\sqrt{2}} = \sqrt{2}^{\sqrt{2}\cdot\sqrt{2}} = \sqrt{2}^{2} = 2,$$

which is rational. Either way the pair exists. □

This proof is airtight and deeply unsatisfying. It proves that a pair exists, and at the end you *still do not know which pair it is*. (It happens that $\sqrt{2}^{\sqrt{2}}$ is irrational, by the Gelfond–Schneider theorem, but the proof above did not need to know that and cannot tell you.)

? A proof that refuses to show its witness

Theorem 63.3 used the law of excluded middle, "x is rational or it is not", to prove existence without producing an example. A school of mathematicians, the intuitionists, rejects such proofs, and for programmers there is a sharp reason to care: a proof of "there exists" that does not contain a recipe cannot be *run*. What kind of logic do you get if every existence proof must carry its witness, and why does that logic turn out to be a programming language? That is the subject of Ch. 66.

Counterexamples: the cheapest proof there is

To refute "for all x, $P(x)$" you need exactly one x that fails. Every example in the hook is a counterexample, and Figure 63.3 shows the first one in full.

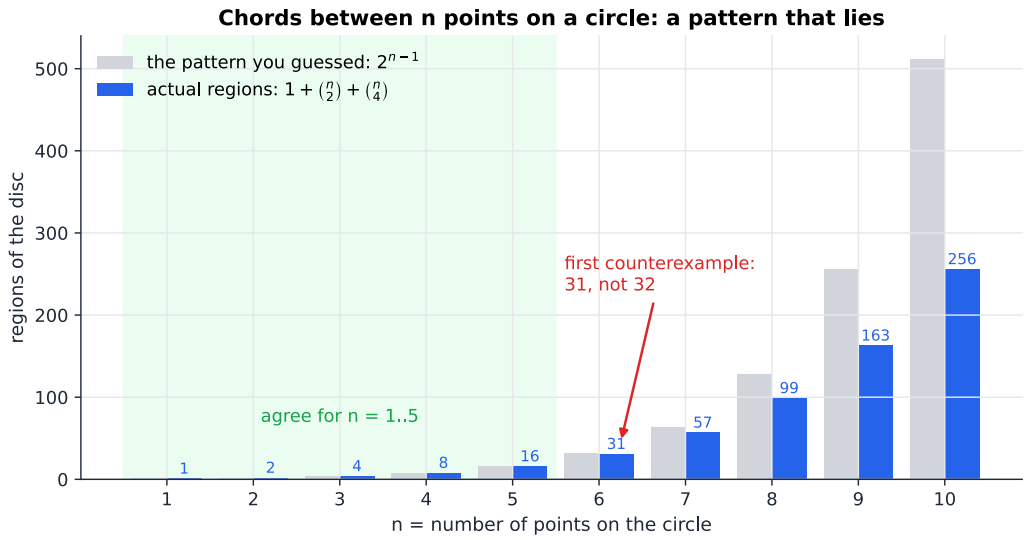


Figure 63.3 — The regions cut out by all chords between n points on a circle (blue) against the doubling pattern (grey). They agree for five values and then separate for good: the true count grows like $n^4/24$, the guess like 2^n.

Why does the true formula $1 + \binom{n}{2} + \binom{n}{4}$ fit five values of 2^{n-1}? Because $2^{n-1} = \sum_i \binom{n-1}{i}$, and for $n \le 5$ that sum and $1 + \binom{n}{2} + \binom{n}{4}$ happen to coincide (both are $\binom{n-1}{0} + \cdots + \binom{n-1}{4}$, the first five terms, by Pascal's rule). At $n = 6$ the doubling sequence picks up a sixth term, $\binom{5}{5} = 1$, and the geometric count does not. The coincidence had a reason, and so did its end. Finding *that* reason, why each new chord crossing adds one region, is what a proof of the formula does; Ch. 28 has the counting tools (Ch. 28).

Three habits make counterexample hunting effective:

1. **Try the edge cases first**: 0, 1, the empty set, negative numbers, equal inputs, degenerate shapes. "Every continuous function is differentiable" dies at $|x|$; "every bounded sequence converges" dies at $(-1)^n$.
2. **Search systematically** when the space is enumerable, and let a computer do it. Euler's conjecture fell to a computer, not to cleverness.
3. **When the hunt keeps failing, ask why.** Each near-miss is information. The reason no counterexample appears is very often the proof.

Counterexamples can hide far out

Pólya conjectured in 1919 that, below any $N \ge 2$, at least half the numbers have an odd number of prime factors. It holds for every N below 906,150,257, where it first fails. The Mertens conjecture was disproved in 1985 without anyone ever finding an explicit counterexample; the best known bound on where the first one lies is around $e^{10^{33}}$. Testing up to a billion proves nothing.

63.4 The algorithm

Two of our five strategies are algorithms in disguise: induction (the proof of Theorem 63.2 is a recursive procedure) and counterexample search (an enumeration with a check). Here is the tromino proof, executed.

```
import numpy as np

def tile(n, hole):
    """Tile a 2^n x 2^n board minus `hole` with L-trominoes (Theorem 63.2)."""
    N = 2 ** n
    board = np.zeros((N, N), dtype=int)
    board[hole] = -1
    count = [0]

    def solve(r, c, size, hr, hc):          # quadrant at (r,c), blocked cell (hr,hc)
        if size == 1:
            return
        count[0] += 1
        t, h = count[0], size // 2
        centre = [(r+h-1, c+h-1), (r+h-1, c+h), (r+h, c+h-1), (r+h, c+h)]
        corners = [(r, c), (r, c+h), (r+h, c), (r+h, c+h)]
        for (qr, qc), (cr, cc) in zip(corners, centre):
            if qr <= hr < qr+h and qc <= hc < qc+h:
                solve(qr, qc, h, hr, hc)      # quadrant with the real hole
            else:
                board[cr, cc] = t             # central tromino covers this cell
                solve(qr, qc, h, cr, cc)      # ...which is now this quadrant's hole
    solve(0, 0, N, *hole)
    return board

print(tile(3, (5, 2)))
```

Each call does constant work and makes four calls on boards a quarter of the size, so the running time satisfies $T(4^n) = 4T(4^{n-1}) + O(1)$, which is $O(4^n)$: linear in the number of squares, the best possible since every square must be covered. The recursion depth is $n = \log_2 N$, so there is no stack problem even for $N = 2^{20}$.

The **counterexample hunter** is simpler still:

```
for x in an enumeration of the domain, in order of "size":
    if not P(x): return x          # refuted
return "no counterexample up to the bound"
```

Its outputs are asymmetric. A returned x is a proof (check it by hand, and you are done forever). The message "none up to the bound" proves nothing about larger x, as the warning above shows. Two engineering points matter:

- **Order of enumeration.** Enumerate by increasing size (sum of the entries, number of nodes, length of the list). Small counterexamples are the common case and the easiest to understand, an idea Ch. 29 calls the small-scope hypothesis.
- **Check P exactly.** Floating point can manufacture a counterexample that is not there, or hide one that is. Here both sides are the 11-digit number $61{,}917{,}364{,}224$, comfortably exact in a double; push the search a little further and the sums pass $2^{53} \approx 9 \times 10^{15}$, where doubles start rounding. Use exact integers.

Finally, proof by exhaustion is legitimate when the cases are finite: the four-colour theorem (1976) and the Kepler conjecture (1998) were proved by reducing an infinite problem to finitely many cases, then checking those by computer. The infinite-to-finite reduction is the human proof; the computer only does the bookkeeping.

63.5 The experiment

The script below is a counterexample hunter with three suspects. Run it, then change the knobs.

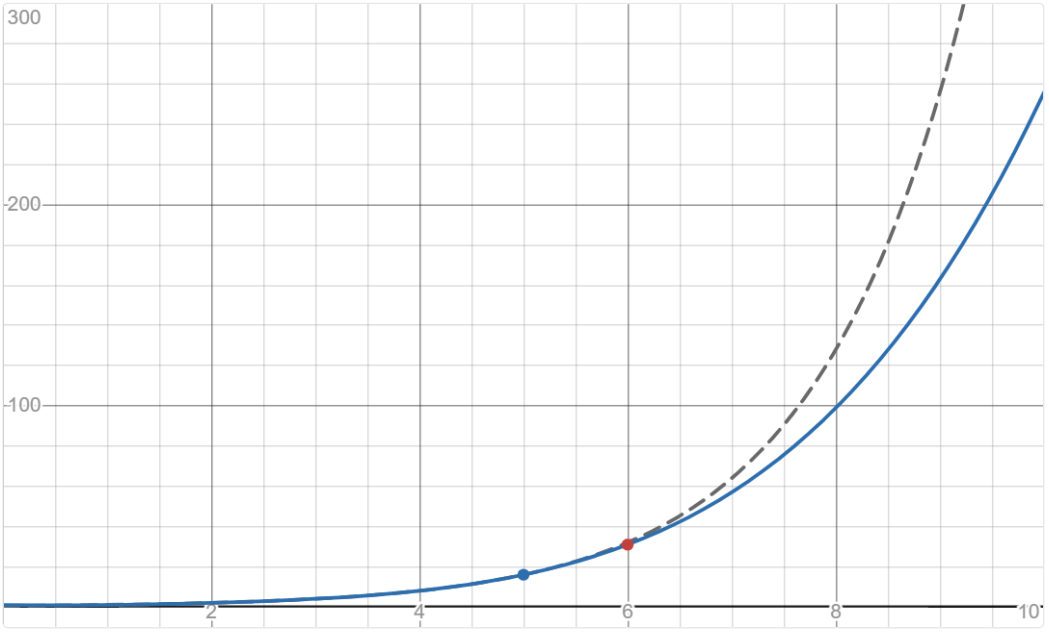
```
import numpy as np
from math import comb, isqrt

def is_prime(n):
    return n >= 2 and all(n % p for p in range(2, isqrt(n) + 1))

# Suspect 1: a prime-generating polynomial  n^2 + A n + B
A, B = 1, 41          # knob: try (-79, 1601), (1, 17), (1, 11)
bad = next(n for n in range(10**4) if not is_prime(n*n + A*n + B))
print(f"n^2+{A}n+{B}: first composite at n={bad}, value {bad*bad + A*bad + B}")

# Suspect 2: circle regions double each time
for n in range(1, 9):
    real, guess = 1 + comb(n, 2) + comb(n, 4), 2 ** (n - 1)
    print(n, real, guess, "" if real == guess else "<-- counterexample")

# Suspect 3: Goldbach -- every even number >= 4 is a sum of two primes
LIMIT = 20_000        # knob: raise it and watch the time grow
sieve = np.ones(LIMIT + 1, bool); sieve[:2] = False
for p in range(2, isqrt(LIMIT) + 1):
    if sieve[p]:
        sieve[p*p::p] = False
primes = np.flatnonzero(sieve)
fails = [m for m in range(4, LIMIT + 1, 2)
         if not any(sieve[m - primes[primes <= m // 2]])]
print("Goldbach failures up to", LIMIT, ":", fails)
```

With the default knobs it reports the first composite at $n = 40$ (value 1681), prints `<-- counterexample` from $n = 6$ on, and finds no Goldbach failure.

Things to try:

1. Set `(A, B) = (-79, 1601)`. This polynomial produces primes for 80 consecutive n (with repeats). Why does *no* non-constant polynomial with integer coefficients produce only primes? Hint: evaluate $f(n + k\,f(n))$ modulo $f(n)$.
2. Raise `LIMIT` to 10^6. Goldbach has been checked by computer up to 4×10^{18} and is still unproved. Count, for each even m, the number of ways to write it as a sum of two primes, and plot that count against m. The cloud of points grows; why does that make a counterexample feel ever less likely, and why is "feels unlikely" not a proof?
3. Change suspect 2 to points placed at the vertices of a *regular* hexagon. Now three chords meet at the centre. Count the regions by drawing them. Which hypothesis of the formula did you violate?

You can also watch the circle-region formula and the doubling guess separate.



The quartic $1 + \binom{x}{2} + \binom{x}{4}$ (blue) and the exponential guess (dashed) touch at $x = 1, \dots, 5$ and part ways. Change the 24 in the constant term to 25 and watch a single coefficient destroy the coincidence.

63.6 The challenge

No answers are given.

1. **Intuition.** Remove two diagonally opposite corners from an ordinary 8×8 chessboard. Can the remaining 62 squares be tiled by 1×2 dominoes? Find a one-sentence argument. Then decide whether your argument is a direct proof, a contradiction, or an invariant, and say why the label matters less than the idea.

2. **Derivation.** Prove by induction that every integer $n \geq 2$ is a product of primes. You will find ordinary induction does not work and strong induction does (§29.3). Then prove *by contradiction, using well-ordering,* that the factorisation is unique up to order, assuming the lemma "if a prime divides ab, it divides a or b".
3. **Implementation.** Extend the tromino code in §63.4 to draw the tiling with matplotlib, one colour per tromino. Then write a brute-force backtracking tiler and compare running times with the recursive one for $n = 2, 3, 4$. Where does brute force give up, and why?
4. **Open investigation.** Take any claim of the form "for all n, $f(n)$ is prime", for polynomials f of degree 2 with small coefficients. Search for the f with the longest initial run of primes. Is there an upper bound on how long the run can be for degree 2? (This touches open questions about prime-producing quadratics, related to class numbers of quadratic fields; nobody has a complete answer.)

63.7 Loose threads

We answered the hook's question, as far as it can be answered. You know a pattern holds for the thousand-and-first case only when you have a finite argument that covers every case, and the logical shape of the claim points to the argument: implications to direct proof, negations to contradiction, statements about all n to induction, existence claims to construction, and suspicious universals to a counterexample hunt. The tromino board of any size is tileable because the strengthened claim carries itself from one size to the next.

But every proof in this chapter rested on things we did not question.

- **What exactly is "valid"?** Each step was justified by "clearly" or "by definition". Could the rules of valid reasoning themselves be written down as a finite list, so precisely that a machine could check every proof? Ch. 65 does exactly that, and Ch. 67 shows engineers using such machines today.
- **What is a natural number?** Induction works on $\mathbb{N}$ because $\mathbb{N}$ has no infinite descending chain. On the integers, the rationals or the reals it fails. Why, and what is the right generalisation to infinite collections? Ch. 64 builds the infinite from sets and extends induction beyond the natural numbers.

> **? True but unprovable?**
>
> Goldbach's conjecture has survived checking up to 4×10^{18}, and the Collatz loop of Ch. 29 has survived up to about 3×10^{20}. Perhaps they are true but no proof exists at all: not hard to find, but *non-existent*, within the rules mathematicians accept. Can that happen to a perfectly ordinary statement about whole numbers? In 1931 Kurt Gödel showed that it can. How, and what it means, is the subject of Ch. 65 and Ch. 68.

> **? Can a machine find the proof for you?**
>
> Checking a written proof is mechanical. Finding one is creative, or is it? Given a statement, is there an algorithm that always finds a proof if one exists, and says "no" otherwise? The answer splits in a strange way: there is a procedure that finds every proof that exists, but it may run forever when none does. Why the "no" half is impossible connects to the halting problem of Ch. 30 and is made precise in Ch. 65.

So proofs are finite arguments about infinitely many cases. The next chapter asks what "infinitely many" means, and finds that there is more than one size of infinity.

Further reading

- D. J. Velleman, *How to Prove It: A Structured Approach*, 3rd ed., Cambridge University Press, 2019. The standard guide to proof strategy, organised by the logical shape of the claim.
- G. Pólya, *How to Solve It*, Princeton University Press, 1945. The classic on heuristics: specialise, generalise, strengthen the hypothesis.
- R. Hammack, *Book of Proof*, 3rd ed., 2018 (free online). Gentle and thorough on direct proof, contrapositive, contradiction and induction.
- I. Lakatos, *Proofs and Refutations*, Cambridge University Press, 1976. A dialogue on how proofs and counterexamples shape each other.

Chapter 64 — Set Theory and the Infinite

> **Level 2 · Track F — Logic, Proofs, and the Foundations of Truth Builds on:** Ch. 1, Ch. 29, Ch. 63 · **Leads to:** Ch. 65, Ch. 68 · **Related:** Ch. 30, Ch. 60, Ch. 19 **Reading time:** ~25 min

64.1 The hook

Imagine a hotel with infinitely many rooms, numbered $1, 2, 3, \ldots$, and every room is occupied. A traveller arrives late at night and asks for a room. In any ordinary hotel the answer is "sorry, we're full". This hotel's manager, David Hilbert in the usual telling, picks up the intercom: *"Would every guest please move from room n to room $n + 1$."* Everyone moves at once, nobody is left without a room, and room 1 is now empty. The traveller checks in.

Next night a coach arrives carrying infinitely many passengers, one per seat, seats numbered $1, 2, 3, \ldots$. The hotel is full again. The manager announces: *"Guest in room n, please move to room $2n$."* All current guests now sit in the even-numbered rooms, and every odd-numbered room is free. The passenger in seat k takes room $2k - 1$. Everyone is housed.

The third night, infinitely many coaches arrive, each with infinitely many passengers. Seat k on coach c is a pair (c, k). Surely this is too much? It is not, and by §64.3 you will be able to house them all yourself.

The fourth night is different. A coach arrives with one passenger for every infinite string of 0s and 1s: one for $0000\ldots$, one for $1010\ldots$, one for every pattern you can imagine. The manager tries every trick in the book, and fails. Not because she is not clever enough. She fails because it is impossible: **however she assigns rooms, some passenger is left in the car park.** This chapter is about why, and it answers a question that sounds as if it should have no answer:

> **Is every infinity the same size, and if not, how could you possibly tell two infinities apart?**

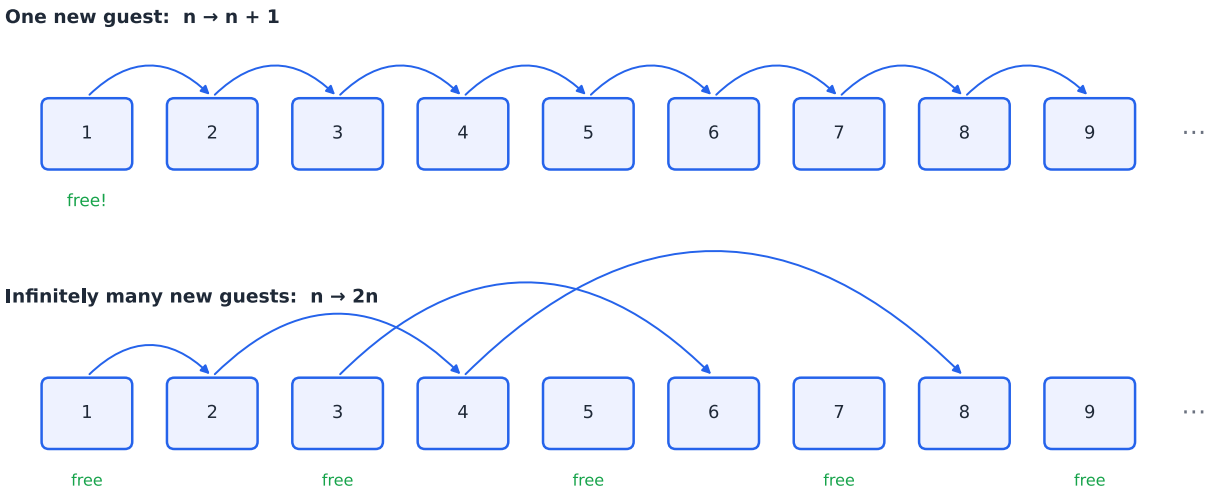


Figure 64.1 — Two of the manager's moves. Top: shifting everyone up one room frees room 1. Bottom: doubling every room number frees all the odd rooms, enough for a whole new infinity of guests. A finite hotel can do neither.

64.2 The big idea

How do you know that a room of chairs has as many chairs as people, without counting either? Ask everyone to sit down. If every person has a chair and every chair has a person, the numbers are equal. You never needed the number itself. **Matching comes before counting.**

Georg Cantor's insight in the 1870s was to take this as the *definition* of "same size", and to apply it fearlessly to infinite collections. Two sets have the same size, or **cardinality**, if their elements can be paired off perfectly, with nothing left over on either side. This one decision has two consequences that look contradictory and are not:

- An infinite set can have the same size as a part of itself. The hotel's first trick pairs $\{1, 2, 3, \ldots\}$ with $\{2, 3, 4, \ldots\}$. This is not a paradox; it is the defining feature of infinity. (Dedekind actually *defined* a set to be infinite if it can be matched with a proper part of itself.)
- Not every pair of infinite sets *can* be matched. Some infinities are strictly larger than others, and there is an infinite ladder of them.

The proof that some infinity is bigger is the **diagonal argument**, one of the most reused ideas in all of mathematics. It reappears in Ch. 30 to show that some problems cannot be solved by any program, and in Ch. 65 to show that some truths cannot be proved. Its form is always the same: *given any list, build something that differs from every entry on the list.*

64.3 The mathematics

We use sets naively for now: a set is a collection of objects, written $\{\dots\}$, with the operations $\cup, \cap, \setminus, \times$ from Ch. 29. §64.3's last part explains why "naively" will not do in the end.

Comparing sizes

Injection, surjection, bijection, cardinality

A function $f : A \to B$ is **injective** (one-to-one) if $f(a) = f(a')$ implies $a = a'$, **surjective** (onto) if every $b \in B$ equals some $f(a)$, and **bijective** if it is both. We write

- $|A| = |B|$ if there is a bijection $A \to B$;
- $|A| \le |B|$ if there is an injection $A \to B$;
- $|A| < |B|$ if $|A| \le |B|$ but $|A| \ne |B|$.

A set is **countable** if it is finite or has the same cardinality as $\mathbb{N} = \{0, 1, 2, \dots\}$. The cardinality of $\mathbb{N}$ is written $\aleph_0$ ("aleph-nought"). An infinite set that is not countable is **uncountable**.

Countable means *listable*: you can write the elements as $a_0, a_1, a_2, \dots$ so that every element appears at some finite position. Hilbert's hotel is a machine for building bijections. $n \mapsto n + 1$ shows $|\mathbb{N}| = |\mathbb{N} \setminus \{0\}|$; $n \mapsto 2n$ shows that the even numbers are exactly as numerous as all the numbers; and the map $n \mapsto n/2$ for even n, $n \mapsto -(n+1)/2$ for odd n lists the integers as $0, -1, 1, -2, 2, \dots$, so $|\mathbb{Z}| = \aleph_0$.

Is $\le$ really an ordering of sizes? If A injects into B and B injects into A, are they the same size? For finite sets obviously. For infinite sets it is true but not obvious.

Schröder–Bernstein theorem

If there are injections $A \to B$ and $B \to A$, then there is a bijection $A \to B$.

We use it without proof; the idea is to follow chains $a \mapsto f(a) \mapsto g(f(a)) \mapsto \cdots$ backwards and forwards and decide, chain by chain, which injection to use. It is a huge labour-saver: to show two sets have the same size, you only need an injection each way, which is usually much easier than a bijection.

The rationals are countable

The rationals look far more numerous than the integers. Between any two of them lies another, so they are **dense**; between 0 and 1 alone there are infinitely many. Surely they cannot be listed?

Theorem 64.1 — $\mathbb{N} \times \mathbb{N}$ and $\mathbb{Q}$ are countable

The map $\pi(p, q) = \dfrac{(p+q)(p+q+1)}{2} + q$ is a bijection $\mathbb{N} \times \mathbb{N} \to \mathbb{N}$. Consequently $|\mathbb{Q}| = \aleph_0$.

Proof. Arrange the pairs (p, q) in a grid and walk the anti-diagonals $p + q = s$ for $s = 0, 1, 2, \dots$ in turn. Diagonal s contains the $s + 1$ pairs $(s, 0), (s-1, 1), \dots, (0, s)$. Before diagonal s begins, we have used up $1 + 2 + \cdots + s = s(s+1)/2$ positions, and within diagonal s the pair (p, q) sits at offset q. So (p, q) lands at position $s(s+1)/2 + q$ with $s = p + q$, which is $\pi(p, q)$. Every position $z \in \mathbb{N}$ is reached exactly once: z lies in exactly one interval $[s(s+1)/2,\ (s+1)(s+2)/2)$, which picks out s, and then $q = z - s(s+1)/2 \in \{0, \dots, s\}$ and $p = s - q$ are forced. So π is a bijection.

For $\mathbb{Q}$: write each positive rational in lowest terms p/q and send it to $\pi(p, q)$. This is an injection from the positive rationals into $\mathbb{N}$, so they are countable; adding 0 and the negatives, as for $\mathbb{Z}$, keeps the set countable. An injection $\mathbb{N} \to \mathbb{Q}$ is obvious ($n \mapsto n$), so by Schröder–Bernstein $|\mathbb{Q}| = \aleph_0$. $\square$

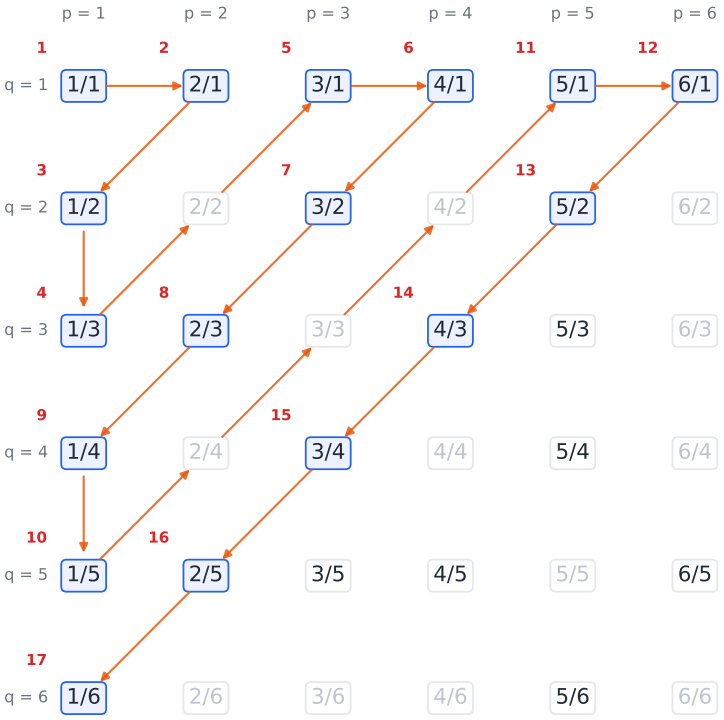


Figure 64.2 — Cantor's zig-zag through the positive fractions, with p across and q down. Each reduced fraction gets a finite position (red numbers); grey fractions such as $2/4$ are skipped because their value already appeared. Every positive rational is reached after finitely many steps.

The same proof houses the hotel's third night: call the guests already in the hotel "coach 0", number coaches and seats from 0, and put seat k of coach c in room $\pi(c, k) + 1$. More generally, a countable union of countable sets is countable, and so are $\mathbb{N}^3$, $\mathbb{N}^{17}$, the set of all finite strings over a finite alphabet, the set of all polynomials with integer coefficients, and the set of all Python programs. That last one will matter.

The diagonal argument

Now the fourth night. Let $\{0,1\}^{\mathbb{N}}$ be the set of infinite binary sequences $s = (s(0), s(1), s(2), \dots)$.

> **Theorem 64.2 — Cantor (1891): binary sequences are uncountable**
>
> No list $s_0, s_1, s_2, \dots$ of binary sequences contains every binary sequence.

Proof. Take any list. Define a new sequence d by flipping the diagonal:

$$d(k) = 1 - s_k(k) \qquad (k = 0, 1, 2, \dots).$$

For each k, d differs from s_k in position k, so $d \neq s_k$. Hence d is not on the list. □

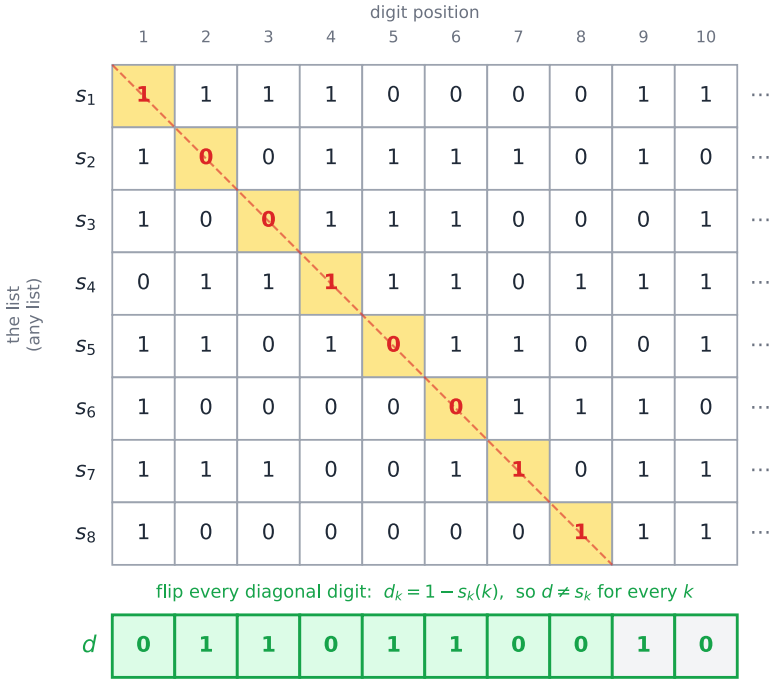


Figure 64.3 — The diagonal argument. Whatever the list, read down the diagonal and flip each digit. The result d disagrees with row k in column k, so it is not any row. The grey tail stands for the digits beyond the drawn part, also flipped from the diagonal of rows further down.

The proof is so short it can feel like a trick. Probe it. "Just add d to the front of the list." Fine: the new list has its own diagonal, and a new d' escapes it. "Then the list was infinite, so perhaps d is at position ∞." There is no such position: every entry of a list sits at a finite index k, and d has been made to differ from the entry at index k, for every k. The argument does not say any particular list is bad; it says *every* list is.

Now transfer the result to the real numbers. Each $x \in (0,1)$ has a decimal expansion $0.x_1x_2x_3\ldots$, unique if we forbid expansions ending in an infinite run of 9s. Given any list $r_1, r_2, \ldots$ of reals in $(0,1)$, define x by its digits: $x_k = 5$ if the k-th digit of r_k is not 5, and $x_k = 4$ if it is. Then x differs from r_k in the k-th digit, and because x uses only the digits 4 and 5, it has no second expansion that could secretly coincide with r_k. So x is not on the list, and:

Corollary — the reals are uncountable

$|\mathbb{N}| < |\mathbb{R}|$. In fact $|\mathbb{R}| = |(0,1)| = |\{0,1\}^{\mathbb{N}}|$, a cardinality written $2^{\aleph_0}$ or $\mathfrak{c}$ (the continuum).

The bijection $(0,1) \to \mathbb{R}$ is $x \mapsto \tan(\pi(x - \tfrac{1}{2}))$, which you can drag around in §64.5; the equality with $\{0,1\}^{\mathbb{N}}$ follows from Schröder–Bernstein using binary and decimal expansions as the two injections. A startling consequence: the rationals are countable, so "almost all" real numbers are irrational. Algebraic numbers (roots of integer polynomials, like $\sqrt{2}$) are countable too, since there are countably many polynomials and each has finitely many roots, so almost all reals are **transcendental**, even though proving that one specific number like π is transcendental is hard. Cantor's argument gives you infinitely many without naming a single one, the non-constructive style of Theorem 63.3 in Ch. 63.

? Numbers nobody can ever name

Every number you can describe, by a formula, a program that prints its digits, or a sentence in English, comes from a description, and descriptions are finite strings: there are only countably many. So all but countably many real numbers can never be specified, computed, or written down by anyone, ever. What does it mean to say such numbers *exist*? Which numbers can a machine compute, and is there a definable number that no machine can compute? Ch. 30 meets the first uncomputable objects, and Ch. 68 asks what "definable" can even mean.

A ladder of infinities

Cantor's argument generalises from $\mathbb{N}$ to any set whatsoever. Write $\mathcal{P}(A)$ for the **power set** of A, the set of all subsets of A. A binary sequence is just a subset of $\mathbb{N}$ in disguise ($s(k) = 1$ means "k is in"), so Theorem 64.2 says $|\mathbb{N}| < |\mathcal{P}(\mathbb{N})|$.

Theorem 64.3 — Cantor's theorem

For every set A, $|A| < |\mathcal{P}(A)|$.

Proof. The map $a \mapsto \{a\}$ is an injection $A \to \mathcal{P}(A)$, so $|A| \le |\mathcal{P}(A)|$. Suppose for contradiction that some $f : A \to \mathcal{P}(A)$ is a bijection; we only need that it is surjective. Define the "diagonal" set

$$D = \{a \in A : \ a \notin f(a)\}.$$

D is a subset of A, so by surjectivity $D = f(d)$ for some $d \in A$. Is $d \in D$? If $d \in D$, then by the definition of D, $d \notin f(d) = D$. If $d \notin D$, then $d \notin f(d)$, so by definition $d \in D$. Either way we contradict ourselves, so no surjection exists. □

So there is no largest infinity:

$$\aleph_0 = |\mathbb{N}| < |\mathcal{P}(\mathbb{N})| = |\mathbb{R}| < |\mathcal{P}(\mathbb{R})| < |\mathcal{P}(\mathcal{P}(\mathbb{R}))| < \cdots$$

The set of all real functions $\mathbb{R} \to \mathbb{R}$ already sits on the third rung: it has cardinality $|\mathcal{P}(\mathbb{R})|$, strictly more than the continuum. The continuous functions, by contrast, are determined by their values on the rationals, so there are only $\mathfrak{c}$ of them. Most functions are wildly discontinuous, which is one reason analysis restricts which functions it talks about (Ch. 60).

> **? The rung that might not exist**
>
> $\aleph_0 < \mathfrak{c}$. Is there any set strictly bigger than $\mathbb{N}$ but strictly smaller than $\mathbb{R}$? Cantor believed not (the **continuum hypothesis**) and spent years failing to prove it. Hilbert put it first on his famous 1900 list of problems. The eventual answer, from Gödel (1940) and Paul Cohen (1963), is one of the strangest in mathematics, and it is not "yes" and it is not "no". What else could it be? See Ch. 68.

Ordinals: counting past infinity

Cardinals answer "how many?". **Ordinals** answer "in what position?". For finite sets the two coincide, but for infinite ones they split. List the natural numbers $0, 1, 2, \ldots$ and then put one more element after all of them:

$$0,\ 1,\ 2,\ 3,\ \ldots\ ;\ \omega.$$

The new element sits at position ω, the first infinite ordinal. Keep going: $\omega + 1,\ \omega + 2,\ \ldots,\ \omega \cdot 2,\ \omega \cdot 2 + 1,\ \ldots,\ \omega \cdot 3,\ \ldots,\ \omega^2,\ \ldots,\ \omega^\omega,\ \ldots$. Each of these is the *order type* of a **well-ordering**: an ordering in which every non-empty subset has a least element. The order $0, 1, 2, \ldots\ ; \omega$ is a different shape from $0, 1, 2, \ldots$ (it has a last element; $\mathbb{N}$ does not), yet as a set it is still countable: it has cardinality $\aleph_0$. So ω and $\omega + 1$ are different ordinals with the same cardinal.

Ordinal arithmetic is arithmetic of arrangements, and it is not commutative. $1 + \omega$ means "one element, then a copy of $\mathbb{N}$", which is just a copy of $\mathbb{N}$ again, so $1 + \omega = \omega$. But $\omega + 1$ means "a copy of $\mathbb{N}$, then one more", which has a last element, so $\omega + 1 \neq \omega$. Hilbert's hotel lives in cardinals, where $1 + \aleph_0 = \aleph_0 + 1 = \aleph_0$; ordinals are what you get when the *queue order* matters.

Why care? Because well-orderings are exactly what induction needs. Ch. 29 proved termination of a loop by a variant that decreases in $\mathbb{N}$ (§29.3). Some programs need more: a process whose "remaining work" is a pair (a, b) ordered lexicographically can make b jump up arbitrarily far each time a drops by one, so no single natural number bounds the number of steps, yet the process must stop because $\omega \cdot \omega = \omega^2$ is well-ordered. **Transfinite induction** extends proof by induction along any ordinal: if a property passes from all smaller ordinals to the next one, it holds for all of them.

Why axioms?

So far a "set" has been any collection we can describe. That freedom is fatal. Consider $R = \{x : x \notin x\}$, the set of all sets that are not members of themselves. Is $R \in R$? The proof of Theorem 64.3 runs again, with f the identity, and both answers contradict themselves. This is **Russell's paradox** (1901), and it showed that "any describable collection is a set" is inconsistent. Its full story, and its cousins, are in Ch. 68.

The repair, settled by Zermelo, Fraenkel and others by the 1920s, is to say precisely which sets exist. The standard list is **ZFC** (Zermelo–Fraenkel with Choice). Roughly:

Axiom	It says
Extensionality	sets with the same elements are equal
Empty set, pairing, union	$\varnothing$, $\{a, b\}$ and $\bigcup A$ exist
Separation	for a set A and a property P, $\{x \in A : P(x)\}$ exists (only *inside* an existing set)
Power set	$\mathcal{P}(A)$ exists
Infinity	an infinite set exists (one containing $\varnothing$ and closed under $x \mapsto x \cup \{x\}$)
Replacement	the image of a set under a definable function is a set
Foundation	no infinite descending chain $\cdots \in x_2 \in x_1 \in x_0$
Choice	from any family of non-empty sets one can choose one element from each

Separation is the key restriction: you may carve a subset out of a set you already have, but not conjure $\{x : P(x)\}$ from nothing. Russell's R is never built. And from these axioms the whole of ordinary mathematics can be constructed: $0 = \varnothing$, $1 = \{0\}$, $2 = \{0, 1\}$, $n + 1 = n \cup \{n\}$, so each natural number is the set of the ones before it, and $\omega = \{0, 1, 2, \ldots\}$ is the set the axiom of infinity provides. Integers are classes of pairs of naturals, rationals classes of pairs of integers, and reals are built from rationals by Dedekind cuts or Cauchy sequences (Ch. 60). Every object in this book, from a phasor to a probability space (Ch. 19), is, at bottom, a set.

64.4 The algorithm

Countability *is* an algorithm: a set is countable exactly when you can write a procedure that, run forever, outputs every element at some finite time. Here are two enumerations of the positive rationals and a diagonaliser.

Cantor pairing and its inverse. The bijection of Theorem 64.1 can be inverted in constant time with an integer square root: given z, the diagonal is $s = \lfloor (\sqrt{8z + 1} - 1)/2 \rfloor$, then $q = z - s(s + 1)/2$, $p = s - q$. For example $\pi(3, 4) = 32$ and $\pi(4, 3) = 31$. Use `math.isqrt`, not floating `sqrt`: for z above about 10^{15} a double cannot hold $8z + 1$ exactly and the floor can land on the wrong diagonal.

The Calkin–Wilf sequence. Skipping non-reduced fractions, as in Figure 64.2, wastes work. A cleverer enumeration visits every positive rational *exactly once* with no gcd tests at all:

$$x_0 = 1, \qquad x_{n+1} = \frac{1}{2\lfloor x_n \rfloor - x_n + 1},$$

giving $1,\ \frac{1}{2},\ 2,\ \frac{1}{3},\ \frac{3}{2},\ \frac{2}{3},\ 3,\ \frac{1}{4},\ \frac{4}{3},\ \frac{3}{5},\ \frac{5}{2},\ \frac{2}{5},\ \frac{5}{3},\ \frac{3}{4},\ 4, \ldots$ (Newman's formula for the Calkin–Wilf tree). Each step costs $O(1)$ operations on exact fractions. Do it with floating point and the sequence drifts after a few dozen terms: the map is exact only on exact rationals.

A diagonaliser. Given *any* enumerator, a function `s(k, i)` returning digit i of the k-th sequence, the procedure `d(i) = 1 - s(i, i)` computes a sequence missing from the enumeration. This is an algorithm that defeats every algorithm of a given kind, and that idea is the engine of Ch. 30: feed it an enumeration of all programs and you get a function no program computes.

64.5 The experiment

The script enumerates rationals by Calkin–Wilf, checks that no value repeats and none is missed, and then diagonalises against a list of your choice.

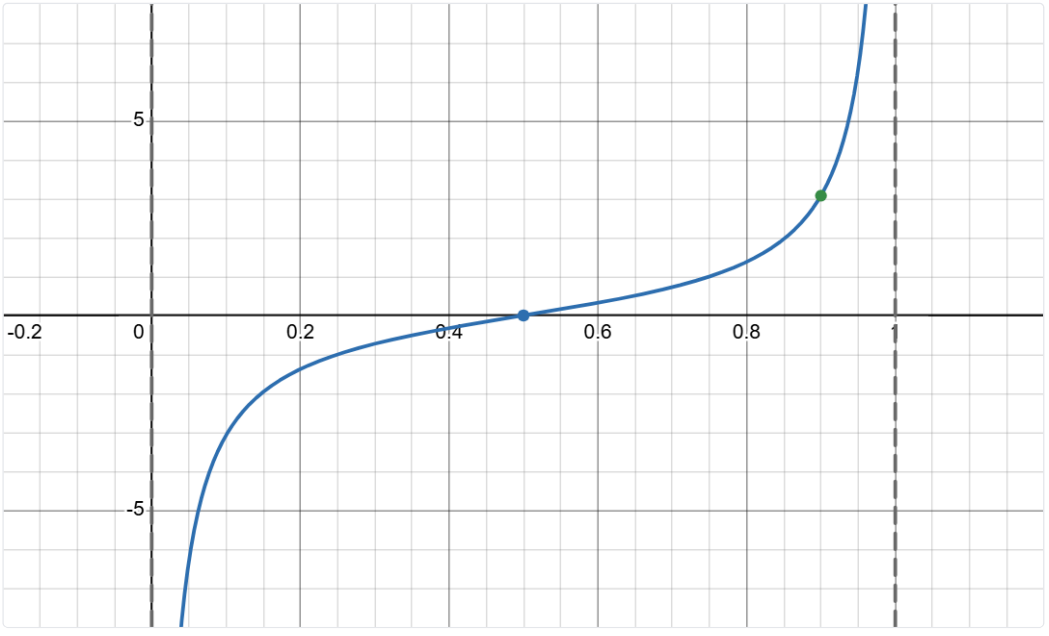
```
from fractions import Fraction
from math import floor, gcd

N = 4100                        # knob: how many Calkin-Wilf terms to generate
M = 12                          # knob: check every p/q with p, q <= M appears

x, seen = Fraction(1), []
for _ in range(N):
    seen.append(x)
    x = 1 / (2 * floor(x) - x + 1)

print("repeats:", len(seen) - len(set(seen)))
wanted = {Fraction(p, q) for p in range(1, M + 1) for q in range(1, M + 1)}
missing = wanted - set(seen)
print(f"p/q with p,q<={M}: {len(wanted)} values, missing from first {N}: {len(missing)}")
print("first 10:", [str(f) for f in seen[:10]])

# Diagonalise: the k-th row is the binary expansion of the k-th rational in (0,1)
ROWS = 16                       # knob: size of the list we defeat
rats = [f for f in seen if f < 1][:ROWS]
def digit(f, i):                # i-th binary digit of f after the point
    return floor(f * 2 ** (i + 1)) % 2
d = [1 - digit(rats[k], k) for k in range(ROWS)]
for k, f in enumerate(rats):
    row = "".join(str(digit(f, i)) for i in range(ROWS))
    print(f"{str(f):>6}  {row}   differs from d at digit {k}")
print("     d  " + "".join(map(str, d)))
```

With the defaults it reports 0 repeats and that all 91 distinct values with $p, q \le 12$ appear among the first 4100 terms (the last to arrive is 12 itself, at position 4094), then prints 16 rows and the escaping sequence d.

Things to try:

1. Lower `N` to 2000 and count what goes missing. Which fractions with $p, q \le 12$ arrive *last*? (Print the positions of $1/M$ and M for several M.) How many terms does Calkin–Wilf need, as a function of M, to reach all of them, and what does that say about using it to enumerate "small" fractions?
2. Replace the Calkin–Wilf step with floating point, `x = 1 / (2*floor(x) - x + 1)` on a `float`. After how many terms does it produce a value that is not a fraction with small denominator, or repeat one?
3. Now the deep one: d is the binary expansion of *some* real number in $[0, 1]$. Increase `ROWS`. Is that number rational? (Is it on the full infinite list?) Explain why the answer must be no, no matter how large `ROWS` is.

The bijection between the interval $(0, 1)$ and the whole real line is easy to see:



The map $x \mapsto \tan(\pi(x - \frac{1}{2}))$ sends the short interval $(0, 1)$ onto all of $\mathbb{R}$, one-to-one. Replace $\tan(\pi(x - 0.5))$ with $\frac{x-0.5}{x(1-x)}$, an algebraic bijection: as many points in a unit interval as on the entire line.

64.6 The challenge

No answers are given.

1. **Intuition.** On the fourth night the manager tries this: "Order the passengers by how many 1s their string has, and give out rooms in that order." Find the exact point at which her plan breaks down, and explain why no repair of the same kind can work.
2. **Derivation.** Prove that $|[0, 1]| = |(0, 1)|$ by constructing an explicit bijection. (Hint: Hilbert's hotel. Pick the countable set $\{\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\}$ inside $(0, 1)$ and shift it to make room for the two endpoints.) Then prove $|\mathbb{R}^2| = |\mathbb{R}|$ by interleaving digits, and deal carefully with the two-expansion problem.
3. **Implementation.** Write a generator that enumerates every finite list of natural numbers exactly once (for example, using Cantor pairing repeatedly, or by ordering lists by "length plus sum"). Check by brute force that your first 10,000 outputs have no repeats and include every list of length ≤ 3 with entries ≤ 5. Then explain why no generator can enumerate every *infinite* list.
4. **Open investigation.** In the Calkin–Wilf sequence, let $n(p/q)$ be the position of the reduced fraction p/q. Plot $\log_2 n(p/q)$ against $p + q$ for all $p, q \leq 200$. The sequence is related to Stern's diatomic sequence and to continued fractions. Find empirically which fractions with a given $p + q$ appear earliest and latest, and try to explain the pattern in terms of the continued-fraction expansion of p/q.

64.7 Loose threads

We have answered the hook. Infinities are compared by matching, not counting. By that measure $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$ and every set of finite descriptions all have the same size $\aleph_0$, while the reals, the binary sequences and the passengers of the fourth night form a strictly larger infinity, because the diagonal argument defeats every attempted list. Cantor's theorem then builds an endless ladder of larger infinities, and ordinals let us count past ω in a well-ordered queue that supports induction.

What we have not done:

- **We used the axioms without asking whether they are safe.** ZFC was built to stop Russell's paradox. Does it stop *every* paradox? Can ZFC prove its own consistency? The answer, which Ch. 68 reveals, is a theorem about all sufficiently strong systems, and it is not the answer anyone hoped for.
- **We used the axiom of choice silently.** "A countable union of countable sets is countable" needs it: you must *choose* an enumeration of each set. Choice also proves that a solid ball can be cut into finitely many pieces and reassembled into two balls of the same size (Banach–Tarski). Should we believe it? See Ch. 68, and Ch. 60 for why the pieces cannot have a volume.
- **We never said what "proof from axioms" means.** ZFC is a list of sentences, but what are the legal moves from them? Is every true statement about sets provable from them? That is what a formal system makes precise, and it is where we go next, in Ch. 65.

A puzzle to carry forward. ZFC is written with countably many symbols and proves "$\mathbb{R}$ is uncountable". Yet logic, as the next chapter shows, guarantees that any consistent set of axioms like ZFC has a model, a mathematical world in which they are all true, whose underlying collection is *countable*. So there is a countable world in which "the reals are uncountable" is true. Is that a contradiction? It is called Skolem's paradox, and it is waiting for you in Ch. 65.

Further reading

- P. R. Halmos, *Naive Set Theory*, Van Nostrand, 1960 (Springer reprint). Short, elegant and still the best first book on sets.
- H. B. Enderton, *Elements of Set Theory*, Academic Press, 1977. Builds $\mathbb{N}$, $\mathbb{R}$, cardinals and ordinals from the ZFC axioms.
- N. Calkin and H. S. Wilf, "Recounting the rationals", *American Mathematical Monthly* 107 (2000), 360–363. The tree behind the enumeration in §64.4.
- R. Rucker, *Infinity and the Mind*, Princeton University Press, 1982. A lively tour of cardinals, ordinals and the philosophy of the infinite.

Chapter 65 — Mathematical Logic and Formal Systems

> **Level 2 · Track F — Logic, Proofs, and the Foundations of Truth Builds on:** Ch. 29, Ch. 63, Ch. 64 · **Leads to:** Ch. 66, Ch. 67, Ch. 68 · **Related:** Ch. 30, Ch. 56, Ch. 1 **Reading time:** ~25 min

65.1 The hook

Here is a game with three letters, M, I and U. You start with the string **MI**. You may apply any of four rules, as often as you like, in any order (here x and y stand for any strings, possibly empty):

Rule	From	You may write	Example
R1	xI	xIU	MI → MIU
R2	Mx	Mxx	MIU → MIUIU
R3	xIIIy	xUy	MIIII → MUI
R4	xUUy	xy	MUUI → MI

Your goal: produce the string **MU**.

Try it for five minutes before reading on. You will double strings with R2 to pile up I's, chop III into U with R3, delete pairs of U's with R4. You will get close, MUI, MUIU, MUUU, and never quite there. (This is Douglas Hofstadter's MU puzzle, from *Gödel, Escher, Bach*.)

Now imagine handing the puzzle to a computer. It can search systematically: all strings one rule away, then two rules away, and so on. If MU is reachable, the search will find it eventually. If MU is *not* reachable, the search runs forever, producing longer and longer strings, and never tells you anything. To know that MU is impossible, you need something the rules themselves do not give you: you need to *step outside the game and reason about it*.

That is the whole of mathematical logic in miniature. Around 1920, David Hilbert proposed to turn all of mathematics into a game like this one: fixed starting strings (axioms), fixed rules (logic), and theorems as whatever strings you can reach. He hoped to show that the game could reach every true statement and never reach a false one. This chapter is about what happened to that hope, and it is a story with a twist:

> **Can a finite set of rules, applied mechanically, produce every truth of mathematics, and nothing else?**

65.2 The big idea

Every formal system lives on two levels at once, and the whole subject is about the bridge between them.

- **Syntax** is symbols and rules. A proof is a finite list of strings, each one an axiom or obtained from earlier ones by a rule. Checking a proof needs no understanding at all: a machine can do it. We write $T \vdash \varphi$ ("T proves φ") when there is a proof of φ from the axioms T.
- **Semantics** is meaning. A **model** of T is a mathematical structure in which every axiom of T is true. We write $T \models \varphi$ ("φ is a consequence of T") when φ is true in *every* model of T.

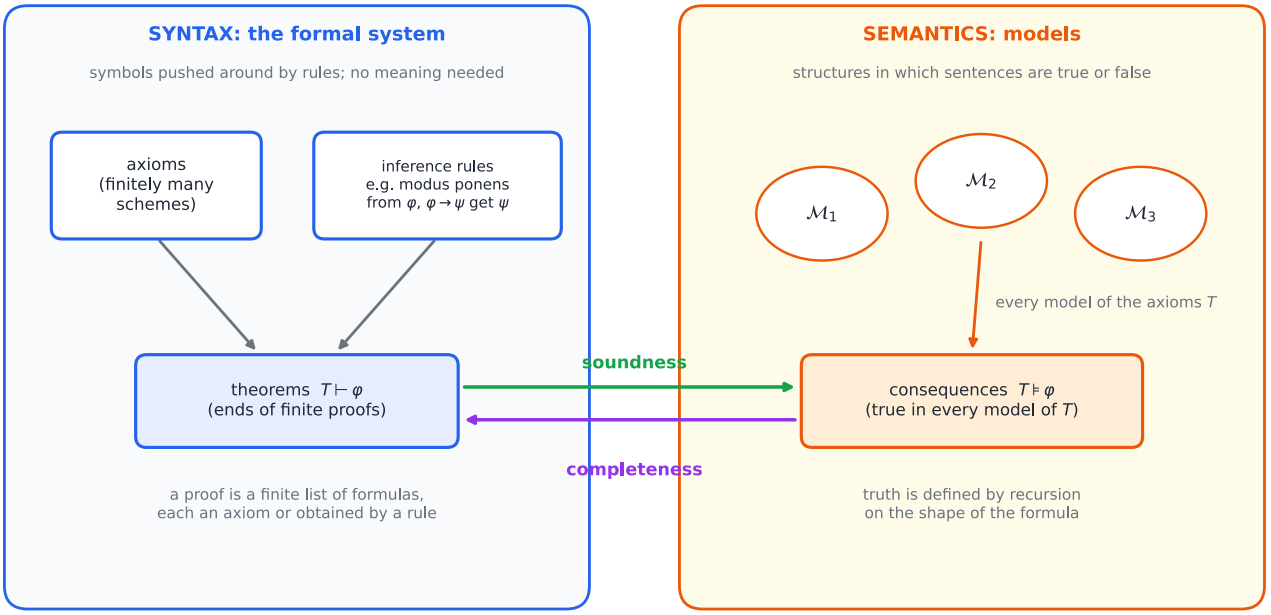


Figure 65.1 — The two worlds of logic. On the left, theorems are produced by pushing symbols; on the right, consequences are determined by truth in models. Soundness says the left never outruns the right; completeness says it always keeps up.

A programmer knows this split already. A type checker works on syntax: it inspects the program text and either accepts or rejects. What the program *does* when run is semantics. A type system is *sound* if every accepted program is safe when run (no false positives), and *complete* if every safe program is accepted (no misses). Real type checkers are sound and incomplete: they reject some perfectly good programs. Logic asks the same two questions of proof rules.

Here is the plot of the chapter in one paragraph. For **first-order logic**, the logic of "for all", "there exists", "and", "not" over any structure, the answer to both questions is yes: the rules are sound and complete, and $\vdash$ and $\models$ coincide exactly (Gödel, 1929). Strange consequences follow, including mathematical worlds with "infinitely large" natural numbers. But when you fix one specific structure, the natural numbers $\mathbb{N}$ with $+$ and $\times$, and ask for rules that prove exactly the sentences *true in* $\mathbb{N}$, the answer flips to no (Gödel again, 1931). Same logician, two years apart, opposite answers.

65.3 The mathematics

First-order languages, briefly

Ch. 29 introduced predicates and quantifiers informally (§29.3). A first-order language fixes the vocabulary.

Language, structure, satisfaction

A **language** L is a set of constant symbols, function symbols and relation symbols, each with an arity. **Terms** are built from variables and constants by applying function symbols; **formulas** are built from atomic formulas ($t_1 = t_2$, or $R(t_1, \dots, t_k)$) using $\neg$, $\to$ (and the other connectives) and $\forall x$, $\exists x$. A **sentence** is a formula with no free variables.

An L**-structure** $\mathcal{M}$ is a non-empty set M (the domain) together with an interpretation of each symbol: an element for each constant, a function $M^k \to M$ for each function symbol, a relation on M^k for each relation symbol. **Truth** $\mathcal{M} \models \varphi$ is defined by recursion on φ: $\mathcal{M} \models \neg\varphi$ iff not $\mathcal{M} \models \varphi$; $\mathcal{M} \models \forall x\, \varphi(x)$ iff $\mathcal{M} \models \varphi(a)$ for every $a \in M$; and so on. $\mathcal{M}$ is a **model** of a set of sentences T if it satisfies every sentence in T.

"First-order" means the quantifiers range over *elements* of the domain, never over subsets or functions. That restriction looks minor. It is the source of everything strange below.

An example. The language of groups has one binary function symbol $\cdot$ and one constant e. The group axioms T_{grp} say that $\cdot$ is associative, e is an identity, and inverses exist (Ch. 56 develops them). Models of T_{grp} are exactly the groups: $(\mathbb{Z}, +, 0)$, the rotations of a square, the permutations of three objects. Consider the sentence $\gamma = \forall x \forall y\ (x \cdot y = y \cdot x)$. It is true in $(\mathbb{Z}, +, 0)$ and false among permutations of three objects. So γ is neither a consequence of T_{grp} nor is $\neg\gamma$. Keep this example in mind: in a moment, soundness will turn it into a proof that γ cannot be proved from the group axioms, without examining a single proof.

A proof system

To prove anything we need rules. Many equivalent systems exist; here is the classical Hilbert system for the propositional part, with $\to$ and $\neg$ as the connectives. Its axioms are all instances (for arbitrary formulas φ, ψ, χ) of

$$\begin{aligned} &\text{A1:} && \varphi \to (\psi \to \varphi) \\ &\text{A2:} && \big(\varphi \to (\psi \to \chi)\big) \to \big((\varphi \to \psi) \to (\varphi \to \chi)\big) \\ &\text{A3:} && (\neg\psi \to \neg\varphi) \to (\varphi \to \psi) \end{aligned}$$

and its only rule is **modus ponens** (MP): from φ and $\varphi \to \psi$, infer ψ. For first-order logic one adds axioms for quantifiers and equality (for example $\forall x\, \varphi(x) \to \varphi(t)$) and the rule of generalisation; we will not need the details.

Proofs in this system are notoriously unnatural. Here is a proof of the obvious-looking $\varphi \to \varphi$:

Line	Formula	Justification
1	$\varphi \to ((\varphi \to \varphi) \to \varphi)$	A1 with $\psi := \varphi \to \varphi$
2	$\big(\varphi \to ((\varphi \to \varphi) \to \varphi)\big) \to \big((\varphi \to (\varphi \to \varphi)) \to (\varphi \to \varphi)\big)$	A2 with $\psi := \varphi \to \varphi$, $\chi := \varphi$
3	$(\varphi \to (\varphi \to \varphi)) \to (\varphi \to \varphi)$	MP from 1, 2
4	$\varphi \to (\varphi \to \varphi)$	A1 with $\psi := \varphi$
5	$\varphi \to \varphi$	MP from 4, 3

Five lines, and every line checkable by pattern-matching, with no idea what φ means. That is the point. Humans write proofs in natural language; logic shows that every such proof *could* be expanded into a list like this one, and Ch. 67 shows engineers doing exactly that with proof assistants.

Soundness: the rules never lie

Theorem 65.1 — Soundness (propositional case)

If $\vdash \varphi$ in the Hilbert system, then φ is a **tautology**: it is true under every assignment of truth values to its variables.

Proof. By strong induction on the length n of a shortest proof of φ; the claim is that every line of every proof is a tautology.

Axioms. Each instance of A1–A3 is a tautology. For A1: if $\varphi \to (\psi \to \varphi)$ were false under some assignment, then φ is true and $\psi \to \varphi$ is false, which needs φ false; contradiction. A2 and A3 are checked the same way, or by an 8-row and a 4-row truth table (the script in §65.4 does it). Substituting formulas for φ, ψ, χ does not matter, because the truth value of the instance depends only on the truth values of the substituted formulas.

Modus ponens. Suppose line k is ψ, obtained from earlier lines φ and $\varphi \to \psi$, both tautologies by the induction hypothesis. Under any assignment, φ is true and $\varphi \to \psi$ is true, so ψ is true. Hence ψ is a tautology.

Every line is an axiom or comes by MP, so every line, in particular the last, is a tautology. □

The first-order version reads: **if** $T \vdash \varphi$ **then** $T \models \varphi$, and the proof is the same induction, with one extra case per quantifier rule. Soundness is a *tool for proving unprovability*. Back to the groups: γ (commutativity) is false in a model of T_{grp}, so $T_{\text{grp}} \not\models \gamma$, so by soundness $T_{\text{grp}} \nvdash \gamma$. No proof of commutativity from the group axioms exists, and we know it without searching. The MU puzzle works the same way: an invariant (the number of I's is never divisible by 3) is a "model" in which every theorem has the property and MU does not.

Completeness: the rules never miss

The converse is the deep one.

> **Theorem 65.2 — Gödel's completeness theorem (1929)**
>
> For every set of first-order sentences T and every sentence φ: if $T \models \varphi$ then $T \vdash \varphi$. Equivalently, every **consistent** T (one that does not prove a contradiction) has a model.

We state it without full proof. The idea, due to Leon Henkin (1949), is a construction worth knowing. Start with a consistent T. Add new constant symbols as "witnesses": for every sentence $\exists x\, \psi(x)$, add a constant c_ψ and the axiom $\exists x\, \psi(x) \to \psi(c_\psi)$. Then extend T, one sentence at a time, to a maximal consistent set, deciding every sentence one way or the other. Finally build a model *out of the symbols themselves*: its elements are the closed terms (modulo provable equality), and a relation holds exactly when the set says it does. The model is made of syntax. And because the language had countably many symbols, the model is countable, which gives a theorem we will need:

> **Löwenheim–Skolem theorem (downward)**
>
> If a set of sentences in a countable language has a model, it has a countable model.

This settles the puzzle left at the end of Ch. 64. If ZFC is consistent, it has a countable model $\mathcal{M}$, and inside $\mathcal{M}$ the sentence "$\mathbb{R}$ is uncountable" is true. There is no contradiction, because "uncountable" means "there is no bijection with $\mathbb{N}$", and inside $\mathcal{M}$ that means "no bijection *that is an element of* $\mathcal{M}$". The bijection that we, outside, can see is simply not one of $\mathcal{M}$'s sets. "Countable" turns out to be relative to the world you are standing in.

Compactness: infinity from finite pieces

Proofs are finite. That trivial fact, combined with completeness, has an astonishing consequence.

> **Theorem 65.3 — Compactness**
>
> A set of first-order sentences T has a model if and only if every finite subset of T has a model.

Proof. If T has a model, so does every subset. Conversely, suppose every finite subset of T has a model, but T does not. Then T is inconsistent by the completeness theorem (in its second form), so there is a proof of a contradiction, say $\chi \wedge \neg\chi$, from T. That proof is a finite list of formulas, so it uses only finitely many axioms of T; call them T_0. Then $T_0 \vdash \chi \wedge \neg\chi$, and by soundness every model of T_0 satisfies $\chi \wedge \neg\chi$, which is impossible, so T_0 has no model. But T_0 is a finite subset of T. Contradiction. □

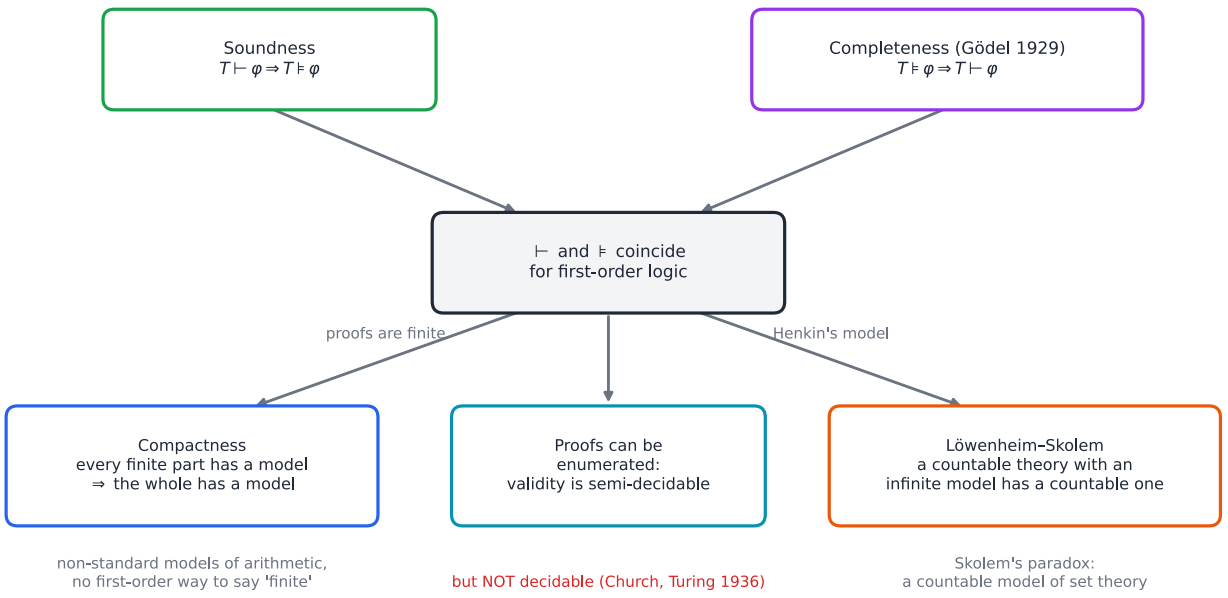


Figure 65.2 — How the big theorems of first-order logic hang together. Soundness and completeness together make provability and truth-in-all-models the same relation; compactness, Löwenheim–Skolem and the mechanical enumerability of theorems all follow, along with some very strange models.

Now watch compactness do something unsettling.

Theorem 65.4 — Non-standard models of arithmetic

Let $\mathrm{Th}(\mathbb{N})$ be the set of *all* first-order sentences in the language $\{0, S, +, \times, <\}$ that are true in the natural numbers. There is a model of $\mathrm{Th}(\mathbb{N})$ containing an element larger than every natural number.

Proof. Add a new constant symbol c, and let $\underline{n}$ be the term $S(S(\cdots S(0)))$ with n copies of S (the numeral for n). Consider

$$T = \mathrm{Th}(\mathbb{N}) \cup \{\, \underline{n} < c \;:\; n = 0, 1, 2, \dots \}.$$

Any finite subset of T mentions only finitely many of the new axioms, say up to $\underline{N} < c$. Interpret c as the number $N + 1$ in the ordinary natural numbers: every sentence of $\mathrm{Th}(\mathbb{N})$ is true there by definition, and so is $\underline{n} < c$ for every $n \le N$. So every finite subset has a model. By compactness, T has a model $\mathcal{M}$. It satisfies every true sentence of arithmetic, and its element $c^{\mathcal{M}}$ is larger than $\underline{n}^{\mathcal{M}}$ for every n. $\square$

Sit with that. $\mathcal{M}$ agrees with $\mathbb{N}$ on *every* first-order sentence: every theorem of number theory you have ever met, every statement about primes, holds in $\mathcal{M}$. Yet $\mathcal{M}$ contains an "infinite" natural number c, and with it $c + 1$, $c - 1$, $2c$, c^2, a whole galaxy of non-standard numbers beyond the standard ones. No first-order sentence can tell $\mathcal{M}$ and $\mathbb{N}$ apart. In particular, "x is a standard number" (equivalently, "the set of things below x is finite") cannot be written in first-order arithmetic at all. Induction itself is the casualty: the first-order induction *scheme* cannot pin down $\mathbb{N}$, because it can only talk about definable properties, and "standard" is not one.

Incompleteness: where the bridge breaks

Completeness says: anything true in *every* model of T is provable from T. It does not say anything is provable that is true in *one particular* model. And for arithmetic, the one model we care about is $\mathbb{N}$.

Peano arithmetic (PA) is the first-order theory with axioms saying 0 is not a successor, S is injective, $+$ and $\times$ obey their recursive definitions ($x + 0 = x$, $x + S(y) = S(x + y)$, $x \cdot 0 = 0$, $x \cdot S(y) = x \cdot y + x$), plus the induction scheme: for every formula $\psi(x)$,

$$\big(\psi(0) \wedge \forall x\, (\psi(x) \to \psi(S(x)))\big) \to \forall x\, \psi(x).$$

PA proves essentially all of elementary number theory. Hilbert hoped that it, or something like it, proves *every* true sentence about $\mathbb{N}$.

Gödel's first incompleteness theorem (1931), stated

Let T be any consistent set of axioms that can be listed by an algorithm and that includes PA (or even a weak fragment of it). Then there is a sentence G_T in the language of arithmetic such that $T \nvdash G_T$ and $T \nvdash \neg G_T$. Moreover G_T is true in $\mathbb{N}$.

The idea, which we only sketch: code formulas and proofs as numbers (**Gödel numbering**), so that "p is the code of a proof of the formula with code q" becomes an arithmetic relation between p and q, checkable by an algorithm and hence expressible in the language of PA. Then build a sentence G_T that says, in effect, "*there is no number that codes a T-proof of me*". If T proved G_T, there would be such a number, and T would prove that too, so T would be inconsistent. So T does not prove G_T, which is exactly what G_T says: it is true.

How does that square with completeness? G_T is true in $\mathbb{N}$ but not provable, so by completeness it must be *false in some other model of T*. That model is non-standard, and in it there is a non-standard number that the model "believes" codes a proof of G_T: an infinitely long proof that only a non-standard world could contain. Completeness and incompleteness are not in conflict. The gap between them is precisely the gap between $\mathbb{N}$ and the strange models of Theorem 65.4.

? The sentence that talks about itself

The sketch above leaned on a sentence that says "I am not provable". But sentences of arithmetic talk about numbers: additions, multiplications, inequalities. How can a sentence about numbers refer to *itself*, without circularity, and without falling into the liar paradox "this sentence is false"? The tool is Gödel's **diagonal lemma**, a cousin of Cantor's diagonal argument and of Turing's halting proof in Ch. 30. Its construction, and what it does to Hilbert's dream, are in Ch. 68.

65.4 The algorithm

Logic is full of algorithms, and the line between what can and cannot be automated is sharp.

Checking a proof is easy and fast: for each line, check that it matches an axiom scheme or follows by a rule from earlier lines. Time is polynomial (essentially linear) in the length of the proof. Here is a checker for the Hilbert system, which also verifies by truth tables that the axioms are tautologies, the base case of Theorem 65.1. Formulas are nested tuples: `('->', a, b)`, `('~', a)`, or a variable name.

```
from itertools import product

def imp(a, b): return ('->', a, b)
def neg(a): return ('~', a)
A1 = lambda p, q: imp(p, imp(q, p))
A2 = lambda p, q, r: imp(imp(p, imp(q, r)), imp(imp(p, q), imp(p, r)))
A3 = lambda p, q: imp(imp(neg(q), neg(p)), imp(p, q))

def value(f, env):
    if isinstance(f, str): return env[f]
    if f[0] == '~': return not value(f[1], env)
    return (not value(f[1], env)) or value(f[2], env)

def atoms(f):
    return {f} if isinstance(f, str) else set().union(*map(atoms, f[1:]))

def tautology(f):
    vs = sorted(atoms(f))
    return all(value(f, dict(zip(vs, bits))) for bits in product([False, True], repeat=len(vs)))

print("A1..A3 tautologies:", tautology(A1('p', 'q')), tautology(A2('p', 'q', 'r')), tautology(A3('p', 'q')))

def check(proof):          # each line: (formula, 'AX') or (formula, ('MP', i, j))
    for k, (f, why) in enumerate(proof):
        if why == 'AX':
            assert tautology(f), f"line {k}: not an axiom instance we trust"
        else:
            _, i, j = why
            assert i < k and j < k and proof[j][0] == imp(proof[i][0], f), f"line {k}: bad MP"
    return proof[-1][0]

p = 'p'
proof = [(A1(p, imp(p, p)), 'AX'), (A2(p, imp(p, p), p), 'AX'), (imp(A1(p, p), imp(p, p)), ('MP', 0, 1)),
         (A1(p, p), 'AX'), (imp(p, p), ('MP', 3, 2))]
print("proved:", check(proof))
```

(For brevity the checker accepts any tautology as an axiom line; a strict one would pattern-match the three schemes, as challenge 3 asks.)

Deciding truth is another matter.

- *Propositional logic* is decidable: a formula with n variables is a tautology iff all 2^n rows of its truth table say true, so `tautology` above always halts, in time $O(2^n \cdot \text{size})$. Whether one can do fundamentally better is the P versus NP question of Ch. 30.
- *First-order logic* is **semi-decidable**. Completeness means that $\models \varphi$ iff some proof of φ exists, and proofs are finite strings, so enumerate all strings in order of length and run the checker on each. If φ is valid, this finds a proof eventually. If φ is not valid, it runs forever, exactly like the search for MU. Church and Turing (1936) proved that nothing better is possible in general: validity of first-order sentences is **undecidable**, by reduction from the halting problem.
- *Truth in* $\mathbb{N}$ is worse still: it is not even semi-decidable. If it were, we could list all true sentences of arithmetic and use that list as an axiom system, which incompleteness forbids.

In practice, automated theorem provers (resolution, superposition, SMT solvers) replace blind enumeration with clever search and succeed astonishingly often. They can never succeed always.

65.5 The experiment

The script runs breadth-first search over the MIU system, the "blind enumeration of theorems" of §65.4, with a cap on string length, and compares what it finds with the invariant.

```
from collections import deque
from itertools import product

CAP = 10          # knob: longest string the search may visit (try 10, 14, 18)
TARGET = "MU"     # knob: try "MUI", "MIIU", "MUIIU", "MIIIIIIU"

def successors(s):
    if s.endswith("I"):
        yield s + "U"                                # R1
    yield "M" + s[1:] * 2                            # R2
    for i in range(1, len(s) - 2):
        if s[i:i + 3] == "III":
            yield s[:i] + "U" + s[i + 3:]            # R3
    for i in range(1, len(s) - 1):
        if s[i:i + 2] == "UU":
            yield s[:i] + s[i + 2:]                  # R4

seen, queue = {"MI": 0}, deque(["MI"])
while queue:
    s = queue.popleft()
    for t in successors(s):
        if len(t) <= CAP and t not in seen:
            seen[t] = seen[s] + 1
            queue.append(t)

print(f"theorems found with length <= {CAP}: {len(seen)}")
print(f"{TARGET}:", f"derived in {seen[TARGET]} steps" if TARGET in seen else "not found")

# The invariant: number of I's is never divisible by 3
print("invariant holds on all found:", all(s.count("I") % 3 != 0 for s in seen))

# Every short string allowed by the invariant: did the search find it?
L = 6
allowed = {"M" + "".join(w) for n in range(L) for w in product("IU", repeat=n)
           if "".join(w).count("I") % 3 != 0}
found = {s for s in seen if len(s) <= L}
print(f"strings of length <= {L} allowed by invariant: {len(allowed)}, found: {len(found)}")
```

With `CAP = 10` it finds 87 theorems, does not find MU, confirms the invariant, and finds only 32 of the 42 short strings the invariant permits.

Things to try:

1. Raise `CAP` to 14. Now all 42 short strings appear (634 theorems in all). Why did some *short* theorems need *long* intermediate strings? Find one and trace its derivation.
2. Set `TARGET = "MIIU"`, then `"MUIIU"`. Record the number of steps. Then raise `CAP` to 18 and check whether the step counts change: the search might find shorter derivations through longer strings.
3. The experiment suggests a conjecture: *a string is a theorem if and only if it is M followed by I's and U's with the number of I's not divisible by 3.* The "only if" half is the invariant. Can you see how to prove the "if" half? (It is true. Think about how R2 doubles the I-count modulo 3, and how R3 and R4 tidy up.)

Notice the asymmetry: raising `CAP` can only *add* theorems, so no run of the search ever establishes that MU is missing. That proof came from outside, from arithmetic modulo 3.

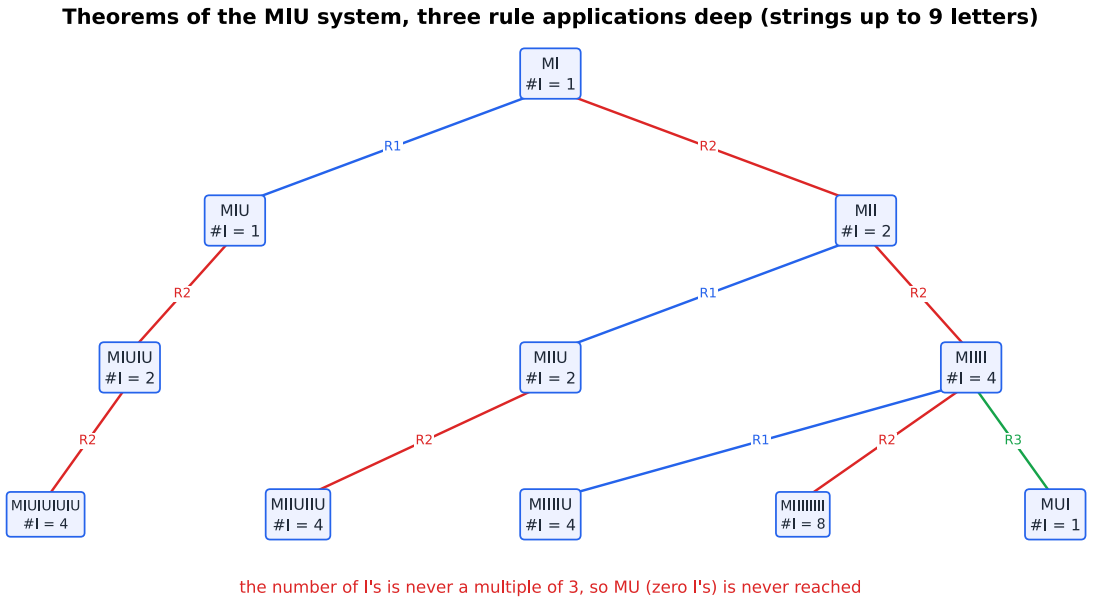


Figure 65.3 — The first three levels of the MIU search tree, coloured by rule, with the count of I's in each box. R2 doubles the count, R3 subtracts 3, R1 and R4 leave it unchanged. Modulo 3, doubling swaps 1 and 2 and subtracting 3 changes nothing, so a count that starts at 1 can never become a multiple of 3.

65.6 The challenge

No answers are given.

1. **Intuition.** In the language with one binary relation $<$, write first-order axioms for a strict linear order (irreflexive, transitive, any two distinct elements comparable). Which of $(\mathbb{N}, <)$, $(\mathbb{Z}, <)$, $(\mathbb{Q}, <)$, $(\mathbb{R}, <)$ are models? Use soundness to show that neither "there is a least element" nor "between any two elements lies a third" can be proved from your axioms, and that neither of their negations can be proved either.
2. **Derivation.** Using compactness, prove that if a first-order theory has arbitrarily large finite models, then it has an infinite model. (Hint: add sentences λ_n saying "there are at least n distinct elements".) Deduce that no set of first-order sentences in the language of groups has exactly the finite groups as its models.
3. **Implementation.** Extend the checker in §65.4 so that axiom lines must match one of the schemes A1, A2, A3 exactly (write a pattern matcher that recovers φ, ψ, χ). Then write out and machine-check a Hilbert-style proof of $\neg\neg p \to p$. Expect it to be long, and record how many lines you needed.
4. **Open investigation.** Let $d(s)$ be the length of the shortest derivation of the MIU theorem s. Using the experiment with a generous `CAP`, tabulate $d(s)$ for all theorems of length ≤ 8. How does the worst case grow with length? Is the cap you need to find the shortest derivation of a length-n string bounded by a simple function of n? (There is no standard answer; you are mapping new ground.)

65.7 Loose threads

We can now answer the hook, and the answer has two halves. For first-order logic in general, **yes**: the Hilbert rules are sound (Theorem 65.1) and complete (Theorem 65.2), so a finite set of mechanical rules produces exactly the sentences true in every model, and nothing else. For the truths of one particular structure, the natural numbers, **no**: every consistent, algorithmically listable axiom system leaves some true sentence unproved (Gödel's first incompleteness theorem). Compactness explains why the two answers can coexist: first-order axioms never pin down $\mathbb{N}$ uniquely, and the unprovable truths are exactly those that fail in some non-standard twin of $\mathbb{N}$.

Three questions remain open, and each one leads onward.

> **? Can mathematics vouch for itself?**
>
> Hilbert's second hope was a proof, inside a system like PA or ZFC, that the system never proves a contradiction. Gödel's *second* incompleteness theorem says what happens if you try, and it is not good news. If our foundations cannot certify their own consistency, what exactly is our confidence in mathematics based on? See Ch. 68, along with the independence of the continuum hypothesis promised in Ch. 64.

> **? What is a proof, computationally?**
>
> Our proof system was classical: it proves $\neg\neg p \to p$ and, with it, existence claims that give no witness, like Theorem 63.3 in Ch. 63. Drop that law, and something remarkable happens: proofs of $\varphi \to \psi$ become *functions* turning evidence for φ into evidence for ψ, and the five-line proof of $\varphi \to \varphi$ in §65.3 becomes the identity function. Why, and how far does the correspondence go? Ch. 66 tells the story.

- **Proof checking is mechanical; can verification of real software be?** Proof assistants such as Lean, Coq and Isabelle are proof checkers like ours, scaled up by decades of engineering. Ch. 67 shows how they are used to prove that compilers, operating-system kernels and cryptographic code meet their specifications.

The MU puzzle had an answer that the rules could not find but a mathematician could. Gödel showed that arithmetic has statements of the same kind, and that a mathematician who writes down new rules to capture them just produces a new system with new unreachable truths. No finite set of rules ever reaches all of them.

Further reading

- H. B. Enderton, *A Mathematical Introduction to Logic*, 2nd ed., Academic Press, 2001. Clear and complete: soundness, completeness, compactness and incompleteness with full proofs.
- D. R. Hofstadter, *Gödel, Escher, Bach: An Eternal Golden Braid*, Basic Books, 1979. The MU puzzle and a playful, deep tour of formal systems and self-reference.
- P. Smith, *An Introduction to Gödel's Theorems*, 2nd ed., Cambridge University Press, 2013. The most readable rigorous account of incompleteness.
- E. Nagel and J. R. Newman, *Gödel's Proof*, NYU Press, 1958 (revised 2001). A short classic for the first encounter.

Chapter 66 — Type Theory and the Curry–Howard Connection

> **Level 2 · Track F — Logic, Proofs, and the Foundations of Truth Builds on:** Ch. 1, Ch. 29, Ch. 63, Ch. 65 · **Leads to:** Ch. 67, Ch. 68 · **Related:** Ch. 30, Ch. 62, Ch. 64 **Reading time:** ~25 min

66.1 The hook

Here is a programming puzzle with an unusual rule. You may write only `lambda`, variables and function calls: no numbers, `if`, loops, exceptions or `None`. And `A`, `B`, `C` are unknown types: integers, images, bank accounts.

Puzzle 1. Write a function with the type

```
(A -> B) -> (B -> C) -> (A -> C)
```

That is easy. Take `f`, take `g`, take `x`, and return `g(f(x))`. It is function composition (Ch. 1, §1.9).

Puzzle 2. Write one of type `A -> (B -> A)`. Also easy: `lambda a: lambda b: a`.

Puzzle 3. Write a function with the type

```
((A -> B) -> A) -> A
```

Try it before you read on. You are handed `h`, which turns "a way to get from `A` to `B`" into an `A`. To call `h` you must supply a function `A -> B`, but you have no `B` and no way to make one. You have no `A` either: getting one is the whole point. Every attempt goes round in a circle.

Now read the three types as statements of logic, with `->` meaning "implies". Puzzle 1 says that if A implies B and B implies C, then A implies C. Puzzle 2 says that if A is true, then B implies A. Puzzle 3 is **Peirce's law**, $((A \to B) \to A) \to A$. Check its truth table (Ch. 29): it is true for every assignment of true and false to A and B. All three are tautologies.

So two tautologies have programs, and one does not, and this is no accident of Python: in any language whose functions always return, no program of type `((A -> B) -> A) -> A` exists for unknown `A` and `B`. This chapter explains why, by showing that "this type has a program" and "this statement has a proof" are the same thing. The missing program is a missing proof, in a logic slightly more demanding than the one you learned at school.

66.2 The big idea

Think of a type as a promise. A value of type `int` is evidence that an integer exists. A value of type `A -> B` is a machine that turns any evidence for `A` into evidence for `B`. That is exactly what a proof of "A implies B" does in everyday mathematics: *assume* A, then reason your way to B.

Push the analogy row by row and nothing breaks. Evidence for "A and B" is a **pair** `(a, b)`. Evidence for "A or B" is one of them *plus a label saying which*, a **tagged union** such as `Left(a)`. "False" has no evidence at all: it is the **empty type**, and a function out of it exists vacuously, so from False anything follows. "Not A" is a function `A -> Empty`.

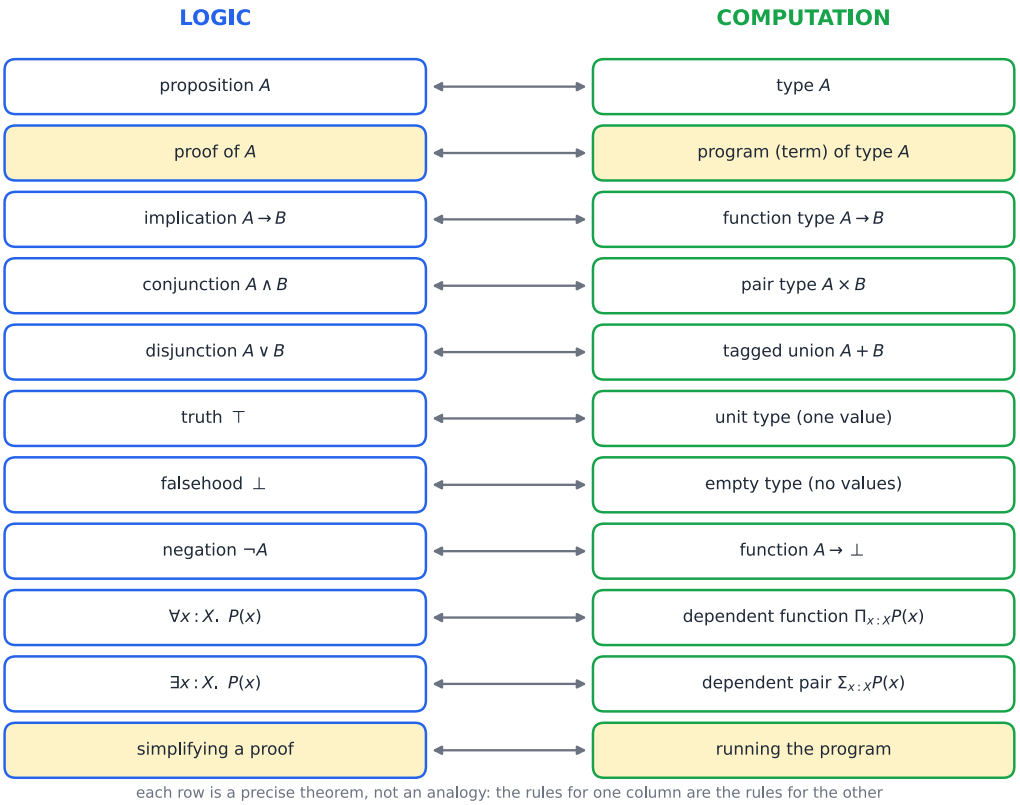


Figure 66.1 — The Curry–Howard dictionary. Propositions are types; a proof is a program of that type; and simplifying a proof is running the program. The last two rows, for quantifiers, need the dependent types of §66.3.

Haskell Curry noticed the first rows in 1934; William Howard wrote the full correspondence down in a manuscript of 1969, published in 1980.

Why does Peirce's law fail? Look at the "or" row. Evidence for $A \vee \neg A$ must say *which* side holds, and for an unknown A you cannot. Truth tables quietly assume every statement already is true or false; evidence does not. The logic you get by insisting that every proof carry its evidence is called **intuitionistic** (or constructive) logic. It is the logic hinted at in Ch. 63, where an existence proof by excluded middle failed to produce its witness. Peirce's law is a disguised form of excluded middle, so it is a classical tautology with no constructive proof.

The analogy has a second half. A proof that proves a lemma $A \to B$, proves A, then applies the lemma, contains a detour: you could have plugged the proof of A straight into the lemma's proof. In programming, that simplification is **running the code**, replacing a call by the function's body with the argument plugged in. Proof simplification and evaluation are one process.

66.3 The mathematics

The simply typed lambda calculus

We need a tiny programming language, small enough to reason about completely. Its **types** are built from atomic types $A, B, C, \ldots$ (unknowns) by

$$T ::= A \mid T_1 \to T_2 \mid T_1 \times T_2 \mid T_1 + T_2 \mid \mathbf{1} \mid \mathbf{0}.$$

Its **terms** are variables x, functions $\lambda x{:}T.\, t$, applications $t\,u$, pairs $\langle t, u\rangle$ with projections $\pi_1 t$, $\pi_2 t$, injections $\mathrm{inl}\, t$, $\mathrm{inr}\, t$ with case analysis $\mathrm{case}\, t$ of $\mathrm{inl}\, x \Rightarrow u \mid \mathrm{inr}\, y \Rightarrow v$, the unit value $\star$, and absurd t, which turns a value of the empty type into anything. We usually drop the annotation on λ.

A **context** $\Gamma = x_1{:}T_1, \ldots, x_n{:}T_n$ lists the variables in scope and their types. A **typing judgement** $\Gamma \vdash t : T$ says that in context Γ, term t has type T. Judgements are derived by rules: premises above a line, conclusion below. The rules for functions are

$$\frac{x{:}T \in \Gamma}{\Gamma \vdash x : T}\ (\mathrm{var}) \qquad \frac{\Gamma,\, x{:}A \vdash t : B}{\Gamma \vdash \lambda x.\, t : A \to B}\ (\mathrm{abs}) \qquad \frac{\Gamma \vdash t : A \to B \qquad \Gamma \vdash u : A}{\Gamma \vdash t\,u : B}\ (\mathrm{app})$$

and for pairs, sums and the empty type

$$\frac{\Gamma \vdash t : A \quad \Gamma \vdash u : B}{\Gamma \vdash \langle t, u\rangle : A \times B} \qquad \frac{\Gamma \vdash t : A \times B}{\Gamma \vdash \pi_1 t : A} \qquad \frac{\Gamma \vdash t : A}{\Gamma \vdash \mathrm{inl}\, t : A + B} \qquad \frac{\Gamma \vdash t : \mathbf{0}}{\Gamma \vdash \mathrm{absurd}\, t : C}$$

with the obvious companions for π_2, inr, $\star : \mathbf{1}$ and case (if $t : A + B$, and $u : C$ assuming $x{:}A$, and $v : C$ assuming $y{:}B$, then the case expression has type C).

Now delete every term and keep only the types. Var becomes "a hypothesis may be used". Abs becomes "if assuming A you can derive B, conclude $A \to B$ and discharge the assumption". App becomes *modus ponens*. Pairing and projection are $\wedge$-introduction and elimination, the injections are $\vee$-introduction, case is proof by cases, and absurd is *ex falso quodlibet*. These are exactly Gentzen's rules of **natural deduction** for intuitionistic logic (Ch. 65), reading $\times, +, \mathbf{1}, \mathbf{0}$ as $\wedge, \vee, \top, \bot$.

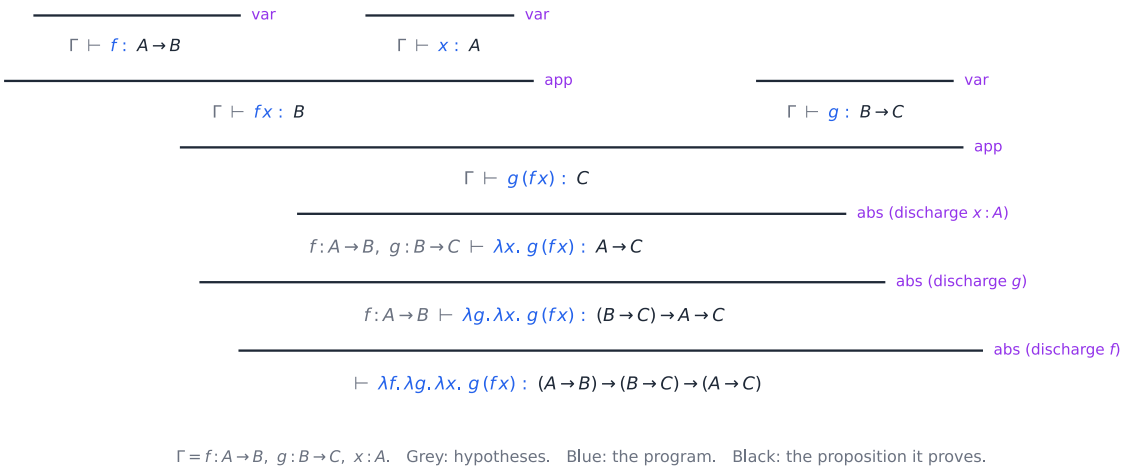


Figure 66.2 — The derivation that gives composition its type. Read upward, it is a type checker's work. Read with the blue terms erased, it is a proof in natural deduction: from $A \to B$, $B \to C$ and A, two uses of modus ponens give C, and three discharges give the theorem.

Theorem 66.1 — Curry–Howard correspondence

Let $\Gamma = x_1{:}A_1, \dots, x_n{:}A_n$. There is a term t with $\Gamma \vdash t : B$ if and only if B is provable from the hypotheses $A_1, \dots, A_n$ in intuitionistic natural deduction. Moreover, typing derivations of t and natural-deduction proofs correspond one to one.

Proof. ($\Rightarrow$) Induction on the typing derivation. Each typing rule, with terms erased, is an instance of a natural-deduction rule with the same premises and conclusion. Erasing terms from a whole derivation therefore leaves a proof.

($\Leftarrow$) Induction on the natural-deduction proof, building t as we go. A hypothesis A_i becomes the variable x_i. If the last step is $\to$-introduction from a proof of B under the extra hypothesis A, the induction hypothesis gives $\Gamma, x{:}A \vdash s : B$ for a fresh x, and abs gives $\lambda x.\, s : A \to B$. If the last step is modus ponens from proofs of $A \to B$ and A, the induction hypothesis gives terms s and u for them, and $s\,u : B$. The other connectives go the same way, one rule at a time. Each step is forced, so the two constructions are inverse to each other. □

So the hook's third puzzle is a question about provability.

Running a program simplifies a proof

Computation in this language is **β-reduction**: a function applied to an argument steps to the body with the argument substituted,

$$(\lambda x.\, t)\, u \;\longrightarrow\; t[u/x],$$

and likewise $\pi_1 \langle t, u \rangle \longrightarrow t$, and a case on $\mathrm{inl}\, u$ steps to its left branch with $x := u$. On the logic side each redex is a **detour**: an introduction rule immediately followed by the matching elimination. For this to make sense, reduction must not change what is being proved.

Theorem 66.2 — Subject reduction (types are preserved by evaluation)

If $\Gamma \vdash t : T$ and $t \longrightarrow t'$, then $\Gamma \vdash t' : T$.

Proof. The heart of it is a **substitution lemma**: if $\Gamma, x{:}A \vdash t : B$ and $\Gamma \vdash u : A$, then $\Gamma \vdash t[u/x] : B$. We prove the lemma by induction on the derivation of t, showing only the function rules.

- t is a variable. If $t = x$, then $B = A$ and $t[u/x] = u$, which has type A. If $t = y \neq x$, the substitution does nothing, and $y{:}B$ is already in Γ.
- $t = t_1\, t_2$ with $t_1 : C \to B$ and $t_2 : C$. By induction $t_1[u/x] : C \to B$ and $t_2[u/x] : C$, so app gives $(t_1 t_2)[u/x] = t_1[u/x]\, t_2[u/x] : B$.
- $t = \lambda y.\, s$ with $B = C \to D$ and $\Gamma, x{:}A, y{:}C \vdash s : D$. Rename y if needed so that it does not occur in u. Adding the unused hypothesis $y{:}C$ to $\Gamma \vdash u : A$ is harmless, so induction gives $\Gamma, y{:}C \vdash s[u/x] : D$, and abs gives $\lambda y.\, s[u/x] : C \to D$.

Now the theorem. If the step is at the top, $t = (\lambda x.\, s)\, u$ with T its type. Its derivation must end with app, whose premises are $\Gamma \vdash \lambda x.s : A \to T$ and $\Gamma \vdash u : A$. The first must come from abs, so $\Gamma, x{:}A \vdash s : T$. The lemma gives $\Gamma \vdash s[u/x] : T$. If the step happens inside a subterm, induction on the derivation applies the same argument there. Pairs and cases are similar. □

In logical terms: removing a detour leaves a proof of the same theorem. Prawitz proved this for natural deduction in 1965, and Gentzen's cut elimination of 1935 is its sequent-calculus cousin.

A term with no redex left is in **normal form**. The next fact is the deep one, and we state it without proof.

Theorem 66.3 — Strong normalization (Tait's method, 1967)

In the simply typed lambda calculus, every well-typed term reaches a normal form, whatever order the reductions are performed in. In particular, every program terminates.

The reason this needs a clever argument is that substitution can make a term *bigger*: $(\lambda x.\ f\ x\ x)\ u$ copies u twice, and the copies may contain new redexes. Untyped, the term $(\lambda x.\ x\ x)(\lambda x.\ x\ x)$ reduces to itself forever. Types forbid $x\ x$, because x would need a type A equal to $A \to B$.

? Why does every typed program halt?

Tait's proof does not count steps; it defines a set of "reducible" terms for each type and shows by induction on types that every well-typed term is reducible. For Girard's System F, where functions can take *types* as arguments, the idea still works, but the proof needs more strength than the system has: normalization of System F implies the consistency of second-order arithmetic. Why a system can never certify its own termination, or its own consistency, is Gödel's second incompleteness theorem, in Ch. 68.

Why excluded middle has no program

Normal forms have a rigid shape, and that shape is what lets us prove a negative. A normal term is either an **introduction** ($\lambda x.\ t$, a pair, $\mathrm{inl}\ t$, $\mathrm{inr}\ t$, $\star$) or **neutral**: a variable with a chain of eliminations applied to it, such as $x\ u_1\ u_2$, $\pi_1(x\ u)$ or $\mathrm{case}\ x\ \ldots$. (If an elimination were applied to an introduction, that would be a redex.) A neutral term has a free variable at its head, so a *closed* normal term, typed in the empty context, must be an introduction.

Theorem 66.4 — Excluded middle is not constructively provable

Let A be an atomic type. There is no closed term of type $A + (A \to \mathbf{0})$. Equivalently, $A \vee \neg A$ has no proof in intuitionistic logic.

Proof. Suppose $\vdash t : A + (A \to \mathbf{0})$. By Theorems 66.2 and 66.3 we may replace t by its normal form, which has the same type and is closed, hence an introduction. For a sum type the only introductions are inl and inr.

Case $t = \mathrm{inl}\ s$ with $\vdash s : A$. Then s is a closed normal term of an atomic type. No introduction rule produces an atomic type, and closed normal terms are not neutral. So no such s exists.

Case $t = \mathrm{inr}\ s$ with $\vdash s : A \to \mathbf{0}$. Then s is an introduction of a function type, $s = \lambda x.\ r$ with $x{:}A \vdash r : \mathbf{0}$ and r normal. No introduction has type $\mathbf{0}$, so r is neutral, and its head variable must be x, the only one in scope. But x has atomic type A: it cannot be applied, projected, case-analysed or passed to absurd, because each of those needs a function, pair, sum or empty type. So $r = x$, whose type is A, not $\mathbf{0}$. Contradiction. □

The proof reasons about what a program *could* look like once it has finished computing. Peirce's law fares the same, because any program for it that works at every type could be turned into a program for excluded middle (challenge problem 2 asks you to write the conversion).

This is the logic keeping a promise. A constructive proof of "A or not A" would be a program that *decides* A, and for "this Turing machine halts" no such program exists (Ch. 30).

Classical reasoning is not lost, only relocated. A program of type $\neg\neg(A \vee \neg A)$, that is $((A + (A \to \mathbf{0})) \to \mathbf{0}) \to \mathbf{0}$, does exist:

$$\lambda k.\ k\ (\mathrm{inr}\ (\lambda a.\ k\ (\mathrm{inl}\ a))).$$

Read it as a con trick. Handed a refutation k of the disjunction, it first claims "not A". If k ever calls that bluff by producing an $a{:}A$, the program rewinds and offers $\mathrm{inl}\ a$ instead. Glivenko showed in 1929 that every classical propositional tautology becomes intuitionistically provable once you put $\neg\neg$ in front of it. In programming, $(A \to R) \to R$ is the **continuation monad**: instead of returning an A, a value takes a callback that consumes one. That answers the question Ch. 62 left about monads and proofs: a monad is a *modality*, "A holds, possibly after some effect", and the effect behind classical logic is the ability to jump back in time.

Dependent types: quantifiers become functions

So far a type cannot mention a value, so we cannot say "for every n, $n + 0 = n$". **Dependent types** lift that restriction. A *type family* $P(x)$ is a type that depends on a value $x{:}X$: for example $\mathrm{Vec}(n)$, the type of lists of length exactly n, or $n + 0 = n$, the type of proofs that this particular equation holds.

- The **dependent function type** $\Pi_{x:X} P(x)$ contains functions sending each $x{:}X$ to something of type $P(x)$. It is $\forall x{:}X.\ P(x)$: a proof of a universal statement is a function from instances to proofs.
- The **dependent pair type** $\Sigma_{x:X} P(x)$ contains pairs $\langle x, p\rangle$ with $p : P(x)$. It is $\exists x{:}X.\ P(x)$: a proof of existence *is* a witness plus evidence that it works.

When P does not depend on x, Π collapses to $X \to P$ and Σ to $X \times P$, recovering the simple types. Induction becomes recursion. Here is a proof assistant (Lean 4) proving that $0 + n = n$ for every natural number, given that addition is defined by recursion on its second argument:

```
theorem zero_add' : ∀ n : Nat, 0 + n = n
  | 0     => rfl                              -- 0 + 0 reduces to 0
  | n + 1 => congrArg Nat.succ (zero_add' n)  -- use the proof for n
```

The second line is a recursive call: the proof for $n + 1$ is built from the proof for n. That this recursion is legal, and that it terminates because n shrinks, is induction on $\mathbb{N}$ (Ch. 29), seen as a program.

One consequence startles set theorists. In type theory the **axiom of choice**, in the form "if for every x there is a y with $R(x, y)$, then there is a function f with $R(x, f(x))$ for every x", is a *theorem*. The proof is one line: given $h : \Pi_x \Sigma_y R(x, y)$, take $f(x) = \pi_1(h\ x)$, and the evidence is $\pi_2(h\ x)$. The existential already contained the choice. In set theory, where existence carries no witness, choice is an independent axiom (Ch. 64, Ch. 68).

Dependent types power modern proof assistants. Coq checked the four colour theorem in 2005 and the Feit–Thompson theorem in 2012; there, "the proof is correct" means "the program type-checks".

66.4 The algorithm

Checking a proof is type checking

Because proofs are programs, a proof checker is a type checker, and a type checker for the simple language of §66.3 fits on a screen. Types are atoms (strings) or tuples `('->', A, B)`; terms are tagged tuples, with λ carrying its argument's type.

```
def typeof(t, ctx):
    tag = t[0]
    if tag == 'var':
        return ctx[t[1]]                              # rule var
    if tag == 'lam':                                  # ('lam', x, A, body): rule abs
        _, x, A, body = t
        return ('->', A, typeof(body, {**ctx, x: A}))
    if tag == 'app':                                  # rule app
        f, a = typeof(t[1], ctx), typeof(t[2], ctx)
        if f[0] != '->' or f[1] != a:
            raise TypeError(f"cannot apply {f} to {a}")
        return f[2]
    raise ValueError(f"unknown term {tag}")           # pairs, sums: challenge 3
```

Feed it composition and it returns `('->', ('->', 'A', 'B'), ('->', ('->', 'B', 'C'), ('->', 'A', 'C')))`, the theorem of Figure 66.2. Feed it the self-application `lam x:A. x x` and it raises `cannot apply A to A`: the term that looped forever in §66.3 is rejected before it runs.

Each branch is one typing rule read upward, and each subterm is visited once, so checking costs about the term size times the size of the largest type. This is the heart of the **de Bruijn criterion** that proof assistants follow: however clever the tools that *find* a proof, the result is checked by a small kernel like this one, so trust rests on a few hundred lines of code rather than on the search.

Finding a proof is finding a program

The reverse direction, **type inhabitation** ("given T, find a t with $\vdash t : T$"), is proof search. It is harder, and how much harder depends on the logic.

Problem	Logic	Status
check that t has type T	any of the systems here	decidable, roughly linear time
infer the type of an unannotated t	ML-style polymorphism (Hindley–Milner)	decidable; worst case exponential, fast in practice
decide whether T is inhabited	intuitionistic propositional logic	decidable, PSPACE-complete (Statman, 1979)
decide whether T is inhabited	first-order or dependent types	undecidable, only semi-decidable

The undecidable row is the theorem of Ch. 65 and Ch. 30 in new clothes: if we could always decide whether a proof exists, we could decide the halting problem. Proof assistants therefore make the human supply the hard steps and automate the rest. The propositional case is decidable because a goal-directed search only ever needs subformulas of the goal, and there are finitely many. The experiment below implements that search.

Failure cases: how a type system stops being a logic

Every type in the correspondence must be honest: an inhabitant must be real evidence. Three features of everyday languages break this.

- **General recursion.** `def loop(): return loop()` can be annotated with any return type. Under Curry–Howard that proves every proposition, including $\bot$. Languages used as logics (Coq, Agda, Lean) accept only recursion they can see terminating, such as structural recursion in `zero_add'`. Since termination is undecidable, such a check must reject some programs that do terminate.
- **Exceptions and null.** `raise` and `None` inhabit every type, so they too "prove" anything.
- **Types of types.** Dependent types need a type of types, so that Vec can map $\mathbb{N}$ to types. The obvious choice, a `Type` that contains every type including itself, is fatal.

> **? The type of all types**
>
> In 1972 Jean-Yves Girard showed that a type theory with `Type` : `Type` is inconsistent: there is a closed term of the empty type, a program that claims to prove False and runs forever. Every proof assistant now uses an infinite tower of universes, $\mathsf{Type}_0 : \mathsf{Type}_1 : \mathsf{Type}_2 : \cdots$, so that no universe contains itself. Why does a universe that contains itself collapse? It is the same trap that destroyed Frege's set theory in 1902, and it is sprung in Ch. 68.

66.5 The experiment

The script checks formulas of implicational logic twice, with `F` for falsehood: **classically** by truth table, and **constructively** by searching for a Python program of that type built from `lambda`, calls and `absurd`. For a goal $A \to B$ it writes a `lambda` and seeks B; for an atomic goal it tries each hypothesis ending in that atom (or in `F`). A loop check makes the search finite, and it finds a program whenever one exists.

```
import itertools

def I(*ts):                     # I(A, B, C) means A -> B -> C (arrows associate right)
    return ts[0] if len(ts) == 1 else ('->', ts[0], I(*ts[1:]))

def show(t): return t if isinstance(t, str) else '(' + show(t[1]) + ' -> ' + show(t[2]) + ')'

def atoms(t): return {t} if isinstance(t, str) else atoms(t[1]) | atoms(t[2])

def tautology(t):               # classical check: a truth table
    def ev(t, v): return v.get(t, 0) if isinstance(t, str) else (not ev(t[1], v)) or ev(t[2], v)
    names = sorted(atoms(t) - {'F'})       # F is always false
    return all(ev(t, dict(zip(names, bits))) for bits in itertools.product([0, 1], repeat=len(names)))

def prove(ctx, goal, seen=frozenset()):   # constructive check: search for a Python program
    if not isinstance(goal, str):                       # goal A -> B: write a lambda
        v = 'x%d' % len(ctx)
        body = prove(ctx + [(v, goal[1])], goal[2], seen)
        return None if body is None else '(lambda %s: %s)' % (v, body)
    key = (frozenset(t for _, t in ctx), goal)
    if key in seen: return None                         # loop check keeps search finite
    for v, t in ctx:                                    # use a hypothesis A1 -> ... -> An -> goal
        args = []
        while not isinstance(t, str): args.append(t[1]); t = t[2]
        if t not in (goal, 'F'): continue               # ...or one ending in F (ex falso)
        subs = [prove(ctx, a, seen | {key}) for a in args]
        if all(s is not None for s in subs):
            r = v + ''.join('(%s)' % s for s in subs)
            return r if t == goal else 'absurd(%s)' % r
    return None

# KNOBS: add your own formulas. 'F' plays the role of falsehood.
tests = {'compose': I(I('A', 'B'), I('B', 'C'), 'A', 'C'),
         'K': I('A', 'B', 'A'),
         'Peirce': I(I(I('A', 'B'), 'A'), 'A'),
         'not-not-intro': I('A', I('A', 'F'), 'F'),
         'not-not-elim': I(I(I('A', 'F'), 'F'), 'A'),
         'not-not-Peirce': I(I(I(I(I('A', 'B'), 'A'), 'A'), 'F'), 'F')}
for name, t in tests.items():
    print(f"{name:14s} {show(t):34s} tautology={tautology(t)!s:5s} program={prove([], t)}")
```

All six are tautologies. Composition, `K` and $A \to \neg\neg A$ get programs; for composition the machine finds `lambda x0: lambda x1: lambda x2: x1(x0(x2))`, the hook's answer. Peirce's law and $\neg\neg A \to A$ get `None`. But put $\neg\neg$ in front of Peirce's law and a program appears, a little time machine of nested callbacks, as Glivenko promised.

Things to change:

1. Add the formula $(A \to B \to C) \to (B \to A \to C)$ (swap the arguments) and predict the program before running.
2. Add **contraposition** $(A \to B) \to (\neg B \to \neg A)$ and its converse $(\neg B \to \neg A) \to (A \to B)$. One of them has a program. Which, and why does the other need classical logic?
3. Delete the `'F'` in `if t not in (goal, 'F')`. You now have **minimal logic**, without *ex falso*. Watch what happens to the double-negated Peirce law.

66.6 The challenge

No answers are given.

1. **Intuition.** The empty type $\mathbf{0}$ has no values, so a function $\mathbf{0} \to C$ "exists" for every C. Explain in plain words why this matches *ex falso quodlibet*. Then explain why a language that lets you write `def f() -> Empty: return f()` cannot be used as a logic, and what that has to do with Theorem 66.3.
2. **Derivation.** Write closed terms for currying, $(A \times B \to C) \to (A \to B \to C)$ and its converse, and for $\neg(A \vee B) \to (\neg A \times \neg B)$. Then show that Peirce's law, assumed for *every* pair of types, yields excluded middle: write a term of type $\neg(A \vee \neg A) \to (A \vee \neg A)$, and use Peirce's law with A replaced by $A \vee \neg A$ and B by $\mathbf{0}$. Why does this, together with Theorem 66.4, show that no polymorphic program for Peirce's law exists?
3. **Implementation.** Extend `typeof` with sums (`inl`, `inr`, `case`) and `absurd`, then extend `prove` from the experiment to conjunction and disjunction. Confirm that $\neg\neg(A \vee \neg A)$ is inhabited and $A \vee \neg A$ is not. Then write a β-reducer and test Theorem 66.2 by re-checking the type after every step.
4. **Open investigation.** The type $(A \to A) \to A \to A$ has infinitely many normal inhabitants: $\lambda f.\lambda x.\, x$, $\lambda f.\lambda x.\, f\, x$, $\lambda f.\lambda x.\, f\,(f\, x)$, and so on, the **Church numerals**. Others have one, finitely many, or none. Write a program that counts normal inhabitants up to a given size, and explore. Can you predict finite versus infinite from the shape of the type alone?

66.7 Loose threads

The hook's third puzzle has no program because Peirce's law has no constructive proof, and Theorem 66.4 shows why from the shape of normal forms: a closed program must commit to an answer, and for an unknown proposition it has nothing to commit to. Proofs are programs, propositions are types, checking is type checking, and dependent types stretch the dictionary to quantifiers.

What we did not tell you:

- **Why typed programs terminate.** Theorem 66.3 was stated, not proved, and its stronger forms cannot be proved inside the systems they describe (Ch. 68).
- **Why universes must be stacked.** Girard's paradox descends from Russell's, which opens Ch. 68.

- **How to use this on real code.** Most software is not written in Lean. How do engineers prove a C loop correct, or a protocol free of deadlock? That is Ch. 67.

? When are two types the same?

A proof of $x = y$ is itself a value, an element of an *identity type*. Can there be two *different* proofs that $x = y$? Around 2006–2009 Vladimir Voevodsky and others saw that identity types behave like paths in a space. His **univalence axiom** says that equivalent types are equal, turning "isomorphic structures are interchangeable" into a rule of logic. Cubical type theory (from 2015) gives univalence a computational meaning, but whether homotopy type theory can serve as a practical foundation for all of mathematics is an open research question.

Further reading

- M. H. Sørensen and P. Urzyczyn, *Lectures on the Curry–Howard Isomorphism*, Elsevier, 2006. The full story, including normalization proofs and the complexity of inhabitation.
- B. C. Pierce, *Types and Programming Languages*, MIT Press, 2002. The standard programmer's introduction to type systems, with implementations.
- P. Wadler, "Propositions as Types", *Communications of the ACM* 58(12), 2015. A short, witty history of the correspondence.
- The Univalent Foundations Program, *Homotopy Type Theory: Univalent Foundations of Mathematics*, Institute for Advanced Study, 2013. Freely available; dependent types taken to their limit.

Chapter 67 — Formal Verification: Proving Programs Correct

> **Level 2 · Track F — Logic, Proofs, and the Foundations of Truth Builds on:** Ch. 24, Ch. 29, Ch. 30, Ch. 65, Ch. 66 · **Leads to:** Ch. 68 · **Related:** Ch. 27, Ch. 34, Ch. 77 **Reading time:** ~25 min

67.1 The hook

In October 1994 a mathematics professor, Thomas Nicely, noticed that his new Pentium computer was getting some divisions wrong. The cleanest example became famous:

$$\frac{4195835}{3145727} = 1.333820449\ldots \quad \text{but the chip said} \quad 1.333739068\ldots$$

The error was in the fourth significant digit. The cause was five missing entries in a lookup table of 1,066 used by the division circuit. Intel had tested the chip intensively, but by its own estimate a random division hit the bad entries about once in nine billion tries. The replacement programme cost the company 475 million US dollars.

Count what exhaustive testing would have needed. A double-precision divider takes two 64-bit inputs, so there are $2^{128} \approx 3.4 \times 10^{38}$ input pairs. At a billion tests a second, checking them all would take about 10^{22} years, a trillion times the age of the universe. Testing samples a vanishing fraction of the input space. It can find bugs, but it cannot show their absence.

Two years later, on 4 June 1996, the first Ariane 5 rocket veered off course about 37 seconds after lift-off and destroyed itself. A 64-bit floating-point velocity had been converted into a 16-bit integer. On the slower Ariane 4, where the code came from, the value always fitted. On Ariane 5 it did not, and the overflow shut down both inertial reference computers, which ran identical software.

Both failures were in components that had been tested and had worked. The question this chapter asks is the one an engineer asks after a disaster: **can we prove, once and for all inputs, that a program or a circuit does what it should?** After 1994 Intel began proving its floating-point units correct with theorem provers. Today Amazon checks its distributed protocols with model checkers, an operating-system kernel and a C compiler have been proved correct line by line, and avionics software is analysed for overflow before it flies. This chapter shows how.

67.2 The big idea

A proof of correctness needs two things that testing does not: a **specification** written as a logical formula, and a way of reasoning about *every* execution at once.

There are three main ways to get the second.

1. **Deductive verification.** Attach logical assertions to points in the program, and prove that each statement carries a true assertion before it to a true assertion after it. Loops are handled by an **invariant**, exactly as in Ch. 29. The proof obligations are ordinary formulas of arithmetic that a solver or a human can discharge. This is **Hoare logic**.
2. **Model checking.** If the system has finitely many states, as a circuit or a protocol does, just explore all of them. A graph search visits every reachable state and either certifies that no bad state is reachable or returns the exact sequence of steps that reaches one.
3. **Interactive theorem proving.** Write the program, its specification and its proof in a proof assistant whose kernel checks every step, using the proofs-as-programs machinery of Ch. 66.

An analogy: testing is spot-checking a bridge by driving trucks over it. Deductive verification is the structural calculation that shows every beam carries its load. Model checking is a stress simulation of every possible load pattern, feasible only because the bridge has finitely many joints. Theorem proving is having the calculation audited by an inspector who accepts nothing on trust.

All three run into the same wall. By Rice's theorem (Ch. 30), no algorithm decides whether an arbitrary program meets an arbitrary specification. Each method gives something up: deductive methods need a human or a heuristic to supply invariants, model checking needs a finite state space, and theorem proving needs human effort.

67.3 The mathematics

Hoare triples

Fix a small imperative language: assignments `x := e`, sequencing `S1; S2`, conditionals `if b then S1 else S2` and loops `while b do S`. A **state** σ gives every variable an integer value. Assertions are formulas of predicate logic about the variables (Ch. 29).

> **Hoare triple**
>
> The triple $\{P\}\ S\ \{Q\}$ holds (in the sense of **partial correctness**) if, whenever S starts in a state satisfying the **precondition** P and terminates, the final state satisfies the **postcondition** Q. It holds in the sense of **total correctness** if, in addition, S always terminates from such states.

Partial correctness says nothing about termination. $\{\top\}$ `while true do skip` $\{\bot\}$ is a valid partial-correctness triple, because the loop never finishes and so never delivers a bad state. Termination is proved separately with a variant, as in Ch. 29.

In 1969 Tony Hoare gave rules that derive valid triples, one per construct:

$$\{Q[e/x]\}\ x := e\ \{Q\}\ \text{(assign)} \qquad \frac{\{P\}\ S_1\ \{R\} \qquad \{R\}\ S_2\ \{Q\}}{\{P\}\ S_1; S_2\ \{Q\}}\ \text{(seq)}$$

$$\frac{\{P \wedge b\}\, S_1\, \{Q\} \qquad \{P \wedge \neg b\}\, S_2\, \{Q\}}{\{P\}\ \texttt{if}\ b\ \texttt{then}\ S_1\ \texttt{else}\ S_2\ \{Q\}}\ (\text{if}) \qquad \frac{\{I \wedge b\}\, S\, \{I\}}{\{I\}\ \texttt{while}\ b\ \texttt{do}\ S\ \{I \wedge \neg b\}}\ (\text{while})$$

$$\frac{P \Rightarrow P' \qquad \{P'\}\, S\, \{Q'\} \qquad Q' \Rightarrow Q}{\{P\}\ S\ \{Q\}}\ (\text{consequence})$$

Here $Q[e/x]$ means Q with every free x replaced by the expression e. The while rule is the loop-invariant principle in compact form: if one pass of the body preserves I, then I survives any number of passes, and on exit the guard is false. The consequence rule is the only place where ordinary mathematics enters: it lets us strengthen a precondition or weaken a postcondition by proving an implication.

The assignment rule runs backwards

The assignment rule looks back to front. To know what must be true *before* `x := x + 1` so that $x > 5$ holds *after*, substitute: $(x > 5)[x + 1/x]$ is $x + 1 > 5$, that is $x > 4$. Reasoning flows from the goal backwards, and the next theorem says this backward step loses nothing.

Weakest precondition

$\mathrm{wp}(S, Q)$ is the assertion satisfied by exactly those states from which S terminates in a state satisfying Q. It is the weakest P for which $\{P\}\ S\ \{Q\}$ holds totally: any valid precondition implies it.

Theorem 67.1 — Assignment is exact

For every assertion Q, $\mathrm{wp}(x := e,\ Q) = Q[e/x]$. In particular the assignment rule is sound, and no weaker precondition would do.

Proof. Write $\sigma[x \mapsto v]$ for the state σ with x changed to v, and $\sigma(e)$ for the value of e in σ. Running $x := e$ from σ always terminates, in $\sigma[x \mapsto \sigma(e)]$. So σ is in $\mathrm{wp}(x := e, Q)$ exactly when $\sigma[x \mapsto \sigma(e)] \models Q$.

It remains to show the **substitution lemma**: $\sigma \models Q[e/x]$ if and only if $\sigma[x \mapsto \sigma(e)] \models Q$. Argue by induction on the structure of Q. For a term t, evaluating $t[e/x]$ in σ gives the same value as evaluating t in $\sigma[x \mapsto \sigma(e)]$: every occurrence of x is replaced by e, whose value in σ is precisely what x holds in the updated state, and the other variables are unchanged. So atomic formulas such as $t_1 < t_2$ agree. The connectives $\neg, \wedge, \vee, \Rightarrow$ preserve agreement directly. For a quantifier $\forall y.\ R$, rename y so that it is neither x nor a variable of e; then substitution passes inside, and the induction hypothesis applies for each value of y. $\square$

Sequencing composes: $\mathrm{wp}(S_1; S_2, Q) = \mathrm{wp}(S_1, \mathrm{wp}(S_2, Q))$, and for conditionals $\mathrm{wp} = (b \Rightarrow \mathrm{wp}(S_1, Q)) \wedge (\neg b \Rightarrow \mathrm{wp}(S_2, Q))$. Straight-line code can therefore be verified mechanically, by computing the weakest precondition and checking one implication. Loops are where this stops. The weakest precondition of a loop is a limit of infinitely many unrollings, and in general no algorithm can write it down. Instead, someone supplies an invariant, and the loop becomes three implications.

A proof of division

Hoare's 1969 paper verified a program that divides x by y using only subtraction:

```
q := 0; r := x;
while r >= y do
    r := r - y; q := q + 1
```

Theorem 67.2 — Division by repeated subtraction is correct

$\{x \geq 0 \wedge y > 0\}$ the program above $\{x = q\,y + r \wedge 0 \leq r < y\}$ holds with total correctness.

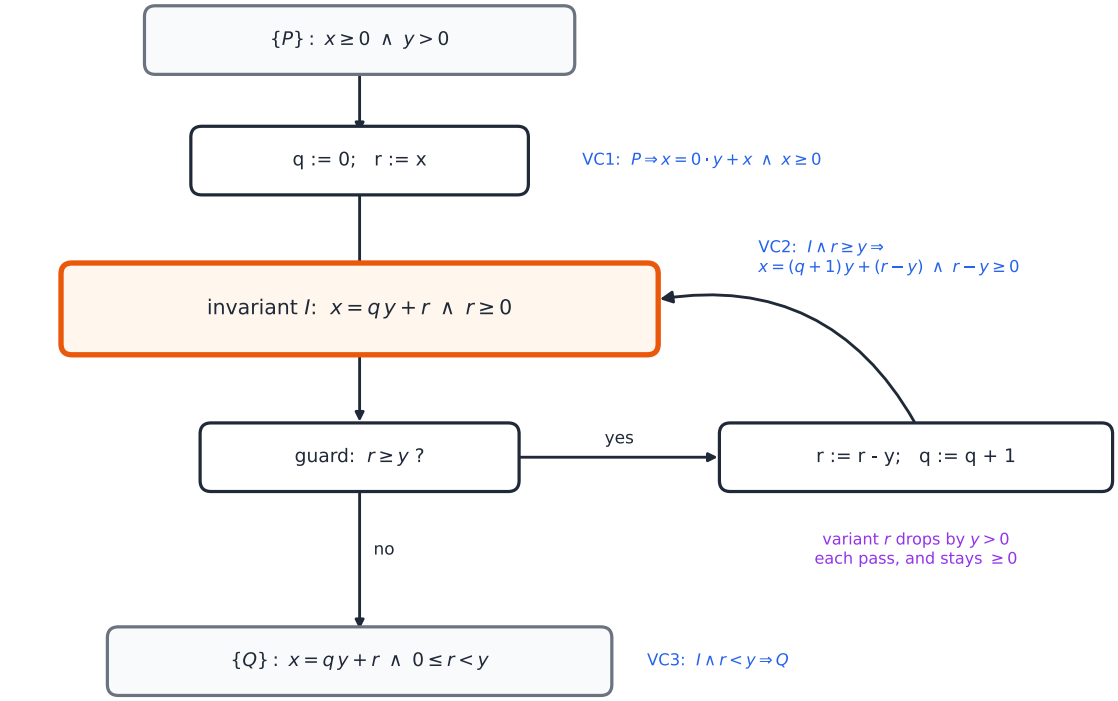
Proof. Take the invariant $I:\ x = q\,y + r \wedge r \geq 0$. Three verification conditions remain, each an implication about integers (Figure 67.1).

VC1, initialisation. By the assignment rule twice, the precondition of `q := 0; r := x` for I is $I[x/r][0/q]$, which is $x = 0 \cdot y + x \wedge x \geq 0$. This follows from $x \geq 0$.

VC2, maintenance. The precondition of the body for I is $I[q + 1/q][r - y/r]$, that is $x = (q + 1)\,y + (r - y) \wedge r - y \geq 0$. Assume I and the guard $r \geq y$. Then $(q + 1)y + (r - y) = qy + r = x$, and $r - y \geq 0$ is the guard. So $\{I \wedge r \geq y\}$ body $\{I\}$, and the while rule gives $\{I\}$ loop $\{I \wedge r < y\}$.

VC3, exit. $I \wedge r < y$ is $x = qy + r \wedge r \geq 0 \wedge r < y$, which is the postcondition word for word.

Termination. Use the variant r. While the guard holds, $r \geq y > 0$, and each pass replaces r by $r - y < r$ with $r - y \geq 0$. A non-negative integer cannot decrease forever (the argument of Ch. 29), so the loop stops. The consequence and sequencing rules stitch the pieces together. $\square$



Three implications of ordinary arithmetic replace infinitely many test runs.

Figure 67.1 — Hoare logic on the division loop. Every path through the program begins and ends at an assertion; the invariant (orange) cuts the loop open so that each path is finite. The three blue verification conditions are all that must be proved. Note how $y > 0$ is used: without it the variant would not decrease.

Notice what has happened. A claim about infinitely many runs, one for each (x, y), has become three facts of high-school algebra. A program called an **SMT solver** (satisfiability modulo theories) can check such facts automatically: it tries to find values making the negation true, and when the search space is exhausted, the implication is proved.

? Is Hoare logic complete?

Every triple derived by Hoare's rules is valid. Is every *valid* triple derivable? Stephen Cook proved in 1978 that it is, *relative* to an oracle for the consequence rule, one that can tell which implications of arithmetic are true. But no proof system can play that oracle. Why there are true statements of integer arithmetic that no consistent, mechanically checkable set of axioms proves is Gödel's first incompleteness theorem, in Ch. 68.

Model checking: search instead of proof

For a system with finitely many states, we can skip invariants and look everywhere.

Transition system and safety

A **transition system** is a finite set of states S, a set of initial states $S_0 \subseteq S$, and a transition relation $\to \subseteq S \times S$. A state is **reachable** if some path $s_0 \to s_1 \to \cdots \to s_k$ leads to it from S_0. A **safety property** names a set of bad states B and asserts that no reachable state is in B. In temporal-logic notation it is written $\mathbf{G}\,\neg \mathit{bad}$, "globally, never bad".

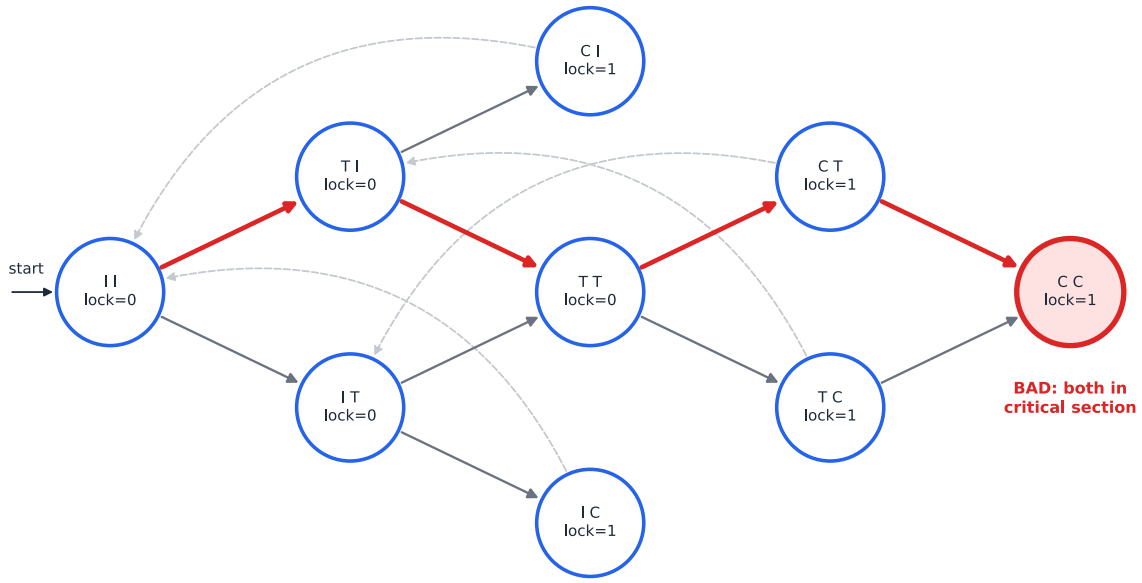
Safety says nothing bad ever happens. Its partner, **liveness**, says something good eventually happens, written $\mathbf{F}\,\mathit{good}$: every request is eventually served, or the system never deadlocks forever. Temporal logics such as LTL and CTL combine $\mathbf{G}$, $\mathbf{F}$ and "until" into a specification language, and Clarke, Emerson and Sifakis shared the 2007 Turing Award for the algorithms that check them.

Theorem 67.3 — Safety checking is a graph search

For a finite transition system, whether some bad state is reachable can be decided in time $O(|S| + |{\to}|)$. If one is reachable, breadth-first search returns a shortest path to it.

Proof. Breadth-first search from S_0 (Ch. 27) visits every reachable state exactly once and examines every transition out of a visited state once, which gives the time bound. It visits states in order of their distance from S_0: the queue always holds states of distance d followed by states of distance $d+1$, by induction on the number of steps. So the first bad state dequeued is a closest one, and following the recorded parent pointers back to S_0 gives a shortest path. If the search ends without dequeuing a bad state, every reachable state has been visited and none is bad. □

The shortest path is the most useful output of all: a **counterexample trace**, a step-by-step replay of the bug. For concurrent programs, which fail only under rare interleavings, it is the difference between a heisenbug and a fix. Figure 67.2 shows a lock that two processes implement by first *testing* a shared flag and then *setting* it. The search finds the four-step trace in which both test before either sets.



I = idle, T = has seen lock free, C = critical. Red: the shortest counterexample found by breadth-first search. Dashed: release steps.

Figure 67.2 — Model checking a test-then-set lock. Each node is a global state: the program counters of both processes and the shared lock bit. Breadth-first search reaches the bad state, both processes in the critical section, in four steps. Four further states beyond the bad one are not drawn.

67.4 The algorithm

Generating verification conditions

Deductive verifiers such as Dafny, Why3 and Frama-C turn an annotated program into formulas by running the weakest-precondition rules backwards, then hand the formulas to an SMT solver. The core fits in a few lines of pseudocode:

```
vcgen(S, Q):                    # returns (precondition, side conditions)
  case S of
    x := e          : return (Q[e/x], [])
    S1; S2          : (P2, C2) = vcgen(S2, Q); (P1, C1) = vcgen(S1, P2)
                      return (P1, C1 + C2)
    if b S1 else S2: (P1, C1) = vcgen(S1, Q); (P2, C2) = vcgen(S2, Q)
                      return ((b => P1) and (not b => P2), C1 + C2)
    while b inv I S: (PB, CB) = vcgen(S, I)
                      return (I, CB + [I and b => PB, I and not b => Q])

verify(Pre, S, Post):
  (P, C) = vcgen(S, Post)
  for each formula F in [Pre => P] + C:  ask the solver whether F is valid
```

Run on the division program, it produces exactly VC1 to VC3 of Theorem 67.2. The cost is linear in the program for straight-line code with care, but naive substitution through a chain of k conditionals can duplicate Q at each branch, giving formulas of size 2^k. Practical tools introduce fresh names for intermediate values to keep the formulas small. The solver's job, deciding whether a formula of arithmetic is valid, is decidable for linear integer arithmetic and undecidable once multiplication of variables is allowed, so the solver may answer "valid", "invalid, here is a counterexample", or "unknown".

Where the invariants come from

The algorithm takes the invariant `I` as input. Ch. 29 asked whether a program could find invariants on its own, and Ch. 30 proved that no program can always succeed. Real tools succeed often anyway, in three ways.

- **The programmer writes them.** Languages like Dafny make `invariant` a keyword, and the verifier checks each one as it is typed. It feels like a type checker that understands arithmetic.
- **Abstract interpretation.** Run the program on *sets* of values instead of values, for example intervals $[a, b]$ for each variable. A loop is iterated until the intervals stop growing, with a "widening" step that jumps to infinity if they would grow forever. The result is an invariant guaranteed to hold, though it may be too weak to prove what you want. The analyser Astrée used this to prove the absence of run-time errors, including overflows of the Ariane kind, in the flight-control software of Airbus aircraft.
- **Learning from failure.** Guess a weak invariant, let the solver produce a counterexample, and refine the guess to exclude it. Techniques called CEGAR and IC3 automate this loop, and they power the hardware model checkers used at chip companies today.

Explicit-state model checking and the state explosion

Theorem 67.3 is the whole algorithm for safety: breadth-first search with a visited set and parent pointers, linear in the size of the state graph. The trouble is that size. A system of n processes, each with k local states, can have up to k^n global states before counting shared variables. Ten processes with ten states each give 10^{10}, and one 32-bit variable multiplies that by four billion. This **state explosion** is the central enemy of model checking, and three ideas fight it.

- **Symbolic model checking** represents sets of states as compact Boolean formulas (binary decision diagrams) and computes the reachable set by manipulating formulas, never listing states. In 1992 this handled systems with more than 10^{20} states.
- **Bounded model checking** asks a SAT solver whether a bad state is reachable within k steps, by unrolling the transition relation k times into one big formula.
- **Partial-order reduction** notices that independent steps of different processes commute, so only one ordering of them needs exploring.

> **? Why does SAT solving work at all?**
>
> Bounded model checking turns verification into Boolean satisfiability, which is NP-complete (Ch. 30). In theory that should be hopeless. In practice, modern solvers routinely settle industrial formulas with millions of variables, and the whole hardware-verification industry rests on this. Why real-world formulas are so much easier than the worst case is not understood. Explaining the success of conflict-driven clause learning on structured instances is an open problem.

Failure cases

A verified system can still fail, and the ways it fails are instructive.

- **The specification is wrong.** A proof shows that the code meets the specification, not that the specification says what you meant. Ariane 5's code met the Ariane 4 assumptions perfectly.
- **The model is not the machine.** Theorem 67.2 used exact integers. On a machine, `q + 1` can overflow. The binary search bug of Ch. 29 and floating-point rounding (Ch. 24) are the same mistake.
- **The trusted base is buggy.** The compiler, the operating system and the hardware are assumed correct. That is why verifying them matters. The seL4 microkernel (2009) is about 8,700 lines of C with roughly 200,000 lines of machine-checked proof. The CompCert C compiler is proved in Coq to preserve the meaning of programs; when researchers bombarded compilers with random programs in 2011, they found hundreds of bugs in GCC and LLVM and none in CompCert's verified core.

67.5 The experiment

The script below is a complete explicit-state model checker in under 40 lines. It explores every interleaving of two processes and either reports how many states it explored and that the property holds (`None`), or prints the shortest trace to a bad state. Two protocols are included: the naive test-then-set lock of Figure 67.2 and **Peterson's algorithm** (1981), which achieves mutual exclusion with two flags and a `turn` variable.

```
from collections import deque

# A state is (pc0, pc1, shared...). Each protocol says what process i may do next.
def naive(s, i):                      # shared: lock. pc: 0 idle, 1 saw lock free, 2 critical
    pc, lock = list(s[:2]), s[2]
    if pc[i] == 0 and lock == 0: pc[i] = 1                # test the lock...
    elif pc[i] == 1: pc[i], lock = 2, 1                   # ...then set it (not atomic!)
    elif pc[i] == 2: pc[i], lock = 0, 0                   # leave, release
    else: return None                                     # blocked
    return (*pc, lock)

def peterson(s, i):                   # shared: flag0, flag1, turn. pc 3 = critical
    pc, flag, turn, j = list(s[:2]), list(s[2:4]), s[4], 1 - i
    if pc[i] == 0: flag[i], pc[i] = 1, 1                  # I want to enter
    elif pc[i] == 1: turn, pc[i] = j, 2                   # but you go first
    elif pc[i] == 2:
        if flag[j] == 0 or turn == i: pc[i] = 3           # enter if you don't want to, or it's my turn
        else: return None                                 # busy-wait
    elif pc[i] == 3: flag[i], pc[i] = 0, 0                # leave
    return (*pc, *flag, turn)

def check(step, start, bad):          # breadth-first search over every interleaving
    parent, queue = {start: None}, deque([start])
    while queue:
        s = queue.popleft()
        if bad(s):                                        # rebuild the shortest path to s
            trace = []
            while s is not None: trace.append(s); s = parent[s]
            return len(parent), trace[::-1]
        for i in (0, 1):
            t = step(s, i)
            if t is not None and t not in parent:
                parent[t] = s; queue.append(t)
    return len(parent), None

# KNOBS: switch protocol; try changing turn = j to turn = i in peterson.
print(check(naive, (0, 0, 0), lambda s: s[0] == 2 and s[1] == 2))
print(check(peterson, (0, 0, 0, 0, 0), lambda s: s[0] == 3 and s[1] == 3))
```

The naive lock fails with the trace `(0,0,0) → (1,0,0) → (1,1,0) → (2,1,1) → (2,2,1)`: process 0 sees the lock free, process 1 sees it free, both enter. Peterson's algorithm explores all 20 of its reachable states and reports `None`: mutual exclusion is *proved* for every interleaving, not tested on some.

Things to change:

1. In `peterson`, replace `turn, pc[i] = j, 2` by `turn, pc[i] = i, 2`, so that each process politely insists on its *own* turn. The checker finds a six-step counterexample among 26 states. Read the trace and explain it in words.

2. Swap the order of the first two steps of Peterson's algorithm (set `turn` before raising the flag). Is it still safe?
3. Add a check for **deadlock**: a reachable state in which neither process can move. Does either protocol have one?

67.6 The challenge

No answers are given.

1. **Intuition.** A bug strikes one random division in nine billion. A test campaign runs a billion random divisions. Estimate the chance that it finds the bug, and then explain why the triple $\{\top\}$ `while true do skip` $\{\bot\}$ is valid. Which of these two facts is more worrying for an engineer, and why?
2. **Derivation.** Prove, with an invariant and a variant, that this program computes a^n for integers a and $n \geq 0$: `p := 1; b := a; k := n; while k > 0 do (if k odd then p := p*b); b := b*b; k := k div 2`. Then compute wp of `x := x + y; y := x - y; x := x - y` for the postcondition $x = Y \wedge y = X$, and say what the result proves.
3. **Implementation.** Generalise `check` to n processes and implement the **filter lock**, Peterson's algorithm for n processes. Measure the number of reachable states for $n = 2, 3, 4$ and fit the growth. How large an n can your laptop verify?
4. **Open investigation.** Run the division program on many random inputs and record (x, y, q, r) at the loop head. From these samples, guess candidate invariants of the form $c_1 x + c_2 q\, y + c_3 r = 0$ or simple inequalities, keeping those that no sample violates (the idea behind the tool Daikon). How often is a guessed invariant actually *inductive*, preserved by one pass of the body? Try other small loops. Is there a reliable way to tell, from samples alone, which guesses will survive a proof?

67.7 Loose threads

The hook asked whether we can prove a program correct for every input. For a large and growing class of programs, yes. Hoare logic turns a loop into three implications once someone supplies the invariant, the weakest-precondition calculus makes the rest mechanical, and model checking settles finite systems outright, with a counterexample when it fails. Proof assistants close the remaining gaps by hand, and kernels, compilers and flight software have been verified that way. What no method does is decide correctness for *all* programs: Rice's theorem forbids it, and every tool is a negotiated settlement with that fact.

What we did not tell you:

- **Where completeness breaks.** The first mystery said Hoare logic is only as complete as arithmetic. Why arithmetic itself cannot be completely axiomatised is Ch. 68.
- **Why the hard problem is easy.** SAT-based verification works far better than complexity theory predicts, an open problem.
- **What happens when the system flips coins.** Every transition in this chapter was certain.

> **? Verifying a coin-flipping protocol**
>
> Ethernet's backoff, randomised consensus protocols and quantum key distribution all make random choices. For them, "a bad state is never reached" is the wrong question; the right one is "with what probability is a bad state reached?" Probabilistic model checkers such as PRISM answer it, and the answer comes from turning the state graph into a Markov chain and solving a linear system for that probability. Where the linear system comes from, and why it has a unique solution, is the first-step analysis of Ch. 34.

Verification also has an adversarial cousin. A cryptographic protocol must be correct not against random inputs but against an attacker who chooses the worst ones, and proving such protocols secure is the work of Ch. 77.

Further reading

- C. A. R. Hoare, "An Axiomatic Basis for Computer Programming", *Communications of the ACM* 12(10), 1969. Six pages that started the field, including the division example.
- C. Baier and J.-P. Katoen, *Principles of Model Checking*, MIT Press, 2008. The standard text on transition systems, temporal logic and state-space methods.
- A. R. Bradley and Z. Manna, *The Calculus of Computation*, Springer, 2007. Decision procedures, SMT and verification conditions, at an introductory level.
- B. C. Pierce et al., *Software Foundations*, online textbook series. Hoare logic and program verification done inside the Coq proof assistant.

Chapter 68 — Paradoxes, Independence, and Mathematical Limits

> **Level 2 · Track F — Logic, Proofs, and the Foundations of Truth Builds on:** Ch. 30, Ch. 64, Ch. 65, Ch. 66 · **Leads to:** Ch. 98, Ch. 62 · **Related:** Ch. 29, Ch. 63, Ch. 67 **Reading time:** ~25 min

68.1 The hook

On 16 June 1902 a young Bertrand Russell wrote a short, polite letter to the logician Gottlob Frege. Frege had spent two decades building arithmetic on pure logic, and the second volume of his masterpiece was at the printer. Russell's letter pointed out, in a few lines, that Frege's system could prove a contradiction. Frege added an appendix to the volume, admitting that the foundation of his building had given way.

The contradiction concerned a single set: the set of all sets that are not members of themselves. Ask whether it is a member of itself, and either answer forces the other.

Thirty years later Kurt Gödel found that the trouble was not confined to careless foundations. Here is a game that shows how strange things got. Take a number, say 4, and write it in "hereditary base 2", where even the exponents are written in base 2:

$$4 = 2^2.$$

Now replace every 2 by a 3, and subtract 1: $3^3 - 1 = 26$. Write 26 in hereditary base 3, $26 = 2 \cdot 3^2 + 2 \cdot 3 + 2$, replace every 3 by a 4, and subtract 1: $2 \cdot 4^2 + 2 \cdot 4 + 2 - 1 = 41$. Keep going, raising the base by one each time:

$$4,\ 26,\ 41,\ 60,\ 83,\ 109,\ 139,\ 173,\ 211,\ 253,\ \ldots$$

The numbers climb. After 2,000 steps the sequence stands at about four million, and it is still climbing. Yet in 1944 Reuben Goodstein proved that **every such sequence, from every starting number, eventually reaches 0**. The one starting at 4 gets there after about $3 \cdot 2^{402653211}$ steps, a number with some 121 million digits.

Here is the twist. In 1982 Laurence Kirby and Jeff Paris proved that Goodstein's theorem, a statement purely about whole numbers, additions and powers, **cannot be proved** from the standard axioms of arithmetic. It is true, and the axioms that every number theorist uses are powerless to show it.

How can a statement about whole numbers be true but unprovable? Which "true" do we mean, if not "provable"? And if the standard axioms miss Goodstein's theorem, what else do they miss? This chapter is about those questions, and about why the answers all go back to Russell's letter.

68.2 The big idea

Every paradox in this chapter has the same skeleton: **an object that talks about itself, with a negation.** "This sentence is false." "The set of all sets not containing themselves." "The program that does the opposite of whatever the halting decider predicts it will do" (Ch. 30). Cantor's proof that the reals are uncountable (Ch. 64) is the same move. List the objects, look down the diagonal, and flip it.

In ordinary language, self-reference produces nonsense. In a precise system it produces a theorem: some object you thought existed cannot exist. Russell's set cannot exist. The halting decider cannot exist. And, in Gödel's version, **a formal system that is consistent, mechanically checkable and strong enough for arithmetic cannot prove every true arithmetic statement.**

Picture a formal system as a machine that prints theorems: it starts from axioms and applies inference rules, forever, printing everything it can derive. We want the machine to be

1. **consistent**: it never prints both a sentence and its negation;
2. **complete**: for every sentence, it eventually prints either the sentence or its negation;
3. **effective**: there is an algorithm to check whether a string is an axiom, so that the machine really is a machine.

For simple subjects all three are possible. For arithmetic of addition alone, Presburger found such a system in 1929. Gödel showed that once multiplication is in, you can have at most two. Since nobody wants an inconsistent or uncheckable system, completeness goes.

A sentence that the system can neither prove nor refute is **independent** of it. Independence is not a vague "too hard". It is a theorem about the system, as definite as any other.

68.3 The mathematics

Russell's paradox and the end of naive sets

Frege's system, like the intuitive set theory most of us carry around, included the principle of **unrestricted comprehension**: for any property $\varphi(x)$, there is a set $\{x : \varphi(x)\}$ of all things with that property.

> **Theorem 68.1 — Russell's paradox**
>
> (a) Unrestricted comprehension is inconsistent. (b) In Zermelo–Fraenkel set theory, for every set A the set $R_A = \{x \in A : x \notin x\}$ is not a member of A. Hence there is no set of all sets.

Proof. (a) Apply comprehension to $\varphi(x) = x \notin x$, giving a set R with $x \in R \iff x \notin x$ for every x. Take $x = R$: then $R \in R \iff R \notin R$. A statement equivalent to its own negation yields a contradiction whichever truth value it has.

(b) Zermelo's **separation** axiom allows only $\{x \in A : \varphi(x)\}$, carving a subset out of a set A that already exists. So R_A exists, and $x \in R_A \iff (x \in A \wedge x \notin x)$. Suppose $R_A \in A$. Then for $x = R_A$ we get $R_A \in R_A \iff R_A \notin R_A$, the same contradiction. So $R_A \notin A$. If there were a set V of all sets, R_V would be a set not in V, which is absurd. □

Part (b) is the repair that became modern set theory (Ch. 64): sets are built in stages from sets already available, and no stage contains everything. The universe of type theory in Ch. 66 is stratified for the same reason, and Girard's paradox about Type : Type is Russell's argument rebuilt with types.

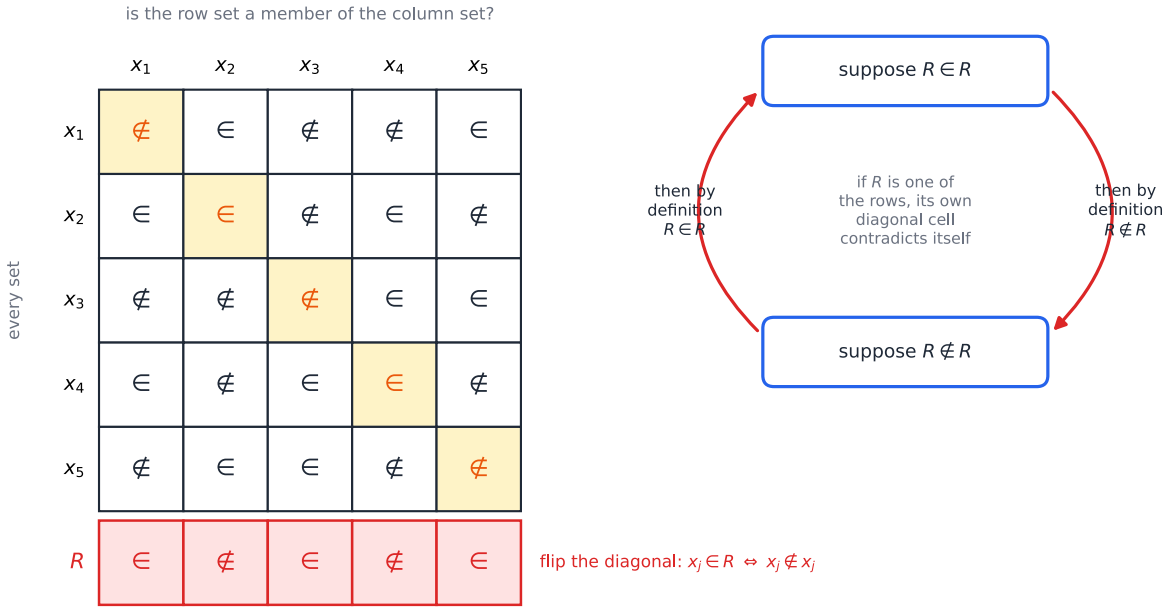


Figure 68.1 — Russell's set is a flipped diagonal. Row x_i, column x_j records whether $x_i \in x_j$. Russell's R contains x_j exactly when the diagonal cell says $x_j \notin x_j$, so R differs from every column on the diagonal. If R were itself one of the sets, its own diagonal cell would have to equal its own negation.

? One theorem behind every diagonal?

Cantor's uncountability proof, Russell's paradox, Turing's halting problem and Gödel's theorem all flip a diagonal. Is that a family resemblance, or a single theorem? In 1969 William Lawvere showed it is one theorem, about functions $A \to (A \to B)$ in a cartesian closed category: if some map $B \to B$ has no fixed point, then no such function can hit every map $A \to B$. Every diagonal argument in this book is an instance, and the language that makes this precise is Ch. 62.

Formal systems, truth and provability

Let T be a theory in the language of arithmetic (symbols 0, S for successor, $+$, $\times$, $<$ and the usual logic). Its standard example is **Peano arithmetic** (PA): basic axioms such as $x + 0 = x$ and $x + S(y) = S(x + y)$, plus the induction schema. T is

- **effectively axiomatised** if an algorithm recognises its axioms. Then its theorems can be listed by a program that enumerates all finite strings and prints those that are correct proofs;
- **consistent** if it proves no sentence together with its negation;
- **sound** if every theorem is *true* in the standard model $\mathbb{N} = \{0, 1, 2, \dots\}$;
- **complete** (or negation-complete) if for every sentence σ it proves σ or $\neg\sigma$.

Do not confuse the last with Gödel's *completeness* theorem of Ch. 65, which says that a sentence true in *every* model of T is provable. Here we ask about truth in the one model we care about, $\mathbb{N}$.

We need one fact from the arithmetisation of computation, which we state without proof. Turing machines and their inputs can be coded by numbers, and "machine M halts on input w" can be written as an arithmetic sentence $H_{M,w}$ of the form $\exists s\, \theta(s)$, where θ says "s codes a halting computation of M on w" and can be checked by bounded search. The sentence is true in $\mathbb{N}$ exactly when M halts on w.

Theorem 68.2 — Incompleteness from the halting problem

Let T be sound and effectively axiomatised. Then there are M and w such that M does not halt on w, but T proves neither $\neg H_{M,w}$ nor $H_{M,w}$. So $\neg H_{M,w}$ is a true sentence that T cannot decide.

Proof. Suppose instead that T proves $\neg H_{M,w}$ for *every* pair (M, w) with M not halting on w. Here is an algorithm for the halting problem. Given (M, w), run two processes side by side, one step of each in turn:

1. simulate M on w; if the simulation halts, answer **halts**;
2. enumerate the theorems of T; if $\neg H_{M,w}$ appears, answer **does not halt**.

The procedure always stops: if M halts, process 1 stops; if not, $\neg H_{M,w}$ is true, so by our assumption T proves it and process 2 finds it. The answer is always correct: process 1 is never wrong, and if process 2 answers, T proved $\neg H_{M,w}$, which is then true by soundness, so M really does not halt. That decides the halting problem, contradicting Theorem 30.2.

So some non-halting pair (M, w) has $T \nvdash \neg H_{M,w}$. And $T \nvdash H_{M,w}$, because $H_{M,w}$ is false and T is sound. □

The proof never wrote the unprovable sentence down. It showed that a complete, sound, effective theory would be a halting decider, and no halting decider exists. Gödel did better: he built the sentence explicitly, and he needed only consistency, not soundness.

Gödel's sentence

Gödel coded formulas as numbers, writing $\ulcorner \varphi \urcorner$ for the code of φ. Checking a proof is mechanical, so there is an arithmetic formula $\mathrm{Prov}_T(x)$ that says "x is the code of a theorem of T". Two standard facts, stated without proof, do the heavy lifting.

- **Representability.** If $T \vdash \varphi$, then $T \vdash \mathrm{Prov}_T(\ulcorner \varphi \urcorner)$: when a proof exists, T can verify it by checking it step by step.
- **The diagonal lemma.** For every formula $\psi(x)$ there is a sentence G with $T \vdash G \leftrightarrow \psi(\ulcorner G \urcorner)$. It is the logical version of a program that prints its own source code (§68.4).

Apply the lemma to $\psi(x) = \neg\,\mathrm{Prov}_T(x)$. The result is a sentence G that T proves equivalent to "G is not provable in T".

Theorem 68.3 — Gödel's first incompleteness theorem (1931)

Let T be consistent, effectively axiomatised and strong enough for representability and the diagonal lemma (Peano arithmetic is more than enough). Then $T \nvdash G$. If T is also sound, G is true and $T \nvdash \neg G$.

Proof. Suppose $T \vdash G$. By representability, $T \vdash \mathrm{Prov}_T(\ulcorner G \urcorner)$. But G is equivalent in T to $\neg\,\mathrm{Prov}_T(\ulcorner G \urcorner)$, so $T \vdash \neg\,\mathrm{Prov}_T(\ulcorner G \urcorner)$ as well. That is a sentence and its negation, contradicting consistency. Hence $T \nvdash G$.

That is exactly what G asserts, so G is true. If T is sound it proves only true sentences, so it cannot prove the false $\neg G$. □

In 1936 Barkley Rosser modified the sentence ("if I am provable, then a shorter refutation of me exists") so that consistency alone shows that neither it nor its negation is provable. Either way, **adding G as a new axiom does not help**: the stronger theory $T + G$ is still consistent and effective, so it has its own Gödel sentence G'.

The second theorem: no system vouches for itself

Let Con_T be the arithmetic sentence $\neg\,\mathrm{Prov}_T(\ulcorner 0 = 1 \urcorner)$, "$T$ does not prove $0 = 1$".

Theorem 68.4 — Gödel's second incompleteness theorem (1931)

If T is consistent, effectively axiomatised and contains Peano arithmetic, then $T \nvdash \mathrm{Con}_T$.

Sketch. The proof of Theorem 68.3 showed, informally, "if T is consistent then G is not provable". Every step of that argument is elementary reasoning about proofs and codes, and with considerable effort one checks that T itself can carry it out. So $T \vdash \mathrm{Con}_T \to \neg\,\mathrm{Prov}_T(\ulcorner G \urcorner)$, that is, $T \vdash \mathrm{Con}_T \to G$. If T proved Con_T, it would prove G, contradicting Theorem 68.3. □

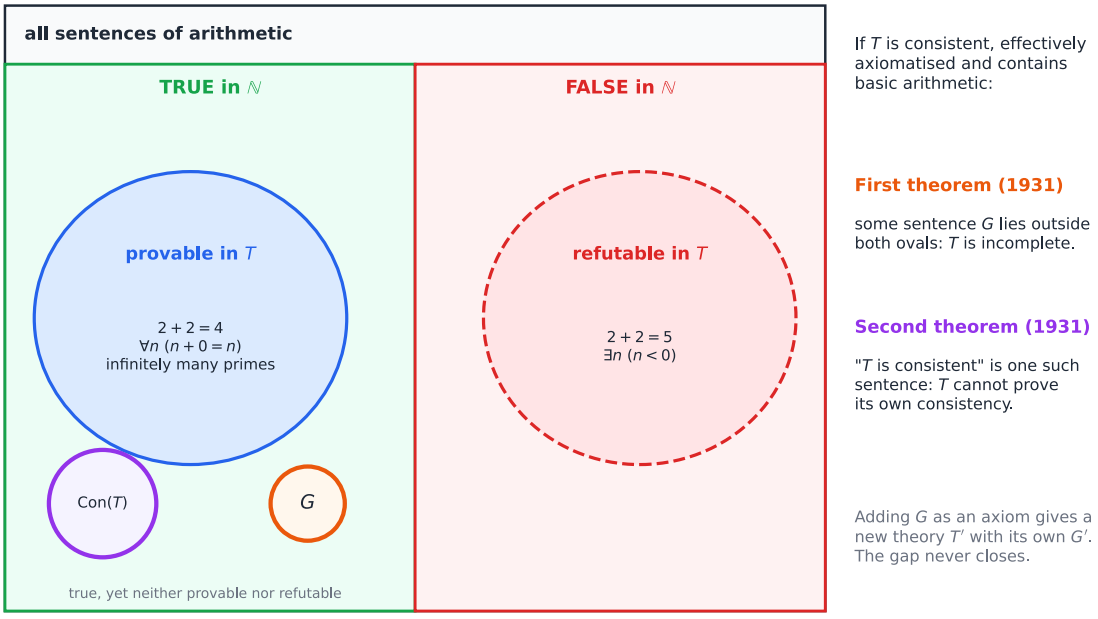


Figure 68.2 — What Gödel's theorems say about a formal system T. Provable sentences (if T is sound) are true, refutable ones are false, but the two ovals never cover everything. The first theorem puts G in the gap; the second puts $\mathrm{Con}(T)$ there too.

This ended Hilbert's programme of the 1920s, which hoped to prove the consistency of all of mathematics by finite, uncontroversial means. Any such proof, formalised in a system T, would be a proof of Con_T inside T. It also explains the first mystery of Ch. 66: proving that every program of System F halts implies the consistency of second-order arithmetic, so that proof can never be carried out *inside* second-order arithmetic.

Independence in the wild

Gödel's G is artificial, built only to say "I am unprovable". The shock of the following decades was that natural questions turned out to be independent too. The method is always the same, and it comes from soundness: if $T \vdash \varphi$, every model of T satisfies φ. So to show $T \nvdash \varphi$, build a model of T in which φ is false. To show independence, build two models, one for each answer. Non-Euclidean geometry proved the parallel postulate independent of Euclid's other axioms in exactly this way, decades before Gödel.

- **Goodstein's theorem** is independent of Peano arithmetic (Kirby and Paris, 1982), though it is provable in set theory. The proof replaces each base by the infinite ordinal ω, so that the sequence becomes a strictly *decreasing* sequence of ordinals below $\varepsilon_0 = \omega^{\omega^{\omega^{\cdots}}}$, which must stop because ordinals are well-ordered (Ch. 64). Peano arithmetic cannot prove that such ordinal descents terminate; in 1936 Gentzen showed that this very fact suffices to prove PA consistent, which by Theorem 68.4 PA cannot do itself.
- **The continuum hypothesis** (CH), Cantor's guess that no set has size strictly between $\mathbb{N}$ and $\mathbb{R}$, is independent of the standard axioms of set theory, ZFC. Gödel built a model of ZFC in which CH holds (1938–40); Paul Cohen invented *forcing* to build one in which it fails (1963), and won the Fields Medal for it.
- **Busy beavers.** The values $\mathrm{BB}(n)$ of Ch. 30 are definite numbers. But in 2016 Adam Yedidia and Scott Aaronson built a 7,918-state Turing machine that halts if and only if ZFC is inconsistent, so ZFC cannot determine its behaviour. Later work cut the size to under 750 states. Somewhere below that, the value of $\mathrm{BB}(n)$ escapes the axioms of all of mathematics.

> **? Is the continuum hypothesis true?**
>
> CH is independent of ZFC, so the axioms do not settle it. Is it then meaningless, or is there a fact of the matter that the axioms simply fail to capture? Set theorists split. Some search for new, natural axioms that decide it: W. Hugh Woodin argued for years that CH is false, then developed an "ultimate L" programme that would make it true. Others hold that there is a *multiverse* of equally good set-theoretic worlds. Which view is right, and whether any new axiom will win acceptance, is an open problem at the foundations of mathematics.

68.4 The algorithm

Self-reference is a programming technique

The diagonal lemma sounds like a logician's trick. It is really an algorithm, and the cleanest way to see it is a program that prints its own source code, a **quine**:

```
s = 's = %r\nprint(s %% s)'
print(s % s)
```

Run it and it prints exactly those two lines. The string `s` is a *template* with a hole; `%r` fills the hole with a quoted copy of the template itself. The diagonal lemma works the same way. Given $\psi(x)$, form the template "ψ(the code of the result of substituting this template into itself)", then substitute the template into itself. The result is a sentence that talks about its own code. Kleene's recursion theorem (1938) says every programming language can do this: any program can be given access to its own source. That is why the halting argument of Ch. 30 works without hand-waving: the machine D really can ask the decider about itself.

Enumerating theorems

The "theorem-printing machine" of §68.2 is an honest algorithm:

```
for n = 1, 2, 3, ...:
    for each string p of length at most n:
        if p is a correctly formed proof in T (check every step against axioms and rules):
            print the last line of p
```

Checking a proof is fast, roughly linear in its length for a well-designed system. The search is hopeless in practice, exponential in the length of the proof sought. But in principle it prints every theorem of T, and only theorems. Theorems of an effective theory are therefore **recursively enumerable**: a machine can list them. Theorem 68.2 used exactly this machine. Incompleteness then says: for arithmetic, the list of theorems can never coincide with the list of true sentences, because the second list cannot be produced by any machine at all.

Computing Goodstein sequences

The bump step in the hook is a recursive algorithm, because exponents must themselves be written in the base:

```
bump(n, b):                         # hereditary base b, then replace b by b+1
    result = 0; power = 0
    while n > 0:
        n, digit = divmod(n, b)
        result = result + digit * (b+1) ^ bump(power, b)
        power = power + 1
    return result
```

Each call handles one level of the hereditary representation, and exponents are tiny compared with n, so one step costs a few big-integer operations. The difficulty is not the cost per step but the number of steps. A program that runs the sequence until it reaches 0 is a program whose termination is Goodstein's theorem, so Peano arithmetic cannot prove that the program halts, even though it does. Such programs are where the independence phenomena of this chapter meet the verification problems of Ch. 67: a verifier built on Peano arithmetic would be unable to certify this loop terminates, whatever invariant you handed it.

Failure cases

- **Self-reference does not always paradox.** "This sentence has five words" is self-referential and simply true. The trouble needs negation, a diagonal that flips.

- **Incompleteness needs strength.** Presburger arithmetic (no multiplication), the theory of real closed fields (Tarski) and plane Euclidean geometry are complete and decidable. Gödel's theorem bites only once a system can encode its own proofs.
- **It is not "everything is unknowable".** Incompleteness says no single effective system captures all arithmetic truth. It does not say any particular open problem is independent, and most never turn out to be.

68.5 The experiment

The script computes a Goodstein sequence, prints the first terms and the term after `STEPS` steps, and plots the number of digits. Set `HEREDITARY = False` to get the **weak Goodstein sequence**, where only the base itself is bumped and the exponents stay as they were.

```
import numpy as np
import matplotlib.pyplot as plt

START, STEPS = 4, 2000      # KNOBS: try START = 3, 5, 6; more STEPS
HEREDITARY = True           # False gives the "weak" Goodstein sequence (exponents not bumped)

def bump(n, b):
    """Write n in base b (hereditarily, if asked), replace every b by b+1, return the value."""
    result, power = 0, 0
    while n:
        n, digit = divmod(n, b)
        exponent = bump(power, b) if HEREDITARY else power
        result += digit * (b + 1) ** exponent
        power += 1
    return result

n, base, values = START, 2, [START]
while n > 0 and len(values) <= STEPS:
    n = bump(n, base) - 1          # bump the base, then subtract one
    base += 1
    values.append(n)

print("first terms:", values[:8])
print("after", len(values) - 1, "steps the value is", values[-1] if values[-1] < 10**12
      else f"a {len(str(values[-1]))}-digit number", "in base", base)
digits = [np.log10(v) if v > 0 else 0 for v in values]
plt.plot(digits)
plt.xlabel("step k"); plt.ylabel("log10 of the term")
plt.title(f"Goodstein sequence from {START} (hereditary={HEREDITARY})")
plt.show()
```

From 3, the sequence is `3, 3, 3, 2, 1, 0`. From 4 it climbs past four million in 2,000 steps, and the plot rises steadily with no sign of turning. Now set `HEREDITARY = False`: from 4, the weak sequence reaches 0 after only 21 steps. The difference between the two versions is exactly the difference between what Peano arithmetic can prove and what it cannot. Weak Goodstein sequences always terminate, and PA proves it.

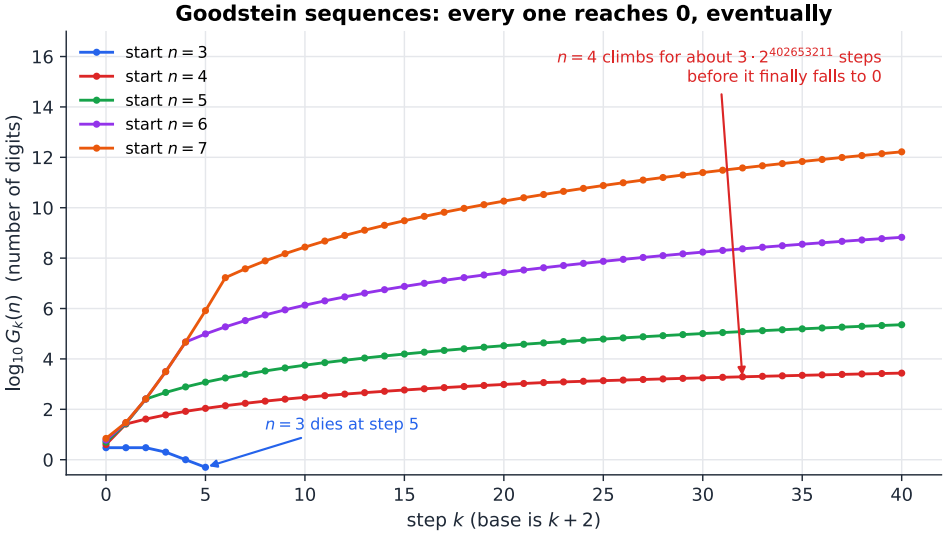


Figure 68.3 — Digits of $G_k(n)$ for the first 40 steps. Only $n = 3$ visibly dies. The others climb, the steeper the larger the start, and every one of them will eventually come back to 0, a fact Peano arithmetic cannot prove.

Things to change:

1. Run `START = 5` and `START = 6` for a few hundred steps. How fast do the digit counts grow? Do not try to run them to the end.
2. With `HEREDITARY = False`, record how many steps the weak sequence takes from $n = 1, \ldots, 8$. Does the step count grow like a polynomial, an exponential, or something worse?
3. Change "subtract one" to "subtract two". Does every sequence still reach 0? Does your reason survive?

68.6 The challenge

No answers are given.

1. **Intuition.** Call an adjective *heterological* if it does not describe itself: "long" is heterological, "short" is not. Is "heterological" heterological? Compare this with the barber who shaves exactly those who do not shave themselves. Why does the barber puzzle merely show that no such barber exists, while Russell's paradox destroyed a whole theory?
2. **Derivation.** Using Theorem 68.4, show that if PA is consistent, then so is the theory $\mathrm{PA} + \neg\,\mathrm{Con}_{\mathrm{PA}}$, which "believes" itself inconsistent. Then explain why this strange theory is consistent but not sound, and what its models must contain that $\mathbb{N}$ does not.
3. **Implementation.** Write a quine in a language other than Python. Then write a Python program that takes a function `halts(src)` as a parameter, reads its own source code using the quine trick, and does the opposite of whatever `halts` predicts about it. Test it against three different (necessarily wrong) `halts` functions and record how each one fails.
4. **Open investigation.** In the **hydra game** of Kirby and Paris, Hercules cuts a head off a tree-shaped hydra, and the hydra regrows copies of the wounded branch, more copies at each step. Every strategy kills every hydra, and this too is unprovable in PA. Simulate it. Investigate how the number of steps depends on Hercules's strategy (always cut the leftmost head, the rightmost, a random one) and on the shape of the starting tree. Is there a strategy that is reliably fastest?

68.7 Loose threads

The hook asked how a statement about whole numbers can be true but unprovable. Now we know. Provability is relative to a formal system, a theorem-printing machine, and Gödel showed that any consistent, effective machine strong enough for arithmetic misses some truths, including the truth of its own consistency. Goodstein's theorem is one of them for Peano arithmetic. It is provable in a stronger system, which has its own blind spots in turn. Russell's paradox started this story by showing that naive self-reference destroys a theory. Gödel finished it by showing that careful self-reference, inside any sufficiently rich theory, reveals limits that cannot be engineered away.

What we did not tell you:

- **How the independence of CH is proved.** Forcing builds new models of set theory by adding a "generic" object to an old one, and it is among the most powerful techniques in logic. We stated Cohen's result without even a sketch.
- **Why the diagonal keeps appearing.** The first mystery points to Lawvere's fixed-point theorem in Ch. 62.
- **What working mathematicians do about it.** Almost nothing, usually, and the reasons are part of how research proceeds: see Ch. 98.

> **? Could P versus NP be independent?**
>
> Ch. 30 left P versus NP open. Could it be open because it is independent of ZFC, like the continuum hypothesis? Some researchers have taken the possibility seriously, and it is known that certain *weak* systems of arithmetic cannot prove some lower bounds of the kind needed. But no one knows how to prove that P versus NP is independent of ZFC, or even of Peano arithmetic, and most experts expect a proof one way or the other to exist. Whether the most famous problem in computer science lies beyond the reach of mathematics itself is an open problem.

Further reading

- E. Nagel and J. R. Newman, *Gödel's Proof*, revised ed., New York University Press, 2001. The classic short account for non-specialists.
- P. Smith, *An Introduction to Gödel's Theorems*, 2nd ed., Cambridge University Press, 2013. Careful, complete proofs, including the second theorem.
- T. Franzén, *Gödel's Theorem: An Incomplete Guide to Its Use and Abuse*, A K Peters, 2005. What the theorems do and do not say, with many misreadings corrected.
- R. M. Smullyan, *Gödel's Incompleteness Theorems*, Oxford University Press, 1992. Elegant proofs built around self-reference and diagonalisation.

Track G — Mathematics for Computer Graphics, Games, and Simulation

> **Level 2 — Specializations.** The geometry, algebra and numerics that turn equations into worlds on a screen.

Suggested reading order for Track G. The grey stations on the left are the core chapters it leans on.

Before you start

This track assumes you are comfortable with these core chapters:

- Ch. 2 — Complex Numbers and Engineering Representation
- Ch. 3 — Trigonometry and Periodic Functions
- Ch. 7 — Vector Calculus
- Ch. 8 — Matrices and Linear Systems
- Ch. 10 — Eigenvalues and Eigenvectors
- Ch. 16 — Fourier Transforms
- Ch. 22 — Numerical Methods

Chapters

Ch.	Chapter	What it explores
69	The Mathematics of 3D Worlds	Coordinate systems, affine transformations, homogeneous coordinates, projections.
70	Quaternions and the Geometry of Rotation	Rotation groups, quaternion algebra, interpolation, rigid-body transforms.
71	Ray Tracing: The Geometry of Light	Rays, intersections, reflection, refraction, sampling, rendering equations.
72	Numerical Simulation and Physics Engines	Numerical integration, collision detection, constraints, stability, error.
73	Fractals: Infinite Complexity from Simple Rules	Iterated functions, self-similarity, fractal dimensions, chaotic sets.
74	Procedural Generation: Creating Worlds with Rules	Noise functions, random fields, cellular automata, grammars, spatial structures.
75	Computational Geometry: Algorithms for Shapes	Convex hulls, triangulation, Voronoi diagrams, spatial indexing.
76	Signal Processing for Audio and Visual Computing	Sampling, filtering, convolution, transforms, spectral analysis.

Every chapter follows the same arc: **the hook** (a puzzle or a failure), **the big idea**, **the mathematics**, **the algorithm**, **the experiment** you can run, **the challenge**, and **loose threads**: the questions the chapter deliberately leaves open.

Chapter 69 — The Mathematics of 3D Worlds

> **Level 2 · Track G — Mathematics for Computer Graphics, Games, and Simulation Builds on:** Ch. 8, Ch. 9, Ch. 7 · **Leads to:** Ch. 70, Ch. 71, Ch. 75 · **Related:** Ch. 22, Ch. 58, Ch. 59 **Reading time:** ~25 min

69.1 The hook

A first-year game programmer is building a toy solar system. She wants the Earth to spin on its axis and also sit 10 units from the Sun. She writes two lines: "rotate the Earth by the day angle" and "move the Earth 10 units out". It runs, and the Earth does something nobody asked for: instead of spinning in place, it swings around the Sun once a day like a stone on a string. She swaps the two lines, and now it spins in place. Same two operations, same numbers, opposite worlds.

Meanwhile her graphics card is doing something that should be impossible. Every frame, sixty times a second, it takes several million corners of triangles, each given as three numbers in the coordinates of its own little model, and lands each one on a pixel of your screen. Along the way every point must be

- **rotated** (the model faces some direction),
- **translated** (the model sits somewhere in the world, the camera sits somewhere else),
- **scaled** (the model was built in centimetres, the world is in metres),
- and **projected**, which means dividing by its distance from the eye, so that far things look small.

Here is the puzzle. A matrix is a *linear* map (see Ch. 9 §9.5): it always sends the zero vector to zero. A translation moves the origin, so no 3×3 matrix can translate anything. And projection divides by depth, which is not even close to linear. Yet the hardware does all of it with one 4×4 matrix per object and a single division per vertex.

How can a matrix with an extra row and column carry a whole world onto a flat screen? And why did swapping two lines move the Earth?

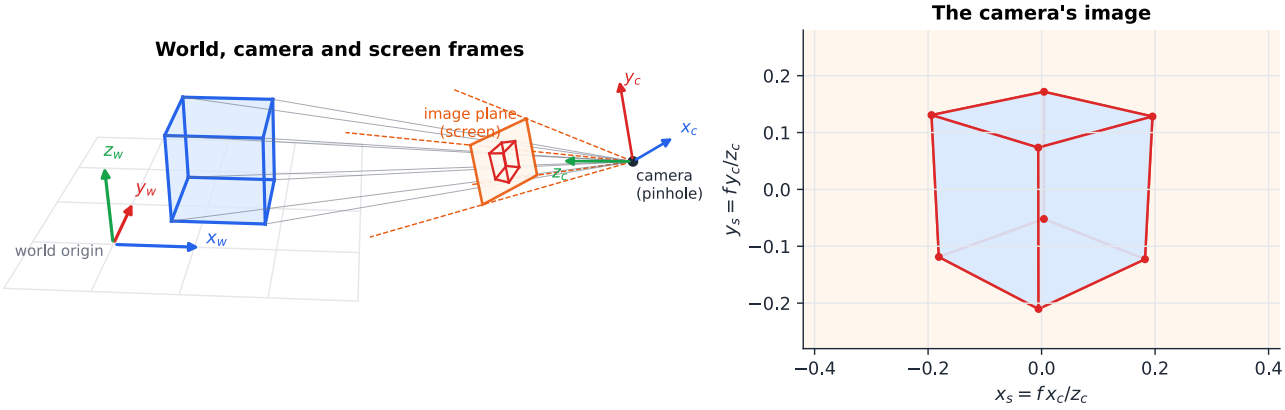
69.2 The big idea

Three ideas do almost all of the work.

Every coordinate is relative to a frame. "The cup is at $(2, 0, 1)$" means nothing until you say *measured from where, along which directions*. A frame is an origin plus three axes. A game has many: each model's own frame (where the artist built it), the world frame, the camera frame, and finally the screen. Rendering is nothing but a chain of changes of frame. The solar-system bug is a question of which frame each operation happens in: rotating *after* moving the Earth out swings it about the Sun's origin; rotating *before* spins it about its own centre.

Add one coordinate, and translation becomes linear. Write a 3-D point as $(x, y, z, 1)$. In four dimensions, the 3-D world is a flat slice at height $w = 1$, and a translation of that slice is a *shear* of 4-D space, which is linear. Directions get $w = 0$, so the shear leaves them alone, exactly as it should: moving a ship does not change which way its nose points.

A camera is a box with a hole. Light from a point passes through a pinhole and hits a screen. By similar triangles the image height is the true height divided by the distance. With the fourth coordinate in hand, we can make a matrix *copy* the depth into w; the hardware's final step "divide by w" then performs the projection.



Three frames in one picture. The world frame sits at the origin; the camera frame (x_c right, y_c up, z_c along the line of sight) sits at the pinhole. Each ray from a cube corner to the pinhole pierces the image plane, and the pierce points (red) form the picture on the right. The left panel was itself drawn by the same projection, from a second camera.

The analogy to hold onto: the GPU is a factory with one machine, "multiply by a 4×4 matrix", and one finishing step, "divide by the last number". Every trick in this chapter is a way of disguising a non-linear wish as a matrix.

69.3 The mathematics

Points, vectors and frames

There are two kinds of 3-D objects, and confusing them causes real bugs. A **point** is a location. A **vector** is a displacement: the arrow from one point to another. Their algebra is different: point minus point is a vector, point plus vector is a point, vector plus vector is a vector, but "point plus point" means nothing (what is Paris plus Tokyo?).

> **Frame**
>
> A **frame** $(\mathbf{o};\ \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$ is an origin point $\mathbf{o}$ and a basis $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ of vectors. The coordinates of a point $\mathbf{p}$ in this frame are the unique numbers (x, y, z) with $\mathbf{p} = \mathbf{o} + x\mathbf{e}_1 + y\mathbf{e}_2 + z\mathbf{e}_3$. The coordinates of a vector $\mathbf{v}$ are the numbers with $\mathbf{v} = x\mathbf{e}_1 + y\mathbf{e}_2 + z\mathbf{e}_3$; the origin plays no part.

When the basis is orthonormal, coordinates are just dot products: $x = (\mathbf{p} - \mathbf{o}) \cdot \mathbf{e}_1$ and so on. For a camera at $\mathbf{c}$ with orthonormal axes $\hat{\mathbf{r}}$ (right), $\hat{\mathbf{u}}$ (up), $\hat{\mathbf{f}}$ (forward), a world point has camera coordinates

$$x_c = (\mathbf{p} - \mathbf{c}) \cdot \hat{\mathbf{r}}, \qquad y_c = (\mathbf{p} - \mathbf{c}) \cdot \hat{\mathbf{u}}, \qquad z_c = (\mathbf{p} - \mathbf{c}) \cdot \hat{\mathbf{f}}.$$

This is a change of basis followed by a shift of origin; the change-of-basis part is Ch. 9 §9.4, and the fact that the inverse of an orthogonal matrix is its transpose is Ch. 9 §9.7. How do we build $\hat{\mathbf{r}}, \hat{\mathbf{u}}, \hat{\mathbf{f}}$ from "camera at $\mathbf{c}$, looking at $\mathbf{t}$, world up is $\hat{\mathbf{z}}$"? Normalise $\mathbf{t} - \mathbf{c}$ to get $\hat{\mathbf{f}}$; then $\hat{\mathbf{r}} = \hat{\mathbf{f}} \times \hat{\mathbf{z}} / \|\hat{\mathbf{f}} \times \hat{\mathbf{z}}\|$ and $\hat{\mathbf{u}} = \hat{\mathbf{r}} \times \hat{\mathbf{f}}$. Two cross products (Ch. 7 §7.2) and you have a camera. (Notice the failure case already: if you look straight up, $\hat{\mathbf{f}} \times \hat{\mathbf{z}} = \mathbf{0}$ and the camera has no idea which way is right.)

Affine maps

> **Affine map**
>
> A map $T : \mathbb{R}^3 \to \mathbb{R}^3$ is **affine** if $T(\mathbf{x}) = \mathbf{A}\mathbf{x} + \mathbf{t}$ for some 3×3 matrix $\mathbf{A}$ and vector $\mathbf{t}$. Rotations, scalings, shears, reflections and translations are all affine. It is **rigid** if $\mathbf{A}$ is a rotation ($\mathbf{A}^{\mathsf{T}}\mathbf{A} = \mathbf{I}$, $\det \mathbf{A} = 1$).

Affine maps are not linear (unless $\mathbf{t} = \mathbf{0}$), but they preserve something almost as good. Call $\sum_i \lambda_i \mathbf{p}_i$ an **affine combination** of points if the weights sum to one, $\sum_i \lambda_i = 1$. Only these combinations make sense for points: $\frac{1}{2}\mathbf{p} + \frac{1}{2}\mathbf{q}$ is the midpoint, wherever the origin is, while $\mathbf{p} + \mathbf{q}$ depends on the arbitrary choice of origin.

> **Result 1: affine maps preserve affine combinations**
>
> If $T(\mathbf{x}) = \mathbf{A}\mathbf{x} + \mathbf{t}$ and $\sum_i \lambda_i = 1$, then $T\left(\sum_i \lambda_i \mathbf{p}_i\right) = \sum_i \lambda_i\, T(\mathbf{p}_i)$.

Proof. Using linearity of $\mathbf{A}$ and then $\sum_i \lambda_i = 1$ to write $\mathbf{t} = \sum_i \lambda_i \mathbf{t}$:

$$T\Big(\sum_i \lambda_i \mathbf{p}_i\Big) = \sum_i \lambda_i \mathbf{A}\mathbf{p}_i + \mathbf{t} = \sum_i \lambda_i \mathbf{A}\mathbf{p}_i + \sum_i \lambda_i \mathbf{t} = \sum_i \lambda_i \left(\mathbf{A}\mathbf{p}_i + \mathbf{t}\right) = \sum_i \lambda_i T(\mathbf{p}_i).$$

The step that used $\sum \lambda_i = 1$ is exactly where a non-affine combination would leave a stray $\mathbf{t}$. $\square$

The consequences are what make 3-D graphics cheap. The segment $\{(1-s)\mathbf{p} + s\mathbf{q}\}$ maps to the segment between $T(\mathbf{p})$ and $T(\mathbf{q})$, so **lines map to lines** and you only ever transform the endpoints. Midpoints map to midpoints, parallel lines stay parallel, and a point inside a triangle with barycentric coordinates $(\lambda_1, \lambda_2, \lambda_3)$ keeps them. A whole triangle with a million interior points is described by three corners. Remember this; projection will break it.

Homogeneous coordinates

Now the trick from §69.2. Attach a fourth coordinate: a point $\mathbf{p}$ becomes $(\mathbf{p}, 1)$ and a vector $\mathbf{v}$ becomes $(\mathbf{v}, 0)$. Then

$$\begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{0}^{\mathsf{T}} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{p} \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{A}\mathbf{p} + \mathbf{t} \\ 1 \end{bmatrix}, \qquad \begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{0}^{\mathsf{T}} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{v} \\ 0 \end{bmatrix} = \begin{bmatrix} \mathbf{A}\mathbf{v} \\ 0 \end{bmatrix}.$$

A 4×4 matrix in this block form performs an affine map on points and its linear part on vectors, with no special cases. The bookkeeping rules for points and vectors fall out of the last coordinate: point minus point has $w = 1 - 1 = 0$ (a vector), point plus vector has $w = 1$ (a point), and point plus point would have $w = 2$, which is not on the slice $w = 1$, the algebra's way of saying "meaningless".

Composition is now matrix multiplication (Ch. 8 §8.2). Applying $T_1(\mathbf{x}) = \mathbf{A}_1\mathbf{x} + \mathbf{t}_1$ and then T_2 gives

$$\begin{bmatrix} \mathbf{A}_2 & \mathbf{t}_2 \\ \mathbf{0}^{\mathsf{T}} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{A}_1 & \mathbf{t}_1 \\ \mathbf{0}^{\mathsf{T}} & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{A}_2\mathbf{A}_1 & \mathbf{A}_2\mathbf{t}_1 + \mathbf{t}_2 \\ \mathbf{0}^{\mathsf{T}} & 1 \end{bmatrix}.$$

Look at the top-right block: the second map *rotates the first map's translation*. That is the solar-system bug, in one formula. Let $\mathbf{R}$ be a rotation by 90° about z and $\mathbf{T}$ a translation by $(10, 0, 0)$. Apply both to the Earth's centre, which starts at the origin:

- **rotate, then translate ($\mathbf{TR}$)**: the centre stays at the origin under $\mathbf{R}$, then moves to $(10, 0, 0)$. The Earth spins in place, out where it belongs.
- **translate, then rotate ($\mathbf{RT}$)**: the centre moves to $(10, 0, 0)$, then the rotation about the Sun's origin carries it to $(0, 10, 0)$. A quarter-day later, the Earth is a quarter of the way around its orbit.

Our programmer's first version applied the day rotation last. The rule of thumb, "matrices act right to left, the rightmost acts first, in the object's own frame", saves more debugging hours in graphics than any other sentence.

Inverting a rigid transform never needs a general inverse. If $\mathbf{x}' = \mathbf{R}\mathbf{x} + \mathbf{t}$, then $\mathbf{x} = \mathbf{R}^{\mathsf{T}}(\mathbf{x}' - \mathbf{t})$, so

$$\begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0}^\mathsf{T} & 1 \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{R}^\mathsf{T} & -\mathbf{R}^\mathsf{T}\mathbf{t} \\ \mathbf{0}^\mathsf{T} & 1 \end{bmatrix}.$$

The **view matrix** of a camera is exactly this: the camera's own placement in the world is a rigid transform $(\mathbf{R}_c, \mathbf{c})$ whose rotation has columns $\hat{\mathbf{r}}, \hat{\mathbf{u}}, \hat{\mathbf{f}}$; the view matrix is its inverse, with rows $\hat{\mathbf{r}}^\mathsf{T}, \hat{\mathbf{u}}^\mathsf{T}, \hat{\mathbf{f}}^\mathsf{T}$, which reproduces the dot-product formulas for x_c, y_c, z_c above. Moving the camera left is the same as moving the world right.

Normals do not transform like points

Here is a trap that catches professionals. A surface's **normal** $\mathbf{n}$ is a vector perpendicular to it, used for lighting. Since normals are vectors, surely they transform by $\mathbf{A}$? Try a non-uniform stretch $\mathbf{A} = \operatorname{diag}(2, 1, 1)$ on the plane $x + y = 0$, which has tangent $\mathbf{s} = (1, -1, 0)$ and normal $\mathbf{n} = (1, 1, 0)$. After stretching, the tangent is $\mathbf{A}\mathbf{s} = (2, -1, 0)$, and the naive normal $\mathbf{A}\mathbf{n} = (2, 1, 0)$ has dot product $4 - 1 = 3$ with it. The lighting is now computed with a normal that is not normal.

> **Result 2: normals transform by the inverse transpose**
>
> If tangent vectors transform by an invertible $\mathbf{A}$, then $\mathbf{n}' = \mathbf{A}^{-\mathsf{T}}\mathbf{n}$ is perpendicular to every transformed tangent.

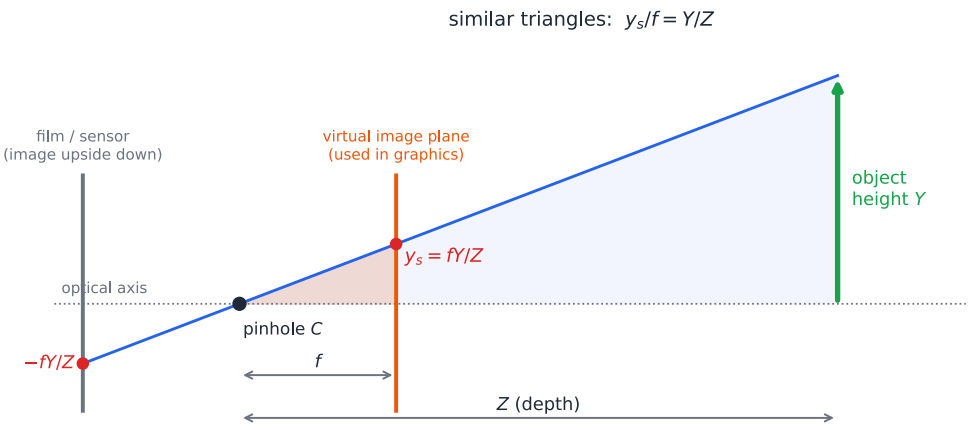
Proof. Perpendicularity is $\mathbf{n}^\mathsf{T}\mathbf{s} = 0$ for every tangent $\mathbf{s}$. Insert $\mathbf{I} = \mathbf{A}^{-1}\mathbf{A}$:

$$0 = \mathbf{n}^\mathsf{T}\mathbf{s} = \mathbf{n}^\mathsf{T}\mathbf{A}^{-1}\mathbf{A}\mathbf{s} = \left(\mathbf{A}^{-\mathsf{T}}\mathbf{n}\right)^\mathsf{T}(\mathbf{A}\mathbf{s}).$$

So $\mathbf{A}^{-\mathsf{T}}\mathbf{n}$ is perpendicular to every transformed tangent $\mathbf{A}\mathbf{s}$. □

In the example, $\mathbf{A}^{-\mathsf{T}}\mathbf{n} = (0.5, 1, 0)$ and $(0.5, 1, 0) \cdot (2, -1, 0) = 1 - 1 = 0$. For a rotation, $\mathbf{R}^{-\mathsf{T}} = \mathbf{R}$, which is why the bug hides until someone scales a model unevenly. The deeper lesson: tangents and normals are different kinds of object. Tangents are vectors; normals are really *covectors*, linear functions that eat vectors, and covectors transform "the other way".

The pinhole and the projection matrix



The pinhole camera. The small shaded triangle (pinhole, f, y_s) and the large one (pinhole, Z, Y) are similar, so $y_s = fY/Z$. A real camera's film sits behind the hole and sees the world upside down; graphics puts a virtual screen in front, which gives the same picture the right way up.

In camera coordinates (looking along $+z_c$), the similar triangles give the **perspective projection**

$$x_s = \frac{f\,x_c}{z_c}, \qquad y_s = \frac{f\,y_c}{z_c}.$$

A point at $(2, 1, 4)$ with $f = 1$ lands at $(0.5, 0.25)$; move it twice as far, to $(4, 2, 8)$, and it lands at the same place: every point on a ray through the pinhole has the same image. That is the geometric content of projection, and it is the same as the defining property of homogeneous coordinates: (x, y, z, w) and $(\lambda x, \lambda y, \lambda z, \lambda w)$ for $\lambda \neq 0$ name the same point $(x/w, y/w, z/w)$. So we let a matrix put z_c into the last slot and let the divide do the rest:

$$\begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & a & b \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_c \\ y_c \\ z_c \\ 1 \end{bmatrix} = \begin{bmatrix} f x_c \\ f y_c \\ a z_c + b \\ z_c \end{bmatrix} \xrightarrow{\div w} \left(\frac{f x_c}{z_c},\ \frac{f y_c}{z_c},\ a + \frac{b}{z_c} \right).$$

The third row keeps a **depth** value for deciding which surface is in front. Choosing a and b so that the near plane $z_c = n$ maps to depth 0 and the far plane $z_c = F$ to depth 1 gives $a = F/(F - n)$ and $b = -nF/(F - n)$. (Conventions differ: OpenGL looks down $-z$ and maps depth to $[-1, 1]$; the mathematics is the same.)

The matrix made the projection look linear, but the divide is not. What survives?

> **Result 3: perspective maps lines to lines, but not midpoints to midpoints**
>
> For points in front of the camera ($z_c > 0$), the projection of a segment is a segment. Ratios along the segment are not preserved.

Proof. Let the 3-D segment be $\mathbf{p}(s) = (1 - s)\mathbf{p}_0 + s\mathbf{p}_1$ with depths $z_0, z_1 > 0$. Its screen image is $\boldsymbol{\sigma}(s) = f\,(x(s), y(s))/z(s)$. Write $\boldsymbol{\sigma}_0, \boldsymbol{\sigma}_1$ for the endpoint images, so $f(x_i, y_i) = z_i\boldsymbol{\sigma}_i$. Then

$$\boldsymbol{\sigma}(s) = \frac{(1-s)z_0\,\boldsymbol{\sigma}_0 + s\,z_1\,\boldsymbol{\sigma}_1}{(1-s)z_0 + s z_1} = (1-u)\,\boldsymbol{\sigma}_0 + u\,\boldsymbol{\sigma}_1, \qquad u = \frac{s z_1}{(1-s)z_0 + s z_1}.$$

As s runs from 0 to 1, u runs monotonically from 0 to 1 (both weights stay positive), so the image is exactly the screen segment from $\boldsymbol{\sigma}_0$ to $\boldsymbol{\sigma}_1$. But $u \neq s$ unless $z_0 = z_1$. $\square$

Take $\mathbf{p}_0 = (-1, 0, 2)$ and $\mathbf{p}_1 = (1, 0, 6)$ with $f = 1$. The endpoints land at $x_s = -0.5$ and $x_s = 1/6$. The 3-D midpoint $(0, 0, 4)$ lands at $x_s = 0$, but the screen midpoint is $-1/6$. The near half of the segment takes up three quarters of its screen length. This is foreshortening, and it is why a texture painted on a floor must not be interpolated linearly across the screen: some mid-1990s game consoles did exactly that, and their floor tiles visibly wobble and swim as you walk.

69.4 The algorithm

Every rasterising renderer, from a phone game to a film previsualisation tool, runs this pipeline on every vertex:

```
for each object:
    M  = model matrix          (object frame  -> world frame; scale, rotate, translate)
    V  = view matrix           (world frame   -> camera frame; inverse of camera placement)
    P  = projection matrix     (camera frame  -> clip space; copies depth into w)
    MVP = P @ V @ M            (compose ONCE per object, 64 multiplies)
    for each vertex v = (x, y, z, 1):
        c = MVP @ v            (16 multiplies, 12 adds)
        clip triangles against the near plane (where w would approach 0)
        ndc = c.xyz / c.w      (the perspective divide)
        pixel = viewport(ndc)  (scale and shift to the window)
    normals: transform by inverse-transpose of the upper-left 3x3 of V @ M
```

Complexity. Composing three matrices costs a few hundred flops once per object; each vertex then costs one matrix–vector product. Precomposing is the whole point: the alternative, applying model, view and projection separately, triples the per-vertex work. With 10^6 vertices at 60 frames per second, that is the difference between 10^9 and 3×10^9 flops per second for this step alone.

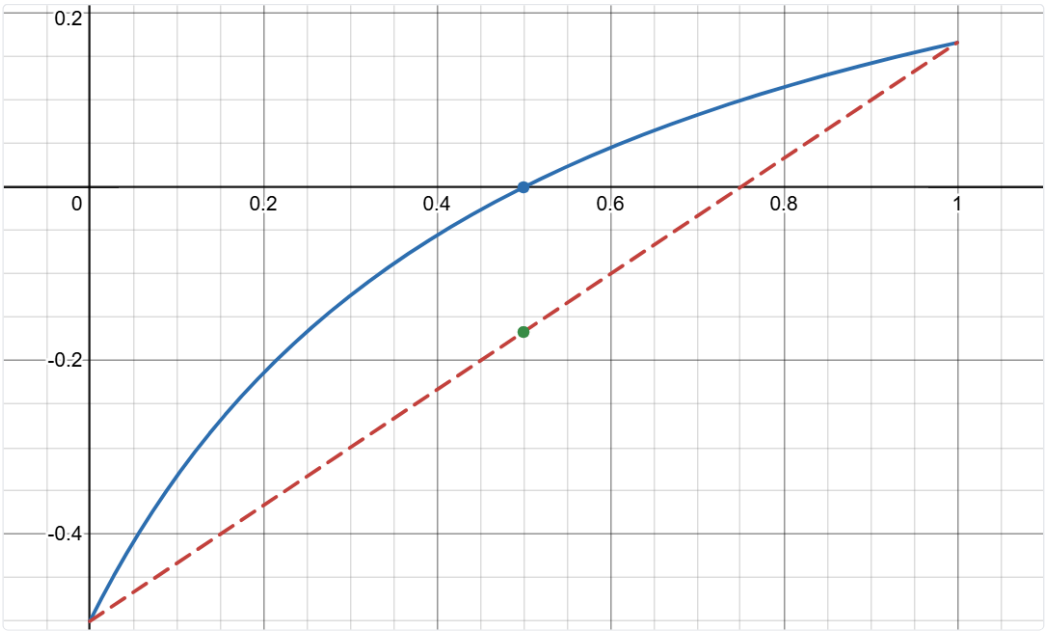
Failure cases.

- *Points behind the eye.* For $z_c < 0$ the divide flips the sign, and a vertex behind you appears in front, upside down. A triangle with one corner behind the camera would be smeared across the whole screen. This is why triangles are **clipped** against the near plane *before* the divide, in homogeneous coordinates where nothing has blown up yet.
- *Degenerate look-at.* Looking straight along "world up" makes $\hat{\mathbf{f}} \times \hat{\mathbf{z}} = \mathbf{0}$. Flight and space games cannot use a fixed world-up vector at all; they need a better description of orientation (Ch. 70).
- *Matrix drift.* An object rotated a little every frame, $\mathbf{R} \leftarrow \mathbf{R}\,\Delta\mathbf{R}$, slowly stops being a rotation in floating point: its columns lose length and orthogonality, and the model visibly shears. The repair (re-orthonormalising, Gram–Schmidt from Ch. 9 §9.7) is a patch; Ch. 70 offers a cure.
- *Depth precision.* The stored depth $a + b/z_c$ is a function of $1/z_c$, not z_c. With $n = 0.1$ and $F = 1000$, depth already reaches 0.50005 at $z_c = 0.2$: half the whole depth range is spent on the first 10 cm in front of your nose. With a 24-bit depth buffer, two surfaces near $z_c = 1000$ must be about 0.6 units apart to be told apart (at $z_c = 100$, about 0.006). Closer than that, they flicker through each other, a defect called **z-fighting**.

> **? The clue we're holding back**
>
> Pushing the near plane out helps z-fighting, but cuts off things close to the eye. Modern engines do something stranger: they flip the depth mapping so the *far* plane is 0 and the near plane is 1, store it as a floating-point number, and the precision problem nearly vanishes. Why does reversing a number line fix it? The answer lies in how floating-point numbers are spaced, in Ch. 22 §22.1.

69.5 The experiment

First, see foreshortening as a graph. A point slides along the 3-D segment from $(-1, 0, 2)$ to $(1, 0, 6)$; the horizontal axis is the 3-D parameter s (here called x) and the curve is its screen position $x_s = (-1 + 2s)/(2 + 4s)$. The dashed line is what naive linear interpolation on the screen would give.



Screen position (blue) against position along the 3-D segment. The blue curve bends: equal steps in the world are unequal on the screen. Change the depth slope `4` in the denominator to `0` (the segment is now parallel to the screen) and the curve becomes the straight line; make it `20` and watch the far half crowd into a sliver.

Second, the whole pipeline in under forty lines. Two wireframe cubes, one near and one far, are carried by a single composed matrix and a perspective divide.

```
import numpy as np
import matplotlib.pyplot as plt

DIST = 6.0          # KNOB: camera distance from the near cube's front face
FOCAL = 0.5 * DIST  # KNOB: focal length; FOCAL proportional to DIST = "dolly zoom"
YAW = 30.0          # KNOB: rotate both cubes about the vertical axis (degrees)
ORTHO = False       # KNOB: True = orthographic (no division by depth)

V = np.array([[x, y, z, 1.0] for x in (-1, 1) for y in (-1, 1) for z in (-1, 1)])
E = [(a, b) for a in range(8) for b in range(a + 1, 8) if np.sum(V[a] != V[b]) == 1]

def T(tx, ty, tz):                      # homogeneous translation
    M = np.eye(4); M[:3, 3] = (tx, ty, tz); return M

def Ry(deg):                            # rotation about the vertical (y) axis
    c, s = np.cos(np.radians(deg)), np.sin(np.radians(deg))
    return np.array([[c, 0, s, 0], [0, 1, 0, 0], [-s, 0, c, 0], [0, 0, 0, 1]])

P = np.array([[FOCAL, 0, 0, 0],         # perspective: w' = z, then divide
              [0, FOCAL, 0, 0],
              [0, 0, 1, 0],
              [0, 0, 1, 0]])
if ORTHO:
    P = np.diag([FOCAL / DIST, FOCAL / DIST, 1.0, 1.0])

view = T(0, 0, DIST + 1)                # camera at origin looking down +z
cubes = {"near cube": T(0, 0, 0) @ Ry(YAW), "far cube": T(2.5, 0, 8) @ Ry(YAW)}
fig, ax = plt.subplots(figsize=(6, 4))
for (name, model), c in zip(cubes.items(), ["tab:blue", "tab:red"]):
    clip = (P @ view @ model @ V.T).T   # one matrix product carries every vertex
    s = clip[:, :2] / clip[:, 3:4]      # the perspective divide
    for a, b in E:
        ax.plot(s[[a, b], 0], s[[a, b], 1], color=c)
    ax.plot([], [], color=c, label=name)
ax.set_aspect("equal"); ax.set_xlim(-1.5, 1.5); ax.set_ylim(-1, 1)
ax.legend(); ax.set_title(f"DIST={DIST}, FOCAL={FOCAL:.2f}, ortho={ORTHO}")
plt.show()
```

Things to try:

- **The dolly zoom.** Because `FOCAL` is tied to `DIST`, the near cube's front face always has half-height 0.5 on screen. Set `DIST = 2.0`: the far cube's front face shrinks to half-height 0.1. Set `DIST = 20.0`: it grows to about 0.36. The subject stays put while the background breathes. Hitchcock used this in *Vertigo* (1958) to make a staircase lurch.
- **Swap the order.** Replace `T(2.5, 0, 8) @ Ry(YAW)` by `Ry(YAW) @ T(2.5, 0, 8)` and vary `YAW`. The far cube now swings around the near cube (the world origin) instead of turning in place: the solar-system bug, reproduced.
- **Go orthographic.** Set `ORTHO = True`. Parallel edges stay parallel and the two cubes are the same size: an engineering drawing, not a photograph.
- **Walk through the eye.** Set `DIST = -1.0` and `FOCAL = 1.0`, so the near cube straddles the camera. Vertices behind the eye get negative w and the drawing explodes. That is the clipping problem of §69.4, in your hands.

69.6 The challenge

1. **Intuition.** Railway tracks are parallel in the world, yet in a photograph they meet at a *vanishing point*. Using Result 3 and the fact that all lines with direction $\mathbf{d}$ share the homogeneous "point" $(\mathbf{d}, 0)$, explain where the vanishing point is on the screen. Why do lines parallel to the image plane have no vanishing point? Where do all the vanishing points of horizontal lines lie?
2. **Derivation.** Show that $1/z_c$ is an *affine* function of the screen coordinates (x_s, y_s) across the image of any planar triangle (start from the plane equation $\alpha x_c + \beta y_c + \gamma z_c = \delta$). Deduce that a texture coordinate q is interpolated correctly on the screen by interpolating q/z_c and $1/z_c$ linearly and dividing.
3. **Implementation.** Extend the experiment script into a tiny **rasteriser**: fill each face of the cube with a flat colour, resolve visibility with a per-pixel depth buffer on a 160×120 grid instead of drawing faces in order, and compute flat shading from face normals transformed by the inverse transpose. Add a non-uniform scale to the model matrix and confirm that the wrong (non-inverse-transpose) normals visibly break the shading.
4. **Open investigation.** Real lenses are not pinholes. Wide-angle lenses bend straight lines near the edges (barrel distortion), often modelled by $r_d = r(1 + k_1 r^2 + k_2 r^4)$ in screen radius r. Distort your rasteriser's output with this model and then try to *undo* it from a photo of a checkerboard, estimating k_1, k_2 by least squares. How many squares do you need? How does the error behave near the corners? Camera calibration is a field in its own right; see how far you get with only this chapter.

69.7 Loose threads

The programmer's Earth now spins in place, because she knows that matrices act right to left and that the rightmost one acts in the object's own frame. The GPU's impossible trick has a name: homogeneous coordinates turn translation into a 4-D shear and projection into "copy depth into w, then divide". Three theorems carry the load: affine maps preserve affine combinations, normals transform by the inverse transpose, and perspective keeps lines straight while distorting distances along them.

What we did not tell you:

- **Rotation is the weak link.** We stored orientation as nine numbers in a matrix, though a rotation has only three degrees of freedom. Nine numbers drift, cannot be averaged, and give no sensible way to turn smoothly from one orientation to another.

> **? Halfway between two orientations**
>
> A camera faces north; a second later it must face straight up and slightly east. Average the two rotation matrices entrywise and you do not get a rotation at all: the result shrinks and shears the world. What *is* halfway between two orientations, and why does the honest answer need a four-dimensional number system invented on a Dublin bridge in 1843? That is Ch. 70.

- **Points at infinity.** The homogeneous points with $w = 0$ behave like genuine points: parallel lines meet there, and a projective transformation can move them into view as vanishing points. Adding them turns 3-D space into *projective space*, where every two distinct lines in a plane meet in exactly one point.

> **? A world with one side**
>
> Collapse each line through the origin of $\mathbb{R}^3$ to a single point and you get the *real projective plane*: the world of homogeneous coordinates (x, y, w) up to scale. It is a closed surface that cannot be built in ordinary 3-D space without crossing itself, and it has only one side. Why, and what does that mean for a creature living on it? The answer is in Ch. 58.

- **The inverse problem.** Projection throws away depth. Recovering it from two or more photographs (stereo, structure from motion) is an inverse problem with its own geometry, the *epipolar* constraint, and its own instabilities; see Ch. 91.
- **The other way to draw.** The pipeline here pushes triangles toward the eye. The opposite approach pulls light backward from the eye into the scene, one ray per pixel, and gets shadows, mirrors and glass almost for free: Ch. 71. Clipping, culling and finding what a ray hits quickly are algorithms on shapes, in Ch. 75.

Further reading

- J. F. Hughes, A. van Dam, M. McGuire, D. F. Sklar, J. D. Foley, S. K. Feiner, K. Akeley, *Computer Graphics: Principles and Practice*, 3rd ed., Addison-Wesley, 2013. Chapters on transformations and the camera, with the point/vector distinction treated carefully.
- S. Marschner, P. Shirley, *Fundamentals of Computer Graphics*, 5th ed., CRC Press, 2021. Compact derivations of the viewing pipeline and perspective-correct interpolation.
- E. Lengyel, *Mathematics for 3D Game Programming and Computer Graphics*, 3rd ed., Cengage, 2011. Homogeneous coordinates, normals and depth precision from a game engine's point of view.
- R. Hartley, A. Zisserman, *Multiple View Geometry in Computer Vision*, 2nd ed., Cambridge University Press, 2004. The pinhole camera, projective geometry and calibration, taken all the way.

Chapter 70 — Quaternions and the Geometry of Rotation

> **Level 2 · Track G — Mathematics for Computer Graphics, Games, and Simulation Builds on:** Ch. 2, Ch. 9, Ch. 10, Ch. 69 · **Leads to:** Ch. 71, Ch. 72 · **Related:** Ch. 56, Ch. 58, Ch. 59 **Reading time:** ~25 min

70.1 The hook

July 1969. Apollo 11 is on its way to the Moon, and the spacecraft's sense of direction lives in a gyroscope-stabilised platform held inside three nested rings, or **gimbals**. Each gimbal lets the spacecraft turn about one axis while the platform stays fixed relative to the stars. Three rings, three angles: yaw, pitch, roll. It sounds complete.

It is not. If the spacecraft turns so that the middle ring is at 90°, the outer and inner rings line up, and their two axes become the same axis. The platform has lost one of its three directions of freedom. A turn in the missing direction now drags the platform with it, and the guidance computer's picture of "which way is up" is gone, to be rebuilt by hand from star sightings. This is **gimbal lock**. The crew were trained to steer clear of it, and the command module pilot, Michael Collins, is often quoted as asking mission control, half joking, for a fourth gimbal for Christmas.

Fifty years later every 3-D game, drone, phone and VR headset faces the same problem in software. Store an orientation as three angles, and there is some orientation where the angles jam. Store it as a 3×3 rotation matrix, as in Ch. 69, and you carry nine numbers for three degrees of freedom; they drift in floating point, and there is no honest way to say what orientation lies *halfway* between two others. A camera that must swing smoothly from "facing north" to "facing straight up" has no good recipe.

The cure was found 126 years before Apollo, on 16 October 1843, by a man walking along a canal in Dublin. William Rowan Hamilton had spent years trying to multiply triples of numbers the way complex numbers multiply pairs. In a flash he saw that triples would never work, but *quadruples* would, and he carved the rule into the stone of Broom Bridge:

$$\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = \mathbf{ijk} = -1.$$

Why should rotating a 3-D world need a 4-D number? And why does the answer involve *half* the rotation angle?

70.2 The big idea

Start with something you already know. In the plane, multiplying by the unit complex number $e^{j\theta}$ rotates every point by θ (Ch. 2 §2.7). Rotations of the plane *are* unit complex numbers; composing rotations is multiplying them; and the halfway point between two orientations is just the point halfway along the arc of the unit circle. One number, no jamming, perfect interpolation.

Quaternions do the same for space, with two twists.

Rotations have axes. Whatever tangle of turns you apply to a rigid body, the net result is a single turn by some angle θ about some fixed axis $\hat{\mathbf{n}}$ (Euler's rotation theorem, proved below). So an orientation is really "an axis and an angle": four numbers with one constraint, which is exactly what a unit quaternion is. The unit quaternions form a 3-D sphere sitting in 4-D space, and **every point on that sphere is an orientation**. There are no poles, no special angles, no place where the coordinates jam.

The angle is halved, and every rotation appears twice. A unit quaternion q does not rotate a vector by multiplying it once; it *sandwiches* it, $\mathbf{v} \mapsto q\mathbf{v}q^{*}$. Each factor contributes half the turn, so q must carry $\theta/2$. The price is that q and $-q$ give the same rotation: the sphere of quaternions wraps twice around the space of rotations.

Interpolation then becomes geometry on that sphere. To swing from orientation q_0 to q_1, walk along the great circle between them at constant speed. That walk is **SLERP**, and it is the reason a game camera can glide where Euler angles would lurch.

The analogy: Euler angles are like describing a location on Earth by latitude and longitude. They work almost everywhere, but at the North Pole longitude is meaningless and a small step can change it by 180°. Quaternions are like describing the location by a point in 3-D space on the globe's surface: no special places at all.

70.3 The mathematics

Rotations and their axes

A **rotation** of space is a 3×3 matrix $\mathbf{R}$ that preserves lengths and handedness: $\mathbf{R}^{\mathsf{T}}\mathbf{R} = \mathbf{I}$ and $\det \mathbf{R} = 1$ (orthogonal matrices are Ch. 9 §9.7). The set of all of them is called $SO(3)$, the *special orthogonal group*. It is a group: products and inverses of rotations are rotations, and $\mathbf{R}^{-1} = \mathbf{R}^{\mathsf{T}}$. It is not commutative: tip a book forward and then turn it left, versus left then forward, and the book ends up in two different poses.

> **Result 1: Euler's rotation theorem**
>
> Every rotation $\mathbf{R} \in SO(3)$ fixes some unit vector $\hat{\mathbf{n}}$: $\mathbf{R}\hat{\mathbf{n}} = \hat{\mathbf{n}}$. So every rotation is a turn about an axis.

Proof. We need 1 to be an eigenvalue of $\mathbf{R}$ (Ch. 10 §10.2), that is, $\det(\mathbf{R} - \mathbf{I}) = 0$. Using $\mathbf{R}\mathbf{R}^{\mathsf{T}} = \mathbf{I}$, $\det \mathbf{R} = 1$ and $\det \mathbf{M}^{\mathsf{T}} = \det \mathbf{M}$:

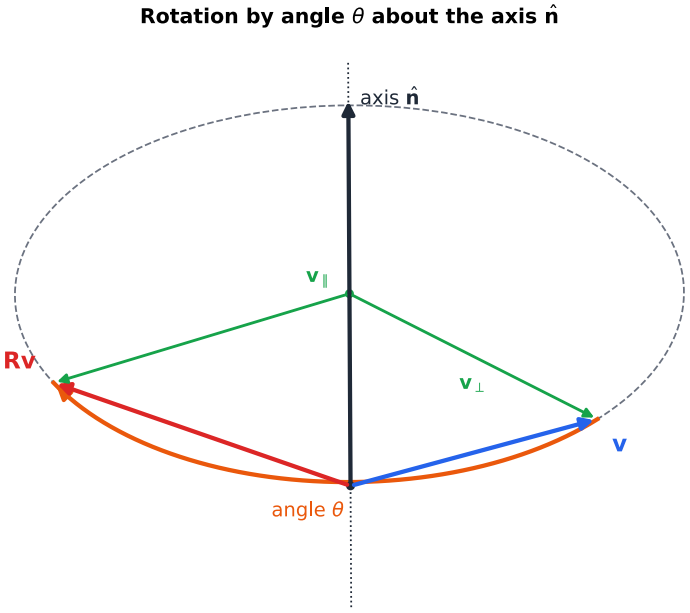
$$\det(\mathbf{R} - \mathbf{I}) = \det\left(\mathbf{R}(\mathbf{I} - \mathbf{R}^{\mathsf{T}})\right) = \det \mathbf{R}\, \det\left((\mathbf{I} - \mathbf{R})^{\mathsf{T}}\right) = \det(\mathbf{I} - \mathbf{R}) = (-1)^3 \det(\mathbf{R} - \mathbf{I}).$$

A number equal to its own negative is zero. The factor $(-1)^3$ is where three dimensions matter: in the plane, $(-1)^2 = 1$ and the argument (rightly) fails, since a planar rotation fixes no direction. □

The axis $\hat{\mathbf{n}}$ is the eigenvector for eigenvalue 1. The other two eigenvalues are $e^{\pm j\theta}$, and θ is the angle of turn; since the trace is the sum of the eigenvalues, $\operatorname{tr}\mathbf{R} = 1 + 2\cos\theta$. Conversely, the rotation by θ about $\hat{\mathbf{n}}$ is given by **Rodrigues' formula**,

$$\mathbf{R}\mathbf{v} = \mathbf{v}\cos\theta + (\hat{\mathbf{n}} \times \mathbf{v})\sin\theta + \hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \mathbf{v})(1 - \cos\theta),$$

which you can read straight off the figure: split $\mathbf{v}$ into its part along the axis, which does not move, and its part perpendicular to the axis, which turns by θ in the plane spanned by $\mathbf{v}_\perp$ and $\hat{\mathbf{n}} \times \mathbf{v}$.



Rotation about an axis. The component $\mathbf{v}_\parallel$ along $\hat{\mathbf{n}}$ stays put; the component $\mathbf{v}_\perp$ swings by θ around the circle. Every rotation of space looks like this, for some axis (Result 1).

Euler angles and gimbal lock

The most popular way to write a rotation is as three turns about fixed coordinate axes, for example

$$\mathbf{R}(\alpha, \beta, \gamma) = \mathbf{R}_z(\alpha)\,\mathbf{R}_y(\beta)\,\mathbf{R}_x(\gamma),$$

with yaw α, pitch β and roll γ. Three numbers that a human can read: an excellent user interface. As a data structure, it has a hole.

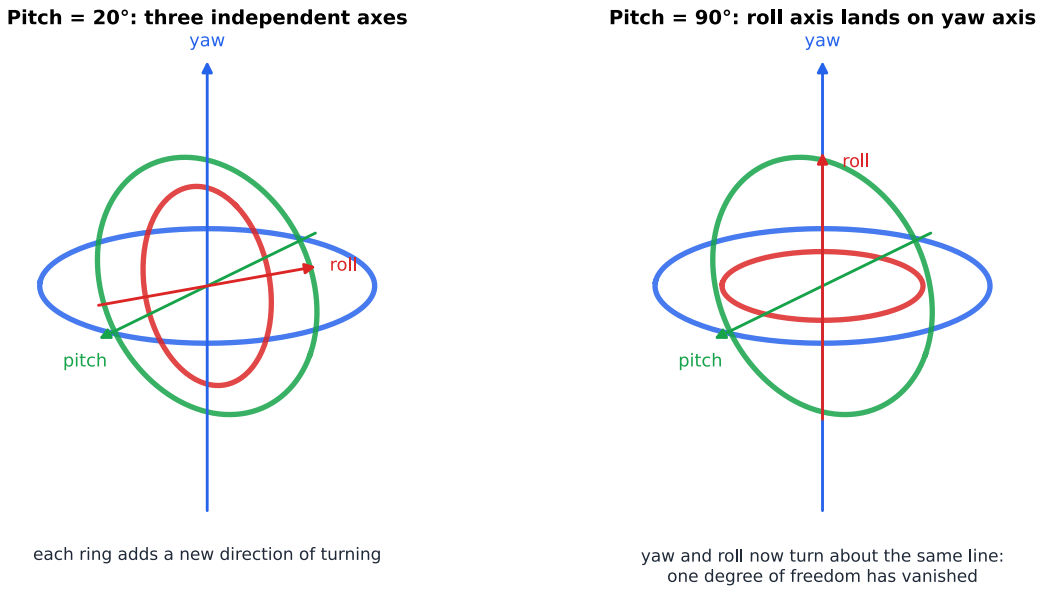
Result 2: gimbal lock

At pitch $\beta = 90°$, the rotation depends only on $\alpha - \gamma$:

$$\mathbf{R}_z(\alpha)\,\mathbf{R}_y(90°)\,\mathbf{R}_x(\gamma) = \mathbf{R}_z(\alpha - \gamma)\,\mathbf{R}_y(90°).$$

Proof. For any rotation $\mathbf{Q}$, the conjugate $\mathbf{Q}\mathbf{R}_{\hat{\mathbf{a}}}(\gamma)\mathbf{Q}^{-1}$ is the rotation by γ about the turned axis $\mathbf{Q}\hat{\mathbf{a}}$ (turn the axis into place, rotate, turn back). Take $\mathbf{Q} = \mathbf{R}_y(90°)$, which sends $\hat{\mathbf{x}}$ to $-\hat{\mathbf{z}}$. Then $\mathbf{R}_y(90°)\mathbf{R}_x(\gamma)\mathbf{R}_y(90°)^{-1}$ is a turn by γ about $-\hat{\mathbf{z}}$, which is $\mathbf{R}_z(-\gamma)$. Hence $\mathbf{R}_y(90°)\mathbf{R}_x(\gamma) = \mathbf{R}_z(-\gamma)\mathbf{R}_y(90°)$, and multiplying on the left by $\mathbf{R}_z(\alpha)$ gives the claim. □

So at that pitch, changing yaw and changing roll do *the same thing*. The map from three angles to rotations, which is smooth and three-to-three almost everywhere, collapses to two dimensions there: its Jacobian (Ch. 6) drops rank. In a mechanical gimbal, that is the moment two rings become coplanar. In software, it shows up as a camera that suddenly spins wildly when it tries to follow a target passing overhead: a small change of orientation demands a huge change of yaw and roll.



Gimbal lock. Each ring turns about its own axis. At 90° of pitch the roll axis lands on the yaw axis, so two rings produce the same motion and no ring can produce the third.

The hole is not a flaw of this particular choice. Any description of all rotations by three smooth angles has such a place somewhere; the reason is topological (see §70.7). The escape is to use more numbers, but cleverly.

Quaternion algebra

Quaternions

A **quaternion** is $q = w + x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ with real w, x, y, z, multiplied using Hamilton's rule $\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = \mathbf{ijk} = -1$, which implies $\mathbf{ij} = \mathbf{k} = -\mathbf{ji}$, $\mathbf{jk} = \mathbf{i} = -\mathbf{kj}$, $\mathbf{ki} = \mathbf{j} = -\mathbf{ik}$. Write $q = (w, \mathbf{v})$ with scalar part w and vector part $\mathbf{v} = (x, y, z)$. The **conjugate** is $q^* = (w, -\mathbf{v})$ and the **norm** is $\|q\| = \sqrt{w^2 + x^2 + y^2 + z^2}$.

(The book's imaginary unit is j; the quaternion units are written bold, $\mathbf{i}, \mathbf{j}, \mathbf{k}$, and each of them squares to -1 like j does. The complex numbers sit inside the quaternions as $w + x\mathbf{i}$.)

Multiplying out with the table and collecting terms gives the whole algebra in one line:

$$(w_1, \mathbf{v}_1)(w_2, \mathbf{v}_2) = \big(w_1 w_2 - \mathbf{v}_1 \cdot \mathbf{v}_2,\ \ w_1\mathbf{v}_2 + w_2\mathbf{v}_1 + \mathbf{v}_1 \times \mathbf{v}_2\big).$$

Both 3-D products appear, the dot and the cross. Hamilton's algebra is in fact where they came from: physicists later split quaternion multiplication into these two pieces and called the result vector analysis. The cross product is why quaternions do not commute, matching the fact that rotations do not.

Three facts follow quickly. $qq^* = (\|q\|^2, \mathbf{0})$, so every non-zero quaternion has an inverse $q^{-1} = q^*/\|q\|^2$. Conjugation reverses products, $(pq)^* = q^*p^*$. And the norm is multiplicative, $\|pq\| = \|p\|\,\|q\|$, since $\|pq\|^2 = pq(pq)^* = p\,qq^*\,p^* = \|q\|^2\,pp^*$. So **products of unit quaternions are unit quaternions**: the unit sphere S^3 in 4-D space is closed under multiplication, just as the unit circle is for complex numbers.

The sandwich

Identify a 3-D vector $\mathbf{v}$ with the "pure" quaternion $(0, \mathbf{v})$.

Result 3: unit quaternions rotate by sandwiching

Let $q = \big(\cos\frac{\theta}{2},\ \sin\frac{\theta}{2}\,\hat{\mathbf{n}}\big)$ with $\hat{\mathbf{n}}$ a unit vector. Then $q\,\mathbf{v}\,q^*$ is a pure quaternion, and it equals the rotation of $\mathbf{v}$ by angle θ about $\hat{\mathbf{n}}$.

Proof. Write $c = \cos\frac{\theta}{2}$, $s = \sin\frac{\theta}{2}$, and split $\mathbf{v} = \mathbf{v}_\parallel + \mathbf{v}_\perp$ along and across $\hat{\mathbf{n}}$. The map $\mathbf{v} \mapsto q\mathbf{v}q^*$ is linear, so treat the two parts separately.

Parallel part. $\mathbf{v}_\parallel = a\hat{\mathbf{n}}$ is a multiple of q's own vector part, so it commutes with q (the cross product term vanishes). Hence $q\mathbf{v}_\parallel q^* = \mathbf{v}_\parallel qq^* = \mathbf{v}_\parallel$: the axis does not move.

Perpendicular part. With $\hat{\mathbf{n}} \cdot \mathbf{v}_\perp = 0$, the product rule gives

$$q\,\mathbf{v}_\perp = (c, s\hat{\mathbf{n}})(0, \mathbf{v}_\perp) = \big(-s\,\hat{\mathbf{n}} \cdot \mathbf{v}_\perp,\ c\,\mathbf{v}_\perp + s\,\hat{\mathbf{n}} \times \mathbf{v}_\perp\big) = \big(0,\ c\,\mathbf{v}_\perp + s\,\hat{\mathbf{n}} \times \mathbf{v}_\perp\big),$$

and the same computation shows $\mathbf{v}_\perp q^* = (0,\ c\,\mathbf{v}_\perp + s\,\hat{\mathbf{n}} \times \mathbf{v}_\perp) = q\mathbf{v}_\perp$. So $q\mathbf{v}_\perp q^* = q\,(q\mathbf{v}_\perp) = q^2\mathbf{v}_\perp$. Now $q^2 = (c^2 - s^2,\ 2cs\,\hat{\mathbf{n}}) = (\cos\theta,\ \sin\theta\,\hat{\mathbf{n}})$ by the double-angle formulas, so applying the first computation again with θ in place of $\theta/2$:

$$q\,\mathbf{v}_\perp\,q^* = \big(0,\ \mathbf{v}_\perp\cos\theta + (\hat{\mathbf{n}} \times \mathbf{v}_\perp)\sin\theta\big).$$

That is $\mathbf{v}_\perp$ turned by θ in its plane. Adding the unmoved $\mathbf{v}_\parallel$ gives exactly Rodrigues' formula. □

Now the half angle is no mystery: one factor q on each side, each turning the perpendicular part by $\theta/2$. A check: rotate $\mathbf{v} = \hat{\mathbf{x}}$ by $90°$ about $\hat{\mathbf{z}}$. Here $q = (\frac{1}{\sqrt{2}}, 0, 0, \frac{1}{\sqrt{2}})$, and two applications of the product rule give $q\hat{\mathbf{x}}q^* = (0, 0, 1, 0)$, the vector $\hat{\mathbf{y}}$. Correct.

Because the sandwich respects products, $(pq)\mathbf{v}(pq)^* = p\,(q\mathbf{v}q^*)\,p^*$, so **composing rotations is multiplying quaternions**, with the rightmost acting first as for matrices. A turn of 90° about $\hat{\mathbf{x}}$ followed by 90° about $\hat{\mathbf{y}}$ is

$$q_y\,q_x = \left(\tfrac{1}{\sqrt{2}},0,\tfrac{1}{\sqrt{2}},0\right)\left(\tfrac{1}{\sqrt{2}},\tfrac{1}{\sqrt{2}},0,0\right) = \left(\tfrac{1}{2},\tfrac{1}{2},\tfrac{1}{2},-\tfrac{1}{2}\right),$$

whose scalar part $\cos\frac{\theta}{2} = \frac{1}{2}$ says $\theta = 120^\circ$, about the axis $(1,1,-1)/\sqrt{3}$. Two quarter-turns about perpendicular axes make a third of a turn about a diagonal: try it with a book.

Finally, since $(-q)\mathbf{v}(-q)^* = q\mathbf{v}q^*$, the antipodal points q and $-q$ on S^3 are the same rotation. Unit quaternions **double cover** $SO(3)$. Rotating by 360° about any axis gives $q = (\cos 180^\circ, \mathbf{0}) = -1$: the identity rotation, but not the identity quaternion. You must turn by 720° to get back to $q = 1$.

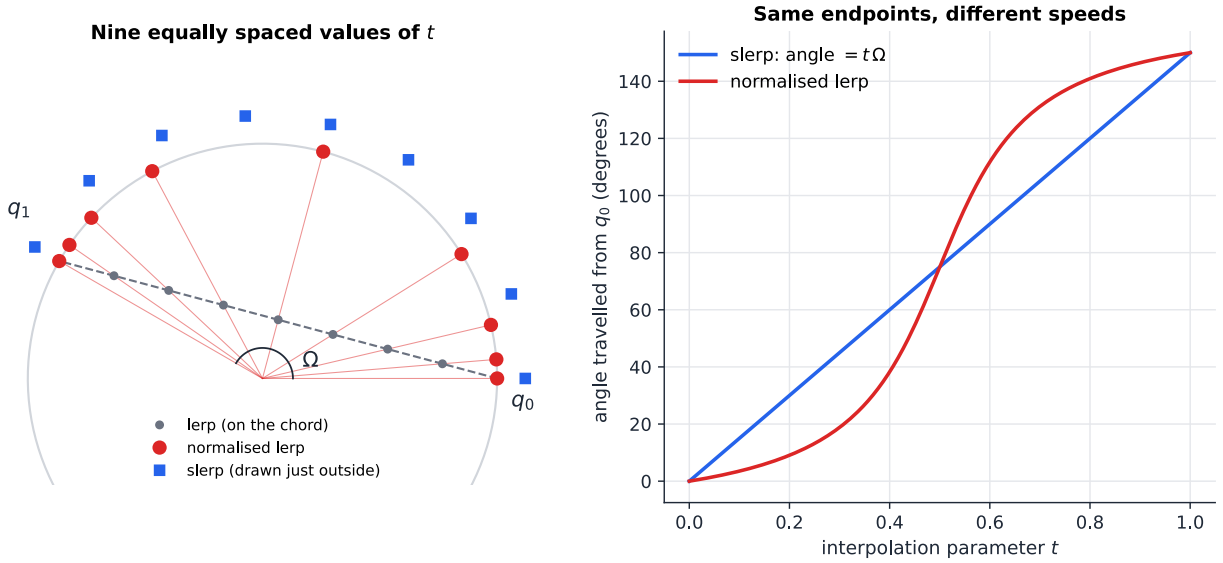
Interpolation: SLERP

Given orientations q_0, q_1 (unit quaternions), what is "the orientation a fraction t of the way from q_0 to q_1"? The natural answer is the point a fraction t of the way along the shortest great-circle arc between them on S^3. Let $\cos\Omega = q_0 \cdot q_1$ (the 4-D dot product). The arc lies in the plane spanned by q_0 and q_1; in that plane, take the unit vector $q_\perp = (q_1 - q_0\cos\Omega)/\sin\Omega$ orthogonal to q_0, so the arc is $q_0\cos(t\Omega) + q_\perp\sin(t\Omega)$. Substituting and using $\sin\Omega\cos t\Omega - \cos\Omega\sin t\Omega = \sin\big((1-t)\Omega\big)$ gives

$$\operatorname{slerp}(q_0,q_1;t) = \frac{\sin\big((1-t)\Omega\big)}{\sin\Omega}\,q_0 + \frac{\sin(t\Omega)}{\sin\Omega}\,q_1.$$

It moves at constant angular speed, and the physical rotation turns at the constant rate 2Ω per unit t about a fixed axis. One detail matters: q_1 and $-q_1$ are the same orientation but different points, and the arc to one is longer than to the other. If $q_0 \cdot q_1 < 0$, replace q_1 by $-q_1$ first, or the camera goes "the long way round".

The cheap alternative, **normalised lerp**, draws the straight chord $(1-t)q_0 + tq_1$ and pushes each point back onto the sphere. It follows the same path but at the wrong speed: slow at the ends, fast in the middle.



SLERP against normalised lerp, on a circle (the 2-D slice of S^3 containing q_0 and q_1), with the angle exaggerated to 150°. Evenly spaced points on the chord project to unevenly spaced points on the arc; at this angle the projected point moves at 2.85 times the slerp speed in the middle and 0.19 times at the ends.

70.4 The algorithm

The core toolkit fits on a page.

```
qmul((w1,v1),(w2,v2)) = (w1*w2 − v1.v2,  w1*v2 + w2*v1 + v1 x v2)   # 16 mul, 12 add
from_axis_angle(n, th) = (cos(th/2), sin(th/2) * n)                  # n must be unit
rotate(q, v)          = vector part of qmul(qmul(q, (0,v)), conj(q))
to_matrix(q)          = 3x3 matrix with entries 1−2(y^2+z^2), 2(xy−wz), ...   # for bulk use
slerp(q0, q1, t):
    d = dot(q0, q1)
    if d < 0: q1, d = −q1, −d            # take the short way round
    if d > 0.9995: return normalize((1−t)*q0 + t*q1)   # nearly equal: avoid 0/0
    W = acos(d)
    return (sin((1−t)*W)*q0 + sin(t*W)*q1) / sin(W)
```

Cost. Composing two rotations costs 16 multiplications as quaternions against 27 as matrices, and a quaternion is 4 numbers against 9. But rotating *one vector* by the sandwich costs more than a matrix–vector product (9 multiplies). So engines store and compose orientations as quaternions, then convert once to a matrix per object per frame before transforming thousands of vertices, exactly as in the pipeline of Ch. 69 §69.4.

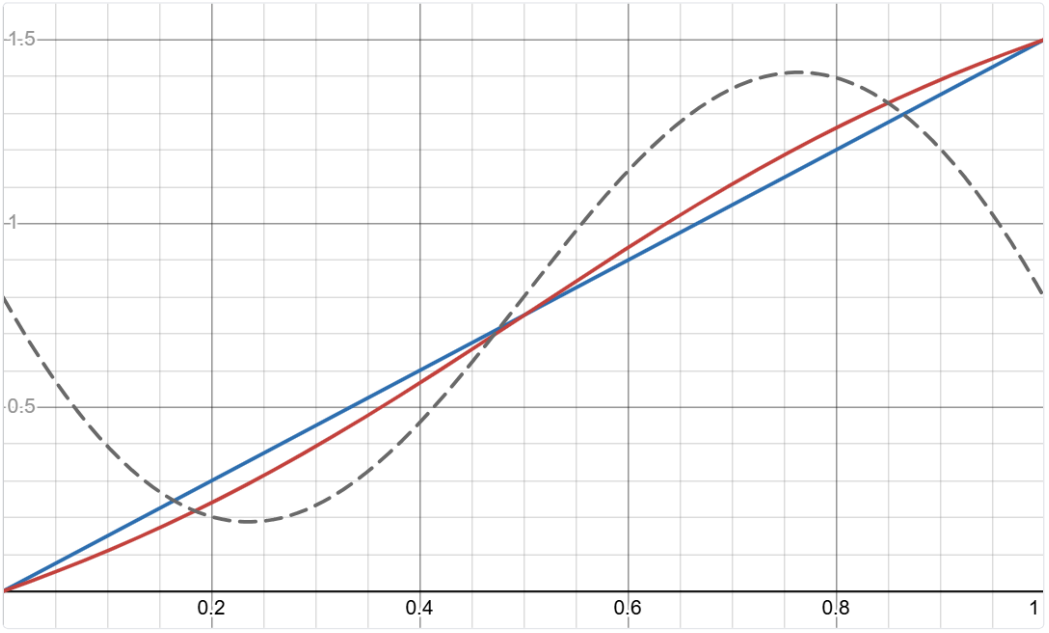
Rigid bodies. A full pose is a rotation and a position, $\mathbf{x} \mapsto q\mathbf{x}q^* + \mathbf{t}$. Poses compose like the block matrices of Ch. 69: $(q_2,\mathbf{t}_2)\circ(q_1,\mathbf{t}_1) = (q_2q_1,\ q_2\mathbf{t}_1q_2^* + \mathbf{t}_2)$, where again the second pose rotates the first one's translation. Skeletal animation stores each bone as such a pair relative to its parent and multiplies down the chain from hip to fingertip. (Blending poses by blending the pairs separately makes twisting joints collapse like a candy wrapper, which is why animators use *dual quaternions*, a variant that interpolates rotation and translation together.)

Failure cases.

- *Drift.* Repeated multiplication in floating point slowly moves q off the unit sphere, and a non-unit q also scales whatever it rotates. The fix is one line, $q \leftarrow q/\|q\|$. A drifting matrix needs a full Gram–Schmidt re-orthonormalisation instead, and you must decide which column to trust.
- *Near-equal endpoints.* In SLERP, $\sin\Omega \to 0$ as $q_0 \to q_1$; switch to normalised lerp, which is then accurate.
- *Sign flips.* Data from sensors or animation files may jump between q and $-q$ from frame to frame. Both are right, but averaging or filtering them naively gives nonsense near zero. Always align signs (by the dot product test) before blending.
- *Conventions.* Some libraries store (x, y, z, w), some (w, x, y, z); some multiply in the opposite order. Most quaternion bugs are convention bugs.

70.5 The experiment

First, see how uneven normalised lerp is. For two quaternions at angle Ω apart, the angle reached by normalised lerp at parameter t (here x) is $\arctan\bigl(t\sin\Omega/(1-t+t\cos\Omega)\bigr)$; slerp reaches $t\Omega$. The graph uses $\Omega = 1.5$ rad, close to the largest angle you meet after the sign test (a rotation of about $172°$).



Angle travelled by slerp (blue) and normalised lerp (red) for $\Omega = 1.5$; the dashed curve is their difference magnified ten times (centred at 0.8). Replace every `1.5` by `0.5`: the curves nearly coincide, which is why games often get away with lerp for small steps. Keep Ω below about 1.57 so the denominator stays positive.

At $\Omega = 1.5$ the red curve starts at 0.665 times the slerp speed and reaches 1.24 times it in the middle; the largest angle error is about $3.5°$, visible as a stutter in a slow camera move.

Second, watch numbers drift. The script below spins a body at constant angular velocity for 10^5 small steps, once by multiplying rotation matrices and once by multiplying quaternions, and measures how far each has wandered from being a true rotation.

```
import numpy as np
import matplotlib.pyplot as plt

DTYPE = np.float32               # KNOB: float32 (a game) or float64 (a simulator)
STEPS = 100_000                  # KNOB: number of frames
OMEGA = np.array([0.3, 1.0, 0.2])  # KNOB: angular velocity (rad/s)
DT = 1e-3                        # KNOB: time step (s)
RENORM_EVERY = 0                 # KNOB: renormalise the quaternion every N steps (0 = never)

def qmul(a, b):                  # Hamilton product, (w, x, y, z)
    w1, v1, w2, v2 = a[0], a[1:], b[0], b[1:]
    return np.r_[w1 * w2 - v1 @ v2, w1 * v2 + w2 * v1 + np.cross(v1, v2)].astype(DTYPE)

def qmat(q):                     # unit quaternion -> rotation matrix
    w, x, y, z = q
    return np.array([[1 - 2*(y*y + z*z), 2*(x*y - w*z), 2*(x*z + w*y)],
                     [2*(x*y + w*z), 1 - 2*(x*x + z*z), 2*(y*z - w*x)],
                     [2*(x*z - w*y), 2*(y*z + w*x), 1 - 2*(x*x + y*y)]], dtype=DTYPE)

th = np.linalg.norm(OMEGA) * DT  # one small step: angle th about OMEGA's direction
dq = np.r_[np.cos(th / 2), np.sin(th / 2) * OMEGA / np.linalg.norm(OMEGA)].astype(DTYPE)
dR = qmat(dq)
q, R = np.array([1, 0, 0, 0], DTYPE), np.eye(3, dtype=DTYPE)
log_t, err_R, err_q = [], [], []
for k in range(1, STEPS + 1):
    R = (R @ dR).astype(DTYPE)   # matrix: 27 multiplies per step
    q = qmul(q, dq)              # quaternion: 16 multiplies per step
    if RENORM_EVERY and k % RENORM_EVERY == 0:
        q = (q / np.linalg.norm(q)).astype(DTYPE)
    if k % 500 == 0:
        log_t.append(k * DT)
        err_R.append(np.abs(R.T @ R - np.eye(3)).max())     # how far from a rotation
        err_q.append(abs(float(np.linalg.norm(q)) - 1.0))  # how far from unit length
print(f"after {STEPS} steps: matrix error {err_R[-1]:.2e}, quaternion error {err_q[-1]:.2e}")
plt.semilogy(log_t, err_R, label="matrix: max |R^T R - I|")
plt.semilogy(log_t, err_q, label="quaternion: | |q| - 1 |")
plt.xlabel("simulated time (s)"); plt.legend(); plt.show()
```

With the defaults (single precision, no renormalisation) both drift by about 10^{-3} after 10^5 steps: the matrix error is 8.6×10^{-4} and the quaternion error 1.8×10^{-3}. Quaternions are *not* magically immune to rounding. The difference is what the repair costs. Now:

- Set `RENORM_EVERY = 100`. One square root and four divisions every hundred frames, and the quaternion error is gone, while the matrix keeps shearing.
- Set `DTYPE = np.float64`. Both errors drop to about 10^{-11}. Simulators can afford this; GPUs and phone sensors often cannot.
- Raise `DT` to `0.05` and compare the final orientations of the two methods. They agree with each other, because each step is an exact rotation. But is each step the *right* rotation when $\boldsymbol{\omega}$ itself changes over time? That question belongs to physics engines.

70.6 The challenge

1. **Intuition.** Hold a cup of water flat on your palm and rotate your hand a full 360° about the vertical, keeping the cup level (your arm will end up twisted). Keep going another 360° in the same direction. What happens to your arm? Relate this to the fact that the rotation by 360° is the quaternion -1, not $+1$.
2. **Derivation.** Show that $\operatorname{tr}\mathbf{R} = 1 + 2\cos\theta$ for the rotation matrix built from $q = (\cos\frac{\theta}{2}, \sin\frac{\theta}{2}\hat{\mathbf{n}})$, and use it to derive a formula recovering q from $\mathbf{R}$ when $w = \cos\frac{\theta}{2}$ is not small. Which entries of $\mathbf{R}$ should you use instead when θ is close to 180°, and why does the naive formula then lose accuracy?
3. **Implementation.** Write a function that converts Euler angles (α, β, γ) to a quaternion and back. Feed it a camera that tracks an aircraft flying straight over your head, and plot the recovered yaw and roll against time. Watch them blow up near pitch 90°, then animate the same flight with SLERP between keyframe quaternions and confirm it stays smooth.
4. **Open investigation.** You have N noisy measurements of the same orientation, as unit quaternions (some with flipped signs). What is their "average"? Try: (a) align signs and normalise the plain sum; (b) take the eigenvector for the largest eigenvalue of $\sum_i q_i q_i^{\mathsf{T}}$ (a 4×4 matrix); (c) iterate "average in the tangent space, then step along the sphere". Compare them as the noise grows from 1° to 90°. When do they disagree, and which one would you trust in a drone?

70.7 Loose threads

Apollo's problem has an answer. Three angles always have a place where they jam (Result 2 shows where for one convention). Rotation matrices never jam but carry nine numbers. Unit quaternions carry four, compose with 16 multiplications, repair their drift with one division, and turn interpolation into walking along a great circle. Rotations have axes (Result 1), and the half-angle is the signature of the sandwich (Result 3).

What we did not tell you:

> **? Why the sphere wraps twice**
>
> Every rotation appears twice on the quaternion sphere, as q and $-q$. That is not a defect of quaternions but a fact about rotations themselves: a full 360° turn traces a loop in $SO(3)$ that cannot be shrunk to nothing, yet going round it *twice* can be (the cup-and-arm trick of challenge 1 is a physical demonstration). This is also why no three smooth angles can describe every rotation without a jam. The tools that make "loops that cannot be shrunk" precise are in Ch. 58.

? Why four, and not three?

Hamilton spent a decade trying to multiply triples and failed. It was not lack of imagination: a number system in which you can multiply, divide and keep the norm multiplicative exists only in dimensions 1, 2, 4 and 8 (the reals, complexes, quaternions and the strange non-associative *octonions*). Why exactly those four dimensions? The algebraic side of the story starts in Ch. 56.

? Rotations that change in time

A spinning body has an angular velocity $\boldsymbol{\omega}(t)$, and its orientation obeys $\dot{q} = \frac{1}{2}(0, \boldsymbol{\omega})\, q$. Integrate that with a textbook method and q leaves the unit sphere; renormalise and you may still get the wrong spin, and a tumbling brick in a game can gain energy from nowhere. Which integrators respect the sphere and the physics is a question for Ch. 72.

- **Lie groups.** $SO(3)$ and S^3 are curved spaces that are also groups. The "tangent space average" of challenge 4 and the exponential map $\boldsymbol{\omega} \mapsto e^{(0,\boldsymbol{\omega})/2}$ are the first steps of Lie theory, the geometry of continuous symmetry; see Ch. 59.
- **Smooth paths through many keys.** SLERP joins two orientations, but a path through five keyframes made of four SLERPs has kinks at the joins. Spherical splines (such as SQUAD) smooth them; making "curvature" meaningful on a sphere is again differential geometry.
- **Light.** We have placed objects and cameras in the world and turned them smoothly. Now the world needs illumination: Ch. 71 follows rays of light, and rotations reappear every time a ray reflects.

Further reading

- J. B. Kuipers, *Quaternions and Rotation Sequences*, Princeton University Press, 1999. A patient, engineering-flavoured treatment, including aerospace sequences and gimbal lock.
- A. J. Hanson, *Visualizing Quaternions*, Morgan Kaufmann, 2006. The geometry of S^3 and the double cover, with many pictures.
- K. Shoemake, "Animating rotation with quaternion curves", *SIGGRAPH* 1985. The paper that brought SLERP to computer graphics.
- J. Stillwell, *Naive Lie Theory*, Springer, 2008. From quaternions and $SO(3)$ to Lie groups, using only undergraduate linear algebra.

Chapter 71 — Ray Tracing: The Geometry of Light

> **Level 2 · Track G — Mathematics for Computer Graphics, Games, and Simulation Builds on:** Ch. 7, Ch. 5, Ch. 19, Ch. 69 · **Leads to:** Ch. 72, Ch. 75, Ch. 76 · **Related:** Ch. 38, Ch. 61, Ch. 22 **Reading time:** ~25 min

71.1 The hook

In 1980 an engineer at Bell Labs named Turner Whitted published a picture of a few spheres floating over a checkerboard, one of them made of glass. The checkerboard is visible *through* the glass sphere, bent and flipped. Another sphere is a mirror, and the board, the sky and its neighbour appear on its surface, curved. Every shadow falls exactly where it should. The image is small and the colours are crude, yet people who saw it knew they were looking at something new: a computer had not been told what a reflection looks like; it had *worked it out*. The picture took well over an hour to compute.

Today a game console does the same kind of computation for two million pixels sixty times a second, and film studios spend thousands of processor-hours on a single frame. Behind both is one question, asked once for every pixel:

> *What light arrives at this point of the camera, from this direction?*

Answering it means following light backward from the eye into the world, finding the first thing it hits, asking what light reaches *that* point, which means following more rays, which hit more things... A mirror sends you on one more bounce. Glass splits your ray in two. A sheet of paper scatters light into every direction at once, so the "one more ray" becomes an integral over a whole hemisphere of directions, each of which needs its own answer.

How can a finite computer evaluate an infinite cascade of bounces? And why is it allowed to trace light backward, from the eye to the lamp, when light really travels the other way?

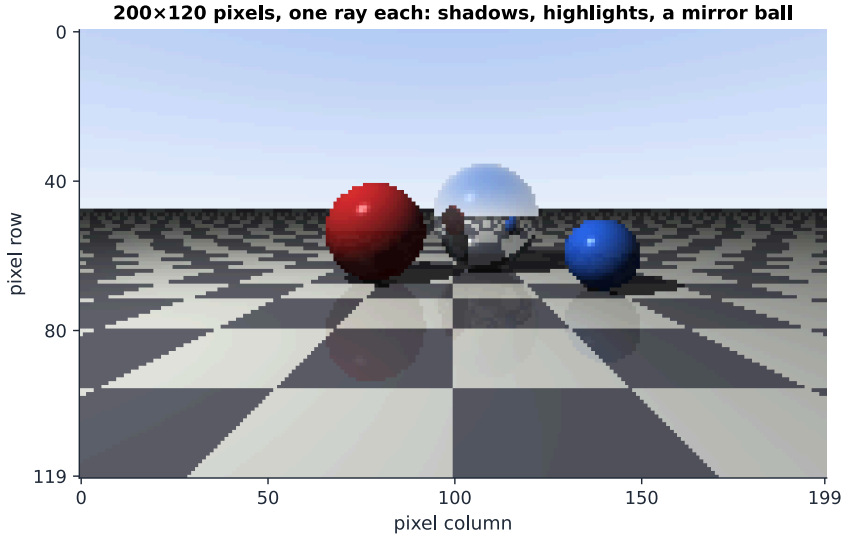
71.2 The big idea

A pixel is a question, a ray is how you ask it. From the camera's pinhole (Ch. 69 §69.3), draw the line through the centre of a pixel and out into the scene. Whatever that line hits first is what the pixel shows. Rasterisation, the method of Ch. 69, asks "where on the screen does this triangle land?"; ray tracing asks "what does this pixel see?". Same geometry, opposite loop order.

Light paths can be run in reverse. The laws of reflection and refraction are symmetric: if light can go from A to B by some path, it can go from B to A by the same path. So instead of firing trillions of photons from the lamp and waiting for the few that happen to land in the camera, we start at the camera and walk backward. When a reverse ray reaches a surface, we ask a second question: can this point see the light? Fire a **shadow ray** toward the lamp. If anything blocks it, the point is in shadow.

Every bounce is high-school geometry plus a rule. Hitting a sphere is solving a quadratic. Bouncing off a mirror is reflecting a vector across the normal. Entering glass is Snell's law. Each rule turns one ray into one or two new rays, and recursion does the rest.

Rough surfaces turn the recursion into an integral, and the integral into dice. A matte surface gathers light from all directions. The colour is an integral over the hemisphere above the point, and we estimate it the way a pollster estimates an election: sample a few random directions and average. The answer is noisy, and the noise falls like $1/\sqrt{N}$ with N samples.



Rendered by this chapter's own tracer (the figure script is about a hundred lines of NumPy): one ray per pixel, a shadow ray to one point light, highlights, and two levels of mirror reflection. The jagged edges are what one sample per pixel looks like.

71.3 The mathematics

Rays and the first hit

Ray

A **ray** is the half-line $\mathbf{p}(t) = \mathbf{o} + t\hat{\mathbf{d}}$, $t > 0$, with origin $\mathbf{o}$ and unit direction $\hat{\mathbf{d}}$. Because $\hat{\mathbf{d}}$ has unit length, t is the distance travelled. Intersecting a ray with a surface means finding the smallest $t > 0$ at which $\mathbf{p}(t)$ lies on the surface.

For a primary ray through pixel (i, j) of a $W \times H$ image, $\mathbf{o}$ is the eye and $\hat{\mathbf{d}}$ is the normalised direction to the pixel's centre on the image plane, built from the camera frame of Ch. 69.

Planes. A plane through $\mathbf{q}$ with normal $\hat{\mathbf{n}}$ is $\{\mathbf{x} : (\mathbf{x} - \mathbf{q}) \cdot \hat{\mathbf{n}} = 0\}$. Substituting the ray gives one linear equation:

$$t = \frac{(\mathbf{q} - \mathbf{o}) \cdot \hat{\mathbf{n}}}{\hat{\mathbf{d}} \cdot \hat{\mathbf{n}}},$$

with no hit if $\hat{\mathbf{d}} \cdot \hat{\mathbf{n}} = 0$ (the ray runs parallel) or $t \le 0$ (the plane is behind). **Triangles**, the workhorse of real scenes, are planes with a test: write the hit point as an affine combination $\lambda_1\mathbf{a} + \lambda_2\mathbf{b} + \lambda_3\mathbf{c}$ (Result 1 of Ch. 69) and accept it when all three barycentric weights are non-negative. The Möller–Trumbore algorithm solves for t and two of the weights at once, as a 3×3 linear system.

Spheres are where the geometry becomes satisfying.

Result 1: ray–sphere intersection

Let the sphere have centre $\mathbf{c}$ and radius r, and put $b = (\mathbf{o} - \mathbf{c}) \cdot \hat{\mathbf{d}}$ and $k = \|\mathbf{o} - \mathbf{c}\|^2 - r^2$. The ray meets the sphere at

$$t_{0,1} = -b \mp \sqrt{b^2 - k},$$

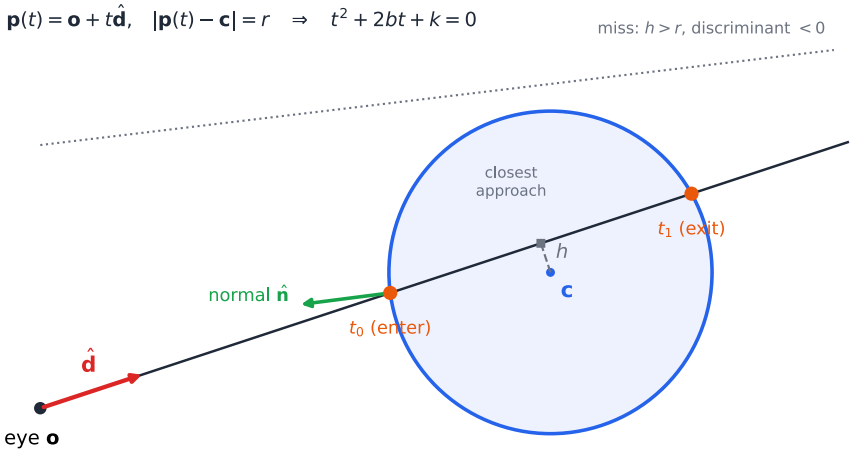
two points if $b^2 > k$, a tangent point if $b^2 = k$, and none if $b^2 < k$.

Proof. The point $\mathbf{p}(t)$ is on the sphere when $\|\mathbf{o} + t\hat{\mathbf{d}} - \mathbf{c}\|^2 = r^2$. Write $\mathbf{m} = \mathbf{o} - \mathbf{c}$ and expand the dot product, using $\hat{\mathbf{d}} \cdot \hat{\mathbf{d}} = 1$:

$$(\mathbf{m} + t\hat{\mathbf{d}}) \cdot (\mathbf{m} + t\hat{\mathbf{d}}) = r^2 \iff t^2 + 2(\mathbf{m} \cdot \hat{\mathbf{d}})\,t + \left(\mathbf{m} \cdot \mathbf{m} - r^2\right) = 0 \iff t^2 + 2bt + k = 0.$$

Completing the square, $(t + b)^2 = b^2 - k$, which gives the roots and the three cases. □

The discriminant has a geometric meaning. The point of the line closest to the centre is at $t = -b$, and its squared distance from $\mathbf{c}$ is $h^2 = \|\mathbf{m}\|^2 - b^2$ (Pythagoras). So $b^2 - k = r^2 - h^2$: the ray hits exactly when it passes within r of the centre, and $\sqrt{r^2 - h^2}$ is half the chord. Example: from $\mathbf{o} = \mathbf{0}$ along $\hat{\mathbf{d}} = \hat{\mathbf{z}}$ to a unit sphere at $(0, 0, 5)$, $b = -5$ and $k = 24$, so $t = 5 \mp 1$: the ray enters at distance 4 and leaves at 6. The surface **normal** at a hit is $\hat{\mathbf{n}} = (\mathbf{p} - \mathbf{c})/r$, free of charge.



Ray–sphere intersection. The ray enters at t_0 and exits at t_1; their midpoint is the closest approach, at distance h from the centre. A ray with $h > r$ misses, and the discriminant $b^2 - k = r^2 - h^2$ is negative.

Reflection

Let $\hat{\mathbf{n}}$ be the unit normal at the hit point and $\hat{\mathbf{d}}$ the incoming direction. A perfect mirror keeps the part of $\hat{\mathbf{d}}$ along the surface and reverses the part along the normal.

Result 2: the mirror direction

The reflected direction is $\hat{\mathbf{r}} = \hat{\mathbf{d}} - 2(\hat{\mathbf{d}} \cdot \hat{\mathbf{n}})\hat{\mathbf{n}}$. It is a unit vector, and it makes the same angle with the normal as the incoming ray (angle of incidence equals angle of reflection).

Proof. Split $\hat{\mathbf{d}} = \mathbf{d}_n + \mathbf{d}_t$ with $\mathbf{d}_n = (\hat{\mathbf{d}} \cdot \hat{\mathbf{n}})\hat{\mathbf{n}}$ along the normal and $\mathbf{d}_t = \hat{\mathbf{d}} - \mathbf{d}_n$ in the surface. The mirror rule "keep $\mathbf{d}_t$, flip $\mathbf{d}_n$" gives $\hat{\mathbf{r}} = \mathbf{d}_t - \mathbf{d}_n = \hat{\mathbf{d}} - 2\mathbf{d}_n$. Since $\mathbf{d}_t \perp \mathbf{d}_n$, $\|\hat{\mathbf{r}}\|^2 = \|\mathbf{d}_t\|^2 + \|\mathbf{d}_n\|^2 = \|\hat{\mathbf{d}}\|^2 = 1$, and $\hat{\mathbf{r}} \cdot \hat{\mathbf{n}} = -\hat{\mathbf{d}} \cdot \hat{\mathbf{n}}$: the outgoing ray has the same cosine with the normal as the incoming one, on the other side. □

The map $\hat{\mathbf{d}} \mapsto \hat{\mathbf{d}} - 2(\hat{\mathbf{d}} \cdot \hat{\mathbf{n}})\hat{\mathbf{n}}$ is the Householder reflection $\mathbf{I} - 2\hat{\mathbf{n}}\hat{\mathbf{n}}^{\mathsf{T}}$, an orthogonal matrix with determinant -1. Two reflections in a row make a rotation: the link back to Ch. 70. The same formula with the light direction in place of $\hat{\mathbf{d}}$ gives the **specular highlight**: a shiny point is bright when the mirror direction of the light points nearly at the eye, and the Phong model raises $\max(0, \hat{\mathbf{r}} \cdot \hat{\mathbf{v}})$ to a power (60 in the figure) to make the highlight tight.

Refraction and Snell's law

When light crosses from a medium of refractive index n_1 (air, $n \approx 1$) into one of index n_2 (glass, about 1.5), it bends according to **Snell's law**,

$$n_1 \sin\theta_1 = n_2 \sin\theta_2,$$

where the angles are measured from the normal. (It follows from Fermat's principle, that light takes the path of least time; a Lagrange-multiplier exercise in the spirit of Ch. 6.) A renderer needs a *direction*, not an angle.

Result 3: the refracted direction

Let $\hat{\mathbf{n}}$ point toward the incoming side, so $c_1 = -\hat{\mathbf{d}} \cdot \hat{\mathbf{n}} = \cos\theta_1 > 0$, and let $\eta = n_1/n_2$. If $1 - \eta^2(1 - c_1^2) \geq 0$, the transmitted direction is

$$\hat{\mathbf{t}} = \eta\,\hat{\mathbf{d}} + \big(\eta c_1 - c_2\big)\hat{\mathbf{n}}, \qquad c_2 = \sqrt{1 - \eta^2\,(1 - c_1^2)} = \cos\theta_2.$$

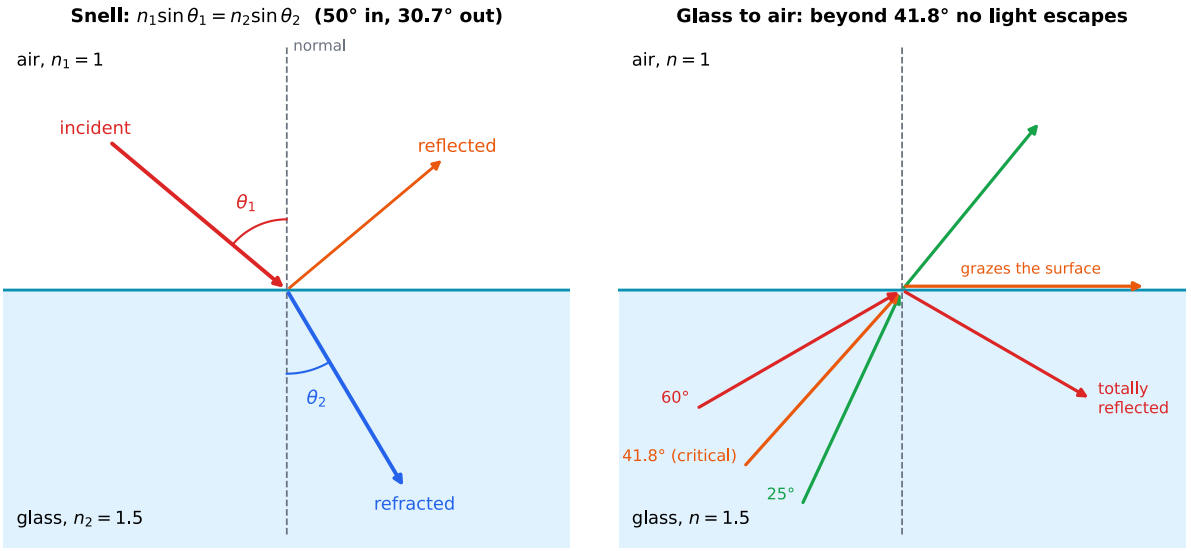
If $1 - \eta^2(1 - c_1^2) < 0$, there is no transmitted ray: **total internal reflection**.

Proof. Split $\hat{\mathbf{d}}$ into its tangential part $\mathbf{d}_t = \hat{\mathbf{d}} + c_1\hat{\mathbf{n}}$ (length $\sin\theta_1$) and its normal part $-c_1\hat{\mathbf{n}}$. The refracted ray stays in the plane of $\hat{\mathbf{d}}$ and $\hat{\mathbf{n}}$, so its tangential part points along $\mathbf{d}_t$; Snell's law fixes its length at $\sin\theta_2 = \eta\sin\theta_1$, so the tangential part is exactly $\eta\,\mathbf{d}_t$. Its normal part points into the second medium, $-\cos\theta_2\,\hat{\mathbf{n}}$, with $\cos\theta_2 = \sqrt{1 - \sin^2\theta_2} = \sqrt{1 - \eta^2(1 - c_1^2)} = c_2$. Adding the parts,

$$\hat{\mathbf{t}} = \eta\big(\hat{\mathbf{d}} + c_1\hat{\mathbf{n}}\big) - c_2\hat{\mathbf{n}} = \eta\hat{\mathbf{d}} + (\eta c_1 - c_2)\hat{\mathbf{n}},$$

a unit vector since $\sin^2\theta_2 + \cos^2\theta_2 = 1$. If $\eta\sin\theta_1 > 1$, no real θ_2 exists and the square root fails: the light cannot leave, and the surface acts as a perfect mirror. □

Numbers: a ray entering glass ($\eta = 1/1.5$) at 50° continues at 30.7°. Going the other way, from glass to air ($\eta = 1.5$), any ray steeper than the **critical angle** $\arcsin(1/1.5) = 41.8°$ is trapped (for water, 48.8°). That is how optical fibres carry light around corners, and why a diamond, with $n \approx 2.4$ and a critical angle under 25°, traps light and sparkles.



Left: reflection and refraction at an air–glass surface. Right: from inside the glass, rays beyond the critical angle of 41.8° cannot get out; the square root in Result 3 has gone negative.

How much light reflects and how much passes through? The exact answer (the Fresnel equations, from electromagnetism) is well approximated by **Schlick's formula** $R(\theta) \approx R_0 + (1 - R_0)(1 - \cos\theta)^5$, with $R_0 = \big((n_1 - n_2)/(n_1 + n_2)\big)^2$. For glass, $R_0 = 0.04$: looking straight at a window, 4% of the light reflects. At 80° the fraction is about 41%, which is why a lake mirrors the far shore but lets you see the stones at your feet.

From rays to the rendering equation

Mirrors and glass send a ray in one or two definite directions. Most surfaces do not: paper, skin, concrete and cloth scatter incoming light over every outgoing direction. The simplest model is the **Lambertian** surface, which looks equally bright from every angle and whose brightness is proportional to $\cos\theta$, the cosine between the normal and the light. (A beam of cross-section A hitting a surface at angle θ spreads over area $A/\cos\theta$.) In our tracer, that is the factor $\max(0, \hat{\mathbf{n}} \cdot \hat{\mathbf{L}})$.

But a matte surface is lit not only by the lamp. Light bouncing off the red sphere tints the floor beside it pink, a ceiling lit by a floor lamp lights the whole room. Accounting for all of it gives the **rendering equation** (Kajiya, 1986): the radiance L_o leaving point $\mathbf{x}$ in direction ω_o is

$$L_o(\mathbf{x}, \omega_o) = L_e(\mathbf{x}, \omega_o) + \int_\Omega f_r(\mathbf{x}, \omega_i, \omega_o)\, L_i(\mathbf{x}, \omega_i)\, \cos\theta_i \, d\omega_i .$$

In words: light leaving = light emitted + light arriving from every direction ω_i in the hemisphere Ω above the surface, weighted by the cosine and by the surface's **BRDF** f_r (how much light from ω_i it sends toward ω_o; for a Lambertian surface of albedo ρ, $f_r = \rho/\pi$). The catch is hidden in L_i: the light arriving at $\mathbf{x}$ from direction ω_i is the light *leaving* some other point in the scene. The unknown function appears on both sides. It is an integral equation, and every bounce of light is one more turn of it.

Integrals over a hemisphere are what Monte Carlo was made for (Ch. 19 §19.9). Draw N random directions ω_k from a density $p(\omega)$ and average:

$$\int_\Omega g(\omega)\, d\omega \;\approx\; \frac{1}{N}\sum_{k=1}^{N} \frac{g(\omega_k)}{p(\omega_k)} .$$

The estimator is **unbiased**, because $E\big[g(\omega)/p(\omega)\big] = \int_\Omega \frac{g}{p}\, p\, d\omega = \int_\Omega g\, d\omega$, and its standard deviation falls as $\sigma/\sqrt{N}$, where σ is the spread of a single sample. A test case: a patch under a uniform white sky of radiance 1 receives $\int_\Omega \cos\theta \, d\omega = \pi$. Sampling directions uniformly ($p = 1/2\pi$) gives single-sample spread $\sigma = 2\pi/\sqrt{12} \approx 1.81$; with 16 samples the estimate has standard error about 0.45, about 14% of π. Quadruple the samples and the error only halves. That $\sqrt{N}$ is why film frames take hours, and why early path-traced images look as if they were printed on sandpaper.

71.4 The algorithm

A Whitted-style tracer (mirrors, glass, shadows) is a short recursion.

```
trace(ray, depth):
    hit = nearest intersection of ray with every object       # Result 1, planes, triangles
    if no hit: return sky(ray.direction)
    p, n, material = hit.point, hit.normal, hit.material
    colour = ambient
    for each light:
        if nearest hit along ray(p + eps*n -> light) is beyond the light:   # shadow ray
            colour += lambert(n, light) + phong(n, light, ray)
    if depth > 0 and material.mirror > 0:
        colour = mix(colour, trace(reflect(ray, n), depth-1), material.mirror)   # Result 2
    if depth > 0 and material.glass:
        R = schlick(cos theta)
        colour = R * trace(reflect) + (1 - R) * trace(refract(ray, n, eta), depth-1)  # Result 3
    return colour
```

A **path tracer** replaces the fixed bounce directions by one random direction per bounce, drawn from the BRDF, and averages many such paths per pixel. It solves the rendering equation; the Whitted tracer only solves the special case of perfect mirrors and point lights.

Complexity. With P pixels, S samples per pixel, D bounces and N objects, a brute-force tracer tests about $PSDN$ intersections. For a modest 1920×1080 frame, one sample, three bounces and a million triangles, that is about 6×10^{12} tests. No hardware does that per frame. Real tracers wrap objects in a hierarchy of bounding boxes, so a ray discards whole groups of objects with one test, and the cost per ray drops from N to roughly $\log N$.

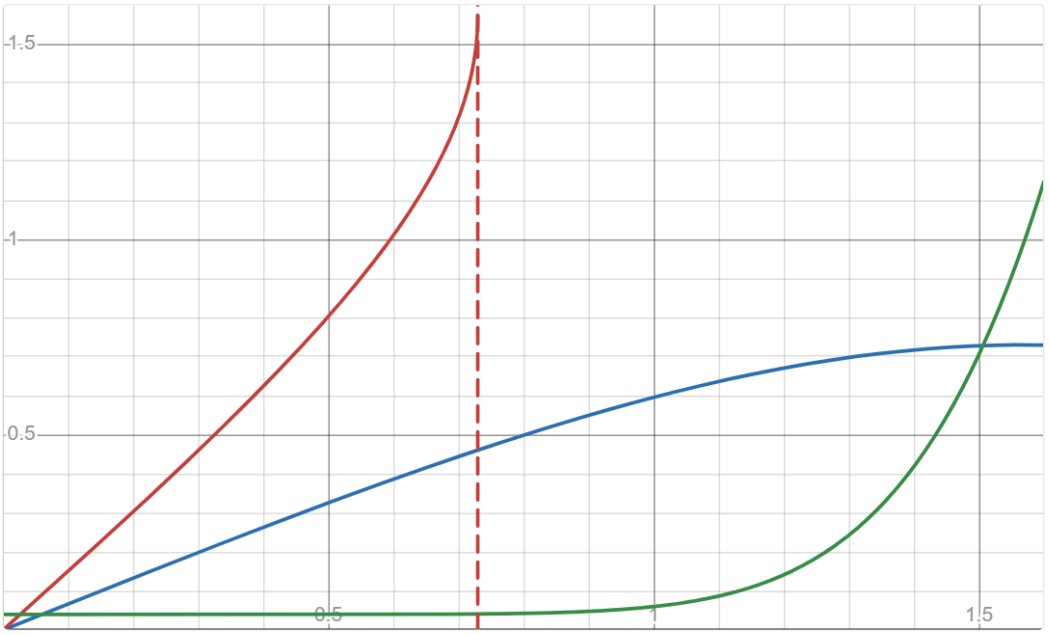
Numerical failure cases.

- *Shadow acne.* A shadow ray starts exactly on the surface; rounding error places its origin slightly *inside*, and it immediately hits the surface it came from. The result is a speckle of false shadow. The fix is to push the origin off the surface by a small ε along the normal, but too large an ε detaches shadows from their objects ("peter-panning"). There is no universally right ε.
- *Cancellation in the quadratic.* When $b^2 \gg k$ (a small or distant sphere), $-b + \sqrt{b^2 - k}$ subtracts two nearly equal numbers and loses digits; compute the larger-magnitude root first and the other from the product of the roots, $t_0 t_1 = k$ (Ch. 22 §22.2).
- *Infinite recursion.* Two facing mirrors bounce a ray forever. A depth limit stops it, which silently darkens the infinite corridor; path tracers instead stop paths at random ("Russian roulette") and reweight the survivors so that the estimate stays unbiased.
- *Aliasing.* One ray per pixel through the pixel centre makes staircase edges, and fine textures in the distance turn into shimmering moiré. More rays per pixel help; *where* inside the pixel you aim them matters more than you would think.

> **? The clue we're holding back**
>
> For the sky-lit patch above, uniform sampling gives a 14% error with 16 rays. A cleverer choice of the density $p(\omega)$ in the same estimator makes the error *exactly zero*, with a single ray. In a real scene the perfect density is unknowable, because it depends on the very answer you are computing, yet renderers get tantalisingly close. How do you choose p, and why is the perfect choice always out of reach? That is importance sampling, in Ch. 38.

71.5 The experiment

First, play with light at a surface. The graph shows, as functions of the incidence angle x in radians, the refracted angle entering glass, the refracted angle *leaving* glass (it stops at the critical angle) and Schlick's reflectance.



Blue: air into glass. Red: glass into air, which ends at the critical angle 0.7297 *rad* $(41.8°$*, dashed). Green: fraction reflected (Schlick). Change* `1.5` *to* `2.4` *(diamond) in both arcsines and watch the red curve end much sooner; the dashed line should move to* $\arcsin(1/2.4) \approx 0.430$*. For the green curve, a diamond's* R_0 *is* $\left(1.4/3.4\right)^2 \approx 0.17$*.*

Second, a complete ray tracer in forty lines. It renders a red ball and a mirror ball on a checkerboard, with a shadow ray per hit and recursive reflections.

```
import numpy as np
import matplotlib.pyplot as plt

W, H = 160, 100                  # KNOB: image size in pixels
LIGHT = np.array([-3.0, 5.0, 1.0])  # KNOB: point light position
MIRROR = 0.7                     # KNOB: reflectivity of the big sphere (0..1)
SPP = 1                          # KNOB: rays per pixel (jittered); try 4 or 16
BOUNCES = 2                      # KNOB: reflection depth
S = [(np.array([-1.0, 0.7, 4.0]), 0.7, np.array([.9, .2, .2]), 0.0),     # centre, radius, colour, mirror
     (np.array([0.7, 1.0, 5.5]), 1.0, np.array([.9, .9, .9]), MIRROR)]

def hit(o, d):                   # nearest hit: t, normal, colour, mirror (ground is the plane y = 0)
    t = np.where(d[:, 1] < -1e-9, -o[:, 1] / np.minimum(d[:, 1], -1e-9), np.inf)
    n = np.tile([0.0, 1.0, 0.0], (len(o), 1)); p = o + np.where(np.isinf(t), 0, t)[:, None] * d
    col = np.where(((np.floor(p[:, 0]) + np.floor(p[:, 2])) % 2 > 0)[:, None], .9, .3) * np.ones(3); m = np.zeros(len(o))
    for c, r, cc, mr in S:       # ray-sphere: t^2 + 2bt + k = 0
        b = np.sum((o - c) * d, 1); k = np.sum((o - c) ** 2, 1) - r * r; disc = b * b - k
        ts = np.where(disc > 0, -b - np.sqrt(np.abs(disc)), np.inf); ts[ts < 1e-4] = np.inf
        closer = ts < t; t = np.where(closer, ts, t)
        n[closer] = (o[closer] + ts[closer, None] * d[closer] - c) / r; col[closer] = cc; m[closer] = mr
    return t, n, col, m

def trace(o, d, depth):
    t, n, col, m = hit(o, d)
    sky = np.array([.5, .7, 1.0]) * (0.6 + 0.4 * d[:, 1:2]); miss = np.isinf(t)
    p = o + np.where(miss, 0, t)[:, None] * d + 1e-4 * n
    L = LIGHT - p; dist = np.linalg.norm(L, axis=1); L /= dist[:, None]
    lit = hit(p, L)[0] > dist    # shadow ray: is anything between p and the light?
    shade = col * (0.1 + 0.9 * np.clip(np.sum(n * L, 1), 0, 1) * lit)[:, None]
    if depth > 0:                # reflection: r = d - 2 (d.n) n
        r = d - 2 * np.sum(d * n, 1)[:, None] * n
        shade = (1 - m[:, None]) * shade + m[:, None] * trace(p, r, depth - 1)
    return np.where(miss[:, None], sky, shade)

ys, xs = np.mgrid[0:H, 0:W]; img = np.zeros((H * W, 3)); rng = np.random.default_rng(0)
for _ in range(SPP):
    jx, jy = (rng.random((2, H * W)) - 0.5) if SPP > 1 else (0, 0)
    d = np.stack([(xs.ravel() + jx - W / 2) / H, (H / 2 - ys.ravel() - jy) / H - 0.1, np.ones(H * W)], 1)
    img += trace(np.tile([0, 1, -1.0], (H * W, 1)), d / np.linalg.norm(d, axis=1)[:, None], BOUNCES)
plt.imshow(np.clip(img / SPP, 0, 1).reshape(H, W, 3)); plt.axis("off"); plt.show()
```

Every ray in the image is processed at once as a row of a NumPy array, so the whole picture takes about a second. Things to try:

- **Move the light.** Set `LIGHT = [3, 2, 2]`: long shadows to the left. Put the light *below* the floor (`y = -1`): everything is in shadow, because every shadow ray hits the floor.
- **Break the epsilon.** Change the offset `1e-4 * n` in `p = ...` to `0.3 * n`: the shadows shrink and come loose from the balls (peter-panning). Then set it, and the `ts < 1e-4` test, to `0`. This little scene is forgiving (double precision, exact spheres, a floor test that ignores upward rays), so look closely for speckle; real renderers working with single-precision triangles are not so lucky.
- **Hall of mirrors.** Set `MIRROR = 1.0` and `BOUNCES = 0, 1, 2, 6`. The reflection of the red ball appears at depth 1; reflections of reflections need more bounces.

- **Antialiasing.** Set `SPP = 4`, then `16`. The staircase edges and the moiré on the distant checkerboard soften: each pixel now averages several jittered rays. Compare the cost with the improvement.

71.6 The challenge

1. **Intuition.** Stand facing a still lake. Using Schlick's formula, explain why the far shore appears mirrored in the water while near your feet you see the bottom. Then explain, with the critical angle of water (48.8°), what a fish looking upward sees: why does the whole sky above it appear squeezed into a circle (Snell's window), and what lies outside that circle?
2. **Derivation.** A triangle has vertices $\mathbf{a}, \mathbf{b}, \mathbf{c}$. Write the hit point as $\mathbf{a} + u(\mathbf{b} - \mathbf{a}) + v(\mathbf{c} - \mathbf{a})$ and equate it to $\mathbf{o} + t\hat{\mathbf{d}}$. Solve the resulting 3×3 system for (t, u, v) with Cramer's rule (Ch. 8 §8.3), and rewrite each determinant as a scalar triple product. This is the Möller–Trumbore algorithm. State the conditions on t, u, v for a hit, and say what happens geometrically when the main determinant is zero.
3. **Implementation.** Add a glass sphere to the experiment script. You will need the refraction direction of Result 3, a sign flip of $\hat{\mathbf{n}}$ and η when the ray is *leaving* the sphere, the far root t_1 for rays that start inside, total internal reflection, and Schlick's weight between the reflected and refracted contributions. Check your result: the checkerboard seen through the ball should appear upside down.
4. **Open investigation.** Replace the shadow ray to a point light by shadow rays to random points on a *disc* light of radius ρ, and render with N samples per pixel. The shadows grow soft edges (penumbras). Measure the pixel noise in the penumbra as a function of N and check the $1/\sqrt{N}$ law. Then try *stratified* sampling: split the disc into $\sqrt{N} \times \sqrt{N}$ cells and take one random point per cell. Does the noise now fall faster than $1/\sqrt{N}$? How fast, and does it depend on how the shadow edge is oriented?

71.7 Loose threads

Whitted's picture is no longer mysterious. Each pixel fires a ray; the first hit is a quadratic (Result 1); the mirror bounce is a reflection across the normal (Result 2); glass is Snell's law in vector form, with total internal reflection where the square root fails (Result 3); shadows are one more ray toward the light. Tracing backward is legitimate because light paths are reversible. And the full problem, light bouncing endlessly between rough surfaces, is an integral equation that we attack with random samples, at the price of noise that fades only as $1/\sqrt{N}$.

What we did not tell you:

> **? Jaggies, moiré and the right place to aim**
>
> One ray per pixel is a *sample* of a continuous image, and a checkerboard receding to the horizon contains detail finer than any pixel. The result is moiré: false, large-scale patterns that are not in the scene. Jittering the rays turns the moiré into noise, which the eye forgives. Why does randomness trade one error for a better-looking one, and what does the spectrum of a sampling pattern have to do with it? The answer is the sampling theorem and its consequences, in Ch. 76.

> **? A million triangles in twenty tests**
>
> Our tracer tests every ray against every object. With a hierarchy of bounding volumes, a ray can find its first hit among a million triangles in a few dozen box tests. How do you build the hierarchy, what makes one tree better than another, and can you prove that log N is the best you can hope for? That is Ch. 75.

- **Solving the integral equation.** Substituting the rendering equation into itself expands light into a sum over paths of 0, 1, 2, ... bounces, a *Neumann series*. When does such a series converge, and how fast? Integral equations of this kind live in the world of Ch. 61.
- **Light that is not a ray.** Ray optics ignores wavelength, so it cannot make rainbows from a thin soap film, or the colours of a peacock's feather. Those need waves and interference, and the wave equation of Ch. 13.
- **Moving worlds.** Every image so far is frozen. Make the balls fall, bounce and roll, and you need to integrate equations of motion without the world exploding: Ch. 72.

Further reading

- P. Shirley, *Ray Tracing in One Weekend* (series of three short books), freely available online, 2016 onward. Build a path tracer from scratch, one chapter at a time.
- M. Pharr, W. Jakob, G. Humphreys, *Physically Based Rendering: From Theory to Implementation*, 4th ed., MIT Press, 2023. The reference: geometry, sampling, Monte Carlo and the rendering equation, with complete code; free online.
- T. Whitted, "An improved illumination model for shaded display", *Communications of the ACM* 23(6), 1980. The recursive ray tracer.
- J. T. Kajiya, "The rendering equation", *SIGGRAPH* 1986. The integral equation behind every modern renderer.

Chapter 72 — Numerical Simulation and Physics Engines

> **Level 2 · Track G — Mathematics for Computer Graphics, Games, and Simulation Builds on:** Ch. 22 (§22.9 Euler and Runge–Kutta, §22.11 stability), Ch. 11, Ch. 6 (Lagrange multipliers), Ch. 69 · **Leads to:** Ch. 75, Ch. 84, Ch. 92 · **Related:** Ch. 70, Ch. 48, Ch. 83 **Reading time:** ~30 min

72.1 The hook

Every game programmer has a version of this story. You write a tiny space game: one sun, one planet, Newton's law of gravity. The planet's state is a position and a velocity, and each frame you do the obvious thing:

```
position += velocity * dt
velocity += acceleration(position) * dt
```

You press play. The planet goes round. It is beautiful. You leave it running while you make coffee, and when you come back the planet is gone. Not crashed into the sun: *gone*, flung into the dark on a widening spiral, as if someone had been quietly feeding it fuel.

Nobody fed it anything. The physics has no engine, no friction, no external force. The real system conserves energy exactly. Your code has manufactured energy out of nothing, a little on every frame.

Now the strange part. Swap the two lines:

```
velocity += acceleration(position) * dt
position += velocity * dt
```

Same cost. Same step size. Same "accuracy" in the textbook sense: both are first-order methods. And now the planet orbits forever. Run it for a million frames and the orbit wobbles a little, but it never spirals out.

The same ghost haunts every game engine. Ragdolls that "explode" when a character falls into a corner. A stack of crates that jitters, then leaps into the air. A bullet that passes through a wall without touching it. A rope that slowly stretches until it is twice its length. None of these are bugs in the usual sense. They are mathematics: the gap between a differential equation and the arithmetic a computer does sixty times per second.

So here is the question this chapter answers:

Why does the obvious update rule create energy from nothing, why does swapping two lines cure it, and what else must a physics engine get right to keep a virtual world from tearing itself apart?

72.2 The big idea

A game is a map, not a flow

The laws of motion are a *flow*: a rule that moves the state continuously in time,

$$\dot{\mathbf{x}} = \mathbf{v}, \qquad \dot{\mathbf{v}} = \mathbf{a}(\mathbf{x}).$$

A game never sees the flow. It sees a **map**: a function Φ_h that takes the state at frame n to the state at frame $n+1$, with h a fixed time step (typically $1/60$ s). Whatever the true physics does between frames is invisible. The only thing that matters is what the map does when it is applied ten thousand times in a row.

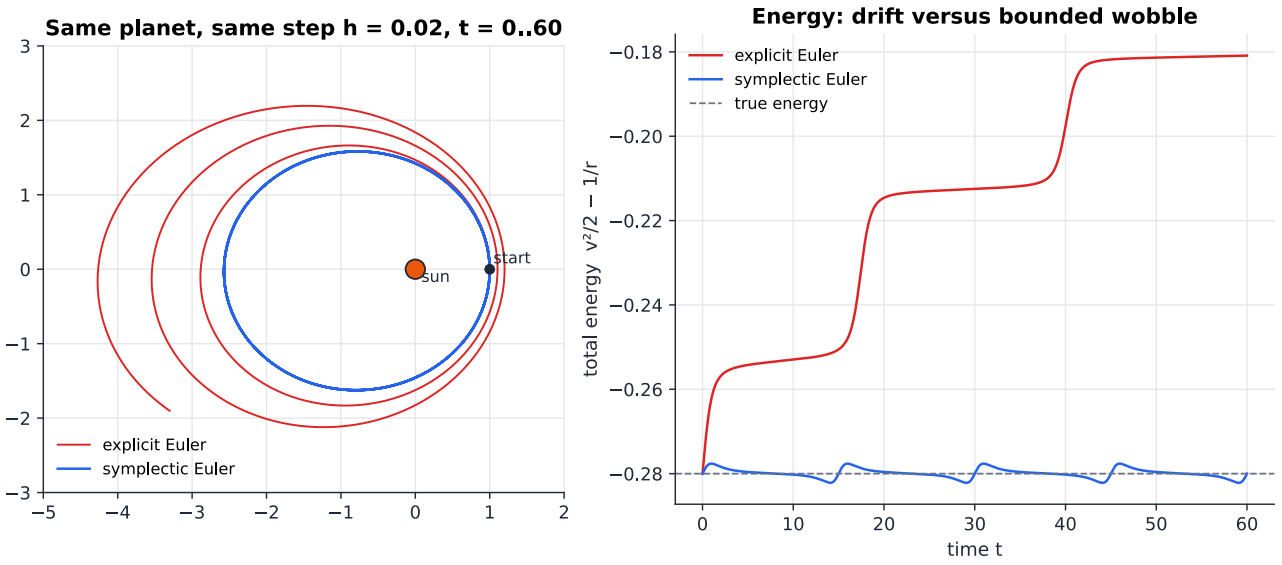
That reframing is the whole chapter. The textbook question about an integrator is "how close is one step to the truth?" (local error, order of accuracy: Ch. 22, §22.9). The game-engine question is "what does the map do *forever*?" A first-order method with the right *structure* beats a fourth-order method with the wrong one, if you run it long enough.

Two lines, two geometries

Think of a frictionless spring, the simplest oscillator. Its state lives in the **phase plane**, position on one axis and velocity on the other, and the true motion goes round an ellipse of constant energy.

Explicit Euler looks at the current state, reads off the tangent to the ellipse, and steps along it. A tangent to a convex curve always points slightly *outside* the curve. So every step lands on a slightly bigger ellipse. Outward, outward, outward: a spiral.

Semi-implicit (symplectic) Euler updates the velocity first and then moves the position using the *new* velocity. The two half-updates err in opposite directions: one pushes outward, the other inward. They do not cancel exactly, but they cancel in a precise geometric sense we will prove: the map preserves **area** in the phase plane. A map that preserves area cannot spiral outward forever, because a spiral would have to sweep a blob of initial conditions into an ever larger region.



The same planet, the same step $h = 0.02$, the same cost per step. Explicit Euler gains energy at each close pass of the sun (from -0.280 to -0.181 in four orbits) and spirals away. Symplectic Euler's energy wobbles near -0.280 and comes back, orbit after orbit.

The three jobs of a physics engine

Integration is one job. A real engine has three, run every frame:

1. **Integrate**: move every body forward by h under forces.
2. **Detect**: find which bodies now touch or overlap, among thousands, in well under a millisecond.
3. **Resolve**: fix the contacts and constraints (bounce, stop, stay attached) without creating or destroying energy you did not intend.

Each job has its own way to fail, and the failures interact.

72.3 The mathematics

The test oscillator

Stability analysis for game physics starts, as in Ch. 22, §22.9, with a linear test problem. Here it is the frictionless spring of angular frequency ω:

$$\dot{x} = v, \qquad \dot{v} = -\omega^2 x, \qquad E = \tfrac{1}{2}\left(v^2 + \omega^2 x^2\right).$$

Every stiff spring, vibrating joint or pendulum near the bottom of its swing looks like this locally, with ω^2 the local stiffness.

Explicit Euler is $x_{n+1} = x_n + h v_n$, $v_{n+1} = v_n - h\omega^2 x_n$. Then

$$v_{n+1}^2 + \omega^2 x_{n+1}^2 = (v_n - h\omega^2 x_n)^2 + \omega^2 (x_n + h v_n)^2 = (1 + h^2\omega^2)\left(v_n^2 + \omega^2 x_n^2\right),$$

because the cross terms $-2h\omega^2 x_n v_n$ and $+2h\omega^2 x_n v_n$ cancel. Energy is multiplied by exactly $1 + h^2\omega^2$ on every step, whatever the state. After time $t = nh$ the factor is $(1 + h^2\omega^2)^{t/h} \approx e^{h\omega^2 t}$. With $\omega = 1$ and $h = 0.1$ that is a factor of 1.85 per period and about 517 after ten periods. Ch. 22 already showed this in its stability-region picture: the eigenvalues $\pm j\omega$ of a lossless oscillator sit on the imaginary axis, outside explicit Euler's region. Making h smaller slows the growth, but never stops it.

Symplectic Euler preserves area

Symplectic (semi-implicit) Euler updates velocity first:

$$v_{n+1} = v_n - h\omega^2 x_n, \qquad x_{n+1} = x_n + h\, v_{n+1}.$$

Substituting the first line into the second gives a linear map $(x_n, v_n) \mapsto (x_{n+1}, v_{n+1})$ with matrix

$$\mathbf{M} = \begin{pmatrix} 1 - h^2\omega^2 & h \\ -h\omega^2 & 1 \end{pmatrix}, \qquad \det \mathbf{M} = (1 - h^2\omega^2) + h^2\omega^2 = 1.$$

Area preservation

Symplectic Euler maps every region of the phase plane to a region of the same area. Explicit Euler multiplies every area by $1 + h^2\omega^2$.

Proof. A linear map multiplies areas by $|\det \mathbf{M}|$. For symplectic Euler $\det \mathbf{M} = 1$, computed above. Explicit Euler's matrix is $\begin{pmatrix} 1 & h \\ -h\omega^2 & 1 \end{pmatrix}$, with determinant $1 + h^2\omega^2$. ∎

For nonlinear forces the same holds with the Jacobian matrix in place of $\mathbf{M}$: the velocity kick $\mathbf{v} \mapsto \mathbf{v} + h\,\mathbf{a}(\mathbf{x})$ is a shear (it changes $\mathbf{v}$ by an amount depending only on $\mathbf{x}$), the drift $\mathbf{x} \mapsto \mathbf{x} + h\mathbf{v}$ is another shear, and shears have determinant 1. The true flow of any Hamiltonian system also preserves phase-space volume (Liouville's theorem). Symplectic Euler is a composition of two exactly solvable sub-flows, and so it inherits that geometry. This is what "symplectic" means at the

level of this chapter.

Area preservation alone forbids an outward spiral, but it does not tell us how big the energy error is. For the oscillator we can do better and find what the method conserves *exactly*.

A conserved "shadow energy"

Under symplectic Euler, the quantity

$$\tilde{E} = \tfrac{1}{2}\left(v^2 + \omega^2 x^2 - h\omega^2\, xv\right)$$

is exactly conserved, and for $h\omega < 2$ the true energy E stays within the band

$$\frac{\tilde{E}_0}{1 + h\omega/2} \le E_n \le \frac{\tilde{E}_0}{1 - h\omega/2}.$$

Proof. Write $x' = x + hv'$ and $v' = v - h\omega^2 x$. First,

$$\begin{aligned} v'^2 + \omega^2 x'^2 - h\omega^2 x'v' &= v'^2 + \omega^2(x^2 + 2hxv' + h^2 v'^2) - h\omega^2(x + hv')v' \\ &= v'^2 + \omega^2 x^2 + h\omega^2 xv'. \end{aligned}$$

Now substitute $v' = v - h\omega^2 x$: $v'^2 = v^2 - 2h\omega^2 xv + h^2\omega^4 x^2$ and $h\omega^2 xv' = h\omega^2 xv - h^2\omega^4 x^2$. Adding, the $h^2\omega^4 x^2$ terms cancel and we are left with $v^2 + \omega^2 x^2 - h\omega^2 xv$. So $\tilde{E}' = \tilde{E}$.

For the band, use $|\omega xv| \le \tfrac{1}{2}(v^2 + \omega^2 x^2) = E$ (the AM–GM inequality). Then $h\omega^2|xv| = h\omega|\omega xv| \le h\omega E$, so $(1 - h\omega/2)E \le \tilde{E} \le (1 + h\omega/2)E$. Since $\tilde{E} = \tilde{E}_0$ for all n, rearranging gives the band. ∎

With $\omega = 1$, $h = 0.1$, $x_0 = 1$, $v_0 = 0$ we get $\tilde{E}_0 = 0.5$, so E must stay between $0.5/1.05 = 0.476$ and $0.5/0.95 = 0.526$. A simulation of 10^4 steps hits 0.4762 and 0.5263: the bound is sharp. The energy error is $O(h)$, as expected for a first-order method, but it is *bounded forever* instead of compounding.

The theorem also gives the stability limit. $\mathbf{M}$ has determinant 1 and trace $2 - h^2\omega^2$, so its eigenvalues are $e^{\pm j\theta}$ (on the unit circle) exactly when $|2 - h^2\omega^2| < 2$, that is when $h\omega < 2$. Beyond that, one eigenvalue leaves the unit circle and the method explodes. For a game: a 1 kg body on a spring of stiffness $k = 10^4$ N/m has $\omega = \sqrt{k/m} = 100$ rad/s, so it needs $h < 0.02$ s. At 60 Hz ($h = 0.0167$ s) it is stable. Drop the physics rate to 30 Hz ($h = 0.033$ s) and the same spring detonates. This is the most common reason a ragdoll explodes: a stiff joint and a long frame.

? The energy the planet secretly conserves

For the linear spring we found the conserved shadow energy $\tilde{E}$ by hand. The planet in the figure obeys a nonlinear force, yet its energy also wobbles and returns, for as long as anyone has run it. Is there a shadow energy for *every* symplectic method and *every* Hamiltonian? Almost: the method exactly follows the flow of a slightly modified Hamiltonian, up to an error that is exponentially small in $1/h$. Why that is true, and what "symplectic" really means as geometry rather than as a determinant, is the story of backward error analysis in Ch. 92.

Verlet: the integrator games actually ship

Add the Taylor expansions of $x(t + h)$ and $x(t - h)$:

$$x(t \pm h) = x \pm h\dot{x} + \tfrac{h^2}{2}\ddot{x} \pm \tfrac{h^3}{6}\dddot{x} + O(h^4).$$

The odd terms cancel, leaving the **Störmer–Verlet** rule

$$x_{n+1} = 2x_n - x_{n-1} + h^2 a(x_n),$$

with local error $O(h^4)$ and global error $O(h^2)$. It has no velocity at all: velocity is implied by the last two positions, $v_n \approx (x_{n+1} - x_{n-1})/(2h)$. That sounds like a flaw and is in fact the feature games love. If a collision or a constraint *moves* a particle, the velocity changes automatically and consistently with the move; there is no separate velocity to forget to fix.

Verlet is also **time-reversible**: swap x_{n+1} and x_{n-1} and the rule is unchanged. Its velocity form, *velocity Verlet* (half kick, full drift, half kick), is symplectic Euler composed with its mirror image, which is why it gains an order of accuracy for free. Runge–Kutta 4 is more accurate per step but neither symplectic nor reversible: its energy drifts, only slowly.

Collision detection: geometry under a deadline

With n bodies there are $n(n-1)/2$ pairs: for 1000 bodies, 499,500 pairs every frame. Engines split detection into a cheap **broad phase** that discards most pairs and an exact **narrow phase** for the survivors.

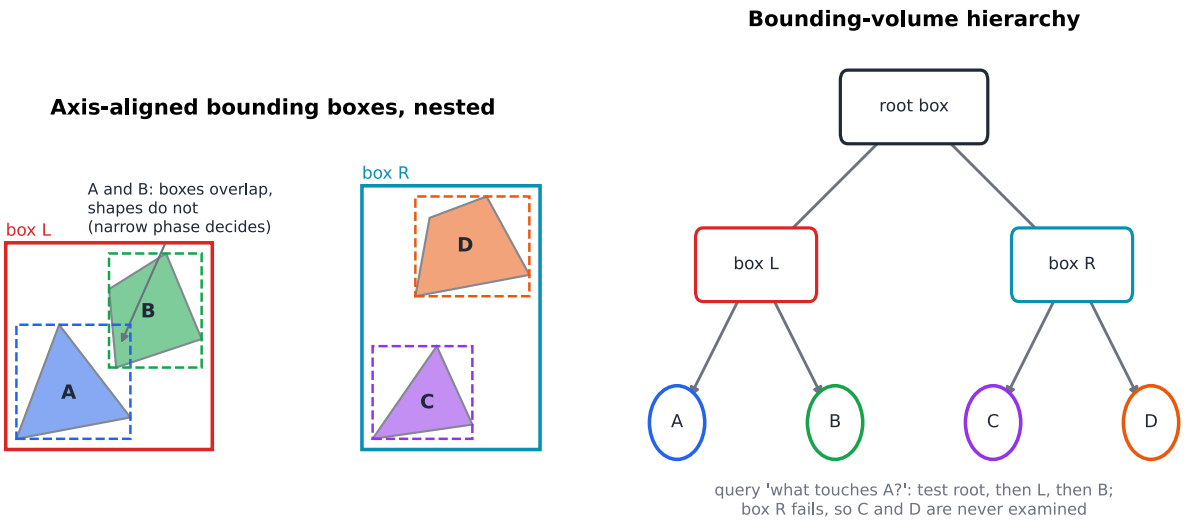
The broad phase wraps each body in a **bounding volume**, usually an axis-aligned bounding box (AABB): the smallest box, with sides parallel to the axes, containing the body.

The AABB overlap test

Two closed intervals $[a_0, a_1]$ and $[b_0, b_1]$ intersect if and only if $a_0 \le b_1$ and $b_0 \le a_1$. Two axis-aligned boxes intersect if and only if their projections onto every coordinate axis intersect.

Proof. If the intervals share a point p, then $a_0 \le p \le b_1$ and $b_0 \le p \le a_1$. Conversely, if both inequalities hold, let $p = \max(a_0, b_0)$. Then $p \le a_1$ (since $a_0 \le a_1$ and $b_0 \le a_1$) and $p \le b_1$ similarly, so p lies in both. For boxes, a box is the Cartesian product of its interval projections; a point $\mathbf{p}$ lies in both boxes exactly when each coordinate p_i lies in both i-th intervals, and such coordinates can be chosen independently axis by axis. ∎

So the test costs six comparisons in 3-D and can exit on the first failure. Boxes are conservative: overlapping boxes do not mean overlapping shapes, only "worth checking".

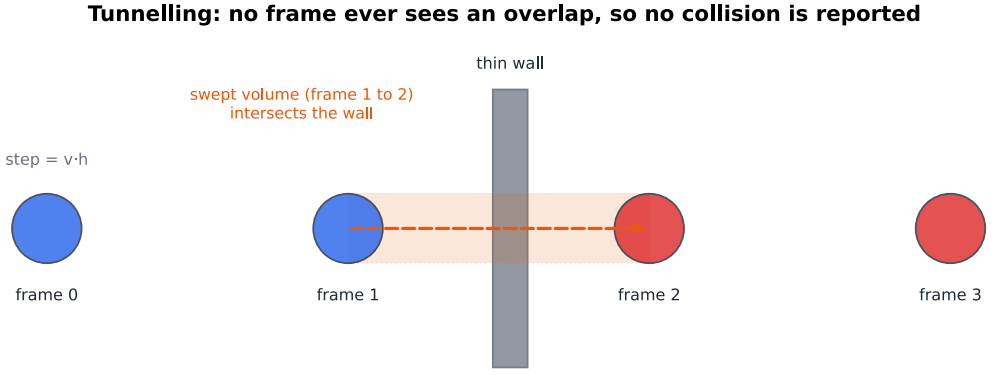


*Bounding boxes nest into a **bounding-volume hierarchy** (BVH). A query descends only into boxes it overlaps, so a body far from everything is rejected after a handful of tests. The overlapping boxes of A and B are a false positive that the narrow phase must reject.*

For the narrow phase on convex shapes, the workhorse is the **separating axis theorem**: two convex polygons (or polyhedra) are disjoint if and only if there is a line onto which their projections are disjoint, and it suffices to try the edge normals of both shapes (plus edge–edge cross products in 3-D). The AABB theorem above is the special case in which the only candidate axes are the coordinate axes. We state the general theorem without proof; it is a consequence of the separating hyperplane theorem for convex sets (Ch. 47).

Tunnelling

Detection that only looks at frames has a blind spot. A ball of radius r moving at speed v jumps vh per frame; if $vh > 2r + w$, where w is the thickness of a wall, there can be a frame before the wall and a frame after it with no overlap in between.



Tunnelling. Each frame's overlap test passes, so no collision is ever reported. Testing the swept volume, the region the body occupies during the step, catches the crossing.

A bullet at 900 m/s simulated at 60 Hz moves 15 m per frame. Engines therefore use **continuous collision detection** for fast objects: solve for the first time $t \in [0, h]$ at which the moving shapes touch. For a sphere against a plane that is a linear equation in t; for two moving spheres it is a quadratic.

Collision response: the impulse

When two bodies touch, the engine changes their velocities instantly with an **impulse** $j\,\hat{\mathbf{n}}$ along the contact normal $\hat{\mathbf{n}}$ (pointing from body 2 to body 1). Body 1 receives $+j\hat{\mathbf{n}}$ and body 2 receives $-j\hat{\mathbf{n}}$, so momentum is conserved automatically.

The collision impulse

Let the relative normal velocity before contact be $u_n = (\mathbf{v}_1 - \mathbf{v}_2)\cdot\hat{\mathbf{n}} < 0$ (approaching), and let the coefficient of restitution $e \in [0,1]$ require $u_n' = -e\,u_n$ afterwards. Then

$$j = \frac{-(1+e)\,u_n}{1/m_1 + 1/m_2}.$$

Proof. After the impulse, $\mathbf{v}_1' = \mathbf{v}_1 + (j/m_1)\hat{\mathbf{n}}$ and $\mathbf{v}_2' = \mathbf{v}_2 - (j/m_2)\hat{\mathbf{n}}$. Dot the difference with $\hat{\mathbf{n}}$ (a unit vector): $u_n' = u_n + j\,(1/m_1 + 1/m_2)$. Setting $u_n' = -eu_n$ gives $j(1/m_1 + 1/m_2) = -(1+e)u_n$. ■

Example: a 1 kg ball at 4 m/s hits a resting 3 kg ball head-on, with $e = 0.5$. Then $u_n = -4$ and $j = 1.5 \times 4/(4/3) = 4.5$ N·s. The first ball rebounds at -0.5 m/s, the second leaves at 1.5 m/s; momentum stays 4 kg·m/s, and kinetic energy falls from 8 J to 3.5 J. With $e = 1$ kinetic energy is conserved exactly; with $e > 1$ the engine would create energy, which is exactly what goes wrong when overlapping bodies are pushed apart too hard.

Constraints: projecting back onto the rules

A rope, a ragdoll limb or a cloth is a set of particles linked by **constraints** such as $C(\mathbf{p}_1, \mathbf{p}_2) = \|\mathbf{p}_1 - \mathbf{p}_2\| - L = 0$. Integration breaks them a little every step; the engine then repairs them. **Position-based dynamics** repairs a violated constraint by the smallest mass-weighted move, which is a Lagrange-multiplier problem (Ch. 6): minimise $\sum_i m_i \|\Delta \mathbf{p}_i\|^2$ subject to the linearised constraint $C + \nabla C \cdot \Delta \mathbf{p} = 0$. The solution moves each particle along the constraint gradient, in proportion to its inverse mass $w_i = 1/m_i$. For the distance constraint, with $\hat{\mathbf{n}} = (\mathbf{p}_1 - \mathbf{p}_2)/\|\mathbf{p}_1 - \mathbf{p}_2\|$,

$$\Delta \mathbf{p}_1 = -\frac{w_1}{w_1 + w_2}\, C\, \hat{\mathbf{n}}, \qquad \Delta \mathbf{p}_2 = +\frac{w_2}{w_1 + w_2}\, C\, \hat{\mathbf{n}}.$$

Two checks. The new separation is $\|\mathbf{p}_1 - \mathbf{p}_2\| - C = L$ exactly. And $m_1 \Delta \mathbf{p}_1 + m_2 \Delta \mathbf{p}_2 = \frac{C\hat{\mathbf{n}}}{w_1 + w_2}(-1 + 1) = \mathbf{0}$: the correction does not move the centre of mass, so it creates no net momentum. A fixed anchor is a particle with $w = 0$; it never moves.

With many constraints, fixing one breaks its neighbours. Engines simply sweep through them a few times per frame (Gauss–Seidel relaxation, the iteration of Ch. 8), accepting a small residual. That residual is why a cheap rope stretches under load, and why adding iterations makes it stiffer.

72.4 The algorithm

The fixed-timestep loop

Rendering rates vary; physics must not, because the stability limit $h\omega < 2$ depends on h. The standard pattern decouples them with an accumulator:

```
h = 1/60                      # physics step: fixed, chosen for stability
acc = 0.0
while running:
    acc += min(frame_time(), 0.25)   # clamp: avoid the "spiral of death"
    while acc >= h:
        integrate(h)          # symplectic Euler or velocity Verlet
        detect_collisions()   # broad phase (BVH or sort-and-sweep), then narrow phase
        resolve(iterations=8)  # impulses and constraint projections
        acc -= h
    render(interpolate(acc / h))     # blend last two physics states for smooth display
```

The clamp matters. If one frame takes too long, the loop runs several physics steps to catch up, which makes the next frame longer still; without a cap, the game freezes.

Verlet with constraints

The particle engine popularised for ragdolls and cloth is a dozen lines:

```
def step(x, x_prev, a, h, constraints, iters=8):
    x_new = 2*x - x_prev + h*h*a            # Verlet: velocity is implicit in x - x_prev
    for _ in range(iters):
        for (i, k, L, wi, wk) in constraints:
            d = x_new[i] - x_new[k]
            dist = np.linalg.norm(d)
            C = dist - L
            n = d / dist
            x_new[i] -= wi/(wi + wk) * C * n
            x_new[k] += wk/(wi + wk) * C * n
        x_new = np.maximum(x_new, floor)    # collision with the ground: project out
    return x_new, x
```

Note the last line inside the loop: collision with the ground is handled as just another constraint (stay above the floor), and because Verlet derives velocity from positions, the bounce is automatic.

Costs and failure modes

Stage	Method	Cost per frame	Typical failure
integrate	symplectic Euler, Verlet	$O(n)$	explodes if $h\omega > 2$ for the stiffest spring
broad phase	brute force	$O(n^2)$	too slow beyond a few hundred bodies
broad phase	sort-and-sweep, BVH	about $O(n \log n + k)$ for k overlapping pairs	degrades when everything is piled together
narrow phase	separating axes	$O(\text{edges})$ per pair	misses fast objects (tunnelling)
resolve	impulses + relaxation	$O(\text{iterations} \times \text{contacts})$	jitter, stretching, stacks that creep

Sort-and-sweep exploits time coherence: bodies barely move between frames, so the sorted list of box endpoints is nearly sorted already, and insertion sort fixes it in close to linear time.

Energy enters through the side door

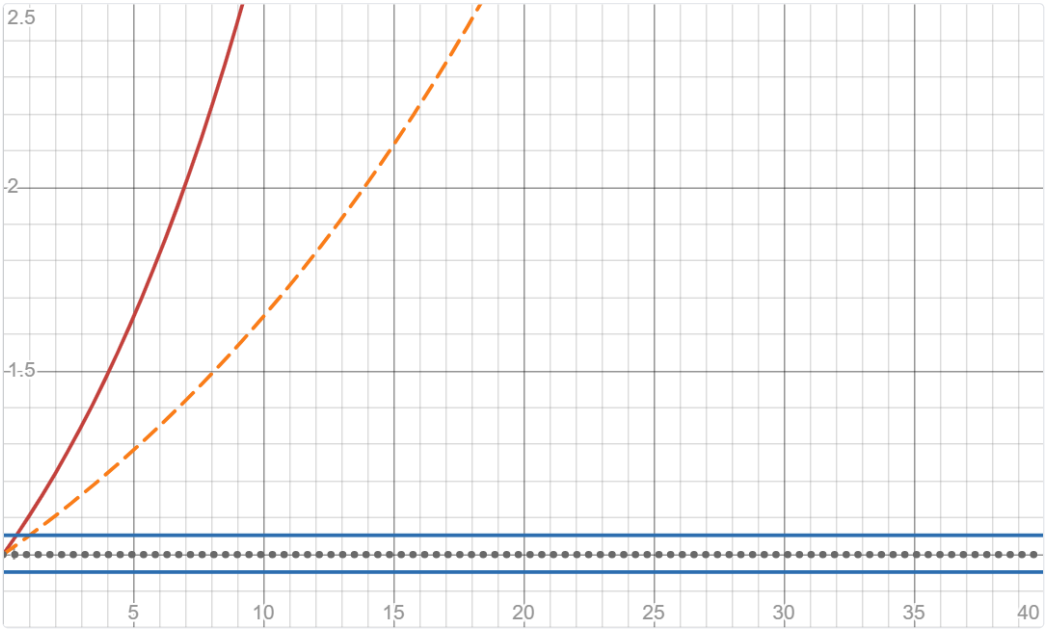
A perfect symplectic integrator still gains energy if collision response shoves overlapping bodies apart too hard. Most "explosions" in shipped games come from resolution, not integration. Plot total energy every frame: it is the cheapest error detector a physics engine has.

? Why a stack of boxes is hard

A single bounce is one impulse. A stack of ten boxes resting on each other is a different problem: every contact must push (never pull), push only when touching, and stop touching only when the push is zero. These "either-or" conditions turn resolution into a **complementarity problem**, and sequential impulses are just one slow way to solve it. Why these conditions are the KKT conditions of an optimisation problem in disguise, and what duality says about solving them, is in Ch. 48.

72.5 The experiment

First, the energy factors themselves. The red curve is explicit Euler's energy, $(1+h^2)^{t/h}$ with $\omega = 1$ and $h = 0.1$, written as 1.01^{10x}. The blue lines are the band that symplectic Euler can never leave, $1/(1 \pm h/2)$ times the shadow energy.



Energy relative to its start, against time. Red: explicit Euler with $h = 0.1$. Orange dashed: explicit Euler with $h = 0.05$, written 1.0025^{20x}, which grows more slowly but still without limit. Blue: the band that bounds symplectic Euler with $h = 0.1$. Change 0.1 to 0.2 (red: 1.04^{5x}; blue: $1/0.9$ and $1/1.1$) and watch the band widen while staying a band.

Now a script. It simulates a nonlinear pendulum, $\ddot{\theta} = -(g/\ell)\sin\theta$, with three integrators and plots the energy of each. Nothing here is linear, so the shadow-energy theorem does not apply directly: you are testing whether its message survives.

```
import numpy as np
import matplotlib.pyplot as plt

g_over_l = 9.81      # (g/l) in 1/s^2: a 1 m pendulum
h = 0.05             # time step in s: try 0.01, 0.1, 0.3, 0.6
T = 60.0             # total time in s
theta0 = 2.5         # starting angle in rad (2.5 is a wild swing; try 0.1)

def acc(th):
    return -g_over_l * np.sin(th)

def energy(th, om):
    return 0.5 * om**2 + g_over_l * (1 - np.cos(th))

def simulate(method):
    th, om = theta0, 0.0
    E = []
    for _ in range(int(T / h)):
        if method == "explicit":
            th, om = th + h * om, om + h * acc(th)
        elif method == "symplectic":
            om = om + h * acc(th)
            th = th + h * om
        else:                                   # velocity Verlet
            om_half = om + 0.5 * h * acc(th)
            th = th + h * om_half
            om = om_half + 0.5 * h * acc(th)
        E.append(energy(th, om))
    return np.array(E)

t = np.arange(int(T / h)) * h
for m in ["explicit", "symplectic", "verlet"]:
    plt.plot(t, simulate(m) / energy(theta0, 0.0), label=m)
plt.ylim(0, 3)
plt.xlabel("time (s)"); plt.ylabel("energy / initial energy")
plt.legend(); plt.show()
```

Things to try. With `h = 0.05` and `theta0 = 2.5` explicit Euler's pendulum gains energy until it goes over the top and spins; the other two hold steady. Shrink `h` to `0.01` and Verlet's wobble shrinks by a factor of about 25 (second order) while symplectic Euler's shrinks by about 5 (first order). Raise `h` until even Verlet fails: the small-angle stability limit is $h\sqrt{g/\ell} < 2$, about $h < 0.64$ s, but the wild swing fails sooner. Find out how much sooner.

72.6 The challenge

1. **Intuition.** A designer complains that springs in your engine "lose bounce" over time and asks you to switch from symplectic Euler to explicit Euler, because she read that explicit Euler "adds energy". Explain, using the phase-plane picture, why that is a bad fix even if it seems to cancel her damping, and suggest where the lost bounce is more likely to be coming from.

2. **Derivation.** For **backward (implicit) Euler** on the oscillator, $x_{n+1} = x_n + hv_{n+1}$, $v_{n+1} = v_n - h\omega^2 x_{n+1}$, find the matrix of the map, show that its determinant is $1/(1+h^2\omega^2)$, and conclude what happens to energy. Then show that alternating one explicit Euler step with one backward Euler step does not give an area-preserving method, but that a half step of each, composed, gives the trapezoidal rule. Is the trapezoidal rule symplectic for this problem?

3. **Implementation.** Build a rope of 20 particles with distance constraints, the top one pinned ($w = 0$), using the Verlet-with-constraints code above. Measure the rope's total length after it settles as a function of the number of relaxation iterations (1, 2, 4, 8, 16, 32). Plot the stretch against iterations on a log–log plot. What power law do you see, and how does it change with the number of particles?

4. **Open investigation.** Stack N identical boxes (in 2-D is fine) using sequential impulses with a fixed number of iterations per frame. For each N, find the minimum number of iterations at which the stack stands still for 60 seconds instead of creeping or collapsing. How does that number grow with N? Try warm starting (reusing last frame's impulses as the first guess) and see whether it changes the growth law. Real engines rely on heuristics here; there is no known clean formula.

72.7 Loose threads

The hook asked why the obvious update creates energy and why swapping two lines cures it. Explicit Euler multiplies every phase-space area, and the oscillator's energy, by $1 + h^2\omega^2$ per step. Symplectic Euler is a composition of two shears, preserves area exactly, and conserves a shadow energy $\tilde{E}$ that pins the true energy inside a band of width $O(h)$ forever. Around the integrator sit bounding volumes, continuous detection, impulses and constraint projection, each with its own way to inject energy.

What the chapter did not tell you:

- **The shadow Hamiltonian in general.** The first mystery: why every symplectic integrator follows a nearby energy-conserving system almost exactly, for exponentially long times. That is backward error analysis and symplectic geometry, in Ch. 92; the geometry of phase space as a manifold starts in Ch. 59.
- **Contact as optimisation.** The second mystery: resting contact and friction form a complementarity problem whose structure is duality, in Ch. 48.
- **Rotation.** Every body here was a point. Rigid bodies also spin, and integrating an orientation without it slowly ceasing to be a rotation needs the quaternions of Ch. 70.
- **Faster broad phases.** Building and updating the BVH, and the alternatives (uniform grids, k-d trees, sweep structures), belong to Ch. 75.
- **Stiffness.** Cloth and stiff joints want large k and large h together, which explicit and symplectic methods cannot deliver. Implicit methods, discussed in Ch. 22, §22.9 under stiffness, buy stability with a linear solve per step.

? Can a simulation of chaos be trusted at all?

A double pendulum is chaotic: two starting states differing by 10^{-12} end up completely different within seconds, and every step of your integrator makes an error far bigger than that. So after a minute your simulated pendulum is certainly not following the true trajectory from its starting point. Yet the motion *looks* right, and its statistics are right. Is the computed orbit following some *other* true trajectory, from a slightly different start? Sometimes provably yes, and the theorem that says when is the shadowing lemma of Ch. 84.

Further reading

- Ernst Hairer, Christian Lubich and Gerhard Wanner, "Geometric numerical integration illustrated by the Störmer–Verlet method", *Acta Numerica* 12 (2003). The clearest account of why Verlet works, including backward error analysis.
- Christer Ericson, *Real-Time Collision Detection* (Morgan Kaufmann, 2005). The standard engineering reference for bounding volumes, separating axes and continuous detection.
- Thomas Jakobsen, "Advanced Character Physics", Game Developers Conference (2001). The short paper that brought Verlet integration and constraint relaxation to games.
- Matthias Müller, Bruno Heidelberger, Marcus Hennix and John Ratcliff, "Position Based Dynamics", *Journal of Visual Communication and Image Representation* 18 (2007). The method behind most real-time cloth and soft bodies.

Chapter 73 — Fractals: Infinite Complexity from Simple Rules

> **Level 2 · Track G — Mathematics for Computer Graphics, Games, and Simulation Builds on:** Ch. 2 (complex multiplication), Ch. 22 (§22.3 fixed-point iteration), Ch. 1 (logarithms, geometric series) · **Leads to:** Ch. 74, Ch. 84, Ch. 97 · **Related:** Ch. 83, Ch. 85, Ch. 64 **Reading time:** ~30 min

73.1 The hook

How long is the coast of Britain?

It sounds like a question with an answer you could look up. In the 1950s the physicist and meteorologist Lewis Fry Richardson tried to look it up, for a study of whether the length of a border between two countries affects how likely they are to go to war. He found that the published figures disagreed wildly. Spain and Portugal quoted their shared border as 987 km and 1214 km. The Netherlands and Belgium disagreed by a similar margin.

Nobody had made an arithmetic mistake. Richardson worked out what was going on by doing the measurement himself, with a pair of dividers on a map. Set the dividers to 100 km and walk them along the coast: you get some total. Set them to 50 km: the dividers now follow bays and headlands the 100 km setting stepped across, and the total goes *up*. Set them to 10 km: up again. Halving the ruler did not converge to a limit. It kept adding length, by roughly the same *factor* each time.

Richardson found that the measured length L followed a power law in the ruler length ε,

$$L(\varepsilon) \approx C\,\varepsilon^{1-D},$$

with an exponent D that was a property of the coastline: about 1.25 for the west coast of Britain, close to 1.0 for the smooth coast of South Africa. A smooth curve has $D = 1$, and then L stops changing as $\varepsilon \to 0$, which is what calculus says should happen (arc length, Ch. 7). The British coast does not.

In 1967 Benoît Mandelbrot read Richardson's forgotten paper and asked the question that founded a field. What if the exponent D is not a nuisance but a *dimension*: a coastline is more than a line and less than an area, and 1.25 measures exactly how much more?

What could it mean for a shape to have dimension 1.25, and how do we compute it?

The answer takes us through a snowflake with infinite perimeter and finite area, a two-line program that draws the most complicated object in mathematics, and a definition of dimension that a ruler can measure.

73.2 The big idea

Rules, not shapes

A smooth curve looks straighter and straighter as you zoom in: that is what having a tangent line means. A **fractal** is a shape that does not simplify under zooming. Zoom into the British coastline and you see bays within bays. Zoom into a fern and you see the same fern.

The deep idea is that such shapes are usually easy to *generate* even though they are hard to *describe*. You do not draw a fern; you write a rule and apply it forever. Three kinds of rule dominate:

- **Replacement rules.** Take a segment, replace it with a bent copy of four smaller segments, and repeat (the Koch curve).
- **Iterated function systems.** Take a set of contracting maps, apply all of them to a picture, overlay the results, and repeat (the Sierpinski triangle, Barnsley's fern).
- **Escape-time rules.** Iterate a simple map like $z \mapsto z^2 + c$ and colour each starting point by what its orbit does (the Mandelbrot and Julia sets).

Each rule is a **feedback loop**: output becomes input. That is why fractals are the natural shapes of iteration, and why they turn up wherever there is feedback: turbulence, river networks, lungs, galaxies, and the chaotic systems of Ch. 84.

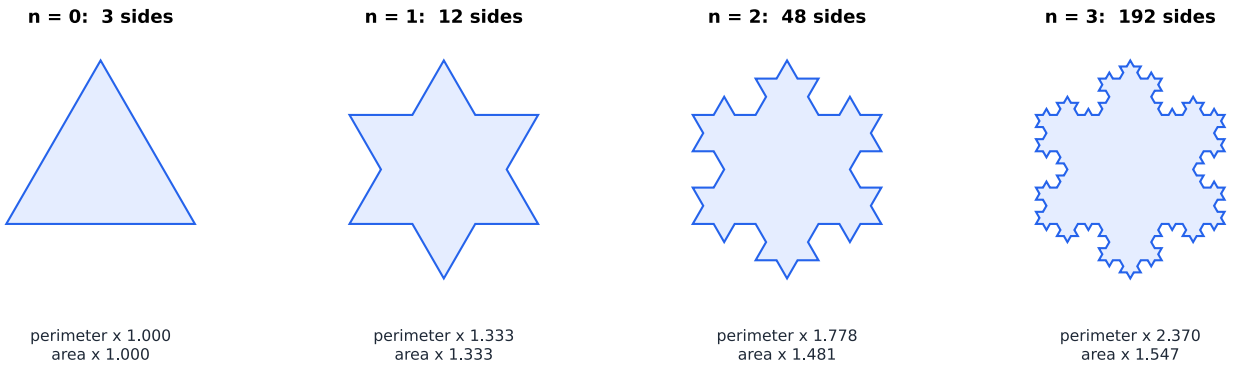
Dimension as a scaling law

Here is the analogy that unlocks fractal dimension. Cut a line segment into pieces scaled by $1/3$: you get 3 pieces. Do it to a square: 9 pieces. To a cube: 27. The number of pieces is 3^D, where D is the dimension. Dimension is the exponent in "how many small copies do I need?"

The Koch curve is built from 4 copies of itself, each scaled by $1/3$. So $3^D = 4$, and

$$D = \frac{\log 4}{\log 3} \approx 1.262.$$

That is the whole idea, and it is not a metaphor: it is a number you can measure with a ruler, exactly as Richardson did, by counting how the number of small boxes needed to cover a shape grows as the boxes shrink.



Koch snowflake, stages 0 to 3. Each stage multiplies the perimeter by $4/3$, so it grows without bound, while the area creeps toward a limit of $8/5$ of the original triangle.

73.3 The mathematics

The Koch snowflake: infinite perimeter, finite area

Start from an equilateral triangle with side s, perimeter $P_0 = 3s$ and area A_0. At each stage, replace every side by four sides one third as long: cut out the middle third and put two sides of a small outward-pointing equilateral triangle in its place.

The Koch snowflake

After n stages the snowflake has $3 \cdot 4^n$ sides, perimeter $P_n = 3s\,(4/3)^n$ and area

$$A_n = A_0 \left(1 + \frac{3}{5}\left(1 - \left(\frac{4}{9}\right)^n\right)\right).$$

So $P_n \to \infty$ while $A_n \to \frac{8}{5}A_0$.

Proof. Each side becomes 4 sides, each a third as long, so the side count is multiplied by 4 and the perimeter by $4/3$ at every stage. For the area: at stage $k \geq 1$ a new small triangle is added on each of the $3 \cdot 4^{k-1}$ sides of stage $k-1$. Its side is $s/3^k$, so its area is $A_0/9^k$ (area scales as length squared). The area added at stage k is

$$3 \cdot 4^{k-1} \cdot \frac{A_0}{9^k} = \frac{3}{4}A_0 \left(\frac{4}{9}\right)^k.$$

Summing the geometric series for $k = 1, \ldots, n$:

$$A_n = A_0 + \frac{3}{4}A_0 \cdot \frac{(4/9)\,(1 - (4/9)^n)}{1 - 4/9} = A_0 \left(1 + \frac{3}{5}\left(1 - \left(\frac{4}{9}\right)^n\right)\right).$$

As $n \to \infty$, $(4/9)^n \to 0$ and $A_n \to A_0(1 + 3/5) = \frac{8}{5}A_0$. ∎

You could paint the snowflake with a finite can of paint, but never fence it. And notice the *reason*: perimeter grows by $4/3 > 1$ per stage, area by additions that shrink like $(4/9)^k$. Length and area scale with different powers of the zoom factor, and a fractal sits between them.

Similarity dimension

Self-similar set and similarity dimension

A set F is **strictly self-similar** if it is the union of N copies of itself, each scaled by the same ratio $r < 1$, that overlap at most in negligible pieces (single points or edges). Its **similarity dimension** is the number D with $N r^D = 1$:

$$D = \frac{\log N}{\log(1/r)}.$$

The formula reproduces ordinary dimensions: a segment is N copies at ratio $1/N$, so $D = 1$; a square is N^2 copies at ratio $1/N$, so $D = 2$. And it gives new ones:

Set	Copies N	Ratio r	$D = \log N / \log(1/r)$
Cantor set (keep outer thirds of $[0,1]$, repeat)	2	$1/3$	0.631
Koch curve	4	$1/3$	1.262
Sierpinski triangle	3	$1/2$	1.585
Sierpinski carpet	8	$1/3$	1.893
Menger sponge	20	$1/3$	2.727

The Cantor set is a good warm-up for the strangeness. It has total length zero (we remove $1/3$, then $2/9$, then $4/27$, ..., which sums to 1), yet it has as many points as the whole interval (Ch. 64). Dimension 0.631 sits between "countable dust" (dimension 0) and "a line" (dimension 1).

When the copies have different ratios $r_1, \dots, r_N$, the same reasoning gives the **Moran equation** $\sum_i r_i^D = 1$, which has exactly one solution because the left side decreases strictly in D.

Box-counting dimension

Most fractals, coastlines above all, are not exactly self-similar. The measurable version of the idea counts boxes.

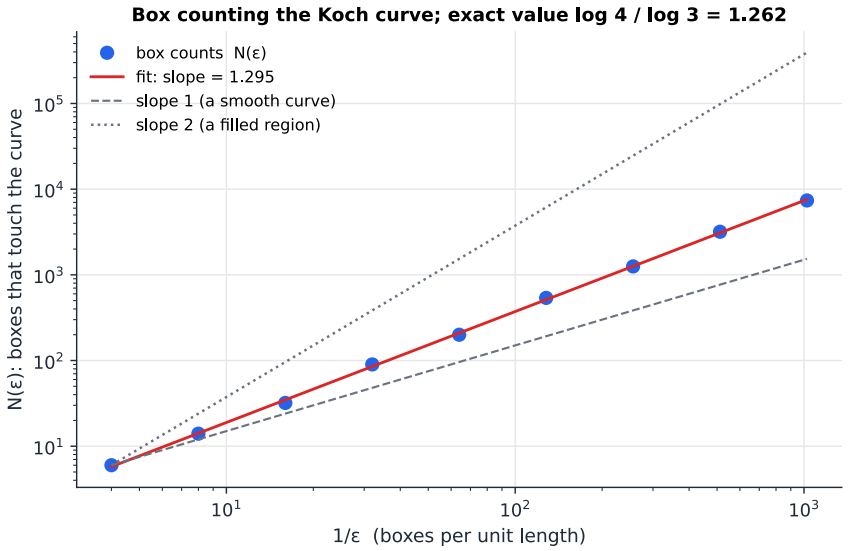
Box-counting dimension

Cover the plane with a grid of squares of side ε and let $N(\varepsilon)$ be the number of squares that meet the set F. If $N(\varepsilon) \approx C\varepsilon^{-D}$ as $\varepsilon \to 0$, then D is the **box-counting dimension**:

$$D = \lim_{\varepsilon \to 0} \frac{\log N(\varepsilon)}{\log(1/\varepsilon)}.$$

In practice, D is the slope of $\log N$ against $\log(1/\varepsilon)$.

For a smooth curve of length L, $N(\varepsilon) \approx L/\varepsilon$ (slope 1); for a region of area A, $N \approx A/\varepsilon^2$ (slope 2). Richardson's dividers are the same thing in disguise: the measured length is (number of steps) $\times\, \varepsilon \approx C\varepsilon^{-D} \cdot \varepsilon = C\varepsilon^{1-D}$, which is exactly his law. The exponent he measured on the British coast was the box-counting dimension.



Box counting the Koch curve with dyadic boxes from $\varepsilon = 1/4$ to $1/1024$. The least-squares slope is 1.295, about 3% above the exact $\log 4/\log 3 = 1.262$. The limit is approached slowly, and the coarse scales bias the fit.

A slope is not a limit

Box counting gives a dimension only in the limit $\varepsilon \to 0$, and every real data set has a smallest scale (pixels, grains of sand, atoms). Fits over a decade or two can be biased by several percent; the local slope between consecutive scales wobbles. Report the range of scales you fitted over, and be suspicious of three significant figures.

Iterated function systems

The Sierpinski triangle is the set left fixed by three maps of the plane. With corners $\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3$, let

$$f_i(\mathbf{x}) = \mathbf{p}_i + \tfrac{1}{2}(\mathbf{x} - \mathbf{p}_i), \qquad i = 1, 2, 3,$$

each a contraction by $1/2$ toward a corner. The triangle S satisfies $S = f_1(S) \cup f_2(S) \cup f_3(S)$: it is three half-size copies of itself.

A family of contractions like this is an **iterated function system** (IFS). Its key property is a fixed-point theorem, the same idea as fixed-point iteration in Ch. 22, §22.3, but where the "points" are whole compact sets and distance between sets is measured by the Hausdorff metric (the largest distance from a point of either set to the other set).

Hutchinson's theorem (stated)

If $f_1, \dots, f_N$ are contractions of the plane, the map on compact sets $\mathcal{H}(K) = f_1(K) \cup \cdots \cup f_N(K)$ is itself a contraction in the Hausdorff metric. So it has exactly one fixed compact set F, the **attractor**, and iterating $\mathcal{H}$ from *any* starting compact set converges to F.

Two consequences matter for computing. First, the starting picture is irrelevant: start from a square, a dot or a photograph of a cat, and the iterates converge to the same Sierpinski triangle. Second, there is a randomized shortcut, the **chaos game**: start at any point, and repeatedly apply one of the f_i chosen at random. After a few steps the point is within a pixel of the attractor and stays there, and the points it visits fill it out. We will use it in the experiment.

The Mandelbrot set

Now the most famous rule of all. Fix a complex number c (Ch. 2) and iterate

$$z_{n+1} = z_n^2 + c, \qquad z_0 = 0.$$

The orbit $0,\ c,\ c^2+c,\ \ldots$ either stays bounded forever or runs off to infinity. The **Mandelbrot set** M is the set of c for which it stays bounded. For $c = 0$ the orbit is $0, 0, 0, \ldots$: in M. For $c = -1$ it is $0, -1, 0, -1, \ldots$: in M. For $c = 1$ it is $0, 1, 2, 5, 26, \ldots$: not in M.

To compute M we need to know when we can stop iterating and declare escape. The answer is a short, exact theorem.

The escape radius

If $|z_n| > 2$ for some n, then $|z_n| \to \infty$. Consequently M lies inside the disc $|c| \le 2$.

Proof. First suppose $|c| > 2$. Then $z_1 = c$ already has $|z_1| > 2$ and $|z_1| \ge |c|$. Otherwise $|c| \le 2$, and any z_n with $|z_n| > 2$ also satisfies $|z_n| > |c|$. So in either case we have some $z = z_n$ with $|z| > 2$ and $|z| \ge |c|$. By the reverse triangle inequality,

$$|z^2 + c| \ge |z|^2 - |c| \ge |z|^2 - |z| = |z|\,(|z| - 1)\,.$$

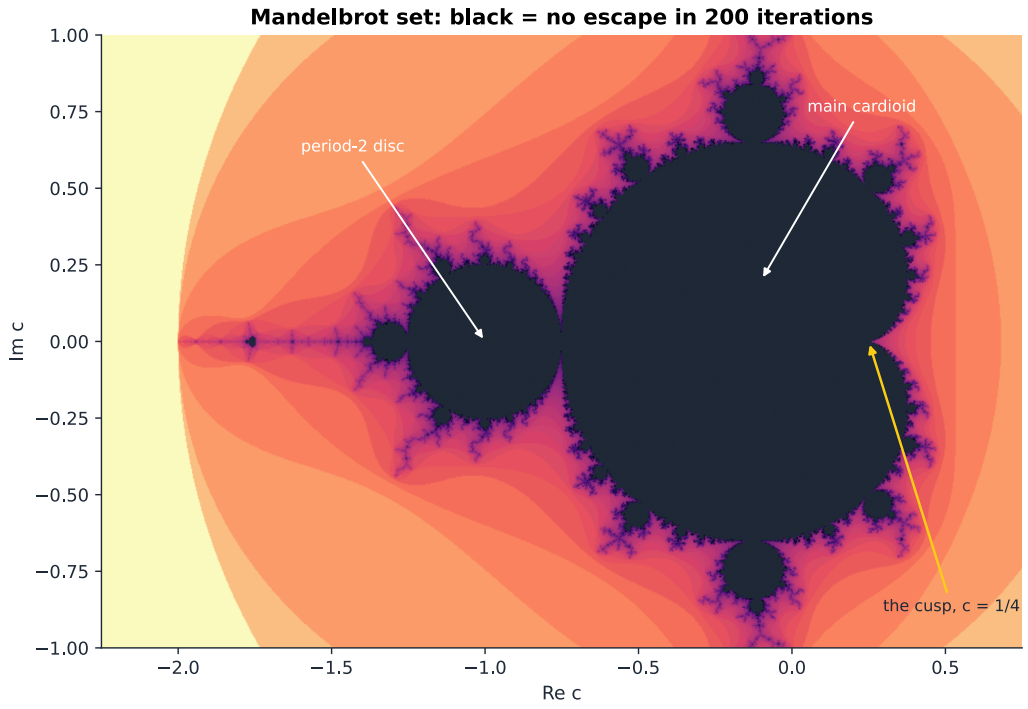
Write $|z| = 2 + \delta$ with $\delta > 0$. Then $|z_{n+1}| \ge (1+\delta)|z_n|$, and $|z_{n+1}| > 2 + \delta$, so the same bound holds at the next step with the same (or larger) δ. By induction $|z_{n+k}| \ge (1+\delta)^k |z_n| \to \infty$. ∎

So a computer can decide escape in finite time for every c outside M: iterate until $|z| > 2$. What it can never do is certify, in finite time, that a point is *inside*: an orbit that has stayed bounded for 200 iterations might escape at iteration 201. Every picture of M is an approximation from outside.

Some of the structure can be derived by hand. The big heart-shaped body, the **main cardioid**, is the set of c for which the map has an attracting fixed point. A fixed point satisfies $z^* = (z^*)^2 + c$, and it attracts when the derivative $|2z^*| < 1$. On the boundary, $2z^* = e^{j\theta}$, and then $c = z^* - (z^*)^2$ gives

$$c(\theta) = \frac{e^{j\theta}}{2} - \frac{e^{2j\theta}}{4},$$

a cardioid with its cusp at $c(0) = 1/4$ and its leftmost point at $c(\pi) = -3/4$. There the fixed point loses stability and an attracting cycle of period 2 takes over; that cycle attracts for $|c + 1| < 1/4$, the disc attached at $-3/4$. Then period 4, period 8, …



The Mandelbrot set computed by the escape-time algorithm of §73.4: 900 × 600 pixels, at most 200 iterations each, coloured by the logarithm of the escape count. The black region never exceeded $|z| = 2$.

? Is the Mandelbrot set in one piece?

Douady and Hubbard proved in 1982 that M is connected: it does not fall apart into islands, even though pictures show apparent islands (they are joined by filaments too thin to render). A much stronger statement, that M is *locally connected* (every point has arbitrarily small connected neighbourhoods), is the **MLC conjecture**. If true, it would give a complete combinatorial description of M and settle whether the hyperbolic components are dense. It is one of the central open problems of complex dynamics; how such problems are attacked is the subject of Ch. 98.

73.4 The algorithm

Escape time, vectorised

The whole Mandelbrot figure above comes from this loop, run on all pixels at once with NumPy:

```
import numpy as np

def mandelbrot(xmin, xmax, ymin, ymax, W=900, H=600, max_iter=200):
    c = np.linspace(xmin, xmax, W)[None, :] + 1j * np.linspace(ymin, ymax, H)[:, None]
    z = np.zeros_like(c)
    count = np.full(c.shape, max_iter)          # max_iter means "never escaped"
    alive = np.ones(c.shape, dtype=bool)
    for n in range(max_iter):
        z[alive] = z[alive]**2 + c[alive]
        escaped = alive & (np.abs(z) > 2)       # the escape-radius theorem
        count[escaped] = n
        alive &= ~escaped
    return count
```

Cost. $O(WHI)$ in the worst case, with I the iteration budget `max_iter`, dominated by points inside M, which never escape and run the full budget. Two cheap tests reject most of the interior before iterating: c is in the main cardioid if $q(q+(x-\tfrac14)) \le \tfrac14 y^2$ with $q=(x-\tfrac14)^2+y^2$, and in the period-2 disc if $(x+1)^2+y^2 \le \tfrac{1}{16}$. Points near the boundary escape slowly, so zooming in demands more iterations: the budget must grow as you zoom.

Smooth colouring. Integer escape counts produce visible bands. Replacing n by $n+1-\log_2\log|z_n|$ (using a large escape radius) gives a continuous value, because for large $|z|$ the map is essentially $z\mapsto z^2$ and $\log\log|z|$ grows by $\log 2$ per step.

Precision. Double-precision numbers have about 16 significant digits. A zoom to a width of 10^{-13} leaves only a few digits to distinguish neighbouring pixels, and the picture dissolves into blocks. Deep-zoom software computes one reference orbit in arbitrary precision and the other pixels as small *perturbations* of it in ordinary floating point: Ch. 22, §22.1 is the reason this is necessary.

The chaos game

```
z = 0j
for i in range(n_points):
    corner = corners[rng.integers(len(corners))]
    z = corner + r * (z - corner)      # apply a random contraction from the IFS
    plot(z)                            # skip the first few points (transient)
```

Cost $O(1)$ per point. The randomness is only a device for visiting the attractor; any sequence of maps that contains every finite pattern of choices works.

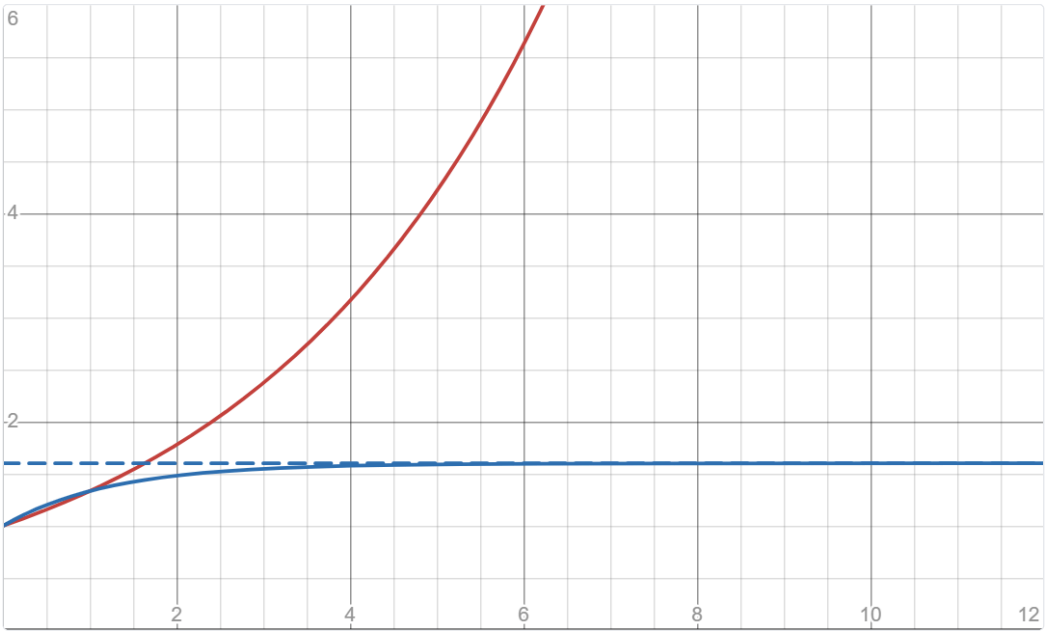
Box counting

```
def box_count(points, eps):
    cells = np.floor(np.column_stack([points.real, points.imag]) / eps)
    return len(np.unique(cells, axis=0))          # number of occupied eps-boxes
# D is the slope of log(box_count) against log(1/eps), over a range of eps
```

Two failure cases. If ε is smaller than the spacing between your sample points, the count saturates at the number of points and the slope falls toward 0. If ε is comparable to the whole set, the count is dominated by its overall shape. The usable range lies between, and it is often only a decade or two wide.

73.5 The experiment

First, the snowflake's two diverging fates. With the stage n on the horizontal axis, the red curve is the perimeter ratio $(4/3)^n$ and the blue curve is the area ratio, which flattens out under its limit $8/5$.



Perimeter (red) and area (blue) of the Koch snowflake relative to the starting triangle. Change the 4 and 3 in the red curve to N and $1/r$ for another replacement rule (try $\frac{5}{3}$ for a rule with five pieces of a third), and ask what happens to the area in that case.

Now play the chaos game and measure the dimension of what it draws. The script uses k corners of a regular polygon and contracts by r toward a random corner each step; then it box-counts the result.

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(1)
k = 3            # number of corners: try 4, 5, 6
r = 0.5          # contraction ratio toward the chosen corner: try 0.4, 0.6, 1/3
n_pts = 200_000  # points in the chaos game

corners = np.exp(2j * np.pi * np.arange(k) / k + 0.5j * np.pi)
z = 0j
pts = np.empty(n_pts, dtype=complex)
for i in range(n_pts):
    c = corners[rng.integers(k)]
    z = c + r * (z - c)          # jump a fraction (1 - r) of the way toward a random corner
    pts[i] = z
pts = pts[100:]                  # discard the transient

# box counting: how many eps-boxes contain at least one point?
eps = 2.0 ** -np.arange(2, 9)
N = [len(np.unique(np.floor(np.column_stack([pts.real, pts.imag]) / e), axis=0)) for e in eps]
D_fit = np.polyfit(np.log(1 / eps), np.log(N), 1)[0]
print(f"box-counting estimate D = {D_fit:.3f};  similarity dimension log k / log(1/r) = {np.log(k) / np.log(1 / r):.3f}")

fig, (a1, a2) = plt.subplots(1, 2, figsize=(10, 4.5))
a1.plot(pts.real, pts.imag, ",", color="navy"); a1.set_aspect("equal"); a1.set_title(f"chaos game, k={k}, r={r}")
a2.loglog(1 / eps, N, "o-"); a2.set_xlabel("1/eps"); a2.set_ylabel("N(eps)"); a2.set_title(f"slope = {D_fit:.3f}")
plt.show()
```

As written ($k = 3, r = 0.5$) it draws the Sierpinski triangle and estimates $D \approx 1.571$ against the exact 1.585. Things to try:

- $k = 4, r = 0.5$: the four copies tile the square exactly, the attractor is the filled square, and the formula says $D = 2$. The estimate is lower (1.914). Is that the fit or the number of points? Raise `n_pts` and find out.
- $k = 3, r = 0.4$: the copies no longer touch; you get a dust of dimension $\log 3 / \log 2.5 = 1.199$. The estimate (1.222) is close.
- $k = 5, r = 0.4$: the formula says 1.756 and the estimate says 1.691. Look at the picture closely, near where neighbouring copies meet. Something in the definition of similarity dimension has been violated. What?

73.6 The challenge

1. **Intuition.** A friend argues: "The Koch curve is made of line segments at every stage, so in the limit it is still a curve, and a curve has dimension 1." Find the flaw. Which property of the limit shape do the finite stages fail to predict, and why does box counting at a fixed ε not care how many stages you drew once the stage's segments are shorter than ε?

2. **Derivation.** The **Vicsek fractal** keeps the centre square and the four corner squares of a 3×3 grid, and repeats. Compute its similarity dimension. Then compute the area remaining after n stages, and the perimeter after n stages, and decide which of them tends to 0, which to infinity, and which to a finite limit.

3. **Implementation.** Extend the Mandelbrot code to draw the **Julia set** of $z \mapsto z^2 + c$ for a fixed c: now c is constant and the *starting point* z_0 varies over the pixels. Draw it for $c = -1$, $c = -0.123 + 0.745j$, and $c = 0.3$. One of these Julia sets is connected and one is a disconnected dust. Check, using your Mandelbrot code, whether each c lies in M, and formulate a conjecture relating the two pictures.

4. **Open investigation.** Box-count a real coastline. Take any outline you can get as a list of coordinates (a country border from an open map data set, or a coastline you trace by hand from a satellite image at several zoom levels), and estimate D over as many scales as the data allow. How stable is your estimate when you shift the grid by half a box, rotate it, or change the range of ε you fit over? Is there a scale range over which the coastline is truly "fractal", or does D drift? Researchers still argue about how to answer this for real data.

73.7 Loose threads

The hook asked what dimension 1.25 could mean. Now we know: a set has dimension D when the number of ε-boxes needed to cover it grows like ε^{-D}, and Richardson's ruler law $L \approx C\varepsilon^{1-D}$ is that statement in disguise. For exactly self-similar sets, $D = \log N / \log(1/r)$; the Koch curve has $D = 1.262$, and its snowflake has infinite perimeter around a finite area. Iterated function systems generate such sets as unique fixed points of a contraction on shapes, and the escape-time rule $z \mapsto z^2 + c$ generates the Mandelbrot set, whose outside a computer can certify and whose inside it never can.

What the chapter left open:

- **Is M locally connected?** The first mystery is the MLC conjecture, still open.
- **The right definition of dimension.** Box counting is easy to compute but has defects: the set of rational numbers in $[0, 1]$ is countable, yet every box meets it, so its box-counting dimension is 1. The **Hausdorff dimension** repairs this using measure theory; see the mystery below.
- **Where the strange attractor's dimension comes from.** Chaotic systems trace out fractal sets, and their dimension is linked to how fast nearby trajectories separate. See the last mystery.
- **Universal numbers.** Along the real axis of M, the period-doubling discs shrink by a ratio that tends to $4.669\ldots$, Feigenbaum's constant. The same number appears in dripping taps and convection cells. Why a single number governs so many unrelated systems is a story about bifurcations and renormalisation, begun in Ch. 85.
- **Fractals as worlds.** Fractal mountains, clouds and coastlines in games are built not from exact self-similarity but from *statistical* self-similarity: noise summed over octaves. That is the subject of Ch. 74.

? A curve of dimension two

The boundary of the Mandelbrot set looks like a curve: it has no interior, it is "one-dimensional" topologically. Yet in 1998 Mitsuhiro Shishikura proved that its Hausdorff dimension is exactly 2, as large as a curve in the plane can possibly be. What is Hausdorff dimension, why can it disagree with box counting, and what does it mean for a boundary to be two-dimensional when nobody yet knows whether it has any area at all? The answer requires building a theory of size for wild sets, which is measure theory, in Ch. 97.

? Where chaos keeps its fractals

Iterate the Hénon map $x_{n+1} = 1 - 1.4x_n^2 + y_n$, $y_{n+1} = 0.3x_n$ from almost any start, and the points settle onto a thin, layered set of box-counting dimension about 1.26, a fractal made by a dynamical system no one designed to make one. Why do chaotic systems produce fractal attractors, and why can their dimension be predicted from the rates at which nearby orbits stretch and fold (the Lyapunov exponents)? That is the Kaplan–Yorke story, in Ch. 84.

Further reading

- Benoît B. Mandelbrot, *The Fractal Geometry of Nature* (W. H. Freeman, 1982). The founding book: idiosyncratic, visual and full of ideas.
- Kenneth Falconer, *Fractal Geometry: Mathematical Foundations and Applications*, 3rd ed. (Wiley, 2014). The standard mathematical text: box counting, Hausdorff dimension, IFS, Julia sets.
- Heinz-Otto Peitgen, Hartmut Jürgens and Dietmar Saupe, *Chaos and Fractals: New Frontiers of Science*, 2nd ed. (Springer, 2004). A long, generous, picture-rich introduction with algorithms.
- Michael F. Barnsley, *Fractals Everywhere*, 2nd ed. (Academic Press, 1993). Iterated function systems and the chaos game, from the person who popularised them.

Chapter 74 — Procedural Generation: Creating Worlds with Rules

> **Level 2 · Track G — Mathematics for Computer Graphics, Games, and Simulation Builds on:** Ch. 21 (random processes, power spectral density), Ch. 10 (dominant eigenvalues), Ch. 22 (§22.5 interpolation), Ch. 73 · **Leads to:** Ch. 75, Ch. 76, Ch. 86 · **Related:** Ch. 81, Ch. 30, Ch. 29 **Reading time:** ~30 min

74.1 The hook

In 1984 two Cambridge undergraduates, David Braben and Ian Bell, released *Elite*, a space-trading game for the BBC Micro. The machine had 32 kilobytes of memory, shared between the program, the screen and everything else. The game contained eight galaxies of 256 star systems each: 2048 worlds, every one with a name, a position, a government, an economy, a population and a short description. Written out as text, that catalogue alone would not have fitted in the machine.

It was never written out. Each galaxy was a few bytes of **seed**. A short, deterministic scrambling routine turned the seed into the next seed, and the bits of each seed were read off as the properties of the next planet. Fly away from Lave and come back, and Lave is still there, with the same name and the same economy, because it is recomputed from the same seed every time.

Thirty-two years later *No Man's Sky* did the same thing at a larger scale: a 64-bit seed space, $2^{64} \approx 1.8 \times 10^{19}$ planets, each with continents, caves, weather and creatures, generated only when a player arrives. Minecraft does it for terrain: give it a seed, and an effectively endless landscape unrolls as you walk, identical for every player who types the same seed.

There is a paradox here. A world worth exploring has to look *designed*: mountains that rise smoothly into ridges, rivers that run downhill, caves that connect, trees that look like trees. Randomness alone gives static, the snow on an untuned television. Order alone gives a chessboard. And whatever you make has to be *reproducible*: the mountain you saw yesterday must still be there today, even though you never stored it.

How can a few bytes and a set of rules produce a world that is random enough to surprise, structured enough to believe, and deterministic enough to revisit?

74.2 The big idea

Replace storage with functions

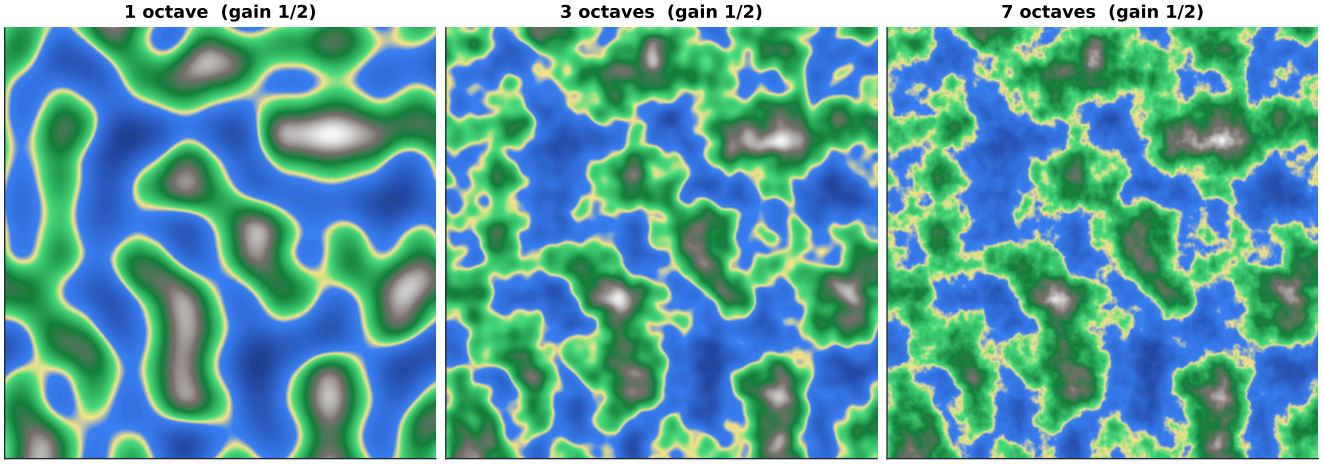
The central trick is to stop thinking of a world as data and start thinking of it as a **function of position**. The height of the terrain at (x, y) is not looked up in a table; it is computed by a function $h(x, y)$ whenever it is needed. If h is a pure function (same input, same output, no hidden state) then the world is automatically consistent. Two players, or the same player on two days, evaluating $h(1000, 2000)$ get the same mountain. Nothing outside the player's view ever needs to exist.

The seed is just one more argument: $h(x, y; \text{seed})$. And the randomness inside h comes from a **hash function**: a deterministic scramble that maps integers to values that look independent, so that `hash(1000, 2000, seed)` and `hash(1001, 2000, seed)` appear unrelated. How a deterministic function can look random enough is a deep question of its own (Ch. 81); here we just use it.

Random, but smooth: noise

Hashing each pixel independently gives television static: every point unrelated to its neighbour. Real terrain is **correlated**: the height a metre away is nearly the same as here, the height a kilometre away is unrelated. In the language of Ch. 21, we want a random field with a chosen autocorrelation, or equivalently a chosen power spectrum.

Ken Perlin's answer, devised in 1983 after he had worked on the film *Tron* and later honoured with an Academy Award for technical achievement, is beautifully cheap. Put random values only at the corners of an integer grid, and interpolate smoothly in between. The result is a hilly landscape with features about one grid cell across. To get features of every size at once, add copies at doubled frequency and halved amplitude, a sum of **octaves**, and the result has the statistical self-similarity of the fractal coastlines of Ch. 73.



Gradient noise summed over 1, 3 and 7 octaves, each octave at double the frequency and half the amplitude of the last, on the same 6×6 grid of lattice cells with the same seed. The large-scale continents are fixed by the first octave; later octaves only add detail.

Three more kinds of rule

Noise makes landscapes. Other structures need other rules, and procedural generation is a toolbox of them:

- **Cellular automata**: a grid where each cell updates from its neighbours by a local rule. A few steps of "become wall if most of your neighbours are walls" turn random static into caves.
- **Grammars**: rewriting rules on strings, like $F \to FF$, whose output is read as drawing instructions. They grow plants, buildings and road networks.
- **Spatial structures**: ways of scattering points so that they are random but not clumped (trees in a forest, stars in a sky), and of carving space into regions around them (biomes, kingdoms).

Each is a small machine for turning a seed into structure. The mathematics is about controlling what comes out.

74.3 The mathematics

Value noise and the fade curve

In one dimension, attach a pseudo-random value a_i to every integer i. For x between i and $i+1$, let $t = x - i \in [0, 1)$ and interpolate:

$$n(x) = a_i + f(t)\,(a_{i+1} - a_i),$$

where f is a **fade curve** with $f(0) = 0$ and $f(1) = 1$. With $f(t) = t$ this is linear interpolation (Ch. 22, §22.5), and the result has corners at every integer: the derivative jumps. For terrain that is fatal, because lighting depends on the surface normal, which depends on the derivative: every kink shows up as a visible crease.

How smooth does f need to be? At an integer the curve from the left cell ends with slope $(a_i - a_{i-1})f'(1)$ and the curve on the right starts with slope $(a_{i+1} - a_i)f'(0)$. These generally differ unless both are zero. The same argument applies to every derivative.

The quintic fade curve

The polynomial of lowest degree with $f(0) = 0$, $f(1) = 1$ and $f'(0) = f'(1) = f''(0) = f''(1) = 0$ is

$$f(t) = 6t^5 - 15t^4 + 10t^3.$$

Interpolating with it makes noise whose first and second derivatives are continuous everywhere.

Proof. We need f' to vanish to first order at $t = 0$ and at $t = 1$ (so that f' and f'' are both zero there). The lowest-degree polynomial with double roots at 0 and 1 is $f'(t) = c\,t^2(1-t)^2$. Integrating from 0,

$$f(t) = c\left(\frac{t^3}{3} - \frac{t^4}{2} + \frac{t^5}{5}\right),$$

and $f(1) = c\,(\tfrac{1}{3} - \tfrac{1}{2} + \tfrac{1}{5}) = c/30$, so $c = 30$, giving $f(t) = 10t^3 - 15t^4 + 6t^5$. Any lower-degree candidate would make f' of degree at most 3 with two double roots, which forces $f' \equiv 0$ and contradicts $f(1) = 1$. Since every cell's interpolant has zero first and second derivatives at both ends, they match across cell boundaries. ∎

Perlin's original 1985 noise used the cubic **smoothstep** $3t^2 - 2t^3$, which has $f'(0) = f'(1) = 0$ but $f''(0) = 6$ and $f''(1) = -6$. The second derivative jumps at every grid line, and on shaded terrain the grid shows through. Perlin switched to the quintic in 2002. You can compare the two curves in the experiment.

Gradient noise

Value noise has a subtler flaw: its features line up with the grid, because the extreme values sit at lattice points. **Gradient noise**, which is what "Perlin noise" usually means, attaches a random *slope* g_i to each lattice point instead of a random value, and blends the linear functions $g_i(x - i)$:

$$n(x) = \big(1 - f(t)\big)\, g_i\, t + f(t)\, g_{i+1}\,(t - 1).$$

At $t = 0$ this is 0, and its derivative there is g_i (because $f(0) = f'(0) = 0$). So gradient noise is *zero at every lattice point* with a random slope through it. The peaks and valleys fall between lattice points, which hides the grid, and the noise has no tendency to drift away from zero over large distances: almost no power at very low frequencies. In 2-D, the slopes become random unit vectors $\mathbf{g}_{ij}$, the linear functions become dot products $\mathbf{g}_{ij} \cdot (\mathbf{x} - \mathbf{x}_{ij})$ at the four corners of the cell, and the blending is the fade curve applied separately in x and y.

Fractal sums of octaves

A single layer of noise has features of one size. Terrain has all sizes. **Fractional Brownian motion** (fBm), in the procedural sense, sums octaves:

$$F(\mathbf{x}) = \sum_{k=0}^{K-1} g^k\, n(2^k \mathbf{x}),$$

with **gain** (or persistence) $g < 1$ and **lacunarity** 2. Two facts make this controllable.

It is bounded. If $|n| \le B$, then $|F| \le B \sum_k g^k = B\,(1 - g^K)/(1 - g) < B/(1-g)$. With $g = 1/2$ and $K = 7$ octaves the bound is $1.98B$: adding octaves can never make mountains taller than twice the first octave's. This is why the continents in the figure do not move as octaves are added.

Its spectrum is a power law. Octave k puts its power, proportional to amplitude squared, g^{2k}, near frequency $\nu_k \propto 2^k$. So power falls off as $\nu^{-\beta}$ with $2^{-\beta k} = g^{2k}$, that is

$$\beta = -2 \log_2 g.$$

The gain $g = 1/2$ gives $\beta = 2$, the spectrum of Brownian motion (Ch. 21). This connects noise to fractals. The exponent $H = -\log_2 g$ (the Hurst exponent) controls how roughness scales, and for $0 < H < 1$ the graph of a one-dimensional profile has fractal dimension $D = 2 - H$. With $g = 0.6$, $H = 0.737$ and $D = 1.263$: a terrain profile with almost exactly the roughness of the Koch curve. We state these relations rather than prove them; the precise statement needs the theory of self-similar random processes.

? The noise that is not quite isotropic

Gradient noise on a square grid is not the same in every direction: look closely at a heightmap and features prefer to align with the axes and diagonals, and the power spectrum, viewed in 2-D, is not a perfect ring. Perlin's later *simplex noise* uses a triangular grid to reduce this, and it is also cheaper in high dimensions. But what exactly is the power spectrum of gradient noise, why is it not band-limited, and what aliasing does that cause when the terrain is sampled onto pixels? Those are sampling-theory questions, answered in Ch. 76.

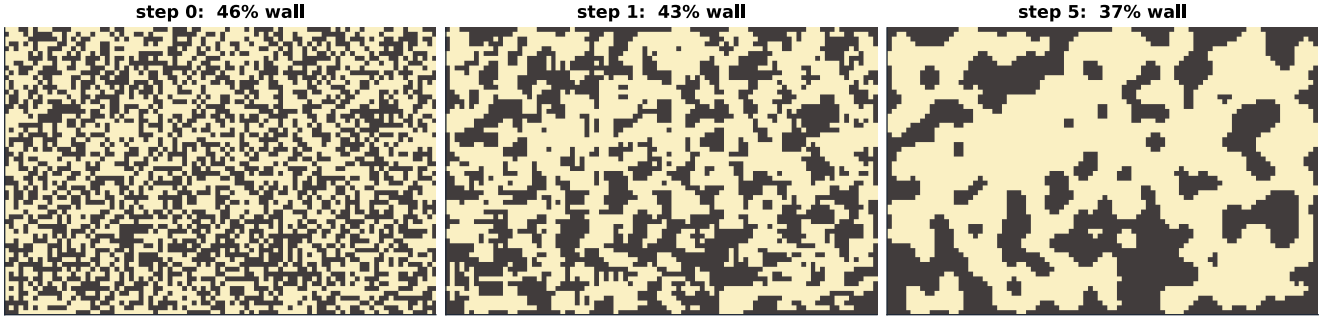
Cellular automata: caves from static

Fill a grid with walls at random, each cell independently a wall with probability p. Then apply the **4–5 rule** a few times: each cell becomes a wall if at least 5 of the 9 cells in its 3×3 block (itself included) are walls, and floor otherwise. It is a local majority vote.

After one step, before any correlations have built up, a cell is a wall with probability

$$P(\text{wall}) = \sum_{k=5}^{9} \binom{9}{k} p^k (1-p)^{9-k}.$$

For $p = 0.45$ this is 0.379: a minority shrinks under majority voting, so the floor grows. But the deeper effect is *spatial*: isolated wall cells vanish, isolated floor cells fill in, and boundaries between wall and floor straighten. After four or five steps the static has become smooth cave walls, and further steps change almost nothing.



Cave generation with the 4–5 rule on a 60×90 grid, cells outside the map counting as wall. One step already clusters the static; by step 5 the caves are smooth and almost stable.

The starting density p is the main control. Too low, and the walls dissolve into one open hall; too high, and the floor breaks into sealed pockets. The useful window is narrow because connectivity of random sets does not change gradually: it switches abruptly at a **percolation threshold**, the density at which a connected floor region spanning the whole map first appears. In practice, generators run a flood fill afterwards and either connect or delete the isolated pockets.

Grammars: L-systems

In 1968 the biologist Aristid Lindenmayer introduced string-rewriting systems to model the growth of algae. An **L-system** has an alphabet, a starting string (the axiom) and rules that replace *every* symbol at once in each generation. Lindenmayer's algae: $A \to AB$, $B \to A$, axiom A:

$$A \to AB \to ABA \to ABAAB \to ABAABABA \to \cdots$$

The lengths are $1, 2, 3, 5, 8, \ldots$: Fibonacci numbers. That is no accident, and the reason gives a general tool.

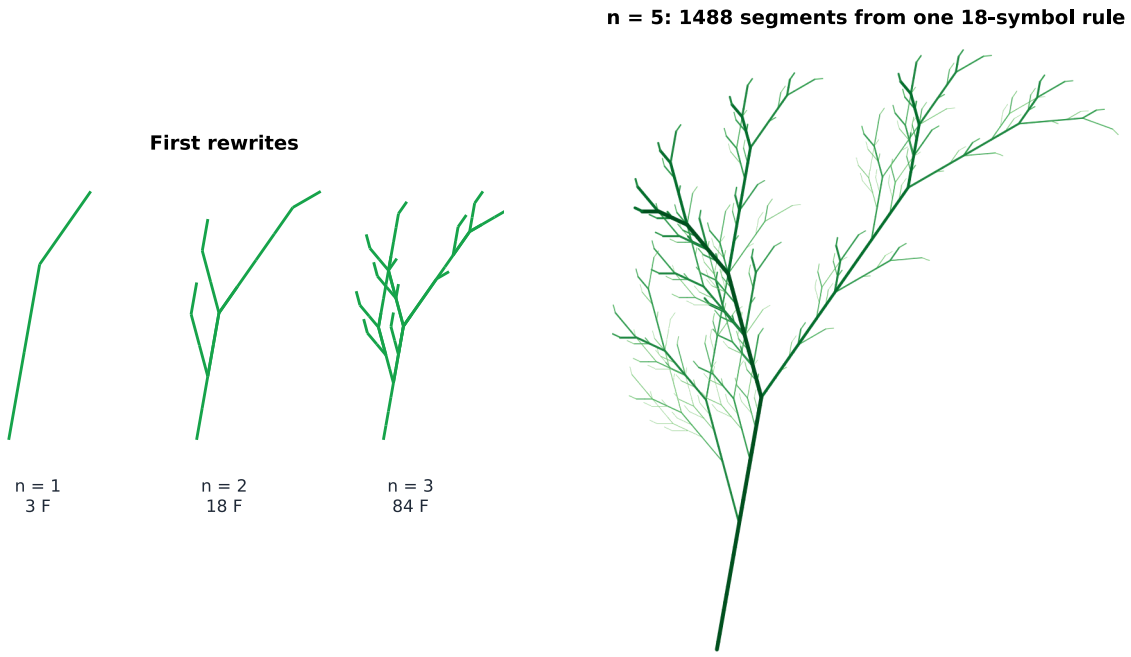
Growth of an L-system

Let c_n be the vector of symbol counts in generation n, and let $\mathbf{M}$ be the matrix whose entry M_{ab} is the number of symbols a produced by one symbol b. Then $c_{n+1} = \mathbf{M} c_n$, so $c_n = \mathbf{M}^n c_0$, and the string length grows like λ^n, where λ is the eigenvalue of $\mathbf{M}$ of largest magnitude (when c_0 has a component along its eigenvector).

Proof. Every symbol is rewritten independently, so the number of a's in generation $n+1$ is $\sum_b M_{ab}\,(c_n)_b$, which is the matrix–vector product. Iterating gives $c_n = \mathbf{M}^n c_0$. Expanding c_0 in eigenvectors of $\mathbf{M}$ (Ch. 10), the component along the largest eigenvalue dominates, exactly as in the power method. ∎

For the algae, $\mathbf{M} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ (columns: what A and B produce), with eigenvalues $\varphi = 1.618$ and -0.618: length grows like the golden ratio to the n-th power.

To draw, read the string as commands for a **turtle**: `F` draws a step forward, `+` and `-` turn by an angle, `[` saves the turtle's position and heading on a stack, `]` restores it. The brackets create branches. The plant in the figure uses $X \to F + [[X] - X] - F[-FX] + X$ and $F \to FF$ with a turn of 25°. The counts obey $f_{n+1} = 2f_n + 3x_n$, $x_{n+1} = 4x_n$, a triangular matrix with eigenvalues 2 and 4, and solving with $f_0 = 0$, $x_0 = 1$ gives $f_n = \frac{3}{2}(4^n - 2^n)$: 1488 segments at $n = 5$.



An L-system plant. One rule, applied five times, gives 1488 branch segments; line width and shade follow the depth of bracket nesting.

Spatial structures

Scatter trees by independent uniform random points and you get clumps and gaps: a Poisson process is clumpier than intuition expects. Forests, stars and village layouts look better with **blue-noise** point sets, where no two points are closer than a radius r but the points are otherwise random. **Poisson-disk sampling** produces them; Bridson's algorithm grows the set outward from a seed point, trying a few candidates in an annulus around each active point and accepting those at least r from all others, using a background grid of cell size $r/\sqrt{2}$ so that each check touches only a few neighbours. It runs in $O(N)$ time.

Given scattered points, the **Voronoi diagram** assigns every location to its nearest point: biomes around seed points, territories around capitals, cells for cracked stone. Computing it efficiently, along with its dual triangulation, is the business of Ch. 75.

74.4 The algorithm

2-D gradient noise

```
def fade(t):
    return t*t*t*(t*(6*t - 15) + 10)            # 6t^5 - 15t^4 + 10t^3

def noise2(x, y, perm, grad):                    # perm: shuffled 0..255 (twice); grad: 256 unit vectors
    i, j = int(np.floor(x)), int(np.floor(y))
    u, v = x - i, y - j                          # position inside the cell
    def corner(di, dj):
        g = grad[perm[perm[(i + di) & 255] + ((j + dj) & 255)]]   # hash the lattice point
        return g[0]*(u - di) + g[1]*(v - dj)     # dot(gradient, offset to corner)
    a, b = fade(u), fade(v)
    bottom = corner(0, 0) + a*(corner(1, 0) - corner(0, 0))
    top    = corner(0, 1) + a*(corner(1, 1) - corner(0, 1))
    return bottom + b*(top - bottom)
```

Cost. Four hashes, four dot products and three interpolations per sample in 2-D, and 2^d corners in d dimensions, which is why simplex noise ($d+1$ corners) wins in 3-D and 4-D. An fBm with K octaves costs K times as much.

Determinism. The permutation table is the seed. Shuffle it with a seeded generator once; never call an unseeded random function inside the noise. The table repeats every 256 cells, which is invisible at the scales most games use and very visible on an infinite map; hashing the full integer coordinates with a good integer hash removes the period.

Precision. At large coordinates (a million cells from the origin) single-precision floats lose the fractional part that the fade curve needs, and the terrain becomes stepped. Infinite worlds work in local coordinates, chunk by chunk.

Caves, plants and points

```
def cave(W, H, p=0.45, steps=5, seed=0):
    rng = np.random.default_rng(seed)
    G = (rng.random((H, W)) < p).astype(int)
    for _ in range(steps):
        P = np.pad(G, 1, constant_values=1)       # border is solid rock
        S = sum(P[1+dy:H+1+dy, 1+dx:W+1+dx] for dy in (-1, 0, 1) for dx in (-1, 0, 1))
        G = (S >= 5).astype(int)                  # 4-5 rule: majority of the 3x3 block
    return G                                      # then flood-fill to find connected caves

def lsystem(axiom, rules, n):
    for _ in range(n):
        axiom = "".join(rules.get(ch, ch) for ch in axiom)
    return axiom                                  # length grows like lambda^n
```

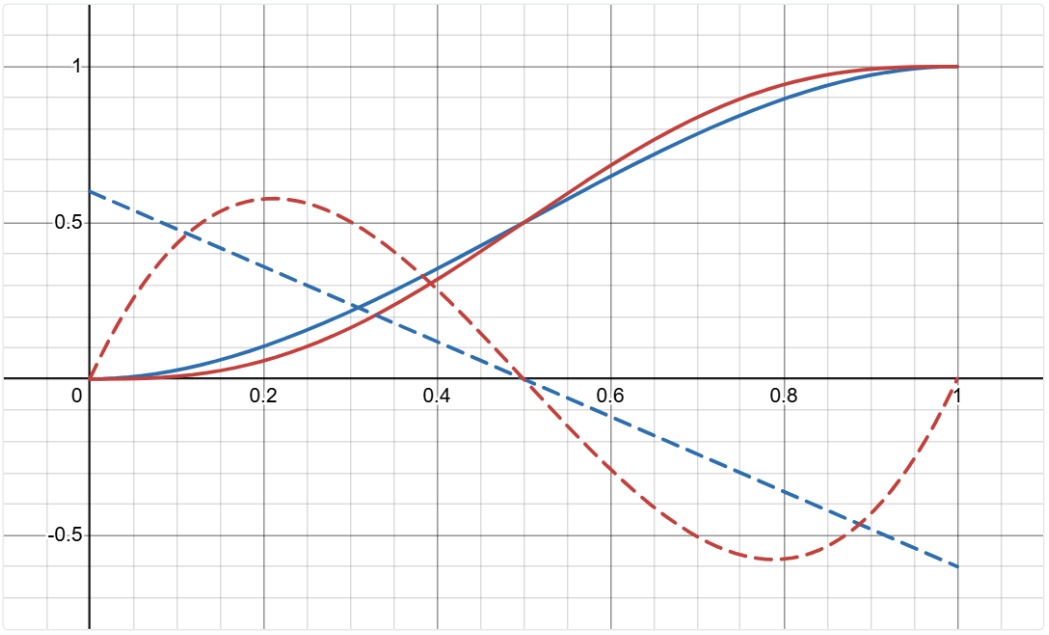
Cave generation costs $O(WH)$ per step and needs only a handful of steps. An L-system's string length grows exponentially, $O(\lambda^n)$, so memory runs out fast; production systems expand lazily, generating each branch only when the camera is close enough to see it, which is the same "compute on demand" principle as noise.

Tool	What it makes	Main knob	Typical failure
gradient noise + fBm	terrain, clouds, textures	gain g, octaves	grid artifacts, repetition, too uniform
cellular automaton	caves, islands, erosion	starting density p, steps	disconnected pockets, one giant hall
L-system	plants, rivers, cities	rules, angle, generations	exponential blow-up, self-intersection
Poisson-disk sampling	forests, stars, loot	minimum spacing r	uniform look at large scale

> **Variety is not the same as interest**
>
> Eighteen quintillion planets generated by one noise function are eighteen quintillion variations on one theme. Players quickly learn the generator's "voice". Shipped games layer rules: noise for the base, automata and erosion for local structure, hand-made pieces placed by grammars, and constraints that guarantee a playable result (a path from start to exit, a key before its lock).

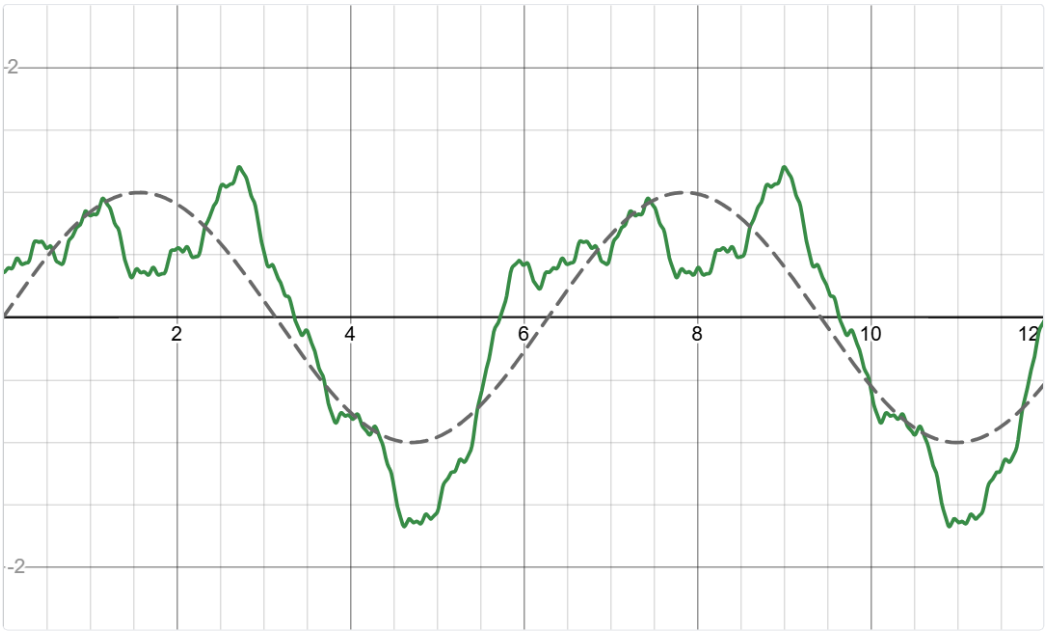
74.5 The experiment

First, see the fade curves and why the quintic is smoother. The blue curve is the cubic smoothstep, the red is the quintic, and the dashed lines are their second derivatives (scaled by $1/10$ to fit).



Cubic smoothstep (blue) and quintic fade (red) on one cell. The blue dashed second derivative is ± 0.6 at the cell edges, so it jumps between cells; the red one is zero at both ends.

Next, a sum of octaves you can edit. Desmos cannot hash, so this uses sines with fixed phases in place of noise, but the octave structure is the same: frequency 2^k, amplitude g^k with $g = 0.5$.



A profile with 7 octaves (green) over its first octave alone (grey dashed). Change the 0.5 to 0.7 for rougher terrain, or to 0.3 for gentle hills, and change the 6 to see how many octaves your eye can tell apart.

Now a real 1-D gradient-noise terrain, with knobs for the seed, the number of octaves and the gain:

```
import numpy as np
import matplotlib.pyplot as plt

seed = 74         # change it: a different world, the same every time you run
octaves = 6       # try 1, 3, 10
gain = 0.5        # amplitude ratio per octave: try 0.3 (smooth) and 0.7 (rough)

rng = np.random.default_rng(seed)
slopes = rng.uniform(-1, 1, 4096)          # one random gradient per lattice point

def fade(t):
    return t**3 * (t * (6 * t - 15) + 10)

def noise(x):
    i = np.floor(x).astype(int)
    t = x - i
    g0, g1 = slopes[i % 4096], slopes[(i + 1) % 4096]
    return (1 - fade(t)) * g0 * t + fade(t) * g1 * (t - 1)   # zero at every integer

def fbm(x):
    return sum(gain**k * noise(2.0**k * x + 17.3 * k) for k in range(octaves))

x = np.linspace(0, 20, 4000)
h = fbm(x)
print(f"range [{h.min():.3f}, {h.max():.3f}],  bound 0.5*(1-g^K)/(1-g) = {0.5 * (1 - gain**octaves) / (1 - gain):.3f}")

plt.fill_between(x, h, h.min() - 0.2, color="sienna")
plt.fill_between(x, np.minimum(h, 0), h.min() - 0.2, where=h < 0, color="steelblue")
plt.title(f"seed {seed}, {octaves} octaves, gain {gain}")
plt.show()
```

The printed bound uses the fact that 1-D gradient noise with slopes in $[-1, 1]$ never exceeds 0.5 in magnitude, so $B = 0.5$. Check it against the printed range for several seeds and gains. Then try `gain = 0.9` with `octaves = 10`: what happens to the picture, and to the gap between the range and the bound?

74.6 The challenge

1. **Intuition.** A level designer generates an island with 8 octaves of fBm, and the beach looks the same at every zoom level, which she likes. The mountains also look the same at every zoom level, which she does not: real mountains are smooth at the top and rough lower down, or the other way round. Explain, without equations, why plain fBm cannot do this, and propose a modification in which the amplitude of each octave depends on the height already accumulated.

2. **Derivation.** For 1-D gradient noise with slopes $g_0, g_1 \in [-1, 1]$ on one cell, show that $|n(x)| \le 1/2$ for all x, and find the slopes and the point t at which the bound is reached. Then find the lowest-degree fade curve whose first, second *and third* derivatives vanish at both ends.

3. **Implementation.** Implement Bridson's Poisson-disk sampling in a unit square with minimum spacing r. Compare its points with the same number of uniform random points by computing, for each set, the histogram of nearest-neighbour distances and the radially averaged power spectrum of the point pattern (place a delta at each point and take a 2-D FFT). Where does the blue-noise spectrum have *less* power than the uniform one, and why does the eye prefer that?

4. **Open investigation.** For the 4–5 cave rule on a large grid, measure, as a function of the starting density p, the probability that after 5 steps the floor contains a connected path from the left edge to the right edge. Locate the value of p at which this probability crosses $1/2$, and see how sharply it switches as the grid grows. Is it the same threshold as for plain random static (site percolation on the 8-connected grid) or is it shifted by the automaton? Nobody has a formula for the shifted threshold.

74.7 Loose threads

The hook asked how a few bytes can produce a believable, revisitable world. The answer has three parts. Make the world a pure function of position and seed, so that it is consistent without being stored. Make the randomness *correlated*: lattice noise interpolated with a fade curve whose first and second derivatives vanish at the cell edges, summed over octaves into a bounded fractal with a power-law spectrum $\beta = -2\log_2 g$. And add structure with other local rules: majority-vote automata that turn static into caves, rewriting grammars whose growth rate is the dominant eigenvalue of a count matrix, and blue-noise point sets carved into Voronoi regions.

What the chapter did not tell you:

- **Spectra and sampling.** The first mystery: what is the exact spectrum of gradient noise, and how should it be filtered before sampling on pixels? That is Ch. 76.
- **What else automata can do.** The 4–5 rule settles down. Other rules, just as simple, never do: Conway's Game of Life and the one-dimensional Rule 110 can carry out any computation. When a local rule produces stable structure, chaos or a universal computer is the subject of Ch. 86.
- **Voronoi and Delaunay.** Computing nearest-point regions and their dual triangulation in $O(N \log N)$ time is Ch. 75.
- **How random is the hash?** Every seed-to-world pipeline relies on a deterministic function behaving "randomly enough". What that means precisely is Ch. 81.

> **? The tiling that no algorithm can finish**
>
> Modern generators such as *Wave Function Collapse* build levels by placing tiles that must match their neighbours' edges, backtracking when they get stuck. It feels like a puzzle that a sufficiently clever program could always solve or reject. It is not. In 1966 Robert Berger proved that, given a finite set of square tiles with coloured edges, *no algorithm* can decide in general whether they tile the infinite plane, and some sets tile it only non-periodically. Why the answer is not merely hard but undecidable is a story about Turing machines hidden inside tiles, told in Ch. 30.

Further reading

- David S. Ebert, F. Kenton Musgrave, Darwyn Peachey, Ken Perlin and Steven Worley, *Texturing and Modeling: A Procedural Approach*, 3rd ed. (Morgan Kaufmann, 2003). The classic: noise, fBm and procedural terrain from the people who invented them.
- Ken Perlin, "Improving Noise", *ACM Transactions on Graphics* 21(3) (SIGGRAPH 2002). Two pages that introduce the quintic fade curve and the modern gradient set.
- Przemysław Prusinkiewicz and Aristid Lindenmayer, *The Algorithmic Beauty of Plants* (Springer, 1990; free online). L-systems and turtle graphics, with stunning figures.
- Noor Shaker, Julian Togelius and Mark J. Nelson, *Procedural Content Generation in Games* (Springer, 2016; free online). A survey of the whole field, from cellular automata to grammars and constraint solving.

Chapter 75 — Computational Geometry: Algorithms for Shapes

> **Level 2 · Track G — Mathematics for Computer Graphics, Games, and Simulation Builds on:** Ch. 8, Ch. 22, Ch. 26 · **Leads to:** Ch. 76, Ch. 89 · **Related:** Ch. 27, Ch. 47, Ch. 71, Ch. 72, Ch. 74 **Reading time:** ~30 min

75.1 The hook

In September 1854 cholera killed more than five hundred people in a few streets of Soho, London, in ten days. The physician John Snow marked every death on a street map, and then drew one extra line on it. The line enclosed every house that was closer, by walking distance, to the water pump on Broad Street than to any other pump. The deaths clustered inside that line. The handle of the pump was removed, and the map became one of the founding documents of epidemiology.

Snow's line is a piece of a **Voronoi diagram**: give every pump the region of the plane that is nearer to it than to any other pump. He drew it by hand for a dozen pumps. Today the same question is asked millions of times a second.

- A phone network asks which tower should serve each handset.
- A game asks which of 20,000 units is nearest to the click of the mouse, sixty times a second.
- A finite-element solver needs the plane cut into well-shaped triangles whose corners are a cloud of measured points.
- A robot wants the route through a crowd of obstacles that stays as far from all of them as possible.

The naive answer to "who is nearest?" is to check every site: n distance computations per query, and n^2 to build the whole diagram. With a million sites that is 10^{12} operations, about a quarter of an hour on a fast core, and it would have to be repeated whenever a site moves.

There is a second, nastier problem. Even the tiniest geometric question ("is this point to the left of that line?") is answered by a computer with rounded arithmetic. Ask it three slightly different ways and you can get three answers that cannot all be true. Geometry programs that are correct on paper crash, loop forever or produce polygons that cross themselves.

So this chapter has two goals:

> **How do you build shapes from points (hulls, triangulations, nearest-site regions) in $O(n \log n)$ time, and how do you make every decision rest on one test you can trust?**

The answer is a small toolkit: one determinant, one circle, one paraboloid and one tree. By the end you will know why Snow's map and a mesh of triangles are the same object drawn twice.

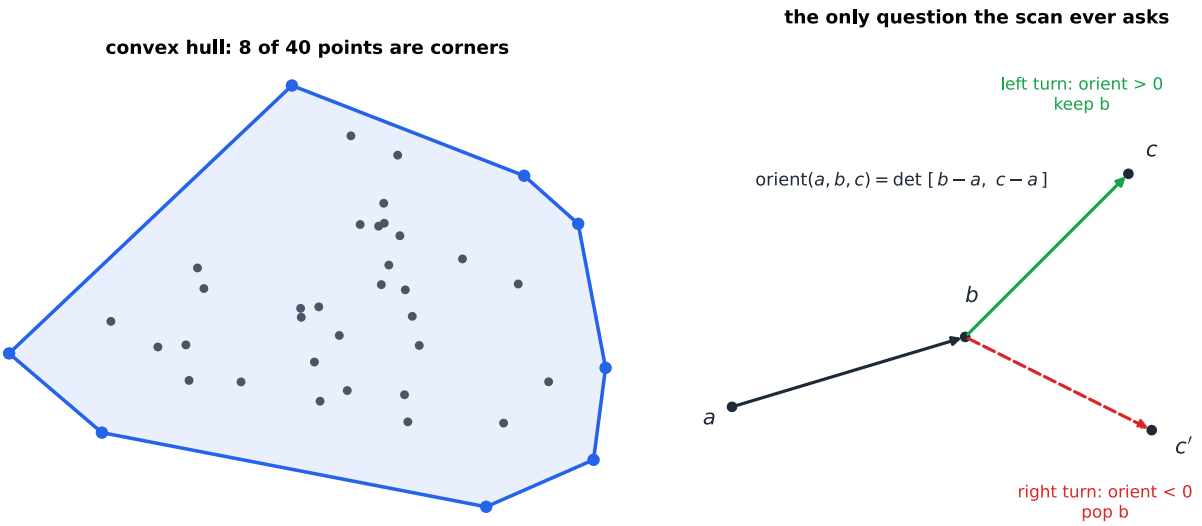
75.2 The big idea

Geometry algorithms are built from tiny yes/no questions. Almost nothing in this chapter computes a length or an angle. Every algorithm asks, over and over, a question with a one-bit answer.

- *Orientation:* do points a, b, c make a left turn, a right turn, or lie on a line?
- *In-circle:* is point d inside the circle through a, b, c?
- *Containment:* does a query box overlap this cell of space?

Each of these is the **sign of a determinant**. The whole art is to arrange the questions cleverly so that you ask as few as possible, and to compute each sign so reliably that the answers never contradict one another.

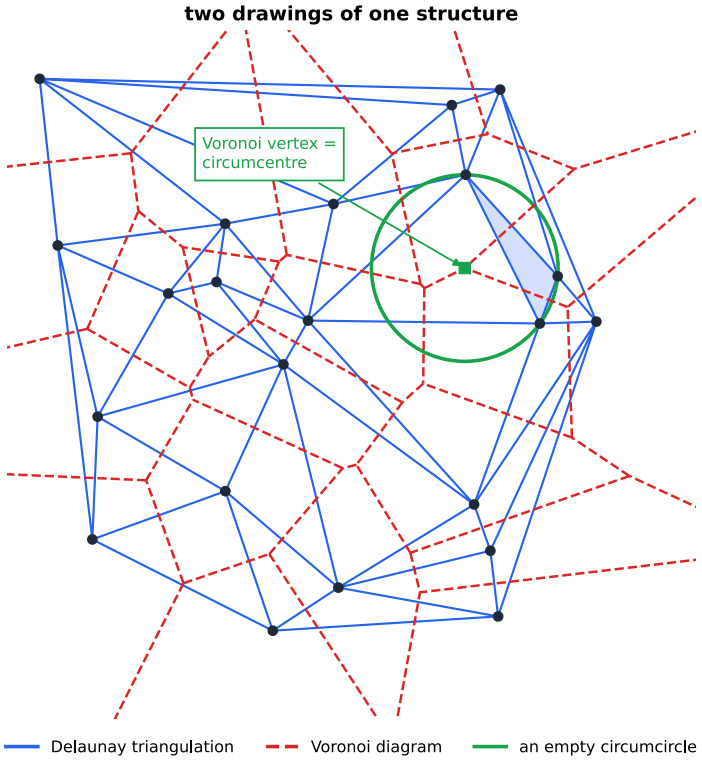
The convex hull is a rubber band. Hammer nails into a board at your points and snap a rubber band around them. The shape it settles into is the **convex hull**: the smallest convex set containing every point (Ch. 47 is the chapter about why convexity matters). Its corners are a subset of your points. To find them you walk around the cloud and keep only the points where the band turns left.



Left: the hull of 40 random points has only 8 corners. Right: the scan's only question is whether the path $a \to b \to c$ turns left (keep b) or right (discard b), and that is the sign of a 2×2 determinant.

Triangulations are the plane cut into pieces you can compute with. Given points, join them with non-crossing segments until no more can be added. You get a triangulation. There are astronomically many. The **Delaunay triangulation** is the one that avoids skinny triangles as far as possible. Its defining rule is a beautifully local one: no point lies inside the circle through the three corners of any triangle.

Voronoi and Delaunay are one structure drawn twice. Draw the Voronoi diagram of the points (Snow's regions). Now join two points whenever their regions share an edge. The result is exactly the Delaunay triangulation. Each Voronoi corner sits at the centre of an empty circle through three sites, which is the circumcircle of a Delaunay triangle.



Delaunay edges (blue) cross the Voronoi edges (red) at right angles. The green circle passes through the three corners of the shaded triangle, contains no other point, and is centred on a Voronoi vertex.

Spatial indexes are filing cabinets for space. To answer "what is near here?" without checking everything, put the points in a tree whose nodes are regions. A **quadtree** cuts a square into four quadrants, and keeps cutting any quadrant that holds too many points. Empty space costs almost nothing, and crowded regions get fine cells. A query descends only into cells that touch the region it cares about.

The analogy to keep in mind is a library. Nobody finds a book by reading every spine. The building is divided into floors, rooms, shelves. Each division answers one yes/no question ("is it in this room?") and discards most of the collection at once. That is where $\log n$ comes from, in sorting (Ch. 26) and in geometry.

75.3 The mathematics

The orientation predicate

Points live in $\mathbb{R}^2$ with coordinates $a = (a_x, a_y)$ and so on.

Orientation

$$\operatorname{orient}(a, b, c) = \det \begin{bmatrix} b_x - a_x & c_x - a_x \\ b_y - a_y & c_y - a_y \end{bmatrix} = (b_x - a_x)(c_y - a_y) - (b_y - a_y)(c_x - a_x).$$

The path $a \to b \to c$ is a **left turn** (counter-clockwise) if orient > 0, a **right turn** if orient < 0, and the three points are **collinear** if it is zero.

Why the sign means what it says: the determinant is twice the signed area of triangle abc (§8.3 interprets a determinant as signed area). It is the z-component of the cross product $(b - a) \times (c - a)$, which is positive exactly when $c - a$ lies counter-clockwise of $b - a$ by an angle between 0 and π. Example: $a = (0, 0)$, $b = (2, 0)$, $c = (1, 1)$ gives $2 \cdot 1 - 0 \cdot 1 = 2 > 0$, a left turn, and the triangle has area 1.

With this one predicate you can already decide whether two segments cross (each segment's endpoints lie on opposite sides of the other's line: four orientation tests) and whether a point is inside a convex polygon (it is left of every edge).

Convex hull by Graham's scan

Sort the points by x (ties by y). Walk left to right building the **lower hull** on a stack. Before pushing a new point p, look at the top two stack points q_1, q_2 (with q_2 on top). While $\operatorname{orient}(q_1, q_2, p) \le 0$, pop q_2. Then push p. Repeat right to left for the **upper hull**, and join the two. (This left-to-right form is Andrew's monotone chain, a variant of Graham's 1972 scan that sorts by angle.)

Graham's scan is correct and runs in $O(n \log n)$

For n points in general position the scan returns the corners of the convex hull in counter-clockwise order. After sorting, it performs at most $2n$ pushes and $2n$ pops, so its total running time is $O(n \log n)$, dominated by the sort.

Proof of correctness (lower hull). Write $p_1, \dots, p_k$ for the first k points in sorted order. We claim that after processing p_k the stack holds the lower hull of $\{p_1, \dots, p_k\}$ from p_1 to p_k. For $k = 2$ the stack is p_1, p_2 and the lower hull of two points is the segment between them. Assume the claim for $k - 1$. The new point p_k has the largest x so far, so it is certainly a corner of the new lower hull. A previous corner q stays on the lower hull exactly when it is not above the segment from its predecessor on the hull to p_k. The pops remove the top point q_2 precisely when $q_1 \to q_2 \to p_k$ fails to turn left, that is, when q_2 lies on or above segment $q_1 p_k$. Once a left turn is found at the top, every point deeper in the stack already makes left turns with its neighbours (induction hypothesis), and since p_k lies to the right of all of them those turns stay left. So the remaining stack plus p_k is a chain that turns left at every corner and has every point on or above it: the lower hull. ■

Proof of the running time. Each point is pushed once per pass and popped at most once per pass, because a popped point never returns. Each orientation test either pops something or ends the while-loop for that p. So the tests number at most (pops) + (pushes) $\le 4n$ over both passes. This is an amortised argument (§26.3): one step can pop many points, but the total over the whole run is linear. ■

Can one do better than $n \log n$? Not in the comparison-style models: if you could compute hulls faster, you could sort faster. Put the numbers $x_1, \dots, x_n$ on the parabola as points (x_i, x_i^2). All of them are hull corners, and reading the hull in order lists the x_i sorted. So the sorting lower bound of §26.3 transfers to hulls.

A hull that pays for its own size

The lower bound used a cruel input where *every* point is a corner. The figure's random cloud has 40 points but only 8 corners. Our simulations of 10^5 uniform points in a square give about 31 corners on average. Is there an algorithm whose cost depends on the *answer* size h, faster than $n \log n$ when h is small? Gift wrapping costs $O(nh)$, which is worse when h is large. Chan's 1996 algorithm achieves $O(n \log h)$ by guessing h and squaring the guess, and a matching lower bound says nobody can beat it. That kind of output-sensitive lower bound is algebraic decision-tree reasoning, the next layer of Ch. 30.

Counting a triangulation

How big is a triangulation? Suppose n points, no three collinear, h of them on the hull. Any triangulation of them, Delaunay or not, has the same size.

Size of a planar triangulation

Every triangulation of n points with h hull corners has exactly $t = 2n - 2 - h$ triangles and $e = 3n - 3 - h$ edges.

Proof. A triangulation is a connected planar graph with $V = n$ vertices. Its faces are the t triangles plus one outer face, so $F = t + 1$. Count edge–face incidences. Each triangle has 3 edges, and the outer face is bounded by the h hull edges. Each edge borders exactly two faces. So

$$3t + h = 2e.$$

Euler's formula $V - E + F = 2$ (Ch. 27) gives $n - e + (t + 1) = 2$, that is $e = n + t - 1$. Substitute into the incidence count: $3t + h = 2n + 2t - 2$, so $t = 2n - 2 - h$ and then $e = 3n - 3 - h$. ■

The 22 points in the Delaunay figure have 7 hull corners, so the formula predicts $2 \cdot 22 - 2 - 7 = 35$ triangles and $3 \cdot 22 - 3 - 7 = 56$ edges. A direct count with `scipy.spatial.Delaunay` agrees. The practical meaning is that a triangulation is **linear in size**: about $2n$ triangles, $3n$ edges, and an average vertex degree just under 6. So does the Voronoi diagram: its vertices are the $2n - 2 - h$ circumcentres.

The empty-circle test, and the paraboloid behind it

The Delaunay triangulation is defined by: *the circumcircle of every triangle contains no other point*. To use that rule we need a test for "is d inside the circle through a, b, c?" The beautiful way to find it is to lift the plane onto a paraboloid.

The lifting map

Send each point $p = (x, y)$ to $\hat{p} = (x,\ y,\ x^2 + y^2)$ on the paraboloid $z = x^2 + y^2$.

Circles become planes

Let C be a circle with centre (c_x, c_y) and radius r. The lifted points of C lie on one non-vertical plane Π. A point d lies strictly inside C if and only if $\hat{d}$ lies strictly below Π.

Proof. Expand the circle's equation:

$$(x - c_x)^2 + (y - c_y)^2 = r^2 \iff x^2 + y^2 = 2c_x x + 2c_y y + \left(r^2 - c_x^2 - c_y^2\right).$$

On the paraboloid $x^2 + y^2 = z$, so the lifted circle satisfies $z = 2c_x x + 2c_y y + (r^2 - c_x^2 - c_y^2)$. That is the plane Π. Now take any point $d = (x, y)$. Its lift has height $x^2 + y^2$, while Π above it has height $2c_x x + 2c_y y + r^2 - c_x^2 - c_y^2$. The difference is

$$(x^2 + y^2) - \left(2c_x x + 2c_y y + r^2 - c_x^2 - c_y^2\right) = (x - c_x)^2 + (y - c_y)^2 - r^2,$$

which is negative exactly when d is inside the circle. ∎

"Is $\hat{d}$ below the plane through $\hat{a}, \hat{b}, \hat{c}$?" is a 3-D orientation question, so the in-circle test is again the sign of a determinant:

$$\operatorname{incircle}(a, b, c, d) = \det \begin{bmatrix} a_x - d_x & a_y - d_y & (a_x - d_x)^2 + (a_y - d_y)^2 \\ b_x - d_x & b_y - d_y & (b_x - d_x)^2 + (b_y - d_y)^2 \\ c_x - d_x & c_y - d_y & (c_x - d_x)^2 + (c_y - d_y)^2 \end{bmatrix},$$

which is positive exactly when d is inside the circle through a, b, c taken counter-clockwise. Check it on $a = (0, 0)$, $b = (4, 0)$, $c = (1, 3)$: their circle has centre $(2, 1)$ and $r^2 = 5$. The point $d = (3.5, 2.5)$ is at squared distance $4.5 < 5$, inside, and the determinant comes out $+6$. Move d to $(4, 3)$, at squared distance 8, and it becomes -36.

The lifting theorem has a stunning corollary. A triangle abc is Delaunay when no lifted point lies below the plane through $\hat{a}, \hat{b}, \hat{c}$. That says the lifted triangle is a face of the **lower convex hull** of the lifted points in 3-D. So:

$$\text{Delaunay triangulation in 2-D} = \text{lower convex hull of the lifted points in 3-D, projected back down.}$$

Hulls and triangulations are the same problem one dimension apart, and any 3-D hull algorithm is a Delaunay algorithm.

Why Delaunay triangles are fat. Take a convex quadrilateral and its two possible diagonals. Exactly one choice makes both circumcircles empty (unless the four points are concyclic), and that choice maximises the smallest of the six angles. Flipping illegal diagonals one at a time therefore never decreases the sorted list of angles, and the process ends at the Delaunay triangulation. So among all triangulations of a point set, Delaunay **maximises the minimum angle**. That is exactly what finite-element and interpolation codes want, because skinny triangles make badly conditioned equations (§8.10).

Voronoi from Delaunay

Voronoi cell

For sites $s_1, \dots, s_n$, the Voronoi cell of s_i is $V_i = \{x : \|x - s_i\| \le \|x - s_j\| \text{ for all } j\}$.

Each condition $\|x - s_i\| \le \|x - s_j\|$ expands to the linear inequality $2(s_j - s_i) \cdot x \le \|s_j\|^2 - \|s_i\|^2$: a half-plane bounded by the perpendicular bisector of $s_i s_j$. So each cell is an intersection of half-planes, and therefore convex. A point x is a Voronoi vertex when it is equidistant from three sites and no site is closer: it is the centre of an empty circle through three sites. That is precisely the circumcentre of a Delaunay triangle, which proves the duality you saw in the figure. Two cells share an edge exactly when there is an empty circle through just those two sites, which is the condition for a Delaunay edge.

Quadtrees: how deep, how fast

A point quadtree with leaf capacity 1 on a square of side s keeps splitting any cell that contains two or more points.

Depth of a quadtree

If the closest pair of points is at distance δ, the quadtree has depth at most $\log_2(s/\delta) + \frac{3}{2}$.

Proof. A cell at depth k has side $s/2^k$ and diagonal $\sqrt{2}\, s/2^k$. A cell is split only when it holds two points, which are at least δ apart, so its diagonal is at least δ: $\sqrt{2}\, s/2^k \ge \delta$, that is $k \le \log_2(s/\delta) + \frac{1}{2}$. The children of the deepest split cell are one level further down. ∎

The depth depends on the **spread** s/δ, not on n. A thousand evenly spaced points give a shallow tree. Two points a nanometre apart in a kilometre-wide map force about 40 levels. The tree adapts to the data, which you can see in the quadtree figure of §75.4: cells subdivide only inside the two clusters, and the number of cells at each depth stays far below the 4^d of a full tree.

75.4 The algorithm

Convex hull in twenty lines

```
def orient(a, b, c):
    return (b[0]-a[0])*(c[1]-a[1]) - (b[1]-a[1])*(c[0]-a[0])

def convex_hull(points):
    pts = sorted(set(map(tuple, points)))      # sort by x, then y
    if len(pts) <= 2:
        return pts
    def chain(seq):
        h = []
        for p in seq:
            while len(h) >= 2 and orient(h[-2], h[-1], p) <= 0:
                h.pop()                        # right turn or straight: drop
            h.append(p)
        return h
    lower, upper = chain(pts), chain(reversed(pts))
    return lower[:-1] + upper[:-1]             # counter-clockwise, no repeats
```

The `<= 0` drops collinear points from the hull. Change it to `< 0` and they are kept, but then exactly collinear duplicates can loop. That edge case is the first taste of the robustness problem below.

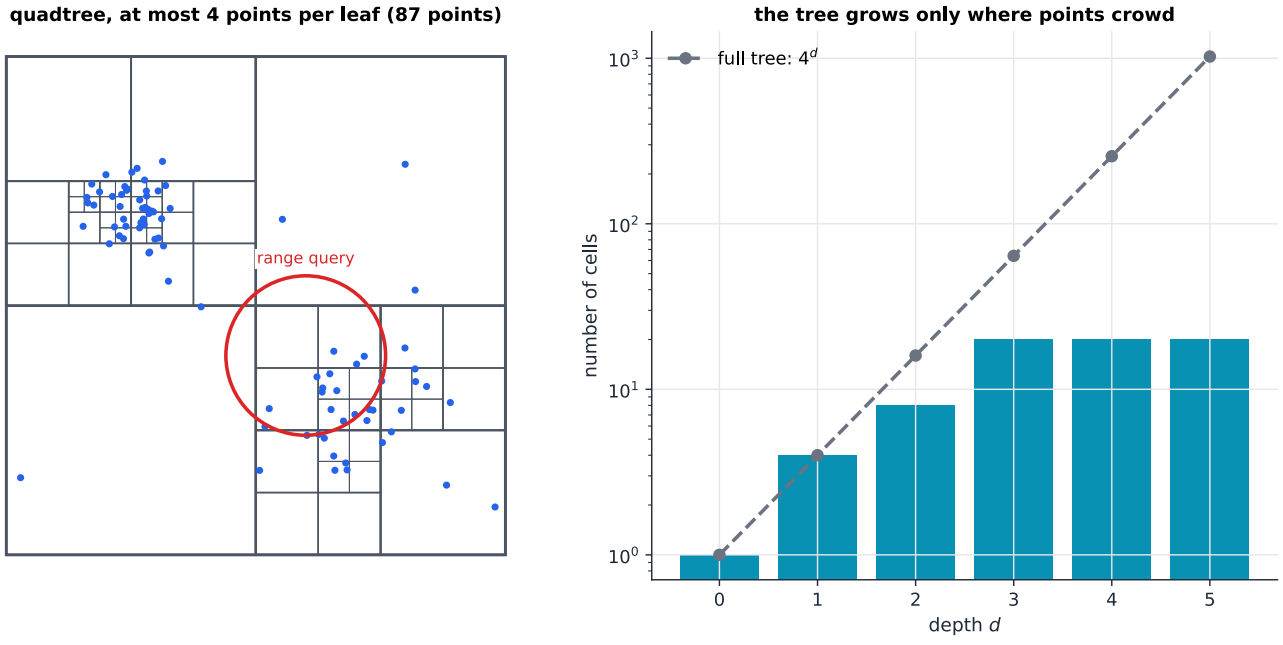
Delaunay by incremental insertion (Bowyer–Watson)

```
start with one huge triangle containing all points
for each point p, in random order:
    find all triangles whose circumcircle contains p     # in-circle test
    delete them: they form a star-shaped hole around p
    join p to every edge of the hole's boundary
remove triangles that touch the huge starting triangle
```

Inserting points in **random order** is the trick. A single insertion can destroy many triangles, but averaged over the random order each insertion creates $O(1)$ new triangles in expectation, and the expected total time is $O(n \log n)$ if a point-location structure finds the first bad triangle quickly. In the worst order (points sorted along a curve, say) the same code takes $O(n^2)$. Fortune's sweep-line algorithm reaches $O(n \log n)$ deterministically, by sweeping a line across the plane and maintaining a "beach line" of parabolic arcs.

Quadtree build and range query

```
insert(cell, p):
    if cell is a leaf with fewer than CAP points: store p; return
    if cell is a leaf: split it into 4 children, push its points down
    insert(child containing p, p)

query(cell, region):
    if cell does not overlap region: return nothing     # prune: one box test
    if cell is a leaf: return its points that lie in region
    return union of query(child, region) over the 4 children
```



Left: cells subdivide only where points crowd. A range query (red) visits only the cells it overlaps. Right: the tree has 20 cells at each of depths 3, 4 and 5, against 64, 256 and 1024 in a full tree.

For well-spread points, a range query costs $O(\log n + k)$ where k is the number of points reported. **kd-trees** do the same job by splitting on alternate coordinates at the median, which guarantees balance (depth $\lceil \log_2 n \rceil$) whatever the spread. **Bounding-volume hierarchies** group objects rather than space, and are how ray tracers (Ch. 71) and physics engines (Ch. 72) avoid testing every ray or pair against everything.

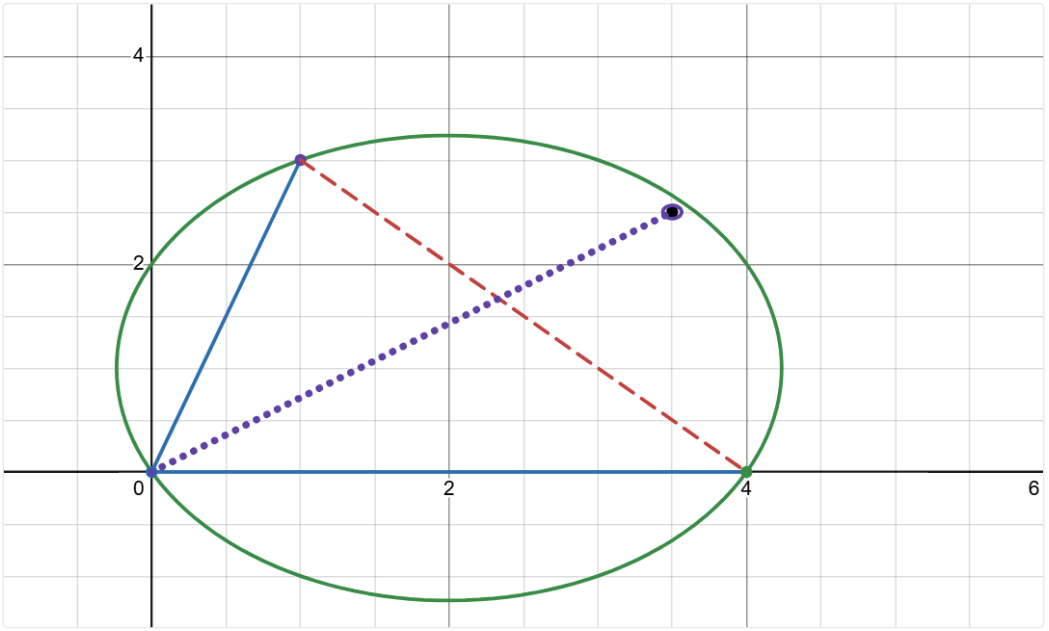
When geometry code fails

- **Degeneracies.** Three collinear points, four concyclic points, duplicate points. Every algorithm above silently assumes "general position". Real data (grids, CAD models, integer pixel coordinates) are full of exact degeneracies.
- **Rounding.** The orientation test subtracts nearly equal numbers. For nearly collinear points the computed sign can be wrong, and two tests that should agree can disagree. A hull algorithm can then pop a point that one test says is outside and another says is inside. In 2004 Kettner and co-authors published small inputs on which textbook hull and Delaunay codes in double precision return non-convex "hulls" or loop forever.
- **The fix is exactness, not tolerance.** Adding an ε ("treat $|\text{orient}| < 10^{-12}$ as zero") makes the problem worse, because "approximately equal" is not transitive. Robust libraries compute the sign exactly: in floating point when the result is clearly far from zero, and with extra-precision arithmetic only in the rare close calls.

> **? The sign you cannot trust**
>
> On a grid of points near $(0.5, 0.5)$ spaced by 2^{-53}, the double-precision orientation test against a fixed line produces a ragged staircase of wrong signs instead of a clean straight boundary. Why does subtracting two nearly equal products lose all the information, and how large can the error be? That is the floating-point error analysis of §22.1. How Shewchuk's *adaptive predicates* return the exact sign while staying almost as fast as the naive formula is a story about error-free transformations of floating-point sums that this chapter does not tell.

75.5 The experiment

Drag the fourth point. The graph shows the triangle $a = (0, 0)$, $b = (4, 0)$, $c = (1, 3)$ and its circumcircle $(x - 2)^2 + (y - 1)^2 = 5$. A fourth point $d = (3.5, 2.5)$ makes the quadrilateral $abdc$, which can be split along the diagonal bc (dashed) or ad (dotted). Because d is inside the circle, triangle abc breaks the empty-circle rule, so Delaunay chooses ad. Change every `3.5` and `2.5` in the last three lines to move d: try $(4, 3)$ (outside, so bc wins), $(3, 2.8)$ (inside) and $(2.5, 3.179)$, which is almost exactly on the circle. That last one is the degenerate case where both diagonals are legal and a tiny rounding error decides.



The green circle passes through a, b, c. When d sits inside it, triangle abc is illegal and the Delaunay triangulation uses the diagonal ad (dotted) instead of bc (dashed).

How many corners does a random hull have? The script draws n random points from three different shapes, computes the hull with the twenty-line scan of §75.4, and prints the average number of corners as n grows. Run it as is first.

```
import numpy as np
import matplotlib.pyplot as plt

SHAPE = "square"          # knob: "square", "disc" or "gauss"
NS = [10, 100, 1000, 10000, 100000]
TRIALS = 20
rng = np.random.default_rng(1)

def orient(a, b, c):
    return (b[0]-a[0])*(c[1]-a[1]) - (b[1]-a[1])*(c[0]-a[0])

def hull(P):
    pts = sorted(map(tuple, P))
    def chain(seq):
        h = []
        for p in seq:
            while len(h) >= 2 and orient(h[-2], h[-1], p) <= 0:
                h.pop()
            h.append(p)
        return h
    return chain(pts)[:-1] + chain(pts[::-1])[:-1]

def sample(n):
    if SHAPE == "square":
        return rng.uniform(-1, 1, (n, 2))
    if SHAPE == "disc":
        r, t = np.sqrt(rng.uniform(0, 1, n)), rng.uniform(0, 2*np.pi, n)
        return np.c_[r*np.cos(t), r*np.sin(t)]
    return rng.normal(size=(n, 2))

means = [np.mean([len(hull(sample(n))) for _ in range(TRIALS)]) for n in NS]
for n, m in zip(NS, means):
    print(f"n = {n:>6}:  average hull size {m:6.1f}")
plt.semilogx(NS, means, "o-"); plt.xlabel("n"); plt.ylabel("hull corners")
plt.title(SHAPE); plt.show()
```

For the square you should see roughly 6, 12, 18, 24 and 31 corners: the hull grows like $\frac{8}{3}\ln n$, so a hundred thousand points have only about thirty corners. Now set `SHAPE = "disc"`. The counts jump to about 15, 33, 74 and 158 for the larger n: growth like $n^{1/3}$. For `"gauss"` they crawl, about 9, 11, 13 and 16. Same algorithm, same n, wildly different answer sizes. Three more things to try:

- Replace the `sample` function with points on a circle, `np.c_[np.cos(t), np.sin(t)]`. Every point is now a corner, which is the worst case of the lower-bound argument.
- Put all the points on an integer grid, `rng.integers(0, 20, (n, 2))`. Now there are many exact collinearities and duplicates. Does the `<= 0` still produce a clean hull? What happens with `< 0`?
- Time the hull for $n = 10^5$ and compare with the time of `sorted` alone. How much of the work is the scan?

75.6 The challenge

1. **Intuition.** Without computing anything, sketch the Voronoi diagram of four points at the corners of a square, and then of the same square with one corner nudged slightly outward. Explain why the Delaunay triangulation of the exact square is not unique, and which of the two diagonals the nudge selects.
2. **Derivation.** Prove the "Delaunay is the lower hull of the lift" corollary carefully: show that if $\hat{a}, \hat{b}, \hat{c}$ span a face of the lower convex hull of the lifted points, then the circumcircle of abc is empty, and conversely. Then use the theorem of §75.3 on triangulation size to show that the Voronoi diagram of n sites has at most $2n - 5$ vertices and $3n - 6$ edges for $n \geq 3$.
3. **Implementation.** Write a Bowyer–Watson Delaunay triangulator from the pseudocode in §75.4, using the in-circle determinant as the only geometric test besides orientation. Check on 500 random points that every triangle's circumcircle is empty and that the triangle count equals $2n - 2 - h$. Then insert the same points sorted by x instead of in random order and measure how the running time scales.
4. **Open investigation.** Build a quadtree and a kd-tree on n points drawn from distributions with very different spreads (uniform; two tight clusters far apart; points on a spiral; a fractal dust from Ch. 73). For each, measure the average number of cells visited by a nearest-neighbour query. Is there a single statistic of the point set (its spread, its fractal dimension, something else) that predicts the query cost for both trees?

75.7 Loose threads

This chapter worked in the flat, clean plane with points in general position. Each of those three words hides a door.

> **? Three dimensions break the count**
>
> In the plane a triangulation of n points always has about $2n$ triangles. In 3-D the analogue fails dramatically: the Delaunay tetrahedralization of n points can have $\Theta(n^2)$ tetrahedra, and even on nice inputs it contains *slivers*, flat tetrahedra whose four corners are nearly coplanar yet whose circumsphere is empty. Slivers wreck finite-element solutions. Why is Euler's formula no longer enough to bound the size, and how do mesh generators get rid of slivers? That is the meshing side of the finite-element method in Ch. 89.

? The curse of the nearest neighbour

A kd-tree answers nearest-neighbour queries in about $\log n$ steps in the plane. In 100 dimensions (image descriptors, word embeddings) the same tree visits almost every leaf, and a linear scan is just as fast. Random points in high dimension are all nearly the same distance apart, so the pruning test almost never prunes. Is there any data structure with polynomial size that answers *exact* nearest-neighbour queries in sub-linear time in high dimension? Conditional lower bounds suggest not, linking the question to the strong exponential time hypothesis of Ch. 30. It is an open problem, and practice has settled for approximate answers.

The robustness mystery of §75.4 is the third thread: geometric algorithms are only as correct as the sign of a determinant, and the exact computation of that sign belongs to the floating-point analysis of Ch. 22. Two quieter threads lead elsewhere in this track. Voronoi cells of randomly placed seeds are a standard way to generate terrain regions, cracked mud and stone textures (Ch. 74). And a triangle mesh is a sampled version of a continuous surface: whether the sampling is fine enough to reconstruct the shape is a cousin of the sampling theorem in Ch. 76.

Further reading

- M. de Berg, O. Cheong, M. van Kreveld and M. Overmars, *Computational Geometry: Algorithms and Applications*, 3rd ed., Springer, 2008. The standard first text: hulls, triangulations, Voronoi diagrams, kd-trees and quadtrees, each with an application.
- J. O'Rourke, *Computational Geometry in C*, 2nd ed., Cambridge University Press, 1998. Implementation-first, with careful attention to degeneracies.
- J. R. Shewchuk, "Adaptive precision floating-point arithmetic and fast robust geometric predicates," *Discrete & Computational Geometry* 18 (1997). The orientation and in-circle tests done exactly.
- L. Kettner, K. Mehlhorn, S. Pion, S. Schirra and C. Yap, "Classroom examples of robustness problems in geometric computations," *Computational Geometry* 40 (2008). Small, vivid failures of textbook code.
- S. Cheng, T. Dey and J. Shewchuk, *Delaunay Mesh Generation*, CRC Press, 2012. Where the 3-D sliver story is told in full.

Chapter 76 — Signal Processing for Audio and Visual Computing

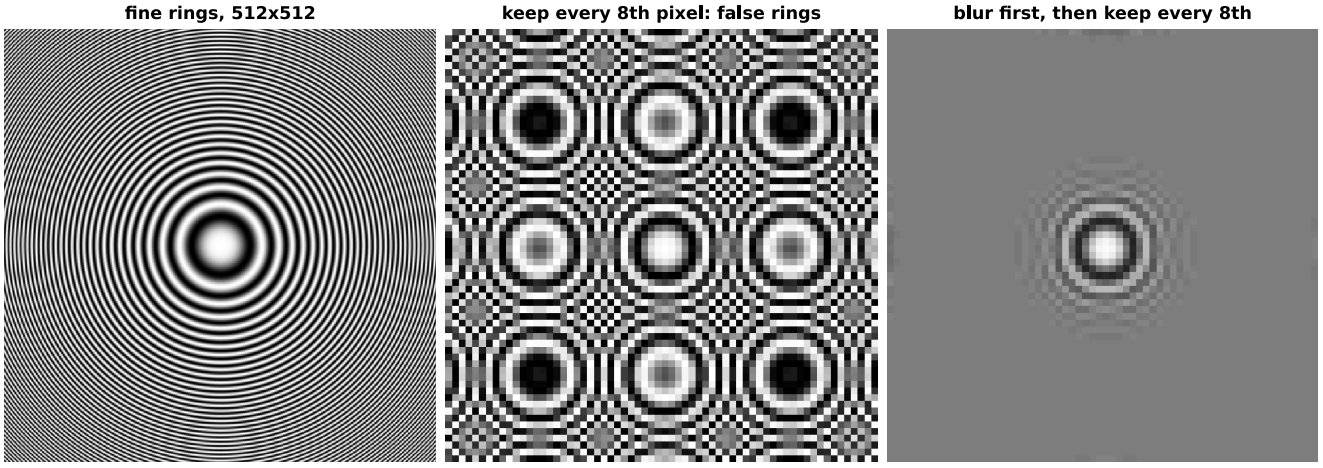
> **Level 2 · Track G — Mathematics for Computer Graphics, Games, and Simulation Builds on:** Ch. 15, Ch. 16, Ch. 17 · **Leads to:** Ch. 80, Ch. 91 · **Related:** Ch. 21, Ch. 33, Ch. 71, Ch. 74, Ch. 75 **Reading time:** ~30 min

76.1 The hook

Television presenters are told never to wear a finely striped shirt or a herringbone jacket. On camera the stripes come alive. Huge, slow, swirling bands of light and dark ripple across the fabric, sometimes with false rainbows, and they crawl whenever the presenter breathes. Nothing like that is on the shirt. The pattern is manufactured by the camera.

The same ghost haunts every screen. In an old video game, a distant chain-link fence or a tiled floor shimmers and boils as the camera moves. In a western, the wagon wheels spin slowly backwards while the wagon races forward. Shrink a photograph of a brick wall in a cheap image editor and the bricks rearrange themselves into giant curved waves. Record a whistle that slides up past the limit of a sound card, and on playback the pitch slides *down*.

Here is a ring pattern that shows the effect at its most dramatic. On the left, fine concentric rings, the rings closer and closer together towards the edge. In the middle, the same picture after keeping only every eighth pixel in each direction, exactly what a careless thumbnail generator does. Whole new sets of rings appear, centred on points where the original has nothing special at all.



Left: rings whose spacing shrinks towards the edge. Centre: keep every 8th pixel and false ring centres appear on a lattice. Right: blur first, then keep every 8th pixel; the false rings are gone and the too-fine detail fades to grey.

The right-hand picture is the cure: blur first, then shrink. It throws information away on purpose and looks *more* faithful. This chapter is about why that works and about the toolkit it opens:

> **Why does sampling invent patterns that were never there, where exactly do the ghosts appear, and how do blurring, sharpening, edge detection, spectrograms and JPEG all follow from one idea about frequencies?**

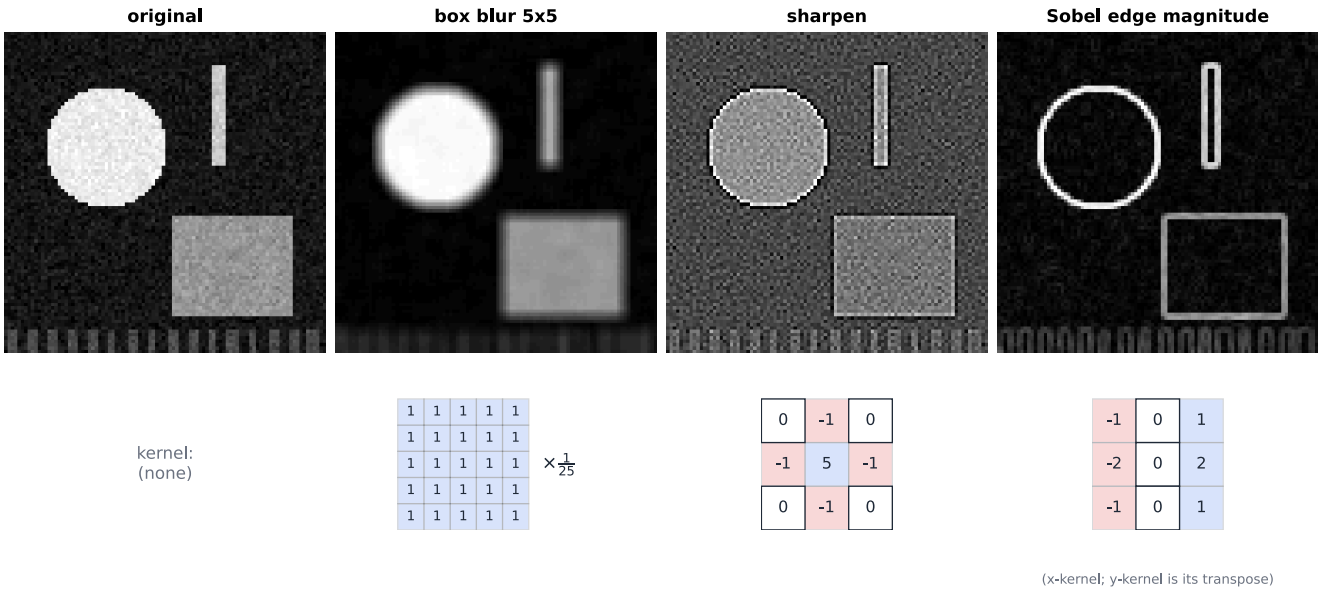
By the end you will be able to predict, from a single formula, where the false ring centres in the middle picture sit, to the pixel.

76.2 The big idea

A picture is a signal in two variables. A sound is a sequence of numbers $x[n]$, one per sample, typically 44,100 or 48,000 per second. A grey-scale image is an array $x[m, n]$ of brightness values. Everything the core chapters proved about one-dimensional signals carries over to images, with one more index.

Frequency in an image means stripes. A sinusoid in time is a pure tone. A sinusoid in an image, $\cos(2\pi(um + vn))$, is a field of parallel stripes. The pair (u, v) is its spatial frequency, in cycles per pixel: its direction is perpendicular to the stripes and its size says how close together they are. Large smooth regions of a photograph are low frequencies. Edges, textures and noise are high frequencies. The Fourier transform (Ch. 16) writes any image as a sum of such stripe fields.

Filtering is a sliding weighted average. Replace each pixel by a weighted sum of its neighbours, using the same small table of weights (a **kernel**) everywhere. Equal positive weights blur. A big positive centre with negative neighbours sharpens. Weights that are negative on one side and positive on the other respond to edges. Each of these is a 2-D convolution, and by the convolution theorem (§16.5) each one simply **multiplies every stripe frequency by a gain**. Blur turns down the high frequencies; sharpen turns them up; an edge detector keeps only a band of them.



One image, three kernels. The box blur softens every edge and wipes out the fine stripes at the bottom. Sharpening amplifies edges and noise alike. The Sobel operator keeps only the outlines. Blue cells are positive weights, red negative.

Sampling folds frequencies. A grid of pixels can only represent stripes up to half a cycle per pixel, the Nyquist limit (§16.8). Any finer stripe does not vanish when sampled. It is **folded back** and impersonates a coarser one. That is the shirt, the wheel and the ring lattice. Blurring before sampling removes the too-fine stripes while they can still be told apart from the coarse ones.

Sound needs a spectrum that changes in time. A single Fourier transform of a whole song tells you which notes occur, but not when. Cut the sound into short overlapping windows and transform each one: you get a **spectrogram**, a picture with time across and frequency up. Musicians, speech recognisers and bat biologists all read these pictures. Their resolution is limited by a trade-off that no cleverness removes.

The analogy to keep in mind is a graphic equaliser on a stereo. Each slider is the gain at one frequency band. Every linear, shift-invariant operation on a sound or an image, however it is described in pixels, is just a setting of those sliders.

76.3 The mathematics

Two-dimensional convolution and its spectrum

The discrete convolution sum of §17.1 gains a second index.

2-D convolution and the 2-D DFT

For an image $x[m,n]$ and a kernel $h[k,l]$,

$$y[m,n] = (h * x)[m,n] = \sum_{k} \sum_{l} h[k,l]\, x[m-k,\, n-l].$$

For an $M \times N$ array, the 2-D DFT is

$$X[p,q] = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} x[m,n]\, e^{-j2\pi(pm/M + qn/N)}.$$

The 2-D DFT is just the 1-D DFT of §15.7 applied to every row and then every column. The central fact is that it turns convolution into multiplication. Here is the proof for **circular** convolution, where indices wrap around modulo M and N (which is what the FFT computes).

The 2-D convolution theorem

If $y[m,n] = \sum_{k,l} h[k,l]\, x[(m-k) \bmod M,\ (n-l) \bmod N]$, then $Y[p,q] = H[p,q]\, X[p,q]$ for every p, q.

Proof. Substitute the definition of y into the DFT and swap the order of summation:

$$Y[p,q] = \sum_{k,l} h[k,l] \sum_{m,n} x[(m-k) \bmod M, (n-l) \bmod N]\, e^{-j2\pi(pm/M + qn/N)}.$$

In the inner sum put $m' = (m-k) \bmod M$ and $n' = (n-l) \bmod N$. As m runs over $0, \dots, M-1$ so does m', and because $e^{-j2\pi p \cdot M/M} = 1$ the exponential is unchanged by the wrap-around: $e^{-j2\pi pm/M} = e^{-j2\pi pm'/M}\, e^{-j2\pi pk/M}$. The inner sum therefore equals $e^{-j2\pi(pk/M + ql/N)} X[p,q]$, and

$$Y[p,q] = X[p,q] \sum_{k,l} h[k,l]\, e^{-j2\pi(pk/M + ql/N)} = H[p,q]\, X[p,q]. \qquad \blacksquare$$

So the **frequency response** of a kernel, its DFT H, is the table of equaliser gains. It also tells you how to compute big blurs fast: transform, multiply, transform back, at $O(MN \log(MN))$ cost whatever the kernel size. For small kernels, direct summation is cheaper. Real images do not wrap around, so practical code pads the borders (by zeros, by copying the edge, or by mirroring) to stop the left edge bleeding into the right.

Reading the three kernels

It is easiest to use the continuous-frequency form $H(\omega_x, \omega_y) = \sum_{k,l} h[k,l] e^{-j(\omega_x k + \omega_y l)}$, with ω in radians per pixel and $\pm\pi$ the Nyquist limit.

Box blur. The $L \times L$ box has weights $1/L^2$, and it is **separable**: it is a 1-D box of length L along the rows followed by the same along the columns. So $H(\omega_x, \omega_y) = B(\omega_x) B(\omega_y)$, with B a geometric series. For odd $L = 2r + 1$, centred,

$$B(\omega) = \frac{1}{L} \sum_{k=-r}^{r} e^{-j\omega k} = \frac{1}{L} \frac{\sin(L\omega/2)}{\sin(\omega/2)}.$$

At $\omega = 0$ the gain is 1: flat regions are untouched. The gain falls to zero at $\omega = 2\pi/L$, so for $L = 5$ stripes with a period of exactly 5 pixels are wiped out completely, and the period-6 stripes at the bottom of the figure, just beyond that zero, keep only a fifth of their contrast. But the gain does not stay small: it comes back negative, and at the Nyquist limit it is $B(\pi) = 1/5$ for $L = 5$ (and $-1/3$ for $L = 3$, a flipped-sign ripple). A box is a crude low-pass filter, and those side lobes are why box-filtered thumbnails still show some moiré. Separability also matters for speed: a 5×5 box costs 25 multiplications per pixel directly, but only $5 + 5 = 10$ as two 1-D passes.

Sharpen. The kernel in the figure is the identity minus the discrete Laplacian ∇^2 (the five-point stencil of §22.10): centre 5, the four neighbours -1. Its response is

$$H(\omega_x, \omega_y) = 5 - 2\cos\omega_x - 2\cos\omega_y.$$

The gain is 1 at $(0, 0)$, rises to 5 for the finest horizontal or vertical stripes, and reaches 9 for the checkerboard at (π, π). Sharpening is a high-frequency boost, and noise lives at high frequency, which is why the sharpened panel looks grainy.

Sobel. The horizontal Sobel kernel is the outer product of a smoothing column $[1, 2, 1]^{\mathsf{T}}$ and a difference row $[-1, 0, 1]$. Its magnitude response is

$$|H(\omega_x, \omega_y)| = 2|\sin\omega_x| \, (2 + 2\cos\omega_y).$$

The $\sin\omega_x$ factor is a derivative: it is zero for flat regions, and for slow variations it grows like ω_x, exactly as d/dx multiplies a frequency by $j\omega$ (§16.3). The $(2 + 2\cos\omega_y)$ factor smooths along the edge and kills the pixel-scale noise at $\omega_y = \pi$. Combining the horizontal and vertical responses into $\sqrt{g_x^2 + g_y^2}$ gives the edge-magnitude image.

Aliasing in two dimensions, and where the ghosts sit

Sampling theory is in §16.8: sampling at rate f_s makes the spectrum periodic with period f_s, so a frequency f is indistinguishable from $f - k f_s$ for every integer k. In audio this is immediate. A 30 kHz tone (inaudible) sampled at 44.1 kHz shows up at $44.1 - 30 = 14.1$ kHz, squarely in the audible band, which is why every sound card has an analogue low-pass filter before its converter. In images the same rule applies to each coordinate separately: a stripe pattern (u, v) sampled on a grid of spacing D pixels aliases to $(u - k/D,\ v - l/D)$ for integers k, l.

Now the ring pattern. The left image is

$$x(s, t) = \tfrac{1}{2} + \tfrac{1}{2} \cos\!\big(2\pi a \, (s^2 + t^2)\big), \qquad a = 110,$$

with s, t measured in image widths from the centre, so $-\tfrac{1}{2} \le s, t \le \tfrac{1}{2}$. Along the s-axis the phase is $2\pi a s^2$. Its local frequency is the derivative of the phase divided by 2π: $2as = 220s$ cycles per image width. The rings get finer linearly with distance from the centre. That is a **chirp** in space.

The middle image keeps every 8th of 512 pixels, so it has 64 samples per image width and its Nyquist limit is 32 cycles per width.

> **False ring centres**
>
> Sampling $\cos(2\pi a s^2)$ at points spaced $1/F$ apart produces exact copies of the central ring pattern centred at $s_0 = kF/(2a)$ for every integer k (and the same in t), up to a constant phase.

Proof. Compare the phase at a sample point $s = n/F$ with the phase of a ring pattern centred at s_0:

$$a\,(s - s_0)^2 = a s^2 - 2a s_0 s + a s_0^2.$$

At the sample points, $2a s_0 s = 2a s_0 \, n/F$. If $s_0 = kF/(2a)$ this equals kn, an integer, which shifts the cosine's phase by a whole number of turns and so changes nothing. The remaining term $a s_0^2$ is the same constant for every sample. So on the sampling grid, the pattern centred at 0 and the pattern centred at s_0 agree, apart from that fixed phase. ■

For our picture $F = 64$ and $a = 110$, so the ghosts sit at $s_0 = 64/220 \approx 0.291$ image widths from the centre, which is about 149 of the original 512 pixels, in each direction: a 3×3 lattice of false centres, exactly as in the middle panel. The constant phase $a s_0^2 \approx 9.31$ turns is not a whole number, so the ghost centres are not as bright as the true one. On the axes the sampled centre value is about 0.33 on the 0-to-1 scale; along the diagonals the phase doubles to about 18.62 turns and the centres are darker still, about 0.12. Aliasing starts where the local frequency crosses Nyquist: $220s = 32$ gives $s \approx 0.145$ widths, about 74 pixels from the centre. Inside that radius the sampled rings are faithful.

The cure in the right panel is a Gaussian blur of standard deviation 4.8 pixels before decimation. A Gaussian's spectrum is again a Gaussian (§16.1), and with that width its gain $e^{-\sigma^2\omega^2/2}$ is about 0.17 at the new Nyquist limit $\omega = \pi/8$ and below 0.001 at twice that frequency. What remains is a faithful picture of the part of the image the coarse grid *can* represent. Ray tracers (Ch. 71) and texture samplers face the same choice with pixels and texels; mipmaps are pre-blurred copies of a texture at each power-of-two size, precisely so that a distant floor is sampled from a version with no stripes finer than the screen can show.

The short-time Fourier transform

STFT and spectrogram

Choose a window $w[n]$ of length L (for example a Hann window) and a hop R. The short-time Fourier transform of $x[n]$ is

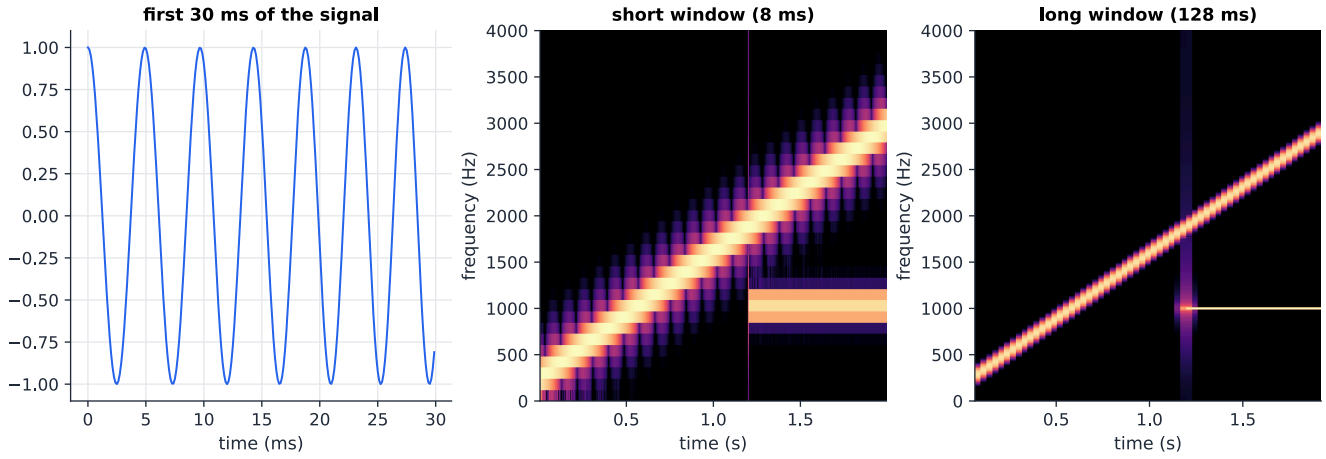
$$X[r,k] = \sum_{n=0}^{L-1} w[n]\, x[rR+n]\, e^{-j2\pi kn/L},$$

the DFT of the r-th windowed frame. The **spectrogram** is $|X[r,k]|^2$, usually drawn in decibels with time rR/f_s across and frequency kf_s/L up.

The window is a short-time version of the leakage story in §16.6: each frame's spectrum is the true local spectrum convolved with the window's spectrum. That leads to the fundamental trade-off. A window of L samples lasts $T = L/f_s$ seconds and has frequency bins spaced $\Delta f = f_s/L$ apart. Their product is

$$T\,\Delta f = 1,$$

whatever the sample rate. Short windows locate events precisely in time and smear them in frequency. Long windows do the opposite.



A chirp sweeping from 200 Hz to 3 kHz in 2 s at $f_s = 8$ kHz, with a 1 kHz tone switched on at 1.2 s. The short window (centre) pins down the onset but blurs every frequency over about 125 Hz. The long window (right) resolves the steady tone to within 8 Hz but smears the onset and thickens the sweeping chirp.

How long should the window be for a chirp? Suppose the frequency rises at κ Hz per second (here $\kappa = 1400$). During a window of length T the chirp sweeps κT hertz, while the bin width is $1/T$. Total frequency blur is roughly $\kappa T + 1/T$, which is smallest when $\kappa T = 1/T$:

$$T_{\text{best}} = \frac{1}{\sqrt{\kappa}} = \frac{1}{\sqrt{1400}}\ \text{s} \approx 26.7\ \text{ms}.$$

The 8 ms window has $1/T = 125$ Hz of bin blur and only about 11 Hz of sweep; the 128 ms window has under 8 Hz of bin blur but sweeps about 179 Hz. Each is far from balanced, and each is bad in the opposite direction.

? A window sharp in both directions?

$T\,\Delta f = 1$ was a statement about DFT bins. Is there some cleverer window, perhaps not a frame at all, that is sharply concentrated in time *and* in frequency? There is a hard limit: the product of the root-mean-square durations in time and frequency can never fall below $1/(4\pi)$, and only the Gaussian attains it. The proof is the same Cauchy–Schwarz argument that gives Heisenberg's uncertainty principle in quantum mechanics, and it is told in Ch. 92. Wavelets, which use long windows for low notes and short windows for high ones, are one way to cheat the trade-off locally; their theory lives in the function spaces of Ch. 61.

The idea behind JPEG

A photograph has strong correlation between neighbouring pixels: most 8×8 blocks are smooth, with a little texture. JPEG exploits this with the **discrete cosine transform** (DCT). For a block of 8 values,

$$c_k = \alpha_k \sum_{n=0}^{7} x[n] \cos\!\left(\frac{\pi(2n+1)k}{16}\right), \qquad \alpha_0 = \sqrt{1/8},\ \alpha_k = \sqrt{2/8}\ (k \ge 1).$$

These eight cosines are orthonormal (§9.7), so the transform is a rotation: no information is lost and energy is preserved. The 2-D version applies it to rows and then columns, giving 64 coefficients per block. Then comes the lossy step. Each coefficient is divided by a step size from a table, larger steps for higher frequencies, and rounded. Most high-frequency coefficients round to zero, and long runs of zeros cost almost nothing to store. A quality setting simply scales the table.

Why cosines rather than the DFT? The DFT treats a block as one period of a periodic signal, so a smooth ramp from dark to light becomes a sawtooth with a jump at the block edge. Jumps make Fourier coefficients decay slowly, like $1/k$ (§15.7). The DCT implicitly extends the block by its mirror image, which is continuous, so coefficients decay like $1/k^2$. On the ramp $x = (0, 1, \dots, 7)$, the first two DCT coefficients carry 99.6% of the energy (all even-index coefficients beyond c_0 are exactly zero). The DFT needs three coefficients (the mean and the ± 1 frequencies) to capture only 89.5%. Audio codecs such as MP3 and AAC use a close relative, the modified DCT on overlapping windows: a spectrogram, quantised according to what the ear cannot hear.

? Why the cosine is nearly optimal

For any family of images you can compute the *best possible* orthogonal transform for compaction: the Karhunen–Loève transform, whose basis vectors are the eigenvectors of the pixel covariance matrix (Ch. 10, Ch. 21). It depends on the data and is expensive. The fixed, data-independent DCT turns out to be almost exactly as good for natural photographs. Why should one universal set of cosines nearly match the eigenvectors of every photograph's statistics? And how many bits per pixel are truly necessary for a given distortion? Both questions are answered by transform coding and rate–distortion theory in Ch. 80.

76.4 The algorithm

Filtering an image

```
input: image x (M x N), kernel h (K x K), border rule
if K is small (say K <= 9):
    if h is separable (h = column * row): filter rows with row, then columns with column
    else: direct sum, K^2 multiply-adds per pixel
else:
    pad x by K pixels on every side using the border rule
    Y = FFT2(x_padded) * FFT2(h, same size)      # convolution theorem
    y = IFFT2(Y), cropped back to M x N
```

Direct filtering costs $O(MNK^2)$, separable filtering $O(MNK)$, and FFT filtering $O(MN \log MN)$ regardless of K. A 1920×1080 frame has about two million pixels; a direct 31×31 blur costs about two billion multiply-adds, while two separable passes cost about 130 million. A Gaussian is separable, which is one reason it is everyone's favourite blur.

Failure cases are mostly about edges and numbers. With no padding, FFT filtering wraps the top of the image into the bottom. With zero padding, blur darkens the borders. Sharpening can push values below 0 or above 255, and integer pipelines must clip. Filters with many large positive and negative weights lose precision in 8-bit arithmetic.

Downsampling without ghosts

```
to shrink an image by an integer factor D:
    blur with a low-pass kernel whose gain is small beyond 1/(2D) cycles per pixel
    keep every D-th pixel in each direction
```

Real resizers fold the two steps into one kernel (bilinear, bicubic, Lanczos), evaluated only at the output pixels. A box of width D is the cheapest acceptable prefilter; as §76.3 showed, its side lobes let some aliasing through. Mipmapping pre-computes the blurred versions at every power-of-two size, and a GPU picks the level whose texel size matches the pixel's footprint on screen.

A spectrogram in ten lines

```
import numpy as np

def spectrogram(x, fs, L=256, hop=64):
    w = np.hanning(L)
    starts = range(0, len(x) - L + 1, hop)
    frames = np.array([w * x[s:s + L] for s in starts])
    S = np.abs(np.fft.rfft(frames, axis=1))**2           # power per frame
    times = (np.array(list(starts)) + L / 2) / fs
    freqs = np.fft.rfftfreq(L, 1 / fs)
    return times, freqs, 10 * np.log10(S.T + 1e-12)      # dB, frequency on rows
```

The cost is one FFT of length L per hop, $O((N/R)\, L \log L)$ for N samples. Choosing the hop $R = L/4$ with a Hann window means the overlapping windows add up to a constant, so the frames can be put back together exactly: you can edit a spectrogram (remove a cough, a hum line, a squeak) and resynthesise sound. Its failure case is the trade-off itself: no setting of L is right for a sound containing both drum hits and sustained bass notes.

The JPEG pipeline

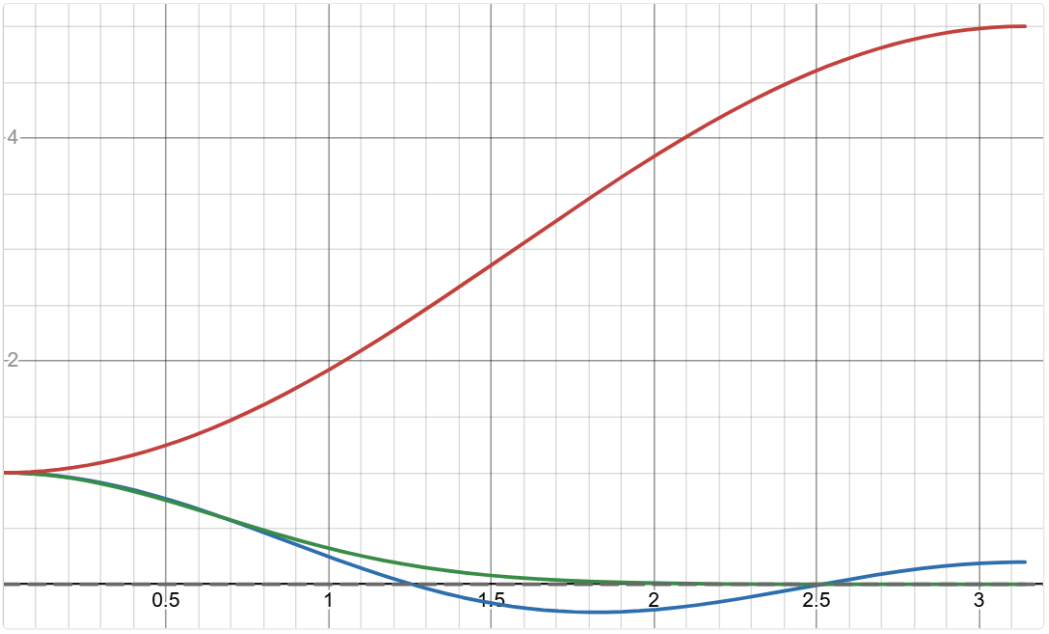
```
convert RGB to luminance + two colour channels; halve the colour resolution
for each 8x8 block of each channel:
    subtract 128; apply the 2-D DCT
    divide each coefficient by its quantisation step, round to an integer
    read coefficients in zig-zag order (low to high frequency)
    run-length encode the zeros, then Huffman-code the result
decoding reverses every step except the rounding
```

Only the rounding loses information. Halving the colour resolution works because the eye is far less sensitive to fine detail in colour than in brightness. The failure case is famous: at high compression the block boundaries become visible ("blocking"), and sharp edges ring with faint ripples, Gibbs-like overshoot (§15.6) from the discarded high frequencies.

76.5 The experiment

Turn the equaliser knobs. The three curves are 1-D frequency responses from $\omega = 0$ to the Nyquist limit $\omega = \pi$: the box blur of length 5 (blue), a Gaussian blur with $\sigma = 1.5$ pixels (green) and the 1-D sharpen kernel $[-1, 3, -1]$ (red). Change the `5`s in the first line to `3` or `9` and watch the zeros of the box move and its negative side lobes appear. Change `1.125` (which is $\sigma^2/2$) to `0.125` or `2` to make the Gaussian narrower or wider. The Gaussian never goes negative, which is

why it never flips the contrast of fine detail the way the box does.



Gain against frequency (radians per pixel). Blurs pass low frequencies and attenuate high ones; sharpening does the opposite, with gain 5 for the finest stripes.

Summon the ghosts, then banish them. The script builds the ring pattern of the hook, samples it naively and with a Gaussian prefilter applied by the convolution theorem, and also sharpens it through the FFT. It prints where the false ring centres should be (149 pixels apart on the original grid) and the sharpen kernel's gains at the two ends of the spectrum (1 and 9).

```
import numpy as np
import matplotlib.pyplot as plt

# --- knobs ---------------------------------------------------------------
A = 110          # ring chirp rate: local frequency is 2*A*r cycles per width
D = 8            # keep every D-th pixel
SIGMA = 0.6 * D  # Gaussian prefilter width in pixels (try 0, 0.3*D, 1.0*D)
KERNEL = np.array([[0, -1, 0], [-1, 5, -1], [0, -1, 0]], float)  # sharpen

M = 512
y, x = (np.mgrid[0:M, 0:M] - M / 2) / M
img = 0.5 + 0.5 * np.cos(2 * np.pi * A * (x**2 + y**2))

def filt(im, H):                       # multiply spectra: convolution theorem
    return np.real(np.fft.ifft2(np.fft.fft2(im) * H))

f = np.fft.fftfreq(M)                  # cycles per pixel
FX, FY = np.meshgrid(f, f)
gauss = np.exp(-2 * (np.pi * SIGMA)**2 * (FX**2 + FY**2))
K = np.zeros((M, M)); K[:3, :3] = KERNEL
K = np.roll(K, (-1, -1), axis=(0, 1))  # centre the kernel on pixel (0, 0)
sharp = np.fft.fft2(K)

naive = img[::D, ::D]
clean = filt(img, gauss)[::D, ::D]
print("false ring centres expected every", round(M * (M / D) / (2 * A)), "pixels")
print("sharpen gain at DC and at (pi, pi):", sharp[0, 0].real, sharp[M//2, M//2].real)

fig, ax = plt.subplots(1, 4, figsize=(13, 3.6))
for a, im, t in zip(ax, [img, naive, clean, filt(img, sharp)],
                    ["original", f"every {D}th pixel", f"blur sigma={SIGMA:.1f}, then sample",
                     "sharpened"]):
    a.imshow(im, cmap="gray", vmin=0, vmax=1); a.set_title(t); a.axis("off")
plt.tight_layout(); plt.show()
```

Things to try:

- Set `D = 4` and predict before running: the ghosts should move twice as far out, to $s_0 = 128/220 \approx 0.58$ widths, which is outside the image. Do any remain?
- Set `SIGMA = 0.3 * D`. The prefilter is now too narrow: how much of the false lattice survives? Then `1.0 * D`: the ghosts are gone, but how much real detail near the centre have you thrown away?
- Replace the Gaussian by a box prefilter of width `D` (build its 2-D transfer function from the formula for $B(\omega)$ in §76.3) and compare. The side lobes should let faint ghosts through.
- Replace `KERNEL` with the Sobel x-kernel `[[-1,0,1],[-2,0,2],[-1,0,1]]` and display `np.abs(...)`. Which ring segments vanish, and why does the formula for $|H|$ predict it?

76.6 The challenge

1. **Intuition.** A camera films a wheel with 12 spokes turning at 2.2 revolutions per second, at 24 frames per second. Without a calculator, decide whether the wheel appears to turn forwards or backwards, and roughly how fast. Then explain in one sentence why a fast-moving fan photographed under a flickering 100 Hz light can appear frozen.
2. **Derivation.** Derive the frequency response of the 3×3 Gaussian-like kernel $\frac{1}{16}[1,2,1]^{\mathsf{T}}[1,2,1]$ and show it is non-negative everywhere, with a zero exactly at the Nyquist limit in each direction. Then prove that the discrete Laplacian kernel (centre -4, neighbours 1) has response $2\cos\omega_x + 2\cos\omega_y - 4$, and use it to explain why "unsharp masking", $x + \lambda(x - \mathrm{blur}(x))$, sharpens.
3. **Implementation.** Implement an 8x8 block DCT codec for a grey-scale image (any photograph you own, or a synthetic one): DCT, a single quantisation step q for all coefficients, rounding, inverse DCT. Plot the fraction of non-zero coefficients and the root-mean-square error against q. Then replace the uniform q with one that grows with frequency and compare the two curves at equal numbers of non-zero coefficients.
4. **Open investigation.** For the chirp of §76.3, the balanced window length was $1/\sqrt{\kappa}$. Real sounds contain several chirps with different rates at once (a bird call, a siren, a speech formant). Design an experiment to measure how readable a spectrogram is (for example, how accurately you can recover the instantaneous frequency of each component) as a function of the window length and the window shape. Is there a single window that is close to the best for most natural sounds, or does each sound need its own?

76.7 Loose threads

The chapter treated sampling as a regular grid, filters as fixed and linear, and images as arbitrary arrays. Each of those assumptions is a door that this chapter leaves closed.

> **? Beating Nyquist**
>
> Sampling theory says you need twice the highest frequency, or ghosts appear. Yet a modern MRI scanner reconstructs a sharp image from a small fraction of the Fourier samples the theorem demands, by sampling them at *random* and assuming the image is sparse in some basis (a few large wavelet coefficients). Random sampling turns structured ghosts into faint noise-like error, and a convex optimization (Ch. 47) then removes it. How few samples are enough, and why does randomness help where a regular grid fails? That is compressed sensing, one of the central stories of Ch. 91.

The two mysteries of §76.3 are the other threads. The time–frequency uncertainty limit and its Gaussian optimum are proved in Ch. 92, and wavelets that sidestep it live in Ch. 61. Why one universal cosine basis nearly matches the eigenvectors of every photograph's statistics, and how many bits a picture truly needs, are questions of transform coding and rate–distortion theory in Ch. 80, built on the entropy of Ch. 33.

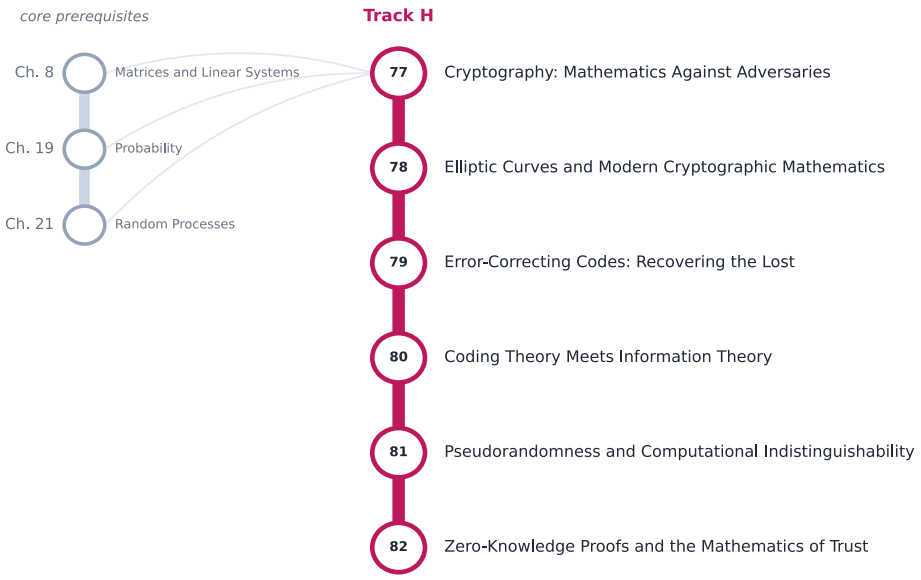
Two quieter connections stay inside this track. Noise textures in procedural generation (Ch. 74) are designed by shaping their power spectrum, the 2-D version of the power spectral density in §21.4. And the irregular sample patterns a ray tracer uses for anti-aliasing (Ch. 71) are chosen, like Poisson-disk points, to push aliasing energy into high frequencies where the eye forgives it. Why the eye forgives it, nobody can yet write down as a theorem.

Further reading

- A. V. Oppenheim and R. W. Schafer, *Discrete-Time Signal Processing*, 3rd ed., Pearson, 2010. The standard reference for sampling, filtering, the DFT and the short-time Fourier transform.
- R. C. Gonzalez and R. E. Woods, *Digital Image Processing*, 4th ed., Pearson, 2018. Spatial and frequency-domain filtering of images, edge detection and compression, with many examples.
- J. O. Smith III, *Spectral Audio Signal Processing*, W3K Publishing, 2011. Windows, the STFT and overlap-add resynthesis for audio. Freely available online.
- G. K. Wallace, "The JPEG still picture compression standard," *Communications of the ACM* 34(4), 1991. A short, readable account of the DCT pipeline by one of its designers.
- D. P. Mitchell and A. N. Netravali, "Reconstruction filters in computer graphics," *SIGGRAPH* 1988. How aliasing and blur trade off when graphics resamples images.

Track H — Cryptography, Coding, and Digital Security

> **Level 2 — Specializations.** Mathematics against adversaries and noise. Correct mathematics alone does not guarantee a secure implementation.



Suggested reading order for Track H. The grey stations on the left are the core chapters it leans on.

Before you start

This track assumes you are comfortable with these core chapters:

- Ch. 8 — Matrices and Linear Systems
- Ch. 19 — Probability
- Ch. 21 — Random Processes

Chapters

Ch.	Chapter	What it explores
77	Cryptography: Mathematics Against Adversaries	Security definitions, symmetric encryption, public-key cryptography.
78	Elliptic Curves and Modern Cryptographic Mathematics	Finite fields, elliptic-curve groups, discrete logarithms.
79	Error-Correcting Codes: Recovering the Lost	Hamming distance, linear codes, cyclic codes, Reed–Solomon codes.
80	Coding Theory Meets Information Theory	Channel capacity, entropy, coding bounds, compression.
81	Pseudorandomness and Computational Indistinguishability	Randomness tests, pseudorandom generators, computational security.
82	Zero-Knowledge Proofs and the Mathematics of Trust	Interactive proofs, commitment schemes, proof systems.

Every chapter follows the same arc: **the hook** (a puzzle or a failure), **the big idea**, **the mathematics**, **the algorithm**, **the experiment** you can run, **the challenge**, and **loose threads**: the questions the chapter deliberately leaves open.

Chapter 77 — Cryptography: Mathematics Against Adversaries

> **Level 2 · Track H — Cryptography, Coding, and Digital Security Builds on:** Ch. 19, Ch. 57, Ch. 30, Ch. 33 · **Leads to:** Ch. 78, Ch. 81, Ch. 82 · **Related:** Ch. 46, Ch. 32, Ch. 56 **Reading time:** ~25 min

77.1 The hook

In the winter of 1942, with German armies outside Moscow, the Soviet office that printed one-time pads ran short of time. A one-time pad is a booklet of random numbers. Sender and receiver hold identical copies; each page is added to one message and then destroyed. Used correctly it is the only cipher in history that has been *proved* unbreakable. Under wartime pressure, some pages were printed twice and issued to different users.

The American codebreakers of the Venona project, working from 1943 on, noticed. Over the next four decades they read parts of nearly three thousand Soviet messages, among them traffic that exposed spies inside the Manhattan Project. The unbreakable cipher broke. Nobody beat its mathematics; someone *used it twice*.

The lesson keeps returning. Ciphers in Wi-Fi routers, payment cards and game consoles are rarely defeated by mathematics; they fall to a reused number, a predictable seed or a timing leak. And yet mathematics is what makes modern security possible at all. Your browser has never met your bank and shares no secret with it, yet over wires anyone may tap, the two agree within milliseconds on a key the eavesdropper cannot compute.

So there are three questions, and this chapter is about all of them:

- What does it even *mean* for a cipher to be secure, stated precisely enough to prove?
- How can a short key protect a long message, when the one provably perfect cipher needs a key as long as the message?
- How can two strangers agree on a secret in public?

77.2 The big idea

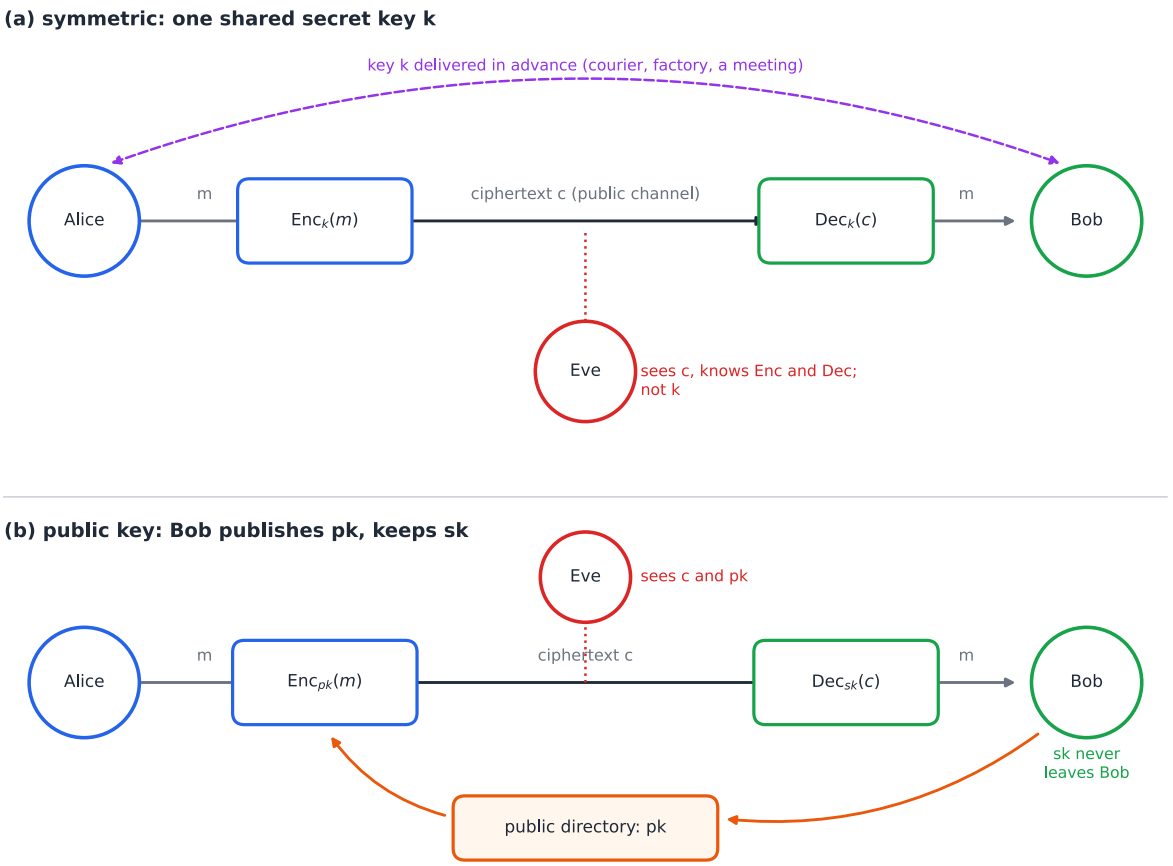
Cryptography is **mathematics with an opponent**. In most of this book the world is indifferent: noise is random, a circuit does not try to fool you. Here the noise is a clever person, Eve, who reads everything you send, knows exactly which algorithm you use, and will exploit any pattern you leave. A security claim is therefore a statement about *every possible strategy* Eve could follow, which is why it has to be framed as a game.

The game has three rules that took cryptographers a century to agree on.

1. **Eve knows the system.** Only the key is secret. This is **Kerckhoffs' principle** (1883), the cryptographer's version of publishing your mixed strategy in Ch. 46. A cipher whose security rests on a secret design is broken the day the design leaks.
2. **Say exactly what Eve can do.** Only listen? Get messages of her choice encrypted? Submit forged ciphertexts and watch your reaction? Each is a different attack model.
3. **Say exactly what "winning" means.** If Eve learns *one bit* about the message, say "buy" versus "sell", she has won.

With the game fixed, there are three levels of security, and they form the spine of this chapter.

- **Perfect secrecy.** The ciphertext carries no information at all about the message, even for an Eve with infinite computing power. Achievable, but the key must be as long as everything you will ever send.
- **Computational security.** Eve could in principle learn something, but only by a computation so large it will not finish before the Sun burns out. This is how short keys protect long messages.
- **Public-key security.** Some functions are easy to compute but hard to invert unless you hold a hidden "trapdoor". Bob publishes the lock and keeps the key; anyone can snap the padlock shut, only Bob can open it.



The two architectures. In symmetric cryptography the hard problem is getting k to Bob in the first place. Public-key cryptography removes that problem by splitting the key in two, at the price of much slower arithmetic; real systems use public keys only to agree on a symmetric key.

The analogy to keep in mind: a symmetric cipher is a **safe with two identical keys**, one for Alice and one for Bob. A public-key cipher is a **mailbox with a slot**: anyone can drop a letter in, but only the owner has the key to the back door.

Never roll your own crypto

This chapter separates two things that are easy to confuse. The *mathematics* of a cryptosystem can be correct, proved and beautiful while an *implementation* of it is trivially broken: by a reused nonce, a weak random-number generator, a non-constant-time comparison, a padding error message, or a key left in memory. Every code sample in this chapter uses toy parameters and textbook constructions chosen for clarity, and every one of them is insecure. For real systems use a vetted, maintained library (libsodium, your platform's TLS stack, a FIPS-validated module) through its highest-level interface, and follow current guidance from bodies such as NIST. Designing ciphers, protocols or even "small tweaks" is a job for specialists, and their work is trusted only after years of public attack.

77.3 The mathematics

Encryption schemes

Symmetric encryption scheme

A scheme is three algorithms. Gen outputs a key k from a key space $\mathcal{K}$. $\mathrm{Enc}_k(m)$ turns a message $m \in \mathcal{M}$ into a ciphertext c; it may use fresh randomness. $\mathrm{Dec}_k(c)$ returns a message. **Correctness** demands $\mathrm{Dec}_k(\mathrm{Enc}_k(m)) = m$ for every k and m.

Treat the message as a random variable M (Eve has some prior belief about what Alice might say) and the key as an independent random variable K, chosen by Gen. Then the ciphertext $C = \mathrm{Enc}_K(M)$ is random too, and we can ask what seeing it does to Eve's beliefs. That is a conditional-probability question, and Ch. 19 has the tools.

Perfect secrecy (Shannon, 1949)

A scheme is **perfectly secret** if for every prior distribution of M, every message m and every ciphertext c with $P(C = c) > 0$,

$$P(M = m \mid C = c) = P(M = m).$$

Seeing the ciphertext does not change Eve's beliefs at all. Equivalently, M and C are independent, so their mutual information (Ch. 33) is zero.

Result 1: the one-time pad is perfectly secret

Let messages, keys and ciphertexts all be n-bit strings, let Gen choose K uniformly from all 2^n strings, and encrypt by bitwise XOR (addition mod 2, written $\oplus$):

$$\mathrm{Enc}_k(m) = m \oplus k, \qquad \mathrm{Dec}_k(c) = c \oplus k.$$

Correctness is one line: $(m \oplus k) \oplus k = m \oplus (k \oplus k) = m \oplus 0 = m$.

The one-time pad is perfectly secret

For any distribution of M independent of the uniform key K, $P(M = m \mid C = c) = P(M = m)$.

Proof. Fix m and c. Given $M = m$, the ciphertext equals c exactly when $K = m \oplus c$, a single key. Because K is uniform and independent of M,

$$P(C = c \mid M = m) = P(K = m \oplus c) = 2^{-n},$$

the same number for every m. By the law of total probability, $P(C = c) = \sum_{m'} P(C = c \mid M = m')P(M = m') = 2^{-n}$. Bayes' theorem then gives

$$P(M = m \mid C = c) = \frac{P(C = c \mid M = m)\,P(M = m)}{P(C = c)} = \frac{2^{-n}\,P(M = m)}{2^{-n}} = P(M = m). \qquad \blacksquare$$

Read the proof again and see where it used its assumptions. It needed K **uniform** (otherwise $P(K = m \oplus c)$ depends on m), **independent of** M, and **used once**. Break the last condition and the pad cancels: if $c_1 = m_1 \oplus k$ and $c_2 = m_2 \oplus k$, then

$$c_1 \oplus c_2 = m_1 \oplus m_2,$$

which no longer involves the key at all. The XOR of two English texts is full of structure, and guessing a common word in one reveals the other at the same position. That is all Venona needed, and it is the experiment in §77.5.

Result 2: perfect secrecy needs a key as long as the message

Could a cleverer scheme achieve perfect secrecy with a shorter key? Shannon proved that it cannot.

Shannon's key-length bound

If a scheme is perfectly secret, then $|\mathcal{K}| \geq |\mathcal{M}|$.

Proof. Take M uniform on $\mathcal{M}$ and pick any ciphertext c that can occur. Let $D(c) = \{\mathrm{Dec}_k(c) : k \in \mathcal{K}\}$ be the set of messages that c could decrypt to under some key. It has at most $|\mathcal{K}|$ elements. Suppose $|\mathcal{K}| < |\mathcal{M}|$. Then some message m^* lies outside $D(c)$, and by correctness c can never be an encryption of m^*, so $P(M = m^* \mid C = c) = 0$. But $P(M = m^*) = 1/|\mathcal{M}| > 0$. This contradicts perfect secrecy. $\blacksquare$

The counting argument covers every cipher anyone will ever invent: a 128-bit key cannot perfectly protect a 129-bit message. That is why the one-time pad lived on only in diplomatic bags and hotlines, and why the rest of cryptography had to change the rules of the game.

Computational security: limiting Eve's budget

The way out is to stop defending against an Eve with unlimited computing power. Instead we ask that every Eve who runs for at most t steps succeeds with probability at most ε above blind guessing. The scheme is then called (t, ε)**-secure**. Asymptotically, with a security parameter λ (roughly the key length), we demand that every adversary running in time polynomial in λ has an advantage that is **negligible**: smaller than $1/\lambda^c$ for every constant c once λ is large enough, as $2^{-\lambda}$ is.

How large is large enough? A 128-bit key has $2^{128} \approx 3.4 \times 10^{38}$ values. A machine testing 10^{18} keys per second, far beyond any that exists, would need about 1.1×10^{13} years to try them all, some 800 times the age of the universe. In 1998 a purpose-built machine searched the 56-bit key space of the old DES standard in 56 hours. Key length is not a detail; each extra bit doubles Eve's work.

Two caveats make computational security a different kind of mathematics from Results 1 and 2.

- **It rests on assumptions.** We cannot prove that any practical cipher needs 2^{128} steps to break, only that "if Eve breaks this scheme, she also solves problem X", where X has resisted the world for decades. This is a **reduction**, the tool Ch. 30 uses to spread undecidability, now spreading hardness.
- **It needs average-case hardness.** NP-hardness (Ch. 30) says that *some* instances are hard. A cipher needs *random* instances, those made by a random key, to be hard almost always. The minimal object that delivers this is a **one-way function**: a function f that is easy to compute, but such that for a random x every efficient algorithm, given $f(x)$, finds some preimage only with negligible probability. If one-way functions exist then $\mathrm{P} \neq \mathrm{NP}$, because inverting f is an NP search problem; the converse is not known. This is the precise sense in which modern cryptography is a bet against $\mathrm{P} = \mathrm{NP}$, and a stronger bet than that.

The engine of symmetric cryptography is the **pseudorandom generator** (PRG): a deterministic function G that stretches a short random seed k into a long string $G(k)$ that no efficient test can tell apart from truly random bits. Replace the one-time pad by $G(k)$ and you have a **stream cipher**: $c = m \oplus G(k)$. Its security proof is a reduction. If some Eve could learn anything about m from c, then running her on $m \oplus s$ for a challenge string s would tell you whether s came from G or from fair coins, because against truly random s we already proved she learns nothing. So a break of the cipher *is* a break of the generator.

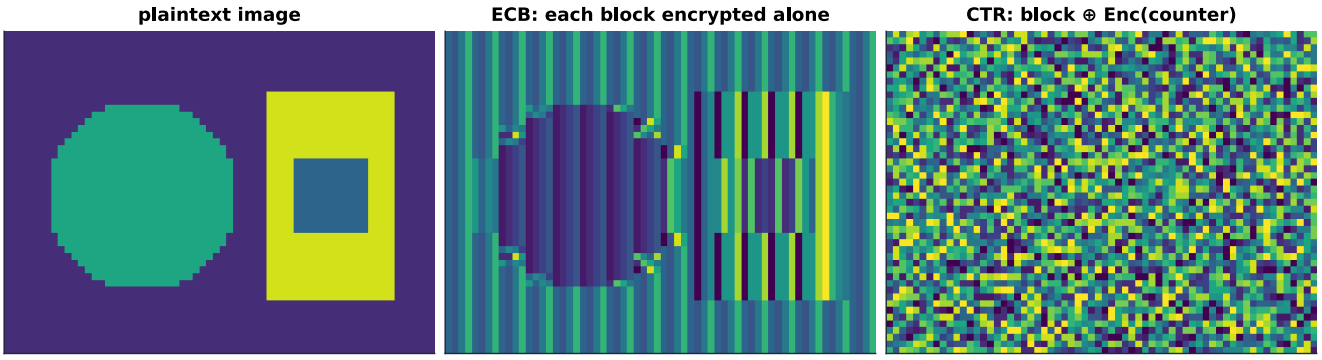
? Do one-way functions exist?

Every computational guarantee in this chapter is conditional. Remarkably, a single assumption would suffice for almost all of symmetric cryptography: Håstad, Impagliazzo, Levin and Luby proved in 1999 that a pseudorandom generator can be built from *any* one-way function. But nobody can prove that a single one-way function exists, and doing so would settle $\mathrm{P} \neq \mathrm{NP}$ along the way. What "indistinguishable from random" means exactly, and how the construction squeezes randomness out of mere one-wayness, is Ch. 81.

Block ciphers, and why encryption must be randomised

Most data is encrypted with a **block cipher**, a keyed permutation E_k of fixed-size blocks. AES, standardised in 2001, permutes 128-bit blocks under a 128-, 192- or 256-bit key. The security goal is that E_k, for a random hidden k, is indistinguishable from a permutation chosen uniformly at random from all $(2^{128})!$ of them, a **pseudorandom permutation**.

A block cipher alone is not an encryption scheme for long messages; you need a **mode of operation**. The obvious one, encrypting each block separately (electronic codebook, ECB), is a famous mistake. Equal plaintext blocks give equal ciphertext blocks, so every repetition in the data survives encryption.



A toy 16-bit "block cipher" (a random permutation of 2^{16} values) applied to an image four pixels at a time. In ECB mode every block of uniform colour maps to the same ciphertext block, so the shapes survive. In counter (CTR) mode each block is XORed with $E_k(\text{counter})$, a fresh pseudorandom pad, and the structure disappears.

The image shows a general fact, and the right definition of security exposes it. In the game of **indistinguishability under chosen-plaintext attack** (IND-CPA), Eve may ask for encryptions of any messages she likes. Then she submits two messages m_0, m_1 of equal length, receives $c^* = \mathrm{Enc}_k(m_b)$ for a secret fair coin b, may ask for more encryptions, and finally guesses b. The scheme is IND-CPA secure if no efficient Eve guesses correctly with probability non-negligibly above $\frac{1}{2}$.

> **Deterministic encryption is never IND-CPA secure**
>
> If Enc_k is a deterministic function of m, there is an Eve who wins the IND-CPA game with probability 1.

Proof. Eve picks any two distinct messages $m_0 \neq m_1$ and asks the encryption oracle for $c_0 = \mathrm{Enc}_k(m_0)$. She submits (m_0, m_1) and receives c^*. Since encryption is deterministic, $c^* = c_0$ if $b = 0$. If $b = 1$, then $c^* = \mathrm{Enc}_k(m_1) \neq c_0$, because by correctness two messages with the same ciphertext under one key would decrypt to the same thing. So Eve outputs 0 if $c^* = c_0$ and 1 otherwise, and is always right. ■

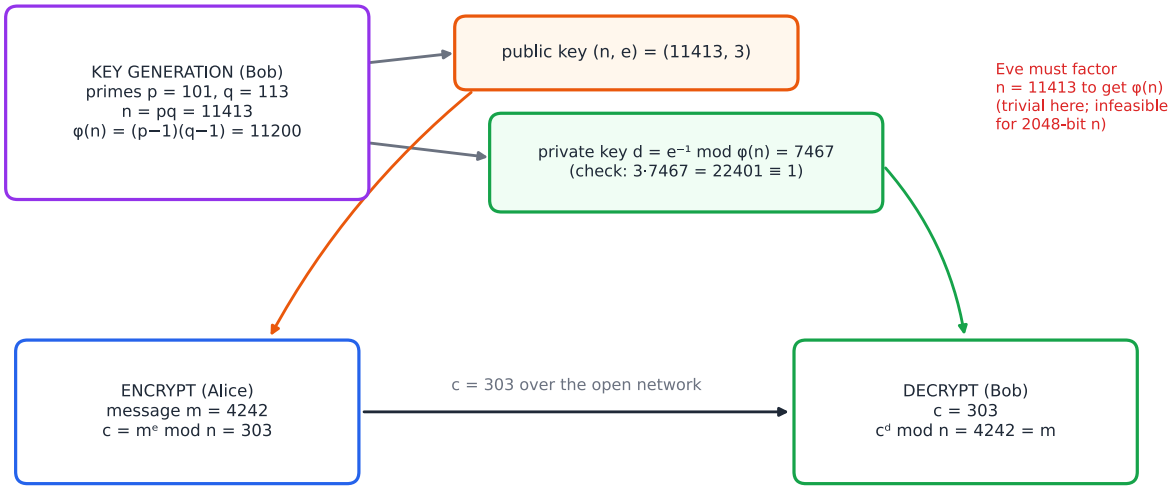
This three-line proof condemns ECB, and it equally condemns the textbook RSA of Ch. 57, where $c = m^e \bmod n$ depends on nothing but m. Secure encryption must use fresh randomness or a never-repeating **nonce** every time. That is why counter mode feeds a counter into the cipher, why real RSA encryption pads the message with random bytes (OAEP), and why, as Venona showed, *reusing* the fresh value destroys everything.

Confidentiality is also not enough. CTR mode is **malleable**: flipping ciphertext bit i flips plaintext bit i, so Eve can turn "pay 100" into "pay 900" blind. Real systems use **authenticated encryption** (AES-GCM, ChaCha20-Poly1305), which adds a tag no forger can compute and rejects tampered ciphertexts.

Public-key cryptography

The key-distribution problem in the figure is solved by functions with a trapdoor. You have met the two classical ones in Ch. 57, which proves why they work; here we look at them as security engineers.

- **Diffie–Hellman** (1976). In a cyclic group with generator g, Alice sends g^a, Bob sends g^b, both compute g^{ab}. Getting g^{ab} from g, g^a, g^b is the **computational Diffie–Hellman** problem; the stronger **decisional** assumption says g^{ab} even *looks* random to Eve. Unauthenticated, the exchange falls to a "man in the middle" who runs one exchange with each party; signatures fix that.
- **RSA** (1977). Bob publishes $n = pq$ and e, keeps $d = e^{-1} \bmod \varphi(n)$. Inverting $m \mapsto m^e \bmod n$ without d is the **RSA problem**; it is no harder than factoring n, and is believed to be no easier. RSA also gives **signatures**: Bob signs a hash h of a document as $s = h^d \bmod n$, and anyone checks $s^e \equiv h$.



Toy RSA from key generation to decryption. Everything Eve needs is public except $\varphi(n)$, and computing $\varphi(n)$ is as hard as factoring n. With 4-digit n that takes microseconds; with a 617-digit n the best known classical algorithm, the number field sieve, would take longer than any realistic budget.

Public-key arithmetic is thousands of times slower than AES, so practice is always **hybrid**: a public-key step agrees on a fresh symmetric key, and authenticated symmetric encryption carries the data. That is what happens in the handshake when your browser opens a bank's page.

Result 3: what a quantum computer would change

Every public-key system above rests on factoring or on discrete logarithms. Both reduce to a single task: finding the **period** of a function. Here is the reduction for factoring, which is completely classical.

> **Factoring reduces to order finding**
>
> Let n be odd and composite, and let a be coprime to n with multiplicative order r (the least $r \geq 1$ with $a^r \equiv 1 \pmod{n}$). If r is even and $a^{r/2} \not\equiv -1 \pmod{n}$, then $\gcd(a^{r/2} - 1,\ n)$ is a non-trivial factor of n.

Proof. Let $x = a^{r/2} \bmod n$. Then $x^2 \equiv 1$, so n divides $x^2 - 1 = (x-1)(x+1)$. Now n does not divide $x - 1$, because $x \equiv 1$ would mean $a^{r/2} \equiv 1$ with $r/2 < r$, contradicting the minimality of r. And n does not divide $x + 1$ by hypothesis. So n divides the product without dividing either factor, which is only possible if n shares a proper factor with each. Hence $1 < \gcd(x-1, n) < n$. ■

For $n = 15$ and $a = 7$: the powers of 7 mod 15 are $7, 4, 13, 1$, so $r = 4$; then $x = 7^2 \bmod 15 = 4$, and $\gcd(3, 15) = 3$, $\gcd(5, 15) = 5$. One can show that when n has at least two distinct odd prime factors, a random a satisfies the hypotheses with probability at least $\frac{1}{2}$; for the toy modulus $n = 11413$ the fraction is about 91 %. So factoring is no harder than order finding.

Classically, order finding is as hard as factoring. But $k \mapsto a^k \bmod n$ is a *periodic sequence*, and finding a period is what Fourier transforms do (Ch. 16). Shor's 1994 algorithm prepares a quantum superposition over all k, computes $a^k \bmod n$ in superposition, and applies a quantum Fourier transform, whose output peaks near multiples of $1/r$; continued fractions then recover r. A large, error-corrected quantum computer would thus break RSA, Diffie–Hellman and the elliptic-curve systems of Ch. 78 in polynomial time. None exists yet, but traffic recorded today can be decrypted later, so the migration has begun: in 2024 NIST standardised **post-quantum** algorithms built on finding short vectors in high-dimensional lattices ("learning with errors"), for which no efficient quantum attack is known. ML-KEM (FIPS 203) replaces Diffie–Hellman key agreement and ML-DSA (FIPS 204) replaces RSA and elliptic-curve signatures; SLH-DSA (FIPS 205) is a conservative backup whose signatures rest only on the security of a hash function. The price is size: post-quantum keys and signatures run to kilobytes instead of tens of bytes. Symmetric ciphers survive with longer keys, since Grover's quantum search only square-roots brute force.

77.4 The algorithm

The code below is a teaching model. It implements textbook RSA and the classical half of Shor's algorithm with numbers small enough to check by hand. Python's built-in three-argument `pow` does modular exponentiation by square-and-multiply (Ch. 57 §57.4), and `pow(e, -1, phi)` computes a modular inverse by the extended Euclidean algorithm.

```
from math import gcd
import random

# --- toy RSA: illustration only, NEVER for real data ---
def keygen(p, q, e=3):
    n, phi = p * q, (p - 1) * (q - 1)
    assert gcd(e, phi) == 1
    return (n, e), pow(e, -1, phi)          # public key, private exponent

def enc(pk, m):  n, e = pk; return pow(m, e, n)
def dec(pk, d, c): n, _ = pk; return pow(c, d, n)

pk, d = keygen(101, 113)
c = enc(pk, 4242)
print(pk, d, c, dec(pk, d, c))              # (11413, 3) 7467 303 4242

# --- factoring from order finding (the classical half of Shor's algorithm) ---
def order(a, n):                            # brute force: the step a quantum computer replaces
    r, x = 1, a % n
    while x != 1:
        x, r = x * a % n, r + 1
    return r

def factor_via_order(n, tries=20):
    for _ in range(tries):
        a = random.randrange(2, n - 1)
        g = gcd(a, n)
        if g > 1:
            return g                        # lucky guess
        r = order(a, n)
        if r % 2 == 0 and pow(a, r // 2, n) != n - 1:
            return gcd(pow(a, r // 2, n) - 1, n)
    return None

print(factor_via_order(15), factor_via_order(11413))   # e.g. 3 101
```

Complexity. Encryption and decryption cost $O(\log d)$ multiplications of λ-bit numbers, roughly $O(\lambda^3)$ bit operations: fast for $\lambda = 2048$. The function `order` may loop r times, and r can be close to $\varphi(n)/2$, around 2^{2047} for a 2048-bit modulus. That one loop is the gap between "encrypt in a millisecond" and "break never". Shor's algorithm replaces exactly that loop, and nothing else changes.

Failure cases, mathematical. `keygen` fails when $\gcd(e, \varphi(n)) > 1$; with $p = 7$, $q = 13$ and $e = 3$, $\varphi(n) = 72$ is divisible by 3 and d does not exist. Decryption fails if $m \geq n$, since only $m \bmod n$ survives. `factor_via_order` fails for prime powers $n = p^k$ (try $n = 9$), which need a separate test.

Failure cases, engineering. These are the ones that break real systems, and the code above has all of them.

- *Determinism.* `enc` is deterministic, so by the theorem of §77.3 it is not IND-CPA secure. With $e = 3$ and a message $m < n^{1/3}$, m^3 never wraps around n, and an ordinary integer cube root recovers m.
- *Malleability.* $\mathrm{Enc}(m_1)\,\mathrm{Enc}(m_2) = \mathrm{Enc}(m_1 m_2) \bmod n$, so Eve can transform ciphertexts meaningfully without the key.
- *Side channels.* `pow` takes time, and draws current, that depends on the bits of d; such measurements have recovered keys from smart cards and remote servers. Production code is constant-time and blinded.
- *Bad randomness.* Devices with poor entropy at boot have produced moduli that share a prime; one $\gcd(n_1, n_2)$ then factors both.
- *Wrong error messages.* A server that reports "bad padding" differently from other errors lets Eve decrypt any ciphertext by adaptive queries (Bleichenbacher, 1998).

None of these is a mistake in the number theory. Each is a gap between a mathematical object and a running program, which is why the warning in §77.2 is not a formality.

> **? Why are elliptic-curve keys so short?**
>
> To get security comparable to a 128-bit symmetric key, current guidance asks for an RSA modulus of about 3072 bits, but an elliptic-curve key of only 256 bits. Both rest on "easy forward, hard backward" arithmetic in a finite group. Why does one group need twelve times fewer bits than the other, and what exactly is the group made of? The answer is a curve, a chord, and the absence of a shortcut, in Ch. 78.

77.5 The experiment

The script below plays Eve against a **two-time pad**. Two messages are encrypted with the same random pad, exactly the Venona mistake. Eve XORs the ciphertexts, which cancels the pad, and then **drags a crib**: she guesses that a common word appears in one message, slides it along $c_1 \oplus c_2$, and at each position XORs it in. Wherever the guess is right, a fragment of the *other* message appears in plain text.

```
import numpy as np
import matplotlib.pyplot as plt

M1 = b"meet the courier at the north gate at dawn"   # KNOB: first secret message
M2 = b"the shipment leaves the harbour on tuesday"   # KNOB: second secret message
CRIB = b" the "                                      # KNOB: a word Eve guesses is in one message
REUSE = True                                         # KNOB: False gives each message a fresh pad

rng = np.random.default_rng(2024)
L = min(len(M1), len(M2))
m1 = np.frombuffer(M1[:L], dtype=np.uint8)
m2 = np.frombuffer(M2[:L], dtype=np.uint8)
k1 = rng.integers(0, 256, L, dtype=np.uint8)
k2 = k1 if REUSE else rng.integers(0, 256, L, dtype=np.uint8)
c1, c2 = m1 ^ k1, m2 ^ k2               # what Eve intercepts
x = c1 ^ c2                             # the pad cancels if it was reused

crib = np.frombuffer(CRIB, dtype=np.uint8)
score = []
for pos in range(L - len(crib) + 1):    # crib dragging: slide the guess along
    frag = x[pos:pos + len(crib)] ^ crib
    ok = np.mean([(32 <= b < 127) and (chr(b).isalpha() or b == 32) for b in frag])
    score.append(ok)
    if ok == 1.0:
        print(f"pos {pos:2d}: other message reads {bytes(frag)!r}")

plt.bar(range(len(score)), score)
plt.xlabel("crib position"); plt.ylabel("fraction of letters/spaces")
plt.title("crib dragging on c1 XOR c2"); plt.show()
```

With the defaults, six positions produce readable-looking fragments. One of them, at position 4, reads `'shipm'`: a genuine piece of the second message, exposed because the first message has " the " there. Another, at position 19, reads `' the '` because *both* messages happen to contain " the " at the same place. The rest are look-alikes such as `'mridr'` that a human rejects at a glance. Things to try:

- **Extend a hit.** Guess that `shipm` continues as `shipment`, set `CRIB = b"shipment"` and look at position 4 again: you now read eight characters of the first message. Real codebreakers bootstrapped whole messages this way, one guessed word at a time.
- **Set `REUSE = False`.** Every position now gives noise, and no fragment passes the filter. Result 1 is at work: with independent pads, $c_1 \oplus c_2$ is itself a uniformly random string, and no guess can be confirmed.
- **Change the language model.** The filter accepts any lowercase letters, which is crude. Score by English letter frequencies instead (English carries only about 1 to 1.5 bits per letter out of a possible $\log_2 26 \approx 4.7$, Ch. 33). How short a crib still finds the true position?

77.6 The challenge

No answers are given.

1. **Intuition.** A friend proposes "double encryption": encrypt with a one-time pad k_1 and then again with an independent pad k_2, for twice the security. Explain in words why this gains nothing. Then explain why encrypting twice with a *block cipher* under two keys (as in "double DES") also gains far less than doubling the key length suggests. (Think of meeting in the middle.)
2. **Derivation.** Prove the converse half of Shannon's theory: if $|\mathcal{K}| = |\mathcal{M}| = |\mathcal{C}|$ and the scheme is perfectly secret, then every key is used with probability $1/|\mathcal{K}|$, and for every pair (m, c) there is exactly one key with $\mathrm{Enc}_k(m) = c$. Then show that the definition of perfect secrecy is equivalent to: for all m_0, m_1, c, $P(C = c \mid M = m_0) = P(C = c \mid M = m_1)$.
3. **Implementation.** Extend the toy RSA code with the Chinese-remainder speed-up for decryption (compute c^d mod p and mod q separately and combine). Measure the speed-up for 512-bit primes generated with a Miller–Rabin test. Then implement a toy version of the common-factor attack: generate 1000 moduli where, through a deliberately weak random generator, a few share a prime, and find them all with pairwise gcds. Keep everything in a sandbox: this is to understand the failure, not to build anything.
4. **Open investigation.** In the experiment, replace the letter filter by a statistical score and automate crib dragging over a dictionary of the 200 most common English words. For two messages of length L, how does the fraction of plaintext you can recover automatically grow with L? Is there a threshold length below which the two-time pad is "practically safe", and how does it depend on the language (try a language with a different letter distribution, or compressed text)?

77.7 Loose threads

The hook asked three questions. **Security** means winning a precisely defined game against every allowed adversary; perfect secrecy and IND-CPA are two such games, and the second showed in three lines that deterministic ciphers fail. **Short keys** protect long messages only against a bounded Eve, through pseudorandom generators and block ciphers resting on unproved assumptions; Shannon's counting shows nothing else is possible. **Strangers agree on a secret** through trapdoor arithmetic whose hardness rests on factoring and discrete logarithms, which a quantum computer would turn into period finding.

The Venona lesson runs under all of it: mathematics proves a scheme secure *under stated conditions*, and practice fails by quietly violating one.

What we did not tell you:

- **Where good randomness comes from**, and what "looks random" must mean so that a stream cipher inherits the one-time pad's security. The first mystery points to Ch. 81.
- **What the elliptic-curve group is.** The second mystery points to Ch. 78, and the finite fields underneath are introduced in Ch. 57.
- **Hash functions.** Signatures, passwords and blockchains all depend on functions where finding two inputs with the same output is infeasible. The birthday bound of Ch. 32 is why a 256-bit hash gives only 128 bits of collision resistance.

- **Why lattice problems resist quantum computers.** Nobody knows. Ajtai (1996) and Regev (2005) showed that random lattice instances are as hard as the worst ones, a guarantee factoring never had, but a quantum algorithm for short lattice vectors has not been ruled out.

? Proving you know a secret without revealing it

A signature convinces Bob that Alice holds a private key, but the signature itself is a lasting piece of evidence that Bob can show to anyone. Is it possible to convince Bob that you know the factorisation of n, or the password, or the solution to a puzzle, so that he learns *nothing* except that the claim is true, not even enough to convince a third party? It is, and the definition of "learns nothing" is one of the strangest ideas in computer science: anything Bob could see, he could have simulated alone. See Ch. 82.

Further reading

- J. Katz and Y. Lindell, *Introduction to Modern Cryptography*, 3rd ed., CRC Press, 2020. The standard text for security definitions, reductions and proofs.
- D. Boneh and V. Shoup, *A Graduate Course in Applied Cryptography*, freely available online. Rigorous, current, and full of real attacks.
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Chapter 78 — Elliptic Curves and Modern Cryptographic Mathematics

Level 2 · Track H — Cryptography, Coding, and Digital Security Builds on: Ch. 57, Ch. 56, Ch. 77, Ch. 1 · **Leads to:** Ch. 79, Ch. 82, Ch. 94 · **Related:** Ch. 32, Ch. 26, Ch. 81 **Reading time:** ~25 min

78.1 The hook

The PlayStation 3 was, for four years, the console nobody could crack. Every program it ran had to carry a digital signature from Sony, made with Sony's private key using the Elliptic Curve Digital Signature Algorithm (ECDSA). The mathematics behind it is the same that protects most of today's secure web connections, and nobody has ever broken it. Without the private key, no unofficial software would run.

In December 2010, a group of hackers calling themselves failoverflow stood on stage at a computer congress in Berlin and showed that they had the key. They had not solved a hard problem. They had noticed that two different signed programs shared one number in their signatures. ECDSA requires a fresh secret random number for every signature, and Sony had used the same one every time. From two signatures and a few lines of algebra, high-school algebra modulo a prime, the private key fell out. Anyone could now sign software as Sony.

Three years later the same mistake emptied Bitcoin wallets: a flaw in a random-number generator on Android phones made some wallets repeat their signing numbers, and thieves who watched the public blockchain for repeats took the coins.

The cryptosystem was not wrong. The mathematics was flawless, the implementation was not. To understand both the strength and the fragility, we need to know what these signatures are made of. The answer is surprising: a *curve*, a rule for "adding" points on it that is built from straight lines, and arithmetic in a finite field where the curve stops looking like a curve at all.

78.2 The big idea

Public-key cryptography needs a group where going forward is easy and going back is hard. In Ch. 57 the group was $\mathbb{Z}_p^\times$: computing g^a is fast, recovering a from g^a (the **discrete logarithm**) is believed slow. Elliptic curves supply a different group with the same shape and, as far as anyone knows, no shortcuts.

The construction is geometric. Draw the cubic curve $y^2 = x^3 + ax + b$. Pick two points on it, draw the straight line through them, and the line hits the curve in exactly **one more point**. Reflect that point in the x-axis: that is the "sum" of the two. To add a point to itself, use the tangent line. Astonishingly, this rule obeys the laws of a group, so you can add a point G to itself again and again, $G,\ 2G,\ 3G, \ldots$, and the multiples behave like the powers $g, g^2, g^3, \ldots$ of a number.

Now do the same arithmetic modulo a prime p. The curve dissolves into a scattered cloud of points on a $p \times p$ grid, but the formulas still work, and the multiples kG hop around the cloud with no visible pattern. Computing kG from k is fast, by doubling and adding. Recovering k from kG is the **elliptic-curve discrete logarithm problem**, and the best known general attack is no better than a sophisticated form of guessing.

An analogy: imagine a billiard table with strange, perfectly deterministic walls. You shoot a ball from a known spot, and it bounces k times and stops. Anyone can simulate k bounces quickly if they know k. But seeing only where the ball stopped, finding k means re-simulating bounce after bounce, because the table remembers nothing about how many bounces got it there.

78.3 The mathematics

The fields underneath

We need a field: a number system where we can add, subtract, multiply and divide by anything nonzero. Over the real numbers we can draw pictures. For cryptography we use the finite field $\mathbb{F}_p = \mathbb{Z}_p$ of integers mod a prime p, where division means multiplying by a modular inverse from the extended Euclidean algorithm. Ch. 57 builds these fields, and the prime-power fields $\mathbb{F}_{2^m}$, and we use them without re-deriving anything. One caution: the formulas below divide by 2 and by 3, so we assume the field's characteristic is not 2 or 3. (Binary curves exist but use a different equation.)

Elliptic curve

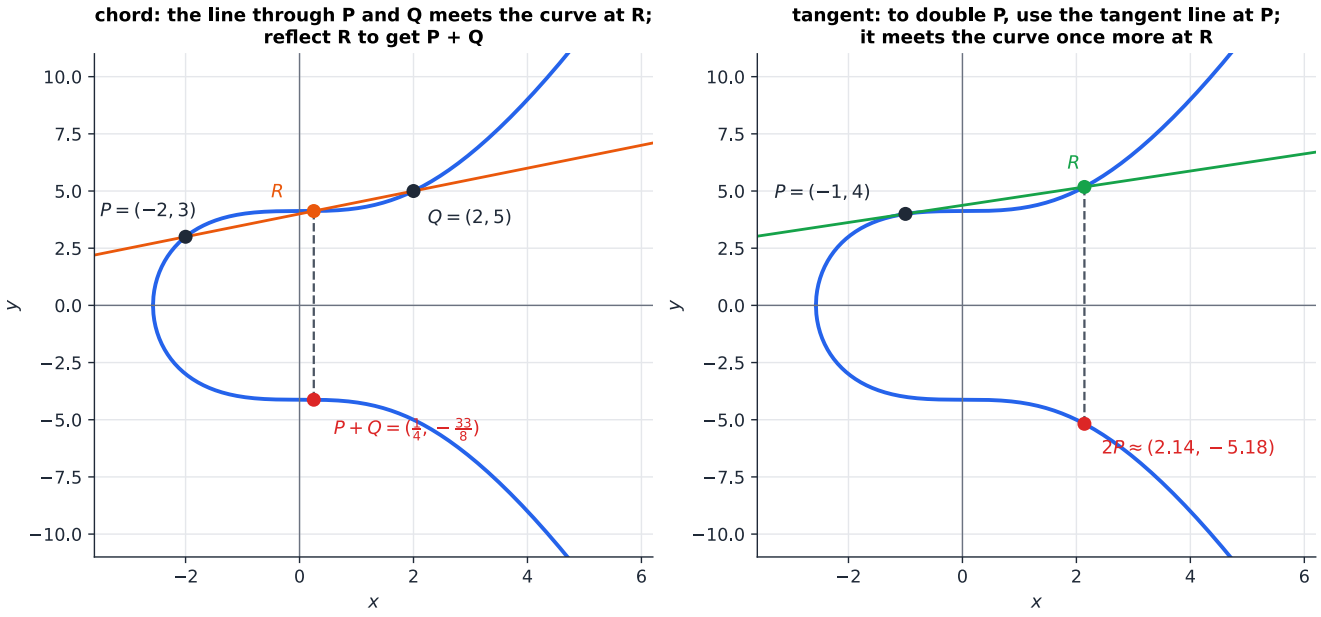
Let K be a field with $1+1 \neq 0$ and $1+1+1 \neq 0$, and let $a, b \in K$ satisfy $4a^3 + 27b^2 \neq 0$. The **elliptic curve** E is the set

$$E(K) = \{(x, y) \in K^2 : \ y^2 = x^3 + ax + b\} \ \cup \ \{\mathcal{O}\},$$

where $\mathcal{O}$ is an extra **point at infinity**.

The condition $4a^3 + 27b^2 \neq 0$ says the cubic $x^3 + ax + b$ has no repeated root, which keeps the curve smooth: no cusps, no self-crossings, and a well-defined tangent everywhere. The point $\mathcal{O}$ is best pictured as sitting infinitely far up (and down) every vertical line; every vertical line meets the curve there.

Result 1: the chord-and-tangent law



Addition and doubling on $y^2 = x^3 + 17$. All four named points have rational coordinates: rational points add to rational points, because the formulas use only field operations.

The addition formulas

Let $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ lie on E with $Q \neq -P$, where $-P = (x_1, -y_1)$. Put

$$\lambda = \begin{cases} \dfrac{y_2 - y_1}{x_2 - x_1}, & P \neq Q, \\ \dfrac{3x_1^2 + a}{2y_1}, & P = Q. \end{cases}$$

Then the line of slope λ through P meets E in a third point $R = (x_3, -y_3)$, and $P + Q := (x_3, y_3)$ is given by

$$x_3 = \lambda^2 - x_1 - x_2, \qquad y_3 = \lambda(x_1 - x_3) - y_1.$$

Proof. The line through P with slope λ is $y = \lambda(x - x_1) + y_1$. In the case $P \neq Q$ it passes through Q by the choice of λ. Substitute it into the curve:

$$\big(\lambda(x - x_1) + y_1\big)^2 = x^3 + ax + b \quad \Longleftrightarrow \quad x^3 - \lambda^2 x^2 + (\cdots)x + (\cdots) = 0.$$

This is a monic cubic in x, and its roots are the x-coordinates where line and curve meet. Two roots are known: x_1 and x_2. A cubic over a field with two roots in the field has a third, because dividing out $(x - x_1)(x - x_2)$ leaves a linear factor. By Vieta's formula the three roots sum to minus the coefficient of x^2:

$$x_1 + x_2 + x_3 = \lambda^2 \quad \Longrightarrow \quad x_3 = \lambda^2 - x_1 - x_2.$$

The third intersection point lies on the line, so its y-coordinate is $\lambda(x_3 - x_1) + y_1$; reflecting changes its sign, giving $y_3 = \lambda(x_1 - x_3) - y_1$. For $P = Q$, implicit differentiation of $y^2 = x^3 + ax + b$ gives $2y\,y' = 3x^2 + a$, so the tangent slope is $(3x_1^2 + a)/(2y_1)$; the tangent touches the curve at P with multiplicity two, so x_1 counts as a double root and the same Vieta argument applies with $x_2 = x_1$. ■

Check it on the figure. For $P = (-2, 3)$, $Q = (2, 5)$ on $y^2 = x^3 + 17$: $\lambda = \frac{2}{4} = \frac{1}{2}$, $x_3 = \frac{1}{4} + 2 - 2 = \frac{1}{4}$, $y_3 = \frac{1}{2}(-2 - \frac{1}{4}) - 3 = -\frac{33}{8}$. And indeed $(\frac{1}{4})^3 + 17 = \frac{1089}{64} = (\frac{33}{8})^2$. Doubling $P = (-1, 4)$: $\lambda = \frac{3}{8}$, and $2P = (\frac{137}{64}, -\frac{2651}{512})$. The denominators grow quickly, which is one reason cryptographers prefer finite fields.

Nothing in the proof used real numbers. It used only $+, -, \times, \div$ and the fact that a cubic with two roots in the field has a third. So **the same formulas define addition on $E(\mathbb{F}_p)$.**

The group

$E(K)$ is an abelian group

With the chord-and-tangent rule, $P + \mathcal{O} = P$, and $P + (-P) = \mathcal{O}$, the set $E(K)$ is an abelian group with identity $\mathcal{O}$.

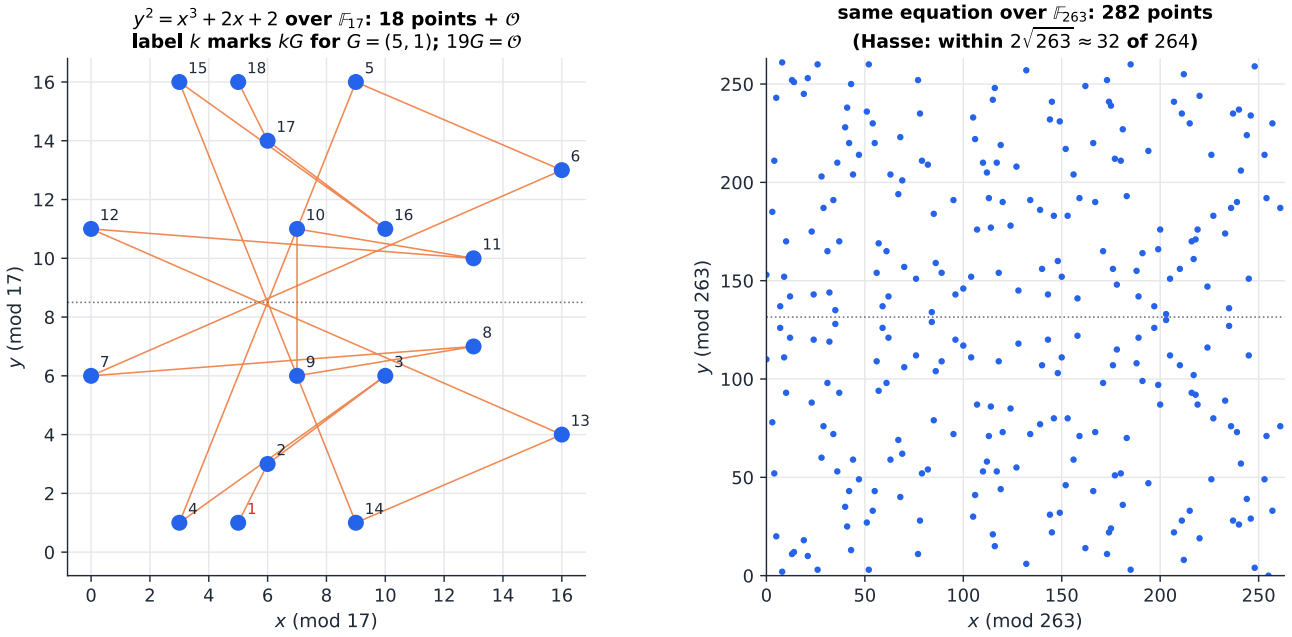
Most axioms are immediate. The identity: the vertical line through P meets the curve at P, $-P$ and $\mathcal{O}$, so "the third point" of P and $\mathcal{O}$ is $-P$, reflected back to P. Inverses: the line through P and $-P$ is vertical, its third point is $\mathcal{O}$. Commutativity: the line through P and Q is the line through Q and P.

Associativity is the miracle. Why should $(P + Q) + R = P + (Q + R)$ for a rule made of lines? One can grind it out from the formulas (a computer algebra system does it in seconds, a human in many pages). A cleaner argument uses a classical fact about cubics: *if two cubic curves meet in exactly nine points, then any cubic passing through eight of them also passes through the ninth* (the Cayley–Bacharach theorem). Multiply the equations of three lines, the line through P and Q, the line through $P + Q$ and R, and the vertical line through $Q + R$, to get a cubic C_1. Likewise the line through Q and R, the line through P and $Q + R$,

and the vertical line through $P+Q$ give a cubic C_2. Each meets E in nine points, and eight are shared: $P, Q, R, \mathcal{O}, \pm(P+Q), \pm(Q+R)$. The ninth intersection of C_1 with E is $-\big((P+Q)+R\big)$, and Cayley–Bacharach forces C_2 through it, which means it equals $-\big(P+(Q+R)\big)$. (The special cases where some of the points coincide need a limiting argument.)

> **? Why lines make a group**
>
> The nine-point argument shows associativity is *true*, but not *why* a cubic curve, of all things, should carry a group law. Conics do not; curves of degree four do not in this simple way. The deep reason is that an elliptic curve is secretly a doughnut: over $\mathbb{C}$ it is the plane modulo a lattice, $\mathbb{C}/\Lambda$, where addition is just addition of complex numbers, and the coordinates x, y are doubly periodic functions of the kind Ch. 18 hints at. Over any field, the group is a shadow of a more general object, the group of "divisors" on a curve of genus one. That story is Ch. 94.

Curves over finite fields



Over a finite field the curve becomes a cloud. The group law still works: the orange path visits all 18 affine points as $G, 2G, \dots, 18G$ and returns to $\mathcal{O}$ at $19G$. Each x gives either no points or a mirror pair $(x, \pm y)$, the only trace left of the real picture.

How many points does $E(\mathbb{F}_p)$ have? Each $x \in \mathbb{F}_p$ gives two points if $x^3 + ax + b$ is a nonzero square mod p, one if it is zero, and none otherwise. Roughly half of the nonzero residues are squares, so we expect about p affine points plus $\mathcal{O}$.

> **Hasse's theorem (1933)**
>
> The number of points satisfies $\big|\,\#E(\mathbb{F}_p) - (p+1)\,\big| \le 2\sqrt{p}$.

On $y^2 = x^3 + 2x + 2$ the count is 19 over $\mathbb{F}_{17}$, within 1 of $p+1 = 18$, and 282 over $\mathbb{F}_{263}$, within 18 of 264; the bound allows 8.2 and 32.4. We will not prove Hasse's theorem here; it is the first case of a vast generalisation (the Weil conjectures) that took most of the twentieth century to prove.

Since 19 is prime, Lagrange's theorem (Ch. 56) leaves every point except $\mathcal{O}$ only one possible order, 19: each one generates the whole group, and $E(\mathbb{F}_{17}) \cong \mathbb{Z}_{19}$. Cryptographic curves are chosen the same way, with a group of prime order n (or a small multiple of it) near 2^{256}.

The discrete logarithm, and how fast it can be attacked

> **Elliptic-curve discrete logarithm problem (ECDLP)**
>
> Given a point G of order n and a point Q in the group it generates, find $d \in \{0, \dots, n-1\}$ with $Q = dG$.

The forward direction is fast. **Double-and-add** reads the bits of d from the top: double the running total at each bit and add G when the bit is 1. That computes dG in at most $2\log_2 d$ group operations, about 510 for a 256-bit d. Brute force in reverse tries $G, 2G, 3G, \dots$ and needs up to n steps. Can we do better than brute force without any special structure? Yes, but only to the square root.

> **Baby-step giant-step (Shanks)**
>
> The ECDLP in a group of order n can be solved with $O(\sqrt{n})$ group operations and $O(\sqrt{n})$ memory.

Proof. Let $m = \lceil \sqrt{n} \rceil$. Every d with $0 \le d < n \le m^2$ can be written uniquely as $d = im + j$ with $0 \le i, j < m$ (division with remainder). Then

$$Q = dG = imG + jG \quad \Longleftrightarrow \quad Q - i(mG) = jG.$$

Baby steps: compute jG for $j = 0, \dots, m-1$ and store them in a hash table (m additions). **Giant steps:** compute $Q,\ Q - mG,\ Q - 2mG, \dots$ and look each one up ($\le m$ additions). Some $i < m$ must produce a hit, because the true d has such a decomposition; the hit gives $d = im + j$. Total work is at most $2m = O(\sqrt{n})$. ■

For a group of order near 2^{256}, $\sqrt{n} \approx 2^{128} \approx 3.4 \times 10^{38}$ operations: the same scale as brute-forcing a 128-bit AES key in Ch. 77. The memory problem is removed by **Pollard's rho** method, which takes a pseudorandom walk through the group and waits for it to collide with itself. By the birthday bound of Ch. 32, a collision arrives after about $\sqrt{\pi n/2}$ steps, using almost no memory.

Two further facts shape how curves are chosen.

- **Pohlig–Hellman.** If the group order n factors into small primes, the ECDLP splits by the Chinese remainder theorem into small problems, one per prime power, so its cost is governed by the *largest prime factor* of n, not n itself. Hence the insistence on prime (or nearly prime) group order.
- **Generic lower bound.** Shoup proved in 1997 that any algorithm that treats the group as a black box, only adding and comparing points, needs $\Omega(\sqrt{n})$ operations. Baby-step giant-step and rho are optimal among such algorithms.

The state of play. For $\mathbb{Z}_p^\times$ there are faster, non-generic attacks: **index calculus** factors many powers g^k over a base of small primes and solves linear equations for their logarithms, giving sub-exponential running time. That is why finite-field Diffie–Hellman and RSA need 3072-bit numbers for 128-bit security. Points on an elliptic curve have no known notion of "small" that a factor base could exploit. So, as far as anyone knows, the best attack on a well-chosen curve is generic, and 256-bit curves give roughly 128-bit security. That is the whole reason elliptic-curve keys are twelve times shorter.

"Well-chosen" carries weight. Some curves *do* have shortcuts: on **supersingular** curves a pairing maps the ECDLP into a finite field where index calculus applies (the MOV attack, 1993), and on **anomalous** curves with exactly p points the ECDLP can be solved in polynomial time (Smart; Satoh–Araki; Semaev, 1998–99). Standard curves are checked against every known weakness, and whether some cleverer index calculus exists for general curves is an open question.

Pairings: a weakness that became a tool

A **pairing** is a map $e : \langle P \rangle \times \langle Q \rangle \to \mathbb{F}_{p^k}^\times$ from pairs of points of prime order n into the multiplicative group of an extension field, which is **bilinear**, $e(aP, bQ) = e(P, Q)^{ab}$, and **non-degenerate**, $e(P, Q) \neq 1$. The Weil and Tate pairings, built from rational functions on the curve, have these properties for every elliptic curve, and Miller's algorithm (1986) evaluates them with $O(\log n)$ operations, in a sense the double-and-add of pairings. The catch is the field they land in. The values are n-th roots of unity, which live in $\mathbb{F}_{p^k}$ only once n divides $p^k - 1$; the least such k is the **embedding degree**. For a random curve k is about as large as n itself, so the target field has some $2^{2^{256}}$ elements and the pairing cannot even be written down. Only special curves have small k: supersingular curves have $k \le 6$, and families such as BN and BLS12 are constructed on purpose to have $k = 12$.

Small k cuts both ways. It is exactly the MOV attack: $e(P, dP) = e(P, P)^d$ moves the discrete logarithm into $\mathbb{F}_{p^k}^\times$, where index calculus works, so pairing-friendly curves must be made larger to stay safe. But it also gives a capability no ordinary group has: anyone can test a *multiplicative* relation between hidden exponents. Given aG, bG and cG, the check $e(aG, bG) = e(G, cG)$ holds exactly when $ab \equiv c \pmod n$, yet reveals none of the three. **BLS signatures** use this directly: the signature on m is $\sigma = d\,H(m)$ for a hash-to-curve point $H(m)$, and it verifies when $e(\sigma, G) = e(H(m), dG)$. Signatures from many signers on many messages can be added into one point, which is why blockchains with thousands of validators use them.

Using the group: key exchange and signatures

ECDH is Diffie–Hellman with points: public G; Alice sends aG, Bob sends bG, and both compute abG. **ECDSA** signs a message hash h (an integer mod n) with private key d and public key $Q = dG$:

1. pick a secret, fresh, uniformly random **nonce** $k \in \{1, \dots, n-1\}$;
2. compute $r = x(kG) \bmod n$ (the x-coordinate of kG, reduced mod n);
3. compute $s = k^{-1}(h + rd) \bmod n$. The signature is (r, s).

To verify, compute $w = s^{-1}$, $u_1 = hw$, $u_2 = rw$ and check that $x(u_1 G + u_2 Q) \bmod n = r$. This works because $u_1 G + u_2 Q = (h + rd)wG = kG$.

Result 3: one repeated nonce reveals the key

Nonce reuse breaks ECDSA

If two signatures (r, s_1) on h_1 and (r, s_2) on $h_2 \neq h_1$ were made with the same nonce k, then anyone can compute

$$k = \frac{h_1 - h_2}{s_1 - s_2} \bmod n, \qquad d = \frac{s_1 k - h_1}{r} \bmod n.$$

Proof. The same k gives the same kG, hence the same r; that is how the repeat is spotted. From $s_i k \equiv h_i + rd \pmod n$ for $i = 1, 2$, subtract: $(s_1 - s_2)k \equiv h_1 - h_2$. Since $h_1 \neq h_2$ and k is invertible, $s_1 \neq s_2$, so divide to get k. Then solve $s_1 k = h_1 + rd$ for d ($r \neq 0$ is required by the standard). Everything happens in the field $\mathbb{Z}_n$, since n is prime. ■

That is the PlayStation 3 key and the Android wallets. The theorem needs no hard problem to be solved; it is two linear equations in two unknowns. Worse, it is not only exact repetition that kills: if the nonces are merely *biased*, say their top few bits are often zero, lattice methods recover the key from a few hundred signatures. This is why modern implementations derive k deterministically from the key and the message (RFC 6979) or use schemes like Ed25519 that do so by design.

? From one multiplication to a whole computation

A pairing checks a single product of hidden exponents, $ab = c$. Yet pairing-based SNARKs convince a verifier, with three curve points, that a computation of a billion steps was carried out correctly, while revealing nothing about its inputs. How does one hidden multiplication grow into a check of an entire program, and what does it mean for a verifier to "learn nothing"? See Ch. 82.

78.4 The algorithm

Here is the whole pipeline, from point addition to the PlayStation-style key recovery, on the 19-point curve of the figure. Every value is small enough to check by hand against the labelled multiples in the left panel.

```
# Toy ECDSA on y^2 = x^3 + 2x + 2 over F_17 (group order 19). NEVER use for real data.
p, a, b = 17, 2, 2
G, n = (5, 1), 19

def add(P, Q):
    if P is None: return Q
    if Q is None: return P
    (x1, y1), (x2, y2) = P, Q
    if x1 == x2 and (y1 + y2) % p == 0: return None            # P + (-P) = O
    if P == Q: lam = (3 * x1 * x1 + a) * pow(2 * y1, -1, p) % p
    else:      lam = (y2 - y1) * pow(x2 - x1, -1, p) % p
    x3 = (lam * lam - x1 - x2) % p
    return (x3, (lam * (x1 - x3) - y1) % p)

def mul(k, P):                                  # double-and-add, NOT constant time
    R = None
    for bit in bin(k)[2:]:
        R = add(R, R)
        if bit == "1": R = add(R, P)
    return R

def sign(d, h, k):                              # k MUST be fresh and secret for every signature
    r = mul(k, G)[0] % n
    s = pow(k, -1, n) * (h + r * d) % n
    return r, s

def verify(Q, h, r, s):
    w = pow(s, -1, n)
    X = add(mul(h * w % n, G), mul(r * w % n, Q))
    return X is not None and X[0] % n == r

d = 11; Q = mul(d, G)                           # private key, public key
r1, s1 = sign(d, 5, k=10)                       # two messages, SAME nonce
r2, s2 = sign(d, 12, k=10)
print(Q, (r1, s1), (r2, s2), verify(Q, 5, r1, s1), verify(Q, 12, r2, s2))

# Eve sees only public data: r1 == r2 betrays the repeated nonce
k = (5 - 12) * pow(s1 - s2, -1, n) % n
d_found = (s1 * k - 5) * pow(r1, -1, n) % n
print("recovered k =", k, " recovered d =", d_found)
```

The output is `(13, 10) (7, 12) (7, 7) True True`, then `recovered k = 10 recovered d = 11`. Both signatures verify, both share $r = 7$, and Result 3 hands Eve the private key.

Complexity. One addition costs one modular inversion and a few multiplications. Real implementations avoid the inversion by working in **projective coordinates** $(X : Y : Z)$, with $x = X/Z$, $y = Y/Z$, which turn each addition into about a dozen multiplications and postpone the single division to the end. Scalar multiplication is $O(\log n)$ additions, so $O(\lambda^3)$ bit operations for a λ-bit curve, on par with RSA at a far smaller λ.

Failure cases, mathematical. If $4a^3 + 27b^2 = 0$, the curve is singular, the group law degenerates, and its "discrete logarithm" collapses into one in the additive or multiplicative group of the field, which is easy or much easier. If n is not prime, Pohlig–Hellman splits the problem into pieces. If a received point is not checked to lie on the curve, an attacker can send a point on a *different* curve with a weak group (the formulas never use b!) and harvest the private key bit by bit. This **invalid-curve attack** has broken real TLS implementations.

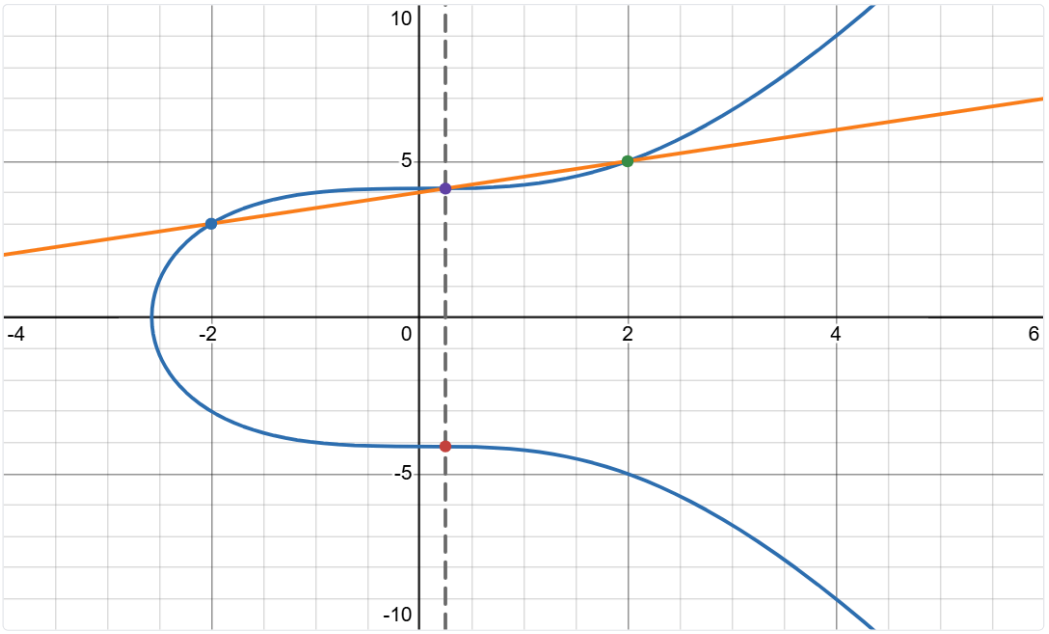
Failure cases, engineering. `mul` branches on the bits of the secret scalar, so its running time and power draw leak them. Production code uses constant-time ladders (the Montgomery ladder does one addition and one doubling for every bit, whatever its value), complete addition formulas without special cases, and libraries such as libsodium or a platform's TLS stack. Nonces must come from a cryptographically secure generator, or be derived deterministically as in RFC 6979 or Ed25519.

> **Never roll your own crypto**
>
> The 30 lines above are correct mathematics and a catastrophically insecure implementation: a 19-element group, a branching scalar multiplication, no point validation, a caller-supplied nonce and no hashing of the message. Production elliptic-curve code is written by specialists, verified, fuzzed, audited and attacked for years. Use a vetted library's high-level API (for example X25519 for key exchange and Ed25519 for signatures), standard curves, and current guidance; never invent curves, nonce schemes or "simplified" protocols.

> **? The generator with a hidden key**
>
> Where do the "random" numbers of cryptography come from? In 2006 a standards body published Dual_EC_DRBG, a random-number generator built from two elliptic-curve points P and Q: each output is roughly the x-coordinate of a multiple of Q, and the internal state advances via P. Researchers soon pointed out that whoever knew a secret e with $P = eQ$ could predict every future output from one sample. The points' origin was never explained, and documents leaked in 2013 suggested that the generator had been deliberately backdoored. How can a generator look perfectly random to everyone but one person, and what definition of "random" would have ruled this out? See Ch. 81.

78.5 The experiment

First, drag the geometry. The block draws $y^2 = x^3 + 17$ with the chord through $P = (-2, 3)$ and $Q = (2, 5)$ and its reflection.



The chord through P and Q on $y^2 = x^3 + 17$ meets the curve at $R = (\frac{1}{4}, \frac{33}{8})$; its mirror image is $P + Q$.

- Replace the orange line by the tangent at $(-1, 4)$, `y=0.375x+4.375`, and find where it meets the curve again (near $x = 2.14$).
- Change the curve to `y^2=x^3-3x+2`. Here $4a^3 + 27b^2 = -108 + 108 = 0$: the curve crosses itself at $(1, 0)$, where no tangent is defined. Then try `y^2=x^3-3x+3` and `y^2=x^3-3x+1`; the second splits into two pieces, an oval and an infinite branch, and the group law still works across both.

Then go finite. The script counts the points of a curve mod p, checks Hasse's bound, and attacks a discrete logarithm with baby-step giant-step.

```
import numpy as np
import matplotlib.pyplot as plt

P_MOD, A, B = 1009, 2, 3      # KNOBS: a prime p > 3 and the curve y^2 = x^3 + A x + B
SECRET = 777                  # KNOB: Alice's private scalar d

sq = {}                                       # y^2 mod p -> list of y
for y in range(P_MOD): sq.setdefault(y * y % P_MOD, []).append(y)
pts = [(x, y) for x in range(P_MOD) for y in sq.get((x**3 + A*x + B) % P_MOD, [])]
N = len(pts) + 1                              # + point at infinity (None)
print(f"N = {N}, p + 1 = {P_MOD + 1}, Hasse bound 2*sqrt(p) = {2*np.sqrt(P_MOD):.1f}")

def add(P, Q):
    if P is None: return Q
    if Q is None: return P
    (x1, y1), (x2, y2) = P, Q
    if x1 == x2 and (y1 + y2) % P_MOD == 0: return None
    lam = ((3*x1*x1 + A) * pow(2*y1, -1, P_MOD) if P == Q else (y2 - y1) * pow(x2 - x1, -1, P_MOD)) % P_MOD
    x3 = (lam * lam - x1 - x2) % P_MOD
    return (x3, (lam * (x1 - x3) - y1) % P_MOD)

def mul(k, P):                                # double-and-add
    R = None
    for bit in bin(k)[2:]:
        R = add(R, R)
        if bit == "1": R = add(R, P)
    return R

G = pts[1]
n = next(k for k in range(1, N + 1) if N % k == 0 and mul(k, G) is None)   # order of G
Q = mul(SECRET % n, G)                        # Alice's public key
m = int(np.ceil(np.sqrt(n)))                  # baby-step giant-step
baby = {}; R = None
for j in range(m): baby.setdefault(R, j); R = add(R, G)
step, R = mul(n - m, G), Q                    # step = -mG
for i in range(m):
    if R in baby: print(f"order of G = {n}; recovered d = {i*m + baby[R]} after {m + i} group ops"); break
    R = add(R, step)
plt.scatter(*np.array(pts).T, s=2); plt.title(f"{N} points mod {P_MOD}"); plt.show()
```

With the defaults the curve has $N = 1068$ points, within 58 of $p + 1 = 1010$ and inside the Hasse bound of 63.5. The chosen G has order only 267, so the script "recovers" $d = 243$: the secret 777 only matters modulo the order, and $777 - 2 \cdot 267 = 243$. Things to try:

- **Factor N.** $1068 = 2^2 \cdot 3 \cdot 89$. A real attacker would not run baby-step giant-step on the full group but on the pieces of size 4, 3 and 89, as Pohlig–Hellman suggests. Can you find A, B for $p = 1009$ that make N prime?
- **Scale p.** Try $p = 10007$ and $p = 100003$, keeping a prime-order or near-prime-order curve. The number of group operations grows like $\sqrt{N}$: it is about 320 for $p = 100003$. Extrapolate to $N \approx 2^{256}$.
- **Break the curve.** Set $A = 0$, $B = 0$, which makes $4A^3 + 27B^2 = 0$. The count comes out as exactly $p + 1$, and the script then crashes while looking for the order of G. Which point is to blame, and what group do the remaining points form?

78.6 The challenge

No answers are given.

1. **Intuition.** On the real curve, a line through two points of an elliptic curve meets it in a third. Why must a line through two points of a *conic* (a circle, say) fail to give a useful "third point", and what replaces the group law there? Then explain, using only the pictures, why the points of order 2 on $y^2 = x^3 + ax + b$ are exactly those with $y = 0$, and why there are at most three of them.
2. **Derivation.** Show that a point $P = (x, y)$ has order 3 exactly when it is an inflection point of the curve. Derive the polynomial whose roots are the x-coordinates of the points of order 3, and check that it has degree 4. Then prove that the map $x \mapsto x^3 + ax + b$ cannot be a square for *every* $x \in \mathbb{F}_p$, so that $\#E(\mathbb{F}_p) < 2p + 1$.
3. **Implementation.** Implement Pollard's rho for the ECDLP (partition the group into three sets by x mod 3 and walk by $R \mapsto R + G$, $2R$, or $R + Q$; detect a cycle with Floyd's tortoise and hare). Measure the number of steps for prime group orders n between 10^3 and 10^7, and compare the average with $\sqrt{\pi n / 2}$.
4. **Open investigation.** For $p = 1009$, compute $N - (p + 1)$ for every pair (a, b) with $4a^3 + 27b^2 \neq 0$, and normalise by $2\sqrt{p}$. Plot the histogram. Does it look like a semicircle? Repeat for other primes and for a fixed curve over many primes. (The Sato–Tate conjecture, proved for most curves only in 2011, predicts a distribution here; find out which one by experiment.)

78.7 Loose threads

The hook asked how one repeated number broke an unbroken system. The **group** is real and strong: the chord-and-tangent law makes the points of a cubic into an abelian group, the same formulas work over $\mathbb{F}_p$, and the best known attack on the discrete logarithm in a well-chosen curve is generic, needing $\sqrt{n}$ steps, which is why 256-bit curves rival 3072-bit RSA. The **signature** is fragile: ECDSA hides the private key behind a fresh secret nonce, and repeating that nonce turns the hard problem into two linear equations. Sony's mathematics was perfect and its implementation reused k.

What we did not tell you:

- **Why the group law exists at all.** The first mystery points to Ch. 94, where curves of genus one, complex tori and divisors live.
- **How to count points on a 256-bit curve.** Brute force is hopeless at that size, yet curve designers need the exact group order. Schoof's algorithm (1985) computes it in polynomial time by computing N modulo many small primes, using the points of small order, then gluing the answers with the Chinese remainder theorem.
- **Whether randomness can be trusted.** The second mystery points to Ch. 81.
- **What pairings make possible.** The third mystery points to Ch. 82.
- **What quantum computers do here.** Shor's period-finding algorithm (Ch. 77) solves the ECDLP too, and with fewer qubits than factoring a comparable RSA modulus. Elliptic curves are not a refuge from quantum computers; lattices may be.
- **Fermat's last theorem.** The proof by Wiles in 1994 runs through elliptic curves over $\mathbb{Q}$ and the modularity of their point counts: the same numbers $N - (p + 1)$ that you histogrammed in the challenge, arranged into a single object.

Further reading

- J. H. Silverman and J. Tate, *Rational Points on Elliptic Curves*, 2nd ed., Springer, 2015. The geometry and arithmetic of the group law, for undergraduates.
- L. C. Washington, *Elliptic Curves: Number Theory and Cryptography*, 2nd ed., CRC Press, 2008. Point counting, discrete-log attacks and pairings, with full proofs.
- D. Hankerson, A. Menezes and S. Vanstone, *Guide to Elliptic Curve Cryptography*, Springer, 2004. The implementation side: coordinates, fast arithmetic, side channels.
- J. Hoffstein, J. Pipher and J. H. Silverman, *An Introduction to Mathematical Cryptography*, 2nd ed., Springer, 2014. Elliptic curves alongside lattices and the rest of public-key cryptography.

Chapter 79 — Error-Correcting Codes: Recovering the Lost

> **Level 2 · Track H — Cryptography, Coding, and Digital Security Builds on:** Ch. 8, Ch. 9, Ch. 57, Ch. 56 · **Leads to:** Ch. 80, Ch. 82 · **Related:** Ch. 19, Ch. 33, Ch. 52, Ch. 60 **Reading time:** ~25 min

79.1 The hook

Look at a QR code on a restaurant menu or a concert poster. Many have a logo pasted right over the middle: a coffee cup, a band's face, a company emblem. Those pixels are simply gone, replaced by something that is not data at all. Point your phone at it anyway. It scans on the first try.

Scratch a music CD with a key and it will usually still play. The disc can lose about 4,000 consecutive bits, some 2.5 mm of track, and reproduce the sound *exactly*, not approximately. In 1989 Voyager 2 sent pictures of Neptune from 4.4 billion kilometres away, with a transmitter of about 20 watts, roughly a refrigerator light bulb. By the time the signal reached Earth it was far weaker than the noise in the receiving antennas, and yet the images arrived essentially without a wrong pixel.

None of these systems has a way to ask for the data again. The QR code cannot re-print itself; Neptune is not taking requests. The missing information is *recovered*, computed from what remains. That sounds impossible: how can you reconstruct what was destroyed?

The answer is that nothing useful was destroyed. Before sending, the data was dressed in carefully designed redundancy, so that any small enough damage leaves a pattern that points back, unambiguously, to the original. This chapter is about how to design that redundancy: what makes it efficient, how the receiver finds and fixes errors it cannot see, and where the limits lie.

79.2 The big idea

The naive way to protect a bit is to repeat it: send `000` for 0 and `111` for 1, and decode by majority vote. One flipped bit is corrected. But you have tripled the cost, and if you want to survive two flips you must send five copies. There must be a better way.

The key idea is **distance**. Think of every possible received word as a point in a space, and the words you actually send, the **codewords**, as a few special points. Noise moves a codeword a short distance. If the codewords are spread so far apart that the little "clouds" of damaged versions around them never overlap, then any received word lies in exactly one cloud, and the receiver simply snaps it back to the centre. Error correction is **nearest-neighbour decoding**, and code design is the art of packing as many non-overlapping clouds as possible into the space.

An analogy: the NATO spelling alphabet. Over a crackling radio, "B" and "D" and "E" sound alike, but "Bravo", "Delta" and "Echo" do not. The longer words carry redundancy, but it is *arranged* so that the words are far apart in sound. A burst of static that blurs one syllable still leaves a word closer to the right one than to any other.

Three questions organise the chapter.

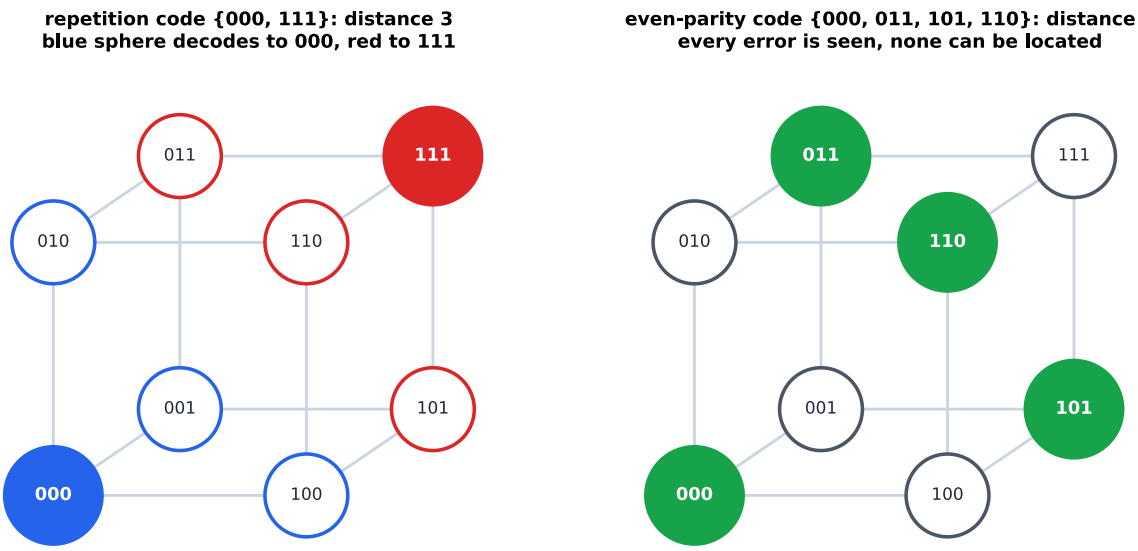
1. **Geometry.** How far apart must codewords be to correct t errors, and how many codewords can fit?
2. **Algebra.** Searching for the nearest of 2^k codewords is hopeless when $k = 200$. Codes built from linear algebra, and then from polynomials, let the receiver compute the error instead of searching for it.
3. **Erasures versus errors.** A logo on a QR code is damage *at a known place*. A scratch the reader cannot detect is damage at an unknown place. Knowing where damage is turns out to be worth exactly a factor of two.

79.3 The mathematics

Hamming distance

> **Codes and distance**
>
> A **block code** of length n over an alphabet A is a set $C \subseteq A^n$ of codewords. The **Hamming distance** $d(x, y)$ between two words is the number of positions where they differ. The **minimum distance** of C is $d = \min\{d(x, y) : x, y \in C,\ x \neq y\}$. A code with $|C| = q^k$ codewords over a q-letter alphabet has **rate** k/n.

Hamming distance is a genuine metric (Ch. 60 lists it among its examples): it is symmetric, zero only for equal words, and satisfies the **triangle inequality** $d(x, z) \leq d(x, y) + d(y, z)$, because any position where x and z differ must be one where x differs from y or y differs from z. The **Hamming ball** of radius t around c is the set of all words within distance t of c: the "cloud" of the big idea.



Codes as points on the cube $\{0,1\}^3$, where neighbours differ in one bit. Left: distance 3; the radius-1 balls around the two codewords tile the cube, so every single error is corrected. Right: distance 2; a single error always lands on a non-codeword (detected), but that word is equally close to three codewords, so it cannot be corrected.

Result 1: distance decides everything

Detection and correction

A code with minimum distance d can **detect** any pattern of up to $d-1$ errors, and **correct** any pattern of up to $t=\lfloor (d-1)/2 \rfloor$ errors by nearest-codeword decoding.

Proof. Detection. If c is sent and between 1 and $d-1$ positions change, the received word r satisfies $0 < d(c,r) \le d-1$. Since distinct codewords are at least d apart, r is not a codeword, so the receiver sees that something is wrong.

Correction. Suppose c is sent and r is received with $d(c,r) \le t$. Let $c' \neq c$ be any other codeword. By the triangle inequality,

$$d \le d(c,c') \le d(c,r) + d(r,c') \quad\Longrightarrow\quad d(r,c') \ge d - t \ge 2t+1-t = t+1 > t \ge d(c,r).$$

So c is strictly the closest codeword to r, and nearest-codeword decoding returns it. ■

Both bounds are sharp. With d errors, c can be turned into another codeword, and nobody can tell. With $t+1$ errors, r can land at distance $t+1$ from c and $d-t-1 \le t+1$ from some c', and the decoder may be fooled. The parity code on the right of the figure has $d=2$, so it detects one error and corrects none; the repetition code on the left has $d=3$ and corrects one.

Result 2: the sphere-packing bound

Large distance costs codewords. How many can a binary code of length n correcting t errors have?

Hamming bound

A binary code of length n that corrects t errors has

$$|C| \sum_{i=0}^{t} \binom{n}{i} \;\le\; 2^n.$$

Proof. A ball of radius t around a word contains the word itself, the $\binom{n}{1}$ words differing in one position, the $\binom{n}{2}$ differing in two, and so on: $V = \sum_{i \le t} \binom{n}{i}$ words. The proof of Result 1 shows that if the code corrects t errors then balls of radius t around distinct codewords are disjoint (a word in two balls would be decoded two ways). So $|C|$ disjoint balls of V words each fit inside the 2^n words of the whole space. ■

A code meeting the bound with equality is **perfect**: its balls tile the space with nothing left over. The repetition code of length 3 is perfect: $2 \cdot (1+3) = 8 = 2^3$. So is the code of the next section, with $16 \cdot (1+7) = 128 = 2^7$, and so is the remarkable binary **Golay code** of length 23 with 2^{12} codewords correcting 3 errors: $2^{12}\left(1+23+253+1771\right) = 2^{12} \cdot 2^{11} = 2^{23}$. Apart from repetition codes and these families, there are no other perfect binary codes; that was a hard theorem of the 1970s.

Linear codes: let the algebra search for you

Use the alphabet $\mathbb{F}_2 = \{0,1\}$ with addition mod 2 (XOR), so that words of length n form the vector space $\mathbb{F}_2^n$ of Ch. 9.

Linear code

A **linear** $[n,k,d]$ **code** is a k-dimensional subspace $C \subseteq \mathbb{F}_2^n$ with minimum distance d. It has a $k \times n$ **generator matrix** $\mathbf{G}$ whose rows form a basis, so the codewords are $\mathbf{mG}$ for messages $\mathbf{m} \in \mathbb{F}_2^k$, and an $(n-k) \times n$ **parity-check matrix** $\mathbf{H}$ whose null space is C: $\mathbf{c} \in C \iff \mathbf{Hc}^{\mathsf{T}} = \mathbf{0}$.

Linearity buys two things at once. First, the **minimum distance equals the minimum weight** (number of 1s) of a nonzero codeword: $d(\mathbf{x},\mathbf{y}) = \mathrm{wt}(\mathbf{x}-\mathbf{y})$, and $\mathbf{x}-\mathbf{y}$ is again a codeword. Second, the **syndrome** of a received word $\mathbf{r} = \mathbf{c} + \mathbf{e}$ depends only on the error:

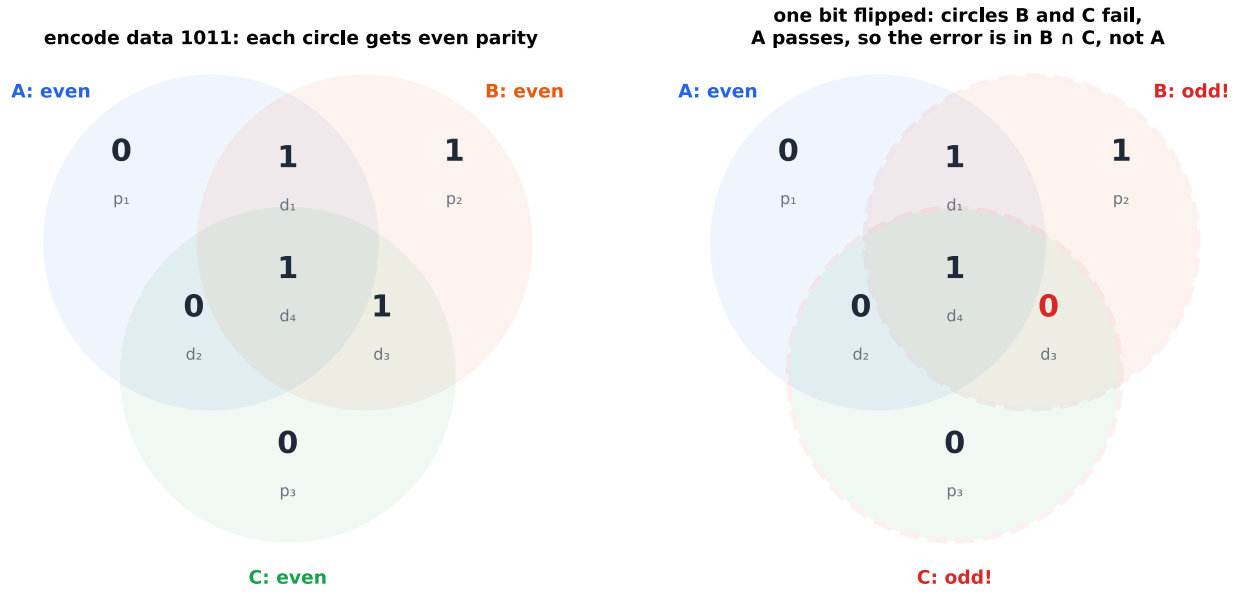
$$\mathbf{s} = \mathbf{H}\mathbf{r}^{\mathsf{T}} = \mathbf{H}\mathbf{c}^{\mathsf{T}} + \mathbf{H}\mathbf{e}^{\mathsf{T}} = \mathbf{H}\mathbf{e}^{\mathsf{T}}.$$

The receiver never needs to know what was sent. If $\mathbf{e}$ has a single 1, in position j, then $\mathbf{s}$ is column j of $\mathbf{H}$. So: if all columns of $\mathbf{H}$ are distinct and nonzero, any single error can be located by reading the syndrome, and $d \geq 3$. (The null-space picture of Ch. 9 §9.6 is exactly what is happening.)

Richard Hamming turned this into the most economical possible design in 1950. Take as the columns of $\mathbf{H}$ *all* the nonzero binary vectors of length 3, in counting order:

$$\mathbf{H} = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{pmatrix}.$$

Column j is the number j in binary (least significant bit on top). This is the **Hamming** $[7,4,3]$ **code**: 7 bits carry 4 data bits, at rate $4/7$ instead of the repetition code's $1/3$, and the syndrome of a single error *is the binary address of the wrong bit*. Positions 1, 2 and 4 (one 1 in their column) hold parity bits p_1, p_2, p_3; positions 3, 5, 6, 7 hold data $d_1, \ldots, d_4$. Each row of $\mathbf{H}$ is one parity rule, and the three rules are three overlapping circles.



The Hamming code as three parity circles. Each circle must contain an even number of 1s. A single flipped bit upsets exactly the circles that contain it, and each of the seven regions lies inside a different combination of circles, so the pattern of failures names the culprit. Here B and C fail and A passes: the error is d_3, at position $6 = 110_2$.

Cyclic codes and CRCs: codes as polynomials

Write a word $(c_0, \ldots, c_{n-1})$ as a polynomial $c(x) = c_0 + c_1 x + \cdots + c_{n-1}x^{n-1}$ over $\mathbb{F}_2$. A **cyclic shift** of the word is multiplication by x, followed by reducing x^n to 1, that is, working in the ring $\mathbb{F}_2[x]/(x^n - 1)$ of Ch. 56. A linear code is **cyclic** if every cyclic shift of a codeword is a codeword. Algebraically, cyclic codes are exactly the ideals of that ring, and every one has the form

$$C = \{\, m(x)\, g(x) \;:\; \deg m < n - \deg g \,\},$$

for a **generator polynomial** $g(x)$ that divides $x^n - 1$. Over $\mathbb{F}_2$, $x^7 + 1 = (x+1)(x^3 + x + 1)(x^3 + x^2 + 1)$, and $g(x) = x^3 + x + 1$ generates a $[7,4,3]$ cyclic code equivalent to Hamming's. Encoding is now polynomial multiplication, which hardware does with a shift register and a few XOR gates.

The same division trick gives the **cyclic redundancy check** (CRC) on every Ethernet frame, USB packet and ZIP file. To protect a message $m(x)$, append the remainder $r(x) = x^r m(x) \bmod g(x)$, where $r = \deg g$; the transmitted word $x^r m(x) + r(x)$ is then divisible by $g(x)$. With $g = x^3 + x + 1$ (bits `1011`) and message `1101`, the remainder of `1101000` is `001`, and `1101001` is sent. The receiver divides by g and accepts only remainder zero. If g has a nonzero constant term, every **burst** of errors confined to r consecutive positions is detected: such an error is $x^i b(x)$ with $\deg b < r$ and $b \neq 0$, and g can divide neither x^i (it is not a power of x) nor b (whose degree is too small). CRC-32 detects every burst up to 32 bits long. It *detects*; it does not correct. For that we need bigger fields.

Reed–Solomon codes: data as a polynomial

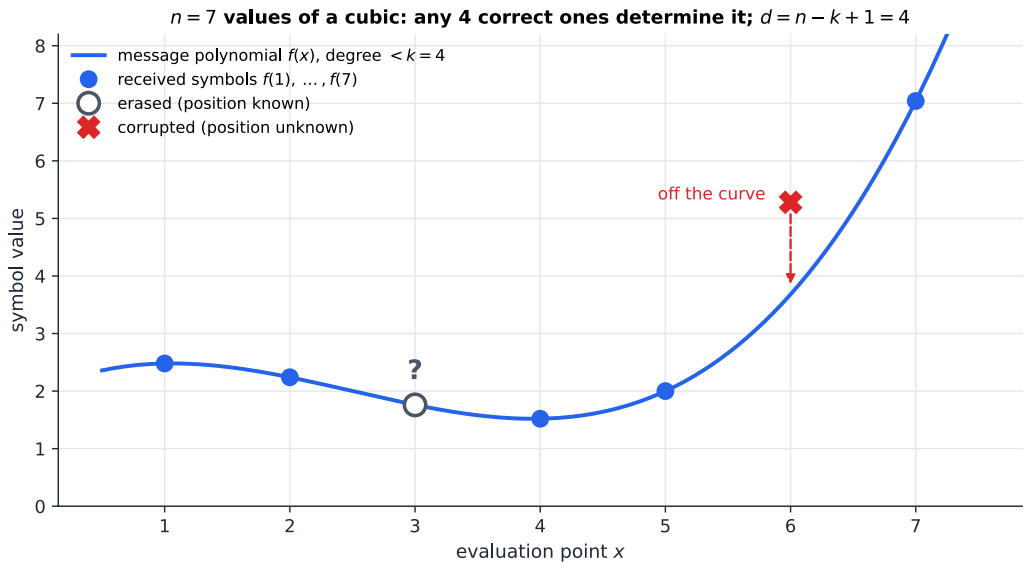
Bits flip one at a time, but scratches, dust and logos destroy whole **symbols**: bytes, or blocks of pixels. Reed and Solomon (1960) chose a large alphabet, a finite field $\mathbb{F}_q$ such as $\mathbb{F}_{2^8}$ whose elements are bytes (Ch. 57), and a beautifully simple idea.

Reed–Solomon code

Fix distinct points $\alpha_1, \ldots, \alpha_n \in \mathbb{F}_q$ (so $n \leq q$) and $k \leq n$. A message $(m_0, \ldots, m_{k-1}) \in \mathbb{F}_q^k$ becomes the polynomial $f(x) = m_0 + m_1 x + \cdots + m_{k-1}x^{k-1}$, and its codeword is the list of values

$$\big(f(\alpha_1),\, f(\alpha_2),\, \ldots,\, f(\alpha_n)\big) \in \mathbb{F}_q^n.$$

Send n samples of a curve that is determined by k of them. The extra $n - k$ samples are the redundancy.



Reed–Solomon, drawn over the real numbers for intuition (real codes use a finite field). The message is a cubic ($k = 4$) and seven values are sent. Any four correct values pin the cubic down, so up to three erasures can be filled in; an undetected wrong value, like the red cross, costs twice as much to fix.

Result 3: Reed–Solomon codes are as good as possible

How large can the minimum distance of *any* code be? There is a simple ceiling.

Singleton bound, and Reed–Solomon meets it

(a) Any code of length n with q^k codewords over a q-letter alphabet has $d \le n - k + 1$. (b) The Reed–Solomon code has $d = n - k + 1$ exactly.

Proof. (a) Delete the last $d - 1$ positions of every codeword. Two distinct codewords differ in at least d places, so they still differ after deleting $d - 1$ of them: the shortened words, of length $n - d + 1$, are all distinct. There are q^k of them and only q^{n-d+1} possible words of that length, so $k \le n - d + 1$.

(b) The code is linear (adding messages adds polynomials and their values), so d is the minimum weight of a nonzero codeword. A nonzero polynomial f of degree at most $k - 1$ has at most $k - 1$ roots in a field, so at most $k - 1$ of the values $f(\alpha_i)$ are zero, and at least $n - k + 1$ are nonzero. Hence every nonzero codeword has weight $\ge n - k + 1$, and with (a), $d = n - k + 1$. ■

Codes meeting the Singleton bound are called **maximum distance separable** (MDS). Combined with Result 1 and a small extension of it to erasures, Result 3 says what a Reed–Solomon decoder can do: with s erasures (positions known) and e errors (positions unknown), it recovers the message whenever

$$2e + s \le n - k.$$

An erasure costs one redundant symbol, an error costs two: knowing *where* damage is halves its price. With only erasures, any k surviving values determine f by Lagrange interpolation, which is the uniqueness of the interpolating polynomial in Ch. 22 §22.5 carried over to a finite field.

That answers the hook. A QR code splits its data into blocks, each a Reed–Solomon codeword over $\mathbb{F}_{2^8}$. At the highest error-correction level, H, the smallest code has 26 bytes of which 17 are redundant, enough to fix 8 wrong bytes, about 30 % of the block. A designer pasting a logo over the centre is spending part of that budget on purpose. The CD uses two Reed–Solomon codes over bytes with the data shuffled between them, so that a long scratch becomes many short, correctable damages spread across many codewords. Deep-space links use a $(255, 223)$ Reed–Solomon code, correcting 16 byte errors in every 255, wrapped around a convolutional code.

? How good can a code be?

The repetition code drives errors to zero only by driving the rate to zero. Hamming's code has rate $4/7$ but corrects only one error in seven. Is there a trade-off you cannot beat? Shannon proved in 1948 that for a channel flipping bits with probability p, every rate below $1 - H_b(p)$ (the binary entropy of Ch. 33) is achievable with error probability *as small as you like*, and no rate above it is. His proof used random codes and was not constructive at all. What that capacity is, why random codes reach it, and how LDPC and polar codes finally got there in practice, is Ch. 80.

79.4 The algorithm

Two decoders, one per idea. The Hamming decoder computes a syndrome and flips one bit. The Reed–Solomon decoder works over the prime field $\mathbb{F}_{929}$ (the field used by the PDF417 barcode on many boarding passes and driving licences) and recovers the message by interpolation.

```
import numpy as np
from itertools import combinations

# --- Hamming(7,4): column j of H is j in binary, positions p1 p2 d1 p3 d2 d3 d4 ---
H = np.array([[(j >> i) & 1 for j in range(1, 8)] for i in range(3)])
DATA_POS = [2, 4, 5, 6]                       # 0-based positions of d1..d4

def ham_encode(d):
    c = np.zeros(7, dtype=int); c[DATA_POS] = d
    for i, pos in enumerate([0, 1, 3]):       # parity bit i fixes row i of H
        c[pos] = H[i] @ c % 2
    return c

def ham_decode(r):
    s = H @ r % 2
    j = s[0] + 2 * s[1] + 4 * s[2]            # syndrome = position of the error
    r = r.copy()
    if j: r[j - 1] ^= 1
    return r[DATA_POS]

c = ham_encode([1, 0, 1, 1]); r = c.copy(); r[5] ^= 1
print(c, r, ham_decode(r))                    # [0 1 1 0 0 1 1] [0 1 1 0 0 0 1] [1 0 1 1]

# --- Reed-Solomon over the prime field GF(929): evaluate, then interpolate ---
P = 929
def rs_encode(msg, n):                        # msg = k coefficients, codeword = f(1..n)
    return [sum(m * pow(x, i, P) for i, m in enumerate(msg)) % P for x in range(1, n + 1)]

def interpolate(pts, x0):                     # Lagrange: value at x0 of the poly through pts
    total = 0
    for xi, yi in pts:
        num = den = 1
        for xj, _ in pts:
            if xj != xi: num, den = num * (x0 - xj) % P, den * (xi - xj) % P
        total += yi * num * pow(den, -1, P)
    return total % P

def rs_decode_bruteforce(word, k):            # try k-subsets; exponential, for illustration only
    n = len(word)
    known = [(x, y) for x, y in zip(range(1, n + 1), word) if y is not None]
    t = (len(known) - k) // 2                 # errors we can afford: 2e + erasures <= n - k
    for sub in combinations(known, k):
        cand = [interpolate(sub, x) for x in range(1, n + 1)]
        if sum(c != y for c, y in zip(cand, word) if y is not None) <= t:
            return cand
    return None

cw = rs_encode([3, 1, 4, 1], n=7)             # k = 4, n = 7, d = 4
print(cw)                                     # [9, 29, 69, 135, 233, 369, 549]
bad = cw.copy(); bad[2] = None; bad[5] = (bad[5] + 100) % P   # one erasure, one error
print(rs_decode_bruteforce(bad, 4) == cw)     # True: 2*1 + 1 <= 7 - 4
```

The Hamming part reproduces the Venn figure: data `1011` becomes `0110011`, the flipped sixth bit gives syndrome 6, and the data comes back. The Reed–Solomon part erases one symbol and corrupts another, a damage pattern with $2e + s = 3 = n - k$, right at the limit, and still recovers the codeword exactly.

Complexity. Hamming decoding is a matrix–vector product: $O(n \log n)$ bit operations for a length-n Hamming code. The Reed–Solomon decoder above is deliberately naive. It tries up to $\binom{n}{k}$ subsets, which is fine for $n = 7$ and absurd for the $(255, 223)$ deep-space code, where $\binom{255}{223} \approx 5 \times 10^{40}$. Real decoders find the errors algebraically. The **Berlekamp–Welch** idea fits in three lines. Let $E(x)$ be the unknown **error-locator** polynomial, monic of degree e, vanishing exactly at the error positions. Then at *every* evaluation point,

$$y_i \, E(\alpha_i) = f(\alpha_i) \, E(\alpha_i) = Q(\alpha_i), \qquad Q := fE, \ \deg Q < k + e,$$

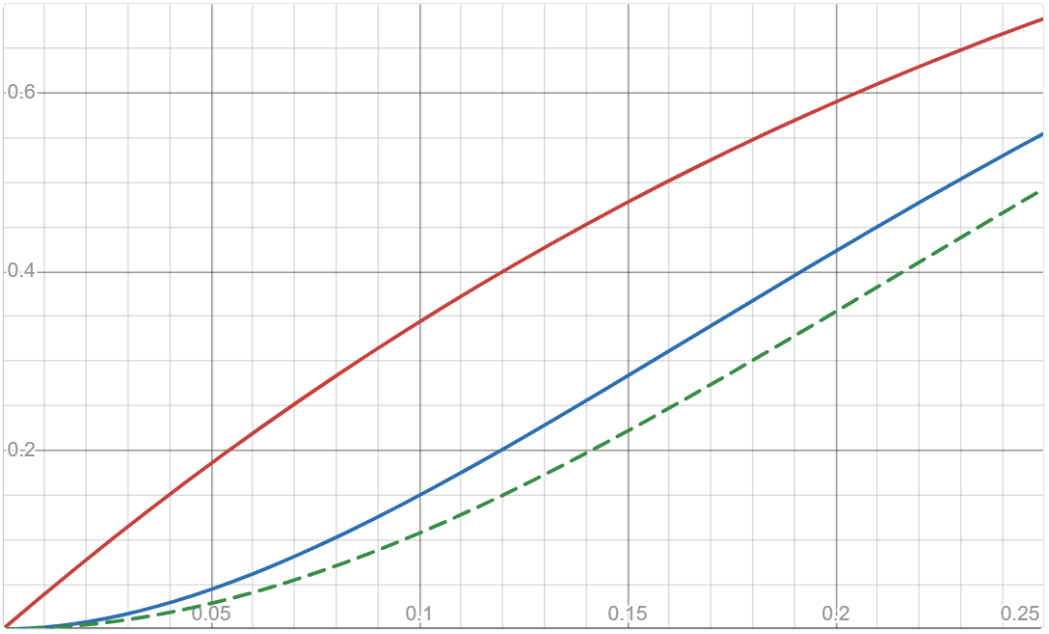
because where y_i is wrong, $E(\alpha_i) = 0$ kills both sides. These are n *linear* equations in the $k + 2e \le n$ unknown coefficients of E and Q. Solve them by Gaussian elimination (Ch. 8 §8.7) in $O(n^3)$, then divide: $f = Q/E$. The nonlinear problem "where are the errors?" has become linear algebra. Production decoders (Berlekamp–Massey with Chien search and Forney's formula) do the same job in $O(n^2)$ or better.

Failure cases. Every decoder has a breaking point, and its behaviour past that point matters.

- *Too many errors.* With $t + 1$ errors, a nearest-codeword decoder may confidently return the *wrong* codeword. For the Hamming code, two errors always produce a nonzero syndrome pointing at a third, innocent bit; the "correction" makes three errors. A system that must know when it failed adds an overall parity bit (the extended $[8, 4, 4]$ code corrects one error *and* detects two).
- *Burst errors.* A scratch destroys many neighbouring bits. Bit-level codes are helpless against it; symbol-level codes plus **interleaving** (spreading each codeword across the medium) turn one long burst into many small, correctable damages.
- *The wrong noise model.* Codes designed for independent flips can do badly on channels where noise comes in bursts or fades. The design starts from the channel, not from the code.
- *Detection is not security.* A CRC catches accidents, but anyone can recompute it after tampering. Protection against an adversary needs the keyed authentication tags of Ch. 77, not a checksum.

79.5 The experiment

First, the formulas. A 4-bit message sent unprotected over a channel that flips each bit with probability x survives only if all four bits survive. Sent as a Hamming codeword, it survives if at most one of the seven bits flips (using the binomial distribution of Ch. 19 §19.6).



Message error probability against bit-flip probability x: uncoded (red), Hamming(7,4) (blue), triple repetition of each bit (green, dashed). At $x = 0.01$ the Hamming code cuts the failure rate from 3.9 % to 0.2 %.

Change the exponents to explore. Replace the blue curve by $1-(1-x)^{15}-15x(1-x)^{14}$, the $[15, 11, 3]$ Hamming code, which carries 11 bits instead of 4; compare it with $1-(1-x)^{11}$. The longer code is cheaper (rate $11/15$ instead of $4/7$), but its curve rises much faster: one correction per 15 bits is spread thinner than one per 7. Where on the x-axis does that difference start to matter?

Then simulate. The script sends many random 4-bit messages through a binary symmetric channel, decodes them three ways, and plots the measured error rates.

```
import numpy as np
import matplotlib.pyplot as plt

FLIP_PROBS = np.linspace(0.001, 0.2, 25)   # KNOB: channel bit-flip probabilities to test
BLOCKS = 20000                             # KNOB: 4-bit messages sent per probability
SEED = 79

rng = np.random.default_rng(SEED)
H = np.array([[(j >> i) & 1 for j in range(1, 8)] for i in range(3)])   # Hamming(7,4)
DATA = [2, 4, 5, 6]
G = np.zeros((4, 7), dtype=int)            # generator: encode each unit data vector
for row, pos in enumerate(DATA):
    c = np.zeros(7, dtype=int); c[pos] = 1
    c[[0, 1, 3]] = H @ c % 2               # parity bits are the syndrome of the data part
    G[row] = c

def run(p):
    d = rng.integers(0, 2, (BLOCKS, 4))
    raw = (rng.random(d.shape) < p).any(axis=1).mean()                    # uncoded: any flip
    rep = d.repeat(3, axis=1) ^ (rng.random((BLOCKS, 12)) < p)            # repetition x3
    rep_dec = rep.reshape(BLOCKS, 4, 3).sum(axis=2) >= 2
    rep_err = (rep_dec != d).any(axis=1).mean()
    r = (d @ G % 2) ^ (rng.random((BLOCKS, 7)) < p)                       # Hamming(7,4)
    j = (H @ r.T % 2).T @ np.array([1, 2, 4])                            # syndrome -> position
    r[np.arange(BLOCKS)[j > 0], j[j > 0] - 1] ^= 1
    ham_err = (r[:, DATA] != d).any(axis=1).mean()
    return raw, rep_err, ham_err

res = np.array([run(p) for p in FLIP_PROBS])
for lab, col in zip(["uncoded (rate 1)", "repetition x3 (rate 1/3)", "Hamming(7,4) (rate 4/7)"], res.T):
    plt.semilogy(FLIP_PROBS, np.maximum(col, 1e-5), "o-", label=lab)
plt.xlabel("bit-flip probability p"); plt.ylabel("message error rate")
plt.legend(); plt.grid(True, which="both"); plt.show()
print("at p = %.3f:" % FLIP_PROBS[5], res[5])
```

At $p \approx 0.042$ the run prints roughly `0.155 0.022 0.031`: uncoded messages fail 15.5 % of the time, Hamming-coded ones 3.1 %, and triple repetition 2.2 %. Repetition wins on reliability, but it sends 12 bits per message where Hamming sends 7. Things to try:

- **Equalise the cost.** Compare codes at the same *rate*, not the same message. Add a $[15, 11]$ Hamming code (column j of $\mathbf{H}$ is j in 4-bit binary) and send 11-bit messages. Which wins per transmitted bit?

- **Add a burst.** Instead of independent flips, flip three *consecutive* bits of each block with probability `p`. Watch the Hamming code fall apart, then interleave: send bit 1 of ten codewords, then bit 2 of each, and so on.
- **Find the wall.** Push `FLIP_PROBS` up to 0.5. All three curves converge on the same failure rate as $p \to \frac{1}{2}$. Why can no code, however clever, help at $p = \frac{1}{2}$, and what happens if you push past it to $p = 0.9$?

79.6 The challenge

No answers are given.

1. **Intuition.** A QR code at level H can survive about 30 % damage, yet a logo covering only 20 % of the symbol can still make it unreadable if placed badly. Using erasures, errors and the structure of the code (data split into several Reed–Solomon blocks, plus fixed finder patterns in three corners), explain why *where* the damage falls matters as much as how much there is.
2. **Derivation.** Prove the **Gilbert–Varshamov bound**: there exists a binary code of length n and minimum distance d with at least $2^n / \sum_{i=0}^{d-1} \binom{n}{i}$ codewords. (Hint: build the code greedily; when you can add no more words, the balls of radius $d-1$ around the codewords must cover the whole space.) Compare with the Hamming bound for $n = 15$, $d = 3$.
3. **Implementation.** Implement Berlekamp–Welch over $\mathbb{F}_{929}$: set up the $n \times (k + 2e)$ linear system, solve it by Gaussian elimination mod 929, and divide polynomials to get f. Test it on the $(7, 4)$ example above, then on $n = 100$, $k = 50$ with 25 random errors, and compare its running time with the brute-force decoder.
4. **Open investigation.** For n up to about 12, search exhaustively (or with a clever backtracking search) for the largest binary code with minimum distance 3, and compare with the Hamming bound and the Gilbert–Varshamov bound. How fast does the exhaustive search blow up? Can a random-restart greedy search match the known optimum for $n = 10$ and $n = 11$ (72 and 144 codewords)?

79.7 Loose threads

The hook asked how a QR code with a logo, a scratched CD or a faint signal from Neptune can recover data that is literally gone. The answer is geometry and algebra. Codewords are placed far apart in Hamming distance, so that damaged words still point home (Result 1), at a cost bounded by sphere packing (Result 2). Linear codes compute the error from a syndrome instead of searching for it. Reed–Solomon codes turn a block of data into a polynomial and send extra samples of it; because a low-degree polynomial has few roots, any k intact samples determine it, and the code meets the Singleton bound exactly (Result 3). Erasures, like a logo at a known place, cost half as much as errors.

What we did not tell you:

- **The limit.** The first mystery points to Ch. 80: Shannon's capacity, and the codes that approach it.
- **Convolutional codes and Viterbi decoding.** Instead of blocks, a convolutional encoder passes an endless bit stream through a shift register. Its decoder finds the most likely path through a trellis of states, a shortest-path dynamic program (Ch. 52).
- **Decoding beyond half the distance.** Result 1 seems to forbid correcting more than $(d-1)/2$ errors. But if the decoder may output a short *list* of candidates, Guruswami and Sudan (1999) showed that Reed–Solomon codes can be decoded far beyond that radius, using curves through the received points.

? Are random codes the best possible?

The Gilbert–Varshamov bound of the challenge says that, as $n \to \infty$ with $d/n = \delta$ fixed, there exist binary codes of rate at least $1 - H_b(\delta)$, and a random linear code achieves it. Seventy years later, nobody has found an explicit family of binary codes that beats it, and nobody has proved that none exists. Upper bounds from linear programming leave a wide gap. Over larger alphabets, codes built from algebraic curves *do* beat the corresponding bound, a stunning result from 1982. For binary codes, the true best rate is an open problem.

? Checking a million symbols by reading twenty

A Reed–Solomon codeword has a rigid structure: its values come from a low-degree polynomial. Suppose a prover sends you a list of a million field values and claims it is (close to) such a codeword. Could you check that claim by reading only a few dozen randomly chosen positions? Surprisingly, yes, and this "proximity test" is the engine of modern succinct proofs, which let a blockchain verify a long computation in milliseconds. How a few random queries can certify a huge object, and how this connects to proving things without revealing them, is Ch. 82.

Further reading

- R. W. Hamming, *Coding and Information Theory*, 2nd ed., Prentice Hall, 1986. The inventor's own gentle introduction.
- R. Roth, *Introduction to Coding Theory*, Cambridge University Press, 2006. Linear, cyclic and Reed–Solomon codes with complete algebraic decoders.
- V. Guruswami, A. Rudra and M. Sudan, *Essential Coding Theory*, draft textbook, freely available online. Bounds, list decoding and the modern algorithmic view.
- F. J. MacWilliams and N. J. A. Sloane, *The Theory of Error-Correcting Codes*, North-Holland, 1977. The classic encyclopedic reference.

Chapter 80 — Coding Theory Meets Information Theory

> **Level 2 · Track H — Cryptography, Coding, and Digital Security Builds on:** Ch. 33, Ch. 79, Ch. 19, Ch. 21 · **Leads to:** Ch. 81, Ch. 77 · **Related:** Ch. 32, Ch. 76, Ch. 6 **Reading time:** ~30 min

80.1 The hook

A wire flips 11 % of the bits you send through it, at random. You want to send a file so that it arrives intact.

The obvious remedy is repetition. Send every bit three times and let the receiver take a majority vote. A bit is now lost only if two of its three copies flip, which happens with probability $3(0.11)^2(0.89) + (0.11)^3 \approx 0.034$. That is three times better, and it costs you two thirds of your throughput. Want one error in a million? Repetition needs each bit sent about 21 times (error 3.5×10^{-6}) and drops the rate to under 5 %. One error in a billion needs 39 copies. Error-free means infinitely many copies and a rate of zero. Every engineer in the 1940s took that trade-off for granted: reliability is paid for with speed, and perfect reliability costs everything.

In 1948 Claude Shannon said this is wrong. For this wire, he claimed, there are codes that send **0.49 bits of information per bit transmitted** with an error probability *as small as you like*: one in a billion, one in 10^{100}, whatever you name. The rate does not have to fall as the reliability rises. There is a single number, the channel's **capacity**, here $C = 1 - H_b(0.11) \approx 0.500$. Below it, reliability is free in the limit of long blocks. Above it, failure is certain.

Ch. 33 stated that claim and then refused to explain it. Its mystery asked two things: why a code chosen *at random* should work, and why it took until the 1990s to build codes that come close. This chapter answers the first question with a complete proof, and goes as far into the second as a single chapter can. On the way you will see where the most famous formula in communications, $C = B\log_2(1 + S/N)$, comes from, and why no radio link anywhere can work with less than -1.59 dB of energy per bit.

80.2 The big idea

Noise is random for one bit and almost deterministic for a million. If you send one bit, it either flips or it doesn't, and you cannot tell which. Send a block of $n = 10\,000$ bits, and the law of large numbers (Ch. 19) guarantees that very nearly 1100 of them flip: the standard deviation is only 31. You still don't know *which* bits flipped, but you know *how many*, to within a few percent. Figure 80.2 (right) shows the fraction of flipped bits concentrating around 0.11 as the block grows.

So the noise draws a ball around each codeword. Picture all 2^n binary words of length n as points in a huge space. Send codeword $\mathbf{x}$, and the received word $\mathbf{y}$ lands, almost surely, at Hamming distance about $0.11n$ from $\mathbf{x}$ (Ch. 79 defined Hamming distance). The set of such words is a "noise ball" around $\mathbf{x}$ containing roughly $2^{nH_b(0.11)}$ words. If the balls around different codewords do not overlap, the receiver just asks which ball $\mathbf{y}$ fell into.

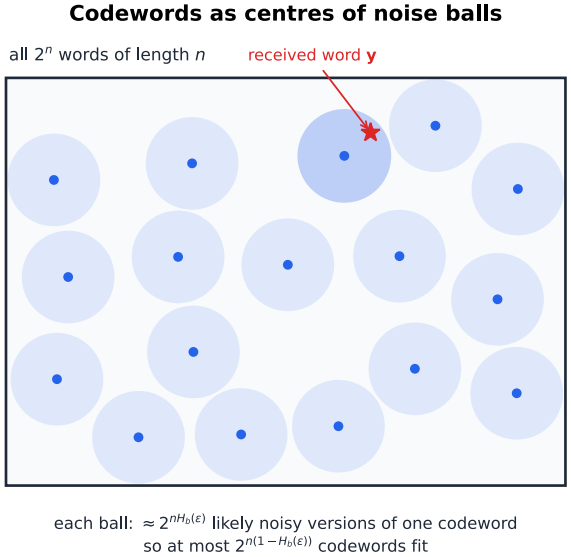
Capacity is a packing count. The space holds 2^n words. Each ball holds about $2^{nH_b(\varepsilon)}$. So at most

$$\frac{2^n}{2^{nH_b(\varepsilon)}} = 2^{n(1-H_b(\varepsilon))}$$

codewords fit, and with 2^{nR} codewords you send nR bits in n channel uses. The rate cannot exceed $1 - H_b(\varepsilon)$. That is the converse, by counting.

And random placement nearly achieves the packing. In a space of a million dimensions, balls scattered at random almost never overlap, as long as there are fewer of them than the count above. You don't need to design a clever arrangement. You need to throw darts. That is why Shannon's proof picks a code at random, and why that proof gives no help at all in building one.

The analogy. A cocktail party where everyone speaks at once is hopeless for one word at a time, but a long sentence from your friend has a statistical fingerprint, the voice, the topic and the rhythm, that no other speaker's sentence will match by accident. Long messages are distinguishable even when every single sound is ambiguous.



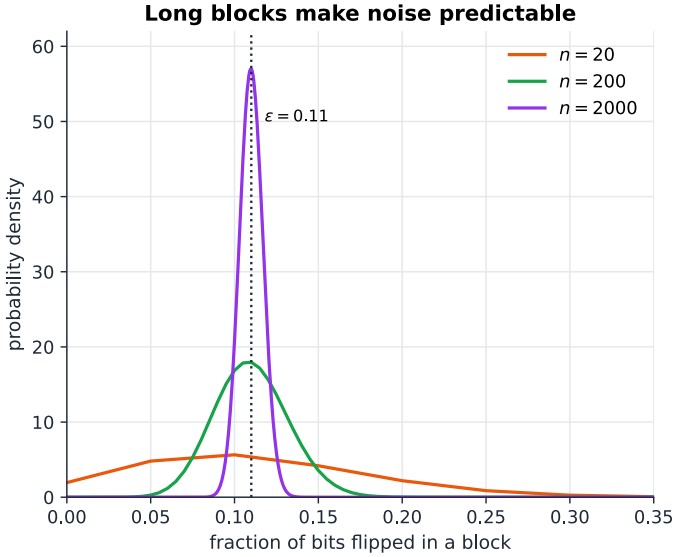


Figure 80.2 — Left: codewords (dots) as centres of noise balls in the space of all 2^n words. A received word falls, with high probability, inside the ball of the codeword that was sent. Right: the fraction of bits flipped in a block of n bits on a channel with $\varepsilon = 0.11$. For long blocks it is pinned to ε, which is what gives the balls a definite size.

80.3 The mathematics

Logarithms are base 2. $H_b(p) = -p\log p - (1-p)\log(1-p)$ is the binary entropy of Ch. 33 §33.3, where entropy, mutual information $I(X;Y)$ and KL divergence are defined. We do not repeat them.

Channels and capacity

Discrete memoryless channel and capacity

A **discrete memoryless channel** (DMC) takes an input symbol x and outputs y with probability $W(y \mid x)$, independently on each use. Its **information capacity** is

$$C = \max_{p_X} I(X;Y),$$

the largest mutual information between input and output over all input distributions.

Three channels carry most of the intuition.

- **Binary symmetric channel** BSC(ε): each bit flips with probability ε. With a uniform input, $I = H(Y) - H(Y \mid X) = 1 - H_b(\varepsilon)$, and no input does better since $H(Y) \le 1$. So $C = 1 - H_b(\varepsilon)$: 0.919 at $\varepsilon = 0.01$, 0.500 at $\varepsilon = 0.11$, zero at $\varepsilon = \tfrac12$.
- **Binary erasure channel** BEC(α): each bit is either delivered or replaced by "?" with probability α. Here $C = 1 - \alpha$, which is exactly the fraction of bits that get through. It is as if the sender knew in advance which bits would be erased. The receiver knows, and remarkably that is enough.
- **Gaussian channel**: $Y = X + Z$ with $Z \sim \mathcal{N}(0, \sigma^2)$ and a power limit $E[X^2] \le P$, treated below.

The definition of C is a maximisation. The theorem that gives it meaning is about *codes*.

Codes, rates and errors

An (n, M) **code** maps each message $w \in \{1, \dots, M\}$ to a codeword $\mathbf{x}(w)$ of n channel symbols. A decoder maps each received word to a guess $\hat{w}$. The **rate** is $R = \frac{\log M}{n}$ bits per channel use. A rate R is **achievable** if there are codes with $M \ge 2^{nR}$ and error probability tending to zero as $n \to \infty$.

Noisy-channel coding theorem (Shannon, 1948)

Every rate $R < C$ is achievable. No rate $R > C$ is achievable: for such rates the error probability stays bounded away from zero, and in fact tends to one.

We prove the achievability half for the BSC, where every step can be done by hand. The same plan works for every DMC, with "typical sets" in place of Hamming balls.

Result 1: the size of a Hamming ball

Let $V(n, r) = \sum_{k=0}^{r} \binom{n}{k}$ be the number of binary words within Hamming distance r of a fixed word.

Volume bound

For $0 < \lambda \le \tfrac12$,

$$V(n, \lambda n) \le 2^{n H_b(\lambda)}.$$

Proof. By the binomial theorem, $1 = (\lambda + (1-\lambda))^n \ge \sum_{k \le \lambda n} \binom{n}{k} \lambda^k (1-\lambda)^{n-k}$. Write each term as $\binom{n}{k}(1-\lambda)^n \left(\frac{\lambda}{1-\lambda}\right)^k$. Because $\lambda \le \tfrac12$, the ratio $\frac{\lambda}{1-\lambda}$ is at most 1, so its k-th power only shrinks as k grows, and for $k \le \lambda n$ it is at least $\left(\frac{\lambda}{1-\lambda}\right)^{\lambda n}$. Hence

$$1 \;\ge\; V(n,\lambda n)\,(1-\lambda)^n \left(\frac{\lambda}{1-\lambda}\right)^{\lambda n} = V(n,\lambda n)\,\lambda^{\lambda n}(1-\lambda)^{(1-\lambda)n} = V(n,\lambda n)\,2^{-n H_b(\lambda)}. \qquad \square$$

The bound is tight in the exponent. For $n = 1000$ and $\lambda = 0.11$, $\log_2 V = 495.5$ while $n H_b(\lambda) = 499.9$: the ball really does hold about 2^{nH_b} words, give or take a factor polynomial in n.

Result 2: random codes achieve every rate below capacity

Random coding on the BSC

Fix $\varepsilon < \tfrac12$, a margin $\delta > 0$ with $\rho = \varepsilon + \delta < \tfrac12$, and a rate $R < 1 - H_b(\rho)$. Then for every n there is a code with $M = 2^{nR}$ codewords whose error probability, averaged over the messages, is at most

$$\frac{\varepsilon(1-\varepsilon)}{n\delta^2} + 2^{-n(1 - H_b(\rho) - R)} \;\xrightarrow[n\to\infty]{}\; 0.$$

Proof. Choose every bit of every codeword independently and uniformly at random. (We have no idea what the code looks like, and we don't need to.) The decoder is simple: given $\mathbf{y}$, output the unique w with $d(\mathbf{x}(w), \mathbf{y}) \le \rho n$. If there is none, or more than one, declare an error.

Suppose message w is sent. The decoder can fail in only two ways.

(a) Too much noise. The number of flips is Binomial(n, ε), with mean $n\varepsilon$ and variance $n\varepsilon(1-\varepsilon)$. Chebyshev's inequality gives

$$P\big(d(\mathbf{x}(w), \mathbf{y}) > \rho n\big) \le P\big(|\text{flips} - n\varepsilon| \ge n\delta\big) \le \frac{\varepsilon(1-\varepsilon)}{n\delta^2}.$$

(b) An impostor. Some other codeword $\mathbf{x}(w')$ lies within ρn of $\mathbf{y}$. Here is the key: $\mathbf{x}(w')$ was drawn independently of $\mathbf{x}(w)$ and of the noise, hence independently of $\mathbf{y}$, and it is uniform on all 2^n words. So the chance that it lands in the ball of radius ρn around $\mathbf{y}$ is the ball's share of the space. By Result 1,

$$P\big(d(\mathbf{x}(w'), \mathbf{y}) \le \rho n\big) = \frac{V(n, \rho n)}{2^n} \le 2^{-n(1-H_b(\rho))}.$$

There are fewer than 2^{nR} impostors, so by the union bound, (b) has probability at most $2^{nR} \cdot 2^{-n(1-H_b(\rho))}$.

Adding (a) and (b) bounds the error probability for each message, *averaged over the random choice of code*. An average over codes is a weighted mean of the averages of individual codes, so at least one code does at least as well as the mean. $\square$

Since δ can be as small as we like, every rate below $1 - H_b(\varepsilon)$ is achievable. Two remarks finish the job. First, "average over messages" can be upgraded to "worst message": throw away the worse half of the codewords. The survivors all have error at most twice the average, and the rate drops by only $1/n$. Second, the proof is an instance of the **probabilistic method** of Ch. 32: it shows that a good code exists by showing that a random one is good on average, and it never exhibits one.

The converse in one line. Fano's inequality says that if the decoder errs with probability P_e, the remaining uncertainty about the message is at most $H(W \mid \mathbf{Y}) \le 1 + P_e\, nR$. And n channel uses can carry at most nC bits of mutual information: $I(W;\mathbf{Y}) \le nC$. A uniform message has $nR = H(W) = I(W;\mathbf{Y}) + H(W \mid \mathbf{Y}) \le nC + 1 + P_e nR$, so

$$P_e \;\ge\; 1 - \frac{C}{R} - \frac{1}{nR}.$$

Above capacity the error probability cannot vanish. This is the counting argument of §80.2, made rigorous. A sharper argument, the **strong converse**, shows that the error probability tends to *one*.

> **? The codes that finally got there**
>
> Result 2 proves that good codes exist, and the experiment in §80.5 will show that finding them by brute force is hopeless: decoding a random code means comparing against 2^{nR} codewords. The codes that reach capacity in practice impose structure. LDPC codes (Gallager, 1960, forgotten until 1996) and turbo codes (1993) come within a fraction of a dB, decoded by passing beliefs along a sparse graph. Polar codes (Arıkan, 2009, now in 5G) are the first *provably* capacity-achieving codes with an $O(n \log n)$ decoder. They work because combining two copies of a channel splits it into one better and one worse channel, and repeating this pushes almost every channel to be perfect or useless. Why must that split polarize, rather than settle somewhere in between? The proof is a martingale convergence argument: see Ch. 97.

Result 3: the Gaussian channel and the Shannon limit

For the channel $Y = X + Z$ with Gaussian noise of variance σ^2 and input power at most P, the entropies are *differential* entropies $h(\cdot)$, the continuous analogue flagged at the end of Ch. 33. Two facts do all the work. Noise independent of the input gives $h(Y \mid X) = h(Z) = \tfrac{1}{2}\log(2\pi e\sigma^2)$. And among all variables with a given variance, the Gaussian has the largest differential entropy, so $h(Y) \le \tfrac{1}{2}\log\big(2\pi e(P+\sigma^2)\big)$, since $\mathrm{Var}(Y) \le P + \sigma^2$. Subtracting,

$$C = \max I(X;Y) = \frac{1}{2}\log_2\!\left(1 + \frac{P}{\sigma^2}\right) \text{ bits per real sample,}$$

with equality for a Gaussian input. A channel of bandwidth B carries $2B$ independent real samples per second (the sampling theorem of Ch. 16), and white noise of spectral density $N_0/2$ (Ch. 21 §21.6) gives noise power $N = N_0 B$. This is the **Shannon–Hartley theorem**:

$$C = B\log_2\!\left(1 + \frac{S}{N}\right) \text{ bits per second.}$$

A 3 kHz telephone line at 30 dB SNR ($S/N = 1000$) has $C = 3000\log_2(1001) \approx 29.9$ kbit/s. The last generation of analogue modems, at 33.6 kbit/s over slightly better lines, ran within a whisker of it. Splitting power among several parallel channels of different quality is the "water-filling" problem solved with Lagrange multipliers in Ch. 6 §6.11.

> **The Shannon limit**
>
> On an AWGN channel, reliable communication at any rate requires
>
> $$\frac{E_b}{N_0} > \ln 2 \approx 0.693 \quad (-1.59 \text{ dB}),$$
>
> where E_b is the energy spent per information bit.

Proof. At rate R bits per real sample, each sample carries energy $E_s = RE_b$, and the noise variance per sample is $\sigma^2 = N_0/2$. So $P/\sigma^2 = 2RE_b/N_0$, and $R < C$ requires

$$R < \frac{1}{2}\log_2\!\left(1 + \frac{2RE_b}{N_0}\right) \quad\Longleftrightarrow\quad \frac{E_b}{N_0} > \frac{2^{2R}-1}{2R}.$$

The right side increases with R. As $R \to 0$, $2^{2R} = e^{2R\ln 2} = 1 + 2R\ln 2 + O(R^2)$, so the bound decreases to $\ln 2$. $\square$

You can buy speed with energy, or save energy by going slowly, but below -1.59 dB no amount of slowness, bandwidth or cleverness helps. Figure 80.1 (right) shows the whole trade-off. Uncoded binary signalling needs 9.6 dB to reach a bit error rate of 10^{-5} at one bit per sample. The Shannon bound at that rate is 1.76 dB. The 7.8 dB between them is the room in which coding theory has worked for seventy years: a factor of six in transmitter power, or in battery life, or in the

distance to a space probe.

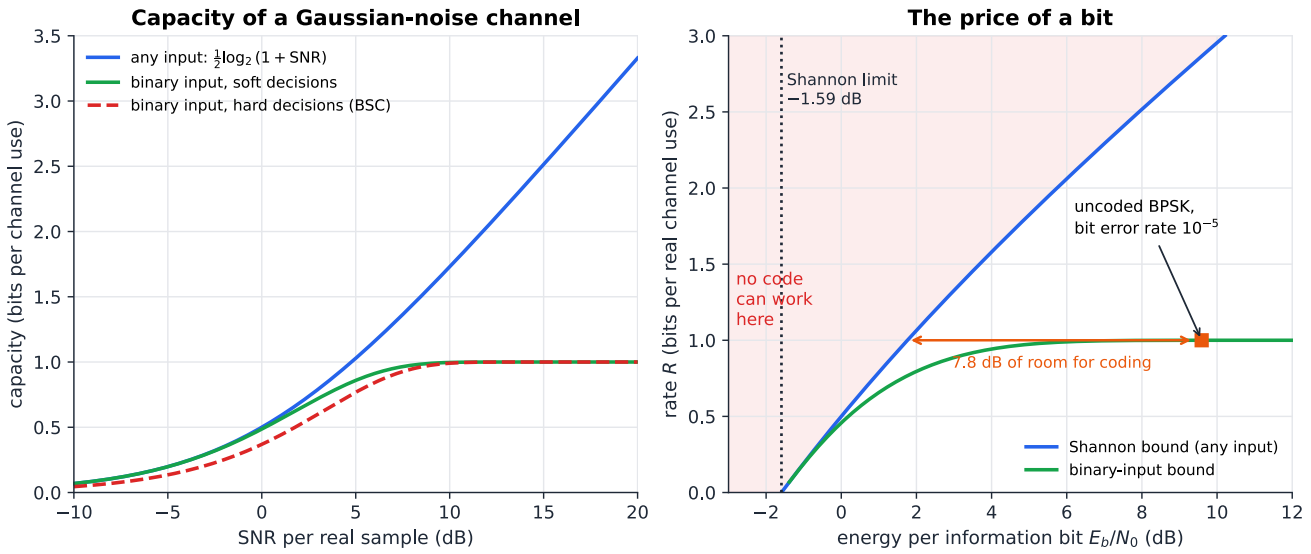


Figure 80.1 — Left: capacity versus SNR. Restricting the input to ± 1 (green) caps the rate at one bit; throwing away the analogue value and keeping only the sign (red, dashed) costs a further fixed loss, about 2 dB at low SNR. Right: the minimum energy per bit at each rate. The shaded region is forbidden by Result 3. The vertical line is the Shannon limit, which no rate escapes.

> **Engineering note: soft decisions**
>
> At low SNR, hard decisions (rounding each received sample to a bit before decoding) multiply the required power by $\pi/2$, a loss of 1.96 dB. That is why modern receivers pass *probabilities*, not bits, to the decoder.

Coding bounds: how far apart can codewords be?

Capacity concerns *random* noise and *average* error. Ch. 79 asked a worst-case question: a code with minimum distance d corrects every pattern of fewer than $d/2$ errors. For long codes write $\delta = d/n$ for the relative distance. What rates are compatible with a given δ? Counting gives an upper and a lower bound.

- **Hamming (sphere-packing) bound.** Balls of radius $\delta n/2$ around codewords are disjoint, so $2^{nR} V(n, \delta n/2) \le 2^n$, which gives asymptotically $R \le 1 - H_b(\delta/2)$.
- **Gilbert–Varshamov bound.** Pick codewords greedily, each at distance at least δn from those chosen so far. You can only get stuck when the balls of radius δn cover the whole space, so you get at least $2^n / V(n, \delta n)$ codewords: $R \ge 1 - H_b(\delta)$ is achievable.
- **Singleton bound.** Delete $d-1$ coordinates and the codewords stay distinct, so $R \le 1 - \delta$. Reed–Solomon codes meet it exactly, but only over large alphabets.

At $\delta = 0.11$, Gilbert–Varshamov guarantees rate 0.50 and the Hamming bound forbids rates above 0.69. The true answer lies somewhere between, and nobody knows where.

> **? Is the greedy code the best code?**
>
> The Gilbert–Varshamov bound comes from the dumbest possible construction, yet for binary codes no one has ever found a family that beats it, and the best upper bound (the "linear programming bound" of McEliece, Rodemich, Rumsey and Welch, 1977) still sits well above it. Over alphabets of size $q \ge 49$, codes built from algebraic curves *do* beat Gilbert–Varshamov, a shock when it was found in 1982. For binary codes, whether $1 - H_b(\delta)$ is the true answer is one of the oldest **open problems** in coding theory.

Compression: the same picture, run backwards

Source coding (Ch. 33 §33.3) and channel coding are mirror images. A source of entropy H produces, in n symbols, about 2^{nH} "typical" sequences that carry nearly all the probability. A compressor indexes them with nH bits and so *removes* redundancy. A channel code *adds* structured redundancy so that noise balls do not overlap. Both rest on the same law of large numbers.

If you will tolerate some error, you can compress below the entropy. The **rate–distortion function** $R(D)$ is the fewest bits per symbol that let you reconstruct with average distortion at most D. For fair random bits with Hamming distortion, $R(D) = 1 - H_b(D)$: allowing 11 % of bits to be wrong halves the file. For a Gaussian source of variance σ^2 with squared error, $R(D) = \frac{1}{2}\log(\sigma^2/D)$, so each extra bit buys 6 dB of quality. This is the theory behind JPEG and MP3 (Ch. 76). Shannon's **separation theorem** says that compressing optimally and then channel-coding optimally loses nothing compared with any joint scheme, as long as the blocks are long. That theorem is why your phone has separate codecs and modems at all.

Correct mathematics is not a secure system

Everything above assumes *random* noise that knows nothing about your code. An adversary is not random. An error-correcting code, or a checksum such as CRC-32, detects accidents but gives no integrity against an attacker. Because CRC is linear, anyone can flip chosen bits of a message *and* fix up the checksum without knowing any key. The original Wi-Fi encryption standard, WEP, relied on a CRC for integrity and was broken in exactly this way in 2001. Integrity against an adversary needs a cryptographic message authentication code (Ch. 77), and resistance to a jammer who knows your code needs secret randomness that the jammer cannot predict (Ch. 81). A capacity theorem is a statement about a probability model. Whether your real channel fits that model is an engineering question that the theorem cannot answer.

80.4 The algorithm

Capacity is defined as a maximum over input distributions. For the BSC symmetry gives the answer, but for a lopsided channel you must compute it. The **Blahut–Arimoto algorithm** (1972) does this by alternating maximisation. It repeatedly reweights each input symbol x by 2^{D_x}, where $D_x = D\big(W(\cdot \mid x) \,\|\, p_Y\big)$ measures how distinctive the outputs of x are compared with the average output.

```
import numpy as np

def blahut_arimoto(W, tol=1e-12, max_iter=10_000):
    """Capacity (bits) of a discrete memoryless channel with transition matrix W[x, y]."""
    q = np.full(W.shape[0], 1.0 / W.shape[0])          # start from the uniform input
    for _ in range(max_iter):
        py = q @ W                                      # output distribution
        with np.errstate(divide="ignore", invalid="ignore"):
            D = np.where(W > 0, W * np.log2(W / py), 0.0).sum(axis=1)   # D(W(.|x) || p_Y)
        lower, upper = q @ D, D.max()                   # I(q) <= C <= max_x D_x
        if upper - lower < tol:
            break
        q = q * np.exp2(D); q /= q.sum()                # reweight inputs by 2^D
    return lower, q

bsc = np.array([[0.89, 0.11], [0.11, 0.89]])
z   = np.array([[1.0, 0.0], [0.5, 0.5]])               # Z-channel: a sent 1 arrives as 0 half the time
print(blahut_arimoto(bsc))   # (0.5000..., [0.5, 0.5])
print(blahut_arimoto(z))     # (0.3219..., [0.6, 0.4])
```

The stopping rule is a certificate, not a guess. For any input q, $I(q) = \sum_x q(x) D_x$ is achievable, and a standard duality argument shows that $C \le \max_x D_x$. When the two meet, you have the capacity to within `tol`. For the Z-channel the optimum sends 1s only 40 % of the time, because 1s are the fragile symbol, and the capacity is $\log_2(1.25) = 0.322$ bits.

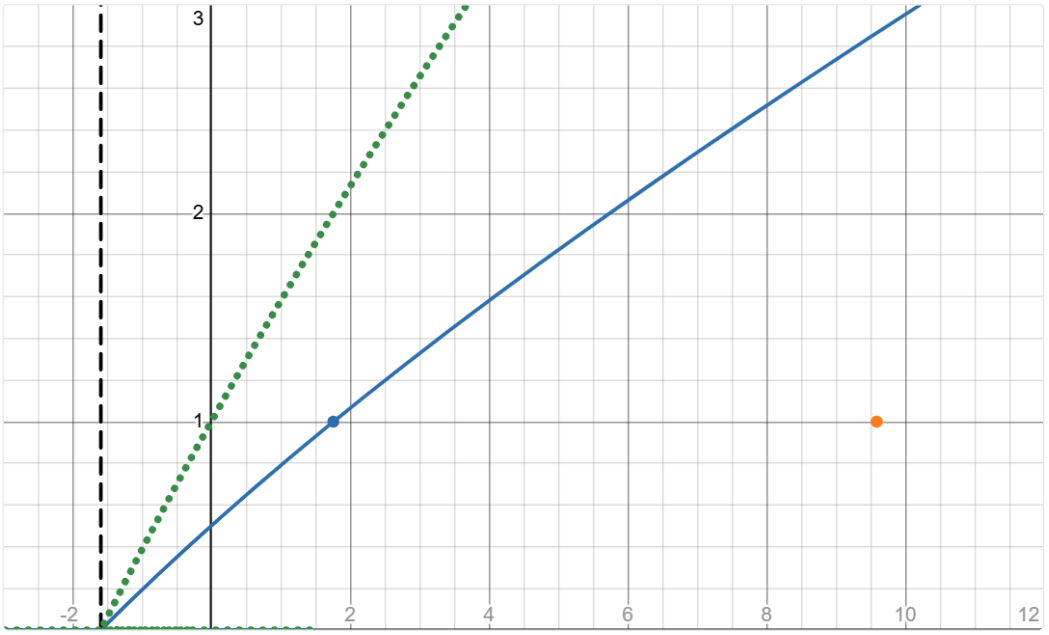
Complexity. Each iteration costs $O(|\mathcal{X}||\mathcal{Y}|)$. Convergence is monotone, and the gap shrinks like $O(1/k)$ in general, much faster when the optimum is unique and in the interior.

Failure cases.

1. *Capacity is not a code.* Blahut–Arimoto tells you the target, not how to hit it. Decoding a random code by brute force costs $n\,2^{nR}$ operations per block. At $n = 1000$ and $R = \frac{1}{2}$ that is 2^{500} comparisons.
2. *Structure helps encoding, not decoding.* Linear codes (Ch. 79) encode in $O(n^2)$ time, and random linear codes still achieve capacity on the BSC. But decoding a general linear code is NP-hard (Berlekamp, McEliece and van Tilborg, 1978). Cryptographers turned that hardness into a weapon: the McEliece cryptosystem hides a decodable code inside a random-looking one.
3. *Finite blocklength.* Capacity is a limit. At $n = 1000$ the best possible rate for a given error probability is noticeably below C, by roughly $\sqrt{V/n}\,Q^{-1}(P_e)$, where V is the "channel dispersion". Short packets, such as control messages in 5G, pay this penalty.

80.5 The experiment

First, drag the Shannon bound around. The curve gives the least E_b/N_0 (in dB, horizontal) needed at each rate R (vertical).



Blue: real channel, $E_b/N_0 > (2^{2R}-1)/(2R)$. Green: the same bound per complex sample (two real dimensions), $(2^R-1)/R$. Both reach -1.59 dB as $R \to 0$. Move the orange point to your own modem's operating point (for example $(4.4, 0.5)$, the classic rate-$\frac{1}{2}$ convolutional code of deep-space links at a bit error rate of 10^{-5}) and read off how much is left to gain. Then add the point $(-1, 0.1)$: any system claiming that operating point is claiming the impossible.

Next, build random codes and watch Result 2 happen, slowly.

```
import numpy as np
rng = np.random.default_rng(1)

EPS = 0.05     # bit-flip probability of the channel (capacity 1 - H(EPS) = 0.714)
R = 0.25       # code rate in bits per channel use: try 0.5, 0.7, 0.8
TRIALS = 1000  # messages sent per block length
MAXBITS = 12   # keeps 2^(nR) codewords manageable

H = lambda p: -p * np.log2(p) - (1 - p) * np.log2(1 - p)
print(f"capacity C = {1 - H(EPS):.3f} bits/use, rate R = {R}")

for n in range(4, int(MAXBITS / R) + 1, max(1, int(2 / R))):
    k = round(n * R)                                # message bits
    M = 2 ** k                                      # codewords
    code = rng.integers(0, 2, (M, n), dtype=np.int8)   # a random codebook
    sent = rng.integers(0, M, TRIALS)
    noise = (rng.random((TRIALS, n)) < EPS).astype(np.int8)
    y = code[sent] ^ noise                          # received words
    dist = (y[:, None, :] ^ code[None, :, :]).sum(axis=2)
    guess = dist.argmin(axis=1)                     # minimum-distance decoding
    err = np.mean(guess != sent)
    print(f"n = {n:3d}  k = {k:2d}  codewords = {M:5d}  block error = {err:.3f}")
```

At $R = 0.25$, a third of capacity, the block error falls from about 4% at $n = 4$ to zero in a thousand trials by $n = 36$. At $R = 0.5$ it hovers around 6–10% up to $n = 24$: the theorem promises it will fall, but only for blocks far longer than your memory allows. At $R = 0.8$, above capacity, it *rises* with n, which is the strong converse in action. Raise `MAXBITS` to 14 or 15 and see how quickly time and memory run out. That wall is the whole reason the codes in the first mystery took fifty years.

80.6 The challenge

1. **Intuition.** On the erasure channel BEC(α) the receiver knows which bits were erased, but the sender does not. Explain in words why capacity is still $1 - \alpha$, as if the sender had known, and why this argument fails on the BSC, where nobody knows which bits flipped.
2. **Derivation.** Prove the lower bound $V(n, \lambda n) \geq \frac{1}{n+1} 2^{n H_b(\lambda)}$ for λn an integer, so that Result 1 is tight up to a polynomial factor. Then use it to prove the sphere-packing converse $R \leq 1 - H_b(\varepsilon)$ for codes that correct *every* pattern of up to εn errors.
3. **Implementation.** Modify the experiment so that the codebook is a random *linear* code: a random $k \times n$ generator matrix over $\mathbb{F}_2$, with codewords $\mathbf{u}\mathbf{G}$. Compare block error rates with the fully random code at the same n and k. Then extend Blahut–Arimoto to a channel with input *cost* (for example, sending a 1 costs energy) and plot capacity against the cost budget.
4. **Open investigation.** For the BSC with $\varepsilon = 0.05$, estimate by simulation the smallest block error achievable at $n = 24$ and $R = 0.5$ by searching over many random codes and keeping the best. How close can you get to the normal approximation $C - \sqrt{V/n}\, Q^{-1}(P_e)$? Exact optimal codes at moderate n are unknown for almost every parameter choice, and your search will not settle it. Where does the improvement curve flatten, and why?

80.7 Loose threads

The question Ch. 33 left open now has its answer. A random code works because, in high dimensions, the noise around each codeword fills a ball of predictable size, and randomly placed balls almost never collide while there are fewer than $2^{n(1-H_b(\varepsilon))}$ of them. The proof is three lines of counting and a law of large numbers. What it does not give is a code you can decode. Bridging that gap took from 1948 to 2009, and the codes that did it are still being refined.

What we did not tell you:

- **How fast the error falls.** Result 2 says the error tends to zero. In fact the best codes drive it down exponentially, $P_e \approx 2^{-nE(R)}$, and the **reliability function** $E(R)$ is known exactly only for rates near capacity. At low rates the best upper and lower bounds disagree, and closing that gap is an open problem tied to the Gilbert–Varshamov mystery above.
- **Feedback.** If the receiver can talk back, capacity does not increase for a memoryless channel (another theorem of Shannon's), but the error exponent improves dramatically and simple schemes become possible.
- **Networks.** With many senders and receivers, the capacity region of even a three-node relay channel is unknown.
- **Secrecy.** Shannon wrote a second great paper in 1949, on secrecy systems. In it he proved that a cipher is *perfectly* secret, revealing nothing about the message to an eavesdropper with unlimited computing power, only if the key carries at least as much entropy as the message. The one-time pad meets this bound, and nothing shorter can.

> **? A short key for a long secret**
>
> By Shannon's 1949 theorem, a 256-bit key can perfectly hide only 256 bits. Yet every web browser encrypts gigabytes with a 256-bit key, and nobody can read the traffic. Both statements are true. The escape is to give up on *perfect* secrecy against *unlimited* adversaries, and to demand only that no *efficient* adversary can tell the ciphertext from random noise. What exactly does "cannot tell" mean, and how can 256 random bits be stretched into a gigabyte of bits that look just as random? See Ch. 81.

Further reading

- T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed., Wiley, 2006. Chapters 7–10 give the channel coding theorem, the Gaussian channel and rate–distortion in full.
- D. J. C. MacKay, *Information Theory, Inference, and Learning Algorithms*, Cambridge University Press, 2003. Free online; outstanding on LDPC codes and message passing.
- T. Richardson and R. Urbanke, *Modern Coding Theory*, Cambridge University Press, 2008. The theory of codes on graphs.
- C. E. Shannon, "A Mathematical Theory of Communication," *Bell System Technical Journal*, 27, 1948, and "Communication Theory of Secrecy Systems," *BSTJ*, 28, 1949.

Chapter 81 — Pseudorandomness and Computational Indistinguishability

> **Level 2 · Track H — Cryptography, Coding, and Digital Security Builds on:** Ch. 19, Ch. 21, Ch. 77, Ch. 80 · **Leads to:** Ch. 82 · **Related:** Ch. 30, Ch. 38, Ch. 57, Ch. 24 **Reading time:** ~30 min

81.1 The hook

1968. George Marsaglia publishes a paper with a blunt title: "Random numbers fall mainly in the planes." The generator he is talking about, IBM's RANDU, had been the standard random number source on mainframes for years. It computes $x_{k+1} = 65539\,x_k \bmod 2^{31}$. Plot its outputs as points (x_k, x_{k+1}, x_{k+2}) in a cube, rotate the cube to the right angle, and the "random" cloud collapses onto just **15 parallel planes**, with nothing at all in between. Physics simulations had been run on those planes for years.

1995. Two Berkeley graduate students, Ian Goldberg and David Wagner, look at how the Netscape browser picks the secret keys that protect credit-card numbers. The keys came from a generator seeded with the time of day and two process IDs. An attacker who could estimate those three numbers was left with a search small enough to finish in minutes on 1995 hardware, so the "128-bit" keys were worth only a few dozen bits.

2008. A Debian developer had removed two lines from the OpenSSL random generator two years earlier, to silence a warning from a code-checking tool. After the change, the only unpredictable input left was the process ID, at most 32,768 values. Every SSH and TLS key generated on Debian and Ubuntu systems in those two years came from a list short enough to print out.

In each case the output *looked* random. RANDU passes the simple frequency tests. The Netscape and Debian keys would pass any statistical test you could name, because each key, on its own, is just a string of bits. What failed was not the look of the numbers but their *unpredictability* to someone who knew how they were made.

So what does it mean for a deterministic program to produce randomness? If an algorithm, a seed and a little arithmetic produce every bit, then each bit is fully determined. In what sense can it be "random" at all? And Ch. 80 left a sharper version of the question. Shannon proved that perfect secrecy needs a key as long as the message, yet a 256-bit key protects gigabytes of web traffic every day. This chapter resolves the contradiction. The answer changes the meaning of "random" itself.

81.2 The big idea

Randomness is in the eye of the observer. A sequence is not random or non-random in itself. It is random *to someone*, if that someone cannot tell it apart from coin flips. A shuffled deck is random to the players and not to the dealer who stacked it. The digits of π are random enough for a dice game and useless as a password, because anyone can compute them.

Make the observer a program with a time limit. Modern cryptography fixes the observer: any algorithm that runs in a reasonable time. It plays a game (Figure 81.2). A challenger secretly flips a coin and hands the algorithm either a truly random string or the output of our generator on a random short seed. The algorithm, the *distinguisher*, must guess which one it got. If no efficient distinguisher guesses right noticeably more often than half the time, the generator is **pseudorandom**. Its output is, for every practical purpose, as good as random, *even though it is not random at all*.

Statistical tests are just distinguishers. A frequency count, a chi-square test or a spectral test is a particular distinguisher. Passing a hundred of them is evidence, not proof: the hundred-and-first may be the one that finds the planes. Cryptographic pseudorandomness asks for something far stronger, namely that *every* efficient test fails.

The analogy. An art forger does not need a painting that *is* a Vermeer. She needs one that no expert can tell from a Vermeer within the time the auction allows. Give the expert unlimited time and an electron microscope and the forgery falls. Bound the expert, and the forgery is, operationally, a Vermeer.

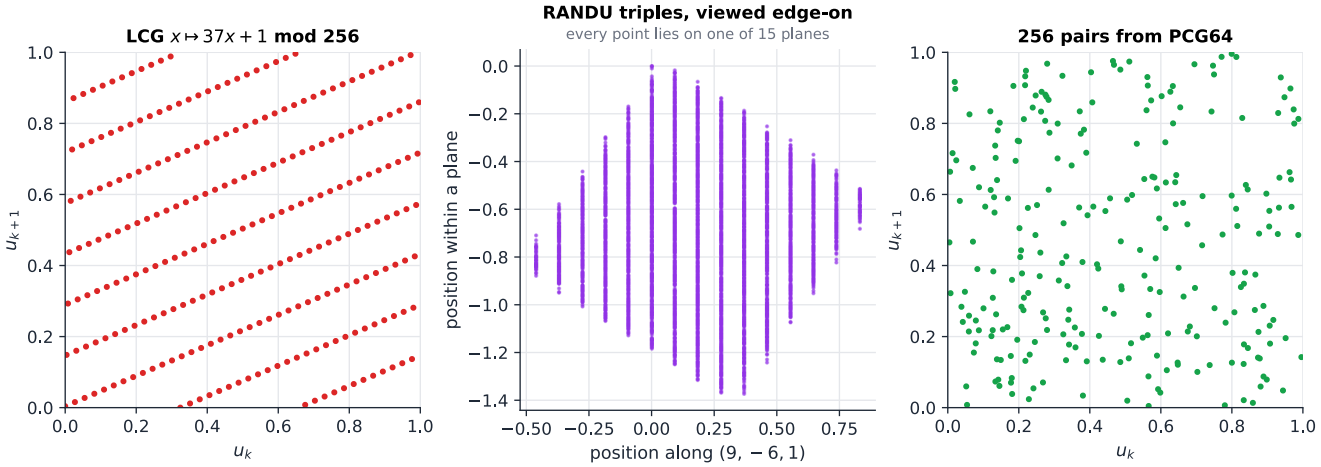


Figure 81.1 — Left: all 256 consecutive pairs of a full-period linear congruential generator lie on a handful of parallel lines. Centre: 6000 RANDU triples projected along the direction $(9, -6, 1)$; every one of them lies on one of 15 planes. Right: pairs from a modern generator (PCG64) show no visible structure. That does not make PCG64 safe for cryptography.

81.3 The mathematics

Throughout, U_ℓ denotes a uniformly random string of ℓ bits, and a "generator" is a deterministic function $G : \{0,1\}^s \to \{0,1\}^\ell$ with $\ell > s$. Its output distribution is $G(U_s)$: the result of feeding it a uniformly random seed.

Result 1: no generator fools an unbounded observer

How different are two distributions P and Q on the same finite set? The natural measure is the **statistical distance**

$$\Delta(P,Q) = \max_A \bigl|P(A) - Q(A)\bigr| = \frac{1}{2}\sum_y \bigl|P(y) - Q(y)\bigr|,$$

the largest difference in probability any event can show. If Δ is tiny, no test whatever, however slow, can tell P from Q with more than tiny advantage.

Generators are statistically far from uniform

For every $G : \{0,1\}^s \to \{0,1\}^\ell$ with $\ell > s$,

$$\Delta\bigl(G(U_s), U_\ell\bigr) \;\geq\; 1 - 2^{s-\ell} \;\geq\; \tfrac{1}{2}.$$

Proof. Let $A = G(\{0,1\}^s)$ be the set of strings the generator can output. There are at most 2^s of them. The generator's output always lies in A, so $P(A) = 1$ for $P = G(U_s)$. A uniform string lands in A with probability $|A|/2^\ell \leq 2^{s-\ell}$. The event A alone therefore shows a difference of at least $1 - 2^{s-\ell}$. □

The distinguisher in this proof is easy to describe ("is y one of the possible outputs?") but it takes 2^s steps to run: try every seed. That is the whole loophole. Statistically, a generator is always a fake. Computationally, checking whether it is a fake may be infeasible.

This is also where Shannon's 1949 theorem from Ch. 80 bites. Perfect secrecy is a statement about statistical distance: the ciphertext's distribution must not depend on the message *at all*. Result 1 says no short key can achieve that. Weaken "not at all" to "not detectably by any efficient algorithm", and the door opens.

Computational indistinguishability

Distinguisher, advantage, pseudorandom generator

A **distinguisher** is an algorithm D that reads a string and outputs 0 or 1. Its **advantage** against G is

$$\mathrm{Adv}(D) = \Bigl|P\bigl(D(G(U_s)) = 1\bigr) - P\bigl(D(U_\ell) = 1\bigr)\Bigr|.$$

G is a (t,ϵ)**-secure pseudorandom generator** if every distinguisher running in time at most t has advantage at most ϵ.

Concrete numbers make the definition real. A generator with a 128-bit seed is considered secure if no attack with t operations has advantage much above $t/2^{128}$. That is the advantage of the trivial attack that guesses seeds at random and checks each guess. Asymptotic cryptography says the same thing with a security parameter κ: G is pseudorandom if every distinguisher running in time polynomial in κ has advantage smaller than every inverse polynomial (a *negligible* function, such as $2^{-\kappa}$).

Two properties make the notion workable.

- **Closure under efficient processing.** If $G(U_s)$ is indistinguishable from U_ℓ, so is $f(G(U_s))$ from $f(U_\ell)$, for any efficient f. Otherwise, composing f with the distinguisher would break G. This is the computational version of the data-processing inequality.
- **Hybrid arguments.** Indistinguishability is "almost transitive": if $X \approx Y$ with advantage ϵ_1 and $Y \approx Z$ with ϵ_2, then $X \approx Z$ with advantage at most $\epsilon_1 + \epsilon_2$, by the triangle inequality. Long chains of small steps are the workhorse of every security proof.

With these, the contradiction of Ch. 80 disappears. Encrypt a long message m as $c = m \oplus G(k)$, where k is a short random key. If G is pseudorandom, c is indistinguishable from $m \oplus U_\ell$, which is a one-time pad and so perfectly random. An efficient eavesdropper learns nothing she could detect. This is the skeleton of every **stream cipher**, from ChaCha20 in your browser to the counter mode of AES (Ch. 77).

Result 2: why RANDU falls on 15 planes

The generator $x_{k+1} = a x_k \bmod m$ with $a = 65539 = 2^{16} + 3$ and $m = 2^{31}$ was chosen because multiplying by $2^{16} + 3$ is fast on binary hardware. That choice is also its downfall.

RANDU's planes

Every RANDU triple satisfies $x_{k+2} \equiv 6x_{k+1} - 9x_k \pmod{2^{31}}$. Consequently the normalised triples $u_k = x_k/2^{31}$ all lie on the 15 planes $9u - 6v + w = j$, $j = -5, \dots, 9$.

Proof. Square the multiplier: $a^2 = 2^{32} + 6 \cdot 2^{16} + 9$. Modulo 2^{31} the first term vanishes, and $6 \cdot 2^{16} = 6(a-3)$, so

$$a^2 \equiv 6(a-3) + 9 = 6a - 9 \pmod{2^{31}}.$$

Since $x_{k+2} \equiv a^2 x_k$ and $x_{k+1} \equiv a x_k$, we get $x_{k+2} \equiv 6x_{k+1} - 9x_k$. Hence $9x_k - 6x_{k+1} + x_{k+2}$ is a multiple of 2^{31}, and dividing by 2^{31} shows that $9u_k - 6u_{k+1} + u_{k+2}$ is an integer. With each u in $[0,1)$, that integer lies strictly between -6 and 10, so it is one of the 15 values $-5, \dots, 9$. □

Every linear congruential generator has some version of this flaw. Marsaglia's theorem says that d-tuples from any LCG with modulus m lie on at most about $(d!\,m)^{1/d}$ hyperplanes. The **spectral test** measures the gap between them, and it is the standard way to grade an LCG multiplier. The Hull–Dobell conditions for a full period (Ch. 57) say nothing about these planes. A generator can visit every value and still be badly structured.

The flaw is a distinguisher in disguise. Compute $9u_k - 6u_{k+1} + u_{k+2}$ for a few consecutive outputs. For RANDU it is always an integer; for true random numbers, essentially never. The advantage is almost exactly 1, and the test costs three multiplications.

Result 3: unpredictability is the same as indistinguishability

A cryptographer's first instinct is a different definition: a good generator is one whose next bit you cannot predict. Given $y_1, \dots, y_{i-1}$, no efficient algorithm should guess y_i with probability noticeably above $\tfrac{1}{2}$. Andrew Yao proved in 1982 that this **next-bit test** is not merely necessary. It is sufficient.

> **Yao's next-bit theorem**
>
> If some distinguisher D running in time t has advantage ϵ against $G : \{0,1\}^s \to \{0,1\}^\ell$, then for some position i there is a predictor A, running in time about t, that guesses y_i from $y_1 \dots y_{i-1}$ with probability at least $\tfrac{1}{2} + \epsilon/\ell$.

Proof. Let $y = G(U_s)$ and define $\ell + 1$ **hybrid** distributions. In H_i the first i bits come from y and the remaining $\ell - i$ bits are fresh coin flips. So $H_0 = U_\ell$ and $H_\ell = G(U_s)$. Let $p_i = P(D(H_i) = 1)$. Replacing D by $1 - D$ if necessary, we may assume $p_\ell - p_0 = \epsilon$. The telescoping sum

$$\epsilon = p_\ell - p_0 = \sum_{i=1}^{\ell} (p_i - p_{i-1})$$

has ℓ terms, so for some i, $p_i - p_{i-1} \geq \epsilon/\ell$. The neighbours H_{i-1} and H_i differ only in bit i: real in H_i, random in H_{i-1}.

The predictor, given $y_1 \dots y_{i-1}$, flips a coin b and fresh coins $r_{i+1}, \dots, r_\ell$, and runs D on $(y_1, \dots, y_{i-1}, b, r_{i+1}, \dots, r_\ell)$. If D says 1 ("looks real"), it predicts $y_i = b$; otherwise it predicts $y_i = 1 - b$.

To analyse it, let q be the probability that D outputs 1 when bit i is set to the *wrong* value $1 - y_i$. Half the time $b = y_i$, and the input to D is distributed exactly as H_i. Half the time $b \neq y_i$. So $p_{i-1} = \tfrac{1}{2}p_i + \tfrac{1}{2}q$, which gives $q = 2p_{i-1} - p_i$. The predictor is right when $b = y_i$ and D says 1, or when $b \neq y_i$ and D says 0:

$$P(\text{correct}) = \tfrac{1}{2}p_i + \tfrac{1}{2}(1 - q) = \tfrac{1}{2} + p_i - p_{i-1} \;\geq\; \tfrac{1}{2} + \frac{\epsilon}{\ell}. \qquad \square$$

Read in reverse, the theorem says that a generator whose bits cannot be predicted passes *every* efficient statistical test, including all the tests nobody has invented yet. That is the strongest guarantee anyone could ask of a sequence, and it follows from a single, natural property.

Where pseudorandom generators come from

A generator needs a source of hardness. The classical recipe has three steps.

1. Start from a **one-way function** f: easy to compute, hard to invert on random inputs. Candidates are multiplying two primes (Ch. 57) and exponentiation in a group (Ch. 78).
2. Find a **hard-core bit**: one bit of x that is as hard to guess from $f(x)$ as x is to find. Goldreich and Levin (1989) showed that every one-way function has one: the parity of a random subset of the bits of x.
3. **Iterate**: output the hard-core bit, apply f, repeat. Yao's theorem, read backwards, proves the output unpredictable.

The **Blum–Blum–Shub** generator is the textbook instance. Pick primes $p \equiv q \equiv 3 \pmod 4$, set $N = pq$, square repeatedly ($x \mapsto x^2 \bmod N$) and output the lowest bit each time. Predicting its bits is provably as hard as factoring N. It is also very slow, which is why practical generators (ChaCha20, AES in counter mode, the DRBGs of NIST SP 800-90A) are built from ciphers whose security is assumed, not proved.

> **? Does pseudorandomness exist at all?**
>
> Everything above is conditional. Håstad, Impagliazzo, Levin and Luby proved in 1999 that pseudorandom generators exist *if and only if* one-way functions exist. But nobody has proved that any one-way function exists, and doing so would prove P $\neq$ NP along the way (Ch. 30). Every stream cipher you have ever used rests on an **open problem**: is there any function that is easy to compute and hard to invert?

81.4 The algorithm

Two short programs show both sides. The first *breaks* a linear congruential generator from three outputs. With a prime modulus and unknown multiplier a and increment c, two consecutive differences satisfy $x_2 - x_1 \equiv a(x_1 - x_0)$, so a is one modular division away. The division uses the inverse from the extended Euclidean algorithm (Ch. 57).

```
def crack_lcg(x0, x1, x2, m):
    """Recover (a, c) of x -> (a*x + c) mod m from three outputs; m prime."""
    a = (x2 - x1) * pow(x1 - x0, -1, m) % m     # modular inverse of (x1 - x0)
    c = (x1 - a * x0) % m
    return a, c

m, a, c = 2**31 - 1, 48271, 12345               # toy secret parameters
xs = [987654]
for _ in range(4):
    xs.append((a * xs[-1] + c) % m)
A, C = crack_lcg(*xs[:3], m)
print(A, C, (A * xs[3] + C) % m == xs[4])       # 48271 12345 True: every future output known
```

That is a next-bit predictor with advantage $\tfrac{1}{2}$: it predicts every bit. By Yao's theorem in its easy direction, it is also a perfect distinguisher. When the modulus is unknown or the outputs are truncated, lattice reduction still recovers the state. The Mersenne Twister, NumPy's legacy default, falls to linear algebra over $\mathbb{F}_2$ after 624 consecutive 32-bit outputs.

The second program is a **toy** Blum–Blum–Shub generator.

```
def bbs_bits(seed, nbits, p=499, q=547):
    """Blum-Blum-Shub with TOY primes p, q = 3 mod 4. Real use needs primes of 1500+ bits."""
    N = p * q
    x = seed * seed % N                          # seed must be coprime to N
    bits = []
    for _ in range(nbits):
        x = x * x % N
        bits.append(x & 1)
    return bits

print("".join(map(str, bbs_bits(2024, 64))))
```

Complexity. An LCG costs one multiply per output. BBS costs one modular squaring of a κ-bit number per output *bit*, about $O(\kappa^2)$ bit operations, so it is thousands of times slower than a cipher-based generator. That is why it lives in textbooks.

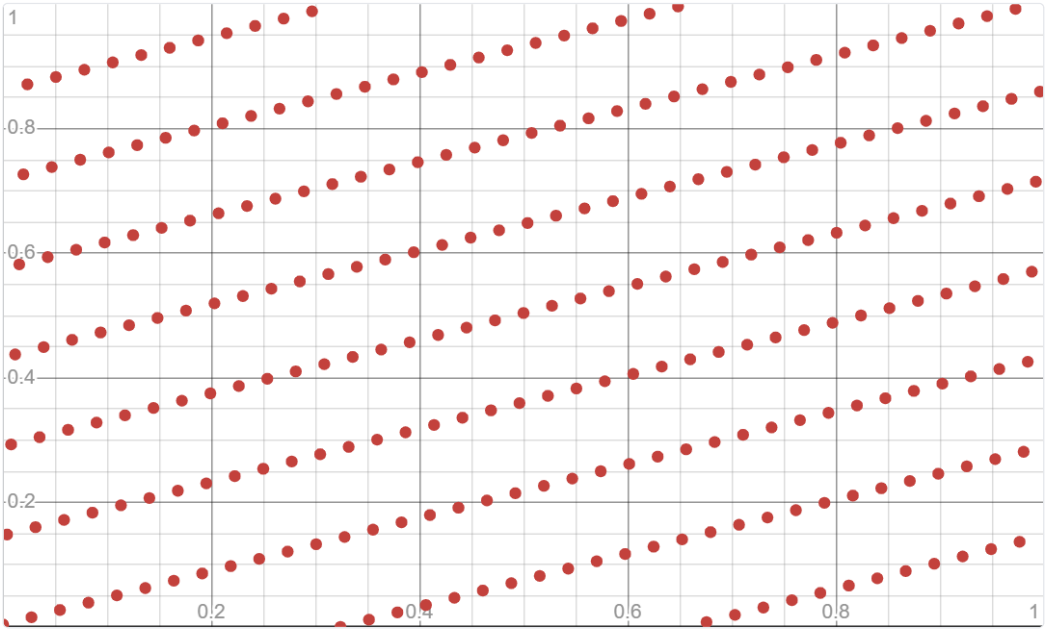
Failure cases.

1. *Toy parameters are toys.* With $p = 499$ and $q = 547$, the seed 2024 enters a cycle of only 492 states, and anyone can factor $N = 272\,953$ instantly. The theorem "as hard as factoring" is worthless when factoring is easy.
2. *The seed is the secret.* A perfect generator on a guessable seed is a guessable generator. Netscape and Debian had sound generators and broken seeds.
3. *Statistical quality is not security.* PCG64 and the Mersenne Twister are excellent for Monte Carlo (Ch. 38) and trivial to predict. Conversely, a CSPRNG (cryptographically secure generator) is fine for simulation, only slower.

> **Correct mathematics is not a secure implementation**
>
> A proof that "G is pseudorandom if factoring is hard" says nothing about where your seed came from, whether two virtual machines cloned from one snapshot now share the same generator state, whether the state leaks through a crash dump or a timing side channel, or whether the standard you followed was designed honestly. (The NIST generator Dual_EC_DRBG was withdrawn in 2014 after evidence that its constants could hide a back door.) In code, use the operating system's generator: `secrets` or `os.urandom` in Python, never `random` or `numpy.random`, for keys, tokens, nonces and passwords. Every generator in this chapter's code is a teaching toy.

81.5 The experiment

First, see the lines. Every point below is a pair (u_k, u_{k+1}) from a full-period LCG with modulus 256.



Change the multiplier 37 *to* 5, 69, 101, 137 *or* 197 *(each keeps the full period, since it is* 1 mod 4*) and count the lines. Which multiplier spreads the points most evenly? That question, asked in higher dimensions, is the spectral test. Then try an even multiplier such as* 38 *and watch the period collapse.*

Now play the distinguishing game of Figure 81.2. Each test below is a distinguisher; the script estimates its advantage against two broken generators.

```
import numpy as np
rng = np.random.default_rng()
N, GAMES = 1200, 400            # numbers per challenge string, rounds of the game

def lcg(a, c, m, seed, n):      # a linear congruential generator, scaled to [0, 1)
    x, out = seed, np.empty(n)
    for i in range(n):
        x = (a * x + c) % m; out[i] = x / m
    return out

GENERATORS = {                  # toy, broken generators: never use these for anything real
    "RANDU":       lambda s: lcg(65539, 0, 2**31, s | 1, N),
    "LCG mod 2^32": lambda s: lcg(1664525, 1013904223, 2**32, s, N),
}
TESTS = {                       # each test is a distinguisher: 1 means "I think this is fake"
    "plane test":  lambda u: np.mean(np.abs(np.round(9*u[:-2] - 6*u[1:-1] + u[2:])
                                    - (9*u[:-2] - 6*u[1:-1] + u[2:])) < 1e-6) > 0.5,
    "low bits":    lambda u: np.all(((u * 2**31).astype(np.int64) & 3)[8:]
                                == ((u * 2**31).astype(np.int64) & 3)[:-8]),
    "pair chi^2":  lambda u: np.sum((np.histogram2d(u[:-1], u[1:], 8)[0] - (N-1)/64)**2
                                / ((N-1)/64)) > 100,
}

for gname, G in GENERATORS.items():
    for tname, D in TESTS.items():
        said_fake, played = [0, 0], [0, 0]
        for _ in range(GAMES):
            b = rng.integers(2)                    # the challenger's secret coin
            u = G(int(rng.integers(2**31))) if b else rng.random(N)
            said_fake[b] += D(u); played[b] += 1
        adv = abs(said_fake[1] / played[1] - said_fake[0] / played[0])
        print(f"{gname:13s} vs {tname:11s} advantage ~ {adv:.2f}")
```

The plane test destroys RANDU (advantage about 1.00) and sees nothing in the other LCG. The low-bits test, which checks whether the two lowest bits repeat with period 8, destroys *both*, because the low bits of any LCG with a power-of-two modulus have tiny periods. The chi-square test on pairs, the kind of check a textbook recommends, has advantage near zero against both. A generator can pass the test you ran and fail the one you didn't. Write your own distinguisher and add it to `TESTS`. Then replace a broken generator by `lambda s: np.random.default_rng(s).random(N)` and try to find *any* test with a visible advantage.

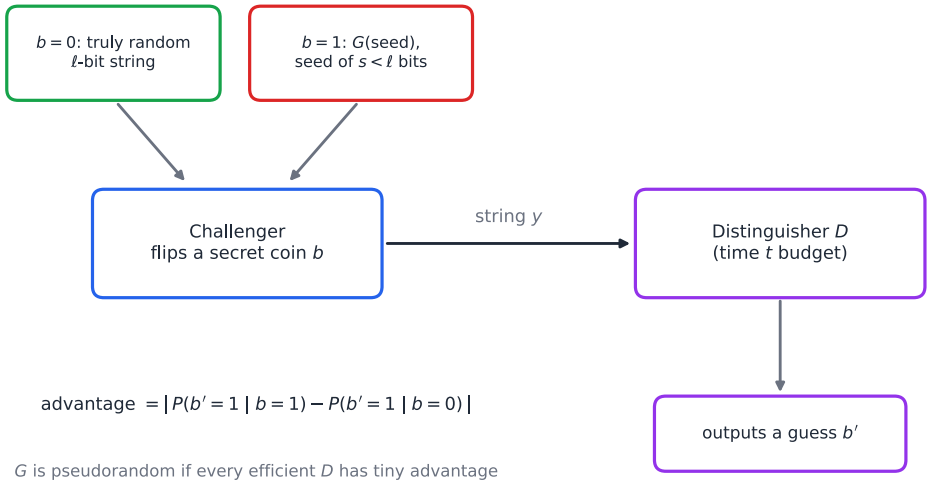


Figure 81.2 — The game that defines pseudorandomness. The challenger hands the distinguisher either true randomness or generator output; the distinguisher's advantage is how much more often it says "1" in one case than in the other. Security means every efficient distinguisher has negligible advantage.

81.6 The challenge

1. **Intuition.** The digits of π are believed to pass every standard statistical test of randomness. Explain why they are nonetheless a terrible pseudorandom generator in the sense of this chapter, and describe a distinguisher with advantage close to 1. What resource does your distinguisher need that a chi-square test does not?
2. **Derivation.** Suppose $G : \{0,1\}^s \to \{0,1\}^{s+1}$ is a (t, ϵ)-secure generator that stretches by just one bit. Define G' by applying G to the seed, outputting the last bit, feeding the first s bits back in, and repeating ℓ times. Use a hybrid argument to show that G' is $(t', \ell\epsilon)$-secure for t' slightly less than t. Identify exactly where the proof uses that each hybrid can be sampled efficiently.
3. **Implementation.** Implement the NIST "runs" test and the "approximate entropy" test, and apply them to the bits of the toy BBS generator, to the lowest bit of an LCG with modulus 2^{32}, and to `os.urandom`. Then write a program that recovers the parameters of an LCG with an *unknown* modulus from eight consecutive outputs. (Hint: the numbers $t_k = x_{k+1} - x_k$ satisfy $t_{k+2}t_k - t_{k+1}^2 \equiv 0 \pmod{m}$.)

4. **Open investigation.** Design the best distinguisher you can against `numpy.random.default_rng` (PCG64) given 10^9 outputs and an hour of laptop time. PCG64 is not a cryptographic generator, and its output function is known, so a distinguisher must exist. How large an advantage can you actually measure? Finding a practical distinguisher, or arguing convincingly that none exists at this budget, is an open experimental question.

81.7 Loose threads

The hook asked how a deterministic program can produce randomness. The answer is that "random" was never a property of the bits alone. It is a relation between a distribution and an observer with limited time. Result 1 shows that no generator fools an unbounded observer. Yao's theorem shows that fooling bounded predictors is the same as fooling bounded tests. And a short key, stretched by a pseudorandom generator, is how a 256-bit secret protects a gigabyte, the answer to the mystery of Ch. 80.

What we did not tell you:

- **Pseudorandom functions.** A generator produces one long string. Many protocols need a random-looking *function*: give it any input and it returns an output that looks fresh. Goldreich, Goldwasser and Micali (1986) built one from any generator, using a binary tree of generator calls. Block ciphers such as AES are modelled as pseudorandom permutations.
- **Extracting randomness.** Seeds come from physical noise (clock jitter, thermal noise in a diode, the timing of keystrokes), which is unpredictable but not uniform. *Randomness extractors* turn such weak sources into nearly uniform bits. Their analysis uses the min-entropy, a cousin of the entropy of Ch. 33.
- **Does randomness help computation?** Ch. 30 and Ch. 38 both asked whether every randomised algorithm can be made deterministic. Pseudorandomness gives a striking conditional answer. Impagliazzo and Wigderson (1997) proved that if some problem solvable in exponential time requires exponential-size circuits, then there are generators strong enough to fool every polynomial-time algorithm, and then $\mathsf{BPP} = \mathsf{P}$. Try all seeds of a logarithmic-length generator and take a majority vote. *Hardness can be traded for randomness*, in both directions.

> **? Is the hardness really there?**
>
> The Impagliazzo–Wigderson theorem turns a lower bound into derandomisation. But no one can prove the lower bound: the best known circuit lower bound for an explicit problem is barely larger than linear. Conversely, Kabanets and Impagliazzo (2003) showed that derandomising even one specific problem (testing whether a polynomial is identically zero) would itself imply new circuit lower bounds. So $\mathsf{BPP} = \mathsf{P}$ and circuit lower bounds rise or fall together. Most researchers believe $\mathsf{BPP} = \mathsf{P}$; proving it is an **open problem** at the heart of complexity theory.

There is one more use for computational indistinguishability, perhaps the most surprising. We used it to say that a ciphertext reveals nothing. It can also say that a *conversation* reveals nothing: that everything you saw while someone convinced you of a fact, you could have produced yourself.

> **? A proof that teaches you nothing**
>
> Suppose a prover convinces you that she knows the password to an account, and the transcript of your conversation is indistinguishable from one you could have generated on your own, without her, without the password. Then you learned nothing except that the claim is true. Can a proof really be both convincing and empty? How do you prove such a property, and why does the proof rely on a simulator that rewinds time? See Ch. 82.

Further reading

- O. Goldreich, *Foundations of Cryptography, Volume 1: Basic Tools*, Cambridge University Press, 2001. Pseudorandom generators, hard-core bits and the hybrid method, done rigorously.
- J. Katz and Y. Lindell, *Introduction to Modern Cryptography*, 3rd ed., CRC Press, 2020. Chapters 3 and 8 connect the definitions to real stream and block ciphers.
- D. E. Knuth, *The Art of Computer Programming, Vol. 2: Seminumerical Algorithms*, 3rd ed., Addison-Wesley, 1997. Chapter 3 is the classic treatment of LCGs and the spectral test.
- S. Vadhan, "Pseudorandomness," *Foundations and Trends in Theoretical Computer Science*, 7, 2012. Extractors, expanders, and hardness versus randomness.

Chapter 82 — Zero-Knowledge Proofs and the Mathematics of Trust

> **Level 2 · Track H — Cryptography, Coding, and Digital Security Builds on:** Ch. 19, Ch. 57, Ch. 77, Ch. 78, Ch. 81 · **Leads to:** Ch. 67, Ch. 94 · **Related:** Ch. 30, Ch. 46, Ch. 63 **Reading time:** ~30 min

82.1 The hook

Every time you log in to a website, you prove who you are by handing over your password. That is a strange way to prove you know a secret: you give it away. If the server is fake, or hacked, or simply logs too much, your secret now belongs to someone else. Could you instead convince the server that you know the password *without revealing anything about it*? Not a hint, not a hash it could attack offline, nothing except the single fact "this person knows it".

That sounds like a contradiction. A proof is supposed to *transfer* understanding. Read a proof that $\sqrt{2}$ is irrational and you can repeat it to someone else. How could a proof convince you and leave you with nothing you could pass on?

Here is a small version. Your friend is red–green colour-blind, and you hold two balls, one red and one green, identical in every other way. You want to convince him that they really are different colours, without telling him which is which. He takes the balls, hides them behind his back, and either swaps them or not, at random. Then he shows them to you and asks: "Did I swap?" You can always answer correctly, because you can see the colours. If the balls were really the same colour, you would be guessing, and you would be right only half the time. After 20 rounds without a mistake, the chance that you were lucky is below one in a million. He is convinced that the colours differ. Yet he has learned nothing about *which* ball is red: every answer you gave was one he already knew, since he did the swapping.

In 1985, Shafi Goldwasser, Silvio Micali and Charles Rackoff turned this idea into mathematics and called it **zero knowledge**. Their theory now protects blockchains worth billions and digital identity documents. This chapter asks the question in the title of their field: how can you prove a statement while revealing nothing beyond its truth? And what, precisely, does "revealing nothing" mean?

82.2 The big idea

A proof can be a conversation. A classical proof is a static text. An *interactive* proof is a dialogue between a **prover** (call her Peggy) and a **verifier** (Victor). Victor asks random questions; Peggy answers. A prover who knows the truth can always answer. A prover who is bluffing will, with high probability, be caught by some question she could not anticipate. Randomness is the verifier's weapon: Peggy cannot prepare for a question that has not been chosen yet.

The cave. A ring-shaped cave has one entrance and a magic door deep inside that blocks the ring (Figure 82.1). Peggy claims to know the word that opens the door. Victor waits outside while she walks in, by path A or path B, and he cannot see which. Then he steps in and shouts a random path: "come out by A!" If Peggy knows the word, she can always comply, passing through the door if necessary. If she does not, she can comply only when Victor happens to name the path she took: probability $\frac{1}{2}$. Thirty rounds drive the bluffing probability to $2^{-30} \approx 10^{-9}$.

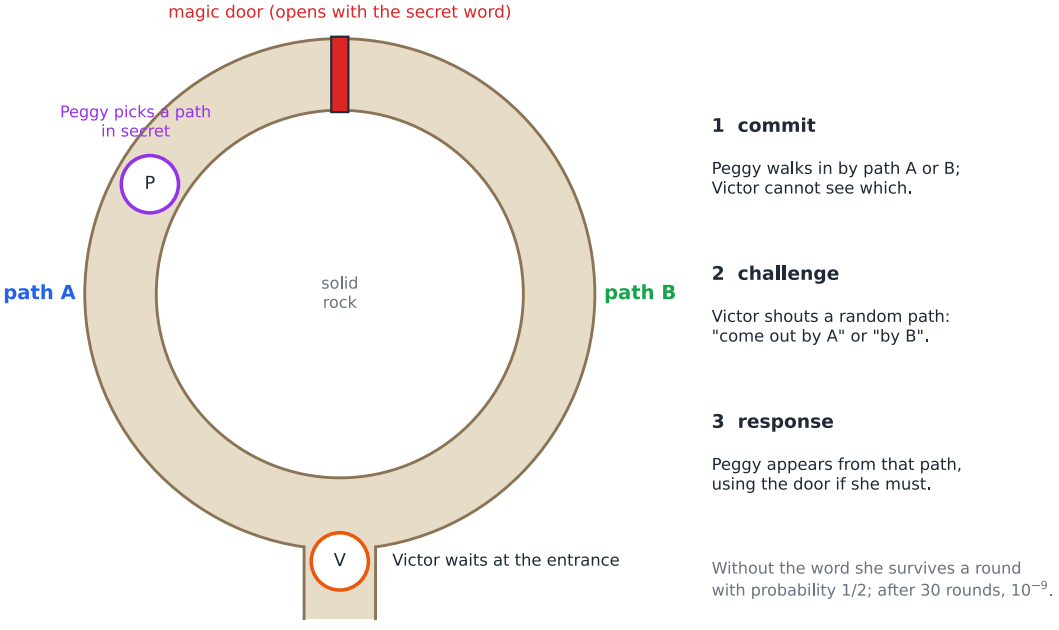


Figure 82.1 — The zero-knowledge cave of Quisquater and colleagues (1989). Commit, challenge, respond: Peggy picks a side in secret, Victor names a side at random, and Peggy must appear from it. Every protocol in this chapter has the same three moves.

Knowledge is what you could not have produced yourself. Victor films the whole session and shows it to a friend. The friend is unimpressed: Peggy and Victor could have agreed in advance which paths would be called, or simply cut out the rounds she failed. A video of a *successful* session can be faked without knowing the word. That observation is the definition. Victor gained nothing from the interaction, because everything he saw, he could have **simulated** alone. What convinced *him* was not the content of the transcript but the fact that he chose the challenges himself, live, after Peggy had committed.

The analogy. A magician performs a card trick at your table. You are sure she has a real method, because you chose the card yourself. But a recording of the trick proves nothing to anyone else, and you leave no closer to knowing how it was done.

82.3 The mathematics

We work in a cyclic group of prime order q, generated by g: for concreteness, the order-q subgroup of $\mathbb{Z}_p^*$ for a prime $p = 2q + 1$, or an elliptic-curve group (Ch. 78). The **discrete logarithm assumption** says that given $y = g^x$ for random x, no efficient algorithm finds x with non-negligible probability (Ch. 57, Ch. 77). Exponents live in $\mathbb{Z}_q$, so they can be added, multiplied and divided modulo q.

Interactive proofs and zero knowledge

Interactive proof

An interactive proof for a claim is a protocol between a prover P and an efficient, randomised verifier V with two properties.

- **Completeness:** if the claim is true and P follows the protocol, V accepts (with probability 1, or close to it).
- **Soundness:** if the claim is false, then *whatever* any cheating prover P^* does, V accepts with probability at most σ, the **soundness error**.

A **proof of knowledge** strengthens soundness: any prover who convinces V with probability noticeably above σ "knows" a witness, in the sense that an efficient **extractor** could compute it by interacting with her.

Repetition shrinks the error: k independent rounds give σ^k.

Zero knowledge

A protocol is **zero-knowledge** if for every efficient verifier V^* there is an efficient **simulator** S that, given only the claim (and no witness), outputs transcripts whose distribution is indistinguishable from that of real conversations between P and V^*. It is *perfect*, *statistical* or *computational* zero-knowledge according to whether the two distributions are identical, statistically close or computationally indistinguishable in the sense of Ch. 81. If this is required only for the honest V, the protocol is **honest-verifier zero-knowledge** (HVZK).

The logic is that of the video. Anything V^* could compute from a real conversation, it could compute from a simulated one, and the simulator needs no secret. So the conversation gave V^* no ability it did not already have.

Commitments: sealed envelopes made of mathematics

Peggy's first move in the cave, entering a path Victor cannot see, is a **commitment**: she fixes a choice now and reveals it later. In digital protocols this is done with a commitment scheme, a mathematical sealed envelope.

Commitment scheme

To commit to a value m, compute $\mathrm{Com}(m; r)$ with fresh randomness r and publish it. To open, reveal (m, r). The scheme is **hiding** if the commitment reveals nothing about m, and **binding** if no one can open it to two different values.

No scheme can be both *perfectly* hiding and *perfectly* binding. If every commitment is consistent with every message (perfect hiding), an unbounded cheater can always find a second opening. One of the two properties must rest on computational hardness.

Pedersen commitment

Let h be a second generator whose discrete logarithm $\log_g h$ nobody knows. Commit to $m \in \mathbb{Z}_q$ as $C = g^m h^r$ with r uniform in $\mathbb{Z}_q$. Then the scheme is perfectly hiding, and any algorithm that breaks binding computes $\log_g h$.

Proof. Hiding. For any fixed m, the map $r \mapsto g^m h^r$ is a bijection from $\mathbb{Z}_q$ onto the group, because h has order q. So with r uniform, C is uniform on the group whatever m is. Its distribution does not depend on m at all.

Binding. Suppose someone produces two openings, $g^m h^r = g^{m'} h^{r'}$ with $m \neq m'$. Then $g^{m-m'} = h^{r'-r}$. Since $m \neq m'$ the left side is not 1, so $r' \neq r$ and $r' - r$ is invertible modulo q. Raising both sides to the power $(r' - r)^{-1}$,

$$h = g^{(m-m')(r'-r)^{-1}}, \qquad \text{so} \qquad \log_g h = (m - m')(r' - r)^{-1} \bmod q.$$

Breaking binding is therefore as hard as computing a discrete logarithm. □

The Schnorr protocol

Peggy's public key is $y = g^x$, and she wants to prove that she knows x. Claus Schnorr's protocol (1989) has the three moves of the cave (Figure 82.2).

1. **Commit.** Peggy picks a random $r \in \mathbb{Z}_q$ and sends $t = g^r$.
2. **Challenge.** Victor picks a random $c \in \{0, \dots, q - 1\}$ and sends it.
3. **Respond.** Peggy sends $s = r + cx \bmod q$. Victor accepts if $g^s = t\, y^c$.

Completeness is one line: $g^s = g^{r+cx} = g^r (g^x)^c = t\, y^c$.

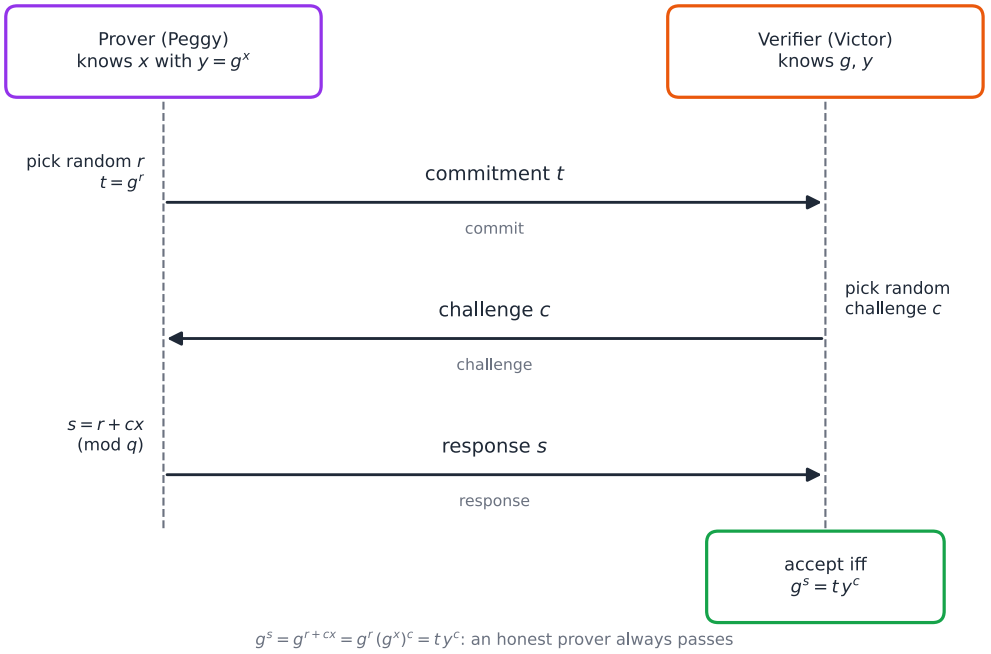


Figure 82.2 — Schnorr's identification protocol. The random r masks the secret x inside s the way a one-time pad masks a message. The challenge c arrives only after t has fixed r.

Special soundness

From two accepting transcripts (t, c, s) and (t, c', s') with the same commitment and $c \neq c'$, the secret is

$$x = (s - s')(c - c')^{-1} \bmod q.$$

Consequently a prover who does not know x can answer at most one challenge per commitment, and convinces Victor with probability at most $1/q$.

Proof. Both transcripts pass the check: $g^s = t y^c$ and $g^{s'} = t y^{c'}$. Dividing eliminates the unknown t: $g^{s-s'} = y^{c-c'} = g^{x(c-c')}$. Since g has prime order q, exponents agree modulo q, so $s - s' \equiv x(c - c')$, and $c - c'$ is invertible. For the second claim, fix any strategy and any commitment t that the strategy sends. If it could answer two different challenges for this t, the formula would produce x, contradicting the assumption that the prover cannot compute it. So for each t at most one of the q equally likely challenges is answerable. □

The rigorous version of the second claim, for a prover who succeeds only sometimes, uses an extractor that **rewinds** her: run her to get t, send a challenge, then rewind to the moment after t and send a different challenge. A forking argument shows that if she succeeds with probability noticeably above $1/q$, two accepting answers for the same t turn up after a few tries. Rewinding is legitimate in a proof, because the extractor is a thought experiment that controls the prover as a program. In the real world nobody can rewind Peggy.

Honest-verifier zero knowledge

The following simulator, which never sees x, outputs transcripts with exactly the same distribution as honest conversations: pick c and s uniformly from $\mathbb{Z}_q$, and set $t = g^s y^{-c}$.

Proof. In a real conversation r and c are independent and uniform. Then $s = r + cx$ is uniform and independent of c, because for each fixed c the map $r \mapsto r + cx$ is a bijection of $\mathbb{Z}_q$. So (c, s) is a uniform pair, and t is the unique value that makes the check pass, $t = g^s y^{-c}$. The simulator produces a uniform pair (c, s) and the same function $t = g^s y^{-c}$ of it. The two distributions are identical. □

The simulator does what Peggy in the cave cannot. It picks the challenge *first* and then builds a commitment to match, exactly like the staged video. Its transcripts are perfectly valid. They only lack the time order that made the real ones convincing.

What about a *dishonest* verifier, whose challenge depends on t? With challenges restricted to $\{0, 1\}$ the simulator can guess c, build t to match, and rewind whenever the verifier asks the other bit. That takes two tries on average, and the protocol is fully zero-knowledge with soundness error $\frac{1}{2}$ per round. With challenges from all of $\mathbb{Z}_q$, the guess succeeds with probability only $1/q$, and no efficient simulator for Schnorr's protocol is known. Large challenges give soundness in one round; small challenges give zero knowledge for every verifier. Getting both at once needs an extra idea, such as having the verifier commit to his challenge in advance.

From conversations to signatures: Fiat–Shamir

Interaction is inconvenient: Victor has to be online. Amos Fiat and Adi Shamir (1986) removed him by replacing his random challenge with a hash of everything said so far, $c = H(g, y, t, \text{message})$. Peggy can now produce (t, s) alone, and anyone can check it. That pair is a **Schnorr signature** on the message. EdDSA (Ed25519, used in SSH and Signal) and Bitcoin's Taproot signatures are Schnorr signatures of this kind.

The security argument models H as a *random oracle*, a truly random function that the prover cannot predict. Then a cheating signer faces the same unpredictable challenge as in the interactive protocol, and the special-soundness extractor works by "rewinding" the oracle.

? The leap of faith in Fiat–Shamir

Real hash functions such as SHA-256 are fixed public programs, not random oracles. Goldwasser and Kalai (2003) built a protocol that is secure as an interactive proof but becomes *insecure* under Fiat–Shamir with **every** concrete hash function. So the transformation is not sound in general. For Schnorr signatures, decades of attacks have found nothing. Which protocols survive the transformation with which hash functions? Recent work using lattice assumptions has proved soundness for some protocols. The general question is an **open problem**.

Zero knowledge for every NP statement

Discrete logarithms are one claim. Goldreich, Micali and Wigderson (1986) showed that *every* claim with a short, checkable certificate (every NP statement, Ch. 30) has a zero-knowledge proof. Since graph 3-colouring is NP-complete, a protocol for it is enough. Peggy knows a proper 3-colouring of a graph with $|E|$ edges.

1. She permutes the three colours at random and commits to the colour of every vertex.
2. Victor picks a random edge.
3. Peggy opens the two commitments at its endpoints. Victor checks that the two colours differ.

A cheater with no valid colouring has at least one bad edge, so she is caught with probability at least $1/|E|$ per round. Soundness error $(1 - 1/|E|)^k$ after k rounds needs about $|E| \ln(1/\sigma)$ rounds: for 20 edges and $\sigma = 10^{-9}$, about 400. Zero knowledge holds because each opening shows two *different random colours*, which the simulator can fake by guessing the edge in advance and rewinding. This is where Ch. 81 enters: the commitments need only computational hiding, and Naor (1991) built them from any pseudorandom generator.

Proof systems beyond NP

Interaction and randomness are astonishingly powerful. Shamir proved in 1990 that IP $=$ PSPACE: interactive proofs can certify any claim checkable with polynomial *memory*, including "the first player has a winning strategy" in any game that lasts polynomially many moves, far beyond NP as far as anyone knows. The proof turns logic into polynomials over a finite field and checks them at random points. The **PCP theorem** (1992) says that any NP proof can be rewritten so that a verifier, reading only a *constant number* of randomly chosen bits, rejects any false claim with probability at least $\frac{1}{2}$.

? A proof shorter than the computation

Modern **SNARKs** (succinct non-interactive arguments of knowledge) prove that a computation of a billion steps was done correctly with a proof of a few hundred bytes, checked in milliseconds, and in zero knowledge. The best-known construction, Groth16, uses three elliptic-curve points. The recipe combines PCP-style polynomial checking with **pairings**, bilinear maps on elliptic curves that let a verifier test a polynomial identity "in the exponent" without seeing the polynomial. What is a pairing, and why do only special curves have efficient ones? See Ch. 78.

Correct mathematics is not a secure implementation

The special-soundness proof *is* an attack. If Peggy ever uses the same random r twice, two signatures share one commitment, and anyone can compute $x = (s - s')(c - c')^{-1}$ by the formula above. In 2010, hackers recovered the master signing key of Sony's PlayStation 3 in exactly this way: its ECDSA signatures used a constant nonce. Even slightly *biased* nonces leak the key through lattice attacks. In 2022, several open-source zero-knowledge libraries were found to compute Fiat–Shamir challenges without hashing all the public inputs ("Frozen Heart"), which allowed forged proofs. A protocol proved secure on paper is secure only when its randomness is fresh, its hash covers everything, and its arithmetic does not leak through timing. The group in this chapter's code has order 1019: a discrete logarithm in it takes microseconds. Real deployments use groups of order about 2^{256} and audited libraries.

82.4 The algorithm

The complete Schnorr protocol, the simulator and the extractor fit in a few lines, using toy parameters. With $p = 2039 = 2 \cdot 1019 + 1$ and $g = 4$, g generates the subgroup of prime order $q = 1019$.

```
import secrets
p, q, g = 2039, 1019, 4          # TOY group, insecure: real q has ~256 bits

def keygen():
    x = secrets.randbelow(q - 1) + 1
    return x, pow(g, x, p)                          # secret x, public y = g^x

def prove_commit():
    r = secrets.randbelow(q)                        # FRESH randomness, every time
    return r, pow(g, r, p)

def prove_respond(x, r, c):
    return (r + c * x) % q

def verify(y, t, c, s):
    return pow(g, s, p) == t * pow(y, c, p) % p

def extract(c1, s1, c2, s2):                        # special soundness
    return (s1 - s2) * pow(c1 - c2, -1, q) % q

x, y = keygen()
r, t = prove_commit(); c = secrets.randbelow(q); s = prove_respond(x, r, c)
print(verify(y, t, c, s))                           # True
```

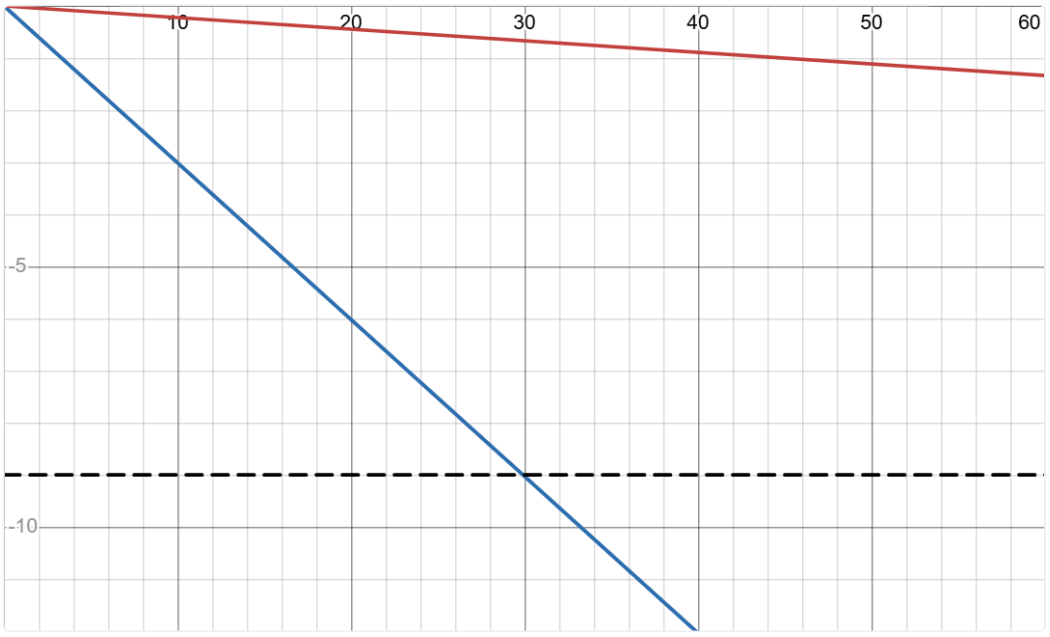
Complexity. The prover does one exponentiation, the verifier two, each $O(\log q)$ group multiplications. With $q \approx 2^{256}$ that is a few hundred multiplications of 256-bit numbers, well under a millisecond. The graph-colouring protocol costs $O(|V|)$ commitments per round and $O(|E|)$ rounds, which is why general-purpose zero knowledge was a theoretical curiosity until SNARKs made proofs short.

Failure cases.

1. *Predictable or repeated r.* As above: the key falls out. The random module of Python, or any generator of Ch. 81 that is not cryptographic, is enough to break it. Deterministic nonces, $r = H(x, \text{message})$, as in EdDSA, remove the dependence on a good runtime generator.
2. *Wrong group.* If g does not have prime order, exponents are not a field, $c - c'$ may not be invertible, and small-subgroup attacks leak bits of x. Every received group element must be checked to lie in the right subgroup.
3. *Challenges not bound to context.* If the Fiat–Shamir hash omits the public key or the statement, a cheater can choose them *after* seeing the challenge. That is the Frozen Heart bug.

82.5 The experiment

How many rounds does trust cost? The curves show $\log_{10}$ of the soundness error after x rounds.



Blue: the cave, or Schnorr with one-bit challenges: error 2^{-x}, below 10^{-9} (the dashed line) after 30 rounds. Red: graph 3-colouring with 20 edges: error $(1 - \frac{1}{20})^x$. Change 20 to 200 and set `right=4000` to see how large graphs blow up the round count.

Now run the protocol, the simulator, a cheater, and the nonce-reuse attack.

```
import random
p, q, g = 2039, 1019, 4          # TOY group: g has prime order q mod p. Real systems use ~256-bit q.
x = random.randrange(1, q)       # Peggy's secret
y = pow(g, x, p)                 # her public key
C = 1019                         # size of the challenge space: try 2

def accept(t, c, s):             # Victor's check: g^s == t * y^c
    return pow(g, s, p) == t * pow(y, c, p) % p

def honest():                    # a real run of the protocol (uses x)
    r = random.randrange(q); t = pow(g, r, p)
    c = random.randrange(C)
    return t, c, (r + c * x) % q

def simulate():                  # a fake transcript made WITHOUT x
    c, s = random.randrange(C), random.randrange(q)
    return pow(g, s, p) * pow(y, -c, p) % p, c, s

def cheat():                     # a prover without x: bet on the challenge in advance
    t, c_bet, s = simulate()
    c = random.randrange(C)      # Victor's real challenge
    return accept(t, c, s) if c == c_bet else accept(t, c, random.randrange(q))

N = 20000
print("honest accepted   ", sum(accept(*honest()) for _ in range(N)) / N)
print("simulated accepted", sum(accept(*simulate()) for _ in range(N)) / N)
print("cheater accepted  ", sum(cheat() for _ in range(N)) / N, " vs 1/C =", 1 / C)
print("three honest:", [honest() for _ in range(3)])
print("three fake:  ", [simulate() for _ in range(3)])

r = random.randrange(q); t = pow(g, r, p)          # the fatal mistake: reuse r
s1, s2 = (r + 5 * x) % q, (r + 9 * x) % q          # two challenges, same commitment
print("secret x =", x, " extracted =", (s1 - s2) * pow(5 - 9, -1, q) % q)
```

Honest and simulated transcripts are both accepted every time. Look at the three of each and try to tell them apart: you cannot, because they have the same distribution. The cheater, who has to bet on the challenge before committing, succeeds about 0.2% of the time: $1/C$, plus a $1/q$ chance that a random guess at s happens to pass. Set `C = 2` and the cheater wins half the time. The last line recovers the secret from two answers to one commitment. (This script uses Python's `random` module only because it is a toy. Real code would use `secrets`.)

82.6 The challenge

1. **Intuition.** In the cave, suppose Victor, instead of choosing at random, always shouts "A". Explain why a Peggy who does not know the word can now fool him every time, and why Victor's recording of a session in which he chose at random is still not evidence for a third party. What exactly is the difference between what convinced Victor and what the recording contains?
2. **Derivation.** Prove that the Pedersen commitment is *additively homomorphic*: $\mathrm{Com}(m_1; r_1) \cdot \mathrm{Com}(m_2; r_2) = \mathrm{Com}(m_1 + m_2; r_1 + r_2)$. Then design a three-move protocol, in the style of Schnorr's, by which Peggy proves she knows an opening (m, r) of a commitment $C = g^m h^r$, and prove that it has special soundness and is honest-verifier zero-knowledge.
3. **Implementation.** Turn the toy Schnorr protocol into a Fiat–Shamir signature: $c = H(y \,\|\, t \,\|\, \text{message}) \bmod q$ with SHA-256, and write `sign` and `verify`. Then deliberately introduce the Frozen Heart bug by leaving y out of the hash, and write a forger that produces a valid-looking signature for a public key of its own choosing without knowing any secret.
4. **Open investigation.** Implement the graph 3-colouring protocol with hash-based commitments, $\mathrm{Com}(m; r) = \text{SHA-256}(r \,\|\, m)$, on random graphs. Measure the total proof size needed for soundness error 2^{-40} as the number of edges grows. How do real systems avoid this cost? Investigate the trade-offs between proof size, prover time and trust assumptions among SNARKs and STARKs. Which assumptions are actually needed for short zero-knowledge proofs is a question still under active research.

82.7 Loose threads

The hook asked how a proof can convince and reveal nothing. The answer has three parts. Randomness in the verifier's challenges makes bluffing detectable. A commitment fixes the prover's choices before she sees those challenges. And "reveals nothing" means "could be simulated": everything the verifier sees, he could have produced alone, so it can carry no knowledge that he lacked. Schnorr's protocol shows all three in four lines of algebra, and the GMW theorem extends them to every statement in NP.

What we did not tell you:

- **Proofs about secrets you share.** Zero knowledge is one tool of *secure multiparty computation*, in which several parties compute a function of their private inputs (an auction, an average salary, a vote count) while learning nothing but the output. The theory again rests on simulators and commitments.
- **Witness indistinguishability.** A weaker property than zero knowledge, but one that survives running many sessions in parallel, which zero knowledge in general does not.
- **Who checks the checker?** A SNARK verifier is a program, and bugs in it have caused real losses. Formally verifying such code, proving the program correct with a proof assistant, is the subject of Ch. 67.
- **Polynomials everywhere.** Modern proof systems reduce every statement to identities between polynomials, which are tested at random points (the Schwartz–Zippel lemma: two different polynomials of degree d agree on at most a fraction d/q of a field of size q). The geometry of polynomial equations is the subject of Ch. 94.

? When the verifier is quantum

Shor's algorithm would compute discrete logarithms on a large quantum computer, breaking Schnorr's assumption outright. There is a subtler problem too. Every zero-knowledge and soundness proof in this chapter **rewinds** a party and runs it twice. A quantum verifier cannot be copied (the no-cloning theorem), and measuring its answer can disturb its state, so rewinding is no longer free. Which zero-knowledge proofs survive against quantum adversaries, and what replaces discrete logarithms? Watrous found the first quantum rewinding technique in 2006, and lattice-based and hash-based proofs are the leading candidates. Post-quantum cryptography is taken up in Ch. 77, and the quantum mechanics behind no-cloning in Ch. 92.

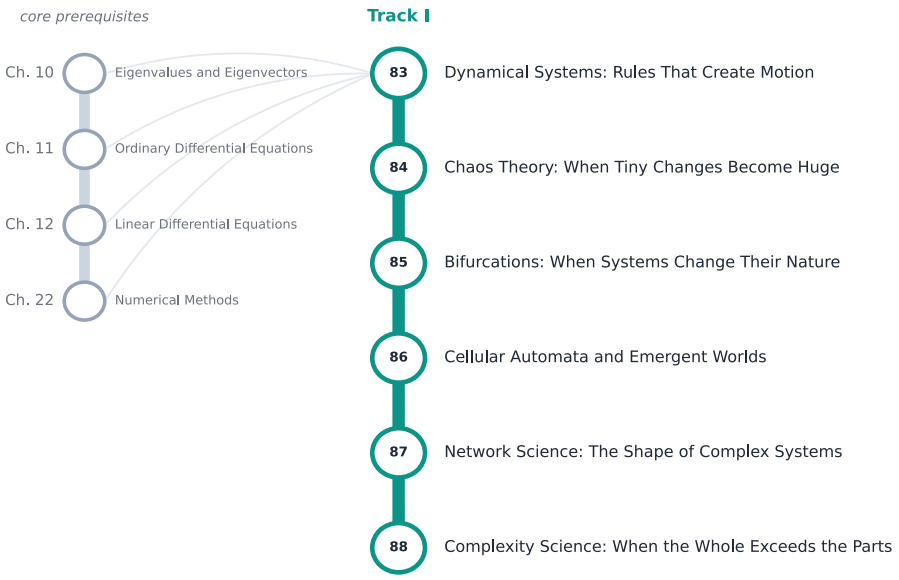
This is the end of Track H. Its six chapters started from codes that defeat noise and ended with proofs that defeat suspicion, and the same tools ran through all of them: probability, finite fields, and the gap between what is true and what can be efficiently checked.

Further reading

- O. Goldreich, *Foundations of Cryptography, Volume 1: Basic Tools*, Cambridge University Press, 2001. Chapter 4 is the definitive treatment of zero-knowledge proofs.
- J. Katz and Y. Lindell, *Introduction to Modern Cryptography*, 3rd ed., CRC Press, 2020. Identification schemes, Schnorr signatures and the Fiat–Shamir transform.
- J.-J. Quisquater et al., "How to Explain Zero-Knowledge Protocols to Your Children," *CRYPTO '89*, Springer, 1990. The cave, in four pages.
- J. Thaler, *Proofs, Arguments, and Zero-Knowledge*, Foundations and Trends in Privacy and Security, 2022. Free online; from interactive proofs to modern SNARKs.

Track I — Dynamical Systems, Chaos, and Complexity

> **Level 2 — Specializations.** Tools for epidemics, ecosystems, power grids, neural networks, traffic, and emergent behavior in games.



Suggested reading order for Track I. The grey stations on the left are the core chapters it leans on.

Before you start

This track assumes you are comfortable with these core chapters:

- Ch. 10 — Eigenvalues and Eigenvectors
- Ch. 11 — Ordinary Differential Equations
- Ch. 12 — Linear Differential Equations
- Ch. 22 — Numerical Methods

Chapters

Ch.	Chapter	What it explores
83	Dynamical Systems: Rules That Create Motion	State spaces, fixed points, stability, phase portraits.
84	Chaos Theory: When Tiny Changes Become Huge	Sensitivity, Lyapunov exponents, strange attractors.
85	Bifurcations: When Systems Change Their Nature	Stability transitions, bifurcation diagrams, nonlinear behavior.
86	Cellular Automata and Emergent Worlds	Local rules, global patterns, computational universality.
87	Network Science: The Shape of Complex Systems	Random networks, scale-free networks, contagion, synchronization.
88	Complexity Science: When the Whole Exceeds the Parts	Emergence, self-organization, feedback, adaptive systems.

Every chapter follows the same arc: **the hook** (a puzzle or a failure), **the big idea**, **the mathematics**, **the algorithm**, **the experiment** you can run, **the challenge**, and **loose threads**: the questions the chapter deliberately leaves open.

Chapter 83 — Dynamical Systems: Rules That Create Motion

> **Level 2 · Track I — Dynamical Systems, Chaos, and Complexity Builds on:** Ch. 11 (§11.7 existence and uniqueness, §11.9 phase plane and linearisation), Ch. 10 (eigenvalues), Ch. 22 (§22.3 fixed-point iteration, §22.9 Runge–Kutta) · **Leads to:** Ch. 84, Ch. 85, Ch. 87 · **Related:** Ch. 45, Ch. 51, Ch. 72, Ch. 25 **Reading time:** ~30 min

83.1 The hook

For almost a century the Hudson's Bay Company kept careful books on the furs its trappers brought in from the Canadian north. When the ecologist Charles Elton went through the ledgers in the 1920s and 1930s he found something nobody had designed and nobody could explain. The number of snowshoe hare pelts rose and fell with a period of roughly ten years. The number of lynx pelts did the same, with the same period, peaking a year or two after the hares.

No calendar sets that ten-year beat. The cycle is made *inside* the system, by two facts: lynx eat hares, and lynx starve without them. Plentiful hares feed more lynx; the lynx eat the hares down, then starve; the hares, relieved, recover. Round and round.

The first serious model of this, written in the 1920s by Alfred Lotka and Vito Volterra independently, is two lines:

$$\dot{x} = \alpha x - \beta x y, \qquad \dot{y} = \delta x y - \gamma y,$$

where x is prey, y is predators, and $\alpha, \beta, \gamma, \delta > 0$. It has an equilibrium where both species coexist, and if you do what Ch. 11, §11.9 taught you, linearise there and compute eigenvalues, you get a pure imaginary pair, $\pm j\sqrt{\alpha\gamma}$. A centre. Oscillation forever, neither growing nor shrinking.

And then §11.9 hands you a warning: a linear centre is exactly the case where linearisation tells you *nothing*. Hartman–Grobman needs every eigenvalue off the imaginary axis. The nonlinear terms you threw away get the deciding vote. They could make the oscillation spiral slowly inward to a quiet balance, or outward until one day the hares crash so low that the lynx die out. A small push either way, accumulated over centuries, decides between eternal cycles, calm and extinction.

So here is the question this chapter will answer: **does the Lotka–Volterra model really cycle forever, and how would you know without solving it?** It has no closed-form solution in elementary functions. You will answer it anyway, in about six lines, by asking not "what is $x(t)$?" but "what *shape* do the solutions have?". That change of question is the founding move of dynamical systems.

83.2 The big idea

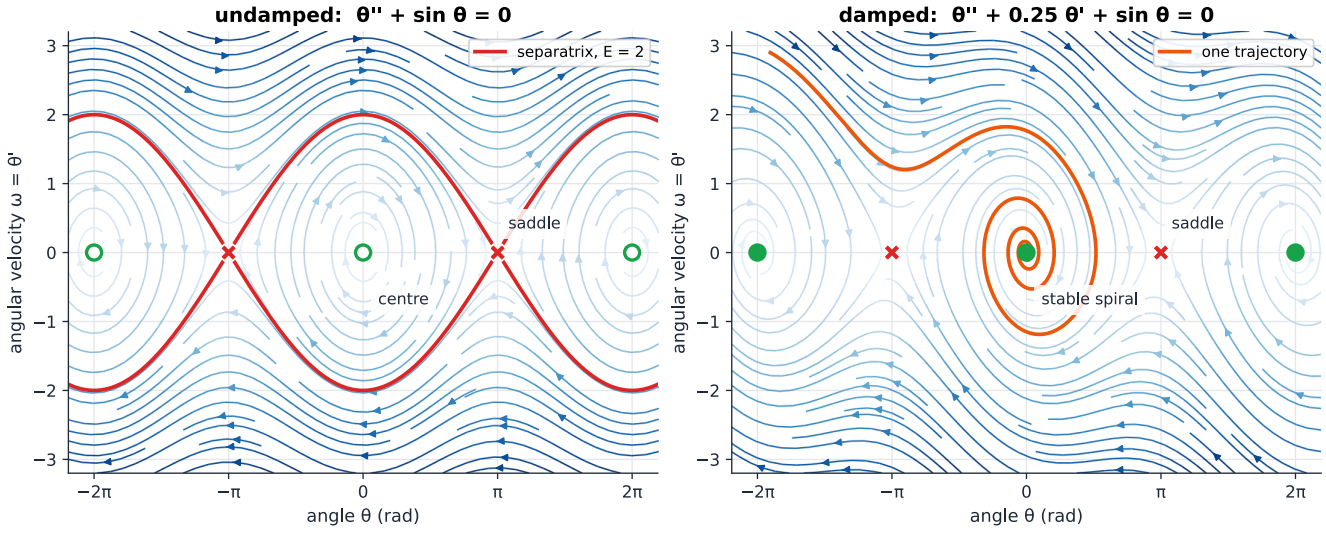
A rule, and the motion it creates

A **dynamical system** is a state and a rule. The state is everything you need to know right now to determine the future: for a pendulum, angle and angular velocity; for the forest, the two populations; for a circuit, capacitor voltages and inductor currents (Ch. 25, §25.2). The rule says how the state changes, either continuously ($\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$, a **flow**) or in jumps ($\mathbf{x}_{n+1} = \mathbf{f}(\mathbf{x}_n)$, a **map**).

The rule is local and dumb: at each point it only says "from here, move that way, this fast". Yet the motion it creates can cycle, settle, switch, and in the next chapter, turn chaotic.

Think in pictures: a river of arrows

Draw the state space as a sheet of paper and, at every point, the arrow $\mathbf{F}(\mathbf{x})$: a picture of a river's surface. A leaf dropped anywhere drifts along a curve tangent to the arrows, a **trajectory** (or **orbit**). Currents plus representative trajectories make the **phase portrait**. Three features matter most: **still water**, points where $\mathbf{F}(\mathbf{x}) = \mathbf{0}$ (**fixed points**); whether nearby leaves drift into still water (**stable**), away from it (**unstable**) or circle it; and **closed loops** of current that attract nearby leaves (**limit cycles**), self-sustained oscillations of a definite size.



Left: the frictionless pendulum. Hanging-down states (green circles) are centres surrounded by closed orbits; the inverted states (red crosses) are saddles. The red separatrix, at energy $E = 2$, divides swinging back and forth from whirling over the top. Right: add friction and the centres become stable spirals; the orange trajectory whirls over the top once, loses energy, and is captured.

In one glance you see every possible motion of the pendulum from every starting state. Henri Poincaré, who invented this way of thinking in the 1880s for planetary orbits, called it the **qualitative theory**: you give up exact numbers and get the shape of all futures.

Why the nonlinear part is the interesting part

Ch. 10 and Ch. 12 solved linear systems completely: every solution is a combination of modes $e^{\lambda t}\mathbf{v}$. A linear system has at most one isolated fixed point, and its oscillations have amplitudes set purely by the initial condition. It cannot choose an amplitude of its own, have two resting states, or switch between them. A flip-flop, an oscillator that grows from noise to one definite amplitude, a pendulum that can hang or whirl: all of these are nonlinear, and the phase portrait is the tool built to see them.

83.3 The mathematics

Flows and the no-crossing rule

Autonomous system, flow, orbit

An **autonomous system** is $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$ with $\mathbf{x} \in \mathbb{R}^n$ (the **state space**) and $\mathbf{F}$ continuously differentiable. Its **flow** $\varphi_t(\mathbf{x}_0)$ is the state at time t of the solution that starts at $\mathbf{x}_0$. The **orbit** of $\mathbf{x}_0$ is the set of all states $\varphi_t(\mathbf{x}_0)$ it passes through.

"Autonomous" means $\mathbf{F}$ does not depend on t: the river's currents do not change with time. A forced system $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}, t)$ can always be made autonomous by adding time as an extra state, $\dot{\tau} = 1$, at the cost of one dimension.

The flow property

For all s, t (for which the solutions exist), $\varphi_{t+s} = \varphi_t \circ \varphi_s$. Consequently two distinct orbits never cross, and an orbit that returns to a state it has visited before is periodic.

Proof. Fix s and $\mathbf{x}_0$, and compare two functions of t: $\mathbf{u}(t) = \varphi_{t+s}(\mathbf{x}_0)$ and $\mathbf{w}(t) = \varphi_t(\varphi_s(\mathbf{x}_0))$. Both satisfy $\dot{\mathbf{u}} = \mathbf{F}(\mathbf{u})$: the first because shifting time does not change an autonomous equation, the second by definition of the flow. Both equal $\varphi_s(\mathbf{x}_0)$ at $t = 0$. Because $\mathbf{F}$ is C^1, it is locally Lipschitz, and the uniqueness theorem of Ch. 11, §11.7 says the two must coincide. If two orbits shared a point $\mathbf{p}$, both would be "the solution through $\mathbf{p}$", so by uniqueness they would be the same orbit. Finally, if $\varphi_T(\mathbf{x}_0) = \mathbf{x}_0$ for some $T > 0$, then $\varphi_{t+T}(\mathbf{x}_0) = \varphi_t(\varphi_T(\mathbf{x}_0)) = \varphi_t(\mathbf{x}_0)$ for every t. ■

In the plane the no-crossing rule is enormously restrictive, and it is the reason two-dimensional flows are tame.

Fixed points of maps

For a map $x_{n+1} = f(x_n)$, a fixed point is x^* with $f(x^*) = x^*$. You have met the numerical side of this in Ch. 22, §22.3, where fixed-point iteration converges when $|g'| < 1$. Read as dynamics, the same statement is a stability theorem:

Stability of a fixed point of a map

Let f be C^1 with $f(x^*) = x^*$. If $|f'(x^*)| < 1$, nearby orbits converge to x^* geometrically. If $|f'(x^*)| > 1$, nearby orbits are pushed away.

Proof. Take $|f'(x^*)| < k < 1$. By continuity, $|f'| \le k$ on some interval $I = [x^* - r, x^* + r]$. For $x_n \in I$ the mean value theorem gives $x_{n+1} - x^* = f(x_n) - f(x^*) = f'(\xi)(x_n - x^*)$ with ξ between them, so $|x_{n+1} - x^*| \le k|x_n - x^*|$. Hence $x_{n+1} \in I$ too, and by induction $|x_n - x^*| \le k^n |x_0 - x^*| \to 0$. The unstable case is the same argument with the inequality reversed. ■

The sign of $f'(x^*)$ matters too: positive means orbits approach from one side, negative means they hop from side to side. That alternation is the seed of the period-doubling cascade of Ch. 85.

What "stable" means

Stability of an equilibrium

An equilibrium $\mathbf{x}^*$ of $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$ is **(Lyapunov) stable** if for every $\varepsilon > 0$ there is a $\delta > 0$ such that $\|\mathbf{x}(0) - \mathbf{x}^*\| < \delta$ implies $\|\mathbf{x}(t) - \mathbf{x}^*\| < \varepsilon$ for all $t \ge 0$. It is **asymptotically stable** if, in addition, $\mathbf{x}(t) \to \mathbf{x}^*$ for all starts close enough.

A centre is stable but not asymptotically stable; a pendulum with friction is both. For a hyperbolic equilibrium the Jacobian decides everything (§11.9). This chapter is about the cases where it does not, and the tool that works anyway.

Lyapunov's direct method: stability without solving

In 1892 Aleksandr Lyapunov proposed generalising the most physical idea in mechanics: if energy never increases, a system at the bottom of a bowl cannot climb out.

Lyapunov's stability theorem

Let $\mathbf{x}^* = \mathbf{0}$ be an equilibrium of $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$, and let V be a C^1 function on a neighbourhood U of $\mathbf{0}$ with $V(\mathbf{0}) = 0$ and $V(\mathbf{x}) > 0$ for $\mathbf{x} \neq \mathbf{0}$. Its derivative along trajectories is

$$\dot{V}(\mathbf{x}) = \nabla V(\mathbf{x}) \cdot \mathbf{F}(\mathbf{x}).$$

If $\dot{V} \leq 0$ on U, the equilibrium is stable. If $\dot{V} < 0$ on $U \setminus \{\mathbf{0}\}$, it is asymptotically stable.

Proof (stability). Given $\varepsilon > 0$ (small enough that the closed ball $\bar{B}_\varepsilon \subset U$), let m be the minimum of V on the sphere $\|\mathbf{x}\| = \varepsilon$; it is positive because $V > 0$ away from the origin and the sphere is compact. By continuity of V and $V(\mathbf{0}) = 0$, choose $\delta < \varepsilon$ with $V < m$ on B_δ. Start at $\mathbf{x}(0) \in B_\delta$. Along the trajectory $\frac{d}{dt} V(\mathbf{x}(t)) = \dot{V} \leq 0$, so $V(\mathbf{x}(t)) \leq V(\mathbf{x}(0)) < m$ for all $t \geq 0$. To leave B_ε the trajectory would have to cross the sphere, where $V \geq m$. It cannot, so $\|\mathbf{x}(t)\| < \varepsilon$ forever.

(Asymptotic stability.) $V(\mathbf{x}(t))$ is decreasing and bounded below by 0, so it tends to a limit $c \geq 0$. If $c > 0$, the trajectory stays in the compact shell $\{\mathbf{x} \in \bar{B}_\varepsilon : V(\mathbf{x}) \geq c\}$, which excludes a neighbourhood of the origin. On that shell $\dot{V}$ is continuous and negative, so $\dot{V} \leq -a$ for some $a > 0$, and $V(\mathbf{x}(t)) \leq V(\mathbf{x}(0)) - at$, which becomes negative: a contradiction. So $c = 0$, and since V is positive away from $\mathbf{0}$, $\mathbf{x}(t) \to \mathbf{0}$. ∎

The theorem needs no solution and no eigenvalues. It works at a centre, where the Jacobian is silent, and it can certify whole regions, not just infinitesimal neighbourhoods.

The damped pendulum, by energy

Take $\dot{\theta} = \omega$, $\dot{\omega} = -\sin\theta - b\omega$ with $b \geq 0$ (time scaled so that $g/L = 1$). Try the mechanical energy $V = \tfrac{1}{2}\omega^2 + (1 - \cos\theta)$, which is zero at $(0, 0)$ and positive nearby. Then

$$\dot{V} = \omega\dot{\omega} + \sin\theta\,\dot{\theta} = \omega(-\sin\theta - b\omega) + \omega\sin\theta = -b\,\omega^2 \leq 0.$$

So the hanging pendulum is stable. But $\dot{V} = 0$ on the whole line $\omega = 0$, so the theorem does not give asymptotic stability, though every simulation shows the damped pendulum coming to rest. The fix, **LaSalle's invariance principle**, notes that a trajectory cannot *stay* on $\omega = 0$ unless $\sin\theta = 0$, so the only motion compatible with $\dot{V} = 0$ forever is the equilibrium itself.

With $b = 0$, energy is **conserved** and the left panel of the pendulum figure is just a contour map of $E = \tfrac{1}{2}\omega^2 + 1 - \cos\theta$; the separatrix is the level $E = 2$ of the inverted pendulum. Its period betrays the nonlinearity: relative to the small-swing period 2π, the swing is 1.066 times longer at amplitude 1 rad, 1.329 times at 2 rad, 2.571 times at 3 rad, and infinitely long as the amplitude approaches π, where the orbit creeps past a saddle.

The answer to the hook: a hidden conserved quantity

Back to the forest. The trick is to look not at $x(t)$ and $y(t)$ but at the orbit, the curve y versus x. Divide the two equations:

$$\frac{dy}{dx} = \frac{y(\delta x - \gamma)}{x(\alpha - \beta y)}.$$

This is separable (Ch. 11, §11.2). Collect the y's on the left and the x's on the right:

$$\frac{\alpha - \beta y}{y}\,dy = \frac{\delta x - \gamma}{x}\,dx \quad\Longrightarrow\quad \alpha \ln y - \beta y = \delta x - \gamma \ln x - C.$$

The Lotka–Volterra invariant

For $x, y > 0$ the function

$$H(x, y) = \delta x - \gamma \ln x + \beta y - \alpha \ln y$$

is constant along every solution. Every orbit in the open quadrant other than the equilibrium $(\gamma/\delta,\ \alpha/\beta)$ is a closed curve, so every solution is periodic.

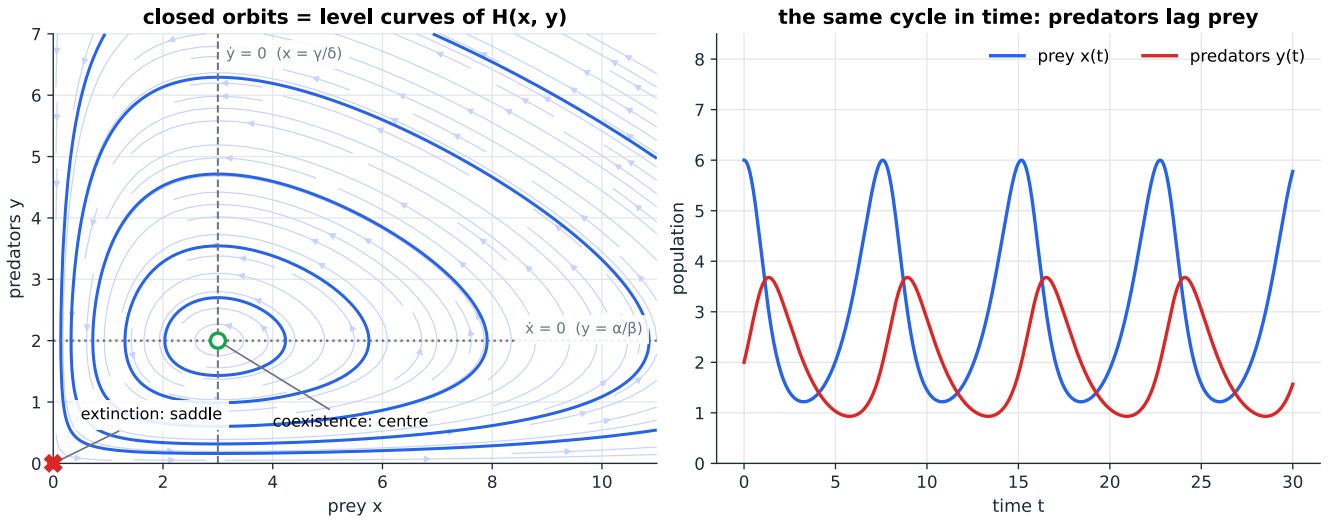
Proof. Constancy is a direct check:

$$\begin{aligned} \dot{H} &= \Big(\delta - \frac{\gamma}{x}\Big)\dot{x} + \Big(\beta - \frac{\alpha}{y}\Big)\dot{y} \\ &= \Big(\delta - \frac{\gamma}{x}\Big)x(\alpha - \beta y) + \Big(\beta - \frac{\alpha}{y}\Big)y(\delta x - \gamma) \\ &= (\delta x - \gamma)(\alpha - \beta y) + (\beta y - \alpha)(\delta x - \gamma) = 0. \end{aligned}$$

Now the shape. The x-part of H, $\delta x - \gamma \ln x$, has second derivative $\gamma/x^2 > 0$: it is strictly convex with a unique minimum at $x = \gamma/\delta$, and tends to $+\infty$ as $x \to 0^+$ and as $x \to \infty$. The y-part is the same with minimum at $y = \alpha/\beta$. So H is a bowl with its lowest point at the coexistence equilibrium and walls rising to infinity at the edges of the quadrant, and every level set $\{H = c\}$ above the minimum is a single closed curve around the equilibrium.

A solution starting on such a curve stays on it, and moves along it with speed $\|\mathbf{F}\|$ bounded below by a positive constant (the curve is compact and contains no zero of $\mathbf{F}$). So it goes all the way round in finite time and returns to its start; by the flow property, it is periodic. ∎

The populations cycle forever, and the size of the cycle is fixed for all time by the initial value of H. You have the complete phase portrait without solving anything.



Left: orbits of the Lotka–Volterra model with $\alpha = 1$, $\beta = 0.5$, $\gamma = 0.75$, $\delta = 0.25$ are the level curves of H, nested around the coexistence centre at $(3, 2)$. The nullclines (dashed and dotted) cross there. Right: the orbit starting at $(6, 2)$ in time. Prey swing between 1.22 and 6, predators between 0.93 and 3.68, with period 7.58, longer than the small-oscillation period $2\pi/\sqrt{\alpha\gamma} = 7.26$. The predator peak lags the prey peak.

The predators' lag, visible in the fur ledgers too, is in the phase plane simply the fact that orbits go round anticlockwise.

A centre is a knife edge

The eternal cycles depend on H being exactly conserved. Give the prey a carrying capacity, $\dot{x} = x(\alpha - \beta y - \varepsilon x)$ with small $\varepsilon > 0$, and the trace of the Jacobian at coexistence becomes $-\varepsilon x^* < 0$: the centre turns into a stable spiral and the cycles die out. A model whose qualitative behaviour changes under arbitrarily small perturbations is **structurally unstable**, and good modellers distrust it. Real lynx–hare cycles persist for reasons the two-line model does not capture.

The invariant reaches beyond ecology. Josef Hofbauer showed in 1981 that the replicator equation of Ch. 45 with $n + 1$ strategies becomes, under the change of variables $y_i = x_i / x_{n+1}$ and a rescaling of time, a Lotka–Volterra system with n species, so the theory of the forest transfers to populations of strategies.

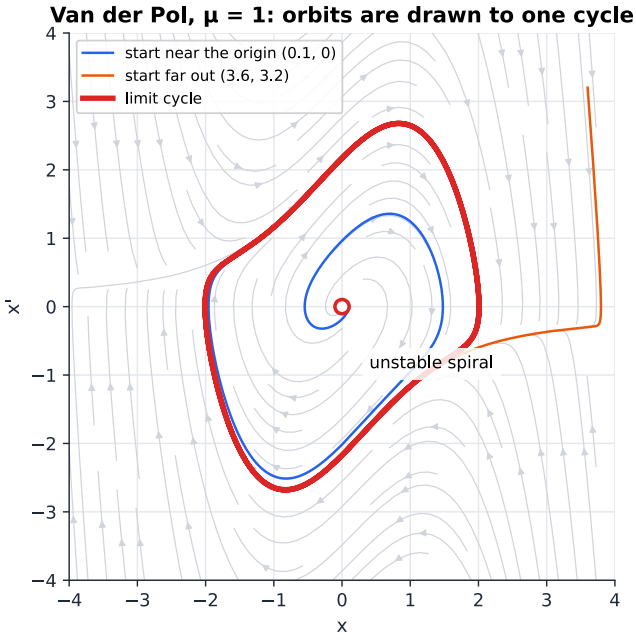
Limit cycles: oscillation with a size of its own

The predator–prey cycles have no preferred amplitude. A clock, a heartbeat or an oscillator circuit should instead settle to *one* amplitude however it was started. That is a limit cycle, and no linear system can have one.

Balthasar van der Pol found the classic example in 1920 while studying triode oscillators:

$$\ddot{x} - \mu(1 - x^2)\dot{x} + x = 0, \qquad \mu > 0.$$

The middle term is a nonlinear damping. For small $|x|$ it is *negative* damping, pumping energy in; for $|x| > 1$ it is positive damping, draining energy out. Small oscillations grow, large ones shrink, and in between there is one closed orbit where the energy pumped in over a cycle exactly balances the energy drained.



The Van der Pol oscillator with $\mu = 1$. A trajectory starting near the unstable spiral at the origin (blue) grows outward; one starting far out (orange) falls inward; both are captured by the same limit cycle (red), with amplitude about 2.01 and period 6.66.

The fact that this picture is typical, not special, is a theorem about the plane:

Poincaré–Bendixson (stated)

Let $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$ be a C^1 system in the **plane**. If a trajectory stays in a closed bounded region that contains no fixed point, it approaches a periodic orbit.

The proof leans on the no-crossing rule and the Jordan curve theorem. For Van der Pol you can build an annulus around the origin whose boundary currents all point inward; it contains no fixed point, so a periodic orbit lives inside. The consequence is strong: in the plane a bounded trajectory approaches a fixed point, a cycle, or a chain of fixed points joined by orbits. There is no fourth option, no wandering forever without repeating.

? Where do limit cycles come from?

Turn the Van der Pol knob μ down through zero and the limit cycle vanishes: for $\mu < 0$ the origin is a stable spiral and every oscillation dies. So at $\mu = 0$ a cycle is *born*. Is it born small, or at full size? Can you predict its amplitude, and does every oscillator start up the same way? That is the Hopf bifurcation of Ch. 85. How many limit cycles a polynomial system in the plane can have is the second half of Hilbert's 16th problem, posed in 1900 and still open even for quadratics.

83.4 The algorithm

A reasonable pipeline for drawing the phase portrait of a planar system $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$:

1. **Find the fixed points.** Run Newton's method for $\mathbf{F}(\mathbf{x}) = \mathbf{0}$ (Ch. 22, §22.4) from a grid of seeds; keep the distinct limits.
2. **Classify each one.** Estimate the Jacobian by central differences, compute its eigenvalues, and read off the type with the §11.9 table. Flag any eigenvalue with real part near zero as *undecided*: that is a case for a Lyapunov function, not for the computer.
3. **Integrate representative trajectories** with RK4 (Ch. 22, §22.9), forward from seeds near unstable points and *backward* in time (integrate $-\mathbf{F}$) to trace stable manifolds of saddles, which are the separatrices that divide basins.
4. **Draw the background current** with a streamline plot.

```
import numpy as np

def F(u):                                   # damped pendulum
    th, w = u
    return np.array([w, -np.sin(th) - 0.25 * w])

def jacobian(F, u, h=1e-6):
    n = len(u); J = np.zeros((n, n))
    for k in range(n):
        e = np.zeros(n); e[k] = h
        J[:, k] = (F(u + e) - F(u - e)) / (2 * h)
    return J

def newton(F, u, iters=50):
    for _ in range(iters):
        u = u - np.linalg.solve(jacobian(F, u), F(u))
    return u

def classify(J, tol=1e-8):
    lam = np.linalg.eigvals(J)
    if np.any(np.abs(lam.real) < tol):  return "non-hyperbolic: undecided", lam
    if lam.real.min() < 0 < lam.real.max(): return "saddle", lam
    kind = "spiral" if np.any(np.abs(lam.imag) > tol) else "node"
    return ("stable " if lam.real.max() < 0 else "unstable ") + kind, lam

seeds = [np.array([a, b]) for a in np.linspace(-4, 4, 9) for b in (-1.0, 0.0, 1.0)]
found = {tuple(np.round(newton(F, s), 6)) for s in seeds}
for p in sorted(found):
    print(p, *classify(jacobian(F, np.array(p))))
```

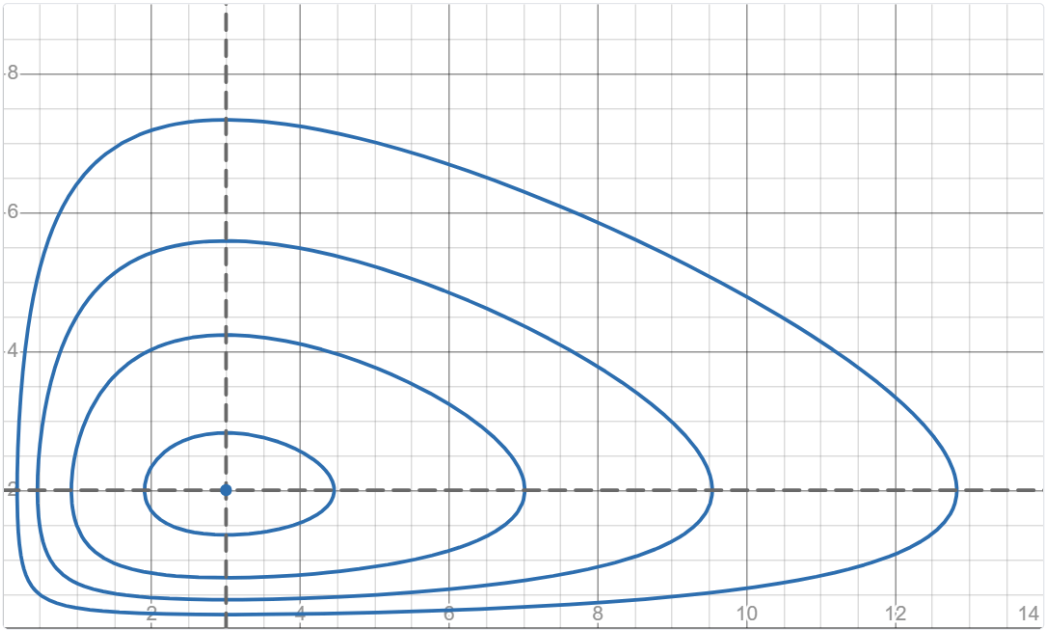
It reports a stable spiral at $(0, 0)$ with eigenvalues $-0.125 \pm 0.992j$ and saddles at $(\pm\pi, 0)$ with eigenvalues 0.883 and -1.133. (Widen the seed grid and it also finds the copies at $\pm 2\pi$, $\pm 3\pi$, and so on.)

Cost and failure modes. Fixed-point finding is cheap; integrating trajectories costs $O(N)$ steps each. The dangerous failures are qualitative:

- **Newton from a grid can miss fixed points** with small Newton basins. There is no general guarantee of finding them all; for polynomial $\mathbf{F}$, algebraic methods do better (Ch. 94).
- **The integrator can change the phase portrait.** Forward Euler turns a centre into an outward spiral: for $\dot{x} = y,\ \dot{y} = -x$ each step multiplies the squared amplitude by exactly $1 + h^2$, so with $h = 0.1$ the amplitude has grown 145-fold by $t = 100$. RK4 is far better but not exact; methods that preserve the conserved quantity by construction (symplectic integrators, Ch. 72) are the principled fix.
- **Non-hyperbolic points** are where step 2 cannot decide, and also where systems change character as parameters move. Treat an "undecided" flag as a signpost to Ch. 85.

83.5 The experiment

First, the forest without solving anything. The curves below are level sets of $H(x, y) = 0.25x - 0.75 \ln x + 0.5y - \ln y$, the invariant for $\alpha = 1$, $\beta = 0.5$, $\gamma = 0.75$, $\delta = 0.25$; its minimum value, at $(3, 2)$, is about 0.233.



Orbits of the predator–prey model as level curves of H. Change the level values (0.3 to 1.6) to see larger and smaller cycles. Then change the coefficient 0.25 of x (that is δ) and move the dashed line to the new $x^* = 0.75/\delta$: how do the cycles shift?

Now draw portraits yourself. The script below integrates a planar system from a ring of starting points with RK4 and overlays streamlines.

```
import numpy as np
import matplotlib.pyplot as plt

b = 0.25          # pendulum damping: try 0, 0.25, 1.0, and negative values
mu = 1.0          # Van der Pol strength: try 0.1, 1, 4
system = "pendulum"   # or "vanderpol"

def F(x, y):
    if system == "pendulum":
        return y, -np.sin(x) - b * y
    return y, mu * (1 - x**2) * y - x

def rk4(x, y, h, n):
    path = [(x, y)]
    for _ in range(n):
        k1 = np.array(F(x, y)); k2 = np.array(F(x + h/2*k1[0], y + h/2*k1[1]))
        k3 = np.array(F(x + h/2*k2[0], y + h/2*k2[1])); k4 = np.array(F(x + h*k3[0], y + h*k3[1]))
        x, y = np.array([x, y]) + h/6 * (k1 + 2*k2 + 2*k3 + k4)
        path.append((x, y))
    return np.array(path)

X, Y = np.meshgrid(np.linspace(-7, 7, 200), np.linspace(-4, 4, 200))
U, V = F(X, Y)
plt.streamplot(X, Y, U, V, color="lightgrey", density=1.2)
for a in np.linspace(0, 2 * np.pi, 12, endpoint=False):
    p = rk4(3 * np.cos(a), 2.5 * np.sin(a), 0.02, 2000)
    plt.plot(p[:, 0], p[:, 1], lw=1)
plt.xlim(-7, 7); plt.ylim(-4, 4); plt.xlabel("x"); plt.ylabel("x'")
plt.title(f"{system}: b={b}, mu={mu}"); plt.show()
```

Things to try:

- Pendulum with `b = 0`, `0.25`, `2.5`. Somewhere in between, the spirals at $\theta = 0$ become nodes and the pendulum stops overshooting. Predict the value ($b = 2$) from the §11.9 table and confirm it.
- Pendulum with `b = -0.1`, a pumped swing. Where do trajectories end up now?
- Van der Pol with `mu = 0.1` and `mu = 4`. The cycle changes from a near-circle to a sharp-cornered **relaxation oscillation**, yet its amplitude stays near 2. Why should it be nearly independent of μ?

83.6 The challenge

1. **Intuition.** A friend claims that $\dot{x} = f(x)$, with x a scalar, can oscillate if f is wiggly enough. Use the no-crossing rule to explain why every bounded solution is monotone and converges to a fixed point. Why does the argument fail for a scalar *map* $x_{n+1} = f(x_n)$?

2. **Derivation.** For the competing-species model $\dot{x} = x(3 - x - 2y)$, $\dot{y} = y(2 - x - y)$, find all four fixed points and classify them with the Jacobian. Then sketch the phase portrait and identify the stable manifold of the saddle: show that it divides the quadrant into two basins, so that one species always wins and which one depends only on the starting point.

3. **Implementation.** Extend the algorithm of §83.4: for each saddle, start two trajectories 10^{-4} either side of it along its stable eigenvector and integrate them *backward* in time. Plot the result for the damped pendulum and for Problem 2's model. You have drawn the separatrices that bound the basins of attraction.

4. **Open investigation.** Lyapunov's theorem needs a function V, and nobody tells you how to find one, or how much of the basin of attraction it can certify. Run the Van der Pol oscillator backward in time: $\dot{x} = -y$, $\dot{y} = x + (x^2 - 1)y$. The origin is now asymptotically stable, and its true basin is the inside of the (now repelling) limit cycle. Show that $V = x^2 + y^2$ certifies the disc $\{V < 1\}$ (with LaSalle's principle). Then search over quadratic forms $V = ax^2 + 2bxy + cy^2$ for the largest certified region $\{V < k\}$, and compare it with the true basin from simulation. How close can quadratics get? What do quartic V buy you? There is no known formula for the best certificate of a given degree.

83.7 Loose threads

The hook asked whether the Lotka–Volterra forest cycles forever, and how you could know without a formula. It does: $H = \delta x - \gamma \ln x + \beta y - \alpha \ln y$ is conserved, its level curves are closed loops, and every orbit is one of them. The same move, finding a function that does something simple along trajectories, gave Lyapunov's theorem. The no-crossing rule, through Poincaré–Bendixson, limits every bounded planar motion to a point, a cycle or a chain of saddles. And the knife-edge warning stands: the forest's cycles die under any perturbation that breaks the conservation law, while self-sustained cycles like Van der Pol's are robust.

What the chapter left open:

- **The birth of cycles.** The first mystery, above, is answered in Ch. 85.
- **Fractal basin boundaries.** In some systems the boundary between two basins is a fractal, and which attractor you reach becomes unpredictable in practice (Ch. 73, Ch. 84).
- **Many interacting units.** Couple thousands of oscillators through a network and they can lock into step, the synchronisation story of Ch. 87.
- **Controlling the flow.** If you may choose part of the rule to steer the state, you are doing optimal control (Ch. 51), where Lyapunov functions reappear as cost-to-go.

? Why the plane cannot be chaotic, and what a third dimension unlocks

Poincaré–Bendixson leaves bounded planar orbits three fates, all eventually regular. Add one state variable and the theorem fails completely: there are three-variable systems with just two quadratic terms whose orbits stay bounded forever, never settle, never repeat, and separate exponentially from their neighbours. What does such an orbit look like, and how do you measure how unpredictable it is? That is Ch. 84.

? Can a computer find the Lyapunov function for you?

Lyapunov's theorem turns stability into a search for a function, and the challenge's Problem 4 shows how unguided that search can be. For systems with polynomial right-hand sides there is a remarkable modern answer: look for V as a *sum of squares* of polynomials, and the conditions $V > 0$, $\dot{V} < 0$ become a convex optimisation problem a computer can solve. Why convexity turns an art into an algorithm is the theme of Ch. 47; how far sum-of-squares certificates can reach, and when they must fail, is an open research area.

Further reading

- Steven H. Strogatz, *Nonlinear Dynamics and Chaos*, 2nd ed. (Westview, 2015). The friendliest introduction to phase portraits, conservative systems and limit cycles.
- Morris W. Hirsch, Stephen Smale and Robert L. Devaney, *Differential Equations, Dynamical Systems, and an Introduction to Chaos*, 3rd ed. (Academic Press, 2013). Careful proofs of the flow property, Lyapunov's theorem and Poincaré–Bendixson.
- Hassan K. Khalil, *Nonlinear Systems*, 3rd ed. (Prentice Hall, 2002). The control engineer's reference for Lyapunov methods, LaSalle and regions of attraction.
- Josef Hofbauer and Karl Sigmund, *Evolutionary Games and Population Dynamics* (Cambridge University Press, 1998). Lotka–Volterra systems and the bridge to the replicator equation.

Chapter 84 — Chaos Theory: When Tiny Changes Become Huge

> **Level 2 · Track I — Dynamical Systems, Chaos, and Complexity Builds on:** Ch. 83 (flows, fixed points, Poincaré–Bendixson), Ch. 7 (divergence), Ch. 22 (§22.1 floating point, §22.9 Runge–Kutta), Ch. 10 · **Leads to:** Ch. 85, Ch. 87, Ch. 88 · **Related:** Ch. 73, Ch. 72, Ch. 45, Ch. 97 **Reading time:** ~30 min

84.1 The hook

In the winter of 1961 the meteorologist Edward Lorenz, at MIT, wanted to look again at a stretch of weather his computer had produced. His model was a dozen equations for a toy atmosphere, run on a Royal McBee LGP-30, a vacuum-tube machine the size of a desk that printed its results line by line. To save time he did not restart from the beginning. He typed in the state from the middle of an earlier printout and went for a coffee.

When he came back, the new run had first copied the old one and then wandered away from it. After a couple of simulated months the two weather histories had nothing to do with each other.

Nothing was broken. The computer stored six digits, the printout showed three. Lorenz had typed 0.506 where the machine had held 0.506127. A difference of one part in four thousand, smaller than any thermometer could measure, had grown until it was as large as the weather itself.

The model was **deterministic**: the same input always gave the same output. Yet it could not be predicted in practice, because the input could never be known exactly. If the real atmosphere behaves like this, Lorenz concluded, long-range forecasting is impossible in principle.

That raises the question this chapter answers: **if a deterministic system amplifies errors, how far ahead can you predict it, and what does it cost to see further?** The answer is a single number per system, and it is not kind. Ten times more precise measurements do not make the forecast ten times longer; they add a fixed amount of time, the same for every digit you pay for.

84.2 The big idea

Stretch, fold, repeat

Watch a baker make puff pastry: roll the dough to twice its length (**stretch**), fold it back on itself (**fold**), repeat. Two flecks of flour a millimetre apart are two millimetres apart after one roll, four after two, and after ten would be a metre apart, except that the fold keeps bringing them back. After a dozen rounds their relative position is effectively random, though every step was deterministic.

That is the mechanism of chaos, in every chaotic system ever found:

- **Stretching** pulls nearby states apart exponentially. This is **sensitive dependence on initial conditions**.
- **Folding** keeps the motion bounded. Without the fold, stretching is just instability: a ball rolling off a hill, which is perfectly predictable.
- **Repetition** mixes. After enough rounds, a tiny region of initial states has been smeared over the whole space.

The rate of stretching is the key number. If errors grow like $\delta(t) \approx \delta_0 e^{\lambda t}$, the error reaches the size Δ of the whole attractor (where your forecast is worthless) at

$$T_{\text{predict}} \approx \frac{1}{\lambda} \ln \frac{\Delta}{\delta_0}.$$

The initial error sits inside a *logarithm*. Divide δ_0 by ten and you gain only $(\ln 10)/\lambda$ extra time units. Exponential growth of error becomes logarithmic return on precision. That is the answer to the hook in outline; the rest of the chapter makes λ precise and computes it.

Why you need room to fold

Ch. 83 proved that a flow in the plane cannot do this: by Poincaré–Bendixson every bounded planar orbit settles on a point, a cycle, or a chain of saddles. Orbits cannot cross, so in two dimensions there is no way to fold a sheet of trajectories back over itself. In three dimensions there is: trajectories can pass over and under one another, the way strands of pastry lie on top of each other after a fold. Three is the minimum for a chaotic flow.

Maps jump, so they need less room: a non-invertible map can be chaotic in one dimension (the logistic map folds an interval onto itself), an invertible one in two (the Hénon map of Ch. 73).

84.3 The mathematics

Sensitivity and the Lyapunov exponent

> **Sensitive dependence**
>
> A map f on a set X has **sensitive dependence on initial conditions** if there is a $\Delta > 0$ such that for every $x \in X$ and every neighbourhood of x, however small, some y in it has an orbit that eventually separates from the orbit of x by more than Δ: $|f^n(x) - f^n(y)| > \Delta$ for some n.

Sensitivity says errors grow; the Lyapunov exponent says how fast. Start with a one-dimensional map $x_{k+1} = f(x_k)$ and two nearby starting points, x_0 and $x_0 + \delta_0$. After n steps, to first order,

$$\delta_n \approx \left(f^n\right)'(x_0)\, \delta_0.$$

The chain rule turns the derivative of the n-fold composition into a product along the orbit:

$$\left(f^n\right)'(x_0) = f'(x_{n-1})\, f'(x_{n-2}) \cdots f'(x_0) = \prod_{k=0}^{n-1} f'(x_k).$$

So the error is multiplied at each step by the local stretching factor $|f'(x_k)|$. Writing $|\delta_n| \approx e^{n\lambda}|\delta_0|$ and solving for λ:

Lyapunov exponent of a one-dimensional map

$$\lambda(x_0) = \lim_{n\to\infty} \frac{1}{n} \ln \left|\left(f^n\right)'(x_0)\right| = \lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} \ln|f'(x_k)|.$$

$\lambda < 0$: nearby orbits converge (a stable fixed point or cycle). $\lambda > 0$ on a bounded orbit: sensitive dependence, the signature of chaos.

The exponent is the **time-average of the logarithm of the stretching**. The logarithm is what makes it an average: stretching factors multiply, their logarithms add. For a stable fixed point x^* the orbit sits at x^*, every term is $\ln|f'(x^*)|$, and $\lambda = \ln|f'(x^*)| < 0$, which is the stability theorem of Ch. 83 in logarithmic dress.

For a flow $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$ the same idea applies to a small separation vector $\boldsymbol{\delta}$, which evolves by the linearised equation $\dot{\boldsymbol{\delta}} = \mathbf{J}(\mathbf{x}(t))\,\boldsymbol{\delta}$ along the orbit. The **largest Lyapunov exponent** is $\lambda_1 = \lim_{t\to\infty} \frac{1}{t} \ln\left(\|\boldsymbol{\delta}(t)\|/\|\boldsymbol{\delta}(0)\|\right)$ for a generic $\boldsymbol{\delta}(0)$. In n dimensions there are n exponents $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$, the growth rates of lengths, areas, volumes and so on, the **Lyapunov spectrum**.

The horizon of §84.2 uses λ_1, and holds while the error is small enough for linearisation.

A chaotic system you can solve exactly: the doubling map

Take $x \in [0, 1)$ and the map

$$D(x) = 2x \bmod 1.$$

Write x in binary, $x = 0.b_1b_2b_3\ldots$. Doubling shifts the binary point one place to the right, and "mod 1" throws away the digit that crosses it:

$$D(0.b_1b_2b_3b_4\ldots) = 0.b_2b_3b_4\ldots$$

Everything about the dynamics can now be read off.

The doubling map is chaotic

(a) D has Lyapunov exponent $\ln 2$ at every point. (b) It has sensitive dependence with $\Delta = 1/4$. (c) Its periodic points are dense in $[0, 1)$. (d) It has orbits that come arbitrarily close to every point.

Proof. (a) $D'(x) = 2$ wherever it is defined, so every term of the average is $\ln 2$.

(b) Take any x and any $\varepsilon > 0$. Choose n with $2^{-n} < \varepsilon$, and let y be x with its binary digit $n+1$ flipped, that is $y = x \pm 2^{-(n+1)}$ with no carry. Then $|y - x| < \varepsilon$, and after n shifts the two points differ only in their first digit, so $|D^n(y) - D^n(x)| = \frac{1}{2} > \frac{1}{4}$.

(c) A point whose binary expansion repeats with period p, such as $0.\overline{011}$, returns to itself after p shifts. Given any x and n, the repeating number made of the first n digits of x is within 2^{-n} of x. So periodic points are arbitrarily close to every point.

(d) Write down every finite binary string, in order of length, and concatenate them into one infinite expansion $z = 0.0\,1\,00\,01\,10\,11\,000\ldots$. Any target x is approximated to n digits by some block of z, and shifting z to that block lands within 2^{-n} of x. ■

Predicting the doubling map n steps ahead requires knowing the initial condition to n binary digits. Each step consumes one bit you had and exposes one you never had: the exponent $\ln 2$ is exactly one bit per step.

Properties (b), (c) and (d) are Robert Devaney's three ingredients of chaos: sensitivity, dense periodic orbits and topological transitivity (some orbit visits everywhere). Notice the paradox in (c): the chaotic set is *packed* with periodic orbits, every one of them unstable, and a typical orbit wanders among them forever, never settling on any.

The logistic map at $r = 4$ is the doubling map in disguise

The logistic map $x_{n+1} = r x_n(1 - x_n)$ on $[0, 1]$ is the textbook example of a non-invertible chaotic map: it stretches the interval and folds it in half. At $r = 4$ its exponent can be found exactly by a change of variables. Set

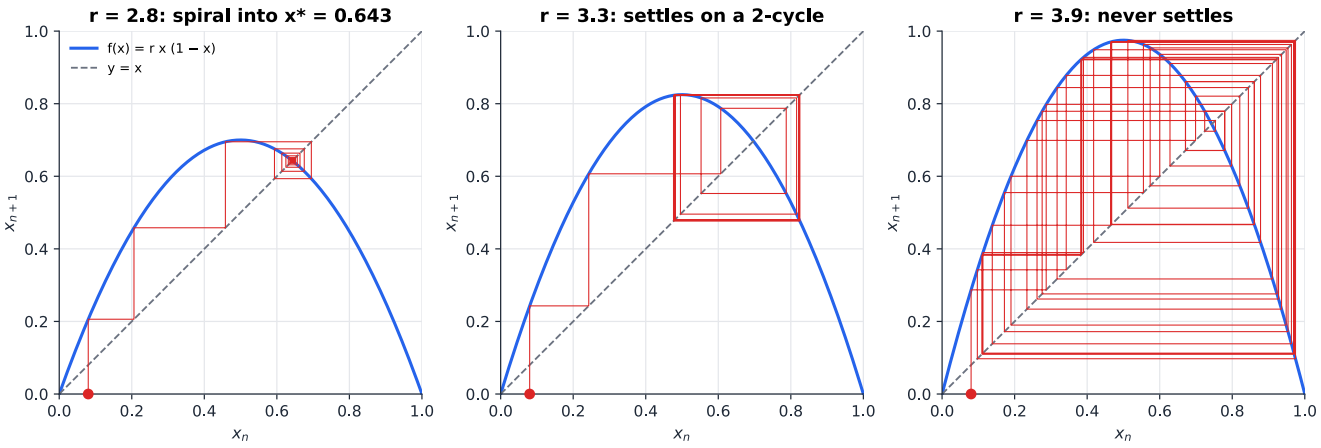
$$x = \sin^2(\pi\theta), \qquad \theta \in [0, 1).$$

Then

$$4x(1 - x) = 4\sin^2(\pi\theta)\cos^2(\pi\theta) = \left(2\sin(\pi\theta)\cos(\pi\theta)\right)^2 = \sin^2(2\pi\theta) = \sin^2\left(\pi\, D(\theta)\right),$$

the last step because $\sin^2$ has period π in its argument, so dropping the integer part of 2θ changes nothing. In words: **if x corresponds to θ, then $4x(1 - x)$ corresponds to $D(\theta)$.** The logistic map at $r = 4$ and the doubling map are the same dynamical system seen through the lens $h(\theta) = \sin^2(\pi\theta)$. Such a change of variables is called a **conjugacy**: $f \circ h = h \circ D$.

A smooth conjugacy cannot change the long-run stretching rate, because h' contributes at most a bounded factor at the start and the end of the orbit, which disappears when divided by n. (There is a fine print: h' vanishes at $\theta = 0$ and $\frac{1}{2}$, so the argument needs typical orbits to avoid lingering there, which they do.) So the logistic map at $r = 4$ has $\lambda = \ln 2 \approx 0.693$ too, and a numerical average of $\ln|4(1 - 2x_k)|$ over 2×10^5 iterates gives 0.6931.



Cobweb diagrams of the logistic map from $x_0 = 0.08$: go up to the curve, across to the diagonal, repeat. At $r = 2.8$ the orbit spirals into the fixed point $x^* = 1 - 1/r = 0.643$, where $f'(x^*) = 2 - r = -0.8$ (the alternation is the negative slope). At $r = 3.3$ it settles onto a cycle of two values. At $r = 3.9$ it never settles; its Lyapunov exponent is about 0.50.

How the logistic map gets from the tidy left panel to the chaotic right one, as r is turned up, is a story of its own.

? The road from order to chaos has a speed limit

Turn r up slowly from 2.8. At $r = 3$ the fixed point gives way to a 2-cycle; at 3.4495 the 2-cycle gives way to a 4-cycle; then 8, 16, 32, faster and faster, until at $r \approx 3.5699$ the period is infinite and chaos begins. The gaps between successive doublings shrink by a ratio that tends to $4.6692\ldots$. Mitchell Feigenbaum found the same ratio in every one-humped map he tried, and experimenters later measured it in convecting fluids and electronic circuits. Why should one number govern systems that have nothing in common? The cascade itself is mapped out in Ch. 85, and the reason for its universality is a renormalisation argument from Ch. 92.

The Lorenz equations: a strange attractor

In 1963 Lorenz boiled a model of a convecting fluid down to three equations:

$$\dot{x} = \sigma(y - x), \qquad \dot{y} = x(\rho - z) - y, \qquad \dot{z} = xy - \beta z,$$

with the classic values $\sigma = 10$, $\rho = 28$, $\beta = 8/3$. Here x measures the strength of the convective roll, y the temperature difference between rising and sinking fluid, and z the distortion of the vertical temperature profile. There are only two nonlinear terms, xz and xy.

Every equilibrium is unstable. The origin has Jacobian eigenvalues 11.83, -2.67, -22.83: a saddle. The other two equilibria are $C_\pm = \big(\pm\sqrt{\beta(\rho-1)},\ \pm\sqrt{\beta(\rho-1)},\ \rho - 1\big) = (\pm 8.49, \pm 8.49, 27)$, each with eigenvalues -13.85 and $0.094 \pm 10.19j$: a spiral that unwinds slowly outward. Orbits spiral away from C_+ until they are flung over to spiral around C_-, then back, with an unpredictable number of turns each time.

Yet orbits stay bounded, and in fact they are squeezed onto a set of zero volume. That is a theorem you can prove in two lines.

Volume contraction in the Lorenz system

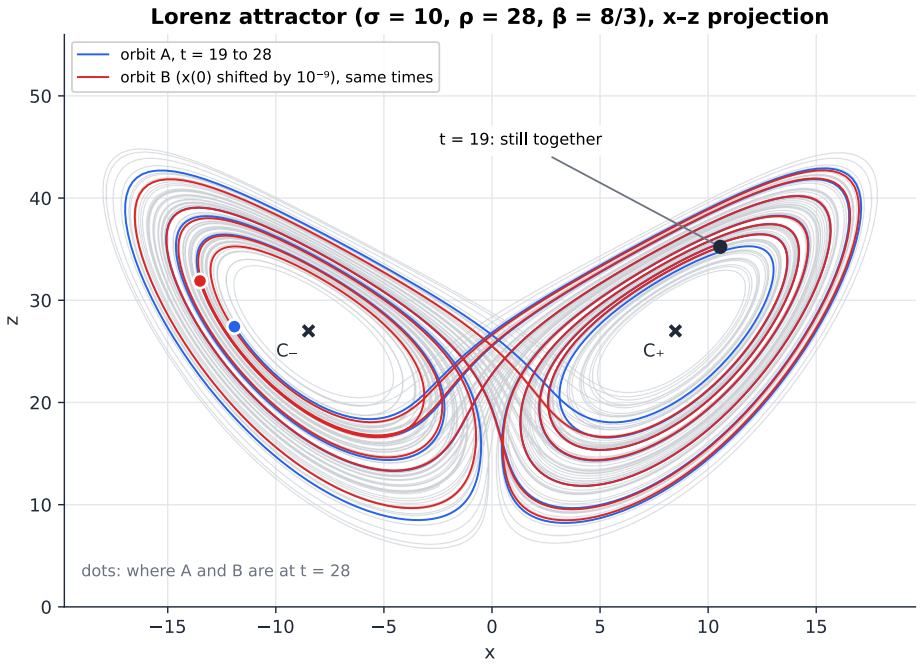
Let $V(t)$ be the volume of a region of initial states carried along by the flow. Then $\dot{V} = \int (\nabla \cdot \mathbf{F})\, dV$, and for the Lorenz equations

$$\nabla \cdot \mathbf{F} = \frac{\partial \dot{x}}{\partial x} + \frac{\partial \dot{y}}{\partial y} + \frac{\partial \dot{z}}{\partial z} = -\sigma - 1 - \beta = -\frac{41}{3} \approx -13.67,$$

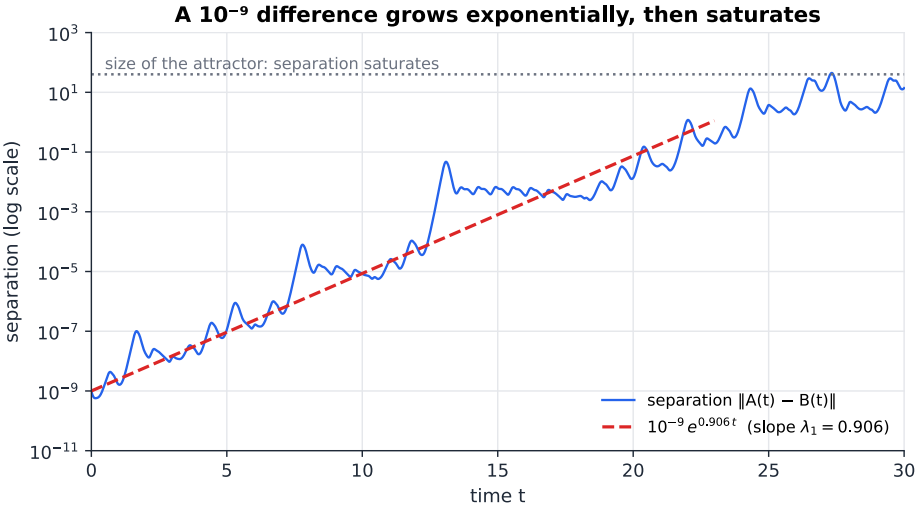
so $V(t) = V(0)\, e^{-13.67\, t}$.

Proof. The rate at which a moving region's volume changes is the flux of the velocity field through its boundary, $\dot{V} = \oint_S \mathbf{F} \cdot \hat{\mathbf{n}}\, dS$, which by the divergence theorem (Ch. 7) equals $\int (\nabla \cdot \mathbf{F})\, dV$. For Lorenz the divergence is the constant $-(\sigma + 1 + \beta)$, so $\dot{V} = -(\sigma + 1 + \beta)V$. ■

In one time unit any blob of states shrinks to $e^{-13.67} \approx 10^{-6}$ of its volume. Combined with boundedness, this means all orbits end on an **attractor** of zero volume. But it cannot be a fixed point (all are unstable), and it cannot be a cycle (nearby orbits separate). It is something new: a set that is locally like a stack of sheets, infinitely many of them, layered like folded pastry. Lorenz called it "an infinite complex of surfaces". It is now called a **strange attractor**: an attractor with sensitive dependence on initial conditions, and with fractal structure.



The Lorenz attractor in the x–z plane (grey), computed with RK4 at step $h = 0.005$. Orbits A and B started 10^{-9} apart. At $t = 19$ they are still indistinguishable on this scale; by $t = 28$ they are on different wings. The crosses are the unstable equilibria $C_{\pm}$.



The distance between orbits A and B on a logarithmic axis. Apart from wiggles (the stretching rate varies around the attractor), it follows a straight line of slope $\lambda_1 = 0.906$ until it reaches the size of the attractor at about $t = 27$, where it saturates: the fold.

This is the hook's answer made quantitative. With $\lambda_1 \approx 0.906$, each factor of ten in initial precision buys $\ln 10 / 0.906 \approx 2.5$ time units. Starting 10^{-9} apart, the forecast fails near $t \approx \ln(40/10^{-9})/0.906 \approx 27$, as plotted. Lorenz's rounding error of about 1.3×10^{-4} would buy about 14 time units in this system; data a millionfold better would add just 15 more.

The full Lyapunov spectrum of the Lorenz attractor, computed by the method of §84.4, is $(0.906,\ 0,\ -14.57)$. The zero exponent belongs to the direction along the flow: shifting a point slightly forward in time along its own orbit neither grows nor shrinks. And the sum is $0.906 + 0 - 14.57 = -13.67$, the divergence, as it must be: the exponents are the growth rates of the principal axes of a small blob, and the volume is their product.

The dimension of a strange attractor

Ch. 73 left a question for this chapter: why should chaotic attractors be fractals, and why can their dimension be predicted from the Lyapunov exponents?

A small ball of states near the attractor stretches at rate $\lambda_1 > 0$, keeps its length along the flow, and contracts at rate $|\lambda_3|$. Stretching makes the attractor at least two-dimensional, contraction makes it less than three-dimensional, and folding stacks the contracted sheets inside each other like the layers of a Cantor set. James Kaplan and James Yorke proposed in 1979 that the dimension balances the expanding directions against the contracting direction that just barely wins:

Kaplan–Yorke (Lyapunov) dimension

If $\lambda_1 + \cdots + \lambda_k \geq 0$ but $\lambda_1 + \cdots + \lambda_{k+1} < 0$, then

$$D_{KY} = k + \frac{\lambda_1 + \cdots + \lambda_k}{|\lambda_{k+1}|}.$$

For Lorenz, $k = 2$ and $D_{KY} = 2 + 0.906/14.57 = 2.06$: a set barely thicker than a surface, which is why it *looks* like two sheets. For the Hénon map of Ch. 73 the exponents are 0.42 and -1.62, so $D_{KY} = 1 + 0.42/1.62 = 1.26$, matching the box-counting dimension measured there. Kaplan and Yorke conjectured that D_{KY} equals the attractor's information dimension for typical systems; it has been proved in some classes of systems and is believed far more widely.

Can a simulation of chaos be trusted?

Every number in this chapter came from a computer, and every computed orbit is wrong. In double precision each RK4 step makes a relative error of about 10^{-16}, and chaos amplifies it by $e^{0.906t}$. By $t \approx 40$ the computed Lorenz orbit has no relation to the true orbit that starts from the same point. So why believe the pictures?

The answer is the **shadowing lemma**, proved by Dmitri Anosov and Rufus Bowen for systems that are *uniformly hyperbolic*: at every point of the invariant set, directions split cleanly into ones that stretch and ones that contract, at rates bounded away from zero.

> **Shadowing lemma (stated)**
>
> On a uniformly hyperbolic invariant set, for every $\varepsilon > 0$ there is a $\delta > 0$ such that every δ**-pseudo-orbit** (a sequence with $\|x_{k+1} - f(x_k)\| < \delta$ at every step, which is what a computer produces) stays within ε of some **true** orbit for all time.

So the computed orbit is not the orbit you asked for, but it is a faithful copy of an orbit from a slightly different start, and its statistics (time averages, exponents, the shape of the attractor) are the statistics of the real system. The doubling map is uniformly hyperbolic, so it is covered. The Lorenz and Hénon attractors are not: Lorenz's contains an equilibrium, and Hénon's has tangencies where stretching and contracting directions line up. For such systems shadowing has been proved only for long but finite stretches of time, with computer-assisted bounds. Warwick Tucker's 1999 proof that the Lorenz attractor really exists, answering Smale's 14th problem, was itself a rigorous computation of this kind.

> **Floating point can kill chaos outright**
>
> Iterate the doubling map in double precision from $x_0 = 0.1234567$ and after 55 steps the computer returns exactly 0, forever. Each doubling shifts a bit out of the 53-bit mantissa and shifts in a zero (Ch. 22, §22.1). Shadowing assumes small random-like errors, not arithmetic that discards information systematically. The logistic map at $r = 4$ survives, because rounding $4x(1-x)$ keeps generating new low-order bits.

Recognising chaos

Ch. 45 asked how you would recognise chaos in a population game; the same question arises for a circuit on a bench or a heartbeat on a monitor. Three tests, each with traps:

- **A positive largest Lyapunov exponent** on a bounded orbit (§84.4). The trap: noise also separates nearby trajectories, so on measured data the estimate must stay positive as you vary the embedding and time scale.
- **A broadband spectrum.** Periodic signals have line spectra (Ch. 15); chaos has a continuous one. The trap: so does noise. The spectrum can rule chaos out, never in.
- **Low-dimensional structure.** Plot a measured signal $s(t)$ against $s(t-\tau)$ and $s(t-2\tau)$. Floris Takens proved in 1981 that these **delay coordinates** reconstruct a faithful copy of a deterministic attractor: chaos appears as a thin folded object, while noise fills the space.

84.4 The algorithm

The definition of λ_1 cannot be used literally: two orbits started close together separate until they saturate at the size of the attractor, after which their distance says nothing. Giancarlo Benettin and colleagues (1976) fixed this by **renormalising**: let the separation grow for a short time, record how much it grew, shrink it back to its original size along the same direction, and repeat.

```
import numpy as np

def lorenz(u, s=10.0, r=28.0, b=8/3):
    x, y, z = u
    return np.array([s*(y - x), x*(r - z) - y, x*y - b*z])

def rk4(f, u, h):
    k1 = f(u); k2 = f(u + h/2*k1); k3 = f(u + h/2*k2); k4 = f(u + h*k3)
    return u + h/6*(k1 + 2*k2 + 2*k3 + k4)

def largest_lyapunov(f, u, h=0.01, d0=1e-8, steps_per_renorm=10, n_renorm=20000):
    for _ in range(2000):                      # discard the transient
        u = rk4(f, u, h)
    v = u + d0 * np.array([1.0, 0, 0])
    total = 0.0
    for _ in range(n_renorm):
        for _ in range(steps_per_renorm):
            u, v = rk4(f, u, h), rk4(f, v, h)
        d = np.linalg.norm(v - u)
        total += np.log(d / d0)                # growth since the last renormalisation
        v = u + (v - u) * (d0 / d)             # shrink back, keep the direction
    return total / (n_renorm * steps_per_renorm * h)

print(largest_lyapunov(lorenz, np.array([1.0, 1.0, 1.0])))   # about 0.91
```

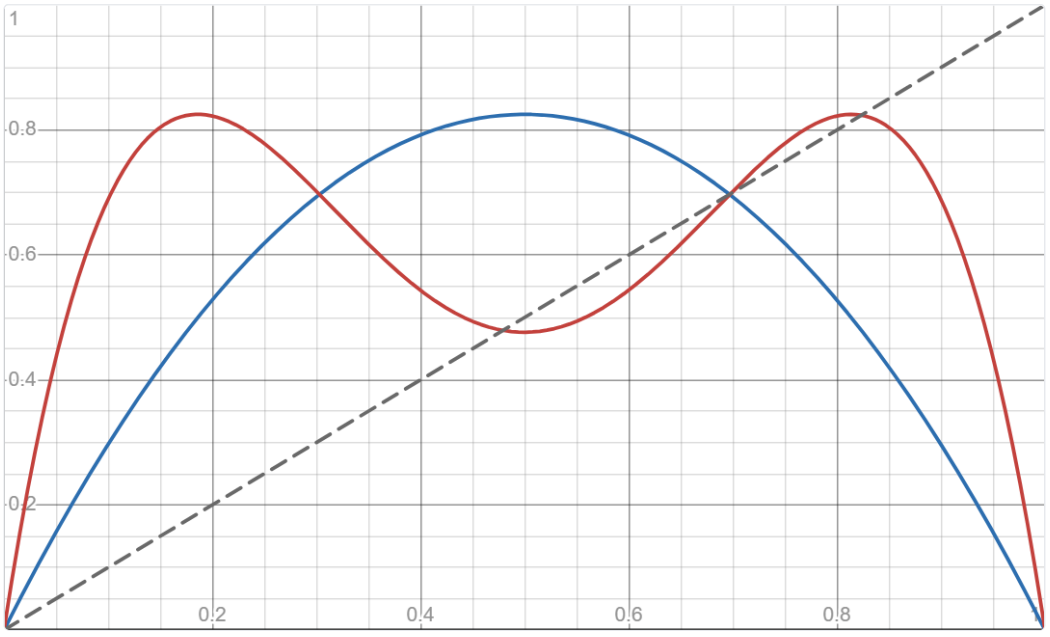
Why it works. The separation stays tiny, so it obeys the linearised dynamics, and its direction swings toward the most-stretched one. The summed logarithms are the logarithm of the product in the definition, computed without ever letting the separation saturate.

Cost and accuracy. Twice the cost of one simulation; the estimate converges roughly like $1/\sqrt{T}$ in the averaging time T. With $T = 2000$ it gives 0.909 against the accepted 0.906. For the full spectrum, evolve n separation vectors with the Jacobian and re-orthonormalise them by QR factorisation (Ch. 24, §24.3); the logarithms of the diagonal of $\mathbf{R}$ accumulate the exponents.

Failure modes. If `d0` is too large the separation leaves the linear regime and the estimate is biased low; if it is near 10^{-16} relative to the state, rounding noise dominates. Too few steps between renormalisations wastes work; too many lets the separation saturate. And a transient that is not discarded measures the approach to the attractor, not the attractor.

84.5 The experiment

First, look at the logistic map and its second iterate. A 2-cycle of f is a pair of fixed points of $f \circ f$ that are not fixed points of f.



The logistic map f (blue) and $f \circ f$ (red) at $r = 3.3$. The red curve crosses the diagonal four times: two of the crossings are the fixed points of f, the other two (near 0.479 and 0.824) are the 2-cycle in the cobweb figure. Change every 3.3 to 2.8, and to 3.9, and watch the crossings appear and the hump steepen.

Now scan the Lyapunov exponent of the logistic map across r. Where it is negative the map settles onto a stable cycle; where it is positive the orbit is chaotic.

```
import numpy as np
import matplotlib.pyplot as plt

r_min, r_max = 2.8, 4.0     # zoom in: try 3.82, 3.86 (the period-3 window)
n_r = 1500                  # number of r values
n_transient = 1000          # iterations thrown away
n_avg = 4000                # iterations averaged

r = np.linspace(r_min, r_max, n_r)
x = np.full(n_r, 0.3)
for _ in range(n_transient):
    x = r * x * (1 - x)
lam = np.zeros(n_r)
for _ in range(n_avg):
    lam += np.log(np.abs(r * (1 - 2 * x)) + 1e-300)
    x = r * x * (1 - x)
lam /= n_avg

plt.plot(r, lam, lw=0.8)
plt.axhline(0, color="k", lw=0.8)
plt.axhline(np.log(2), color="r", ls="--", lw=0.8, label="ln 2")
plt.xlabel("r"); plt.ylabel("Lyapunov exponent λ"); plt.legend()
plt.ylim(-2.5, 1); plt.title("Logistic map: λ < 0 periodic, λ > 0 chaotic"); plt.show()
for rv in (2.8, 3.5, 3.7, 3.83, 3.9, 4.0):
    print(rv, round(float(np.interp(rv, r, lam)), 3))
```

Things to notice and try:

- At $r = 2.8$ the exponent is $\ln 0.8 = -0.223$, the stable fixed point's slope. At $r = 4$ it reaches $\ln 2$.
- The plot touches zero from below at $r = 3$ and $r = 3.4495$ (where cycles double) and dives toward $-\infty$ in between, at **superstable** values where the cycle passes through $x = \tfrac{1}{2}$ and $f' = 0$.
- Above 3.5699, chaos is interrupted by **windows** of negative exponent, the widest near $r = 3.83$ (period 3, with $\lambda \approx -0.37$). Zoom in on one. How many windows can you find, and how does their width change as you go?

84.6 The challenge

1. **Intuition.** A weather service improves its observations so that initial errors fall by a factor of 1000. If the forecast horizon was 7 days and errors grow exponentially with a doubling time of 2 days, how much longer is the horizon now? Explain why a forecaster would rather halve the doubling time's effect than buy better thermometers, and what that would even mean.

2. **Derivation.** The **tent map** is $T(x) = 2x$ for $x \le \frac{1}{2}$ and $2 - 2x$ for $x > \frac{1}{2}$. Compute its Lyapunov exponent. Then show that T is conjugate to the logistic map at $r = 4$ via $x = \sin^2(\pi y/2)$, and explain why the conjugacy is consistent with both exponents being $\ln 2$.

3. **Implementation.** Extend the algorithm of §84.4 to the full Lyapunov spectrum of the Lorenz system using QR re-orthonormalisation. Check that the exponents sum to $-41/3$ and that the middle one tends to 0. Then compute D_{KY} as a function of ρ from 25 to 40.

4. **Open investigation.** Take a chaotic signal you did not generate: a dripping tap timed with a microphone, a double pendulum filmed with a phone, or a published series of sunspot numbers. Build delay coordinates and estimate the largest Lyapunov exponent. How sensitive is your estimate to the delay, the embedding dimension, and noise? Can you convince a sceptic that the signal is chaos and not noise? For short, noisy data there is no agreed method.

84.7 Loose threads

The hook asked how far ahead an error-amplifying deterministic system can be predicted. The answer is $T \approx \lambda_1^{-1} \ln(\Delta/\delta_0)$: a horizon that grows only logarithmically with precision. For Lorenz's equations each extra digit buys 2.5 time units, no more. The mechanism is stretching plus folding; the result, a strange attractor of zero volume and dimension 2.06. The doubling map showed chaos as the loss of one bit per step, and shadowing explained why computed chaos still tells the truth about statistics.

What the chapter left open:

- **The route into chaos.** The Feigenbaum mystery above is mapped out in Ch. 85.
- **Chaos and fractals.** Why D_{KY} matches a box-counting dimension is a theorem only in special cases; the measure-theoretic notions of dimension it needs are in Ch. 97.
- **Chaos with many parts.** Turbulence and weather have millions of positive exponents; how such systems can still have predictable large-scale statistics is a theme of Ch. 88.

? Why does the average exist at all?

The Lyapunov exponent is a limit of averages along a single orbit. Why should that limit exist, and why should almost every starting point give the *same* number, when the orbits themselves are wildly different? For the doubling map you can almost see it: the binary digits of a typical number behave like fair coin flips. In general, the answer is the ergodic theorem, which equates time averages with averages over the attractor, and Oseledets' theorem, which guarantees the whole spectrum. Both are theorems of measure theory, in Ch. 97.

? Two chaotic circuits in lockstep

Chaos amplifies every difference, so two chaotic systems, however alike, should drift apart. Yet Louis Pecora and Thomas Carroll showed in 1990 that two chaotic circuits, one driving the other through a single wire, can fall into perfect step and stay there, both still chaotic. How can coupling beat exponential divergence, how strong must it be, and what happens in a network of thousands of chaotic units? That is the synchronisation story of Ch. 87.

Further reading

- James Gleick, *Chaos: Making a New Science* (Viking, 1987). The history, including Lorenz's printout and Feigenbaum's numbers, told for a general reader.
- Edward N. Lorenz, "Deterministic nonperiodic flow", *Journal of the Atmospheric Sciences* 20 (1963), 130–141. The original paper, still readable.
- Steven H. Strogatz, *Nonlinear Dynamics and Chaos*, 2nd ed. (Westview, 2015), Chapters 9–10. The Lorenz equations and one-dimensional maps.
- Kathleen T. Alligood, Tim D. Sauer and James A. Yorke, *Chaos: An Introduction to Dynamical Systems* (Springer, 1996). Lyapunov exponents, conjugacies, fractal dimension and shadowing at an undergraduate level.
- Holger Kantz and Thomas Schreiber, *Nonlinear Time Series Analysis*, 2nd ed. (Cambridge University Press, 2004). Delay embedding and exponents from measured data, with all the traps.

Chapter 85 — Bifurcations: When Systems Change Their Nature

> **Level 2 · Track I — Dynamical Systems, Chaos, and Complexity Builds on:** Ch. 83 (fixed points, limit cycles), Ch. 84 (logistic map, Lyapunov exponents), Ch. 6 (implicit function theorem, Taylor series), Ch. 22 (§22.4 Newton's method) · **Leads to:** Ch. 86, Ch. 88, Ch. 89, Ch. 92 · **Related:** Ch. 73, Ch. 25, Ch. 11 **Reading time:** ~30 min

85.1 The hook

A power line feeds a city. As evening comes and air conditioners switch on, the load rises, and the voltage at the city end sags a little, as it always does: current through the line's reactance drops some of the voltage before it arrives. Operators watch the voltage. At 50% of the line's maximum transfer it is at 97% of the sending voltage. At 90% it is still at 85%. Nothing alarming.

Then the load rises a few percent more, and the voltage does not sag a little more. It slides, and keeps sliding, and the protection trips. Voltage collapse of this kind has blacked out whole regions: France in December 1978 and Tokyo on a hot July afternoon in 1987 are textbook cases.

Right up to the edge, the system looked healthy: voltage acceptable, every eigenvalue in the left half-plane. Nothing in a snapshot of the state said "you are 2% from the cliff", because the cliff was not in the state. It was in the *parameter*, the load: past a critical value, the operating point simply ceased to exist.

The model is short. A source of voltage E feeds a resistive load drawing power P through a line of reactance X. The load voltage V satisfies $E^2 = V^2 + (XP/V)^2$ (the line drop XI is at right angles to V, and $I = P/V$), which rearranges to

$$V^4 - E^2V^2 + X^2P^2 = 0 \quad\Longrightarrow\quad \frac{V^2}{E^2} = \frac{1 \pm \sqrt{1 - (P/P_{\max})^2}}{2}, \qquad P_{\max} = \frac{E^2}{2X}.$$

For $P < P_{\max}$ there are two operating voltages, a high one (normal operation) and a low one. At $P = P_{\max}$ they merge at $V = E/\sqrt{2}$. Above it there are none. Plotted, this is the **nose curve** of power engineering, and it is the subject of this chapter in disguise.

The question: **can you see the cliff coming?** If the state looks fine right up to the edge, is there *any* measurable symptom of approaching it? There is, and it follows from one line of algebra about how equilibria are born and die.

85.2 The big idea

Parameters move slowly; states move fast

Every model in Chapters 83 and 84 had knobs: damping b, Van der Pol's μ, the logistic r, Lorenz's ρ. Turn a knob slowly and usually the phase portrait just deforms. Occasionally, at special values, it changes its *nature*: two fixed points collide and annihilate, a stable point throws off a cycle, a cycle doubles. These events are **bifurcations**.

Picture a ball in a valley of a landscape you are slowly tilting. Mostly the ball just shifts. But if the valley gets shallower as you tilt, its floor and rim eventually merge, the valley vanishes, and the ball rolls somewhere completely different, and tilting back a little will not undo it.

The warning sign is in the restoring force

Just before the valley vanishes it is very flat, and a nudged ball rolls back slowly. That is the symptom: **recovery from small disturbances slows down as a bifurcation approaches**, called **critical slowing down**. Measure not where the state *is*, but how quickly it comes back when kicked. The recovery rate is an eigenvalue of the linearisation, and bifurcations happen where an eigenvalue reaches zero or a pair reaches the imaginary axis: the "non-hyperbolic, inconclusive" cases that Ch. 11, §11.9 set aside.

85.3 The mathematics

Where bifurcations can happen

Consider a one-parameter family $\dot{x} = f(x, \mu)$ with f smooth, and an equilibrium x_0 at μ_0: $f(x_0, \mu_0) = 0$.

> **Hyperbolic equilibria persist**
>
> If $f_x(x_0, \mu_0) \neq 0$, then for μ near μ_0 there is exactly one equilibrium $x^*(\mu)$ near x_0, it depends smoothly on μ, and it has the same stability as x_0.

Proof. This is the implicit function theorem (Ch. 6): $f(x, \mu) = 0$ with $f_x \neq 0$ can be solved for x as a smooth function of μ near (x_0, μ_0), uniquely in a neighbourhood, with slope $dx^*/d\mu = -f_\mu / f_x$. Stability is decided by the sign of $f_x(x^*(\mu), \mu)$, which is continuous in μ and nonzero at μ_0, so it keeps its sign nearby. ■

The same holds in n dimensions with "$f_x \neq 0$" replaced by "the Jacobian has no eigenvalue on the imaginary axis". So **local bifurcations of equilibria happen only where the linearisation has an eigenvalue with zero real part.** There are two ways to get there: a real eigenvalue passing through 0, or a complex pair crossing the imaginary axis. The first gives saddle-nodes, transcritical and pitchfork bifurcations; the second gives Hopf bifurcations.

The saddle-node: creation from nothing

The simplest way for f_x to vanish is the **saddle-node** (or **fold**) bifurcation, with normal form

$$\dot{x} = \mu - x^2.$$

For $\mu < 0$ there are no equilibria; $\dot{x} < 0$ everywhere and every state drifts off to $-\infty$. For $\mu > 0$ there are two, $x^* = \pm\sqrt{\mu}$, and since $f_x = -2x^*$, the upper one is stable (eigenvalue $-2\sqrt{\mu}$) and the lower one unstable (eigenvalue $+2\sqrt{\mu}$). At $\mu = 0$ they merge into one half-stable point and annihilate.

Why should this cartoon matter? Because it is what *every* fold looks like up close. Suppose $f(x_0, \mu_0) = 0$, $f_x = 0$, but $f_{xx} \neq 0$ and $f_\mu \neq 0$ there. Taylor expanding,

$$f(x, \mu) \approx f_\mu\,(\mu - \mu_0) + \tfrac{1}{2} f_{xx}\,(x - x_0)^2,$$

and after shifting and scaling x and μ, this is $\mu - x^2$ (or $-\mu + x^2$). The nose curve has exactly this structure at $P = P_{\max}$: near the tip, V differs from $E/\sqrt{2}$ by an amount proportional to $\pm\sqrt{P_{\max} - P}$. The square root is the fingerprint of a fold.

Critical slowing down and the bottleneck

For $\dot{x} = \mu - x^2$: (a) for $\mu > 0$, perturbations of the stable equilibrium decay like $e^{-t/\tau}$ with recovery time $\tau = 1/(2\sqrt{\mu})$, which diverges as $\mu \to 0^+$; (b) for $\mu < 0$, a state takes time $\pi/\sqrt{-\mu}$ to pass from $x = +\infty$ to $x = -\infty$, almost all of it spent near $x = 0$, the "ghost" of the vanished equilibria.

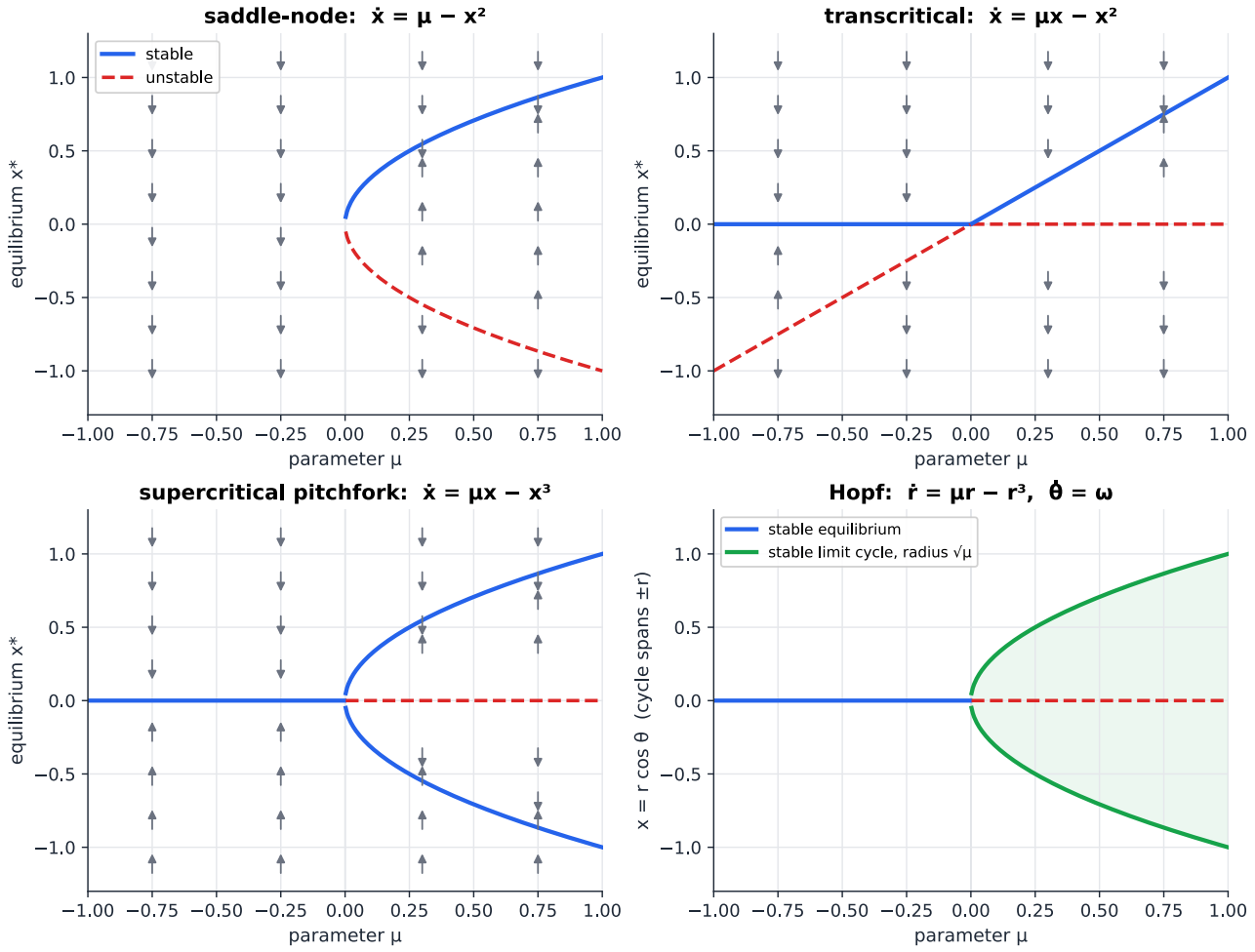
Proof. (a) The linearisation at $x^* = \sqrt{\mu}$ is $\dot{\eta} = f_x\,\eta = -2\sqrt{\mu}\,\eta$, so $\eta(t) = \eta(0)e^{-2\sqrt{\mu}\,t}$.

(b) Separate variables, writing $\mu = -a^2$:

$$T = \int_{+\infty}^{-\infty} \frac{dx}{\dot{x}} = \int_{+\infty}^{-\infty} \frac{dx}{-(a^2 + x^2)} = \int_{-\infty}^{\infty} \frac{dx}{a^2 + x^2} = \frac{1}{a}\Big[\arctan\frac{x}{a}\Big]_{-\infty}^{\infty} = \frac{\pi}{a} = \frac{\pi}{\sqrt{-\mu}}.$$

The integrand is largest near $x = 0$, where $\dot{x} = \mu$ is tiny. Half the passage time is spent in $|x| < a$, an interval that shrinks to nothing as $\mu \to 0^-$. ■

Part (a) answers the hook. At $\mu = 0.01$ the recovery time is 5; at $\mu = 0.0001$ it is 50. The equilibrium moves only from 0.1 to 0.01, easy to miss, while the sluggishness grows tenfold, easy to measure from how the system recovers from the random kicks that loads give it all the time. Voltage-stability monitors use this idea in the form of the smallest eigenvalue or singular value of the power-flow Jacobian, which tends to zero at the nose. Part (b) says that just past the cliff the system lingers in the bottleneck, one reason a collapse can take seconds to minutes to unfold: a window for emergency action.



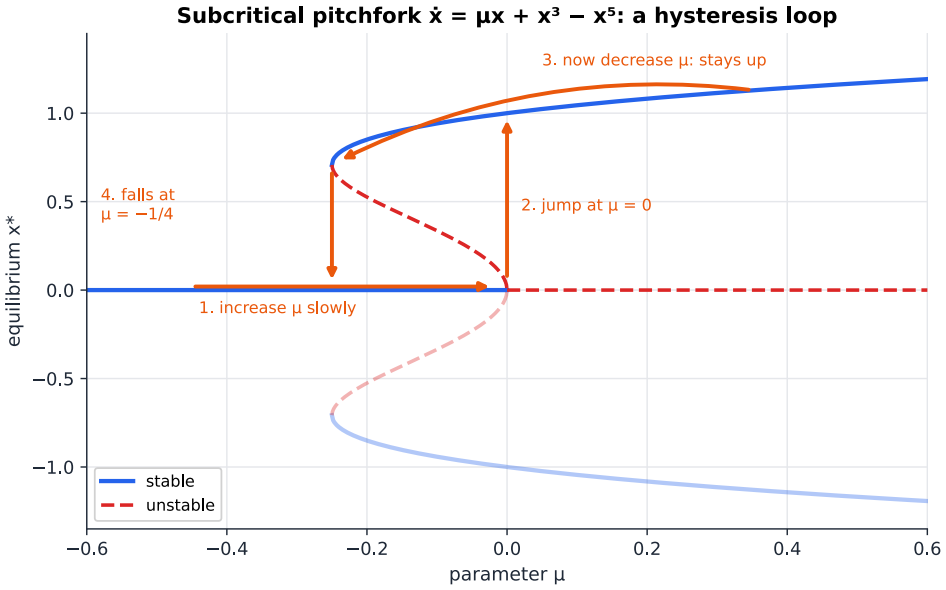
Bifurcation diagrams: equilibria (and, for Hopf, the cycle amplitude) against the parameter μ. Solid blue: stable; dashed red: unstable; grey arrows: the direction of flow. Saddle-node: a stable–unstable pair is created from nothing. Transcritical: two branches cross and exchange stability. Pitchfork: a symmetric equilibrium loses stability to two new stable ones. Hopf: a stable spiral gives birth to a limit cycle whose radius grows like $\sqrt{\mu}$.

Transcritical and pitchfork: when something forbids creation from nothing

Sometimes an equilibrium cannot disappear, because the model forbids it. Zero population, zero current or a perfectly upright beam are equilibria for *every* parameter value. Then the zero eigenvalue produces a different event.

- **Transcritical**, $\dot{x} = \mu x - x^2$: the branches $x = 0$ and $x = \mu$ cross at $\mu = 0$ and exchange stability. Laser and epidemic ($R_0 = 1$) thresholds have this form: below, "nothing happening" is stable; above, the nonzero state takes over.
- **Pitchfork**, $\dot{x} = \mu x - x^3$, needs a symmetry $x \to -x$. For $\mu > 0$ the symmetric state becomes unstable and two mirror-image stable states $\pm\sqrt{\mu}$ appear. A column buckles left or right, a powered-up flip-flop settles to 0 or 1, and imperfections too small to see decide which.

The cubic term $-x^3$ makes the pitchfork **supercritical**: the new states start at zero amplitude and grow smoothly. Flip its sign and the bifurcation turns **subcritical** and dangerous. The standard model is $\dot{x} = \mu x + x^3 - x^5$, in which the quintic term restrains the growth at large amplitude:



*The subcritical pitchfork $\dot{x} = \mu x + x^3 - x^5$. Raising μ past 0 destabilises $x = 0$ with no nearby stable state to go to, so the system jumps to the large-amplitude branch. Lowering μ again does not bring it back at 0: it stays up until the fold at $\mu = -1/4$, then falls. Between $-1/4$ and 0 both states are stable: a **hysteresis** loop.*

The nonzero equilibria satisfy $\mu + x^2 - x^4 = 0$, so $x^2 = \left(1 \pm \sqrt{1 + 4\mu}\right)/2$, which exist only for $\mu \geq -1/4$: the jump-back point is itself a saddle-node. Hysteresis is a memory: in the overlap, the state depends on history, not just on the present value of the knob. A Schmitt trigger is built to have it; a buckling shell or a collapsing grid has it whether you like it or not.

Hopf: the birth of a cycle

Ch. 83 left a mystery: as Van der Pol's μ is turned down through zero the limit cycle vanishes, so how is it born? When a complex pair of eigenvalues crosses the imaginary axis, the linearisation turns from a stable spiral to an unstable one. The nonlinear terms decide what happens to the outward-spiralling orbits. In polar coordinates the normal form is

$$\dot{r} = \mu r - r^3, \qquad \dot{\theta} = \omega.$$

The angle just turns at rate ω. The radius obeys a pitchfork-like equation on $r \geq 0$: for $\mu < 0$, $r = 0$ is the only equilibrium and it is stable; for $\mu > 0$ it is unstable, and $r^* = \sqrt{\mu}$ is a stable equilibrium of the radius, that is, a circular **limit cycle** of radius $\sqrt{\mu}$. Its stability follows from $\frac{d}{dr}(\mu r - r^3) = \mu - 3r^2 = -2\mu < 0$ at $r^* = \sqrt{\mu}$. Hopf's theorem (1942) says this is generic: when a pair crosses the axis with nonzero speed and a certain cubic coefficient (the **first Lyapunov coefficient**) is negative, a stable cycle of amplitude proportional to $\sqrt{\mu - \mu_c}$ is born, with period near $2\pi/\omega$.

> **Oscillator start-up, and the Van der Pol mystery solved**
>
> Take $\ddot{x} - (\mu - x^2)\dot{x} + x = 0$: the damping $-(\mu - x^2)$ is negative for small x once $\mu > 0$. The linearisation at the origin has eigenvalues $\mu/2 \pm j\sqrt{1 - \mu^2/4}$, crossing the imaginary axis at $\mu = 0$ with $\omega = 1$. Averaging the energy balance over one cycle of $x \approx a\cos t$ gives $\dot{a} = \frac{a}{2}\left(\mu - a^2/4\right)$, a Hopf normal form with stable amplitude
>
> $$a^* = 2\sqrt{\mu}.$$
>
> Numerically, the cycle amplitude is 0.200 at $\mu = 0.01$, 0.400 at $\mu = 0.04$ and 0.632 at $\mu = 0.1$, against $2\sqrt{\mu} = 0.2,\ 0.4,\ 0.632$.
>
> Now substitute $x = \sqrt{\mu}\,X$ (for $\mu > 0$). Every term picks up a factor $\sqrt{\mu}$, and the equation becomes $\ddot{X} - \mu(1 - X^2)\dot{X} + X = 0$: exactly the Van der Pol equation of Ch. 83. So Van der Pol's amplitude of 2 in X is the amplitude $2\sqrt{\mu}$ in x. The cycle is born at zero size at $\mu = 0$ and grows like the square root; the way Van der Pol wrote the equation simply hides this by measuring amplitude in units that shrink with μ.

This is how an electronic oscillator starts: past the gain where net damping changes sign, the amplitude grows from noise to a value proportional to the square root of the excess gain. Linear analysis tells you *whether* it starts, never *at what amplitude*, as the engineering note of Ch. 11, §11.9 warned.

If the first Lyapunov coefficient is positive the Hopf bifurcation is **subcritical**: an *unstable* cycle shrinks onto the equilibrium and the state jumps to whatever large-amplitude motion is available. Aircraft wing flutter can behave this way. So does the Lorenz system: at $\rho_H = \sigma(\sigma+\beta+3)/(\sigma-\beta-1) \approx 24.74$ the equilibria $C_\pm$ lose stability in a subcritical Hopf bifurcation, and there is nothing nearby to catch the orbit except the strange attractor of Ch. 84.

Period doubling: the bifurcation that only maps can do

A map $x_{n+1} = f(x_n, r)$ has one more way to lose stability. Its fixed point is stable while $|f'(x^*)| < 1$ (Ch. 83), so stability can be lost at $f' = +1$ (a saddle-node or relative, as for flows) or at $f' = -1$, where orbits overshoot with ever-growing alternation. That is a **flip** or **period-doubling bifurcation**, and it creates a 2-cycle.

The first period doubling of the logistic map

For $f(x) = rx(1-x)$, the fixed point $x^* = 1 - 1/r$ is stable for $1 < r < 3$ and loses stability at $r = 3$ with $f'(x^*) = -1$. For $r > 3$ there is a 2-cycle

$$p, q = \frac{r+1 \pm \sqrt{(r-3)(r+1)}}{2r},$$

which is stable for $3 < r < 1+\sqrt{6} \approx 3.4495$.

Proof. At the fixed point $f'(x^*) = r(1-2x^*) = r\big(1-2+2/r\big) = 2-r$, which lies in $(-1, 1)$ exactly when $1 < r < 3$, and equals -1 at $r = 3$.

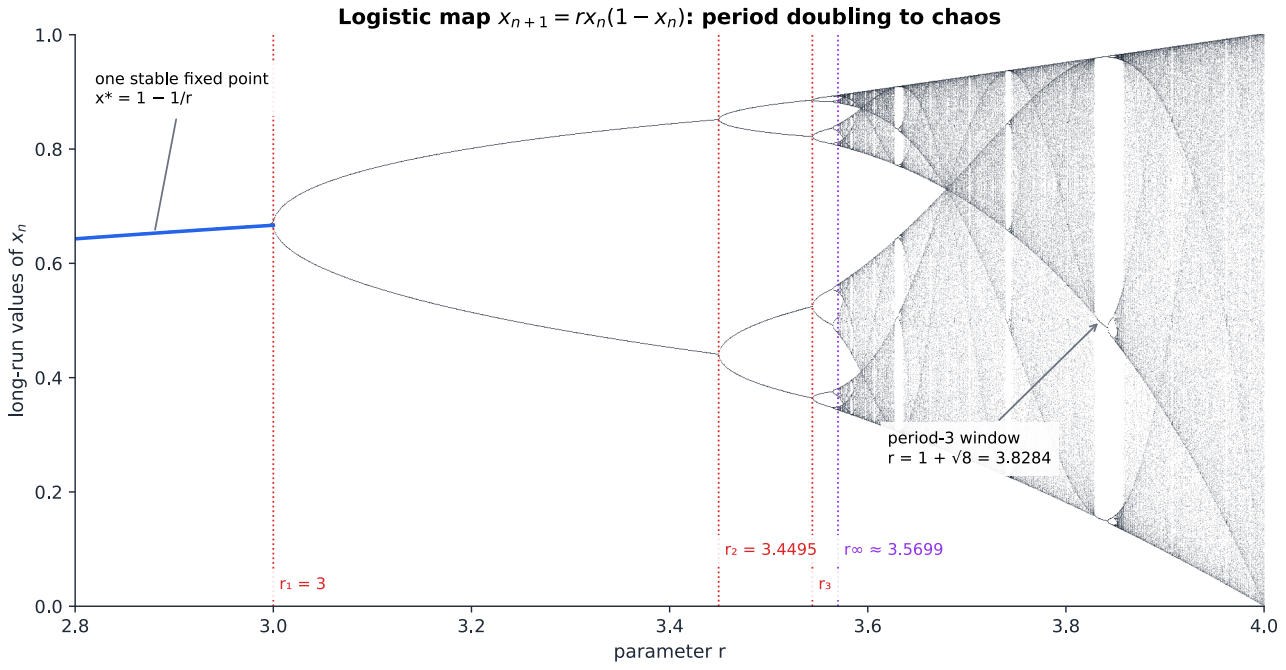
A 2-cycle is a pair with $f(p) = q$, $f(q) = p$, $p \neq q$. Subtracting $q = rp(1-p)$ from $p = rq(1-q)$ gives $p - q = r(q-p)(1-p-q)$, and dividing by $p - q \neq 0$ gives $p + q = (r+1)/r$. Adding the two equations gives $p + q = r(p+q) - r(p^2+q^2)$; substituting $p^2 + q^2 = (p+q)^2 - 2pq$ and $p + q = (r+1)/r$ gives $pq = (r+1)/r^2$. So p and q are the roots of $r^2 z^2 - r(r+1)z + (r+1) = 0$, which is the formula above. They are real and distinct exactly when $(r-3)(r+1) > 0$, that is, $r > 3$.

The cycle is a fixed point of $f \circ f$, with multiplier $(f\circ f)'(p) = f'(q)f'(p) = r^2(1-2p)(1-2q)$ by the chain rule. Expanding with the values of $p+q$ and pq:

$$r^2\big(1 - 2(p+q) + 4pq\big) = r^2 - 2r(r+1) + 4(r+1) = -r^2 + 2r + 4.$$

This equals $+1$ at $r = 3$ (the cycle is born from the fixed point) and decreases as r grows; it reaches -1 when $r^2 - 2r - 5 = 0$, at $r = 1+\sqrt{6}$. In between, its absolute value is below 1 and the cycle is stable. ■

At $r = 1+\sqrt{6}$ the 2-cycle's own multiplier hits -1, so it flips in turn and gives birth to a 4-cycle. Then 8, 16, 32: the **period-doubling cascade**.



The long-run values of the logistic map for each r (300 iterates plotted after 1000 discarded). One stable fixed point until $r_1 = 3$, a 2-cycle until $r_2 = 1+\sqrt{6}$, then 4, 8, ..., accumulating at $r_\infty \approx 3.5699$. Beyond, chaos, interrupted by windows of periodic behaviour; the widest is the period-3 window that opens at $r = 1+\sqrt{8} \approx 3.8284$.

The doublings come faster and faster. The first few happen at

n	r_n (period 2^n born)	$\frac{r_n - r_{n-1}}{r_{n+1} - r_n}$
1	3.000000	
2	3.449490	4.751
3	3.544090	4.656
4	3.564407	4.668
5	3.568759	4.665
∞	3.569946	$\delta = 4.6692\ldots$

(The last ratio uses $r_6 = 3.569692$.) The gaps shrink geometrically with ratio approaching **Feigenbaum's constant** $\delta = 4.6692$, so the infinitely many doublings fit into a finite interval and the period becomes infinite at r_∞. Past r_∞ the Lyapunov exponent of Ch. 84 turns positive (except in windows), and the orbit is chaotic.

Why does the cascade repeat? Look at $f \circ f$ near one point of the 2-cycle, just after it is born. It has a hump and a fixed point, and, flipped over and magnified, it looks like a small copy of f itself near its own fixed point. As r increases, the small copy goes through the same story as the original: its fixed point flips, which for f is the birth of the 4-cycle. Then f^4 contains a smaller copy again, and so on. Each copy is the previous one rescaled by a factor that tends to $\alpha \approx -2.5029$ in x and $\delta \approx 4.669$ in r. Feigenbaum turned this "copy of a copy" into an equation for the shape of the limiting map. This is where the story that Ch. 73 pointed to begins.

The period-3 window shows a different birth. At $r = 1 + \sqrt{8}$, the third iterate f^3 develops three new pairs of fixed points by saddle-node bifurcations: a stable 3-cycle and an unstable one, out of nothing, inside chaos. Just below $1 + \sqrt{8}$ the orbit lingers near the ghost of the 3-cycle for long stretches, then bursts out into chaos and back: **intermittency**, the bottleneck of part (b) of the saddle-node theorem, seen in a map.

> **? Why the same 4.669 everywhere?**
>
> The logistic map is not special. Any smooth map with a single quadratic hump, sin, $xe^{r(1-x)}$, a fitted model of a dripping tap, produces the same ratio $\delta = 4.6692\ldots$ and the same rescaling $\alpha = -2.5029\ldots$. Albert Libchaber measured period doubling in convecting liquid helium in 1980 and found ratios consistent with δ. The universality is explained by treating "take $f \circ f$ and rescale" as an operator on the space of all maps, and showing that it has a fixed point whose unstable eigenvalue is δ. That renormalisation idea, borrowed from the physics of phase transitions, is the subject of Ch. 92.

85.4 The algorithm

Two computational tools find bifurcations, and they answer different questions.

Brute force (the bifurcation diagram above) iterates for each parameter value, discards a transient and plots what remains. It shows attractors, chaotic ones included, but never unstable branches, misses attractors your start is not attracted to, and blurs near bifurcations, where transients become very long.

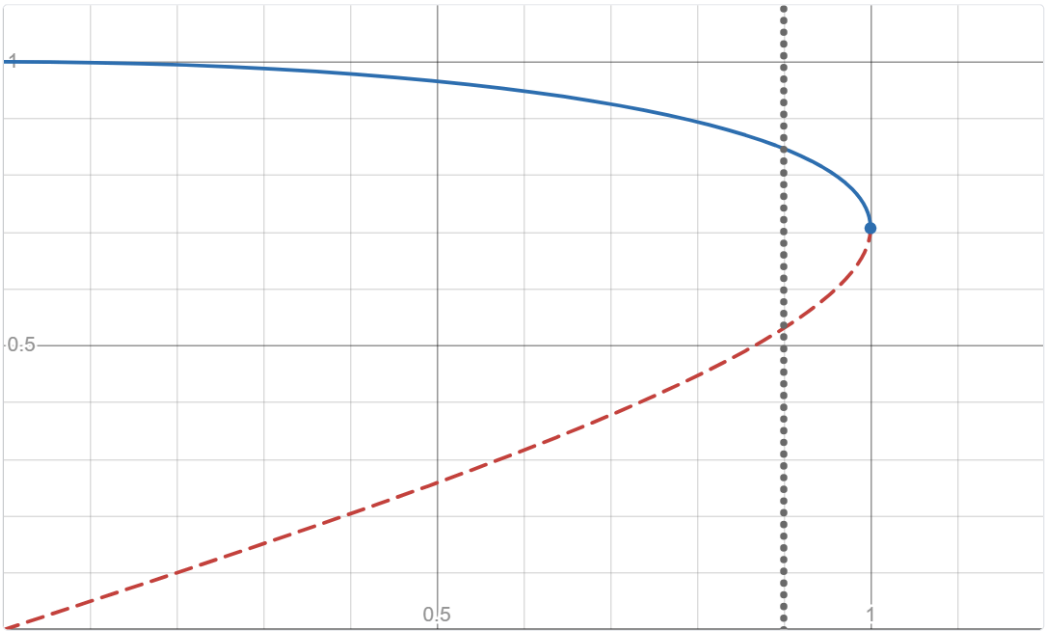
Continuation follows a branch of equilibria $g(\mathbf{x}, \mu) = \mathbf{0}$, stable or not, by stepping μ and correcting with Newton's method (Ch. 22, §22.4), monitoring the Jacobian's eigenvalues for sign changes. Stepping μ naively fails at a fold, where the branch turns back. **Pseudo-arclength continuation** parametrises the curve by its arclength s and treats μ as one more unknown. The code below traces the whole nose curve of §85.1, round the tip and down the lower branch, with $E = X = 1$ and load $p = P/P_{\max}$, so that $g(V, p) = V^4 - V^2 + p^2/4$.

```
import numpy as np

def g(V, p):                       # equilibrium condition of the line + load, E = X = 1
    return V**4 - V**2 + p**2 / 4

def grad(V, p):
    return np.array([4*V**3 - 2*V, p / 2])        # (dg/dV, dg/dp)

def continuation(V, p, ds=0.02, steps=80):
    t = np.array([0.0, 1.0])                      # initial tangent (dV, dp): increase load
    out = [(p, V)]
    for _ in range(steps):
        Vp, pp = V + ds * t[0], p + ds * t[1]      # predictor: step along the tangent
        for _ in range(20):                       # corrector: Newton on g = 0 plus arclength
            G = np.array([g(Vp, pp), t[0]*(Vp - V) + t[1]*(pp - p) - ds])
            J = np.array([grad(Vp, pp), t])
            dV, dp = np.linalg.solve(J, -G)
            Vp, pp = Vp + dV, pp + dp
        gV, gp = grad(Vp, pp)
        t_new = np.array([-gp, gV]); t_new /= np.linalg.norm(t_new)
        if t_new @ t < 0: t_new = -t_new           # keep moving the same way round the curve
        if t_new[1] * t[1] < 0:
            print(f"fold: load p = {pp:.4f}, voltage V = {Vp:.4f}")
        V, p, t = Vp, pp, t_new
        out.append((p, V))
    return np.array(out)

curve = continuation(V=1.0, p=0.0)
print(curve[::10].round(3))
```

It reports the fold one step after passing it, at $p = 0.9986$, $V = 0.688$ (exact: $p = 1$, $V = 0.707$); bisecting on the sign change of dp/ds pins it down. The augmented Jacobian (the gradient stacked on the tangent) stays invertible at the fold even though $\partial g/\partial V = 0$ there, which is the whole point. Each step costs one small linear solve per Newton iteration; for a grid model with 10^5 buses the same method uses sparse solvers (Ch. 24, §24.5). Packages such as AUTO and MatCont add automatic detection of Hopf points and continuation of periodic orbits.

85.5 The experiment

First the nose curve itself, both branches, with load on the horizontal axis as a fraction of $P_{\max}$ and voltage as a fraction of E.



The nose curve: normal operation (blue) and the low-voltage branch (red) merge at the fold $p = 1$. Move the dotted line from 0.9 toward 1 and watch how little the blue voltage changes compared with how fast the gap to the red branch closes; that gap is proportional to $\sqrt{1-p}$ near the tip.

Now draw the logistic bifurcation diagram yourself, and zoom.

```
import numpy as np
import matplotlib.pyplot as plt

r_min, r_max = 2.8, 4.0      # zoom: try 3.54, 3.575 or 3.82, 3.86
x_min, x_max = 0.0, 1.0      # and the matching x-range, e.g. 0.88, 0.90
n_r = 2000                   # parameter values
n_transient = 2000           # iterations discarded (raise it near bifurcations)
n_plot = 400                 # iterations plotted

r = np.linspace(r_min, r_max, n_r)
x = np.full(n_r, 0.5)
for _ in range(n_transient):
    x = r * x * (1 - x)
R, X = [], []
for _ in range(n_plot):
    x = r * x * (1 - x)
    keep = (x >= x_min) & (x <= x_max)
    R.append(r[keep]); X.append(x[keep])
plt.plot(np.concatenate(R), np.concatenate(X), ",k", alpha=0.3)
plt.xlim(r_min, r_max); plt.ylim(x_min, x_max)
plt.xlabel("r"); plt.ylabel("x"); plt.title("logistic map"); plt.show()
```

Things to try:

- Zoom into $3.54 \le r \le 3.575$, $0.88 \le x \le 0.90$. The picture repeats: a shrunken copy of the whole diagram. That is renormalisation made visible.
- Zoom into the period-3 window, $3.82 \le r \le 3.86$. Inside it, find a second period-doubling cascade (periods 6, 12, 24) and measure the first two ratios of its gaps. Are they close to 4.669?
- Set `n_transient = 50`. Near $r = 3$ and $r = 3.4495$ the branches smear out: critical slowing, seen as a slow-to-settle transient.

85.6 The challenge

1. **Intuition.** A thermostat with hysteresis switches the heater on at $19\,^\circ$C and off at $21\,^\circ$C. Sketch its "bifurcation diagram" (heater state against temperature) and explain why the two thresholds play the role of the two folds in the subcritical pitchfork figure. What would happen with no hysteresis?

2. **Derivation.** For $\dot{x} = \mu - x^2 + \varepsilon x^3$ with small $\varepsilon > 0$, show that a second fold appears far from the origin, find its location to leading order in ε, and sketch the resulting S-shaped curve. Then show that the pitchfork $\dot{x} = \mu x - x^3$ with a small imperfection h, $\dot{x} = h + \mu x - x^3$, no longer has a pitchfork at all but a smooth branch and a separate fold, and find the curve in the (μ, h) plane where folds occur.

3. **Implementation.** Modify the continuation code to trace the equilibria of $\dot{x} = \mu x + x^3 - x^5$ (equilibrium condition only, as a curve in (μ, x)) starting from $(\mu, x) = (0.5, 1.2)$ and moving toward smaller μ. Detect both folds and record the sign of $\partial f/\partial x$ on each part of the curve. Compare with the hysteresis figure.

4. **Open investigation.** Critical slowing should be visible in noisy data as rising variance and rising lag-1 autocorrelation as a fold approaches. Simulate $dx = (\mu - x^2)\,dt + \sigma\,dW$ with μ decreasing slowly toward 0 (Euler–Maruyama, Ch. 36), and compute both indicators in a sliding window. How early, and how reliably, do they warn? How do false alarms depend on the window length, the noise level and the speed at which μ changes? For real ecosystems and climate records this remains an open, much-debated question.

85.7 Loose threads

The hook asked whether an approaching cliff can be detected while the state still looks healthy. It can: near a saddle-node the equilibrium moves only like $\sqrt{\mu}$, but recovery time grows like $1/(2\sqrt{\mu})$. Bifurcations of equilibria happen only where an eigenvalue reaches the imaginary axis: a zero eigenvalue gives folds, transcritical and pitchfork bifurcations; a crossing pair gives Hopf's cycle of amplitude $\sqrt{\mu}$, which solved the Van der Pol mystery of Ch. 83. Maps can also flip, and the logistic map's flips cascade at a universal rate into chaos.

What the chapter left open:

- **Global bifurcations.** Cycles can also die by colliding with distant saddles (homoclinic bifurcations), which is how the Lorenz system's chaos is first assembled near $\rho \approx 13.9$; see Guckenheimer and Holmes below.
- **Two knobs at once.** With two parameters, curves of folds can meet at a **cusp**, the organising centre of hysteresis, and Hopf and fold curves can meet at points where cycles, tori and chaos are born together.
- **From local rules to global patterns.** In Ch. 86, the rule is discrete in space too, and changing one bit of the rule changes the world it creates.

? Tipping points in a noisy world

Critical slowing is clean in a deterministic model, but real systems are noisy and their parameters drift while you watch. Ecologists have reported rising variance and autocorrelation before lakes flipped from clear to turbid, and climate scientists have searched for the same signals before abrupt changes in ice-core records. When do such signals warn reliably, when do they cry wolf, and can noise tip a system over the rim before the valley vanishes, with no warning at all? These questions are taken up in Ch. 88.

? Bifurcations that draw pictures

Every bifurcation in this chapter happened to a handful of numbers. Let the state be a whole field, a chemical concentration over a surface, and an instability can choose a *shape*: Alan Turing showed in 1952 that two chemicals that react and diffuse at different rates can bifurcate from a uniform grey into spots or stripes of a definite wavelength. Why does diffusion, which smooths everything, create patterns, and how is the wavelength selected? That is a bifurcation of a partial differential equation, in Ch. 89.

Further reading

- Steven H. Strogatz, *Nonlinear Dynamics and Chaos*, 2nd ed. (Westview, 2015), Chapters 3, 8 and 10. Saddle-node, pitchfork and Hopf bifurcations and the logistic map, at the level of this chapter.
- Yuri A. Kuznetsov, *Elements of Applied Bifurcation Theory*, 4th ed. (Springer, 2023). Normal forms, the Hopf theorem with the Lyapunov coefficient, and numerical continuation.
- John Guckenheimer and Philip Holmes, *Nonlinear Oscillations, Dynamical Systems, and Bifurcations of Vector Fields* (Springer, 1983). The classic graduate text, including global bifurcations and the Lorenz system.
- Thierry Van Cutsem and Costas Vournas, *Voltage Stability of Electric Power Systems* (Springer, 1998). Nose curves, saddle-node bifurcations and collapse mechanisms in real grids.
- Marten Scheffer et al., "Early-warning signals for critical transitions", *Nature* 461 (2009), 53–59. Critical slowing down as a warning in ecology and climate.

Chapter 86 — Cellular Automata and Emergent Worlds

> **Level 2 · Track I — Dynamical Systems, Chaos, and Complexity Builds on:** Ch. 17, Ch. 22, Ch. 24, Ch. 83 · **Leads to:** Ch. 87, Ch. 88 · **Related:** Ch. 30, Ch. 73, Ch. 74, Ch. 81, Ch. 84 **Reading time:** ~25 min

86.1 The hook

Take a row of light bulbs, each either on or off. Once a second, every bulb looks at itself and its two immediate neighbours, and decides whether to be on or off in the next second. All bulbs use the same rule. There is no conductor, no memory beyond one second, no randomness, and nothing any bulb can see except three bits.

There are only $2^3 = 8$ possible neighbourhoods, and a rule is just a choice of on or off for each, so there are exactly $2^8 = 256$ such rules. You could print every one on a single sheet of paper. Surely, with so little to work with, the behaviour must be boring: everything dies out, or everything lights up, or some simple stripe repeats forever.

Start with a single bulb lit and run three of those rules. One draws a perfect fractal, the Sierpinski triangle, with no sign that anyone asked for it. One produces a pattern so disordered that its centre column was used for years as the random-number generator inside a commercial computer-algebra system. And one, which differs from its neighbours in the table by only a couple of bits, was proved in 2004 to be able to compute *anything any computer can compute*. Three bits of input per cell, and somewhere among the 256 rules sits a universal computer.

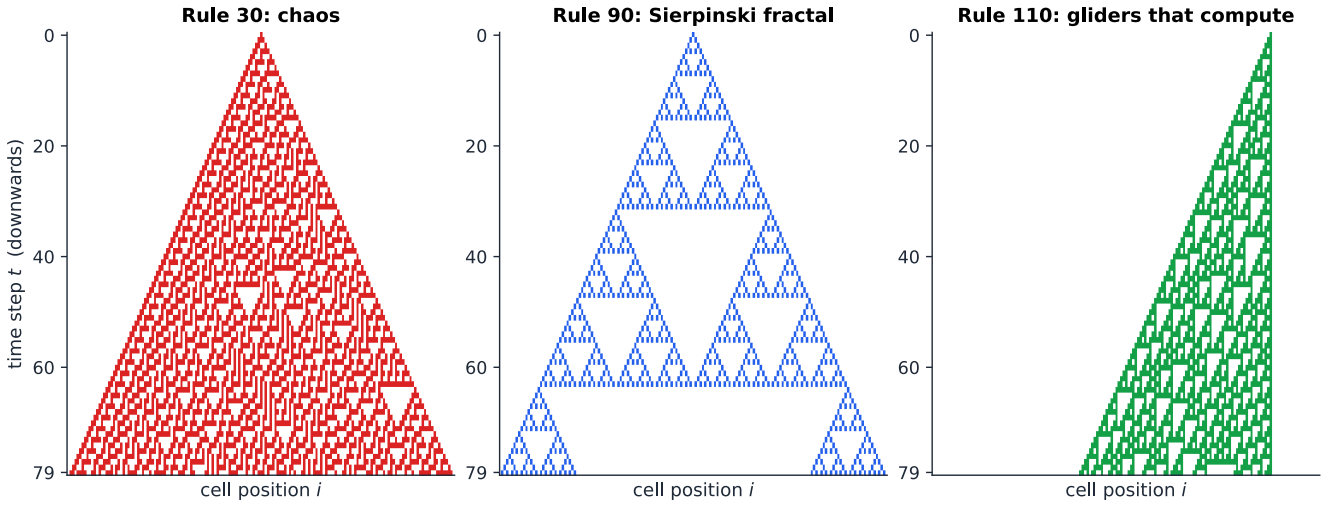
That is the paradox this chapter is about. How do rules that are purely local, tiny and deterministic produce patterns that are global, enormous and, in a precise sense, unpredictable? And if the rule is that simple, why can't we just *solve* it, the way we solve a linear difference equation?

86.2 The big idea

A **cellular automaton** (CA) is a dynamical system (Ch. 83) stripped to the bone. Space is a grid of cells. Time is a tick of a clock. Each cell holds one of a few states, and at every tick all cells update *simultaneously* by the same local rule, reading only the states in a small neighbourhood. That's the whole definition.

The analogy worth carrying is a stadium wave. Nobody in the crowd sees the wave. Each person follows one local rule ("stand up shortly after the person on your left stands, then sit down"), and a large-scale object, the wave, moves around the stadium at a definite speed, with a definite width, and survives collisions with stragglers. The wave isn't written anywhere in the rule. It is **emergent**: a pattern that belongs to the collective and has properties (speed, shape, persistence) that no individual has.

Cellular automata let us study emergence with nothing hidden. In a fluid or a brain we never know whether the interesting behaviour comes from details we left out. In a CA there are no details to leave out. If something interesting appears, the rule alone did it.



Three elementary cellular automata started from one live cell. Each row is the whole world at one instant; time runs downwards. Rule 30 (left) looks random, rule 90 (centre) is the Sierpinski triangle, and rule 110 (right) grows a lattice of interacting structures that can be made to compute.

Look at the figure the way a physicist looks at a bubble chamber. A one-dimensional CA leaves its entire history as a two-dimensional picture, the **space–time diagram**. Every pattern in that picture is something the rule did. The triangles in rule 90 nest inside one another forever; the speckle in rule 30 never settles; the diagonal stripes in rule 110 are moving particles, sometimes called *gliders*, that collide and either annihilate or create new ones.

Three intuitions carry the rest of the chapter.

1. **Locality limits speed.** A cell can only hear from its neighbours, so information moves at most one cell per tick. Every CA has a built-in speed of light, and the edges of the triangles above are exactly that light cone.
2. **Linearity makes things solvable.** Rule 90 says "new state = left XOR right". XOR is addition modulo 2, so rule 90 is a *linear* system, and linear systems obey superposition. That's why we will be able to predict its every cell exactly, in the same way we solve a linear difference equation with the Z-transform (Ch. 17).

3. **Nonlinearity plus locality breeds complexity.** Rule 30 is "left XOR (centre OR right)". That one OR breaks linearity, and with it every shortcut. Rule 110 goes further: its gliders are rich enough to store and process information.

86.3 The mathematics

Elementary cellular automata

Elementary cellular automaton

An **elementary cellular automaton** (ECA) is a map on configurations $x = (x_i)_{i \in \mathbb{Z}}$ with $x_i \in \{0,1\}$, given by

$$x_i^{t+1} = f\left(x_{i-1}^t,\, x_i^t,\, x_{i+1}^t\right)$$

for a fixed **local rule** $f : \{0,1\}^3 \to \{0,1\}$, applied to every i at once. On a finite ring of n cells, indices are read modulo n (periodic boundary).

Wolfram's numbering turns the rule into a byte. List the neighbourhoods in descending binary order, $111, 110, \dots, 000$, write the output bit under each, and read the eight output bits as a binary number. For rule 30:

neighbourhood (x_{i-1}, x_i, x_{i+1})	**111**	**110**	**101**	**100**	**011**	**010**	**001**	**000**
new x_i	0	0	0	1	1	1	1	0

and $00011110_2 = 30$. Equivalently, bit k of the rule number is the output for the neighbourhood whose binary value is k; that is exactly how the code in §86.4 reads it. In Boolean form, writing $\oplus$ for XOR,

$$\begin{aligned} \text{rule 30:}\quad & x_i^{t+1} = x_{i-1} \oplus (x_i \vee x_{i+1}), \\ \text{rule 90:}\quad & x_i^{t+1} = x_{i-1} \oplus x_{i+1}, \\ \text{rule 110:}\quad & x_i^{t+1} = (x_i \oplus x_{i+1}) \vee (x_i \wedge \neg x_{i-1}), \end{aligned}$$

each of which you can check against its eight-entry table in a minute (all three were checked by brute force).

Two symmetries cut the 256 rules down. Mirror reflection (swap left and right) and complementation (swap 0 and 1) map rules to rules with "the same" behaviour, and the 256 rules fall into 88 essentially different classes. Rule 110, for instance, is the mirror image of rule 124.

Result 1: the speed of light

Light cone

If two configurations x and y agree on the cells $a \le i \le b$ at time 0, then under any ECA they agree on $a + t \le i \le b - t$ at time t. Equivalently, the state of cell i at time t depends only on cells $i - t, \dots, i + t$ at time 0.

Proof. Induction on t. The case $t = 0$ is the hypothesis. If x^t and y^t agree on $[a + t,\ b - t]$, take any i with $a + t + 1 \le i \le b - t - 1$. Its neighbours $i - 1, i, i + 1$ all lie in $[a + t,\ b - t]$, so the local rule sees identical inputs and $x_i^{t+1} = y_i^{t+1}$. $\blacksquare$

That is why every pattern in the figure lives inside a triangle of slope one: a single live cell can't influence anything outside its cone. Rule 30 fills its cone entirely; rule 110 fills only the left half, because its rule happens to send information leftwards faster than rightwards. The same idea, a finite propagation speed set by the stencil, is the CFL condition for explicit PDE solvers (Ch. 22). A CA is, in a sense, a PDE solver that forgot to have a PDE.

Result 2: rule 90 is Pascal's triangle in disguise

Rule 90 is linear over the two-element field $\mathbb{F}_2 = \{0,1\}$ with addition $\oplus$. If x and y are two configurations, then applying rule 90 to $x \oplus y$ gives the XOR of the two separate results, because $(x_{i-1} \oplus y_{i-1}) \oplus (x_{i+1} \oplus y_{i+1}) = (x_{i-1} \oplus x_{i+1}) \oplus (y_{i-1} \oplus y_{i+1})$. Superposition holds, so it is enough to understand one live cell.

Encode a configuration as a Laurent polynomial with coefficients in $\mathbb{F}_2$, $X(z) = \sum_i x_i z^i$, the same bookkeeping trick as a generating function or a Z-transform. Shifting left and right is multiplication by z and z^{-1}, so one step of rule 90 is

$$X^{t+1}(z) = \left(z + z^{-1}\right) X^t(z), \qquad \text{hence} \qquad X^t(z) = \left(z + z^{-1}\right)^t X^0(z).$$

The single-seed solution of rule 90

Starting from one live cell at position 0, cell k is alive at time t if and only if $t + k$ is even and $\binom{t}{(t+k)/2}$ is odd. The number of live cells at time t is exactly

$$N(t) = 2^{s(t)},$$

where $s(t)$ is the number of 1s in the binary expansion of t.

Proof. With $X^0 = 1$, the binomial theorem gives $X^t(z) = \sum_{m=0}^{t} \binom{t}{m} z^{2m-t}$ with coefficients read modulo 2. Setting $k = 2m - t$ gives the first claim.

For the count, use the "freshman's dream", which is true in characteristic 2: $(a + b)^2 = a^2 + 2ab + b^2 = a^2 + b^2$, and by repeating it, $(a + b)^{2^m} = a^{2^m} + b^{2^m}$. Write $t = 2^{m_1} + 2^{m_2} + \cdots + 2^{m_s}$ with distinct $m_1 > m_2 > \cdots > m_s$. Then

$$\left(z + z^{-1}\right)^t = \prod_{r=1}^{s} \left(z^{2^{m_r}} + z^{-2^{m_r}}\right).$$

Expanding the product picks one of two signs from each factor, giving 2^s terms $z^{\pm 2^{m_1} \pm 2^{m_2} \pm \cdots \pm 2^{m_s}}$. These exponents are all distinct: two different sign choices first differ at some largest power 2^{m_r}, which changes the sum by $2 \cdot 2^{m_r} = 2^{m_r+1}$, while all smaller powers together can change it by at most $2(2^{m_r} - 1) < 2^{m_r+1}$. So no two terms cancel, and there are exactly $2^{s(t)}$ live cells. ■

So at $t = 63 = 111111_2$ the row has $2^6 = 64$ live cells, and one tick later, at $t = 64 = 1000000_2$, it collapses to just 2: the two lonely seeds at the bottom of the centre panel. Summing over the first 2^n rows gives 3^n live cells in total (729 for the 64 rows drawn), which is precisely the self-similar counting of the Sierpinski triangle, whose fractal dimension $\log 3 / \log 2 \approx 1.585$ you meet in Ch. 73. The fractal is the shadow of a binomial theorem.

> **Engineering note**
>
> Linear CAs over $\mathbb{F}_2$ are the spatial cousins of linear-feedback shift registers. Built-in self-test circuits in VLSI chips use rule-90/150 hybrid automata as pattern generators, because each cell needs only one XOR gate and neighbours are physically adjacent, so the wiring is short.

Result 3: worlds with no past

Run a CA forward and you get a future. Can you also run it backward? A configuration with no predecessor at all is called a **Garden of Eden**: it can be a starting state, but the rule can never *produce* it.

> **Gardens of Eden for rule 90 on a ring**
>
> On a ring of $n \geq 3$ cells, rule 90 is not invertible. When n is odd, exactly half of all 2^n configurations are Gardens of Eden; when n is even, exactly three quarters are.

Proof. On the ring, rule 90 is a linear map $\mathbf{A} : \mathbb{F}_2^n \to \mathbb{F}_2^n$, $(\mathbf{A}x)_i = x_{i-1} \oplus x_{i+1}$. Its kernel is the set of x with $x_{i-1} = x_{i+1}$ for all i, so the cells two apart agree. If n is odd, stepping by 2 around the ring visits every cell, so all cells are equal: the kernel is $\{00\ldots0,\ 11\ldots1\}$, of dimension 1. If n is even, the even-indexed cells must all agree and the odd-indexed cells must all agree, independently, which gives four kernel elements, dimension 2. By rank–nullity (Ch. 9), the image has dimension $n-1$ or $n-2$, so it contains 2^{n-1} or 2^{n-2} configurations. Everything outside the image is a Garden of Eden, a fraction $1/2$ or $3/4$. ■

Notice the logic: rule 90 is non-injective (two different states, like all-zeros and all-ones, both map to all-zeros), and *therefore* non-surjective. On a finite set that is trivial. The deep fact is that it stays true on infinite grids for every CA, linear or not:

> **Moore–Myhill Garden-of-Eden theorem (stated)**
>
> A cellular automaton on $\mathbb{Z}^d$ has Garden-of-Eden configurations if and only if it has two configurations that differ in finitely many cells but have the same image.

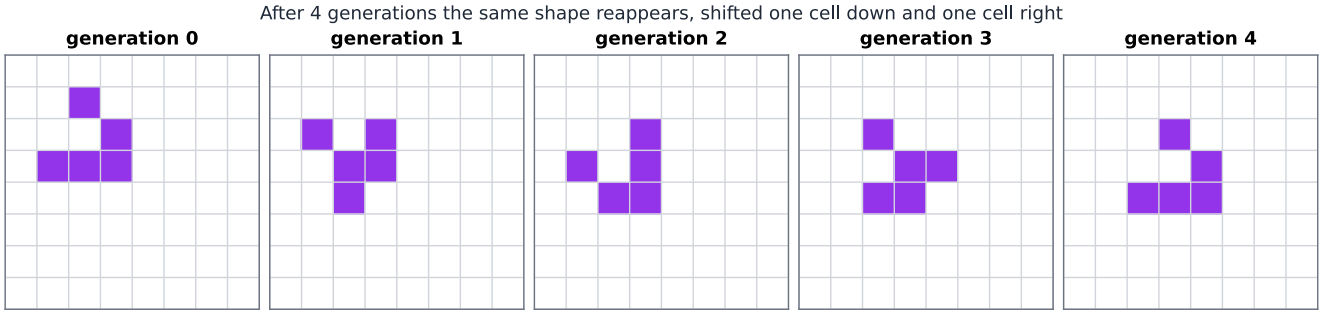
Conway's Game of Life has both, so it has Gardens of Eden, and people have found explicit ones. Information destruction (two pasts, one future) forces impossibility (some futures with no past). That trade-off is the discrete seed of the thermodynamic arrow of time.

Two dimensions: the Game of Life

In 1970 John Conway designed a two-dimensional CA on the square grid, with each cell reading its eight surrounding neighbours. The rule, written B3/S23, is:

- a dead cell with exactly 3 live neighbours is **born**;
- a live cell with 2 or 3 live neighbours **survives**;
- every other cell is dead at the next tick.

Out of this come still lifes (a 2×2 block never changes), oscillators (three in a row flips between horizontal and vertical with period 2), and the famous **glider**: five cells that, after four generations, reproduce their own shape shifted one cell diagonally.



A glider in Conway's Game of Life. The intermediate shapes are different, but generation 4 is generation 0 translated by one cell diagonally, so the glider travels at a quarter of the speed of light, $c/4$.

Gliders carry information across the grid. A **glider gun** (found by Bill Gosper in 1970) emits one every 30 generations, and colliding streams of gliders can be arranged to behave like AND, OR and NOT gates, wires and memory. Assemble enough of them and Life simulates a universal Turing machine. People have since built, inside Life, a working digital clock, a Life simulator running Life, and a pattern that computes the primes.

Wolfram's four classes and universality

Running all 256 elementary rules from random starts, Stephen Wolfram sorted their behaviour into four qualitative classes:

class	behaviour	examples
I	everything converges to a uniform state	rule 0, rule 32
II	settles into stable or periodic stripes	rule 4, rule 108
III	chaotic, aperiodic, random-looking	rule 30, rule 45
IV	localised structures that move and interact	rule 110, Life

Classes I–III map onto ideas from Ch. 83 and Ch. 84: fixed points, periodic orbits and chaos. Class III rules show sensitive dependence: flip one cell in rule 30 and the difference spreads rightwards at the full speed of light (one cell per tick) and leftwards at roughly a fifth of it, never healing. Class IV is new. It sits between order and chaos, and it is the class where computation lives.

Universality of rule 110 (Cook, 2004; stated)

With a suitable periodic background and finite initial pattern, rule 110 can simulate any Turing machine.

The proof shows that rule 110's gliders can emulate a *cyclic tag system*, a simple string-rewriting machine already known to be universal. It is long and ingenious, and its consequence is drastic. Universality plus the halting problem (Ch. 30) means that questions like "will this rule-110 pattern ever produce glider type X?" are **undecidable**: no algorithm answers them for all inputs. For such systems, there is in general no shortcut. The fastest way to learn what the automaton does in a million steps is to run it for a million steps. Wolfram calls this *computational irreducibility*.

That answers the hook's second question. We can solve rule 90 because it is linear: a closed form exists, and $N(t) = 2^{s(t)}$ lets you jump to step 10^{100} instantly. We cannot solve rule 110, and not merely because nobody has been clever enough. If a general closed form existed, it would decide the halting problem.

? Is rule 30 really random?

Rule 30's centre column starts $1, 1, 0, 1, 1, 1, 0, 0, 1, 1, \ldots$ and over the first 1000 steps it is 1 about 48% of the time, as a fair coin would be. Every statistical test anyone has run on it passes. Yet nobody has proved that the column is not eventually periodic, that 0s and 1s each occur with limiting frequency $\tfrac{1}{2}$, or that computing the nth bit requires about n steps of work. Wolfram offered prizes for each of the three in 2019, and they remain unclaimed. What it would even *mean* to prove "random-looking" is the subject of Ch. 81, where randomness is defined by what no efficient algorithm can tell apart.

When the averages lie: mean-field theory

Here is a tempting shortcut. Pretend each cell of Life is independently alive with probability p. Then a dead cell is born with probability $\binom{8}{3}p^3(1-p)^5$, a live cell survives with probability $\binom{8}{2}p^2(1-p)^6 + \binom{8}{3}p^3(1-p)^5$, and the density obeys a one-dimensional map

$$p_{t+1} = F(p_t) = 84\,p_t^3(1-p_t)^6 + 56\,p_t^4(1-p_t)^5.$$

This **mean-field** map has a stable fixed point at $p^* \approx 0.370$ (with $F'(p^*) \approx -0.08$, so convergence is fast) and an unstable threshold at $p \approx 0.192$ below which it predicts extinction. Now actually run Life from a random 35% soup. The density does not settle near 37%; it falls below 10% within about a hundred generations and keeps sinking for thousands more, to roughly 3% (on a 300×300 torus): scattered blocks, blinkers and the odd glider. Mean-field theory is wrong by an order of magnitude.

The error is the independence assumption. Live cells in Life are strongly *correlated*: they cluster into small structures with empty space between them. The whole point of emergence is that the parts are not independent, so any theory that averages them away throws away the phenomenon. You will meet the same failure, and the same partial rescue, in network epidemics (Ch. 87) and in critical phenomena (Ch. 88).

86.4 The algorithm

Updating a whole world at once

The only subtle point in simulating a CA is the word *simultaneously*. If you update cells one by one in a loop, overwriting as you go, cell $i+1$ sees the *new* value of cell i and you are simulating a different system. Always compute the next configuration from a frozen copy of the current one.

With NumPy the freeze comes for free: build the neighbourhood index for every cell in one vectorised expression and look the answers up in the rule table.

```
import numpy as np

def eca_step(x, rule):
    table = np.array([(rule >> k) & 1 for k in range(8)], dtype=np.uint8)
    left, right = np.roll(x, 1), np.roll(x, -1)      # periodic ring
    return table[4 * left + 2 * x + right]           # neighbourhood -> bit k of rule

def life_step(G):
    n = sum(np.roll(np.roll(G, a, 0), b, 1)
            for a in (-1, 0, 1) for b in (-1, 0, 1)) - G   # live-neighbour count
    return ((n == 3) | ((G == 1) & (n == 2))).astype(np.uint8)
```

`4*left + 2*x + right` is the neighbourhood read as a 3-bit binary number k, so `table[k]` is bit k of the rule number, exactly Wolfram's convention. For Life, the neighbour count is a sum of eight shifted copies of the grid; equivalently, a 2-D convolution with a 3×3 kernel of ones whose centre is zero, which is the same stencil idea as a finite-difference Laplacian.

Cost and tricks

One step costs $O(N)$ for N cells, so T steps cost $O(NT)$, and there is no way around this for irreducible rules. Engineers still find huge constant factors:

- **Bit packing.** Store 64 cells in one machine word and apply the Boolean formula for the rule with word-wide AND, OR and XOR. Rule 90 becomes `(x << 1) ^ (x >> 1)` and processes 64 cells per instruction.
- **Activity lists.** In Life most of the grid is empty and stays empty. Only cells next to a change can change, so track the active region.
- **HashLife.** Bill Gosper's algorithm memoises the evolution of quadtree blocks: a $2^k \times 2^k$ block determines its own centre 2^{k-2} generations later (light cone again!), and since repetitive patterns reuse the same blocks, it can leap billions of generations ahead for structured patterns. On random soup it gains nothing, which is computational irreducibility showing its teeth.

Failure cases

> **The boundary is part of the rule**
>
> A periodic ring of n cells is a *different dynamical system* from the infinite line. Once the light cone wraps around (after about $n/2$ steps for a single seed), the two sides interfere and the picture changes. Rule 90 on a ring of 2^m cells dies completely at $t = 2^{m-1}$: by the freshman's dream, $(z + z^{-1})^{2^{m-1}} = z^{2^{m-1}} + z^{-2^{m-1}}$, and on the ring $z^{2^m} = 1$, so the two terms are the same cell and cancel. Fixed-zero boundaries give yet another system. Always state the boundary condition, and make the grid much wider than the light cone if you want the infinite-line behaviour.

Other pitfalls: using signed `int8` and overflowing neighbour counts (fine for Life, since the maximum is 8, but not for automata with many states); and drawing conclusions from a single random seed, when classes III and IV can look alike for a few hundred steps.

86.5 The experiment

Part 1: your own space–time diagrams

Run this script, then change the knobs at the top.

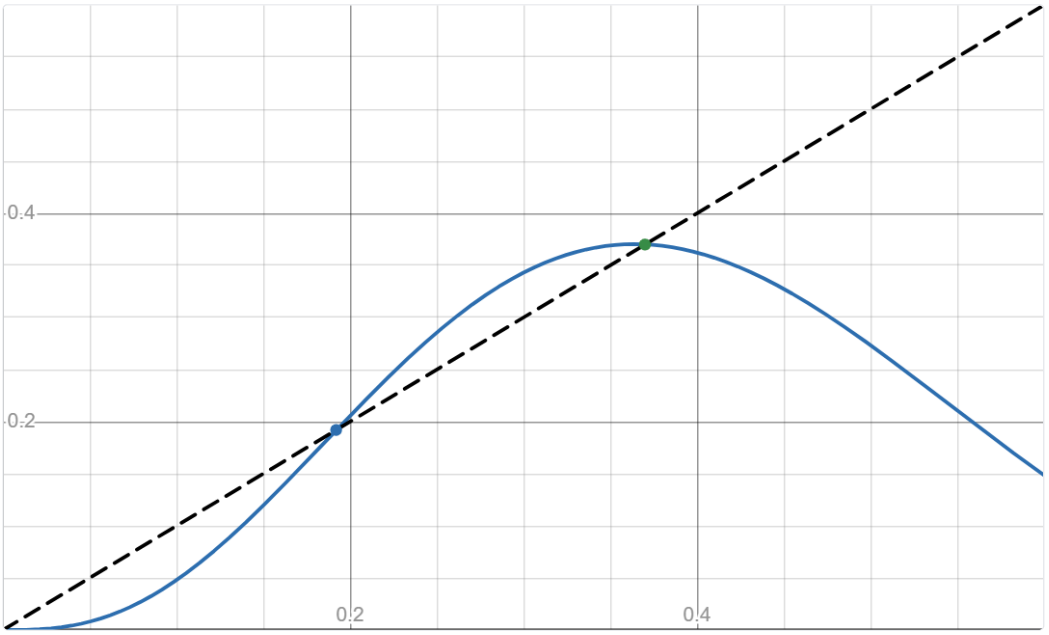
```
import numpy as np
import matplotlib.pyplot as plt

RULE   = 30        # try 90, 110, 184, 54, 73, 45, 150
WIDTH  = 301       # number of cells (periodic ring)
STEPS  = 200       # number of time steps
RANDOM = False     # True: random start; False: single live cell
SEED   = 1

table = np.array([(RULE >> k) & 1 for k in range(8)], dtype=np.uint8)
rng = np.random.default_rng(SEED)
x = rng.integers(0, 2, WIDTH).astype(np.uint8) if RANDOM else np.zeros(WIDTH, np.uint8)
if not RANDOM:
    x[WIDTH // 2] = 1

rows = [x]
for _ in range(STEPS - 1):
    x = table[4 * np.roll(x, 1) + 2 * x + np.roll(x, -1)]
    rows.append(x)
grid = np.array(rows)

centre = grid[:, WIDTH // 2]
print(f"rule {RULE}: final density {grid[-1].mean():.3f}, "
      f"centre column mean {centre.mean():.3f}")

plt.figure(figsize=(8, 5))
plt.imshow(grid, cmap="Greys", interpolation="nearest", aspect="auto")
plt.title(f"Rule {RULE}")
plt.xlabel("cell")
plt.ylabel("time")
plt.show()
```

Things to try:

- Set `RULE = 90` and `WIDTH = 256` and watch the Sierpinski triangle wrap around and annihilate at $t = 128$, as the warning above predicts.
- Set `RANDOM = True` and compare rules 30 and 110. Rule 110's background settles into a periodic texture with gliders sliding over it; rule 30 never settles.
- Rule 184 is a traffic model: a 1 is a car that moves right if the cell ahead is empty. Start random with density below and above $\frac{1}{2}$ and look for traffic jams that move *backwards*.
- Flip one bit of a random start (`x[WIDTH//2] ^= 1` on a second copy) and plot the XOR of the two runs: that is the CA's version of the butterfly effect from Ch. 84.

Part 2: the mean-field map for Life

The density map from §86.3 is a one-dimensional iterated map, so a cobweb picture tells you everything the mean-field theory believes.



Mean-field density map $p_{t+1} = F(p_t)$ for the Game of Life (blue) against the diagonal. The blue curve crosses the diagonal at the two fixed points marked. The real Life soup ends near 3%, far below the stable point the curve predicts.

The polynomial is "born with exactly 3" plus "survive with 2 or 3". Change the rule and watch the fixed points move. HighLife (B36/S23) adds birth on 6 neighbours: add the term `28x^{6}(1-x)^{3}`, which is the probability $(1-x)$ that the cell was dead times $\binom{8}{6}x^6(1-x)^2$. Or try "Seeds" (B2/S, every live cell dies), whose map is just $(1-x)\binom{8}{2}x^2(1-x)^6 = 28x^2(1-x)^7$. Ask each time: does the mean-field prediction match what the real automaton does? Run the Life step from §86.4 to find out.

86.6 The challenge

1. **Intuition.** Without running any code, predict what rule 0, rule 255, rule 204 and rule 51 do from any starting configuration. (Hint: write out the eight-entry tables.) Then explain in one sentence why rule 184, the traffic rule, must conserve the number of 1s, and why no rule that conserves 1s can ever be in Wolfram's class I unless the density is 0 or 1.
2. **Derivation.** Rule 150 is $x_i^{t+1} = x_{i-1} \oplus x_i \oplus x_{i+1}$. Show that it is linear over $\mathbb{F}_2$, write its single-seed evolution as $\left(z^{-1}+1+z\right)^t X^0(z)$, and prove that the number of live cells at time 2^m is exactly 3. Then decide, for rings of n cells, for which n rule 150 is invertible, by computing its kernel as in Result 3.
3. **Implementation.** Write a program that classifies all 88 inequivalent elementary rules automatically. From 20 random starts on a ring of 400 cells, run 1000 steps and compute (a) the final density, (b) whether the configuration has become periodic in time, and (c) the spreading speed of a single flipped bit. Use these numbers to sort the rules into Wolfram's four classes, and list the rules your program cannot confidently place.
4. **Open investigation.** Search the space of two-dimensional "Life-like" rules (each is a pair of sets B/S of neighbour counts, so there are $2^{18} = 262144$ of them) for rules that support a glider. Starting from random soups of density 0.3, how often do spontaneously moving objects appear? Is there a simple statistic of the rule (the number of birth conditions, the mean-field fixed point, the rate of density decay) that predicts whether gliders exist? Nobody has a clean criterion.

86.7 Loose threads

The chapter showed that tiny local rules can be solvable (rule 90), random-looking (rule 30) or universal (rule 110 and Life), and that mean-field averaging fails precisely where things get interesting. It skipped three big questions on purpose.

? Does computation live at the edge of chaos?

Class IV sits between frozen order (classes I and II) and seething disorder (class III). In 1990 Chris Langton parametrised rules by λ, the fraction of neighbourhoods that map to a live state, and claimed that complex, computation-capable behaviour clusters near a critical λ, the way a magnet is most responsive at its critical temperature. Others reran the experiments and disputed the conclusion. Is there a phase transition in "rule space", and is it where adaptive systems, from cells to brains, tune themselves? The idea of self-tuned criticality returns in Ch. 88, and whether it holds up is still argued.

? Can a cellular automaton be a fluid?

Put particles on a hexagonal lattice, let them hop and collide by a few local rules that conserve number and momentum, and average over large blocks. In 1986 Frisch, Hasslacher and Pomeau showed that the averages obey the Navier–Stokes equations of fluid dynamics, even though nothing in the rules mentions viscosity or pressure. Why do the conservation laws and the lattice symmetry alone fix the continuum equations, and why does a square lattice give the *wrong* answer? That passage from discrete microscopic rules to continuum PDEs is a theme of Ch. 92.

Other threads:

- **Undecidability, made precise.** The claim that rule 110 has undecidable long-term behaviour rests on the halting problem and on reductions between machines, which Ch. 30 builds from scratch.

- **Procedural worlds.** Game designers use CAs (for instance, "a wall survives if 4 or more of its 8 neighbours are walls") to carve caves from noise in a few passes: Ch. 74.
- **From grids to networks.** A CA lives on a lattice where every cell has the same neighbours. Real systems, from power grids to social networks, have wildly uneven neighbourhoods, and the same local-rule thinking on an irregular graph is the subject of Ch. 87.
- **Reversible worlds.** Some CAs are invertible, so no information is ever lost. Physics at the microscopic level is reversible too. Whether reversible CAs can still be universal (they can) and what that has to do with the energy cost of computation is a story about Landauer's principle that Ch. 33 starts.

Further reading

- S. Wolfram, *A New Kind of Science*, Wolfram Media, 2002 (free online). The exhaustive, opinionated catalogue of elementary rules and the four classes.
- M. Cook, "Universality in elementary cellular automata," *Complex Systems* 15 (2004), 1–40. The rule-110 proof.
- E. R. Berlekamp, J. H. Conway and R. K. Guy, *Winning Ways for Your Mathematical Plays*, vol. 4, 2nd ed., A K Peters, 2004. Includes the construction of a universal computer in Life.
- O. Martin, A. M. Odlyzko and S. Wolfram, "Algebraic properties of cellular automata," *Communications in Mathematical Physics* 93 (1984), 219–258. Rule 90 and its relatives done properly with polynomials over $\mathbb{F}_2$.
- N. Johnston and D. Greene, *Conway's Game of Life: Mathematics and Construction*, 2022 (free online). A modern, rigorous tour of Life's engineering.

Chapter 87 — Network Science: The Shape of Complex Systems

> **Level 2 · Track I — Dynamical Systems, Chaos, and Complexity Builds on:** Ch. 11, Ch. 19, Ch. 27, Ch. 34 · **Leads to:** Ch. 88, Ch. 90 · **Related:** Ch. 10, Ch. 45, Ch. 86 **Reading time:** ~25 min

87.1 The hook

On average, your friends have more friends than you do.

This is not a statement about you. It is true for most people in almost every social network ever measured, from Facebook to high-school friendships to sexual-contact surveys. It sounds like a statistical insult, and it is a theorem, one we will prove in five lines.

Here is a second puzzle with the same root. Two cities each have a million people, and in both, the average person has six close contacts. A new virus arrives in each, with identical biology: same chance of passing per contact, same recovery time. In one city the outbreak fizzles. In the other it infects nearly half the population. Nothing differs except *who is connected to whom*.

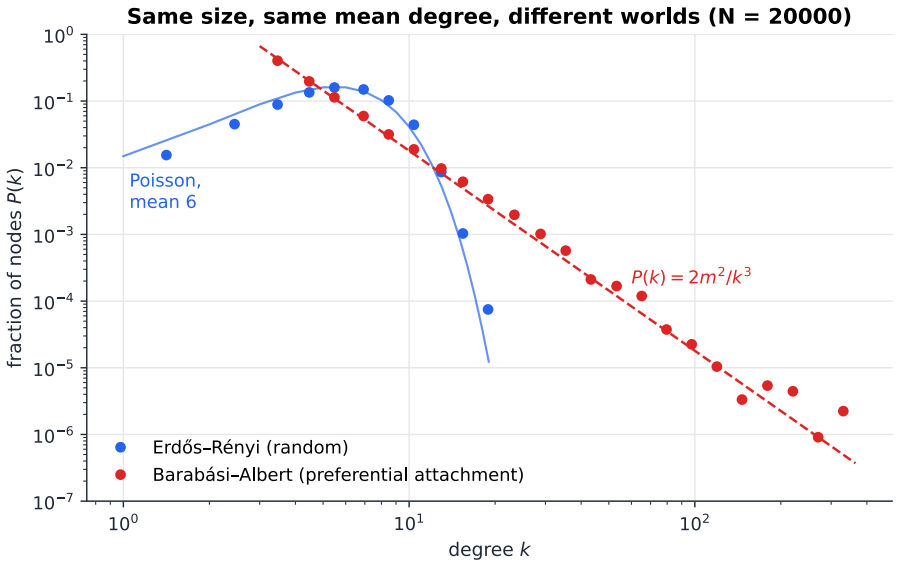
And a third. On a summer evening in Southeast Asia, thousands of male fireflies in a single tree begin to flash in unison. Nobody conducts them. Each firefly only adjusts its rhythm slightly toward the flashes it sees. Below a certain level of mutual influence, nothing happens; the tree twinkles at random. Above it, synchrony appears suddenly and spreads through the tree.

Three phenomena, one lesson. Once many parts interact, the **pattern of connections** is itself a variable, often the most important one. The question for this chapter: what can mathematics say about a system when you know the shape of its network but not the details of its parts?

87.2 The big idea

Chapter 27 treated a graph as a fixed object to be searched: find the shortest path, the best matching, the communities. Network science asks a different question. Real networks have millions of nodes, nobody designed them, and they grow and change. So instead of studying one graph, we study **families of random graphs** with the right statistics, and ask which large-scale properties are almost certain.

The single most revealing statistic is the **degree distribution** $P(k)$: the fraction of nodes with exactly k links. Two networks can share size and mean degree and still be different worlds.



Two networks of 20,000 nodes, both with mean degree 6. In the random (Erdős–Rényi) network, degrees cluster tightly around 6 and nobody has more than 18 links. In the preferential-attachment (Barabási–Albert) network, the distribution is a straight line on log–log axes, a power law, and the best-connected node has 365 links.

Picture the two as road maps. The Erdős–Rényi network is like rural roads: every village has a handful of connections and no village is special. The Barabási–Albert network is like an airline map: most airports have a few routes, and a few hubs have hundreds. In the second kind there is no "typical" scale, which is why it is called **scale-free**.

Three intuitions follow and organise the rest of the chapter.

1. **Thresholds are everywhere.** Add links to a sparse random network one by one and nothing much happens, until suddenly a single connected "giant component" spans the network. The same arithmetic governs whether a disease, a rumour or a power-grid failure spreads: an epidemic is a giant component of infections.
2. **Hubs dominate.** If you reach a node by following a link, you are more likely to land on a hub, simply because hubs have more links pointing at them. That one bias explains the friendship paradox and why scale-free networks are so vulnerable to epidemics.
3. **Coupling creates collective rhythm.** Oscillators that each nudge each other weakly can, above a critical coupling, lock into synchrony. Like the giant component, it appears abruptly at a threshold.

87.3 The mathematics

Random networks and the giant component

Erdős–Rényi random graph

$G(n,p)$ has n nodes, and each of the $\binom{n}{2}$ possible edges is present independently with probability p. Its mean degree is $c=(n-1)p$. For large n with c fixed, the degree of a node is Binomial$(n-1,p)\approx$ Poisson(c):

$$P(k)=e^{-c}\frac{c^k}{k!}.$$

The Poisson limit is the law of rare events from Ch. 19. Its variance equals its mean, so with $c=6$ almost all degrees lie between 1 and 13.

The key fact about sparse random graphs is that they are **locally tree-like**: short loops are rare, because closing a loop of length ℓ needs an unlikely extra edge, with probability about c/n. Near any node, the network looks like a branching family tree. That turns global connectivity into a branching-process calculation.

The giant component

In $G(n,c/(n-1))$ with $n\to\infty$, let S be the fraction of nodes in the largest connected component. Then S is the largest solution in $[0,1]$ of

$$S=1-e^{-cS}.$$

For $c\le 1$ the only solution is $S=0$ (all components are small); for $c>1$ there is a unique solution with $S>0$, a **giant component** containing a finite fraction of all nodes.

Derivation. Let u be the probability that a node is *not* in the giant component. A node avoids the giant component exactly when every one of its neighbours does, and in a tree-like network the neighbours' fates are independent. A node of degree k therefore avoids it with probability u^k, and averaging over the Poisson degree,

$$u=\sum_{k=0}^{\infty}e^{-c}\frac{c^k}{k!}u^k=e^{-c}e^{cu}=e^{-c(1-u)}.$$

(For Poisson degrees, the number of *further* neighbours of a node reached along an edge is again Poisson(c), so the same u appears on both sides.) Writing $S=1-u$ gives $S=1-e^{-cS}$.

Now look at the right-hand side $h(S)=1-e^{-cS}$. It is increasing and concave, with $h(0)=0$, $h'(0)=c$ and $h(1)<1$. If $c\le 1$, concavity puts $h(S)<cS\le S$ for every $S>0$, so the curve lies below the diagonal and only $S=0$ works. If $c>1$, the curve starts *above* the diagonal (slope $c>1$) and ends below it, so it crosses exactly once more in $(0,1)$. ■

The solution rises steeply: $S\approx 0.583$ at $c=1.5$, 0.797 at $c=2$, and 0.997 at $c=6$. (A simulation of $n=100{,}000$ nodes at $c=2$ gives 0.797.) The transition at $c=1$ is a genuine **phase transition**, like water freezing: as c crosses 1, the size of the largest cluster jumps from a vanishing fraction to a finite one. You will watch it happen in the experiment.

The tree-like picture also explains why random networks are **small worlds**. From a typical node, about c nodes are one step away, c^2 two steps away, and c^ℓ at distance ℓ. These run out of network when $c^\ell\approx n$, so typical distances are

$$\ell\approx\frac{\ln n}{\ln c}.$$

For $n=20{,}000$ and $c=6$ this gives 5.5, and the network in the figure has mean distance 5.8. Distances grow only *logarithmically* with size: that is the "six degrees of separation" phenomenon, explained by nothing more than branching.

Result 2: the friendship paradox

Pick a random node and look at its degree: on average, $\langle k\rangle$. Now pick a random *edge* and follow it to one of its ends. A node with k links is the end of k edges, so it is reached with probability proportional to k. The degree of a node reached along an edge has distribution $kP(k)/\langle k\rangle$, and its mean is

$$\langle k\rangle_{\text{neighbour}}=\sum_k k\cdot\frac{kP(k)}{\langle k\rangle}=\frac{\langle k^2\rangle}{\langle k\rangle}.$$

Friendship paradox

In any network, the mean degree of a node reached by following a random edge is at least the mean degree of a random node:

$$\frac{\langle k^2\rangle}{\langle k\rangle}=\langle k\rangle+\frac{\mathrm{Var}(k)}{\langle k\rangle}\ \ge\ \langle k\rangle,$$

with equality only when every node has the same degree.

Proof. $\langle k^2\rangle=\langle k\rangle^2+\mathrm{Var}(k)$ (Ch. 19). Divide by $\langle k\rangle>0$, and use $\mathrm{Var}(k)\ge 0$, with equality only for constant k. ■

The size of the paradox is the variance. For the random network in the figure, $\langle k\rangle=6$ and $\langle k^2\rangle/\langle k\rangle=7.0$: your friends have one more friend than you. For the scale-free network with the *same* mean degree, $\langle k^2\rangle/\langle k\rangle=19.5$: your friends have more than three times as many friends as you. The hubs are hiding in everyone's friend list.

Engineering note

The paradox is a sampling tool. To find well-connected nodes without knowing the network, pick random nodes and ask each for *a random neighbour*. During outbreaks, monitoring the friends of randomly chosen people detects epidemics earlier than monitoring the people themselves, and "acquaintance immunisation" (vaccinate a random friend of a random person) protects a scale-free population far better than random vaccination.

Result 3: where power laws come from

The Erdős–Rényi model can never produce a hub with 365 links: the Poisson tail falls faster than exponentially. In 1999 Barabási and Albert proposed a mechanism with two ingredients that real networks (the web, citations, airline routes) clearly have: **growth** and **preferential attachment**, "the rich get richer".

Barabási–Albert model

Start with a few connected nodes. At each time step add one new node with m edges, and attach each edge to an existing node i with probability proportional to its degree, $k_i / \sum_j k_j$.

Degree exponent of the BA model

In the continuum (mean-field) approximation, the BA model produces a degree distribution $P(k) \approx 2m^2/k^3$ for $k \geq m$: a power law with exponent 3.

Derivation. After t steps there are about t nodes and mt edges, so $\sum_j k_j = 2mt$. Treat k_i as a continuous function of time. Node i gains links from each of the m new edges with probability $k_i/(2mt)$, so

$$\frac{dk_i}{dt} = m \cdot \frac{k_i}{2mt} = \frac{k_i}{2t}, \qquad k_i(t_i) = m,$$

a separable ODE with solution $k_i(t) = m\sqrt{t/t_i}$, where t_i is the node's birth time. Old nodes are big; that's the whole mechanism. A node has degree below k exactly when it was born late enough: $k_i(t) < k \iff t_i > m^2 t/k^2$. Birth times are uniform on $[0, t]$, so

$$P(k_i < k) = 1 - \frac{m^2}{k^2}, \qquad P(k) = \frac{d}{dk} P(k_i < k) = \frac{2m^2}{k^3}.$$

■

The dashed line in the figure is $2m^2/k^3$ with $m = 3$, and the simulated network sits right on it. The important consequence is the second moment. With $P(k) \propto k^{-3}$, $\langle k^2 \rangle \approx \int_m^{k_{\max}} k^2 \cdot 2m^2 k^{-3}\, dk = 2m^2 \ln(k_{\max}/m)$, which grows without bound as the network grows. For exponents between 2 and 3, common in real networks, it grows like a power of n. **The variance of a scale-free network is effectively infinite.** Every result below that depends on $\langle k^2 \rangle$ changes character.

? Are real networks actually scale-free?

Power laws are easy to see and hard to prove. A straight line on a log–log plot is weak evidence: log-normal and stretched-exponential distributions mimic it over two decades. In 2019 Broido and Clauset fitted nearly a thousand real networks with careful likelihood tests and found strong power laws in only a small minority, setting off a fierce debate that is not settled. How would you even test the claim? The tools (maximum-likelihood fits, goodness-of-fit and likelihood-ratio tests) are in Ch. 20; the verdict on the world's networks is still an open question.

Contagion: an epidemic is a giant component

The simplest epidemic model on a network is **SIR**: each node is Susceptible, Infected or Recovered. An infected node passes the infection along each of its edges with some probability per unit time, and recovers (permanently immune) at some rate. The well-mixed version, where everyone meets everyone, is the classic system of ODEs $\dot{S} = -\beta SI$, $\dot{I} = \beta SI - \gamma I$, $\dot{R} = \gamma I$ (Ch. 11), with its famous threshold $R_0 = \beta/\gamma > 1$. On a network, the threshold moves.

Collapse the timing into one number: the **transmissibility** T, the probability that an infected node infects a given neighbour before recovering. Then an outbreak is just a random subgraph: keep each edge with probability T, and the people eventually infected are the component of the kept-edge network that contains patient zero. A large epidemic is possible exactly when the kept-edge network has a giant component.

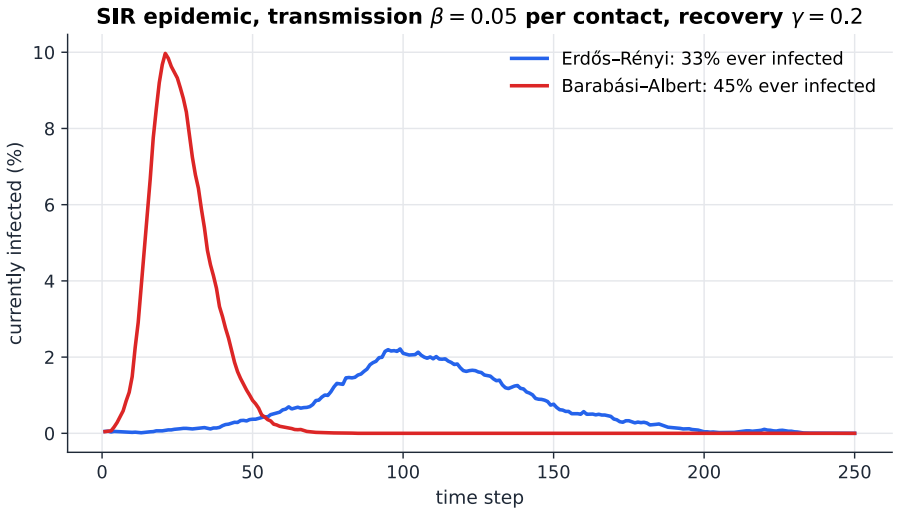
Now use the friendship-paradox calculation. Follow the infection down an edge to a newly infected node. That node was reached along an edge, so its mean degree is $\langle k^2 \rangle / \langle k \rangle$; one of its edges leads back to where the infection came from, which leaves $\langle k^2 \rangle / \langle k \rangle - 1$ onward edges, each transmitting with probability T. In a tree-like network the infection is a branching process with mean offspring

$$R = T\, \frac{\langle k^2 \rangle - \langle k \rangle}{\langle k \rangle},$$

and a branching process survives with positive probability exactly when its mean offspring exceeds 1. So the **epidemic threshold** is

$$T_c = \frac{\langle k \rangle}{\langle k^2 \rangle - \langle k \rangle}.$$

For a Poisson network with mean c, $\langle k^2 \rangle = c^2 + c$, so $T_c = 1/c$, which is the giant-component condition again with c replaced by Tc. For the two networks in the figure: $T_c \approx 0.167$ for the random network and $T_c \approx 0.054$ for the scale-free one.



The same disease on the two networks of the first figure (ten random initial cases, one realisation each). Per step, an infected node infects each susceptible neighbour with probability 0.05 and recovers with probability 0.2, giving transmissibility $T \approx 0.208$. On the random network $R \approx 1.25$ and the outbreak smoulders; on the scale-free network $R \approx 3.9$ and it explodes through the hubs, peaks around step 20, and reaches 45% of nodes.

That is the answer to the two-cities puzzle in the hook. Same biology, same average contacts, different variance. And for a scale-free network with exponent at most 3, $\langle k^2 \rangle \to \infty$ as the network grows, so $T_c \to 0$: **there is no epidemic threshold at all**. However weakly a pathogen spreads, a large enough scale-free network can sustain it, because the hubs act as amplifiers. Pastor-Satorras and Vespignani pointed this out in 2001 for computer viruses, which had puzzled everyone by persisting for years at low prevalence. The cure follows from the same algebra: target the hubs, the nodes that make $\langle k^2 \rangle$ large.

? The threshold is an eigenvalue

The formula $T_c = \langle k \rangle / (\langle k^2 \rangle - \langle k \rangle)$ only uses the degree distribution, as if the network were otherwise random. For a *particular* network, with its clustering, communities and correlations, a different number controls the threshold: the largest eigenvalue $\lambda_{\max}$ of the adjacency matrix $\mathbf{A}$, with the threshold near $T\lambda_{\max} = 1$. For the random network above $\lambda_{\max} \approx 7.17$, close to $\langle k^2 \rangle / \langle k \rangle = 7.0$; for the scale-free one $\lambda_{\max} \approx 22.6$ versus 19.5. Why should the top eigenvalue of a matrix of 0s and 1s decide whether a disease dies out, and what does its eigenvector say about *where* to vaccinate? The Perron–Frobenius theory behind it is in Ch. 90.

Synchronisation: the Kuramoto model

Now replace "infected or not" by a phase. Give each node an oscillator with angle θ_i and its own natural frequency ω_i, drawn from a distribution $g(\omega)$ that is symmetric about 0 (we can always rotate to a frame where the mean frequency is 0). Each oscillator speeds up or slows down toward the others:

$$\frac{d\theta_i}{dt} = \omega_i + \frac{K}{N} \sum_{j=1}^{N} \sin\left(\theta_j - \theta_i\right).$$

This is Kuramoto's model (1975), here on the complete graph; on a general network the sum runs over neighbours. To measure synchrony, average the oscillators as unit phasors in the complex plane, exactly as in a phasor diagram for AC circuits:

$$r\, e^{j\psi} = \frac{1}{N} \sum_{j=1}^{N} e^{j\theta_j}.$$

The **order parameter** $r \in [0, 1]$ is the length of the average phasor: $r \approx 0$ when phases are scattered, $r = 1$ when all oscillators point the same way.

Multiply the definition by $e^{-j\theta_i}$ and take imaginary parts: $r \sin(\psi - \theta_i) = \frac{1}{N} \sum_j \sin(\theta_j - \theta_i)$. So the model becomes

$$\frac{d\theta_i}{dt} = \omega_i + Kr \sin\left(\psi - \theta_i\right).$$

Each oscillator feels *only the mean field*: a pull of strength Kr toward the collective phase ψ. This is the feedback at the heart of synchrony. More order means a stronger pull, and a stronger pull creates more order.

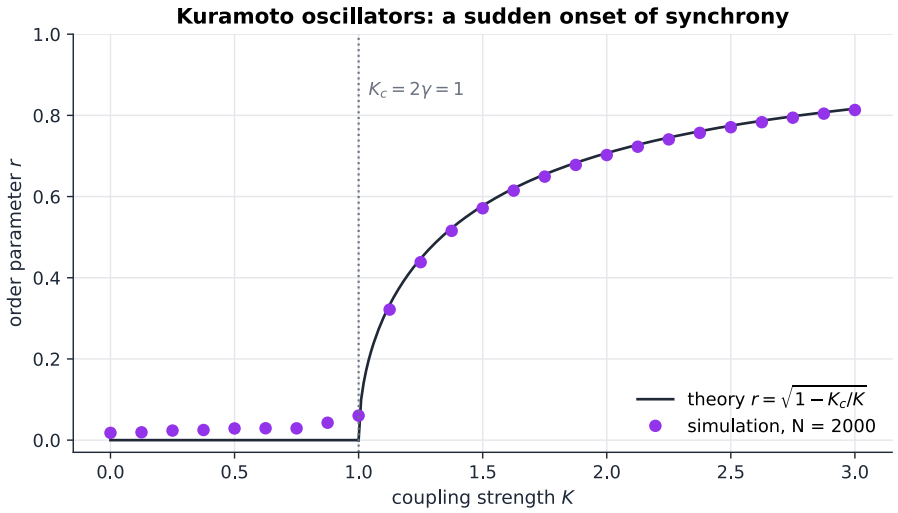
Suppose a steady state exists with constant r and $\psi = 0$. An oscillator with $|\omega_i| \le Kr$ can sit at the fixed point $\sin\theta_i = \omega_i / (Kr)$: it is **locked**. One with $|\omega_i| > Kr$ drifts forever. By symmetry of g the drifting ones contribute nothing to the average phasor, while a locked oscillator with $\omega = Kr \sin\theta$ contributes $\cos\theta$. Changing variables from ω to θ gives the **self-consistency equation**

$$r = \int_{-\pi/2}^{\pi/2} \cos\theta \; g\left(Kr \sin\theta\right) Kr \cos\theta \, d\theta = Kr \int_{-\pi/2}^{\pi/2} \cos^2\theta \; g\left(Kr \sin\theta\right) d\theta.$$

$r = 0$ always solves it (incoherence). A nonzero solution branches off when, dividing by r and letting $r \to 0$, $1 = Kg(0) \int_{-\pi/2}^{\pi/2} \cos^2\theta \, d\theta = Kg(0)\pi/2$. That is

$$K_c = \frac{2}{\pi g(0)}.$$

For the Lorentzian distribution $g(\omega) = \gamma / \left(\pi(\omega^2 + \gamma^2)\right)$, $g(0) = 1/(\pi\gamma)$, so $K_c = 2\gamma$, and the integral can be done exactly to give $r = \sqrt{1 - K_c/K}$ for $K > K_c$.



Simulated order parameter (purple, 2,000 oscillators with Lorentzian frequencies, $\gamma = 0.5$) against the exact large-N theory (black). Below $K_c = 1$ the phases stay scattered and r is only finite-size noise of order $1/\sqrt{N}$; above it, synchrony switches on with the square-root shape of a pitchfork bifurcation (Ch. 85).

The fireflies, the pacemaker cells of your heart, the generators of a power grid and the pedestrians who made London's Millennium Bridge sway on its opening day in 2000 all follow this script: a population of slightly different oscillators, a coupling that grows with the order it produces, and a threshold.

Self-consistency is not stability

The argument above found a *candidate* steady state. It did not show that the incoherent state $r = 0$ is stable below K_c, nor that the partially locked state is stable above it, and for $N = \infty$ both questions turn out to be surprisingly delicate. The simulation agrees with the theory, but agreement is evidence, not proof.

87.4 The algorithm

Building random networks without $O(n^2)$ work

Naively, $G(n, p)$ needs a coin flip for each of $\binom{n}{2}$ pairs, which for $n = 10^6$ is 5×10^{11} flips. Instead, draw the *number* of edges $M \sim \text{Binomial}(\binom{n}{2}, p)$, then draw M random pairs. This costs $O(n + M)$, produces a handful of self-loops and duplicate edges (which you discard; there are $O(1)$ of them in expectation), and is how the figure's networks were made.

For preferential attachment, the trick is a list in which every node appears once per unit of degree. Choosing a uniformly random entry of that list *is* choosing a node with probability proportional to degree, so each new edge costs $O(1)$.

```
def barabasi_albert(n, m, rng):
    repeated, edges, targets = [], [], list(range(m))
    for new in range(m, n):
        edges += [(new, t) for t in targets]
        repeated += targets + [new] * m          # one entry per unit of degree
        chosen = set()
        while len(chosen) < m:                   # m distinct targets
            chosen.add(repeated[rng.integers(len(repeated))])
        targets = list(chosen)
    return edges
```

The **configuration model** is the general tool: choose any degree sequence $k_1, \dots, k_n$, give node i exactly k_i "stubs", shuffle all stubs and pair them up. It produces a random network with exactly the degrees you asked for and nothing else, which makes it the natural null model for testing whether a real network's structure is more than its degrees.

Simulating dynamics on networks

Store the adjacency matrix in sparse form (only the $2M$ nonzeros). One step of discrete-time SIR is then one sparse matrix–vector product: $\mathbf{A}\mathbf{i}$ counts each node's infected neighbours n_i, and a susceptible node is infected with probability $1 - (1 - \beta)^{n_i}$. Cost: $O(n + M)$ per step. For exact continuous-time dynamics, treat the epidemic as a continuous-time Markov chain (Ch. 36) and use the Gillespie algorithm, which draws an exponential waiting time and jumps directly from one event to the next.

For Kuramoto on the complete graph, never evaluate the double sum, which costs $O(N^2)$. Compute the order parameter $re^{j\psi}$ in $O(N)$ and use the mean-field form; one Euler step is

```
z = np.exp(1j * theta).mean()                    # r * exp(j psi)
theta += dt * (omega + K * np.abs(z) * np.sin(np.angle(z) - theta))
```

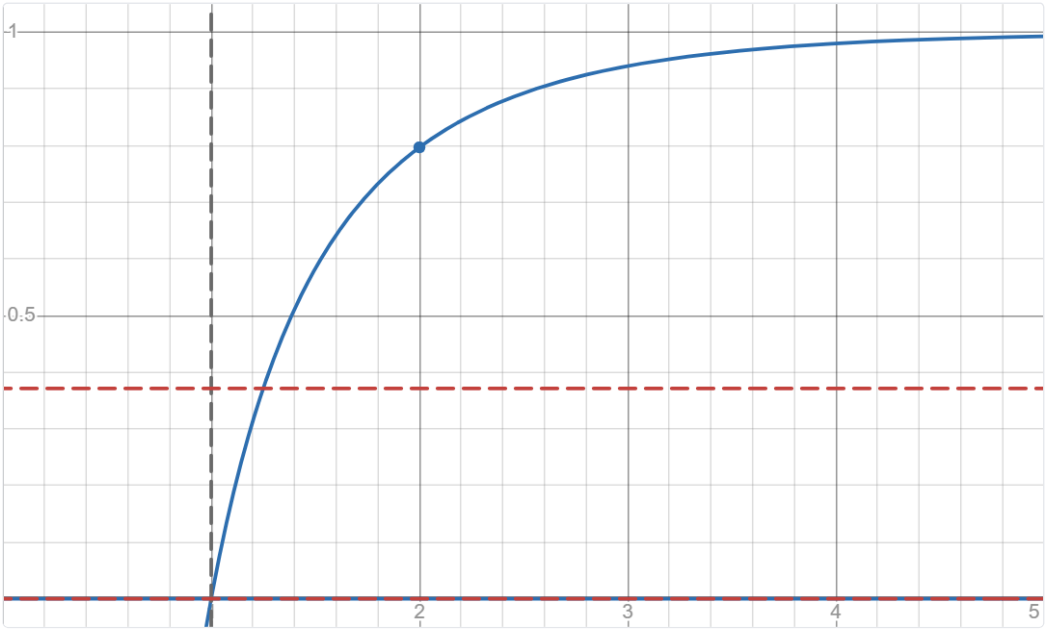
Failure cases

- **Stochastic die-out.** Even above threshold, an epidemic started from one case can die in the first few generations by bad luck; for a branching process the extinction probability is the smallest root of a generating-function equation. Always average over many seeds and report the *conditional* size of large outbreaks separately.
- **Finite-size rounding.** At $c = 1$ or $T = T_c$, the "transition" in a finite network is smeared over a window that shrinks only like a power of n. Theory is a limit; simulations live at finite n.
- **Stiffness and time step.** Oscillators with huge natural frequencies (the Lorentzian has heavy tails) rotate fast; an Euler step that is fine for the locked core can alias for them. They don't affect r much because they drift anyway, but check by halving `dt`.
- **The hub cutoff.** In a scale-free network of size n the largest degree grows like $\sqrt{n}$ (for exponent 3), so $\langle k^2 \rangle$ grows like $\ln n$, slowly. A scale-free network of a few thousand nodes can look as if it has a threshold.

87.5 The experiment

Part 1: watch the giant component appear

The equation $S = 1 - e^{-cS}$ is an implicit curve in the (c, S) plane, and Desmos draws implicit curves directly. Here x plays the role of the mean degree c and y the giant-component fraction S.



The blue curve is the solution set of $S = 1 - e^{-cS}$: the trivial branch $S = 0$ and the giant branch that leaves it at $c = 1$. The red dashed line asks the same question at the single value $c = 1.25$, the Tc of the random-network epidemic: its height, about 0.37, is the predicted final epidemic size there.

Change `1.25` in the red line to `0.9` (it collapses onto $y = 0$: no outbreak), then to `3`. Replace the blue equation by `y=1-e^{-2xy}` and explain why the transition moves to $x = 0.5$.

Part 2: an epidemic on a network you design

This script builds a configuration-model network with either Poisson or power-law degrees, then runs SIR on it. It uses only NumPy.

```
import numpy as np
import matplotlib.pyplot as plt

N, BETA, GAMMA, STEPS = 20000, 0.05, 0.2, 300
DEGREES = "powerlaw"      # "poisson" or "powerlaw"
rng = np.random.default_rng(0)

if DEGREES == "poisson":
    k = rng.poisson(6, N)
else:                     # Pareto tail, exponent ~3, minimum degree 3
    k = np.floor(3 * (1 - rng.random(N)) ** (-1 / 2)).astype(int)
k[0] += k.sum() % 2       # total number of stubs must be even
stubs = rng.permutation(np.repeat(np.arange(N), k))
u, v = stubs[0::2], stubs[1::2]
keep = u != v             # drop self-loops
u, v = u[keep], v[keep]
print(f"mean degree {k.mean():.2f}, <k^2>/<k> {np.mean(k**2)/k.mean():.1f}, max {k.max()}")

S = np.ones(N, bool); I = np.zeros(N, bool)
first = rng.choice(N, 10, replace=False); S[first] = False; I[first] = True
curve = []
for t in range(STEPS):
    n_inf = np.bincount(u[I[v]], minlength=N) + np.bincount(v[I[u]], minlength=N)
    new = S & (rng.random(N) < 1 - (1 - BETA) ** n_inf)
    rec = I & (rng.random(N) < GAMMA)
    S &= ~new
    I = (I | new) & ~rec
    curve.append(I.mean())

print(f"ever infected: {1 - S.mean():.1%}")
plt.plot(100 * np.array(curve))
plt.xlabel("time step"); plt.ylabel("infected (%)")
plt.title(f"SIR on a {DEGREES} network"); plt.show()
```

Things to try:

- Switch `DEGREES` between `"poisson"` and `"powerlaw"`. The printout shows $\langle k^2\rangle/\langle k\rangle$; predict from T_c which network will have an outbreak before you look at the plot.
- Lower `BETA` to 0.02. Transmissibility drops to about 0.09, below the Poisson threshold $1/6$: the Poisson network should stay clean while the power-law one may still burn.
- **Vaccinate the hubs.** Before the loop, set `S[np.argsort(k)[-N//20:]] = False` to immunise the best-connected 5%. Then try 5% chosen at random. Then try "acquaintance immunisation": pick 5% of random nodes and vaccinate one random *neighbour* of each. Which works, and why does the friendship paradox predict it?
- Change the Pareto exponent: `-1/2` gives $P(k) \propto k^{-3}$; `-1/1.5` gives $k^{-2.5}$, with an even fatter tail.

87.6 The challenge

1. **Intuition.** Ten people stand in a ring, each holding hands with the two people next to them, so everyone has degree 2. Is there a friendship paradox? Now add one person in the middle who holds a ribbon to all ten. Without calculating, say whether the paradox appears, who feels it, and how strongly. Then explain why random vaccination is a poor strategy in this network.
2. **Derivation.** Generalise the giant-component argument to any degree distribution. Let $G_0(x) = \sum_k P(k)x^k$ and $G_1(x) = G_0'(x)/G_0'(1)$. Show that the probability u that an edge does not lead to the giant component satisfies $u = G_1(u)$, that $S = 1 - G_0(u)$, and that a giant component exists exactly when $\langle k^2\rangle > 2\langle k\rangle$ (the Molloy–Reed criterion). Check that you recover $S = 1 - e^{-cS}$ for Poisson degrees.
3. **Implementation.** Simulate the Kuramoto model on an Erdős–Rényi network (the sum runs over neighbours only, divided by the mean degree) with $N = 1000$ and Lorentzian frequencies of width $\gamma = 0.5$. Plot r against K for mean degrees 4, 10 and 50, and compare with the complete-graph curve. How sparse can the network be before synchrony needs much more coupling?
4. **Open investigation.** Take a scale-free network and give each oscillator a natural frequency *equal to its degree*. Increase K slowly from 0, record r, then decrease K slowly back to 0. Reports in the literature describe "explosive synchronisation", a sudden jump with hysteresis, in this setup. Can you reproduce it? How does the jump depend on the degree exponent and on the network size? Is there a correlation between frequency and degree weaker than equality that still produces it? There is no complete theory.

87.7 Loose threads

This chapter used two ideas over and over: networks look like trees locally, so branching processes predict thresholds; and following edges biases you toward hubs, so the second moment $\langle k^2\rangle$ controls everything. We took the degree distribution as the whole story and the complete graph as the arena for synchrony. Both choices hid a lot.

> **? Why does incoherence stay incoherent?**
>
> Below K_c the Kuramoto order parameter decays to zero. Yet if you linearise the infinite-N model around the incoherent state, the spectrum has no decaying eigenvalues at all, only a continuous spectrum sitting exactly on the imaginary axis. Something other than ordinary exponential stability is pulling r to zero: a phase-mixing effect first found for waves in plasmas, called Landau damping. How can a system relax without any damping mechanism, and why do the same mathematics describe galaxies and plasmas? That is a story for Ch. 92.

Other threads:

- **Communities and centrality.** Real networks are not uniformly random: they have dense communities and important nodes. Detecting communities with the eigenvectors of the graph Laplacian, and measuring importance by eigenvector centrality and PageRank, is in Ch. 27 and Ch. 34.
- **Robustness and attack.** Remove random nodes from a scale-free network and the giant component survives almost to the end; remove the hubs and it shatters after a few per cent. This is percolation on networks, and the Molloy–Reed machinery from challenge 2 predicts both numbers.
- **Clustering and small worlds.** Social networks have many triangles (your friends know each other), which random graphs lack. Watts and Strogatz showed in 1998 that rewiring a few links of a lattice keeps the triangles but collapses the distances. Triangles break the tree-like assumption behind every threshold in this chapter.
- **Games on networks.** Cooperation survives in the prisoner's dilemma far more easily when players interact only with neighbours on a network: Ch. 45.
- **From networks to complexity.** Thresholds, feedback through a mean field and the sudden appearance of collective order recur everywhere in Ch. 88, which asks whether they share one explanation.

Further reading

- M. E. J. Newman, *Networks*, 2nd ed., Oxford University Press, 2018. The standard textbook; generating functions, epidemics and the configuration model done carefully.
- A.-L. Barabási, *Network Science*, Cambridge University Press, 2016 (free online). Visual and readable, from the author of preferential attachment.
- R. Pastor-Satorras, C. Castellano, P. Van Mieghem and A. Vespignani, "Epidemic processes in complex networks," *Reviews of Modern Physics* 87 (2015), 925–979.
- S. H. Strogatz, "From Kuramoto to Crawford: exploring the onset of synchronization in populations of coupled oscillators," *Physica D* 143 (2000), 1–20.
- S. H. Strogatz, *Sync: The Emerging Science of Spontaneous Order*, Hyperion, 2003. Fireflies, hearts and bridges for a general reader.
- A. D. Broido and A. Clauset, "Scale-free networks are rare," *Nature Communications* 10 (2019), 1017.

Chapter 88 — Complexity Science: When the Whole Exceeds the Parts

> **Level 2 · Track I — Dynamical Systems, Chaos, and Complexity Builds on:** Ch. 14, Ch. 25, Ch. 34, Ch. 86, Ch. 87 · **Leads to:** Track J, Ch. 92, Ch. 98 · **Related:** Ch. 45, Ch. 51, Ch. 85 **Reading time:** ~25 min

88.1 The hook

Shortly after 3 p.m. on 14 August 2003, a heavily loaded transmission line in northern Ohio sagged in the summer heat and touched a tree that should have been trimmed. The line tripped, exactly as its protection was designed to do. Its current shifted to neighbouring lines, which grew hotter and sagged in turn. An alarm program in the local control room had silently stalled, so the operators did not see what was happening. Over the next hour more lines tripped, each one pushing load onto the rest, until shortly after 4 p.m. the cascade swept across the north-eastern United States and Ontario in about seven minutes. Some 50 million people lost power.

Nothing in that story is exotic. Every component did what it was built to do. The grid had redundancy precisely so that losing one line would not matter, and usually it doesn't. Almost every day, somewhere, a line trips and nothing happens. Very occasionally, the same small event takes down a continent.

Engineers who collected decades of blackout records found something stranger still. Plot how often blackouts of each size occur and there is no "typical" blackout. Small ones are common and big ones are rare, but the frequency falls off slowly, roughly like a power of the size, so that the catastrophic events are far more likely than a bell curve would ever allow. Earthquakes, forest fires, stock-market crashes and extinctions in the fossil record show the same heavy-tailed signature.

Why would systems as different as power grids and the Earth's crust share the statistics of their disasters? And why would a system built to be stable organise itself, without anyone intending it, to the brink of collapse?

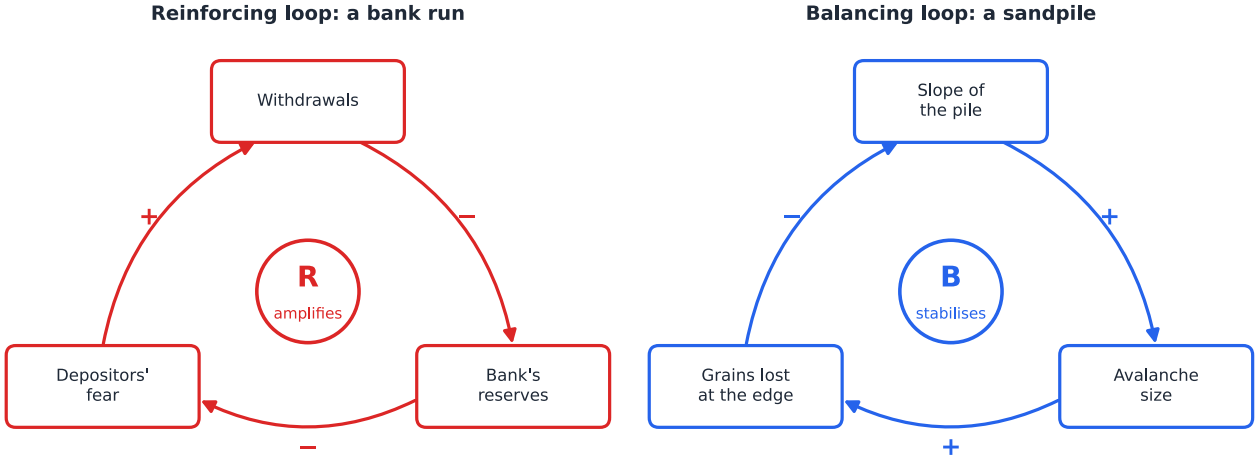
88.2 The big idea

"More is different," wrote the physicist Philip Anderson in 1972. Knowing everything about one water molecule tells you nothing about waves, whirlpools or ice. Knowing one neuron tells you nothing about a thought. **Complexity science** studies systems of many interacting parts whose collective behaviour has properties the parts don't have. Three words organise the field.

Emergence is the appearance of large-scale patterns from local interactions, as with the gliders of Ch. 86 and the synchrony of Ch. 87. An emergent pattern can be described by its own laws (a glider moves at $c/4$; a flock has a mean heading) without reference to the parts.

Self-organisation is emergence without a designer or a leader. A flock of starlings has no commander, an ant colony has no architect, and a market has no central planner, yet each produces coordinated structure. The organising force is the interactions themselves.

Feedback is the mechanism. When an effect circles back to influence its own cause, the loop either amplifies (positive, or *reinforcing*, feedback) or damps (negative, or *balancing*, feedback). Engineers know feedback from control theory (Ch. 14), where a balancing loop holds a motor at its set speed. In complex systems the loops are not designed, there are many of them, and they interact.



Loop polarity = product of the signs on its arrows: $(-)(-)(+) = +$ reinforces; $(+)(+)(-) = -$ balances

Causal-loop diagrams. Each arrow carries a sign: $+$ if more of the cause means more of the effect, $-$ if it means less. In a bank run (left), withdrawals drain reserves, falling reserves raise fear, and fear causes more withdrawals: two minus signs and a plus multiply to a reinforcing loop. In a sandpile (right), a steeper pile makes larger avalanches, which spill more grains off the edge, which lowers the slope: a balancing loop.

The analogy for the whole chapter is a sandpile on a table. Drop sand grains one at a time on a pile. At first the pile just grows. As it steepens, grains start to slide, small avalanches at first, occasionally large ones that spill sand over the edge. Eventually the pile reaches a slope it cannot exceed: add more sand and it sheds, on average, exactly what you add. At that point the pile is **critical**. Most grains do nothing; a few trigger slides that run across the entire pile. Nobody tuned the slope. The balancing loop in the figure found it.

That is the big idea of **self-organised criticality** (Bak, Tang and Wiesenfeld, 1987): slowly driven systems with thresholds and dissipation can drive *themselves* to a critical state, where events of all sizes occur and the size distribution is a power law. It is a candidate answer to the hook. The grid is loaded by growing demand (the slow drive), protected by trips (the threshold), and repaired and upgraded after failures (the dissipation). Its own feedback loops may push it to the edge.

88.3 The mathematics

Feedback in one line

Take a quantity x and a loop that returns a fraction g of any change to the input on the next round, so that a small deviation evolves as $\delta_{t+1} = g\,\delta_t$, and $\delta_t = g^t \delta_0$. The **loop gain** g is the product of the gains around the loop, which is why the *signs* on a causal-loop diagram multiply. If $g > 0$ the loop is reinforcing, if $g < 0$ it is balancing, and the deviation dies out exactly when $|g| < 1$. (For a balancing loop with $g < -1$, the correction overshoots by more than the error, and the system oscillates with growing amplitude: the mechanism behind the bullwhip effect in supply chains and an over-tuned thermostat.) This is the scalar case of the stability theory in Ch. 17.

In a complex system the gain is not fixed. It depends on the state, and the state depends on the gain. That loop-of-loops is what lets a system *find* its own critical point, and the cleanest place to see it is a pile of sand.

The Abelian sandpile

Bak–Tang–Wiesenfeld (Abelian) sandpile

Each site of an $L \times L$ grid holds $z_i \in \{0, 1, 2, \dots\}$ grains. A site with $z_i \geq 4$ is **unstable** and **topples**: it loses 4 grains and each of its four neighbours gains one. Grains sent off the edge of the grid are lost. A configuration with all $z_i \leq 3$ is **stable**. The dynamics: add one grain at a random site, then topple unstable sites until the configuration is stable. The total number of topplings is the **avalanche size** s.

Two things need proof before this is even a well-defined model. Does an avalanche always stop? And when several sites are unstable at once, does it matter which one topples first?

Avalanches are finite

On a finite grid with grains lost at the boundary, every avalanche ends after finitely many topplings.

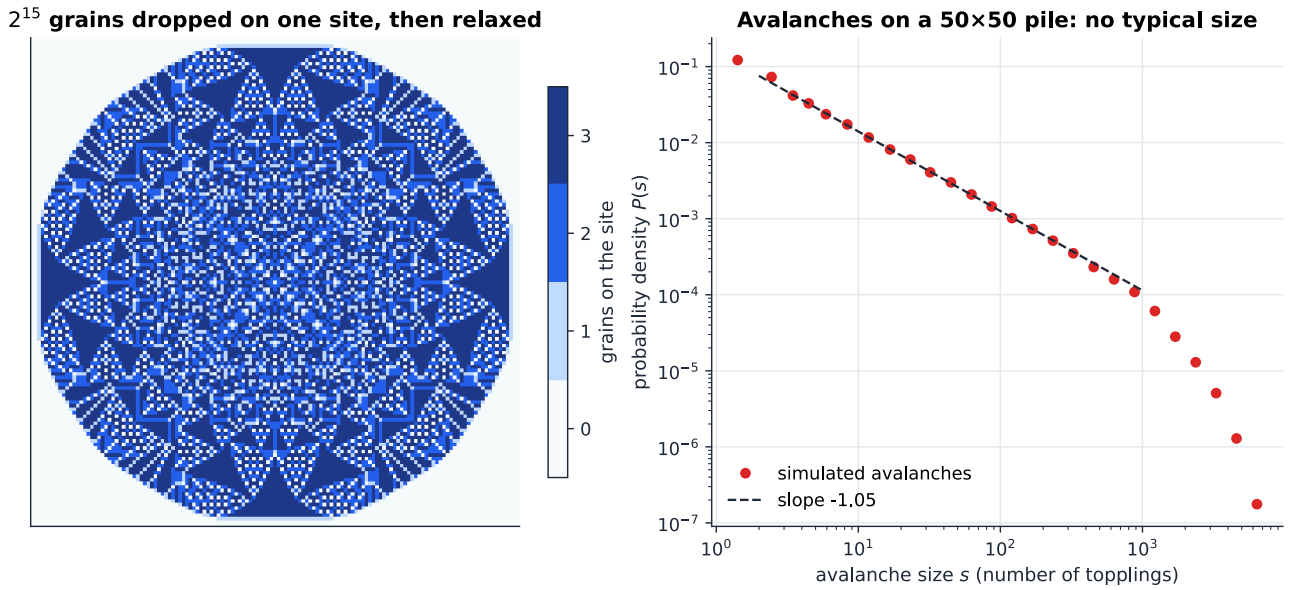
Proof. Suppose some avalanche never ends. There are finitely many sites, so some site a topples infinitely often. Each of its topplings gives a grain to each neighbour, so every neighbour of a receives infinitely many grains; a site that topples only finitely often can only gain grains from some point on, never lose them, so its height would grow without bound and it would have to topple again. Hence every neighbour of a also topples infinitely often, and by repeating the argument, so does every site of the connected grid, including a boundary site. But each toppling of a boundary site throws at least one grain off the grid, permanently, while the total number of grains is finite and only decreases during an avalanche. Infinitely many such losses are impossible. ■

The Abelian property

The final stable configuration, and the number of times each site topples, do not depend on the order in which unstable sites are toppled.

Proof sketch. The local step is that two topplings **commute**. Suppose sites a and b are both unstable. Toppling a can only *add* grains to b, so b is still unstable afterwards, and vice versa. Toppling a site is the same as subtracting a fixed vector Δ_a (minus 4 at a, plus 1 at each neighbour) from the configuration, and vector subtraction commutes: toppling a then b and toppling b then a both give $z - \Delta_a - \Delta_b$. So any two legal toppling sequences can be rearranged into one another by swapping neighbouring steps, provided both are finite. That each order is finite is the previous theorem. The full argument (a "least action" principle: any legal toppling sequence topples each site at most as often as any other sequence that reaches a stable configuration) turns the local swap into the global statement. ■

The name *Abelian* comes from this commutativity. Write S_a for "add a grain at a and stabilise". Then $S_a S_b = S_b S_a$: the order in which grains arrive doesn't matter either. It is a startlingly strong property for a model of disasters, and it is the key to every exact result about the sandpile.



Left: $2^{15} = 32{,}768$ grains placed on one site of a large grid and toppled to stability. The result is a disc of radius about 70 sites filled with intricate, self-similar structure, though the rule is just "four grains, topple". Right: dropping grains one at a time at random sites of a 50×50 pile, after it has reached its steady state, gives avalanches whose sizes follow a straight line on log–log axes for three decades, until the finite pile cuts them off.

Two facts in the figure deserve emphasis. In the driven pile, the mean height settles at about 2.1 grains per site, *whatever the initial condition*. Start empty, and the pile fills up until it gets there; start overfull, and the first avalanches shed the excess. That is the balancing loop made precise. And at that height the avalanches have no typical size: on this pile the distribution falls roughly like $s^{-1.05}$, and the largest avalanche toppled sites more than seven thousand times.

Result 3: why criticality means power laws

What does "critical" mean? Think of an avalanche as a **branching process**: each toppling sends grains to neighbours, and each of them may topple in turn. If each toppling triggers on average μ further topplings, avalanches die out quickly for $\mu < 1$ (subcritical), can run forever for $\mu > 1$ (supercritical), and sit on the knife edge for $\mu = 1$. The pile's feedback loop pushes μ toward 1 from both sides: below it, grains accumulate and the pile steepens, raising μ; above it, huge avalanches dump grains off the edge, lowering μ.

The simplest critical branching process can be solved exactly, and it shows where the power law comes from.

Total size of a critical binary branching process

Each event independently has either 0 or 2 offspring events, with probability $\frac{1}{2}$ each. The probability that the total number of events is $s = 2n + 1$ is

$$P(s) = \frac{C_n}{2^{2n+1}}, \qquad C_n = \frac{1}{n+1}\binom{2n}{n},$$

and for large s,

$$P(s) \approx \sqrt{\frac{2}{\pi}}\, s^{-3/2}.$$

Proof. A finished avalanche is a tree in which every node has 0 or 2 children: a *full binary tree*. A full binary tree with n internal nodes (those with two children) has $n + 1$ leaves, so $2n + 1$ nodes in all. Each internal node has probability $\frac{1}{2}$ (it chose "2") and each leaf has probability $\frac{1}{2}$ (it chose "0"), independently, so each particular tree has probability $2^{-(2n+1)}$. The number of full binary trees with n internal nodes is the Catalan number C_n (a standard count: splitting at the root gives $C_n = \sum_{i=0}^{n-1} C_i C_{n-1-i}$, the recurrence that defines the Catalan numbers, solved with a generating function in the style of Ch. 28). Multiplying gives $P(s) = C_n 2^{-(2n+1)}$.

For the asymptotics, Stirling's formula gives $\binom{2n}{n} \approx 4^n/\sqrt{\pi n}$, so $C_n \approx 4^n/(\sqrt{\pi}\, n^{3/2})$ and

$$P(s) \approx \frac{1}{2\sqrt{\pi}\, n^{3/2}} = \frac{1}{2\sqrt{\pi}} \left(\frac{s}{2}\right)^{-3/2} = \sqrt{\frac{2}{\pi}}\, s^{-3/2},$$

using $n \approx s/2$. ■

(Check: at $s = 101$ the exact value is 7.80×10^{-4} and the asymptotic formula gives 7.86×10^{-4}.) Two features of this answer are worth staring at. The probabilities add up to 1, so every avalanche ends; but the mean avalanche size $\sum_s s\, P(s)$ diverges, because $s \cdot s^{-3/2} = s^{-1/2}$ is not summable. *Every avalanche is finite and the average avalanche is infinite.* That is the mathematical signature of criticality, and the reason risk calculations based on averages fail for heavy-tailed systems.

Move off criticality and the power law acquires a cutoff. With "2 offspring" probability $p \neq \frac{1}{2}$, the same counting gives $P(s) = C_n p^n (1-p)^{n+1}$, and since $C_n \approx 4^n/(\sqrt{\pi} n^{3/2})$, the factor $(4p(1-p))^n$ appears. Because $4p(1-p) < 1$ whenever $p \neq \frac{1}{2}$, it decays exponentially: small avalanches still look like a power law, but there is now a typical largest size. Only at the critical point does the cutoff disappear. You will drag this cutoff around in the experiment.

The exponent $\frac{3}{2}$ is the **mean-field** value, exact for branching on a tree, or for sandpiles in high dimensions. The two-dimensional sandpile has different exponents, because avalanches there fold back and revisit sites. Simulations put them near the 1.05 in the figure for small piles, drifting with size, and nobody has computed them exactly.

? What is the sandpile's hidden algebra?

The Abelian property says that adding grains are commuting operations. Restrict them to the *recurrent* configurations, the ones the driven pile keeps returning to, and they form a finite **group**, with an identity element that is itself a striking fractal picture. The number of elements of this group equals the number of spanning trees of the grid, which in turn is the determinant of a reduced graph Laplacian. Why should a toy model of avalanches know how many spanning trees a graph has? The group structure is the language of Ch. 56, and the Laplacian is the matrix of Ch. 27; putting them together to count trees is a theorem you'll have to hunt down.

Self-organisation without a leader: flocking

A sandpile organises itself in *time*, toward a critical state. Flocks organise themselves in *space and direction*. In 1995 Tamás Vicsek and colleagues proposed the simplest model of collective motion.

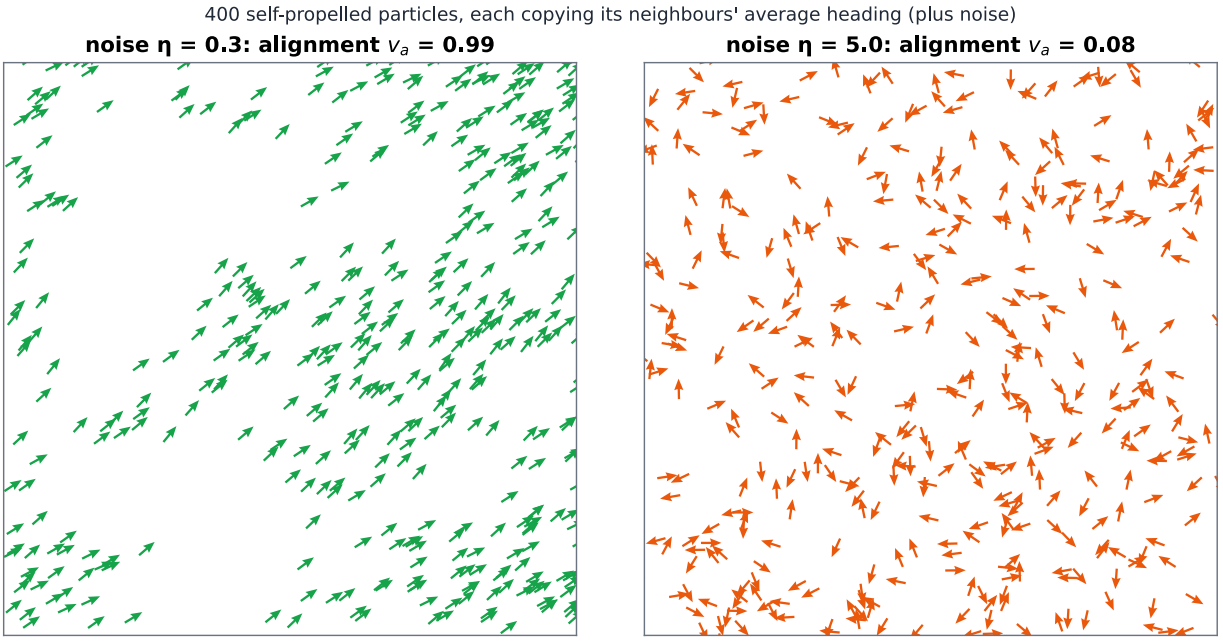
Vicsek model

N particles move in a periodic box at constant speed v. At each step, particle i adopts the average heading of all particles within distance r of it (itself included), plus noise:

$$\theta_i(t+1) = \operatorname{Arg}\left(\sum_{\|\mathbf{x}_j - \mathbf{x}_i\| < r} e^{j\theta_j(t)} \right) + \eta\,\xi_i(t), \qquad \mathbf{x}_i(t+1) = \mathbf{x}_i(t) + v \begin{pmatrix} \cos\theta_i(t+1) \\ \sin\theta_i(t+1) \end{pmatrix},$$

with ξ_i independent and uniform on $\left[-\frac{1}{2}, \frac{1}{2}\right]$. The **alignment** is $v_a = \left| \frac{1}{N} \sum_i e^{j\theta_i} \right|$.

The alignment v_a is the same average phasor as the Kuramoto order parameter of Ch. 87, and the logic is similar: a feedback between individual headings and the local average. The new ingredient is that the network of who-sees-whom is not fixed; it is rebuilt every step by the motion itself. Aligned neighbours travel together and stay neighbours, which strengthens their alignment. That is a reinforcing loop, and the noise is the dissipation that opposes it.



The Vicsek model with 400 particles in a 10×10 box (interaction radius 1, speed 0.03) after 600 steps. At low noise (left) a single flock forms with almost perfect alignment, $v_a = 0.99$, from random initial headings. At high noise (right) headings stay disordered and $v_a = 0.08$, close to the $1/\sqrt{N} = 0.05$ expected for random directions.

Between the two panels lies a phase transition at a critical noise η_c that depends on the density. Nobody in the flock knows the heading of the flock, and no particle is special. Order is a collective property, decided by a competition between the reinforcing loop of alignment and the randomising effect of noise.

Adaptive systems

In a sandpile and a flock the rules are fixed. In an **adaptive** system the parts change their rules in response to what the collective does, and the collective is made of those changing rules. Markets, immune systems, ecosystems and traffic are all of this kind.

The classic toy is Brian Arthur's El Farol bar problem (1994). One hundred people decide independently each week whether to go to a bar. The evening is pleasant if fewer than 60 go, and unpleasant otherwise. There is no correct prediction that everybody can follow: if everyone believes "few will go", everyone goes, and the belief is falsified. Arthur gave each agent a small bag of rules of thumb ("same as last week", "average of the last four weeks", "mirror image of last week around 50") and let each agent use whichever rule had recently predicted best. Attendance never settles, but it fluctuates around 60. The comfortable capacity is found by nobody and enforced by nobody: it emerges from a balancing loop between collective attendance and individual strategy choice.

The mathematics of adaptive populations, where strategies that do well become more common, is evolutionary game theory and the replicator equation of Ch. 45. Its phenomena are new: no equilibrium may exist, the system may cycle forever, and the "rules of the game" are themselves an outcome of the game.

> **Emergence is not magic**
>
> It is tempting to treat "emergent" as an explanation. It is a description. The gliders of Life, the power law of the sandpile and the order of the flock are all *consequences* of their rules and can be derived, simulated or bounded. What makes them emergent is that they are not visible in any single part's rule and are best described in a new vocabulary. Whenever you read that something "emerges", ask: from which interactions, through which feedback, and what would make it fail?

88.4 The algorithm

Toppling a sandpile

There are two ways to relax a sandpile, and the Abelian property says they give the same answer.

Parallel toppling topples every unstable site at once, and can even topple a site several times in one sweep: a site with z grains topples $\lfloor z/4 \rfloor$ times. With NumPy this is a few array operations per sweep:

```
def stabilise(z):
    """Relax a sandpile; grains falling off the edge are lost. Returns grid and avalanche size."""
    total = 0
    while True:
        t = z // 4                    # topplings at each site this sweep
        if not t.any():
            return z, total
        total += int(t.sum())
        z = z - 4 * t
        z[1:, :] += t[:-1, :]         # each toppling sends one grain to each neighbour
        z[:-1, :] += t[1:, :]
        z[:, 1:] += t[:, :-1]
        z[:, :-1] += t[:, 1:]
```

Sequential toppling with a **stack** of unstable sites does the same work site by site, and is faster when avalanches are small. Either way, the cost of an avalanche is proportional to its size, which for a critical pile is heavy-tailed: most grains cost nothing, and a few cost a large fraction of the pile. Placing N grains on one site is much worse than N separate drops, because the resulting disc has radius about $\sqrt{N}$ and every grain must be passed outward across it. For the figure's 2^{15} grains that is about 19 million single topplings, which the parallel version performs in about 6,000 sweeps.

Flocks in linear time

A naive Vicsek step compares every pair of particles, $O(N^2)$. Because the interaction radius r is fixed, divide the box into cells of side r and only compare particles in the same or adjacent cells. Each step then costs $O(N)$ at fixed density, which is the same **cell-list** idea used in molecular dynamics and physics engines (Ch. 72).

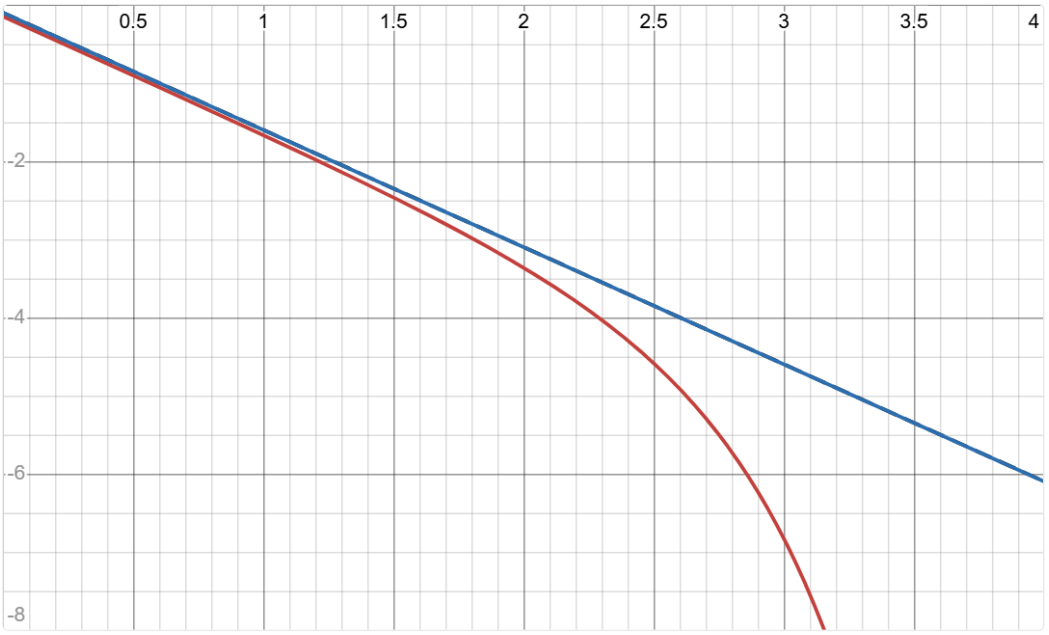
Failure cases

- **Transients.** A sandpile started empty spends thousands of grains just filling up. Measuring avalanches before it reaches its steady state (the figure discarded the first 10,000 grains) mixes two different statistics.
- **Fitting power laws by straight lines.** A least-squares line through a log–log histogram is biased, depends on the binning and ignores the cutoff. It is good enough for a picture, not for a claim. Use maximum-likelihood estimators and compare against alternative distributions.
- **Finite size.** Every simulated power law ends at a cutoff set by the system size. The honest test of criticality is **finite-size scaling**: check that the cutoff moves as a power of L as you simulate $L = 25, 50, 100, 200$.

88.5 The experiment

Part 1: the knife edge of criticality

On log–log axes, the size distribution of a branching process with "2 offspring" probability p is a straight line of slope $-\frac{3}{2}$ times an exponential cutoff. With $x = \log_{10} s$ and $y = \log_{10} P(s)$, the large-s approximation from §88.3 reads



Log–log avalanche-size distributions. Blue (critical, $p = 0.5$) lies exactly on the dashed power law of slope -1.5; red ($p = 0.55$, so each event triggers on average 1.1 more) bends down sharply after a few hundred events. A supercritical avalanche that has grown that large almost never stops, so large finite avalanches are rare; about 18% of these avalanches run forever.

The formula is $P(s) \approx (1-p)\,2^{1.5}\pi^{-1/2}s^{-3/2}\,(4p(1-p))^{s/2}$. Change `0.55` and `0.45` in the red line together to `0.51` and `0.49` and watch the cutoff move out to the right; the typical largest avalanche grows like $1/(p-\frac{1}{2})^2$. Then swap the two numbers in the red line, giving $p = 0.45$ and $1-p = 0.55$ (a subcritical process). The curve is nearly the same: on a log–log plot, sub- and supercritical processes look alike, and only the critical one has no scale at all.

Part 2: grow your own critical pile

```
import numpy as np
import matplotlib.pyplot as plt

L, GRAINS, BURN = 40, 40000, 8000      # pile size, grains to drop, grains to discard
rng = np.random.default_rng(0)
z = np.zeros((L, L), dtype=int)
sizes, heights = [], []

for g in range(GRAINS):
    i, j = rng.integers(0, L, 2)
    z[i, j] += 1
    s = 0
    while (t := z // 4).any():         # parallel toppling until stable
        s += t.sum()
        z -= 4 * t
        z[1:, :] += t[:-1, :]; z[:-1, :] += t[1:, :]
        z[:, 1:] += t[:, :-1]; z[:, :-1] += t[:, 1:]
    heights.append(z.mean())
    if g >= BURN and s > 0:
        sizes.append(s)

sizes = np.array(sizes)
print(f"final mean height {z.mean():.3f}, avalanches {len(sizes)}, largest {sizes.max()}")
bins = np.logspace(0, np.log10(sizes.max()), 25)
h, e = np.histogram(sizes, bins=bins)
c = np.sqrt(e[1:] * e[:-1]); ok = h > 0
fig, (a, b) = plt.subplots(1, 2, figsize=(10, 4))
a.plot(heights); a.set_xlabel("grains dropped"); a.set_ylabel("mean height")
b.loglog(c[ok], h[ok] / np.diff(e)[ok] / len(sizes), "o")
b.set_xlabel("avalanche size s"); b.set_ylabel("P(s)")
plt.show()
```

Things to try:

- The left plot shows the pile filling up and then *stopping* at a height near 2.1 by itself. Start instead from `z = 3 * np.ones((L, L), dtype=int)` (overfull and fragile). Where does it end up?
- Double `L` and compare the largest avalanche. A critical system's cutoff should grow with its size; a non-critical one would not.
- **Break the criticality.** Add dissipation inside the pile: after each toppling sweep, remove each grain with a small probability (for example `z -= (rng.random((L, L)) < 0.001) & (z > 0)` inside the `while` loop). The balancing loop now has a second exit, the pile settles lower, and the power law acquires a cutoff that no longer depends on `L`.
- Make sites topple only when they hold 5 grains while still sending out 4 (change `z // 4` to `z // 5` but keep `z -= 4 * t`). Where does the height settle now, and is the avalanche distribution still a power law?

88.6 The challenge

1. **Intuition.** A thermostat heats a room when it is below 20 °C, but the heater takes ten minutes to warm up and ten minutes to cool down. Draw the causal-loop diagram, mark the signs, and explain why the room temperature oscillates instead of settling, even though the loop is balancing. Then name one real complex system in which a delay turns a balancing loop into a source of boom-and-bust cycles.
2. **Derivation.** For the branching process with offspring 0 or 2 (probability $1-p$ and p), let q be the probability that an avalanche is finite. Show that $q = (1-p) + pq^2$, solve it, and prove that $q = 1$ for $p \le \frac{1}{2}$ and $q = (1-p)/p$ for $p > \frac{1}{2}$. Then show that the mean size of a finite avalanche is $1/(1-2p)$ for $p < \frac{1}{2}$, and explain what its divergence at $p = \frac{1}{2}$ means for anyone trying to estimate risk from past averages.
3. **Implementation.** Use the experiment's sandpile code to test the Abelian property directly. Start from one stable configuration, add 50 grains at 50 chosen sites, and stabilise after each grain. Then add the same 50 grains in a random different order, and finally add all 50 at once and stabilise once. Check that all three final configurations are identical and that the *total* number of topplings is the same in all three cases. Then break the property: make toppling send grains only to the right and down neighbours, and see whether order now matters.
4. **Open investigation.** Build the El Farol bar model: 100 agents, threshold 60, each with k random predictors chosen from a pool (for example weighted averages of the last few weeks), each using its recently most accurate predictor. Measure the mean and variance of attendance as you change k, the pool and the memory. Does attendance always hover near 60? When does it fluctuate wildly, and is there a "critical" amount of strategic diversity at which the fluctuations are largest? There is no complete theory.

88.7 Loose threads

This chapter offered a three-part vocabulary (emergence, self-organisation, feedback), one exactly solvable mechanism (the critical branching process and its $s^{-3/2}$ law), and two famous models (the sandpile and the flock). The real excitement is in what it left unresolved.

? How can a flock beat the theorem that forbids it?

In two dimensions, a magnet made of spins that align with their neighbours cannot have true long-range order at any positive temperature: thermal fluctuations always win at long distances (the Mermin–Wagner theorem). The Vicsek flock looks like the same system, alignment plus noise, yet simulations of huge flocks show genuine long-range order: particles at opposite ends of a very large flock agree on a heading. The difference is that Vicsek particles *move*, and moving carries information about orientation across the flock faster than diffusion could. In 1995 John Toner and Yuhai Tu built a continuum theory of flocks, the "hydrodynamics of active matter", to explain it. How motion defeats an equilibrium theorem is a question for Ch. 92.

? Is the brain poised at criticality?

In 2003 John Beggs and Dietmar Plenz recorded bursts of activity in slices of rat cortex and found "neuronal avalanches" whose sizes followed a power law with exponent close to $\frac{3}{2}$, the critical branching value derived in this chapter. A critical brain would have maximal dynamic range and information transmission, which makes the hypothesis attractive. But power laws can arise in many ways that have nothing to do with criticality, subsampling a network can create or destroy them, and twenty years of experiments have not settled whether the brain tunes itself to a critical point, hovers near one, or neither. This is an open problem; how researchers go about attacking such a question is the subject of Ch. 98.

Other threads:

- **The edge of chaos, revisited.** Ch. 86 asked whether computation concentrates at a transition between order and chaos. Self-organised criticality is a mechanism by which a system could *put itself* there. Whether adaptive systems in fact evolve toward criticality is part of the same unresolved debate.
- **From feedback loops to control.** A causal-loop diagram is a qualitative transfer-function block diagram. Making it quantitative, with gains, delays and stability margins, is the engineering of Ch. 14; choosing the feedback to optimise performance is Ch. 51.
- **Cascades on networks.** The 2003 blackout spread along a network, not a grid. Combining the thresholds of this chapter with the degree distributions of Ch. 87 gives models of cascading failure in which hubs are both the network's strength and its weak point.

Further reading

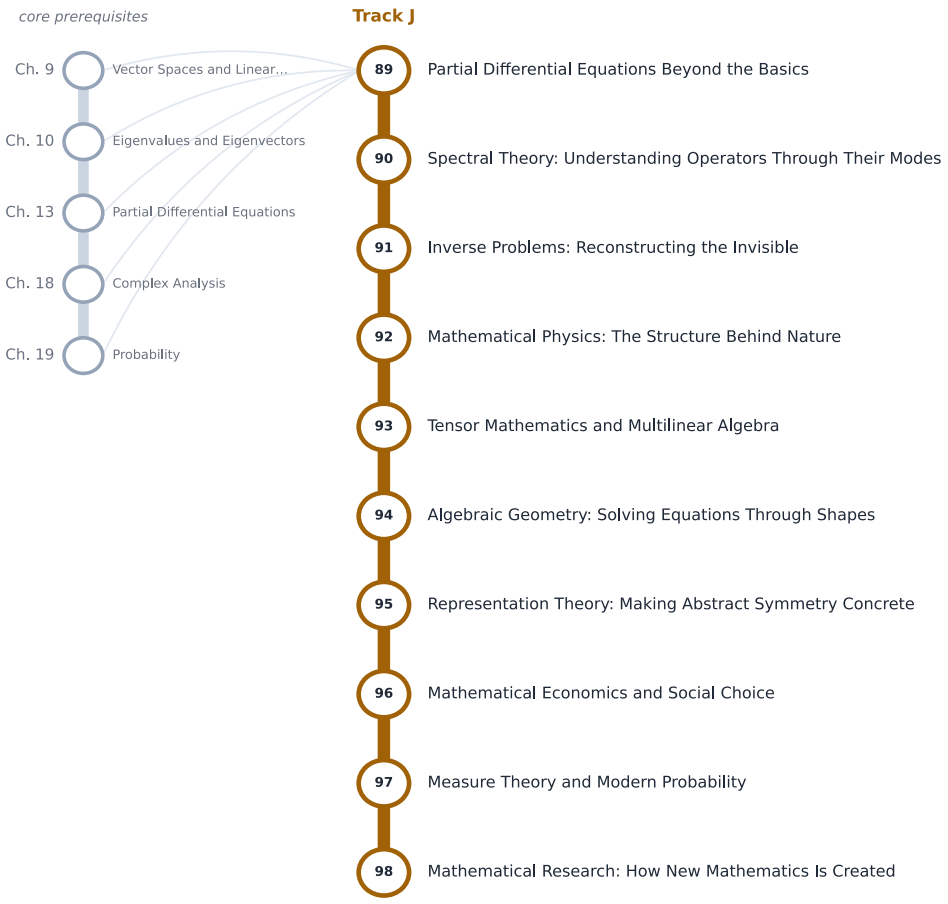
- P. Bak, *How Nature Works: The Science of Self-Organized Criticality*, Copernicus, 1996. The founder's passionate, contestable case.
- P. Bak, C. Tang and K. Wiesenfeld, "Self-organized criticality: an explanation of the 1/f noise," *Physical Review Letters* 59 (1987), 381–384.
- D. Dhar, "Theoretical studies of self-organized criticality," *Physica A* 369 (2006), 29–70. The Abelian sandpile, its group and its exact results.
- T. Vicsek, A. Czirók, E. Ben-Jacob, I. Cohen and O. Shochet, "Novel type of phase transition in a system of self-driven particles," *Physical Review Letters* 75 (1995), 1226–1229.
- M. Mitchell, *Complexity: A Guided Tour*, Oxford University Press, 2009. A clear, broad and honest introduction.
- W. B. Arthur, "Inductive reasoning and bounded rationality," *American Economic Review* 84 (1994), 406–411. The El Farol bar problem.
- I. Dobson, B. A. Carreras, V. E. Lynch and D. E. Newman, "Complex systems analysis of series of blackouts: cascading failure, critical points, and self-organization," *Chaos* 17 (2007), 026103.

Level 3 — Research Frontier

Track J. Deeper territories to explore as your interests develop.

Track J — Mathematics at the Frontier

> **Level 3 — Research Frontier.** Not prerequisites for every programmer: the deeper territories to explore as your interests develop. The frontier is open-ended.



Suggested reading order for Track J. The grey stations on the left are the core chapters it leans on.

Before you start

This track assumes you are comfortable with these core chapters:

- Ch. 9 — Vector Spaces and Linear Transformations
- Ch. 10 — Eigenvalues and Eigenvectors
- Ch. 13 — Partial Differential Equations
- Ch. 18 — Complex Analysis
- Ch. 19 — Probability

Chapters

Ch.	Chapter	What it explores
89	Partial Differential Equations Beyond the Basics	Weak solutions, boundary-value problems, regularity.
90	Spectral Theory: Understanding Operators Through Their Modes	Eigenfunction expansions, operator spectra, applications to PDEs.
91	Inverse Problems: Reconstructing the Invisible	Parameter estimation, regularization, ill-posed problems.
92	Mathematical Physics: The Structure Behind Nature	Variational principles, symmetries, Hamiltonian mechanics.
93	Tensor Mathematics and Multilinear Algebra	Tensor products, multilinear maps, tensor decompositions.
94	Algebraic Geometry: Solving Equations Through Shapes	Polynomial varieties, ideals, coordinate rings.
95	Representation Theory: Making Abstract Symmetry Concrete	Group representations, characters, irreducible structures.
96	Mathematical Economics and Social Choice	Equilibrium, preference relations, welfare, voting theory.

Ch.	Chapter	What it explores
97	Measure Theory and Modern Probability	Sigma-algebras, measures, Lebesgue integration, convergence.
98	Mathematical Research: How New Mathematics Is Created	Conjectures, counterexamples, proof strategy, literature exploration.

Every chapter follows the same arc: **the hook** (a puzzle or a failure), **the big idea**, **the mathematics**, **the algorithm**, **the experiment** you can run, **the challenge**, and **loose threads**: the questions the chapter deliberately leaves open.

Chapter 89 — Partial Differential Equations Beyond the Basics

> **Level 3 · Track J — Mathematics at the Frontier Builds on:** Ch. 13, Ch. 22, Ch. 61 · **Leads to:** Ch. 90, Ch. 91, Ch. 92 · **Related:** Ch. 36, Ch. 52, Ch. 60, Ch. 75 **Reading time:** ~30 min

89.1 The hook

Stretch a guitar string between two pegs and press it down with one fingertip. The string takes the shape of a tent: two straight segments meeting at a sharp corner under your finger. Every physicist in the world agrees this is the correct shape. Now ask what equation it solves.

The textbook answer is the equation of a loaded string, $-Tu''(x) = f(x)$, with tension T and load density f. But the tent has no second derivative at the corner. The slope jumps from one value to another, so u'' simply does not exist there. And the load of a fingertip is not a function at all: it is all concentrated at one point, a Dirac delta. The "solution" fails to be twice differentiable, the "data" fails to be a function, and yet nothing about the physics is in doubt.

This is not a curiosity. Look around an engineering office:

- A bar made of two materials glued together, copper on the left and steel on the right, carrying heat. The temperature is continuous, but its slope jumps at the glue line because the conductivity jumps. The heat equation $-(ku')' = f$ of §13.4 has no classical solution.
- An L-shaped room, or an L-shaped bracket under load. Near the inside corner the stress, which is a derivative of the displacement, is *infinite*. The solution exists; its gradient does not stay bounded.
- A shock wave in a supersonic flow: the density jumps across a surface. A traffic jam forms on a motorway: the car density jumps. Neither "solution" is even continuous.

Meanwhile, every serious engineering simulation code (structural analysis, electromagnetics, chip thermal design, crash testing) solves PDEs with exactly these features, on millions of unknowns, every day, and gets the right answer. What do those codes know that §13.3's separation of variables doesn't?

> **What does it mean for a function that can't be differentiated to *solve* a differential equation, and how can that weaker meaning be the foundation of the most successful numerical method in engineering?**

The answer is a shift of viewpoint that took mathematics from Riemann (1850s) to Sobolev (1930s) to Lax, Milgram and the engineers of the 1950s and 60s to complete. It is also the doorway to some of the hardest open problems in mathematics.

89.2 The big idea

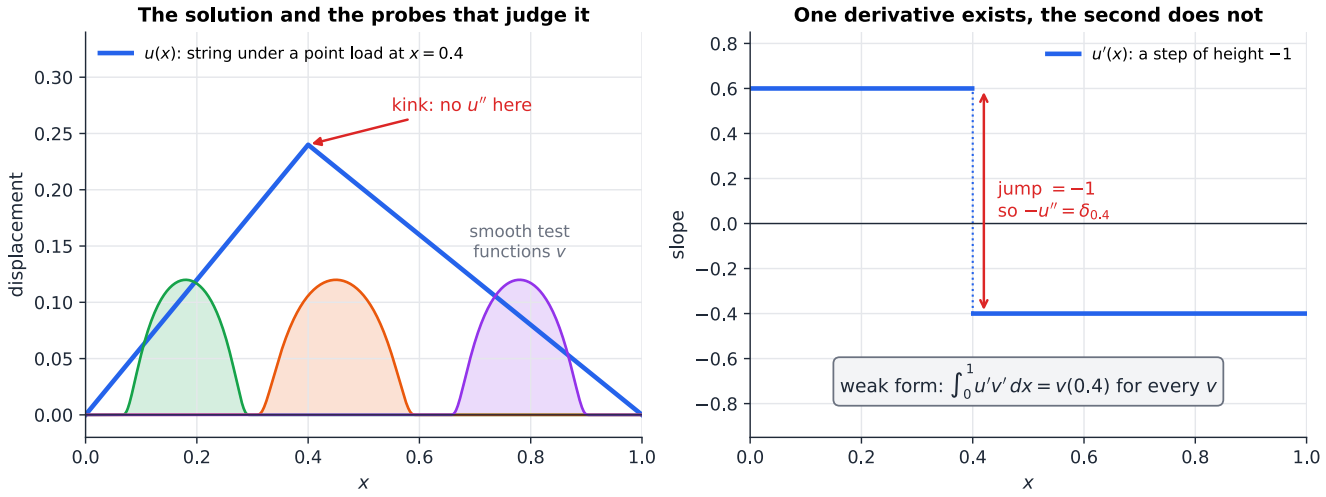
Judge a function by what it does to probes, not by its value at points. You cannot evaluate u'' at the corner of the tent. But you can ask a gentler question. Take any smooth "probe" function $v(x)$, a bump that vanishes at both ends of the string, and compute the integral $\int u''v\,dx$. Integrate by parts once:

$$-\int_0^1 u''\,v\,dx = \int_0^1 u'\,v'\,dx \qquad (v(0) = v(1) = 0).$$

The right-hand side only needs **one** derivative of u, and the tent has one: its slope is a step function, perfectly integrable. So we *define* "u solves $-u'' = f$" to mean

$$\int_0^1 u'\,v'\,dx = \int_0^1 f\,v\,dx \quad \text{for every probe } v.$$

The derivative has been moved off the rough unknown and onto the smooth probe. For the point load at $x = a$ the right side is just $v(a)$, and the tent passes this test. A function that passes it is a **weak solution**.



A weak solution is tested by integrals against smooth probes v, never at single points. The tent has a slope everywhere except one point, and that is enough to make $\int u'v' = v(0.4)$ meaningful and true.

An analogy: a referee who only watches replays averaged over a few seconds. A classical solution must be correct at every instant. A weak solution must be correct on average over every window, however small. A function can fail at isolated instants and still pass every averaged check, and for physics that is exactly right, because no instrument measures at a mathematical point.

The second idea: the solution is the cheapest shape. The weak equation is also the condition for a minimum. The string settles into the shape that minimizes its total energy, stretching energy minus the work done by the load:

$$E(v) = \tfrac{1}{2}\int_0^1 (v')^2\,dx - \int_0^1 f v\,dx.$$

The energy only involves v', never v''. So the natural home for the problem is the space of functions with **finite energy**: square-integrable functions with square-integrable first derivatives. That space, the **Sobolev space** H^1, was promised in Ch. 61's loose threads. It is a Hilbert space, so all the geometry of projection and orthogonality is available. Solving the PDE becomes "find the closest point" in a Hilbert space.

The third idea: regularity is earned, not assumed. Once a weak solution exists, you can ask how smooth it actually is. Often it is perfectly smooth and the whole detour was a formality. Sometimes, near a corner, a material interface or a shock, it is not, and knowing exactly *how* rough it is tells you how fast your simulation will converge. The modern theory of PDEs is largely this two-step programme: **first existence of a weak solution, then regularity**.

89.3 The mathematics

We work with the model boundary-value problem: find u on $(0,1)$ with

$$-(k(x)\,u')' = f \ \text{ on } (0,1), \qquad u(0) = u(1) = 0,$$

where the conductivity (or stiffness) satisfies $0 < k_{\min} \le k(x) \le k_{\max}$. Everything below extends, with harder analysis but the same ideas, to $-\nabla\cdot(k\nabla u) = f$ on a domain $\Omega \subset \mathbb{R}^d$.

Sobolev spaces

The space $H_0^1(0,1)$

A function $u \in L^2(0,1)$ has a **weak derivative** $w \in L^2(0,1)$ if $\int_0^1 u\,\varphi'\,dx = -\int_0^1 w\,\varphi\,dx$ for every smooth φ vanishing near 0 and 1. We write $u' = w$. The **Sobolev space** $H^1(0,1)$ is the set of $u \in L^2$ with a weak derivative in L^2, with inner product $\langle u, v\rangle_{H^1} = \int (uv + u'v')\,dx$. The subspace H_0^1 contains those with $u(0) = u(1) = 0$.

For the tent, the weak derivative is the step function, and the definition just says integration by parts still works when one factor has a corner. A function with a *jump* has no weak derivative in L^2: its derivative would have to be a delta. So H^1 functions in one dimension can have corners but not jumps. In fact they are always continuous, and here is the one-line reason. For $x < y$, by Cauchy–Schwarz (§9.8),

$$|u(y) - u(x)| = \left|\int_x^y u'\,dt\right| \le \sqrt{y-x}\;\|u'\|_{L^2}.$$

Finite energy forces Hölder continuity with exponent $\tfrac{1}{2}$. This is the simplest **Sobolev embedding theorem**. It is also where dimension starts to matter: in two dimensions the function $u = \ln\ln(1/r)$ near the origin has finite energy, $\int|\nabla u|^2 = 2\pi\int dr/(r\ln^2(1/r)) < \infty$, yet it is unbounded. A finite-energy membrane can have an infinitely tall spike. And conversely, the 2-D point-load solution $\ln(1/r)$ has *infinite* energy, $2\pi\int dr/r = \infty$: in a plane, a pin-point load is not an H^1 problem at all.

The weak formulation

Multiply the PDE by $v \in H_0^1$, integrate, and move one derivative across. The boundary term $[ku'v]_0^1$ vanishes because v does. This gives the problem we will actually solve.

Weak solution

$u \in H_0^1(0,1)$ is a **weak solution** if

$$a(u,v) = F(v) \quad \text{for all } v \in H_0^1, \qquad a(u,v) = \int_0^1 k\,u'v'\,dx, \quad F(v) = \int_0^1 f v\,dx.$$

Two quick facts justify the name. If u is a classical C^2 solution, it is a weak solution: that is the integration by parts we just did. Conversely, if a weak solution happens to be C^2 (and k is C^1), then integrating back gives $\int(-(ku')' - f)\,v = 0$ for all v, and a continuous function orthogonal to every probe is zero. So nothing is lost when solutions are smooth, and something is gained when they are not. For the two-material bar, the weak form automatically enforces the physically correct interface condition: the *flux* ku' is continuous across the glue line even though u' jumps.

Existence and uniqueness

Existence and uniqueness of the weak solution

If $0 < k_{\min} \le k \le k_{\max}$ and $f \in L^2(0,1)$, there is exactly one $u \in H_0^1$ with $a(u,v) = F(v)$ for all $v \in H_0^1$. It is also the unique minimizer of $E(v) = \tfrac{1}{2}a(v,v) - F(v)$ over H_0^1, and $\|u'\|_{L^2} \le \|f\|_{L^2}/k_{\min}$.

Proof. **Step 1, a Poincaré inequality.** For $v \in H^1_0$, $v(0) = 0$, so the embedding estimate with $x = 0$ gives $|v(y)| \le \sqrt{y}\,\|v'\| \le \|v'\|$. Squaring and integrating over $y \in (0,1)$:

$$\|v\|_{L^2} \le \|v'\|_{L^2}.$$

(The sharp constant is $1/\pi$; you will see why in Ch. 90.)

Step 2, a new inner product. Define $\langle u, v\rangle_a = a(u,v) = \int k u'v'$. It is symmetric and bilinear, and $\langle v,v\rangle_a \ge k_{\min}\|v'\|^2 \ge \frac{k_{\min}}{2}\|v\|^2_{H^1}$ by Step 1, so it is positive definite. It is also bounded above by $k_{\max}\|v\|^2_{H^1}$, so the "energy norm" $\|v\|_a = \sqrt{a(v,v)}$ is equivalent to the H^1 norm and H^1_0 is complete under it: a Hilbert space.

Step 3, Riesz. F is linear, and bounded in the energy norm: $|F(v)| \le \|f\|\,\|v\| \le \|f\|\,\|v'\| \le \|f\|\,\|v\|_a/\sqrt{k_{\min}}$. By the Riesz representation theorem (Ch. 61), every bounded linear functional on a Hilbert space is an inner product with a unique element: there is exactly one u with $\langle u, v\rangle_a = F(v)$ for all v. That is the weak equation. Taking $v = u$ gives $k_{\min}\|u'\|^2 \le \|f\|\|u'\|$, the stated bound.

Step 4, minimum energy. For any $w \in H^1_0$,

$$E(u+w) = E(u) + \underbrace{\big[a(u,w) - F(w)\big]}_{=0} + \tfrac{1}{2}a(w,w) \;\ge\; E(u),$$

with equality only if $a(w,w) = 0$, that is $w = 0$. ∎

Notice what did the work: not a formula, but geometry. The solution is the Riesz representer of the load, equivalently the point of H^1_0 that minimizes energy. Existence came from **completeness**. This is exactly where Dirichlet and Riemann went wrong in the 1850s. They "proved" existence for Laplace's equation by saying an energy bounded below must attain its minimum. Weierstrass gave a counterexample in 1870, an energy whose infimum is never attained in the space of smooth functions, and the "Dirichlet principle" was discredited for thirty years until Hilbert rescued it in 1900. The fix is to minimize over a complete space, H^1, and then prove the minimizer is smooth. That second step is the regularity theory.

For non-symmetric problems (add a convection term bu') the form a is no longer an inner product, and the **Lax–Milgram theorem** (1954) replaces Riesz: if a is bounded and coercive, $a(v,v) \ge \alpha\|v\|^2$, the weak problem is still uniquely solvable.

The point load, and Green's function

For $f = \delta_a$ the load is $F(v) = v(a)$. It is bounded on H^1_0 in one dimension, since $|v(a)| \le \|v'\|$, so the theorem applies even though $\delta_a \notin L^2$. The solution is the tent, which is the Green's function of §11.8:

$$G(x,a) = \begin{cases} x(1-a), & x \le a, \\ a(1-x), & x \ge a. \end{cases}$$

Check: G' is $1-a$ to the left of a and $-a$ to the right, a jump of -1. Using $v(0) = v(1) = 0$,

$$\int_0^1 G'v'\,dx = (1-a)\big[v(a) - v(0)\big] - a\big[v(1) - v(a)\big] = (1-a)v(a) + a\,v(a) = v(a).$$

The tent passes every test.

Boundary conditions: essential and natural

The weak form sorts boundary conditions into two kinds, a distinction that classical theory (§13.2) never makes.

Suppose the right end is not clamped but has a prescribed flux, $k(1)u'(1) = g$. Do *not* require $v(1) = 0$. The boundary term from integration by parts is now $k(1)u'(1)v(1) = g\,v(1)$, and the weak problem becomes: find u with $u(0) = 0$ such that

$$\int_0^1 k u'v'\,dx = \int_0^1 f v\,dx + g\,v(1) \quad \text{for all } v \text{ with } v(0) = 0.$$

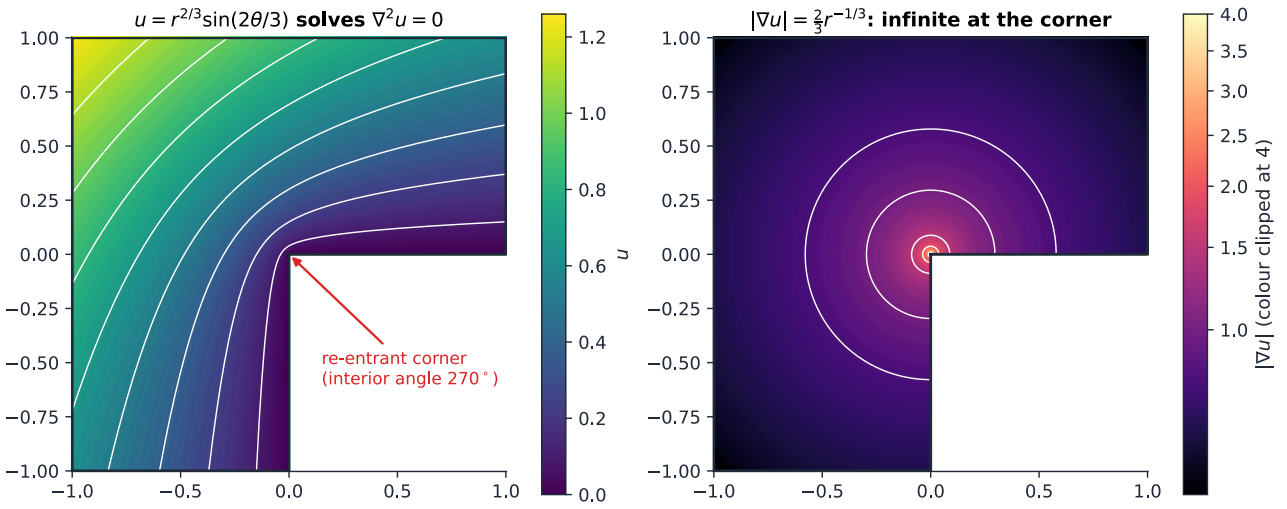
The Dirichlet condition $u(0) = 0$ had to be built into the space: it is **essential**. The Neumann condition never appears as a constraint at all; a weak solution that is smooth satisfies it automatically, because any mismatch $k(1)u'(1) - g$ would show up against probes with $v(1) \neq 0$. It is **natural**. In mechanics, displacements are essential and forces are natural; in electrostatics, potentials are essential and surface charges are natural.

Regularity: how smooth is the weak solution?

> **Elliptic regularity (stated)**
>
> If k is smooth and $f \in H^m$ (m derivatives in L^2), the weak solution lies in H^{m+2}: it gains two derivatives over the data. On a domain $\Omega \subset \mathbb{R}^d$ the same holds in the interior, and up to the boundary if the boundary is smooth or the domain is convex.

In one dimension you can see it directly: $ku' = -\int f + C$ is one derivative smoother than f, and u one more. The interesting cases are where the hypotheses fail:

- **Jumping coefficients.** The two-material bar has $u \in H^1$ but $u \notin H^2$: u' jumps at the interface. Regularity is limited by the material, not by the load.
- **Corners.** On the L-shaped domain, $u = r^{2/3}\sin(2\theta/3)$ is harmonic, vanishes on the two edges meeting at the re-entrant corner, and has $|\nabla u| = \frac{2}{3}r^{-1/3}$, which blows up. It lies in H^1 but not H^2. Generally, a corner of interior angle ω produces a singularity $r^{\pi/\omega}$, which is rougher than smooth exactly when $\omega > \pi$.



A perfectly smooth boundary value (zero) and a perfectly smooth equation (Laplace) still give an infinite gradient, purely because of geometry. Structural engineers know this as stress concentration at an inside corner, which is why such corners are rounded off with a fillet.

? Two derivatives for free, but only in the linear world

For $-\nabla \cdot (k\nabla u) = f$ with smooth k the theorem gives two derivatives for free. What if $k(x)$ is merely bounded and measurable, with no continuity at all, like a random composite? Then no derivatives are free, and in 1957–58 De Giorgi and Nash independently proved that weak solutions are *still* Hölder continuous, solving Hilbert's 19th problem. For *systems* of equations, such as elasticity, De Giorgi himself found a counterexample in 1968. For the 3-D Navier–Stokes equations, Leray proved in 1934 that weak solutions exist for all time. Whether they stay smooth, or whether a smooth flow can develop a singularity in finite time, is one of the seven Clay Millennium Problems, and it is open.

Other ways to be a solution

Integration by parts needs the equation to be in "divergence form", so that a derivative can be moved onto v. Two important families are not.

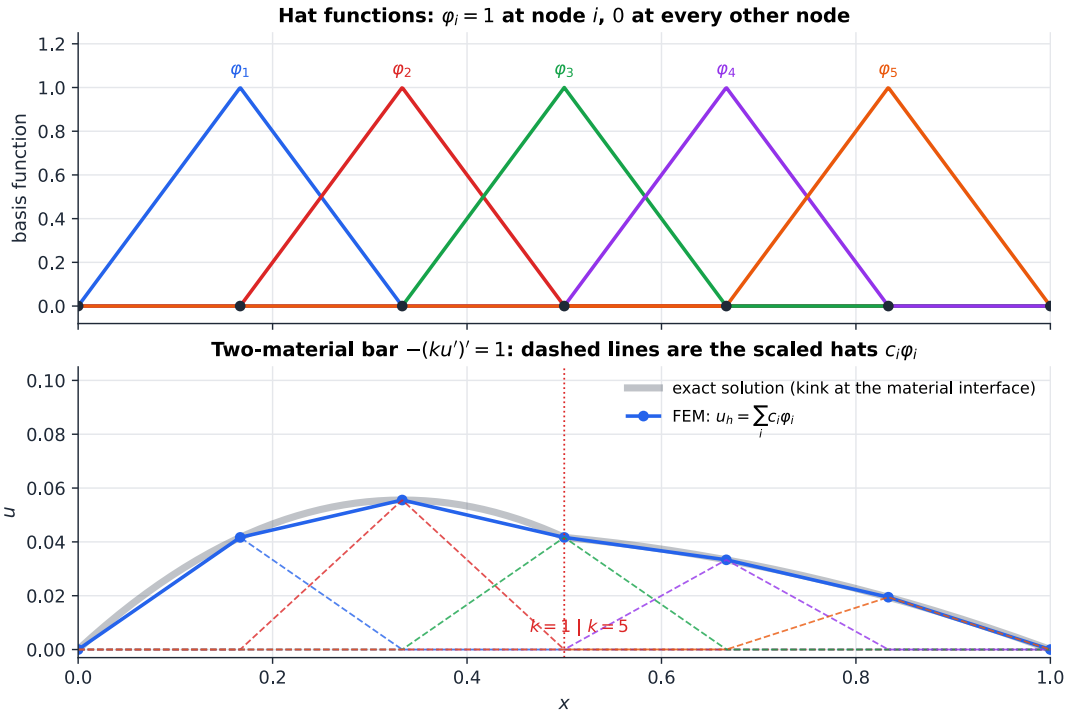
Viscosity solutions. The Hamilton–Jacobi–Bellman equation of Ch. 52 is nonlinear in u'. The simplest example is $|u'| = 1$ on $(-1, 1)$ with $u(\pm 1) = 0$: the travel time to the nearest exit, walking at unit speed. The true answer is $u = 1 - |x|$, with a kink in the middle. But the zigzag $\frac{1}{2} - \left||x| - \frac{1}{2}\right|$ also has $|u'| = 1$ everywhere except at three kinks, and so do infinitely many others. "Almost everywhere" is far too weak. Crandall and Lions (1983) found the right test: at a kink, compare u against smooth functions φ that touch it from above or from below, and demand an inequality on φ' there. The kink of $1 - |x|$ points upward and can only be touched from above, where the test passes. The zigzag's downward kink at 0 is touched from below by $\varphi \equiv 0$, where $|\varphi'| - 1 = -1 < 0$ violates the required inequality. So among all the almost-everywhere solutions exactly one survives, and it is the one obtained as the limit of $|u'| - 1 = \varepsilon u''$ as the "viscosity" $\varepsilon \to 0$.

Probabilistic solutions. The harmonic function with boundary values g is $u(\mathbf{x}) = E[g(\mathbf{W}_\tau)]$: start a Brownian motion at $\mathbf{x}$ and average g over the point where it first exits (Ch. 36). This needs no derivatives of u at all, and it is the seed of Monte Carlo PDE solvers that work in hundreds of dimensions, where no grid could ever fit.

89.4 The algorithm

The weak formulation is not only a theory. Replace "for all v in an infinite-dimensional space" by "for all v in a finite-dimensional subspace" and you have an algorithm. This is the **Galerkin method**, and with the right subspace it is the **finite element method** (FEM).

Step 1: choose a subspace. Split $[0, 1]$ into elements $[x_{i-1}, x_i]$ of width h. Let V_h be the continuous functions that are linear on every element and vanish at the ends. Its natural basis is the **hat functions**: φ_i equals 1 at node x_i, 0 at every other node, and is linear in between. Each hat lives on just two elements.



The FEM approximation is a sum of hats, $u_h = \sum_i c_i\varphi_i$, so its coefficients are its nodal values. With six elements the polygon already sits on the exact curve at every node.

Step 2: ask the weak equation only of the probes in V_h. Write $u_h = \sum_j c_j\varphi_j$ and demand $a(u_h, \varphi_i) = F(\varphi_i)$ for each i. That is a linear system

$$\mathbf{A}\mathbf{c} = \mathbf{b}, \qquad A_{ij} = \int_0^1 k\,\varphi_i'\varphi_j'\,dx, \qquad b_i = \int_0^1 f\,\varphi_i\,dx.$$

The **stiffness matrix A** is symmetric positive definite, because it is the Gram matrix of the energy inner product (§10.6). It is also sparse: $A_{ij} = 0$ unless the hats i and j overlap. For $k = 1$ and uniform h it is $\frac{1}{h}\,\mathrm{tridiag}(-1, 2, -1)$, the same matrix as the finite-difference scheme of §22.10. The two methods agree on easy problems and part company on hard ones: FEM handles jumping coefficients, curved boundaries, unstructured meshes and Neumann conditions with no special effort.

Step 3: assemble element by element. Real codes never form A_{ij} by thinking about pairs of hats. They loop over elements, compute a small element matrix, and add it into the global one:

```
import numpy as np

def fem_1d(k, f, n):
    """Linear finite elements for -(k u')' = f on (0,1), u(0) = u(1) = 0."""
    x = np.linspace(0, 1, n + 1)
    A = np.zeros((n + 1, n + 1)); b = np.zeros(n + 1)
    for e in range(n):                       # loop over elements [x_e, x_{e+1}]
        h = x[e + 1] - x[e]; mid = 0.5 * (x[e] + x[e + 1])
        Ke = k(mid) / h * np.array([[1, -1], [-1, 1]])   # element stiffness
        Fe = f(mid) * h / 2 * np.array([1, 1])           # element load (midpoint rule)
        idx = [e, e + 1]
        A[np.ix_(idx, idx)] += Ke; b[idx] += Fe
    inner = slice(1, n)                      # Dirichlet: drop the boundary nodes
    u = np.zeros(n + 1)
    u[inner] = np.linalg.solve(A[inner, inner], b[inner])
    return x, u

k = lambda x: 1.0 if x < 0.5 else 5.0       # two materials, interface at x = 0.5
f = lambda x: 1.0
x, u = fem_1d(k, f, 8)
print(np.round(u, 5))
```

With eight elements the output at $x = 0.5$ is 0.04167, which is the exact value $1/24$. In fact all nine nodal values are exact. That is not luck, and the reason is worth a proof.

> **Galerkin orthogonality and Céa's lemma**
>
> The FEM solution satisfies $a(u - u_h, v_h) = 0$ for every $v_h \in V_h$, and therefore
>
> $$\|u - u_h\|_a = \min_{v_h \in V_h} \|u - v_h\|_a.$$

Proof. Subtract $a(u_h, v_h) = F(v_h)$ from $a(u, v_h) = F(v_h)$, which holds because $V_h \subset H_0^1$. For any v_h, split $u - u_h = (u - v_h) + (v_h - u_h)$, where the second piece lies in V_h:

$$\|u-u_h\|_a^2 = a(u-u_h, u-v_h) + \underbrace{a(u-u_h, v_h-u_h)}_{=0} \le \|u-u_h\|_a \, \|u-v_h\|_a .$$

Divide by $\|u-u_h\|_a$. ■

So FEM is an **orthogonal projection** in the energy inner product: it finds the best approximation the subspace can offer, in exactly the sense of §9.9. The error is then a pure approximation question: how well can piecewise-linear functions approximate u? For $u \in H^2$ the interpolant has energy error $O(h)$, so FEM converges like h. For a rougher u it converges more slowly, and regularity theory tells you precisely how slowly.

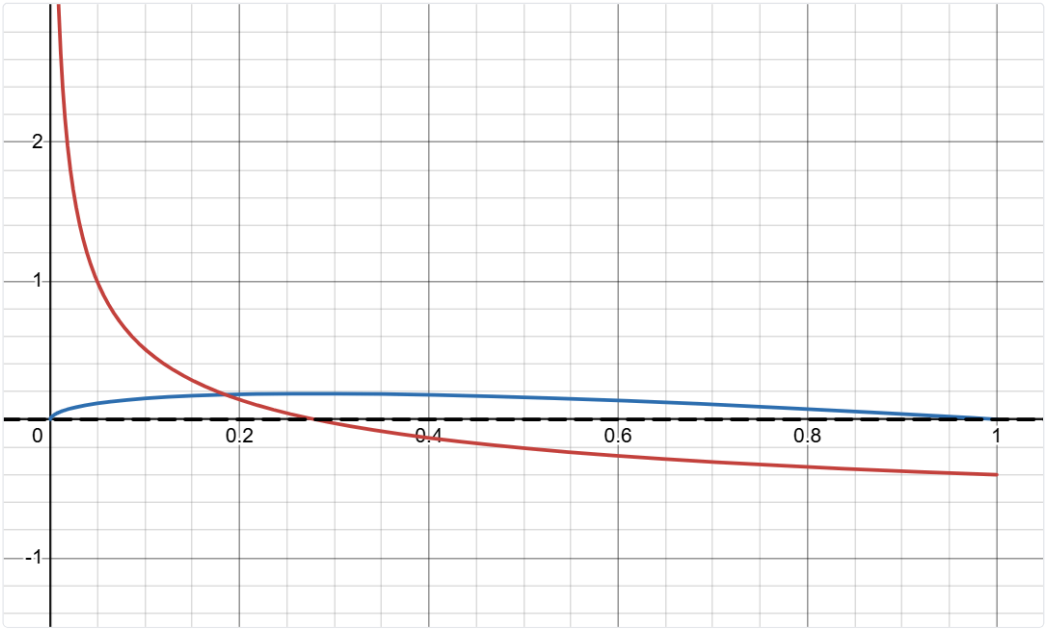
The nodal exactness follows too, for $k=1$. The Green's function $G_i = G(\cdot, x_i)$ is a tent with its kink at a node, so it lies in V_h. Since $a(G_i, w) = w(x_i)$ for every w, Galerkin orthogonality gives $u(x_i) - u_h(x_i) = a(G_i, u-u_h) = 0$. In one dimension FEM is exact at the nodes (when the load integrals are exact). In two dimensions Green's function has a logarithmic spike and no mesh contains it, so this miracle is lost.

Cost and failure modes.

- Assembly costs $O(n)$. The sparse SPD system can be solved by banded Cholesky in $O(n)$ in 1-D, and by preconditioned conjugate gradients or multigrid (§24.4) in higher dimensions. Multigrid is optimal: $O(n)$ for n unknowns.
- The condition number of $\mathbf{A}$ grows like h^{-2}. Fine meshes need good iterative solvers, not better pivoting.
- **Mesh quality.** In 2-D and 3-D the constant in "error $\le C\,h$" depends on element shape. Triangles with an angle near 180°, or the flat "sliver" tetrahedra of Ch. 75, make C enormous: the gradient of a linear function on a nearly degenerate element is wildly sensitive to its nodal values (the maximum-angle condition of Babuška and Aziz, 1976). That is why mesh generators fight so hard to remove slivers.
- **Locking.** For nearly incompressible materials (rubber) or thin plates, low-order elements become artificially stiff and the computed displacement is far too small, however fine the mesh. Coercivity holds, but with a constant that degenerates with the physical parameter. Mixed and discontinuous Galerkin methods were invented to escape this.

89.5 The experiment

When does a corner have finite energy? The function $u = x^\alpha - x$ vanishes at 0 and 1. Its energy $\int_0^1 (u')^2\,dx$ is finite exactly when $\int x^{2\alpha-2}\,dx$ converges at 0, that is when $\alpha > \tfrac12$. Change the exponent 0.6 in both lines to 0.51, then 0.5, then 0.4, and watch the derivative (red) shoot up near $x=0$.



$u = x^{0.6} - x$ (blue) is continuous and vanishes at both ends; its slope (red) is infinite at $x=0$ but square-integrable. Below $\alpha = \tfrac12$ the energy is infinite and u leaves H^1.

Regularity controls convergence. This script solves $-u'' = f$ by FEM with the exact solution $u = x^\alpha - x$, then measures the energy error $\|u' - u_h'\|$ as the mesh is halved. To keep the code short, the load integrals are computed exactly by integrating by parts against the known u.

```
import numpy as np

alpha = 2 / 3                       # knob: try 3, 1.5, 0.75, 0.6, 0.51
u  = lambda x: x**alpha - x         # exact solution, u(0) = u(1) = 0
du = lambda x: alpha * x**(alpha - 1) - 1

def fem(n):
    """Linear FEM for -u'' = f on (0,1); returns nodes and nodal values."""
    x = np.linspace(0, 1, n + 1); h = 1 / n
    A = (2 * np.eye(n - 1) - np.eye(n - 1, k=1) - np.eye(n - 1, k=-1)) / h
    # load b_i = int f phi_i = int u' phi_i'  (exact, by integration by parts)
    b = (2 * u(x[1:-1]) - u(x[:-2]) - u(x[2:])) / h
    U = np.zeros(n + 1); U[1:-1] = np.linalg.solve(A, b)
    return x, U

def energy_error(x, U, m=400):
    """|| u' - u_h' ||_L2, with a graded quadrature in every element."""
    err2 = 0.0
    t = (np.arange(m) + 0.5) / m
    for i in range(len(x) - 1):
        s = x[i] + (x[i + 1] - x[i]) * t**3          # cluster points near x[i]
        w = (x[i + 1] - x[i]) * 3 * t**2 / m
        slope = (U[i + 1] - U[i]) / (x[i + 1] - x[i])
        err2 += np.sum((du(s) - slope)**2 * w)
    return np.sqrt(err2)

prev = None
print("   n    energy error   rate")
for n in [8, 16, 32, 64, 128, 256]:
    x, U = fem(n)
    e = energy_error(x, U)
    rate = "" if prev is None else f"{np.log2(prev / e):6.3f}"
    print(f"{n:4d}   {e:.4e}   {rate}")
    prev = e
print("nodal error:", np.abs(U - u(x)).max())   # a 1-D surprise
```

For the smooth case $\alpha = 3$ the rate column reads 1.000: halving h halves the error. For $\alpha = \frac{2}{3}$ (the L-shaped corner exponent) it reads 0.167: halving h cuts the error by only 11%, and going from 8 to 256 elements improves it from 0.41 to 0.23. The rate is $\alpha - \frac{1}{2}$ exactly, a number set by the regularity of the solution and by nothing else. Meanwhile the last line prints a nodal error around 10^{-15}: the FEM solution is exact at the nodes, as proved in §89.4, and all of the error lives between them. Knobs to turn:

- Set `alpha` to 0.51. The method still converges, at a rate of 0.01. Would you notice convergence at all?
- Replace the uniform mesh by a graded one, `x = np.linspace(0, 1, n + 1)**3`, which crowds nodes toward the singularity (you will need to rewrite `A` and `b` element by element, as in §89.4). How much of the optimal rate comes back?
- Try `alpha = 1.5`. The solution is in H^1 but not quite H^2. What rate do you observe, and does the column settle?

89.6 The challenge

1. **Intuition.** Which of these belong to $H^1(-1,1)$: the tent $1 - |x|$; the step $\operatorname{sign}(x)$; $|x|^{1/3}$; $|x|^{2/3}$? For each one that fails, say in a sentence which physical quantity would be infinite if it were the displacement of a string.
2. **Derivation.** Write the weak form of the pure Neumann problem $-u'' = f$ on $(0,1)$, $u'(0) = u'(1) = 0$, over the whole of $H^1(0,1)$. Show that a solution can exist only if $\int_0^1 f\,dx = 0$, and that it is then unique up to an added constant. Interpret the condition for a heat-conducting rod with insulated ends.
3. **Implementation.** Solve $\nabla^2 u = 0$ on the L-shaped domain of §89.3 with boundary values taken from $u = r^{2/3}\sin(2\theta/3)$, using linear triangles or the five-point stencil of §22.10. Measure the maximum error and the energy error for $h = 2^{-3}, \dots, 2^{-7}$ and report the observed rates. Then refine only near the corner and see whether the rate recovers.
4. **Open investigation.** For $u = x^\alpha - x$, design an *adaptive* method: after each solve, estimate the error on every element (for instance, from the jump in u_h' between neighbours) and split the worst elements. Plot error against the number of elements N for uniform refinement and for your adaptive rule, for several α. Does your rule reach the rate N^{-1} of a smooth problem? Is there a grading that is optimal for every α at once, or must the mesh know the singularity?

89.7 Loose threads

The weak formulation turned a PDE into a question of Hilbert-space geometry, existence into the Riesz theorem, and numerical solution into orthogonal projection. Regularity theory told us when the weak solution is secretly classical, and when corners and material interfaces leave it rough. Two more clues are still on the table.

> **? Frequencies that are always too high**
>
> The same hat-function machinery computes vibration frequencies: replace $\mathbf{Ac} = \mathbf{b}$ by the generalized eigenproblem $\mathbf{Ac} = \lambda \mathbf{Mc}$, with the mass matrix $M_{ij} = \int \varphi_i \varphi_j$. For a string with ten elements the first three computed eigenvalues are 9.95, 40.79 and 95.58, against the exact $\pi^2 = 9.87$, $4\pi^2 = 39.48$ and $9\pi^2 = 88.83$. Every one is too high, and they stay too high on every mesh you try. Why can a Galerkin eigenvalue never undershoot? The answer is the min–max characterization of eigenvalues in Ch. 90, which is also where the Poincaré constant $1/\pi$ comes from.

? Running the problem backwards

This chapter assumed you know the material, $k(x)$, and want the temperature u. A thermal camera gives you the opposite: u measured on the surface, with noise, and the material unknown. Can k be recovered at all? In one dimension the answer is a qualified yes. For a 3-D body probed only by voltages and currents on its skin, which is electrical impedance tomography, it is the **Calderón problem**, and how unstable its solution is turns out to be the central difficulty. That is where Ch. 91 begins.

Further pointers:

- **Variational principles.** The energy $E(v)$ was a special case of an action functional. Why physics is so full of quantities that nature minimizes, and what symmetries of the functional tell you, is in Ch. 92.
- **Weak derivatives need Lebesgue.** "Square-integrable" and "almost everywhere" quietly used the Lebesgue integral. The full construction of Sobolev spaces as complete spaces needs Ch. 97.
- **High dimensions.** The random-walker formula of §89.3 has no grid. Its error depends on the number of samples, not the dimension, which is why Monte Carlo (Ch. 38) and, more recently, neural-network PDE solvers are being pushed into dimensions where FEM is hopeless. Proving when those neural solvers converge is active research.

Further reading

- L. C. Evans, *Partial Differential Equations*, 2nd ed., AMS, 2010. The standard graduate text: Sobolev spaces, weak solutions, regularity and viscosity solutions in one coherent story.
- S. C. Brenner and L. R. Scott, *The Mathematical Theory of Finite Element Methods*, 3rd ed., Springer, 2008. Céa's lemma, interpolation estimates and mesh conditions, done carefully.
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Chapter 90 — Spectral Theory: Understanding Operators Through Their Modes

> **Level 3 · Track J — Mathematics at the Frontier Builds on:** Ch. 10, Ch. 13, Ch. 61, Ch. 89 · **Leads to:** Ch. 91, Ch. 92, Ch. 95 · **Related:** Ch. 27, Ch. 34, Ch. 38, Ch. 99 **Reading time:** ~30 min

90.1 The hook

In 1910 the physicist Hendrik Lorentz gave a series of lectures in Göttingen. He posed a problem that came out of the theory of black-body radiation: take a cavity of any shape, count how many standing electromagnetic waves it supports below a given frequency, and show that for high frequencies the count depends only on the cavity's *volume*, not its shape. David Hilbert, in the audience, is said to have predicted that he would not live to see it proved. His student Hermann Weyl proved it in 1911.

Fifty-five years later, Mark Kac turned Weyl's theorem into a question that every mathematician now knows by its title: *Can one hear the shape of a drum?* A drum is a membrane clamped along its rim. When struck, it vibrates in a superposition of standing modes, each with its own frequency. An ideal ear hears the complete list of frequencies, the **spectrum**, and nothing else. Weyl showed that this list reveals the drum's area. Later work showed it also reveals the perimeter, and the number of holes. So how much more does it reveal? Is the shape itself encoded in the sound?

This is not just a riddle about drums. The same question, "what does the list of eigenvalues tell me about the object?", sits underneath a remarkable range of things:

- A **graph's** Laplacian eigenvalues tell you whether the network has a bottleneck (Ch. 27), and Google-style ranking and spectral clustering are built on them.
- A **Markov chain's** second eigenvalue tells you how fast it forgets its starting point, which is how long a Monte Carlo simulation must run (Ch. 34, Ch. 38).
- An **atom's** spectrum, the frequencies of light it emits, was decoded into the energy levels of a Schrödinger operator. Spectroscopy reads the structure of matter from a list of numbers.
- A **bridge's** or **aircraft wing's** lowest natural frequencies decide whether wind or engines can shake it apart, which is the modal analysis of §10.8.

> **How much of an operator, and of the shape it lives on, can you recover from its spectrum alone? And when an operator has no eigenvectors at all, what is its spectrum made of?**

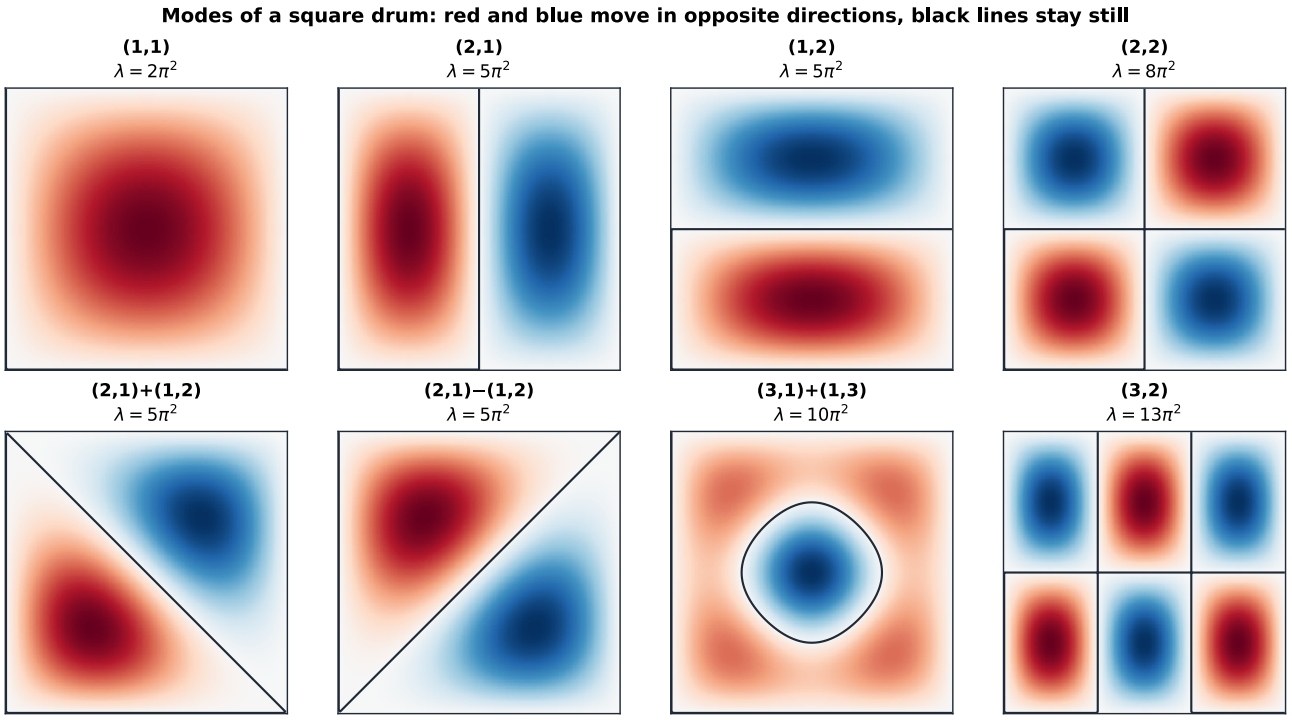
The second half of that question is where matrices stop being a good guide. The first half was answered for drums in 1992, and the answer is a surprise.

90.2 The big idea

An operator is an orchestra of modes. For a symmetric matrix the spectral theorem says: there is an orthonormal basis of eigenvectors, and in that basis the matrix just multiplies each coordinate by its eigenvalue (§10.5). Spectral theory is the same statement for operators on functions. For the Laplacian on a drum the "eigenvectors" are mode shapes, the **eigenfunctions** $\phi_k(x, y)$, and the eigenvalues are squared frequencies:

$$-\nabla^2 \phi_k = \lambda_k \phi_k \ \text{ inside the drum}, \qquad \phi_k = 0 \ \text{ on the rim}.$$

Once you know the modes, any question about the operator becomes a question about numbers. Heat flow multiplies mode k by $e^{-\lambda_k t}$. A struck membrane oscillates mode k as $\cos(\sqrt{\lambda_k}\, t)$. Solving $-\nabla^2 u = f$ divides mode k of f by λ_k. Every one of these is "apply a function to the eigenvalues". That is exactly why separation of variables works (§13.3).



Modes of the unit square, $\phi_{mn} = \sin(m\pi x)\sin(n\pi y)$ with $\lambda = \pi^2(m^2+n^2)$. Because the square is symmetric, $(2,1)$ and $(1,2)$ share an eigenvalue, and every combination of them is also a mode: the bottom row shows diagonal nodal lines and even a closed nodal curve, none of which appears in any single product of sines.

The low notes know the big picture; the high notes know the details. The lowest eigenvalue λ_1 is a single number that summarizes the drum's overall size and shape: how hard it is to bend. High eigenvalues come in a dense crowd, and their *density*, how many there are below λ, is controlled by the area. Between these extremes lies the subtle information: the perimeter, the corners, the holes, and possibly the shape.

When there are no modes. On a bounded drum the modes are a discrete list. On an infinite string there are no standing waves at all: a wave $e^{j\omega x}$ has the right shape but infinite energy, so it is not an eigenfunction. The spectrum does not disappear; it smears into a continuum. Spectral theory's deepest result says that even then, the operator is still "multiplication by a number" in a suitable sense, and the Fourier transform of Ch. 16 is the prototype.

90.3 The mathematics

Throughout, Ω is a bounded region in the plane, and $-\nabla^2$ acts on functions that vanish on its boundary $\partial\Omega$ (a **Dirichlet** drum). The inner product is $\langle u, v\rangle = \iint_\Omega u\, v\, dA$.

Self-adjointness: why the modes are real and orthogonal

The Dirichlet Laplacian is symmetric and positive

For smooth u, v vanishing on $\partial\Omega$: $\langle -\nabla^2 u, v\rangle = \langle u, -\nabla^2 v\rangle = \iint_\Omega \nabla u \cdot \nabla v\, dA$. Consequently its eigenvalues are real and positive, and eigenfunctions with different eigenvalues are orthogonal.

Proof. Green's first identity (§7.11) gives $\iint (-\nabla^2 u)v\, dA = \iint \nabla u\cdot\nabla v\, dA - \oint v\,\partial_n u\, ds$, and the boundary term vanishes because $v = 0$ there. The result is symmetric in u and v. If $-\nabla^2\phi = \lambda\phi$ with $\phi \neq 0$, take $u = v = \phi$: $\lambda\|\phi\|^2 = \iint |\nabla\phi|^2 > 0$, since a function with zero gradient that vanishes on the boundary is zero. If $-\nabla^2\phi = \lambda\phi$ and $-\nabla^2\psi = \mu\psi$, then $\lambda\langle\phi,\psi\rangle = \langle -\nabla^2\phi, \psi\rangle = \langle \phi, -\nabla^2\psi\rangle = \mu\langle\phi,\psi\rangle$, so $\lambda \neq \mu$ forces $\langle\phi,\psi\rangle = 0$. ∎

This is the matrix proof of §10.5 with the transpose replaced by integration by parts. What it does *not* give is existence: a matrix always has eigenvalues, an operator need not. For the drum, existence comes from Ch. 61's spectral theorem for compact self-adjoint operators, applied to the inverse $(-\nabla^2)^{-1}$, which solves the weak problem of Ch. 89. That inverse is compact because finite-energy functions on a bounded domain cannot oscillate arbitrarily fast without paying for it (Rellich's theorem). The conclusion:

Eigenfunction expansion (stated)

On a bounded domain, the Dirichlet Laplacian has eigenvalues $0 < \lambda_1 \le \lambda_2 \le \cdots \to \infty$, each of finite multiplicity, with an orthonormal basis of eigenfunctions ϕ_k of $L^2(\Omega)$. Every $f \in L^2(\Omega)$ satisfies $f = \sum_k \langle f, \phi_k\rangle\phi_k$, with convergence in the L^2 norm.

Applied to the heat equation. If $u_t = \nabla^2 u$ with $u = 0$ on the boundary and $u(\cdot,0) = f$, then

$$u(\cdot,t) = \sum_k e^{-\lambda_k t}\langle f,\phi_k\rangle\,\phi_k, \qquad \|u(\cdot,t)\| \le e^{-\lambda_1 t}\|f\|.$$

The inequality follows from Parseval: every coefficient shrinks by at least $e^{-\lambda_1 t}$. So λ_1 is the cooling rate, and in the long run $u \approx e^{-\lambda_1 t}\langle f,\phi_1\rangle\phi_1$: everything except the fundamental mode is forgotten. For an insulated (Neumann) body, $\lambda_1 = 0$ is the constant mode, the final uniform temperature, and the approach to it is governed by the **spectral gap** λ_2. Replace "temperature" by "probability" and this is the convergence of a Markov chain to equilibrium, which is why mixing-time

bounds in Ch. 38 are spectral-gap bounds.

The variational principle

Rayleigh quotient and min–max

Let $R(v) = \dfrac{\iint |\nabla v|^2 \, dA}{\iint v^2 \, dA}$ for nonzero v in $H_0^1(\Omega)$. Then

$$\lambda_1 = \min_{v \neq 0} R(v), \qquad \lambda_k = \min_{\dim S = k} \; \max_{v \in S,\, v \neq 0} R(v),$$

where S ranges over k-dimensional subspaces of $H_0^1(\Omega)$ (Courant–Fischer).

Proof of the first statement. Expand $v = \sum_k c_k \phi_k$. By orthogonality, $\iint v^2 = \sum c_k^2$. Because $\iint \nabla \phi_i \cdot \nabla \phi_k = \langle -\nabla^2 \phi_i, \phi_k \rangle = \lambda_i \delta_{ik}$, also $\iint |\nabla v|^2 = \sum \lambda_k c_k^2$. Hence

$$R(v) = \frac{\sum_k \lambda_k c_k^2}{\sum_k c_k^2} \;\geq\; \frac{\sum_k \lambda_1 c_k^2}{\sum_k c_k^2} = \lambda_1,$$

with equality for $v = \phi_1$. ■ The min–max formula is proved the same way: any k-dimensional S contains a nonzero v orthogonal to $\phi_1, \dots, \phi_{k-1}$ (a dimension count), and for that v the same estimate gives $R(v) \geq \lambda_k$; the subspace spanned by $\phi_1, \dots, \phi_k$ attains it.

This one principle settles several questions at once.

- **Poincaré's constant.** On $(0,1)$, $\lambda_1 = \pi^2$, so $\int v^2 \leq \pi^{-2} \int (v')^2$ for every $v \in H_0^1$: the sharp constant $1/\pi$ that Ch. 89 promised.
- **Cheap estimates.** Any trial shape gives an upper bound. For $v = x(1-x)$ on $(0,1)$, $R(v) = \frac{1/3}{1/30} = 10$, within 1.3% of $\pi^2 = 9.8696$, with no differential equation solved.
- **Galerkin eigenvalues are always too high.** A finite-element method restricts the subspaces S in the min–max formula to subspaces of V_h. Minimizing over fewer candidates cannot give a smaller minimum, so $\lambda_k^h \geq \lambda_k$ for every k and every mesh. That is the clue left open in Ch. 89, where ten elements gave $9.95 > \pi^2$.
- **Domain monotonicity.** If $\Omega_1 \subset \Omega_2$, every trial function for Ω_1 (extended by zero) is a trial function for Ω_2, so $\lambda_k(\Omega_2) \leq \lambda_k(\Omega_1)$. Bigger drums sound lower.
- **Bottlenecks.** For a dumbbell, two rooms joined by a thin corridor, a Neumann trial function that is $+1$ in one room and -1 in the other has tiny gradient energy, so λ_2 is tiny. Cheeger's inequality (1970) makes this two-sided: $\lambda_2 \geq h^2/4$, where h is the smallest ratio of "cut length" to "area cut off". A small gap *forces* a bottleneck. The graph version is the clue left open in Ch. 27.

Weyl's law: hearing the area

Let $N(\lambda)$ be the number of eigenvalues $\leq \lambda$, counted with multiplicity.

Weyl's law

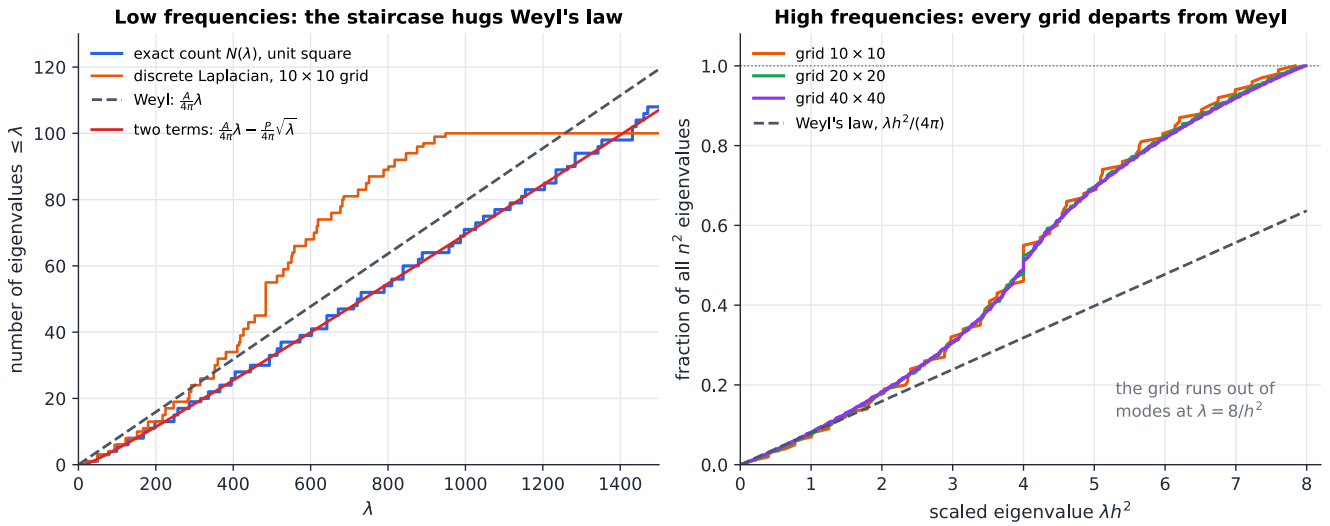
For a bounded plane domain of area A, $N(\lambda) \sim \dfrac{A}{4\pi}\,\lambda$ as $\lambda \to \infty$. In d dimensions, $N(\lambda) \sim \dfrac{\omega_d \, |\Omega|}{(2\pi)^d} \lambda^{d/2}$, where ω_d is the volume of the unit ball.

Proof for a rectangle $[0,a] \times [0,b]$. The eigenfunctions are $\sin(m\pi x/a)\sin(n\pi y/b)$ with $\lambda = \pi^2(m^2/a^2 + n^2/b^2)$, $m, n \geq 1$. So $N(\lambda)$ counts lattice points (m,n) with $m, n \geq 1$ inside the quarter-ellipse $E = \{m^2/a^2 + n^2/b^2 \leq \lambda/\pi^2,\ m, n \geq 0\}$, whose area is $\frac{\pi}{4} \cdot \frac{a\sqrt{\lambda}}{\pi} \cdot \frac{b\sqrt{\lambda}}{\pi} = \frac{ab}{4\pi}\lambda$. Attach to each counted point the unit square $[m-1, m] \times [n-1, n]$ that has it as its top-right corner. These squares do not overlap, and each lies inside E (moving down or left only decreases $m^2/a^2 + n^2/b^2$). So $N(\lambda) \leq \text{area}(E) = \frac{ab}{4\pi}\lambda$. Conversely, a point of E with positive coordinates lies in the square whose top-right corner is the lattice point just above and to the right of it. If the point is more than $\sqrt{2}$ from the curved edge, that corner is in E and is counted. So the squares cover all of E except a strip of width $\sqrt{2}$ along the curved edge, whose area is $O(\sqrt{\lambda})$. Hence $N(\lambda) = \frac{ab}{4\pi}\lambda + O(\sqrt{\lambda})$, and the leading term is $A\lambda/(4\pi)$. ■

The general case glues rectangles together, using domain monotonicity to squeeze the count from above and below. The $O(\sqrt{\lambda})$ correction is not random. If lattice points on the two axes were allowed, the count would overshoot the area by about half of the points lying on those axes, and there are about $\frac{(a+b)\sqrt{\lambda}}{\pi}$ of them. Excluding them ($m, n \geq 1$) leaves a net deficit of about $\frac{(a+b)\sqrt{\lambda}}{2\pi}$, which is the second term of Weyl's conjecture,

$$N(\lambda) \approx \frac{A}{4\pi}\lambda - \frac{P}{4\pi}\sqrt{\lambda},$$

where P is the perimeter. For the unit square at $\lambda = 10{,}000$ the true count is 764; one term predicts 795.8, two terms predict 763.9.



Left: the eigenvalue count of a square drum follows Weyl's law, and the two-term version almost exactly. A 10×10 finite-difference grid agrees at first, then overcounts and stops at 100 modes. Right: every grid, rescaled, follows the same curve, which bends away from Weyl long before its last mode at $\lambda = 8/h^2$. A discretization only hears the low notes correctly.

A cleaner way to hear the geometry is the **heat trace** $Z(t) = \sum_k e^{-\lambda_k t}$, which is determined by the spectrum. For a domain with a smooth boundary and g holes, as $t \to 0^+$,

$$Z(t) = \frac{A}{4\pi t} - \frac{P}{8\sqrt{\pi t}} + \frac{1-g}{6} + O(\sqrt{t}).$$

Area, perimeter and the number of holes can all be heard. (Corners add their own constants: for the square the constant is $\frac{1}{4}$, which you can check from $Z = \left(\sum_m e^{-\pi^2 m^2 t}\right)^2$.)

And the shape? In 1992 Carolyn Gordon, David Webb and Scott Wolpert built two different polygons, each assembled from seven congruent half-squares, with exactly the same spectrum. One cannot hear the shape of a drum. The proof is a transplantation: cut an eigenfunction on one drum into its seven pieces and reassemble them, with sign changes, on the other.

? Can one hear the shape of a *smooth* drum?

The Gordon–Webb–Wolpert drums have corners. Nobody knows whether two different smooth convex drums can share a spectrum. Zelditch (2009) proved that among analytic domains with one mirror symmetry, the spectrum does determine the shape, and Hezari and Zelditch later extended the result to nearly circular ellipses. The general question, which Kac really asked, is an open problem. The tools that might settle it come from billiards: the heat trace and its cousin, the wave trace, carry the lengths of periodic billiard paths inside the drum.

? The second term of Weyl's law

The two-term formula above is called Weyl's *conjecture*. Ivrii proved it in 1980, but only under a condition on billiards: the periodic trajectories bouncing around inside the drum must form a set of measure zero. Whether every domain satisfies this condition is unknown. It is open even for convex domains with smooth boundary in the plane. A question about counting musical notes is waiting on a question about pool tables.

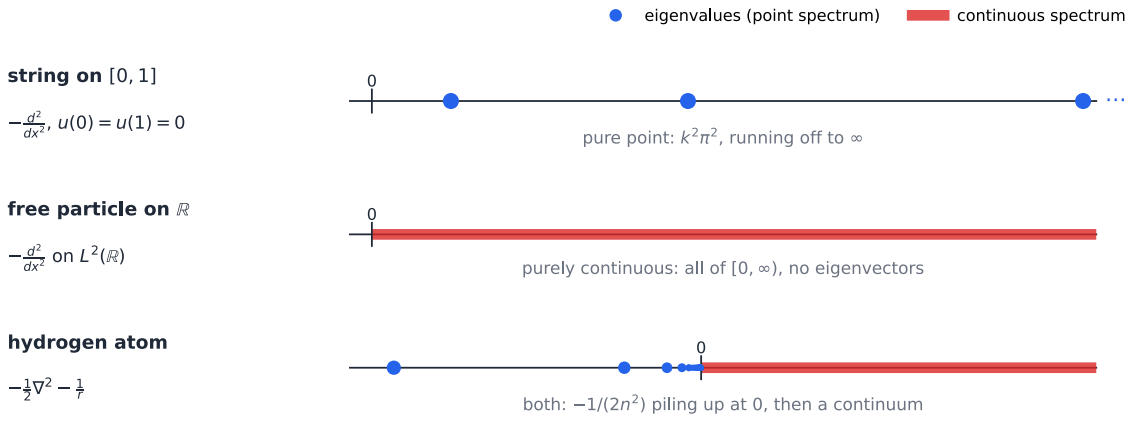
When there are no eigenvectors: continuous spectrum

On the whole line, $-d^2/dx^2$ acting on $L^2(\mathbb{R})$ has no eigenfunctions: the solutions of $-u'' = \lambda u$ are $e^{\pm j\sqrt{\lambda}x}$, or exponentials, none of them square-integrable. But the Fourier transform (§16.3) turns differentiation into multiplication, $\mathcal{F}\{-u''\} = \omega^2\,\mathcal{F}\{u\}$. In the frequency domain the operator is just "multiply by ω^2": diagonal, with a continuum of diagonal entries $[0, \infty)$ instead of a list. This is the general picture, and it answers the clue of Ch. 61.

Spectral theorem for self-adjoint operators, multiplication form (stated)

For every self-adjoint operator T on a Hilbert space $\mathcal{H}$, possibly unbounded, there is a measure space (M, μ), a real function g on M and a unitary map $U : \mathcal{H} \to L^2(M, \mu)$ with $UTU^{-1} =$ multiplication by g. Equivalently, $T = \int \lambda\, dE(\lambda)$ for a projection-valued measure E.

Eigenvalues are the atoms of the measure, where a single value of g carries positive weight; continuous spectrum is the rest. The physicist's "eigenstates" $|x_0\rangle$ and plane waves $e^{j\omega x}$ are not vectors in $\mathcal{H}$; they are the kernel of U, a coordinate system for the multiplication picture. Real operators often mix both kinds:



Point, continuous and mixed spectra. Hydrogen's bound states $-1/(2n^2)$ (in atomic units) are the spectral lines of the Balmer and Lyman series; the continuum above 0 is an electron that has escaped.

Beyond self-adjoint. For non-symmetric operators, such as convection-diffusion or the linearized equations of fluid flow, eigenvalues can be dangerously misleading. A flow whose eigenvalues all predict decay can still amplify a disturbance by a factor of thousands before decaying, which is one route to turbulence. The robust object is the **ε-pseudospectrum**, the set of z where $\|(T - zI)^{-1}\| > 1/\varepsilon$, introduced in Ch. 99's loose threads. For a normal operator it is just the ε-neighbourhood of the spectrum. For a non-normal one it can be enormously larger, and it, not the spectrum, predicts the transient growth.

90.4 The algorithm

Discretize a drum on an $n \times n$ grid and the Laplacian becomes a sparse symmetric matrix of size $N = n^2$. For a square it is a **Kronecker sum**, $\mathbf{A} = \mathbf{I} \otimes \mathbf{T} + \mathbf{T} \otimes \mathbf{I}$, where $\mathbf{T} = h^{-2}\,\mathrm{tridiag}(-1, 2, -1)$ is the 1-D second difference. Its eigenvalues are known exactly:

$$\lambda^h_{mn} = \frac{4}{h^2}\Big(\sin^2\frac{m\pi h}{2} + \sin^2\frac{n\pi h}{2}\Big), \qquad 1 \le m, n \le n_{\text{grid}}.$$

For small mh this is $\pi^2(m^2 + n^2)$ minus a correction of order h^2, so finite differences *underestimate* the low eigenvalues, the opposite of finite elements. For large m the formula saturates at $8/h^2$: the grid has no notes above that. This is why the right panel of the Weyl figure bends away.

You never want *all* N eigenvalues of a large drum. A dense symmetric eigensolver costs $O(N^3)$ time and $O(N^2)$ memory, so $N = 40{,}000$ already needs about 13 GB just to store the matrix. Engineers want the lowest few modes. The workhorse is **inverse iteration with a shift**: factor $\mathbf{A} - \sigma\mathbf{I}$ once, then repeatedly solve with it. Each solve multiplies the component along mode k by $1/(\lambda_k - \sigma)$, so the mode nearest σ wins.

```
import numpy as np
import scipy.sparse as sp
from scipy.sparse.linalg import splu, eigsh

n = 199; h = 1.0 / (n + 1)                      # interior grid of the unit square
T = sp.diags([-1.0, 2.0, -1.0], [-1, 0, 1], shape=(n, n)) / h**2
I = sp.identity(n)
A = (sp.kron(I, T) + sp.kron(T, I)).tocsc()     # 2-D Laplacian: a Kronecker sum

def inverse_iteration(A, sigma=0.0, iters=50, tol=1e-12):
    """Eigenpair of A nearest the shift sigma."""
    lu = splu(A - sigma * sp.identity(A.shape[0], format="csc"))   # factor once
    v = np.random.default_rng(1).standard_normal(A.shape[0])
    lam = 0.0
    for _ in range(iters):
        w = lu.solve(v)                         # w = (A - sigma I)^{-1} v
        v = w / np.linalg.norm(w)
        lam_new = v @ (A @ v)                   # Rayleigh quotient
        if abs(lam_new - lam) < tol * abs(lam_new): break
        lam = lam_new
    return lam_new, v

lam1, v1 = inverse_iteration(A)
print("inverse iteration:", lam1, " exact 2*pi^2 =", 2 * np.pi**2)
print("Lanczos, 6 smallest:", np.round(eigsh(A, k=6, sigma=0, which="LM")[0], 3))
```

On this 39,601-unknown problem, inverse iteration returns $\lambda^h_1 = 19.7388$ against the exact $2\pi^2 = 19.7392$. The last line uses the **Lanczos** method in shift-invert mode, which builds the best approximations from the whole Krylov space $\{\mathbf{v}, \mathbf{A}^{-1}\mathbf{v}, \mathbf{A}^{-2}\mathbf{v}, \dots\}$ instead of keeping only the latest vector. It finds $19.739,\ 49.345,\ 49.345,\ 78.95,\ 98.679,\ 98.679$: the double eigenvalues $5\pi^2$ and $10\pi^2$ appear twice, as they should.

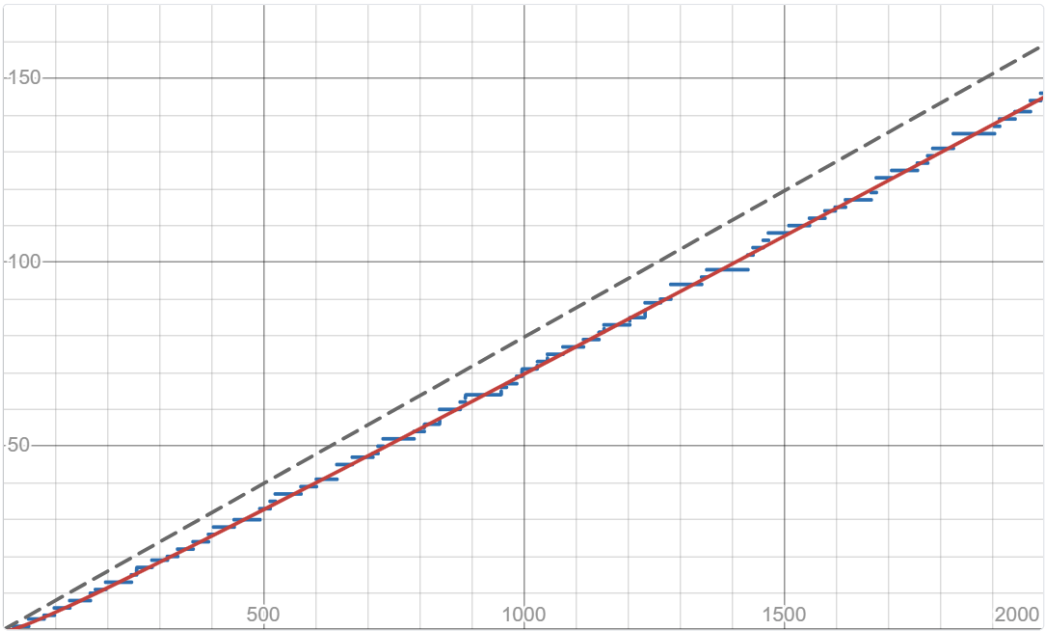
Cost and failure modes.

- Inverse iteration converges with ratio $|\lambda_1 - \sigma|/|\lambda_2 - \sigma|$ per step. A good shift makes it fast; a shift halfway between two eigenvalues makes it stall. Rayleigh-quotient iteration, which updates σ each step, converges cubically, at the price of refactoring every step.
- **Degenerate eigenvalues.** For $5\pi^2$ the solver returns *some* orthonormal pair from a two-dimensional eigenspace, and which pair depends on rounding and on the random start. Comparing "mode 2" between two runs, or between two meshes, can fail for this reason alone. Only the eigenspace is well defined.

- **Non-symmetric matrices.** For convection-dominated problems, eigenvalues of the matrix can sit far from those of the operator, and tiny perturbations move them a lot (the pseudospectra of §90.3). Check them, don't trust them.
- **Spectral pollution.** For operators with continuous spectrum, or for Maxwell's equations discretized with the wrong elements, Galerkin methods can produce eigenvalues that converge to nothing at all, sitting in a gap of the true spectrum. The min–max principle protects the bottom of the spectrum, never the interior. Detecting and avoiding pollution is current research.

90.5 The experiment

Hear the area and the perimeter. For an $a \times b$ rectangle the counting function is a finite sum of floor functions. With $a = b = 1$, the blue staircase is the exact $N(\lambda)$ and the red curve is the two-term Weyl law. Then make the drum long and thin with the same area: set $a = 4$, $b = 0.25$. In the sum, b is the factor in front of the square root and a is the base of `1^{2}`; in the Weyl curves, ab is `1\cdot1` and the perimeter $2(a + b)$ is `2(1+1)`. The one-term line does not move, because the area is unchanged; the staircase drops (at $\lambda = 2000$, from 146 modes to 131), and the perimeter term explains by how much.



Exact eigenvalue count of a rectangular drum (blue) against Weyl's law with one term (dashed) and two terms (red). The same area gives the same slope; the perimeter sets the offset.

Your own drum. This script builds the five-point Laplacian on any shape you can describe by a mask, computes all its eigenvalues with a dense solver, and compares the count with Weyl's law. The shapes live in $[-1, 1]^2$: a square of side 2, the L-shape of Ch. 89 (area 3), and the unit disk.

```
import numpy as np

n = 30                                  # knob: grid points per unit length
shape = "L"                             # knob: "square", "L" or "disk"
h = 1.0 / n
g = (np.arange(-n, n) + 0.5) * h        # cell centres covering [-1, 1]
X, Y = np.meshgrid(g, g)
masks = {"square": (np.abs(X) < 1) & (np.abs(Y) < 1),
         "L":      ~((X > 0) & (Y < 0)),
         "disk":   X**2 + Y**2 < 1}
area_perim = {"square": (4, 8), "L": (3, 8), "disk": (np.pi, 2 * np.pi)}
inside = masks[shape]
idx = -np.ones(X.shape, int); idx[inside] = np.arange(inside.sum())
N = inside.sum()

# five-point Dirichlet Laplacian on the cells inside the drum
A = np.zeros((N, N))
for i, j in zip(*np.nonzero(inside)):
    p = idx[i, j]; A[p, p] = 4 / h**2
    for di, dj in [(1, 0), (-1, 0), (0, 1), (0, -1)]:
        a, b = i + di, j + dj
        if 0 <= a < 2 * n and 0 <= b < 2 * n and inside[a, b]:
            A[p, idx[a, b]] = -1 / h**2
lam = np.linalg.eigvalsh(A)             # N x N dense: keep N below ~4000

print(f"{shape}: {N} unknowns; first eigenvalues:", np.round(lam[:6], 2))
Area, P = area_perim[shape]
print("   lambda   count   Weyl 1-term   Weyl 2-term")
for L in [50, 100, 200, 400, 800]:
    w1 = Area * L / (4 * np.pi); w2 = w1 - P * np.sqrt(L) / (4 * np.pi)
    print(f"{L:8d} {np.sum(lam <= L):7d} {w1:12.1f} {w2:12.1f}")
```

For the L-shape with $n = 30$ the first eigenvalue comes out as 9.08. The true value is 9.6397: this is the famous L-shaped membrane whose first mode is the MATLAB logo, computed to high accuracy by Fox, Henrici and Moler in 1967. The grid is crude, and because the grid's drum is effectively half a cell larger than the true one, its eigenvalues sit below the truth. At $\lambda = 400$ the count is 90, between the one-term (95.5) and two-term (82.8) predictions. Knobs to turn:

- Raise `n` to 35 (about 3,700 unknowns) and watch λ_1 creep toward 9.64. How fast? Compare with the square, whose exact λ_1 is $\pi^2/2 \approx 4.935$, and explain the difference using the corner singularity of Ch. 89.
- Look at `lam[1]` and `lam[2]` for the square and the disk. Which are degenerate, and which symmetry is responsible?
- Add your own mask: an annulus, a dumbbell, two disks joined by a thin channel. For the dumbbell, rerun with Neumann conditions (set each diagonal entry to the number of inside neighbours divided by h^2) and watch λ_2 collapse as the channel narrows.

90.6 The challenge

1. **Intuition.** Without computing, rank these drums of equal area by their fundamental frequency $\sqrt{\lambda_1}$: a disk, a square, an equilateral triangle, a $4 \times \frac{1}{4}$ rectangle. Which principle of §90.3 helps, and which pairs does it fail to decide?
2. **Derivation.** For the equilateral triangle and the square of the same area, compute the two-term Weyl predictions and say which drum has more modes below a large λ. Then verify the heat-trace constant $\frac{1}{4}$ for the unit square from $Z(t) = \left(\sum_{m\geq 1} e^{-\pi^2 m^2 t}\right)^2$, using $\sum_{m\geq 1} e^{-\pi^2 m^2 t} \approx \frac{1}{2\sqrt{\pi t}} - \frac{1}{2}$ for small t.
3. **Implementation.** Build the Gordon–Webb–Wolpert isospectral pair from seven half-squares each (look up their coordinates), discretize both with the script of §90.5 on triangular-friendly grids, and compare their first 20 eigenvalues. How close do two genuinely isospectral drums look after discretization, and can you tell them apart from a *non*-isospectral pair with the same area and perimeter?
4. **Open investigation.** Compute the first 2,000 eigenvalues of a rectangle with irrational aspect ratio and of a "stadium" (a rectangle with semicircular ends). Unfold each spectrum by subtracting the two-term Weyl law, form the gaps between consecutive eigenvalues, and plot their histograms. One histogram should look like e^{-s}; the other should vanish at $s = 0$. Which is which, and why might the shape of a billiard table decide whether eigenvalues repel each other?

90.7 Loose threads

Spectral theory turned operators into lists of numbers and the lists into geometry. The bottom of the spectrum measured stiffness, cooling rates and bottlenecks through the min–max principle. The density of the spectrum measured area, and its first correction, the perimeter. And when eigenvectors disappeared, the spectral theorem survived as "multiplication by a function". The challenge's last problem points at the most mysterious clue of all.

> **? Why do some spectra repel?**
>
> Unfold the eigenvalues of a rectangle and their gaps look like independent random numbers: a Poisson process, with many near-collisions. Do the same for a stadium, whose billiard paths are chaotic, and the gaps follow the statistics of eigenvalues of large random symmetric matrices, the Wigner surmise $\frac{\pi s}{2} e^{-\pi s^2/4}$, which vanishes at $s = 0$. Bohigas, Giannoni and Schmit conjectured in 1984 that this holds for every chaotic system, and the same statistics appear in heavy nuclei and in the zeros of the Riemann zeta function. Nobody has proved it for a single drum. The physics of it is in Ch. 92; the proof is an open problem.

Further pointers:

- **Inverse problems.** Singular values are the spectral theory of $\mathbf{K}^{\mathsf{T}}\mathbf{K}$, and the speed at which they decay decides whether an inverse problem is mildly or catastrophically unstable: Ch. 91.
- **Symmetry and degeneracy.** The double eigenvalue $5\pi^2$ is the square's rotation symmetry in disguise. In general, the multiplicities of a symmetric operator are the dimensions of the irreducible representations of its symmetry group: Ch. 95.
- **Graphs.** Everything here has a discrete twin. The graph Laplacian has a min–max principle, a Cheeger inequality and even isospectral pairs, and spectral clustering uses its second eigenvector to cut networks (Ch. 27, Ch. 87).

Further reading

- M. Kac, "Can one hear the shape of a drum?", *American Mathematical Monthly* 73 (1966), 1–23. Still the best invitation to the subject.
- C. Gordon, D. Webb and S. Wolpert, "One cannot hear the shape of a drum", *Bulletin of the AMS* 27 (1992), 134–138. Four pages, with pictures of the two drums.
- D. S. Grebenkov and B.-T. Nguyen, "Geometrical structure of Laplacian eigenfunctions", *SIAM Review* 55 (2013), 601–667. A survey of what is known about modes, nodal lines and localization, with many figures.
- E. B. Davies, *Spectral Theory and Differential Operators*, Cambridge University Press, 1995. A rigorous course on the spectral theorem and its use for PDEs.
- L. N. Trefethen and M. Embree, *Spectra and Pseudospectra*, Princeton University Press, 2005. What eigenvalues do not tell you about non-normal matrices and operators.
- V. Ivrii, "100 years of Weyl's law", *Bulletin of Mathematical Sciences* 6 (2016), 379–452. The history and current state of eigenvalue asymptotics, by the person who proved the two-term formula.

Chapter 91 — Inverse Problems: Reconstructing the Invisible

> **Level 3 · Track J — Mathematics at the Frontier Builds on:** Ch. 9, Ch. 10, Ch. 13, Ch. 61, Ch. 90 · **Leads to:** Ch. 92, Ch. 97 · **Related:** Ch. 24, Ch. 35, Ch. 76, Ch. 89 **Reading time:** ~30 min

91.1 The hook

On 24 April 1990 the Space Shuttle *Discovery* carried the Hubble Space Telescope into orbit. Its 2.4-metre primary mirror was the most precisely polished large mirror ever made. Within weeks the first images came back blurred. Stars that should have been sharp points were surrounded by a wide, faint halo. The investigation found that the mirror had been ground to the wrong shape: near its edge it was too flat by about 2.2 micrometres, roughly a fiftieth of the width of a human hair. A testing instrument used during polishing had been assembled with one lens about a millimetre out of place.

The error was tiny, but it was also *perfectly known*. Engineers could compute exactly how the faulty mirror spread the light of a point star, the **point-spread function**. Every Hubble image was therefore the true sky convolved with a known kernel. In principle, undoing the blur is just solving a linear system: the blurred image $\mathbf{g}$ equals $\mathbf{K}\mathbf{f}$, with $\mathbf{K}$ known, so $\mathbf{f} = \mathbf{K}^{-1}\mathbf{g}$. Astronomers spent three years doing exactly this kind of computation, until a servicing mission in December 1993 fitted corrective optics. They learned the hard way that the direct inverse is useless. Reconstructions came out as fields of violent speckle, with every pixel of sensor noise magnified into garbage.

Here is the same experience in one dimension. Take a signal with a box, a ramp and a bump; blur it with a narrow Gaussian; add noise of 1%, smaller than the line width on a plot. The blurring matrix is perfectly invertible. Its exact inverse returns a signal whose error is **273,000 times** the size of the signal itself.

And this is not an astronomy problem. It is the generic situation whenever you measure effects and want causes:

- **CT and MRI**: X-ray attenuations along lines, or Fourier samples, in; a tissue map out.
- **Seismic imaging**: echoes recorded at the surface in; rock layers kilometres down out.
- **Electrical impedance tomography**: voltages on the skin in; the conductivity of the lungs out.
- **Parameter estimation**: a step response in; the time constants of a circuit or a thermal model out (§25.8).

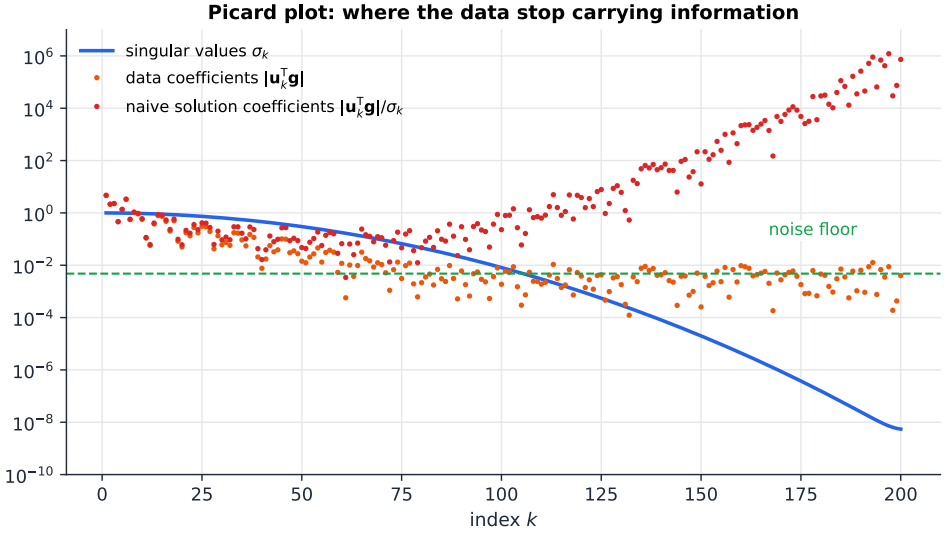
> **If you know the forward process exactly, why can't you just run it backwards? And if the exact inverse is worthless, what are you allowed to compute instead, and how do you know it isn't fiction?**

91.2 The big idea

Forward processes forget. Blurring averages neighbouring values together. Diffusion smears heat. A CT projection sums along a line. Physics is mostly smoothing, and smoothing destroys fine detail: two very different signals, one of them with a high-frequency wiggle added, produce almost identical measurements. The forward map is stable because it forgets. That same forgetting makes the inverse unstable: to recover the wiggle, the inverse must amplify the tiny trace it left in the data, and it amplifies the noise sitting in the same place just as much.

In the language of Ch. 13, the forward problem is well posed and the inverse problem is **ill posed**: its solution does not depend continuously on the data. Hadamard thought such problems were physically meaningless. A century later, they are most of medical imaging.

The singular value decomposition shows exactly where the forgetting happens. Write the forward matrix as $\mathbf{K} = \sum_k \sigma_k \mathbf{u}_k \mathbf{v}_k^{\mathsf{T}}$ (§10.7). Each input pattern $\mathbf{v}_k$ is mapped to output pattern $\mathbf{u}_k$, shrunk by σ_k. For a blur the $\mathbf{v}_k$ are oscillations of increasing frequency and the σ_k fall off a cliff. The exact inverse divides by σ_k, and in the components where σ_k is below the noise level it is dividing noise by almost nothing.



The Picard plot. Where the data coefficients (orange) still track the singular values (blue), the data carry information about $\mathbf{f}$. Beyond about $k = 60$ they hit the noise floor and carry only noise, and dividing by σ_k (red) inflates that noise by up to a million.

The cure is to want less. Since some components of $\mathbf{f}$ cannot be recovered, stop trying. Replace "find $\mathbf{f}$ with $\mathbf{Kf} = \mathbf{g}$" by "find a *reasonable* $\mathbf{f}$ that *nearly* fits the data". Reasonable means extra information you supply: the signal is not huge, or is smooth, or has few edges. That extra information is called **regularization**, and the art is in how much of it to add. An analogy: reading a smudged handwritten word. You cannot recover the ink exactly. You recover it by combining what is legible with what words usually look like, and if you trust your expectations too much you read the word you expected instead of the one that was written.

91.3 The mathematics

Throughout, the data are $\mathbf{g} = \mathbf{Kf} + \mathbf{e}$ with noise $\|\mathbf{e}\| \le \delta$, and $\mathbf{K} = \sum_k \sigma_k \mathbf{u}_k \mathbf{v}_k^{\mathsf{T}}$ with $\sigma_1 \ge \sigma_2 \ge \cdots > 0$. The same formulas hold for compact operators between Hilbert spaces, where $\sigma_k \to 0$ is forced (Ch. 61) and the inverse is unbounded rather than merely large.

Why the naive inverse explodes

Noise amplification

The naive solution is $\mathbf{K}^{-1}\mathbf{g} = \mathbf{f} + \sum_k \dfrac{\mathbf{u}_k^{\mathsf{T}}\mathbf{e}}{\sigma_k}\mathbf{v}_k$. Its worst-case error over all noise with $\|\mathbf{e}\| \le \delta$ is $\delta/\sigma_{\min}$. If the noise is white with variance η^2 per component, the expected squared error is $\eta^2 \sum_k \sigma_k^{-2}$.

Proof. Since $\{\mathbf{u}_k\}$ is orthonormal, $\mathbf{K}^{-1} = \sum_k \sigma_k^{-1}\mathbf{v}_k\mathbf{u}_k^{\mathsf{T}}$, and applying it to $\mathbf{Kf} + \mathbf{e}$ gives the formula. The error has coordinates $\mathbf{u}_k^{\mathsf{T}}\mathbf{e}/\sigma_k$ in the orthonormal basis $\{\mathbf{v}_k\}$, so its squared norm is $\sum_k (\mathbf{u}_k^{\mathsf{T}}\mathbf{e})^2/\sigma_k^2 \le \sigma_{\min}^{-2}\sum_k (\mathbf{u}_k^{\mathsf{T}}\mathbf{e})^2 = \delta^2/\sigma_{\min}^2$, with equality for $\mathbf{e} = \delta\mathbf{u}_{\min}$. For white noise, $E[(\mathbf{u}_k^{\mathsf{T}}\mathbf{e})^2] = \eta^2$ for each k. ∎

In the hook the condition number of $\mathbf{K}$ is 1.8×10^8, so 1% noise can become an error of over a million times the signal; the observed 2.7×10^5 is the typical case rather than the worst. Making the grid finer does not help. It adds more components with tiny σ_k, and in the continuum limit $\sigma_{\min} \to 0$.

How bad is bad? The decay of σ_k grades the difficulty. If $\sigma_k \sim k^{-p}$ the problem is **mildly ill posed**: differentiation is $p = 1$ (the Volterra operator of §61.4 has $\sigma_k = 2/((2k-1)\pi)$), and the CT scanner's Radon transform behaves like $p = \frac{1}{2}$ in two dimensions. If σ_k decays exponentially, the problem is **severely ill posed**: a Gaussian blur damps frequency ω by $e^{-s^2\omega^2/2}$, and running the heat equation backwards for time T requires undoing $e^{-k^2\pi^2 T}$. In a severely ill-posed problem, halving the noise buys only a few more recoverable components. That is why CT images are sharp and why nobody reconstructs the temperature history of a rod from its present state.

The Picard condition

A solution exists in the continuum only if $\sum_k (\mathbf{u}_k^{\mathsf{T}}\mathbf{g})^2/\sigma_k^2 < \infty$: the data coefficients must decay faster than the singular values. Exact data always satisfy it. Noisy data never do, because noise coefficients stop decaying. The Picard plot shows the index where the condition breaks, and that index is roughly how many components any method can hope to recover.

Tikhonov regularization

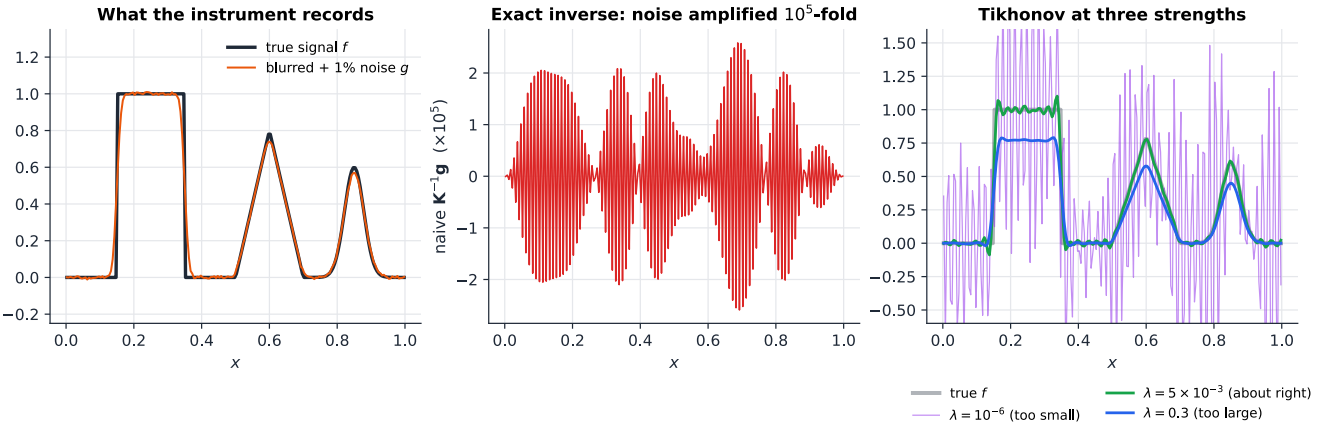
Andrey Tikhonov's 1963 proposal: instead of solving $\mathbf{Kf} = \mathbf{g}$, minimize

$$J_\lambda(\mathbf{f}) = \|\mathbf{Kf} - \mathbf{g}\|^2 + \lambda\|\mathbf{f}\|^2, \qquad \lambda > 0.$$

The first term asks for data fit, the second penalizes size, and λ sets the exchange rate. Setting the gradient to zero gives the regularized normal equations $(\mathbf{K}^{\mathsf{T}}\mathbf{K} + \lambda\mathbf{I})\mathbf{f}_\lambda = \mathbf{K}^{\mathsf{T}}\mathbf{g}$, which always have a unique solution because $\mathbf{K}^{\mathsf{T}}\mathbf{K} + \lambda\mathbf{I}$ is positive definite. It is ordinary least squares (§9.11) for the stacked system $\begin{bmatrix}\mathbf{K}\\ \sqrt{\lambda}\mathbf{I}\end{bmatrix}\mathbf{f} \approx \begin{bmatrix}\mathbf{g}\\ \mathbf{0}\end{bmatrix}$. In the SVD basis, $\mathbf{K}^{\mathsf{T}}\mathbf{K} + \lambda\mathbf{I}$ is diagonal with entries $\sigma_k^2 + \lambda$, so

$$\mathbf{f}_\lambda = \sum_k \underbrace{\frac{\sigma_k^2}{\sigma_k^2 + \lambda}}_{\text{filter factor}} \frac{\mathbf{u}_k^{\mathsf{T}}\mathbf{g}}{\sigma_k}\mathbf{v}_k.$$

Where $\sigma_k^2 \gg \lambda$ the filter factor is nearly 1 and the component is kept. Where $\sigma_k^2 \ll \lambda$ it is nearly 0 and the component is suppressed. Tikhonov is a smooth version of truncating the SVD, which Ch. 61 tried, with the cutoff at $\sigma_k \approx \sqrt{\lambda}$.



Deblurring a signal. The measurement (left) looks like the truth, yet the exact inverse (centre) is pure amplified noise. Tikhonov (right) with λ too small is still noise-dominated (relative error 116%); with $\lambda = 5 \times 10^{-3}$ the error is 9.8%; with $\lambda = 0.3$ the noise is gone but so are the corners and the peak heights (error 27%).

How good can it get?

The regularized solution has two errors with opposite appetites for λ, and the theorem below makes their trade exact. It needs one assumption about the truth, because without some smoothness of $\mathbf{f}$ no method can do better than arbitrarily badly.

Tikhonov error bound

Suppose $\mathbf{f} = \mathbf{K}^{\mathsf{T}}\mathbf{w}$ for some $\mathbf{w}$ (a **source condition**: $\mathbf{f}$ is smoother than a generic vector, in the sense the forward map defines). Then

$$\|\mathbf{f}_\lambda - \mathbf{f}\| \le \frac{\delta}{2\sqrt{\lambda}} + \frac{\sqrt{\lambda}}{2}\|\mathbf{w}\|,$$

and choosing $\lambda = \delta/\|\mathbf{w}\|$ gives $\|\mathbf{f}_\lambda - \mathbf{f}\| \le \sqrt{\delta\,\|\mathbf{w}\|}$.

Proof. Insert $\mathbf{g} = \mathbf{K}\mathbf{f} + \mathbf{e}$ into the filtered formula and use $\mathbf{u}_k^{\mathsf{T}}\mathbf{K}\mathbf{f} = \sigma_k \mathbf{v}_k^{\mathsf{T}}\mathbf{f}$:

$$\mathbf{f}_\lambda - \mathbf{f} = \sum_k \frac{\sigma_k}{\sigma_k^2 + \lambda}(\mathbf{u}_k^{\mathsf{T}}\mathbf{e})\,\mathbf{v}_k \;-\; \sum_k \frac{\lambda}{\sigma_k^2 + \lambda}(\mathbf{v}_k^{\mathsf{T}}\mathbf{f})\,\mathbf{v}_k.$$

Noise term. By the AM–GM inequality, $\sigma^2 + \lambda \ge 2\sigma\sqrt{\lambda}$, so every coefficient $\sigma_k/(\sigma_k^2+\lambda) \le 1/(2\sqrt{\lambda})$. The first sum therefore has norm at most $\|\mathbf{e}\|/(2\sqrt{\lambda}) \le \delta/(2\sqrt{\lambda})$.

Bias term. The source condition gives $\mathbf{v}_k^{\mathsf{T}}\mathbf{f} = \mathbf{v}_k^{\mathsf{T}}\mathbf{K}^{\mathsf{T}}\mathbf{w} = \sigma_k\,\mathbf{u}_k^{\mathsf{T}}\mathbf{w}$. So the second sum has coefficients $\lambda\sigma_k(\mathbf{u}_k^{\mathsf{T}}\mathbf{w})/(\sigma_k^2+\lambda)$, and by the same inequality $\lambda\sigma/(\sigma^2+\lambda) \le \sqrt{\lambda}/2$. Its norm is at most $\frac{\sqrt{\lambda}}{2}\|\mathbf{w}\|$.

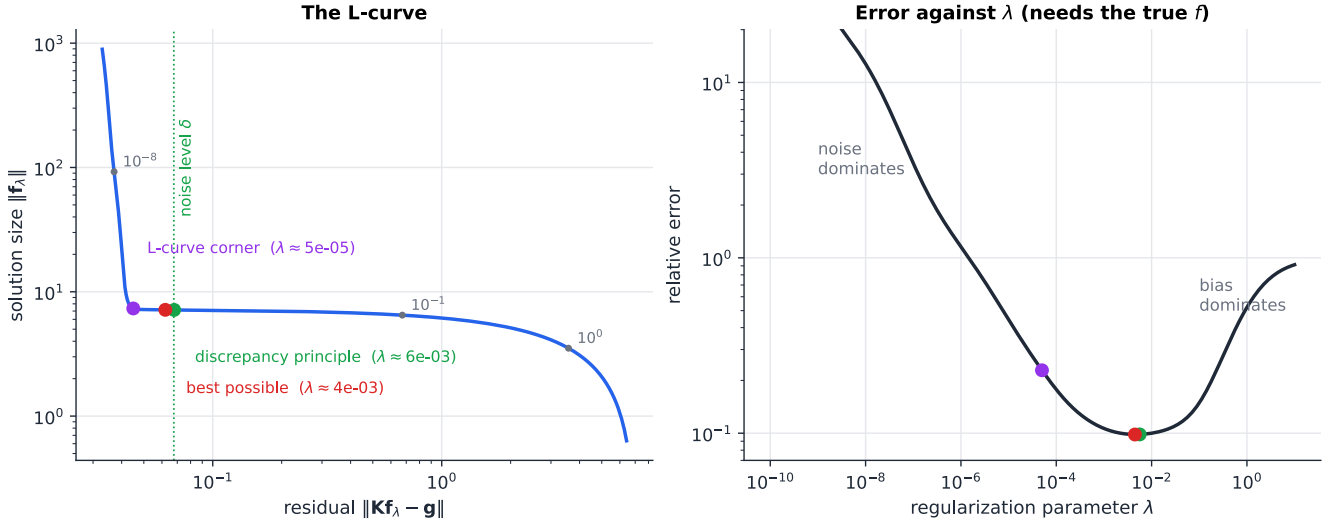
Balance. The sum $a/\sqrt{\lambda} + b\sqrt{\lambda}$ with $a = \delta/2$, $b = \|\mathbf{w}\|/2$ is minimized at $\lambda = a/b = \delta/\|\mathbf{w}\|$, where it equals $2\sqrt{ab} = \sqrt{\delta\|\mathbf{w}\|}$. ∎

Read the result carefully, because it is the central fact of the subject. With exact data ($\delta \to 0$) and λ chosen proportionally to δ, the reconstruction converges to the truth, so the method is a genuine **regularization**. But the error goes like $\sqrt{\delta}$, not δ: to gain one more correct digit you need two more digits of data accuracy. Stronger smoothness assumptions improve the exponent up to $\delta^{2/3}$ and no further for Tikhonov (its **saturation**). Ill-posedness is never fully cured; it is only negotiated.

Choosing λ from the data

The optimal λ in the theorem depends on $\|\mathbf{w}\|$, which describes the unknown truth. In practice λ must come from what you have.

- **The discrepancy principle** (Morozov, 1966). Do not fit the data better than the noise deserves: pick λ so that $\|\mathbf{K}\mathbf{f}_\lambda - \mathbf{g}\| = \tau\delta$ with $\tau \ge 1$. It needs the noise level δ, and with it, it provably achieves the optimal rate $O(\sqrt{\delta})$ under the source condition. In the hook's example it picks $\lambda \approx 5.6 \times 10^{-3}$ and an error of 9.8%, essentially the best possible.
- **The L-curve** (Hansen, 1992). Plot the solution size against the residual on log–log axes for all λ. Small λ gives large solutions with small residuals; large λ, the reverse. The curve has a corner where neither can be reduced without a large increase in the other, and the corner is the choice. It needs no knowledge of δ.
- **Generalized cross-validation** chooses λ so that the fit predicts each data point well when that point is left out. It also ignores δ.



Three ways of choosing λ on the same problem. The discrepancy principle, which knows δ, lands at the bottom of the error valley. The L-curve corner, which doesn't, picks $\lambda \approx 5 \times 10^{-5}$, about 100 times too small, and more than doubles the error to 23%.

Is there a rule that never needs δ and always works? No, and the reason is a theorem. **Bakushinskii's veto** (1984): for an ill-posed problem, any regularization method whose parameter is chosen from the data $\mathbf{g}$ alone, without knowledge of δ, fails to converge for some true solutions as $\delta \to 0$. The noise level is information that cannot be manufactured from the data. Heuristic rules like the L-curve and GCV still work very well in most practical cases, and understanding exactly when is an active line of research.

The Bayesian reading

Model the unknown as random, with a Gaussian prior $\mathbf{f} \sim \mathcal{N}(\mathbf{0}, \tau^2 \mathbf{I})$, and the noise as $\mathbf{e} \sim \mathcal{N}(\mathbf{0}, \eta^2 \mathbf{I})$. By Bayes' rule (Ch. 35) the posterior density is proportional to

$$\exp\Big(-\frac{\|\mathbf{K}\mathbf{f}-\mathbf{g}\|^2}{2\eta^2} - \frac{\|\mathbf{f}\|^2}{2\tau^2}\Big),$$

and its maximizer is exactly the Tikhonov solution with $\lambda = \eta^2/\tau^2$. The regularization parameter is a ratio of noise variance to signal variance, and the penalty $\|\mathbf{f}\|^2$ is a belief about what signals look like. Changing the penalty changes the belief: $\|\mathbf{D}\mathbf{f}\|^2$ with a difference matrix $\mathbf{D}$ says "smooth", the total variation $\|\mathbf{D}\mathbf{f}\|_1$ says "piecewise constant, with sharp edges", and $\|\mathbf{f}\|_1$ in a wavelet basis says "sparse", the idea behind compressed sensing (Ch. 76).

The Bayesian view gives something Tikhonov alone does not: an uncertainty. The posterior covariance $(\mathbf{K}^{\mathsf{T}}\mathbf{K}/\eta^2 + \mathbf{I}/\tau^2)^{-1}$ says which features of the reconstruction are determined by the data and which are only the prior talking. But it also creates the tension Ch. 61 warned about. In the Bayesian view λ is not tuned; it is fixed by η and τ. And the prior $\mathcal{N}(\mathbf{0}, \tau^2\mathbf{I})$ says each pixel is independent, so as the grid is refined, a "typical" signal under the prior becomes white noise, nothing like the box and bump we were reconstructing.

> **? A prior that survives refinement**
>
> A good prior should describe the same beliefs whether the signal is stored on 200 points or 200 million. That requires a probability measure on a space of *functions*, and on an infinite-dimensional space there is no uniform "Lebesgue" measure to write densities against. Gaussian measures with carefully decaying covariances work. In 2004 Lassas and Siltanen showed that the natural discretized total-variation prior, the edge-preserving favourite of image processing, does not: as the grid is refined, its reconstructions converge either to something blurry or to something meaningless, depending on how the prior is scaled. Which priors are "discretization invariant", and why, is a question about measures on function spaces: Ch. 97.

Nonlinear inverse problems and parameter estimation

Most real inverse problems are nonlinear: the data $\mathbf{g} = F(\mathbf{p}) + \mathbf{e}$ depend nonlinearly on parameters $\mathbf{p}$, such as the conductivity $k(x)$ of the rod in Ch. 89 or the time constants of a model. The workhorse is to linearize and regularize at each step. Gauss–Newton (§24.7) solves $\mathbf{J}\,\Delta\mathbf{p} \approx \mathbf{g} - F(\mathbf{p})$ with the Jacobian $\mathbf{J}$, and the Levenberg–Marquardt modification adds $\mu\|\Delta\mathbf{p}\|^2$: it is Tikhonov regularization applied to every linearized step. A new difficulty appears, **identifiability**: different parameters can produce *identical* data, not just similar data, and then no amount of regularization can tell them apart.

The model problem of the field is **Calderón's problem** (1980): recover the conductivity $\gamma(\mathbf{x})$ inside a body from all possible pairs of boundary voltages and boundary currents, the Dirichlet-to-Neumann map. Uniqueness was proved by Sylvester and Uhlmann in 1987 for smooth conductivities in three or more dimensions, and by Astala and Päivärinta in 2006 for any bounded conductivity in the plane. Stability is another matter: Mandache proved in 2001 that it can be at best *logarithmic*, so that improving the data accuracy from δ to δ^2 may only double the reconstruction accuracy. Impedance tomography is exponentially ill posed, however clever the algorithm.

> **? How rough can a conductivity be and still be heard?**
>
> In three dimensions uniqueness for Calderón's problem is known for Lipschitz conductivities (Caro and Rogers, 2016) and somewhat beyond, but not for merely bounded ones, which is what real tissue boundaries and composite materials look like. Whether two different bounded conductivities in 3-D can produce exactly the same boundary measurements is an open problem. So is a satisfying answer to the practical question: given logarithmic stability, how much prior information buys how much resolution in a real impedance scanner?

91.4 The algorithm

For a moderate problem, compute the SVD once, at $O(n^3)$ cost. After that, $\mathbf{f}_\lambda$ for any λ costs only $O(n^2)$, so scanning a hundred values of λ is cheap. That is what the experiment below does. Without an SVD, the stable way to compute one Tikhonov solution is the stacked least-squares problem, solved by QR rather than by forming $\mathbf{K}^{\mathsf{T}}\mathbf{K}$, which would square the condition number (§24.3). The discrepancy principle then becomes a one-dimensional root-finding problem in $\log\lambda$, because the residual $\|\mathbf{K}\mathbf{f}_\lambda - \mathbf{g}\|$ increases monotonically with λ.

```
import numpy as np

def tikhonov(K, g, lam):
    """argmin ||K f - g||^2 + lam ||f||^2, via a stacked least-squares problem (QR inside)."""
    n = K.shape[1]
    A = np.vstack([K, np.sqrt(lam) * np.eye(n)])
    b = np.concatenate([g, np.zeros(n)])
    return np.linalg.lstsq(A, b, rcond=None)[0]

def discrepancy(K, g, delta, tau=1.0, lo=1e-12, hi=1e2, steps=40):
    """Largest lam with ||K f_lam - g|| <= tau * delta, by bisection in log(lam)."""
    for _ in range(steps):
        mid = np.sqrt(lo * hi)
        if np.linalg.norm(K @ tikhonov(K, g, mid) - g) > tau * delta:
            hi = mid                    # residual too big: regularizing too much
        else:
            lo = mid
    return lo
```

On the hook's problem, `discrepancy` returns $\lambda = 5.6 \times 10^{-3}$ and a reconstruction with relative error 9.8%, matching the SVD scan.

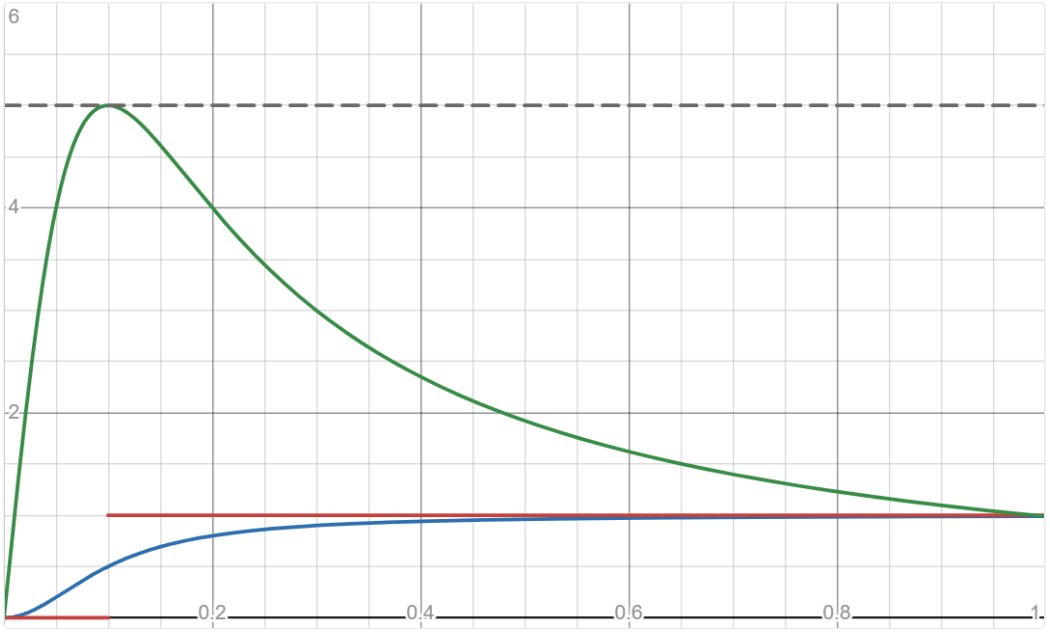
Large problems. A CT image of 512×512 pixels has $n \approx 2.6 \times 10^5$ unknowns, and $\mathbf{K}$ is never stored as a dense matrix: it is a routine that computes projections. Then the method of choice is an iterative least-squares solver such as CGLS or LSQR, which needs only products with $\mathbf{K}$ and $\mathbf{K}^{\mathsf{T}}$. These methods regularize by themselves. Early iterations pick up the large-σ_k components; later ones reach the small-σ_k, noise-dominated components. So the error first falls and then rises again, which is called **semi-convergence**, and the iteration count plays the role of $1/\lambda$. Stopping by the discrepancy principle is standard.

Failure modes.

- **The inverse crime.** If you generate synthetic data with the same discretized $\mathbf{K}$ that you then invert, the model is perfect and the reconstruction looks far better than it will on real data. Honest tests generate data on a finer grid, or with a different method, than the one used for inversion. The experiment below commits this crime on purpose; one of its knobs removes it.
- **Wrong noise model.** Astronomical images have Poisson photon noise, not Gaussian noise. The least-squares misfit is then the wrong measure, and the Richardson–Lucy iteration, which maximizes the Poisson likelihood and keeps intensities positive, was the tool astronomers actually used on Hubble's flawed images.
- **Wrong prior.** The penalty $\|\mathbf{f}\|^2$ rounds every edge, as the $\lambda = 0.3$ curve shows. Edge-preserving penalties such as total variation are better, but they are non-smooth and need convex-optimization algorithms (Ch. 47).
- **Boundary effects.** Near the ends of the interval, the blur averages over values the model assumes are zero. If the true scene continues beyond the edge, the reconstruction invents artefacts there.

91.5 The experiment

See the filter. The blue curve is Tikhonov's filter factor $\sigma^2/(\sigma^2+\lambda)$ with $\lambda = 0.01$, plotted against the singular value σ; the red step is truncated SVD with cutoff $\sigma = \sqrt{\lambda} = 0.1$. The green curve, $\sigma/(\sigma^2+\lambda)$, is how much Tikhonov amplifies noise in a component with singular value σ, and the dashed line is the bound $1/(2\sqrt{\lambda}) = 5$ from §91.3. Change `0.01` to `0.0001` everywhere (and the cutoff `0.1` to `0.01`, the bound `5` to `50`) and watch the filter sharpen while the noise amplification grows tenfold.



Tikhonov's filter (blue) keeps components with $\sigma \gg \sqrt{\lambda}$ and smoothly suppresses the rest, unlike the hard cut of truncated SVD (red). The amplification of noise (green) peaks exactly at $\sigma = \sqrt{\lambda}$, at the height the error bound uses.

Deblur it yourself. This script rebuilds the hook's example, prints the error for a range of λ, and lets the discrepancy principle choose.

```
import numpy as np

n, width, noise_level = 200, 0.01, 0.01   # knobs: grid, blur width, relative noise
rng = np.random.default_rng(0)
x = (np.arange(n) + 0.5) / n
K = np.exp(-(x[:, None] - x[None, :])**2 / (2 * width**2)) / (np.sqrt(2 * np.pi) * width * n)
f = ((x > 0.15) & (x < 0.35)) + 0.8 * np.maximum(0, 1 - np.abs(x - 0.6) / 0.1) \
    + 0.6 * np.exp(-((x - 0.85) / 0.04)**2)          # box, tent, bump
g_clean = K @ f
e = noise_level * np.linalg.norm(g_clean) / np.sqrt(n) * rng.standard_normal(n)
g = g_clean + e
delta = np.linalg.norm(e)

U, s, Vt = np.linalg.svd(K)
print(f"condition number {s[0] / s[-1]:.2e}")
rel = lambda fr: np.linalg.norm(fr - f) / np.linalg.norm(f)
print(f"naive inverse: relative error {rel(np.linalg.solve(K, g)):.2e}")

def tik(lam):                                       # Tikhonov through the SVD filter
    return Vt.T @ (s / (s**2 + lam) * (U.T @ g))

print("   lambda    residual/delta   rel. error")
for lam in [1e-8, 1e-6, 1e-4, 1e-3, 5e-3, 1e-2, 1e-1, 1]:
    fl = tik(lam)
    print(f"{lam:9.0e}   {np.linalg.norm(K @ fl - g) / delta:10.2f}   {rel(fl):10.3f}")

lams = np.logspace(-10, 1, 400)                     # discrepancy principle
res = np.array([np.linalg.norm(K @ tik(l) - g) for l in lams])
lam_d = lams[res <= delta].max()
print(f"discrepancy principle picks lambda = {lam_d:.1e}, error {rel(tik(lam_d)):.3f}")
```

With the default knobs the condition number is 1.83×10^{8} and the naive error is 2.73×10^{5}. The table's error column runs 12.8, 1.16, 0.178, 0.106, 0.098, 0.100, 0.149, 0.522 as λ grows: a valley with a flat floor between about 10^{-3} and 10^{-2}. The discrepancy principle picks $\lambda = 5.2 \times 10^{-3}$, with error 0.098. Knobs to turn:

- Set `noise_level` to 10^{-3} and then 10^{-4}. By how much does the best error improve each time? Compare with the $\sqrt{\delta}$ rate of the theorem: is this signal smooth enough to satisfy the source condition? (The box has jumps.)
- Double `width`. The problem becomes far more severely ill posed. How many singular values are now above the noise floor?
- Commit no crime: build the data with a finer kernel matrix (say $3n$ points, then average each group of three), and invert with the n-point `K`. How much worse is the best achievable error?
- Replace the penalty $\|\mathbf{f}\|^2$ by $\|\mathbf{D}\mathbf{f}\|^2$ with `D = np.diff(np.eye(n), axis=0)`, solving the normal equations $(\mathbf{K}^{\mathsf{T}}\mathbf{K} + \lambda \mathbf{D}^{\mathsf{T}}\mathbf{D})\mathbf{f} = \mathbf{K}^{\mathsf{T}}\mathbf{g}$ directly. Which features improve, and which get worse?

91.6 The challenge

1. **Intuition.** A thermometer at the midpoint of a rod records the temperature for one hour. Explain without equations why estimating the rod's *initial* temperature profile from this record is severely ill posed, while estimating a single unknown heat-loss coefficient from it is well posed. What is different about the number of unknowns compared with the number of components the data can carry?
2. **Derivation.** For truncated SVD with cutoff index r, $\mathbf{f}_r = \sum_{k \le r} \sigma_k^{-1} (\mathbf{u}_k^{\mathsf{T}}\mathbf{g})\mathbf{v}_k$, derive the analogue of the Tikhonov error bound under the source condition $\mathbf{f} = \mathbf{K}^{\mathsf{T}}\mathbf{w}$, and show that the best choice of r again gives an error of order $\sqrt{\delta}\|\mathbf{w}\|$. Then show that under the stronger condition $\mathbf{f} = (\mathbf{K}^{\mathsf{T}}\mathbf{K})\mathbf{z}$, truncated SVD reaches $O(\delta^{2/3})$ and keeps improving under still stronger conditions, while Tikhonov stalls at $\delta^{2/3}$. Find the step of the Tikhonov argument that causes the stall.
3. **Implementation.** Implement total-variation deblurring for the hook's signal: minimize $\|\mathbf{K}\mathbf{f} - \mathbf{g}\|^2 + \lambda \sum_i \sqrt{(f_{i+1} - f_i)^2 + \epsilon^2}$ by gradient descent or by lagged diffusivity (re-solve a weighted least-squares problem at each step). Compare its reconstruction of the box with Tikhonov's at the discrepancy-principle λ, and report both errors.
4. **Open investigation.** Train the simplest possible learned reconstructor: a linear map $\mathbf{f} \approx \mathbf{B}\mathbf{g}$ fitted by least squares on a thousand random training signals made of boxes and bumps. Compare it with Tikhonov on new signals of the same kind, then on signals of a different kind (a sine wave, a single spike). Then search for a small perturbation of the data, $\|\Delta\mathbf{g}\| \le \delta$, that maximizes the change in $\mathbf{B}\mathbf{g}$. Is the learned map more or less stable than Tikhonov, and does it trade stability for accuracy in any predictable way?

91.7 Loose threads

An inverse problem is a forward problem that forgot something. The SVD shows exactly which components were forgotten, regularization fills them in with assumptions, and the Tikhonov bound shows the price: errors that shrink like $\sqrt{\delta}$ at best, and a parameter whose choice needs information, the noise level, that the data cannot supply. The Bayesian reading turns the assumptions into explicit probability and buys honest error bars. Calderón's problem and the priors that survive refinement are still open. One more clue comes from the present day.

> **? Reconstructions that hallucinate**
>
> Since about 2017, neural networks trained on large image collections have produced MRI and CT reconstructions sharper than any regularization formula, from fewer measurements. In 2020 Antun, Renna, Poon, Adcock and Hansen showed that several such networks are unstable in a way Tikhonov never is. A tiny, carefully chosen perturbation of the measurements can make a lesion vanish or a false structure appear, and adding *more* measurements can make the result worse. Their follow-up work showed that, for some inverse problems, stable and accurate neural networks exist but no training algorithm can be guaranteed to find them. When a learned reconstructor can be trusted, and how to certify it on a particular scan, is an open problem with patients waiting on the answer.

Further pointers:

- **The physics behind the operators.** Most forward maps in this chapter, from blur to radiation transport to wave scattering, come from variational principles and conservation laws, and the symmetries of the physics decide which parameters can be identified: Ch. 92.
- **Uncertainty at scale.** Turning a posterior over a million unknowns into error bars needs sampling methods that work in high dimension, such as the preconditioned Crank–Nicolson MCMC, whose design comes straight from the infinite-dimensional view: Ch. 38.
- **Geometry from images.** Recovering depth from photographs, the inverse problem left open in Ch. 69, is nonlinear, and its ill-posedness shows up as near-degenerate camera configurations.

Further reading

- P. C. Hansen, *Discrete Inverse Problems: Insight and Algorithms*, SIAM, 2010. The Picard plot, the L-curve and Tikhonov regularization in practice, with MATLAB code throughout.
- J. L. Mueller and S. Siltanen, *Linear and Nonlinear Inverse Problems with Practical Applications*, SIAM, 2012. From deblurring and CT to electrical impedance tomography, with real data.
- H. W. Engl, M. Hanke and A. Neubauer, *Regularization of Inverse Problems*, Kluwer, 1996. The rigorous theory: source conditions, convergence rates, saturation and parameter choice.
- J. Kaipio and E. Somersalo, *Statistical and Computational Inverse Problems*, Springer, 2005. The Bayesian approach, including priors, error modelling and the inverse crime.
- A. M. Stuart, "Inverse problems: a Bayesian perspective", *Acta Numerica* 19 (2010), 451–559. Bayesian inversion formulated directly on function spaces.
- V. Antun, F. Renna, C. Poon, B. Adcock and A. C. Hansen, "On instabilities of deep learning in image reconstruction and the potential costs of AI", *PNAS* 117 (2020), 30088–30095. The paper behind the last mystery.

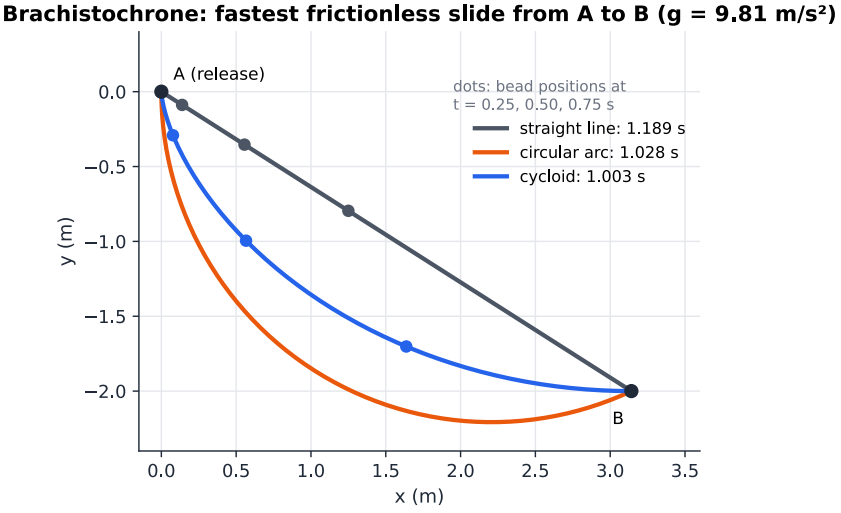
Chapter 92 — Mathematical Physics: The Structure Behind Nature

> **Level 3 · Track J — Mathematics at the Frontier Builds on:** Ch. 6 (§6.11 Lagrange multipliers), Ch. 11, Ch. 51, Ch. 59 · **Leads to:** Ch. 93, Ch. 95, Ch. 90 · **Related:** Ch. 72, Ch. 38, Ch. 56, Ch. 61, Ch. 84 **Reading time:** ~30 min

92.1 The hook

In June 1696 Johann Bernoulli published a challenge "to the sharpest mathematicians in the world". Two points A and B sit at different heights, with B lower and off to the side. A bead slides without friction along a wire from A to B, pulled only by gravity. What shape of wire gets it there fastest?

The straight line is the shortest path, but it is not the fastest. Take B to be π metres to the right of A and 2 metres below it. On the straight wire the bead takes **1.189 s**. Bend the wire into a circular arc that starts vertically and the bead wins, at **1.028 s**: it plunges first, builds up speed, and spends that speed on the longer route. The best wire of all takes **1.003 s**. It is a **cycloid**, the curve traced by a point on the rim of a rolling wheel. Newton is said to have solved it overnight and replied anonymously; Bernoulli recognised the lion "by his claw".



Three wires from A to B. The dots mark where each bead is after 0.25, 0.50 and 0.75 s. The cycloid bead drops steeply, gains speed early, and arrives first even though its wire is longer than the straight line.

The puzzle is old, but the idea inside it shaped the next three centuries of physics. The answer to "which *curve* is best?" is a differential equation, and nature seems to answer such questions all the time. Light crossing from air into water bends so as to minimise its travel time (Fermat). A planet follows the orbit that makes a certain integral, the **action**, stationary (Maupertuis, Euler, Lagrange, Hamilton). A clock in free fall follows the path through spacetime along which it ticks the most (Einstein).

Then, in 1918, Emmy Noether found something that is still astonishing. Once a law of motion is written as "make this integral stationary", every **continuous symmetry** of the integral hands you a **conserved quantity** for free. Laws that don't change from one day to the next force energy to be conserved. Laws that are the same everywhere force momentum to be conserved. Laws that look the same in every direction force angular momentum to be conserved.

> **Why should a symmetry, a statement about what does *not* matter, produce a number that never changes?**

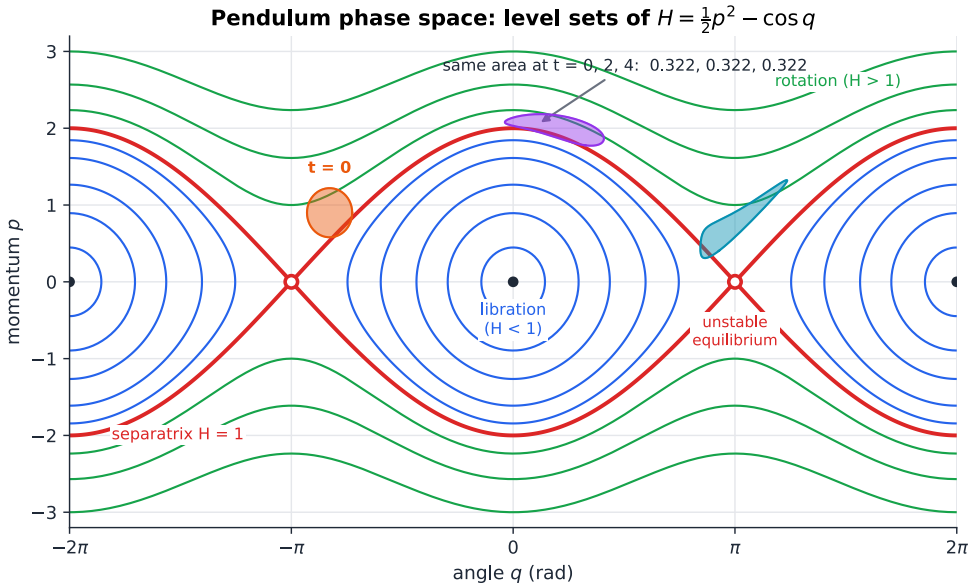
This chapter answers that question. It also follows the same structure into phase space, where it explains why some numerical integrators run stably for billions of steps, and to the edge of quantum mechanics.

92.2 The big idea

Idea 1: optimise over whole paths, not numbers. In calculus you minimise a function of a few numbers by setting a derivative to zero (§6.11). In the calculus of variations the unknown is a *whole function* $y(x)$, and the thing to minimise is a **functional**, a number computed from the entire curve, such as the bead's travel time. The trick is the same as in ordinary calculus. Wiggle the curve slightly, by $\varepsilon\,\eta(x)$, and demand that the first-order change vanish for *every* wiggle η. Because the wiggle is arbitrary, the condition holds at each point separately and becomes a differential equation.

Idea 2: a symmetry is a wiggle that costs nothing. Suppose moving the entire experiment 1 cm to the left changes nothing. That shift is a wiggle of the path that leaves the integral exactly unchanged, at every instant and not only to first order. Feed this special wiggle into the stationarity condition and it does not produce an equation for the path. It produces a statement that some combination of positions and velocities has zero time derivative. The symmetry is spent on buying a conservation law.

Idea 3: phase space is a place where volume is sacred. Hamilton rewrote mechanics using positions q *and* momenta p as equal partners. A state of the system is then a point in **phase space**, and the motion is a flow of that space. For a pendulum the flow runs along the level curves of the energy $H(q,p)$: closed ovals while it swings, wavy lines while it spins over the top, and the **separatrix** between them.



Pendulum phase space. Every trajectory runs along a level set of H. A disc of initial states (orange) is carried by the exact flow and sheared into crescents (purple at $t = 2$, teal at $t = 4$), but its area stays at 0.322 to six digits. This is Liouville's theorem.

An analogy: phase-space flow is like stirring paint into honey. You can stretch a blob into an arbitrarily thin thread, but never compress it. Facts as different as "a Hamiltonian system has no attractors" and the success of Hamiltonian Monte Carlo follow from this one incompressibility.

92.3 The mathematics

The Euler–Lagrange equation

Let $q(t) \in \mathbb{R}^n$ be a path with fixed endpoints $q(t_0) = q_0$ and $q(t_1) = q_1$, and let

$$S[q] = \int_{t_0}^{t_1} L\big(q(t), \dot q(t), t\big)\, dt$$

for a smooth function L, the **Lagrangian**.

Euler–Lagrange equations

If q makes S stationary among all smooth paths with the same endpoints, then for each $i = 1, \dots, n$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot q_i} - \frac{\partial L}{\partial q_i} = 0.$$

Proof. Take any smooth $\eta(t)$ with $\eta(t_0) = \eta(t_1) = 0$ and compare q with $q + \varepsilon\eta$. Differentiating under the integral sign,

$$\frac{d}{d\varepsilon} S[q + \varepsilon\eta]\Big|_{\varepsilon = 0} = \int_{t_0}^{t_1} \sum_i \Big(\frac{\partial L}{\partial q_i}\eta_i + \frac{\partial L}{\partial \dot q_i}\dot\eta_i \Big)\, dt.$$

Integrate the second term by parts. The boundary term $\big[\sum_i \frac{\partial L}{\partial \dot q_i}\eta_i\big]_{t_0}^{t_1}$ vanishes because η vanishes at both ends, which leaves

$$0 = \int_{t_0}^{t_1} \sum_i \Big(\frac{\partial L}{\partial q_i} - \frac{d}{dt}\frac{\partial L}{\partial \dot q_i} \Big)\eta_i\, dt \quad \text{for every admissible } \eta.$$

If the bracket were nonzero at some time t^* in some component, say positive, it would stay positive on a small interval around t^*. Choose η_i to be a smooth bump supported inside that interval and every other component zero: the integral would then be positive, a contradiction. (This step is called the *fundamental lemma of the calculus of variations*.) ∎

Newton from a single function. For a particle of mass m in a potential $V(q)$, take $L = T - V = \tfrac{1}{2}m\|\dot q\|^2 - V(q)$. Then $\partial L/\partial \dot q = m\dot q$ and $\partial L/\partial q = -\nabla V$, so the Euler–Lagrange equation reads $m\ddot q = -\nabla V$: Newton's second law. The **principle of stationary action** packs mechanics into one scalar function, and it works in *any* coordinates. Write L in polar angles or a robot arm's joint angles, and the same recipe gives correct equations of motion without resolving a single force.

The brachistochrone, briefly. With y measured downward, a bead starting from rest has speed $\sqrt{2gy}$ by energy conservation, so the travel time is

$$T[y] = \int_0^X \frac{\sqrt{1 + y'^2}}{\sqrt{2gy}}\, dx.$$

The integrand $F(y, y')$ does not depend on x. For such integrands the Euler–Lagrange equation has a first integral, the **Beltrami identity** $F - y'\,\partial F/\partial y' = $ const, which here reduces to $y(1 + y'^2) = 2r$ for a constant r. Substituting $x = r(\theta - \sin\theta)$, $y = r(1 - \cos\theta)$ solves it: a cycloid. The bead's parameter turns out to be proportional to time, $\theta = t\sqrt{g/r}$, so reaching the bottom of an arch takes $\pi\sqrt{r/g}$, which is 1.003 s for $r = 1$ m. The Beltrami identity is our first conservation law from a symmetry. The integrand did not care about x, and something was conserved.

Noether's theorem

Now make that observation general. A **one-parameter symmetry** of L is a smooth family of transformations $q \mapsto q + \varepsilon K(q) + O(\varepsilon^2)$, generated by a vector field K, under which the Lagrangian does not change to first order:

$$\frac{d}{d\varepsilon} L\big(q + \varepsilon K(q),\ \dot q + \varepsilon \tfrac{d}{dt} K(q)\big)\Big|_{\varepsilon = 0} = \sum_i \Big(\frac{\partial L}{\partial q_i} K_i + \frac{\partial L}{\partial \dot q_i} \frac{dK_i}{dt} \Big) = 0.$$

Define the **momentum** conjugate to q_i as $p_i = \partial L / \partial \dot q_i$.

Noether's theorem (point symmetries)

If K generates a symmetry of L, then along every solution of the Euler–Lagrange equations the quantity

$$Q = \sum_i p_i\, K_i(q) = \sum_i \frac{\partial L}{\partial \dot q_i} K_i(q)$$

is constant in time.

Proof. Differentiate Q along a solution and use the product rule:

$$\frac{dQ}{dt} = \sum_i \Big(\frac{d}{dt}\frac{\partial L}{\partial \dot q_i} \Big) K_i + \sum_i \frac{\partial L}{\partial \dot q_i} \frac{dK_i}{dt}.$$

By the Euler–Lagrange equation the first bracket equals $\partial L / \partial q_i$. What remains is exactly the left-hand side of the symmetry condition, which is zero. ∎

That is the whole proof. The answer to the hook's question is now visible. The Euler–Lagrange equation says *forces equal the rate of change of momentum*. The symmetry condition says *the force along the direction K is cancelled by the way momentum feeds into K*. Put together, the momentum along K has nothing left to change it.

The three classical conservation laws

- **Translation.** N particles with $L = \sum_a \tfrac12 m_a \|\dot{\mathbf{x}}_a\|^2 - V$, where V depends only on differences $\mathbf{x}_a - \mathbf{x}_b$. Shifting every particle by $\varepsilon \hat{\mathbf{e}}$ leaves L unchanged, so $K = (\hat{\mathbf{e}}, \dots, \hat{\mathbf{e}})$ and $Q = \hat{\mathbf{e}} \cdot \sum_a m_a \dot{\mathbf{x}}_a$. Total momentum is conserved.
- **Rotation.** In the plane, rotating by ε moves (x, y) by $\varepsilon(-y, x)$. If V depends only on distances, $Q = p_x(-y) + p_y x = m(x\dot y - y\dot x)$. Angular momentum is conserved, which is Kepler's equal-area law.
- **Time translation.** If L has no explicit t, define the **energy function** $h = \sum_i p_i \dot q_i - L$. Then $dh/dt = -\partial L/\partial t = 0$ (differentiate and use Euler–Lagrange, exactly as above). For $L = T - V$ with T quadratic in $\dot q$, $h = T + V$. Energy is conserved.

Time translation needs the slightly larger version of the theorem that allows t to move too, but the computation is the same in spirit. The theorem also runs in reverse: in Hamiltonian form below, a conserved quantity *generates* a symmetry. The dictionary is two-way.

Not every conservation law is Noether's

Noether needs an action. Friction has none, which is why a damped oscillator has a time-independent law and still loses energy. Discrete symmetries such as reflections give selection rules rather than conserved numbers (Ch. 95).

Hamilton's equations and the incompressible flow

The **Legendre transform** trades velocities for momenta. Assuming $p = \partial L / \partial \dot q$ can be solved for $\dot q$, define the **Hamiltonian**

$$H(q, p) = \sum_i p_i \dot q_i - L(q, \dot q),$$

which is the energy function h written in terms of (q, p). Computing dH in two ways, and using the Euler–Lagrange equation, gives **Hamilton's equations**:

$$\dot q_i = \frac{\partial H}{\partial p_i}, \qquad \dot p_i = -\frac{\partial H}{\partial q_i}.$$

For the pendulum, with unit mass, length and g, $L = \tfrac12 \dot q^2 + \cos q$, $p = \dot q$, and $H = \tfrac12 p^2 - \cos q$. Two immediate consequences follow. First, $dH/dt = \sum_i (H_{q_i}\dot q_i + H_{p_i}\dot p_i) = \sum_i (H_{q_i}H_{p_i} - H_{p_i}H_{q_i}) = 0$, so trajectories stay on level sets, as in the figure. Second, and deeper:

Liouville's theorem

Let Φ_t be the flow of Hamilton's equations on $\mathbb{R}^{2n}$. For any region D of phase space, $\operatorname{vol}(\Phi_t(D)) = \operatorname{vol}(D)$ for all t.

Proof. Write the equations as $\dot z = F(z)$ with $z = (q, p)$ and $F = (\partial H/\partial p,\ -\partial H/\partial q)$. Its divergence is

$$\nabla \cdot F = \sum_i \Big(\frac{\partial^2 H}{\partial q_i \partial p_i} - \frac{\partial^2 H}{\partial p_i \partial q_i} \Big) = 0.$$

Let $J(t) = \partial \Phi_t(z_0)/\partial z_0$ be the Jacobian matrix of the flow. Differentiating $\dot z = F(z)$ with respect to the initial condition gives the linear equation $\dot J = DF(z(t))\, J$ with $J(0) = \mathbf{I}$. For any matrix solution of such an equation, $\frac{d}{dt} \det J = \operatorname{tr}(DF) \det J$ (Jacobi's formula; it follows from $\det(\mathbf{I} + h\mathbf{A}) = 1 + h \operatorname{tr}\mathbf{A} + O(h^2)$). Since $\operatorname{tr} DF = \nabla \cdot F = 0$, $\det J(t) = 1$ for all t, and the change-of-variables formula gives $\operatorname{vol}(\Phi_t(D)) = \int_D |\det J|\, dz_0 = \operatorname{vol}(D)$. ∎

The Hamiltonian flow preserves more than volume. Its Jacobian satisfies $J^{\mathsf{T}}\boldsymbol{\Omega} J = \boldsymbol{\Omega}$, with $\boldsymbol{\Omega} = \begin{pmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{I} & \mathbf{0} \end{pmatrix}$. Such a map is called **symplectic**. Geometrically, it preserves the sum of the oriented areas that any small 2-dimensional patch projects onto the n planes (q_i, p_i). In one degree of freedom "symplectic" and "area-preserving" coincide, which is the version you met in §72.3. In more dimensions it is far stronger: Gromov's **non-squeezing theorem** (1985) forbids a symplectic map from squeezing a ball of radius r into a cylinder $\{q_1^2 + p_1^2 < R^2\}$ with $R < r$, which a merely volume-preserving map could do.

Brackets, and the doorway to quantum mechanics

For any observable $f(q,p)$, the chain rule and Hamilton's equations give $\dot f = \{f, H\}$, where the **Poisson bracket** is

$$\{f, g\} = \sum_i \left(\frac{\partial f}{\partial q_i}\frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i}\frac{\partial g}{\partial q_i} \right), \qquad \{q_i, p_j\} = \delta_{ij}.$$

Noether's theorem now becomes beautifully symmetric. $\{Q, H\} = 0$ says both that Q is conserved along the flow of H, and (because the bracket is antisymmetric) that H is unchanged along the flow generated by Q, which is the symmetry. Momentum generates translations; angular momentum generates rotations; the Hamiltonian generates time evolution.

Dirac noticed that quantum mechanics keeps this algebra but replaces numbers by operators. The bracket $\{f,g\}$ becomes the commutator $[\hat f, \hat g]/(j\hbar)$, so $\{q,p\} = 1$ becomes

$$\hat X \hat P - \hat P \hat X = j\hbar\, \hat I.$$

Ch. 61 showed that no pair of *matrices* satisfies this, by taking traces. Could bounded operators on an infinite-dimensional Hilbert space do it? **Wielandt's argument** (1949) says no, in four lines. Suppose $XP - PX = c\,I$ with $c \neq 0$ and X, P bounded. By induction, $XP^n - P^nX = nc\,P^{n-1}$ for every $n \geq 1$. (Check it: $XP^{n+1} - P^{n+1}X = (XP^n - P^nX)P + P^n(XP - PX) = nc\,P^n + c\,P^n$.) Taking norms,

$$n|c|\,\|P^{n-1}\| \leq 2\|X\|\,\|P\|\,\|P^{n-1}\|.$$

If $P^{n-1} \neq 0$ for every n, dividing gives $n \leq 2\|X\|\|P\|/|c|$ for all n, which is absurd. If instead $P^m = 0$ for some m, the identity with $n = m$ gives $mc\,P^{m-1} = 0$, so $P^{m-1} = 0$; descending one power at a time eventually gives $c\,I = 0$, again absurd. ■ Position and momentum *must* be unbounded operators, with all the delicacy of domains that Ch. 61 warned about. The **Stone–von Neumann theorem** (1931) then says that, once the relation is stated in the robust exponentiated form $e^{ja\hat X}e^{jb\hat P} = e^{-jab\hbar}e^{jb\hat P}e^{ja\hat X}$, every irreducible solution is unitarily equivalent to the familiar one: $\hat X$ multiplies $\psi(x)$ by x and $\hat P = -j\hbar\, d/dx$. In this sense quantum mechanics of a particle is *unique*.

The uncertainty principle is Cauchy–Schwarz. Here is the signal-processing version promised in Ch. 76. Let $x(t)$ be a smooth signal with $\int |x|^2 dt = 1$, centred so that its mean time and mean frequency are zero, and decaying fast enough that $t|x(t)|^2 \to 0$ as $|t| \to \infty$. Define $\sigma_t^2 = \int t^2 |x|^2 dt$ and $\sigma_\omega^2 = \frac{1}{2\pi}\int \omega^2 |X(\omega)|^2 d\omega$, which equals $\int |x'(t)|^2 dt$ by Parseval's theorem (§16.7), since x' has transform $j\omega X$. Integrating by parts,

$$1 = \int |x|^2 dt = -\int t\,\frac{d}{dt}|x|^2\,dt = -2\,\mathrm{Re}\int t\,x^* x'\,dt \leq 2\Big(\int t^2|x|^2dt\Big)^{1/2}\Big(\int |x'|^2 dt\Big)^{1/2}.$$

So $\sigma_t\sigma_\omega \geq \tfrac12$, or $\sigma_t\sigma_f \geq 1/(4\pi)$ with $f = \omega/2\pi$. Equality in Cauchy–Schwarz needs $x' = -\lambda t\, x$ for a real $\lambda > 0$, whose solution is the Gaussian $e^{-\lambda t^2/2}$. Replace ω by $p/\hbar$ (a wavefunction's frequency is its momentum) and the same lines give Heisenberg's $\sigma_x\sigma_p \geq \hbar/2$. The physics is the Fourier transform. The quantum part is only the claim that a particle *is* a wave.

> **? Why does nature extremise anything?**
>
> A thrown ball does not try out paths and pick the best one, so why is mechanics a variational problem at all? Feynman's answer is that in quantum mechanics a particle takes *every* path, each weighted by the phase $e^{jS/\hbar}$. Paths far from stationary action cancel each other out, like a rapidly oscillating integrand, and only the neighbourhood of the classical path survives as $\hbar \to 0$. The trouble is that the "integral over all paths" has never been defined as a genuine integral in most cases of physical interest. For four-dimensional gauge theories, making it rigorous is the Yang–Mills existence and mass gap problem, one of the Clay Millennium Prize problems, and it is still **open**.

92.4 The algorithm

Liouville's theorem says the exact flow preserves area. An integrator that shares this property inherits much of the flow's long-term behaviour. The **Störmer–Verlet** or **leapfrog** method, for $H = \tfrac12 p^2 + V(q)$, is three shears: half a kick, a drift, and half a kick.

```
import numpy as np

def H(q, p):                      # pendulum, g/l = 1, unit mass and length
    return 0.5 * p**2 - np.cos(q)

def euler(q, p, h):               # explicit Euler: uses old q and old p
    return q + h * p, p - h * np.sin(q)

def leapfrog(q, p, h):            # Stormer-Verlet: kick, drift, kick
    p = p - 0.5 * h * np.sin(q)
    q = q + h * p
    p = p - 0.5 * h * np.sin(q)
    return q, p

for step in (euler, leapfrog):
    q, p, h = 1.0, 0.0, 0.1
    E0, worst = H(q, p), 0.0
    for n in range(100_000):      # 10,000 time units, about 1,500 swings
        q, p = step(q, p, h)
        worst = max(worst, abs(H(q, p) - E0))
    print(f"{step.__name__:9s} max |H - H0| = {worst:.2e}, final q = {q:.3f}")
```

Both methods cost about one force evaluation per step, since leapfrog's closing half-kick can be reused as the next step's opening one. Explicit Euler gains energy on every swing. Within four swings the pendulum is spinning over the top, and by the end the maximum energy error is about 330 and q has wound round to -1.9×10^{5} rad. Leapfrog's energy error never exceeds 1.1×10^{-3} over the whole run. It does not drift; it oscillates.

Why it works. Each substep is a shear of phase space: a kick changes p by an amount that depends only on q, and a drift changes q by an amount that depends only on p. Each shear has Jacobian determinant 1 and is symplectic, so their composition is too. The scheme is also **time-reversible**: run it with $-h$ from the end and you retrace the steps exactly. **Backward error analysis** (Benettin–Giorgilli 1994, Hairer–Lubich 1997) then gives the deep explanation of the shadow energy from Ch. 72. A symplectic method applied to an analytic Hamiltonian H is, up to an error of order $e^{-c/h}$, the *exact* flow of a modified Hamiltonian $\tilde{H} = H + h^2 H_2 + h^4 H_4 + \dots$, and the error stays exponentially small for times up to order $e^{c/h}$. The method conserves $\tilde{H}$ almost exactly, so it stays within $O(h^2)$ of the true energy surface for exponentially long times. For the harmonic oscillator, $H = \tfrac{1}{2}(p^2 + q^2)$, the series sums in closed form: leapfrog conserves $\tilde{H} = \tfrac{1}{2}p^2 + \tfrac{1}{2}(1 - h^2/4)q^2$ *exactly*, which you can check in two lines.

Hamiltonian Monte Carlo. This answers the question left open in Ch. 38. To sample from a density $\pi(q) \propto e^{-U(q)}$, add a fake Gaussian momentum and set $H = U(q) + \tfrac{1}{2}\|p\|^2$:

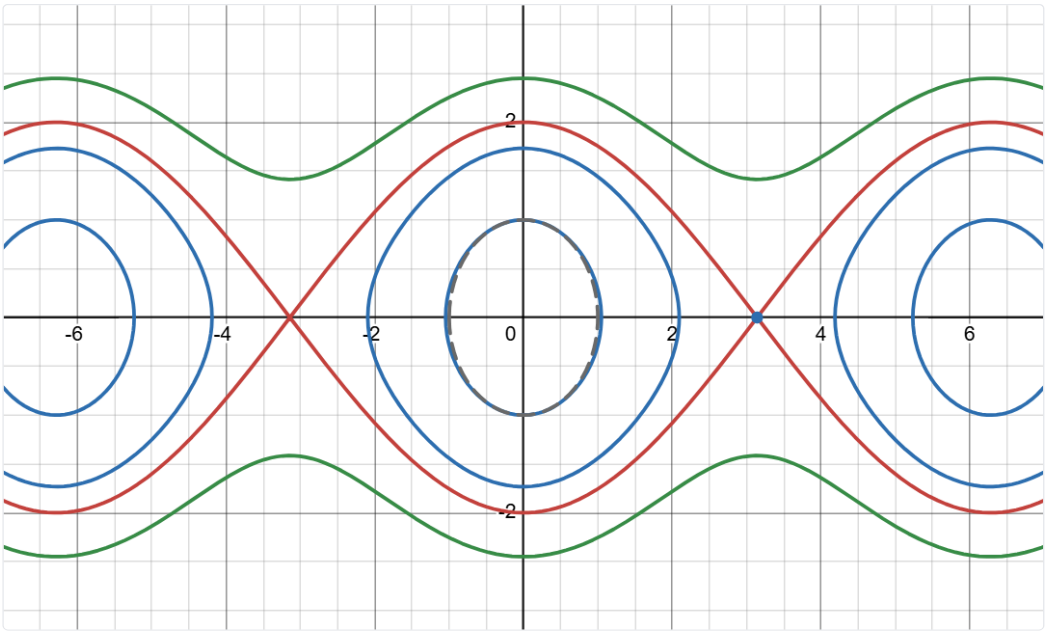
```
HMC step from q:
    draw p ~ N(0, I)
    (q*, p*) = L leapfrog steps of size h from (q, p) on H
    accept q* with probability min(1, exp(H(q, p) - H(q*, p*)))
```

Three properties of leapfrog make this rule exactly correct. *Volume preservation* removes any Jacobian factor from the acceptance ratio. *Reversibility* (flip p at the end) makes the proposal symmetric, as Metropolis needs. And the *small energy error* keeps acceptance near 1 even for long trajectories that cross the whole distribution. RK4 would need a Jacobian correction it cannot compute cheaply. For a product target in d dimensions, the leapfrog energy error adds up over the coordinates, and keeping the acceptance rate fixed requires $h \propto d^{-1/4}$ (Beskos et al., 2013). That is the $d^{1/4}$ cost quoted in Ch. 38.

Failure cases. Leapfrog is only second order and needs $h\omega < 2$ for an oscillator of frequency ω. Naive adaptive step sizes destroy its symplectic structure, and stiff or dissipative systems need other tools (§22.11).

92.5 The experiment

Phase portraits by hand. Every curve below is a level set $\tfrac{1}{2}p^2 - \cos q = E$ of the pendulum Hamiltonian, with x for q and y for p. The dashed grey ellipse is the small-angle approximation $\tfrac{1}{2}p^2 + \tfrac{1}{2}q^2 - 1 = E$ at the same energy $E = -0.5$.



Level sets of the pendulum Hamiltonian. Blue: swinging; red: the separatrix; green: spinning over the top. The dashed ellipse is the linearised pendulum.

Things to change:

- Change both -0.5 values to -0.9, then to 0.5. The ellipse and the true orbit agree for small swings and come apart as the energy approaches the separatrix. This is why a pendulum clock's period depends on its amplitude.
- Change the red level from 1 to 0.99 and to 1.01. A tiny change of energy changes the *kind* of motion. Near the separatrix the period becomes infinite, and this is where chaos first appears once the pendulum is pushed periodically.

Find the brachistochrone by brute force. This script never mentions cycloids or Euler–Lagrange. It cuts the wire into 40 straight segments, computes the exact travel time down each one (constant acceleration, so the time is length divided by mean speed), and moves the heights of the joints downhill in travel time.

```
import numpy as np

g, X, Y, N = 9.81, np.pi, -2.0, 40       # gravity, end point (X, Y), number of segments
x = np.linspace(0, X, N + 1)

def travel_time(y_inner):                # y_inner: (..., N-1) heights of the inner nodes
    y_inner = np.atleast_2d(y_inner)
    ends = np.ones((len(y_inner), 1))
    y = np.hstack([0 * ends, y_inner, Y * ends])
    v = np.sqrt(2 * g * np.maximum(-y, 0))           # speed from energy conservation
    L = np.hypot(np.diff(x), np.diff(y, axis=1))     # segment lengths
    return np.sum(2 * L / (v[:, :-1] + v[:, 1:]), axis=1)   # constant acceleration per segment

y = np.linspace(0, Y, N + 1)[1:-1]                   # start from the straight line
print(f"straight line : {travel_time(y)[0]:.4f} s")
eps, E = 1e-6, np.eye(N - 1)
for it in range(20000):                              # plain gradient descent
    grad = (travel_time(y + eps * E) - travel_time(y - eps * E)) / (2 * eps)
    y -= 0.02 * grad / max(1.0, np.abs(grad).max())
print(f"optimised path: {travel_time(y)[0]:.4f} s, lowest point y = {y.min():.3f}")
print(f"cycloid theory: {np.pi * np.sqrt(1 / g):.4f} s (radius 1)")
```

It starts at 1.1890 s on the straight line and settles at 1.0051 s, within 0.2% of the cycloid's 1.0030 s. The remaining gap is the cost of using 40 straight pieces. Things to change:

- Set `X = 2*np.pi`, so that B is far to the side. The optimiser now dips *below* B, to about $y = -2.49$, and climbs back up to it, in 1.4846 s. The exact cycloid (radius 1.275 m) bottoms out at -2.55 and takes 1.4804 s. Going downhill past your destination can be the fastest way to reach it.
- Increase `N` to 80 and watch the gap to theory shrink.
- Replace `2 * L / (v[:, :-1] + v[:, 1:])` by `L` and the same code finds the *shortest* path, the straight line. One functional per question.

92.6 The challenge

1. **Intuition.** A bead slides on a frictionless *helical* wire wound around a vertical axis, under gravity. Which of energy, vertical momentum and angular momentum about the axis are conserved? Now let the helix rotate about its axis at a constant rate. Which symmetry has been lost, and what conserved quantity takes the place of energy?
2. **Derivation.** For a particle in a central potential, $L = \tfrac{1}{2}m(\dot r^2 + r^2\dot\phi^2) - V(r)$, derive the Euler–Lagrange equations, use Noether to eliminate $\dot\phi$, and reduce the motion to a one-dimensional problem with an effective potential. Then derive the Beltrami identity $F - y'F_{y'} = \text{const}$ for an integrand $F(y, y')$ and show that it gives $y(1 + y'^2) = \text{const}$ for the brachistochrone.

3. **Implementation.** Write a leapfrog integrator for the Kepler problem, $H = \frac{1}{2}\|\mathbf{p}\|^2 - 1/\|\mathbf{q}\|$, starting on an orbit of eccentricity 0.6. Over 1,000 orbits, plot the energy error and the angular-momentum error for leapfrog and for RK4 at matched cost. Explain why leapfrog conserves angular momentum *exactly* (to rounding), while its energy merely oscillates.
4. **Open investigation.** Build the HMC sampler of §92.4 for a d-dimensional standard Gaussian. For $d = 10, 100, 1000$, find the step size h that gives an acceptance rate of about 65%, and test how h scales with d. Then repeat for a strongly correlated target such as Neal's funnel, where no single step size works everywhere. Can you design a position-dependent step or mass matrix that keeps the sampler exactly correct? (Riemannian-manifold HMC is one answer, and its cost is a live research question.)

92.7 Loose threads

The clock in orbit. The GPS mystery of Ch. 59 is a variational principle too. A free-falling clock follows the path through spacetime that makes its own elapsed time, the **proper time** τ, stationary. In a weak gravitational potential Φ and at speed $v \ll c$,

$$d\tau \approx \Big(1 + \frac{\Phi}{c^2} - \frac{v^2}{2c^2}\Big)dt, \qquad \text{so} \qquad -mc^2\int d\tau \approx \int \Big(\tfrac{1}{2}mv^2 - m\Phi\Big)dt - mc^2\int dt.$$

Stationary proper time is stationary Newtonian action, with $L = T - V$. The same formula also gives the GPS clock rates. With $GM = 3.986 \times 10^{14}\ \mathrm{m^3/s^2}$, Earth's radius 6,371 km and the orbital radius 26,561 km, the potential term makes the satellite clock gain 45.7 μs per day, and its orbital speed of 3.87 km/s makes it lose 7.2 μs per day. The net 38.5 µs per day is what the satellites correct for. As for *which* curvature matter controls: the Einstein equations $G_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}$ tie the **Ricci** part of the curvature, which governs how a small ball of test particles changes its *volume*, to energy and momentum. The remaining **Weyl** part is free: it carries tides and gravitational waves through empty space. "Why that one" has a Noether answer. Energy–momentum conservation, $\nabla_\mu T^{\mu\nu} = 0$, comes from the symmetry of physics under shifts of spacetime position. It can only be consistent if the geometric side is automatically divergence-free too. Lovelock proved (1971) that in four dimensions $G_{\mu\nu}$, plus a cosmological-constant term, is the *only* divergence-free symmetric tensor built from the metric and its first two derivatives. It is also what you get by applying Euler–Lagrange to the Einstein–Hilbert action $\int R\sqrt{-g}\,d^4x$. The curvature tensors themselves, and why they are tensors at all, are in Ch. 93.

Itô or Stratonovich? This is the physics answer to the mystery from Ch. 36. If the noise's correlation time is the *only* small timescale, the white-noise limit is Stratonovich (Wong–Zakai), natural for a circuit driven by thermal noise. Itô is right when the noise is non-anticipating by construction: a trader cannot react to tomorrow's price change. With *two* small timescales, such as a Brownian particle's inertia and its noise's correlation time, the limit depends on their ratio and can be neither convention (Kupferman, Pavliotis and Stuart, 2004); colloids diffusing near a wall show the resulting drift, matching the "anti-Itô" convention (Volpe et al., 2010).

? Do the tori survive?

A pendulum, a planet around a single sun, and any system with as many independent conserved quantities as degrees of freedom is **integrable**. Its phase space is foliated by invariant tori, and motion on each torus is perfectly regular. Add Jupiter's pull on the Earth and the extra conserved quantities are gone. Does the solar system then stay regular? Kolmogorov, Arnold and Moser proved that *most* tori survive a small enough perturbation, the ones whose frequencies are "sufficiently irrational", while the rest dissolve into thin layers of chaos. How irrational is sufficient, and what happens between the survivors, is the subject of Ch. 84.

? What is an electron, mathematically?

Noether turned each continuous symmetry into a conserved number. Wigner (1939) went much further. In quantum theory, the symmetry group of spacetime acts on the space of states by unitary operators, and an *elementary particle* is an **irreducible** piece of that action. Mass and spin, including the half-integer spin no classical object has, then come out as labels of the irreducible pieces, not as extra assumptions. Why the rotation group has a "double cover" that allows spin $\frac{1}{2}$, and how to classify irreducible representations at all, are the subject of Ch. 95.

Why a quantum verifier cannot be rewound. The zero-knowledge proofs of Ch. 82 copy a party's state and run it twice. Quantum states cannot be copied, and the reason is one line of linear algebra. Quantum evolution is a unitary, hence linear and inner-product-preserving, map U. Suppose some U cloned every state, $U(\psi \otimes \chi_0) = \psi \otimes \psi$, using a fixed blank state χ_0. Taking inner products of the results for two unit states ψ, φ gives $\langle\varphi,\psi\rangle = \langle\varphi,\psi\rangle^2$. So $\langle\varphi,\psi\rangle$ is 0 or 1: only states that are identical or perfectly distinguishable can be cloned, never a general superposition (Wootters–Zurek and Dieks, 1982). A classical memory can be copied freely, because a classical state is a point, not a vector in a space of superpositions. In quantum theory every state is such a vector, which is what makes copying impossible and measurement disturbing.

Two other threads go elsewhere. Pontryagin's costate in Ch. 51 is the momentum of a variational problem, and the Hamilton–Jacobi–Bellman equation of Ch. 52 is the Hamilton–Jacobi equation of mechanics in disguise. Replacing paths $q(t)$ by fields $u(x,t)$ turns Euler–Lagrange into the wave and Maxwell equations, whose spectra are the business of Ch. 90.

Further reading

- V. I. Arnold, *Mathematical Methods of Classical Mechanics*, 2nd ed., Springer, 1989. The classic geometric treatment: variational principles, symplectic manifolds, KAM theory. Demanding and unforgettable.
- D. E. Neuenschwander, *Emmy Noether's Wonderful Theorem*, 2nd ed., Johns Hopkins University Press, 2017. Noether's theorem from first principles, with history and many worked examples.
- E. Hairer, C. Lubich and G. Wanner, *Geometric Numerical Integration*, 2nd ed., Springer, 2006. The definitive book on symplectic integrators and backward error analysis.
- R. M. Neal, "MCMC using Hamiltonian dynamics", in *Handbook of Markov Chain Monte Carlo*, CRC Press, 2011. The clearest introduction to HMC, with practical advice.
- B. C. Hall, *Quantum Theory for Mathematicians*, Springer, 2013. Unbounded operators, the Stone–von Neumann theorem and the uncertainty principle, done properly.

Chapter 93 — Tensor Mathematics and Multilinear Algebra

> **Level 3 · Track J — Mathematics at the Frontier Builds on:** Ch. 10 (§10.7 SVD), Ch. 99, Ch. 59 · **Leads to:** Ch. 94, Ch. 95, Ch. 98 · **Related:** Ch. 26, Ch. 92, Ch. 62 **Reading time:** ~30 min

93.1 The hook

Multiply two 2×2 matrices the way you learned at school:

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} = \begin{pmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{pmatrix}.$$

That takes eight multiplications. For more than a century nobody seriously asked whether eight was necessary. In 1969 Volker Strassen, trying to *prove* that it was, found seven:

$$\begin{aligned} M_1 &= (a_{11} + a_{22})(b_{11} + b_{22}), & M_2 &= (a_{21} + a_{22})\,b_{11}, & M_3 &= a_{11}(b_{12} - b_{22}), \\ M_4 &= a_{22}(b_{21} - b_{11}), & M_5 &= (a_{11} + a_{12})\,b_{22}, & M_6 &= (a_{21} - a_{11})(b_{11} + b_{12}), \\ M_7 &= (a_{12} - a_{22})(b_{21} + b_{22}), \end{aligned}$$

and then $c_{11} = M_1 + M_4 - M_5 + M_7$, $c_{12} = M_3 + M_5$, $c_{21} = M_2 + M_4$, $c_{22} = M_1 - M_2 + M_3 + M_6$. Expand and check: every unwanted cross term cancels. Applied recursively to blocks, saving one multiplication in eight turns n^3 into $n^{2.807}$ (Ch. 26).

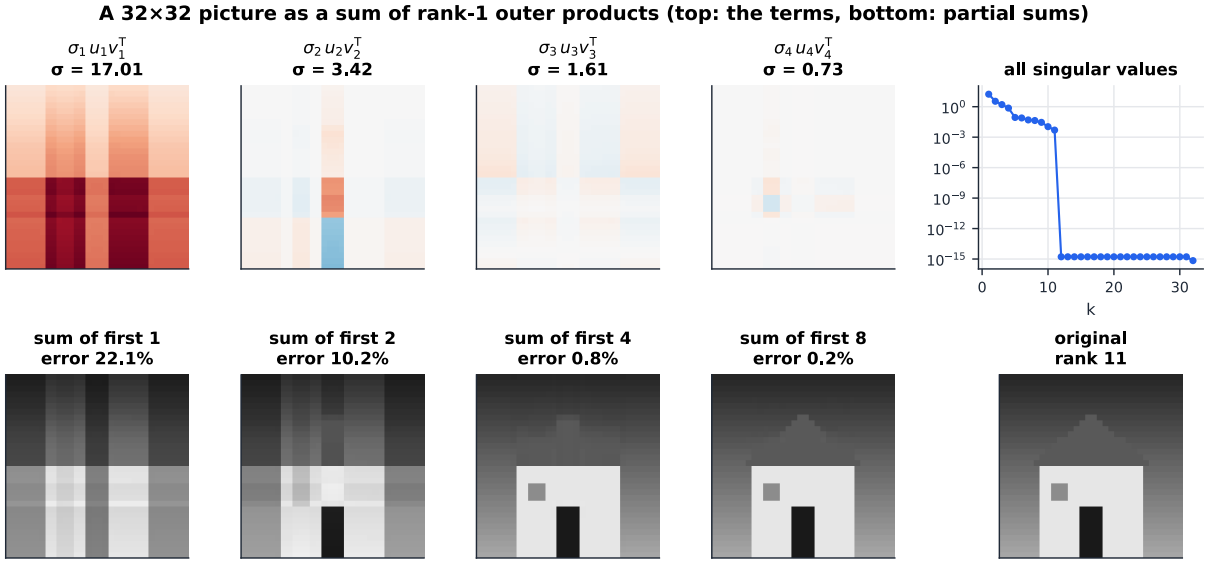
Two years later Winograd proved that seven is optimal for 2×2. For 3×3 matrices the best known algorithm (Laderman, 1976) uses 23 multiplications, the best lower bound is 19, and **nobody knows the true answer**. It is a question about 81 numbers that half a century of effort, and a reinforcement-learning system, has not settled.

The surprise this chapter builds on is that "how many multiplications?" is not really a question about algorithms. It asks for the **rank** of a certain three-dimensional array of zeros and ones. For a two-dimensional array, a matrix, rank is the most tractable number in linear algebra: Gaussian elimination computes it in a fraction of a second. Add a third index and computing rank becomes NP-hard, best approximations can fail to exist, and the answer can depend on whether you allow complex numbers.

> **Why does one extra index turn the easiest problem in linear algebra into one of the hardest?**

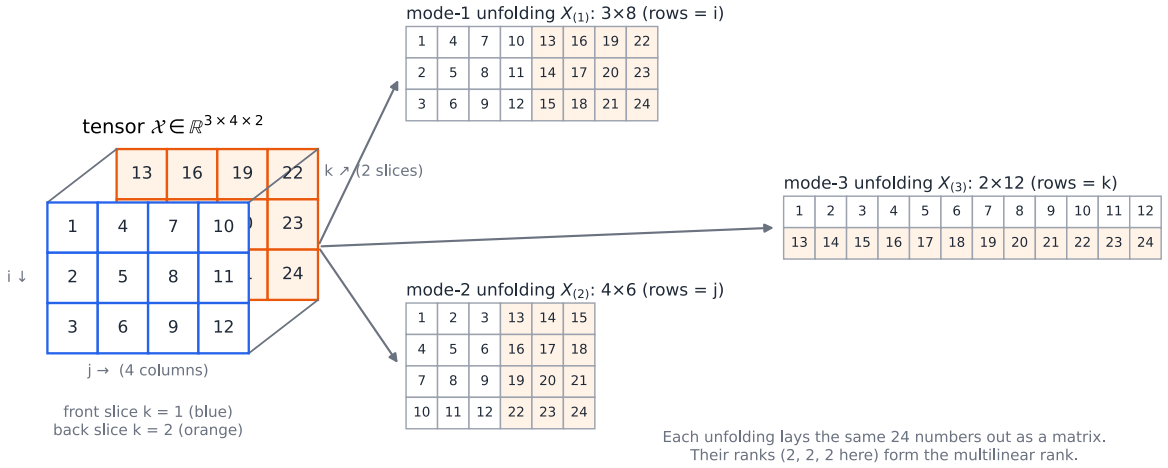
93.2 The big idea

A matrix is a sum of outer products. The singular value decomposition (§10.7) writes any matrix as $\mathbf{M} = \sum_k \sigma_k \mathbf{u}_k \mathbf{v}_k^{\mathsf{T}}$. Each term $\mathbf{u}\mathbf{v}^{\mathsf{T}}$ is the simplest possible matrix: every row is a multiple of the same row. The **rank** is the fewest such terms you need. Truncating the sum after r terms gives the best rank-r approximation (Eckart–Young), which is how a picture can be stored as a handful of vectors.



A 32×32 picture is a sum of outer products. Each term (top) is a "plaid" of one column pattern times one row pattern. Four of them reproduce the house to within 0.8%, and eleven reproduce it exactly: its rank is 11.

A tensor is a cube of numbers, and the same idea applies. An array $\mathcal{X}$ with entries x_{ijk} is a third-order tensor. The simplest cube is an outer product of three vectors, $(\mathbf{a} \otimes \mathbf{b} \otimes \mathbf{c})_{ijk} = a_i b_j c_k$. The **rank** of $\mathcal{X}$ is again the fewest such terms that sum to it. You can also slice the cube and lay the slices side by side, in three different ways. These **unfoldings** are ordinary matrices, and matrix tools apply to them.



A $3 \times 4 \times 2$ tensor and its three unfoldings. Each unfolding keeps one index as the row index and merges the other two into the column index.

The catch is that no unfolding tells you the tensor's rank. Each unfolding is a different "shadow" of the cube, and the cube can be far more complicated than any of its shadows.

An analogy for the tensor product. Suppose you want to multiply a vector by a vector and get *something*, demanding nothing except that the product be linear in each factor. The tensor product $\mathbf{v} \otimes \mathbf{w}$ is the most honest answer: it remembers that bilinearity and nothing else. Every other bilinear product (the dot product, the cross product, matrix multiplication, the determinant of a pair of columns) is obtained from it by one linear map. It is the universal "blank form" on which every bilinear operation is written.

93.3 The mathematics

Multilinear maps and the tensor product

Let V and W be vector spaces over a field K with bases $e_1, \dots, e_m$ and $f_1, \dots, f_n$. A map $B : V \times W \to X$ is **bilinear** if it is linear in each argument when the other is held fixed. Bilinear maps are *not* linear maps on $V \times W$: for example $B(2v, 2w) = 4B(v, w)$, not $2B(v, w)$. Ch. 99 asked for a single space that converts them into linear maps. Here it is.

Tensor product (concrete form)

$V \otimes W$ is the vector space with basis the mn symbols $e_i \otimes f_j$. For $v = \sum_i v_i e_i$ and $w = \sum_j w_j f_j$, define $v \otimes w = \sum_{i,j} v_i w_j \, e_i \otimes f_j$.

The map $(v, w) \mapsto v \otimes w$ is bilinear by construction. In coordinates an element of $V \otimes W$ is an $m \times n$ array, and $v \otimes w$ is the outer product $\mathbf{v}\mathbf{w}^{\mathsf{T}}$.

Universal property

For every bilinear map $B : V \times W \to X$ there is exactly one *linear* map $\tilde{B} : V \otimes W \to X$ with $\tilde{B}(v \otimes w) = B(v, w)$ for all v, w.

Proof. Existence: a linear map may be prescribed freely on a basis, so define $\tilde{B}(e_i \otimes f_j) = B(e_i, f_j)$ and extend linearly. Then, by bilinearity of B,

$$\tilde{B}(v \otimes w) = \sum_{i,j} v_i w_j \, \tilde{B}(e_i \otimes f_j) = \sum_{i,j} v_i w_j \, B(e_i, f_j) = B\Big(\sum_i v_i e_i, \sum_j w_j f_j\Big) = B(v, w).$$

Uniqueness: any such $\tilde{B}$ is forced to take the value $B(e_i, f_j)$ on each $e_i \otimes f_j$, and these elements form a basis. ■

The theorem also explains why the construction does not really depend on the bases. Any two spaces with the universal property are identified by a unique isomorphism that respects $\otimes$, by the standard argument of Ch. 62. Two consequences matter constantly.

Most tensors are not products. In coordinates $v \otimes w$ is a rank-1 matrix, so a general element of $V \otimes W$ is a *sum* $\sum_k v_k \otimes w_k$. The fewest terms needed is the matrix rank. In quantum mechanics, where the state space of two systems is $\mathcal{H}_1 \otimes \mathcal{H}_2$, a state that is not a single product is called **entangled**.

Linear maps are tensors, and upper and lower indices are vectors and covectors. The map $\varphi \otimes w \mapsto \big(v \mapsto \varphi(v)\, w\big)$ is an isomorphism $V^* \otimes W \cong \mathrm{Hom}(V, W)$, so a matrix *is* a tensor with one covector slot and one vector slot. Now change basis in V by $e'_a = \sum_i A^i{}_a e_i$. Components of a vector $v = \sum_i v^i e_i = \sum_a v'^a e'_a$ must transform by the *inverse* matrix, $v'^a = \sum_i (A^{-1})^a{}_i v^i$. Components of a covector, $\varphi_a = \varphi(e'_a) = \sum_i A^i{}_a \varphi_i$, transform by A itself. Physicists record the difference by writing vector indices up and covector indices down. An element of $V^{\otimes r} \otimes (V^*)^{\otimes s}$ has r upper and s lower indices, and its components pick up one factor of A^{-1} per upper index and one factor of A per lower index. On a manifold, with A the Jacobian of a coordinate change, this is exactly the "one Jacobian factor per index" rule for the metric in Ch. 59. A tensor in the physicist's sense is an element of a tensor product, and the transformation law is what that looks like in coordinates.

Rank, and matrix multiplication as a tensor

For three spaces, $U \otimes V \otimes W$ has basis $e_i \otimes f_j \otimes g_k$, its elements are $I \times J \times K$ arrays, and the **rank** of $\mathcal{T}$ is the least R with

$$\mathcal{T} = \sum_{r=1}^{R} \mathbf{u}_r \otimes \mathbf{v}_r \otimes \mathbf{w}_r .$$

This is the **CP decomposition** (canonical polyadic, also called CANDECOMP/PARAFAC). Write $\operatorname{vec}\mathbf{A}$ for the n^2 entries of $\mathbf{A}$ as one vector. Matrix multiplication $\mathbf{C} = \mathbf{A}\mathbf{B}$ is a bilinear map $K^{n^2} \times K^{n^2} \to K^{n^2}$, and by the universal property it corresponds to a tensor $\mathcal{M}_n$ with $n^2 \times n^2 \times n^2$ entries. Its entry is 1 at position $\big((a,b),(b,c),(a,c)\big)$, because $a_{ab}b_{bc}$ contributes to c_{ac}, and 0 elsewhere.

> **Multiplications = rank**
>
> $n \times n$ matrices can be multiplied by a bilinear algorithm that uses R multiplications if and only if $\operatorname{rank}\mathcal{M}_n \le R$.

Proof. A bilinear algorithm forms R products $m_r = (\mathbf{u}_r \cdot \operatorname{vec}\mathbf{A})(\mathbf{v}_r \cdot \operatorname{vec}\mathbf{B})$ and outputs $\operatorname{vec}\mathbf{C} = \sum_r m_r \mathbf{w}_r$. Read off the coefficient of $a_\alpha b_\beta$ in the γ-th output. The algorithm computes $c_\gamma = \sum_{\alpha,\beta} (\mathcal{M}_n)_{\alpha\beta\gamma} a_\alpha b_\beta$ exactly when

$$(\mathcal{M}_n)_{\alpha\beta\gamma} = \sum_{r=1}^{R} (\mathbf{u}_r)_\alpha (\mathbf{v}_r)_\beta (\mathbf{w}_r)_\gamma \quad \text{for all } \alpha,\beta,\gamma,$$

that is, when $\mathcal{M}_n = \sum_r \mathbf{u}_r \otimes \mathbf{v}_r \otimes \mathbf{w}_r$. Conversely, any such decomposition *is* an algorithm, with the vectors as its coefficients. ∎

The schoolbook method is the decomposition of $\mathcal{M}_2$ into its eight nonzero entries. Strassen's is a decomposition into seven rank-1 terms. For instance M_1 is $\mathbf{u}_1 = \mathbf{v}_1 = (1,0,0,1)$ (that is, $a_{11}+a_{22}$ and $b_{11}+b_{22}$), and since M_1 appears in c_{11} and c_{22}, $\mathbf{w}_1 = (1,0,0,1)$. Winograd's theorem says $\operatorname{rank}\mathcal{M}_2 = 7$. How such a finite rank becomes the asymptotic exponent $\omega \le \log_n R$ is proved in Ch. 98.

Where matrix intuition breaks

For matrices, the set of matrices of rank at most r is closed: a limit of rank-r matrices has rank at most r, because rank is detected by the vanishing of $(r+1)\times(r+1)$ minors, which are continuous functions. For tensors this fails.

> **A tensor that rank-2 tensors approximate arbitrarily well**
>
> Let $\mathbf{a},\mathbf{b}$ be linearly independent and $\mathcal{W} = \mathbf{a}\otimes\mathbf{a}\otimes\mathbf{b} + \mathbf{a}\otimes\mathbf{b}\otimes\mathbf{a} + \mathbf{b}\otimes\mathbf{a}\otimes\mathbf{a}$. Then $\mathcal{W}$ has rank 3, yet it is the limit of the rank-2 tensors
>
> $$\mathcal{T}_\varepsilon = \frac{1}{\varepsilon}\Big[(\mathbf{a}+\varepsilon\mathbf{b})^{\otimes 3} - \mathbf{a}^{\otimes 3}\Big] \qquad \text{as } \varepsilon \to 0.$$

Proof of the limit. Expand $(\mathbf{a}+\varepsilon\mathbf{b})^{\otimes 3}$ by multilinearity into eight terms. The ε^0 term is $\mathbf{a}^{\otimes 3}$, which cancels. The three ε^1 terms, divided by ε, are exactly $\mathcal{W}$. What remains is $\varepsilon(\mathbf{a}\otimes\mathbf{b}\otimes\mathbf{b} + \mathbf{b}\otimes\mathbf{a}\otimes\mathbf{b} + \mathbf{b}\otimes\mathbf{b}\otimes\mathbf{a}) + \varepsilon^2\mathbf{b}^{\otimes 3} \to 0$. With orthonormal $\mathbf{a},\mathbf{b}$, $\|\mathcal{T}_\varepsilon - \mathcal{W}\| = \varepsilon\sqrt{3+\varepsilon^2}$. That $\mathcal{W}$ itself has rank 3, not 2, takes a short but careful argument; see de Silva and Lim (2008). ∎

Look at the price of the approximation. The two rank-1 terms have norm about $1/\varepsilon$ each, and they nearly cancel. This is the **degeneracy** that makes CP fitting so treacherous. If you ask for the best rank-2 approximation of $\mathcal{W}$, there is none. The error can be made as small as you like, but never zero, and the factors blow up as you try. The least r such that $\mathcal{T}$ is a limit of rank-r tensors is called its **border rank**. The record exponents for matrix multiplication since 1979 have all come from border rank.

The other differences from matrices are just as stark:

property	matrices	tensors of order ≥ 3
computing rank	Gaussian elimination, polynomial time	NP-hard (Håstad 1990)
best rank-r approximation	always exists (Eckart–Young)	may not exist
rank over $\mathbb{R}$ vs $\mathbb{C}$	the same	can differ
rank of a random array	full, $\min(m,n)$, with probability 1	several "typical" ranks may each have positive probability
maximum rank	$\min(m,n)$	can exceed every side length

For instance, a random real $2\times2\times2$ tensor with Gaussian entries has rank 2 with probability $\pi/4 \approx 0.785$ and rank 3 otherwise. Over $\mathbb{C}$ it has rank 2 with probability 1. The dividing line is the sign of a single polynomial in the eight entries, Cayley's hyperdeterminant. Its exact role is a matter for Ch. 94.

> **? Which tensors have border rank r?**
>
> The tensors of border rank at most r form a closed set cut out by polynomial equations: it is an algebraic variety, called a **secant variety**. Its equations certify lower bounds. If such a polynomial does not vanish on $\mathcal{M}_n$, then $\mathcal{M}_n$ needs more than r terms, even approximately. Finding those polynomials is how Landsberg proved in 2006 that the border rank of $\mathcal{M}_2$ is also 7. For general n, nobody knows the equations, and proving that $\mathcal{M}_n$ has border rank much larger than n^2 is wide open. The language of varieties and ideals that this program needs starts in Ch. 94.

Answer from Chapter 59: the Riemann curvature tensor

Ch. 59 asked how an ant can compute the Gaussian curvature $K = \kappa_1\kappa_2$ from the metric alone. Here is the tensor answer. Carry a tangent vector $\mathbf{v}$ by parallel transport around a tiny loop spanned by two coordinate directions μ,ν, with coordinate area δA. It returns changed by $\delta v^\rho \approx -R^\rho{}_{\sigma\mu\nu} v^\sigma \, \delta A$ (the sign depends on the orientation convention), where

$$R^{\rho}{}_{\sigma\mu\nu} = \partial_\mu \Gamma^{\rho}_{\nu\sigma} - \partial_\nu \Gamma^{\rho}_{\mu\sigma} + \sum_{\lambda} \left(\Gamma^{\rho}_{\mu\lambda}\Gamma^{\lambda}_{\nu\sigma} - \Gamma^{\rho}_{\nu\lambda}\Gamma^{\lambda}_{\mu\sigma} \right).$$

Three facts carry the argument.

1. **It is a tensor.** The Christoffel symbols Γ are *not* tensors: they can be made zero at any one point by a clever choice of coordinates. Yet the combination above transforms with one Jacobian factor per index. It can be written without coordinates as $R(X,Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z$, a failure of covariant derivatives to commute. So if it vanishes in one chart it vanishes in all, and it is a property of the geometry, not of the coordinates.
2. **It is built from the metric alone.** The Γ's come from first derivatives of g_{ij}, so R needs nothing but g and its first two derivatives. An insider can compute it.
3. **On a surface it is one number.** With all indices lowered, $R_{\rho\sigma\mu\nu}$ is antisymmetric in (ρ,σ) and in (μ,ν), symmetric under swapping the two pairs, and satisfies a cyclic identity. In n dimensions these symmetries leave $n^2(n^2-1)/12$ independent components, which is 1, 6 and 20 for $n = 2, 3, 4$. On a surface the single component is $R_{1212} = K\,(g_{11}g_{22} - g_{12}^2)$, so

$$K = \frac{R_{1212}}{\det g},$$

which depends only on g. That is the Theorema Egregium, and the reason behind Gauss's "page of brute calculation". The six components in three dimensions and the twenty in four are why one number no longer suffices there, and why Einstein's equations constrain only part of the curvature (Ch. 92).

93.4 The algorithm

Alternating least squares (CP-ALS). Fitting $\mathcal{T} \approx \sum_{r} \mathbf{u}_r \otimes \mathbf{v}_r \otimes \mathbf{w}_r$ is a nonconvex problem in all the factors at once. If you freeze $\mathbf{V} = [\mathbf{v}_1 \cdots \mathbf{v}_R]$ and $\mathbf{W}$, however, the model is *linear* in $\mathbf{U}$, and minimising the squared error over $\mathbf{U}$ is an ordinary least-squares problem (§9.11). Its normal equations have a pleasantly small matrix:

$$\mathbf{U}\left[(\mathbf{V}^{\mathsf T}\mathbf{V}) * (\mathbf{W}^{\mathsf T}\mathbf{W})\right] = \mathbf{G}, \qquad G_{ir} = \sum_{j,k} T_{ijk} V_{jr} W_{kr},$$

where $*$ is the entrywise product of two $R \times R$ matrices. (The left side is $\mathbf{U}$ times the Gram matrix of the columns of the Khatri–Rao product of $\mathbf{W}$ and $\mathbf{V}$.) Cycle through $\mathbf{U}$, $\mathbf{V}$, $\mathbf{W}$ and repeat:

```
import numpy as np

def cp_als(T, R, iters=500, lam=0.0, seed=0):
    rng = np.random.default_rng(seed)
    U, V, W = (rng.standard_normal((s, R)) for s in T.shape)
    reg = lam * np.eye(R)
    for _ in range(iters):
        U = np.einsum('ijk,jr,kr->ir', T, V, W) @ np.linalg.inv((V.T @ V) * (W.T @ W) + reg)
        V = np.einsum('ijk,ir,kr->jr', T, U, W) @ np.linalg.inv((U.T @ U) * (W.T @ W) + reg)
        W = np.einsum('ijk,ir,jr->kr', T, U, V) @ np.linalg.inv((U.T @ U) * (V.T @ V) + reg)
    return U, V, W
```

Each sweep costs $O(IJKR)$ for the three contractions plus $O(R^3)$ for the small solves. The error never increases from one update to the next, because each update is an exact minimisation over its block.

The Tucker form and the HOSVD. A different decomposition gives up the minimal number of terms in exchange for stability. Take the SVD of each unfolding, keep the leading r_1, r_2, r_3 left singular vectors as $\mathbf{Q}_1, \mathbf{Q}_2, \mathbf{Q}_3$, and project: $\mathcal{G} = \mathcal{T} \times_1 \mathbf{Q}_1^{\mathsf T} \times_2 \mathbf{Q}_2^{\mathsf T} \times_3 \mathbf{Q}_3^{\mathsf T}$, so that $\mathcal{T} \approx \mathcal{G} \times_1 \mathbf{Q}_1 \times_2 \mathbf{Q}_2 \times_3 \mathbf{Q}_3$. This is the **higher-order SVD** (De Lathauwer, De Moor and Vandewalle, 2000). It is computed by three ordinary SVDs, is never degenerate, and its error is within a factor $\sqrt{3}$ of the best possible for those sizes. The triple (r_1, r_2, r_3) of unfolding ranks is the **multilinear rank**, computable in polynomial time, unlike the CP rank. In the figure of §93.2 it is $(2, 2, 2)$. The price is a dense core $\mathcal{G}$ of size $r_1 r_2 r_3$, which grows exponentially with the number of indices. Tensor trains, also called matrix product states, are the modern fix for that.

Failure cases.

- *Wrong R.* Nothing in ALS tells you the rank. Too small and the fit stalls; too large and the factors become non-unique and can drift.
- *Degeneracy.* When the best rank-R approximation does not exist, as for $\mathcal{W}$, ALS drives pairs of factors to infinity with opposite signs while the error creeps down. The small ridge term `lam` in the code is the usual cure: it penalises large factors and keeps the problem well posed.
- *Swamps.* ALS can spend thousands of sweeps on a plateau where the error barely moves, then suddenly converge. Line searches, damped Gauss–Newton (Levenberg–Marquardt) and random restarts all help, but no method is guaranteed to work.
- *Uniqueness, the good news.* Unlike a matrix factorisation, which can always be rotated ($\mathbf{U}\mathbf{V}^{\mathsf T} = (\mathbf{U}\mathbf{Q})(\mathbf{V}\mathbf{Q})^{\mathsf T}$), a CP decomposition is usually *unique* up to scaling and reordering of its terms. Kruskal's theorem (1977) gives a sufficient condition. This is why chemists can separate the spectra of overlapping fluorescent compounds from a three-way array, and why psychometricians invented PARAFAC in the first place.

93.5 The experiment

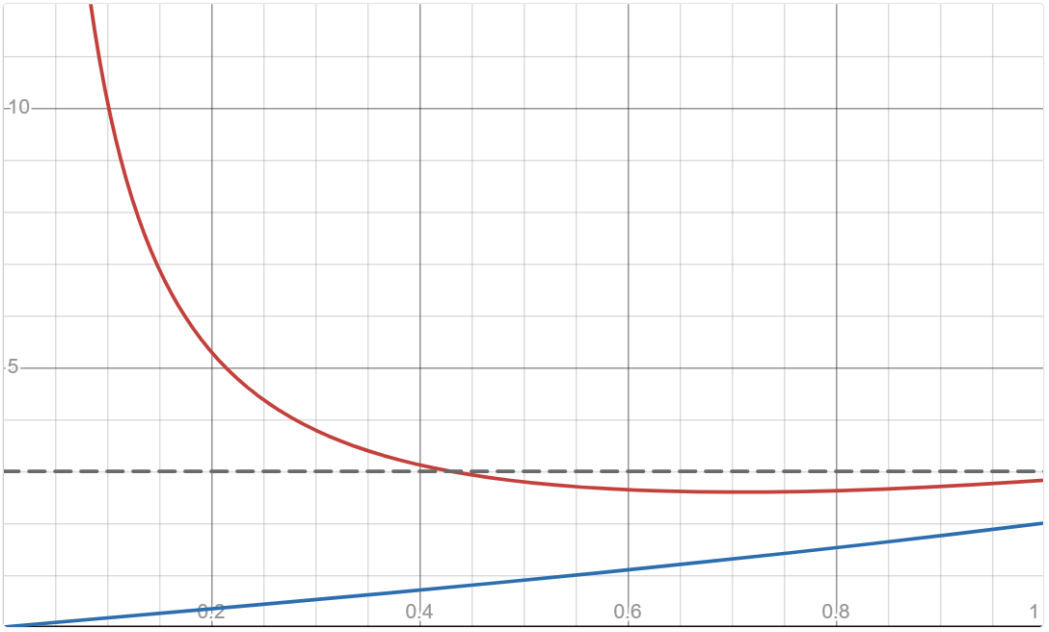
Rediscover Strassen. This script builds the $4 \times 4 \times 4$ tensor $\mathcal{M}_2$ and asks CP-ALS for decompositions of rank 8, 7 and 6, from ten random starting points each. The regulariser starts at 10^{-3} and fades out.

```
import numpy as np

n = 2                                          # multiply n x n matrices
T = np.zeros((n * n, n * n, n * n))            # matrix multiplication tensor
for a in range(n):
    for b in range(n):
        for c in range(n):
            T[a * n + b, b * n + c, a * n + c] = 1   # c_ac += a_ab * b_bc

def cp_als(T, R, iters=3000, lam=1e-3, seed=0):
    rng = np.random.default_rng(seed)
    U, V, W = (rng.standard_normal((s, R)) for s in T.shape)
    for _ in range(iters):                     # each update is a linear least-squares solve
        U = np.einsum('ijk,jr,kr->ir', T, V, W) @ np.linalg.inv((V.T @ V) * (W.T @ W) + lam * np.eye(R))
        V = np.einsum('ijk,ir,kr->jr', T, U, W) @ np.linalg.inv((U.T @ U) * (W.T @ W) + lam * np.eye(R))
        W = np.einsum('ijk,ir,jr->kr', T, U, V) @ np.linalg.inv((U.T @ U) * (V.T @ V) + lam * np.eye(R))
        lam *= 0.995                           # fade out the regulariser
    approx = np.einsum('ir,jr,kr->ijk', U, V, W)
    return np.linalg.norm(T - approx)

for R in (8, 7, 6):
    errs = [cp_als(T, R, seed=s) for s in range(10)]
    print(f"rank {R}: best residual over 10 starts = {min(errs):.2e}, "
          f"successes = {sum(e < 1e-3 for e in errs)}/10")
```

On a typical run, rank 8 and rank 7 succeed from all ten starts, with residuals around 3×10^{-9}. Rank 6 fails every time, and the residual sticks at about 1.0. By de Groote's theorem (1978), every rank-7 decomposition is Strassen's algorithm in disguise, transformed by the symmetries of $\mathcal{M}_2$, so the script keeps rediscovering Strassen with messier real coefficients. Things to change:

- Print the factor matrices of a rank-7 solution and compute $\mathbf{U}^{\mathsf{T}}\mathbf{U}$. The columns are not orthogonal, unlike the SVD's. Can you rescale them to recover entries near $0, \pm 1$?
- Set `n = 3` and try `R = 23`, Laderman's count. Plain ALS usually stalls with a residual of a few percent. Add restarts, longer runs or a slower fade of `lam`, and see whether you can reach Laderman. Nobody knows whether `R = 22` is possible.
- Set `lam = 0` from the start and watch some rank-7 runs blow up or crawl: that is degeneracy and swamps in action.

Border rank in one picture. For $\mathcal{T}_\varepsilon$ in §93.3 with orthonormal $\mathbf{a}, \mathbf{b}$, the blue curve is the approximation error $\varepsilon\sqrt{3+\varepsilon^2}$ and the red curve is the size $(1+\varepsilon^2)^{3/2}/\varepsilon$ of each rank-1 term, with x playing the role of ε.



As $\varepsilon \to 0$ the error (blue) vanishes while the two rank-1 terms (red) blow up. The dashed line at 3 is the total size of the honest rank-3 decomposition: three unit terms.

Change the blue curve to $y = x^2$ to see what a second-order error would look like. Then try to invent a family whose error falls like ε^2 while its terms grow only like $1/\varepsilon$. (Hint: combine $(\mathbf{a} + \varepsilon\mathbf{b})^{\otimes 3}$ and $(\mathbf{a} - \varepsilon\mathbf{b})^{\otimes 3}$.)

93.6 The challenge

1. **Intuition.** Explain without computation why every unfolding of a rank-R tensor has matrix rank at most R. Then explain why the converse fails: the three unfoldings of $\mathcal{W}$ all have rank 2, yet $\mathcal{W}$ has rank 3.
2. **Derivation.** Prove that $V^* \otimes W \to \operatorname{Hom}(V, W)$, $\varphi \otimes w \mapsto (v \mapsto \varphi(v)w)$, is an isomorphism for finite-dimensional spaces, and that under it the rank of a tensor equals the rank of the linear map. Then show that the contraction $V^* \otimes V \to K$, $\varphi \otimes v \mapsto \varphi(v)$, corresponds to the trace, which gives a basis-free proof that the trace is basis-independent.
3. **Implementation.** Write the HOSVD of §93.4 and apply it to a $64 \times 64 \times 3$ colour image, or to a stack of 30 frames of a slowly changing synthetic video. Compare its storage and error with a separate SVD of each frame and with CP-ALS at equal storage. Which wins, and why?

4. **Open investigation.** For random real $2 \times 2 \times 2$ tensors the probability of rank 2 is $\pi/4$. Estimate by simulation the probabilities of each typical rank for random real $3 \times 3 \times 2$, $2 \times 2 \times 3$ and $3 \times 3 \times 3$ tensors, by deciding rank with a numerical test of your choice. How reliable is your test near the boundary between ranks, and can you find a closed form for any of these probabilities?

93.7 Loose threads

Two more mysteries lead beyond this chapter.

? What do the symmetries of $V^{\otimes n}$ decompose into?

The space $V \otimes V$ splits into symmetric tensors ($T_{ij} = T_{ji}$) and antisymmetric ones ($T_{ij} = -T_{ji}$). The determinant forms of Ch. 99 live in the fully antisymmetric part of $V^{\otimes n}$. For three or more factors these two pieces no longer account for everything, and the leftover parts are indexed by **Young diagrams**. Schur–Weyl duality says that the general linear group acting on each factor and the permutation group shuffling the factors split $V^{\otimes n}$ into irreducible pieces in perfectly matched pairs. Why the pieces are counted by partitions of n, and why those same pieces label the particle states of Ch. 92, is the subject of Ch. 95.

? Twenty-two multiplications?

The rank of $\mathcal{M}_3$ lies between 19 and 23, For 4×4 the classic count is Strassen applied twice, 49; AlphaTensor found 47 in arithmetic modulo 2, and the true rank is unknown. A single tensor of $9 \times 9 \times 9 = 729$ zeros and ones has resisted fifty years of human ingenuity and machine search. Settling its rank is an **open problem**. How finite gadgets like this feed the exponent ω, and how researchers attack such problems, is in Ch. 98.

The tensors of physics have also not been exhausted. The stress tensor of elasticity, the electromagnetic field tensor $F_{\mu\nu}$ and the curvature tensors above are all elements of tensor products of a tangent space and its dual. In each case, the symmetries of the tensor (symmetric, antisymmetric, the curvature identities) decide how many independent numbers the physics contains. Machine learning uses "tensor" more loosely, for any multi-index array, but its successes with low-rank structure are real: CP and tensor-train factorisations compress the weight arrays of neural networks, and attention layers are bilinear maps in disguise. Whether *generic* data have low tensor rank, and why the hard cases of the table in §93.3 rarely bite in practice, is an active research area with few theorems.

Further reading

- T. G. Kolda and B. W. Bader, "Tensor decompositions and applications", *SIAM Review* 51 (2009), 455–500. The standard survey: notation, unfoldings, CP, Tucker and algorithms. Start here.
- J. M. Landsberg, *Tensors: Geometry and Applications*, AMS, 2012. Rank, border rank and secant varieties, with matrix multiplication as the running example.
- V. de Silva and L.-H. Lim, "Tensor rank and the ill-posedness of the best low-rank approximation problem", *SIAM J. Matrix Anal. Appl.* 30 (2008). The theorem behind $\mathcal{W}$.
- C. J. Hillar and L.-H. Lim, "Most tensor problems are NP-hard", *Journal of the ACM* 60 (2013).
- A. Fawzi et al., "Discovering faster matrix multiplication algorithms with reinforcement learning", *Nature* 610 (2022). AlphaTensor.
- P. Bürgisser, M. Clausen and M. A. Shokrollahi, *Algebraic Complexity Theory*, Springer, 1997. The definitive treatment of bilinear complexity.

Chapter 94 — Algebraic Geometry: Solving Equations Through Shapes

> **Level 3 · Track J — Mathematics at the Frontier Builds on:** Ch. 1 (§1.4 polynomials), Ch. 56 (rings and fields), Ch. 18 · **Leads to:** Ch. 95, Ch. 78, Ch. 98 · **Related:** Ch. 57, Ch. 75, Ch. 82, Ch. 93 **Reading time:** ~30 min

94.1 The hook

A planar robot arm has its shoulder bolted to the origin, an upper arm of length 2 and a forearm of length 1. You want its hand at the point $(2, 1)$. Where must the elbow go?

The elbow (u, v) is at distance 2 from the shoulder and distance 1 from the hand:

$$u^2 + v^2 = 4, \qquad (u-2)^2 + (v-1)^2 = 1.$$

Subtract the two equations and the squares cancel, leaving the line $2u + v = 4$. Substitute $v = 4 - 2u$ back into the first equation to get $5u^2 - 16u + 12 = 0$, so $u = 2$ or $u = 1.2$. The two elbows are $(2, 0)$ ("elbow down") and $(1.2, 1.6)$ ("elbow up"). Every robotics engineer knows this answer.

Now look at it as a mathematician would. Each equation has degree 2: each describes a **conic**. There is a famous theorem, stated by Newton and Bézout and proved properly much later, that two curves of degrees d and e meet in **exactly** de **points**. For two conics that means four. An ellipse and a hyperbola really do meet in four points. But two different circles *never* meet in more than two. You can slide them about as you like and you will never find a third intersection.

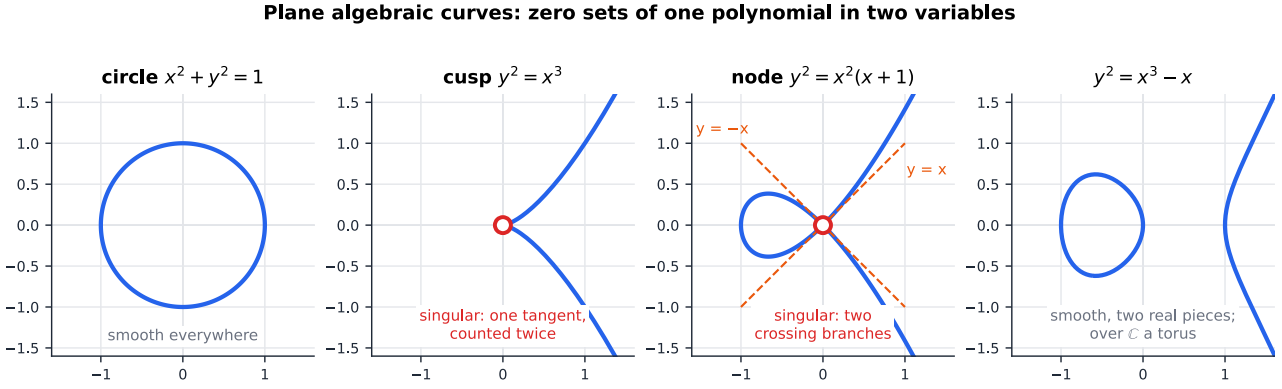
So either Bézout's theorem is false, or two intersection points of every pair of circles are hiding somewhere.

> **Where are the two missing intersection points of two circles?**

Answering it means changing what "point" means. Allow complex coordinates and add points at infinity, and polynomial equations stop behaving erratically and start obeying exact counting laws. That change of viewpoint, from equations to the *shapes* they define and back, is algebraic geometry. It now underlies robot kinematics, computer-aided design, elliptic-curve cryptography and the proof of Fermat's Last Theorem.

94.2 The big idea

Idea 1: an equation is a shape. The set of points where a polynomial vanishes is a geometric object: $x^2 + y^2 - 1 = 0$ is a circle, $y^2 - x^3 = 0$ is a curve with a sharp point. Such zero sets are called **varieties**. Their geometry, such as how many pieces they have, where they are pointed or self-crossing, and how they meet, tells you about the equations. In return, algebra about the polynomials, such as factoring, substituting and eliminating, tells you about the shapes.



Four plane curves. The cusp and the node have singular points, circled in red, where the curve fails to look like a smooth line. The last curve is an elliptic curve. Its real points form two pieces, but its complex points, together with one point at infinity, form a single torus.

Idea 2: the shape doesn't care which equations you wrote. The circle is described by $x^2 + y^2 - 1 = 0$, but equally by $(x^2 + y^2 - 1)(x^2 + 1) = 0$, or by any system of equations whose consequences are the same. What actually determines the shape is the set of *all* polynomials that follow from the equations by adding and multiplying. That set is an **ideal**, and ideals are the algebraic shadows of shapes.

Idea 3: functions on the shape form a ring, and the ring knows everything. On the circle, x^2 and $1 - y^2$ are the same function, because they agree at every point. Identify polynomials that agree on the shape and you get the **coordinate ring**: the algebra of polynomial functions on the variety. Two shapes are "the same" exactly when their coordinate rings are the same ring. Geometry turns into ring theory, and the ring theory of Ch. 56 gets a picture.

An analogy: a variety is a building, and its coordinate ring is the complete set of measurements you could take inside it. Two buildings that give identical answers to every possible measurement are, for every practical purpose, the same building.

94.3 The mathematics

Fix a field k, usually $\mathbb{R}$ or $\mathbb{C}$, and the polynomial ring $k[x_1, \ldots, x_n]$.

Affine variety and ideal

For polynomials $f_1, \dots, f_m$, the **affine variety** they define is

$$V(f_1, \dots, f_m) = \{\mathbf{a} \in k^n : f_1(\mathbf{a}) = \cdots = f_m(\mathbf{a}) = 0\}.$$

The **ideal** they generate is $\langle f_1, \dots, f_m \rangle = \{h_1 f_1 + \cdots + h_m f_m : h_i \in k[x_1, \dots, x_n]\}$.

Every element of the ideal vanishes wherever all the f_i do. So $V(f_1, \dots, f_m)$ depends only on the ideal, and any other generators of the same ideal describe the same set. The robot arm's trick of subtracting equations was a change of generators: $\langle f_1, f_2 \rangle = \langle f_1,\ f_1 - f_2 \rangle$. Two set operations have one-line algebraic counterparts: $V(f) \cup V(g) = V(fg)$ (a product vanishes when a factor does) and $V(f) \cap V(g) = V(f, g)$.

Going the other way, from a set $X \subseteq k^n$ to polynomials, gives the ideal $I(X)$ of all polynomials vanishing on X. Two classical theorems make the dictionary work.

- **Hilbert's basis theorem (1890).** Every ideal of $k[x_1, \dots, x_n]$ is generated by finitely many polynomials. So every variety, even one described by infinitely many equations, is cut out by finitely many.
- **Hilbert's Nullstellensatz (1893).** Over $\mathbb{C}$, or any algebraically closed field: $V(J) = \varnothing$ only if $1 \in J$, and in general $I(V(J))$ is the **radical** $\sqrt{J} = \{f : f^r \in J \text{ for some } r\}$.

The Nullstellensatz fails over $\mathbb{R}$: $x^2 + y^2 + 1$ has no real zeros, yet $1 \notin \langle x^2 + y^2 + 1 \rangle$. This is the first sign that the reals are the wrong place to count solutions, just as the real numbers were the wrong place to count roots of $x^2 + 1$ (Ch. 1).

Smooth and singular points

A point $\mathbf{a}$ of the plane curve $V(f)$ is **singular** if $\nabla f(\mathbf{a}) = \mathbf{0}$. At a non-singular point the implicit function theorem (Ch. 6) makes the curve locally the graph of a smooth function. At a singular point it can fail.

- **Cusp**, $f = y^2 - x^3$. Here $\nabla f = (-3x^2, 2y)$, which vanishes only at the origin, and the origin lies on the curve. Near it, the lowest-order part of f is y^2: one tangent line $y = 0$, counted twice. The curve arrives along it from both sides and turns back.
- **Node**, $f = y^2 - x^2(x+1) = y^2 - x^2 - x^3$. Here $\nabla f = (-2x - 3x^2,\ 2y)$, which vanishes at $(0, 0)$ and at $(-\tfrac{2}{3}, 0)$. Only the origin lies on the curve. The lowest-order part, $y^2 - x^2 = (y - x)(y + x)$, gives two distinct tangent lines $y = \pm x$: two branches cross.

For the family $y^2 = x^3 + ax + b$, the curve is singular exactly when the cubic $x^3 + ax + b$ has a repeated root, that is, when $4a^3 + 27b^2 = 0$. The cusp is $a = b = 0$. For $a = -3$, $b = 2$ the discriminant is $4(-3)^3 + 27 \cdot 2^2 = -108 + 108 = 0$, and indeed $x^3 - 3x + 2 = (x - 1)^2(x + 2)$: the curve $y^2 = x^3 - 3x + 2$ has a node at $(1, 0)$, just like the one in the figure. When the discriminant is nonzero the curve is smooth, and it is an **elliptic curve**, the object behind Ch. 78.

The coordinate ring

Coordinate ring

The **coordinate ring** of a variety $X \subseteq k^n$ is $k[X] = k[x_1, \dots, x_n]/I(X)$: polynomials, with two identified when they agree at every point of X.

A **polynomial map** $\phi : X \to Y$ between varieties is one given by polynomials in the coordinates. It induces the **pullback** $\phi^* : k[Y] \to k[X]$, $g \mapsto g \circ \phi$, a ring homomorphism. The arrows reverse direction. The central fact, which we state rather than prove, is that this correspondence is perfect: polynomial maps $X \to Y$ correspond exactly to k-algebra homomorphisms $k[Y] \to k[X]$. In particular X and Y are **isomorphic**, meaning there are polynomial maps both ways that are mutually inverse, if and only if $k[X] \cong k[Y]$.

The parabola is a line

The parabola $P = V(y - x^2)$ and the line k are isomorphic via $t \mapsto (t, t^2)$ and its inverse $(x, y) \mapsto x$, both polynomial. Correspondingly $k[x, y]/\langle y - x^2 \rangle \cong k[t]$: replace each y by x^2.

The cusp looks just as tame at first. The map $t \mapsto (t^2, t^3)$ is a bijection from the line onto $V(y^2 - x^3)$, and it is polynomial. Is the cusp a line too?

The cusp is not a line

There is no polynomial map from the cusp $C = V(y^2 - x^3)$ to the line that inverts $\phi(t) = (t^2, t^3)$. In fact C is not isomorphic to the line by any polynomial maps.

Proof of the first claim. An inverse would be a polynomial $h(x, y)$ with $h(t^2, t^3) = t$ for all t. But every monomial $x^a y^b$ becomes t^{2a+3b}, and $2a + 3b$ takes the values $0, 2, 3, 4, 5, \dots$ and *never* the value 1. So $h(t^2, t^3)$ has no t^1 term and cannot equal t. ∎

The same computation shows what the coordinate ring of the cusp *is*. Pulling back along ϕ identifies $k[C]$ with $k[t^2, t^3]$, the polynomials in t with no linear term. That ring is not $k[t]$. The quickest invariant that tells them apart is the dimension of the tangent space $\mathfrak{m}/\mathfrak{m}^2$ at the origin: it is 2 for the cusp, spanned by x and y, and 1 at every point of a line. So no isomorphism of any kind exists. The singularity is visible in the algebra as a "missing" function, $t = y/x$, which is defined on the cusp away from the origin but is not a polynomial.

Bézout's theorem, and the missing points

Two lines can be parallel, a circle and a line can miss each other, and a tangent line meets a parabola only once. To count intersections exactly, three repairs are needed.

1. **Complex points.** A circle and a line that "miss" meet at complex conjugate points.

2. **Points at infinity.** Parallel lines meet at infinity. Formally, the **projective plane** $\mathbb{P}^2$ is the set of nonzero triples $(x : y : z)$ up to scaling. The ordinary point (x, y) is $(x : y : 1)$, and the points with $z = 0$ form the line at infinity. A curve $f(x, y) = 0$ of degree d becomes the homogeneous equation $F(x, y, z) = z^d f(x/z, y/z) = 0$.
3. **Multiplicity.** A tangency counts twice, just as a double root of a polynomial counts twice.

Bézout's theorem

Two projective plane curves of degrees d and e over $\mathbb{C}$ with no common component meet in exactly de points, counted with multiplicity.

The general proof needs the resultant or intersection theory. The case $d = 1$ is a short argument that shows exactly why the three repairs are needed.

Proof for a line and a curve. Parametrise the projective line through two distinct points P and Q as $sP + tQ$, with $(s : t) \in \mathbb{P}^1$. Substituting into the homogeneous F of degree d gives $g(s, t) = F(sP + tQ)$, a homogeneous polynomial of degree d in (s, t). If $g \equiv 0$, the line is a component of the curve. Otherwise g factors over $\mathbb{C}$ into d linear factors $\prod_{i=1}^{d}(\beta_i s - \alpha_i t)$, by the fundamental theorem of algebra (Ch. 18) applied to $g(s, 1)$ together with the factors of t that account for roots "at $s = \infty$". Each factor is a point $(\alpha_i : \beta_i)$ of the line lying on the curve, and repeated factors are tangencies. So there are exactly d intersection points counted with multiplicity. ■

Without complex numbers, g might have no real factors. Without the projective line, the factors of t would be roots "at infinity" and would be lost. Without multiplicity, repeated factors would be undercounted. The theorem is clean only after all three repairs.

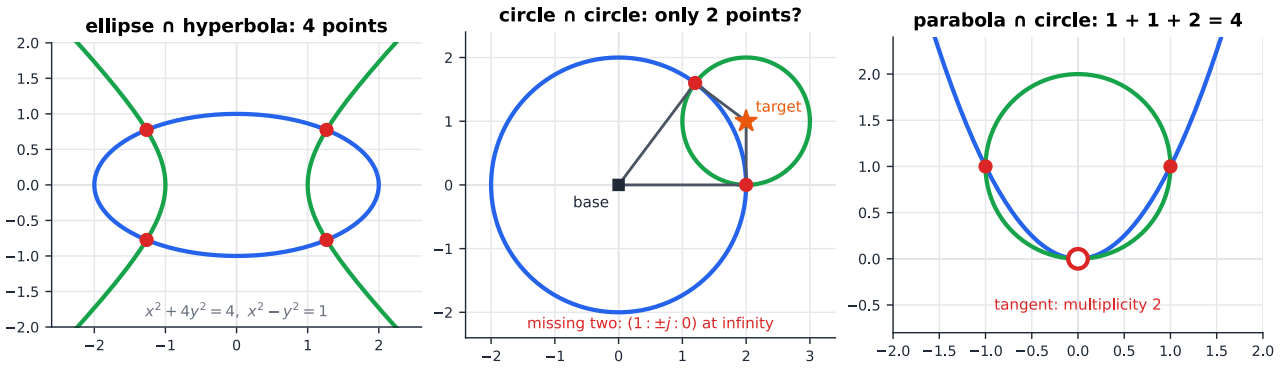
The answer to the hook. Any circle has equation $x^2 + y^2 + Dx + Ey + F = 0$, which homogenises to

$$x^2 + y^2 + Dxz + Eyz + Fz^2 = 0.$$

On the line at infinity, $z = 0$, this reduces to $x^2 + y^2 = 0$, that is, $(x + jy)(x - jy) = 0$. So *every circle*, whatever its centre and radius, passes through the two points

$$I = (1 : j : 0), \qquad J = (1 : -j : 0),$$

the **circular points at infinity**. Two circles share these two points, which leaves $4 - 2 = 2$ finite intersections. That is why the robot arm has at most two elbow positions, and why subtracting the two circle equations killed both squares: the difference of two circle equations has no $x^2 + y^2$ term, so it is a line, the line through the two finite intersections. Conversely, a conic through I and J is a circle. That is how projective geometers define a circle.



Bézout's count for conics. Left: four ordinary points. Middle: the robot arm's two circles meet at two real points; the other two are the circular points at infinity, shared by every circle. Right: $y = x^2$ and $x^2 + (y - 1)^2 = 1$ touch at the origin with multiplicity 2 and cross at $(\pm 1, 1)$, so $2 + 1 + 1 = 4$.

? Why is the chord-and-tangent rule associative?

Bézout says a line meets a cubic curve in exactly three points. On an elliptic curve that lets you "add" two points P and Q: draw the line through them, find the third intersection, and reflect it. Commutativity is obvious, since the line through P and Q is the line through Q and P. But why should $(P + Q) + R = P + (Q + R)$? Checked by brute algebra, it is a famously ugly computation. The conceptual proof uses Bézout again, this time for two cubics meeting in nine points: if eight of those points lie on a third cubic, so does the ninth. Why that is true, and how the group law becomes the backbone of modern cryptography, is in Ch. 78.

94.4 The algorithm

Gröbner bases: Gaussian elimination for polynomials. For linear equations, elimination (§8.7) turns any system into a triangular one: the last equation involves one unknown, the one before it two, and so on. A **Gröbner basis** does the same for polynomial systems. Fix a *monomial order*, a rule for which term of a polynomial is "leading". In **lex order** with $u > v$, any term containing a higher power of u wins. Then:

```
Buchberger(F):                    # F = list of polynomials generating the ideal
    G = F
    repeat:
        for each pair (f, g) in G:
            m = lcm(LT(f), LT(g))              # least common multiple of the leading terms
            S = (m / LT(f)) * f − (m / LT(g)) * g    # the S-polynomial cancels the leading terms
            r = remainder of S on division by G
            if r != 0: add r to G
    until no pair produces a new remainder
    return G
```

Buchberger's theorem (1965) says the loop always stops, by Hilbert's basis theorem, and that the result G generates the same ideal. With G in hand, every polynomial has a *unique* remainder on division by G. So $f \in \langle F \rangle$ if and only if its remainder is 0, and the ideal membership problem is solved. In lex order, the **elimination theorem** says the elements of G that involve only the last variable generate all consequences in that variable alone: the system has been triangularised.

The robot arm, by Buchberger. With $f_1 = u^2 + v^2 - 4$ and $f_2 = (u-2)^2 + (v-1)^2 - 1 = u^2 - 4u + v^2 - 2v + 4$, both leading terms are u^2. The S-polynomial is $f_1 - f_2 = 4u + 2v - 8$, which we halve to $g = 2u + v - 4$. Dividing f_1 by g (substitute $u = (4 - v)/2$) leaves the remainder $\tfrac{1}{4}(5v^2 - 8v)$. The reduced Gröbner basis is

$$\{\, 2u + v - 4,\;\; 5v^2 - 8v \,\}:$$

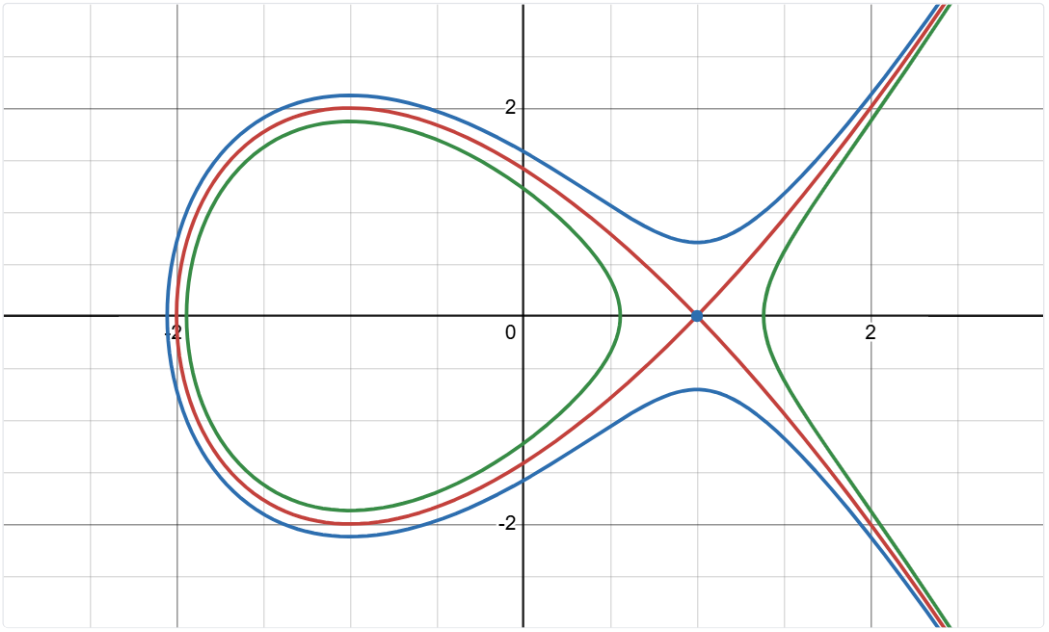
a quadratic in v alone, $v(5v - 8) = 0$, and then a linear equation for u. These are the two elbows, $(2, 0)$ and $(1.2, 1.6)$, found by the same steps a person would take, but now as an algorithm that works for any number of equations and unknowns.

Costs and failure modes. Gröbner bases are exact and complete, and computer algebra systems compute them routinely for kinematics and geometric constraint problems. Their worst case is brutal, though: the degrees in a Gröbner basis can be doubly exponential in the number of variables. Over floating-point numbers they are numerically fragile, because whether a coefficient is exactly zero changes the answer. Engineers who need *numbers* rather than formulas often use **resultants** instead (§94.5), or **homotopy continuation**. That method starts from a simple system whose $de\cdots$ solutions are known, by Bézout, and deforms it into the target system while tracking every solution path with an ODE solver (§22.9). The general six-joint robot arm has 16 inverse-kinematics solutions, a count found this way and then proved.

> **? Doubly exponential, yet it works**
>
> Mayr and Meyer (1982) built ideals for which *any* algorithm deciding ideal membership needs space exponential in the number of variables, and Gröbner bases with doubly exponential degrees. Yet the systems that come from engineering are solved in seconds. Is the worst case rare, or are real problems special? How problems can be provably hard in the worst case and easy in practice is a theme of Ch. 30.

94.5 The experiment

Watch singularities form. Desmos plots implicit equations directly. The curve $y^2 = x^3 - 3x + b$ is shown for $b = 2$, where it has a node, together with two nearby smooth members of the same family.



At $b = 2$ (red) the loop and the branch touch at the node $(1, 0)$. Slightly above (blue, $b = 2.5$) they merge into one smooth piece. Slightly below (green, $b = 1.5$) they separate into an oval and a branch.

Things to change:

- Replace $-3x$ by $0x$ and set all three constants to 0, 0.3 and -0.3. The red curve becomes the cusp $y^2 = x^3$. Its neighbours are smooth, but they do not split into two pieces.

- The singular curves are those with $4a^3 + 27b^2 = 0$. Use this to predict, then check, which b gives a node when $a = -12$.
- Add the lines `x^2+y^2=4`, `(x-2)^2+(y-1)^2=1` and `2x+y=4` to see the robot arm: the two circles and the line through their two finite intersections.

Count every intersection. This script finds *all* common points of two conics $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$, complex ones included. It eliminates y with the **resultant** of the two quadratics in y, a polynomial in x alone that vanishes exactly where they share a root, and then recovers y.

```
import numpy as np
P = np.poly1d

def intersect(c1, c2, tol=1e-4):
    """Common points of two conics A x^2 + B xy + C y^2 + D x + E y + F = 0."""
    def in_y(c):                                # coefficients of y^2, y^1, y^0 as polys in x
        A, B, C, D, E, F = c
        return P([C]), P([B, E]), P([A, D, F])
    a2, a1, a0 = in_y(c1)
    b2, b1, b0 = in_y(c2)
    res = (a2 * b0 - a0 * b2)**2 - (a2 * b1 - a1 * b2) * (a1 * b0 - a0 * b1)   # resultant in y
    res = P(np.where(abs(res.coeffs) < 1e-12, 0, res.coeffs))
    print("  resultant has degree", res.order, "in x")
    pts = []
    for x in np.unique(np.round(res.roots, 5)):   # each x may carry several points
        for y in np.unique(np.round(np.roots([a2(x), a1(x), a0(x)]), 5)):
            if abs(b2(x) * y**2 + b1(x) * y + b0(x)) < tol * (1 + abs(x)**2 + abs(y)**2):
                pts.append((complex(x), complex(y)))
    return pts

ellipse   = (1, 0, 4, 0, 0, -4)        # x^2 + 4y^2 = 4
hyperbola = (1, 0, -1, 0, 0, -1)       # x^2 - y^2 = 1
circle1   = (1, 0, 1, 0, 0, -4)        # x^2 + y^2 = 4
circle2   = (1, 0, 1, -4, -2, 4)       # (x-2)^2 + (y-1)^2 = 1

for name, c1, c2 in [("ellipse & hyperbola", ellipse, hyperbola),
                     ("two circles", circle1, circle2)]:
    print(name)
    for x, y in intersect(c1, c2):
        print(f"   x = {x.real:+.4f}{x.imag:+.4f}j,  y = {y.real:+.4f}{y.imag:+.4f}j")
```

For the ellipse and hyperbola the resultant has degree 4 and the script prints the four points $(\pm 1.2649, \pm 0.7746)$, which are $(\pm\sqrt{8/5}, \pm\sqrt{3/5})$. For the two circles the resultant collapses to degree 2: two solutions, the elbows $(1.2, 1.6)$ and $(2, 0)$. The lost degree *is* the pair of circular points at infinity. Things to change:

- Replace the hyperbola by the circle $(x-4)^2 + y^2 = 1$, `(1, 0, 1, -8, 0, 15)`, which misses the ellipse entirely in the real plane. The script still finds four points: $(8/3, \pm 0.8819j)$ and $(8, \pm 3.8730j)$.
- Give `circle2` a different x^2 coefficient, for instance `(1.1, 0, 1, -4, -2, 4)`. It is now an ellipse, it no longer passes through $(1 : \pm j : 0)$, and the resultant degree jumps back to 4. The two new solutions are huge, about $(38.4 \mp 20.9j,\ -20.9 \mp 38.3j)$, with $y/x \approx \mp j$: the points that were at infinity have come in from $(1 : \mp j : 0)$. Try `1.01` and watch them recede.
- Make two conics tangent, for instance $y = x^2$ and $x^2 + (y-1)^2 = 1$, `(1, 0, 0, 0, -1, 0)` and `(1, 0, 1, 0, -2, 0)`. Only three points print. Print `res.roots` to see the fourth: $x = 0$ appears twice.

94.6 The challenge

1. **Intuition.** Three generic quadric surfaces in space meet in $2 \cdot 2 \cdot 2 = 8$ points. Three spheres, as in GPS trilateration, meet in at most 2. Where are the other six? (Hint: what does every sphere have in common at infinity?)
2. **Derivation.** Over $\mathbb{C}$, show that the substitution $u = x + jy$, $v = x - jy$ turns the circle $x^2 + y^2 = 1$ into the hyperbola $uv = 1$, and deduce that its coordinate ring is $\mathbb{C}[u, u^{-1}]$. Then use the lines through $(-1, 0)$ to derive the rational parametrisation $x = (1 - t^2)/(1 + t^2)$, $y = 2t/(1 + t^2)$, and show that it gives every Pythagorean triple.
3. **Implementation.** Extend the resultant script to two *cubic* curves, using the 6×6 Sylvester determinant (evaluated at enough values of x and interpolated) to get a resultant of degree 9. Verify Bézout's count of 9 on random cubics, then find a pair of cubics for which the degree drops, and identify the common points at infinity.
4. **Open investigation.** Draw the six coefficients of each of two conics independently from a standard Gaussian. Estimate by simulation the probabilities that they meet in 0, 2 or 4 real points. Then try two random curves of degree d for $d = 2, \dots, 6$ and plot the average number of real intersection points against d. For one special distribution (Kostlan's) the expected count is exactly $\sqrt{de}$ (Shub and Smale, 1993). How does your distribution compare, and can you guess its growth rate?

94.7 Loose threads

Why SNARKs trust random points. The zero-knowledge proof systems of Ch. 82 encode a whole computation as an identity between polynomials and then check it at one random point. The geometric fact behind this is that a hypersurface is *thin*. The **Schwartz–Zippel lemma** says that if $f \neq 0$ has total degree d and $\mathbf{r}$ is drawn uniformly from S^n for a finite set S, then $P(f(\mathbf{r}) = 0) \leq d/|S|$. The proof is a short induction. For $n = 1$, a nonzero polynomial of degree d has at most d roots. For $n > 1$, write $f = \sum_{i \leq k} x_1^i f_i(x_2, \dots, x_n)$ with $f_k \neq 0$. By induction, $f_k(r_2, \dots, r_n) = 0$ with probability at most $(d-k)/|S|$. Otherwise f is a nonzero polynomial of degree k in x_1, which vanishes at r_1 with probability at most $k/|S|$. Adding the two gives $d/|S|$. Over a field with $q \approx 2^{255}$ elements, a false identity of degree a million survives a random test with probability below 2^{-235}. Seen this way, a variety is a set of measure almost zero, and a random point almost surely lies off it.

Fermat's Last Theorem. Ch. 57 asked how geometry enters Fermat's problem. A solution of $a^n + b^n = c^n$ in positive integers is a rational point $(a/c, b/c)$ on the curve $x^n + y^n = 1$. The **genus** of a smooth curve, the number of holes in its complex points viewed as a surface, controls how many rational points it can have. Genus 0 curves, like the circle, have none or infinitely many, which is why there are infinitely many Pythagorean triples. Genus 1 curves are elliptic curves, whose rational points form a finitely generated group. Curves of genus at least 2 have only *finitely* many rational points (Faltings, 1983). The Fermat curve has genus $(n-1)(n-2)/2$, which is at least 3 once $n \geq 4$, so Faltings already gives finitely many solutions. Wiles' proof (1995) that there are *none* goes through an elliptic curve, Frey's curve $y^2 = x(x - a^n)(x + b^n)$, which would be too strange to exist if a solution did.

? Can every singularity be smoothed away?

The cusp can be repaired. Substitute $y = tx$ into $y^2 = x^3$ and divide by x^2: the result is $t^2 = x$, a smooth parabola. This "blow-up" replaces the singular point by the directions of approach to it. Hironaka proved in 1964, in a 200-page paper that earned a Fields Medal, that every variety over a field of characteristic 0, such as $\mathbb{R}$ or $\mathbb{C}$, can be made smooth by repeated blow-ups. Over fields of characteristic p, the setting of cryptography and coding theory, the same statement is known only in dimension at most 3. In higher dimensions it is an **open problem**, and one of the most sought-after in the field.

The other threads lead into the rest of the track. The variety of tensors of border rank at most r from Ch. 93 is a secant variety, and finding its equations is algebraic geometry at the research frontier. Symmetry groups acting on varieties, and on the coordinate rings that describe them, bring in the representation theory of Ch. 95. Grothendieck's reformulation of the whole subject, which replaces varieties by "schemes" so that any commutative ring has a geometry, shows how mathematicians build new language when the old one runs out (Ch. 98).

Further reading

- D. A. Cox, J. Little and D. O'Shea, *Ideals, Varieties, and Algorithms*, 4th ed., Springer, 2015. The best first book: Gröbner bases, elimination and the Nullstellensatz, with robotics examples.
- M. Reid, *Undergraduate Algebraic Geometry*, Cambridge University Press, 1988. Short, witty and geometric: conics, cubics and the dictionary.
- F. Kirwan, *Complex Algebraic Curves*, Cambridge University Press, 1992. Bézout's theorem, singularities and the topology of complex curves.
- W. Fulton, *Algebraic Curves*, freely available online. A classic, concise treatment of intersection multiplicity and Bézout.
- A. J. Sommese and C. W. Wampler, *The Numerical Solution of Systems of Polynomials Arising in Engineering and Science*, World Scientific, 2005. Homotopy continuation and robot kinematics.

Chapter 95 — Representation Theory: Making Abstract Symmetry Concrete

> **Level 3 · Track J — Mathematics at the Frontier Builds on:** Ch. 9, Ch. 10, Ch. 15, Ch. 56, Ch. 99 · **Leads to:** Ch. 92, Ch. 93, Ch. 94 · **Related:** Ch. 62, Ch. 90, Ch. 79 **Reading time:** ~30 min

95.1 The hook

In 1978 John McKay, a group theorist at Concordia University, was reading a paper on number theory and noticed a coefficient. The paper was about the **j-function**, a function of a complex variable studied since the nineteenth century, which has a series expansion beginning

$$j(q) = \frac{1}{q} + 744 + 196884\,q + 21493760\,q^2 + 864299970\,q^3 + \cdots$$

McKay knew the number 196883 from a completely different world. It was the smallest dimension in which the **Monster**, the largest sporadic finite simple group (Ch. 56), with about 8×10^{53} elements, can act by rotations without acting trivially. He wrote to John Thompson, who checked the next coefficient:

$$\begin{aligned} 196884 &= 1 + 196883, \\ 21493760 &= 1 + 196883 + 21296876, \\ 864299970 &= 2 \cdot 1 + 2 \cdot 196883 + 21296876 + 842609326. \end{aligned}$$

Every number on the right is the dimension of one of the Monster's "irreducible representations". A function from complex analysis, which knows nothing about groups, was spelling out the building blocks of the biggest exceptional symmetry object in mathematics. John Conway called this **monstrous moonshine**, meaning crazy talk.

To even say what these numbers *are* you need one idea. A group is an abstract table of how symmetries combine. A **representation** turns each symmetry into a matrix, so that combining symmetries becomes multiplying matrices. Then the full power of linear algebra (eigenvalues, traces, change of basis) can be aimed at the group. And just as a signal splits into sine waves that cannot be split further, a representation splits into irreducible pieces. The Monster has exactly 194 of them, and the smallest non-trivial one has dimension 196883.

You've already met a representation without being told. The DFT of Ch. 15 turns every circular convolution into a diagonal matrix. Why? Not because of some property of $e^{j\theta}$. It's because convolution commutes with shifting, and the shifts form a group. This chapter explains that sentence, and it shows that the same reasoning works for the symmetries of a square, of a molecule, or of the Monster.

95.2 The big idea

Take a symmetric object, say four equal masses at the corners of a square joined by identical springs. You want its vibration modes. The honest way is to write the 8×8 stiffness matrix and find its eigenvectors (§10.8). The clever way is to notice that the square has eight symmetries (four rotations, four reflections). The stiffness matrix commutes with all of them, because rotating the square and then letting it vibrate is the same as letting it vibrate and then rotating it.

Here is the key fact. **If a matrix commutes with a group of symmetries, then it cannot mix vectors that transform in genuinely different ways under the group.** So you first sort the space of motions into "transformation types", the irreducible representations, and the matrix becomes block-diagonal *before you compute anything about it*. The symmetry does the work.

The analogy is a prism. White light enters, and the prism sorts it by frequency because a prism treats every frequency differently but each one consistently. A group splits a vector space by "symmetry type" in the same way. For the cyclic group of shifts the types are exactly the frequencies, which is why Fourier analysis is the representation theory of $\mathbb{Z}_n$.

Two more ideas make the theory practical:

1. **Characters.** The trace of a matrix doesn't change under change of basis. So the function $g \mapsto \operatorname{tr} \rho(g)$, the **character**, is a basis-free fingerprint of a representation. Remarkably, it determines the representation completely.
2. **Orthogonality.** The characters of the irreducible representations are orthonormal vectors, in the same way that sampled sinusoids are. So to find how many times a type appears, you take an inner product, just like computing a Fourier coefficient.

95.3 The mathematics

Throughout, G is a finite group and vector spaces are over $\mathbb{C}$. Groups, homomorphisms and conjugacy were set up in Ch. 56; invariant subspaces and direct sums in Ch. 99.

Representation, subrepresentation, irreducible

A **representation** of G on a finite-dimensional space V is a homomorphism $\rho : G \to \mathrm{GL}(V)$, so $\rho(gh) = \rho(g)\rho(h)$ and $\rho(e) = \mathbf{I}$. Its **dimension** is $\dim V$. A subspace $W \subseteq V$ is **invariant** if $\rho(g)W \subseteq W$ for all g. The representation is **irreducible** if its only invariant subspaces are $\{0\}$ and V. Two representations are **equivalent** if some invertible $\mathbf{P}$ satisfies $\mathbf{P}\rho(g)\mathbf{P}^{-1} = \rho'(g)$ for every g, the same change of basis for all of them at once. The **character** is $\chi_\rho(g) = \operatorname{tr} \rho(g)$. It is constant on conjugacy classes, since $\operatorname{tr}(\mathbf{ABA}^{-1}) = \operatorname{tr} \mathbf{B}$.

Three examples to hold on to:

- **The trivial representation:** $\rho(g) = 1$ for all g, on $\mathbb{C}^1$.
- **Permutation representations:** if G permutes a set X, let $\rho(g)$ be the permutation matrix on $\mathbb{C}^X$. Its character $\chi(g)$ counts the points of X that g fixes. The **regular representation** is the case $X = G$, with G acting on itself by left multiplication.
- **Characters of $\mathbb{Z}_n$:** for each $k = 0, \dots, n-1$, the map $\chi_k(m) = e^{2\pi jkm/n}$ is a homomorphism $\mathbb{Z}_n \to \mathbb{C}^\times$, a one-dimensional representation.

Result 1: everything breaks into irreducibles

> **Maschke's theorem**
>
> If W is an invariant subspace of a representation V of a finite group, then W has an invariant complement W', with $V = W \oplus W'$. Hence every representation is a direct sum of irreducible ones.

Proof. Start with any inner product $\langle \cdot, \cdot \rangle_0$ on V and average it over the group:

$$\langle \mathbf{u}, \mathbf{v} \rangle = \frac{1}{|G|} \sum_{g \in G} \langle \rho(g)\mathbf{u}, \rho(g)\mathbf{v} \rangle_0.$$

This is again an inner product (a positive sum of inner products). It is **invariant**: replacing $\mathbf{u}, \mathbf{v}$ by $\rho(h)\mathbf{u}, \rho(h)\mathbf{v}$ turns $\rho(g)$ into $\rho(gh)$, and as g runs over G so does gh, so the sum is unchanged. Every $\rho(h)$ is therefore unitary for $\langle \cdot, \cdot \rangle$.

Let $W' = W^\perp$ for this inner product. If $\mathbf{w}' \in W'$ and $\mathbf{w} \in W$, then

$$\langle \rho(h)\mathbf{w}', \mathbf{w} \rangle = \langle \mathbf{w}', \rho(h)^{-1}\mathbf{w} \rangle = \langle \mathbf{w}', \rho(h^{-1})\mathbf{w} \rangle = 0,$$

since $\rho(h^{-1})\mathbf{w} \in W$. So $\rho(h)\mathbf{w}' \in W'$, and W' is invariant. Repeat inside W and W'. Dimensions drop each time, so the process ends with irreducible pieces. ∎

The averaging trick is the whole proof, and it is also where finiteness was used: we divided by $|G|$. For the group $\mathbb{Z}$ acting on $\mathbb{C}^2$ by powers of $\left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]$, the line spanned by $(1, 0)$ is invariant and has no invariant complement. That is the Jordan block of Ch. 99 refusing to split.

Result 2: Schur's lemma, the engine

> **Schur's lemma**
>
> Let V, W be irreducible representations and $\mathbf{T} : V \to W$ a linear map with $\mathbf{T}\rho_V(g) = \rho_W(g)\mathbf{T}$ for all g (an **intertwiner**). (a) Either $\mathbf{T} = 0$ or $\mathbf{T}$ is invertible; so if V and W are not equivalent, $\mathbf{T} = 0$. (b) If $V = W$, then $\mathbf{T} = \lambda \mathbf{I}$ for some scalar λ.

Proof. (a) The kernel of $\mathbf{T}$ is invariant: if $\mathbf{T}\mathbf{v} = 0$ then $\mathbf{T}\rho_V(g)\mathbf{v} = \rho_W(g)\mathbf{T}\mathbf{v} = 0$. By irreducibility it is $\{0\}$ or V. Likewise the image is invariant in W, so it is $\{0\}$ or W. If $\mathbf{T} \neq 0$ the kernel is $\{0\}$ and the image is W, so $\mathbf{T}$ is invertible.

(b) Over $\mathbb{C}$, $\mathbf{T}$ has an eigenvalue λ (Ch. 10). Then $\mathbf{T} - \lambda\mathbf{I}$ is also an intertwiner, and it isn't invertible. By (a) it is zero. ∎

Here is the engineering payoff, the precise form of the big idea:

> **Symmetric operators are block-diagonal**
>
> Suppose $V = V_1 \oplus \cdots \oplus V_r$ with the V_i irreducible and pairwise inequivalent, and $\mathbf{K}$ commutes with every $\rho(g)$. Then $\mathbf{K}$ maps each V_i into itself and acts on it as a scalar λ_i. In other words, the V_i are eigenspaces of $\mathbf{K}$, found without knowing $\mathbf{K}$.

Proof. Let π_i be the projection onto V_i along the other pieces. It commutes with $\rho(g)$ because each piece is invariant. The map $\pi_i \mathbf{K}|_{V_k} : V_k \to V_i$ is an intertwiner. By Schur (a) it vanishes for $i \neq k$, so $\mathbf{K}V_k \subseteq V_k$. By Schur (b) it is a scalar for $i = k$. ∎

When some irreducible type appears m times, the same argument gives an $m \times m$ block, which is still a huge saving. The square of springs has an 8-dimensional configuration space. Symmetry splits it into pieces of dimension at most 2, with the 2-dimensional type appearing twice. You are left with one 2×2 eigenproblem and a few scalars, instead of an 8×8 one.

Result 3: Fourier analysis is the case $G = \mathbb{Z}_n$

> **The DFT diagonalises every shift-invariant operator**
>
> Let $\mathbf{S}$ be the cyclic shift on $\mathbb{C}^n$ and $\mathbf{C}$ any matrix with $\mathbf{CS} = \mathbf{SC}$. Then each vector $\mathbf{f}_k = (e^{2\pi jkm/n})_{m=0}^{n-1}$ is an eigenvector of $\mathbf{C}$. The matrices commuting with $\mathbf{S}$ are exactly the circular convolutions $\mathbf{x} \mapsto h \circledast \mathbf{x}$, and the eigenvalue on $\mathbf{f}_k$ is the DFT coefficient $\hat{h}[k] = \sum_m h[m] e^{-2\pi jkm/n}$.

Proof. Write $(\mathbf{S}\mathbf{x})[m] = x[m-1]$, indices mod n. Then $\mathbf{S}\mathbf{f}_k = e^{-2\pi jk/n}\mathbf{f}_k$. The n eigenvalues $e^{-2\pi jk/n}$ are distinct, so each eigenspace of $\mathbf{S}$ is the line through $\mathbf{f}_k$. These lines are the n one-dimensional representations of $\mathbb{Z}_n$, pairwise inequivalent, so the block-diagonal theorem says $\mathbf{C}\mathbf{f}_k = \lambda_k \mathbf{f}_k$. (You can also see it directly: $\mathbf{CS}\mathbf{f}_k = \mathbf{SC}\mathbf{f}_k$ says $\mathbf{C}\mathbf{f}_k$ is again an eigenvector of $\mathbf{S}$ with the same eigenvalue.)

If $\mathbf{C}$ commutes with $\mathbf{S}$, set $h = \mathbf{C}\boldsymbol{\delta}$, the image of the unit impulse. Every $\mathbf{x} = \sum_k x[k]\mathbf{S}^k\boldsymbol{\delta}$, so $\mathbf{C}\mathbf{x} = \sum_k x[k]\mathbf{S}^k h = h \circledast \mathbf{x}$. Finally, $(h \circledast \mathbf{f}_k)[m] = \sum_i h[i] e^{2\pi jk(m-i)/n} = \hat{h}[k]\, f_k[m]$. ∎

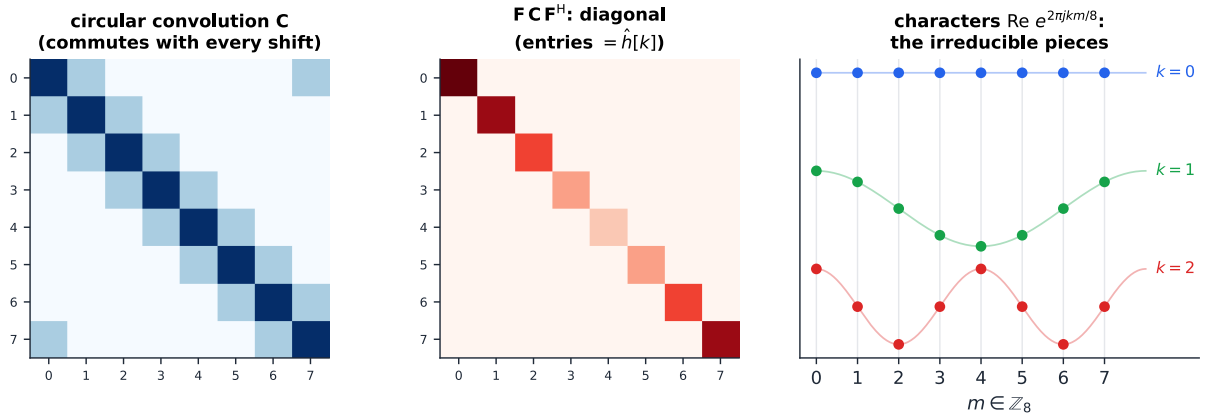


Figure 95.1 — Convolution on $\mathbb{Z}_8$ commutes with shifts, so in the basis of characters (right) it becomes diagonal (middle). The convolution theorem of §16.5 is Schur's lemma for an abelian group.

For an abelian group every irreducible representation is one-dimensional (by Schur (b), each $\rho(g)$ commutes with everything, so it is a scalar, and every line is invariant). For a non-abelian group some irreducibles must be bigger, and "frequencies" become small matrices. That is the only genuinely new phenomenon, and it is where the theory gets its depth.

Characters and the character table

The following facts follow from Schur's lemma by averaging intertwiners over the group. We state them; Serre's book in the reading list proves them in about ten pages.

Character theory of a finite group

Define $\langle \chi, \psi \rangle = \frac{1}{|G|} \sum_{g} \chi(g)\overline{\psi(g)}$.

1. The irreducible characters $\chi_1, \dots, \chi_r$ are orthonormal: $\langle \chi_i, \chi_k \rangle = \delta_{ik}$.
2. Their number r equals the number of conjugacy classes of G.
3. If $V \cong \bigoplus_i V_i^{\oplus m_i}$, then $m_i = \langle \chi_V, \chi_i \rangle$. So two representations with the same character are equivalent.
4. The regular representation contains each V_i exactly $d_i = \dim V_i$ times, so $\sum_i d_i^2 = |G|$.

Fact 4 is a quick consequence of fact 3. The regular character is $|G|$ at e and 0 elsewhere (only the identity fixes a group element under left multiplication). So $m_i = \frac{1}{|G|}|G|\,\overline{\chi_i(e)} = d_i$, and comparing dimensions gives $|G| = \sum_i d_i \cdot d_i$.

Example: S_3. The symmetric group on three letters has 6 elements in three classes: the identity, three transpositions, two 3-cycles. So it has three irreducibles, and $d_1^2 + d_2^2 + d_3^2 = 6$ forces dimensions $1, 1, 2$. Two are easy: the trivial character, and the sign of a permutation. For the third, S_3 permutes the corners of a triangle, giving a 3-dimensional permutation representation with character $(3, 1, 0)$ (fixed points). It contains the trivial representation once (the all-ones vector), and what remains is the 2-dimensional rotations-and-reflections of the triangle's plane, with character $(3, 1, 0) - (1, 1, 1) = (2, 0, -1)$.

irreducible	e (1 element)	(1 2) (3 elements)	(1 2 3) (2 elements)
trivial χ_1	1	1	1
sign χ_2	1	-1	1
standard χ_3	2	0	-1

class sizes weight the sum

rows are orthogonal: $\langle \chi_2, \chi_3 \rangle = \frac{1}{6}(1 \cdot 2 \cdot 1 + 3 \cdot (-1) \cdot 0 + 2 \cdot 1 \cdot (-1)) = 0$
first column: $1^2 + 1^2 + 2^2 = 6 = |S_3|$

Figure 95.2 — The character table of S_3. Rows are orthonormal once each column is weighted by its class size, and the first column's squares add up to the group order.

Example: the square, D_4. The eight symmetries fall into five classes: $\{e\}$, $\{r^2\}$, $\{r, r^3\}$, the two edge flips, and the two diagonal flips. So there are five irreducibles, and $8 = 1 + 1 + 1 + 1 + 2^2$. Let D_4 permute the four corners. A diagonal flip fixes two corners and everything else except e fixes none, so the character is 4 at e, 2 on diagonal flips and 0 elsewhere. Taking inner products with the five irreducible characters gives multiplicities $1, 0, 1, 0, 1$: the corner space is trivial $\oplus$ one sign-like line $\oplus$ the 2-dimensional irreducible.

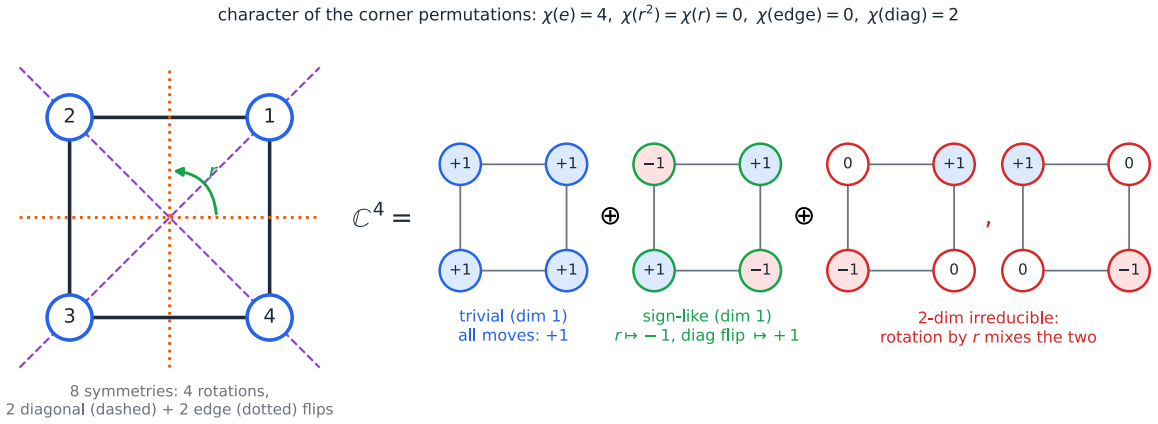


Figure 95.3 — The eight symmetries of the square act on functions of its corners. The space splits into three pieces. Any symmetric operator, such as a spring network's stiffness matrix, must respect this split.

Why one matrix is tame and two are wild

Ch. 99 left a puzzle. One linear map T on V, up to change of basis, is classified by Jordan form. Two maps S, T changed by the *same* basis have no such list. Representation theory explains the difference by changing what we represent. A single T is a representation of the polynomial ring $\mathbb{C}[x]$ (let x act as T). That ring is a principal ideal domain, and the structure theorem for its modules is exactly the Jordan form: the indecomposable pieces are the Jordan blocks, one family per eigenvalue and size.

A pair (S, T) is a representation of the free algebra $\mathbb{C}\langle x, y\rangle$, in which $xy \neq yx$. Every finite-dimensional algebra is a quotient of a free algebra, and with a little coding one can embed the representations of *any* finite-dimensional algebra (triples, quadruples of matrices, anything) into pairs. A classification of pairs would therefore contain every other classification problem of linear algebra. That is what **wild** means. Drozd proved in 1979 that every finite-dimensional algebra is either *tame* (indecomposables come in finitely many one-parameter families in each dimension) or wild, never both.

The sharpest picture uses **quivers**, directed graphs with a vector space at each vertex and a matrix on each arrow. Gabriel's theorem (1972) says that a connected quiver has only finitely many indecomposable representations exactly when its underlying graph is one of the Dynkin diagrams A_n, D_n, E_6, E_7, E_8. Two vertices joined by two arrows, Kronecker's "pencils of matrices" $\lambda\mathbf{A} + \mu\mathbf{B}$, are tame. One vertex with two loops, our pair (S, T), is wild.

> **? The same five letters, everywhere**
>
> The list A_n, D_n, E_6, E_7, E_8 that decides which quivers are classifiable is also the list of simple Lie algebras with one root length, of the finite subgroups of $\mathrm{SU}(2)$ (the symmetries of the Platonic solids, doubled), and of the simplest singular points of surfaces, such as $x^2 + y^3 + z^5 = 0$. The **McKay correspondence** links the last two through representations. Why should a polynomial singularity know about the icosahedron? The geometry of resolving singularities is in Ch. 94.

95.4 The algorithm

Given a representation as a list of matrices, how do you split it? Character theory turns the job into inner products and projections.

```
decompose(G, rho, irreducible characters chi_1..chi_r):
    chi = [trace(rho(g)) for g in G]                # O(|G| d)
    for each i:
        m_i = (1/|G|) * sum_g chi(g) * conj(chi_i(g))  # multiplicity
        P_i = (d_i/|G|) * sum_g conj(chi_i(g)) * rho(g) # projector, O(|G| d^2)
        V_i-part = column space of P_i              # dimension m_i * d_i
```

The projector formula $\mathbf{P}_i = \frac{d_i}{|G|} \sum_g \overline{\chi_i(g)}\, \rho(g)$ comes from the same averaging as Maschke's proof. It commutes with every $\rho(h)$, so by Schur it is a scalar on each irreducible piece, and orthogonality makes that scalar 1 on type i and 0 on every other type. For $G = \mathbb{Z}_n$ this is exactly the DFT: $\mathbf{P}_k$ projects onto the k-th frequency.

Costs and pitfalls:

- **Group size.** The sums run over $|G|$ elements. For S_{10} that is 3.6 million matrices. Summing over conjugacy classes (character values are constant on them) helps for multiplicities, though not for projectors.
- **Fast transforms.** The FFT computes all n projections of $\mathbb{Z}_n$ in $O(n \log n)$ by exploiting the subgroup chain $\mathbb{Z}_n \supset \mathbb{Z}_{n/2} \supset \cdots$. Clausen, Diaconis and Rockmore built analogous fast Fourier transforms for S_n and other non-abelian groups, and they are used to analyse ranked data such as elections and preference surveys.
- **Finding the character table.** If you don't know the χ_i, the Burnside–Dixon–Schneider algorithm finds them as simultaneous eigenvectors of class-multiplication matrices, which is linear algebra again. Computer algebra systems (GAP, Magma) store tables for thousands of groups, including the Monster's 194×194 table.
- **Numerical stability.** Floating-point projectors have rank only approximately $m_i d_i$. Use an SVD with a tolerance (§10.7) rather than Gaussian elimination to extract the subspace.

95.5 The experiment

The script builds the eight 4×4 permutation matrices of D_4 acting on corners, computes the character, and finds multiplicities and projector ranks for all five irreducibles. Then it checks that the DFT diagonalises a random circulant matrix.

```
import numpy as np

# D4 acting on the 4 corners of a square (corners 0,1,2,3 counter-clockwise)
def perm_matrix(p):                      # p[i] = where corner i goes
    P = np.zeros((4, 4)); P[p, range(4)] = 1; return P

r = perm_matrix([1, 2, 3, 0])            # rotate by 90 degrees
s = perm_matrix([0, 3, 2, 1])            # flip in the diagonal through corner 0
group = [np.linalg.matrix_power(r, k) @ f for k in range(4) for f in (np.eye(4), s)]

chi = np.array([np.trace(g) for g in group])          # character of this rep
print("character of corner rep:", chi)

# characters of the 5 irreducibles, element by element (same order as `group`)
signs_r = np.array([(-1) ** k for k in range(4) for _ in (0, 1)])
is_flip = np.array([f for _ in range(4) for f in (0, 1)])
diag = np.array([k % 2 == 0 for k in range(4) for _ in (0, 1)]) & (is_flip == 1)
irreps = {
    "A1 trivial": np.ones(8),
    "A2 (flips -1)": np.where(is_flip, -1, 1),
    "B1 (r -1, diag +1)": np.where(is_flip, np.where(diag, 1, -1), signs_r),
    "B2 (r -1, edge +1)": np.where(is_flip, np.where(diag, -1, 1), signs_r),
    "E  (2-dim)": np.array([np.trace(np.linalg.matrix_power(np.array([[0, -1], [1, 0]]), k)) * (1 - f)
                            for k in range(4) for f in (0, 1)]),
}
for name, x in irreps.items():
    mult = np.dot(chi, x) / len(group)                # <chi, chi_i>
    P = sum(xi * g for xi, g in zip(x, group)) * x[0] / len(group)
    print(f"{name:20s} multiplicity {mult:+.0f}   projector rank {np.linalg.matrix_rank(P)}")

# Fourier on Z_n: any matrix commuting with the cyclic shift is diagonalised by the DFT
n = 8
h = np.random.default_rng(1).normal(size=n)          # try your own filter here
C = np.array([[h[(i - k) % n] for k in range(n)] for i in range(n)])
F = np.fft.fft(np.eye(n)) / np.sqrt(n)
D = F @ C @ F.conj().T
print("largest off-diagonal entry:", np.abs(D - np.diag(np.diag(D))).max())
print("diagonal equals fft(h):", np.allclose(np.diag(D), np.fft.fft(h)))
```

Running it prints multiplicities $1, 0, 1, 0, 1$ with projector ranks $1, 0, 1, 0, 2$, and an off-diagonal residue around 10^{-16}. Things to change:

- **Act on edges instead of corners.** Replace `r` and `s` by the permutations of the four edges (number them so edge i joins corners i and $i+1$). Predict the character, and therefore the decomposition, before you run it. Which irreducible swaps in?
- **Act on motions.** Build the 8×8 representation on corner displacements $(\Delta x_i, \Delta y_i)$: each symmetry permutes the corners *and* rotates each displacement vector, so it is a Kronecker product of a permutation and a 2×2 rotation. Show that its character is the product of the two characters, and decompose it. You have just predicted the mode shapes of the spring square of §95.2.
- **Break the symmetry.** Make one entry of `h` depend on position (a non-circulant matrix). Watch the off-diagonal residue jump from 10^{-16} to order one.

95.6 The challenge

1. **Intuition.** A matrix commutes with all eight symmetries of the square. Without computing anything, how many distinct eigenvalues can it have on the corner space $\mathbb{C}^4$ at most? Why might it have fewer?
2. **Derivation.** Show that the number of one-dimensional representations of a finite group G equals $|G/[G,G]|$, where $[G,G]$ is the subgroup generated by all commutators $ghg^{-1}h^{-1}$. Check your formula on S_3 and D_4.
3. **Implementation.** Write code that, given generating permutations of a group, produces all its elements and its conjugacy classes, and then finds the dimensions of its irreducibles from $\sum d_i^2 = |G|$ together with the number of classes. Run it on the rotation group of the cube (order 24), and decide whether these two constraints alone always pin down the dimensions.
4. **Open investigation.** Take a random walk on S_n by repeatedly swapping two random cards of a deck. Diaconis and Shahshahani used the characters of S_n to show that the walk mixes after about $\frac{1}{2} n \ln n$ swaps. Simulate the distance to uniform for $n = 6, 8, 10$ and look for a sudden "cutoff". Is there a general way to predict from a group's character table alone when a random walk on it shows a cutoff? This is a research question, and no general answer is known.

95.7 Loose threads

The hook asked why a function from complex analysis spells out the dimensions of the Monster's representations. We've built the dictionary for even asking. A representation is a group written in matrices. Maschke's averaging splits every representation into irreducibles. Schur's lemma says symmetric operators are scalars on each irreducible type, which is why Fourier analysis diagonalises convolution and why symmetry predicts mode shapes. And characters, orthogonal like sinusoids, count the pieces.

Moonshine itself needed far more. Conway and Norton conjectured in 1979 that the j-function is the "graded dimension" of an infinite-dimensional representation $V^\natural = V_0 \oplus V_1 \oplus V_2 \oplus \cdots$ of the Monster, and that replacing dimensions by traces of other elements gives other classical functions. Frenkel, Lepowsky and Meurman constructed $V^\natural$ in 1988 from the Leech lattice. Richard Borcherds proved the conjecture in 1992 using a "no-ghost theorem" borrowed from string theory, and won a Fields Medal.

? Why does string theory know about the Monster?

Borcherds' proof works, but many mathematicians feel it explains *how* rather than *why*. $V^\natural$ is the state space of a two-dimensional quantum field theory, and the Monster is its symmetry group. Newer "umbral" and "Mathieu" moonshines link other sporadic groups to other functions (one discovered in 2010 from the K3 surfaces of string theory), and for most of them the representations have been shown to exist without anyone writing them down. A conceptual reason why sporadic groups and modular functions are linked remains an open problem. Symmetry in physics, the doorway to this story, is in Ch. 92.

We also stayed with finite groups. For the circle group $\mathbb{R}/2\pi\mathbb{Z}$ the irreducibles are $e^{jk\theta}$, and decomposing functions into them is the Fourier series of Ch. 15. For the line $\mathbb{R}$ there is a continuum of characters $e^{j\omega t}$, and you get the Fourier transform. For the rotation group $\mathrm{SO}(3)$ the irreducibles have dimensions $1, 3, 5, 7, \ldots$, and they are the spherical harmonics behind atomic orbitals and antenna radiation patterns.

? Fourier analysis on groups that do not commute

For a compact group such as $\mathrm{SO}(3)$, the Peter–Weyl theorem says the matrix entries of the irreducible representations form an orthonormal basis of $L^2(G)$, a non-commutative Fourier series. For non-compact, non-abelian groups such as the Lorentz group of relativity, the unitary irreducibles (the ones quantum mechanics needs) are infinite-dimensional and the "spectrum" can have continuous parts. How a spectrum can be partly a list and partly a continuum, and what an operator's spectrum even is, is the business of Ch. 90.

Further out lies the **Langlands program**, a web of conjectures, proved only in pieces, predicting that representations of symmetry groups of number fields correspond to automorphic functions generalising the j-function. Wiles' proof of Fermat's Last Theorem was one corner of it. And tensor products of representations, which is how two spin-$\frac{1}{2}$ particles combine into spin 0 and spin 1, are taken up in Ch. 93.

Further reading

- J.-P. Serre, *Linear Representations of Finite Groups*, Springer GTM 42, 1977. Part I proves all of character theory in about thirty pages.
- W. Fulton and J. Harris, *Representation Theory: A First Course*, Springer GTM 129, 1991. Examples first, from S_3 to Lie groups.
- P. Diaconis, *Group Representations in Probability and Statistics*, IMS Lecture Notes 11, 1988. Card shuffling, ranked data and fast Fourier transforms on groups.
- M. Ronan, *Symmetry and the Monster*, Oxford University Press, 2006. A popular history of the classification and moonshine.
- T. Gannon, *Moonshine beyond the Monster*, Cambridge University Press, 2006. The research-level survey of moonshine and its links to physics.

Chapter 96 — Mathematical Economics and Social Choice

> **Level 3 · Track J — Mathematics at the Frontier Builds on:** Ch. 6, Ch. 23, Ch. 42, Ch. 44 · **Leads to:** Ch. 97, Ch. 98 · **Related:** Ch. 39, Ch. 41, Ch. 43, Ch. 47, Ch. 30 **Reading time:** ~30 min

96.1 The hook

In 1951 a young economist named Kenneth Arrow published two results that seem to contradict each other.

The first was a mathematical version of Adam Smith's "invisible hand". Let millions of selfish people trade at market prices, each caring only about their own basket of goods. If prices settle where supply meets demand, the outcome is **Pareto efficient**: no central planner, however wise, could rearrange the goods to make one person better off without making someone else worse off. Selfish choices, aggregated by prices, give a collectively optimal result.

The second was his doctoral thesis, *Social Choice and Individual Values*. It asked whether a society could aggregate its members' preferences by *voting*, and it said no. Any rule that turns individual rankings of three or more options into a social ranking, respects unanimity, and judges each pair of options by individual opinions on that pair alone, must be a **dictatorship**: one voter's ranking decides everything.

So markets aggregate preferences perfectly and ballots cannot aggregate them at all. Arrow shared the 1972 Nobel prize largely for these two results, and it is not obvious that they can both be true. Hasn't a market just counted votes in dollars?

This chapter proves both theorems, and on the way it settles several debts left by the game-theory track: why Gibbard–Satterthwaite forces a dictator (Ch. 42), why no trading rule can be efficient and balanced at once, and why haggling ends at the Nash bargaining solution (Ch. 44). The answer to the hook turns out to be about *information*. A price carries more than a vote does, and it asks for less.

96.2 The big idea

Start with one person. Economics doesn't assume we can measure happiness. It assumes only that you can *compare*: given two bundles, you prefer one, or the other, or you're indifferent. A utility function is merely a way of writing those comparisons as numbers, and any increasing relabelling of it ($u \mapsto u^3$, $u \mapsto \ln u$) describes the same person. Utility is **ordinal**.

Now two people, Ann and Bob, and two goods, apples and bread. Draw a rectangle whose width is the total number of apples and whose height is the total bread. A point inside is a way to split everything: measure Ann's holdings from the bottom-left corner and Bob's from the top-right. This is the **Edgeworth box**, and every question about two-person exchange becomes geometry inside it.

Through the starting point (the **endowment**) runs one of Ann's indifference curves and one of Bob's. Between them is a lens of allocations that *both* prefer. Trading inside the lens helps both. Trading stops being mutually helpful only when the two indifference curves touch without crossing, that is, when their slopes (the rates at which each would swap apples for bread) are equal. The locus of all such tangencies is the **contract curve**, the set of Pareto-efficient splits.

A price line through the endowment is a budget: at price ratio p, Ann can sell apples for bread along it. Each person picks their favourite point on the line, so each sets their own rate of substitution equal to the *same* price. That is why markets hit the contract curve. Everyone's private trade-off is aligned to one public number, and nobody needs to know anyone else's preferences.

Voting fails to do this because a ranking carries no rate of exchange. Arrow's independence axiom says society's verdict on a versus b may depend only on who prefers a to b, not on *how much*. Prices express intensity, and ballots by construction don't. Both theorems are about how much information an aggregation rule is allowed to use.

96.3 The mathematics

Preferences and utility

> **Preference relation**
>
> A **preference** on a set X of bundles is a relation $\succsim$ ("at least as good as") that is **complete** (for all x, y: $x \succsim y$ or $y \succsim x$) and **transitive**. A function $u : X \to \mathbb{R}$ **represents** it if $x \succsim y \iff u(x) \geq u(y)$.

On a finite set every preference has a utility (rank the bundles and number them). On $\mathbb{R}^2$ the natural-looking **lexicographic** order, "more apples is better, and bread only breaks ties", has none.

Proof. Suppose u represents it. For each apple quantity $x \in \mathbb{R}$, $(x, 1) \succ (x, 0)$, so the interval $I_x = (u(x,0), u(x,1))$ is non-empty and contains a rational q_x. If $x < x'$ then $(x, 1) \prec (x', 0)$, so I_x lies entirely below $I_{x'}$ and $q_x \neq q_{x'}$. That gives an injection from $\mathbb{R}$ into $\mathbb{Q}$, which is impossible because $\mathbb{R}$ is uncountable (Ch. 64). ∎

Debreu proved in 1954 that **continuity** (strict-preference sets are open) is enough to guarantee a continuous utility, which is what licenses calculus in economics.

Exchange, efficiency and equilibrium

Let consumers $i = 1, \dots, I$ have endowments $\boldsymbol{\omega}_i \in \mathbb{R}^L_{\geq 0}$ of L goods. An **allocation** $(\mathbf{x}_i)$ is **feasible** if $\sum_i \mathbf{x}_i = \sum_i \boldsymbol{\omega}_i$. It is **Pareto efficient** if no feasible allocation makes everyone weakly better off and someone strictly better off, the multi-objective optimality of §23.9.

Walrasian (competitive) equilibrium

A price vector $\mathbf{p}$ and a feasible allocation $(\mathbf{x}_i^*)$ such that each $\mathbf{x}_i^*$ is a best bundle for i in the budget set $\{\mathbf{x} \geq 0 : \mathbf{p}\cdot\mathbf{x} \leq \mathbf{p}\cdot\boldsymbol{\omega}_i\}$.

Ann and Bob

Ten apples and ten loaves in total. Ann starts with $(8,2)$ and has $u_A = x^{0.6}y^{0.4}$; Bob starts with $(2,8)$ and has $u_B = x^{0.4}y^{0.6}$. Fix the bread price at 1 and the apple price at p. Maximising $x^a y^{1-a}$ on a budget I by a Lagrange multiplier (§6.11) gives the Cobb–Douglas rule "spend the fraction a on apples": $x = aI/p$. Ann's income is $8p+2$ and Bob's is $2p+8$. Apples clear when

$$\frac{0.6(8p+2)+0.4(2p+8)}{p} = 10 \iff 5.6p + 4.4 = 10p \iff p = 1.$$

So Ann ends with $(6,4)$ and Bob with $(4,6)$. Equal rates of substitution, $\frac{0.6}{0.4}\frac{y}{x} = \frac{0.4}{0.6}\frac{10-y}{10-x}$, give the contract curve $y = 8x/(18-x)$, and $(6,4)$ lies on it: $48/12 = 4$.

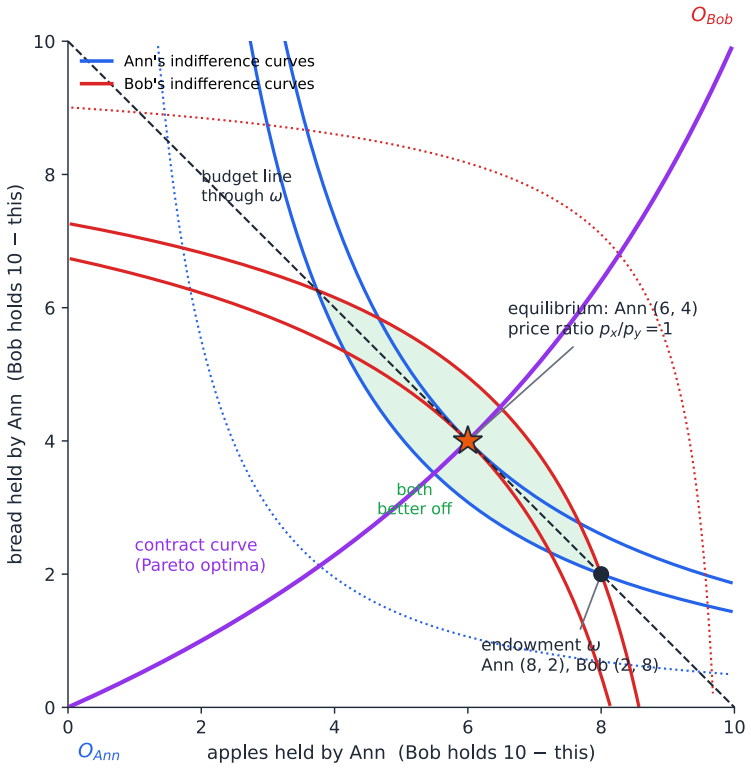


Figure 96.1 — The Edgeworth box. Trades in the green lens help both. The price line through the endowment meets the contract curve exactly where both indifference curves are tangent to it.

First fundamental theorem of welfare economics

If preferences are **locally non-satiated** (near any bundle there is a strictly better one), every Walrasian equilibrium allocation is Pareto efficient.

Proof. Let $(\mathbf{p}, \mathbf{x}^*)$ be an equilibrium and suppose a feasible $(\mathbf{y}_i)$ Pareto-dominates it. If $\mathbf{y}_j \succ_j \mathbf{x}_j^*$, then $\mathbf{y}_j$ was unaffordable, so $\mathbf{p}\cdot\mathbf{y}_j > \mathbf{p}\cdot\boldsymbol{\omega}_j$. If $\mathbf{y}_i \succsim_i \mathbf{x}_i^*$, then $\mathbf{p}\cdot\mathbf{y}_i \geq \mathbf{p}\cdot\boldsymbol{\omega}_i$. Otherwise $\mathbf{y}_i$ would cost strictly less than the budget, a slightly better bundle nearby would still be affordable, and it would beat $\mathbf{x}_i^*$, contradicting optimality. Summing over all consumers, with at least one strict inequality,

$$\mathbf{p}\cdot\sum_i \mathbf{y}_i > \mathbf{p}\cdot\sum_i \boldsymbol{\omega}_i,$$

which contradicts feasibility $\sum_i \mathbf{y}_i = \sum_i \boldsymbol{\omega}_i$. ∎

Notice what the proof did *not* use: convexity, differentiability, or anyone knowing anyone else's tastes. Prices did all the coordinating. Two companions complete the picture. The **second welfare theorem** says that with convex preferences *every* Pareto optimum is an equilibrium after suitable lump-sum transfers. Its proof is a separating hyperplane, whose normal vector *is* the price (Ch. 47, Ch. 48). **Existence** of an equilibrium (Arrow and Debreu, 1954) is a fixed-point theorem: the map "raise prices of over-demanded goods" sends the price simplex continuously into itself, the same kind of fixed-point argument (Brouwer's theorem, or Kakutani's extension of it) that gives Nash equilibria (Ch. 39).

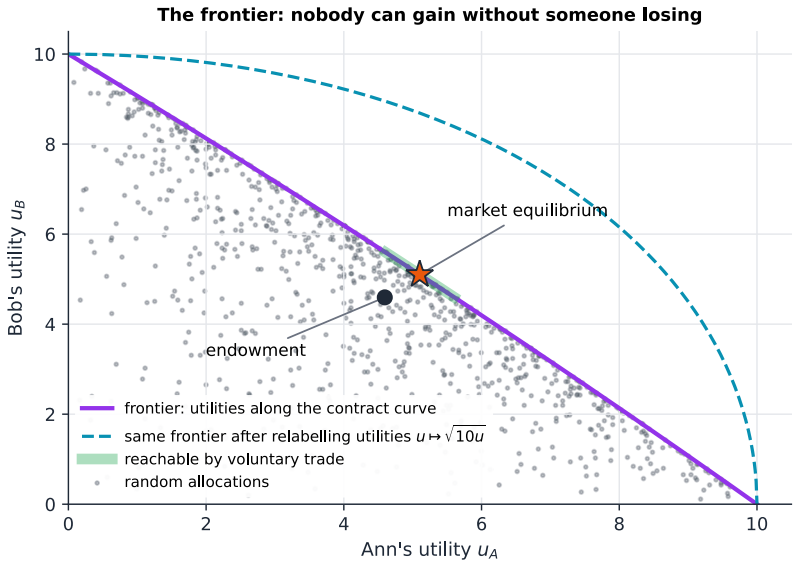


Figure 96.2 — The utility-possibility frontier. The market lands on it, but only somewhere in the short green segment the endowment allows. Its shape is not even meaningful: relabelling utilities (same preferences) bends a straight line into a circle.

Figure 96.2 already hints at the hook's answer. Efficiency is a very weak demand. Every point of the frontier is efficient, including "Ann gets everything". Choosing *among* efficient points is a question of justice, and it requires comparing Ann's welfare with Bob's, a comparison that ordinal utility forbids.

Arrow's theorem, proved

Let A be a set of at least three alternatives and $N = \{1, \dots, n\}$ the voters. A **social welfare function** F maps each profile of strict individual rankings to a strict social ranking. The axioms (stated in §42.3) are **unanimity** (if all rank a above b, so does society) and **independence of irrelevant alternatives** (IIA: society's a-versus-b verdict depends only on each voter's a-versus-b verdict).

Arrow (1951)

If $|A| \geq 3$ and F satisfies unanimity and IIA, some voter n^* is a dictator: society ranks a above c whenever n^* does.

Proof (following Geanakoplos, 2005).

Step 1: extremes stay extreme. Suppose every voter puts alternative b at the very top or very bottom of their ranking. We claim society does too. If not, there are a, c with $a \succ b \succ c$ socially. Change every voter's ranking by moving c just above a. No voter's a-versus-b or b-versus-c comparison changes, because b is extreme. By IIA society still has $a \succ b$ and $b \succ c$, so $a \succ c$ by transitivity. But now every voter has c above a, and unanimity demands $c \succ a$. Contradiction.

Step 2: find the pivot. Start with a profile where everyone ranks b last, so society does (unanimity). Move b to the top of voter 1's ranking, then voter 2's, and so on. At the end everyone ranks b first and so does society. Let n^* be the first voter whose move changes society's ranking of b. By Step 1 it jumps from bottom to top. Call the profiles just before and just after this move I and II.

Step 3: n^* dictates every pair not involving b. Take $a, c \neq b$ and any profile in which n^* ranks a above c. Rearrange it, without changing anyone's a-versus-c order, so that n^* ranks $a \succ b \succ c$, voters before n^* put b first and voters after put b last. On the pair $\{a, b\}$ every voter now agrees with profile I, where b was socially last, so by IIA society has $a \succ b$. On $\{b, c\}$ every voter agrees with profile II, where b was socially first, so $b \succ c$. Hence society has $a \succ c$. Everyone except n^* ranked a and c arbitrarily, so by IIA society follows n^* on $\{a, c\}$.

Step 4: n^* dictates pairs involving b too. Repeat Steps 2–3 with a third alternative c in the role of b. This produces a voter who dictates every pair not involving c, including $\{a, b\}$. But going from profile I to profile II, only n^* changed her ranking, and society's verdict on a versus b flipped. So nobody other than n^* can dictate $\{a, b\}$, and the new dictator is n^* again. ∎

Kinship with Gibbard–Satterthwaite. Given a strategy-proof voting rule f that picks a winner and can elect anyone, define a social ranking by "a above b if f elects a when every voter lifts a and b (in their own order) to the top of their ballot". Strategy-proofness implies that this ranking satisfies IIA and is transitive, and ontoness gives unanimity. Arrow then supplies a dictator, and that dictator's favourite always wins under f. The pivotal voter of Step 2 is the "voter whose ranking flips society's choice" that Ch. 42 promised.

Why markets escape: an exchange economy isn't a social welfare function on arbitrary rankings. Preferences are restricted (more is better, convexity), the outcome is an allocation rather than a ranking, and prices use the *marginal rates* that IIA throws away. Restrict preferences for voting too, for example to **single-peaked** ones (each voter has an ideal point on a left–right line), and majority rule becomes transitive again (Black, 1948).

? How often can a voter cheat?

Gibbard–Satterthwaite says every non-dictatorial rule is manipulable at *some* profile. Is manipulation common or vanishingly rare? Friedgut, Kalai and Nisan (2008), and then Mossel and Rácz (2015) for any number of alternatives, proved that for any rule far from every dictatorship, a random profile is manipulable with probability at least an inverse polynomial in n and the number of alternatives. The proofs use Fourier analysis on the group $\{0,1\}^n$, which is the representation theory of Ch. 95. What the *sharp* bound is remains open.

Bargaining as a game: Rubinstein

Two players split a pie of size 1. Player 1 offers a split, and player 2 accepts or rejects. After a rejection, one period passes and player 2 makes an offer, and so on forever. A payoff received t periods from now is worth δ^t of its face value, with $0 < \delta < 1$.

> **Rubinstein (1982)**
>
> The game has a unique subgame-perfect equilibrium. The first proposer gets $\dfrac{1}{1+\delta}$, the responder accepts at once, and nothing is wasted.

Proof (Shaked and Sutton). Every subgame where a given player proposes is the same game. Let M and m be the supremum and infimum of the proposer's payoff over all subgame-perfect equilibria. A responder who rejects becomes the proposer next period, worth at most δM today. So any offer giving her more than δM is accepted, and the proposer can secure nearly $1-\delta M$: $m \ge 1-\delta M$. Rejecting is worth at least δm to the responder, so she never accepts less, and the proposer gets at most $1-\delta m$ (a rejected offer gives the proposer at most $\delta(1-m) \le 1-\delta m$): $M \le 1-\delta m$. Combining,

$$M \le 1-\delta m \le 1-\delta(1-\delta M) \implies M(1-\delta^2) \le 1-\delta \implies M \le \frac{1}{1+\delta},$$

and then $m \ge 1-\delta M \ge \frac{1}{1+\delta}$. So $M = m = \frac{1}{1+\delta}$. ∎

The first-mover advantage $\frac{1}{1+\delta} - \frac{\delta}{1+\delta}$ vanishes as offers come faster. With different patience rates, $\delta_i = e^{-r_i\Delta}$ for a period of length Δ, the stationary equations $x_1 = 1-\delta_2 x_2$, $x_2 = 1-\delta_1 x_1$ give

$$x_1 = \frac{1-\delta_2}{1-\delta_1\delta_2} = \frac{1-e^{-r_2\Delta}}{1-e^{-(r_1+r_2)\Delta}} \longrightarrow \frac{r_2}{r_1+r_2} \quad (\Delta \to 0).$$

That is precisely the maximiser of the weighted Nash product $x^{1/r_1}(1-x)^{1/r_2}$. With equal patience it is the symmetric Nash solution $\frac{1}{2}$ of Ch. 44. Nash's axioms are what impatience looks like in the limit, and patience is bargaining power.

Two more debts: Shapley and Myerson–Satterthwaite

Why Shapley's axioms leave no freedom. Every game v on N is uniquely a combination $v = \sum_{T \ne \varnothing} c_T\, u_T$ of **unanimity games** $u_T(S) = 1$ if $T \subseteq S$, else 0 (solve for the "dividends" c_T by inclusion–exclusion). In $c\,u_T$, players outside T are null and those inside are symmetric, so efficiency forces $c/|T|$ each. Additivity then fixes the value of every game: $\phi_i(v) = \sum_{T \ni i} c_T/|T|$. For the microgrid of Ch. 44 the dividends are $c_A = 10$, $c_B = 20$, $c_C = 0$, $c_{AB} = 10$, $c_{AC} = c_{BC} = 40$, $c_{ABC} = -30$, and A gets $10 + 5 + 20 - 10 = 25$, as before. Drop additivity and you get the nucleolus, the Banzhaf index and other values, each defensible and each different.

Why efficient trade can't balance its budget. A seller with cost c and a buyer with value v, both independent and uniform on $[0,1]$, want to trade whenever $v > c$. For any incentive-compatible mechanism that both agree to join, the payment identity of Ch. 42 (Myerson's lemma, applied to each side) shows that an exactly balanced budget requires expected **virtual surplus**

$$E\Big[\big(\underbrace{2v-1}_{\text{buyer's virtual value}} - \underbrace{2c}_{\text{seller's virtual cost}}\big)\, q(v,c)\Big] \;\ge\; 0,$$

where q is the probability of trade. For efficient trade, $q = \mathbf{1}\{v > c\}$, and the left side is $\int_0^1 \int_0^v (2v-1-2c)\,dc\,dv = \int_0^1 (v^2 - v)\,dv = -\frac{1}{6} < 0$. So no such mechanism exists (Myerson and Satterthwaite, 1983). The best achievable rule trades only when $v - c \ge \frac{1}{4}$. It makes the integral exactly 0 and captures $\frac{9}{64}$ of the $\frac{1}{6}$ available gains, about 84%.

96.4 The algorithm

Arrow's theorem is small enough to check by brute force. With two voters and three alternatives, IIA means a rule is just one function per pair from "(voter 1 prefers x?, voter 2 prefers x?)" to "society prefers x?". Unanimity fixes two of the four inputs, so each pair has two free bits and there are $2^6 = 64$ candidate rules. Keep those that never produce a cycle on any of the $6^2 = 36$ profiles:

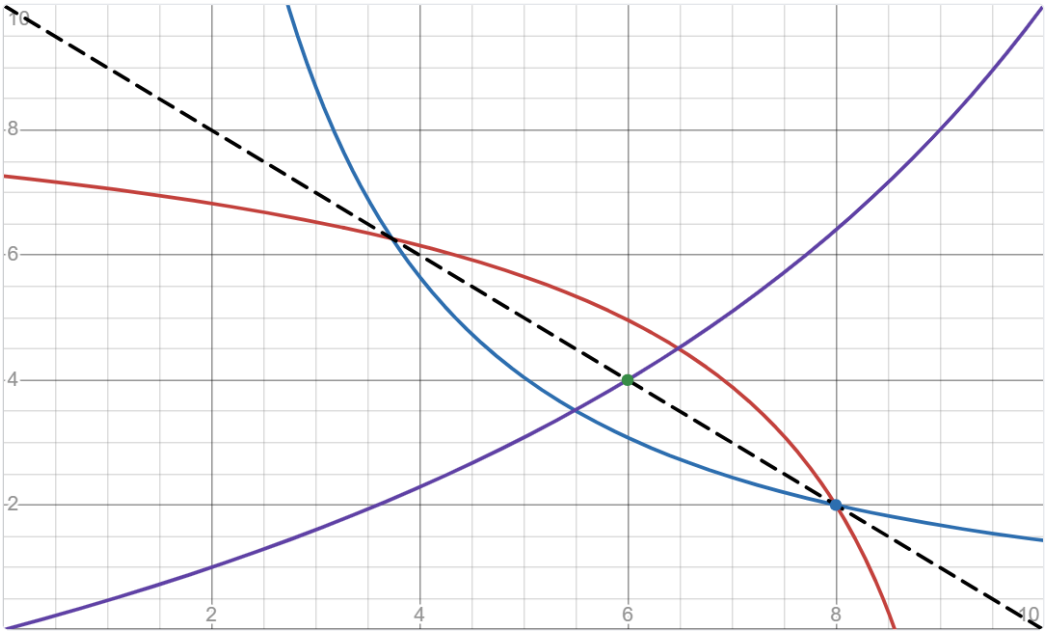
```
from itertools import permutations, product, combinations

alts = "abc"
rankings = list(permutations(alts))                 # 6 strict rankings
pairs = list(combinations(alts, 2))                 # (a,b), (a,c), (b,c)
prefers = lambda r, x, y: r.index(x) < r.index(y)

survivors = []
for bits in product([False, True], repeat=6):
    rule = {p: {(True, True): True, (False, False): False,
                (True, False): bits[2 * i], (False, True): bits[2 * i + 1]} for i, p in enumerate(pairs)}
    def social(profile, x, y):                      # does society rank x above y?
        if (x, y) in rule:
            return rule[(x, y)][tuple(prefers(r, x, y) for r in profile)]
        return not rule[(y, x)][tuple(prefers(r, y, x) for r in profile)]
    ok = all(not (social(P, x, y) and social(P, y, z) and social(P, z, x))
             for P in product(rankings, repeat=2) for x, y, z in permutations(alts))
    if ok:
        survivors.append(bits)
print(len(survivors), "transitive rules out of 64:", survivors)
```

Exactly two rules survive: "voter 1 decides every pair" and "voter 2 decides every pair". For three voters the same search has 2^{18} candidates, and it is still instant. Brute force grows doubly exponentially, though: n voters give $2^n - 2$ free bits per pair. Computer proofs of impossibility theorems (Tang and Lin, 2009, using a SAT solver) work by reducing to such a small base case and proving an induction step by hand.

Computing a market equilibrium is harder than checking one. The natural algorithm is **tâtonnement** ("groping"), Walras' own idea: raise the price of every good in excess demand, lower it for goods in excess supply, and repeat. Each step costs one evaluation of the excess-demand function $\mathbf{z}(\mathbf{p}) = \sum_i \big(\mathbf{x}_i(\mathbf{p}) - \boldsymbol{\omega}_i\big)$. For economies where goods are **gross substitutes** (raising one price raises demand for every other good), tâtonnement converges. In general it need not, and Scarf's algorithm (1967), which follows a path through a triangulated price simplex, was the first method guaranteed to find an approximate fixed point.

96.5 The experiment

First, drag the Edgeworth box. Ann's weight on apples is 0.6 and Bob's is 0.4; the endowment is $(8, 2)$.



Ann's (blue) and Bob's (red) indifference curves through the endowment, the contract curve (purple) and the budget line at price ratio 1.

Change the endowment $(8, 2)$ to $(9, 5)$ (replace both $8^{0.6} \cdot 2^{0.4}$ and $2^{0.4} \cdot 8^{0.6}$ by the new holdings, Ann's in the first line and Bob's $(1, 5)$ in the second). The lens moves, but the contract curve does not, because it depends only on preferences. Then change the weights 0.6 and 0.4 to equal values 0.5 and watch the contract curve straighten into the diagonal.

Then run tâtonnement, first on the box economy and then on Herbert Scarf's 1960 economy. In Scarf's economy three consumers each own one good and want two goods in fixed proportions.

```
import numpy as np

# Tatonnement: raise the price of whatever is over-demanded, lower it if under-demanded.
a, b = 0.6, 0.4                 # Cobb-Douglas weights on apples for Ann and Bob
wA = np.array([8.0, 2.0])       # Ann's endowment (apples, bread); Bob has the rest
wB = np.array([10.0, 10.0]) - wA

def excess_box(p):              # p = (p_apple, p_bread)
    IA, IB = p @ wA, p @ wB
    xA = np.array([a * IA / p[0], (1 - a) * IA / p[1]])
    xB = np.array([b * IB / p[0], (1 - b) * IB / p[1]])
    return xA + xB - wA - wB

def excess_scarf(p):            # Scarf (1960): consumer i owns good i, wants goods i and i+1 in fixed 1:1 ratio
    z = -np.ones(3)
    for i in range(3):
        k = (i + 1) % 3
        q = p[i] / (p[i] + p[k])          # consumer i's income buys q units of each of goods i and k
        z[i] += q; z[k] += q
    return z

def tatonnement(excess, p, h=0.05, steps=4000):
    path = [p.copy()]
    for _ in range(steps):
        p = np.maximum(p + h * excess(p), 1e-9)
        p = p / p.sum()                   # only relative prices matter
        path.append(p.copy())
    return np.array(path)

box = tatonnement(excess_box, np.array([0.8, 0.2]))
print("box economy: final prices", box[-1].round(4), " excess demand", excess_box(box[-1]).round(6))

scarf = tatonnement(excess_scarf, np.array([0.5, 0.3, 0.2]))
dist = np.linalg.norm(scarf - 1 / 3, axis=1)
print("Scarf economy: distance from equilibrium (1/3,1/3,1/3) at steps 0, 1000, 2000, 4000:",
      dist[[0, 1000, 2000, 4000]].round(4))
```

The box economy settles at equal prices $(0.5, 0.5)$, our $p = 1$. Scarf's economy has a unique equilibrium at equal prices, yet the distance from it *grows* from 0.22 to about 0.5, and prices circle the equilibrium without ever approaching it. Things to try: shrink the step `h` to 0.005 (does the spiral stop growing?), start Scarf's prices at exactly $(1/3, 1/3, 1/3)$ plus a tiny nudge, and plot `scarf[:, 0]` against `scarf[:, 1]` to see the orbit.

? The market that never settles

In continuous time Scarf's prices move on closed orbits around the equilibrium forever. His economy is perfectly well behaved (unique equilibrium, continuous demand), yet the invisible hand never finds it. Whether an equilibrium is *stable* is a question about a dynamical system $\dot{\mathbf{p}} = \mathbf{z}(\mathbf{p})$, and centres, spirals and limit cycles are classified in Ch. 83.

96.6 The challenge

1. **Intuition.** In the Edgeworth box, why must the contract curve pass through both corners? Is the equilibrium's position on the contract curve determined by preferences, by endowments, or by both?
2. **Derivation.** Borda count (Ch. 44) satisfies unanimity and isn't a dictatorship, so by Arrow it violates IIA. Construct an explicit three-voter, three-candidate pair of profiles in which no voter changes her a-versus-b order but Borda's social a-versus-b order flips.
3. **Implementation.** Extend the brute-force Arrow search to three voters (2^{18} rules). Then drop unanimity (all 4 inputs free per pair, 2^{24} rules for three voters) and list the transitive IIA rules that survive. Wilson's theorem (1972) predicts what they are: describe them.
4. **Open investigation.** Run tâtonnement on random exchange economies with three goods and Cobb–Douglas, Leontief or CES consumers. Estimate how often it fails to converge as preferences move from substitutes towards complements. No general characterisation of the economies on which simple price dynamics converge is known.

96.7 Loose threads

The hook asked how markets can aggregate preferences perfectly while votes cannot aggregate them at all. Now we can say. The first welfare theorem asks for very little (efficiency only, which even "Ann gets everything" satisfies), and it gets to use prices, which encode marginal rates of substitution. Arrow asks for much more, a full social ranking that works on every possible profile, while IIA forbids it to use intensity. Prices carry information that ballots throw away, and markets promise something that ballots are asked to exceed. Rubinstein added that even raw haggling, given impatience, ends at Nash's axiomatic split. Myerson–Satterthwaite added a limit on what any market can do when people hold private information.

What we didn't tell you:

- **Common values and the winner's curse.** In the auctions of Ch. 43, when an item is worth the same unknown amount to everyone, winning means you were the most optimistic. Milgrom and Weber (1982) made "affiliated" signals precise: high signals make other high signals more likely. They proved the **linkage principle**, which says that formats tying the price more closely to other bidders' information raise expected revenue. So the English auction beats the second-price auction, which beats the first-price. The 2020 Nobel prize was awarded for the spectrum auctions built on this.
- **Which equilibrium is reasonable?** The Intuitive Criterion of Ch. 41 kills implausible beliefs in signalling games, but stronger refinements (divinity, stability in the sense of Kohlberg and Mertens) can disagree with it, and experiments find people often play the "unreasonable" pooling equilibria. Whether refinement theory describes behaviour or only consistency is still argued.
- **Large markets.** Debreu and Scarf (1963) proved that as an economy is replicated, the **core** (allocations no coalition can block) shrinks to the set of Walrasian equilibria. With many traders, competitive prices are the only allocations that survive coalitional bargaining.

? Anything can happen, and finding it is hard

Individual demand functions have strong structure, but the Sonnenschein–Mantel–Debreu theorem (1972–74) says aggregate excess demand can be essentially *any* continuous function satisfying Walras' law. So economic theory alone predicts almost nothing about the number, location or stability of equilibria. And finding one is, in general, as hard as finding a Brouwer fixed point: computing equilibria of exchange economies is PPAD-complete. What PPAD is, and why guaranteed existence doesn't make search easy, is in Ch. 30.

Each of these is a doorway into the economics research literature. The proofs you have just seen, an inequality summed over consumers, a pivot hunted down by moving one voter at a time, a sup and an inf squeezed together, are typical of how that literature works: short arguments around a clever choice of what to compare. How such arguments are found is the subject of Ch. 98.

Further reading

- A. Mas-Colell, M. D. Whinston, J. R. Green, *Microeconomic Theory*, Oxford University Press, 1995. Chapters 1–3, 15–17 and 21–23 cover everything here at graduate level.
- K. J. Arrow, *Social Choice and Individual Values*, 3rd ed., Yale University Press, 2012. The original thesis, with later reflections.
- J. Geanakoplos, "Three brief proofs of Arrow's impossibility theorem", *Economic Theory* 26 (2005) 211–215. The proof given above, and two others.
- M. J. Osborne and A. Rubinstein, *Bargaining and Markets*, Academic Press, 1990 (free online). Rubinstein's model and its connection to Nash and to markets.
- H. Moulin, *Fair Division and Collective Welfare*, MIT Press, 2003. Shapley values, bargaining and fair division in one framework.

Chapter 97 — Measure Theory and Modern Probability

> **Level 3 · Track J — Mathematics at the Frontier Builds on:** Ch. 5, Ch. 19, Ch. 31, Ch. 36, Ch. 60 · **Leads to:** Ch. 98, Ch. 89 · **Related:** Ch. 35, Ch. 55, Ch. 61, Ch. 73, Ch. 80 **Reading time:** ~30 min

97.1 The hook

Throw an ideal dart at the interval $[0, 1]$, so that every point is equally likely. What is the probability that it lands *exactly* on 0.5?

It must be zero. If it were some $\varepsilon > 0$, the same would be true for every point, and a million points would already carry total probability $10^6\varepsilon$, which exceeds 1 once $\varepsilon > 10^{-6}$. So every single point has probability 0, and yet the dart lands *somewhere*, with probability 1. Zero plus zero plus zero, uncountably many times, makes one.

Fine, you say: probabilities add for countably many disjoint events, not uncountably many. Then the rationals in $[0, 1]$, which are countable, have total probability $0 + 0 + \cdots = 0$. The dart almost surely misses every rational, even though rationals are dense and lie between any two points it could hit.

Now two harder questions.

Can every set of points be given a probability? Not "every set you can describe", but *every* subset of $[0, 1]$. Surely the dart either lands in a set or it doesn't, so "the probability it lands in S" should mean something for every S.

Can a fair game be beaten? Bet 1 on a fair coin. If you lose, double the stake; stop at the first win. You stop with probability 1, and when you stop you are up exactly 1. Your fortune is a fair game (a martingale, Ch. 36), so its expectation should stay 0. Yet it ends at 1 with certainty.

The first question has the answer **no**, and the proof is half a page. The second has the answer "only if you have infinite credit", but saying precisely *why* requires rebuilding probability. In 1933 Andrei Kolmogorov did exactly that, by declaring that probability *is* a measure: a rule that assigns sizes to a carefully chosen family of sets. This chapter shows why the family has to be chosen, and what the rebuilding buys: a strong law of large numbers that needs only a mean, a precise version of "you can't beat a fair game", distributions with no density and no jumps, and a notion of dimension for wild sets.

97.2 The big idea

The Riemann integral of Ch. 5 slices the *domain* into thin strips and adds height × width. Lebesgue sliced the *range* instead (§60.3). For each height y, ask "how big is the set of x where $f(x) > y$?", and add up those sizes over all heights:

$$\int f\,dm = \int_0^\infty m\{x : f(x) > y\}\,dy \qquad (f \geq 0).$$

A banker counting a pile of coins does the same: rather than going through the coins in the order they lie (Riemann), she sorts them by denomination and multiplies each value by how many coins have it (Lebesgue). The order of the coins no longer matters, only the *size* of each set $\{f > y\}$. That is why the Lebesgue integral copes with functions that jump about wildly.

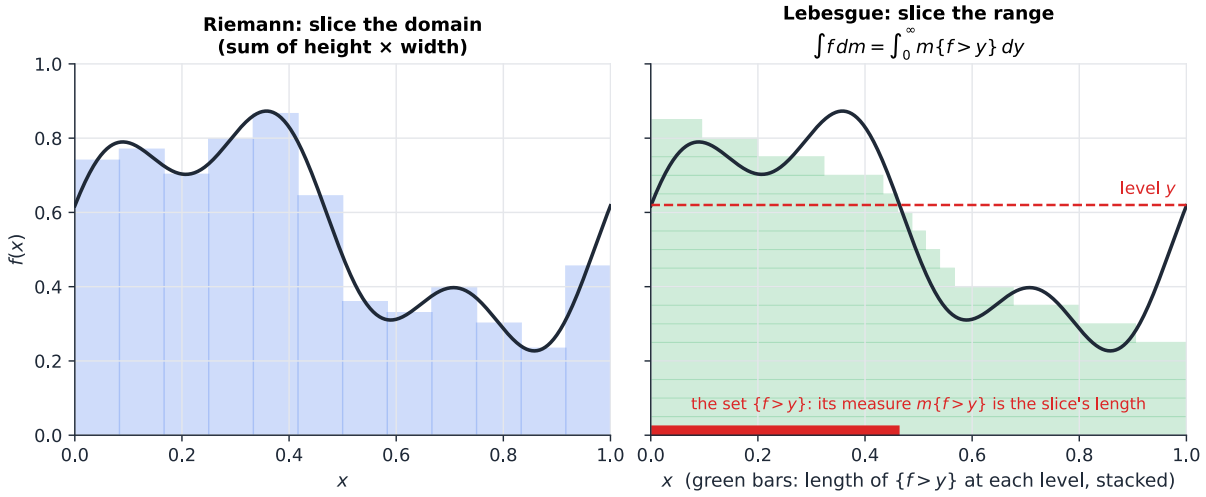


Figure 97.1 — Riemann versus Lebesgue. The Lebesgue integral needs only one thing, the size $m\{f > y\}$ of each "level set", however scattered that set is.

So everything rests on measuring sets. A **measure** is a size function that adds up correctly over countably many disjoint pieces. A **probability** is a measure with total size 1. A **random variable** is a function whose level sets can be measured, and its **expectation** is its Lebesgue integral. The family of sets we are allowed to measure, a **σ-algebra**, turns out to be more than a technicality: in probability it represents *information*. "The sets I can decide after seeing the first n coin tosses" is a σ-algebra, and a gambler's strategy may only use that one. This is how measure theory solves the doubling puzzle.

97.3 The mathematics

σ-algebra, measure, probability space

A **σ-algebra** on a set Ω is a family $\mathcal{F}$ of subsets that contains Ω and is closed under complements and countable unions. A **measure** is a map $\mu : \mathcal{F} \to [0,\infty]$ with $\mu(\varnothing) = 0$ and **countable additivity**: $\mu\big(\bigcup_k A_k\big) = \sum_k \mu(A_k)$ for disjoint $A_1, A_2, \ldots \in \mathcal{F}$. A **probability space** $(\Omega, \mathcal{F}, P)$ is a measure space with $P(\Omega) = 1$ (Kolmogorov, 1933). A **random variable** is a function $X : \Omega \to \mathbb{R}$ with $\{X \le t\} \in \mathcal{F}$ for every t, and $E[X] = \int_\Omega X \, dP$.

Countable additivity gives continuity for free. If $A_1 \subseteq A_2 \subseteq \cdots$, then writing $\bigcup A_n$ as the disjoint union of the rings $A_n \setminus A_{n-1}$ gives $\mu(\bigcup A_n) = \lim \mu(A_n)$. Almost every limit theorem in probability starts from this one line.

Result 1: you cannot measure every set

Vitali (1905)

There is no function μ defined on *all* subsets of the circle $[0,1)$ (with addition mod 1) that is countably additive, invariant under rotation, and has $\mu([0,1)) = 1$.

Proof. Call $x \sim y$ if $x - y$ is rational. This splits $[0,1)$ into equivalence classes, each a rotated copy of $\mathbb{Q} \cap [0,1)$. Using the axiom of choice, pick exactly one point from each class, and call the resulting set V. For each rational $q \in [0,1)$ let $V_q = V + q \pmod 1$.

The sets V_q are disjoint: if $v + q \equiv v' + q'$ with $v, v' \in V$, then $v \sim v'$, so $v = v'$ (one point per class) and $q = q'$. They cover $[0,1)$: any x is equivalent to some $v \in V$, and $x - v \equiv q$ for a rational q. So $[0,1)$ is the disjoint union of countably many rotated copies of V. If μ existed, rotation invariance would make all $\mu(V_q)$ equal to $\mu(V)$, and countable additivity would give

$$1 = \mu([0,1)) = \sum_{q} \mu(V_q) = \sum_{q} \mu(V) \in \{0, \infty\}.$$

Contradiction. ■

This is why probability needs σ-algebras: we must give up measuring *some* sets. The good news (Carathéodory's extension theorem) is that we lose nothing we care about. Length on intervals extends uniquely to a countably additive measure on the **Borel** σ-algebra, the smallest one containing all intervals, which includes every set you can build from intervals by countably many unions, intersections and complements. Vitali's set needed the axiom of choice to exist at all, which answers the question Ch. 60 left open. Solovay showed in 1970 that, if one gives up full choice (and assumes a large cardinal is consistent), it is consistent that *every* set of reals is Lebesgue measurable.

The same extension builds infinite coin-tossing. Write the dart's position in binary, $x = 0.b_1 b_2 b_3 \ldots$. Under Lebesgue measure the digits b_k are independent fair coin flips. So "an infinite sequence of fair tosses" and "a uniform point of $[0,1]$" are the same probability space. The strong law of large numbers for coins becomes **Borel's normal number theorem** (1909): Lebesgue-almost every real number has, in its binary expansion, half its digits equal to 1. Kolmogorov's **extension theorem** generalises this. Any consistent family of finite-dimensional distributions comes from one probability measure on the space of whole paths, and that is how Brownian motion is constructed. The Gaussian finite-dimensional distributions of Ch. 36 define a measure on paths, and a further argument (Kolmogorov's continuity criterion, or Lévy's explicit series of tent functions) shows that the paths can be taken continuous. The resulting **Wiener measure** puts probability 1 on paths that are differentiable nowhere.

? One ball becomes two

Vitali's set is a mild warning. In 1924 Banach and Tarski cut a solid ball into five pieces and, using only rotations and translations, reassembled them into *two* balls each identical to the original. The pieces are non-measurable, so no volume is violated, because none is assigned. In one or two dimensions this is impossible: a finitely additive, motion-invariant "area" exists on every subset of the plane. The difference is that the rotation group of space contains a free group, and the plane's doesn't. Why that algebraic fact breaks geometry, and what it says about the axiom of choice, is in Ch. 68.

Result 2: big sets of size zero, and the dimension between

Build the **Cantor set** C: from $[0,1]$ remove the open middle third $(\tfrac13, \tfrac23)$, then the middle third of each remaining piece, and so on forever. After n steps 2^n intervals of length 3^{-n} remain.

The Cantor set is uncountable and has measure zero

$m(C) = 0$, yet C has as many points as $[0,1]$. Its Hausdorff dimension is at most $\log 2 / \log 3 \approx 0.6309$ (in fact exactly this).

Proof. C is contained in the n-th stage, which has length $(2/3)^n \to 0$, so $m(C) = 0$. A point lies in C exactly when it has a ternary expansion $0.t_1 t_2 t_3 \ldots$ using only the digits 0 and 2. Mapping $t_k \mapsto t_k/2$ turns that into a binary expansion, a surjection from C onto $[0,1]$. So C is uncountable.

For dimension, define the s-dimensional **Hausdorff measure** $\mathcal{H}^s(A) = \lim_{\delta \to 0} \inf \sum_i (\operatorname{diam} U_i)^s$, the infimum taken over countable covers of A by sets U_i of diameter at most δ. The **Hausdorff dimension** is the value of s at which $\mathcal{H}^s$ drops from ∞ to 0. Cover C by its 2^n stage-n intervals:

$$\mathcal{H}^s_{3^{-n}}(C) \le 2^n \, (3^{-n})^s = \left(\frac{2}{3^s}\right)^n.$$

For $s > \log 2 / \log 3$ the base is below 1 and the bound tends to 0, so $\mathcal{H}^s(C) = 0$. Hence $\dim_H C \le \log 2 / \log 3$. The matching lower bound spreads mass evenly over C and shows that no cover can do much better. ■

So Hausdorff measure interpolates between counting ($s=0$), length ($s=1$) and area ($s=2$). It also answers Ch. 73. A set can have zero area and still have $\mathcal{H}^s > 0$ for every $s < 2$, so dimension 2 does not require area. Shishikura proved that the boundary of the Mandelbrot set has Hausdorff dimension 2. Whether it has positive area as well is still an open problem. Box counting uses covers by equal-sized boxes. Hausdorff dimension lets the sizes vary, so it can be smaller, for instance 0 for the countable set $\{0, 1, \frac{1}{2}, \frac{1}{3}, \dots\}$, whose box dimension is $\frac{1}{2}$.

Now spread probability evenly over C. Choose fair bits b_k and set $X = \sum_k 2b_k/3^k$. Its CDF F is the **Cantor function** (Figure 97.2). It is continuous, so X has no atoms: $P(X = x) = 0$ for every x. But F is flat on every removed interval, so $F' = 0$ almost everywhere and X has no density either, since $\int F' = 0 \neq 1$. Discrete or continuous with a density, the dichotomy of §19.7, is not the whole story.

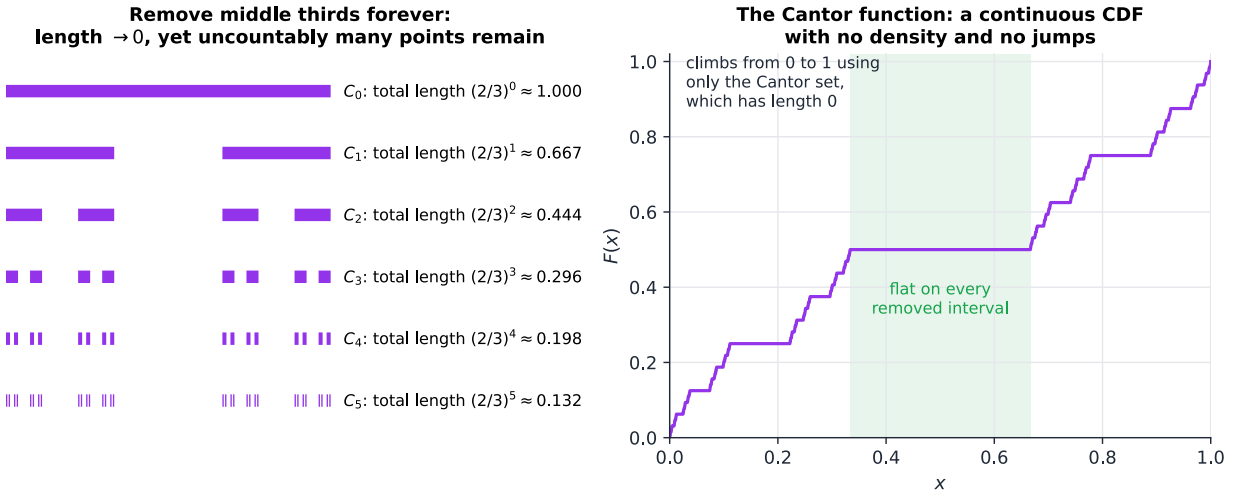


Figure 97.2 — The Cantor set and its distribution function. All the probability lives on a set of length zero, but no single point carries any of it.

The **Lebesgue decomposition theorem** says this is the only other possibility. Every probability distribution on $\mathbb{R}$ is uniquely a mixture of a discrete part, an absolutely continuous part (one with a density), and a **singular continuous** part like the Cantor distribution. The **Radon–Nikodym theorem** says when a density exists: μ has a density $d\mu/dm$ exactly when $m(A) = 0$ forces $\mu(A) = 0$. Likelihood ratios in statistics, change of measure in finance and the Wasserstein couplings of Ch. 55 are all Radon–Nikodym derivatives.

Convergence, made precise

The monotone and dominated convergence theorems were stated in §60.3. Measure theory adds an important distinction between kinds of convergence of random variables:

- **almost surely:** $P(X_n \to X) = 1$;
- **in probability:** $P(|X_n - X| > \varepsilon) \to 0$ for every ε;
- **in L^1:** $E|X_n - X| \to 0$;
- **in distribution:** CDFs converge at continuity points (the central limit theorem's kind).

Almost-sure convergence implies convergence in probability, but not conversely. Sweep an indicator of width $1/k$ repeatedly across $[0,1]$ (widths $1, \frac{1}{2}, \frac{1}{2}, \frac{1}{3}, \frac{1}{3}, \frac{1}{3}, \dots$). The probability of being hit tends to 0, yet every point is hit infinitely often, so the sequence converges at *no* point.

Kolmogorov's strong law. Let $X_1, X_2, \dots$ be independent with the same distribution. Then $\bar{X}_n \to \mu$ almost surely **if and only if** $E|X_1| < \infty$ and $\mu = E[X_1]$. If $E|X_1| = \infty$, then $\limsup |\bar{X}_n| = \infty$ almost surely. Ch. 31 proved the law with four moments. The proof with one moment **truncates**: let $Y_k = X_k \mathbf{1}\{|X_k| \le k\}$. Then

$$\sum_k P(X_k \neq Y_k) = \sum_k P(|X_1| > k) \le E|X_1| < \infty,$$

so by Borel–Cantelli only finitely many X_k differ from Y_k, and it suffices to study the Y_k. These are bounded, with $\sum_k \operatorname{Var}(Y_k)/k^2 < \infty$ (a short computation), and a maximal inequality then gives convergence of $\sum (Y_k - EY_k)/k$, which turns into convergence of averages by Kronecker's lemma. Etemadi (1981) found a proof that needs only pairwise independence. Everything happens on the single probability space of infinite sequences, which is why the measure-theoretic setting is not optional.

Result 3: you cannot beat a fair game, unless you can

Information enters through σ-algebras. A **filtration** $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \cdots$ lists what is known at each time. For coin tossing, $\mathcal{F}_n$ consists of the events decided by the first n tosses.

> **Conditional expectation, martingale, stopping time**
>
> Given X with $E|X| < \infty$ and a sub-σ-algebra $\mathcal{G}$, the **conditional expectation** $E[X \mid \mathcal{G}]$ is the (almost surely unique) $\mathcal{G}$-measurable Y with $E[Y\mathbf{1}_A] = E[X\mathbf{1}_A]$ for every $A \in \mathcal{G}$. It exists by Radon–Nikodym. When $E[X^2] < \infty$ it is the orthogonal projection of X onto the $\mathcal{G}$-measurable variables (§9.9), the best guess of X from the information in $\mathcal{G}$. A **martingale** is a sequence M_n, each $\mathcal{F}_n$-measurable and integrable, with $E[M_{n+1} \mid \mathcal{F}_n] = M_n$. A **stopping time** is a random time τ with $\{\tau \le n\} \in \mathcal{F}_n$: you decide to stop using only the past.

Optional stopping (bounded version)

If M is a martingale and $\tau \le N$ is a bounded stopping time, then $E[M_\tau] = E[M_0]$.

Proof. Write the stopped value as a sum of the increments you were present for:

$$M_\tau = M_0 + \sum_{k=1}^{N} \mathbf{1}\{\tau \ge k\}\,(M_k - M_{k-1}).$$

The event $\{\tau \ge k\}$ is the complement of $\{\tau \le k-1\}$, so it lies in $\mathcal{F}_{k-1}$. Conditioning on $\mathcal{F}_{k-1}$, the indicator is known and comes outside:

$$E\big[\mathbf{1}\{\tau \ge k\}(M_k - M_{k-1})\big] = E\big[\mathbf{1}\{\tau \ge k\}\,E[M_k - M_{k-1} \mid \mathcal{F}_{k-1}]\big] = 0.$$

Sum the finitely many terms. ■

The doubling strategy has an **unbounded** stopping time, and that is the only place it slips through. Its stopped fortune $M_{n\wedge\tau}$ is 1 with probability $1 - 2^{-n}$ and $-(2^n - 1)$ with probability 2^{-n}. That has mean exactly 0 for every n, by the theorem. As $n \to \infty$ it converges almost surely to 1, yet its mean stays 0. This is precisely the escaping-mass spike of §60.3: a loss of about 2^n on an event of probability 2^{-n}, which no integrable function dominates. The general theorem says $E[M_\tau] = E[M_0]$ whenever $\tau < \infty$ almost surely and the stopped process $\{M_{n\wedge\tau}\}$ is **uniformly integrable** (for example, dominated by one integrable variable). A sufficient condition is $E[\tau] < \infty$ together with bounded increments. Doubling has $E[\tau] = 2$, but its increments are unbounded. With any finite credit line, the strategy is a bounded stopping time and the game is fair again.

Martingale convergence (Doob)

If a martingale satisfies $\sup_n E|M_n| < \infty$, then M_n converges almost surely to a finite limit M_∞. If the martingale is uniformly integrable (for instance, bounded), then $M_n \to M_\infty$ in L^1 as well, and $M_n = E[M_\infty \mid \mathcal{F}_n]$.

The idea of Doob's proof is that a martingale which oscillated forever would have to cross some band $[a, b]$ infinitely often. Buying at a and selling at b would then be a winning strategy against a fair game, and an "upcrossing inequality" bounds the expected number of crossings by $\sup E|M_n|/(b-a)$. The doubling fortune is a warning here too: $E|M_{n\wedge\tau}| < 2$, so it converges almost surely (to 1), but not in L^1.

Martingale convergence is how polar codes (Ch. 80) are proved to work. Follow a random path down Arıkan's recursion, choosing the "better" or "worse" synthetic channel by a fair coin at each level. The capacity $I_n \in [0,1]$ of the channel you reach is a bounded martingale, because each split preserves average capacity. So I_n converges almost surely, and in particular $I_{n+1} - I_n \to 0$. But a split changes the capacity by an amount that is zero *only* when the capacity is already 0 or 1. (For the erasure channel this is exact: the split sends I to I^2 or $2I - I^2$.) So the limit is 0 or 1 almost surely: almost every channel polarises to useless or perfect, and by the martingale property the fraction that become perfect is the original capacity.

Bernstein–von Mises, stated properly. For a smooth, identifiable model with finitely many parameters, true parameter θ_0 and a prior with continuous positive density near θ_0,

$$\big\| \Pi(\cdot \mid X_1, \dots, X_n) - \mathcal{N}\big(\hat\theta_n,\, (nI(\theta_0))^{-1}\big) \big\|_{\mathrm{TV}} \;\to\; 0 \quad \text{in } P_{\theta_0}\text{-probability},$$

where $\hat\theta_n$ is the maximum-likelihood estimate and TV is total variation distance between measures. Every word in it ("in probability", "total variation", the conditional law Π) is measure-theoretic. Under misspecification, which was the question of Ch. 35, Kleijn and van der Vaart (2012) showed that the posterior still becomes Gaussian, centred at the KL-closest parameter θ^*. But its covariance is the inverse Hessian of the KL divergence, not the "sandwich" covariance of $\hat\theta_n$, so credible sets have the wrong frequentist coverage.

Carleson's theorem, finally, now has its precise form. If $\int_0^{2\pi} |f|^2 < \infty$, the Fourier partial sums $S_N f(x)$ converge to $f(x)$ for all x outside a Lebesgue-null set (Carleson, 1966; Hunt extended it to L^p, $p > 1$). For L^1 it fails badly: Kolmogorov built an integrable function whose Fourier series diverges almost everywhere (1923), and then one where it diverges everywhere (1926).

? Carleson in two dimensions

For a function on the torus $[0, 2\pi)^2$ there is no single "partial sum": you must choose which frequencies (k_1, k_2) to keep. Keep a growing square, $\max(|k_1|, |k_2|) \le N$, and Carleson's theorem extends (Fefferman, Sjölin, 1971). Keep a growing disc, $k_1^2 + k_2^2 \le R^2$, the most natural choice, and nobody knows whether the partial sums of every square-integrable function converge almost everywhere. Fefferman's 1971 theorem that the disc is a bad Fourier multiplier on L^p for $p \neq 2$ shows how strange discs are, but the L^2 question is **open**.

97.4 The algorithm

A computer never sees a non-measurable set, so measure theory's algorithms are about integrals, and the Lebesgue view gives a practical one. To integrate $f \ge 0$ against a measure μ that you can sample from, draw $X_1, \dots, X_n \sim \mu$, sort the values $f(X_i)$, and integrate the layer-cake formula $\int_0^\infty \mu\{f > y\}\,dy$ using the empirical fractions. Sorting and summing collapses this to the ordinary Monte Carlo average:

```
layer_cake(f, sample, n):
    v = sort([f(sample()) for _ in range(n)])            # O(n log n)
    # mu{f > y} is estimated by the fraction of v above y; integrate over y
    total = 0; prev = 0
    for i, y in enumerate(v):                            # between consecutive values
        total += (y - prev) * (n - i) / n                # height of slice * measure of level set
        prev = y
    return total                                         # equals mean(v) exactly
```

The telescoping identity "slice integral = sample mean" is why Monte Carlo (Ch. 38) is Lebesgue integration in disguise. It never needs f to be continuous, only measurable, and its error is $\sigma/\sqrt{n}$ whatever the dimension, whereas a Riemann grid's error degrades with dimension.

Two more computational facts:

- **Sampling singular distributions.** The Cantor variable is generated digit by digit, $X = \sum_k 2b_k/3^k$. With k digits the error is at most 3^{-k}, so 40 digits already reach the limit of double precision. Its CDF, the Cantor function, is computed the same way: read ternary digits until the first 1, then convert the 2s before it to binary 1s.
- **Computing Hausdorff dimension.** There is no general algorithm. For self-similar sets satisfying a separation condition, Moran's equation $\sum_i r_i^s = 1$ gives the dimension (for C: $2 \cdot 3^{-s} = 1$). For sets like the Mandelbrot boundary, numerical box counting gives only an estimate, and here it estimates a different quantity. Rigorous computer-assisted bounds on Hausdorff dimensions, for example of Julia sets, are an active research area.

97.5 The experiment

The script below tests three claims of this chapter: doubling with a credit limit is a fair game, the Cantor variable has no density and no atoms, and a bounded stopping time cannot beat a random walk.

```
import numpy as np
rng = np.random.default_rng(0)

# 1. The doubling gambler with a credit limit: at most K losses in a row can be covered.
K, trials = 10, 200_000                  # try K = 3, 10, 20
flips = rng.random((trials, K)) < 0.5    # True = win
first_win = np.where(flips.any(axis=1), flips.argmax(axis=1), K)
# stake 2^t at round t; win at round t nets +1, losing all K rounds nets -(2^K - 1)
gain = np.where(first_win < K, 1.0, -(2.0 ** K - 1))
print(f"K={K}: P(ruin) = {np.mean(gain < 0):.5f}, mean gain = {gain.mean():+.4f} "
      f"(+-{gain.std() / np.sqrt(trials):.4f})")

# 2. A random variable with no density and no atoms: X = sum 2*b_k / 3^k, b_k fair bits.
depth, n = 40, 200_000
bits = rng.integers(0, 2, size=(n, depth))
X = (2 * bits / 3.0 ** np.arange(1, depth + 1)).sum(axis=1)
print("mean, variance:", X.mean().round(4), X.var().round(4), " (theory 1/2 and 1/8)")
print("fraction landing in the removed middle third (1/3, 2/3):", np.mean((X > 1/3) & (X < 2/3)))
for L in [1, 3, 6]:                      # cover the support by 2^L intervals of length 3^-L
    cells = np.unique(np.floor(X * 3 ** L))
    print(f"level {L}: occupied cells {len(cells)} = 2^{L}, total length {len(cells) / 3 ** L:.4f}")

# 3. Optional stopping: a bounded stopping time cannot beat a fair game.
walk = np.cumsum(rng.choice([-1, 1], size=(trials, 50)), axis=1)
hit = np.argmax(walk >= 3, axis=1)                      # first time the walk reaches +3 ...
tau = np.where((walk >= 3).any(axis=1), hit, 49)        # ... or step 50, whichever comes first
print("E[M_tau] with tau <= 50:", walk[np.arange(trials), tau].mean().round(3))
```

With the seed above, $K = 10$ gives a ruin frequency near $2^{-10} \approx 0.00098$ and a mean gain of -0.08 ± 0.07, consistent with 0. The Cantor variable has mean 0.5004 and variance 0.1254 (theory $\frac{1}{2}$ and $\frac{1}{8}$), never lands in the removed middle third, and occupies exactly 2^L of the 3^L cells at level L. The stopped walk averages -0.011, again consistent with 0. Things to change:

- **Raise K to 20.** Ruin becomes a one-in-a-million event. Run it several times: most runs show a mean gain of nearly $+1$, and the rare run that contains a ruin shows a huge loss. The expectation is 0, but the *typical* outcome is $+1$. That gap is exactly the failure of uniform integrability.
- **Remove the cap of 50 steps** in part 3 (let the walk run 5000 steps). The proportion of walks that reach $+3$ climbs towards 1, the mean of M_τ climbs towards 3, and optional stopping stops applying.
- **Change the Cantor rule** to digits from $\{0, 1, 2\}$ chosen with probabilities $(p, 0, 1-p)$, or to base 5 keeping two digits. Predict the dimension of the support from Moran's equation, and estimate it from the cell counts as $\log(\text{cells})/(L \log 3)$.

97.6 The challenge

1. **Intuition.** A friend says: "Probability zero means impossible." Using the dart, the rationals and the Cantor set, explain why measure theory distinguishes "probability zero" from "impossible", and why it has to.
2. **Derivation.** Show that the Cantor distribution has mean $\frac{1}{2}$ and variance $\frac{1}{8}$, using only its self-similarity: $X \overset{d}{=} \frac{2b}{3} + \frac{X'}{3}$, where b is a fair bit independent of the copy X'.
3. **Implementation.** Simulate the "sweeping indicator" sequence of §97.3 on a grid of 10^4 points. Show numerically that $P(|X_n| > 0) \to 0$ while every grid point is hit infinitely often. Then find a subsequence that converges almost surely, as a theorem of Riesz guarantees.
4. **Open investigation.** Take a random fractal: start with $[0, 1]$, split every interval into thirds, and keep each third independently with probability p. For which p does the limit set survive, and what is its dimension when it does? Then repeat in 2-D with squares (Mandelbrot's percolation), and explore numerically for which p the surviving set contains a path crossing the square. The exact critical p for crossing is not known.

97.7 Loose threads

The hook asked whether every set can be given a probability, and whether a fair game can be beaten. Measure theory says no to the first, because Vitali's set breaks any countably additive, translation-invariant size. So probability lives on a σ-algebra, and that sacrifice costs nothing we can describe. To the second it says "not with finite credit". A fair game stays fair at every bounded stopping time. The doubling strategy wins only by putting a loss of 2^n on an event of

probability 2^{-n}, the same escaping mass that breaks $\lim \int = \int \lim$. Along the way the σ-algebra turned into information, the Cantor set gave a distribution with neither density nor atoms, and Hausdorff measure gave dimensions between the integers.

What we did not tell you:

- **Stochastic integrals.** Brownian paths have infinite length on every interval, so $\int H_t \, dB_t$ cannot be a Riemann–Stieltjes integral. Itô defined it as an L^2 limit, and its isometry $E[(\int H \, dB)^2] = E \int H^2 \, dt$ is a statement about measures on path space. The resulting calculus, and its link between diffusions and PDEs, is taken up in Ch. 89.
- **L^p spaces.** Identify functions that agree almost everywhere, and the integrable functions form a complete normed space L^1, the square-integrable ones a Hilbert space L^2. This is the quotient promised in Ch. 99, and the setting of Ch. 61.
- **Weak convergence.** The central limit theorem and the Wasserstein distances of Ch. 55 are about convergence of *measures*, not of random variables. Prokhorov's theorem says when a family of measures has convergent subsequences.

? When the parameter is a whole function

Bernstein–von Mises makes Bayesians and frequentists agree, but only for finitely many parameters. In 1999 David Freedman showed that in a simple infinite-dimensional model (estimating a whole function observed in white noise), the posterior can be asymptotically Gaussian yet give credible sets with frequentist coverage tending to zero. Castillo and Nickl (2013–14) recovered versions of the theorem in carefully chosen weaker norms. Which nonparametric questions a posterior answers honestly, and how to choose priors so that it does, is a current research frontier, of exactly the kind Ch. 98 is about.

Further reading

- J. S. Rosenthal, *A First Look at Rigorous Probability Theory*, 2nd ed., World Scientific, 2006. Short and friendly: measure-theoretic probability for readers who have seen calculus.
- D. Williams, *Probability with Martingales*, Cambridge University Press, 1991. Measure theory built towards martingale convergence and optional stopping.
- R. Durrett, *Probability: Theory and Examples*, 5th ed., Cambridge University Press, 2019 (free online). The standard graduate text, including Kolmogorov's and Etemadi's strong laws.
- K. Falconer, *Fractal Geometry: Mathematical Foundations and Applications*, 3rd ed., Wiley, 2014. Hausdorff measure and dimension, with the Cantor set worked in full.
- A. W. van der Vaart, *Asymptotic Statistics*, Cambridge University Press, 1998. Chapter 10 states and proves Bernstein–von Mises.

Chapter 98 — Mathematical Research: How New Mathematics Is Created

> **Level 3 · Track J — Mathematics at the Frontier Builds on:** Ch. 63, Ch. 28, Ch. 26, Ch. 57 · **Leads to:** every open thread in this book (Figure 98.3), and the ones nobody has written down yet · **Related:** Ch. 27, Ch. 30, Ch. 32, Ch. 67, Ch. 68, Ch. 73 **Reading time:** ~30 min

98.1 The hook

Line up the odd primes and split them into two teams. Team Three takes the primes that leave remainder 3 when divided by 4: $3, 7, 11, 19, 23, 31, \ldots$ Team One takes those that leave remainder 1: $5, 13, 17, 29, 37, 41, \ldots$ Now run a race. At each x, count how many primes up to x each team has. Call the counts $\pi(x; 4, 3)$ and $\pi(x; 4, 1)$.

Team Three is ahead at $x = 3$, and still ahead or level at $x = 100$, 1000, 10,000. Chebyshev noticed this in 1853. An experimental scientist would be entitled, by $x = 20{,}000$, to announce a law of nature: *the $4k + 3$ primes always lead.*

The law fails at $x = 26{,}861$, a prime of the form $4k + 1$, where Team One goes ahead by a single prime, for two integers. Then Team Three is back in front until $x = 616{,}841$. In the first ten million integers, Team One is in the lead at only 3,406 values of x. And yet in 1914 J. E. Littlewood proved that the lead changes hands **infinitely often**.

A second race is even crueller. The prime-counting function $\pi(x)$ is approximated by the logarithmic integral $\mathrm{li}(x) = \int_0^x dt/\ln t$, and in every table anyone has ever computed, $\pi(x) < \mathrm{li}(x)$. At $x = 10^7$ the gap is 339 primes. The inequality has been checked by computer for every x up to 10^{19}. Littlewood's same 1914 paper proved that it is false: $\pi(x) - \mathrm{li}(x)$ also changes sign infinitely often. Nobody has ever seen a single crossing. The first one is known to lie somewhere between 10^{19} and about 1.4×10^{316}.

So here is the question this last chapter is about. If a pattern can survive ten quintillion checks and still be false, **how does anyone ever come to know anything new in mathematics?** Where do conjectures come from, how do you decide which ones to believe, how do you hunt for the counterexample that would kill one, and what does it take to turn one into a theorem?

Chapter after chapter of this book ended with a `[!mystery]` pointing here: the exponent ω of Ch. 26, $R(5, 5)$ in Ch. 28 and Ch. 32, the four-colour proof of Ch. 27, the Mandelbrot set of Ch. 73, the zeta zeros of Ch. 57. Nobody can resolve them. This chapter shows how each is being attacked, with methods you could use yourself.

98.2 The big idea

Research is a loop, not a ladder. Textbooks, this one included, present mathematics as a ladder of definitions, theorems and proofs, each rung resting on the last. That is the shape of finished mathematics. Mathematics in the making has the shape of Figure 98.1.

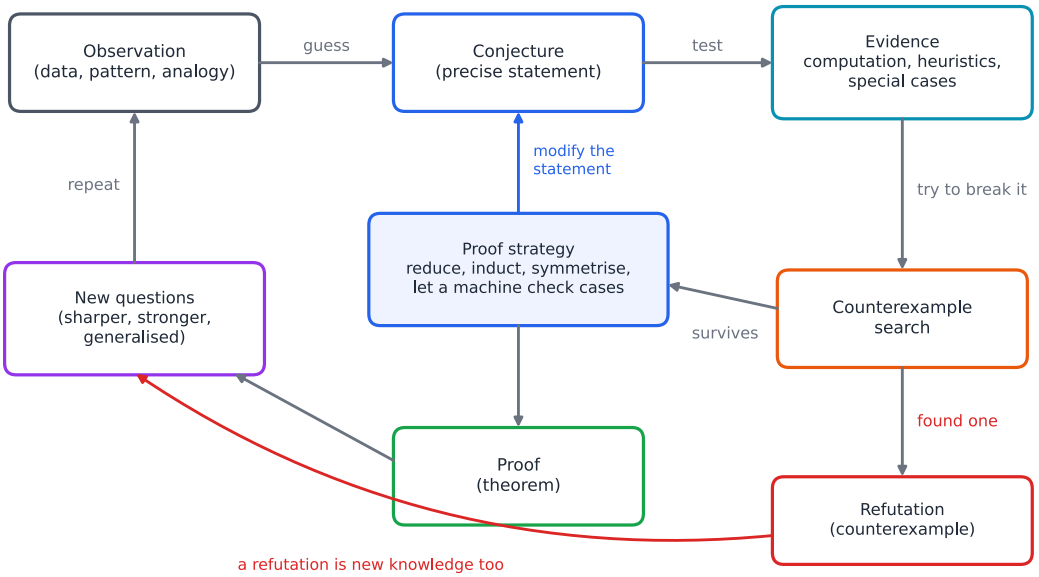


Figure 98.1 — The conjecture lifecycle. Both exits, proof and refutation, lead back to new questions; the loop has no final state.

1. **Observation.** A pattern in a table, a simulation, an analogy between fields, or a proof that works "too well".
2. **Conjecture.** The pattern stated precisely enough to be false. "Primes like remainder 3" is a mood; "$\pi(x; 4, 3) \geq \pi(x; 4, 1)$ for all x" is a conjecture.
3. **Evidence.** Computation, special cases, heuristic models, consistency with other conjectures.
4. **Counterexample search.** You try hard to break it. Most conjectures die here, and that is a success: a counterexample is a theorem too.
5. **Proof strategy.** If it survives, look for a reason: reduce, strengthen for induction, exploit symmetry, or cut it to finitely many cases for a machine.
6. **New questions.** A proof shows what else is true; a refutation asks what the *right* statement was.

Evidence is not uniform. Think like a detective. Ten thousand witnesses who all saw the suspect in the same café prove less than one fingerprint, because the witnesses are correlated. The checks of $\pi(x) < \mathrm{li}(x)$ up to 10^{19} are ten quintillion correlated witnesses, all dominated by the same slowly varying terms. A **heuristic model**, which explains *why* a pattern should hold and how large the deviations should be, is worth more than any number of checks; Result 3 shows one predicting a counterexample nobody has seen.

The frontier is a literature. Before you spend a month on a pattern, you find out whether it is known (OEIS), who has worked on it (MathSciNet, zbMATH) and what the latest attempt looks like (arXiv); when you have a proof, you may ask a machine to check it (Lean, Coq, Isabelle). Section 98.4 makes this a procedure.

In this chapter the `[!mystery]` callouts change meaning. Elsewhere they pointed to a chapter holding the answer. Here the honest answer is: **nobody knows yet.**

98.3 The mathematics

Four kinds of statement

The vocabulary of research

- A **conjecture** is a precise, falsifiable statement believed but not proved.
- A **counterexample** to $\forall n\, P(n)$ is a single n with $\neg P(n)$. One settles the matter forever.
- A **heuristic** is an argument that would be a proof if an unproved modelling assumption held, typically "these quantities behave like independent random variables".
- A **certificate** is a finite object, cheap to check, that implies the claim: a colouring, a factorization, a proof file. Finding it may be hard; checking it is easy (the NP idea of Ch. 30, as a research method).

Research is largely the art of converting one kind of statement into another. Each result below shows one conversion, on a problem the book left open.

Result 1: symmetry turns a search into a proof

Ch. 28 introduced Ramsey numbers: $R(s,t)$ is the smallest N such that every red/blue colouring of the edges of the complete graph K_N contains a red K_s or a blue K_t. It stated $R(4,4) = 18$. To prove $R(4,4) > 17$ you must exhibit a colouring of K_{17} with no monochromatic K_4: a certificate. A blind search is hopeless, since there are 2^{136} colourings. But the right colouring can be found by thinking instead of searching, and then *checked* by thinking too.

The Paley graph P_{17}

The **quadratic residues** modulo 17 are the non-zero squares: $Q = \{1, 2, 4, 8, 9, 13, 15, 16\}$, exactly half of the 16 non-zero residues. Join $u, v \in \{0, 1, \ldots, 16\}$ by a red edge if $u - v \in Q$, and by a blue edge otherwise.

The colouring is well defined because $-1 \equiv 16 = 4^2$ is a square, so $u - v \in Q$ if and only if $v - u \in Q$.

Theorem 98.1 (Greenwood–Gleason, 1955)

P_{17} contains no red K_4 and no blue K_4. Hence $R(4,4) \geq 18$.

Proof. We use three facts about Q, all checkable from the list: Q is closed under multiplication; the product of two non-residues is a residue; and $3 \notin Q$. (The non-zero residues mod 17 form a cyclic group of order 16, and Q is its subgroup of index 2, which gives the first two facts.)

Step 1: the graph looks the same from every red edge. For $a \in Q$ and any b, the map $f(x) = ax + b$ sends a red edge to a red edge, because $f(u) - f(v) = a(u - v)$ and Q is closed under multiplication. Now take any red edge $\{u, v\}$. Since $v - u \in Q$, so is its inverse, and $g(x) = (x - u)/(v - u)$ is such a map, with $g(u) = 0$ and $g(v) = 1$. So if a red K_4 exists, there is one containing the vertices 0 and 1.

Step 2: the two other vertices of that K_4 are red neighbours of both 0 and 1. Those are the x with $x \in Q$ and $x - 1 \in Q$. Running through Q: $x = 2, 9, 16$ qualify (as $1, 8, 15 \in Q$), while $1, 4, 8, 13, 15$ do not (as $0, 3, 7, 12, 14 \notin Q$). So the candidates are $\{2, 9, 16\}$.

Step 3: no two candidates are joined in red. Their differences are $9 - 2 = 7$, $16 - 9 = 7$ and $16 - 2 = 14$, none of which lies in Q. So there is no red K_4.

Step 4: the blue graph is a copy of the red one. Multiply every vertex by the non-residue 3. A blue edge has $u - v \notin Q$, so $3(u - v)$ is a product of two non-residues, hence in Q: blue edges become red. A blue K_4 would become a red K_4, which Step 3 excluded. ■

A brute-force check of all $\binom{17}{4} = 2380$ four-vertex subsets agrees. But the proof never examined 2380 cases: symmetry collapsed them to *one* ("cliques through 0 and 1"), then to a three-element set. Combined with the recursive bound $R(4,4) \leq R(3,4) + R(4,3) = 9 + 9 = 18$, the value $R(4,4) = 18$ is settled.

Why the same idea stalls at $R(5,5)$. The best known colouring for the next case, found by Exoo in 1989, is a colouring of K_{42} with no monochromatic K_5, so $R(5,5) \geq 43$. No Paley graph does as well, and no one has found structure that gives K_{43}. On the other side, proving $R(5,5) \leq N$ means showing that *every* one of the $2^{\binom{N}{2}}$ colourings fails; for $N = 43$ that is 2^{903}, a 272-digit number. The upper bound came down from 49 (1995) to 48 (2018) to 46 (Angeltveit and McKay, 2024), each step mixing linear-programming bounds with huge, carefully organised computer searches. McKay and Radziszowski found 656 colourings of K_{42} with no monochromatic K_5, found no more, and *conjectured* that $R(5,5) = 43$: a conjecture built on a search that came up empty, exactly the kind of evidence the hook warned about.

Paul Erdős liked to say that if hostile aliens demanded $R(5,5)$ on pain of destroying the Earth, we should marshal every computer and mathematician and try to find it. If they demanded $R(6,6)$ (known only to lie between 102 and the 160s), we should try to destroy the aliens.

? Is $R(5,5)=43$?

Everyone expects it; nobody can prove it. Asymptotically, $(\sqrt{2})^k < R(k,k)$ by Ch. 32's random colourings, and $R(k,k) < (4-\varepsilon)^k$ only since 2023 (Campos, Griffiths, Morris and Sahasrabudhe), later about 3.8^k. Whether the true base is $\sqrt{2}$, near 4, or between is an **open problem**. No chapter of any book contains the answer.

Result 2: one finite gadget, an asymptotic theorem

Ch. 26 left a number hanging: the exponent ω of matrix multiplication, defined as the infimum of all τ such that two $n\times n$ matrices can be multiplied in $O(n^{\tau})$ arithmetic operations. Reading the input forces $\omega \ge 2$; the schoolbook method gives $\omega \le 3$. The key to every improvement since 1969 is the following conversion of a *finite* fact into an *asymptotic* bound.

Bilinear algorithm

A **bilinear algorithm** for $k\times k$ matrices with r multiplications computes r products

$$m_\ell = \Big(\sum_{i,j} \alpha_{ij}^{(\ell)} a_{ij}\Big)\Big(\sum_{i,j} \beta_{ij}^{(\ell)} b_{ij}\Big), \qquad \ell = 1,\dots,r,$$

and then each entry c_{ij} of $\mathbf{C}=\mathbf{AB}$ as a fixed linear combination of $m_1,\dots,m_r$, with constant coefficients α,β,γ. It must never use $a_{ij}b_{kl}=b_{kl}a_{ij}$.

Strassen's 1969 algorithm is a bilinear algorithm with $k=2$, $r=7$.

Theorem 98.2

If there is a bilinear algorithm for $k\times k$ matrices with r multiplications, then $\omega \le \log_k r$.

Proof. Because the algorithm never commutes a factor of $\mathbf{A}$ past a factor of $\mathbf{B}$, it is an identity in any ring, in particular in the ring of $m\times m$ matrices. So an $n\times n$ product with $n=k^t$ can be computed by splitting each matrix into a $k\times k$ grid of blocks of size n/k, forming the r block products recursively, and combining them with a fixed number c of block additions, each costing $(n/k)^2$ operations. The operation count satisfies

$$T(n) = r\,T(n/k) + c\,(n/k)^2, \qquad T(1)=1.$$

Unrolling it, as in the recursion tree of Ch. 26:

$$T(k^t) = r^t + c\sum_{i=1}^{t} r^{\,i-1}\,k^{2(t-i)} = r^t + \frac{c}{r}\,k^{2t}\sum_{i=1}^{t}\Big(\frac{r}{k^2}\Big)^{i}.$$

If $r > k^2$ the geometric sum is at most $\frac{r/k^2}{r/k^2-1}(r/k^2)^t$, so $T(k^t) \le C\,r^t = C\,n^{\log_k r}$ for a constant C. (If $r \le k^2$ the bound is $O(n^2\log n)$, which is even better; in fact $r<k^2$ is impossible.) For general n, pad the matrices with zeros up to the next power k^t with $n \le k^t < kn$; this multiplies the cost by at most $k^{\log_k r}=r$, a constant. Hence $T(n)=O(n^{\log_k r})$, and $\omega \le \log_k r$. $\blacksquare$

With $k=2$, $r=7$ this gives $\omega \le \log_2 7 \approx 2.807$. The theorem turns an asymptotic question into *finite* puzzles: for each k, what is the smallest r? For $k=3$ the record is Laderman's $r=23$ (1976), giving only $\log_3 23 \approx 2.854$; beating Strassen needs $r \le 21$, since $3^{2.807}\approx 21.85$. The best lower bound is 19, and nobody knows whether 20, 21 or 22 are possible. In 2022 a reinforcement-learning search (AlphaTensor) found a 4×4 algorithm with 47 multiplications, valid in arithmetic modulo 2: $\log_4 47 \approx 2.777$, a certificate humans had missed for fifty years.

The records come from elsewhere: since Coppersmith and Winograd (1990, $\omega < 2.376$), from approximate ("border rank") versions of Theorem 98.2 applied to large, cleverly chosen tensors. The 2024 record is $\omega < 2.3714$, improvements now arriving in the fourth decimal place.

? Is $\omega = 2$?

Many experts believe so, partly because the group-theoretic approach of Cohn and Umans (2003) would give $\omega=2$ if certain combinatorial objects exist. But the method behind every record since 1990, applied to the tensors used so far, provably cannot go below about 2.3. Reaching 2 needs an idea nobody has had: an **open problem**, answered in no chapter.

Result 3: a heuristic that predicted a counterexample nobody has seen

Return to the hook. Why should anyone believe Littlewood rather than the tables? The cleanest illustration is a third race, from Ch. 63: the **Mertens function**

$$M(x) = \sum_{n\le x} \mu(n),$$

where the Möbius function $\mu(n)$ is 0 if n has a repeated prime factor, and $(-1)^k$ if n is a product of k distinct primes. So $\mu(6)=1$, $\mu(30)=-1$, $\mu(12)=0$. In 1897 Mertens conjectured $|M(x)| < \sqrt{x}$ for all $x>1$. Figure 98.2 shows why: our computation gives $\max |M(x)|/\sqrt{x} = 0.472$ over $1000 \le x \le 10^7$, and the inequality has been verified up to 10^{16}.

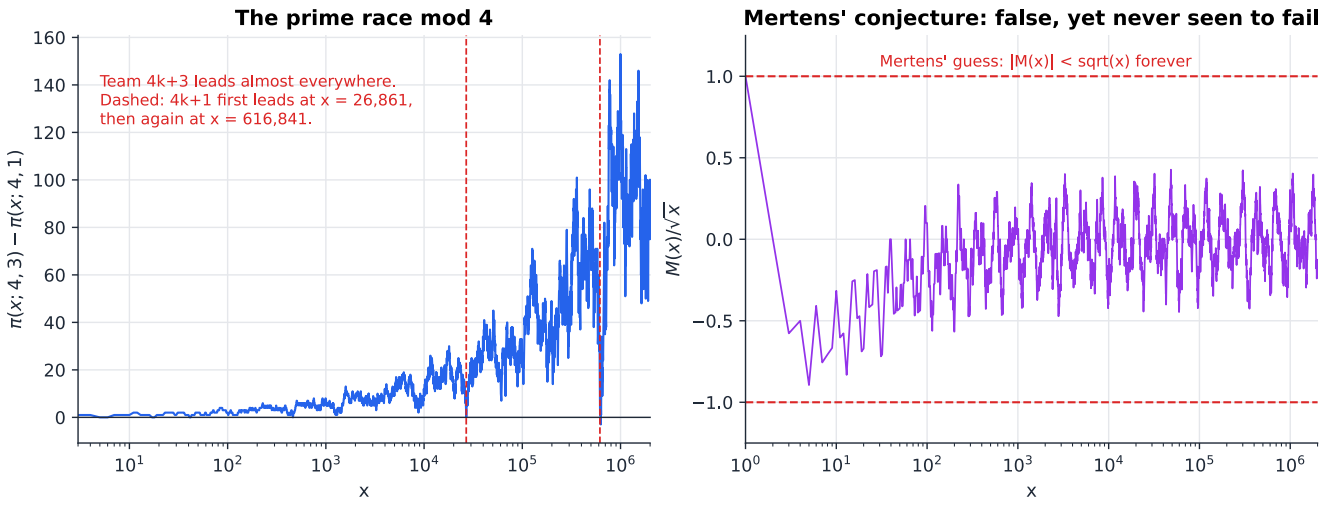


Figure 98.2 — Two patterns that survive every table and are still false. Left: $\pi(x;4,3)-\pi(x;4,1)$, first negative at $x = 26{,}861$. Right: $M(x)/\sqrt{x}$, which never comes near the dashed lines below 10^{16}, yet must cross them eventually.

Here is the heuristic. Pretend that, on the squarefree integers, the signs $\mu(n) = \pm 1$ are independent fair coin flips. Then $M(x)$ is a random walk (Ch. 34) with one step for each squarefree $n \le x$. About $6/\pi^2 \approx 0.608$ of all integers are squarefree, so

$$E\big[M(x)\big] = 0, \qquad \operatorname{Var}\big[M(x)\big] \approx \frac{6}{\pi^2}\,x, \qquad \text{so} \qquad \frac{M(x)}{\sqrt{x}} \text{ is typically of size } \sqrt{6/\pi^2} \approx 0.78.$$

So far the model agrees with Mertens: deviations of order $\sqrt{x}$. But a random walk does more than fluctuate at scale $\sqrt{n}$. The **law of the iterated logarithm** says that a walk S_n of fair ± 1 steps satisfies, with probability one,

$$\limsup_{n\to\infty} \frac{S_n}{\sqrt{2n \ln\ln n}} = 1.$$

Since $\sqrt{\ln\ln n} \to \infty$, the ratio $S_n/\sqrt{n}$ is unbounded. In the model, $|M(x)|/\sqrt{x}$ eventually exceeds 1, and 10, and any bound you name. The heuristic predicts that Mertens is wrong, and why the tables cannot see it: $\sqrt{\ln\ln x}$ is 1.90 at $x = 10^{16}$ and only 2.28 at $x = 10^{80}$.

The heuristic is not a proof (μ is not random), but it pointed the right way. In 1985 Odlyzko and te Riele proved $\limsup M(x)/\sqrt{x} > 1.06$ without finding any x: they computed 2000 zeros of the zeta function to about 100 digits and showed, through an explicit formula, that $M(x)/\sqrt{x}$ must swing beyond ± 1 somewhere. The first counterexample is known only to be smaller than roughly $e^{10^{33}}$.

The same random model gives a sharper statement that *is* believed, and that is equivalent to the biggest open problem in the book:

Theorem 98.3 (von Koch, 1901; Littlewood, 1912; stated without proof)

The Riemann hypothesis is true if and only if $M(x) = O\big(x^{1/2+\varepsilon}\big)$ for every $\varepsilon > 0$. Equivalently, if and only if $|\pi(x) - \operatorname{li}(x)| = O\big(\sqrt{x}\ln x\big)$.

So the Riemann hypothesis, a statement about complex zeros (Ch. 57, Ch. 18), says the Möbius signs cancel like coin flips, up to a factor x^{ε}. Mertens asked for too much (no x^{ε}, constant exactly 1) and died; the coin-flip version survives. The prime race is governed by similar zeros: Rubinstein and Sarnak (1994) showed, assuming a generalised Riemann hypothesis and a hypothesis on the zeros, that Team Three leads on a set of logarithmic density 0.9959. So strong a bias is why the counterexample came late; a bias short of 1 is why it came at all.

? Where is the first crossing of $\pi(x)$ and $\operatorname{li}(x)$?

Littlewood proved it exists. Skewes (1933) gave the first explicit bound, about $10^{10^{10^{34}}}$, assuming the Riemann hypothesis. Modern work has located a region of crossings below about 1.4×10^{316}, but whether an earlier one exists between 10^{19} and there is **unknown**. So is the Riemann hypothesis itself: more than ten trillion zeros lie on the line $\operatorname{Re}\{s\} = \frac{1}{2}$, and no proof is in sight.

Proofs no human reads

Results 1 and 2 used symmetry and recursion to *avoid* computation. When it cannot be avoided, a new question arises: what does it mean to trust a proof?

- **The four-colour theorem** (Ch. 27), Appel and Haken, 1976, has the shape *unavoidability plus reducibility*: every planar map contains one of a finite list of configurations, and each one on the list can be removed and its colouring restored. The list had nearly two thousand entries, and checking them took about 1200 hours of computer time. Robertson, Sanders, Seymour and Thomas cut it to 633 in 1996; Gonthier checked the entire proof in the Coq proof assistant in 2005.
- **The Kepler conjecture** on sphere packing (Hales, 1998) left its referees "99% certain"; Hales's Flyspeck project answered with a formal proof in 2014.
- **Boolean Pythagorean triples.** Heule, Kullmann and Marek (2016) showed by a SAT solver (Ch. 30) that $\{1,\dots,7824\}$ can be 2-coloured with no monochromatic $a^2 + b^2 = c^2$ and $\{1,\dots,7825\}$ cannot. The impossibility certificate was 200 terabytes, checked by a formally verified checker.

The trusted part shrinks: nobody trusts a 200-terabyte search, but people trust a small checker, which trusts a small set of logical rules (Ch. 65, Ch. 66). In 2022 the Lean community completed Peter Scholze's *Liquid Tensor Experiment*, formalising a cutting-edge theorem; in 2023 the proof of the polynomial Freiman–Ruzsa conjecture was formalised about three weeks after it appeared. Formal verification (Ch. 67) now checks new theorems, not just old ones.

98.4 The algorithm

A counterexample hunter

Most of research is not proof. It is disciplined searching. Here is the skeleton every serious conjecture-tester shares.

```
HUNT(P, N):                      # test "P(n) holds for all n <= N"
    build the data incrementally   # sieve, recurrence, running sum: never recompute from scratch
    for n = 1 .. N:
        update the running quantity Q(n)
        if not P(n):
            output certificate (n and the data that refutes P); stop
        if Q(n) sets a record (max, min, closest approach):
            log (n, Q(n))          # the records are the real result
    report: "no counterexample below N"; plus the record table
```

The record table matters more than the verdict. "No counterexample below 10^7" is weak; "the closest approach was 0.472 at $x = 2803$, and records grow like $\sqrt{\ln \ln x}$" is a measurement you can compare with a heuristic.

Complexity. The sieve of Eratosthenes (and the Möbius sieve below) costs $O(N \ln \ln N)$ time and $O(N)$ memory, and memory runs out first. A **segmented sieve** works in blocks of about $\sqrt{N}$, cutting memory to $O(\sqrt{N})$. The great records ($\pi(x) < \mathrm{li}(x)$ to 10^{19}, Goldbach to 4×10^{18}) add distributed computing and analytic methods that count primes without listing them.

Failure cases.

- *Floating point.* $\mathrm{li}(10^{19}) \approx 2.3 \times 10^{17}$ carries an absolute rounding error of tens in double precision. Harmless while $\mathrm{li}(x) - \pi(x) \approx 10^8$, fatal near a crossing, where rounding decides the sign. Use exact integers, or interval arithmetic.
- *Unrepresentative ranges.* Biases like the prime race's live on a logarithmic scale: going from 10^4 to 10^5 is one more decade, not "ten times the evidence".
- *The law of small numbers* (Richard Guy): there are not enough small numbers for the demands made of them. Chords joining n points on a circle cut it into $1, 2, 4, 8, 16$ regions for $n = 1, \ldots, 5$, and then 31.

Exploring the literature

A procedure, in the order a working mathematician uses it:

1. **OEIS** (oeis.org). If your problem produces integers, paste in the first ten. The On-Line Encyclopedia of Integer Sequences holds more than 370,000 sequences with formulas, references and code. Type $1, 2, 4, 8, 16, 31$ and you get A000127, its closed form $\binom{n}{4} + \binom{n}{2} + 1$, and a century of references. Many "new" patterns turn out to be known within a minute.
2. **Name the objects.** Search with the field's names, not yours: "Paley graph", "tensor rank", "Chebyshev bias". Surveys give them to you.
3. **MathSciNet** (Mathematical Reviews, from 1940) and **zbMATH Open** (free since 2021) review nearly every research paper, classified by subject (MSC 05 combinatorics, 11 number theory, 37 dynamics). Their real power is the *citation graph*: find one good paper, read who cited it, and walk forward in time to the current state of the problem.
4. **arXiv** (arxiv.org), since 1991, holds preprints, often years before publication: `math.CO`, `math.NT`, `math.DS`, `cs.CC`. Every record in this chapter since 2018 appeared there first. Preprints are not refereed: strong evidence, not theorems.
5. **Proof assistants.** Lean's `mathlib`, Coq and Isabelle hold hundreds of thousands of formalised results: what is truly established, in exact form, ready to build on.
6. **Ask people.** MathOverflow, a seminar, an email to the author of the closest paper. A precise, well-prepared question usually gets an answer.

98.5 The experiment

The script below runs the hook's prime race and the Mertens test side by side, with the record-keeping of the hunter. It reproduces the numbers quoted in this chapter in a few seconds.

```
import numpy as np
import matplotlib.pyplot as plt

N = 10**7           # how far to look (try 10**6, then 10**8 if you have the memory)
q, a, b = 4, 3, 1   # race between primes = a (mod q) and primes = b (mod q); try (3,2,1), (8,3,1)

sieve = np.ones(N + 1, dtype=bool); sieve[:2] = False
for i in range(2, int(N**0.5) + 1):
    if sieve[i]:
        sieve[i*i::i] = False
p = np.nonzero(sieve)[0]

step = np.zeros(N + 1, dtype=np.int64)
step[p[p % q == a]] += 1
step[p[p % q == b]] -= 1
lead = np.cumsum(step)                      # pi(x;q,a) - pi(x;q,b)
losing = np.nonzero(lead < 0)[0]
print(f"team {a} (mod {q}) ahead or level at {N + 1 - len(losing):,} of {N + 1:,} values of x")
print("first x where team", b, "leads:", losing[0] if len(losing) else "none below N")

mu = np.ones(N + 1, dtype=np.int64); mu[0] = 0   # Moebius function by a sieve
for r in p:
    mu[r::r] *= -1
    if r * r <= N:
        mu[r*r::r*r] = 0
M = np.cumsum(mu)
x = np.arange(1, N + 1)
ratio = M[1:] / np.sqrt(x)
big = np.abs(ratio[999:]).argmax() + 1000
print(f"max |M(x)|/sqrt(x) for 1000 <= x <= N: {abs(ratio[big - 1]):.3f} at x = {big:,}")

xs = np.unique(np.geomspace(10, N, 5000).astype(int))
fig, ax = plt.subplots(1, 2, figsize=(11, 4))
ax[0].semilogx(xs, lead[xs]); ax[0].axhline(0, color="k", lw=0.8)
ax[0].set_title(f"pi(x;{q},{a}) - pi(x;{q},{b})")
ax[1].semilogx(xs, M[xs] / np.sqrt(xs)); ax[1].axhline(1, ls="--", color="r"); ax[1].axhline(-1, ls="--", color="r")
ax[1].set_title("M(x)/sqrt(x)")
plt.show()
```

With the default knobs it prints that Team Three is ahead or level at 9,996,595 of the 10,000,001 values $0 \le x \le 10^7$, that Team One first leads at 26,861, and that $\max |M(x)|/\sqrt{x} = 0.472$ at $x = 2803$. Things to change:

1. **Set `q, a, b = 3, 2, 1`.** No lead change below 10^7. Conclude nothing: the first one (Bays and Hudson, 1978) is at $x = 608{,}981{,}813{,}029$, beyond any array your laptop can hold.
2. **Set `q, a, b = 8, 3, 1`**, then `(8, 5, 1)` and `(8, 7, 1)`. Chebyshev's bias says the non-squares $3, 5, 7$ should beat the square 1. By how much?
3. **Change `N`** between 10^6 and 10^8 (the last needs about 6 GB). The Mertens record barely moves; compare it with $\sqrt{\ln \ln N}$.
4. **Replace the Möbius signs by coin flips**: put `mu[mu != 0] = np.random.choice([-1, 1], size=np.count_nonzero(mu))` just before `M = np.cumsum(mu)`. Over twenty seeds, how often does the walk cross ± 1 below 10^7? That is Result 3's heuristic, made empirical.

98.6 The challenge

1. **Intuition.** $\pi(x) < \mathrm{li}(x)$ holds up to 10^{19} and is false; ten trillion zeta zeros lie on the critical line, and most mathematicians believe the Riemann hypothesis. Explain, without formulas, why the second body of evidence counts for more. What would make you stop believing it?
2. **Derivation.** (a) Show that the Paley graph P_{13}, built from the quadratic residues modulo 13, has no red K_4 and no blue K_4, by adapting Steps 1 to 4 of Theorem 98.1. What does that prove about $R(4,4)$, and why is it weaker than Theorem 98.1? (b) In Theorem 98.2, take $k = 3$. Show that a bilinear algorithm with r multiplications improves on Strassen's exponent if and only if $r \le 21$.
3. **Implementation.** Write a colouring hunter: start from a random red/blue colouring of K_n and repeatedly flip an edge lying in the most monochromatic K_4s, with occasional random flips to escape local minima. For $n = 17$, is what it finds a relabelling of P_{17}? Then hunt for colourings with no monochromatic K_5. How far below 42 do you get stuck?
4. **Open investigation.** Pick one problem from Figure 98.3. Find its OEIS entry (if any), its latest arXiv preprint, and the zbMATH or MathSciNet review of the paper that posed it. Design a computation that *could* find a counterexample below a bound you can afford, run it with a record table, and write one paragraph on what your negative result does and does not tell you.

98.7 Loose threads

In the last chapter, the loose threads point to no other chapter. They are the edge of what anyone knows.

Open problems this book has pointed at, by track

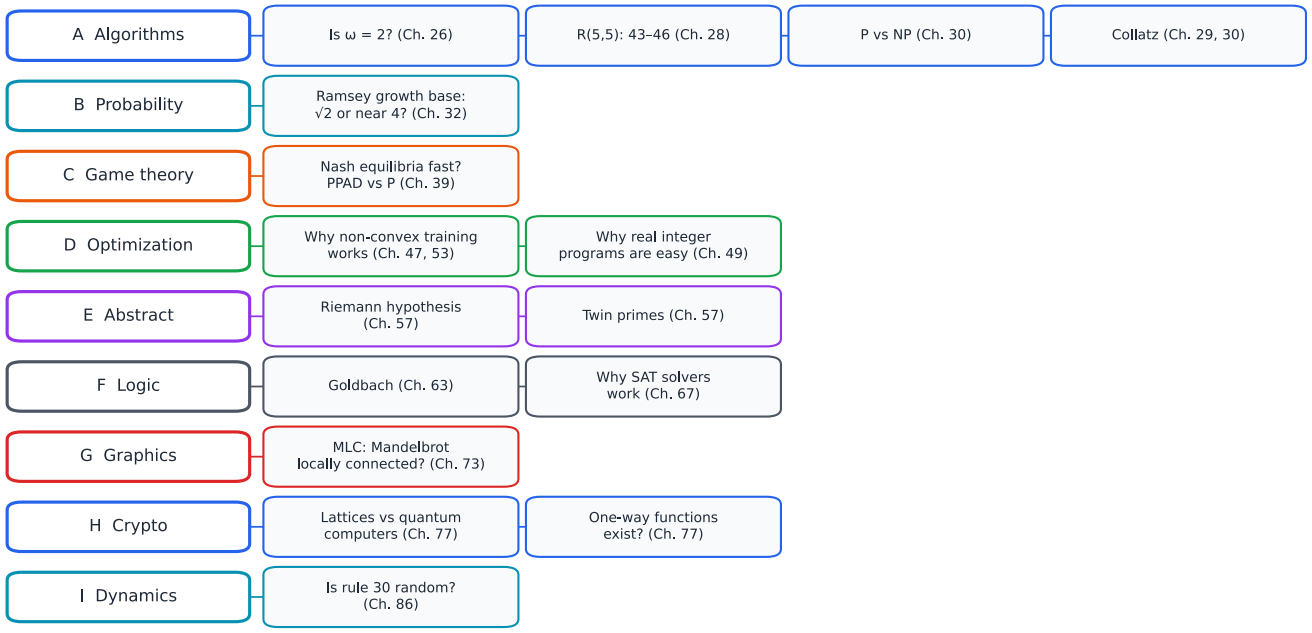


Figure 98.3 — The open problems this book has pointed at, by track. Every box is a question that nobody, anywhere, can currently answer.

- **Is $\omega = 2$?** Unknown. The best bound is $\omega < 2.3714$ (Ch. 26). Even the smallest finite version, whether 3×3 matrices can be multiplied with 22 or fewer multiplications, is open.
- **What is $R(5,5)$?** Somewhere in $\{43, 44, 45, 46\}$ (Ch. 28); the asymptotic base of $R(k,k)$ is somewhere between $\sqrt{2}$ and about 3.8 (Ch. 32).
- **The four-colour theorem** is closed: true, and formally verified. What remains open is softer: is there a proof a human can read in an afternoon? (Ch. 27)
- **Is the Mandelbrot set locally connected?** Open (Ch. 73). Yoccoz proved it at a large class of parameters in the 1990s; the "infinitely renormalizable" ones are where the difficulty lives. MLC would imply that hyperbolic components are dense, also open.
- **The Riemann hypothesis** (Ch. 57) is open, and with it the size of the prime races' deviations and the place where $\pi(x)$ first passes $\mathrm{li}(x)$.
- **P versus NP** (Ch. 30), one-way functions (Ch. 77), why SAT and integer-programming solvers work so well (Ch. 67, Ch. 49), why gradient descent trains non-convex networks (Ch. 47): all open, and probably connected in ways nobody yet sees.
- **Goldbach, twin primes, Collatz, rule 30** (Ch. 63, Ch. 57, Ch. 30, Ch. 86): checked to enormous bounds, explained by heuristics, unproved. Ch. 65 and Ch. 68 explain why some such statement might have no proof at all.

> **The thread you pick**
>
> Most of this book's unanswered questions have answers in other chapters. The ones in Figure 98.3 do not. They are waiting for someone to notice a pattern, to find the symmetry that collapses a search, the gadget that implies an asymptotic bound, or the heuristic that predicts where the counterexample hides. The methods are in this chapter, and the tools (a sieve, a SAT solver, the OEIS, arXiv, a proof assistant) are free. **Which thread will you pull?**

This book has given you a map, incomplete on purpose, because the real one is too. Pick a thread, run the hunter, read the last three papers on it, and write down a conjecture precise enough to be wrong. That is how all of this mathematics was made, and how the next chapter, the one nobody has written yet, will be.

Further reading

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- I. Lakatos, *Proofs and Refutations: The Logic of Mathematical Discovery*, Cambridge University Press, 1976. A dialogue on how counterexamples reshape definitions and theorems, built around Euler's $V - E + F = 2$.
- R. K. Guy, "The strong law of small numbers", *American Mathematical Monthly* 95 (1988), 697–712. Thirty-five patterns that look true; some are.
- A. Granville and G. Martin, "Prime number races", *American Mathematical Monthly* 113 (2006), 1–33. The hook's races, Littlewood's theorem and Chebyshev's bias, accessibly.
- A. M. Odlyzko and H. J. J. te Riele, "Disproof of the Mertens conjecture", *Journal für die reine und angewandte Mathematik* 357 (1985), 138–160.
- S. P. Radziszowski, "Small Ramsey numbers", *Electronic Journal of Combinatorics*, Dynamic Survey DS1 (regularly revised). The live scoreboard for $R(5,5)$ and its relatives.
- T. C. Hales, "Formal proof", *Notices of the AMS* 55 (2008), 1370–1380, and G. Gonthier, "Formal proof — the four-color theorem", *Notices of the AMS* 55 (2008), 1382–1393.
- K. Buzzard, "The rise of formalism in mathematics", *Proceedings of the ICM 2022*, and the Lean community's *Mathematics in Lean* (free online). Where proof assistants are going, and how to start.
- The OEIS (oeis.org), arXiv (arxiv.org), zbMATH Open (zbmath.org) and MathOverflow (mathoverflow.net). The working mathematician's library, all free.

Appendix — Derivations and Proofs

Appendix A — Calculus and Complex Numbers

This appendix gives complete proofs of the single-variable calculus results, the complex-exponential results and the two multivariable results that the chapters of Parts I and II state with only a sketch. The results are proved in logical order, not in the order the chapters use them: the mean value theorem comes first because almost everything else rests on it, and Euler's formula comes after Taylor's theorem because one of its proofs needs the Taylor series of cos and sin.

What is assumed. The real numbers are complete (every non-empty set bounded above has a least upper bound). From completeness we take, without proof, three standard facts: the **extreme value theorem** (a continuous function on a closed bounded interval $[a,b]$ attains a maximum and a minimum), the **intermediate value theorem** (a continuous function on $[a,b]$ takes every value between $f(a)$ and $f(b)$), and the **Riemann integrability of continuous functions** on $[a,b]$, together with the elementary properties of the integral (linearity, additivity over intervals, and monotonicity: $f \le g \Rightarrow \int f \le \int g$). We also use the differentiation rules (sum, product, quotient, chain) of Ch. 4 §4.2–4.3, the derivatives $\sin' = \cos$, $\cos' = -\sin$, and the identity $\cos^2\theta + \sin^2\theta = 1$. The implicit function theorem is stated precisely and used, but not proved.

Level of rigour. Every hypothesis is stated; every step of every proof is justified from the assumptions above or from an earlier result of this appendix. Where a hypothesis cannot be dropped, a counterexample is given. Proofs end with ■.

Complex-valued functions. Several results are applied to functions $F : [a,b] \to \mathbb{C}$ of a real variable, such as $e^{j\omega t}$. For these, $F = U + jV$ with $U = \mathrm{Re}\{F\}$, $V = \mathrm{Im}\{F\}$ real, and limits, continuity, derivatives and integrals are taken component by component: $F' = U' + jV'$, $\int F = \int U + j\int V$. Because complex addition and multiplication obey the same algebraic laws as real ones and $|z|$ obeys the triangle inequality, the sum, product, quotient and chain rules (for a real inner function) carry over word for word.

A.1 Rolle's theorem, the mean value theorem and Cauchy's mean value theorem

The mean value theorem (MVT) is the bridge between local information (a derivative at a point) and global information (a change over an interval). It is the tool used to prove the FTC, Taylor's theorem, L'Hôpital's rule, the uniqueness half of Euler's formula and Clairaut's theorem below.

We first need the fact that a derivative vanishes at an interior extremum.

Lemma A.1 (Fermat)

Let f be defined on an open interval I, let $c \in I$, and suppose $f(c) \ge f(x)$ for all $x \in I$ (or $f(c) \le f(x)$ for all $x \in I$). If $f'(c)$ exists, then $f'(c) = 0$.

Proof. Take the maximum case; the minimum case follows by applying it to $-f$. For $h > 0$ small enough that $c + h \in I$, $f(c+h) - f(c) \le 0$, so

$$\frac{f(c+h)-f(c)}{h} \le 0 \quad (h>0), \qquad \frac{f(c+h)-f(c)}{h} \ge 0 \quad (h<0).$$

The derivative is the limit of these quotients as $h \to 0$; the limit of the right-hand quotients ($h \to 0^+$) is ≤ 0 and that of the left-hand ones ($h \to 0^-$) is ≥ 0 (limits preserve non-strict inequalities). Both one-sided limits equal $f'(c)$, so $0 \le f'(c) \le 0$. ■

Theorem A.1 (Rolle, MVT, Cauchy MVT)

Let f and g be real-valued, continuous on $[a,b]$ and differentiable on (a,b), with $a < b$.

1. **(Rolle)** If $f(a) = f(b)$, there is $c \in (a,b)$ with $f'(c) = 0$.
2. **(Mean value theorem)** There is $c \in (a,b)$ with
$$f'(c) = \frac{f(b)-f(a)}{b-a}.$$
3. **(Cauchy's mean value theorem)** There is $c \in (a,b)$ with
$$\big[f(b)-f(a)\big]\,g'(c) = \big[g(b)-g(a)\big]\,f'(c).$$

If in addition $g'(x) \neq 0$ on (a,b), then $g(b) \neq g(a)$ and $\dfrac{f(b)-f(a)}{g(b)-g(a)} = \dfrac{f'(c)}{g'(c)}$.

Proof.

Rolle. By the extreme value theorem, f attains a maximum value M and a minimum value m on $[a,b]$.

- If $M = m$, f is constant on $[a,b]$, so $f'(c) = 0$ for every $c \in (a,b)$.
- If $M > m$, at least one of M, m differs from the common endpoint value $f(a) = f(b)$. Say $M \neq f(a)$ (the other case is symmetric). Then the maximum is attained at some c with $c \neq a$, $c \neq b$, i.e. $c \in (a,b)$. On the open interval (a,b), $f(c) \ge f(x)$ for all x, and $f'(c)$ exists, so $f'(c) = 0$ by Lemma A.1.

MVT. Let $s = \dfrac{f(b)-f(a)}{b-a}$ (the slope of the chord) and define

$$\varphi(x) = f(x) - f(a) - s\,(x-a).$$

φ is continuous on $[a,b]$ and differentiable on (a,b) (a difference of such functions), with $\varphi(a) = 0$ and $\varphi(b) = f(b) - f(a) - s(b-a) = 0$. Rolle gives $c \in (a,b)$ with $0 = \varphi'(c) = f'(c) - s$.

Cauchy MVT. Define

$$h(x) = \big[f(b) - f(a)\big]g(x) - \big[g(b) - g(a)\big]f(x).$$

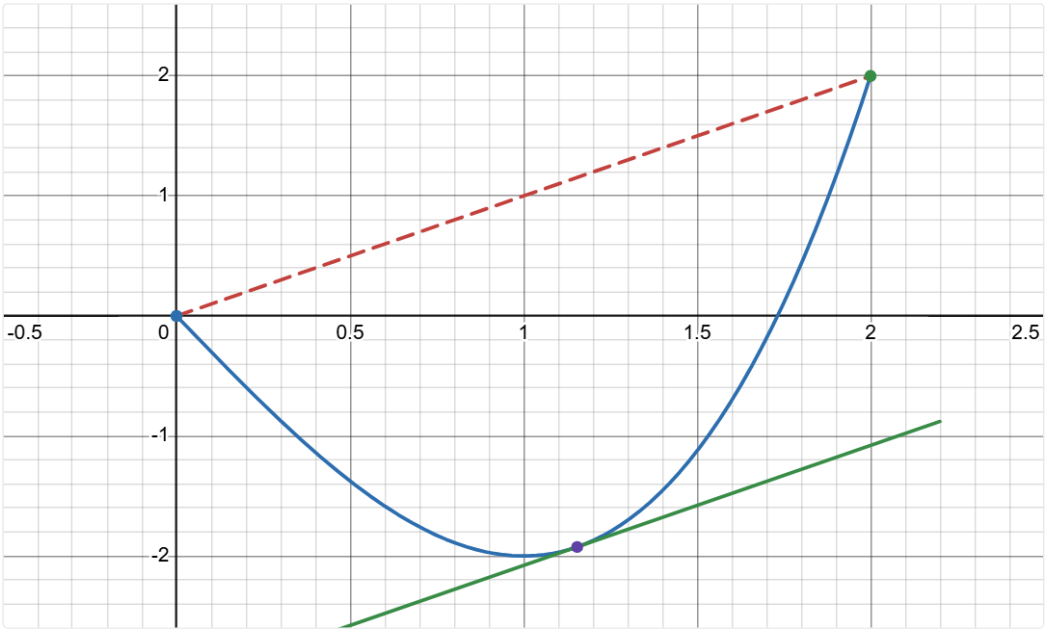
h is continuous on $[a, b]$, differentiable on (a, b), and

$$h(a) = f(b)g(a) - f(a)g(a) - g(b)f(a) + g(a)f(a) = f(b)g(a) - g(b)f(a),$$

$$h(b) = f(b)g(b) - f(a)g(b) - g(b)f(b) + g(a)f(b) = f(b)g(a) - g(b)f(a).$$

So $h(a) = h(b)$, and Rolle gives c with $h'(c) = [f(b) - f(a)]g'(c) - [g(b) - g(a)]f'(c) = 0$, which is the claim. Finally, if $g' \neq 0$ on (a, b) then $g(b) \neq g(a)$, for otherwise Rolle applied to g would produce a zero of g'; dividing by $[g(b) - g(a)]g'(c) \neq 0$ gives the quotient form. ■

The MVT is the special case $g(x) = x$ of Cauchy's theorem. Geometrically, Cauchy's theorem says that a parametric curve $t \mapsto (g(t), f(t))$ has, somewhere, a tangent parallel to the chord joining its ends.



Mean value theorem for $f(x) = x^3 - 3x$ on $[0, 2]$: the chord (red, dashed) has slope 1, and the tangent at $c = 2/\sqrt{3} \approx 1.1547$ (green) is parallel to it, since $f'(c) = 3c^2 - 3 = 1$.

Corollary A.1 (consequences of the MVT)

Let I be an interval and f differentiable on I.

1. If $f'(x) = 0$ for all $x \in I$, then f is constant on I. This holds also for complex-valued f.
2. If $f' = g'$ on I, then $f - g$ is constant on I.
3. If $f' > 0$ on I then f is strictly increasing on I; if $f' \geq 0$, non-decreasing. (Similarly for $<$, $\leq$.)
4. If $|f'(x)| \leq M$ on I, then $|f(x_2) - f(x_1)| \leq M|x_2 - x_1|$ for all $x_1, x_2 \in I$.

Proof. For $x_1 < x_2$ in I, f is continuous on $[x_1, x_2]$ (differentiable implies continuous) and differentiable on (x_1, x_2), so $f(x_2) - f(x_1) = f'(c)(x_2 - x_1)$ for some $c \in (x_1, x_2)$. Items 1, 3 and 4 for real f read off directly from this formula. For complex $f = U + jV$, $f' = 0$ means $U' = V' = 0$, so U and V are each constant, hence so is f. Item 2 is item 1 applied to $f - g$. ■

Remarks.

- *Differentiability in the interior is essential.* $f(x) = |x|$ on $[-1, 1]$ is continuous with $f(-1) = f(1)$, yet $f'(x) = \pm 1$ wherever it exists: there is no c with $f'(c) = 0$. The failure is the single point $x = 0$ where f' does not exist.
- *Continuity at the endpoints is essential.* $f(x) = x$ on $[0, 1)$ with $f(1) = 0$ is differentiable on $(0, 1)$ and has $f(0) = f(1)$, but $f' \equiv 1$.
- *The MVT itself fails for complex-valued functions.* $F(t) = e^{jt}$ on $[0, 2\pi]$ has $F(2\pi) - F(0) = 0$, but $F'(t) = je^{jt}$ has modulus 1 everywhere, so no c gives $F'(c) \cdot 2\pi = 0$. (Euler's formula, used here only as an illustration, is proved in §A.6 without this example.) What survives for complex functions is Corollary A.1(1), proved through real and imaginary parts, and the inequality form (4), which can be proved from the FTC (§A.2) as $|F(x_2) - F(x_1)| = \left|\int_{x_1}^{x_2} F'\right| \leq M|x_2 - x_1|$ when F' is continuous. This is why the proof of Euler's formula in §A.6 is arranged around "zero derivative implies constant".
- *Where it is used.* The MVT and its corollaries are stated in Ch. 4 §4.2; Corollary A.1(2) is the justification of the "$+C$" of indefinite integration in Ch. 5 §5.1.4, and item 4 is the Lipschitz estimate behind the existence–uniqueness theory of Ch. 11 §11.7 and the error analysis of fixed-point and Newton iterations in Ch. 22.

A.2 The Fundamental Theorem of Calculus

The FTC connects the two halves of calculus: the integral, defined as a limit of Riemann sums (Ch. 5 §5.1.2), and the derivative. We recall the definition so that the proof of Part 2 can refer to it.

Riemann integral

A **partition** P of $[a,b]$ is a finite set of points $a = x_0 < x_1 < \cdots < x_n = b$, with mesh $\|P\| = \max_i (x_i - x_{i-1})$. A **Riemann sum** for f is $\sum_{i=1}^{n} f(c_i)\,\Delta x_i$ with $\Delta x_i = x_i - x_{i-1}$ and any tags $c_i \in [x_{i-1}, x_i]$. A bounded function f is **Riemann integrable** on $[a,b]$, with integral $\int_a^b f$, if for every $\varepsilon > 0$ there is $\delta > 0$ such that every Riemann sum with $\|P\| < \delta$ (and any choice of tags) lies within ε of $\int_a^b f$. We set $\int_b^a f = -\int_a^b f$ and $\int_a^a f = 0$.

We first need the mean value theorem for integrals.

Lemma A.2 (Mean value theorem for integrals)

If f is continuous on the closed interval with end-points $p \neq q$, there is c between p and q (inclusive) with

$$\frac{1}{q-p}\int_p^q f(t)\,dt = f(c).$$

Proof. Suppose $p < q$ (otherwise swap them; both sides of the formula are unchanged). By the extreme value theorem f has a minimum $m = f(t_1)$ and a maximum $M = f(t_2)$ on $[p,q]$. Monotonicity of the integral applied to $m \le f \le M$ gives $m(q-p) \le \int_p^q f \le M(q-p)$, so the average $\mu = \frac{1}{q-p}\int_p^q f$ satisfies $m \le \mu \le M$. By the intermediate value theorem applied to f on the closed interval between t_1 and t_2, there is c in that interval, hence in $[p,q]$, with $f(c) = \mu$. ∎

Theorem A.2 (Fundamental Theorem of Calculus)

Part 1. Let f be Riemann integrable on $[a,b]$ and define $A(x) = \displaystyle\int_a^x f(t)\,dt$ for $x \in [a,b]$. Then A is continuous on $[a,b]$. If moreover f is continuous at a point $x_0 \in [a,b]$, then A is differentiable at x_0 (one-sidedly if x_0 is an end-point) and $A'(x_0) = f(x_0)$. In particular, if f is continuous on $[a,b]$, then $A' = f$ on $[a,b]$: every continuous function has an antiderivative.

Part 2. Let F be continuous on $[a,b]$ and differentiable on (a,b), and suppose $F' = f$ on (a,b), where f is Riemann integrable on $[a,b]$. Then

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

Proof of Part 1. Since f is integrable it is bounded, say $|f| \le K$. For $x, x+h \in [a,b]$, additivity of the integral gives

$$A(x+h) - A(x) = \int_x^{x+h} f(t)\,dt,$$

and monotonicity gives $|A(x+h) - A(x)| \le K|h|$, so A is (Lipschitz) continuous.

Now let f be continuous at x_0 and let $\varepsilon > 0$. Choose $\delta > 0$ with $|f(t) - f(x_0)| < \varepsilon$ whenever $|t - x_0| < \delta$, $t \in [a,b]$. For $0 < |h| < \delta$ with $x_0 + h \in [a,b]$, write $f(x_0) = \frac{1}{h}\int_{x_0}^{x_0+h} f(x_0)\,dt$; then

$$\left| \frac{A(x_0+h) - A(x_0)}{h} - f(x_0) \right| = \left| \frac{1}{h}\int_{x_0}^{x_0+h} \big[f(t) - f(x_0)\big]\,dt \right| \le \frac{1}{|h|}\cdot |h|\,\varepsilon = \varepsilon,$$

because every t between x_0 and $x_0 + h$ satisfies $|t - x_0| < \delta$. Hence the difference quotient tends to $f(x_0)$. (When f is continuous on the whole interval one can instead quote Lemma A.2: the difference quotient equals $f(c_h)$ with c_h between x_0 and $x_0 + h$, and $f(c_h) \to f(x_0)$ — this is the version in Ch. 5.) ∎

Proof of Part 2. Let $\varepsilon > 0$ and choose δ as in the definition of $\int_a^b f$. Take any partition $a = x_0 < \cdots < x_n = b$ with mesh $< \delta$. On each sub-interval $[x_{i-1}, x_i]$, F is continuous, and differentiable in the interior, so by the MVT (Theorem A.1) there is $c_i \in (x_{i-1}, x_i)$ with $F(x_i) - F(x_{i-1}) = F'(c_i)\Delta x_i = f(c_i)\Delta x_i$. Summing, the left side telescopes:

$$F(b) - F(a) = \sum_{i=1}^{n} \big[F(x_i) - F(x_{i-1})\big] = \sum_{i=1}^{n} f(c_i)\,\Delta x_i.$$

The right side is a Riemann sum with mesh $< \delta$, so it is within ε of $\int_a^b f$. Thus $|F(b) - F(a) - \int_a^b f| < \varepsilon$ for every $\varepsilon > 0$, i.e. the two are equal. ∎

The proof of Part 2 does not use Part 1. For continuous f there is also the two-line proof of Ch. 5: A and F both have derivative f on (a,b), so $F = A + C$ by Corollary A.1(2); both are continuous on $[a,b]$, so the identity extends to the end-points, and $F(b) - F(a) = A(b) - A(a) = \int_a^b f$. The version above is stronger: it needs only integrability of F', not continuity.

Remarks.

- *Part 1 needs continuity at the point.* With $f(t) = -1$ for $t < 0$ and $f(t) = +1$ for $t \ge 0$ on $[-1,1]$, $A(x) = \int_{-1}^{x} f = |x| - 1$, which is continuous (as Part 1 promises for any integrable f) but not differentiable at 0, the jump of f. In circuit language: the charge $q(t) = \int i\,dt$ on a capacitor is continuous even when the current switches abruptly, but dq/dt fails to exist at the switching instant.
- *Part 2 needs $F' = f$ on the whole interval.* The formal computation $\int_{-1}^{1} \frac{dx}{x^2} = \big[-\frac{1}{x}\big]_{-1}^{1} = -2$ is absurd for a positive integrand. Here $F = -1/x$ is not continuous (not even defined) on $[-1,1]$, and $1/x^2$ is unbounded, hence not Riemann integrable; the integral is improper and divergent (Ch. 5 §5.3).
- *Part 2 needs F' to be integrable.* $F(x) = x^2 \sin(1/x^2)$, $F(0) = 0$, is differentiable at every point of $[0,1]$, but $F'(x) = 2x\sin(1/x^2) - \frac{2}{x}\cos(1/x^2)$ for $x \neq 0$ is unbounded near 0, so the Riemann integral of F' over $[0,1]$ does not exist even though $F(1) - F(0) = \sin 1$ is perfectly finite.
- *Complex-valued integrands.* Applying both parts to the real and imaginary parts shows that they hold verbatim for $f : [a,b] \to \mathbb{C}$; for example $\int_0^T e^{j\omega t}\,dt = \big[e^{j\omega t}/(j\omega)\big]_0^T$ for $\omega \neq 0$, because $\frac{d}{dt}e^{j\omega t} = j\omega e^{j\omega t}$ (§A.6).
- *Where it is used.* The FTC is stated in Ch. 5 §5.1.5. Part 1 underlies every "accumulated quantity" of Ch. 5 §5.8 (charge from current, energy from power) and the integrating-factor method of Ch. 11 §11.3; Part 2 is the evaluation rule used in every chapter.

A.3 Integration by parts and integration by substitution

Each of the two main integration techniques is a differentiation rule read backwards through FTC Part 2: parts reverses the product rule, substitution reverses the chain rule. The proofs are short, but the hypotheses are where students (and computer algebra systems) go wrong.

Theorem A.3 (Integration by parts)

Let u and v be continuously differentiable (C^1) on $[a, b]$, real- or complex-valued. Then

$$\int_a^b u(x)\, v'(x)\, dx = \Big[u(x)\, v(x)\Big]_a^b - \int_a^b u'(x)\, v(x)\, dx.$$

On any interval where u, v are C^1, the indefinite form $\int u\, v'\, dx = uv - \int u'v\, dx$ holds (each side an antiderivative of uv', up to an additive constant).

Proof. By the product rule, $(uv)' = u'v + uv'$ on $[a, b]$ (one-sided at the end-points), and the right-hand side is continuous because u, v, u', v' are. Continuous functions are integrable, so FTC Part 2 applied to $F = uv$ gives

$$\Big[uv\Big]_a^b = \int_a^b (uv)'\, dx = \int_a^b u'v\, dx + \int_a^b uv'\, dx,$$

using linearity of the integral. Rearranging gives the formula. For the indefinite form: if G is an antiderivative of $u'v$ (it exists by FTC Part 1), then $(uv - G)' = u'v + uv' - u'v = uv'$, so $uv - G$ is an antiderivative of uv'. For complex-valued u, v the product rule holds by the same algebra, and FTC applies componentwise. ∎

Theorem A.4 (Integration by substitution)

Let $g : [\alpha, \beta] \to \mathbb{R}$ be C^1, and let f be continuous on an interval I that contains $g([\alpha, \beta])$. Then

$$\int_\alpha^\beta f\big(g(t)\big)\, g'(t)\, dt = \int_{g(\alpha)}^{g(\beta)} f(x)\, dx.$$

g need **not** be one-to-one.

Inverse form. If in addition g is strictly monotonic on $[\alpha, \beta]$, then for $a = g(\alpha)$, $b = g(\beta)$ every integral $\int_a^b f(x)\, dx$ may be computed as $\int_{g^{-1}(a)}^{g^{-1}(b)} f(g(t))\, g'(t)\, dt$, i.e. by putting $x = g(t)$, $dx = g'(t)\, dt$ and transforming the limits with g^{-1}.

Proof. Fix $x_0 \in I$ and let $F(x) = \int_{x_0}^x f(s)\, ds$ for $x \in I$. Since f is continuous on the interval I, FTC Part 1 gives $F' = f$ on I. The composite $F \circ g$ is defined on $[\alpha, \beta]$ because g takes values in I, and by the chain rule

$$\frac{d}{dt} F\big(g(t)\big) = F'\big(g(t)\big)\, g'(t) = f\big(g(t)\big)\, g'(t),$$

which is continuous on $[\alpha, \beta]$ (composition and product of continuous functions). FTC Part 2, applied twice, gives

$$\int_\alpha^\beta f\big(g(t)\big) g'(t)\, dt = F\big(g(\beta)\big) - F\big(g(\alpha)\big) = \int_{g(\alpha)}^{g(\beta)} f(x)\, dx.$$

The inverse form is the same identity read from right to left; monotonicity is needed only so that $t = g^{-1}(x)$ is a well-defined function and the new limits $g^{-1}(a) = \alpha$, $g^{-1}(b) = \beta$ are determined uniquely. ∎

Remarks.

- *Non-injective substitutions are fine in the forward direction.* With $g(t) = t^2$ on $[-1, 2]$ (which is not one-to-one: it folds $[-1, 0]$ back over $[0, 1]$),

$$\int_{-1}^{2} 2t \cos(t^2)\, dt = \int_1^4 \cos x\, dx = \sin 4 - \sin 1 \approx -1.59827,$$

which agrees with direct numerical quadrature to all printed digits. The folded part is traversed twice in opposite directions and cancels — the theorem accounts for orientation automatically, which is why no absolute value $|g'|$ appears (contrast the multivariable change of variables of Ch. 6 §6.8, which uses $|\det \mathbf{J}|$ and does require injectivity).

- *The hypothesis that g is C^1 on the whole interval is essential.* The tangent half-angle substitution $x = \tan(t/2)$ of Ch. 5 §5.6.5 turns $\int \frac{dt}{2+\cos t}$ into $\int \frac{2\,dx}{3+x^2}$. Applied blindly to $\int_0^{2\pi}$, it maps both limits to $x = \tan 0 = \tan \pi = 0$ and gives 0, which is impossible for a positive integrand. The true value is $2\pi/\sqrt{3} \approx 3.62760$ (checked by numerical quadrature). The error: $\tan(t/2)$ is not even defined at $t = \pi$, so the hypotheses fail. The correct procedure is to split at $t = \pi$; each half maps onto a half-line of x and gives an improper integral equal to $\pi/\sqrt{3}$.
- *The hypothesis on f concerns the whole range of g.* f must be continuous on an interval containing all values $g(t)$, not merely on the interval between $g(\alpha)$ and $g(\beta)$. If g wanders outside that interval into a region where f is singular, the left side may diverge while the right side is finite.
- *Integration by parts with a jump.* If u or v has a jump discontinuity inside $[a, b]$, split the interval at the jump; the boundary terms then contribute the jump values. This is the "jump method" for Fourier coefficients in Ch. 15 §15.7.
- *Where they are used.* The rules are stated in Ch. 5 §5.5–5.6. Parts gives the Laplace-transform derivative rule $\mathcal{L}\{x'\} = sX(s) - x(0^-)$ (Ch. 14), the decay of Fourier coefficients with smoothness (Ch. 15), the symmetry of Sturm–Liouville operators (Ch. 11 §11.8) and the integral form of Taylor's remainder (§A.4). Linear substitution $x = at + b$ gives the time-shift and time-scaling theorems of every transform in Part V.

A.4 Taylor's theorem with Lagrange and integral remainders

Taylor's theorem quantifies how well the Taylor polynomial

$$p_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k = f(a) + f'(a)(x-a) + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n$$

approximates f near a (Ch. 4 §4.7). The **remainder** is $R_n(x) = f(x) - p_n(x)$. Two exact formulas for it are proved here: Lagrange's, which needs the weakest hypotheses and gives the familiar error bound, and the integral form, which is exact without an unknown intermediate point.

Theorem A.5 (Taylor's theorem)

Let I be an open interval, $a, x \in I$, and $n \geq 0$ an integer.

(a) Lagrange remainder. If f is $n+1$ times differentiable on I, then there is a point c strictly between a and x (take $c = a$ if $x = a$) such that

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}.$$

Consequently, if $|f^{(n+1)}| \leq M$ on the interval between a and x, then $|R_n(x)| \leq \dfrac{M\,|x-a|^{n+1}}{(n+1)!}$.

(b) Integral remainder. If f is C^{n+1} on I (that is, $f^{(n+1)}$ exists and is continuous), then

$$R_n(x) = \frac{1}{n!}\int_a^x f^{(n+1)}(t)\,(x-t)^n\,dt.$$

Proof of (a). The case $x = a$ is trivial ($R_n(a) = 0$), so fix $x \neq a$. Define the constant K by

$$f(x) = p_n(x) + \frac{K}{(n+1)!}(x-a)^{n+1},$$

which is possible because $(x-a)^{n+1} \neq 0$. We must show $K = f^{(n+1)}(c)$ for some c between a and x. The trick is to treat the expansion point as the variable: for $t \in I$ define

$$G(t) = f(x) - \sum_{k=0}^{n} \frac{f^{(k)}(t)}{k!}(x-t)^k - \frac{K}{(n+1)!}(x-t)^{n+1}.$$

Values at the ends. At $t = x$ every term with $k \geq 1$ and the last term vanish, and the $k = 0$ term is $f(x)$, so $G(x) = 0$. At $t = a$, the sum is $p_n(x)$, so $G(a) = f(x) - p_n(x) - \frac{K}{(n+1)!}(x-a)^{n+1} = 0$ by the choice of K.

Derivative. Each $f^{(k)}$ with $k \leq n$ is differentiable on I, so G is differentiable on I (hence continuous). By the product rule, for $k \geq 1$,

$$\frac{d}{dt}\left[\frac{f^{(k)}(t)}{k!}(x-t)^k\right] = \frac{f^{(k+1)}(t)}{k!}(x-t)^k - \frac{f^{(k)}(t)}{(k-1)!}(x-t)^{k-1},$$

and for $k = 0$ the derivative is $f'(t)$, which is the first term of the same formula with no second term. Summing over $k = 0, \dots, n$, the second part of term k cancels the first part of term $k-1$; the sum telescopes to its last surviving piece, $\frac{f^{(n+1)}(t)}{n!}(x-t)^n$. Therefore

$$G'(t) = -\frac{f^{(n+1)}(t)}{n!}(x-t)^n + \frac{K}{n!}(x-t)^n = \frac{(x-t)^n}{n!}\Big[K - f^{(n+1)}(t)\Big].$$

Rolle. G is continuous on the closed interval between a and x, differentiable inside it, and vanishes at both ends. By Rolle's theorem (Theorem A.1) there is c strictly between a and x with $G'(c) = 0$. Since $c \neq x$, $(x-c)^n \neq 0$, so $K = f^{(n+1)}(c)$. The bound follows by taking absolute values. ∎

Proof of (b). Induction on n.

Base case $n = 0$. The claim is $f(x) - f(a) = \int_a^x f'(t)\,dt$, which is FTC Part 2 (Theorem A.2) applied to the continuous function f'.

Inductive step. Suppose the formula holds for $n-1$ (with $n \geq 1$) and that f is C^{n+1}; then in particular f is C^n and

$$R_{n-1}(x) = \frac{1}{(n-1)!}\int_a^x f^{(n)}(t)(x-t)^{n-1}\,dt.$$

Integrate by parts (Theorem A.3) with $u(t) = f^{(n)}(t)$ and $v(t) = -\dfrac{(x-t)^n}{n!}$, so that $v'(t) = \dfrac{(x-t)^{n-1}}{(n-1)!}$; both u and v are C^1 because f is C^{n+1}. Then

$$R_{n-1}(x) = \left[-f^{(n)}(t)\frac{(x-t)^n}{n!}\right]_{t=a}^{t=x} + \int_a^x f^{(n+1)}(t)\frac{(x-t)^n}{n!}\,dt = \frac{f^{(n)}(a)}{n!}(x-a)^n + \frac{1}{n!}\int_a^x f^{(n+1)}(t)(x-t)^n\,dt.$$

Since $R_n(x) = R_{n-1}(x) - \frac{f^{(n)}(a)}{n!}(x-a)^n$ (the degree-n polynomial has one more term), this is exactly the formula for R_n. ∎

From (b) back to (a). Under the stronger hypothesis $f \in C^{n+1}$, (a) also follows from (b) via the weighted mean value theorem for integrals: $(x-t)^n$ does not change sign between a and x, so $\int_a^x f^{(n+1)}(t)(x-t)^n\,dt = f^{(n+1)}(c)\int_a^x (x-t)^n\,dt = f^{(n+1)}(c)\frac{(x-a)^{n+1}}{n+1}$ for some c (the proof is that of Lemma A.2, with $m \leq f^{(n+1)} \leq M$ multiplied by the one-signed weight). Dividing by $n!$ gives Lagrange's form.

Corollary A.5 (The exponential, sine and cosine series)

For every real x,

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}, \qquad \cos x = \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m}}{(2m)!}, \qquad \sin x = \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m+1}}{(2m+1)!}.$$

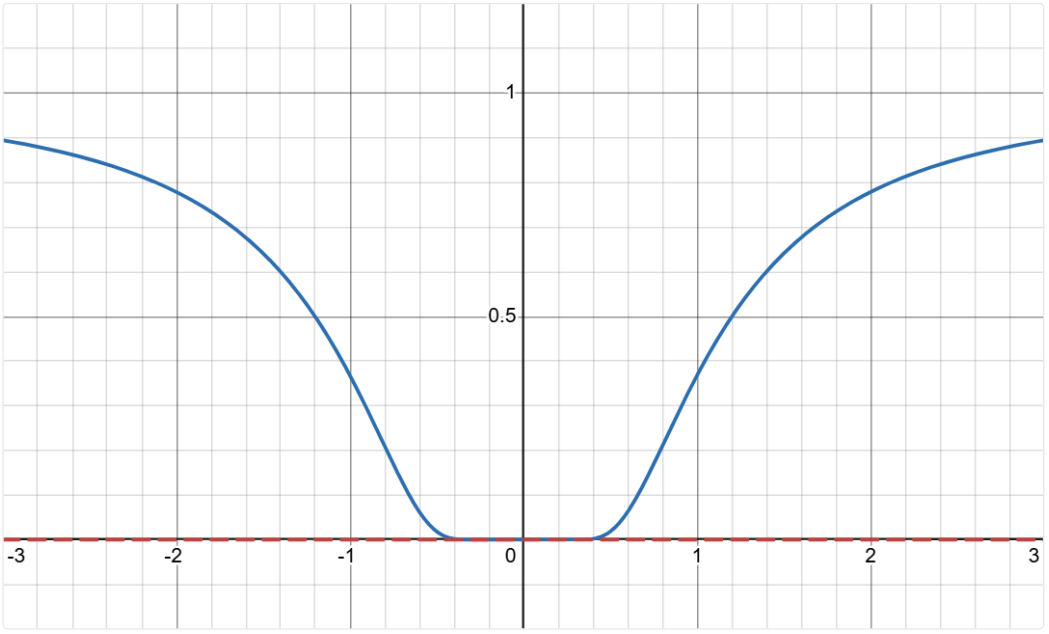
Proof. First, $r^n/n! \to 0$ for every fixed $r \geq 0$: choose an integer $N \geq 2r$; for $n > N$, each factor r/k with $k > N$ is at most $\tfrac{1}{2}$, so $\frac{r^n}{n!} = \frac{r^N}{N!}\prod_{k=N+1}^{n}\frac{r}{k} \leq \frac{r^N}{N!}2^{-(n-N)} \to 0$.

For cos and sin, every derivative is one of $\pm\sin, \pm\cos$, so $|f^{(n+1)}| \leq 1$ everywhere; by Theorem A.5(a) with $a = 0$, $|R_n(x)| \leq |x|^{n+1}/(n+1)! \to 0$. The Taylor coefficients at 0 follow from the cycle $\cos,\ -\sin,\ -\cos,\ \sin$ evaluated at 0: $1, 0, -1, 0, \ldots$ for cos and $0, 1, 0, -1, \ldots$ for sin. For e^x every derivative is e^x, so on the interval between 0 and x, $|f^{(n+1)}| \leq e^{|x|}$ and $|R_n(x)| \leq e^{|x|}|x|^{n+1}/(n+1)! \to 0$. In each case $p_n(x) \to f(x)$, which is what convergence of the series to f means. ∎

Numerical check. For $e^{0.5}$ with $n = 3$: $p_3(0.5) = 1 + 0.5 + 0.125 + 0.020\overline{83} = 1.645\overline{83}$, and $e^{0.5} = 1.6487213$, so the actual error is 0.0028879. Lagrange's bound with $M = e^{0.5}$ is $1.6487 \times 0.5^4/24 = 0.0042935$, which is indeed larger. For $\cos\pi$ with the degree-6 polynomial (which equals p_7, since the x^7 coefficient is zero), $p_6(\pi) = -1.21135$, error 0.21135, and the bound $\pi^8/8! = 0.23533$ holds.

Remarks.

- *A smooth function need not equal its Taylor series.* Let $\varphi(x) = e^{-1/x^2}$ for $x \neq 0$ and $\varphi(0) = 0$. By induction, for $x \neq 0$, $\varphi^{(k)}(x) = P_k(1/x)\,e^{-1/x^2}$ with P_k a polynomial (differentiate and use the chain rule). Since $u^m e^{-u^2} \to 0$ as $|u| \to \infty$ for every m, the difference quotient $\varphi^{(k)}(h)/h = \frac{1}{h}P_k(\frac{1}{h})e^{-1/h^2} \to 0$, so $\varphi^{(k+1)}(0) = 0$ for every k. The Taylor series of φ at 0 is identically zero, yet $\varphi(x) > 0$ for $x \neq 0$: here $R_n(x) = \varphi(x)$ for every n, and Theorem A.5 is not contradicted — the derivatives of φ near 0 grow so fast that the bound does not tend to 0. Functions equal to their Taylor series near every point are called **analytic**; in complex analysis (Ch. 18) differentiability in the complex sense forces analyticity, which is one reason that theory is so rigid.
- *Lagrange's form does not hold for complex-valued functions* (it is a mean value statement; see the remark in §A.1). The integral form (b) does hold for complex-valued f, by applying it to real and imaginary parts, and gives the bound $|R_n(x)| \leq M|x-a|^{n+1}/(n+1)!$ with $M = \max|f^{(n+1)}|$, since $|\int F| \leq \int|F|$.
- *Peano form.* If f is C^{n+1} near a, part (a) gives $R_n(x) = O(|x-a|^{n+1})$: halving the step reduces the error of a degree-n approximation by about 2^{n+1}. This is the source of the error orders of the finite-difference formulas in Ch. 4 §4.7, of the trapezoidal and Simpson rules in Ch. 5 §5.7, and of Runge–Kutta methods in Ch. 22.
- *Where it is used.* Linearization and small-signal models (Ch. 4 §4.8), the second-derivative test and Newton's method in optimization (Ch. 23 §23.4–23.6) and the two-variable Taylor theorem of Ch. 6 §6.3 (obtained by applying Theorem A.5 to $t \mapsto f(\mathbf{a} + t\mathbf{h})$).



The function e^{-1/x^2} (blue) is infinitely differentiable, and every derivative vanishes at 0, so its Maclaurin series is the zero function (red, dashed) — which agrees with it only at the origin.

A.5 L'Hôpital's rule (the $0/0$ case)

Theorem A.6 (L'Hôpital's rule, $0/0$ form)

Let $-\infty \leq a < b \leq \infty$ and let f, g be differentiable on (a, b) with $g'(x) \neq 0$ for all $x \in (a, b)$. Suppose

$$\lim_{x \to a^+} f(x) = \lim_{x \to a^+} g(x) = 0 \qquad \text{and} \qquad \lim_{x \to a^+} \frac{f'(x)}{g'(x)} = L,$$

where L is a real number, $+\infty$ or $-\infty$. Then $g(x) \neq 0$ on (a, b) and

$$\lim_{x \to a^+} \frac{f(x)}{g(x)} = L.$$

The same holds for left-hand limits $x \to b^-$ (including $b = +\infty$), and hence, by combining the two sides, for two-sided limits $x \to a$ when the hypotheses hold on a punctured neighbourhood of a.

Proof for finite a. Extend f and g to $[a, b)$ by setting $f(a) = g(a) = 0$. Because the limits at a^+ are 0, the extended functions are continuous on $[a, x]$ for every $x \in (a, b)$, and they are differentiable on (a, x).

Step 1: g does not vanish. If $g(x) = 0$ for some $x \in (a, b)$, then $g(a) = g(x) = 0$ and Rolle's theorem would give a point of (a, x) where $g' = 0$, contrary to hypothesis.

Step 2: Cauchy's MVT. Fix $x \in (a, b)$. By Theorem A.1(3) on $[a, x]$ (where $g' \neq 0$ on (a, x)), there is $\xi_x \in (a, x)$ with

$$\frac{f(x)}{g(x)} = \frac{f(x) - f(a)}{g(x) - g(a)} = \frac{f'(\xi_x)}{g'(\xi_x)}.$$

Step 3: pass to the limit. Let V be any neighbourhood of L (an interval $(L - \varepsilon, L + \varepsilon)$ if L is finite, or (K, ∞) if $L = +\infty$, or $(-\infty, -K)$ if $L = -\infty$). By hypothesis there is $\eta > 0$ with $f'(s)/g'(s) \in V$ for all $s \in (a, a + \eta)$. If $x \in (a, a + \eta)$, then $\xi_x \in (a, x) \subset (a, a + \eta)$, so $f(x)/g(x) = f'(\xi_x)/g'(\xi_x) \in V$. Since V was arbitrary, $f(x)/g(x) \to L$. ∎

Proof for $a = -\infty$ (and, symmetrically, for $b = +\infty$). Take $b = +\infty$ for definiteness, with the hypotheses holding on (M, ∞), $M > 0$, as $x \to +\infty$. Put $F(t) = f(1/t)$ and $G(t) = g(1/t)$ for $t \in (0, 1/M)$. By the chain rule $F'(t) = -f'(1/t)/t^2$ and $G'(t) = -g'(1/t)/t^2 \neq 0$, and

$$\lim_{t \to 0^+} F(t) = \lim_{x \to \infty} f(x) = 0, \quad \lim_{t \to 0^+} G(t) = 0, \quad \frac{F'(t)}{G'(t)} = \frac{f'(1/t)}{g'(1/t)} \xrightarrow[t \to 0^+]{} L.$$

The finite case at 0^+ gives $F(t)/G(t) \to L$, i.e. $f(x)/g(x) \to L$ as $x = 1/t \to \infty$. ∎

Example. $\lim_{x \to 0} \dfrac{1 - \cos x}{x^2}$: numerator and denominator tend to 0, and $g'(x) = 2x \neq 0$ for $x \neq 0$. The ratio of derivatives $\frac{\sin x}{2x}$ is again $0/0$ with $(2x)' = 2 \neq 0$, and $\frac{\cos x}{2} \to \frac{1}{2}$. Applying Theorem A.6 twice (inner limit first) gives $\frac{1}{2}$. The Taylor series give the same answer at once: $1 - \cos x = \frac{x^2}{2} - \frac{x^4}{24} + \dots$. Similarly $\lim_{x \to 0} \frac{\sin x - x}{x^3} = -\frac{1}{6}$.

Remarks.

- *The Ch. 4 sketch versus this proof.* The sketch in Ch. 4 §4.2 assumes f, g are differentiable *at* a with $g'(a) \neq 0$. The proof above assumes nothing at a itself, which is what is needed when the rule must be applied repeatedly (after one application, f' and g' are typically $0/0$ at a again) and for limits at infinity.
- *The rule is one-directional.* $f(x) = x^2 \sin(1/x)$, $g(x) = x$ at $a = 0$: $f/g = x \sin(1/x) \to 0$ by the squeeze theorem, but $f'/g' = 2x \sin(1/x) - \cos(1/x)$ has no limit. If $\lim f'/g'$ does not exist, the rule says nothing.
- *The hypothesis $g' \neq 0$ is not a formality.* It is needed in Step 1 and in Cauchy's theorem. The classical example of Stolz, in the ∞/∞ form, shows what goes wrong: $f(x) = x + \sin x \cos x$, $g(x) = e^{\sin x} f(x)$ as $x \to \infty$. Here $f' = 2\cos^2 x$ and $g' = e^{\sin x} \cos x \, (f + 2\cos x)$, so after cancelling the common factor $\cos x$, $\frac{f'}{g'} = \frac{2e^{-\sin x} \cos x}{f + 2\cos x} \to 0$; but $f/g = e^{-\sin x}$ oscillates between e^{-1} and e. The culprit: g' vanishes at every zero of $\cos x$, arbitrarily far out, and the "cancellation" divided by zero.
- *The ∞/∞ form.* If instead $|g(x)| \to \infty$ as $x \to a^+$ (no condition on f), the conclusion again holds. The proof applies Cauchy's MVT on $[x, y]$ with y fixed close to a and $x \to a^+$, writing $\frac{f(x)}{g(x)} = \frac{f(x) - f(y)}{g(x) - g(y)} \cdot \frac{g(x) - g(y)}{g(x)} + \frac{f(y)}{g(x)}$ and noting that the last two factors tend to 1 and 0; see Rudin's *Principles*, Thm 5.13.
- *Where it is used.* Limits in Ch. 4 §4.1–4.2 (exponentials dominate powers), initial and final value theorems and the residue at a simple pole $\lim_{s \to p}(s - p)N(s)/D(s) = N(p)/D'(p)$ (Ch. 14, Ch. 18), and the value $\operatorname{sinc}(0) = 1$ used throughout Ch. 16.

A.6 Euler's formula

Ch. 2 §2.4 derives $e^{j\theta} = \cos\theta + j\sin\theta$ in three ways. Here we make two of them rigorous. The first question is what $e^{j\theta}$ *means*: the real exponential is not defined for imaginary arguments, so we must choose a definition and then prove the formula from it.

The complex exponential

For $z \in \mathbb{C}$,

$$e^z = \exp(z) = \sum_{n=0}^{\infty} \frac{z^n}{n!} = \lim_{N \to \infty} \sum_{n=0}^{N} \frac{z^n}{n!}.$$

The series converges for every z: its real and imaginary parts are real series whose terms are bounded in modulus by $|z|^n/n!$, and $\sum |z|^n/n! = e^{|z|}$ converges (Corollary A.5), so both are absolutely convergent and hence convergent. For real z this agrees with the real exponential, by Corollary A.5.

Throughout this section write

$$c(\theta) = \cos\theta + j\sin\theta, \qquad \bar{c}(\theta) = \cos\theta - j\sin\theta \qquad (\theta \in \mathbb{R}),$$

so that $c'(\theta) = -\sin\theta + j\cos\theta = j\,c(\theta)$, $\bar{c}'(\theta) = -\sin\theta - j\cos\theta = -j\,\bar{c}(\theta)$, and $c(\theta)\,\bar{c}(\theta) = \cos^2\theta + \sin^2\theta = 1$.

Theorem A.7 (Euler's formula)

For every real θ,

$$e^{j\theta} = \cos\theta + j\sin\theta.$$

Moreover $\dfrac{d}{d\theta} e^{j\theta} = j\,e^{j\theta}$, and $e^{j\theta}$ is the **only** differentiable function $y : \mathbb{R} \to \mathbb{C}$ with $y' = jy$ and $y(0) = 1$.

We give two independent proofs. The first uses the Taylor series of cos and sin; the second uses only their derivatives.

Proof 1 — power series

Let $S_N(\theta) = \sum_{n=0}^{N} \frac{(j\theta)^n}{n!}$. The powers of j cycle with period 4: $j^{2m} = (j^2)^m = (-1)^m$ and $j^{2m+1} = j\,(-1)^m$. Split the partial sum with an odd number of terms, S_{2N+1}, into even and odd n:

$$S_{2N+1}(\theta) = \sum_{m=0}^{N} \frac{(-1)^m \theta^{2m}}{(2m)!} + j \sum_{m=0}^{N} \frac{(-1)^m \theta^{2m+1}}{(2m+1)!}.$$

No rearrangement of an infinite series is involved: this is a regrouping of finitely many terms. The real part is the N-th partial sum of the cosine series and the imaginary part is the N-th partial sum of the sine series, so by Corollary A.5,

$$\operatorname{Re}\{S_{2N+1}(\theta)\} \to \cos\theta, \qquad \operatorname{Im}\{S_{2N+1}(\theta)\} \to \sin\theta \qquad (N \to \infty).$$

A sequence of complex numbers converges if and only if its real and imaginary parts converge, so $S_{2N+1}(\theta) \to \cos\theta + j\sin\theta$. But the whole sequence $S_N(\theta)$ converges to $e^{j\theta}$ by definition, and every subsequence of a convergent sequence has the same limit. Hence $e^{j\theta} = \cos\theta + j\sin\theta$. ■

Proof 2 — the differential equation and uniqueness

This proof has two steps: the series satisfies the initial-value problem $y' = jy$, $y(0) = 1$, and that problem has only one solution.

> **Lemma A.7a (Uniqueness)**
>
> Let I be an interval containing 0 and $y : I \to \mathbb{C}$ differentiable with $y'(\theta) = j\,y(\theta)$ on I. Then $y(\theta) = y(0)\,c(\theta)$ for all $\theta \in I$.

Proof. Let $w(\theta) = y(\theta)\,\bar{c}(\theta)$. By the product rule (valid for complex-valued functions),

$$w' = y'\,\bar{c} + y\,\bar{c}' = (jy)\,\bar{c} + y\,(-j\bar{c}) = 0.$$

By Corollary A.1(1) for complex-valued functions, w is constant on I, equal to $w(0) = y(0)\bar{c}(0) = y(0)$. Multiplying $y\bar{c} = y(0)$ by c and using $c\bar{c} = 1$ gives $y = y(0)\,c$. ■

> **Lemma A.7b (Derivative of the series)**
>
> $E(\theta) = \sum_{n=0}^{\infty} \frac{(j\theta)^n}{n!}$ is differentiable on $\mathbb{R}$ with $E'(\theta) = j\,E(\theta)$.

Proof. Fix θ and let $0 < |h| \le 1$. For each $n \ge 2$, Theorem A.5(a) applied to $t \mapsto t^n$ at $a = \theta$ with $n = 1$ gives

$$(\theta + h)^n - \theta^n - n\theta^{n-1}h = \frac{n(n-1)\gamma^{n-2}}{2}h^2$$

for some γ between θ and $\theta + h$, so that $|\gamma| \le |\theta| + 1 =: r$. For $n = 0, 1$ the left side is 0. The series $\sum_n \frac{j^n n\theta^{n-1}}{n!} = j\sum_{n\ge1}\frac{(j\theta)^{n-1}}{(n-1)!} = jE(\theta)$ converges, so we may combine three convergent series term by term:

$$E(\theta+h) - E(\theta) - h\,jE(\theta) = \sum_{n=2}^{\infty} \frac{j^n}{n!}\Big[(\theta+h)^n - \theta^n - n\theta^{n-1}h\Big].$$

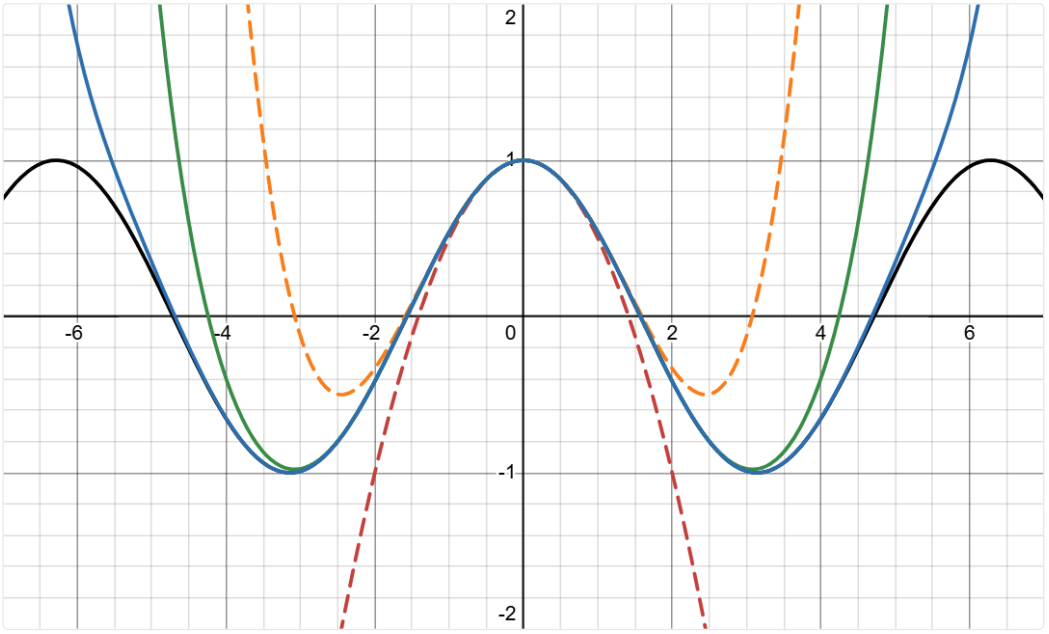
Taking moduli and using $|j^n| = 1$,

$$\Big|E(\theta+h) - E(\theta) - h\,jE(\theta)\Big| \le \frac{h^2}{2}\sum_{n=2}^{\infty}\frac{n(n-1)}{n!}r^{n-2} = \frac{h^2}{2}\sum_{m=0}^{\infty}\frac{r^m}{m!} = \frac{h^2}{2}e^r.$$

Dividing by $|h|$: $\left|\frac{E(\theta+h)-E(\theta)}{h} - jE(\theta)\right| \le \frac{|h|}{2}e^{|\theta|+1} \to 0$ as $h \to 0$. ■

(The identity $\sum_{n\ge2}\frac{n(n-1)}{n!}r^{n-2} = e^r$ used in the last step was also checked numerically, e.g. both sides equal 9.97418 at $r = 2.3$.)

Conclusion of Proof 2. By Lemma A.7b, $y = E$ satisfies $y' = jy$ on $\mathbb{R}$ with $y(0) = 1$. By Lemma A.7a, $E(\theta) = c(\theta) = \cos\theta + j\sin\theta$. The uniqueness statement of the theorem is Lemma A.7a with $y(0) = 1$. ■



The real part of the partial sums of $e^{j\theta}$ (degrees 2, 4, 8, 12) against $\cos\theta$ (black). Each polynomial hugs the cosine over a wider interval; by Corollary A.5 the agreement extends to any bounded interval as the degree grows.

Corollary A.7 (Consequences of Euler's formula)

For all real α, β, θ:

1. $|e^{j\theta}| = 1$ and $\overline{e^{j\theta}} = e^{-j\theta} = 1/e^{j\theta}$.
2. $e^{j(\alpha+\beta)} = e^{j\alpha}e^{j\beta}$; taking real and imaginary parts gives $\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$ and $\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$.
3. $\cos\theta = \dfrac{e^{j\theta} + e^{-j\theta}}{2}$ and $\sin\theta = \dfrac{e^{j\theta} - e^{-j\theta}}{2j}$.
4. $e^{j\theta} = 1$ if and only if $\theta = 2\pi m$ for some integer m; hence $e^{j\theta}$ is periodic with fundamental period 2π.
5. For real ω, $\dfrac{d}{dt}e^{j\omega t} = j\omega\, e^{j\omega t}$.

Proof. (1) $|e^{j\theta}|^2 = \cos^2\theta + \sin^2\theta = 1$; $\overline{c(\theta)} = \cos\theta - j\sin\theta = \cos(-\theta) + j\sin(-\theta) = c(-\theta)$ because cos is even and sin odd; and $c(\theta)c(-\theta) = c(\theta)\bar{c}(\theta) = 1$. (2) Fix α and let $y(\theta) = e^{j(\alpha+\theta)} = c(\alpha+\theta)$. By the chain rule $y'(\theta) = c'(\alpha+\theta) = j\,y(\theta)$, and $y(0) = e^{j\alpha}$. Lemma A.7a gives $y(\theta) = e^{j\alpha}e^{j\theta}$. Expanding $c(\alpha)c(\beta)$ and comparing real and imaginary parts gives the compound-angle formulas — so these formulas are a *consequence* of Euler's formula, not an input to it (Proof 2 used only $\sin' = \cos$, $\cos' = -\sin$ and $\cos^2 + \sin^2 = 1$). (3) Add and subtract $e^{j\theta} = \cos\theta + j\sin\theta$ and $e^{-j\theta} = \cos\theta - j\sin\theta$. (4) $e^{j\theta} = 1$ means $\cos\theta = 1$ and $\sin\theta = 0$; and $\cos\theta = 1$ exactly when $\theta \in 2\pi\mathbb{Z}$, which is the statement that the point at arc length θ on the unit circle returns to $(1,0)$ only after whole turns (the unit-circle definition of cos, Ch. 3 §3.1); conversely $\theta = 2\pi m$ gives $\cos\theta = 1$, $\sin\theta = 0$. Then $e^{j(\theta+T)} = e^{j\theta}$ for all θ iff $e^{jT} = 1$ (by (2)) iff $T \in 2\pi\mathbb{Z}$. (5) Chain rule with Theorem A.7: $\frac{d}{dt}c(\omega t) = \omega\, c'(\omega t) = j\omega\, c(\omega t)$. ∎

Remarks.

- *Why two proofs.* Proof 1 explains where the formula "comes from" — the even and odd parts of the exponential series are the cosine and sine series — but it leans on Taylor's theorem for cos and sin. Proof 2 is the more robust argument: it shows that *any* reasonable definition of $e^{j\theta}$ — anything differentiable that satisfies $y' = jy$, $y(0) = 1$ — is forced to equal $\cos\theta + j\sin\theta$. It is the simplest instance of the uniqueness theorem for linear ODEs (Ch. 11 §11.7), and the same "multiply by the inverse solution and show the derivative vanishes" trick proves that $e^{\mathbf{A}t}$ is the unique solution of $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ (Ch. 11 §11.9).
- *Radians are essential.* The proofs use $\sin' = \cos$, which holds only when angles are measured in radians (Ch. 3 §3.1). In degrees, $\frac{d}{d\theta}\sin\theta^\circ = \frac{\pi}{180}\cos\theta^\circ$, and $e^{j\theta}$ would have to be replaced by $e^{j\pi\theta/180}$.
- *The general exponential.* For $z = \sigma + j\omega$, the identity $e^z = e^\sigma e^{j\omega} = e^\sigma(\cos\omega + j\sin\omega)$ requires the addition law $e^{z+w} = e^z e^w$ for complex z, w, which follows from the Cauchy product of two absolutely convergent series (or from the uniqueness argument in the complex-variable setting of Ch. 18). It is the basis of the s-plane picture of e^{st}: σ sets growth or decay and ω sets oscillation.
- *Where it is used.* Phasors and impedance (Ch. 2 §2.8–2.9); all trigonometric identities by the "master method" (Ch. 3 §3.6); complex Fourier series and transforms (Ch. 15, Ch. 16); characteristic-root solutions of linear ODEs (Ch. 12).

A.7 De Moivre's theorem and the n-th roots of unity

We use the polar form of Ch. 2 §2.1: every $z \neq 0$ can be written $z = r\,e^{j\varphi}$ with $r = |z| > 0$ and φ real (an argument of z), and φ is determined up to an additive multiple of 2π (Corollary A.7(4)).

Theorem A.8 (De Moivre)

For every integer n (positive, zero or negative) and every real θ,

$$(\cos\theta + j\sin\theta)^n = \cos n\theta + j\sin n\theta, \qquad \text{i.e.} \qquad \left(e^{j\theta}\right)^n = e^{jn\theta},$$

and for $r > 0$, $(re^{j\theta})^n = r^n e^{jn\theta}$.

Proof. *$n \geq 0$, by induction.* For $n = 0$ both sides equal 1. If $(e^{j\theta})^n = e^{jn\theta}$, then by Corollary A.7(2),

$$\left(e^{j\theta}\right)^{n+1} = e^{jn\theta}\,e^{j\theta} = e^{j(n\theta+\theta)} = e^{j(n+1)\theta}.$$

$n = -m < 0$. By Corollary A.7(1), $(e^{j\theta})^{-1} = e^{-j\theta}$, so $(e^{j\theta})^{-m} = \left((e^{j\theta})^{-1}\right)^m = (e^{j(-\theta)})^m = e^{-jm\theta}$ by the case already proved. The last statement follows because real and complex powers of a product factor: $(r\,e^{j\theta})^n = r^n(e^{j\theta})^n$ for integer n (commutativity of multiplication). ∎

n-th roots and roots of unity

For an integer $n \geq 1$ and $z \in \mathbb{C}$, an **n-th root** of z is any $w \in \mathbb{C}$ with $w^n = z$. The n-th roots of 1 are the **n-th roots of unity**. Write $W = W_n = e^{j2\pi/n}$.

Theorem A.9 (n-th roots)

Let $n \geq 1$ and $z = re^{j\varphi} \neq 0$. Then $w^n = z$ has **exactly** n solutions,

$$w_k = r^{1/n}\, e^{j(\varphi + 2\pi k)/n}, \qquad k = 0, 1, \dots, n-1,$$

where $r^{1/n}$ is the positive real n-th root. They are $w_k = w_0 W^k$, equally spaced on the circle of radius $r^{1/n}$. In particular the n-th roots of unity are $1, W, W^2, \dots, W^{n-1}$, and:

1. $w^n - 1 = \prod_{k=0}^{n-1} \left(w - W^k\right)$ for all $w \in \mathbb{C}$;
2. for every integer m,

$$\sum_{k=0}^{n-1} W^{mk} = \begin{cases} n, & n \mid m, \\ 0, & \text{otherwise.} \end{cases}$$

Proof. *Each w_k is a root.* By Theorem A.8, $w_k^n = r\, e^{j(\varphi+2\pi k)} = r\, e^{j\varphi} e^{j2\pi k} = z$, using Corollary A.7(2) and $e^{j2\pi k} = 1$.

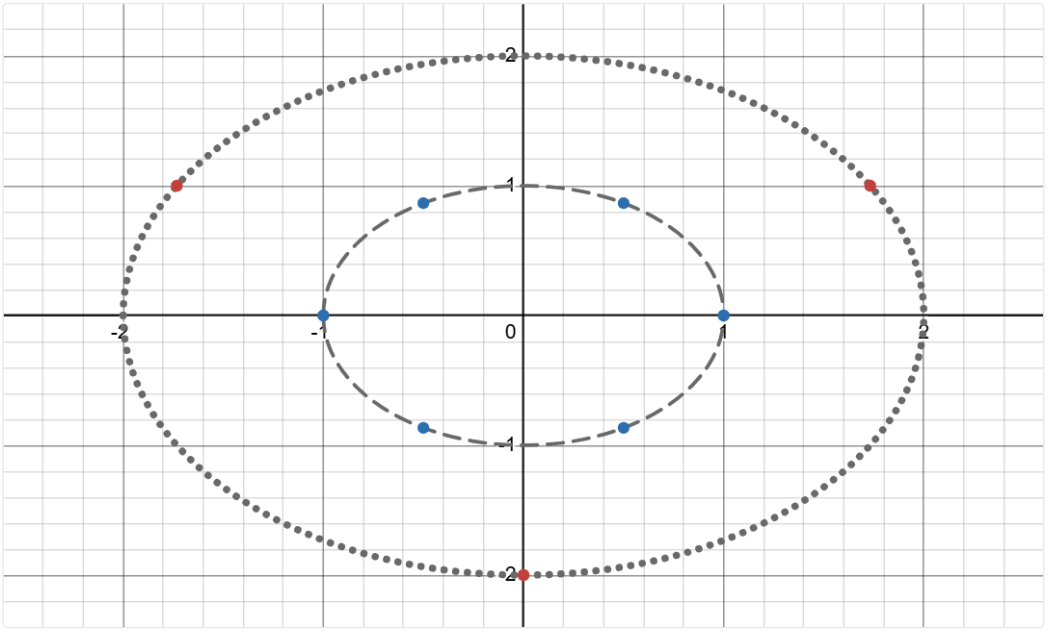
Every root is some w_k. Let $w^n = z$. Then $w \neq 0$ (else $z = 0$), so $w = \rho e^{j\psi}$ with $\rho > 0$. By Theorem A.8, $\rho^n e^{jn\psi} = re^{j\varphi}$. Taking moduli (Corollary A.7(1)) gives $\rho^n = r$; since $t \mapsto t^n$ is strictly increasing on $(0, \infty)$ (its derivative nt^{n-1} is positive, Corollary A.1(3)), $\rho = r^{1/n}$. Dividing out, $e^{j(n\psi - \varphi)} = 1$, so by Corollary A.7(4), $n\psi - \varphi = 2\pi m$ for some integer m, i.e. $\psi = (\varphi + 2\pi m)/n$. By the division algorithm $m = qn + k$ with $q \in \mathbb{Z}$ and $0 \leq k \leq n-1$; then $e^{j\psi} = e^{j(\varphi+2\pi k)/n}\, e^{j2\pi q} = e^{j(\varphi+2\pi k)/n}$, so $w = w_k$.

The w_k are distinct. If $w_k = w_l$ with $0 \leq l < k \leq n-1$, dividing gives $e^{j2\pi(k-l)/n} = 1$, so $(k-l)/n$ is an integer by Corollary A.7(4). But $0 < k - l < n$, a contradiction. Hence there are exactly n roots. The form $w_k = w_0 W^k$ is Corollary A.7(2).

(1) Factorisation. Let $p(w) = w^n - 1$. If $p(\alpha) = 0$, polynomial division gives $p(w) = (w - \alpha)q(w)$ with q monic of degree $n-1$ (the factor theorem). Apply this with $\alpha = W^0 = 1$; each remaining W^k, $k \geq 1$, satisfies $0 = p(W^k) = (W^k - 1)q(W^k)$ with $W^k - 1 \neq 0$ (distinctness), so W^k is a root of q. Repeating n times leaves a monic polynomial of degree 0, i.e. the constant 1.

(2) Sums. If $n \mid m$, then $W^m = e^{j2\pi(m/n)} = 1$ and every term is 1: the sum is n. Otherwise $\rho = W^m \neq 1$ (Corollary A.7(4)) and $\rho^n = (W^n)^m = 1$. The finite geometric sum gives $\sum_{k=0}^{n-1} \rho^k = \frac{1-\rho^n}{1-\rho} = 0$. ∎

Example. The cube roots of $z = 8j$: $r = 8$, $\varphi = \pi/2$, so $w_k = 2e^{j(\pi/6 + 2\pi k/3)}$, i.e. $w_0 = \sqrt{3} + j$, $w_1 = -\sqrt{3} + j$, $w_2 = -2j$. Each cubes to $8j$ (checked numerically), and their sum is 0, as part 2 with $m = 1$ predicts for $w_0(1 + W + W^2)$.



The sixth roots of unity (blue) form a regular hexagon on the unit circle; the cube roots of $8j$ (red) form an equilateral triangle on the circle of radius $8^{1/3} = 2$, rotated to angle $\varphi/n = 30°$.

Remarks.

- *Why exactly n and not infinitely many.* The infinitely many arguments $\varphi + 2\pi m$ of z give infinitely many candidate angles $(\varphi + 2\pi m)/n$, but they repeat with period n in m. This is the precise sense in which $z^{1/n}$ is "multi-valued" (Ch. 2 §2.6, Ch. 18).
- *De Moivre is for integer exponents.* Applied naively to $n = \frac{1}{2}$ it would give a contradiction: $e^{j0} = e^{j2\pi} = 1$, yet "$\left(e^{j2\pi}\right)^{1/2} = e^{j\pi} = -1$" while $\left(e^{j0}\right)^{1/2} = 1$. Likewise $(z^a)^b = z^{ab}$ fails for non-integer exponents: $\left((-1)^2\right)^{1/2} = 1$ but $\left((-1)^{1/2}\right)^2 = -1$ for either choice of $(-1)^{1/2} = \pm j$.
- *The hypothesis $z \neq 0$.* For $z = 0$ the only root is $w = 0$, with multiplicity n; the polar argument is undefined.
- *Where it is used.* Part 2 is the orthogonality relation of the DFT, $\sum_{k=0}^{N-1} e^{j2\pi(m-l)k/N} = N\delta_{ml}$ for $0 \leq m, l < N$ (Ch. 15 §15.7, Ch. 24), and with $n = 3$ it is the identity $1 + a + a^2 = 0$, $a = e^{j2\pi/3}$, which makes the phasors of a balanced three-phase set sum to zero (Ch. 3 §3.3). Part 1 underlies root-finding and factoring in Ch. 2 §2.6, the poles of Butterworth filters (equally spaced on a circle in the s-plane), and the N-point FFT's recursive splitting of W_N into $W_{N/2}$.

A.8 RMS of a sinusoid and of a sum of sinusoids of different frequencies

The statement "powers of components at different frequencies add" is used constantly — for harmonic distortion, for noise, for Parseval's theorem. Here it is proved in two settings: exactly over a common period when the frequencies are harmonics of one fundamental, and as a long-time average when they are not (for example $\cos t + \cos \sqrt{2}\,t$, which is not periodic at all).

Mean, mean-square and RMS

For x continuous on $[t_0, t_0+T]$, $T>0$, the **mean** and **RMS value over the window** are

$$\bar{x}_{[t_0,T]} = \frac{1}{T}\int_{t_0}^{t_0+T} x(t)\,dt, \qquad X_{\mathrm{rms},[t_0,T]} = \sqrt{\frac{1}{T}\int_{t_0}^{t_0+T} x(t)^2\,dt}.$$

For a T-periodic x the window is one period. For an arbitrary bounded x, the **long-time mean-square value** (average power in $1\ \Omega$) is $\langle x^2\rangle = \lim_{T\to\infty}\frac{1}{T}\int_{t_0}^{t_0+T} x^2\,dt$, when the limit exists.

Lemma A.8 (Averages over a period do not depend on the starting point; averages of sinusoids)

1. If x is continuous and T-periodic, then $\int_{t_0}^{t_0+T} x\,dt$ is the same for every t_0.
2. For $\Omega \neq 0$ and any real ψ, t_0, $T>0$,

$$\left|\int_{t_0}^{t_0+T} \cos(\Omega t+\psi)\,dt\right| \le \frac{2}{|\Omega|},$$

and the integral is exactly 0 when T is an integer multiple of $2\pi/|\Omega|$.

Proof. (1) Let $I(t_0) = \int_{t_0}^{t_0+T} x\,dt = \int_0^{t_0+T} x\,dt - \int_0^{t_0} x\,dt$. By FTC Part 1 (Theorem A.2) and the chain rule, $I'(t_0) = x(t_0+T) - x(t_0) = 0$, so I is constant by Corollary A.1(1). (2) An antiderivative of $\cos(\Omega t+\psi)$ is $\sin(\Omega t+\psi)/\Omega$, so by FTC Part 2 the integral equals $\big[\sin(\Omega(t_0+T)+\psi) - \sin(\Omega t_0+\psi)\big]/\Omega$, whose modulus is at most $2/|\Omega|$. If $T = 2\pi q/|\Omega|$ with q an integer, the two sines are equal because sin has period 2π. ∎

Theorem A.10 (RMS of sinusoids)

(a) Single sinusoid. Let $x(t) = C + A\cos(\omega t+\phi)$ with $\omega>0$ and A, C, ϕ real, and $T = 2\pi/\omega$. Over any window of length T,

$$\bar{x} = C, \qquad X_{\mathrm{rms}} = \sqrt{C^2 + \tfrac{1}{2}A^2}; \qquad \text{for } C=0:\ X_{\mathrm{rms}} = \frac{|A|}{\sqrt{2}}.$$

(b) Different frequencies, long-time average. Let

$$x(t) = A_0 + \sum_{k=1}^{N} A_k\cos(\omega_k t+\phi_k),$$

with real A_k, ϕ_k and **distinct** positive frequencies $\omega_1,\dots,\omega_N$ (no commensurability assumed). Then, uniformly in t_0,

$$\left|\frac{1}{T}\int_{t_0}^{t_0+T} x^2\,dt - P\right| \le \frac{B}{T}, \qquad P = A_0^2 + \sum_{k=1}^{N}\frac{A_k^2}{2},$$

with the constant

$$B = \sum_{k=1}^{N}\left(\frac{4|A_0A_k|}{\omega_k} + \frac{A_k^2}{2\omega_k}\right) + \sum_{1\le k<l\le N} 2|A_kA_l|\left(\frac{1}{|\omega_k-\omega_l|} + \frac{1}{\omega_k+\omega_l}\right).$$

Hence $\langle x^2\rangle = P$ and the long-time RMS value is $\sqrt{A_0^2 + \sum_k A_k^2/2}$, independent of the phases ϕ_k.

(c) Harmonics. If $\omega_k = n_k\omega_0$ for distinct positive integers n_k, then x is periodic with period $T_0 = 2\pi/\omega_0$, and the mean-square value over **any** window of length T_0 equals P exactly.

Proof. Write $\theta_k = \omega_k t + \phi_k$. Expanding the square of a finite sum,

$$x^2 = A_0^2 + 2A_0\sum_k A_k\cos\theta_k + \sum_k A_k^2\cos^2\theta_k + 2\sum_{k<l} A_kA_l\cos\theta_k\cos\theta_l.$$

Use the power-reduction and product-to-sum identities (both follow from Corollary A.7(2)–(3)):

$$\cos^2\theta_k = \tfrac{1}{2} + \tfrac{1}{2}\cos 2\theta_k, \qquad 2\cos\theta_k\cos\theta_l = \cos(\theta_k-\theta_l) + \cos(\theta_k+\theta_l).$$

So

$$x^2 = \underbrace{A_0^2 + \sum_k \frac{A_k^2}{2}}_{P} + 2A_0\sum_k A_k\cos\theta_k + \sum_k \frac{A_k^2}{2}\cos 2\theta_k + \sum_{k<l} A_kA_l\Big[\cos(\theta_k-\theta_l) + \cos(\theta_k+\theta_l)\Big].$$

Every term after P is a constant times $\cos(\Omega t+\psi)$ with angular frequency Ω equal to one of ω_k, $2\omega_k$, $\omega_k-\omega_l$ or $\omega_k+\omega_l$ ($k<l$). **None of these is zero**: the ω_k are positive, and $\omega_k-\omega_l\neq 0$ because the frequencies are distinct. This is the only place the hypothesis is used.

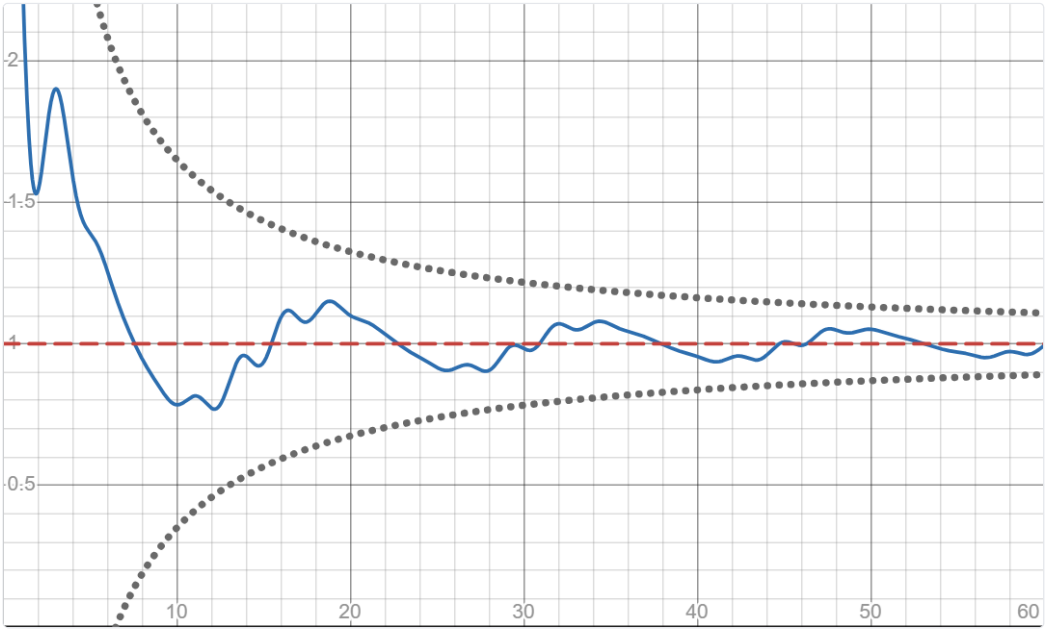
(b) Integrate over $[t_0, t_0+T]$, divide by T and apply Lemma A.8(2) term by term with the triangle inequality: the terms contribute at most $\frac{1}{T}\cdot 2|A_0A_k|\frac{2}{\omega_k}$, $\frac{1}{T}\cdot\frac{A_k^2}{2}\cdot\frac{2}{2\omega_k}$, and $\frac{1}{T}|A_kA_l|\big(\frac{2}{|\omega_k-\omega_l|}+\frac{2}{\omega_k+\omega_l}\big)$. Their sum is B/T. Letting $T\to\infty$ gives $\langle x^2\rangle = P$.

(a) is the case $N=1$, $\omega_1=\omega$, $A_0=C$ of the expansion: the oscillating terms have frequencies ω and 2ω, and $T=2\pi/\omega$ is a multiple of both $2\pi/\omega$ and $2\pi/(2\omega)$, so by Lemma A.8(2) their integrals over the window vanish exactly; the mean-square is $C^2+A^2/2$. The mean is C by the same lemma applied to x itself.

(c) Each Ω above is now a non-zero integer multiple of ω_0: $n_k\omega_0$, $2n_k\omega_0$, $(n_k-n_l)\omega_0$ or $(n_k+n_l)\omega_0$. So T_0 is an integer multiple of every $2\pi/|\Omega|$, every oscillating term integrates to exactly 0 over any window of length T_0 (Lemma A.8(2)), and the mean-square is exactly P. ■

Numerical checks.

- Ch. 3 Worked Example 3.29: $v = 10 + 8\cos\omega t + 6\cos(3\omega t + 40^\circ)$ gives $P = 100 + 32 + 18 = 150\ \mathrm{V}^2$, $V_{\mathrm{rms}} = 12.2474\ \mathrm{V}$; averaging v^2 over one period at 200 000 points gives 12.247449 V.
- The non-periodic signal $x = \cos t + \cos\sqrt{2}\,t$ has $P = 1$ and, from the formula, $B = \frac{1}{2}+\frac{1}{2\sqrt{2}}+2\big(\frac{1}{\sqrt{2}-1}+\frac{1}{\sqrt{2}+1}\big) = 6.51$. Numerical integration gives windowed mean-squares over $[0,T]$ of $0.7848,\ 0.98667,\ 0.99969,\ 1.00028$ for $T = 10, 100, 1000, 10\,000$; the deviations $0.215,\ 0.0133,\ 0.0003,\ 0.0003$ respect the bound $B/T = 0.65,\ 0.065,\ 0.0065,\ 0.00065$.



Running mean-square of $\cos t + \cos\sqrt{2}\,t$ (blue). It never settles over a "period" (there is none) but converges to $P = \frac{1}{2}+\frac{1}{2} = 1$ (red), staying inside the envelope $1 \pm B/T$ of Theorem A.10(b) (grey).

Remarks.

- *Distinct frequencies are essential.* For $x = \cos\omega t + \cos(\omega t + \pi) \equiv 0$ the formula would predict $\langle x^2\rangle = \frac{1}{2}+\frac{1}{2} = 1$, but the true value is 0; for $\cos\omega t + \cos\omega t$ it would predict 1 against the true 2. The cross term $\cos(\theta_k - \theta_l)$ then has frequency 0 and does not average out. Components at the *same* frequency must first be combined with phasors: $A\cos(\omega t+\alpha) + B\cos(\omega t+\beta) = \mathrm{Re}\{(Ae^{j\alpha}+Be^{j\beta})e^{j\omega t}\}$ has mean-square $\frac{1}{2}|Ae^{j\alpha}+Be^{j\beta}|^2$ (Ch. 3 §3.3).
- *The $1/T$ convergence rate matters for instruments.* A true-RMS meter averaging over a window T has a reading error of order B/T; the constant B is dominated by the closest pair of frequencies, $1/|\omega_k-\omega_l|$. Two tones 1 Hz apart need an averaging time of many seconds — the beat period (Ch. 3 §3.7) — before their powers can be read as adding.
- *Power adds, amplitude does not.* The peak value of x depends strongly on the phases; only the mean-square value is phase-independent.
- *Extension.* The same proof with $N\to\infty$ (under $\sum A_k^2 < \infty$ and an argument for exchanging limit and sum) gives Parseval's theorem for Fourier series (Ch. 15 §15.8); the statistical analogue — the variances of uncorrelated random signals add — is in Ch. 21.
- *Where it is used.* Ch. 3 §3.2 and §3.7, Ch. 5 §5.9 (the DC-plus-AC rule $V_{\mathrm{rms}}^2 = V_{\mathrm{dc}}^2 + V_{\mathrm{ac,rms}}^2$ is the case of a constant plus a zero-mean periodic part) and total harmonic distortion (Ch. 15 §15.8).

A.9 Equality of mixed partial derivatives (Clairaut–Schwarz)

Notation follows Ch. 6 §6.2: $f_{xy} = \dfrac{\partial}{\partial y}\Big(\dfrac{\partial f}{\partial x}\Big)$ (first x, then y) and $f_{yx} = \dfrac{\partial}{\partial x}\Big(\dfrac{\partial f}{\partial y}\Big)$.

Theorem A.11 (Clairaut, Schwarz)

Let $U \subset \mathbb{R}^2$ be open, $(a,b)\in U$, and $f: U\to\mathbb{R}$.

(a) Clairaut. If f_x, f_y, f_{xy} and f_{yx} exist on U and f_{xy}, f_{yx} are continuous at (a,b), then $f_{xy}(a,b) = f_{yx}(a,b)$.

(b) Schwarz (sharper form). If f_x, f_y and f_{xy} exist on U and f_{xy} is continuous at (a,b), then $f_{yx}(a,b)$ exists and equals $f_{xy}(a,b)$.

In particular, if $f\in C^2(U)$, then $f_{xy} = f_{yx}$ throughout U.

Proof. Choose $\delta > 0$ such that the closed square $|x-a|\le\delta$, $|y-b|\le\delta$ lies in U. For $0<|h|,|k|<\delta$ form the **second difference**

$$\Delta(h,k) = f(a+h,b+k) - f(a+h,b) - f(a,b+k) + f(a,b),$$

the alternating sum of f over the corners of a small rectangle.

Step 1: Δ via f_{xy}. Let $\varphi(x) = f(x,b+k) - f(x,b)$ for x between a and $a+h$. Then $\Delta = \varphi(a+h) - \varphi(a)$. Since f_x exists on U, φ is differentiable with $\varphi'(x) = f_x(x,b+k) - f_x(x,b)$ (and therefore continuous). By the MVT there is $\theta_1 \in (0,1)$ with

$$\Delta = h\,\varphi'(a+\theta_1 h) = h\Big[f_x(a+\theta_1 h,\, b+k) - f_x(a+\theta_1 h,\, b)\Big].$$

Now fix $x_1 = a+\theta_1 h$ and consider $y \mapsto f_x(x_1, y)$; it is differentiable with derivative $f_{xy}(x_1,y)$ because f_{xy} exists on U. By the MVT again, there is $\theta_2 \in (0,1)$ with

$$\Delta(h,k) = hk\, f_{xy}(a+\theta_1 h,\; b+\theta_2 k).$$

Step 2 (for (a)): Δ via f_{yx}. Grouping the same four terms the other way, $\Delta = \psi(b+k) - \psi(b)$ with $\psi(y) = f(a+h,y) - f(a,y)$. The same two applications of the MVT, first in y (using f_y) and then in x (using f_{yx}), give $\theta_3, \theta_4 \in (0,1)$ with

$$\Delta(h,k) = hk\, f_{yx}(a+\theta_4 h,\; b+\theta_3 k).$$

Conclusion of (a). Comparing, $f_{xy}(a+\theta_1 h, b+\theta_2 k) = f_{yx}(a+\theta_4 h, b+\theta_3 k)$ for all small non-zero h,k. Put $k = h$ and let $h \to 0$: both points tend to (a,b), and continuity of f_{xy} and f_{yx} at (a,b) gives $f_{xy}(a,b) = f_{yx}(a,b)$. ∎

Proof of (b). Only Step 1 is used. Let $\varepsilon > 0$. By continuity of f_{xy} at (a,b) choose $\delta' \in (0,\delta]$ with $|f_{xy}(p,q) - f_{xy}(a,b)| < \varepsilon$ whenever $|p-a| < \delta'$, $|q-b| < \delta'$. The point in Step 1 satisfies this, so

$$\left|\frac{\Delta(h,k)}{hk} - f_{xy}(a,b)\right| < \varepsilon \qquad \text{for all } 0 < |h|, |k| < \delta'.$$

Fix h and let $k \to 0$. Since f_y exists at $(a+h,b)$ and at (a,b),

$$\frac{\Delta(h,k)}{k} = \frac{f(a+h,b+k) - f(a+h,b)}{k} - \frac{f(a,b+k) - f(a,b)}{k} \;\longrightarrow\; f_y(a+h,b) - f_y(a,b).$$

A non-strict inequality survives the limit, so

$$\left|\frac{f_y(a+h,b) - f_y(a,b)}{h} - f_{xy}(a,b)\right| \le \varepsilon \qquad \text{for all } 0 < |h| < \delta'.$$

This says precisely that the difference quotient defining $f_{yx}(a,b) = \frac{\partial}{\partial x} f_y\big|_{(a,b)}$ converges to $f_{xy}(a,b)$. ∎

The counterexample. Let

$$f(x,y) = \begin{cases} \dfrac{xy\,(x^2 - y^2)}{x^2 + y^2}, & (x,y) \neq (0,0), \\[2mm] 0, & (x,y) = (0,0). \end{cases}$$

In polar coordinates $f = \tfrac{1}{4} r^2 \sin 4\phi$, so f is continuous and small near the origin. For $(x,y) \neq (0,0)$ the quotient rule gives (checked with a computer algebra system)

$$f_x = \frac{y\,(x^4 + 4x^2y^2 - y^4)}{(x^2+y^2)^2}, \qquad f_y = \frac{x\,(x^4 - 4x^2y^2 - y^4)}{(x^2+y^2)^2}.$$

At the origin, $f(x,0) = 0$ and $f(0,y) = 0$ give $f_x(0,0) = f_y(0,0) = 0$. Since $|x^4 \pm 4x^2y^2 - y^4| \le 2(x^2+y^2)^2$, we have $|f_x| \le 2|y|$ and $|f_y| \le 2|x|$, so f_x, f_y are continuous everywhere: f is C^1. On the axes, $f_x(0,y) = -y$ and $f_y(x,0) = x$ (including at 0). Therefore

$$f_{xy}(0,0) = \lim_{k\to 0} \frac{f_x(0,k) - f_x(0,0)}{k} = -1, \qquad f_{yx}(0,0) = \lim_{h\to 0} \frac{f_y(h,0) - f_y(0,0)}{h} = +1.$$

The mixed partials exist everywhere but differ at the origin. Away from the origin both equal

$$f_{xy} = f_{yx} = \frac{(x^2-y^2)(x^4 + 10x^2y^2 + y^4)}{(x^2+y^2)^3} = \cos 2\phi\,\big(1 + 2\sin^2 2\phi\big),$$

which depends only on the direction ϕ: it is $+1$ along the x-axis and -1 along the y-axis, so it has no limit at the origin. The continuity hypothesis of Theorem A.11 fails exactly there, and the conclusion fails with it.

Remarks.

- *Higher order.* If $f \in C^k$ on an open set in $\mathbb{R}^n$, any two k-th order partial derivatives that differentiate the same number of times in each variable are equal. Proof: any ordering can be turned into any other by swapping adjacent differentiations, and each swap is Theorem A.11 applied to a $(k-2)$-th order partial (which is C^2) as a function of the two variables being swapped, the others held fixed.
- *Why engineers may swap orders freely — and when not.* Potentials and fields from smooth physical models are C^2 away from sources. At a point charge or on a conductor edge they are not, and statements such as $\nabla \times \nabla\varphi = \mathbf{0}$ must be checked separately (or interpreted in the sense of distributions).
- *Where it is used.* Symmetry of the Hessian (Ch. 6 §6.2, §6.10; Ch. 23 §23.4); the exactness test $M_y = N_x$ for $M\,dx + N\,dy$ (Ch. 6 §6.3, Ch. 11 §11.1); $\nabla \times \nabla\varphi = \mathbf{0}$ and $\nabla \cdot (\nabla \times \mathbf{F}) = 0$ (Ch. 7); the Cauchy–Riemann equations and harmonicity of $\operatorname{Re}\{f\}$ (Ch. 18).

A.10 The Lagrange multiplier condition

The method of Lagrange multipliers (Ch. 6 §6.11) rests on one geometric fact: at a constrained extremum, ∇f is perpendicular to the constraint surface. To make "surface" precise we need the implicit function theorem, which says that near a regular point the constraint set is the graph of a smooth function. Write points of $\mathbb{R}^n$ as $\mathbf{x} = (\mathbf{u}, \mathbf{v})$ with $\mathbf{u} \in \mathbb{R}^{n-m}$, $\mathbf{v} \in \mathbb{R}^m$, and Jacobian blocks $D_{\mathbf{u}}$, $D_{\mathbf{v}}$; gradients are column vectors and $Df = \nabla f^{\mathsf{T}}$ is a row.

Implicit function theorem (used without proof)

Let $\Omega \subset \mathbb{R}^{n-m} \times \mathbb{R}^m$ be open, $\mathbf{G} : \Omega \to \mathbb{R}^m$ of class C^1, and $(\mathbf{u}^*, \mathbf{v}^*) \in \Omega$ with $\mathbf{G}(\mathbf{u}^*, \mathbf{v}^*) = \mathbf{0}$ and the $m \times m$ matrix $D_{\mathbf{v}}\mathbf{G}(\mathbf{u}^*, \mathbf{v}^*)$ invertible. Then there are open sets $\mathcal{U} \ni \mathbf{u}^*$ and $\mathcal{V} \ni \mathbf{v}^*$ with $\mathcal{U} \times \mathcal{V} \subset \Omega$, and a C^1 function $\boldsymbol{\varphi} : \mathcal{U} \to \mathcal{V}$ with $\boldsymbol{\varphi}(\mathbf{u}^*) = \mathbf{v}^*$, such that

$$\{(\mathbf{u}, \mathbf{v}) \in \mathcal{U} \times \mathcal{V} : \ \mathbf{G}(\mathbf{u}, \mathbf{v}) = \mathbf{0}\} = \{(\mathbf{u}, \boldsymbol{\varphi}(\mathbf{u})) : \ \mathbf{u} \in \mathcal{U}\}.$$

(Proof: Rudin, *Principles*, Thm 9.28, via the contraction mapping principle. The derivative formula $D\boldsymbol{\varphi} = -(D_{\mathbf{v}}\mathbf{G})^{-1} D_{\mathbf{u}}\mathbf{G}$ quoted in Ch. 6 §6.6 is re-derived in Step 3 below.)

Theorem A.12 (Lagrange multiplier condition)

Let $U \subset \mathbb{R}^n$ be open, $1 \le m < n$, and $f, g_1, \dots, g_m : U \to \mathbb{R}$ of class C^1. Let $S = \{\mathbf{x} \in U : \ g_i(\mathbf{x}) = c_i, \ i = 1, \dots, m\}$ and suppose $\mathbf{x}^* \in S$ is a **local extremum of f on S**: there is $\rho > 0$ with $f(\mathbf{x}) \le f(\mathbf{x}^*)$ for all $\mathbf{x} \in S$ with $\|\mathbf{x} - \mathbf{x}^*\| < \rho$ (or $\ge$ for all such $\mathbf{x}$). Suppose the **regularity condition** holds: $\nabla g_1(\mathbf{x}^*), \dots, \nabla g_m(\mathbf{x}^*)$ are linearly independent. Then there are unique scalars $\lambda_1, \dots, \lambda_m$ with

$$\nabla f(\mathbf{x}^*) = \lambda_1 \nabla g_1(\mathbf{x}^*) + \cdots + \lambda_m \nabla g_m(\mathbf{x}^*).$$

For a single constraint ($m = 1$) the condition reads $\nabla f(\mathbf{x}^*) = \lambda \nabla g(\mathbf{x}^*)$ with $\nabla g(\mathbf{x}^*) \neq \mathbf{0}$.

Proof. Treat the maximum case (for a minimum replace f by $-f$, which replaces $\boldsymbol{\lambda}$ by $-\boldsymbol{\lambda}$). Let $\mathbf{g} = (g_1, \dots, g_m)^{\mathsf{T}}$ and $\mathbf{c} = (c_1, \dots, c_m)^{\mathsf{T}}$.

Step 0: choose coordinates. The $m \times n$ Jacobian $D\mathbf{g}(\mathbf{x}^*)$ has linearly independent rows, so its rank is m and it has m linearly independent columns. Renumber the coordinates (which permutes the components of every gradient in the same way and does not affect the conclusion) so that these are the last m columns. Writing $\mathbf{x} = (\mathbf{u}, \mathbf{v})$ and $\mathbf{x}^* = (\mathbf{u}^*, \mathbf{v}^*)$,

$$D\mathbf{g}(\mathbf{x}^*) = \begin{bmatrix} \mathbf{A} & \mathbf{B} \end{bmatrix}, \qquad \mathbf{A} = D_{\mathbf{u}}\mathbf{g}(\mathbf{x}^*) \ (m \times (n-m)), \quad \mathbf{B} = D_{\mathbf{v}}\mathbf{g}(\mathbf{x}^*) \ (m \times m, \ \text{invertible}).$$

Step 1: the constraint set is a graph. Apply the implicit function theorem to $\mathbf{G} = \mathbf{g} - \mathbf{c}$ on $\Omega = U$. We get open $\mathcal{U} \ni \mathbf{u}^*$, $\mathcal{V} \ni \mathbf{v}^*$ and a C^1 map $\boldsymbol{\varphi} : \mathcal{U} \to \mathcal{V}$ with $\boldsymbol{\varphi}(\mathbf{u}^*) = \mathbf{v}^*$ such that the points of S in $\mathcal{U} \times \mathcal{V}$ are exactly the points $(\mathbf{u}, \boldsymbol{\varphi}(\mathbf{u}))$.

Step 2: an unconstrained problem. Define $F(\mathbf{u}) = f(\mathbf{u}, \boldsymbol{\varphi}(\mathbf{u}))$ on $\mathcal{U}$; it is C^1 by the chain rule. Because $\boldsymbol{\varphi}$ is continuous, there is a neighbourhood $\mathcal{U}' \subset \mathcal{U}$ of $\mathbf{u}^*$ on which $\|(\mathbf{u}, \boldsymbol{\varphi}(\mathbf{u})) - \mathbf{x}^*\| < \rho$. For $\mathbf{u} \in \mathcal{U}'$ the point $(\mathbf{u}, \boldsymbol{\varphi}(\mathbf{u}))$ lies in S and within ρ of $\mathbf{x}^*$, so $F(\mathbf{u}) \le f(\mathbf{x}^*) = F(\mathbf{u}^*)$: F has an unconstrained local maximum at $\mathbf{u}^*$. For each $i = 1, \dots, n-m$, the one-variable function $t \mapsto F(\mathbf{u}^* + t\mathbf{e}_i)$ is defined on an open interval around 0, has a maximum at $t = 0$ and is differentiable there, so by Fermat's lemma (Lemma A.1) $\partial F / \partial u_i(\mathbf{u}^*) = 0$. By the chain rule,

$$\mathbf{0}^{\mathsf{T}} = DF(\mathbf{u}^*) = D_{\mathbf{u}}f(\mathbf{x}^*) + D_{\mathbf{v}}f(\mathbf{x}^*)\, D\boldsymbol{\varphi}(\mathbf{u}^*).$$

Step 3: the slope of the constraint set. Differentiating the identity $\mathbf{g}(\mathbf{u}, \boldsymbol{\varphi}(\mathbf{u})) = \mathbf{c}$ on $\mathcal{U}$ at $\mathbf{u}^*$ (chain rule),

$$\mathbf{A} + \mathbf{B}\, D\boldsymbol{\varphi}(\mathbf{u}^*) = \mathbf{0} \qquad \Longrightarrow \qquad D\boldsymbol{\varphi}(\mathbf{u}^*) = -\mathbf{B}^{-1}\mathbf{A}.$$

Step 4: define the multipliers. Let $\boldsymbol{\lambda}^{\mathsf{T}} = D_{\mathbf{v}}f(\mathbf{x}^*)\, \mathbf{B}^{-1}$, a $1 \times m$ row vector. Then $\boldsymbol{\lambda}^{\mathsf{T}}\mathbf{B} = D_{\mathbf{v}}f(\mathbf{x}^*)$ by construction, and by Steps 2 and 3,

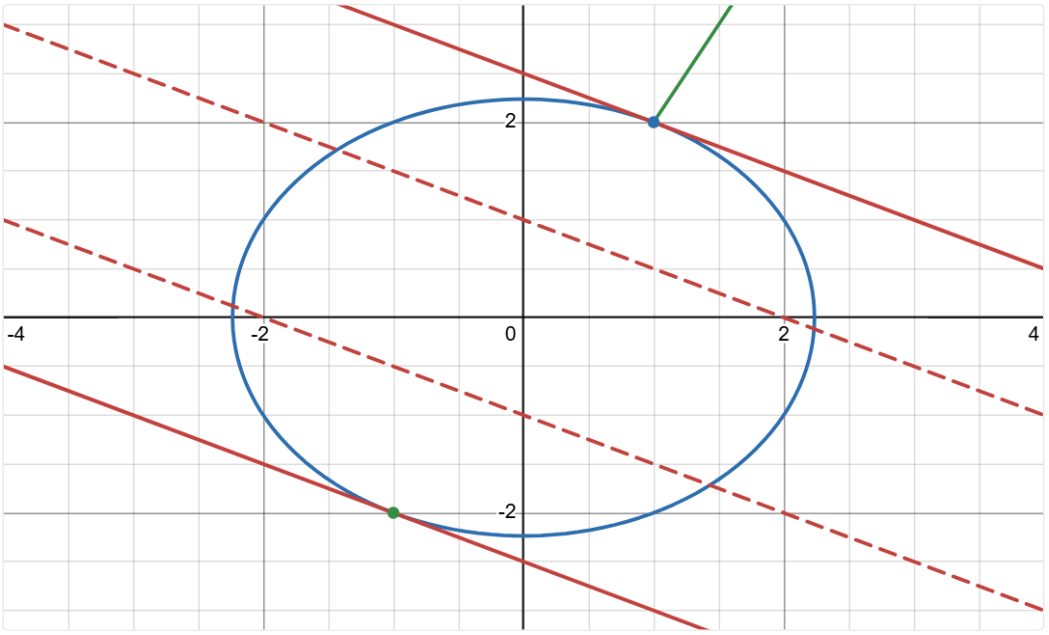
$$\boldsymbol{\lambda}^{\mathsf{T}}\mathbf{A} = D_{\mathbf{v}}f\, \mathbf{B}^{-1}\mathbf{A} = -D_{\mathbf{v}}f\, D\boldsymbol{\varphi}(\mathbf{u}^*) = D_{\mathbf{u}}f(\mathbf{x}^*).$$

Putting the two blocks side by side, $\boldsymbol{\lambda}^{\mathsf{T}}\begin{bmatrix} \mathbf{A} & \mathbf{B} \end{bmatrix} = \begin{bmatrix} D_{\mathbf{u}}f & D_{\mathbf{v}}f \end{bmatrix}$, i.e. $Df(\mathbf{x}^*) = \boldsymbol{\lambda}^{\mathsf{T}} D\mathbf{g}(\mathbf{x}^*)$. Transposing, $\nabla f(\mathbf{x}^*) = D\mathbf{g}(\mathbf{x}^*)^{\mathsf{T}}\boldsymbol{\lambda} = \sum_i \lambda_i \nabla g_i(\mathbf{x}^*)$.

Uniqueness. If also $\nabla f = \sum_i \mu_i \nabla g_i$, then $\sum_i (\lambda_i - \mu_i) \nabla g_i(\mathbf{x}^*) = \mathbf{0}$, and linear independence forces $\lambda_i = \mu_i$. $\blacksquare$

For one constraint ($m = 1$), after renumbering so that $g_{x_n}(\mathbf{x}^*) \neq 0$, the proof gives the explicit multiplier $\lambda = f_{x_n}(\mathbf{x}^*) / g_{x_n}(\mathbf{x}^*)$, and Step 3 is the familiar implicit-differentiation formula $\partial x_n / \partial x_i = -g_{x_i} / g_{x_n}$ of Ch. 4 §4.4.

Geometric reading. The columns of $\begin{bmatrix} \mathbf{I} \\ D\boldsymbol{\varphi}(\mathbf{u}^*) \end{bmatrix}$ span the tangent space $T = \ker D\mathbf{g}(\mathbf{x}^*)$ of S at $\mathbf{x}^*$ (an $(n-m)$-dimensional subspace, by Step 3 and the rank–nullity theorem of Ch. 9). Step 2 says $Df(\mathbf{x}^*)$ annihilates T: $\nabla f \perp T$. The orthogonal complement of $\ker D\mathbf{g}$ is the row space of $D\mathbf{g}$, spanned by the ∇g_i — which is Step 4.



Extremising $f = x + 2y$ on the circle $x^2 + y^2 = 5$ (Ch. 6, Worked Example 1). Level lines of f (red) cross the circle where f can still increase along it; at the extremes $(\pm 1, \pm 2)$ they are tangent, so $\nabla f = (1, 2)$ (green, at the maximum) is parallel to $\nabla g = (2x, 2y)$, with $\lambda = \frac{1}{2}$ at $(1, 2)$.

Remarks.

- *Regularity cannot be dropped.* Minimise $f(x, y) = x$ on $S: \; y^2 - x^3 = 0$. On S, $x^3 = y^2 \geq 0$, so $x \geq 0$ and the minimum is 0, at the origin. But $\nabla g = (-3x^2, \; 2y)$ vanishes there, while $\nabla f = (1, 0)$; no λ satisfies $(1, 0) = \lambda(0, 0)$. The cusp of S at the origin is not the graph of a C^1 function in either variable, so Step 1 fails. (The weaker **Fritz John** condition $\lambda_0 \nabla f = \lambda \nabla g$ with $(\lambda_0, \lambda) \neq (0, 0)$ survives, with $\lambda_0 = 0$; it carries no information about f.)
- *The condition is necessary, not sufficient.* $f = x^3 + y$ on $g = y = 0$: at the origin $\nabla f = (0, 1) = 1 \cdot \nabla g$, yet on the constraint $f = x^3$ has no extremum at 0 . Candidates must be compared, or tested with the bordered Hessian (Ch. 6 §6.11).
- *The meaning of λ.* Suppose the constrained optimum $\mathbf{x}^*(c)$ and its multiplier $\lambda(c)$ depend differentiably on the level c of a single constraint, and let $f^*(c) = f(\mathbf{x}^*(c))$. Then, by the chain rule and the theorem,

$$\frac{df^*}{dc} = \nabla f \cdot \frac{d\mathbf{x}^*}{dc} = \lambda \, \nabla g \cdot \frac{d\mathbf{x}^*}{dc} = \lambda \frac{d}{dc} g\big(\mathbf{x}^*(c)\big) = \lambda \cdot \frac{d}{dc} c = \lambda.$$

The multiplier is the sensitivity of the optimal value to the constraint level — a "shadow price". Check: on $x^2 + y^2 = c$, $\max(x + 2y) = \sqrt{5c}$, whose derivative $\sqrt{5}/(2\sqrt{c})$ equals $\frac{1}{2} = \lambda$ at $c = 5$.

- *Where it is used.* Ch. 6 §6.11 (the method), Ch. 23 §23.4 (Lagrange and KKT conditions — KKT is this theorem plus sign conditions for active inequality constraints, under an analogous regularity hypothesis), equality-constrained least squares and economic dispatch (equal incremental cost $= \lambda$) in Ch. 23 §23.8, §23.10, and the variational characterisation of eigenvalues: maximising $\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x}$ for symmetric $\mathbf{A}$ subject to $\mathbf{x}^{\mathsf{T}}\mathbf{x} = 1$ gives $2\mathbf{A}\mathbf{x} = \lambda \, 2\mathbf{x}$, the eigenvalue equation, with the multiplier equal to the maximum value (Ch. 10).

Further reading

- T. M. Apostol, *Calculus*, Vols. I–II, 2nd ed., Wiley, 1967–1969 — rigorous proofs of the MVT, FTC, Taylor's theorem and the multivariable results at the level of this appendix.
- M. Spivak, *Calculus*, 4th ed., Publish or Perish, 2008 — the cleanest treatment of the single-variable results, including the counterexamples used here.
- W. Rudin, *Principles of Mathematical Analysis*, 3rd ed., McGraw-Hill, 1976 — L'Hôpital's rule in full generality (Thm 5.13), the inverse and implicit function theorems (Ch. 9), and the exponential function defined by its series (Ch. 8).
- T. Needham, *Visual Complex Analysis*, Oxford University Press, 1997 — the geometric meaning of $e^{j\theta}$, De Moivre's theorem and roots of unity.

Appendix B — Vector Calculus and Field Theorems

What is proved here. The results that make vector calculus usable in field theory: the change-of-variables formula and the Jacobian as an area scale factor; the gradient as the direction of steepest ascent; the theorems of Green, Gauss (divergence) and Stokes; the equivalence "curl-free ⇔ conservative" on simply connected domains, with its failure on the punctured plane; the standard vector identities by index notation; grad, div, curl and Laplacian in cylindrical and spherical coordinates via scale factors; and the passage from Maxwell's integral laws to their differential form, the continuity equation and the vacuum wave equation.

Prerequisites. Partial derivatives, the multivariable chain rule, Clairaut's theorem on mixed partials and Fubini's theorem (Ch. 6); determinants (Ch. 8); line, surface and volume integrals as defined in Ch. 7.

Level of rigour. Every theorem is stated with explicit hypotheses. The proofs are complete for the "elementary" regions on which the integral theorems are really proved (regions bounded by graphs). The step from elementary regions to general regions — cutting a region into finitely many elementary pieces — is carried out as an argument about cancelling boundary contributions, and the topological facts it uses (that such a decomposition exists for the piecewise-smooth regions met in engineering) are stated, not proved. In the change-of-variables theorem the linear-algebra step is proved in full and the passage to nonlinear maps is a careful sketch. Throughout, "smooth" means C^1 (continuous first partial derivatives) unless C^2 is stated; regions are bounded and their boundaries piecewise smooth.

Notation follows the book (Notation and Conventions): cylindrical coordinates are (r, ϕ, z), spherical coordinates are (R, θ, ϕ) with θ the polar angle from $+z$; ∂_i abbreviates $\partial/\partial x_i$ with $(x_1, x_2, x_3) = (x, y, z)$.

B.1 Change of variables and the Jacobian as area scale factor

The change-of-variables formula is used every time the book writes $dA = r\,dr\,d\phi$ or $dV = R^2 \sin\theta\, dR\, d\theta\, d\phi$ (§6.8, §6.9), and every time a probability density is transformed (Ch. 19). Its heart is a statement about linear maps, which we prove exactly; the nonlinear statement then follows because a differentiable map is, on a small enough square, almost linear.

Jacobian matrix and determinant

For a map $\mathbf{T}(u, v) = \big(x(u, v),\ y(u, v)\big)$ with continuous first partials, the **Jacobian matrix** and **Jacobian determinant** at (u, v) are

$$D\mathbf{T} = \begin{bmatrix} x_u & x_v \\ y_u & y_v \end{bmatrix}, \qquad \frac{\partial(x, y)}{\partial(u, v)} = \det D\mathbf{T} = x_u y_v - x_v y_u.$$

Theorem B.1a — Linear maps scale area by $|\det|$

Let $\mathbf{A}$ be a real 2×2 matrix and $S \subset \mathbb{R}^2$ a bounded region with an area (for instance a finite union of rectangles, or any region bounded by finitely many smooth curves). Then $\mathbf{A}(S) = \{\mathbf{A}\mathbf{s} : \mathbf{s} \in S\}$ has an area and

$$\operatorname{area}\big(\mathbf{A}(S)\big) = |\det \mathbf{A}|\ \operatorname{area}(S).$$

Theorem B.1b — Change of variables in a double integral

Let $U \subset \mathbb{R}^2$ be open, $\mathbf{T} : U \to \mathbb{R}^2$ one-to-one with continuous first partial derivatives, and $\det D\mathbf{T} \neq 0$ on U. Let S be a closed bounded region in U with piecewise-smooth boundary, and $R = \mathbf{T}(S)$. If f is continuous on R, then

$$\iint_R f(x, y)\, dx\, dy = \iint_S f\big(\mathbf{T}(u, v)\big) \left|\frac{\partial(x, y)}{\partial(u, v)}\right| du\, dv.$$

The same holds in $\mathbb{R}^3$ with $|\partial(x, y, z)/\partial(u, v, w)|$, and the hypothesis $\det D\mathbf{T} \neq 0$ may fail on a set of zero area (such as the origin in polar coordinates) without changing the conclusion.

Proof of B.1a.

Step 1 — reduction to rectangles. If the formula holds for every axis-parallel rectangle, it holds for finite unions of non-overlapping rectangles (areas add, and the images of non-overlapping rectangles under an invertible linear map are non-overlapping). A region with an area is, by definition of area (the Riemann/Jordan construction behind the double integral in §6.7), squeezed between an inner union of grid rectangles S_- and an outer union S_+ with $\operatorname{area}(S_+) - \operatorname{area}(S_-) < \varepsilon$. Then $\mathbf{A}(S_-) \subset \mathbf{A}(S) \subset \mathbf{A}(S_+)$ and the two outer sets have areas $|\det \mathbf{A}|\operatorname{area}(S_\pm)$ differing by at most $|\det \mathbf{A}|\varepsilon$. Letting $\varepsilon \to 0$ gives the formula for S. If $\det \mathbf{A} = 0$, $\mathbf{A}$ maps the plane into a line (or a point), and any bounded subset of a line has zero area, which is what the formula says. So assume $\det \mathbf{A} \neq 0$ and S a rectangle $[a, b] \times [c, d]$.

Step 2 — factor $\mathbf{A}$ into elementary matrices. Gaussian elimination (§8.7) reduces any invertible matrix to the identity by row operations, each of which is left-multiplication by an elementary matrix. Hence $\mathbf{A} = \mathbf{E}_1 \mathbf{E}_2 \cdots \mathbf{E}_k$ with each $\mathbf{E}_i$ of one of three types:

$$\text{scaling } \begin{bmatrix} \lambda & 0 \\ 0 & 1 \end{bmatrix} \text{ or } \begin{bmatrix} 1 & 0 \\ 0 & \lambda \end{bmatrix}\ (\lambda \neq 0), \qquad \text{swap } \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \qquad \text{shear } \begin{bmatrix} 1 & c \\ 0 & 1 \end{bmatrix} \text{ or } \begin{bmatrix} 1 & 0 \\ c & 1 \end{bmatrix}.$$

Since $\det(\mathbf{A}) = \prod \det \mathbf{E}_i$ and $\mathbf{A}(S) = \mathbf{E}_1(\mathbf{E}_2(\cdots \mathbf{E}_k(S)))$, it suffices to prove $\operatorname{area}(\mathbf{E}(S)) = |\det \mathbf{E}|\operatorname{area}(S)$ for each elementary $\mathbf{E}$ and every region S with an area — not just rectangles, because the intermediate images $\mathbf{E}_k(S)$, $\mathbf{E}_{k-1}\mathbf{E}_k(S)$, … are parallelograms. By Step 1 (applied to the single matrix $\mathbf{E}$) the general case follows from the case of rectangles, so it is enough to check rectangles for each type.

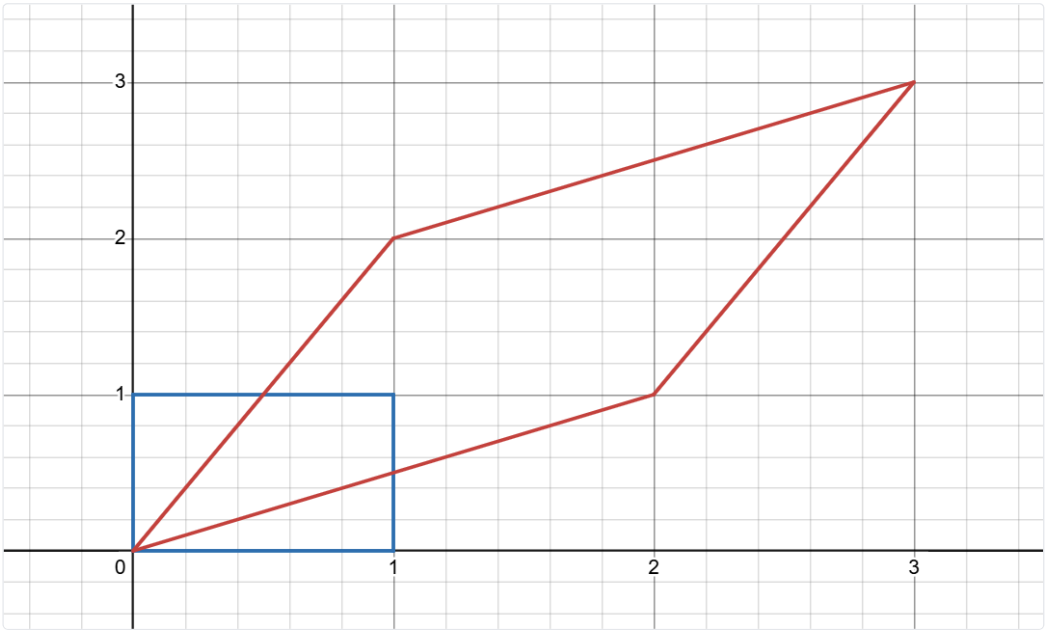
Step 3 — the three types.

- *Scaling* $(u, v) \mapsto (\lambda u, v)$ sends $[a, b] \times [c, d]$ to a rectangle with side lengths $|\lambda|(b - a)$ and $d - c$: area multiplied by $|\lambda| = |\det \mathbf{E}|$.
- *Swap* $(u, v) \mapsto (v, u)$ sends a rectangle to a congruent rectangle: area unchanged, and $|\det \mathbf{E}| = |-1| = 1$.
- *Shear* $(u, v) \mapsto (u + cv,\ v)$ sends $[a, b] \times [c_1, d_1]$ to the parallelogram $P = \{(x, y) : c_1 \le y \le d_1,\ a + cy \le x \le b + cy\}$. By Fubini, slicing horizontally,

$$\operatorname{area}(P) = \int_{c_1}^{d_1} \big[(b + cy) - (a + cy)\big]\, dy = (b - a)(d_1 - c_1),$$

so the area is unchanged, and $\det \mathbf{E} = 1$. (This is Cavalieri's principle: every horizontal slice keeps its length.) The lower shear is the same with the roles of x and y exchanged.

Multiplying the factors gives $\operatorname{area}(\mathbf{A}(S)) = \prod_i |\det \mathbf{E}_i| \operatorname{area}(S) = |\det \mathbf{A}| \operatorname{area}(S)$. ■



The unit square (blue) and its image under $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ (red): a parallelogram spanned by $(2, 1)$ and $(1, 2)$, of area $|\det \mathbf{A}| = 3$.

Proof sketch of B.1b.

Step 1 — local linearity, uniformly. Fix a compact set $K \subset U$ containing S in its interior. Because the partial derivatives of $\mathbf{T}$ are continuous on K they are uniformly continuous there, so for every $\varepsilon > 0$ there is $h_0 > 0$ such that for every square $Q \subset K$ of side $h < h_0$ with centre $\mathbf{q}$ and every $\mathbf{u} \in Q$,

$$\big\|\mathbf{T}(\mathbf{u}) - \mathbf{T}(\mathbf{q}) - D\mathbf{T}(\mathbf{q})(\mathbf{u} - \mathbf{q})\big\| \le \varepsilon h.$$

(Apply the mean value theorem to each component along the segment from $\mathbf{q}$ to $\mathbf{u}$.) So $\mathbf{T}(Q)$ lies within distance εh of the parallelogram $\mathbf{T}(\mathbf{q}) + D\mathbf{T}(\mathbf{q})(Q - \mathbf{q})$, whose area by Theorem B.1a is $|\det D\mathbf{T}(\mathbf{q})| h^2$.

Step 2 — area of the image of a small square. The εh-neighbourhood of that parallelogram's boundary has area at most (perimeter)$\times 2\varepsilon h + \pi(\varepsilon h)^2 \le C\varepsilon h^2$, where C depends only on a bound for $\|D\mathbf{T}\|$ on K. Since $\mathbf{T}$ is one-to-one with invertible derivative, it maps the boundary of Q onto the boundary of $\mathbf{T}(Q)$ (invariance of domain, here a consequence of the inverse function theorem, §6.6), and therefore $\mathbf{T}(Q)$ differs from the parallelogram only inside that thin neighbourhood. Hence

$$\Big|\operatorname{area}\big(\mathbf{T}(Q)\big) - |\det D\mathbf{T}(\mathbf{q})|\, h^2\Big| \le C\varepsilon h^2.$$

Step 3 — Riemann sums. Cover S by a grid of squares Q_k of side h with centres $\mathbf{q}_k$. The squares meeting ∂S have total area $\to 0$ as $h \to 0$ (a piecewise-smooth curve has zero area), and so do their images (Step 2 bounds image areas by a constant times h^2). For the interior squares, the images $\mathbf{T}(Q_k)$ tile R up to a set of small area, and

$$\sum_k f\big(\mathbf{T}(\mathbf{q}_k)\big) \operatorname{area}\big(\mathbf{T}(Q_k)\big) = \sum_k f\big(\mathbf{T}(\mathbf{q}_k)\big) |\det D\mathbf{T}(\mathbf{q}_k)| h^2 + O\big(\varepsilon \max|f| \operatorname{area}(S)\big)\,.$$

The left side is a Riemann sum for $\iint_R f\, dA$ (over a partition of R into the curvilinear pieces $\mathbf{T}(Q_k)$, whose diameters $\to 0$ uniformly), and the first sum on the right is a Riemann sum for $\iint_S f(\mathbf{T}) |\det D\mathbf{T}|\, du\, dv$. Let $h \to 0$ and then $\varepsilon \to 0$. If $\det D\mathbf{T} = 0$ or injectivity fails only on a set Z of zero area, apply the result on S minus a small neighbourhood of Z and let the neighbourhood shrink; both integrals change by arbitrarily little. In three dimensions the same argument runs with cubes and parallelepipeds, using the 3×3 elementary-matrix factorisation. ■

Remarks.

- *Why the absolute value.* An integral of a positive function is positive, but $\det D\mathbf{T}$ is negative for orientation-reversing maps (e.g. the swap). The absolute value is the price of measuring unsigned area. In the integral theorems (B.3–B.5) orientation matters and the sign of the Jacobian reappears as the rule for orienting boundaries.
- *Why injectivity.* If $\mathbf{T}$ covers part of R twice the right side counts that part twice. Example: $\mathbf{T}(r, \phi) = (r\cos\phi, r\sin\phi)$ on $0 \le r \le 1$, $0 \le \phi \le 4\pi$ gives $\iint r\, dr\, d\phi = 2\pi$, twice the area of the unit disc.

- *Scale factors.* For an orthogonal coordinate system the columns of $D\mathbf{T}$ are perpendicular with lengths h_1, h_2, h_3, so $|\det D\mathbf{T}| = h_1 h_2 h_3$: this gives r (cylindrical) and $R^2 \sin\theta$ (spherical) without expanding a determinant, as used in §6.9 and again in B.8 below.
- *Used in:* polar, cylindrical and spherical integrals (§6.8–6.9); the volume elements of §7.8; the transformation of densities $f_Y(\mathbf{y}) = f_X(\mathbf{x})/|\det D\mathbf{T}|$ in Ch. 19.

B.2 The gradient as the direction of steepest ascent

§6.4–6.5 define the gradient by its components and state that it points uphill. The proof rests on one fact — that for a differentiable function every directional derivative is a dot product with ∇f — and on the Cauchy–Schwarz inequality. The point of stating the hypotheses carefully is that the existence of partial derivatives alone is **not** enough.

Differentiability and directional derivative

$f : U \to \mathbb{R}$ ($U \subset \mathbb{R}^n$ open) is **differentiable** at $\mathbf{a}$ if there is a vector $\mathbf{g}$ with

$$f(\mathbf{a}+\mathbf{h}) = f(\mathbf{a}) + \mathbf{g}\cdot\mathbf{h} + o(\|\mathbf{h}\|) \quad (\mathbf{h}\to\mathbf{0}).$$

Then necessarily $\mathbf{g} = \nabla f(\mathbf{a}) = (\partial_1 f, \dots, \partial_n f)(\mathbf{a})$. For a unit vector $\hat{\mathbf{u}}$ the **directional derivative** is $D_{\hat{\mathbf{u}}} f(\mathbf{a}) = \lim_{s\to 0^+} \big[f(\mathbf{a}+s\hat{\mathbf{u}}) - f(\mathbf{a})\big]/s$.

Theorem B.2 — Steepest ascent and normality to level sets

Let f be differentiable at $\mathbf{a}$.

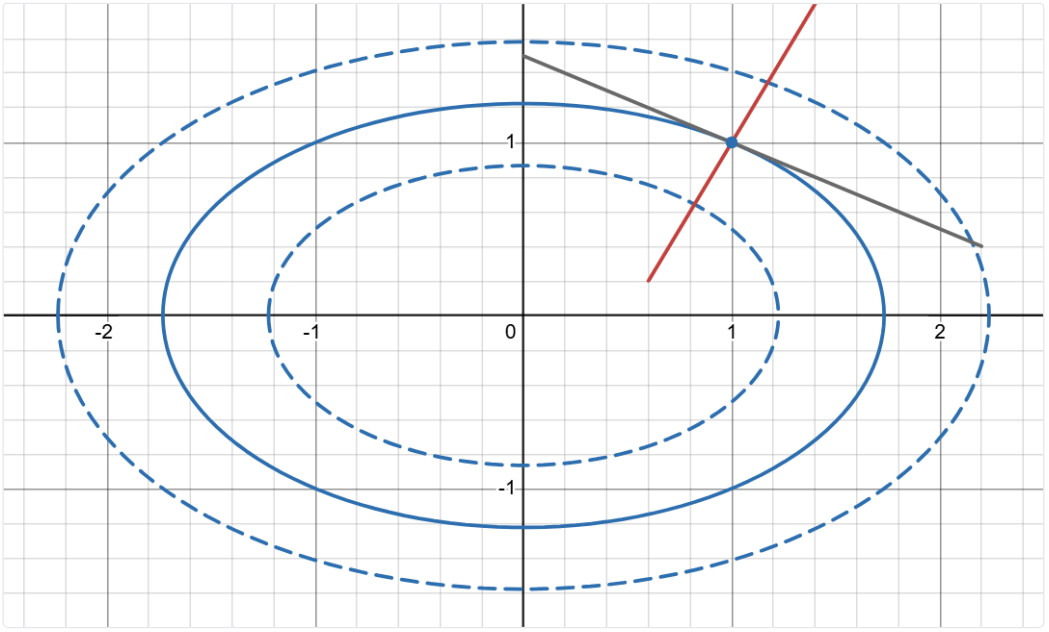
1. For every unit vector $\hat{\mathbf{u}}$, $D_{\hat{\mathbf{u}}} f(\mathbf{a}) = \nabla f(\mathbf{a})\cdot\hat{\mathbf{u}}$.
2. If $\nabla f(\mathbf{a}) \neq \mathbf{0}$, then $\max_{\|\hat{\mathbf{u}}\|=1} D_{\hat{\mathbf{u}}} f(\mathbf{a}) = \|\nabla f(\mathbf{a})\|$, attained **only** at $\hat{\mathbf{u}} = \nabla f(\mathbf{a})/\|\nabla f(\mathbf{a})\|$; the minimum $-\|\nabla f(\mathbf{a})\|$ is attained only in the opposite direction; and $D_{\hat{\mathbf{u}}} f(\mathbf{a}) = 0$ exactly for $\hat{\mathbf{u}} \perp \nabla f(\mathbf{a})$.
3. If $\mathbf{r}(t)$ is a curve, differentiable at $t = 0$, lying in the level set $\{f = f(\mathbf{a})\}$ with $\mathbf{r}(0) = \mathbf{a}$, then $\nabla f(\mathbf{a})\cdot\mathbf{r}'(0) = 0$.

Proof.

Part 1. Put $\mathbf{h} = s\hat{\mathbf{u}}$ in the definition: $f(\mathbf{a}+s\hat{\mathbf{u}}) - f(\mathbf{a}) = s\,\nabla f(\mathbf{a})\cdot\hat{\mathbf{u}} + o(s)$. Divide by $s > 0$ and let $s \to 0^+$; the remainder tends to 0 because $o(\|\mathbf{h}\|)/\|\mathbf{h}\| \to 0$ and $\|\mathbf{h}\| = s$.

Part 2. By Cauchy–Schwarz, $|\nabla f\cdot\hat{\mathbf{u}}| \le \|\nabla f\|\,\|\hat{\mathbf{u}}\| = \|\nabla f\|$, so $-\|\nabla f\| \le D_{\hat{\mathbf{u}}} f \le \|\nabla f\|$. Equality in Cauchy–Schwarz holds if and only if $\hat{\mathbf{u}}$ is a scalar multiple of ∇f; with $\|\hat{\mathbf{u}}\| = 1$ the only multiples are $\pm\nabla f/\|\nabla f\|$, which give $\pm\|\nabla f\|$ respectively. Writing $\nabla f\cdot\hat{\mathbf{u}} = \|\nabla f\|\cos\alpha$ shows the whole picture: the rate of increase is the projection of the gradient onto the direction of travel, and it is zero exactly when $\alpha = 90^\circ$.

Part 3. The function $g(t) = f(\mathbf{r}(t))$ is constant, so $g'(0) = 0$. By the chain rule (valid because f is differentiable at $\mathbf{a}$ and $\mathbf{r}$ at 0), $g'(0) = \nabla f(\mathbf{a})\cdot\mathbf{r}'(0)$. ∎



Level curves of $f = x^2 + 2y^2$. At $\mathbf{a} = (1,1)$ the gradient direction $(2,4)$ (red) is normal to the level curve $f = 3$ and to its tangent line $x + 2y = 3$ (grey): it crosses the contours by the shortest route.

Remarks.

- *Differentiability is essential.* Let $f(x,y) = x^2 y/(x^2+y^2)$ for $(x,y) \neq \mathbf{0}$ and $f(\mathbf{0}) = 0$. Both partials at the origin are 0 (the function vanishes on both axes), so the would-be gradient is $\mathbf{0}$; yet along $\hat{\mathbf{u}} = (1,1)/\sqrt{2}$, $f(s/\sqrt{2}, s/\sqrt{2})/s = 1/(2\sqrt{2}) \neq 0$. The formula $D_{\hat{\mathbf{u}}} f = \nabla f\cdot\hat{\mathbf{u}}$ fails: this f is continuous but not differentiable at $\mathbf{0}$. A convenient sufficient condition for differentiability is that the partials exist near $\mathbf{a}$ and are continuous at $\mathbf{a}$ (§6.3).
- *Units.* $\|\nabla f\|$ has units of f per metre, which is why $\|\nabla V\|$ is a field strength in V/m. The relation $\mathbf{E} = -\nabla V$ says that $\mathbf{E}$ points in the direction of steepest **descent** of potential and is normal to the equipotentials (Part 3).

- *"Steepest" depends on the metric.* Steepest means per unit Euclidean distance. If distance is measured with a weighted norm $\|\mathbf{h}\|_{\mathbf{M}} = \sqrt{\mathbf{h}^{\mathsf{T}}\mathbf{M}\mathbf{h}}$ ($\mathbf{M}$ symmetric positive definite), the same Cauchy–Schwarz argument in the $\mathbf{M}$-inner product shows that the steepest direction is $\mathbf{M}^{-1}\nabla f$ — the idea behind preconditioned and Newton-type methods in Ch. 23.
- *Used in:* gradient descent and the stationarity condition $\nabla f = \mathbf{0}$ (§6.4, Ch. 23); Lagrange multipliers, where $\nabla f \parallel \nabla g$ follows from Part 3 applied to the constraint surface (§6.11); field lines orthogonal to equipotentials (§7.5).

B.3 Green's theorem

Green's theorem is the two-dimensional prototype of all the integral theorems (§7.9). We prove it completely for regions that are simultaneously of "type I" and "type II", then extend it by cutting.

Elementary plane regions and positive orientation

A closed bounded region D is **type I** (vertically simple) if $D = \{(x,y) : a \le x \le b,\ g_1(x) \le y \le g_2(x)\}$ with $g_1 \le g_2$ continuous and piecewise C^1 on $[a,b]$; it is **type II** (horizontally simple) if $D = \{(x,y) : c \le y \le d,\ h_1(y) \le x \le h_2(y)\}$ with the analogous conditions. The boundary ∂D is **positively oriented** if D lies to the left as one traverses it: counter-clockwise on an outer boundary, clockwise on the boundary of a hole.

Theorem B.3 — Green's theorem

Let D be a closed bounded region whose boundary ∂D consists of finitely many piecewise-smooth simple closed curves, positively oriented. Let P, Q have continuous first partial derivatives on an open set containing D. Then

$$\oint_{\partial D} (P\,dx + Q\,dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA.$$

Proof. Both sides are linear in (P, Q), so it suffices to prove separately

$$\oint_{\partial D} P\,dx = -\iint_D \frac{\partial P}{\partial y}\,dA \qquad \text{and} \qquad \oint_{\partial D} Q\,dy = \iint_D \frac{\partial Q}{\partial x}\,dA.$$

Step 1 — the P identity on a type I region. Since $\partial P/\partial y$ is continuous, Fubini's theorem and the fundamental theorem of calculus in y give

$$-\iint_D \frac{\partial P}{\partial y}\,dA = -\int_a^b \left[\int_{g_1(x)}^{g_2(x)} \frac{\partial P}{\partial y}(x,y)\,dy\right] dx = \int_a^b \Big[P\big(x, g_1(x)\big) - P\big(x, g_2(x)\big)\Big]\,dx.$$

The positively oriented boundary of a type I region consists of (at most) four arcs:

- C_1, the bottom graph, $\mathbf{r}(t) = (t, g_1(t))$, $a \le t \le b$, traversed left to right. Here $dx = dt$, so $\int_{C_1} P\,dx = \int_a^b P(t, g_1(t))\,dt$.
- C_2, the right side $x = b$, $g_1(b) \le y \le g_2(b)$ (possibly a single point). Here x is constant, so $dx = 0$ and $\int_{C_2} P\,dx = 0$.
- C_3, the top graph, traversed right to left. Parametrising by $(t, g_2(t))$ and reversing direction, $\int_{C_3} P\,dx = -\int_a^b P(t, g_2(t))\,dt$.
- C_4, the left side $x = a$; again $\int_{C_4} P\,dx = 0$.

Adding, $\oint_{\partial D} P\,dx = \int_a^b \big[P(t, g_1(t)) - P(t, g_2(t))\big]\,dt$, which is the expression above.

Step 2 — the Q identity on a type II region. With $D = \{c \le y \le d,\ h_1(y) \le x \le h_2(y)\}$,

$$\iint_D \frac{\partial Q}{\partial x}\,dA = \int_c^d \Big[Q\big(h_2(y), y\big) - Q\big(h_1(y), y\big)\Big]\,dy.$$

On the positively oriented boundary the right graph $x = h_2(y)$ is traversed upward, contributing $+\int_c^d Q(h_2(y), y)\,dy$; the left graph $x = h_1(y)$ is traversed downward, contributing $-\int_c^d Q(h_1(y), y)\,dy$; the horizontal top and bottom segments contribute nothing because $dy = 0$ on them. The asymmetry between Steps 1 and 2 is the origin of the minus sign in $Q_x - P_y$: going counter-clockwise, the *bottom* is traversed in the $+x$ direction but the *right* side in the $+y$ direction.

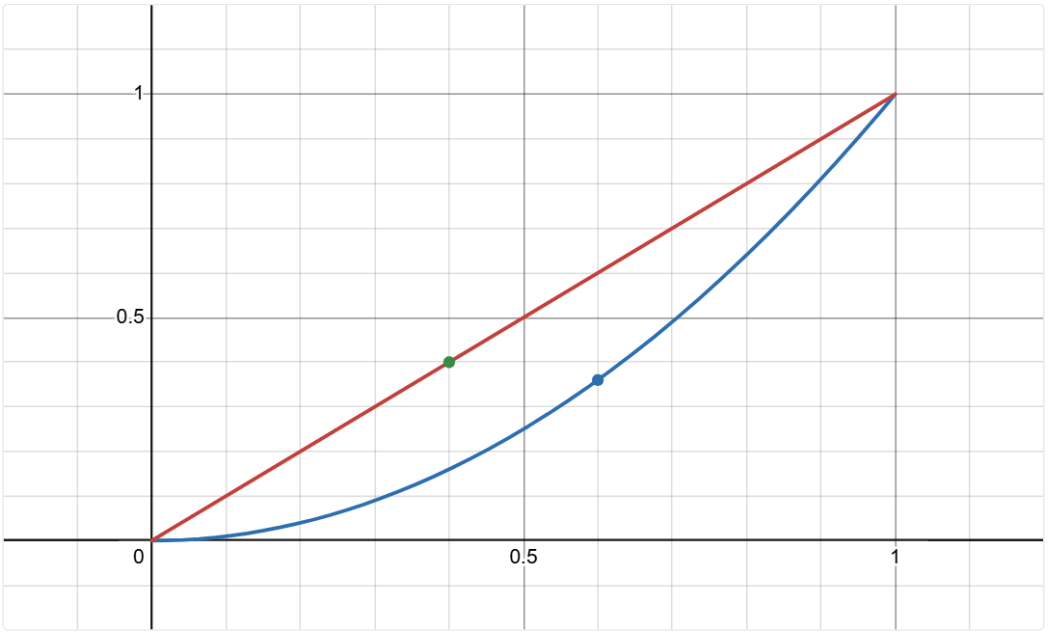
Step 3 — regions of both types. If D is both type I and type II (a disc, a rectangle, a triangle, any convex region), Steps 1 and 2 apply to the same oriented boundary, and adding them proves the theorem for D.

Step 4 — general regions by decomposition. Suppose D can be cut by finitely many piecewise-smooth arcs into regions $D_1, \dots, D_m$, each of both types, overlapping only along the cuts. (For regions bounded by finitely many piecewise-smooth curves such a decomposition exists; in practice one draws a few vertical and horizontal cuts.) Apply Step 3 to each D_k with its own positive orientation and add:

$$\sum_{k=1}^{m} \oint_{\partial D_k} (P\,dx + Q\,dy) = \sum_{k=1}^{m} \iint_{D_k} (Q_x - P_y)\,dA = \iint_D (Q_x - P_y)\,dA,$$

the last equality because the cuts have zero area. On the left, each cut arc is a boundary arc of exactly two pieces, which lie on opposite sides of it. "Keep the region on your left" therefore traverses the arc in opposite directions for the two pieces, and since reversing a curve negates a line integral, the two contributions cancel. What remains are the arcs of ∂D, each traversed with D on its left — that is, ∂D with positive orientation. This proves the theorem for D.

Step 5 — holes. A region with holes, such as the annulus $1 \le x^2 + y^2 \le 4$, is handled by the same cutting: two radial cuts split the annulus into two pieces, each a finite union of elementary regions. After cancellation the outer circle is traversed counter-clockwise and the inner circle clockwise — exactly the positive orientation of the definition. ∎



The region $x^2 \le y \le x$, $0 \le x \le 1$, which is both type I and type II. For $P = x - y^2$, $Q = xy$ both sides of Green's theorem equal $\iint_D 3y\,dA = \frac{1}{5}$ (checked symbolically): the bottom arc C_1 (blue, left to right) contributes $\frac{7}{10}$, the top arc C_3 (red, right to left) contributes $-\frac{1}{2}$.

Remarks.

- *The C^1 hypothesis cannot fail even at one point.* For $P = -y/(x^2+y^2)$, $Q = x/(x^2+y^2)$ on the closed unit disc, $Q_x - P_y = 0$ wherever defined, yet $\oint_{|\mathbf{r}|=1}(P\,dx + Q\,dy) = 2\pi$. Green's theorem is not violated: P and Q are not defined at the origin, which lies in D. Applied instead to the annulus $\varepsilon \le |\mathbf{r}| \le 1$ it gives the correct statement — the counter-clockwise circulations around the two circles are equal. This example is the subject of B.6.
- *Area formula.* $P = -y/2$, $Q = x/2$ gives $\operatorname{area}(D) = \frac{1}{2}\oint_{\partial D}(x\,dy - y\,dx)$, the principle of the planimeter and of the "shoelace" polygon-area formula (§7.9).
- *Flux form.* Replacing (P, Q) by $(-F_y, F_x)$ gives $\oint_{\partial D}\mathbf{F}\cdot\hat{\mathbf{n}}\,ds = \iint_D \nabla\cdot\mathbf{F}\,dA$ with $\hat{\mathbf{n}}$ the outward normal: the 2-D divergence theorem, which B.4 lifts to three dimensions.
- *Used in:* the proof of Stokes' theorem (B.5); Cauchy's integral theorem in Ch. 18, where Green applied to $f = u + jv$ together with the Cauchy–Riemann equations gives $\oint f\,dz = 0$ (under the extra hypothesis that f' is continuous, which Goursat's proof removes).

B.4 The divergence theorem

The divergence theorem (Gauss's theorem) converts the flux through a closed surface into a volume integral of the divergence (§7.11). The proof has exactly the structure of B.3: one Cartesian component at a time, the fundamental theorem of calculus along one axis, and a cut-and-cancel argument for general regions.

Flux and outward normal

For a piecewise-smooth surface S with a chosen continuous unit normal $\hat{\mathbf{n}}$, the **flux** of $\mathbf{F}$ is $\iint_S \mathbf{F}\cdot d\mathbf{S} = \iint_S \mathbf{F}\cdot\hat{\mathbf{n}}\,dS$. If $S = \mathbf{r}(D)$ with $\mathbf{r}_u \times \mathbf{r}_v$ pointing along $\hat{\mathbf{n}}$, then $\iint_S \mathbf{F}\cdot d\mathbf{S} = \iint_D \mathbf{F}(\mathbf{r}(u,v))\cdot(\mathbf{r}_u\times\mathbf{r}_v)\,du\,dv$ (§7.7). For the boundary ∂V of a solid V, $\hat{\mathbf{n}}$ is always the **outward** normal.

Elementary solids

A closed bounded solid V is **z-simple** if $V = \{(x,y,z) : (x,y)\in D_{xy},\ g_1(x,y) \le z \le g_2(x,y)\}$ for a plane region D_{xy} with piecewise-smooth boundary and $g_1 \le g_2$ continuous on D_{xy} and C^1 on its interior; **x-simple** and **y-simple** are defined in the same way with the roles of the coordinates permuted. V is **elementary** if it is simple in all three directions (balls, boxes, tetrahedra, any convex solid with piecewise-smooth boundary).

Theorem B.4 — Divergence theorem

Let V be a closed bounded solid that can be cut into finitely many elementary solids, with piecewise-smooth boundary ∂V oriented by the outward normal. Let $\mathbf{F} = (F_1, F_2, F_3)$ have continuous first partial derivatives on an open set containing V. Then

$$\oint_{\partial V}\mathbf{F}\cdot d\mathbf{S} = \iiint_V \nabla\cdot\mathbf{F}\,dV, \qquad \nabla\cdot\mathbf{F} = \partial_1 F_1 + \partial_2 F_2 + \partial_3 F_3.$$

Proof.

Step 1 — a box. Let $V = [a_1,b_1]\times[a_2,b_2]\times[a_3,b_3]$. Take the third term first. By Fubini and the fundamental theorem of calculus in z,

$$\iiint_V \frac{\partial F_3}{\partial z}\,dV = \int_{a_1}^{b_1}\int_{a_2}^{b_2}\Big[F_3(x,y,b_3) - F_3(x,y,a_3)\Big]\,dy\,dx.$$

Now compute the flux of the single component $F_3\hat{\mathbf{z}}$ out of the six faces. On the top face $z = b_3$ the outward normal is $+\hat{\mathbf{z}}$ and $dS = dx\,dy$, giving $\iint F_3(x,y,b_3)\,dx\,dy$; on the bottom face $z = a_3$ it is $-\hat{\mathbf{z}}$, giving $-\iint F_3(x,y,a_3)\,dx\,dy$; on the four side faces the normal is $\pm\hat{\mathbf{x}}$ or $\pm\hat{\mathbf{y}}$, perpendicular to $\hat{\mathbf{z}}$, so they contribute nothing. The two expressions agree. The same computation in x and in y gives the corresponding identities for $F_1\hat{\mathbf{x}}$ and $F_2\hat{\mathbf{y}}$; adding the three proves

the theorem for the box. This case already contains the physics: the net outflow in the z direction is the difference between what leaves through the top and what enters through the bottom, and $\partial_z F_3$ is that difference per unit volume.

Step 2 — the F_3 term on a z-simple solid. Let $V = \{(x,y) \in D_{xy},\ g_1 \le z \le g_2\}$. As in Step 1,

$$\iiint_V \frac{\partial F_3}{\partial z}\, dV = \iint_{D_{xy}} \Big[F_3\big(x,y,g_2(x,y)\big) - F_3\big(x,y,g_1(x,y)\big)\Big]\, dA.$$

The boundary ∂V has three parts.

- *Top S_2*: parametrise by $\mathbf{r}(x,y) = (x,y,g_2(x,y))$, $(x,y) \in D_{xy}$. Then $\mathbf{r}_x \times \mathbf{r}_y = (1,0,g_{2,x}) \times (0,1,g_{2,y}) = (-g_{2,x},\ -g_{2,y},\ 1)$, whose z-component is positive, so it points upward — out of V. Hence $\iint_{S_2} F_3\hat{\mathbf{z}} \cdot d\mathbf{S} = \iint_{D_{xy}} F_3(x,y,g_2)\,(\hat{\mathbf{z}} \cdot (-g_{2,x}, -g_{2,y}, 1))\, dA = \iint_{D_{xy}} F_3(x,y,g_2)\, dA$.
- *Bottom S_1*: the same parametrisation with g_1 gives an upward normal, but outward is now downward, so the sign flips: $\iint_{S_1} F_3\hat{\mathbf{z}} \cdot d\mathbf{S} = -\iint_{D_{xy}} F_3(x,y,g_1)\, dA$.
- *Lateral wall S_3* (the vertical surface over ∂D_{xy}, possibly empty): it is made of vertical lines, so its normal is horizontal and $F_3\hat{\mathbf{z}} \cdot \hat{\mathbf{n}} = 0$.

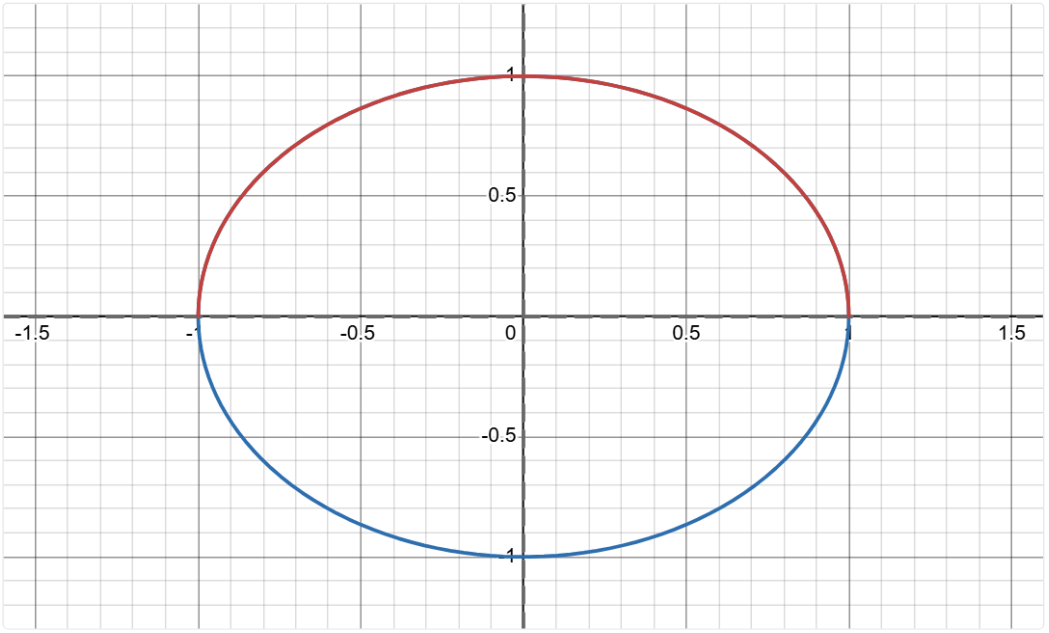
Adding the three parts reproduces the volume integral: $\oint_{\partial V} F_3\hat{\mathbf{z}} \cdot d\mathbf{S} = \iiint_V \partial_z F_3\, dV$. Note how cleanly the tilt of the top surface disappears: the factor $\sqrt{1+g_x^2+g_y^2}$ in dS is exactly cancelled by the factor $1/\sqrt{1+g_x^2+g_y^2}$ in $\hat{\mathbf{n}} \cdot \hat{\mathbf{z}}$ — the flux through a tilted patch equals the flux through its shadow on the xy-plane.

Step 3 — elementary solids. If V is x-, y- and z-simple, Step 2 and its two permuted versions give

$$\oint_{\partial V} F_1\hat{\mathbf{x}} \cdot d\mathbf{S} = \iiint_V \partial_x F_1\, dV, \quad \oint_{\partial V} F_2\hat{\mathbf{y}} \cdot d\mathbf{S} = \iiint_V \partial_y F_2\, dV, \quad \oint_{\partial V} F_3\hat{\mathbf{z}} \cdot d\mathbf{S} = \iiint_V \partial_z F_3\, dV,$$

and adding the three proves the theorem for V.

Step 4 — cutting. If $V = V_1 \cup \cdots \cup V_m$ with elementary pieces meeting only along (piecewise-smooth) faces, add the theorem for each piece. The volume integrals add to $\iiint_V \nabla \cdot \mathbf{F}\, dV$ since the faces have zero volume. Each internal face belongs to exactly two pieces, and the outward normal of one is the inward normal of the other, so its two flux contributions are equal and opposite and cancel. The remaining faces make up ∂V, each with the outward normal of V. ■



Cross-section of the unit ball, an elementary solid: it is z-simple with $g_{1,2} = \mp\sqrt{1-x^2-y^2}$ (top hemisphere red). Cuts along coordinate planes (grey) split non-elementary solids, such as a hollow shell, into elementary pieces; the flux through each cut cancels in pairs. For $\mathbf{F} = (x^3, y^3, z^3)$ both sides of the theorem on the unit ball equal $12\pi/5$ (checked symbolically).

Remarks.

- *Why C^1 on all of V.* For the point-charge field $\mathbf{F} = \hat{\mathbf{R}}/R^2$, $\nabla \cdot \mathbf{F} = \frac{1}{R^2}\partial_R(R^2 \cdot R^{-2}) = 0$ for $R > 0$, yet the flux through any sphere centred at the origin is $\frac{1}{a^2} \cdot 4\pi a^2 = 4\pi$. The theorem does not apply to a ball containing the origin, where $\mathbf{F}$ is singular. Applied to the shell $\varepsilon \le R \le a$, whose boundary includes the inner sphere with **inward-pointing** normal (outward from the shell), it correctly gives $4\pi - 4\pi = 0$. The physicist's repair, $\nabla \cdot (\hat{\mathbf{R}}/R^2) = 4\pi\delta^3(\mathbf{R})$, is discussed in §7.11.
- *Coordinate-free divergence.* Applying the theorem to a small ball $B_\varepsilon(\mathbf{a})$ and using continuity of $\nabla \cdot \mathbf{F}$ (mean value theorem for integrals) gives $\nabla \cdot \mathbf{F}(\mathbf{a}) = \lim_{\varepsilon \to 0} \frac{1}{\mathrm{vol}(B_\varepsilon)} \oint_{\partial B_\varepsilon} \mathbf{F} \cdot d\mathbf{S}$: the divergence is outward flux per unit volume, independent of coordinates. This is the starting point of B.8.
- *Green's identities.* Applying the theorem to $\mathbf{F} = u\nabla w$ and using the product rule $\nabla \cdot (u\nabla w) = \nabla u \cdot \nabla w + u\nabla^2 w$ (B.7) gives Green's first identity $\oint_{\partial V} u\, \partial_n w\, dS = \iiint_V (\nabla u \cdot \nabla w + u\nabla^2 w)\, dV$, from which the uniqueness of solutions of Laplace's and Poisson's equations follows (§7.11, §13.6).
- *Used in:* Gauss's law in differential form and the continuity equation (B.9); the energy method for the heat equation (§13.4); finite-volume discretisations (Ch. 22).

B.5 Stokes' theorem

Stokes' theorem lifts Green's theorem from a flat region to a curved surface in space (§7.10). The cleanest proof does not tile the surface at all: it pulls the whole problem back to the parameter domain, applies Green's theorem there, and recognises the integrand. All the geometry is concentrated in one algebraic lemma.

Parametrised surface and induced orientation

A **smooth parametrised surface** is $S = \mathbf{r}(D)$ where D is a plane region to which Green's theorem applies (B.3), $\mathbf{r}$ is one-to-one on D and of class C^2 on an open set containing D, and $\mathbf{r}_u \times \mathbf{r}_v \neq \mathbf{0}$ on D. It is oriented by $\hat{\mathbf{n}} = \mathbf{r}_u \times \mathbf{r}_v / \|\mathbf{r}_u \times \mathbf{r}_v\|$. Its boundary curve $C = \partial S$ is $\mathbf{r}(\partial D)$, traversed in the direction inherited from the positive orientation of ∂D; this is the **right-hand rule**: with the thumb along $\hat{\mathbf{n}}$, the fingers curl along C.

Lemma B.5 — The antisymmetric part of the Jacobian is a cross product

Let $\mathbf{F}$ be C^1, $\mathbf{J} = D\mathbf{F}$ its Jacobian matrix ($J_{ij} = \partial_j F_i$), and $\boldsymbol{\omega} = \nabla \times \mathbf{F}$. Then for all vectors $\mathbf{a}, \mathbf{b}$

$$(\mathbf{J} - \mathbf{J}^{\mathsf{T}})\,\mathbf{a} = \boldsymbol{\omega} \times \mathbf{a}, \qquad \text{hence} \qquad \mathbf{b} \cdot (\mathbf{J}\mathbf{a}) - \mathbf{a} \cdot (\mathbf{J}\mathbf{b}) = \boldsymbol{\omega} \cdot (\mathbf{a} \times \mathbf{b}).$$

Proof of the lemma. The first component of $(\mathbf{J} - \mathbf{J}^{\mathsf{T}})\mathbf{a}$ is $(\partial_2 F_1 - \partial_1 F_2)a_2 + (\partial_3 F_1 - \partial_1 F_3)a_3 = -\omega_3 a_2 + \omega_2 a_3$, which is the first component of $\boldsymbol{\omega} \times \mathbf{a} = (\omega_2 a_3 - \omega_3 a_2, \dots)$. The other two components follow by cyclic permutation $1 \to 2 \to 3 \to 1$. For the second identity, $\mathbf{a} \cdot (\mathbf{J}\mathbf{b}) = \mathbf{b} \cdot (\mathbf{J}^{\mathsf{T}}\mathbf{a})$, so the left side is $\mathbf{b} \cdot \big((\mathbf{J} - \mathbf{J}^{\mathsf{T}})\mathbf{a}\big) = \mathbf{b} \cdot (\boldsymbol{\omega} \times \mathbf{a}) = \boldsymbol{\omega} \cdot (\mathbf{a} \times \mathbf{b})$ by the cyclic property of the scalar triple product (§7.2). ∎

Theorem B.5 — Stokes' theorem

Let $S = \mathbf{r}(D)$ be a smooth parametrised surface with boundary C oriented by the right-hand rule, or a finite union of such surfaces glued along boundary arcs with compatible orientations. If $\mathbf{F}$ has continuous first partial derivatives on an open set containing S, then

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}.$$

Proof.

Step 1 — pull the line integral back to D. Let $(u(t), v(t))$ trace ∂D positively. Then $\mathbf{r}(u(t), v(t))$ traces C, and by the chain rule $\frac{d}{dt}\mathbf{r} = \mathbf{r}_u \dot{u} + \mathbf{r}_v \dot{v}$. Hence

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_{\partial D} \big(\tilde{P}\,du + \tilde{Q}\,dv\big), \qquad \tilde{P}(u,v) = \mathbf{F}(\mathbf{r}) \cdot \mathbf{r}_u, \quad \tilde{Q}(u,v) = \mathbf{F}(\mathbf{r}) \cdot \mathbf{r}_v.$$

Because $\mathbf{F}$ is C^1 and $\mathbf{r}$ is C^2, $\tilde{P}$ and $\tilde{Q}$ are C^1, so Green's theorem applies.

Step 2 — Green's theorem in the (u,v)-plane.

$$\oint_{\partial D} \big(\tilde{P}\,du + \tilde{Q}\,dv\big) = \iint_D \left(\frac{\partial \tilde{Q}}{\partial u} - \frac{\partial \tilde{P}}{\partial v}\right) du\,dv.$$

Step 3 — differentiate. By the chain rule $\partial_u\big[\mathbf{F}(\mathbf{r}(u,v))\big] = \mathbf{J}\mathbf{r}_u$ and $\partial_v\big[\mathbf{F}(\mathbf{r}(u,v))\big] = \mathbf{J}\mathbf{r}_v$, with $\mathbf{J}$ evaluated at $\mathbf{r}(u,v)$. The product rule gives

$$\frac{\partial \tilde{Q}}{\partial u} = (\mathbf{J}\mathbf{r}_u) \cdot \mathbf{r}_v + \mathbf{F} \cdot \mathbf{r}_{vu}, \qquad \frac{\partial \tilde{P}}{\partial v} = (\mathbf{J}\mathbf{r}_v) \cdot \mathbf{r}_u + \mathbf{F} \cdot \mathbf{r}_{uv}.$$

Since $\mathbf{r}$ is C^2, Clairaut's theorem gives $\mathbf{r}_{uv} = \mathbf{r}_{vu}$, so the second-derivative terms cancel on subtraction. This is the only place C^2 is used.

Step 4 — recognise the curl. By Lemma B.5 with $\mathbf{a} = \mathbf{r}_u$, $\mathbf{b} = \mathbf{r}_v$,

$$\frac{\partial \tilde{Q}}{\partial u} - \frac{\partial \tilde{P}}{\partial v} = \mathbf{r}_v \cdot (\mathbf{J}\mathbf{r}_u) - \mathbf{r}_u \cdot (\mathbf{J}\mathbf{r}_v) = (\nabla \times \mathbf{F})\big(\mathbf{r}(u,v)\big) \cdot \big(\mathbf{r}_u \times \mathbf{r}_v\big).$$

Step 5 — identify the flux. Combining Steps 1–4,

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_D (\nabla \times \mathbf{F})(\mathbf{r}) \cdot (\mathbf{r}_u \times \mathbf{r}_v)\,du\,dv = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S},$$

the last equality being the definition of flux with the orientation $\hat{\mathbf{n}} \parallel \mathbf{r}_u \times \mathbf{r}_v$.

Step 6 — glued surfaces. If $S = S_1 \cup \cdots \cup S_m$ with each S_k a smooth parametrised surface and the orientations agreeing across shared edges, add the theorem for each piece. The flux integrals add. A shared edge is traversed in opposite directions by the induced orientations of its two neighbours (compatibility of orientation means precisely this), so its line integrals cancel, leaving the outer boundary C. This covers closed-up shapes such as a hemisphere parametrised in two patches, a cylinder with a lid, or a polyhedral surface. ∎

Check. For $\mathbf{F} = (z, x, y)$, $\nabla \times \mathbf{F} = (1,1,1)$. On the upper unit hemisphere with outward normal, the boundary is the unit circle in the xy-plane, counter-clockwise; there $\mathbf{F} = (0, \cos t, \sin t)$, $d\mathbf{r} = (-\sin t, \cos t, 0)\,dt$, and $\oint \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} \cos^2 t\,dt = \pi$. The flux of the constant vector $(1,1,1)$ through the hemisphere equals its flux through the flat unit disc (both surfaces share the boundary — see the remark below), which is $1 \cdot \pi = \pi$. ✓

Remarks.

- *Green is the special case.* With $\mathbf{r}(u,v) = (u, v, 0)$ we get $\mathbf{r}_u \times \mathbf{r}_v = \hat{\mathbf{z}}$ and $(\nabla \times \mathbf{F}) \cdot \hat{\mathbf{z}} = \partial_x F_y - \partial_y F_x$: Theorem B.3. The graph case $z = g(x,y)$ sketched in §7.10 is $\mathbf{r}(x,y) = (x, y, g(x,y))$, $\mathbf{r}_x \times \mathbf{r}_y = (-g_x, -g_y, 1)$.
- *Orientation is a hypothesis.* On a non-orientable surface such as the Möbius strip no continuous $\hat{\mathbf{n}}$ exists and the theorem has no meaning; on an orientable one, reversing $\hat{\mathbf{n}}$ reverses C and both sides change sign together.

- *Independence of the surface.* If S and S' have the same boundary C with the same induced orientation, the theorem gives the same flux of $\nabla\times\mathbf{F}$ through both. This is consistent with B.4: $S\cup(-S')$ is a closed surface, and the outward flux of a curl through a closed surface is $\iiint \nabla\cdot(\nabla\times\mathbf{F})\,dV = 0$ by B.7. It is exactly the consistency requirement that forced Maxwell's displacement current (B.9).
- *Coordinate-free curl.* For a small flat disc of radius ε centred at $\mathbf{a}$ with normal $\hat{\mathbf{n}}$, continuity of $\nabla\times\mathbf{F}$ gives $(\nabla\times\mathbf{F})(\mathbf{a})\cdot\hat{\mathbf{n}} = \lim_{\varepsilon\to 0}\frac{1}{\pi\varepsilon^2}\oint \mathbf{F}\cdot d\mathbf{r}$: curl is circulation per unit area, as in §7.5.
- *Used in:* Faraday's and Ampère's laws in differential form (B.9); the curl test for conservative fields (B.6); EMF in moving and stationary loops (§7.12).

B.6 Curl-free versus conservative fields

§7.6 states that a field with zero curl is the gradient of a potential *if the domain is simply connected*. This section proves the equivalence of the three characterisations of a conservative field, proves the curl test on star-shaped and then simply connected domains, and shows by the standard counterexample that the topological hypothesis cannot be removed.

Conservative, star-shaped, simply connected

Let $D\subset\mathbb{R}^n$ ($n=2$ or 3) be open and connected. A continuous field $\mathbf{F}$ on D is **conservative** if $\mathbf{F}=\nabla f$ for some C^1 function f on D (a **potential**; in electrostatics the potential is $V=-f$). D is **star-shaped** about $\mathbf{a}\in D$ if the segment from $\mathbf{a}$ to every $\mathbf{x}\in D$ lies in D (convex sets, all of $\mathbb{R}^3$, a half-space). D is **simply connected** if every closed curve in D can be continuously shrunk to a point without leaving D: there is a continuous $\mathbf{H}:[0,1]\times[0,1]\to D$ with $\mathbf{H}(s,0)=\boldsymbol{\gamma}(s)$, $\mathbf{H}(s,1)=$ a constant point, and $\mathbf{H}(0,t)=\mathbf{H}(1,t)$ for all t.

Theorem B.6 — Conservative fields and the curl test

Let $\mathbf{F}$ be continuous on an open connected set $D\subset\mathbb{R}^3$ (for $D\subset\mathbb{R}^2$ read $\nabla\times\mathbf{F}$ as the scalar $\partial_x F_y-\partial_y F_x$).

1. The following are equivalent: (a) $\mathbf{F}$ is conservative; (b) $\int_C \mathbf{F}\cdot d\mathbf{r}$ depends only on the endpoints of the piecewise-smooth curve $C\subset D$; (c) $\oint_C \mathbf{F}\cdot d\mathbf{r}=0$ for every piecewise-smooth closed curve $C\subset D$.
2. If $\mathbf{F}$ is C^1 and conservative, then $\nabla\times\mathbf{F}=\mathbf{0}$ on D.
3. (**Poincaré lemma**) If D is star-shaped about $\mathbf{a}$, $\mathbf{F}$ is C^1 and $\nabla\times\mathbf{F}=\mathbf{0}$ on D, then $\mathbf{F}=\nabla f$ with
$$f(\mathbf{x}) = \int_0^1 \mathbf{F}\big(\mathbf{a}+t(\mathbf{x}-\mathbf{a})\big)\cdot(\mathbf{x}-\mathbf{a})\,dt.$$
4. If D is simply connected, $\mathbf{F}$ is C^1 and $\nabla\times\mathbf{F}=\mathbf{0}$ on D, then $\mathbf{F}$ is conservative.

Proof.

Part 1, (a)⇒(b). If $\mathbf{F}=\nabla f$ and C is parametrised by $\mathbf{r}(t)$, $\alpha\le t\le\beta$, then by the chain rule $\frac{d}{dt}f(\mathbf{r}(t)) = \nabla f\cdot\mathbf{r}'(t)$ on each smooth piece, and the fundamental theorem of calculus (summed over the pieces, the intermediate values telescoping) gives $\int_C \mathbf{F}\cdot d\mathbf{r} = f(\mathbf{r}(\beta))-f(\mathbf{r}(\alpha))$.

(b)⇒(c). A closed curve starts and ends at the same point, so it has the same endpoints as the constant path, whose integral is 0.

(c)⇒(b). If C_1, C_2 run from $\mathbf{p}$ to $\mathbf{q}$, then C_1 followed by C_2 reversed is closed, so $0=\int_{C_1}-\int_{C_2}$.

(b)⇒(a). Fix $\mathbf{p}_0\in D$ and define $f(\mathbf{x})=\int_{\mathbf{p}_0}^{\mathbf{x}}\mathbf{F}\cdot d\mathbf{r}$ along any piecewise-smooth path in D; one exists because an open connected set is path-connected by polygonal paths, and the value is path-independent by (b). For $\mathbf{x}\in D$ choose $\delta>0$ with the ball $B_\delta(\mathbf{x})\subset D$. For $0<|h|<\delta$, reach $\mathbf{x}+h\hat{\mathbf{x}}$ by going first to $\mathbf{x}$ and then along the straight segment $\mathbf{x}+s\hat{\mathbf{x}}$, $0\le s\le h$:
$$\frac{f(\mathbf{x}+h\hat{\mathbf{x}})-f(\mathbf{x})}{h} = \frac{1}{h}\int_0^h F_1(\mathbf{x}+s\hat{\mathbf{x}})\,ds \xrightarrow{\;h\to 0\;} F_1(\mathbf{x})$$
by continuity of F_1. So $\partial_x f = F_1$, and likewise $\partial_y f = F_2$, $\partial_z f = F_3$. The partials are continuous, so f is C^1 and $\nabla f=\mathbf{F}$.

Part 2. If $\mathbf{F}=\nabla f$ with $\mathbf{F}$ of class C^1, then f is C^2 and $(\nabla\times\nabla f)_3 = \partial_x\partial_y f-\partial_y\partial_x f=0$ by Clairaut; similarly for the other components (see also B.7).

Part 3. Take $\mathbf{a}=\mathbf{0}$ (translate). Put $\mathbf{y}=t\mathbf{x}$. Since $\mathbf{F}$ is C^1 and $[0,1]$ is compact, we may differentiate under the integral sign:
$$\partial_i f(\mathbf{x}) = \int_0^1 \Big[F_i(t\mathbf{x}) + \sum_{j} x_j\, t\,(\partial_i F_j)(t\mathbf{x})\Big]dt.$$
The curl condition says $\partial_i F_j=\partial_j F_i$ for all i,j (each such equation is one component of $\nabla\times\mathbf{F}=\mathbf{0}$). Hence $\sum_j x_j\, t\,\partial_i F_j(t\mathbf{x}) = t\sum_j x_j\,\partial_j F_i(t\mathbf{x}) = t\,\frac{d}{dt}\big[F_i(t\mathbf{x})\big]$, by the chain rule. Therefore
$$\partial_i f(\mathbf{x}) = \int_0^1 \frac{d}{dt}\Big[t\,F_i(t\mathbf{x})\Big]dt = 1\cdot F_i(\mathbf{x}) - 0 = F_i(\mathbf{x}),$$
using the product rule $\frac{d}{dt}[tF_i] = F_i + t\frac{d}{dt}F_i$. Star-shapedness was used exactly once: to guarantee $t\mathbf{x}\in D$ for $t\in[0,1]$.

Part 4. By Part 1 it suffices to show $\oint_\gamma \mathbf{F}\cdot d\mathbf{r}=0$ for every closed piecewise-smooth $\boldsymbol{\gamma}$ in D. Let $\mathbf{H}$ be a contraction of $\boldsymbol{\gamma}$ as in the definition. A continuous homotopy inside an open set can be replaced by a C^2 one with the same bottom edge, staying inside D (a standard smoothing fact, which we assume; it is proved by convolving $\mathbf{H}$ with a smooth bump and using that $\mathbf{H}([0,1]^2)$ is a compact subset of the open set D). With $\mathbf{H}$ of class C^2, Steps 1–4 of the proof of B.5 apply verbatim to $\mathbf{r}=\mathbf{H}$ on the square $[0,1]^2$ — they used neither injectivity nor $\mathbf{r}_u\times\mathbf{r}_v\neq\mathbf{0}$ — and give
$$\oint_{\partial[0,1]^2} \mathbf{F}(\mathbf{H})\cdot\big(\mathbf{H}_s\,ds+\mathbf{H}_t\,dt\big) = \iint_{[0,1]^2} (\nabla\times\mathbf{F})(\mathbf{H})\cdot(\mathbf{H}_s\times\mathbf{H}_t)\,ds\,dt = 0.$$

Traverse the boundary of the square counter-clockwise. The bottom edge $t = 0$ gives $\oint_\gamma \mathbf{F} \cdot d\mathbf{r}$. The top edge $t = 1$ is mapped to a single point, so $\mathbf{H}_s = \mathbf{0}$ there and it contributes 0. The right edge $s = 1$ (upward) and the left edge $s = 0$ (downward) are the same path $t \mapsto \mathbf{H}(0, t) = \mathbf{H}(1, t)$ traversed in opposite directions, so they cancel. Hence $\oint_\gamma \mathbf{F} \cdot d\mathbf{r} = 0$. ■

The counterexample on the punctured plane

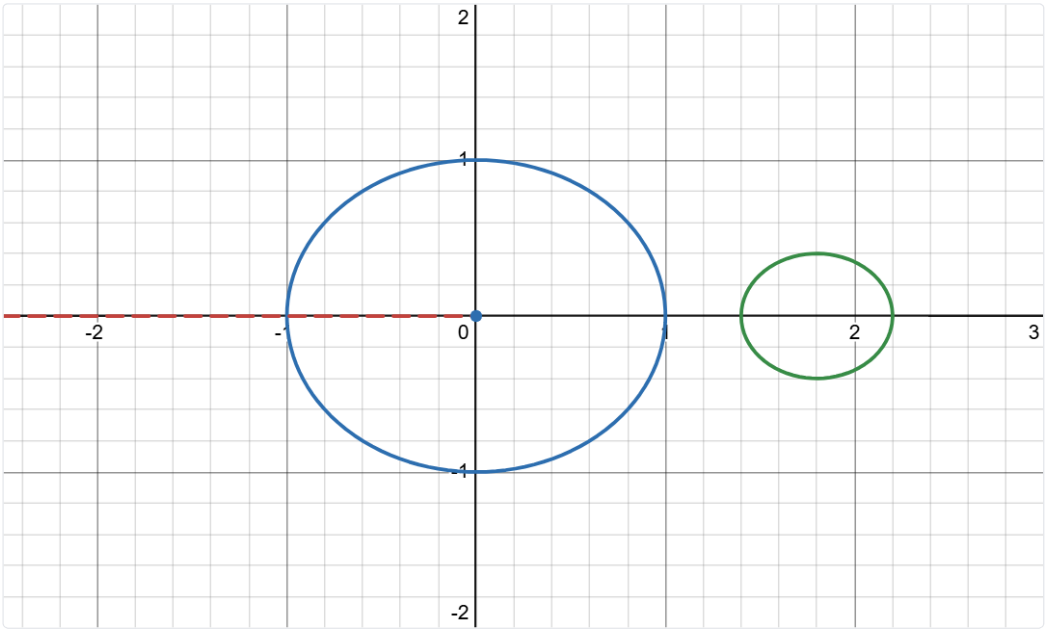
On $D = \mathbb{R}^2 \setminus \{\mathbf{0}\}$ let

$$\mathbf{F}(x, y) = \frac{-y\,\hat{\mathbf{x}} + x\,\hat{\mathbf{y}}}{x^2 + y^2} = \frac{1}{r}\hat{\boldsymbol{\phi}}.$$

Curl. $\partial_x \dfrac{x}{x^2+y^2} = \dfrac{y^2 - x^2}{(x^2+y^2)^2}$ and $\partial_y \dfrac{-y}{x^2+y^2} = \dfrac{y^2 - x^2}{(x^2+y^2)^2}$, so $\partial_x F_y - \partial_y F_x = 0$ everywhere on D; $\mathbf{F}$ is C^∞ there.

Circulation. On the unit circle $\mathbf{r} = (\cos t, \sin t)$, $\mathbf{F} = (-\sin t, \cos t) = \mathbf{r}'(t)$, so $\oint \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} 1\, dt = 2\pi \neq 0$. By Part 1, $\mathbf{F}$ is **not** conservative on D.

What goes wrong. D is not simply connected: the unit circle cannot be shrunk to a point without passing through the origin. Locally $\mathbf{F} = \nabla\phi$ with ϕ the polar angle, and on the slit plane $D \setminus \{(x, 0) : x \leq 0\}$ — which is star-shaped about $(1, 0)$ — $\phi \in (-\pi, \pi)$ is a genuine potential, as Part 3 guarantees. But ϕ cannot be extended continuously across the slit: it jumps by 2π. For any closed curve in D, the circulation is 2π times its winding number about the origin; a loop that does not encircle the origin has zero circulation.



The punctured plane. Circulation of $\hat{\boldsymbol{\phi}}/r$ round the blue circle is 2π; round the green circle, which does not enclose the origin, it is 0. Removing the red slit makes the domain star-shaped and the polar angle a single-valued potential.

Remarks.

- *Electromagnetic meaning.* The field of a long straight wire on the z-axis is $\mathbf{H} = \dfrac{I}{2\pi r}\hat{\boldsymbol{\phi}}$: curl-free in the region $r > 0$ but with circulation I around the wire (Ampère's law). The magnetic scalar potential $\psi = -I\phi/(2\pi)$ with $\mathbf{H} = -\nabla\psi$ is therefore multivalued, a fact exploited in magnetostatics by introducing a "barrier" surface — the slit above.
- *Holes of different dimension.* $\mathbb{R}^3$ minus a line is not simply connected (loops around the line), but $\mathbb{R}^3$ minus a point **is**: every loop can be slid off the point. So a C^1 curl-free field on $\mathbb{R}^3 \setminus \{\mathbf{0}\}$, such as the Coulomb field, is always conservative, which is why the electrostatic potential $q/(4\pi\varepsilon_0 R)$ exists globally. (The analogous statement for divergence-free fields fails there: $\hat{\mathbf{R}}/R^2$ has zero divergence on $\mathbb{R}^3 \setminus \{\mathbf{0}\}$ but is not a curl, since its flux through a sphere is $4\pi \neq 0$ while the flux of a curl through a closed surface vanishes by B.5.)
- *Kirchhoff's voltage law.* KVL is statement (c) for $\mathbf{E}$. Faraday's law (B.9) says $\nabla \times \mathbf{E} = -\partial_t \mathbf{B}$, so KVL holds only where $\partial_t \mathbf{B}$ is negligible — see the engineering note in §7.6.
- *Used in:* finding potentials (§7.6); exact differential equations, where $M\,dx + N\,dy$ is exact if $M_y = N_x$ on a simply connected domain (Ch. 11); the existence of analytic logarithms on simply connected domains and the role of the winding number in the residue theorem (Ch. 18).

B.7 Vector identities by index notation

The identities of §7.5 can each be verified by writing out components, but index notation proves them all with two tools: the Levi-Civita symbol and one contraction identity.

Summation convention, Kronecker delta, Levi-Civita symbol

Indices $i, j, k, \ldots$ run over $1, 2, 3$, and a **repeated index is summed**: $a_i b_i = \sum_{i=1}^{3} a_i b_i$. The **Kronecker delta** is $\delta_{ij} = 1$ if $i = j$ and 0 otherwise. The **Levi-Civita symbol** ε_{ijk} is $+1$ if (i, j, k) is an even permutation of $(1, 2, 3)$, -1 if odd, and 0 if any two indices coincide. Then, in Cartesian components,

$$\mathbf{a} \cdot \mathbf{b} = a_i b_i, \quad (\mathbf{a} \times \mathbf{b})_i = \varepsilon_{ijk} a_j b_k, \quad \nabla \cdot \mathbf{F} = \partial_i F_i, \quad (\nabla \times \mathbf{F})_i = \varepsilon_{ijk} \partial_j F_k, \quad \nabla^2 f = \partial_i \partial_i f.$$

(For instance $(\mathbf{a}\times\mathbf{b})_1 = \varepsilon_{123}a_2b_3 + \varepsilon_{132}a_3b_2 = a_2b_3 - a_3b_2$.) Three properties of ε_{ijk} are used repeatedly: it is **cyclic**, $\varepsilon_{ijk} = \varepsilon_{jki} = \varepsilon_{kij}$; it is **antisymmetric** under swapping any two indices; and it annihilates symmetric objects: if $S_{jk} = S_{kj}$ then $\varepsilon_{ijk}S_{jk} = 0$ (rename $j \leftrightarrow k$ to get $\varepsilon_{ikj}S_{kj} = -\varepsilon_{ijk}S_{jk}$, so the sum equals its own negative).

Lemma B.7 — The ε–δ identity

$$\varepsilon_{kij}\,\varepsilon_{kmn} = \delta_{im}\delta_{jn} - \delta_{in}\delta_{jm}.$$

Proof. Fix i, j, m, n. On the left, a term $\varepsilon_{kij}\varepsilon_{kmn}$ is non-zero only if $i \neq j$, $m \neq n$, and k is the third index different from both i, j and from both m, n; so the left side vanishes unless $i \neq j$ and $\{m, n\} = \{i, j\}$, and then exactly one k contributes. If $(m, n) = (i, j)$ the two symbols are equal and the product is $+1$; if $(m, n) = (j, i)$ they differ by one swap and the product is -1. On the right: if $i \neq j$ and $(m, n) = (i, j)$ we get $1 - 0 = 1$; if $(m, n) = (j, i)$, $0 - 1 = -1$; if $i = j$ the two terms are $\delta_{im}\delta_{in} - \delta_{in}\delta_{im} = 0$; and if $\{m, n\} \neq \{i, j\}$ with $i \neq j$, some index of $\{i, j\}$ is missing from $\{m, n\}$, which kills both terms. The two sides agree in every case. ∎

Contracting once more gives $\varepsilon_{kij}\varepsilon_{kmj} = 3\delta_{im} - \delta_{im} = 2\delta_{im}$; the BAC–CAB rule follows at once: $[\mathbf{a}\times(\mathbf{b}\times\mathbf{c})]_i = \varepsilon_{ijk}a_j\varepsilon_{kmn}b_mc_n = \varepsilon_{kij}\varepsilon_{kmn}a_jb_mc_n = b_i(a_jc_j) - c_i(a_jb_j)$, i.e. $\mathbf{a}\times(\mathbf{b}\times\mathbf{c}) = \mathbf{b}(\mathbf{a}\cdot\mathbf{c}) - \mathbf{c}(\mathbf{a}\cdot\mathbf{b})$.

Theorem B.7 — Vector-calculus identities

Let f, g be scalar and $\mathbf{A}, \mathbf{B}$ vector fields on an open set in $\mathbb{R}^3$. If they are C^1:

1. $\nabla(fg) = f\nabla g + g\nabla f$;
2. $\nabla\cdot(f\mathbf{A}) = f\,\nabla\cdot\mathbf{A} + \mathbf{A}\cdot\nabla f$;
3. $\nabla\times(f\mathbf{A}) = f\,\nabla\times\mathbf{A} + \nabla f\times\mathbf{A}$;
4. $\nabla\cdot(\mathbf{A}\times\mathbf{B}) = \mathbf{B}\cdot(\nabla\times\mathbf{A}) - \mathbf{A}\cdot(\nabla\times\mathbf{B})$;
5. $\nabla\times(\mathbf{A}\times\mathbf{B}) = \mathbf{A}(\nabla\cdot\mathbf{B}) - \mathbf{B}(\nabla\cdot\mathbf{A}) + (\mathbf{B}\cdot\nabla)\mathbf{A} - (\mathbf{A}\cdot\nabla)\mathbf{B}$;
6. $\nabla(\mathbf{A}\cdot\mathbf{B}) = (\mathbf{A}\cdot\nabla)\mathbf{B} + (\mathbf{B}\cdot\nabla)\mathbf{A} + \mathbf{A}\times(\nabla\times\mathbf{B}) + \mathbf{B}\times(\nabla\times\mathbf{A})$.

If they are C^2:

7. $\nabla\times(\nabla f) = \mathbf{0}$;
8. $\nabla\cdot(\nabla\times\mathbf{A}) = 0$;
9. $\nabla\times(\nabla\times\mathbf{A}) = \nabla(\nabla\cdot\mathbf{A}) - \nabla^2\mathbf{A}$, where $\nabla^2\mathbf{A}$ is the vector whose Cartesian components are $\nabla^2 A_i$.

Here $(\mathbf{B}\cdot\nabla)\mathbf{A}$ has components $B_j\partial_j A_i$.

Proof. Every step is the ordinary product rule $\partial_j(uv) = (\partial_j u)v + u\,\partial_j v$ for scalar components, plus the algebra of ε and δ.

1. $\partial_i(fg) = f\,\partial_i g + g\,\partial_i f$.
2. $\partial_i(fA_i) = f\,\partial_i A_i + A_i\,\partial_i f$.
3. $\varepsilon_{ijk}\partial_j(fA_k) = f\,\varepsilon_{ijk}\partial_j A_k + \varepsilon_{ijk}(\partial_j f)A_k$; the second term is $(\nabla f\times\mathbf{A})_i$.
4. $\partial_i(\varepsilon_{ijk}A_jB_k) = \varepsilon_{ijk}(\partial_i A_j)B_k + \varepsilon_{ijk}A_j(\partial_i B_k)$. By cyclicity $\varepsilon_{ijk} = \varepsilon_{kij}$, the first term is $B_k\,\varepsilon_{kij}\partial_i A_j = \mathbf{B}\cdot(\nabla\times\mathbf{A})$. By one swap $\varepsilon_{ijk} = -\varepsilon_{jik}$, the second is $-A_j\,\varepsilon_{jik}\partial_i B_k = -\mathbf{A}\cdot(\nabla\times\mathbf{B})$.
5. $[\nabla\times(\mathbf{A}\times\mathbf{B})]_i = \varepsilon_{ijk}\partial_j(\varepsilon_{kmn}A_mB_n) = \varepsilon_{kij}\varepsilon_{kmn}\partial_j(A_mB_n) = (\delta_{im}\delta_{jn} - \delta_{in}\delta_{jm})\big[(\partial_j A_m)B_n + A_m\partial_j B_n\big]$. The four resulting terms are $B_j\partial_j A_i + A_i\partial_j B_j - B_i\partial_j A_j - A_j\partial_j B_i$, which is the i-th component of the right side.
6. Compute $[\mathbf{A}\times(\nabla\times\mathbf{B})]_i = \varepsilon_{kij}\varepsilon_{kmn}A_j\partial_m B_n = A_j\partial_i B_j - A_j\partial_j B_i$ by Lemma B.7. Hence $A_j\partial_i B_j = [\mathbf{A}\times(\nabla\times\mathbf{B})]_i + [(\mathbf{A}\cdot\nabla)\mathbf{B}]_i$. Add the same identity with $\mathbf{A}\leftrightarrow\mathbf{B}$; the left sides add to $\partial_i(A_jB_j)$.
7. $(\nabla\times\nabla f)_i = \varepsilon_{ijk}\partial_j\partial_k f = 0$, because $\partial_j\partial_k f$ is symmetric in j, k (Clairaut, using C^2) and ε_{ijk} annihilates symmetric objects.
8. $\partial_i(\varepsilon_{ijk}\partial_j A_k) = \varepsilon_{ijk}\,\partial_i\partial_j A_k = 0$, for the same reason with the symmetric pair (i, j).
9. $[\nabla\times(\nabla\times\mathbf{A})]_i = \varepsilon_{ijk}\partial_j(\varepsilon_{kmn}\partial_m A_n) = \varepsilon_{kij}\varepsilon_{kmn}\partial_j\partial_m A_n = (\delta_{im}\delta_{jn} - \delta_{in}\delta_{jm})\partial_j\partial_m A_n = \partial_i(\partial_j A_j) - \partial_j\partial_j A_i$. ∎

All nine were also checked symbolically for arbitrary component functions.

Remarks.

- *Smoothness matters for 7–9.* They rest on Clairaut's theorem. At a singularity they can fail in the distributional sense: the polar angle ϕ of B.6 has $\nabla\phi = \hat{\boldsymbol{\phi}}/r$, whose curl is zero for $r > 0$ but equals $2\pi\delta(x)\delta(y)\hat{\mathbf{z}}$ as a distribution (the circulation 2π concentrated on the axis). Likewise $\nabla^2(1/R) = -4\pi\delta^3(\mathbf{R})$.
- *Identity 9 in curvilinear coordinates.* Only in Cartesian coordinates is $\nabla^2\mathbf{A}$ obtained by applying the scalar Laplacian to each component, because only there are the unit vectors constant. In cylindrical or spherical coordinates the **vector Laplacian is defined** by identity 9, $\nabla^2\mathbf{A} := \nabla(\nabla\cdot\mathbf{A}) - \nabla\times(\nabla\times\mathbf{A})$, evaluated with the formulas of B.8; e.g. $(\nabla^2\mathbf{A})_r = \nabla^2 A_r - A_r/r^2 - (2/r^2)\partial_\phi A_\phi$, not $\nabla^2 A_r$.
- *Identity 5 is not symmetric in appearance* but is antisymmetric under $\mathbf{A}\leftrightarrow\mathbf{B}$, as it must be: a quick check catches sign errors.
- *Used in:* the wave equation (identity 9, B.9); the necessity of the displacement current (identity 8, B.9); Poynting's theorem, $\nabla\cdot(\mathbf{E}\times\mathbf{H}) = \mathbf{H}\cdot(\nabla\times\mathbf{E}) - \mathbf{E}\cdot(\nabla\times\mathbf{H})$ (identity 4, §7.12); Green's identities (identity 2, B.4); potentials $\mathbf{B} = \nabla\times\mathbf{A}$, $\mathbf{E} = -\nabla V - \partial_t\mathbf{A}$ (identities 7–8).

B.8 Gradient, divergence, curl and Laplacian in curvilinear coordinates

The cylindrical and spherical formulas tabulated in §7.5 and used throughout Ch. 13 are often "derived" by drawing a small curved box and estimating fluxes. Here they are derived exactly, for every orthogonal coordinate system at once, using only the chain rule and the identities of B.7. The only geometric input is the set of three scale factors.

Orthogonal curvilinear coordinates and scale factors

Let $\mathbf{x} = \mathbf{x}(u_1, u_2, u_3)$ be a one-to-one C^2 map from an open set of (u_1, u_2, u_3)-space onto an open set $\Omega \subset \mathbb{R}^3$, with non-zero Jacobian determinant. The coordinates are **orthogonal** if the three tangent vectors $\partial \mathbf{x}/\partial u_i$ are mutually perpendicular at every point. The **scale factors** and **unit vectors** are

$$h_i = \left\| \frac{\partial \mathbf{x}}{\partial u_i} \right\|, \qquad \hat{\mathbf{u}}_i = \frac{1}{h_i}\frac{\partial \mathbf{x}}{\partial u_i} \qquad (i = 1, 2, 3, \text{ no sum}),$$

and we assume the system is right-handed: $\hat{\mathbf{u}}_1 \times \hat{\mathbf{u}}_2 = \hat{\mathbf{u}}_3$ (and cyclically). A field is written $\mathbf{F} = F_1\hat{\mathbf{u}}_1 + F_2\hat{\mathbf{u}}_2 + F_3\hat{\mathbf{u}}_3$, and ∂_i now means $\partial/\partial u_i$.

For cylindrical coordinates $(u_1, u_2, u_3) = (r, \phi, z)$, $\mathbf{x} = (r\cos\phi, r\sin\phi, z)$:

$$\frac{\partial \mathbf{x}}{\partial r} = (\cos\phi, \sin\phi, 0), \quad \frac{\partial \mathbf{x}}{\partial \phi} = r(-\sin\phi, \cos\phi, 0), \quad \frac{\partial \mathbf{x}}{\partial z} = (0, 0, 1), \qquad (h_r, h_\phi, h_z) = (1,\ r,\ 1).$$

For spherical coordinates $(u_1, u_2, u_3) = (R, \theta, \phi)$, $\mathbf{x} = (R\sin\theta\cos\phi,\ R\sin\theta\sin\phi,\ R\cos\theta)$:

$$\frac{\partial \mathbf{x}}{\partial R} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta), \quad \frac{\partial \mathbf{x}}{\partial \theta} = R(\cos\theta\cos\phi, \cos\theta\sin\phi, -\sin\theta), \quad \frac{\partial \mathbf{x}}{\partial \phi} = R\sin\theta(-\sin\phi, \cos\phi, 0),$$

so $(h_R, h_\theta, h_\phi) = (1,\ R,\ R\sin\theta)$. Pairwise dot products are zero by inspection, and both systems are right-handed ($\hat{\mathbf{r}} \times \hat{\boldsymbol{\phi}} = \hat{\mathbf{z}}$, $\hat{\mathbf{R}} \times \hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\phi}}$). The maps are one-to-one with non-zero Jacobian away from the z-axis ($r = 0$, respectively $\sin\theta = 0$); the formulas below hold there.

Lemma B.8 — Gradients of the coordinate functions

Regard u_i as functions of $\mathbf{x}$ (via the inverse map, which is C^2 by the inverse function theorem). Then

$$\nabla u_i = \frac{\hat{\mathbf{u}}_i}{h_i}, \qquad \text{and consequently} \qquad \hat{\mathbf{u}}_1 = h_2 h_3\, \nabla u_2 \times \nabla u_3, \quad \hat{\mathbf{u}}_1 = h_1 \nabla u_1 \quad \text{(and cyclically)}.$$

Proof. The Jacobian matrix of $\mathbf{x}(\mathbf{u})$ has columns $h_i\hat{\mathbf{u}}_i$, so it factors as $\mathbf{Q}\,\mathrm{diag}(h_1, h_2, h_3)$ with $\mathbf{Q} = [\hat{\mathbf{u}}_1\ \hat{\mathbf{u}}_2\ \hat{\mathbf{u}}_3]$ orthogonal. The Jacobian of the inverse map $\mathbf{u}(\mathbf{x})$ is the inverse matrix, $\mathrm{diag}(1/h_1, 1/h_2, 1/h_3)\,\mathbf{Q}^{\mathsf{T}}$, whose i-th row is $\hat{\mathbf{u}}_i^{\mathsf{T}}/h_i$. But the i-th row of the Jacobian of $\mathbf{u}(\mathbf{x})$ is $(\nabla u_i)^{\mathsf{T}}$. The consequences follow from $\nabla u_2 \times \nabla u_3 = \hat{\mathbf{u}}_2 \times \hat{\mathbf{u}}_3/(h_2h_3) = \hat{\mathbf{u}}_1/(h_2h_3)$. ∎

Theorem B.8 — Differential operators in orthogonal coordinates

For a C^1 scalar f and C^1 field $\mathbf{F} = \sum F_i\hat{\mathbf{u}}_i$ (and C^2 f for the Laplacian), with $h = h_1h_2h_3$:

$$\begin{aligned}
\nabla f &= \frac{1}{h_1}\frac{\partial f}{\partial u_1}\hat{\mathbf{u}}_1 + \frac{1}{h_2}\frac{\partial f}{\partial u_2}\hat{\mathbf{u}}_2 + \frac{1}{h_3}\frac{\partial f}{\partial u_3}\hat{\mathbf{u}}_3, \\
\nabla \cdot \mathbf{F} &= \frac{1}{h}\left[\frac{\partial}{\partial u_1}(h_2h_3F_1) + \frac{\partial}{\partial u_2}(h_3h_1F_2) + \frac{\partial}{\partial u_3}(h_1h_2F_3)\right], \\
\nabla \times \mathbf{F} &= \frac{1}{h}\det\begin{bmatrix} h_1\hat{\mathbf{u}}_1 & h_2\hat{\mathbf{u}}_2 & h_3\hat{\mathbf{u}}_3 \\ \partial_1 & \partial_2 & \partial_3 \\ h_1F_1 & h_2F_2 & h_3F_3 \end{bmatrix}, \\
\nabla^2 f &= \frac{1}{h}\left[\frac{\partial}{\partial u_1}\left(\frac{h_2h_3}{h_1}\frac{\partial f}{\partial u_1}\right) + \frac{\partial}{\partial u_2}\left(\frac{h_3h_1}{h_2}\frac{\partial f}{\partial u_2}\right) + \frac{\partial}{\partial u_3}\left(\frac{h_1h_2}{h_3}\frac{\partial f}{\partial u_3}\right)\right].
\end{aligned}$$

Proof.

Gradient. By the chain rule, $\partial f/\partial x_j = \sum_i (\partial f/\partial u_i)(\partial u_i/\partial x_j)$, i.e. $\nabla f = \sum_i \partial_i f\, \nabla u_i$. Insert Lemma B.8.

Divergence. It suffices (by linearity) to treat $F_1\hat{\mathbf{u}}_1$. Write it as $(h_2h_3F_1)\,(\nabla u_2 \times \nabla u_3)$ using Lemma B.8. By identity 4 of B.7 and identity 7 (the u_i are C^2),

$$\nabla \cdot (\nabla u_2 \times \nabla u_3) = \nabla u_3 \cdot (\nabla \times \nabla u_2) - \nabla u_2 \cdot (\nabla \times \nabla u_3) = 0.$$

Then identity 2 of B.7 gives

$$\nabla \cdot (F_1\hat{\mathbf{u}}_1) = \nabla(h_2h_3F_1) \cdot (\nabla u_2 \times \nabla u_3) = \nabla(h_2h_3F_1) \cdot \frac{\hat{\mathbf{u}}_1}{h_2h_3} = \frac{1}{h_1h_2h_3}\frac{\partial}{\partial u_1}(h_2h_3F_1),$$

the last step by the gradient formula (only the $\hat{\mathbf{u}}_1$-component survives the dot product). The other two terms follow cyclically.

Curl. Again treat $F_1\hat{\mathbf{u}}_1 = (h_1F_1)\nabla u_1$. By identity 3 of B.7 and identity 7,

$$\nabla \times \big((h_1F_1)\nabla u_1\big) = \nabla(h_1F_1) \times \nabla u_1 = \left[\sum_k \frac{1}{h_k}\partial_k(h_1F_1)\hat{\mathbf{u}}_k\right] \times \frac{\hat{\mathbf{u}}_1}{h_1} = \frac{1}{h_3h_1}\partial_3(h_1F_1)\,\hat{\mathbf{u}}_2 - \frac{1}{h_1h_2}\partial_2(h_1F_1)\,\hat{\mathbf{u}}_3,$$

using $\hat{\mathbf{u}}_3 \times \hat{\mathbf{u}}_1 = \hat{\mathbf{u}}_2$ and $\hat{\mathbf{u}}_2 \times \hat{\mathbf{u}}_1 = -\hat{\mathbf{u}}_3$. Adding the cyclic analogues for $F_2\hat{\mathbf{u}}_2$ and $F_3\hat{\mathbf{u}}_3$ and collecting the coefficient of each unit vector gives, for example,

$$(\nabla \times \mathbf{F})_1 = \frac{1}{h_2h_3}\Big[\partial_2(h_3F_3) - \partial_3(h_2F_2)\Big],$$

which is exactly the expansion of the stated determinant along its first row.

Laplacian. $\nabla^2 f = \nabla \cdot (\nabla f)$; apply the divergence formula to $F_i = \frac{1}{h_i}\partial_i f$. ∎

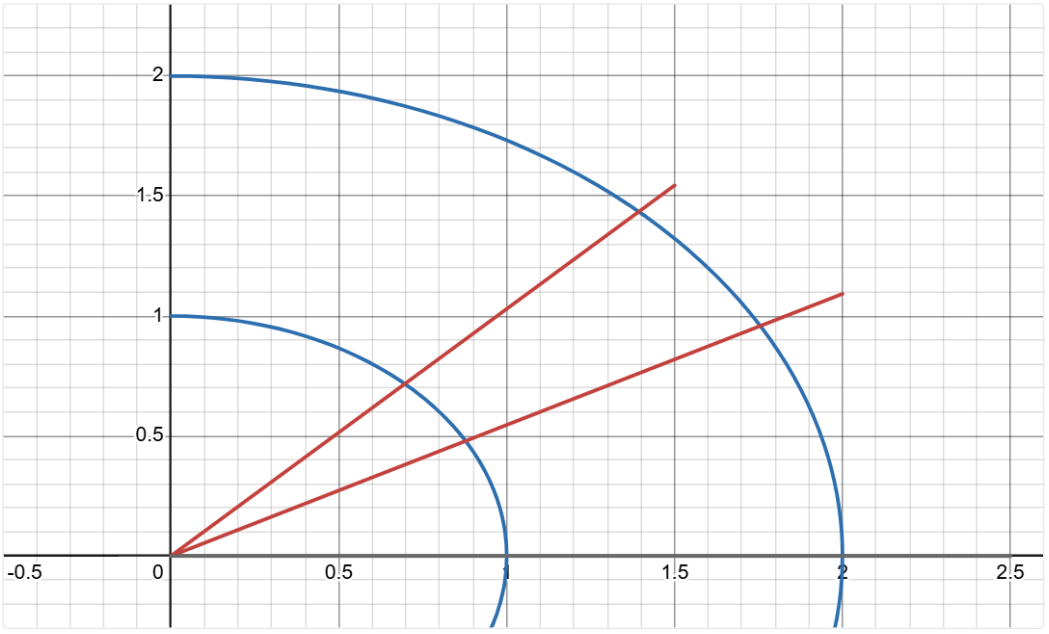
Specialising (all formulas were verified symbolically against the Cartesian operators on test fields):

Cylindrical coordinates (r,ϕ,z), $h=(1,r,1)$

$$\nabla f=\frac{\partial f}{\partial r}\hat{\mathbf{r}}+\frac{1}{r}\frac{\partial f}{\partial \phi}\hat{\boldsymbol{\phi}}+\frac{\partial f}{\partial z}\hat{\mathbf{z}},\qquad \nabla\cdot\mathbf{F}=\frac{1}{r}\frac{\partial}{\partial r}(rF_r)+\frac{1}{r}\frac{\partial F_\phi}{\partial \phi}+\frac{\partial F_z}{\partial z},$$
$$\nabla\times\mathbf{F}=\left(\frac{1}{r}\frac{\partial F_z}{\partial \phi}-\frac{\partial F_\phi}{\partial z}\right)\hat{\mathbf{r}}+\left(\frac{\partial F_r}{\partial z}-\frac{\partial F_z}{\partial r}\right)\hat{\boldsymbol{\phi}}+\frac{1}{r}\left(\frac{\partial}{\partial r}(rF_\phi)-\frac{\partial F_r}{\partial \phi}\right)\hat{\mathbf{z}},$$
$$\nabla^2 f=\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial f}{\partial r}\right)+\frac{1}{r^2}\frac{\partial^2 f}{\partial \phi^2}+\frac{\partial^2 f}{\partial z^2}.$$

Spherical coordinates (R,θ,ϕ), $h=(1,R,R\sin\theta)$

$$\nabla f=\frac{\partial f}{\partial R}\hat{\mathbf{R}}+\frac{1}{R}\frac{\partial f}{\partial \theta}\hat{\boldsymbol{\theta}}+\frac{1}{R\sin\theta}\frac{\partial f}{\partial \phi}\hat{\boldsymbol{\phi}},$$
$$\nabla\cdot\mathbf{F}=\frac{1}{R^2}\frac{\partial}{\partial R}(R^2F_R)+\frac{1}{R\sin\theta}\frac{\partial}{\partial \theta}(\sin\theta\,F_\theta)+\frac{1}{R\sin\theta}\frac{\partial F_\phi}{\partial \phi},$$
$$\nabla\times\mathbf{F}=\frac{1}{R\sin\theta}\left(\frac{\partial}{\partial \theta}(\sin\theta\,F_\phi)-\frac{\partial F_\theta}{\partial \phi}\right)\hat{\mathbf{R}}+\frac{1}{R}\left(\frac{1}{\sin\theta}\frac{\partial F_R}{\partial \phi}-\frac{\partial}{\partial R}(RF_\phi)\right)\hat{\boldsymbol{\theta}}+\frac{1}{R}\left(\frac{\partial}{\partial R}(RF_\theta)-\frac{\partial F_R}{\partial \theta}\right)\hat{\boldsymbol{\phi}},$$
$$\nabla^2 f=\frac{1}{R^2}\frac{\partial}{\partial R}\left(R^2\frac{\partial f}{\partial R}\right)+\frac{1}{R^2\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta\frac{\partial f}{\partial \theta}\right)+\frac{1}{R^2\sin^2\theta}\frac{\partial^2 f}{\partial \phi^2}.$$



Cylindrical (polar) coordinate curves. The patch between $r=1,2$ and $\phi=0.5,0.8$ has radial sides of length Δr but arcs of length $r\,\Delta\phi$ — longer on the outer circle. That growth is the scale factor $h_\phi=r$, and it is why $\partial_r(rF_r)$, not $\partial_r F_r$, appears in the divergence.

Remarks.

- *Orthogonality is essential.* For non-orthogonal coordinates ∇u_i is no longer parallel to $\partial\mathbf{x}/\partial u_i$ and the formulas need the full metric tensor $g_{ij}=\partial_i\mathbf{x}\cdot\partial_j\mathbf{x}$; e.g. $\nabla\cdot\mathbf{F}=\frac{1}{\sqrt{g}}\partial_i(\sqrt{g}\,F^i)$ with contravariant components. All coordinate systems in this book are orthogonal.
- *Check by a known field.* For the Coulomb field $F_R=1/R^2$: $\nabla\cdot\mathbf{F}=\frac{1}{R^2}\partial_R(1)=0$ for $R>0$, consistent with B.4. For $f=1/R$: $\nabla^2 f=\frac{1}{R^2}\partial_R(R^2\cdot(-R^{-2}))=0$. For the wire field $F_\phi=1/r$: $(\nabla\times\mathbf{F})_z=\frac{1}{r}\partial_r(r\cdot\frac{1}{r})=0$, consistent with B.6.
- *A pitfall.* Do not differentiate unit vectors as if constant: $\nabla\cdot\hat{\mathbf{r}}=\frac{1}{r}\partial_r(r)=\frac{1}{r}\neq 0$, even though "$F_r=1$ is constant". The scale factors encode exactly the variation of $\hat{\mathbf{u}}_i$ that a naive componentwise derivative misses.
- *Axis singularities.* On $r=0$ (or $\sin\theta=0$) the coordinates fail to be a valid chart; smooth Cartesian fields still have finite operators there, and the apparently singular terms such as $\frac{1}{r^2}\partial_\phi^2 f$ combine into finite limits.
- *Used in:* separation of variables for Laplace's equation in cylinders and spheres (Bessel and Legendre equations, §13.6); radiation and waveguide fields (§7.12); the gradient in polar coordinates (§6.4).

B.9 Maxwell's equations: from integral to differential form, continuity, and waves

§7.12 derives the differential form of Maxwell's equations in a few lines. This section supplies the two analytical facts those lines rely on — localisation of an integral identity, and differentiation of a flux under the integral sign — and states the smoothness hypotheses under which the differential form, the continuity equation and the wave equation follow.

Lemma B.9a — Localisation

Let $\Omega \subset \mathbb{R}^3$ be open.

1. If g is continuous on Ω and $\iiint_B g\,dV = 0$ for every closed ball $B \subset \Omega$, then $g \equiv 0$ on Ω.
2. If $\mathbf{G}$ is continuous on Ω and $\iint_S \mathbf{G}\cdot d\mathbf{S} = 0$ for every flat disc $S \subset \Omega$ (of every orientation), then $\mathbf{G} \equiv \mathbf{0}$ on Ω.

Proof. (1) Suppose $g(\mathbf{a}) > 0$ for some $\mathbf{a}$ (the case < 0 is symmetric). By continuity there is $\delta > 0$ with $g > g(\mathbf{a})/2$ on the ball $B_\delta(\mathbf{a}) \subset \Omega$, so $\iiint_{B_\delta} g\,dV \geq \frac{g(\mathbf{a})}{2}\cdot\frac{4}{3}\pi\delta^3 > 0$, a contradiction. (2) Suppose $\mathbf{G}(\mathbf{a}) \neq \mathbf{0}$ and put $\hat{\mathbf{n}} = \mathbf{G}(\mathbf{a})/\|\mathbf{G}(\mathbf{a})\|$. The continuous function $\mathbf{G}\cdot\hat{\mathbf{n}}$ equals $\|\mathbf{G}(\mathbf{a})\| > 0$ at $\mathbf{a}$, so it exceeds half that on a ball around $\mathbf{a}$; the flat disc through $\mathbf{a}$ with normal $\hat{\mathbf{n}}$ and small radius δ lies in that ball and has flux at least $\tfrac12\|\mathbf{G}(\mathbf{a})\|\pi\delta^2 > 0$, a contradiction. ∎

Lemma B.9b — Differentiating a flux through a fixed surface

Let $S = \mathbf{r}(D)$ be a fixed (time-independent) smooth parametrised surface and $\mathbf{B}(\mathbf{x},t)$ a field such that $\mathbf{B}$ and $\partial\mathbf{B}/\partial t$ are continuous on $S \times I$ for an open time interval I. Then

$$\frac{d}{dt}\iint_S \mathbf{B}\cdot d\mathbf{S} = \iint_S \frac{\partial \mathbf{B}}{\partial t}\cdot d\mathbf{S}.$$

The same holds for volume integrals $\frac{d}{dt}\iiint_V \rho\,dV = \iiint_V \partial_t\rho\,dV$ over a fixed V.

Proof. In parameters the flux is $\Phi(t) = \iint_D \mathbf{B}(\mathbf{r}(u,v),t)\cdot\mathbf{n}(u,v)\,du\,dv$ with $\mathbf{n} = \mathbf{r}_u \times \mathbf{r}_v$ independent of t. The integrand and its t-derivative $\partial_t\mathbf{B}(\mathbf{r},t)\cdot\mathbf{n}$ are continuous on the compact set $D \times [t_0-\eta, t_0+\eta]$, hence uniformly continuous. By the mean value theorem, $\big[\Phi(t_0+\tau)-\Phi(t_0)\big]/\tau = \iint_D \partial_t\mathbf{B}(\mathbf{r},t_0+\vartheta\tau)\cdot\mathbf{n}\,du\,dv$ for some $\vartheta = \vartheta(u,v) \in (0,1)$, and uniform continuity makes this converge to $\iint_D \partial_t\mathbf{B}(\mathbf{r},t_0)\cdot\mathbf{n}\,du\,dv$ as $\tau \to 0$. The volume case is identical. ∎

Theorem B.9 — Differential form of Maxwell's equations

Let $\Omega \subset \mathbb{R}^3$ be open and I an open time interval. Suppose $\mathbf{D},\mathbf{B},\mathbf{E},\mathbf{H}$ are C^1 in $(\mathbf{x},t)$ on $\Omega \times I$ and $\rho, \mathbf{J}$ are continuous. Then the integral laws

$$\oint_{\partial V}\mathbf{D}\cdot d\mathbf{S} = \iiint_V \rho\,dV, \qquad \oint_{\partial V}\mathbf{B}\cdot d\mathbf{S} = 0,$$
$$\oint_{\partial S}\mathbf{E}\cdot d\mathbf{l} = -\frac{d}{dt}\iint_S \mathbf{B}\cdot d\mathbf{S}, \qquad \oint_{\partial S}\mathbf{H}\cdot d\mathbf{l} = \iint_S \mathbf{J}\cdot d\mathbf{S} + \frac{d}{dt}\iint_S \mathbf{D}\cdot d\mathbf{S},$$

holding for every solid $V \subset \Omega$ to which B.4 applies and every fixed oriented surface $S \subset \Omega$ to which B.5 applies, at every $t \in I$, are **equivalent** to the differential laws

$$\nabla\cdot\mathbf{D} = \rho, \qquad \nabla\cdot\mathbf{B} = 0, \qquad \nabla\times\mathbf{E} = -\frac{\partial\mathbf{B}}{\partial t}, \qquad \nabla\times\mathbf{H} = \mathbf{J} + \frac{\partial\mathbf{D}}{\partial t} \qquad \text{on } \Omega\times I.$$

Proof.

Integral ⇒ differential. (Gauss.) By B.4, $\oint_{\partial V}\mathbf{D}\cdot d\mathbf{S} = \iiint_V \nabla\cdot\mathbf{D}\,dV$, so $\iiint_V(\nabla\cdot\mathbf{D}-\rho)\,dV = 0$ for every ball $V \subset \Omega$. The integrand is continuous ($\mathbf{D}$ is C^1, ρ continuous), so Lemma B.9a(1) gives $\nabla\cdot\mathbf{D} = \rho$. The same argument with $\rho = 0$ gives $\nabla\cdot\mathbf{B} = 0$.

(Faraday.) For a fixed surface, Lemma B.9b moves the time derivative inside, and B.5 converts the circulation:

$$\iint_S\left(\nabla\times\mathbf{E} + \frac{\partial\mathbf{B}}{\partial t}\right)\cdot d\mathbf{S} = 0 \qquad \text{for every flat disc } S \subset \Omega.$$

The bracket is continuous, so Lemma B.9a(2) gives $\nabla\times\mathbf{E} = -\partial_t\mathbf{B}$. (Ampère–Maxwell.) Identically, $\iint_S(\nabla\times\mathbf{H} - \mathbf{J} - \partial_t\mathbf{D})\cdot d\mathbf{S} = 0$ for every disc, whence $\nabla\times\mathbf{H} = \mathbf{J} + \partial_t\mathbf{D}$.

Differential ⇒ integral. Integrate each differential law over V or S and apply B.4, B.5 and Lemma B.9b in the reverse direction. ∎

Corollary B.9c — The continuity equation

(a) If charge is conserved in the integral sense, $\frac{d}{dt}\iiint_V \rho\,dV = -\oint_{\partial V}\mathbf{J}\cdot d\mathbf{S}$ for every fixed $V \subset \Omega$, with ρ of class C^1 and $\mathbf{J}$ of class C^1, then

$$\frac{\partial\rho}{\partial t} + \nabla\cdot\mathbf{J} = 0.$$

(b) Conversely, the continuity equation is a **consequence** of Maxwell's equations whenever $\mathbf{H}$ and $\mathbf{D}$ are C^2 and $\mathbf{J}$ is C^1.

Proof. (a) Lemma B.9b and B.4 give $\iiint_V(\partial_t\rho + \nabla\cdot\mathbf{J})\,dV = 0$ for every ball; apply Lemma B.9a(1). (b) Take the divergence of the Ampère–Maxwell law. By identity 8 of B.7, $\nabla\cdot(\nabla\times\mathbf{H}) = 0$, so $0 = \nabla\cdot\mathbf{J} + \nabla\cdot\partial_t\mathbf{D}$. Because $\mathbf{D}$ is C^2, the space and time derivatives commute (Clairaut), $\nabla\cdot\partial_t\mathbf{D} = \partial_t(\nabla\cdot\mathbf{D}) = \partial_t\rho$ by Gauss's law. ∎

Part (b) is the precise form of Maxwell's argument for the displacement current (§7.12): without the term $\partial_t\mathbf{D}$ the same computation would force $\nabla\cdot\mathbf{J} = 0$, contradicting charge accumulation on a capacitor plate. In the same way the divergence of Faraday's law gives $\partial_t(\nabla\cdot\mathbf{B}) = 0$, so $\nabla\cdot\mathbf{B} = 0$ is preserved in time once it holds initially: the two divergence equations are constraints compatible with the two curl equations.

Corollary B.9d — The wave equation in vacuum

In a region with $\rho = 0$, $\mathbf{J} = \mathbf{0}$, $\mathbf{D} = \varepsilon_0 \mathbf{E}$, $\mathbf{B} = \mu_0 \mathbf{H}$, and fields of class C^2,

$$\nabla^2 \mathbf{E} = \mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}, \qquad \nabla^2 \mathbf{B} = \mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2}, \qquad c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}.$$

Proof. Take the curl of Faraday's law. Since $\mathbf{B}$ is C^2, $\nabla \times \partial_t \mathbf{B} = \partial_t (\nabla \times \mathbf{B})$, and by Ampère–Maxwell $\nabla \times \mathbf{B} = \mu_0 \nabla \times \mathbf{H} = \mu_0 \varepsilon_0 \, \partial_t \mathbf{E}$. Hence

$$\nabla \times (\nabla \times \mathbf{E}) = -\partial_t (\nabla \times \mathbf{B}) = -\mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}.$$

By identity 9 of B.7 (in Cartesian components), $\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} = -\nabla^2 \mathbf{E}$, since $\nabla \cdot \mathbf{E} = \rho / \varepsilon_0 = 0$. This is the first equation. For $\mathbf{B}$, take the curl of Ampère–Maxwell: $\nabla \times (\nabla \times \mathbf{B}) = \mu_0 \varepsilon_0 \partial_t (\nabla \times \mathbf{E}) = -\mu_0 \varepsilon_0 \partial_t^2 \mathbf{B}$, and $\nabla \cdot \mathbf{B} = 0$. $\blacksquare$

Numbers and a check. With $\mu_0 = 4\pi \times 10^{-7}$ H/m and $\varepsilon_0 = 8.8541878 \times 10^{-12}$ F/m, $1/\sqrt{\mu_0 \varepsilon_0} = 2.99792458 \times 10^8$ m/s (computed), the measured speed of light. Dimensionally, $[\mu_0 \varepsilon_0] = (\mathrm{H/m})(\mathrm{F/m}) = \mathrm{s}^2/\mathrm{m}^2$ because $[LC] = \mathrm{s}^2$, so $1/\sqrt{\mu_0 \varepsilon_0}$ is a speed. The plane wave $\mathbf{E} = E_0 \cos(\omega t - kz)\, \hat{\mathbf{x}}$ satisfies the equation exactly when $k^2 = \mu_0 \varepsilon_0 \omega^2$, i.e. $\omega / k = c$; it also satisfies $\nabla \cdot \mathbf{E} = \partial_x E_x = 0$, and Faraday's law then gives $\mathbf{B} = (E_0 / c) \cos(\omega t - kz)\, \hat{\mathbf{y}}$ — transverse and in phase, as solved in §13.5.

Remarks.

- *Why the integral form is more fundamental.* The integral laws make sense for fields that are discontinuous across material interfaces, sheets of charge or surface currents, where the differential form fails (the derivatives do not exist). Applied to thin pillboxes and loops straddling an interface they give the boundary conditions $\hat{\mathbf{n}} \cdot (\mathbf{D}_1 - \mathbf{D}_2) = \rho_s$, $\hat{\mathbf{n}} \times (\mathbf{E}_1 - \mathbf{E}_2) = \mathbf{0}$ (§7.12). The equivalence in Theorem B.9 holds precisely in regions where the C^1 hypothesis is met.
- *Fixed versus moving surfaces.* Lemma B.9b requires S to be stationary. For a moving loop, $\frac{d}{dt} \iint_{S(t)} \mathbf{B} \cdot d\mathbf{S}$ picks up an extra term $-\oint_{\partial S} (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{l}$ (using $\nabla \cdot \mathbf{B} = 0$); this is the motional EMF of generators. The differential law $\nabla \times \mathbf{E} = -\partial_t \mathbf{B}$ is unaffected.
- *Is the wave equation all of Maxwell?* No. Every solution of Maxwell's equations in vacuum solves the wave equation, but not conversely: a solution of $\nabla^2 \mathbf{E} = c^{-2} \partial_t^2 \mathbf{E}$ must also satisfy $\nabla \cdot \mathbf{E} = 0$ (transversality), and $\mathbf{B}$ is then determined by Faraday's law up to a static field. Longitudinal solutions such as $\cos(\omega t - kz) \hat{\mathbf{z}}$ satisfy the wave equation but are excluded by Gauss's law.
- *Used in:* the wave equation and Helmholtz equation (§7.12, §13.5); Poisson's equation $\nabla^2 V = -\rho/\varepsilon$ from $\mathbf{E} = -\nabla V$, $\nabla \cdot \mathbf{D} = \rho$ (§13.6, §13.8); the phasor form $\partial_t \to j\omega$ used with the Fourier transform (Ch. 16).

Further reading

- J. E. Marsden and A. Tromba, *Vector Calculus*, 6th ed., W. H. Freeman — proofs of Green, Gauss and Stokes for elementary regions, in the style used here.
- T. M. Apostol, *Calculus*, Vol. II, 2nd ed., Wiley — change of variables and the integral theorems with full rigour.
- M. Spivak, *Calculus on Manifolds*, Benjamin — the general Stokes theorem for differential forms, of which B.3–B.5 are special cases, and the Poincaré lemma.
- D. K. Cheng, *Field and Wave Electromagnetics*, 2nd ed., Addison-Wesley — curvilinear operators via scale factors and Maxwell's equations in the conventions of this book.
- H. M. Schey, *Div, Grad, Curl, and All That*, 4th ed., Norton — an informal companion that motivates the coordinate-free definitions.

Appendix C — Linear Algebra

This appendix gives complete proofs of the central theorems of Part III (Ch. 8, Ch. 9, Ch. 10), which the chapters state with only sketches, together with the two results from numerical analysis and dynamics that lean on them most heavily: the condition-number error bound (§8.10, Ch. 22) and the eigenvalue test for stability of $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ (§10.9, §12.7).

Prerequisites. Matrix algebra, row reduction and echelon form (Ch. 8); vector spaces, span, independence, basis and dimension (Ch. 9), in particular the facts that every independent set in a finite-dimensional space can be extended to a basis and that any n independent vectors in an n-dimensional space form a basis. The fundamental theorem of algebra (every non-constant complex polynomial has a complex root) is used without proof.

Rigor. Hypotheses are stated exactly and every step is justified. Scalars are $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. Where a result holds for complex matrices with T replaced by H, we say so; proofs are written in whichever setting is the more natural. The sections are ordered so that each proof uses only earlier ones.

C.1 Determinants: the product rule and invertibility

Chapter 8 defines $\det \mathbf{A}$ by cofactor expansion and notes that it equals the Leibniz sum. Here we take the Leibniz sum as the working definition, because the proofs are cleanest from it.

Permutations and their sign

A **permutation** of $\{1, \dots, n\}$ is a bijection σ of that set to itself; the set of all $n!$ of them is S_n. A **transposition** swaps two elements and fixes the rest. Every permutation is a product of transpositions, and although the factorisation is not unique, the **parity** of the number of transpositions is; $\operatorname{sgn}\sigma = +1$ for an even number, -1 for an odd number. Hence $\operatorname{sgn}(\sigma\tau) = \operatorname{sgn}\sigma \, \operatorname{sgn}\tau$ and $\operatorname{sgn}\sigma^{-1} = \operatorname{sgn}\sigma$.

Determinant (Leibniz form)

For $\mathbf{A} = [a_{ij}] \in \mathbb{F}^{n\times n}$,

$$\det \mathbf{A} = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma)\, a_{1\sigma(1)} a_{2\sigma(2)} \cdots a_{n\sigma(n)}.$$

We regard det as a function $D(\mathbf{r}_1, \dots, \mathbf{r}_n)$ of the rows $\mathbf{r}_i$ of $\mathbf{A}$.

Multilinear, alternating

A function f of n row vectors in $\mathbb{F}^n$ is **multilinear** if it is linear in each row when the others are held fixed, and **alternating** if $f = 0$ whenever two rows are equal.

Theorem C.1 — Product rule and invertibility

Let $\mathbf{A}, \mathbf{B} \in \mathbb{F}^{n\times n}$. Then

1. $\det(\mathbf{AB}) = \det \mathbf{A} \, \det \mathbf{B}$;
2. $\det \mathbf{A}^{\mathsf{T}} = \det \mathbf{A}$;
3. $\mathbf{A}$ is invertible (there is $\mathbf{X}$ with $\mathbf{AX} = \mathbf{XA} = \mathbf{I}$) if and only if $\det \mathbf{A} \neq 0$, and then $\det \mathbf{A}^{-1} = 1/\det \mathbf{A}$.

Proof

Step 1: det is multilinear, alternating, and $\det \mathbf{I} = 1$. Each Leibniz term contains exactly one factor $a_{i\sigma(i)}$ from row i, so each term — and hence the sum — is linear in row i. For $\mathbf{I}$, the only non-zero term is $\sigma = \mathrm{id}$, giving 1. For the alternating property, suppose rows $p \neq q$ are equal and let τ be the transposition swapping p and q. The map $\sigma \mapsto \sigma\tau$ pairs the permutations into pairs $\{\sigma, \sigma\tau\}$ (it is its own inverse and has no fixed points). In the pair, the terms are

$$\operatorname{sgn}(\sigma) \prod_i a_{i\sigma(i)} \quad \text{and} \quad \operatorname{sgn}(\sigma\tau) \prod_i a_{i\sigma\tau(i)} = -\operatorname{sgn}(\sigma) \prod_i a_{i\sigma\tau(i)}.$$

The two products differ only in the factors for $i = p, q$: the first has $a_{p\sigma(p)} a_{q\sigma(q)}$, the second $a_{p\sigma(q)} a_{q\sigma(p)}$. Since rows p and q are equal, $a_{pk} = a_{qk}$ for all k, so the products are equal and the pair cancels. Hence $\det \mathbf{A} = 0$.

Step 2: alternating implies antisymmetric. If f is multilinear and alternating, swapping two rows negates f. Indeed, put $\mathbf{u} + \mathbf{v}$ in both positions p and q (other rows fixed) and expand by multilinearity:

$$0 = f(\dots, \mathbf{u}+\mathbf{v}, \dots, \mathbf{u}+\mathbf{v}, \dots) = f(\dots,\mathbf{u},\dots,\mathbf{u},\dots) + f(\dots,\mathbf{u},\dots,\mathbf{v},\dots) + f(\dots,\mathbf{v},\dots,\mathbf{u},\dots) + f(\dots,\mathbf{v},\dots,\mathbf{v},\dots).$$

The first and last terms vanish (equal rows), so $f(\dots,\mathbf{u},\dots,\mathbf{v},\dots) = -f(\dots,\mathbf{v},\dots,\mathbf{u},\dots)$.

Step 3: uniqueness lemma. *If f is multilinear and alternating in the rows of an $n \times n$ matrix, then $f(\mathbf{A}) = f(\mathbf{I}) \det \mathbf{A}$ for every $\mathbf{A}$.* Write row i as $\mathbf{r}_i = \sum_k a_{ik}\mathbf{e}_k^{\mathsf{T}}$ and expand f by multilinearity in all n rows:

$$f(\mathbf{A}) = \sum_{k_1=1}^{n} \cdots \sum_{k_n=1}^{n} a_{1k_1} \cdots a_{nk_n}\, f(\mathbf{e}_{k_1}^{\mathsf{T}}, \dots, \mathbf{e}_{k_n}^{\mathsf{T}}).$$

If two indices $k_p = k_q$ coincide, the corresponding f value has two equal rows and vanishes. The surviving index tuples are exactly those with $k_i = \sigma(i)$ for a permutation σ. For such a tuple, the rows $\mathbf{e}_{\sigma(1)}^{\mathsf{T}}, \dots, \mathbf{e}_{\sigma(n)}^{\mathsf{T}}$ can be sorted into $\mathbf{I}$ by a sequence of row swaps whose number has the parity of σ; by Step 2 each swap flips the sign, so $f(\mathbf{e}_{\sigma(1)}^{\mathsf{T}}, \dots, \mathbf{e}_{\sigma(n)}^{\mathsf{T}}) = \operatorname{sgn}(\sigma) f(\mathbf{I})$. Therefore

$$f(\mathbf{A}) = f(\mathbf{I}) \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{1\sigma(1)} \cdots a_{n\sigma(n)} = f(\mathbf{I}) \det \mathbf{A}.$$

Step 4: product rule. Fix $\mathbf{B}$ and define $f(\mathbf{A}) = \det(\mathbf{AB})$. Row i of $\mathbf{AB}$ is $\mathbf{r}_i\mathbf{B}$, a linear function of $\mathbf{r}_i$ alone. Hence f is multilinear in the rows of $\mathbf{A}$ (composition of the linear map $\mathbf{r}_i \mapsto \mathbf{r}_i\mathbf{B}$ with the multilinear det), and alternating (equal rows of $\mathbf{A}$ give equal rows of $\mathbf{AB}$). By Step 3, $\det(\mathbf{AB}) = f(\mathbf{I}) \det \mathbf{A} = \det \mathbf{B} \det \mathbf{A}$. This proves (1).

Step 5: transpose. $\det \mathbf{A}^{\mathsf{T}} = \sum_\sigma \operatorname{sgn}\sigma \prod_i a_{\sigma(i)i}$. Re-index each product by $k = \sigma(i)$, i.e. $i = \sigma^{-1}(k)$: $\prod_i a_{\sigma(i)i} = \prod_k a_{k\sigma^{-1}(k)}$. As σ runs over S_n so does $\pi = \sigma^{-1}$, with $\operatorname{sgn}\pi = \operatorname{sgn}\sigma$. So $\det \mathbf{A}^{\mathsf{T}} = \sum_\pi \operatorname{sgn}\pi \prod_k a_{k\pi(k)} = \det \mathbf{A}$. This proves (2). In particular det is also multilinear and alternating in the **columns**.

Step 6: invertible $\Rightarrow \det \neq 0$. If $\mathbf{AX} = \mathbf{I}$ then by (1), $\det \mathbf{A} \det \mathbf{X} = \det \mathbf{I} = 1$, so $\det \mathbf{A} \neq 0$ and $\det \mathbf{A}^{-1} = 1/\det \mathbf{A}$.

Step 7: $\det \neq 0 \Rightarrow$ invertible. First, *if the columns $\mathbf{a}_1, \dots, \mathbf{a}_n$ of $\mathbf{A}$ are dependent, then* $\det \mathbf{A} = 0$. Some column is a combination of the others, say $\mathbf{a}_k = \sum_{j \neq k} c_j \mathbf{a}_j$. By column multilinearity,

$$\det \mathbf{A} = \sum_{j \neq k} c_j \det[\mathbf{a}_1, \dots, \underbrace{\mathbf{a}_j}_{\text{position } k}, \dots, \mathbf{a}_n],$$

and each determinant on the right has column $\mathbf{a}_j$ in two positions, so vanishes. Contrapositively, $\det \mathbf{A} \neq 0$ implies the n columns are independent, hence a basis of $\mathbb{F}^n$ (Ch. 9). So each $\mathbf{e}_i$ is a combination $\mathbf{A}\mathbf{x}_i$, and $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_n]$ satisfies $\mathbf{AX} = \mathbf{I}$. By Step 6, $\det \mathbf{X} = 1/\det \mathbf{A} \neq 0$, so by the same argument $\mathbf{X}$ has a right inverse $\mathbf{Y}$, $\mathbf{XY} = \mathbf{I}$. Then

$$\mathbf{A} = \mathbf{A}(\mathbf{XY}) = (\mathbf{AX})\mathbf{Y} = \mathbf{Y},$$

so $\mathbf{XA} = \mathbf{XY} = \mathbf{I}$ as well: $\mathbf{X}$ is a two-sided inverse. This proves (3). ■

Remarks

- **What the proof really shows.** Step 3 says the determinant is the *only* multilinear alternating function with $f(\mathbf{I}) = 1$. That characterisation, not the $n!$-term formula, is what makes the determinant useful: it is why row replacement leaves det unchanged, row swaps negate it, and $\det \mathbf{A} = (-1)^s u_{11} \cdots u_{nn}$ from elimination (§8.3). Cofactor expansion along row 1 is also multilinear, alternating and equals 1 at $\mathbf{I}$, which is the cleanest proof that the two definitions in Ch. 8 agree.
- **Square matrices only.** For rectangular $\mathbf{A}$ ($m \times n$) and $\mathbf{B}$ ($n \times m$) with $m < n$, $\mathbf{AB}$ can be invertible though neither factor has a determinant; the replacement is the Cauchy–Binet formula. For $m > n$, $\det(\mathbf{AB}) = 0$ always, since $\operatorname{rank}(\mathbf{AB}) \leq n < m$.
- **Sum rule fails.** $\det(\mathbf{A} + \mathbf{B}) \neq \det \mathbf{A} + \det \mathbf{B}$ in general: $\mathbf{A} = \mathbf{B} = \mathbf{I}_2$ gives $4 \neq 2$.
- **Step 7 is the theoretical half of the invertible matrix theorem** (§8.4): it also shows that a one-sided inverse of a square matrix is automatically two-sided. Numerically, though, $\det \mathbf{A}$ is a poor measure of "nearly singular" — $\det(10^{-1}\mathbf{I}_{30}) = 10^{-30}$ for a perfectly conditioned matrix — which is why §C.11 uses the condition number instead.
- **Uses.** The characteristic polynomial $\det(\lambda\mathbf{I} - \mathbf{A})$ and the fact that λ is an eigenvalue iff $\mathbf{A} - \lambda\mathbf{I}$ is singular (§10.1) rest on part 3; similarity invariance $\det(\mathbf{S}^{-1}\mathbf{AS}) = \det \mathbf{A}$ and $\det \mathbf{A} = \prod \lambda_i$ rest on part 1; Cramer's rule and the Jacobian change of variables (Ch. 6) use both.

C.2 The rank–nullity theorem

Kernel, range, rank, nullity

For a linear map $T : V \to W$, the **kernel** (null space) is $\ker T = \{\mathbf{v} \in V : T\mathbf{v} = \mathbf{0}\}$ and the **range** is $\operatorname{range} T = \{T\mathbf{v} : \mathbf{v} \in V\}$. Both are subspaces. Their dimensions are the **nullity** and the **rank**. For a matrix $\mathbf{A} \in \mathbb{F}^{m \times n}$ acting by $\mathbf{x} \mapsto \mathbf{Ax}$, the kernel is $\operatorname{null}(\mathbf{A})$ and the range is the column space $\operatorname{col}(\mathbf{A})$.

Theorem C.2 — Rank–nullity

Let V be finite-dimensional and $T : V \to W$ linear. Then

$$\dim \operatorname{range} T + \dim \ker T = \dim V.$$

For $\mathbf{A} \in \mathbb{F}^{m \times n}$: $\operatorname{rank} \mathbf{A} + \dim \operatorname{null} \mathbf{A} = n$. Moreover the row rank equals the column rank, $\dim \operatorname{row} \mathbf{A} = \dim \operatorname{col} \mathbf{A}$, and consequently $\dim \operatorname{null} \mathbf{A}^{\mathsf{T}} = m - \operatorname{rank} \mathbf{A}$.

Proof

Rank–nullity. Let $n = \dim V$. The kernel is a subspace of V, hence finite-dimensional; choose a basis $\mathbf{k}_1, \dots, \mathbf{k}_q$ of it ($q = 0$ if $\ker T = \{\mathbf{0}\}$). An independent set extends to a basis of V, so there are $\mathbf{v}_1, \dots, \mathbf{v}_s$ with $\mathbf{k}_1, \dots, \mathbf{k}_q, \mathbf{v}_1, \dots, \mathbf{v}_s$ a basis of V; thus $q + s = n$. We claim that $T\mathbf{v}_1, \dots, T\mathbf{v}_s$ is a basis of $\operatorname{range} T$, which gives $\dim \operatorname{range} T = s = n - q$.

Spanning. Any $\mathbf{w} \in \operatorname{range} T$ is $T\mathbf{v}$ for some $\mathbf{v} = \sum_i a_i \mathbf{k}_i + \sum_j b_j \mathbf{v}_j$. By linearity and $T\mathbf{k}_i = \mathbf{0}$, $\mathbf{w} = \sum_j b_j T\mathbf{v}_j$.

Independence. Suppose $\sum_j c_j T\mathbf{v}_j = \mathbf{0}$. Then $T\big(\sum_j c_j \mathbf{v}_j\big) = \mathbf{0}$, so $\sum_j c_j \mathbf{v}_j \in \ker T$ and equals some $\sum_i d_i \mathbf{k}_i$. The relation $\sum_j c_j \mathbf{v}_j - \sum_i d_i \mathbf{k}_i = \mathbf{0}$ among basis vectors of V forces all $c_j = 0$ (and all $d_i = 0$).

Row rank equals column rank. Let $r = \dim \operatorname{col} \mathbf{A}$ and let $\mathbf{c}_1, \dots, \mathbf{c}_r \in \mathbb{F}^m$ be a basis of $\operatorname{col} \mathbf{A}$; put $\mathbf{C} = [\mathbf{c}_1, \dots, \mathbf{c}_r]$ ($m \times r$). Each column $\mathbf{a}_j$ of $\mathbf{A}$ is a combination $\mathbf{a}_j = \mathbf{C}\mathbf{f}_j$; collecting the coefficient vectors into the $r \times n$ matrix $\mathbf{F} = [\mathbf{f}_1, \dots, \mathbf{f}_n]$ gives the **rank factorisation**

$$\mathbf{A} = \mathbf{C}\mathbf{F}.$$

Read row-wise, row i of $\mathbf{A}$ equals $\sum_{k=1}^{r} c_{ik}$ (row k of $\mathbf{F}$): every row of $\mathbf{A}$ lies in the span of the r rows of $\mathbf{F}$. Hence $\dim \operatorname{row} \mathbf{A} \le r = \dim \operatorname{col} \mathbf{A}$. Applying this inequality to $\mathbf{A}^{\mathsf{T}}$, whose rows are the columns of $\mathbf{A}$, gives the reverse inequality. So the two are equal.

Left null space. Apply the matrix form of rank–nullity to $\mathbf{A}^{\mathsf{T}} \in \mathbb{F}^{n \times m}$, which has m columns and rank $\dim \operatorname{col} \mathbf{A}^{\mathsf{T}} = \dim \operatorname{row} \mathbf{A} = r$: $\dim \operatorname{null} \mathbf{A}^{\mathsf{T}} = m - r$. ∎

Remarks

- **Finite dimension of V is essential; of W it is not.** On the infinite-dimensional space of polynomials, differentiation D has a one-dimensional kernel (the constants) and is onto, so the identity would read $\infty = \infty + 1$ and carries no information. Worse, a map with zero kernel need not be onto there: multiplication by t is one-to-one but misses the constants. For square matrices, by contrast, rank–nullity gives "one-to-one $\iff$ onto", the backbone of the invertible matrix theorem (§8.4).
- **Nothing depends on row reduction.** The chapter proof (§9.6) counts pivot and free columns of the reduced echelon form; the proof above shows the result is a statement about the map, independent of any algorithm or basis.
- **Uses.** Existence and uniqueness for $\mathbf{A}\mathbf{x} = \mathbf{b}$ (§8.6): the solution set is $\mathbf{x}_p + \operatorname{null} \mathbf{A}$, an affine set of dimension $n - r$. In circuit theory, for a connected graph with N nodes and B branches the reduced incidence matrix has rank $N - 1$, so the loop-current space has dimension $B - N + 1$, the number of independent meshes (§8.5). Sections C.6 and C.10 need $\operatorname{rank}(\mathbf{A}^{\mathsf{T}}\mathbf{A}) = \operatorname{rank} \mathbf{A}$, which follows from rank–nullity once the null spaces are shown equal.

C.3 Existence and uniqueness of the LU factorisation

Leading principal submatrices and minors

For $\mathbf{A} \in \mathbb{F}^{n \times n}$, $\mathbf{A}_k = \mathbf{A}_{1:k,1:k}$ is the **leading principal submatrix** of order k, and $\Delta_k = \det \mathbf{A}_k$ the **leading principal minor**; set $\Delta_0 = 1$. A **unit lower triangular** matrix is lower triangular with ones on the diagonal. A **permutation matrix** is $\mathbf{I}$ with its rows reordered.

Three facts about triangular matrices are used throughout. Products and inverses of (unit) lower triangular matrices are (unit) lower triangular, and likewise for upper. A triangular matrix is invertible iff its diagonal entries are non-zero, its determinant being their product (§8.3). And the **leading-block rule**: if $\mathbf{L}$ is lower and $\mathbf{U}$ upper triangular then

$$(\mathbf{L}\mathbf{U})_k = \mathbf{L}_k \mathbf{U}_k,$$

because $(\mathbf{L}\mathbf{U})_{ij} = \sum_p \ell_{ip} u_{pj}$ and for $i, j \le k$ every non-zero term has $p \le \min(i, j) \le k$.

Theorem C.3 — LU without pivoting

Let $\mathbf{A} \in \mathbb{F}^{n \times n}$.

1. If $\Delta_1, \dots, \Delta_{n-1}$ are all non-zero, then $\mathbf{A} = \mathbf{L}\mathbf{U}$ with $\mathbf{L}$ unit lower triangular and $\mathbf{U}$ upper triangular, and this factorisation is **unique**. The pivots are $u_{kk} = \Delta_k / \Delta_{k-1}$.
2. Conversely, if $\mathbf{A}$ is nonsingular and has such a factorisation, then all $\Delta_k \neq 0$, $k = 1, \dots, n$.

Thus a nonsingular $\mathbf{A}$ has an LU factorisation (Gaussian elimination runs without row interchanges) if and only if every leading principal minor is non-zero.

Theorem C.4 — LU with pivoting

For every nonsingular $\mathbf{A} \in \mathbb{F}^{n \times n}$ there is a permutation matrix $\mathbf{P}$ such that $\mathbf{P}\mathbf{A} = \mathbf{L}\mathbf{U}$, with $\mathbf{L}$ unit lower triangular and $\mathbf{U}$ upper triangular and nonsingular. $\mathbf{P}$ can be chosen (partial pivoting) so that $|\ell_{ik}| \le 1$ for all i, k.

Proof of Theorem C.3

Existence, by induction on n. For $n = 1$, $[a] = [1][a]$. Suppose the result holds for order $n - 1$ and let $\Delta_1, \dots, \Delta_{n-1} \neq 0$. Partition

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_{n-1} & \mathbf{b} \\ \mathbf{c}^{\mathsf{T}} & d \end{bmatrix}.$$

The matrix $\mathbf{A}_{n-1}$ has leading minors $\Delta_1, \dots, \Delta_{n-2}$, all non-zero, so by hypothesis $\mathbf{A}_{n-1} = \mathbf{L}'\mathbf{U}'$. Since $\det \mathbf{U}' = \det \mathbf{L}' \det \mathbf{U}' = \Delta_{n-1} \neq 0$, $\mathbf{U}'$ is invertible, and $\mathbf{L}'$ is invertible as a unit triangular matrix. Seek

$$\mathbf{A} = \begin{bmatrix} \mathbf{L}' & \mathbf{0} \\ \boldsymbol{\ell}^{\mathsf{T}} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{U}' & \mathbf{u} \\ \mathbf{0}^{\mathsf{T}} & u_{nn} \end{bmatrix} = \begin{bmatrix} \mathbf{L}'\mathbf{U}' & \mathbf{L}'\mathbf{u} \\ \boldsymbol{\ell}^{\mathsf{T}}\mathbf{U}' & \boldsymbol{\ell}^{\mathsf{T}}\mathbf{u} + u_{nn} \end{bmatrix}.$$

Matching blocks requires

$$\mathbf{L}'\mathbf{u} = \mathbf{b}, \qquad \boldsymbol{\ell}^{\mathsf{T}}\mathbf{U}' = \mathbf{c}^{\mathsf{T}}, \qquad u_{nn} = d - \boldsymbol{\ell}^{\mathsf{T}}\mathbf{u},$$

which have the solutions $\mathbf{u} = \mathbf{L}'^{-1}\mathbf{b}$, $\boldsymbol{\ell}^{\mathsf{T}} = \mathbf{c}^{\mathsf{T}}\mathbf{U}'^{-1}$, $u_{nn} = d - \mathbf{c}^{\mathsf{T}}\mathbf{A}_{n-1}^{-1}\mathbf{b}$. The two factors are unit lower and upper triangular, completing the induction. Note that Δ_n is not needed: u_{nn} may be zero.

Uniqueness, by the same induction. Suppose $\mathbf{A} = \mathbf{L}\mathbf{U}$ is any such factorisation. By the leading-block rule $\mathbf{A}_{n-1} = \mathbf{L}_{n-1}\mathbf{U}_{n-1}$, so by uniqueness at order $n - 1$, $\mathbf{L}_{n-1} = \mathbf{L}'$ and $\mathbf{U}_{n-1} = \mathbf{U}'$. The remaining blocks must then satisfy the three displayed equations, each of which has exactly one solution because $\mathbf{L}'$ and $\mathbf{U}'$ are invertible. So $\mathbf{L}$, $\mathbf{U}$ are determined.

Pivots. By the leading-block rule, $\Delta_k = \det \mathbf{L}_k \det \mathbf{U}_k = u_{11} u_{22} \cdots u_{kk}$ for every k. Dividing consecutive minors gives $u_{kk} = \Delta_k / \Delta_{k-1}$.

Converse. If $\mathbf{A} = \mathbf{L}\mathbf{U}$ is nonsingular then $0 \neq \det \mathbf{A} = \prod_{i=1}^{n} u_{ii}$, so every $u_{ii} \neq 0$ and $\Delta_k = \prod_{i \le k} u_{ii} \neq 0$ for every k. ∎

Proof of Theorem C.4

Induction on n; $n = 1$ is trivial with $\mathbf{P} = [1]$. Let $\mathbf{A}$ be nonsingular of order n. Its first column is non-zero (otherwise $\det \mathbf{A} = 0$ by column linearity), so choose a row p with $|a_{p1}| = \max_i |a_{i1}| > 0$ and let $\mathbf{P}_1$ swap rows 1 and p. Write

$$\mathbf{P}_1 \mathbf{A} = \begin{bmatrix} \alpha & \mathbf{w}^{\mathsf{T}} \\ \mathbf{v} & \mathbf{A}' \end{bmatrix} = \begin{bmatrix} 1 & \mathbf{0}^{\mathsf{T}} \\ \mathbf{v}/\alpha & \mathbf{I} \end{bmatrix} \begin{bmatrix} \alpha & \mathbf{w}^{\mathsf{T}} \\ \mathbf{0} & \mathbf{S} \end{bmatrix}, \qquad \mathbf{S} = \mathbf{A}' - \frac{\mathbf{v}\mathbf{w}^{\mathsf{T}}}{\alpha},$$

which is verified by multiplying out. The entries of $\mathbf{v}/\alpha$ have modulus at most 1 by the choice of pivot. Taking determinants, $\pm \det \mathbf{A} = \alpha \det \mathbf{S}$, so the **Schur complement** $\mathbf{S}$, of order $n-1$, is nonsingular. By the induction hypothesis $\mathbf{P}_2 \mathbf{S} = \mathbf{L}_2 \mathbf{U}_2$ with multipliers of modulus at most 1. Put

$$\mathbf{P} = \begin{bmatrix} 1 & \mathbf{0}^{\mathsf{T}} \\ \mathbf{0} & \mathbf{P}_2 \end{bmatrix} \mathbf{P}_1, \qquad \mathbf{L} = \begin{bmatrix} 1 & \mathbf{0}^{\mathsf{T}} \\ \mathbf{P}_2 \mathbf{v}/\alpha & \mathbf{L}_2 \end{bmatrix}, \qquad \mathbf{U} = \begin{bmatrix} \alpha & \mathbf{w}^{\mathsf{T}} \\ \mathbf{0} & \mathbf{U}_2 \end{bmatrix}.$$

Then

$$\mathbf{P}\mathbf{A} = \begin{bmatrix} 1 & \mathbf{0}^{\mathsf{T}} \\ \mathbf{0} & \mathbf{P}_2 \end{bmatrix} \begin{bmatrix} \alpha & \mathbf{w}^{\mathsf{T}} \\ \mathbf{v} & \mathbf{A}' \end{bmatrix} = \begin{bmatrix} \alpha & \mathbf{w}^{\mathsf{T}} \\ \mathbf{P}_2 \mathbf{v} & \mathbf{P}_2 \mathbf{A}' \end{bmatrix}, \qquad \mathbf{L}\mathbf{U} = \begin{bmatrix} \alpha & \mathbf{w}^{\mathsf{T}} \\ \mathbf{P}_2 \mathbf{v} & \mathbf{P}_2 \mathbf{v}\mathbf{w}^{\mathsf{T}}/\alpha + \mathbf{L}_2 \mathbf{U}_2 \end{bmatrix},$$

and the bottom-right blocks agree because $\mathbf{L}_2 \mathbf{U}_2 = \mathbf{P}_2 \mathbf{S} = \mathbf{P}_2 \mathbf{A}' - \mathbf{P}_2 \mathbf{v}\mathbf{w}^{\mathsf{T}}/\alpha$. $\mathbf{P}$ is a product of permutation matrices, hence a permutation matrix; $\mathbf{L}$ is unit lower triangular with all multipliers of modulus at most 1 (the permutation $\mathbf{P}_2$ only reorders the entries of $\mathbf{v}/\alpha$); and $\mathbf{U}$ is upper triangular with $\det \mathbf{U} = \alpha \det \mathbf{U}_2 \neq 0$. $\blacksquare$

Remarks

- **The hypothesis cannot be dropped.** $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ is nonsingular but $\Delta_1 = 0$, and it has no LU factorisation: $u_{11} = a_{11} = 0$ and $\ell_{21} u_{11} = a_{21} = 1$ are incompatible. One row swap fixes it, as Theorem C.4 guarantees.
- **Uniqueness needs $\Delta_1, \dots, \Delta_{n-1} \neq 0$.** $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \ell & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ for every ℓ. Without the unit-diagonal normalisation of $\mathbf{L}$ uniqueness also fails, since $(\mathbf{L}\mathbf{D}^{-1})(\mathbf{D}\mathbf{U})$ is another factorisation for any nonsingular diagonal $\mathbf{D}$.
- **Existence is not stability.** $\mathbf{A} = \begin{bmatrix} \varepsilon & 1 \\ 1 & 1 \end{bmatrix}$ with $0 < |\varepsilon| \ll 1$ has $\Delta_1 = \varepsilon \neq 0$, so LU without pivoting exists, but $\ell_{21} = 1/\varepsilon$ and $u_{22} = 1 - 1/\varepsilon$ are huge and rounding destroys the answer (see the pivoting discussion in §8.7). The bound $|\ell_{ik}| \le 1$ in Theorem C.4 keeps the computed factors representative of $\mathbf{A}$; growth in $\mathbf{U}$ can still occur in contrived cases but is rare in practice (§24.3).
- **A check.** For $\mathbf{A} = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{bmatrix}$: $\Delta_1 = 2$, $\Delta_2 = 2$, $\Delta_3 = 4$, so the pivots without pivoting are $2,\ 1,\ 2$ and $\det \mathbf{A} = 4$. Partial pivoting instead brings row 3 to the top and produces pivots $8,\ -0.75,\ -0.\overline{6}$ (product 4, since the row permutation used is a cyclic shift with determinant $+1$) and multipliers $0.25,\ 0.5,\ 0.\overline{6}$, all at most 1.
- **Uses.** Solving nodal equations (§8.9); the pivot formula $u_{kk} = \Delta_k / \Delta_{k-1}$ is the bridge to Sylvester's criterion in §C.9; the Schur complement $\mathbf{S}$ is exactly Kron reduction of a network (§8.4).

C.4 Cauchy–Schwarz and triangle inequalities

We use the book's convention (§9.8): an inner product on a vector space V over $\mathbb{F}$ is linear in its first argument, conjugate-symmetric, $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle^*$, and positive, $\langle \mathbf{u}, \mathbf{u} \rangle > 0$ for $\mathbf{u} \neq \mathbf{0}$. It follows that $\langle \mathbf{u}, a\mathbf{v} \rangle = a^* \langle \mathbf{u}, \mathbf{v} \rangle$. The induced norm is $\|\mathbf{u}\| = \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle}$. No finite-dimensionality is assumed in this section: the results hold verbatim for signals with $\langle x, y \rangle = \int x(t) y^*(t)\, dt$.

Theorem C.5 — Cauchy–Schwarz inequality

For all $\mathbf{u}, \mathbf{v}$ in an inner-product space,

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \le \|\mathbf{u}\| \, \|\mathbf{v}\|,$$

with equality if and only if $\mathbf{u}$ and $\mathbf{v}$ are linearly dependent (one is a scalar multiple of the other).

Theorem C.6 — Triangle inequality

For all $\mathbf{u}, \mathbf{v}$, $\|\mathbf{u} + \mathbf{v}\| \le \|\mathbf{u}\| + \|\mathbf{v}\|$, with equality if and only if one vector is a non-negative real multiple of the other. Consequently the induced norm is a norm, and $d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$ is a distance.

Proof of Theorem C.5

If $\mathbf{v} = \mathbf{0}$ both sides are 0 (since $\langle \mathbf{u}, \mathbf{0} \rangle = \langle \mathbf{u}, 0 \cdot \mathbf{0} \rangle = 0$) and the vectors are dependent, so the claim holds. Assume $\mathbf{v} \neq \mathbf{0}$. For any scalar s, positivity gives

$$0 \le \|\mathbf{u} - s\mathbf{v}\|^2 = \langle \mathbf{u} - s\mathbf{v}, \mathbf{u} - s\mathbf{v} \rangle = \|\mathbf{u}\|^2 - s^* \langle \mathbf{u}, \mathbf{v} \rangle - s \langle \mathbf{v}, \mathbf{u} \rangle + |s|^2 \|\mathbf{v}\|^2,$$

where we expanded using linearity in the first slot and conjugate-linearity in the second. Choose $s = \langle \mathbf{u}, \mathbf{v} \rangle / \|\mathbf{v}\|^2$, the coefficient of the projection of $\mathbf{u}$ onto $\mathbf{v}$. Writing $c = \langle \mathbf{u}, \mathbf{v} \rangle$, so that $\langle \mathbf{v}, \mathbf{u} \rangle = c^*$, the three last terms are

$$s^* c = \frac{c^* c}{\|\mathbf{v}\|^2}, \qquad s\, c^* = \frac{c\, c^*}{\|\mathbf{v}\|^2}, \qquad |s|^2 \|\mathbf{v}\|^2 = \frac{|c|^2}{\|\mathbf{v}\|^2},$$

so

$$0 \le \|\mathbf{u} - s\mathbf{v}\|^2 = \|\mathbf{u}\|^2 - \frac{|\langle \mathbf{u}, \mathbf{v}\rangle|^2}{\|\mathbf{v}\|^2}.$$

Multiplying by $\|\mathbf{v}\|^2 > 0$ and taking square roots gives the inequality.

Equality. With $\mathbf{v} \neq \mathbf{0}$, equality holds iff $\|\mathbf{u} - s\mathbf{v}\| = 0$, i.e. (by positivity) $\mathbf{u} = s\mathbf{v}$. Conversely, if $\mathbf{u} = a\mathbf{v}$ then $|\langle \mathbf{u}, \mathbf{v}\rangle| = |a|\|\mathbf{v}\|^2 = \|\mathbf{u}\|\|\mathbf{v}\|$. ■

Proof of Theorem C.6

Expand:

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \langle \mathbf{u}, \mathbf{v}\rangle + \langle \mathbf{v}, \mathbf{u}\rangle + \|\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + 2\,\mathrm{Re}\{\langle \mathbf{u}, \mathbf{v}\rangle\} + \|\mathbf{v}\|^2,$$

since $z + z^* = 2\,\mathrm{Re}\{z\}$. Now $\mathrm{Re}\{z\} \le |z|$ and Cauchy–Schwarz give

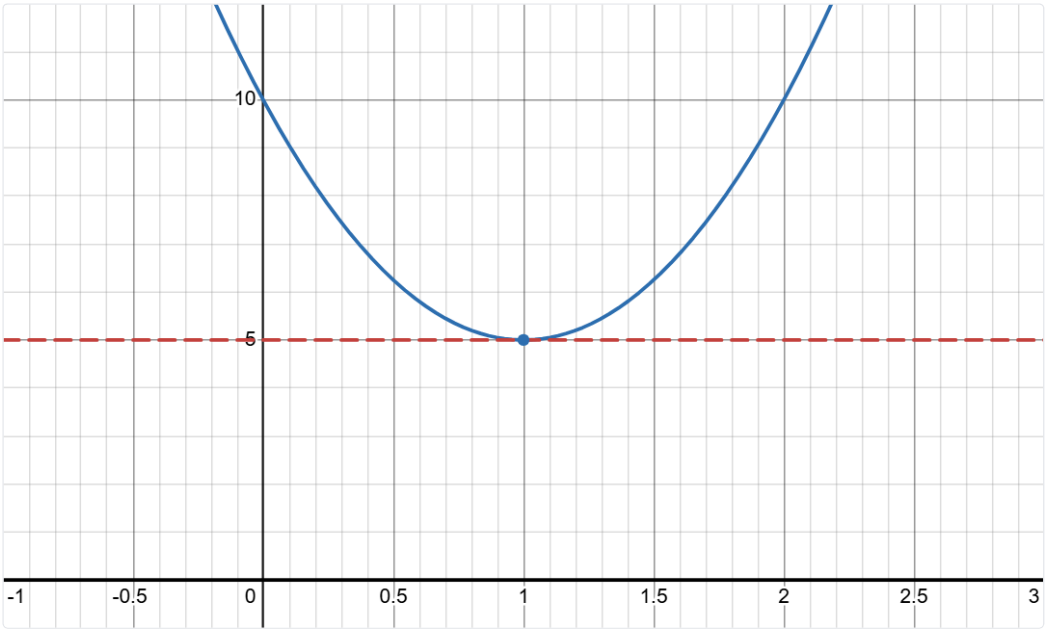
$$\|\mathbf{u} + \mathbf{v}\|^2 \le \|\mathbf{u}\|^2 + 2\|\mathbf{u}\|\|\mathbf{v}\| + \|\mathbf{v}\|^2 = \big(\|\mathbf{u}\| + \|\mathbf{v}\|\big)^2.$$

Take square roots (both sides are non-negative).

Equality requires equality in both steps: $\mathrm{Re}\{\langle \mathbf{u}, \mathbf{v}\rangle\} = |\langle \mathbf{u}, \mathbf{v}\rangle|$, i.e. $\langle \mathbf{u}, \mathbf{v}\rangle$ is real and ≥ 0, and dependence. If $\mathbf{v} \neq \mathbf{0}$ then $\mathbf{u} = a\mathbf{v}$ and $\langle \mathbf{u}, \mathbf{v}\rangle = a\|\mathbf{v}\|^2 \ge 0$ forces $a \ge 0$; the case $\mathbf{v} = \mathbf{0}$ is trivial. Conversely $\mathbf{u} = a\mathbf{v}$, $a \ge 0$, gives $\|(1 + a)\mathbf{v}\| = (1 + a)\|\mathbf{v}\|$.

Norm axioms. $\|\mathbf{u}\| \ge 0$ with equality only at $\mathbf{0}$ is positivity; $\|a\mathbf{u}\| = \sqrt{aa^*\langle \mathbf{u}, \mathbf{u}\rangle} = |a|\|\mathbf{u}\|$; the triangle inequality was just proved. For the distance, symmetry is $\|-\mathbf{w}\| = \|\mathbf{w}\|$ and the triangle inequality for d is Theorem C.6 applied to $(\mathbf{u} - \mathbf{w}) + (\mathbf{w} - \mathbf{v})$. ■

The proof is easiest to see in the real case, where $q(t) = \|\mathbf{u} - t\mathbf{v}\|^2 = \|\mathbf{u}\|^2 - 2t\langle \mathbf{u}, \mathbf{v}\rangle + t^2\|\mathbf{v}\|^2$ is a parabola that never goes below zero, so its discriminant $4\langle \mathbf{u}, \mathbf{v}\rangle^2 - 4\|\mathbf{u}\|^2\|\mathbf{v}\|^2$ is ≤ 0. For $\mathbf{u} = (3, 1)$, $\mathbf{v} = (1, 2)$: $\|\mathbf{u}\|^2 = 10$, $\langle \mathbf{u}, \mathbf{v}\rangle = 5$, $\|\mathbf{v}\|^2 = 5$, so $q(t) = 10 - 10t + 5t^2$ with minimum $q(1) = 5 = 10 - 5^2/5$.



The real Cauchy–Schwarz parabola $q(t) = \|\mathbf{u} - t\mathbf{v}\|^2$ for $\mathbf{u} = (3, 1)$, $\mathbf{v} = (1, 2)$: it stays above the axis, its minimum value is the squared distance from $\mathbf{u}$ to the line through $\mathbf{v}$, and it touches zero only when $\mathbf{u}$ is a multiple of $\mathbf{v}$.

Remarks

- **Positivity is the whole proof.** For an indefinite "inner product" such as the Minkowski form $\langle \mathbf{u}, \mathbf{v}\rangle = u_1v_1 - u_2v_2$ on $\mathbb{R}^2$, Cauchy–Schwarz fails: $\mathbf{u} = (1, 0)$, $\mathbf{v} = (1, 1)$ give $|\langle \mathbf{u}, \mathbf{v}\rangle| = 1$ but $\langle \mathbf{v}, \mathbf{v}\rangle = 0$. Positive *semi*-definiteness is enough for the inequality (the proof goes through after a limiting argument) but not for the equality case.
- **The conjugate is essential in the complex case.** Without it, $\langle (1, j), (1, j)\rangle = 1 + j^2 = 0$ and positivity fails (§9.8).
- **Uses.** Cauchy–Schwarz defines angles, $\cos\theta = \langle \mathbf{u}, \mathbf{v}\rangle/(\|\mathbf{u}\|\|\mathbf{v}\|) \in [-1, 1]$ (§9.7); it proves the **matched filter** is optimal, because the output SNR is proportional to $|\langle s, h\rangle|^2/\|h\|^2 \le \|s\|^2$ with equality iff $h \propto s$ (§9.8, Ch. 21); and it gives $|\rho_{XY}| \le 1$ for the correlation coefficient, with $\langle X, Y\rangle = E[XY]$ on zero-mean random variables (Ch. 19). The triangle inequality is what makes $\|\cdot\|$ usable as an error measure everywhere in Part VII.

C.5 Gram–Schmidt orthonormalisation and the QR factorisation

Theorem C.7 — Gram–Schmidt

Let $\mathbf{a}_1, \dots, \mathbf{a}_n$ be linearly independent vectors in an inner-product space. Define recursively, for $k = 1, \dots, n$,

$$\mathbf{w}_k = \mathbf{a}_k - \sum_{i=1}^{k-1} \langle \mathbf{a}_k, \mathbf{q}_i \rangle \, \mathbf{q}_i, \qquad \mathbf{q}_k = \frac{\mathbf{w}_k}{\|\mathbf{w}_k\|}.$$

Then every $\mathbf{w}_k \neq \mathbf{0}$ (so the division is legitimate), the vectors $\mathbf{q}_1, \dots, \mathbf{q}_n$ are orthonormal, and for every k

$$\operatorname{span}\{\mathbf{q}_1, \dots, \mathbf{q}_k\} = \operatorname{span}\{\mathbf{a}_1, \dots, \mathbf{a}_k\}.$$

In particular every finite-dimensional inner-product space has an orthonormal basis, and every orthonormal set in it extends to an orthonormal basis.

Theorem C.8 — QR factorisation

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ have linearly independent columns ($m \geq n$, $\operatorname{rank} \mathbf{A} = n$). Then $\mathbf{A} = \mathbf{Q}\mathbf{R}$ where $\mathbf{Q} \in \mathbb{R}^{m \times n}$ has orthonormal columns ($\mathbf{Q}^{\mathsf{T}}\mathbf{Q} = \mathbf{I}_n$) and $\mathbf{R} \in \mathbb{R}^{n \times n}$ is upper triangular with positive diagonal. This factorisation is unique. There is also a **full** QR factorisation $\mathbf{A} = \widetilde{\mathbf{Q}} \begin{bmatrix} \mathbf{R} \\ \mathbf{0} \end{bmatrix}$ with $\widetilde{\mathbf{Q}}$ $m \times m$ orthogonal.

(Complex case: $\mathsf{T} \to \mathsf{H}$, orthogonal $\to$ unitary.)

Proof of Theorem C.7

Induction on k, with the statement S_k: "$\mathbf{w}_1, \dots, \mathbf{w}_k \neq \mathbf{0}$, $\mathbf{q}_1, \dots, \mathbf{q}_k$ are orthonormal, and $\operatorname{span}\{\mathbf{q}_1, \dots, \mathbf{q}_j\} = \operatorname{span}\{\mathbf{a}_1, \dots, \mathbf{a}_j\}$ for $j \leq k$".

Base. $\mathbf{w}_1 = \mathbf{a}_1 \neq \mathbf{0}$, since an independent set contains no zero vector. $\mathbf{q}_1$ is a unit vector and spans the same line as $\mathbf{a}_1$.

Step. Assume S_{k-1}.

1. $\mathbf{w}_k \neq \mathbf{0}$. If $\mathbf{w}_k = \mathbf{0}$, then $\mathbf{a}_k = \sum_{i<k} \langle \mathbf{a}_k, \mathbf{q}_i \rangle \mathbf{q}_i \in \operatorname{span}\{\mathbf{q}_1, \dots, \mathbf{q}_{k-1}\} = \operatorname{span}\{\mathbf{a}_1, \dots, \mathbf{a}_{k-1}\}$, contradicting independence of the $\mathbf{a}$'s.
2. **Orthogonality.** For $l < k$, using linearity in the first slot and $\langle \mathbf{q}_i, \mathbf{q}_l \rangle = \delta_{il}$ for $i, l < k$,

$$\langle \mathbf{w}_k, \mathbf{q}_l \rangle = \langle \mathbf{a}_k, \mathbf{q}_l \rangle - \sum_{i<k} \langle \mathbf{a}_k, \mathbf{q}_i \rangle \langle \mathbf{q}_i, \mathbf{q}_l \rangle = \langle \mathbf{a}_k, \mathbf{q}_l \rangle - \langle \mathbf{a}_k, \mathbf{q}_l \rangle = 0.$$

Dividing by the positive real $\|\mathbf{w}_k\|$ keeps this, and makes $\|\mathbf{q}_k\| = 1$. 3. **Spans.** By the defining formula $\mathbf{q}_k$ is a combination of $\mathbf{a}_k$ and $\mathbf{q}_1, \dots, \mathbf{q}_{k-1}$, which lie in $\operatorname{span}\{\mathbf{a}_1, \dots, \mathbf{a}_{k-1}\}$; so $\operatorname{span}\{\mathbf{q}_1, \dots, \mathbf{q}_k\} \subseteq \operatorname{span}\{\mathbf{a}_1, \dots, \mathbf{a}_k\}$. Conversely $\mathbf{a}_k = \|\mathbf{w}_k\| \mathbf{q}_k + \sum_{i<k} \langle \mathbf{a}_k, \mathbf{q}_i \rangle \mathbf{q}_i$ lies in the left-hand span. Both spans are therefore equal.

This proves S_n. For the final assertions: apply the process to any basis of the space; and to extend an orthonormal set $\mathbf{q}_1, \dots, \mathbf{q}_p$, extend it to a basis $\mathbf{q}_1, \dots, \mathbf{q}_p, \mathbf{a}_{p+1}, \dots, \mathbf{a}_N$ and run the process — it leaves the first p vectors unchanged, because for orthonormal input $\mathbf{w}_k = \mathbf{q}_k$. $\blacksquare$

Proof of Theorem C.8

Existence. Run Gram–Schmidt on the columns $\mathbf{a}_1, \dots, \mathbf{a}_n$ with the standard inner product. Rearranging the definition of $\mathbf{w}_k$,

$$\mathbf{a}_k = \sum_{i=1}^{k-1} r_{ik} \mathbf{q}_i + r_{kk} \mathbf{q}_k, \qquad r_{ik} = \langle \mathbf{a}_k, \mathbf{q}_i \rangle = \mathbf{q}_i^{\mathsf{T}} \mathbf{a}_k \ (i < k), \qquad r_{kk} = \|\mathbf{w}_k\| > 0.$$

Setting $r_{ik} = 0$ for $i > k$, this says column k of $\mathbf{A}$ is column k of $\mathbf{Q}\mathbf{R}$, with $\mathbf{Q} = [\mathbf{q}_1, \dots, \mathbf{q}_n]$ and $\mathbf{R} = [r_{ik}]$ upper triangular with positive diagonal. Orthonormality of the $\mathbf{q}_i$ is $\mathbf{Q}^{\mathsf{T}}\mathbf{Q} = \mathbf{I}$.

Uniqueness. Suppose $\mathbf{Q}_1\mathbf{R}_1 = \mathbf{Q}_2\mathbf{R}_2$ are two such factorisations. Both $\mathbf{R}_i$ are invertible (positive diagonal). Then $\mathbf{Q}_1 = \mathbf{Q}_2\mathbf{M}$ with $\mathbf{M} = \mathbf{R}_2\mathbf{R}_1^{-1}$ upper triangular with positive diagonal. Also

$$\mathbf{I} = \mathbf{Q}_1^{\mathsf{T}}\mathbf{Q}_1 = \mathbf{M}^{\mathsf{T}}\mathbf{Q}_2^{\mathsf{T}}\mathbf{Q}_2\mathbf{M} = \mathbf{M}^{\mathsf{T}}\mathbf{M},$$

so $\mathbf{M}^{-1} = \mathbf{M}^{\mathsf{T}}$. The left side is upper triangular and the right side lower triangular, so $\mathbf{M}$ is diagonal; then $\mathbf{M}^2 = \mathbf{I}$ with positive diagonal forces $\mathbf{M} = \mathbf{I}$. Hence $\mathbf{R}_1 = \mathbf{R}_2$ and $\mathbf{Q}_1 = \mathbf{Q}_2$.

Full QR. Extend $\mathbf{q}_1, \dots, \mathbf{q}_n$ to an orthonormal basis $\mathbf{q}_1, \dots, \mathbf{q}_m$ of $\mathbb{R}^m$ (Theorem C.7) and put $\widetilde{\mathbf{Q}} = [\mathbf{q}_1, \dots, \mathbf{q}_m]$; the extra columns multiply the zero block. $\blacksquare$

Check. For $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$: $\mathbf{q}_1 = (1,1,0)/\sqrt{2}$, $r_{11} = \sqrt{2}$; $r_{12} = \langle \mathbf{a}_2, \mathbf{q}_1 \rangle = 1/\sqrt{2}$; $\mathbf{w}_2 = (1,0,1) - \tfrac{1}{2}(1,1,0) = (\tfrac{1}{2}, -\tfrac{1}{2}, 1)$, $r_{22} = \sqrt{3/2} = 1.2247$. A library QR returns the same $|r_{ik}|$, with signs flipped where it does not enforce a positive diagonal.

Remarks

- **Independence is necessary.** If $\mathbf{a}_k \in \operatorname{span}\{\mathbf{a}_1, \dots, \mathbf{a}_{k-1}\}$ the process produces $\mathbf{w}_k = \mathbf{0}$ and stops. A rank-deficient $\mathbf{A}$ still has a factorisation $\mathbf{A} = \mathbf{Q}\mathbf{R}$ (skip such columns, or use column pivoting, $\mathbf{A}\mathbf{\Pi} = \mathbf{Q}\mathbf{R}$), but $\mathbf{R}$ then has zero diagonal entries and uniqueness is lost.
- **Exact versus floating-point.** The theorem is about exact arithmetic. In floating point, classical Gram–Schmidt loses orthogonality roughly in proportion to $\kappa(\mathbf{A})^2$ times the unit round-off; the modified version (subtract each projection from the running vector) and Householder reflections behave far better (§9.10, §24.3). All three produce the same $\mathbf{Q}$ and $\mathbf{R}$ in exact arithmetic, by the uniqueness just proved.

- **Uses.** Orthogonal polynomials (Legendre, Chebyshev) are Gram–Schmidt applied to $1, t, t^2, \dots$ under an integral inner product (§9.10, §22.6). QR is the stable route to least squares (§C.6) and the engine of the QR eigenvalue algorithm. The extension statement is used in the induction step of the spectral theorem (§C.8) and in building $\mathbf{U}$ in the SVD (§C.10).

C.6 The orthogonal projection theorem and least squares

Orthogonal complement and projection

For a subspace W of an inner-product space V, $W^{\perp} = \{\mathbf{z} \in V : \langle \mathbf{z}, \mathbf{w} \rangle = 0 \text{ for all } \mathbf{w} \in W\}$. A vector $\mathbf{p} \in W$ with $\mathbf{b} - \mathbf{p} \in W^{\perp}$ is called an **orthogonal projection** of $\mathbf{b}$ onto W.

Theorem C.9 — Orthogonal projection theorem

Let W be a **finite-dimensional** subspace of an inner-product space V, and $\mathbf{b} \in V$. Then:

1. $\mathbf{b}$ has exactly one orthogonal projection $\mathbf{p}$ onto W; if $\mathbf{q}_1, \dots, \mathbf{q}_k$ is an orthonormal basis of W,
$$\mathbf{p} = \sum_{i=1}^{k} \langle \mathbf{b}, \mathbf{q}_i \rangle \, \mathbf{q}_i;$$
2. $\mathbf{p}$ is the unique closest point of W to $\mathbf{b}$: for every $\mathbf{w} \in W$,
$$\|\mathbf{b} - \mathbf{w}\|^2 = \|\mathbf{b} - \mathbf{p}\|^2 + \|\mathbf{p} - \mathbf{w}\|^2;$$
3. (Bessel) $\|\mathbf{p}\|^2 = \sum_i |\langle \mathbf{b}, \mathbf{q}_i \rangle|^2 \le \|\mathbf{b}\|^2$.

Theorem C.10 — Normal equations

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$. A vector $\hat{\mathbf{x}}$ minimises $J(\mathbf{x}) = \|\mathbf{b} - \mathbf{A}\mathbf{x}\|^2$ if and only if
$$\mathbf{A}^{\mathsf{T}}\mathbf{A}\,\hat{\mathbf{x}} = \mathbf{A}^{\mathsf{T}}\mathbf{b}.$$
The normal equations are always consistent, $\mathbf{A}\hat{\mathbf{x}}$ is the same vector (the projection of $\mathbf{b}$ onto $\operatorname{col}\mathbf{A}$) for every minimiser, and the minimiser is **unique if and only if** $\operatorname{rank}\mathbf{A} = n$ (independent columns), in which case $\hat{\mathbf{x}} = (\mathbf{A}^{\mathsf{T}}\mathbf{A})^{-1}\mathbf{A}^{\mathsf{T}}\mathbf{b} = \mathbf{R}^{-1}\mathbf{Q}^{\mathsf{T}}\mathbf{b}$ with $\mathbf{A} = \mathbf{Q}\mathbf{R}$. (Complex case: $\mathsf{T} \to \mathsf{H}$.)

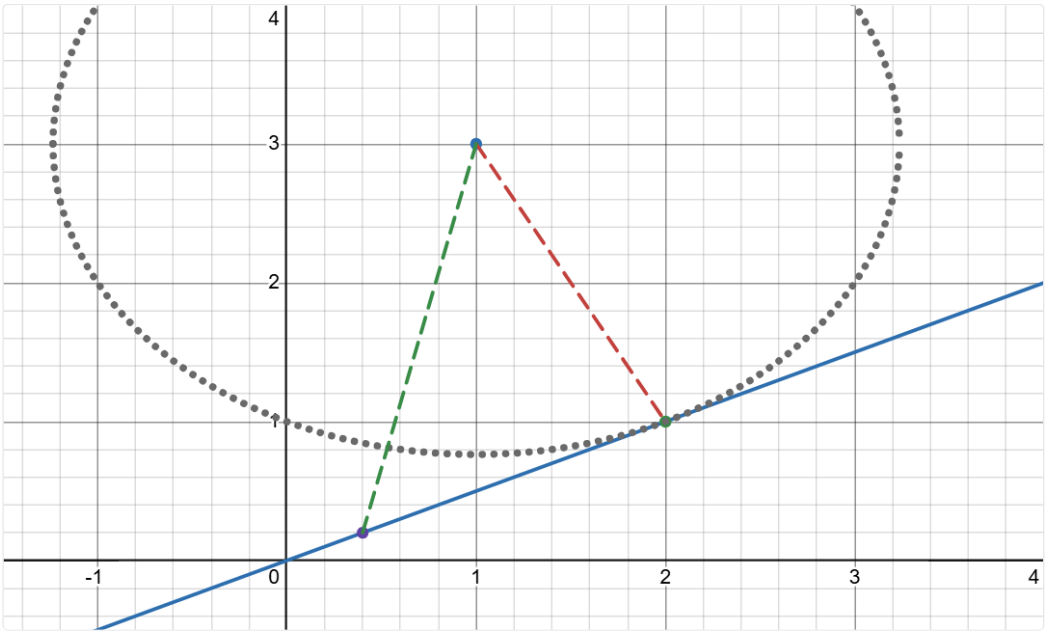
Proof of Theorem C.9

Existence. W has an orthonormal basis $\mathbf{q}_1, \dots, \mathbf{q}_k$ by Theorem C.7. Define $\mathbf{p}$ by the formula. It lies in W, and for each l
$$\langle \mathbf{b} - \mathbf{p}, \mathbf{q}_l \rangle = \langle \mathbf{b}, \mathbf{q}_l \rangle - \sum_i \langle \mathbf{b}, \mathbf{q}_i \rangle \langle \mathbf{q}_i, \mathbf{q}_l \rangle = 0.$$
An arbitrary $\mathbf{w} \in W$ is $\sum_l c_l \mathbf{q}_l$, and $\langle \mathbf{b} - \mathbf{p}, \mathbf{w} \rangle = \sum_l c_l^{*} \langle \mathbf{b} - \mathbf{p}, \mathbf{q}_l \rangle = 0$. So $\mathbf{b} - \mathbf{p} \in W^{\perp}$.

Uniqueness. If $\mathbf{p}, \mathbf{p}'$ are both orthogonal projections, then $\mathbf{p} - \mathbf{p}' = (\mathbf{b} - \mathbf{p}') - (\mathbf{b} - \mathbf{p})$ lies in W (difference of two elements of W) and in $W^{\perp}$ (difference of two elements of the subspace $W^{\perp}$). A vector orthogonal to itself has $\|\mathbf{p} - \mathbf{p}'\|^2 = 0$, so $\mathbf{p} = \mathbf{p}'$.

Best approximation. For $\mathbf{w} \in W$, write $\mathbf{b} - \mathbf{w} = (\mathbf{b} - \mathbf{p}) + (\mathbf{p} - \mathbf{w})$. The first summand is in $W^{\perp}$ and the second in W, so their inner products vanish and the expansion of $\|\mathbf{b} - \mathbf{w}\|^2$ (as in the proof of C.6) reduces to Pythagoras:
$$\|\mathbf{b} - \mathbf{w}\|^2 = \|\mathbf{b} - \mathbf{p}\|^2 + \|\mathbf{p} - \mathbf{w}\|^2 \;\ge\; \|\mathbf{b} - \mathbf{p}\|^2,$$
with equality iff $\|\mathbf{p} - \mathbf{w}\| = 0$, i.e. $\mathbf{w} = \mathbf{p}$.

Bessel. Pythagoras for orthonormal vectors gives $\|\mathbf{p}\|^2 = \sum_i |\langle \mathbf{b}, \mathbf{q}_i \rangle|^2$, and the identity above with $\mathbf{w} = \mathbf{0}$ gives $\|\mathbf{b}\|^2 = \|\mathbf{b} - \mathbf{p}\|^2 + \|\mathbf{p}\|^2 \ge \|\mathbf{p}\|^2$. ∎



Projection of $\mathbf{b} = (1, 3)$ onto the line W spanned by $(2, 1)$: $\mathbf{p} = \frac{5}{5}(2, 1) = (2, 1)$ and the residual $(-1, 2)$ is perpendicular to W. The circle of radius $\|\mathbf{b} - \mathbf{p}\| = \sqrt{5}$ about $\mathbf{b}$ touches W only at $\mathbf{p}$, so any other point $\mathbf{w}$ of W (green segment) is farther away — Pythagoras in picture form.

Proof of Theorem C.10

Let $W = \operatorname{col}\mathbf{A} = \{\mathbf{A}\mathbf{x} : \mathbf{x} \in \mathbb{R}^n\}$, a subspace of $\mathbb{R}^m$ of dimension $\le n$. By Theorem C.9 there is a unique $\mathbf{p} \in W$ minimising $\|\mathbf{b} - \mathbf{w}\|$ over $\mathbf{w} \in W$. Since $J(\mathbf{x}) = \|\mathbf{b} - \mathbf{A}\mathbf{x}\|^2$ and $\mathbf{A}\mathbf{x}$ ranges over exactly W, $\hat{\mathbf{x}}$ minimises J iff $\mathbf{A}\hat{\mathbf{x}} = \mathbf{p}$. By part 1 of C.9 (uniqueness of the orthogonal projection), this holds iff $\mathbf{b} - \mathbf{A}\hat{\mathbf{x}} \perp W$, i.e. iff $\langle \mathbf{b} - \mathbf{A}\hat{\mathbf{x}}, \mathbf{a}_j \rangle = 0$ for each column $\mathbf{a}_j$ (orthogonality to a spanning set suffices, by linearity). Stacking these n conditions,

$$\mathbf{A}^{\mathsf{T}}(\mathbf{b} - \mathbf{A}\hat{\mathbf{x}}) = \mathbf{0} \iff \mathbf{A}^{\mathsf{T}}\mathbf{A}\hat{\mathbf{x}} = \mathbf{A}^{\mathsf{T}}\mathbf{b}.$$

Consistency. $\mathbf{p} \in W$, so $\mathbf{p} = \mathbf{A}\mathbf{x}_0$ for some $\mathbf{x}_0$, which then solves the normal equations. All minimisers share $\mathbf{A}\hat{\mathbf{x}} = \mathbf{p}$.

Uniqueness criterion. First, $\operatorname{null}(\mathbf{A}^{\mathsf{T}}\mathbf{A}) = \operatorname{null}\mathbf{A}$: clearly $\mathbf{A}\mathbf{x} = \mathbf{0} \Rightarrow \mathbf{A}^{\mathsf{T}}\mathbf{A}\mathbf{x} = \mathbf{0}$; conversely $\mathbf{A}^{\mathsf{T}}\mathbf{A}\mathbf{x} = \mathbf{0}$ implies $0 = \mathbf{x}^{\mathsf{T}}\mathbf{A}^{\mathsf{T}}\mathbf{A}\mathbf{x} = \|\mathbf{A}\mathbf{x}\|^2$, so $\mathbf{A}\mathbf{x} = \mathbf{0}$. The solution set of the consistent system $\mathbf{A}^{\mathsf{T}}\mathbf{A}\mathbf{x} = \mathbf{A}^{\mathsf{T}}\mathbf{b}$ is $\mathbf{x}_0 + \operatorname{null}(\mathbf{A}^{\mathsf{T}}\mathbf{A}) = \mathbf{x}_0 + \operatorname{null}\mathbf{A}$. It is a single point iff $\operatorname{null}\mathbf{A} = \{\mathbf{0}\}$, i.e. by rank–nullity (C.2) iff $\operatorname{rank}\mathbf{A} = n$. In that case the $n \times n$ matrix $\mathbf{A}^{\mathsf{T}}\mathbf{A}$ has trivial null space, hence is invertible, and $\hat{\mathbf{x}} = (\mathbf{A}^{\mathsf{T}}\mathbf{A})^{-1}\mathbf{A}^{\mathsf{T}}\mathbf{b}$.

QR form. With $\mathbf{A} = \mathbf{Q}\mathbf{R}$ (Theorem C.8), $\mathbf{A}^{\mathsf{T}}\mathbf{A} = \mathbf{R}^{\mathsf{T}}\mathbf{R}$ and $\mathbf{A}^{\mathsf{T}}\mathbf{b} = \mathbf{R}^{\mathsf{T}}\mathbf{Q}^{\mathsf{T}}\mathbf{b}$; cancelling the invertible $\mathbf{R}^{\mathsf{T}}$ gives $\mathbf{R}\hat{\mathbf{x}} = \mathbf{Q}^{\mathsf{T}}\mathbf{b}$. $\blacksquare$

Check. Fitting $y = c_0 + c_1 x$ to $(0,1), (1,2), (2,2), (3,4)$: $\mathbf{A}^{\mathsf{T}}\mathbf{A} = \begin{bmatrix} 4 & 6 \\ 6 & 14 \end{bmatrix}$, $\mathbf{A}^{\mathsf{T}}\mathbf{b} = (9, 18)^{\mathsf{T}}$, so $c_0 = c_1 = 0.9$. The residual $(0.1,\ 0.2,\ -0.7,\ 0.4)$ sums to zero and has zero inner product with $(0, 1, 2, 3)$: it is orthogonal to both columns, exactly as the proof requires.

Remarks

- **Finite dimension of W (or completeness) is needed.** In the space of continuous functions on $[-1, 1]$ with $\langle f, g \rangle = \int fg$, the subspace W of polynomials is infinite-dimensional and dense, so a non-polynomial b such as $|t|$ has no closest polynomial: every $\varepsilon > 0$ is beaten by some polynomial, but 0 is not attained. In Hilbert spaces the theorem holds for every *closed* subspace; that is the setting of Fourier series (Ch. 15), where Bessel's inequality becomes Parseval's identity for a complete orthonormal set.
- **Rank deficiency.** If $\operatorname{rank}\mathbf{A} < n$ the fitted values $\mathbf{A}\hat{\mathbf{x}}$ are still unique but the parameters are not; the conventional choice is the minimum-norm solution $\mathbf{A}^{+}\mathbf{b}$ given by the SVD (§10.7). In parameter estimation this is non-identifiability (§24.7).
- **Normal equations square the condition number.** $\kappa_2(\mathbf{A}^{\mathsf{T}}\mathbf{A}) = \kappa_2(\mathbf{A})^2$ (from the SVD, §C.10), which is why QR or SVD is preferred numerically (§23.8, §24.3).
- **The projector.** With full column rank, $\mathbf{P} = \mathbf{A}(\mathbf{A}^{\mathsf{T}}\mathbf{A})^{-1}\mathbf{A}^{\mathsf{T}} = \mathbf{Q}\mathbf{Q}^{\mathsf{T}}$ satisfies $\mathbf{P}^2 = \mathbf{P} = \mathbf{P}^{\mathsf{T}}$; conversely any symmetric idempotent matrix is the orthogonal projector onto its column space.
- **Uses.** Least-squares curve fitting and calibration (§9.11, §22.6), linear regression (Ch. 20), the orthogonality principle of Wiener filtering (Ch. 21), and Gauss–Newton steps (§24.7).

C.7 Independence of eigenvectors and diagonalizability

Eigenpairs, multiplicities, diagonalizability

$\lambda \in \mathbb{C}$ is an **eigenvalue** of $\mathbf{A} \in \mathbb{C}^{n \times n}$ with **eigenvector** $\mathbf{v} \neq \mathbf{0}$ if $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$; equivalently (Theorem C.1) $\det(\lambda\mathbf{I} - \mathbf{A}) = 0$. The **algebraic multiplicity** $a(\lambda)$ is the multiplicity of λ as a root of the characteristic polynomial $p(s) = \det(s\mathbf{I} - \mathbf{A})$; the **geometric multiplicity** $g(\lambda) = \dim\operatorname{null}(\mathbf{A} - \lambda\mathbf{I})$. $\mathbf{A}$ is **diagonalizable** if $\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$ for some invertible $\mathbf{V}$ and diagonal $\mathbf{\Lambda}$.

Theorem C.11 — Eigenvectors for distinct eigenvalues are independent

If $\mathbf{v}_1, \dots, \mathbf{v}_k$ are eigenvectors of $\mathbf{A}$ for **pairwise distinct** eigenvalues $\lambda_1, \dots, \lambda_k$, then $\mathbf{v}_1, \dots, \mathbf{v}_k$ are linearly independent. More generally, if for each distinct eigenvalue λ_i we take a basis of its eigenspace, the union of these bases is independent.

Theorem C.12 — Diagonalizability

For $\mathbf{A} \in \mathbb{C}^{n\times n}$:

1. $\mathbf{A}$ is diagonalizable if and only if it has n linearly independent eigenvectors; then the columns of $\mathbf{V}$ are such eigenvectors and $\mathbf{\Lambda}$ holds the corresponding eigenvalues.
2. $1 \le g(\lambda) \le a(\lambda)$ for every eigenvalue, and $\mathbf{A}$ is diagonalizable iff $g(\lambda) = a(\lambda)$ for every eigenvalue.
3. In particular, if $\mathbf{A}$ has n distinct eigenvalues, it is diagonalizable.

Proof of Theorem C.11

Suppose not, and among all dependent sets of eigenvectors with distinct eigenvalues choose one with the fewest vectors, $\mathbf{v}_1, \dots, \mathbf{v}_k$; then $k \ge 2$ (a single eigenvector is non-zero, hence independent), and there is a relation

$$c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k = \mathbf{0}$$

in which every $c_i \neq 0$ — if some c_i were zero, the remaining $k-1$ vectors would be a smaller dependent set. Apply $\mathbf{A}$, and separately multiply by λ_k:

$$c_1\lambda_1\mathbf{v}_1 + \cdots + c_k\lambda_k\mathbf{v}_k = \mathbf{0}, \qquad c_1\lambda_k\mathbf{v}_1 + \cdots + c_k\lambda_k\mathbf{v}_k = \mathbf{0}.$$

Subtracting eliminates $\mathbf{v}_k$:

$$c_1(\lambda_1 - \lambda_k)\mathbf{v}_1 + \cdots + c_{k-1}(\lambda_{k-1} - \lambda_k)\mathbf{v}_{k-1} = \mathbf{0}.$$

The coefficients $c_i(\lambda_i - \lambda_k)$ are non-zero because $c_i \neq 0$ and the eigenvalues are distinct. So $\mathbf{v}_1, \dots, \mathbf{v}_{k-1}$ is a smaller dependent set, a contradiction.

General form. Let $\mathcal{B}_i$ be a basis of the eigenspace $E_i = \operatorname{null}(\mathbf{A} - \lambda_i\mathbf{I})$ and suppose a combination of all the vectors in $\bigcup_i \mathcal{B}_i$ vanishes. Grouping terms by eigenvalue, it reads $\mathbf{z}_1 + \cdots + \mathbf{z}_k = \mathbf{0}$ with $\mathbf{z}_i \in E_i$. Those $\mathbf{z}_i$ that are non-zero are eigenvectors for distinct eigenvalues summing to zero, which the first part forbids; so every $\mathbf{z}_i = \mathbf{0}$. Each $\mathbf{z}_i$ is a combination of the independent set $\mathcal{B}_i$, so all its coefficients vanish. ■

Proof of Theorem C.12

Part 1. Write $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_n]$ and $\mathbf{\Lambda} = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$. Column j of $\mathbf{AV}$ is $\mathbf{A}\mathbf{v}_j$ and column j of $\mathbf{V\Lambda}$ is $\lambda_j\mathbf{v}_j$, so

$$\mathbf{AV} = \mathbf{V\Lambda} \iff \mathbf{A}\mathbf{v}_j = \lambda_j\mathbf{v}_j \ \text{ for all } j.$$

If the $\mathbf{v}_j$ are n independent eigenvectors, $\mathbf{V}$ is invertible (Theorem C.1, Step 7 and rank–nullity) and $\mathbf{A} = \mathbf{V\Lambda V}^{-1}$. Conversely, if $\mathbf{A} = \mathbf{V\Lambda V}^{-1}$ then $\mathbf{AV} = \mathbf{V\Lambda}$, so the columns of $\mathbf{V}$ — non-zero and independent because $\mathbf{V}$ is invertible — are eigenvectors.

Part 2. $g(\lambda) \ge 1$ because $\mathbf{A} - \lambda\mathbf{I}$ is singular. Let $g = g(\lambda)$, choose a basis $\mathbf{v}_1, \dots, \mathbf{v}_g$ of the eigenspace, extend it to a basis of $\mathbb{C}^n$ and let $\mathbf{S}$ be the invertible matrix with these columns. Since $\mathbf{A}\mathbf{v}_i = \lambda\mathbf{v}_i$ for $i \le g$, the first g columns of $\mathbf{S}^{-1}\mathbf{AS}$ are $\lambda\mathbf{e}_1, \dots, \lambda\mathbf{e}_g$:

$$\mathbf{S}^{-1}\mathbf{AS} = \begin{bmatrix} \lambda\mathbf{I}_g & \mathbf{B} \\ \mathbf{0} & \mathbf{C} \end{bmatrix}.$$

Similar matrices have the same characteristic polynomial, by the product rule: $\det(s\mathbf{I} - \mathbf{S}^{-1}\mathbf{AS}) = \det\left(\mathbf{S}^{-1}(s\mathbf{I} - \mathbf{A})\mathbf{S}\right) = \det(s\mathbf{I} - \mathbf{A})$. The determinant of a block upper-triangular matrix is the product of the determinants of its diagonal blocks (expand down the first g columns, or apply the uniqueness lemma of C.1 to the lower block), so

$$p(s) = (s - \lambda)^g \det(s\mathbf{I} - \mathbf{C}),$$

and $(s-\lambda)^g$ divides p: $a(\lambda) \ge g$.

Now let $\mu_1, \dots, \mu_k$ be the distinct eigenvalues. By the fundamental theorem of algebra $p(s) = \prod_i (s - \mu_i)^{a(\mu_i)}$, so $\sum_i a(\mu_i) = n$. By Theorem C.11 the union of eigenspace bases is an independent set of $\sum_i g(\mu_i)$ eigenvectors, and every eigenvector lies in some eigenspace, so the largest number of independent eigenvectors is exactly $\sum_i g(\mu_i)$. Hence, by Part 1,

$$\mathbf{A} \text{ diagonalizable} \iff \sum_i g(\mu_i) = n = \sum_i a(\mu_i) \iff g(\mu_i) = a(\mu_i) \text{ for all } i,$$

the last step because $g(\mu_i) \le a(\mu_i)$ term by term.

Part 3. With n distinct eigenvalues each has $a = 1$, so $1 \le g \le a = 1$ forces $g = a$. (Directly: one eigenvector per eigenvalue gives n independent eigenvectors by C.11.) ■

Remarks

- **Distinctness is sufficient, not necessary.** $\mathbf{I}$ has a single eigenvalue of multiplicity n and is diagonal. Symmetric matrices are always diagonalizable however their eigenvalues repeat (§C.8).
- **The standard counterexample.** $\mathbf{N} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ has $a(1) = 2$ but $\operatorname{null}(\mathbf{N} - \mathbf{I}) = \operatorname{span}\{\mathbf{e}_1\}$, so $g(1) = 1 < 2$: it is **defective** and not diagonalizable. Its $\dot{\mathbf{x}} = (\mathbf{N} - 2\mathbf{I})\mathbf{x}$ solutions contain te^{-t} — the critically damped RLC response (§12.2).

- **Generic behaviour.** Matrices with a repeated eigenvalue form a set of measure zero (the discriminant of p vanishes), so almost every matrix met in practice is diagonalizable — but *nearly* defective matrices have ill-conditioned eigenvector matrices $\mathbf{V}$, with consequences in §C.12.
- **Field matters.** The rotation $\begin{bmatrix}0 & -1\\ 1 & 0\end{bmatrix}$ has eigenvalues $\pm j$; it is diagonalizable over $\mathbb{C}$ but not over $\mathbb{R}$. The real block form of §10.3 is the real substitute.
- **Uses.** Diagonalization (§10.3), matrix powers and $e^{\mathbf{A}t}$ (§10.4), modal decomposition of coupled circuits and vibrating systems (§10.8, §12.6), and the stability theorem of §C.12.

C.8 The spectral theorem for real symmetric matrices

Theorem C.13 — Spectral theorem

Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ be symmetric, $\mathbf{A}^{\mathsf{T}} = \mathbf{A}$. Then

1. every eigenvalue of $\mathbf{A}$ is real, and to each there corresponds a real eigenvector;
2. eigenvectors belonging to distinct eigenvalues are orthogonal;
3. there is an orthogonal matrix $\mathbf{Q}$ ($\mathbf{Q}^{\mathsf{T}}\mathbf{Q} = \mathbf{I}$) and a real diagonal $\mathbf{\Lambda}$ with

$$\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^{\mathsf{T}} = \sum_{i=1}^{n} \lambda_i \, \mathbf{q}_i \mathbf{q}_i^{\mathsf{T}}.$$

Equivalently, $\mathbb{R}^n$ has an orthonormal basis of eigenvectors of $\mathbf{A}$. For complex Hermitian $\mathbf{A} = \mathbf{A}^{\mathsf{H}}$ the same holds with a unitary $\mathbf{U}$: $\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^{\mathsf{H}}$, $\mathbf{\Lambda}$ real.

Proof

Part 1: real eigenvalues. By the fundamental theorem of algebra the characteristic polynomial has a root $\lambda \in \mathbb{C}$, with an eigenvector $\mathbf{x} \in \mathbb{C}^n$, $\mathbf{x} \neq \mathbf{0}$. Consider the scalar $\mathbf{x}^{\mathsf{H}}\mathbf{A}\mathbf{x}$. Since $\mathbf{A}$ is real and symmetric, $\mathbf{A}^{\mathsf{H}} = \mathbf{A}$, and the conjugate of this 1×1 matrix is its conjugate transpose:

$$(\mathbf{x}^{\mathsf{H}}\mathbf{A}\mathbf{x})^{*} = (\mathbf{x}^{\mathsf{H}}\mathbf{A}\mathbf{x})^{\mathsf{H}} = \mathbf{x}^{\mathsf{H}}\mathbf{A}^{\mathsf{H}}\mathbf{x} = \mathbf{x}^{\mathsf{H}}\mathbf{A}\mathbf{x},$$

so $\mathbf{x}^{\mathsf{H}}\mathbf{A}\mathbf{x}$ is real. But $\mathbf{x}^{\mathsf{H}}\mathbf{A}\mathbf{x} = \lambda\,\mathbf{x}^{\mathsf{H}}\mathbf{x} = \lambda\|\mathbf{x}\|^2$ with $\|\mathbf{x}\|^2 > 0$ real. Hence $\lambda = \mathbf{x}^{\mathsf{H}}\mathbf{A}\mathbf{x}/\|\mathbf{x}\|^2$ is real.

Real eigenvector. $\mathbf{A} - \lambda\mathbf{I}$ is now a real matrix with determinant zero. By Theorem C.1 applied over $\mathbb{R}$, it is not invertible as a real matrix, so (Step 7 of that proof, contrapositive) its columns are dependent over $\mathbb{R}$: there is a real $\mathbf{q} \neq \mathbf{0}$ with $(\mathbf{A} - \lambda\mathbf{I})\mathbf{q} = \mathbf{0}$. (Alternatively, write $\mathbf{x} = \mathbf{y} + j\mathbf{z}$ with $\mathbf{y}, \mathbf{z}$ real; since $\mathbf{A}$ and λ are real, $\mathbf{A}\mathbf{y} = \lambda\mathbf{y}$ and $\mathbf{A}\mathbf{z} = \lambda\mathbf{z}$, and $\mathbf{y}, \mathbf{z}$ are not both zero.)

Part 2: orthogonality. Let $\mathbf{A}\mathbf{u} = \lambda\mathbf{u}$, $\mathbf{A}\mathbf{v} = \mu\mathbf{v}$ with real $\lambda \neq \mu$ and real vectors. Using symmetry,

$$\lambda\,\mathbf{v}^{\mathsf{T}}\mathbf{u} = \mathbf{v}^{\mathsf{T}}\mathbf{A}\mathbf{u} = (\mathbf{A}^{\mathsf{T}}\mathbf{v})^{\mathsf{T}}\mathbf{u} = (\mathbf{A}\mathbf{v})^{\mathsf{T}}\mathbf{u} = \mu\,\mathbf{v}^{\mathsf{T}}\mathbf{u},$$

so $(\lambda - \mu)\mathbf{v}^{\mathsf{T}}\mathbf{u} = 0$ and $\mathbf{v}^{\mathsf{T}}\mathbf{u} = 0$.

Part 3: orthogonal diagonalization, by induction on n. For $n = 1$ take $\mathbf{Q} = [1]$. Assume the statement for symmetric matrices of order $n-1$, and let $\mathbf{A}$ be symmetric of order n.

Step (a): split off one eigenvector. By Part 1 there is a real eigenvalue λ_1 with a real eigenvector, which we normalise to $\|\mathbf{q}_1\| = 1$. By Theorem C.7 extend $\mathbf{q}_1$ to an orthonormal basis $\mathbf{q}_1, \mathbf{p}_2, \dots, \mathbf{p}_n$ of $\mathbb{R}^n$, and set $\mathbf{P} = [\mathbf{p}_2, \dots, \mathbf{p}_n]$ ($n \times (n-1)$) and $\mathbf{Q}_1 = [\mathbf{q}_1\ \mathbf{P}]$, an orthogonal matrix.

Step (b): the similarity is block diagonal. Compute

$$\mathbf{Q}_1^{\mathsf{T}}\mathbf{A}\mathbf{Q}_1 = \begin{bmatrix} \mathbf{q}_1^{\mathsf{T}}\mathbf{A}\mathbf{q}_1 & \mathbf{q}_1^{\mathsf{T}}\mathbf{A}\mathbf{P} \\ \mathbf{P}^{\mathsf{T}}\mathbf{A}\mathbf{q}_1 & \mathbf{P}^{\mathsf{T}}\mathbf{A}\mathbf{P} \end{bmatrix}.$$

Here $\mathbf{q}_1^{\mathsf{T}}\mathbf{A}\mathbf{q}_1 = \lambda_1\mathbf{q}_1^{\mathsf{T}}\mathbf{q}_1 = \lambda_1$; $\mathbf{P}^{\mathsf{T}}\mathbf{A}\mathbf{q}_1 = \lambda_1\mathbf{P}^{\mathsf{T}}\mathbf{q}_1 = \mathbf{0}$ because each $\mathbf{p}_i \perp \mathbf{q}_1$; and $\mathbf{q}_1^{\mathsf{T}}\mathbf{A}\mathbf{P} = (\mathbf{P}^{\mathsf{T}}\mathbf{A}^{\mathsf{T}}\mathbf{q}_1)^{\mathsf{T}} = (\mathbf{P}^{\mathsf{T}}\mathbf{A}\mathbf{q}_1)^{\mathsf{T}} = \mathbf{0}^{\mathsf{T}}$ — **this is where symmetry is used a second time.** So

$$\mathbf{Q}_1^{\mathsf{T}}\mathbf{A}\mathbf{Q}_1 = \begin{bmatrix} \lambda_1 & \mathbf{0}^{\mathsf{T}} \\ \mathbf{0} & \mathbf{B} \end{bmatrix}, \qquad \mathbf{B} = \mathbf{P}^{\mathsf{T}}\mathbf{A}\mathbf{P},$$

and $\mathbf{B}^{\mathsf{T}} = \mathbf{P}^{\mathsf{T}}\mathbf{A}^{\mathsf{T}}\mathbf{P} = \mathbf{B}$ is symmetric of order $n-1$.

Step (c): induct. By hypothesis $\mathbf{B} = \mathbf{Q}_2\mathbf{\Lambda}_2\mathbf{Q}_2^{\mathsf{T}}$ with $\mathbf{Q}_2$ orthogonal and $\mathbf{\Lambda}_2$ real diagonal. Put

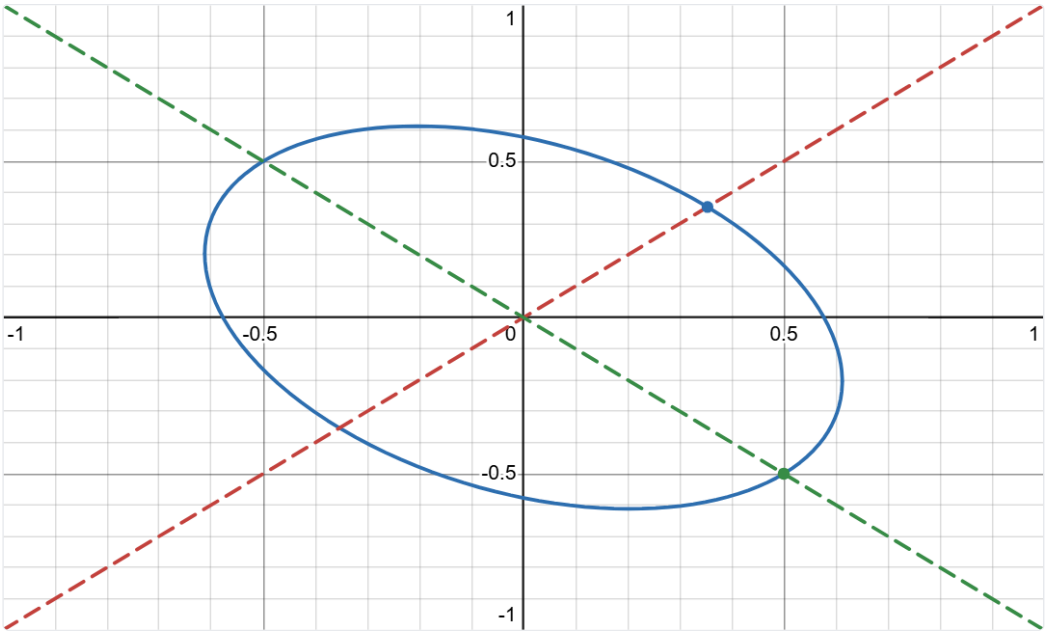
$$\mathbf{Q} = \mathbf{Q}_1 \begin{bmatrix} 1 & \mathbf{0}^{\mathsf{T}} \\ \mathbf{0} & \mathbf{Q}_2 \end{bmatrix}, \qquad \mathbf{\Lambda} = \begin{bmatrix} \lambda_1 & \mathbf{0}^{\mathsf{T}} \\ \mathbf{0} & \mathbf{\Lambda}_2 \end{bmatrix}.$$

$\mathbf{Q}$ is a product of orthogonal matrices, hence orthogonal ($(\mathbf{Q}_a\mathbf{Q}_b)^{\mathsf{T}}\mathbf{Q}_a\mathbf{Q}_b = \mathbf{Q}_b^{\mathsf{T}}\mathbf{Q}_b = \mathbf{I}$), and

$$\mathbf{Q}^{\mathsf{T}}\mathbf{A}\mathbf{Q} = \begin{bmatrix} 1 & \mathbf{0}^{\mathsf{T}} \\ \mathbf{0} & \mathbf{Q}_2^{\mathsf{T}} \end{bmatrix} \begin{bmatrix} \lambda_1 & \mathbf{0}^{\mathsf{T}} \\ \mathbf{0} & \mathbf{B} \end{bmatrix} \begin{bmatrix} 1 & \mathbf{0}^{\mathsf{T}} \\ \mathbf{0} & \mathbf{Q}_2 \end{bmatrix} = \begin{bmatrix} \lambda_1 & \mathbf{0}^{\mathsf{T}} \\ \mathbf{0} & \mathbf{Q}_2^{\mathsf{T}}\mathbf{B}\mathbf{Q}_2 \end{bmatrix} = \mathbf{\Lambda}.$$

Hence $\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^{\mathsf{T}}$. Writing the product column by column gives $\mathbf{A} = \sum_i \lambda_i \mathbf{q}_i\mathbf{q}_i^{\mathsf{T}}$, and $\mathbf{A}\mathbf{Q} = \mathbf{Q}\mathbf{\Lambda}$ shows the columns $\mathbf{q}_i$ are orthonormal eigenvectors.

Hermitian case. Replace T by H and "orthogonal" by "unitary" throughout; Part 1 was already proved for $\mathbf{A}^{\mathsf{H}} = \mathbf{A}$, and Gram–Schmidt works in $\mathbb{C}^n$. $\blacksquare$



Level curve $\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} = 1$ for $\mathbf{A} = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$. The eigenvectors $(1,1)/\sqrt{2}$ (eigenvalue 4) and $(1,-1)/\sqrt{2}$ (eigenvalue 2) are orthogonal and lie along the principal axes, with semi-axes $1/\sqrt{\lambda_i} = 0.5$ and 0.707: in the rotated coordinates $\mathbf{y} = \mathbf{Q}^{\mathsf{T}}\mathbf{x}$ the form is $4y_1^2 + 2y_2^2$, with no cross term.

Remarks

- **Repeated eigenvalues are harmless here.** The induction never assumes distinct eigenvalues: it produces n orthonormal eigenvectors regardless. Contrast Theorem C.12, where distinctness was the easy sufficient condition. Within an eigenspace of dimension > 1 the orthonormal eigenvectors are not unique (any orthonormal basis of the eigenspace will do), but the spectral projectors $\sum_{\lambda_i = \lambda} \mathbf{q}_i \mathbf{q}_i^{\mathsf{T}}$ are.
- **Symmetry is essential, twice.** Without it eigenvalues can be complex ($\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$), eigenvectors can be non-orthogonal ($\begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$ has eigenvectors $(1,0)$ and $(1,1)$), and the matrix can be defective ($\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$). In the induction, symmetry is what kills the top-right block in Step (b); for a general matrix the same construction gives only a block upper-triangular form, which iterated yields the **Schur decomposition** $\mathbf{A} = \mathbf{U}\mathbf{T}\mathbf{U}^{\mathsf{H}}$ with $\mathbf{T}$ upper triangular. A matrix is unitarily diagonalizable iff it is **normal**, $\mathbf{A}\mathbf{A}^{\mathsf{H}} = \mathbf{A}^{\mathsf{H}}\mathbf{A}$.
- **Rayleigh quotient.** From $\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} = \sum \lambda_i y_i^2$ with $\mathbf{y} = \mathbf{Q}^{\mathsf{T}}\mathbf{x}$, $\|\mathbf{y}\| = \|\mathbf{x}\|$, one gets $\lambda_{\min} \le \mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x}/\mathbf{x}^{\mathsf{T}}\mathbf{x} \le \lambda_{\max}$, with the extremes attained at the corresponding eigenvectors (§10.5).
- **Uses.** Principal axes of quadratic forms and energy functions (§10.5); normal modes of LC networks and mass–spring systems, where $\mathbf{M}^{-1/2}\mathbf{K}\mathbf{M}^{-1/2}$ is symmetric (§10.8); principal component analysis and Karhunen–Loève expansions of covariance matrices (Ch. 20, Ch. 21); the convergence analysis of gradient descent on quadratics (§23.5). It is the foundation of §C.9 and §C.10.

C.9 Characterisations of positive-definite matrices

Positive definite

A real symmetric $\mathbf{A} \in \mathbb{R}^{n \times n}$ is **positive definite** ($\mathbf{A} \succ 0$) if $\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} > 0$ for every $\mathbf{x} \neq \mathbf{0}$, and **positive semidefinite** ($\mathbf{A} \succeq 0$) if $\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} \ge 0$ for every $\mathbf{x}$.

Theorem C.14 — Tests for positive definiteness

For real symmetric $\mathbf{A} \in \mathbb{R}^{n \times n}$ the following are equivalent:

1. (energy) $\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$;
2. (eigenvalues) every eigenvalue of $\mathbf{A}$ is positive;
3. (Sylvester's criterion) every leading principal minor is positive: $\Delta_k = \det \mathbf{A}_k > 0$, $k = 1, \dots, n$;
4. (pivots) Gaussian elimination without row interchanges runs to completion with all pivots positive;
5. (Cholesky) $\mathbf{A} = \mathbf{L}\mathbf{L}^{\mathsf{T}}$ with $\mathbf{L}$ lower triangular with positive diagonal entries; this $\mathbf{L}$ is unique;
6. (Gram) $\mathbf{A} = \mathbf{R}^{\mathsf{T}}\mathbf{R}$ for some (possibly rectangular) $\mathbf{R}$ with linearly independent columns.

Proof

We prove $1 \Leftrightarrow 2$, then the cycle $1 \Rightarrow 3 \Rightarrow 4 \Rightarrow 5 \Rightarrow 6 \Rightarrow 1$.

$1 \Leftrightarrow 2$. By Theorem C.13, $\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^{\mathsf{T}}$. Substituting $\mathbf{y} = \mathbf{Q}^{\mathsf{T}}\mathbf{x}$, which is a bijection of $\mathbb{R}^n$ mapping non-zero vectors to non-zero vectors,

$$\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} = \mathbf{y}^{\mathsf{T}}\mathbf{\Lambda}\mathbf{y} = \sum_{i=1}^{n} \lambda_i y_i^2.$$

If all $\lambda_i > 0$, this is positive whenever $\mathbf{y} \neq \mathbf{0}$. Conversely, if 1 holds, take $\mathbf{x} = \mathbf{q}_i$ (i.e. $\mathbf{y} = \mathbf{e}_i$): $\lambda_i = \mathbf{q}_i^{\mathsf{T}}\mathbf{A}\mathbf{q}_i > 0$.

$1 \Rightarrow 3$. Fix k. For $\mathbf{z} \in \mathbb{R}^k$, $\mathbf{z} \neq \mathbf{0}$, pad it with $n-k$ zeros to $\mathbf{x} = (\mathbf{z}, \mathbf{0})$; then $\mathbf{z}^{\mathsf{T}}\mathbf{A}_k\mathbf{z} = \mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} > 0$. So the symmetric matrix $\mathbf{A}_k$ satisfies 1, hence (by $1 \Rightarrow 2$ applied to $\mathbf{A}_k$) has positive eigenvalues $\mu_1, \dots, \mu_k$. By the spectral theorem and the product rule, $\Delta_k = \det(\mathbf{Q}_k\mathbf{M}\mathbf{Q}_k^{\mathsf{T}}) = \det \mathbf{M} = \mu_1 \cdots \mu_k > 0$, where $\det \mathbf{Q}_k \det \mathbf{Q}_k^{\mathsf{T}} = \det(\mathbf{Q}_k\mathbf{Q}_k^{\mathsf{T}}) = 1$.

$3 \Rightarrow 4$. All $\Delta_k \neq 0$, so Theorem C.3 gives $\mathbf{A} = \mathbf{L}_1\mathbf{U}$ (elimination without interchanges succeeds), with pivots $u_{kk} = \Delta_k/\Delta_{k-1} > 0$.

$4 \Rightarrow 5$. Let $\mathbf{A} = \mathbf{L}_1\mathbf{U}$ with $\mathbf{L}_1$ unit lower triangular and pivots $d_k = u_{kk} > 0$. Put $\mathbf{D} = \operatorname{diag}(d_1, \dots, d_n)$ and $\mathbf{U}_1 = \mathbf{D}^{-1}\mathbf{U}$, which is unit upper triangular. Then $\mathbf{A} = \mathbf{L}_1\mathbf{D}\mathbf{U}_1$ and, by symmetry,

$$\mathbf{A} = \mathbf{A}^{\mathsf{T}} = \mathbf{U}_1^{\mathsf{T}}\,(\mathbf{D}\mathbf{L}_1^{\mathsf{T}}).$$

This is a second factorisation of $\mathbf{A}$ into a unit lower triangular matrix $\mathbf{U}_1^{\mathsf{T}}$ and an upper triangular matrix $\mathbf{D}\mathbf{L}_1^{\mathsf{T}}$. Since the pivots are non-zero, $\Delta_k = d_1 \cdots d_k \neq 0$ and Theorem C.3 says the LU factorisation is unique; hence $\mathbf{U}_1^{\mathsf{T}} = \mathbf{L}_1$, i.e.

$$\mathbf{A} = \mathbf{L}_1\mathbf{D}\mathbf{L}_1^{\mathsf{T}}.$$

With $\mathbf{D}^{1/2} = \operatorname{diag}(\sqrt{d_k})$ — real because $d_k > 0$ — set $\mathbf{L} = \mathbf{L}_1\mathbf{D}^{1/2}$: lower triangular with diagonal $\sqrt{d_k} > 0$, and $\mathbf{L}\mathbf{L}^{\mathsf{T}} = \mathbf{L}_1\mathbf{D}\mathbf{L}_1^{\mathsf{T}} = \mathbf{A}$.

Uniqueness of $\mathbf{L}$. If $\mathbf{L}\mathbf{L}^{\mathsf{T}} = \mathbf{G}\mathbf{G}^{\mathsf{T}}$ with both factors lower triangular with positive diagonal, then $\mathbf{M} = \mathbf{G}^{-1}\mathbf{L} = \mathbf{G}^{\mathsf{T}}\mathbf{L}^{-\mathsf{T}}$. The left expression is lower triangular, the right upper triangular, so $\mathbf{M}$ is diagonal with positive diagonal; and $\mathbf{M}\mathbf{M}^{\mathsf{T}} = \mathbf{G}^{-1}\mathbf{L}\mathbf{L}^{\mathsf{T}}\mathbf{G}^{-\mathsf{T}} = \mathbf{I}$ gives $m_{ii}^2 = 1$, so $\mathbf{M} = \mathbf{I}$ and $\mathbf{L} = \mathbf{G}$.

$5 \Rightarrow 6$. Take $\mathbf{R} = \mathbf{L}^{\mathsf{T}}$; it is triangular with non-zero diagonal, hence invertible, so its columns are independent.

$6 \Rightarrow 1$. $\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} = \mathbf{x}^{\mathsf{T}}\mathbf{R}^{\mathsf{T}}\mathbf{R}\mathbf{x} = \|\mathbf{R}\mathbf{x}\|^2 \geq 0$, and it vanishes only if $\mathbf{R}\mathbf{x} = \mathbf{0}$, which by independence of the columns of $\mathbf{R}$ means $\mathbf{x} = \mathbf{0}$. ∎

Check. $\mathbf{A} = \begin{bmatrix} 4 & 2 & 0 \\ 2 & 5 & 3 \\ 0 & 3 & 10 \end{bmatrix}$ has $\Delta_1 = 4$, $\Delta_2 = 16$, $\Delta_3 = 124$, so it is positive definite with pivots $4,\ 4,\ 7.75$. The Cholesky factor is $\mathbf{L} = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 2 & 0 \\ 0 & 1.5 & \sqrt{7.75} \end{bmatrix}$ with $\sqrt{7.75} = 2.7839$ (its diagonal is the square roots of the pivots, as the proof predicts), and the eigenvalues $1.941,\ 5.553,\ 11.507$ are indeed all positive with product 124.

Remarks

- **Symmetry is part of the hypothesis.** For non-symmetric $\mathbf{A}$, only the symmetric part $\tfrac{1}{2}(\mathbf{A} + \mathbf{A}^{\mathsf{T}})$ enters $\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x}$, and the tests come apart: $\begin{bmatrix} 1 & -4 \\ 0 & 1 \end{bmatrix}$ has positive eigenvalues and positive leading minors, yet $\mathbf{x} = (1, 1)$ gives $\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} = -2$.
- **Semidefinite: leading minors are not enough.** $\mathbf{A} = \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}$ has $\Delta_1 = \Delta_2 = 0 \geq 0$ but is negative semidefinite. The correct semidefinite version of Sylvester's criterion requires **all** principal minors (every subset of rows with the same columns) to be ≥ 0; the eigenvalue and Gram tests carry over with $>$ replaced by $\geq$ and "independent columns" dropped. The proof of $3 \Rightarrow 1$ fails exactly because a zero pivot halts elimination.
- **Negative definite** is not "all $\Delta_k < 0$" but alternating signs, $(-1)^k\Delta_k > 0$, from applying the theorem to $-\mathbf{A}$ ($\det(-\mathbf{A}_k) = (-1)^k\Delta_k$).
- **Engineering reading.** For a passive RLC network, the inductance and capacitance matrices are positive definite because stored magnetic and electric energy $\tfrac{1}{2}\mathbf{i}^{\mathsf{T}}\mathbf{L}\mathbf{i}$, $\tfrac{1}{2}\mathbf{v}^{\mathsf{T}}\mathbf{C}\mathbf{v}$ is positive for any non-zero state. Nodal conductance matrices of connected resistive networks with a ground path are positive definite too, which is why Cholesky (half the work of LU, and stable without pivoting) can solve them (§8.9).
- **Uses.** Second-derivative test and convexity: a C^2 function is strictly convex where its Hessian is positive definite (§23.4, §23.7); covariance matrices are positive semidefinite (Ch. 19); the normal-equation matrix $\mathbf{A}^{\mathsf{T}}\mathbf{A}$ is positive definite exactly when $\mathbf{A}$ has full column rank, by test 6 (§C.6); weighted inner products need $\mathbf{W} \succ 0$ (§9.8); Lyapunov stability, $\mathbf{A}^{\mathsf{T}}\mathbf{P} + \mathbf{P}\mathbf{A} \prec 0$ with $\mathbf{P} \succ 0$ (§10.9).

C.10 Existence of the singular value decomposition

Theorem C.15 — Singular value decomposition

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ have rank r. There are orthogonal $\mathbf{U} \in \mathbb{R}^{m \times m}$, $\mathbf{V} \in \mathbb{R}^{n \times n}$ and numbers $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$ such that

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\mathsf{T}} = \sum_{i=1}^{r} \sigma_i\, \mathbf{u}_i\mathbf{v}_i^{\mathsf{T}}, \qquad \mathbf{\Sigma} = \begin{bmatrix} \operatorname{diag}(\sigma_1, \dots, \sigma_r) & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \in \mathbb{R}^{m \times n}.$$

The σ_i^2 are the non-zero eigenvalues of $\mathbf{A}^{\mathsf{T}}\mathbf{A}$ (and of $\mathbf{A}\mathbf{A}^{\mathsf{T}}$), so the singular values are uniquely determined by $\mathbf{A}$. Moreover

$$\|\mathbf{A}\|_2 = \max_{\mathbf{x} \neq \mathbf{0}} \frac{\|\mathbf{A}\mathbf{x}\|_2}{\|\mathbf{x}\|_2} = \sigma_1,$$

and if $\mathbf{A}$ is square and invertible ($r = n = m$), $\|\mathbf{A}^{-1}\|_2 = 1/\sigma_n$. (Complex case: $\mathsf{T} \to \mathsf{H}$, orthogonal $\to$ unitary; the σ_i remain real and positive.)

Proof

Step 1: eigen-decompose the Gram matrix. $\mathbf{A}^{\mathsf{T}}\mathbf{A}$ is $n \times n$ and symmetric, so by Theorem C.13 it has orthonormal eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ with real eigenvalues $\mu_1, \dots, \mu_n$, which we order so that $\mu_1 \geq \mu_2 \geq \cdots \geq \mu_n$. Each is non-negative:

$$\mu_i = \mu_i\mathbf{v}_i^{\mathsf{T}}\mathbf{v}_i = \mathbf{v}_i^{\mathsf{T}}\mathbf{A}^{\mathsf{T}}\mathbf{A}\mathbf{v}_i = \|\mathbf{A}\mathbf{v}_i\|^2 \geq 0.$$

Step 2: exactly r of them are positive. In §C.6 we showed $\operatorname{null}(\mathbf{A}^{\mathsf{T}}\mathbf{A}) = \operatorname{null}\mathbf{A}$, so by rank–nullity $\operatorname{rank}(\mathbf{A}^{\mathsf{T}}\mathbf{A}) = n - \dim\operatorname{null}\mathbf{A} = r$. The rank of the diagonalizable matrix $\mathbf{A}^{\mathsf{T}}\mathbf{A} = \mathbf{V}\mathbf{M}\mathbf{V}^{\mathsf{T}}$ equals the rank of $\mathbf{M} = \operatorname{diag}(\mu_i)$ (multiplying by invertible matrices does not change rank), which is the number of non-zero μ_i. So $\mu_1 \geq \cdots \geq \mu_r > 0 = \mu_{r+1} = \cdots = \mu_n$. Define $\sigma_i = \sqrt{\mu_i}$ for $i \leq r$.

Step 3: build orthonormal left vectors. For $i \leq r$ put $\mathbf{u}_i = \mathbf{A}\mathbf{v}_i/\sigma_i \in \mathbb{R}^m$. Then for $i, k \leq r$

$$\mathbf{u}_i^{\mathsf{T}}\mathbf{u}_k = \frac{\mathbf{v}_i^{\mathsf{T}}\mathbf{A}^{\mathsf{T}}\mathbf{A}\mathbf{v}_k}{\sigma_i\sigma_k} = \frac{\mu_k\,\mathbf{v}_i^{\mathsf{T}}\mathbf{v}_k}{\sigma_i\sigma_k} = \frac{\sigma_k^2\,\delta_{ik}}{\sigma_i\sigma_k} = \delta_{ik},$$

so $\mathbf{u}_1,\dots,\mathbf{u}_r$ are orthonormal. By Theorem C.7 extend them to an orthonormal basis $\mathbf{u}_1,\dots,\mathbf{u}_m$ of $\mathbb{R}^m$ and let $\mathbf{U}=[\mathbf{u}_1,\dots,\mathbf{u}_m]$, $\mathbf{V}=[\mathbf{v}_1,\dots,\mathbf{v}_n]$; both are orthogonal.

Step 4: verify $\mathbf{AV}=\mathbf{U\Sigma}$. Column i of $\mathbf{U\Sigma}$ is $\sigma_i\mathbf{u}_i$ for $i\le r$ and $\mathbf{0}$ for $i>r$. Column i of $\mathbf{AV}$ is $\mathbf{A}\mathbf{v}_i$, which equals $\sigma_i\mathbf{u}_i$ for $i\le r$ by definition, and for $i>r$ satisfies $\|\mathbf{A}\mathbf{v}_i\|^2=\mu_i=0$, so $\mathbf{A}\mathbf{v}_i=\mathbf{0}$. Hence $\mathbf{AV}=\mathbf{U\Sigma}$; multiplying on the right by $\mathbf{V}^{\mathsf{T}}=\mathbf{V}^{-1}$ gives $\mathbf{A}=\mathbf{U\Sigma V}^{\mathsf{T}}$. Expanding the product by columns gives the rank-one sum.

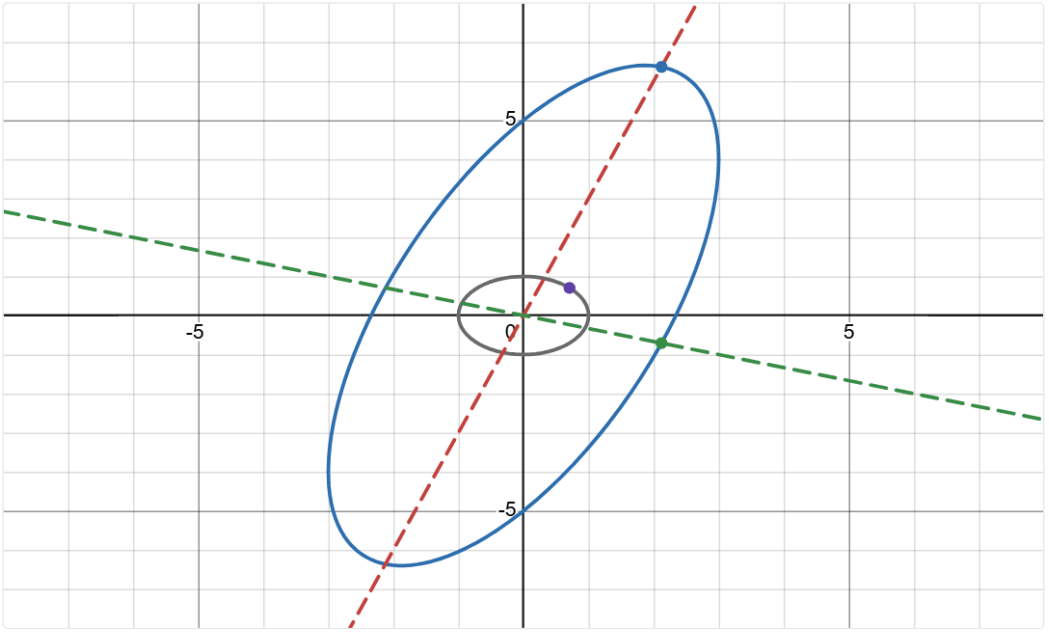
Step 5: uniqueness of the σ_i. From any SVD, $\mathbf{A}^{\mathsf{T}}\mathbf{A}=\mathbf{V\Sigma}^{\mathsf{T}}\mathbf{\Sigma V}^{\mathsf{T}}$, an orthogonal diagonalization with diagonal $\sigma_1^2,\dots,\sigma_r^2,0,\dots$; similarly $\mathbf{AA}^{\mathsf{T}}=\mathbf{U\Sigma\Sigma}^{\mathsf{T}}\mathbf{U}^{\mathsf{T}}$. So the σ_i^2 are the non-zero eigenvalues of either product, which depend only on $\mathbf{A}$.

Step 6: the 2-norm. For any $\mathbf{x}$ let $\mathbf{y}=\mathbf{V}^{\mathsf{T}}\mathbf{x}$; orthogonal matrices preserve length ($\|\mathbf{Q}\mathbf{z}\|^2=\mathbf{z}^{\mathsf{T}}\mathbf{Q}^{\mathsf{T}}\mathbf{Q}\mathbf{z}=\|\mathbf{z}\|^2$), so $\|\mathbf{y}\|=\|\mathbf{x}\|$ and

$$\|\mathbf{A}\mathbf{x}\|^2=\|\mathbf{U\Sigma}\mathbf{y}\|^2=\|\mathbf{\Sigma}\mathbf{y}\|^2=\sum_{i=1}^{r}\sigma_i^2y_i^2\le\sigma_1^2\|\mathbf{y}\|^2=\sigma_1^2\|\mathbf{x}\|^2,$$

with equality at $\mathbf{x}=\mathbf{v}_1$. Hence $\|\mathbf{A}\|_2=\sigma_1$. If $\mathbf{A}$ is square and invertible, $\mathbf{A}^{-1}=\mathbf{V\Sigma}^{-1}\mathbf{U}^{\mathsf{T}}$ is an SVD (after reversing the order of the columns) with singular values $1/\sigma_n\ge\cdots\ge1/\sigma_1$, so $\|\mathbf{A}^{-1}\|_2=1/\sigma_n$. ■

Check. $\mathbf{A}=\begin{bmatrix}3&0\\4&5\end{bmatrix}$: $\mathbf{A}^{\mathsf{T}}\mathbf{A}=\begin{bmatrix}25&20\\20&25\end{bmatrix}$ has eigenvalues 45 and 5 with eigenvectors $(1,1)/\sqrt{2}$ and $(1,-1)/\sqrt{2}$, so $\sigma_1=3\sqrt{5}=6.708$, $\sigma_2=\sqrt{5}=2.236$, and $\mathbf{u}_1=\mathbf{A}\mathbf{v}_1/\sigma_1=(1,3)/\sqrt{10}$, $\mathbf{u}_2=(3,-1)/\sqrt{10}$, which are orthogonal as Step 3 promises. Note $\sigma_1\sigma_2=15=|\det\mathbf{A}|$.



The SVD as geometry: $\mathbf{A}=\begin{bmatrix}3&0\\4&5\end{bmatrix}$ maps the unit circle onto the ellipse $\mathbf{y}^{\mathsf{T}}(\mathbf{AA}^{\mathsf{T}})^{-1}\mathbf{y}=1$, i.e. $41x^2-24xy+9y^2=225$. The right singular vector $\mathbf{v}_1$ on the circle lands on the long semi-axis $\sigma_1\mathbf{u}_1$ (length 6.708); $\mathbf{v}_2$ lands on the short one, $\sigma_2\mathbf{u}_2$ (length 2.236); the axes are perpendicular.

Remarks

- **No hypothesis at all.** Unlike diagonalization, the SVD exists for every matrix — rectangular, singular, defective — because it uses two different orthonormal bases and always works through the symmetric matrix $\mathbf{A}^{\mathsf{T}}\mathbf{A}$. The defective $\begin{bmatrix}1&1\\0&1\end{bmatrix}$, with no eigenvector basis, has singular values $(1\pm\sqrt{5})/2$ in absolute value, i.e. 1.618 and 0.618.
- **Singular values are not eigenvalue magnitudes** except for normal matrices. For $\begin{bmatrix}0&100\\0&0\end{bmatrix}$ both eigenvalues are 0 but $\sigma_1=100$. This gap between $|\lambda|$ and σ is exactly the transient-growth phenomenon of §C.12.
- **Do not compute the SVD this way.** Forming $\mathbf{A}^{\mathsf{T}}\mathbf{A}$ squares the condition number and loses small singular values below $\sqrt{u}\,\sigma_1$ (with u the unit round-off). Practical algorithms bidiagonalise $\mathbf{A}$ with orthogonal transformations (§24.3); the proof above is an existence proof.
- **Uses.** Rank and the four fundamental subspaces, the pseudoinverse $\mathbf{A}^{+}=\mathbf{V\Sigma}^{+}\mathbf{U}^{\mathsf{T}}$ and minimum-norm least squares (§10.7); best rank-k approximation (Eckart–Young) behind PCA and data compression (Ch. 20); numerical rank (§8.5); and $\kappa_2=\sigma_1/\sigma_n$ in §C.11.

C.11 The condition-number error bound

Induced norm and condition number

Given a vector norm $\|\cdot\|$ on $\mathbb{F}^n$, the induced matrix norm is $\|\mathbf{A}\|=\max_{\mathbf{x}\neq\mathbf{0}}\|\mathbf{A}\mathbf{x}\|/\|\mathbf{x}\|$, and for nonsingular $\mathbf{A}$ the **condition number** is $\kappa(\mathbf{A})=\|\mathbf{A}\|\,\|\mathbf{A}^{-1}\|$ (§8.10).

The maximum exists because $\mathbf{x} \mapsto \|\mathbf{A}\mathbf{x}\|$ is continuous and the unit sphere of a finite-dimensional space is compact; by homogeneity the ratio takes the same values on the sphere as on all of $\mathbb{F}^n \setminus \{\mathbf{0}\}$. Two consequences are immediate from the definition: $\|\mathbf{A}\mathbf{x}\| \le \|\mathbf{A}\|\|\mathbf{x}\|$ for every $\mathbf{x}$, and $\|\mathbf{A}\mathbf{B}\| \le \|\mathbf{A}\|\|\mathbf{B}\|$ (apply the first twice). With $\mathbf{B} = \mathbf{A}^{-1}$ this gives $\kappa(\mathbf{A}) \ge \|\mathbf{I}\| = 1$.

Theorem C.16 — Perturbation bounds for $\mathbf{A}\mathbf{x} = \mathbf{b}$

Let $\mathbf{A}$ be nonsingular, $\mathbf{b} \neq \mathbf{0}$, and $\mathbf{A}\mathbf{x} = \mathbf{b}$.

1. (right-hand side) If $\mathbf{A}(\mathbf{x} + \delta\mathbf{x}) = \mathbf{b} + \delta\mathbf{b}$, then

$$\frac{\|\delta\mathbf{x}\|}{\|\mathbf{x}\|} \le \kappa(\mathbf{A})\,\frac{\|\delta\mathbf{b}\|}{\|\mathbf{b}\|}.$$

In the 2-norm the bound is sharp: there exist $\mathbf{b}$, $\delta\mathbf{b}$ attaining equality. 2. (residual) For any approximate solution $\hat{\mathbf{x}}$ with residual $\mathbf{r} = \mathbf{b} - \mathbf{A}\hat{\mathbf{x}}$: $\|\mathbf{x} - \hat{\mathbf{x}}\|/\|\mathbf{x}\| \le \kappa(\mathbf{A})\|\mathbf{r}\|/\|\mathbf{b}\|$. 3. (matrix) If $(\mathbf{A} + \delta\mathbf{A})(\mathbf{x} + \delta\mathbf{x}) = \mathbf{b}$, then $\dfrac{\|\delta\mathbf{x}\|}{\|\mathbf{x} + \delta\mathbf{x}\|} \le \kappa(\mathbf{A})\dfrac{\|\delta\mathbf{A}\|}{\|\mathbf{A}\|}$.

Proof

Part 1. Subtracting $\mathbf{A}\mathbf{x} = \mathbf{b}$ from the perturbed equation gives $\mathbf{A}\,\delta\mathbf{x} = \delta\mathbf{b}$, so $\delta\mathbf{x} = \mathbf{A}^{-1}\delta\mathbf{b}$ and

$$\|\delta\mathbf{x}\| \le \|\mathbf{A}^{-1}\|\,\|\delta\mathbf{b}\|.$$

Also $\|\mathbf{b}\| = \|\mathbf{A}\mathbf{x}\| \le \|\mathbf{A}\|\|\mathbf{x}\|$; since $\mathbf{b} \neq \mathbf{0}$ we have $\mathbf{x} \neq \mathbf{0}$, and

$$\frac{1}{\|\mathbf{x}\|} \le \frac{\|\mathbf{A}\|}{\|\mathbf{b}\|}.$$

Multiplying the two inequalities (both sides non-negative) gives the bound.

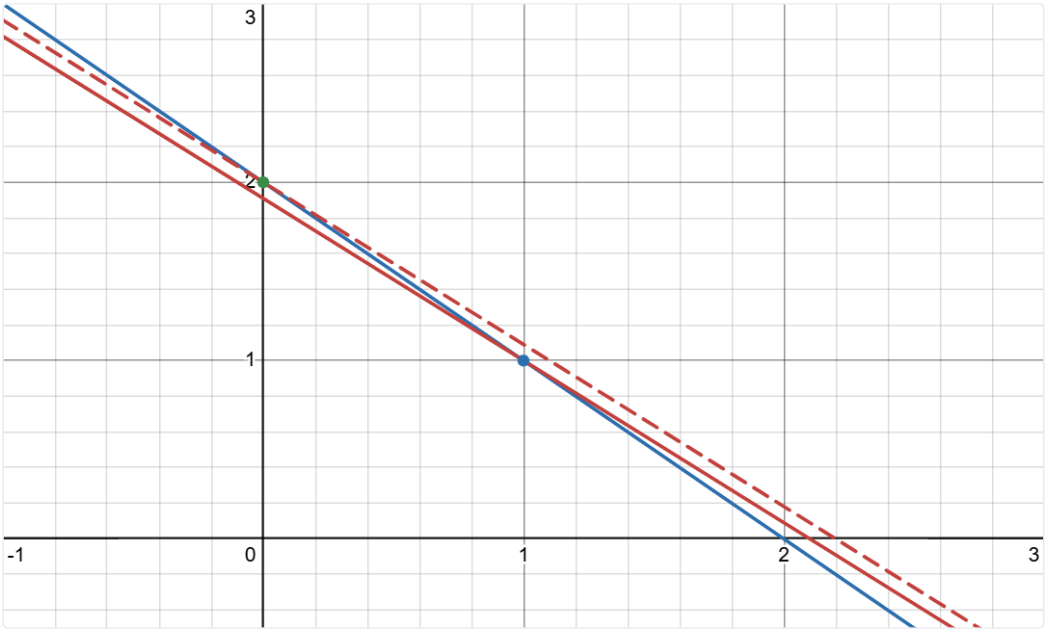
Sharpness in the 2-norm. Take the SVD of $\mathbf{A}$ (Theorem C.15), square of order n, with $\sigma_n > 0$. Choose $\mathbf{x} = \mathbf{v}_1$, so $\mathbf{b} = \sigma_1\mathbf{u}_1$ and $\|\mathbf{A}\mathbf{x}\| = \|\mathbf{A}\|_2\|\mathbf{x}\|$; and $\delta\mathbf{b} = \varepsilon\mathbf{u}_n$, so $\delta\mathbf{x} = \varepsilon\mathbf{v}_n/\sigma_n$ and $\|\delta\mathbf{x}\| = \|\mathbf{A}^{-1}\|_2\|\delta\mathbf{b}\|$. Both inequalities are then equalities:

$$\frac{\|\delta\mathbf{x}\|}{\|\mathbf{x}\|} = \frac{\varepsilon}{\sigma_n}, \qquad \kappa_2(\mathbf{A})\frac{\|\delta\mathbf{b}\|}{\|\mathbf{b}\|} = \frac{\sigma_1}{\sigma_n}\cdot\frac{\varepsilon}{\sigma_1} = \frac{\varepsilon}{\sigma_n}.$$

Part 2. $\hat{\mathbf{x}}$ solves $\mathbf{A}\hat{\mathbf{x}} = \mathbf{b} - \mathbf{r}$ exactly; apply Part 1 with $\delta\mathbf{b} = -\mathbf{r}$ and $\delta\mathbf{x} = \hat{\mathbf{x}} - \mathbf{x}$.

Part 3. Expanding, $\mathbf{A}\mathbf{x} + \mathbf{A}\,\delta\mathbf{x} + \delta\mathbf{A}(\mathbf{x} + \delta\mathbf{x}) = \mathbf{b}$, and $\mathbf{A}\mathbf{x} = \mathbf{b}$, so $\delta\mathbf{x} = -\mathbf{A}^{-1}\delta\mathbf{A}(\mathbf{x} + \delta\mathbf{x})$. Taking norms, $\|\delta\mathbf{x}\| \le \|\mathbf{A}^{-1}\|\|\delta\mathbf{A}\|\|\mathbf{x} + \delta\mathbf{x}\|$; divide by $\|\mathbf{x} + \delta\mathbf{x}\|$ (if it is zero, then $\mathbf{b} = \mathbf{0}$, excluded) and write $\|\mathbf{A}^{-1}\| = \kappa(\mathbf{A})/\|\mathbf{A}\|$. ■

Check. $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1.1 \end{bmatrix}$, $\mathbf{b} = (2, 2.1)$, $\mathbf{x} = (1, 1)$. Changing b_2 to 2.2 ($\|\delta\mathbf{b}\|_2/\|\mathbf{b}\|_2 = 0.1/2.900 = 0.0345$) moves the solution to $(0, 2)$, a relative change $\|(-1, 1)\|/\|(1, 1)\| = 1$: an amplification of 29. The theorem allows up to $\kappa_2(\mathbf{A}) = 42.08$, i.e. a relative change of 1.45, so the bound holds and is of the right order. Making the $(2, 2)$ entry 1.0001 instead raises κ_2 to about 4.0×10^4.



Ill-conditioning seen geometrically: the two equations of $\mathbf{A}\mathbf{x} = \mathbf{b}$ are nearly parallel lines (blue and red). Shifting the red line by 0.1 in b_2 (dashed) slides the intersection from $(1, 1)$ to $(0, 2)$ — a 3.4% data change producing a 100% solution change.

Remarks

- **Sharpness in other norms.** For $\|\cdot\|_1$ and $\|\cdot\|_\infty$ the bound is also attained for suitable $\mathbf{b}$, $\delta\mathbf{b}$ (pick a vector achieving $\|\mathbf{A}\|$ and one achieving $\|\mathbf{A}^{-1}\|$; any induced norm attains its maximum). For a particular $\mathbf{b}$ the actual amplification can be much smaller — the bound is worst case over the data.

- **Part 3 is "first order".** Bounding $\|\delta\mathbf{x}\|/\|\mathbf{x}\|$ rather than $/\|\mathbf{x}+\delta\mathbf{x}\|$ requires $\kappa(\mathbf{A})\|\delta\mathbf{A}\|/\|\mathbf{A}\| < 1$, which also guarantees $\mathbf{A}+\delta\mathbf{A}$ is nonsingular; then the bound is $\kappa\epsilon_A/(1-\kappa\epsilon_A)$ with $\epsilon_A = \|\delta\mathbf{A}\|/\|\mathbf{A}\|$ (§8.10). Without that condition there is no bound at all: $\mathbf{A}+\delta\mathbf{A}$ may be singular.
- **Small residual does not mean small error.** Part 2 shows a residual at round-off level only guarantees about $\kappa(\mathbf{A})\cdot u$ relative accuracy. This is the rule of thumb "you lose $\log_{10}\kappa$ digits" (§22.2), and combined with backward stability of LU with pivoting it is the basis of all error estimates for linear solvers (§24.3).
- **Determinant is not the measure.** $\det \mathbf{A} = 0.1$ above, but scaling $\mathbf{A}$ by 10 changes the determinant by 100 and leaves κ unchanged; κ is scale- and unit-free, the determinant is neither.
- **Uses.** Sensitivity of nodal analysis to component tolerances, error analysis of least squares ($\kappa(\mathbf{A}^{\mathsf{T}}\mathbf{A}) = \kappa_2(\mathbf{A})^2$), convergence rates of iterative solvers and gradient descent, which scale with κ or $\sqrt{\kappa}$ (§23.5, §24.4).

C.12 Stability of $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$

For $\mathbf{A}\in\mathbb{C}^{n\times n}$, the initial-value problem $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$, $\mathbf{x}(0) = \mathbf{x}_0$ has the unique solution $\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}_0$, where $e^{\mathbf{A}t} = \sum_{k\ge 0}\mathbf{A}^k t^k/k!$ (§10.4, §12.7). We use two properties of the series: $e^{\mathbf{S}\mathbf{B}\mathbf{S}^{-1}t} = \mathbf{S}e^{\mathbf{B}t}\mathbf{S}^{-1}$ (because $(\mathbf{S}\mathbf{B}\mathbf{S}^{-1})^k = \mathbf{S}\mathbf{B}^k\mathbf{S}^{-1}$ term by term), and $e^{(\lambda\mathbf{I}+\mathbf{N})t} = e^{\lambda t}e^{\mathbf{N}t}$ (because $\lambda\mathbf{I}$ commutes with $\mathbf{N}$). The **spectral abscissa** is $\alpha(\mathbf{A}) = \max_i \mathrm{Re}\{\lambda_i\}$. Stability is defined as in §10.9: asymptotically stable if every solution tends to $\mathbf{0}$, marginally stable if every solution is bounded but not all tend to $\mathbf{0}$, unstable if some solution is unbounded.

Theorem C.17 — Eigenvalue test, diagonalizable case

Let $\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$ be diagonalizable. Then for all $t \ge 0$

$$\|e^{\mathbf{A}t}\mathbf{x}_0\|_2 \le \kappa_2(\mathbf{V})\, e^{\alpha(\mathbf{A})t}\,\|\mathbf{x}_0\|_2,$$

and

1. $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ is asymptotically stable $\iff$ $\mathrm{Re}\{\lambda_i\} < 0$ for all i;
2. every solution is bounded on $[0,\infty)$ $\iff$ $\mathrm{Re}\{\lambda_i\} \le 0$ for all i.

If $\mathbf{A}$ is real, the same equivalences hold when only real initial conditions are allowed.

Theorem C.18 — General case (Jordan form)

For arbitrary $\mathbf{A}\in\mathbb{C}^{n\times n}$:

1. $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ is asymptotically stable $\iff$ $\mathrm{Re}\{\lambda_i\} < 0$ for all i; in that case, for every β with $\alpha(\mathbf{A}) < \beta < 0$ there is C with $\|e^{\mathbf{A}t}\| \le Ce^{\beta t}$ for $t \ge 0$;
2. all solutions are bounded $\iff$ $\mathrm{Re}\{\lambda_i\} \le 0$ for all i and every eigenvalue with $\mathrm{Re}\{\lambda_i\} = 0$ is **semisimple** ($g = a$, i.e. all its Jordan blocks are 1×1).

Proof of Theorem C.17

The bound. By the similarity property, $e^{\mathbf{A}t} = \mathbf{V}e^{\mathbf{\Lambda}t}\mathbf{V}^{-1}$, and $e^{\mathbf{\Lambda}t} = \mathrm{diag}(e^{\lambda_1 t},\dots,e^{\lambda_n t})$ since powers of a diagonal matrix are diagonal. Equivalently, in modal coordinates $\mathbf{y} = \mathbf{V}^{-1}\mathbf{x}$ the system decouples into $\dot y_i = \lambda_i y_i$, with $y_i(t) = e^{\lambda_i t}y_i(0)$. The 2-norm of a diagonal matrix is its largest entry modulus (Theorem C.15, or directly), and $|e^{\lambda_i t}| = e^{\mathrm{Re}\{\lambda_i\}t} \le e^{\alpha t}$ for $t \ge 0$. Submultiplicativity gives

$$\|e^{\mathbf{A}t}\mathbf{x}_0\| \le \|\mathbf{V}\|\,\|e^{\mathbf{\Lambda}t}\|\,\|\mathbf{V}^{-1}\|\,\|\mathbf{x}_0\| \le \kappa_2(\mathbf{V})\,e^{\alpha t}\|\mathbf{x}_0\|.$$

Sufficiency. If all $\mathrm{Re}\{\lambda_i\} < 0$ then $\alpha < 0$, so the right-hand side tends to 0 for every $\mathbf{x}_0$. If all $\mathrm{Re}\{\lambda_i\} \le 0$ then $\alpha \le 0$ and every solution is bounded by $\kappa_2(\mathbf{V})\|\mathbf{x}_0\|$.

Necessity. Suppose some eigenvalue has $\mathrm{Re}\{\lambda_k\} \ge 0$, with eigenvector $\mathbf{v}_k$. The solution from $\mathbf{x}_0 = \mathbf{v}_k$ is $e^{\lambda_k t}\mathbf{v}_k$ (it satisfies the equation and the initial condition, and solutions are unique), whose norm $e^{\mathrm{Re}\{\lambda_k\}t}\|\mathbf{v}_k\| \ge \|\mathbf{v}_k\| > 0$ does not tend to 0. So asymptotic stability forces all $\mathrm{Re}\{\lambda_i\} < 0$. Likewise if $\mathrm{Re}\{\lambda_k\} > 0$ this solution is unbounded, so boundedness forces all $\mathrm{Re}\{\lambda_i\} \le 0$.

Real initial conditions. Let $\mathbf{A}$ be real and $\mathbf{v}_k = \mathbf{a} + j\mathbf{b}$ with $\mathbf{a},\mathbf{b}$ real. Because $\mathbf{A}$ is real, $e^{\mathbf{A}t}$ is real, so $e^{\lambda_k t}\mathbf{v}_k = e^{\mathbf{A}t}\mathbf{a} + j\,e^{\mathbf{A}t}\mathbf{b}$ splits into the two real solutions from $\mathbf{a}$ and from $\mathbf{b}$. If both tended to $\mathbf{0}$ (or both were bounded), so would their complex combination, contradicting the previous paragraph. Hence one of the real initial conditions $\mathbf{a}$, $\mathbf{b}$ already exhibits the failure. ■

Proof of Theorem C.18

We use the Jordan form (§10.3): $\mathbf{A} = \mathbf{S}\mathbf{J}\mathbf{S}^{-1}$ with $\mathbf{J}$ block diagonal, each block $\mathbf{J}_m = \lambda\mathbf{I}_m + \mathbf{N}_m$ of size m, where $\mathbf{N}_m$ has ones on the superdiagonal and $\mathbf{N}_m^m = \mathbf{0}$. Since $e^{\mathbf{A}t} = \mathbf{S}e^{\mathbf{J}t}\mathbf{S}^{-1}$ and $e^{\mathbf{J}t}$ is block diagonal with blocks

$$e^{\mathbf{J}_m t} = e^{\lambda t}\sum_{k=0}^{m-1}\frac{t^k}{k!}\mathbf{N}_m^k = e^{\lambda t}\begin{bmatrix} 1 & t & \frac{t^2}{2} & \cdots & \frac{t^{m-1}}{(m-1)!} \\ & 1 & t & \cdots & \vdots \\ & & \ddots & \ddots & \frac{t^2}{2} \\ & & & 1 & t \\ & & & & 1 \end{bmatrix}$$

(the series for $e^{\mathbf{N}_m t}$ terminates), every entry of $e^{\mathbf{A}t}$ is a finite combination of functions $t^k e^{\lambda t}$, where $k < m$ for an eigenvalue whose largest Jordan block has size m.

Part 1, sufficiency and the rate. Let $\alpha < \beta < 0$. For each term, $|t^k e^{\lambda t}| = t^k e^{\mathrm{Re}\{\lambda\}t} = e^{\beta t}\cdot t^k e^{-(\beta-\mathrm{Re}\{\lambda\})t}$, and since $\beta - \mathrm{Re}\{\lambda\} > 0$ the factor $t^k e^{-(\beta-\mathrm{Re}\{\lambda\})t}$ is continuous on $[0,\infty)$ and tends to 0 (exponentials beat polynomials), hence is bounded. A finite sum of such bounds, combined with $\|\mathbf{S}\|\|\mathbf{S}^{-1}\|$, gives $\|e^{\mathbf{A}t}\| \le Ce^{\beta t} \to 0$.

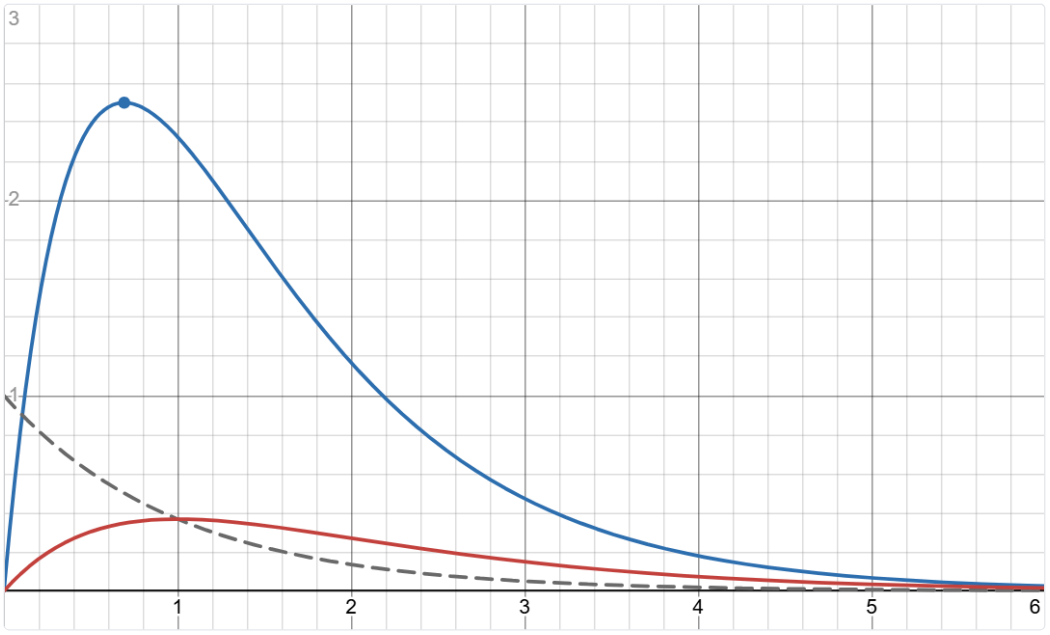
Part 1, necessity. As in C.17: an eigenvector $\mathbf{v}$ with $\mathrm{Re}\{\lambda\} \geq 0$ gives the non-decaying solution $e^{\lambda t}\mathbf{v}$.

Part 2, sufficiency. If $\mathrm{Re}\{\lambda\} < 0$, the block terms are bounded by Part 1's argument. If $\mathrm{Re}\{\lambda\} = 0$ and λ is semisimple, its blocks are 1×1 and contribute $e^{\lambda t}$, of modulus 1. So all entries of $e^{\mathbf{J}t}$, and hence of $e^{\mathbf{A}t}$, are bounded.

Part 2, necessity. If some $\mathrm{Re}\{\lambda\} > 0$, the eigenvector solution is unbounded. If $\mathrm{Re}\{\lambda\} = 0$ and λ has a Jordan block of size $m \geq 2$, let $\mathbf{s}_1, \mathbf{s}_2$ be the columns of $\mathbf{S}$ corresponding to the first two positions of that block, so $(\mathbf{A} - \lambda\mathbf{I})\mathbf{s}_1 = \mathbf{0}$ and $(\mathbf{A} - \lambda\mathbf{I})\mathbf{s}_2 = \mathbf{s}_1$. From the block formula, the solution with $\mathbf{x}_0 = \mathbf{s}_2$ is

$$\mathbf{x}(t) = e^{\lambda t}(\mathbf{s}_2 + t\,\mathbf{s}_1),$$

which can be verified directly: $\dot{\mathbf{x}} = \lambda\mathbf{x} + e^{\lambda t}\mathbf{s}_1$ and $\mathbf{A}\mathbf{x} = e^{\lambda t}(\lambda\mathbf{s}_2 + \mathbf{s}_1 + t\lambda\mathbf{s}_1) = \lambda\mathbf{x} + e^{\lambda t}\mathbf{s}_1$. Its norm is $\|\mathbf{s}_2 + t\mathbf{s}_1\| \geq t\|\mathbf{s}_1\| - \|\mathbf{s}_2\| \to \infty$ (triangle inequality, C.6). For real $\mathbf{A}$, real initial conditions suffice by the splitting argument of C.17. ■



Asymptotic stability says nothing about the size of the transient. For $\mathbf{A} = \begin{bmatrix} -1 & 10 \\ 0 & -2 \end{bmatrix}$ (eigenvalues $-1, -2$, eigenvector matrix with $\kappa_2(\mathbf{V}) \approx 102$), the first state component from $\mathbf{x}_0 = (0, 1)$ is $10(e^{-t} - e^{-2t})$ (blue): it rises to 2.5 before decaying at the eigenvalue rate e^{-t} (grey). The red curve te^{-t} is the extra polynomial factor carried by a 2×2 Jordan block at -1: still decaying, but more slowly than e^{-t}.

Remarks

- **Where the hypotheses bite.** The single criterion "$\mathrm{Re}\{\lambda\} < 0$" decides asymptotic stability whether or not $\mathbf{A}$ is diagonalizable; Jordan structure matters only on the boundary $\mathrm{Re}\{\lambda\} = 0$. The double integrator $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ has both eigenvalues at 0 but $x_1(t) = x_1(0) + t\,x_2(0)$ is unbounded: a defective imaginary-axis eigenvalue, exactly the counterexample of Theorem C.18(2). An undamped LC tank, $\begin{bmatrix} 0 & 1 \\ -\omega_0^2 & 0 \end{bmatrix}$, has simple eigenvalues $\pm j\omega_0$ and is marginally stable. The same dichotomy appears for scalar ODEs as "repeated imaginary-axis roots are unstable" (§12.2).
- **Transient growth and non-normality.** The constant $\kappa_2(\mathbf{V})$ in Theorem C.17 is not decoration. For the matrix in the graph, $\max_{t \geq 0}\|e^{\mathbf{A}t}\|_2 \approx 2.56$ (near $t = 0.66$) even though every eigenvalue is negative; for highly non-normal matrices the peak can be orders of magnitude, which matters for saturating amplifiers and in fluid and power-system models. For normal (e.g. symmetric) $\mathbf{A}$, $\mathbf{V}$ can be taken unitary, $\kappa_2(\mathbf{V}) = 1$, and $\|e^{\mathbf{A}t}\|_2 = e^{\alpha t}$ exactly.
- **Discrete time.** The same proofs, with $e^{\lambda t}$ replaced by λ^k and $|e^{\lambda t}| = e^{\mathrm{Re}\{\lambda\}t}$ by $|\lambda^k| = |\lambda|^k$, show that $\mathbf{x}[k+1] = \mathbf{A}\mathbf{x}[k]$ is asymptotically stable iff all $|\lambda_i| < 1$ (§10.4, Ch. 17); this is also the convergence condition $\rho(\mathbf{G}) < 1$ for stationary iterative solvers (§24.4).
- **Nonlinear systems.** By Lyapunov's indirect method, an equilibrium of $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$ is locally asymptotically stable if the Jacobian there has all $\mathrm{Re}\{\lambda\} < 0$, and unstable if some $\mathrm{Re}\{\lambda\} > 0$; the boundary case is inconclusive (§10.9, Ch. 11).
- **Uses.** Every stability statement in the book for LTI state models: natural response of RLC circuits (§12.8), pole locations and BIBO stability (Ch. 14), and modal analysis (§10.8).

Further reading

- G. Strang, *Linear Algebra and Its Applications*, 4th ed., Cengage, 2006 — the four subspaces, LU, least squares and the SVD from the same geometric viewpoint as Part III.
- S. Axler, *Linear Algebra Done Right*, 4th ed., Springer, 2024 — determinant-free proofs of the spectral theorem and of rank–nullity in abstract form.
- R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013 — the definitive reference for positive-definite matrices, Sylvester's criterion, Schur and Jordan forms, and norms.
- L. N. Trefethen and D. Bau, *Numerical Linear Algebra*, SIAM, 1997 — QR, SVD, conditioning and the stability of Gaussian elimination, with the floating-point view of §C.3, §C.5 and §C.11.
- C. D. Meyer, *Matrix Analysis and Applied Linear Algebra*, 2nd ed., SIAM, 2023 — readable proofs throughout, including the matrix exponential and stability.

Appendix D — Transforms, ODEs and Complex Analysis

This appendix gives complete proofs of the results that Chapters 11 and 14–18 state with only a sketch: existence and uniqueness for initial-value problems, the analytic foundations of the Laplace, Fourier and Z-transforms, the convergence theory of Fourier series (Bessel, Parseval, Dirichlet, Gibbs), the sampling theorem, and the core of complex analysis (Cauchy–Riemann, Cauchy's theorem and integral formula, residues, Jordan's lemma).

Prerequisites. Real analysis at the level of a first course: uniform convergence and the fact that a uniform limit of continuous functions is continuous, the Weierstrass M-test, Riemann (or Lebesgue) integration including improper integrals, and the interchange theorems of Fubini–Tonelli and dominated convergence. These are quoted, not proved. Green's theorem is taken from Ch. 7; Laurent's theorem from Ch. 18, §18.6.

Level of rigor. Every hypothesis is stated precisely and every nontrivial step is justified. Where the cleanest general proof needs measure theory (Plancherel on all of L^2, the density of continuous functions in L^2), the proof is given in full for a large class that covers every engineering signal, and the extension is described and cited. Notation follows the Notation and Conventions: $j^2=-1$, ω in rad/s, and the Fourier transform is $X(\omega)=\int_{-\infty}^{\infty} x(t)e^{-j\omega t}\,dt$ with inverse $x(t)=\frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)e^{j\omega t}\,d\omega$, as in Ch. 16.

D.1 Picard–Lindelöf existence and uniqueness

Lipschitz condition

A function $f(t,y)$ defined on a set $R\subset\mathbb{R}^2$ is **Lipschitz in** y on R with constant $K\ge 0$ if

$$|f(t,y_1)-f(t,y_2)|\le K|y_1-y_2| \qquad \text{for all } (t,y_1),(t,y_2)\in R.$$

Picard–Lindelöf

Let $R=\{(t,y):|t-t_0|\le a,\ |y-y_0|\le b\}$ with $a,b>0$. Suppose f is continuous on R and Lipschitz in y on R with constant K. Let $M=\max_R|f|$ (if $M=0$ put $h=a$) and $h=\min(a,\ b/M)$. Then the initial-value problem

$$y'(t)=f(t,y(t)),\qquad y(t_0)=y_0$$

has exactly one solution y on $I=[t_0-h,\ t_0+h]$, and the Picard iterates $y_{k+1}(t)=y_0+\int_{t_0}^{t} f(s,y_k(s))\,ds$, $y_0(t)\equiv y_0$, converge to it uniformly on I.

The proof uses the contraction-mapping principle, which we prove first.

Banach fixed-point theorem

Let $(\mathcal{X},d)$ be a non-empty complete metric space and $\Phi:\mathcal{X}\to\mathcal{X}$ a **contraction**: $d(\Phi u,\Phi v)\le q\,d(u,v)$ for some $q<1$ and all u,v. Then Φ has exactly one fixed point $u^\star$, and for any u_0 the iterates $u_{k+1}=\Phi u_k$ converge to $u^\star$ with $d(u_k,u^\star)\le \dfrac{q^k}{1-q}\,d(u_1,u_0)$.

Proof of the fixed-point theorem. By induction $d(u_{k+1},u_k)\le q^k d(u_1,u_0)$. For $m>k$, the triangle inequality and the geometric series give

$$d(u_m,u_k)\le\sum_{i=k}^{m-1} d(u_{i+1},u_i)\le d(u_1,u_0)\sum_{i=k}^{\infty} q^i=\frac{q^k}{1-q}\,d(u_1,u_0)\xrightarrow[k\to\infty]{}0,$$

so (u_k) is Cauchy and, by completeness, converges to some $u^\star$. A contraction is continuous, so $\Phi u^\star=\lim \Phi u_k=\lim u_{k+1}=u^\star$. Letting $m\to\infty$ in the display gives the error bound. If $u^\star,v^\star$ are both fixed, $d(u^\star,v^\star)=d(\Phi u^\star,\Phi v^\star)\le q\,d(u^\star,v^\star)$, which forces $d(u^\star,v^\star)=0$. $\blacksquare$

Proof of Picard–Lindelöf.

Step 1 — integral equation. A continuous function y on I with graph in R solves the IVP if and only if

$$y(t)=y_0+\int_{t_0}^{t} f(s,y(s))\,ds \qquad (t\in I).$$

Indeed, if y solves the IVP then y' is continuous (a composition of continuous functions) and integrating from t_0 gives the equation; conversely, the right-hand side is differentiable by the fundamental theorem of calculus, because $s\mapsto f(s,y(s))$ is continuous, and differentiating gives $y'=f(t,y)$ with $y(t_0)=y_0$.

Step 2 — the space. Let $\mathcal{X}$ be the set of continuous $y:I\to\mathbb{R}$ with $|y(t)-y_0|\le b$ on I. Equip it with the **weighted (Bielecki) metric**

$$d(y,z)=\max_{t\in I} e^{-2K|t-t_0|}\,|y(t)-z(t)|.$$

Since $e^{-2Kh}\le e^{-2K|t-t_0|}\le 1$, this metric is equivalent to the sup metric: $e^{-2Kh}\|y-z\|_\infty\le d(y,z)\le\|y-z\|_\infty$. A sequence is Cauchy for d iff it is uniformly Cauchy, and a uniform limit of continuous functions is continuous and still satisfies $|y-y_0|\le b$. Hence $(\mathcal{X},d)$ is complete. It is non-empty (it contains the constant y_0).

Step 3 — Φ maps $\mathcal{X}$ into itself. Define $(\Phi y)(t)=y_0+\int_{t_0}^{t} f(s,y(s))\,ds$. For $y\in\mathcal{X}$ the integrand is continuous, so Φy is continuous, and

$$|(\Phi y)(t)-y_0|\le M|t-t_0|\le Mh\le b.$$

This is exactly why $h\le b/M$: the iterates cannot leave the rectangle where the bounds on f hold.

Step 4 — Φ is a contraction. Take $t\ge t_0$ (the case $t<t_0$ is symmetric). By the Lipschitz condition and the definition of d,

$$|(\Phi y)(t) - (\Phi z)(t)| \le \int_{t_0}^{t} K|y(s)-z(s)|\,ds \le K\,d(y,z)\int_{t_0}^{t} e^{2K(s-t_0)}\,ds$$
$$= \frac{d(y,z)}{2}\bigl(e^{2K(t-t_0)}-1\bigr) < \frac{1}{2}\,e^{2K(t-t_0)}\,d(y,z).$$

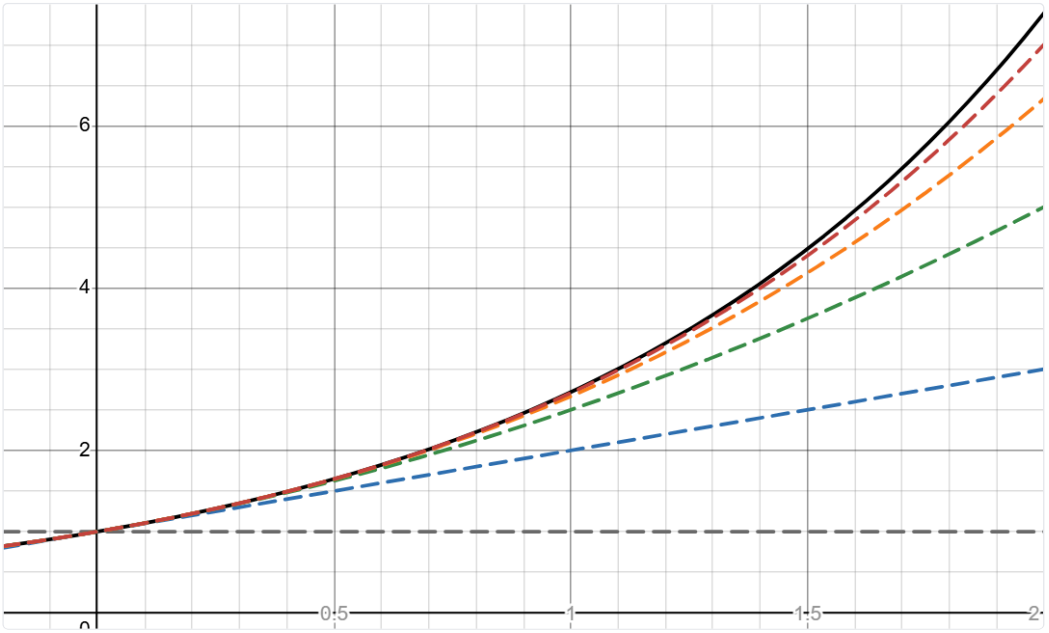
Multiplying by $e^{-2K(t-t_0)}$ and maximising over t gives $d(\Phi y,\Phi z) \le \tfrac12\, d(y,z)$. (If $K=0$ the integral is zero and the bound is trivial.)

Step 5 — existence. By the fixed-point theorem Φ has a fixed point $y \in \mathcal{X}$, which by Step 1 solves the IVP on I. The iterates $y_k = \Phi^k y_0$ converge to it in d, hence uniformly.

Step 6 — uniqueness among all solutions. The fixed-point theorem gives uniqueness only inside $\mathcal{X}$, so we must show that *every* solution z on I has its graph in R. Suppose z leaves the strip $|y-y_0| \le b$ at some $t > t_0$ in I, and let $\tau = \inf\{t > t_0 : |z(t)-y_0| > b\}$. On $[t_0,\tau]$ the graph is in R, so $|z(\tau)-y_0| \le M(\tau-t_0)$; by continuity $|z(\tau)-y_0| = b$. Hence $\tau - t_0 \ge b/M \ge h$, so $\tau \ge t_0 + h$ and z does not leave the strip inside I after all — a contradiction. The case $t < t_0$ is symmetric. So $z \in \mathcal{X}$ is a fixed point of Φ, and $z = y$. $\blacksquare$

Remarks.

1. *The Lipschitz condition is what buys uniqueness.* For $y' = 3y^{2/3}$, $y(0)=0$, the function f is continuous but not Lipschitz at $y=0$, and both $y \equiv 0$ and $y = t^3$ are solutions (indeed infinitely many, $y = (t-c)^3$ for $t \ge c$ and 0 before). Continuity alone still gives *existence* (Peano's theorem, proved by compactness rather than contraction), but not uniqueness.
2. *The result is local.* For $y' = y^2$, $y(0)=1$, f is Lipschitz on every bounded rectangle, yet the solution $y = 1/(1-t)$ blows up at $t=1$. The theorem's $h = \min(a, b/M)$ with $M = (1+b)^2$ is maximised at $b=1$, $a \ge 1/4$, giving $h = 1/4$; repeated application extends the solution as long as it stays bounded, never past $t=1$.
3. *Why the weighted metric.* With the plain sup metric Step 4 gives the factor Kh, so the proof would need $h < 1/K$ in addition. The weight $e^{-2K|t-t_0|}$ absorbs the exponential growth that Lipschitz feedback allows (the same growth that appears in Grönwall's inequality) and gives a contraction on the full interval.
4. *Systems.* The proof works verbatim for $\mathbf{y}' = \mathbf{f}(t,\mathbf{y})$ with $|\cdot|$ replaced by a vector norm; this covers every nth-order equation written as a first-order system, and hence the state equations of any lumped circuit. For a linear system $\mathbf{y}' = \mathbf{A}(t)\mathbf{y} + \mathbf{g}(t)$ with continuous coefficients, f is globally Lipschitz and a solution exists on the whole interval of continuity (Ch. 11, §11.9).
5. *Where it is used.* The theorem justifies the statement in Ch. 11, §11.7 that an RLC circuit's response is fixed by its initial capacitor voltages and inductor currents, and underlies the error analysis of the numerical ODE solvers of Ch. 22.



Picard iterates $y_0, \dots, y_4$ for $y' = y$, $y(0)=1$ (dashed) approaching the fixed point e^t (black); the error of y_k is bounded by $|t|^{k+1}e^{|t|}/(k+1)!$, the factorial being the hallmark of the contraction argument.

D.2 Existence of the Laplace transform

Piecewise continuity and exponential order

A function $x : [0,\infty) \to \mathbb{C}$ is **piecewise continuous** if every bounded interval $[0,\tau]$ contains only finitely many discontinuities of x, each a finite jump. It is of **exponential order** α if there are constants $M > 0$ and $T_0 \ge 0$ with $|x(t)| \le M e^{\alpha t}$ for all $t \ge T_0$.

Existence of the Laplace transform

If x is piecewise continuous on $[0,\infty)$ and of exponential order α, then

$$X(s)=\int_0^{\infty} x(t)\,e^{-st}\,dt$$

converges absolutely for every s with $\operatorname{Re}\{s\}>\alpha$. Moreover the convergence is uniform on every half-plane $\operatorname{Re}\{s\}\ge\sigma_1>\alpha$, X is analytic for $\operatorname{Re}\{s\}>\alpha$, and $X(s)\to 0$ as $\operatorname{Re}\{s\}\to+\infty$.

Proof. Write $s=\sigma+j\omega$, so $|e^{-st}|=e^{-\sigma t}$.

Absolute convergence. On $[0,T_0]$ the function x is piecewise continuous, hence bounded, say $|x|\le B$, and Riemann integrable; so $\int_0^{T_0}|x(t)|e^{-\sigma t}dt\le BT_0\max(1,e^{-\sigma T_0})<\infty$. On $[T_0,\infty)$,

$$\int_{T_0}^{\tau}|x(t)|e^{-\sigma t}\,dt\le M\int_{T_0}^{\tau}e^{-(\sigma-\alpha)t}\,dt\le\frac{M\,e^{-(\sigma-\alpha)T_0}}{\sigma-\alpha}\qquad(\sigma>\alpha),$$

a bound independent of τ. A non-negative integrand with bounded partial integrals has a convergent improper integral, so $\int_0^{\infty}|x|e^{-\sigma t}dt<\infty$, and absolute convergence implies convergence.

Uniformity. If $\sigma\ge\sigma_1>\alpha$ then $|x(t)e^{-st}|\le|x(t)|e^{-\sigma_1 t}$, an integrable majorant independent of s. Hence the tails satisfy $\left|\int_{\tau}^{\infty}xe^{-st}dt\right|\le\int_{\tau}^{\infty}|x|e^{-\sigma_1 t}dt\to 0$ uniformly in s (the Weierstrass M-test for integrals).

Analyticity. Each truncation $X_\tau(s)=\int_0^{\tau}x(t)e^{-st}dt$ is entire: differentiation under the integral sign over a bounded interval is permitted because $\partial_s(xe^{-st})=-t\,x\,e^{-st}$ is continuous in s and bounded on compact s-sets. By the uniformity just proved, $X_\tau\to X$ uniformly on each half-plane $\operatorname{Re}\{s\}\ge\sigma_1$, and a locally uniform limit of analytic functions is analytic (each X_τ integrates to zero around every triangle by §D.14, the limit passes inside the contour integral, and Morera's theorem applies).

Decay. For $\sigma>\max(\alpha,0)$,

$$|X(s)|\le B\int_0^{T_0}e^{-\sigma t}dt+M\int_{T_0}^{\infty}e^{-(\sigma-\alpha)t}dt\le\frac{B}{\sigma}+\frac{M}{\sigma-\alpha}\xrightarrow[\sigma\to\infty]{}0.$$

∎

Remarks.

1. *Sufficient, not necessary.* $x(t)=t^{-1/2}$ is not piecewise continuous at 0 (it is unbounded), yet its transform $\sqrt{\pi/s}$ exists because the singularity is integrable. The theorem extends to any x that is absolutely integrable on each $[0,\tau]$ and of exponential order at infinity, with the same proof.
2. *Growth control is essential.* $x(t)=e^{t^2}$ has no Laplace transform for any s: for real s the integrand $e^{t(t-s)}\to\infty$. More generally, the set of s where the integral converges absolutely is a right half-plane $\operatorname{Re}\{s\}>\sigma_a$ (possibly with its boundary line), because absolute convergence at σ implies it at every $\sigma'>\sigma$ by comparison.
3. *Consequence for rational transforms.* The decay $X(s)\to 0$ shows that a polynomial in s, or $s/(s+1)$, cannot be the transform of an ordinary function; transforms of ordinary functions with rational X are strictly proper. Constants and higher powers of s correspond to impulses and their derivatives at $t=0$.
4. Used in Ch. 14, §14.1.

D.3 The Laplace derivative theorem

Transform of a derivative

Let x be continuous on $[0,\infty)$ and of exponential order α, and let $\dot{x}$ be piecewise continuous on $[0,\infty)$ (at finitely many points in each bounded interval $\dot{x}$ may fail to exist; there x has a corner). Then $\mathcal{L}\{\dot{x}\}$ exists for $\operatorname{Re}\{s\}>\alpha$ and

$$\mathcal{L}\{\dot{x}(t)\}=sX(s)-x(0).$$

If $x,\dot{x},\dots,x^{(n-1)}$ are continuous and of exponential order α and $x^{(n)}$ is piecewise continuous, then

$$\mathcal{L}\{x^{(n)}\}=s^nX(s)-s^{n-1}x(0)-s^{n-2}\dot{x}(0)-\cdots-x^{(n-1)}(0).$$

Proof. Fix s with $\sigma=\operatorname{Re}\{s\}>\alpha$ and $\tau>0$. Let $0=t_0<t_1<\cdots<t_m=\tau$ include all points of $[0,\tau]$ where $\dot{x}$ is discontinuous or undefined. On each $[t_{i-1},t_i]$ the function x is continuously differentiable (with one-sided derivatives at the ends), so integration by parts is valid:

$$\int_{t_{i-1}}^{t_i}\dot{x}(t)e^{-st}\,dt=\Big[x(t)e^{-st}\Big]_{t_{i-1}}^{t_i}+s\int_{t_{i-1}}^{t_i}x(t)e^{-st}\,dt.$$

Summing over i, the boundary terms telescope **because x is continuous** at each interior t_i:

$$\int_0^{\tau}\dot{x}(t)e^{-st}\,dt=x(\tau)e^{-s\tau}-x(0)+s\int_0^{\tau}x(t)e^{-st}\,dt.$$

Now let $\tau\to\infty$. For $\tau\ge T_0$, $|x(\tau)e^{-s\tau}|\le Me^{-(\sigma-\alpha)\tau}\to 0$, and the last integral tends to $X(s)$ by §D.2. The right-hand side therefore has the limit $sX(s)-x(0)$, so the improper integral on the left converges and equals it. The nth-order formula follows by induction: applying the first-order result to $x^{(n-1)}$ gives $\mathcal{L}\{x^{(n)}\}=s\,\mathcal{L}\{x^{(n-1)}\}-x^{(n-1)}(0)$, and substituting the formula for $\mathcal{L}\{x^{(n-1)}\}$ yields the stated sum. ∎

Remarks.

1. *Continuity matters.* If x has a jump of size J at $t_1 > 0$, the boundary terms fail to telescope and an extra term $-Je^{-st_1}$ appears: the classical derivative misses the impulse $J\delta(t-t_1)$ that the generalized derivative contains. For a jump at the origin the book uses the 0^- convention, $\mathcal{L}\{\dot{x}\} = sX(s) - x(0^-)$, in which $\dot{x}$ includes the impulse $[x(0^+) - x(0^-)]\delta(t)$; the proof above with $x(0)$ replaced by $x(0^+)$ gives the 0^+ form, and the two differ exactly by that impulse (Ch. 14, §14.2).
2. *$\dot{x}$ need not be of exponential order.* The proof never bounds $\dot{x}$; the existence of $\mathcal{L}\{\dot{x}\}$, possibly as a conditionally convergent integral, is a conclusion. Example: $x(t) = \sin(e^{t^2})$ is bounded, so $\mathcal{L}\{x\}$ exists for $\mathrm{Re}\{s\} > 0$, while $\dot{x}(t) = 2te^{t^2}\cos(e^{t^2})$ grows faster than any exponential; nevertheless $\int_0^\infty \dot{x}\, e^{-st}dt$ converges for $\mathrm{Re}\{s\} > 0$.
3. This is the result that turns a linear constant-coefficient ODE with initial conditions into an algebraic equation in s (Ch. 14, §14.8).

D.4 The Laplace convolution theorem

Convolution theorem (Laplace)

Let f and g be piecewise continuous on $[0,\infty)$ and of exponential order α, and define

$$(f*g)(t) = \int_0^t f(\tau)\, g(t-\tau)\, d\tau \qquad (t \geq 0).$$

Then $f*g$ is continuous, of exponential order $\alpha + \varepsilon$ for every $\varepsilon > 0$, and for $\mathrm{Re}\{s\} > \alpha$

$$\mathcal{L}\{f*g\}(s) = F(s)\, G(s).$$

Proof. *Growth of $f*g$.* Enlarging M if necessary, we may take $|f(t)|, |g(t)| \leq Me^{\alpha t}$ for **all** $t \geq 0$ (on $[0,T_0]$ both functions are bounded and $e^{\alpha t}$ is bounded below). Then

$$|(f*g)(t)| \leq M^2 \int_0^t e^{\alpha\tau} e^{\alpha(t-\tau)} d\tau = M^2\, t\, e^{\alpha t} \leq \frac{M^2}{\varepsilon e}\, e^{(\alpha+\varepsilon)t},$$

using $te^{-\varepsilon t} \leq 1/(\varepsilon e)$. For continuity, fix $\tau_0 > 0$ and $t, t' \in [0,\tau_0]$; extending g by 0 on $t<0$, $(f*g)(t') - (f*g)(t) = \int_0^{\tau_0} f(\tau)[g(t'-\tau) - g(t-\tau)]d\tau$. As $t' \to t$ the bracket tends to 0 for all but finitely many τ (the points where $t-\tau$ is a jump of g) and is bounded, so the integral tends to 0 by dominated convergence. Hence $\mathcal{L}\{f*g\}$ exists for $\mathrm{Re}\{s\} > \alpha$ (choose $\varepsilon < \mathrm{Re}\{s\} - \alpha$).

The product formula. Fix s with $\sigma = \mathrm{Re}\{s\} > \alpha$. With $g = 0$ on $t<0$, $(f*g)(t) = \int_0^\infty f(\tau)g(t-\tau)d\tau$, so

$$\mathcal{L}\{f*g\}(s) = \int_0^\infty \int_0^\infty f(\tau)\, g(t-\tau)\, e^{-st}\, d\tau\, dt.$$

For the absolute integrand, Tonelli's theorem (iterated integrals of a non-negative function agree in either order) gives

$$\int_0^\infty \int_0^\infty |f(\tau)|\, |g(t-\tau)|\, e^{-\sigma t}\, dt\, d\tau = \int_0^\infty |f(\tau)| e^{-\sigma\tau}\, d\tau \cdot \int_0^\infty |g(u)| e^{-\sigma u}\, du < \infty,$$

where we substituted $u = t - \tau$ in the inner integral and used the absolute convergence of §D.2 for both factors. So Fubini's theorem allows the order of integration to be exchanged in the signed integral, and the same substitution gives

$$\mathcal{L}\{f*g\}(s) = \int_0^\infty f(\tau) e^{-s\tau} \left(\int_0^\infty g(u) e^{-su}\, du \right) d\tau = F(s)\, G(s).$$

∎

Remarks.

1. *Region of validity.* The formula holds on the intersection of the half-planes of absolute convergence of F and G. Absolute convergence is exactly the hypothesis that licenses Fubini; with only conditional convergence the exchange of integrals can fail.
2. *Why it matters.* With $g = h$ the impulse response of a causal LTI system and f its input, the theorem says the output transform is $H(s)F(s)$ — the definition of the transfer function. Geometrically, the region $0 \leq \tau \leq t$ of the (τ, t) plane can be swept by vertical lines (convolve first) or horizontal lines (transform first). See Ch. 14, §14.6.

D.5 Initial- and final-value theorems

Initial-value theorem (IVT)

Let x be continuous on $(0,\infty)$ with a finite right limit $x(0^+)$ and of exponential order, and let $\dot{x}$ be piecewise continuous and of exponential order α on $(0,\infty)$. Then, along real $s \to +\infty$,

$$\lim_{s \to +\infty} sX(s) = x(0^+).$$

Final-value theorem (FVT)

(a) Abelian form. If x is piecewise continuous on $[0,\infty)$ and the limit $x(\infty)=\lim_{t\to\infty}x(t)$ exists and is finite, then $X(s)$ exists for $\operatorname{Re}\{s\}>0$ and

$$\lim_{s\to 0^+} sX(s) = x(\infty) \qquad (s \text{ real}).$$

(b) Rational form. If $X(s)$ is a strictly proper rational function and **every pole of $sX(s)$ lies in the open left half-plane** $\operatorname{Re}\{s\}<0$, then $x(\infty)$ exists and equals $\lim_{s\to 0} sX(s)$.

Proof of the IVT. Redefine $x(0)=x(0^+)$; then x is continuous on $[0,\infty)$ and §D.3 gives, for real $s>\alpha$,

$$sX(s)-x(0^+)=\int_0^\infty \dot{x}(t)\,e^{-st}\,dt=\mathcal{L}\{\dot{x}\}(s).$$

Because $\dot{x}$ is piecewise continuous and of exponential order, §D.2 applies to $\dot{x}$ itself and gives $\mathcal{L}\{\dot{x}\}(s)\to 0$ as $s\to+\infty$. ■

Proof of FVT (a). Write $L=x(\infty)$. Since x converges it is bounded for large t, and it is bounded on any $[0,T]$ by piecewise continuity; so it is of exponential order 0 and $X(s)$ exists for $\operatorname{Re}\{s\}>0$. Using $s\int_0^\infty e^{-st}dt=1$ for $s>0$,

$$sX(s)-L=s\int_0^\infty \big(x(t)-L\big)e^{-st}\,dt.$$

Given $\varepsilon>0$ pick T with $|x(t)-L|<\varepsilon$ for $t>T$, and let $B=\sup_{[0,T]}|x-L|<\infty$. Then for $s>0$

$$|sX(s)-L|\le s\int_0^T B\,dt + s\int_T^\infty \varepsilon\,e^{-st}dt \le sBT+\varepsilon.$$

Letting $s\to 0^+$ gives $\limsup|sX(s)-L|\le\varepsilon$, and ε was arbitrary. ■

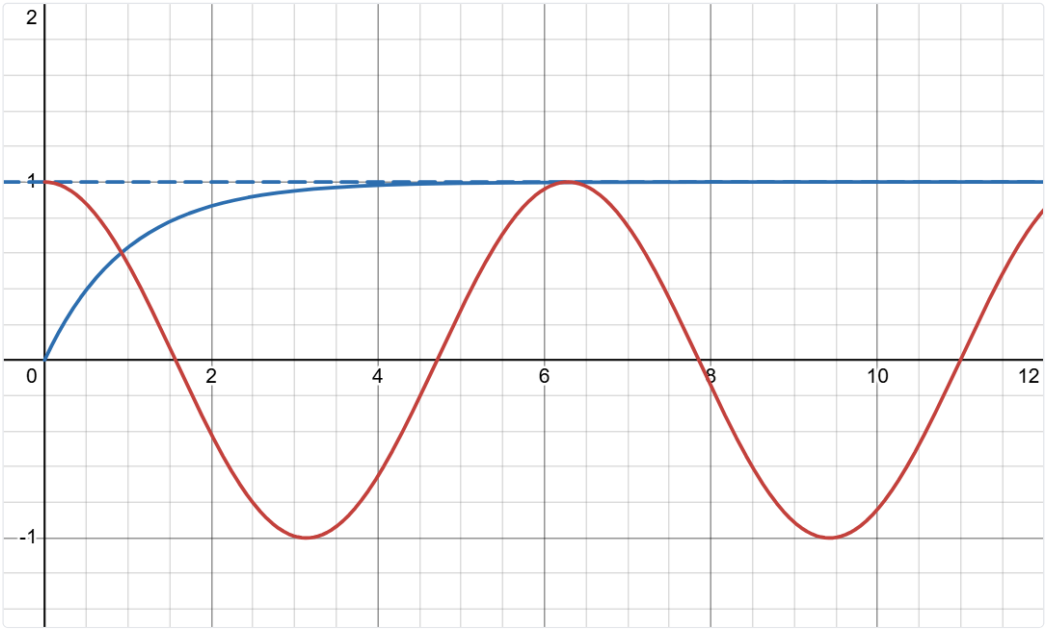
Proof of FVT (b). Let $X=N/D$ be strictly proper with distinct poles $p_1,\dots,p_r$ of multiplicities $m_1,\dots,m_r$. The partial-fraction expansion (Ch. 14, §14.5) and uniqueness of the inverse transform give

$$x(t)=\sum_{i=1}^{r}\sum_{k=1}^{m_i}\frac{A_{ik}}{(k-1)!}\,t^{k-1}e^{p_i t} \qquad (t>0).$$

The poles of $sX(s)$ are those of X, except that a pole of X at 0 has its order lowered by one (and disappears if it was simple). The hypothesis therefore says: every $p_i\neq 0$ has $\operatorname{Re}\{p_i\}<0$, and 0 is at most a *simple* pole of X. Each term with $\operatorname{Re}\{p_i\}<0$ tends to zero, since $t^{k-1}e^{\operatorname{Re}\{p_i\}t}\to 0$. If 0 is a simple pole, the remaining term is the constant $A_0=\lim_{s\to0}sX(s)$ (the residue of X at 0); if 0 is not a pole, there is no such term and $\lim_{s\to0}sX(s)=0\cdot X(0)=0$. In either case $x(t)\to\lim_{s\to0}sX(s)$. ■

Remarks — when the FVT fails.

1. *Poles on the imaginary axis.* $x(t)=\cos t$ has $X(s)=s/(s^2+1)$ and $sX(s)=s^2/(s^2+1)\to 0$ as $s\to 0$, but $\cos t$ has no limit. Hypothesis (b) fails because $sX(s)$ has poles at $\pm j$. The Abelian form (a) is not contradicted: it assumes the limit exists.
2. *Poles in the right half-plane.* $x(t)=e^{t}$ has $sX(s)=s/(s-1)\to 0$ as $s\to 0$, a meaningless "final value" for a signal that diverges. The Laplace integral does not converge near $s=0$; the formula $s/(s-1)$ there is an analytic continuation, which is why blind substitution misleads.
3. *Double pole at the origin.* $x(t)=t$ gives $sX(s)=1/s\to\infty$, correctly signalling divergence; hypothesis (b) fails because $sX(s)$ has a pole at 0.
4. *The converse of (a) is false.* That $sX(s)$ has a limit as $s\to0^+$ does not imply that $x(t)$ converges (Remark 1). Converse results need extra "Tauberian" hypotheses — for instance $t\,\dot{x}(t)$ bounded below (Hardy–Littlewood); for rational X, hypothesis (b) plays this role.
5. *IVT hypotheses.* If x contains an impulse at 0 then X is not strictly proper (e.g. $X=1+1/(s+1)$) and $sX(s)\to\infty$; the IVT applies to the regular part only. The limit in the IVT may be taken along any path with $\operatorname{Re}\{s\}\to\infty$, by the uniform estimate of §D.2.
6. Used in Ch. 14, §14.7 for steady-state errors of control loops and DC gains.



Blue: $x = 1 - e^{-t}$, $X = 1/[s(s+1)]$, whose sX has its only pole at -1, so the FVT correctly gives $x(\infty) = 1$. Red: $\cos t$, for which $sX(s) \to 0$ although no final value exists — the poles $\pm j$ violate the hypothesis.

D.6 Fourier coefficients from orthogonality; Bessel's inequality

Throughout §§D.6–D.9, x is T_0-periodic, $\omega_0 = 2\pi/T_0$, and $\int_{T_0}$ denotes integration over any interval of length T_0. For functions square-integrable over a period we use the normalised inner product and norm of Ch. 15, §15.1,

$$\langle x, y \rangle = \frac{1}{T_0}\int_{T_0} x(t)\, y^{*}(t)\, dt, \qquad \|x\|^2 = \frac{1}{T_0}\int_{T_0} |x(t)|^2 dt,$$

so $\|x\|^2$ is the mean-square value (the power) of x. Write $\varphi_n(t) = e^{jn\omega_0 t}$.

Orthonormality and the coefficient formula

(i) $\langle \varphi_m, \varphi_n \rangle = \delta_{mn}$ for all integers m, n.

(ii) If a series $\sum_{n=-\infty}^{\infty} d_n \varphi_n(t)$ converges **uniformly** on a period to $x(t)$, then necessarily

$$d_n = c_n := \langle x, \varphi_n \rangle = \frac{1}{T_0}\int_{T_0} x(t)\, e^{-jn\omega_0 t}\, dt,$$

and for real x the trigonometric coefficients are $a_n = 2\,\mathrm{Re}\{c_n\} = \frac{2}{T_0}\int_{T_0} x\cos n\omega_0 t\, dt$, $b_n = -2\,\mathrm{Im}\{c_n\} = \frac{2}{T_0}\int_{T_0} x \sin n\omega_0 t\, dt$.

Bessel's inequality

If $\int_{T_0}|x|^2 dt < \infty$ and $S_N = \sum_{n=-N}^{N} c_n \varphi_n$, then for every N

$$\|x - S_N\|^2 = \|x\|^2 - \sum_{n=-N}^{N} |c_n|^2 \;\geq 0, \qquad \text{hence} \qquad \sum_{n=-\infty}^{\infty} |c_n|^2 \leq \frac{1}{T_0}\int_{T_0} |x(t)|^2 dt.$$

In particular $c_n \to 0$ as $|n| \to \infty$. Moreover S_N is the unique trigonometric polynomial of degree $\leq N$ closest to x in mean square.

Proof. (i) For $m = n$ the integrand is 1. For $k = m - n \neq 0$, $\int_{t_1}^{t_1+T_0} e^{jk\omega_0 t} dt = e^{jk\omega_0 t_1}\big(e^{j2\pi k} - 1\big)/(jk\omega_0) = 0$.

(ii) Multiply $x(t) = \sum_m d_m \varphi_m(t)$ by the bounded function $\varphi_n^{*}(t)$; the product series still converges uniformly, so it may be integrated term by term (the integral of a uniform limit over a bounded interval is the limit of the integrals). By (i), $\langle x, \varphi_n\rangle = \sum_m d_m \langle \varphi_m, \varphi_n\rangle = d_n$. The real formulas follow from $\cos n\omega_0 t = \tfrac{1}{2}(\varphi_n + \varphi_{-n})$ and $\sin n\omega_0 t = \tfrac{1}{2j}(\varphi_n - \varphi_{-n})$.

Bessel. Let $p = \sum_{|n|\leq N} d_n \varphi_n$ be any trigonometric polynomial. Expanding the inner product and using (i),

$$\begin{aligned} \|x - p\|^2 &= \|x\|^2 - \sum_{|n|\leq N}\big(d_n^{*} c_n + d_n c_n^{*}\big) + \sum_{|n|\leq N} |d_n|^2 \\ &= \|x\|^2 - \sum_{|n|\leq N} |c_n|^2 + \sum_{|n|\leq N} |d_n - c_n|^2, \end{aligned}$$

the second line by completing the square, $|d_n - c_n|^2 = |d_n|^2 - d_n^{*}c_n - d_n c_n^{*} + |c_n|^2$. The last sum is ≥ 0 and vanishes only for $d_n = c_n$; this proves the best-approximation property. Taking $d_n = c_n$ gives the identity $\|x - S_N\|^2 = \|x\|^2 - \sum_{|n|\leq N}|c_n|^2$, whose left side is non-negative. The partial sums of the non-negative series $\sum |c_n|^2$ are therefore bounded by $\|x\|^2$, so the series converges with sum $\leq \|x\|^2$, and its terms tend to zero. $\blacksquare$

Remarks.

1. Part (ii) explains the *form* of the coefficient formula; the hypothesis of uniform convergence is there only so that the term-by-term integration is legitimate. For an arbitrary square-integrable x we *define* c_n by the formula and ask afterwards whether the series represents x — the subject of §§D.7–D.8.
2. Bessel's inequality is a statement about geometry, not Fourier analysis: it holds for any orthonormal set in any inner-product space, because S_N is the orthogonal projection of x on the span of $\varphi_{-N}, \dots, \varphi_N$ and projection cannot increase length. What is special about the trigonometric system is that equality holds (completeness), proved next.
3. The decay $c_n \to 0$ for square-integrable x is the Fourier-series form of the Riemann–Lebesgue lemma; §D.8 proves it for merely absolutely integrable x.
4. Used in Ch. 15, §15.1 and §15.7.

D.7 Parseval's theorem for Fourier series

Parseval's theorem

If x is T_0-periodic and $\int_{T_0}|x|^2 dt < \infty$, then $\|x - S_N\| \to 0$ as $N \to \infty$ and

$$\frac{1}{T_0}\int_{T_0} |x(t)|^2\, dt = \sum_{n=-\infty}^{\infty} |c_n|^2.$$

More generally $\langle x, y\rangle = \sum_n c_n\, d_n^{*}$ for two such functions with coefficients c_n, d_n.

By the identity in Bessel's inequality, Parseval's equality is *equivalent* to $\|x - S_N\| \to 0$. We prove that by approximating x first by a continuous function and then by Fejér means, which converge uniformly.

Fejér means and Fejér kernel

Take $T_0 = 2\pi$ (the general case follows by the change of variable $t \mapsto \omega_0 t$). The **Fejér mean** is the average of the first $N+1$ partial sums, $\sigma_N = \frac{1}{N+1}\sum_{k=0}^{N} S_k$. Since $S_k(t) = \frac{1}{2\pi}\int_{-\pi}^{\pi} x(t-u)D_k(u)\,du$ with the Dirichlet kernel $D_k(u) = \sum_{n=-k}^{k} e^{jnu}$ (§D.8),

$$\sigma_N(t) = \frac{1}{2\pi}\int_{-\pi}^{\pi} x(t-u)\,F_N(u)\,du, \qquad F_N(u) = \frac{1}{N+1}\sum_{k=0}^{N} D_k(u) = \frac{1}{N+1}\left(\frac{\sin\big((N+1)u/2\big)}{\sin(u/2)}\right)^2.$$

Lemma (closed form and properties of F_N). (a) the closed form above holds; (b) $F_N \ge 0$; (c) $\frac{1}{2\pi}\int_{-\pi}^{\pi} F_N = 1$; (d) for $0 < \delta \le |u| \le \pi$, $F_N(u) \le \dfrac{1}{(N+1)\sin^2(\delta/2)}$.

Proof. From §D.8, $D_k(u) = \sin\big((k+\tfrac12)u\big)/\sin(u/2)$. Using $2\sin(u/2)\sin\big((k+\tfrac12)u\big) = \cos ku - \cos(k+1)u$ and telescoping,

$$\sum_{k=0}^{N} D_k(u) = \frac{1-\cos(N+1)u}{2\sin^2(u/2)} = \frac{\sin^2\big((N+1)u/2\big)}{\sin^2(u/2)},$$

which is (a), and (b) is evident. Each D_k has mean value 1 over a period (only the $n=0$ term survives integration), so (c) follows. For (d), the numerator is at most 1 and $\sin^2(u/2) \ge \sin^2(\delta/2)$ on $\delta \le |u| \le \pi$. $\square$

Fejér's theorem. *If x is continuous and 2π-periodic, then $\sigma_N \to x$ uniformly.*

Proof. By (c), $\sigma_N(t) - x(t) = \frac{1}{2\pi}\int_{-\pi}^{\pi}\big[x(t-u)-x(t)\big]F_N(u)\,du$. A continuous periodic function is bounded, $|x| \le B$, and uniformly continuous: given $\varepsilon > 0$ there is $\delta > 0$ with $|x(t-u)-x(t)| < \varepsilon$ whenever $|u| < \delta$, for all t. Splitting the integral and using (b), (c), (d),

$$|\sigma_N(t) - x(t)| \le \frac{\varepsilon}{2\pi}\int_{|u|<\delta} F_N + \frac{2B}{2\pi}\int_{\delta\le|u|\le\pi} F_N \le \varepsilon + \frac{2B}{(N+1)\sin^2(\delta/2)},$$

uniformly in t. The last term is $< \varepsilon$ for N large. $\square$

Proof of Parseval's theorem.

Step 1 — continuous x. σ_N is a trigonometric polynomial of degree $\le N$. By the best-approximation property of §D.6 and Fejér's theorem,

$$\|x - S_N\| \le \|x - \sigma_N\| \le \max_t |x(t) - \sigma_N(t)| \xrightarrow[N\to\infty]{} 0,$$

the middle inequality because a mean-square value never exceeds the square of the maximum.

Step 2 — general square-integrable x. Continuous periodic functions are dense in mean square: for every $\varepsilon > 0$ there is a continuous periodic g with $\|x - g\| < \varepsilon$. (For Riemann-integrable x this is elementary: choose a step function ψ from a lower Darboux sum with $\frac{1}{T_0}\int|x-\psi|$ small and $|\psi| \le \sup|x|$, so that $\|x-\psi\|^2 \le 2\sup|x| \cdot \frac{1}{T_0}\int|x-\psi|$ is small, then replace each jump of ψ by a steep linear ramp, which changes $\|\cdot\|$ by an arbitrarily small amount; unbounded x are first truncated at a large level. For Lebesgue-square-integrable x it is a standard theorem of measure theory.) Let S_N^g be the partial sums of g. Since $x \mapsto S_N$ is an orthogonal projection, $\|S_N - S_N^g\| \le \|x - g\|$. Then

$$\|x - S_N\| \le \|x - g\| + \|g - S_N^g\| + \|S_N^g - S_N\| < 2\varepsilon + \|g - S_N^g\|,$$

and the last term tends to zero by Step 1. Hence $\limsup_N \|x - S_N\| \le 2\varepsilon$ for every ε.

Step 3 — the identity. By §D.6, $\|x\|^2 - \sum_{|n|\le N}|c_n|^2 = \|x - S_N\|^2 \to 0$.

Step 4 — the inner-product form. Apply the identity to $x + y$, $x - y$, $x + jy$, $x - jy$ and combine with the polarization identity $\langle x, y\rangle = \frac14\sum_{k=0}^{3} j^k\|x + j^k y\|^2$; the coefficient sequences combine the same way. ■

Remarks.

1. *Physical reading.* The left side is the average power of x; the right side distributes it over harmonics, $|c_n|^2 + |c_{-n}|^2$ being the power at frequency $n\omega_0$. For real x this is $\big(\frac{a_0}{2}\big)^2 + \frac12\sum_{n\ge1}(a_n^2 + b_n^2)$. Check: the unit square wave (±1) has $b_n = 4/(n\pi)$ for odd n and mean-square value 1; numerically $\frac12\sum_{n\ \text{odd}} 16/(n\pi)^2 = 1.0000$ (summing 10^5 terms gives 0.999998), equivalently $\sum_{n\ \text{odd}} 1/n^2 = \pi^2/8$.
2. *What Parseval does not say.* Mean-square convergence does not give convergence at any particular point: there are continuous functions whose Fourier series diverge at a point (du Bois-Reymond, 1873). Pointwise statements need the extra regularity of §D.8.
3. *Fejér versus Dirichlet.* F_N is non-negative while D_N oscillates in sign with $\frac{1}{2\pi}\int|D_N| \sim \frac{4}{\pi^2}\ln N \to \infty$; this is exactly why Cesàro averaging cures the Gibbs overshoot (Ch. 15, §15.6, "Taming Gibbs").
4. Used for RMS values, THD and power factor in Ch. 15, §15.8, and in §D.11 below.

D.8 Dirichlet's pointwise convergence theorem

Dirichlet kernel

For $T_0 = 2\pi$ and $N \ge 0$,

$$D_N(u) = \sum_{n=-N}^{N} e^{jnu} = \frac{\sin\big((N+\tfrac12)u\big)}{\sin(u/2)} \quad (u \notin 2\pi\mathbb{Z}), \qquad D_N(2\pi k) = 2N+1.$$

The closed form follows from the geometric sum: $(e^{ju/2} - e^{-ju/2})D_N(u) = e^{j(N+\frac12)u} - e^{-j(N+\frac12)u}$. D_N is even, 2π-periodic, and $\frac{1}{2\pi}\int_{-\pi}^{\pi} D_N(u)\,du = 1$, so $\frac{1}{2\pi}\int_0^{\pi} D_N = \frac12$ by evenness.

Riemann–Lebesgue lemma

If g is absolutely integrable on an interval (a, b) (bounded or not, Riemann-improper or Lebesgue), then

$$\lim_{\lambda \to \pm\infty} \int_a^b g(u)\, e^{j\lambda u}\, du = 0,$$

and hence also $\int_a^b g(u) \sin \lambda u \, du \to 0$ and $\int_a^b g(u) \cos \lambda u \, du \to 0$.

Proof of Riemann–Lebesgue. *Step functions.* For the indicator of an interval $[\alpha, \beta]$, $\left| \int_\alpha^\beta e^{j\lambda u} du \right| = |e^{j\lambda\beta} - e^{j\lambda\alpha}|/|\lambda| \le 2/|\lambda| \to 0$; by linearity the result holds for every step function (finite combination of indicators).

Approximation. Given $\varepsilon > 0$ there is a step function ψ with $\int_a^b |g - \psi| < \varepsilon$. (For a Riemann-integrable g on a bounded interval take ψ from a lower Darboux sum close to the integral, treating real and imaginary parts separately; for an improper integral first cut off a neighbourhood of each singularity and of infinity where $\int |g| < \varepsilon/2$; for Lebesgue-integrable g density of step functions is standard.) Then

$$\left| \int_a^b g\, e^{j\lambda u} du \right| \le \int_a^b |g - \psi|\, du + \left| \int_a^b \psi\, e^{j\lambda u} du \right| < \varepsilon + \left| \int_a^b \psi\, e^{j\lambda u} du \right|,$$

and the last term tends to zero. So the lim sup is at most ε for every ε. The sine and cosine versions are the imaginary and real parts when g is real, and linear combinations of λ and $-\lambda$ in general. ∎

Dirichlet's theorem (pointwise convergence)

Let x be 2π-periodic and absolutely integrable over a period. Fix t and suppose the one-sided limits $x(t^+), x(t^-)$ exist and that x satisfies one-sided Lipschitz (Hölder-type) bounds at t: there are $C, \delta > 0$ with

$$|x(t+u) - x(t^+)| \le C u, \qquad |x(t-u) - x(t^-)| \le C u \qquad (0 < u < \delta).$$

Then

$$\lim_{N \to \infty} S_N(t) = \frac{x(t^+) + x(t^-)}{2}.$$

In particular this holds at every point when x is piecewise smooth (piecewise C^1 with one-sided limits of x and x' at the break points), which covers the Dirichlet conditions as used in Ch. 15, §15.6 for all practical waveforms.

Proof. *Step 1 — the partial sum as a convolution.* Inserting the coefficient formula into S_N and exchanging the finite sum with the integral,

$$S_N(t) = \sum_{n=-N}^{N} \frac{1}{2\pi} \int_{-\pi}^{\pi} x(\tau) e^{-jn\tau} d\tau \; e^{jnt} = \frac{1}{2\pi} \int_{-\pi}^{\pi} x(\tau)\, D_N(t - \tau)\, d\tau = \frac{1}{2\pi} \int_{-\pi}^{\pi} x(t+u)\, D_N(u)\, du,$$

where we substituted $\tau = t + u$, used the evenness of D_N, and used periodicity to keep the interval $[-\pi, \pi]$.

Step 2 — fold onto $(0, \pi)$. Splitting at $u = 0$ and replacing u by $-u$ on $(-\pi, 0)$,

$$S_N(t) = \frac{1}{2\pi} \int_0^{\pi} \big[x(t+u) + x(t-u) \big] D_N(u)\, du.$$

Since $\frac{1}{2\pi} \int_0^\pi D_N = \frac{1}{2}$, subtracting $\frac{1}{2}[x(t^+) + x(t^-)]$ gives

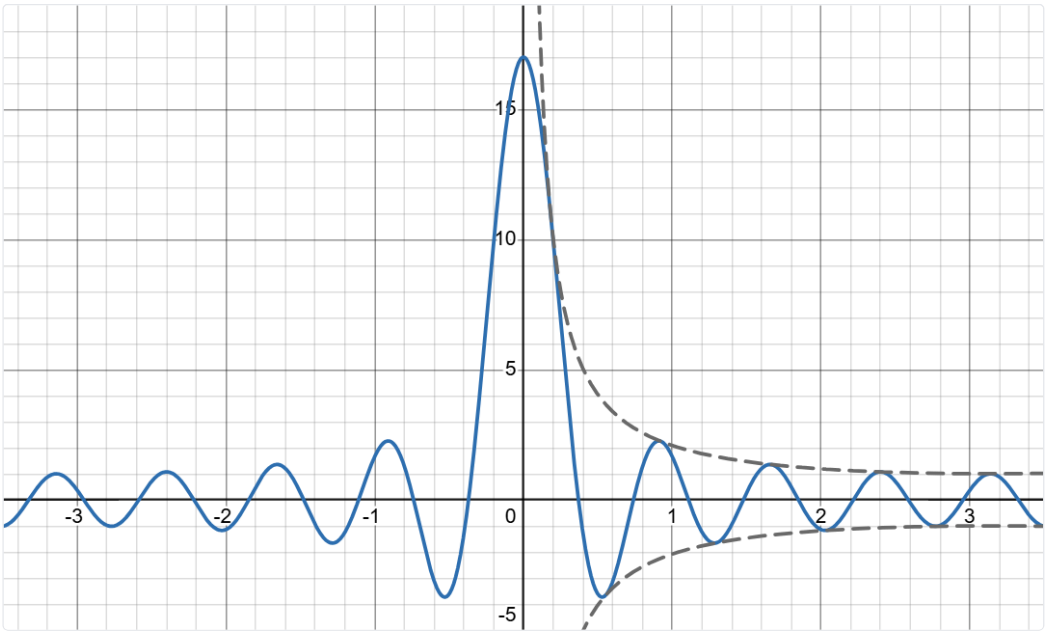
$$S_N(t) - \frac{x(t^+) + x(t^-)}{2} = \frac{1}{2\pi} \int_0^{\pi} g(u)\, \sin\big((N + \tfrac{1}{2})u\big)\, du, \qquad g(u) = \frac{x(t+u) - x(t^+) + x(t-u) - x(t^-)}{\sin(u/2)}.$$

Step 3 — g is absolutely integrable on $(0, \pi)$. On $(0, \delta)$, the numerator is at most $2Cu$ in modulus and $\sin(u/2) \ge u/\pi$ (Jordan's inequality, §D.16), so $|g(u)| \le 2\pi C$: bounded. On $[\delta, \pi]$, $\sin(u/2) \ge \sin(\delta/2) > 0$, so $|g| \le \big(|x(t+u)| + |x(t-u)| + |x(t^+)| + |x(t^-)|\big)/\sin(\delta/2)$, which is integrable because x is absolutely integrable over a period.

Step 4 — conclude. By the Riemann–Lebesgue lemma with $\lambda = N + \frac{1}{2} \to \infty$, the integral tends to zero. ∎

Remarks.

1. *Localisation.* The proof shows that convergence at t depends only on the behaviour of x in an arbitrarily small neighbourhood of t (the part of the integral over $[\delta, \pi]$ always tends to zero). Two functions that agree near t have Fourier series that converge or diverge together at t, even though their coefficients may be completely different.
2. *Why a regularity hypothesis is needed.* Continuity alone is not enough: there are continuous periodic functions whose Fourier series diverges at a point (Remark 2 of §D.7). The Lipschitz bound can be weakened to Dini's condition $\int_0^\delta |x(t \pm u) - x(t^\pm)|\, du/u < \infty$, or replaced by bounded variation near t (the Dirichlet–Jordan theorem, the version stated in Ch. 15 with "finitely many maxima and minima"); both proofs follow the same three steps.
3. *Midpoint value.* At a jump the series converges to the average because D_N is even: it weights the two sides of the jump equally.
4. *Uniform convergence.* If x is continuous and piecewise smooth, then $\sum |c_n| < \infty$ (since $c_n = c'_n/(jn)$ where c'_n are the coefficients of x', and Cauchy–Schwarz with Bessel for x' gives $\sum |c_n| \le |c_0| + (\sum |c'_n|^2)^{1/2} (2 \sum n^{-2})^{1/2}$), so by the M-test the series converges uniformly — necessarily to x, by Dirichlet's theorem.



The Dirichlet kernel $D_8(u) = \sin(8.5u)/\sin(u/2)$: a central peak of height $2N+1 = 17$ and width $\approx 2\pi/(N+\frac{1}{2})$, with side-lobes confined by the envelope $\pm 1/\sin(u/2)$ (dashed). Its area is fixed but its absolute area grows like $\ln N$ — the source of both localisation and Gibbs.

D.9 The Gibbs overshoot

Sine integral

$\operatorname{Si}(x) = \displaystyle\int_0^x \frac{\sin v}{v}\,dv$. Numerically $\operatorname{Si}(\pi) = 1.851937\ldots$ and $\operatorname{Si}(\infty) = \pi/2$.

Gibbs phenomenon for the square wave

Let x be the 2π-periodic square wave equal to 1 on $(0,\pi)$ and -1 on $(-\pi,0)$, a jump of size $J = 2$ at $t = 0$. Its Fourier series is $\frac{4}{\pi}\sum_{k\ge 1}\frac{\sin(2k-1)t}{2k-1}$; let S_m denote the sum of the first m terms. Then:

1. $S_m(t) = \dfrac{2}{\pi}\displaystyle\int_0^t \frac{\sin 2mu}{\sin u}\,du$;
2. on $(0,\pi)$ the maximum of S_m is attained at $t_m = \dfrac{\pi}{2m}$;
3. $\displaystyle\lim_{m\to\infty} S_m(t_m) = \frac{2}{\pi}\operatorname{Si}(\pi) = 1.178980\ldots$

So the overshoot above the value 1 tends to $\left(\frac{1}{\pi}\operatorname{Si}(\pi) - \frac{1}{2}\right)J = 0.0894899\,J$, about **8.95% of the jump**, while its location $t_m \to 0$.

Proof. *1.* Differentiating term by term (a finite sum),

$$S_m'(t) = \frac{4}{\pi}\sum_{k=1}^{m}\cos(2k-1)t = \frac{2}{\pi}\,\frac{\sin 2mt}{\sin t} \qquad (t \notin \pi\mathbb{Z}),$$

because $2\sin t\cos(2k-1)t = \sin 2kt - \sin(2k-2)t$, which telescopes to $\sin 2mt$. The quotient extends continuously to $t = 0$ (value $4m/\pi$), and $S_m(0) = 0$, so integration from 0 gives the formula.

2. On $(0,\pi)$ the critical points are the zeros of $\sin 2mt$, $t = k\pi/(2m)$, $k = 1,\ldots,2m-1$. Since $\sin t > 0$ there, S_m' has the sign of $\sin 2mt$: positive on $(0,\pi/2m)$, then alternating. So the local maxima are at $\tau_i = (2i-1)\pi/(2m)$. Between consecutive maxima,

$$S_m(\tau_{i+1}) - S_m(\tau_i) = \frac{2}{\pi}\left(\int_{\tau_i}^{i\pi/m} + \int_{i\pi/m}^{\tau_{i+1}}\right)\frac{\sin 2mu}{\sin u}\,du.$$

On the first sub-interval $\sin 2mu \le 0$, on the second $\sin 2mu \ge 0$, and the two half-lobes of $|\sin 2mu|$ are translates of each other by $\pi/(2m)$. If $\tau_{i+1} \le \pi/2$, the weight $1/\sin u$ is strictly decreasing over both, so the negative half-lobe outweighs the positive one and $S_m(\tau_{i+1}) < S_m(\tau_i)$. Thus the maxima decrease on $(0,\pi/2]$. Finally $S_m(\pi - t) = S_m(t)$, because $\sin\big((2k-1)(\pi-t)\big) = \sin(2k-1)t$, so the maxima on $[\pi/2,\pi)$ mirror those on $(0,\pi/2]$. Hence the largest is $S_m(\tau_1) = S_m(\pi/2m)$.

3. Substitute $v = 2mu$:

$$S_m\Big(\frac{\pi}{2m}\Big) = \frac{2}{\pi}\int_0^{\pi}\frac{\sin v}{2m\sin\left(v/2m\right)}\,dv.$$

For $0 < v \le \pi$, $2m\sin(v/2m) \to v$ as $m\to\infty$, and by Jordan's inequality $2m\sin(v/2m) \ge 2m\cdot\frac{2}{\pi}\cdot\frac{v}{2m} = \frac{2v}{\pi}$, so the integrand is bounded by $\frac{\pi}{2}\frac{\sin v}{v} \le \frac{\pi}{2}$. Dominated convergence (or, elementarily, uniform convergence of the integrand on $[0,\pi]$) gives

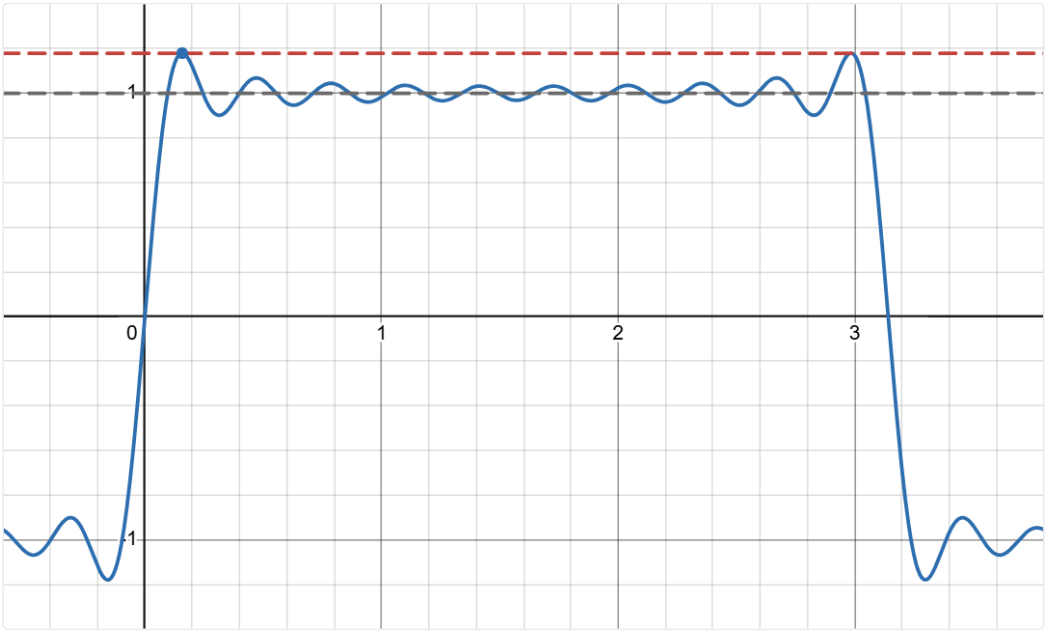
$$\lim_{m\to\infty} S_m\Big(\frac{\pi}{2m}\Big) = \frac{2}{\pi}\int_0^{\pi}\frac{\sin v}{v}\,dv = \frac{2}{\pi}\,\mathrm{Si}(\pi).$$

Numerically $\frac{2}{\pi}\,\mathrm{Si}(\pi) = 1.1789797$, an overshoot of $0.1789797 = 0.0894899 \times J$ with $J = 2$. ■

Numerical check. Direct summation gives $S_{10}(\pi/20) = 1.179814$, $S_{100}(\pi/200) = 1.178988$, $S_{1000}(\pi/2000) = 1.178980$; a dense grid search on $(0,\pi)$ for $m = 10$ confirms the global maximum at $t = 0.15708 = \pi/20$.

Remarks.

1. *General jumps.* If x is piecewise smooth with a jump $J = x(t_0^+) - x(t_0^-)$ at t_0, then $x - \frac{J}{2}\,q(t - t_0)$, with q the square wave above, is continuous and piecewise smooth near t_0; its Fourier series converges uniformly there (§D.8, Remark 4), so it contributes no overshoot, and the overshoot of x is $\frac{J}{2} \times 0.1789797 = 0.0895\,J$ on each side.
2. *Consistency with convergence theorems.* The peak moves to the jump as $m \to \infty$, so at each fixed $t \neq t_0$ the series converges (Dirichlet), and the overshoot region has width $O(1/m)$, so its contribution to the mean-square error vanishes (Parseval). What fails is *uniform* convergence: $\sup_t |S_m - x| \not\to 0$. No series of continuous functions can converge uniformly to a discontinuous limit.
3. *Engineering reading.* Truncating a spectrum with a brick-wall window causes 9% ringing on every edge, whatever the bandwidth — the same constant governs the step response of an ideal low-pass filter, whose peak is $\frac{1}{2} + \frac{1}{\pi}\,\mathrm{Si}(\pi) = 1.0895$ of the step. See Ch. 15, §15.6 and Ch. 16, §16.5.



Partial sum S_{10} of the square wave. The first peak at $t = \pi/20$ reaches 1.1798; the red line is the limiting value $\frac{2}{\pi}\,\mathrm{Si}(\pi) = 1.1790$, which is not reduced by taking more terms.

D.10 Fourier-transform convolution theorem and Plancherel–Parseval

Write L^1 for functions with $\int_{-\infty}^{\infty}|x|dt < \infty$ and L^2 for those with finite energy $\int|x|^2dt < \infty$. For $x \in L^1$ the transform $X(\omega) = \int x(t)e^{-j\omega t}dt$ exists for every ω, is bounded by $\|x\|_1$ and is continuous (Ch. 16, §16.1).

Convolution theorem (Fourier)

If $x, h \in L^1$, then $y(t) = (x * h)(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\,d\tau$ converges absolutely for almost every t (for every t if one of the two is also bounded), $y \in L^1$ with $\|y\|_1 \le \|x\|_1\|h\|_1$, and

$$Y(\omega) = X(\omega)\,H(\omega) \qquad \text{for every } \omega.$$

Proof. By Tonelli's theorem and the substitution $u = t - \tau$,

$$\iint |x(\tau)|\,|h(t-\tau)|\,d\tau\,dt = \int |x(\tau)|\left(\int |h(u)|du\right)d\tau = \|x\|_1\|h\|_1 < \infty.$$

Hence the inner integral $\int|x(\tau)h(t-\tau)|d\tau$ is finite for almost every t, which defines $y(t)$, and $\int|y|dt \le \|x\|_1\|h\|_1$. (If $|h| \le B$, then $\int|x(\tau)h(t-\tau)|d\tau \le B\|x\|_1$ for every t.) Since $|e^{-j\omega t}| = 1$, the function $(\tau, t) \mapsto x(\tau)h(t-\tau)e^{-j\omega t}$ is absolutely integrable on $\mathbb{R}^2$, so Fubini's theorem allows

$$Y(\omega) = \int\left(\int x(\tau)h(t-\tau)\,d\tau\right)e^{-j\omega t}dt = \int x(\tau)e^{-j\omega\tau}\left(\int h(t-\tau)e^{-j\omega(t-\tau)}dt\right)d\tau = H(\omega)X(\omega).$$

■

Plancherel–Parseval

If $x \in L^1 \cap L^2$, then $X \in L^2$ and

$$\int_{-\infty}^{\infty} |x(t)|^2\,dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} |X(\omega)|^2\,d\omega.$$

For $x, y \in L^1 \cap L^2$, $\displaystyle\int x(t)\,y^*(t)\,dt = \frac{1}{2\pi}\int X(\omega)\,Y^*(\omega)\,d\omega$. The map $x \mapsto X/\sqrt{2\pi}$ therefore extends uniquely to a norm-preserving map of all of L^2 onto L^2 (Plancherel's theorem).

Proof. *Step 1 — the autocorrelation.* Let $\tilde{x}(t) = x^*(-t)$ and $g = x * \tilde{x}$, i.e.

$$g(t) = \int x(\tau)\,x^*(\tau - t)\,d\tau.$$

Because $x \in L^2$, this integral converges absolutely for **every** t by Cauchy–Schwarz, with $|g(t)| \le \|x\|_2^2 = g(0)$. Moreover g is continuous: $|g(t) - g(t')| \le \|x\|_2\,\big(\int |x(\tau - t) - x(\tau - t')|^2 d\tau\big)^{1/2}$, and translation is continuous in L^2 (true for continuous compactly supported functions by uniform continuity, and for general x by density). Since $x \in L^1$, the convolution theorem gives $g \in L^1$ and, because $\tilde{X}(\omega) = \int x^*(-t)e^{-j\omega t}dt = X^*(\omega)$,

$$G(\omega) = X(\omega)X^*(\omega) = |X(\omega)|^2 \;\ge 0.$$

So the claim is $g(0) = \frac{1}{2\pi}\int G(\omega)\,d\omega$: an inversion formula at the single point $t = 0$ for a function whose transform is non-negative.

Step 2 — Gaussian regularisation. For $\varepsilon > 0$ the Gaussian $e^{-\varepsilon\omega^2}$ is the transform of

$$\phi_\varepsilon(t) = \frac{1}{2\pi}\int e^{-\varepsilon\omega^2}e^{j\omega t}d\omega = \frac{1}{\sqrt{4\pi\varepsilon}}\,e^{-t^2/(4\varepsilon)},$$

the standard Gaussian integral of Ch. 16 (complete the square, or use $\mathcal{F}\{e^{-at^2}\} = \sqrt{\pi/a}\,e^{-\omega^2/4a}$). Note that $\phi_\varepsilon > 0$, $\int \phi_\varepsilon = 1$, and ϕ_ε concentrates at 0 as $\varepsilon \to 0$. Because $g \in L^1$ and $e^{-\varepsilon\omega^2} \in L^1$, the double integral below is absolutely convergent and Fubini gives

$$\frac{1}{2\pi}\int G(\omega)e^{-\varepsilon\omega^2}d\omega = \frac{1}{2\pi}\iint g(t)e^{-j\omega t}e^{-\varepsilon\omega^2}\,dt\,d\omega = \int g(t)\,\phi_\varepsilon(-t)\,dt = \int g(t)\,\phi_\varepsilon(t)\,dt.$$

Step 3 — the limit $\varepsilon \to 0$. On the right: $\int g\phi_\varepsilon - g(0) = \int [g(t) - g(0)]\phi_\varepsilon(t)dt$. Given $\eta > 0$ choose δ with $|g(t) - g(0)| < \eta$ for $|t| < \delta$; the rest is bounded by $2g(0)\int_{|t|\ge\delta}\phi_\varepsilon \to 0$. So the right side tends to $g(0) = \|x\|_2^2$. On the left: $G\,e^{-\varepsilon\omega^2}$ increases to G as $\varepsilon \downarrow 0$ because $G \ge 0$, so by the monotone convergence theorem the left side tends to $\frac{1}{2\pi}\int G$ — which is therefore finite. Equating the limits gives $\|x\|_2^2 = \frac{1}{2\pi}\int |X|^2 d\omega$, and in particular $X \in L^2$.

Step 4 — inner products and extension. Polarization, as in §D.7 Step 4, gives the two-signal form. Since $L^1 \cap L^2$ is dense in L^2 (truncate to $[-T, T]$), the isometry $x \mapsto X/\sqrt{2\pi}$ extends by continuity to all of L^2; the extended transform is the mean-square limit of the truncated transforms $X_T(\omega) = \int_{-T}^{T} x(t)e^{-j\omega t}dt$, which is how Ch. 16, §16.1 defines it. That the extension is *onto* follows from applying the same argument to the inverse transform. ∎

Remarks.

1. *Why non-negativity is the trick.* The inversion formula can fail pointwise for general L^1 functions (the transform need not be integrable). Working with $G = |X|^2 \ge 0$ lets monotone convergence do the work that would otherwise require knowing $G \in L^1$ in advance.
2. *Check.* $x(t) = e^{-t}u(t)$ has energy $\int_0^\infty e^{-2t}dt = \frac{1}{2}$ and $X(\omega) = 1/(1 + j\omega)$; numerically $\frac{1}{2\pi}\int \frac{d\omega}{1+\omega^2} = 0.5000$.
3. *Convention.* The factor $1/(2\pi)$ belongs to the angular-frequency convention used in this book; in hertz, $\int |x|^2 dt = \int |X_f(f)|^2 df$.
4. Used for energy spectral density in Ch. 16, §16.7 and for LTI filtering in §16.5.

D.11 The Nyquist–Shannon sampling theorem

Notation follows Ch. 16, §16.8: sampling interval T, $\omega_s = 2\pi/T$, and $\operatorname{sinc}(u) = \sin(\pi u)/(\pi u)$.

Poisson summation (band-limited form)

Let $X \in L^1$ vanish outside a bounded interval and be square-integrable, and let $x(t) = \frac{1}{2\pi}\int X(\omega)e^{j\omega t}d\omega$ (a continuous function). Then the ω_s-periodic function $P(\omega) = \sum_{k=-\infty}^{\infty} X(\omega - k\omega_s)$ — a finite sum at each ω — has the mean-square convergent Fourier series

$$\sum_{k=-\infty}^{\infty} X(\omega - k\omega_s) = T\sum_{n=-\infty}^{\infty} x(nT)\,e^{-j\omega nT}.$$

Sampling theorem (Nyquist–Shannon)

Let x be as above with $X(\omega) = 0$ for $|\omega| > \omega_M$, and let the sampling rate satisfy $\omega_s > 2\omega_M$. Then x is uniquely determined by its samples $x(nT)$, the samples are square-summable, and for every real t

$$x(t) = \sum_{n=-\infty}^{\infty} x(nT)\,\operatorname{sinc}\!\left(\frac{t - nT}{T}\right),$$

the symmetric partial sums converging uniformly in t on all of $\mathbb{R}$.

Proof of Poisson summation. P is ω_s-periodic. On one period $I = [-\omega_s/2, \omega_s/2)$ only finitely many shifts $X(\omega - k\omega_s)$ are non-zero (the support of X is bounded), so P is square-integrable over I. Its Fourier series in the variable ω, with period ω_s and basis $e^{-j\omega nT}$ (note $\omega_s T = 2\pi$), has coefficients

$$d_n = \frac{1}{\omega_s}\int_I P(\omega)e^{j\omega nT}d\omega = \frac{1}{\omega_s}\sum_k \int_I X(\omega - k\omega_s)e^{j\omega nT}d\omega = \frac{1}{\omega_s}\int_{-\infty}^{\infty} X(\nu)e^{j\nu nT}d\nu = \frac{2\pi}{\omega_s}\,x(nT) = T\,x(nT).$$

In the third step we substituted $\nu = \omega - k\omega_s$, used $e^{jk\omega_s nT} = e^{j2\pi kn} = 1$, and observed that the translates $I + k\omega_s$ tile the line. By Parseval's theorem (§D.7, which applies to any square-integrable periodic function), the series $\sum_n d_n e^{-j\omega nT}$ converges to P in mean square over a period, and $\frac{1}{\omega_s}\int_I |P|^2 = \sum_n |d_n|^2$. ∎

Proof of the sampling theorem. *Step 1 — no aliasing.* Since $\omega_M < \omega_s/2$, for $\omega \in I = (-\omega_s/2, \omega_s/2)$ every shift with $k \neq 0$ is evaluated at $|\omega - k\omega_s| > \omega_s/2 > \omega_M$, where X vanishes. So $P = X$ on I, and Poisson summation gives, in mean square on I,

$$X(\omega) = T\sum_{n=-\infty}^{\infty} x(nT)\,e^{-j\omega nT} \qquad (|\omega| < \omega_s/2).$$

This already shows that the samples determine X on I (and $X = 0$ outside), hence determine x; and Parseval gives $\sum_n |x(nT)|^2 = \frac{1}{T^2\omega_s}\int_I |X|^2 < \infty$.

Step 2 — inversion term by term. Since $X = 0$ outside I,

$$x(t) = \frac{1}{2\pi}\int_I X(\omega)\,e^{j\omega t}\,d\omega = \frac{\omega_s}{2\pi}\,\langle X,\ e_t\rangle_I, \qquad e_t(\omega) = e^{-j\omega t},$$

with $\langle F, G\rangle_I = \frac{1}{\omega_s}\int_I F\,G^*$. An inner product with a fixed square-integrable function is continuous with respect to mean-square convergence (Cauchy–Schwarz), so we may insert the series of Step 1 and integrate term by term:

$$x(t) = \sum_n x(nT)\,\frac{T}{2\pi}\int_{-\omega_s/2}^{\omega_s/2} e^{j\omega(t-nT)}d\omega = \sum_n x(nT)\,\frac{T}{2\pi}\cdot\frac{2\sin\big(\omega_s(t-nT)/2\big)}{t-nT}.$$

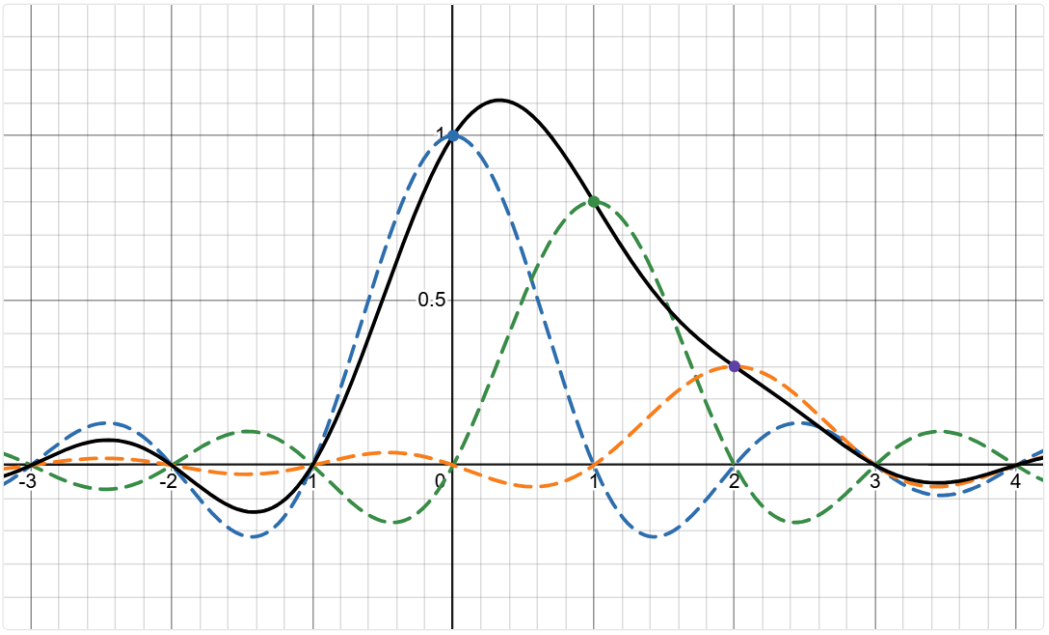
With $\omega_s/2 = \pi/T$, the coefficient is $\dfrac{\sin\big(\pi(t-nT)/T\big)}{\pi(t-nT)/T} = \operatorname{sinc}\big((t-nT)/T\big)$ (value 1 at $t = nT$).

Step 3 — uniform convergence. Let x_N be the partial sum over $|n| \le N$. By the same computation, $x(t) - x_N(t) = \frac{\omega_s}{2\pi}\langle X - X_N, e_t\rangle_I$, where X_N is the corresponding partial Fourier sum. Since $\|e_t\|_I = 1$ for every t, Cauchy–Schwarz gives $|x(t) - x_N(t)| \le \frac{\omega_s}{2\pi}\|X - X_N\|_I \to 0$, uniformly in t. ∎

Numerical check. $x(t) = \operatorname{sinc}^2(t/2)$ has a triangular spectrum supported on $|\omega| \le \pi$. With $T = 0.8$ ($\omega_s = 7.85 > 2\pi$) and $|n| \le 20000$, the reconstruction at $t = 0.37$ gives 0.8923548471, equal to $x(0.37)$ to ten digits.

Remarks.

1. *Why $\omega_s > 2\omega_M$.* If $\omega_s < 2\omega_M$, the shifted copies in P overlap on I, so $P \neq X$ there: the samples determine only the folded spectrum P, and different signals share the same samples (aliasing). For example $\cos(\omega_1 t)$ and $\cos\big((\omega_s - \omega_1)t\big)$ agree at every $t = nT$.
2. *The boundary case.* With $\omega_s = 2\omega_M$ the proof goes through whenever $X \in L^2$ is supported on $[-\omega_M, \omega_M]$, because the endpoints have measure zero. But it fails for spectra containing impulses at $\pm\omega_M$: $x(t) = \sin(\omega_M t)$ sampled at $T = \pi/\omega_M$ gives all-zero samples. This is why the theorem is stated with strict inequality.
3. *Interpretation.* The sinc functions $\operatorname{sinc}\big((t-nT)/T\big)$ are orthogonal and are the impulse responses of the ideal low-pass filter $H_r(\omega) = T\,\mathrm{rect}(\omega/\omega_s)$ shifted to the sampling instants. Reconstruction is "filter the impulse-sampled signal", exactly as derived with impulses in Ch. 16, §16.8; the proof here avoids generalized functions by working with Fourier series in ω.
4. *Practicality.* The sinc decays only like $1/|t|$, so truncating the series converges slowly, and real converters oversample and use smoother interpolation kernels.



Sinc reconstruction with $T = 1$ and three non-zero samples: each dashed sinc equals its sample at its own instant and vanishes at all the others, so the black sum passes exactly through every sample.

D.12 Z-transform: the ROC is an annulus; BIBO stability

Z-transform and ROC

For a sequence $x[n]$, $n \in \mathbb{Z}$, the Z-transform is $X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$. Its **region of convergence** R_x is the set of $z \in \mathbb{C} \setminus \{0\}$ (with $z = \infty$ allowed when $x[n] = 0$ for $n < 0$, and $z = 0$ allowed when $x[n] = 0$ for $n > 0$) at which the series converges **absolutely**. A discrete-time LTI system with impulse response $h[n]$ is **BIBO stable** if every bounded input $|x[n]| \le B_x$ produces a bounded output $y = h * x$.

Shape of the ROC

Let

$$r_1 = \limsup_{n\to\infty} |x[n]|^{1/n}, \qquad \frac{1}{r_2} = \limsup_{m\to\infty} |x[-m]|^{1/m}$$

(with $1/0 = \infty$, $1/\infty = 0$). Then the series converges absolutely for $r_1 < |z| < r_2$ and diverges for $|z| < r_1$ and for $|z| > r_2$. Consequently R_x is an annulus $\{r_1 < |z| < r_2\}$, together possibly with some or all points of the boundary circles, and is empty if $r_1 > r_2$. On the open annulus X is analytic.

Stability and the unit circle

An LTI system is BIBO stable if and only if $\sum_n |h[n]| < \infty$, and this holds if and only if the unit circle $|z| = 1$ lies in R_h. A causal system with rational $H(z)$ is therefore stable if and only if all its poles satisfy $|p| < 1$.

Proof of the ROC theorem. Put $r = |z|$ and split $X(z) = \sum_{n\ge 0} x[n] z^{-n} + \sum_{m\ge 1} x[-m] z^{m}$. Absolute convergence of the whole series is equivalent to absolute convergence of both parts, and depends only on r because $|x[n] z^{-n}| = |x[n]| r^{-n}$.

Causal part. Apply the root test to $\sum_{n\ge 0} |x[n]| r^{-n}$: $\limsup_n \left(|x[n]| r^{-n}\right)^{1/n} = r_1/r$. If $r > r_1$ this is < 1 and the series converges (choose q with $r_1/r < q < 1$; then $|x[n]| r^{-n} \le q^n$ for large n, and compare with the geometric series). If $r < r_1$ it is > 1, so $|x[n]| r^{-n} \ge 1$ for infinitely many n, the terms do not tend to zero, and the series diverges.

Anticausal part. The same test on $\sum_{m\ge 1} |x[-m]| r^{m}$ gives $\limsup \left(|x[-m]| r^{m}\right)^{1/m} = r/r_2$: convergence for $r < r_2$, divergence for $r > r_2$.

Both parts converge for $r_1 < r < r_2$ and at least one diverges outside $[r_1, r_2]$; on the circles $r = r_1, r = r_2$ the test is inconclusive, which is why boundary points may or may not belong to R_x. Analyticity: on any closed sub-annulus $r_1 < \rho_1 \le |z| \le \rho_2 < r_2$ the terms are bounded by $|x[n]| \rho_1^{-n}$ ($n \ge 0$) and $|x[-m]| \rho_2^{m}$ ($m \ge 1$), which form convergent series by the above, so the Laurent series converges uniformly there (M-test) and its sum is analytic, each term being analytic on $\mathbb{C} \setminus \{0\}$. ∎

Proof of the stability theorem. *Sufficiency.* If $S = \sum |h[k]| < \infty$ and $|x[n]| \le B_x$, then

$$|y[n]| = \left| \sum_k h[k] x[n-k] \right| \le B_x \sum_k |h[k]| = B_x S,$$

and the series defining $y[n]$ converges absolutely for each n.

Necessity. Suppose $\sum |h[k]| = \infty$. Define the bounded input $x[n] = h^*[-n] / |h[-n]|$ where $h[-n] \neq 0$ and $x[n] = 0$ otherwise, so $|x[n]| \le 1$. Then

$$y[0] = \sum_k h[k] x[-k] = \sum_k \frac{h[k] h^*[k]}{|h[k]|} = \sum_k |h[k]| = \infty,$$

so this bounded input produces an output that is not even finite at $n = 0$, and the system is not BIBO stable. (To avoid a divergent defining sum, use instead the truncated inputs x_N, equal to x for $|n| \le N$ and 0 otherwise. Each satisfies $|x_N| \le 1$ and has a well-defined output with $y_N[0] = \sum_{|k|\le N} |h[k]|$. If the system were stable, the map $x \mapsto y$, being linear and time-invariant, would satisfy a bound $|y[n]| \le C \sup|x|$ — for a linear system, boundedness of outputs over the unit ball of inputs is equivalent to such a bound by the uniform boundedness principle — whereas $y_N[0] \to \infty$.)

The unit circle. On $|z| = 1$, $\sum_n |h[n] z^{-n}| = \sum_n |h[n]|$, so the unit circle lies in R_h (with absolute convergence, as in the definition) if and only if h is absolutely summable.

Causal rational case. If h is causal, $h[-m] = 0$ for $m \ge 1$, so $r_2 = \infty$ and $R_h \supseteq \{|z| > r_1\}$. For rational H, partial fractions give $h[n]$ as a finite sum of terms $n^k p^n u[n]$ (plus finitely many impulses), and $r_1 = \max|p|$ over the poles; X cannot be analytic at a pole, so no pole lies in the open ROC. Thus $R_h = \{|z| > \max|p|\}$ (including ∞), and it contains the unit circle iff $\max|p| < 1$. A pole exactly on the circle gives a term $n^k p^n$ with $|p| = 1$, which is not absolutely summable, so such a system is unstable even though the circle is the ROC's boundary. ∎

Remarks.

1. *The role of absolute convergence.* With a definition of the ROC by mere convergence, boundary points could be included for non-absolutely-summable sequences (e.g. $h[n] = u[n-1]/n$ converges on $|z| = 1$ except at $z = 1$), and the clean equivalence with stability would fail. Within the open annulus the two definitions agree.
2. *Analogy.* The continuous-time statement is that a causal LTI system is BIBO stable iff $\int |h| < \infty$ iff the $j\omega$-axis lies in the ROC of $H(s)$; the map $z = e^{sT}$ carries the axis onto the unit circle and the left half-plane into the disc (Ch. 17, §17.7).
3. Used in Ch. 17, §17.3 and §17.6.

D.13 Cauchy–Riemann equations: necessity and sufficiency

Complex differentiability

Let $f = u + jv$ be defined on an open set $D \subseteq \mathbb{C}$, with $z = x + jy$ and u, v real. f is **complex differentiable** at $z_0 \in D$ if

$$f'(z_0) = \lim_{\Delta z \to 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}$$

exists, the limit being taken as $\Delta z \to 0$ through **all** complex values. f is **analytic** on D if it is complex differentiable at every point of D.

Cauchy–Riemann equations

(Necessity.) If f is complex differentiable at $z_0 = x_0 + jy_0$, then the four partial derivatives u_x, u_y, v_x, v_y exist at (x_0, y_0) and satisfy

$$u_x = v_y, \qquad u_y = -v_x, \qquad \text{and} \qquad f'(z_0) = u_x + jv_x = v_y - ju_y.$$

(Sufficiency.) Conversely, if u and v are (real-)differentiable at (x_0, y_0) — in particular, if their first partials exist near (x_0, y_0) and are continuous at it — and satisfy the Cauchy–Riemann equations there, then f is complex differentiable at z_0.

Proof of necessity. Let $L = f'(z_0)$. Since the limit exists along every path, take $\Delta z = h$ real:

$$L = \lim_{h \to 0} \frac{u(x_0 + h, y_0) - u(x_0, y_0)}{h} + j\,\frac{v(x_0 + h, y_0) - v(x_0, y_0)}{h}.$$

A complex limit exists iff the limits of its real and imaginary parts exist, so u_x, v_x exist and $L = u_x + jv_x$. Now take $\Delta z = jh$, h real:

$$L = \lim_{h \to 0} \frac{u(x_0, y_0 + h) - u(x_0, y_0) + j\,[v(x_0, y_0 + h) - v(x_0, y_0)]}{jh} = \frac{1}{j}\,(u_y + jv_y) = v_y - ju_y.$$

Equating real and imaginary parts of the two expressions for L gives $u_x = v_y$ and $v_x = -u_y$. ∎

Proof of sufficiency. Real differentiability of u at (x_0, y_0) means that, with $\Delta z = h + jk$,

$$u(x_0 + h, y_0 + k) - u(x_0, y_0) = u_x h + u_y k + \varepsilon_1(h, k)\,|\Delta z|, \qquad \varepsilon_1 \to 0 \text{ as } \Delta z \to 0,$$

and similarly for v with a remainder ε_2. (If the partials are continuous near the point, this follows from the mean-value theorem applied separately in x and in y.) Using $v_x = -u_y$ and $v_y = u_x$,

$$\begin{aligned} f(z_0 + \Delta z) - f(z_0) &= (u_x h + u_y k) + j(v_x h + v_y k) + (\varepsilon_1 + j\varepsilon_2)|\Delta z| \\ &= (u_x h + u_y k) + j(-u_y h + u_x k) + (\varepsilon_1 + j\varepsilon_2)|\Delta z| \\ &= (u_x - ju_y)(h + jk) + (\varepsilon_1 + j\varepsilon_2)|\Delta z|. \end{aligned}$$

(Check the last line: $(u_x - ju_y)(h + jk) = u_x h + u_y k + j(u_x k - u_y h)$.) Dividing by Δz,

$$\frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} = (u_x - ju_y) + (\varepsilon_1 + j\varepsilon_2)\,\frac{|\Delta z|}{\Delta z},$$

and the last term has modulus $|\varepsilon_1 + j\varepsilon_2| \to 0$. So $f'(z_0) = u_x - ju_y = u_x + jv_x$. ∎

Remarks.

1. *Geometric meaning.* Real differentiability says the map $(x, y) \mapsto (u, v)$ is approximated by a 2×2 Jacobian $\begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix}$. The Cauchy–Riemann equations say this matrix has the form $\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$, i.e. it is multiplication by the complex number $a + jb$: a rotation combined with a scaling. That is why analytic maps are conformal where $f' \neq 0$.
2. *CR alone is not sufficient.* Let $f(z) = \sqrt{|xy|}$ (so $u = \sqrt{|xy|}$, $v = 0$). On the axes $u = 0$, so $u_x(0,0) = u_y(0,0) = 0 = v_x = v_y$ and the Cauchy–Riemann equations hold at the origin. But along $z = t(1 + j)$, $t > 0$, $\big(f(z) - f(0)\big)/z = t/\big(t(1 + j)\big) = 1/(1 + j)$, whereas along the real axis the quotient is 0; the limit does not exist. Here u is not real-differentiable at 0, which is the hypothesis that fails.
3. *Differentiability alone is not sufficient.* $f(z) = \bar{z}$ has $u = x$, $v = -y$, smooth, but $u_x = 1 \neq v_y = -1$; it is nowhere complex differentiable (the difference quotient is $\overline{\Delta z}/\Delta z$, which takes every value on the unit circle).
4. Used in Ch. 18, §18.3, and, through $\nabla^2 u = 0$, for the harmonic potentials of electrostatics.

D.14 Cauchy's integral theorem and integral formula

Contours and positive orientation

A **contour** is a continuous, piecewise continuously differentiable curve $z(\tau)$, $a \le \tau \le b$, and $\int_C f(z)\,dz = \int_a^b f(z(\tau))\,z'(\tau)\,d\tau$. A **simple closed** contour bounds a bounded region Ω (Jordan curve theorem); it is **positively oriented** when Ω lies to its left. A domain D is **simply connected** if the interior of every simple closed contour in D lies in D.

Cauchy's integral theorem (C^1 version)

Let f be analytic on a simply connected domain D, with f' continuous on D. Then for every simple closed contour C in D

$$\oint_C f(z)\,dz = 0.$$

More generally, if Ω is a bounded region whose boundary $\partial\Omega$ consists of finitely many disjoint simple closed contours, oriented with Ω to the left, and f is analytic with continuous f' on an open set containing $\overline{\Omega} = \Omega \cup \partial\Omega$, then $\oint_{\partial\Omega} f\,dz = 0$.

Cauchy's integral formula

Under the hypotheses of the first statement, for every z_0 inside C

$$f(z_0) = \frac{1}{2\pi j}\oint_C \frac{f(z)}{z - z_0}\,dz, \qquad \text{and} \qquad f^{(n)}(z_0) = \frac{n!}{2\pi j}\oint_C \frac{f(z)}{(z - z_0)^{n+1}}\,dz \quad (n \geq 1).$$

Proof of Cauchy's theorem. Write $f = u + jv$ and $dz = dx + j\,dy$. Then

$$\oint_C f\,dz = \oint_C (u\,dx - v\,dy) + j\oint_C (v\,dx + u\,dy).$$

Since $f' = u_x + jv_x = v_y - ju_y$ is continuous, the partials of u and v are continuous on D, and $\overline{\Omega}$ (the region bounded by C, together with C) lies in D by simple connectivity. Green's theorem $\oint_{\partial\Omega}(P\,dx + Q\,dy) = \iint_\Omega (Q_x - P_y)\,dx\,dy$ (Ch. 7), applied to each real line integral, gives

$$\oint_C f\,dz = \iint_\Omega (-v_x - u_y)\,dx\,dy + j\iint_\Omega (u_x - v_y)\,dx\,dy = 0,$$

since both integrands vanish identically by the Cauchy–Riemann equations (§D.13). Green's theorem holds verbatim for regions with finitely many holes (cut the region into simply connected pieces; the integrals along the cuts cancel in pairs), which proves the general statement. ∎

Proof of the integral formula. Fix z_0 inside C and choose $r > 0$ so small that the closed disc $|z - z_0| \leq r$ lies inside C. Let C_r be the circle $|z - z_0| = r$, positively oriented, and Ω_r the region between C and C_r. The function $g(z) = f(z)/(z - z_0)$ is analytic with continuous derivative on a neighbourhood of $\overline{\Omega_r}$, whose boundary is C (positive) together with C_r (negative). By the general form of Cauchy's theorem,

$$\oint_C \frac{f(z)}{z - z_0}\,dz = \oint_{C_r} \frac{f(z)}{z - z_0}\,dz \qquad \text{for every small } r > 0.$$

Now split the right side:

$$\oint_{C_r} \frac{f(z)}{z - z_0}\,dz = f(z_0)\oint_{C_r} \frac{dz}{z - z_0} + \oint_{C_r} \frac{f(z) - f(z_0)}{z - z_0}\,dz.$$

With $z = z_0 + re^{j\theta}$, $dz = jre^{j\theta}d\theta$, the first integral is $\int_0^{2\pi} j\,d\theta = 2\pi j$. The second is bounded, by the ML inequality (Ch. 18, §18.4), by $2\pi r \cdot \max_{C_r}|f(z) - f(z_0)|/r = 2\pi \max_{C_r}|f(z) - f(z_0)|$, which tends to 0 as $r \to 0$ by continuity of f. Since the left side does not depend on r, it equals $2\pi j\,f(z_0)$.

Derivatives. Let $F(w) = \frac{1}{2\pi j}\oint_C \frac{f(z)}{z-w}dz$ for w inside C; we have shown $F = f$ there. For w at distance $\geq d > 0$ from C, the difference quotient of the integrand,

$$\frac{1}{h}\left(\frac{1}{z - w - h} - \frac{1}{z - w}\right) = \frac{1}{(z - w - h)(z - w)} \longrightarrow \frac{1}{(z - w)^2}$$

uniformly for $z \in C$ as $h \to 0$ (for $|h| < d/2$ the denominators are bounded below by $d^2/2$, and the difference from the limit is $h/[(z - w - h)(z - w)^2]$, bounded by $2|h|/d^3$). Hence one may differentiate under the integral sign, and by induction $f^{(n)}(w) = \frac{n!}{2\pi j}\oint_C \frac{f(z)}{(z-w)^{n+1}}dz$. ∎

Remarks.

1. *Removing the C^1 hypothesis.* The Cauchy–Goursat theorem proves $\oint_C f\,dz = 0$ assuming only that f' exists at every point, by subdividing triangles and estimating — no continuity of f' and no Green's theorem. Combined with the derivative formula just proved, it shows that **an analytic function is automatically infinitely differentiable**, so the C^1 hypothesis used here costs no generality *a posteriori*. See Ch. 18, §18.5.
2. *Simple connectivity is essential.* $f(z) = 1/z$ is analytic on $\mathbb{C} \setminus \{0\}$, but $\oint_{|z|=1} dz/z = 2\pi j \neq 0$. The punctured plane is not simply connected — the hole at 0 is exactly what the integral detects, and the residue theorem turns this into a computational tool.
3. *Mean-value property.* With C a circle of radius r about z_0, the formula reads $f(z_0) = \frac{1}{2\pi}\int_0^{2\pi} f(z_0 + re^{j\theta})\,d\theta$: an analytic function (and hence its real part, a harmonic function) equals its average on every circle — the basis of the maximum-modulus principle and of Liouville's theorem in Ch. 18.
4. *Consequence for the Laplace transform.* The Bromwich inversion integral of Ch. 14, §14.4 is a contour integral of $X(s)e^{st}$; Cauchy's theorem is what allows the vertical line to be moved freely within the ROC.

D.15 The residue theorem

Residue

If f is analytic in a punctured disc $0 < |z - z_k| < \rho$, it has a Laurent expansion $f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_k)^n$ there, converging uniformly on every circle $|z - z_k| = r$, $0 < r < \rho$ (Ch. 18, §18.6). The **residue** of f at z_k is $\operatorname{Res}_{z=z_k} f = a_{-1}$.

Residue theorem

Let C be a positively oriented simple closed contour, and let f be analytic on an open set containing C and its interior, except at finitely many points $z_1, \dots, z_K$ inside C. Then

$$\oint_C f(z)\,dz = 2\pi j \sum_{k=1}^{K} \operatorname{Res}_{z=z_k} f.$$

Proof. *Step 1 — one circle.* Let C_k be the circle $|z - z_k| = r$, positively oriented, with r small. On C_k the Laurent series converges uniformly, so it may be integrated term by term. For each integer n, with $z = z_k + re^{j\theta}$,

$$\oint_{C_k} (z - z_k)^n\,dz = \int_0^{2\pi} r^n e^{jn\theta} \cdot jre^{j\theta}\,d\theta = jr^{n+1}\int_0^{2\pi} e^{j(n+1)\theta}\,d\theta = \begin{cases} 2\pi j, & n = -1, \\ 0, & n \neq -1. \end{cases}$$

Hence $\oint_{C_k} f\,dz = 2\pi j\, a_{-1} = 2\pi j \operatorname{Res}_{z=z_k} f$. Only the $(z - z_k)^{-1}$ term survives, because every other power has a single-valued antiderivative on the punctured disc.

Step 2 — deformation. Choose r so small that the closed discs $|z - z_k| \le r$ are disjoint and lie inside C. Let Ω be the region inside C and outside all the discs. Its boundary is C (positively oriented) together with the circles C_k traversed negatively, and f is analytic on a neighbourhood of $\overline{\Omega}$ (with continuous derivative, by Remark 1 of §D.14). By the general form of Cauchy's theorem,

$$\oint_C f\,dz - \sum_{k=1}^{K} \oint_{C_k} f\,dz = 0.$$

Substituting Step 1 completes the proof. ∎

Remarks.

1. *Computing residues.* At a simple pole, $\operatorname{Res} = \lim_{z\to z_k}(z - z_k)f(z)$; for $f = N/D$ with $D(z_k) = 0 \neq D'(z_k)$ this is $N(z_k)/D'(z_k)$; at a pole of order m, $\operatorname{Res} = \frac{1}{(m-1)!}\lim_{z\to z_k}\frac{d^{m-1}}{dz^{m-1}}\big[(z - z_k)^m f(z)\big]$. Each follows by reading off a_{-1} from the Laurent series (Ch. 18, §18.7).
2. *Hypotheses.* The singularities must be isolated and finite in number inside C, and none may lie *on* C. A pole on the contour makes the integral improper; one then uses indentations and principal values, each small semicircle contributing πj times the residue in the limit.
3. *Where it is used.* Inverse Laplace transforms by closing the Bromwich contour to the left (the Heaviside expansion is the sum of residues of $X(s)e^{st}$), inverse Z-transforms $x[n] = \frac{1}{2\pi j}\oint X(z)z^{n-1}dz$, real integrals (Ch. 18, §18.8), and the argument principle behind the Nyquist criterion (§18.9).

D.16 Jordan's lemma

Jordan's inequality and Jordan's lemma

(Inequality.) For $0 \le \theta \le \pi/2$, $\dfrac{2\theta}{\pi} \le \sin\theta \le \theta$.

(Lemma.) Let $\omega > 0$, let C_R be the upper semicircle $z = Re^{j\theta}$, $0 \le \theta \le \pi$, and let g be continuous on C_R for all large R with $M_R = \max_{C_R}|g| \to 0$ as $R \to \infty$. Then

$$\left|\int_{C_R} g(z)\,e^{j\omega z}\,dz\right| \le \frac{\pi}{\omega}\,M_R\big(1 - e^{-\omega R}\big) \le \frac{\pi M_R}{\omega} \xrightarrow[R\to\infty]{} 0.$$

Proof of the inequality. $\sin\theta \le \theta$ because $\frac{d}{d\theta}(\theta - \sin\theta) = 1 - \cos\theta \ge 0$ and equality holds at 0. For the lower bound, sin is concave on $[0, \pi/2]$ (its second derivative $-\sin\theta \le 0$), so its graph lies above the chord joining $(0, 0)$ and $(\pi/2, 1)$, which is the line $2\theta/\pi$. ∎

Proof of the lemma. On C_R, $z = R(\cos\theta + j\sin\theta)$, so $|e^{j\omega z}| = e^{-\omega R\sin\theta}$ and $|dz| = R\,d\theta$. Hence

$$\left|\int_{C_R} g\,e^{j\omega z}dz\right| \le M_R\,R\int_0^{\pi} e^{-\omega R\sin\theta}\,d\theta = 2M_R R\int_0^{\pi/2} e^{-\omega R\sin\theta}\,d\theta,$$

using the symmetry $\sin(\pi - \theta) = \sin\theta$. By Jordan's inequality $e^{-\omega R\sin\theta} \le e^{-2\omega R\theta/\pi}$ on $[0, \pi/2]$, so

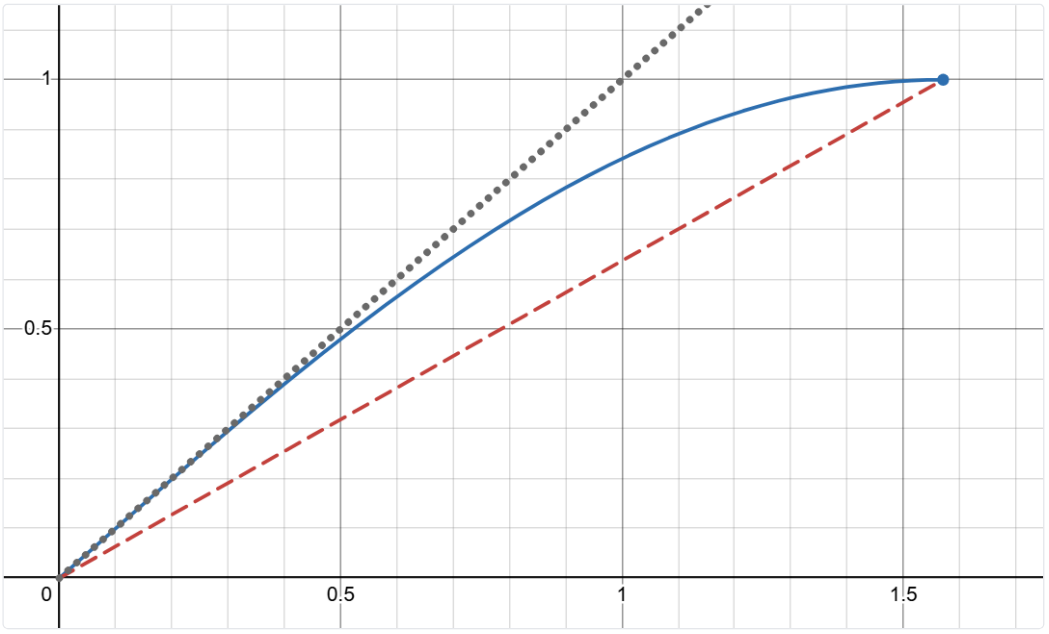
$$2M_R R\int_0^{\pi/2} e^{-2\omega R\theta/\pi}\,d\theta = 2M_R R\cdot\frac{\pi}{2\omega R}\big(1 - e^{-\omega R}\big) = \frac{\pi M_R}{\omega}\big(1 - e^{-\omega R}\big).$$

∎

Remarks.

1. *Why the lemma is sharper than ML.* The ML inequality gives only $|\int| \le \pi R M_R$, which tends to zero only if $M_R = o(1/R)$, e.g. $g = 1/z^2$. Jordan's lemma needs only $M_R \to 0$, e.g. $g = 1/z$: the factor R from the arc length is cancelled because $e^{j\omega z}$ is exponentially small except within an angle of order $1/(\omega R)$ of the real axis. Numerically, for $\omega = 1$ the key integral $\int_0^{\pi} e^{-R\sin\theta}d\theta$ equals 0.4208 at $R = 5$ and 0.0400 at $R = 50$, against Jordan's bounds $\frac{\pi}{R}(1 - e^{-R}) = 0.6241$ and 0.0628: both decay like $1/R$, as the proof predicts.
2. *The sign of ω dictates the half-plane.* For $\omega < 0$ the factor $e^{j\omega z}$ is small in the *lower* half-plane, and the contour must be closed there (clockwise, introducing a minus sign). The same rule closes the Bromwich contour for $X(s)e^{st}$: to the left for $t > 0$, to the right for $t < 0$, which is why causal signals have all their poles to the left of the ROC.

3. *Typical use.* For $\int_{-\infty}^{\infty} \frac{\cos x}{x^2+1}dx$, integrate $e^{jz}/(z^2+1)$ over $[-R,R]$ plus C_R. The only enclosed pole is $z=j$, with residue $e^{-1}/(2j)$; the arc vanishes by Jordan's lemma (here even ML suffices); so the integral is $2\pi j \cdot e^{-1}/(2j) = \pi/e = 1.15573$, which agrees with numerical quadrature. For $\int_{-\infty}^{\infty} \frac{\sin x}{x}dx = \pi$, the case $g = 1/z$, ML fails and Jordan's lemma is indispensable. See Ch. 18, §18.8 and the Fourier inversion of rational spectra in Ch. 16, §16.2.



Jordan's inequality: on $[0,\pi/2]$ the concave curve $\sin\theta$ (blue) lies between its chord $2\theta/\pi$ (red, dashed) and its tangent θ (grey). The lower bound is what turns $e^{-\omega R\sin\theta}$ into an integrable exponential.

Further reading

- E. A. Coddington and N. Levinson, *Theory of Ordinary Differential Equations*, McGraw-Hill — existence, uniqueness and dependence on data (§D.1).
- W. Rudin, *Real and Complex Analysis*, 3rd ed., McGraw-Hill — Fourier analysis on L^1 and L^2, Plancherel's theorem, Cauchy's theorem in full generality (§§D.7, D.10, D.14).
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Appendix E — Probability, Statistics and Numerical Methods

This appendix supplies complete proofs of the central results of Part VI (Chapters 19–21) and of the numerical chapters of Part VII (Chapters 22 and 24) that the chapters state with only a sketch. The probability results run from the elementary (Bayes' theorem, linearity of expectation) through the limit theorems (the weak law of large numbers and the central limit theorem) to the second-order theory of random processes (Wiener–Khinchin, filtering of noise). The statistics results are the unbiasedness of the sample variance, least squares and the Gauss–Markov theorem, and the GUM law of propagation of uncertainty. The numerical results are the convergence theorems for bisection, Newton–Raphson, the trapezoidal and Simpson rules, Euler's method and Monte Carlo integration.

Prerequisites. Calculus with Taylor's theorem and its remainder (Ch. 4–6), matrix algebra (Ch. 8), Fourier transforms (Ch. 16), and the probability vocabulary of Ch. 19.

Level of rigour. Every hypothesis is stated precisely, and every step is justified from those hypotheses. A few deep tools of measure-theoretic probability and analysis are *quoted* with a reference rather than proved: the linearity and monotonicity of the Lebesgue integral, Fubini's theorem, the dominated convergence theorem, Lévy's continuity theorem, and Fourier inversion. Each is named where it is used. Random variables are real-valued and defined on a probability space $(\Omega, \mathcal{F}, P)$ (Section 19.1). "i.i.d." means independent and identically distributed.

E.1 The law of total probability and Bayes' theorem

Conditional probability and partitions

For events A, B with $P(B) > 0$, the conditional probability of A given B is

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

A finite or countable family of events $B_1, B_2, \ldots$ is a **partition** of Ω if the B_i are pairwise disjoint ($B_i \cap B_k = \varnothing$ for $i \neq k$) and $\bigcup_i B_i = \Omega$.

Statement

Law of total probability and Bayes' theorem

Let $B_1, B_2, \ldots$ be a finite or countable partition of Ω with $P(B_i) > 0$ for every i, and let A be any event.

1. **Total probability:**

$$P(A) = \sum_i P(A \mid B_i)\, P(B_i).$$

2. **Bayes' theorem:** if in addition $P(A) > 0$, then for every k

$$P(B_k \mid A) = \frac{P(A \mid B_k)\, P(B_k)}{\sum_i P(A \mid B_i)\, P(B_i)}.$$

Proof

Step 1 (decompose A). Because the B_i cover Ω,

$$A = A \cap \Omega = A \cap \Big(\bigcup_i B_i\Big) = \bigcup_i (A \cap B_i),$$

by the distributive law of intersection over union.

Step 2 (the pieces are disjoint). For $i \neq k$, $(A \cap B_i) \cap (A \cap B_k) \subseteq B_i \cap B_k = \varnothing$.

Step 3 (additivity). By the countable additivity axiom (Section 19.1), applied to the disjoint family of Step 2,

$$P(A) = \sum_i P(A \cap B_i).$$

Step 4 (multiplication rule). Since $P(B_i) > 0$, the definition of conditional probability rearranges to $P(A \cap B_i) = P(A \mid B_i) P(B_i)$. Substituting into Step 3 proves part 1.

Step 5 (Bayes). Since $P(A) > 0$ and $P(B_k) > 0$, the definition gives two expressions for the same joint probability:

$$P(B_k \mid A)\, P(A) = P(A \cap B_k) = P(A \mid B_k)\, P(B_k).$$

Divide by $P(A)$ and replace $P(A)$ by the sum from part 1. ■

Remarks

- *Why the hypotheses matter.* If the B_i overlap, Step 2 fails and the sum in part 1 counts the overlap twice. For example, with $B_1 = B_2 = \Omega$ the formula would give $P(A) = 2P(A)$. If the B_i do not cover Ω, the sum misses $A \setminus \bigcup_i B_i$. The condition $P(B_i) > 0$ is needed only so that $P(A \mid B_i)$ is defined. Cells with $P(B_i) = 0$ may simply be dropped, because they contribute $P(A \cap B_i) = 0$.

- *Base-rate effect.* Suppose a fault detector flags a faulty unit with probability 0.99, flags a healthy unit with probability 0.05, and 1% of units are faulty. Bayes' theorem gives

$$P(\text{faulty} \mid \text{flag}) = \frac{0.99 \cdot 0.01}{0.99 \cdot 0.01 + 0.05 \cdot 0.99} = \frac{1}{6} \approx 0.167.$$

So five out of six alarms are false, because the small prior is weighted against a comparatively large false-alarm rate. The chapter treats this in Section 19.3.

- *Continuous form.* If X has density f_X and the conditional densities exist, the same argument with integrals in place of sums gives $f_{X\mid A}(x) = P(A \mid X = x) f_X(x) / \int P(A \mid X = u) f_X(u)\, du$. Strictly, conditioning on the null event $\{X = x\}$ needs the measure-theoretic definition of conditional probability. The formula used in Section 19.7 is the limit of the discrete one over shrinking cells $\{x < X \le x + \Delta\}$.
- *Where used.* Total probability appears in Section 19.2. Bayes' theorem underlies the binary channel and MAP/ML detection in Section 19.3, and likelihood ratio tests in Section 20.8.

E.2 Linearity of expectation and the variance of a sum

Expectation, variance, covariance

A random variable X is **integrable** if $E|X| < \infty$. Its expectation is $E[X] = \int_\Omega X\, dP$. This equals $\sum_x x\, p_X(x)$ for a pmf and $\int x f_X(x)\, dx$ for a pdf (Section 19.9). If $E[X^2] < \infty$, the **variance** is $\operatorname{Var}(X) = E[(X - \mu_X)^2]$ with $\mu_X = E[X]$. For X, Y with finite second moments, the **covariance** is $\operatorname{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$.

Statement

Linearity of expectation; variance of a sum

1. If X and Y are integrable and $a, b \in \mathbb{R}$, then $aX + bY$ is integrable and

$$E[aX + bY] = a\,E[X] + b\,E[Y].$$

No independence is assumed. 2. If $E[X^2] < \infty$ and $E[Y^2] < \infty$, then $\operatorname{Cov}(X, Y)$ is finite and

$$\operatorname{Var}(X + Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\operatorname{Cov}(X, Y).$$

3. More generally, for a random vector $\mathbf{X} = (X_1, \dots, X_n)^{\mathsf{T}}$ with finite second moments, covariance matrix $\boldsymbol{\Sigma}$ with entries $\Sigma_{ik} = \operatorname{Cov}(X_i, X_k)$, and $\mathbf{a} \in \mathbb{R}^n$,

$$\operatorname{Var}(\mathbf{a}^{\mathsf{T}}\mathbf{X}) = \sum_{i=1}^{n}\sum_{k=1}^{n} a_i a_k \Sigma_{ik} = \mathbf{a}^{\mathsf{T}}\boldsymbol{\Sigma}\,\mathbf{a},$$

and $\boldsymbol{\Sigma}$ is symmetric positive semidefinite.

Proof

Part 1. Integrability follows from the triangle inequality, $|aX + bY| \le |a||X| + |b||Y|$, together with monotonicity of the integral. The identity is the linearity of the Lebesgue integral $\int_\Omega (\cdot)\, dP$, which we quote. The two cases used in this book can also be checked directly.

- *Joint pmf $p(x, y)$.* By the law of the unconscious statistician (Section 19.9),

$$E[aX + bY] = \sum_{x,y} (ax + by)\, p(x, y) = a \sum_x x \sum_y p(x, y) + b \sum_y y \sum_x p(x, y) = a \sum_x x\, p_X(x) + b \sum_y y\, p_Y(y).$$

The rearrangement is legitimate because the double series converges absolutely: $\sum |ax + by|\, p \le |a| E|X| + |b| E|Y| < \infty$.

- *Joint pdf $f(x, y)$.* The same computation with $\iint$ in place of $\sum\sum$ works, using Fubini's theorem (justified by the same absolute bound) and the marginalisation $\int f(x, y)\, dy = f_X(x)$.

Only the marginals p_X and p_Y survive the computation. The dependence between X and Y, which is encoded in $p(x, y)$, has been summed out. That is why independence is not needed.

Part 2. First, finiteness. From $(|x| - |y|)^2 \ge 0$ we get $|xy| \le \tfrac{1}{2}(x^2 + y^2)$, so $E|XY| \le \tfrac{1}{2}(E[X^2] + E[Y^2]) < \infty$. Likewise $E|X| \le \tfrac{1}{2}(1 + E[X^2]) < \infty$. All the expectations below therefore exist. Put $\tilde{X} = X - \mu_X$ and $\tilde{Y} = Y - \mu_Y$. By part 1, $E[X + Y] = \mu_X + \mu_Y$, so $X + Y - E[X + Y] = \tilde{X} + \tilde{Y}$ and

$$\operatorname{Var}(X + Y) = E\big[(\tilde{X} + \tilde{Y})^2\big] = E\big[\tilde{X}^2 + \tilde{Y}^2 + 2\tilde{X}\tilde{Y}\big] = E[\tilde{X}^2] + E[\tilde{Y}^2] + 2E[\tilde{X}\tilde{Y}],$$

where the last equality is part 1 again. The three terms are $\operatorname{Var} X$, $\operatorname{Var} Y$ and $\operatorname{Cov}(X, Y)$.

Part 3. With $\tilde{X}_i = X_i - \mu_i$, we have $\mathbf{a}^{\mathsf{T}}\mathbf{X} - E[\mathbf{a}^{\mathsf{T}}\mathbf{X}] = \sum_i a_i \tilde{X}_i$. Squaring the finite sum and applying part 1 term by term gives

$$E\Big[\Big(\sum_i a_i \tilde{X}_i\Big)^2\Big] = \sum_i \sum_k a_i a_k E[\tilde{X}_i \tilde{X}_k] = \mathbf{a}^{\mathsf{T}}\boldsymbol{\Sigma}\,\mathbf{a}.$$

Symmetry of $\boldsymbol{\Sigma}$ is clear from the definition. Positive semidefiniteness holds because $\mathbf{a}^{\mathsf{T}}\boldsymbol{\Sigma}\mathbf{a}$ is a variance, the expectation of a square, and so is ≥ 0 for every $\mathbf{a}$. $\blacksquare$

Remarks

- *Independent or uncorrelated terms.* If the X_i are pairwise **uncorrelated** ($\Sigma_{ik} = 0$ for $i \ne k$), part 3 reduces to $\operatorname{Var}(\sum a_i X_i) = \sum a_i^2 \operatorname{Var} X_i$. Independence implies uncorrelatedness (Section 19.9) but is not needed. This is the "quadrature addition" of tolerances and noise powers.

- *Counterexample to "variances always add".* Take $Y = -X$. Then $\operatorname{Var}(X+Y) = 0$, while $\operatorname{Var} X + \operatorname{Var} Y = 2\operatorname{Var} X$. The covariance term $-2\operatorname{Var} X$ accounts for the difference exactly. Differential signalling exploits this. Common-mode noise is positively correlated on the two wires, so it cancels in the difference.
- *Finite moments are essential.* If X is Cauchy, $E|X| = \infty$, and $E[X] + E[-X]$ is the meaningless $\infty - \infty$, even though $X + (-X) = 0$.
- *Where used.* Section 19.9 (sums, tolerance stack-up), Section 20.3 (covariance matrix), and every later section of this appendix.

E.3 Markov and Chebyshev inequalities

Statement

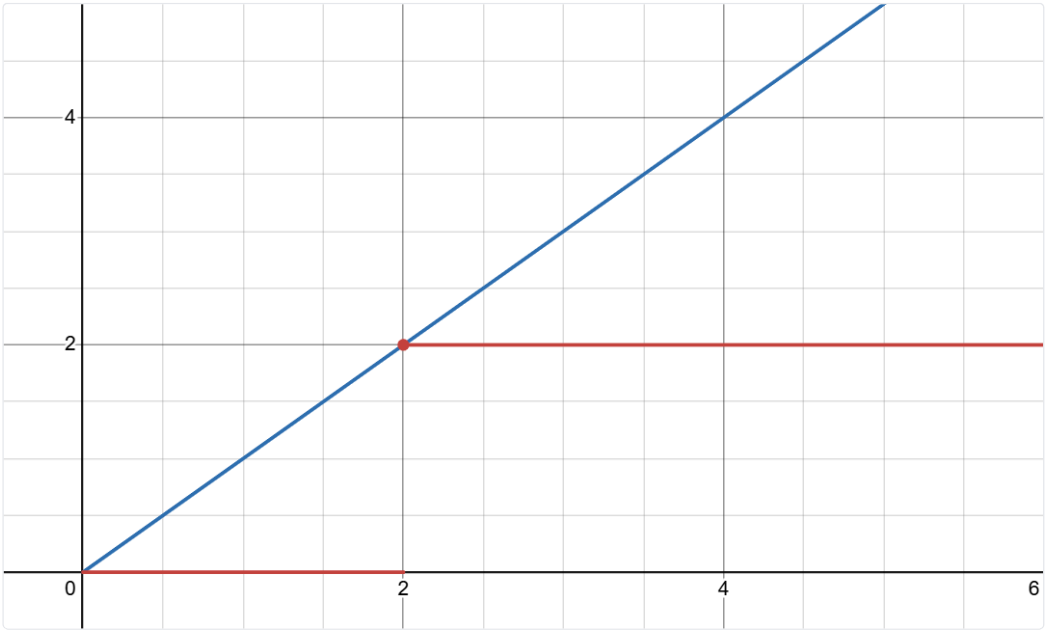
Markov and Chebyshev inequalities

1. **Markov.** If $X \geq 0$ and $a > 0$, then
$$P(X \geq a) \leq \frac{E[X]}{a}.$$
2. **Chebyshev.** If $E[X^2] < \infty$, $\mu = E[X]$, $\sigma^2 = \operatorname{Var} X$ and $\varepsilon > 0$, then
$$P(|X-\mu| \geq \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2}, \qquad \text{equivalently} \qquad P(|X-\mu| \geq k\sigma) \leq \frac{1}{k^2} \quad (k>0,\ \sigma>0).$$

Proof

Markov. Let $I = \mathbf{1}_{\{X \geq a\}}$ be the indicator of the event $\{X \geq a\}$. We claim that $aI \leq X$ at every outcome ω. If $X(\omega) \geq a$, then $aI(\omega) = a \leq X(\omega)$. Otherwise $aI(\omega) = 0 \leq X(\omega)$, because $X \geq 0$. Expectation is monotone: if $U \leq V$, then $E[V] - E[U] = E[V-U] \geq 0$ by E.2 and the non-negativity of the integral of a non-negative function. Hence $E[aI] \leq E[X]$. But $E[aI] = a\,P(X \geq a)$. Divide by $a > 0$. If $E[X] = \infty$, there is nothing to prove.

Chebyshev. The variable $Y = (X-\mu)^2$ is non-negative with $E[Y] = \sigma^2$. The events $\{|X-\mu| \geq \varepsilon\}$ and $\{Y \geq \varepsilon^2\}$ are identical, because $t \mapsto t^2$ is increasing on $t \geq 0$. Markov with $a = \varepsilon^2$ gives $P(Y \geq \varepsilon^2) \leq \sigma^2/\varepsilon^2$. Setting $\varepsilon = k\sigma$ gives the second form. ■



The pointwise inequality behind Markov's bound for $a = 2$: the step $a\,\mathbf{1}_{\{x \geq a\}}$ (red) never exceeds x (blue) on $x \geq 0$. Taking expectations of both sides gives $aP(X \geq a) \leq E[X]$.

Remarks

- *Sharpness.* Both inequalities are attained, so they cannot be improved without more information about the distribution. For Markov, let $X = a$ with probability p and $X = 0$ otherwise. Then $E[X]/a = p = P(X \geq a)$. For Chebyshev with $k \geq 1$, take the three-point law $P(X = \pm k\sigma) = \frac{1}{2k^2}$, $P(X = 0) = 1 - \frac{1}{k^2}$. It has mean 0, variance σ^2 and $P(|X| \geq k\sigma) = 1/k^2$.
- *Non-negativity matters for Markov.* For $X = \pm 1$ with equal probability, $E[X] = 0$ but $P(X \geq 1) = \frac{1}{2} > 0$. The proof breaks at $aI \leq X$ when $X(\omega) = -1$.
- *Generalisations.* Applying Markov to $|X-\mu|^r$ gives $P(|X-\mu| \geq \varepsilon) \leq E|X-\mu|^r/\varepsilon^r$. Applying it to e^{sX} gives the Chernoff bound $P(X \geq a) \leq e^{-sa}M_X(s)$ for $s > 0$, the basis of the exponential error bounds of coding theory.
- *Where used.* Distribution-free yield guarantees (Section 19.9, Example 19.43), the weak law in E.4, and the non-asymptotic Monte Carlo error bound in E.14.

E.4 The weak law of large numbers

Convergence in probability and in distribution

$Z_n \to c$ **in probability** if $P(|Z_n - c| \geq \varepsilon) \to 0$ as $n \to \infty$ for every $\varepsilon > 0$. $Z_n \to Z$ **in distribution** if $F_{Z_n}(z) \to F_Z(z)$ at every point z where F_Z is continuous.

Statement

Weak law of large numbers

Let $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$.

1. **(L^2 version.)** If $X_1, X_2, \ldots$ are pairwise uncorrelated with common mean μ and variances $\operatorname{Var} X_i \leq \sigma^2 < \infty$, then for every $\varepsilon > 0$

$$P\big(|\bar{X}_n - \mu| \geq \varepsilon\big) \leq \frac{\sigma^2}{n\varepsilon^2} \;\longrightarrow\; 0.$$

2. **(Khinchin's version.)** If $X_1, X_2, \ldots$ are i.i.d. with $E|X_1| < \infty$ and $\mu = E[X_1]$, then $\bar{X}_n \to \mu$ in probability. No variance is assumed.

Proof

Part 1. By linearity (E.2), $E[\bar{X}_n] = \frac{1}{n}\sum E[X_i] = \mu$. Because the X_i are pairwise uncorrelated, E.2 part 3 with $a_i = 1/n$ gives

$$\operatorname{Var}(\bar{X}_n) = \frac{1}{n^2}\sum_{i=1}^{n} \operatorname{Var} X_i \leq \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}.$$

Chebyshev's inequality (E.3) applied to $\bar{X}_n$ gives $P(|\bar{X}_n - \mu| \geq \varepsilon) \leq \operatorname{Var}(\bar{X}_n)/\varepsilon^2 \leq \sigma^2/(n\varepsilon^2)$.

Part 2. We use characteristic functions $\varphi_X(\omega) = E[e^{j\omega X}]$ (Section 19.9), Lévy's continuity theorem (stated in E.5), the product Lemma E.5a (proved in E.5), and one elementary fact:

- (ii) If $Z_n \to c$ in distribution and c is a constant, then $Z_n \to c$ in probability.

For (ii), note that the limit CDF $\mathbf{1}_{\{z \geq c\}}$ is continuous at $c - \varepsilon$ and at $c + \varepsilon/2$. So $P(|Z_n - c| \geq \varepsilon) \leq F_{Z_n}(c - \varepsilon) + 1 - F_{Z_n}(c + \varepsilon/2) \to 0 + 1 - 1 = 0$.

Step 1 (first-order expansion). Let $\varphi = \varphi_{X_1}$. For real x, $|e^{jx} - 1| \leq \min(|x|, 2)$, and hence $|e^{jx} - 1 - jx| \leq \min(x^2/2,\, 2|x|)$. The bounds $|x|$ and $x^2/2$ come from writing $e^{jx} - 1 = \int_0^x j e^{ju}\,du$ and $e^{jx} - 1 - jx = \int_0^x j(e^{ju} - 1)\,du$ and using $|e^{ju}| = 1$. The bounds 2 and $2|x|$ come from the triangle inequality. Hence

$$\big|\varphi(\omega) - 1 - j\omega\mu\big| = \big|E\big[e^{j\omega X_1} - 1 - j\omega X_1\big]\big| \leq |\omega|\, E\Big[\min\Big(\tfrac{|\omega| X_1^2}{2},\, 2|X_1|\Big)\Big].$$

The random variable inside the last expectation tends to 0 pointwise as $\omega \to 0$ and is dominated by $2|X_1|$, which is integrable. By dominated convergence, the expectation tends to 0. Therefore $\varphi(\omega) = 1 + j\omega\mu + \omega\, r(\omega)$ with $r(\omega) \to 0$ as $\omega \to 0$.

Step 2 (product). By independence, the characteristic function of a sum is the product of the characteristic functions, so $\varphi_{\bar{X}_n}(\omega) = \varphi(\omega/n)^n$. Fix ω. Lemma E.5a needs factors of modulus at most 1. Characteristic functions satisfy $|\varphi| \leq 1$, and so does $e^{j\omega\mu/n}$. Using Step 1 and $|e^{jx} - 1 - jx| \leq x^2/2$,

$$\big|\varphi(\omega/n) - e^{j\omega\mu/n}\big| \leq \frac{|\omega|}{n}\,|r(\omega/n)| + \frac{\omega^2\mu^2}{2n^2}.$$

Lemma E.5a with n equal factors then gives

$$\big|\varphi(\omega/n)^n - e^{j\omega\mu}\big| \leq n\,\big|\varphi(\omega/n) - e^{j\omega\mu/n}\big| \leq |\omega|\,|r(\omega/n)| + \frac{\omega^2\mu^2}{2n} \;\longrightarrow\; 0.$$

Step 3 (conclusion). $e^{j\omega\mu}$ is the characteristic function of the constant μ. By Lévy's theorem $\bar{X}_n \to \mu$ in distribution, and by (ii) in probability. ∎

Remarks

- *What the law says and does not say.* Part 1 gives a *rate*, $P(|\bar{X}_n - \mu| \geq \varepsilon) = O(1/n)$, with an explicit constant. It justifies the relative-frequency interpretation of Section 19.1. With $X_i = \mathbf{1}_A$ on independent trials, $\bar{X}_n = N_A/n \to P(A)$. It says nothing about a *single* infinite run of experiments converging. That is the strong law ($\bar{X}_n \to \mu$ with probability one), whose proof is harder and which is quoted in Section 24.6.
- *Counterexample without a mean.* If the X_i are i.i.d. standard Cauchy, then $\varphi(\omega) = e^{-|\omega|}$ and $\varphi(\omega/n)^n = e^{-|\omega|}$. So $\bar{X}_n$ is again standard Cauchy for every n and does not concentrate at all. Averaging does not help against impulsive noise of this kind. Median filters do.
- *Correlation matters.* If $X_i = X_1$ for all i (perfect correlation), then $\bar{X}_n = X_1$ and nothing converges. Part 1 needs the variance of the average to vanish. The same principle, with correlation that decays in time, gives the mean-ergodic theorem of Section 21.7.
- *Where used.* Consistency of estimators (Section 20.6), Monte Carlo (E.14), ergodicity (Section 21.7).

E.5 The central limit theorem

Statement

Central limit theorem (Lindeberg–Lévy)

Let $X_1, X_2, \dots$ be i.i.d. with mean μ and finite variance $\sigma^2 > 0$. Put

$$Z_n = \frac{\sum_{i=1}^{n} X_i - n\mu}{\sigma\sqrt{n}} = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}}.$$

Then $Z_n \to \mathcal{N}(0,1)$ in distribution, i.e. $P(Z_n \le z) \to \Phi(z)$ for every $z \in \mathbb{R}$.

The proof rests on one quoted theorem and two lemmas.

Lévy's continuity theorem (quoted)

Let Z_n have characteristic functions φ_n. If $\varphi_n(\omega) \to \psi(\omega)$ for every $\omega \in \mathbb{R}$, and ψ is continuous at $\omega = 0$, then ψ is the characteristic function of some random variable Z, and $Z_n \to Z$ in distribution.

For a proof, see Billingsley (*Probability and Measure*, §26) or Durrett (*Probability: Theory and Examples*, §3.3). It is the probabilistic counterpart of the uniqueness of the Fourier transform (Ch. 16). The characteristic function is the Fourier transform of the pdf, with the sign of the exponent reversed.

Lemma E.5a (products of numbers in the unit disc)

If $a_1, \dots, a_n, b_1, \dots, b_n \in \mathbb{C}$ all have modulus at most 1, then

$$\Big|\prod_{i=1}^{n} a_i - \prod_{i=1}^{n} b_i\Big| \le \sum_{i=1}^{n} |a_i - b_i|.$$

Proof. Use induction on n. The case $n = 1$ is trivial. For the inductive step, write $A_m = \prod_{i \le m} a_i$ and $B_m = \prod_{i \le m} b_i$, both of modulus ≤ 1. Then

$$A_n - B_n = a_n(A_{n-1} - B_{n-1}) + (a_n - b_n)B_{n-1},$$

so $|A_n - B_n| \le |A_{n-1} - B_{n-1}| + |a_n - b_n|$, which together with the induction hypothesis proves the claim. ∎

Lemma E.5b (second-order expansion of a characteristic function)

If $E[Y] = 0$ and $E[Y^2] = 1$, then $\varphi_Y(\omega) = 1 - \frac{\omega^2}{2} + \omega^2 r(\omega)$ with $r(\omega) \to 0$ as $\omega \to 0$.

Proof. For real x,

$$\Big|e^{jx} - 1 - jx + \frac{x^2}{2}\Big| \le \min\Big(\frac{|x|^3}{6},\, x^2\Big).$$

For the first bound, note that $e^{jx} - 1 - jx + \frac{x^2}{2} = \int_0^x j\big(e^{ju} - 1 - ju\big)\,du$. The integrand has modulus at most $u^2/2$ (E.4, Step 1), which integrates to $|x|^3/6$. For the second bound, $|e^{jx} - 1 - jx| \le x^2/2$ plus $|x^2/2|$ gives x^2. Now set $x = \omega Y$ and take expectations. Using $E[Y] = 0$ and $E[Y^2] = 1$,

$$\Big|\varphi_Y(\omega) - 1 + \frac{\omega^2}{2}\Big| \le E\Big[\min\Big(\frac{|\omega|^3|Y|^3}{6},\, \omega^2 Y^2\Big)\Big] = \omega^2\, E\Big[\min\Big(\frac{|\omega|\,|Y|^3}{6},\, Y^2\Big)\Big].$$

Define $r(\omega) = \big(\varphi_Y(\omega) - 1 + \omega^2/2\big)/\omega^2$ for $\omega \neq 0$. The display says that $|r(\omega)|$ is at most the last expectation. The variable inside that expectation tends to 0 pointwise as $\omega \to 0$ and is dominated by the integrable Y^2, so by dominated convergence $r(\omega) \to 0$. ∎

Proof of the theorem

Step 1 (standardise). Let $Y_i = (X_i - \mu)/\sigma$. The Y_i are i.i.d. with $E[Y_i] = 0$ and $E[Y_i^2] = 1$, and $Z_n = n^{-1/2}\sum_{i=1}^{n} Y_i$. Let φ be the common characteristic function of the Y_i.

Step 2 (characteristic function of Z_n). By independence, the expectation of a product of functions of independent variables factorises (Section 19.9). Hence

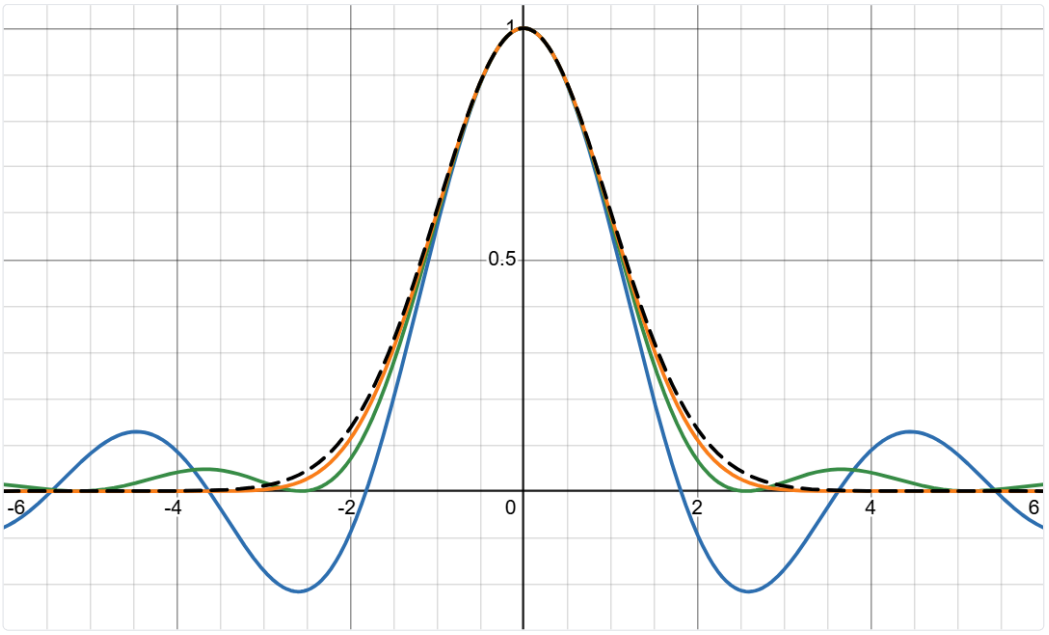
$$\varphi_{Z_n}(\omega) = E\Big[\prod_{i=1}^{n} e^{j\omega Y_i/\sqrt{n}}\Big] = \prod_{i=1}^{n} E\big[e^{j\omega Y_i/\sqrt{n}}\big] = \varphi\Big(\frac{\omega}{\sqrt{n}}\Big)^n.$$

Step 3 (compare with a simple product). Fix ω and let $n \ge \omega^2/2$, so that $b_n = 1 - \frac{\omega^2}{2n} \in [0,1]$. Every characteristic function satisfies $|\varphi| \le E|e^{j\omega Y}| = 1$. Lemma E.5a with n equal factors and Lemma E.5b give

$$\Big|\varphi\Big(\frac{\omega}{\sqrt{n}}\Big)^n - b_n^n\Big| \le n\,\Big|\varphi\Big(\frac{\omega}{\sqrt{n}}\Big) - 1 + \frac{\omega^2}{2n}\Big| = n\cdot\frac{\omega^2}{n}\Big|r\Big(\frac{\omega}{\sqrt{n}}\Big)\Big| = \omega^2\Big|r\Big(\frac{\omega}{\sqrt{n}}\Big)\Big| \;\longrightarrow\; 0.$$

Step 4 (the limit). $b_n^n = \big(1 - \frac{\omega^2}{2n}\big)^n \to e^{-\omega^2/2}$, the standard limit $(1 + c/n)^n \to e^c$ (Ch. 4). Together with Step 3, $\varphi_{Z_n}(\omega) \to e^{-\omega^2/2}$ for every ω.

Step 5 (identify and conclude). $e^{-\omega^2/2}$ is continuous and is the characteristic function of $\mathcal{N}(0,1)$ (Section 19.9). By Lévy's continuity theorem, $Z_n \to \mathcal{N}(0,1)$ in distribution. Since Φ is continuous everywhere, $P(Z_n \le z) \to \Phi(z)$ at every z. ∎



The proof in pictures. The characteristic function $\varphi(\omega) = \sin(\sqrt{3}\,\omega)/(\sqrt{3}\,\omega)$ of a unit-variance uniform variable (blue) and $\varphi(\omega/\sqrt{n})^n$ for $n = 2$ (green) and $n = 5$ (orange) close in on $e^{-\omega^2/2}$ (dashed). At $\omega = 1.5$ the values are $0.199,\ 0.276,\ 0.307$, and 0.320 for $n = 20$, against 0.325.

Remarks

- *Finite variance is essential.* For i.i.d. standard Cauchy variables, $\varphi(\omega) = e^{-|\omega|}$ and $\varphi(\omega/\sqrt{n})^n = e^{-\sqrt{n}|\omega|}$. This tends to 0 for $\omega \neq 0$ and to 1 at $\omega = 0$, a limit that is not continuous at 0, so Lévy's theorem does not apply. Indeed Z_n (with $\mu = 0$, $\sigma = 1$ inserted formally) is Cauchy with scale $\sqrt{n}$, and its mass escapes to infinity. The correct normalisation for Cauchy sums is $1/n$, under which $\bar{X}_n$ stays Cauchy (E.4). Lemma E.5b is exactly where $E[Y^2] < \infty$ is used.
- *Independence is essential.* If $X_i = X_1$ for all i, then $Z_n = \sqrt{n}\,(X_1 - \mu)/\sigma$, which never becomes Gaussian unless X_1 already is. Weak dependence is allowed in versions for mixing sequences, which matter for correlated noise samples (Section 21.7).
- *Identical distribution can be relaxed.* The Lindeberg condition allows independent, non-identical summands provided no single term dominates the variance. This is why the sum of many small, independent physical contributions (electron motions in a resistor, Section 21.8) is Gaussian.
- *Rate.* The proof gives no rate. The Berry–Esseen theorem, $\sup_z |P(Z_n \le z) - \Phi(z)| \le C\,E|Y|^3/\sqrt{n}$, needs a third moment (Section 20.5).
- *Where used.* Sampling distributions and confidence intervals (Sections 20.5 and 20.7), Gaussian noise models (Section 21.5), and Monte Carlo error bars (E.14).

E.6 The sample variance with divisor $n - 1$ is unbiased

Statement

Unbiasedness of S^2

Let $X_1, \dots, X_n$ ($n \ge 2$) be pairwise uncorrelated with common mean μ and common finite variance σ^2. Let $\bar{X} = \frac{1}{n}\sum X_i$ and

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

Then $E[S^2] = \sigma^2$.

Proof

Step 1 (algebraic identity). Write $X_i - \bar{X} = (X_i - \mu) - (\bar{X} - \mu)$ and expand the square:

$$\sum_{i=1}^{n}(X_i - \bar{X})^2 = \sum_{i=1}^{n}(X_i - \mu)^2 - 2(\bar{X} - \mu)\sum_{i=1}^{n}(X_i - \mu) + n(\bar{X} - \mu)^2.$$

Since $\sum_i (X_i - \mu) = n(\bar{X} - \mu)$, the middle term is $-2n(\bar{X} - \mu)^2$. Therefore

$$\sum_{i=1}^{n}(X_i - \bar{X})^2 = \sum_{i=1}^{n}(X_i - \mu)^2 - n(\bar{X} - \mu)^2.$$

Step 2 (expectations). $E[(X_i - \mu)^2] = \sigma^2$ by definition. $E[\bar{X}] = \mu$ by linearity, so $E[(\bar{X} - \mu)^2] = \operatorname{Var}\bar{X} = \sigma^2/n$ by E.2, part 3, using pairwise uncorrelatedness.

Step 3 (combine). By linearity,

$$E\Big[\sum_{i=1}^{n}(X_i - \bar{X})^2\Big] = n\sigma^2 - n\cdot\frac{\sigma^2}{n} = (n-1)\sigma^2,$$

and dividing by $n - 1$ gives $E[S^2] = \sigma^2$. ∎

Remarks

- *The divisor n is biased.* $\hat\sigma^2 = \frac{1}{n}\sum (X_i - \bar X)^2$ has $E[\hat\sigma^2] = \frac{n-1}{n}\sigma^2$, an underestimate by σ^2/n. A simulation with $n = 5$ standard Gaussian samples (2×10^5 replications) gives mean values 0.999 for S^2 and 0.799 for $\hat\sigma^2$, against the predicted 1 and 0.8. The bias comes from Step 1. The data sit closer to their own mean $\bar X$ than to μ, and the deficit is exactly $n(\bar X - \mu)^2$, with expectation σ^2.
- *Counterexample: correlated samples.* If the X_i are equicorrelated, $\operatorname{Cov}(X_i, X_k) = \rho\sigma^2$ for $i \neq k$, then $\operatorname{Var}\bar X = \sigma^2\big(1 + (n-1)\rho\big)/n$. Step 3 becomes $n\sigma^2 - \sigma^2\big(1 + (n-1)\rho\big) = (n-1)(1-\rho)\sigma^2$, so $E[S^2] = (1-\rho)\sigma^2$. Positively correlated samples, such as noise sampled faster than its correlation time (Section 21.3), make S^2 underestimate the variance. Slow drift is invisible to it.
- *S is biased for σ.* $\sqrt{\cdot}$ is strictly concave, so Jensen's inequality gives $E[S] < \sqrt{E[S^2]} = \sigma$ unless S is almost surely constant.
- *Where used.* Section 20.2 ("Why divide by $n-1$?"), the χ^2_{n-1} sampling distribution in Section 20.5, Type A uncertainty in Section 20.10, and the Monte Carlo standard error in Section 24.6. The regression analogue, divisor $n - p$, is proved in E.7.

E.7 Least-squares regression and the Gauss–Markov theorem

Linear model

The **linear model** is $\mathbf{y} = \mathbf{X}\boldsymbol\beta + \boldsymbol\varepsilon$. Here $\mathbf{y} \in \mathbb{R}^n$ is observed, the **design matrix** $\mathbf{X} \in \mathbb{R}^{n \times p}$ is known and deterministic, $\boldsymbol\beta \in \mathbb{R}^p$ is unknown, and the errors satisfy $E[\boldsymbol\varepsilon] = \mathbf{0}$ and $\operatorname{Cov}(\boldsymbol\varepsilon) = \sigma^2\mathbf{I}$, i.e. they are uncorrelated with equal variance (*homoscedastic*). Simple linear regression $Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ is the case $p = 2$, where row i of $\mathbf{X}$ is $(1,\ x_i)$.

Statement

Least squares and its sampling properties

Suppose $\operatorname{rank}\mathbf{X} = p$ (so $n \geq p$).

1. $\mathbf{X}^{\mathsf T}\mathbf{X}$ is invertible, and $S(\boldsymbol\beta) = \|\mathbf{y} - \mathbf{X}\boldsymbol\beta\|^2$ has the **unique** minimiser
$$\hat{\boldsymbol\beta} = (\mathbf{X}^{\mathsf T}\mathbf{X})^{-1}\mathbf{X}^{\mathsf T}\mathbf{y}.$$

The residual $\mathbf{e} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol\beta}$ satisfies $\mathbf{X}^{\mathsf T}\mathbf{e} = \mathbf{0}$. For simple regression, the rank condition says that the x_i are not all equal ($S_{xx} > 0$), and

$$\hat\beta_1 = \frac{S_{xy}}{S_{xx}}, \qquad \hat\beta_0 = \bar y - \hat\beta_1 \bar x, \qquad S_{xx} = \sum (x_i - \bar x)^2,\ \ S_{xy} = \sum (x_i - \bar x)(y_i - \bar y).$$

2. Under the linear model, $E[\hat{\boldsymbol\beta}] = \boldsymbol\beta$, $\operatorname{Cov}(\hat{\boldsymbol\beta}) = \sigma^2(\mathbf{X}^{\mathsf T}\mathbf{X})^{-1}$, and $E[\mathrm{SSE}] = (n-p)\sigma^2$, where $\mathrm{SSE} = \|\mathbf{e}\|^2$. In simple regression, $\operatorname{Var}\hat\beta_1 = \sigma^2/S_{xx}$ and $\operatorname{Var}\hat\beta_0 = \sigma^2\sum x_i^2/(nS_{xx})$.
3. **Gauss–Markov.** Let $\tilde{\boldsymbol\beta} = \mathbf{C}\mathbf{y}$, with $\mathbf{C} \in \mathbb{R}^{p \times n}$ deterministic, be any linear estimator that is unbiased for every $\boldsymbol\beta$. Then $\operatorname{Cov}(\tilde{\boldsymbol\beta}) - \operatorname{Cov}(\hat{\boldsymbol\beta})$ is positive semidefinite. In particular, $\operatorname{Var}(\mathbf{a}^{\mathsf T}\tilde{\boldsymbol\beta}) \geq \operatorname{Var}(\mathbf{a}^{\mathsf T}\hat{\boldsymbol\beta})$ for every $\mathbf{a} \in \mathbb{R}^p$. Least squares is the **best linear unbiased estimator** (BLUE).

Proof

Part 1 (invertibility). If $\mathbf{X}^{\mathsf T}\mathbf{X}\mathbf{v} = \mathbf{0}$, then $0 = \mathbf{v}^{\mathsf T}\mathbf{X}^{\mathsf T}\mathbf{X}\mathbf{v} = \|\mathbf{X}\mathbf{v}\|^2$, so $\mathbf{X}\mathbf{v} = \mathbf{0}$. Full column rank then forces $\mathbf{v} = \mathbf{0}$. A square matrix with trivial null space is invertible (Ch. 8).

Part 1 (minimiser). Define $\hat{\boldsymbol\beta}$ by the formula. Then $\mathbf{X}^{\mathsf T}\mathbf{X}\hat{\boldsymbol\beta} = \mathbf{X}^{\mathsf T}\mathbf{y}$ (the **normal equations**), i.e. $\mathbf{X}^{\mathsf T}\mathbf{e} = \mathbf{0}$. For any $\boldsymbol\beta$, write $\mathbf{y} - \mathbf{X}\boldsymbol\beta = \mathbf{e} + \mathbf{X}(\hat{\boldsymbol\beta} - \boldsymbol\beta)$ and expand:

$$S(\boldsymbol\beta) = \|\mathbf{e}\|^2 + 2(\hat{\boldsymbol\beta} - \boldsymbol\beta)^{\mathsf T}\mathbf{X}^{\mathsf T}\mathbf{e} + \|\mathbf{X}(\hat{\boldsymbol\beta} - \boldsymbol\beta)\|^2 = S(\hat{\boldsymbol\beta}) + \|\mathbf{X}(\hat{\boldsymbol\beta} - \boldsymbol\beta)\|^2.$$

The last term is ≥ 0, with equality iff $\mathbf{X}(\hat{\boldsymbol\beta} - \boldsymbol\beta) = \mathbf{0}$, iff $\boldsymbol\beta = \hat{\boldsymbol\beta}$ (full rank again). So $\hat{\boldsymbol\beta}$ is the unique minimiser. This argument needs no calculus. It is Pythagoras' theorem for the orthogonal projection $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol\beta}$ of $\mathbf{y}$ onto the column space of $\mathbf{X}$ (Ch. 9).

Part 1 (simple regression). The first row of $\mathbf{X}^{\mathsf T}\mathbf{e} = \mathbf{0}$ is $\sum e_i = 0$, i.e. $\bar y = \hat\beta_0 + \hat\beta_1 \bar x$. The second row is $\sum x_i e_i = 0$. Subtract $\bar x$ times the first row to get $\sum (x_i - \bar x)e_i = 0$. Substituting $e_i = (y_i - \bar y) - \hat\beta_1(x_i - \bar x)$ gives $S_{xy} - \hat\beta_1 S_{xx} = 0$. The two columns $(1, \dots, 1)^{\mathsf T}$ and $(x_1, \dots, x_n)^{\mathsf T}$ are linearly independent iff the x_i are not all equal, iff $S_{xx} > 0$.

Part 2. Let $\mathbf{A} = (\mathbf{X}^{\mathsf T}\mathbf{X})^{-1}\mathbf{X}^{\mathsf T}$, so that $\mathbf{A}\mathbf{X} = \mathbf{I}_p$ and $\hat{\boldsymbol\beta} = \mathbf{A}\mathbf{y} = \boldsymbol\beta + \mathbf{A}\boldsymbol\varepsilon$. Linearity gives $E[\hat{\boldsymbol\beta}] = \boldsymbol\beta$. For any deterministic matrix $\mathbf{M}$, $\operatorname{Cov}(\mathbf{M}\boldsymbol\varepsilon) = \mathbf{M}\operatorname{Cov}(\boldsymbol\varepsilon)\mathbf{M}^{\mathsf T}$. This is E.2, part 3, applied to each linear combination. So

$$\operatorname{Cov}(\hat{\boldsymbol\beta}) = \sigma^2\mathbf{A}\mathbf{A}^{\mathsf T} = \sigma^2(\mathbf{X}^{\mathsf T}\mathbf{X})^{-1}\mathbf{X}^{\mathsf T}\mathbf{X}(\mathbf{X}^{\mathsf T}\mathbf{X})^{-1} = \sigma^2(\mathbf{X}^{\mathsf T}\mathbf{X})^{-1}.$$

For the residuals, let $\mathbf{H} = \mathbf{X}\mathbf{A}$ (the hat matrix). It is symmetric and idempotent ($\mathbf{H}^2 = \mathbf{H}$), and $(\mathbf{I} - \mathbf{H})\mathbf{X} = \mathbf{0}$. Hence $\mathbf{e} = (\mathbf{I} - \mathbf{H})\mathbf{y} = (\mathbf{I} - \mathbf{H})\boldsymbol\varepsilon$ and $\mathrm{SSE} = \boldsymbol\varepsilon^{\mathsf T}(\mathbf{I} - \mathbf{H})\boldsymbol\varepsilon$. For any square $\mathbf{M}$, $E[\boldsymbol\varepsilon^{\mathsf T}\mathbf{M}\boldsymbol\varepsilon] = \sum_{i,k} M_{ik}E[\varepsilon_i\varepsilon_k] = \sigma^2 \operatorname{tr}\mathbf{M}$. Using $\operatorname{tr}(\mathbf{P}\mathbf{Q}) = \operatorname{tr}(\mathbf{Q}\mathbf{P})$,

$$E[\mathrm{SSE}] = \sigma^2 \operatorname{tr}(\mathbf{I}_n - \mathbf{H}) = \sigma^2\big(n - \operatorname{tr}\left((\mathbf{X}^{\mathsf T}\mathbf{X})^{-1}\mathbf{X}^{\mathsf T}\mathbf{X}\right)\big) = \sigma^2(n - \operatorname{tr}\mathbf{I}_p) = (n-p)\sigma^2.$$

For simple regression, $\mathbf{X}^{\mathsf T}\mathbf{X} = \begin{pmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix}$ has determinant $n\sum x_i^2 - (\sum x_i)^2 = nS_{xx}$. Its inverse is $\frac{1}{nS_{xx}}\begin{pmatrix} \sum x_i^2 & -\sum x_i \\ -\sum x_i & n \end{pmatrix}$, whose diagonal gives the two variances.

Part 3 (Gauss–Markov). Unbiasedness for every $\boldsymbol\beta$ means $E[\mathbf{C}\mathbf{y}] = \mathbf{C}\mathbf{X}\boldsymbol\beta = \boldsymbol\beta$ for all $\boldsymbol\beta$, i.e. $\mathbf{C}\mathbf{X} = \mathbf{I}_p$. Put $\mathbf{D} = \mathbf{C} - \mathbf{A}$. Then $\mathbf{D}\mathbf{X} = \mathbf{I}_p - \mathbf{I}_p = \mathbf{0}$, and therefore

$$\mathbf{A}\mathbf{D}^{\mathsf T} = (\mathbf{X}^{\mathsf T}\mathbf{X})^{-1}(\mathbf{D}\mathbf{X})^{\mathsf T} = \mathbf{0}, \qquad \mathbf{D}\mathbf{A}^{\mathsf T} = (\mathbf{A}\mathbf{D}^{\mathsf T})^{\mathsf T} = \mathbf{0}.$$

Since $\tilde{\boldsymbol\beta} = \mathbf{C}\mathbf{X}\boldsymbol\beta + \mathbf{C}\boldsymbol\varepsilon$,

$$\operatorname{Cov}(\tilde{\boldsymbol{\beta}}) = \sigma^2 \mathbf{C}\mathbf{C}^{\mathsf{T}} = \sigma^2 (\mathbf{A}+\mathbf{D})(\mathbf{A}+\mathbf{D})^{\mathsf{T}} = \sigma^2 \left(\mathbf{A}\mathbf{A}^{\mathsf{T}} + \mathbf{D}\mathbf{D}^{\mathsf{T}}\right) = \operatorname{Cov}(\hat{\boldsymbol{\beta}}) + \sigma^2 \mathbf{D}\mathbf{D}^{\mathsf{T}}.$$

Finally, $\mathbf{a}^{\mathsf{T}}\mathbf{D}\mathbf{D}^{\mathsf{T}}\mathbf{a} = \|\mathbf{D}^{\mathsf{T}}\mathbf{a}\|^2 \geq 0$, so the difference is positive semidefinite, and $\operatorname{Var}(\mathbf{a}^{\mathsf{T}}\tilde{\boldsymbol{\beta}}) - \operatorname{Var}(\mathbf{a}^{\mathsf{T}}\hat{\boldsymbol{\beta}}) = \sigma^2 \|\mathbf{D}^{\mathsf{T}}\mathbf{a}\|^2 \geq 0$. Equality for every $\mathbf{a}$ holds iff $\mathbf{D} = \mathbf{0}$, so least squares is the *unique* BLUE. ■

Remarks

- *No Gaussian assumption.* Gauss–Markov uses only the first two moments of $\boldsymbol{\varepsilon}$. With Gaussian errors, $\hat{\boldsymbol{\beta}}$ is also the maximum-likelihood estimator and attains the Cramér–Rao bound, so it is best among *all* unbiased estimators, linear or not (Section 20.6). Without Gaussianity, nonlinear estimators can win. For heavy-tailed (e.g. Laplacian) errors, least-absolute-deviation fitting has smaller variance than least squares.
- *Each hypothesis matters.*
  - **Equal variance.** If $\operatorname{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{V}$ with known $\mathbf{V} \neq \mathbf{I}$, then ordinary least squares is still unbiased but no longer best. Applying the theorem to the whitened model $\mathbf{V}^{-1/2}\mathbf{y} = \mathbf{V}^{-1/2}\mathbf{X}\boldsymbol{\beta} + \mathbf{V}^{-1/2}\boldsymbol{\varepsilon}$ shows that the BLUE is generalised least squares, $(\mathbf{X}^{\mathsf{T}}\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{X}^{\mathsf{T}}\mathbf{V}^{-1}\mathbf{y}$. With diagonal $\mathbf{V}$ this is the weighted least squares of Section 20.9.
  - **Unbiasedness.** Biased linear estimators can have smaller mean-square error. Ridge regression $(\mathbf{X}^{\mathsf{T}}\mathbf{X} + \lambda \mathbf{I})^{-1}\mathbf{X}^{\mathsf{T}}\mathbf{y}$ does, for suitable $\lambda > 0$, when $\mathbf{X}^{\mathsf{T}}\mathbf{X}$ is ill-conditioned.
  - **Full rank.** With collinear regressors, $\boldsymbol{\beta}$ is not identifiable. Many $\boldsymbol{\beta}$ give the same $\mathbf{X}\boldsymbol{\beta}$ (Section 25.8).
- *Divisor $n-p$.* Part 2 shows that $s^2 = \mathrm{SSE}/(n-p)$ is unbiased for σ^2. E.6 is the special case $p = 1$, $\mathbf{X} = (1, \dots, 1)^{\mathsf{T}}$, $\hat{\beta} = \bar{y}$.
- *Numerics.* The formula $(\mathbf{X}^{\mathsf{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathsf{T}}\mathbf{y}$ is for analysis. Compute $\hat{\boldsymbol{\beta}}$ by QR factorisation (Section 24.3), because $\kappa(\mathbf{X}^{\mathsf{T}}\mathbf{X}) = \kappa(\mathbf{X})^2$.
- *Where used.* Section 20.9, discrete least squares in Section 22.6, and parameter identification in Section 25.8.

E.8 The law of propagation of uncertainty (GUM)

Measurement model

A measurand is modelled as $Y = f(\mathbf{X})$, where $\mathbf{X} = (X_1, \dots, X_N)^{\mathsf{T}}$ is a random vector of input quantities. The input **estimates** are $\mathbf{x} = E[\mathbf{X}]$, the **standard uncertainties** are $u(x_i) = \sqrt{\operatorname{Var} X_i}$, and the **covariances** are $u(x_i, x_k) = \operatorname{Cov}(X_i, X_k)$. Together they form the input covariance matrix $\boldsymbol{\Sigma}_x$. The **sensitivity coefficients** are $c_i = \partial f / \partial x_i$ evaluated at $\mathbf{x}$, and $\mathbf{c} = \nabla f(\mathbf{x})$.

Statement

First-order propagation of uncertainty and its accuracy

Let $f : \mathbb{R}^N \to \mathbb{R}$ be twice continuously differentiable, with Hessian bounded in the spectral norm, $\|\nabla^2 f(\boldsymbol{\xi})\|_2 \leq M$ for all $\boldsymbol{\xi}$. Let $\mathbf{X}$ have mean $\mathbf{x}$, covariance $\boldsymbol{\Sigma}_x$, and finite fourth moments, $m_4 = E\|\mathbf{X} - \mathbf{x}\|^4 < \infty$. Define the linearised model $Y_{\mathrm{lin}} = f(\mathbf{x}) + \mathbf{c}^{\mathsf{T}}(\mathbf{X} - \mathbf{x})$ and the **combined standard uncertainty**

$$u_c^2(y) = \mathbf{c}^{\mathsf{T}} \boldsymbol{\Sigma}_x \mathbf{c} = \sum_{i=1}^{N} c_i^2 u^2(x_i) + 2 \sum_{i=1}^{N-1} \sum_{k=i+1}^{N} c_i c_k \, u(x_i, x_k).$$

Then:

1. $E[Y_{\mathrm{lin}}] = f(\mathbf{x})$ and $\operatorname{Var}(Y_{\mathrm{lin}}) = u_c^2(y)$ **exactly**;
2. $|Y - Y_{\mathrm{lin}}| \leq \tfrac{M}{2}\|\mathbf{X} - \mathbf{x}\|^2$ at every outcome, hence $|E[Y] - f(\mathbf{x})| \leq \tfrac{M}{2} \operatorname{tr} \boldsymbol{\Sigma}_x$;
3. the true standard deviation σ_Y of Y satisfies

$$\left|\sigma_Y - u_c(y)\right| \leq \frac{M}{2} \sqrt{m_4}.$$

If all input deviations scale with a small parameter u ($\mathbf{X} - \mathbf{x} = u\mathbf{W}$ with $\mathbf{W}$ fixed), then $u_c = O(u)$ while the errors in parts 2 and 3 are $O(u^2)$. The relative error of the GUM formula is therefore $O(u)$, provided $\mathbf{c} \neq \mathbf{0}$.

Proof

Part 1. Y_{lin} is the constant $f(\mathbf{x}) - \mathbf{c}^{\mathsf{T}}\mathbf{x}$ plus the linear form $\mathbf{c}^{\mathsf{T}}\mathbf{X}$. By linearity of expectation, $E[Y_{\mathrm{lin}}] = f(\mathbf{x}) + \mathbf{c}^{\mathsf{T}}(E\mathbf{X} - \mathbf{x}) = f(\mathbf{x})$. Adding a constant does not change a variance, so E.2, part 3, gives $\operatorname{Var}(Y_{\mathrm{lin}}) = \mathbf{c}^{\mathsf{T}}\boldsymbol{\Sigma}_x\mathbf{c}$. Written out, the double sum $\sum_{i,k} c_i c_k (\boldsymbol{\Sigma}_x)_{ik}$ has diagonal terms $c_i^2 u^2(x_i)$ and each off-diagonal pair (i,k), (k,i) appearing twice. That is the displayed formula.

Part 2 (Taylor's theorem). Fix an outcome and put $\mathbf{d} = \mathbf{X} - \mathbf{x}$ and $g(t) = f(\mathbf{x} + t\mathbf{d})$ for $t \in [0,1]$. Then $g \in C^2$, $g'(t) = \nabla f(\mathbf{x} + t\mathbf{d})^{\mathsf{T}}\mathbf{d}$ and $g''(t) = \mathbf{d}^{\mathsf{T}} \nabla^2 f(\mathbf{x} + t\mathbf{d})\, \mathbf{d}$ (chain rule, Ch. 6). Taylor's theorem with Lagrange remainder (Ch. 4) gives $g(1) = g(0) + g'(0) + \tfrac{1}{2} g''(\theta)$ for some $\theta \in (0,1)$, i.e.

$$R := Y - Y_{\mathrm{lin}} = \tfrac{1}{2}\, \mathbf{d}^{\mathsf{T}} \nabla^2 f(\mathbf{x} + \theta \mathbf{d})\, \mathbf{d}, \qquad |R| \leq \tfrac{1}{2}\|\nabla^2 f\|_2 \|\mathbf{d}\|^2 \leq \tfrac{M}{2}\|\mathbf{X} - \mathbf{x}\|^2.$$

The quadratic-form bound $|\mathbf{d}^{\mathsf{T}}\mathbf{B}\mathbf{d}| \leq \|\mathbf{B}\|_2 \|\mathbf{d}\|^2$ is the Cauchy–Schwarz inequality (Ch. 9). Taking expectations, $|E[Y] - f(\mathbf{x})| = |E[R]| \leq E|R| \leq \tfrac{M}{2} E\|\mathbf{X} - \mathbf{x}\|^2 = \tfrac{M}{2} \operatorname{tr} \boldsymbol{\Sigma}_x$.

Part 3. First, Y has a finite second moment, because $|Y| \leq |f(\mathbf{x})| + \|\mathbf{c}\|\|\mathbf{d}\| + \tfrac{M}{2}\|\mathbf{d}\|^2$ and $m_4 < \infty$. Write $\|Z\|_2 = \sqrt{E[Z^2]}$, which is a norm on random variables with finite second moment (Minkowski's inequality). Then $\sigma_Y = \|Y - EY\|_2$ and $u_c = \|Y_{\mathrm{lin}} - f(\mathbf{x})\|_2$, and

$$Y - E[Y] = \big(Y_{\mathrm{lin}} - f(\mathbf{x})\big) + \big(R - E[R]\big).$$

By the reverse triangle inequality for the norm,

$$|\sigma_Y - u_c| \leq \|R - E[R]\|_2 = \sqrt{\operatorname{Var} R} \leq \sqrt{E[R^2]} \leq \frac{M}{2} \sqrt{E\|\mathbf{X} - \mathbf{x}\|^4}.$$

Scaling. With $\mathbf{d} = u\mathbf{W}$, $u_c = u\sqrt{\mathbf{c}^{\mathsf{T}} \operatorname{Cov}(\mathbf{W})\, \mathbf{c}}$ is proportional to u, while $\operatorname{tr} \boldsymbol{\Sigma}_x$ and $\sqrt{m_4}$ are proportional to u^2. ■

Remarks

- *Products and quotients.* For $Y = A\,X_1^{p_1}\cdots X_N^{p_N}$ with $x_i \neq 0$, $c_i = p_i y/x_i$. For uncorrelated inputs, part 1 becomes the relative-uncertainty rule $(u_c/y)^2 = \sum (p_i u(x_i)/x_i)^2$ of Section 20.10. Such f do not have a globally bounded Hessian. The theorem then applies with M the bound over a region that contains $\mathbf{X}$ with overwhelming probability, and the part of the distribution outside that region must be handled separately. This is the usual situation in practice.
- *Counterexample: vanishing sensitivity.* Let $Y = X^2$ with $X \sim \mathcal{N}(x, u^2)$. Then exactly $\operatorname{Var} Y = 4x^2u^2 + 2u^4$, while $u_c = 2|x|u$. At $x = 1$, $u = 0.1$: $\sigma_Y = 0.20050$ against $u_c = 0.2$, well inside the bound $\frac{M}{2}\sqrt{m_4} = \sqrt{3}\,u^2 = 0.0173$ ($M = 2$, $m_4 = 3u^4$). At $x = 0$, however, $c = 0$ and $u_c = 0$, yet $\sigma_Y = \sqrt{2}\,u^2$. The first-order term has vanished, and the "$O(u^2)$ error" is the whole answer. The GUM then requires second-order terms or Monte Carlo propagation (Section 24.8).
- *Second-order mean correction.* The proof also shows that $E[Y] = f(\mathbf{x}) + \frac{1}{2}\operatorname{tr}\left(\nabla^2 f(\mathbf{x})\boldsymbol{\Sigma}_x\right) + o(u^2)$ when $\nabla^2 f$ is continuous. A curved model *biases* the output. For example, $E[X^2] = x^2 + u^2$ exactly.
- *Correlation.* Part 1 is an identity for the linearised model, so the covariance terms are not optional. With $r = \pm 1$ they turn root-sum-of-squares addition into linear addition or cancellation, as the chapter discusses.
- *Where used.* Section 20.10, propagation of errors in Section 22.2, and linearised uncertainty propagation in Section 24.8.

E.9 The Wiener–Khinchin theorem and filtered noise

Power spectral density

Let $X(t)$ be a real wide-sense stationary (WSS) process with autocorrelation $R_{XX}(\tau) = E[X(t+\tau)X(t)]$. Assume X is **mean-square continuous** (R_{XX} is continuous at 0, hence everywhere, Section 21.3) and has jointly measurable sample paths. For $T > 0$, let $X_T(\omega) = \int_{-T}^{T} X(t)e^{-j\omega t}\,dt$. The **power spectral density** is

$$S_{XX}(\omega) = \lim_{T\to\infty} \frac{E\left[|X_T(\omega)|^2\right]}{2T},$$

whenever the limit exists.

Statement

Wiener–Khinchin

Under the assumptions of the definition, suppose $R_{XX} \in L^1(\mathbb{R})$, i.e. $\int_{-\infty}^{\infty}|R_{XX}(\tau)|\,d\tau < \infty$. Then for every ω the limit defining $S_{XX}(\omega)$ exists and

$$S_{XX}(\omega) = \int_{-\infty}^{\infty} R_{XX}(\tau)\, e^{-j\omega\tau}\,d\tau.$$

S_{XX} is real, even, non-negative, continuous and bounded by $\|R_{XX}\|_1$. If also $S_{XX} \in L^1$, then for every τ

$$R_{XX}(\tau) = \frac{1}{2\pi}\int_{-\infty}^{\infty} S_{XX}(\omega)e^{j\omega\tau}\,d\omega, \qquad E[X^2(t)] = \frac{1}{2\pi}\int_{-\infty}^{\infty} S_{XX}(\omega)\,d\omega.$$

Lemma E.9a (diagonal integration)

If g is integrable on $[-2T, 2T]$, then

$$\int_{-T}^{T}\int_{-T}^{T} g(t-s)\,dt\,ds = \int_{-2T}^{2T} \big(2T - |\tau|\big)\,g(\tau)\,d\tau.$$

Proof. In the inner integral substitute $\tau = t - s$ for fixed s. This gives $\int_{-T}^{T}\int_{-T-s}^{T-s} g(\tau)\,d\tau\,ds$, the integral of $g(\tau)$ over the region $\{(\tau, s) : |s| \le T,\ |s+\tau| \le T\}$. By Fubini's theorem, which applies because $\iint |g(t-s)|\,dt\,ds \le 2T\int_{-2T}^{2T}|g| < \infty$, we may integrate over s first. For fixed τ with $|\tau| \le 2T$, the admissible s form the interval $[\max(-T, -T-\tau),\ \min(T, T-\tau)]$. Its length is $2T - |\tau|$. ∎

Proof of the theorem

Step 1 (X_T exists and $E|X_T|^2$ is a double integral). By Cauchy–Schwarz, $E|X(t)X(s)| \le \sqrt{E[X^2(t)]E[X^2(s)]} = R_{XX}(0)$. Therefore

$$E\int_{-T}^{T}\int_{-T}^{T}|X(t)X(s)|\,dt\,ds \le 4T^2 R_{XX}(0) < \infty.$$

So, with probability one, X is integrable on $[-T, T]$, $X_T(\omega)$ exists, and Fubini's theorem allows expectation and integration to be interchanged:

$$E|X_T(\omega)|^2 = E\Big[\int_{-T}^{T}\int_{-T}^{T} X(t)X(s)e^{-j\omega(t-s)}\,dt\,ds\Big] = \int_{-T}^{T}\int_{-T}^{T} R_{XX}(t-s)\,e^{-j\omega(t-s)}\,dt\,ds.$$

Here $|X_T|^2 = X_T\overline{X_T}$ was used, and X is real.

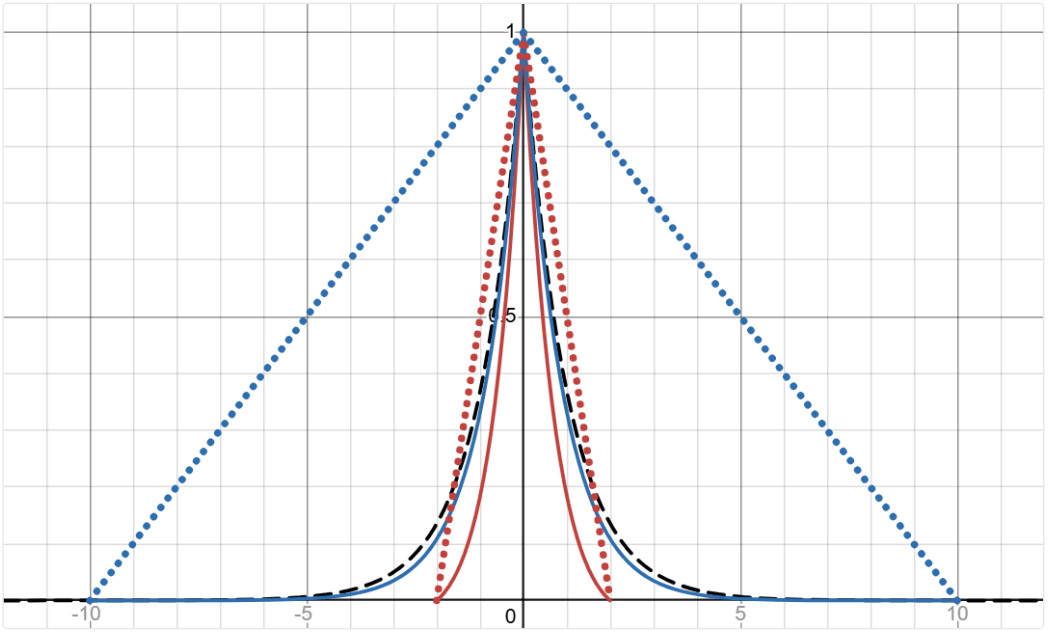
Step 2 (reduce to one integral). Lemma E.9a with $g(\tau) = R_{XX}(\tau)e^{-j\omega\tau}$ gives

$$\frac{E|X_T(\omega)|^2}{2T} = \int_{-\infty}^{\infty} w_T(\tau)\,R_{XX}(\tau)e^{-j\omega\tau}\,d\tau, \qquad w_T(\tau) = \Big(1 - \frac{|\tau|}{2T}\Big)\mathbf{1}_{\{|\tau|\le 2T\}}.$$

Step 3 (limit). For every τ, $w_T(\tau) \to 1$ as $T \to \infty$, and $|w_T R_{XX} e^{-j\omega\tau}| \le |R_{XX}(\tau)|$, which is integrable. The dominated convergence theorem gives the stated limit.

Step 4 (properties). R_{XX} is even, since $R_{XX}(-\tau) = E[X(t-\tau)X(t)] = R_{XX}(\tau)$ by stationarity. So $S_{XX}(\omega) = \int R_{XX}(\tau)\cos\omega\tau\, d\tau$ is real and even. It is non-negative as a limit of the non-negative numbers $E|X_T|^2/(2T)$. Also $|S_{XX}| \le \|R_{XX}\|_1$, and continuity in ω follows by dominated convergence again.

Step 5 (inversion). If $S_{XX} \in L^1$, the Fourier inversion theorem for L^1 functions with L^1 transforms (Ch. 16) gives $R_{XX}(\tau) = \frac{1}{2\pi}\int S_{XX} e^{j\omega\tau} d\omega$ for almost every τ. Both sides are continuous in τ (the right by dominated convergence), so equality holds for every τ. At $\tau = 0$ we get $E[X^2] = R_{XX}(0)$. ■



Step 2 in pictures for $R_{XX}(\tau) = e^{-|\tau|}$ (dashed): a record of length $2T$ sees the autocorrelation through the triangular window w_T (dotted), giving the products shown for $T = 1$ (red) and $T = 5$ (blue). As $T \to \infty$ the window flattens to 1. At $\omega = 0.5$, $E|X_T|^2/2T = 1.082,\ 1.504,\ 1.590$ for $T = 1, 5, 50$, against $S_{XX}(0.5) = 2/(1+0.25) = 1.6$.

Random signals through LTI systems

Let X satisfy the hypotheses of the Wiener–Khinchin theorem. Let h be real with $h \in L^1(\mathbb{R})$, i.e. the system is BIBO stable (Ch. 14), and let $H(\omega) = \int h(t)e^{-j\omega t}\, dt$. Define $Y(t) = \int_{-\infty}^{\infty} h(\alpha)X(t-\alpha)\, d\alpha$. Then:

1. Y is a well-defined WSS process with $E[Y] = \mu_X H(0)$ and

$$R_{YY}(\tau) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} h(\alpha)h(\beta)\, R_{XX}(\tau - \alpha + \beta)\, d\alpha\, d\beta,$$

a continuous function in L^1 with $\|R_{YY}\|_1 \le \|h\|_1^2\|R_{XX}\|_1$; 2. $S_{YY}(\omega) = |H(\omega)|^2 S_{XX}(\omega)$ for every ω; 3. if $S_{XX} \in L^1$, then $E[Y^2] = \frac{1}{2\pi}\int_{-\infty}^{\infty} |H(\omega)|^2 S_{XX}(\omega)\, d\omega$.

Proof

Part 1. By Cauchy–Schwarz, $E|X(t)| \le \sqrt{R_{XX}(0)}$, so $E\int|h(\alpha)||X(t-\alpha)|\, d\alpha \le \|h\|_1\sqrt{R_{XX}(0)} < \infty$. Hence the integral defining $Y(t)$ converges absolutely with probability one. By Fubini, $E[Y(t)] = \int h(\alpha)\mu_X\, d\alpha = \mu_X H(0)$. For the correlation, the integrand of

$$E[Y(t+\tau)Y(t)] = E\Big[\iint h(\alpha)h(\beta)X(t+\tau-\alpha)X(t-\beta)\, d\alpha\, d\beta\Big]$$

is dominated in expectation by $|h(\alpha)h(\beta)|R_{XX}(0)$, whose double integral is $\|h\|_1^2 R_{XX}(0) < \infty$. So Fubini again moves E inside, and $E[X(t+\tau-\alpha)X(t-\beta)] = R_{XX}(\tau-\alpha+\beta)$. The result depends on τ only, and $E[Y^2] = R_{YY}(0) \le \|h\|_1^2 R_{XX}(0) < \infty$, so Y is WSS. Integrating $|R_{YY}|$ over τ and using Tonelli's theorem with the substitution $u = \tau - \alpha + \beta$ gives $\|R_{YY}\|_1 \le \|h\|_1^2\|R_{XX}\|_1$. Continuity of R_{YY} follows from continuity of R_{XX}, the bound $|R_{XX}| \le R_{XX}(0)$, and dominated convergence.

Part 2. By part 1, Y satisfies the hypotheses of the Wiener–Khinchin theorem, so $S_{YY}(\omega) = \int R_{YY}(\tau)e^{-j\omega\tau}\, d\tau$. Insert the double integral. The triple integral converges absolutely (the bound in part 1), so Fubini permits any order of integration. Integrate over τ first with $u = \tau - \alpha + \beta$, noting $e^{-j\omega\tau} = e^{-j\omega u}e^{-j\omega\alpha}e^{j\omega\beta}$:

$$S_{YY}(\omega) = \int h(\alpha)e^{-j\omega\alpha}d\alpha \cdot \int h(\beta)e^{j\omega\beta}d\beta \cdot \int R_{XX}(u)e^{-j\omega u}du = H(\omega)\,\overline{H(\omega)}\,S_{XX}(\omega).$$

The middle factor is $\overline{H(\omega)}$ because h is real. Hence $S_{YY} = |H|^2 S_{XX}$.

Part 3. $|H(\omega)| \le \|h\|_1$, so $S_{YY} \le \|h\|_1^2 S_{XX} \in L^1$. The inversion part of the Wiener–Khinchin theorem at $\tau = 0$ applies to Y. ■

Remarks

- *Why $R_{XX} \in L^1$.* The hypothesis excludes components with a discrete spectrum. For the random-phase sinusoid $X(t) = A\cos(\omega_0 t + \Theta)$, $\Theta \sim \mathcal{U}(0, 2\pi)$, we have $R_{XX} = \frac{A^2}{2}\cos\omega_0\tau \notin L^1$. Then $E|X_T(\omega_0)|^2/(2T)$ grows like T, and the PSD exists only as the generalised function $\frac{\pi A^2}{2}[\delta(\omega-\omega_0) + \delta(\omega+\omega_0)]$. A non-zero mean behaves the same way, contributing $2\pi\mu_X^2\delta(\omega)$. The chapter's table of transform pairs (Section 21.4) uses this distributional extension. The chapter's sufficient condition $\int|\tau R_{XX}| < \infty$ (together with $R_{XX} \in L^1$) gives the stronger conclusion that the convergence is $O(1/T)$, because $\big|E|X_T|^2/(2T) - S_{XX}\big| \le \frac{1}{2T}\int|\tau R_{XX}|\, d\tau + \int_{|\tau|>2T}|R_{XX}|\, d\tau$.

- *Non-negativity for free.* Step 4 shows that the Fourier transform of a valid autocorrelation is non-negative. This is the easy half of Bochner's theorem. For example, a rectangular "autocorrelation" $\mathbf{1}_{\{|\tau|\le 1\}}$ has transform $2\sin\omega/\omega$, which takes negative values, so no WSS process has it.
- *White noise.* $R = \frac{N_0}{2}\delta$ is not a function, so white noise lies outside the theorem. Its role is as a limit: band-limited white noise satisfies the hypotheses, and letting the bandwidth grow in part 2 gives $S_{YY} = \frac{N_0}{2}|H|^2$ for every stable H with $\int |H|^2 < \infty$. For the RC low-pass $H = 1/(1 + j\omega RC)$, part 3 gives $E[Y^2] = \frac{1}{2\pi}\int \frac{N_0/2}{1+\omega^2R^2C^2}\,d\omega = \frac{N_0}{4RC}$, confirmed by numerical quadrature. This matches the chapter's worked example.
- *Where used.* Section 21.4 (PSD, LTI systems with random inputs), noise-equivalent bandwidth and coloured noise in Section 21.6, and the matched filter in Section 21.8.

E.10 Quadratic convergence of Newton–Raphson

Newton's iteration for $f(x) = 0$ is $x_{k+1} = x_k - f(x_k)/f'(x_k)$ (Section 22.4). Write $e_k = x_k - x^*$ for the error.

Statement

Local quadratic convergence of Newton's method

Let $f(x^*) = 0$, and let f be twice continuously differentiable on $I = [x^* - r,\ x^* + r]$ with $f' \neq 0$ on I (a **simple root**; if only $f'(x^*) \neq 0$ is known, shrink r using continuity of f'). Put

$$m = \min_I |f'| > 0, \qquad M = \max_I |f''|, \qquad K = \frac{M}{2m}, \qquad \delta = \min\left(r,\ \frac{1}{2K}\right) \quad (\delta = r \text{ if } M = 0).$$

If $|x_0 - x^*| \le \delta$, then all iterates are defined and lie in I, and

1. $|e_{k+1}| \le K\,e_k^2$ and $|e_{k+1}| \le \tfrac{1}{2}|e_k|$ for every k, so $x_k \to x^*$;
2. $K|e_k| \le \big(K|e_0|\big)^{2^k} \le 2^{-2^k}$: the number of correct digits roughly doubles at every step;
3. if $e_k \neq 0$ for all k, then $\lim\limits_{k\to\infty} \dfrac{e_{k+1}}{e_k^2} = \dfrac{f''(x^*)}{2f'(x^*)}$.

Proof

Step 1 (one-step error identity). Suppose $x_k \in I$. Then $f'(x_k) \neq 0$, so x_{k+1} is defined. Taylor's theorem with Lagrange remainder (Ch. 4), expanding about x_k and evaluating at x^*, gives

$$0 = f(x^*) = f(x_k) + f'(x_k)(x^* - x_k) + \tfrac{1}{2}f''(\xi_k)(x^* - x_k)^2$$

for some ξ_k between x_k and x^*, hence $\xi_k \in I$. Divide by $f'(x_k)$ and use $f(x_k)/f'(x_k) = x_k - x_{k+1}$:

$$0 = (x_k - x_{k+1}) + (x^* - x_k) + \frac{f''(\xi_k)}{2f'(x_k)}e_k^2 \quad\Longrightarrow\quad e_{k+1} = \frac{f''(\xi_k)}{2f'(x_k)}\,e_k^2.$$

So $|e_{k+1}| \le K e_k^2$.

Step 2 (induction: the iterates stay in I). Suppose $|e_k| \le \delta$, which holds for $k = 0$. Then $x_k \in I$, and by Step 1, $|e_{k+1}| \le (K|e_k|)|e_k| \le (K\delta)|e_k| \le \tfrac{1}{2}|e_k| \le \delta$. By induction every iterate satisfies $|e_k| \le \delta$, and $|e_k| \le 2^{-k}|e_0| \to 0$. This proves part 1. (If some $e_k = 0$, all later errors are 0, since $f(x^*) = 0$.)

Step 3 (doubling of digits). Let $q_k = K|e_k|$. Step 1 gives $q_{k+1} \le q_k^2$, so by induction $q_k \le q_0^{2^k}$, and $q_0 \le K\delta \le \tfrac{1}{2}$.

Step 4 (the limit). As $k \to \infty$, $x_k \to x^*$ and $\xi_k \to x^*$ (it lies between them). By continuity of f'' and f', the factor $f''(\xi_k)/(2f'(x_k))$ tends to $f''(x^*)/(2f'(x^*))$. ∎

Remarks

- *Numerical check.* For $f(x) = x^3 - x - 2$ (Worked Example 22.14), the root is $x^* = 1.5213797$. The predicted constant is $f''(x^*)/(2f'(x^*)) = 6x^*/(2(3x^{*2} - 1)) = 0.768$, matching the observed ratios $0.79,\ 0.77,\ 0.77$ in the chapter's table.
- *Counterexample 1: multiple roots.* For $f(x) = x^2$ (double root, $f'(0) = 0$) Newton gives $x_{k+1} = x_k/2$. The convergence is only linear with rate $\tfrac{1}{2}$. In general, a root of multiplicity p gives the linear rate $1 - 1/p$ (Section 22.4, "Multiple roots"). The hypothesis $f'(x^*) \neq 0$ is what makes $m > 0$.
- *Counterexample 2: the result is local.* For $f(x) = \arctan x$, the iteration is $x_{k+1} = x_k - (1 + x_k^2)\arctan x_k$. It converges for $|x_0| < x_c$ and diverges with growing oscillation for $|x_0| > x_c$. Here $x_c = 1.39175$ solves $2x = (1 + x^2)\arctan x$. From $x_0 = 1.5$ the iterates are $-1.694,\ 2.321,\ -5.114,\ 32.3, \ldots$. From $x_0 = 1.3$ they are $-1.162,\ 0.859,\ -0.374,\ 0.034,\ -2.6 \times 10^{-5},\ 1.2 \times 10^{-14}$. The theorem's δ is a safe radius, not the exact basin.
- *Smoothness.* Suppose that instead of $f \in C^2$ we only know that f' is Hölder continuous with exponent $\alpha \in (0, 1]$. Writing the Taylor remainder as $\int_{x_k}^{x^*}(f'(s) - f'(x_k))\,ds = O(|e_k|^{1+\alpha})$, Step 1 gives order $1 + \alpha$ instead of 2. At the other extreme, if $f''(x^*) = 0$ and $f \in C^3$, the same expansion carried one order further gives cubic convergence.
- *Where used.* Section 22.4 (including diode circuit analysis), Gauss–Newton in Section 24.7, and Newton steps inside implicit ODE solvers (Section 22.9).

E.11 The bisection error bound

Bisection starts from $[a_0, b_0]$ with $f(a_0)f(b_0) < 0$. At step k it sets $m_k = \tfrac{1}{2}(a_k + b_k)$. It stops if $f(m_k) = 0$. Otherwise it keeps the half on which f changes sign (Section 22.4).

Statement

Convergence of bisection

Let f be continuous on $[a_0, b_0]$ with $f(a_0)f(b_0) < 0$. If the algorithm never stops at an exact zero, then:

1. $a_0 \le a_1 \le \cdots \le a_k < b_k \le \cdots \le b_1 \le b_0$, $f(a_k)f(b_k) < 0$, and $b_k - a_k = (b_0 - a_0)/2^k$;
2. a_k and b_k converge to a common limit x^* with $f(x^*) = 0$, and $a_k \le x^* \le b_k$ for every k;
3. $|m_k - x^*| \le \dfrac{b_0 - a_0}{2^{k+1}}$ for every k. Hence $|m_k - x^*| \le \tau$ is guaranteed as soon as $k \ge \log_2 \dfrac{b_0 - a_0}{\tau} - 1$.

Proof

Part 1 (induction). The case $k = 0$ is the hypothesis. Suppose part 1 holds at step k. Since $f(m_k) \ne 0$ and $f(a_k)f(b_k) < 0$, exactly one of $f(a_k)f(m_k) < 0$ and $f(m_k)f(b_k) < 0$ holds: $f(m_k)$ has the sign of one endpoint value and the opposite sign of the other. The algorithm keeps that half. So $[a_{k+1}, b_{k+1}]$ is either $[a_k, m_k]$ or $[m_k, b_k]$. In both cases the sign condition holds, the new interval is nested in the old one, and its width is $\frac{1}{2}(b_k - a_k) = (b_0 - a_0)/2^{k+1}$.

Part 2. (a_k) is non-decreasing and bounded above by b_0. (b_k) is non-increasing and bounded below by a_0. Monotone bounded sequences converge (completeness of $\mathbb{R}$, Ch. 4). Since $b_k - a_k \to 0$, the limits coincide; call the common limit x^*. Monotonicity gives $a_k \le x^* \le b_k$. By continuity, $f(a_k)f(b_k) \to f(x^*)^2$. Each term is negative, so the limit satisfies $f(x^*)^2 \le 0$, i.e. $f(x^*) = 0$.

Part 3. x^* and m_k both lie in $[a_k, b_k]$, and m_k is its centre. So $|m_k - x^*| \le \frac{1}{2}(b_k - a_k) = (b_0 - a_0)/2^{k+1}$. The iteration count follows by solving $(b_0 - a_0)/2^{k+1} \le \tau$ for k. ∎

Remarks

- *Stronger than "some root".* Part 2 identifies one root x^* that is the limit and that satisfies the bound at *every* step. The intermediate value theorem by itself would only give a root in each bracket, possibly a different one each time.
- *Counterexample: continuity is essential.* $f(x) = 1/x$ on $[-1, 2]$ has $f(-1)f(2) < 0$ but no root. Bisection keeps the sign change and converges to the pole $x = 0$ (brackets $[-1, 0.5]$, $[-0.25, 0.5]$, $[-0.25, 0.125]$, ...). Part 2 fails exactly at the step "by continuity".
- *Sharpness.* The bound cannot be improved in general. If the root sits arbitrarily close to an endpoint of the current bracket, the midpoint is almost half a width away.
- *Floating point.* In finite precision the bracket stops shrinking at about one unit in the last place of x^*. Correct sign evaluation is all that is needed for the bound to hold until then (Section 22.1).
- *Where used.* Section 22.4 (Example 22.13). It is also the safeguard inside hybrid root finders that fall back to bisection when Newton or secant steps leave the bracket.

E.12 Error terms of the trapezoidal and Simpson rules

Throughout, the error of a rule Q for $I = \int_a^b f\,dx$ is defined as $E = I - Q$.

Statement

Trapezoidal and Simpson error terms

1. **Trapezoidal rule.** If $f \in C^2[a, b]$, then for some $\xi \in (a, b)$
$$\int_a^b f\,dx - \frac{b-a}{2}\big(f(a) + f(b)\big) = -\frac{(b-a)^3}{12} f''(\xi).$$
2. **Simpson's rule.** If $f \in C^4[a, b]$, $h = \frac{b-a}{2}$ and $m = \frac{a+b}{2}$, then for some $\xi \in (a, b)$
$$\int_a^b f\,dx - \frac{h}{3}\big(f(a) + 4f(m) + f(b)\big) = -\frac{h^5}{90} f^{(4)}(\xi).$$
3. **Composite rules.** With $h = (b-a)/n$ and nodes $x_i = a + ih$, for some $\xi, \xi' \in (a, b)$
$$I - T_n = -\frac{(b-a)h^2}{12} f''(\xi) \quad (f \in C^2), \qquad I - S_n = -\frac{(b-a)h^4}{180} f^{(4)}(\xi') \quad (f \in C^4,\ n \text{ even}).$$

Two lemmas carry the proof.

Lemma E.12a (weighted mean value theorem)

Let ϕ be continuous on $[a, b]$, and let $w \le 0$ be integrable on $[a, b]$. Let q be a function with $q\,w$ integrable and $\min \phi \le q(x) \le \max \phi$ wherever $w(x) \ne 0$. Then $\int_a^b q\,w\,dx = \phi(\xi) \int_a^b w\,dx$ for some $\xi \in [a, b]$.

Proof. Let $\underline{\phi} = \min \phi$ and $\overline{\phi} = \max \phi$. Since $w \le 0$, we have $\overline{\phi}\,w \le q\,w \le \underline{\phi}\,w$ pointwise. Integrating gives $\overline{\phi} W \le \int qw \le \underline{\phi} W$ with $W = \int w \le 0$. If $W = 0$, then $w = 0$ almost everywhere, both sides vanish, and any ξ works. Otherwise divide by $W < 0$, which reverses the inequalities: $\underline{\phi} \le \int qw / W \le \overline{\phi}$. The intermediate value theorem then gives ξ with $\phi(\xi) = \int qw / W$. ∎

Lemma E.12b (interpolation errors)

(i) If $f \in C^2[a,b]$ and p_1 is the linear interpolant at a, b, then for each $x \in (a,b)$ there is $\eta_x \in (a,b)$ with $f(x) - p_1(x) = \tfrac{1}{2} f''(\eta_x)(x-a)(x-b)$.

(ii) If $f \in C^4[a,b]$, there is a unique cubic p_3 with $p_3 = f$ at a, m, b and $p_3'(m) = f'(m)$. For each $x \in (a,b) \setminus \{m\}$ there is $\eta_x \in (a,b)$ with $f(x) - p_3(x) = \tfrac{1}{24} f^{(4)}(\eta_x)(x-a)(x-m)^2(x-b)$.

Proof. (ii) *Existence.* Let p_2 interpolate f at a, m, b, and put $p_3 = p_2 + c\,(x-a)(x-m)(x-b)$. The cubic term vanishes at the three nodes, and its derivative at m is $(m-a)(m-b) = -h^2 \neq 0$. So exactly one c makes $p_3'(m) = f'(m)$. Uniqueness follows because the difference of two such cubics would have a double zero at m and simple zeros at a, b, which is four zeros counted with multiplicity, so it is identically zero.

Error. Fix $x \notin \{a, m, b\}$ and let $\omega(t) = (t-a)(t-m)^2(t-b)$, so $\omega(x) \neq 0$. Choose κ with $F(t) = f(t) - p_3(t) - \kappa\,\omega(t)$ vanishing at $t = x$. Then F vanishes at the four distinct points a, m, b, x, and also $F'(m) = 0$. Rolle's theorem on the three gaps between consecutive points gives three zeros of F', each strictly inside a gap and therefore different from m. Together with m itself, F' has at least four distinct zeros. Repeating Rolle, F'' has three, F''' two, and $F^{(4)}$ at least one, $\eta_x \in (a,b)$. Since $p_3^{(4)} = 0$ and $\omega^{(4)} = 24$, we get $0 = f^{(4)}(\eta_x) - 24\kappa$. (i) is the same argument with $\omega(t) = (t-a)(t-b)$, three zeros a, x, b, and two differentiations. ∎

Proof of the theorem

Part 1. By Lemma E.12b(i), $I - Q = \int_a^b (f - p_1)\,dx$, because the trapezoidal rule is the exact integral of p_1. Apply Lemma E.12a with $\phi = \tfrac{1}{2} f''$, $w = (x-a)(x-b) \leq 0$ on $[a,b]$, and $q = (f - p_1)/w$ on (a,b). This q is continuous on (a,b), takes values $\tfrac{1}{2} f''(\eta_x)$ between $\min \phi$ and $\max \phi$, and $qw = f - p_1$ is integrable. Since $\int_a^b (x-a)(x-b)\,dx = -\tfrac{(b-a)^3}{6}$,

$$I - Q = \tfrac{1}{2} f''(\xi) \cdot \Big(-\frac{(b-a)^3}{6} \Big) = -\frac{(b-a)^3}{12} f''(\xi).$$

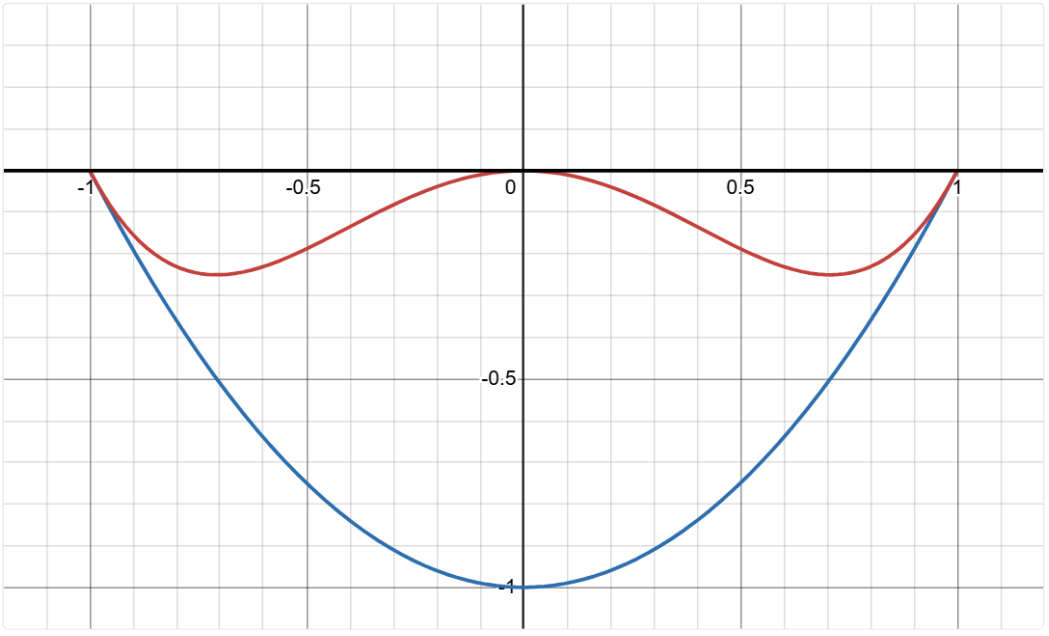
If ξ lands on an endpoint, the intermediate value theorem applied on (a,b) to the continuous f'' moves it inside.

Part 2. **Simpson's rule is exact for cubics.** By linearity it suffices to check the basis $1,\ (x-m),\ (x-m)^2,\ (x-m)^3$. The exact integrals over $[m-h, m+h]$ are $2h,\ 0,\ \tfrac{2}{3}h^3,\ 0$. The rule gives $\tfrac{h}{3}(1+4+1) = 2h$, $\tfrac{h}{3}(-h+0+h) = 0$, $\tfrac{h}{3}(h^2+0+h^2) = \tfrac{2}{3}h^3$ and $\tfrac{h}{3}(-h^3+0+h^3) = 0$. **Reduction.** The rule uses only $f(a), f(m), f(b)$, where $f = p_3$. So $Q(f) = Q(p_3) = \int_a^b p_3\,dx$, and $I - Q = \int_a^b (f - p_3)\,dx$. **Weighted mean value.** Lemma E.12a applies with $\phi = f^{(4)}/24$ and $w = (x-a)(x-m)^2(x-b) \leq 0$ on $[a,b]$. With $x = m + t$,

$$\int_a^b w\,dx = \int_{-h}^{h} (t+h)\,t^2\,(t-h)\,dt = \int_{-h}^{h} \big(t^4 - h^2 t^2\big)\,dt = \frac{2h^5}{5} - \frac{2h^5}{3} = -\frac{4h^5}{15},$$

so $I - Q = \tfrac{f^{(4)}(\xi)}{24} \cdot \big(-\tfrac{4h^5}{15} \big) = -\tfrac{h^5}{90} f^{(4)}(\xi)$.

Part 3. The composite trapezoidal error is the sum of the n panel errors, each $-\tfrac{h^3}{12} f''(\xi_i)$ by part 1. The average $\tfrac{1}{n}\sum_i f''(\xi_i)$ lies between $\min f''$ and $\max f''$, so it equals $f''(\xi)$ for some ξ by the intermediate value theorem. The total is $-\tfrac{nh^3}{12} f''(\xi) = -\tfrac{(b-a)h^2}{12} f''(\xi)$, since $nh = b - a$. For Simpson, the $n/2$ double panels each contribute $-\tfrac{h^5}{90} f^{(4)}(\xi_i)$ by part 2, and the same averaging gives $-\tfrac{n}{2} \cdot \tfrac{h^5}{90} f^{(4)}(\xi') = -\tfrac{(b-a)h^4}{180} f^{(4)}(\xi')$. ∎



The error kernels on $[-1,1]$: $(x-a)(x-b)$ for the trapezoidal rule (blue) and $(x-a)(x-m)^2(x-b)$ for Simpson's rule (red). Neither changes sign, which is exactly what Lemma E.12a needs. Their integrals, $-\tfrac{4}{3}$ and $-\tfrac{4}{15}$, produce the constants $\tfrac{1}{12}$ and $\tfrac{1}{90}$.

Remarks

- *Numerical check.* For $\int_0^1 e^x\,dx = e - 1$, the single trapezoid has error $-0.14086 = -\tfrac{1}{12} e^{\xi}$, giving $\xi = 0.525 \in (0,1)$. Single-panel Simpson has error $-5.79 \times 10^{-4} = -\tfrac{(0.5)^5}{90} e^{\xi}$, giving $\xi = 0.512$. Both are consistent with the theorem.

- *Counterexample: smoothness matters.* For $f(x) = \sqrt{x}$ on $[0, 1]$, f'' is unbounded and part 3 does not apply. The composite trapezoidal errors for $n = 8, 16, 32, 64$ are $8.5 \times 10^{-3},\ 3.1 \times 10^{-3},\ 1.1 \times 10^{-3},\ 4.0 \times 10^{-4}$. They shrink by about $2.8 \approx 2^{1.5}$ per halving, an $O(h^{1.5})$ rate instead of $O(h^2)$. A singularity at an endpoint degrades the order. Subtracting the singularity, changing variables, or Gaussian quadrature restores it (Section 22.8).
- *The bound is not always sharp.* For smooth periodic integrands over a full period, the trapezoidal rule converges far faster than h^2. Every term of the Euler–Maclaurin expansion vanishes (Section 22.8). The theorem gives a guarantee, not the true error in every case.
- *Where used.* Section 22.8 (Newton–Cotes and composite rules, Richardson/Romberg). The same Rolle-counting proof gives the general interpolation error formula of Section 22.5.

E.13 Euler's method: global $O(h)$ convergence and absolute stability

Consider the initial-value problem $y' = f(t, y)$, $y(t_0) = y_0$, on $[t_0, T]$. Euler's method with step $h > 0$ is $y_{n+1} = y_n + h f(t_n, y_n)$ with $t_n = t_0 + nh$ (Section 22.9). The global error is $e_n = y(t_n) - y_n$.

Statement

Convergence of Euler's method

Assume that f is continuous on $[t_0, T] \times \mathbb{R}$ and **Lipschitz** in y, i.e. $|f(t, u) - f(t, v)| \le L|u - v|$ for all t, u, v, with $L > 0$. Assume also that the exact solution y is twice continuously differentiable with $|y''| \le M$ on $[t_0, T]$. If the method is started from y_0 with initial error e_0, then for every n with $t_n \le T$

$$|e_n| \le e^{L(t_n - t_0)}|e_0| + \frac{Mh}{2L}\Big(e^{L(t_n - t_0)} - 1\Big).$$

In particular, with $e_0 = 0$, $\max_{t_n \le T}|e_n| \le C\,h$ with $C = \frac{M}{2L}(e^{L(T - t_0)} - 1)$ independent of h. Euler's method is convergent of order one.

Lemma E.13a (discrete Gronwall inequality)

If $a_n \ge 0$ and $a_{n+1} \le (1 + hL)a_n + B$ for $n \ge 0$, with $h, L > 0$ and $B \ge 0$, then

$$a_n \le (1 + hL)^n a_0 + B\,\frac{(1 + hL)^n - 1}{hL} \le e^{nhL}a_0 + \frac{B}{hL}\big(e^{nhL} - 1\big).$$

Proof. Induction gives $a_n \le (1 + hL)^n a_0 + B\sum_{k=0}^{n-1}(1 + hL)^k$. The geometric sum equals $\big((1 + hL)^n - 1\big)/(hL)$. Finally, $1 + x \le e^x$ for all real x, because e^x is convex and $1 + x$ is its tangent at 0. Hence $(1 + hL)^n \le e^{nhL}$. ∎

Proof of the theorem

Step 1 (local truncation error). By Taylor's theorem with Lagrange remainder and $y'(t_n) = f(t_n, y(t_n))$,

$$y(t_{n+1}) = y(t_n) + h f\big(t_n, y(t_n)\big) + \tau_{n+1}, \qquad \tau_{n+1} = \tfrac{h^2}{2}y''(\xi_n), \qquad |\tau_{n+1}| \le \tfrac{M}{2}h^2.$$

Step 2 (error recursion). Subtract the Euler step $y_{n+1} = y_n + h f(t_n, y_n)$:

$$e_{n+1} = e_n + h\big[f(t_n, y(t_n)) - f(t_n, y_n)\big] + \tau_{n+1}.$$

By the Lipschitz condition, the bracket is at most $L|e_n|$ in modulus. So $|e_{n+1}| \le (1 + hL)|e_n| + \frac{M}{2}h^2$.

Step 3 (accumulate). Apply Lemma E.13a with $a_n = |e_n|$ and $B = \frac{M}{2}h^2$. Then $B/(hL) = Mh/(2L)$, and $nh = t_n - t_0$. ∎

The structure of the proof is the general principle "consistency + stability ⟹ convergence" of Section 22.11. There are $O(1/h)$ local errors, each $O(h^2)$. Each is amplified by at most $e^{L(T - t_0)}$ by the time it reaches T.

Statement (absolute stability)

Absolute stability of Euler's method

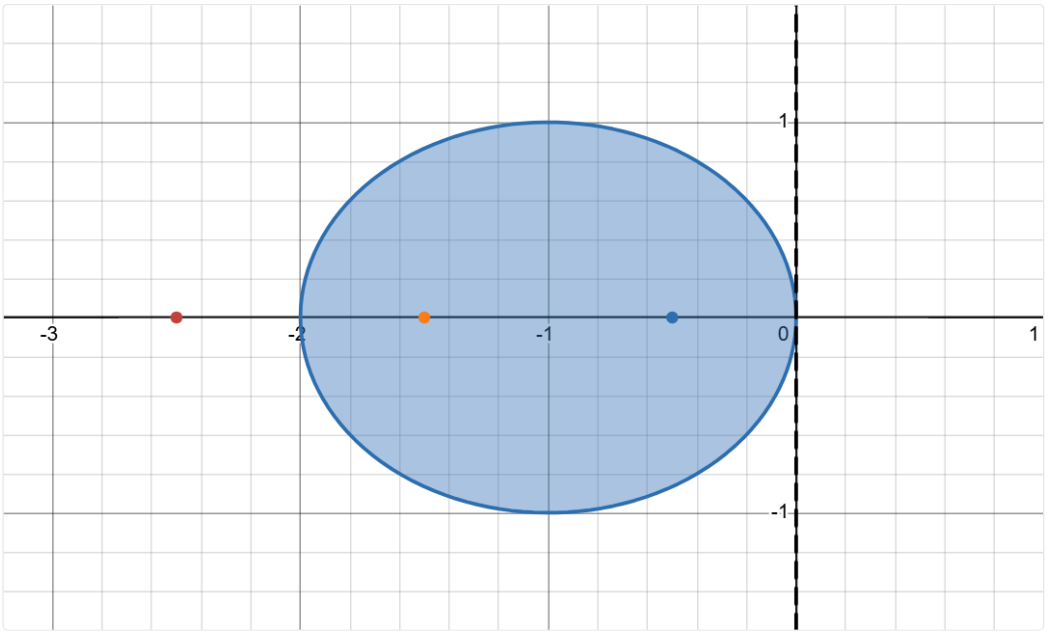
1. For the test equation $y' = \lambda y$, $\lambda \in \mathbb{C}$, Euler's method gives $y_n = (1 + h\lambda)^n y_0$. The numerical solution is bounded for every y_0 iff $|1 + h\lambda| \le 1$, and tends to 0 for every y_0 iff $|1 + h\lambda| < 1$. The region of absolute stability $\{z = h\lambda : |1 + z| \le 1\}$ is the closed disc of radius 1 centred at -1.
2. For real $\lambda < 0$ the condition is $0 < h \le 2/|\lambda|$. The numerical solution decays without sign changes iff $h \le 1/|\lambda|$.
3. For $\mathbf{y}' = \mathbf{A}\mathbf{y}$ with $\mathbf{A}$ diagonalisable, with eigenvalues $\lambda_1, \dots, \lambda_N$, the Euler iterates $\mathbf{y}_n = (\mathbf{I} + h\mathbf{A})^n \mathbf{y}_0$ are bounded for every $\mathbf{y}_0$ iff $|1 + h\lambda_i| \le 1$ for every i.

Proof

Part 1. With $f(t, y) = \lambda y$ the Euler step is $y_{n+1} = (1 + h\lambda)y_n$, so by induction $y_n = (1 + h\lambda)^n y_0$ and $|y_n| = |1 + h\lambda|^n |y_0|$. For $y_0 \ne 0$, a geometric sequence ρ^n with $\rho \ge 0$ is bounded iff $\rho \le 1$ and tends to 0 iff $\rho < 1$. Writing $z = h\lambda = x + jy$, the condition $|1 + z| \le 1$ reads $(x + 1)^2 + y^2 \le 1$.

Part 2. For real $z = h\lambda < 0$, $|1 + z| \le 1$ means $-1 \le 1 + z \le 1$, i.e. $-2 \le z \le 0$, i.e. $h \le 2/|\lambda|$. The factor $1 + z$ is non-negative, so the iterates keep the sign of y_0, iff $z \ge -1$.

Part 3. Write $\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$ with $\mathbf{V}$ invertible. Then $\mathbf{I} + h\mathbf{A} = \mathbf{V}(\mathbf{I} + h\mathbf{\Lambda})\mathbf{V}^{-1}$ and $(\mathbf{I} + h\mathbf{A})^n = \mathbf{V}(\mathbf{I} + h\mathbf{\Lambda})^n\mathbf{V}^{-1}$. In the coordinates $\mathbf{w} = \mathbf{V}^{-1}\mathbf{y}$, the iteration decouples into $w_{i,n} = (1 + h\lambda_i)^n w_{i,0}$, and part 1 applies to each component. Since $\|\mathbf{V}^{-1}\|^{-1}\|\mathbf{y}_n\| \le \|\mathbf{w}_n\| \le \|\mathbf{V}^{-1}\|\|\mathbf{y}_n\|$, the sequence $\mathbf{y}_n$ is bounded iff $\mathbf{w}_n$ is. ∎



Region of absolute stability of explicit Euler in the complex $z = h\lambda$ plane: the disc $|1 + z| \le 1$. The three points are the step sizes of the chapter's figure for $y' = -2y$. $z = -0.5$ gives monotone decay, $z = -1.5$ oscillating decay, and $z = -2.5$ lies outside the disc, so the iterates grow.

Remarks

- *Numerical check of the convergence bound.* For $y' = y$, $y(0) = 1$ on $[0, 1]$, take $L = 1$ and $M = e$. At $t = 1$ the bound is $\frac{eh}{2}(e - 1) = 2.335h$. The actual errors $e - (1 + h)^{1/h}$ are $0.1245,\ 0.0650,\ 0.01347$ for $h = 0.1,\ 0.05,\ 0.01$. Each is below the bound, and the ratio error$/h$ tends to $e/2 = 1.359$, confirming first order.
- *Counterexample: the Lipschitz hypothesis.* $y' = \sqrt{|y|}$, $y(0) = 0$, is not Lipschitz at $y = 0$. It has the solutions $y \equiv 0$ and $y = t^2/4$ (among others). Euler from $y_0 = 0$ produces $y_n \equiv 0$ and converges to one solution. If $y_0 = \varepsilon > 0$, it follows the other branch instead. Without uniqueness, "the error" is not even well defined.
- *Convergence versus stability.* The convergence theorem is about $h \to 0$ at fixed T. Its constant $e^{L(T - t_0)}$ can be astronomically large. The stability theorem is about fixed h and $n \to \infty$. For a stiff system with $\lambda = -1000$, part 2 forces $h \le 2 \times 10^{-3}$ for as long as the integration runs, even after the fast mode has decayed. That is the motivation for implicit methods (Section 22.9, "Stiffness").
- *Counterexample: diagonalisability in part 3.* For $\mathbf{A} = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$ with $|1 + h\lambda| = 1$, $(\mathbf{I} + h\mathbf{A})^n = \begin{pmatrix} r^n & nhr^{n-1} \\ 0 & r^n \end{pmatrix}$ with $r = 1 + h\lambda$. It grows linearly in n although every eigenvalue lies on the boundary of the stability disc. Strict inequality $|1 + h\lambda_i| < 1$ suffices in general.
- *Rounding.* If each step also commits a rounding error of size ϵ, then $B = \frac{M}{2}h^2 + \epsilon$ in Step 3, and the bound becomes $\left(\frac{Mh}{2} + \frac{\epsilon}{h}\right)\frac{e^{L(T - t_0)} - 1}{L}$. It is minimised near $h = \sqrt{2\epsilon/M}$. Making h smaller beyond this point makes the answer worse.
- *Where used.* Euler's method and absolute stability in Section 22.9, zero-stability and the Lax equivalence theorem in Section 22.11, and solver choice in Section 25.9.

E.14 The Monte Carlo $1/\sqrt{N}$ error rate

Statement

Error of the Monte Carlo estimator

Let $\mathbf{X}_1, \dots, \mathbf{X}_N$ be i.i.d. copies of a random vector $\mathbf{X}$, and let g be a function with $\mu = E[g(\mathbf{X})]$ and $\sigma^2 = \operatorname{Var} g(\mathbf{X}) < \infty$. Let

$$\hat{\mu}_N = \frac{1}{N}\sum_{i=1}^{N} g(\mathbf{X}_i), \qquad s_N^2 = \frac{1}{N - 1}\sum_{i=1}^{N} \big(g(\mathbf{X}_i) - \hat{\mu}_N\big)^2.$$

Then:

1. $E[\hat{\mu}_N] = \mu$, and the root-mean-square error is **exactly** $\sqrt{E[(\hat{\mu}_N - \mu)^2]} = \sigma/\sqrt{N}$;
2. for every N and every $k > 0$, $P\big(|\hat{\mu}_N - \mu| \ge k\sigma/\sqrt{N}\big) \le 1/k^2$;
3. if $\sigma > 0$, then for every $z > 0$, $P\big(|\hat{\mu}_N - \mu| \le z\sigma/\sqrt{N}\big) \to 2\Phi(z) - 1$ as $N \to \infty$;
4. $E[s_N^2] = \sigma^2$ and $s_N \to \sigma$ in probability. Consequently, part 3 also holds with σ replaced by s_N, and $\hat{\mu}_N \pm 1.96\, s_N/\sqrt{N}$ is an asymptotic 95% confidence interval.

Proof

Part 1. The $Y_i = g(\mathbf{X}_i)$ are i.i.d. with mean μ and variance σ^2. By E.4 (part 1 of the proof), $E[\hat{\mu}_N] = \mu$ and $\operatorname{Var}\hat{\mu}_N = \sigma^2/N$. Independence implies uncorrelatedness, so the variance identity is exact, not just a bound. Since $E[\hat{\mu}_N] = \mu$, the mean-square error equals the variance.

Part 2. Chebyshev's inequality (E.3) applied to $\hat{\mu}_N$, whose standard deviation is $\sigma/\sqrt{N}$.

Part 3. By the central limit theorem (E.5) applied to the Y_i, $Z_N = \sqrt{N}(\hat{\mu}_N - \mu)/\sigma \to \mathcal{N}(0,1)$ in distribution. Because Φ is continuous, $P(-z \le Z_N \le z) = F_{Z_N}(z) - F_{Z_N}(-z) + P(Z_N = -z)$. The first two terms converge to $\Phi(z) - \Phi(-z)$. For the atom term, $0 \le P(Z_N = -z) \le F_{Z_N}(-z) - F_{Z_N}(-z-\eta) \to \Phi(-z) - \Phi(-z-\eta)$ for every $\eta > 0$, which can be made arbitrarily small. So $P(|Z_N| \le z) \to 2\Phi(z) - 1$.

Part 4. Unbiasedness is E.6. For consistency, expand as in E.6, Step 1:

$$s_N^2 = \frac{N}{N-1}\Big[\frac{1}{N}\sum_{i=1}^{N} Y_i^2 - \hat{\mu}_N^2\Big].$$

By Khinchin's weak law (E.4, part 2), which applies because $E|Y_i^2| = \sigma^2 + \mu^2 < \infty$, we have $\frac{1}{N}\sum Y_i^2 \to \sigma^2 + \mu^2$ and $\hat{\mu}_N \to \mu$ in probability. Continuous functions preserve convergence in probability (the continuous mapping theorem, quoted). So $s_N^2 \to (\sigma^2 + \mu^2) - \mu^2 = \sigma^2$, and $s_N = \sqrt{s_N^2} \to \sigma$. Slutsky's theorem (quoted: if $Z_N \to Z$ in distribution and $C_N \to c \neq 0$ in probability, then $Z_N/C_N \to Z/c$ in distribution) applied with $C_N = s_N/\sigma \to 1$ shows that $\sqrt{N}(\hat{\mu}_N - \mu)/s_N \to \mathcal{N}(0,1)$. The interval statement follows as in part 3. ■

Remarks

- *The rate is exact.* Part 1 is an identity, not a bound. For plain Monte Carlo with i.i.d. samples, the RMS error is $\sigma N^{-1/2}$ for every N and cannot be improved by any choice of N. Halving the error costs four times the samples, and one more decimal digit costs a hundred times. The only levers are σ (variance reduction, which replaces g by an estimator with the same mean and smaller variance) and abandoning independence (quasi-Monte Carlo) (Section 24.6).
- *No dimension in the rate.* Neither the dimension of $\mathbf{X}$ nor the smoothness of g appears in the theorem. Only σ does. By contrast, a product trapezoidal rule in d dimensions has error $O(N^{-2/d})$ (E.12, part 3, in each coordinate). This is why Monte Carlo wins for $d > 4$.
- *Numerical check.* For $\int_0^1 e^x\,dx$ with uniform samples, $\sigma = \sqrt{\frac{1}{2}(e^2-1) - (e-1)^2} = 0.49197$. Over 2000 independent repetitions, the observed RMS errors were $0.0496,\ 0.0152,\ 0.00481$ for $N = 10^2,\ 10^3,\ 10^4$. The predicted values $\sigma/\sqrt{N}$ are $0.0492,\ 0.0156,\ 0.00492$.
- *Counterexample: infinite variance.* For $g(x) = x^{-1/2}$ with $X \sim \mathcal{U}(0,1)$, $\mu = 2$ is finite but $E[g^2] = \int_0^1 x^{-1}dx = \infty$. Khinchin's law still gives $\hat{\mu}_N \to 2$ in probability, but parts 1–4 fail. The error decays more slowly than $N^{-1/2}$ (here like $N^{-1/2}\sqrt{\log N}$), and s_N systematically misreports it. Importance sampling with a density $\propto x^{-1/2}$ removes the singularity and gives zero variance.
- *Chebyshev versus CLT.* Part 2 holds for every N but is conservative. With $k = 4.47$ it guarantees 95% coverage. Part 3 needs only $z = 1.96$, but it is asymptotic, and for skewed integrands with small N the actual coverage can fall short (Berry–Esseen, E.5).
- *Where used.* Section 24.6 (the basic estimator and its error bar), Monte Carlo tolerance analysis and uncertainty propagation (Sections 24.6 and 24.8), and GUM Supplement 1 in Section 20.10.

Further reading

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